

**CHEMISTRY 110 –Midterm Exam****Thursday November 7, 2024, 6:30 – 8:30 pm**

Examiners: Pallavi Sirjoosingh and Ashok Kakkar

First name (please write as legibly as possible within the boxes)

--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--

Last name

--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--

Student ID number

--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--

This booklet has 8 questions on 10 pages (The last page can be used as a scrap page).

The staple must NOT be removed from the question booklet. Your intact question booklet must be returned at the end of the exam.

A separate booklet contains a periodic table, and datasheet.

Write your answers directly under the questions in the space provided. Do not write on the QR codes or near the top edges (as it may be cut off).

Please write **legibly and dark enough** for the scan to be readable. For students who use English as a second language, dictionaries are permitted. Calculators are allowed.

DO NOT ASK QUESTIONS OF THE INVIGILATORS about interpretations of the exam questions. If you have a concern about a question, please summarize it at the top of this page. All concerns will be evaluated after the exam. GOOD LUCK!

**Question 1 8 points** (*Assume all transitions are possible*)

a. 6 points Calculate the **energy (in kJ/mol)** required to ionize (completely remove) the electron from the **$n=3$** level of **He^+ ion** (He: Helium). Show your work including equations. (3 sig. figs.)

b. 2 points

(I) Select the largest (longest) wavelength from the three choices below:

- i. Wavelength of photon to *completely remove* an electron from *1s orbital of Hydrogen atom*
- ii. Wavelength of photon to *completely remove* an electron from *1s orbital of Boron atom (B)*
- iii. Wavelength of photon to *completely remove* an electron from *1s orbital of Boron cation (B^+)*

i or ii or iii

(II) Select the smallest (shortest) wavelength from the three choices below (*for H atom*):

- i. Wavelength of photon *emitted* for an electron transitioning from *2p to 1s orbital*
- ii. Wavelength of photon *absorbed* for an electron transitioning from *2s to 4p orbital*
- iii. Wavelength of photon *absorbed* for an electron transitioning from *1s to 3p orbital*

i or ii or iii



Question 2 8 points



A Chem110 student plots the graph shown on the left, from data obtained for the kinetic energy (eV) of the ejected electron versus frequency (s^{-1}) of the incident light, for an unknown metal. ($1\text{eV} = 1.6022 \times 10^{-19} \text{ J}$)

Based on the plot, answer the questions **a to c** below:

a. 1 point On the plot above, label the **threshold frequency**.

b. 2 points What is the value of the work potential (in J) of the unknown metal?

c. 3 points What is the *longest* wavelength (in nm) required to eject an electron from this metal's surface? Show your work including equation(s) used. (3 sig figs)

d. 2 points In the space provided below, plot the following

(Assume frequency is *above* threshold frequency):

- i. *Kinetic energy* of an ejected electron versus the *intensity* of the incident light (left)
- ii. *Number of electrons* ejected versus the *frequency* of the incident light (right)



**Question 3** **6 points****a. 2 points**

In the ground state of Ge (Germanium; $Z = 32$) atom, **name** the following orbitals:

i. Orbital containing the electron that requires the maximum energy to be removed.

ii. Orbital containing the electron that experiences the minimum effective nuclear charge.

b. 2 points

An unknown anion (A^-) has the following *excited* electronic configuration: $[Ar] 4s^2 3d^3 4p^1$

(Ar: Argon; $Z = 18$)

Give the **condensed ground state electronic configuration** of the atom, **A** and the cation, **A^{2+}** .
Show your work. (Assume all electronic transitions are possible)

Electronic Configuration of A:

Electronic Configuration of A^{2+} :

c. 2 points

In a single atom, determine how many electrons can have the following set of quantum numbers?

(i) $n = 5; l = 3$

In a single atom, determine how many electrons can have the following set of quantum numbers?

(i) $n = 3; l = 2; m_s = +1/2$

**Question 4** **8 points**

Rank the following from 1 to 4 in **increasing** order. Provide the ranking in the boxes provided.

Note: Atomic number (Z) are provided for relevant atoms

a) Increasing (lowest to highest) order of *ionization energy*: (1= lowest; 4 = highest)

Mg

Mg: Z = 12

Rb

Rb: Z = 37

Cs

Cs: Z = 55

K

K : Z = 19

b) Increasing (lowest to highest) order of *bond angle*: (1= lowest; 4 = highest)

H₂OCCl₄XeF₄

HCN

c) Increasing (smallest to largest) order of *ionic size* (1= smallest; 4 = largest)

S²⁻

S: Z = 16

O²⁻

O: Z = 8

Be²⁺

Be: Z = 4

Na⁺

Na: Z = 11

d) Increasing (lowest to highest) order of *bond energy* (1= lowest; 4 = highest)

P-F

P-Cl

P-I

P-Br

**Question 5** **12 points**

a. 9 points For the molecules HClO_4 and TeF_6 , draw the most stable *Lewis structure* showing appropriate shape and *dashed/wedged bonds*, based on VSEPR. Determine the **molecular geometry** around the central atom. For HClO_4 , label the *most polar bond*.



b. 3 points (Note: Lone pairs are not shown in structures below)

(i) For the three *resonance* structures (1, 2 and 3) shown below, answer the following question:



(I) Circle the *most contributing* resonance structure.

Structure 1

Structure 2

Structure 3

(ii) For the three structures given (4, 5, and 6) below, answer the following questions:



(I) Circle all the structures that represent *resonance structures* of each other:

Structure 4

Structure 5

Structure 6

(II) Among the resonance structures, circle the *most contributing* resonance structure.

Structure 4

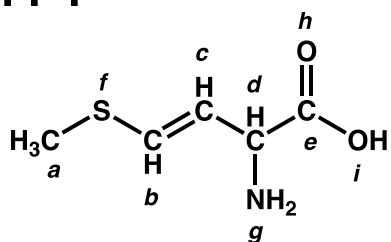
Structure 5

Structure 6

**Question 6** **6 points**

Using the data given below, calculate the **lattice energy** for SrO(s).

	ΔH° (kJ/mol)
$\text{Sr}(s) \longrightarrow \text{Sr}(g)$	+180
$\text{Sr}(g) \longrightarrow \text{Sr}^{2+}(g) + 2e^-$	+1468
$\text{O}_2(g) \longrightarrow 2\text{O}(g)$	+500
$\text{O}(g) + 2e^- \longrightarrow \text{O}^{2-}(g)$	+600
$\text{Sr}(s) + \frac{1}{2}\text{O}_2(g) \longrightarrow \text{SrO}(s)$	-550

**Question 7 10 points**

Consider the molecule given on the left, and answer questions (i) to (vi) using chemical bonding theories that we have learnt in class (valence bond theory, hybridization).

Note that the lone pairs are not shown in the structure:

i. Consider the carbon atoms marked in as ***a***, ***b***, ***c***, ***d***, and ***e*** in the molecule, and determine **hybridization** at each of the carbon centers. (2.5 points)

Atom	Hybridization
Carbon (<i>a</i>)	
Carbon (<i>b</i>)	
Carbon (<i>c</i>)	
Carbon (<i>d</i>)	
Carbon (<i>e</i>)	

ii. Now, consider the sulfur atom (S) marked as ***f***, nitrogen (N) atom (***g***), and oxygen(O) atoms (***h*** and ***i***), and determine their **hybridization**. (2 points)

Atom	Hybridization
Sulfur (<i>f</i>)	
Nitrogen (<i>g</i>)	
Oxygen (<i>h</i>)	
Oxygen (<i>i</i>)	

iii. **Describe** all the bonds formed by the carbon atoms ***a*** and ***b***, using sigma and pi designation as appropriate. (2 points)

Atom	Bonds formed (use sigma and pi as appropriate)
Carbon (<i>a</i>)	
Carbon (<i>b</i>)	

iv. How many total sp^3 -s sigma bonds are in the molecule? (1 point)

v. Determine **electronic** and **molecular geometry** at carbon (C) labeled ***e*** and sulfur (S) labeled ***f*** centres. (2 points)

Atom	Electronic Geometry	Molecular Geometry
Carbon (<i>e</i>)		
Sulfur (<i>f</i>)		

vi. Circle the value from the choices given below, that describes the **C-S-C bond angle at sulfur** (S) marked ***f***: (0.5 points)

>90° to <109.5°

109.5°

>109.5 to <120°

120°

>120° to <180°


Question 8 12 points

Using the **valence shell orbitals only** draw a **complete** molecular orbital (MO) diagram (with no s-p mixing) for the molecule **PS**, in the space provided below. **(6 points)**

Assume that 3s and 3p atomic orbitals will form molecular orbitals like those formed by the combination of 2s and 2p atomic orbitals that we learnt in class, and there is no s-p mixing.

Show **energy arrow**, **atomic orbitals** involved, **atom labels**, the **molecular orbitals** that are formed, **filled electrons** in the MO diagram etc. Make sure to label all the atomic and molecular orbitals.

Using the MO diagram above, answer the following questions:

- (i) Which one of the following will have the highest bond order? Circle the correct choice
 A) PS^{1-} B) PS C) PS^{1+}
- (ii) Which one of the following will have the longest bond length? Circle the correct choice
 A) PS^{1-} B) PS C) PS^{1+}
- (iii) Which of the following will be diamagnetic? Circle the correct choice.
 A) PS^{1-} B) PS C) PS^{1+}
- (iv) Draw Lewis dot structure of the molecule with the shortest bond length (based on the MO diagram) from among the three (PS^{1-} , PS, PS^{1+}), and indicate the bond order calculated from the Lewis dot structure. **(3 points)**

Lewis Dot Structure

Bond Order



594D8334-FA63-4C7F-9451-9605F1F2A5C1

midterm-exam-93388

#1 Page 10 of 10

SCRAP PAGE

**CHEMISTRY 110 –Midterm Exam****Thursday November 7, 2024, 6:30 – 8:30 pm****Examiners: Pallavi Sirjoosingh and Ashok Kakkar**

First name (please write as legibly as possible within the boxes)

--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--

Last name

--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--

Student ID number

--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--

This booklet has 8 questions on 10 pages (The last page can be used as a scrap page).

The staple must NOT be removed from the question booklet. Your intact question booklet must be returned at the end of the exam.

A separate booklet contains a periodic table, and datasheet.

Write your answers directly under the questions in the space provided. Do not write on the QR codes or near the top edges (as it may be cut off).

Please write **legibly and dark enough** for the scan to be readable. For students who use English as a second language, dictionaries are permitted. Calculators are allowed.

DO NOT ASK QUESTIONS OF THE INVIGILATORS about interpretations of the exam questions. If you have a concern about a question, please summarize it at the top of this page. All concerns will be evaluated after the exam. GOOD LUCK!

**Question 1 8 points** (*Assume all transitions are possible*)

a. 6 points Calculate the **energy (in kJ/mol)** required to ionize (completely remove) the electron from the **n=3** level of **He⁺ ion** (He: Helium). Show your work including equations. (3 sig. figs.)

b. 2 points

(I) Select the largest (longest) wavelength from the three choices below:

- i. Wavelength of photon to *completely remove* an electron from *1s orbital of Hydrogen atom*
- ii. Wavelength of photon to *completely remove* an electron from *1s orbital of Boron atom (B)*
- iii. Wavelength of photon to *completely remove* an electron from *1s orbital of Boron cation (B⁺)*

i or ii or iii

(II) Select the smallest (shortest) wavelength from the three choices below (*for H atom*):

- i. Wavelength of photon *emitted* for an electron transitioning from *2p to 1s orbital*
- ii. Wavelength of photon *absorbed* for an electron transitioning from *2s to 4p orbital*
- iii. Wavelength of photon *absorbed* for an electron transitioning from *1s to 3p orbital*

i or ii or iii



Question 2 8 points



A Chem110 student plots the graph shown on the left, from data obtained for the kinetic energy (eV) of the ejected electron versus frequency (s^{-1}) of the incident light, for an unknown metal. ($1\text{eV} = 1.6022 \times 10^{-19} \text{ J}$)

Based on the plot, answer the questions **a to c** below:

a. 1 point On the plot above, label the **threshold frequency**.

b. 2 points What is the value of the work potential (in J) of the unknown metal?

c. 3 points What is the *longest* wavelength (in nm) required to eject an electron from this metal's surface? Show your work including equation(s) used. (3 sig figs)

d. 2 points In the space provided below, plot the following

(Assume frequency is *above* threshold frequency):

- Kinetic energy* of an ejected electron versus the *intensity* of the incident light (left)
- Number of electrons* ejected versus the *frequency* of the incident light (right)



**Question 3** **6 points****a. 2 points**

In the ground state of Ge (Germanium; $Z = 32$) atom, **name** the following orbitals:

i. Orbital containing the electron that requires the maximum energy to be removed.

ii. Orbital containing the electron that experiences the minimum effective nuclear charge.

b. 2 points

An unknown anion (A^-) has the following *excited* electronic configuration: $[Ar] 4s^2 3d^3 4p^1$

(Ar: Argon; $Z = 18$)

Give the **condensed ground state electronic configuration** of the atom, **A** and the cation, A^{2+} .
Show your work. (Assume all electronic transitions are possible)

Electronic Configuration of A:

Electronic Configuration of A^{2+} :

c. 2 points

In a single atom, determine how many electrons can have the following set of quantum numbers?

(i) $n = 5; l = 3$

In a single atom, determine how many electrons can have the following set of quantum numbers?

(i) $n = 3; l = 2; m_s = +1/2$

**Question 4** **8 points**

Rank the following from 1 to 4 in **increasing** order. Provide the ranking in the boxes provided.

Note: Atomic number (Z) are provided for relevant atoms

a) Increasing (lowest to highest) order of *ionization energy*: (1= lowest; 4 = highest)

Mg

Mg: Z = 12

Rb

Rb: Z = 37

Cs

Cs: Z = 55

K

K : Z = 19

b) Increasing (lowest to highest) order of *bond angle*: (1= lowest; 4 = highest)

H₂OCCl₄XeF₄

HCN

c) Increasing (smallest to largest) order of *ionic size* (1= smallest; 4 = largest)

S²⁻

S: Z = 16

O²⁻

O: Z = 8

Be²⁺

Be: Z = 4

Na⁺

Na: Z = 11

d) Increasing (lowest to highest) order of *bond energy* (1= lowest; 4 = highest)

P-F

P-Cl

P-I

P-Br

**Question 5** **12 points**

a. 9 points For the molecules HClO_4 and TeF_6 , draw the most stable *Lewis structure* showing appropriate shape and *dashed/wedged bonds*, based on VSEPR. Determine the **molecular geometry** around the central atom. For HClO_4 , label the *most polar bond*.



b. 3 points (Note: Lone pairs are not shown in structures below)

(i) For the three *resonance* structures (1, 2 and 3) shown below, answer the following question:



(I) Circle the *most contributing* resonance structure.

Structure 1

Structure 2

Structure 3

(ii) For the three structures given (4, 5, and 6) below, answer the following questions:



(I) Circle all the structures that represent *resonance structures* of each other:

Structure 4

Structure 5

Structure 6

(II) Among the resonance structures, circle the *most contributing* resonance structure.

Structure 4

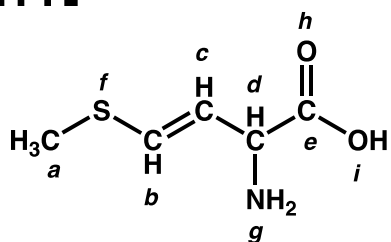
Structure 5

Structure 6

**Question 6** **6 points**

Using the data given below, calculate the **lattice energy** for SrO(s).

	ΔH° (kJ/mol)
$\text{Sr}(s) \longrightarrow \text{Sr}(g)$	+180
$\text{Sr}(g) \longrightarrow \text{Sr}^{2+}(g) + 2e^-$	+1468
$\text{O}_2(g) \longrightarrow 2\text{O}(g)$	+500
$\text{O}(g) + 2e^- \longrightarrow \text{O}^{2-}(g)$	+600
$\text{Sr}(s) + \frac{1}{2}\text{O}_2(g) \longrightarrow \text{SrO}(s)$	-550

**Question 7 10 points**

Consider the molecule given on the left, and answer questions (i) to (vi) using chemical bonding theories that we have learnt in class (valence bond theory, hybridization).

Note that the lone pairs are not shown in the structure:

i. Consider the carbon atoms marked in as ***a***, ***b***, ***c***, ***d***, and ***e*** in the molecule, and determine **hybridization** at each of the carbon centers. (2.5 points)

Atom	Hybridization
Carbon (<i>a</i>)	
Carbon (<i>b</i>)	
Carbon (<i>c</i>)	
Carbon (<i>d</i>)	
Carbon (<i>e</i>)	

ii. Now, consider the sulfur atom (S) marked as ***f***, nitrogen (N) atom (***g***), and oxygen(O) atoms (***h*** and ***i***), and determine their **hybridization**. (2 points)

Atom	Hybridization
Sulfur (<i>f</i>)	
Nitrogen (<i>g</i>)	
Oxygen (<i>h</i>)	
Oxygen (<i>i</i>)	

iii. **Describe** all the bonds formed by the carbon atoms ***a*** and ***b***, using sigma and pi designation as appropriate. (2 points)

Atom	Bonds formed (use sigma and pi as appropriate)
Carbon (<i>a</i>)	
Carbon (<i>b</i>)	

iv. How many total sp^3 -s sigma bonds are in the molecule? (1 point)

v. Determine **electronic** and **molecular geometry** at carbon (C) labeled ***e*** and sulfur (S) labeled ***f*** centres. (2 points)

Atom	Electronic Geometry	Molecular Geometry
Carbon (<i>e</i>)		
Sulfur (<i>f</i>)		

vi. Circle the value from the choices given below, that describes the **C-S-C bond angle at sulfur** (S) marked ***f***: (0.5 points)

$>90^\circ$ to $<109.5^\circ$ 109.5° >109.5 to $<120^\circ$ 120° $>120^\circ$ to $<180^\circ$

**Question 8 12 points**

Using the **valence shell orbitals only** draw a **complete** molecular orbital (MO) diagram (with no s-p mixing) for the molecule **PS**, in the space provided below. **(6 points)**

Assume that 3s and 3p atomic orbitals will form molecular orbitals like those formed by the combination of 2s and 2p atomic orbitals that we learnt in class, and there is no s-p mixing.

Show **energy arrow**, **atomic orbitals** involved, **atom labels**, the **molecular orbitals** that are formed, **filled electrons** in the MO diagram etc. Make sure to label all the atomic and molecular orbitals.

Using the MO diagram above, answer the following questions:

- (i) Which one of the following will have the highest bond order? Circle the correct choice
A) PS^{1-} B) PS C) PS^{1+}
- (ii) Which one of the following will have the longest bond length? Circle the correct choice
A) PS^{1-} B) PS C) PS^{1+}
- (iii) Which of the following will be diamagnetic? Circle the correct choice.
A) PS^{1-} B) PS C) PS^{1+}
- (iv) Draw Lewis dot structure of the molecule with the shortest bond length (based on the MO diagram) from among the three (PS^{1-} , PS, PS^{1+}), and indicate the bond order calculated from the Lewis dot structure. **(3 points)**

Lewis Dot Structure

Bond Order



1D8F87DE-D447-4B2D-9BE3-D5021847C002

midterm-exam-93388

#2 Page 10 of 10

SCRAP PAGE

**CHEMISTRY 110 –Midterm Exam****Thursday November 7, 2024, 6:30 – 8:30 pm**

Examiners: Pallavi Sirjoosingh and Ashok Kakkar

First name (please write as legibly as possible within the boxes)

--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--

Last name

--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--

Student ID number

--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--

This booklet has 8 questions on 10 pages (The last page can be used as a scrap page).

The staple must NOT be removed from the question booklet. Your intact question booklet must be returned at the end of the exam.

A separate booklet contains a periodic table, and datasheet.

Write your answers directly under the questions in the space provided. Do not write on the QR codes or near the top edges (as it may be cut off).

Please write **legibly and dark enough** for the scan to be readable. For students who use English as a second language, dictionaries are permitted. Calculators are allowed.

DO NOT ASK QUESTIONS OF THE INVIGILATORS about interpretations of the exam questions. If you have a concern about a question, please summarize it at the top of this page. All concerns will be evaluated after the exam. GOOD LUCK!

**Question 1 8 points** (*Assume all transitions are possible*)

a. 6 points Calculate the **energy (in kJ/mol)** required to ionize (completely remove) the electron from the **n=3** level of **He⁺ ion** (He: Helium). Show your work including equations. (3 sig. figs.)

b. 2 points

(I) Select the largest (longest) wavelength from the three choices below:

- i. Wavelength of photon to *completely remove* an electron from *1s orbital of Hydrogen atom*
- ii. Wavelength of photon to *completely remove* an electron from *1s orbital of Boron atom (B)*
- iii. Wavelength of photon to *completely remove* an electron from *1s orbital of Boron cation (B⁺)*

i or ii or iii

(II) Select the smallest (shortest) wavelength from the three choices below (*for H atom*):

- i. Wavelength of photon *emitted* for an electron transitioning from *2p to 1s orbital*
- ii. Wavelength of photon *absorbed* for an electron transitioning from *2s to 4p orbital*
- iii. Wavelength of photon *absorbed* for an electron transitioning from *1s to 3p orbital*

i or ii or iii



Question 2 8 points



A Chem110 student plots the graph shown on the left, from data obtained for the kinetic energy (eV) of the ejected electron versus frequency (s^{-1}) of the incident light, for an unknown metal. ($1\text{eV} = 1.6022 \times 10^{-19} \text{ J}$)

Based on the plot, answer the questions **a to c** below:

a. 1 point On the plot above, label the **threshold frequency**.

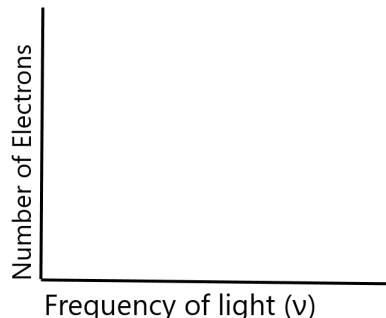
b. 2 points What is the value of the work potential (in J) of the unknown metal?

c. 3 points What is the *longest* wavelength (in nm) required to eject an electron from this metal's surface? Show your work including equation(s) used. (3 sig figs)

d. 2 points In the space provided below, plot the following

(Assume frequency is *above* threshold frequency):

- i. *Kinetic energy* of an ejected electron versus the *intensity* of the incident light (left)
- ii. *Number of electrons* ejected versus the *frequency* of the incident light (right)



**Question 3** **6 points****a. 2 points**

In the ground state of Ge (Germanium; $Z = 32$) atom, **name** the following orbitals:

i. Orbital containing the electron that requires the maximum energy to be removed.

ii. Orbital containing the electron that experiences the minimum effective nuclear charge.

b. 2 points

An unknown anion (A^-) has the following *excited* electronic configuration: $[Ar] 4s^2 3d^3 4p^1$

(Ar: Argon; $Z = 18$)

Give the **condensed ground state electronic configuration** of the atom, **A** and the cation, **A^{2+}** .
Show your work. (Assume all electronic transitions are possible)

Electronic Configuration of A:

Electronic Configuration of A^{2+} :

c. 2 points

In a single atom, determine how many electrons can have the following set of quantum numbers?

(i) $n = 5; l = 3$

In a single atom, determine how many electrons can have the following set of quantum numbers?

(i) $n = 3; l = 2; m_s = +1/2$

**Question 4** **8 points**

Rank the following from 1 to 4 in **increasing** order. Provide the ranking in the boxes provided.

Note: Atomic number (Z) are provided for relevant atoms

a) Increasing (lowest to highest) order of *ionization energy*: (1= lowest; 4 = highest)

Mg

Mg: Z = 12

Rb

Rb: Z = 37

Cs

Cs: Z = 55

K

K : Z = 19

b) Increasing (lowest to highest) order of *bond angle*: (1= lowest; 4 = highest)

H₂OCCl₄XeF₄

HCN

c) Increasing (smallest to largest) order of *ionic size* (1= smallest; 4 = largest)

S²⁻

S: Z = 16

O²⁻

O: Z = 8

Be²⁺

Be: Z = 4

Na⁺

Na: Z = 11

d) Increasing (lowest to highest) order of *bond energy* (1= lowest; 4 = highest)

P-F

P-Cl

P-I

P-Br

**Question 5** **12 points**

a. 9 points For the molecules HClO_4 and TeF_6 , draw the most stable *Lewis structure* showing appropriate shape and *dashed/wedged bonds*, based on VSEPR. Determine the **molecular geometry** around the central atom. For HClO_4 , label the *most polar bond*.



b. 3 points (Note: Lone pairs are not shown in structures below)

(i) For the three *resonance* structures (1, 2 and 3) shown below, answer the following question:



(I) Circle the *most contributing* resonance structure.

Structure 1

Structure 2

Structure 3

(ii) For the three structures given (4, 5, and 6) below, answer the following questions:



(I) Circle all the structures that represent *resonance structures* of each other:

Structure 4

Structure 5

Structure 6

(II) Among the resonance structures, circle the *most contributing* resonance structure.

Structure 4

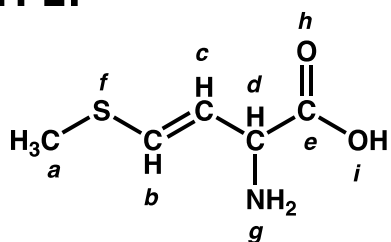
Structure 5

Structure 6

**Question 6** **6 points**

Using the data given below, calculate the **lattice energy** for SrO(s).

	ΔH° (kJ/mol)
$\text{Sr}(s) \longrightarrow \text{Sr}(g)$	+180
$\text{Sr}(g) \longrightarrow \text{Sr}^{2+}(g) + 2e^-$	+1468
$\text{O}_2(g) \longrightarrow 2\text{O}(g)$	+500
$\text{O}(g) + 2e^- \longrightarrow \text{O}^{2-}(g)$	+600
$\text{Sr}(s) + \frac{1}{2}\text{O}_2(g) \longrightarrow \text{SrO}(s)$	-550

**Question 7 10 points**

Consider the molecule given on the left, and answer questions (i) to (vi) using chemical bonding theories that we have learnt in class (valence bond theory, hybridization).

Note that the lone pairs are not shown in the structure:

i. Consider the carbon atoms marked in as ***a***, ***b***, ***c***, ***d***, and ***e*** in the molecule, and determine **hybridization** at each of the carbon centers. (2.5 points)

Atom	Hybridization
Carbon (<i>a</i>)	
Carbon (<i>b</i>)	
Carbon (<i>c</i>)	
Carbon (<i>d</i>)	
Carbon (<i>e</i>)	

ii. Now, consider the sulfur atom (S) marked as ***f***, nitrogen (N) atom (***g***), and oxygen(O) atoms (***h*** and ***i***), and determine their **hybridization**. (2 points)

Atom	Hybridization
Sulfur (<i>f</i>)	
Nitrogen (<i>g</i>)	
Oxygen (<i>h</i>)	
Oxygen (<i>i</i>)	

iii. **Describe** all the bonds formed by the carbon atoms ***a*** and ***b***, using sigma and pi designation as appropriate. (2 points)

Atom	Bonds formed (use sigma and pi as appropriate)
Carbon (<i>a</i>)	
Carbon (<i>b</i>)	

iv. How many total sp^3 -s sigma bonds are in the molecule? (1 point)

v. Determine **electronic** and **molecular geometry** at carbon (C) labeled ***e*** and sulfur (S) labeled ***f*** centres. (2 points)

Atom	Electronic Geometry	Molecular Geometry
Carbon (<i>e</i>)		
Sulfur (<i>f</i>)		

vi. Circle the value from the choices given below, that describes the **C-S-C bond angle at sulfur** (S) marked ***f***: (0.5 points)

>90° to <109.5° 109.5° >109.5 to <120° 120° >120° to <180°


Question 8 12 points

Using the **valence shell orbitals only** draw a **complete** molecular orbital (MO) diagram (with no s-p mixing) for the molecule **PS**, in the space provided below. **(6 points)**

Assume that 3s and 3p atomic orbitals will form molecular orbitals like those formed by the combination of 2s and 2p atomic orbitals that we learnt in class, and there is no s-p mixing.

Show **energy arrow**, **atomic orbitals** involved, **atom labels**, the **molecular orbitals** that are formed, **filled electrons** in the MO diagram etc. Make sure to label all the atomic and molecular orbitals.

Using the MO diagram above, answer the following questions:

- (i) Which one of the following will have the highest bond order? Circle the correct choice
 A) PS^{1-} B) PS C) PS^{1+}
- (ii) Which one of the following will have the longest bond length? Circle the correct choice
 A) PS^{1-} B) PS C) PS^{1+}
- (iii) Which of the following will be diamagnetic? Circle the correct choice.
 A) PS^{1-} B) PS C) PS^{1+}
- (iv) Draw Lewis dot structure of the molecule with the shortest bond length (based on the MO diagram) from among the three (PS^{1-} , PS, PS^{1+}), and indicate the bond order calculated from the Lewis dot structure. **(3 points)**

Lewis Dot Structure

Bond Order



0197C99D-00FD-48DA-8877-6402EC95339C

midterm-exam-93388

#3 Page 10 of 10

SCRAP PAGE

**CHEMISTRY 110 –Midterm Exam****Thursday November 7, 2024, 6:30 – 8:30 pm**

Examiners: Pallavi Sirjoosingh and Ashok Kakkar

First name (please write as legibly as possible within the boxes)

--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--

Last name

--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--

Student ID number

--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--

This booklet has 8 questions on 10 pages (The last page can be used as a scrap page).

The staple must NOT be removed from the question booklet. Your intact question booklet must be returned at the end of the exam.

A separate booklet contains a periodic table, and datasheet.

Write your answers directly under the questions in the space provided. Do not write on the QR codes or near the top edges (as it may be cut off).

Please write **legibly and dark enough** for the scan to be readable. For students who use English as a second language, dictionaries are permitted. Calculators are allowed.

DO NOT ASK QUESTIONS OF THE INVIGILATORS about interpretations of the exam questions. If you have a concern about a question, please summarize it at the top of this page. All concerns will be evaluated after the exam. GOOD LUCK!

**Question 1 8 points** (*Assume all transitions are possible*)

a. 6 points Calculate the **energy (in kJ/mol)** required to ionize (completely remove) the electron from the **n=3** level of **He⁺ ion** (He: Helium). Show your work including equations. (3 sig. figs.)

b. 2 points

(I) Select the largest (longest) wavelength from the three choices below:

- i. Wavelength of photon to *completely remove* an electron from *1s orbital of Hydrogen atom*
- ii. Wavelength of photon to *completely remove* an electron from *1s orbital of Boron atom (B)*
- iii. Wavelength of photon to *completely remove* an electron from *1s orbital of Boron cation (B⁺)*

i or ii or iii

(II) Select the smallest (shortest) wavelength from the three choices below (*for H atom*):

- i. Wavelength of photon *emitted* for an electron transitioning from *2p to 1s orbital*
- ii. Wavelength of photon *absorbed* for an electron transitioning from *2s to 4p orbital*
- iii. Wavelength of photon *absorbed* for an electron transitioning from *1s to 3p orbital*

i or ii or iii



Question 2 8 points



A Chem110 student plots the graph shown on the left, from data obtained for the kinetic energy (eV) of the ejected electron versus frequency (s^{-1}) of the incident light, for an unknown metal. ($1\text{eV} = 1.6022 \times 10^{-19} \text{ J}$)

Based on the plot, answer the questions **a to c** below:

a. 1 point On the plot above, label the **threshold frequency**.

b. 2 points What is the value of the work potential (in J) of the unknown metal?

c. 3 points What is the *longest* wavelength (in nm) required to eject an electron from this metal's surface? Show your work including equation(s) used. (3 sig figs)

d. 2 points In the space provided below, plot the following

(Assume frequency is *above* threshold frequency):

- Kinetic energy* of an ejected electron versus the *intensity* of the incident light (left)
- Number of electrons* ejected versus the *frequency* of the incident light (right)



**Question 3** **6 points****a. 2 points**

In the ground state of Ge (Germanium; $Z = 32$) atom, **name** the following orbitals:

i. Orbital containing the electron that requires the maximum energy to be removed.

ii. Orbital containing the electron that experiences the minimum effective nuclear charge.

b. 2 points

An unknown anion (A^-) has the following *excited* electronic configuration: $[Ar] 4s^2 3d^3 4p^1$

(Ar: Argon; $Z = 18$)

Give the **condensed ground state electronic configuration** of the atom, **A** and the cation, **A^{2+}** .
Show your work. (Assume all electronic transitions are possible)

Electronic Configuration of A:

Electronic Configuration of A^{2+} :

c. 2 points

In a single atom, determine how many electrons can have the following set of quantum numbers?

(i) $n = 5; l = 3$

In a single atom, determine how many electrons can have the following set of quantum numbers?

(i) $n = 3; l = 2; m_s = +1/2$

**Question 4** **8 points**

Rank the following from 1 to 4 in **increasing** order. Provide the ranking in the boxes provided.

Note: Atomic number (Z) are provided for relevant atoms

a) Increasing (lowest to highest) order of *ionization energy*: (1= lowest; 4 = highest)

Mg

Mg: Z = 12

Rb

Rb: Z = 37

Cs

Cs: Z = 55

K

K : Z = 19

b) Increasing (lowest to highest) order of *bond angle*: (1= lowest; 4 = highest)

H₂OCCl₄XeF₄

HCN

c) Increasing (smallest to largest) order of *ionic size* (1= smallest; 4 = largest)

S²⁻

S: Z = 16

O²⁻

O: Z = 8

Be²⁺

Be: Z = 4

Na⁺

Na: Z = 11

d) Increasing (lowest to highest) order of *bond energy* (1= lowest; 4 = highest)

P-F

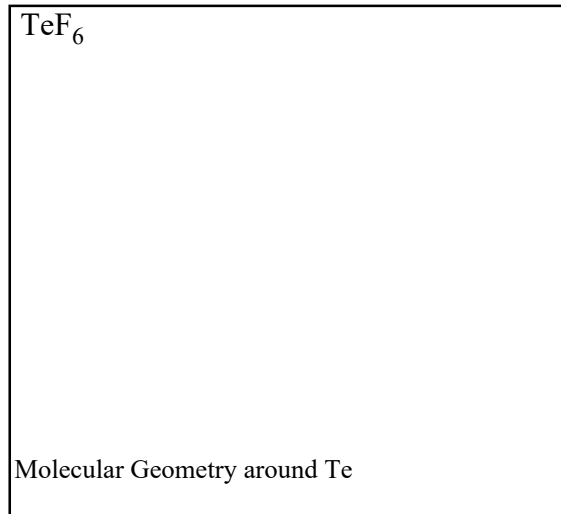
P-Cl

P-I

P-Br

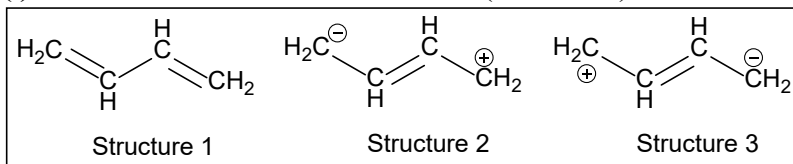
**Question 5** **12 points**

a. 9 points For the molecules HClO_4 and TeF_6 , draw the most stable *Lewis structure* showing appropriate shape and *dashed/wedged bonds*, based on VSEPR. Determine the **molecular geometry** around the central atom. For HClO_4 , label the *most polar bond*.



b. 3 points (Note: Lone pairs are not shown in structures below)

(i) For the three *resonance* structures (1, 2 and 3) shown below, answer the following question:



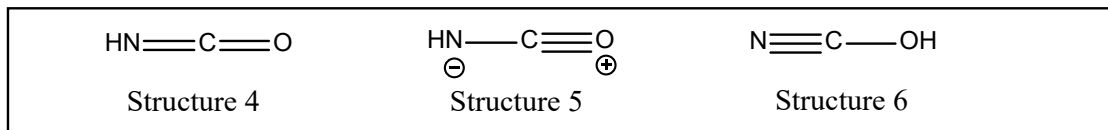
(I) Circle the *most contributing* resonance structure.

Structure 1

Structure 2

Structure 3

(ii) For the three structures given (4, 5, and 6) below, answer the following questions:



(I) Circle all the structures that represent *resonance structures* of each other:

Structure 4

Structure 5

Structure 6

(II) Among the resonance structures, circle the *most contributing* resonance structure.

Structure 4

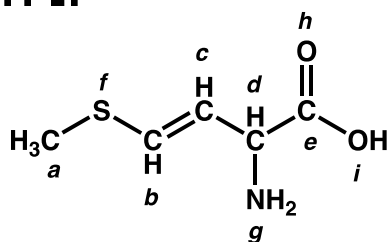
Structure 5

Structure 6

**Question 6** **6 points**

Using the data given below, calculate the **lattice energy** for SrO(s).

	ΔH° (kJ/mol)
$\text{Sr}(s) \longrightarrow \text{Sr}(g)$	+180
$\text{Sr}(g) \longrightarrow \text{Sr}^{2+}(g) + 2e^-$	+1468
$\text{O}_2(g) \longrightarrow 2\text{O}(g)$	+500
$\text{O}(g) + 2e^- \longrightarrow \text{O}^{2-}(g)$	+600
$\text{Sr}(s) + \frac{1}{2}\text{O}_2(g) \longrightarrow \text{SrO}(s)$	-550

**Question 7 10 points**

Consider the molecule given on the left, and answer questions (i) to (vi) using chemical bonding theories that we have learnt in class (valence bond theory, hybridization).

Note that the lone pairs are not shown in the structure:

i. Consider the carbon atoms marked in as ***a***, ***b***, ***c***, ***d***, and ***e*** in the molecule, and determine **hybridization** at each of the carbon centers. (2.5 points)

Atom	Hybridization
Carbon (<i>a</i>)	
Carbon (<i>b</i>)	
Carbon (<i>c</i>)	
Carbon (<i>d</i>)	
Carbon (<i>e</i>)	

ii. Now, consider the sulfur atom (S) marked as ***f***, nitrogen (N) atom (***g***), and oxygen(O) atoms (***h*** and ***i***), and determine their **hybridization**. (2 points)

Atom	Hybridization
Sulfur (<i>f</i>)	
Nitrogen (<i>g</i>)	
Oxygen (<i>h</i>)	
Oxygen (<i>i</i>)	

iii. **Describe** all the bonds formed by the carbon atoms ***a*** and ***b***, using sigma and pi designation as appropriate. (2 points)

Atom	Bonds formed (use sigma and pi as appropriate)
Carbon (<i>a</i>)	
Carbon (<i>b</i>)	

iv. How many total sp^3 -s sigma bonds are in the molecule? (1 point)

v. Determine **electronic** and **molecular geometry** at carbon (C) labeled ***e*** and sulfur (S) labeled ***f*** centres. (2 points)

Atom	Electronic Geometry	Molecular Geometry
Carbon (<i>e</i>)		
Sulfur (<i>f</i>)		

vi. Circle the value from the choices given below, that describes the **C-S-C bond angle at sulfur** (S) marked ***f***: (0.5 points)

>90° to <109.5° 109.5° >109.5 to <120° 120° >120° to <180°


Question 8 12 points

Using the **valence shell orbitals only** draw a **complete** molecular orbital (MO) diagram (with no s-p mixing) for the molecule **PS**, in the space provided below. **(6 points)**

Assume that 3s and 3p atomic orbitals will form molecular orbitals like those formed by the combination of 2s and 2p atomic orbitals that we learnt in class, and there is no s-p mixing.

Show **energy arrow**, **atomic orbitals** involved, **atom labels**, the **molecular orbitals** that are formed, **filled electrons** in the MO diagram etc. Make sure to label all the atomic and molecular orbitals.

Using the MO diagram above, answer the following questions:

- (i) Which one of the following will have the highest bond order? Circle the correct choice
 A) PS^{1-} B) PS C) PS^{1+}
- (ii) Which one of the following will have the longest bond length? Circle the correct choice
 A) PS^{1-} B) PS C) PS^{1+}
- (iii) Which of the following will be diamagnetic? Circle the correct choice.
 A) PS^{1-} B) PS C) PS^{1+}
- (iv) Draw Lewis dot structure of the molecule with the shortest bond length (based on the MO diagram) from among the three (PS^{1-} , PS, PS^{1+}), and indicate the bond order calculated from the Lewis dot structure. **(3 points)**

Lewis Dot Structure

Bond Order



32FB70F7-093A-44D9-B995-0431E28AD115

midterm-exam-93388

#4 Page 10 of 10

SCRAP PAGE

**CHEMISTRY 110 –Midterm Exam****Thursday November 7, 2024, 6:30 – 8:30 pm**

Examiners: Pallavi Sirjoosingh and Ashok Kakkar

First name (please write as legibly as possible within the boxes)

--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--

Last name

--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--

Student ID number

--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--

This booklet has 8 questions on 10 pages (The last page can be used as a scrap page).

The staple must NOT be removed from the question booklet. Your intact question booklet must be returned at the end of the exam.

A separate booklet contains a periodic table, and datasheet.

Write your answers directly under the questions in the space provided. Do not write on the QR codes or near the top edges (as it may be cut off).

Please write **legibly and dark enough** for the scan to be readable. For students who use English as a second language, dictionaries are permitted. Calculators are allowed.

DO NOT ASK QUESTIONS OF THE INVIGILATORS about interpretations of the exam questions. If you have a concern about a question, please summarize it at the top of this page. All concerns will be evaluated after the exam. GOOD LUCK!

**Question 1 8 points** (*Assume all transitions are possible*)

a. 6 points Calculate the **energy (in kJ/mol)** required to ionize (completely remove) the electron from the **n=3** level of **He⁺ ion** (He: Helium). Show your work including equations. (3 sig. figs.)

b. 2 points

(I) Select the largest (longest) wavelength from the three choices below:

- i. Wavelength of photon to *completely remove* an electron from *1s orbital of Hydrogen atom*
- ii. Wavelength of photon to *completely remove* an electron from *1s orbital of Boron atom (B)*
- iii. Wavelength of photon to *completely remove* an electron from *1s orbital of Boron cation (B⁺)*

i or ii or iii

(II) Select the smallest (shortest) wavelength from the three choices below (*for H atom*):

- i. Wavelength of photon *emitted* for an electron transitioning from *2p to 1s orbital*
- ii. Wavelength of photon *absorbed* for an electron transitioning from *2s to 4p orbital*
- iii. Wavelength of photon *absorbed* for an electron transitioning from *1s to 3p orbital*

i or ii or iii



Question 2 8 points



A Chem110 student plots the graph shown on the left, from data obtained for the kinetic energy (eV) of the ejected electron versus frequency (s^{-1}) of the incident light, for an unknown metal. ($1\text{eV} = 1.6022 \times 10^{-19} \text{ J}$)

Based on the plot, answer the questions **a to c** below:

a. 1 point On the plot above, label the **threshold frequency**.

b. 2 points What is the value of the work potential (in J) of the unknown metal?

c. 3 points What is the *longest* wavelength (in nm) required to eject an electron from this metal's surface? Show your work including equation(s) used. (3 sig figs)

d. 2 points In the space provided below, plot the following

(Assume frequency is *above* threshold frequency):

- i. *Kinetic energy* of an ejected electron versus the *intensity* of the incident light (left)
- ii. *Number of electrons* ejected versus the *frequency* of the incident light (right)



**Question 3** **6 points****a. 2 points**

In the ground state of Ge (Germanium; $Z = 32$) atom, **name** the following orbitals:

i. Orbital containing the electron that requires the maximum energy to be removed.

ii. Orbital containing the electron that experiences the minimum effective nuclear charge.

b. 2 points

An unknown anion (A^-) has the following *excited* electronic configuration: $[Ar] 4s^2 3d^3 4p^1$

(Ar: Argon; $Z = 18$)

Give the **condensed ground state electronic configuration** of the atom, **A** and the cation, **A^{2+}** .
Show your work. (Assume all electronic transitions are possible)

Electronic Configuration of A:

Electronic Configuration of A^{2+} :

c. 2 points

In a single atom, determine how many electrons can have the following set of quantum numbers?

(i) $n = 5; l = 3$

In a single atom, determine how many electrons can have the following set of quantum numbers?

(i) $n = 3; l = 2; m_s = +1/2$

**Question 4** **8 points**

Rank the following from 1 to 4 in **increasing** order. Provide the ranking in the boxes provided.

Note: Atomic number (Z) are provided for relevant atoms

a) Increasing (lowest to highest) order of *ionization energy*: (1= lowest; 4 = highest)

Mg

Mg: Z = 12

Rb

Rb: Z = 37

Cs

Cs: Z = 55

K

K : Z = 19

b) Increasing (lowest to highest) order of *bond angle*: (1= lowest; 4 = highest)

H₂OCCl₄XeF₄

HCN

c) Increasing (smallest to largest) order of *ionic size* (1= smallest; 4 = largest)

S²⁻

S: Z = 16

O²⁻

O: Z = 8

Be²⁺

Be: Z = 4

Na⁺

Na: Z = 11

d) Increasing (lowest to highest) order of *bond energy* (1= lowest; 4 = highest)

P-F

P-Cl

P-I

P-Br

**Question 5** **12 points**

a. 9 points For the molecules HClO_4 and TeF_6 , draw the most stable *Lewis structure* showing appropriate shape and *dashed/wedged bonds*, based on VSEPR. Determine the **molecular geometry** around the central atom. For HClO_4 , label the *most polar bond*.



b. 3 points (Note: Lone pairs are not shown in structures below)

(i) For the three *resonance* structures (1, 2 and 3) shown below, answer the following question:



(I) Circle the *most contributing* resonance structure.

Structure 1

Structure 2

Structure 3

(ii) For the three structures given (4, 5, and 6) below, answer the following questions:



(I) Circle all the structures that represent *resonance structures* of each other:

Structure 4

Structure 5

Structure 6

(II) Among the resonance structures, circle the *most contributing* resonance structure.

Structure 4

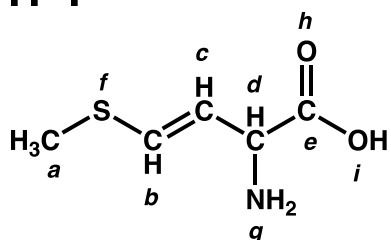
Structure 5

Structure 6

**Question 6** **6 points**

Using the data given below, calculate the **lattice energy** for SrO(s).

	ΔH° (kJ/mol)
$\text{Sr}(s) \longrightarrow \text{Sr}(g)$	+180
$\text{Sr}(g) \longrightarrow \text{Sr}^{2+}(g) + 2e^-$	+1468
$\text{O}_2(g) \longrightarrow 2\text{O}(g)$	+500
$\text{O}(g) + 2e^- \longrightarrow \text{O}^{2-}(g)$	+600
$\text{Sr}(s) + \frac{1}{2}\text{O}_2(g) \longrightarrow \text{SrO}(s)$	-550

**Question 7 10 points**

Consider the molecule given on the left, and answer questions (i) to (vi) using chemical bonding theories that we have learnt in class (valence bond theory, hybridization).

Note that the lone pairs are not shown in the structure:

i. Consider the carbon atoms marked in as ***a***, ***b***, ***c***, ***d***, and ***e*** in the molecule, and determine **hybridization** at each of the carbon centers. (2.5 points)

Atom	Hybridization
Carbon (<i>a</i>)	
Carbon (<i>b</i>)	
Carbon (<i>c</i>)	
Carbon (<i>d</i>)	
Carbon (<i>e</i>)	

ii. Now, consider the sulfur atom (S) marked as ***f***, nitrogen (N) atom (***g***), and oxygen(O) atoms (***h*** and ***i***), and determine their **hybridization**. (2 points)

Atom	Hybridization
Sulfur (<i>f</i>)	
Nitrogen (<i>g</i>)	
Oxygen (<i>h</i>)	
Oxygen (<i>i</i>)	

iii. **Describe** all the bonds formed by the carbon atoms ***a*** and ***b***, using sigma and pi designation as appropriate. (2 points)

Atom	Bonds formed (use sigma and pi as appropriate)
Carbon (<i>a</i>)	
Carbon (<i>b</i>)	

iv. How many total sp^3 -s sigma bonds are in the molecule? (1 point)

v. Determine **electronic** and **molecular geometry** at carbon (C) labeled ***e*** and sulfur (S) labeled ***f*** centres. (2 points)

Atom	Electronic Geometry	Molecular Geometry
Carbon (<i>e</i>)		
Sulfur (<i>f</i>)		

vi. Circle the value from the choices given below, that describes the **C-S-C bond angle at sulfur** (S) marked ***f***: (0.5 points)

$>90^\circ$ to $<109.5^\circ$ 109.5° >109.5 to $<120^\circ$ 120° $>120^\circ$ to $<180^\circ$


Question 8 12 points

Using the **valence shell orbitals only** draw a **complete** molecular orbital (MO) diagram (with no s-p mixing) for the molecule **PS**, in the space provided below. **(6 points)**

Assume that 3s and 3p atomic orbitals will form molecular orbitals like those formed by the combination of 2s and 2p atomic orbitals that we learnt in class, and there is no s-p mixing.

Show **energy arrow**, **atomic orbitals** involved, **atom labels**, the **molecular orbitals** that are formed, **filled electrons** in the MO diagram etc. Make sure to label all the atomic and molecular orbitals.

Using the MO diagram above, answer the following questions:

- (i) Which one of the following will have the highest bond order? Circle the correct choice
 A) PS^{1-} B) PS C) PS^{1+}
- (ii) Which one of the following will have the longest bond length? Circle the correct choice
 A) PS^{1-} B) PS C) PS^{1+}
- (iii) Which of the following will be diamagnetic? Circle the correct choice.
 A) PS^{1-} B) PS C) PS^{1+}
- (iv) Draw Lewis dot structure of the molecule with the shortest bond length (based on the MO diagram) from among the three (PS^{1-} , PS, PS^{1+}), and indicate the bond order calculated from the Lewis dot structure. **(3 points)**

Lewis Dot Structure

Bond Order



54B18E33-5EBB-49CE-862E-241AF2695F43

midterm-exam-93388

#5 Page 10 of 10

SCRAP PAGE

**CHEMISTRY 110 –Midterm Exam****Thursday November 7, 2024, 6:30 – 8:30 pm**

Examiners: Pallavi Sirjoosingh and Ashok Kakkar

First name (please write as legibly as possible within the boxes)

--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--

Last name

--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--

Student ID number

--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--

This booklet has 8 questions on 10 pages (The last page can be used as a scrap page).

The staple must NOT be removed from the question booklet. Your intact question booklet must be returned at the end of the exam.

A separate booklet contains a periodic table, and datasheet.

Write your answers directly under the questions in the space provided. Do not write on the QR codes or near the top edges (as it may be cut off).

Please write **legibly and dark enough** for the scan to be readable. For students who use English as a second language, dictionaries are permitted. Calculators are allowed.

DO NOT ASK QUESTIONS OF THE INVIGILATORS about interpretations of the exam questions. If you have a concern about a question, please summarize it at the top of this page. All concerns will be evaluated after the exam. GOOD LUCK!

**Question 1 8 points** (*Assume all transitions are possible*)

a. 6 points Calculate the **energy (in kJ/mol)** required to ionize (completely remove) the electron from the **$n=3$** level of **He^+ ion** (He: Helium). Show your work including equations. (3 sig. figs.)

b. 2 points

(I) Select the largest (longest) wavelength from the three choices below:

- i. Wavelength of photon to *completely remove* an electron from *1s orbital of Hydrogen atom*
- ii. Wavelength of photon to *completely remove* an electron from *1s orbital of Boron atom (B)*
- iii. Wavelength of photon to *completely remove* an electron from *1s orbital of Boron cation (B^+)*

i or ii or iii

(II) Select the smallest (shortest) wavelength from the three choices below (*for H atom*):

- i. Wavelength of photon *emitted* for an electron transitioning from *2p to 1s orbital*
- ii. Wavelength of photon *absorbed* for an electron transitioning from *2s to 4p orbital*
- iii. Wavelength of photon *absorbed* for an electron transitioning from *1s to 3p orbital*

i or ii or iii



Question 2 8 points



A Chem110 student plots the graph shown on the left, from data obtained for the kinetic energy (eV) of the ejected electron versus frequency (s^{-1}) of the incident light, for an unknown metal. ($1\text{eV} = 1.6022 \times 10^{-19} \text{ J}$)

Based on the plot, answer the questions **a to c** below:

a. 1 point On the plot above, label the **threshold frequency**.

b. 2 points What is the value of the work potential (in J) of the unknown metal?

c. 3 points What is the *longest* wavelength (in nm) required to eject an electron from this metal's surface? Show your work including equation(s) used. (3 sig figs)

d. 2 points In the space provided below, plot the following

(Assume frequency is *above* threshold frequency):

- i. *Kinetic energy* of an ejected electron versus the *intensity* of the incident light (left)
- ii. *Number of electrons* ejected versus the *frequency* of the incident light (right)



**Question 3** **6 points****a. 2 points**

In the ground state of Ge (Germanium; $Z = 32$) atom, **name** the following orbitals:

i. Orbital containing the electron that requires the maximum energy to be removed.

ii. Orbital containing the electron that experiences the minimum effective nuclear charge.

b. 2 points

An unknown anion (A^-) has the following *excited* electronic configuration: $[Ar] 4s^2 3d^3 4p^1$

(Ar: Argon; $Z = 18$)

Give the **condensed ground state electronic configuration** of the atom, **A** and the cation, **A^{2+}** .
Show your work. (Assume all electronic transitions are possible)

Electronic Configuration of A:

Electronic Configuration of A^{2+} :

c. 2 points

In a single atom, determine how many electrons can have the following set of quantum numbers?

(i) $n = 5; l = 3$

In a single atom, determine how many electrons can have the following set of quantum numbers?

(i) $n = 3; l = 2; m_s = +1/2$

**Question 4** **8 points**

Rank the following from 1 to 4 in **increasing** order. Provide the ranking in the boxes provided.

Note: Atomic number (Z) are provided for relevant atoms

a) Increasing (lowest to highest) order of *ionization energy*: (1= lowest; 4 = highest)

Mg

Mg: Z = 12

Rb

Rb: Z = 37

Cs

Cs: Z = 55

K

K : Z = 19

b) Increasing (lowest to highest) order of *bond angle*: (1= lowest; 4 = highest)

H₂OCCl₄XeF₄

HCN

c) Increasing (smallest to largest) order of *ionic size* (1= smallest; 4 = largest)

S²⁻

S: Z = 16

O²⁻

O: Z = 8

Be²⁺

Be: Z = 4

Na⁺

Na: Z = 11

d) Increasing (lowest to highest) order of *bond energy* (1= lowest; 4 = highest)

P-F

P-Cl

P-I

P-Br

**Question 5** **12 points**

a. 9 points For the molecules HClO_4 and TeF_6 , draw the most stable *Lewis structure* showing appropriate shape and *dashed/wedged bonds*, based on VSEPR. Determine the **molecular geometry** around the central atom. For HClO_4 , label the *most polar bond*.



b. 3 points (Note: Lone pairs are not shown in structures below)

(i) For the three *resonance* structures (1, 2 and 3) shown below, answer the following question:



(I) Circle the *most contributing* resonance structure.

Structure 1

Structure 2

Structure 3

(ii) For the three structures given (4, 5, and 6) below, answer the following questions:



(I) Circle all the structures that represent *resonance structures* of each other:

Structure 4

Structure 5

Structure 6

(II) Among the resonance structures, circle the *most contributing* resonance structure.

Structure 4

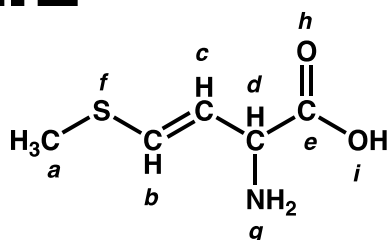
Structure 5

Structure 6

**Question 6** **6 points**

Using the data given below, calculate the **lattice energy** for SrO(s).

	ΔH° (kJ/mol)
$\text{Sr}(s) \longrightarrow \text{Sr}(g)$	+180
$\text{Sr}(g) \longrightarrow \text{Sr}^{2+}(g) + 2e^-$	+1468
$\text{O}_2(g) \longrightarrow 2\text{O}(g)$	+500
$\text{O}(g) + 2e^- \longrightarrow \text{O}^{2-}(g)$	+600
$\text{Sr}(s) + \frac{1}{2}\text{O}_2(g) \longrightarrow \text{SrO}(s)$	-550

**Question 7 10 points**

Consider the molecule given on the left, and answer questions (i) to (vi) using chemical bonding theories that we have learnt in class (valence bond theory, hybridization).

Note that the lone pairs are not shown in the structure:

i. Consider the carbon atoms marked in as ***a***, ***b***, ***c***, ***d***, and ***e*** in the molecule, and determine **hybridization** at each of the carbon centers. (2.5 points)

Atom	Hybridization
Carbon (<i>a</i>)	
Carbon (<i>b</i>)	
Carbon (<i>c</i>)	
Carbon (<i>d</i>)	
Carbon (<i>e</i>)	

ii. Now, consider the sulfur atom (S) marked as ***f***, nitrogen (N) atom (***g***), and oxygen(O) atoms (***h*** and ***i***), and determine their **hybridization**. (2 points)

Atom	Hybridization
Sulfur (<i>f</i>)	
Nitrogen (<i>g</i>)	
Oxygen (<i>h</i>)	
Oxygen (<i>i</i>)	

iii. **Describe** all the bonds formed by the carbon atoms ***a*** and ***b***, using sigma and pi designation as appropriate. (2 points)

Atom	Bonds formed (use sigma and pi as appropriate)
Carbon (<i>a</i>)	
Carbon (<i>b</i>)	

iv. How many total sp^3 -s sigma bonds are in the molecule? (1 point)

v. Determine **electronic** and **molecular geometry** at carbon (C) labeled ***e*** and sulfur (S) labeled ***f*** centres. (2 points)

Atom	Electronic Geometry	Molecular Geometry
Carbon (<i>e</i>)		
Sulfur (<i>f</i>)		

vi. Circle the value from the choices given below, that describes the **C-S-C bond angle at sulfur** (S) marked ***f***: (0.5 points)

>90° to <109.5° 109.5° >109.5 to <120° 120° >120° to <180°


Question 8 12 points

Using the **valence shell orbitals only** draw a **complete** molecular orbital (MO) diagram (with no s-p mixing) for the molecule **PS**, in the space provided below. **(6 points)**

Assume that 3s and 3p atomic orbitals will form molecular orbitals like those formed by the combination of 2s and 2p atomic orbitals that we learnt in class, and there is no s-p mixing.

Show **energy arrow**, **atomic orbitals** involved, **atom labels**, the **molecular orbitals** that are formed, **filled electrons** in the MO diagram etc. Make sure to label all the atomic and molecular orbitals.

Using the MO diagram above, answer the following questions:

- (i) Which one of the following will have the highest bond order? Circle the correct choice
 A) PS^{1-} B) PS C) PS^{1+}
- (ii) Which one of the following will have the longest bond length? Circle the correct choice
 A) PS^{1-} B) PS C) PS^{1+}
- (iii) Which of the following will be diamagnetic? Circle the correct choice.
 A) PS^{1-} B) PS C) PS^{1+}
- (iv) Draw Lewis dot structure of the molecule with the shortest bond length (based on the MO diagram) from among the three (PS^{1-} , PS, PS^{1+}), and indicate the bond order calculated from the Lewis dot structure. **(3 points)**

Lewis Dot Structure

Bond Order



0BF361DB-07B5-4A27-AE7F-F6FE14103D12

midterm-exam-93388

#6 Page 10 of 10

SCRAP PAGE

**CHEMISTRY 110 –Midterm Exam****Thursday November 7, 2024, 6:30 – 8:30 pm**

Examiners: Pallavi Sirjoosingh and Ashok Kakkar

First name (please write as legibly as possible within the boxes)

--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--

Last name

--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--

Student ID number

--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--

This booklet has 8 questions on 10 pages (The last page can be used as a scrap page).

The staple must NOT be removed from the question booklet. Your intact question booklet must be returned at the end of the exam.

A separate booklet contains a periodic table, and datasheet.

Write your answers directly under the questions in the space provided. Do not write on the QR codes or near the top edges (as it may be cut off).

Please write **legibly and dark enough** for the scan to be readable. For students who use English as a second language, dictionaries are permitted. Calculators are allowed.

DO NOT ASK QUESTIONS OF THE INVIGILATORS about interpretations of the exam questions. If you have a concern about a question, please summarize it at the top of this page. All concerns will be evaluated after the exam. GOOD LUCK!

**Question 1 8 points** (*Assume all transitions are possible*)

a. 6 points Calculate the **energy (in kJ/mol)** required to ionize (completely remove) the electron from the **$n=3$** level of **He^+ ion** (He: Helium). Show your work including equations. (3 sig. figs.)

b. 2 points

(I) Select the largest (longest) wavelength from the three choices below:

- i. Wavelength of photon to *completely remove* an electron from *1s orbital of Hydrogen atom*
- ii. Wavelength of photon to *completely remove* an electron from *1s orbital of Boron atom (B)*
- iii. Wavelength of photon to *completely remove* an electron from *1s orbital of Boron cation (B^+)*

i or ii or iii

(II) Select the smallest (shortest) wavelength from the three choices below (*for H atom*):

- i. Wavelength of photon *emitted* for an electron transitioning from *2p to 1s orbital*
- ii. Wavelength of photon *absorbed* for an electron transitioning from *2s to 4p orbital*
- iii. Wavelength of photon *absorbed* for an electron transitioning from *1s to 3p orbital*

i or ii or iii



Question 2 8 points



A Chem110 student plots the graph shown on the left, from data obtained for the kinetic energy (eV) of the ejected electron versus frequency (s^{-1}) of the incident light, for an unknown metal. ($1\text{eV} = 1.6022 \times 10^{-19} \text{ J}$)

Based on the plot, answer the questions **a to c** below:

a. 1 point On the plot above, label the **threshold frequency**.

b. 2 points What is the value of the work potential (in J) of the unknown metal?

c. 3 points What is the *longest* wavelength (in nm) required to eject an electron from this metal's surface? Show your work including equation(s) used. (3 sig figs)

d. 2 points In the space provided below, plot the following

(Assume frequency is *above* threshold frequency):

- i. *Kinetic energy* of an ejected electron versus the *intensity* of the incident light (left)
- ii. *Number of electrons* ejected versus the *frequency* of the incident light (right)



**Question 3** **6 points****a. 2 points**

In the ground state of Ge (Germanium; $Z = 32$) atom, **name** the following orbitals:

i. Orbital containing the electron that requires the maximum energy to be removed.

ii. Orbital containing the electron that experiences the minimum effective nuclear charge.

b. 2 points

An unknown anion (A^-) has the following *excited* electronic configuration: $[Ar] 4s^2 3d^3 4p^1$

(Ar: Argon; $Z = 18$)

Give the **condensed ground state electronic configuration** of the atom, **A** and the cation, **A^{2+}** .
Show your work. (Assume all electronic transitions are possible)

Electronic Configuration of A:

Electronic Configuration of A^{2+} :

c. 2 points

In a single atom, determine how many electrons can have the following set of quantum numbers?

(i) $n = 5; l = 3$

In a single atom, determine how many electrons can have the following set of quantum numbers?

(i) $n = 3; l = 2; m_s = +1/2$

**Question 4** **8 points**

Rank the following from 1 to 4 in **increasing** order. Provide the ranking in the boxes provided.

Note: Atomic number (Z) are provided for relevant atoms

a) Increasing (lowest to highest) order of *ionization energy*: (1= lowest; 4 = highest)

Mg

Mg: Z = 12

Rb

Rb: Z = 37

Cs

Cs: Z = 55

K

K : Z = 19

b) Increasing (lowest to highest) order of *bond angle*: (1= lowest; 4 = highest)

H₂OCCl₄XeF₄

HCN

c) Increasing (smallest to largest) order of *ionic size* (1= smallest; 4 = largest)

S²⁻

S: Z = 16

O²⁻

O: Z = 8

Be²⁺

Be: Z = 4

Na⁺

Na: Z = 11

d) Increasing (lowest to highest) order of *bond energy* (1= lowest; 4 = highest)

P-F

P-Cl

P-I

P-Br

**Question 5** **12 points**

a. 9 points For the molecules HClO_4 and TeF_6 , draw the most stable *Lewis structure* showing appropriate shape and *dashed/wedged bonds*, based on VSEPR. Determine the **molecular geometry** around the central atom. For HClO_4 , label the *most polar bond*.



b. 3 points (Note: Lone pairs are not shown in structures below)

(i) For the three *resonance* structures (1, 2 and 3) shown below, answer the following question:



(I) Circle the *most contributing* resonance structure.

Structure 1

Structure 2

Structure 3

(ii) For the three structures given (4, 5, and 6) below, answer the following questions:



(I) Circle all the structures that represent *resonance structures* of each other:

Structure 4

Structure 5

Structure 6

(II) Among the resonance structures, circle the *most contributing* resonance structure.

Structure 4

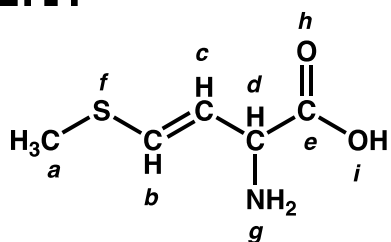
Structure 5

Structure 6

**Question 6** **6 points**

Using the data given below, calculate the **lattice energy** for SrO(s).

	ΔH° (kJ/mol)
$\text{Sr}(s) \longrightarrow \text{Sr}(g)$	+180
$\text{Sr}(g) \longrightarrow \text{Sr}^{2+}(g) + 2e^-$	+1468
$\text{O}_2(g) \longrightarrow 2\text{O}(g)$	+500
$\text{O}(g) + 2e^- \longrightarrow \text{O}^{2-}(g)$	+600
$\text{Sr}(s) + \frac{1}{2} \text{O}_2(g) \longrightarrow \text{SrO}(s)$	-550

**Question 7 10 points**

Consider the molecule given on the left, and answer questions (i) to (vi) using chemical bonding theories that we have learnt in class (valence bond theory, hybridization).

Note that the lone pairs are not shown in the structure:

i. Consider the carbon atoms marked in as ***a***, ***b***, ***c***, ***d***, and ***e*** in the molecule, and determine **hybridization** at each of the carbon centers. (2.5 points)

Atom	Hybridization
Carbon (<i>a</i>)	
Carbon (<i>b</i>)	
Carbon (<i>c</i>)	
Carbon (<i>d</i>)	
Carbon (<i>e</i>)	

ii. Now, consider the sulfur atom (S) marked as ***f***, nitrogen (N) atom (***g***), and oxygen(O) atoms (***h*** and ***i***), and determine their **hybridization**. (2 points)

Atom	Hybridization
Sulfur (<i>f</i>)	
Nitrogen (<i>g</i>)	
Oxygen (<i>h</i>)	
Oxygen (<i>i</i>)	

iii. **Describe** all the bonds formed by the carbon atoms ***a*** and ***b***, using sigma and pi designation as appropriate. (2 points)

Atom	Bonds formed (use sigma and pi as appropriate)
Carbon (<i>a</i>)	
Carbon (<i>b</i>)	

iv. How many total sp^3 -s sigma bonds are in the molecule? (1 point)

v. Determine **electronic** and **molecular geometry** at carbon (C) labeled ***e*** and sulfur (S) labeled ***f*** centres. (2 points)

Atom	Electronic Geometry	Molecular Geometry
Carbon (<i>e</i>)		
Sulfur (<i>f</i>)		

vi. Circle the value from the choices given below, that describes the **C-S-C bond angle at sulfur** (S) marked ***f***: (0.5 points)

>90° to <109.5° 109.5° >109.5 to <120° 120° >120° to <180°


Question 8 12 points

Using the **valence shell orbitals only** draw a **complete** molecular orbital (MO) diagram (with no s-p mixing) for the molecule **PS**, in the space provided below. **(6 points)**

Assume that 3s and 3p atomic orbitals will form molecular orbitals like those formed by the combination of 2s and 2p atomic orbitals that we learnt in class, and there is no s-p mixing.

Show **energy arrow**, **atomic orbitals** involved, **atom labels**, the **molecular orbitals** that are formed, **filled electrons** in the MO diagram etc. Make sure to label all the atomic and molecular orbitals.

Using the MO diagram above, answer the following questions:

- (i) Which one of the following will have the highest bond order? Circle the correct choice
 A) PS^{1-} B) PS C) PS^{1+}
- (ii) Which one of the following will have the longest bond length? Circle the correct choice
 A) PS^{1-} B) PS C) PS^{1+}
- (iii) Which of the following will be diamagnetic? Circle the correct choice.
 A) PS^{1-} B) PS C) PS^{1+}
- (iv) Draw Lewis dot structure of the molecule with the shortest bond length (based on the MO diagram) from among the three (PS^{1-} , PS, PS^{1+}), and indicate the bond order calculated from the Lewis dot structure. **(3 points)**

Lewis Dot Structure

Bond Order



CD622485-64B8-4085-9F34-8F37C854F604

midterm-exam-93388

#7 Page 10 of 10

SCRAP PAGE

**CHEMISTRY 110 –Midterm Exam****Thursday November 7, 2024, 6:30 – 8:30 pm****Examiners: Pallavi Sirjoosingh and Ashok Kakkar**

First name (please write as legibly as possible within the boxes)

--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--

Last name

--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--

Student ID number

--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--

This booklet has 8 questions on 10 pages (The last page can be used as a scrap page).

The staple must NOT be removed from the question booklet. Your intact question booklet must be returned at the end of the exam.

A separate booklet contains a periodic table, and datasheet.

Write your answers directly under the questions in the space provided. Do not write on the QR codes or near the top edges (as it may be cut off).

Please write **legibly and dark enough** for the scan to be readable. For students who use English as a second language, dictionaries are permitted. Calculators are allowed.

DO NOT ASK QUESTIONS OF THE INVIGILATORS about interpretations of the exam questions. If you have a concern about a question, please summarize it at the top of this page. All concerns will be evaluated after the exam. GOOD LUCK!

**Question 1 8 points** (*Assume all transitions are possible*)

a. 6 points Calculate the **energy (in kJ/mol)** required to ionize (completely remove) the electron from the **$n=3$** level of **He^+ ion** (He: Helium). Show your work including equations. (3 sig. figs.)

b. 2 points

(I) Select the largest (longest) wavelength from the three choices below:

- i. Wavelength of photon to *completely remove* an electron from *1s orbital of Hydrogen atom*
- ii. Wavelength of photon to *completely remove* an electron from *1s orbital of Boron atom (B)*
- iii. Wavelength of photon to *completely remove* an electron from *1s orbital of Boron cation (B^+)*

i or ii or iii

(II) Select the smallest (shortest) wavelength from the three choices below (*for H atom*):

- i. Wavelength of photon *emitted* for an electron transitioning from *2p to 1s orbital*
- ii. Wavelength of photon *absorbed* for an electron transitioning from *2s to 4p orbital*
- iii. Wavelength of photon *absorbed* for an electron transitioning from *1s to 3p orbital*

i or ii or iii



Question 2 8 points



A Chem110 student plots the graph shown on the left, from data obtained for the kinetic energy (eV) of the ejected electron versus frequency (s^{-1}) of the incident light, for an unknown metal. ($1\text{eV} = 1.6022 \times 10^{-19} \text{ J}$)

Based on the plot, answer the questions **a to c** below:

a. 1 point On the plot above, label the **threshold frequency**.

b. 2 points What is the value of the work potential (in J) of the unknown metal?

c. 3 points What is the *longest* wavelength (in nm) required to eject an electron from this metal's surface? Show your work including equation(s) used. (3 sig figs)

d. 2 points In the space provided below, plot the following

(Assume frequency is *above* threshold frequency):

- i. *Kinetic energy* of an ejected electron versus the *intensity* of the incident light (left)
- ii. *Number of electrons* ejected versus the *frequency* of the incident light (right)



**Question 3** **6 points****a. 2 points**

In the ground state of Ge (Germanium; $Z = 32$) atom, **name** the following orbitals:

i. Orbital containing the electron that requires the maximum energy to be removed.

ii. Orbital containing the electron that experiences the minimum effective nuclear charge.

b. 2 points

An unknown anion (A^-) has the following *excited* electronic configuration: $[Ar] 4s^2 3d^3 4p^1$

(Ar: Argon; $Z = 18$)

Give the **condensed ground state electronic configuration** of the atom, **A** and the cation, **A^{2+}** .
Show your work. (Assume all electronic transitions are possible)

Electronic Configuration of A:

Electronic Configuration of A^{2+} :

c. 2 points

In a single atom, determine how many electrons can have the following set of quantum numbers?

(i) $n = 5; l = 3$

In a single atom, determine how many electrons can have the following set of quantum numbers?

(i) $n = 3; l = 2; m_s = +1/2$

**Question 4** **8 points**

Rank the following from 1 to 4 in **increasing** order. Provide the ranking in the boxes provided.

Note: Atomic number (Z) are provided for relevant atoms

a) Increasing (lowest to highest) order of *ionization energy*: (1= lowest; 4 = highest)

Mg

Mg: Z = 12

Rb

Rb: Z = 37

Cs

Cs: Z = 55

K

K : Z = 19

b) Increasing (lowest to highest) order of *bond angle*: (1= lowest; 4 = highest)

H₂OCCl₄XeF₄

HCN

c) Increasing (smallest to largest) order of *ionic size* (1= smallest; 4 = largest)

S²⁻

S: Z = 16

O²⁻

O: Z = 8

Be²⁺

Be: Z = 4

Na⁺

Na: Z = 11

d) Increasing (lowest to highest) order of *bond energy* (1= lowest; 4 = highest)

P-F

P-Cl

P-I

P-Br

**Question 5** **12 points**

a. 9 points For the molecules HClO_4 and TeF_6 , draw the most stable *Lewis structure* showing appropriate shape and *dashed/wedged bonds*, based on VSEPR. Determine the **molecular geometry** around the central atom. For HClO_4 , label the *most polar bond*.

HClO_4 (there are no O-O bonds)

Label the most polar bond

Molecular Geometry around Cl

TeF_6

Molecular Geometry around Te

b. 3 points (Note: Lone pairs are not shown in structures below)

(i) For the three *resonance* structures (1, 2 and 3) shown below, answer the following question:



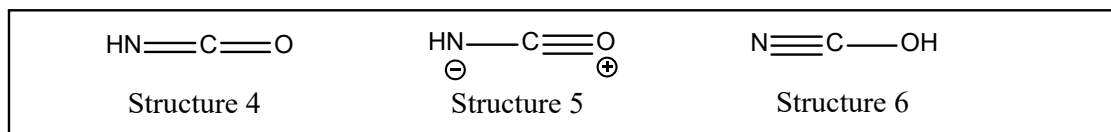
(I) Circle the *most contributing* resonance structure.

Structure 1

Structure 2

Structure 3

(ii) For the three structures given (4, 5, and 6) below, answer the following questions:



(I) Circle all the structures that represent *resonance structures* of each other:

Structure 4

Structure 5

Structure 6

(II) Among the resonance structures, circle the *most contributing* resonance structure.

Structure 4

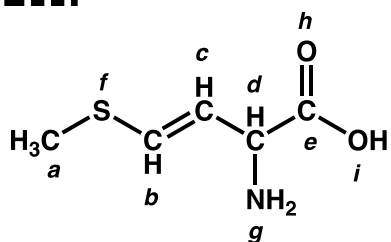
Structure 5

Structure 6

**Question 6** **6 points**

Using the data given below, calculate the **lattice energy** for SrO(s).

	ΔH° (kJ/mol)
$\text{Sr}(s) \longrightarrow \text{Sr}(g)$	+180
$\text{Sr}(g) \longrightarrow \text{Sr}^{2+}(g) + 2e^-$	+1468
$\text{O}_2(g) \longrightarrow 2\text{O}(g)$	+500
$\text{O}(g) + 2e^- \longrightarrow \text{O}^{2-}(g)$	+600
$\text{Sr}(s) + \frac{1}{2}\text{O}_2(g) \longrightarrow \text{SrO}(s)$	-550

**Question 7 10 points**

Consider the molecule given on the left, and answer questions (i) to (vi) using chemical bonding theories that we have learnt in class (valence bond theory, hybridization).

Note that the lone pairs are not shown in the structure:

i. Consider the carbon atoms marked in as ***a***, ***b***, ***c***, ***d***, and ***e*** in the molecule, and determine **hybridization** at each of the carbon centers. (2.5 points)

Atom	Hybridization
Carbon (<i>a</i>)	
Carbon (<i>b</i>)	
Carbon (<i>c</i>)	
Carbon (<i>d</i>)	
Carbon (<i>e</i>)	

ii. Now, consider the sulfur atom (S) marked as ***f***, nitrogen (N) atom (***g***), and oxygen(O) atoms (***h*** and ***i***), and determine their **hybridization**. (2 points)

Atom	Hybridization
Sulfur (<i>f</i>)	
Nitrogen (<i>g</i>)	
Oxygen (<i>h</i>)	
Oxygen (<i>i</i>)	

iii. **Describe** all the bonds formed by the carbon atoms ***a*** and ***b***, using sigma and pi designation as appropriate. (2 points)

Atom	Bonds formed (use sigma and pi as appropriate)
Carbon (<i>a</i>)	
Carbon (<i>b</i>)	

iv. How many total sp^3 -s sigma bonds are in the molecule? (1 point)

v. Determine **electronic** and **molecular geometry** at carbon (C) labeled ***e*** and sulfur (S) labeled ***f*** centres. (2 points)

Atom	Electronic Geometry	Molecular Geometry
Carbon (<i>e</i>)		
Sulfur (<i>f</i>)		

vi. Circle the value from the choices given below, that describes the **C-S-C bond angle at sulfur** (S) marked ***f***: (0.5 points)

$>90^\circ$ to $<109.5^\circ$ 109.5° >109.5 to $<120^\circ$ 120° $>120^\circ$ to $<180^\circ$


Question 8 12 points

Using the **valence shell orbitals only** draw a **complete** molecular orbital (MO) diagram (with no s-p mixing) for the molecule **PS**, in the space provided below. **(6 points)**

Assume that 3s and 3p atomic orbitals will form molecular orbitals like those formed by the combination of 2s and 2p atomic orbitals that we learnt in class, and there is no s-p mixing.

Show **energy arrow**, **atomic orbitals** involved, **atom labels**, the **molecular orbitals** that are formed, **filled electrons** in the MO diagram etc. Make sure to label all the atomic and molecular orbitals.

Using the MO diagram above, answer the following questions:

- (i) Which one of the following will have the highest bond order? Circle the correct choice
 A) PS^{1-} B) PS C) PS^{1+}
- (ii) Which one of the following will have the longest bond length? Circle the correct choice
 A) PS^{1-} B) PS C) PS^{1+}
- (iii) Which of the following will be diamagnetic? Circle the correct choice.
 A) PS^{1-} B) PS C) PS^{1+}
- (iv) Draw Lewis dot structure of the molecule with the shortest bond length (based on the MO diagram) from among the three (PS^{1-} , PS, PS^{1+}), and indicate the bond order calculated from the Lewis dot structure. **(3 points)**

Lewis Dot Structure

Bond Order



78BC6001-48BD-4F34-B74D-C6E083249BBA

midterm-exam-93388

#8 Page 10 of 10

SCRAP PAGE

**CHEMISTRY 110 –Midterm Exam****Thursday November 7, 2024, 6:30 – 8:30 pm**

Examiners: Pallavi Sirjoosingh and Ashok Kakkar

First name (please write as legibly as possible within the boxes)

--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--

Last name

--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--

Student ID number

--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--

This booklet has 8 questions on 10 pages (The last page can be used as a scrap page).

The staple must NOT be removed from the question booklet. Your intact question booklet must be returned at the end of the exam.

A separate booklet contains a periodic table, and datasheet.

Write your answers directly under the questions in the space provided. Do not write on the QR codes or near the top edges (as it may be cut off).

Please write **legibly and dark enough** for the scan to be readable. For students who use English as a second language, dictionaries are permitted. Calculators are allowed.

DO NOT ASK QUESTIONS OF THE INVIGILATORS about interpretations of the exam questions. If you have a concern about a question, please summarize it at the top of this page. All concerns will be evaluated after the exam. GOOD LUCK!

**Question 1 8 points** (*Assume all transitions are possible*)

a. 6 points Calculate the **energy (in kJ/mol)** required to ionize (completely remove) the electron from the **$n=3$** level of **He^+ ion** (He: Helium). Show your work including equations. (3 sig. figs.)

b. 2 points

(I) Select the largest (longest) wavelength from the three choices below:

- i. Wavelength of photon to *completely remove* an electron from *1s orbital of Hydrogen atom*
- ii. Wavelength of photon to *completely remove* an electron from *1s orbital of Boron atom (B)*
- iii. Wavelength of photon to *completely remove* an electron from *1s orbital of Boron cation (B^+)*

i or ii or iii

(II) Select the smallest (shortest) wavelength from the three choices below (*for H atom*):

- i. Wavelength of photon *emitted* for an electron transitioning from *2p to 1s orbital*
- ii. Wavelength of photon *absorbed* for an electron transitioning from *2s to 4p orbital*
- iii. Wavelength of photon *absorbed* for an electron transitioning from *1s to 3p orbital*

i or ii or iii



Question 2 8 points



A Chem110 student plots the graph shown on the left, from data obtained for the kinetic energy (eV) of the ejected electron versus frequency (s^{-1}) of the incident light, for an unknown metal. ($1\text{eV} = 1.6022 \times 10^{-19} \text{ J}$)

Based on the plot, answer the questions **a to c** below:

a. 1 point On the plot above, label the **threshold frequency**.

b. 2 points What is the value of the work potential (in J) of the unknown metal?

c. 3 points What is the *longest* wavelength (in nm) required to eject an electron from this metal's surface? Show your work including equation(s) used. (3 sig figs)

d. 2 points In the space provided below, plot the following

(Assume frequency is *above* threshold frequency):

- i. *Kinetic energy* of an ejected electron versus the *intensity* of the incident light (left)
- ii. *Number of electrons* ejected versus the *frequency* of the incident light (right)



**Question 3** **6 points****a. 2 points**

In the ground state of Ge (Germanium; $Z = 32$) atom, **name** the following orbitals:

i. Orbital containing the electron that requires the maximum energy to be removed.

ii. Orbital containing the electron that experiences the minimum effective nuclear charge.

b. 2 points

An unknown anion (A^-) has the following *excited* electronic configuration: $[Ar] 4s^2 3d^3 4p^1$

(Ar: Argon; $Z = 18$)

Give the **condensed ground state electronic configuration** of the atom, **A** and the cation, **A^{2+}** .
Show your work. (Assume all electronic transitions are possible)

Electronic Configuration of A:

Electronic Configuration of A^{2+} :

c. 2 points

In a single atom, determine how many electrons can have the following set of quantum numbers?

(i) $n = 5; l = 3$

In a single atom, determine how many electrons can have the following set of quantum numbers?

(i) $n = 3; l = 2; m_s = +1/2$

**Question 4** **8 points**

Rank the following from 1 to 4 in **increasing** order. Provide the ranking in the boxes provided.

Note: Atomic number (Z) are provided for relevant atoms

a) Increasing (lowest to highest) order of *ionization energy*: (1= lowest; 4 = highest)

Mg

Mg: Z = 12

Rb

Rb: Z = 37

Cs

Cs: Z = 55

K

K : Z = 19

b) Increasing (lowest to highest) order of *bond angle*: (1= lowest; 4 = highest)

H₂OCCl₄XeF₄

HCN

c) Increasing (smallest to largest) order of *ionic size* (1= smallest; 4 = largest)

S²⁻

S: Z = 16

O²⁻

O: Z = 8

Be²⁺

Be: Z = 4

Na⁺

Na: Z = 11

d) Increasing (lowest to highest) order of *bond energy* (1= lowest; 4 = highest)

P-F

P-Cl

P-I

P-Br

**Question 5** **12 points**

a. 9 points For the molecules HClO_4 and TeF_6 , draw the most stable *Lewis structure* showing appropriate shape and *dashed/wedged bonds*, based on VSEPR. Determine the **molecular geometry** around the central atom. For HClO_4 , label the *most polar bond*.



b. 3 points (Note: Lone pairs are not shown in structures below)

(i) For the three *resonance* structures (1, 2 and 3) shown below, answer the following question:



(I) Circle the *most contributing* resonance structure.

Structure 1

Structure 2

Structure 3

(ii) For the three structures given (4, 5, and 6) below, answer the following questions:



(I) Circle all the structures that represent *resonance structures* of each other:

Structure 4

Structure 5

Structure 6

(II) Among the resonance structures, circle the *most contributing* resonance structure.

Structure 4

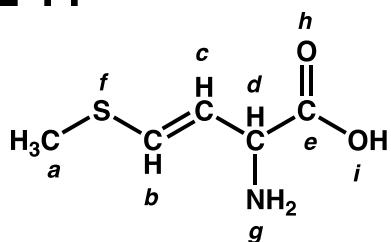
Structure 5

Structure 6

**Question 6** **6 points**

Using the data given below, calculate the **lattice energy** for SrO(s).

	ΔH° (kJ/mol)
$\text{Sr}(s) \longrightarrow \text{Sr}(g)$	+180
$\text{Sr}(g) \longrightarrow \text{Sr}^{2+}(g) + 2e^-$	+1468
$\text{O}_2(g) \longrightarrow 2\text{O}(g)$	+500
$\text{O}(g) + 2e^- \longrightarrow \text{O}^{2-}(g)$	+600
$\text{Sr}(s) + \frac{1}{2}\text{O}_2(g) \longrightarrow \text{SrO}(s)$	-550

**Question 7 10 points**

Consider the molecule given on the left, and answer questions (i) to (vi) using chemical bonding theories that we have learnt in class (valence bond theory, hybridization).

Note that the lone pairs are not shown in the structure:

i. Consider the carbon atoms marked in as ***a***, ***b***, ***c***, ***d***, and ***e*** in the molecule, and determine **hybridization** at each of the carbon centers. (2.5 points)

Atom	Hybridization
Carbon (<i>a</i>)	
Carbon (<i>b</i>)	
Carbon (<i>c</i>)	
Carbon (<i>d</i>)	
Carbon (<i>e</i>)	

ii. Now, consider the sulfur atom (S) marked as ***f***, nitrogen (N) atom (***g***), and oxygen(O) atoms (***h*** and ***i***), and determine their **hybridization**. (2 points)

Atom	Hybridization
Sulfur (<i>f</i>)	
Nitrogen (<i>g</i>)	
Oxygen (<i>h</i>)	
Oxygen (<i>i</i>)	

iii. **Describe** all the bonds formed by the carbon atoms ***a*** and ***b***, using sigma and pi designation as appropriate. (2 points)

Atom	Bonds formed (use sigma and pi as appropriate)
Carbon (<i>a</i>)	
Carbon (<i>b</i>)	

iv. How many total sp^3 -s sigma bonds are in the molecule? (1 point)

v. Determine **electronic** and **molecular geometry** at carbon (C) labeled ***e*** and sulfur (S) labeled ***f*** centres. (2 points)

Atom	Electronic Geometry	Molecular Geometry
Carbon (<i>e</i>)		
Sulfur (<i>f</i>)		

vi. Circle the value from the choices given below, that describes the **C-S-C bond angle at sulfur** (S) marked ***f***: (0.5 points)

$>90^\circ$ to $<109.5^\circ$ 109.5° >109.5 to $<120^\circ$ 120° $>120^\circ$ to $<180^\circ$


Question 8 12 points

Using the **valence shell orbitals only** draw a **complete** molecular orbital (MO) diagram (with no s-p mixing) for the molecule **PS**, in the space provided below. **(6 points)**

Assume that 3s and 3p atomic orbitals will form molecular orbitals like those formed by the combination of 2s and 2p atomic orbitals that we learnt in class, and there is no s-p mixing.

Show **energy arrow**, **atomic orbitals** involved, **atom labels**, the **molecular orbitals** that are formed, **filled electrons** in the MO diagram etc. Make sure to label all the atomic and molecular orbitals.

Using the MO diagram above, answer the following questions:

- (i) Which one of the following will have the highest bond order? Circle the correct choice
 A) PS^{1-} B) PS C) PS^{1+}
- (ii) Which one of the following will have the longest bond length? Circle the correct choice
 A) PS^{1-} B) PS C) PS^{1+}
- (iii) Which of the following will be diamagnetic? Circle the correct choice.
 A) PS^{1-} B) PS C) PS^{1+}
- (iv) Draw Lewis dot structure of the molecule with the shortest bond length (based on the MO diagram) from among the three (PS^{1-} , PS, PS^{1+}), and indicate the bond order calculated from the Lewis dot structure. **(3 points)**

Lewis Dot Structure

Bond Order



3FEF1ABE-BF3D-4FF3-B0B0-84992AE77903

midterm-exam-93388

#9 Page 10 of 10

SCRAP PAGE

**CHEMISTRY 110 –Midterm Exam****Thursday November 7, 2024, 6:30 – 8:30 pm**

Examiners: Pallavi Sirjoosingh and Ashok Kakkar

First name (please write as legibly as possible within the boxes)

--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--

Last name

--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--

Student ID number

--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--

This booklet has 8 questions on 10 pages (The last page can be used as a scrap page).

The staple must NOT be removed from the question booklet. Your intact question booklet must be returned at the end of the exam.

A separate booklet contains a periodic table, and datasheet.

Write your answers directly under the questions in the space provided. Do not write on the QR codes or near the top edges (as it may be cut off).

Please write **legibly and dark enough** for the scan to be readable. For students who use English as a second language, dictionaries are permitted. Calculators are allowed.

DO NOT ASK QUESTIONS OF THE INVIGILATORS about interpretations of the exam questions. If you have a concern about a question, please summarize it at the top of this page. All concerns will be evaluated after the exam. GOOD LUCK!

**Question 1 8 points** (*Assume all transitions are possible*)

a. 6 points Calculate the **energy (in kJ/mol)** required to ionize (completely remove) the electron from the **n=3** level of **He⁺ ion** (He: Helium). Show your work including equations. (3 sig. figs.)

b. 2 points

(I) **Select** the **largest (longest)** wavelength from the three choices below:

- i. Wavelength of photon to *completely remove* an electron from *1s orbital of Hydrogen atom*
- ii. Wavelength of photon to *completely remove* an electron from *1s orbital of Boron atom (B)*
- iii. Wavelength of photon to *completely remove* an electron from *1s orbital of Boron cation (B⁺)*

i **or** **ii** **or** **iii**

(II) **Select** the **smallest (shortest)** wavelength from the three choices below (*for H atom*):

- i. Wavelength of photon *emitted* for an electron transitioning from *2p to 1s orbital*
- ii. Wavelength of photon *absorbed* for an electron transitioning from *2s to 4p orbital*
- iii. Wavelength of photon *absorbed* for an electron transitioning from *1s to 3p orbital*

i **or** **ii** **or** **iii**



Question 2 8 points



A Chem110 student plots the graph shown on the left, from data obtained for the kinetic energy (eV) of the ejected electron versus frequency (s^{-1}) of the incident light, for an unknown metal. ($1\text{eV} = 1.6022 \times 10^{-19} \text{ J}$)

Based on the plot, answer the questions **a to c** below:

a. 1 point On the plot above, label the **threshold frequency**.

b. 2 points What is the value of the work potential (in J) of the unknown metal?

c. 3 points What is the *longest* wavelength (in nm) required to eject an electron from this metal's surface? Show your work including equation(s) used. (3 sig figs)

d. 2 points In the space provided below, plot the following

(Assume frequency is *above* threshold frequency):

- i. *Kinetic energy* of an ejected electron versus the *intensity* of the incident light (left)
- ii. *Number of electrons* ejected versus the *frequency* of the incident light (right)



**Question 3** **6 points****a. 2 points**

In the ground state of Ge (Germanium; $Z = 32$) atom, **name** the following orbitals:

i. Orbital containing the electron that requires the maximum energy to be removed.

ii. Orbital containing the electron that experiences the minimum effective nuclear charge.

b. 2 points

An unknown anion (A^-) has the following *excited* electronic configuration: $[Ar] 4s^2 3d^3 4p^1$

(Ar: Argon; $Z = 18$)

Give the **condensed ground state electronic configuration** of the atom, **A** and the cation, **A^{2+}** .
Show your work. (Assume all electronic transitions are possible)

Electronic Configuration of A:

Electronic Configuration of A^{2+} :

c. 2 points

In a single atom, determine how many electrons can have the following set of quantum numbers?

(i) $n = 5; l = 3$

In a single atom, determine how many electrons can have the following set of quantum numbers?

(i) $n = 3; l = 2; m_s = +1/2$

**Question 4** **8 points**

Rank the following from 1 to 4 in **increasing** order. Provide the ranking in the boxes provided.

Note: Atomic number (Z) are provided for relevant atoms

a) Increasing (lowest to highest) order of *ionization energy*: (1= lowest; 4 = highest)

Mg

Mg: Z = 12

Rb

Rb: Z = 37

Cs

Cs: Z = 55

K

K : Z = 19

b) Increasing (lowest to highest) order of *bond angle*: (1= lowest; 4 = highest)

H₂OCCl₄XeF₄

HCN

c) Increasing (smallest to largest) order of *ionic size* (1= smallest; 4 = largest)

S²⁻

S: Z = 16

O²⁻

O: Z = 8

Be²⁺

Be: Z = 4

Na⁺

Na: Z = 11

d) Increasing (lowest to highest) order of *bond energy* (1= lowest; 4 = highest)

P-F

P-Cl

P-I

P-Br

**Question 5** **12 points**

a. 9 points For the molecules HClO_4 and TeF_6 , draw the most stable *Lewis structure* showing appropriate shape and *dashed/wedged bonds*, based on VSEPR. Determine the **molecular geometry** around the central atom. For HClO_4 , label the *most polar bond*.



b. 3 points (Note: Lone pairs are not shown in structures below)

(i) For the three *resonance* structures (1, 2 and 3) shown below, answer the following question:



(I) Circle the *most contributing* resonance structure.

Structure 1

Structure 2

Structure 3

(ii) For the three structures given (4, 5, and 6) below, answer the following questions:



(I) Circle all the structures that represent *resonance structures* of each other:

Structure 4

Structure 5

Structure 6

(II) Among the resonance structures, circle the *most contributing* resonance structure.

Structure 4

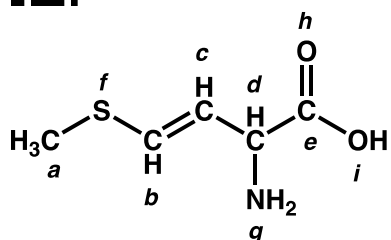
Structure 5

Structure 6

**Question 6** **6 points**

Using the data given below, calculate the **lattice energy** for SrO(s).

	ΔH° (kJ/mol)
$\text{Sr}(s) \longrightarrow \text{Sr}(g)$	+180
$\text{Sr}(g) \longrightarrow \text{Sr}^{2+}(g) + 2e^-$	+1468
$\text{O}_2(g) \longrightarrow 2\text{O}(g)$	+500
$\text{O}(g) + 2e^- \longrightarrow \text{O}^{2-}(g)$	+600
$\text{Sr}(s) + \frac{1}{2}\text{O}_2(g) \longrightarrow \text{SrO}(s)$	-550

**Question 7 10 points**

Consider the molecule given on the left, and answer questions (i) to (vi) using chemical bonding theories that we have learnt in class (valence bond theory, hybridization).

Note that the lone pairs are not shown in the structure:

i. Consider the carbon atoms marked in as ***a***, ***b***, ***c***, ***d***, and ***e*** in the molecule, and determine **hybridization** at each of the carbon centers. (2.5 points)

Atom	Hybridization
Carbon (<i>a</i>)	
Carbon (<i>b</i>)	
Carbon (<i>c</i>)	
Carbon (<i>d</i>)	
Carbon (<i>e</i>)	

ii. Now, consider the sulfur atom (S) marked as ***f***, nitrogen (N) atom (***g***), and oxygen(O) atoms (***h*** and ***i***), and determine their **hybridization**. (2 points)

Atom	Hybridization
Sulfur (<i>f</i>)	
Nitrogen (<i>g</i>)	
Oxygen (<i>h</i>)	
Oxygen (<i>i</i>)	

iii. **Describe** all the bonds formed by the carbon atoms ***a*** and ***b***, using sigma and pi designation as appropriate. (2 points)

Atom	Bonds formed (use sigma and pi as appropriate)
Carbon (<i>a</i>)	
Carbon (<i>b</i>)	

iv. How many total sp^3 -s sigma bonds are in the molecule? (1 point)

v. Determine **electronic** and **molecular geometry** at carbon (C) labeled ***e*** and sulfur (S) labeled ***f*** centres. (2 points)

Atom	Electronic Geometry	Molecular Geometry
Carbon (<i>e</i>)		
Sulfur (<i>f</i>)		

vi. Circle the value from the choices given below, that describes the **C-S-C bond angle at sulfur** (S) marked ***f***: (0.5 points)

>90° to <109.5°

109.5°

>109.5 to <120°

120°

>120° to <180°


Question 8 12 points

Using the **valence shell orbitals only** draw a **complete** molecular orbital (MO) diagram (with no s-p mixing) for the molecule **PS**, in the space provided below. **(6 points)**

Assume that 3s and 3p atomic orbitals will form molecular orbitals like those formed by the combination of 2s and 2p atomic orbitals that we learnt in class, and there is no s-p mixing.

Show **energy arrow**, **atomic orbitals** involved, **atom labels**, the **molecular orbitals** that are formed, **filled electrons** in the MO diagram etc. Make sure to label all the atomic and molecular orbitals.

Using the MO diagram above, answer the following questions:

- (i) Which one of the following will have the highest bond order? Circle the correct choice
 A) PS^{1-} B) PS C) PS^{1+}
- (ii) Which one of the following will have the longest bond length? Circle the correct choice
 A) PS^{1-} B) PS C) PS^{1+}
- (iii) Which of the following will be diamagnetic? Circle the correct choice.
 A) PS^{1-} B) PS C) PS^{1+}
- (iv) Draw Lewis dot structure of the molecule with the shortest bond length (based on the MO diagram) from among the three (PS^{1-} , PS, PS^{1+}), and indicate the bond order calculated from the Lewis dot structure. **(3 points)**

Lewis Dot Structure

Bond Order



2C4C8874-8779-4144-B7B1-293746F48D53

midterm-exam-93388

#10 Page 10 of 10

SCRAP PAGE

**CHEMISTRY 110 –Midterm Exam****Thursday November 7, 2024, 6:30 – 8:30 pm**

Examiners: Pallavi Sirjoosingh and Ashok Kakkar

First name (please write as legibly as possible within the boxes)

--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--

Last name

--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--

Student ID number

--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--

This booklet has 8 questions on 10 pages (The last page can be used as a scrap page).

The staple must NOT be removed from the question booklet. Your intact question booklet must be returned at the end of the exam.

A separate booklet contains a periodic table, and datasheet.

Write your answers directly under the questions in the space provided. Do not write on the QR codes or near the top edges (as it may be cut off).

Please write **legibly and dark enough** for the scan to be readable. For students who use English as a second language, dictionaries are permitted. Calculators are allowed.

DO NOT ASK QUESTIONS OF THE INVIGILATORS about interpretations of the exam questions. If you have a concern about a question, please summarize it at the top of this page. All concerns will be evaluated after the exam. GOOD LUCK!

**Question 1 8 points** (*Assume all transitions are possible*)

a. 6 points Calculate the **energy (in kJ/mol)** required to ionize (completely remove) the electron from the **n=3** level of **He⁺ ion** (He: Helium). Show your work including equations. (3 sig. figs.)

b. 2 points

(I) Select the largest (longest) wavelength from the three choices below:

- i. Wavelength of photon to *completely remove* an electron from *1s orbital of Hydrogen atom*
- ii. Wavelength of photon to *completely remove* an electron from *1s orbital of Boron atom (B)*
- iii. Wavelength of photon to *completely remove* an electron from *1s orbital of Boron cation (B⁺)*

i or ii or iii

(II) Select the smallest (shortest) wavelength from the three choices below (*for H atom*):

- i. Wavelength of photon *emitted* for an electron transitioning from *2p to 1s orbital*
- ii. Wavelength of photon *absorbed* for an electron transitioning from *2s to 4p orbital*
- iii. Wavelength of photon *absorbed* for an electron transitioning from *1s to 3p orbital*

i or ii or iii



Question 2 8 points



A Chem110 student plots the graph shown on the left, from data obtained for the kinetic energy (eV) of the ejected electron versus frequency (s^{-1}) of the incident light, for an unknown metal. ($1\text{eV} = 1.6022 \times 10^{-19} \text{ J}$)

Based on the plot, answer the questions **a to c** below:

a. 1 point On the plot above, label the **threshold frequency**.

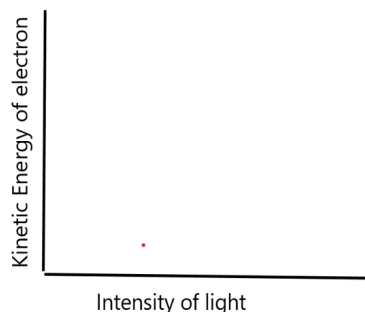
b. 2 points What is the value of the work potential (in J) of the unknown metal?

c. 3 points What is the *longest* wavelength (in nm) required to eject an electron from this metal's surface? Show your work including equation(s) used. (3 sig figs)

d. 2 points In the space provided below, plot the following

(Assume frequency is *above* threshold frequency):

- Kinetic energy* of an ejected electron versus the *intensity* of the incident light (left)
- Number of electrons* ejected versus the *frequency* of the incident light (right)



**Question 3** **6 points****a. 2 points**

In the ground state of Ge (Germanium; $Z = 32$) atom, **name** the following orbitals:

i. Orbital containing the electron that requires the maximum energy to be removed.

ii. Orbital containing the electron that experiences the minimum effective nuclear charge.

b. 2 points

An unknown anion (A^-) has the following *excited* electronic configuration: $[Ar] 4s^2 3d^3 4p^1$

(Ar: Argon; $Z = 18$)

Give the **condensed ground state electronic configuration** of the atom, **A** and the cation, **A^{2+}** .
Show your work. (Assume all electronic transitions are possible)

Electronic Configuration of A:

Electronic Configuration of A^{2+} :

c. 2 points

In a single atom, determine how many electrons can have the following set of quantum numbers?

(i) $n = 5; l = 3$

In a single atom, determine how many electrons can have the following set of quantum numbers?

(i) $n = 3; l = 2; m_s = +1/2$

**Question 4** **8 points**

Rank the following from 1 to 4 in **increasing** order. Provide the ranking in the boxes provided.

Note: Atomic number (Z) are provided for relevant atoms

a) Increasing (lowest to highest) order of *ionization energy*: (1= lowest; 4 = highest)

Mg

Mg: Z = 12

Rb

Rb: Z = 37

Cs

Cs: Z = 55

K

K : Z = 19

b) Increasing (lowest to highest) order of *bond angle*: (1= lowest; 4 = highest)

H₂OCCl₄XeF₄

HCN

c) Increasing (smallest to largest) order of *ionic size* (1= smallest; 4 = largest)

S²⁻

S: Z = 16

O²⁻

O: Z = 8

Be²⁺

Be: Z = 4

Na⁺

Na: Z = 11

d) Increasing (lowest to highest) order of *bond energy* (1= lowest; 4 = highest)

P-F

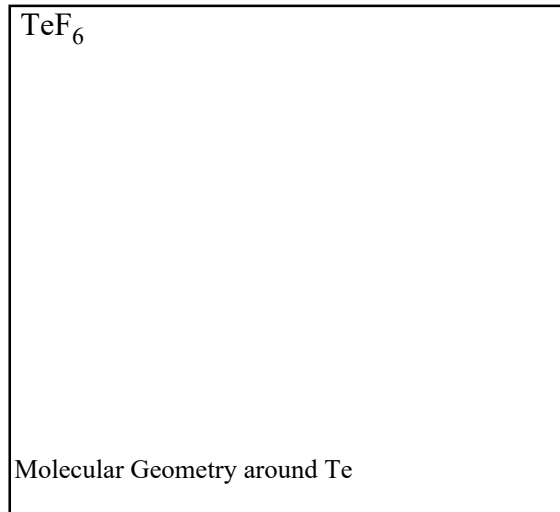
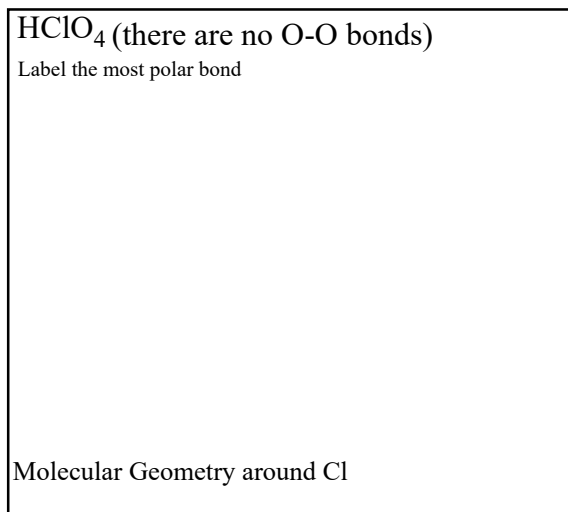
P-Cl

P-I

P-Br

**Question 5** **12 points**

a. 9 points For the molecules HClO_4 and TeF_6 , draw the most stable *Lewis structure* showing appropriate shape and *dashed/wedged bonds*, based on VSEPR. Determine the **molecular geometry** around the central atom. For HClO_4 , label the *most polar bond*.



b. 3 points (Note: Lone pairs are not shown in structures below)

(i) For the three *resonance* structures (1, 2 and 3) shown below, answer the following question:



(I) Circle the *most contributing* resonance structure.

Structure 1

Structure 2

Structure 3

(ii) For the three structures given (4, 5, and 6) below, answer the following questions:



(I) Circle all the structures that represent *resonance structures* of each other:

Structure 4

Structure 5

Structure 6

(II) Among the resonance structures, circle the *most contributing* resonance structure.

Structure 4

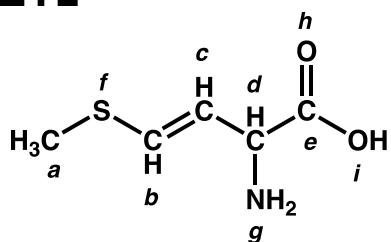
Structure 5

Structure 6

**Question 6** **6 points**

Using the data given below, calculate the **lattice energy** for SrO(s).

	ΔH° (kJ/mol)
$\text{Sr}(s) \longrightarrow \text{Sr}(g)$	+180
$\text{Sr}(g) \longrightarrow \text{Sr}^{2+}(g) + 2e^-$	+1468
$\text{O}_2(g) \longrightarrow 2\text{O}(g)$	+500
$\text{O}(g) + 2e^- \longrightarrow \text{O}^{2-}(g)$	+600
$\text{Sr}(s) + \frac{1}{2}\text{O}_2(g) \longrightarrow \text{SrO}(s)$	-550

**Question 7 10 points**

Consider the molecule given on the left, and answer questions (i) to (vi) using chemical bonding theories that we have learnt in class (valence bond theory, hybridization).

Note that the lone pairs are not shown in the structure:

i. Consider the carbon atoms marked in as ***a***, ***b***, ***c***, ***d***, and ***e*** in the molecule, and determine **hybridization** at each of the carbon centers. (2.5 points)

Atom	Hybridization
Carbon (<i>a</i>)	
Carbon (<i>b</i>)	
Carbon (<i>c</i>)	
Carbon (<i>d</i>)	
Carbon (<i>e</i>)	

ii. Now, consider the sulfur atom (S) marked as ***f***, nitrogen (N) atom (***g***), and oxygen(O) atoms (***h*** and ***i***), and determine their **hybridization**. (2 points)

Atom	Hybridization
Sulfur (<i>f</i>)	
Nitrogen (<i>g</i>)	
Oxygen (<i>h</i>)	
Oxygen (<i>i</i>)	

iii. **Describe** all the bonds formed by the carbon atoms ***a*** and ***b***, using sigma and pi designation as appropriate. (2 points)

Atom	Bonds formed (use sigma and pi as appropriate)
Carbon (<i>a</i>)	
Carbon (<i>b</i>)	

iv. How many total sp^3 -s sigma bonds are in the molecule? (1 point)

v. Determine **electronic** and **molecular geometry** at carbon (C) labeled ***e*** and sulfur (S) labeled ***f*** centres. (2 points)

Atom	Electronic Geometry	Molecular Geometry
Carbon (<i>e</i>)		
Sulfur (<i>f</i>)		

vi. Circle the value from the choices given below, that describes the **C-S-C bond angle at sulfur** (S) marked ***f***: (0.5 points)

$>90^\circ$ to $<109.5^\circ$ 109.5° >109.5 to $<120^\circ$ 120° $>120^\circ$ to $<180^\circ$

**Question 8 12 points**

Using the **valence shell orbitals only** draw a **complete** molecular orbital (MO) diagram (with no s-p mixing) for the molecule **PS**, in the space provided below. **(6 points)**

Assume that 3s and 3p atomic orbitals will form molecular orbitals like those formed by the combination of 2s and 2p atomic orbitals that we learnt in class, and there is no s-p mixing.

Show **energy arrow**, **atomic orbitals** involved, **atom labels**, the **molecular orbitals** that are formed, **filled electrons** in the MO diagram etc. Make sure to label all the atomic and molecular orbitals.

Using the MO diagram above, answer the following questions:

- (i) Which one of the following will have the highest bond order? Circle the correct choice
A) PS^{1-} B) PS C) PS^{1+}
- (ii) Which one of the following will have the longest bond length? Circle the correct choice
A) PS^{1-} B) PS C) PS^{1+}
- (iii) Which of the following will be diamagnetic? Circle the correct choice.
A) PS^{1-} B) PS C) PS^{1+}
- (iv) Draw Lewis dot structure of the molecule with the shortest bond length (based on the MO diagram) from among the three (PS^{1-} , PS, PS^{1+}), and indicate the bond order calculated from the Lewis dot structure. **(3 points)**

Lewis Dot Structure

Bond Order



68B47489-EF58-46A1-9C96-5413B6070FC4

midterm-exam-93388

#11 Page 10 of 10

SCRAP PAGE

**CHEMISTRY 110 –Midterm Exam****Thursday November 7, 2024, 6:30 – 8:30 pm****Examiners: Pallavi Sirjoosingh and Ashok Kakkar**

First name (please write as legibly as possible within the boxes)

--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--

Last name

--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--

Student ID number

--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--

This booklet has 8 questions on 10 pages (The last page can be used as a scrap page).

The staple must NOT be removed from the question booklet. Your intact question booklet must be returned at the end of the exam.

A separate booklet contains a periodic table, and datasheet.

Write your answers directly under the questions in the space provided. Do not write on the QR codes or near the top edges (as it may be cut off).

Please write **legibly and dark enough** for the scan to be readable. For students who use English as a second language, dictionaries are permitted. Calculators are allowed.

DO NOT ASK QUESTIONS OF THE INVIGILATORS about interpretations of the exam questions. If you have a concern about a question, please summarize it at the top of this page. All concerns will be evaluated after the exam. GOOD LUCK!

**Question 1 8 points** (*Assume all transitions are possible*)

a. 6 points Calculate the **energy (in kJ/mol)** required to ionize (completely remove) the electron from the **n=3** level of **He⁺ ion** (He: Helium). Show your work including equations. (3 sig. figs.)

b. 2 points

(I) Select the largest (longest) wavelength from the three choices below:

- i. Wavelength of photon to *completely remove* an electron from *1s orbital of Hydrogen atom*
- ii. Wavelength of photon to *completely remove* an electron from *1s orbital of Boron atom (B)*
- iii. Wavelength of photon to *completely remove* an electron from *1s orbital of Boron cation (B⁺)*

i or ii or iii

(II) Select the smallest (shortest) wavelength from the three choices below (*for H atom*):

- i. Wavelength of photon *emitted* for an electron transitioning from *2p to 1s orbital*
- ii. Wavelength of photon *absorbed* for an electron transitioning from *2s to 4p orbital*
- iii. Wavelength of photon *absorbed* for an electron transitioning from *1s to 3p orbital*

i or ii or iii



Question 2 8 points



A Chem110 student plots the graph shown on the left, from data obtained for the kinetic energy (eV) of the ejected electron versus frequency (s^{-1}) of the incident light, for an unknown metal. ($1\text{eV} = 1.6022 \times 10^{-19} \text{ J}$)

Based on the plot, answer the questions **a to c** below:

a. 1 point On the plot above, label the **threshold frequency**.

b. 2 points What is the value of the work potential (in J) of the unknown metal?

c. 3 points What is the *longest* wavelength (in nm) required to eject an electron from this metal's surface? Show your work including equation(s) used. (3 sig figs)

d. 2 points In the space provided below, plot the following

(Assume frequency is *above* threshold frequency):

- Kinetic energy* of an ejected electron versus the *intensity* of the incident light (left)
- Number of electrons* ejected versus the *frequency* of the incident light (right)



**Question 3** **6 points****a. 2 points**

In the ground state of Ge (Germanium; $Z = 32$) atom, **name** the following orbitals:

i. Orbital containing the electron that requires the maximum energy to be removed.

ii. Orbital containing the electron that experiences the minimum effective nuclear charge.

b. 2 points

An unknown anion (A^-) has the following *excited* electronic configuration: $[Ar] 4s^2 3d^3 4p^1$

(Ar: Argon; $Z = 18$)

Give the **condensed ground state electronic configuration** of the atom, **A** and the cation, **A^{2+}** .
Show your work. (Assume all electronic transitions are possible)

Electronic Configuration of A:

Electronic Configuration of A^{2+} :

c. 2 points

In a single atom, determine how many electrons can have the following set of quantum numbers?

(i) $n = 5; l = 3$

In a single atom, determine how many electrons can have the following set of quantum numbers?

(i) $n = 3; l = 2; m_s = +1/2$

**Question 4** **8 points**

Rank the following from 1 to 4 in **increasing** order. Provide the ranking in the boxes provided.

Note: Atomic number (Z) are provided for relevant atoms

a) Increasing (lowest to highest) order of *ionization energy*: (1= lowest; 4 = highest)

Mg

Mg: Z = 12

Rb

Rb: Z = 37

Cs

Cs: Z = 55

K

K : Z = 19

b) Increasing (lowest to highest) order of *bond angle*: (1= lowest; 4 = highest)

H₂OCCl₄XeF₄

HCN

c) Increasing (smallest to largest) order of *ionic size* (1= smallest; 4 = largest)

S²⁻

S: Z = 16

O²⁻

O: Z = 8

Be²⁺

Be: Z = 4

Na⁺

Na: Z = 11

d) Increasing (lowest to highest) order of *bond energy* (1= lowest; 4 = highest)

P-F

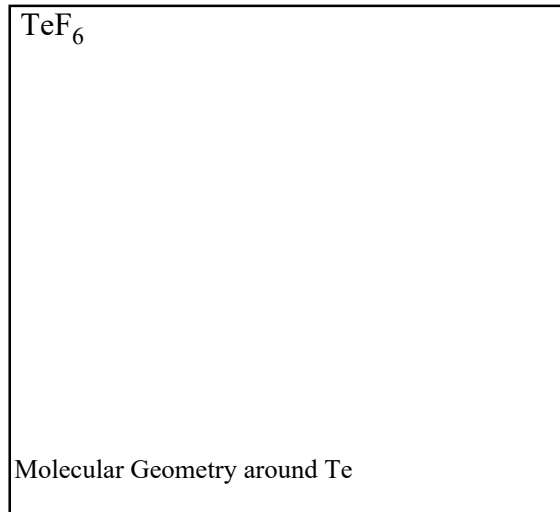
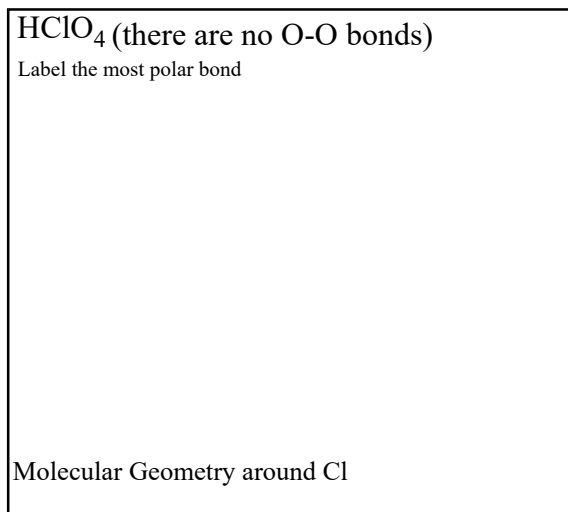
P-Cl

P-I

P-Br

**Question 5** **12 points**

a. 9 points For the molecules HClO_4 and TeF_6 , draw the most stable *Lewis structure* showing appropriate shape and *dashed/wedged bonds*, based on VSEPR. Determine the **molecular geometry** around the central atom. For HClO_4 , label the *most polar bond*.



b. 3 points (Note: Lone pairs are not shown in structures below)

(i) For the three *resonance* structures (1, 2 and 3) shown below, answer the following question:



(I) Circle the *most contributing* resonance structure.

Structure 1

Structure 2

Structure 3

(ii) For the three structures given (4, 5, and 6) below, answer the following questions:



(I) Circle all the structures that represent *resonance structures* of each other:

Structure 4

Structure 5

Structure 6

(II) Among the resonance structures, circle the *most contributing* resonance structure.

Structure 4

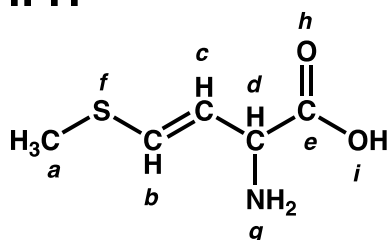
Structure 5

Structure 6

**Question 6** **6 points**

Using the data given below, calculate the **lattice energy** for SrO(s).

	ΔH° (kJ/mol)
$\text{Sr}(s) \longrightarrow \text{Sr}(g)$	+180
$\text{Sr}(g) \longrightarrow \text{Sr}^{2+}(g) + 2e^-$	+1468
$\text{O}_2(g) \longrightarrow 2\text{O}(g)$	+500
$\text{O}(g) + 2e^- \longrightarrow \text{O}^{2-}(g)$	+600
$\text{Sr}(s) + \frac{1}{2}\text{O}_2(g) \longrightarrow \text{SrO}(s)$	-550

**Question 7 10 points**

Consider the molecule given on the left, and answer questions (i) to (vi) using chemical bonding theories that we have learnt in class (valence bond theory, hybridization).

Note that the lone pairs are not shown in the structure:

i. Consider the carbon atoms marked in as ***a***, ***b***, ***c***, ***d***, and ***e*** in the molecule, and determine **hybridization** at each of the carbon centers. (2.5 points)

Atom	Hybridization
Carbon (<i>a</i>)	
Carbon (<i>b</i>)	
Carbon (<i>c</i>)	
Carbon (<i>d</i>)	
Carbon (<i>e</i>)	

ii. Now, consider the sulfur atom (S) marked as ***f***, nitrogen (N) atom (***g***), and oxygen(O) atoms (***h*** and ***i***), and determine their **hybridization**. (2 points)

Atom	Hybridization
Sulfur (<i>f</i>)	
Nitrogen (<i>g</i>)	
Oxygen (<i>h</i>)	
Oxygen (<i>i</i>)	

iii. **Describe** all the bonds formed by the carbon atoms ***a*** and ***b***, using sigma and pi designation as appropriate. (2 points)

Atom	Bonds formed (use sigma and pi as appropriate)
Carbon (<i>a</i>)	
Carbon (<i>b</i>)	

iv. How many total sp^3 -s sigma bonds are in the molecule? (1 point)

v. Determine **electronic** and **molecular geometry** at carbon (C) labeled ***e*** and sulfur (S) labeled ***f*** centres. (2 points)

Atom	Electronic Geometry	Molecular Geometry
Carbon (<i>e</i>)		
Sulfur (<i>f</i>)		

vi. Circle the value from the choices given below, that describes the **C-S-C bond angle at sulfur** (S) marked ***f***: (0.5 points)

>90° to <109.5° 109.5° >109.5 to <120° 120° >120° to <180°



Question 8 12 points

Using the **valence shell orbitals only** draw a **complete** molecular orbital (MO) diagram (with no s-p mixing) for the molecule **PS**, in the space provided below. **(6 points)**

Assume that 3s and 3p atomic orbitals will form molecular orbitals like those formed by the combination of 2s and 2p atomic orbitals that we learnt in class, and there is no s-p mixing.

Show **energy arrow**, **atomic orbitals** involved, **atom labels**, the **molecular orbitals** that are formed, **filled electrons** in the MO diagram etc. Make sure to label all the atomic and molecular orbitals.

Using the MO diagram above, answer the following questions:

- (i) Which one of the following will have the highest bond order? Circle the correct choice
 A) PS^{1-} B) PS C) PS^{1+}
- (ii) Which one of the following will have the longest bond length? Circle the correct choice
 A) PS^{1-} B) PS C) PS^{1+}
- (iii) Which of the following will be diamagnetic? Circle the correct choice.
 A) PS^{1-} B) PS C) PS^{1+}
- (iv) Draw Lewis dot structure of the molecule with the shortest bond length (based on the MO diagram) from among the three (PS^{1-} , PS, PS^{1+}), and indicate the bond order calculated from the Lewis dot structure. **(3 points)**

Lewis Dot Structure

Bond Order



10871FC4-105B-4D99-A66C-C097B124E742

midterm-exam-93388

#12 Page 10 of 10

SCRAP PAGE

**CHEMISTRY 110 –Midterm Exam****Thursday November 7, 2024, 6:30 – 8:30 pm**

Examiners: Pallavi Sirjoosingh and Ashok Kakkar

First name (please write as legibly as possible within the boxes)

--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--

Last name

--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--

Student ID number

--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--

This booklet has 8 questions on 10 pages (The last page can be used as a scrap page).

The staple must NOT be removed from the question booklet. Your intact question booklet must be returned at the end of the exam.

A separate booklet contains a periodic table, and datasheet.

Write your answers directly under the questions in the space provided. Do not write on the QR codes or near the top edges (as it may be cut off).

Please write **legibly and dark enough** for the scan to be readable. For students who use English as a second language, dictionaries are permitted. Calculators are allowed.

DO NOT ASK QUESTIONS OF THE INVIGILATORS about interpretations of the exam questions. If you have a concern about a question, please summarize it at the top of this page. All concerns will be evaluated after the exam. GOOD LUCK!

**Question 1 8 points** (*Assume all transitions are possible*)

a. 6 points Calculate the **energy (in kJ/mol)** required to ionize (completely remove) the electron from the **n=3** level of **He⁺ ion** (He: Helium). Show your work including equations. (3 sig. figs.)

b. 2 points

(I) Select the largest (longest) wavelength from the three choices below:

- i. Wavelength of photon to *completely remove* an electron from *1s orbital of Hydrogen atom*
- ii. Wavelength of photon to *completely remove* an electron from *1s orbital of Boron atom (B)*
- iii. Wavelength of photon to *completely remove* an electron from *1s orbital of Boron cation (B⁺)*

i or ii or iii

(II) Select the smallest (shortest) wavelength from the three choices below (*for H atom*):

- i. Wavelength of photon *emitted* for an electron transitioning from *2p to 1s orbital*
- ii. Wavelength of photon *absorbed* for an electron transitioning from *2s to 4p orbital*
- iii. Wavelength of photon *absorbed* for an electron transitioning from *1s to 3p orbital*

i or ii or iii



Question 2 8 points



A Chem110 student plots the graph shown on the left, from data obtained for the kinetic energy (eV) of the ejected electron versus frequency (s^{-1}) of the incident light, for an unknown metal. ($1\text{eV} = 1.6022 \times 10^{-19} \text{ J}$)

Based on the plot, answer the questions **a to c** below:

a. 1 point On the plot above, label the **threshold frequency**.

b. 2 points What is the value of the work potential (in J) of the unknown metal?

c. 3 points What is the *longest* wavelength (in nm) required to eject an electron from this metal's surface? Show your work including equation(s) used. (3 sig figs)

d. 2 points In the space provided below, plot the following

(Assume frequency is *above* threshold frequency):

- i. *Kinetic energy* of an ejected electron versus the *intensity* of the incident light (left)
- ii. *Number of electrons* ejected versus the *frequency* of the incident light (right)



**Question 3** **6 points****a. 2 points**

In the ground state of Ge (Germanium; $Z = 32$) atom, **name** the following orbitals:

i. Orbital containing the electron that requires the maximum energy to be removed.

ii. Orbital containing the electron that experiences the minimum effective nuclear charge.

b. 2 points

An unknown anion (A^-) has the following *excited* electronic configuration: $[Ar] 4s^2 3d^3 4p^1$

(Ar: Argon; $Z = 18$)

Give the **condensed ground state electronic configuration** of the atom, **A** and the cation, **A^{2+}** .
Show your work. (Assume all electronic transitions are possible)

Electronic Configuration of A:

Electronic Configuration of A^{2+} :

c. 2 points

In a single atom, determine how many electrons can have the following set of quantum numbers?

(i) $n = 5; l = 3$

In a single atom, determine how many electrons can have the following set of quantum numbers?

(i) $n = 3; l = 2; m_s = +1/2$

**Question 4** **8 points**

Rank the following from 1 to 4 in **increasing** order. Provide the ranking in the boxes provided.

Note: Atomic number (Z) are provided for relevant atoms

a) Increasing (lowest to highest) order of *ionization energy*: (1= lowest; 4 = highest)

Mg

Mg: Z = 12

Rb

Rb: Z = 37

Cs

Cs: Z = 55

K

K : Z = 19

b) Increasing (lowest to highest) order of *bond angle*: (1= lowest; 4 = highest)

H₂OCCl₄XeF₄

HCN

c) Increasing (smallest to largest) order of *ionic size* (1= smallest; 4 = largest)

S²⁻

S: Z = 16

O²⁻

O: Z = 8

Be²⁺

Be: Z = 4

Na⁺

Na: Z = 11

d) Increasing (lowest to highest) order of *bond energy* (1= lowest; 4 = highest)

P-F

P-Cl

P-I

P-Br

**Question 5** **12 points**

a. 9 points For the molecules HClO_4 and TeF_6 , draw the most stable *Lewis structure* showing appropriate shape and *dashed/wedged bonds*, based on VSEPR. Determine the **molecular geometry** around the central atom. For HClO_4 , label the *most polar bond*.



b. 3 points (Note: Lone pairs are not shown in structures below)

(i) For the three *resonance* structures (1, 2 and 3) shown below, answer the following question:



(I) Circle the *most contributing* resonance structure.

Structure 1

Structure 2

Structure 3

(ii) For the three structures given (4, 5, and 6) below, answer the following questions:



(I) Circle all the structures that represent *resonance structures* of each other:

Structure 4

Structure 5

Structure 6

(II) Among the resonance structures, circle the *most contributing* resonance structure.

Structure 4

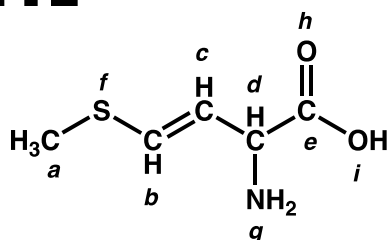
Structure 5

Structure 6

**Question 6** **6 points**

Using the data given below, calculate the **lattice energy** for SrO(s).

	ΔH° (kJ/mol)
$\text{Sr}(s) \longrightarrow \text{Sr}(g)$	+180
$\text{Sr}(g) \longrightarrow \text{Sr}^{2+}(g) + 2e^-$	+1468
$\text{O}_2(g) \longrightarrow 2\text{O}(g)$	+500
$\text{O}(g) + 2e^- \longrightarrow \text{O}^{2-}(g)$	+600
$\text{Sr}(s) + \frac{1}{2}\text{O}_2(g) \longrightarrow \text{SrO}(s)$	-550

**Question 7 10 points**

Consider the molecule given on the left, and answer questions (i) to (vi) using chemical bonding theories that we have learnt in class (valence bond theory, hybridization).

Note that the lone pairs are not shown in the structure:

i. Consider the carbon atoms marked in as ***a***, ***b***, ***c***, ***d***, and ***e*** in the molecule, and determine **hybridization** at each of the carbon centers. (2.5 points)

Atom	Hybridization
Carbon (<i>a</i>)	
Carbon (<i>b</i>)	
Carbon (<i>c</i>)	
Carbon (<i>d</i>)	
Carbon (<i>e</i>)	

ii. Now, consider the sulfur atom (S) marked as ***f***, nitrogen (N) atom (***g***), and oxygen(O) atoms (***h*** and ***i***), and determine their **hybridization**. (2 points)

Atom	Hybridization
Sulfur (<i>f</i>)	
Nitrogen (<i>g</i>)	
Oxygen (<i>h</i>)	
Oxygen (<i>i</i>)	

iii. **Describe** all the bonds formed by the carbon atoms ***a*** and ***b***, using sigma and pi designation as appropriate. (2 points)

Atom	Bonds formed (use sigma and pi as appropriate)
Carbon (<i>a</i>)	
Carbon (<i>b</i>)	

iv. How many total sp^3 -s sigma bonds are in the molecule? (1 point)

v. Determine **electronic** and **molecular geometry** at carbon (C) labeled ***e*** and sulfur (S) labeled ***f*** centres. (2 points)

Atom	Electronic Geometry	Molecular Geometry
Carbon (<i>e</i>)		
Sulfur (<i>f</i>)		

vi. Circle the value from the choices given below, that describes the **C-S-C bond angle at sulfur** (S) marked ***f***: (0.5 points)

>90° to <109.5° 109.5° >109.5 to <120° 120° >120° to <180°


Question 8 12 points

Using the **valence shell orbitals only** draw a **complete** molecular orbital (MO) diagram (with no s-p mixing) for the molecule **PS**, in the space provided below. **(6 points)**

Assume that 3s and 3p atomic orbitals will form molecular orbitals like those formed by the combination of 2s and 2p atomic orbitals that we learnt in class, and there is no s-p mixing.

Show **energy arrow**, **atomic orbitals** involved, **atom labels**, the **molecular orbitals** that are formed, **filled electrons** in the MO diagram etc. Make sure to label all the atomic and molecular orbitals.

Using the MO diagram above, answer the following questions:

- (i) Which one of the following will have the highest bond order? Circle the correct choice
 A) PS^{1-} B) PS C) PS^{1+}
- (ii) Which one of the following will have the longest bond length? Circle the correct choice
 A) PS^{1-} B) PS C) PS^{1+}
- (iii) Which of the following will be diamagnetic? Circle the correct choice.
 A) PS^{1-} B) PS C) PS^{1+}
- (iv) Draw Lewis dot structure of the molecule with the shortest bond length (based on the MO diagram) from among the three (PS^{1-} , PS, PS^{1+}), and indicate the bond order calculated from the Lewis dot structure. **(3 points)**

Lewis Dot Structure

Bond Order



AD18C21B-D445-4A2B-9558-80743AC918C3

midterm-exam-93388

#13 Page 10 of 10

SCRAP PAGE

**CHEMISTRY 110 –Midterm Exam****Thursday November 7, 2024, 6:30 – 8:30 pm**

Examiners: Pallavi Sirjoosingh and Ashok Kakkar

First name (please write as legibly as possible within the boxes)

--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--

Last name

--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--

Student ID number

--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--

This booklet has 8 questions on 10 pages (The last page can be used as a scrap page).

The staple must NOT be removed from the question booklet. Your intact question booklet must be returned at the end of the exam.

A separate booklet contains a periodic table, and datasheet.

Write your answers directly under the questions in the space provided. Do not write on the QR codes or near the top edges (as it may be cut off).

Please write **legibly and dark enough** for the scan to be readable. For students who use English as a second language, dictionaries are permitted. Calculators are allowed.

DO NOT ASK QUESTIONS OF THE INVIGILATORS about interpretations of the exam questions. If you have a concern about a question, please summarize it at the top of this page. All concerns will be evaluated after the exam. GOOD LUCK!

**Question 1 8 points** (*Assume all transitions are possible*)

a. 6 points Calculate the **energy (in kJ/mol)** required to ionize (completely remove) the electron from the **$n=3$** level of **He^+ ion** (He: Helium). Show your work including equations. (3 sig. figs.)

b. 2 points

(I) Select the largest (longest) wavelength from the three choices below:

- i. Wavelength of photon to *completely remove* an electron from *1s orbital of Hydrogen atom*
- ii. Wavelength of photon to *completely remove* an electron from *1s orbital of Boron atom (B)*
- iii. Wavelength of photon to *completely remove* an electron from *1s orbital of Boron cation (B^+)*

i or ii or iii

(II) Select the smallest (shortest) wavelength from the three choices below (*for H atom*):

- i. Wavelength of photon *emitted* for an electron transitioning from *2p to 1s orbital*
- ii. Wavelength of photon *absorbed* for an electron transitioning from *2s to 4p orbital*
- iii. Wavelength of photon *absorbed* for an electron transitioning from *1s to 3p orbital*

i or ii or iii



Question 2 8 points



A Chem110 student plots the graph shown on the left, from data obtained for the kinetic energy (eV) of the ejected electron versus frequency (s^{-1}) of the incident light, for an unknown metal. ($1\text{eV} = 1.6022 \times 10^{-19} \text{ J}$)

Based on the plot, answer the questions **a to c** below:

a. 1 point On the plot above, label the **threshold frequency**.

b. 2 points What is the value of the work potential (in J) of the unknown metal?

c. 3 points What is the *longest* wavelength (in nm) required to eject an electron from this metal's surface? Show your work including equation(s) used. (3 sig figs)

d. 2 points In the space provided below, plot the following

(Assume frequency is *above* threshold frequency):

- i. *Kinetic energy* of an ejected electron versus the *intensity* of the incident light (left)
- ii. *Number of electrons* ejected versus the *frequency* of the incident light (right)



**Question 3** **6 points****a. 2 points**

In the ground state of Ge (Germanium; $Z = 32$) atom, **name** the following orbitals:

i. Orbital containing the electron that requires the maximum energy to be removed.

ii. Orbital containing the electron that experiences the minimum effective nuclear charge.

b. 2 points

An unknown anion (A^-) has the following *excited* electronic configuration: $[Ar] 4s^2 3d^3 4p^1$

(Ar: Argon; $Z = 18$)

Give the **condensed ground state electronic configuration** of the atom, **A** and the cation, **A^{2+}** .
Show your work. (Assume all electronic transitions are possible)

Electronic Configuration of A:

Electronic Configuration of A^{2+} :

c. 2 points

In a single atom, determine how many electrons can have the following set of quantum numbers?

(i) $n = 5; l = 3$

In a single atom, determine how many electrons can have the following set of quantum numbers?

(i) $n = 3; l = 2; m_s = +1/2$

**Question 4** **8 points**

Rank the following from 1 to 4 in **increasing** order. Provide the ranking in the boxes provided.

Note: Atomic number (Z) are provided for relevant atoms

a) Increasing (lowest to highest) order of *ionization energy*: (1= lowest; 4 = highest)

Mg

Mg: Z = 12

Rb

Rb: Z = 37

Cs

Cs: Z = 55

K

K : Z = 19

b) Increasing (lowest to highest) order of *bond angle*: (1= lowest; 4 = highest)

H₂OCCl₄XeF₄

HCN

c) Increasing (smallest to largest) order of *ionic size* (1= smallest; 4 = largest)

S²⁻

S: Z = 16

O²⁻

O: Z = 8

Be²⁺

Be: Z = 4

Na⁺

Na: Z = 11

d) Increasing (lowest to highest) order of *bond energy* (1= lowest; 4 = highest)

P-F

P-Cl

P-I

P-Br

**Question 5** **12 points**

a. 9 points For the molecules HClO_4 and TeF_6 , draw the most stable *Lewis structure* showing appropriate shape and *dashed/wedged bonds*, based on VSEPR. Determine the **molecular geometry** around the central atom. For HClO_4 , label the *most polar bond*.



b. 3 points (Note: Lone pairs are not shown in structures below)

(i) For the three *resonance* structures (1, 2 and 3) shown below, answer the following question:



(I) Circle the *most contributing* resonance structure.

Structure 1

Structure 2

Structure 3

(ii) For the three structures given (4, 5, and 6) below, answer the following questions:



(I) Circle all the structures that represent *resonance structures* of each other:

Structure 4

Structure 5

Structure 6

(II) Among the resonance structures, circle the *most contributing* resonance structure.

Structure 4

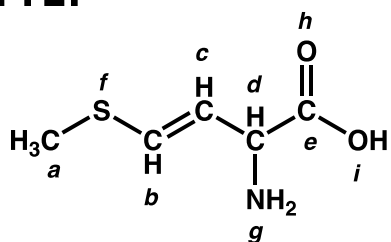
Structure 5

Structure 6

**Question 6** **6 points**

Using the data given below, calculate the **lattice energy** for SrO(s).

	ΔH° (kJ/mol)
$\text{Sr}(s) \longrightarrow \text{Sr}(g)$	+180
$\text{Sr}(g) \longrightarrow \text{Sr}^{2+}(g) + 2e^-$	+1468
$\text{O}_2(g) \longrightarrow 2\text{O}(g)$	+500
$\text{O}(g) + 2e^- \longrightarrow \text{O}^{2-}(g)$	+600
$\text{Sr}(s) + \frac{1}{2}\text{O}_2(g) \longrightarrow \text{SrO}(s)$	-550

**Question 7 10 points**

Consider the molecule given on the left, and answer questions (i) to (vi) using chemical bonding theories that we have learnt in class (valence bond theory, hybridization).

Note that the lone pairs are not shown in the structure:

i. Consider the carbon atoms marked in as ***a***, ***b***, ***c***, ***d***, and ***e*** in the molecule, and determine **hybridization** at each of the carbon centers. (2.5 points)

Atom	Hybridization
Carbon (<i>a</i>)	
Carbon (<i>b</i>)	
Carbon (<i>c</i>)	
Carbon (<i>d</i>)	
Carbon (<i>e</i>)	

ii. Now, consider the sulfur atom (S) marked as ***f***, nitrogen (N) atom (***g***), and oxygen(O) atoms (***h*** and ***i***), and determine their **hybridization**. (2 points)

Atom	Hybridization
Sulfur (<i>f</i>)	
Nitrogen (<i>g</i>)	
Oxygen (<i>h</i>)	
Oxygen (<i>i</i>)	

iii. **Describe** all the bonds formed by the carbon atoms ***a*** and ***b***, using sigma and pi designation as appropriate. (2 points)

Atom	Bonds formed (use sigma and pi as appropriate)
Carbon (<i>a</i>)	
Carbon (<i>b</i>)	

iv. How many total sp^3 -s sigma bonds are in the molecule? (1 point)

v. Determine **electronic** and **molecular geometry** at carbon (C) labeled ***e*** and sulfur (S) labeled ***f*** centres. (2 points)

Atom	Electronic Geometry	Molecular Geometry
Carbon (<i>e</i>)		
Sulfur (<i>f</i>)		

vi. Circle the value from the choices given below, that describes the **C-S-C bond angle at sulfur** (S) marked ***f***: (0.5 points)

>90° to <109.5° 109.5° >109.5 to <120° 120° >120° to <180°


Question 8 12 points

Using the **valence shell orbitals only** draw a **complete** molecular orbital (MO) diagram (with no s-p mixing) for the molecule **PS**, in the space provided below. **(6 points)**

Assume that 3s and 3p atomic orbitals will form molecular orbitals like those formed by the combination of 2s and 2p atomic orbitals that we learnt in class, and there is no s-p mixing.

Show **energy arrow**, **atomic orbitals** involved, **atom labels**, the **molecular orbitals** that are formed, **filled electrons** in the MO diagram etc. Make sure to label all the atomic and molecular orbitals.

Using the MO diagram above, answer the following questions:

- (i) Which one of the following will have the highest bond order? Circle the correct choice
 A) PS^{1-} B) PS C) PS^{1+}
- (ii) Which one of the following will have the longest bond length? Circle the correct choice
 A) PS^{1-} B) PS C) PS^{1+}
- (iii) Which of the following will be diamagnetic? Circle the correct choice.
 A) PS^{1-} B) PS C) PS^{1+}
- (iv) Draw Lewis dot structure of the molecule with the shortest bond length (based on the MO diagram) from among the three (PS^{1-} , PS, PS^{1+}), and indicate the bond order calculated from the Lewis dot structure. **(3 points)**

Lewis Dot Structure

Bond Order



964EB078-E987-4398-B929-F2F2712B85B0

midterm-exam-93388

#14 Page 10 of 10

SCRAP PAGE

**CHEMISTRY 110 –Midterm Exam****Thursday November 7, 2024, 6:30 – 8:30 pm****Examiners: Pallavi Sirjoosingh and Ashok Kakkar**

First name (please write as legibly as possible within the boxes)

--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--

Last name

--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--

Student ID number

--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--

This booklet has 8 questions on 10 pages (The last page can be used as a scrap page).

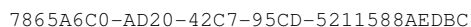
The staple must NOT be removed from the question booklet. Your intact question booklet must be returned at the end of the exam.

A separate booklet contains a periodic table, and datasheet.

Write your answers directly under the questions in the space provided. Do not write on the QR codes or near the top edges (as it may be cut off).

Please write **legibly and dark enough** for the scan to be readable. For students who use English as a second language, dictionaries are permitted. Calculators are allowed.

DO NOT ASK QUESTIONS OF THE INVIGILATORS about interpretations of the exam questions. If you have a concern about a question, please summarize it at the top of this page. All concerns will be evaluated after the exam. GOOD LUCK!



midterm-exam-93388

#15 Page 2 of 10

a. 6 points Calculate the **energy (in kJ/mol)** required to ionize (completely remove) the electron from the **n=3** level of **He⁺ ion** (He: Helium). Show your work including equations. (3 sig. figs.)

(I) Select the largest (longest) wavelength from the three choices below:

- Wavelength of photon to *completely remove* an electron from *1s orbital of Hydrogen atom*
- Wavelength of photon to *completely remove* an electron from *1s orbital of Boron atom (B)*
- Wavelength of photon to *completely remove* an electron from *1s orbital of Boron cation (B⁺)*

i or **ii** or **iii**

(II) Select the smallest (shortest) wavelength from the three choices below (*for H atom*):

- Wavelength of photon *emitted* for an electron transitioning from $2p$ to $1s$ orbital
- Wavelength of photon *absorbed* for an electron transitioning from $2s$ to $4p$ orbital
- Wavelength of photon *absorbed* for an electron transitioning from $1s$ to $3p$ orbital

i
or
ii
or
iii



Question 2 8 points



A Chem110 student plots the graph shown on the left, from data obtained for the kinetic energy (eV) of the ejected electron versus frequency (s^{-1}) of the incident light, for an unknown metal. ($1\text{eV} = 1.6022 \times 10^{-19} \text{ J}$)

Based on the plot, answer the questions **a to c** below:

a. 1 point On the plot above, label the **threshold frequency**.

b. 2 points What is the value of the work potential (in J) of the unknown metal?

c. 3 points What is the *longest* wavelength (in nm) required to eject an electron from this metal's surface? Show your work including equation(s) used. (3 sig figs)

d. 2 points In the space provided below, plot the following

(Assume frequency is *above* threshold frequency):

- i. *Kinetic energy* of an ejected electron versus the *intensity* of the incident light (left)
- ii. *Number of electrons* ejected versus the *frequency* of the incident light (right)



**Question 3** **6 points****a. 2 points**

In the ground state of Ge (Germanium; $Z = 32$) atom, **name** the following orbitals:

i. Orbital containing the electron that requires the maximum energy to be removed.

ii. Orbital containing the electron that experiences the minimum effective nuclear charge.

b. 2 points

An unknown anion (A^-) has the following *excited* electronic configuration: $[Ar] 4s^2 3d^3 4p^1$

(Ar: Argon; $Z = 18$)

Give the **condensed ground state electronic configuration** of the atom, **A** and the cation, **A^{2+}** .
Show your work. (Assume all electronic transitions are possible)

Electronic Configuration of A:

Electronic Configuration of A^{2+} :

c. 2 points

In a single atom, determine how many electrons can have the following set of quantum numbers?

(i) $n = 5; l = 3$

In a single atom, determine how many electrons can have the following set of quantum numbers?

(i) $n = 3; l = 2; m_s = +1/2$

**Question 4** **8 points**

Rank the following from 1 to 4 in **increasing** order. Provide the ranking in the boxes provided.

Note: Atomic number (Z) are provided for relevant atoms

a) Increasing (lowest to highest) order of *ionization energy*: (1= lowest; 4 = highest)

Mg

Mg: Z = 12

Rb

Rb: Z = 37

Cs

Cs: Z = 55

K

K : Z = 19

b) Increasing (lowest to highest) order of *bond angle*: (1= lowest; 4 = highest)

H₂OCCl₄XeF₄

HCN

c) Increasing (smallest to largest) order of *ionic size* (1= smallest; 4 = largest)

S²⁻

S: Z = 16

O²⁻

O: Z = 8

Be²⁺

Be: Z = 4

Na⁺

Na: Z = 11

d) Increasing (lowest to highest) order of *bond energy* (1= lowest; 4 = highest)

P-F

P-Cl

P-I

P-Br

**Question 5** **12 points**

a. 9 points For the molecules HClO_4 and TeF_6 , draw the most stable *Lewis structure* showing appropriate shape and *dashed/wedged bonds*, based on VSEPR. Determine the **molecular geometry** around the central atom. For HClO_4 , label the *most polar bond*.



b. 3 points (Note: Lone pairs are not shown in structures below)

(i) For the three *resonance* structures (1, 2 and 3) shown below, answer the following question:



(I) Circle the *most contributing* resonance structure.

Structure 1

Structure 2

Structure 3

(ii) For the three structures given (4, 5, and 6) below, answer the following questions:



(I) Circle all the structures that represent *resonance structures* of each other:

Structure 4

Structure 5

Structure 6

(II) Among the resonance structures, circle the *most contributing* resonance structure.

Structure 4

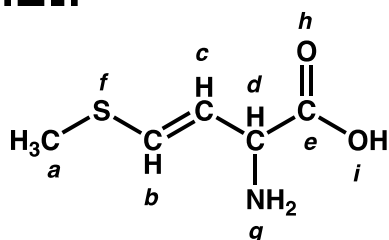
Structure 5

Structure 6

**Question 6** **6 points**

Using the data given below, calculate the **lattice energy** for SrO(s).

	ΔH° (kJ/mol)
$\text{Sr}(s) \longrightarrow \text{Sr}(g)$	+180
$\text{Sr}(g) \longrightarrow \text{Sr}^{2+}(g) + 2e^-$	+1468
$\text{O}_2(g) \longrightarrow 2\text{O}(g)$	+500
$\text{O}(g) + 2e^- \longrightarrow \text{O}^{2-}(g)$	+600
$\text{Sr}(s) + \frac{1}{2}\text{O}_2(g) \longrightarrow \text{SrO}(s)$	-550

**Question 7 10 points**

Consider the molecule given on the left, and answer questions (i) to (vi) using chemical bonding theories that we have learnt in class (valence bond theory, hybridization).

Note that the lone pairs are not shown in the structure:

i. Consider the carbon atoms marked in as ***a***, ***b***, ***c***, ***d***, and ***e*** in the molecule, and determine **hybridization** at each of the carbon centers. (2.5 points)

Atom	Hybridization
Carbon (<i>a</i>)	
Carbon (<i>b</i>)	
Carbon (<i>c</i>)	
Carbon (<i>d</i>)	
Carbon (<i>e</i>)	

ii. Now, consider the sulfur atom (S) marked as ***f***, nitrogen (N) atom (***g***), and oxygen(O) atoms (***h*** and ***i***), and determine their **hybridization**. (2 points)

Atom	Hybridization
Sulfur (<i>f</i>)	
Nitrogen (<i>g</i>)	
Oxygen (<i>h</i>)	
Oxygen (<i>i</i>)	

iii. **Describe** all the bonds formed by the carbon atoms ***a*** and ***b***, using sigma and pi designation as appropriate. (2 points)

Atom	Bonds formed (use sigma and pi as appropriate)
Carbon (<i>a</i>)	
Carbon (<i>b</i>)	

iv. How many total sp^3 -s sigma bonds are in the molecule? (1 point)

v. Determine **electronic** and **molecular geometry** at carbon (C) labeled ***e*** and sulfur (S) labeled ***f*** centres. (2 points)

Atom	Electronic Geometry	Molecular Geometry
Carbon (<i>e</i>)		
Sulfur (<i>f</i>)		

vi. Circle the value from the choices given below, that describes the **C-S-C bond angle at sulfur** (S) marked ***f***: (0.5 points)

>90° to <109.5° 109.5° >109.5 to <120° 120° >120° to <180°


Question 8 12 points

Using the **valence shell orbitals only** draw a **complete** molecular orbital (MO) diagram (with no s-p mixing) for the molecule **PS**, in the space provided below. **(6 points)**

Assume that 3s and 3p atomic orbitals will form molecular orbitals like those formed by the combination of 2s and 2p atomic orbitals that we learnt in class, and there is no s-p mixing.

Show **energy arrow**, **atomic orbitals** involved, **atom labels**, the **molecular orbitals** that are formed, **filled electrons** in the MO diagram etc. Make sure to label all the atomic and molecular orbitals.

Using the MO diagram above, answer the following questions:

- (i) Which one of the following will have the highest bond order? Circle the correct choice
 A) PS^{1-} B) PS C) PS^{1+}
- (ii) Which one of the following will have the longest bond length? Circle the correct choice
 A) PS^{1-} B) PS C) PS^{1+}
- (iii) Which of the following will be diamagnetic? Circle the correct choice.
 A) PS^{1-} B) PS C) PS^{1+}
- (iv) Draw Lewis dot structure of the molecule with the shortest bond length (based on the MO diagram) from among the three (PS^{1-} , PS, PS^{1+}), and indicate the bond order calculated from the Lewis dot structure. **(3 points)**

Lewis Dot Structure

Bond Order



F5D69F5F-A853-4B70-B702-DE09433900DF

midterm-exam-93388

#15 Page 10 of 10

SCRAP PAGE

**CHEMISTRY 110 –Midterm Exam****Thursday November 7, 2024, 6:30 – 8:30 pm****Examiners: Pallavi Sirjoosingh and Ashok Kakkar**

First name (please write as legibly as possible within the boxes)

--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--

Last name

--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--

Student ID number

--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--

This booklet has 8 questions on 10 pages (The last page can be used as a scrap page).

The staple must NOT be removed from the question booklet. Your intact question booklet must be returned at the end of the exam.

A separate booklet contains a periodic table, and datasheet.

Write your answers directly under the questions in the space provided. Do not write on the QR codes or near the top edges (as it may be cut off).

Please write **legibly and dark enough** for the scan to be readable. For students who use English as a second language, dictionaries are permitted. Calculators are allowed.

DO NOT ASK QUESTIONS OF THE INVIGILATORS about interpretations of the exam questions. If you have a concern about a question, please summarize it at the top of this page. All concerns will be evaluated after the exam. GOOD LUCK!

**Question 1 8 points** (*Assume all transitions are possible*)

a. 6 points Calculate the **energy (in kJ/mol)** required to ionize (completely remove) the electron from the **$n=3$** level of **He^+ ion** (He: Helium). Show your work including equations. (3 sig. figs.)

b. 2 points

(I) Select the largest (longest) wavelength from the three choices below:

- i. Wavelength of photon to *completely remove* an electron from *1s orbital of Hydrogen atom*
- ii. Wavelength of photon to *completely remove* an electron from *1s orbital of Boron atom (B)*
- iii. Wavelength of photon to *completely remove* an electron from *1s orbital of Boron cation (B^+)*

i or ii or iii

(II) Select the smallest (shortest) wavelength from the three choices below (*for H atom*):

- i. Wavelength of photon *emitted* for an electron transitioning from *2p to 1s orbital*
- ii. Wavelength of photon *absorbed* for an electron transitioning from *2s to 4p orbital*
- iii. Wavelength of photon *absorbed* for an electron transitioning from *1s to 3p orbital*

i or ii or iii



Question 2 8 points



A Chem110 student plots the graph shown on the left, from data obtained for the kinetic energy (eV) of the ejected electron versus frequency (s^{-1}) of the incident light, for an unknown metal. ($1\text{eV} = 1.6022 \times 10^{-19} \text{ J}$)

Based on the plot, answer the questions **a to c** below:

a. 1 point On the plot above, label the **threshold frequency**.

b. 2 points What is the value of the work potential (in J) of the unknown metal?

c. 3 points What is the *longest* wavelength (in nm) required to eject an electron from this metal's surface? Show your work including equation(s) used. (3 sig figs)

d. 2 points In the space provided below, plot the following

(Assume frequency is *above* threshold frequency):

- i. *Kinetic energy* of an ejected electron versus the *intensity* of the incident light (left)
- ii. *Number of electrons* ejected versus the *frequency* of the incident light (right)



**Question 3** **6 points****a. 2 points**

In the ground state of Ge (Germanium; $Z = 32$) atom, **name** the following orbitals:

i. Orbital containing the electron that requires the maximum energy to be removed.

ii. Orbital containing the electron that experiences the minimum effective nuclear charge.

b. 2 points

An unknown anion (A^-) has the following *excited* electronic configuration: $[Ar] 4s^2 3d^3 4p^1$

(Ar: Argon; $Z = 18$)

Give the **condensed ground state electronic configuration** of the atom, **A** and the cation, **A^{2+}** .
Show your work. (Assume all electronic transitions are possible)

Electronic Configuration of A:

Electronic Configuration of A^{2+} :

c. 2 points

In a single atom, determine how many electrons can have the following set of quantum numbers?

(i) $n = 5; l = 3$

In a single atom, determine how many electrons can have the following set of quantum numbers?

(i) $n = 3; l = 2; m_s = +1/2$

**Question 4** **8 points**

Rank the following from 1 to 4 in **increasing** order. Provide the ranking in the boxes provided.

Note: Atomic number (Z) are provided for relevant atoms

a) Increasing (lowest to highest) order of *ionization energy*: (1= lowest; 4 = highest)

Mg

Mg: Z = 12

Rb

Rb: Z = 37

Cs

Cs: Z = 55

K

K : Z = 19

b) Increasing (lowest to highest) order of *bond angle*: (1= lowest; 4 = highest)

H₂OCCl₄XeF₄

HCN

c) Increasing (smallest to largest) order of *ionic size* (1= smallest; 4 = largest)

S²⁻

S: Z = 16

O²⁻

O: Z = 8

Be²⁺

Be: Z = 4

Na⁺

Na: Z = 11

d) Increasing (lowest to highest) order of *bond energy* (1= lowest; 4 = highest)

P-F

P-Cl

P-I

P-Br

**Question 5** **12 points**

a. 9 points For the molecules HClO_4 and TeF_6 , draw the most stable *Lewis structure* showing appropriate shape and *dashed/wedged bonds*, based on VSEPR. Determine the **molecular geometry** around the central atom. For HClO_4 , label the *most polar bond*.

HClO_4 (there are no O-O bonds)

Label the most polar bond

Molecular Geometry around Cl

TeF_6

Molecular Geometry around Te

b. 3 points (Note: Lone pairs are not shown in structures below)

(i) For the three *resonance* structures (1, 2 and 3) shown below, answer the following question:



(I) Circle the *most contributing* resonance structure.

Structure 1

Structure 2

Structure 3

(ii) For the three structures given (4, 5, and 6) below, answer the following questions:



(I) Circle all the structures that represent *resonance structures* of each other:

Structure 4

Structure 5

Structure 6

(II) Among the resonance structures, circle the *most contributing* resonance structure.

Structure 4

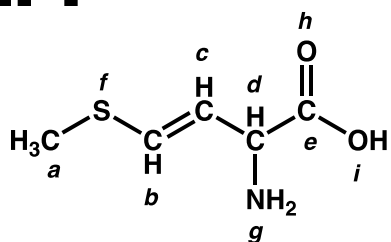
Structure 5

Structure 6

**Question 6** **6 points**

Using the data given below, calculate the **lattice energy** for SrO(s).

	ΔH° (kJ/mol)
$\text{Sr}(s) \longrightarrow \text{Sr}(g)$	+180
$\text{Sr}(g) \longrightarrow \text{Sr}^{2+}(g) + 2e^-$	+1468
$\text{O}_2(g) \longrightarrow 2\text{O}(g)$	+500
$\text{O}(g) + 2e^- \longrightarrow \text{O}^{2-}(g)$	+600
$\text{Sr}(s) + \frac{1}{2}\text{O}_2(g) \longrightarrow \text{SrO}(s)$	-550

**Question 7 10 points**

Consider the molecule given on the left, and answer questions (i) to (vi) using chemical bonding theories that we have learnt in class (valence bond theory, hybridization).

Note that the lone pairs are not shown in the structure:

i. Consider the carbon atoms marked in as ***a***, ***b***, ***c***, ***d***, and ***e*** in the molecule, and determine **hybridization** at each of the carbon centers. (2.5 points)

Atom	Hybridization
Carbon (<i>a</i>)	
Carbon (<i>b</i>)	
Carbon (<i>c</i>)	
Carbon (<i>d</i>)	
Carbon (<i>e</i>)	

ii. Now, consider the sulfur atom (S) marked as ***f***, nitrogen (N) atom (***g***), and oxygen(O) atoms (***h*** and ***i***), and determine their **hybridization**. (2 points)

Atom	Hybridization
Sulfur (<i>f</i>)	
Nitrogen (<i>g</i>)	
Oxygen (<i>h</i>)	
Oxygen (<i>i</i>)	

iii. **Describe** all the bonds formed by the carbon atoms ***a*** and ***b***, using sigma and pi designation as appropriate. (2 points)

Atom	Bonds formed (use sigma and pi as appropriate)
Carbon (<i>a</i>)	
Carbon (<i>b</i>)	

iv. How many total sp^3 -s sigma bonds are in the molecule? (1 point)

v. Determine **electronic** and **molecular geometry** at carbon (C) labeled ***e*** and sulfur (S) labeled ***f*** centres. (2 points)

Atom	Electronic Geometry	Molecular Geometry
Carbon (<i>e</i>)		
Sulfur (<i>f</i>)		

vi. Circle the value from the choices given below, that describes the **C-S-C bond angle at sulfur** (S) marked ***f***: (0.5 points)

>90° to <109.5°

109.5°

>109.5 to <120°

120°

>120° to <180°

**Question 8 12 points**

Using the **valence shell orbitals only** draw a **complete** molecular orbital (MO) diagram (with no s-p mixing) for the molecule **PS**, in the space provided below. **(6 points)**

Assume that 3s and 3p atomic orbitals will form molecular orbitals like those formed by the combination of 2s and 2p atomic orbitals that we learnt in class, and there is no s-p mixing.

Show **energy arrow**, **atomic orbitals** involved, **atom labels**, the **molecular orbitals** that are formed, **filled electrons** in the MO diagram etc. Make sure to label all the atomic and molecular orbitals.

Using the MO diagram above, answer the following questions:

- (i) Which one of the following will have the highest bond order? Circle the correct choice
A) PS^{1-} B) PS C) PS^{1+}
- (ii) Which one of the following will have the longest bond length? Circle the correct choice
A) PS^{1-} B) PS C) PS^{1+}
- (iii) Which of the following will be diamagnetic? Circle the correct choice.
A) PS^{1-} B) PS C) PS^{1+}
- (iv) Draw Lewis dot structure of the molecule with the shortest bond length (based on the MO diagram) from among the three (PS^{1-} , PS, PS^{1+}), and indicate the bond order calculated from the Lewis dot structure. **(3 points)**

Lewis Dot Structure

Bond Order



3699E4D5-973B-4471-94EA-A271806127EE

midterm-exam-93388

#16 Page 10 of 10

SCRAP PAGE

**CHEMISTRY 110 –Midterm Exam****Thursday November 7, 2024, 6:30 – 8:30 pm****Examiners: Pallavi Sirjoosingh and Ashok Kakkar**

First name (please write as legibly as possible within the boxes)

--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--

Last name

--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--

Student ID number

--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--

This booklet has 8 questions on 10 pages (The last page can be used as a scrap page).

The staple must NOT be removed from the question booklet. Your intact question booklet must be returned at the end of the exam.

A separate booklet contains a periodic table, and datasheet.

Write your answers directly under the questions in the space provided. Do not write on the QR codes or near the top edges (as it may be cut off).

Please write **legibly and dark enough** for the scan to be readable. For students who use English as a second language, dictionaries are permitted. Calculators are allowed.

DO NOT ASK QUESTIONS OF THE INVIGILATORS about interpretations of the exam questions. If you have a concern about a question, please summarize it at the top of this page. All concerns will be evaluated after the exam. GOOD LUCK!

**Question 1 8 points** (*Assume all transitions are possible*)

a. 6 points Calculate the **energy (in kJ/mol)** required to ionize (completely remove) the electron from the **n=3** level of **He⁺ ion** (He: Helium). Show your work including equations. (3 sig. figs.)

b. 2 points

(I) Select the largest (longest) wavelength from the three choices below:

- i. Wavelength of photon to *completely remove* an electron from *1s orbital of Hydrogen atom*
- ii. Wavelength of photon to *completely remove* an electron from *1s orbital of Boron atom (B)*
- iii. Wavelength of photon to *completely remove* an electron from *1s orbital of Boron cation (B⁺)*

i or ii or iii

(II) Select the smallest (shortest) wavelength from the three choices below (*for H atom*):

- i. Wavelength of photon *emitted* for an electron transitioning from *2p to 1s orbital*
- ii. Wavelength of photon *absorbed* for an electron transitioning from *2s to 4p orbital*
- iii. Wavelength of photon *absorbed* for an electron transitioning from *1s to 3p orbital*

i or ii or iii



Question 2 8 points



A Chem110 student plots the graph shown on the left, from data obtained for the kinetic energy (eV) of the ejected electron versus frequency (s^{-1}) of the incident light, for an unknown metal. ($1\text{eV} = 1.6022 \times 10^{-19} \text{ J}$)

Based on the plot, answer the questions **a to c** below:

a. 1 point On the plot above, label the **threshold frequency**.

b. 2 points What is the value of the work potential (in J) of the unknown metal?

c. 3 points What is the *longest* wavelength (in nm) required to eject an electron from this metal's surface? Show your work including equation(s) used. (3 sig figs)

d. 2 points In the space provided below, plot the following

(Assume frequency is *above* threshold frequency):

- Kinetic energy* of an ejected electron versus the *intensity* of the incident light (left)
- Number of electrons* ejected versus the *frequency* of the incident light (right)



**Question 3** **6 points****a. 2 points**

In the ground state of Ge (Germanium; $Z = 32$) atom, **name** the following orbitals:

i. Orbital containing the electron that requires the maximum energy to be removed.

ii. Orbital containing the electron that experiences the minimum effective nuclear charge.

b. 2 points

An unknown anion (A^-) has the following *excited* electronic configuration: $[Ar] 4s^2 3d^3 4p^1$

(Ar: Argon; $Z = 18$)

Give the **condensed ground state electronic configuration** of the atom, **A** and the cation, **A^{2+}** .
Show your work. (Assume all electronic transitions are possible)

Electronic Configuration of A:

Electronic Configuration of A^{2+} :

c. 2 points

In a single atom, determine how many electrons can have the following set of quantum numbers?

(i) $n = 5; l = 3$

In a single atom, determine how many electrons can have the following set of quantum numbers?

(i) $n = 3; l = 2; m_s = +1/2$

**Question 4** **8 points**

Rank the following from 1 to 4 in **increasing** order. Provide the ranking in the boxes provided.

Note: Atomic number (Z) are provided for relevant atoms

a) Increasing (lowest to highest) order of *ionization energy*: (1= lowest; 4 = highest)

Mg

Mg: Z = 12

Rb

Rb: Z = 37

Cs

Cs: Z = 55

K

K : Z = 19

b) Increasing (lowest to highest) order of *bond angle*: (1= lowest; 4 = highest)

H₂OCCl₄XeF₄

HCN

c) Increasing (smallest to largest) order of *ionic size* (1= smallest; 4 = largest)

S²⁻

S: Z = 16

O²⁻

O: Z = 8

Be²⁺

Be: Z = 4

Na⁺

Na: Z = 11

d) Increasing (lowest to highest) order of *bond energy* (1= lowest; 4 = highest)

P-F

P-Cl

P-I

P-Br

**Question 5** **12 points**

a. 9 points For the molecules HClO_4 and TeF_6 , draw the most stable *Lewis structure* showing appropriate shape and *dashed/wedged bonds*, based on VSEPR. Determine the **molecular geometry** around the central atom. For HClO_4 , label the *most polar bond*.



b. 3 points (Note: Lone pairs are not shown in structures below)

(i) For the three *resonance* structures (1, 2 and 3) shown below, answer the following question:



(I) Circle the *most contributing* resonance structure.

Structure 1

Structure 2

Structure 3

(ii) For the three structures given (4, 5, and 6) below, answer the following questions:



(I) Circle all the structures that represent *resonance structures* of each other:

Structure 4

Structure 5

Structure 6

(II) Among the resonance structures, circle the *most contributing* resonance structure.

Structure 4

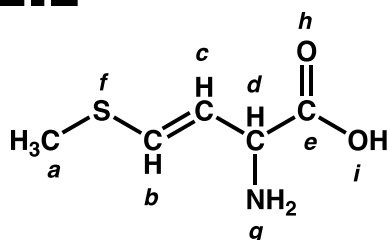
Structure 5

Structure 6

**Question 6** **6 points**

Using the data given below, calculate the **lattice energy** for SrO(s).

	ΔH° (kJ/mol)
$\text{Sr}(s) \longrightarrow \text{Sr}(g)$	+180
$\text{Sr}(g) \longrightarrow \text{Sr}^{2+}(g) + 2e^-$	+1468
$\text{O}_2(g) \longrightarrow 2\text{O}(g)$	+500
$\text{O}(g) + 2e^- \longrightarrow \text{O}^{2-}(g)$	+600
$\text{Sr}(s) + \frac{1}{2}\text{O}_2(g) \longrightarrow \text{SrO}(s)$	-550

**Question 7 10 points**

Consider the molecule given on the left, and answer questions (i) to (vi) using chemical bonding theories that we have learnt in class (valence bond theory, hybridization).

Note that the lone pairs are not shown in the structure:

i. Consider the carbon atoms marked in as ***a***, ***b***, ***c***, ***d***, and ***e*** in the molecule, and determine **hybridization** at each of the carbon centers. (2.5 points)

Atom	Hybridization
Carbon (<i>a</i>)	
Carbon (<i>b</i>)	
Carbon (<i>c</i>)	
Carbon (<i>d</i>)	
Carbon (<i>e</i>)	

ii. Now, consider the sulfur atom (S) marked as ***f***, nitrogen (N) atom (***g***), and oxygen(O) atoms (***h*** and ***i***), and determine their **hybridization**. (2 points)

Atom	Hybridization
Sulfur (<i>f</i>)	
Nitrogen (<i>g</i>)	
Oxygen (<i>h</i>)	
Oxygen (<i>i</i>)	

iii. **Describe** all the bonds formed by the carbon atoms ***a*** and ***b***, using sigma and pi designation as appropriate. (2 points)

Atom	Bonds formed (use sigma and pi as appropriate)
Carbon (<i>a</i>)	
Carbon (<i>b</i>)	

iv. How many total sp^3 -s sigma bonds are in the molecule? (1 point)

v. Determine **electronic** and **molecular geometry** at carbon (C) labeled ***e*** and sulfur (S) labeled ***f*** centres. (2 points)

Atom	Electronic Geometry	Molecular Geometry
Carbon (<i>e</i>)		
Sulfur (<i>f</i>)		

vi. Circle the value from the choices given below, that describes the **C-S-C bond angle at sulfur** (S) marked ***f***: (0.5 points)

>90° to <109.5°

109.5°

>109.5 to <120°

120°

>120° to <180°


Question 8 12 points

Using the **valence shell orbitals only** draw a **complete** molecular orbital (MO) diagram (with no s-p mixing) for the molecule **PS**, in the space provided below. **(6 points)**

Assume that 3s and 3p atomic orbitals will form molecular orbitals like those formed by the combination of 2s and 2p atomic orbitals that we learnt in class, and there is no s-p mixing.

Show **energy arrow**, **atomic orbitals** involved, **atom labels**, the **molecular orbitals** that are formed, **filled electrons** in the MO diagram etc. Make sure to label all the atomic and molecular orbitals.

Using the MO diagram above, answer the following questions:

- (i) Which one of the following will have the highest bond order? Circle the correct choice
 A) PS^{1-} B) PS C) PS^{1+}
- (ii) Which one of the following will have the longest bond length? Circle the correct choice
 A) PS^{1-} B) PS C) PS^{1+}
- (iii) Which of the following will be diamagnetic? Circle the correct choice.
 A) PS^{1-} B) PS C) PS^{1+}
- (iv) Draw Lewis dot structure of the molecule with the shortest bond length (based on the MO diagram) from among the three (PS^{1-} , PS, PS^{1+}), and indicate the bond order calculated from the Lewis dot structure. **(3 points)**

Lewis Dot Structure

Bond Order



A1852B1F-05E8-47B4-95F4-F46D96D108BC

midterm-exam-93388

#17 Page 10 of 10

SCRAP PAGE

**CHEMISTRY 110 –Midterm Exam****Thursday November 7, 2024, 6:30 – 8:30 pm**

Examiners: Pallavi Sirjoosingh and Ashok Kakkar

First name (please write as legibly as possible within the boxes)

--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--

Last name

--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--

Student ID number

--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--

This booklet has 8 questions on 10 pages (The last page can be used as a scrap page).

The staple must NOT be removed from the question booklet. Your intact question booklet must be returned at the end of the exam.

A separate booklet contains a periodic table, and datasheet.

Write your answers directly under the questions in the space provided. Do not write on the QR codes or near the top edges (as it may be cut off).

Please write **legibly and dark enough** for the scan to be readable. For students who use English as a second language, dictionaries are permitted. Calculators are allowed.

DO NOT ASK QUESTIONS OF THE INVIGILATORS about interpretations of the exam questions. If you have a concern about a question, please summarize it at the top of this page. All concerns will be evaluated after the exam. GOOD LUCK!

**Question 1 8 points** (*Assume all transitions are possible*)

a. 6 points Calculate the **energy (in kJ/mol)** required to ionize (completely remove) the electron from the **n=3** level of **He⁺ ion** (He: Helium). Show your work including equations. (3 sig. figs.)

b. 2 points

(I) Select the largest (longest) wavelength from the three choices below:

- i. Wavelength of photon to *completely remove* an electron from *1s orbital of Hydrogen atom*
- ii. Wavelength of photon to *completely remove* an electron from *1s orbital of Boron atom (B)*
- iii. Wavelength of photon to *completely remove* an electron from *1s orbital of Boron cation (B⁺)*

i or ii or iii

(II) Select the smallest (shortest) wavelength from the three choices below (*for H atom*):

- i. Wavelength of photon *emitted* for an electron transitioning from *2p to 1s orbital*
- ii. Wavelength of photon *absorbed* for an electron transitioning from *2s to 4p orbital*
- iii. Wavelength of photon *absorbed* for an electron transitioning from *1s to 3p orbital*

i or ii or iii



Question 2 8 points



A Chem110 student plots the graph shown on the left, from data obtained for the kinetic energy (eV) of the ejected electron versus frequency (s^{-1}) of the incident light, for an unknown metal. ($1\text{eV} = 1.6022 \times 10^{-19} \text{ J}$)

Based on the plot, answer the questions **a to c** below:

a. 1 point On the plot above, label the **threshold frequency**.

b. 2 points What is the value of the work potential (in J) of the unknown metal?

c. 3 points What is the *longest* wavelength (in nm) required to eject an electron from this metal's surface? Show your work including equation(s) used. (3 sig figs)

d. 2 points In the space provided below, plot the following

(Assume frequency is *above* threshold frequency):

- Kinetic energy* of an ejected electron versus the *intensity* of the incident light (left)
- Number of electrons* ejected versus the *frequency* of the incident light (right)



**Question 3** **6 points****a. 2 points**

In the ground state of Ge (Germanium; $Z = 32$) atom, **name** the following orbitals:

i. Orbital containing the electron that requires the maximum energy to be removed.

ii. Orbital containing the electron that experiences the minimum effective nuclear charge.

b. 2 points

An unknown anion (A^-) has the following *excited* electronic configuration: $[Ar] 4s^2 3d^3 4p^1$

(Ar: Argon; $Z = 18$)

Give the **condensed ground state electronic configuration** of the atom, **A** and the cation, **A^{2+}** .
Show your work. (Assume all electronic transitions are possible)

Electronic Configuration of A:

Electronic Configuration of A^{2+} :

c. 2 points

In a single atom, determine how many electrons can have the following set of quantum numbers?

(i) $n = 5; l = 3$

In a single atom, determine how many electrons can have the following set of quantum numbers?

(i) $n = 3; l = 2; m_s = +1/2$

**Question 4** **8 points**

Rank the following from 1 to 4 in **increasing** order. Provide the ranking in the boxes provided.

Note: Atomic number (Z) are provided for relevant atoms

a) Increasing (lowest to highest) order of *ionization energy*: (1= lowest; 4 = highest)

Mg

Mg: Z = 12

Rb

Rb: Z = 37

Cs

Cs: Z = 55

K

K : Z = 19

b) Increasing (lowest to highest) order of *bond angle*: (1= lowest; 4 = highest)

H₂OCCl₄XeF₄

HCN

c) Increasing (smallest to largest) order of *ionic size* (1= smallest; 4 = largest)

S²⁻

S: Z = 16

O²⁻

O: Z = 8

Be²⁺

Be: Z = 4

Na⁺

Na: Z = 11

d) Increasing (lowest to highest) order of *bond energy* (1= lowest; 4 = highest)

P-F

P-Cl

P-I

P-Br

**Question 5** **12 points**

a. 9 points For the molecules HClO_4 and TeF_6 , draw the most stable *Lewis structure* showing appropriate shape and *dashed/wedged bonds*, based on VSEPR. Determine the **molecular geometry** around the central atom. For HClO_4 , label the *most polar bond*.



b. 3 points (Note: Lone pairs are not shown in structures below)

(i) For the three *resonance* structures (1, 2 and 3) shown below, answer the following question:



(I) Circle the *most contributing* resonance structure.

Structure 1

Structure 2

Structure 3

(ii) For the three structures given (4, 5, and 6) below, answer the following questions:



(I) Circle all the structures that represent *resonance structures* of each other:

Structure 4

Structure 5

Structure 6

(II) Among the resonance structures, circle the *most contributing* resonance structure.

Structure 4

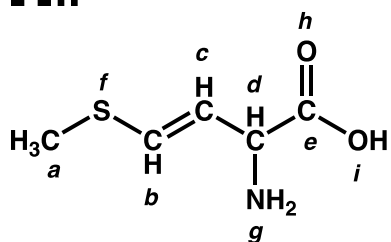
Structure 5

Structure 6

**Question 6** **6 points**

Using the data given below, calculate the **lattice energy** for SrO(s).

	ΔH° (kJ/mol)
$\text{Sr}(s) \longrightarrow \text{Sr}(g)$	+180
$\text{Sr}(g) \longrightarrow \text{Sr}^{2+}(g) + 2e^-$	+1468
$\text{O}_2(g) \longrightarrow 2\text{O}(g)$	+500
$\text{O}(g) + 2e^- \longrightarrow \text{O}^{2-}(g)$	+600
$\text{Sr}(s) + \frac{1}{2}\text{O}_2(g) \longrightarrow \text{SrO}(s)$	-550

**Question 7 10 points**

Consider the molecule given on the left, and answer questions (i) to (vi) using chemical bonding theories that we have learnt in class (valence bond theory, hybridization).

Note that the lone pairs are not shown in the structure:

i. Consider the carbon atoms marked in as ***a***, ***b***, ***c***, ***d***, and ***e*** in the molecule, and determine **hybridization** at each of the carbon centers. (2.5 points)

Atom	Hybridization
Carbon (<i>a</i>)	
Carbon (<i>b</i>)	
Carbon (<i>c</i>)	
Carbon (<i>d</i>)	
Carbon (<i>e</i>)	

ii. Now, consider the sulfur atom (S) marked as ***f***, nitrogen (N) atom (***g***), and oxygen(O) atoms (***h*** and ***i***), and determine their **hybridization**. (2 points)

Atom	Hybridization
Sulfur (<i>f</i>)	
Nitrogen (<i>g</i>)	
Oxygen (<i>h</i>)	
Oxygen (<i>i</i>)	

iii. **Describe** all the bonds formed by the carbon atoms ***a*** and ***b***, using sigma and pi designation as appropriate. (2 points)

Atom	Bonds formed (use sigma and pi as appropriate)
Carbon (<i>a</i>)	
Carbon (<i>b</i>)	

iv. How many total sp^3 -s sigma bonds are in the molecule? (1 point)

v. Determine **electronic** and **molecular geometry** at carbon (C) labeled ***e*** and sulfur (S) labeled ***f*** centres. (2 points)

Atom	Electronic Geometry	Molecular Geometry
Carbon (<i>e</i>)		
Sulfur (<i>f</i>)		

vi. Circle the value from the choices given below, that describes the **C-S-C bond angle at sulfur** (S) marked ***f***: (0.5 points)

>90° to <109.5°

109.5°

>109.5 to <120°

120°

>120° to <180°


Question 8 12 points

Using the **valence shell orbitals only** draw a **complete** molecular orbital (MO) diagram (with no s-p mixing) for the molecule **PS**, in the space provided below. **(6 points)**

Assume that 3s and 3p atomic orbitals will form molecular orbitals like those formed by the combination of 2s and 2p atomic orbitals that we learnt in class, and there is no s-p mixing.

Show **energy arrow**, **atomic orbitals** involved, **atom labels**, the **molecular orbitals** that are formed, **filled electrons** in the MO diagram etc. Make sure to label all the atomic and molecular orbitals.

Using the MO diagram above, answer the following questions:

- (i) Which one of the following will have the highest bond order? Circle the correct choice
 A) PS^{1-} B) PS C) PS^{1+}
- (ii) Which one of the following will have the longest bond length? Circle the correct choice
 A) PS^{1-} B) PS C) PS^{1+}
- (iii) Which of the following will be diamagnetic? Circle the correct choice.
 A) PS^{1-} B) PS C) PS^{1+}
- (iv) Draw Lewis dot structure of the molecule with the shortest bond length (based on the MO diagram) from among the three (PS^{1-} , PS, PS^{1+}), and indicate the bond order calculated from the Lewis dot structure. **(3 points)**

Lewis Dot Structure

Bond Order



40C52A41-554F-4C8D-989B-DDB58F8C3574

midterm-exam-93388

#18 Page 10 of 10

SCRAP PAGE

**CHEMISTRY 110 –Midterm Exam****Thursday November 7, 2024, 6:30 – 8:30 pm**

Examiners: Pallavi Sirjoosingh and Ashok Kakkar

First name (please write as legibly as possible within the boxes)

--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--

Last name

--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--

Student ID number

--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--

This booklet has 8 questions on 10 pages (The last page can be used as a scrap page).

The staple must NOT be removed from the question booklet. Your intact question booklet must be returned at the end of the exam.

A separate booklet contains a periodic table, and datasheet.

Write your answers directly under the questions in the space provided. Do not write on the QR codes or near the top edges (as it may be cut off).

Please write **legibly and dark enough** for the scan to be readable. For students who use English as a second language, dictionaries are permitted. Calculators are allowed.

DO NOT ASK QUESTIONS OF THE INVIGILATORS about interpretations of the exam questions. If you have a concern about a question, please summarize it at the top of this page. All concerns will be evaluated after the exam. GOOD LUCK!

**Question 1 8 points** (*Assume all transitions are possible*)

a. 6 points Calculate the **energy (in kJ/mol)** required to ionize (completely remove) the electron from the **n=3** level of **He⁺ ion** (He: Helium). Show your work including equations. (3 sig. figs.)

b. 2 points

(I) Select the largest (longest) wavelength from the three choices below:

- i. Wavelength of photon to *completely remove* an electron from *1s orbital of Hydrogen atom*
- ii. Wavelength of photon to *completely remove* an electron from *1s orbital of Boron atom (B)*
- iii. Wavelength of photon to *completely remove* an electron from *1s orbital of Boron cation (B⁺)*

i or ii or iii

(II) Select the smallest (shortest) wavelength from the three choices below (*for H atom*):

- i. Wavelength of photon *emitted* for an electron transitioning from *2p to 1s orbital*
- ii. Wavelength of photon *absorbed* for an electron transitioning from *2s to 4p orbital*
- iii. Wavelength of photon *absorbed* for an electron transitioning from *1s to 3p orbital*

i or ii or iii



Question 2 8 points



A Chem110 student plots the graph shown on the left, from data obtained for the kinetic energy (eV) of the ejected electron versus frequency (s^{-1}) of the incident light, for an unknown metal. ($1\text{eV} = 1.6022 \times 10^{-19} \text{ J}$)

Based on the plot, answer the questions **a to c** below:

a. 1 point On the plot above, label the **threshold frequency**.

b. 2 points What is the value of the work potential (in J) of the unknown metal?

c. 3 points What is the *longest* wavelength (in nm) required to eject an electron from this metal's surface? Show your work including equation(s) used. (3 sig figs)

d. 2 points In the space provided below, plot the following

(Assume frequency is *above* threshold frequency):

- i. *Kinetic energy* of an ejected electron versus the *intensity* of the incident light (left)
- ii. *Number of electrons* ejected versus the *frequency* of the incident light (right)



**Question 3** **6 points****a. 2 points**

In the ground state of Ge (Germanium; $Z = 32$) atom, **name** the following orbitals:

i. Orbital containing the electron that requires the maximum energy to be removed.

ii. Orbital containing the electron that experiences the minimum effective nuclear charge.

b. 2 points

An unknown anion (A^-) has the following *excited* electronic configuration: $[Ar] 4s^2 3d^3 4p^1$

(Ar: Argon; $Z = 18$)

Give the **condensed ground state electronic configuration** of the atom, **A** and the cation, **A^{2+}** .
Show your work. (Assume all electronic transitions are possible)

Electronic Configuration of A:

Electronic Configuration of A^{2+} :

c. 2 points

In a single atom, determine how many electrons can have the following set of quantum numbers?

(i) $n = 5; l = 3$

In a single atom, determine how many electrons can have the following set of quantum numbers?

(i) $n = 3; l = 2; m_s = +1/2$

**Question 4** **8 points**

Rank the following from 1 to 4 in **increasing** order. Provide the ranking in the boxes provided.

Note: Atomic number (Z) are provided for relevant atoms

a) Increasing (lowest to highest) order of *ionization energy*: (1= lowest; 4 = highest)

Mg

Mg: Z = 12

Rb

Rb: Z = 37

Cs

Cs: Z = 55

K

K : Z = 19

b) Increasing (lowest to highest) order of *bond angle*: (1= lowest; 4 = highest)

H₂OCCl₄XeF₄

HCN

c) Increasing (smallest to largest) order of *ionic size* (1= smallest; 4 = largest)

S²⁻

S: Z = 16

O²⁻

O: Z = 8

Be²⁺

Be: Z = 4

Na⁺

Na: Z = 11

d) Increasing (lowest to highest) order of *bond energy* (1= lowest; 4 = highest)

P-F

P-Cl

P-I

P-Br

**Question 5** **12 points**

a. 9 points For the molecules HClO_4 and TeF_6 , draw the most stable *Lewis structure* showing appropriate shape and *dashed/wedged bonds*, based on VSEPR. Determine the **molecular geometry** around the central atom. For HClO_4 , label the *most polar bond*.



b. 3 points (Note: Lone pairs are not shown in structures below)

(i) For the three *resonance* structures (1, 2 and 3) shown below, answer the following question:



(I) Circle the *most contributing* resonance structure.

Structure 1

Structure 2

Structure 3

(ii) For the three structures given (4, 5, and 6) below, answer the following questions:



(I) Circle all the structures that represent *resonance structures* of each other:

Structure 4

Structure 5

Structure 6

(II) Among the resonance structures, circle the *most contributing* resonance structure.

Structure 4

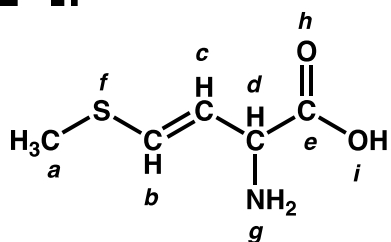
Structure 5

Structure 6

**Question 6** **6 points**

Using the data given below, calculate the **lattice energy** for SrO(s).

	ΔH° (kJ/mol)
$\text{Sr}(s) \longrightarrow \text{Sr}(g)$	+180
$\text{Sr}(g) \longrightarrow \text{Sr}^{2+}(g) + 2e^-$	+1468
$\text{O}_2(g) \longrightarrow 2\text{O}(g)$	+500
$\text{O}(g) + 2e^- \longrightarrow \text{O}^{2-}(g)$	+600
$\text{Sr}(s) + \frac{1}{2}\text{O}_2(g) \longrightarrow \text{SrO}(s)$	-550

**Question 7 10 points**

Consider the molecule given on the left, and answer questions (i) to (vi) using chemical bonding theories that we have learnt in class (valence bond theory, hybridization).

Note that the lone pairs are not shown in the structure:

i. Consider the carbon atoms marked in as ***a***, ***b***, ***c***, ***d***, and ***e*** in the molecule, and determine **hybridization** at each of the carbon centers. (2.5 points)

Atom	Hybridization
Carbon (<i>a</i>)	
Carbon (<i>b</i>)	
Carbon (<i>c</i>)	
Carbon (<i>d</i>)	
Carbon (<i>e</i>)	

ii. Now, consider the sulfur atom (S) marked as ***f***, nitrogen (N) atom (***g***), and oxygen(O) atoms (***h*** and ***i***), and determine their **hybridization**. (2 points)

Atom	Hybridization
Sulfur (<i>f</i>)	
Nitrogen (<i>g</i>)	
Oxygen (<i>h</i>)	
Oxygen (<i>i</i>)	

iii. **Describe** all the bonds formed by the carbon atoms ***a*** and ***b***, using sigma and pi designation as appropriate. (2 points)

Atom	Bonds formed (use sigma and pi as appropriate)
Carbon (<i>a</i>)	
Carbon (<i>b</i>)	

iv. How many total sp^3 -s sigma bonds are in the molecule? (1 point)

v. Determine **electronic** and **molecular geometry** at carbon (C) labeled ***e*** and sulfur (S) labeled ***f*** centres. (2 points)

Atom	Electronic Geometry	Molecular Geometry
Carbon (<i>e</i>)		
Sulfur (<i>f</i>)		

vi. Circle the value from the choices given below, that describes the **C-S-C bond angle at sulfur** (S) marked ***f***: (0.5 points)

>90° to <109.5° 109.5° >109.5 to <120° 120° >120° to <180°


Question 8 12 points

Using the **valence shell orbitals only** draw a **complete** molecular orbital (MO) diagram (with no s-p mixing) for the molecule **PS**, in the space provided below. **(6 points)**

Assume that 3s and 3p atomic orbitals will form molecular orbitals like those formed by the combination of 2s and 2p atomic orbitals that we learnt in class, and there is no s-p mixing.

Show **energy arrow**, **atomic orbitals** involved, **atom labels**, the **molecular orbitals** that are formed, **filled electrons** in the MO diagram etc. Make sure to label all the atomic and molecular orbitals.

Using the MO diagram above, answer the following questions:

- (i) Which one of the following will have the highest bond order? Circle the correct choice
 A) PS^{1-} B) PS C) PS^{1+}
- (ii) Which one of the following will have the longest bond length? Circle the correct choice
 A) PS^{1-} B) PS C) PS^{1+}
- (iii) Which of the following will be diamagnetic? Circle the correct choice.
 A) PS^{1-} B) PS C) PS^{1+}
- (iv) Draw Lewis dot structure of the molecule with the shortest bond length (based on the MO diagram) from among the three (PS^{1-} , PS, PS^{1+}), and indicate the bond order calculated from the Lewis dot structure. **(3 points)**

Lewis Dot Structure

Bond Order



0B6E542C-1F63-4CE2-8161-4090680C6B7F

midterm-exam-93388

#19 Page 10 of 10

SCRAP PAGE

**CHEMISTRY 110 –Midterm Exam****Thursday November 7, 2024, 6:30 – 8:30 pm**

Examiners: Pallavi Sirjoosingh and Ashok Kakkar

First name (please write as legibly as possible within the boxes)

--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--

Last name

--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--

Student ID number

--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--

This booklet has 8 questions on 10 pages (The last page can be used as a scrap page).

The staple must NOT be removed from the question booklet. Your intact question booklet must be returned at the end of the exam.

A separate booklet contains a periodic table, and datasheet.

Write your answers directly under the questions in the space provided. Do not write on the QR codes or near the top edges (as it may be cut off).

Please write **legibly and dark enough** for the scan to be readable. For students who use English as a second language, dictionaries are permitted. Calculators are allowed.

DO NOT ASK QUESTIONS OF THE INVIGILATORS about interpretations of the exam questions. If you have a concern about a question, please summarize it at the top of this page. All concerns will be evaluated after the exam. GOOD LUCK!

**Question 1 8 points** (*Assume all transitions are possible*)

a. 6 points Calculate the **energy (in kJ/mol)** required to ionize (completely remove) the electron from the **n=3** level of **He⁺ ion** (He: Helium). Show your work including equations. (3 sig. figs.)

b. 2 points

(I) Select the largest (longest) wavelength from the three choices below:

- i. Wavelength of photon to *completely remove* an electron from *1s orbital of Hydrogen atom*
- ii. Wavelength of photon to *completely remove* an electron from *1s orbital of Boron atom (B)*
- iii. Wavelength of photon to *completely remove* an electron from *1s orbital of Boron cation (B⁺)*

i or ii or iii

(II) Select the smallest (shortest) wavelength from the three choices below (*for H atom*):

- i. Wavelength of photon *emitted* for an electron transitioning from *2p to 1s orbital*
- ii. Wavelength of photon *absorbed* for an electron transitioning from *2s to 4p orbital*
- iii. Wavelength of photon *absorbed* for an electron transitioning from *1s to 3p orbital*

i or ii or iii



Question 2 8 points



A Chem110 student plots the graph shown on the left, from data obtained for the kinetic energy (eV) of the ejected electron versus frequency (s^{-1}) of the incident light, for an unknown metal. ($1\text{eV} = 1.6022 \times 10^{-19} \text{ J}$)

Based on the plot, answer the questions **a to c** below:

a. 1 point On the plot above, label the **threshold frequency**.

b. 2 points What is the value of the work potential (in J) of the unknown metal?

c. 3 points What is the *longest* wavelength (in nm) required to eject an electron from this metal's surface? Show your work including equation(s) used. (3 sig figs)

d. 2 points In the space provided below, plot the following

(Assume frequency is *above* threshold frequency):

- i. *Kinetic energy* of an ejected electron versus the *intensity* of the incident light (left)
- ii. *Number of electrons* ejected versus the *frequency* of the incident light (right)



**Question 3** **6 points****a. 2 points**

In the ground state of Ge (Germanium; $Z = 32$) atom, **name** the following orbitals:

i. Orbital containing the electron that requires the maximum energy to be removed.

ii. Orbital containing the electron that experiences the minimum effective nuclear charge.

b. 2 points

An unknown anion (A^-) has the following *excited* electronic configuration: $[Ar] 4s^2 3d^3 4p^1$

(Ar: Argon; $Z = 18$)

Give the **condensed ground state electronic configuration** of the atom, **A** and the cation, **A^{2+}** .
Show your work. (Assume all electronic transitions are possible)

Electronic Configuration of A:

Electronic Configuration of A^{2+} :

c. 2 points

In a single atom, determine how many electrons can have the following set of quantum numbers?

(i) $n = 5; l = 3$

In a single atom, determine how many electrons can have the following set of quantum numbers?

(i) $n = 3; l = 2; m_s = +1/2$

**Question 4** **8 points**

Rank the following from 1 to 4 in **increasing** order. Provide the ranking in the boxes provided.

Note: Atomic number (Z) are provided for relevant atoms

a) Increasing (lowest to highest) order of *ionization energy*: (1= lowest; 4 = highest)

Mg

Mg: Z = 12

Rb

Rb: Z = 37

Cs

Cs: Z = 55

K

K : Z = 19

b) Increasing (lowest to highest) order of *bond angle*: (1= lowest; 4 = highest)

H₂OCCl₄XeF₄

HCN

c) Increasing (smallest to largest) order of *ionic size* (1= smallest; 4 = largest)

S²⁻

S: Z = 16

O²⁻

O: Z = 8

Be²⁺

Be: Z = 4

Na⁺

Na: Z = 11

d) Increasing (lowest to highest) order of *bond energy* (1= lowest; 4 = highest)

P-F

P-Cl

P-I

P-Br

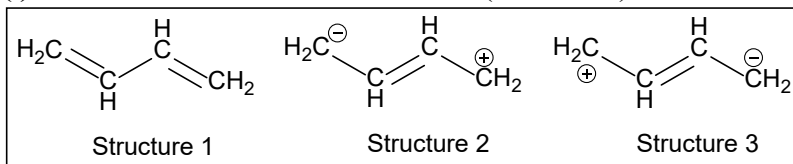
**Question 5** **12 points**

a. 9 points For the molecules HClO_4 and TeF_6 , draw the most stable *Lewis structure* showing appropriate shape and *dashed/wedged bonds*, based on VSEPR. Determine the **molecular geometry** around the central atom. For HClO_4 , label the *most polar bond*.



b. 3 points (Note: Lone pairs are not shown in structures below)

(i) For the three *resonance* structures (1, 2 and 3) shown below, answer the following question:



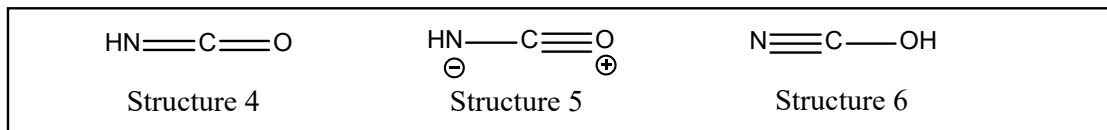
(I) Circle the *most contributing* resonance structure.

Structure 1

Structure 2

Structure 3

(ii) For the three structures given (4, 5, and 6) below, answer the following questions:



(I) Circle all the structures that represent *resonance structures* of each other:

Structure 4

Structure 5

Structure 6

(II) Among the resonance structures, circle the *most contributing* resonance structure.

Structure 4

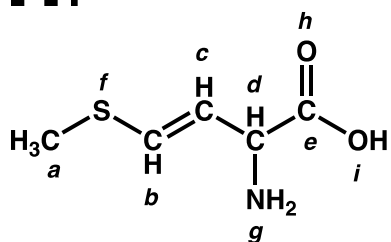
Structure 5

Structure 6

**Question 6** **6 points**

Using the data given below, calculate the **lattice energy** for SrO(s).

	ΔH° (kJ/mol)
$\text{Sr}(s) \longrightarrow \text{Sr}(g)$	+180
$\text{Sr}(g) \longrightarrow \text{Sr}^{2+}(g) + 2e^-$	+1468
$\text{O}_2(g) \longrightarrow 2\text{O}(g)$	+500
$\text{O}(g) + 2e^- \longrightarrow \text{O}^{2-}(g)$	+600
$\text{Sr}(s) + \frac{1}{2}\text{O}_2(g) \longrightarrow \text{SrO}(s)$	-550

**Question 7 10 points**

Consider the molecule given on the left, and answer questions (i) to (vi) using chemical bonding theories that we have learnt in class (valence bond theory, hybridization).

Note that the lone pairs are not shown in the structure:

i. Consider the carbon atoms marked in as ***a***, ***b***, ***c***, ***d***, and ***e*** in the molecule, and determine **hybridization** at each of the carbon centers. (2.5 points)

Atom	Hybridization
Carbon (<i>a</i>)	
Carbon (<i>b</i>)	
Carbon (<i>c</i>)	
Carbon (<i>d</i>)	
Carbon (<i>e</i>)	

ii. Now, consider the sulfur atom (S) marked as ***f***, nitrogen (N) atom (***g***), and oxygen(O) atoms (***h*** and ***i***), and determine their **hybridization**. (2 points)

Atom	Hybridization
Sulfur (<i>f</i>)	
Nitrogen (<i>g</i>)	
Oxygen (<i>h</i>)	
Oxygen (<i>i</i>)	

iii. **Describe** all the bonds formed by the carbon atoms ***a*** and ***b***, using sigma and pi designation as appropriate. (2 points)

Atom	Bonds formed (use sigma and pi as appropriate)
Carbon (<i>a</i>)	
Carbon (<i>b</i>)	

iv. How many total sp^3 -s sigma bonds are in the molecule? (1 point)

v. Determine **electronic** and **molecular geometry** at carbon (C) labeled ***e*** and sulfur (S) labeled ***f*** centres. (2 points)

Atom	Electronic Geometry	Molecular Geometry
Carbon (<i>e</i>)		
Sulfur (<i>f</i>)		

vi. Circle the value from the choices given below, that describes the **C-S-C bond angle at sulfur** (S) marked ***f***: (0.5 points)

>90° to <109.5° 109.5° >109.5 to <120° 120° >120° to <180°


Question 8 12 points

Using the **valence shell orbitals only** draw a **complete** molecular orbital (MO) diagram (with no s-p mixing) for the molecule **PS**, in the space provided below. **(6 points)**

Assume that 3s and 3p atomic orbitals will form molecular orbitals like those formed by the combination of 2s and 2p atomic orbitals that we learnt in class, and there is no s-p mixing.

Show **energy arrow**, **atomic orbitals** involved, **atom labels**, the **molecular orbitals** that are formed, **filled electrons** in the MO diagram etc. Make sure to label all the atomic and molecular orbitals.

Using the MO diagram above, answer the following questions:

- (i) Which one of the following will have the highest bond order? Circle the correct choice
 A) PS^{1-} B) PS C) PS^{1+}
- (ii) Which one of the following will have the longest bond length? Circle the correct choice
 A) PS^{1-} B) PS C) PS^{1+}
- (iii) Which of the following will be diamagnetic? Circle the correct choice.
 A) PS^{1-} B) PS C) PS^{1+}
- (iv) Draw Lewis dot structure of the molecule with the shortest bond length (based on the MO diagram) from among the three (PS^{1-} , PS, PS^{1+}), and indicate the bond order calculated from the Lewis dot structure. **(3 points)**

Lewis Dot Structure

Bond Order



BA43F17B-CD36-484E-8246-5805C96C03F9

midterm-exam-93388

#20 Page 10 of 10

SCRAP PAGE

**CHEMISTRY 110 –Midterm Exam****Thursday November 7, 2024, 6:30 – 8:30 pm**

Examiners: Pallavi Sirjoosingh and Ashok Kakkar

First name (please write as legibly as possible within the boxes)

--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--

Last name

--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--

Student ID number

--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--

This booklet has 8 questions on 10 pages (The last page can be used as a scrap page).

The staple must NOT be removed from the question booklet. Your intact question booklet must be returned at the end of the exam.

A separate booklet contains a periodic table, and datasheet.

Write your answers directly under the questions in the space provided. Do not write on the QR codes or near the top edges (as it may be cut off).

Please write **legibly and dark enough** for the scan to be readable. For students who use English as a second language, dictionaries are permitted. Calculators are allowed.

DO NOT ASK QUESTIONS OF THE INVIGILATORS about interpretations of the exam questions. If you have a concern about a question, please summarize it at the top of this page. All concerns will be evaluated after the exam. GOOD LUCK!

**Question 1 8 points** (*Assume all transitions are possible*)

a. 6 points Calculate the **energy (in kJ/mol)** required to ionize (completely remove) the electron from the **n=3** level of **He⁺ ion** (He: Helium). Show your work including equations. (3 sig. figs.)

b. 2 points

(I) Select the largest (longest) wavelength from the three choices below:

- i. Wavelength of photon to *completely remove* an electron from *1s orbital of Hydrogen atom*
- ii. Wavelength of photon to *completely remove* an electron from *1s orbital of Boron atom (B)*
- iii. Wavelength of photon to *completely remove* an electron from *1s orbital of Boron cation (B⁺)*

i or ii or iii

(II) Select the smallest (shortest) wavelength from the three choices below (*for H atom*):

- i. Wavelength of photon *emitted* for an electron transitioning from *2p to 1s orbital*
- ii. Wavelength of photon *absorbed* for an electron transitioning from *2s to 4p orbital*
- iii. Wavelength of photon *absorbed* for an electron transitioning from *1s to 3p orbital*

i or ii or iii



Question 2 8 points



A Chem110 student plots the graph shown on the left, from data obtained for the kinetic energy (eV) of the ejected electron versus frequency (s^{-1}) of the incident light, for an unknown metal. ($1\text{eV} = 1.6022 \times 10^{-19} \text{ J}$)

Based on the plot, answer the questions **a to c** below:

a. 1 point On the plot above, label the **threshold frequency**.

b. 2 points What is the value of the work potential (in J) of the unknown metal?

c. 3 points What is the *longest* wavelength (in nm) required to eject an electron from this metal's surface? Show your work including equation(s) used. (3 sig figs)

d. 2 points In the space provided below, plot the following

(Assume frequency is *above* threshold frequency):

- Kinetic energy* of an ejected electron versus the *intensity* of the incident light (left)
- Number of electrons* ejected versus the *frequency* of the incident light (right)



**Question 3** **6 points****a. 2 points**

In the ground state of Ge (Germanium; $Z = 32$) atom, **name** the following orbitals:

i. Orbital containing the electron that requires the maximum energy to be removed.

ii. Orbital containing the electron that experiences the minimum effective nuclear charge.

b. 2 points

An unknown anion (A^-) has the following *excited* electronic configuration: $[Ar] 4s^2 3d^3 4p^1$

(Ar: Argon; $Z = 18$)

Give the **condensed ground state electronic configuration** of the atom, **A** and the cation, **A^{2+}** .
Show your work. (Assume all electronic transitions are possible)

Electronic Configuration of A:

Electronic Configuration of A^{2+} :

c. 2 points

In a single atom, determine how many electrons can have the following set of quantum numbers?

(i) $n = 5; l = 3$

In a single atom, determine how many electrons can have the following set of quantum numbers?

(i) $n = 3; l = 2; m_s = +1/2$

**Question 4** **8 points**

Rank the following from 1 to 4 in **increasing** order. Provide the ranking in the boxes provided.

Note: Atomic number (Z) are provided for relevant atoms

a) Increasing (lowest to highest) order of *ionization energy*: (1= lowest; 4 = highest)

Mg

Mg: Z = 12

Rb

Rb: Z = 37

Cs

Cs: Z = 55

K

K : Z = 19

b) Increasing (lowest to highest) order of *bond angle*: (1= lowest; 4 = highest)

H₂OCCl₄XeF₄

HCN

c) Increasing (smallest to largest) order of *ionic size* (1= smallest; 4 = largest)

S²⁻

S: Z = 16

O²⁻

O: Z = 8

Be²⁺

Be: Z = 4

Na⁺

Na: Z = 11

d) Increasing (lowest to highest) order of *bond energy* (1= lowest; 4 = highest)

P-F

P-Cl

P-I

P-Br

**Question 5** **12 points**

a. 9 points For the molecules HClO_4 and TeF_6 , draw the most stable *Lewis structure* showing appropriate shape and *dashed/wedged bonds*, based on VSEPR. Determine the **molecular geometry** around the central atom. For HClO_4 , label the *most polar bond*.



b. 3 points (Note: Lone pairs are not shown in structures below)

(i) For the three *resonance* structures (1, 2 and 3) shown below, answer the following question:



(I) Circle the *most contributing* resonance structure.

Structure 1

Structure 2

Structure 3

(ii) For the three structures given (4, 5, and 6) below, answer the following questions:



(I) Circle all the structures that represent *resonance structures* of each other:

Structure 4

Structure 5

Structure 6

(II) Among the resonance structures, circle the *most contributing* resonance structure.

Structure 4

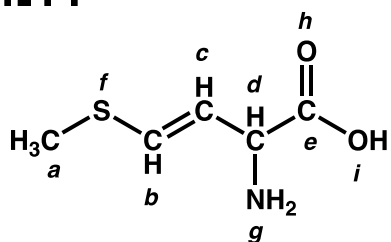
Structure 5

Structure 6

**Question 6** **6 points**

Using the data given below, calculate the **lattice energy** for SrO(s).

	ΔH° (kJ/mol)
$\text{Sr}(s) \longrightarrow \text{Sr}(g)$	+180
$\text{Sr}(g) \longrightarrow \text{Sr}^{2+}(g) + 2e^-$	+1468
$\text{O}_2(g) \longrightarrow 2\text{O}(g)$	+500
$\text{O}(g) + 2e^- \longrightarrow \text{O}^{2-}(g)$	+600
$\text{Sr}(s) + \frac{1}{2}\text{O}_2(g) \longrightarrow \text{SrO}(s)$	-550

**Question 7 10 points**

Consider the molecule given on the left, and answer questions (i) to (vi) using chemical bonding theories that we have learnt in class (valence bond theory, hybridization).

Note that the lone pairs are not shown in the structure:

i. Consider the carbon atoms marked in as ***a***, ***b***, ***c***, ***d***, and ***e*** in the molecule, and determine **hybridization** at each of the carbon centers. (2.5 points)

Atom	Hybridization
Carbon (<i>a</i>)	
Carbon (<i>b</i>)	
Carbon (<i>c</i>)	
Carbon (<i>d</i>)	
Carbon (<i>e</i>)	

ii. Now, consider the sulfur atom (S) marked as ***f***, nitrogen (N) atom (***g***), and oxygen(O) atoms (***h*** and ***i***), and determine their **hybridization**. (2 points)

Atom	Hybridization
Sulfur (<i>f</i>)	
Nitrogen (<i>g</i>)	
Oxygen (<i>h</i>)	
Oxygen (<i>i</i>)	

iii. **Describe** all the bonds formed by the carbon atoms ***a*** and ***b***, using sigma and pi designation as appropriate. (2 points)

Atom	Bonds formed (use sigma and pi as appropriate)
Carbon (<i>a</i>)	
Carbon (<i>b</i>)	

iv. How many total sp^3 -s sigma bonds are in the molecule? (1 point)

v. Determine **electronic** and **molecular geometry** at carbon (C) labeled ***e*** and sulfur (S) labeled ***f*** centres. (2 points)

Atom	Electronic Geometry	Molecular Geometry
Carbon (<i>e</i>)		
Sulfur (<i>f</i>)		

vi. Circle the value from the choices given below, that describes the **C-S-C bond angle at sulfur** (S) marked ***f***: (0.5 points)

$>90^\circ$ to $<109.5^\circ$

109.5°

>109.5 to $<120^\circ$

120°

$>120^\circ$ to $<180^\circ$


Question 8 12 points

Using the **valence shell orbitals only** draw a **complete** molecular orbital (MO) diagram (with no s-p mixing) for the molecule **PS**, in the space provided below. **(6 points)**

Assume that 3s and 3p atomic orbitals will form molecular orbitals like those formed by the combination of 2s and 2p atomic orbitals that we learnt in class, and there is no s-p mixing.

Show **energy arrow**, **atomic orbitals** involved, **atom labels**, the **molecular orbitals** that are formed, **filled electrons** in the MO diagram etc. Make sure to label all the atomic and molecular orbitals.

Using the MO diagram above, answer the following questions:

- (i) Which one of the following will have the highest bond order? Circle the correct choice
 A) PS^{1-} B) PS C) PS^{1+}
- (ii) Which one of the following will have the longest bond length? Circle the correct choice
 A) PS^{1-} B) PS C) PS^{1+}
- (iii) Which of the following will be diamagnetic? Circle the correct choice.
 A) PS^{1-} B) PS C) PS^{1+}
- (iv) Draw Lewis dot structure of the molecule with the shortest bond length (based on the MO diagram) from among the three (PS^{1-} , PS, PS^{1+}), and indicate the bond order calculated from the Lewis dot structure. **(3 points)**

Lewis Dot Structure

Bond Order



613F53C7-5B6A-4CE9-8357-0275C54441CD

midterm-exam-93388

#21 Page 10 of 10

SCRAP PAGE

**CHEMISTRY 110 –Midterm Exam****Thursday November 7, 2024, 6:30 – 8:30 pm**

Examiners: Pallavi Sirjoosingh and Ashok Kakkar

First name (please write as legibly as possible within the boxes)

--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--

Last name

--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--

Student ID number

--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--

This booklet has 8 questions on 10 pages (The last page can be used as a scrap page).

The staple must NOT be removed from the question booklet. Your intact question booklet must be returned at the end of the exam.

A separate booklet contains a periodic table, and datasheet.

Write your answers directly under the questions in the space provided. Do not write on the QR codes or near the top edges (as it may be cut off).

Please write **legibly and dark enough** for the scan to be readable. For students who use English as a second language, dictionaries are permitted. Calculators are allowed.

DO NOT ASK QUESTIONS OF THE INVIGILATORS about interpretations of the exam questions. If you have a concern about a question, please summarize it at the top of this page. All concerns will be evaluated after the exam. GOOD LUCK!

**Question 1 8 points** (*Assume all transitions are possible*)

a. 6 points Calculate the **energy (in kJ/mol)** required to ionize (completely remove) the electron from the **n=3** level of **He⁺ ion** (He: Helium). Show your work including equations. (3 sig. figs.)

b. 2 points

(I) Select the largest (longest) wavelength from the three choices below:

- i. Wavelength of photon to *completely remove* an electron from *1s orbital of Hydrogen atom*
- ii. Wavelength of photon to *completely remove* an electron from *1s orbital of Boron atom (B)*
- iii. Wavelength of photon to *completely remove* an electron from *1s orbital of Boron cation (B⁺)*

i or ii or iii

(II) Select the smallest (shortest) wavelength from the three choices below (*for H atom*):

- i. Wavelength of photon *emitted* for an electron transitioning from *2p to 1s orbital*
- ii. Wavelength of photon *absorbed* for an electron transitioning from *2s to 4p orbital*
- iii. Wavelength of photon *absorbed* for an electron transitioning from *1s to 3p orbital*

i or ii or iii



Question 2 8 points



A Chem110 student plots the graph shown on the left, from data obtained for the kinetic energy (eV) of the ejected electron versus frequency (s^{-1}) of the incident light, for an unknown metal. ($1\text{eV} = 1.6022 \times 10^{-19} \text{ J}$)

Based on the plot, answer the questions **a to c** below:

a. 1 point On the plot above, label the **threshold frequency**.

b. 2 points What is the value of the work potential (in J) of the unknown metal?

c. 3 points What is the *longest* wavelength (in nm) required to eject an electron from this metal's surface? Show your work including equation(s) used. (3 sig figs)

d. 2 points In the space provided below, plot the following

(Assume frequency is *above* threshold frequency):

- Kinetic energy* of an ejected electron versus the *intensity* of the incident light (left)
- Number of electrons* ejected versus the *frequency* of the incident light (right)



**Question 3** **6 points****a. 2 points**

In the ground state of Ge (Germanium; $Z = 32$) atom, **name** the following orbitals:

i. Orbital containing the electron that requires the maximum energy to be removed.

ii. Orbital containing the electron that experiences the minimum effective nuclear charge.

b. 2 points

An unknown anion (A^-) has the following *excited* electronic configuration: $[Ar] 4s^2 3d^3 4p^1$

(Ar: Argon; $Z = 18$)

Give the **condensed ground state electronic configuration** of the atom, **A** and the cation, **A^{2+}** .
Show your work. (Assume all electronic transitions are possible)

Electronic Configuration of A:

Electronic Configuration of A^{2+} :

c. 2 points

In a single atom, determine how many electrons can have the following set of quantum numbers?

(i) $n = 5; l = 3$

In a single atom, determine how many electrons can have the following set of quantum numbers?

(i) $n = 3; l = 2; m_s = +1/2$

**Question 4** **8 points**

Rank the following from 1 to 4 in **increasing** order. Provide the ranking in the boxes provided.

Note: Atomic number (Z) are provided for relevant atoms

a) Increasing (lowest to highest) order of *ionization energy*: (1= lowest; 4 = highest)

Mg

Mg: Z = 12

Rb

Rb: Z = 37

Cs

Cs: Z = 55

K

K : Z = 19

b) Increasing (lowest to highest) order of *bond angle*: (1= lowest; 4 = highest)

H₂OCCl₄XeF₄

HCN

c) Increasing (smallest to largest) order of *ionic size* (1= smallest; 4 = largest)

S²⁻

S: Z = 16

O²⁻

O: Z = 8

Be²⁺

Be: Z = 4

Na⁺

Na: Z = 11

d) Increasing (lowest to highest) order of *bond energy* (1= lowest; 4 = highest)

P-F

P-Cl

P-I

P-Br

**Question 5** **12 points**

a. 9 points For the molecules HClO_4 and TeF_6 , draw the most stable *Lewis structure* showing appropriate shape and *dashed/wedged bonds*, based on VSEPR. Determine the **molecular geometry** around the central atom. For HClO_4 , label the *most polar bond*.



b. 3 points (Note: Lone pairs are not shown in structures below)

(i) For the three *resonance* structures (1, 2 and 3) shown below, answer the following question:



(I) Circle the *most contributing* resonance structure.

Structure 1

Structure 2

Structure 3

(ii) For the three structures given (4, 5, and 6) below, answer the following questions:



(I) Circle all the structures that represent *resonance structures* of each other:

Structure 4

Structure 5

Structure 6

(II) Among the resonance structures, circle the *most contributing* resonance structure.

Structure 4

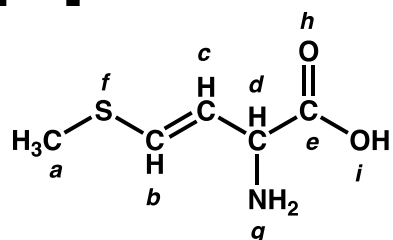
Structure 5

Structure 6

**Question 6** **6 points**

Using the data given below, calculate the **lattice energy** for SrO(s).

	ΔH° (kJ/mol)
$\text{Sr}(s) \longrightarrow \text{Sr}(g)$	+180
$\text{Sr}(g) \longrightarrow \text{Sr}^{2+}(g) + 2e^-$	+1468
$\text{O}_2(g) \longrightarrow 2\text{O}(g)$	+500
$\text{O}(g) + 2e^- \longrightarrow \text{O}^{2-}(g)$	+600
$\text{Sr}(s) + \frac{1}{2}\text{O}_2(g) \longrightarrow \text{SrO}(s)$	-550

**Question 7 10 points**

Consider the molecule given on the left, and answer questions (i) to (vi) using chemical bonding theories that we have learnt in class (valence bond theory, hybridization).

Note that the lone pairs are not shown in the structure:

i. Consider the carbon atoms marked in as ***a***, ***b***, ***c***, ***d***, and ***e*** in the molecule, and determine **hybridization** at each of the carbon centers. (2.5 points)

Atom	Hybridization
Carbon (<i>a</i>)	
Carbon (<i>b</i>)	
Carbon (<i>c</i>)	
Carbon (<i>d</i>)	
Carbon (<i>e</i>)	

ii. Now, consider the sulfur atom (S) marked as ***f***, nitrogen (N) atom (***g***), and oxygen(O) atoms (***h*** and ***i***), and determine their **hybridization**. (2 points)

Atom	Hybridization
Sulfur (<i>f</i>)	
Nitrogen (<i>g</i>)	
Oxygen (<i>h</i>)	
Oxygen (<i>i</i>)	

iii. **Describe** all the bonds formed by the carbon atoms ***a*** and ***b***, using sigma and pi designation as appropriate. (2 points)

Atom	Bonds formed (use sigma and pi as appropriate)
Carbon (<i>a</i>)	
Carbon (<i>b</i>)	

iv. How many total sp^3 -s sigma bonds are in the molecule? (1 point)

v. Determine **electronic** and **molecular geometry** at carbon (C) labeled ***e*** and sulfur (S) labeled ***f*** centres. (2 points)

Atom	Electronic Geometry	Molecular Geometry
Carbon (<i>e</i>)		
Sulfur (<i>f</i>)		

vi. Circle the value from the choices given below, that describes the **C-S-C bond angle at sulfur** (S) marked ***f***: (0.5 points)

>90° to <109.5° 109.5° >109.5 to <120° 120° >120° to <180°



Question 8 12 points

Using the **valence shell orbitals only** draw a **complete** molecular orbital (MO) diagram (with no s-p mixing) for the molecule **PS**, in the space provided below. **(6 points)**

Assume that 3s and 3p atomic orbitals will form molecular orbitals like those formed by the combination of 2s and 2p atomic orbitals that we learnt in class, and there is no s-p mixing.

Show **energy arrow**, **atomic orbitals** involved, **atom labels**, the **molecular orbitals** that are formed, **filled electrons** in the MO diagram etc. Make sure to label all the atomic and molecular orbitals.

Using the MO diagram above, answer the following questions:

- (i) Which one of the following will have the highest bond order? Circle the correct choice
 A) PS^{1-} B) PS C) PS^{1+}
- (ii) Which one of the following will have the longest bond length? Circle the correct choice
 A) PS^{1-} B) PS C) PS^{1+}
- (iii) Which of the following will be diamagnetic? Circle the correct choice.
 A) PS^{1-} B) PS C) PS^{1+}
- (iv) Draw Lewis dot structure of the molecule with the shortest bond length (based on the MO diagram) from among the three (PS^{1-} , PS, PS^{1+}), and indicate the bond order calculated from the Lewis dot structure. **(3 points)**

Lewis Dot Structure

Bond Order



77F95ED0-4369-41AF-B703-CE62BCF75EFC

midterm-exam-93388

#22 Page 10 of 10

SCRAP PAGE

**CHEMISTRY 110 –Midterm Exam****Thursday November 7, 2024, 6:30 – 8:30 pm****Examiners: Pallavi Sirjoosingh and Ashok Kakkar**

First name (please write as legibly as possible within the boxes)

--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--

Last name

--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--

Student ID number

--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--

This booklet has 8 questions on 10 pages (The last page can be used as a scrap page).

The staple must NOT be removed from the question booklet. Your intact question booklet must be returned at the end of the exam.

A separate booklet contains a periodic table, and datasheet.

Write your answers directly under the questions in the space provided. Do not write on the QR codes or near the top edges (as it may be cut off).

Please write **legibly and dark enough** for the scan to be readable. For students who use English as a second language, dictionaries are permitted. Calculators are allowed.

DO NOT ASK QUESTIONS OF THE INVIGILATORS about interpretations of the exam questions. If you have a concern about a question, please summarize it at the top of this page. All concerns will be evaluated after the exam. GOOD LUCK!

**Question 1 8 points** (*Assume all transitions are possible*)

a. 6 points Calculate the **energy (in kJ/mol)** required to ionize (completely remove) the electron from the **n=3** level of **He⁺ ion** (He: Helium). Show your work including equations. (3 sig. figs.)

b. 2 points

(I) Select the largest (longest) wavelength from the three choices below:

- i. Wavelength of photon to *completely remove* an electron from *1s orbital of Hydrogen atom*
- ii. Wavelength of photon to *completely remove* an electron from *1s orbital of Boron atom (B)*
- iii. Wavelength of photon to *completely remove* an electron from *1s orbital of Boron cation (B⁺)*

i or ii or iii

(II) Select the smallest (shortest) wavelength from the three choices below (*for H atom*):

- i. Wavelength of photon *emitted* for an electron transitioning from *2p to 1s orbital*
- ii. Wavelength of photon *absorbed* for an electron transitioning from *2s to 4p orbital*
- iii. Wavelength of photon *absorbed* for an electron transitioning from *1s to 3p orbital*

i or ii or iii



Question 2 8 points



A Chem110 student plots the graph shown on the left, from data obtained for the kinetic energy (eV) of the ejected electron versus frequency (s^{-1}) of the incident light, for an unknown metal. ($1\text{eV} = 1.6022 \times 10^{-19} \text{ J}$)

Based on the plot, answer the questions **a to c** below:

a. 1 point On the plot above, label the **threshold frequency**.

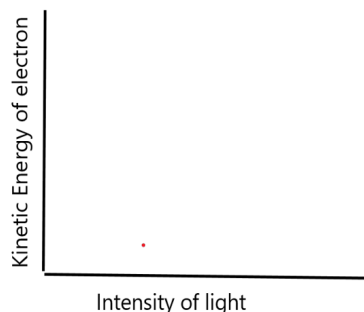
b. 2 points What is the value of the work potential (in J) of the unknown metal?

c. 3 points What is the *longest* wavelength (in nm) required to eject an electron from this metal's surface? Show your work including equation(s) used. (3 sig figs)

d. 2 points In the space provided below, plot the following

(Assume frequency is *above* threshold frequency):

- i. *Kinetic energy* of an ejected electron versus the *intensity* of the incident light (left)
- ii. *Number of electrons* ejected versus the *frequency* of the incident light (right)



**Question 3** **6 points****a. 2 points**

In the ground state of Ge (Germanium; $Z = 32$) atom, **name** the following orbitals:

i. Orbital containing the electron that requires the maximum energy to be removed.

ii. Orbital containing the electron that experiences the minimum effective nuclear charge.

b. 2 points

An unknown anion (A^-) has the following *excited* electronic configuration: $[Ar] 4s^2 3d^3 4p^1$

(Ar: Argon; $Z = 18$)

Give the **condensed ground state electronic configuration** of the atom, **A** and the cation, **A^{2+}** .
Show your work. (Assume all electronic transitions are possible)

Electronic Configuration of A:

Electronic Configuration of A^{2+} :

c. 2 points

In a single atom, determine how many electrons can have the following set of quantum numbers?

(i) $n = 5; l = 3$

In a single atom, determine how many electrons can have the following set of quantum numbers?

(i) $n = 3; l = 2; m_s = +1/2$

**Question 4** **8 points**

Rank the following from 1 to 4 in **increasing** order. Provide the ranking in the boxes provided.

Note: Atomic number (Z) are provided for relevant atoms

a) Increasing (lowest to highest) order of *ionization energy*: (1= lowest; 4 = highest)

Mg

Mg: Z = 12

Rb

Rb: Z = 37

Cs

Cs: Z = 55

K

K : Z = 19

b) Increasing (lowest to highest) order of *bond angle*: (1= lowest; 4 = highest)

H₂OCCl₄XeF₄

HCN

c) Increasing (smallest to largest) order of *ionic size* (1= smallest; 4 = largest)

S²⁻

S: Z = 16

O²⁻

O: Z = 8

Be²⁺

Be: Z = 4

Na⁺

Na: Z = 11

d) Increasing (lowest to highest) order of *bond energy* (1= lowest; 4 = highest)

P-F

P-Cl

P-I

P-Br

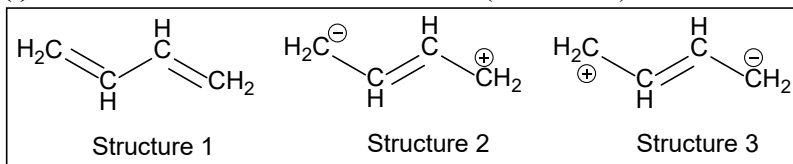
**Question 5** **12 points**

a. 9 points For the molecules HClO_4 and TeF_6 , draw the most stable *Lewis structure* showing appropriate shape and *dashed/wedged bonds*, based on VSEPR. Determine the **molecular geometry** around the central atom. For HClO_4 , label the *most polar bond*.



b. 3 points (Note: Lone pairs are not shown in structures below)

(i) For the three *resonance* structures (1, 2 and 3) shown below, answer the following question:



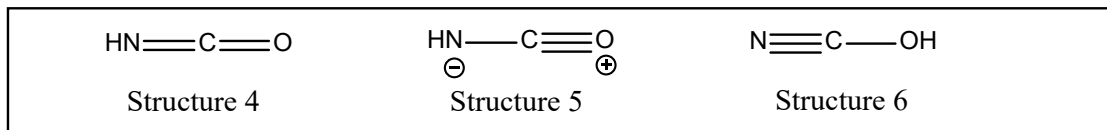
(I) Circle the *most contributing* resonance structure.

Structure 1

Structure 2

Structure 3

(ii) For the three structures given (4, 5, and 6) below, answer the following questions:



(I) Circle all the structures that represent *resonance structures* of each other:

Structure 4

Structure 5

Structure 6

(II) Among the resonance structures, circle the *most contributing* resonance structure.

Structure 4

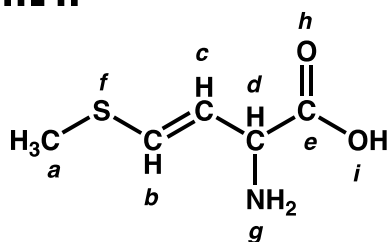
Structure 5

Structure 6

**Question 6** **6 points**

Using the data given below, calculate the **lattice energy** for SrO(s).

	ΔH° (kJ/mol)
$\text{Sr}(s) \longrightarrow \text{Sr}(g)$	+180
$\text{Sr}(g) \longrightarrow \text{Sr}^{2+}(g) + 2e^-$	+1468
$\text{O}_2(g) \longrightarrow 2\text{O}(g)$	+500
$\text{O}(g) + 2e^- \longrightarrow \text{O}^{2-}(g)$	+600
$\text{Sr}(s) + \frac{1}{2}\text{O}_2(g) \longrightarrow \text{SrO}(s)$	-550

**Question 7 10 points**

Consider the molecule given on the left, and answer questions (i) to (vi) using chemical bonding theories that we have learnt in class (valence bond theory, hybridization).

Note that the lone pairs are not shown in the structure:

i. Consider the carbon atoms marked in as ***a***, ***b***, ***c***, ***d***, and ***e*** in the molecule, and determine **hybridization** at each of the carbon centers. (2.5 points)

Atom	Hybridization
Carbon (<i>a</i>)	
Carbon (<i>b</i>)	
Carbon (<i>c</i>)	
Carbon (<i>d</i>)	
Carbon (<i>e</i>)	

ii. Now, consider the sulfur atom (S) marked as ***f***, nitrogen (N) atom (***g***), and oxygen(O) atoms (***h*** and ***i***), and determine their **hybridization**. (2 points)

Atom	Hybridization
Sulfur (<i>f</i>)	
Nitrogen (<i>g</i>)	
Oxygen (<i>h</i>)	
Oxygen (<i>i</i>)	

iii. **Describe** all the bonds formed by the carbon atoms ***a*** and ***b***, using sigma and pi designation as appropriate. (2 points)

Atom	Bonds formed (use sigma and pi as appropriate)
Carbon (<i>a</i>)	
Carbon (<i>b</i>)	

iv. How many total sp^3 -s sigma bonds are in the molecule? (1 point)

v. Determine **electronic** and **molecular geometry** at carbon (C) labeled ***e*** and sulfur (S) labeled ***f*** centres. (2 points)

Atom	Electronic Geometry	Molecular Geometry
Carbon (<i>e</i>)		
Sulfur (<i>f</i>)		

vi. Circle the value from the choices given below, that describes the **C-S-C bond angle at sulfur** (S) marked ***f***: (0.5 points)

>90° to <109.5° 109.5° >109.5 to <120° 120° >120° to <180°



Question 8 12 points

Using the **valence shell orbitals only** draw a **complete** molecular orbital (MO) diagram (with no s-p mixing) for the molecule **PS**, in the space provided below. **(6 points)**

Assume that 3s and 3p atomic orbitals will form molecular orbitals like those formed by the combination of 2s and 2p atomic orbitals that we learnt in class, and there is no s-p mixing.

Show **energy arrow**, **atomic orbitals** involved, **atom labels**, the **molecular orbitals** that are formed, **filled electrons** in the MO diagram etc. Make sure to label all the atomic and molecular orbitals.

Using the MO diagram above, answer the following questions:

- (i) Which one of the following will have the highest bond order? Circle the correct choice
 A) PS^{1-} B) PS C) PS^{1+}
- (ii) Which one of the following will have the longest bond length? Circle the correct choice
 A) PS^{1-} B) PS C) PS^{1+}
- (iii) Which of the following will be diamagnetic? Circle the correct choice.
 A) PS^{1-} B) PS C) PS^{1+}
- (iv) Draw Lewis dot structure of the molecule with the shortest bond length (based on the MO diagram) from among the three (PS^{1-} , PS, PS^{1+}), and indicate the bond order calculated from the Lewis dot structure. **(3 points)**

Lewis Dot Structure

Bond Order



7911EAF2-E00B-423E-826A-1E3003542754

midterm-exam-93388

#23 Page 10 of 10

SCRAP PAGE

**CHEMISTRY 110 –Midterm Exam****Thursday November 7, 2024, 6:30 – 8:30 pm**

Examiners: Pallavi Sirjoosingh and Ashok Kakkar

First name (please write as legibly as possible within the boxes)

--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--

Last name

--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--

Student ID number

--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--

This booklet has 8 questions on 10 pages (The last page can be used as a scrap page).

The staple must NOT be removed from the question booklet. Your intact question booklet must be returned at the end of the exam.

A separate booklet contains a periodic table, and datasheet.

Write your answers directly under the questions in the space provided. Do not write on the QR codes or near the top edges (as it may be cut off).

Please write **legibly and dark enough** for the scan to be readable. For students who use English as a second language, dictionaries are permitted. Calculators are allowed.

DO NOT ASK QUESTIONS OF THE INVIGILATORS about interpretations of the exam questions. If you have a concern about a question, please summarize it at the top of this page. All concerns will be evaluated after the exam. GOOD LUCK!

**Question 1 8 points** (*Assume all transitions are possible*)

a. 6 points Calculate the **energy (in kJ/mol)** required to ionize (completely remove) the electron from the **$n=3$** level of **He^+ ion** (He: Helium). Show your work including equations. (3 sig. figs.)

b. 2 points

(I) Select the largest (longest) wavelength from the three choices below:

- i. Wavelength of photon to *completely remove* an electron from *1s orbital of Hydrogen atom*
- ii. Wavelength of photon to *completely remove* an electron from *1s orbital of Boron atom (B)*
- iii. Wavelength of photon to *completely remove* an electron from *1s orbital of Boron cation (B^+)*

i or ii or iii

(II) Select the smallest (shortest) wavelength from the three choices below (*for H atom*):

- i. Wavelength of photon *emitted* for an electron transitioning from *2p to 1s orbital*
- ii. Wavelength of photon *absorbed* for an electron transitioning from *2s to 4p orbital*
- iii. Wavelength of photon *absorbed* for an electron transitioning from *1s to 3p orbital*

i or ii or iii



Question 2 8 points



A Chem110 student plots the graph shown on the left, from data obtained for the kinetic energy (eV) of the ejected electron versus frequency (s^{-1}) of the incident light, for an unknown metal. ($1\text{eV} = 1.6022 \times 10^{-19} \text{ J}$)

Based on the plot, answer the questions **a to c** below:

a. 1 point On the plot above, label the **threshold frequency**.

b. 2 points What is the value of the work potential (in J) of the unknown metal?

c. 3 points What is the *longest* wavelength (in nm) required to eject an electron from this metal's surface? Show your work including equation(s) used. (3 sig figs)

d. 2 points In the space provided below, plot the following

(Assume frequency is *above* threshold frequency):

- Kinetic energy* of an ejected electron versus the *intensity* of the incident light (left)
- Number of electrons* ejected versus the *frequency* of the incident light (right)



**Question 3** **6 points****a. 2 points**

In the ground state of Ge (Germanium; $Z = 32$) atom, **name** the following orbitals:

i. Orbital containing the electron that requires the maximum energy to be removed.

ii. Orbital containing the electron that experiences the minimum effective nuclear charge.

b. 2 points

An unknown anion (A^-) has the following *excited* electronic configuration: $[Ar] 4s^2 3d^3 4p^1$

(Ar: Argon; $Z = 18$)

Give the **condensed ground state electronic configuration** of the atom, **A** and the cation, **A^{2+}** .
Show your work. (Assume all electronic transitions are possible)

Electronic Configuration of A:

Electronic Configuration of A^{2+} :

c. 2 points

In a single atom, determine how many electrons can have the following set of quantum numbers?

(i) $n = 5; l = 3$

In a single atom, determine how many electrons can have the following set of quantum numbers?

(i) $n = 3; l = 2; m_s = +1/2$

**Question 4** **8 points**

Rank the following from 1 to 4 in **increasing** order. Provide the ranking in the boxes provided.

Note: Atomic number (Z) are provided for relevant atoms

a) Increasing (lowest to highest) order of *ionization energy*: (1= lowest; 4 = highest)

Mg

Mg: Z = 12

Rb

Rb: Z = 37

Cs

Cs: Z = 55

K

K : Z = 19

b) Increasing (lowest to highest) order of *bond angle*: (1= lowest; 4 = highest)

H₂OCCl₄XeF₄

HCN

c) Increasing (smallest to largest) order of *ionic size* (1= smallest; 4 = largest)

S²⁻

S: Z = 16

O²⁻

O: Z = 8

Be²⁺

Be: Z = 4

Na⁺

Na: Z = 11

d) Increasing (lowest to highest) order of *bond energy* (1= lowest; 4 = highest)

P-F

P-Cl

P-I

P-Br

**Question 5** **12 points**

a. 9 points For the molecules HClO_4 and TeF_6 , draw the most stable *Lewis structure* showing appropriate shape and *dashed/wedged bonds*, based on VSEPR. Determine the **molecular geometry** around the central atom. For HClO_4 , label the *most polar bond*.



b. 3 points (Note: Lone pairs are not shown in structures below)

(i) For the three *resonance* structures (1, 2 and 3) shown below, answer the following question:



(I) Circle the *most contributing* resonance structure.

Structure 1

Structure 2

Structure 3

(ii) For the three structures given (4, 5, and 6) below, answer the following questions:



(I) Circle all the structures that represent *resonance structures* of each other:

Structure 4

Structure 5

Structure 6

(II) Among the resonance structures, circle the *most contributing* resonance structure.

Structure 4

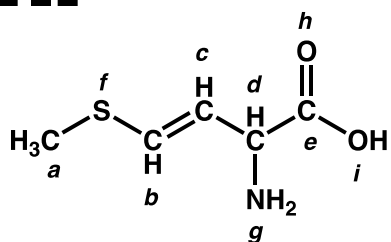
Structure 5

Structure 6

**Question 6** **6 points**

Using the data given below, calculate the **lattice energy** for SrO(s).

	ΔH° (kJ/mol)
$\text{Sr}(s) \longrightarrow \text{Sr}(g)$	+180
$\text{Sr}(g) \longrightarrow \text{Sr}^{2+}(g) + 2e^-$	+1468
$\text{O}_2(g) \longrightarrow 2\text{O}(g)$	+500
$\text{O}(g) + 2e^- \longrightarrow \text{O}^{2-}(g)$	+600
$\text{Sr}(s) + \frac{1}{2}\text{O}_2(g) \longrightarrow \text{SrO}(s)$	-550

**Question 7 10 points**

Consider the molecule given on the left, and answer questions (i) to (vi) using chemical bonding theories that we have learnt in class (valence bond theory, hybridization).

Note that the lone pairs are not shown in the structure:

i. Consider the carbon atoms marked in as ***a***, ***b***, ***c***, ***d***, and ***e*** in the molecule, and determine **hybridization** at each of the carbon centers. (2.5 points)

Atom	Hybridization
Carbon (<i>a</i>)	
Carbon (<i>b</i>)	
Carbon (<i>c</i>)	
Carbon (<i>d</i>)	
Carbon (<i>e</i>)	

ii. Now, consider the sulfur atom (S) marked as ***f***, nitrogen (N) atom (***g***), and oxygen(O) atoms (***h*** and ***i***), and determine their **hybridization**. (2 points)

Atom	Hybridization
Sulfur (<i>f</i>)	
Nitrogen (<i>g</i>)	
Oxygen (<i>h</i>)	
Oxygen (<i>i</i>)	

iii. **Describe** all the bonds formed by the carbon atoms ***a*** and ***b***, using sigma and pi designation as appropriate. (2 points)

Atom	Bonds formed (use sigma and pi as appropriate)
Carbon (<i>a</i>)	
Carbon (<i>b</i>)	

iv. How many total sp^3 -s sigma bonds are in the molecule? (1 point)

v. Determine **electronic** and **molecular geometry** at carbon (C) labeled ***e*** and sulfur (S) labeled ***f*** centres. (2 points)

Atom	Electronic Geometry	Molecular Geometry
Carbon (<i>e</i>)		
Sulfur (<i>f</i>)		

vi. Circle the value from the choices given below, that describes the **C-S-C bond angle at sulfur** (S) marked ***f***: (0.5 points)

$>90^\circ$ to $<109.5^\circ$

109.5°

>109.5 to $<120^\circ$

120°

$>120^\circ$ to $<180^\circ$


Question 8 12 points

Using the **valence shell orbitals only** draw a **complete** molecular orbital (MO) diagram (with no s-p mixing) for the molecule **PS**, in the space provided below. **(6 points)**

Assume that 3s and 3p atomic orbitals will form molecular orbitals like those formed by the combination of 2s and 2p atomic orbitals that we learnt in class, and there is no s-p mixing.

Show **energy arrow**, **atomic orbitals** involved, **atom labels**, the **molecular orbitals** that are formed, **filled electrons** in the MO diagram etc. Make sure to label all the atomic and molecular orbitals.

Using the MO diagram above, answer the following questions:

- (i) Which one of the following will have the highest bond order? Circle the correct choice
 A) PS^{1-} B) PS C) PS^{1+}
- (ii) Which one of the following will have the longest bond length? Circle the correct choice
 A) PS^{1-} B) PS C) PS^{1+}
- (iii) Which of the following will be diamagnetic? Circle the correct choice.
 A) PS^{1-} B) PS C) PS^{1+}
- (iv) Draw Lewis dot structure of the molecule with the shortest bond length (based on the MO diagram) from among the three (PS^{1-} , PS, PS^{1+}), and indicate the bond order calculated from the Lewis dot structure. **(3 points)**

Lewis Dot Structure

Bond Order



67552D43-28F5-4757-B08D-0A10A0FCB816

midterm-exam-93388

#24 Page 10 of 10

SCRAP PAGE

**CHEMISTRY 110 –Midterm Exam****Thursday November 7, 2024, 6:30 – 8:30 pm****Examiners: Pallavi Sirjoosingh and Ashok Kakkar**

First name (please write as legibly as possible within the boxes)

--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--

Last name

--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--

Student ID number

--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--

This booklet has 8 questions on 10 pages (The last page can be used as a scrap page).

The staple must NOT be removed from the question booklet. Your intact question booklet must be returned at the end of the exam.

A separate booklet contains a periodic table, and datasheet.

Write your answers directly under the questions in the space provided. Do not write on the QR codes or near the top edges (as it may be cut off).

Please write **legibly and dark enough** for the scan to be readable. For students who use English as a second language, dictionaries are permitted. Calculators are allowed.

DO NOT ASK QUESTIONS OF THE INVIGILATORS about interpretations of the exam questions. If you have a concern about a question, please summarize it at the top of this page. All concerns will be evaluated after the exam. GOOD LUCK!

**Question 1 8 points** (*Assume all transitions are possible*)

a. 6 points Calculate the **energy (in kJ/mol)** required to ionize (completely remove) the electron from the **n=3** level of **He⁺ ion** (He: Helium). Show your work including equations. (3 sig. figs.)

b. 2 points

(I) Select the largest (longest) wavelength from the three choices below:

- i. Wavelength of photon to *completely remove* an electron from *1s orbital of Hydrogen atom*
- ii. Wavelength of photon to *completely remove* an electron from *1s orbital of Boron atom (B)*
- iii. Wavelength of photon to *completely remove* an electron from *1s orbital of Boron cation (B⁺)*

i or ii or iii

(II) Select the smallest (shortest) wavelength from the three choices below (*for H atom*):

- i. Wavelength of photon *emitted* for an electron transitioning from *2p to 1s orbital*
- ii. Wavelength of photon *absorbed* for an electron transitioning from *2s to 4p orbital*
- iii. Wavelength of photon *absorbed* for an electron transitioning from *1s to 3p orbital*

i or ii or iii



Question 2 8 points



A Chem110 student plots the graph shown on the left, from data obtained for the kinetic energy (eV) of the ejected electron versus frequency (s^{-1}) of the incident light, for an unknown metal. ($1\text{eV} = 1.6022 \times 10^{-19} \text{ J}$)

Based on the plot, answer the questions **a to c** below:

a. 1 point On the plot above, label the **threshold frequency**.

b. 2 points What is the value of the work potential (in J) of the unknown metal?

c. 3 points What is the *longest* wavelength (in nm) required to eject an electron from this metal's surface? Show your work including equation(s) used. (3 sig figs)

d. 2 points In the space provided below, plot the following

(Assume frequency is *above* threshold frequency):

- i. *Kinetic energy* of an ejected electron versus the *intensity* of the incident light (left)
- ii. *Number of electrons* ejected versus the *frequency* of the incident light (right)



**Question 3** **6 points****a. 2 points**

In the ground state of Ge (Germanium; $Z = 32$) atom, **name** the following orbitals:

i. Orbital containing the electron that requires the maximum energy to be removed.

ii. Orbital containing the electron that experiences the minimum effective nuclear charge.

b. 2 points

An unknown anion (A^-) has the following *excited* electronic configuration: $[Ar] 4s^2 3d^3 4p^1$

(Ar: Argon; $Z = 18$)

Give the **condensed ground state electronic configuration** of the atom, **A** and the cation, **A^{2+}** .
Show your work. (Assume all electronic transitions are possible)

Electronic Configuration of A:

Electronic Configuration of A^{2+} :

c. 2 points

In a single atom, determine how many electrons can have the following set of quantum numbers?

(i) $n = 5; l = 3$

In a single atom, determine how many electrons can have the following set of quantum numbers?

(i) $n = 3; l = 2; m_s = +1/2$

**Question 4** **8 points**

Rank the following from 1 to 4 in **increasing** order. Provide the ranking in the boxes provided.

Note: Atomic number (Z) are provided for relevant atoms

a) Increasing (lowest to highest) order of *ionization energy*: (1= lowest; 4 = highest)

Mg

Mg: Z = 12

Rb

Rb: Z = 37

Cs

Cs: Z = 55

K

K : Z = 19

b) Increasing (lowest to highest) order of *bond angle*: (1= lowest; 4 = highest)

H₂OCCl₄XeF₄

HCN

c) Increasing (smallest to largest) order of *ionic size* (1= smallest; 4 = largest)

S²⁻

S: Z = 16

O²⁻

O: Z = 8

Be²⁺

Be: Z = 4

Na⁺

Na: Z = 11

d) Increasing (lowest to highest) order of *bond energy* (1= lowest; 4 = highest)

P-F

P-Cl

P-I

P-Br

**Question 5** **12 points**

a. 9 points For the molecules HClO_4 and TeF_6 , draw the most stable *Lewis structure* showing appropriate shape and *dashed/wedged bonds*, based on VSEPR. Determine the **molecular geometry** around the central atom. For HClO_4 , label the *most polar bond*.



b. 3 points (Note: Lone pairs are not shown in structures below)

(i) For the three *resonance* structures (1, 2 and 3) shown below, answer the following question:



(I) Circle the *most contributing* resonance structure.

Structure 1

Structure 2

Structure 3

(ii) For the three structures given (4, 5, and 6) below, answer the following questions:



(I) Circle all the structures that represent *resonance structures* of each other:

Structure 4

Structure 5

Structure 6

(II) Among the resonance structures, circle the *most contributing* resonance structure.

Structure 4

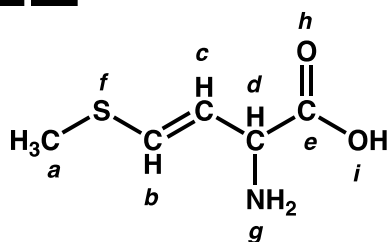
Structure 5

Structure 6

**Question 6** **6 points**

Using the data given below, calculate the **lattice energy** for SrO(s).

	ΔH° (kJ/mol)
$\text{Sr}(s) \longrightarrow \text{Sr}(g)$	+180
$\text{Sr}(g) \longrightarrow \text{Sr}^{2+}(g) + 2e^-$	+1468
$\text{O}_2(g) \longrightarrow 2\text{O}(g)$	+500
$\text{O}(g) + 2e^- \longrightarrow \text{O}^{2-}(g)$	+600
$\text{Sr}(s) + \frac{1}{2}\text{O}_2(g) \longrightarrow \text{SrO}(s)$	-550

**Question 7 10 points**

Consider the molecule given on the left, and answer questions (i) to (vi) using chemical bonding theories that we have learnt in class (valence bond theory, hybridization).

Note that the lone pairs are not shown in the structure:

i. Consider the carbon atoms marked in as ***a***, ***b***, ***c***, ***d***, and ***e*** in the molecule, and determine **hybridization** at each of the carbon centers. (2.5 points)

Atom	Hybridization
Carbon (<i>a</i>)	
Carbon (<i>b</i>)	
Carbon (<i>c</i>)	
Carbon (<i>d</i>)	
Carbon (<i>e</i>)	

ii. Now, consider the sulfur atom (S) marked as ***f***, nitrogen (N) atom (***g***), and oxygen(O) atoms (***h*** and ***i***), and determine their **hybridization**. (2 points)

Atom	Hybridization
Sulfur (<i>f</i>)	
Nitrogen (<i>g</i>)	
Oxygen (<i>h</i>)	
Oxygen (<i>i</i>)	

iii. **Describe** all the bonds formed by the carbon atoms ***a*** and ***b***, using sigma and pi designation as appropriate. (2 points)

Atom	Bonds formed (use sigma and pi as appropriate)
Carbon (<i>a</i>)	
Carbon (<i>b</i>)	

iv. How many total sp^3 -s sigma bonds are in the molecule? (1 point)

v. Determine **electronic** and **molecular geometry** at carbon (C) labeled ***e*** and sulfur (S) labeled ***f*** centres. (2 points)

Atom	Electronic Geometry	Molecular Geometry
Carbon (<i>e</i>)		
Sulfur (<i>f</i>)		

vi. Circle the value from the choices given below, that describes the **C-S-C bond angle at sulfur** (S) marked ***f***: (0.5 points)

>90° to <109.5° 109.5° >109.5 to <120° 120° >120° to <180°

**Question 8 12 points**

Using the **valence shell orbitals only** draw a **complete** molecular orbital (MO) diagram (with no s-p mixing) for the molecule **PS**, in the space provided below. **(6 points)**

Assume that 3s and 3p atomic orbitals will form molecular orbitals like those formed by the combination of 2s and 2p atomic orbitals that we learnt in class, and there is no s-p mixing.

Show **energy arrow**, **atomic orbitals** involved, **atom labels**, the **molecular orbitals** that are formed, **filled electrons** in the MO diagram etc. Make sure to label all the atomic and molecular orbitals.

Using the MO diagram above, answer the following questions:

- (i) Which one of the following will have the highest bond order? Circle the correct choice
A) PS^{1-} B) PS C) PS^{1+}
- (ii) Which one of the following will have the longest bond length? Circle the correct choice
A) PS^{1-} B) PS C) PS^{1+}
- (iii) Which of the following will be diamagnetic? Circle the correct choice.
A) PS^{1-} B) PS C) PS^{1+}
- (iv) Draw Lewis dot structure of the molecule with the shortest bond length (based on the MO diagram) from among the three (PS^{1-} , PS, PS^{1+}), and indicate the bond order calculated from the Lewis dot structure. **(3 points)**

Lewis Dot Structure

Bond Order



693F4801-75ED-406E-BB88-94F53CD8B852

midterm-exam-93388

#25 Page 10 of 10

SCRAP PAGE

**CHEMISTRY 110 –Midterm Exam****Thursday November 7, 2024, 6:30 – 8:30 pm**

Examiners: Pallavi Sirjoosingh and Ashok Kakkar

First name (please write as legibly as possible within the boxes)

--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--

Last name

--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--

Student ID number

--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--

This booklet has 8 questions on 10 pages (The last page can be used as a scrap page).

The staple must NOT be removed from the question booklet. Your intact question booklet must be returned at the end of the exam.

A separate booklet contains a periodic table, and datasheet.

Write your answers directly under the questions in the space provided. Do not write on the QR codes or near the top edges (as it may be cut off).

Please write **legibly and dark enough** for the scan to be readable. For students who use English as a second language, dictionaries are permitted. Calculators are allowed.

DO NOT ASK QUESTIONS OF THE INVIGILATORS about interpretations of the exam questions. If you have a concern about a question, please summarize it at the top of this page. All concerns will be evaluated after the exam. GOOD LUCK!

**Question 1 8 points** (*Assume all transitions are possible*)

a. 6 points Calculate the **energy (in kJ/mol)** required to ionize (completely remove) the electron from the **n=3** level of **He⁺ ion** (He: Helium). Show your work including equations. (3 sig. figs.)

b. 2 points

(I) Select the largest (longest) wavelength from the three choices below:

- i. Wavelength of photon to *completely remove* an electron from *1s orbital of Hydrogen atom*
- ii. Wavelength of photon to *completely remove* an electron from *1s orbital of Boron atom (B)*
- iii. Wavelength of photon to *completely remove* an electron from *1s orbital of Boron cation (B⁺)*

i or ii or iii

(II) Select the smallest (shortest) wavelength from the three choices below (*for H atom*):

- i. Wavelength of photon *emitted* for an electron transitioning from *2p to 1s orbital*
- ii. Wavelength of photon *absorbed* for an electron transitioning from *2s to 4p orbital*
- iii. Wavelength of photon *absorbed* for an electron transitioning from *1s to 3p orbital*

i or ii or iii



Question 2 8 points



A Chem110 student plots the graph shown on the left, from data obtained for the kinetic energy (eV) of the ejected electron versus frequency (s^{-1}) of the incident light, for an unknown metal. ($1\text{eV} = 1.6022 \times 10^{-19} \text{ J}$)

Based on the plot, answer the questions **a to c** below:

a. 1 point On the plot above, label the **threshold frequency**.

b. 2 points What is the value of the work potential (in J) of the unknown metal?

c. 3 points What is the *longest* wavelength (in nm) required to eject an electron from this metal's surface? Show your work including equation(s) used. (3 sig figs)

d. 2 points In the space provided below, plot the following

(Assume frequency is *above* threshold frequency):

- i. *Kinetic energy* of an ejected electron versus the *intensity* of the incident light (left)
- ii. *Number of electrons* ejected versus the *frequency* of the incident light (right)



**Question 3** **6 points****a. 2 points**

In the ground state of Ge (Germanium; $Z = 32$) atom, **name** the following orbitals:

i. Orbital containing the electron that requires the maximum energy to be removed.

ii. Orbital containing the electron that experiences the minimum effective nuclear charge.

b. 2 points

An unknown anion (A^-) has the following *excited* electronic configuration: $[Ar] 4s^2 3d^3 4p^1$

(Ar: Argon; $Z = 18$)

Give the **condensed ground state electronic configuration** of the atom, **A** and the cation, **A^{2+}** .
Show your work. (Assume all electronic transitions are possible)

Electronic Configuration of A:

Electronic Configuration of A^{2+} :

c. 2 points

In a single atom, determine how many electrons can have the following set of quantum numbers?

(i) $n = 5; l = 3$

In a single atom, determine how many electrons can have the following set of quantum numbers?

(i) $n = 3; l = 2; m_s = +1/2$

**Question 4** **8 points**

Rank the following from 1 to 4 in **increasing** order. Provide the ranking in the boxes provided.

Note: Atomic number (Z) are provided for relevant atoms

a) Increasing (lowest to highest) order of *ionization energy*: (1= lowest; 4 = highest)

Mg

Mg: Z = 12

Rb

Rb: Z = 37

Cs

Cs: Z = 55

K

K : Z = 19

b) Increasing (lowest to highest) order of *bond angle*: (1= lowest; 4 = highest)

H₂OCCl₄XeF₄

HCN

c) Increasing (smallest to largest) order of *ionic size* (1= smallest; 4 = largest)

S²⁻

S: Z = 16

O²⁻

O: Z = 8

Be²⁺

Be: Z = 4

Na⁺

Na: Z = 11

d) Increasing (lowest to highest) order of *bond energy* (1= lowest; 4 = highest)

P-F

P-Cl

P-I

P-Br

**Question 5** **12 points**

a. 9 points For the molecules HClO_4 and TeF_6 , draw the most stable *Lewis structure* showing appropriate shape and *dashed/wedged bonds*, based on VSEPR. Determine the **molecular geometry** around the central atom. For HClO_4 , label the *most polar bond*.



b. 3 points (Note: Lone pairs are not shown in structures below)

(i) For the three *resonance* structures (1, 2 and 3) shown below, answer the following question:



(I) Circle the *most contributing* resonance structure.

Structure 1

Structure 2

Structure 3

(ii) For the three structures given (4, 5, and 6) below, answer the following questions:



(I) Circle all the structures that represent *resonance structures* of each other:

Structure 4

Structure 5

Structure 6

(II) Among the resonance structures, circle the *most contributing* resonance structure.

Structure 4

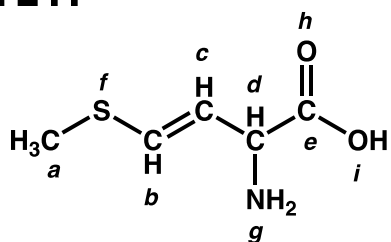
Structure 5

Structure 6

**Question 6** **6 points**

Using the data given below, calculate the **lattice energy** for SrO(s).

	ΔH° (kJ/mol)
$\text{Sr}(s) \longrightarrow \text{Sr}(g)$	+180
$\text{Sr}(g) \longrightarrow \text{Sr}^{2+}(g) + 2e^-$	+1468
$\text{O}_2(g) \longrightarrow 2\text{O}(g)$	+500
$\text{O}(g) + 2e^- \longrightarrow \text{O}^{2-}(g)$	+600
$\text{Sr}(s) + \frac{1}{2}\text{O}_2(g) \longrightarrow \text{SrO}(s)$	-550

**Question 7 10 points**

Consider the molecule given on the left, and answer questions (i) to (vi) using chemical bonding theories that we have learnt in class (valence bond theory, hybridization).

Note that the lone pairs are not shown in the structure:

i. Consider the carbon atoms marked in as ***a***, ***b***, ***c***, ***d***, and ***e*** in the molecule, and determine **hybridization** at each of the carbon centers. (2.5 points)

Atom	Hybridization
Carbon (<i>a</i>)	
Carbon (<i>b</i>)	
Carbon (<i>c</i>)	
Carbon (<i>d</i>)	
Carbon (<i>e</i>)	

ii. Now, consider the sulfur atom (S) marked as ***f***, nitrogen (N) atom (***g***), and oxygen(O) atoms (***h*** and ***i***), and determine their **hybridization**. (2 points)

Atom	Hybridization
Sulfur (<i>f</i>)	
Nitrogen (<i>g</i>)	
Oxygen (<i>h</i>)	
Oxygen (<i>i</i>)	

iii. **Describe** all the bonds formed by the carbon atoms ***a*** and ***b***, using sigma and pi designation as appropriate. (2 points)

Atom	Bonds formed (use sigma and pi as appropriate)
Carbon (<i>a</i>)	
Carbon (<i>b</i>)	

iv. How many total sp^3 -s sigma bonds are in the molecule? (1 point)

v. Determine **electronic** and **molecular geometry** at carbon (C) labeled ***e*** and sulfur (S) labeled ***f*** centres. (2 points)

Atom	Electronic Geometry	Molecular Geometry
Carbon (<i>e</i>)		
Sulfur (<i>f</i>)		

vi. Circle the value from the choices given below, that describes the **C-S-C bond angle at sulfur** (S) marked ***f***: (0.5 points)

>90° to <109.5°

109.5°

>109.5 to <120°

120°

>120° to <180°



Question 8 12 points

Using the **valence shell orbitals only** draw a **complete** molecular orbital (MO) diagram (with no s-p mixing) for the molecule **PS**, in the space provided below. **(6 points)**

Assume that 3s and 3p atomic orbitals will form molecular orbitals like those formed by the combination of 2s and 2p atomic orbitals that we learnt in class, and there is no s-p mixing.

Show **energy arrow**, **atomic orbitals** involved, **atom labels**, the **molecular orbitals** that are formed, **filled electrons** in the MO diagram etc. Make sure to label all the atomic and molecular orbitals.

Using the MO diagram above, answer the following questions:

- (i) Which one of the following will have the highest bond order? Circle the correct choice
 A) PS^{1-} B) PS C) PS^{1+}
- (ii) Which one of the following will have the longest bond length? Circle the correct choice
 A) PS^{1-} B) PS C) PS^{1+}
- (iii) Which of the following will be diamagnetic? Circle the correct choice.
 A) PS^{1-} B) PS C) PS^{1+}
- (iv) Draw Lewis dot structure of the molecule with the shortest bond length (based on the MO diagram) from among the three (PS^{1-} , PS, PS^{1+}), and indicate the bond order calculated from the Lewis dot structure. **(3 points)**

Lewis Dot Structure

Bond Order



07F9E8B3-D40D-4103-8852-327AB68D18F6

midterm-exam-93388

#26 Page 10 of 10

SCRAP PAGE

**CHEMISTRY 110 –Midterm Exam****Thursday November 7, 2024, 6:30 – 8:30 pm****Examiners: Pallavi Sirjoosingh and Ashok Kakkar**

First name (please write as legibly as possible within the boxes)

--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--

Last name

--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--

Student ID number

--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--

This booklet has 8 questions on 10 pages (The last page can be used as a scrap page).

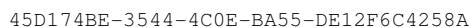
The staple must NOT be removed from the question booklet. Your intact question booklet must be returned at the end of the exam.

A separate booklet contains a periodic table, and datasheet.

Write your answers directly under the questions in the space provided. Do not write on the QR codes or near the top edges (as it may be cut off).

Please write **legibly and dark enough** for the scan to be readable. For students who use English as a second language, dictionaries are permitted. Calculators are allowed.

DO NOT ASK QUESTIONS OF THE INVIGILATORS about interpretations of the exam questions. If you have a concern about a question, please summarize it at the top of this page. All concerns will be evaluated after the exam. GOOD LUCK!



midterm-exam-93388

#27 Page 2 of 10

a. 6 points Calculate the **energy (in kJ/mol)** required to ionize (completely remove) the electron from the **n=3** level of **He⁺ ion** (He: Helium). Show your work including equations. (3 sig. figs.)

(I) Select the largest (longest) wavelength from the three choices below:

- Wavelength of photon to *completely remove* an electron from *1s orbital of Hydrogen atom*
- Wavelength of photon to *completely remove* an electron from *1s orbital of Boron atom (B)*
- Wavelength of photon to *completely remove* an electron from *1s orbital of Boron cation (B⁺)*

i or ii or iii

(II) Select the smallest (shortest) wavelength from the three choices below (*for H atom*):

- Wavelength of photon *emitted* for an electron transitioning from $2p$ to $1s$ orbital
- Wavelength of photon *absorbed* for an electron transitioning from $2s$ to $4p$ orbital
- Wavelength of photon *absorbed* for an electron transitioning from $1s$ to $3p$ orbital

i
or
ii
or
iii



Question 2 8 points



A Chem110 student plots the graph shown on the left, from data obtained for the kinetic energy (eV) of the ejected electron versus frequency (s^{-1}) of the incident light, for an unknown metal. ($1\text{eV} = 1.6022 \times 10^{-19} \text{ J}$)

Based on the plot, answer the questions **a to c** below:

a. 1 point On the plot above, label the **threshold frequency**.

b. 2 points What is the value of the work potential (in J) of the unknown metal?

c. 3 points What is the *longest* wavelength (in nm) required to eject an electron from this metal's surface? Show your work including equation(s) used. (3 sig figs)

d. 2 points In the space provided below, plot the following

(Assume frequency is *above* threshold frequency):

- Kinetic energy* of an ejected electron versus the *intensity* of the incident light (left)
- Number of electrons* ejected versus the *frequency* of the incident light (right)



**Question 3** **6 points****a. 2 points**

In the ground state of Ge (Germanium; $Z = 32$) atom, **name** the following orbitals:

i. Orbital containing the electron that requires the maximum energy to be removed.

ii. Orbital containing the electron that experiences the minimum effective nuclear charge.

b. 2 points

An unknown anion (A^-) has the following *excited* electronic configuration: $[Ar] 4s^2 3d^3 4p^1$

(Ar: Argon; $Z = 18$)

Give the **condensed ground state electronic configuration** of the atom, **A** and the cation, **A^{2+}** .
Show your work. (Assume all electronic transitions are possible)

Electronic Configuration of A:

Electronic Configuration of A^{2+} :

c. 2 points

In a single atom, determine how many electrons can have the following set of quantum numbers?

(i) $n = 5; l = 3$

In a single atom, determine how many electrons can have the following set of quantum numbers?

(i) $n = 3; l = 2; m_s = +1/2$

**Question 4** **8 points**

Rank the following from 1 to 4 in **increasing** order. Provide the ranking in the boxes provided.

Note: Atomic number (Z) are provided for relevant atoms

a) Increasing (lowest to highest) order of *ionization energy*: (1= lowest; 4 = highest)

Mg

Mg: Z = 12

Rb

Rb: Z = 37

Cs

Cs: Z = 55

K

K : Z = 19

b) Increasing (lowest to highest) order of *bond angle*: (1= lowest; 4 = highest)

H₂OCCl₄XeF₄

HCN

c) Increasing (smallest to largest) order of *ionic size* (1= smallest; 4 = largest)

S²⁻

S: Z = 16

O²⁻

O: Z = 8

Be²⁺

Be: Z = 4

Na⁺

Na: Z = 11

d) Increasing (lowest to highest) order of *bond energy* (1= lowest; 4 = highest)

P-F

P-Cl

P-I

P-Br

**Question 5** **12 points**

a. 9 points For the molecules HClO_4 and TeF_6 , draw the most stable *Lewis structure* showing appropriate shape and *dashed/wedged bonds*, based on VSEPR. Determine the **molecular geometry** around the central atom. For HClO_4 , label the *most polar bond*.



b. 3 points (Note: Lone pairs are not shown in structures below)

(i) For the three *resonance* structures (1, 2 and 3) shown below, answer the following question:



(I) Circle the *most contributing* resonance structure.

Structure 1

Structure 2

Structure 3

(ii) For the three structures given (4, 5, and 6) below, answer the following questions:



(I) Circle all the structures that represent *resonance structures* of each other:

Structure 4

Structure 5

Structure 6

(II) Among the resonance structures, circle the *most contributing* resonance structure.

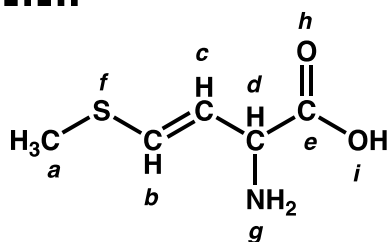
Structure 4

Structure 5

Structure 6

**Question 6** **6 points**Using the data given below, calculate the **lattice energy** for SrO(s).

	ΔH° (kJ/mol)
$\text{Sr}(s) \longrightarrow \text{Sr}(g)$	+180
$\text{Sr}(g) \longrightarrow \text{Sr}^{2+}(g) + 2e^-$	+1468
$\text{O}_2(g) \longrightarrow 2\text{O}(g)$	+500
$\text{O}(g) + 2e^- \longrightarrow \text{O}^{2-}(g)$	+600
$\text{Sr}(s) + \frac{1}{2}\text{O}_2(g) \longrightarrow \text{SrO}(s)$	-550

**Question 7 10 points**

Consider the molecule given on the left, and answer questions (i) to (vi) using chemical bonding theories that we have learnt in class (valence bond theory, hybridization).

Note that the lone pairs are not shown in the structure:

i. Consider the carbon atoms marked in as ***a***, ***b***, ***c***, ***d***, and ***e*** in the molecule, and determine **hybridization** at each of the carbon centers. (2.5 points)

Atom	Hybridization
Carbon (<i>a</i>)	
Carbon (<i>b</i>)	
Carbon (<i>c</i>)	
Carbon (<i>d</i>)	
Carbon (<i>e</i>)	

ii. Now, consider the sulfur atom (S) marked as ***f***, nitrogen (N) atom (***g***), and oxygen(O) atoms (***h*** and ***i***), and determine their **hybridization**. (2 points)

Atom	Hybridization
Sulfur (<i>f</i>)	
Nitrogen (<i>g</i>)	
Oxygen (<i>h</i>)	
Oxygen (<i>i</i>)	

iii. **Describe** all the bonds formed by the carbon atoms ***a*** and ***b***, using sigma and pi designation as appropriate. (2 points)

Atom	Bonds formed (use sigma and pi as appropriate)
Carbon (<i>a</i>)	
Carbon (<i>b</i>)	

iv. How many total sp^3 -s sigma bonds are in the molecule? (1 point)

v. Determine **electronic** and **molecular geometry** at carbon (C) labeled ***e*** and sulfur (S) labeled ***f*** centres. (2 points)

Atom	Electronic Geometry	Molecular Geometry
Carbon (<i>e</i>)		
Sulfur (<i>f</i>)		

vi. Circle the value from the choices given below, that describes the **C-S-C bond angle at sulfur** (S) marked ***f***: (0.5 points)

$>90^\circ$ to $<109.5^\circ$ 109.5° >109.5 to $<120^\circ$ 120° $>120^\circ$ to $<180^\circ$

**Question 8 12 points**

Using the **valence shell orbitals only** draw a **complete** molecular orbital (MO) diagram (with no s-p mixing) for the molecule **PS**, in the space provided below. **(6 points)**

Assume that 3s and 3p atomic orbitals will form molecular orbitals like those formed by the combination of 2s and 2p atomic orbitals that we learnt in class, and there is no s-p mixing.

Show **energy arrow**, **atomic orbitals** involved, **atom labels**, the **molecular orbitals** that are formed, **filled electrons** in the MO diagram etc. Make sure to label all the atomic and molecular orbitals.

Using the MO diagram above, answer the following questions:

- (i) Which one of the following will have the highest bond order? Circle the correct choice
A) PS^{1-} B) PS C) PS^{1+}
- (ii) Which one of the following will have the longest bond length? Circle the correct choice
A) PS^{1-} B) PS C) PS^{1+}
- (iii) Which of the following will be diamagnetic? Circle the correct choice.
A) PS^{1-} B) PS C) PS^{1+}
- (iv) Draw Lewis dot structure of the molecule with the shortest bond length (based on the MO diagram) from among the three (PS^{1-} , PS, PS^{1+}), and indicate the bond order calculated from the Lewis dot structure. **(3 points)**

Lewis Dot Structure

Bond Order



EDEB968D-131F-490D-B54C-A54BC558E93F

midterm-exam-93388

#27 Page 10 of 10

SCRAP PAGE

**CHEMISTRY 110 –Midterm Exam****Thursday November 7, 2024, 6:30 – 8:30 pm**

Examiners: Pallavi Sirjoosingh and Ashok Kakkar

First name (please write as legibly as possible within the boxes)

--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--

Last name

--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--

Student ID number

--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--

This booklet has 8 questions on 10 pages (The last page can be used as a scrap page).

The staple must NOT be removed from the question booklet. Your intact question booklet must be returned at the end of the exam.

A separate booklet contains a periodic table, and datasheet.

Write your answers directly under the questions in the space provided. Do not write on the QR codes or near the top edges (as it may be cut off).

Please write **legibly and dark enough** for the scan to be readable. For students who use English as a second language, dictionaries are permitted. Calculators are allowed.

DO NOT ASK QUESTIONS OF THE INVIGILATORS about interpretations of the exam questions. If you have a concern about a question, please summarize it at the top of this page. All concerns will be evaluated after the exam. GOOD LUCK!

**Question 1 8 points** (*Assume all transitions are possible*)

a. 6 points Calculate the **energy (in kJ/mol)** required to ionize (completely remove) the electron from the **$n=3$** level of **He^+ ion** (He: Helium). Show your work including equations. (3 sig. figs.)

b. 2 points

(I) Select the largest (longest) wavelength from the three choices below:

- i. Wavelength of photon to *completely remove* an electron from *1s orbital of Hydrogen atom*
- ii. Wavelength of photon to *completely remove* an electron from *1s orbital of Boron atom (B)*
- iii. Wavelength of photon to *completely remove* an electron from *1s orbital of Boron cation (B^+)*

i or ii or iii

(II) Select the smallest (shortest) wavelength from the three choices below (*for H atom*):

- i. Wavelength of photon *emitted* for an electron transitioning from *2p to 1s orbital*
- ii. Wavelength of photon *absorbed* for an electron transitioning from *2s to 4p orbital*
- iii. Wavelength of photon *absorbed* for an electron transitioning from *1s to 3p orbital*

i or ii or iii



Question 2 8 points



A Chem110 student plots the graph shown on the left, from data obtained for the kinetic energy (eV) of the ejected electron versus frequency (s^{-1}) of the incident light, for an unknown metal. ($1\text{eV} = 1.6022 \times 10^{-19} \text{ J}$)

Based on the plot, answer the questions **a to c** below:

a. 1 point On the plot above, label the **threshold frequency**.

b. 2 points What is the value of the work potential (in J) of the unknown metal?

c. 3 points What is the *longest* wavelength (in nm) required to eject an electron from this metal's surface? Show your work including equation(s) used. (3 sig figs)

d. 2 points In the space provided below, plot the following

(Assume frequency is *above* threshold frequency):

- i. *Kinetic energy* of an ejected electron versus the *intensity* of the incident light (left)
- ii. *Number of electrons* ejected versus the *frequency* of the incident light (right)



**Question 3** **6 points****a. 2 points**

In the ground state of Ge (Germanium; $Z = 32$) atom, **name** the following orbitals:

i. Orbital containing the electron that requires the maximum energy to be removed.

ii. Orbital containing the electron that experiences the minimum effective nuclear charge.

b. 2 points

An unknown anion (A^-) has the following *excited* electronic configuration: $[Ar] 4s^2 3d^3 4p^1$

(Ar: Argon; $Z = 18$)

Give the **condensed ground state electronic configuration** of the atom, **A** and the cation, **A^{2+}** .
Show your work. (Assume all electronic transitions are possible)

Electronic Configuration of A:

Electronic Configuration of A^{2+} :

c. 2 points

In a single atom, determine how many electrons can have the following set of quantum numbers?

(i) $n = 5; l = 3$

In a single atom, determine how many electrons can have the following set of quantum numbers?

(i) $n = 3; l = 2; m_s = +1/2$

**Question 4** **8 points**

Rank the following from 1 to 4 in **increasing** order. Provide the ranking in the boxes provided.

Note: Atomic number (Z) are provided for relevant atoms

a) Increasing (lowest to highest) order of *ionization energy*: (1= lowest; 4 = highest)

Mg

Mg: Z = 12

Rb

Rb: Z = 37

Cs

Cs: Z = 55

K

K : Z = 19

b) Increasing (lowest to highest) order of *bond angle*: (1= lowest; 4 = highest)

H₂OCCl₄XeF₄

HCN

c) Increasing (smallest to largest) order of *ionic size* (1= smallest; 4 = largest)

S²⁻

S: Z = 16

O²⁻

O: Z = 8

Be²⁺

Be: Z = 4

Na⁺

Na: Z = 11

d) Increasing (lowest to highest) order of *bond energy* (1= lowest; 4 = highest)

P-F

P-Cl

P-I

P-Br

**Question 5** **12 points**

a. 9 points For the molecules HClO_4 and TeF_6 , draw the most stable *Lewis structure* showing appropriate shape and *dashed/wedged bonds*, based on VSEPR. Determine the **molecular geometry** around the central atom. For HClO_4 , label the *most polar bond*.

HClO_4 (there are no O-O bonds)

Label the most polar bond

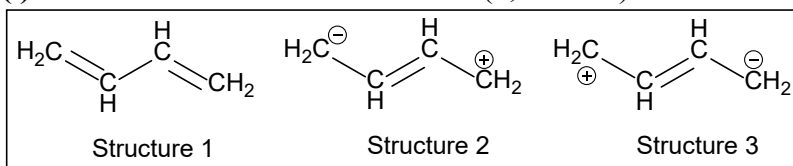
Molecular Geometry around Cl

TeF_6

Molecular Geometry around Te

b. 3 points (Note: Lone pairs are not shown in structures below)

(i) For the three *resonance* structures (1, 2 and 3) shown below, answer the following question:



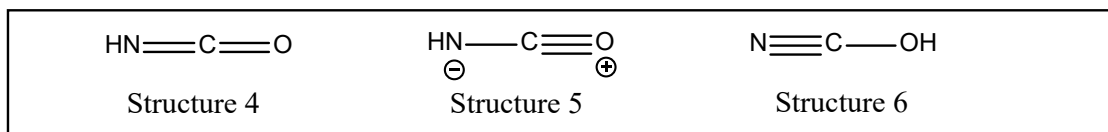
(I) Circle the *most contributing* resonance structure.

Structure 1

Structure 2

Structure 3

(ii) For the three structures given (4, 5, and 6) below, answer the following questions:



(I) Circle all the structures that represent *resonance structures* of each other:

Structure 4

Structure 5

Structure 6

(II) Among the resonance structures, circle the *most contributing* resonance structure.

Structure 4

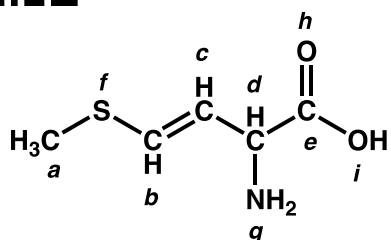
Structure 5

Structure 6

**Question 6** **6 points**

Using the data given below, calculate the **lattice energy** for SrO(s).

	ΔH° (kJ/mol)
$\text{Sr}(s) \longrightarrow \text{Sr}(g)$	+180
$\text{Sr}(g) \longrightarrow \text{Sr}^{2+}(g) + 2e^-$	+1468
$\text{O}_2(g) \longrightarrow 2\text{O}(g)$	+500
$\text{O}(g) + 2e^- \longrightarrow \text{O}^{2-}(g)$	+600
$\text{Sr}(s) + \frac{1}{2}\text{O}_2(g) \longrightarrow \text{SrO}(s)$	-550

**Question 7 10 points**

Consider the molecule given on the left, and answer questions (i) to (vi) using chemical bonding theories that we have learnt in class (valence bond theory, hybridization).

Note that the lone pairs are not shown in the structure:

i. Consider the carbon atoms marked in as ***a***, ***b***, ***c***, ***d***, and ***e*** in the molecule, and determine **hybridization** at each of the carbon centers. (2.5 points)

Atom	Hybridization
Carbon (<i>a</i>)	
Carbon (<i>b</i>)	
Carbon (<i>c</i>)	
Carbon (<i>d</i>)	
Carbon (<i>e</i>)	

ii. Now, consider the sulfur atom (S) marked as ***f***, nitrogen (N) atom (***g***), and oxygen(O) atoms (***h*** and ***i***), and determine their **hybridization**. (2 points)

Atom	Hybridization
Sulfur (<i>f</i>)	
Nitrogen (<i>g</i>)	
Oxygen (<i>h</i>)	
Oxygen (<i>i</i>)	

iii. **Describe** all the bonds formed by the carbon atoms ***a*** and ***b***, using sigma and pi designation as appropriate. (2 points)

Atom	Bonds formed (use sigma and pi as appropriate)
Carbon (<i>a</i>)	
Carbon (<i>b</i>)	

iv. How many total sp^3 -s sigma bonds are in the molecule? (1 point)

v. Determine **electronic** and **molecular geometry** at carbon (C) labeled ***e*** and sulfur (S) labeled ***f*** centres. (2 points)

Atom	Electronic Geometry	Molecular Geometry
Carbon (<i>e</i>)		
Sulfur (<i>f</i>)		

vi. Circle the value from the choices given below, that describes the **C-S-C bond angle at sulfur** (S) marked ***f***: (0.5 points)

>90° to <109.5° 109.5° >109.5 to <120° 120° >120° to <180°


Question 8 12 points

Using the **valence shell orbitals only** draw a **complete** molecular orbital (MO) diagram (with no s-p mixing) for the molecule **PS**, in the space provided below. **(6 points)**

Assume that 3s and 3p atomic orbitals will form molecular orbitals like those formed by the combination of 2s and 2p atomic orbitals that we learnt in class, and there is no s-p mixing.

Show **energy arrow**, **atomic orbitals** involved, **atom labels**, the **molecular orbitals** that are formed, **filled electrons** in the MO diagram etc. Make sure to label all the atomic and molecular orbitals.

Using the MO diagram above, answer the following questions:

- (i) Which one of the following will have the highest bond order? Circle the correct choice
 A) PS^{1-} B) PS C) PS^{1+}
- (ii) Which one of the following will have the longest bond length? Circle the correct choice
 A) PS^{1-} B) PS C) PS^{1+}
- (iii) Which of the following will be diamagnetic? Circle the correct choice.
 A) PS^{1-} B) PS C) PS^{1+}
- (iv) Draw Lewis dot structure of the molecule with the shortest bond length (based on the MO diagram) from among the three (PS^{1-} , PS, PS^{1+}), and indicate the bond order calculated from the Lewis dot structure. **(3 points)**

Lewis Dot Structure

Bond Order



071CA439-41AC-46EB-B94A-74CD7D76D4F7

midterm-exam-93388

#28 Page 10 of 10

SCRAP PAGE

**CHEMISTRY 110 –Midterm Exam****Thursday November 7, 2024, 6:30 – 8:30 pm**

Examiners: Pallavi Sirjoosingh and Ashok Kakkar

First name (please write as legibly as possible within the boxes)

--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--

Last name

--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--

Student ID number

--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--

This booklet has 8 questions on 10 pages (The last page can be used as a scrap page).

The staple must NOT be removed from the question booklet. Your intact question booklet must be returned at the end of the exam.

A separate booklet contains a periodic table, and datasheet.

Write your answers directly under the questions in the space provided. Do not write on the QR codes or near the top edges (as it may be cut off).

Please write **legibly and dark enough** for the scan to be readable. For students who use English as a second language, dictionaries are permitted. Calculators are allowed.

DO NOT ASK QUESTIONS OF THE INVIGILATORS about interpretations of the exam questions. If you have a concern about a question, please summarize it at the top of this page. All concerns will be evaluated after the exam. GOOD LUCK!

**Question 1 8 points** (*Assume all transitions are possible*)

a. 6 points Calculate the **energy (in kJ/mol)** required to ionize (completely remove) the electron from the **n=3** level of **He⁺ ion** (He: Helium). Show your work including equations. (3 sig. figs.)

b. 2 points

(I) Select the largest (longest) wavelength from the three choices below:

- i. Wavelength of photon to *completely remove* an electron from *1s orbital of Hydrogen atom*
- ii. Wavelength of photon to *completely remove* an electron from *1s orbital of Boron atom (B)*
- iii. Wavelength of photon to *completely remove* an electron from *1s orbital of Boron cation (B⁺)*

i or ii or iii

(II) Select the smallest (shortest) wavelength from the three choices below (*for H atom*):

- i. Wavelength of photon *emitted* for an electron transitioning from *2p to 1s orbital*
- ii. Wavelength of photon *absorbed* for an electron transitioning from *2s to 4p orbital*
- iii. Wavelength of photon *absorbed* for an electron transitioning from *1s to 3p orbital*

i or ii or iii

**Question 2 8 points**

A Chem110 student plots the graph shown on the left, from data obtained for the kinetic energy (eV) of the ejected electron versus frequency (s^{-1}) of the incident light, for an unknown metal. ($1\text{eV} = 1.6022 \times 10^{-19} \text{ J}$)

Based on the plot, answer the questions **a to c** below:

a. 1 point On the plot above, label the **threshold frequency**.

b. 2 points What is the value of the work potential (in J) of the unknown metal?

c. 3 points What is the *longest* wavelength (in nm) required to eject an electron from this metal's surface? Show your work including equation(s) used. (3 sig figs)

d. 2 points In the space provided below, plot the following

(Assume frequency is *above* threshold frequency):

- Kinetic energy* of an ejected electron versus the *intensity* of the incident light (left)
- Number of electrons* ejected versus the *frequency* of the incident light (right)



**Question 3** **6 points****a. 2 points**

In the ground state of Ge (Germanium; $Z = 32$) atom, **name** the following orbitals:

i. Orbital containing the electron that requires the maximum energy to be removed.

ii. Orbital containing the electron that experiences the minimum effective nuclear charge.

b. 2 points

An unknown anion (A^-) has the following *excited* electronic configuration: $[Ar] 4s^2 3d^3 4p^1$

(Ar: Argon; $Z = 18$)

Give the **condensed ground state electronic configuration** of the atom, **A** and the cation, **A^{2+}** .
Show your work. (Assume all electronic transitions are possible)

Electronic Configuration of A:

Electronic Configuration of A^{2+} :

c. 2 points

In a single atom, determine how many electrons can have the following set of quantum numbers?

(i) $n = 5; l = 3$

In a single atom, determine how many electrons can have the following set of quantum numbers?

(i) $n = 3; l = 2; m_s = +1/2$

**Question 4** **8 points**

Rank the following from 1 to 4 in **increasing** order. Provide the ranking in the boxes provided.

Note: Atomic number (Z) are provided for relevant atoms

a) Increasing (lowest to highest) order of *ionization energy*: (1= lowest; 4 = highest)

Mg

Mg: Z = 12

Rb

Rb: Z = 37

Cs

Cs: Z = 55

K

K : Z = 19

b) Increasing (lowest to highest) order of *bond angle*: (1= lowest; 4 = highest)

H₂OCCl₄XeF₄

HCN

c) Increasing (smallest to largest) order of *ionic size* (1= smallest; 4 = largest)

S²⁻

S: Z = 16

O²⁻

O: Z = 8

Be²⁺

Be: Z = 4

Na⁺

Na: Z = 11

d) Increasing (lowest to highest) order of *bond energy* (1= lowest; 4 = highest)

P-F

P-Cl

P-I

P-Br

**Question 5** **12 points**

a. 9 points For the molecules HClO_4 and TeF_6 , draw the most stable *Lewis structure* showing appropriate shape and *dashed/wedged bonds*, based on VSEPR. Determine the **molecular geometry** around the central atom. For HClO_4 , label the *most polar bond*.



b. 3 points (Note: Lone pairs are not shown in structures below)

(i) For the three *resonance* structures (1, 2 and 3) shown below, answer the following question:



(I) Circle the *most contributing* resonance structure.

Structure 1

Structure 2

Structure 3

(ii) For the three structures given (4, 5, and 6) below, answer the following questions:



(I) Circle all the structures that represent *resonance structures* of each other:

Structure 4

Structure 5

Structure 6

(II) Among the resonance structures, circle the *most contributing* resonance structure.

Structure 4

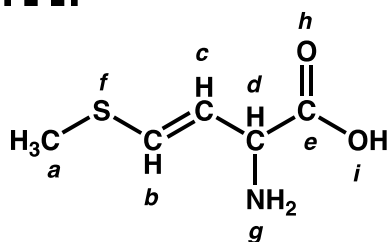
Structure 5

Structure 6

**Question 6** **6 points**

Using the data given below, calculate the **lattice energy** for SrO(s).

	ΔH° (kJ/mol)
$\text{Sr}(s) \longrightarrow \text{Sr}(g)$	+180
$\text{Sr}(g) \longrightarrow \text{Sr}^{2+}(g) + 2e^-$	+1468
$\text{O}_2(g) \longrightarrow 2\text{O}(g)$	+500
$\text{O}(g) + 2e^- \longrightarrow \text{O}^{2-}(g)$	+600
$\text{Sr}(s) + \frac{1}{2}\text{O}_2(g) \longrightarrow \text{SrO}(s)$	-550

**Question 7 10 points**

Consider the molecule given on the left, and answer questions (i) to (vi) using chemical bonding theories that we have learnt in class (valence bond theory, hybridization).

Note that the lone pairs are not shown in the structure:

i. Consider the carbon atoms marked in as ***a***, ***b***, ***c***, ***d***, and ***e*** in the molecule, and determine **hybridization** at each of the carbon centers. (2.5 points)

Atom	Hybridization
Carbon (<i>a</i>)	
Carbon (<i>b</i>)	
Carbon (<i>c</i>)	
Carbon (<i>d</i>)	
Carbon (<i>e</i>)	

ii. Now, consider the sulfur atom (S) marked as ***f***, nitrogen (N) atom (***g***), and oxygen(O) atoms (***h*** and ***i***), and determine their **hybridization**. (2 points)

Atom	Hybridization
Sulfur (<i>f</i>)	
Nitrogen (<i>g</i>)	
Oxygen (<i>h</i>)	
Oxygen (<i>i</i>)	

iii. **Describe** all the bonds formed by the carbon atoms ***a*** and ***b***, using sigma and pi designation as appropriate. (2 points)

Atom	Bonds formed (use sigma and pi as appropriate)
Carbon (<i>a</i>)	
Carbon (<i>b</i>)	

iv. How many total sp^3 -s sigma bonds are in the molecule? (1 point)

v. Determine **electronic** and **molecular geometry** at carbon (C) labeled ***e*** and sulfur (S) labeled ***f*** centres. (2 points)

Atom	Electronic Geometry	Molecular Geometry
Carbon (<i>e</i>)		
Sulfur (<i>f</i>)		

vi. Circle the value from the choices given below, that describes the **C-S-C bond angle at sulfur** (S) marked ***f***: (0.5 points)

>90° to <109.5° 109.5° >109.5 to <120° 120° >120° to <180°


Question 8 12 points

Using the **valence shell orbitals only** draw a **complete** molecular orbital (MO) diagram (with no s-p mixing) for the molecule **PS**, in the space provided below. **(6 points)**

Assume that 3s and 3p atomic orbitals will form molecular orbitals like those formed by the combination of 2s and 2p atomic orbitals that we learnt in class, and there is no s-p mixing.

Show **energy arrow**, **atomic orbitals** involved, **atom labels**, the **molecular orbitals** that are formed, **filled electrons** in the MO diagram etc. Make sure to label all the atomic and molecular orbitals.

Using the MO diagram above, answer the following questions:

- (i) Which one of the following will have the highest bond order? Circle the correct choice
 A) PS^{1-} B) PS C) PS^{1+}
- (ii) Which one of the following will have the longest bond length? Circle the correct choice
 A) PS^{1-} B) PS C) PS^{1+}
- (iii) Which of the following will be diamagnetic? Circle the correct choice.
 A) PS^{1-} B) PS C) PS^{1+}
- (iv) Draw Lewis dot structure of the molecule with the shortest bond length (based on the MO diagram) from among the three (PS^{1-} , PS, PS^{1+}), and indicate the bond order calculated from the Lewis dot structure. **(3 points)**

Lewis Dot Structure

Bond Order



5F4B877D-E3B5-4C4B-B29D-9F91D3E4679F

midterm-exam-93388

#29 Page 10 of 10

SCRAP PAGE

**CHEMISTRY 110 –Midterm Exam****Thursday November 7, 2024, 6:30 – 8:30 pm**

Examiners: Pallavi Sirjoosingh and Ashok Kakkar

First name (please write as legibly as possible within the boxes)

--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--

Last name

--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--

Student ID number

--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--

This booklet has 8 questions on 10 pages (The last page can be used as a scrap page).

The staple must NOT be removed from the question booklet. Your intact question booklet must be returned at the end of the exam.

A separate booklet contains a periodic table, and datasheet.

Write your answers directly under the questions in the space provided. Do not write on the QR codes or near the top edges (as it may be cut off).

Please write **legibly and dark enough** for the scan to be readable. For students who use English as a second language, dictionaries are permitted. Calculators are allowed.

DO NOT ASK QUESTIONS OF THE INVIGILATORS about interpretations of the exam questions. If you have a concern about a question, please summarize it at the top of this page. All concerns will be evaluated after the exam. GOOD LUCK!

**Question 1 8 points** (*Assume all transitions are possible*)

a. 6 points Calculate the **energy (in kJ/mol)** required to ionize (completely remove) the electron from the **n=3** level of **He⁺ ion** (He: Helium). Show your work including equations. (3 sig. figs.)

b. 2 points

(I) Select the largest (longest) wavelength from the three choices below:

- i. Wavelength of photon to *completely remove* an electron from *1s orbital of Hydrogen atom*
- ii. Wavelength of photon to *completely remove* an electron from *1s orbital of Boron atom (B)*
- iii. Wavelength of photon to *completely remove* an electron from *1s orbital of Boron cation (B⁺)*

i or ii or iii

(II) Select the smallest (shortest) wavelength from the three choices below (*for H atom*):

- i. Wavelength of photon *emitted* for an electron transitioning from *2p to 1s orbital*
- ii. Wavelength of photon *absorbed* for an electron transitioning from *2s to 4p orbital*
- iii. Wavelength of photon *absorbed* for an electron transitioning from *1s to 3p orbital*

i or ii or iii



Question 2 8 points



A Chem110 student plots the graph shown on the left, from data obtained for the kinetic energy (eV) of the ejected electron versus frequency (s^{-1}) of the incident light, for an unknown metal. ($1\text{eV} = 1.6022 \times 10^{-19} \text{ J}$)

Based on the plot, answer the questions **a to c** below:

a. 1 point On the plot above, label the **threshold frequency**.

b. 2 points What is the value of the work potential (in J) of the unknown metal?

c. 3 points What is the *longest* wavelength (in nm) required to eject an electron from this metal's surface? Show your work including equation(s) used. (3 sig figs)

d. 2 points In the space provided below, plot the following

(Assume frequency is *above* threshold frequency):

- i. *Kinetic energy* of an ejected electron versus the *intensity* of the incident light (left)
- ii. *Number of electrons* ejected versus the *frequency* of the incident light (right)



**Question 3** **6 points****a. 2 points**

In the ground state of Ge (Germanium; $Z = 32$) atom, **name** the following orbitals:

i. Orbital containing the electron that requires the maximum energy to be removed.

ii. Orbital containing the electron that experiences the minimum effective nuclear charge.

b. 2 points

An unknown anion (A^-) has the following *excited* electronic configuration: $[Ar] 4s^2 3d^3 4p^1$

(Ar: Argon; $Z = 18$)

Give the **condensed ground state electronic configuration** of the atom, **A** and the cation, **A^{2+}** .
Show your work. (Assume all electronic transitions are possible)

Electronic Configuration of A:

Electronic Configuration of A^{2+} :

c. 2 points

In a single atom, determine how many electrons can have the following set of quantum numbers?

(i) $n = 5; l = 3$

In a single atom, determine how many electrons can have the following set of quantum numbers?

(i) $n = 3; l = 2; m_s = +1/2$

**Question 4** **8 points**

Rank the following from 1 to 4 in **increasing** order. Provide the ranking in the boxes provided.

Note: Atomic number (Z) are provided for relevant atoms

a) Increasing (lowest to highest) order of *ionization energy*: (1= lowest; 4 = highest)

Mg

Mg: Z = 12

Rb

Rb: Z = 37

Cs

Cs: Z = 55

K

K : Z = 19

b) Increasing (lowest to highest) order of *bond angle*: (1= lowest; 4 = highest)

H₂OCCl₄XeF₄

HCN

c) Increasing (smallest to largest) order of *ionic size* (1= smallest; 4 = largest)

S²⁻

S: Z = 16

O²⁻

O: Z = 8

Be²⁺

Be: Z = 4

Na⁺

Na: Z = 11

d) Increasing (lowest to highest) order of *bond energy* (1= lowest; 4 = highest)

P-F

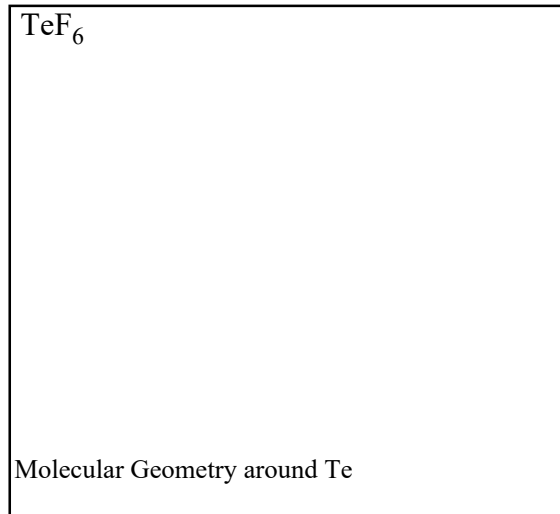
P-Cl

P-I

P-Br

**Question 5** **12 points**

a. 9 points For the molecules HClO_4 and TeF_6 , draw the most stable *Lewis structure* showing appropriate shape and *dashed/wedged bonds*, based on VSEPR. Determine the **molecular geometry** around the central atom. For HClO_4 , label the *most polar bond*.



b. 3 points (Note: Lone pairs are not shown in structures below)

(i) For the three *resonance* structures (1, 2 and 3) shown below, answer the following question:



(I) Circle the *most contributing* resonance structure.

Structure 1

Structure 2

Structure 3

(ii) For the three structures given (4, 5, and 6) below, answer the following questions:



(I) Circle all the structures that represent *resonance structures* of each other:

Structure 4

Structure 5

Structure 6

(II) Among the resonance structures, circle the *most contributing* resonance structure.

Structure 4

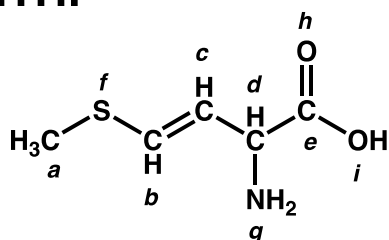
Structure 5

Structure 6

**Question 6** **6 points**

Using the data given below, calculate the **lattice energy** for SrO(s).

	ΔH° (kJ/mol)
$\text{Sr}(s) \longrightarrow \text{Sr}(g)$	+180
$\text{Sr}(g) \longrightarrow \text{Sr}^{2+}(g) + 2e^-$	+1468
$\text{O}_2(g) \longrightarrow 2\text{O}(g)$	+500
$\text{O}(g) + 2e^- \longrightarrow \text{O}^{2-}(g)$	+600
$\text{Sr}(s) + \frac{1}{2}\text{O}_2(g) \longrightarrow \text{SrO}(s)$	-550

**Question 7 10 points**

Consider the molecule given on the left, and answer questions (i) to (vi) using chemical bonding theories that we have learnt in class (valence bond theory, hybridization).

Note that the lone pairs are not shown in the structure:

i. Consider the carbon atoms marked in as ***a***, ***b***, ***c***, ***d***, and ***e*** in the molecule, and determine **hybridization** at each of the carbon centers. (2.5 points)

Atom	Hybridization
Carbon (<i>a</i>)	
Carbon (<i>b</i>)	
Carbon (<i>c</i>)	
Carbon (<i>d</i>)	
Carbon (<i>e</i>)	

ii. Now, consider the sulfur atom (S) marked as ***f***, nitrogen (N) atom (***g***), and oxygen(O) atoms (***h*** and ***i***), and determine their **hybridization**. (2 points)

Atom	Hybridization
Sulfur (<i>f</i>)	
Nitrogen (<i>g</i>)	
Oxygen (<i>h</i>)	
Oxygen (<i>i</i>)	

iii. **Describe** all the bonds formed by the carbon atoms ***a*** and ***b***, using sigma and pi designation as appropriate. (2 points)

Atom	Bonds formed (use sigma and pi as appropriate)
Carbon (<i>a</i>)	
Carbon (<i>b</i>)	

iv. How many total sp^3 -s sigma bonds are in the molecule? (1 point)

v. Determine **electronic** and **molecular geometry** at carbon (C) labeled ***e*** and sulfur (S) labeled ***f*** centres. (2 points)

Atom	Electronic Geometry	Molecular Geometry
Carbon (<i>e</i>)		
Sulfur (<i>f</i>)		

vi. Circle the value from the choices given below, that describes the **C-S-C bond angle at sulfur** (S) marked ***f***: (0.5 points)

>90° to <109.5° 109.5° >109.5 to <120° 120° >120° to <180°


Question 8 12 points

Using the **valence shell orbitals only** draw a **complete** molecular orbital (MO) diagram (with no s-p mixing) for the molecule **PS**, in the space provided below. **(6 points)**

Assume that 3s and 3p atomic orbitals will form molecular orbitals like those formed by the combination of 2s and 2p atomic orbitals that we learnt in class, and there is no s-p mixing.

Show **energy arrow**, **atomic orbitals** involved, **atom labels**, the **molecular orbitals** that are formed, **filled electrons** in the MO diagram etc. Make sure to label all the atomic and molecular orbitals.

Using the MO diagram above, answer the following questions:

- (i) Which one of the following will have the highest bond order? Circle the correct choice
 A) PS^{1-} B) PS C) PS^{1+}
- (ii) Which one of the following will have the longest bond length? Circle the correct choice
 A) PS^{1-} B) PS C) PS^{1+}
- (iii) Which of the following will be diamagnetic? Circle the correct choice.
 A) PS^{1-} B) PS C) PS^{1+}
- (iv) Draw Lewis dot structure of the molecule with the shortest bond length (based on the MO diagram) from among the three (PS^{1-} , PS, PS^{1+}), and indicate the bond order calculated from the Lewis dot structure. **(3 points)**

Lewis Dot Structure

Bond Order



B6FEBDE2-7938-4568-85E8-249B5D7221F1

midterm-exam-93388

#30 Page 10 of 10

SCRAP PAGE

**CHEMISTRY 110 –Midterm Exam****Thursday November 7, 2024, 6:30 – 8:30 pm**

Examiners: Pallavi Sirjoosingh and Ashok Kakkar

First name (please write as legibly as possible within the boxes)

--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--

Last name

--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--

Student ID number

--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--

This booklet has 8 questions on 10 pages (The last page can be used as a scrap page).

The staple must NOT be removed from the question booklet. Your intact question booklet must be returned at the end of the exam.

A separate booklet contains a periodic table, and datasheet.

Write your answers directly under the questions in the space provided. Do not write on the QR codes or near the top edges (as it may be cut off).

Please write **legibly and dark enough** for the scan to be readable. For students who use English as a second language, dictionaries are permitted. Calculators are allowed.

DO NOT ASK QUESTIONS OF THE INVIGILATORS about interpretations of the exam questions. If you have a concern about a question, please summarize it at the top of this page. All concerns will be evaluated after the exam. GOOD LUCK!

**Question 1 8 points** (*Assume all transitions are possible*)

a. 6 points Calculate the **energy (in kJ/mol)** required to ionize (completely remove) the electron from the **n=3** level of **He⁺ ion** (He: Helium). Show your work including equations. (3 sig. figs.)

b. 2 points

(I) Select the largest (longest) wavelength from the three choices below:

- i. Wavelength of photon to *completely remove* an electron from *1s orbital of Hydrogen atom*
- ii. Wavelength of photon to *completely remove* an electron from *1s orbital of Boron atom (B)*
- iii. Wavelength of photon to *completely remove* an electron from *1s orbital of Boron cation (B⁺)*

i or ii or iii

(II) Select the smallest (shortest) wavelength from the three choices below (*for H atom*):

- i. Wavelength of photon *emitted* for an electron transitioning from *2p to 1s orbital*
- ii. Wavelength of photon *absorbed* for an electron transitioning from *2s to 4p orbital*
- iii. Wavelength of photon *absorbed* for an electron transitioning from *1s to 3p orbital*

i or ii or iii



Question 2 8 points



A Chem110 student plots the graph shown on the left, from data obtained for the kinetic energy (eV) of the ejected electron versus frequency (s^{-1}) of the incident light, for an unknown metal. ($1\text{eV} = 1.6022 \times 10^{-19} \text{ J}$)

Based on the plot, answer the questions **a to c** below:

a. 1 point On the plot above, label the **threshold frequency**.

b. 2 points What is the value of the work potential (in J) of the unknown metal?

c. 3 points What is the *longest* wavelength (in nm) required to eject an electron from this metal's surface? Show your work including equation(s) used. (3 sig figs)

d. 2 points In the space provided below, plot the following

(Assume frequency is *above* threshold frequency):

- i. *Kinetic energy* of an ejected electron versus the *intensity* of the incident light (left)
- ii. *Number of electrons* ejected versus the *frequency* of the incident light (right)



**Question 3** **6 points****a. 2 points**

In the ground state of Ge (Germanium; $Z = 32$) atom, **name** the following orbitals:

i. Orbital containing the electron that requires the maximum energy to be removed.

ii. Orbital containing the electron that experiences the minimum effective nuclear charge.

b. 2 points

An unknown anion (A^-) has the following *excited* electronic configuration: $[Ar] 4s^2 3d^3 4p^1$

(Ar: Argon; $Z = 18$)

Give the **condensed ground state electronic configuration** of the atom, **A** and the cation, **A^{2+}** .
Show your work. (Assume all electronic transitions are possible)

Electronic Configuration of A:

Electronic Configuration of A^{2+} :

c. 2 points

In a single atom, determine how many electrons can have the following set of quantum numbers?

(i) $n = 5; l = 3$

In a single atom, determine how many electrons can have the following set of quantum numbers?

(i) $n = 3; l = 2; m_s = +1/2$

**Question 4** **8 points**

Rank the following from 1 to 4 in **increasing** order. Provide the ranking in the boxes provided.

Note: Atomic number (Z) are provided for relevant atoms

a) Increasing (lowest to highest) order of *ionization energy*: (1= lowest; 4 = highest)

Mg

Mg: Z = 12

Rb

Rb: Z = 37

Cs

Cs: Z = 55

K

K : Z = 19

b) Increasing (lowest to highest) order of *bond angle*: (1= lowest; 4 = highest)

H₂OCCl₄XeF₄

HCN

c) Increasing (smallest to largest) order of *ionic size* (1= smallest; 4 = largest)

S²⁻

S: Z = 16

O²⁻

O: Z = 8

Be²⁺

Be: Z = 4

Na⁺

Na: Z = 11

d) Increasing (lowest to highest) order of *bond energy* (1= lowest; 4 = highest)

P-F

P-Cl

P-I

P-Br

**Question 5** **12 points**

a. 9 points For the molecules HClO_4 and TeF_6 , draw the most stable *Lewis structure* showing appropriate shape and *dashed/wedged bonds*, based on VSEPR. Determine the **molecular geometry** around the central atom. For HClO_4 , label the *most polar bond*.



b. 3 points (Note: Lone pairs are not shown in structures below)

(i) For the three *resonance* structures (1, 2 and 3) shown below, answer the following question:



(I) Circle the *most contributing* resonance structure.

Structure 1

Structure 2

Structure 3

(ii) For the three structures given (4, 5, and 6) below, answer the following questions:



(I) Circle all the structures that represent *resonance structures* of each other:

Structure 4

Structure 5

Structure 6

(II) Among the resonance structures, circle the *most contributing* resonance structure.

Structure 4

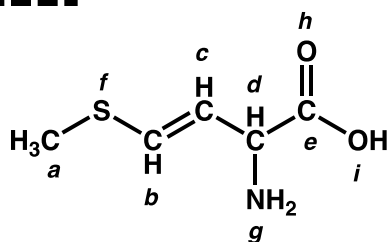
Structure 5

Structure 6

**Question 6** **6 points**

Using the data given below, calculate the **lattice energy** for SrO(s).

	ΔH° (kJ/mol)
$\text{Sr}(s) \longrightarrow \text{Sr}(g)$	+180
$\text{Sr}(g) \longrightarrow \text{Sr}^{2+}(g) + 2e^-$	+1468
$\text{O}_2(g) \longrightarrow 2\text{O}(g)$	+500
$\text{O}(g) + 2e^- \longrightarrow \text{O}^{2-}(g)$	+600
$\text{Sr}(s) + \frac{1}{2}\text{O}_2(g) \longrightarrow \text{SrO}(s)$	-550

**Question 7 10 points**

Consider the molecule given on the left, and answer questions (i) to (vi) using chemical bonding theories that we have learnt in class (valence bond theory, hybridization).

Note that the lone pairs are not shown in the structure:

i. Consider the carbon atoms marked in as ***a***, ***b***, ***c***, ***d***, and ***e*** in the molecule, and determine **hybridization** at each of the carbon centers. (2.5 points)

Atom	Hybridization
Carbon (<i>a</i>)	
Carbon (<i>b</i>)	
Carbon (<i>c</i>)	
Carbon (<i>d</i>)	
Carbon (<i>e</i>)	

ii. Now, consider the sulfur atom (S) marked as ***f***, nitrogen (N) atom (***g***), and oxygen(O) atoms (***h*** and ***i***), and determine their **hybridization**. (2 points)

Atom	Hybridization
Sulfur (<i>f</i>)	
Nitrogen (<i>g</i>)	
Oxygen (<i>h</i>)	
Oxygen (<i>i</i>)	

iii. **Describe** all the bonds formed by the carbon atoms ***a*** and ***b***, using sigma and pi designation as appropriate. (2 points)

Atom	Bonds formed (use sigma and pi as appropriate)
Carbon (<i>a</i>)	
Carbon (<i>b</i>)	

iv. How many total sp^3 -s sigma bonds are in the molecule? (1 point)

v. Determine **electronic** and **molecular geometry** at carbon (C) labeled ***e*** and sulfur (S) labeled ***f*** centres. (2 points)

Atom	Electronic Geometry	Molecular Geometry
Carbon (<i>e</i>)		
Sulfur (<i>f</i>)		

vi. Circle the value from the choices given below, that describes the **C-S-C bond angle at sulfur** (S) marked ***f***: (0.5 points)

$>90^\circ$ to $<109.5^\circ$ 109.5° >109.5 to $<120^\circ$ 120° $>120^\circ$ to $<180^\circ$


Question 8 12 points

Using the **valence shell orbitals only** draw a **complete** molecular orbital (MO) diagram (with no s-p mixing) for the molecule **PS**, in the space provided below. **(6 points)**

Assume that 3s and 3p atomic orbitals will form molecular orbitals like those formed by the combination of 2s and 2p atomic orbitals that we learnt in class, and there is no s-p mixing.

Show **energy arrow**, **atomic orbitals** involved, **atom labels**, the **molecular orbitals** that are formed, **filled electrons** in the MO diagram etc. Make sure to label all the atomic and molecular orbitals.

Using the MO diagram above, answer the following questions:

- (i) Which one of the following will have the highest bond order? Circle the correct choice
 A) PS^{1-} B) PS C) PS^{1+}
- (ii) Which one of the following will have the longest bond length? Circle the correct choice
 A) PS^{1-} B) PS C) PS^{1+}
- (iii) Which of the following will be diamagnetic? Circle the correct choice.
 A) PS^{1-} B) PS C) PS^{1+}
- (iv) Draw Lewis dot structure of the molecule with the shortest bond length (based on the MO diagram) from among the three (PS^{1-} , PS, PS^{1+}), and indicate the bond order calculated from the Lewis dot structure. **(3 points)**

Lewis Dot Structure

Bond Order



4BA369DE-FC20-408A-A1E3-9308DEEC65D4

midterm-exam-93388

#31 Page 10 of 10

SCRAP PAGE

**CHEMISTRY 110 –Midterm Exam****Thursday November 7, 2024, 6:30 – 8:30 pm****Examiners: Pallavi Sirjoosingh and Ashok Kakkar**

First name (please write as legibly as possible within the boxes)

--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--

Last name

--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--

Student ID number

--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--

This booklet has 8 questions on 10 pages (The last page can be used as a scrap page).

The staple must NOT be removed from the question booklet. Your intact question booklet must be returned at the end of the exam.

A separate booklet contains a periodic table, and datasheet.

Write your answers directly under the questions in the space provided. Do not write on the QR codes or near the top edges (as it may be cut off).

Please write **legibly and dark enough** for the scan to be readable. For students who use English as a second language, dictionaries are permitted. Calculators are allowed.

DO NOT ASK QUESTIONS OF THE INVIGILATORS about interpretations of the exam questions. If you have a concern about a question, please summarize it at the top of this page. All concerns will be evaluated after the exam. GOOD LUCK!

**Question 1 8 points** (*Assume all transitions are possible*)

a. 6 points Calculate the **energy (in kJ/mol)** required to ionize (completely remove) the electron from the **n=3** level of **He⁺ ion** (He: Helium). Show your work including equations. (3 sig. figs.)

b. 2 points

(I) Select the largest (longest) wavelength from the three choices below:

- i. Wavelength of photon to *completely remove* an electron from *1s orbital of Hydrogen atom*
- ii. Wavelength of photon to *completely remove* an electron from *1s orbital of Boron atom (B)*
- iii. Wavelength of photon to *completely remove* an electron from *1s orbital of Boron cation (B⁺)*

i or ii or iii

(II) Select the smallest (shortest) wavelength from the three choices below (*for H atom*):

- i. Wavelength of photon *emitted* for an electron transitioning from *2p to 1s orbital*
- ii. Wavelength of photon *absorbed* for an electron transitioning from *2s to 4p orbital*
- iii. Wavelength of photon *absorbed* for an electron transitioning from *1s to 3p orbital*

i or ii or iii



Question 2 8 points



A Chem110 student plots the graph shown on the left, from data obtained for the kinetic energy (eV) of the ejected electron versus frequency (s^{-1}) of the incident light, for an unknown metal. ($1\text{eV} = 1.6022 \times 10^{-19} \text{ J}$)

Based on the plot, answer the questions **a to c** below:

a. 1 point On the plot above, label the **threshold frequency**.

b. 2 points What is the value of the work potential (in J) of the unknown metal?

c. 3 points What is the *longest* wavelength (in nm) required to eject an electron from this metal's surface? Show your work including equation(s) used. (3 sig figs)

d. 2 points In the space provided below, plot the following

(Assume frequency is *above* threshold frequency):

- Kinetic energy* of an ejected electron versus the *intensity* of the incident light (left)
- Number of electrons* ejected versus the *frequency* of the incident light (right)



**Question 3** **6 points****a. 2 points**

In the ground state of Ge (Germanium; $Z = 32$) atom, **name** the following orbitals:

i. Orbital containing the electron that requires the maximum energy to be removed.

ii. Orbital containing the electron that experiences the minimum effective nuclear charge.

b. 2 points

An unknown anion (A^-) has the following *excited* electronic configuration: $[Ar] 4s^2 3d^3 4p^1$

(Ar: Argon; $Z = 18$)

Give the **condensed ground state electronic configuration** of the atom, **A** and the cation, **A^{2+}** .
Show your work. (Assume all electronic transitions are possible)

Electronic Configuration of A:

Electronic Configuration of A^{2+} :

c. 2 points

In a single atom, determine how many electrons can have the following set of quantum numbers?

(i) $n = 5; l = 3$

In a single atom, determine how many electrons can have the following set of quantum numbers?

(i) $n = 3; l = 2; m_s = +1/2$


Question 4 **8 points**

Rank the following from 1 to 4 in **increasing** order. Provide the ranking in the boxes provided.

Note: Atomic number (Z) are provided for relevant atoms

a) Increasing (lowest to highest) order of *ionization energy*: (1= lowest; 4 = highest)

Mg

Mg: Z = 12

Rb

Rb: Z = 37

Cs

Cs: Z = 55

K

K : Z = 19

b) Increasing (lowest to highest) order of *bond angle*: (1= lowest; 4 = highest)

H₂O
CCl₄

XeF₄

HCN

c) Increasing (smallest to largest) order of *ionic size* (1= smallest; 4 = largest)

S²⁻

S: Z = 16

O²⁻

O: Z = 8

Be²⁺

Be: Z = 4

Na⁺

Na: Z = 11

d) Increasing (lowest to highest) order of *bond energy* (1= lowest; 4 = highest)

P-F

P-Cl

P-I

P-Br

**Question 5** **12 points**

a. 9 points For the molecules HClO_4 and TeF_6 , draw the most stable *Lewis structure* showing appropriate shape and *dashed/wedged bonds*, based on VSEPR. Determine the **molecular geometry** around the central atom. For HClO_4 , label the *most polar bond*.



b. 3 points (Note: Lone pairs are not shown in structures below)

(i) For the three *resonance* structures (1, 2 and 3) shown below, answer the following question:



(I) Circle the *most contributing* resonance structure.

Structure 1

Structure 2

Structure 3

(ii) For the three structures given (4, 5, and 6) below, answer the following questions:



(I) Circle all the structures that represent *resonance structures* of each other:

Structure 4

Structure 5

Structure 6

(II) Among the resonance structures, circle the *most contributing* resonance structure.

Structure 4

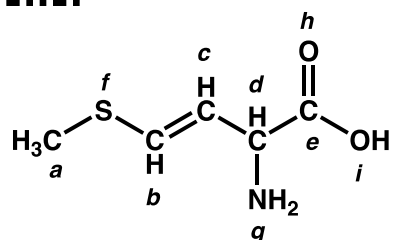
Structure 5

Structure 6

**Question 6** **6 points**

Using the data given below, calculate the **lattice energy** for SrO(s).

	ΔH° (kJ/mol)
$\text{Sr}(s) \longrightarrow \text{Sr}(g)$	+180
$\text{Sr}(g) \longrightarrow \text{Sr}^{2+}(g) + 2e^-$	+1468
$\text{O}_2(g) \longrightarrow 2\text{O}(g)$	+500
$\text{O}(g) + 2e^- \longrightarrow \text{O}^{2-}(g)$	+600
$\text{Sr}(s) + \frac{1}{2}\text{O}_2(g) \longrightarrow \text{SrO}(s)$	-550

**Question 7 10 points**

Consider the molecule given on the left, and answer questions (i) to (vi) using chemical bonding theories that we have learnt in class (valence bond theory, hybridization).

Note that the lone pairs are not shown in the structure:

i. Consider the carbon atoms marked in as ***a***, ***b***, ***c***, ***d***, and ***e*** in the molecule, and determine **hybridization** at each of the carbon centers. (2.5 points)

Atom	Hybridization
Carbon (<i>a</i>)	
Carbon (<i>b</i>)	
Carbon (<i>c</i>)	
Carbon (<i>d</i>)	
Carbon (<i>e</i>)	

ii. Now, consider the sulfur atom (S) marked as ***f***, nitrogen (N) atom (***g***), and oxygen(O) atoms (***h*** and ***i***), and determine their **hybridization**. (2 points)

Atom	Hybridization
Sulfur (<i>f</i>)	
Nitrogen (<i>g</i>)	
Oxygen (<i>h</i>)	
Oxygen (<i>i</i>)	

iii. **Describe** all the bonds formed by the carbon atoms ***a*** and ***b***, using sigma and pi designation as appropriate. (2 points)

Atom	Bonds formed (use sigma and pi as appropriate)
Carbon (<i>a</i>)	
Carbon (<i>b</i>)	

iv. How many total sp^3 -s sigma bonds are in the molecule? (1 point)

v. Determine **electronic** and **molecular geometry** at carbon (C) labeled ***e*** and sulfur (S) labeled ***f*** centres. (2 points)

Atom	Electronic Geometry	Molecular Geometry
Carbon (<i>e</i>)		
Sulfur (<i>f</i>)		

vi. Circle the value from the choices given below, that describes the **C-S-C bond angle at sulfur** (S) marked ***f***: (0.5 points)

>90° to <109.5° 109.5° >109.5 to <120° 120° >120° to <180°


Question 8 12 points

Using the **valence shell orbitals only** draw a **complete** molecular orbital (MO) diagram (with no s-p mixing) for the molecule **PS**, in the space provided below. **(6 points)**

Assume that 3s and 3p atomic orbitals will form molecular orbitals like those formed by the combination of 2s and 2p atomic orbitals that we learnt in class, and there is no s-p mixing.

Show **energy arrow**, **atomic orbitals** involved, **atom labels**, the **molecular orbitals** that are formed, **filled electrons** in the MO diagram etc. Make sure to label all the atomic and molecular orbitals.

Using the MO diagram above, answer the following questions:

- (i) Which one of the following will have the highest bond order? Circle the correct choice
 A) PS^{1-} B) PS C) PS^{1+}
- (ii) Which one of the following will have the longest bond length? Circle the correct choice
 A) PS^{1-} B) PS C) PS^{1+}
- (iii) Which of the following will be diamagnetic? Circle the correct choice.
 A) PS^{1-} B) PS C) PS^{1+}
- (iv) Draw Lewis dot structure of the molecule with the shortest bond length (based on the MO diagram) from among the three (PS^{1-} , PS, PS^{1+}), and indicate the bond order calculated from the Lewis dot structure. **(3 points)**

Lewis Dot Structure

Bond Order



C9E67C45-1855-4DEC-A28F-0AAAA4133E08

midterm-exam-93388

#32 Page 10 of 10

SCRAP PAGE

**CHEMISTRY 110 –Midterm Exam****Thursday November 7, 2024, 6:30 – 8:30 pm****Examiners: Pallavi Sirjoosingh and Ashok Kakkar**

First name (please write as legibly as possible within the boxes)

--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--

Last name

--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--

Student ID number

--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--

This booklet has 8 questions on 10 pages (The last page can be used as a scrap page).

The staple must NOT be removed from the question booklet. Your intact question booklet must be returned at the end of the exam.

A separate booklet contains a periodic table, and datasheet.

Write your answers directly under the questions in the space provided. Do not write on the QR codes or near the top edges (as it may be cut off).

Please write **legibly and dark enough** for the scan to be readable. For students who use English as a second language, dictionaries are permitted. Calculators are allowed.

DO NOT ASK QUESTIONS OF THE INVIGILATORS about interpretations of the exam questions. If you have a concern about a question, please summarize it at the top of this page. All concerns will be evaluated after the exam. GOOD LUCK!

**Question 1 8 points** (*Assume all transitions are possible*)

a. 6 points Calculate the **energy (in kJ/mol)** required to ionize (completely remove) the electron from the **n=3** level of **He⁺ ion** (He: Helium). Show your work including equations. (3 sig. figs.)

b. 2 points

(I) Select the largest (longest) wavelength from the three choices below:

- i. Wavelength of photon to *completely remove* an electron from *1s orbital of Hydrogen atom*
- ii. Wavelength of photon to *completely remove* an electron from *1s orbital of Boron atom (B)*
- iii. Wavelength of photon to *completely remove* an electron from *1s orbital of Boron cation (B⁺)*

i or ii or iii

(II) Select the smallest (shortest) wavelength from the three choices below (*for H atom*):

- i. Wavelength of photon *emitted* for an electron transitioning from *2p to 1s orbital*
- ii. Wavelength of photon *absorbed* for an electron transitioning from *2s to 4p orbital*
- iii. Wavelength of photon *absorbed* for an electron transitioning from *1s to 3p orbital*

i or ii or iii



Question 2 8 points



A Chem110 student plots the graph shown on the left, from data obtained for the kinetic energy (eV) of the ejected electron versus frequency (s^{-1}) of the incident light, for an unknown metal. ($1\text{eV} = 1.6022 \times 10^{-19} \text{ J}$)

Based on the plot, answer the questions **a to c** below:

a. 1 point On the plot above, label the **threshold frequency**.

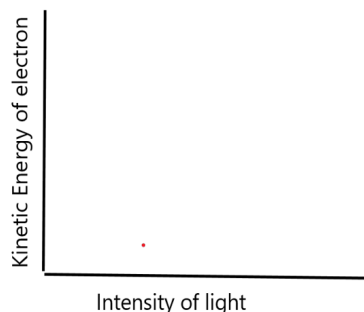
b. 2 points What is the value of the work potential (in J) of the unknown metal?

c. 3 points What is the *longest* wavelength (in nm) required to eject an electron from this metal's surface? Show your work including equation(s) used. (3 sig figs)

d. 2 points In the space provided below, plot the following

(Assume frequency is *above* threshold frequency):

- Kinetic energy* of an ejected electron versus the *intensity* of the incident light (left)
- Number of electrons* ejected versus the *frequency* of the incident light (right)



**Question 3** **6 points****a. 2 points**

In the ground state of Ge (Germanium; $Z = 32$) atom, **name** the following orbitals:

i. Orbital containing the electron that requires the maximum energy to be removed.

ii. Orbital containing the electron that experiences the minimum effective nuclear charge.

b. 2 points

An unknown anion (A^-) has the following *excited* electronic configuration: $[Ar] 4s^2 3d^3 4p^1$

(Ar: Argon; $Z = 18$)

Give the **condensed ground state electronic configuration** of the atom, **A** and the cation, **A^{2+}** .
Show your work. (Assume all electronic transitions are possible)

Electronic Configuration of A:

Electronic Configuration of A^{2+} :

c. 2 points

In a single atom, determine how many electrons can have the following set of quantum numbers?

(i) $n = 5; l = 3$

In a single atom, determine how many electrons can have the following set of quantum numbers?

(i) $n = 3; l = 2; m_s = +1/2$

**Question 4** **8 points**

Rank the following from 1 to 4 in **increasing** order. Provide the ranking in the boxes provided.

Note: Atomic number (Z) are provided for relevant atoms

a) Increasing (lowest to highest) order of *ionization energy*: (1= lowest; 4 = highest)

Mg

Mg: Z = 12

Rb

Rb: Z = 37

Cs

Cs: Z = 55

K

K : Z = 19

b) Increasing (lowest to highest) order of *bond angle*: (1= lowest; 4 = highest)

H₂OCCl₄XeF₄

HCN

c) Increasing (smallest to largest) order of *ionic size* (1= smallest; 4 = largest)

S²⁻

S: Z = 16

O²⁻

O: Z = 8

Be²⁺

Be: Z = 4

Na⁺

Na: Z = 11

d) Increasing (lowest to highest) order of *bond energy* (1= lowest; 4 = highest)

P-F

P-Cl

P-I

P-Br

**Question 5** **12 points**

a. 9 points For the molecules HClO_4 and TeF_6 , draw the most stable *Lewis structure* showing appropriate shape and *dashed/wedged bonds*, based on VSEPR. Determine the **molecular geometry** around the central atom. For HClO_4 , label the *most polar bond*.



b. 3 points (Note: Lone pairs are not shown in structures below)

(i) For the three *resonance* structures (1, 2 and 3) shown below, answer the following question:



(I) Circle the *most contributing* resonance structure.

Structure 1

Structure 2

Structure 3

(ii) For the three structures given (4, 5, and 6) below, answer the following questions:



(I) Circle all the structures that represent *resonance structures* of each other:

Structure 4

Structure 5

Structure 6

(II) Among the resonance structures, circle the *most contributing* resonance structure.

Structure 4

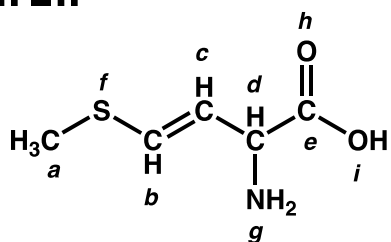
Structure 5

Structure 6

**Question 6** **6 points**

Using the data given below, calculate the **lattice energy** for SrO(s).

	ΔH° (kJ/mol)
$\text{Sr}(s) \longrightarrow \text{Sr}(g)$	+180
$\text{Sr}(g) \longrightarrow \text{Sr}^{2+}(g) + 2e^-$	+1468
$\text{O}_2(g) \longrightarrow 2\text{O}(g)$	+500
$\text{O}(g) + 2e^- \longrightarrow \text{O}^{2-}(g)$	+600
$\text{Sr}(s) + \frac{1}{2}\text{O}_2(g) \longrightarrow \text{SrO}(s)$	-550

**Question 7 10 points**

Consider the molecule given on the left, and answer questions (i) to (vi) using chemical bonding theories that we have learnt in class (valence bond theory, hybridization).

Note that the lone pairs are not shown in the structure:

i. Consider the carbon atoms marked in as ***a***, ***b***, ***c***, ***d***, and ***e*** in the molecule, and determine **hybridization** at each of the carbon centers. (2.5 points)

Atom	Hybridization
Carbon (<i>a</i>)	
Carbon (<i>b</i>)	
Carbon (<i>c</i>)	
Carbon (<i>d</i>)	
Carbon (<i>e</i>)	

ii. Now, consider the sulfur atom (S) marked as ***f***, nitrogen (N) atom (***g***), and oxygen(O) atoms (***h*** and ***i***), and determine their **hybridization**. (2 points)

Atom	Hybridization
Sulfur (<i>f</i>)	
Nitrogen (<i>g</i>)	
Oxygen (<i>h</i>)	
Oxygen (<i>i</i>)	

iii. **Describe** all the bonds formed by the carbon atoms ***a*** and ***b***, using sigma and pi designation as appropriate. (2 points)

Atom	Bonds formed (use sigma and pi as appropriate)
Carbon (<i>a</i>)	
Carbon (<i>b</i>)	

iv. How many total sp^3 -s sigma bonds are in the molecule? (1 point)

v. Determine **electronic** and **molecular geometry** at carbon (C) labeled ***e*** and sulfur (S) labeled ***f*** centres. (2 points)

Atom	Electronic Geometry	Molecular Geometry
Carbon (<i>e</i>)		
Sulfur (<i>f</i>)		

vi. Circle the value from the choices given below, that describes the **C-S-C bond angle at sulfur** (S) marked ***f***: (0.5 points)

>90° to <109.5° 109.5° >109.5 to <120° 120° >120° to <180°


Question 8 12 points

Using the **valence shell orbitals only** draw a **complete** molecular orbital (MO) diagram (with no s-p mixing) for the molecule **PS**, in the space provided below. **(6 points)**

Assume that 3s and 3p atomic orbitals will form molecular orbitals like those formed by the combination of 2s and 2p atomic orbitals that we learnt in class, and there is no s-p mixing.

Show **energy arrow**, **atomic orbitals** involved, **atom labels**, the **molecular orbitals** that are formed, **filled electrons** in the MO diagram etc. Make sure to label all the atomic and molecular orbitals.

Using the MO diagram above, answer the following questions:

- (i) Which one of the following will have the highest bond order? Circle the correct choice
 A) PS^{1-} B) PS C) PS^{1+}
- (ii) Which one of the following will have the longest bond length? Circle the correct choice
 A) PS^{1-} B) PS C) PS^{1+}
- (iii) Which of the following will be diamagnetic? Circle the correct choice.
 A) PS^{1-} B) PS C) PS^{1+}
- (iv) Draw Lewis dot structure of the molecule with the shortest bond length (based on the MO diagram) from among the three (PS^{1-} , PS, PS^{1+}), and indicate the bond order calculated from the Lewis dot structure. **(3 points)**

Lewis Dot Structure

Bond Order



432F866C-BAF4-437B-8306-5C8CF18258E9

midterm-exam-93388

#33 Page 10 of 10

SCRAP PAGE

**CHEMISTRY 110 –Midterm Exam****Thursday November 7, 2024, 6:30 – 8:30 pm**

Examiners: Pallavi Sirjoosingh and Ashok Kakkar

First name (please write as legibly as possible within the boxes)

--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--

Last name

--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--

Student ID number

--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--

This booklet has 8 questions on 10 pages (The last page can be used as a scrap page).

The staple must NOT be removed from the question booklet. Your intact question booklet must be returned at the end of the exam.

A separate booklet contains a periodic table, and datasheet.

Write your answers directly under the questions in the space provided. Do not write on the QR codes or near the top edges (as it may be cut off).

Please write **legibly and dark enough** for the scan to be readable. For students who use English as a second language, dictionaries are permitted. Calculators are allowed.

DO NOT ASK QUESTIONS OF THE INVIGILATORS about interpretations of the exam questions. If you have a concern about a question, please summarize it at the top of this page. All concerns will be evaluated after the exam. GOOD LUCK!

**Question 1 8 points** (*Assume all transitions are possible*)

a. 6 points Calculate the **energy (in kJ/mol)** required to ionize (completely remove) the electron from the **$n=3$** level of **He^+ ion** (He: Helium). Show your work including equations. (3 sig. figs.)

b. 2 points

(I) Select the largest (longest) wavelength from the three choices below:

- i. Wavelength of photon to *completely remove* an electron from *1s orbital of Hydrogen atom*
- ii. Wavelength of photon to *completely remove* an electron from *1s orbital of Boron atom (B)*
- iii. Wavelength of photon to *completely remove* an electron from *1s orbital of Boron cation (B^+)*

i or ii or iii

(II) Select the smallest (shortest) wavelength from the three choices below (*for H atom*):

- i. Wavelength of photon *emitted* for an electron transitioning from *2p to 1s orbital*
- ii. Wavelength of photon *absorbed* for an electron transitioning from *2s to 4p orbital*
- iii. Wavelength of photon *absorbed* for an electron transitioning from *1s to 3p orbital*

i or ii or iii



Question 2 8 points



A Chem110 student plots the graph shown on the left, from data obtained for the kinetic energy (eV) of the ejected electron versus frequency (s^{-1}) of the incident light, for an unknown metal. ($1\text{eV} = 1.6022 \times 10^{-19} \text{ J}$)

Based on the plot, answer the questions **a to c** below:

a. 1 point On the plot above, label the **threshold frequency**.

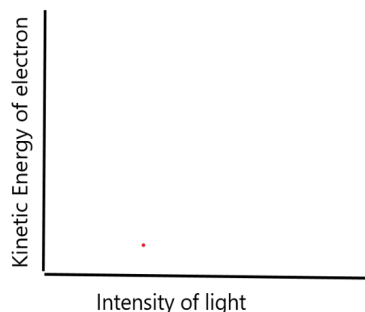
b. 2 points What is the value of the work potential (in J) of the unknown metal?

c. 3 points What is the *longest* wavelength (in nm) required to eject an electron from this metal's surface? Show your work including equation(s) used. (3 sig figs)

d. 2 points In the space provided below, plot the following

(Assume frequency is *above* threshold frequency):

- i. *Kinetic energy* of an ejected electron versus the *intensity* of the incident light (left)
- ii. *Number of electrons* ejected versus the *frequency* of the incident light (right)



**Question 3** **6 points****a. 2 points**

In the ground state of Ge (Germanium; $Z = 32$) atom, **name** the following orbitals:

i. Orbital containing the electron that requires the maximum energy to be removed.

ii. Orbital containing the electron that experiences the minimum effective nuclear charge.

b. 2 points

An unknown anion (A^-) has the following *excited* electronic configuration: $[Ar] 4s^2 3d^3 4p^1$

(Ar: Argon; $Z = 18$)

Give the **condensed ground state electronic configuration** of the atom, **A** and the cation, **A^{2+}** .
Show your work. (Assume all electronic transitions are possible)

Electronic Configuration of A:

Electronic Configuration of A^{2+} :

c. 2 points

In a single atom, determine how many electrons can have the following set of quantum numbers?

(i) $n = 5; l = 3$

In a single atom, determine how many electrons can have the following set of quantum numbers?

(i) $n = 3; l = 2; m_s = +1/2$

**Question 4** **8 points**

Rank the following from 1 to 4 in **increasing** order. Provide the ranking in the boxes provided.

Note: Atomic number (Z) are provided for relevant atoms

a) Increasing (lowest to highest) order of *ionization energy*: (1= lowest; 4 = highest)

Mg

Mg: Z = 12

Rb

Rb: Z = 37

Cs

Cs: Z = 55

K

K : Z = 19

b) Increasing (lowest to highest) order of *bond angle*: (1= lowest; 4 = highest)

H₂OCCl₄XeF₄

HCN

c) Increasing (smallest to largest) order of *ionic size* (1= smallest; 4 = largest)

S²⁻

S: Z = 16

O²⁻

O: Z = 8

Be²⁺

Be: Z = 4

Na⁺

Na: Z = 11

d) Increasing (lowest to highest) order of *bond energy* (1= lowest; 4 = highest)

P-F

P-Cl

P-I

P-Br

**Question 5** **12 points**

a. 9 points For the molecules HClO_4 and TeF_6 , draw the most stable *Lewis structure* showing appropriate shape and *dashed/wedged bonds*, based on VSEPR. Determine the **molecular geometry** around the central atom. For HClO_4 , label the *most polar bond*.



b. 3 points (Note: Lone pairs are not shown in structures below)

(i) For the three *resonance* structures (1, 2 and 3) shown below, answer the following question:



(I) Circle the *most contributing* resonance structure.

Structure 1

Structure 2

Structure 3

(ii) For the three structures given (4, 5, and 6) below, answer the following questions:



(I) Circle all the structures that represent *resonance structures* of each other:

Structure 4

Structure 5

Structure 6

(II) Among the resonance structures, circle the *most contributing* resonance structure.

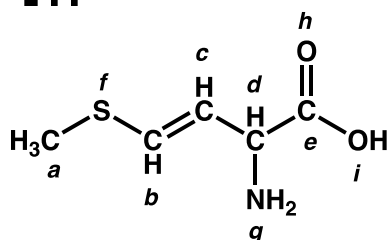
Structure 4

Structure 5

Structure 6

**Question 6** **6 points**Using the data given below, calculate the **lattice energy** for SrO(s).

	ΔH° (kJ/mol)
$\text{Sr}(s) \longrightarrow \text{Sr}(g)$	+180
$\text{Sr}(g) \longrightarrow \text{Sr}^{2+}(g) + 2e^-$	+1468
$\text{O}_2(g) \longrightarrow 2\text{O}(g)$	+500
$\text{O}(g) + 2e^- \longrightarrow \text{O}^{2-}(g)$	+600
$\text{Sr}(s) + \frac{1}{2}\text{O}_2(g) \longrightarrow \text{SrO}(s)$	-550

**Question 7 10 points**

Consider the molecule given on the left, and answer questions (i) to (vi) using chemical bonding theories that we have learnt in class (valence bond theory, hybridization).

Note that the lone pairs are not shown in the structure:

i. Consider the carbon atoms marked in as ***a***, ***b***, ***c***, ***d***, and ***e*** in the molecule, and determine **hybridization** at each of the carbon centers. (2.5 points)

Atom	Hybridization
Carbon (<i>a</i>)	
Carbon (<i>b</i>)	
Carbon (<i>c</i>)	
Carbon (<i>d</i>)	
Carbon (<i>e</i>)	

ii. Now, consider the sulfur atom (S) marked as ***f***, nitrogen (N) atom (***g***), and oxygen(O) atoms (***h*** and ***i***), and determine their **hybridization**. (2 points)

Atom	Hybridization
Sulfur (<i>f</i>)	
Nitrogen (<i>g</i>)	
Oxygen (<i>h</i>)	
Oxygen (<i>i</i>)	

iii. **Describe** all the bonds formed by the carbon atoms ***a*** and ***b***, using sigma and pi designation as appropriate. (2 points)

Atom	Bonds formed (use sigma and pi as appropriate)
Carbon (<i>a</i>)	
Carbon (<i>b</i>)	

iv. How many total sp^3 -s sigma bonds are in the molecule? (1 point)

v. Determine **electronic** and **molecular geometry** at carbon (C) labeled ***e*** and sulfur (S) labeled ***f*** centres. (2 points)

Atom	Electronic Geometry	Molecular Geometry
Carbon (<i>e</i>)		
Sulfur (<i>f</i>)		

vi. Circle the value from the choices given below, that describes the **C-S-C bond angle at sulfur** (S) marked ***f***: (0.5 points)

$>90^\circ$ to $<109.5^\circ$ 109.5° >109.5 to $<120^\circ$ 120° $>120^\circ$ to $<180^\circ$

**Question 8 12 points**

Using the **valence shell orbitals only** draw a **complete** molecular orbital (MO) diagram (with no s-p mixing) for the molecule **PS**, in the space provided below. **(6 points)**

Assume that 3s and 3p atomic orbitals will form molecular orbitals like those formed by the combination of 2s and 2p atomic orbitals that we learnt in class, and there is no s-p mixing.

Show **energy arrow**, **atomic orbitals** involved, **atom labels**, the **molecular orbitals** that are formed, **filled electrons** in the MO diagram etc. Make sure to label all the atomic and molecular orbitals.

Using the MO diagram above, answer the following questions:

- (i) Which one of the following will have the highest bond order? Circle the correct choice
A) PS^{1-} B) PS C) PS^{1+}
- (ii) Which one of the following will have the longest bond length? Circle the correct choice
A) PS^{1-} B) PS C) PS^{1+}
- (iii) Which of the following will be diamagnetic? Circle the correct choice.
A) PS^{1-} B) PS C) PS^{1+}
- (iv) Draw Lewis dot structure of the molecule with the shortest bond length (based on the MO diagram) from among the three (PS^{1-} , PS, PS^{1+}), and indicate the bond order calculated from the Lewis dot structure. **(3 points)**

Lewis Dot Structure

Bond Order



93A9B2DE-04E7-43EC-BAB8-E19B2B45E9C6

midterm-exam-93388

#34 Page 10 of 10

SCRAP PAGE

**CHEMISTRY 110 –Midterm Exam****Thursday November 7, 2024, 6:30 – 8:30 pm**

Examiners: Pallavi Sirjoosingh and Ashok Kakkar

First name (please write as legibly as possible within the boxes)

--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--

Last name

--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--

Student ID number

--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--

This booklet has 8 questions on 10 pages (The last page can be used as a scrap page).

The staple must NOT be removed from the question booklet. Your intact question booklet must be returned at the end of the exam.

A separate booklet contains a periodic table, and datasheet.

Write your answers directly under the questions in the space provided. Do not write on the QR codes or near the top edges (as it may be cut off).

Please write **legibly and dark enough** for the scan to be readable. For students who use English as a second language, dictionaries are permitted. Calculators are allowed.

DO NOT ASK QUESTIONS OF THE INVIGILATORS about interpretations of the exam questions. If you have a concern about a question, please summarize it at the top of this page. All concerns will be evaluated after the exam. GOOD LUCK!

**Question 1 8 points** (*Assume all transitions are possible*)

a. 6 points Calculate the **energy (in kJ/mol)** required to ionize (completely remove) the electron from the **$n=3$** level of **He^+ ion** (He: Helium). Show your work including equations. (3 sig. figs.)

b. 2 points

(I) Select the largest (longest) wavelength from the three choices below:

- i. Wavelength of photon to *completely remove* an electron from *1s orbital of Hydrogen atom*
- ii. Wavelength of photon to *completely remove* an electron from *1s orbital of Boron atom (B)*
- iii. Wavelength of photon to *completely remove* an electron from *1s orbital of Boron cation (B^+)*

i or ii or iii

(II) Select the smallest (shortest) wavelength from the three choices below (*for H atom*):

- i. Wavelength of photon *emitted* for an electron transitioning from *2p to 1s orbital*
- ii. Wavelength of photon *absorbed* for an electron transitioning from *2s to 4p orbital*
- iii. Wavelength of photon *absorbed* for an electron transitioning from *1s to 3p orbital*

i or ii or iii



Question 2 8 points



A Chem110 student plots the graph shown on the left, from data obtained for the kinetic energy (eV) of the ejected electron versus frequency (s^{-1}) of the incident light, for an unknown metal. ($1\text{eV} = 1.6022 \times 10^{-19} \text{ J}$)

Based on the plot, answer the questions **a to c** below:

a. 1 point On the plot above, label the **threshold frequency**.

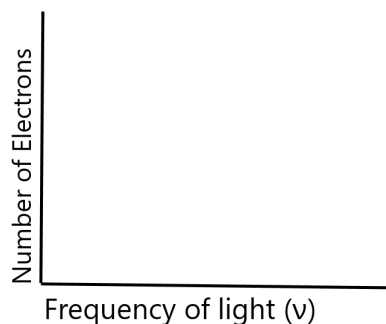
b. 2 points What is the value of the work potential (in J) of the unknown metal?

c. 3 points What is the *longest* wavelength (in nm) required to eject an electron from this metal's surface? Show your work including equation(s) used. (3 sig figs)

d. 2 points In the space provided below, plot the following

(Assume frequency is *above* threshold frequency):

- i. *Kinetic energy* of an ejected electron versus the *intensity* of the incident light (left)
- ii. *Number of electrons* ejected versus the *frequency* of the incident light (right)



**Question 3** **6 points****a. 2 points**

In the ground state of Ge (Germanium; $Z = 32$) atom, **name** the following orbitals:

i. Orbital containing the electron that requires the maximum energy to be removed.

ii. Orbital containing the electron that experiences the minimum effective nuclear charge.

b. 2 points

An unknown anion (A^-) has the following *excited* electronic configuration: $[Ar] 4s^2 3d^3 4p^1$

(Ar: Argon; $Z = 18$)

Give the **condensed ground state electronic configuration** of the atom, **A** and the cation, **A^{2+}** .
Show your work. (Assume all electronic transitions are possible)

Electronic Configuration of A:

Electronic Configuration of A^{2+} :

c. 2 points

In a single atom, determine how many electrons can have the following set of quantum numbers?

(i) $n = 5; l = 3$

In a single atom, determine how many electrons can have the following set of quantum numbers?

(i) $n = 3; l = 2; m_s = +1/2$

**Question 4** **8 points**

Rank the following from 1 to 4 in **increasing** order. Provide the ranking in the boxes provided.

Note: Atomic number (Z) are provided for relevant atoms

a) Increasing (lowest to highest) order of *ionization energy*: (1= lowest; 4 = highest)

Mg

Mg: Z = 12

Rb

Rb: Z = 37

Cs

Cs: Z = 55

K

K : Z = 19

b) Increasing (lowest to highest) order of *bond angle*: (1= lowest; 4 = highest)

H₂OCCl₄XeF₄

HCN

c) Increasing (smallest to largest) order of *ionic size* (1= smallest; 4 = largest)

S²⁻

S: Z = 16

O²⁻

O: Z = 8

Be²⁺

Be: Z = 4

Na⁺

Na: Z = 11

d) Increasing (lowest to highest) order of *bond energy* (1= lowest; 4 = highest)

P-F

P-Cl

P-I

P-Br

**Question 5** **12 points**

a. 9 points For the molecules HClO_4 and TeF_6 , draw the most stable *Lewis structure* showing appropriate shape and *dashed/wedged bonds*, based on VSEPR. Determine the **molecular geometry** around the central atom. For HClO_4 , label the *most polar bond*.



b. 3 points (Note: Lone pairs are not shown in structures below)

(i) For the three *resonance* structures (1, 2 and 3) shown below, answer the following question:



(I) Circle the *most contributing* resonance structure.

Structure 1

Structure 2

Structure 3

(ii) For the three structures given (4, 5, and 6) below, answer the following questions:



(I) Circle all the structures that represent *resonance structures* of each other:

Structure 4

Structure 5

Structure 6

(II) Among the resonance structures, circle the *most contributing* resonance structure.

Structure 4

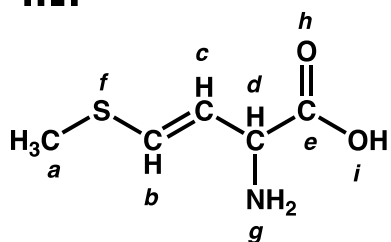
Structure 5

Structure 6

**Question 6** **6 points**

Using the data given below, calculate the **lattice energy** for SrO(s).

	ΔH° (kJ/mol)
$\text{Sr}(s) \longrightarrow \text{Sr}(g)$	+180
$\text{Sr}(g) \longrightarrow \text{Sr}^{2+}(g) + 2e^-$	+1468
$\text{O}_2(g) \longrightarrow 2\text{O}(g)$	+500
$\text{O}(g) + 2e^- \longrightarrow \text{O}^{2-}(g)$	+600
$\text{Sr}(s) + \frac{1}{2}\text{O}_2(g) \longrightarrow \text{SrO}(s)$	-550

**Question 7 10 points**

Consider the molecule given on the left, and answer questions (i) to (vi) using chemical bonding theories that we have learnt in class (valence bond theory, hybridization).

Note that the lone pairs are not shown in the structure:

i. Consider the carbon atoms marked in as ***a***, ***b***, ***c***, ***d***, and ***e*** in the molecule, and determine **hybridization** at each of the carbon centers. (2.5 points)

Atom	Hybridization
Carbon (<i>a</i>)	
Carbon (<i>b</i>)	
Carbon (<i>c</i>)	
Carbon (<i>d</i>)	
Carbon (<i>e</i>)	

ii. Now, consider the sulfur atom (S) marked as ***f***, nitrogen (N) atom (***g***), and oxygen(O) atoms (***h*** and ***i***), and determine their **hybridization**. (2 points)

Atom	Hybridization
Sulfur (<i>f</i>)	
Nitrogen (<i>g</i>)	
Oxygen (<i>h</i>)	
Oxygen (<i>i</i>)	

iii. **Describe** all the bonds formed by the carbon atoms ***a*** and ***b***, using sigma and pi designation as appropriate. (2 points)

Atom	Bonds formed (use sigma and pi as appropriate)
Carbon (<i>a</i>)	
Carbon (<i>b</i>)	

iv. How many total sp^3 -s sigma bonds are in the molecule? (1 point)

v. Determine **electronic** and **molecular geometry** at carbon (C) labeled ***e*** and sulfur (S) labeled ***f*** centres. (2 points)

Atom	Electronic Geometry	Molecular Geometry
Carbon (<i>e</i>)		
Sulfur (<i>f</i>)		

vi. Circle the value from the choices given below, that describes the **C-S-C bond angle at sulfur** (S) marked ***f***: (0.5 points)

>90° to <109.5°

109.5°

>109.5 to <120°

120°

>120° to <180°


Question 8 12 points

Using the **valence shell orbitals only** draw a **complete** molecular orbital (MO) diagram (with no s-p mixing) for the molecule **PS**, in the space provided below. **(6 points)**

Assume that 3s and 3p atomic orbitals will form molecular orbitals like those formed by the combination of 2s and 2p atomic orbitals that we learnt in class, and there is no s-p mixing.

Show **energy arrow**, **atomic orbitals** involved, **atom labels**, the **molecular orbitals** that are formed, **filled electrons** in the MO diagram etc. Make sure to **label all the atomic and molecular orbitals**.

Using the MO diagram above, answer the following questions:

- (i) Which one of the following will have the highest bond order? Circle the correct choice
 A) PS^{1-} B) PS C) PS^{1+}
- (ii) Which one of the following will have the longest bond length? Circle the correct choice
 A) PS^{1-} B) PS C) PS^{1+}
- (iii) Which of the following will be diamagnetic? Circle the correct choice.
 A) PS^{1-} B) PS C) PS^{1+}
- (iv) Draw Lewis dot structure of the molecule with the shortest bond length (based on the MO diagram) from among the three (PS^{1-} , PS, PS^{1+}), and indicate the bond order calculated from the Lewis dot structure. **(3 points)**

Lewis Dot Structure

Bond Order



B778C7F7-0B72-4B70-9793-EA3891E58D4B

midterm-exam-93388

#35 Page 10 of 10

SCRAP PAGE

**CHEMISTRY 110 –Midterm Exam****Thursday November 7, 2024, 6:30 – 8:30 pm**

Examiners: Pallavi Sirjoosingh and Ashok Kakkar

First name (please write as legibly as possible within the boxes)

--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--

Last name

--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--

Student ID number

--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--

This booklet has 8 questions on 10 pages (The last page can be used as a scrap page).

The staple must NOT be removed from the question booklet. Your intact question booklet must be returned at the end of the exam.

A separate booklet contains a periodic table, and datasheet.

Write your answers directly under the questions in the space provided. Do not write on the QR codes or near the top edges (as it may be cut off).

Please write **legibly and dark enough** for the scan to be readable. For students who use English as a second language, dictionaries are permitted. Calculators are allowed.

DO NOT ASK QUESTIONS OF THE INVIGILATORS about interpretations of the exam questions. If you have a concern about a question, please summarize it at the top of this page. All concerns will be evaluated after the exam. GOOD LUCK!

**Question 1 8 points** (*Assume all transitions are possible*)

a. 6 points Calculate the **energy (in kJ/mol)** required to ionize (completely remove) the electron from the **n=3** level of **He⁺ ion** (He: Helium). Show your work including equations. (3 sig. figs.)

b. 2 points

(I) Select the largest (longest) wavelength from the three choices below:

- i. Wavelength of photon to *completely remove* an electron from *1s orbital of Hydrogen atom*
- ii. Wavelength of photon to *completely remove* an electron from *1s orbital of Boron atom (B)*
- iii. Wavelength of photon to *completely remove* an electron from *1s orbital of Boron cation (B⁺)*

i or ii or iii

(II) Select the smallest (shortest) wavelength from the three choices below (*for H atom*):

- i. Wavelength of photon *emitted* for an electron transitioning from *2p to 1s orbital*
- ii. Wavelength of photon *absorbed* for an electron transitioning from *2s to 4p orbital*
- iii. Wavelength of photon *absorbed* for an electron transitioning from *1s to 3p orbital*

i or ii or iii



Question 2 8 points



A Chem110 student plots the graph shown on the left, from data obtained for the kinetic energy (eV) of the ejected electron versus frequency (s^{-1}) of the incident light, for an unknown metal. ($1\text{eV} = 1.6022 \times 10^{-19} \text{ J}$)

Based on the plot, answer the questions **a to c** below:

a. 1 point On the plot above, label the **threshold frequency**.

b. 2 points What is the value of the work potential (in J) of the unknown metal?

c. 3 points What is the *longest* wavelength (in nm) required to eject an electron from this metal's surface? Show your work including equation(s) used. (3 sig figs)

d. 2 points In the space provided below, plot the following

(Assume frequency is *above* threshold frequency):

- Kinetic energy* of an ejected electron versus the *intensity* of the incident light (left)
- Number of electrons* ejected versus the *frequency* of the incident light (right)



**Question 3** **6 points****a. 2 points**

In the ground state of Ge (Germanium; $Z = 32$) atom, **name** the following orbitals:

i. Orbital containing the electron that requires the maximum energy to be removed.

ii. Orbital containing the electron that experiences the minimum effective nuclear charge.

b. 2 points

An unknown anion (A^-) has the following *excited* electronic configuration: $[Ar] 4s^2 3d^3 4p^1$

(Ar: Argon; $Z = 18$)

Give the **condensed ground state electronic configuration** of the atom, **A** and the cation, **A^{2+}** .
Show your work. (Assume all electronic transitions are possible)

Electronic Configuration of A:

Electronic Configuration of A^{2+} :

c. 2 points

In a single atom, determine how many electrons can have the following set of quantum numbers?

(i) $n = 5; l = 3$

In a single atom, determine how many electrons can have the following set of quantum numbers?

(i) $n = 3; l = 2; m_s = +1/2$

**Question 4** **8 points**

Rank the following from 1 to 4 in **increasing** order. Provide the ranking in the boxes provided.

Note: Atomic number (Z) are provided for relevant atoms

a) Increasing (lowest to highest) order of *ionization energy*: (1= lowest; 4 = highest)

Mg

Mg: Z = 12

Rb

Rb: Z = 37

Cs

Cs: Z = 55

K

K : Z = 19

b) Increasing (lowest to highest) order of *bond angle*: (1= lowest; 4 = highest)

H₂OCCl₄XeF₄

HCN

c) Increasing (smallest to largest) order of *ionic size* (1= smallest; 4 = largest)

S²⁻

S: Z = 16

O²⁻

O: Z = 8

Be²⁺

Be: Z = 4

Na⁺

Na: Z = 11

d) Increasing (lowest to highest) order of *bond energy* (1= lowest; 4 = highest)

P-F

P-Cl

P-I

P-Br

**Question 5** **12 points**

a. 9 points For the molecules HClO_4 and TeF_6 , draw the most stable *Lewis structure* showing appropriate shape and *dashed/wedged bonds*, based on VSEPR. Determine the **molecular geometry** around the central atom. For HClO_4 , label the *most polar bond*.



b. 3 points (Note: Lone pairs are not shown in structures below)

(i) For the three *resonance* structures (1, 2 and 3) shown below, answer the following question:



(I) Circle the *most contributing* resonance structure.

Structure 1

Structure 2

Structure 3

(ii) For the three structures given (4, 5, and 6) below, answer the following questions:



(I) Circle all the structures that represent *resonance structures* of each other:

Structure 4

Structure 5

Structure 6

(II) Among the resonance structures, circle the *most contributing* resonance structure.

Structure 4

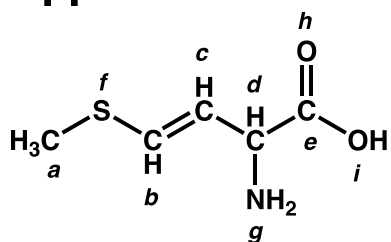
Structure 5

Structure 6

**Question 6** **6 points**

Using the data given below, calculate the **lattice energy** for SrO(s).

	ΔH° (kJ/mol)
$\text{Sr}(s) \longrightarrow \text{Sr}(g)$	+180
$\text{Sr}(g) \longrightarrow \text{Sr}^{2+}(g) + 2e^-$	+1468
$\text{O}_2(g) \longrightarrow 2\text{O}(g)$	+500
$\text{O}(g) + 2e^- \longrightarrow \text{O}^{2-}(g)$	+600
$\text{Sr}(s) + \frac{1}{2}\text{O}_2(g) \longrightarrow \text{SrO}(s)$	-550

**Question 7 10 points**

Consider the molecule given on the left, and answer questions (i) to (vi) using chemical bonding theories that we have learnt in class (valence bond theory, hybridization).

Note that the lone pairs are not shown in the structure:

i. Consider the carbon atoms marked in as ***a***, ***b***, ***c***, ***d***, and ***e*** in the molecule, and determine **hybridization** at each of the carbon centers. (2.5 points)

Atom	Hybridization
Carbon (<i>a</i>)	
Carbon (<i>b</i>)	
Carbon (<i>c</i>)	
Carbon (<i>d</i>)	
Carbon (<i>e</i>)	

ii. Now, consider the sulfur atom (S) marked as ***f***, nitrogen (N) atom (***g***), and oxygen(O) atoms (***h*** and ***i***), and determine their **hybridization**. (2 points)

Atom	Hybridization
Sulfur (<i>f</i>)	
Nitrogen (<i>g</i>)	
Oxygen (<i>h</i>)	
Oxygen (<i>i</i>)	

iii. **Describe** all the bonds formed by the carbon atoms ***a*** and ***b***, using sigma and pi designation as appropriate. (2 points)

Atom	Bonds formed (use sigma and pi as appropriate)
Carbon (<i>a</i>)	
Carbon (<i>b</i>)	

iv. How many total sp^3 -s sigma bonds are in the molecule? (1 point)

v. Determine **electronic** and **molecular geometry** at carbon (C) labeled ***e*** and sulfur (S) labeled ***f*** centres. (2 points)

Atom	Electronic Geometry	Molecular Geometry
Carbon (<i>e</i>)		
Sulfur (<i>f</i>)		

vi. Circle the value from the choices given below, that describes the **C-S-C bond angle at sulfur** (S) marked ***f***: (0.5 points)

>90° to <109.5° 109.5° >109.5 to <120° 120° >120° to <180°


Question 8 12 points

Using the **valence shell orbitals only** draw a **complete** molecular orbital (MO) diagram (with no s-p mixing) for the molecule **PS**, in the space provided below. **(6 points)**

Assume that 3s and 3p atomic orbitals will form molecular orbitals like those formed by the combination of 2s and 2p atomic orbitals that we learnt in class, and there is no s-p mixing.

Show **energy arrow**, **atomic orbitals** involved, **atom labels**, the **molecular orbitals** that are formed, **filled electrons** in the MO diagram etc. Make sure to label all the atomic and molecular orbitals.

Using the MO diagram above, answer the following questions:

- (i) Which one of the following will have the highest bond order? Circle the correct choice
 A) PS^{1-} B) PS C) PS^{1+}
- (ii) Which one of the following will have the longest bond length? Circle the correct choice
 A) PS^{1-} B) PS C) PS^{1+}
- (iii) Which of the following will be diamagnetic? Circle the correct choice.
 A) PS^{1-} B) PS C) PS^{1+}
- (iv) Draw Lewis dot structure of the molecule with the shortest bond length (based on the MO diagram) from among the three (PS^{1-} , PS, PS^{1+}), and indicate the bond order calculated from the Lewis dot structure. **(3 points)**

Lewis Dot Structure

Bond Order



C7E448F8-2116-4C09-A52E-7F53429CF9DA

midterm-exam-93388

#36 Page 10 of 10

SCRAP PAGE

**CHEMISTRY 110 –Midterm Exam****Thursday November 7, 2024, 6:30 – 8:30 pm****Examiners: Pallavi Sirjoosingh and Ashok Kakkar**

First name (please write as legibly as possible within the boxes)

--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--

Last name

--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--

Student ID number

--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--

This booklet has 8 questions on 10 pages (The last page can be used as a scrap page).

The staple must NOT be removed from the question booklet. Your intact question booklet must be returned at the end of the exam.

A separate booklet contains a periodic table, and datasheet.

Write your answers directly under the questions in the space provided. Do not write on the QR codes or near the top edges (as it may be cut off).

Please write **legibly and dark enough** for the scan to be readable. For students who use English as a second language, dictionaries are permitted. Calculators are allowed.

DO NOT ASK QUESTIONS OF THE INVIGILATORS about interpretations of the exam questions. If you have a concern about a question, please summarize it at the top of this page. All concerns will be evaluated after the exam. GOOD LUCK!

**Question 1 8 points** (*Assume all transitions are possible*)

a. 6 points Calculate the **energy (in kJ/mol)** required to ionize (completely remove) the electron from the **n=3** level of **He⁺ ion** (He: Helium). Show your work including equations. (3 sig. figs.)

b. 2 points

(I) Select the largest (longest) wavelength from the three choices below:

- i. Wavelength of photon to *completely remove* an electron from *1s orbital of Hydrogen atom*
- ii. Wavelength of photon to *completely remove* an electron from *1s orbital of Boron atom (B)*
- iii. Wavelength of photon to *completely remove* an electron from *1s orbital of Boron cation (B⁺)*

i or ii or iii

(II) Select the smallest (shortest) wavelength from the three choices below (*for H atom*):

- i. Wavelength of photon *emitted* for an electron transitioning from *2p to 1s orbital*
- ii. Wavelength of photon *absorbed* for an electron transitioning from *2s to 4p orbital*
- iii. Wavelength of photon *absorbed* for an electron transitioning from *1s to 3p orbital*

i or ii or iii



Question 2 8 points



A Chem110 student plots the graph shown on the left, from data obtained for the kinetic energy (eV) of the ejected electron versus frequency (s^{-1}) of the incident light, for an unknown metal. ($1\text{eV} = 1.6022 \times 10^{-19} \text{ J}$)

Based on the plot, answer the questions **a to c** below:

a. 1 point On the plot above, label the **threshold frequency**.

b. 2 points What is the value of the work potential (in J) of the unknown metal?

c. 3 points What is the *longest* wavelength (in nm) required to eject an electron from this metal's surface? Show your work including equation(s) used. (3 sig figs)

d. 2 points In the space provided below, plot the following

(Assume frequency is *above* threshold frequency):

- Kinetic energy* of an ejected electron versus the *intensity* of the incident light (left)
- Number of electrons* ejected versus the *frequency* of the incident light (right)



**Question 3** **6 points****a. 2 points**

In the ground state of Ge (Germanium; $Z = 32$) atom, **name** the following orbitals:

i. Orbital containing the electron that requires the maximum energy to be removed.

ii. Orbital containing the electron that experiences the minimum effective nuclear charge.

b. 2 points

An unknown anion (A^-) has the following *excited* electronic configuration: $[Ar] 4s^2 3d^3 4p^1$

(Ar: Argon; $Z = 18$)

Give the **condensed ground state electronic configuration** of the atom, **A** and the cation, **A^{2+}** .
Show your work. (Assume all electronic transitions are possible)

Electronic Configuration of A:

Electronic Configuration of A^{2+} :

c. 2 points

In a single atom, determine how many electrons can have the following set of quantum numbers?

(i) $n = 5; l = 3$

In a single atom, determine how many electrons can have the following set of quantum numbers?

(i) $n = 3; l = 2; m_s = +1/2$

**Question 4** **8 points**

Rank the following from 1 to 4 in **increasing** order. Provide the ranking in the boxes provided.

Note: Atomic number (Z) are provided for relevant atoms

a) Increasing (lowest to highest) order of *ionization energy*: (1= lowest; 4 = highest)

Mg

Mg: Z = 12

Rb

Rb: Z = 37

Cs

Cs: Z = 55

K

K : Z = 19

b) Increasing (lowest to highest) order of *bond angle*: (1= lowest; 4 = highest)

H₂OCCl₄XeF₄

HCN

c) Increasing (smallest to largest) order of *ionic size* (1= smallest; 4 = largest)

S²⁻

S: Z = 16

O²⁻

O: Z = 8

Be²⁺

Be: Z = 4

Na⁺

Na: Z = 11

d) Increasing (lowest to highest) order of *bond energy* (1= lowest; 4 = highest)

P-F

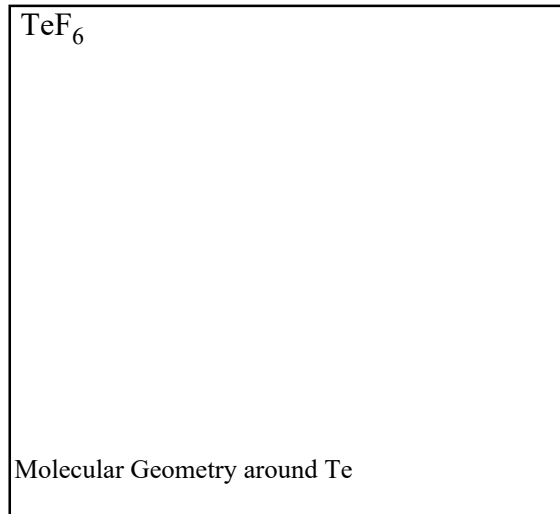
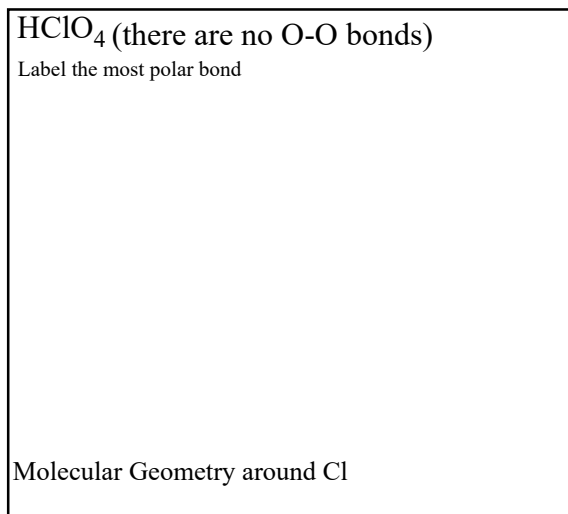
P-Cl

P-I

P-Br

**Question 5** **12 points**

a. 9 points For the molecules HClO_4 and TeF_6 , draw the most stable *Lewis structure* showing appropriate shape and *dashed/wedged bonds*, based on VSEPR. Determine the **molecular geometry** around the central atom. For HClO_4 , label the *most polar bond*.



b. 3 points (Note: Lone pairs are not shown in structures below)

(i) For the three *resonance* structures (1, 2 and 3) shown below, answer the following question:



(I) Circle the *most contributing* resonance structure.

Structure 1

Structure 2

Structure 3

(ii) For the three structures given (4, 5, and 6) below, answer the following questions:



(I) Circle all the structures that represent *resonance structures* of each other:

Structure 4

Structure 5

Structure 6

(II) Among the resonance structures, circle the *most contributing* resonance structure.

Structure 4

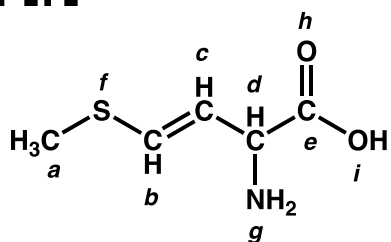
Structure 5

Structure 6

**Question 6** **6 points**

Using the data given below, calculate the **lattice energy** for SrO(s).

	ΔH° (kJ/mol)
$\text{Sr}(s) \longrightarrow \text{Sr}(g)$	+180
$\text{Sr}(g) \longrightarrow \text{Sr}^{2+}(g) + 2e^-$	+1468
$\text{O}_2(g) \longrightarrow 2\text{O}(g)$	+500
$\text{O}(g) + 2e^- \longrightarrow \text{O}^{2-}(g)$	+600
$\text{Sr}(s) + \frac{1}{2}\text{O}_2(g) \longrightarrow \text{SrO}(s)$	-550

**Question 7 10 points**

Consider the molecule given on the left, and answer questions (i) to (vi) using chemical bonding theories that we have learnt in class (valence bond theory, hybridization).

Note that the lone pairs are not shown in the structure:

i. Consider the carbon atoms marked in as ***a***, ***b***, ***c***, ***d***, and ***e*** in the molecule, and determine **hybridization** at each of the carbon centers. (2.5 points)

Atom	Hybridization
Carbon (<i>a</i>)	
Carbon (<i>b</i>)	
Carbon (<i>c</i>)	
Carbon (<i>d</i>)	
Carbon (<i>e</i>)	

ii. Now, consider the sulfur atom (S) marked as ***f***, nitrogen (N) atom (***g***), and oxygen(O) atoms (***h*** and ***i***), and determine their **hybridization**. (2 points)

Atom	Hybridization
Sulfur (<i>f</i>)	
Nitrogen (<i>g</i>)	
Oxygen (<i>h</i>)	
Oxygen (<i>i</i>)	

iii. **Describe** all the bonds formed by the carbon atoms ***a*** and ***b***, using sigma and pi designation as appropriate. (2 points)

Atom	Bonds formed (use sigma and pi as appropriate)
Carbon (<i>a</i>)	
Carbon (<i>b</i>)	

iv. How many total sp^3 -s sigma bonds are in the molecule? (1 point)

v. Determine **electronic** and **molecular geometry** at carbon (C) labeled ***e*** and sulfur (S) labeled ***f*** centres. (2 points)

Atom	Electronic Geometry	Molecular Geometry
Carbon (<i>e</i>)		
Sulfur (<i>f</i>)		

vi. Circle the value from the choices given below, that describes the **C-S-C bond angle at sulfur** (S) marked ***f***: (0.5 points)

>90° to <109.5° 109.5° >109.5 to <120° 120° >120° to <180°

**Question 8 12 points**

Using the **valence shell orbitals only** draw a **complete** molecular orbital (MO) diagram (with no s-p mixing) for the molecule **PS**, in the space provided below. **(6 points)**

Assume that 3s and 3p atomic orbitals will form molecular orbitals like those formed by the combination of 2s and 2p atomic orbitals that we learnt in class, and there is no s-p mixing.

Show **energy arrow**, **atomic orbitals** involved, **atom labels**, the **molecular orbitals** that are formed, **filled electrons** in the MO diagram etc. Make sure to label all the atomic and molecular orbitals.

Using the MO diagram above, answer the following questions:

- (i) Which one of the following will have the highest bond order? Circle the correct choice
A) PS^{1-} B) PS C) PS^{1+}
- (ii) Which one of the following will have the longest bond length? Circle the correct choice
A) PS^{1-} B) PS C) PS^{1+}
- (iii) Which of the following will be diamagnetic? Circle the correct choice.
A) PS^{1-} B) PS C) PS^{1+}
- (iv) Draw Lewis dot structure of the molecule with the shortest bond length (based on the MO diagram) from among the three (PS^{1-} , PS, PS^{1+}), and indicate the bond order calculated from the Lewis dot structure. **(3 points)**

Lewis Dot Structure

Bond Order



7F99A688-5A74-4211-9CF0-7A6B188FCC2A

midterm-exam-93388

#37 Page 10 of 10

SCRAP PAGE

**CHEMISTRY 110 –Midterm Exam****Thursday November 7, 2024, 6:30 – 8:30 pm**

Examiners: Pallavi Sirjoosingh and Ashok Kakkar

First name (please write as legibly as possible within the boxes)

--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--

Last name

--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--

Student ID number

--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--

This booklet has 8 questions on 10 pages (The last page can be used as a scrap page).

The staple must NOT be removed from the question booklet. Your intact question booklet must be returned at the end of the exam.

A separate booklet contains a periodic table, and datasheet.

Write your answers directly under the questions in the space provided. Do not write on the QR codes or near the top edges (as it may be cut off).

Please write **legibly and dark enough** for the scan to be readable. For students who use English as a second language, dictionaries are permitted. Calculators are allowed.

DO NOT ASK QUESTIONS OF THE INVIGILATORS about interpretations of the exam questions. If you have a concern about a question, please summarize it at the top of this page. All concerns will be evaluated after the exam. GOOD LUCK!

**Question 1 8 points** (*Assume all transitions are possible*)

a. 6 points Calculate the **energy (in kJ/mol)** required to ionize (completely remove) the electron from the **n=3** level of **He⁺ ion** (He: Helium). Show your work including equations. (3 sig. figs.)

b. 2 points

(I) Select the largest (longest) wavelength from the three choices below:

- i. Wavelength of photon to *completely remove* an electron from *1s orbital of Hydrogen atom*
- ii. Wavelength of photon to *completely remove* an electron from *1s orbital of Boron atom (B)*
- iii. Wavelength of photon to *completely remove* an electron from *1s orbital of Boron cation (B⁺)*

i or ii or iii

(II) Select the smallest (shortest) wavelength from the three choices below (*for H atom*):

- i. Wavelength of photon *emitted* for an electron transitioning from *2p to 1s orbital*
- ii. Wavelength of photon *absorbed* for an electron transitioning from *2s to 4p orbital*
- iii. Wavelength of photon *absorbed* for an electron transitioning from *1s to 3p orbital*

i or ii or iii



Question 2 8 points



A Chem110 student plots the graph shown on the left, from data obtained for the kinetic energy (eV) of the ejected electron versus frequency (s^{-1}) of the incident light, for an unknown metal. ($1\text{eV} = 1.6022 \times 10^{-19} \text{ J}$)

Based on the plot, answer the questions **a to c** below:

a. 1 point On the plot above, label the **threshold frequency**.

b. 2 points What is the value of the work potential (in J) of the unknown metal?

c. 3 points What is the *longest* wavelength (in nm) required to eject an electron from this metal's surface? Show your work including equation(s) used. (3 sig figs)

d. 2 points In the space provided below, plot the following

(Assume frequency is *above* threshold frequency):

- i. *Kinetic energy* of an ejected electron versus the *intensity* of the incident light (left)
- ii. *Number of electrons* ejected versus the *frequency* of the incident light (right)



**Question 3** **6 points****a. 2 points**

In the ground state of Ge (Germanium; $Z = 32$) atom, **name** the following orbitals:

i. Orbital containing the electron that requires the maximum energy to be removed.

ii. Orbital containing the electron that experiences the minimum effective nuclear charge.

b. 2 points

An unknown anion (A^-) has the following *excited* electronic configuration: $[Ar] 4s^2 3d^3 4p^1$

(Ar: Argon; $Z = 18$)

Give the **condensed ground state electronic configuration** of the atom, **A** and the cation, **A^{2+}** .
Show your work. (Assume all electronic transitions are possible)

Electronic Configuration of A:

Electronic Configuration of A^{2+} :

c. 2 points

In a single atom, determine how many electrons can have the following set of quantum numbers?

(i) $n = 5; l = 3$

In a single atom, determine how many electrons can have the following set of quantum numbers?

(i) $n = 3; l = 2; m_s = +1/2$

**Question 4** **8 points**

Rank the following from 1 to 4 in **increasing** order. Provide the ranking in the boxes provided.

Note: Atomic number (Z) are provided for relevant atoms

a) Increasing (lowest to highest) order of *ionization energy*: (1= lowest; 4 = highest)

Mg

Mg: Z = 12

Rb

Rb: Z = 37

Cs

Cs: Z = 55

K

K : Z = 19

b) Increasing (lowest to highest) order of *bond angle*: (1= lowest; 4 = highest)

H₂OCCl₄XeF₄

HCN

c) Increasing (smallest to largest) order of *ionic size* (1= smallest; 4 = largest)

S²⁻

S: Z = 16

O²⁻

O: Z = 8

Be²⁺

Be: Z = 4

Na⁺

Na: Z = 11

d) Increasing (lowest to highest) order of *bond energy* (1= lowest; 4 = highest)

P-F

P-Cl

P-I

P-Br

**Question 5** **12 points**

a. 9 points For the molecules HClO_4 and TeF_6 , draw the most stable *Lewis structure* showing appropriate shape and *dashed/wedged bonds*, based on VSEPR. Determine the **molecular geometry** around the central atom. For HClO_4 , label the *most polar bond*.



b. 3 points (Note: Lone pairs are not shown in structures below)

(i) For the three *resonance* structures (1, 2 and 3) shown below, answer the following question:



(I) Circle the *most contributing* resonance structure.

Structure 1

Structure 2

Structure 3

(ii) For the three structures given (4, 5, and 6) below, answer the following questions:



(I) Circle all the structures that represent *resonance structures* of each other:

Structure 4

Structure 5

Structure 6

(II) Among the resonance structures, circle the *most contributing* resonance structure.

Structure 4

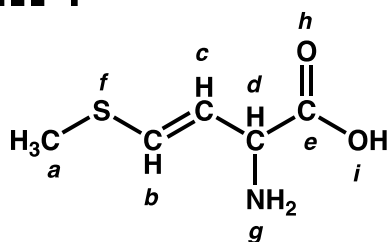
Structure 5

Structure 6

**Question 6** **6 points**

Using the data given below, calculate the **lattice energy** for SrO(s).

	ΔH° (kJ/mol)
$\text{Sr}(s) \longrightarrow \text{Sr}(g)$	+180
$\text{Sr}(g) \longrightarrow \text{Sr}^{2+}(g) + 2e^-$	+1468
$\text{O}_2(g) \longrightarrow 2\text{O}(g)$	+500
$\text{O}(g) + 2e^- \longrightarrow \text{O}^{2-}(g)$	+600
$\text{Sr}(s) + \frac{1}{2}\text{O}_2(g) \longrightarrow \text{SrO}(s)$	-550

**Question 7 10 points**

Consider the molecule given on the left, and answer questions (i) to (vi) using chemical bonding theories that we have learnt in class (valence bond theory, hybridization).

Note that the lone pairs are not shown in the structure:

i. Consider the carbon atoms marked in as ***a***, ***b***, ***c***, ***d***, and ***e*** in the molecule, and determine **hybridization** at each of the carbon centers. (2.5 points)

Atom	Hybridization
Carbon (<i>a</i>)	
Carbon (<i>b</i>)	
Carbon (<i>c</i>)	
Carbon (<i>d</i>)	
Carbon (<i>e</i>)	

ii. Now, consider the sulfur atom (S) marked as ***f***, nitrogen (N) atom (***g***), and oxygen(O) atoms (***h*** and ***i***), and determine their **hybridization**. (2 points)

Atom	Hybridization
Sulfur (<i>f</i>)	
Nitrogen (<i>g</i>)	
Oxygen (<i>h</i>)	
Oxygen (<i>i</i>)	

iii. **Describe** all the bonds formed by the carbon atoms ***a*** and ***b***, using sigma and pi designation as appropriate. (2 points)

Atom	Bonds formed (use sigma and pi as appropriate)
Carbon (<i>a</i>)	
Carbon (<i>b</i>)	

iv. How many total sp^3 -s sigma bonds are in the molecule? (1 point)

v. Determine **electronic** and **molecular geometry** at carbon (C) labeled ***e*** and sulfur (S) labeled ***f*** centres. (2 points)

Atom	Electronic Geometry	Molecular Geometry
Carbon (<i>e</i>)		
Sulfur (<i>f</i>)		

vi. Circle the value from the choices given below, that describes the **C-S-C bond angle at sulfur** (S) marked ***f***: (0.5 points)

>90° to <109.5° 109.5° >109.5 to <120° 120° >120° to <180°


Question 8 12 points

Using the **valence shell orbitals only** draw a **complete** molecular orbital (MO) diagram (with no s-p mixing) for the molecule **PS**, in the space provided below. **(6 points)**

Assume that 3s and 3p atomic orbitals will form molecular orbitals like those formed by the combination of 2s and 2p atomic orbitals that we learnt in class, and there is no s-p mixing.

Show **energy arrow**, **atomic orbitals** involved, **atom labels**, the **molecular orbitals** that are formed, **filled electrons** in the MO diagram etc. Make sure to label all the atomic and molecular orbitals.

Using the MO diagram above, answer the following questions:

- (i) Which one of the following will have the highest bond order? Circle the correct choice
 A) PS^{1-} B) PS C) PS^{1+}
- (ii) Which one of the following will have the longest bond length? Circle the correct choice
 A) PS^{1-} B) PS C) PS^{1+}
- (iii) Which of the following will be diamagnetic? Circle the correct choice.
 A) PS^{1-} B) PS C) PS^{1+}
- (iv) Draw Lewis dot structure of the molecule with the shortest bond length (based on the MO diagram) from among the three (PS^{1-} , PS, PS^{1+}), and indicate the bond order calculated from the Lewis dot structure. **(3 points)**

Lewis Dot Structure

Bond Order



503792D2-6945-4367-BF79-7D0FB22EFD4D

midterm-exam-93388

#38 Page 10 of 10

SCRAP PAGE

**CHEMISTRY 110 –Midterm Exam****Thursday November 7, 2024, 6:30 – 8:30 pm**

Examiners: Pallavi Sirjoosingh and Ashok Kakkar

First name (please write as legibly as possible within the boxes)

--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--

Last name

--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--

Student ID number

--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--

This booklet has 8 questions on 10 pages (The last page can be used as a scrap page).

The staple must NOT be removed from the question booklet. Your intact question booklet must be returned at the end of the exam.

A separate booklet contains a periodic table, and datasheet.

Write your answers directly under the questions in the space provided. Do not write on the QR codes or near the top edges (as it may be cut off).

Please write **legibly and dark enough** for the scan to be readable. For students who use English as a second language, dictionaries are permitted. Calculators are allowed.

DO NOT ASK QUESTIONS OF THE INVIGILATORS about interpretations of the exam questions. If you have a concern about a question, please summarize it at the top of this page. All concerns will be evaluated after the exam. GOOD LUCK!

**Question 1 8 points** (*Assume all transitions are possible*)

a. 6 points Calculate the **energy (in kJ/mol)** required to ionize (completely remove) the electron from the **$n=3$** level of **He^+ ion** (He: Helium). Show your work including equations. (3 sig. figs.)

b. 2 points

(I) Select the largest (longest) wavelength from the three choices below:

- i. Wavelength of photon to *completely remove* an electron from *1s orbital of Hydrogen atom*
- ii. Wavelength of photon to *completely remove* an electron from *1s orbital of Boron atom (B)*
- iii. Wavelength of photon to *completely remove* an electron from *1s orbital of Boron cation (B^+)*

i or ii or iii

(II) Select the smallest (shortest) wavelength from the three choices below (*for H atom*):

- i. Wavelength of photon *emitted* for an electron transitioning from *2p to 1s orbital*
- ii. Wavelength of photon *absorbed* for an electron transitioning from *2s to 4p orbital*
- iii. Wavelength of photon *absorbed* for an electron transitioning from *1s to 3p orbital*

i or ii or iii



Question 2 8 points



A Chem110 student plots the graph shown on the left, from data obtained for the kinetic energy (eV) of the ejected electron versus frequency (s^{-1}) of the incident light, for an unknown metal. ($1\text{eV} = 1.6022 \times 10^{-19} \text{ J}$)

Based on the plot, answer the questions **a to c** below:

a. 1 point On the plot above, label the **threshold frequency**.

b. 2 points What is the value of the work potential (in J) of the unknown metal?

c. 3 points What is the *longest* wavelength (in nm) required to eject an electron from this metal's surface? Show your work including equation(s) used. (3 sig figs)

d. 2 points In the space provided below, plot the following

(Assume frequency is *above* threshold frequency):

- Kinetic energy* of an ejected electron versus the *intensity* of the incident light (left)
- Number of electrons* ejected versus the *frequency* of the incident light (right)



**Question 3** **6 points****a. 2 points**

In the ground state of Ge (Germanium; $Z = 32$) atom, **name** the following orbitals:

i. Orbital containing the electron that requires the maximum energy to be removed.

ii. Orbital containing the electron that experiences the minimum effective nuclear charge.

b. 2 points

An unknown anion (A^-) has the following *excited* electronic configuration: $[Ar] 4s^2 3d^3 4p^1$

(Ar: Argon; $Z = 18$)

Give the **condensed ground state electronic configuration** of the atom, **A** and the cation, **A^{2+}** .
Show your work. (Assume all electronic transitions are possible)

Electronic Configuration of A:

Electronic Configuration of A^{2+} :

c. 2 points

In a single atom, determine how many electrons can have the following set of quantum numbers?

(i) $n = 5; l = 3$

In a single atom, determine how many electrons can have the following set of quantum numbers?

(i) $n = 3; l = 2; m_s = +1/2$

**Question 4** **8 points**

Rank the following from 1 to 4 in **increasing** order. Provide the ranking in the boxes provided.

Note: Atomic number (Z) are provided for relevant atoms

a) Increasing (lowest to highest) order of *ionization energy*: (1= lowest; 4 = highest)

Mg

Mg: Z = 12

Rb

Rb: Z = 37

Cs

Cs: Z = 55

K

K : Z = 19

b) Increasing (lowest to highest) order of *bond angle*: (1= lowest; 4 = highest)

H₂OCCl₄XeF₄

HCN

c) Increasing (smallest to largest) order of *ionic size* (1= smallest; 4 = largest)

S²⁻

S: Z = 16

O²⁻

O: Z = 8

Be²⁺

Be: Z = 4

Na⁺

Na: Z = 11

d) Increasing (lowest to highest) order of *bond energy* (1= lowest; 4 = highest)

P-F

P-Cl

P-I

P-Br

**Question 5** **12 points**

a. 9 points For the molecules HClO_4 and TeF_6 , draw the most stable *Lewis structure* showing appropriate shape and *dashed/wedged bonds*, based on VSEPR. Determine the **molecular geometry** around the central atom. For HClO_4 , label the *most polar bond*.



b. 3 points (Note: Lone pairs are not shown in structures below)

(i) For the three *resonance* structures (1, 2 and 3) shown below, answer the following question:



(I) Circle the *most contributing* resonance structure.

Structure 1

Structure 2

Structure 3

(ii) For the three structures given (4, 5, and 6) below, answer the following questions:



(I) Circle all the structures that represent *resonance structures* of each other:

Structure 4

Structure 5

Structure 6

(II) Among the resonance structures, circle the *most contributing* resonance structure.

Structure 4

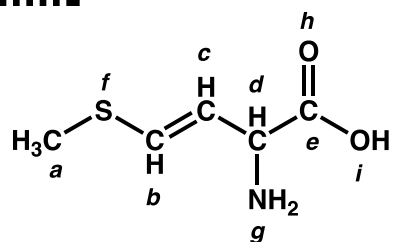
Structure 5

Structure 6

**Question 6** **6 points**

Using the data given below, calculate the **lattice energy** for SrO(s).

	ΔH° (kJ/mol)
$\text{Sr}(s) \longrightarrow \text{Sr}(g)$	+180
$\text{Sr}(g) \longrightarrow \text{Sr}^{2+}(g) + 2e^-$	+1468
$\text{O}_2(g) \longrightarrow 2\text{O}(g)$	+500
$\text{O}(g) + 2e^- \longrightarrow \text{O}^{2-}(g)$	+600
$\text{Sr}(s) + \frac{1}{2}\text{O}_2(g) \longrightarrow \text{SrO}(s)$	-550

**Question 7 10 points**

Consider the molecule given on the left, and answer questions (i) to (vi) using chemical bonding theories that we have learnt in class (valence bond theory, hybridization).

Note that the lone pairs are not shown in the structure:

i. Consider the carbon atoms marked in as ***a***, ***b***, ***c***, ***d***, and ***e*** in the molecule, and determine **hybridization** at each of the carbon centers. (2.5 points)

Atom	Hybridization
Carbon (<i>a</i>)	
Carbon (<i>b</i>)	
Carbon (<i>c</i>)	
Carbon (<i>d</i>)	
Carbon (<i>e</i>)	

ii. Now, consider the sulfur atom (S) marked as ***f***, nitrogen (N) atom (***g***), and oxygen(O) atoms (***h*** and ***i***), and determine their **hybridization**. (2 points)

Atom	Hybridization
Sulfur (<i>f</i>)	
Nitrogen (<i>g</i>)	
Oxygen (<i>h</i>)	
Oxygen (<i>i</i>)	

iii. **Describe** all the bonds formed by the carbon atoms ***a*** and ***b***, using sigma and pi designation as appropriate. (2 points)

Atom	Bonds formed (use sigma and pi as appropriate)
Carbon (<i>a</i>)	
Carbon (<i>b</i>)	

iv. How many total sp^3 -s sigma bonds are in the molecule? (1 point)

v. Determine **electronic** and **molecular geometry** at carbon (C) labeled ***e*** and sulfur (S) labeled ***f*** centres. (2 points)

Atom	Electronic Geometry	Molecular Geometry
Carbon (<i>e</i>)		
Sulfur (<i>f</i>)		

vi. Circle the value from the choices given below, that describes the **C-S-C bond angle at sulfur** (S) marked ***f***: (0.5 points)

$>90^\circ$ to $<109.5^\circ$ 109.5° >109.5 to $<120^\circ$ 120° $>120^\circ$ to $<180^\circ$


Question 8 12 points

Using the **valence shell orbitals only** draw a **complete** molecular orbital (MO) diagram (with no s-p mixing) for the molecule **PS**, in the space provided below. **(6 points)**

Assume that 3s and 3p atomic orbitals will form molecular orbitals like those formed by the combination of 2s and 2p atomic orbitals that we learnt in class, and there is no s-p mixing.

Show **energy arrow**, **atomic orbitals** involved, **atom labels**, the **molecular orbitals** that are formed, **filled electrons** in the MO diagram etc. Make sure to label all the atomic and molecular orbitals.

Using the MO diagram above, answer the following questions:

- (i) Which one of the following will have the highest bond order? Circle the correct choice
 A) PS^{1-} B) PS C) PS^{1+}
- (ii) Which one of the following will have the longest bond length? Circle the correct choice
 A) PS^{1-} B) PS C) PS^{1+}
- (iii) Which of the following will be diamagnetic? Circle the correct choice.
 A) PS^{1-} B) PS C) PS^{1+}
- (iv) Draw Lewis dot structure of the molecule with the shortest bond length (based on the MO diagram) from among the three (PS^{1-} , PS, PS^{1+}), and indicate the bond order calculated from the Lewis dot structure. **(3 points)**

Lewis Dot Structure

Bond Order



D21C2E87-051A-4F8E-A7D6-57AF9E07375D

midterm-exam-93388

#39 Page 10 of 10

SCRAP PAGE

**CHEMISTRY 110 –Midterm Exam****Thursday November 7, 2024, 6:30 – 8:30 pm****Examiners: Pallavi Sirjoosingh and Ashok Kakkar**

First name (please write as legibly as possible within the boxes)

--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--

Last name

--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--

Student ID number

--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--

This booklet has 8 questions on 10 pages (The last page can be used as a scrap page).

The staple must NOT be removed from the question booklet. Your intact question booklet must be returned at the end of the exam.

A separate booklet contains a periodic table, and datasheet.

Write your answers directly under the questions in the space provided. Do not write on the QR codes or near the top edges (as it may be cut off).

Please write **legibly and dark enough** for the scan to be readable. For students who use English as a second language, dictionaries are permitted. Calculators are allowed.

DO NOT ASK QUESTIONS OF THE INVIGILATORS about interpretations of the exam questions. If you have a concern about a question, please summarize it at the top of this page. All concerns will be evaluated after the exam. GOOD LUCK!

**Question 1 8 points** (*Assume all transitions are possible*)

a. 6 points Calculate the **energy (in kJ/mol)** required to ionize (completely remove) the electron from the **n=3** level of **He⁺ ion** (He: Helium). Show your work including equations. (3 sig. figs.)

b. 2 points

(I) Select the largest (longest) wavelength from the three choices below:

- i. Wavelength of photon to *completely remove* an electron from *1s orbital of Hydrogen atom*
- ii. Wavelength of photon to *completely remove* an electron from *1s orbital of Boron atom (B)*
- iii. Wavelength of photon to *completely remove* an electron from *1s orbital of Boron cation (B⁺)*

i or ii or iii

(II) Select the smallest (shortest) wavelength from the three choices below (*for H atom*):

- i. Wavelength of photon *emitted* for an electron transitioning from *2p to 1s orbital*
- ii. Wavelength of photon *absorbed* for an electron transitioning from *2s to 4p orbital*
- iii. Wavelength of photon *absorbed* for an electron transitioning from *1s to 3p orbital*

i or ii or iii



Question 2 8 points



A Chem110 student plots the graph shown on the left, from data obtained for the kinetic energy (eV) of the ejected electron versus frequency (s^{-1}) of the incident light, for an unknown metal. ($1\text{eV} = 1.6022 \times 10^{-19} \text{ J}$)

Based on the plot, answer the questions **a to c** below:

a. 1 point On the plot above, label the **threshold frequency**.

b. 2 points What is the value of the work potential (in J) of the unknown metal?

c. 3 points What is the *longest* wavelength (in nm) required to eject an electron from this metal's surface? Show your work including equation(s) used. (3 sig figs)

d. 2 points In the space provided below, plot the following

(Assume frequency is *above* threshold frequency):

- Kinetic energy* of an ejected electron versus the *intensity* of the incident light (left)
- Number of electrons* ejected versus the *frequency* of the incident light (right)



**Question 3** **6 points****a. 2 points**

In the ground state of Ge (Germanium; $Z = 32$) atom, **name** the following orbitals:

i. Orbital containing the electron that requires the maximum energy to be removed.

ii. Orbital containing the electron that experiences the minimum effective nuclear charge.

b. 2 points

An unknown anion (A^-) has the following *excited* electronic configuration: $[Ar] 4s^2 3d^3 4p^1$

(Ar: Argon; $Z = 18$)

Give the **condensed ground state electronic configuration** of the atom, **A** and the cation, **A^{2+}** .
Show your work. (Assume all electronic transitions are possible)

Electronic Configuration of A:

Electronic Configuration of A^{2+} :

c. 2 points

In a single atom, determine how many electrons can have the following set of quantum numbers?

(i) $n = 5; l = 3$

In a single atom, determine how many electrons can have the following set of quantum numbers?

(i) $n = 3; l = 2; m_s = +1/2$

**Question 4** **8 points**

Rank the following from 1 to 4 in **increasing** order. Provide the ranking in the boxes provided.

Note: Atomic number (Z) are provided for relevant atoms

a) Increasing (lowest to highest) order of *ionization energy*: (1= lowest; 4 = highest)

Mg

Mg: Z = 12

Rb

Rb: Z = 37

Cs

Cs: Z = 55

K

K : Z = 19

b) Increasing (lowest to highest) order of *bond angle*: (1= lowest; 4 = highest)

H₂OCCl₄XeF₄

HCN

c) Increasing (smallest to largest) order of *ionic size* (1= smallest; 4 = largest)

S²⁻

S: Z = 16

O²⁻

O: Z = 8

Be²⁺

Be: Z = 4

Na⁺

Na: Z = 11

d) Increasing (lowest to highest) order of *bond energy* (1= lowest; 4 = highest)

P-F

P-Cl

P-I

P-Br

**Question 5** **12 points**

a. 9 points For the molecules HClO_4 and TeF_6 , draw the most stable *Lewis structure* showing appropriate shape and *dashed/wedged bonds*, based on VSEPR. Determine the **molecular geometry** around the central atom. For HClO_4 , label the *most polar bond*.



b. 3 points (Note: Lone pairs are not shown in structures below)

(i) For the three *resonance* structures (1, 2 and 3) shown below, answer the following question:



(I) Circle the *most contributing* resonance structure.

Structure 1

Structure 2

Structure 3

(ii) For the three structures given (4, 5, and 6) below, answer the following questions:



(I) Circle all the structures that represent *resonance structures* of each other:

Structure 4

Structure 5

Structure 6

(II) Among the resonance structures, circle the *most contributing* resonance structure.

Structure 4

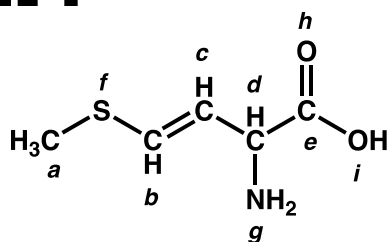
Structure 5

Structure 6

**Question 6** **6 points**

Using the data given below, calculate the **lattice energy** for SrO(s).

	ΔH° (kJ/mol)
$\text{Sr}(s) \longrightarrow \text{Sr}(g)$	+180
$\text{Sr}(g) \longrightarrow \text{Sr}^{2+}(g) + 2e^-$	+1468
$\text{O}_2(g) \longrightarrow 2\text{O}(g)$	+500
$\text{O}(g) + 2e^- \longrightarrow \text{O}^{2-}(g)$	+600
$\text{Sr}(s) + \frac{1}{2}\text{O}_2(g) \longrightarrow \text{SrO}(s)$	-550

**Question 7 10 points**

Consider the molecule given on the left, and answer questions (i) to (vi) using chemical bonding theories that we have learnt in class (valence bond theory, hybridization).

Note that the lone pairs are not shown in the structure:

i. Consider the carbon atoms marked in as ***a***, ***b***, ***c***, ***d***, and ***e*** in the molecule, and determine **hybridization** at each of the carbon centers. (2.5 points)

Atom	Hybridization
Carbon (<i>a</i>)	
Carbon (<i>b</i>)	
Carbon (<i>c</i>)	
Carbon (<i>d</i>)	
Carbon (<i>e</i>)	

ii. Now, consider the sulfur atom (S) marked as ***f***, nitrogen (N) atom (***g***), and oxygen(O) atoms (***h*** and ***i***), and determine their **hybridization**. (2 points)

Atom	Hybridization
Sulfur (<i>f</i>)	
Nitrogen (<i>g</i>)	
Oxygen (<i>h</i>)	
Oxygen (<i>i</i>)	

iii. **Describe** all the bonds formed by the carbon atoms ***a*** and ***b***, using sigma and pi designation as appropriate. (2 points)

Atom	Bonds formed (use sigma and pi as appropriate)
Carbon (<i>a</i>)	
Carbon (<i>b</i>)	

iv. How many total sp^3 -s sigma bonds are in the molecule? (1 point)

v. Determine **electronic** and **molecular geometry** at carbon (C) labeled ***e*** and sulfur (S) labeled ***f*** centres. (2 points)

Atom	Electronic Geometry	Molecular Geometry
Carbon (<i>e</i>)		
Sulfur (<i>f</i>)		

vi. Circle the value from the choices given below, that describes the **C-S-C bond angle at sulfur** (S) marked ***f***: (0.5 points)

>90° to <109.5° 109.5° >109.5 to <120° 120° >120° to <180°


Question 8 12 points

Using the **valence shell orbitals only** draw a **complete** molecular orbital (MO) diagram (with no s-p mixing) for the molecule **PS**, in the space provided below. **(6 points)**

Assume that 3s and 3p atomic orbitals will form molecular orbitals like those formed by the combination of 2s and 2p atomic orbitals that we learnt in class, and there is no s-p mixing.

Show **energy arrow**, **atomic orbitals** involved, **atom labels**, the **molecular orbitals** that are formed, **filled electrons** in the MO diagram etc. Make sure to label all the atomic and molecular orbitals.

Using the MO diagram above, answer the following questions:

- (i) Which one of the following will have the highest bond order? Circle the correct choice
 A) PS^{1-} B) PS C) PS^{1+}
- (ii) Which one of the following will have the longest bond length? Circle the correct choice
 A) PS^{1-} B) PS C) PS^{1+}
- (iii) Which of the following will be diamagnetic? Circle the correct choice.
 A) PS^{1-} B) PS C) PS^{1+}
- (iv) Draw Lewis dot structure of the molecule with the shortest bond length (based on the MO diagram) from among the three (PS^{1-} , PS, PS^{1+}), and indicate the bond order calculated from the Lewis dot structure. **(3 points)**

Lewis Dot Structure

Bond Order



809F589B-8F6C-4D61-8007-101B62F77BA2

midterm-exam-93388

#40 Page 10 of 10

SCRAP PAGE

**CHEMISTRY 110 –Midterm Exam****Thursday November 7, 2024, 6:30 – 8:30 pm**

Examiners: Pallavi Sirjoosingh and Ashok Kakkar

First name (please write as legibly as possible within the boxes)

--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--

Last name

--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--

Student ID number

--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--

This booklet has 8 questions on 10 pages (The last page can be used as a scrap page).

The staple must NOT be removed from the question booklet. Your intact question booklet must be returned at the end of the exam.

A separate booklet contains a periodic table, and datasheet.

Write your answers directly under the questions in the space provided. Do not write on the QR codes or near the top edges (as it may be cut off).

Please write **legibly and dark enough** for the scan to be readable. For students who use English as a second language, dictionaries are permitted. Calculators are allowed.

DO NOT ASK QUESTIONS OF THE INVIGILATORS about interpretations of the exam questions. If you have a concern about a question, please summarize it at the top of this page. All concerns will be evaluated after the exam. GOOD LUCK!

**Question 1 8 points** (*Assume all transitions are possible*)

a. 6 points Calculate the **energy (in kJ/mol)** required to ionize (completely remove) the electron from the **n=3** level of **He⁺ ion** (He: Helium). Show your work including equations. (3 sig. figs.)

b. 2 points

(I) Select the largest (longest) wavelength from the three choices below:

- i. Wavelength of photon to *completely remove* an electron from *1s orbital of Hydrogen atom*
- ii. Wavelength of photon to *completely remove* an electron from *1s orbital of Boron atom (B)*
- iii. Wavelength of photon to *completely remove* an electron from *1s orbital of Boron cation (B⁺)*

i or ii or iii

(II) Select the smallest (shortest) wavelength from the three choices below (*for H atom*):

- i. Wavelength of photon *emitted* for an electron transitioning from *2p to 1s orbital*
- ii. Wavelength of photon *absorbed* for an electron transitioning from *2s to 4p orbital*
- iii. Wavelength of photon *absorbed* for an electron transitioning from *1s to 3p orbital*

i or ii or iii



Question 2 8 points



A Chem110 student plots the graph shown on the left, from data obtained for the kinetic energy (eV) of the ejected electron versus frequency (s^{-1}) of the incident light, for an unknown metal. ($1\text{eV} = 1.6022 \times 10^{-19} \text{ J}$)

Based on the plot, answer the questions **a to c** below:

a. 1 point On the plot above, label the **threshold frequency**.

b. 2 points What is the value of the work potential (in J) of the unknown metal?

c. 3 points What is the *longest* wavelength (in nm) required to eject an electron from this metal's surface? Show your work including equation(s) used. (3 sig figs)

d. 2 points In the space provided below, plot the following

(Assume frequency is *above* threshold frequency):

- i. *Kinetic energy* of an ejected electron versus the *intensity* of the incident light (left)
- ii. *Number of electrons* ejected versus the *frequency* of the incident light (right)



**Question 3** **6 points****a. 2 points**

In the ground state of Ge (Germanium; $Z = 32$) atom, **name** the following orbitals:

i. Orbital containing the electron that requires the maximum energy to be removed.

ii. Orbital containing the electron that experiences the minimum effective nuclear charge.

b. 2 points

An unknown anion (A^-) has the following *excited* electronic configuration: $[Ar] 4s^2 3d^3 4p^1$

(Ar: Argon; $Z = 18$)

Give the **condensed ground state electronic configuration** of the atom, **A** and the cation, **A^{2+}** .
Show your work. (Assume all electronic transitions are possible)

Electronic Configuration of A:

Electronic Configuration of A^{2+} :

c. 2 points

In a single atom, determine how many electrons can have the following set of quantum numbers?

(i) $n = 5; l = 3$

In a single atom, determine how many electrons can have the following set of quantum numbers?

(i) $n = 3; l = 2; m_s = +1/2$

**Question 4** **8 points**

Rank the following from 1 to 4 in **increasing** order. Provide the ranking in the boxes provided.

Note: Atomic number (Z) are provided for relevant atoms

a) Increasing (lowest to highest) order of *ionization energy*: (1= lowest; 4 = highest)

Mg

Mg: Z = 12

Rb

Rb: Z = 37

Cs

Cs: Z = 55

K

K : Z = 19

b) Increasing (lowest to highest) order of *bond angle*: (1= lowest; 4 = highest)

H₂OCCl₄XeF₄

HCN

c) Increasing (smallest to largest) order of *ionic size* (1= smallest; 4 = largest)

S²⁻

S: Z = 16

O²⁻

O: Z = 8

Be²⁺

Be: Z = 4

Na⁺

Na: Z = 11

d) Increasing (lowest to highest) order of *bond energy* (1= lowest; 4 = highest)

P-F

P-Cl

P-I

P-Br

**Question 5** **12 points**

a. 9 points For the molecules HClO_4 and TeF_6 , draw the most stable *Lewis structure* showing appropriate shape and *dashed/wedged bonds*, based on VSEPR. Determine the **molecular geometry** around the central atom. For HClO_4 , label the *most polar bond*.



b. 3 points (Note: Lone pairs are not shown in structures below)

(i) For the three *resonance* structures (1, 2 and 3) shown below, answer the following question:



(I) Circle the *most contributing* resonance structure.

Structure 1

Structure 2

Structure 3

(ii) For the three structures given (4, 5, and 6) below, answer the following questions:



(I) Circle all the structures that represent *resonance structures* of each other:

Structure 4

Structure 5

Structure 6

(II) Among the resonance structures, circle the *most contributing* resonance structure.

Structure 4

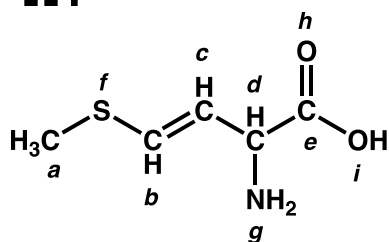
Structure 5

Structure 6

**Question 6** **6 points**

Using the data given below, calculate the **lattice energy** for SrO(s).

	ΔH° (kJ/mol)
$\text{Sr}(s) \longrightarrow \text{Sr}(g)$	+180
$\text{Sr}(g) \longrightarrow \text{Sr}^{2+}(g) + 2e^-$	+1468
$\text{O}_2(g) \longrightarrow 2\text{O}(g)$	+500
$\text{O}(g) + 2e^- \longrightarrow \text{O}^{2-}(g)$	+600
$\text{Sr}(s) + \frac{1}{2}\text{O}_2(g) \longrightarrow \text{SrO}(s)$	-550

**Question 7 10 points**

Consider the molecule given on the left, and answer questions (i) to (vi) using chemical bonding theories that we have learnt in class (valence bond theory, hybridization).

Note that the lone pairs are not shown in the structure:

i. Consider the carbon atoms marked in as ***a***, ***b***, ***c***, ***d***, and ***e*** in the molecule, and determine **hybridization** at each of the carbon centers. (2.5 points)

Atom	Hybridization
Carbon (<i>a</i>)	
Carbon (<i>b</i>)	
Carbon (<i>c</i>)	
Carbon (<i>d</i>)	
Carbon (<i>e</i>)	

ii. Now, consider the sulfur atom (S) marked as ***f***, nitrogen (N) atom (***g***), and oxygen(O) atoms (***h*** and ***i***), and determine their **hybridization**. (2 points)

Atom	Hybridization
Sulfur (<i>f</i>)	
Nitrogen (<i>g</i>)	
Oxygen (<i>h</i>)	
Oxygen (<i>i</i>)	

iii. **Describe** all the bonds formed by the carbon atoms ***a*** and ***b***, using sigma and pi designation as appropriate. (2 points)

Atom	Bonds formed (use sigma and pi as appropriate)
Carbon (<i>a</i>)	
Carbon (<i>b</i>)	

iv. How many total sp^3 -s sigma bonds are in the molecule? (1 point)

v. Determine **electronic** and **molecular geometry** at carbon (C) labeled ***e*** and sulfur (S) labeled ***f*** centres. (2 points)

Atom	Electronic Geometry	Molecular Geometry
Carbon (<i>e</i>)		
Sulfur (<i>f</i>)		

vi. Circle the value from the choices given below, that describes the **C-S-C bond angle at sulfur** (S) marked ***f***: (0.5 points)

>90° to <109.5° 109.5° >109.5 to <120° 120° >120° to <180°


Question 8 12 points

Using the **valence shell orbitals only** draw a **complete** molecular orbital (MO) diagram (with no s-p mixing) for the molecule **PS**, in the space provided below. **(6 points)**

Assume that 3s and 3p atomic orbitals will form molecular orbitals like those formed by the combination of 2s and 2p atomic orbitals that we learnt in class, and there is no s-p mixing.

Show **energy arrow**, **atomic orbitals** involved, **atom labels**, the **molecular orbitals** that are formed, **filled electrons** in the MO diagram etc. Make sure to label all the atomic and molecular orbitals.

Using the MO diagram above, answer the following questions:

- (i) Which one of the following will have the highest bond order? Circle the correct choice
 A) PS^{1-} B) PS C) PS^{1+}
- (ii) Which one of the following will have the longest bond length? Circle the correct choice
 A) PS^{1-} B) PS C) PS^{1+}
- (iii) Which of the following will be diamagnetic? Circle the correct choice.
 A) PS^{1-} B) PS C) PS^{1+}
- (iv) Draw Lewis dot structure of the molecule with the shortest bond length (based on the MO diagram) from among the three (PS^{1-} , PS, PS^{1+}), and indicate the bond order calculated from the Lewis dot structure. **(3 points)**

Lewis Dot Structure

Bond Order



BA2D9D11-97F4-477C-A88E-DCB953C5DF0D

midterm-exam-93388

#41 Page 10 of 10

SCRAP PAGE

**CHEMISTRY 110 –Midterm Exam****Thursday November 7, 2024, 6:30 – 8:30 pm**

Examiners: Pallavi Sirjoosingh and Ashok Kakkar

First name (please write as legibly as possible within the boxes)

--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--

Last name

--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--

Student ID number

--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--

This booklet has 8 questions on 10 pages (The last page can be used as a scrap page).

The staple must NOT be removed from the question booklet. Your intact question booklet must be returned at the end of the exam.

A separate booklet contains a periodic table, and datasheet.

Write your answers directly under the questions in the space provided. Do not write on the QR codes or near the top edges (as it may be cut off).

Please write **legibly and dark enough** for the scan to be readable. For students who use English as a second language, dictionaries are permitted. Calculators are allowed.

DO NOT ASK QUESTIONS OF THE INVIGILATORS about interpretations of the exam questions. If you have a concern about a question, please summarize it at the top of this page. All concerns will be evaluated after the exam. GOOD LUCK!

**Question 1 8 points** (*Assume all transitions are possible*)

a. 6 points Calculate the **energy (in kJ/mol)** required to ionize (completely remove) the electron from the **n=3** level of **He⁺ ion** (He: Helium). Show your work including equations. (3 sig. figs.)

b. 2 points

(I) Select the largest (longest) wavelength from the three choices below:

- i. Wavelength of photon to *completely remove* an electron from *1s orbital of Hydrogen atom*
- ii. Wavelength of photon to *completely remove* an electron from *1s orbital of Boron atom (B)*
- iii. Wavelength of photon to *completely remove* an electron from *1s orbital of Boron cation (B⁺)*

i or ii or iii

(II) Select the smallest (shortest) wavelength from the three choices below (*for H atom*):

- i. Wavelength of photon *emitted* for an electron transitioning from *2p to 1s orbital*
- ii. Wavelength of photon *absorbed* for an electron transitioning from *2s to 4p orbital*
- iii. Wavelength of photon *absorbed* for an electron transitioning from *1s to 3p orbital*

i or ii or iii



Question 2 8 points



A Chem110 student plots the graph shown on the left, from data obtained for the kinetic energy (eV) of the ejected electron versus frequency (s^{-1}) of the incident light, for an unknown metal. ($1\text{eV} = 1.6022 \times 10^{-19} \text{ J}$)

Based on the plot, answer the questions **a to c** below:

a. 1 point On the plot above, label the **threshold frequency**.

b. 2 points What is the value of the work potential (in J) of the unknown metal?

c. 3 points What is the *longest* wavelength (in nm) required to eject an electron from this metal's surface? Show your work including equation(s) used. (3 sig figs)

d. 2 points In the space provided below, plot the following

(Assume frequency is *above* threshold frequency):

- Kinetic energy* of an ejected electron versus the *intensity* of the incident light (left)
- Number of electrons* ejected versus the *frequency* of the incident light (right)



**Question 3** **6 points****a. 2 points**

In the ground state of Ge (Germanium; $Z = 32$) atom, **name** the following orbitals:

i. Orbital containing the electron that requires the maximum energy to be removed.

ii. Orbital containing the electron that experiences the minimum effective nuclear charge.

b. 2 points

An unknown anion (A^-) has the following *excited* electronic configuration: $[Ar] 4s^2 3d^3 4p^1$

(Ar: Argon; $Z = 18$)

Give the **condensed ground state electronic configuration** of the atom, **A** and the cation, **A^{2+}** .
Show your work. (Assume all electronic transitions are possible)

Electronic Configuration of A:

Electronic Configuration of A^{2+} :

c. 2 points

In a single atom, determine how many electrons can have the following set of quantum numbers?

(i) $n = 5; l = 3$

In a single atom, determine how many electrons can have the following set of quantum numbers?

(i) $n = 3; l = 2; m_s = +1/2$

**Question 4** **8 points**

Rank the following from 1 to 4 in **increasing** order. Provide the ranking in the boxes provided.

Note: Atomic number (Z) are provided for relevant atoms

a) Increasing (lowest to highest) order of *ionization energy*: (1= lowest; 4 = highest)

Mg

Mg: Z = 12

Rb

Rb: Z = 37

Cs

Cs: Z = 55

K

K : Z = 19

b) Increasing (lowest to highest) order of *bond angle*: (1= lowest; 4 = highest)

H₂OCCl₄XeF₄

HCN

c) Increasing (smallest to largest) order of *ionic size* (1= smallest; 4 = largest)

S²⁻

S: Z = 16

O²⁻

O: Z = 8

Be²⁺

Be: Z = 4

Na⁺

Na: Z = 11

d) Increasing (lowest to highest) order of *bond energy* (1= lowest; 4 = highest)

P-F

P-Cl

P-I

P-Br

**Question 5** **12 points**

a. 9 points For the molecules HClO_4 and TeF_6 , draw the most stable *Lewis structure* showing appropriate shape and *dashed/wedged bonds*, based on VSEPR. Determine the **molecular geometry** around the central atom. For HClO_4 , label the *most polar bond*.



b. 3 points (Note: Lone pairs are not shown in structures below)

(i) For the three *resonance* structures (1, 2 and 3) shown below, answer the following question:



(I) Circle the *most contributing* resonance structure.

Structure 1

Structure 2

Structure 3

(ii) For the three structures given (4, 5, and 6) below, answer the following questions:



(I) Circle all the structures that represent *resonance structures* of each other:

Structure 4

Structure 5

Structure 6

(II) Among the resonance structures, circle the *most contributing* resonance structure.

Structure 4

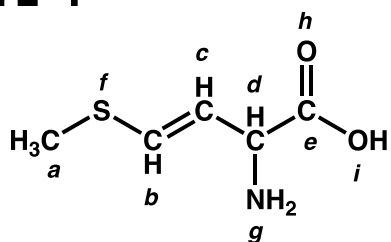
Structure 5

Structure 6

**Question 6** **6 points**

Using the data given below, calculate the **lattice energy** for SrO(s).

	ΔH° (kJ/mol)
$\text{Sr}(s) \longrightarrow \text{Sr}(g)$	+180
$\text{Sr}(g) \longrightarrow \text{Sr}^{2+}(g) + 2e^-$	+1468
$\text{O}_2(g) \longrightarrow 2\text{O}(g)$	+500
$\text{O}(g) + 2e^- \longrightarrow \text{O}^{2-}(g)$	+600
$\text{Sr}(s) + \frac{1}{2}\text{O}_2(g) \longrightarrow \text{SrO}(s)$	-550

**Question 7 10 points**

Consider the molecule given on the left, and answer questions (i) to (vi) using chemical bonding theories that we have learnt in class (valence bond theory, hybridization).

Note that the lone pairs are not shown in the structure:

i. Consider the carbon atoms marked in as ***a***, ***b***, ***c***, ***d***, and ***e*** in the molecule, and determine **hybridization** at each of the carbon centers. (2.5 points)

Atom	Hybridization
Carbon (<i>a</i>)	
Carbon (<i>b</i>)	
Carbon (<i>c</i>)	
Carbon (<i>d</i>)	
Carbon (<i>e</i>)	

ii. Now, consider the sulfur atom (S) marked as ***f***, nitrogen (N) atom (***g***), and oxygen(O) atoms (***h*** and ***i***), and determine their **hybridization**. (2 points)

Atom	Hybridization
Sulfur (<i>f</i>)	
Nitrogen (<i>g</i>)	
Oxygen (<i>h</i>)	
Oxygen (<i>i</i>)	

iii. **Describe** all the bonds formed by the carbon atoms ***a*** and ***b***, using sigma and pi designation as appropriate. (2 points)

Atom	Bonds formed (use sigma and pi as appropriate)
Carbon (<i>a</i>)	
Carbon (<i>b</i>)	

iv. How many total sp^3 -s sigma bonds are in the molecule? (1 point)

v. Determine **electronic** and **molecular geometry** at carbon (C) labeled ***e*** and sulfur (S) labeled ***f*** centres. (2 points)

Atom	Electronic Geometry	Molecular Geometry
Carbon (<i>e</i>)		
Sulfur (<i>f</i>)		

vi. Circle the value from the choices given below, that describes the **C-S-C bond angle at sulfur** (S) marked ***f***: (0.5 points)

>90° to <109.5° 109.5° >109.5 to <120° 120° >120° to <180°


Question 8 12 points

Using the **valence shell orbitals only** draw a **complete** molecular orbital (MO) diagram (with no s-p mixing) for the molecule **PS**, in the space provided below. **(6 points)**

Assume that 3s and 3p atomic orbitals will form molecular orbitals like those formed by the combination of 2s and 2p atomic orbitals that we learnt in class, and there is no s-p mixing.

Show **energy arrow**, **atomic orbitals** involved, **atom labels**, the **molecular orbitals** that are formed, **filled electrons** in the MO diagram etc. Make sure to label all the atomic and molecular orbitals.

Using the MO diagram above, answer the following questions:

- (i) Which one of the following will have the highest bond order? Circle the correct choice
 A) PS^{1-} B) PS C) PS^{1+}
- (ii) Which one of the following will have the longest bond length? Circle the correct choice
 A) PS^{1-} B) PS C) PS^{1+}
- (iii) Which of the following will be diamagnetic? Circle the correct choice.
 A) PS^{1-} B) PS C) PS^{1+}
- (iv) Draw Lewis dot structure of the molecule with the shortest bond length (based on the MO diagram) from among the three (PS^{1-} , PS, PS^{1+}), and indicate the bond order calculated from the Lewis dot structure. **(3 points)**

Lewis Dot Structure

Bond Order



3605FB1F-743B-4BDA-AA77-1E88F6ADB188

midterm-exam-93388

#42 Page 10 of 10

SCRAP PAGE

**CHEMISTRY 110 –Midterm Exam****Thursday November 7, 2024, 6:30 – 8:30 pm**

Examiners: Pallavi Sirjoosingh and Ashok Kakkar

First name (please write as legibly as possible within the boxes)

--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--

Last name

--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--

Student ID number

--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--

This booklet has 8 questions on 10 pages (The last page can be used as a scrap page).

The staple must NOT be removed from the question booklet. Your intact question booklet must be returned at the end of the exam.

A separate booklet contains a periodic table, and datasheet.

Write your answers directly under the questions in the space provided. Do not write on the QR codes or near the top edges (as it may be cut off).

Please write **legibly and dark enough** for the scan to be readable. For students who use English as a second language, dictionaries are permitted. Calculators are allowed.

DO NOT ASK QUESTIONS OF THE INVIGILATORS about interpretations of the exam questions. If you have a concern about a question, please summarize it at the top of this page. All concerns will be evaluated after the exam. GOOD LUCK!

**Question 1 8 points** (*Assume all transitions are possible*)

a. 6 points Calculate the **energy (in kJ/mol)** required to ionize (completely remove) the electron from the **n=3** level of **He⁺ ion** (He: Helium). Show your work including equations. (3 sig. figs.)

b. 2 points

(I) Select the largest (longest) wavelength from the three choices below:

- i. Wavelength of photon to *completely remove* an electron from *1s orbital of Hydrogen atom*
- ii. Wavelength of photon to *completely remove* an electron from *1s orbital of Boron atom (B)*
- iii. Wavelength of photon to *completely remove* an electron from *1s orbital of Boron cation (B⁺)*

i or ii or iii

(II) Select the smallest (shortest) wavelength from the three choices below (*for H atom*):

- i. Wavelength of photon *emitted* for an electron transitioning from *2p to 1s orbital*
- ii. Wavelength of photon *absorbed* for an electron transitioning from *2s to 4p orbital*
- iii. Wavelength of photon *absorbed* for an electron transitioning from *1s to 3p orbital*

i or ii or iii



Question 2 8 points



A Chem110 student plots the graph shown on the left, from data obtained for the kinetic energy (eV) of the ejected electron versus frequency (s^{-1}) of the incident light, for an unknown metal. ($1\text{eV} = 1.6022 \times 10^{-19} \text{ J}$)

Based on the plot, answer the questions **a to c** below:

a. 1 point On the plot above, label the **threshold frequency**.

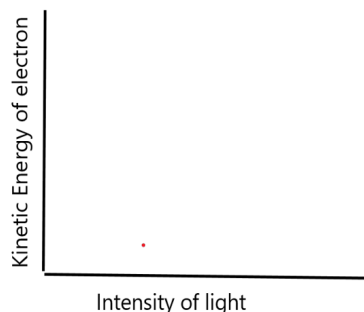
b. 2 points What is the value of the work potential (in J) of the unknown metal?

c. 3 points What is the *longest* wavelength (in nm) required to eject an electron from this metal's surface? Show your work including equation(s) used. (3 sig figs)

d. 2 points In the space provided below, plot the following

(Assume frequency is *above* threshold frequency):

- i. *Kinetic energy* of an ejected electron versus the *intensity* of the incident light (left)
- ii. *Number of electrons* ejected versus the *frequency* of the incident light (right)



**Question 3** **6 points****a. 2 points**

In the ground state of Ge (Germanium; $Z = 32$) atom, **name** the following orbitals:

i. Orbital containing the electron that requires the maximum energy to be removed.

ii. Orbital containing the electron that experiences the minimum effective nuclear charge.

b. 2 points

An unknown anion (A^-) has the following *excited* electronic configuration: $[Ar] 4s^2 3d^3 4p^1$

(Ar: Argon; $Z = 18$)

Give the **condensed ground state electronic configuration** of the atom, **A** and the cation, **A^{2+}** .
Show your work. (Assume all electronic transitions are possible)

Electronic Configuration of A:

Electronic Configuration of A^{2+} :

c. 2 points

In a single atom, determine how many electrons can have the following set of quantum numbers?

(i) $n = 5; l = 3$

In a single atom, determine how many electrons can have the following set of quantum numbers?

(i) $n = 3; l = 2; m_s = +1/2$

**Question 4** **8 points**

Rank the following from 1 to 4 in **increasing** order. Provide the ranking in the boxes provided.

Note: Atomic number (Z) are provided for relevant atoms

a) Increasing (lowest to highest) order of *ionization energy*: (1= lowest; 4 = highest)

Mg

Mg: Z = 12

Rb

Rb: Z = 37

Cs

Cs: Z = 55

K

K : Z = 19

b) Increasing (lowest to highest) order of *bond angle*: (1= lowest; 4 = highest)

H₂OCCl₄XeF₄

HCN

c) Increasing (smallest to largest) order of *ionic size* (1= smallest; 4 = largest)

S²⁻

S: Z = 16

O²⁻

O: Z = 8

Be²⁺

Be: Z = 4

Na⁺

Na: Z = 11

d) Increasing (lowest to highest) order of *bond energy* (1= lowest; 4 = highest)

P-F

P-Cl

P-I

P-Br

**Question 5** **12 points**

a. 9 points For the molecules HClO_4 and TeF_6 , draw the most stable *Lewis structure* showing appropriate shape and *dashed/wedged bonds*, based on VSEPR. Determine the **molecular geometry** around the central atom. For HClO_4 , label the *most polar bond*.



b. 3 points (Note: Lone pairs are not shown in structures below)

(i) For the three *resonance* structures (1, 2 and 3) shown below, answer the following question:



(I) Circle the *most contributing* resonance structure.

Structure 1

Structure 2

Structure 3

(ii) For the three structures given (4, 5, and 6) below, answer the following questions:



(I) Circle all the structures that represent *resonance structures* of each other:

Structure 4

Structure 5

Structure 6

(II) Among the resonance structures, circle the *most contributing* resonance structure.

Structure 4

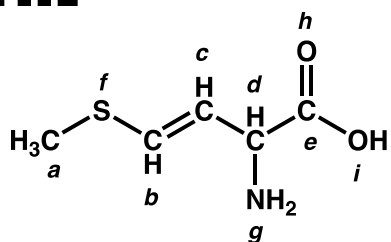
Structure 5

Structure 6

**Question 6** **6 points**

Using the data given below, calculate the **lattice energy** for SrO(s).

	ΔH° (kJ/mol)
$\text{Sr}(s) \longrightarrow \text{Sr}(g)$	+180
$\text{Sr}(g) \longrightarrow \text{Sr}^{2+}(g) + 2e^-$	+1468
$\text{O}_2(g) \longrightarrow 2\text{O}(g)$	+500
$\text{O}(g) + 2e^- \longrightarrow \text{O}^{2-}(g)$	+600
$\text{Sr}(s) + \frac{1}{2}\text{O}_2(g) \longrightarrow \text{SrO}(s)$	-550

**Question 7 10 points**

Consider the molecule given on the left, and answer questions (i) to (vi) using chemical bonding theories that we have learnt in class (valence bond theory, hybridization).

Note that the lone pairs are not shown in the structure:

i. Consider the carbon atoms marked in as ***a***, ***b***, ***c***, ***d***, and ***e*** in the molecule, and determine **hybridization** at each of the carbon centers. (2.5 points)

Atom	Hybridization
Carbon (<i>a</i>)	
Carbon (<i>b</i>)	
Carbon (<i>c</i>)	
Carbon (<i>d</i>)	
Carbon (<i>e</i>)	

ii. Now, consider the sulfur atom (S) marked as ***f***, nitrogen (N) atom (***g***), and oxygen(O) atoms (***h*** and ***i***), and determine their **hybridization**. (2 points)

Atom	Hybridization
Sulfur (<i>f</i>)	
Nitrogen (<i>g</i>)	
Oxygen (<i>h</i>)	
Oxygen (<i>i</i>)	

iii. **Describe** all the bonds formed by the carbon atoms ***a*** and ***b***, using sigma and pi designation as appropriate. (2 points)

Atom	Bonds formed (use sigma and pi as appropriate)
Carbon (<i>a</i>)	
Carbon (<i>b</i>)	

iv. How many total sp^3 -s sigma bonds are in the molecule? (1 point)

v. Determine **electronic** and **molecular geometry** at carbon (C) labeled ***e*** and sulfur (S) labeled ***f*** centres. (2 points)

Atom	Electronic Geometry	Molecular Geometry
Carbon (<i>e</i>)		
Sulfur (<i>f</i>)		

vi. Circle the value from the choices given below, that describes the **C-S-C bond angle at sulfur** (S) marked ***f***: (0.5 points)

>90° to <109.5° 109.5° >109.5 to <120° 120° >120° to <180°


Question 8 12 points

Using the **valence shell orbitals only** draw a **complete** molecular orbital (MO) diagram (with no s-p mixing) for the molecule **PS**, in the space provided below. **(6 points)**

Assume that 3s and 3p atomic orbitals will form molecular orbitals like those formed by the combination of 2s and 2p atomic orbitals that we learnt in class, and there is no s-p mixing.

Show **energy arrow**, **atomic orbitals** involved, **atom labels**, the **molecular orbitals** that are formed, **filled electrons** in the MO diagram etc. Make sure to label all the atomic and molecular orbitals.

Using the MO diagram above, answer the following questions:

- (i) Which one of the following will have the highest bond order? Circle the correct choice
 A) PS^{1-} B) PS C) PS^{1+}
- (ii) Which one of the following will have the longest bond length? Circle the correct choice
 A) PS^{1-} B) PS C) PS^{1+}
- (iii) Which of the following will be diamagnetic? Circle the correct choice.
 A) PS^{1-} B) PS C) PS^{1+}
- (iv) Draw Lewis dot structure of the molecule with the shortest bond length (based on the MO diagram) from among the three (PS^{1-} , PS, PS^{1+}), and indicate the bond order calculated from the Lewis dot structure. **(3 points)**

Lewis Dot Structure

Bond Order



330934DA-839F-46AE-9ADB-A09BB2A99DF2

midterm-exam-93388

#43 Page 10 of 10

SCRAP PAGE

**CHEMISTRY 110 –Midterm Exam****Thursday November 7, 2024, 6:30 – 8:30 pm**

Examiners: Pallavi Sirjoosingh and Ashok Kakkar

First name (please write as legibly as possible within the boxes)

--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--

Last name

--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--

Student ID number

--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--

This booklet has 8 questions on 10 pages (The last page can be used as a scrap page).

The staple must NOT be removed from the question booklet. Your intact question booklet must be returned at the end of the exam.

A separate booklet contains a periodic table, and datasheet.

Write your answers directly under the questions in the space provided. Do not write on the QR codes or near the top edges (as it may be cut off).

Please write **legibly and dark enough** for the scan to be readable. For students who use English as a second language, dictionaries are permitted. Calculators are allowed.

DO NOT ASK QUESTIONS OF THE INVIGILATORS about interpretations of the exam questions. If you have a concern about a question, please summarize it at the top of this page. All concerns will be evaluated after the exam. GOOD LUCK!

**Question 1 8 points** (*Assume all transitions are possible*)

a. 6 points Calculate the **energy (in kJ/mol)** required to ionize (completely remove) the electron from the **n=3** level of **He⁺ ion** (He: Helium). Show your work including equations. (3 sig. figs.)

b. 2 points

(I) Select the largest (longest) wavelength from the three choices below:

- i. Wavelength of photon to *completely remove* an electron from *1s orbital of Hydrogen atom*
- ii. Wavelength of photon to *completely remove* an electron from *1s orbital of Boron atom (B)*
- iii. Wavelength of photon to *completely remove* an electron from *1s orbital of Boron cation (B⁺)*

i or ii or iii

(II) Select the smallest (shortest) wavelength from the three choices below (*for H atom*):

- i. Wavelength of photon *emitted* for an electron transitioning from *2p to 1s orbital*
- ii. Wavelength of photon *absorbed* for an electron transitioning from *2s to 4p orbital*
- iii. Wavelength of photon *absorbed* for an electron transitioning from *1s to 3p orbital*

i or ii or iii



Question 2 8 points



A Chem110 student plots the graph shown on the left, from data obtained for the kinetic energy (eV) of the ejected electron versus frequency (s^{-1}) of the incident light, for an unknown metal. ($1\text{eV} = 1.6022 \times 10^{-19} \text{ J}$)

Based on the plot, answer the questions **a to c** below:

a. 1 point On the plot above, label the **threshold frequency**.

b. 2 points What is the value of the work potential (in J) of the unknown metal?

c. 3 points What is the *longest* wavelength (in nm) required to eject an electron from this metal's surface? Show your work including equation(s) used. (3 sig figs)

d. 2 points In the space provided below, plot the following

(Assume frequency is *above* threshold frequency):

- Kinetic energy* of an ejected electron versus the *intensity* of the incident light (left)
- Number of electrons* ejected versus the *frequency* of the incident light (right)



**Question 3** **6 points****a. 2 points**

In the ground state of Ge (Germanium; $Z = 32$) atom, **name** the following orbitals:

i. Orbital containing the electron that requires the maximum energy to be removed.

ii. Orbital containing the electron that experiences the minimum effective nuclear charge.

b. 2 points

An unknown anion (A^-) has the following *excited* electronic configuration: $[Ar] 4s^2 3d^3 4p^1$

(Ar: Argon; $Z = 18$)

Give the **condensed ground state electronic configuration** of the atom, **A** and the cation, **A^{2+}** .
Show your work. (Assume all electronic transitions are possible)

Electronic Configuration of A:

Electronic Configuration of A^{2+} :

c. 2 points

In a single atom, determine how many electrons can have the following set of quantum numbers?

(i) $n = 5; l = 3$

In a single atom, determine how many electrons can have the following set of quantum numbers?

(i) $n = 3; l = 2; m_s = +1/2$

**Question 4** **8 points**

Rank the following from 1 to 4 in **increasing** order. Provide the ranking in the boxes provided.

Note: Atomic number (Z) are provided for relevant atoms

a) Increasing (lowest to highest) order of *ionization energy*: (1= lowest; 4 = highest)

Mg

Mg: Z = 12

Rb

Rb: Z = 37

Cs

Cs: Z = 55

K

K : Z = 19

b) Increasing (lowest to highest) order of *bond angle*: (1= lowest; 4 = highest)

H₂OCCl₄XeF₄

HCN

c) Increasing (smallest to largest) order of *ionic size* (1= smallest; 4 = largest)

S²⁻

S: Z = 16

O²⁻

O: Z = 8

Be²⁺

Be: Z = 4

Na⁺

Na: Z = 11

d) Increasing (lowest to highest) order of *bond energy* (1= lowest; 4 = highest)

P-F

P-Cl

P-I

P-Br

**Question 5** **12 points**

a. 9 points For the molecules HClO_4 and TeF_6 , draw the most stable *Lewis structure* showing appropriate shape and *dashed/wedged bonds*, based on VSEPR. Determine the **molecular geometry** around the central atom. For HClO_4 , label the *most polar bond*.



b. 3 points (Note: Lone pairs are not shown in structures below)

(i) For the three *resonance* structures (1, 2 and 3) shown below, answer the following question:



(I) Circle the *most contributing* resonance structure.

Structure 1

Structure 2

Structure 3

(ii) For the three structures given (4, 5, and 6) below, answer the following questions:



(I) Circle all the structures that represent *resonance structures* of each other:

Structure 4

Structure 5

Structure 6

(II) Among the resonance structures, circle the *most contributing* resonance structure.

Structure 4

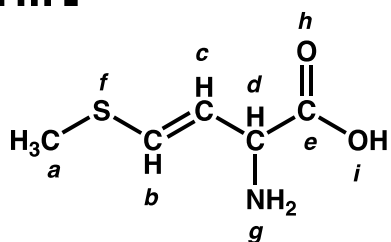
Structure 5

Structure 6

**Question 6** **6 points**

Using the data given below, calculate the **lattice energy** for SrO(s).

	ΔH° (kJ/mol)
$\text{Sr}(s) \longrightarrow \text{Sr}(g)$	+180
$\text{Sr}(g) \longrightarrow \text{Sr}^{2+}(g) + 2e^-$	+1468
$\text{O}_2(g) \longrightarrow 2\text{O}(g)$	+500
$\text{O}(g) + 2e^- \longrightarrow \text{O}^{2-}(g)$	+600
$\text{Sr}(s) + \frac{1}{2}\text{O}_2(g) \longrightarrow \text{SrO}(s)$	-550

**Question 7 10 points**

Consider the molecule given on the left, and answer questions (i) to (vi) using chemical bonding theories that we have learnt in class (valence bond theory, hybridization).

Note that the lone pairs are not shown in the structure:

i. Consider the carbon atoms marked in as ***a***, ***b***, ***c***, ***d***, and ***e*** in the molecule, and determine **hybridization** at each of the carbon centers. (2.5 points)

Atom	Hybridization
Carbon (<i>a</i>)	
Carbon (<i>b</i>)	
Carbon (<i>c</i>)	
Carbon (<i>d</i>)	
Carbon (<i>e</i>)	

ii. Now, consider the sulfur atom (S) marked as ***f***, nitrogen (N) atom (***g***), and oxygen(O) atoms (***h*** and ***i***), and determine their **hybridization**. (2 points)

Atom	Hybridization
Sulfur (<i>f</i>)	
Nitrogen (<i>g</i>)	
Oxygen (<i>h</i>)	
Oxygen (<i>i</i>)	

iii. **Describe** all the bonds formed by the carbon atoms ***a*** and ***b***, using sigma and pi designation as appropriate. (2 points)

Atom	Bonds formed (use sigma and pi as appropriate)
Carbon (<i>a</i>)	
Carbon (<i>b</i>)	

iv. How many total sp^3 -s sigma bonds are in the molecule? (1 point)

v. Determine **electronic** and **molecular geometry** at carbon (C) labeled ***e*** and sulfur (S) labeled ***f*** centres. (2 points)

Atom	Electronic Geometry	Molecular Geometry
Carbon (<i>e</i>)		
Sulfur (<i>f</i>)		

vi. Circle the value from the choices given below, that describes the **C-S-C bond angle at sulfur** (S) marked ***f***: (0.5 points)

$>90^\circ$ to $<109.5^\circ$ 109.5° >109.5 to $<120^\circ$ 120° $>120^\circ$ to $<180^\circ$


Question 8 12 points

Using the **valence shell orbitals only** draw a **complete** molecular orbital (MO) diagram (with no s-p mixing) for the molecule **PS**, in the space provided below. **(6 points)**

Assume that 3s and 3p atomic orbitals will form molecular orbitals like those formed by the combination of 2s and 2p atomic orbitals that we learnt in class, and there is no s-p mixing.

Show **energy arrow**, **atomic orbitals** involved, **atom labels**, the **molecular orbitals** that are formed, **filled electrons** in the MO diagram etc. Make sure to label all the atomic and molecular orbitals.

Using the MO diagram above, answer the following questions:

- (i) Which one of the following will have the highest bond order? Circle the correct choice
 A) PS^{1-} B) PS C) PS^{1+}
- (ii) Which one of the following will have the longest bond length? Circle the correct choice
 A) PS^{1-} B) PS C) PS^{1+}
- (iii) Which of the following will be diamagnetic? Circle the correct choice.
 A) PS^{1-} B) PS C) PS^{1+}
- (iv) Draw Lewis dot structure of the molecule with the shortest bond length (based on the MO diagram) from among the three (PS^{1-} , PS, PS^{1+}), and indicate the bond order calculated from the Lewis dot structure. **(3 points)**

Lewis Dot Structure

Bond Order



40AD9579-582B-4ECD-BA9B-3F06802DD392

midterm-exam-93388

#44 Page 10 of 10

SCRAP PAGE

**CHEMISTRY 110 –Midterm Exam****Thursday November 7, 2024, 6:30 – 8:30 pm**

Examiners: Pallavi Sirjoosingh and Ashok Kakkar

First name (please write as legibly as possible within the boxes)

--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--

Last name

--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--

Student ID number

--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--

This booklet has 8 questions on 10 pages (The last page can be used as a scrap page).

The staple must NOT be removed from the question booklet. Your intact question booklet must be returned at the end of the exam.

A separate booklet contains a periodic table, and datasheet.

Write your answers directly under the questions in the space provided. Do not write on the QR codes or near the top edges (as it may be cut off).

Please write **legibly and dark enough** for the scan to be readable. For students who use English as a second language, dictionaries are permitted. Calculators are allowed.

DO NOT ASK QUESTIONS OF THE INVIGILATORS about interpretations of the exam questions. If you have a concern about a question, please summarize it at the top of this page. All concerns will be evaluated after the exam. GOOD LUCK!

**Question 1 8 points** (*Assume all transitions are possible*)

a. 6 points Calculate the **energy (in kJ/mol)** required to ionize (completely remove) the electron from the **n=3** level of **He⁺ ion** (He: Helium). Show your work including equations. (3 sig. figs.)

b. 2 points

(I) Select the largest (longest) wavelength from the three choices below:

- i. Wavelength of photon to *completely remove* an electron from *1s orbital of Hydrogen atom*
- ii. Wavelength of photon to *completely remove* an electron from *1s orbital of Boron atom (B)*
- iii. Wavelength of photon to *completely remove* an electron from *1s orbital of Boron cation (B⁺)*

i or ii or iii

(II) Select the smallest (shortest) wavelength from the three choices below (*for H atom*):

- i. Wavelength of photon *emitted* for an electron transitioning from *2p to 1s orbital*
- ii. Wavelength of photon *absorbed* for an electron transitioning from *2s to 4p orbital*
- iii. Wavelength of photon *absorbed* for an electron transitioning from *1s to 3p orbital*

i or ii or iii



Question 2 8 points



A Chem110 student plots the graph shown on the left, from data obtained for the kinetic energy (eV) of the ejected electron versus frequency (s^{-1}) of the incident light, for an unknown metal. ($1\text{eV} = 1.6022 \times 10^{-19} \text{ J}$)

Based on the plot, answer the questions **a to c** below:

a. 1 point On the plot above, label the **threshold frequency**.

b. 2 points What is the value of the work potential (in J) of the unknown metal?

c. 3 points What is the *longest* wavelength (in nm) required to eject an electron from this metal's surface? Show your work including equation(s) used. (3 sig figs)

d. 2 points In the space provided below, plot the following

(Assume frequency is *above* threshold frequency):

- Kinetic energy* of an ejected electron versus the *intensity* of the incident light (left)
- Number of electrons* ejected versus the *frequency* of the incident light (right)



**Question 3** **6 points****a. 2 points**

In the ground state of Ge (Germanium; $Z = 32$) atom, **name** the following orbitals:

i. Orbital containing the electron that requires the maximum energy to be removed.

ii. Orbital containing the electron that experiences the minimum effective nuclear charge.

b. 2 points

An unknown anion (A^-) has the following *excited* electronic configuration: $[Ar] 4s^2 3d^3 4p^1$

(Ar: Argon; $Z = 18$)

Give the **condensed ground state electronic configuration** of the atom, **A** and the cation, **A^{2+}** .
Show your work. (Assume all electronic transitions are possible)

Electronic Configuration of A:

Electronic Configuration of A^{2+} :

c. 2 points

In a single atom, determine how many electrons can have the following set of quantum numbers?

(i) $n = 5; l = 3$

In a single atom, determine how many electrons can have the following set of quantum numbers?

(i) $n = 3; l = 2; m_s = +1/2$

**Question 4** **8 points**

Rank the following from 1 to 4 in **increasing** order. Provide the ranking in the boxes provided.

Note: Atomic number (Z) are provided for relevant atoms

a) Increasing (lowest to highest) order of *ionization energy*: (1= lowest; 4 = highest)

Mg

Mg: Z = 12

Rb

Rb: Z = 37

Cs

Cs: Z = 55

K

K : Z = 19

b) Increasing (lowest to highest) order of *bond angle*: (1= lowest; 4 = highest)

H₂OCCl₄XeF₄

HCN

c) Increasing (smallest to largest) order of *ionic size* (1= smallest; 4 = largest)

S²⁻

S: Z = 16

O²⁻

O: Z = 8

Be²⁺

Be: Z = 4

Na⁺

Na: Z = 11

d) Increasing (lowest to highest) order of *bond energy* (1= lowest; 4 = highest)

P-F

P-Cl

P-I

P-Br

**Question 5** **12 points**

a. 9 points For the molecules HClO_4 and TeF_6 , draw the most stable *Lewis structure* showing appropriate shape and *dashed/wedged bonds*, based on VSEPR. Determine the **molecular geometry** around the central atom. For HClO_4 , label the *most polar bond*.



b. 3 points (Note: Lone pairs are not shown in structures below)

(i) For the three *resonance* structures (1, 2 and 3) shown below, answer the following question:



(I) Circle the *most contributing* resonance structure.

Structure 1

Structure 2

Structure 3

(ii) For the three structures given (4, 5, and 6) below, answer the following questions:



(I) Circle all the structures that represent *resonance structures* of each other:

Structure 4

Structure 5

Structure 6

(II) Among the resonance structures, circle the *most contributing* resonance structure.

Structure 4

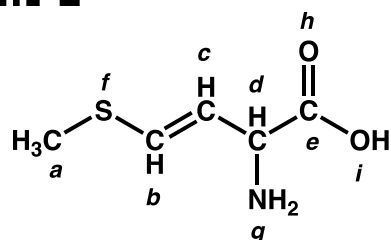
Structure 5

Structure 6

**Question 6** **6 points**

Using the data given below, calculate the **lattice energy** for SrO(s).

	ΔH° (kJ/mol)
$\text{Sr}(s) \longrightarrow \text{Sr}(g)$	+180
$\text{Sr}(g) \longrightarrow \text{Sr}^{2+}(g) + 2e^-$	+1468
$\text{O}_2(g) \longrightarrow 2\text{O}(g)$	+500
$\text{O}(g) + 2e^- \longrightarrow \text{O}^{2-}(g)$	+600
$\text{Sr}(s) + \frac{1}{2}\text{O}_2(g) \longrightarrow \text{SrO}(s)$	-550

**Question 7 10 points**

Consider the molecule given on the left, and answer questions (i) to (vi) using chemical bonding theories that we have learnt in class (valence bond theory, hybridization).

Note that the lone pairs are not shown in the structure:

i. Consider the carbon atoms marked in as ***a***, ***b***, ***c***, ***d***, and ***e*** in the molecule, and determine **hybridization** at each of the carbon centers. (2.5 points)

Atom	Hybridization
Carbon (<i>a</i>)	
Carbon (<i>b</i>)	
Carbon (<i>c</i>)	
Carbon (<i>d</i>)	
Carbon (<i>e</i>)	

ii. Now, consider the sulfur atom (S) marked as ***f***, nitrogen (N) atom (***g***), and oxygen(O) atoms (***h*** and ***i***), and determine their **hybridization**. (2 points)

Atom	Hybridization
Sulfur (<i>f</i>)	
Nitrogen (<i>g</i>)	
Oxygen (<i>h</i>)	
Oxygen (<i>i</i>)	

iii. **Describe** all the bonds formed by the carbon atoms ***a*** and ***b***, using sigma and pi designation as appropriate. (2 points)

Atom	Bonds formed (use sigma and pi as appropriate)
Carbon (<i>a</i>)	
Carbon (<i>b</i>)	

iv. How many total sp^3 -s sigma bonds are in the molecule? (1 point)

v. Determine **electronic** and **molecular geometry** at carbon (C) labeled ***e*** and sulfur (S) labeled ***f*** centres. (2 points)

Atom	Electronic Geometry	Molecular Geometry
Carbon (<i>e</i>)		
Sulfur (<i>f</i>)		

vi. Circle the value from the choices given below, that describes the **C-S-C bond angle at sulfur** (S) marked ***f***: (0.5 points)

>90° to <109.5°

109.5°

>109.5 to <120°

120°

>120° to <180°


Question 8 12 points

Using the **valence shell orbitals only** draw a **complete** molecular orbital (MO) diagram (with no s-p mixing) for the molecule **PS**, in the space provided below. **(6 points)**

Assume that 3s and 3p atomic orbitals will form molecular orbitals like those formed by the combination of 2s and 2p atomic orbitals that we learnt in class, and there is no s-p mixing.

Show **energy arrow**, **atomic orbitals** involved, **atom labels**, the **molecular orbitals** that are formed, **filled electrons** in the MO diagram etc. Make sure to label all the atomic and molecular orbitals.

Using the MO diagram above, answer the following questions:

- (i) Which one of the following will have the highest bond order? Circle the correct choice
 A) PS^{1-} B) PS C) PS^{1+}
- (ii) Which one of the following will have the longest bond length? Circle the correct choice
 A) PS^{1-} B) PS C) PS^{1+}
- (iii) Which of the following will be diamagnetic? Circle the correct choice.
 A) PS^{1-} B) PS C) PS^{1+}
- (iv) Draw Lewis dot structure of the molecule with the shortest bond length (based on the MO diagram) from among the three (PS^{1-} , PS, PS^{1+}), and indicate the bond order calculated from the Lewis dot structure. **(3 points)**

Lewis Dot Structure

Bond Order



C68342B6-A11B-4414-80CC-B430F99A7EE7

midterm-exam-93388

#45 Page 10 of 10

SCRAP PAGE

**CHEMISTRY 110 –Midterm Exam****Thursday November 7, 2024, 6:30 – 8:30 pm**

Examiners: Pallavi Sirjoosingh and Ashok Kakkar

First name (please write as legibly as possible within the boxes)

--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--

Last name

--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--

Student ID number

--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--

This booklet has 8 questions on 10 pages (The last page can be used as a scrap page).

The staple must NOT be removed from the question booklet. Your intact question booklet must be returned at the end of the exam.

A separate booklet contains a periodic table, and datasheet.

Write your answers directly under the questions in the space provided. Do not write on the QR codes or near the top edges (as it may be cut off).

Please write **legibly and dark enough** for the scan to be readable. For students who use English as a second language, dictionaries are permitted. Calculators are allowed.

DO NOT ASK QUESTIONS OF THE INVIGILATORS about interpretations of the exam questions. If you have a concern about a question, please summarize it at the top of this page. All concerns will be evaluated after the exam. GOOD LUCK!

**Question 1 8 points** (*Assume all transitions are possible*)

a. 6 points Calculate the **energy (in kJ/mol)** required to ionize (completely remove) the electron from the **n=3** level of **He⁺ ion** (He: Helium). Show your work including equations. (3 sig. figs.)

b. 2 points

(I) Select the largest (longest) wavelength from the three choices below:

- i. Wavelength of photon to *completely remove* an electron from *1s orbital of Hydrogen atom*
- ii. Wavelength of photon to *completely remove* an electron from *1s orbital of Boron atom (B)*
- iii. Wavelength of photon to *completely remove* an electron from *1s orbital of Boron cation (B⁺)*

i or ii or iii

(II) Select the smallest (shortest) wavelength from the three choices below (*for H atom*):

- i. Wavelength of photon *emitted* for an electron transitioning from *2p to 1s orbital*
- ii. Wavelength of photon *absorbed* for an electron transitioning from *2s to 4p orbital*
- iii. Wavelength of photon *absorbed* for an electron transitioning from *1s to 3p orbital*

i or ii or iii



Question 2 8 points



A Chem110 student plots the graph shown on the left, from data obtained for the kinetic energy (eV) of the ejected electron versus frequency (s^{-1}) of the incident light, for an unknown metal. ($1\text{eV} = 1.6022 \times 10^{-19} \text{ J}$)

Based on the plot, answer the questions **a to c** below:

a. 1 point On the plot above, label the **threshold frequency**.

b. 2 points What is the value of the work potential (in J) of the unknown metal?

c. 3 points What is the *longest* wavelength (in nm) required to eject an electron from this metal's surface? Show your work including equation(s) used. (3 sig figs)

d. 2 points In the space provided below, plot the following

(Assume frequency is *above* threshold frequency):

- i. *Kinetic energy* of an ejected electron versus the *intensity* of the incident light (left)
- ii. *Number of electrons* ejected versus the *frequency* of the incident light (right)



**Question 3** **6 points****a. 2 points**

In the ground state of Ge (Germanium; $Z = 32$) atom, **name** the following orbitals:

i. Orbital containing the electron that requires the maximum energy to be removed.

ii. Orbital containing the electron that experiences the minimum effective nuclear charge.

b. 2 points

An unknown anion (A^-) has the following *excited* electronic configuration: $[Ar] 4s^2 3d^3 4p^1$

(Ar: Argon; $Z = 18$)

Give the **condensed ground state electronic configuration** of the atom, **A** and the cation, **A^{2+}** .
Show your work. (Assume all electronic transitions are possible)

Electronic Configuration of A:

Electronic Configuration of A^{2+} :

c. 2 points

In a single atom, determine how many electrons can have the following set of quantum numbers?

(i) $n = 5; l = 3$

In a single atom, determine how many electrons can have the following set of quantum numbers?

(i) $n = 3; l = 2; m_s = +1/2$

**Question 4** **8 points**

Rank the following from 1 to 4 in **increasing** order. Provide the ranking in the boxes provided.

Note: Atomic number (Z) are provided for relevant atoms

a) Increasing (lowest to highest) order of *ionization energy*: (1= lowest; 4 = highest)

Mg

Mg: Z = 12

Rb

Rb: Z = 37

Cs

Cs: Z = 55

K

K : Z = 19

b) Increasing (lowest to highest) order of *bond angle*: (1= lowest; 4 = highest)

H₂OCCl₄XeF₄

HCN

c) Increasing (smallest to largest) order of *ionic size* (1= smallest; 4 = largest)

S²⁻

S: Z = 16

O²⁻

O: Z = 8

Be²⁺

Be: Z = 4

Na⁺

Na: Z = 11

d) Increasing (lowest to highest) order of *bond energy* (1= lowest; 4 = highest)

P-F

P-Cl

P-I

P-Br

**Question 5** **12 points**

a. 9 points For the molecules HClO_4 and TeF_6 , draw the most stable *Lewis structure* showing appropriate shape and *dashed/wedged bonds*, based on VSEPR. Determine the **molecular geometry** around the central atom. For HClO_4 , label the *most polar bond*.



b. 3 points (Note: Lone pairs are not shown in structures below)

(i) For the three *resonance* structures (1, 2 and 3) shown below, answer the following question:



(I) Circle the *most contributing* resonance structure.

Structure 1

Structure 2

Structure 3

(ii) For the three structures given (4, 5, and 6) below, answer the following questions:



(I) Circle all the structures that represent *resonance structures* of each other:

Structure 4

Structure 5

Structure 6

(II) Among the resonance structures, circle the *most contributing* resonance structure.

Structure 4

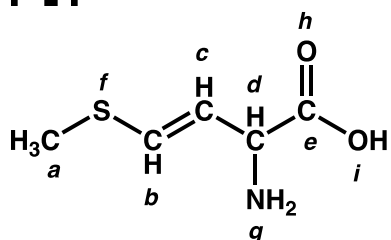
Structure 5

Structure 6

**Question 6** **6 points**

Using the data given below, calculate the **lattice energy** for SrO(s).

	ΔH° (kJ/mol)
$\text{Sr}(s) \longrightarrow \text{Sr}(g)$	+180
$\text{Sr}(g) \longrightarrow \text{Sr}^{2+}(g) + 2e^-$	+1468
$\text{O}_2(g) \longrightarrow 2\text{O}(g)$	+500
$\text{O}(g) + 2e^- \longrightarrow \text{O}^{2-}(g)$	+600
$\text{Sr}(s) + \frac{1}{2}\text{O}_2(g) \longrightarrow \text{SrO}(s)$	-550

**Question 7 10 points**

Consider the molecule given on the left, and answer questions (i) to (vi) using chemical bonding theories that we have learnt in class (valence bond theory, hybridization).

Note that the lone pairs are not shown in the structure:

i. Consider the carbon atoms marked in as ***a***, ***b***, ***c***, ***d***, and ***e*** in the molecule, and determine **hybridization** at each of the carbon centers. (2.5 points)

Atom	Hybridization
Carbon (<i>a</i>)	
Carbon (<i>b</i>)	
Carbon (<i>c</i>)	
Carbon (<i>d</i>)	
Carbon (<i>e</i>)	

ii. Now, consider the sulfur atom (S) marked as ***f***, nitrogen (N) atom (***g***), and oxygen(O) atoms (***h*** and ***i***), and determine their **hybridization**. (2 points)

Atom	Hybridization
Sulfur (<i>f</i>)	
Nitrogen (<i>g</i>)	
Oxygen (<i>h</i>)	
Oxygen (<i>i</i>)	

iii. **Describe** all the bonds formed by the carbon atoms ***a*** and ***b***, using sigma and pi designation as appropriate. (2 points)

Atom	Bonds formed (use sigma and pi as appropriate)
Carbon (<i>a</i>)	
Carbon (<i>b</i>)	

iv. How many total sp^3 -s sigma bonds are in the molecule? (1 point)

v. Determine **electronic** and **molecular geometry** at carbon (C) labeled ***e*** and sulfur (S) labeled ***f*** centres. (2 points)

Atom	Electronic Geometry	Molecular Geometry
Carbon (<i>e</i>)		
Sulfur (<i>f</i>)		

vi. Circle the value from the choices given below, that describes the **C-S-C bond angle at sulfur** (S) marked ***f***: (0.5 points)

>90° to <109.5° 109.5° >109.5 to <120° 120° >120° to <180°

**Question 8 12 points**

Using the **valence shell orbitals only** draw a **complete** molecular orbital (MO) diagram (with no s-p mixing) for the molecule **PS**, in the space provided below. **(6 points)**

Assume that 3s and 3p atomic orbitals will form molecular orbitals like those formed by the combination of 2s and 2p atomic orbitals that we learnt in class, and there is no s-p mixing.

Show **energy arrow**, **atomic orbitals** involved, **atom labels**, the **molecular orbitals** that are formed, **filled electrons** in the MO diagram etc. Make sure to label all the atomic and molecular orbitals.

Using the MO diagram above, answer the following questions:

- (i) Which one of the following will have the highest bond order? Circle the correct choice
A) PS^{1-} B) PS C) PS^{1+}
- (ii) Which one of the following will have the longest bond length? Circle the correct choice
A) PS^{1-} B) PS C) PS^{1+}
- (iii) Which of the following will be diamagnetic? Circle the correct choice.
A) PS^{1-} B) PS C) PS^{1+}
- (iv) Draw Lewis dot structure of the molecule with the shortest bond length (based on the MO diagram) from among the three (PS^{1-} , PS, PS^{1+}), and indicate the bond order calculated from the Lewis dot structure. **(3 points)**

Lewis Dot Structure

Bond Order



560ABAA3-F55F-448B-B8BE-6B27B06F1BD9

midterm-exam-93388

#46 Page 10 of 10

SCRAP PAGE

**CHEMISTRY 110 –Midterm Exam****Thursday November 7, 2024, 6:30 – 8:30 pm**

Examiners: Pallavi Sirjoosingh and Ashok Kakkar

First name (please write as legibly as possible within the boxes)

--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--

Last name

--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--

Student ID number

--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--

This booklet has 8 questions on 10 pages (The last page can be used as a scrap page).

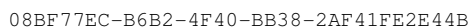
The staple must NOT be removed from the question booklet. Your intact question booklet must be returned at the end of the exam.

A separate booklet contains a periodic table, and datasheet.

Write your answers directly under the questions in the space provided. Do not write on the QR codes or near the top edges (as it may be cut off).

Please write **legibly and dark enough** for the scan to be readable. For students who use English as a second language, dictionaries are permitted. Calculators are allowed.

DO NOT ASK QUESTIONS OF THE INVIGILATORS about interpretations of the exam questions. If you have a concern about a question, please summarize it at the top of this page. All concerns will be evaluated after the exam. GOOD LUCK!



midterm-exam-93388

#47 Page 2 of 10

a. 6 points Calculate the **energy (in kJ/mol)** required to ionize (completely remove) the electron from the **n=3** level of **He⁺ ion** (He: Helium). Show your work including equations. (3 sig. figs.)

(I) Select the largest (longest) wavelength from the three choices below:

- Wavelength of photon to *completely remove* an electron from *1s orbital of Hydrogen atom*
- Wavelength of photon to *completely remove* an electron from *1s orbital of Boron atom (B)*
- Wavelength of photon to *completely remove* an electron from *1s orbital of Boron cation (B⁺)*

i or ii or iii

(II) Select the smallest (shortest) wavelength from the three choices below (*for H atom*):

- Wavelength of photon *emitted* for an electron transitioning from $2p$ to $1s$ orbital
- Wavelength of photon *absorbed* for an electron transitioning from $2s$ to $4p$ orbital
- Wavelength of photon *absorbed* for an electron transitioning from $1s$ to $3p$ orbital

i **or** **ii** **or** **iii**



Question 2 8 points



A Chem110 student plots the graph shown on the left, from data obtained for the kinetic energy (eV) of the ejected electron versus frequency (s^{-1}) of the incident light, for an unknown metal. ($1\text{eV} = 1.6022 \times 10^{-19} \text{ J}$)

Based on the plot, answer the questions **a to c** below:

a. 1 point On the plot above, label the **threshold frequency**.

b. 2 points What is the value of the work potential (in J) of the unknown metal?

c. 3 points What is the *longest* wavelength (in nm) required to eject an electron from this metal's surface? Show your work including equation(s) used. (3 sig figs)

d. 2 points In the space provided below, plot the following

(Assume frequency is *above* threshold frequency):

- i. *Kinetic energy* of an ejected electron versus the *intensity* of the incident light (left)
- ii. *Number of electrons* ejected versus the *frequency* of the incident light (right)



**Question 3** **6 points****a. 2 points**

In the ground state of Ge (Germanium; $Z = 32$) atom, **name** the following orbitals:

i. Orbital containing the electron that requires the maximum energy to be removed.

ii. Orbital containing the electron that experiences the minimum effective nuclear charge.

b. 2 points

An unknown anion (A^-) has the following *excited* electronic configuration: $[Ar] 4s^2 3d^3 4p^1$

(Ar: Argon; $Z = 18$)

Give the **condensed ground state electronic configuration** of the atom, **A** and the cation, **A^{2+}** .
Show your work. (Assume all electronic transitions are possible)

Electronic Configuration of A:

Electronic Configuration of A^{2+} :

c. 2 points

In a single atom, determine how many electrons can have the following set of quantum numbers?

(i) $n = 5; l = 3$

In a single atom, determine how many electrons can have the following set of quantum numbers?

(i) $n = 3; l = 2; m_s = +1/2$

**Question 4** **8 points**

Rank the following from 1 to 4 in **increasing** order. Provide the ranking in the boxes provided.

Note: Atomic number (Z) are provided for relevant atoms

a) Increasing (lowest to highest) order of *ionization energy*: (1= lowest; 4 = highest)

Mg

Mg: Z = 12

Rb

Rb: Z = 37

Cs

Cs: Z = 55

K

K : Z = 19

b) Increasing (lowest to highest) order of *bond angle*: (1= lowest; 4 = highest)

H₂OCCl₄XeF₄

HCN

c) Increasing (smallest to largest) order of *ionic size* (1= smallest; 4 = largest)

S²⁻

S: Z = 16

O²⁻

O: Z = 8

Be²⁺

Be: Z = 4

Na⁺

Na: Z = 11

d) Increasing (lowest to highest) order of *bond energy* (1= lowest; 4 = highest)

P-F

P-Cl

P-I

P-Br

**Question 5** **12 points**

a. 9 points For the molecules HClO_4 and TeF_6 , draw the most stable *Lewis structure* showing appropriate shape and *dashed/wedged bonds*, based on VSEPR. Determine the **molecular geometry** around the central atom. For HClO_4 , label the *most polar bond*.



b. 3 points (Note: Lone pairs are not shown in structures below)

(i) For the three *resonance* structures (1, 2 and 3) shown below, answer the following question:



(I) Circle the *most contributing* resonance structure.

Structure 1

Structure 2

Structure 3

(ii) For the three structures given (4, 5, and 6) below, answer the following questions:



(I) Circle all the structures that represent *resonance structures* of each other:

Structure 4

Structure 5

Structure 6

(II) Among the resonance structures, circle the *most contributing* resonance structure.

Structure 4

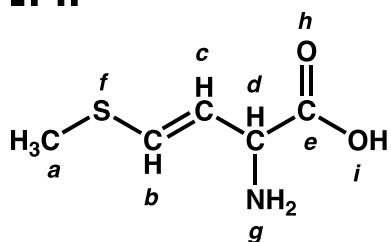
Structure 5

Structure 6

**Question 6** **6 points**

Using the data given below, calculate the **lattice energy** for SrO(s).

	ΔH° (kJ/mol)
$\text{Sr}(s) \longrightarrow \text{Sr}(g)$	+180
$\text{Sr}(g) \longrightarrow \text{Sr}^{2+}(g) + 2e^-$	+1468
$\text{O}_2(g) \longrightarrow 2\text{O}(g)$	+500
$\text{O}(g) + 2e^- \longrightarrow \text{O}^{2-}(g)$	+600
$\text{Sr}(s) + \frac{1}{2}\text{O}_2(g) \longrightarrow \text{SrO}(s)$	-550

**Question 7 10 points**

Consider the molecule given on the left, and answer questions (i) to (vi) using chemical bonding theories that we have learnt in class (valence bond theory, hybridization).

Note that the lone pairs are not shown in the structure:

i. Consider the carbon atoms marked in as ***a***, ***b***, ***c***, ***d***, and ***e*** in the molecule, and determine **hybridization** at each of the carbon centers. (2.5 points)

Atom	Hybridization
Carbon (<i>a</i>)	
Carbon (<i>b</i>)	
Carbon (<i>c</i>)	
Carbon (<i>d</i>)	
Carbon (<i>e</i>)	

ii. Now, consider the sulfur atom (S) marked as ***f***, nitrogen (N) atom (***g***), and oxygen(O) atoms (***h*** and ***i***), and determine their **hybridization**. (2 points)

Atom	Hybridization
Sulfur (<i>f</i>)	
Nitrogen (<i>g</i>)	
Oxygen (<i>h</i>)	
Oxygen (<i>i</i>)	

iii. **Describe** all the bonds formed by the carbon atoms ***a*** and ***b***, using sigma and pi designation as appropriate. (2 points)

Atom	Bonds formed (use sigma and pi as appropriate)
Carbon (<i>a</i>)	
Carbon (<i>b</i>)	

iv. How many total sp^3 -s sigma bonds are in the molecule? (1 point)

v. Determine **electronic** and **molecular geometry** at carbon (C) labeled ***e*** and sulfur (S) labeled ***f*** centres. (2 points)

Atom	Electronic Geometry	Molecular Geometry
Carbon (<i>e</i>)		
Sulfur (<i>f</i>)		

vi. Circle the value from the choices given below, that describes the **C-S-C bond angle at sulfur** (S) marked ***f***: (0.5 points)

>90° to <109.5° 109.5° >109.5 to <120° 120° >120° to <180°



Question 8 12 points

Using the **valence shell orbitals only** draw a **complete** molecular orbital (MO) diagram (with no s-p mixing) for the molecule **PS**, in the space provided below. **(6 points)**

Assume that 3s and 3p atomic orbitals will form molecular orbitals like those formed by the combination of 2s and 2p atomic orbitals that we learnt in class, and there is no s-p mixing.

Show **energy arrow**, **atomic orbitals** involved, **atom labels**, the **molecular orbitals** that are formed, **filled electrons** in the MO diagram etc. Make sure to label all the atomic and molecular orbitals.

Using the MO diagram above, answer the following questions:

- (i) Which one of the following will have the highest bond order? Circle the correct choice
 A) PS^{1-} B) PS C) PS^{1+}
- (ii) Which one of the following will have the longest bond length? Circle the correct choice
 A) PS^{1-} B) PS C) PS^{1+}
- (iii) Which of the following will be diamagnetic? Circle the correct choice.
 A) PS^{1-} B) PS C) PS^{1+}
- (iv) Draw Lewis dot structure of the molecule with the shortest bond length (based on the MO diagram) from among the three (PS^{1-} , PS, PS^{1+}), and indicate the bond order calculated from the Lewis dot structure. **(3 points)**

Lewis Dot Structure

Bond Order



990E3577-6A97-4EBF-AEE3-E071A660E62E

midterm-exam-93388

#47 Page 10 of 10

SCRAP PAGE

**CHEMISTRY 110 –Midterm Exam****Thursday November 7, 2024, 6:30 – 8:30 pm**

Examiners: Pallavi Sirjoosingh and Ashok Kakkar

First name (please write as legibly as possible within the boxes)

--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--

Last name

--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--

Student ID number

--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--

This booklet has 8 questions on 10 pages (The last page can be used as a scrap page).

The staple must NOT be removed from the question booklet. Your intact question booklet must be returned at the end of the exam.

A separate booklet contains a periodic table, and datasheet.

Write your answers directly under the questions in the space provided. Do not write on the QR codes or near the top edges (as it may be cut off).

Please write **legibly and dark enough** for the scan to be readable. For students who use English as a second language, dictionaries are permitted. Calculators are allowed.

DO NOT ASK QUESTIONS OF THE INVIGILATORS about interpretations of the exam questions. If you have a concern about a question, please summarize it at the top of this page. All concerns will be evaluated after the exam. GOOD LUCK!

**Question 1 8 points** (*Assume all transitions are possible*)

a. 6 points Calculate the **energy (in kJ/mol)** required to ionize (completely remove) the electron from the **n=3** level of **He⁺ ion** (He: Helium). Show your work including equations. (3 sig. figs.)

b. 2 points

(I) Select the largest (longest) wavelength from the three choices below:

- i. Wavelength of photon to *completely remove* an electron from *1s orbital of Hydrogen atom*
- ii. Wavelength of photon to *completely remove* an electron from *1s orbital of Boron atom (B)*
- iii. Wavelength of photon to *completely remove* an electron from *1s orbital of Boron cation (B⁺)*

i or ii or iii

(II) Select the smallest (shortest) wavelength from the three choices below (*for H atom*):

- i. Wavelength of photon *emitted* for an electron transitioning from *2p to 1s orbital*
- ii. Wavelength of photon *absorbed* for an electron transitioning from *2s to 4p orbital*
- iii. Wavelength of photon *absorbed* for an electron transitioning from *1s to 3p orbital*

i or ii or iii



Question 2 8 points



A Chem110 student plots the graph shown on the left, from data obtained for the kinetic energy (eV) of the ejected electron versus frequency (s^{-1}) of the incident light, for an unknown metal. ($1\text{eV} = 1.6022 \times 10^{-19} \text{ J}$)

Based on the plot, answer the questions **a to c** below:

a. 1 point On the plot above, label the **threshold frequency**.

b. 2 points What is the value of the work potential (in J) of the unknown metal?

c. 3 points What is the *longest* wavelength (in nm) required to eject an electron from this metal's surface? Show your work including equation(s) used. (3 sig figs)

d. 2 points In the space provided below, plot the following

(Assume frequency is *above* threshold frequency):

- Kinetic energy* of an ejected electron versus the *intensity* of the incident light (left)
- Number of electrons* ejected versus the *frequency* of the incident light (right)



**Question 3** **6 points****a. 2 points**

In the ground state of Ge (Germanium; $Z = 32$) atom, **name** the following orbitals:

i. Orbital containing the electron that requires the maximum energy to be removed.

ii. Orbital containing the electron that experiences the minimum effective nuclear charge.

b. 2 points

An unknown anion (A^-) has the following *excited* electronic configuration: $[Ar] 4s^2 3d^3 4p^1$

(Ar: Argon; $Z = 18$)

Give the **condensed ground state electronic configuration** of the atom, **A** and the cation, **A^{2+}** .
Show your work. (Assume all electronic transitions are possible)

Electronic Configuration of A:

Electronic Configuration of A^{2+} :

c. 2 points

In a single atom, determine how many electrons can have the following set of quantum numbers?

(i) $n = 5; l = 3$

In a single atom, determine how many electrons can have the following set of quantum numbers?

(i) $n = 3; l = 2; m_s = +1/2$

**Question 4** **8 points**

Rank the following from 1 to 4 in **increasing** order. Provide the ranking in the boxes provided.

Note: Atomic number (Z) are provided for relevant atoms

a) Increasing (lowest to highest) order of *ionization energy*: (1= lowest; 4 = highest)

Mg

Mg: Z = 12

Rb

Rb: Z = 37

Cs

Cs: Z = 55

K

K : Z = 19

b) Increasing (lowest to highest) order of *bond angle*: (1= lowest; 4 = highest)

H₂OCCl₄XeF₄

HCN

c) Increasing (smallest to largest) order of *ionic size* (1= smallest; 4 = largest)

S²⁻

S: Z = 16

O²⁻

O: Z = 8

Be²⁺

Be: Z = 4

Na⁺

Na: Z = 11

d) Increasing (lowest to highest) order of *bond energy* (1= lowest; 4 = highest)

P-F

P-Cl

P-I

P-Br

**Question 5** **12 points**

a. 9 points For the molecules HClO_4 and TeF_6 , draw the most stable *Lewis structure* showing appropriate shape and *dashed/wedged bonds*, based on VSEPR. Determine the **molecular geometry** around the central atom. For HClO_4 , label the *most polar bond*.



b. 3 points (Note: Lone pairs are not shown in structures below)

(i) For the three *resonance* structures (1, 2 and 3) shown below, answer the following question:



(I) Circle the *most contributing* resonance structure.

Structure 1

Structure 2

Structure 3

(ii) For the three structures given (4, 5, and 6) below, answer the following questions:



(I) Circle all the structures that represent *resonance structures* of each other:

Structure 4

Structure 5

Structure 6

(II) Among the resonance structures, circle the *most contributing* resonance structure.

Structure 4

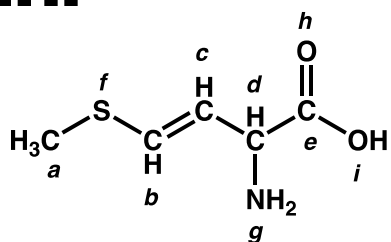
Structure 5

Structure 6

**Question 6** **6 points**

Using the data given below, calculate the **lattice energy** for SrO(s).

	ΔH° (kJ/mol)
$\text{Sr}(s) \longrightarrow \text{Sr}(g)$	+180
$\text{Sr}(g) \longrightarrow \text{Sr}^{2+}(g) + 2e^-$	+1468
$\text{O}_2(g) \longrightarrow 2\text{O}(g)$	+500
$\text{O}(g) + 2e^- \longrightarrow \text{O}^{2-}(g)$	+600
$\text{Sr}(s) + \frac{1}{2}\text{O}_2(g) \longrightarrow \text{SrO}(s)$	-550

**Question 7 10 points**

Consider the molecule given on the left, and answer questions (i) to (vi) using chemical bonding theories that we have learnt in class (valence bond theory, hybridization).

Note that the lone pairs are not shown in the structure:

i. Consider the carbon atoms marked in as ***a***, ***b***, ***c***, ***d***, and ***e*** in the molecule, and determine **hybridization** at each of the carbon centers. (2.5 points)

Atom	Hybridization
Carbon (<i>a</i>)	
Carbon (<i>b</i>)	
Carbon (<i>c</i>)	
Carbon (<i>d</i>)	
Carbon (<i>e</i>)	

ii. Now, consider the sulfur atom (S) marked as ***f***, nitrogen (N) atom (***g***), and oxygen(O) atoms (***h*** and ***i***), and determine their **hybridization**. (2 points)

Atom	Hybridization
Sulfur (<i>f</i>)	
Nitrogen (<i>g</i>)	
Oxygen (<i>h</i>)	
Oxygen (<i>i</i>)	

iii. **Describe** all the bonds formed by the carbon atoms ***a*** and ***b***, using sigma and pi designation as appropriate. (2 points)

Atom	Bonds formed (use sigma and pi as appropriate)
Carbon (<i>a</i>)	
Carbon (<i>b</i>)	

iv. How many total sp^3 -s sigma bonds are in the molecule? (1 point)

v. Determine **electronic** and **molecular geometry** at carbon (C) labeled ***e*** and sulfur (S) labeled ***f*** centres. (2 points)

Atom	Electronic Geometry	Molecular Geometry
Carbon (<i>e</i>)		
Sulfur (<i>f</i>)		

vi. Circle the value from the choices given below, that describes the **C-S-C bond angle at sulfur** (S) marked ***f***: (0.5 points)

>90° to <109.5° 109.5° >109.5 to <120° 120° >120° to <180°


Question 8 12 points

Using the **valence shell orbitals only** draw a **complete** molecular orbital (MO) diagram (with no s-p mixing) for the molecule **PS**, in the space provided below. **(6 points)**

Assume that 3s and 3p atomic orbitals will form molecular orbitals like those formed by the combination of 2s and 2p atomic orbitals that we learnt in class, and there is no s-p mixing.

Show **energy arrow**, **atomic orbitals** involved, **atom labels**, the **molecular orbitals** that are formed, **filled electrons** in the MO diagram etc. Make sure to label all the atomic and molecular orbitals.

Using the MO diagram above, answer the following questions:

- (i) Which one of the following will have the highest bond order? Circle the correct choice
 A) PS^{1-} B) PS C) PS^{1+}
- (ii) Which one of the following will have the longest bond length? Circle the correct choice
 A) PS^{1-} B) PS C) PS^{1+}
- (iii) Which of the following will be diamagnetic? Circle the correct choice.
 A) PS^{1-} B) PS C) PS^{1+}
- (iv) Draw Lewis dot structure of the molecule with the shortest bond length (based on the MO diagram) from among the three (PS^{1-} , PS, PS^{1+}), and indicate the bond order calculated from the Lewis dot structure. **(3 points)**

Lewis Dot Structure

Bond Order



A44663DC-3E40-4C0D-B67A-E1A192382B11

midterm-exam-93388

#48 Page 10 of 10

SCRAP PAGE

**CHEMISTRY 110 –Midterm Exam****Thursday November 7, 2024, 6:30 – 8:30 pm****Examiners: Pallavi Sirjoosingh and Ashok Kakkar**

First name (please write as legibly as possible within the boxes)

--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--

Last name

--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--

Student ID number

--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--

This booklet has 8 questions on 10 pages (The last page can be used as a scrap page).

The staple must NOT be removed from the question booklet. Your intact question booklet must be returned at the end of the exam.

A separate booklet contains a periodic table, and datasheet.

Write your answers directly under the questions in the space provided. Do not write on the QR codes or near the top edges (as it may be cut off).

Please write **legibly and dark enough** for the scan to be readable. For students who use English as a second language, dictionaries are permitted. Calculators are allowed.

DO NOT ASK QUESTIONS OF THE INVIGILATORS about interpretations of the exam questions. If you have a concern about a question, please summarize it at the top of this page. All concerns will be evaluated after the exam. GOOD LUCK!

**Question 1 8 points** (*Assume all transitions are possible*)

a. 6 points Calculate the **energy (in kJ/mol)** required to ionize (completely remove) the electron from the **n=3** level of **He⁺ ion** (He: Helium). Show your work including equations. (3 sig. figs.)

b. 2 points

(I) Select the largest (longest) wavelength from the three choices below:

- i. Wavelength of photon to *completely remove* an electron from *1s orbital of Hydrogen atom*
- ii. Wavelength of photon to *completely remove* an electron from *1s orbital of Boron atom (B)*
- iii. Wavelength of photon to *completely remove* an electron from *1s orbital of Boron cation (B⁺)*

i or ii or iii

(II) Select the smallest (shortest) wavelength from the three choices below (*for H atom*):

- i. Wavelength of photon *emitted* for an electron transitioning from *2p to 1s orbital*
- ii. Wavelength of photon *absorbed* for an electron transitioning from *2s to 4p orbital*
- iii. Wavelength of photon *absorbed* for an electron transitioning from *1s to 3p orbital*

i or ii or iii



Question 2 8 points



A Chem110 student plots the graph shown on the left, from data obtained for the kinetic energy (eV) of the ejected electron versus frequency (s^{-1}) of the incident light, for an unknown metal. ($1\text{eV} = 1.6022 \times 10^{-19} \text{ J}$)

Based on the plot, answer the questions **a to c** below:

a. 1 point On the plot above, label the **threshold frequency**.

b. 2 points What is the value of the work potential (in J) of the unknown metal?

c. 3 points What is the *longest* wavelength (in nm) required to eject an electron from this metal's surface? Show your work including equation(s) used. (3 sig figs)

d. 2 points In the space provided below, plot the following

(Assume frequency is *above* threshold frequency):

- i. *Kinetic energy* of an ejected electron versus the *intensity* of the incident light (left)
- ii. *Number of electrons* ejected versus the *frequency* of the incident light (right)



**Question 3** **6 points****a. 2 points**

In the ground state of Ge (Germanium; $Z = 32$) atom, **name** the following orbitals:

i. Orbital containing the electron that requires the maximum energy to be removed.

ii. Orbital containing the electron that experiences the minimum effective nuclear charge.

b. 2 points

An unknown anion (A^-) has the following *excited* electronic configuration: $[Ar] 4s^2 3d^3 4p^1$

(Ar: Argon; $Z = 18$)

Give the **condensed ground state electronic configuration** of the atom, **A** and the cation, **A^{2+}** .
Show your work. (Assume all electronic transitions are possible)

Electronic Configuration of A:

Electronic Configuration of A^{2+} :

c. 2 points

In a single atom, determine how many electrons can have the following set of quantum numbers?

(i) $n = 5; l = 3$

In a single atom, determine how many electrons can have the following set of quantum numbers?

(i) $n = 3; l = 2; m_s = +1/2$

**Question 4** **8 points**

Rank the following from 1 to 4 in **increasing** order. Provide the ranking in the boxes provided.

Note: Atomic number (Z) are provided for relevant atoms

a) Increasing (lowest to highest) order of *ionization energy*: (1= lowest; 4 = highest)

Mg

Mg: Z = 12

Rb

Rb: Z = 37

Cs

Cs: Z = 55

K

K : Z = 19

b) Increasing (lowest to highest) order of *bond angle*: (1= lowest; 4 = highest)

H₂OCCl₄XeF₄

HCN

c) Increasing (smallest to largest) order of *ionic size* (1= smallest; 4 = largest)

S²⁻

S: Z = 16

O²⁻

O: Z = 8

Be²⁺

Be: Z = 4

Na⁺

Na: Z = 11

d) Increasing (lowest to highest) order of *bond energy* (1= lowest; 4 = highest)

P-F

P-Cl

P-I

P-Br

**Question 5** **12 points**

a. 9 points For the molecules HClO_4 and TeF_6 , draw the most stable *Lewis structure* showing appropriate shape and *dashed/wedged bonds*, based on VSEPR. Determine the **molecular geometry** around the central atom. For HClO_4 , label the *most polar bond*.

HClO_4 (there are no O-O bonds)

Label the most polar bond

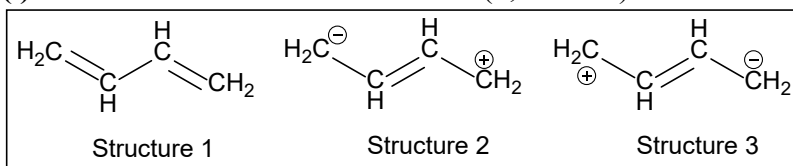
Molecular Geometry around Cl

TeF_6

Molecular Geometry around Te

b. 3 points (Note: Lone pairs are not shown in structures below)

(i) For the three *resonance* structures (1, 2 and 3) shown below, answer the following question:



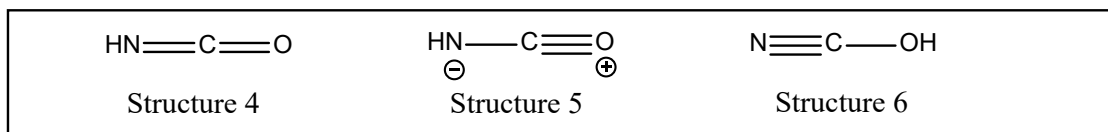
(I) Circle the *most contributing* resonance structure.

Structure 1

Structure 2

Structure 3

(ii) For the three structures given (4, 5, and 6) below, answer the following questions:



(I) Circle all the structures that represent *resonance structures* of each other:

Structure 4

Structure 5

Structure 6

(II) Among the resonance structures, circle the *most contributing* resonance structure.

Structure 4

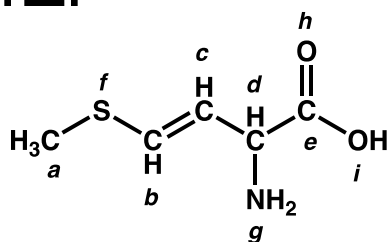
Structure 5

Structure 6

**Question 6** **6 points**

Using the data given below, calculate the **lattice energy** for SrO(s).

	ΔH° (kJ/mol)
$\text{Sr}(s) \longrightarrow \text{Sr}(g)$	+180
$\text{Sr}(g) \longrightarrow \text{Sr}^{2+}(g) + 2e^-$	+1468
$\text{O}_2(g) \longrightarrow 2\text{O}(g)$	+500
$\text{O}(g) + 2e^- \longrightarrow \text{O}^{2-}(g)$	+600
$\text{Sr}(s) + \frac{1}{2}\text{O}_2(g) \longrightarrow \text{SrO}(s)$	-550

**Question 7 10 points**

Consider the molecule given on the left, and answer questions (i) to (vi) using chemical bonding theories that we have learnt in class (valence bond theory, hybridization).

Note that the lone pairs are not shown in the structure:

i. Consider the carbon atoms marked in as ***a***, ***b***, ***c***, ***d***, and ***e*** in the molecule, and determine **hybridization** at each of the carbon centers. (2.5 points)

Atom	Hybridization
Carbon (<i>a</i>)	
Carbon (<i>b</i>)	
Carbon (<i>c</i>)	
Carbon (<i>d</i>)	
Carbon (<i>e</i>)	

ii. Now, consider the sulfur atom (S) marked as ***f***, nitrogen (N) atom (***g***), and oxygen(O) atoms (***h*** and ***i***), and determine their **hybridization**. (2 points)

Atom	Hybridization
Sulfur (<i>f</i>)	
Nitrogen (<i>g</i>)	
Oxygen (<i>h</i>)	
Oxygen (<i>i</i>)	

iii. **Describe** all the bonds formed by the carbon atoms ***a*** and ***b***, using sigma and pi designation as appropriate. (2 points)

Atom	Bonds formed (use sigma and pi as appropriate)
Carbon (<i>a</i>)	
Carbon (<i>b</i>)	

iv. How many total sp^3 -s sigma bonds are in the molecule? (1 point)

v. Determine **electronic** and **molecular geometry** at carbon (C) labeled ***e*** and sulfur (S) labeled ***f*** centres. (2 points)

Atom	Electronic Geometry	Molecular Geometry
Carbon (<i>e</i>)		
Sulfur (<i>f</i>)		

vi. Circle the value from the choices given below, that describes the **C-S-C bond angle at sulfur** (S) marked ***f***: (0.5 points)

>90° to <109.5°

109.5°

>109.5 to <120°

120°

>120° to <180°


Question 8 12 points

Using the **valence shell orbitals only** draw a **complete** molecular orbital (MO) diagram (with no s-p mixing) for the molecule **PS**, in the space provided below. **(6 points)**

Assume that 3s and 3p atomic orbitals will form molecular orbitals like those formed by the combination of 2s and 2p atomic orbitals that we learnt in class, and there is no s-p mixing.

Show **energy arrow**, **atomic orbitals** involved, **atom labels**, the **molecular orbitals** that are formed, **filled electrons** in the MO diagram etc. Make sure to label all the atomic and molecular orbitals.

Using the MO diagram above, answer the following questions:

- (i) Which one of the following will have the highest bond order? Circle the correct choice
 A) PS^{1-} B) PS C) PS^{1+}
- (ii) Which one of the following will have the longest bond length? Circle the correct choice
 A) PS^{1-} B) PS C) PS^{1+}
- (iii) Which of the following will be diamagnetic? Circle the correct choice.
 A) PS^{1-} B) PS C) PS^{1+}
- (iv) Draw Lewis dot structure of the molecule with the shortest bond length (based on the MO diagram) from among the three (PS^{1-} , PS, PS^{1+}), and indicate the bond order calculated from the Lewis dot structure. **(3 points)**

Lewis Dot Structure

Bond Order



076784CB-6118-4F07-BA9C-732045B14343

midterm-exam-93388

#49 Page 10 of 10

SCRAP PAGE

**CHEMISTRY 110 –Midterm Exam****Thursday November 7, 2024, 6:30 – 8:30 pm**

Examiners: Pallavi Sirjoosingh and Ashok Kakkar

First name (please write as legibly as possible within the boxes)

--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--

Last name

--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--

Student ID number

--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--

This booklet has 8 questions on 10 pages (The last page can be used as a scrap page).

The staple must NOT be removed from the question booklet. Your intact question booklet must be returned at the end of the exam.

A separate booklet contains a periodic table, and datasheet.

Write your answers directly under the questions in the space provided. Do not write on the QR codes or near the top edges (as it may be cut off).

Please write **legibly and dark enough** for the scan to be readable. For students who use English as a second language, dictionaries are permitted. Calculators are allowed.

DO NOT ASK QUESTIONS OF THE INVIGILATORS about interpretations of the exam questions. If you have a concern about a question, please summarize it at the top of this page. All concerns will be evaluated after the exam. GOOD LUCK!

**Question 1 8 points** (*Assume all transitions are possible*)

a. 6 points Calculate the **energy (in kJ/mol)** required to ionize (completely remove) the electron from the **n=3** level of **He⁺ ion** (He: Helium). Show your work including equations. (3 sig. figs.)

b. 2 points

(I) Select the largest (longest) wavelength from the three choices below:

- i. Wavelength of photon to *completely remove* an electron from *1s orbital of Hydrogen atom*
- ii. Wavelength of photon to *completely remove* an electron from *1s orbital of Boron atom (B)*
- iii. Wavelength of photon to *completely remove* an electron from *1s orbital of Boron cation (B⁺)*

i

or

ii

or

iii

(II) Select the smallest (shortest) wavelength from the three choices below (*for H atom*):

- i. Wavelength of photon *emitted* for an electron transitioning from *2p to 1s orbital*
- ii. Wavelength of photon *absorbed* for an electron transitioning from *2s to 4p orbital*
- iii. Wavelength of photon *absorbed* for an electron transitioning from *1s to 3p orbital*

i

or

ii

or

iii



Question 2 8 points



A Chem110 student plots the graph shown on the left, from data obtained for the kinetic energy (eV) of the ejected electron versus frequency (s^{-1}) of the incident light, for an unknown metal. ($1\text{eV} = 1.6022 \times 10^{-19} \text{ J}$)

Based on the plot, answer the questions **a to c** below:

a. 1 point On the plot above, label the **threshold frequency**.

b. 2 points What is the value of the work potential (in J) of the unknown metal?

c. 3 points What is the *longest* wavelength (in nm) required to eject an electron from this metal's surface? Show your work including equation(s) used. (3 sig figs)

d. 2 points In the space provided below, plot the following

(Assume frequency is *above* threshold frequency):

- i. *Kinetic energy* of an ejected electron versus the *intensity* of the incident light (left)
- ii. *Number of electrons* ejected versus the *frequency* of the incident light (right)



**Question 3** **6 points****a. 2 points**

In the ground state of Ge (Germanium; $Z = 32$) atom, **name** the following orbitals:

i. Orbital containing the electron that requires the maximum energy to be removed.

ii. Orbital containing the electron that experiences the minimum effective nuclear charge.

b. 2 points

An unknown anion (A^-) has the following *excited* electronic configuration: $[Ar] 4s^2 3d^3 4p^1$

(Ar: Argon; $Z = 18$)

Give the **condensed ground state electronic configuration** of the atom, **A** and the cation, **A^{2+}** .
Show your work. (Assume all electronic transitions are possible)

Electronic Configuration of A:

Electronic Configuration of A^{2+} :

c. 2 points

In a single atom, determine how many electrons can have the following set of quantum numbers?

(i) $n = 5; l = 3$

In a single atom, determine how many electrons can have the following set of quantum numbers?

(i) $n = 3; l = 2; m_s = +1/2$

**Question 4** **8 points**

Rank the following from 1 to 4 in **increasing** order. Provide the ranking in the boxes provided.

Note: Atomic number (Z) are provided for relevant atoms

a) Increasing (lowest to highest) order of *ionization energy*: (1= lowest; 4 = highest)

Mg

Mg: Z = 12

Rb

Rb: Z = 37

Cs

Cs: Z = 55

K

K : Z = 19

b) Increasing (lowest to highest) order of *bond angle*: (1= lowest; 4 = highest)

H₂OCCl₄XeF₄

HCN

c) Increasing (smallest to largest) order of *ionic size* (1= smallest; 4 = largest)

S²⁻

S: Z = 16

O²⁻

O: Z = 8

Be²⁺

Be: Z = 4

Na⁺

Na: Z = 11

d) Increasing (lowest to highest) order of *bond energy* (1= lowest; 4 = highest)

P-F

P-Cl

P-I

P-Br

**Question 5** **12 points**

a. 9 points For the molecules HClO_4 and TeF_6 , draw the most stable *Lewis structure* showing appropriate shape and *dashed/wedged bonds*, based on VSEPR. Determine the **molecular geometry** around the central atom. For HClO_4 , label the *most polar bond*.



b. 3 points (Note: Lone pairs are not shown in structures below)

(i) For the three *resonance* structures (1, 2 and 3) shown below, answer the following question:



(I) Circle the *most contributing* resonance structure.

Structure 1

Structure 2

Structure 3

(ii) For the three structures given (4, 5, and 6) below, answer the following questions:



(I) Circle all the structures that represent *resonance structures* of each other:

Structure 4

Structure 5

Structure 6

(II) Among the resonance structures, circle the *most contributing* resonance structure.

Structure 4

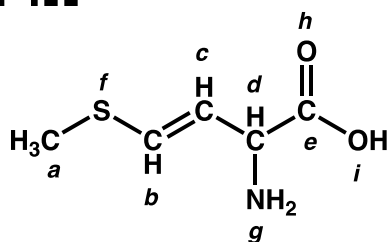
Structure 5

Structure 6

**Question 6** **6 points**

Using the data given below, calculate the **lattice energy** for SrO(s).

	ΔH° (kJ/mol)
$\text{Sr}(s) \longrightarrow \text{Sr}(g)$	+180
$\text{Sr}(g) \longrightarrow \text{Sr}^{2+}(g) + 2e^-$	+1468
$\text{O}_2(g) \longrightarrow 2\text{O}(g)$	+500
$\text{O}(g) + 2e^- \longrightarrow \text{O}^{2-}(g)$	+600
$\text{Sr}(s) + \frac{1}{2}\text{O}_2(g) \longrightarrow \text{SrO}(s)$	-550

**Question 7 10 points**

Consider the molecule given on the left, and answer questions (i) to (vi) using chemical bonding theories that we have learnt in class (valence bond theory, hybridization).

Note that the lone pairs are not shown in the structure:

i. Consider the carbon atoms marked in as ***a***, ***b***, ***c***, ***d***, and ***e*** in the molecule, and determine **hybridization** at each of the carbon centers. (2.5 points)

Atom	Hybridization
Carbon (<i>a</i>)	
Carbon (<i>b</i>)	
Carbon (<i>c</i>)	
Carbon (<i>d</i>)	
Carbon (<i>e</i>)	

ii. Now, consider the sulfur atom (S) marked as ***f***, nitrogen (N) atom (***g***), and oxygen(O) atoms (***h*** and ***i***), and determine their **hybridization**. (2 points)

Atom	Hybridization
Sulfur (<i>f</i>)	
Nitrogen (<i>g</i>)	
Oxygen (<i>h</i>)	
Oxygen (<i>i</i>)	

iii. **Describe** all the bonds formed by the carbon atoms ***a*** and ***b***, using sigma and pi designation as appropriate. (2 points)

Atom	Bonds formed (use sigma and pi as appropriate)
Carbon (<i>a</i>)	
Carbon (<i>b</i>)	

iv. How many total sp^3 -s sigma bonds are in the molecule? (1 point)

v. Determine **electronic** and **molecular geometry** at carbon (C) labeled ***e*** and sulfur (S) labeled ***f*** centres. (2 points)

Atom	Electronic Geometry	Molecular Geometry
Carbon (<i>e</i>)		
Sulfur (<i>f</i>)		

vi. Circle the value from the choices given below, that describes the **C-S-C bond angle at sulfur** (S) marked ***f***: (0.5 points)

>90° to <109.5° 109.5° >109.5 to <120° 120° >120° to <180°


Question 8 12 points

Using the **valence shell orbitals only** draw a **complete** molecular orbital (MO) diagram (with no s-p mixing) for the molecule **PS**, in the space provided below. **(6 points)**

Assume that 3s and 3p atomic orbitals will form molecular orbitals like those formed by the combination of 2s and 2p atomic orbitals that we learnt in class, and there is no s-p mixing.

Show **energy arrow**, **atomic orbitals** involved, **atom labels**, the **molecular orbitals** that are formed, **filled electrons** in the MO diagram etc. Make sure to label all the atomic and molecular orbitals.

Using the MO diagram above, answer the following questions:

- (i) Which one of the following will have the highest bond order? Circle the correct choice
 A) PS^{1-} B) PS C) PS^{1+}
- (ii) Which one of the following will have the longest bond length? Circle the correct choice
 A) PS^{1-} B) PS C) PS^{1+}
- (iii) Which of the following will be diamagnetic? Circle the correct choice.
 A) PS^{1-} B) PS C) PS^{1+}
- (iv) Draw Lewis dot structure of the molecule with the shortest bond length (based on the MO diagram) from among the three (PS^{1-} , PS, PS^{1+}), and indicate the bond order calculated from the Lewis dot structure. **(3 points)**

Lewis Dot Structure

Bond Order



EA312201-38C1-404F-B7DD-55E574378BA9

midterm-exam-93388

#50 Page 10 of 10

SCRAP PAGE

**CHEMISTRY 110 –Midterm Exam****Thursday November 7, 2024, 6:30 – 8:30 pm**

Examiners: Pallavi Sirjoosingh and Ashok Kakkar

First name (please write as legibly as possible within the boxes)

--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--

Last name

--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--

Student ID number

--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--

This booklet has 8 questions on 10 pages (The last page can be used as a scrap page).

The staple must NOT be removed from the question booklet. Your intact question booklet must be returned at the end of the exam.

A separate booklet contains a periodic table, and datasheet.

Write your answers directly under the questions in the space provided. Do not write on the QR codes or near the top edges (as it may be cut off).

Please write **legibly and dark enough** for the scan to be readable. For students who use English as a second language, dictionaries are permitted. Calculators are allowed.

DO NOT ASK QUESTIONS OF THE INVIGILATORS about interpretations of the exam questions. If you have a concern about a question, please summarize it at the top of this page. All concerns will be evaluated after the exam. GOOD LUCK!

**Question 1 8 points** (*Assume all transitions are possible*)

a. 6 points Calculate the **energy (in kJ/mol)** required to ionize (completely remove) the electron from the **n=3** level of **He⁺ ion** (He: Helium). Show your work including equations. (3 sig. figs.)

b. 2 points

(I) Select the largest (longest) wavelength from the three choices below:

- i. Wavelength of photon to *completely remove* an electron from *1s orbital of Hydrogen atom*
- ii. Wavelength of photon to *completely remove* an electron from *1s orbital of Boron atom (B)*
- iii. Wavelength of photon to *completely remove* an electron from *1s orbital of Boron cation (B⁺)*

i or ii or iii

(II) Select the smallest (shortest) wavelength from the three choices below (*for H atom*):

- i. Wavelength of photon *emitted* for an electron transitioning from *2p to 1s orbital*
- ii. Wavelength of photon *absorbed* for an electron transitioning from *2s to 4p orbital*
- iii. Wavelength of photon *absorbed* for an electron transitioning from *1s to 3p orbital*

i or ii or iii



Question 2 8 points



A Chem110 student plots the graph shown on the left, from data obtained for the kinetic energy (eV) of the ejected electron versus frequency (s^{-1}) of the incident light, for an unknown metal. ($1\text{eV} = 1.6022 \times 10^{-19} \text{ J}$)

Based on the plot, answer the questions **a to c** below:

a. 1 point On the plot above, label the **threshold frequency**.

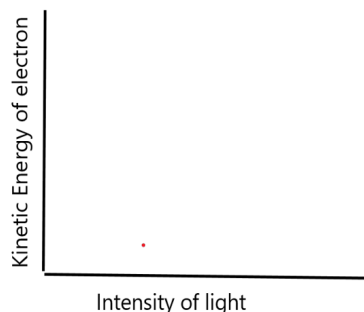
b. 2 points What is the value of the work potential (in J) of the unknown metal?

c. 3 points What is the *longest* wavelength (in nm) required to eject an electron from this metal's surface? Show your work including equation(s) used. (3 sig figs)

d. 2 points In the space provided below, plot the following

(Assume frequency is *above* threshold frequency):

- Kinetic energy* of an ejected electron versus the *intensity* of the incident light (left)
- Number of electrons* ejected versus the *frequency* of the incident light (right)



**Question 3** **6 points****a. 2 points**

In the ground state of Ge (Germanium; $Z = 32$) atom, **name** the following orbitals:

i. Orbital containing the electron that requires the maximum energy to be removed.

ii. Orbital containing the electron that experiences the minimum effective nuclear charge.

b. 2 points

An unknown anion (A^-) has the following *excited* electronic configuration: $[Ar] 4s^2 3d^3 4p^1$

(Ar: Argon; $Z = 18$)

Give the **condensed ground state electronic configuration** of the atom, **A** and the cation, **A^{2+}** .
Show your work. (Assume all electronic transitions are possible)

Electronic Configuration of A:

Electronic Configuration of A^{2+} :

c. 2 points

In a single atom, determine how many electrons can have the following set of quantum numbers?

(i) $n = 5; l = 3$

In a single atom, determine how many electrons can have the following set of quantum numbers?

(i) $n = 3; l = 2; m_s = +1/2$

**Question 4** **8 points**

Rank the following from 1 to 4 in **increasing** order. Provide the ranking in the boxes provided.

Note: Atomic number (Z) are provided for relevant atoms

a) Increasing (lowest to highest) order of *ionization energy*: (1= lowest; 4 = highest)

Mg

Mg: Z = 12

Rb

Rb: Z = 37

Cs

Cs: Z = 55

K

K : Z = 19

b) Increasing (lowest to highest) order of *bond angle*: (1= lowest; 4 = highest)

H₂OCCl₄XeF₄

HCN

c) Increasing (smallest to largest) order of *ionic size* (1= smallest; 4 = largest)

S²⁻

S: Z = 16

O²⁻

O: Z = 8

Be²⁺

Be: Z = 4

Na⁺

Na: Z = 11

d) Increasing (lowest to highest) order of *bond energy* (1= lowest; 4 = highest)

P-F

P-Cl

P-I

P-Br

**Question 5** **12 points**

a. 9 points For the molecules HClO_4 and TeF_6 , draw the most stable *Lewis structure* showing appropriate shape and *dashed/wedged bonds*, based on VSEPR. Determine the **molecular geometry** around the central atom. For HClO_4 , label the *most polar bond*.



b. 3 points (Note: Lone pairs are not shown in structures below)

(i) For the three *resonance* structures (1, 2 and 3) shown below, answer the following question:



(I) Circle the *most contributing* resonance structure.

Structure 1

Structure 2

Structure 3

(ii) For the three structures given (4, 5, and 6) below, answer the following questions:



(I) Circle all the structures that represent *resonance structures* of each other:

Structure 4

Structure 5

Structure 6

(II) Among the resonance structures, circle the *most contributing* resonance structure.

Structure 4

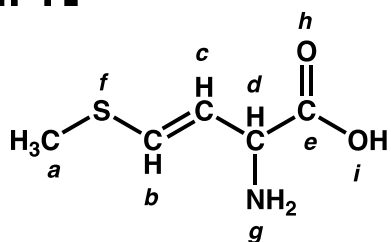
Structure 5

Structure 6

**Question 6** **6 points**

Using the data given below, calculate the **lattice energy** for SrO(s).

	ΔH° (kJ/mol)
$\text{Sr}(s) \longrightarrow \text{Sr}(g)$	+180
$\text{Sr}(g) \longrightarrow \text{Sr}^{2+}(g) + 2e^-$	+1468
$\text{O}_2(g) \longrightarrow 2\text{O}(g)$	+500
$\text{O}(g) + 2e^- \longrightarrow \text{O}^{2-}(g)$	+600
$\text{Sr}(s) + \frac{1}{2}\text{O}_2(g) \longrightarrow \text{SrO}(s)$	-550

**Question 7 10 points**

Consider the molecule given on the left, and answer questions (i) to (vi) using chemical bonding theories that we have learnt in class (valence bond theory, hybridization).

Note that the lone pairs are not shown in the structure:

i. Consider the carbon atoms marked in as ***a***, ***b***, ***c***, ***d***, and ***e*** in the molecule, and determine **hybridization** at each of the carbon centers. (2.5 points)

Atom	Hybridization
Carbon (<i>a</i>)	
Carbon (<i>b</i>)	
Carbon (<i>c</i>)	
Carbon (<i>d</i>)	
Carbon (<i>e</i>)	

ii. Now, consider the sulfur atom (S) marked as ***f***, nitrogen (N) atom (***g***), and oxygen(O) atoms (***h*** and ***i***), and determine their **hybridization**. (2 points)

Atom	Hybridization
Sulfur (<i>f</i>)	
Nitrogen (<i>g</i>)	
Oxygen (<i>h</i>)	
Oxygen (<i>i</i>)	

iii. **Describe** all the bonds formed by the carbon atoms ***a*** and ***b***, using sigma and pi designation as appropriate. (2 points)

Atom	Bonds formed (use sigma and pi as appropriate)
Carbon (<i>a</i>)	
Carbon (<i>b</i>)	

iv. How many total sp^3 -s sigma bonds are in the molecule? (1 point)

v. Determine **electronic** and **molecular geometry** at carbon (C) labeled ***e*** and sulfur (S) labeled ***f*** centres. (2 points)

Atom	Electronic Geometry	Molecular Geometry
Carbon (<i>e</i>)		
Sulfur (<i>f</i>)		

vi. Circle the value from the choices given below, that describes the **C-S-C bond angle at sulfur** (S) marked ***f***: (0.5 points)

>90° to <109.5° 109.5° >109.5 to <120° 120° >120° to <180°

**Question 8 12 points**

Using the **valence shell orbitals only** draw a **complete** molecular orbital (MO) diagram (with no s-p mixing) for the molecule **PS**, in the space provided below. **(6 points)**

Assume that 3s and 3p atomic orbitals will form molecular orbitals like those formed by the combination of 2s and 2p atomic orbitals that we learnt in class, and there is no s-p mixing.

Show **energy arrow**, **atomic orbitals** involved, **atom labels**, the **molecular orbitals** that are formed, **filled electrons** in the MO diagram etc. Make sure to label all the atomic and molecular orbitals.

Using the MO diagram above, answer the following questions:

- (i) Which one of the following will have the highest bond order? Circle the correct choice
A) PS^{1-} B) PS C) PS^{1+}
- (ii) Which one of the following will have the longest bond length? Circle the correct choice
A) PS^{1-} B) PS C) PS^{1+}
- (iii) Which of the following will be diamagnetic? Circle the correct choice.
A) PS^{1-} B) PS C) PS^{1+}
- (iv) Draw Lewis dot structure of the molecule with the shortest bond length (based on the MO diagram) from among the three (PS^{1-} , PS, PS^{1+}), and indicate the bond order calculated from the Lewis dot structure. **(3 points)**

Lewis Dot Structure

Bond Order



86932F12-B8E6-4E3D-8A08-F9CA07D7D4B0

midterm-exam-93388

#51 Page 10 of 10

SCRAP PAGE

**CHEMISTRY 110 –Midterm Exam****Thursday November 7, 2024, 6:30 – 8:30 pm****Examiners: Pallavi Sirjoosingh and Ashok Kakkar**

First name (please write as legibly as possible within the boxes)

--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--

Last name

--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--

Student ID number

--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--

This booklet has 8 questions on 10 pages (The last page can be used as a scrap page).

The staple must NOT be removed from the question booklet. Your intact question booklet must be returned at the end of the exam.

A separate booklet contains a periodic table, and datasheet.

Write your answers directly under the questions in the space provided. Do not write on the QR codes or near the top edges (as it may be cut off).

Please write **legibly and dark enough** for the scan to be readable. For students who use English as a second language, dictionaries are permitted. Calculators are allowed.

DO NOT ASK QUESTIONS OF THE INVIGILATORS about interpretations of the exam questions. If you have a concern about a question, please summarize it at the top of this page. All concerns will be evaluated after the exam. GOOD LUCK!

**Question 1 8 points** (*Assume all transitions are possible*)

a. 6 points Calculate the **energy (in kJ/mol)** required to ionize (completely remove) the electron from the **$n=3$** level of **He^+ ion** (He: Helium). Show your work including equations. (3 sig. figs.)

b. 2 points

(I) Select the largest (longest) wavelength from the three choices below:

- i. Wavelength of photon to *completely remove* an electron from *1s orbital of Hydrogen atom*
- ii. Wavelength of photon to *completely remove* an electron from *1s orbital of Boron atom (B)*
- iii. Wavelength of photon to *completely remove* an electron from *1s orbital of Boron cation (B^+)*

i or ii or iii

(II) Select the smallest (shortest) wavelength from the three choices below (*for H atom*):

- i. Wavelength of photon *emitted* for an electron transitioning from *2p to 1s orbital*
- ii. Wavelength of photon *absorbed* for an electron transitioning from *2s to 4p orbital*
- iii. Wavelength of photon *absorbed* for an electron transitioning from *1s to 3p orbital*

i or ii or iii



Question 2 8 points



A Chem110 student plots the graph shown on the left, from data obtained for the kinetic energy (eV) of the ejected electron versus frequency (s^{-1}) of the incident light, for an unknown metal. ($1\text{eV} = 1.6022 \times 10^{-19} \text{ J}$)

Based on the plot, answer the questions **a to c** below:

a. 1 point On the plot above, label the **threshold frequency**.

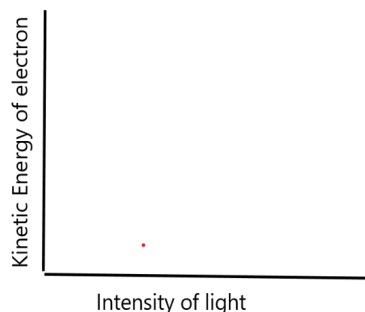
b. 2 points What is the value of the work potential (in J) of the unknown metal?

c. 3 points What is the *longest* wavelength (in nm) required to eject an electron from this metal's surface? Show your work including equation(s) used. (3 sig figs)

d. 2 points In the space provided below, plot the following

(Assume frequency is *above* threshold frequency):

- Kinetic energy* of an ejected electron versus the *intensity* of the incident light (left)
- Number of electrons* ejected versus the *frequency* of the incident light (right)



**Question 3** **6 points****a. 2 points**

In the ground state of Ge (Germanium; $Z = 32$) atom, **name** the following orbitals:

i. Orbital containing the electron that requires the maximum energy to be removed.

ii. Orbital containing the electron that experiences the minimum effective nuclear charge.

b. 2 points

An unknown anion (A^-) has the following *excited* electronic configuration: $[Ar] 4s^2 3d^3 4p^1$

(Ar: Argon; $Z = 18$)

Give the **condensed ground state electronic configuration** of the atom, **A** and the cation, **A^{2+}** .
Show your work. (Assume all electronic transitions are possible)

Electronic Configuration of A:

Electronic Configuration of A^{2+} :

c. 2 points

In a single atom, determine how many electrons can have the following set of quantum numbers?

(i) $n = 5; l = 3$

In a single atom, determine how many electrons can have the following set of quantum numbers?

(i) $n = 3; l = 2; m_s = +1/2$

**Question 4** **8 points**

Rank the following from 1 to 4 in **increasing** order. Provide the ranking in the boxes provided.

Note: Atomic number (Z) are provided for relevant atoms

a) Increasing (lowest to highest) order of *ionization energy*: (1= lowest; 4 = highest)

Mg

Mg: Z = 12

Rb

Rb: Z = 37

Cs

Cs: Z = 55

K

K : Z = 19

b) Increasing (lowest to highest) order of *bond angle*: (1= lowest; 4 = highest)

H₂OCCl₄XeF₄

HCN

c) Increasing (smallest to largest) order of *ionic size* (1= smallest; 4 = largest)

S²⁻

S: Z = 16

O²⁻

O: Z = 8

Be²⁺

Be: Z = 4

Na⁺

Na: Z = 11

d) Increasing (lowest to highest) order of *bond energy* (1= lowest; 4 = highest)

P-F

P-Cl

P-I

P-Br

**Question 5** **12 points**

a. 9 points For the molecules HClO_4 and TeF_6 , draw the most stable *Lewis structure* showing appropriate shape and *dashed/wedged bonds*, based on VSEPR. Determine the **molecular geometry** around the central atom. For HClO_4 , label the *most polar bond*.



b. 3 points (Note: Lone pairs are not shown in structures below)

(i) For the three *resonance* structures (1, 2 and 3) shown below, answer the following question:



(I) Circle the *most contributing* resonance structure.

Structure 1

Structure 2

Structure 3

(ii) For the three structures given (4, 5, and 6) below, answer the following questions:



(I) Circle all the structures that represent *resonance structures* of each other:

Structure 4

Structure 5

Structure 6

(II) Among the resonance structures, circle the *most contributing* resonance structure.

Structure 4

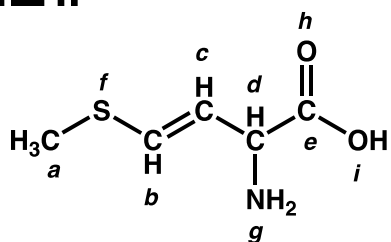
Structure 5

Structure 6

**Question 6** **6 points**

Using the data given below, calculate the **lattice energy** for SrO(s).

	ΔH° (kJ/mol)
$\text{Sr}(s) \longrightarrow \text{Sr}(g)$	+180
$\text{Sr}(g) \longrightarrow \text{Sr}^{2+}(g) + 2e^-$	+1468
$\text{O}_2(g) \longrightarrow 2\text{O}(g)$	+500
$\text{O}(g) + 2e^- \longrightarrow \text{O}^{2-}(g)$	+600
$\text{Sr}(s) + \frac{1}{2}\text{O}_2(g) \longrightarrow \text{SrO}(s)$	-550

**Question 7 10 points**

Consider the molecule given on the left, and answer questions (i) to (vi) using chemical bonding theories that we have learnt in class (valence bond theory, hybridization).

Note that the lone pairs are not shown in the structure:

i. Consider the carbon atoms marked in as ***a***, ***b***, ***c***, ***d***, and ***e*** in the molecule, and determine **hybridization** at each of the carbon centers. (2.5 points)

Atom	Hybridization
Carbon (<i>a</i>)	
Carbon (<i>b</i>)	
Carbon (<i>c</i>)	
Carbon (<i>d</i>)	
Carbon (<i>e</i>)	

ii. Now, consider the sulfur atom (S) marked as ***f***, nitrogen (N) atom (***g***), and oxygen(O) atoms (***h*** and ***i***), and determine their **hybridization**. (2 points)

Atom	Hybridization
Sulfur (<i>f</i>)	
Nitrogen (<i>g</i>)	
Oxygen (<i>h</i>)	
Oxygen (<i>i</i>)	

iii. **Describe** all the bonds formed by the carbon atoms ***a*** and ***b***, using sigma and pi designation as appropriate. (2 points)

Atom	Bonds formed (use sigma and pi as appropriate)
Carbon (<i>a</i>)	
Carbon (<i>b</i>)	

iv. How many total sp^3 -s sigma bonds are in the molecule? (1 point)

v. Determine **electronic** and **molecular geometry** at carbon (C) labeled ***e*** and sulfur (S) labeled ***f*** centres. (2 points)

Atom	Electronic Geometry	Molecular Geometry
Carbon (<i>e</i>)		
Sulfur (<i>f</i>)		

vi. Circle the value from the choices given below, that describes the **C-S-C bond angle at sulfur** (S) marked ***f***: (0.5 points)

>90° to <109.5° 109.5° >109.5 to <120° 120° >120° to <180°


Question 8 12 points

Using the **valence shell orbitals only** draw a **complete** molecular orbital (MO) diagram (with no s-p mixing) for the molecule **PS**, in the space provided below. **(6 points)**

Assume that 3s and 3p atomic orbitals will form molecular orbitals like those formed by the combination of 2s and 2p atomic orbitals that we learnt in class, and there is no s-p mixing.

Show **energy arrow**, **atomic orbitals** involved, **atom labels**, the **molecular orbitals** that are formed, **filled electrons** in the MO diagram etc. Make sure to label all the atomic and molecular orbitals.

Using the MO diagram above, answer the following questions:

- (i) Which one of the following will have the highest bond order? Circle the correct choice
 A) PS^{1-} B) PS C) PS^{1+}
- (ii) Which one of the following will have the longest bond length? Circle the correct choice
 A) PS^{1-} B) PS C) PS^{1+}
- (iii) Which of the following will be diamagnetic? Circle the correct choice.
 A) PS^{1-} B) PS C) PS^{1+}
- (iv) Draw Lewis dot structure of the molecule with the shortest bond length (based on the MO diagram) from among the three (PS^{1-} , PS, PS^{1+}), and indicate the bond order calculated from the Lewis dot structure. **(3 points)**

Lewis Dot Structure

Bond Order



604EAB84-2586-47E1-B17B-1173463DFE23

midterm-exam-93388

#52 Page 10 of 10

SCRAP PAGE

**CHEMISTRY 110 –Midterm Exam****Thursday November 7, 2024, 6:30 – 8:30 pm**

Examiners: Pallavi Sirjoosingh and Ashok Kakkar

First name (please write as legibly as possible within the boxes)

--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--

Last name

--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--

Student ID number

--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--

This booklet has 8 questions on 10 pages (The last page can be used as a scrap page).

The staple must NOT be removed from the question booklet. Your intact question booklet must be returned at the end of the exam.

A separate booklet contains a periodic table, and datasheet.

Write your answers directly under the questions in the space provided. Do not write on the QR codes or near the top edges (as it may be cut off).

Please write **legibly and dark enough** for the scan to be readable. For students who use English as a second language, dictionaries are permitted. Calculators are allowed.

DO NOT ASK QUESTIONS OF THE INVIGILATORS about interpretations of the exam questions. If you have a concern about a question, please summarize it at the top of this page. All concerns will be evaluated after the exam. GOOD LUCK!

**Question 1 8 points** (*Assume all transitions are possible*)

a. 6 points Calculate the **energy (in kJ/mol)** required to ionize (completely remove) the electron from the **$n=3$** level of **He^+ ion** (He: Helium). Show your work including equations. (3 sig. figs.)

b. 2 points

(I) Select the largest (longest) wavelength from the three choices below:

- i. Wavelength of photon to *completely remove* an electron from *1s orbital of Hydrogen atom*
- ii. Wavelength of photon to *completely remove* an electron from *1s orbital of Boron atom (B)*
- iii. Wavelength of photon to *completely remove* an electron from *1s orbital of Boron cation (B^+)*

i or ii or iii

(II) Select the smallest (shortest) wavelength from the three choices below (*for H atom*):

- i. Wavelength of photon *emitted* for an electron transitioning from *2p to 1s orbital*
- ii. Wavelength of photon *absorbed* for an electron transitioning from *2s to 4p orbital*
- iii. Wavelength of photon *absorbed* for an electron transitioning from *1s to 3p orbital*

i or ii or iii



Question 2 8 points



A Chem110 student plots the graph shown on the left, from data obtained for the kinetic energy (eV) of the ejected electron versus frequency (s^{-1}) of the incident light, for an unknown metal. ($1\text{eV} = 1.6022 \times 10^{-19} \text{ J}$)

Based on the plot, answer the questions **a to c** below:

a. 1 point On the plot above, label the **threshold frequency**.

b. 2 points What is the value of the work potential (in J) of the unknown metal?

c. 3 points What is the *longest* wavelength (in nm) required to eject an electron from this metal's surface? Show your work including equation(s) used. (3 sig figs)

d. 2 points In the space provided below, plot the following

(Assume frequency is *above* threshold frequency):

- i. *Kinetic energy* of an ejected electron versus the *intensity* of the incident light (left)
- ii. *Number of electrons* ejected versus the *frequency* of the incident light (right)



**Question 3** **6 points****a. 2 points**

In the ground state of Ge (Germanium; $Z = 32$) atom, **name** the following orbitals:

i. Orbital containing the electron that requires the maximum energy to be removed.

ii. Orbital containing the electron that experiences the minimum effective nuclear charge.

b. 2 points

An unknown anion (A^-) has the following *excited* electronic configuration: $[Ar] 4s^2 3d^3 4p^1$

(Ar: Argon; $Z = 18$)

Give the **condensed ground state electronic configuration** of the atom, **A** and the cation, **A^{2+}** .
Show your work. (Assume all electronic transitions are possible)

Electronic Configuration of A:

Electronic Configuration of A^{2+} :

c. 2 points

In a single atom, determine how many electrons can have the following set of quantum numbers?

(i) $n = 5; l = 3$

In a single atom, determine how many electrons can have the following set of quantum numbers?

(i) $n = 3; l = 2; m_s = +1/2$

**Question 4** **8 points**

Rank the following from 1 to 4 in **increasing** order. Provide the ranking in the boxes provided.

Note: Atomic number (Z) are provided for relevant atoms

a) Increasing (lowest to highest) order of *ionization energy*: (1= lowest; 4 = highest)

Mg

Mg: Z = 12

Rb

Rb: Z = 37

Cs

Cs: Z = 55

K

K : Z = 19

b) Increasing (lowest to highest) order of *bond angle*: (1= lowest; 4 = highest)

H₂OCCl₄XeF₄

HCN

c) Increasing (smallest to largest) order of *ionic size* (1= smallest; 4 = largest)

S²⁻

S: Z = 16

O²⁻

O: Z = 8

Be²⁺

Be: Z = 4

Na⁺

Na: Z = 11

d) Increasing (lowest to highest) order of *bond energy* (1= lowest; 4 = highest)

P-F

P-Cl

P-I

P-Br

**Question 5** **12 points**

a. 9 points For the molecules HClO_4 and TeF_6 , draw the most stable *Lewis structure* showing appropriate shape and *dashed/wedged bonds*, based on VSEPR. Determine the **molecular geometry** around the central atom. For HClO_4 , label the *most polar bond*.

HClO_4 (there are no O-O bonds)

Label the most polar bond

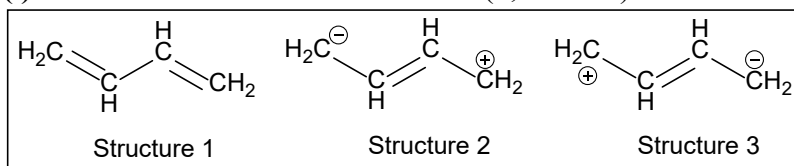
Molecular Geometry around Cl

TeF_6

Molecular Geometry around Te

b. 3 points (Note: Lone pairs are not shown in structures below)

(i) For the three *resonance* structures (1, 2 and 3) shown below, answer the following question:



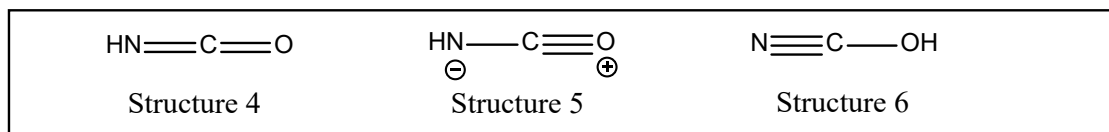
(I) Circle the *most contributing* resonance structure.

Structure 1

Structure 2

Structure 3

(ii) For the three structures given (4, 5, and 6) below, answer the following questions:



(I) Circle all the structures that represent *resonance structures* of each other:

Structure 4

Structure 5

Structure 6

(II) Among the resonance structures, circle the *most contributing* resonance structure.

Structure 4

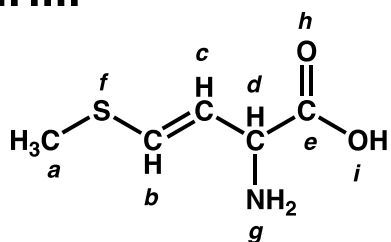
Structure 5

Structure 6

**Question 6** **6 points**

Using the data given below, calculate the **lattice energy** for SrO(s).

	ΔH° (kJ/mol)
$\text{Sr}(s) \longrightarrow \text{Sr}(g)$	+180
$\text{Sr}(g) \longrightarrow \text{Sr}^{2+}(g) + 2e^-$	+1468
$\text{O}_2(g) \longrightarrow 2\text{O}(g)$	+500
$\text{O}(g) + 2e^- \longrightarrow \text{O}^{2-}(g)$	+600
$\text{Sr}(s) + \frac{1}{2}\text{O}_2(g) \longrightarrow \text{SrO}(s)$	-550

**Question 7 10 points**

Consider the molecule given on the left, and answer questions (i) to (vi) using chemical bonding theories that we have learnt in class (valence bond theory, hybridization).

Note that the lone pairs are not shown in the structure:

i. Consider the carbon atoms marked in as ***a***, ***b***, ***c***, ***d***, and ***e*** in the molecule, and determine **hybridization** at each of the carbon centers. (2.5 points)

Atom	Hybridization
Carbon (<i>a</i>)	
Carbon (<i>b</i>)	
Carbon (<i>c</i>)	
Carbon (<i>d</i>)	
Carbon (<i>e</i>)	

ii. Now, consider the sulfur atom (S) marked as ***f***, nitrogen (N) atom (***g***), and oxygen(O) atoms (***h*** and ***i***), and determine their **hybridization**. (2 points)

Atom	Hybridization
Sulfur (<i>f</i>)	
Nitrogen (<i>g</i>)	
Oxygen (<i>h</i>)	
Oxygen (<i>i</i>)	

iii. **Describe** all the bonds formed by the carbon atoms ***a*** and ***b***, using sigma and pi designation as appropriate. (2 points)

Atom	Bonds formed (use sigma and pi as appropriate)
Carbon (<i>a</i>)	
Carbon (<i>b</i>)	

iv. How many total sp^3 -s sigma bonds are in the molecule? (1 point)

v. Determine **electronic** and **molecular geometry** at carbon (C) labeled ***e*** and sulfur (S) labeled ***f*** centres. (2 points)

Atom	Electronic Geometry	Molecular Geometry
Carbon (<i>e</i>)		
Sulfur (<i>f</i>)		

vi. Circle the value from the choices given below, that describes the **C-S-C bond angle at sulfur** (S) marked ***f***: (0.5 points)

>90° to <109.5° 109.5° >109.5 to <120° 120° >120° to <180°


Question 8 12 points

Using the **valence shell orbitals only** draw a **complete** molecular orbital (MO) diagram (with no s-p mixing) for the molecule **PS**, in the space provided below. **(6 points)**

Assume that 3s and 3p atomic orbitals will form molecular orbitals like those formed by the combination of 2s and 2p atomic orbitals that we learnt in class, and there is no s-p mixing.

Show **energy arrow**, **atomic orbitals** involved, **atom labels**, the **molecular orbitals** that are formed, **filled electrons** in the MO diagram etc. Make sure to label all the atomic and molecular orbitals.

Using the MO diagram above, answer the following questions:

- (i) Which one of the following will have the highest bond order? Circle the correct choice
 A) PS^{1-} B) PS C) PS^{1+}
- (ii) Which one of the following will have the longest bond length? Circle the correct choice
 A) PS^{1-} B) PS C) PS^{1+}
- (iii) Which of the following will be diamagnetic? Circle the correct choice.
 A) PS^{1-} B) PS C) PS^{1+}
- (iv) Draw Lewis dot structure of the molecule with the shortest bond length (based on the MO diagram) from among the three (PS^{1-} , PS, PS^{1+}), and indicate the bond order calculated from the Lewis dot structure. **(3 points)**

Lewis Dot Structure

Bond Order



C5EDD6FB-697C-40F1-94B4-F47AB8927CED

midterm-exam-93388

#53 Page 10 of 10

SCRAP PAGE

**CHEMISTRY 110 –Midterm Exam****Thursday November 7, 2024, 6:30 – 8:30 pm**

Examiners: Pallavi Sirjoosingh and Ashok Kakkar

First name (please write as legibly as possible within the boxes)

--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--

Last name

--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--

Student ID number

--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--

This booklet has 8 questions on 10 pages (The last page can be used as a scrap page).

The staple must NOT be removed from the question booklet. Your intact question booklet must be returned at the end of the exam.

A separate booklet contains a periodic table, and datasheet.

Write your answers directly under the questions in the space provided. Do not write on the QR codes or near the top edges (as it may be cut off).

Please write **legibly and dark enough** for the scan to be readable. For students who use English as a second language, dictionaries are permitted. Calculators are allowed.

DO NOT ASK QUESTIONS OF THE INVIGILATORS about interpretations of the exam questions. If you have a concern about a question, please summarize it at the top of this page. All concerns will be evaluated after the exam. GOOD LUCK!

**Question 1 8 points** (*Assume all transitions are possible*)

a. 6 points Calculate the **energy (in kJ/mol)** required to ionize (completely remove) the electron from the **n=3** level of **He⁺ ion** (He: Helium). Show your work including equations. (3 sig. figs.)

b. 2 points

(I) Select the largest (longest) wavelength from the three choices below:

- i. Wavelength of photon to *completely remove* an electron from *1s orbital of Hydrogen atom*
- ii. Wavelength of photon to *completely remove* an electron from *1s orbital of Boron atom (B)*
- iii. Wavelength of photon to *completely remove* an electron from *1s orbital of Boron cation (B⁺)*

i or ii or iii

(II) Select the smallest (shortest) wavelength from the three choices below (*for H atom*):

- i. Wavelength of photon *emitted* for an electron transitioning from *2p to 1s orbital*
- ii. Wavelength of photon *absorbed* for an electron transitioning from *2s to 4p orbital*
- iii. Wavelength of photon *absorbed* for an electron transitioning from *1s to 3p orbital*

i or ii or iii



Question 2 8 points



A Chem110 student plots the graph shown on the left, from data obtained for the kinetic energy (eV) of the ejected electron versus frequency (s^{-1}) of the incident light, for an unknown metal. ($1\text{eV} = 1.6022 \times 10^{-19} \text{ J}$)

Based on the plot, answer the questions **a to c** below:

a. 1 point On the plot above, label the **threshold frequency**.

b. 2 points What is the value of the work potential (in J) of the unknown metal?

c. 3 points What is the *longest* wavelength (in nm) required to eject an electron from this metal's surface? Show your work including equation(s) used. (3 sig figs)

d. 2 points In the space provided below, plot the following

(Assume frequency is *above* threshold frequency):

- Kinetic energy* of an ejected electron versus the *intensity* of the incident light (left)
- Number of electrons* ejected versus the *frequency* of the incident light (right)



**Question 3** **6 points****a. 2 points**

In the ground state of Ge (Germanium; $Z = 32$) atom, **name** the following orbitals:

i. Orbital containing the electron that requires the maximum energy to be removed.

ii. Orbital containing the electron that experiences the minimum effective nuclear charge.

b. 2 points

An unknown anion (A^-) has the following *excited* electronic configuration: $[Ar] 4s^2 3d^3 4p^1$

(Ar: Argon; $Z = 18$)

Give the **condensed ground state electronic configuration** of the atom, **A** and the cation, **A^{2+}** .
Show your work. (Assume all electronic transitions are possible)

Electronic Configuration of A:

Electronic Configuration of A^{2+} :

c. 2 points

In a single atom, determine how many electrons can have the following set of quantum numbers?

(i) $n = 5; l = 3$

In a single atom, determine how many electrons can have the following set of quantum numbers?

(i) $n = 3; l = 2; m_s = +1/2$

**Question 4** **8 points**

Rank the following from 1 to 4 in **increasing** order. Provide the ranking in the boxes provided.

Note: Atomic number (Z) are provided for relevant atoms

a) Increasing (lowest to highest) order of *ionization energy*: (1= lowest; 4 = highest)

Mg

Mg: Z = 12

Rb

Rb: Z = 37

Cs

Cs: Z = 55

K

K : Z = 19

b) Increasing (lowest to highest) order of *bond angle*: (1= lowest; 4 = highest)

H₂OCCl₄XeF₄

HCN

c) Increasing (smallest to largest) order of *ionic size* (1= smallest; 4 = largest)

S²⁻

S: Z = 16

O²⁻

O: Z = 8

Be²⁺

Be: Z = 4

Na⁺

Na: Z = 11

d) Increasing (lowest to highest) order of *bond energy* (1= lowest; 4 = highest)

P-F

P-Cl

P-I

P-Br

**Question 5** **12 points**

a. 9 points For the molecules HClO_4 and TeF_6 , draw the most stable *Lewis structure* showing appropriate shape and *dashed/wedged bonds*, based on VSEPR. Determine the **molecular geometry** around the central atom. For HClO_4 , label the *most polar bond*.



b. 3 points (Note: Lone pairs are not shown in structures below)

(i) For the three *resonance* structures (1, 2 and 3) shown below, answer the following question:



(I) Circle the *most contributing* resonance structure.

Structure 1

Structure 2

Structure 3

(ii) For the three structures given (4, 5, and 6) below, answer the following questions:



(I) Circle all the structures that represent *resonance structures* of each other:

Structure 4

Structure 5

Structure 6

(II) Among the resonance structures, circle the *most contributing* resonance structure.

Structure 4

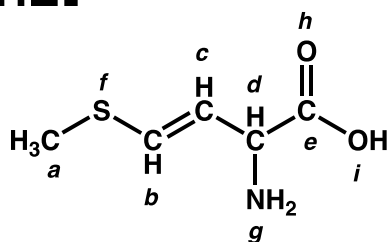
Structure 5

Structure 6

**Question 6** **6 points**

Using the data given below, calculate the **lattice energy** for SrO(s).

	ΔH° (kJ/mol)
$\text{Sr}(s) \longrightarrow \text{Sr}(g)$	+180
$\text{Sr}(g) \longrightarrow \text{Sr}^{2+}(g) + 2e^-$	+1468
$\text{O}_2(g) \longrightarrow 2\text{O}(g)$	+500
$\text{O}(g) + 2e^- \longrightarrow \text{O}^{2-}(g)$	+600
$\text{Sr}(s) + \frac{1}{2}\text{O}_2(g) \longrightarrow \text{SrO}(s)$	-550

**Question 7 10 points**

Consider the molecule given on the left, and answer questions (i) to (vi) using chemical bonding theories that we have learnt in class (valence bond theory, hybridization).

Note that the lone pairs are not shown in the structure:

i. Consider the carbon atoms marked in as ***a***, ***b***, ***c***, ***d***, and ***e*** in the molecule, and determine **hybridization** at each of the carbon centers. (2.5 points)

Atom	Hybridization
Carbon (<i>a</i>)	
Carbon (<i>b</i>)	
Carbon (<i>c</i>)	
Carbon (<i>d</i>)	
Carbon (<i>e</i>)	

ii. Now, consider the sulfur atom (S) marked as ***f***, nitrogen (N) atom (***g***), and oxygen(O) atoms (***h*** and ***i***), and determine their **hybridization**. (2 points)

Atom	Hybridization
Sulfur (<i>f</i>)	
Nitrogen (<i>g</i>)	
Oxygen (<i>h</i>)	
Oxygen (<i>i</i>)	

iii. **Describe** all the bonds formed by the carbon atoms ***a*** and ***b***, using sigma and pi designation as appropriate. (2 points)

Atom	Bonds formed (use sigma and pi as appropriate)
Carbon (<i>a</i>)	
Carbon (<i>b</i>)	

iv. How many total sp^3 -s sigma bonds are in the molecule? (1 point)

v. Determine **electronic** and **molecular geometry** at carbon (C) labeled ***e*** and sulfur (S) labeled ***f*** centres. (2 points)

Atom	Electronic Geometry	Molecular Geometry
Carbon (<i>e</i>)		
Sulfur (<i>f</i>)		

vi. Circle the value from the choices given below, that describes the **C-S-C bond angle at sulfur** (S) marked ***f***: (0.5 points)

$>90^\circ$ to $<109.5^\circ$ 109.5° >109.5 to $<120^\circ$ 120° $>120^\circ$ to $<180^\circ$

**Question 8 12 points**

Using the **valence shell orbitals only** draw a **complete** molecular orbital (MO) diagram (with no s-p mixing) for the molecule **PS**, in the space provided below. **(6 points)**

Assume that 3s and 3p atomic orbitals will form molecular orbitals like those formed by the combination of 2s and 2p atomic orbitals that we learnt in class, and there is no s-p mixing.

Show **energy arrow**, **atomic orbitals** involved, **atom labels**, the **molecular orbitals** that are formed, **filled electrons** in the MO diagram etc. Make sure to label all the atomic and molecular orbitals.

Using the MO diagram above, answer the following questions:

- (i) Which one of the following will have the highest bond order? Circle the correct choice
A) PS^{1-} B) PS C) PS^{1+}
- (ii) Which one of the following will have the longest bond length? Circle the correct choice
A) PS^{1-} B) PS C) PS^{1+}
- (iii) Which of the following will be diamagnetic? Circle the correct choice.
A) PS^{1-} B) PS C) PS^{1+}
- (iv) Draw Lewis dot structure of the molecule with the shortest bond length (based on the MO diagram) from among the three (PS^{1-} , PS, PS^{1+}), and indicate the bond order calculated from the Lewis dot structure. **(3 points)**

Lewis Dot Structure

Bond Order



FA912273-0F3D-4684-8B10-DEA399A60FD7

midterm-exam-93388

#54 Page 10 of 10

SCRAP PAGE

**CHEMISTRY 110 –Midterm Exam****Thursday November 7, 2024, 6:30 – 8:30 pm**

Examiners: Pallavi Sirjoosingh and Ashok Kakkar

First name (please write as legibly as possible within the boxes)

--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--

Last name

--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--

Student ID number

--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--

This booklet has 8 questions on 10 pages (The last page can be used as a scrap page).

The staple must NOT be removed from the question booklet. Your intact question booklet must be returned at the end of the exam.

A separate booklet contains a periodic table, and datasheet.

Write your answers directly under the questions in the space provided. Do not write on the QR codes or near the top edges (as it may be cut off).

Please write **legibly and dark enough** for the scan to be readable. For students who use English as a second language, dictionaries are permitted. Calculators are allowed.

DO NOT ASK QUESTIONS OF THE INVIGILATORS about interpretations of the exam questions. If you have a concern about a question, please summarize it at the top of this page. All concerns will be evaluated after the exam. GOOD LUCK!

**Question 1 8 points** (*Assume all transitions are possible*)

a. 6 points Calculate the **energy (in kJ/mol)** required to ionize (completely remove) the electron from the **$n=3$** level of **He^+ ion** (He: Helium). Show your work including equations. (3 sig. figs.)

b. 2 points

(I) Select the largest (longest) wavelength from the three choices below:

- i. Wavelength of photon to *completely remove* an electron from *1s orbital of Hydrogen atom*
- ii. Wavelength of photon to *completely remove* an electron from *1s orbital of Boron atom (B)*
- iii. Wavelength of photon to *completely remove* an electron from *1s orbital of Boron cation (B^+)*

i or ii or iii

(II) Select the smallest (shortest) wavelength from the three choices below (*for H atom*):

- i. Wavelength of photon *emitted* for an electron transitioning from *2p to 1s orbital*
- ii. Wavelength of photon *absorbed* for an electron transitioning from *2s to 4p orbital*
- iii. Wavelength of photon *absorbed* for an electron transitioning from *1s to 3p orbital*

i or ii or iii



Question 2 8 points



A Chem110 student plots the graph shown on the left, from data obtained for the kinetic energy (eV) of the ejected electron versus frequency (s^{-1}) of the incident light, for an unknown metal. ($1\text{eV} = 1.6022 \times 10^{-19} \text{ J}$)

Based on the plot, answer the questions **a to c** below:

a. 1 point On the plot above, label the **threshold frequency**.

b. 2 points What is the value of the work potential (in J) of the unknown metal?

c. 3 points What is the *longest* wavelength (in nm) required to eject an electron from this metal's surface? Show your work including equation(s) used. (3 sig figs)

d. 2 points In the space provided below, plot the following

(Assume frequency is *above* threshold frequency):

- i. *Kinetic energy* of an ejected electron versus the *intensity* of the incident light (left)
- ii. *Number of electrons* ejected versus the *frequency* of the incident light (right)



**Question 3** **6 points****a. 2 points**

In the ground state of Ge (Germanium; $Z = 32$) atom, **name** the following orbitals:

i. Orbital containing the electron that requires the maximum energy to be removed.

ii. Orbital containing the electron that experiences the minimum effective nuclear charge.

b. 2 points

An unknown anion (A^-) has the following *excited* electronic configuration: $[Ar] 4s^2 3d^3 4p^1$

(Ar: Argon; $Z = 18$)

Give the **condensed ground state electronic configuration** of the atom, **A** and the cation, **A^{2+}** .
Show your work. (Assume all electronic transitions are possible)

Electronic Configuration of A:

Electronic Configuration of A^{2+} :

c. 2 points

In a single atom, determine how many electrons can have the following set of quantum numbers?

(i) $n = 5; l = 3$

In a single atom, determine how many electrons can have the following set of quantum numbers?

(i) $n = 3; l = 2; m_s = +1/2$

**Question 4** **8 points**

Rank the following from 1 to 4 in **increasing** order. Provide the ranking in the boxes provided.

Note: Atomic number (Z) are provided for relevant atoms

a) Increasing (lowest to highest) order of *ionization energy*: (1= lowest; 4 = highest)

Mg

Mg: Z = 12

Rb

Rb: Z = 37

Cs

Cs: Z = 55

K

K : Z = 19

b) Increasing (lowest to highest) order of *bond angle*: (1= lowest; 4 = highest)

H₂OCCl₄XeF₄

HCN

c) Increasing (smallest to largest) order of *ionic size* (1= smallest; 4 = largest)

S²⁻

S: Z = 16

O²⁻

O: Z = 8

Be²⁺

Be: Z = 4

Na⁺

Na: Z = 11

d) Increasing (lowest to highest) order of *bond energy* (1= lowest; 4 = highest)

P-F

P-Cl

P-I

P-Br

**Question 5** **12 points**

a. 9 points For the molecules HClO_4 and TeF_6 , draw the most stable *Lewis structure* showing appropriate shape and *dashed/wedged bonds*, based on VSEPR. Determine the **molecular geometry** around the central atom. For HClO_4 , label the *most polar bond*.



b. 3 points (Note: Lone pairs are not shown in structures below)

(i) For the three *resonance* structures (1, 2 and 3) shown below, answer the following question:



(I) Circle the *most contributing* resonance structure.

Structure 1

Structure 2

Structure 3

(ii) For the three structures given (4, 5, and 6) below, answer the following questions:



(I) Circle all the structures that represent *resonance structures* of each other:

Structure 4

Structure 5

Structure 6

(II) Among the resonance structures, circle the *most contributing* resonance structure.

Structure 4

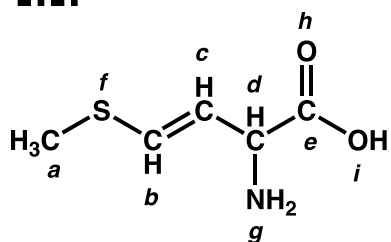
Structure 5

Structure 6

**Question 6** **6 points**

Using the data given below, calculate the **lattice energy** for SrO(s).

	ΔH° (kJ/mol)
$\text{Sr}(s) \longrightarrow \text{Sr}(g)$	+180
$\text{Sr}(g) \longrightarrow \text{Sr}^{2+}(g) + 2e^-$	+1468
$\text{O}_2(g) \longrightarrow 2\text{O}(g)$	+500
$\text{O}(g) + 2e^- \longrightarrow \text{O}^{2-}(g)$	+600
$\text{Sr}(s) + \frac{1}{2}\text{O}_2(g) \longrightarrow \text{SrO}(s)$	-550

**Question 7 10 points**

Consider the molecule given on the left, and answer questions (i) to (vi) using chemical bonding theories that we have learnt in class (valence bond theory, hybridization).

Note that the lone pairs are not shown in the structure:

i. Consider the carbon atoms marked in as ***a***, ***b***, ***c***, ***d***, and ***e*** in the molecule, and determine **hybridization** at each of the carbon centers. (2.5 points)

Atom	Hybridization
Carbon (<i>a</i>)	
Carbon (<i>b</i>)	
Carbon (<i>c</i>)	
Carbon (<i>d</i>)	
Carbon (<i>e</i>)	

ii. Now, consider the sulfur atom (S) marked as ***f***, nitrogen (N) atom (***g***), and oxygen(O) atoms (***h*** and ***i***), and determine their **hybridization**. (2 points)

Atom	Hybridization
Sulfur (<i>f</i>)	
Nitrogen (<i>g</i>)	
Oxygen (<i>h</i>)	
Oxygen (<i>i</i>)	

iii. **Describe** all the bonds formed by the carbon atoms ***a*** and ***b***, using sigma and pi designation as appropriate. (2 points)

Atom	Bonds formed (use sigma and pi as appropriate)
Carbon (<i>a</i>)	
Carbon (<i>b</i>)	

iv. How many total sp^3 -s sigma bonds are in the molecule? (1 point)

v. Determine **electronic** and **molecular geometry** at carbon (C) labeled ***e*** and sulfur (S) labeled ***f*** centres. (2 points)

Atom	Electronic Geometry	Molecular Geometry
Carbon (<i>e</i>)		
Sulfur (<i>f</i>)		

vi. Circle the value from the choices given below, that describes the **C-S-C bond angle at sulfur** (S) marked ***f***: (0.5 points)

$>90^\circ$ to $<109.5^\circ$ 109.5° >109.5 to $<120^\circ$ 120° $>120^\circ$ to $<180^\circ$



Question 8 12 points

Using the **valence shell orbitals only** draw a **complete** molecular orbital (MO) diagram (with no s-p mixing) for the molecule **PS**, in the space provided below. **(6 points)**

Assume that 3s and 3p atomic orbitals will form molecular orbitals like those formed by the combination of 2s and 2p atomic orbitals that we learnt in class, and there is no s-p mixing.

Show **energy arrow**, **atomic orbitals** involved, **atom labels**, the **molecular orbitals** that are formed, **filled electrons** in the MO diagram etc. Make sure to label all the atomic and molecular orbitals.

Using the MO diagram above, answer the following questions:

- (i) Which one of the following will have the highest bond order? Circle the correct choice
 A) PS^{1-} B) PS C) PS^{1+}
- (ii) Which one of the following will have the longest bond length? Circle the correct choice
 A) PS^{1-} B) PS C) PS^{1+}
- (iii) Which of the following will be diamagnetic? Circle the correct choice.
 A) PS^{1-} B) PS C) PS^{1+}
- (iv) Draw Lewis dot structure of the molecule with the shortest bond length (based on the MO diagram) from among the three (PS^{1-} , PS, PS^{1+}), and indicate the bond order calculated from the Lewis dot structure. **(3 points)**

Lewis Dot Structure

Bond Order



A92189D7-6ECD-4BA7-8F4D-6384C38405C0

midterm-exam-93388

#55 Page 10 of 10

SCRAP PAGE

**CHEMISTRY 110 –Midterm Exam****Thursday November 7, 2024, 6:30 – 8:30 pm****Examiners: Pallavi Sirjoosingh and Ashok Kakkar**

First name (please write as legibly as possible within the boxes)

--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--

Last name

--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--

Student ID number

--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--

This booklet has 8 questions on 10 pages (The last page can be used as a scrap page).

The staple must NOT be removed from the question booklet. Your intact question booklet must be returned at the end of the exam.

A separate booklet contains a periodic table, and datasheet.

Write your answers directly under the questions in the space provided. Do not write on the QR codes or near the top edges (as it may be cut off).

Please write **legibly and dark enough** for the scan to be readable. For students who use English as a second language, dictionaries are permitted. Calculators are allowed.

DO NOT ASK QUESTIONS OF THE INVIGILATORS about interpretations of the exam questions. If you have a concern about a question, please summarize it at the top of this page. All concerns will be evaluated after the exam. GOOD LUCK!

**Question 1 8 points** (*Assume all transitions are possible*)

a. 6 points Calculate the **energy (in kJ/mol)** required to ionize (completely remove) the electron from the **n=3** level of **He⁺ ion** (He: Helium). Show your work including equations. (3 sig. figs.)

b. 2 points

(I) Select the largest (longest) wavelength from the three choices below:

- i. Wavelength of photon to *completely remove* an electron from *1s orbital of Hydrogen atom*
- ii. Wavelength of photon to *completely remove* an electron from *1s orbital of Boron atom (B)*
- iii. Wavelength of photon to *completely remove* an electron from *1s orbital of Boron cation (B⁺)*

i or ii or iii

(II) Select the smallest (shortest) wavelength from the three choices below (*for H atom*):

- i. Wavelength of photon *emitted* for an electron transitioning from *2p to 1s orbital*
- ii. Wavelength of photon *absorbed* for an electron transitioning from *2s to 4p orbital*
- iii. Wavelength of photon *absorbed* for an electron transitioning from *1s to 3p orbital*

i or ii or iii



Question 2 8 points



A Chem110 student plots the graph shown on the left, from data obtained for the kinetic energy (eV) of the ejected electron versus frequency (s^{-1}) of the incident light, for an unknown metal. ($1\text{eV} = 1.6022 \times 10^{-19} \text{ J}$)

Based on the plot, answer the questions **a to c** below:

a. 1 point On the plot above, label the **threshold frequency**.

b. 2 points What is the value of the work potential (in J) of the unknown metal?

c. 3 points What is the *longest* wavelength (in nm) required to eject an electron from this metal's surface? Show your work including equation(s) used. (3 sig figs)

d. 2 points In the space provided below, plot the following

(Assume frequency is *above* threshold frequency):

- i. *Kinetic energy* of an ejected electron versus the *intensity* of the incident light (left)
- ii. *Number of electrons* ejected versus the *frequency* of the incident light (right)



**Question 3** **6 points****a. 2 points**

In the ground state of Ge (Germanium; $Z = 32$) atom, **name** the following orbitals:

i. Orbital containing the electron that requires the maximum energy to be removed.

ii. Orbital containing the electron that experiences the minimum effective nuclear charge.

b. 2 points

An unknown anion (A^-) has the following *excited* electronic configuration: $[Ar] 4s^2 3d^3 4p^1$

(Ar: Argon; $Z = 18$)

Give the **condensed ground state electronic configuration** of the atom, **A** and the cation, **A^{2+}** .
Show your work. (Assume all electronic transitions are possible)

Electronic Configuration of A:

Electronic Configuration of A^{2+} :

c. 2 points

In a single atom, determine how many electrons can have the following set of quantum numbers?

(i) $n = 5; l = 3$

In a single atom, determine how many electrons can have the following set of quantum numbers?

(i) $n = 3; l = 2; m_s = +1/2$

**Question 4** **8 points**

Rank the following from 1 to 4 in **increasing** order. Provide the ranking in the boxes provided.

Note: Atomic number (Z) are provided for relevant atoms

a) Increasing (lowest to highest) order of *ionization energy*: (1= lowest; 4 = highest)

Mg

Mg: Z = 12

Rb

Rb: Z = 37

Cs

Cs: Z = 55

K

K : Z = 19

b) Increasing (lowest to highest) order of *bond angle*: (1= lowest; 4 = highest)

H₂OCCl₄XeF₄

HCN

c) Increasing (smallest to largest) order of *ionic size* (1= smallest; 4 = largest)

S²⁻

S: Z = 16

O²⁻

O: Z = 8

Be²⁺

Be: Z = 4

Na⁺

Na: Z = 11

d) Increasing (lowest to highest) order of *bond energy* (1= lowest; 4 = highest)

P-F

P-Cl

P-I

P-Br

**Question 5** **12 points**

a. 9 points For the molecules HClO_4 and TeF_6 , draw the most stable *Lewis structure* showing appropriate shape and *dashed/wedged bonds*, based on VSEPR. Determine the **molecular geometry** around the central atom. For HClO_4 , label the *most polar bond*.



b. 3 points (Note: Lone pairs are not shown in structures below)

(i) For the three *resonance* structures (1, 2 and 3) shown below, answer the following question:



(I) Circle the *most contributing* resonance structure.

Structure 1

Structure 2

Structure 3

(ii) For the three structures given (4, 5, and 6) below, answer the following questions:



(I) Circle all the structures that represent *resonance structures* of each other:

Structure 4

Structure 5

Structure 6

(II) Among the resonance structures, circle the *most contributing* resonance structure.

Structure 4

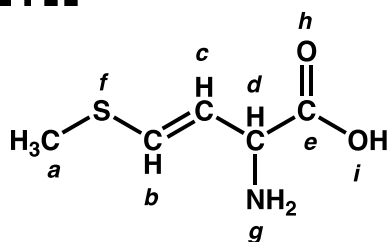
Structure 5

Structure 6

**Question 6** **6 points**

Using the data given below, calculate the **lattice energy** for SrO(s).

	ΔH° (kJ/mol)
$\text{Sr}(s) \longrightarrow \text{Sr}(g)$	+180
$\text{Sr}(g) \longrightarrow \text{Sr}^{2+}(g) + 2e^-$	+1468
$\text{O}_2(g) \longrightarrow 2\text{O}(g)$	+500
$\text{O}(g) + 2e^- \longrightarrow \text{O}^{2-}(g)$	+600
$\text{Sr}(s) + \frac{1}{2}\text{O}_2(g) \longrightarrow \text{SrO}(s)$	-550

**Question 7 10 points**

Consider the molecule given on the left, and answer questions (i) to (vi) using chemical bonding theories that we have learnt in class (valence bond theory, hybridization).

Note that the lone pairs are not shown in the structure:

i. Consider the carbon atoms marked in as ***a***, ***b***, ***c***, ***d***, and ***e*** in the molecule, and determine **hybridization** at each of the carbon centers. (2.5 points)

Atom	Hybridization
Carbon (<i>a</i>)	
Carbon (<i>b</i>)	
Carbon (<i>c</i>)	
Carbon (<i>d</i>)	
Carbon (<i>e</i>)	

ii. Now, consider the sulfur atom (S) marked as ***f***, nitrogen (N) atom (***g***), and oxygen(O) atoms (***h*** and ***i***), and determine their **hybridization**. (2 points)

Atom	Hybridization
Sulfur (<i>f</i>)	
Nitrogen (<i>g</i>)	
Oxygen (<i>h</i>)	
Oxygen (<i>i</i>)	

iii. **Describe** all the bonds formed by the carbon atoms ***a*** and ***b***, using sigma and pi designation as appropriate. (2 points)

Atom	Bonds formed (use sigma and pi as appropriate)
Carbon (<i>a</i>)	
Carbon (<i>b</i>)	

iv. How many total sp^3 -s sigma bonds are in the molecule? (1 point)

v. Determine **electronic** and **molecular geometry** at carbon (C) labeled ***e*** and sulfur (S) labeled ***f*** centres. (2 points)

Atom	Electronic Geometry	Molecular Geometry
Carbon (<i>e</i>)		
Sulfur (<i>f</i>)		

vi. Circle the value from the choices given below, that describes the **C-S-C bond angle at sulfur** (S) marked ***f***: (0.5 points)

>90° to <109.5° 109.5° >109.5 to <120° 120° >120° to <180°



Question 8 12 points

Using the **valence shell orbitals only** draw a **complete** molecular orbital (MO) diagram (with no s-p mixing) for the molecule **PS**, in the space provided below. **(6 points)**

Assume that 3s and 3p atomic orbitals will form molecular orbitals like those formed by the combination of 2s and 2p atomic orbitals that we learnt in class, and there is no s-p mixing.

Show **energy arrow**, **atomic orbitals** involved, **atom labels**, the **molecular orbitals** that are formed, **filled electrons** in the MO diagram etc. Make sure to label all the atomic and molecular orbitals.

Using the MO diagram above, answer the following questions:

- (i) Which one of the following will have the highest bond order? Circle the correct choice
 A) PS^{1-} B) PS C) PS^{1+}
- (ii) Which one of the following will have the longest bond length? Circle the correct choice
 A) PS^{1-} B) PS C) PS^{1+}
- (iii) Which of the following will be diamagnetic? Circle the correct choice.
 A) PS^{1-} B) PS C) PS^{1+}
- (iv) Draw Lewis dot structure of the molecule with the shortest bond length (based on the MO diagram) from among the three (PS^{1-} , PS, PS^{1+}), and indicate the bond order calculated from the Lewis dot structure. **(3 points)**

Lewis Dot Structure

Bond Order



0A6A7FF6-F5BE-4F05-9B59-BB898756C5BD

midterm-exam-93388

#56 Page 10 of 10

SCRAP PAGE

**CHEMISTRY 110 –Midterm Exam****Thursday November 7, 2024, 6:30 – 8:30 pm**

Examiners: Pallavi Sirjoosingh and Ashok Kakkar

First name (please write as legibly as possible within the boxes)

--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--

Last name

--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--

Student ID number

--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--

This booklet has 8 questions on 10 pages (The last page can be used as a scrap page).

The staple must NOT be removed from the question booklet. Your intact question booklet must be returned at the end of the exam.

A separate booklet contains a periodic table, and datasheet.

Write your answers directly under the questions in the space provided. Do not write on the QR codes or near the top edges (as it may be cut off).

Please write **legibly and dark enough** for the scan to be readable. For students who use English as a second language, dictionaries are permitted. Calculators are allowed.

DO NOT ASK QUESTIONS OF THE INVIGILATORS about interpretations of the exam questions. If you have a concern about a question, please summarize it at the top of this page. All concerns will be evaluated after the exam. GOOD LUCK!

**Question 1 8 points** (*Assume all transitions are possible*)

a. 6 points Calculate the **energy (in kJ/mol)** required to ionize (completely remove) the electron from the **n=3** level of **He⁺ ion** (He: Helium). Show your work including equations. (3 sig. figs.)

b. 2 points

(I) Select the largest (longest) wavelength from the three choices below:

- i. Wavelength of photon to *completely remove* an electron from *1s orbital of Hydrogen atom*
- ii. Wavelength of photon to *completely remove* an electron from *1s orbital of Boron atom (B)*
- iii. Wavelength of photon to *completely remove* an electron from *1s orbital of Boron cation (B⁺)*

i or ii or iii

(II) Select the smallest (shortest) wavelength from the three choices below (*for H atom*):

- i. Wavelength of photon *emitted* for an electron transitioning from *2p to 1s orbital*
- ii. Wavelength of photon *absorbed* for an electron transitioning from *2s to 4p orbital*
- iii. Wavelength of photon *absorbed* for an electron transitioning from *1s to 3p orbital*

i or ii or iii



Question 2 8 points



A Chem110 student plots the graph shown on the left, from data obtained for the kinetic energy (eV) of the ejected electron versus frequency (s^{-1}) of the incident light, for an unknown metal. ($1\text{eV} = 1.6022 \times 10^{-19} \text{ J}$)

Based on the plot, answer the questions **a to c** below:

a. 1 point On the plot above, label the **threshold frequency**.

b. 2 points What is the value of the work potential (in J) of the unknown metal?

c. 3 points What is the *longest* wavelength (in nm) required to eject an electron from this metal's surface? Show your work including equation(s) used. (3 sig figs)

d. 2 points In the space provided below, plot the following

(Assume frequency is *above* threshold frequency):

- Kinetic energy* of an ejected electron versus the *intensity* of the incident light (left)
- Number of electrons* ejected versus the *frequency* of the incident light (right)



**Question 3** **6 points****a. 2 points**

In the ground state of Ge (Germanium; $Z = 32$) atom, **name** the following orbitals:

i. Orbital containing the electron that requires the maximum energy to be removed.

ii. Orbital containing the electron that experiences the minimum effective nuclear charge.

b. 2 points

An unknown anion (A^-) has the following *excited* electronic configuration: $[Ar] 4s^2 3d^3 4p^1$

(Ar: Argon; $Z = 18$)

Give the **condensed ground state electronic configuration** of the atom, **A** and the cation, **A^{2+}** .
Show your work. (Assume all electronic transitions are possible)

Electronic Configuration of A:

Electronic Configuration of A^{2+} :

c. 2 points

In a single atom, determine how many electrons can have the following set of quantum numbers?

(i) $n = 5; l = 3$

In a single atom, determine how many electrons can have the following set of quantum numbers?

(i) $n = 3; l = 2; m_s = +1/2$

**Question 4** **8 points**

Rank the following from 1 to 4 in **increasing** order. Provide the ranking in the boxes provided.

Note: Atomic number (Z) are provided for relevant atoms

a) Increasing (lowest to highest) order of *ionization energy*: (1= lowest; 4 = highest)

Mg

Mg: Z = 12

Rb

Rb: Z = 37

Cs

Cs: Z = 55

K

K : Z = 19

b) Increasing (lowest to highest) order of *bond angle*: (1= lowest; 4 = highest)

H₂OCCl₄XeF₄

HCN

c) Increasing (smallest to largest) order of *ionic size* (1= smallest; 4 = largest)

S²⁻

S: Z = 16

O²⁻

O: Z = 8

Be²⁺

Be: Z = 4

Na⁺

Na: Z = 11

d) Increasing (lowest to highest) order of *bond energy* (1= lowest; 4 = highest)

P-F

P-Cl

P-I

P-Br

**Question 5** **12 points**

a. 9 points For the molecules HClO_4 and TeF_6 , draw the most stable *Lewis structure* showing appropriate shape and *dashed/wedged bonds*, based on VSEPR. Determine the **molecular geometry** around the central atom. For HClO_4 , label the *most polar bond*.

HClO_4 (there are no O-O bonds)

Label the most polar bond

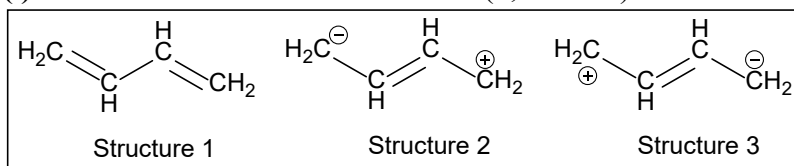
Molecular Geometry around Cl

TeF_6

Molecular Geometry around Te

b. 3 points (Note: Lone pairs are not shown in structures below)

(i) For the three *resonance* structures (1, 2 and 3) shown below, answer the following question:



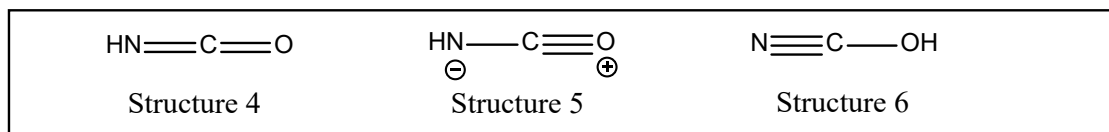
(I) Circle the *most contributing* resonance structure.

Structure 1

Structure 2

Structure 3

(ii) For the three structures given (4, 5, and 6) below, answer the following questions:



(I) Circle all the structures that represent *resonance structures* of each other:

Structure 4

Structure 5

Structure 6

(II) Among the resonance structures, circle the *most contributing* resonance structure.

Structure 4

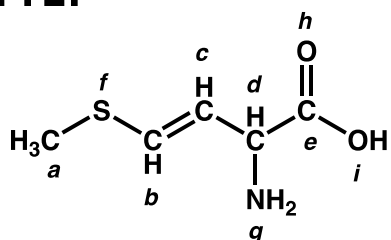
Structure 5

Structure 6

**Question 6** **6 points**

Using the data given below, calculate the **lattice energy** for SrO(s).

	ΔH° (kJ/mol)
$\text{Sr}(s) \longrightarrow \text{Sr}(g)$	+180
$\text{Sr}(g) \longrightarrow \text{Sr}^{2+}(g) + 2e^-$	+1468
$\text{O}_2(g) \longrightarrow 2\text{O}(g)$	+500
$\text{O}(g) + 2e^- \longrightarrow \text{O}^{2-}(g)$	+600
$\text{Sr}(s) + \frac{1}{2}\text{O}_2(g) \longrightarrow \text{SrO}(s)$	-550

**Question 7 10 points**

Consider the molecule given on the left, and answer questions (i) to (vi) using chemical bonding theories that we have learnt in class (valence bond theory, hybridization).

Note that the lone pairs are not shown in the structure:

i. Consider the carbon atoms marked in as ***a***, ***b***, ***c***, ***d***, and ***e*** in the molecule, and determine **hybridization** at each of the carbon centers. (2.5 points)

Atom	Hybridization
Carbon (<i>a</i>)	
Carbon (<i>b</i>)	
Carbon (<i>c</i>)	
Carbon (<i>d</i>)	
Carbon (<i>e</i>)	

ii. Now, consider the sulfur atom (S) marked as ***f***, nitrogen (N) atom (***g***), and oxygen(O) atoms (***h*** and ***i***), and determine their **hybridization**. (2 points)

Atom	Hybridization
Sulfur (<i>f</i>)	
Nitrogen (<i>g</i>)	
Oxygen (<i>h</i>)	
Oxygen (<i>i</i>)	

iii. **Describe** all the bonds formed by the carbon atoms ***a*** and ***b***, using sigma and pi designation as appropriate. (2 points)

Atom	Bonds formed (use sigma and pi as appropriate)
Carbon (<i>a</i>)	
Carbon (<i>b</i>)	

iv. How many total sp^3 -s sigma bonds are in the molecule? (1 point)

v. Determine **electronic** and **molecular geometry** at carbon (C) labeled ***e*** and sulfur (S) labeled ***f*** centres. (2 points)

Atom	Electronic Geometry	Molecular Geometry
Carbon (<i>e</i>)		
Sulfur (<i>f</i>)		

vi. Circle the value from the choices given below, that describes the **C-S-C bond angle at sulfur** (S) marked ***f***: (0.5 points)

>90° to <109.5° 109.5° >109.5 to <120° 120° >120° to <180°

**Question 8 12 points**

Using the **valence shell orbitals only** draw a **complete** molecular orbital (MO) diagram (with no s-p mixing) for the molecule **PS**, in the space provided below. **(6 points)**

Assume that 3s and 3p atomic orbitals will form molecular orbitals like those formed by the combination of 2s and 2p atomic orbitals that we learnt in class, and there is no s-p mixing.

Show **energy arrow**, **atomic orbitals** involved, **atom labels**, the **molecular orbitals** that are formed, **filled electrons** in the MO diagram etc. Make sure to label all the atomic and molecular orbitals.

Using the MO diagram above, answer the following questions:

- (i) Which one of the following will have the highest bond order? Circle the correct choice
A) PS^{1-} B) PS C) PS^{1+}
- (ii) Which one of the following will have the longest bond length? Circle the correct choice
A) PS^{1-} B) PS C) PS^{1+}
- (iii) Which of the following will be diamagnetic? Circle the correct choice.
A) PS^{1-} B) PS C) PS^{1+}
- (iv) Draw Lewis dot structure of the molecule with the shortest bond length (based on the MO diagram) from among the three (PS^{1-} , PS, PS^{1+}), and indicate the bond order calculated from the Lewis dot structure. **(3 points)**

Lewis Dot Structure

Bond Order



EB7170A4-C0C5-4CC0-8C78-82711BAF3326

midterm-exam-93388

#57 Page 10 of 10

SCRAP PAGE

**CHEMISTRY 110 –Midterm Exam****Thursday November 7, 2024, 6:30 – 8:30 pm**

Examiners: Pallavi Sirjoosingh and Ashok Kakkar

First name (please write as legibly as possible within the boxes)

--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--

Last name

--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--

Student ID number

--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--

This booklet has 8 questions on 10 pages (The last page can be used as a scrap page).

The staple must NOT be removed from the question booklet. Your intact question booklet must be returned at the end of the exam.

A separate booklet contains a periodic table, and datasheet.

Write your answers directly under the questions in the space provided. Do not write on the QR codes or near the top edges (as it may be cut off).

Please write **legibly and dark enough** for the scan to be readable. For students who use English as a second language, dictionaries are permitted. Calculators are allowed.

DO NOT ASK QUESTIONS OF THE INVIGILATORS about interpretations of the exam questions. If you have a concern about a question, please summarize it at the top of this page. All concerns will be evaluated after the exam. GOOD LUCK!

**Question 1 8 points** (*Assume all transitions are possible*)

a. 6 points Calculate the **energy (in kJ/mol)** required to ionize (completely remove) the electron from the **n=3** level of **He⁺ ion** (He: Helium). Show your work including equations. (3 sig. figs.)

b. 2 points

(I) Select the largest (longest) wavelength from the three choices below:

- i. Wavelength of photon to *completely remove* an electron from *1s orbital of Hydrogen atom*
- ii. Wavelength of photon to *completely remove* an electron from *1s orbital of Boron atom (B)*
- iii. Wavelength of photon to *completely remove* an electron from *1s orbital of Boron cation (B⁺)*

i or ii or iii

(II) Select the smallest (shortest) wavelength from the three choices below (*for H atom*):

- i. Wavelength of photon *emitted* for an electron transitioning from *2p to 1s orbital*
- ii. Wavelength of photon *absorbed* for an electron transitioning from *2s to 4p orbital*
- iii. Wavelength of photon *absorbed* for an electron transitioning from *1s to 3p orbital*

i or ii or iii



Question 2 8 points



A Chem110 student plots the graph shown on the left, from data obtained for the kinetic energy (eV) of the ejected electron versus frequency (s^{-1}) of the incident light, for an unknown metal. ($1\text{eV} = 1.6022 \times 10^{-19} \text{ J}$)

Based on the plot, answer the questions **a to c** below:

a. 1 point On the plot above, label the **threshold frequency**.

b. 2 points What is the value of the work potential (in J) of the unknown metal?

c. 3 points What is the *longest* wavelength (in nm) required to eject an electron from this metal's surface? Show your work including equation(s) used. (3 sig figs)

d. 2 points In the space provided below, plot the following

(Assume frequency is *above* threshold frequency):

- Kinetic energy* of an ejected electron versus the *intensity* of the incident light (left)
- Number of electrons* ejected versus the *frequency* of the incident light (right)



**Question 3** **6 points****a. 2 points**

In the ground state of Ge (Germanium; $Z = 32$) atom, **name** the following orbitals:

i. Orbital containing the electron that requires the maximum energy to be removed.

ii. Orbital containing the electron that experiences the minimum effective nuclear charge.

b. 2 points

An unknown anion (A^-) has the following *excited* electronic configuration: $[Ar] 4s^2 3d^3 4p^1$

(Ar: Argon; $Z = 18$)

Give the **condensed ground state electronic configuration** of the atom, **A** and the cation, **A^{2+}** .
Show your work. (Assume all electronic transitions are possible)

Electronic Configuration of A:

Electronic Configuration of A^{2+} :

c. 2 points

In a single atom, determine how many electrons can have the following set of quantum numbers?

(i) $n = 5; l = 3$

In a single atom, determine how many electrons can have the following set of quantum numbers?

(i) $n = 3; l = 2; m_s = +1/2$

**Question 4** **8 points**

Rank the following from 1 to 4 in **increasing** order. Provide the ranking in the boxes provided.

Note: Atomic number (Z) are provided for relevant atoms

a) Increasing (lowest to highest) order of *ionization energy*: (1= lowest; 4 = highest)

Mg

Mg: Z = 12

Rb

Rb: Z = 37

Cs

Cs: Z = 55

K

K : Z = 19

b) Increasing (lowest to highest) order of *bond angle*: (1= lowest; 4 = highest)

H₂OCCl₄XeF₄

HCN

c) Increasing (smallest to largest) order of *ionic size* (1= smallest; 4 = largest)

S²⁻

S: Z = 16

O²⁻

O: Z = 8

Be²⁺

Be: Z = 4

Na⁺

Na: Z = 11

d) Increasing (lowest to highest) order of *bond energy* (1= lowest; 4 = highest)

P-F

P-Cl

P-I

P-Br

**Question 5** **12 points**

a. 9 points For the molecules HClO_4 and TeF_6 , draw the most stable *Lewis structure* showing appropriate shape and *dashed/wedged bonds*, based on VSEPR. Determine the **molecular geometry** around the central atom. For HClO_4 , label the *most polar bond*.

HClO_4 (there are no O-O bonds)

Label the most polar bond

Molecular Geometry around Cl

TeF_6

Molecular Geometry around Te

b. 3 points (Note: Lone pairs are not shown in structures below)

(i) For the three *resonance* structures (1, 2 and 3) shown below, answer the following question:



(I) Circle the *most contributing* resonance structure.

Structure 1

Structure 2

Structure 3

(ii) For the three structures given (4, 5, and 6) below, answer the following questions:



(I) Circle all the structures that represent *resonance structures* of each other:

Structure 4

Structure 5

Structure 6

(II) Among the resonance structures, circle the *most contributing* resonance structure.

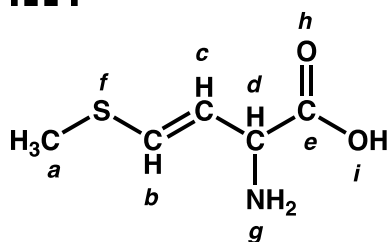
Structure 4

Structure 5

Structure 6

**Question 6** **6 points**Using the data given below, calculate the **lattice energy** for SrO(s).

	ΔH° (kJ/mol)
$\text{Sr}(s) \longrightarrow \text{Sr}(g)$	+180
$\text{Sr}(g) \longrightarrow \text{Sr}^{2+}(g) + 2e^-$	+1468
$\text{O}_2(g) \longrightarrow 2\text{O}(g)$	+500
$\text{O}(g) + 2e^- \longrightarrow \text{O}^{2-}(g)$	+600
$\text{Sr}(s) + \frac{1}{2}\text{O}_2(g) \longrightarrow \text{SrO}(s)$	-550

**Question 7 10 points**

Consider the molecule given on the left, and answer questions (i) to (vi) using chemical bonding theories that we have learnt in class (valence bond theory, hybridization).

Note that the lone pairs are not shown in the structure:

i. Consider the carbon atoms marked in as ***a***, ***b***, ***c***, ***d***, and ***e*** in the molecule, and determine **hybridization** at each of the carbon centers. (2.5 points)

Atom	Hybridization
Carbon (<i>a</i>)	
Carbon (<i>b</i>)	
Carbon (<i>c</i>)	
Carbon (<i>d</i>)	
Carbon (<i>e</i>)	

ii. Now, consider the sulfur atom (S) marked as ***f***, nitrogen (N) atom (***g***), and oxygen(O) atoms (***h*** and ***i***), and determine their **hybridization**. (2 points)

Atom	Hybridization
Sulfur (<i>f</i>)	
Nitrogen (<i>g</i>)	
Oxygen (<i>h</i>)	
Oxygen (<i>i</i>)	

iii. **Describe** all the bonds formed by the carbon atoms ***a*** and ***b***, using sigma and pi designation as appropriate. (2 points)

Atom	Bonds formed (use sigma and pi as appropriate)
Carbon (<i>a</i>)	
Carbon (<i>b</i>)	

iv. How many total sp^3 -s sigma bonds are in the molecule? (1 point)

v. Determine **electronic** and **molecular geometry** at carbon (C) labeled ***e*** and sulfur (S) labeled ***f*** centres. (2 points)

Atom	Electronic Geometry	Molecular Geometry
Carbon (<i>e</i>)		
Sulfur (<i>f</i>)		

vi. Circle the value from the choices given below, that describes the **C-S-C bond angle at sulfur** (S) marked ***f***: (0.5 points)

>90° to <109.5° 109.5° >109.5 to <120° 120° >120° to <180°


Question 8 12 points

Using the **valence shell orbitals only** draw a **complete** molecular orbital (MO) diagram (with no s-p mixing) for the molecule **PS**, in the space provided below. **(6 points)**

Assume that 3s and 3p atomic orbitals will form molecular orbitals like those formed by the combination of 2s and 2p atomic orbitals that we learnt in class, and there is no s-p mixing.

Show **energy arrow**, **atomic orbitals** involved, **atom labels**, the **molecular orbitals** that are formed, **filled electrons** in the MO diagram etc. Make sure to label all the atomic and molecular orbitals.

Using the MO diagram above, answer the following questions:

- (i) Which one of the following will have the highest bond order? Circle the correct choice
 A) PS^{1-} B) PS C) PS^{1+}
- (ii) Which one of the following will have the longest bond length? Circle the correct choice
 A) PS^{1-} B) PS C) PS^{1+}
- (iii) Which of the following will be diamagnetic? Circle the correct choice.
 A) PS^{1-} B) PS C) PS^{1+}
- (iv) Draw Lewis dot structure of the molecule with the shortest bond length (based on the MO diagram) from among the three (PS^{1-} , PS, PS^{1+}), and indicate the bond order calculated from the Lewis dot structure. **(3 points)**

Lewis Dot Structure

Bond Order



57C42520-92ED-41BE-9D8A-41BC7DDB3934

midterm-exam-93388

#58 Page 10 of 10

SCRAP PAGE

**CHEMISTRY 110 –Midterm Exam****Thursday November 7, 2024, 6:30 – 8:30 pm**

Examiners: Pallavi Sirjoosingh and Ashok Kakkar

First name (please write as legibly as possible within the boxes)

--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--

Last name

--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--

Student ID number

--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--

This booklet has 8 questions on 10 pages (The last page can be used as a scrap page).

The staple must NOT be removed from the question booklet. Your intact question booklet must be returned at the end of the exam.

A separate booklet contains a periodic table, and datasheet.

Write your answers directly under the questions in the space provided. Do not write on the QR codes or near the top edges (as it may be cut off).

Please write **legibly and dark enough** for the scan to be readable. For students who use English as a second language, dictionaries are permitted. Calculators are allowed.

DO NOT ASK QUESTIONS OF THE INVIGILATORS about interpretations of the exam questions. If you have a concern about a question, please summarize it at the top of this page. All concerns will be evaluated after the exam. GOOD LUCK!

**Question 1 8 points** (*Assume all transitions are possible*)

a. 6 points Calculate the **energy (in kJ/mol)** required to ionize (completely remove) the electron from the **n=3** level of **He⁺ ion** (He: Helium). Show your work including equations. (3 sig. figs.)

b. 2 points

(I) Select the largest (longest) wavelength from the three choices below:

- i. Wavelength of photon to *completely remove* an electron from *1s orbital of Hydrogen atom*
- ii. Wavelength of photon to *completely remove* an electron from *1s orbital of Boron atom (B)*
- iii. Wavelength of photon to *completely remove* an electron from *1s orbital of Boron cation (B⁺)*

i or ii or iii

(II) Select the smallest (shortest) wavelength from the three choices below (*for H atom*):

- i. Wavelength of photon *emitted* for an electron transitioning from *2p to 1s orbital*
- ii. Wavelength of photon *absorbed* for an electron transitioning from *2s to 4p orbital*
- iii. Wavelength of photon *absorbed* for an electron transitioning from *1s to 3p orbital*

i or ii or iii



Question 2 8 points



A Chem110 student plots the graph shown on the left, from data obtained for the kinetic energy (eV) of the ejected electron versus frequency (s^{-1}) of the incident light, for an unknown metal. ($1\text{eV} = 1.6022 \times 10^{-19} \text{ J}$)

Based on the plot, answer the questions **a to c** below:

a. 1 point On the plot above, label the **threshold frequency**.

b. 2 points What is the value of the work potential (in J) of the unknown metal?

c. 3 points What is the *longest* wavelength (in nm) required to eject an electron from this metal's surface? Show your work including equation(s) used. (3 sig figs)

d. 2 points In the space provided below, plot the following

(Assume frequency is *above* threshold frequency):

- Kinetic energy* of an ejected electron versus the *intensity* of the incident light (left)
- Number of electrons* ejected versus the *frequency* of the incident light (right)



**Question 3** **6 points****a. 2 points**

In the ground state of Ge (Germanium; $Z = 32$) atom, **name** the following orbitals:

i. Orbital containing the electron that requires the maximum energy to be removed.

ii. Orbital containing the electron that experiences the minimum effective nuclear charge.

b. 2 points

An unknown anion (A^-) has the following *excited* electronic configuration: $[Ar] 4s^2 3d^3 4p^1$

(Ar: Argon; $Z = 18$)

Give the **condensed ground state electronic configuration** of the atom, **A** and the cation, **A^{2+}** .
Show your work. (Assume all electronic transitions are possible)

Electronic Configuration of A:

Electronic Configuration of A^{2+} :

c. 2 points

In a single atom, determine how many electrons can have the following set of quantum numbers?

(i) $n = 5; l = 3$

In a single atom, determine how many electrons can have the following set of quantum numbers?

(i) $n = 3; l = 2; m_s = +1/2$

**Question 4** **8 points**

Rank the following from 1 to 4 in **increasing** order. Provide the ranking in the boxes provided.

Note: Atomic number (Z) are provided for relevant atoms

a) Increasing (lowest to highest) order of *ionization energy*: (1= lowest; 4 = highest)

Mg

Mg: Z = 12

Rb

Rb: Z = 37

Cs

Cs: Z = 55

K

K : Z = 19

b) Increasing (lowest to highest) order of *bond angle*: (1= lowest; 4 = highest)

H₂OCCl₄XeF₄

HCN

c) Increasing (smallest to largest) order of *ionic size* (1= smallest; 4 = largest)

S²⁻

S: Z = 16

O²⁻

O: Z = 8

Be²⁺

Be: Z = 4

Na⁺

Na: Z = 11

d) Increasing (lowest to highest) order of *bond energy* (1= lowest; 4 = highest)

P-F

P-Cl

P-I

P-Br

**Question 5** **12 points**

a. 9 points For the molecules HClO_4 and TeF_6 , draw the most stable *Lewis structure* showing appropriate shape and *dashed/wedged bonds*, based on VSEPR. Determine the **molecular geometry** around the central atom. For HClO_4 , label the *most polar bond*.

HClO_4 (there are no O-O bonds)

Label the most polar bond

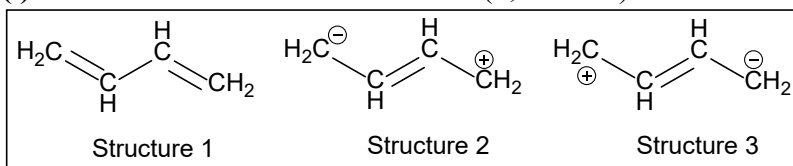
Molecular Geometry around Cl

TeF_6

Molecular Geometry around Te

b. 3 points (Note: Lone pairs are not shown in structures below)

(i) For the three *resonance* structures (1, 2 and 3) shown below, answer the following question:



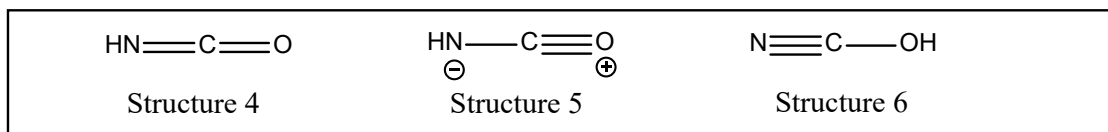
(I) Circle the *most contributing* resonance structure.

Structure 1

Structure 2

Structure 3

(ii) For the three structures given (4, 5, and 6) below, answer the following questions:



(I) Circle all the structures that represent *resonance structures* of each other:

Structure 4

Structure 5

Structure 6

(II) Among the resonance structures, circle the *most contributing* resonance structure.

Structure 4

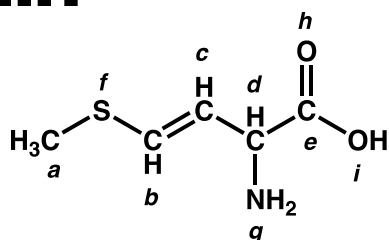
Structure 5

Structure 6

**Question 6** **6 points**

Using the data given below, calculate the **lattice energy** for SrO(s).

	ΔH° (kJ/mol)
$\text{Sr}(s) \longrightarrow \text{Sr}(g)$	+180
$\text{Sr}(g) \longrightarrow \text{Sr}^{2+}(g) + 2e^-$	+1468
$\text{O}_2(g) \longrightarrow 2\text{O}(g)$	+500
$\text{O}(g) + 2e^- \longrightarrow \text{O}^{2-}(g)$	+600
$\text{Sr}(s) + \frac{1}{2}\text{O}_2(g) \longrightarrow \text{SrO}(s)$	-550

**Question 7 10 points**

Consider the molecule given on the left, and answer questions (i) to (vi) using chemical bonding theories that we have learnt in class (valence bond theory, hybridization).

Note that the lone pairs are not shown in the structure:

i. Consider the carbon atoms marked in as ***a***, ***b***, ***c***, ***d***, and ***e*** in the molecule, and determine **hybridization** at each of the carbon centers. (2.5 points)

Atom	Hybridization
Carbon (<i>a</i>)	
Carbon (<i>b</i>)	
Carbon (<i>c</i>)	
Carbon (<i>d</i>)	
Carbon (<i>e</i>)	

ii. Now, consider the sulfur atom (S) marked as ***f***, nitrogen (N) atom (***g***), and oxygen(O) atoms (***h*** and ***i***), and determine their **hybridization**. (2 points)

Atom	Hybridization
Sulfur (<i>f</i>)	
Nitrogen (<i>g</i>)	
Oxygen (<i>h</i>)	
Oxygen (<i>i</i>)	

iii. **Describe** all the bonds formed by the carbon atoms ***a*** and ***b***, using sigma and pi designation as appropriate. (2 points)

Atom	Bonds formed (use sigma and pi as appropriate)
Carbon (<i>a</i>)	
Carbon (<i>b</i>)	

iv. How many total sp^3 -s sigma bonds are in the molecule? (1 point)

v. Determine **electronic** and **molecular geometry** at carbon (C) labeled ***e*** and sulfur (S) labeled ***f*** centres. (2 points)

Atom	Electronic Geometry	Molecular Geometry
Carbon (<i>e</i>)		
Sulfur (<i>f</i>)		

vi. Circle the value from the choices given below, that describes the **C-S-C bond angle at sulfur** (S) marked ***f***: (0.5 points)

>90° to <109.5°

109.5°

>109.5 to <120°

120°

>120° to <180°

**Question 8 12 points**

Using the **valence shell orbitals only** draw a **complete** molecular orbital (MO) diagram (with no s-p mixing) for the molecule **PS**, in the space provided below. **(6 points)**

Assume that 3s and 3p atomic orbitals will form molecular orbitals like those formed by the combination of 2s and 2p atomic orbitals that we learnt in class, and there is no s-p mixing.

Show **energy arrow**, **atomic orbitals** involved, **atom labels**, the **molecular orbitals** that are formed, **filled electrons** in the MO diagram etc. Make sure to label all the atomic and molecular orbitals.

Using the MO diagram above, answer the following questions:

- (i) Which one of the following will have the highest bond order? Circle the correct choice
A) PS^{1-} B) PS C) PS^{1+}
- (ii) Which one of the following will have the longest bond length? Circle the correct choice
A) PS^{1-} B) PS C) PS^{1+}
- (iii) Which of the following will be diamagnetic? Circle the correct choice.
A) PS^{1-} B) PS C) PS^{1+}
- (iv) Draw Lewis dot structure of the molecule with the shortest bond length (based on the MO diagram) from among the three (PS^{1-} , PS, PS^{1+}), and indicate the bond order calculated from the Lewis dot structure. **(3 points)**

Lewis Dot Structure

Bond Order



B58EB604-3FD1-46C7-849A-2B62D5CCE3E2

midterm-exam-93388

#59 Page 10 of 10

SCRAP PAGE

**CHEMISTRY 110 –Midterm Exam****Thursday November 7, 2024, 6:30 – 8:30 pm****Examiners: Pallavi Sirjoosingh and Ashok Kakkar**

First name (please write as legibly as possible within the boxes)

--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--

Last name

--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--

Student ID number

--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--

This booklet has 8 questions on 10 pages (The last page can be used as a scrap page).

The staple must NOT be removed from the question booklet. Your intact question booklet must be returned at the end of the exam.

A separate booklet contains a periodic table, and datasheet.

Write your answers directly under the questions in the space provided. Do not write on the QR codes or near the top edges (as it may be cut off).

Please write **legibly and dark enough** for the scan to be readable. For students who use English as a second language, dictionaries are permitted. Calculators are allowed.

DO NOT ASK QUESTIONS OF THE INVIGILATORS about interpretations of the exam questions. If you have a concern about a question, please summarize it at the top of this page. All concerns will be evaluated after the exam. GOOD LUCK!

**Question 1 8 points** (*Assume all transitions are possible*)

a. 6 points Calculate the **energy (in kJ/mol)** required to ionize (completely remove) the electron from the **n=3** level of **He⁺ ion** (He: Helium). Show your work including equations. (3 sig. figs.)

b. 2 points

(I) Select the largest (longest) wavelength from the three choices below:

- i. Wavelength of photon to *completely remove* an electron from *1s orbital of Hydrogen atom*
- ii. Wavelength of photon to *completely remove* an electron from *1s orbital of Boron atom (B)*
- iii. Wavelength of photon to *completely remove* an electron from *1s orbital of Boron cation (B⁺)*

i or ii or iii

(II) Select the smallest (shortest) wavelength from the three choices below (*for H atom*):

- i. Wavelength of photon *emitted* for an electron transitioning from *2p to 1s orbital*
- ii. Wavelength of photon *absorbed* for an electron transitioning from *2s to 4p orbital*
- iii. Wavelength of photon *absorbed* for an electron transitioning from *1s to 3p orbital*

i or ii or iii



Question 2 8 points



A Chem110 student plots the graph shown on the left, from data obtained for the kinetic energy (eV) of the ejected electron versus frequency (s^{-1}) of the incident light, for an unknown metal. ($1\text{eV} = 1.6022 \times 10^{-19} \text{ J}$)

Based on the plot, answer the questions **a to c** below:

a. 1 point On the plot above, label the **threshold frequency**.

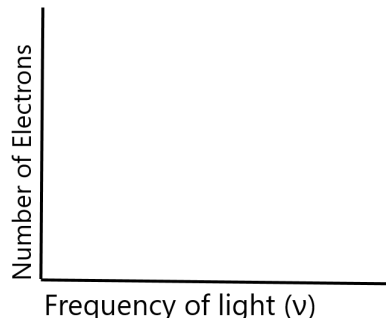
b. 2 points What is the value of the work potential (in J) of the unknown metal?

c. 3 points What is the *longest* wavelength (in nm) required to eject an electron from this metal's surface? Show your work including equation(s) used. (3 sig figs)

d. 2 points In the space provided below, plot the following

(Assume frequency is *above* threshold frequency):

- i. *Kinetic energy* of an ejected electron versus the *intensity* of the incident light (left)
- ii. *Number of electrons* ejected versus the *frequency* of the incident light (right)



**Question 3** **6 points****a. 2 points**

In the ground state of Ge (Germanium; $Z = 32$) atom, **name** the following orbitals:

i. Orbital containing the electron that requires the maximum energy to be removed.

ii. Orbital containing the electron that experiences the minimum effective nuclear charge.

b. 2 points

An unknown anion (A^-) has the following *excited* electronic configuration: $[Ar] 4s^2 3d^3 4p^1$

(Ar: Argon; $Z = 18$)

Give the **condensed ground state electronic configuration** of the atom, **A** and the cation, **A^{2+}** .
Show your work. (Assume all electronic transitions are possible)

Electronic Configuration of A:

Electronic Configuration of A^{2+} :

c. 2 points

In a single atom, determine how many electrons can have the following set of quantum numbers?

(i) $n = 5; l = 3$

In a single atom, determine how many electrons can have the following set of quantum numbers?

(i) $n = 3; l = 2; m_s = +1/2$

**Question 4** **8 points**

Rank the following from 1 to 4 in **increasing** order. Provide the ranking in the boxes provided.

Note: Atomic number (Z) are provided for relevant atoms

a) Increasing (lowest to highest) order of *ionization energy*: (1= lowest; 4 = highest)

Mg

Mg: Z = 12

Rb

Rb: Z = 37

Cs

Cs: Z = 55

K

K : Z = 19

b) Increasing (lowest to highest) order of *bond angle*: (1= lowest; 4 = highest)

H₂OCCl₄XeF₄

HCN

c) Increasing (smallest to largest) order of *ionic size* (1= smallest; 4 = largest)

S²⁻

S: Z = 16

O²⁻

O: Z = 8

Be²⁺

Be: Z = 4

Na⁺

Na: Z = 11

d) Increasing (lowest to highest) order of *bond energy* (1= lowest; 4 = highest)

P-F

P-Cl

P-I

P-Br

**Question 5** **12 points**

a. 9 points For the molecules HClO_4 and TeF_6 , draw the most stable *Lewis structure* showing appropriate shape and *dashed/wedged bonds*, based on VSEPR. Determine the **molecular geometry** around the central atom. For HClO_4 , label the *most polar bond*.



b. 3 points (Note: Lone pairs are not shown in structures below)

(i) For the three *resonance* structures (1, 2 and 3) shown below, answer the following question:



(I) Circle the *most contributing* resonance structure.

Structure 1

Structure 2

Structure 3

(ii) For the three structures given (4, 5, and 6) below, answer the following questions:



(I) Circle all the structures that represent *resonance structures* of each other:

Structure 4

Structure 5

Structure 6

(II) Among the resonance structures, circle the *most contributing* resonance structure.

Structure 4

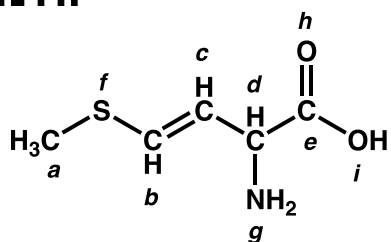
Structure 5

Structure 6

**Question 6** **6 points**

Using the data given below, calculate the **lattice energy** for SrO(s).

	ΔH° (kJ/mol)
$\text{Sr}(s) \longrightarrow \text{Sr}(g)$	+180
$\text{Sr}(g) \longrightarrow \text{Sr}^{2+}(g) + 2e^-$	+1468
$\text{O}_2(g) \longrightarrow 2\text{O}(g)$	+500
$\text{O}(g) + 2e^- \longrightarrow \text{O}^{2-}(g)$	+600
$\text{Sr}(s) + \frac{1}{2}\text{O}_2(g) \longrightarrow \text{SrO}(s)$	-550

**Question 7 10 points**

Consider the molecule given on the left, and answer questions (i) to (vi) using chemical bonding theories that we have learnt in class (valence bond theory, hybridization).

Note that the lone pairs are not shown in the structure:

i. Consider the carbon atoms marked in as ***a***, ***b***, ***c***, ***d***, and ***e*** in the molecule, and determine **hybridization** at each of the carbon centers. (2.5 points)

Atom	Hybridization
Carbon (<i>a</i>)	
Carbon (<i>b</i>)	
Carbon (<i>c</i>)	
Carbon (<i>d</i>)	
Carbon (<i>e</i>)	

ii. Now, consider the sulfur atom (S) marked as ***f***, nitrogen (N) atom (***g***), and oxygen(O) atoms (***h*** and ***i***), and determine their **hybridization**. (2 points)

Atom	Hybridization
Sulfur (<i>f</i>)	
Nitrogen (<i>g</i>)	
Oxygen (<i>h</i>)	
Oxygen (<i>i</i>)	

iii. **Describe** all the bonds formed by the carbon atoms ***a*** and ***b***, using sigma and pi designation as appropriate. (2 points)

Atom	Bonds formed (use sigma and pi as appropriate)
Carbon (<i>a</i>)	
Carbon (<i>b</i>)	

iv. How many total sp^3 -s sigma bonds are in the molecule? (1 point)

v. Determine **electronic** and **molecular geometry** at carbon (C) labeled ***e*** and sulfur (S) labeled ***f*** centres. (2 points)

Atom	Electronic Geometry	Molecular Geometry
Carbon (<i>e</i>)		
Sulfur (<i>f</i>)		

vi. Circle the value from the choices given below, that describes the **C-S-C bond angle at sulfur** (S) marked ***f***: (0.5 points)

>90° to <109.5° 109.5° >109.5 to <120° 120° >120° to <180°


Question 8 12 points

Using the **valence shell orbitals only** draw a **complete** molecular orbital (MO) diagram (with no s-p mixing) for the molecule **PS**, in the space provided below. **(6 points)**

Assume that 3s and 3p atomic orbitals will form molecular orbitals like those formed by the combination of 2s and 2p atomic orbitals that we learnt in class, and there is no s-p mixing.

Show **energy arrow**, **atomic orbitals** involved, **atom labels**, the **molecular orbitals** that are formed, **filled electrons** in the MO diagram etc. Make sure to label all the atomic and molecular orbitals.

Using the MO diagram above, answer the following questions:

- (i) Which one of the following will have the highest bond order? Circle the correct choice
 A) PS^{1-} B) PS C) PS^{1+}
- (ii) Which one of the following will have the longest bond length? Circle the correct choice
 A) PS^{1-} B) PS C) PS^{1+}
- (iii) Which of the following will be diamagnetic? Circle the correct choice.
 A) PS^{1-} B) PS C) PS^{1+}
- (iv) Draw Lewis dot structure of the molecule with the shortest bond length (based on the MO diagram) from among the three (PS^{1-} , PS, PS^{1+}), and indicate the bond order calculated from the Lewis dot structure. **(3 points)**

Lewis Dot Structure

Bond Order



1E0AD12A-73A3-4FB6-AA16-E053A45E152D

midterm-exam-93388

#60 Page 10 of 10

SCRAP PAGE

**CHEMISTRY 110 –Midterm Exam****Thursday November 7, 2024, 6:30 – 8:30 pm****Examiners: Pallavi Sirjoosingh and Ashok Kakkar**

First name (please write as legibly as possible within the boxes)

--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--

Last name

--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--

Student ID number

--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--

This booklet has 8 questions on 10 pages (The last page can be used as a scrap page).

The staple must NOT be removed from the question booklet. Your intact question booklet must be returned at the end of the exam.

A separate booklet contains a periodic table, and datasheet.

Write your answers directly under the questions in the space provided. Do not write on the QR codes or near the top edges (as it may be cut off).

Please write **legibly and dark enough** for the scan to be readable. For students who use English as a second language, dictionaries are permitted. Calculators are allowed.

DO NOT ASK QUESTIONS OF THE INVIGILATORS about interpretations of the exam questions. If you have a concern about a question, please summarize it at the top of this page. All concerns will be evaluated after the exam. GOOD LUCK!

**Question 1 8 points** (*Assume all transitions are possible*)

a. 6 points Calculate the **energy (in kJ/mol)** required to ionize (completely remove) the electron from the **n=3** level of **He⁺ ion** (He: Helium). Show your work including equations. (3 sig. figs.)

b. 2 points

(I) Select the largest (longest) wavelength from the three choices below:

- i. Wavelength of photon to *completely remove* an electron from *1s orbital of Hydrogen atom*
- ii. Wavelength of photon to *completely remove* an electron from *1s orbital of Boron atom (B)*
- iii. Wavelength of photon to *completely remove* an electron from *1s orbital of Boron cation (B⁺)*

i or ii or iii

(II) Select the smallest (shortest) wavelength from the three choices below (*for H atom*):

- i. Wavelength of photon *emitted* for an electron transitioning from *2p to 1s orbital*
- ii. Wavelength of photon *absorbed* for an electron transitioning from *2s to 4p orbital*
- iii. Wavelength of photon *absorbed* for an electron transitioning from *1s to 3p orbital*

i or ii or iii



Question 2 8 points



A Chem110 student plots the graph shown on the left, from data obtained for the kinetic energy (eV) of the ejected electron versus frequency (s^{-1}) of the incident light, for an unknown metal. ($1\text{eV} = 1.6022 \times 10^{-19} \text{ J}$)

Based on the plot, answer the questions **a to c** below:

a. 1 point On the plot above, label the **threshold frequency**.

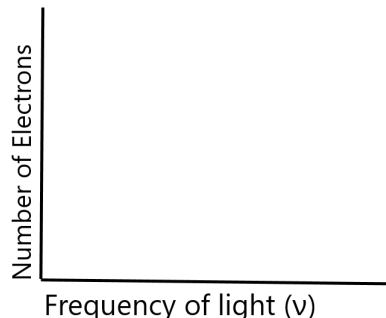
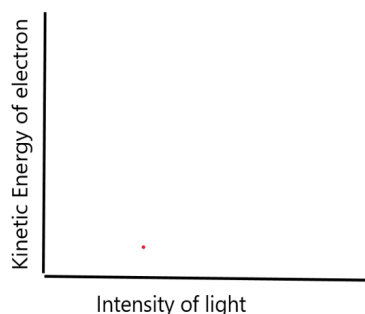
b. 2 points What is the value of the work potential (in J) of the unknown metal?

c. 3 points What is the *longest* wavelength (in nm) required to eject an electron from this metal's surface? Show your work including equation(s) used. (3 sig figs)

d. 2 points In the space provided below, plot the following

(Assume frequency is *above* threshold frequency):

- i. *Kinetic energy* of an ejected electron versus the *intensity* of the incident light (left)
- ii. *Number of electrons* ejected versus the *frequency* of the incident light (right)



**Question 3** **6 points****a. 2 points**

In the ground state of Ge (Germanium; $Z = 32$) atom, **name** the following orbitals:

i. Orbital containing the electron that requires the maximum energy to be removed.

ii. Orbital containing the electron that experiences the minimum effective nuclear charge.

b. 2 points

An unknown anion (A^-) has the following *excited* electronic configuration: $[Ar] 4s^2 3d^3 4p^1$

(Ar: Argon; $Z = 18$)

Give the **condensed ground state electronic configuration** of the atom, **A** and the cation, **A^{2+}** .
Show your work. (Assume all electronic transitions are possible)

Electronic Configuration of A:

Electronic Configuration of A^{2+} :

c. 2 points

In a single atom, determine how many electrons can have the following set of quantum numbers?

(i) $n = 5; l = 3$

In a single atom, determine how many electrons can have the following set of quantum numbers?

(i) $n = 3; l = 2; m_s = +1/2$

**Question 4** **8 points**

Rank the following from 1 to 4 in **increasing** order. Provide the ranking in the boxes provided.

Note: Atomic number (Z) are provided for relevant atoms

a) Increasing (lowest to highest) order of *ionization energy*: (1= lowest; 4 = highest)

Mg

Mg: Z = 12

Rb

Rb: Z = 37

Cs

Cs: Z = 55

K

K : Z = 19

b) Increasing (lowest to highest) order of *bond angle*: (1= lowest; 4 = highest)

H₂OCCl₄XeF₄

HCN

c) Increasing (smallest to largest) order of *ionic size* (1= smallest; 4 = largest)

S²⁻

S: Z = 16

O²⁻

O: Z = 8

Be²⁺

Be: Z = 4

Na⁺

Na: Z = 11

d) Increasing (lowest to highest) order of *bond energy* (1= lowest; 4 = highest)

P-F

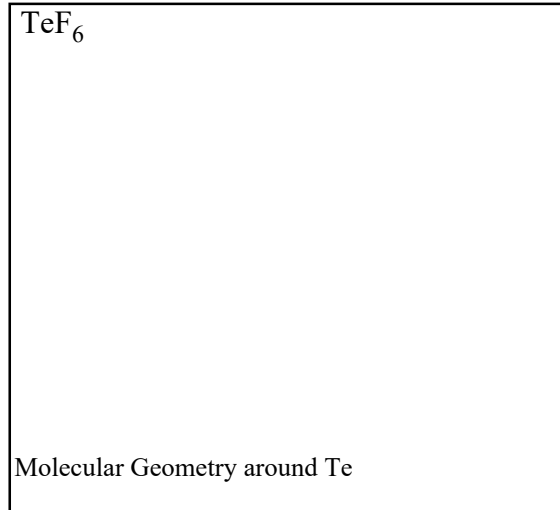
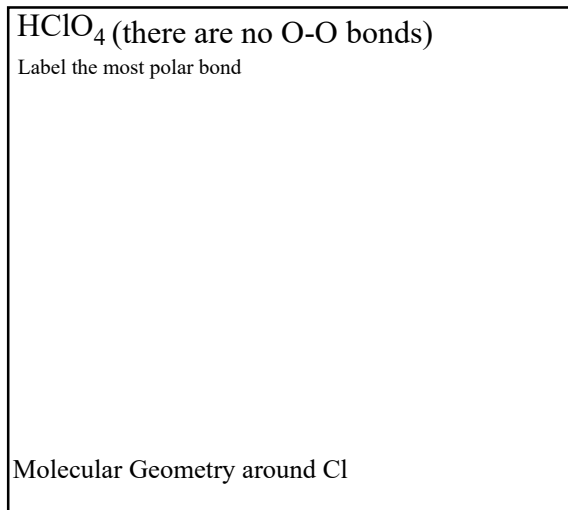
P-Cl

P-I

P-Br

**Question 5** **12 points**

a. 9 points For the molecules HClO_4 and TeF_6 , draw the most stable *Lewis structure* showing appropriate shape and *dashed/wedged bonds*, based on VSEPR. Determine the **molecular geometry** around the central atom. For HClO_4 , label the *most polar bond*.



b. 3 points (Note: Lone pairs are not shown in structures below)

(i) For the three *resonance* structures (1, 2 and 3) shown below, answer the following question:



(I) Circle the *most contributing* resonance structure.

Structure 1

Structure 2

Structure 3

(ii) For the three structures given (4, 5, and 6) below, answer the following questions:



(I) Circle all the structures that represent *resonance structures* of each other:

Structure 4

Structure 5

Structure 6

(II) Among the resonance structures, circle the *most contributing* resonance structure.

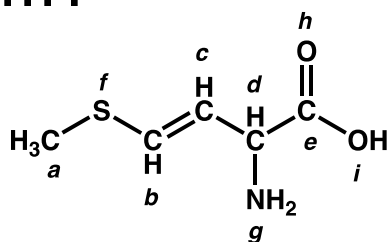
Structure 4

Structure 5

Structure 6

**Question 6** **6 points**Using the data given below, calculate the **lattice energy** for SrO(s).

	ΔH° (kJ/mol)
$\text{Sr}(s) \longrightarrow \text{Sr}(g)$	+180
$\text{Sr}(g) \longrightarrow \text{Sr}^{2+}(g) + 2e^-$	+1468
$\text{O}_2(g) \longrightarrow 2\text{O}(g)$	+500
$\text{O}(g) + 2e^- \longrightarrow \text{O}^{2-}(g)$	+600
$\text{Sr}(s) + \frac{1}{2}\text{O}_2(g) \longrightarrow \text{SrO}(s)$	-550

**Question 7 10 points**

Consider the molecule given on the left, and answer questions (i) to (vi) using chemical bonding theories that we have learnt in class (valence bond theory, hybridization).

Note that the lone pairs are not shown in the structure:

i. Consider the carbon atoms marked in as ***a***, ***b***, ***c***, ***d***, and ***e*** in the molecule, and determine **hybridization** at each of the carbon centers. (2.5 points)

Atom	Hybridization
Carbon (<i>a</i>)	
Carbon (<i>b</i>)	
Carbon (<i>c</i>)	
Carbon (<i>d</i>)	
Carbon (<i>e</i>)	

ii. Now, consider the sulfur atom (S) marked as ***f***, nitrogen (N) atom (***g***), and oxygen(O) atoms (***h*** and ***i***), and determine their **hybridization**. (2 points)

Atom	Hybridization
Sulfur (<i>f</i>)	
Nitrogen (<i>g</i>)	
Oxygen (<i>h</i>)	
Oxygen (<i>i</i>)	

iii. **Describe** all the bonds formed by the carbon atoms ***a*** and ***b***, using sigma and pi designation as appropriate. (2 points)

Atom	Bonds formed (use sigma and pi as appropriate)
Carbon (<i>a</i>)	
Carbon (<i>b</i>)	

iv. How many total sp^3 -s sigma bonds are in the molecule? (1 point)

v. Determine **electronic** and **molecular geometry** at carbon (C) labeled ***e*** and sulfur (S) labeled ***f*** centres. (2 points)

Atom	Electronic Geometry	Molecular Geometry
Carbon (<i>e</i>)		
Sulfur (<i>f</i>)		

vi. Circle the value from the choices given below, that describes the **C-S-C bond angle at sulfur** (S) marked ***f***: (0.5 points)

>90° to <109.5°

109.5°

>109.5 to <120°

120°

>120° to <180°


Question 8 12 points

Using the **valence shell orbitals only** draw a **complete** molecular orbital (MO) diagram (with no s-p mixing) for the molecule **PS**, in the space provided below. **(6 points)**

Assume that 3s and 3p atomic orbitals will form molecular orbitals like those formed by the combination of 2s and 2p atomic orbitals that we learnt in class, and there is no s-p mixing.

Show **energy arrow**, **atomic orbitals** involved, **atom labels**, the **molecular orbitals** that are formed, **filled electrons** in the MO diagram etc. Make sure to label all the atomic and molecular orbitals.

Using the MO diagram above, answer the following questions:

- (i) Which one of the following will have the highest bond order? Circle the correct choice
 A) PS^{1-} B) PS C) PS^{1+}
- (ii) Which one of the following will have the longest bond length? Circle the correct choice
 A) PS^{1-} B) PS C) PS^{1+}
- (iii) Which of the following will be diamagnetic? Circle the correct choice.
 A) PS^{1-} B) PS C) PS^{1+}
- (iv) Draw Lewis dot structure of the molecule with the shortest bond length (based on the MO diagram) from among the three (PS^{1-} , PS, PS^{1+}), and indicate the bond order calculated from the Lewis dot structure. **(3 points)**

Lewis Dot Structure

Bond Order



1CCBE0AF-C67D-4556-A38E-1FFA48A2F609

midterm-exam-93388

#61 Page 10 of 10

SCRAP PAGE

**CHEMISTRY 110 –Midterm Exam****Thursday November 7, 2024, 6:30 – 8:30 pm**

Examiners: Pallavi Sirjoosingh and Ashok Kakkar

First name (please write as legibly as possible within the boxes)

--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--

Last name

--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--

Student ID number

--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--

This booklet has 8 questions on 10 pages (The last page can be used as a scrap page).

The staple must NOT be removed from the question booklet. Your intact question booklet must be returned at the end of the exam.

A separate booklet contains a periodic table, and datasheet.

Write your answers directly under the questions in the space provided. Do not write on the QR codes or near the top edges (as it may be cut off).

Please write **legibly and dark enough** for the scan to be readable. For students who use English as a second language, dictionaries are permitted. Calculators are allowed.

DO NOT ASK QUESTIONS OF THE INVIGILATORS about interpretations of the exam questions. If you have a concern about a question, please summarize it at the top of this page. All concerns will be evaluated after the exam. GOOD LUCK!

**Question 1 8 points** (*Assume all transitions are possible*)

a. 6 points Calculate the **energy (in kJ/mol)** required to ionize (completely remove) the electron from the **n=3** level of **He⁺ ion** (He: Helium). Show your work including equations. (3 sig. figs.)

b. 2 points

(I) Select the largest (longest) wavelength from the three choices below:

- i. Wavelength of photon to *completely remove* an electron from *1s orbital of Hydrogen atom*
- ii. Wavelength of photon to *completely remove* an electron from *1s orbital of Boron atom (B)*
- iii. Wavelength of photon to *completely remove* an electron from *1s orbital of Boron cation (B⁺)*

i or ii or iii

(II) Select the smallest (shortest) wavelength from the three choices below (*for H atom*):

- i. Wavelength of photon *emitted* for an electron transitioning from *2p to 1s orbital*
- ii. Wavelength of photon *absorbed* for an electron transitioning from *2s to 4p orbital*
- iii. Wavelength of photon *absorbed* for an electron transitioning from *1s to 3p orbital*

i or ii or iii



Question 2 8 points



A Chem110 student plots the graph shown on the left, from data obtained for the kinetic energy (eV) of the ejected electron versus frequency (s^{-1}) of the incident light, for an unknown metal. ($1\text{eV} = 1.6022 \times 10^{-19} \text{ J}$)

Based on the plot, answer the questions **a to c** below:

a. 1 point On the plot above, label the **threshold frequency**.

b. 2 points What is the value of the work potential (in J) of the unknown metal?

c. 3 points What is the *longest* wavelength (in nm) required to eject an electron from this metal's surface? Show your work including equation(s) used. (3 sig figs)

d. 2 points In the space provided below, plot the following

(Assume frequency is *above* threshold frequency):

- i. *Kinetic energy* of an ejected electron versus the *intensity* of the incident light (left)
- ii. *Number of electrons* ejected versus the *frequency* of the incident light (right)



**Question 3** **6 points****a. 2 points**

In the ground state of Ge (Germanium; $Z = 32$) atom, **name** the following orbitals:

i. Orbital containing the electron that requires the maximum energy to be removed.

ii. Orbital containing the electron that experiences the minimum effective nuclear charge.

b. 2 points

An unknown anion (A^-) has the following *excited* electronic configuration: $[Ar] 4s^2 3d^3 4p^1$

(Ar: Argon; $Z = 18$)

Give the **condensed ground state electronic configuration** of the atom, **A** and the cation, **A^{2+}** .
Show your work. (Assume all electronic transitions are possible)

Electronic Configuration of A:

Electronic Configuration of A^{2+} :

c. 2 points

In a single atom, determine how many electrons can have the following set of quantum numbers?

(i) $n = 5; l = 3$

In a single atom, determine how many electrons can have the following set of quantum numbers?

(i) $n = 3; l = 2; m_s = +1/2$

**Question 4** **8 points**

Rank the following from 1 to 4 in **increasing** order. Provide the ranking in the boxes provided.

Note: Atomic number (Z) are provided for relevant atoms

a) Increasing (lowest to highest) order of *ionization energy*: (1= lowest; 4 = highest)

Mg

Mg: Z = 12

Rb

Rb: Z = 37

Cs

Cs: Z = 55

K

K : Z = 19

b) Increasing (lowest to highest) order of *bond angle*: (1= lowest; 4 = highest)

H₂OCCl₄XeF₄

HCN

c) Increasing (smallest to largest) order of *ionic size* (1= smallest; 4 = largest)

S²⁻

S: Z = 16

O²⁻

O: Z = 8

Be²⁺

Be: Z = 4

Na⁺

Na: Z = 11

d) Increasing (lowest to highest) order of *bond energy* (1= lowest; 4 = highest)

P-F

P-Cl

P-I

P-Br

**Question 5** **12 points**

a. 9 points For the molecules HClO_4 and TeF_6 , draw the most stable *Lewis structure* showing appropriate shape and *dashed/wedged bonds*, based on VSEPR. Determine the **molecular geometry** around the central atom. For HClO_4 , label the *most polar bond*.



b. 3 points (Note: Lone pairs are not shown in structures below)

(i) For the three *resonance* structures (1, 2 and 3) shown below, answer the following question:



(I) Circle the *most contributing* resonance structure.

Structure 1

Structure 2

Structure 3

(ii) For the three structures given (4, 5, and 6) below, answer the following questions:



(I) Circle all the structures that represent *resonance structures* of each other:

Structure 4

Structure 5

Structure 6

(II) Among the resonance structures, circle the *most contributing* resonance structure.

Structure 4

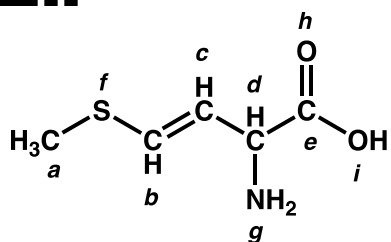
Structure 5

Structure 6

**Question 6** **6 points**

Using the data given below, calculate the **lattice energy** for SrO(s).

	ΔH° (kJ/mol)
$\text{Sr}(s) \longrightarrow \text{Sr}(g)$	+180
$\text{Sr}(g) \longrightarrow \text{Sr}^{2+}(g) + 2e^-$	+1468
$\text{O}_2(g) \longrightarrow 2\text{O}(g)$	+500
$\text{O}(g) + 2e^- \longrightarrow \text{O}^{2-}(g)$	+600
$\text{Sr}(s) + \frac{1}{2}\text{O}_2(g) \longrightarrow \text{SrO}(s)$	-550

**Question 7 10 points**

Consider the molecule given on the left, and answer questions (i) to (vi) using chemical bonding theories that we have learnt in class (valence bond theory, hybridization).

Note that the lone pairs are not shown in the structure:

i. Consider the carbon atoms marked in as ***a***, ***b***, ***c***, ***d***, and ***e*** in the molecule, and determine **hybridization** at each of the carbon centers. (2.5 points)

Atom	Hybridization
Carbon (<i>a</i>)	
Carbon (<i>b</i>)	
Carbon (<i>c</i>)	
Carbon (<i>d</i>)	
Carbon (<i>e</i>)	

ii. Now, consider the sulfur atom (S) marked as ***f***, nitrogen (N) atom (***g***), and oxygen(O) atoms (***h*** and ***i***), and determine their **hybridization**. (2 points)

Atom	Hybridization
Sulfur (<i>f</i>)	
Nitrogen (<i>g</i>)	
Oxygen (<i>h</i>)	
Oxygen (<i>i</i>)	

iii. **Describe** all the bonds formed by the carbon atoms ***a*** and ***b***, using sigma and pi designation as appropriate. (2 points)

Atom	Bonds formed (use sigma and pi as appropriate)
Carbon (<i>a</i>)	
Carbon (<i>b</i>)	

iv. How many total sp^3 -s sigma bonds are in the molecule? (1 point)

v. Determine **electronic** and **molecular geometry** at carbon (C) labeled ***e*** and sulfur (S) labeled ***f*** centres. (2 points)

Atom	Electronic Geometry	Molecular Geometry
Carbon (<i>e</i>)		
Sulfur (<i>f</i>)		

vi. Circle the value from the choices given below, that describes the **C-S-C bond angle at sulfur** (S) marked ***f***: (0.5 points)

>90° to <109.5° 109.5° >109.5 to <120° 120° >120° to <180°


Question 8 12 points

Using the **valence shell orbitals only** draw a **complete** molecular orbital (MO) diagram (with no s-p mixing) for the molecule **PS**, in the space provided below. **(6 points)**

Assume that 3s and 3p atomic orbitals will form molecular orbitals like those formed by the combination of 2s and 2p atomic orbitals that we learnt in class, and there is no s-p mixing.

Show **energy arrow**, **atomic orbitals** involved, **atom labels**, the **molecular orbitals** that are formed, **filled electrons** in the MO diagram etc. Make sure to label all the atomic and molecular orbitals.

Using the MO diagram above, answer the following questions:

- (i) Which one of the following will have the highest bond order? Circle the correct choice
 A) PS^{1-} B) PS C) PS^{1+}
- (ii) Which one of the following will have the longest bond length? Circle the correct choice
 A) PS^{1-} B) PS C) PS^{1+}
- (iii) Which of the following will be diamagnetic? Circle the correct choice.
 A) PS^{1-} B) PS C) PS^{1+}
- (iv) Draw Lewis dot structure of the molecule with the shortest bond length (based on the MO diagram) from among the three (PS^{1-} , PS, PS^{1+}), and indicate the bond order calculated from the Lewis dot structure. **(3 points)**

Lewis Dot Structure

Bond Order



40AFAADF-17AF-4A6C-9591-90D88A6EEFDB

midterm-exam-93388

#62 Page 10 of 10

SCRAP PAGE

**CHEMISTRY 110 –Midterm Exam****Thursday November 7, 2024, 6:30 – 8:30 pm****Examiners: Pallavi Sirjoosingh and Ashok Kakkar**

First name (please write as legibly as possible within the boxes)

--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--

Last name

--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--

Student ID number

--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--

This booklet has 8 questions on 10 pages (The last page can be used as a scrap page).

The staple must NOT be removed from the question booklet. Your intact question booklet must be returned at the end of the exam.

A separate booklet contains a periodic table, and datasheet.

Write your answers directly under the questions in the space provided. Do not write on the QR codes or near the top edges (as it may be cut off).

Please write **legibly and dark enough** for the scan to be readable. For students who use English as a second language, dictionaries are permitted. Calculators are allowed.

DO NOT ASK QUESTIONS OF THE INVIGILATORS about interpretations of the exam questions. If you have a concern about a question, please summarize it at the top of this page. All concerns will be evaluated after the exam. GOOD LUCK!

**Question 1 8 points** (*Assume all transitions are possible*)

a. 6 points Calculate the **energy (in kJ/mol)** required to ionize (completely remove) the electron from the **n=3** level of **He⁺ ion** (He: Helium). Show your work including equations. (3 sig. figs.)

b. 2 points

(I) Select the largest (longest) wavelength from the three choices below:

- i. Wavelength of photon to *completely remove* an electron from *1s orbital of Hydrogen atom*
- ii. Wavelength of photon to *completely remove* an electron from *1s orbital of Boron atom (B)*
- iii. Wavelength of photon to *completely remove* an electron from *1s orbital of Boron cation (B⁺)*

i or ii or iii

(II) Select the smallest (shortest) wavelength from the three choices below (*for H atom*):

- i. Wavelength of photon *emitted* for an electron transitioning from *2p to 1s orbital*
- ii. Wavelength of photon *absorbed* for an electron transitioning from *2s to 4p orbital*
- iii. Wavelength of photon *absorbed* for an electron transitioning from *1s to 3p orbital*

i or ii or iii



Question 2 8 points



A Chem110 student plots the graph shown on the left, from data obtained for the kinetic energy (eV) of the ejected electron versus frequency (s^{-1}) of the incident light, for an unknown metal. ($1\text{eV} = 1.6022 \times 10^{-19} \text{ J}$)

Based on the plot, answer the questions **a to c** below:

a. 1 point On the plot above, label the **threshold frequency**.

b. 2 points What is the value of the work potential (in J) of the unknown metal?

c. 3 points What is the *longest* wavelength (in nm) required to eject an electron from this metal's surface? Show your work including equation(s) used. (3 sig figs)

d. 2 points In the space provided below, plot the following

(Assume frequency is *above* threshold frequency):

- Kinetic energy* of an ejected electron versus the *intensity* of the incident light (left)
- Number of electrons* ejected versus the *frequency* of the incident light (right)



**Question 3** **6 points****a. 2 points**

In the ground state of Ge (Germanium; $Z = 32$) atom, **name** the following orbitals:

i. Orbital containing the electron that requires the maximum energy to be removed.

ii. Orbital containing the electron that experiences the minimum effective nuclear charge.

b. 2 points

An unknown anion (A^-) has the following *excited* electronic configuration: $[Ar] 4s^2 3d^3 4p^1$

(Ar: Argon; $Z = 18$)

Give the **condensed ground state electronic configuration** of the atom, **A** and the cation, **A^{2+}** .
Show your work. (Assume all electronic transitions are possible)

Electronic Configuration of A:

Electronic Configuration of A^{2+} :

c. 2 points

In a single atom, determine how many electrons can have the following set of quantum numbers?

(i) $n = 5; l = 3$

In a single atom, determine how many electrons can have the following set of quantum numbers?

(i) $n = 3; l = 2; m_s = +1/2$

**Question 4** **8 points**

Rank the following from 1 to 4 in **increasing** order. Provide the ranking in the boxes provided.

Note: Atomic number (Z) are provided for relevant atoms

a) Increasing (lowest to highest) order of *ionization energy*: (1= lowest; 4 = highest)

Mg

Mg: Z = 12

Rb

Rb: Z = 37

Cs

Cs: Z = 55

K

K : Z = 19

b) Increasing (lowest to highest) order of *bond angle*: (1= lowest; 4 = highest)

H₂OCCl₄XeF₄

HCN

c) Increasing (smallest to largest) order of *ionic size* (1= smallest; 4 = largest)

S²⁻

S: Z = 16

O²⁻

O: Z = 8

Be²⁺

Be: Z = 4

Na⁺

Na: Z = 11

d) Increasing (lowest to highest) order of *bond energy* (1= lowest; 4 = highest)

P-F

P-Cl

P-I

P-Br

**Question 5** **12 points**

a. 9 points For the molecules HClO_4 and TeF_6 , draw the most stable *Lewis structure* showing appropriate shape and *dashed/wedged bonds*, based on VSEPR. Determine the **molecular geometry** around the central atom. For HClO_4 , label the *most polar bond*.



b. 3 points (Note: Lone pairs are not shown in structures below)

(i) For the three *resonance* structures (1, 2 and 3) shown below, answer the following question:



(I) Circle the *most contributing* resonance structure.

Structure 1

Structure 2

Structure 3

(ii) For the three structures given (4, 5, and 6) below, answer the following questions:



(I) Circle all the structures that represent *resonance structures* of each other:

Structure 4

Structure 5

Structure 6

(II) Among the resonance structures, circle the *most contributing* resonance structure.

Structure 4

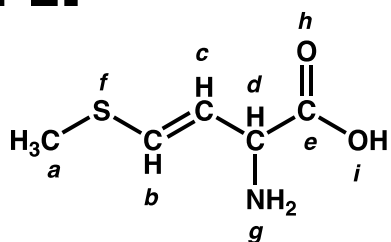
Structure 5

Structure 6

**Question 6** **6 points**

Using the data given below, calculate the **lattice energy** for SrO(s).

	ΔH° (kJ/mol)
$\text{Sr}(s) \longrightarrow \text{Sr}(g)$	+180
$\text{Sr}(g) \longrightarrow \text{Sr}^{2+}(g) + 2e^-$	+1468
$\text{O}_2(g) \longrightarrow 2\text{O}(g)$	+500
$\text{O}(g) + 2e^- \longrightarrow \text{O}^{2-}(g)$	+600
$\text{Sr}(s) + \frac{1}{2}\text{O}_2(g) \longrightarrow \text{SrO}(s)$	-550

**Question 7 10 points**

Consider the molecule given on the left, and answer questions (i) to (vi) using chemical bonding theories that we have learnt in class (valence bond theory, hybridization).

Note that the lone pairs are not shown in the structure:

i. Consider the carbon atoms marked in as ***a***, ***b***, ***c***, ***d***, and ***e*** in the molecule, and determine **hybridization** at each of the carbon centers. (2.5 points)

Atom	Hybridization
Carbon (<i>a</i>)	
Carbon (<i>b</i>)	
Carbon (<i>c</i>)	
Carbon (<i>d</i>)	
Carbon (<i>e</i>)	

ii. Now, consider the sulfur atom (S) marked as ***f***, nitrogen (N) atom (***g***), and oxygen(O) atoms (***h*** and ***i***), and determine their **hybridization**. (2 points)

Atom	Hybridization
Sulfur (<i>f</i>)	
Nitrogen (<i>g</i>)	
Oxygen (<i>h</i>)	
Oxygen (<i>i</i>)	

iii. **Describe** all the bonds formed by the carbon atoms ***a*** and ***b***, using sigma and pi designation as appropriate. (2 points)

Atom	Bonds formed (use sigma and pi as appropriate)
Carbon (<i>a</i>)	
Carbon (<i>b</i>)	

iv. How many total sp^3 -s sigma bonds are in the molecule? (1 point)

v. Determine **electronic** and **molecular geometry** at carbon (C) labeled ***e*** and sulfur (S) labeled ***f*** centres. (2 points)

Atom	Electronic Geometry	Molecular Geometry
Carbon (<i>e</i>)		
Sulfur (<i>f</i>)		

vi. Circle the value from the choices given below, that describes the **C-S-C bond angle at sulfur** (S) marked ***f***: (0.5 points)

>90° to <109.5° 109.5° >109.5 to <120° 120° >120° to <180°



Question 8 12 points

Using the **valence shell orbitals only** draw a **complete** molecular orbital (MO) diagram (with no s-p mixing) for the molecule **PS**, in the space provided below. **(6 points)**

Assume that 3s and 3p atomic orbitals will form molecular orbitals like those formed by the combination of 2s and 2p atomic orbitals that we learnt in class, and there is no s-p mixing.

Show **energy arrow**, **atomic orbitals** involved, **atom labels**, the **molecular orbitals** that are formed, **filled electrons** in the MO diagram etc. Make sure to label all the atomic and molecular orbitals.

Using the MO diagram above, answer the following questions:

- (i) Which one of the following will have the highest bond order? Circle the correct choice
 A) PS^{1-} B) PS C) PS^{1+}
- (ii) Which one of the following will have the longest bond length? Circle the correct choice
 A) PS^{1-} B) PS C) PS^{1+}
- (iii) Which of the following will be diamagnetic? Circle the correct choice.
 A) PS^{1-} B) PS C) PS^{1+}
- (iv) Draw Lewis dot structure of the molecule with the shortest bond length (based on the MO diagram) from among the three (PS^{1-} , PS, PS^{1+}), and indicate the bond order calculated from the Lewis dot structure. **(3 points)**

Lewis Dot Structure

Bond Order



7AB5C9E1-3F2C-4EC7-8E2B-9E8B766CB35B

midterm-exam-93388

#63 Page 10 of 10

SCRAP PAGE

**CHEMISTRY 110 –Midterm Exam****Thursday November 7, 2024, 6:30 – 8:30 pm**

Examiners: Pallavi Sirjoosingh and Ashok Kakkar

First name (please write as legibly as possible within the boxes)

--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--

Last name

--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--

Student ID number

--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--

This booklet has 8 questions on 10 pages (The last page can be used as a scrap page).

The staple must NOT be removed from the question booklet. Your intact question booklet must be returned at the end of the exam.

A separate booklet contains a periodic table, and datasheet.

Write your answers directly under the questions in the space provided. Do not write on the QR codes or near the top edges (as it may be cut off).

Please write **legibly and dark enough** for the scan to be readable. For students who use English as a second language, dictionaries are permitted. Calculators are allowed.

DO NOT ASK QUESTIONS OF THE INVIGILATORS about interpretations of the exam questions. If you have a concern about a question, please summarize it at the top of this page. All concerns will be evaluated after the exam. GOOD LUCK!

**Question 1 8 points** (*Assume all transitions are possible*)

a. 6 points Calculate the **energy (in kJ/mol)** required to ionize (completely remove) the electron from the **n=3** level of **He⁺ ion** (He: Helium). Show your work including equations. (3 sig. figs.)

b. 2 points

(I) Select the largest (longest) wavelength from the three choices below:

- i. Wavelength of photon to *completely remove* an electron from *1s orbital of Hydrogen atom*
- ii. Wavelength of photon to *completely remove* an electron from *1s orbital of Boron atom (B)*
- iii. Wavelength of photon to *completely remove* an electron from *1s orbital of Boron cation (B⁺)*

i or ii or iii

(II) Select the smallest (shortest) wavelength from the three choices below (*for H atom*):

- i. Wavelength of photon *emitted* for an electron transitioning from *2p to 1s orbital*
- ii. Wavelength of photon *absorbed* for an electron transitioning from *2s to 4p orbital*
- iii. Wavelength of photon *absorbed* for an electron transitioning from *1s to 3p orbital*

i or ii or iii



Question 2 8 points



A Chem110 student plots the graph shown on the left, from data obtained for the kinetic energy (eV) of the ejected electron versus frequency (s^{-1}) of the incident light, for an unknown metal. ($1\text{eV} = 1.6022 \times 10^{-19} \text{ J}$)

Based on the plot, answer the questions **a to c** below:

a. 1 point On the plot above, label the **threshold frequency**.

b. 2 points What is the value of the work potential (in J) of the unknown metal?

c. 3 points What is the *longest* wavelength (in nm) required to eject an electron from this metal's surface? Show your work including equation(s) used. (3 sig figs)

d. 2 points In the space provided below, plot the following

(Assume frequency is *above* threshold frequency):

- i. *Kinetic energy* of an ejected electron versus the *intensity* of the incident light (left)
- ii. *Number of electrons* ejected versus the *frequency* of the incident light (right)



**Question 3** **6 points****a. 2 points**

In the ground state of Ge (Germanium; $Z = 32$) atom, **name** the following orbitals:

i. Orbital containing the electron that requires the maximum energy to be removed.

ii. Orbital containing the electron that experiences the minimum effective nuclear charge.

b. 2 points

An unknown anion (A^-) has the following *excited* electronic configuration: $[Ar] 4s^2 3d^3 4p^1$

(Ar: Argon; $Z = 18$)

Give the **condensed ground state electronic configuration** of the atom, **A** and the cation, **A^{2+}** .
Show your work. (Assume all electronic transitions are possible)

Electronic Configuration of A:

Electronic Configuration of A^{2+} :

c. 2 points

In a single atom, determine how many electrons can have the following set of quantum numbers?

(i) $n = 5; l = 3$

In a single atom, determine how many electrons can have the following set of quantum numbers?

(i) $n = 3; l = 2; m_s = +1/2$

**Question 4** **8 points**

Rank the following from 1 to 4 in **increasing** order. Provide the ranking in the boxes provided.

Note: Atomic number (Z) are provided for relevant atoms

a) Increasing (lowest to highest) order of *ionization energy*: (1= lowest; 4 = highest)

Mg

Mg: Z = 12

Rb

Rb: Z = 37

Cs

Cs: Z = 55

K

K : Z = 19

b) Increasing (lowest to highest) order of *bond angle*: (1= lowest; 4 = highest)

H₂OCCl₄XeF₄

HCN

c) Increasing (smallest to largest) order of *ionic size* (1= smallest; 4 = largest)

S²⁻

S: Z = 16

O²⁻

O: Z = 8

Be²⁺

Be: Z = 4

Na⁺

Na: Z = 11

d) Increasing (lowest to highest) order of *bond energy* (1= lowest; 4 = highest)

P-F

P-Cl

P-I

P-Br

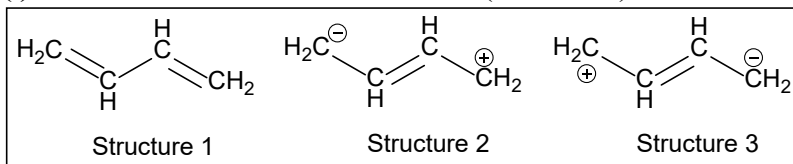
**Question 5** **12 points**

a. 9 points For the molecules HClO_4 and TeF_6 , draw the most stable *Lewis structure* showing appropriate shape and *dashed/wedged bonds*, based on VSEPR. Determine the **molecular geometry** around the central atom. For HClO_4 , label the *most polar bond*.



b. 3 points (Note: Lone pairs are not shown in structures below)

(i) For the three *resonance* structures (1, 2 and 3) shown below, answer the following question:



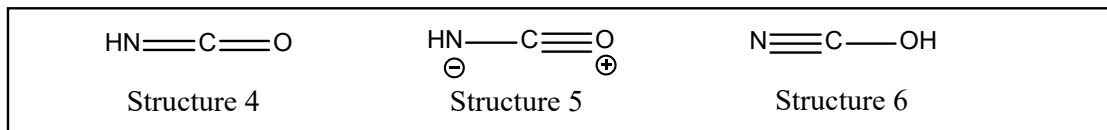
(I) Circle the *most contributing* resonance structure.

Structure 1

Structure 2

Structure 3

(ii) For the three structures given (4, 5, and 6) below, answer the following questions:



(I) Circle all the structures that represent *resonance structures* of each other:

Structure 4

Structure 5

Structure 6

(II) Among the resonance structures, circle the *most contributing* resonance structure.

Structure 4

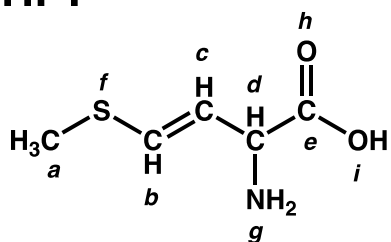
Structure 5

Structure 6

**Question 6** **6 points**

Using the data given below, calculate the **lattice energy** for SrO(s).

	ΔH° (kJ/mol)
$\text{Sr}(s) \longrightarrow \text{Sr}(g)$	+180
$\text{Sr}(g) \longrightarrow \text{Sr}^{2+}(g) + 2e^-$	+1468
$\text{O}_2(g) \longrightarrow 2\text{O}(g)$	+500
$\text{O}(g) + 2e^- \longrightarrow \text{O}^{2-}(g)$	+600
$\text{Sr}(s) + \frac{1}{2}\text{O}_2(g) \longrightarrow \text{SrO}(s)$	-550

**Question 7 10 points**

Consider the molecule given on the left, and answer questions (i) to (vi) using chemical bonding theories that we have learnt in class (valence bond theory, hybridization).

Note that the lone pairs are not shown in the structure:

i. Consider the carbon atoms marked in as ***a***, ***b***, ***c***, ***d***, and ***e*** in the molecule, and determine **hybridization** at each of the carbon centers. (2.5 points)

Atom	Hybridization
Carbon (<i>a</i>)	
Carbon (<i>b</i>)	
Carbon (<i>c</i>)	
Carbon (<i>d</i>)	
Carbon (<i>e</i>)	

ii. Now, consider the sulfur atom (S) marked as ***f***, nitrogen (N) atom (***g***), and oxygen(O) atoms (***h*** and ***i***), and determine their **hybridization**. (2 points)

Atom	Hybridization
Sulfur (<i>f</i>)	
Nitrogen (<i>g</i>)	
Oxygen (<i>h</i>)	
Oxygen (<i>i</i>)	

iii. **Describe** all the bonds formed by the carbon atoms ***a*** and ***b***, using sigma and pi designation as appropriate. (2 points)

Atom	Bonds formed (use sigma and pi as appropriate)
Carbon (<i>a</i>)	
Carbon (<i>b</i>)	

iv. How many total sp^3 -s sigma bonds are in the molecule? (1 point)

v. Determine **electronic** and **molecular geometry** at carbon (C) labeled ***e*** and sulfur (S) labeled ***f*** centres. (2 points)

Atom	Electronic Geometry	Molecular Geometry
Carbon (<i>e</i>)		
Sulfur (<i>f</i>)		

vi. Circle the value from the choices given below, that describes the **C-S-C bond angle at sulfur** (S) marked ***f***: (0.5 points)

$>90^\circ$ to $<109.5^\circ$

109.5°

>109.5 to $<120^\circ$

120°

$>120^\circ$ to $<180^\circ$


Question 8 12 points

Using the **valence shell orbitals only** draw a **complete** molecular orbital (MO) diagram (with no s-p mixing) for the molecule **PS**, in the space provided below. **(6 points)**

Assume that 3s and 3p atomic orbitals will form molecular orbitals like those formed by the combination of 2s and 2p atomic orbitals that we learnt in class, and there is no s-p mixing.

Show **energy arrow**, **atomic orbitals** involved, **atom labels**, the **molecular orbitals** that are formed, **filled electrons** in the MO diagram etc. Make sure to **label all the atomic and molecular orbitals**.

Using the MO diagram above, answer the following questions:

- (i) Which one of the following will have the highest bond order? Circle the correct choice
 A) PS^{1-} B) PS C) PS^{1+}
- (ii) Which one of the following will have the longest bond length? Circle the correct choice
 A) PS^{1-} B) PS C) PS^{1+}
- (iii) Which of the following will be diamagnetic? Circle the correct choice.
 A) PS^{1-} B) PS C) PS^{1+}
- (iv) Draw Lewis dot structure of the molecule with the shortest bond length (based on the MO diagram) from among the three (PS^{1-} , PS, PS^{1+}), and indicate the bond order calculated from the Lewis dot structure. **(3 points)**

Lewis Dot Structure

Bond Order



5B7BCE87-46F9-4B27-840B-2018E2DC72B5

midterm-exam-93388

#64 Page 10 of 10

SCRAP PAGE

**CHEMISTRY 110 –Midterm Exam****Thursday November 7, 2024, 6:30 – 8:30 pm**

Examiners: Pallavi Sirjoosingh and Ashok Kakkar

First name (please write as legibly as possible within the boxes)

--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--

Last name

--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--

Student ID number

--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--

This booklet has 8 questions on 10 pages (The last page can be used as a scrap page).

The staple must NOT be removed from the question booklet. Your intact question booklet must be returned at the end of the exam.

A separate booklet contains a periodic table, and datasheet.

Write your answers directly under the questions in the space provided. Do not write on the QR codes or near the top edges (as it may be cut off).

Please write **legibly and dark enough** for the scan to be readable. For students who use English as a second language, dictionaries are permitted. Calculators are allowed.

DO NOT ASK QUESTIONS OF THE INVIGILATORS about interpretations of the exam questions. If you have a concern about a question, please summarize it at the top of this page. All concerns will be evaluated after the exam. GOOD LUCK!

**Question 1 8 points** (*Assume all transitions are possible*)

a. 6 points Calculate the **energy (in kJ/mol)** required to ionize (completely remove) the electron from the **n=3** level of **He⁺ ion** (He: Helium). Show your work including equations. (3 sig. figs.)

b. 2 points

(I) Select the largest (longest) wavelength from the three choices below:

- i. Wavelength of photon to *completely remove* an electron from *1s orbital of Hydrogen atom*
- ii. Wavelength of photon to *completely remove* an electron from *1s orbital of Boron atom (B)*
- iii. Wavelength of photon to *completely remove* an electron from *1s orbital of Boron cation (B⁺)*

i or ii or iii

(II) Select the smallest (shortest) wavelength from the three choices below (*for H atom*):

- i. Wavelength of photon *emitted* for an electron transitioning from *2p to 1s orbital*
- ii. Wavelength of photon *absorbed* for an electron transitioning from *2s to 4p orbital*
- iii. Wavelength of photon *absorbed* for an electron transitioning from *1s to 3p orbital*

i or ii or iii



Question 2 8 points



A Chem110 student plots the graph shown on the left, from data obtained for the kinetic energy (eV) of the ejected electron versus frequency (s^{-1}) of the incident light, for an unknown metal. ($1\text{eV} = 1.6022 \times 10^{-19} \text{ J}$)

Based on the plot, answer the questions **a to c** below:

a. 1 point On the plot above, label the **threshold frequency**.

b. 2 points What is the value of the work potential (in J) of the unknown metal?

c. 3 points What is the *longest* wavelength (in nm) required to eject an electron from this metal's surface? Show your work including equation(s) used. (3 sig figs)

d. 2 points In the space provided below, plot the following

(Assume frequency is *above* threshold frequency):

- i. *Kinetic energy* of an ejected electron versus the *intensity* of the incident light (left)
- ii. *Number of electrons* ejected versus the *frequency* of the incident light (right)



**Question 3** **6 points****a. 2 points**

In the ground state of Ge (Germanium; $Z = 32$) atom, **name** the following orbitals:

i. Orbital containing the electron that requires the maximum energy to be removed.

ii. Orbital containing the electron that experiences the minimum effective nuclear charge.

b. 2 points

An unknown anion (A^-) has the following *excited* electronic configuration: $[Ar] 4s^2 3d^3 4p^1$

(Ar: Argon; $Z = 18$)

Give the **condensed ground state electronic configuration** of the atom, **A** and the cation, **A^{2+}** .
Show your work. (Assume all electronic transitions are possible)

Electronic Configuration of A:

Electronic Configuration of A^{2+} :

c. 2 points

In a single atom, determine how many electrons can have the following set of quantum numbers?

(i) $n = 5; l = 3$

In a single atom, determine how many electrons can have the following set of quantum numbers?

(i) $n = 3; l = 2; m_s = +1/2$

**Question 4** **8 points**

Rank the following from 1 to 4 in **increasing** order. Provide the ranking in the boxes provided.

Note: Atomic number (Z) are provided for relevant atoms

a) Increasing (lowest to highest) order of *ionization energy*: (1= lowest; 4 = highest)

Mg

Mg: Z = 12

Rb

Rb: Z = 37

Cs

Cs: Z = 55

K

K : Z = 19

b) Increasing (lowest to highest) order of *bond angle*: (1= lowest; 4 = highest)

H₂OCCl₄XeF₄

HCN

c) Increasing (smallest to largest) order of *ionic size* (1= smallest; 4 = largest)

S²⁻

S: Z = 16

O²⁻

O: Z = 8

Be²⁺

Be: Z = 4

Na⁺

Na: Z = 11

d) Increasing (lowest to highest) order of *bond energy* (1= lowest; 4 = highest)

P-F

P-Cl

P-I

P-Br

**Question 5** **12 points**

a. 9 points For the molecules HClO_4 and TeF_6 , draw the most stable *Lewis structure* showing appropriate shape and *dashed/wedged bonds*, based on VSEPR. Determine the **molecular geometry** around the central atom. For HClO_4 , label the *most polar bond*.

HClO_4 (there are no O-O bonds)

Label the most polar bond

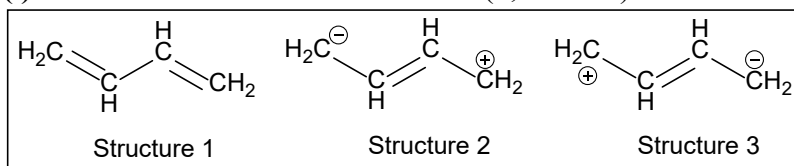
Molecular Geometry around Cl

TeF_6

Molecular Geometry around Te

b. 3 points (Note: Lone pairs are not shown in structures below)

(i) For the three *resonance* structures (1, 2 and 3) shown below, answer the following question:



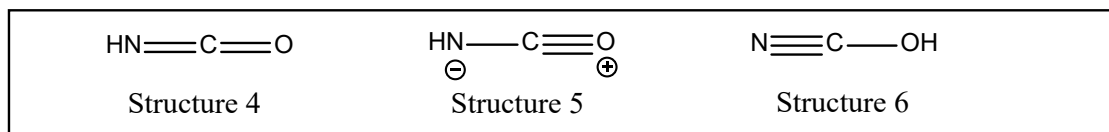
(I) Circle the *most contributing* resonance structure.

Structure 1

Structure 2

Structure 3

(ii) For the three structures given (4, 5, and 6) below, answer the following questions:



(I) Circle all the structures that represent *resonance structures* of each other:

Structure 4

Structure 5

Structure 6

(II) Among the resonance structures, circle the *most contributing* resonance structure.

Structure 4

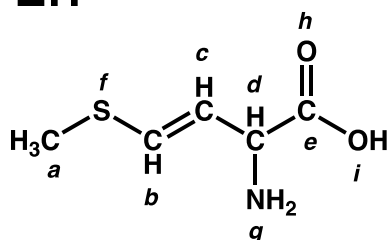
Structure 5

Structure 6

**Question 6** **6 points**

Using the data given below, calculate the **lattice energy** for SrO(s).

	ΔH° (kJ/mol)
$\text{Sr}(s) \longrightarrow \text{Sr}(g)$	+180
$\text{Sr}(g) \longrightarrow \text{Sr}^{2+}(g) + 2e^-$	+1468
$\text{O}_2(g) \longrightarrow 2\text{O}(g)$	+500
$\text{O}(g) + 2e^- \longrightarrow \text{O}^{2-}(g)$	+600
$\text{Sr}(s) + \frac{1}{2}\text{O}_2(g) \longrightarrow \text{SrO}(s)$	-550

**Question 7 10 points**

Consider the molecule given on the left, and answer questions (i) to (vi) using chemical bonding theories that we have learnt in class (valence bond theory, hybridization).

Note that the lone pairs are not shown in the structure:

i. Consider the carbon atoms marked in as ***a***, ***b***, ***c***, ***d***, and ***e*** in the molecule, and determine **hybridization** at each of the carbon centers. (2.5 points)

Atom	Hybridization
Carbon (<i>a</i>)	
Carbon (<i>b</i>)	
Carbon (<i>c</i>)	
Carbon (<i>d</i>)	
Carbon (<i>e</i>)	

ii. Now, consider the sulfur atom (S) marked as ***f***, nitrogen (N) atom (***g***), and oxygen(O) atoms (***h*** and ***i***), and determine their **hybridization**. (2 points)

Atom	Hybridization
Sulfur (<i>f</i>)	
Nitrogen (<i>g</i>)	
Oxygen (<i>h</i>)	
Oxygen (<i>i</i>)	

iii. **Describe** all the bonds formed by the carbon atoms ***a*** and ***b***, using sigma and pi designation as appropriate. (2 points)

Atom	Bonds formed (use sigma and pi as appropriate)
Carbon (<i>a</i>)	
Carbon (<i>b</i>)	

iv. How many total sp^3 -s sigma bonds are in the molecule? (1 point)

v. Determine **electronic** and **molecular geometry** at carbon (C) labeled ***e*** and sulfur (S) labeled ***f*** centres. (2 points)

Atom	Electronic Geometry	Molecular Geometry
Carbon (<i>e</i>)		
Sulfur (<i>f</i>)		

vi. Circle the value from the choices given below, that describes the **C-S-C bond angle at sulfur** (S) marked ***f***: (0.5 points)

$>90^\circ$ to $<109.5^\circ$ 109.5° >109.5 to $<120^\circ$ 120° $>120^\circ$ to $<180^\circ$



Question 8 12 points

Using the **valence shell orbitals only** draw a **complete** molecular orbital (MO) diagram (with no s-p mixing) for the molecule **PS**, in the space provided below. **(6 points)**

Assume that 3s and 3p atomic orbitals will form molecular orbitals like those formed by the combination of 2s and 2p atomic orbitals that we learnt in class, and there is no s-p mixing.

Show **energy arrow**, **atomic orbitals** involved, **atom labels**, the **molecular orbitals** that are formed, **filled electrons** in the MO diagram etc. Make sure to label all the atomic and molecular orbitals.

Using the MO diagram above, answer the following questions:

- (i) Which one of the following will have the highest bond order? Circle the correct choice
 A) PS^{1-} B) PS C) PS^{1+}
- (ii) Which one of the following will have the longest bond length? Circle the correct choice
 A) PS^{1-} B) PS C) PS^{1+}
- (iii) Which of the following will be diamagnetic? Circle the correct choice.
 A) PS^{1-} B) PS C) PS^{1+}
- (iv) Draw Lewis dot structure of the molecule with the shortest bond length (based on the MO diagram) from among the three (PS^{1-} , PS, PS^{1+}), and indicate the bond order calculated from the Lewis dot structure. **(3 points)**

Lewis Dot Structure

Bond Order



82F3E895-81CE-49B2-86E4-A50B09275C37

midterm-exam-93388

#65 Page 10 of 10

SCRAP PAGE

**CHEMISTRY 110 –Midterm Exam****Thursday November 7, 2024, 6:30 – 8:30 pm**

Examiners: Pallavi Sirjoosingh and Ashok Kakkar

First name (please write as legibly as possible within the boxes)

--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--

Last name

--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--

Student ID number

--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--

This booklet has 8 questions on 10 pages (The last page can be used as a scrap page).

The staple must NOT be removed from the question booklet. Your intact question booklet must be returned at the end of the exam.

A separate booklet contains a periodic table, and datasheet.

Write your answers directly under the questions in the space provided. Do not write on the QR codes or near the top edges (as it may be cut off).

Please write **legibly and dark enough** for the scan to be readable. For students who use English as a second language, dictionaries are permitted. Calculators are allowed.

DO NOT ASK QUESTIONS OF THE INVIGILATORS about interpretations of the exam questions. If you have a concern about a question, please summarize it at the top of this page. All concerns will be evaluated after the exam. GOOD LUCK!

**Question 1 8 points** (*Assume all transitions are possible*)

a. 6 points Calculate the **energy (in kJ/mol)** required to ionize (completely remove) the electron from the **n=3** level of **He⁺ ion** (He: Helium). Show your work including equations. (3 sig. figs.)

b. 2 points

(I) Select the largest (longest) wavelength from the three choices below:

- i. Wavelength of photon to *completely remove* an electron from *1s orbital of Hydrogen atom*
- ii. Wavelength of photon to *completely remove* an electron from *1s orbital of Boron atom (B)*
- iii. Wavelength of photon to *completely remove* an electron from *1s orbital of Boron cation (B⁺)*

i or ii or iii

(II) Select the smallest (shortest) wavelength from the three choices below (*for H atom*):

- i. Wavelength of photon *emitted* for an electron transitioning from *2p to 1s orbital*
- ii. Wavelength of photon *absorbed* for an electron transitioning from *2s to 4p orbital*
- iii. Wavelength of photon *absorbed* for an electron transitioning from *1s to 3p orbital*

i or ii or iii



Question 2 8 points



A Chem110 student plots the graph shown on the left, from data obtained for the kinetic energy (eV) of the ejected electron versus frequency (s^{-1}) of the incident light, for an unknown metal. ($1\text{eV} = 1.6022 \times 10^{-19} \text{ J}$)

Based on the plot, answer the questions **a to c** below:

a. 1 point On the plot above, label the **threshold frequency**.

b. 2 points What is the value of the work potential (in J) of the unknown metal?

c. 3 points What is the *longest* wavelength (in nm) required to eject an electron from this metal's surface? Show your work including equation(s) used. (3 sig figs)

d. 2 points In the space provided below, plot the following

(Assume frequency is *above* threshold frequency):

- Kinetic energy* of an ejected electron versus the *intensity* of the incident light (left)
- Number of electrons* ejected versus the *frequency* of the incident light (right)



**Question 3** **6 points****a. 2 points**

In the ground state of Ge (Germanium; $Z = 32$) atom, **name** the following orbitals:

i. Orbital containing the electron that requires the maximum energy to be removed.

ii. Orbital containing the electron that experiences the minimum effective nuclear charge.

b. 2 points

An unknown anion (A^-) has the following *excited* electronic configuration: $[Ar] 4s^2 3d^3 4p^1$

(Ar: Argon; $Z = 18$)

Give the **condensed ground state electronic configuration** of the atom, **A** and the cation, **A^{2+}** .
Show your work. (Assume all electronic transitions are possible)

Electronic Configuration of A:

Electronic Configuration of A^{2+} :

c. 2 points

In a single atom, determine how many electrons can have the following set of quantum numbers?

(i) $n = 5; l = 3$

In a single atom, determine how many electrons can have the following set of quantum numbers?

(i) $n = 3; l = 2; m_s = +1/2$

**Question 4** **8 points**

Rank the following from 1 to 4 in **increasing** order. Provide the ranking in the boxes provided.

Note: Atomic number (Z) are provided for relevant atoms

a) Increasing (lowest to highest) order of *ionization energy*: (1= lowest; 4 = highest)

Mg

Mg: Z = 12

Rb

Rb: Z = 37

Cs

Cs: Z = 55

K

K : Z = 19

b) Increasing (lowest to highest) order of *bond angle*: (1= lowest; 4 = highest)

H₂OCCl₄XeF₄

HCN

c) Increasing (smallest to largest) order of *ionic size* (1= smallest; 4 = largest)

S²⁻

S: Z = 16

O²⁻

O: Z = 8

Be²⁺

Be: Z = 4

Na⁺

Na: Z = 11

d) Increasing (lowest to highest) order of *bond energy* (1= lowest; 4 = highest)

P-F

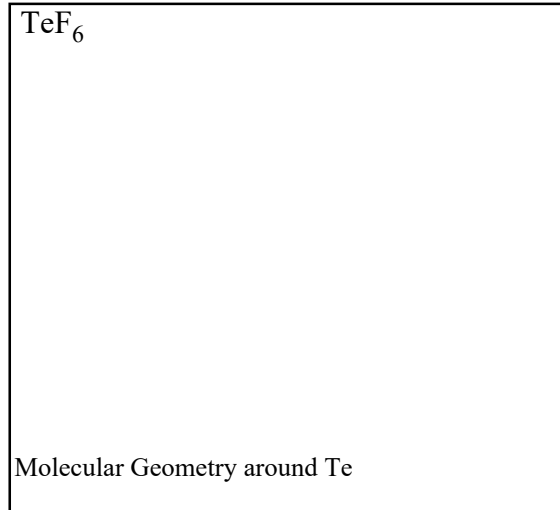
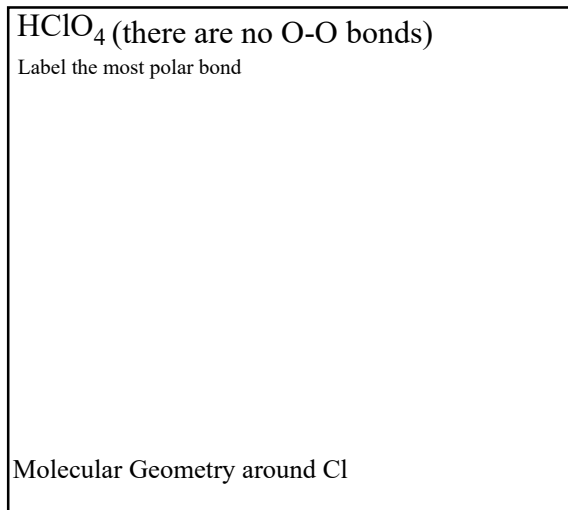
P-Cl

P-I

P-Br

**Question 5** **12 points**

a. 9 points For the molecules HClO_4 and TeF_6 , draw the most stable *Lewis structure* showing appropriate shape and *dashed/wedged bonds*, based on VSEPR. Determine the **molecular geometry** around the central atom. For HClO_4 , label the *most polar bond*.



b. 3 points (Note: Lone pairs are not shown in structures below)

(i) For the three *resonance* structures (1, 2 and 3) shown below, answer the following question:



(I) Circle the *most contributing* resonance structure.

Structure 1

Structure 2

Structure 3

(ii) For the three structures given (4, 5, and 6) below, answer the following questions:



(I) Circle all the structures that represent *resonance structures* of each other:

Structure 4

Structure 5

Structure 6

(II) Among the resonance structures, circle the *most contributing* resonance structure.

Structure 4

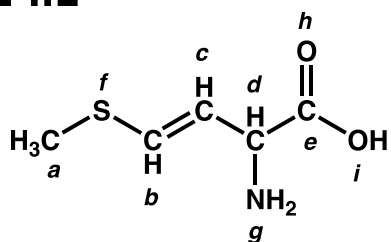
Structure 5

Structure 6

**Question 6** **6 points**

Using the data given below, calculate the **lattice energy** for SrO(s).

	ΔH° (kJ/mol)
$\text{Sr}(s) \longrightarrow \text{Sr}(g)$	+180
$\text{Sr}(g) \longrightarrow \text{Sr}^{2+}(g) + 2e^-$	+1468
$\text{O}_2(g) \longrightarrow 2\text{O}(g)$	+500
$\text{O}(g) + 2e^- \longrightarrow \text{O}^{2-}(g)$	+600
$\text{Sr}(s) + \frac{1}{2}\text{O}_2(g) \longrightarrow \text{SrO}(s)$	-550

**Question 7 10 points**

Consider the molecule given on the left, and answer questions (i) to (vi) using chemical bonding theories that we have learnt in class (valence bond theory, hybridization).

Note that the lone pairs are not shown in the structure:

i. Consider the carbon atoms marked in as ***a***, ***b***, ***c***, ***d***, and ***e*** in the molecule, and determine **hybridization** at each of the carbon centers. (2.5 points)

Atom	Hybridization
Carbon (<i>a</i>)	
Carbon (<i>b</i>)	
Carbon (<i>c</i>)	
Carbon (<i>d</i>)	
Carbon (<i>e</i>)	

ii. Now, consider the sulfur atom (S) marked as ***f***, nitrogen (N) atom (***g***), and oxygen(O) atoms (***h*** and ***i***), and determine their **hybridization**. (2 points)

Atom	Hybridization
Sulfur (<i>f</i>)	
Nitrogen (<i>g</i>)	
Oxygen (<i>h</i>)	
Oxygen (<i>i</i>)	

iii. **Describe** all the bonds formed by the carbon atoms ***a*** and ***b***, using sigma and pi designation as appropriate. (2 points)

Atom	Bonds formed (use sigma and pi as appropriate)
Carbon (<i>a</i>)	
Carbon (<i>b</i>)	

iv. How many total sp^3 -s sigma bonds are in the molecule? (1 point)

v. Determine **electronic** and **molecular geometry** at carbon (C) labeled ***e*** and sulfur (S) labeled ***f*** centres. (2 points)

Atom	Electronic Geometry	Molecular Geometry
Carbon (<i>e</i>)		
Sulfur (<i>f</i>)		

vi. Circle the value from the choices given below, that describes the **C-S-C bond angle at sulfur** (S) marked ***f***: (0.5 points)

>90° to <109.5°

109.5°

>109.5 to <120°

120°

>120° to <180°

**Question 8 12 points**

Using the **valence shell orbitals only** draw a **complete** molecular orbital (MO) diagram (with no s-p mixing) for the molecule **PS**, in the space provided below. **(6 points)**

Assume that 3s and 3p atomic orbitals will form molecular orbitals like those formed by the combination of 2s and 2p atomic orbitals that we learnt in class, and there is no s-p mixing.

Show **energy arrow**, **atomic orbitals** involved, **atom labels**, the **molecular orbitals** that are formed, **filled electrons** in the MO diagram etc. Make sure to label all the atomic and molecular orbitals.

Using the MO diagram above, answer the following questions:

- (i) Which one of the following will have the highest bond order? Circle the correct choice
A) PS^{1-} B) PS C) PS^{1+}
- (ii) Which one of the following will have the longest bond length? Circle the correct choice
A) PS^{1-} B) PS C) PS^{1+}
- (iii) Which of the following will be diamagnetic? Circle the correct choice.
A) PS^{1-} B) PS C) PS^{1+}
- (iv) Draw Lewis dot structure of the molecule with the shortest bond length (based on the MO diagram) from among the three (PS^{1-} , PS, PS^{1+}), and indicate the bond order calculated from the Lewis dot structure. **(3 points)**

Lewis Dot Structure

Bond Order



0DB02878-5396-4488-8DBB-1F9C9DFF16C9

midterm-exam-93388

#66 Page 10 of 10

SCRAP PAGE

**CHEMISTRY 110 –Midterm Exam****Thursday November 7, 2024, 6:30 – 8:30 pm**

Examiners: Pallavi Sirjoosingh and Ashok Kakkar

First name (please write as legibly as possible within the boxes)

--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--

Last name

--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--

Student ID number

--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--

This booklet has 8 questions on 10 pages (The last page can be used as a scrap page).

The staple must NOT be removed from the question booklet. Your intact question booklet must be returned at the end of the exam.

A separate booklet contains a periodic table, and datasheet.

Write your answers directly under the questions in the space provided. Do not write on the QR codes or near the top edges (as it may be cut off).

Please write **legibly and dark enough** for the scan to be readable. For students who use English as a second language, dictionaries are permitted. Calculators are allowed.

DO NOT ASK QUESTIONS OF THE INVIGILATORS about interpretations of the exam questions. If you have a concern about a question, please summarize it at the top of this page. All concerns will be evaluated after the exam. GOOD LUCK!

**Question 1 8 points** (*Assume all transitions are possible*)

a. 6 points Calculate the **energy (in kJ/mol)** required to ionize (completely remove) the electron from the **n=3** level of **He⁺ ion** (He: Helium). Show your work including equations. (3 sig. figs.)

b. 2 points

(I) Select the largest (longest) wavelength from the three choices below:

- i. Wavelength of photon to *completely remove* an electron from *1s orbital of Hydrogen atom*
- ii. Wavelength of photon to *completely remove* an electron from *1s orbital of Boron atom (B)*
- iii. Wavelength of photon to *completely remove* an electron from *1s orbital of Boron cation (B⁺)*

i or ii or iii

(II) Select the smallest (shortest) wavelength from the three choices below (*for H atom*):

- i. Wavelength of photon *emitted* for an electron transitioning from *2p to 1s orbital*
- ii. Wavelength of photon *absorbed* for an electron transitioning from *2s to 4p orbital*
- iii. Wavelength of photon *absorbed* for an electron transitioning from *1s to 3p orbital*

i or ii or iii



Question 2 8 points



A Chem110 student plots the graph shown on the left, from data obtained for the kinetic energy (eV) of the ejected electron versus frequency (s^{-1}) of the incident light, for an unknown metal. ($1\text{eV} = 1.6022 \times 10^{-19} \text{ J}$)

Based on the plot, answer the questions **a to c** below:

a. 1 point On the plot above, label the **threshold frequency**.

b. 2 points What is the value of the work potential (in J) of the unknown metal?

c. 3 points What is the *longest* wavelength (in nm) required to eject an electron from this metal's surface? Show your work including equation(s) used. (3 sig figs)

d. 2 points In the space provided below, plot the following

(Assume frequency is *above* threshold frequency):

- i. *Kinetic energy* of an ejected electron versus the *intensity* of the incident light (left)
- ii. *Number of electrons* ejected versus the *frequency* of the incident light (right)



**Question 3** **6 points****a. 2 points**

In the ground state of Ge (Germanium; $Z = 32$) atom, **name** the following orbitals:

i. Orbital containing the electron that requires the maximum energy to be removed.

ii. Orbital containing the electron that experiences the minimum effective nuclear charge.

b. 2 points

An unknown anion (A^-) has the following *excited* electronic configuration: $[Ar] 4s^2 3d^3 4p^1$

(Ar: Argon; $Z = 18$)

Give the **condensed ground state electronic configuration** of the atom, **A** and the cation, **A^{2+}** .
Show your work. (Assume all electronic transitions are possible)

Electronic Configuration of A:

Electronic Configuration of A^{2+} :

c. 2 points

In a single atom, determine how many electrons can have the following set of quantum numbers?

(i) $n = 5; l = 3$

In a single atom, determine how many electrons can have the following set of quantum numbers?

(i) $n = 3; l = 2; m_s = +1/2$

**Question 4** **8 points**

Rank the following from 1 to 4 in **increasing** order. Provide the ranking in the boxes provided.

Note: Atomic number (Z) are provided for relevant atoms

a) Increasing (lowest to highest) order of *ionization energy*: (1= lowest; 4 = highest)

Mg

Mg: Z = 12

Rb

Rb: Z = 37

Cs

Cs: Z = 55

K

K : Z = 19

b) Increasing (lowest to highest) order of *bond angle*: (1= lowest; 4 = highest)

H₂OCCl₄XeF₄

HCN

c) Increasing (smallest to largest) order of *ionic size* (1= smallest; 4 = largest)

S²⁻

S: Z = 16

O²⁻

O: Z = 8

Be²⁺

Be: Z = 4

Na⁺

Na: Z = 11

d) Increasing (lowest to highest) order of *bond energy* (1= lowest; 4 = highest)

P-F

P-Cl

P-I

P-Br

**Question 5** **12 points**

a. 9 points For the molecules HClO_4 and TeF_6 , draw the most stable *Lewis structure* showing appropriate shape and *dashed/wedged bonds*, based on VSEPR. Determine the **molecular geometry** around the central atom. For HClO_4 , label the *most polar bond*.

HClO_4 (there are no O-O bonds)

Label the most polar bond

Molecular Geometry around Cl

TeF_6

Molecular Geometry around Te

b. 3 points (Note: Lone pairs are not shown in structures below)

(i) For the three *resonance* structures (1, 2 and 3) shown below, answer the following question:



(I) Circle the *most contributing* resonance structure.

Structure 1

Structure 2

Structure 3

(ii) For the three structures given (4, 5, and 6) below, answer the following questions:



(I) Circle all the structures that represent *resonance structures* of each other:

Structure 4

Structure 5

Structure 6

(II) Among the resonance structures, circle the *most contributing* resonance structure.

Structure 4

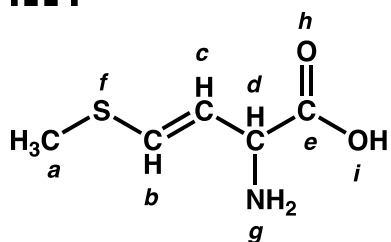
Structure 5

Structure 6

**Question 6** **6 points**

Using the data given below, calculate the **lattice energy** for SrO(s).

	ΔH° (kJ/mol)
$\text{Sr}(s) \longrightarrow \text{Sr}(g)$	+180
$\text{Sr}(g) \longrightarrow \text{Sr}^{2+}(g) + 2e^-$	+1468
$\text{O}_2(g) \longrightarrow 2\text{O}(g)$	+500
$\text{O}(g) + 2e^- \longrightarrow \text{O}^{2-}(g)$	+600
$\text{Sr}(s) + \frac{1}{2}\text{O}_2(g) \longrightarrow \text{SrO}(s)$	-550

**Question 7 10 points**

Consider the molecule given on the left, and answer questions (i) to (vi) using chemical bonding theories that we have learnt in class (valence bond theory, hybridization).

Note that the lone pairs are not shown in the structure:

i. Consider the carbon atoms marked in as ***a***, ***b***, ***c***, ***d***, and ***e*** in the molecule, and determine **hybridization** at each of the carbon centers. (2.5 points)

Atom	Hybridization
Carbon (<i>a</i>)	
Carbon (<i>b</i>)	
Carbon (<i>c</i>)	
Carbon (<i>d</i>)	
Carbon (<i>e</i>)	

ii. Now, consider the sulfur atom (S) marked as ***f***, nitrogen (N) atom (***g***), and oxygen(O) atoms (***h*** and ***i***), and determine their **hybridization**. (2 points)

Atom	Hybridization
Sulfur (<i>f</i>)	
Nitrogen (<i>g</i>)	
Oxygen (<i>h</i>)	
Oxygen (<i>i</i>)	

iii. **Describe** all the bonds formed by the carbon atoms ***a*** and ***b***, using sigma and pi designation as appropriate. (2 points)

Atom	Bonds formed (use sigma and pi as appropriate)
Carbon (<i>a</i>)	
Carbon (<i>b</i>)	

iv. How many total sp^3 -s sigma bonds are in the molecule? (1 point)

v. Determine **electronic** and **molecular geometry** at carbon (C) labeled ***e*** and sulfur (S) labeled ***f*** centres. (2 points)

Atom	Electronic Geometry	Molecular Geometry
Carbon (<i>e</i>)		
Sulfur (<i>f</i>)		

vi. Circle the value from the choices given below, that describes the **C-S-C bond angle at sulfur** (S) marked ***f***: (0.5 points)

>90° to <109.5° 109.5° >109.5 to <120° 120° >120° to <180°



Question 8 12 points

Using the **valence shell orbitals only** draw a **complete** molecular orbital (MO) diagram (with no s-p mixing) for the molecule **PS**, in the space provided below. **(6 points)**

Assume that 3s and 3p atomic orbitals will form molecular orbitals like those formed by the combination of 2s and 2p atomic orbitals that we learnt in class, and there is no s-p mixing.

Show **energy arrow**, **atomic orbitals** involved, **atom labels**, the **molecular orbitals** that are formed, **filled electrons** in the MO diagram etc. Make sure to label all the atomic and molecular orbitals.

Using the MO diagram above, answer the following questions:

- (i) Which one of the following will have the highest bond order? Circle the correct choice
 A) PS^{1-} B) PS C) PS^{1+}
- (ii) Which one of the following will have the longest bond length? Circle the correct choice
 A) PS^{1-} B) PS C) PS^{1+}
- (iii) Which of the following will be diamagnetic? Circle the correct choice.
 A) PS^{1-} B) PS C) PS^{1+}
- (iv) Draw Lewis dot structure of the molecule with the shortest bond length (based on the MO diagram) from among the three (PS^{1-} , PS, PS^{1+}), and indicate the bond order calculated from the Lewis dot structure. **(3 points)**

Lewis Dot Structure

Bond Order



20A1F69E-7F59-414C-B19E-B7D80C4AC794

midterm-exam-93388

#67 Page 10 of 10

SCRAP PAGE

**CHEMISTRY 110 –Midterm Exam****Thursday November 7, 2024, 6:30 – 8:30 pm**

Examiners: Pallavi Sirjoosingh and Ashok Kakkar

First name (please write as legibly as possible within the boxes)

--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--

Last name

--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--

Student ID number

--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--

This booklet has 8 questions on 10 pages (The last page can be used as a scrap page).

The staple must NOT be removed from the question booklet. Your intact question booklet must be returned at the end of the exam.

A separate booklet contains a periodic table, and datasheet.

Write your answers directly under the questions in the space provided. Do not write on the QR codes or near the top edges (as it may be cut off).

Please write **legibly and dark enough** for the scan to be readable. For students who use English as a second language, dictionaries are permitted. Calculators are allowed.

DO NOT ASK QUESTIONS OF THE INVIGILATORS about interpretations of the exam questions. If you have a concern about a question, please summarize it at the top of this page. All concerns will be evaluated after the exam. GOOD LUCK!

**Question 1 8 points** (*Assume all transitions are possible*)

a. 6 points Calculate the **energy (in kJ/mol)** required to ionize (completely remove) the electron from the **n=3** level of **He⁺ ion** (He: Helium). Show your work including equations. (3 sig. figs.)

b. 2 points

(I) Select the largest (longest) wavelength from the three choices below:

- i. Wavelength of photon to *completely remove* an electron from *1s orbital of Hydrogen atom*
- ii. Wavelength of photon to *completely remove* an electron from *1s orbital of Boron atom (B)*
- iii. Wavelength of photon to *completely remove* an electron from *1s orbital of Boron cation (B⁺)*

i or ii or iii

(II) Select the smallest (shortest) wavelength from the three choices below (*for H atom*):

- i. Wavelength of photon *emitted* for an electron transitioning from *2p to 1s orbital*
- ii. Wavelength of photon *absorbed* for an electron transitioning from *2s to 4p orbital*
- iii. Wavelength of photon *absorbed* for an electron transitioning from *1s to 3p orbital*

i or ii or iii



Question 2 8 points



A Chem110 student plots the graph shown on the left, from data obtained for the kinetic energy (eV) of the ejected electron versus frequency (s^{-1}) of the incident light, for an unknown metal. ($1\text{eV} = 1.6022 \times 10^{-19} \text{ J}$)

Based on the plot, answer the questions **a to c** below:

a. 1 point On the plot above, label the **threshold frequency**.

b. 2 points What is the value of the work potential (in J) of the unknown metal?

c. 3 points What is the *longest* wavelength (in nm) required to eject an electron from this metal's surface? Show your work including equation(s) used. (3 sig figs)

d. 2 points In the space provided below, plot the following

(Assume frequency is *above* threshold frequency):

- Kinetic energy* of an ejected electron versus the *intensity* of the incident light (left)
- Number of electrons* ejected versus the *frequency* of the incident light (right)



**Question 3** **6 points****a. 2 points**

In the ground state of Ge (Germanium; $Z = 32$) atom, **name** the following orbitals:

i. Orbital containing the electron that requires the maximum energy to be removed.

ii. Orbital containing the electron that experiences the minimum effective nuclear charge.

b. 2 points

An unknown anion (A^-) has the following *excited* electronic configuration: $[Ar] 4s^2 3d^3 4p^1$

(Ar: Argon; $Z = 18$)

Give the **condensed ground state electronic configuration** of the atom, **A** and the cation, **A^{2+}** .
Show your work. (Assume all electronic transitions are possible)

Electronic Configuration of A:

Electronic Configuration of A^{2+} :

c. 2 points

In a single atom, determine how many electrons can have the following set of quantum numbers?

(i) $n = 5; l = 3$

In a single atom, determine how many electrons can have the following set of quantum numbers?

(i) $n = 3; l = 2; m_s = +1/2$

**Question 4** **8 points**

Rank the following from 1 to 4 in **increasing** order. Provide the ranking in the boxes provided.

Note: Atomic number (Z) are provided for relevant atoms

a) Increasing (lowest to highest) order of *ionization energy*: (1= lowest; 4 = highest)

Mg

Mg: Z = 12

Rb

Rb: Z = 37

Cs

Cs: Z = 55

K

K : Z = 19

b) Increasing (lowest to highest) order of *bond angle*: (1= lowest; 4 = highest)

H₂OCCl₄XeF₄

HCN

c) Increasing (smallest to largest) order of *ionic size* (1= smallest; 4 = largest)

S²⁻

S: Z = 16

O²⁻

O: Z = 8

Be²⁺

Be: Z = 4

Na⁺

Na: Z = 11

d) Increasing (lowest to highest) order of *bond energy* (1= lowest; 4 = highest)

P-F

P-Cl

P-I

P-Br

**Question 5** **12 points**

a. 9 points For the molecules HClO_4 and TeF_6 , draw the most stable *Lewis structure* showing appropriate shape and *dashed/wedged bonds*, based on VSEPR. Determine the **molecular geometry** around the central atom. For HClO_4 , label the *most polar bond*.



b. 3 points (Note: Lone pairs are not shown in structures below)

(i) For the three *resonance* structures (1, 2 and 3) shown below, answer the following question:



(I) Circle the *most contributing* resonance structure.

Structure 1

Structure 2

Structure 3

(ii) For the three structures given (4, 5, and 6) below, answer the following questions:



(I) Circle all the structures that represent *resonance structures* of each other:

Structure 4

Structure 5

Structure 6

(II) Among the resonance structures, circle the *most contributing* resonance structure.

Structure 4

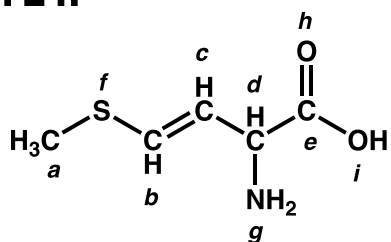
Structure 5

Structure 6

**Question 6** **6 points**

Using the data given below, calculate the **lattice energy** for SrO(s).

	ΔH° (kJ/mol)
$\text{Sr}(s) \longrightarrow \text{Sr}(g)$	+180
$\text{Sr}(g) \longrightarrow \text{Sr}^{2+}(g) + 2e^-$	+1468
$\text{O}_2(g) \longrightarrow 2\text{O}(g)$	+500
$\text{O}(g) + 2e^- \longrightarrow \text{O}^{2-}(g)$	+600
$\text{Sr}(s) + \frac{1}{2}\text{O}_2(g) \longrightarrow \text{SrO}(s)$	-550

**Question 7 10 points**

Consider the molecule given on the left, and answer questions (i) to (vi) using chemical bonding theories that we have learnt in class (valence bond theory, hybridization).

Note that the lone pairs are not shown in the structure:

i. Consider the carbon atoms marked in as ***a***, ***b***, ***c***, ***d***, and ***e*** in the molecule, and determine **hybridization** at each of the carbon centers. (2.5 points)

Atom	Hybridization
Carbon (<i>a</i>)	
Carbon (<i>b</i>)	
Carbon (<i>c</i>)	
Carbon (<i>d</i>)	
Carbon (<i>e</i>)	

ii. Now, consider the sulfur atom (S) marked as ***f***, nitrogen (N) atom (***g***), and oxygen(O) atoms (***h*** and ***i***), and determine their **hybridization**. (2 points)

Atom	Hybridization
Sulfur (<i>f</i>)	
Nitrogen (<i>g</i>)	
Oxygen (<i>h</i>)	
Oxygen (<i>i</i>)	

iii. **Describe** all the bonds formed by the carbon atoms ***a*** and ***b***, using sigma and pi designation as appropriate. (2 points)

Atom	Bonds formed (use sigma and pi as appropriate)
Carbon (<i>a</i>)	
Carbon (<i>b</i>)	

iv. How many total sp^3 -s sigma bonds are in the molecule? (1 point)

v. Determine **electronic** and **molecular geometry** at carbon (C) labeled ***e*** and sulfur (S) labeled ***f*** centres. (2 points)

Atom	Electronic Geometry	Molecular Geometry
Carbon (<i>e</i>)		
Sulfur (<i>f</i>)		

vi. Circle the value from the choices given below, that describes the **C-S-C bond angle at sulfur** (S) marked ***f***: (0.5 points)

>90° to <109.5° 109.5° >109.5 to <120° 120° >120° to <180°


Question 8 12 points

Using the **valence shell orbitals only** draw a **complete** molecular orbital (MO) diagram (with no s-p mixing) for the molecule **PS**, in the space provided below. **(6 points)**

Assume that 3s and 3p atomic orbitals will form molecular orbitals like those formed by the combination of 2s and 2p atomic orbitals that we learnt in class, and there is no s-p mixing.

Show **energy arrow**, **atomic orbitals** involved, **atom labels**, the **molecular orbitals** that are formed, **filled electrons** in the MO diagram etc. Make sure to label all the atomic and molecular orbitals.

Using the MO diagram above, answer the following questions:

- (i) Which one of the following will have the highest bond order? Circle the correct choice
 A) PS^{1-} B) PS C) PS^{1+}
- (ii) Which one of the following will have the longest bond length? Circle the correct choice
 A) PS^{1-} B) PS C) PS^{1+}
- (iii) Which of the following will be diamagnetic? Circle the correct choice.
 A) PS^{1-} B) PS C) PS^{1+}
- (iv) Draw Lewis dot structure of the molecule with the shortest bond length (based on the MO diagram) from among the three (PS^{1-} , PS, PS^{1+}), and indicate the bond order calculated from the Lewis dot structure. **(3 points)**

Lewis Dot Structure

Bond Order



345177A9-757C-47F0-A635-219DFC9BFF4B

midterm-exam-93388

#68 Page 10 of 10

SCRAP PAGE

**CHEMISTRY 110 –Midterm Exam****Thursday November 7, 2024, 6:30 – 8:30 pm****Examiners: Pallavi Sirjoosingh and Ashok Kakkar**

First name (please write as legibly as possible within the boxes)

--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--

Last name

--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--

Student ID number

--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--

This booklet has 8 questions on 10 pages (The last page can be used as a scrap page).

The staple must NOT be removed from the question booklet. Your intact question booklet must be returned at the end of the exam.

A separate booklet contains a periodic table, and datasheet.

Write your answers directly under the questions in the space provided. Do not write on the QR codes or near the top edges (as it may be cut off).

Please write **legibly and dark enough** for the scan to be readable. For students who use English as a second language, dictionaries are permitted. Calculators are allowed.

DO NOT ASK QUESTIONS OF THE INVIGILATORS about interpretations of the exam questions. If you have a concern about a question, please summarize it at the top of this page. All concerns will be evaluated after the exam. GOOD LUCK!

**Question 1 8 points** (*Assume all transitions are possible*)

a. 6 points Calculate the **energy (in kJ/mol)** required to ionize (completely remove) the electron from the **n=3** level of **He⁺ ion** (He: Helium). Show your work including equations. (3 sig. figs.)

b. 2 points

(I) Select the largest (longest) wavelength from the three choices below:

- i. Wavelength of photon to *completely remove* an electron from *1s orbital of Hydrogen atom*
- ii. Wavelength of photon to *completely remove* an electron from *1s orbital of Boron atom (B)*
- iii. Wavelength of photon to *completely remove* an electron from *1s orbital of Boron cation (B⁺)*

i or ii or iii

(II) Select the smallest (shortest) wavelength from the three choices below (*for H atom*):

- i. Wavelength of photon *emitted* for an electron transitioning from *2p to 1s orbital*
- ii. Wavelength of photon *absorbed* for an electron transitioning from *2s to 4p orbital*
- iii. Wavelength of photon *absorbed* for an electron transitioning from *1s to 3p orbital*

i or ii or iii



Question 2 8 points



A Chem110 student plots the graph shown on the left, from data obtained for the kinetic energy (eV) of the ejected electron versus frequency (s^{-1}) of the incident light, for an unknown metal. ($1\text{eV} = 1.6022 \times 10^{-19} \text{ J}$)

Based on the plot, answer the questions **a to c** below:

a. 1 point On the plot above, label the **threshold frequency**.

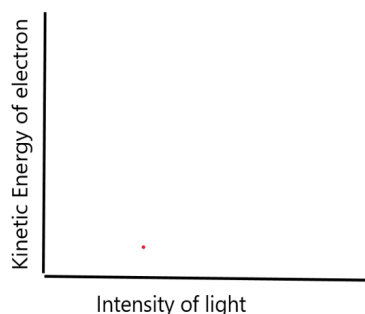
b. 2 points What is the value of the work potential (in J) of the unknown metal?

c. 3 points What is the *longest* wavelength (in nm) required to eject an electron from this metal's surface? Show your work including equation(s) used. (3 sig figs)

d. 2 points In the space provided below, plot the following

(Assume frequency is *above* threshold frequency):

- i. *Kinetic energy* of an ejected electron versus the *intensity* of the incident light (left)
- ii. *Number of electrons* ejected versus the *frequency* of the incident light (right)



**Question 3** **6 points****a. 2 points**

In the ground state of Ge (Germanium; $Z = 32$) atom, **name** the following orbitals:

i. Orbital containing the electron that requires the maximum energy to be removed.

ii. Orbital containing the electron that experiences the minimum effective nuclear charge.

b. 2 points

An unknown anion (A^-) has the following *excited* electronic configuration: $[Ar] 4s^2 3d^3 4p^1$

(Ar: Argon; $Z = 18$)

Give the **condensed ground state electronic configuration** of the atom, **A** and the cation, **A^{2+}** .
Show your work. (Assume all electronic transitions are possible)

Electronic Configuration of A:

Electronic Configuration of A^{2+} :

c. 2 points

In a single atom, determine how many electrons can have the following set of quantum numbers?

(i) $n = 5; l = 3$

In a single atom, determine how many electrons can have the following set of quantum numbers?

(i) $n = 3; l = 2; m_s = +1/2$

**Question 4** **8 points**

Rank the following from 1 to 4 in **increasing** order. Provide the ranking in the boxes provided.

Note: Atomic number (Z) are provided for relevant atoms

a) Increasing (lowest to highest) order of *ionization energy*: (1= lowest; 4 = highest)

Mg

Mg: Z = 12

Rb

Rb: Z = 37

Cs

Cs: Z = 55

K

K : Z = 19

b) Increasing (lowest to highest) order of *bond angle*: (1= lowest; 4 = highest)

H₂OCCl₄XeF₄

HCN

c) Increasing (smallest to largest) order of *ionic size* (1= smallest; 4 = largest)

S²⁻

S: Z = 16

O²⁻

O: Z = 8

Be²⁺

Be: Z = 4

Na⁺

Na: Z = 11

d) Increasing (lowest to highest) order of *bond energy* (1= lowest; 4 = highest)

P-F

P-Cl

P-I

P-Br

**Question 5** **12 points**

a. 9 points For the molecules HClO_4 and TeF_6 , draw the most stable *Lewis structure* showing appropriate shape and *dashed/wedged bonds*, based on VSEPR. Determine the **molecular geometry** around the central atom. For HClO_4 , label the *most polar bond*.



b. 3 points (Note: Lone pairs are not shown in structures below)

(i) For the three *resonance* structures (1, 2 and 3) shown below, answer the following question:



(I) Circle the *most contributing* resonance structure.

Structure 1

Structure 2

Structure 3

(ii) For the three structures given (4, 5, and 6) below, answer the following questions:



(I) Circle all the structures that represent *resonance structures* of each other:

Structure 4

Structure 5

Structure 6

(II) Among the resonance structures, circle the *most contributing* resonance structure.

Structure 4

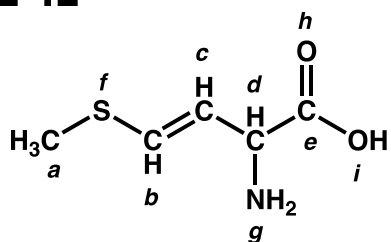
Structure 5

Structure 6

**Question 6** **6 points**

Using the data given below, calculate the **lattice energy** for SrO(s).

	ΔH° (kJ/mol)
$\text{Sr}(s) \longrightarrow \text{Sr}(g)$	+180
$\text{Sr}(g) \longrightarrow \text{Sr}^{2+}(g) + 2e^-$	+1468
$\text{O}_2(g) \longrightarrow 2\text{O}(g)$	+500
$\text{O}(g) + 2e^- \longrightarrow \text{O}^{2-}(g)$	+600
$\text{Sr}(s) + \frac{1}{2}\text{O}_2(g) \longrightarrow \text{SrO}(s)$	-550

**Question 7 10 points**

Consider the molecule given on the left, and answer questions (i) to (vi) using chemical bonding theories that we have learnt in class (valence bond theory, hybridization).

Note that the lone pairs are not shown in the structure:

i. Consider the carbon atoms marked in as ***a***, ***b***, ***c***, ***d***, and ***e*** in the molecule, and determine **hybridization** at each of the carbon centers. (2.5 points)

Atom	Hybridization
Carbon (<i>a</i>)	
Carbon (<i>b</i>)	
Carbon (<i>c</i>)	
Carbon (<i>d</i>)	
Carbon (<i>e</i>)	

ii. Now, consider the sulfur atom (S) marked as ***f***, nitrogen (N) atom (***g***), and oxygen(O) atoms (***h*** and ***i***), and determine their **hybridization**. (2 points)

Atom	Hybridization
Sulfur (<i>f</i>)	
Nitrogen (<i>g</i>)	
Oxygen (<i>h</i>)	
Oxygen (<i>i</i>)	

iii. **Describe** all the bonds formed by the carbon atoms ***a*** and ***b***, using sigma and pi designation as appropriate. (2 points)

Atom	Bonds formed (use sigma and pi as appropriate)
Carbon (<i>a</i>)	
Carbon (<i>b</i>)	

iv. How many total sp^3 -s sigma bonds are in the molecule? (1 point)

v. Determine **electronic** and **molecular geometry** at carbon (C) labeled ***e*** and sulfur (S) labeled ***f*** centres. (2 points)

Atom	Electronic Geometry	Molecular Geometry
Carbon (<i>e</i>)		
Sulfur (<i>f</i>)		

vi. Circle the value from the choices given below, that describes the **C-S-C bond angle at sulfur** (S) marked ***f***: (0.5 points)

>90° to <109.5° 109.5° >109.5 to <120° 120° >120° to <180°

**Question 8 12 points**

Using the **valence shell orbitals only** draw a **complete** molecular orbital (MO) diagram (with no s-p mixing) for the molecule **PS**, in the space provided below. **(6 points)**

Assume that 3s and 3p atomic orbitals will form molecular orbitals like those formed by the combination of 2s and 2p atomic orbitals that we learnt in class, and there is no s-p mixing.

Show **energy arrow**, **atomic orbitals** involved, **atom labels**, the **molecular orbitals** that are formed, **filled electrons** in the MO diagram etc. Make sure to label all the atomic and molecular orbitals.

Using the MO diagram above, answer the following questions:

- (i) Which one of the following will have the highest bond order? Circle the correct choice
A) PS^{1-} B) PS C) PS^{1+}
- (ii) Which one of the following will have the longest bond length? Circle the correct choice
A) PS^{1-} B) PS C) PS^{1+}
- (iii) Which of the following will be diamagnetic? Circle the correct choice.
A) PS^{1-} B) PS C) PS^{1+}
- (iv) Draw Lewis dot structure of the molecule with the shortest bond length (based on the MO diagram) from among the three (PS^{1-} , PS, PS^{1+}), and indicate the bond order calculated from the Lewis dot structure. **(3 points)**

Lewis Dot Structure

Bond Order



E45A1F1E-B250-476E-8DC3-2E5CEB9F496A

midterm-exam-93388

#69 Page 10 of 10

SCRAP PAGE

**CHEMISTRY 110 –Midterm Exam****Thursday November 7, 2024, 6:30 – 8:30 pm****Examiners: Pallavi Sirjoosingh and Ashok Kakkar**

First name (please write as legibly as possible within the boxes)

--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--

Last name

--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--

Student ID number

--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--

This booklet has 8 questions on 10 pages (The last page can be used as a scrap page).

The staple must NOT be removed from the question booklet. Your intact question booklet must be returned at the end of the exam.

A separate booklet contains a periodic table, and datasheet.

Write your answers directly under the questions in the space provided. Do not write on the QR codes or near the top edges (as it may be cut off).

Please write **legibly and dark enough** for the scan to be readable. For students who use English as a second language, dictionaries are permitted. Calculators are allowed.

DO NOT ASK QUESTIONS OF THE INVIGILATORS about interpretations of the exam questions. If you have a concern about a question, please summarize it at the top of this page. All concerns will be evaluated after the exam. GOOD LUCK!

**Question 1 8 points** (*Assume all transitions are possible*)

a. 6 points Calculate the **energy (in kJ/mol)** required to ionize (completely remove) the electron from the **n=3** level of **He⁺ ion** (He: Helium). Show your work including equations. (3 sig. figs.)

b. 2 points

(I) Select the largest (longest) wavelength from the three choices below:

- i. Wavelength of photon to *completely remove* an electron from *1s orbital of Hydrogen atom*
- ii. Wavelength of photon to *completely remove* an electron from *1s orbital of Boron atom (B)*
- iii. Wavelength of photon to *completely remove* an electron from *1s orbital of Boron cation (B⁺)*

i or ii or iii

(II) Select the smallest (shortest) wavelength from the three choices below (*for H atom*):

- i. Wavelength of photon *emitted* for an electron transitioning from *2p to 1s orbital*
- ii. Wavelength of photon *absorbed* for an electron transitioning from *2s to 4p orbital*
- iii. Wavelength of photon *absorbed* for an electron transitioning from *1s to 3p orbital*

i or ii or iii



Question 2 8 points



A Chem110 student plots the graph shown on the left, from data obtained for the kinetic energy (eV) of the ejected electron versus frequency (s^{-1}) of the incident light, for an unknown metal. ($1\text{eV} = 1.6022 \times 10^{-19} \text{ J}$)

Based on the plot, answer the questions **a to c** below:

a. 1 point On the plot above, label the **threshold frequency**.

b. 2 points What is the value of the work potential (in J) of the unknown metal?

c. 3 points What is the *longest* wavelength (in nm) required to eject an electron from this metal's surface? Show your work including equation(s) used. (3 sig figs)

d. 2 points In the space provided below, plot the following

(Assume frequency is *above* threshold frequency):

- i. *Kinetic energy* of an ejected electron versus the *intensity* of the incident light (left)
- ii. *Number of electrons* ejected versus the *frequency* of the incident light (right)



**Question 3** **6 points****a. 2 points**

In the ground state of Ge (Germanium; $Z = 32$) atom, **name** the following orbitals:

i. Orbital containing the electron that requires the maximum energy to be removed.

ii. Orbital containing the electron that experiences the minimum effective nuclear charge.

b. 2 points

An unknown anion (A^-) has the following *excited* electronic configuration: $[Ar] 4s^2 3d^3 4p^1$

(Ar: Argon; $Z = 18$)

Give the **condensed ground state electronic configuration** of the atom, **A** and the cation, **A^{2+}** .
Show your work. (Assume all electronic transitions are possible)

Electronic Configuration of A:

Electronic Configuration of A^{2+} :

c. 2 points

In a single atom, determine how many electrons can have the following set of quantum numbers?

(i) $n = 5; l = 3$

In a single atom, determine how many electrons can have the following set of quantum numbers?

(i) $n = 3; l = 2; m_s = +1/2$

**Question 4** **8 points**

Rank the following from 1 to 4 in **increasing** order. Provide the ranking in the boxes provided.

Note: Atomic number (Z) are provided for relevant atoms

a) Increasing (lowest to highest) order of *ionization energy*: (1= lowest; 4 = highest)

Mg

Mg: Z = 12

Rb

Rb: Z = 37

Cs

Cs: Z = 55

K

K : Z = 19

b) Increasing (lowest to highest) order of *bond angle*: (1= lowest; 4 = highest)

H₂OCCl₄XeF₄

HCN

c) Increasing (smallest to largest) order of *ionic size* (1= smallest; 4 = largest)

S²⁻

S: Z = 16

O²⁻

O: Z = 8

Be²⁺

Be: Z = 4

Na⁺

Na: Z = 11

d) Increasing (lowest to highest) order of *bond energy* (1= lowest; 4 = highest)

P-F

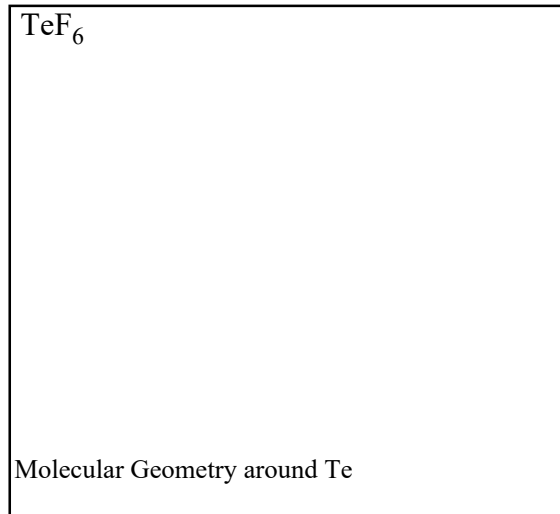
P-Cl

P-I

P-Br

**Question 5** **12 points**

a. 9 points For the molecules HClO_4 and TeF_6 , draw the most stable *Lewis structure* showing appropriate shape and *dashed/wedged bonds*, based on VSEPR. Determine the **molecular geometry** around the central atom. For HClO_4 , label the *most polar bond*.



b. 3 points (Note: Lone pairs are not shown in structures below)

(i) For the three *resonance* structures (1, 2 and 3) shown below, answer the following question:



(I) Circle the *most contributing* resonance structure.

Structure 1

Structure 2

Structure 3

(ii) For the three structures given (4, 5, and 6) below, answer the following questions:



(I) Circle all the structures that represent *resonance structures* of each other:

Structure 4

Structure 5

Structure 6

(II) Among the resonance structures, circle the *most contributing* resonance structure.

Structure 4

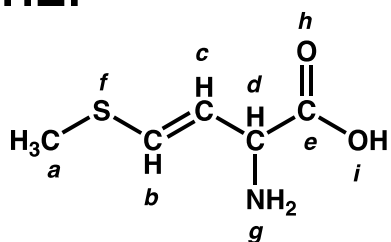
Structure 5

Structure 6

**Question 6** **6 points**

Using the data given below, calculate the **lattice energy** for SrO(s).

	ΔH° (kJ/mol)
$\text{Sr}(s) \longrightarrow \text{Sr}(g)$	+180
$\text{Sr}(g) \longrightarrow \text{Sr}^{2+}(g) + 2e^-$	+1468
$\text{O}_2(g) \longrightarrow 2\text{O}(g)$	+500
$\text{O}(g) + 2e^- \longrightarrow \text{O}^{2-}(g)$	+600
$\text{Sr}(s) + \frac{1}{2}\text{O}_2(g) \longrightarrow \text{SrO}(s)$	-550

**Question 7 10 points**

Consider the molecule given on the left, and answer questions (i) to (vi) using chemical bonding theories that we have learnt in class (valence bond theory, hybridization).

Note that the lone pairs are not shown in the structure:

i. Consider the carbon atoms marked in as ***a***, ***b***, ***c***, ***d***, and ***e*** in the molecule, and determine **hybridization** at each of the carbon centers. (2.5 points)

Atom	Hybridization
Carbon (<i>a</i>)	
Carbon (<i>b</i>)	
Carbon (<i>c</i>)	
Carbon (<i>d</i>)	
Carbon (<i>e</i>)	

ii. Now, consider the sulfur atom (S) marked as ***f***, nitrogen (N) atom (***g***), and oxygen(O) atoms (***h*** and ***i***), and determine their **hybridization**. (2 points)

Atom	Hybridization
Sulfur (<i>f</i>)	
Nitrogen (<i>g</i>)	
Oxygen (<i>h</i>)	
Oxygen (<i>i</i>)	

iii. **Describe** all the bonds formed by the carbon atoms ***a*** and ***b***, using sigma and pi designation as appropriate. (2 points)

Atom	Bonds formed (use sigma and pi as appropriate)
Carbon (<i>a</i>)	
Carbon (<i>b</i>)	

iv. How many total sp^3 -s sigma bonds are in the molecule? (1 point)

v. Determine **electronic** and **molecular geometry** at carbon (C) labeled ***e*** and sulfur (S) labeled ***f*** centres. (2 points)

Atom	Electronic Geometry	Molecular Geometry
Carbon (<i>e</i>)		
Sulfur (<i>f</i>)		

vi. Circle the value from the choices given below, that describes the **C-S-C bond angle at sulfur** (S) marked ***f***: (0.5 points)

>90° to <109.5°

109.5°

>109.5 to <120°

120°

>120° to <180°

**Question 8 12 points**

Using the **valence shell orbitals only** draw a **complete** molecular orbital (MO) diagram (with no s-p mixing) for the molecule **PS**, in the space provided below. **(6 points)**

Assume that 3s and 3p atomic orbitals will form molecular orbitals like those formed by the combination of 2s and 2p atomic orbitals that we learnt in class, and there is no s-p mixing.

Show **energy arrow**, **atomic orbitals** involved, **atom labels**, the **molecular orbitals** that are formed, **filled electrons** in the MO diagram etc. Make sure to label all the atomic and molecular orbitals.

Using the MO diagram above, answer the following questions:

- (i) Which one of the following will have the highest bond order? Circle the correct choice
A) PS^{1-} B) PS C) PS^{1+}
- (ii) Which one of the following will have the longest bond length? Circle the correct choice
A) PS^{1-} B) PS C) PS^{1+}
- (iii) Which of the following will be diamagnetic? Circle the correct choice.
A) PS^{1-} B) PS C) PS^{1+}
- (iv) Draw Lewis dot structure of the molecule with the shortest bond length (based on the MO diagram) from among the three (PS^{1-} , PS, PS^{1+}), and indicate the bond order calculated from the Lewis dot structure. **(3 points)**

Lewis Dot Structure

Bond Order



26E851D8-6842-4DEF-A613-9BB3B0D6D495

midterm-exam-93388

#70 Page 10 of 10

SCRAP PAGE

**CHEMISTRY 110 –Midterm Exam****Thursday November 7, 2024, 6:30 – 8:30 pm**

Examiners: Pallavi Sirjoosingh and Ashok Kakkar

First name (please write as legibly as possible within the boxes)

--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--

Last name

--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--

Student ID number

--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--

This booklet has 8 questions on 10 pages (The last page can be used as a scrap page).

The staple must NOT be removed from the question booklet. Your intact question booklet must be returned at the end of the exam.

A separate booklet contains a periodic table, and datasheet.

Write your answers directly under the questions in the space provided. Do not write on the QR codes or near the top edges (as it may be cut off).

Please write **legibly and dark enough** for the scan to be readable. For students who use English as a second language, dictionaries are permitted. Calculators are allowed.

DO NOT ASK QUESTIONS OF THE INVIGILATORS about interpretations of the exam questions. If you have a concern about a question, please summarize it at the top of this page. All concerns will be evaluated after the exam. GOOD LUCK!

**Question 1 8 points** (*Assume all transitions are possible*)

a. 6 points Calculate the **energy (in kJ/mol)** required to ionize (completely remove) the electron from the **$n=3$** level of **He^+ ion** (He: Helium). Show your work including equations. (3 sig. figs.)

b. 2 points

(I) Select the largest (longest) wavelength from the three choices below:

- i. Wavelength of photon to *completely remove* an electron from *1s orbital of Hydrogen atom*
- ii. Wavelength of photon to *completely remove* an electron from *1s orbital of Boron atom (B)*
- iii. Wavelength of photon to *completely remove* an electron from *1s orbital of Boron cation (B^+)*

i or ii or iii

(II) Select the smallest (shortest) wavelength from the three choices below (*for H atom*):

- i. Wavelength of photon *emitted* for an electron transitioning from *2p to 1s orbital*
- ii. Wavelength of photon *absorbed* for an electron transitioning from *2s to 4p orbital*
- iii. Wavelength of photon *absorbed* for an electron transitioning from *1s to 3p orbital*

i or ii or iii



Question 2 8 points



A Chem110 student plots the graph shown on the left, from data obtained for the kinetic energy (eV) of the ejected electron versus frequency (s^{-1}) of the incident light, for an unknown metal. ($1\text{eV} = 1.6022 \times 10^{-19} \text{ J}$)

Based on the plot, answer the questions **a to c** below:

a. 1 point On the plot above, label the **threshold frequency**.

b. 2 points What is the value of the work potential (in J) of the unknown metal?

c. 3 points What is the *longest* wavelength (in nm) required to eject an electron from this metal's surface? Show your work including equation(s) used. (3 sig figs)

d. 2 points In the space provided below, plot the following

(Assume frequency is *above* threshold frequency):

- i. *Kinetic energy* of an ejected electron versus the *intensity* of the incident light (left)
- ii. *Number of electrons* ejected versus the *frequency* of the incident light (right)



**Question 3** **6 points****a. 2 points**

In the ground state of Ge (Germanium; $Z = 32$) atom, **name** the following orbitals:

i. Orbital containing the electron that requires the maximum energy to be removed.

ii. Orbital containing the electron that experiences the minimum effective nuclear charge.

b. 2 points

An unknown anion (A^-) has the following *excited* electronic configuration: $[Ar] 4s^2 3d^3 4p^1$

(Ar: Argon; $Z = 18$)

Give the **condensed ground state electronic configuration** of the atom, **A** and the cation, A^{2+} .
Show your work. (Assume all electronic transitions are possible)

Electronic Configuration of A:

Electronic Configuration of A^{2+} :

c. 2 points

In a single atom, determine how many electrons can have the following set of quantum numbers?

(i) $n = 5; l = 3$

In a single atom, determine how many electrons can have the following set of quantum numbers?

(i) $n = 3; l = 2; m_s = +1/2$

**Question 4** **8 points**

Rank the following from 1 to 4 in **increasing** order. Provide the ranking in the boxes provided.

Note: Atomic number (Z) are provided for relevant atoms

a) Increasing (lowest to highest) order of *ionization energy*: (1= lowest; 4 = highest)

Mg

Mg: Z = 12

Rb

Rb: Z = 37

Cs

Cs: Z = 55

K

K : Z = 19

b) Increasing (lowest to highest) order of *bond angle*: (1= lowest; 4 = highest)

H₂OCCl₄XeF₄

HCN

c) Increasing (smallest to largest) order of *ionic size* (1= smallest; 4 = largest)

S²⁻

S: Z = 16

O²⁻

O: Z = 8

Be²⁺

Be: Z = 4

Na⁺

Na: Z = 11

d) Increasing (lowest to highest) order of *bond energy* (1= lowest; 4 = highest)

P-F

P-Cl

P-I

P-Br

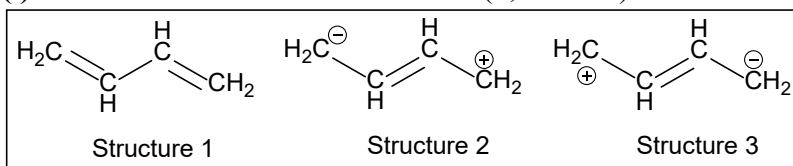
**Question 5** **12 points**

a. 9 points For the molecules HClO_4 and TeF_6 , draw the most stable *Lewis structure* showing appropriate shape and *dashed/wedged bonds*, based on VSEPR. Determine the **molecular geometry** around the central atom. For HClO_4 , label the *most polar bond*.



b. 3 points (Note: Lone pairs are not shown in structures below)

(i) For the three *resonance* structures (1, 2 and 3) shown below, answer the following question:



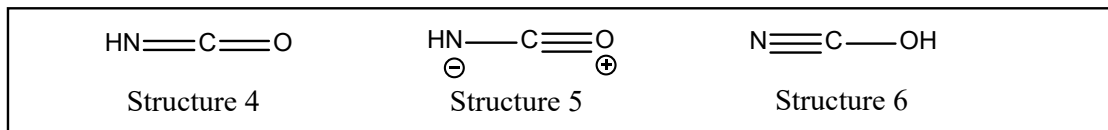
(I) Circle the *most contributing* resonance structure.

Structure 1

Structure 2

Structure 3

(ii) For the three structures given (4, 5, and 6) below, answer the following questions:



(I) Circle all the structures that represent *resonance structures* of each other:

Structure 4

Structure 5

Structure 6

(II) Among the resonance structures, circle the *most contributing* resonance structure.

Structure 4

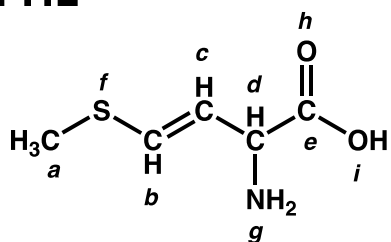
Structure 5

Structure 6

**Question 6** **6 points**

Using the data given below, calculate the **lattice energy** for SrO(s).

	ΔH° (kJ/mol)
$\text{Sr}(s) \longrightarrow \text{Sr}(g)$	+180
$\text{Sr}(g) \longrightarrow \text{Sr}^{2+}(g) + 2e^-$	+1468
$\text{O}_2(g) \longrightarrow 2\text{O}(g)$	+500
$\text{O}(g) + 2e^- \longrightarrow \text{O}^{2-}(g)$	+600
$\text{Sr}(s) + \frac{1}{2}\text{O}_2(g) \longrightarrow \text{SrO}(s)$	-550

**Question 7 10 points**

Consider the molecule given on the left, and answer questions (i) to (vi) using chemical bonding theories that we have learnt in class (valence bond theory, hybridization).

Note that the lone pairs are not shown in the structure:

i. Consider the carbon atoms marked in as ***a***, ***b***, ***c***, ***d***, and ***e*** in the molecule, and determine **hybridization** at each of the carbon centers. (2.5 points)

Atom	Hybridization
Carbon (<i>a</i>)	
Carbon (<i>b</i>)	
Carbon (<i>c</i>)	
Carbon (<i>d</i>)	
Carbon (<i>e</i>)	

ii. Now, consider the sulfur atom (S) marked as ***f***, nitrogen (N) atom (***g***), and oxygen(O) atoms (***h*** and ***i***), and determine their **hybridization**. (2 points)

Atom	Hybridization
Sulfur (<i>f</i>)	
Nitrogen (<i>g</i>)	
Oxygen (<i>h</i>)	
Oxygen (<i>i</i>)	

iii. **Describe** all the bonds formed by the carbon atoms ***a*** and ***b***, using sigma and pi designation as appropriate. (2 points)

Atom	Bonds formed (use sigma and pi as appropriate)
Carbon (<i>a</i>)	
Carbon (<i>b</i>)	

iv. How many total sp^3 -s sigma bonds are in the molecule? (1 point)

v. Determine **electronic** and **molecular geometry** at carbon (C) labeled ***e*** and sulfur (S) labeled ***f*** centres. (2 points)

Atom	Electronic Geometry	Molecular Geometry
Carbon (<i>e</i>)		
Sulfur (<i>f</i>)		

vi. Circle the value from the choices given below, that describes the **C-S-C bond angle at sulfur** (S) marked ***f***: (0.5 points)

>90° to <109.5°

109.5°

>109.5 to <120°

120°

>120° to <180°



Question 8 12 points

Using the **valence shell orbitals only** draw a **complete** molecular orbital (MO) diagram (with no s-p mixing) for the molecule **PS**, in the space provided below. **(6 points)**

Assume that 3s and 3p atomic orbitals will form molecular orbitals like those formed by the combination of 2s and 2p atomic orbitals that we learnt in class, and there is no s-p mixing.

Show **energy arrow**, **atomic orbitals** involved, **atom labels**, the **molecular orbitals** that are formed, **filled electrons** in the MO diagram etc. Make sure to label all the atomic and molecular orbitals.

Using the MO diagram above, answer the following questions:

- (i) Which one of the following will have the highest bond order? Circle the correct choice
 A) PS^{1-} B) PS C) PS^{1+}
- (ii) Which one of the following will have the longest bond length? Circle the correct choice
 A) PS^{1-} B) PS C) PS^{1+}
- (iii) Which of the following will be diamagnetic? Circle the correct choice.
 A) PS^{1-} B) PS C) PS^{1+}
- (iv) Draw Lewis dot structure of the molecule with the shortest bond length (based on the MO diagram) from among the three (PS^{1-} , PS, PS^{1+}), and indicate the bond order calculated from the Lewis dot structure. **(3 points)**

Lewis Dot Structure

Bond Order



14E62AE0-74C0-40BE-9872-EE635224AA3C

midterm-exam-93388

#71 Page 10 of 10

SCRAP PAGE

**CHEMISTRY 110 –Midterm Exam****Thursday November 7, 2024, 6:30 – 8:30 pm**

Examiners: Pallavi Sirjoosingh and Ashok Kakkar

First name (please write as legibly as possible within the boxes)

--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--

Last name

--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--

Student ID number

--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--

This booklet has 8 questions on 10 pages (The last page can be used as a scrap page).

The staple must NOT be removed from the question booklet. Your intact question booklet must be returned at the end of the exam.

A separate booklet contains a periodic table, and datasheet.

Write your answers directly under the questions in the space provided. Do not write on the QR codes or near the top edges (as it may be cut off).

Please write **legibly and dark enough** for the scan to be readable. For students who use English as a second language, dictionaries are permitted. Calculators are allowed.

DO NOT ASK QUESTIONS OF THE INVIGILATORS about interpretations of the exam questions. If you have a concern about a question, please summarize it at the top of this page. All concerns will be evaluated after the exam. GOOD LUCK!

**Question 1 8 points** (*Assume all transitions are possible*)

a. 6 points Calculate the **energy (in kJ/mol)** required to ionize (completely remove) the electron from the **n=3** level of **He⁺ ion** (He: Helium). Show your work including equations. (3 sig. figs.)

b. 2 points

(I) Select the largest (longest) wavelength from the three choices below:

- i. Wavelength of photon to *completely remove* an electron from *1s orbital of Hydrogen atom*
- ii. Wavelength of photon to *completely remove* an electron from *1s orbital of Boron atom (B)*
- iii. Wavelength of photon to *completely remove* an electron from *1s orbital of Boron cation (B⁺)*

i or ii or iii

(II) Select the smallest (shortest) wavelength from the three choices below (*for H atom*):

- i. Wavelength of photon *emitted* for an electron transitioning from *2p to 1s orbital*
- ii. Wavelength of photon *absorbed* for an electron transitioning from *2s to 4p orbital*
- iii. Wavelength of photon *absorbed* for an electron transitioning from *1s to 3p orbital*

i or ii or iii



Question 2 8 points



A Chem110 student plots the graph shown on the left, from data obtained for the kinetic energy (eV) of the ejected electron versus frequency (s^{-1}) of the incident light, for an unknown metal. ($1\text{eV} = 1.6022 \times 10^{-19} \text{ J}$)

Based on the plot, answer the questions **a to c** below:

a. 1 point On the plot above, label the **threshold frequency**.

b. 2 points What is the value of the work potential (in J) of the unknown metal?

c. 3 points What is the *longest* wavelength (in nm) required to eject an electron from this metal's surface? Show your work including equation(s) used. (3 sig figs)

d. 2 points In the space provided below, plot the following

(Assume frequency is *above* threshold frequency):

- i. *Kinetic energy* of an ejected electron versus the *intensity* of the incident light (left)
- ii. *Number of electrons* ejected versus the *frequency* of the incident light (right)



**Question 3** **6 points****a. 2 points**

In the ground state of Ge (Germanium; $Z = 32$) atom, **name** the following orbitals:

i. Orbital containing the electron that requires the maximum energy to be removed.

ii. Orbital containing the electron that experiences the minimum effective nuclear charge.

b. 2 points

An unknown anion (A^-) has the following *excited* electronic configuration: $[Ar] 4s^2 3d^3 4p^1$

(Ar: Argon; $Z = 18$)

Give the **condensed ground state electronic configuration** of the atom, **A** and the cation, **A^{2+}** .
Show your work. (Assume all electronic transitions are possible)

Electronic Configuration of A:

Electronic Configuration of A^{2+} :

c. 2 points

In a single atom, determine how many electrons can have the following set of quantum numbers?

(i) $n = 5; l = 3$

In a single atom, determine how many electrons can have the following set of quantum numbers?

(i) $n = 3; l = 2; m_s = +1/2$

**Question 4** **8 points**

Rank the following from 1 to 4 in **increasing** order. Provide the ranking in the boxes provided.

Note: Atomic number (Z) are provided for relevant atoms

a) Increasing (lowest to highest) order of *ionization energy*: (1= lowest; 4 = highest)

Mg

Mg: Z = 12

Rb

Rb: Z = 37

Cs

Cs: Z = 55

K

K : Z = 19

b) Increasing (lowest to highest) order of *bond angle*: (1= lowest; 4 = highest)

H₂OCCl₄XeF₄

HCN

c) Increasing (smallest to largest) order of *ionic size* (1= smallest; 4 = largest)

S²⁻

S: Z = 16

O²⁻

O: Z = 8

Be²⁺

Be: Z = 4

Na⁺

Na: Z = 11

d) Increasing (lowest to highest) order of *bond energy* (1= lowest; 4 = highest)

P-F

P-Cl

P-I

P-Br

**Question 5** **12 points**

a. 9 points For the molecules HClO_4 and TeF_6 , draw the most stable *Lewis structure* showing appropriate shape and *dashed/wedged bonds*, based on VSEPR. Determine the **molecular geometry** around the central atom. For HClO_4 , label the *most polar bond*.



b. 3 points (Note: Lone pairs are not shown in structures below)

(i) For the three *resonance* structures (1, 2 and 3) shown below, answer the following question:



(I) Circle the *most contributing* resonance structure.

Structure 1

Structure 2

Structure 3

(ii) For the three structures given (4, 5, and 6) below, answer the following questions:



(I) Circle all the structures that represent *resonance structures* of each other:

Structure 4

Structure 5

Structure 6

(II) Among the resonance structures, circle the *most contributing* resonance structure.

Structure 4

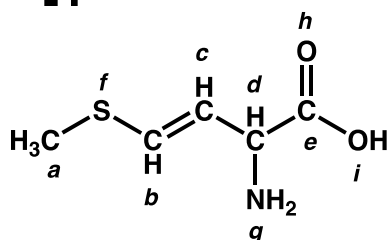
Structure 5

Structure 6

**Question 6** **6 points**

Using the data given below, calculate the **lattice energy** for SrO(s).

	ΔH° (kJ/mol)
$\text{Sr}(s) \longrightarrow \text{Sr}(g)$	+180
$\text{Sr}(g) \longrightarrow \text{Sr}^{2+}(g) + 2e^-$	+1468
$\text{O}_2(g) \longrightarrow 2\text{O}(g)$	+500
$\text{O}(g) + 2e^- \longrightarrow \text{O}^{2-}(g)$	+600
$\text{Sr}(s) + \frac{1}{2}\text{O}_2(g) \longrightarrow \text{SrO}(s)$	-550

**Question 7 10 points**

Consider the molecule given on the left, and answer questions (i) to (vi) using chemical bonding theories that we have learnt in class (valence bond theory, hybridization).

Note that the lone pairs are not shown in the structure:

i. Consider the carbon atoms marked in as ***a***, ***b***, ***c***, ***d***, and ***e*** in the molecule, and determine **hybridization** at each of the carbon centers. (2.5 points)

Atom	Hybridization
Carbon (<i>a</i>)	
Carbon (<i>b</i>)	
Carbon (<i>c</i>)	
Carbon (<i>d</i>)	
Carbon (<i>e</i>)	

ii. Now, consider the sulfur atom (S) marked as ***f***, nitrogen (N) atom (***g***), and oxygen(O) atoms (***h*** and ***i***), and determine their **hybridization**. (2 points)

Atom	Hybridization
Sulfur (<i>f</i>)	
Nitrogen (<i>g</i>)	
Oxygen (<i>h</i>)	
Oxygen (<i>i</i>)	

iii. **Describe** all the bonds formed by the carbon atoms ***a*** and ***b***, using sigma and pi designation as appropriate. (2 points)

Atom	Bonds formed (use sigma and pi as appropriate)
Carbon (<i>a</i>)	
Carbon (<i>b</i>)	

iv. How many total sp^3 -s sigma bonds are in the molecule? (1 point)

v. Determine **electronic** and **molecular geometry** at carbon (C) labeled ***e*** and sulfur (S) labeled ***f*** centres. (2 points)

Atom	Electronic Geometry	Molecular Geometry
Carbon (<i>e</i>)		
Sulfur (<i>f</i>)		

vi. Circle the value from the choices given below, that describes the **C-S-C bond angle at sulfur** (S) marked ***f***: (0.5 points)

$>90^\circ$ to $<109.5^\circ$ 109.5° >109.5 to $<120^\circ$ 120° $>120^\circ$ to $<180^\circ$

**Question 8 12 points**

Using the **valence shell orbitals only** draw a **complete** molecular orbital (MO) diagram (with no s-p mixing) for the molecule **PS**, in the space provided below. **(6 points)**

Assume that 3s and 3p atomic orbitals will form molecular orbitals like those formed by the combination of 2s and 2p atomic orbitals that we learnt in class, and there is no s-p mixing.

Show **energy arrow**, **atomic orbitals** involved, **atom labels**, the **molecular orbitals** that are formed, **filled electrons** in the MO diagram etc. Make sure to label all the atomic and molecular orbitals.

Using the MO diagram above, answer the following questions:

- (i) Which one of the following will have the highest bond order? Circle the correct choice
A) PS^{1-} B) PS C) PS^{1+}
- (ii) Which one of the following will have the longest bond length? Circle the correct choice
A) PS^{1-} B) PS C) PS^{1+}
- (iii) Which of the following will be diamagnetic? Circle the correct choice.
A) PS^{1-} B) PS C) PS^{1+}
- (iv) Draw Lewis dot structure of the molecule with the shortest bond length (based on the MO diagram) from among the three (PS^{1-} , PS, PS^{1+}), and indicate the bond order calculated from the Lewis dot structure. **(3 points)**

Lewis Dot Structure

Bond Order



E603C55A-B9ED-417B-BAF2-A4CEDC9E50D0

midterm-exam-93388

#72 Page 10 of 10

SCRAP PAGE

**CHEMISTRY 110 –Midterm Exam****Thursday November 7, 2024, 6:30 – 8:30 pm****Examiners: Pallavi Sirjoosingh and Ashok Kakkar**

First name (please write as legibly as possible within the boxes)

--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--

Last name

--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--

Student ID number

--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--

This booklet has 8 questions on 10 pages (The last page can be used as a scrap page).

The staple must NOT be removed from the question booklet. Your intact question booklet must be returned at the end of the exam.

A separate booklet contains a periodic table, and datasheet.

Write your answers directly under the questions in the space provided. Do not write on the QR codes or near the top edges (as it may be cut off).

Please write **legibly and dark enough** for the scan to be readable. For students who use English as a second language, dictionaries are permitted. Calculators are allowed.

DO NOT ASK QUESTIONS OF THE INVIGILATORS about interpretations of the exam questions. If you have a concern about a question, please summarize it at the top of this page. All concerns will be evaluated after the exam. GOOD LUCK!

**Question 1 8 points** (*Assume all transitions are possible*)

a. 6 points Calculate the **energy (in kJ/mol)** required to ionize (completely remove) the electron from the **n=3** level of **He⁺ ion** (He: Helium). Show your work including equations. (3 sig. figs.)

b. 2 points

(I) Select the largest (longest) wavelength from the three choices below:

- i. Wavelength of photon to *completely remove* an electron from *1s orbital of Hydrogen atom*
- ii. Wavelength of photon to *completely remove* an electron from *1s orbital of Boron atom (B)*
- iii. Wavelength of photon to *completely remove* an electron from *1s orbital of Boron cation (B⁺)*

i or ii or iii

(II) Select the smallest (shortest) wavelength from the three choices below (*for H atom*):

- i. Wavelength of photon *emitted* for an electron transitioning from *2p to 1s orbital*
- ii. Wavelength of photon *absorbed* for an electron transitioning from *2s to 4p orbital*
- iii. Wavelength of photon *absorbed* for an electron transitioning from *1s to 3p orbital*

i or ii or iii



Question 2 8 points



A Chem110 student plots the graph shown on the left, from data obtained for the kinetic energy (eV) of the ejected electron versus frequency (s^{-1}) of the incident light, for an unknown metal. ($1\text{eV} = 1.6022 \times 10^{-19} \text{ J}$)

Based on the plot, answer the questions **a to c** below:

a. 1 point On the plot above, label the **threshold frequency**.

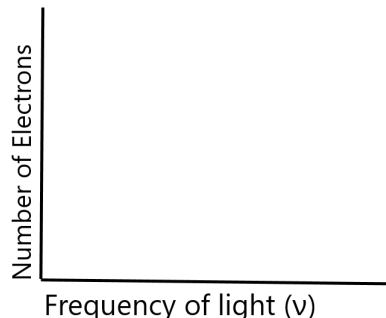
b. 2 points What is the value of the work potential (in J) of the unknown metal?

c. 3 points What is the *longest* wavelength (in nm) required to eject an electron from this metal's surface? Show your work including equation(s) used. (3 sig figs)

d. 2 points In the space provided below, plot the following

(Assume frequency is *above* threshold frequency):

- i. *Kinetic energy* of an ejected electron versus the *intensity* of the incident light (left)
- ii. *Number of electrons* ejected versus the *frequency* of the incident light (right)



**Question 3** **6 points****a. 2 points**

In the ground state of Ge (Germanium; $Z = 32$) atom, **name** the following orbitals:

i. Orbital containing the electron that requires the maximum energy to be removed.

ii. Orbital containing the electron that experiences the minimum effective nuclear charge.

b. 2 points

An unknown anion (A^-) has the following *excited* electronic configuration: $[Ar] 4s^2 3d^3 4p^1$

(Ar: Argon; $Z = 18$)

Give the **condensed ground state electronic configuration** of the atom, **A** and the cation, **A^{2+}** .
Show your work. (Assume all electronic transitions are possible)

Electronic Configuration of A:

Electronic Configuration of A^{2+} :

c. 2 points

In a single atom, determine how many electrons can have the following set of quantum numbers?

(i) $n = 5; l = 3$

In a single atom, determine how many electrons can have the following set of quantum numbers?

(i) $n = 3; l = 2; m_s = +1/2$

**Question 4** **8 points**

Rank the following from 1 to 4 in **increasing** order. Provide the ranking in the boxes provided.

Note: Atomic number (Z) are provided for relevant atoms

a) Increasing (lowest to highest) order of *ionization energy*: (1= lowest; 4 = highest)

Mg

Mg: Z = 12

Rb

Rb: Z = 37

Cs

Cs: Z = 55

K

K : Z = 19

b) Increasing (lowest to highest) order of *bond angle*: (1= lowest; 4 = highest)

H₂OCCl₄XeF₄

HCN

c) Increasing (smallest to largest) order of *ionic size* (1= smallest; 4 = largest)

S²⁻

S: Z = 16

O²⁻

O: Z = 8

Be²⁺

Be: Z = 4

Na⁺

Na: Z = 11

d) Increasing (lowest to highest) order of *bond energy* (1= lowest; 4 = highest)

P-F

P-Cl

P-I

P-Br

**Question 5** **12 points**

a. 9 points For the molecules HClO_4 and TeF_6 , draw the most stable *Lewis structure* showing appropriate shape and *dashed/wedged bonds*, based on VSEPR. Determine the **molecular geometry** around the central atom. For HClO_4 , label the *most polar bond*.



b. 3 points (Note: Lone pairs are not shown in structures below)

(i) For the three *resonance* structures (1, 2 and 3) shown below, answer the following question:



(I) Circle the *most contributing* resonance structure.

Structure 1

Structure 2

Structure 3

(ii) For the three structures given (4, 5, and 6) below, answer the following questions:



(I) Circle all the structures that represent *resonance structures* of each other:

Structure 4

Structure 5

Structure 6

(II) Among the resonance structures, circle the *most contributing* resonance structure.

Structure 4

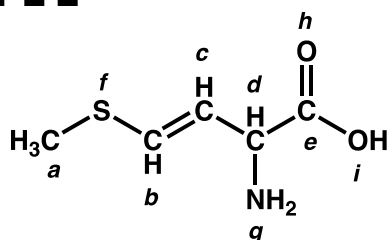
Structure 5

Structure 6

**Question 6** **6 points**

Using the data given below, calculate the **lattice energy** for SrO(s).

	ΔH° (kJ/mol)
$\text{Sr}(s) \longrightarrow \text{Sr}(g)$	+180
$\text{Sr}(g) \longrightarrow \text{Sr}^{2+}(g) + 2e^-$	+1468
$\text{O}_2(g) \longrightarrow 2\text{O}(g)$	+500
$\text{O}(g) + 2e^- \longrightarrow \text{O}^{2-}(g)$	+600
$\text{Sr}(s) + \frac{1}{2}\text{O}_2(g) \longrightarrow \text{SrO}(s)$	-550

**Question 7 10 points**

Consider the molecule given on the left, and answer questions (i) to (vi) using chemical bonding theories that we have learnt in class (valence bond theory, hybridization).

Note that the lone pairs are not shown in the structure:

i. Consider the carbon atoms marked in as ***a***, ***b***, ***c***, ***d***, and ***e*** in the molecule, and determine **hybridization** at each of the carbon centers. (2.5 points)

Atom	Hybridization
Carbon (<i>a</i>)	
Carbon (<i>b</i>)	
Carbon (<i>c</i>)	
Carbon (<i>d</i>)	
Carbon (<i>e</i>)	

ii. Now, consider the sulfur atom (S) marked as ***f***, nitrogen (N) atom (***g***), and oxygen(O) atoms (***h*** and ***i***), and determine their **hybridization**. (2 points)

Atom	Hybridization
Sulfur (<i>f</i>)	
Nitrogen (<i>g</i>)	
Oxygen (<i>h</i>)	
Oxygen (<i>i</i>)	

iii. **Describe** all the bonds formed by the carbon atoms ***a*** and ***b***, using sigma and pi designation as appropriate. (2 points)

Atom	Bonds formed (use sigma and pi as appropriate)
Carbon (<i>a</i>)	
Carbon (<i>b</i>)	

iv. How many total sp^3 -s sigma bonds are in the molecule? (1 point)

v. Determine **electronic** and **molecular geometry** at carbon (C) labeled ***e*** and sulfur (S) labeled ***f*** centres. (2 points)

Atom	Electronic Geometry	Molecular Geometry
Carbon (<i>e</i>)		
Sulfur (<i>f</i>)		

vi. Circle the value from the choices given below, that describes the **C-S-C bond angle at sulfur** (S) marked ***f***: (0.5 points)

>90° to <109.5° 109.5° >109.5 to <120° 120° >120° to <180°


Question 8 12 points

Using the **valence shell orbitals only** draw a **complete** molecular orbital (MO) diagram (with no s-p mixing) for the molecule **PS**, in the space provided below. **(6 points)**

Assume that 3s and 3p atomic orbitals will form molecular orbitals like those formed by the combination of 2s and 2p atomic orbitals that we learnt in class, and there is no s-p mixing.

Show **energy arrow**, **atomic orbitals** involved, **atom labels**, the **molecular orbitals** that are formed, **filled electrons** in the MO diagram etc. Make sure to label all the atomic and molecular orbitals.

Using the MO diagram above, answer the following questions:

- (i) Which one of the following will have the highest bond order? Circle the correct choice
 A) PS^{1-} B) PS C) PS^{1+}
- (ii) Which one of the following will have the longest bond length? Circle the correct choice
 A) PS^{1-} B) PS C) PS^{1+}
- (iii) Which of the following will be diamagnetic? Circle the correct choice.
 A) PS^{1-} B) PS C) PS^{1+}
- (iv) Draw Lewis dot structure of the molecule with the shortest bond length (based on the MO diagram) from among the three (PS^{1-} , PS, PS^{1+}), and indicate the bond order calculated from the Lewis dot structure. **(3 points)**

Lewis Dot Structure

Bond Order



C490A42C-5734-4C52-BFCA-91D87B6463F4

midterm-exam-93388

#73 Page 10 of 10

SCRAP PAGE

**CHEMISTRY 110 –Midterm Exam****Thursday November 7, 2024, 6:30 – 8:30 pm**

Examiners: Pallavi Sirjoosingh and Ashok Kakkar

First name (please write as legibly as possible within the boxes)

--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--

Last name

--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--

Student ID number

--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--

This booklet has 8 questions on 10 pages (The last page can be used as a scrap page).

The staple must NOT be removed from the question booklet. Your intact question booklet must be returned at the end of the exam.

A separate booklet contains a periodic table, and datasheet.

Write your answers directly under the questions in the space provided. Do not write on the QR codes or near the top edges (as it may be cut off).

Please write **legibly and dark enough** for the scan to be readable. For students who use English as a second language, dictionaries are permitted. Calculators are allowed.

DO NOT ASK QUESTIONS OF THE INVIGILATORS about interpretations of the exam questions. If you have a concern about a question, please summarize it at the top of this page. All concerns will be evaluated after the exam. GOOD LUCK!

**Question 1 8 points** (*Assume all transitions are possible*)

a. 6 points Calculate the **energy (in kJ/mol)** required to ionize (completely remove) the electron from the **n=3** level of **He⁺ ion** (He: Helium). Show your work including equations. (3 sig. figs.)

b. 2 points

(I) Select the largest (longest) wavelength from the three choices below:

- i. Wavelength of photon to *completely remove* an electron from *1s orbital of Hydrogen atom*
- ii. Wavelength of photon to *completely remove* an electron from *1s orbital of Boron atom (B)*
- iii. Wavelength of photon to *completely remove* an electron from *1s orbital of Boron cation (B⁺)*

i or ii or iii

(II) Select the smallest (shortest) wavelength from the three choices below (*for H atom*):

- i. Wavelength of photon *emitted* for an electron transitioning from *2p to 1s orbital*
- ii. Wavelength of photon *absorbed* for an electron transitioning from *2s to 4p orbital*
- iii. Wavelength of photon *absorbed* for an electron transitioning from *1s to 3p orbital*

i or ii or iii



Question 2 8 points



A Chem110 student plots the graph shown on the left, from data obtained for the kinetic energy (eV) of the ejected electron versus frequency (s^{-1}) of the incident light, for an unknown metal. ($1\text{eV} = 1.6022 \times 10^{-19} \text{ J}$)

Based on the plot, answer the questions **a to c** below:

a. 1 point On the plot above, label the **threshold frequency**.

b. 2 points What is the value of the work potential (in J) of the unknown metal?

c. 3 points What is the *longest* wavelength (in nm) required to eject an electron from this metal's surface? Show your work including equation(s) used. (3 sig figs)

d. 2 points In the space provided below, plot the following

(Assume frequency is *above* threshold frequency):

- Kinetic energy* of an ejected electron versus the *intensity* of the incident light (left)
- Number of electrons* ejected versus the *frequency* of the incident light (right)



**Question 3** **6 points****a. 2 points**

In the ground state of Ge (Germanium; $Z = 32$) atom, **name** the following orbitals:

i. Orbital containing the electron that requires the maximum energy to be removed.

ii. Orbital containing the electron that experiences the minimum effective nuclear charge.

b. 2 points

An unknown anion (A^-) has the following *excited* electronic configuration: $[Ar] 4s^2 3d^3 4p^1$

(Ar: Argon; $Z = 18$)

Give the **condensed ground state electronic configuration** of the atom, **A** and the cation, **A^{2+}** .
Show your work. (Assume all electronic transitions are possible)

Electronic Configuration of A:

Electronic Configuration of A^{2+} :

c. 2 points

In a single atom, determine how many electrons can have the following set of quantum numbers?

(i) $n = 5; l = 3$

In a single atom, determine how many electrons can have the following set of quantum numbers?

(i) $n = 3; l = 2; m_s = +1/2$

**Question 4** **8 points**

Rank the following from 1 to 4 in **increasing** order. Provide the ranking in the boxes provided.

Note: Atomic number (Z) are provided for relevant atoms

a) Increasing (lowest to highest) order of *ionization energy*: (1= lowest; 4 = highest)

Mg

Mg: Z = 12

Rb

Rb: Z = 37

Cs

Cs: Z = 55

K

K : Z = 19

b) Increasing (lowest to highest) order of *bond angle*: (1= lowest; 4 = highest)

H₂OCCl₄XeF₄

HCN

c) Increasing (smallest to largest) order of *ionic size* (1= smallest; 4 = largest)

S²⁻

S: Z = 16

O²⁻

O: Z = 8

Be²⁺

Be: Z = 4

Na⁺

Na: Z = 11

d) Increasing (lowest to highest) order of *bond energy* (1= lowest; 4 = highest)

P-F

P-Cl

P-I

P-Br

**Question 5** **12 points**

a. 9 points For the molecules HClO_4 and TeF_6 , draw the most stable *Lewis structure* showing appropriate shape and *dashed/wedged bonds*, based on VSEPR. Determine the **molecular geometry** around the central atom. For HClO_4 , label the *most polar bond*.



b. 3 points (Note: Lone pairs are not shown in structures below)

(i) For the three *resonance* structures (1, 2 and 3) shown below, answer the following question:



(I) Circle the *most contributing* resonance structure.

Structure 1

Structure 2

Structure 3

(ii) For the three structures given (4, 5, and 6) below, answer the following questions:



(I) Circle all the structures that represent *resonance structures* of each other:

Structure 4

Structure 5

Structure 6

(II) Among the resonance structures, circle the *most contributing* resonance structure.

Structure 4

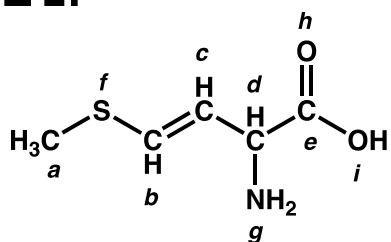
Structure 5

Structure 6

**Question 6** **6 points**

Using the data given below, calculate the **lattice energy** for SrO(s).

	ΔH° (kJ/mol)
$\text{Sr}(s) \longrightarrow \text{Sr}(g)$	+180
$\text{Sr}(g) \longrightarrow \text{Sr}^{2+}(g) + 2e^-$	+1468
$\text{O}_2(g) \longrightarrow 2\text{O}(g)$	+500
$\text{O}(g) + 2e^- \longrightarrow \text{O}^{2-}(g)$	+600
$\text{Sr}(s) + \frac{1}{2}\text{O}_2(g) \longrightarrow \text{SrO}(s)$	-550

**Question 7 10 points**

Consider the molecule given on the left, and answer questions (i) to (vi) using chemical bonding theories that we have learnt in class (valence bond theory, hybridization).

Note that the lone pairs are not shown in the structure:

i. Consider the carbon atoms marked in as ***a***, ***b***, ***c***, ***d***, and ***e*** in the molecule, and determine **hybridization** at each of the carbon centers. (2.5 points)

Atom	Hybridization
Carbon (<i>a</i>)	
Carbon (<i>b</i>)	
Carbon (<i>c</i>)	
Carbon (<i>d</i>)	
Carbon (<i>e</i>)	

ii. Now, consider the sulfur atom (S) marked as ***f***, nitrogen (N) atom (***g***), and oxygen(O) atoms (***h*** and ***i***), and determine their **hybridization**. (2 points)

Atom	Hybridization
Sulfur (<i>f</i>)	
Nitrogen (<i>g</i>)	
Oxygen (<i>h</i>)	
Oxygen (<i>i</i>)	

iii. **Describe** all the bonds formed by the carbon atoms ***a*** and ***b***, using sigma and pi designation as appropriate. (2 points)

Atom	Bonds formed (use sigma and pi as appropriate)
Carbon (<i>a</i>)	
Carbon (<i>b</i>)	

iv. How many total sp^3 -s sigma bonds are in the molecule? (1 point)

v. Determine **electronic** and **molecular geometry** at carbon (C) labeled ***e*** and sulfur (S) labeled ***f*** centres. (2 points)

Atom	Electronic Geometry	Molecular Geometry
Carbon (<i>e</i>)		
Sulfur (<i>f</i>)		

vi. Circle the value from the choices given below, that describes the **C-S-C bond angle at sulfur** (S) marked ***f***: (0.5 points)

$>90^\circ$ to $<109.5^\circ$ 109.5° >109.5 to $<120^\circ$ 120° $>120^\circ$ to $<180^\circ$


Question 8 12 points

Using the **valence shell orbitals only** draw a **complete** molecular orbital (MO) diagram (with no s-p mixing) for the molecule **PS**, in the space provided below. **(6 points)**

Assume that 3s and 3p atomic orbitals will form molecular orbitals like those formed by the combination of 2s and 2p atomic orbitals that we learnt in class, and there is no s-p mixing.

Show **energy arrow**, **atomic orbitals** involved, **atom labels**, the **molecular orbitals** that are formed, **filled electrons** in the MO diagram etc. Make sure to **label all the atomic and molecular orbitals**.

Using the MO diagram above, answer the following questions:

- (i) Which one of the following will have the highest bond order? Circle the correct choice
 A) PS^{1-} B) PS C) PS^{1+}
- (ii) Which one of the following will have the longest bond length? Circle the correct choice
 A) PS^{1-} B) PS C) PS^{1+}
- (iii) Which of the following will be diamagnetic? Circle the correct choice.
 A) PS^{1-} B) PS C) PS^{1+}
- (iv) Draw Lewis dot structure of the molecule with the shortest bond length (based on the MO diagram) from among the three (PS^{1-} , PS, PS^{1+}), and indicate the bond order calculated from the Lewis dot structure. **(3 points)**

Lewis Dot Structure

Bond Order



6DF2F427-318C-4742-AB57-7B01BFD8171E

midterm-exam-93388

#74 Page 10 of 10

SCRAP PAGE

**CHEMISTRY 110 –Midterm Exam****Thursday November 7, 2024, 6:30 – 8:30 pm****Examiners: Pallavi Sirjoosingh and Ashok Kakkar**

First name (please write as legibly as possible within the boxes)

--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--

Last name

--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--

Student ID number

--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--

This booklet has 8 questions on 10 pages (The last page can be used as a scrap page).

The staple must NOT be removed from the question booklet. Your intact question booklet must be returned at the end of the exam.

A separate booklet contains a periodic table, and datasheet.

Write your answers directly under the questions in the space provided. Do not write on the QR codes or near the top edges (as it may be cut off).

Please write **legibly and dark enough** for the scan to be readable. For students who use English as a second language, dictionaries are permitted. Calculators are allowed.

DO NOT ASK QUESTIONS OF THE INVIGILATORS about interpretations of the exam questions. If you have a concern about a question, please summarize it at the top of this page. All concerns will be evaluated after the exam. GOOD LUCK!

**Question 1 8 points** (*Assume all transitions are possible*)

a. 6 points Calculate the **energy (in kJ/mol)** required to ionize (completely remove) the electron from the **$n=3$** level of **He^+ ion** (He: Helium). Show your work including equations. (3 sig. figs.)

b. 2 points

(I) Select the largest (longest) wavelength from the three choices below:

- i. Wavelength of photon to *completely remove* an electron from *1s orbital of Hydrogen atom*
- ii. Wavelength of photon to *completely remove* an electron from *1s orbital of Boron atom (B)*
- iii. Wavelength of photon to *completely remove* an electron from *1s orbital of Boron cation (B^+)*

i or ii or iii

(II) Select the smallest (shortest) wavelength from the three choices below (*for H atom*):

- i. Wavelength of photon *emitted* for an electron transitioning from *2p to 1s orbital*
- ii. Wavelength of photon *absorbed* for an electron transitioning from *2s to 4p orbital*
- iii. Wavelength of photon *absorbed* for an electron transitioning from *1s to 3p orbital*

i or ii or iii



Question 2 8 points



A Chem110 student plots the graph shown on the left, from data obtained for the kinetic energy (eV) of the ejected electron versus frequency (s^{-1}) of the incident light, for an unknown metal. ($1\text{eV} = 1.6022 \times 10^{-19} \text{ J}$)

Based on the plot, answer the questions **a to c** below:

a. 1 point On the plot above, label the **threshold frequency**.

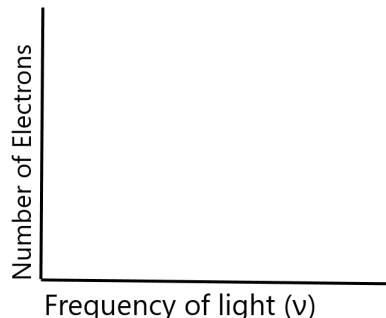
b. 2 points What is the value of the work potential (in J) of the unknown metal?

c. 3 points What is the *longest* wavelength (in nm) required to eject an electron from this metal's surface? Show your work including equation(s) used. (3 sig figs)

d. 2 points In the space provided below, plot the following

(Assume frequency is *above* threshold frequency):

- Kinetic energy* of an ejected electron versus the *intensity* of the incident light (left)
- Number of electrons* ejected versus the *frequency* of the incident light (right)



**Question 3** **6 points****a. 2 points**

In the ground state of Ge (Germanium; $Z = 32$) atom, **name** the following orbitals:

i. Orbital containing the electron that requires the maximum energy to be removed.

ii. Orbital containing the electron that experiences the minimum effective nuclear charge.

b. 2 points

An unknown anion (A^-) has the following *excited* electronic configuration: $[Ar] 4s^2 3d^3 4p^1$

(Ar: Argon; $Z = 18$)

Give the **condensed ground state electronic configuration** of the atom, **A** and the cation, **A^{2+}** .
Show your work. (Assume all electronic transitions are possible)

Electronic Configuration of A:

Electronic Configuration of A^{2+} :

c. 2 points

In a single atom, determine how many electrons can have the following set of quantum numbers?

(i) $n = 5; l = 3$

In a single atom, determine how many electrons can have the following set of quantum numbers?

(i) $n = 3; l = 2; m_s = +1/2$

**Question 4** **8 points**

Rank the following from 1 to 4 in **increasing** order. Provide the ranking in the boxes provided.

Note: Atomic number (Z) are provided for relevant atoms

a) Increasing (lowest to highest) order of *ionization energy*: (1= lowest; 4 = highest)

Mg

Mg: Z = 12

Rb

Rb: Z = 37

Cs

Cs: Z = 55

K

K : Z = 19

b) Increasing (lowest to highest) order of *bond angle*: (1= lowest; 4 = highest)

H₂OCCl₄XeF₄

HCN

c) Increasing (smallest to largest) order of *ionic size* (1= smallest; 4 = largest)

S²⁻

S: Z = 16

O²⁻

O: Z = 8

Be²⁺

Be: Z = 4

Na⁺

Na: Z = 11

d) Increasing (lowest to highest) order of *bond energy* (1= lowest; 4 = highest)

P-F

P-Cl

P-I

P-Br

**Question 5** **12 points**

a. 9 points For the molecules HClO_4 and TeF_6 , draw the most stable *Lewis structure* showing appropriate shape and *dashed/wedged bonds*, based on VSEPR. Determine the **molecular geometry** around the central atom. For HClO_4 , label the *most polar bond*.

HClO_4 (there are no O-O bonds)

Label the most polar bond

Molecular Geometry around Cl

TeF_6

Molecular Geometry around Te

b. 3 points (Note: Lone pairs are not shown in structures below)

(i) For the three *resonance* structures (1, 2 and 3) shown below, answer the following question:



(I) Circle the *most contributing* resonance structure.

Structure 1

Structure 2

Structure 3

(ii) For the three structures given (4, 5, and 6) below, answer the following questions:



(I) Circle all the structures that represent *resonance structures* of each other:

Structure 4

Structure 5

Structure 6

(II) Among the resonance structures, circle the *most contributing* resonance structure.

Structure 4

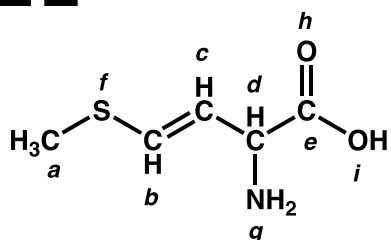
Structure 5

Structure 6

**Question 6** **6 points**

Using the data given below, calculate the **lattice energy** for SrO(s).

	ΔH° (kJ/mol)
$\text{Sr}(s) \longrightarrow \text{Sr}(g)$	+180
$\text{Sr}(g) \longrightarrow \text{Sr}^{2+}(g) + 2e^-$	+1468
$\text{O}_2(g) \longrightarrow 2\text{O}(g)$	+500
$\text{O}(g) + 2e^- \longrightarrow \text{O}^{2-}(g)$	+600
$\text{Sr}(s) + \frac{1}{2}\text{O}_2(g) \longrightarrow \text{SrO}(s)$	-550

**Question 7 10 points**

Consider the molecule given on the left, and answer questions (i) to (vi) using chemical bonding theories that we have learnt in class (valence bond theory, hybridization).

Note that the lone pairs are not shown in the structure:

i. Consider the carbon atoms marked in as ***a***, ***b***, ***c***, ***d***, and ***e*** in the molecule, and determine **hybridization** at each of the carbon centers. (2.5 points)

Atom	Hybridization
Carbon (<i>a</i>)	
Carbon (<i>b</i>)	
Carbon (<i>c</i>)	
Carbon (<i>d</i>)	
Carbon (<i>e</i>)	

ii. Now, consider the sulfur atom (S) marked as ***f***, nitrogen (N) atom (***g***), and oxygen(O) atoms (***h*** and ***i***), and determine their **hybridization**. (2 points)

Atom	Hybridization
Sulfur (<i>f</i>)	
Nitrogen (<i>g</i>)	
Oxygen (<i>h</i>)	
Oxygen (<i>i</i>)	

iii. **Describe** all the bonds formed by the carbon atoms ***a*** and ***b***, using sigma and pi designation as appropriate. (2 points)

Atom	Bonds formed (use sigma and pi as appropriate)
Carbon (<i>a</i>)	
Carbon (<i>b</i>)	

iv. How many total sp^3 -s sigma bonds are in the molecule? (1 point)

v. Determine **electronic** and **molecular geometry** at carbon (C) labeled ***e*** and sulfur (S) labeled ***f*** centres. (2 points)

Atom	Electronic Geometry	Molecular Geometry
Carbon (<i>e</i>)		
Sulfur (<i>f</i>)		

vi. Circle the value from the choices given below, that describes the **C-S-C bond angle at sulfur** (S) marked ***f***: (0.5 points)

>90° to <109.5°

109.5°

>109.5 to <120°

120°

>120° to <180°

**Question 8 12 points**

Using the **valence shell orbitals only** draw a **complete** molecular orbital (MO) diagram (with no s-p mixing) for the molecule **PS**, in the space provided below. **(6 points)**

Assume that 3s and 3p atomic orbitals will form molecular orbitals like those formed by the combination of 2s and 2p atomic orbitals that we learnt in class, and there is no s-p mixing.

Show **energy arrow**, **atomic orbitals** involved, **atom labels**, the **molecular orbitals** that are formed, **filled electrons** in the MO diagram etc. Make sure to label all the atomic and molecular orbitals.

Using the MO diagram above, answer the following questions:

- (i) Which one of the following will have the highest bond order? Circle the correct choice
A) PS^{1-} B) PS C) PS^{1+}
- (ii) Which one of the following will have the longest bond length? Circle the correct choice
A) PS^{1-} B) PS C) PS^{1+}
- (iii) Which of the following will be diamagnetic? Circle the correct choice.
A) PS^{1-} B) PS C) PS^{1+}
- (iv) Draw Lewis dot structure of the molecule with the shortest bond length (based on the MO diagram) from among the three (PS^{1-} , PS, PS^{1+}), and indicate the bond order calculated from the Lewis dot structure. **(3 points)**

Lewis Dot Structure

Bond Order



4B7B6858-8AF2-4F63-9450-474BFCEC99B5

midterm-exam-93388

#75 Page 10 of 10

SCRAP PAGE

**CHEMISTRY 110 –Midterm Exam****Thursday November 7, 2024, 6:30 – 8:30 pm****Examiners: Pallavi Sirjoosingh and Ashok Kakkar**

First name (please write as legibly as possible within the boxes)

--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--

Last name

--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--

Student ID number

--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--

This booklet has 8 questions on 10 pages (The last page can be used as a scrap page).

The staple must NOT be removed from the question booklet. Your intact question booklet must be returned at the end of the exam.

A separate booklet contains a periodic table, and datasheet.

Write your answers directly under the questions in the space provided. Do not write on the QR codes or near the top edges (as it may be cut off).

Please write **legibly and dark enough** for the scan to be readable. For students who use English as a second language, dictionaries are permitted. Calculators are allowed.

DO NOT ASK QUESTIONS OF THE INVIGILATORS about interpretations of the exam questions. If you have a concern about a question, please summarize it at the top of this page. All concerns will be evaluated after the exam. GOOD LUCK!

**Question 1 8 points** (*Assume all transitions are possible*)

a. 6 points Calculate the **energy (in kJ/mol)** required to ionize (completely remove) the electron from the **n=3** level of **He⁺ ion** (He: Helium). Show your work including equations. (3 sig. figs.)

b. 2 points

(I) Select the largest (longest) wavelength from the three choices below:

- i. Wavelength of photon to *completely remove* an electron from *1s orbital of Hydrogen atom*
- ii. Wavelength of photon to *completely remove* an electron from *1s orbital of Boron atom (B)*
- iii. Wavelength of photon to *completely remove* an electron from *1s orbital of Boron cation (B⁺)*

i or ii or iii

(II) Select the smallest (shortest) wavelength from the three choices below (*for H atom*):

- i. Wavelength of photon *emitted* for an electron transitioning from *2p to 1s orbital*
- ii. Wavelength of photon *absorbed* for an electron transitioning from *2s to 4p orbital*
- iii. Wavelength of photon *absorbed* for an electron transitioning from *1s to 3p orbital*

i or ii or iii



Question 2 8 points



A Chem110 student plots the graph shown on the left, from data obtained for the kinetic energy (eV) of the ejected electron versus frequency (s^{-1}) of the incident light, for an unknown metal. ($1\text{eV} = 1.6022 \times 10^{-19} \text{ J}$)

Based on the plot, answer the questions **a to c** below:

a. 1 point On the plot above, label the **threshold frequency**.

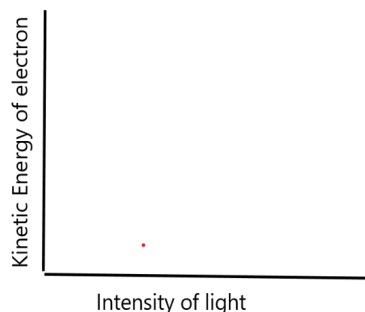
b. 2 points What is the value of the work potential (in J) of the unknown metal?

c. 3 points What is the *longest* wavelength (in nm) required to eject an electron from this metal's surface? Show your work including equation(s) used. (3 sig figs)

d. 2 points In the space provided below, plot the following

(Assume frequency is *above* threshold frequency):

- Kinetic energy* of an ejected electron versus the *intensity* of the incident light (left)
- Number of electrons* ejected versus the *frequency* of the incident light (right)



**Question 3** **6 points****a. 2 points**

In the ground state of Ge (Germanium; $Z = 32$) atom, **name** the following orbitals:

i. Orbital containing the electron that requires the maximum energy to be removed.

ii. Orbital containing the electron that experiences the minimum effective nuclear charge.

b. 2 points

An unknown anion (A^-) has the following *excited* electronic configuration: $[Ar] 4s^2 3d^3 4p^1$

(Ar: Argon; $Z = 18$)

Give the **condensed ground state electronic configuration** of the atom, **A** and the cation, **A^{2+}** .
Show your work. (Assume all electronic transitions are possible)

Electronic Configuration of A:

Electronic Configuration of A^{2+} :

c. 2 points

In a single atom, determine how many electrons can have the following set of quantum numbers?

(i) $n = 5; l = 3$

In a single atom, determine how many electrons can have the following set of quantum numbers?

(i) $n = 3; l = 2; m_s = +1/2$

**Question 4** **8 points**

Rank the following from 1 to 4 in **increasing** order. Provide the ranking in the boxes provided.

Note: Atomic number (Z) are provided for relevant atoms

a) Increasing (lowest to highest) order of *ionization energy*: (1= lowest; 4 = highest)

Mg

Mg: Z = 12

Rb

Rb: Z = 37

Cs

Cs: Z = 55

K

K : Z = 19

b) Increasing (lowest to highest) order of *bond angle*: (1= lowest; 4 = highest)

H₂OCCl₄XeF₄

HCN

c) Increasing (smallest to largest) order of *ionic size* (1= smallest; 4 = largest)

S²⁻

S: Z = 16

O²⁻

O: Z = 8

Be²⁺

Be: Z = 4

Na⁺

Na: Z = 11

d) Increasing (lowest to highest) order of *bond energy* (1= lowest; 4 = highest)

P-F

P-Cl

P-I

P-Br

**Question 5** **12 points**

a. 9 points For the molecules HClO_4 and TeF_6 , draw the most stable *Lewis structure* showing appropriate shape and *dashed/wedged bonds*, based on VSEPR. Determine the **molecular geometry** around the central atom. For HClO_4 , label the *most polar bond*.



b. 3 points (Note: Lone pairs are not shown in structures below)

(i) For the three *resonance* structures (1, 2 and 3) shown below, answer the following question:



(I) Circle the *most contributing* resonance structure.

Structure 1

Structure 2

Structure 3

(ii) For the three structures given (4, 5, and 6) below, answer the following questions:



(I) Circle all the structures that represent *resonance structures* of each other:

Structure 4

Structure 5

Structure 6

(II) Among the resonance structures, circle the *most contributing* resonance structure.

Structure 4

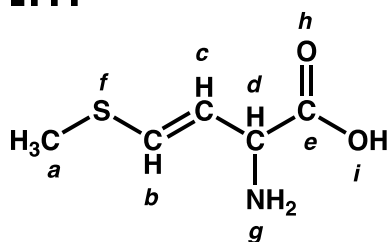
Structure 5

Structure 6

**Question 6** **6 points**

Using the data given below, calculate the **lattice energy** for SrO(s).

	ΔH° (kJ/mol)
$\text{Sr}(s) \longrightarrow \text{Sr}(g)$	+180
$\text{Sr}(g) \longrightarrow \text{Sr}^{2+}(g) + 2e^-$	+1468
$\text{O}_2(g) \longrightarrow 2\text{O}(g)$	+500
$\text{O}(g) + 2e^- \longrightarrow \text{O}^{2-}(g)$	+600
$\text{Sr}(s) + \frac{1}{2}\text{O}_2(g) \longrightarrow \text{SrO}(s)$	-550

**Question 7 10 points**

Consider the molecule given on the left, and answer questions (i) to (vi) using chemical bonding theories that we have learnt in class (valence bond theory, hybridization).

Note that the lone pairs are not shown in the structure:

i. Consider the carbon atoms marked in as ***a***, ***b***, ***c***, ***d***, and ***e*** in the molecule, and determine **hybridization** at each of the carbon centers. (2.5 points)

Atom	Hybridization
Carbon (<i>a</i>)	
Carbon (<i>b</i>)	
Carbon (<i>c</i>)	
Carbon (<i>d</i>)	
Carbon (<i>e</i>)	

ii. Now, consider the sulfur atom (S) marked as ***f***, nitrogen (N) atom (***g***), and oxygen(O) atoms (***h*** and ***i***), and determine their **hybridization**. (2 points)

Atom	Hybridization
Sulfur (<i>f</i>)	
Nitrogen (<i>g</i>)	
Oxygen (<i>h</i>)	
Oxygen (<i>i</i>)	

iii. **Describe** all the bonds formed by the carbon atoms ***a*** and ***b***, using sigma and pi designation as appropriate. (2 points)

Atom	Bonds formed (use sigma and pi as appropriate)
Carbon (<i>a</i>)	
Carbon (<i>b</i>)	

iv. How many total sp^3 -s sigma bonds are in the molecule? (1 point)

v. Determine **electronic** and **molecular geometry** at carbon (C) labeled ***e*** and sulfur (S) labeled ***f*** centres. (2 points)

Atom	Electronic Geometry	Molecular Geometry
Carbon (<i>e</i>)		
Sulfur (<i>f</i>)		

vi. Circle the value from the choices given below, that describes the **C-S-C bond angle at sulfur** (S) marked ***f***: (0.5 points)

>90° to <109.5° 109.5° >109.5 to <120° 120° >120° to <180°


Question 8 12 points

Using the **valence shell orbitals only** draw a **complete** molecular orbital (MO) diagram (with no s-p mixing) for the molecule **PS**, in the space provided below. **(6 points)**

Assume that 3s and 3p atomic orbitals will form molecular orbitals like those formed by the combination of 2s and 2p atomic orbitals that we learnt in class, and there is no s-p mixing.

Show **energy arrow**, **atomic orbitals** involved, **atom labels**, the **molecular orbitals** that are formed, **filled electrons** in the MO diagram etc. Make sure to label all the atomic and molecular orbitals.

Using the MO diagram above, answer the following questions:

- (i) Which one of the following will have the highest bond order? Circle the correct choice
 A) PS^{1-} B) PS C) PS^{1+}
- (ii) Which one of the following will have the longest bond length? Circle the correct choice
 A) PS^{1-} B) PS C) PS^{1+}
- (iii) Which of the following will be diamagnetic? Circle the correct choice.
 A) PS^{1-} B) PS C) PS^{1+}
- (iv) Draw Lewis dot structure of the molecule with the shortest bond length (based on the MO diagram) from among the three (PS^{1-} , PS, PS^{1+}), and indicate the bond order calculated from the Lewis dot structure. **(3 points)**

Lewis Dot Structure

Bond Order



4389E068-D635-4AD4-A0B2-FD9DD0867448

midterm-exam-93388

#76 Page 10 of 10

SCRAP PAGE

**CHEMISTRY 110 –Midterm Exam****Thursday November 7, 2024, 6:30 – 8:30 pm**

Examiners: Pallavi Sirjoosingh and Ashok Kakkar

First name (please write as legibly as possible within the boxes)

--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--

Last name

--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--

Student ID number

--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--

This booklet has 8 questions on 10 pages (The last page can be used as a scrap page).

The staple must NOT be removed from the question booklet. Your intact question booklet must be returned at the end of the exam.

A separate booklet contains a periodic table, and datasheet.

Write your answers directly under the questions in the space provided. Do not write on the QR codes or near the top edges (as it may be cut off).

Please write **legibly and dark enough** for the scan to be readable. For students who use English as a second language, dictionaries are permitted. Calculators are allowed.

DO NOT ASK QUESTIONS OF THE INVIGILATORS about interpretations of the exam questions. If you have a concern about a question, please summarize it at the top of this page. All concerns will be evaluated after the exam. GOOD LUCK!

**Question 1 8 points** (*Assume all transitions are possible*)

a. 6 points Calculate the **energy (in kJ/mol)** required to ionize (completely remove) the electron from the **n=3** level of **He⁺ ion** (He: Helium). Show your work including equations. (3 sig. figs.)

b. 2 points

(I) Select the largest (longest) wavelength from the three choices below:

- i. Wavelength of photon to *completely remove* an electron from *1s orbital of Hydrogen atom*
- ii. Wavelength of photon to *completely remove* an electron from *1s orbital of Boron atom (B)*
- iii. Wavelength of photon to *completely remove* an electron from *1s orbital of Boron cation (B⁺)*

i or ii or iii

(II) Select the smallest (shortest) wavelength from the three choices below (*for H atom*):

- i. Wavelength of photon *emitted* for an electron transitioning from *2p to 1s orbital*
- ii. Wavelength of photon *absorbed* for an electron transitioning from *2s to 4p orbital*
- iii. Wavelength of photon *absorbed* for an electron transitioning from *1s to 3p orbital*

i or ii or iii



Question 2 8 points



A Chem110 student plots the graph shown on the left, from data obtained for the kinetic energy (eV) of the ejected electron versus frequency (s^{-1}) of the incident light, for an unknown metal. ($1\text{eV} = 1.6022 \times 10^{-19} \text{ J}$)

Based on the plot, answer the questions **a to c** below:

a. 1 point On the plot above, label the **threshold frequency**.

b. 2 points What is the value of the work potential (in J) of the unknown metal?

c. 3 points What is the *longest* wavelength (in nm) required to eject an electron from this metal's surface? Show your work including equation(s) used. (3 sig figs)

d. 2 points In the space provided below, plot the following

(Assume frequency is *above* threshold frequency):

- Kinetic energy* of an ejected electron versus the *intensity* of the incident light (left)
- Number of electrons* ejected versus the *frequency* of the incident light (right)



**Question 3** **6 points****a. 2 points**

In the ground state of Ge (Germanium; $Z = 32$) atom, **name** the following orbitals:

i. Orbital containing the electron that requires the maximum energy to be removed.

ii. Orbital containing the electron that experiences the minimum effective nuclear charge.

b. 2 points

An unknown anion (A^-) has the following *excited* electronic configuration: $[Ar] 4s^2 3d^3 4p^1$

(Ar: Argon; $Z = 18$)

Give the **condensed ground state electronic configuration** of the atom, **A** and the cation, **A^{2+}** .
Show your work. (Assume all electronic transitions are possible)

Electronic Configuration of A:

Electronic Configuration of A^{2+} :

c. 2 points

In a single atom, determine how many electrons can have the following set of quantum numbers?

(i) $n = 5; l = 3$

In a single atom, determine how many electrons can have the following set of quantum numbers?

(i) $n = 3; l = 2; m_s = +1/2$

**Question 4** **8 points**

Rank the following from 1 to 4 in **increasing** order. Provide the ranking in the boxes provided.

Note: Atomic number (Z) are provided for relevant atoms

a) Increasing (lowest to highest) order of *ionization energy*: (1= lowest; 4 = highest)

Mg

Mg: Z = 12

Rb

Rb: Z = 37

Cs

Cs: Z = 55

K

K : Z = 19

b) Increasing (lowest to highest) order of *bond angle*: (1= lowest; 4 = highest)

H₂OCCl₄XeF₄

HCN

c) Increasing (smallest to largest) order of *ionic size* (1= smallest; 4 = largest)

S²⁻

S: Z = 16

O²⁻

O: Z = 8

Be²⁺

Be: Z = 4

Na⁺

Na: Z = 11

d) Increasing (lowest to highest) order of *bond energy* (1= lowest; 4 = highest)

P-F

P-Cl

P-I

P-Br

**Question 5** **12 points**

a. 9 points For the molecules HClO_4 and TeF_6 , draw the most stable *Lewis structure* showing appropriate shape and *dashed/wedged bonds*, based on VSEPR. Determine the **molecular geometry** around the central atom. For HClO_4 , label the *most polar bond*.



b. 3 points (Note: Lone pairs are not shown in structures below)

(i) For the three *resonance* structures (1, 2 and 3) shown below, answer the following question:



(I) Circle the *most contributing* resonance structure.

Structure 1

Structure 2

Structure 3

(ii) For the three structures given (4, 5, and 6) below, answer the following questions:



(I) Circle all the structures that represent *resonance structures* of each other:

Structure 4

Structure 5

Structure 6

(II) Among the resonance structures, circle the *most contributing* resonance structure.

Structure 4

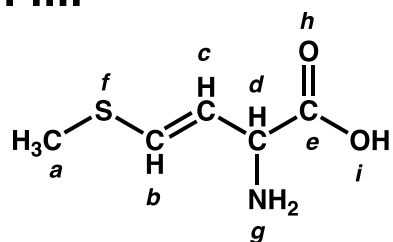
Structure 5

Structure 6

**Question 6** **6 points**

Using the data given below, calculate the **lattice energy** for SrO(s).

	ΔH° (kJ/mol)
$\text{Sr}(s) \longrightarrow \text{Sr}(g)$	+180
$\text{Sr}(g) \longrightarrow \text{Sr}^{2+}(g) + 2e^-$	+1468
$\text{O}_2(g) \longrightarrow 2\text{O}(g)$	+500
$\text{O}(g) + 2e^- \longrightarrow \text{O}^{2-}(g)$	+600
$\text{Sr}(s) + \frac{1}{2}\text{O}_2(g) \longrightarrow \text{SrO}(s)$	-550

**Question 7 10 points**

Consider the molecule given on the left, and answer questions (i) to (vi) using chemical bonding theories that we have learnt in class (valence bond theory, hybridization).

Note that the lone pairs are not shown in the structure:

i. Consider the carbon atoms marked in as ***a***, ***b***, ***c***, ***d***, and ***e*** in the molecule, and determine **hybridization** at each of the carbon centers. (2.5 points)

Atom	Hybridization
Carbon (<i>a</i>)	
Carbon (<i>b</i>)	
Carbon (<i>c</i>)	
Carbon (<i>d</i>)	
Carbon (<i>e</i>)	

ii. Now, consider the sulfur atom (S) marked as ***f***, nitrogen (N) atom (***g***), and oxygen(O) atoms (***h*** and ***i***), and determine their **hybridization**. (2 points)

Atom	Hybridization
Sulfur (<i>f</i>)	
Nitrogen (<i>g</i>)	
Oxygen (<i>h</i>)	
Oxygen (<i>i</i>)	

iii. **Describe** all the bonds formed by the carbon atoms ***a*** and ***b***, using sigma and pi designation as appropriate. (2 points)

Atom	Bonds formed (use sigma and pi as appropriate)
Carbon (<i>a</i>)	
Carbon (<i>b</i>)	

iv. How many total sp^3 -s sigma bonds are in the molecule? (1 point)

v. Determine **electronic** and **molecular geometry** at carbon (C) labeled ***e*** and sulfur (S) labeled ***f*** centres. (2 points)

Atom	Electronic Geometry	Molecular Geometry
Carbon (<i>e</i>)		
Sulfur (<i>f</i>)		

vi. Circle the value from the choices given below, that describes the **C-S-C bond angle at sulfur** (S) marked ***f***: (0.5 points)

>90° to <109.5° 109.5° >109.5 to <120° 120° >120° to <180°


Question 8 12 points

Using the **valence shell orbitals only** draw a **complete** molecular orbital (MO) diagram (with no s-p mixing) for the molecule **PS**, in the space provided below. **(6 points)**

Assume that 3s and 3p atomic orbitals will form molecular orbitals like those formed by the combination of 2s and 2p atomic orbitals that we learnt in class, and there is no s-p mixing.

Show **energy arrow**, **atomic orbitals** involved, **atom labels**, the **molecular orbitals** that are formed, **filled electrons** in the MO diagram etc. Make sure to label all the atomic and molecular orbitals.

Using the MO diagram above, answer the following questions:

- (i) Which one of the following will have the highest bond order? Circle the correct choice
 A) PS^{1-} B) PS C) PS^{1+}
- (ii) Which one of the following will have the longest bond length? Circle the correct choice
 A) PS^{1-} B) PS C) PS^{1+}
- (iii) Which of the following will be diamagnetic? Circle the correct choice.
 A) PS^{1-} B) PS C) PS^{1+}
- (iv) Draw Lewis dot structure of the molecule with the shortest bond length (based on the MO diagram) from among the three (PS^{1-} , PS, PS^{1+}), and indicate the bond order calculated from the Lewis dot structure. **(3 points)**

Lewis Dot Structure

Bond Order



3611B656-AAF4-4FAD-86BA-99947B65A1B1

midterm-exam-93388

#77 Page 10 of 10

SCRAP PAGE

**CHEMISTRY 110 –Midterm Exam****Thursday November 7, 2024, 6:30 – 8:30 pm****Examiners: Pallavi Sirjoosingh and Ashok Kakkar**

First name (please write as legibly as possible within the boxes)

--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--

Last name

--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--

Student ID number

--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--

This booklet has 8 questions on 10 pages (The last page can be used as a scrap page).

The staple must NOT be removed from the question booklet. Your intact question booklet must be returned at the end of the exam.

A separate booklet contains a periodic table, and datasheet.

Write your answers directly under the questions in the space provided. Do not write on the QR codes or near the top edges (as it may be cut off).

Please write **legibly and dark enough** for the scan to be readable. For students who use English as a second language, dictionaries are permitted. Calculators are allowed.

DO NOT ASK QUESTIONS OF THE INVIGILATORS about interpretations of the exam questions. If you have a concern about a question, please summarize it at the top of this page. All concerns will be evaluated after the exam. GOOD LUCK!

**Question 1 8 points** (*Assume all transitions are possible*)

a. 6 points Calculate the **energy (in kJ/mol)** required to ionize (completely remove) the electron from the **n=3** level of **He⁺ ion** (He: Helium). Show your work including equations. (3 sig. figs.)

b. 2 points

(I) Select the largest (longest) wavelength from the three choices below:

- i. Wavelength of photon to *completely remove* an electron from *1s orbital of Hydrogen atom*
- ii. Wavelength of photon to *completely remove* an electron from *1s orbital of Boron atom (B)*
- iii. Wavelength of photon to *completely remove* an electron from *1s orbital of Boron cation (B⁺)*

i or ii or iii

(II) Select the smallest (shortest) wavelength from the three choices below (*for H atom*):

- i. Wavelength of photon *emitted* for an electron transitioning from *2p to 1s orbital*
- ii. Wavelength of photon *absorbed* for an electron transitioning from *2s to 4p orbital*
- iii. Wavelength of photon *absorbed* for an electron transitioning from *1s to 3p orbital*

i or ii or iii



Question 2 8 points



A Chem110 student plots the graph shown on the left, from data obtained for the kinetic energy (eV) of the ejected electron versus frequency (s^{-1}) of the incident light, for an unknown metal. ($1\text{eV} = 1.6022 \times 10^{-19} \text{ J}$)

Based on the plot, answer the questions **a to c** below:

a. 1 point On the plot above, label the **threshold frequency**.

b. 2 points What is the value of the work potential (in J) of the unknown metal?

c. 3 points What is the *longest* wavelength (in nm) required to eject an electron from this metal's surface? Show your work including equation(s) used. (3 sig figs)

d. 2 points In the space provided below, plot the following

(Assume frequency is *above* threshold frequency):

- i. *Kinetic energy* of an ejected electron versus the *intensity* of the incident light (left)
- ii. *Number of electrons* ejected versus the *frequency* of the incident light (right)



**Question 3** **6 points****a. 2 points**

In the ground state of Ge (Germanium; $Z = 32$) atom, **name** the following orbitals:

i. Orbital containing the electron that requires the maximum energy to be removed.

ii. Orbital containing the electron that experiences the minimum effective nuclear charge.

b. 2 points

An unknown anion (A^-) has the following *excited* electronic configuration: $[Ar] 4s^2 3d^3 4p^1$

(Ar: Argon; $Z = 18$)

Give the **condensed ground state electronic configuration** of the atom, **A** and the cation, **A^{2+}** .
Show your work. (Assume all electronic transitions are possible)

Electronic Configuration of A:

Electronic Configuration of A^{2+} :

c. 2 points

In a single atom, determine how many electrons can have the following set of quantum numbers?

(i) $n = 5; l = 3$

In a single atom, determine how many electrons can have the following set of quantum numbers?

(i) $n = 3; l = 2; m_s = +1/2$

**Question 4** **8 points**

Rank the following from 1 to 4 in **increasing** order. Provide the ranking in the boxes provided.

Note: Atomic number (Z) are provided for relevant atoms

a) Increasing (lowest to highest) order of *ionization energy*: (1= lowest; 4 = highest)

Mg

Mg: Z = 12

Rb

Rb: Z = 37

Cs

Cs: Z = 55

K

K : Z = 19

b) Increasing (lowest to highest) order of *bond angle*: (1= lowest; 4 = highest)

H₂OCCl₄XeF₄

HCN

c) Increasing (smallest to largest) order of *ionic size* (1= smallest; 4 = largest)

S²⁻

S: Z = 16

O²⁻

O: Z = 8

Be²⁺

Be: Z = 4

Na⁺

Na: Z = 11

d) Increasing (lowest to highest) order of *bond energy* (1= lowest; 4 = highest)

P-F

P-Cl

P-I

P-Br

**Question 5** **12 points**

a. 9 points For the molecules HClO_4 and TeF_6 , draw the most stable *Lewis structure* showing appropriate shape and *dashed/wedged bonds*, based on VSEPR. Determine the **molecular geometry** around the central atom. For HClO_4 , label the *most polar bond*.



b. 3 points (Note: Lone pairs are not shown in structures below)

(i) For the three *resonance* structures (1, 2 and 3) shown below, answer the following question:



(I) Circle the *most contributing* resonance structure.

Structure 1

Structure 2

Structure 3

(ii) For the three structures given (4, 5, and 6) below, answer the following questions:



(I) Circle all the structures that represent *resonance structures* of each other:

Structure 4

Structure 5

Structure 6

(II) Among the resonance structures, circle the *most contributing* resonance structure.

Structure 4

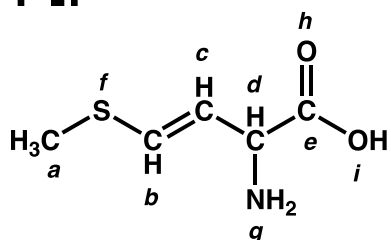
Structure 5

Structure 6

**Question 6** **6 points**

Using the data given below, calculate the **lattice energy** for SrO(s).

	ΔH° (kJ/mol)
$\text{Sr}(s) \longrightarrow \text{Sr}(g)$	+180
$\text{Sr}(g) \longrightarrow \text{Sr}^{2+}(g) + 2e^-$	+1468
$\text{O}_2(g) \longrightarrow 2\text{O}(g)$	+500
$\text{O}(g) + 2e^- \longrightarrow \text{O}^{2-}(g)$	+600
$\text{Sr}(s) + \frac{1}{2}\text{O}_2(g) \longrightarrow \text{SrO}(s)$	-550

**Question 7 10 points**

Consider the molecule given on the left, and answer questions (i) to (vi) using chemical bonding theories that we have learnt in class (valence bond theory, hybridization).

Note that the lone pairs are not shown in the structure:

i. Consider the carbon atoms marked in as ***a***, ***b***, ***c***, ***d***, and ***e*** in the molecule, and determine **hybridization** at each of the carbon centers. (2.5 points)

Atom	Hybridization
Carbon (<i>a</i>)	
Carbon (<i>b</i>)	
Carbon (<i>c</i>)	
Carbon (<i>d</i>)	
Carbon (<i>e</i>)	

ii. Now, consider the sulfur atom (S) marked as ***f***, nitrogen (N) atom (***g***), and oxygen(O) atoms (***h*** and ***i***), and determine their **hybridization**. (2 points)

Atom	Hybridization
Sulfur (<i>f</i>)	
Nitrogen (<i>g</i>)	
Oxygen (<i>h</i>)	
Oxygen (<i>i</i>)	

iii. **Describe** all the bonds formed by the carbon atoms ***a*** and ***b***, using sigma and pi designation as appropriate. (2 points)

Atom	Bonds formed (use sigma and pi as appropriate)
Carbon (<i>a</i>)	
Carbon (<i>b</i>)	

iv. How many total sp^3 -s sigma bonds are in the molecule? (1 point)

v. Determine **electronic** and **molecular geometry** at carbon (C) labeled ***e*** and sulfur (S) labeled ***f*** centres. (2 points)

Atom	Electronic Geometry	Molecular Geometry
Carbon (<i>e</i>)		
Sulfur (<i>f</i>)		

vi. Circle the value from the choices given below, that describes the **C-S-C bond angle at sulfur** (S) marked ***f***: (0.5 points)

$>90^\circ$ to $<109.5^\circ$ 109.5° >109.5 to $<120^\circ$ 120° $>120^\circ$ to $<180^\circ$


Question 8 12 points

Using the **valence shell orbitals only** draw a **complete** molecular orbital (MO) diagram (with no s-p mixing) for the molecule **PS**, in the space provided below. **(6 points)**

Assume that 3s and 3p atomic orbitals will form molecular orbitals like those formed by the combination of 2s and 2p atomic orbitals that we learnt in class, and there is no s-p mixing.

Show **energy arrow**, **atomic orbitals** involved, **atom labels**, the **molecular orbitals** that are formed, **filled electrons** in the MO diagram etc. Make sure to label all the atomic and molecular orbitals.

Using the MO diagram above, answer the following questions:

- (i) Which one of the following will have the highest bond order? Circle the correct choice
 A) PS^{1-} B) PS C) PS^{1+}
- (ii) Which one of the following will have the longest bond length? Circle the correct choice
 A) PS^{1-} B) PS C) PS^{1+}
- (iii) Which of the following will be diamagnetic? Circle the correct choice.
 A) PS^{1-} B) PS C) PS^{1+}
- (iv) Draw Lewis dot structure of the molecule with the shortest bond length (based on the MO diagram) from among the three (PS^{1-} , PS, PS^{1+}), and indicate the bond order calculated from the Lewis dot structure. **(3 points)**

Lewis Dot Structure

Bond Order



1866C965-4F17-484E-8258-D74D2E2FDB55

midterm-exam-93388

#78 Page 10 of 10

SCRAP PAGE

**CHEMISTRY 110 –Midterm Exam****Thursday November 7, 2024, 6:30 – 8:30 pm**

Examiners: Pallavi Sirjoosingh and Ashok Kakkar

First name (please write as legibly as possible within the boxes)

--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--

Last name

--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--

Student ID number

--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--

This booklet has 8 questions on 10 pages (The last page can be used as a scrap page).

The staple must NOT be removed from the question booklet. Your intact question booklet must be returned at the end of the exam.

A separate booklet contains a periodic table, and datasheet.

Write your answers directly under the questions in the space provided. Do not write on the QR codes or near the top edges (as it may be cut off).

Please write **legibly and dark enough** for the scan to be readable. For students who use English as a second language, dictionaries are permitted. Calculators are allowed.

DO NOT ASK QUESTIONS OF THE INVIGILATORS about interpretations of the exam questions. If you have a concern about a question, please summarize it at the top of this page. All concerns will be evaluated after the exam. GOOD LUCK!

**Question 1 8 points** (*Assume all transitions are possible*)

a. 6 points Calculate the **energy (in kJ/mol)** required to ionize (completely remove) the electron from the **n=3** level of **He⁺ ion** (He: Helium). Show your work including equations. (3 sig. figs.)

b. 2 points

(I) Select the largest (longest) wavelength from the three choices below:

- i. Wavelength of photon to *completely remove* an electron from *1s orbital of Hydrogen atom*
- ii. Wavelength of photon to *completely remove* an electron from *1s orbital of Boron atom (B)*
- iii. Wavelength of photon to *completely remove* an electron from *1s orbital of Boron cation (B⁺)*

i or ii or iii

(II) Select the smallest (shortest) wavelength from the three choices below (*for H atom*):

- i. Wavelength of photon *emitted* for an electron transitioning from *2p to 1s orbital*
- ii. Wavelength of photon *absorbed* for an electron transitioning from *2s to 4p orbital*
- iii. Wavelength of photon *absorbed* for an electron transitioning from *1s to 3p orbital*

i or ii or iii



Question 2 8 points



A Chem110 student plots the graph shown on the left, from data obtained for the kinetic energy (eV) of the ejected electron versus frequency (s^{-1}) of the incident light, for an unknown metal. ($1\text{eV} = 1.6022 \times 10^{-19} \text{ J}$)

Based on the plot, answer the questions **a to c** below:

a. 1 point On the plot above, label the **threshold frequency**.

b. 2 points What is the value of the work potential (in J) of the unknown metal?

c. 3 points What is the *longest* wavelength (in nm) required to eject an electron from this metal's surface? Show your work including equation(s) used. (3 sig figs)

d. 2 points In the space provided below, plot the following

(Assume frequency is *above* threshold frequency):

- i. *Kinetic energy* of an ejected electron versus the *intensity* of the incident light (left)
- ii. *Number of electrons* ejected versus the *frequency* of the incident light (right)



**Question 3** **6 points****a. 2 points**

In the ground state of Ge (Germanium; $Z = 32$) atom, **name** the following orbitals:

i. Orbital containing the electron that requires the maximum energy to be removed.

ii. Orbital containing the electron that experiences the minimum effective nuclear charge.

b. 2 points

An unknown anion (A^-) has the following *excited* electronic configuration: $[Ar] 4s^2 3d^3 4p^1$

(Ar: Argon; $Z = 18$)

Give the **condensed ground state electronic configuration** of the atom, **A** and the cation, **A^{2+}** .
Show your work. (Assume all electronic transitions are possible)

Electronic Configuration of A:

Electronic Configuration of A^{2+} :

c. 2 points

In a single atom, determine how many electrons can have the following set of quantum numbers?

(i) $n = 5; l = 3$

In a single atom, determine how many electrons can have the following set of quantum numbers?

(i) $n = 3; l = 2; m_s = +1/2$

**Question 4** **8 points**

Rank the following from 1 to 4 in **increasing** order. Provide the ranking in the boxes provided.

Note: Atomic number (Z) are provided for relevant atoms

a) Increasing (lowest to highest) order of *ionization energy*: (1= lowest; 4 = highest)

Mg

Mg: Z = 12

Rb

Rb: Z = 37

Cs

Cs: Z = 55

K

K : Z = 19

b) Increasing (lowest to highest) order of *bond angle*: (1= lowest; 4 = highest)

H₂OCCl₄XeF₄

HCN

c) Increasing (smallest to largest) order of *ionic size* (1= smallest; 4 = largest)

S²⁻

S: Z = 16

O²⁻

O: Z = 8

Be²⁺

Be: Z = 4

Na⁺

Na: Z = 11

d) Increasing (lowest to highest) order of *bond energy* (1= lowest; 4 = highest)

P-F

P-Cl

P-I

P-Br

**Question 5** **12 points**

a. 9 points For the molecules HClO_4 and TeF_6 , draw the most stable *Lewis structure* showing appropriate shape and *dashed/wedged bonds*, based on VSEPR. Determine the **molecular geometry** around the central atom. For HClO_4 , label the *most polar bond*.



b. 3 points (Note: Lone pairs are not shown in structures below)

(i) For the three *resonance* structures (1, 2 and 3) shown below, answer the following question:



(I) Circle the *most contributing* resonance structure.

Structure 1

Structure 2

Structure 3

(ii) For the three structures given (4, 5, and 6) below, answer the following questions:



(I) Circle all the structures that represent *resonance structures* of each other:

Structure 4

Structure 5

Structure 6

(II) Among the resonance structures, circle the *most contributing* resonance structure.

Structure 4

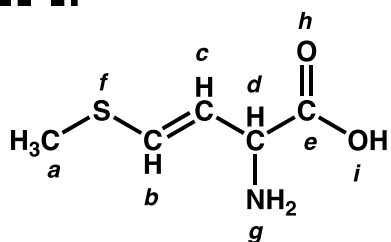
Structure 5

Structure 6

**Question 6** **6 points**

Using the data given below, calculate the **lattice energy** for SrO(s).

	ΔH° (kJ/mol)
$\text{Sr}(s) \longrightarrow \text{Sr}(g)$	+180
$\text{Sr}(g) \longrightarrow \text{Sr}^{2+}(g) + 2e^-$	+1468
$\text{O}_2(g) \longrightarrow 2\text{O}(g)$	+500
$\text{O}(g) + 2e^- \longrightarrow \text{O}^{2-}(g)$	+600
$\text{Sr}(s) + \frac{1}{2}\text{O}_2(g) \longrightarrow \text{SrO}(s)$	-550

**Question 7 10 points**

Consider the molecule given on the left, and answer questions (i) to (vi) using chemical bonding theories that we have learnt in class (valence bond theory, hybridization).

Note that the lone pairs are not shown in the structure:

i. Consider the carbon atoms marked in as ***a***, ***b***, ***c***, ***d***, and ***e*** in the molecule, and determine **hybridization** at each of the carbon centers. (2.5 points)

Atom	Hybridization
Carbon (<i>a</i>)	
Carbon (<i>b</i>)	
Carbon (<i>c</i>)	
Carbon (<i>d</i>)	
Carbon (<i>e</i>)	

ii. Now, consider the sulfur atom (S) marked as ***f***, nitrogen (N) atom (***g***), and oxygen(O) atoms (***h*** and ***i***), and determine their **hybridization**. (2 points)

Atom	Hybridization
Sulfur (<i>f</i>)	
Nitrogen (<i>g</i>)	
Oxygen (<i>h</i>)	
Oxygen (<i>i</i>)	

iii. **Describe** all the bonds formed by the carbon atoms ***a*** and ***b***, using sigma and pi designation as appropriate. (2 points)

Atom	Bonds formed (use sigma and pi as appropriate)
Carbon (<i>a</i>)	
Carbon (<i>b</i>)	

iv. How many total sp^3 -s sigma bonds are in the molecule? (1 point)

v. Determine **electronic** and **molecular geometry** at carbon (C) labeled ***e*** and sulfur (S) labeled ***f*** centres. (2 points)

Atom	Electronic Geometry	Molecular Geometry
Carbon (<i>e</i>)		
Sulfur (<i>f</i>)		

vi. Circle the value from the choices given below, that describes the **C-S-C bond angle at sulfur** (S) marked ***f***: (0.5 points)

>90° to <109.5°

109.5°

>109.5 to <120°

120°

>120° to <180°


Question 8 12 points

Using the **valence shell orbitals only** draw a **complete** molecular orbital (MO) diagram (with no s-p mixing) for the molecule **PS**, in the space provided below. **(6 points)**

Assume that 3s and 3p atomic orbitals will form molecular orbitals like those formed by the combination of 2s and 2p atomic orbitals that we learnt in class, and there is no s-p mixing.

Show **energy arrow**, **atomic orbitals** involved, **atom labels**, the **molecular orbitals** that are formed, **filled electrons** in the MO diagram etc. Make sure to label all the atomic and molecular orbitals.

Using the MO diagram above, answer the following questions:

- (i) Which one of the following will have the highest bond order? Circle the correct choice
 A) PS^{1-} B) PS C) PS^{1+}
- (ii) Which one of the following will have the longest bond length? Circle the correct choice
 A) PS^{1-} B) PS C) PS^{1+}
- (iii) Which of the following will be diamagnetic? Circle the correct choice.
 A) PS^{1-} B) PS C) PS^{1+}
- (iv) Draw Lewis dot structure of the molecule with the shortest bond length (based on the MO diagram) from among the three (PS^{1-} , PS, PS^{1+}), and indicate the bond order calculated from the Lewis dot structure. **(3 points)**

Lewis Dot Structure

Bond Order



2186EE93-35BF-4E3D-BE93-9B18A29CC79D

midterm-exam-93388

#79 Page 10 of 10

SCRAP PAGE

**CHEMISTRY 110 –Midterm Exam****Thursday November 7, 2024, 6:30 – 8:30 pm**

Examiners: Pallavi Sirjoosingh and Ashok Kakkar

First name (please write as legibly as possible within the boxes)

--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--

Last name

--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--

Student ID number

--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--

This booklet has 8 questions on 10 pages (The last page can be used as a scrap page).

The staple must NOT be removed from the question booklet. Your intact question booklet must be returned at the end of the exam.

A separate booklet contains a periodic table, and datasheet.

Write your answers directly under the questions in the space provided. Do not write on the QR codes or near the top edges (as it may be cut off).

Please write **legibly and dark enough** for the scan to be readable. For students who use English as a second language, dictionaries are permitted. Calculators are allowed.

DO NOT ASK QUESTIONS OF THE INVIGILATORS about interpretations of the exam questions. If you have a concern about a question, please summarize it at the top of this page. All concerns will be evaluated after the exam. GOOD LUCK!

**Question 1 8 points** (*Assume all transitions are possible*)

a. 6 points Calculate the **energy (in kJ/mol)** required to ionize (completely remove) the electron from the **n=3** level of **He⁺ ion** (He: Helium). Show your work including equations. (3 sig. figs.)

b. 2 points

(I) Select the largest (longest) wavelength from the three choices below:

- i. Wavelength of photon to *completely remove* an electron from *1s orbital of Hydrogen atom*
- ii. Wavelength of photon to *completely remove* an electron from *1s orbital of Boron atom (B)*
- iii. Wavelength of photon to *completely remove* an electron from *1s orbital of Boron cation (B⁺)*

i or ii or iii

(II) Select the smallest (shortest) wavelength from the three choices below (*for H atom*):

- i. Wavelength of photon *emitted* for an electron transitioning from *2p to 1s orbital*
- ii. Wavelength of photon *absorbed* for an electron transitioning from *2s to 4p orbital*
- iii. Wavelength of photon *absorbed* for an electron transitioning from *1s to 3p orbital*

i or ii or iii



Question 2 8 points



A Chem110 student plots the graph shown on the left, from data obtained for the kinetic energy (eV) of the ejected electron versus frequency (s^{-1}) of the incident light, for an unknown metal. ($1\text{eV} = 1.6022 \times 10^{-19} \text{ J}$)

Based on the plot, answer the questions **a to c** below:

a. 1 point On the plot above, label the **threshold frequency**.

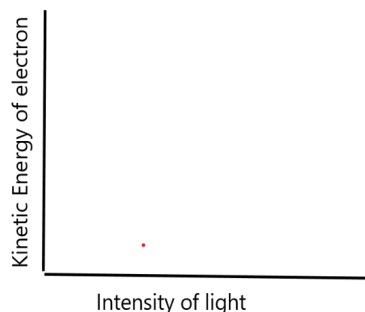
b. 2 points What is the value of the work potential (in J) of the unknown metal?

c. 3 points What is the *longest* wavelength (in nm) required to eject an electron from this metal's surface? Show your work including equation(s) used. (3 sig figs)

d. 2 points In the space provided below, plot the following

(Assume frequency is *above* threshold frequency):

- Kinetic energy* of an ejected electron versus the *intensity* of the incident light (left)
- Number of electrons* ejected versus the *frequency* of the incident light (right)



**Question 3** **6 points****a. 2 points**

In the ground state of Ge (Germanium; $Z = 32$) atom, **name** the following orbitals:

i. Orbital containing the electron that requires the maximum energy to be removed.

ii. Orbital containing the electron that experiences the minimum effective nuclear charge.

b. 2 points

An unknown anion (A^-) has the following *excited* electronic configuration: $[Ar] 4s^2 3d^3 4p^1$

(Ar: Argon; $Z = 18$)

Give the **condensed ground state electronic configuration** of the atom, **A** and the cation, A^{2+} .
Show your work. (Assume all electronic transitions are possible)

Electronic Configuration of A:

Electronic Configuration of A^{2+} :

c. 2 points

In a single atom, determine how many electrons can have the following set of quantum numbers?

(i) $n = 5; l = 3$

In a single atom, determine how many electrons can have the following set of quantum numbers?

(i) $n = 3; l = 2; m_s = +1/2$

**Question 4** **8 points**

Rank the following from 1 to 4 in **increasing** order. Provide the ranking in the boxes provided.

Note: Atomic number (Z) are provided for relevant atoms

a) Increasing (lowest to highest) order of *ionization energy*: (1= lowest; 4 = highest)

Mg

Mg: Z = 12

Rb

Rb: Z = 37

Cs

Cs: Z = 55

K

K : Z = 19

b) Increasing (lowest to highest) order of *bond angle*: (1= lowest; 4 = highest)

H₂OCCl₄XeF₄

HCN

c) Increasing (smallest to largest) order of *ionic size* (1= smallest; 4 = largest)

S²⁻

S: Z = 16

O²⁻

O: Z = 8

Be²⁺

Be: Z = 4

Na⁺

Na: Z = 11

d) Increasing (lowest to highest) order of *bond energy* (1= lowest; 4 = highest)

P-F

P-Cl

P-I

P-Br

**Question 5** **12 points**

a. 9 points For the molecules HClO_4 and TeF_6 , draw the most stable *Lewis structure* showing appropriate shape and *dashed/wedged bonds*, based on VSEPR. Determine the **molecular geometry** around the central atom. For HClO_4 , label the *most polar bond*.



b. 3 points (Note: Lone pairs are not shown in structures below)

(i) For the three *resonance* structures (1, 2 and 3) shown below, answer the following question:



(I) Circle the *most contributing* resonance structure.

Structure 1

Structure 2

Structure 3

(ii) For the three structures given (4, 5, and 6) below, answer the following questions:



(I) Circle all the structures that represent *resonance structures* of each other:

Structure 4

Structure 5

Structure 6

(II) Among the resonance structures, circle the *most contributing* resonance structure.

Structure 4

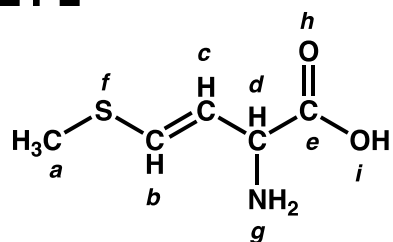
Structure 5

Structure 6

**Question 6** **6 points**

Using the data given below, calculate the **lattice energy** for SrO(s).

	ΔH° (kJ/mol)
$\text{Sr}(s) \longrightarrow \text{Sr}(g)$	+180
$\text{Sr}(g) \longrightarrow \text{Sr}^{2+}(g) + 2e^-$	+1468
$\text{O}_2(g) \longrightarrow 2\text{O}(g)$	+500
$\text{O}(g) + 2e^- \longrightarrow \text{O}^{2-}(g)$	+600
$\text{Sr}(s) + \frac{1}{2}\text{O}_2(g) \longrightarrow \text{SrO}(s)$	-550

**Question 7 10 points**

Consider the molecule given on the left, and answer questions (i) to (vi) using chemical bonding theories that we have learnt in class (valence bond theory, hybridization).

Note that the lone pairs are not shown in the structure:

i. Consider the carbon atoms marked in as ***a***, ***b***, ***c***, ***d***, and ***e*** in the molecule, and determine **hybridization** at each of the carbon centers. (2.5 points)

Atom	Hybridization
Carbon (<i>a</i>)	
Carbon (<i>b</i>)	
Carbon (<i>c</i>)	
Carbon (<i>d</i>)	
Carbon (<i>e</i>)	

ii. Now, consider the sulfur atom (S) marked as ***f***, nitrogen (N) atom (***g***), and oxygen(O) atoms (***h*** and ***i***), and determine their **hybridization**. (2 points)

Atom	Hybridization
Sulfur (<i>f</i>)	
Nitrogen (<i>g</i>)	
Oxygen (<i>h</i>)	
Oxygen (<i>i</i>)	

iii. **Describe** all the bonds formed by the carbon atoms ***a*** and ***b***, using sigma and pi designation as appropriate. (2 points)

Atom	Bonds formed (use sigma and pi as appropriate)
Carbon (<i>a</i>)	
Carbon (<i>b</i>)	

iv. How many total sp^3 -s sigma bonds are in the molecule? (1 point)

v. Determine **electronic** and **molecular geometry** at carbon (C) labeled ***e*** and sulfur (S) labeled ***f*** centres. (2 points)

Atom	Electronic Geometry	Molecular Geometry
Carbon (<i>e</i>)		
Sulfur (<i>f</i>)		

vi. Circle the value from the choices given below, that describes the **C-S-C bond angle at sulfur** (S) marked ***f***: (0.5 points)

>90° to <109.5° 109.5° >109.5 to <120° 120° >120° to <180°


Question 8 12 points

Using the **valence shell orbitals only** draw a **complete** molecular orbital (MO) diagram (with no s-p mixing) for the molecule **PS**, in the space provided below. **(6 points)**

Assume that 3s and 3p atomic orbitals will form molecular orbitals like those formed by the combination of 2s and 2p atomic orbitals that we learnt in class, and there is no s-p mixing.

Show **energy arrow**, **atomic orbitals** involved, **atom labels**, the **molecular orbitals** that are formed, **filled electrons** in the MO diagram etc. Make sure to label all the atomic and molecular orbitals.

Using the MO diagram above, answer the following questions:

- (i) Which one of the following will have the highest bond order? Circle the correct choice
 A) PS^{1-} B) PS C) PS^{1+}
- (ii) Which one of the following will have the longest bond length? Circle the correct choice
 A) PS^{1-} B) PS C) PS^{1+}
- (iii) Which of the following will be diamagnetic? Circle the correct choice.
 A) PS^{1-} B) PS C) PS^{1+}
- (iv) Draw Lewis dot structure of the molecule with the shortest bond length (based on the MO diagram) from among the three (PS^{1-} , PS, PS^{1+}), and indicate the bond order calculated from the Lewis dot structure. **(3 points)**

Lewis Dot Structure

Bond Order



59B96916-7A2E-4323-8A30-E9A4905FBB7F

midterm-exam-93388

#80 Page 10 of 10

SCRAP PAGE

**CHEMISTRY 110 –Midterm Exam****Thursday November 7, 2024, 6:30 – 8:30 pm**

Examiners: Pallavi Sirjoosingh and Ashok Kakkar

First name (please write as legibly as possible within the boxes)

--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--

Last name

--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--

Student ID number

--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--

This booklet has 8 questions on 10 pages (The last page can be used as a scrap page).

The staple must NOT be removed from the question booklet. Your intact question booklet must be returned at the end of the exam.

A separate booklet contains a periodic table, and datasheet.

Write your answers directly under the questions in the space provided. Do not write on the QR codes or near the top edges (as it may be cut off).

Please write **legibly and dark enough** for the scan to be readable. For students who use English as a second language, dictionaries are permitted. Calculators are allowed.

DO NOT ASK QUESTIONS OF THE INVIGILATORS about interpretations of the exam questions. If you have a concern about a question, please summarize it at the top of this page. All concerns will be evaluated after the exam. GOOD LUCK!

**Question 1 8 points** (*Assume all transitions are possible*)

a. 6 points Calculate the **energy (in kJ/mol)** required to ionize (completely remove) the electron from the **n=3** level of **He⁺ ion** (He: Helium). Show your work including equations. (3 sig. figs.)

b. 2 points

(I) Select the largest (longest) wavelength from the three choices below:

- i. Wavelength of photon to *completely remove* an electron from *1s orbital of Hydrogen atom*
- ii. Wavelength of photon to *completely remove* an electron from *1s orbital of Boron atom (B)*
- iii. Wavelength of photon to *completely remove* an electron from *1s orbital of Boron cation (B⁺)*

i or ii or iii

(II) Select the smallest (shortest) wavelength from the three choices below (*for H atom*):

- i. Wavelength of photon *emitted* for an electron transitioning from *2p to 1s orbital*
- ii. Wavelength of photon *absorbed* for an electron transitioning from *2s to 4p orbital*
- iii. Wavelength of photon *absorbed* for an electron transitioning from *1s to 3p orbital*

i or ii or iii



Question 2 8 points



A Chem110 student plots the graph shown on the left, from data obtained for the kinetic energy (eV) of the ejected electron versus frequency (s^{-1}) of the incident light, for an unknown metal. ($1\text{eV} = 1.6022 \times 10^{-19} \text{ J}$)

Based on the plot, answer the questions **a to c** below:

a. 1 point On the plot above, label the **threshold frequency**.

b. 2 points What is the value of the work potential (in J) of the unknown metal?

c. 3 points What is the *longest* wavelength (in nm) required to eject an electron from this metal's surface? Show your work including equation(s) used. (3 sig figs)

d. 2 points In the space provided below, plot the following

(Assume frequency is *above* threshold frequency):

- i. *Kinetic energy* of an ejected electron versus the *intensity* of the incident light (left)
- ii. *Number of electrons* ejected versus the *frequency* of the incident light (right)



**Question 3** **6 points****a. 2 points**

In the ground state of Ge (Germanium; $Z = 32$) atom, **name** the following orbitals:

i. Orbital containing the electron that requires the maximum energy to be removed.

ii. Orbital containing the electron that experiences the minimum effective nuclear charge.

b. 2 points

An unknown anion (A^-) has the following *excited* electronic configuration: $[Ar] 4s^2 3d^3 4p^1$

(Ar: Argon; $Z = 18$)

Give the **condensed ground state electronic configuration** of the atom, **A** and the cation, **A^{2+}** .
Show your work. (Assume all electronic transitions are possible)

Electronic Configuration of A:

Electronic Configuration of A^{2+} :

c. 2 points

In a single atom, determine how many electrons can have the following set of quantum numbers?

(i) $n = 5; l = 3$

In a single atom, determine how many electrons can have the following set of quantum numbers?

(i) $n = 3; l = 2; m_s = +1/2$

**Question 4** **8 points**

Rank the following from 1 to 4 in **increasing** order. Provide the ranking in the boxes provided.

Note: Atomic number (Z) are provided for relevant atoms

a) Increasing (lowest to highest) order of *ionization energy*: (1= lowest; 4 = highest)

Mg

Mg: Z = 12

Rb

Rb: Z = 37

Cs

Cs: Z = 55

K

K : Z = 19

b) Increasing (lowest to highest) order of *bond angle*: (1= lowest; 4 = highest)

H₂OCCl₄XeF₄

HCN

c) Increasing (smallest to largest) order of *ionic size* (1= smallest; 4 = largest)

S²⁻

S: Z = 16

O²⁻

O: Z = 8

Be²⁺

Be: Z = 4

Na⁺

Na: Z = 11

d) Increasing (lowest to highest) order of *bond energy* (1= lowest; 4 = highest)

P-F

P-Cl

P-I

P-Br

**Question 5** **12 points**

a. 9 points For the molecules HClO_4 and TeF_6 , draw the most stable *Lewis structure* showing appropriate shape and *dashed/wedged bonds*, based on VSEPR. Determine the **molecular geometry** around the central atom. For HClO_4 , label the *most polar bond*.

HClO_4 (there are no O-O bonds)

Label the most polar bond

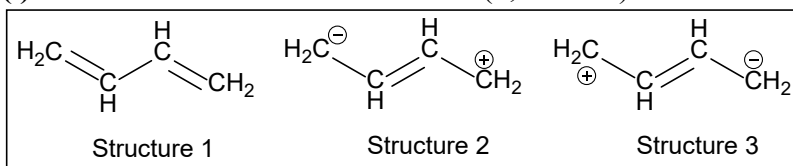
Molecular Geometry around Cl

TeF_6

Molecular Geometry around Te

b. 3 points (Note: Lone pairs are not shown in structures below)

(i) For the three *resonance* structures (1, 2 and 3) shown below, answer the following question:



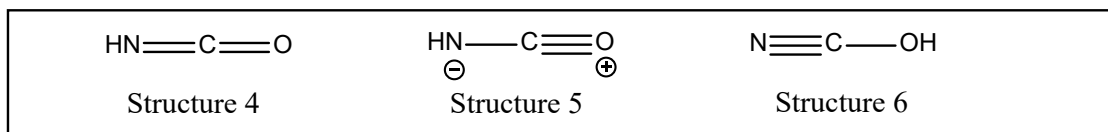
(I) Circle the *most contributing* resonance structure.

Structure 1

Structure 2

Structure 3

(ii) For the three structures given (4, 5, and 6) below, answer the following questions:



(I) Circle all the structures that represent *resonance structures* of each other:

Structure 4

Structure 5

Structure 6

(II) Among the resonance structures, circle the *most contributing* resonance structure.

Structure 4

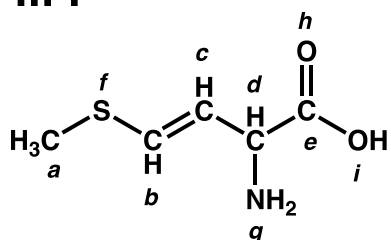
Structure 5

Structure 6

**Question 6** **6 points**

Using the data given below, calculate the **lattice energy** for SrO(s).

	ΔH° (kJ/mol)
$\text{Sr}(s) \longrightarrow \text{Sr}(g)$	+180
$\text{Sr}(g) \longrightarrow \text{Sr}^{2+}(g) + 2e^-$	+1468
$\text{O}_2(g) \longrightarrow 2\text{O}(g)$	+500
$\text{O}(g) + 2e^- \longrightarrow \text{O}^{2-}(g)$	+600
$\text{Sr}(s) + \frac{1}{2}\text{O}_2(g) \longrightarrow \text{SrO}(s)$	-550

**Question 7 10 points**

Consider the molecule given on the left, and answer questions (i) to (vi) using chemical bonding theories that we have learnt in class (valence bond theory, hybridization).

Note that the lone pairs are not shown in the structure:

i. Consider the carbon atoms marked in as ***a***, ***b***, ***c***, ***d***, and ***e*** in the molecule, and determine **hybridization** at each of the carbon centers. (2.5 points)

Atom	Hybridization
Carbon (<i>a</i>)	
Carbon (<i>b</i>)	
Carbon (<i>c</i>)	
Carbon (<i>d</i>)	
Carbon (<i>e</i>)	

ii. Now, consider the sulfur atom (S) marked as ***f***, nitrogen (N) atom (***g***), and oxygen(O) atoms (***h*** and ***i***), and determine their **hybridization**. (2 points)

Atom	Hybridization
Sulfur (<i>f</i>)	
Nitrogen (<i>g</i>)	
Oxygen (<i>h</i>)	
Oxygen (<i>i</i>)	

iii. **Describe** all the bonds formed by the carbon atoms ***a*** and ***b***, using sigma and pi designation as appropriate. (2 points)

Atom	Bonds formed (use sigma and pi as appropriate)
Carbon (<i>a</i>)	
Carbon (<i>b</i>)	

iv. How many total sp^3 -s sigma bonds are in the molecule? (1 point)

v. Determine **electronic** and **molecular geometry** at carbon (C) labeled ***e*** and sulfur (S) labeled ***f*** centres. (2 points)

Atom	Electronic Geometry	Molecular Geometry
Carbon (<i>e</i>)		
Sulfur (<i>f</i>)		

vi. Circle the value from the choices given below, that describes the **C-S-C bond angle at sulfur** (S) marked ***f***: (0.5 points)

>90° to <109.5°

109.5°

>109.5 to <120°

120°

>120° to <180°


Question 8 12 points

Using the **valence shell orbitals only** draw a **complete** molecular orbital (MO) diagram (with no s-p mixing) for the molecule **PS**, in the space provided below. **(6 points)**

Assume that 3s and 3p atomic orbitals will form molecular orbitals like those formed by the combination of 2s and 2p atomic orbitals that we learnt in class, and there is no s-p mixing.

Show **energy arrow**, **atomic orbitals** involved, **atom labels**, the **molecular orbitals** that are formed, **filled electrons** in the MO diagram etc. Make sure to label all the atomic and molecular orbitals.

Using the MO diagram above, answer the following questions:

- (i) Which one of the following will have the highest bond order? Circle the correct choice
 A) PS^{1-} B) PS C) PS^{1+}
- (ii) Which one of the following will have the longest bond length? Circle the correct choice
 A) PS^{1-} B) PS C) PS^{1+}
- (iii) Which of the following will be diamagnetic? Circle the correct choice.
 A) PS^{1-} B) PS C) PS^{1+}
- (iv) Draw Lewis dot structure of the molecule with the shortest bond length (based on the MO diagram) from among the three (PS^{1-} , PS, PS^{1+}), and indicate the bond order calculated from the Lewis dot structure. **(3 points)**

Lewis Dot Structure

Bond Order



5B148651-7060-4E35-B18F-73E369567D64

midterm-exam-93388

#81 Page 10 of 10

SCRAP PAGE

**CHEMISTRY 110 –Midterm Exam****Thursday November 7, 2024, 6:30 – 8:30 pm**

Examiners: Pallavi Sirjoosingh and Ashok Kakkar

First name (please write as legibly as possible within the boxes)

--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--

Last name

--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--

Student ID number

--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--

This booklet has 8 questions on 10 pages (The last page can be used as a scrap page).

The staple must NOT be removed from the question booklet. Your intact question booklet must be returned at the end of the exam.

A separate booklet contains a periodic table, and datasheet.

Write your answers directly under the questions in the space provided. Do not write on the QR codes or near the top edges (as it may be cut off).

Please write **legibly and dark enough** for the scan to be readable. For students who use English as a second language, dictionaries are permitted. Calculators are allowed.

DO NOT ASK QUESTIONS OF THE INVIGILATORS about interpretations of the exam questions. If you have a concern about a question, please summarize it at the top of this page. All concerns will be evaluated after the exam. GOOD LUCK!

**Question 1 8 points** (*Assume all transitions are possible*)

a. 6 points Calculate the **energy (in kJ/mol)** required to ionize (completely remove) the electron from the **n=3** level of **He⁺ ion** (He: Helium). Show your work including equations. (3 sig. figs.)

b. 2 points

(I) Select the largest (longest) wavelength from the three choices below:

- i. Wavelength of photon to *completely remove* an electron from *1s orbital of Hydrogen atom*
- ii. Wavelength of photon to *completely remove* an electron from *1s orbital of Boron atom (B)*
- iii. Wavelength of photon to *completely remove* an electron from *1s orbital of Boron cation (B⁺)*

i or ii or iii

(II) Select the smallest (shortest) wavelength from the three choices below (*for H atom*):

- i. Wavelength of photon *emitted* for an electron transitioning from *2p to 1s orbital*
- ii. Wavelength of photon *absorbed* for an electron transitioning from *2s to 4p orbital*
- iii. Wavelength of photon *absorbed* for an electron transitioning from *1s to 3p orbital*

i or ii or iii



Question 2 8 points



A Chem110 student plots the graph shown on the left, from data obtained for the kinetic energy (eV) of the ejected electron versus frequency (s^{-1}) of the incident light, for an unknown metal. ($1\text{eV} = 1.6022 \times 10^{-19} \text{ J}$)

Based on the plot, answer the questions **a to c** below:

a. 1 point On the plot above, label the **threshold frequency**.

b. 2 points What is the value of the work potential (in J) of the unknown metal?

c. 3 points What is the *longest* wavelength (in nm) required to eject an electron from this metal's surface? Show your work including equation(s) used. (3 sig figs)

d. 2 points In the space provided below, plot the following

(Assume frequency is *above* threshold frequency):

- i. *Kinetic energy* of an ejected electron versus the *intensity* of the incident light (left)
- ii. *Number of electrons* ejected versus the *frequency* of the incident light (right)



**Question 3** **6 points****a. 2 points**

In the ground state of Ge (Germanium; $Z = 32$) atom, **name** the following orbitals:

i. Orbital containing the electron that requires the maximum energy to be removed.

ii. Orbital containing the electron that experiences the minimum effective nuclear charge.

b. 2 points

An unknown anion (A^-) has the following *excited* electronic configuration: $[Ar] 4s^2 3d^3 4p^1$

(Ar: Argon; $Z = 18$)

Give the **condensed ground state electronic configuration** of the atom, **A** and the cation, **A^{2+}** .
Show your work. (Assume all electronic transitions are possible)

Electronic Configuration of A:

Electronic Configuration of A^{2+} :

c. 2 points

In a single atom, determine how many electrons can have the following set of quantum numbers?

(i) $n = 5; l = 3$

In a single atom, determine how many electrons can have the following set of quantum numbers?

(i) $n = 3; l = 2; m_s = +1/2$

**Question 4** **8 points**

Rank the following from 1 to 4 in **increasing** order. Provide the ranking in the boxes provided.

Note: Atomic number (Z) are provided for relevant atoms

a) Increasing (lowest to highest) order of *ionization energy*: (1= lowest; 4 = highest)

Mg

Mg: Z = 12

Rb

Rb: Z = 37

Cs

Cs: Z = 55

K

K : Z = 19

b) Increasing (lowest to highest) order of *bond angle*: (1= lowest; 4 = highest)

H₂OCCl₄XeF₄

HCN

c) Increasing (smallest to largest) order of *ionic size* (1= smallest; 4 = largest)

S²⁻

S: Z = 16

O²⁻

O: Z = 8

Be²⁺

Be: Z = 4

Na⁺

Na: Z = 11

d) Increasing (lowest to highest) order of *bond energy* (1= lowest; 4 = highest)

P-F

P-Cl

P-I

P-Br

**Question 5** **12 points**

a. 9 points For the molecules HClO_4 and TeF_6 , draw the most stable *Lewis structure* showing appropriate shape and *dashed/wedged bonds*, based on VSEPR. Determine the **molecular geometry** around the central atom. For HClO_4 , label the *most polar bond*.



b. 3 points (Note: Lone pairs are not shown in structures below)

(i) For the three *resonance* structures (1, 2 and 3) shown below, answer the following question:



(I) Circle the *most contributing* resonance structure.

Structure 1

Structure 2

Structure 3

(ii) For the three structures given (4, 5, and 6) below, answer the following questions:



(I) Circle all the structures that represent *resonance structures* of each other:

Structure 4

Structure 5

Structure 6

(II) Among the resonance structures, circle the *most contributing* resonance structure.

Structure 4

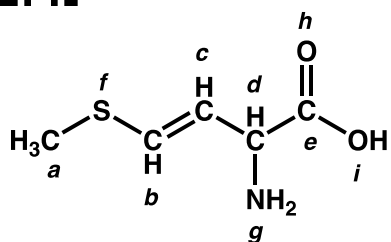
Structure 5

Structure 6

**Question 6** **6 points**

Using the data given below, calculate the **lattice energy** for SrO(s).

	ΔH° (kJ/mol)
$\text{Sr}(s) \longrightarrow \text{Sr}(g)$	+180
$\text{Sr}(g) \longrightarrow \text{Sr}^{2+}(g) + 2e^-$	+1468
$\text{O}_2(g) \longrightarrow 2\text{O}(g)$	+500
$\text{O}(g) + 2e^- \longrightarrow \text{O}^{2-}(g)$	+600
$\text{Sr}(s) + \frac{1}{2}\text{O}_2(g) \longrightarrow \text{SrO}(s)$	-550

**Question 7 10 points**

Consider the molecule given on the left, and answer questions (i) to (vi) using chemical bonding theories that we have learnt in class (valence bond theory, hybridization).

Note that the lone pairs are not shown in the structure:

i. Consider the carbon atoms marked in as ***a***, ***b***, ***c***, ***d***, and ***e*** in the molecule, and determine **hybridization** at each of the carbon centers. (2.5 points)

Atom	Hybridization
Carbon (<i>a</i>)	
Carbon (<i>b</i>)	
Carbon (<i>c</i>)	
Carbon (<i>d</i>)	
Carbon (<i>e</i>)	

ii. Now, consider the sulfur atom (S) marked as ***f***, nitrogen (N) atom (***g***), and oxygen(O) atoms (***h*** and ***i***), and determine their **hybridization**. (2 points)

Atom	Hybridization
Sulfur (<i>f</i>)	
Nitrogen (<i>g</i>)	
Oxygen (<i>h</i>)	
Oxygen (<i>i</i>)	

iii. **Describe** all the bonds formed by the carbon atoms ***a*** and ***b***, using sigma and pi designation as appropriate. (2 points)

Atom	Bonds formed (use sigma and pi as appropriate)
Carbon (<i>a</i>)	
Carbon (<i>b</i>)	

iv. How many total sp^3 -s sigma bonds are in the molecule? (1 point)

v. Determine **electronic** and **molecular geometry** at carbon (C) labeled ***e*** and sulfur (S) labeled ***f*** centres. (2 points)

Atom	Electronic Geometry	Molecular Geometry
Carbon (<i>e</i>)		
Sulfur (<i>f</i>)		

vi. Circle the value from the choices given below, that describes the **C-S-C bond angle at sulfur** (S) marked ***f***: (0.5 points)

>90° to <109.5° 109.5° >109.5 to <120° 120° >120° to <180°


Question 8 12 points

Using the **valence shell orbitals only** draw a **complete** molecular orbital (MO) diagram (with no s-p mixing) for the molecule **PS**, in the space provided below. **(6 points)**

Assume that 3s and 3p atomic orbitals will form molecular orbitals like those formed by the combination of 2s and 2p atomic orbitals that we learnt in class, and there is no s-p mixing.

Show **energy arrow**, **atomic orbitals** involved, **atom labels**, the **molecular orbitals** that are formed, **filled electrons** in the MO diagram etc. Make sure to label all the atomic and molecular orbitals.

Using the MO diagram above, answer the following questions:

- (i) Which one of the following will have the highest bond order? Circle the correct choice
 A) PS^{1-} B) PS C) PS^{1+}
- (ii) Which one of the following will have the longest bond length? Circle the correct choice
 A) PS^{1-} B) PS C) PS^{1+}
- (iii) Which of the following will be diamagnetic? Circle the correct choice.
 A) PS^{1-} B) PS C) PS^{1+}
- (iv) Draw Lewis dot structure of the molecule with the shortest bond length (based on the MO diagram) from among the three (PS^{1-} , PS, PS^{1+}), and indicate the bond order calculated from the Lewis dot structure. **(3 points)**

Lewis Dot Structure

Bond Order



C56AB15B-178E-4E43-A6BD-FE7DB7A306B5

midterm-exam-93388

#82 Page 10 of 10

SCRAP PAGE

**CHEMISTRY 110 –Midterm Exam****Thursday November 7, 2024, 6:30 – 8:30 pm****Examiners: Pallavi Sirjoosingh and Ashok Kakkar**

First name (please write as legibly as possible within the boxes)

--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--

Last name

--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--

Student ID number

--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--

This booklet has 8 questions on 10 pages (The last page can be used as a scrap page).

The staple must NOT be removed from the question booklet. Your intact question booklet must be returned at the end of the exam.

A separate booklet contains a periodic table, and datasheet.

Write your answers directly under the questions in the space provided. Do not write on the QR codes or near the top edges (as it may be cut off).

Please write **legibly and dark enough** for the scan to be readable. For students who use English as a second language, dictionaries are permitted. Calculators are allowed.

DO NOT ASK QUESTIONS OF THE INVIGILATORS about interpretations of the exam questions. If you have a concern about a question, please summarize it at the top of this page. All concerns will be evaluated after the exam. GOOD LUCK!

**Question 1 8 points** (*Assume all transitions are possible*)

a. 6 points Calculate the **energy (in kJ/mol)** required to ionize (completely remove) the electron from the **$n=3$** level of **He^+ ion** (He: Helium). Show your work including equations. (3 sig. figs.)

b. 2 points

(I) Select the largest (longest) wavelength from the three choices below:

- i. Wavelength of photon to *completely remove* an electron from *1s orbital of Hydrogen atom*
- ii. Wavelength of photon to *completely remove* an electron from *1s orbital of Boron atom (B)*
- iii. Wavelength of photon to *completely remove* an electron from *1s orbital of Boron cation (B^+)*

i or ii or iii

(II) Select the smallest (shortest) wavelength from the three choices below (*for H atom*):

- i. Wavelength of photon *emitted* for an electron transitioning from *2p to 1s orbital*
- ii. Wavelength of photon *absorbed* for an electron transitioning from *2s to 4p orbital*
- iii. Wavelength of photon *absorbed* for an electron transitioning from *1s to 3p orbital*

i or ii or iii



Question 2 8 points



A Chem110 student plots the graph shown on the left, from data obtained for the kinetic energy (eV) of the ejected electron versus frequency (s^{-1}) of the incident light, for an unknown metal. ($1\text{eV} = 1.6022 \times 10^{-19} \text{ J}$)

Based on the plot, answer the questions **a to c** below:

a. 1 point On the plot above, label the **threshold frequency**.

b. 2 points What is the value of the work potential (in J) of the unknown metal?

c. 3 points What is the *longest* wavelength (in nm) required to eject an electron from this metal's surface? Show your work including equation(s) used. (3 sig figs)

d. 2 points In the space provided below, plot the following

(Assume frequency is *above* threshold frequency):

- Kinetic energy* of an ejected electron versus the *intensity* of the incident light (left)
- Number of electrons* ejected versus the *frequency* of the incident light (right)



**Question 3** **6 points****a. 2 points**

In the ground state of Ge (Germanium; $Z = 32$) atom, **name** the following orbitals:

i. Orbital containing the electron that requires the maximum energy to be removed.

ii. Orbital containing the electron that experiences the minimum effective nuclear charge.

b. 2 points

An unknown anion (A^-) has the following *excited* electronic configuration: $[Ar] 4s^2 3d^3 4p^1$

(Ar: Argon; $Z = 18$)

Give the **condensed ground state electronic configuration** of the atom, **A** and the cation, **A^{2+}** .
Show your work. (Assume all electronic transitions are possible)

Electronic Configuration of A:

Electronic Configuration of A^{2+} :

c. 2 points

In a single atom, determine how many electrons can have the following set of quantum numbers?

(i) $n = 5; l = 3$

In a single atom, determine how many electrons can have the following set of quantum numbers?

(i) $n = 3; l = 2; m_s = +1/2$

**Question 4** **8 points**

Rank the following from 1 to 4 in **increasing** order. Provide the ranking in the boxes provided.

Note: Atomic number (Z) are provided for relevant atoms

a) Increasing (lowest to highest) order of *ionization energy*: (1= lowest; 4 = highest)

Mg

Mg: Z = 12

Rb

Rb: Z = 37

Cs

Cs: Z = 55

K

K : Z = 19

b) Increasing (lowest to highest) order of *bond angle*: (1= lowest; 4 = highest)

H₂OCCl₄XeF₄

HCN

c) Increasing (smallest to largest) order of *ionic size* (1= smallest; 4 = largest)

S²⁻

S: Z = 16

O²⁻

O: Z = 8

Be²⁺

Be: Z = 4

Na⁺

Na: Z = 11

d) Increasing (lowest to highest) order of *bond energy* (1= lowest; 4 = highest)

P-F

P-Cl

P-I

P-Br

**Question 5** **12 points**

a. 9 points For the molecules HClO_4 and TeF_6 , draw the most stable *Lewis structure* showing appropriate shape and *dashed/wedged bonds*, based on VSEPR. Determine the **molecular geometry** around the central atom. For HClO_4 , label the *most polar bond*.



b. 3 points (Note: Lone pairs are not shown in structures below)

(i) For the three *resonance* structures (1, 2 and 3) shown below, answer the following question:



(I) Circle the *most contributing* resonance structure.

Structure 1

Structure 2

Structure 3

(ii) For the three structures given (4, 5, and 6) below, answer the following questions:



(I) Circle all the structures that represent *resonance structures* of each other:

Structure 4

Structure 5

Structure 6

(II) Among the resonance structures, circle the *most contributing* resonance structure.

Structure 4

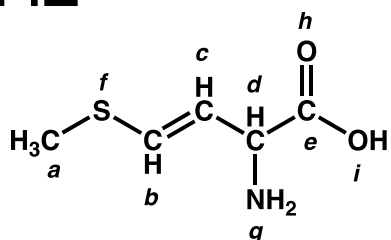
Structure 5

Structure 6

**Question 6** **6 points**

Using the data given below, calculate the **lattice energy** for SrO(s).

	ΔH° (kJ/mol)
$\text{Sr}(s) \longrightarrow \text{Sr}(g)$	+180
$\text{Sr}(g) \longrightarrow \text{Sr}^{2+}(g) + 2e^-$	+1468
$\text{O}_2(g) \longrightarrow 2\text{O}(g)$	+500
$\text{O}(g) + 2e^- \longrightarrow \text{O}^{2-}(g)$	+600
$\text{Sr}(s) + \frac{1}{2}\text{O}_2(g) \longrightarrow \text{SrO}(s)$	-550

**Question 7 10 points**

Consider the molecule given on the left, and answer questions (i) to (vi) using chemical bonding theories that we have learnt in class (valence bond theory, hybridization).

Note that the lone pairs are not shown in the structure:

i. Consider the carbon atoms marked in as ***a***, ***b***, ***c***, ***d***, and ***e*** in the molecule, and determine **hybridization** at each of the carbon centers. (2.5 points)

Atom	Hybridization
Carbon (<i>a</i>)	
Carbon (<i>b</i>)	
Carbon (<i>c</i>)	
Carbon (<i>d</i>)	
Carbon (<i>e</i>)	

ii. Now, consider the sulfur atom (S) marked as ***f***, nitrogen (N) atom (***g***), and oxygen(O) atoms (***h*** and ***i***), and determine their **hybridization**. (2 points)

Atom	Hybridization
Sulfur (<i>f</i>)	
Nitrogen (<i>g</i>)	
Oxygen (<i>h</i>)	
Oxygen (<i>i</i>)	

iii. **Describe** all the bonds formed by the carbon atoms ***a*** and ***b***, using sigma and pi designation as appropriate. (2 points)

Atom	Bonds formed (use sigma and pi as appropriate)
Carbon (<i>a</i>)	
Carbon (<i>b</i>)	

iv. How many total sp^3 -s sigma bonds are in the molecule? (1 point)

v. Determine **electronic** and **molecular geometry** at carbon (C) labeled ***e*** and sulfur (S) labeled ***f*** centres. (2 points)

Atom	Electronic Geometry	Molecular Geometry
Carbon (<i>e</i>)		
Sulfur (<i>f</i>)		

vi. Circle the value from the choices given below, that describes the **C-S-C bond angle at sulfur** (S) marked ***f***: (0.5 points)

>90° to <109.5° 109.5° >109.5 to <120° 120° >120° to <180°

**Question 8 12 points**

Using the **valence shell orbitals only** draw a **complete** molecular orbital (MO) diagram (with no s-p mixing) for the molecule **PS**, in the space provided below. **(6 points)**

Assume that 3s and 3p atomic orbitals will form molecular orbitals like those formed by the combination of 2s and 2p atomic orbitals that we learnt in class, and there is no s-p mixing.

Show **energy arrow**, **atomic orbitals** involved, **atom labels**, the **molecular orbitals** that are formed, **filled electrons** in the MO diagram etc. Make sure to label all the atomic and molecular orbitals.

Using the MO diagram above, answer the following questions:

- (i) Which one of the following will have the highest bond order? Circle the correct choice
A) PS^{1-} B) PS C) PS^{1+}
- (ii) Which one of the following will have the longest bond length? Circle the correct choice
A) PS^{1-} B) PS C) PS^{1+}
- (iii) Which of the following will be diamagnetic? Circle the correct choice.
A) PS^{1-} B) PS C) PS^{1+}
- (iv) Draw Lewis dot structure of the molecule with the shortest bond length (based on the MO diagram) from among the three (PS^{1-} , PS, PS^{1+}), and indicate the bond order calculated from the Lewis dot structure. **(3 points)**

Lewis Dot Structure

Bond Order



26A80054-83B8-40F6-A670-4A265A69A567

midterm-exam-93388

#83 Page 10 of 10

SCRAP PAGE

**CHEMISTRY 110 –Midterm Exam****Thursday November 7, 2024, 6:30 – 8:30 pm**

Examiners: Pallavi Sirjoosingh and Ashok Kakkar

First name (please write as legibly as possible within the boxes)

--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--

Last name

--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--

Student ID number

--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--

This booklet has 8 questions on 10 pages (The last page can be used as a scrap page).

The staple must NOT be removed from the question booklet. Your intact question booklet must be returned at the end of the exam.

A separate booklet contains a periodic table, and datasheet.

Write your answers directly under the questions in the space provided. Do not write on the QR codes or near the top edges (as it may be cut off).

Please write **legibly and dark enough** for the scan to be readable. For students who use English as a second language, dictionaries are permitted. Calculators are allowed.

DO NOT ASK QUESTIONS OF THE INVIGILATORS about interpretations of the exam questions. If you have a concern about a question, please summarize it at the top of this page. All concerns will be evaluated after the exam. GOOD LUCK!

**Question 1 8 points** (*Assume all transitions are possible*)

a. 6 points Calculate the **energy (in kJ/mol)** required to ionize (completely remove) the electron from the **$n=3$** level of **He^+ ion** (He: Helium). Show your work including equations. (3 sig. figs.)

b. 2 points

(I) Select the largest (longest) wavelength from the three choices below:

- i. Wavelength of photon to *completely remove* an electron from *1s orbital of Hydrogen atom*
- ii. Wavelength of photon to *completely remove* an electron from *1s orbital of Boron atom (B)*
- iii. Wavelength of photon to *completely remove* an electron from *1s orbital of Boron cation (B^+)*

i or ii or iii

(II) Select the smallest (shortest) wavelength from the three choices below (*for H atom*):

- i. Wavelength of photon *emitted* for an electron transitioning from *2p to 1s orbital*
- ii. Wavelength of photon *absorbed* for an electron transitioning from *2s to 4p orbital*
- iii. Wavelength of photon *absorbed* for an electron transitioning from *1s to 3p orbital*

i or ii or iii



Question 2 8 points



A Chem110 student plots the graph shown on the left, from data obtained for the kinetic energy (eV) of the ejected electron versus frequency (s^{-1}) of the incident light, for an unknown metal. ($1\text{eV} = 1.6022 \times 10^{-19} \text{ J}$)

Based on the plot, answer the questions **a to c** below:

a. 1 point On the plot above, label the **threshold frequency**.

b. 2 points What is the value of the work potential (in J) of the unknown metal?

c. 3 points What is the *longest* wavelength (in nm) required to eject an electron from this metal's surface? Show your work including equation(s) used. (3 sig figs)

d. 2 points In the space provided below, plot the following

(Assume frequency is *above* threshold frequency):

- i. *Kinetic energy* of an ejected electron versus the *intensity* of the incident light (left)
- ii. *Number of electrons* ejected versus the *frequency* of the incident light (right)



**Question 3** **6 points****a. 2 points**

In the ground state of Ge (Germanium; $Z = 32$) atom, **name** the following orbitals:

i. Orbital containing the electron that requires the maximum energy to be removed.

ii. Orbital containing the electron that experiences the minimum effective nuclear charge.

b. 2 points

An unknown anion (A^-) has the following *excited* electronic configuration: $[Ar] 4s^2 3d^3 4p^1$

(Ar: Argon; $Z = 18$)

Give the **condensed ground state electronic configuration** of the atom, **A** and the cation, **A^{2+}** .
Show your work. (Assume all electronic transitions are possible)

Electronic Configuration of A:

Electronic Configuration of A^{2+} :

c. 2 points

In a single atom, determine how many electrons can have the following set of quantum numbers?

(i) $n = 5; l = 3$

In a single atom, determine how many electrons can have the following set of quantum numbers?

(i) $n = 3; l = 2; m_s = +1/2$

**Question 4** **8 points**

Rank the following from 1 to 4 in **increasing** order. Provide the ranking in the boxes provided.

Note: Atomic number (Z) are provided for relevant atoms

a) Increasing (lowest to highest) order of *ionization energy*: (1= lowest; 4 = highest)

Mg

Mg: Z = 12

Rb

Rb: Z = 37

Cs

Cs: Z = 55

K

K : Z = 19

b) Increasing (lowest to highest) order of *bond angle*: (1= lowest; 4 = highest)

H₂OCCl₄XeF₄

HCN

c) Increasing (smallest to largest) order of *ionic size* (1= smallest; 4 = largest)

S²⁻

S: Z = 16

O²⁻

O: Z = 8

Be²⁺

Be: Z = 4

Na⁺

Na: Z = 11

d) Increasing (lowest to highest) order of *bond energy* (1= lowest; 4 = highest)

P-F

P-Cl

P-I

P-Br

**Question 5** **12 points**

a. 9 points For the molecules HClO_4 and TeF_6 , draw the most stable *Lewis structure* showing appropriate shape and *dashed/wedged bonds*, based on VSEPR. Determine the **molecular geometry** around the central atom. For HClO_4 , label the *most polar bond*.



b. 3 points (Note: Lone pairs are not shown in structures below)

(i) For the three *resonance* structures (1, 2 and 3) shown below, answer the following question:



(I) Circle the *most contributing* resonance structure.

Structure 1

Structure 2

Structure 3

(ii) For the three structures given (4, 5, and 6) below, answer the following questions:



(I) Circle all the structures that represent *resonance structures* of each other:

Structure 4

Structure 5

Structure 6

(II) Among the resonance structures, circle the *most contributing* resonance structure.

Structure 4

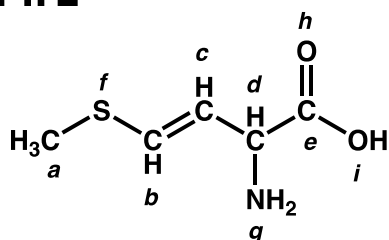
Structure 5

Structure 6

**Question 6** **6 points**

Using the data given below, calculate the **lattice energy** for SrO(s).

	ΔH° (kJ/mol)
$\text{Sr}(s) \longrightarrow \text{Sr}(g)$	+180
$\text{Sr}(g) \longrightarrow \text{Sr}^{2+}(g) + 2e^-$	+1468
$\text{O}_2(g) \longrightarrow 2\text{O}(g)$	+500
$\text{O}(g) + 2e^- \longrightarrow \text{O}^{2-}(g)$	+600
$\text{Sr}(s) + \frac{1}{2}\text{O}_2(g) \longrightarrow \text{SrO}(s)$	-550

**Question 7 10 points**

Consider the molecule given on the left, and answer questions (i) to (vi) using chemical bonding theories that we have learnt in class (valence bond theory, hybridization).

Note that the lone pairs are not shown in the structure:

i. Consider the carbon atoms marked in as ***a***, ***b***, ***c***, ***d***, and ***e*** in the molecule, and determine **hybridization** at each of the carbon centers. (2.5 points)

Atom	Hybridization
Carbon (<i>a</i>)	
Carbon (<i>b</i>)	
Carbon (<i>c</i>)	
Carbon (<i>d</i>)	
Carbon (<i>e</i>)	

ii. Now, consider the sulfur atom (S) marked as ***f***, nitrogen (N) atom (***g***), and oxygen(O) atoms (***h*** and ***i***), and determine their **hybridization**. (2 points)

Atom	Hybridization
Sulfur (<i>f</i>)	
Nitrogen (<i>g</i>)	
Oxygen (<i>h</i>)	
Oxygen (<i>i</i>)	

iii. **Describe** all the bonds formed by the carbon atoms ***a*** and ***b***, using sigma and pi designation as appropriate. (2 points)

Atom	Bonds formed (use sigma and pi as appropriate)
Carbon (<i>a</i>)	
Carbon (<i>b</i>)	

iv. How many total sp^3 -s sigma bonds are in the molecule? (1 point)

v. Determine **electronic** and **molecular geometry** at carbon (C) labeled ***e*** and sulfur (S) labeled ***f*** centres. (2 points)

Atom	Electronic Geometry	Molecular Geometry
Carbon (<i>e</i>)		
Sulfur (<i>f</i>)		

vi. Circle the value from the choices given below, that describes the **C-S-C bond angle at sulfur** (S) marked ***f***: (0.5 points)

>90° to <109.5° 109.5° >109.5 to <120° 120° >120° to <180°


Question 8 12 points

Using the **valence shell orbitals only** draw a **complete** molecular orbital (MO) diagram (with no s-p mixing) for the molecule **PS**, in the space provided below. **(6 points)**

Assume that 3s and 3p atomic orbitals will form molecular orbitals like those formed by the combination of 2s and 2p atomic orbitals that we learnt in class, and there is no s-p mixing.

Show **energy arrow**, **atomic orbitals** involved, **atom labels**, the **molecular orbitals** that are formed, **filled electrons** in the MO diagram etc. Make sure to **label all the atomic and molecular orbitals**.

Using the MO diagram above, answer the following questions:

- (i) Which one of the following will have the highest bond order? Circle the correct choice
 A) PS^{1-} B) PS C) PS^{1+}
- (ii) Which one of the following will have the longest bond length? Circle the correct choice
 A) PS^{1-} B) PS C) PS^{1+}
- (iii) Which of the following will be diamagnetic? Circle the correct choice.
 A) PS^{1-} B) PS C) PS^{1+}
- (iv) Draw Lewis dot structure of the molecule with the shortest bond length (based on the MO diagram) from among the three (PS^{1-} , PS, PS^{1+}), and indicate the bond order calculated from the Lewis dot structure. **(3 points)**

Lewis Dot Structure

Bond Order



20FE37A6-1326-48B1-88D5-2230C9CCE59A

midterm-exam-93388

#84 Page 10 of 10

SCRAP PAGE

**CHEMISTRY 110 –Midterm Exam****Thursday November 7, 2024, 6:30 – 8:30 pm**

Examiners: Pallavi Sirjoosingh and Ashok Kakkar

First name (please write as legibly as possible within the boxes)

--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--

Last name

--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--

Student ID number

--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--

This booklet has 8 questions on 10 pages (The last page can be used as a scrap page).

The staple must NOT be removed from the question booklet. Your intact question booklet must be returned at the end of the exam.

A separate booklet contains a periodic table, and datasheet.

Write your answers directly under the questions in the space provided. Do not write on the QR codes or near the top edges (as it may be cut off).

Please write **legibly and dark enough** for the scan to be readable. For students who use English as a second language, dictionaries are permitted. Calculators are allowed.

DO NOT ASK QUESTIONS OF THE INVIGILATORS about interpretations of the exam questions. If you have a concern about a question, please summarize it at the top of this page. All concerns will be evaluated after the exam. GOOD LUCK!

**Question 1 8 points** (*Assume all transitions are possible*)

a. 6 points Calculate the **energy (in kJ/mol)** required to ionize (completely remove) the electron from the **n=3** level of **He⁺ ion** (He: Helium). Show your work including equations. (3 sig. figs.)

b. 2 points

(I) Select the largest (longest) wavelength from the three choices below:

- i. Wavelength of photon to *completely remove* an electron from *1s orbital of Hydrogen atom*
- ii. Wavelength of photon to *completely remove* an electron from *1s orbital of Boron atom (B)*
- iii. Wavelength of photon to *completely remove* an electron from *1s orbital of Boron cation (B⁺)*

i or ii or iii

(II) Select the smallest (shortest) wavelength from the three choices below (*for H atom*):

- i. Wavelength of photon *emitted* for an electron transitioning from *2p to 1s orbital*
- ii. Wavelength of photon *absorbed* for an electron transitioning from *2s to 4p orbital*
- iii. Wavelength of photon *absorbed* for an electron transitioning from *1s to 3p orbital*

i or ii or iii



Question 2 8 points



A Chem110 student plots the graph shown on the left, from data obtained for the kinetic energy (eV) of the ejected electron versus frequency (s^{-1}) of the incident light, for an unknown metal. ($1\text{eV} = 1.6022 \times 10^{-19} \text{ J}$)

Based on the plot, answer the questions **a to c** below:

a. 1 point On the plot above, label the **threshold frequency**.

b. 2 points What is the value of the work potential (in J) of the unknown metal?

c. 3 points What is the *longest* wavelength (in nm) required to eject an electron from this metal's surface? Show your work including equation(s) used. (3 sig figs)

d. 2 points In the space provided below, plot the following

(Assume frequency is *above* threshold frequency):

- i. *Kinetic energy* of an ejected electron versus the *intensity* of the incident light (left)
- ii. *Number of electrons* ejected versus the *frequency* of the incident light (right)



**Question 3** **6 points****a. 2 points**

In the ground state of Ge (Germanium; $Z = 32$) atom, **name** the following orbitals:

i. Orbital containing the electron that requires the maximum energy to be removed.

ii. Orbital containing the electron that experiences the minimum effective nuclear charge.

b. 2 points

An unknown anion (A^-) has the following *excited* electronic configuration: $[Ar] 4s^2 3d^3 4p^1$

(Ar: Argon; $Z = 18$)

Give the **condensed ground state electronic configuration** of the atom, **A** and the cation, **A^{2+}** .
Show your work. (Assume all electronic transitions are possible)

Electronic Configuration of A:

Electronic Configuration of A^{2+} :

c. 2 points

In a single atom, determine how many electrons can have the following set of quantum numbers?

(i) $n = 5; l = 3$

In a single atom, determine how many electrons can have the following set of quantum numbers?

(i) $n = 3; l = 2; m_s = +1/2$

**Question 4** **8 points**

Rank the following from 1 to 4 in **increasing** order. Provide the ranking in the boxes provided.

Note: Atomic number (Z) are provided for relevant atoms

a) Increasing (lowest to highest) order of *ionization energy*: (1= lowest; 4 = highest)

Mg

Mg: Z = 12

Rb

Rb: Z = 37

Cs

Cs: Z = 55

K

K : Z = 19

b) Increasing (lowest to highest) order of *bond angle*: (1= lowest; 4 = highest)

H₂OCCl₄XeF₄

HCN

c) Increasing (smallest to largest) order of *ionic size* (1= smallest; 4 = largest)

S²⁻

S: Z = 16

O²⁻

O: Z = 8

Be²⁺

Be: Z = 4

Na⁺

Na: Z = 11

d) Increasing (lowest to highest) order of *bond energy* (1= lowest; 4 = highest)

P-F

P-Cl

P-I

P-Br

**Question 5** **12 points**

a. 9 points For the molecules HClO_4 and TeF_6 , draw the most stable *Lewis structure* showing appropriate shape and *dashed/wedged bonds*, based on VSEPR. Determine the **molecular geometry** around the central atom. For HClO_4 , label the *most polar bond*.



b. 3 points (Note: Lone pairs are not shown in structures below)

(i) For the three *resonance* structures (1, 2 and 3) shown below, answer the following question:



(I) Circle the *most contributing* resonance structure.

Structure 1

Structure 2

Structure 3

(ii) For the three structures given (4, 5, and 6) below, answer the following questions:



(I) Circle all the structures that represent *resonance structures* of each other:

Structure 4

Structure 5

Structure 6

(II) Among the resonance structures, circle the *most contributing* resonance structure.

Structure 4

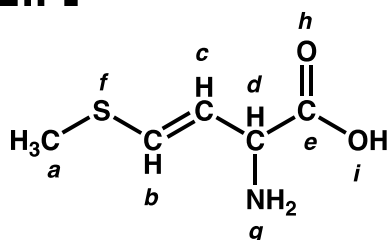
Structure 5

Structure 6

**Question 6** **6 points**

Using the data given below, calculate the **lattice energy** for SrO(s).

	ΔH° (kJ/mol)
$\text{Sr}(s) \longrightarrow \text{Sr}(g)$	+180
$\text{Sr}(g) \longrightarrow \text{Sr}^{2+}(g) + 2e^-$	+1468
$\text{O}_2(g) \longrightarrow 2\text{O}(g)$	+500
$\text{O}(g) + 2e^- \longrightarrow \text{O}^{2-}(g)$	+600
$\text{Sr}(s) + \frac{1}{2}\text{O}_2(g) \longrightarrow \text{SrO}(s)$	-550

**Question 7 10 points**

Consider the molecule given on the left, and answer questions (i) to (vi) using chemical bonding theories that we have learnt in class (valence bond theory, hybridization).

Note that the lone pairs are not shown in the structure:

i. Consider the carbon atoms marked in as ***a***, ***b***, ***c***, ***d***, and ***e*** in the molecule, and determine **hybridization** at each of the carbon centers. (2.5 points)

Atom	Hybridization
Carbon (<i>a</i>)	
Carbon (<i>b</i>)	
Carbon (<i>c</i>)	
Carbon (<i>d</i>)	
Carbon (<i>e</i>)	

ii. Now, consider the sulfur atom (S) marked as ***f***, nitrogen (N) atom (***g***), and oxygen(O) atoms (***h*** and ***i***), and determine their **hybridization**. (2 points)

Atom	Hybridization
Sulfur (<i>f</i>)	
Nitrogen (<i>g</i>)	
Oxygen (<i>h</i>)	
Oxygen (<i>i</i>)	

iii. **Describe** all the bonds formed by the carbon atoms ***a*** and ***b***, using sigma and pi designation as appropriate. (2 points)

Atom	Bonds formed (use sigma and pi as appropriate)
Carbon (<i>a</i>)	
Carbon (<i>b</i>)	

iv. How many total sp^3 -s sigma bonds are in the molecule? (1 point)

v. Determine **electronic** and **molecular geometry** at carbon (C) labeled ***e*** and sulfur (S) labeled ***f*** centres. (2 points)

Atom	Electronic Geometry	Molecular Geometry
Carbon (<i>e</i>)		
Sulfur (<i>f</i>)		

vi. Circle the value from the choices given below, that describes the **C-S-C bond angle at sulfur** (S) marked ***f***: (0.5 points)

>90° to <109.5°

109.5°

>109.5 to <120°

120°

>120° to <180°

**Question 8 12 points**

Using the **valence shell orbitals only** draw a **complete** molecular orbital (MO) diagram (with no s-p mixing) for the molecule **PS**, in the space provided below. **(6 points)**

Assume that 3s and 3p atomic orbitals will form molecular orbitals like those formed by the combination of 2s and 2p atomic orbitals that we learnt in class, and there is no s-p mixing.

Show **energy arrow**, **atomic orbitals** involved, **atom labels**, the **molecular orbitals** that are formed, **filled electrons** in the MO diagram etc. Make sure to label all the atomic and molecular orbitals.

Using the MO diagram above, answer the following questions:

- (i) Which one of the following will have the highest bond order? Circle the correct choice
A) PS^{1-} B) PS C) PS^{1+}
- (ii) Which one of the following will have the longest bond length? Circle the correct choice
A) PS^{1-} B) PS C) PS^{1+}
- (iii) Which of the following will be diamagnetic? Circle the correct choice.
A) PS^{1-} B) PS C) PS^{1+}
- (iv) Draw Lewis dot structure of the molecule with the shortest bond length (based on the MO diagram) from among the three (PS^{1-} , PS, PS^{1+}), and indicate the bond order calculated from the Lewis dot structure. **(3 points)**

Lewis Dot Structure

Bond Order



C9DE4A17-0F9F-4099-B0F7-19F4CA23BEE6

midterm-exam-93388

#85 Page 10 of 10

SCRAP PAGE

**CHEMISTRY 110 –Midterm Exam****Thursday November 7, 2024, 6:30 – 8:30 pm****Examiners: Pallavi Sirjoosingh and Ashok Kakkar**

First name (please write as legibly as possible within the boxes)

--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--

Last name

--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--

Student ID number

--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--

This booklet has 8 questions on 10 pages (The last page can be used as a scrap page).

The staple must NOT be removed from the question booklet. Your intact question booklet must be returned at the end of the exam.

A separate booklet contains a periodic table, and datasheet.

Write your answers directly under the questions in the space provided. Do not write on the QR codes or near the top edges (as it may be cut off).

Please write **legibly and dark enough** for the scan to be readable. For students who use English as a second language, dictionaries are permitted. Calculators are allowed.

DO NOT ASK QUESTIONS OF THE INVIGILATORS about interpretations of the exam questions. If you have a concern about a question, please summarize it at the top of this page. All concerns will be evaluated after the exam. GOOD LUCK!

**Question 1 8 points** (*Assume all transitions are possible*)

a. 6 points Calculate the **energy (in kJ/mol)** required to ionize (completely remove) the electron from the **n=3** level of **He⁺ ion** (He: Helium). Show your work including equations. (3 sig. figs.)

b. 2 points

(I) Select the largest (longest) wavelength from the three choices below:

- i. Wavelength of photon to *completely remove* an electron from *1s orbital of Hydrogen atom*
- ii. Wavelength of photon to *completely remove* an electron from *1s orbital of Boron atom (B)*
- iii. Wavelength of photon to *completely remove* an electron from *1s orbital of Boron cation (B⁺)*

i or ii or iii

(II) Select the smallest (shortest) wavelength from the three choices below (*for H atom*):

- i. Wavelength of photon *emitted* for an electron transitioning from *2p to 1s orbital*
- ii. Wavelength of photon *absorbed* for an electron transitioning from *2s to 4p orbital*
- iii. Wavelength of photon *absorbed* for an electron transitioning from *1s to 3p orbital*

i or ii or iii



Question 2 8 points



A Chem110 student plots the graph shown on the left, from data obtained for the kinetic energy (eV) of the ejected electron versus frequency (s^{-1}) of the incident light, for an unknown metal. ($1\text{eV} = 1.6022 \times 10^{-19} \text{ J}$)

Based on the plot, answer the questions **a to c** below:

a. 1 point On the plot above, label the **threshold frequency**.

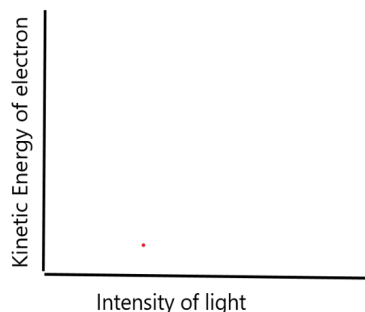
b. 2 points What is the value of the work potential (in J) of the unknown metal?

c. 3 points What is the *longest* wavelength (in nm) required to eject an electron from this metal's surface? Show your work including equation(s) used. (3 sig figs)

d. 2 points In the space provided below, plot the following

(Assume frequency is *above* threshold frequency):

- Kinetic energy* of an ejected electron versus the *intensity* of the incident light (left)
- Number of electrons* ejected versus the *frequency* of the incident light (right)



**Question 3** **6 points****a. 2 points**

In the ground state of Ge (Germanium; $Z = 32$) atom, **name** the following orbitals:

i. Orbital containing the electron that requires the maximum energy to be removed.

ii. Orbital containing the electron that experiences the minimum effective nuclear charge.

b. 2 points

An unknown anion (A^-) has the following *excited* electronic configuration: $[Ar] 4s^2 3d^3 4p^1$

(Ar: Argon; $Z = 18$)

Give the **condensed ground state electronic configuration** of the atom, **A** and the cation, **A^{2+}** .
Show your work. (Assume all electronic transitions are possible)

Electronic Configuration of A:

Electronic Configuration of A^{2+} :

c. 2 points

In a single atom, determine how many electrons can have the following set of quantum numbers?

(i) $n = 5; l = 3$

In a single atom, determine how many electrons can have the following set of quantum numbers?

(i) $n = 3; l = 2; m_s = +1/2$

**Question 4** **8 points**

Rank the following from 1 to 4 in **increasing** order. Provide the ranking in the boxes provided.

Note: Atomic number (Z) are provided for relevant atoms

a) Increasing (lowest to highest) order of *ionization energy*: (1= lowest; 4 = highest)

Mg

Mg: Z = 12

Rb

Rb: Z = 37

Cs

Cs: Z = 55

K

K : Z = 19

b) Increasing (lowest to highest) order of *bond angle*: (1= lowest; 4 = highest)

H₂OCCl₄XeF₄

HCN

c) Increasing (smallest to largest) order of *ionic size* (1= smallest; 4 = largest)

S²⁻

S: Z = 16

O²⁻

O: Z = 8

Be²⁺

Be: Z = 4

Na⁺

Na: Z = 11

d) Increasing (lowest to highest) order of *bond energy* (1= lowest; 4 = highest)

P-F

P-Cl

P-I

P-Br

**Question 5** **12 points**

a. 9 points For the molecules HClO_4 and TeF_6 , draw the most stable *Lewis structure* showing appropriate shape and *dashed/wedged bonds*, based on VSEPR. Determine the **molecular geometry** around the central atom. For HClO_4 , label the *most polar bond*.

HClO_4 (there are no O-O bonds)

Label the most polar bond

Molecular Geometry around Cl

TeF_6

Molecular Geometry around Te

b. 3 points (Note: Lone pairs are not shown in structures below)

(i) For the three *resonance* structures (1, 2 and 3) shown below, answer the following question:



(I) Circle the *most contributing* resonance structure.

Structure 1

Structure 2

Structure 3

(ii) For the three structures given (4, 5, and 6) below, answer the following questions:



(I) Circle all the structures that represent *resonance structures* of each other:

Structure 4

Structure 5

Structure 6

(II) Among the resonance structures, circle the *most contributing* resonance structure.

Structure 4

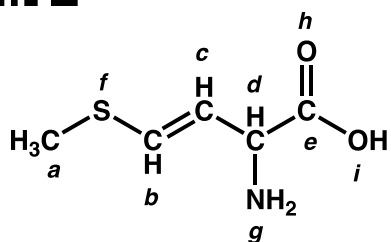
Structure 5

Structure 6

**Question 6** **6 points**

Using the data given below, calculate the **lattice energy** for SrO(s).

	ΔH° (kJ/mol)
$\text{Sr}(s) \longrightarrow \text{Sr}(g)$	+180
$\text{Sr}(g) \longrightarrow \text{Sr}^{2+}(g) + 2e^-$	+1468
$\text{O}_2(g) \longrightarrow 2\text{O}(g)$	+500
$\text{O}(g) + 2e^- \longrightarrow \text{O}^{2-}(g)$	+600
$\text{Sr}(s) + \frac{1}{2}\text{O}_2(g) \longrightarrow \text{SrO}(s)$	-550

**Question 7 10 points**

Consider the molecule given on the left, and answer questions (i) to (vi) using chemical bonding theories that we have learnt in class (valence bond theory, hybridization).

Note that the lone pairs are not shown in the structure:

i. Consider the carbon atoms marked in as ***a***, ***b***, ***c***, ***d***, and ***e*** in the molecule, and determine **hybridization** at each of the carbon centers. (2.5 points)

Atom	Hybridization
Carbon (<i>a</i>)	
Carbon (<i>b</i>)	
Carbon (<i>c</i>)	
Carbon (<i>d</i>)	
Carbon (<i>e</i>)	

ii. Now, consider the sulfur atom (S) marked as ***f***, nitrogen (N) atom (***g***), and oxygen(O) atoms (***h*** and ***i***), and determine their **hybridization**. (2 points)

Atom	Hybridization
Sulfur (<i>f</i>)	
Nitrogen (<i>g</i>)	
Oxygen (<i>h</i>)	
Oxygen (<i>i</i>)	

iii. **Describe** all the bonds formed by the carbon atoms ***a*** and ***b***, using sigma and pi designation as appropriate. (2 points)

Atom	Bonds formed (use sigma and pi as appropriate)
Carbon (<i>a</i>)	
Carbon (<i>b</i>)	

iv. How many total sp^3 -s sigma bonds are in the molecule? (1 point)

v. Determine **electronic** and **molecular geometry** at carbon (C) labeled ***e*** and sulfur (S) labeled ***f*** centres. (2 points)

Atom	Electronic Geometry	Molecular Geometry
Carbon (<i>e</i>)		
Sulfur (<i>f</i>)		

vi. Circle the value from the choices given below, that describes the **C-S-C bond angle at sulfur** (S) marked ***f***: (0.5 points)

$>90^\circ$ to $<109.5^\circ$ 109.5° >109.5 to $<120^\circ$ 120° $>120^\circ$ to $<180^\circ$


Question 8 12 points

Using the **valence shell orbitals only** draw a **complete** molecular orbital (MO) diagram (with no s-p mixing) for the molecule **PS**, in the space provided below. **(6 points)**

Assume that 3s and 3p atomic orbitals will form molecular orbitals like those formed by the combination of 2s and 2p atomic orbitals that we learnt in class, and there is no s-p mixing.

Show **energy arrow**, **atomic orbitals** involved, **atom labels**, the **molecular orbitals** that are formed, **filled electrons** in the MO diagram etc. Make sure to label all the atomic and molecular orbitals.

Using the MO diagram above, answer the following questions:

- (i) Which one of the following will have the highest bond order? Circle the correct choice
 A) PS^{1-} B) PS C) PS^{1+}
- (ii) Which one of the following will have the longest bond length? Circle the correct choice
 A) PS^{1-} B) PS C) PS^{1+}
- (iii) Which of the following will be diamagnetic? Circle the correct choice.
 A) PS^{1-} B) PS C) PS^{1+}
- (iv) Draw Lewis dot structure of the molecule with the shortest bond length (based on the MO diagram) from among the three (PS^{1-} , PS, PS^{1+}), and indicate the bond order calculated from the Lewis dot structure. **(3 points)**

Lewis Dot Structure

Bond Order



326DF6EB-9998-457B-9BB4-7187F1F1BCB5

midterm-exam-93388

#86 Page 10 of 10

SCRAP PAGE

**CHEMISTRY 110 –Midterm Exam****Thursday November 7, 2024, 6:30 – 8:30 pm**

Examiners: Pallavi Sirjoosingh and Ashok Kakkar

First name (please write as legibly as possible within the boxes)

--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--

Last name

--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--

Student ID number

--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--

This booklet has 8 questions on 10 pages (The last page can be used as a scrap page).

The staple must NOT be removed from the question booklet. Your intact question booklet must be returned at the end of the exam.

A separate booklet contains a periodic table, and datasheet.

Write your answers directly under the questions in the space provided. Do not write on the QR codes or near the top edges (as it may be cut off).

Please write **legibly and dark enough** for the scan to be readable. For students who use English as a second language, dictionaries are permitted. Calculators are allowed.

DO NOT ASK QUESTIONS OF THE INVIGILATORS about interpretations of the exam questions. If you have a concern about a question, please summarize it at the top of this page. All concerns will be evaluated after the exam. GOOD LUCK!

**Question 1 8 points** (*Assume all transitions are possible*)

a. 6 points Calculate the **energy (in kJ/mol)** required to ionize (completely remove) the electron from the **n=3** level of **He⁺ ion** (He: Helium). Show your work including equations. (3 sig. figs.)

b. 2 points

(I) Select the largest (longest) wavelength from the three choices below:

- i. Wavelength of photon to *completely remove* an electron from *1s orbital of Hydrogen atom*
- ii. Wavelength of photon to *completely remove* an electron from *1s orbital of Boron atom (B)*
- iii. Wavelength of photon to *completely remove* an electron from *1s orbital of Boron cation (B⁺)*

i or ii or iii

(II) Select the smallest (shortest) wavelength from the three choices below (*for H atom*):

- i. Wavelength of photon *emitted* for an electron transitioning from *2p to 1s orbital*
- ii. Wavelength of photon *absorbed* for an electron transitioning from *2s to 4p orbital*
- iii. Wavelength of photon *absorbed* for an electron transitioning from *1s to 3p orbital*

i or ii or iii



Question 2 8 points



A Chem110 student plots the graph shown on the left, from data obtained for the kinetic energy (eV) of the ejected electron versus frequency (s^{-1}) of the incident light, for an unknown metal. ($1\text{eV} = 1.6022 \times 10^{-19} \text{ J}$)

Based on the plot, answer the questions **a to c** below:

a. 1 point On the plot above, label the **threshold frequency**.

b. 2 points What is the value of the work potential (in J) of the unknown metal?

c. 3 points What is the *longest* wavelength (in nm) required to eject an electron from this metal's surface? Show your work including equation(s) used. (3 sig figs)

d. 2 points In the space provided below, plot the following

(Assume frequency is *above* threshold frequency):

- Kinetic energy* of an ejected electron versus the *intensity* of the incident light (left)
- Number of electrons* ejected versus the *frequency* of the incident light (right)



**Question 3** **6 points****a. 2 points**

In the ground state of Ge (Germanium; $Z = 32$) atom, **name** the following orbitals:

i. Orbital containing the electron that requires the maximum energy to be removed.

ii. Orbital containing the electron that experiences the minimum effective nuclear charge.

b. 2 points

An unknown anion (A^-) has the following *excited* electronic configuration: $[Ar] 4s^2 3d^3 4p^1$

(Ar: Argon; $Z = 18$)

Give the **condensed ground state electronic configuration** of the atom, **A** and the cation, **A^{2+}** .
Show your work. (Assume all electronic transitions are possible)

Electronic Configuration of A:

Electronic Configuration of A^{2+} :

c. 2 points

In a single atom, determine how many electrons can have the following set of quantum numbers?

(i) $n = 5; l = 3$

In a single atom, determine how many electrons can have the following set of quantum numbers?

(i) $n = 3; l = 2; m_s = +1/2$

**Question 4** **8 points**

Rank the following from 1 to 4 in **increasing** order. Provide the ranking in the boxes provided.

Note: Atomic number (Z) are provided for relevant atoms

a) Increasing (lowest to highest) order of *ionization energy*: (1= lowest; 4 = highest)

Mg

Mg: Z = 12

Rb

Rb: Z = 37

Cs

Cs: Z = 55

K

K : Z = 19

b) Increasing (lowest to highest) order of *bond angle*: (1= lowest; 4 = highest)

H₂OCCl₄XeF₄

HCN

c) Increasing (smallest to largest) order of *ionic size* (1= smallest; 4 = largest)

S²⁻

S: Z = 16

O²⁻

O: Z = 8

Be²⁺

Be: Z = 4

Na⁺

Na: Z = 11

d) Increasing (lowest to highest) order of *bond energy* (1= lowest; 4 = highest)

P-F

P-Cl

P-I

P-Br

**Question 5** **12 points**

a. 9 points For the molecules HClO_4 and TeF_6 , draw the most stable *Lewis structure* showing appropriate shape and *dashed/wedged bonds*, based on VSEPR. Determine the **molecular geometry** around the central atom. For HClO_4 , label the *most polar bond*.



b. 3 points (Note: Lone pairs are not shown in structures below)

(i) For the three *resonance* structures (1, 2 and 3) shown below, answer the following question:



(I) Circle the *most contributing* resonance structure.

Structure 1

Structure 2

Structure 3

(ii) For the three structures given (4, 5, and 6) below, answer the following questions:



(I) Circle all the structures that represent *resonance structures* of each other:

Structure 4

Structure 5

Structure 6

(II) Among the resonance structures, circle the *most contributing* resonance structure.

Structure 4

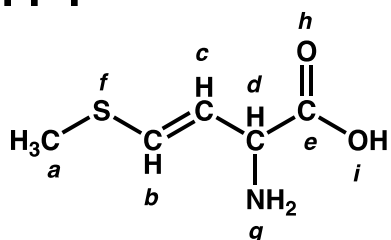
Structure 5

Structure 6

**Question 6** **6 points**

Using the data given below, calculate the **lattice energy** for SrO(s).

	ΔH° (kJ/mol)
$\text{Sr}(s) \longrightarrow \text{Sr}(g)$	+180
$\text{Sr}(g) \longrightarrow \text{Sr}^{2+}(g) + 2e^-$	+1468
$\text{O}_2(g) \longrightarrow 2\text{O}(g)$	+500
$\text{O}(g) + 2e^- \longrightarrow \text{O}^{2-}(g)$	+600
$\text{Sr}(s) + \frac{1}{2}\text{O}_2(g) \longrightarrow \text{SrO}(s)$	-550

**Question 7 10 points**

Consider the molecule given on the left, and answer questions (i) to (vi) using chemical bonding theories that we have learnt in class (valence bond theory, hybridization).

Note that the lone pairs are not shown in the structure:

i. Consider the carbon atoms marked in as ***a***, ***b***, ***c***, ***d***, and ***e*** in the molecule, and determine **hybridization** at each of the carbon centers. (2.5 points)

Atom	Hybridization
Carbon (<i>a</i>)	
Carbon (<i>b</i>)	
Carbon (<i>c</i>)	
Carbon (<i>d</i>)	
Carbon (<i>e</i>)	

ii. Now, consider the sulfur atom (S) marked as ***f***, nitrogen (N) atom (***g***), and oxygen(O) atoms (***h*** and ***i***), and determine their **hybridization**. (2 points)

Atom	Hybridization
Sulfur (<i>f</i>)	
Nitrogen (<i>g</i>)	
Oxygen (<i>h</i>)	
Oxygen (<i>i</i>)	

iii. **Describe** all the bonds formed by the carbon atoms ***a*** and ***b***, using sigma and pi designation as appropriate. (2 points)

Atom	Bonds formed (use sigma and pi as appropriate)
Carbon (<i>a</i>)	
Carbon (<i>b</i>)	

iv. How many total sp^3 -s sigma bonds are in the molecule? (1 point)

v. Determine **electronic** and **molecular geometry** at carbon (C) labeled ***e*** and sulfur (S) labeled ***f*** centres. (2 points)

Atom	Electronic Geometry	Molecular Geometry
Carbon (<i>e</i>)		
Sulfur (<i>f</i>)		

vi. Circle the value from the choices given below, that describes the **C-S-C bond angle at sulfur** (S) marked ***f***: (0.5 points)

>90° to <109.5° 109.5° >109.5 to <120° 120° >120° to <180°


Question 8 12 points

Using the **valence shell orbitals only** draw a **complete** molecular orbital (MO) diagram (with no s-p mixing) for the molecule **PS**, in the space provided below. **(6 points)**

Assume that 3s and 3p atomic orbitals will form molecular orbitals like those formed by the combination of 2s and 2p atomic orbitals that we learnt in class, and there is no s-p mixing.

Show **energy arrow**, **atomic orbitals** involved, **atom labels**, the **molecular orbitals** that are formed, **filled electrons** in the MO diagram etc. Make sure to label all the atomic and molecular orbitals.

Using the MO diagram above, answer the following questions:

- (i) Which one of the following will have the highest bond order? Circle the correct choice
 A) PS^{1-} B) PS C) PS^{1+}
- (ii) Which one of the following will have the longest bond length? Circle the correct choice
 A) PS^{1-} B) PS C) PS^{1+}
- (iii) Which of the following will be diamagnetic? Circle the correct choice.
 A) PS^{1-} B) PS C) PS^{1+}
- (iv) Draw Lewis dot structure of the molecule with the shortest bond length (based on the MO diagram) from among the three (PS^{1-} , PS, PS^{1+}), and indicate the bond order calculated from the Lewis dot structure. **(3 points)**

Lewis Dot Structure

Bond Order



1A539F7E-967C-4ED7-BA17-DB48C1D4D263

midterm-exam-93388

#87 Page 10 of 10

SCRAP PAGE

**CHEMISTRY 110 –Midterm Exam****Thursday November 7, 2024, 6:30 – 8:30 pm**

Examiners: Pallavi Sirjoosingh and Ashok Kakkar

First name (please write as legibly as possible within the boxes)

--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--

Last name

--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--

Student ID number

--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--

This booklet has 8 questions on 10 pages (The last page can be used as a scrap page).

The staple must NOT be removed from the question booklet. Your intact question booklet must be returned at the end of the exam.

A separate booklet contains a periodic table, and datasheet.

Write your answers directly under the questions in the space provided. Do not write on the QR codes or near the top edges (as it may be cut off).

Please write **legibly and dark enough** for the scan to be readable. For students who use English as a second language, dictionaries are permitted. Calculators are allowed.

DO NOT ASK QUESTIONS OF THE INVIGILATORS about interpretations of the exam questions. If you have a concern about a question, please summarize it at the top of this page. All concerns will be evaluated after the exam. GOOD LUCK!

**Question 1 8 points** (*Assume all transitions are possible*)

a. 6 points Calculate the **energy (in kJ/mol)** required to ionize (completely remove) the electron from the **n=3** level of **He⁺ ion** (He: Helium). Show your work including equations. (3 sig. figs.)

b. 2 points

(I) Select the largest (longest) wavelength from the three choices below:

- i. Wavelength of photon to *completely remove* an electron from *1s orbital of Hydrogen atom*
- ii. Wavelength of photon to *completely remove* an electron from *1s orbital of Boron atom (B)*
- iii. Wavelength of photon to *completely remove* an electron from *1s orbital of Boron cation (B⁺)*

i or ii or iii

(II) Select the smallest (shortest) wavelength from the three choices below (*for H atom*):

- i. Wavelength of photon *emitted* for an electron transitioning from *2p to 1s orbital*
- ii. Wavelength of photon *absorbed* for an electron transitioning from *2s to 4p orbital*
- iii. Wavelength of photon *absorbed* for an electron transitioning from *1s to 3p orbital*

i or ii or iii



Question 2 8 points



A Chem110 student plots the graph shown on the left, from data obtained for the kinetic energy (eV) of the ejected electron versus frequency (s^{-1}) of the incident light, for an unknown metal. ($1\text{eV} = 1.6022 \times 10^{-19} \text{ J}$)

Based on the plot, answer the questions **a to c** below:

a. 1 point On the plot above, label the **threshold frequency**.

b. 2 points What is the value of the work potential (in J) of the unknown metal?

c. 3 points What is the *longest* wavelength (in nm) required to eject an electron from this metal's surface? Show your work including equation(s) used. (3 sig figs)

d. 2 points In the space provided below, plot the following

(Assume frequency is *above* threshold frequency):

- i. *Kinetic energy* of an ejected electron versus the *intensity* of the incident light (left)
- ii. *Number of electrons* ejected versus the *frequency* of the incident light (right)



**Question 3** **6 points****a. 2 points**

In the ground state of Ge (Germanium; $Z = 32$) atom, **name** the following orbitals:

i. Orbital containing the electron that requires the maximum energy to be removed.

ii. Orbital containing the electron that experiences the minimum effective nuclear charge.

b. 2 points

An unknown anion (A^-) has the following *excited* electronic configuration: $[Ar] 4s^2 3d^3 4p^1$

(Ar: Argon; $Z = 18$)

Give the **condensed ground state electronic configuration** of the atom, **A** and the cation, **A^{2+}** .
Show your work. (Assume all electronic transitions are possible)

Electronic Configuration of A:

Electronic Configuration of A^{2+} :

c. 2 points

In a single atom, determine how many electrons can have the following set of quantum numbers?

(i) $n = 5; l = 3$

In a single atom, determine how many electrons can have the following set of quantum numbers?

(i) $n = 3; l = 2; m_s = +1/2$

**Question 4** **8 points**

Rank the following from 1 to 4 in **increasing** order. Provide the ranking in the boxes provided.

Note: Atomic number (Z) are provided for relevant atoms

a) Increasing (lowest to highest) order of *ionization energy*: (1= lowest; 4 = highest)

Mg

Mg: Z = 12

Rb

Rb: Z = 37

Cs

Cs: Z = 55

K

K : Z = 19

b) Increasing (lowest to highest) order of *bond angle*: (1= lowest; 4 = highest)

H₂OCCl₄XeF₄

HCN

c) Increasing (smallest to largest) order of *ionic size* (1= smallest; 4 = largest)

S²⁻

S: Z = 16

O²⁻

O: Z = 8

Be²⁺

Be: Z = 4

Na⁺

Na: Z = 11

d) Increasing (lowest to highest) order of *bond energy* (1= lowest; 4 = highest)

P-F

P-Cl

P-I

P-Br

**Question 5** **12 points**

a. 9 points For the molecules HClO_4 and TeF_6 , draw the most stable *Lewis structure* showing appropriate shape and *dashed/wedged bonds*, based on VSEPR. Determine the **molecular geometry** around the central atom. For HClO_4 , label the *most polar bond*.



b. 3 points (Note: Lone pairs are not shown in structures below)

(i) For the three *resonance* structures (1, 2 and 3) shown below, answer the following question:



(I) Circle the *most contributing* resonance structure.

Structure 1

Structure 2

Structure 3

(ii) For the three structures given (4, 5, and 6) below, answer the following questions:



(I) Circle all the structures that represent *resonance structures* of each other:

Structure 4

Structure 5

Structure 6

(II) Among the resonance structures, circle the *most contributing* resonance structure.

Structure 4

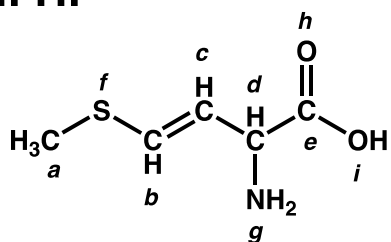
Structure 5

Structure 6

**Question 6** **6 points**

Using the data given below, calculate the **lattice energy** for SrO(s).

	ΔH° (kJ/mol)
$\text{Sr}(s) \longrightarrow \text{Sr}(g)$	+180
$\text{Sr}(g) \longrightarrow \text{Sr}^{2+}(g) + 2e^-$	+1468
$\text{O}_2(g) \longrightarrow 2\text{O}(g)$	+500
$\text{O}(g) + 2e^- \longrightarrow \text{O}^{2-}(g)$	+600
$\text{Sr}(s) + \frac{1}{2}\text{O}_2(g) \longrightarrow \text{SrO}(s)$	-550

**Question 7 10 points**

Consider the molecule given on the left, and answer questions (i) to (vi) using chemical bonding theories that we have learnt in class (valence bond theory, hybridization).

Note that the lone pairs are not shown in the structure:

i. Consider the carbon atoms marked in as ***a***, ***b***, ***c***, ***d***, and ***e*** in the molecule, and determine **hybridization** at each of the carbon centers. (2.5 points)

Atom	Hybridization
Carbon (<i>a</i>)	
Carbon (<i>b</i>)	
Carbon (<i>c</i>)	
Carbon (<i>d</i>)	
Carbon (<i>e</i>)	

ii. Now, consider the sulfur atom (S) marked as ***f***, nitrogen (N) atom (***g***), and oxygen(O) atoms (***h*** and ***i***), and determine their **hybridization**. (2 points)

Atom	Hybridization
Sulfur (<i>f</i>)	
Nitrogen (<i>g</i>)	
Oxygen (<i>h</i>)	
Oxygen (<i>i</i>)	

iii. **Describe** all the bonds formed by the carbon atoms ***a*** and ***b***, using sigma and pi designation as appropriate. (2 points)

Atom	Bonds formed (use sigma and pi as appropriate)
Carbon (<i>a</i>)	
Carbon (<i>b</i>)	

iv. How many total sp^3 -s sigma bonds are in the molecule? (1 point)

v. Determine **electronic** and **molecular geometry** at carbon (C) labeled ***e*** and sulfur (S) labeled ***f*** centres. (2 points)

Atom	Electronic Geometry	Molecular Geometry
Carbon (<i>e</i>)		
Sulfur (<i>f</i>)		

vi. Circle the value from the choices given below, that describes the **C-S-C bond angle at sulfur** (S) marked ***f***: (0.5 points)

>90° to <109.5° 109.5° >109.5 to <120° 120° >120° to <180°



Question 8 12 points

Using the **valence shell orbitals only** draw a **complete** molecular orbital (MO) diagram (with no s-p mixing) for the molecule **PS**, in the space provided below. **(6 points)**

Assume that 3s and 3p atomic orbitals will form molecular orbitals like those formed by the combination of 2s and 2p atomic orbitals that we learnt in class, and there is no s-p mixing.

Show **energy arrow**, **atomic orbitals** involved, **atom labels**, the **molecular orbitals** that are formed, **filled electrons** in the MO diagram etc. Make sure to label all the atomic and molecular orbitals.

Using the MO diagram above, answer the following questions:

- (i) Which one of the following will have the highest bond order? Circle the correct choice
 A) PS^{1-} B) PS C) PS^{1+}
- (ii) Which one of the following will have the longest bond length? Circle the correct choice
 A) PS^{1-} B) PS C) PS^{1+}
- (iii) Which of the following will be diamagnetic? Circle the correct choice.
 A) PS^{1-} B) PS C) PS^{1+}
- (iv) Draw Lewis dot structure of the molecule with the shortest bond length (based on the MO diagram) from among the three (PS^{1-} , PS, PS^{1+}), and indicate the bond order calculated from the Lewis dot structure. **(3 points)**

Lewis Dot Structure

Bond Order



4C0B8DE0-70DD-43BB-82C9-C4267802158B

midterm-exam-93388

#88 Page 10 of 10

SCRAP PAGE

**CHEMISTRY 110 –Midterm Exam****Thursday November 7, 2024, 6:30 – 8:30 pm**

Examiners: Pallavi Sirjoosingh and Ashok Kakkar

First name (please write as legibly as possible within the boxes)

--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--

Last name

--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--

Student ID number

--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--

This booklet has 8 questions on 10 pages (The last page can be used as a scrap page).

The staple must NOT be removed from the question booklet. Your intact question booklet must be returned at the end of the exam.

A separate booklet contains a periodic table, and datasheet.

Write your answers directly under the questions in the space provided. Do not write on the QR codes or near the top edges (as it may be cut off).

Please write **legibly and dark enough** for the scan to be readable. For students who use English as a second language, dictionaries are permitted. Calculators are allowed.

DO NOT ASK QUESTIONS OF THE INVIGILATORS about interpretations of the exam questions. If you have a concern about a question, please summarize it at the top of this page. All concerns will be evaluated after the exam. GOOD LUCK!

**Question 1 8 points** (*Assume all transitions are possible*)

a. 6 points Calculate the **energy (in kJ/mol)** required to ionize (completely remove) the electron from the **n=3** level of **He⁺ ion** (He: Helium). Show your work including equations. (3 sig. figs.)

b. 2 points

(I) Select the largest (longest) wavelength from the three choices below:

- i. Wavelength of photon to *completely remove* an electron from *1s orbital of Hydrogen atom*
- ii. Wavelength of photon to *completely remove* an electron from *1s orbital of Boron atom (B)*
- iii. Wavelength of photon to *completely remove* an electron from *1s orbital of Boron cation (B⁺)*

i or ii or iii

(II) Select the smallest (shortest) wavelength from the three choices below (*for H atom*):

- i. Wavelength of photon *emitted* for an electron transitioning from *2p to 1s orbital*
- ii. Wavelength of photon *absorbed* for an electron transitioning from *2s to 4p orbital*
- iii. Wavelength of photon *absorbed* for an electron transitioning from *1s to 3p orbital*

i or ii or iii



Question 2 8 points



A Chem110 student plots the graph shown on the left, from data obtained for the kinetic energy (eV) of the ejected electron versus frequency (s^{-1}) of the incident light, for an unknown metal. ($1\text{eV} = 1.6022 \times 10^{-19} \text{ J}$)

Based on the plot, answer the questions **a to c** below:

a. 1 point On the plot above, label the **threshold frequency**.

b. 2 points What is the value of the work potential (in J) of the unknown metal?

c. 3 points What is the *longest* wavelength (in nm) required to eject an electron from this metal's surface? Show your work including equation(s) used. (3 sig figs)

d. 2 points In the space provided below, plot the following

(Assume frequency is *above* threshold frequency):

- Kinetic energy* of an ejected electron versus the *intensity* of the incident light (left)
- Number of electrons* ejected versus the *frequency* of the incident light (right)



**Question 3** **6 points****a. 2 points**

In the ground state of Ge (Germanium; $Z = 32$) atom, **name** the following orbitals:

i. Orbital containing the electron that requires the maximum energy to be removed.

ii. Orbital containing the electron that experiences the minimum effective nuclear charge.

b. 2 points

An unknown anion (A^-) has the following *excited* electronic configuration: $[Ar] 4s^2 3d^3 4p^1$

(Ar: Argon; $Z = 18$)

Give the **condensed ground state electronic configuration** of the atom, **A** and the cation, **A^{2+}** .
Show your work. (Assume all electronic transitions are possible)

Electronic Configuration of A:

Electronic Configuration of A^{2+} :

c. 2 points

In a single atom, determine how many electrons can have the following set of quantum numbers?

(i) $n = 5; l = 3$

In a single atom, determine how many electrons can have the following set of quantum numbers?

(i) $n = 3; l = 2; m_s = +1/2$

**Question 4** **8 points**

Rank the following from 1 to 4 in **increasing** order. Provide the ranking in the boxes provided.

Note: Atomic number (Z) are provided for relevant atoms

a) Increasing (lowest to highest) order of *ionization energy*: (1= lowest; 4 = highest)

Mg

Mg: Z = 12

Rb

Rb: Z = 37

Cs

Cs: Z = 55

K

K : Z = 19

b) Increasing (lowest to highest) order of *bond angle*: (1= lowest; 4 = highest)

H₂OCCl₄XeF₄

HCN

c) Increasing (smallest to largest) order of *ionic size* (1= smallest; 4 = largest)

S²⁻

S: Z = 16

O²⁻

O: Z = 8

Be²⁺

Be: Z = 4

Na⁺

Na: Z = 11

d) Increasing (lowest to highest) order of *bond energy* (1= lowest; 4 = highest)

P-F

P-Cl

P-I

P-Br

**Question 5** **12 points**

a. 9 points For the molecules HClO_4 and TeF_6 , draw the most stable *Lewis structure* showing appropriate shape and *dashed/wedged bonds*, based on VSEPR. Determine the **molecular geometry** around the central atom. For HClO_4 , label the *most polar bond*.



b. 3 points (Note: Lone pairs are not shown in structures below)

(i) For the three *resonance* structures (1, 2 and 3) shown below, answer the following question:



(I) Circle the *most contributing* resonance structure.

Structure 1

Structure 2

Structure 3

(ii) For the three structures given (4, 5, and 6) below, answer the following questions:



(I) Circle all the structures that represent *resonance structures* of each other:

Structure 4

Structure 5

Structure 6

(II) Among the resonance structures, circle the *most contributing* resonance structure.

Structure 4

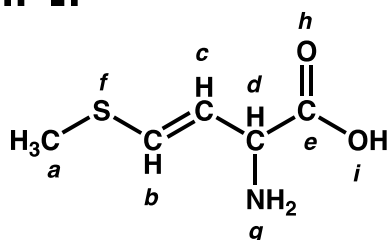
Structure 5

Structure 6

**Question 6** **6 points**

Using the data given below, calculate the **lattice energy** for SrO(s).

	ΔH° (kJ/mol)
$\text{Sr}(s) \longrightarrow \text{Sr}(g)$	+180
$\text{Sr}(g) \longrightarrow \text{Sr}^{2+}(g) + 2e^-$	+1468
$\text{O}_2(g) \longrightarrow 2\text{O}(g)$	+500
$\text{O}(g) + 2e^- \longrightarrow \text{O}^{2-}(g)$	+600
$\text{Sr}(s) + \frac{1}{2}\text{O}_2(g) \longrightarrow \text{SrO}(s)$	-550

**Question 7 10 points**

Consider the molecule given on the left, and answer questions (i) to (vi) using chemical bonding theories that we have learnt in class (valence bond theory, hybridization).

Note that the lone pairs are not shown in the structure:

i. Consider the carbon atoms marked in as ***a***, ***b***, ***c***, ***d***, and ***e*** in the molecule, and determine **hybridization** at each of the carbon centers. (2.5 points)

Atom	Hybridization
Carbon (<i>a</i>)	
Carbon (<i>b</i>)	
Carbon (<i>c</i>)	
Carbon (<i>d</i>)	
Carbon (<i>e</i>)	

ii. Now, consider the sulfur atom (S) marked as ***f***, nitrogen (N) atom (***g***), and oxygen(O) atoms (***h*** and ***i***), and determine their **hybridization**. (2 points)

Atom	Hybridization
Sulfur (<i>f</i>)	
Nitrogen (<i>g</i>)	
Oxygen (<i>h</i>)	
Oxygen (<i>i</i>)	

iii. **Describe** all the bonds formed by the carbon atoms ***a*** and ***b***, using sigma and pi designation as appropriate. (2 points)

Atom	Bonds formed (use sigma and pi as appropriate)
Carbon (<i>a</i>)	
Carbon (<i>b</i>)	

iv. How many total sp^3 -s sigma bonds are in the molecule? (1 point)

v. Determine **electronic** and **molecular geometry** at carbon (C) labeled ***e*** and sulfur (S) labeled ***f*** centres. (2 points)

Atom	Electronic Geometry	Molecular Geometry
Carbon (<i>e</i>)		
Sulfur (<i>f</i>)		

vi. Circle the value from the choices given below, that describes the **C-S-C bond angle at sulfur** (S) marked ***f***: (0.5 points)

>90° to <109.5° 109.5° >109.5 to <120° 120° >120° to <180°

**Question 8 12 points**

Using the **valence shell orbitals only** draw a **complete** molecular orbital (MO) diagram (with no s-p mixing) for the molecule **PS**, in the space provided below. **(6 points)**

Assume that 3s and 3p atomic orbitals will form molecular orbitals like those formed by the combination of 2s and 2p atomic orbitals that we learnt in class, and there is no s-p mixing.

Show **energy arrow**, **atomic orbitals** involved, **atom labels**, the **molecular orbitals** that are formed, **filled electrons** in the MO diagram etc. Make sure to label all the atomic and molecular orbitals.

Using the MO diagram above, answer the following questions:

- (i) Which one of the following will have the highest bond order? Circle the correct choice
A) PS^{1-} B) PS C) PS^{1+}
- (ii) Which one of the following will have the longest bond length? Circle the correct choice
A) PS^{1-} B) PS C) PS^{1+}
- (iii) Which of the following will be diamagnetic? Circle the correct choice.
A) PS^{1-} B) PS C) PS^{1+}
- (iv) Draw Lewis dot structure of the molecule with the shortest bond length (based on the MO diagram) from among the three (PS^{1-} , PS, PS^{1+}), and indicate the bond order calculated from the Lewis dot structure. **(3 points)**

Lewis Dot Structure

Bond Order



FC52DFA2-8F23-4FFC-BBCC-77A4C21C8AC2

midterm-exam-93388

#89 Page 10 of 10

SCRAP PAGE

**CHEMISTRY 110 –Midterm Exam****Thursday November 7, 2024, 6:30 – 8:30 pm****Examiners: Pallavi Sirjoosingh and Ashok Kakkar**

First name (please write as legibly as possible within the boxes)

--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--

Last name

--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--

Student ID number

--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--

This booklet has 8 questions on 10 pages (The last page can be used as a scrap page).

The staple must NOT be removed from the question booklet. Your intact question booklet must be returned at the end of the exam.

A separate booklet contains a periodic table, and datasheet.

Write your answers directly under the questions in the space provided. Do not write on the QR codes or near the top edges (as it may be cut off).

Please write **legibly and dark enough** for the scan to be readable. For students who use English as a second language, dictionaries are permitted. Calculators are allowed.

DO NOT ASK QUESTIONS OF THE INVIGILATORS about interpretations of the exam questions. If you have a concern about a question, please summarize it at the top of this page. All concerns will be evaluated after the exam. GOOD LUCK!

**Question 1 8 points** (*Assume all transitions are possible*)

a. 6 points Calculate the **energy (in kJ/mol)** required to ionize (completely remove) the electron from the **n=3** level of **He⁺ ion** (He: Helium). Show your work including equations. (3 sig. figs.)

b. 2 points

(I) Select the largest (longest) wavelength from the three choices below:

- i. Wavelength of photon to *completely remove* an electron from *1s orbital of Hydrogen atom*
- ii. Wavelength of photon to *completely remove* an electron from *1s orbital of Boron atom (B)*
- iii. Wavelength of photon to *completely remove* an electron from *1s orbital of Boron cation (B⁺)*

i or ii or iii

(II) Select the smallest (shortest) wavelength from the three choices below (*for H atom*):

- i. Wavelength of photon *emitted* for an electron transitioning from *2p to 1s orbital*
- ii. Wavelength of photon *absorbed* for an electron transitioning from *2s to 4p orbital*
- iii. Wavelength of photon *absorbed* for an electron transitioning from *1s to 3p orbital*

i or ii or iii



Question 2 8 points



A Chem110 student plots the graph shown on the left, from data obtained for the kinetic energy (eV) of the ejected electron versus frequency (s^{-1}) of the incident light, for an unknown metal. ($1\text{eV} = 1.6022 \times 10^{-19} \text{ J}$)

Based on the plot, answer the questions **a to c** below:

a. 1 point On the plot above, label the **threshold frequency**.

b. 2 points What is the value of the work potential (in J) of the unknown metal?

c. 3 points What is the *longest* wavelength (in nm) required to eject an electron from this metal's surface? Show your work including equation(s) used. (3 sig figs)

d. 2 points In the space provided below, plot the following

(Assume frequency is *above* threshold frequency):

- i. *Kinetic energy* of an ejected electron versus the *intensity* of the incident light (left)
- ii. *Number of electrons* ejected versus the *frequency* of the incident light (right)



**Question 3** **6 points****a. 2 points**

In the ground state of Ge (Germanium; $Z = 32$) atom, **name** the following orbitals:

i. Orbital containing the electron that requires the maximum energy to be removed.

ii. Orbital containing the electron that experiences the minimum effective nuclear charge.

b. 2 points

An unknown anion (A^-) has the following *excited* electronic configuration: $[Ar] 4s^2 3d^3 4p^1$

(Ar: Argon; $Z = 18$)

Give the **condensed ground state electronic configuration** of the atom, **A** and the cation, **A^{2+}** .
Show your work. (Assume all electronic transitions are possible)

Electronic Configuration of A:

Electronic Configuration of A^{2+} :

c. 2 points

In a single atom, determine how many electrons can have the following set of quantum numbers?

(i) $n = 5; l = 3$

In a single atom, determine how many electrons can have the following set of quantum numbers?

(i) $n = 3; l = 2; m_s = +1/2$

**Question 4** **8 points**

Rank the following from 1 to 4 in **increasing** order. Provide the ranking in the boxes provided.

Note: Atomic number (Z) are provided for relevant atoms

a) Increasing (lowest to highest) order of *ionization energy*: (1= lowest; 4 = highest)

Mg

Mg: Z = 12

Rb

Rb: Z = 37

Cs

Cs: Z = 55

K

K : Z = 19

b) Increasing (lowest to highest) order of *bond angle*: (1= lowest; 4 = highest)

H₂OCCl₄XeF₄

HCN

c) Increasing (smallest to largest) order of *ionic size* (1= smallest; 4 = largest)

S²⁻

S: Z = 16

O²⁻

O: Z = 8

Be²⁺

Be: Z = 4

Na⁺

Na: Z = 11

d) Increasing (lowest to highest) order of *bond energy* (1= lowest; 4 = highest)

P-F

P-Cl

P-I

P-Br

**Question 5** **12 points**

a. 9 points For the molecules HClO_4 and TeF_6 , draw the most stable *Lewis structure* showing appropriate shape and *dashed/wedged bonds*, based on VSEPR. Determine the **molecular geometry** around the central atom. For HClO_4 , label the *most polar bond*.



b. 3 points (Note: Lone pairs are not shown in structures below)

(i) For the three *resonance* structures (1, 2 and 3) shown below, answer the following question:



(I) Circle the *most contributing* resonance structure.

Structure 1

Structure 2

Structure 3

(ii) For the three structures given (4, 5, and 6) below, answer the following questions:



(I) Circle all the structures that represent *resonance structures* of each other:

Structure 4

Structure 5

Structure 6

(II) Among the resonance structures, circle the *most contributing* resonance structure.

Structure 4

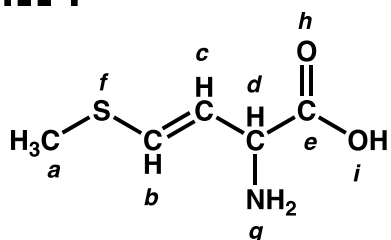
Structure 5

Structure 6

**Question 6** **6 points**

Using the data given below, calculate the **lattice energy** for SrO(s).

	ΔH° (kJ/mol)
$\text{Sr}(s) \longrightarrow \text{Sr}(g)$	+180
$\text{Sr}(g) \longrightarrow \text{Sr}^{2+}(g) + 2e^-$	+1468
$\text{O}_2(g) \longrightarrow 2\text{O}(g)$	+500
$\text{O}(g) + 2e^- \longrightarrow \text{O}^{2-}(g)$	+600
$\text{Sr}(s) + \frac{1}{2}\text{O}_2(g) \longrightarrow \text{SrO}(s)$	-550

**Question 7 10 points**

Consider the molecule given on the left, and answer questions (i) to (vi) using chemical bonding theories that we have learnt in class (valence bond theory, hybridization).

Note that the lone pairs are not shown in the structure:

i. Consider the carbon atoms marked in as ***a***, ***b***, ***c***, ***d***, and ***e*** in the molecule, and determine **hybridization** at each of the carbon centers. (2.5 points)

Atom	Hybridization
Carbon (<i>a</i>)	
Carbon (<i>b</i>)	
Carbon (<i>c</i>)	
Carbon (<i>d</i>)	
Carbon (<i>e</i>)	

ii. Now, consider the sulfur atom (S) marked as ***f***, nitrogen (N) atom (***g***), and oxygen(O) atoms (***h*** and ***i***), and determine their **hybridization**. (2 points)

Atom	Hybridization
Sulfur (<i>f</i>)	
Nitrogen (<i>g</i>)	
Oxygen (<i>h</i>)	
Oxygen (<i>i</i>)	

iii. **Describe** all the bonds formed by the carbon atoms ***a*** and ***b***, using sigma and pi designation as appropriate. (2 points)

Atom	Bonds formed (use sigma and pi as appropriate)
Carbon (<i>a</i>)	
Carbon (<i>b</i>)	

iv. How many total sp^3 -s sigma bonds are in the molecule? (1 point)

v. Determine **electronic** and **molecular geometry** at carbon (C) labeled ***e*** and sulfur (S) labeled ***f*** centres. (2 points)

Atom	Electronic Geometry	Molecular Geometry
Carbon (<i>e</i>)		
Sulfur (<i>f</i>)		

vi. Circle the value from the choices given below, that describes the **C-S-C bond angle at sulfur** (S) marked ***f***: (0.5 points)

$>90^\circ$ to $<109.5^\circ$ 109.5° >109.5 to $<120^\circ$ 120° $>120^\circ$ to $<180^\circ$


Question 8 12 points

Using the **valence shell orbitals only** draw a **complete** molecular orbital (MO) diagram (with no s-p mixing) for the molecule **PS**, in the space provided below. **(6 points)**

Assume that 3s and 3p atomic orbitals will form molecular orbitals like those formed by the combination of 2s and 2p atomic orbitals that we learnt in class, and there is no s-p mixing.

Show **energy arrow**, **atomic orbitals** involved, **atom labels**, the **molecular orbitals** that are formed, **filled electrons** in the MO diagram etc. Make sure to label all the atomic and molecular orbitals.

Using the MO diagram above, answer the following questions:

- (i) Which one of the following will have the highest bond order? Circle the correct choice
 A) PS^{1-} B) PS C) PS^{1+}
- (ii) Which one of the following will have the longest bond length? Circle the correct choice
 A) PS^{1-} B) PS C) PS^{1+}
- (iii) Which of the following will be diamagnetic? Circle the correct choice.
 A) PS^{1-} B) PS C) PS^{1+}
- (iv) Draw Lewis dot structure of the molecule with the shortest bond length (based on the MO diagram) from among the three (PS^{1-} , PS, PS^{1+}), and indicate the bond order calculated from the Lewis dot structure. **(3 points)**

Lewis Dot Structure

Bond Order



EB398F33-BF15-42AD-8420-2EB157E543CA

midterm-exam-93388

#90 Page 10 of 10

SCRAP PAGE

**CHEMISTRY 110 –Midterm Exam****Thursday November 7, 2024, 6:30 – 8:30 pm**

Examiners: Pallavi Sirjoosingh and Ashok Kakkar

First name (please write as legibly as possible within the boxes)

--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--

Last name

--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--

Student ID number

--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--

This booklet has 8 questions on 10 pages (The last page can be used as a scrap page).

The staple must NOT be removed from the question booklet. Your intact question booklet must be returned at the end of the exam.

A separate booklet contains a periodic table, and datasheet.

Write your answers directly under the questions in the space provided. Do not write on the QR codes or near the top edges (as it may be cut off).

Please write **legibly and dark enough** for the scan to be readable. For students who use English as a second language, dictionaries are permitted. Calculators are allowed.

DO NOT ASK QUESTIONS OF THE INVIGILATORS about interpretations of the exam questions. If you have a concern about a question, please summarize it at the top of this page. All concerns will be evaluated after the exam. GOOD LUCK!

**Question 1 8 points** (*Assume all transitions are possible*)

a. 6 points Calculate the **energy (in kJ/mol)** required to ionize (completely remove) the electron from the **n=3** level of **He⁺ ion** (He: Helium). Show your work including equations. (3 sig. figs.)

b. 2 points

(I) Select the largest (longest) wavelength from the three choices below:

- i. Wavelength of photon to *completely remove* an electron from *1s orbital of Hydrogen atom*
- ii. Wavelength of photon to *completely remove* an electron from *1s orbital of Boron atom (B)*
- iii. Wavelength of photon to *completely remove* an electron from *1s orbital of Boron cation (B⁺)*

i or ii or iii

(II) Select the smallest (shortest) wavelength from the three choices below (*for H atom*):

- i. Wavelength of photon *emitted* for an electron transitioning from *2p to 1s orbital*
- ii. Wavelength of photon *absorbed* for an electron transitioning from *2s to 4p orbital*
- iii. Wavelength of photon *absorbed* for an electron transitioning from *1s to 3p orbital*

i or ii or iii



Question 2 8 points



A Chem110 student plots the graph shown on the left, from data obtained for the kinetic energy (eV) of the ejected electron versus frequency (s^{-1}) of the incident light, for an unknown metal. ($1\text{eV} = 1.6022 \times 10^{-19} \text{ J}$)

Based on the plot, answer the questions **a to c** below:

a. 1 point On the plot above, label the **threshold frequency**.

b. 2 points What is the value of the work potential (in J) of the unknown metal?

c. 3 points What is the *longest* wavelength (in nm) required to eject an electron from this metal's surface? Show your work including equation(s) used. (3 sig figs)

d. 2 points In the space provided below, plot the following

(Assume frequency is *above* threshold frequency):

- Kinetic energy* of an ejected electron versus the *intensity* of the incident light (left)
- Number of electrons* ejected versus the *frequency* of the incident light (right)



**Question 3** **6 points****a. 2 points**

In the ground state of Ge (Germanium; $Z = 32$) atom, **name** the following orbitals:

i. Orbital containing the electron that requires the maximum energy to be removed.

ii. Orbital containing the electron that experiences the minimum effective nuclear charge.

b. 2 points

An unknown anion (A^-) has the following *excited* electronic configuration: $[Ar] 4s^2 3d^3 4p^1$

(Ar: Argon; $Z = 18$)

Give the **condensed ground state electronic configuration** of the atom, **A** and the cation, **A^{2+}** .
Show your work. (Assume all electronic transitions are possible)

Electronic Configuration of A:

Electronic Configuration of A^{2+} :

c. 2 points

In a single atom, determine how many electrons can have the following set of quantum numbers?

(i) $n = 5; l = 3$

In a single atom, determine how many electrons can have the following set of quantum numbers?

(i) $n = 3; l = 2; m_s = +1/2$

**Question 4** **8 points**

Rank the following from 1 to 4 in **increasing** order. Provide the ranking in the boxes provided.

Note: Atomic number (Z) are provided for relevant atoms

a) Increasing (lowest to highest) order of *ionization energy*: (1= lowest; 4 = highest)

Mg

Mg: Z = 12

Rb

Rb: Z = 37

Cs

Cs: Z = 55

K

K : Z = 19

b) Increasing (lowest to highest) order of *bond angle*: (1= lowest; 4 = highest)

H₂OCCl₄XeF₄

HCN

c) Increasing (smallest to largest) order of *ionic size* (1= smallest; 4 = largest)

S²⁻

S: Z = 16

O²⁻

O: Z = 8

Be²⁺

Be: Z = 4

Na⁺

Na: Z = 11

d) Increasing (lowest to highest) order of *bond energy* (1= lowest; 4 = highest)

P-F

P-Cl

P-I

P-Br

**Question 5** **12 points**

a. 9 points For the molecules HClO_4 and TeF_6 , draw the most stable *Lewis structure* showing appropriate shape and *dashed/wedged bonds*, based on VSEPR. Determine the **molecular geometry** around the central atom. For HClO_4 , label the *most polar bond*.



b. 3 points (Note: Lone pairs are not shown in structures below)

(i) For the three *resonance* structures (1, 2 and 3) shown below, answer the following question:



(I) Circle the *most contributing* resonance structure.

Structure 1

Structure 2

Structure 3

(ii) For the three structures given (4, 5, and 6) below, answer the following questions:



(I) Circle all the structures that represent *resonance structures* of each other:

Structure 4

Structure 5

Structure 6

(II) Among the resonance structures, circle the *most contributing* resonance structure.

Structure 4

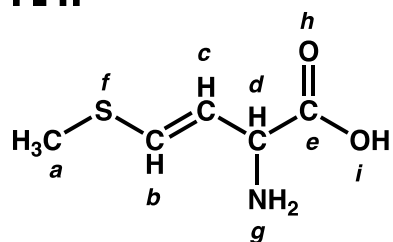
Structure 5

Structure 6

**Question 6** **6 points**

Using the data given below, calculate the **lattice energy** for SrO(s).

	ΔH° (kJ/mol)
$\text{Sr}(s) \longrightarrow \text{Sr}(g)$	+180
$\text{Sr}(g) \longrightarrow \text{Sr}^{2+}(g) + 2e^-$	+1468
$\text{O}_2(g) \longrightarrow 2\text{O}(g)$	+500
$\text{O}(g) + 2e^- \longrightarrow \text{O}^{2-}(g)$	+600
$\text{Sr}(s) + \frac{1}{2}\text{O}_2(g) \longrightarrow \text{SrO}(s)$	-550

**Question 7 10 points**

Consider the molecule given on the left, and answer questions (i) to (vi) using chemical bonding theories that we have learnt in class (valence bond theory, hybridization).

Note that the lone pairs are not shown in the structure:

i. Consider the carbon atoms marked in as ***a***, ***b***, ***c***, ***d***, and ***e*** in the molecule, and determine **hybridization** at each of the carbon centers. (2.5 points)

Atom	Hybridization
Carbon (<i>a</i>)	
Carbon (<i>b</i>)	
Carbon (<i>c</i>)	
Carbon (<i>d</i>)	
Carbon (<i>e</i>)	

ii. Now, consider the sulfur atom (S) marked as ***f***, nitrogen (N) atom (***g***), and oxygen(O) atoms (***h*** and ***i***), and determine their **hybridization**. (2 points)

Atom	Hybridization
Sulfur (<i>f</i>)	
Nitrogen (<i>g</i>)	
Oxygen (<i>h</i>)	
Oxygen (<i>i</i>)	

iii. **Describe** all the bonds formed by the carbon atoms ***a*** and ***b***, using sigma and pi designation as appropriate. (2 points)

Atom	Bonds formed (use sigma and pi as appropriate)
Carbon (<i>a</i>)	
Carbon (<i>b</i>)	

iv. How many total sp^3 -s sigma bonds are in the molecule? (1 point)

v. Determine **electronic** and **molecular geometry** at carbon (C) labeled ***e*** and sulfur (S) labeled ***f*** centres. (2 points)

Atom	Electronic Geometry	Molecular Geometry
Carbon (<i>e</i>)		
Sulfur (<i>f</i>)		

vi. Circle the value from the choices given below, that describes the **C-S-C bond angle at sulfur** (S) marked ***f***: (0.5 points)

>90° to <109.5°

109.5°

>109.5 to <120°

120°

>120° to <180°


Question 8 12 points

Using the **valence shell orbitals only** draw a **complete** molecular orbital (MO) diagram (with no s-p mixing) for the molecule **PS**, in the space provided below. **(6 points)**

Assume that 3s and 3p atomic orbitals will form molecular orbitals like those formed by the combination of 2s and 2p atomic orbitals that we learnt in class, and there is no s-p mixing.

Show **energy arrow**, **atomic orbitals** involved, **atom labels**, the **molecular orbitals** that are formed, **filled electrons** in the MO diagram etc. Make sure to label all the atomic and molecular orbitals.

Using the MO diagram above, answer the following questions:

- (i) Which one of the following will have the highest bond order? Circle the correct choice
 A) PS^{1-} B) PS C) PS^{1+}
- (ii) Which one of the following will have the longest bond length? Circle the correct choice
 A) PS^{1-} B) PS C) PS^{1+}
- (iii) Which of the following will be diamagnetic? Circle the correct choice.
 A) PS^{1-} B) PS C) PS^{1+}
- (iv) Draw Lewis dot structure of the molecule with the shortest bond length (based on the MO diagram) from among the three (PS^{1-} , PS, PS^{1+}), and indicate the bond order calculated from the Lewis dot structure. **(3 points)**

Lewis Dot Structure

Bond Order



5BCE38E0-C6E3-4D3C-A0AE-38E72E3C7AA9

midterm-exam-93388

#91 Page 10 of 10

SCRAP PAGE

**CHEMISTRY 110 –Midterm Exam****Thursday November 7, 2024, 6:30 – 8:30 pm**

Examiners: Pallavi Sirjoosingh and Ashok Kakkar

First name (please write as legibly as possible within the boxes)

--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--

Last name

--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--

Student ID number

--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--

This booklet has 8 questions on 10 pages (The last page can be used as a scrap page).

The staple must NOT be removed from the question booklet. Your intact question booklet must be returned at the end of the exam.

A separate booklet contains a periodic table, and datasheet.

Write your answers directly under the questions in the space provided. Do not write on the QR codes or near the top edges (as it may be cut off).

Please write **legibly and dark enough** for the scan to be readable. For students who use English as a second language, dictionaries are permitted. Calculators are allowed.

DO NOT ASK QUESTIONS OF THE INVIGILATORS about interpretations of the exam questions. If you have a concern about a question, please summarize it at the top of this page. All concerns will be evaluated after the exam. GOOD LUCK!

**Question 1 8 points** (*Assume all transitions are possible*)

a. 6 points Calculate the **energy (in kJ/mol)** required to ionize (completely remove) the electron from the **$n=3$** level of **He^+ ion** (He: Helium). Show your work including equations. (3 sig. figs.)

b. 2 points

(I) Select the largest (longest) wavelength from the three choices below:

- i. Wavelength of photon to *completely remove* an electron from *1s orbital of Hydrogen atom*
- ii. Wavelength of photon to *completely remove* an electron from *1s orbital of Boron atom (B)*
- iii. Wavelength of photon to *completely remove* an electron from *1s orbital of Boron cation (B^+)*

i or ii or iii

(II) Select the smallest (shortest) wavelength from the three choices below (*for H atom*):

- i. Wavelength of photon *emitted* for an electron transitioning from *2p to 1s orbital*
- ii. Wavelength of photon *absorbed* for an electron transitioning from *2s to 4p orbital*
- iii. Wavelength of photon *absorbed* for an electron transitioning from *1s to 3p orbital*

i or ii or iii



Question 2 8 points



A Chem110 student plots the graph shown on the left, from data obtained for the kinetic energy (eV) of the ejected electron versus frequency (s^{-1}) of the incident light, for an unknown metal. ($1\text{eV} = 1.6022 \times 10^{-19} \text{ J}$)

Based on the plot, answer the questions **a to c** below:

a. 1 point On the plot above, label the **threshold frequency**.

b. 2 points What is the value of the work potential (in J) of the unknown metal?

c. 3 points What is the *longest* wavelength (in nm) required to eject an electron from this metal's surface? Show your work including equation(s) used. (3 sig figs)

d. 2 points In the space provided below, plot the following

(Assume frequency is *above* threshold frequency):

- i. *Kinetic energy* of an ejected electron versus the *intensity* of the incident light (left)
- ii. *Number of electrons* ejected versus the *frequency* of the incident light (right)



**Question 3** **6 points****a. 2 points**

In the ground state of Ge (Germanium; $Z = 32$) atom, **name** the following orbitals:

i. Orbital containing the electron that requires the maximum energy to be removed.

ii. Orbital containing the electron that experiences the minimum effective nuclear charge.

b. 2 points

An unknown anion (A^-) has the following *excited* electronic configuration: $[Ar] 4s^2 3d^3 4p^1$

(Ar: Argon; $Z = 18$)

Give the **condensed ground state electronic configuration** of the atom, **A** and the cation, **A^{2+}** .
Show your work. (Assume all electronic transitions are possible)

Electronic Configuration of A:

Electronic Configuration of A^{2+} :

c. 2 points

In a single atom, determine how many electrons can have the following set of quantum numbers?

(i) $n = 5; l = 3$

In a single atom, determine how many electrons can have the following set of quantum numbers?

(i) $n = 3; l = 2; m_s = +1/2$

**Question 4** **8 points**

Rank the following from 1 to 4 in **increasing** order. Provide the ranking in the boxes provided.

Note: Atomic number (Z) are provided for relevant atoms

a) Increasing (lowest to highest) order of *ionization energy*: (1= lowest; 4 = highest)

Mg

Mg: Z = 12

Rb

Rb: Z = 37

Cs

Cs: Z = 55

K

K : Z = 19

b) Increasing (lowest to highest) order of *bond angle*: (1= lowest; 4 = highest)

H₂OCCl₄XeF₄

HCN

c) Increasing (smallest to largest) order of *ionic size* (1= smallest; 4 = largest)

S²⁻

S: Z = 16

O²⁻

O: Z = 8

Be²⁺

Be: Z = 4

Na⁺

Na: Z = 11

d) Increasing (lowest to highest) order of *bond energy* (1= lowest; 4 = highest)

P-F

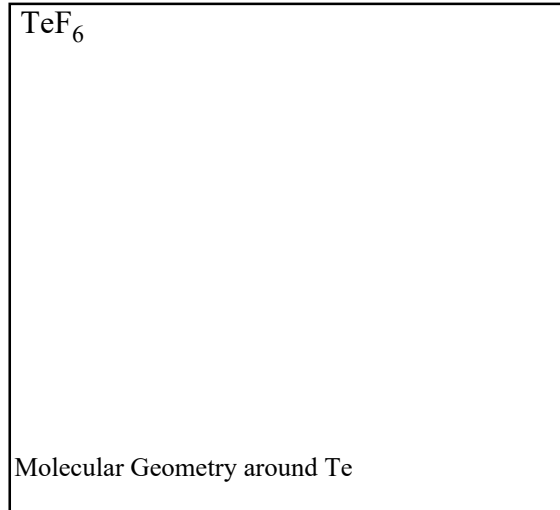
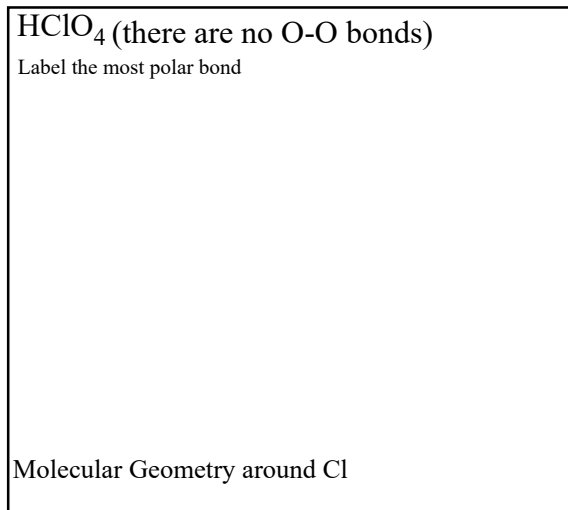
P-Cl

P-I

P-Br

**Question 5** **12 points**

a. 9 points For the molecules HClO_4 and TeF_6 , draw the most stable *Lewis structure* showing appropriate shape and *dashed/wedged bonds*, based on VSEPR. Determine the **molecular geometry** around the central atom. For HClO_4 , label the *most polar bond*.



b. 3 points (Note: Lone pairs are not shown in structures below)

(i) For the three *resonance* structures (1, 2 and 3) shown below, answer the following question:



(I) Circle the *most contributing* resonance structure.

Structure 1

Structure 2

Structure 3

(ii) For the three structures given (4, 5, and 6) below, answer the following questions:



(I) Circle all the structures that represent *resonance structures* of each other:

Structure 4

Structure 5

Structure 6

(II) Among the resonance structures, circle the *most contributing* resonance structure.

Structure 4

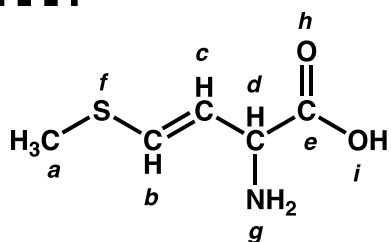
Structure 5

Structure 6

**Question 6** **6 points**

Using the data given below, calculate the **lattice energy** for SrO(s).

	ΔH° (kJ/mol)
$\text{Sr}(s) \longrightarrow \text{Sr}(g)$	+180
$\text{Sr}(g) \longrightarrow \text{Sr}^{2+}(g) + 2e^-$	+1468
$\text{O}_2(g) \longrightarrow 2\text{O}(g)$	+500
$\text{O}(g) + 2e^- \longrightarrow \text{O}^{2-}(g)$	+600
$\text{Sr}(s) + \frac{1}{2}\text{O}_2(g) \longrightarrow \text{SrO}(s)$	-550

**Question 7 10 points**

Consider the molecule given on the left, and answer questions (i) to (vi) using chemical bonding theories that we have learnt in class (valence bond theory, hybridization).

Note that the lone pairs are not shown in the structure:

i. Consider the carbon atoms marked in as ***a***, ***b***, ***c***, ***d***, and ***e*** in the molecule, and determine **hybridization** at each of the carbon centers. (2.5 points)

Atom	Hybridization
Carbon (<i>a</i>)	
Carbon (<i>b</i>)	
Carbon (<i>c</i>)	
Carbon (<i>d</i>)	
Carbon (<i>e</i>)	

ii. Now, consider the sulfur atom (S) marked as ***f***, nitrogen (N) atom (***g***), and oxygen(O) atoms (***h*** and ***i***), and determine their **hybridization**. (2 points)

Atom	Hybridization
Sulfur (<i>f</i>)	
Nitrogen (<i>g</i>)	
Oxygen (<i>h</i>)	
Oxygen (<i>i</i>)	

iii. **Describe** all the bonds formed by the carbon atoms ***a*** and ***b***, using sigma and pi designation as appropriate. (2 points)

Atom	Bonds formed (use sigma and pi as appropriate)
Carbon (<i>a</i>)	
Carbon (<i>b</i>)	

iv. How many total sp^3 -s sigma bonds are in the molecule? (1 point)

v. Determine **electronic** and **molecular geometry** at carbon (C) labeled ***e*** and sulfur (S) labeled ***f*** centres. (2 points)

Atom	Electronic Geometry	Molecular Geometry
Carbon (<i>e</i>)		
Sulfur (<i>f</i>)		

vi. Circle the value from the choices given below, that describes the **C-S-C bond angle at sulfur** (S) marked ***f***: (0.5 points)

>90° to <109.5° 109.5° >109.5 to <120° 120° >120° to <180°

**Question 8 12 points**

Using the **valence shell orbitals only** draw a **complete** molecular orbital (MO) diagram (with no s-p mixing) for the molecule **PS**, in the space provided below. **(6 points)**

Assume that 3s and 3p atomic orbitals will form molecular orbitals like those formed by the combination of 2s and 2p atomic orbitals that we learnt in class, and there is no s-p mixing.

Show **energy arrow**, **atomic orbitals** involved, **atom labels**, the **molecular orbitals** that are formed, **filled electrons** in the MO diagram etc. Make sure to label all the atomic and molecular orbitals.

Using the MO diagram above, answer the following questions:

- (i) Which one of the following will have the highest bond order? Circle the correct choice
A) PS^{1-} B) PS C) PS^{1+}
- (ii) Which one of the following will have the longest bond length? Circle the correct choice
A) PS^{1-} B) PS C) PS^{1+}
- (iii) Which of the following will be diamagnetic? Circle the correct choice.
A) PS^{1-} B) PS C) PS^{1+}
- (iv) Draw Lewis dot structure of the molecule with the shortest bond length (based on the MO diagram) from among the three (PS^{1-} , PS, PS^{1+}), and indicate the bond order calculated from the Lewis dot structure. **(3 points)**

Lewis Dot Structure

Bond Order



AB5BD668-C62A-4C28-82D2-188BC733700E

midterm-exam-93388

#92 Page 10 of 10

SCRAP PAGE

**CHEMISTRY 110 –Midterm Exam****Thursday November 7, 2024, 6:30 – 8:30 pm**

Examiners: Pallavi Sirjoosingh and Ashok Kakkar

First name (please write as legibly as possible within the boxes)

--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--

Last name

--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--

Student ID number

--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--

This booklet has 8 questions on 10 pages (The last page can be used as a scrap page).

The staple must NOT be removed from the question booklet. Your intact question booklet must be returned at the end of the exam.

A separate booklet contains a periodic table, and datasheet.

Write your answers directly under the questions in the space provided. Do not write on the QR codes or near the top edges (as it may be cut off).

Please write **legibly and dark enough** for the scan to be readable. For students who use English as a second language, dictionaries are permitted. Calculators are allowed.

DO NOT ASK QUESTIONS OF THE INVIGILATORS about interpretations of the exam questions. If you have a concern about a question, please summarize it at the top of this page. All concerns will be evaluated after the exam. GOOD LUCK!

**Question 1 8 points** (*Assume all transitions are possible*)

a. 6 points Calculate the **energy (in kJ/mol)** required to ionize (completely remove) the electron from the **$n=3$** level of **He^+ ion** (He: Helium). Show your work including equations. (3 sig. figs.)

b. 2 points

(I) Select the largest (longest) wavelength from the three choices below:

- i. Wavelength of photon to *completely remove* an electron from *1s orbital of Hydrogen atom*
- ii. Wavelength of photon to *completely remove* an electron from *1s orbital of Boron atom (B)*
- iii. Wavelength of photon to *completely remove* an electron from *1s orbital of Boron cation (B^+)*

i or ii or iii

(II) Select the smallest (shortest) wavelength from the three choices below (*for H atom*):

- i. Wavelength of photon *emitted* for an electron transitioning from *2p to 1s orbital*
- ii. Wavelength of photon *absorbed* for an electron transitioning from *2s to 4p orbital*
- iii. Wavelength of photon *absorbed* for an electron transitioning from *1s to 3p orbital*

i or ii or iii



Question 2 8 points



A Chem110 student plots the graph shown on the left, from data obtained for the kinetic energy (eV) of the ejected electron versus frequency (s^{-1}) of the incident light, for an unknown metal. ($1\text{eV} = 1.6022 \times 10^{-19} \text{ J}$)

Based on the plot, answer the questions **a to c** below:

a. 1 point On the plot above, label the **threshold frequency**.

b. 2 points What is the value of the work potential (in J) of the unknown metal?

c. 3 points What is the *longest* wavelength (in nm) required to eject an electron from this metal's surface? Show your work including equation(s) used. (3 sig figs)

d. 2 points In the space provided below, plot the following

(Assume frequency is *above* threshold frequency):

- Kinetic energy* of an ejected electron versus the *intensity* of the incident light (left)
- Number of electrons* ejected versus the *frequency* of the incident light (right)



**Question 3** **6 points****a. 2 points**

In the ground state of Ge (Germanium; $Z = 32$) atom, **name** the following orbitals:

i. Orbital containing the electron that requires the maximum energy to be removed.

ii. Orbital containing the electron that experiences the minimum effective nuclear charge.

b. 2 points

An unknown anion (A^-) has the following *excited* electronic configuration: $[Ar] 4s^2 3d^3 4p^1$

(Ar: Argon; $Z = 18$)

Give the **condensed ground state electronic configuration** of the atom, **A** and the cation, **A^{2+}** .
Show your work. (Assume all electronic transitions are possible)

Electronic Configuration of A:

Electronic Configuration of A^{2+} :

c. 2 points

In a single atom, determine how many electrons can have the following set of quantum numbers?

(i) $n = 5; l = 3$

In a single atom, determine how many electrons can have the following set of quantum numbers?

(i) $n = 3; l = 2; m_s = +1/2$

**Question 4** **8 points**

Rank the following from 1 to 4 in **increasing** order. Provide the ranking in the boxes provided.

Note: Atomic number (Z) are provided for relevant atoms

a) Increasing (lowest to highest) order of *ionization energy*: (1= lowest; 4 = highest)

Mg

Mg: Z = 12

Rb

Rb: Z = 37

Cs

Cs: Z = 55

K

K : Z = 19

b) Increasing (lowest to highest) order of *bond angle*: (1= lowest; 4 = highest)

H₂OCCl₄XeF₄

HCN

c) Increasing (smallest to largest) order of *ionic size* (1= smallest; 4 = largest)

S²⁻

S: Z = 16

O²⁻

O: Z = 8

Be²⁺

Be: Z = 4

Na⁺

Na: Z = 11

d) Increasing (lowest to highest) order of *bond energy* (1= lowest; 4 = highest)

P-F

P-Cl

P-I

P-Br

**Question 5** **12 points**

a. 9 points For the molecules HClO_4 and TeF_6 , draw the most stable *Lewis structure* showing appropriate shape and *dashed/wedged bonds*, based on VSEPR. Determine the **molecular geometry** around the central atom. For HClO_4 , label the *most polar bond*.



b. 3 points (Note: Lone pairs are not shown in structures below)

(i) For the three *resonance* structures (1, 2 and 3) shown below, answer the following question:



(I) Circle the *most contributing* resonance structure.

Structure 1

Structure 2

Structure 3

(ii) For the three structures given (4, 5, and 6) below, answer the following questions:



(I) Circle all the structures that represent *resonance structures* of each other:

Structure 4

Structure 5

Structure 6

(II) Among the resonance structures, circle the *most contributing* resonance structure.

Structure 4

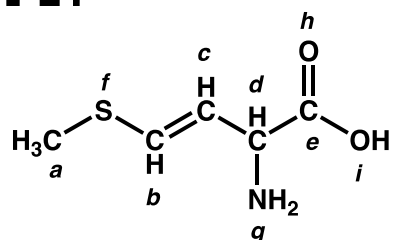
Structure 5

Structure 6

**Question 6** **6 points**

Using the data given below, calculate the **lattice energy** for SrO(s).

	ΔH° (kJ/mol)
$\text{Sr}(s) \longrightarrow \text{Sr}(g)$	+180
$\text{Sr}(g) \longrightarrow \text{Sr}^{2+}(g) + 2e^-$	+1468
$\text{O}_2(g) \longrightarrow 2\text{O}(g)$	+500
$\text{O}(g) + 2e^- \longrightarrow \text{O}^{2-}(g)$	+600
$\text{Sr}(s) + \frac{1}{2}\text{O}_2(g) \longrightarrow \text{SrO}(s)$	-550

**Question 7 10 points**

Consider the molecule given on the left, and answer questions (i) to (vi) using chemical bonding theories that we have learnt in class (valence bond theory, hybridization).

Note that the lone pairs are not shown in the structure:

i. Consider the carbon atoms marked in as ***a***, ***b***, ***c***, ***d***, and ***e*** in the molecule, and determine **hybridization** at each of the carbon centers. (2.5 points)

Atom	Hybridization
Carbon (<i>a</i>)	
Carbon (<i>b</i>)	
Carbon (<i>c</i>)	
Carbon (<i>d</i>)	
Carbon (<i>e</i>)	

ii. Now, consider the sulfur atom (S) marked as ***f***, nitrogen (N) atom (***g***), and oxygen(O) atoms (***h*** and ***i***), and determine their **hybridization**. (2 points)

Atom	Hybridization
Sulfur (<i>f</i>)	
Nitrogen (<i>g</i>)	
Oxygen (<i>h</i>)	
Oxygen (<i>i</i>)	

iii. **Describe** all the bonds formed by the carbon atoms ***a*** and ***b***, using sigma and pi designation as appropriate. (2 points)

Atom	Bonds formed (use sigma and pi as appropriate)
Carbon (<i>a</i>)	
Carbon (<i>b</i>)	

iv. How many total sp^3 -s sigma bonds are in the molecule? (1 point)

v. Determine **electronic** and **molecular geometry** at carbon (C) labeled ***e*** and sulfur (S) labeled ***f*** centres. (2 points)

Atom	Electronic Geometry	Molecular Geometry
Carbon (<i>e</i>)		
Sulfur (<i>f</i>)		

vi. Circle the value from the choices given below, that describes the **C-S-C bond angle at sulfur** (S) marked ***f***: (0.5 points)

>90° to <109.5°

109.5°

>109.5 to <120°

120°

>120° to <180°


Question 8 12 points

Using the **valence shell orbitals only** draw a **complete** molecular orbital (MO) diagram (with no s-p mixing) for the molecule **PS**, in the space provided below. **(6 points)**

Assume that 3s and 3p atomic orbitals will form molecular orbitals like those formed by the combination of 2s and 2p atomic orbitals that we learnt in class, and there is no s-p mixing.

Show **energy arrow**, **atomic orbitals** involved, **atom labels**, the **molecular orbitals** that are formed, **filled electrons** in the MO diagram etc. Make sure to label all the atomic and molecular orbitals.

Using the MO diagram above, answer the following questions:

- (i) Which one of the following will have the highest bond order? Circle the correct choice
 A) PS^{1-} B) PS C) PS^{1+}
- (ii) Which one of the following will have the longest bond length? Circle the correct choice
 A) PS^{1-} B) PS C) PS^{1+}
- (iii) Which of the following will be diamagnetic? Circle the correct choice.
 A) PS^{1-} B) PS C) PS^{1+}
- (iv) Draw Lewis dot structure of the molecule with the shortest bond length (based on the MO diagram) from among the three (PS^{1-} , PS, PS^{1+}), and indicate the bond order calculated from the Lewis dot structure. **(3 points)**

Lewis Dot Structure

Bond Order



7D9C0884-D620-4B5D-81E7-78F7C08DC376

midterm-exam-93388

#93 Page 10 of 10

SCRAP PAGE

**CHEMISTRY 110 –Midterm Exam****Thursday November 7, 2024, 6:30 – 8:30 pm**

Examiners: Pallavi Sirjoosingh and Ashok Kakkar

First name (please write as legibly as possible within the boxes)

--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--

Last name

--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--

Student ID number

--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--

This booklet has 8 questions on 10 pages (The last page can be used as a scrap page).

The staple must NOT be removed from the question booklet. Your intact question booklet must be returned at the end of the exam.

A separate booklet contains a periodic table, and datasheet.

Write your answers directly under the questions in the space provided. Do not write on the QR codes or near the top edges (as it may be cut off).

Please write **legibly and dark enough** for the scan to be readable. For students who use English as a second language, dictionaries are permitted. Calculators are allowed.

DO NOT ASK QUESTIONS OF THE INVIGILATORS about interpretations of the exam questions. If you have a concern about a question, please summarize it at the top of this page. All concerns will be evaluated after the exam. GOOD LUCK!

Question 1 **8 points** *(Assume all transitions are possible)*

a. 6 points Calculate the **energy (in kJ/mol)** required to ionize (completely remove) the electron from the **n=3** level of **He⁺ ion** (He: Helium). Show your work including equations. (3 sig. figs.)

b. 2 points

(I) Select the largest (longest) wavelength from the three choices below:

- Wavelength of photon to *completely remove* an electron from *1s orbital of Hydrogen atom*
- Wavelength of photon to *completely remove* an electron from *1s orbital of Boron atom (B)*
- Wavelength of photon to *completely remove* an electron from *1s orbital of Boron cation (B⁺)*

i or ii or iii

(II) Select the smallest (shortest) wavelength from the three choices below (*for H atom*):

- Wavelength of photon *emitted* for an electron transitioning from $2p$ to $1s$ orbital
- Wavelength of photon *absorbed* for an electron transitioning from $2s$ to $4p$ orbital
- Wavelength of photon *absorbed* for an electron transitioning from $1s$ to $3p$ orbital

i or ii or iii



Question 2 8 points



A Chem110 student plots the graph shown on the left, from data obtained for the kinetic energy (eV) of the ejected electron versus frequency (s^{-1}) of the incident light, for an unknown metal. ($1\text{eV} = 1.6022 \times 10^{-19} \text{ J}$)

Based on the plot, answer the questions **a to c** below:

a. 1 point On the plot above, label the **threshold frequency**.

b. 2 points What is the value of the work potential (in J) of the unknown metal?

c. 3 points What is the *longest* wavelength (in nm) required to eject an electron from this metal's surface? Show your work including equation(s) used. (3 sig figs)

d. 2 points In the space provided below, plot the following

(Assume frequency is *above* threshold frequency):

- i. *Kinetic energy* of an ejected electron versus the *intensity* of the incident light (left)
- ii. *Number of electrons* ejected versus the *frequency* of the incident light (right)



**Question 3** **6 points****a. 2 points**

In the ground state of Ge (Germanium; $Z = 32$) atom, **name** the following orbitals:

i. Orbital containing the electron that requires the maximum energy to be removed.

ii. Orbital containing the electron that experiences the minimum effective nuclear charge.

b. 2 points

An unknown anion (A^-) has the following *excited* electronic configuration: $[Ar] 4s^2 3d^3 4p^1$

(Ar: Argon; $Z = 18$)

Give the **condensed ground state electronic configuration** of the atom, **A** and the cation, **A^{2+}** .
Show your work. (Assume all electronic transitions are possible)

Electronic Configuration of A:

Electronic Configuration of A^{2+} :

c. 2 points

In a single atom, determine how many electrons can have the following set of quantum numbers?

(i) $n = 5; l = 3$

In a single atom, determine how many electrons can have the following set of quantum numbers?

(i) $n = 3; l = 2; m_s = +1/2$

**Question 4** **8 points**

Rank the following from 1 to 4 in **increasing** order. Provide the ranking in the boxes provided.

Note: Atomic number (Z) are provided for relevant atoms

a) Increasing (lowest to highest) order of *ionization energy*: (1= lowest; 4 = highest)

Mg

Mg: Z = 12

Rb

Rb: Z = 37

Cs

Cs: Z = 55

K

K : Z = 19

b) Increasing (lowest to highest) order of *bond angle*: (1= lowest; 4 = highest)

H₂OCCl₄XeF₄

HCN

c) Increasing (smallest to largest) order of *ionic size* (1= smallest; 4 = largest)

S²⁻

S: Z = 16

O²⁻

O: Z = 8

Be²⁺

Be: Z = 4

Na⁺

Na: Z = 11

d) Increasing (lowest to highest) order of *bond energy* (1= lowest; 4 = highest)

P-F

P-Cl

P-I

P-Br

**Question 5** **12 points**

a. 9 points For the molecules HClO_4 and TeF_6 , draw the most stable *Lewis structure* showing appropriate shape and *dashed/wedged bonds*, based on VSEPR. Determine the **molecular geometry** around the central atom. For HClO_4 , label the *most polar bond*.



b. 3 points (Note: Lone pairs are not shown in structures below)

(i) For the three *resonance* structures (1, 2 and 3) shown below, answer the following question:



(I) Circle the *most contributing* resonance structure.

Structure 1

Structure 2

Structure 3

(ii) For the three structures given (4, 5, and 6) below, answer the following questions:



(I) Circle all the structures that represent *resonance structures* of each other:

Structure 4

Structure 5

Structure 6

(II) Among the resonance structures, circle the *most contributing* resonance structure.

Structure 4

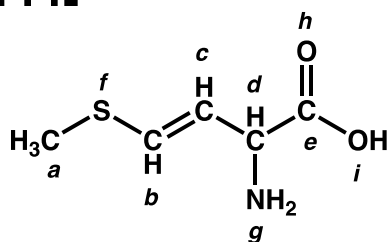
Structure 5

Structure 6

**Question 6** **6 points**

Using the data given below, calculate the **lattice energy** for SrO(s).

	ΔH° (kJ/mol)
$\text{Sr}(s) \longrightarrow \text{Sr}(g)$	+180
$\text{Sr}(g) \longrightarrow \text{Sr}^{2+}(g) + 2e^-$	+1468
$\text{O}_2(g) \longrightarrow 2\text{O}(g)$	+500
$\text{O}(g) + 2e^- \longrightarrow \text{O}^{2-}(g)$	+600
$\text{Sr}(s) + \frac{1}{2}\text{O}_2(g) \longrightarrow \text{SrO}(s)$	-550

**Question 7 10 points**

Consider the molecule given on the left, and answer questions (i) to (vi) using chemical bonding theories that we have learnt in class (valence bond theory, hybridization).

Note that the lone pairs are not shown in the structure:

i. Consider the carbon atoms marked in as ***a***, ***b***, ***c***, ***d***, and ***e*** in the molecule, and determine **hybridization** at each of the carbon centers. (2.5 points)

Atom	Hybridization
Carbon (<i>a</i>)	
Carbon (<i>b</i>)	
Carbon (<i>c</i>)	
Carbon (<i>d</i>)	
Carbon (<i>e</i>)	

ii. Now, consider the sulfur atom (S) marked as ***f***, nitrogen (N) atom (***g***), and oxygen(O) atoms (***h*** and ***i***), and determine their **hybridization**. (2 points)

Atom	Hybridization
Sulfur (<i>f</i>)	
Nitrogen (<i>g</i>)	
Oxygen (<i>h</i>)	
Oxygen (<i>i</i>)	

iii. **Describe** all the bonds formed by the carbon atoms ***a*** and ***b***, using sigma and pi designation as appropriate. (2 points)

Atom	Bonds formed (use sigma and pi as appropriate)
Carbon (<i>a</i>)	
Carbon (<i>b</i>)	

iv. How many total sp^3 -s sigma bonds are in the molecule? (1 point)

v. Determine **electronic** and **molecular geometry** at carbon (C) labeled ***e*** and sulfur (S) labeled ***f*** centres. (2 points)

Atom	Electronic Geometry	Molecular Geometry
Carbon (<i>e</i>)		
Sulfur (<i>f</i>)		

vi. Circle the value from the choices given below, that describes the **C-S-C bond angle at sulfur** (S) marked ***f***: (0.5 points)

>90° to <109.5° 109.5° >109.5 to <120° 120° >120° to <180°

**Question 8 12 points**

Using the **valence shell orbitals only** draw a **complete** molecular orbital (MO) diagram (with no s-p mixing) for the molecule **PS**, in the space provided below. **(6 points)**

Assume that 3s and 3p atomic orbitals will form molecular orbitals like those formed by the combination of 2s and 2p atomic orbitals that we learnt in class, and there is no s-p mixing.

Show **energy arrow**, **atomic orbitals** involved, **atom labels**, the **molecular orbitals** that are formed, **filled electrons** in the MO diagram etc. Make sure to label all the atomic and molecular orbitals.

Using the MO diagram above, answer the following questions:

- (i) Which one of the following will have the highest bond order? Circle the correct choice
A) PS^{1-} B) PS C) PS^{1+}
- (ii) Which one of the following will have the longest bond length? Circle the correct choice
A) PS^{1-} B) PS C) PS^{1+}
- (iii) Which of the following will be diamagnetic? Circle the correct choice.
A) PS^{1-} B) PS C) PS^{1+}
- (iv) Draw Lewis dot structure of the molecule with the shortest bond length (based on the MO diagram) from among the three (PS^{1-} , PS, PS^{1+}), and indicate the bond order calculated from the Lewis dot structure. **(3 points)**

Lewis Dot Structure

Bond Order



C9E6F471-94FF-46B9-AE69-4F168830290F

midterm-exam-93388

#94 Page 10 of 10

SCRAP PAGE

**CHEMISTRY 110 –Midterm Exam****Thursday November 7, 2024, 6:30 – 8:30 pm**

Examiners: Pallavi Sirjoosingh and Ashok Kakkar

First name (please write as legibly as possible within the boxes)

--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--

Last name

--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--

Student ID number

--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--

This booklet has 8 questions on 10 pages (The last page can be used as a scrap page).

The staple must NOT be removed from the question booklet. Your intact question booklet must be returned at the end of the exam.

A separate booklet contains a periodic table, and datasheet.

Write your answers directly under the questions in the space provided. Do not write on the QR codes or near the top edges (as it may be cut off).

Please write **legibly and dark enough** for the scan to be readable. For students who use English as a second language, dictionaries are permitted. Calculators are allowed.

DO NOT ASK QUESTIONS OF THE INVIGILATORS about interpretations of the exam questions. If you have a concern about a question, please summarize it at the top of this page. All concerns will be evaluated after the exam. GOOD LUCK!

**Question 1 8 points** (*Assume all transitions are possible*)

a. 6 points Calculate the **energy (in kJ/mol)** required to ionize (completely remove) the electron from the **n=3** level of **He⁺ ion** (He: Helium). Show your work including equations. (3 sig. figs.)

b. 2 points

(I) Select the largest (longest) wavelength from the three choices below:

- i. Wavelength of photon to *completely remove* an electron from *1s orbital of Hydrogen atom*
- ii. Wavelength of photon to *completely remove* an electron from *1s orbital of Boron atom (B)*
- iii. Wavelength of photon to *completely remove* an electron from *1s orbital of Boron cation (B⁺)*

i or ii or iii

(II) Select the smallest (shortest) wavelength from the three choices below (*for H atom*):

- i. Wavelength of photon *emitted* for an electron transitioning from *2p to 1s orbital*
- ii. Wavelength of photon *absorbed* for an electron transitioning from *2s to 4p orbital*
- iii. Wavelength of photon *absorbed* for an electron transitioning from *1s to 3p orbital*

i or ii or iii



Question 2 8 points



A Chem110 student plots the graph shown on the left, from data obtained for the kinetic energy (eV) of the ejected electron versus frequency (s^{-1}) of the incident light, for an unknown metal. ($1\text{eV} = 1.6022 \times 10^{-19} \text{ J}$)

Based on the plot, answer the questions **a to c** below:

a. 1 point On the plot above, label the **threshold frequency**.

b. 2 points What is the value of the work potential (in J) of the unknown metal?

c. 3 points What is the *longest* wavelength (in nm) required to eject an electron from this metal's surface? Show your work including equation(s) used. (3 sig figs)

d. 2 points In the space provided below, plot the following

(Assume frequency is *above* threshold frequency):

- Kinetic energy* of an ejected electron versus the *intensity* of the incident light (left)
- Number of electrons* ejected versus the *frequency* of the incident light (right)



**Question 3** **6 points****a. 2 points**

In the ground state of Ge (Germanium; $Z = 32$) atom, **name** the following orbitals:

i. Orbital containing the electron that requires the maximum energy to be removed.

ii. Orbital containing the electron that experiences the minimum effective nuclear charge.

b. 2 points

An unknown anion (A^-) has the following *excited* electronic configuration: $[Ar] 4s^2 3d^3 4p^1$

(Ar: Argon; $Z = 18$)

Give the **condensed ground state electronic configuration** of the atom, **A** and the cation, **A^{2+}** .
Show your work. (Assume all electronic transitions are possible)

Electronic Configuration of A:

Electronic Configuration of A^{2+} :

c. 2 points

In a single atom, determine how many electrons can have the following set of quantum numbers?

(i) $n = 5; l = 3$

In a single atom, determine how many electrons can have the following set of quantum numbers?

(i) $n = 3; l = 2; m_s = +1/2$

**Question 4** **8 points**

Rank the following from 1 to 4 in **increasing** order. Provide the ranking in the boxes provided.

Note: Atomic number (Z) are provided for relevant atoms

a) Increasing (lowest to highest) order of *ionization energy*: (1= lowest; 4 = highest)

Mg

Mg: Z = 12

Rb

Rb: Z = 37

Cs

Cs: Z = 55

K

K : Z = 19

b) Increasing (lowest to highest) order of *bond angle*: (1= lowest; 4 = highest)

H₂OCCl₄XeF₄

HCN

c) Increasing (smallest to largest) order of *ionic size* (1= smallest; 4 = largest)

S²⁻

S: Z = 16

O²⁻

O: Z = 8

Be²⁺

Be: Z = 4

Na⁺

Na: Z = 11

d) Increasing (lowest to highest) order of *bond energy* (1= lowest; 4 = highest)

P-F

P-Cl

P-I

P-Br

**Question 5** **12 points**

a. 9 points For the molecules HClO_4 and TeF_6 , draw the most stable *Lewis structure* showing appropriate shape and *dashed/wedged bonds*, based on VSEPR. Determine the **molecular geometry** around the central atom. For HClO_4 , label the *most polar bond*.



b. 3 points (Note: Lone pairs are not shown in structures below)

(i) For the three *resonance* structures (1, 2 and 3) shown below, answer the following question:



(I) Circle the *most contributing* resonance structure.

Structure 1

Structure 2

Structure 3

(ii) For the three structures given (4, 5, and 6) below, answer the following questions:



(I) Circle all the structures that represent *resonance structures* of each other:

Structure 4

Structure 5

Structure 6

(II) Among the resonance structures, circle the *most contributing* resonance structure.

Structure 4

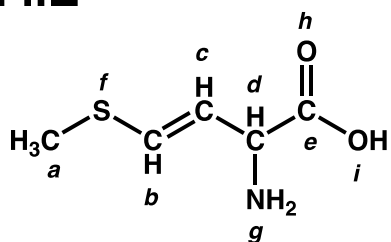
Structure 5

Structure 6

**Question 6** **6 points**

Using the data given below, calculate the **lattice energy** for SrO(s).

	ΔH° (kJ/mol)
$\text{Sr}(s) \longrightarrow \text{Sr}(g)$	+180
$\text{Sr}(g) \longrightarrow \text{Sr}^{2+}(g) + 2e^-$	+1468
$\text{O}_2(g) \longrightarrow 2\text{O}(g)$	+500
$\text{O}(g) + 2e^- \longrightarrow \text{O}^{2-}(g)$	+600
$\text{Sr}(s) + \frac{1}{2}\text{O}_2(g) \longrightarrow \text{SrO}(s)$	-550

**Question 7 10 points**

Consider the molecule given on the left, and answer questions (i) to (vi) using chemical bonding theories that we have learnt in class (valence bond theory, hybridization).

Note that the lone pairs are not shown in the structure:

i. Consider the carbon atoms marked in as ***a***, ***b***, ***c***, ***d***, and ***e*** in the molecule, and determine **hybridization** at each of the carbon centers. (2.5 points)

Atom	Hybridization
Carbon (<i>a</i>)	
Carbon (<i>b</i>)	
Carbon (<i>c</i>)	
Carbon (<i>d</i>)	
Carbon (<i>e</i>)	

ii. Now, consider the sulfur atom (S) marked as ***f***, nitrogen (N) atom (***g***), and oxygen(O) atoms (***h*** and ***i***), and determine their **hybridization**. (2 points)

Atom	Hybridization
Sulfur (<i>f</i>)	
Nitrogen (<i>g</i>)	
Oxygen (<i>h</i>)	
Oxygen (<i>i</i>)	

iii. **Describe** all the bonds formed by the carbon atoms ***a*** and ***b***, using sigma and pi designation as appropriate. (2 points)

Atom	Bonds formed (use sigma and pi as appropriate)
Carbon (<i>a</i>)	
Carbon (<i>b</i>)	

iv. How many total sp^3 -s sigma bonds are in the molecule? (1 point)

v. Determine **electronic** and **molecular geometry** at carbon (C) labeled ***e*** and sulfur (S) labeled ***f*** centres. (2 points)

Atom	Electronic Geometry	Molecular Geometry
Carbon (<i>e</i>)		
Sulfur (<i>f</i>)		

vi. Circle the value from the choices given below, that describes the **C-S-C bond angle at sulfur** (S) marked ***f***: (0.5 points)

>90° to <109.5° 109.5° >109.5 to <120° 120° >120° to <180°


Question 8 12 points

Using the **valence shell orbitals only** draw a **complete** molecular orbital (MO) diagram (with no s-p mixing) for the molecule **PS**, in the space provided below. **(6 points)**

Assume that 3s and 3p atomic orbitals will form molecular orbitals like those formed by the combination of 2s and 2p atomic orbitals that we learnt in class, and there is no s-p mixing.

Show **energy arrow**, **atomic orbitals** involved, **atom labels**, the **molecular orbitals** that are formed, **filled electrons** in the MO diagram etc. Make sure to label all the atomic and molecular orbitals.

Using the MO diagram above, answer the following questions:

- (i) Which one of the following will have the highest bond order? Circle the correct choice
 A) PS^{1-} B) PS C) PS^{1+}
- (ii) Which one of the following will have the longest bond length? Circle the correct choice
 A) PS^{1-} B) PS C) PS^{1+}
- (iii) Which of the following will be diamagnetic? Circle the correct choice.
 A) PS^{1-} B) PS C) PS^{1+}
- (iv) Draw Lewis dot structure of the molecule with the shortest bond length (based on the MO diagram) from among the three (PS^{1-} , PS, PS^{1+}), and indicate the bond order calculated from the Lewis dot structure. **(3 points)**

Lewis Dot Structure

Bond Order



2E86BBE4-37CD-4472-A306-C030A9AF7970

midterm-exam-93388

#95 Page 10 of 10

SCRAP PAGE

**CHEMISTRY 110 –Midterm Exam****Thursday November 7, 2024, 6:30 – 8:30 pm**

Examiners: Pallavi Sirjoosingh and Ashok Kakkar

First name (please write as legibly as possible within the boxes)

--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--

Last name

--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--

Student ID number

--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--

This booklet has 8 questions on 10 pages (The last page can be used as a scrap page).

The staple must NOT be removed from the question booklet. Your intact question booklet must be returned at the end of the exam.

A separate booklet contains a periodic table, and datasheet.

Write your answers directly under the questions in the space provided. Do not write on the QR codes or near the top edges (as it may be cut off).

Please write **legibly and dark enough** for the scan to be readable. For students who use English as a second language, dictionaries are permitted. Calculators are allowed.

DO NOT ASK QUESTIONS OF THE INVIGILATORS about interpretations of the exam questions. If you have a concern about a question, please summarize it at the top of this page. All concerns will be evaluated after the exam. GOOD LUCK!

**Question 1 8 points** (*Assume all transitions are possible*)

a. 6 points Calculate the **energy (in kJ/mol)** required to ionize (completely remove) the electron from the **n=3** level of **He⁺ ion** (He: Helium). Show your work including equations. (3 sig. figs.)

b. 2 points

(I) Select the largest (longest) wavelength from the three choices below:

- i. Wavelength of photon to *completely remove* an electron from *1s orbital of Hydrogen atom*
- ii. Wavelength of photon to *completely remove* an electron from *1s orbital of Boron atom (B)*
- iii. Wavelength of photon to *completely remove* an electron from *1s orbital of Boron cation (B⁺)*

i or ii or iii

(II) Select the smallest (shortest) wavelength from the three choices below (*for H atom*):

- i. Wavelength of photon *emitted* for an electron transitioning from *2p to 1s orbital*
- ii. Wavelength of photon *absorbed* for an electron transitioning from *2s to 4p orbital*
- iii. Wavelength of photon *absorbed* for an electron transitioning from *1s to 3p orbital*

i or ii or iii



Question 2 8 points



A Chem110 student plots the graph shown on the left, from data obtained for the kinetic energy (eV) of the ejected electron versus frequency (s^{-1}) of the incident light, for an unknown metal. ($1\text{eV} = 1.6022 \times 10^{-19} \text{ J}$)

Based on the plot, answer the questions **a to c** below:

a. 1 point On the plot above, label the **threshold frequency**.

b. 2 points What is the value of the work potential (in J) of the unknown metal?

c. 3 points What is the *longest* wavelength (in nm) required to eject an electron from this metal's surface? Show your work including equation(s) used. (3 sig figs)

d. 2 points In the space provided below, plot the following

(Assume frequency is *above* threshold frequency):

- i. *Kinetic energy* of an ejected electron versus the *intensity* of the incident light (left)
- ii. *Number of electrons* ejected versus the *frequency* of the incident light (right)



**Question 3** **6 points****a. 2 points**

In the ground state of Ge (Germanium; $Z = 32$) atom, **name** the following orbitals:

i. Orbital containing the electron that requires the maximum energy to be removed.

ii. Orbital containing the electron that experiences the minimum effective nuclear charge.

b. 2 points

An unknown anion (A^-) has the following *excited* electronic configuration: $[Ar] 4s^2 3d^3 4p^1$

(Ar: Argon; $Z = 18$)

Give the **condensed ground state electronic configuration** of the atom, **A** and the cation, **A^{2+}** .
Show your work. (Assume all electronic transitions are possible)

Electronic Configuration of A:

Electronic Configuration of A^{2+} :

c. 2 points

In a single atom, determine how many electrons can have the following set of quantum numbers?

(i) $n = 5; l = 3$

In a single atom, determine how many electrons can have the following set of quantum numbers?

(i) $n = 3; l = 2; m_s = +1/2$

**Question 4** **8 points**

Rank the following from 1 to 4 in **increasing** order. Provide the ranking in the boxes provided.

Note: Atomic number (Z) are provided for relevant atoms

a) Increasing (lowest to highest) order of *ionization energy*: (1= lowest; 4 = highest)

Mg

Mg: Z = 12

Rb

Rb: Z = 37

Cs

Cs: Z = 55

K

K : Z = 19

b) Increasing (lowest to highest) order of *bond angle*: (1= lowest; 4 = highest)

H₂OCCl₄XeF₄

HCN

c) Increasing (smallest to largest) order of *ionic size* (1= smallest; 4 = largest)

S²⁻

S: Z = 16

O²⁻

O: Z = 8

Be²⁺

Be: Z = 4

Na⁺

Na: Z = 11

d) Increasing (lowest to highest) order of *bond energy* (1= lowest; 4 = highest)

P-F

P-Cl

P-I

P-Br

**Question 5** **12 points**

a. 9 points For the molecules HClO_4 and TeF_6 , draw the most stable *Lewis structure* showing appropriate shape and *dashed/wedged bonds*, based on VSEPR. Determine the **molecular geometry** around the central atom. For HClO_4 , label the *most polar bond*.



b. 3 points (Note: Lone pairs are not shown in structures below)

(i) For the three *resonance* structures (1, 2 and 3) shown below, answer the following question:



(I) Circle the *most contributing* resonance structure.

Structure 1

Structure 2

Structure 3

(ii) For the three structures given (4, 5, and 6) below, answer the following questions:



(I) Circle all the structures that represent *resonance structures* of each other:

Structure 4

Structure 5

Structure 6

(II) Among the resonance structures, circle the *most contributing* resonance structure.

Structure 4

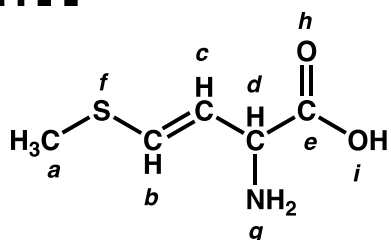
Structure 5

Structure 6

**Question 6** **6 points**

Using the data given below, calculate the **lattice energy** for SrO(s).

	ΔH° (kJ/mol)
$\text{Sr}(s) \longrightarrow \text{Sr}(g)$	+180
$\text{Sr}(g) \longrightarrow \text{Sr}^{2+}(g) + 2e^-$	+1468
$\text{O}_2(g) \longrightarrow 2\text{O}(g)$	+500
$\text{O}(g) + 2e^- \longrightarrow \text{O}^{2-}(g)$	+600
$\text{Sr}(s) + \frac{1}{2}\text{O}_2(g) \longrightarrow \text{SrO}(s)$	-550

**Question 7 10 points**

Consider the molecule given on the left, and answer questions (i) to (vi) using chemical bonding theories that we have learnt in class (valence bond theory, hybridization).

Note that the lone pairs are not shown in the structure:

i. Consider the carbon atoms marked in as ***a***, ***b***, ***c***, ***d***, and ***e*** in the molecule, and determine **hybridization** at each of the carbon centers. (2.5 points)

Atom	Hybridization
Carbon (<i>a</i>)	
Carbon (<i>b</i>)	
Carbon (<i>c</i>)	
Carbon (<i>d</i>)	
Carbon (<i>e</i>)	

ii. Now, consider the sulfur atom (S) marked as ***f***, nitrogen (N) atom (***g***), and oxygen(O) atoms (***h*** and ***i***), and determine their **hybridization**. (2 points)

Atom	Hybridization
Sulfur (<i>f</i>)	
Nitrogen (<i>g</i>)	
Oxygen (<i>h</i>)	
Oxygen (<i>i</i>)	

iii. **Describe** all the bonds formed by the carbon atoms ***a*** and ***b***, using sigma and pi designation as appropriate. (2 points)

Atom	Bonds formed (use sigma and pi as appropriate)
Carbon (<i>a</i>)	
Carbon (<i>b</i>)	

iv. How many total sp^3 -s sigma bonds are in the molecule? (1 point)

v. Determine **electronic** and **molecular geometry** at carbon (C) labeled ***e*** and sulfur (S) labeled ***f*** centres. (2 points)

Atom	Electronic Geometry	Molecular Geometry
Carbon (<i>e</i>)		
Sulfur (<i>f</i>)		

vi. Circle the value from the choices given below, that describes the **C-S-C bond angle at sulfur** (S) marked ***f***: (0.5 points)

>90° to <109.5° 109.5° >109.5 to <120° 120° >120° to <180°

**Question 8 12 points**

Using the **valence shell orbitals only** draw a **complete** molecular orbital (MO) diagram (with no s-p mixing) for the molecule **PS**, in the space provided below. **(6 points)**

Assume that 3s and 3p atomic orbitals will form molecular orbitals like those formed by the combination of 2s and 2p atomic orbitals that we learnt in class, and there is no s-p mixing.

Show **energy arrow**, **atomic orbitals** involved, **atom labels**, the **molecular orbitals** that are formed, **filled electrons** in the MO diagram etc. Make sure to label all the atomic and molecular orbitals.

Using the MO diagram above, answer the following questions:

- (i) Which one of the following will have the highest bond order? Circle the correct choice
A) PS^{1-} B) PS C) PS^{1+}
- (ii) Which one of the following will have the longest bond length? Circle the correct choice
A) PS^{1-} B) PS C) PS^{1+}
- (iii) Which of the following will be diamagnetic? Circle the correct choice.
A) PS^{1-} B) PS C) PS^{1+}
- (iv) Draw Lewis dot structure of the molecule with the shortest bond length (based on the MO diagram) from among the three (PS^{1-} , PS, PS^{1+}), and indicate the bond order calculated from the Lewis dot structure. **(3 points)**

Lewis Dot Structure

Bond Order



688A7D3F-8631-4CE9-936E-9DCD1956B3CB

midterm-exam-93388

#96 Page 10 of 10

SCRAP PAGE

**CHEMISTRY 110 –Midterm Exam****Thursday November 7, 2024, 6:30 – 8:30 pm**

Examiners: Pallavi Sirjoosingh and Ashok Kakkar

First name (please write as legibly as possible within the boxes)

--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--

Last name

--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--

Student ID number

--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--

This booklet has 8 questions on 10 pages (The last page can be used as a scrap page).

The staple must NOT be removed from the question booklet. Your intact question booklet must be returned at the end of the exam.

A separate booklet contains a periodic table, and datasheet.

Write your answers directly under the questions in the space provided. Do not write on the QR codes or near the top edges (as it may be cut off).

Please write **legibly and dark enough** for the scan to be readable. For students who use English as a second language, dictionaries are permitted. Calculators are allowed.

DO NOT ASK QUESTIONS OF THE INVIGILATORS about interpretations of the exam questions. If you have a concern about a question, please summarize it at the top of this page. All concerns will be evaluated after the exam. GOOD LUCK!

**Question 1 8 points** (*Assume all transitions are possible*)

a. 6 points Calculate the **energy (in kJ/mol)** required to ionize (completely remove) the electron from the **n=3** level of **He⁺ ion** (He: Helium). Show your work including equations. (3 sig. figs.)

b. 2 points

(I) Select the largest (longest) wavelength from the three choices below:

- i. Wavelength of photon to *completely remove* an electron from *1s orbital of Hydrogen atom*
- ii. Wavelength of photon to *completely remove* an electron from *1s orbital of Boron atom (B)*
- iii. Wavelength of photon to *completely remove* an electron from *1s orbital of Boron cation (B⁺)*

i or ii or iii

(II) Select the smallest (shortest) wavelength from the three choices below (*for H atom*):

- i. Wavelength of photon *emitted* for an electron transitioning from *2p to 1s orbital*
- ii. Wavelength of photon *absorbed* for an electron transitioning from *2s to 4p orbital*
- iii. Wavelength of photon *absorbed* for an electron transitioning from *1s to 3p orbital*

i or ii or iii



Question 2 8 points



A Chem110 student plots the graph shown on the left, from data obtained for the kinetic energy (eV) of the ejected electron versus frequency (s^{-1}) of the incident light, for an unknown metal. ($1\text{eV} = 1.6022 \times 10^{-19} \text{ J}$)

Based on the plot, answer the questions **a to c** below:

a. 1 point On the plot above, label the **threshold frequency**.

b. 2 points What is the value of the work potential (in J) of the unknown metal?

c. 3 points What is the *longest* wavelength (in nm) required to eject an electron from this metal's surface? Show your work including equation(s) used. (3 sig figs)

d. 2 points In the space provided below, plot the following

(Assume frequency is *above* threshold frequency):

- Kinetic energy* of an ejected electron versus the *intensity* of the incident light (left)
- Number of electrons* ejected versus the *frequency* of the incident light (right)



**Question 3** **6 points****a. 2 points**

In the ground state of Ge (Germanium; $Z = 32$) atom, **name** the following orbitals:

i. Orbital containing the electron that requires the maximum energy to be removed.

ii. Orbital containing the electron that experiences the minimum effective nuclear charge.

b. 2 points

An unknown anion (A^-) has the following *excited* electronic configuration: $[Ar] 4s^2 3d^3 4p^1$

(Ar: Argon; $Z = 18$)

Give the **condensed ground state electronic configuration** of the atom, **A** and the cation, **A^{2+}** .
Show your work. (Assume all electronic transitions are possible)

Electronic Configuration of A:

Electronic Configuration of A^{2+} :

c. 2 points

In a single atom, determine how many electrons can have the following set of quantum numbers?

(i) $n = 5; l = 3$

In a single atom, determine how many electrons can have the following set of quantum numbers?

(i) $n = 3; l = 2; m_s = +1/2$

**Question 4** **8 points**

Rank the following from 1 to 4 in **increasing** order. Provide the ranking in the boxes provided.

Note: Atomic number (Z) are provided for relevant atoms

a) Increasing (lowest to highest) order of *ionization energy*: (1= lowest; 4 = highest)

Mg

Mg: Z = 12

Rb

Rb: Z = 37

Cs

Cs: Z = 55

K

K : Z = 19

b) Increasing (lowest to highest) order of *bond angle*: (1= lowest; 4 = highest)

H₂OCCl₄XeF₄

HCN

c) Increasing (smallest to largest) order of *ionic size* (1= smallest; 4 = largest)

S²⁻

S: Z = 16

O²⁻

O: Z = 8

Be²⁺

Be: Z = 4

Na⁺

Na: Z = 11

d) Increasing (lowest to highest) order of *bond energy* (1= lowest; 4 = highest)

P-F

P-Cl

P-I

P-Br

**Question 5** **12 points**

a. 9 points For the molecules HClO_4 and TeF_6 , draw the most stable *Lewis structure* showing appropriate shape and *dashed/wedged bonds*, based on VSEPR. Determine the **molecular geometry** around the central atom. For HClO_4 , label the *most polar bond*.



b. 3 points (Note: Lone pairs are not shown in structures below)

(i) For the three *resonance* structures (1, 2 and 3) shown below, answer the following question:



(I) Circle the *most contributing* resonance structure.

Structure 1

Structure 2

Structure 3

(ii) For the three structures given (4, 5, and 6) below, answer the following questions:



(I) Circle all the structures that represent *resonance structures* of each other:

Structure 4

Structure 5

Structure 6

(II) Among the resonance structures, circle the *most contributing* resonance structure.

Structure 4

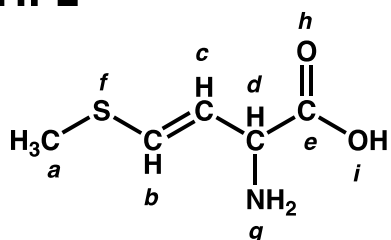
Structure 5

Structure 6

**Question 6** **6 points**

Using the data given below, calculate the **lattice energy** for SrO(s).

	ΔH° (kJ/mol)
$\text{Sr}(s) \longrightarrow \text{Sr}(g)$	+180
$\text{Sr}(g) \longrightarrow \text{Sr}^{2+}(g) + 2e^-$	+1468
$\text{O}_2(g) \longrightarrow 2\text{O}(g)$	+500
$\text{O}(g) + 2e^- \longrightarrow \text{O}^{2-}(g)$	+600
$\text{Sr}(s) + \frac{1}{2}\text{O}_2(g) \longrightarrow \text{SrO}(s)$	-550

**Question 7 10 points**

Consider the molecule given on the left, and answer questions (i) to (vi) using chemical bonding theories that we have learnt in class (valence bond theory, hybridization).

Note that the lone pairs are not shown in the structure:

i. Consider the carbon atoms marked in as ***a***, ***b***, ***c***, ***d***, and ***e*** in the molecule, and determine **hybridization** at each of the carbon centers. (2.5 points)

Atom	Hybridization
Carbon (<i>a</i>)	
Carbon (<i>b</i>)	
Carbon (<i>c</i>)	
Carbon (<i>d</i>)	
Carbon (<i>e</i>)	

ii. Now, consider the sulfur atom (S) marked as ***f***, nitrogen (N) atom (***g***), and oxygen(O) atoms (***h*** and ***i***), and determine their **hybridization**. (2 points)

Atom	Hybridization
Sulfur (<i>f</i>)	
Nitrogen (<i>g</i>)	
Oxygen (<i>h</i>)	
Oxygen (<i>i</i>)	

iii. **Describe** all the bonds formed by the carbon atoms ***a*** and ***b***, using sigma and pi designation as appropriate. (2 points)

Atom	Bonds formed (use sigma and pi as appropriate)
Carbon (<i>a</i>)	
Carbon (<i>b</i>)	

iv. How many total sp^3 -s sigma bonds are in the molecule? (1 point)

v. Determine **electronic** and **molecular geometry** at carbon (C) labeled ***e*** and sulfur (S) labeled ***f*** centres. (2 points)

Atom	Electronic Geometry	Molecular Geometry
Carbon (<i>e</i>)		
Sulfur (<i>f</i>)		

vi. Circle the value from the choices given below, that describes the **C-S-C bond angle at sulfur** (S) marked ***f***: (0.5 points)

>90° to <109.5° 109.5° >109.5 to <120° 120° >120° to <180°


Question 8 12 points

Using the **valence shell orbitals only** draw a **complete** molecular orbital (MO) diagram (with no s-p mixing) for the molecule **PS**, in the space provided below. **(6 points)**

Assume that 3s and 3p atomic orbitals will form molecular orbitals like those formed by the combination of 2s and 2p atomic orbitals that we learnt in class, and there is no s-p mixing.

Show **energy arrow**, **atomic orbitals** involved, **atom labels**, the **molecular orbitals** that are formed, **filled electrons** in the MO diagram etc. Make sure to label all the atomic and molecular orbitals.

Using the MO diagram above, answer the following questions:

- (i) Which one of the following will have the highest bond order? Circle the correct choice
 A) PS^{1-} B) PS C) PS^{1+}
- (ii) Which one of the following will have the longest bond length? Circle the correct choice
 A) PS^{1-} B) PS C) PS^{1+}
- (iii) Which of the following will be diamagnetic? Circle the correct choice.
 A) PS^{1-} B) PS C) PS^{1+}
- (iv) Draw Lewis dot structure of the molecule with the shortest bond length (based on the MO diagram) from among the three (PS^{1-} , PS, PS^{1+}), and indicate the bond order calculated from the Lewis dot structure. **(3 points)**

Lewis Dot Structure

Bond Order



867D4476-D836-4B86-8750-8F5C5D80AE31

midterm-exam-93388

#97 Page 10 of 10

SCRAP PAGE

**CHEMISTRY 110 –Midterm Exam****Thursday November 7, 2024, 6:30 – 8:30 pm**

Examiners: Pallavi Sirjoosingh and Ashok Kakkar

First name (please write as legibly as possible within the boxes)

--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--

Last name

--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--

Student ID number

--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--

This booklet has 8 questions on 10 pages (The last page can be used as a scrap page).

The staple must NOT be removed from the question booklet. Your intact question booklet must be returned at the end of the exam.

A separate booklet contains a periodic table, and datasheet.

Write your answers directly under the questions in the space provided. Do not write on the QR codes or near the top edges (as it may be cut off).

Please write **legibly and dark enough** for the scan to be readable. For students who use English as a second language, dictionaries are permitted. Calculators are allowed.

DO NOT ASK QUESTIONS OF THE INVIGILATORS about interpretations of the exam questions. If you have a concern about a question, please summarize it at the top of this page. All concerns will be evaluated after the exam. GOOD LUCK!

**Question 1 8 points** (*Assume all transitions are possible*)

a. 6 points Calculate the **energy (in kJ/mol)** required to ionize (completely remove) the electron from the **n=3** level of **He⁺ ion** (He: Helium). Show your work including equations. (3 sig. figs.)

b. 2 points

(I) Select the largest (longest) wavelength from the three choices below:

- i. Wavelength of photon to *completely remove* an electron from *1s orbital of Hydrogen atom*
- ii. Wavelength of photon to *completely remove* an electron from *1s orbital of Boron atom (B)*
- iii. Wavelength of photon to *completely remove* an electron from *1s orbital of Boron cation (B⁺)*

i or ii or iii

(II) Select the smallest (shortest) wavelength from the three choices below (*for H atom*):

- i. Wavelength of photon *emitted* for an electron transitioning from *2p to 1s orbital*
- ii. Wavelength of photon *absorbed* for an electron transitioning from *2s to 4p orbital*
- iii. Wavelength of photon *absorbed* for an electron transitioning from *1s to 3p orbital*

i or ii or iii



Question 2 8 points



A Chem110 student plots the graph shown on the left, from data obtained for the kinetic energy (eV) of the ejected electron versus frequency (s^{-1}) of the incident light, for an unknown metal. ($1\text{eV} = 1.6022 \times 10^{-19} \text{ J}$)

Based on the plot, answer the questions **a to c** below:

a. 1 point On the plot above, label the **threshold frequency**.

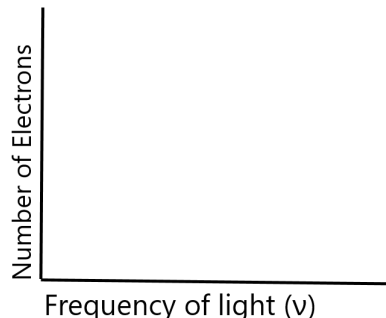
b. 2 points What is the value of the work potential (in J) of the unknown metal?

c. 3 points What is the *longest* wavelength (in nm) required to eject an electron from this metal's surface? Show your work including equation(s) used. (3 sig figs)

d. 2 points In the space provided below, plot the following

(Assume frequency is *above* threshold frequency):

- i. *Kinetic energy* of an ejected electron versus the *intensity* of the incident light (left)
- ii. *Number of electrons* ejected versus the *frequency* of the incident light (right)



**Question 3** **6 points****a. 2 points**

In the ground state of Ge (Germanium; $Z = 32$) atom, **name** the following orbitals:

i. Orbital containing the electron that requires the maximum energy to be removed.

ii. Orbital containing the electron that experiences the minimum effective nuclear charge.

b. 2 points

An unknown anion (A^-) has the following *excited* electronic configuration: $[Ar] 4s^2 3d^3 4p^1$

(Ar: Argon; $Z = 18$)

Give the **condensed ground state electronic configuration** of the atom, **A** and the cation, **A^{2+}** .
Show your work. (Assume all electronic transitions are possible)

Electronic Configuration of A:

Electronic Configuration of A^{2+} :

c. 2 points

In a single atom, determine how many electrons can have the following set of quantum numbers?

(i) $n = 5; l = 3$

In a single atom, determine how many electrons can have the following set of quantum numbers?

(i) $n = 3; l = 2; m_s = +1/2$

**Question 4** **8 points**

Rank the following from 1 to 4 in **increasing** order. Provide the ranking in the boxes provided.

Note: Atomic number (Z) are provided for relevant atoms

a) Increasing (lowest to highest) order of *ionization energy*: (1= lowest; 4 = highest)

Mg

Mg: Z = 12

Rb

Rb: Z = 37

Cs

Cs: Z = 55

K

K : Z = 19

b) Increasing (lowest to highest) order of *bond angle*: (1= lowest; 4 = highest)

H₂OCCl₄XeF₄

HCN

c) Increasing (smallest to largest) order of *ionic size* (1= smallest; 4 = largest)

S²⁻

S: Z = 16

O²⁻

O: Z = 8

Be²⁺

Be: Z = 4

Na⁺

Na: Z = 11

d) Increasing (lowest to highest) order of *bond energy* (1= lowest; 4 = highest)

P-F

P-Cl

P-I

P-Br

**Question 5** **12 points**

a. 9 points For the molecules HClO_4 and TeF_6 , draw the most stable *Lewis structure* showing appropriate shape and *dashed/wedged bonds*, based on VSEPR. Determine the **molecular geometry** around the central atom. For HClO_4 , label the *most polar bond*.



b. 3 points (Note: Lone pairs are not shown in structures below)

(i) For the three *resonance* structures (1, 2 and 3) shown below, answer the following question:



(I) Circle the *most contributing* resonance structure.

Structure 1

Structure 2

Structure 3

(ii) For the three structures given (4, 5, and 6) below, answer the following questions:



(I) Circle all the structures that represent *resonance structures* of each other:

Structure 4

Structure 5

Structure 6

(II) Among the resonance structures, circle the *most contributing* resonance structure.

Structure 4

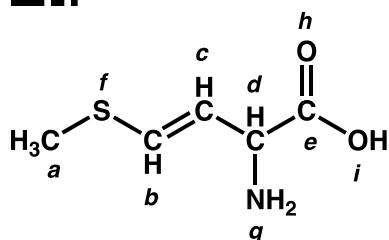
Structure 5

Structure 6

**Question 6** **6 points**

Using the data given below, calculate the **lattice energy** for SrO(s).

	ΔH° (kJ/mol)
$\text{Sr}(s) \longrightarrow \text{Sr}(g)$	+180
$\text{Sr}(g) \longrightarrow \text{Sr}^{2+}(g) + 2e^-$	+1468
$\text{O}_2(g) \longrightarrow 2\text{O}(g)$	+500
$\text{O}(g) + 2e^- \longrightarrow \text{O}^{2-}(g)$	+600
$\text{Sr}(s) + \frac{1}{2}\text{O}_2(g) \longrightarrow \text{SrO}(s)$	-550

**Question 7 10 points**

Consider the molecule given on the left, and answer questions (i) to (vi) using chemical bonding theories that we have learnt in class (valence bond theory, hybridization).

Note that the lone pairs are not shown in the structure:

i. Consider the carbon atoms marked in as ***a***, ***b***, ***c***, ***d***, and ***e*** in the molecule, and determine **hybridization** at each of the carbon centers. (2.5 points)

Atom	Hybridization
Carbon (<i>a</i>)	
Carbon (<i>b</i>)	
Carbon (<i>c</i>)	
Carbon (<i>d</i>)	
Carbon (<i>e</i>)	

ii. Now, consider the sulfur atom (S) marked as ***f***, nitrogen (N) atom (***g***), and oxygen(O) atoms (***h*** and ***i***), and determine their **hybridization**. (2 points)

Atom	Hybridization
Sulfur (<i>f</i>)	
Nitrogen (<i>g</i>)	
Oxygen (<i>h</i>)	
Oxygen (<i>i</i>)	

iii. **Describe** all the bonds formed by the carbon atoms ***a*** and ***b***, using sigma and pi designation as appropriate. (2 points)

Atom	Bonds formed (use sigma and pi as appropriate)
Carbon (<i>a</i>)	
Carbon (<i>b</i>)	

iv. How many total sp^3 -s sigma bonds are in the molecule? (1 point)

v. Determine **electronic** and **molecular geometry** at carbon (C) labeled ***e*** and sulfur (S) labeled ***f*** centres. (2 points)

Atom	Electronic Geometry	Molecular Geometry
Carbon (<i>e</i>)		
Sulfur (<i>f</i>)		

vi. Circle the value from the choices given below, that describes the **C-S-C bond angle at sulfur** (S) marked ***f***: (0.5 points)

>90° to <109.5° 109.5° >109.5 to <120° 120° >120° to <180°


Question 8 12 points

Using the **valence shell orbitals only** draw a **complete** molecular orbital (MO) diagram (with no s-p mixing) for the molecule **PS**, in the space provided below. **(6 points)**

Assume that 3s and 3p atomic orbitals will form molecular orbitals like those formed by the combination of 2s and 2p atomic orbitals that we learnt in class, and there is no s-p mixing.

Show **energy arrow**, **atomic orbitals** involved, **atom labels**, the **molecular orbitals** that are formed, **filled electrons** in the MO diagram etc. Make sure to label all the atomic and molecular orbitals.

Using the MO diagram above, answer the following questions:

- (i) Which one of the following will have the highest bond order? Circle the correct choice
 A) PS^{1-} B) PS C) PS^{1+}
- (ii) Which one of the following will have the longest bond length? Circle the correct choice
 A) PS^{1-} B) PS C) PS^{1+}
- (iii) Which of the following will be diamagnetic? Circle the correct choice.
 A) PS^{1-} B) PS C) PS^{1+}
- (iv) Draw Lewis dot structure of the molecule with the shortest bond length (based on the MO diagram) from among the three (PS^{1-} , PS, PS^{1+}), and indicate the bond order calculated from the Lewis dot structure. **(3 points)**

Lewis Dot Structure

Bond Order



7380D063-C8CB-4C8C-959E-30FC087EFF99

midterm-exam-93388

#98 Page 10 of 10

SCRAP PAGE

**CHEMISTRY 110 –Midterm Exam****Thursday November 7, 2024, 6:30 – 8:30 pm**

Examiners: Pallavi Sirjoosingh and Ashok Kakkar

First name (please write as legibly as possible within the boxes)

--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--

Last name

--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--

Student ID number

--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--

This booklet has 8 questions on 10 pages (The last page can be used as a scrap page).

The staple must NOT be removed from the question booklet. Your intact question booklet must be returned at the end of the exam.

A separate booklet contains a periodic table, and datasheet.

Write your answers directly under the questions in the space provided. Do not write on the QR codes or near the top edges (as it may be cut off).

Please write **legibly and dark enough** for the scan to be readable. For students who use English as a second language, dictionaries are permitted. Calculators are allowed.

DO NOT ASK QUESTIONS OF THE INVIGILATORS about interpretations of the exam questions. If you have a concern about a question, please summarize it at the top of this page. All concerns will be evaluated after the exam. GOOD LUCK!

**Question 1 8 points** (*Assume all transitions are possible*)

a. 6 points Calculate the **energy (in kJ/mol)** required to ionize (completely remove) the electron from the **$n=3$** level of **He^+ ion** (He: Helium). Show your work including equations. (3 sig. figs.)

b. 2 points

(I) Select the largest (longest) wavelength from the three choices below:

- i. Wavelength of photon to *completely remove* an electron from *1s orbital of Hydrogen atom*
- ii. Wavelength of photon to *completely remove* an electron from *1s orbital of Boron atom (B)*
- iii. Wavelength of photon to *completely remove* an electron from *1s orbital of Boron cation (B^+)*

i or ii or iii

(II) Select the smallest (shortest) wavelength from the three choices below (*for H atom*):

- i. Wavelength of photon *emitted* for an electron transitioning from *2p to 1s orbital*
- ii. Wavelength of photon *absorbed* for an electron transitioning from *2s to 4p orbital*
- iii. Wavelength of photon *absorbed* for an electron transitioning from *1s to 3p orbital*

i or ii or iii



Question 2 8 points



A Chem110 student plots the graph shown on the left, from data obtained for the kinetic energy (eV) of the ejected electron versus frequency (s^{-1}) of the incident light, for an unknown metal. ($1\text{eV} = 1.6022 \times 10^{-19} \text{ J}$)

Based on the plot, answer the questions **a to c** below:

a. 1 point On the plot above, label the **threshold frequency**.

b. 2 points What is the value of the work potential (in J) of the unknown metal?

c. 3 points What is the *longest* wavelength (in nm) required to eject an electron from this metal's surface? Show your work including equation(s) used. (3 sig figs)

d. 2 points In the space provided below, plot the following

(Assume frequency is *above* threshold frequency):

- Kinetic energy* of an ejected electron versus the *intensity* of the incident light (left)
- Number of electrons* ejected versus the *frequency* of the incident light (right)



**Question 3** **6 points****a. 2 points**

In the ground state of Ge (Germanium; $Z = 32$) atom, **name** the following orbitals:

i. Orbital containing the electron that requires the maximum energy to be removed.

ii. Orbital containing the electron that experiences the minimum effective nuclear charge.

b. 2 points

An unknown anion (A^-) has the following *excited* electronic configuration: $[Ar] 4s^2 3d^3 4p^1$

(Ar: Argon; $Z = 18$)

Give the **condensed ground state electronic configuration** of the atom, **A** and the cation, **A^{2+}** .
Show your work. (Assume all electronic transitions are possible)

Electronic Configuration of A:

Electronic Configuration of A^{2+} :

c. 2 points

In a single atom, determine how many electrons can have the following set of quantum numbers?

(i) $n = 5; l = 3$

In a single atom, determine how many electrons can have the following set of quantum numbers?

(i) $n = 3; l = 2; m_s = +1/2$

**Question 4** **8 points**

Rank the following from 1 to 4 in **increasing** order. Provide the ranking in the boxes provided.

Note: Atomic number (Z) are provided for relevant atoms

a) Increasing (lowest to highest) order of *ionization energy*: (1= lowest; 4 = highest)

Mg

Mg: Z = 12

Rb

Rb: Z = 37

Cs

Cs: Z = 55

K

K : Z = 19

b) Increasing (lowest to highest) order of *bond angle*: (1= lowest; 4 = highest)

H₂OCCl₄XeF₄

HCN

c) Increasing (smallest to largest) order of *ionic size* (1= smallest; 4 = largest)

S²⁻

S: Z = 16

O²⁻

O: Z = 8

Be²⁺

Be: Z = 4

Na⁺

Na: Z = 11

d) Increasing (lowest to highest) order of *bond energy* (1= lowest; 4 = highest)

P-F

P-Cl

P-I

P-Br

**Question 5** **12 points**

a. 9 points For the molecules HClO_4 and TeF_6 , draw the most stable *Lewis structure* showing appropriate shape and *dashed/wedged bonds*, based on VSEPR. Determine the **molecular geometry** around the central atom. For HClO_4 , label the *most polar bond*.



b. 3 points (Note: Lone pairs are not shown in structures below)

(i) For the three *resonance* structures (1, 2 and 3) shown below, answer the following question:



(I) Circle the *most contributing* resonance structure.

Structure 1

Structure 2

Structure 3

(ii) For the three structures given (4, 5, and 6) below, answer the following questions:



(I) Circle all the structures that represent *resonance structures* of each other:

Structure 4

Structure 5

Structure 6

(II) Among the resonance structures, circle the *most contributing* resonance structure.

Structure 4

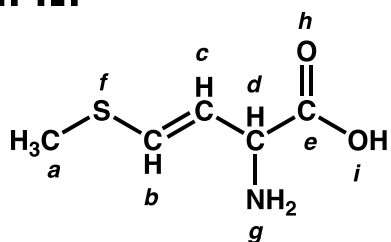
Structure 5

Structure 6

**Question 6** **6 points**

Using the data given below, calculate the **lattice energy** for SrO(s).

	ΔH° (kJ/mol)
$\text{Sr}(s) \longrightarrow \text{Sr}(g)$	+180
$\text{Sr}(g) \longrightarrow \text{Sr}^{2+}(g) + 2e^-$	+1468
$\text{O}_2(g) \longrightarrow 2\text{O}(g)$	+500
$\text{O}(g) + 2e^- \longrightarrow \text{O}^{2-}(g)$	+600
$\text{Sr}(s) + \frac{1}{2}\text{O}_2(g) \longrightarrow \text{SrO}(s)$	-550

**Question 7 10 points**

Consider the molecule given on the left, and answer questions (i) to (vi) using chemical bonding theories that we have learnt in class (valence bond theory, hybridization).

Note that the lone pairs are not shown in the structure:

i. Consider the carbon atoms marked in as ***a***, ***b***, ***c***, ***d***, and ***e*** in the molecule, and determine **hybridization** at each of the carbon centers. (2.5 points)

Atom	Hybridization
Carbon (<i>a</i>)	
Carbon (<i>b</i>)	
Carbon (<i>c</i>)	
Carbon (<i>d</i>)	
Carbon (<i>e</i>)	

ii. Now, consider the sulfur atom (S) marked as ***f***, nitrogen (N) atom (***g***), and oxygen(O) atoms (***h*** and ***i***), and determine their **hybridization**. (2 points)

Atom	Hybridization
Sulfur (<i>f</i>)	
Nitrogen (<i>g</i>)	
Oxygen (<i>h</i>)	
Oxygen (<i>i</i>)	

iii. **Describe** all the bonds formed by the carbon atoms ***a*** and ***b***, using sigma and pi designation as appropriate. (2 points)

Atom	Bonds formed (use sigma and pi as appropriate)
Carbon (<i>a</i>)	
Carbon (<i>b</i>)	

iv. How many total sp^3 -s sigma bonds are in the molecule? (1 point)

v. Determine **electronic** and **molecular geometry** at carbon (C) labeled ***e*** and sulfur (S) labeled ***f*** centres. (2 points)

Atom	Electronic Geometry	Molecular Geometry
Carbon (<i>e</i>)		
Sulfur (<i>f</i>)		

vi. Circle the value from the choices given below, that describes the **C-S-C bond angle at sulfur** (S) marked ***f***: (0.5 points)

>90° to <109.5° 109.5° >109.5 to <120° 120° >120° to <180°


Question 8 12 points

Using the **valence shell orbitals only** draw a **complete** molecular orbital (MO) diagram (with no s-p mixing) for the molecule **PS**, in the space provided below. **(6 points)**

Assume that 3s and 3p atomic orbitals will form molecular orbitals like those formed by the combination of 2s and 2p atomic orbitals that we learnt in class, and there is no s-p mixing.

Show **energy arrow**, **atomic orbitals** involved, **atom labels**, the **molecular orbitals** that are formed, **filled electrons** in the MO diagram etc. Make sure to label all the atomic and molecular orbitals.

Using the MO diagram above, answer the following questions:

- (i) Which one of the following will have the highest bond order? Circle the correct choice
 A) PS^{1-} B) PS C) PS^{1+}
- (ii) Which one of the following will have the longest bond length? Circle the correct choice
 A) PS^{1-} B) PS C) PS^{1+}
- (iii) Which of the following will be diamagnetic? Circle the correct choice.
 A) PS^{1-} B) PS C) PS^{1+}
- (iv) Draw Lewis dot structure of the molecule with the shortest bond length (based on the MO diagram) from among the three (PS^{1-} , PS, PS^{1+}), and indicate the bond order calculated from the Lewis dot structure. **(3 points)**

Lewis Dot Structure

Bond Order



6F3A891B-CE9E-4974-8BA3-C007285A01EF

midterm-exam-93388

#99 Page 10 of 10

SCRAP PAGE

**CHEMISTRY 110 –Midterm Exam****Thursday November 7, 2024, 6:30 – 8:30 pm****Examiners: Pallavi Sirjoosingh and Ashok Kakkar**

First name (please write as legibly as possible within the boxes)

--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--

Last name

--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--

Student ID number

--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--

This booklet has 8 questions on 10 pages (The last page can be used as a scrap page).

The staple must NOT be removed from the question booklet. Your intact question booklet must be returned at the end of the exam.

A separate booklet contains a periodic table, and datasheet.

Write your answers directly under the questions in the space provided. Do not write on the QR codes or near the top edges (as it may be cut off).

Please write **legibly and dark enough** for the scan to be readable. For students who use English as a second language, dictionaries are permitted. Calculators are allowed.

DO NOT ASK QUESTIONS OF THE INVIGILATORS about interpretations of the exam questions. If you have a concern about a question, please summarize it at the top of this page. All concerns will be evaluated after the exam. GOOD LUCK!

**Question 1 8 points** (*Assume all transitions are possible*)

a. 6 points Calculate the **energy (in kJ/mol)** required to ionize (completely remove) the electron from the **n=3** level of **He⁺ ion** (He: Helium). Show your work including equations. (3 sig. figs.)

b. 2 points

(I) Select the largest (longest) wavelength from the three choices below:

- i. Wavelength of photon to *completely remove* an electron from *1s orbital of Hydrogen atom*
- ii. Wavelength of photon to *completely remove* an electron from *1s orbital of Boron atom (B)*
- iii. Wavelength of photon to *completely remove* an electron from *1s orbital of Boron cation (B⁺)*

i or ii or iii

(II) Select the smallest (shortest) wavelength from the three choices below (*for H atom*):

- i. Wavelength of photon *emitted* for an electron transitioning from *2p to 1s orbital*
- ii. Wavelength of photon *absorbed* for an electron transitioning from *2s to 4p orbital*
- iii. Wavelength of photon *absorbed* for an electron transitioning from *1s to 3p orbital*

i or ii or iii



Question 2 8 points



A Chem110 student plots the graph shown on the left, from data obtained for the kinetic energy (eV) of the ejected electron versus frequency (s^{-1}) of the incident light, for an unknown metal. ($1\text{eV} = 1.6022 \times 10^{-19} \text{ J}$)

Based on the plot, answer the questions **a to c** below:

a. 1 point On the plot above, label the **threshold frequency**.

b. 2 points What is the value of the work potential (in J) of the unknown metal?

c. 3 points What is the *longest* wavelength (in nm) required to eject an electron from this metal's surface? Show your work including equation(s) used. (3 sig figs)

d. 2 points In the space provided below, plot the following

(Assume frequency is *above* threshold frequency):

- Kinetic energy* of an ejected electron versus the *intensity* of the incident light (left)
- Number of electrons* ejected versus the *frequency* of the incident light (right)



**Question 3** **6 points****a. 2 points**

In the ground state of Ge (Germanium; $Z = 32$) atom, **name** the following orbitals:

i. Orbital containing the electron that requires the maximum energy to be removed.

ii. Orbital containing the electron that experiences the minimum effective nuclear charge.

b. 2 points

An unknown anion (A^-) has the following *excited* electronic configuration: $[Ar] 4s^2 3d^3 4p^1$

(Ar: Argon; $Z = 18$)

Give the **condensed ground state electronic configuration** of the atom, **A** and the cation, **A^{2+}** .
Show your work. (Assume all electronic transitions are possible)

Electronic Configuration of A:

Electronic Configuration of A^{2+} :

c. 2 points

In a single atom, determine how many electrons can have the following set of quantum numbers?

(i) $n = 5; l = 3$

In a single atom, determine how many electrons can have the following set of quantum numbers?

(i) $n = 3; l = 2; m_s = +1/2$

**Question 4** **8 points**

Rank the following from 1 to 4 in **increasing** order. Provide the ranking in the boxes provided.

Note: Atomic number (Z) are provided for relevant atoms

a) Increasing (lowest to highest) order of *ionization energy*: (1= lowest; 4 = highest)

Mg

Mg: Z = 12

Rb

Rb: Z = 37

Cs

Cs: Z = 55

K

K : Z = 19

b) Increasing (lowest to highest) order of *bond angle*: (1= lowest; 4 = highest)

H₂OCCl₄XeF₄

HCN

c) Increasing (smallest to largest) order of *ionic size* (1= smallest; 4 = largest)

S²⁻

S: Z = 16

O²⁻

O: Z = 8

Be²⁺

Be: Z = 4

Na⁺

Na: Z = 11

d) Increasing (lowest to highest) order of *bond energy* (1= lowest; 4 = highest)

P-F

P-Cl

P-I

P-Br

**Question 5** **12 points**

a. 9 points For the molecules HClO_4 and TeF_6 , draw the most stable *Lewis structure* showing appropriate shape and *dashed/wedged bonds*, based on VSEPR. Determine the **molecular geometry** around the central atom. For HClO_4 , label the *most polar bond*.

HClO_4 (there are no O-O bonds)

Label the most polar bond

Molecular Geometry around Cl

TeF_6

Molecular Geometry around Te

b. 3 points (Note: Lone pairs are not shown in structures below)

(i) For the three *resonance* structures (1, 2 and 3) shown below, answer the following question:



(I) Circle the *most contributing* resonance structure.

Structure 1

Structure 2

Structure 3

(ii) For the three structures given (4, 5, and 6) below, answer the following questions:



(I) Circle all the structures that represent *resonance structures* of each other:

Structure 4

Structure 5

Structure 6

(II) Among the resonance structures, circle the *most contributing* resonance structure.

Structure 4

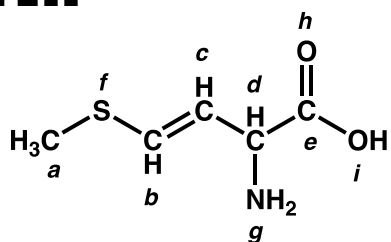
Structure 5

Structure 6

**Question 6** **6 points**

Using the data given below, calculate the **lattice energy** for SrO(s).

	ΔH° (kJ/mol)
$\text{Sr}(s) \longrightarrow \text{Sr}(g)$	+180
$\text{Sr}(g) \longrightarrow \text{Sr}^{2+}(g) + 2e^-$	+1468
$\text{O}_2(g) \longrightarrow 2\text{O}(g)$	+500
$\text{O}(g) + 2e^- \longrightarrow \text{O}^{2-}(g)$	+600
$\text{Sr}(s) + \frac{1}{2}\text{O}_2(g) \longrightarrow \text{SrO}(s)$	-550

**Question 7 10 points**

Consider the molecule given on the left, and answer questions (i) to (vi) using chemical bonding theories that we have learnt in class (valence bond theory, hybridization).

Note that the lone pairs are not shown in the structure:

i. Consider the carbon atoms marked in as ***a***, ***b***, ***c***, ***d***, and ***e*** in the molecule, and determine **hybridization** at each of the carbon centers. (2.5 points)

Atom	Hybridization
Carbon (<i>a</i>)	
Carbon (<i>b</i>)	
Carbon (<i>c</i>)	
Carbon (<i>d</i>)	
Carbon (<i>e</i>)	

ii. Now, consider the sulfur atom (S) marked as ***f***, nitrogen (N) atom (***g***), and oxygen(O) atoms (***h*** and ***i***), and determine their **hybridization**. (2 points)

Atom	Hybridization
Sulfur (<i>f</i>)	
Nitrogen (<i>g</i>)	
Oxygen (<i>h</i>)	
Oxygen (<i>i</i>)	

iii. **Describe** all the bonds formed by the carbon atoms ***a*** and ***b***, using sigma and pi designation as appropriate. (2 points)

Atom	Bonds formed (use sigma and pi as appropriate)
Carbon (<i>a</i>)	
Carbon (<i>b</i>)	

iv. How many total sp^3 -s sigma bonds are in the molecule? (1 point)

v. Determine **electronic** and **molecular geometry** at carbon (C) labeled ***e*** and sulfur (S) labeled ***f*** centres. (2 points)

Atom	Electronic Geometry	Molecular Geometry
Carbon (<i>e</i>)		
Sulfur (<i>f</i>)		

vi. Circle the value from the choices given below, that describes the **C-S-C bond angle at sulfur** (S) marked ***f***: (0.5 points)

>90° to <109.5° 109.5° >109.5 to <120° 120° >120° to <180°


Question 8 12 points

Using the **valence shell orbitals only** draw a **complete** molecular orbital (MO) diagram (with no s-p mixing) for the molecule **PS**, in the space provided below. **(6 points)**

Assume that 3s and 3p atomic orbitals will form molecular orbitals like those formed by the combination of 2s and 2p atomic orbitals that we learnt in class, and there is no s-p mixing.

Show **energy arrow**, **atomic orbitals** involved, **atom labels**, the **molecular orbitals** that are formed, **filled electrons** in the MO diagram etc. Make sure to **label all the atomic and molecular orbitals**.

Using the MO diagram above, answer the following questions:

- (i) Which one of the following will have the highest bond order? Circle the correct choice
 A) PS^{1-} B) PS C) PS^{1+}
- (ii) Which one of the following will have the longest bond length? Circle the correct choice
 A) PS^{1-} B) PS C) PS^{1+}
- (iii) Which of the following will be diamagnetic? Circle the correct choice.
 A) PS^{1-} B) PS C) PS^{1+}
- (iv) Draw Lewis dot structure of the molecule with the shortest bond length (based on the MO diagram) from among the three (PS^{1-} , PS, PS^{1+}), and indicate the bond order calculated from the Lewis dot structure. **(3 points)**

Lewis Dot Structure

Bond Order



DFADF362-9FCC-403D-8621-2CE8E88BB877

midterm-exam-93388

#100 Page 10 of 10

SCRAP PAGE

**CHEMISTRY 110 –Midterm Exam****Thursday November 7, 2024, 6:30 – 8:30 pm**

Examiners: Pallavi Sirjoosingh and Ashok Kakkar

First name (please write as legibly as possible within the boxes)

--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--

Last name

--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--

Student ID number

--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--

This booklet has 8 questions on 10 pages (The last page can be used as a scrap page).

The staple must NOT be removed from the question booklet. Your intact question booklet must be returned at the end of the exam.

A separate booklet contains a periodic table, and datasheet.

Write your answers directly under the questions in the space provided. Do not write on the QR codes or near the top edges (as it may be cut off).

Please write **legibly and dark enough** for the scan to be readable. For students who use English as a second language, dictionaries are permitted. Calculators are allowed.

DO NOT ASK QUESTIONS OF THE INVIGILATORS about interpretations of the exam questions. If you have a concern about a question, please summarize it at the top of this page. All concerns will be evaluated after the exam. GOOD LUCK!

**Question 1 8 points** (*Assume all transitions are possible*)

a. 6 points Calculate the **energy (in kJ/mol)** required to ionize (completely remove) the electron from the **$n=3$** level of **He^+ ion** (He: Helium). Show your work including equations. (3 sig. figs.)

b. 2 points

(I) Select the largest (longest) wavelength from the three choices below:

- i. Wavelength of photon to *completely remove* an electron from *1s orbital of Hydrogen atom*
- ii. Wavelength of photon to *completely remove* an electron from *1s orbital of Boron atom (B)*
- iii. Wavelength of photon to *completely remove* an electron from *1s orbital of Boron cation (B^+)*

i or ii or iii

(II) Select the smallest (shortest) wavelength from the three choices below (*for H atom*):

- i. Wavelength of photon *emitted* for an electron transitioning from *2p to 1s orbital*
- ii. Wavelength of photon *absorbed* for an electron transitioning from *2s to 4p orbital*
- iii. Wavelength of photon *absorbed* for an electron transitioning from *1s to 3p orbital*

i or ii or iii



Question 2 8 points



A Chem110 student plots the graph shown on the left, from data obtained for the kinetic energy (eV) of the ejected electron versus frequency (s^{-1}) of the incident light, for an unknown metal. ($1\text{eV} = 1.6022 \times 10^{-19} \text{ J}$)

Based on the plot, answer the questions **a to c** below:

a. 1 point On the plot above, label the **threshold frequency**.

b. 2 points What is the value of the work potential (in J) of the unknown metal?

c. 3 points What is the *longest* wavelength (in nm) required to eject an electron from this metal's surface? Show your work including equation(s) used. (3 sig figs)

d. 2 points In the space provided below, plot the following

(Assume frequency is *above* threshold frequency):

- i. *Kinetic energy* of an ejected electron versus the *intensity* of the incident light (left)
- ii. *Number of electrons* ejected versus the *frequency* of the incident light (right)



**Question 3** **6 points****a. 2 points**

In the ground state of Ge (Germanium; $Z = 32$) atom, **name** the following orbitals:

i. Orbital containing the electron that requires the maximum energy to be removed.

ii. Orbital containing the electron that experiences the minimum effective nuclear charge.

b. 2 points

An unknown anion (A^-) has the following *excited* electronic configuration: $[Ar] 4s^2 3d^3 4p^1$

(Ar: Argon; $Z = 18$)

Give the **condensed ground state electronic configuration** of the atom, **A** and the cation, **A^{2+}** .
Show your work. (Assume all electronic transitions are possible)

Electronic Configuration of A:

Electronic Configuration of A^{2+} :

c. 2 points

In a single atom, determine how many electrons can have the following set of quantum numbers?

(i) $n = 5; l = 3$

In a single atom, determine how many electrons can have the following set of quantum numbers?

(i) $n = 3; l = 2; m_s = +1/2$

**Question 4** **8 points**

Rank the following from 1 to 4 in **increasing** order. Provide the ranking in the boxes provided.

Note: Atomic number (Z) are provided for relevant atoms

a) Increasing (lowest to highest) order of *ionization energy*: (1= lowest; 4 = highest)

Mg

Mg: Z = 12

Rb

Rb: Z = 37

Cs

Cs: Z = 55

K

K : Z = 19

b) Increasing (lowest to highest) order of *bond angle*: (1= lowest; 4 = highest)

H₂OCCl₄XeF₄

HCN

c) Increasing (smallest to largest) order of *ionic size* (1= smallest; 4 = largest)

S²⁻

S: Z = 16

O²⁻

O: Z = 8

Be²⁺

Be: Z = 4

Na⁺

Na: Z = 11

d) Increasing (lowest to highest) order of *bond energy* (1= lowest; 4 = highest)

P-F

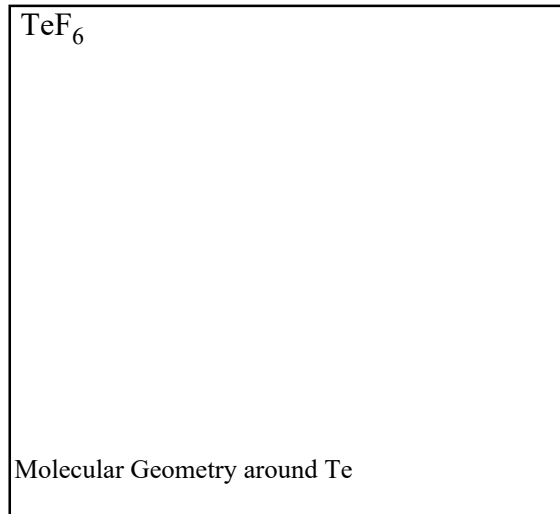
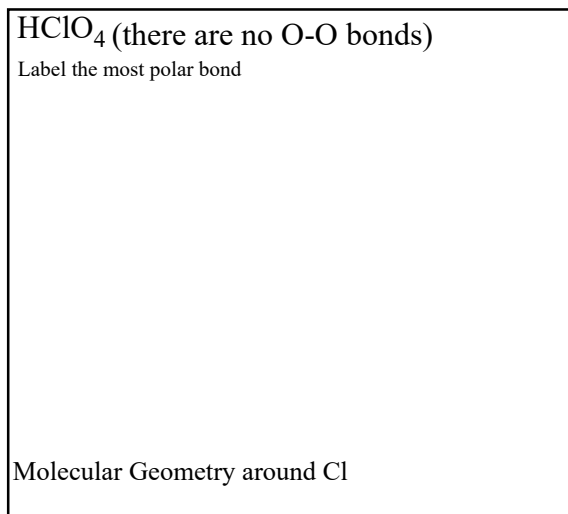
P-Cl

P-I

P-Br

**Question 5** **12 points**

a. 9 points For the molecules HClO_4 and TeF_6 , draw the most stable *Lewis structure* showing appropriate shape and *dashed/wedged bonds*, based on VSEPR. Determine the **molecular geometry** around the central atom. For HClO_4 , label the *most polar bond*.



b. 3 points (Note: Lone pairs are not shown in structures below)

(i) For the three *resonance* structures (1, 2 and 3) shown below, answer the following question:



(I) Circle the *most contributing* resonance structure.

Structure 1

Structure 2

Structure 3

(ii) For the three structures given (4, 5, and 6) below, answer the following questions:



(I) Circle all the structures that represent *resonance structures* of each other:

Structure 4

Structure 5

Structure 6

(II) Among the resonance structures, circle the *most contributing* resonance structure.

Structure 4

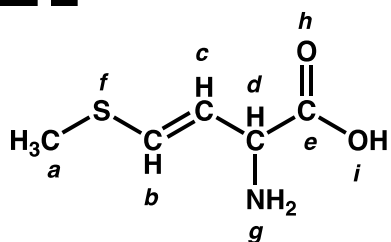
Structure 5

Structure 6

**Question 6** **6 points**

Using the data given below, calculate the **lattice energy** for SrO(s).

	ΔH° (kJ/mol)
$\text{Sr}(s) \longrightarrow \text{Sr}(g)$	+180
$\text{Sr}(g) \longrightarrow \text{Sr}^{2+}(g) + 2e^-$	+1468
$\text{O}_2(g) \longrightarrow 2\text{O}(g)$	+500
$\text{O}(g) + 2e^- \longrightarrow \text{O}^{2-}(g)$	+600
$\text{Sr}(s) + \frac{1}{2}\text{O}_2(g) \longrightarrow \text{SrO}(s)$	-550

**Question 7 10 points**

Consider the molecule given on the left, and answer questions (i) to (vi) using chemical bonding theories that we have learnt in class (valence bond theory, hybridization).

Note that the lone pairs are not shown in the structure:

i. Consider the carbon atoms marked in as ***a***, ***b***, ***c***, ***d***, and ***e*** in the molecule, and determine **hybridization** at each of the carbon centers. (2.5 points)

Atom	Hybridization
Carbon (<i>a</i>)	
Carbon (<i>b</i>)	
Carbon (<i>c</i>)	
Carbon (<i>d</i>)	
Carbon (<i>e</i>)	

ii. Now, consider the sulfur atom (S) marked as ***f***, nitrogen (N) atom (***g***), and oxygen(O) atoms (***h*** and ***i***), and determine their **hybridization**. (2 points)

Atom	Hybridization
Sulfur (<i>f</i>)	
Nitrogen (<i>g</i>)	
Oxygen (<i>h</i>)	
Oxygen (<i>i</i>)	

iii. **Describe** all the bonds formed by the carbon atoms ***a*** and ***b***, using sigma and pi designation as appropriate. (2 points)

Atom	Bonds formed (use sigma and pi as appropriate)
Carbon (<i>a</i>)	
Carbon (<i>b</i>)	

iv. How many total sp^3 -s sigma bonds are in the molecule? (1 point)

v. Determine **electronic** and **molecular geometry** at carbon (C) labeled ***e*** and sulfur (S) labeled ***f*** centres. (2 points)

Atom	Electronic Geometry	Molecular Geometry
Carbon (<i>e</i>)		
Sulfur (<i>f</i>)		

vi. Circle the value from the choices given below, that describes the **C-S-C bond angle at sulfur** (S) marked ***f***: (0.5 points)

>90° to <109.5° 109.5° >109.5 to <120° 120° >120° to <180°


Question 8 12 points

Using the **valence shell orbitals only** draw a **complete** molecular orbital (MO) diagram (with no s-p mixing) for the molecule **PS**, in the space provided below. **(6 points)**

Assume that 3s and 3p atomic orbitals will form molecular orbitals like those formed by the combination of 2s and 2p atomic orbitals that we learnt in class, and there is no s-p mixing.

Show **energy arrow**, **atomic orbitals** involved, **atom labels**, the **molecular orbitals** that are formed, **filled electrons** in the MO diagram etc. Make sure to label all the atomic and molecular orbitals.

Using the MO diagram above, answer the following questions:

- (i) Which one of the following will have the highest bond order? Circle the correct choice
 A) PS^{1-} B) PS C) PS^{1+}
- (ii) Which one of the following will have the longest bond length? Circle the correct choice
 A) PS^{1-} B) PS C) PS^{1+}
- (iii) Which of the following will be diamagnetic? Circle the correct choice.
 A) PS^{1-} B) PS C) PS^{1+}
- (iv) Draw Lewis dot structure of the molecule with the shortest bond length (based on the MO diagram) from among the three (PS^{1-} , PS, PS^{1+}), and indicate the bond order calculated from the Lewis dot structure. **(3 points)**

Lewis Dot Structure

Bond Order



68AAA0F0-0DC2-4B76-A8EA-02616C50981B

midterm-exam-93388

#101 Page 10 of 10

SCRAP PAGE

**CHEMISTRY 110 –Midterm Exam****Thursday November 7, 2024, 6:30 – 8:30 pm**

Examiners: Pallavi Sirjoosingh and Ashok Kakkar

First name (please write as legibly as possible within the boxes)

--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--

Last name

--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--

Student ID number

--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--

This booklet has 8 questions on 10 pages (The last page can be used as a scrap page).

The staple must NOT be removed from the question booklet. Your intact question booklet must be returned at the end of the exam.

A separate booklet contains a periodic table, and datasheet.

Write your answers directly under the questions in the space provided. Do not write on the QR codes or near the top edges (as it may be cut off).

Please write **legibly and dark enough** for the scan to be readable. For students who use English as a second language, dictionaries are permitted. Calculators are allowed.

DO NOT ASK QUESTIONS OF THE INVIGILATORS about interpretations of the exam questions. If you have a concern about a question, please summarize it at the top of this page. All concerns will be evaluated after the exam. GOOD LUCK!

**Question 1 8 points** (*Assume all transitions are possible*)

a. 6 points Calculate the **energy (in kJ/mol)** required to ionize (completely remove) the electron from the **n=3** level of **He⁺ ion** (He: Helium). Show your work including equations. (3 sig. figs.)

b. 2 points

(I) Select the largest (longest) wavelength from the three choices below:

- i. Wavelength of photon to *completely remove* an electron from *1s orbital of Hydrogen atom*
- ii. Wavelength of photon to *completely remove* an electron from *1s orbital of Boron atom (B)*
- iii. Wavelength of photon to *completely remove* an electron from *1s orbital of Boron cation (B⁺)*

i or ii or iii

(II) Select the smallest (shortest) wavelength from the three choices below (*for H atom*):

- i. Wavelength of photon *emitted* for an electron transitioning from *2p to 1s orbital*
- ii. Wavelength of photon *absorbed* for an electron transitioning from *2s to 4p orbital*
- iii. Wavelength of photon *absorbed* for an electron transitioning from *1s to 3p orbital*

i or ii or iii



Question 2 8 points



A Chem110 student plots the graph shown on the left, from data obtained for the kinetic energy (eV) of the ejected electron versus frequency (s^{-1}) of the incident light, for an unknown metal. ($1\text{eV} = 1.6022 \times 10^{-19} \text{ J}$)

Based on the plot, answer the questions **a to c** below:

a. 1 point On the plot above, label the **threshold frequency**.

b. 2 points What is the value of the work potential (in J) of the unknown metal?

c. 3 points What is the *longest* wavelength (in nm) required to eject an electron from this metal's surface? Show your work including equation(s) used. (3 sig figs)

d. 2 points In the space provided below, plot the following

(Assume frequency is *above* threshold frequency):

- Kinetic energy* of an ejected electron versus the *intensity* of the incident light (left)
- Number of electrons* ejected versus the *frequency* of the incident light (right)



**Question 3** **6 points****a. 2 points**

In the ground state of Ge (Germanium; $Z = 32$) atom, **name** the following orbitals:

i. Orbital containing the electron that requires the maximum energy to be removed.

ii. Orbital containing the electron that experiences the minimum effective nuclear charge.

b. 2 points

An unknown anion (A^-) has the following *excited* electronic configuration: $[Ar] 4s^2 3d^3 4p^1$

(Ar: Argon; $Z = 18$)

Give the **condensed ground state electronic configuration** of the atom, **A** and the cation, **A^{2+}** .
Show your work. (Assume all electronic transitions are possible)

Electronic Configuration of A:

Electronic Configuration of A^{2+} :

c. 2 points

In a single atom, determine how many electrons can have the following set of quantum numbers?

(i) $n = 5; l = 3$

In a single atom, determine how many electrons can have the following set of quantum numbers?

(i) $n = 3; l = 2; m_s = +1/2$

**Question 4** **8 points**

Rank the following from 1 to 4 in **increasing** order. Provide the ranking in the boxes provided.

Note: Atomic number (Z) are provided for relevant atoms

a) Increasing (lowest to highest) order of *ionization energy*: (1= lowest; 4 = highest)

Mg

Mg: Z = 12

Rb

Rb: Z = 37

Cs

Cs: Z = 55

K

K : Z = 19

b) Increasing (lowest to highest) order of *bond angle*: (1= lowest; 4 = highest)

H₂OCCl₄XeF₄

HCN

c) Increasing (smallest to largest) order of *ionic size* (1= smallest; 4 = largest)

S²⁻

S: Z = 16

O²⁻

O: Z = 8

Be²⁺

Be: Z = 4

Na⁺

Na: Z = 11

d) Increasing (lowest to highest) order of *bond energy* (1= lowest; 4 = highest)

P-F

P-Cl

P-I

P-Br

**Question 5** **12 points**

a. 9 points For the molecules HClO_4 and TeF_6 , draw the most stable *Lewis structure* showing appropriate shape and *dashed/wedged bonds*, based on VSEPR. Determine the **molecular geometry** around the central atom. For HClO_4 , label the *most polar bond*.

HClO_4 (there are no O-O bonds)

Label the most polar bond

Molecular Geometry around Cl

TeF_6

Molecular Geometry around Te

b. 3 points (Note: Lone pairs are not shown in structures below)

(i) For the three *resonance* structures (1, 2 and 3) shown below, answer the following question:



(I) Circle the *most contributing* resonance structure.

Structure 1

Structure 2

Structure 3

(ii) For the three structures given (4, 5, and 6) below, answer the following questions:



(I) Circle all the structures that represent *resonance structures* of each other:

Structure 4

Structure 5

Structure 6

(II) Among the resonance structures, circle the *most contributing* resonance structure.

Structure 4

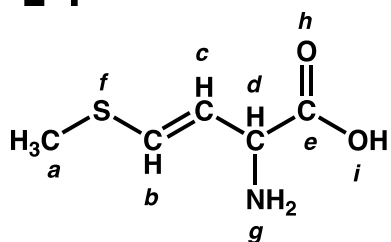
Structure 5

Structure 6

**Question 6** **6 points**

Using the data given below, calculate the **lattice energy** for SrO(s).

	ΔH° (kJ/mol)
$\text{Sr}(s) \longrightarrow \text{Sr}(g)$	+180
$\text{Sr}(g) \longrightarrow \text{Sr}^{2+}(g) + 2e^-$	+1468
$\text{O}_2(g) \longrightarrow 2\text{O}(g)$	+500
$\text{O}(g) + 2e^- \longrightarrow \text{O}^{2-}(g)$	+600
$\text{Sr}(s) + \frac{1}{2}\text{O}_2(g) \longrightarrow \text{SrO}(s)$	-550

**Question 7 10 points**

Consider the molecule given on the left, and answer questions (i) to (vi) using chemical bonding theories that we have learnt in class (valence bond theory, hybridization).

Note that the lone pairs are not shown in the structure:

i. Consider the carbon atoms marked in as ***a***, ***b***, ***c***, ***d***, and ***e*** in the molecule, and determine **hybridization** at each of the carbon centers. (2.5 points)

Atom	Hybridization
Carbon (<i>a</i>)	
Carbon (<i>b</i>)	
Carbon (<i>c</i>)	
Carbon (<i>d</i>)	
Carbon (<i>e</i>)	

ii. Now, consider the sulfur atom (S) marked as ***f***, nitrogen (N) atom (***g***), and oxygen(O) atoms (***h*** and ***i***), and determine their **hybridization**. (2 points)

Atom	Hybridization
Sulfur (<i>f</i>)	
Nitrogen (<i>g</i>)	
Oxygen (<i>h</i>)	
Oxygen (<i>i</i>)	

iii. **Describe** all the bonds formed by the carbon atoms ***a*** and ***b***, using sigma and pi designation as appropriate. (2 points)

Atom	Bonds formed (use sigma and pi as appropriate)
Carbon (<i>a</i>)	
Carbon (<i>b</i>)	

iv. How many total sp^3 -s sigma bonds are in the molecule? (1 point)

v. Determine **electronic** and **molecular geometry** at carbon (C) labeled ***e*** and sulfur (S) labeled ***f*** centres. (2 points)

Atom	Electronic Geometry	Molecular Geometry
Carbon (<i>e</i>)		
Sulfur (<i>f</i>)		

vi. Circle the value from the choices given below, that describes the **C-S-C bond angle at sulfur** (S) marked ***f***: (0.5 points)

$>90^\circ$ to $<109.5^\circ$ 109.5° >109.5 to $<120^\circ$ 120° $>120^\circ$ to $<180^\circ$


Question 8 12 points

Using the **valence shell orbitals only** draw a **complete** molecular orbital (MO) diagram (with no s-p mixing) for the molecule **PS**, in the space provided below. **(6 points)**

Assume that 3s and 3p atomic orbitals will form molecular orbitals like those formed by the combination of 2s and 2p atomic orbitals that we learnt in class, and there is no s-p mixing.

Show **energy arrow**, **atomic orbitals** involved, **atom labels**, the **molecular orbitals** that are formed, **filled electrons** in the MO diagram etc. Make sure to label all the atomic and molecular orbitals.

Using the MO diagram above, answer the following questions:

- (i) Which one of the following will have the highest bond order? Circle the correct choice
 A) PS^{1-} B) PS C) PS^{1+}
- (ii) Which one of the following will have the longest bond length? Circle the correct choice
 A) PS^{1-} B) PS C) PS^{1+}
- (iii) Which of the following will be diamagnetic? Circle the correct choice.
 A) PS^{1-} B) PS C) PS^{1+}
- (iv) Draw Lewis dot structure of the molecule with the shortest bond length (based on the MO diagram) from among the three (PS^{1-} , PS, PS^{1+}), and indicate the bond order calculated from the Lewis dot structure. **(3 points)**

Lewis Dot Structure

Bond Order



4CF60D47-70CA-4F5E-84A7-5D6B418079E2

midterm-exam-93388

#102 Page 10 of 10

SCRAP PAGE

**CHEMISTRY 110 –Midterm Exam****Thursday November 7, 2024, 6:30 – 8:30 pm****Examiners: Pallavi Sirjoosingh and Ashok Kakkar**

First name (please write as legibly as possible within the boxes)

--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--

Last name

--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--

Student ID number

--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--

This booklet has 8 questions on 10 pages (The last page can be used as a scrap page).

The staple must NOT be removed from the question booklet. Your intact question booklet must be returned at the end of the exam.

A separate booklet contains a periodic table, and datasheet.

Write your answers directly under the questions in the space provided. Do not write on the QR codes or near the top edges (as it may be cut off).

Please write **legibly and dark enough** for the scan to be readable. For students who use English as a second language, dictionaries are permitted. Calculators are allowed.

DO NOT ASK QUESTIONS OF THE INVIGILATORS about interpretations of the exam questions. If you have a concern about a question, please summarize it at the top of this page. All concerns will be evaluated after the exam. GOOD LUCK!

**Question 1 8 points** (*Assume all transitions are possible*)

a. 6 points Calculate the **energy (in kJ/mol)** required to ionize (completely remove) the electron from the **$n=3$** level of **He^+ ion** (He: Helium). Show your work including equations. (3 sig. figs.)

b. 2 points

(I) Select the largest (longest) wavelength from the three choices below:

- i. Wavelength of photon to *completely remove* an electron from *1s orbital of Hydrogen atom*
- ii. Wavelength of photon to *completely remove* an electron from *1s orbital of Boron atom (B)*
- iii. Wavelength of photon to *completely remove* an electron from *1s orbital of Boron cation (B^+)*

i or ii or iii

(II) Select the smallest (shortest) wavelength from the three choices below (*for H atom*):

- i. Wavelength of photon *emitted* for an electron transitioning from *2p to 1s orbital*
- ii. Wavelength of photon *absorbed* for an electron transitioning from *2s to 4p orbital*
- iii. Wavelength of photon *absorbed* for an electron transitioning from *1s to 3p orbital*

i or ii or iii



Question 2 8 points



A Chem110 student plots the graph shown on the left, from data obtained for the kinetic energy (eV) of the ejected electron versus frequency (s^{-1}) of the incident light, for an unknown metal. ($1\text{eV} = 1.6022 \times 10^{-19} \text{ J}$)

Based on the plot, answer the questions **a to c** below:

a. 1 point On the plot above, label the **threshold frequency**.

b. 2 points What is the value of the work potential (in J) of the unknown metal?

c. 3 points What is the *longest* wavelength (in nm) required to eject an electron from this metal's surface? Show your work including equation(s) used. (3 sig figs)

d. 2 points In the space provided below, plot the following

(Assume frequency is *above* threshold frequency):

- i. *Kinetic energy* of an ejected electron versus the *intensity* of the incident light (left)
- ii. *Number of electrons* ejected versus the *frequency* of the incident light (right)



**Question 3** **6 points****a. 2 points**

In the ground state of Ge (Germanium; $Z = 32$) atom, **name** the following orbitals:

i. Orbital containing the electron that requires the maximum energy to be removed.

ii. Orbital containing the electron that experiences the minimum effective nuclear charge.

b. 2 points

An unknown anion (A^-) has the following *excited* electronic configuration: $[Ar] 4s^2 3d^3 4p^1$

(Ar: Argon; $Z = 18$)

Give the **condensed ground state electronic configuration** of the atom, **A** and the cation, **A^{2+}** .
Show your work. (Assume all electronic transitions are possible)

Electronic Configuration of A:

Electronic Configuration of A^{2+} :

c. 2 points

In a single atom, determine how many electrons can have the following set of quantum numbers?

(i) $n = 5; l = 3$

In a single atom, determine how many electrons can have the following set of quantum numbers?

(i) $n = 3; l = 2; m_s = +1/2$

**Question 4** **8 points**

Rank the following from 1 to 4 in **increasing** order. Provide the ranking in the boxes provided.

Note: Atomic number (Z) are provided for relevant atoms

a) Increasing (lowest to highest) order of *ionization energy*: (1= lowest; 4 = highest)

Mg

Mg: Z = 12

Rb

Rb: Z = 37

Cs

Cs: Z = 55

K

K : Z = 19

b) Increasing (lowest to highest) order of *bond angle*: (1= lowest; 4 = highest)

H₂OCCl₄XeF₄

HCN

c) Increasing (smallest to largest) order of *ionic size* (1= smallest; 4 = largest)

S²⁻

S: Z = 16

O²⁻

O: Z = 8

Be²⁺

Be: Z = 4

Na⁺

Na: Z = 11

d) Increasing (lowest to highest) order of *bond energy* (1= lowest; 4 = highest)

P-F

P-Cl

P-I

P-Br

**Question 5** **12 points**

a. 9 points For the molecules HClO_4 and TeF_6 , draw the most stable *Lewis structure* showing appropriate shape and *dashed/wedged bonds*, based on VSEPR. Determine the **molecular geometry** around the central atom. For HClO_4 , label the *most polar bond*.



b. 3 points (Note: Lone pairs are not shown in structures below)

(i) For the three *resonance* structures (1, 2 and 3) shown below, answer the following question:



(I) Circle the *most contributing* resonance structure.

Structure 1

Structure 2

Structure 3

(ii) For the three structures given (4, 5, and 6) below, answer the following questions:



(I) Circle all the structures that represent *resonance structures* of each other:

Structure 4

Structure 5

Structure 6

(II) Among the resonance structures, circle the *most contributing* resonance structure.

Structure 4

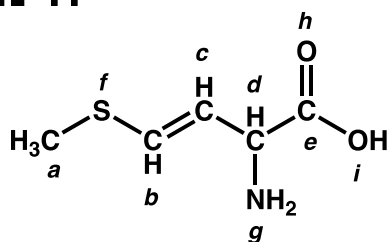
Structure 5

Structure 6

**Question 6** **6 points**

Using the data given below, calculate the **lattice energy** for SrO(s).

	ΔH° (kJ/mol)
$\text{Sr}(s) \longrightarrow \text{Sr}(g)$	+180
$\text{Sr}(g) \longrightarrow \text{Sr}^{2+}(g) + 2e^-$	+1468
$\text{O}_2(g) \longrightarrow 2\text{O}(g)$	+500
$\text{O}(g) + 2e^- \longrightarrow \text{O}^{2-}(g)$	+600
$\text{Sr}(s) + \frac{1}{2}\text{O}_2(g) \longrightarrow \text{SrO}(s)$	-550

**Question 7 10 points**

Consider the molecule given on the left, and answer questions (i) to (vi) using chemical bonding theories that we have learnt in class (valence bond theory, hybridization).

Note that the lone pairs are not shown in the structure:

i. Consider the carbon atoms marked in as ***a***, ***b***, ***c***, ***d***, and ***e*** in the molecule, and determine **hybridization** at each of the carbon centers. (2.5 points)

Atom	Hybridization
Carbon (<i>a</i>)	
Carbon (<i>b</i>)	
Carbon (<i>c</i>)	
Carbon (<i>d</i>)	
Carbon (<i>e</i>)	

ii. Now, consider the sulfur atom (S) marked as ***f***, nitrogen (N) atom (***g***), and oxygen(O) atoms (***h*** and ***i***), and determine their **hybridization**. (2 points)

Atom	Hybridization
Sulfur (<i>f</i>)	
Nitrogen (<i>g</i>)	
Oxygen (<i>h</i>)	
Oxygen (<i>i</i>)	

iii. **Describe** all the bonds formed by the carbon atoms ***a*** and ***b***, using sigma and pi designation as appropriate. (2 points)

Atom	Bonds formed (use sigma and pi as appropriate)
Carbon (<i>a</i>)	
Carbon (<i>b</i>)	

iv. How many total sp^3 -s sigma bonds are in the molecule? (1 point)

v. Determine **electronic** and **molecular geometry** at carbon (C) labeled ***e*** and sulfur (S) labeled ***f*** centres. (2 points)

Atom	Electronic Geometry	Molecular Geometry
Carbon (<i>e</i>)		
Sulfur (<i>f</i>)		

vi. Circle the value from the choices given below, that describes the **C-S-C bond angle at sulfur** (S) marked ***f***: (0.5 points)

>90° to <109.5° 109.5° >109.5 to <120° 120° >120° to <180°

**Question 8 12 points**

Using the **valence shell orbitals only** draw a **complete** molecular orbital (MO) diagram (with no s-p mixing) for the molecule **PS**, in the space provided below. **(6 points)**

Assume that 3s and 3p atomic orbitals will form molecular orbitals like those formed by the combination of 2s and 2p atomic orbitals that we learnt in class, and there is no s-p mixing.

Show **energy arrow**, **atomic orbitals** involved, **atom labels**, the **molecular orbitals** that are formed, **filled electrons** in the MO diagram etc. Make sure to label all the atomic and molecular orbitals.

Using the MO diagram above, answer the following questions:

- (i) Which one of the following will have the highest bond order? Circle the correct choice
A) PS^{1-} B) PS C) PS^{1+}
- (ii) Which one of the following will have the longest bond length? Circle the correct choice
A) PS^{1-} B) PS C) PS^{1+}
- (iii) Which of the following will be diamagnetic? Circle the correct choice.
A) PS^{1-} B) PS C) PS^{1+}
- (iv) Draw Lewis dot structure of the molecule with the shortest bond length (based on the MO diagram) from among the three (PS^{1-} , PS, PS^{1+}), and indicate the bond order calculated from the Lewis dot structure. **(3 points)**

Lewis Dot Structure

Bond Order



ABFF7B6F-DDD1-4A20-A793-CE1C8B3A43F8

midterm-exam-93388

#103 Page 10 of 10

SCRAP PAGE

**CHEMISTRY 110 –Midterm Exam****Thursday November 7, 2024, 6:30 – 8:30 pm****Examiners: Pallavi Sirjoosingh and Ashok Kakkar**

First name (please write as legibly as possible within the boxes)

--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--

Last name

--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--

Student ID number

--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--

This booklet has 8 questions on 10 pages (The last page can be used as a scrap page).

The staple must NOT be removed from the question booklet. Your intact question booklet must be returned at the end of the exam.

A separate booklet contains a periodic table, and datasheet.

Write your answers directly under the questions in the space provided. Do not write on the QR codes or near the top edges (as it may be cut off).

Please write **legibly and dark enough** for the scan to be readable. For students who use English as a second language, dictionaries are permitted. Calculators are allowed.

DO NOT ASK QUESTIONS OF THE INVIGILATORS about interpretations of the exam questions. If you have a concern about a question, please summarize it at the top of this page. All concerns will be evaluated after the exam. GOOD LUCK!

**Question 1 8 points** (*Assume all transitions are possible*)

a. 6 points Calculate the **energy (in kJ/mol)** required to ionize (completely remove) the electron from the **n=3** level of **He⁺ ion** (He: Helium). Show your work including equations. (3 sig. figs.)

b. 2 points

(I) Select the largest (longest) wavelength from the three choices below:

- i. Wavelength of photon to *completely remove* an electron from *1s orbital of Hydrogen atom*
- ii. Wavelength of photon to *completely remove* an electron from *1s orbital of Boron atom (B)*
- iii. Wavelength of photon to *completely remove* an electron from *1s orbital of Boron cation (B⁺)*

i or ii or iii

(II) Select the smallest (shortest) wavelength from the three choices below (*for H atom*):

- i. Wavelength of photon *emitted* for an electron transitioning from *2p to 1s orbital*
- ii. Wavelength of photon *absorbed* for an electron transitioning from *2s to 4p orbital*
- iii. Wavelength of photon *absorbed* for an electron transitioning from *1s to 3p orbital*

i or ii or iii



Question 2 8 points



A Chem110 student plots the graph shown on the left, from data obtained for the kinetic energy (eV) of the ejected electron versus frequency (s^{-1}) of the incident light, for an unknown metal. ($1\text{eV} = 1.6022 \times 10^{-19} \text{ J}$)

Based on the plot, answer the questions **a to c** below:

a. 1 point On the plot above, label the **threshold frequency**.

b. 2 points What is the value of the work potential (in J) of the unknown metal?

c. 3 points What is the *longest* wavelength (in nm) required to eject an electron from this metal's surface? Show your work including equation(s) used. (3 sig figs)

d. 2 points In the space provided below, plot the following

(Assume frequency is *above* threshold frequency):

- Kinetic energy* of an ejected electron versus the *intensity* of the incident light (left)
- Number of electrons* ejected versus the *frequency* of the incident light (right)



**Question 3** **6 points****a. 2 points**

In the ground state of Ge (Germanium; $Z = 32$) atom, **name** the following orbitals:

i. Orbital containing the electron that requires the maximum energy to be removed.

ii. Orbital containing the electron that experiences the minimum effective nuclear charge.

b. 2 points

An unknown anion (A^-) has the following *excited* electronic configuration: $[Ar] 4s^2 3d^3 4p^1$

(Ar: Argon; $Z = 18$)

Give the **condensed ground state electronic configuration** of the atom, **A** and the cation, **A^{2+}** .
Show your work. (Assume all electronic transitions are possible)

Electronic Configuration of A:

Electronic Configuration of A^{2+} :

c. 2 points

In a single atom, determine how many electrons can have the following set of quantum numbers?

(i) $n = 5; l = 3$

In a single atom, determine how many electrons can have the following set of quantum numbers?

(i) $n = 3; l = 2; m_s = +1/2$

**Question 4** **8 points**

Rank the following from 1 to 4 in **increasing** order. Provide the ranking in the boxes provided.

Note: Atomic number (Z) are provided for relevant atoms

a) Increasing (lowest to highest) order of *ionization energy*: (1= lowest; 4 = highest)

Mg

Mg: Z = 12

Rb

Rb: Z = 37

Cs

Cs: Z = 55

K

K : Z = 19

b) Increasing (lowest to highest) order of *bond angle*: (1= lowest; 4 = highest)

H₂OCCl₄XeF₄

HCN

c) Increasing (smallest to largest) order of *ionic size* (1= smallest; 4 = largest)

S²⁻

S: Z = 16

O²⁻

O: Z = 8

Be²⁺

Be: Z = 4

Na⁺

Na: Z = 11

d) Increasing (lowest to highest) order of *bond energy* (1= lowest; 4 = highest)

P-F

P-Cl

P-I

P-Br

**Question 5** **12 points**

a. 9 points For the molecules HClO_4 and TeF_6 , draw the most stable *Lewis structure* showing appropriate shape and *dashed/wedged bonds*, based on VSEPR. Determine the **molecular geometry** around the central atom. For HClO_4 , label the *most polar bond*.



b. 3 points (Note: Lone pairs are not shown in structures below)

(i) For the three *resonance* structures (1, 2 and 3) shown below, answer the following question:



(I) Circle the *most contributing* resonance structure.

Structure 1

Structure 2

Structure 3

(ii) For the three structures given (4, 5, and 6) below, answer the following questions:



(I) Circle all the structures that represent *resonance structures* of each other:

Structure 4

Structure 5

Structure 6

(II) Among the resonance structures, circle the *most contributing* resonance structure.

Structure 4

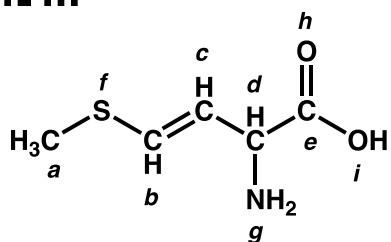
Structure 5

Structure 6

**Question 6** **6 points**

Using the data given below, calculate the **lattice energy** for SrO(s).

	ΔH° (kJ/mol)
$\text{Sr}(s) \longrightarrow \text{Sr}(g)$	+180
$\text{Sr}(g) \longrightarrow \text{Sr}^{2+}(g) + 2e^-$	+1468
$\text{O}_2(g) \longrightarrow 2\text{O}(g)$	+500
$\text{O}(g) + 2e^- \longrightarrow \text{O}^{2-}(g)$	+600
$\text{Sr}(s) + \frac{1}{2}\text{O}_2(g) \longrightarrow \text{SrO}(s)$	-550

**Question 7 10 points**

Consider the molecule given on the left, and answer questions (i) to (vi) using chemical bonding theories that we have learnt in class (valence bond theory, hybridization).

Note that the lone pairs are not shown in the structure:

i. Consider the carbon atoms marked in as ***a***, ***b***, ***c***, ***d***, and ***e*** in the molecule, and determine **hybridization** at each of the carbon centers. (2.5 points)

Atom	Hybridization
Carbon (<i>a</i>)	
Carbon (<i>b</i>)	
Carbon (<i>c</i>)	
Carbon (<i>d</i>)	
Carbon (<i>e</i>)	

ii. Now, consider the sulfur atom (S) marked as ***f***, nitrogen (N) atom (***g***), and oxygen(O) atoms (***h*** and ***i***), and determine their **hybridization**. (2 points)

Atom	Hybridization
Sulfur (<i>f</i>)	
Nitrogen (<i>g</i>)	
Oxygen (<i>h</i>)	
Oxygen (<i>i</i>)	

iii. **Describe** all the bonds formed by the carbon atoms ***a*** and ***b***, using sigma and pi designation as appropriate. (2 points)

Atom	Bonds formed (use sigma and pi as appropriate)
Carbon (<i>a</i>)	
Carbon (<i>b</i>)	

iv. How many total sp^3 -s sigma bonds are in the molecule? (1 point)

v. Determine **electronic** and **molecular geometry** at carbon (C) labeled ***e*** and sulfur (S) labeled ***f*** centres. (2 points)

Atom	Electronic Geometry	Molecular Geometry
Carbon (<i>e</i>)		
Sulfur (<i>f</i>)		

vi. Circle the value from the choices given below, that describes the **C-S-C bond angle at sulfur** (S) marked ***f***: (0.5 points)

>90° to <109.5°

109.5°

>109.5 to <120°

120°

>120° to <180°


Question 8 12 points

Using the **valence shell orbitals only** draw a **complete** molecular orbital (MO) diagram (with no s-p mixing) for the molecule **PS**, in the space provided below. **(6 points)**

Assume that 3s and 3p atomic orbitals will form molecular orbitals like those formed by the combination of 2s and 2p atomic orbitals that we learnt in class, and there is no s-p mixing.

Show **energy arrow**, **atomic orbitals** involved, **atom labels**, the **molecular orbitals** that are formed, **filled electrons** in the MO diagram etc. Make sure to label all the atomic and molecular orbitals.

Using the MO diagram above, answer the following questions:

- (i) Which one of the following will have the highest bond order? Circle the correct choice
 A) PS^{1-} B) PS C) PS^{1+}
- (ii) Which one of the following will have the longest bond length? Circle the correct choice
 A) PS^{1-} B) PS C) PS^{1+}
- (iii) Which of the following will be diamagnetic? Circle the correct choice.
 A) PS^{1-} B) PS C) PS^{1+}
- (iv) Draw Lewis dot structure of the molecule with the shortest bond length (based on the MO diagram) from among the three (PS^{1-} , PS, PS^{1+}), and indicate the bond order calculated from the Lewis dot structure. **(3 points)**

Lewis Dot Structure

Bond Order



4AABCAF3-67DA-43BA-B826-1165E09147D7

midterm-exam-93388

#104 Page 10 of 10

SCRAP PAGE

**CHEMISTRY 110 –Midterm Exam****Thursday November 7, 2024, 6:30 – 8:30 pm**

Examiners: Pallavi Sirjoosingh and Ashok Kakkar

First name (please write as legibly as possible within the boxes)

--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--

Last name

--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--

Student ID number

--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--

This booklet has 8 questions on 10 pages (The last page can be used as a scrap page).

The staple must NOT be removed from the question booklet. Your intact question booklet must be returned at the end of the exam.

A separate booklet contains a periodic table, and datasheet.

Write your answers directly under the questions in the space provided. Do not write on the QR codes or near the top edges (as it may be cut off).

Please write **legibly and dark enough** for the scan to be readable. For students who use English as a second language, dictionaries are permitted. Calculators are allowed.

DO NOT ASK QUESTIONS OF THE INVIGILATORS about interpretations of the exam questions. If you have a concern about a question, please summarize it at the top of this page. All concerns will be evaluated after the exam. GOOD LUCK!

**Question 1 8 points** (*Assume all transitions are possible*)

a. 6 points Calculate the **energy (in kJ/mol)** required to ionize (completely remove) the electron from the **n=3** level of **He⁺ ion** (He: Helium). Show your work including equations. (3 sig. figs.)

b. 2 points

(I) Select the largest (longest) wavelength from the three choices below:

- i. Wavelength of photon to *completely remove* an electron from *1s orbital of Hydrogen atom*
- ii. Wavelength of photon to *completely remove* an electron from *1s orbital of Boron atom (B)*
- iii. Wavelength of photon to *completely remove* an electron from *1s orbital of Boron cation (B⁺)*

i or ii or iii

(II) Select the smallest (shortest) wavelength from the three choices below (*for H atom*):

- i. Wavelength of photon *emitted* for an electron transitioning from *2p to 1s orbital*
- ii. Wavelength of photon *absorbed* for an electron transitioning from *2s to 4p orbital*
- iii. Wavelength of photon *absorbed* for an electron transitioning from *1s to 3p orbital*

i or ii or iii



Question 2 8 points



A Chem110 student plots the graph shown on the left, from data obtained for the kinetic energy (eV) of the ejected electron versus frequency (s^{-1}) of the incident light, for an unknown metal. ($1\text{eV} = 1.6022 \times 10^{-19} \text{ J}$)

Based on the plot, answer the questions **a to c** below:

a. 1 point On the plot above, label the **threshold frequency**.

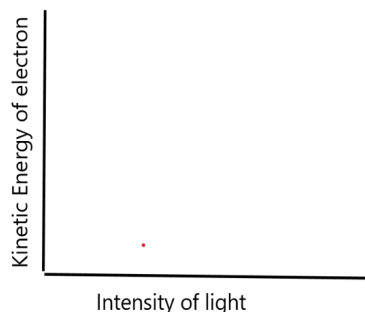
b. 2 points What is the value of the work potential (in J) of the unknown metal?

c. 3 points What is the *longest* wavelength (in nm) required to eject an electron from this metal's surface? Show your work including equation(s) used. (3 sig figs)

d. 2 points In the space provided below, plot the following

(Assume frequency is *above* threshold frequency):

- i. *Kinetic energy* of an ejected electron versus the *intensity* of the incident light (left)
- ii. *Number of electrons* ejected versus the *frequency* of the incident light (right)



**Question 3** **6 points****a. 2 points**

In the ground state of Ge (Germanium; $Z = 32$) atom, **name** the following orbitals:

i. Orbital containing the electron that requires the maximum energy to be removed.

ii. Orbital containing the electron that experiences the minimum effective nuclear charge.

b. 2 points

An unknown anion (A^-) has the following *excited* electronic configuration: $[Ar] 4s^2 3d^3 4p^1$

(Ar: Argon; $Z = 18$)

Give the **condensed ground state electronic configuration** of the atom, **A** and the cation, **A^{2+}** .
Show your work. (Assume all electronic transitions are possible)

Electronic Configuration of A:

Electronic Configuration of A^{2+} :

c. 2 points

In a single atom, determine how many electrons can have the following set of quantum numbers?

(i) $n = 5; l = 3$

In a single atom, determine how many electrons can have the following set of quantum numbers?

(i) $n = 3; l = 2; m_s = +1/2$

**Question 4** **8 points**

Rank the following from 1 to 4 in **increasing** order. Provide the ranking in the boxes provided.

Note: Atomic number (Z) are provided for relevant atoms

a) Increasing (lowest to highest) order of *ionization energy*: (1= lowest; 4 = highest)

Mg

Mg: Z = 12

Rb

Rb: Z = 37

Cs

Cs: Z = 55

K

K : Z = 19

b) Increasing (lowest to highest) order of *bond angle*: (1= lowest; 4 = highest)

H₂OCCl₄XeF₄

HCN

c) Increasing (smallest to largest) order of *ionic size* (1= smallest; 4 = largest)

S²⁻

S: Z = 16

O²⁻

O: Z = 8

Be²⁺

Be: Z = 4

Na⁺

Na: Z = 11

d) Increasing (lowest to highest) order of *bond energy* (1= lowest; 4 = highest)

P-F

P-Cl

P-I

P-Br

**Question 5** **12 points**

a. 9 points For the molecules HClO_4 and TeF_6 , draw the most stable *Lewis structure* showing appropriate shape and *dashed/wedged bonds*, based on VSEPR. Determine the **molecular geometry** around the central atom. For HClO_4 , label the *most polar bond*.

HClO_4 (there are no O-O bonds)

Label the most polar bond

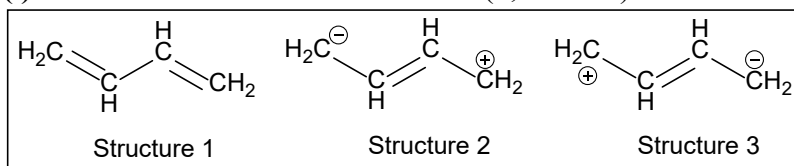
Molecular Geometry around Cl

TeF_6

Molecular Geometry around Te

b. 3 points (Note: Lone pairs are not shown in structures below)

(i) For the three *resonance* structures (1, 2 and 3) shown below, answer the following question:



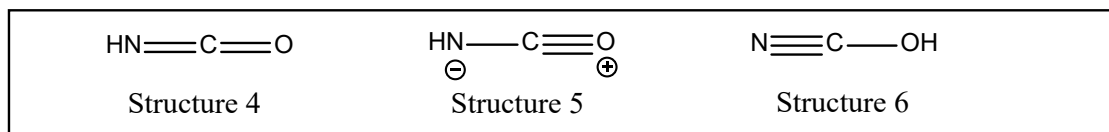
(I) Circle the *most contributing* resonance structure.

Structure 1

Structure 2

Structure 3

(ii) For the three structures given (4, 5, and 6) below, answer the following questions:



(I) Circle all the structures that represent *resonance structures* of each other:

Structure 4

Structure 5

Structure 6

(II) Among the resonance structures, circle the *most contributing* resonance structure.

Structure 4

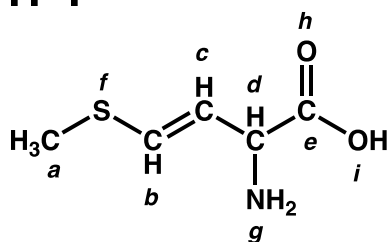
Structure 5

Structure 6

**Question 6** **6 points**

Using the data given below, calculate the **lattice energy** for SrO(s).

	ΔH° (kJ/mol)
$\text{Sr}(s) \longrightarrow \text{Sr}(g)$	+180
$\text{Sr}(g) \longrightarrow \text{Sr}^{2+}(g) + 2e^-$	+1468
$\text{O}_2(g) \longrightarrow 2\text{O}(g)$	+500
$\text{O}(g) + 2e^- \longrightarrow \text{O}^{2-}(g)$	+600
$\text{Sr}(s) + \frac{1}{2}\text{O}_2(g) \longrightarrow \text{SrO}(s)$	-550

**Question 7 10 points**

Consider the molecule given on the left, and answer questions (i) to (vi) using chemical bonding theories that we have learnt in class (valence bond theory, hybridization).

Note that the lone pairs are not shown in the structure:

i. Consider the carbon atoms marked in as ***a***, ***b***, ***c***, ***d***, and ***e*** in the molecule, and determine **hybridization** at each of the carbon centers. (2.5 points)

Atom	Hybridization
Carbon (<i>a</i>)	
Carbon (<i>b</i>)	
Carbon (<i>c</i>)	
Carbon (<i>d</i>)	
Carbon (<i>e</i>)	

ii. Now, consider the sulfur atom (S) marked as ***f***, nitrogen (N) atom (***g***), and oxygen(O) atoms (***h*** and ***i***), and determine their **hybridization**. (2 points)

Atom	Hybridization
Sulfur (<i>f</i>)	
Nitrogen (<i>g</i>)	
Oxygen (<i>h</i>)	
Oxygen (<i>i</i>)	

iii. **Describe** all the bonds formed by the carbon atoms ***a*** and ***b***, using sigma and pi designation as appropriate. (2 points)

Atom	Bonds formed (use sigma and pi as appropriate)
Carbon (<i>a</i>)	
Carbon (<i>b</i>)	

iv. How many total sp^3 -s sigma bonds are in the molecule? (1 point)

v. Determine **electronic** and **molecular geometry** at carbon (C) labeled ***e*** and sulfur (S) labeled ***f*** centres. (2 points)

Atom	Electronic Geometry	Molecular Geometry
Carbon (<i>e</i>)		
Sulfur (<i>f</i>)		

vi. Circle the value from the choices given below, that describes the **C-S-C bond angle at sulfur** (S) marked ***f***: (0.5 points)

>90° to <109.5° 109.5° >109.5 to <120° 120° >120° to <180°


Question 8 12 points

Using the **valence shell orbitals only** draw a **complete** molecular orbital (MO) diagram (with no s-p mixing) for the molecule **PS**, in the space provided below. **(6 points)**

Assume that 3s and 3p atomic orbitals will form molecular orbitals like those formed by the combination of 2s and 2p atomic orbitals that we learnt in class, and there is no s-p mixing.

Show **energy arrow**, **atomic orbitals** involved, **atom labels**, the **molecular orbitals** that are formed, **filled electrons** in the MO diagram etc. Make sure to label all the atomic and molecular orbitals.

Using the MO diagram above, answer the following questions:

- (i) Which one of the following will have the highest bond order? Circle the correct choice
 A) PS^{1-} B) PS C) PS^{1+}
- (ii) Which one of the following will have the longest bond length? Circle the correct choice
 A) PS^{1-} B) PS C) PS^{1+}
- (iii) Which of the following will be diamagnetic? Circle the correct choice.
 A) PS^{1-} B) PS C) PS^{1+}
- (iv) Draw Lewis dot structure of the molecule with the shortest bond length (based on the MO diagram) from among the three (PS^{1-} , PS, PS^{1+}), and indicate the bond order calculated from the Lewis dot structure. **(3 points)**

Lewis Dot Structure

Bond Order



B5B695EC-FD3A-4AD6-9EEF-8E51FC88E342

midterm-exam-93388

#105 Page 10 of 10

SCRAP PAGE

**CHEMISTRY 110 –Midterm Exam****Thursday November 7, 2024, 6:30 – 8:30 pm****Examiners: Pallavi Sirjoosingh and Ashok Kakkar**

First name (please write as legibly as possible within the boxes)

--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--

Last name

--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--

Student ID number

--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--

This booklet has 8 questions on 10 pages (The last page can be used as a scrap page).

The staple must NOT be removed from the question booklet. Your intact question booklet must be returned at the end of the exam.

A separate booklet contains a periodic table, and datasheet.

Write your answers directly under the questions in the space provided. Do not write on the QR codes or near the top edges (as it may be cut off).

Please write **legibly and dark enough** for the scan to be readable. For students who use English as a second language, dictionaries are permitted. Calculators are allowed.

DO NOT ASK QUESTIONS OF THE INVIGILATORS about interpretations of the exam questions. If you have a concern about a question, please summarize it at the top of this page. All concerns will be evaluated after the exam. GOOD LUCK!

**Question 1 8 points** (*Assume all transitions are possible*)

a. 6 points Calculate the **energy (in kJ/mol)** required to ionize (completely remove) the electron from the **n=3** level of **He⁺ ion** (He: Helium). Show your work including equations. (3 sig. figs.)

b. 2 points

(I) Select the largest (longest) wavelength from the three choices below:

- i. Wavelength of photon to *completely remove* an electron from *1s orbital of Hydrogen atom*
- ii. Wavelength of photon to *completely remove* an electron from *1s orbital of Boron atom (B)*
- iii. Wavelength of photon to *completely remove* an electron from *1s orbital of Boron cation (B⁺)*

i

or

ii

or

iii

(II) Select the smallest (shortest) wavelength from the three choices below (*for H atom*):

- i. Wavelength of photon *emitted* for an electron transitioning from *2p to 1s orbital*
- ii. Wavelength of photon *absorbed* for an electron transitioning from *2s to 4p orbital*
- iii. Wavelength of photon *absorbed* for an electron transitioning from *1s to 3p orbital*

i

or

ii

or

iii



Question 2 8 points



A Chem110 student plots the graph shown on the left, from data obtained for the kinetic energy (eV) of the ejected electron versus frequency (s^{-1}) of the incident light, for an unknown metal. ($1\text{eV} = 1.6022 \times 10^{-19} \text{ J}$)

Based on the plot, answer the questions **a to c** below:

a. 1 point On the plot above, label the **threshold frequency**.

b. 2 points What is the value of the work potential (in J) of the unknown metal?

c. 3 points What is the *longest* wavelength (in nm) required to eject an electron from this metal's surface? Show your work including equation(s) used. (3 sig figs)

d. 2 points In the space provided below, plot the following

(Assume frequency is *above* threshold frequency):

- i. *Kinetic energy* of an ejected electron versus the *intensity* of the incident light (left)
- ii. *Number of electrons* ejected versus the *frequency* of the incident light (right)



**Question 3** **6 points****a. 2 points**

In the ground state of Ge (Germanium; $Z = 32$) atom, **name** the following orbitals:

i. Orbital containing the electron that requires the maximum energy to be removed.

ii. Orbital containing the electron that experiences the minimum effective nuclear charge.

b. 2 points

An unknown anion (A^-) has the following *excited* electronic configuration: $[Ar] 4s^2 3d^3 4p^1$

(Ar: Argon; $Z = 18$)

Give the **condensed ground state electronic configuration** of the atom, **A** and the cation, **A^{2+}** .
Show your work. (Assume all electronic transitions are possible)

Electronic Configuration of A:

Electronic Configuration of A^{2+} :

c. 2 points

In a single atom, determine how many electrons can have the following set of quantum numbers?

(i) $n = 5; l = 3$

In a single atom, determine how many electrons can have the following set of quantum numbers?

(i) $n = 3; l = 2; m_s = +1/2$

**Question 4** **8 points**

Rank the following from 1 to 4 in **increasing** order. Provide the ranking in the boxes provided.

Note: Atomic number (Z) are provided for relevant atoms

a) Increasing (lowest to highest) order of *ionization energy*: (1= lowest; 4 = highest)

Mg

Mg: Z = 12

Rb

Rb: Z = 37

Cs

Cs: Z = 55

K

K : Z = 19

b) Increasing (lowest to highest) order of *bond angle*: (1= lowest; 4 = highest)

H₂OCCl₄XeF₄

HCN

c) Increasing (smallest to largest) order of *ionic size* (1= smallest; 4 = largest)

S²⁻

S: Z = 16

O²⁻

O: Z = 8

Be²⁺

Be: Z = 4

Na⁺

Na: Z = 11

d) Increasing (lowest to highest) order of *bond energy* (1= lowest; 4 = highest)

P-F

P-Cl

P-I

P-Br

**Question 5** **12 points**

a. 9 points For the molecules HClO_4 and TeF_6 , draw the most stable *Lewis structure* showing appropriate shape and *dashed/wedged bonds*, based on VSEPR. Determine the **molecular geometry** around the central atom. For HClO_4 , label the *most polar bond*.



b. 3 points (Note: Lone pairs are not shown in structures below)

(i) For the three *resonance* structures (1, 2 and 3) shown below, answer the following question:



(I) Circle the *most contributing* resonance structure.

Structure 1

Structure 2

Structure 3

(ii) For the three structures given (4, 5, and 6) below, answer the following questions:



(I) Circle all the structures that represent *resonance structures* of each other:

Structure 4

Structure 5

Structure 6

(II) Among the resonance structures, circle the *most contributing* resonance structure.

Structure 4

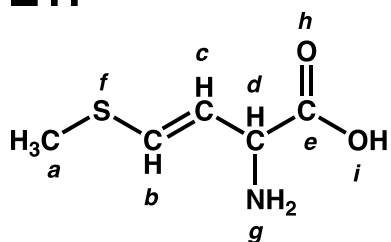
Structure 5

Structure 6

**Question 6** **6 points**

Using the data given below, calculate the **lattice energy** for SrO(s).

	ΔH° (kJ/mol)
$\text{Sr}(s) \longrightarrow \text{Sr}(g)$	+180
$\text{Sr}(g) \longrightarrow \text{Sr}^{2+}(g) + 2e^-$	+1468
$\text{O}_2(g) \longrightarrow 2\text{O}(g)$	+500
$\text{O}(g) + 2e^- \longrightarrow \text{O}^{2-}(g)$	+600
$\text{Sr}(s) + \frac{1}{2}\text{O}_2(g) \longrightarrow \text{SrO}(s)$	-550

**Question 7 10 points**

Consider the molecule given on the left, and answer questions (i) to (vi) using chemical bonding theories that we have learnt in class (valence bond theory, hybridization).

Note that the lone pairs are not shown in the structure:

i. Consider the carbon atoms marked in as ***a***, ***b***, ***c***, ***d***, and ***e*** in the molecule, and determine **hybridization** at each of the carbon centers. (2.5 points)

Atom	Hybridization
Carbon (<i>a</i>)	
Carbon (<i>b</i>)	
Carbon (<i>c</i>)	
Carbon (<i>d</i>)	
Carbon (<i>e</i>)	

ii. Now, consider the sulfur atom (S) marked as ***f***, nitrogen (N) atom (***g***), and oxygen(O) atoms (***h*** and ***i***), and determine their **hybridization**. (2 points)

Atom	Hybridization
Sulfur (<i>f</i>)	
Nitrogen (<i>g</i>)	
Oxygen (<i>h</i>)	
Oxygen (<i>i</i>)	

iii. **Describe** all the bonds formed by the carbon atoms ***a*** and ***b***, using sigma and pi designation as appropriate. (2 points)

Atom	Bonds formed (use sigma and pi as appropriate)
Carbon (<i>a</i>)	
Carbon (<i>b</i>)	

iv. How many total sp^3 -s sigma bonds are in the molecule? (1 point)

v. Determine **electronic** and **molecular geometry** at carbon (C) labeled ***e*** and sulfur (S) labeled ***f*** centres. (2 points)

Atom	Electronic Geometry	Molecular Geometry
Carbon (<i>e</i>)		
Sulfur (<i>f</i>)		

vi. Circle the value from the choices given below, that describes the **C-S-C bond angle at sulfur** (S) marked ***f***: (0.5 points)

>90° to <109.5° 109.5° >109.5 to <120° 120° >120° to <180°

**Question 8 12 points**

Using the **valence shell orbitals only** draw a **complete** molecular orbital (MO) diagram (with no s-p mixing) for the molecule **PS**, in the space provided below. **(6 points)**

Assume that 3s and 3p atomic orbitals will form molecular orbitals like those formed by the combination of 2s and 2p atomic orbitals that we learnt in class, and there is no s-p mixing.

Show **energy arrow**, **atomic orbitals** involved, **atom labels**, the **molecular orbitals** that are formed, **filled electrons** in the MO diagram etc. Make sure to label all the atomic and molecular orbitals.

Using the MO diagram above, answer the following questions:

- (i) Which one of the following will have the highest bond order? Circle the correct choice
A) PS^{1-} B) PS C) PS^{1+}
- (ii) Which one of the following will have the longest bond length? Circle the correct choice
A) PS^{1-} B) PS C) PS^{1+}
- (iii) Which of the following will be diamagnetic? Circle the correct choice.
A) PS^{1-} B) PS C) PS^{1+}
- (iv) Draw Lewis dot structure of the molecule with the shortest bond length (based on the MO diagram) from among the three (PS^{1-} , PS, PS^{1+}), and indicate the bond order calculated from the Lewis dot structure. **(3 points)**

Lewis Dot Structure

Bond Order



181E5EFD-C050-4E3E-8986-DF88810B97F6

midterm-exam-93388

#106 Page 10 of 10

SCRAP PAGE

**CHEMISTRY 110 –Midterm Exam****Thursday November 7, 2024, 6:30 – 8:30 pm****Examiners: Pallavi Sirjoosingh and Ashok Kakkar**

First name (please write as legibly as possible within the boxes)

--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--

Last name

--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--

Student ID number

--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--

This booklet has 8 questions on 10 pages (The last page can be used as a scrap page).

The staple must NOT be removed from the question booklet. Your intact question booklet must be returned at the end of the exam.

A separate booklet contains a periodic table, and datasheet.

Write your answers directly under the questions in the space provided. Do not write on the QR codes or near the top edges (as it may be cut off).

Please write **legibly and dark enough** for the scan to be readable. For students who use English as a second language, dictionaries are permitted. Calculators are allowed.

DO NOT ASK QUESTIONS OF THE INVIGILATORS about interpretations of the exam questions. If you have a concern about a question, please summarize it at the top of this page. All concerns will be evaluated after the exam. GOOD LUCK!

**Question 1 8 points** (*Assume all transitions are possible*)

a. 6 points Calculate the **energy (in kJ/mol)** required to ionize (completely remove) the electron from the **$n=3$** level of **He^+ ion** (He: Helium). Show your work including equations. (3 sig. figs.)

b. 2 points

(I) Select the largest (longest) wavelength from the three choices below:

- i. Wavelength of photon to *completely remove* an electron from *1s orbital of Hydrogen atom*
- ii. Wavelength of photon to *completely remove* an electron from *1s orbital of Boron atom (B)*
- iii. Wavelength of photon to *completely remove* an electron from *1s orbital of Boron cation (B^+)*

i or ii or iii

(II) Select the smallest (shortest) wavelength from the three choices below (*for H atom*):

- i. Wavelength of photon *emitted* for an electron transitioning from *2p to 1s orbital*
- ii. Wavelength of photon *absorbed* for an electron transitioning from *2s to 4p orbital*
- iii. Wavelength of photon *absorbed* for an electron transitioning from *1s to 3p orbital*

i or ii or iii



Question 2 8 points



A Chem110 student plots the graph shown on the left, from data obtained for the kinetic energy (eV) of the ejected electron versus frequency (s^{-1}) of the incident light, for an unknown metal. ($1\text{eV} = 1.6022 \times 10^{-19} \text{ J}$)

Based on the plot, answer the questions **a to c** below:

a. 1 point On the plot above, label the **threshold frequency**.

b. 2 points What is the value of the work potential (in J) of the unknown metal?

c. 3 points What is the *longest* wavelength (in nm) required to eject an electron from this metal's surface? Show your work including equation(s) used. (3 sig figs)

d. 2 points In the space provided below, plot the following

(Assume frequency is *above* threshold frequency):

- Kinetic energy* of an ejected electron versus the *intensity* of the incident light (left)
- Number of electrons* ejected versus the *frequency* of the incident light (right)



**Question 3** **6 points****a. 2 points**

In the ground state of Ge (Germanium; $Z = 32$) atom, **name** the following orbitals:

i. Orbital containing the electron that requires the maximum energy to be removed.

ii. Orbital containing the electron that experiences the minimum effective nuclear charge.

b. 2 points

An unknown anion (A^-) has the following *excited* electronic configuration: $[Ar] 4s^2 3d^3 4p^1$

(Ar: Argon; $Z = 18$)

Give the **condensed ground state electronic configuration** of the atom, **A** and the cation, **A^{2+}** .
Show your work. (Assume all electronic transitions are possible)

Electronic Configuration of A:

Electronic Configuration of A^{2+} :

c. 2 points

In a single atom, determine how many electrons can have the following set of quantum numbers?

(i) $n = 5; l = 3$

In a single atom, determine how many electrons can have the following set of quantum numbers?

(i) $n = 3; l = 2; m_s = +1/2$

**Question 4** **8 points**

Rank the following from 1 to 4 in **increasing** order. Provide the ranking in the boxes provided.

Note: Atomic number (Z) are provided for relevant atoms

a) Increasing (lowest to highest) order of *ionization energy*: (1= lowest; 4 = highest)

Mg

Mg: Z = 12

Rb

Rb: Z = 37

Cs

Cs: Z = 55

K

K : Z = 19

b) Increasing (lowest to highest) order of *bond angle*: (1= lowest; 4 = highest)

H₂OCCl₄XeF₄

HCN

c) Increasing (smallest to largest) order of *ionic size* (1= smallest; 4 = largest)

S²⁻

S: Z = 16

O²⁻

O: Z = 8

Be²⁺

Be: Z = 4

Na⁺

Na: Z = 11

d) Increasing (lowest to highest) order of *bond energy* (1= lowest; 4 = highest)

P-F

P-Cl

P-I

P-Br

**Question 5** **12 points**

a. 9 points For the molecules HClO_4 and TeF_6 , draw the most stable *Lewis structure* showing appropriate shape and *dashed/wedged bonds*, based on VSEPR. Determine the **molecular geometry** around the central atom. For HClO_4 , label the *most polar bond*.



b. 3 points (Note: Lone pairs are not shown in structures below)

(i) For the three *resonance* structures (1, 2 and 3) shown below, answer the following question:



(I) Circle the *most contributing* resonance structure.

Structure 1

Structure 2

Structure 3

(ii) For the three structures given (4, 5, and 6) below, answer the following questions:



(I) Circle all the structures that represent *resonance structures* of each other:

Structure 4

Structure 5

Structure 6

(II) Among the resonance structures, circle the *most contributing* resonance structure.

Structure 4

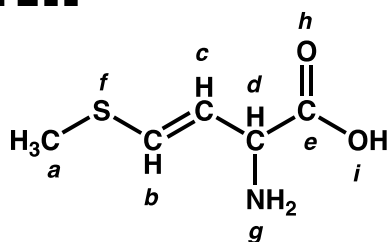
Structure 5

Structure 6

**Question 6** **6 points**

Using the data given below, calculate the **lattice energy** for SrO(s).

	ΔH° (kJ/mol)
$\text{Sr}(s) \longrightarrow \text{Sr}(g)$	+180
$\text{Sr}(g) \longrightarrow \text{Sr}^{2+}(g) + 2e^-$	+1468
$\text{O}_2(g) \longrightarrow 2\text{O}(g)$	+500
$\text{O}(g) + 2e^- \longrightarrow \text{O}^{2-}(g)$	+600
$\text{Sr}(s) + \frac{1}{2}\text{O}_2(g) \longrightarrow \text{SrO}(s)$	-550

**Question 7 10 points**

Consider the molecule given on the left, and answer questions (i) to (vi) using chemical bonding theories that we have learnt in class (valence bond theory, hybridization).

Note that the lone pairs are not shown in the structure:

i. Consider the carbon atoms marked in as ***a***, ***b***, ***c***, ***d***, and ***e*** in the molecule, and determine **hybridization** at each of the carbon centers. (2.5 points)

Atom	Hybridization
Carbon (<i>a</i>)	
Carbon (<i>b</i>)	
Carbon (<i>c</i>)	
Carbon (<i>d</i>)	
Carbon (<i>e</i>)	

ii. Now, consider the sulfur atom (S) marked as ***f***, nitrogen (N) atom (***g***), and oxygen(O) atoms (***h*** and ***i***), and determine their **hybridization**. (2 points)

Atom	Hybridization
Sulfur (<i>f</i>)	
Nitrogen (<i>g</i>)	
Oxygen (<i>h</i>)	
Oxygen (<i>i</i>)	

iii. **Describe** all the bonds formed by the carbon atoms ***a*** and ***b***, using sigma and pi designation as appropriate. (2 points)

Atom	Bonds formed (use sigma and pi as appropriate)
Carbon (<i>a</i>)	
Carbon (<i>b</i>)	

iv. How many total sp^3 -s sigma bonds are in the molecule? (1 point)

v. Determine **electronic** and **molecular geometry** at carbon (C) labeled ***e*** and sulfur (S) labeled ***f*** centres. (2 points)

Atom	Electronic Geometry	Molecular Geometry
Carbon (<i>e</i>)		
Sulfur (<i>f</i>)		

vi. Circle the value from the choices given below, that describes the **C-S-C bond angle at sulfur** (S) marked ***f***: (0.5 points)

>90° to <109.5° 109.5° >109.5 to <120° 120° >120° to <180°


Question 8 12 points

Using the **valence shell orbitals only** draw a **complete** molecular orbital (MO) diagram (with no s-p mixing) for the molecule **PS**, in the space provided below. **(6 points)**

Assume that 3s and 3p atomic orbitals will form molecular orbitals like those formed by the combination of 2s and 2p atomic orbitals that we learnt in class, and there is no s-p mixing.

Show **energy arrow**, **atomic orbitals** involved, **atom labels**, the **molecular orbitals** that are formed, **filled electrons** in the MO diagram etc. Make sure to label all the atomic and molecular orbitals.

Using the MO diagram above, answer the following questions:

- (i) Which one of the following will have the highest bond order? Circle the correct choice
 A) PS^{1-} B) PS C) PS^{1+}
- (ii) Which one of the following will have the longest bond length? Circle the correct choice
 A) PS^{1-} B) PS C) PS^{1+}
- (iii) Which of the following will be diamagnetic? Circle the correct choice.
 A) PS^{1-} B) PS C) PS^{1+}
- (iv) Draw Lewis dot structure of the molecule with the shortest bond length (based on the MO diagram) from among the three (PS^{1-} , PS, PS^{1+}), and indicate the bond order calculated from the Lewis dot structure. **(3 points)**

Lewis Dot Structure

Bond Order



050E7AE8-32AE-41B8-8E6B-1A1B5E379E76

midterm-exam-93388

#107 Page 10 of 10

SCRAP PAGE

**CHEMISTRY 110 –Midterm Exam****Thursday November 7, 2024, 6:30 – 8:30 pm**

Examiners: Pallavi Sirjoosingh and Ashok Kakkar

First name (please write as legibly as possible within the boxes)

--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--

Last name

--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--

Student ID number

--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--

This booklet has 8 questions on 10 pages (The last page can be used as a scrap page).

The staple must NOT be removed from the question booklet. Your intact question booklet must be returned at the end of the exam.

A separate booklet contains a periodic table, and datasheet.

Write your answers directly under the questions in the space provided. Do not write on the QR codes or near the top edges (as it may be cut off).

Please write **legibly and dark enough** for the scan to be readable. For students who use English as a second language, dictionaries are permitted. Calculators are allowed.

DO NOT ASK QUESTIONS OF THE INVIGILATORS about interpretations of the exam questions. If you have a concern about a question, please summarize it at the top of this page. All concerns will be evaluated after the exam. GOOD LUCK!

**Question 1 8 points** (*Assume all transitions are possible*)

a. 6 points Calculate the **energy (in kJ/mol)** required to ionize (completely remove) the electron from the **n=3** level of **He⁺ ion** (He: Helium). Show your work including equations. (3 sig. figs.)

b. 2 points

(I) Select the largest (longest) wavelength from the three choices below:

- i. Wavelength of photon to *completely remove* an electron from *1s orbital of Hydrogen atom*
- ii. Wavelength of photon to *completely remove* an electron from *1s orbital of Boron atom (B)*
- iii. Wavelength of photon to *completely remove* an electron from *1s orbital of Boron cation (B⁺)*

i or ii or iii

(II) Select the smallest (shortest) wavelength from the three choices below (*for H atom*):

- i. Wavelength of photon *emitted* for an electron transitioning from *2p to 1s orbital*
- ii. Wavelength of photon *absorbed* for an electron transitioning from *2s to 4p orbital*
- iii. Wavelength of photon *absorbed* for an electron transitioning from *1s to 3p orbital*

i or ii or iii



Question 2 8 points



A Chem110 student plots the graph shown on the left, from data obtained for the kinetic energy (eV) of the ejected electron versus frequency (s^{-1}) of the incident light, for an unknown metal. ($1\text{eV} = 1.6022 \times 10^{-19} \text{ J}$)

Based on the plot, answer the questions **a to c** below:

a. 1 point On the plot above, label the **threshold frequency**.

b. 2 points What is the value of the work potential (in J) of the unknown metal?

c. 3 points What is the *longest* wavelength (in nm) required to eject an electron from this metal's surface? Show your work including equation(s) used. (3 sig figs)

d. 2 points In the space provided below, plot the following

(Assume frequency is *above* threshold frequency):

- i. *Kinetic energy* of an ejected electron versus the *intensity* of the incident light (left)
- ii. *Number of electrons* ejected versus the *frequency* of the incident light (right)



**Question 3** **6 points****a. 2 points**

In the ground state of Ge (Germanium; $Z = 32$) atom, **name** the following orbitals:

i. Orbital containing the electron that requires the maximum energy to be removed.

ii. Orbital containing the electron that experiences the minimum effective nuclear charge.

b. 2 points

An unknown anion (A^-) has the following *excited* electronic configuration: $[Ar] 4s^2 3d^3 4p^1$

(Ar: Argon; $Z = 18$)

Give the **condensed ground state electronic configuration** of the atom, **A** and the cation, **A^{2+}** .
Show your work. (Assume all electronic transitions are possible)

Electronic Configuration of A:

Electronic Configuration of A^{2+} :

c. 2 points

In a single atom, determine how many electrons can have the following set of quantum numbers?

(i) $n = 5; l = 3$

In a single atom, determine how many electrons can have the following set of quantum numbers?

(i) $n = 3; l = 2; m_s = +1/2$

**Question 4** **8 points**

Rank the following from 1 to 4 in **increasing** order. Provide the ranking in the boxes provided.

Note: Atomic number (Z) are provided for relevant atoms

a) Increasing (lowest to highest) order of *ionization energy*: (1= lowest; 4 = highest)

Mg

Mg: Z = 12

Rb

Rb: Z = 37

Cs

Cs: Z = 55

K

K : Z = 19

b) Increasing (lowest to highest) order of *bond angle*: (1= lowest; 4 = highest)

H₂OCCl₄XeF₄

HCN

c) Increasing (smallest to largest) order of *ionic size* (1= smallest; 4 = largest)

S²⁻

S: Z = 16

O²⁻

O: Z = 8

Be²⁺

Be: Z = 4

Na⁺

Na: Z = 11

d) Increasing (lowest to highest) order of *bond energy* (1= lowest; 4 = highest)

P-F

P-Cl

P-I

P-Br

**Question 5** **12 points**

a. 9 points For the molecules HClO_4 and TeF_6 , draw the most stable *Lewis structure* showing appropriate shape and *dashed/wedged bonds*, based on VSEPR. Determine the **molecular geometry** around the central atom. For HClO_4 , label the *most polar bond*.



b. 3 points (Note: Lone pairs are not shown in structures below)

(i) For the three *resonance* structures (1, 2 and 3) shown below, answer the following question:



(I) Circle the *most contributing* resonance structure.

Structure 1

Structure 2

Structure 3

(ii) For the three structures given (4, 5, and 6) below, answer the following questions:



(I) Circle all the structures that represent *resonance structures* of each other:

Structure 4

Structure 5

Structure 6

(II) Among the resonance structures, circle the *most contributing* resonance structure.

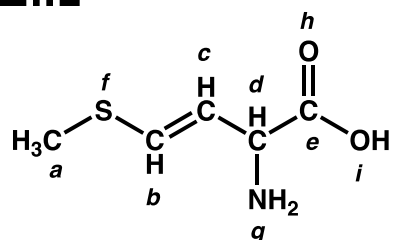
Structure 4

Structure 5

Structure 6

**Question 6** **6 points**Using the data given below, calculate the **lattice energy** for SrO(s).

	ΔH° (kJ/mol)
$\text{Sr}(s) \longrightarrow \text{Sr}(g)$	+180
$\text{Sr}(g) \longrightarrow \text{Sr}^{2+}(g) + 2e^-$	+1468
$\text{O}_2(g) \longrightarrow 2\text{O}(g)$	+500
$\text{O}(g) + 2e^- \longrightarrow \text{O}^{2-}(g)$	+600
$\text{Sr}(s) + \frac{1}{2}\text{O}_2(g) \longrightarrow \text{SrO}(s)$	-550

**Question 7 10 points**

Consider the molecule given on the left, and answer questions (i) to (vi) using chemical bonding theories that we have learnt in class (valence bond theory, hybridization).

Note that the lone pairs are not shown in the structure:

i. Consider the carbon atoms marked in as ***a***, ***b***, ***c***, ***d***, and ***e*** in the molecule, and determine **hybridization** at each of the carbon centers. (2.5 points)

Atom	Hybridization
Carbon (<i>a</i>)	
Carbon (<i>b</i>)	
Carbon (<i>c</i>)	
Carbon (<i>d</i>)	
Carbon (<i>e</i>)	

ii. Now, consider the sulfur atom (S) marked as ***f***, nitrogen (N) atom (***g***), and oxygen(O) atoms (***h*** and ***i***), and determine their **hybridization**. (2 points)

Atom	Hybridization
Sulfur (<i>f</i>)	
Nitrogen (<i>g</i>)	
Oxygen (<i>h</i>)	
Oxygen (<i>i</i>)	

iii. **Describe** all the bonds formed by the carbon atoms ***a*** and ***b***, using sigma and pi designation as appropriate. (2 points)

Atom	Bonds formed (use sigma and pi as appropriate)
Carbon (<i>a</i>)	
Carbon (<i>b</i>)	

iv. How many total sp^3 -s sigma bonds are in the molecule? (1 point)

v. Determine **electronic** and **molecular geometry** at carbon (C) labeled ***e*** and sulfur (S) labeled ***f*** centres. (2 points)

Atom	Electronic Geometry	Molecular Geometry
Carbon (<i>e</i>)		
Sulfur (<i>f</i>)		

vi. Circle the value from the choices given below, that describes the **C-S-C bond angle at sulfur** (S) marked ***f***: (0.5 points)

>90° to <109.5° 109.5° >109.5 to <120° 120° >120° to <180°

**Question 8 12 points**

Using the **valence shell orbitals only** draw a **complete** molecular orbital (MO) diagram (with no s-p mixing) for the molecule **PS**, in the space provided below. **(6 points)**

Assume that 3s and 3p atomic orbitals will form molecular orbitals like those formed by the combination of 2s and 2p atomic orbitals that we learnt in class, and there is no s-p mixing.

Show **energy arrow**, **atomic orbitals** involved, **atom labels**, the **molecular orbitals** that are formed, **filled electrons** in the MO diagram etc. Make sure to label all the atomic and molecular orbitals.

Using the MO diagram above, answer the following questions:

- (i) Which one of the following will have the highest bond order? Circle the correct choice
A) PS^{1-} B) PS C) PS^{1+}
- (ii) Which one of the following will have the longest bond length? Circle the correct choice
A) PS^{1-} B) PS C) PS^{1+}
- (iii) Which of the following will be diamagnetic? Circle the correct choice.
A) PS^{1-} B) PS C) PS^{1+}
- (iv) Draw Lewis dot structure of the molecule with the shortest bond length (based on the MO diagram) from among the three (PS^{1-} , PS, PS^{1+}), and indicate the bond order calculated from the Lewis dot structure. **(3 points)**

Lewis Dot Structure

Bond Order



CBE1C5D9-C11B-4CD6-822B-D1D79B367B4C

midterm-exam-93388

#108 Page 10 of 10

SCRAP PAGE

**CHEMISTRY 110 –Midterm Exam****Thursday November 7, 2024, 6:30 – 8:30 pm**

Examiners: Pallavi Sirjoosingh and Ashok Kakkar

First name (please write as legibly as possible within the boxes)

--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--

Last name

--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--

Student ID number

--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--

This booklet has 8 questions on 10 pages (The last page can be used as a scrap page).

The staple must NOT be removed from the question booklet. Your intact question booklet must be returned at the end of the exam.

A separate booklet contains a periodic table, and datasheet.

Write your answers directly under the questions in the space provided. Do not write on the QR codes or near the top edges (as it may be cut off).

Please write **legibly and dark enough** for the scan to be readable. For students who use English as a second language, dictionaries are permitted. Calculators are allowed.

DO NOT ASK QUESTIONS OF THE INVIGILATORS about interpretations of the exam questions. If you have a concern about a question, please summarize it at the top of this page. All concerns will be evaluated after the exam. GOOD LUCK!

**Question 1 8 points** (*Assume all transitions are possible*)

a. 6 points Calculate the **energy (in kJ/mol)** required to ionize (completely remove) the electron from the **n=3** level of **He⁺ ion** (He: Helium). Show your work including equations. (3 sig. figs.)

b. 2 points

(I) Select the largest (longest) wavelength from the three choices below:

- i. Wavelength of photon to *completely remove* an electron from *1s orbital of Hydrogen atom*
- ii. Wavelength of photon to *completely remove* an electron from *1s orbital of Boron atom (B)*
- iii. Wavelength of photon to *completely remove* an electron from *1s orbital of Boron cation (B⁺)*

i or ii or iii

(II) Select the smallest (shortest) wavelength from the three choices below (*for H atom*):

- i. Wavelength of photon *emitted* for an electron transitioning from *2p to 1s orbital*
- ii. Wavelength of photon *absorbed* for an electron transitioning from *2s to 4p orbital*
- iii. Wavelength of photon *absorbed* for an electron transitioning from *1s to 3p orbital*

i or ii or iii



Question 2 8 points



A Chem110 student plots the graph shown on the left, from data obtained for the kinetic energy (eV) of the ejected electron versus frequency (s^{-1}) of the incident light, for an unknown metal. ($1\text{eV} = 1.6022 \times 10^{-19} \text{ J}$)

Based on the plot, answer the questions **a to c** below:

a. 1 point On the plot above, label the **threshold frequency**.

b. 2 points What is the value of the work potential (in J) of the unknown metal?

c. 3 points What is the *longest* wavelength (in nm) required to eject an electron from this metal's surface? Show your work including equation(s) used. (3 sig figs)

d. 2 points In the space provided below, plot the following

(Assume frequency is *above* threshold frequency):

- i. *Kinetic energy* of an ejected electron versus the *intensity* of the incident light (left)
- ii. *Number of electrons* ejected versus the *frequency* of the incident light (right)



**Question 3** **6 points****a. 2 points**

In the ground state of Ge (Germanium; $Z = 32$) atom, **name** the following orbitals:

i. Orbital containing the electron that requires the maximum energy to be removed.

ii. Orbital containing the electron that experiences the minimum effective nuclear charge.

b. 2 points

An unknown anion (A^-) has the following *excited* electronic configuration: $[Ar] 4s^2 3d^3 4p^1$

(Ar: Argon; $Z = 18$)

Give the **condensed ground state electronic configuration** of the atom, **A** and the cation, **A^{2+}** .
Show your work. (Assume all electronic transitions are possible)

Electronic Configuration of A:

Electronic Configuration of A^{2+} :

c. 2 points

In a single atom, determine how many electrons can have the following set of quantum numbers?

(i) $n = 5; l = 3$

In a single atom, determine how many electrons can have the following set of quantum numbers?

(i) $n = 3; l = 2; m_s = +1/2$

**Question 4** **8 points**

Rank the following from 1 to 4 in **increasing** order. Provide the ranking in the boxes provided.

Note: Atomic number (Z) are provided for relevant atoms

a) Increasing (lowest to highest) order of *ionization energy*: (1= lowest; 4 = highest)

Mg

Mg: Z = 12

Rb

Rb: Z = 37

Cs

Cs: Z = 55

K

K : Z = 19

b) Increasing (lowest to highest) order of *bond angle*: (1= lowest; 4 = highest)

H₂OCCl₄XeF₄

HCN

c) Increasing (smallest to largest) order of *ionic size* (1= smallest; 4 = largest)

S²⁻

S: Z = 16

O²⁻

O: Z = 8

Be²⁺

Be: Z = 4

Na⁺

Na: Z = 11

d) Increasing (lowest to highest) order of *bond energy* (1= lowest; 4 = highest)

P-F

P-Cl

P-I

P-Br

**Question 5** **12 points**

a. 9 points For the molecules HClO_4 and TeF_6 , draw the most stable *Lewis structure* showing appropriate shape and *dashed/wedged bonds*, based on VSEPR. Determine the **molecular geometry** around the central atom. For HClO_4 , label the *most polar bond*.



b. 3 points (Note: Lone pairs are not shown in structures below)

(i) For the three *resonance* structures (1, 2 and 3) shown below, answer the following question:



(I) Circle the *most contributing* resonance structure.

Structure 1

Structure 2

Structure 3

(ii) For the three structures given (4, 5, and 6) below, answer the following questions:



(I) Circle all the structures that represent *resonance structures* of each other:

Structure 4

Structure 5

Structure 6

(II) Among the resonance structures, circle the *most contributing* resonance structure.

Structure 4

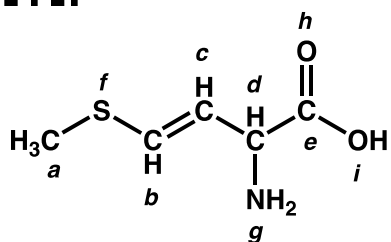
Structure 5

Structure 6

**Question 6** **6 points**

Using the data given below, calculate the **lattice energy** for SrO(s).

	ΔH° (kJ/mol)
$\text{Sr}(s) \longrightarrow \text{Sr}(g)$	+180
$\text{Sr}(g) \longrightarrow \text{Sr}^{2+}(g) + 2e^-$	+1468
$\text{O}_2(g) \longrightarrow 2\text{O}(g)$	+500
$\text{O}(g) + 2e^- \longrightarrow \text{O}^{2-}(g)$	+600
$\text{Sr}(s) + \frac{1}{2}\text{O}_2(g) \longrightarrow \text{SrO}(s)$	-550

**Question 7 10 points**

Consider the molecule given on the left, and answer questions (i) to (vi) using chemical bonding theories that we have learnt in class (valence bond theory, hybridization).

Note that the lone pairs are not shown in the structure:

i. Consider the carbon atoms marked in as ***a***, ***b***, ***c***, ***d***, and ***e*** in the molecule, and determine **hybridization** at each of the carbon centers. (2.5 points)

Atom	Hybridization
Carbon (<i>a</i>)	
Carbon (<i>b</i>)	
Carbon (<i>c</i>)	
Carbon (<i>d</i>)	
Carbon (<i>e</i>)	

ii. Now, consider the sulfur atom (S) marked as ***f***, nitrogen (N) atom (***g***), and oxygen(O) atoms (***h*** and ***i***), and determine their **hybridization**. (2 points)

Atom	Hybridization
Sulfur (<i>f</i>)	
Nitrogen (<i>g</i>)	
Oxygen (<i>h</i>)	
Oxygen (<i>i</i>)	

iii. **Describe** all the bonds formed by the carbon atoms ***a*** and ***b***, using sigma and pi designation as appropriate. (2 points)

Atom	Bonds formed (use sigma and pi as appropriate)
Carbon (<i>a</i>)	
Carbon (<i>b</i>)	

iv. How many total sp^3 -s sigma bonds are in the molecule? (1 point)

v. Determine **electronic** and **molecular geometry** at carbon (C) labeled ***e*** and sulfur (S) labeled ***f*** centres. (2 points)

Atom	Electronic Geometry	Molecular Geometry
Carbon (<i>e</i>)		
Sulfur (<i>f</i>)		

vi. Circle the value from the choices given below, that describes the **C-S-C bond angle at sulfur** (S) marked ***f***: (0.5 points)

>90° to <109.5° 109.5° >109.5 to <120° 120° >120° to <180°


Question 8 12 points

Using the **valence shell orbitals only** draw a **complete** molecular orbital (MO) diagram (with no s-p mixing) for the molecule **PS**, in the space provided below. **(6 points)**

Assume that 3s and 3p atomic orbitals will form molecular orbitals like those formed by the combination of 2s and 2p atomic orbitals that we learnt in class, and there is no s-p mixing.

Show **energy arrow**, **atomic orbitals** involved, **atom labels**, the **molecular orbitals** that are formed, **filled electrons** in the MO diagram etc. Make sure to label all the atomic and molecular orbitals.

Using the MO diagram above, answer the following questions:

- (i) Which one of the following will have the highest bond order? Circle the correct choice
 A) PS^{1-} B) PS C) PS^{1+}
- (ii) Which one of the following will have the longest bond length? Circle the correct choice
 A) PS^{1-} B) PS C) PS^{1+}
- (iii) Which of the following will be diamagnetic? Circle the correct choice.
 A) PS^{1-} B) PS C) PS^{1+}
- (iv) Draw Lewis dot structure of the molecule with the shortest bond length (based on the MO diagram) from among the three (PS^{1-} , PS, PS^{1+}), and indicate the bond order calculated from the Lewis dot structure. **(3 points)**

Lewis Dot Structure

Bond Order



D2F7FF3A-43E9-4142-B910-CFAC0BFF0FE7

midterm-exam-93388

#109 Page 10 of 10

SCRAP PAGE

**CHEMISTRY 110 –Midterm Exam****Thursday November 7, 2024, 6:30 – 8:30 pm****Examiners: Pallavi Sirjoosingh and Ashok Kakkar**

First name (please write as legibly as possible within the boxes)

--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--

Last name

--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--

Student ID number

--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--

This booklet has 8 questions on 10 pages (The last page can be used as a scrap page).

The staple must NOT be removed from the question booklet. Your intact question booklet must be returned at the end of the exam.

A separate booklet contains a periodic table, and datasheet.

Write your answers directly under the questions in the space provided. Do not write on the QR codes or near the top edges (as it may be cut off).

Please write **legibly and dark enough** for the scan to be readable. For students who use English as a second language, dictionaries are permitted. Calculators are allowed.

DO NOT ASK QUESTIONS OF THE INVIGILATORS about interpretations of the exam questions. If you have a concern about a question, please summarize it at the top of this page. All concerns will be evaluated after the exam. GOOD LUCK!

**Question 1 8 points** (*Assume all transitions are possible*)

a. 6 points Calculate the **energy (in kJ/mol)** required to ionize (completely remove) the electron from the **n=3** level of **He⁺ ion** (He: Helium). Show your work including equations. (3 sig. figs.)

b. 2 points

(I) Select the largest (longest) wavelength from the three choices below:

- i. Wavelength of photon to *completely remove* an electron from *1s orbital of Hydrogen atom*
- ii. Wavelength of photon to *completely remove* an electron from *1s orbital of Boron atom (B)*
- iii. Wavelength of photon to *completely remove* an electron from *1s orbital of Boron cation (B⁺)*

i or ii or iii

(II) Select the smallest (shortest) wavelength from the three choices below (*for H atom*):

- i. Wavelength of photon *emitted* for an electron transitioning from *2p to 1s orbital*
- ii. Wavelength of photon *absorbed* for an electron transitioning from *2s to 4p orbital*
- iii. Wavelength of photon *absorbed* for an electron transitioning from *1s to 3p orbital*

i or ii or iii



Question 2 8 points



A Chem110 student plots the graph shown on the left, from data obtained for the kinetic energy (eV) of the ejected electron versus frequency (s^{-1}) of the incident light, for an unknown metal. ($1\text{eV} = 1.6022 \times 10^{-19} \text{ J}$)

Based on the plot, answer the questions **a to c** below:

a. 1 point On the plot above, label the **threshold frequency**.

b. 2 points What is the value of the work potential (in J) of the unknown metal?

c. 3 points What is the *longest* wavelength (in nm) required to eject an electron from this metal's surface? Show your work including equation(s) used. (3 sig figs)

d. 2 points In the space provided below, plot the following

(Assume frequency is *above* threshold frequency):

- Kinetic energy* of an ejected electron versus the *intensity* of the incident light (left)
- Number of electrons* ejected versus the *frequency* of the incident light (right)



**Question 3** **6 points****a. 2 points**

In the ground state of Ge (Germanium; $Z = 32$) atom, **name** the following orbitals:

i. Orbital containing the electron that requires the maximum energy to be removed.

ii. Orbital containing the electron that experiences the minimum effective nuclear charge.

b. 2 points

An unknown anion (A^-) has the following *excited* electronic configuration: $[Ar] 4s^2 3d^3 4p^1$

(Ar: Argon; $Z = 18$)

Give the **condensed ground state electronic configuration** of the atom, **A** and the cation, **A^{2+}** .
Show your work. (Assume all electronic transitions are possible)

Electronic Configuration of A:

Electronic Configuration of A^{2+} :

c. 2 points

In a single atom, determine how many electrons can have the following set of quantum numbers?

(i) $n = 5; l = 3$

In a single atom, determine how many electrons can have the following set of quantum numbers?

(i) $n = 3; l = 2; m_s = +1/2$

**Question 4** **8 points**

Rank the following from 1 to 4 in **increasing** order. Provide the ranking in the boxes provided.

Note: Atomic number (Z) are provided for relevant atoms

a) Increasing (lowest to highest) order of *ionization energy*: (1= lowest; 4 = highest)

Mg

Mg: Z = 12

Rb

Rb: Z = 37

Cs

Cs: Z = 55

K

K : Z = 19

b) Increasing (lowest to highest) order of *bond angle*: (1= lowest; 4 = highest)

H₂OCCl₄XeF₄

HCN

c) Increasing (smallest to largest) order of *ionic size* (1= smallest; 4 = largest)

S²⁻

S: Z = 16

O²⁻

O: Z = 8

Be²⁺

Be: Z = 4

Na⁺

Na: Z = 11

d) Increasing (lowest to highest) order of *bond energy* (1= lowest; 4 = highest)

P-F

P-Cl

P-I

P-Br

**Question 5** **12 points**

a. 9 points For the molecules HClO_4 and TeF_6 , draw the most stable *Lewis structure* showing appropriate shape and *dashed/wedged bonds*, based on VSEPR. Determine the **molecular geometry** around the central atom. For HClO_4 , label the *most polar bond*.



b. 3 points (Note: Lone pairs are not shown in structures below)

(i) For the three *resonance* structures (1, 2 and 3) shown below, answer the following question:



(I) Circle the *most contributing* resonance structure.

Structure 1

Structure 2

Structure 3

(ii) For the three structures given (4, 5, and 6) below, answer the following questions:



(I) Circle all the structures that represent *resonance structures* of each other:

Structure 4

Structure 5

Structure 6

(II) Among the resonance structures, circle the *most contributing* resonance structure.

Structure 4

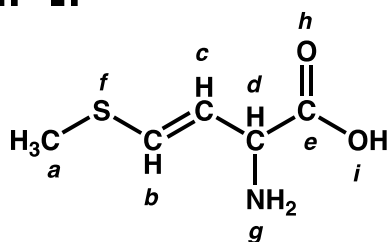
Structure 5

Structure 6

**Question 6** **6 points**

Using the data given below, calculate the **lattice energy** for SrO(s).

	ΔH° (kJ/mol)
$\text{Sr}(s) \longrightarrow \text{Sr}(g)$	+180
$\text{Sr}(g) \longrightarrow \text{Sr}^{2+}(g) + 2e^-$	+1468
$\text{O}_2(g) \longrightarrow 2\text{O}(g)$	+500
$\text{O}(g) + 2e^- \longrightarrow \text{O}^{2-}(g)$	+600
$\text{Sr}(s) + \frac{1}{2}\text{O}_2(g) \longrightarrow \text{SrO}(s)$	-550

**Question 7 10 points**

Consider the molecule given on the left, and answer questions (i) to (vi) using chemical bonding theories that we have learnt in class (valence bond theory, hybridization).

Note that the lone pairs are not shown in the structure:

i. Consider the carbon atoms marked in as ***a***, ***b***, ***c***, ***d***, and ***e*** in the molecule, and determine **hybridization** at each of the carbon centers. (2.5 points)

Atom	Hybridization
Carbon (<i>a</i>)	
Carbon (<i>b</i>)	
Carbon (<i>c</i>)	
Carbon (<i>d</i>)	
Carbon (<i>e</i>)	

ii. Now, consider the sulfur atom (S) marked as ***f***, nitrogen (N) atom (***g***), and oxygen(O) atoms (***h*** and ***i***), and determine their **hybridization**. (2 points)

Atom	Hybridization
Sulfur (<i>f</i>)	
Nitrogen (<i>g</i>)	
Oxygen (<i>h</i>)	
Oxygen (<i>i</i>)	

iii. **Describe** all the bonds formed by the carbon atoms ***a*** and ***b***, using sigma and pi designation as appropriate. (2 points)

Atom	Bonds formed (use sigma and pi as appropriate)
Carbon (<i>a</i>)	
Carbon (<i>b</i>)	

iv. How many total sp^3 -s sigma bonds are in the molecule? (1 point)

v. Determine **electronic** and **molecular geometry** at carbon (C) labeled ***e*** and sulfur (S) labeled ***f*** centres. (2 points)

Atom	Electronic Geometry	Molecular Geometry
Carbon (<i>e</i>)		
Sulfur (<i>f</i>)		

vi. Circle the value from the choices given below, that describes the **C-S-C bond angle at sulfur** (S) marked ***f***: (0.5 points)

>90° to <109.5° 109.5° >109.5 to <120° 120° >120° to <180°

**Question 8 12 points**

Using the **valence shell orbitals only** draw a **complete** molecular orbital (MO) diagram (with no s-p mixing) for the molecule **PS**, in the space provided below. **(6 points)**

Assume that 3s and 3p atomic orbitals will form molecular orbitals like those formed by the combination of 2s and 2p atomic orbitals that we learnt in class, and there is no s-p mixing.

Show **energy arrow**, **atomic orbitals** involved, **atom labels**, the **molecular orbitals** that are formed, **filled electrons** in the MO diagram etc. Make sure to label all the atomic and molecular orbitals.

Using the MO diagram above, answer the following questions:

- (i) Which one of the following will have the highest bond order? Circle the correct choice
A) PS^{1-} B) PS C) PS^{1+}
- (ii) Which one of the following will have the longest bond length? Circle the correct choice
A) PS^{1-} B) PS C) PS^{1+}
- (iii) Which of the following will be diamagnetic? Circle the correct choice.
A) PS^{1-} B) PS C) PS^{1+}
- (iv) Draw Lewis dot structure of the molecule with the shortest bond length (based on the MO diagram) from among the three (PS^{1-} , PS, PS^{1+}), and indicate the bond order calculated from the Lewis dot structure. **(3 points)**

Lewis Dot Structure

Bond Order



731097C4-E33A-4023-96AF-028EE2B2D1E8

midterm-exam-93388

#110 Page 10 of 10

SCRAP PAGE

**CHEMISTRY 110 –Midterm Exam****Thursday November 7, 2024, 6:30 – 8:30 pm****Examiners: Pallavi Sirjoosingh and Ashok Kakkar**

First name (please write as legibly as possible within the boxes)

--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--

Last name

--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--

Student ID number

--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--

This booklet has 8 questions on 10 pages (The last page can be used as a scrap page).

The staple must NOT be removed from the question booklet. Your intact question booklet must be returned at the end of the exam.

A separate booklet contains a periodic table, and datasheet.

Write your answers directly under the questions in the space provided. Do not write on the QR codes or near the top edges (as it may be cut off).

Please write **legibly and dark enough** for the scan to be readable. For students who use English as a second language, dictionaries are permitted. Calculators are allowed.

DO NOT ASK QUESTIONS OF THE INVIGILATORS about interpretations of the exam questions. If you have a concern about a question, please summarize it at the top of this page. All concerns will be evaluated after the exam. GOOD LUCK!

**Question 1 8 points** (*Assume all transitions are possible*)

a. 6 points Calculate the **energy (in kJ/mol)** required to ionize (completely remove) the electron from the **n=3** level of **He⁺ ion** (He: Helium). Show your work including equations. (3 sig. figs.)

b. 2 points

(I) Select the largest (longest) wavelength from the three choices below:

- i. Wavelength of photon to *completely remove* an electron from *1s orbital of Hydrogen atom*
- ii. Wavelength of photon to *completely remove* an electron from *1s orbital of Boron atom (B)*
- iii. Wavelength of photon to *completely remove* an electron from *1s orbital of Boron cation (B⁺)*

i or ii or iii

(II) Select the smallest (shortest) wavelength from the three choices below (*for H atom*):

- i. Wavelength of photon *emitted* for an electron transitioning from *2p to 1s orbital*
- ii. Wavelength of photon *absorbed* for an electron transitioning from *2s to 4p orbital*
- iii. Wavelength of photon *absorbed* for an electron transitioning from *1s to 3p orbital*

i or ii or iii



Question 2 8 points



A Chem110 student plots the graph shown on the left, from data obtained for the kinetic energy (eV) of the ejected electron versus frequency (s^{-1}) of the incident light, for an unknown metal. ($1\text{eV} = 1.6022 \times 10^{-19} \text{ J}$)

Based on the plot, answer the questions **a to c** below:

a. 1 point On the plot above, label the **threshold frequency**.

b. 2 points What is the value of the work potential (in J) of the unknown metal?

c. 3 points What is the *longest* wavelength (in nm) required to eject an electron from this metal's surface? Show your work including equation(s) used. (3 sig figs)

d. 2 points In the space provided below, plot the following

(Assume frequency is *above* threshold frequency):

- Kinetic energy* of an ejected electron versus the *intensity* of the incident light (left)
- Number of electrons* ejected versus the *frequency* of the incident light (right)



**Question 3** **6 points****a. 2 points**

In the ground state of Ge (Germanium; $Z = 32$) atom, **name** the following orbitals:

i. Orbital containing the electron that requires the maximum energy to be removed.

ii. Orbital containing the electron that experiences the minimum effective nuclear charge.

b. 2 points

An unknown anion (A^-) has the following *excited* electronic configuration: $[Ar] 4s^2 3d^3 4p^1$

(Ar: Argon; $Z = 18$)

Give the **condensed ground state electronic configuration** of the atom, **A** and the cation, **A^{2+}** .
Show your work. (Assume all electronic transitions are possible)

Electronic Configuration of A:

Electronic Configuration of A^{2+} :

c. 2 points

In a single atom, determine how many electrons can have the following set of quantum numbers?

(i) $n = 5; l = 3$

In a single atom, determine how many electrons can have the following set of quantum numbers?

(i) $n = 3; l = 2; m_s = +1/2$

**Question 4** **8 points**

Rank the following from 1 to 4 in **increasing** order. Provide the ranking in the boxes provided.

Note: Atomic number (Z) are provided for relevant atoms

a) Increasing (lowest to highest) order of *ionization energy*: (1= lowest; 4 = highest)

Mg

Mg: Z = 12

Rb

Rb: Z = 37

Cs

Cs: Z = 55

K

K : Z = 19

b) Increasing (lowest to highest) order of *bond angle*: (1= lowest; 4 = highest)

H₂OCCl₄XeF₄

HCN

c) Increasing (smallest to largest) order of *ionic size* (1= smallest; 4 = largest)

S²⁻

S: Z = 16

O²⁻

O: Z = 8

Be²⁺

Be: Z = 4

Na⁺

Na: Z = 11

d) Increasing (lowest to highest) order of *bond energy* (1= lowest; 4 = highest)

P-F

P-Cl

P-I

P-Br

**Question 5** **12 points**

a. 9 points For the molecules HClO_4 and TeF_6 , draw the most stable *Lewis structure* showing appropriate shape and *dashed/wedged bonds*, based on VSEPR. Determine the **molecular geometry** around the central atom. For HClO_4 , label the *most polar bond*.



b. 3 points (Note: Lone pairs are not shown in structures below)

(i) For the three *resonance* structures (1, 2 and 3) shown below, answer the following question:



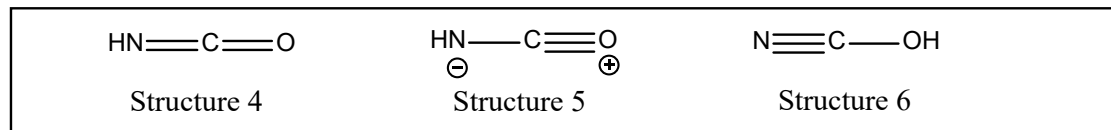
(I) Circle the *most contributing* resonance structure.

Structure 1

Structure 2

Structure 3

(ii) For the three structures given (4, 5, and 6) below, answer the following questions:



(I) Circle all the structures that represent *resonance structures* of each other:

Structure 4

Structure 5

Structure 6

(II) Among the resonance structures, circle the *most contributing* resonance structure.

Structure 4

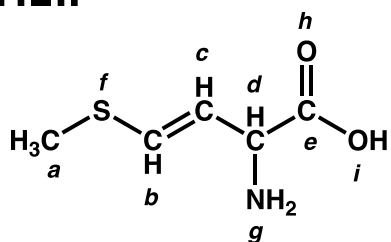
Structure 5

Structure 6

**Question 6** **6 points**

Using the data given below, calculate the **lattice energy** for SrO(s).

	ΔH° (kJ/mol)
$\text{Sr}(s) \longrightarrow \text{Sr}(g)$	+180
$\text{Sr}(g) \longrightarrow \text{Sr}^{2+}(g) + 2e^-$	+1468
$\text{O}_2(g) \longrightarrow 2\text{O}(g)$	+500
$\text{O}(g) + 2e^- \longrightarrow \text{O}^{2-}(g)$	+600
$\text{Sr}(s) + \frac{1}{2}\text{O}_2(g) \longrightarrow \text{SrO}(s)$	-550

**Question 7 10 points**

Consider the molecule given on the left, and answer questions (i) to (vi) using chemical bonding theories that we have learnt in class (valence bond theory, hybridization).

Note that the lone pairs are not shown in the structure:

i. Consider the carbon atoms marked in as ***a***, ***b***, ***c***, ***d***, and ***e*** in the molecule, and determine **hybridization** at each of the carbon centers. (2.5 points)

Atom	Hybridization
Carbon (<i>a</i>)	
Carbon (<i>b</i>)	
Carbon (<i>c</i>)	
Carbon (<i>d</i>)	
Carbon (<i>e</i>)	

ii. Now, consider the sulfur atom (S) marked as ***f***, nitrogen (N) atom (***g***), and oxygen(O) atoms (***h*** and ***i***), and determine their **hybridization**. (2 points)

Atom	Hybridization
Sulfur (<i>f</i>)	
Nitrogen (<i>g</i>)	
Oxygen (<i>h</i>)	
Oxygen (<i>i</i>)	

iii. **Describe** all the bonds formed by the carbon atoms ***a*** and ***b***, using sigma and pi designation as appropriate. (2 points)

Atom	Bonds formed (use sigma and pi as appropriate)
Carbon (<i>a</i>)	
Carbon (<i>b</i>)	

iv. How many total sp^3 -s sigma bonds are in the molecule? (1 point)

v. Determine **electronic** and **molecular geometry** at carbon (C) labeled ***e*** and sulfur (S) labeled ***f*** centres. (2 points)

Atom	Electronic Geometry	Molecular Geometry
Carbon (<i>e</i>)		
Sulfur (<i>f</i>)		

vi. Circle the value from the choices given below, that describes the **C-S-C bond angle at sulfur** (S) marked ***f***: (0.5 points)

>90° to <109.5° 109.5° >109.5 to <120° 120° >120° to <180°

**Question 8 12 points**

Using the **valence shell orbitals only** draw a **complete** molecular orbital (MO) diagram (with no s-p mixing) for the molecule **PS**, in the space provided below. **(6 points)**

Assume that 3s and 3p atomic orbitals will form molecular orbitals like those formed by the combination of 2s and 2p atomic orbitals that we learnt in class, and there is no s-p mixing.

Show **energy arrow**, **atomic orbitals** involved, **atom labels**, the **molecular orbitals** that are formed, **filled electrons** in the MO diagram etc. Make sure to label all the atomic and molecular orbitals.

Using the MO diagram above, answer the following questions:

- (i) Which one of the following will have the highest bond order? Circle the correct choice
A) PS^{1-} B) PS C) PS^{1+}
- (ii) Which one of the following will have the longest bond length? Circle the correct choice
A) PS^{1-} B) PS C) PS^{1+}
- (iii) Which of the following will be diamagnetic? Circle the correct choice.
A) PS^{1-} B) PS C) PS^{1+}
- (iv) Draw Lewis dot structure of the molecule with the shortest bond length (based on the MO diagram) from among the three (PS^{1-} , PS, PS^{1+}), and indicate the bond order calculated from the Lewis dot structure. **(3 points)**

Lewis Dot Structure

Bond Order



327E6A06-493B-42D4-AB49-16F370034438

midterm-exam-93388

#111 Page 10 of 10

SCRAP PAGE

**CHEMISTRY 110 –Midterm Exam****Thursday November 7, 2024, 6:30 – 8:30 pm****Examiners: Pallavi Sirjoosingh and Ashok Kakkar**

First name (please write as legibly as possible within the boxes)

--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--

Last name

--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--

Student ID number

--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--

This booklet has 8 questions on 10 pages (The last page can be used as a scrap page).

The staple must NOT be removed from the question booklet. Your intact question booklet must be returned at the end of the exam.

A separate booklet contains a periodic table, and datasheet.

Write your answers directly under the questions in the space provided. Do not write on the QR codes or near the top edges (as it may be cut off).

Please write **legibly and dark enough** for the scan to be readable. For students who use English as a second language, dictionaries are permitted. Calculators are allowed.

DO NOT ASK QUESTIONS OF THE INVIGILATORS about interpretations of the exam questions. If you have a concern about a question, please summarize it at the top of this page. All concerns will be evaluated after the exam. GOOD LUCK!

**Question 1 8 points** (*Assume all transitions are possible*)

a. 6 points Calculate the **energy (in kJ/mol)** required to ionize (completely remove) the electron from the **n=3** level of **He⁺ ion** (He: Helium). Show your work including equations. (3 sig. figs.)

b. 2 points

(I) Select the largest (longest) wavelength from the three choices below:

- i. Wavelength of photon to *completely remove* an electron from *1s orbital of Hydrogen atom*
- ii. Wavelength of photon to *completely remove* an electron from *1s orbital of Boron atom (B)*
- iii. Wavelength of photon to *completely remove* an electron from *1s orbital of Boron cation (B⁺)*

i or ii or iii

(II) Select the smallest (shortest) wavelength from the three choices below (*for H atom*):

- i. Wavelength of photon *emitted* for an electron transitioning from *2p to 1s orbital*
- ii. Wavelength of photon *absorbed* for an electron transitioning from *2s to 4p orbital*
- iii. Wavelength of photon *absorbed* for an electron transitioning from *1s to 3p orbital*

i or ii or iii



Question 2 8 points



A Chem110 student plots the graph shown on the left, from data obtained for the kinetic energy (eV) of the ejected electron versus frequency (s^{-1}) of the incident light, for an unknown metal. ($1\text{eV} = 1.6022 \times 10^{-19} \text{ J}$)

Based on the plot, answer the questions **a to c** below:

a. 1 point On the plot above, label the **threshold frequency**.

b. 2 points What is the value of the work potential (in J) of the unknown metal?

c. 3 points What is the *longest* wavelength (in nm) required to eject an electron from this metal's surface? Show your work including equation(s) used. (3 sig figs)

d. 2 points In the space provided below, plot the following

(Assume frequency is *above* threshold frequency):

- Kinetic energy* of an ejected electron versus the *intensity* of the incident light (left)
- Number of electrons* ejected versus the *frequency* of the incident light (right)



**Question 3** **6 points****a. 2 points**

In the ground state of Ge (Germanium; $Z = 32$) atom, **name** the following orbitals:

i. Orbital containing the electron that requires the maximum energy to be removed.

ii. Orbital containing the electron that experiences the minimum effective nuclear charge.

b. 2 points

An unknown anion (A^-) has the following *excited* electronic configuration: $[Ar] 4s^2 3d^3 4p^1$

(Ar: Argon; $Z = 18$)

Give the **condensed ground state electronic configuration** of the atom, **A** and the cation, **A^{2+}** .
Show your work. (Assume all electronic transitions are possible)

Electronic Configuration of A:

Electronic Configuration of A^{2+} :

c. 2 points

In a single atom, determine how many electrons can have the following set of quantum numbers?

(i) $n = 5; l = 3$

In a single atom, determine how many electrons can have the following set of quantum numbers?

(i) $n = 3; l = 2; m_s = +1/2$

**Question 4** **8 points**

Rank the following from 1 to 4 in **increasing** order. Provide the ranking in the boxes provided.

Note: Atomic number (Z) are provided for relevant atoms

a) Increasing (lowest to highest) order of *ionization energy*: (1= lowest; 4 = highest)

Mg

Mg: Z = 12

Rb

Rb: Z = 37

Cs

Cs: Z = 55

K

K : Z = 19

b) Increasing (lowest to highest) order of *bond angle*: (1= lowest; 4 = highest)

H₂OCCl₄XeF₄

HCN

c) Increasing (smallest to largest) order of *ionic size* (1= smallest; 4 = largest)

S²⁻

S: Z = 16

O²⁻

O: Z = 8

Be²⁺

Be: Z = 4

Na⁺

Na: Z = 11

d) Increasing (lowest to highest) order of *bond energy* (1= lowest; 4 = highest)

P-F

P-Cl

P-I

P-Br

**Question 5** **12 points**

a. 9 points For the molecules HClO_4 and TeF_6 , draw the most stable *Lewis structure* showing appropriate shape and *dashed/wedged bonds*, based on VSEPR. Determine the **molecular geometry** around the central atom. For HClO_4 , label the *most polar bond*.



b. 3 points (Note: Lone pairs are not shown in structures below)

(i) For the three *resonance* structures (1, 2 and 3) shown below, answer the following question:



(I) Circle the *most contributing* resonance structure.

Structure 1

Structure 2

Structure 3

(ii) For the three structures given (4, 5, and 6) below, answer the following questions:



(I) Circle all the structures that represent *resonance structures* of each other:

Structure 4

Structure 5

Structure 6

(II) Among the resonance structures, circle the *most contributing* resonance structure.

Structure 4

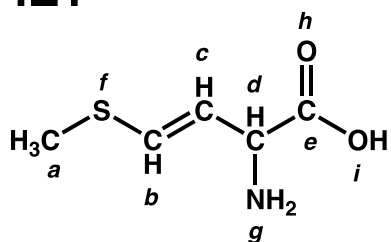
Structure 5

Structure 6

**Question 6** **6 points**

Using the data given below, calculate the **lattice energy** for SrO(s).

	ΔH° (kJ/mol)
$\text{Sr}(s) \longrightarrow \text{Sr}(g)$	+180
$\text{Sr}(g) \longrightarrow \text{Sr}^{2+}(g) + 2e^-$	+1468
$\text{O}_2(g) \longrightarrow 2\text{O}(g)$	+500
$\text{O}(g) + 2e^- \longrightarrow \text{O}^{2-}(g)$	+600
$\text{Sr}(s) + \frac{1}{2}\text{O}_2(g) \longrightarrow \text{SrO}(s)$	-550

**Question 7 10 points**

Consider the molecule given on the left, and answer questions (i) to (vi) using chemical bonding theories that we have learnt in class (valence bond theory, hybridization).

Note that the lone pairs are not shown in the structure:

i. Consider the carbon atoms marked in as ***a***, ***b***, ***c***, ***d***, and ***e*** in the molecule, and determine **hybridization** at each of the carbon centers. (2.5 points)

Atom	Hybridization
Carbon (<i>a</i>)	
Carbon (<i>b</i>)	
Carbon (<i>c</i>)	
Carbon (<i>d</i>)	
Carbon (<i>e</i>)	

ii. Now, consider the sulfur atom (S) marked as ***f***, nitrogen (N) atom (***g***), and oxygen(O) atoms (***h*** and ***i***), and determine their **hybridization**. (2 points)

Atom	Hybridization
Sulfur (<i>f</i>)	
Nitrogen (<i>g</i>)	
Oxygen (<i>h</i>)	
Oxygen (<i>i</i>)	

iii. **Describe** all the bonds formed by the carbon atoms ***a*** and ***b***, using sigma and pi designation as appropriate. (2 points)

Atom	Bonds formed (use sigma and pi as appropriate)
Carbon (<i>a</i>)	
Carbon (<i>b</i>)	

iv. How many total sp^3 -s sigma bonds are in the molecule? (1 point)

v. Determine **electronic** and **molecular geometry** at carbon (C) labeled ***e*** and sulfur (S) labeled ***f*** centres. (2 points)

Atom	Electronic Geometry	Molecular Geometry
Carbon (<i>e</i>)		
Sulfur (<i>f</i>)		

vi. Circle the value from the choices given below, that describes the **C-S-C bond angle at sulfur** (S) marked ***f***: (0.5 points)

>90° to <109.5° 109.5° >109.5 to <120° 120° >120° to <180°


Question 8 12 points

Using the **valence shell orbitals only** draw a **complete** molecular orbital (MO) diagram (with no s-p mixing) for the molecule **PS**, in the space provided below. **(6 points)**

Assume that 3s and 3p atomic orbitals will form molecular orbitals like those formed by the combination of 2s and 2p atomic orbitals that we learnt in class, and there is no s-p mixing.

Show **energy arrow**, **atomic orbitals** involved, **atom labels**, the **molecular orbitals** that are formed, **filled electrons** in the MO diagram etc. Make sure to label all the atomic and molecular orbitals.

Using the MO diagram above, answer the following questions:

- (i) Which one of the following will have the highest bond order? Circle the correct choice
 A) PS^{1-} B) PS C) PS^{1+}
- (ii) Which one of the following will have the longest bond length? Circle the correct choice
 A) PS^{1-} B) PS C) PS^{1+}
- (iii) Which of the following will be diamagnetic? Circle the correct choice.
 A) PS^{1-} B) PS C) PS^{1+}
- (iv) Draw Lewis dot structure of the molecule with the shortest bond length (based on the MO diagram) from among the three (PS^{1-} , PS, PS^{1+}), and indicate the bond order calculated from the Lewis dot structure. **(3 points)**

Lewis Dot Structure

Bond Order



2442B668-836D-4118-B38B-3498BDEAD6DC

midterm-exam-93388

#112 Page 10 of 10

SCRAP PAGE

**CHEMISTRY 110 –Midterm Exam****Thursday November 7, 2024, 6:30 – 8:30 pm**

Examiners: Pallavi Sirjoosingh and Ashok Kakkar

First name (please write as legibly as possible within the boxes)

--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--

Last name

--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--

Student ID number

--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--

This booklet has 8 questions on 10 pages (The last page can be used as a scrap page).

The staple must NOT be removed from the question booklet. Your intact question booklet must be returned at the end of the exam.

A separate booklet contains a periodic table, and datasheet.

Write your answers directly under the questions in the space provided. Do not write on the QR codes or near the top edges (as it may be cut off).

Please write **legibly and dark enough** for the scan to be readable. For students who use English as a second language, dictionaries are permitted. Calculators are allowed.

DO NOT ASK QUESTIONS OF THE INVIGILATORS about interpretations of the exam questions. If you have a concern about a question, please summarize it at the top of this page. All concerns will be evaluated after the exam. GOOD LUCK!

**Question 1 8 points** (*Assume all transitions are possible*)

a. 6 points Calculate the **energy (in kJ/mol)** required to ionize (completely remove) the electron from the **$n=3$** level of **He^+ ion** (He: Helium). Show your work including equations. (3 sig. figs.)

b. 2 points

(I) Select the largest (longest) wavelength from the three choices below:

- i. Wavelength of photon to *completely remove* an electron from *1s orbital of Hydrogen atom*
- ii. Wavelength of photon to *completely remove* an electron from *1s orbital of Boron atom (B)*
- iii. Wavelength of photon to *completely remove* an electron from *1s orbital of Boron cation (B^+)*

i or ii or iii

(II) Select the smallest (shortest) wavelength from the three choices below (*for H atom*):

- i. Wavelength of photon *emitted* for an electron transitioning from *2p to 1s orbital*
- ii. Wavelength of photon *absorbed* for an electron transitioning from *2s to 4p orbital*
- iii. Wavelength of photon *absorbed* for an electron transitioning from *1s to 3p orbital*

i or ii or iii



Question 2 8 points



A Chem110 student plots the graph shown on the left, from data obtained for the kinetic energy (eV) of the ejected electron versus frequency (s^{-1}) of the incident light, for an unknown metal. ($1\text{eV} = 1.6022 \times 10^{-19} \text{ J}$)

Based on the plot, answer the questions **a to c** below:

a. 1 point On the plot above, label the **threshold frequency**.

b. 2 points What is the value of the work potential (in J) of the unknown metal?

c. 3 points What is the *longest* wavelength (in nm) required to eject an electron from this metal's surface? Show your work including equation(s) used. (3 sig figs)

d. 2 points In the space provided below, plot the following

(Assume frequency is *above* threshold frequency):

- Kinetic energy* of an ejected electron versus the *intensity* of the incident light (left)
- Number of electrons* ejected versus the *frequency* of the incident light (right)



**Question 3** **6 points****a. 2 points**

In the ground state of Ge (Germanium; $Z = 32$) atom, **name** the following orbitals:

i. Orbital containing the electron that requires the maximum energy to be removed.

ii. Orbital containing the electron that experiences the minimum effective nuclear charge.

b. 2 points

An unknown anion (A^-) has the following *excited* electronic configuration: $[Ar] 4s^2 3d^3 4p^1$

(Ar: Argon; $Z = 18$)

Give the **condensed ground state electronic configuration** of the atom, **A** and the cation, **A^{2+}** .
Show your work. (Assume all electronic transitions are possible)

Electronic Configuration of A:

Electronic Configuration of A^{2+} :

c. 2 points

In a single atom, determine how many electrons can have the following set of quantum numbers?

(i) $n = 5; l = 3$

In a single atom, determine how many electrons can have the following set of quantum numbers?

(i) $n = 3; l = 2; m_s = +1/2$

**Question 4** **8 points**

Rank the following from 1 to 4 in **increasing** order. Provide the ranking in the boxes provided.

Note: Atomic number (Z) are provided for relevant atoms

a) Increasing (lowest to highest) order of *ionization energy*: (1= lowest; 4 = highest)

Mg

Mg: Z = 12

Rb

Rb: Z = 37

Cs

Cs: Z = 55

K

K : Z = 19

b) Increasing (lowest to highest) order of *bond angle*: (1= lowest; 4 = highest)

H₂OCCl₄XeF₄

HCN

c) Increasing (smallest to largest) order of *ionic size* (1= smallest; 4 = largest)

S²⁻

S: Z = 16

O²⁻

O: Z = 8

Be²⁺

Be: Z = 4

Na⁺

Na: Z = 11

d) Increasing (lowest to highest) order of *bond energy* (1= lowest; 4 = highest)

P-F

P-Cl

P-I

P-Br

**Question 5** **12 points**

a. 9 points For the molecules HClO_4 and TeF_6 , draw the most stable *Lewis structure* showing appropriate shape and *dashed/wedged bonds*, based on VSEPR. Determine the **molecular geometry** around the central atom. For HClO_4 , label the *most polar bond*.



b. 3 points (Note: Lone pairs are not shown in structures below)

(i) For the three *resonance* structures (1, 2 and 3) shown below, answer the following question:



(I) Circle the *most contributing* resonance structure.

Structure 1

Structure 2

Structure 3

(ii) For the three structures given (4, 5, and 6) below, answer the following questions:



(I) Circle all the structures that represent *resonance structures* of each other:

Structure 4

Structure 5

Structure 6

(II) Among the resonance structures, circle the *most contributing* resonance structure.

Structure 4

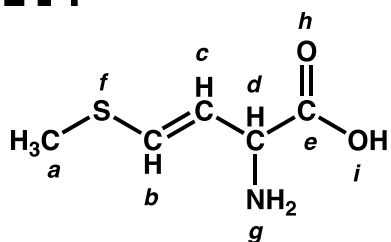
Structure 5

Structure 6

**Question 6** **6 points**

Using the data given below, calculate the **lattice energy** for SrO(s).

	ΔH° (kJ/mol)
$\text{Sr}(s) \longrightarrow \text{Sr}(g)$	+180
$\text{Sr}(g) \longrightarrow \text{Sr}^{2+}(g) + 2e^-$	+1468
$\text{O}_2(g) \longrightarrow 2\text{O}(g)$	+500
$\text{O}(g) + 2e^- \longrightarrow \text{O}^{2-}(g)$	+600
$\text{Sr}(s) + \frac{1}{2}\text{O}_2(g) \longrightarrow \text{SrO}(s)$	-550

**Question 7 10 points**

Consider the molecule given on the left, and answer questions (i) to (vi) using chemical bonding theories that we have learnt in class (valence bond theory, hybridization).

Note that the lone pairs are not shown in the structure:

i. Consider the carbon atoms marked in as ***a***, ***b***, ***c***, ***d***, and ***e*** in the molecule, and determine **hybridization** at each of the carbon centers. (2.5 points)

Atom	Hybridization
Carbon (<i>a</i>)	
Carbon (<i>b</i>)	
Carbon (<i>c</i>)	
Carbon (<i>d</i>)	
Carbon (<i>e</i>)	

ii. Now, consider the sulfur atom (S) marked as ***f***, nitrogen (N) atom (***g***), and oxygen(O) atoms (***h*** and ***i***), and determine their **hybridization**. (2 points)

Atom	Hybridization
Sulfur (<i>f</i>)	
Nitrogen (<i>g</i>)	
Oxygen (<i>h</i>)	
Oxygen (<i>i</i>)	

iii. **Describe** all the bonds formed by the carbon atoms ***a*** and ***b***, using sigma and pi designation as appropriate. (2 points)

Atom	Bonds formed (use sigma and pi as appropriate)
Carbon (<i>a</i>)	
Carbon (<i>b</i>)	

iv. How many total sp^3 -s sigma bonds are in the molecule? (1 point)

v. Determine **electronic** and **molecular geometry** at carbon (C) labeled ***e*** and sulfur (S) labeled ***f*** centres. (2 points)

Atom	Electronic Geometry	Molecular Geometry
Carbon (<i>e</i>)		
Sulfur (<i>f</i>)		

vi. Circle the value from the choices given below, that describes the **C-S-C bond angle at sulfur** (S) marked ***f***: (0.5 points)

>90° to <109.5° 109.5° >109.5 to <120° 120° >120° to <180°


Question 8 12 points

Using the **valence shell orbitals only** draw a **complete** molecular orbital (MO) diagram (with no s-p mixing) for the molecule **PS**, in the space provided below. **(6 points)**

Assume that 3s and 3p atomic orbitals will form molecular orbitals like those formed by the combination of 2s and 2p atomic orbitals that we learnt in class, and there is no s-p mixing.

Show **energy arrow**, **atomic orbitals** involved, **atom labels**, the **molecular orbitals** that are formed, **filled electrons** in the MO diagram etc. Make sure to label all the atomic and molecular orbitals.

Using the MO diagram above, answer the following questions:

- (i) Which one of the following will have the highest bond order? Circle the correct choice
 A) PS^{1-} B) PS C) PS^{1+}
- (ii) Which one of the following will have the longest bond length? Circle the correct choice
 A) PS^{1-} B) PS C) PS^{1+}
- (iii) Which of the following will be diamagnetic? Circle the correct choice.
 A) PS^{1-} B) PS C) PS^{1+}
- (iv) Draw Lewis dot structure of the molecule with the shortest bond length (based on the MO diagram) from among the three (PS^{1-} , PS, PS^{1+}), and indicate the bond order calculated from the Lewis dot structure. **(3 points)**

Lewis Dot Structure

Bond Order



980B3B18-6C91-4390-A987-90341E044931

midterm-exam-93388

#113 Page 10 of 10

SCRAP PAGE

**CHEMISTRY 110 –Midterm Exam****Thursday November 7, 2024, 6:30 – 8:30 pm**

Examiners: Pallavi Sirjoosingh and Ashok Kakkar

First name (please write as legibly as possible within the boxes)

--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--

Last name

--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--

Student ID number

--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--

This booklet has 8 questions on 10 pages (The last page can be used as a scrap page).

The staple must NOT be removed from the question booklet. Your intact question booklet must be returned at the end of the exam.

A separate booklet contains a periodic table, and datasheet.

Write your answers directly under the questions in the space provided. Do not write on the QR codes or near the top edges (as it may be cut off).

Please write **legibly and dark enough** for the scan to be readable. For students who use English as a second language, dictionaries are permitted. Calculators are allowed.

DO NOT ASK QUESTIONS OF THE INVIGILATORS about interpretations of the exam questions. If you have a concern about a question, please summarize it at the top of this page. All concerns will be evaluated after the exam. GOOD LUCK!

**Question 1 8 points** (*Assume all transitions are possible*)

a. 6 points Calculate the **energy (in kJ/mol)** required to ionize (completely remove) the electron from the **$n=3$** level of **He^+ ion** (He: Helium). Show your work including equations. (3 sig. figs.)

b. 2 points

(I) Select the largest (longest) wavelength from the three choices below:

- i. Wavelength of photon to *completely remove* an electron from *1s orbital of Hydrogen atom*
- ii. Wavelength of photon to *completely remove* an electron from *1s orbital of Boron atom (B)*
- iii. Wavelength of photon to *completely remove* an electron from *1s orbital of Boron cation (B^+)*

i or ii or iii

(II) Select the smallest (shortest) wavelength from the three choices below (*for H atom*):

- i. Wavelength of photon *emitted* for an electron transitioning from *2p to 1s orbital*
- ii. Wavelength of photon *absorbed* for an electron transitioning from *2s to 4p orbital*
- iii. Wavelength of photon *absorbed* for an electron transitioning from *1s to 3p orbital*

i or ii or iii



Question 2 8 points



A Chem110 student plots the graph shown on the left, from data obtained for the kinetic energy (eV) of the ejected electron versus frequency (s^{-1}) of the incident light, for an unknown metal. ($1\text{eV} = 1.6022 \times 10^{-19} \text{ J}$)

Based on the plot, answer the questions **a to c** below:

a. 1 point On the plot above, label the **threshold frequency**.

b. 2 points What is the value of the work potential (in J) of the unknown metal?

c. 3 points What is the *longest* wavelength (in nm) required to eject an electron from this metal's surface? Show your work including equation(s) used. (3 sig figs)

d. 2 points In the space provided below, plot the following

(Assume frequency is *above* threshold frequency):

- Kinetic energy* of an ejected electron versus the *intensity* of the incident light (left)
- Number of electrons* ejected versus the *frequency* of the incident light (right)



**Question 3** **6 points****a. 2 points**

In the ground state of Ge (Germanium; $Z = 32$) atom, **name** the following orbitals:

i. Orbital containing the electron that requires the maximum energy to be removed.

ii. Orbital containing the electron that experiences the minimum effective nuclear charge.

b. 2 points

An unknown anion (A^-) has the following *excited* electronic configuration: $[Ar] 4s^2 3d^3 4p^1$

(Ar: Argon; $Z = 18$)

Give the **condensed ground state electronic configuration** of the atom, **A** and the cation, **A^{2+}** .
Show your work. (Assume all electronic transitions are possible)

Electronic Configuration of A:

Electronic Configuration of A^{2+} :

c. 2 points

In a single atom, determine how many electrons can have the following set of quantum numbers?

(i) $n = 5; l = 3$

In a single atom, determine how many electrons can have the following set of quantum numbers?

(i) $n = 3; l = 2; m_s = +1/2$

**Question 4** **8 points**

Rank the following from 1 to 4 in **increasing** order. Provide the ranking in the boxes provided.

Note: Atomic number (Z) are provided for relevant atoms

a) Increasing (lowest to highest) order of *ionization energy*: (1= lowest; 4 = highest)

Mg

Mg: Z = 12

Rb

Rb: Z = 37

Cs

Cs: Z = 55

K

K : Z = 19

b) Increasing (lowest to highest) order of *bond angle*: (1= lowest; 4 = highest)

H₂OCCl₄XeF₄

HCN

c) Increasing (smallest to largest) order of *ionic size* (1= smallest; 4 = largest)

S²⁻

S: Z = 16

O²⁻

O: Z = 8

Be²⁺

Be: Z = 4

Na⁺

Na: Z = 11

d) Increasing (lowest to highest) order of *bond energy* (1= lowest; 4 = highest)

P-F

P-Cl

P-I

P-Br

**Question 5** **12 points**

a. 9 points For the molecules HClO_4 and TeF_6 , draw the most stable *Lewis structure* showing appropriate shape and *dashed/wedged bonds*, based on VSEPR. Determine the **molecular geometry** around the central atom. For HClO_4 , label the *most polar bond*.



b. 3 points (Note: Lone pairs are not shown in structures below)

(i) For the three *resonance* structures (1, 2 and 3) shown below, answer the following question:



(I) Circle the *most contributing* resonance structure.

Structure 1

Structure 2

Structure 3

(ii) For the three structures given (4, 5, and 6) below, answer the following questions:



(I) Circle all the structures that represent *resonance structures* of each other:

Structure 4

Structure 5

Structure 6

(II) Among the resonance structures, circle the *most contributing* resonance structure.

Structure 4

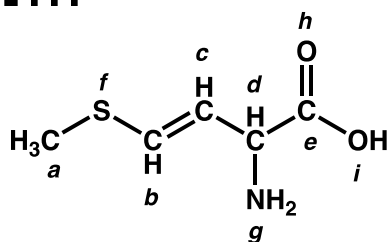
Structure 5

Structure 6

**Question 6** **6 points**

Using the data given below, calculate the **lattice energy** for SrO(s).

	ΔH° (kJ/mol)
$\text{Sr}(s) \longrightarrow \text{Sr}(g)$	+180
$\text{Sr}(g) \longrightarrow \text{Sr}^{2+}(g) + 2e^-$	+1468
$\text{O}_2(g) \longrightarrow 2\text{O}(g)$	+500
$\text{O}(g) + 2e^- \longrightarrow \text{O}^{2-}(g)$	+600
$\text{Sr}(s) + \frac{1}{2}\text{O}_2(g) \longrightarrow \text{SrO}(s)$	-550

**Question 7 10 points**

Consider the molecule given on the left, and answer questions (i) to (vi) using chemical bonding theories that we have learnt in class (valence bond theory, hybridization).

Note that the lone pairs are not shown in the structure:

i. Consider the carbon atoms marked in as ***a***, ***b***, ***c***, ***d***, and ***e*** in the molecule, and determine **hybridization** at each of the carbon centers. (2.5 points)

Atom	Hybridization
Carbon (<i>a</i>)	
Carbon (<i>b</i>)	
Carbon (<i>c</i>)	
Carbon (<i>d</i>)	
Carbon (<i>e</i>)	

ii. Now, consider the sulfur atom (S) marked as ***f***, nitrogen (N) atom (***g***), and oxygen(O) atoms (***h*** and ***i***), and determine their **hybridization**. (2 points)

Atom	Hybridization
Sulfur (<i>f</i>)	
Nitrogen (<i>g</i>)	
Oxygen (<i>h</i>)	
Oxygen (<i>i</i>)	

iii. **Describe** all the bonds formed by the carbon atoms ***a*** and ***b***, using sigma and pi designation as appropriate. (2 points)

Atom	Bonds formed (use sigma and pi as appropriate)
Carbon (<i>a</i>)	
Carbon (<i>b</i>)	

iv. How many total sp^3 -s sigma bonds are in the molecule? (1 point)

v. Determine **electronic** and **molecular geometry** at carbon (C) labeled ***e*** and sulfur (S) labeled ***f*** centres. (2 points)

Atom	Electronic Geometry	Molecular Geometry
Carbon (<i>e</i>)		
Sulfur (<i>f</i>)		

vi. Circle the value from the choices given below, that describes the **C-S-C bond angle at sulfur** (S) marked ***f***: (0.5 points)

>90° to <109.5° 109.5° >109.5 to <120° 120° >120° to <180°

**Question 8 12 points**

Using the **valence shell orbitals only** draw a **complete** molecular orbital (MO) diagram (with no s-p mixing) for the molecule **PS**, in the space provided below. **(6 points)**

Assume that 3s and 3p atomic orbitals will form molecular orbitals like those formed by the combination of 2s and 2p atomic orbitals that we learnt in class, and there is no s-p mixing.

Show **energy arrow**, **atomic orbitals** involved, **atom labels**, the **molecular orbitals** that are formed, **filled electrons** in the MO diagram etc. Make sure to label all the atomic and molecular orbitals.

Using the MO diagram above, answer the following questions:

- (i) Which one of the following will have the highest bond order? Circle the correct choice
A) PS^{1-} B) PS C) PS^{1+}
- (ii) Which one of the following will have the longest bond length? Circle the correct choice
A) PS^{1-} B) PS C) PS^{1+}
- (iii) Which of the following will be diamagnetic? Circle the correct choice.
A) PS^{1-} B) PS C) PS^{1+}
- (iv) Draw Lewis dot structure of the molecule with the shortest bond length (based on the MO diagram) from among the three (PS^{1-} , PS, PS^{1+}), and indicate the bond order calculated from the Lewis dot structure. **(3 points)**

Lewis Dot Structure

Bond Order



25605EED-2E2C-4906-B047-8C3E4BD2FE96

midterm-exam-93388

#114 Page 10 of 10

SCRAP PAGE

**CHEMISTRY 110 –Midterm Exam****Thursday November 7, 2024, 6:30 – 8:30 pm**

Examiners: Pallavi Sirjoosingh and Ashok Kakkar

First name (please write as legibly as possible within the boxes)

--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--

Last name

--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--

Student ID number

--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--

This booklet has 8 questions on 10 pages (The last page can be used as a scrap page).

The staple must NOT be removed from the question booklet. Your intact question booklet must be returned at the end of the exam.

A separate booklet contains a periodic table, and datasheet.

Write your answers directly under the questions in the space provided. Do not write on the QR codes or near the top edges (as it may be cut off).

Please write **legibly and dark enough** for the scan to be readable. For students who use English as a second language, dictionaries are permitted. Calculators are allowed.

DO NOT ASK QUESTIONS OF THE INVIGILATORS about interpretations of the exam questions. If you have a concern about a question, please summarize it at the top of this page. All concerns will be evaluated after the exam. GOOD LUCK!

**Question 1 8 points** (*Assume all transitions are possible*)

a. 6 points Calculate the **energy (in kJ/mol)** required to ionize (completely remove) the electron from the **n=3** level of **He⁺ ion** (He: Helium). Show your work including equations. (3 sig. figs.)

b. 2 points

(I) Select the largest (longest) wavelength from the three choices below:

- i. Wavelength of photon to *completely remove* an electron from *1s orbital of Hydrogen atom*
- ii. Wavelength of photon to *completely remove* an electron from *1s orbital of Boron atom (B)*
- iii. Wavelength of photon to *completely remove* an electron from *1s orbital of Boron cation (B⁺)*

i or ii or iii

(II) Select the smallest (shortest) wavelength from the three choices below (*for H atom*):

- i. Wavelength of photon *emitted* for an electron transitioning from *2p to 1s orbital*
- ii. Wavelength of photon *absorbed* for an electron transitioning from *2s to 4p orbital*
- iii. Wavelength of photon *absorbed* for an electron transitioning from *1s to 3p orbital*

i or ii or iii



Question 2 8 points



A Chem110 student plots the graph shown on the left, from data obtained for the kinetic energy (eV) of the ejected electron versus frequency (s^{-1}) of the incident light, for an unknown metal. ($1\text{eV} = 1.6022 \times 10^{-19} \text{ J}$)

Based on the plot, answer the questions **a to c** below:

a. 1 point On the plot above, label the **threshold frequency**.

b. 2 points What is the value of the work potential (in J) of the unknown metal?

c. 3 points What is the *longest* wavelength (in nm) required to eject an electron from this metal's surface? Show your work including equation(s) used. (3 sig figs)

d. 2 points In the space provided below, plot the following

(Assume frequency is *above* threshold frequency):

- i. *Kinetic energy* of an ejected electron versus the *intensity* of the incident light (left)
- ii. *Number of electrons* ejected versus the *frequency* of the incident light (right)



**Question 3** **6 points****a. 2 points**

In the ground state of Ge (Germanium; $Z = 32$) atom, **name** the following orbitals:

i. Orbital containing the electron that requires the maximum energy to be removed.

ii. Orbital containing the electron that experiences the minimum effective nuclear charge.

b. 2 points

An unknown anion (A^-) has the following *excited* electronic configuration: $[Ar] 4s^2 3d^3 4p^1$

(Ar: Argon; $Z = 18$)

Give the **condensed ground state electronic configuration** of the atom, **A** and the cation, **A^{2+}** .
Show your work. (Assume all electronic transitions are possible)

Electronic Configuration of A:

Electronic Configuration of A^{2+} :

c. 2 points

In a single atom, determine how many electrons can have the following set of quantum numbers?

(i) $n = 5; l = 3$

In a single atom, determine how many electrons can have the following set of quantum numbers?

(i) $n = 3; l = 2; m_s = +1/2$

**Question 4** **8 points**

Rank the following from 1 to 4 in **increasing** order. Provide the ranking in the boxes provided.

Note: Atomic number (Z) are provided for relevant atoms

a) Increasing (lowest to highest) order of *ionization energy*: (1= lowest; 4 = highest)

Mg

Mg: Z = 12

Rb

Rb: Z = 37

Cs

Cs: Z = 55

K

K : Z = 19

b) Increasing (lowest to highest) order of *bond angle*: (1= lowest; 4 = highest)

H₂OCCl₄XeF₄

HCN

c) Increasing (smallest to largest) order of *ionic size* (1= smallest; 4 = largest)

S²⁻

S: Z = 16

O²⁻

O: Z = 8

Be²⁺

Be: Z = 4

Na⁺

Na: Z = 11

d) Increasing (lowest to highest) order of *bond energy* (1= lowest; 4 = highest)

P-F

P-Cl

P-I

P-Br

**Question 5** **12 points**

a. 9 points For the molecules HClO_4 and TeF_6 , draw the most stable *Lewis structure* showing appropriate shape and *dashed/wedged bonds*, based on VSEPR. Determine the **molecular geometry** around the central atom. For HClO_4 , label the *most polar bond*.

HClO_4 (there are no O-O bonds)

Label the most polar bond

Molecular Geometry around Cl

TeF_6

Molecular Geometry around Te

b. 3 points (Note: Lone pairs are not shown in structures below)

(i) For the three *resonance* structures (1, 2 and 3) shown below, answer the following question:



(I) Circle the *most contributing* resonance structure.

Structure 1

Structure 2

Structure 3

(ii) For the three structures given (4, 5, and 6) below, answer the following questions:



(I) Circle all the structures that represent *resonance structures* of each other:

Structure 4

Structure 5

Structure 6

(II) Among the resonance structures, circle the *most contributing* resonance structure.

Structure 4

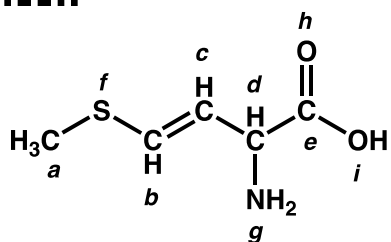
Structure 5

Structure 6

**Question 6** **6 points**

Using the data given below, calculate the **lattice energy** for SrO(s).

	ΔH° (kJ/mol)
$\text{Sr}(s) \longrightarrow \text{Sr}(g)$	+180
$\text{Sr}(g) \longrightarrow \text{Sr}^{2+}(g) + 2e^-$	+1468
$\text{O}_2(g) \longrightarrow 2\text{O}(g)$	+500
$\text{O}(g) + 2e^- \longrightarrow \text{O}^{2-}(g)$	+600
$\text{Sr}(s) + \frac{1}{2}\text{O}_2(g) \longrightarrow \text{SrO}(s)$	-550

**Question 7 10 points**

Consider the molecule given on the left, and answer questions (i) to (vi) using chemical bonding theories that we have learnt in class (valence bond theory, hybridization).

Note that the lone pairs are not shown in the structure:

i. Consider the carbon atoms marked in as ***a***, ***b***, ***c***, ***d***, and ***e*** in the molecule, and determine **hybridization** at each of the carbon centers. (2.5 points)

Atom	Hybridization
Carbon (<i>a</i>)	
Carbon (<i>b</i>)	
Carbon (<i>c</i>)	
Carbon (<i>d</i>)	
Carbon (<i>e</i>)	

ii. Now, consider the sulfur atom (S) marked as ***f***, nitrogen (N) atom (***g***), and oxygen(O) atoms (***h*** and ***i***), and determine their **hybridization**. (2 points)

Atom	Hybridization
Sulfur (<i>f</i>)	
Nitrogen (<i>g</i>)	
Oxygen (<i>h</i>)	
Oxygen (<i>i</i>)	

iii. **Describe** all the bonds formed by the carbon atoms ***a*** and ***b***, using sigma and pi designation as appropriate. (2 points)

Atom	Bonds formed (use sigma and pi as appropriate)
Carbon (<i>a</i>)	
Carbon (<i>b</i>)	

iv. How many total sp^3 -s sigma bonds are in the molecule? (1 point)

v. Determine **electronic** and **molecular geometry** at carbon (C) labeled ***e*** and sulfur (S) labeled ***f*** centres. (2 points)

Atom	Electronic Geometry	Molecular Geometry
Carbon (<i>e</i>)		
Sulfur (<i>f</i>)		

vi. Circle the value from the choices given below, that describes the **C-S-C bond angle at sulfur** (S) marked ***f***: (0.5 points)

>90° to <109.5°

109.5°

>109.5 to <120°

120°

>120° to <180°


Question 8 12 points

Using the **valence shell orbitals only** draw a **complete** molecular orbital (MO) diagram (with no s-p mixing) for the molecule **PS**, in the space provided below. **(6 points)**

Assume that 3s and 3p atomic orbitals will form molecular orbitals like those formed by the combination of 2s and 2p atomic orbitals that we learnt in class, and there is no s-p mixing.

Show **energy arrow**, **atomic orbitals** involved, **atom labels**, the **molecular orbitals** that are formed, **filled electrons** in the MO diagram etc. Make sure to label all the atomic and molecular orbitals.

Using the MO diagram above, answer the following questions:

- (i) Which one of the following will have the highest bond order? Circle the correct choice
 A) PS^{1-} B) PS C) PS^{1+}
- (ii) Which one of the following will have the longest bond length? Circle the correct choice
 A) PS^{1-} B) PS C) PS^{1+}
- (iii) Which of the following will be diamagnetic? Circle the correct choice.
 A) PS^{1-} B) PS C) PS^{1+}
- (iv) Draw Lewis dot structure of the molecule with the shortest bond length (based on the MO diagram) from among the three (PS^{1-} , PS, PS^{1+}), and indicate the bond order calculated from the Lewis dot structure. **(3 points)**

Lewis Dot Structure

Bond Order



0AFE3BA3-300D-41F7-B1B4-0241EB6C2344

midterm-exam-93388

#115 Page 10 of 10

SCRAP PAGE

**CHEMISTRY 110 –Midterm Exam****Thursday November 7, 2024, 6:30 – 8:30 pm**

Examiners: Pallavi Sirjoosingh and Ashok Kakkar

First name (please write as legibly as possible within the boxes)

--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--

Last name

--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--

Student ID number

--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--

This booklet has 8 questions on 10 pages (The last page can be used as a scrap page).

The staple must NOT be removed from the question booklet. Your intact question booklet must be returned at the end of the exam.

A separate booklet contains a periodic table, and datasheet.

Write your answers directly under the questions in the space provided. Do not write on the QR codes or near the top edges (as it may be cut off).

Please write **legibly and dark enough** for the scan to be readable. For students who use English as a second language, dictionaries are permitted. Calculators are allowed.

DO NOT ASK QUESTIONS OF THE INVIGILATORS about interpretations of the exam questions. If you have a concern about a question, please summarize it at the top of this page. All concerns will be evaluated after the exam. GOOD LUCK!

**Question 1 8 points** (*Assume all transitions are possible*)

a. 6 points Calculate the **energy (in kJ/mol)** required to ionize (completely remove) the electron from the **n=3** level of **He⁺ ion** (He: Helium). Show your work including equations. (3 sig. figs.)

b. 2 points

(I) Select the largest (longest) wavelength from the three choices below:

- i. Wavelength of photon to *completely remove* an electron from *1s orbital of Hydrogen atom*
- ii. Wavelength of photon to *completely remove* an electron from *1s orbital of Boron atom (B)*
- iii. Wavelength of photon to *completely remove* an electron from *1s orbital of Boron cation (B⁺)*

i or ii or iii

(II) Select the smallest (shortest) wavelength from the three choices below (*for H atom*):

- i. Wavelength of photon *emitted* for an electron transitioning from *2p to 1s orbital*
- ii. Wavelength of photon *absorbed* for an electron transitioning from *2s to 4p orbital*
- iii. Wavelength of photon *absorbed* for an electron transitioning from *1s to 3p orbital*

i or ii or iii



Question 2 8 points



A Chem110 student plots the graph shown on the left, from data obtained for the kinetic energy (eV) of the ejected electron versus frequency (s^{-1}) of the incident light, for an unknown metal. ($1\text{eV} = 1.6022 \times 10^{-19} \text{ J}$)

Based on the plot, answer the questions **a to c** below:

a. 1 point On the plot above, label the **threshold frequency**.

b. 2 points What is the value of the work potential (in J) of the unknown metal?

c. 3 points What is the *longest* wavelength (in nm) required to eject an electron from this metal's surface? Show your work including equation(s) used. (3 sig figs)

d. 2 points In the space provided below, plot the following

(Assume frequency is *above* threshold frequency):

- Kinetic energy* of an ejected electron versus the *intensity* of the incident light (left)
- Number of electrons* ejected versus the *frequency* of the incident light (right)



**Question 3** **6 points****a. 2 points**

In the ground state of Ge (Germanium; $Z = 32$) atom, **name** the following orbitals:

i. Orbital containing the electron that requires the maximum energy to be removed.

ii. Orbital containing the electron that experiences the minimum effective nuclear charge.

b. 2 points

An unknown anion (A^-) has the following *excited* electronic configuration: $[Ar] 4s^2 3d^3 4p^1$

(Ar: Argon; $Z = 18$)

Give the **condensed ground state electronic configuration** of the atom, **A** and the cation, **A^{2+}** .
Show your work. (Assume all electronic transitions are possible)

Electronic Configuration of A:

Electronic Configuration of A^{2+} :

c. 2 points

In a single atom, determine how many electrons can have the following set of quantum numbers?

(i) $n = 5; l = 3$

In a single atom, determine how many electrons can have the following set of quantum numbers?

(i) $n = 3; l = 2; m_s = +1/2$

**Question 4** **8 points**

Rank the following from 1 to 4 in **increasing** order. Provide the ranking in the boxes provided.

Note: Atomic number (Z) are provided for relevant atoms

a) Increasing (lowest to highest) order of *ionization energy*: (1= lowest; 4 = highest)

Mg

Mg: Z = 12

Rb

Rb: Z = 37

Cs

Cs: Z = 55

K

K : Z = 19

b) Increasing (lowest to highest) order of *bond angle*: (1= lowest; 4 = highest)

H₂OCCl₄XeF₄

HCN

c) Increasing (smallest to largest) order of *ionic size* (1= smallest; 4 = largest)

S²⁻

S: Z = 16

O²⁻

O: Z = 8

Be²⁺

Be: Z = 4

Na⁺

Na: Z = 11

d) Increasing (lowest to highest) order of *bond energy* (1= lowest; 4 = highest)

P-F

P-Cl

P-I

P-Br

**Question 5** **12 points**

a. 9 points For the molecules HClO_4 and TeF_6 , draw the most stable *Lewis structure* showing appropriate shape and *dashed/wedged bonds*, based on VSEPR. Determine the **molecular geometry** around the central atom. For HClO_4 , label the *most polar bond*.

HClO_4 (there are no O-O bonds)

Label the most polar bond

Molecular Geometry around Cl

TeF_6

Molecular Geometry around Te

b. 3 points (Note: Lone pairs are not shown in structures below)

(i) For the three *resonance* structures (1, 2 and 3) shown below, answer the following question:



(I) Circle the *most contributing* resonance structure.

Structure 1

Structure 2

Structure 3

(ii) For the three structures given (4, 5, and 6) below, answer the following questions:



(I) Circle all the structures that represent *resonance structures* of each other:

Structure 4

Structure 5

Structure 6

(II) Among the resonance structures, circle the *most contributing* resonance structure.

Structure 4

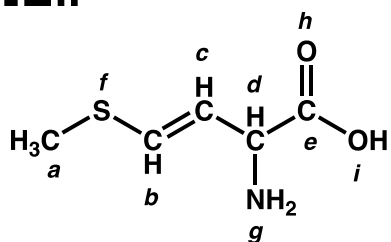
Structure 5

Structure 6

**Question 6** **6 points**

Using the data given below, calculate the **lattice energy** for SrO(s).

	ΔH° (kJ/mol)
$\text{Sr}(s) \longrightarrow \text{Sr}(g)$	+180
$\text{Sr}(g) \longrightarrow \text{Sr}^{2+}(g) + 2e^-$	+1468
$\text{O}_2(g) \longrightarrow 2\text{O}(g)$	+500
$\text{O}(g) + 2e^- \longrightarrow \text{O}^{2-}(g)$	+600
$\text{Sr}(s) + \frac{1}{2}\text{O}_2(g) \longrightarrow \text{SrO}(s)$	-550

**Question 7 10 points**

Consider the molecule given on the left, and answer questions (i) to (vi) using chemical bonding theories that we have learnt in class (valence bond theory, hybridization).

Note that the lone pairs are not shown in the structure:

i. Consider the carbon atoms marked in as ***a***, ***b***, ***c***, ***d***, and ***e*** in the molecule, and determine **hybridization** at each of the carbon centers. (2.5 points)

Atom	Hybridization
Carbon (<i>a</i>)	
Carbon (<i>b</i>)	
Carbon (<i>c</i>)	
Carbon (<i>d</i>)	
Carbon (<i>e</i>)	

ii. Now, consider the sulfur atom (S) marked as ***f***, nitrogen (N) atom (***g***), and oxygen(O) atoms (***h*** and ***i***), and determine their **hybridization**. (2 points)

Atom	Hybridization
Sulfur (<i>f</i>)	
Nitrogen (<i>g</i>)	
Oxygen (<i>h</i>)	
Oxygen (<i>i</i>)	

iii. **Describe** all the bonds formed by the carbon atoms ***a*** and ***b***, using sigma and pi designation as appropriate. (2 points)

Atom	Bonds formed (use sigma and pi as appropriate)
Carbon (<i>a</i>)	
Carbon (<i>b</i>)	

iv. How many total sp^3 -s sigma bonds are in the molecule? (1 point)

v. Determine **electronic** and **molecular geometry** at carbon (C) labeled ***e*** and sulfur (S) labeled ***f*** centres. (2 points)

Atom	Electronic Geometry	Molecular Geometry
Carbon (<i>e</i>)		
Sulfur (<i>f</i>)		

vi. Circle the value from the choices given below, that describes the **C-S-C bond angle at sulfur** (S) marked ***f***: (0.5 points)

>90° to <109.5° 109.5° >109.5 to <120° 120° >120° to <180°


Question 8 12 points

Using the **valence shell orbitals only** draw a **complete** molecular orbital (MO) diagram (with no s-p mixing) for the molecule **PS**, in the space provided below. **(6 points)**

Assume that 3s and 3p atomic orbitals will form molecular orbitals like those formed by the combination of 2s and 2p atomic orbitals that we learnt in class, and there is no s-p mixing.

Show **energy arrow**, **atomic orbitals** involved, **atom labels**, the **molecular orbitals** that are formed, **filled electrons** in the MO diagram etc. Make sure to label all the atomic and molecular orbitals.

Using the MO diagram above, answer the following questions:

- (i) Which one of the following will have the highest bond order? Circle the correct choice
 A) PS^{1-} B) PS C) PS^{1+}
- (ii) Which one of the following will have the longest bond length? Circle the correct choice
 A) PS^{1-} B) PS C) PS^{1+}
- (iii) Which of the following will be diamagnetic? Circle the correct choice.
 A) PS^{1-} B) PS C) PS^{1+}
- (iv) Draw Lewis dot structure of the molecule with the shortest bond length (based on the MO diagram) from among the three (PS^{1-} , PS, PS^{1+}), and indicate the bond order calculated from the Lewis dot structure. **(3 points)**

Lewis Dot Structure

Bond Order



2C4B10DE-2508-4996-A503-D2AEEE1AB161

midterm-exam-93388

#116 Page 10 of 10

SCRAP PAGE

**CHEMISTRY 110 –Midterm Exam****Thursday November 7, 2024, 6:30 – 8:30 pm**

Examiners: Pallavi Sirjoosingh and Ashok Kakkar

First name (please write as legibly as possible within the boxes)

--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--

Last name

--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--

Student ID number

--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--

This booklet has 8 questions on 10 pages (The last page can be used as a scrap page).

The staple must NOT be removed from the question booklet. Your intact question booklet must be returned at the end of the exam.

A separate booklet contains a periodic table, and datasheet.

Write your answers directly under the questions in the space provided. Do not write on the QR codes or near the top edges (as it may be cut off).

Please write **legibly and dark enough** for the scan to be readable. For students who use English as a second language, dictionaries are permitted. Calculators are allowed.

DO NOT ASK QUESTIONS OF THE INVIGILATORS about interpretations of the exam questions. If you have a concern about a question, please summarize it at the top of this page. All concerns will be evaluated after the exam. GOOD LUCK!

**Question 1 8 points** (*Assume all transitions are possible*)

a. 6 points Calculate the **energy (in kJ/mol)** required to ionize (completely remove) the electron from the **$n=3$** level of **He^+ ion** (He: Helium). Show your work including equations. (3 sig. figs.)

b. 2 points

(I) Select the largest (longest) wavelength from the three choices below:

- i. Wavelength of photon to *completely remove* an electron from *1s orbital of Hydrogen atom*
- ii. Wavelength of photon to *completely remove* an electron from *1s orbital of Boron atom (B)*
- iii. Wavelength of photon to *completely remove* an electron from *1s orbital of Boron cation (B^+)*

i or ii or iii

(II) Select the smallest (shortest) wavelength from the three choices below (*for H atom*):

- i. Wavelength of photon *emitted* for an electron transitioning from *2p to 1s orbital*
- ii. Wavelength of photon *absorbed* for an electron transitioning from *2s to 4p orbital*
- iii. Wavelength of photon *absorbed* for an electron transitioning from *1s to 3p orbital*

i or ii or iii



Question 2 8 points



A Chem110 student plots the graph shown on the left, from data obtained for the kinetic energy (eV) of the ejected electron versus frequency (s^{-1}) of the incident light, for an unknown metal. ($1\text{eV} = 1.6022 \times 10^{-19} \text{ J}$)

Based on the plot, answer the questions **a to c** below:

a. 1 point On the plot above, label the **threshold frequency**.

b. 2 points What is the value of the work potential (in J) of the unknown metal?

c. 3 points What is the *longest* wavelength (in nm) required to eject an electron from this metal's surface? Show your work including equation(s) used. (3 sig figs)

d. 2 points In the space provided below, plot the following

(Assume frequency is *above* threshold frequency):

- Kinetic energy* of an ejected electron versus the *intensity* of the incident light (left)
- Number of electrons* ejected versus the *frequency* of the incident light (right)



**Question 3** **6 points****a. 2 points**

In the ground state of Ge (Germanium; $Z = 32$) atom, **name** the following orbitals:

i. Orbital containing the electron that requires the maximum energy to be removed.

ii. Orbital containing the electron that experiences the minimum effective nuclear charge.

b. 2 points

An unknown anion (A^-) has the following *excited* electronic configuration: $[Ar] 4s^2 3d^3 4p^1$

(Ar: Argon; $Z = 18$)

Give the **condensed ground state electronic configuration** of the atom, **A** and the cation, A^{2+} .
Show your work. (Assume all electronic transitions are possible)

Electronic Configuration of A:

Electronic Configuration of A^{2+} :

c. 2 points

In a single atom, determine how many electrons can have the following set of quantum numbers?

(i) $n = 5; l = 3$

In a single atom, determine how many electrons can have the following set of quantum numbers?

(i) $n = 3; l = 2; m_s = +1/2$

**Question 4** **8 points**

Rank the following from 1 to 4 in **increasing** order. Provide the ranking in the boxes provided.

Note: Atomic number (Z) are provided for relevant atoms

a) Increasing (lowest to highest) order of *ionization energy*: (1= lowest; 4 = highest)

Mg

Mg: Z = 12

Rb

Rb: Z = 37

Cs

Cs: Z = 55

K

K : Z = 19

b) Increasing (lowest to highest) order of *bond angle*: (1= lowest; 4 = highest)

H₂OCCl₄XeF₄

HCN

c) Increasing (smallest to largest) order of *ionic size* (1= smallest; 4 = largest)

S²⁻

S: Z = 16

O²⁻

O: Z = 8

Be²⁺

Be: Z = 4

Na⁺

Na: Z = 11

d) Increasing (lowest to highest) order of *bond energy* (1= lowest; 4 = highest)

P-F

P-Cl

P-I

P-Br

**Question 5** **12 points**

a. 9 points For the molecules HClO_4 and TeF_6 , draw the most stable *Lewis structure* showing appropriate shape and *dashed/wedged bonds*, based on VSEPR. Determine the **molecular geometry** around the central atom. For HClO_4 , label the *most polar bond*.



b. 3 points (Note: Lone pairs are not shown in structures below)

(i) For the three *resonance* structures (1, 2 and 3) shown below, answer the following question:



(I) Circle the *most contributing* resonance structure.

Structure 1

Structure 2

Structure 3

(ii) For the three structures given (4, 5, and 6) below, answer the following questions:



(I) Circle all the structures that represent *resonance structures* of each other:

Structure 4

Structure 5

Structure 6

(II) Among the resonance structures, circle the *most contributing* resonance structure.

Structure 4

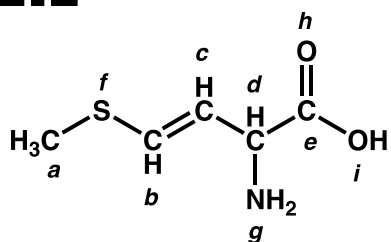
Structure 5

Structure 6

**Question 6** **6 points**

Using the data given below, calculate the **lattice energy** for SrO(s).

	ΔH° (kJ/mol)
$\text{Sr}(s) \longrightarrow \text{Sr}(g)$	+180
$\text{Sr}(g) \longrightarrow \text{Sr}^{2+}(g) + 2e^-$	+1468
$\text{O}_2(g) \longrightarrow 2\text{O}(g)$	+500
$\text{O}(g) + 2e^- \longrightarrow \text{O}^{2-}(g)$	+600
$\text{Sr}(s) + \frac{1}{2}\text{O}_2(g) \longrightarrow \text{SrO}(s)$	-550

**Question 7 10 points**

Consider the molecule given on the left, and answer questions (i) to (vi) using chemical bonding theories that we have learnt in class (valence bond theory, hybridization).

Note that the lone pairs are not shown in the structure:

i. Consider the carbon atoms marked in as ***a***, ***b***, ***c***, ***d***, and ***e*** in the molecule, and determine **hybridization** at each of the carbon centers. (2.5 points)

Atom	Hybridization
Carbon (<i>a</i>)	
Carbon (<i>b</i>)	
Carbon (<i>c</i>)	
Carbon (<i>d</i>)	
Carbon (<i>e</i>)	

ii. Now, consider the sulfur atom (S) marked as ***f***, nitrogen (N) atom (***g***), and oxygen(O) atoms (***h*** and ***i***), and determine their **hybridization**. (2 points)

Atom	Hybridization
Sulfur (<i>f</i>)	
Nitrogen (<i>g</i>)	
Oxygen (<i>h</i>)	
Oxygen (<i>i</i>)	

iii. **Describe** all the bonds formed by the carbon atoms ***a*** and ***b***, using sigma and pi designation as appropriate. (2 points)

Atom	Bonds formed (use sigma and pi as appropriate)
Carbon (<i>a</i>)	
Carbon (<i>b</i>)	

iv. How many total sp^3 -s sigma bonds are in the molecule? (1 point)

v. Determine **electronic** and **molecular geometry** at carbon (C) labeled ***e*** and sulfur (S) labeled ***f*** centres. (2 points)

Atom	Electronic Geometry	Molecular Geometry
Carbon (<i>e</i>)		
Sulfur (<i>f</i>)		

vi. Circle the value from the choices given below, that describes the **C-S-C bond angle at sulfur** (S) marked ***f***: (0.5 points)

>90° to <109.5°

109.5°

>109.5 to <120°

120°

>120° to <180°


Question 8 12 points

Using the **valence shell orbitals only** draw a **complete** molecular orbital (MO) diagram (with no s-p mixing) for the molecule **PS**, in the space provided below. **(6 points)**

Assume that 3s and 3p atomic orbitals will form molecular orbitals like those formed by the combination of 2s and 2p atomic orbitals that we learnt in class, and there is no s-p mixing.

Show **energy arrow**, **atomic orbitals** involved, **atom labels**, the **molecular orbitals** that are formed, **filled electrons** in the MO diagram etc. Make sure to label all the atomic and molecular orbitals.

Using the MO diagram above, answer the following questions:

- (i) Which one of the following will have the highest bond order? Circle the correct choice
 A) PS^{1-} B) PS C) PS^{1+}
- (ii) Which one of the following will have the longest bond length? Circle the correct choice
 A) PS^{1-} B) PS C) PS^{1+}
- (iii) Which of the following will be diamagnetic? Circle the correct choice.
 A) PS^{1-} B) PS C) PS^{1+}
- (iv) Draw Lewis dot structure of the molecule with the shortest bond length (based on the MO diagram) from among the three (PS^{1-} , PS, PS^{1+}), and indicate the bond order calculated from the Lewis dot structure. **(3 points)**

Lewis Dot Structure

Bond Order



365E6DD9-7675-4831-87E9-8FCAA99ED8FB

midterm-exam-93388

#117 Page 10 of 10

SCRAP PAGE

**CHEMISTRY 110 –Midterm Exam****Thursday November 7, 2024, 6:30 – 8:30 pm****Examiners: Pallavi Sirjoosingh and Ashok Kakkar**

First name (please write as legibly as possible within the boxes)

--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--

Last name

--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--

Student ID number

--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--

This booklet has 8 questions on 10 pages (The last page can be used as a scrap page).

The staple must NOT be removed from the question booklet. Your intact question booklet must be returned at the end of the exam.

A separate booklet contains a periodic table, and datasheet.

Write your answers directly under the questions in the space provided. Do not write on the QR codes or near the top edges (as it may be cut off).

Please write **legibly and dark enough** for the scan to be readable. For students who use English as a second language, dictionaries are permitted. Calculators are allowed.

DO NOT ASK QUESTIONS OF THE INVIGILATORS about interpretations of the exam questions. If you have a concern about a question, please summarize it at the top of this page. All concerns will be evaluated after the exam. GOOD LUCK!

**Question 1 8 points** (*Assume all transitions are possible*)

a. 6 points Calculate the **energy (in kJ/mol)** required to ionize (completely remove) the electron from the **n=3** level of **He⁺ ion** (He: Helium). Show your work including equations. (3 sig. figs.)

b. 2 points

(I) Select the largest (longest) wavelength from the three choices below:

- i. Wavelength of photon to *completely remove* an electron from *1s orbital of Hydrogen atom*
- ii. Wavelength of photon to *completely remove* an electron from *1s orbital of Boron atom (B)*
- iii. Wavelength of photon to *completely remove* an electron from *1s orbital of Boron cation (B⁺)*

i or ii or iii

(II) Select the smallest (shortest) wavelength from the three choices below (*for H atom*):

- i. Wavelength of photon *emitted* for an electron transitioning from *2p to 1s orbital*
- ii. Wavelength of photon *absorbed* for an electron transitioning from *2s to 4p orbital*
- iii. Wavelength of photon *absorbed* for an electron transitioning from *1s to 3p orbital*

i or ii or iii



Question 2 8 points



A Chem110 student plots the graph shown on the left, from data obtained for the kinetic energy (eV) of the ejected electron versus frequency (s^{-1}) of the incident light, for an unknown metal. ($1\text{eV} = 1.6022 \times 10^{-19} \text{ J}$)

Based on the plot, answer the questions **a to c** below:

a. 1 point On the plot above, label the **threshold frequency**.

b. 2 points What is the value of the work potential (in J) of the unknown metal?

c. 3 points What is the *longest* wavelength (in nm) required to eject an electron from this metal's surface? Show your work including equation(s) used. (3 sig figs)

d. 2 points In the space provided below, plot the following

(Assume frequency is *above* threshold frequency):

- i. *Kinetic energy* of an ejected electron versus the *intensity* of the incident light (left)
- ii. *Number of electrons* ejected versus the *frequency* of the incident light (right)



**Question 3** **6 points****a. 2 points**

In the ground state of Ge (Germanium; $Z = 32$) atom, **name** the following orbitals:

i. Orbital containing the electron that requires the maximum energy to be removed.

ii. Orbital containing the electron that experiences the minimum effective nuclear charge.

b. 2 points

An unknown anion (A^-) has the following *excited* electronic configuration: $[Ar] 4s^2 3d^3 4p^1$

(Ar: Argon; $Z = 18$)

Give the **condensed ground state electronic configuration** of the atom, **A** and the cation, **A^{2+}** .
Show your work. (Assume all electronic transitions are possible)

Electronic Configuration of A:

Electronic Configuration of A^{2+} :

c. 2 points

In a single atom, determine how many electrons can have the following set of quantum numbers?

(i) $n = 5; l = 3$

In a single atom, determine how many electrons can have the following set of quantum numbers?

(i) $n = 3; l = 2; m_s = +1/2$

**Question 4** **8 points**

Rank the following from 1 to 4 in **increasing** order. Provide the ranking in the boxes provided.

Note: Atomic number (Z) are provided for relevant atoms

a) Increasing (lowest to highest) order of *ionization energy*: (1= lowest; 4 = highest)

Mg

Mg: Z = 12

Rb

Rb: Z = 37

Cs

Cs: Z = 55

K

K : Z = 19

b) Increasing (lowest to highest) order of *bond angle*: (1= lowest; 4 = highest)

H₂OCCl₄XeF₄

HCN

c) Increasing (smallest to largest) order of *ionic size* (1= smallest; 4 = largest)

S²⁻

S: Z = 16

O²⁻

O: Z = 8

Be²⁺

Be: Z = 4

Na⁺

Na: Z = 11

d) Increasing (lowest to highest) order of *bond energy* (1= lowest; 4 = highest)

P-F

P-Cl

P-I

P-Br

**Question 5** **12 points**

a. 9 points For the molecules HClO_4 and TeF_6 , draw the most stable *Lewis structure* showing appropriate shape and *dashed/wedged bonds*, based on VSEPR. Determine the **molecular geometry** around the central atom. For HClO_4 , label the *most polar bond*.



b. 3 points (Note: Lone pairs are not shown in structures below)

(i) For the three *resonance* structures (1, 2 and 3) shown below, answer the following question:



(I) Circle the *most contributing* resonance structure.

Structure 1

Structure 2

Structure 3

(ii) For the three structures given (4, 5, and 6) below, answer the following questions:



(I) Circle all the structures that represent *resonance structures* of each other:

Structure 4

Structure 5

Structure 6

(II) Among the resonance structures, circle the *most contributing* resonance structure.

Structure 4

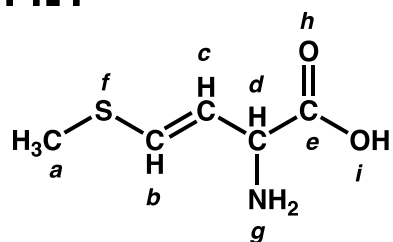
Structure 5

Structure 6

**Question 6** **6 points**

Using the data given below, calculate the **lattice energy** for SrO(s).

	ΔH° (kJ/mol)
$\text{Sr}(s) \longrightarrow \text{Sr}(g)$	+180
$\text{Sr}(g) \longrightarrow \text{Sr}^{2+}(g) + 2e^-$	+1468
$\text{O}_2(g) \longrightarrow 2\text{O}(g)$	+500
$\text{O}(g) + 2e^- \longrightarrow \text{O}^{2-}(g)$	+600
$\text{Sr}(s) + \frac{1}{2}\text{O}_2(g) \longrightarrow \text{SrO}(s)$	-550

**Question 7 10 points**

Consider the molecule given on the left, and answer questions (i) to (vi) using chemical bonding theories that we have learnt in class (valence bond theory, hybridization).

Note that the lone pairs are not shown in the structure:

i. Consider the carbon atoms marked in as ***a***, ***b***, ***c***, ***d***, and ***e*** in the molecule, and determine **hybridization** at each of the carbon centers. (2.5 points)

Atom	Hybridization
Carbon (<i>a</i>)	
Carbon (<i>b</i>)	
Carbon (<i>c</i>)	
Carbon (<i>d</i>)	
Carbon (<i>e</i>)	

ii. Now, consider the sulfur atom (S) marked as ***f***, nitrogen (N) atom (***g***), and oxygen(O) atoms (***h*** and ***i***), and determine their **hybridization**. (2 points)

Atom	Hybridization
Sulfur (<i>f</i>)	
Nitrogen (<i>g</i>)	
Oxygen (<i>h</i>)	
Oxygen (<i>i</i>)	

iii. **Describe** all the bonds formed by the carbon atoms ***a*** and ***b***, using sigma and pi designation as appropriate. (2 points)

Atom	Bonds formed (use sigma and pi as appropriate)
Carbon (<i>a</i>)	
Carbon (<i>b</i>)	

iv. How many total sp^3 -s sigma bonds are in the molecule? (1 point)

v. Determine **electronic** and **molecular geometry** at carbon (C) labeled ***e*** and sulfur (S) labeled ***f*** centres. (2 points)

Atom	Electronic Geometry	Molecular Geometry
Carbon (<i>e</i>)		
Sulfur (<i>f</i>)		

vi. Circle the value from the choices given below, that describes the **C-S-C bond angle at sulfur** (S) marked ***f***: (0.5 points)

$>90^\circ$ to $<109.5^\circ$ 109.5° >109.5 to $<120^\circ$ 120° $>120^\circ$ to $<180^\circ$


Question 8 12 points

Using the **valence shell orbitals only** draw a **complete** molecular orbital (MO) diagram (with no s-p mixing) for the molecule **PS**, in the space provided below. **(6 points)**

Assume that 3s and 3p atomic orbitals will form molecular orbitals like those formed by the combination of 2s and 2p atomic orbitals that we learnt in class, and there is no s-p mixing.

Show **energy arrow**, **atomic orbitals** involved, **atom labels**, the **molecular orbitals** that are formed, **filled electrons** in the MO diagram etc. Make sure to label all the atomic and molecular orbitals.

Using the MO diagram above, answer the following questions:

- (i) Which one of the following will have the highest bond order? Circle the correct choice
 A) PS^{1-} B) PS C) PS^{1+}
- (ii) Which one of the following will have the longest bond length? Circle the correct choice
 A) PS^{1-} B) PS C) PS^{1+}
- (iii) Which of the following will be diamagnetic? Circle the correct choice.
 A) PS^{1-} B) PS C) PS^{1+}
- (iv) Draw Lewis dot structure of the molecule with the shortest bond length (based on the MO diagram) from among the three (PS^{1-} , PS, PS^{1+}), and indicate the bond order calculated from the Lewis dot structure. **(3 points)**

Lewis Dot Structure

Bond Order



5F9B44BA-0675-4926-BB9C-8594766C5031

midterm-exam-93388

#118 Page 10 of 10

SCRAP PAGE

**CHEMISTRY 110 –Midterm Exam****Thursday November 7, 2024, 6:30 – 8:30 pm**

Examiners: Pallavi Sirjoosingh and Ashok Kakkar

First name (please write as legibly as possible within the boxes)

--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--

Last name

--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--

Student ID number

--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--

This booklet has 8 questions on 10 pages (The last page can be used as a scrap page).

The staple must NOT be removed from the question booklet. Your intact question booklet must be returned at the end of the exam.

A separate booklet contains a periodic table, and datasheet.

Write your answers directly under the questions in the space provided. Do not write on the QR codes or near the top edges (as it may be cut off).

Please write **legibly and dark enough** for the scan to be readable. For students who use English as a second language, dictionaries are permitted. Calculators are allowed.

DO NOT ASK QUESTIONS OF THE INVIGILATORS about interpretations of the exam questions. If you have a concern about a question, please summarize it at the top of this page. All concerns will be evaluated after the exam. GOOD LUCK!

**Question 1 8 points** (*Assume all transitions are possible*)

a. 6 points Calculate the **energy (in kJ/mol)** required to ionize (completely remove) the electron from the **$n=3$** level of **He^+ ion** (He: Helium). Show your work including equations. (3 sig. figs.)

b. 2 points

(I) Select the largest (longest) wavelength from the three choices below:

- i. Wavelength of photon to *completely remove* an electron from *1s orbital of Hydrogen atom*
- ii. Wavelength of photon to *completely remove* an electron from *1s orbital of Boron atom (B)*
- iii. Wavelength of photon to *completely remove* an electron from *1s orbital of Boron cation (B^+)*

i or ii or iii

(II) Select the smallest (shortest) wavelength from the three choices below (*for H atom*):

- i. Wavelength of photon *emitted* for an electron transitioning from *2p to 1s orbital*
- ii. Wavelength of photon *absorbed* for an electron transitioning from *2s to 4p orbital*
- iii. Wavelength of photon *absorbed* for an electron transitioning from *1s to 3p orbital*

i or ii or iii



Question 2 8 points



A Chem110 student plots the graph shown on the left, from data obtained for the kinetic energy (eV) of the ejected electron versus frequency (s^{-1}) of the incident light, for an unknown metal. ($1\text{eV} = 1.6022 \times 10^{-19} \text{ J}$)

Based on the plot, answer the questions **a to c** below:

a. 1 point On the plot above, label the **threshold frequency**.

b. 2 points What is the value of the work potential (in J) of the unknown metal?

c. 3 points What is the *longest* wavelength (in nm) required to eject an electron from this metal's surface? Show your work including equation(s) used. (3 sig figs)

d. 2 points In the space provided below, plot the following

(Assume frequency is *above* threshold frequency):

- i. *Kinetic energy* of an ejected electron versus the *intensity* of the incident light (left)
- ii. *Number of electrons* ejected versus the *frequency* of the incident light (right)



**Question 3** **6 points****a. 2 points**

In the ground state of Ge (Germanium; $Z = 32$) atom, **name** the following orbitals:

i. Orbital containing the electron that requires the maximum energy to be removed.

ii. Orbital containing the electron that experiences the minimum effective nuclear charge.

b. 2 points

An unknown anion (A^-) has the following *excited* electronic configuration: $[Ar] 4s^2 3d^3 4p^1$

(Ar: Argon; $Z = 18$)

Give the **condensed ground state electronic configuration** of the atom, **A** and the cation, A^{2+} .
Show your work. (Assume all electronic transitions are possible)

Electronic Configuration of A:

Electronic Configuration of A^{2+} :

c. 2 points

In a single atom, determine how many electrons can have the following set of quantum numbers?

(i) $n = 5; l = 3$

In a single atom, determine how many electrons can have the following set of quantum numbers?

(i) $n = 3; l = 2; m_s = +1/2$

**Question 4** **8 points**

Rank the following from 1 to 4 in **increasing** order. Provide the ranking in the boxes provided.

Note: Atomic number (Z) are provided for relevant atoms

a) Increasing (lowest to highest) order of *ionization energy*: (1= lowest; 4 = highest)

Mg

Mg: Z = 12

Rb

Rb: Z = 37

Cs

Cs: Z = 55

K

K : Z = 19

b) Increasing (lowest to highest) order of *bond angle*: (1= lowest; 4 = highest)

H₂OCCl₄XeF₄

HCN

c) Increasing (smallest to largest) order of *ionic size* (1= smallest; 4 = largest)

S²⁻

S: Z = 16

O²⁻

O: Z = 8

Be²⁺

Be: Z = 4

Na⁺

Na: Z = 11

d) Increasing (lowest to highest) order of *bond energy* (1= lowest; 4 = highest)

P-F

P-Cl

P-I

P-Br

**Question 5** **12 points**

a. 9 points For the molecules HClO_4 and TeF_6 , draw the most stable *Lewis structure* showing appropriate shape and *dashed/wedged bonds*, based on VSEPR. Determine the **molecular geometry** around the central atom. For HClO_4 , label the *most polar bond*.



b. 3 points (Note: Lone pairs are not shown in structures below)

(i) For the three *resonance* structures (1, 2 and 3) shown below, answer the following question:



(I) Circle the *most contributing* resonance structure.

Structure 1

Structure 2

Structure 3

(ii) For the three structures given (4, 5, and 6) below, answer the following questions:



(I) Circle all the structures that represent *resonance structures* of each other:

Structure 4

Structure 5

Structure 6

(II) Among the resonance structures, circle the *most contributing* resonance structure.

Structure 4

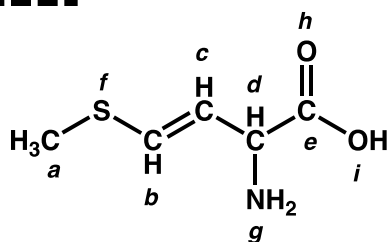
Structure 5

Structure 6

**Question 6** **6 points**

Using the data given below, calculate the **lattice energy** for SrO(s).

	ΔH° (kJ/mol)
$\text{Sr}(s) \longrightarrow \text{Sr}(g)$	+180
$\text{Sr}(g) \longrightarrow \text{Sr}^{2+}(g) + 2e^-$	+1468
$\text{O}_2(g) \longrightarrow 2\text{O}(g)$	+500
$\text{O}(g) + 2e^- \longrightarrow \text{O}^{2-}(g)$	+600
$\text{Sr}(s) + \frac{1}{2}\text{O}_2(g) \longrightarrow \text{SrO}(s)$	-550

**Question 7 10 points**

Consider the molecule given on the left, and answer questions (i) to (vi) using chemical bonding theories that we have learnt in class (valence bond theory, hybridization).

Note that the lone pairs are not shown in the structure:

i. Consider the carbon atoms marked in as ***a***, ***b***, ***c***, ***d***, and ***e*** in the molecule, and determine **hybridization** at each of the carbon centers. (2.5 points)

Atom	Hybridization
Carbon (<i>a</i>)	
Carbon (<i>b</i>)	
Carbon (<i>c</i>)	
Carbon (<i>d</i>)	
Carbon (<i>e</i>)	

ii. Now, consider the sulfur atom (S) marked as ***f***, nitrogen (N) atom (***g***), and oxygen(O) atoms (***h*** and ***i***), and determine their **hybridization**. (2 points)

Atom	Hybridization
Sulfur (<i>f</i>)	
Nitrogen (<i>g</i>)	
Oxygen (<i>h</i>)	
Oxygen (<i>i</i>)	

iii. **Describe** all the bonds formed by the carbon atoms ***a*** and ***b***, using sigma and pi designation as appropriate. (2 points)

Atom	Bonds formed (use sigma and pi as appropriate)
Carbon (<i>a</i>)	
Carbon (<i>b</i>)	

iv. How many total sp^3 -s sigma bonds are in the molecule? (1 point)

v. Determine **electronic** and **molecular geometry** at carbon (C) labeled ***e*** and sulfur (S) labeled ***f*** centres. (2 points)

Atom	Electronic Geometry	Molecular Geometry
Carbon (<i>e</i>)		
Sulfur (<i>f</i>)		

vi. Circle the value from the choices given below, that describes the **C-S-C bond angle at sulfur** (S) marked ***f***: (0.5 points)

>90° to <109.5° 109.5° >109.5 to <120° 120° >120° to <180°


Question 8 12 points

Using the **valence shell orbitals only** draw a **complete** molecular orbital (MO) diagram (with no s-p mixing) for the molecule **PS**, in the space provided below. **(6 points)**

Assume that 3s and 3p atomic orbitals will form molecular orbitals like those formed by the combination of 2s and 2p atomic orbitals that we learnt in class, and there is no s-p mixing.

Show **energy arrow**, **atomic orbitals** involved, **atom labels**, the **molecular orbitals** that are formed, **filled electrons** in the MO diagram etc. Make sure to **label all the atomic and molecular orbitals**.

Using the MO diagram above, answer the following questions:

- (i) Which one of the following will have the highest bond order? Circle the correct choice
 A) PS^{1-} B) PS C) PS^{1+}
- (ii) Which one of the following will have the longest bond length? Circle the correct choice
 A) PS^{1-} B) PS C) PS^{1+}
- (iii) Which of the following will be diamagnetic? Circle the correct choice.
 A) PS^{1-} B) PS C) PS^{1+}
- (iv) Draw Lewis dot structure of the molecule with the shortest bond length (based on the MO diagram) from among the three (PS^{1-} , PS, PS^{1+}), and indicate the bond order calculated from the Lewis dot structure. **(3 points)**

Lewis Dot Structure

Bond Order



27A78E1F-1AFF-4A58-A6FA-0B2C7895D1CC

midterm-exam-93388

#119 Page 10 of 10

SCRAP PAGE

**CHEMISTRY 110 –Midterm Exam****Thursday November 7, 2024, 6:30 – 8:30 pm**

Examiners: Pallavi Sirjoosingh and Ashok Kakkar

First name (please write as legibly as possible within the boxes)

--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--

Last name

--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--

Student ID number

--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--

This booklet has 8 questions on 10 pages (The last page can be used as a scrap page).

The staple must NOT be removed from the question booklet. Your intact question booklet must be returned at the end of the exam.

A separate booklet contains a periodic table, and datasheet.

Write your answers directly under the questions in the space provided. Do not write on the QR codes or near the top edges (as it may be cut off).

Please write **legibly and dark enough** for the scan to be readable. For students who use English as a second language, dictionaries are permitted. Calculators are allowed.

DO NOT ASK QUESTIONS OF THE INVIGILATORS about interpretations of the exam questions. If you have a concern about a question, please summarize it at the top of this page. All concerns will be evaluated after the exam. GOOD LUCK!

**Question 1 8 points** (*Assume all transitions are possible*)

a. 6 points Calculate the **energy (in kJ/mol)** required to ionize (completely remove) the electron from the **$n=3$** level of **He^+ ion** (He: Helium). Show your work including equations. (3 sig. figs.)

b. 2 points

(I) Select the largest (longest) wavelength from the three choices below:

- i. Wavelength of photon to *completely remove* an electron from *1s orbital of Hydrogen atom*
- ii. Wavelength of photon to *completely remove* an electron from *1s orbital of Boron atom (B)*
- iii. Wavelength of photon to *completely remove* an electron from *1s orbital of Boron cation (B^+)*

i or ii or iii

(II) Select the smallest (shortest) wavelength from the three choices below (*for H atom*):

- i. Wavelength of photon *emitted* for an electron transitioning from *2p to 1s orbital*
- ii. Wavelength of photon *absorbed* for an electron transitioning from *2s to 4p orbital*
- iii. Wavelength of photon *absorbed* for an electron transitioning from *1s to 3p orbital*

i or ii or iii



Question 2 8 points



A Chem110 student plots the graph shown on the left, from data obtained for the kinetic energy (eV) of the ejected electron versus frequency (s^{-1}) of the incident light, for an unknown metal. ($1\text{eV} = 1.6022 \times 10^{-19} \text{ J}$)

Based on the plot, answer the questions **a to c** below:

a. 1 point On the plot above, label the **threshold frequency**.

b. 2 points What is the value of the work potential (in J) of the unknown metal?

c. 3 points What is the *longest* wavelength (in nm) required to eject an electron from this metal's surface? Show your work including equation(s) used. (3 sig figs)

d. 2 points In the space provided below, plot the following

(Assume frequency is *above* threshold frequency):

- i. *Kinetic energy* of an ejected electron versus the *intensity* of the incident light (left)
- ii. *Number of electrons* ejected versus the *frequency* of the incident light (right)



**Question 3** **6 points****a. 2 points**

In the ground state of Ge (Germanium; $Z = 32$) atom, **name** the following orbitals:

i. Orbital containing the electron that requires the maximum energy to be removed.

ii. Orbital containing the electron that experiences the minimum effective nuclear charge.

b. 2 points

An unknown anion (A^-) has the following *excited* electronic configuration: $[Ar] 4s^2 3d^3 4p^1$

(Ar: Argon; $Z = 18$)

Give the **condensed ground state electronic configuration** of the atom, **A** and the cation, **A^{2+}** .
Show your work. (Assume all electronic transitions are possible)

Electronic Configuration of A:

Electronic Configuration of A^{2+} :

c. 2 points

In a single atom, determine how many electrons can have the following set of quantum numbers?

(i) $n = 5; l = 3$

In a single atom, determine how many electrons can have the following set of quantum numbers?

(i) $n = 3; l = 2; m_s = +1/2$

**Question 4** **8 points**

Rank the following from 1 to 4 in **increasing** order. Provide the ranking in the boxes provided.

Note: Atomic number (Z) are provided for relevant atoms

a) Increasing (lowest to highest) order of *ionization energy*: (1= lowest; 4 = highest)

Mg

Mg: Z = 12

Rb

Rb: Z = 37

Cs

Cs: Z = 55

K

K : Z = 19

b) Increasing (lowest to highest) order of *bond angle*: (1= lowest; 4 = highest)

H₂OCCl₄XeF₄

HCN

c) Increasing (smallest to largest) order of *ionic size* (1= smallest; 4 = largest)

S²⁻

S: Z = 16

O²⁻

O: Z = 8

Be²⁺

Be: Z = 4

Na⁺

Na: Z = 11

d) Increasing (lowest to highest) order of *bond energy* (1= lowest; 4 = highest)

P-F

P-Cl

P-I

P-Br

**Question 5** **12 points**

a. 9 points For the molecules HClO_4 and TeF_6 , draw the most stable *Lewis structure* showing appropriate shape and *dashed/wedged bonds*, based on VSEPR. Determine the **molecular geometry** around the central atom. For HClO_4 , label the *most polar bond*.



b. 3 points (Note: Lone pairs are not shown in structures below)

(i) For the three *resonance* structures (1, 2 and 3) shown below, answer the following question:



(I) Circle the *most contributing* resonance structure.

Structure 1

Structure 2

Structure 3

(ii) For the three structures given (4, 5, and 6) below, answer the following questions:



(I) Circle all the structures that represent *resonance structures* of each other:

Structure 4

Structure 5

Structure 6

(II) Among the resonance structures, circle the *most contributing* resonance structure.

Structure 4

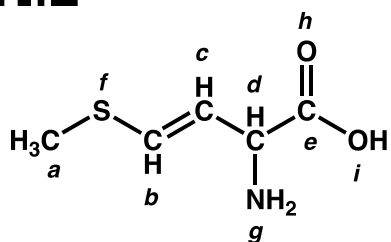
Structure 5

Structure 6

**Question 6** **6 points**

Using the data given below, calculate the **lattice energy** for SrO(s).

	ΔH° (kJ/mol)
$\text{Sr}(s) \longrightarrow \text{Sr}(g)$	+180
$\text{Sr}(g) \longrightarrow \text{Sr}^{2+}(g) + 2e^-$	+1468
$\text{O}_2(g) \longrightarrow 2\text{O}(g)$	+500
$\text{O}(g) + 2e^- \longrightarrow \text{O}^{2-}(g)$	+600
$\text{Sr}(s) + \frac{1}{2}\text{O}_2(g) \longrightarrow \text{SrO}(s)$	-550

**Question 7 10 points**

Consider the molecule given on the left, and answer questions (i) to (vi) using chemical bonding theories that we have learnt in class (valence bond theory, hybridization).

Note that the lone pairs are not shown in the structure:

i. Consider the carbon atoms marked in as ***a***, ***b***, ***c***, ***d***, and ***e*** in the molecule, and determine **hybridization** at each of the carbon centers. (2.5 points)

Atom	Hybridization
Carbon (<i>a</i>)	
Carbon (<i>b</i>)	
Carbon (<i>c</i>)	
Carbon (<i>d</i>)	
Carbon (<i>e</i>)	

ii. Now, consider the sulfur atom (S) marked as ***f***, nitrogen (N) atom (***g***), and oxygen(O) atoms (***h*** and ***i***), and determine their **hybridization**. (2 points)

Atom	Hybridization
Sulfur (<i>f</i>)	
Nitrogen (<i>g</i>)	
Oxygen (<i>h</i>)	
Oxygen (<i>i</i>)	

iii. **Describe** all the bonds formed by the carbon atoms ***a*** and ***b***, using sigma and pi designation as appropriate. (2 points)

Atom	Bonds formed (use sigma and pi as appropriate)
Carbon (<i>a</i>)	
Carbon (<i>b</i>)	

iv. How many total sp^3 -s sigma bonds are in the molecule? (1 point)

v. Determine **electronic** and **molecular geometry** at carbon (C) labeled ***e*** and sulfur (S) labeled ***f*** centres. (2 points)

Atom	Electronic Geometry	Molecular Geometry
Carbon (<i>e</i>)		
Sulfur (<i>f</i>)		

vi. Circle the value from the choices given below, that describes the **C-S-C bond angle at sulfur** (S) marked ***f***: (0.5 points)

>90° to <109.5° 109.5° >109.5 to <120° 120° >120° to <180°


Question 8 12 points

Using the **valence shell orbitals only** draw a **complete** molecular orbital (MO) diagram (with no s-p mixing) for the molecule **PS**, in the space provided below. **(6 points)**

Assume that 3s and 3p atomic orbitals will form molecular orbitals like those formed by the combination of 2s and 2p atomic orbitals that we learnt in class, and there is no s-p mixing.

Show **energy arrow**, **atomic orbitals** involved, **atom labels**, the **molecular orbitals** that are formed, **filled electrons** in the MO diagram etc. Make sure to label all the atomic and molecular orbitals.

Using the MO diagram above, answer the following questions:

- (i) Which one of the following will have the highest bond order? Circle the correct choice
 A) PS^{1-} B) PS C) PS^{1+}
- (ii) Which one of the following will have the longest bond length? Circle the correct choice
 A) PS^{1-} B) PS C) PS^{1+}
- (iii) Which of the following will be diamagnetic? Circle the correct choice.
 A) PS^{1-} B) PS C) PS^{1+}
- (iv) Draw Lewis dot structure of the molecule with the shortest bond length (based on the MO diagram) from among the three (PS^{1-} , PS, PS^{1+}), and indicate the bond order calculated from the Lewis dot structure. **(3 points)**

Lewis Dot Structure

Bond Order



10DEC768-F07D-4C5D-82DA-3013E8D0BE01

midterm-exam-93388

#120 Page 10 of 10

SCRAP PAGE

**CHEMISTRY 110 –Midterm Exam****Thursday November 7, 2024, 6:30 – 8:30 pm**

Examiners: Pallavi Sirjoosingh and Ashok Kakkar

First name (please write as legibly as possible within the boxes)

--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--

Last name

--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--

Student ID number

--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--

This booklet has 8 questions on 10 pages (The last page can be used as a scrap page).

The staple must NOT be removed from the question booklet. Your intact question booklet must be returned at the end of the exam.

A separate booklet contains a periodic table, and datasheet.

Write your answers directly under the questions in the space provided. Do not write on the QR codes or near the top edges (as it may be cut off).

Please write **legibly and dark enough** for the scan to be readable. For students who use English as a second language, dictionaries are permitted. Calculators are allowed.

DO NOT ASK QUESTIONS OF THE INVIGILATORS about interpretations of the exam questions. If you have a concern about a question, please summarize it at the top of this page. All concerns will be evaluated after the exam. GOOD LUCK!

**Question 1 8 points** (*Assume all transitions are possible*)

a. 6 points Calculate the **energy (in kJ/mol)** required to ionize (completely remove) the electron from the **n=3** level of **He⁺ ion** (He: Helium). Show your work including equations. (3 sig. figs.)

b. 2 points

(I) Select the largest (longest) wavelength from the three choices below:

- i. Wavelength of photon to *completely remove* an electron from *1s orbital of Hydrogen atom*
- ii. Wavelength of photon to *completely remove* an electron from *1s orbital of Boron atom (B)*
- iii. Wavelength of photon to *completely remove* an electron from *1s orbital of Boron cation (B⁺)*

i or ii or iii

(II) Select the smallest (shortest) wavelength from the three choices below (*for H atom*):

- i. Wavelength of photon *emitted* for an electron transitioning from *2p to 1s orbital*
- ii. Wavelength of photon *absorbed* for an electron transitioning from *2s to 4p orbital*
- iii. Wavelength of photon *absorbed* for an electron transitioning from *1s to 3p orbital*

i or ii or iii



Question 2 8 points



A Chem110 student plots the graph shown on the left, from data obtained for the kinetic energy (eV) of the ejected electron versus frequency (s^{-1}) of the incident light, for an unknown metal. ($1\text{eV} = 1.6022 \times 10^{-19} \text{ J}$)

Based on the plot, answer the questions **a to c** below:

a. 1 point On the plot above, label the **threshold frequency**.

b. 2 points What is the value of the work potential (in J) of the unknown metal?

c. 3 points What is the *longest* wavelength (in nm) required to eject an electron from this metal's surface? Show your work including equation(s) used. (3 sig figs)

d. 2 points In the space provided below, plot the following

(Assume frequency is *above* threshold frequency):

- Kinetic energy* of an ejected electron versus the *intensity* of the incident light (left)
- Number of electrons* ejected versus the *frequency* of the incident light (right)



**Question 3** **6 points****a. 2 points**

In the ground state of Ge (Germanium; $Z = 32$) atom, **name** the following orbitals:

i. Orbital containing the electron that requires the maximum energy to be removed.

ii. Orbital containing the electron that experiences the minimum effective nuclear charge.

b. 2 points

An unknown anion (A^-) has the following *excited* electronic configuration: $[Ar] 4s^2 3d^3 4p^1$

(Ar: Argon; $Z = 18$)

Give the **condensed ground state electronic configuration** of the atom, **A** and the cation, **A^{2+}** .
Show your work. (Assume all electronic transitions are possible)

Electronic Configuration of A:

Electronic Configuration of A^{2+} :

c. 2 points

In a single atom, determine how many electrons can have the following set of quantum numbers?

(i) $n = 5; l = 3$

In a single atom, determine how many electrons can have the following set of quantum numbers?

(i) $n = 3; l = 2; m_s = +1/2$

**Question 4** **8 points**

Rank the following from 1 to 4 in **increasing** order. Provide the ranking in the boxes provided.

Note: Atomic number (Z) are provided for relevant atoms

a) Increasing (lowest to highest) order of *ionization energy*: (1= lowest; 4 = highest)

Mg

Mg: Z = 12

Rb

Rb: Z = 37

Cs

Cs: Z = 55

K

K : Z = 19

b) Increasing (lowest to highest) order of *bond angle*: (1= lowest; 4 = highest)

H₂OCCl₄XeF₄

HCN

c) Increasing (smallest to largest) order of *ionic size* (1= smallest; 4 = largest)

S²⁻

S: Z = 16

O²⁻

O: Z = 8

Be²⁺

Be: Z = 4

Na⁺

Na: Z = 11

d) Increasing (lowest to highest) order of *bond energy* (1= lowest; 4 = highest)

P-F

P-Cl

P-I

P-Br

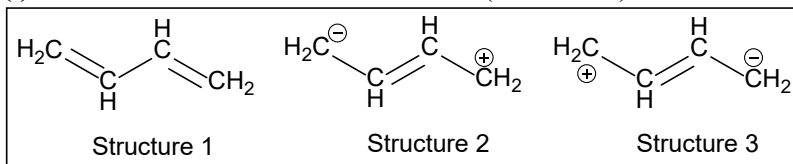
**Question 5** **12 points**

a. 9 points For the molecules HClO_4 and TeF_6 , draw the most stable *Lewis structure* showing appropriate shape and *dashed/wedged bonds*, based on VSEPR. Determine the **molecular geometry** around the central atom. For HClO_4 , label the *most polar bond*.



b. 3 points (Note: Lone pairs are not shown in structures below)

(i) For the three *resonance* structures (1, 2 and 3) shown below, answer the following question:



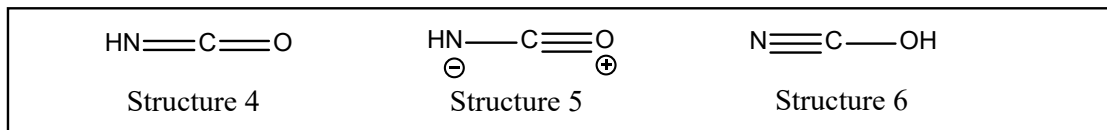
(I) Circle the *most contributing* resonance structure.

Structure 1

Structure 2

Structure 3

(ii) For the three structures given (4, 5, and 6) below, answer the following questions:



(I) Circle all the structures that represent *resonance structures* of each other:

Structure 4

Structure 5

Structure 6

(II) Among the resonance structures, circle the *most contributing* resonance structure.

Structure 4

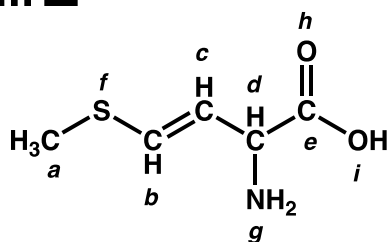
Structure 5

Structure 6

**Question 6** **6 points**

Using the data given below, calculate the **lattice energy** for SrO(s).

	ΔH° (kJ/mol)
$\text{Sr}(s) \longrightarrow \text{Sr}(g)$	+180
$\text{Sr}(g) \longrightarrow \text{Sr}^{2+}(g) + 2e^-$	+1468
$\text{O}_2(g) \longrightarrow 2\text{O}(g)$	+500
$\text{O}(g) + 2e^- \longrightarrow \text{O}^{2-}(g)$	+600
$\text{Sr}(s) + \frac{1}{2}\text{O}_2(g) \longrightarrow \text{SrO}(s)$	-550

**Question 7 10 points**

Consider the molecule given on the left, and answer questions (i) to (vi) using chemical bonding theories that we have learnt in class (valence bond theory, hybridization).

Note that the lone pairs are not shown in the structure:

i. Consider the carbon atoms marked in as ***a***, ***b***, ***c***, ***d***, and ***e*** in the molecule, and determine **hybridization** at each of the carbon centers. (2.5 points)

Atom	Hybridization
Carbon (<i>a</i>)	
Carbon (<i>b</i>)	
Carbon (<i>c</i>)	
Carbon (<i>d</i>)	
Carbon (<i>e</i>)	

ii. Now, consider the sulfur atom (S) marked as ***f***, nitrogen (N) atom (***g***), and oxygen(O) atoms (***h*** and ***i***), and determine their **hybridization**. (2 points)

Atom	Hybridization
Sulfur (<i>f</i>)	
Nitrogen (<i>g</i>)	
Oxygen (<i>h</i>)	
Oxygen (<i>i</i>)	

iii. **Describe** all the bonds formed by the carbon atoms ***a*** and ***b***, using sigma and pi designation as appropriate. (2 points)

Atom	Bonds formed (use sigma and pi as appropriate)
Carbon (<i>a</i>)	
Carbon (<i>b</i>)	

iv. How many total sp^3 -s sigma bonds are in the molecule? (1 point)

v. Determine **electronic** and **molecular geometry** at carbon (C) labeled ***e*** and sulfur (S) labeled ***f*** centres. (2 points)

Atom	Electronic Geometry	Molecular Geometry
Carbon (<i>e</i>)		
Sulfur (<i>f</i>)		

vi. Circle the value from the choices given below, that describes the **C-S-C bond angle at sulfur** (S) marked ***f***: (0.5 points)

>90° to <109.5° 109.5° >109.5 to <120° 120° >120° to <180°

**Question 8 12 points**

Using the **valence shell orbitals only** draw a **complete** molecular orbital (MO) diagram (with no s-p mixing) for the molecule **PS**, in the space provided below. **(6 points)**

Assume that 3s and 3p atomic orbitals will form molecular orbitals like those formed by the combination of 2s and 2p atomic orbitals that we learnt in class, and there is no s-p mixing.

Show **energy arrow**, **atomic orbitals** involved, **atom labels**, the **molecular orbitals** that are formed, **filled electrons** in the MO diagram etc. Make sure to label all the atomic and molecular orbitals.

Using the MO diagram above, answer the following questions:

- (i) Which one of the following will have the highest bond order? Circle the correct choice
A) PS^{1-} B) PS C) PS^{1+}
- (ii) Which one of the following will have the longest bond length? Circle the correct choice
A) PS^{1-} B) PS C) PS^{1+}
- (iii) Which of the following will be diamagnetic? Circle the correct choice.
A) PS^{1-} B) PS C) PS^{1+}
- (iv) Draw Lewis dot structure of the molecule with the shortest bond length (based on the MO diagram) from among the three (PS^{1-} , PS, PS^{1+}), and indicate the bond order calculated from the Lewis dot structure. **(3 points)**

Lewis Dot Structure

Bond Order



2E74EE05-013E-42AE-83F8-83B0F74C11E4

midterm-exam-93388

#121 Page 10 of 10

SCRAP PAGE

**CHEMISTRY 110 –Midterm Exam****Thursday November 7, 2024, 6:30 – 8:30 pm**

Examiners: Pallavi Sirjoosingh and Ashok Kakkar

First name (please write as legibly as possible within the boxes)

--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--

Last name

--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--

Student ID number

--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--

This booklet has 8 questions on 10 pages (The last page can be used as a scrap page).

The staple must NOT be removed from the question booklet. Your intact question booklet must be returned at the end of the exam.

A separate booklet contains a periodic table, and datasheet.

Write your answers directly under the questions in the space provided. Do not write on the QR codes or near the top edges (as it may be cut off).

Please write **legibly and dark enough** for the scan to be readable. For students who use English as a second language, dictionaries are permitted. Calculators are allowed.

DO NOT ASK QUESTIONS OF THE INVIGILATORS about interpretations of the exam questions. If you have a concern about a question, please summarize it at the top of this page. All concerns will be evaluated after the exam. GOOD LUCK!

**Question 1 8 points** (*Assume all transitions are possible*)

a. 6 points Calculate the **energy (in kJ/mol)** required to ionize (completely remove) the electron from the **n=3** level of **He⁺ ion** (He: Helium). Show your work including equations. (3 sig. figs.)

b. 2 points

(I) Select the largest (longest) wavelength from the three choices below:

- i. Wavelength of photon to *completely remove* an electron from *1s orbital of Hydrogen atom*
- ii. Wavelength of photon to *completely remove* an electron from *1s orbital of Boron atom (B)*
- iii. Wavelength of photon to *completely remove* an electron from *1s orbital of Boron cation (B⁺)*

i or ii or iii

(II) Select the smallest (shortest) wavelength from the three choices below (*for H atom*):

- i. Wavelength of photon *emitted* for an electron transitioning from *2p to 1s orbital*
- ii. Wavelength of photon *absorbed* for an electron transitioning from *2s to 4p orbital*
- iii. Wavelength of photon *absorbed* for an electron transitioning from *1s to 3p orbital*

i or ii or iii



Question 2 8 points



A Chem110 student plots the graph shown on the left, from data obtained for the kinetic energy (eV) of the ejected electron versus frequency (s^{-1}) of the incident light, for an unknown metal. ($1\text{eV} = 1.6022 \times 10^{-19} \text{ J}$)

Based on the plot, answer the questions **a to c** below:

a. 1 point On the plot above, label the **threshold frequency**.

b. 2 points What is the value of the work potential (in J) of the unknown metal?

c. 3 points What is the *longest* wavelength (in nm) required to eject an electron from this metal's surface? Show your work including equation(s) used. (3 sig figs)

d. 2 points In the space provided below, plot the following

(Assume frequency is *above* threshold frequency):

- i. *Kinetic energy* of an ejected electron versus the *intensity* of the incident light (left)
- ii. *Number of electrons* ejected versus the *frequency* of the incident light (right)



**Question 3** **6 points****a. 2 points**

In the ground state of Ge (Germanium; $Z = 32$) atom, **name** the following orbitals:

i. Orbital containing the electron that requires the maximum energy to be removed.

ii. Orbital containing the electron that experiences the minimum effective nuclear charge.

b. 2 points

An unknown anion (A^-) has the following *excited* electronic configuration: $[Ar] 4s^2 3d^3 4p^1$

(Ar: Argon; $Z = 18$)

Give the **condensed ground state electronic configuration** of the atom, **A** and the cation, **A^{2+}** .
Show your work. (Assume all electronic transitions are possible)

Electronic Configuration of A:

Electronic Configuration of A^{2+} :

c. 2 points

In a single atom, determine how many electrons can have the following set of quantum numbers?

(i) $n = 5; l = 3$

In a single atom, determine how many electrons can have the following set of quantum numbers?

(i) $n = 3; l = 2; m_s = +1/2$

**Question 4** **8 points**

Rank the following from 1 to 4 in **increasing** order. Provide the ranking in the boxes provided.

Note: Atomic number (Z) are provided for relevant atoms

a) Increasing (lowest to highest) order of *ionization energy*: (1= lowest; 4 = highest)

Mg

Mg: Z = 12

Rb

Rb: Z = 37

Cs

Cs: Z = 55

K

K : Z = 19

b) Increasing (lowest to highest) order of *bond angle*: (1= lowest; 4 = highest)

H₂OCCl₄XeF₄

HCN

c) Increasing (smallest to largest) order of *ionic size* (1= smallest; 4 = largest)

S²⁻

S: Z = 16

O²⁻

O: Z = 8

Be²⁺

Be: Z = 4

Na⁺

Na: Z = 11

d) Increasing (lowest to highest) order of *bond energy* (1= lowest; 4 = highest)

P-F

P-Cl

P-I

P-Br

**Question 5** **12 points**

a. 9 points For the molecules HClO_4 and TeF_6 , draw the most stable *Lewis structure* showing appropriate shape and *dashed/wedged bonds*, based on VSEPR. Determine the **molecular geometry** around the central atom. For HClO_4 , label the *most polar bond*.



b. 3 points (Note: Lone pairs are not shown in structures below)

(i) For the three *resonance* structures (1, 2 and 3) shown below, answer the following question:



(I) Circle the *most contributing* resonance structure.

Structure 1

Structure 2

Structure 3

(ii) For the three structures given (4, 5, and 6) below, answer the following questions:



(I) Circle all the structures that represent *resonance structures* of each other:

Structure 4

Structure 5

Structure 6

(II) Among the resonance structures, circle the *most contributing* resonance structure.

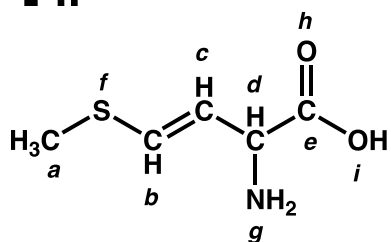
Structure 4

Structure 5

Structure 6

**Question 6** **6 points**Using the data given below, calculate the **lattice energy** for SrO(s).

	ΔH° (kJ/mol)
$\text{Sr}(s) \longrightarrow \text{Sr}(g)$	+180
$\text{Sr}(g) \longrightarrow \text{Sr}^{2+}(g) + 2e^-$	+1468
$\text{O}_2(g) \longrightarrow 2\text{O}(g)$	+500
$\text{O}(g) + 2e^- \longrightarrow \text{O}^{2-}(g)$	+600
$\text{Sr}(s) + \frac{1}{2}\text{O}_2(g) \longrightarrow \text{SrO}(s)$	-550

**Question 7 10 points**

Consider the molecule given on the left, and answer questions (i) to (vi) using chemical bonding theories that we have learnt in class (valence bond theory, hybridization).

Note that the lone pairs are not shown in the structure:

i. Consider the carbon atoms marked in as ***a***, ***b***, ***c***, ***d***, and ***e*** in the molecule, and determine **hybridization** at each of the carbon centers. (2.5 points)

Atom	Hybridization
Carbon (<i>a</i>)	
Carbon (<i>b</i>)	
Carbon (<i>c</i>)	
Carbon (<i>d</i>)	
Carbon (<i>e</i>)	

ii. Now, consider the sulfur atom (S) marked as ***f***, nitrogen (N) atom (***g***), and oxygen(O) atoms (***h*** and ***i***), and determine their **hybridization**. (2 points)

Atom	Hybridization
Sulfur (<i>f</i>)	
Nitrogen (<i>g</i>)	
Oxygen (<i>h</i>)	
Oxygen (<i>i</i>)	

iii. **Describe** all the bonds formed by the carbon atoms ***a*** and ***b***, using sigma and pi designation as appropriate. (2 points)

Atom	Bonds formed (use sigma and pi as appropriate)
Carbon (<i>a</i>)	
Carbon (<i>b</i>)	

iv. How many total sp^3 -s sigma bonds are in the molecule? (1 point)

v. Determine **electronic** and **molecular geometry** at carbon (C) labeled ***e*** and sulfur (S) labeled ***f*** centres. (2 points)

Atom	Electronic Geometry	Molecular Geometry
Carbon (<i>e</i>)		
Sulfur (<i>f</i>)		

vi. Circle the value from the choices given below, that describes the **C-S-C bond angle at sulfur** (S) marked ***f***: (0.5 points)

>90° to <109.5° 109.5° >109.5 to <120° 120° >120° to <180°


Question 8 12 points

Using the **valence shell orbitals only** draw a **complete** molecular orbital (MO) diagram (with no s-p mixing) for the molecule **PS**, in the space provided below. **(6 points)**

Assume that 3s and 3p atomic orbitals will form molecular orbitals like those formed by the combination of 2s and 2p atomic orbitals that we learnt in class, and there is no s-p mixing.

Show **energy arrow**, **atomic orbitals** involved, **atom labels**, the **molecular orbitals** that are formed, **filled electrons** in the MO diagram etc. Make sure to label all the atomic and molecular orbitals.

Using the MO diagram above, answer the following questions:

- (i) Which one of the following will have the highest bond order? Circle the correct choice
 A) PS^{1-} B) PS C) PS^{1+}
- (ii) Which one of the following will have the longest bond length? Circle the correct choice
 A) PS^{1-} B) PS C) PS^{1+}
- (iii) Which of the following will be diamagnetic? Circle the correct choice.
 A) PS^{1-} B) PS C) PS^{1+}
- (iv) Draw Lewis dot structure of the molecule with the shortest bond length (based on the MO diagram) from among the three (PS^{1-} , PS, PS^{1+}), and indicate the bond order calculated from the Lewis dot structure. **(3 points)**

Lewis Dot Structure

Bond Order



60B902CD-43A9-4F7D-85B4-6E11DDF2BADC

midterm-exam-93388

#122 Page 10 of 10

SCRAP PAGE

**CHEMISTRY 110 –Midterm Exam****Thursday November 7, 2024, 6:30 – 8:30 pm****Examiners: Pallavi Sirjoosingh and Ashok Kakkar**

First name (please write as legibly as possible within the boxes)

--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--

Last name

--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--

Student ID number

--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--

This booklet has 8 questions on 10 pages (The last page can be used as a scrap page).

The staple must NOT be removed from the question booklet. Your intact question booklet must be returned at the end of the exam.

A separate booklet contains a periodic table, and datasheet.

Write your answers directly under the questions in the space provided. Do not write on the QR codes or near the top edges (as it may be cut off).

Please write **legibly and dark enough** for the scan to be readable. For students who use English as a second language, dictionaries are permitted. Calculators are allowed.

DO NOT ASK QUESTIONS OF THE INVIGILATORS about interpretations of the exam questions. If you have a concern about a question, please summarize it at the top of this page. All concerns will be evaluated after the exam. GOOD LUCK!

**Question 1 8 points** (*Assume all transitions are possible*)

a. 6 points Calculate the **energy (in kJ/mol)** required to ionize (completely remove) the electron from the **$n=3$** level of **He^+ ion** (He: Helium). Show your work including equations. (3 sig. figs.)

b. 2 points

(I) Select the largest (longest) wavelength from the three choices below:

- i. Wavelength of photon to *completely remove* an electron from *1s orbital of Hydrogen atom*
- ii. Wavelength of photon to *completely remove* an electron from *1s orbital of Boron atom (B)*
- iii. Wavelength of photon to *completely remove* an electron from *1s orbital of Boron cation (B^+)*

i or ii or iii

(II) Select the smallest (shortest) wavelength from the three choices below (*for H atom*):

- i. Wavelength of photon *emitted* for an electron transitioning from *2p to 1s orbital*
- ii. Wavelength of photon *absorbed* for an electron transitioning from *2s to 4p orbital*
- iii. Wavelength of photon *absorbed* for an electron transitioning from *1s to 3p orbital*

i or ii or iii



Question 2 8 points



A Chem110 student plots the graph shown on the left, from data obtained for the kinetic energy (eV) of the ejected electron versus frequency (s^{-1}) of the incident light, for an unknown metal. ($1\text{eV} = 1.6022 \times 10^{-19} \text{ J}$)

Based on the plot, answer the questions **a to c** below:

a. 1 point On the plot above, label the **threshold frequency**.

b. 2 points What is the value of the work potential (in J) of the unknown metal?

c. 3 points What is the *longest* wavelength (in nm) required to eject an electron from this metal's surface? Show your work including equation(s) used. (3 sig figs)

d. 2 points In the space provided below, plot the following

(Assume frequency is *above* threshold frequency):

- Kinetic energy* of an ejected electron versus the *intensity* of the incident light (left)
- Number of electrons* ejected versus the *frequency* of the incident light (right)



**Question 3** **6 points****a. 2 points**

In the ground state of Ge (Germanium; $Z = 32$) atom, **name** the following orbitals:

i. Orbital containing the electron that requires the maximum energy to be removed.

ii. Orbital containing the electron that experiences the minimum effective nuclear charge.

b. 2 points

An unknown anion (A^-) has the following *excited* electronic configuration: $[Ar] 4s^2 3d^3 4p^1$

(Ar: Argon; $Z = 18$)

Give the **condensed ground state electronic configuration** of the atom, **A** and the cation, **A^{2+}** .
Show your work. (Assume all electronic transitions are possible)

Electronic Configuration of A:

Electronic Configuration of A^{2+} :

c. 2 points

In a single atom, determine how many electrons can have the following set of quantum numbers?

(i) $n = 5; l = 3$

In a single atom, determine how many electrons can have the following set of quantum numbers?

(i) $n = 3; l = 2; m_s = +1/2$

**Question 4** **8 points**

Rank the following from 1 to 4 in **increasing** order. Provide the ranking in the boxes provided.

Note: Atomic number (Z) are provided for relevant atoms

a) Increasing (lowest to highest) order of *ionization energy*: (1= lowest; 4 = highest)

Mg

Mg: Z = 12

Rb

Rb: Z = 37

Cs

Cs: Z = 55

K

K : Z = 19

b) Increasing (lowest to highest) order of *bond angle*: (1= lowest; 4 = highest)

H₂OCCl₄XeF₄

HCN

c) Increasing (smallest to largest) order of *ionic size* (1= smallest; 4 = largest)

S²⁻

S: Z = 16

O²⁻

O: Z = 8

Be²⁺

Be: Z = 4

Na⁺

Na: Z = 11

d) Increasing (lowest to highest) order of *bond energy* (1= lowest; 4 = highest)

P-F

P-Cl

P-I

P-Br

**Question 5** **12 points**

a. 9 points For the molecules HClO_4 and TeF_6 , draw the most stable *Lewis structure* showing appropriate shape and *dashed/wedged bonds*, based on VSEPR. Determine the **molecular geometry** around the central atom. For HClO_4 , label the *most polar bond*.

HClO_4 (there are no O-O bonds)

Label the most polar bond

Molecular Geometry around Cl

TeF_6

Molecular Geometry around Te

b. 3 points (Note: Lone pairs are not shown in structures below)

(i) For the three *resonance* structures (1, 2 and 3) shown below, answer the following question:



(I) Circle the *most contributing* resonance structure.

Structure 1

Structure 2

Structure 3

(ii) For the three structures given (4, 5, and 6) below, answer the following questions:



(I) Circle all the structures that represent *resonance structures* of each other:

Structure 4

Structure 5

Structure 6

(II) Among the resonance structures, circle the *most contributing* resonance structure.

Structure 4

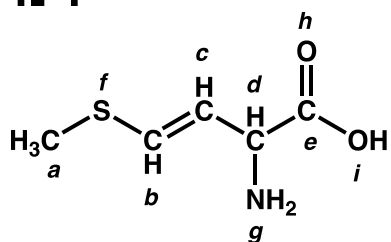
Structure 5

Structure 6

**Question 6** **6 points**

Using the data given below, calculate the **lattice energy** for SrO(s).

	ΔH° (kJ/mol)
$\text{Sr}(s) \longrightarrow \text{Sr}(g)$	+180
$\text{Sr}(g) \longrightarrow \text{Sr}^{2+}(g) + 2e^-$	+1468
$\text{O}_2(g) \longrightarrow 2\text{O}(g)$	+500
$\text{O}(g) + 2e^- \longrightarrow \text{O}^{2-}(g)$	+600
$\text{Sr}(s) + \frac{1}{2}\text{O}_2(g) \longrightarrow \text{SrO}(s)$	-550

**Question 7 10 points**

Consider the molecule given on the left, and answer questions (i) to (vi) using chemical bonding theories that we have learnt in class (valence bond theory, hybridization).

Note that the lone pairs are not shown in the structure:

i. Consider the carbon atoms marked in as ***a***, ***b***, ***c***, ***d***, and ***e*** in the molecule, and determine **hybridization** at each of the carbon centers. (2.5 points)

Atom	Hybridization
Carbon (<i>a</i>)	
Carbon (<i>b</i>)	
Carbon (<i>c</i>)	
Carbon (<i>d</i>)	
Carbon (<i>e</i>)	

ii. Now, consider the sulfur atom (S) marked as ***f***, nitrogen (N) atom (***g***), and oxygen(O) atoms (***h*** and ***i***), and determine their **hybridization**. (2 points)

Atom	Hybridization
Sulfur (<i>f</i>)	
Nitrogen (<i>g</i>)	
Oxygen (<i>h</i>)	
Oxygen (<i>i</i>)	

iii. **Describe** all the bonds formed by the carbon atoms ***a*** and ***b***, using sigma and pi designation as appropriate. (2 points)

Atom	Bonds formed (use sigma and pi as appropriate)
Carbon (<i>a</i>)	
Carbon (<i>b</i>)	

iv. How many total sp^3 -s sigma bonds are in the molecule? (1 point)

v. Determine **electronic** and **molecular geometry** at carbon (C) labeled ***e*** and sulfur (S) labeled ***f*** centres. (2 points)

Atom	Electronic Geometry	Molecular Geometry
Carbon (<i>e</i>)		
Sulfur (<i>f</i>)		

vi. Circle the value from the choices given below, that describes the **C-S-C bond angle at sulfur** (S) marked ***f***: (0.5 points)

>90° to <109.5° 109.5° >109.5 to <120° 120° >120° to <180°


Question 8 12 points

Using the **valence shell orbitals only** draw a **complete** molecular orbital (MO) diagram (with no s-p mixing) for the molecule **PS**, in the space provided below. **(6 points)**

Assume that 3s and 3p atomic orbitals will form molecular orbitals like those formed by the combination of 2s and 2p atomic orbitals that we learnt in class, and there is no s-p mixing.

Show **energy arrow**, **atomic orbitals** involved, **atom labels**, the **molecular orbitals** that are formed, **filled electrons** in the MO diagram etc. Make sure to label all the atomic and molecular orbitals.

Using the MO diagram above, answer the following questions:

- (i) Which one of the following will have the highest bond order? Circle the correct choice
 A) PS^{1-} B) PS C) PS^{1+}
- (ii) Which one of the following will have the longest bond length? Circle the correct choice
 A) PS^{1-} B) PS C) PS^{1+}
- (iii) Which of the following will be diamagnetic? Circle the correct choice.
 A) PS^{1-} B) PS C) PS^{1+}
- (iv) Draw Lewis dot structure of the molecule with the shortest bond length (based on the MO diagram) from among the three (PS^{1-} , PS, PS^{1+}), and indicate the bond order calculated from the Lewis dot structure. **(3 points)**

Lewis Dot Structure

Bond Order



DFBFB683-428C-4727-ACC2-A2DEB2B656D1

midterm-exam-93388

#123 Page 10 of 10

SCRAP PAGE

**CHEMISTRY 110 –Midterm Exam****Thursday November 7, 2024, 6:30 – 8:30 pm****Examiners: Pallavi Sirjoosingh and Ashok Kakkar**

First name (please write as legibly as possible within the boxes)

--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--

Last name

--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--

Student ID number

--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--

This booklet has 8 questions on 10 pages (The last page can be used as a scrap page).

The staple must NOT be removed from the question booklet. Your intact question booklet must be returned at the end of the exam.

A separate booklet contains a periodic table, and datasheet.

Write your answers directly under the questions in the space provided. Do not write on the QR codes or near the top edges (as it may be cut off).

Please write **legibly and dark enough** for the scan to be readable. For students who use English as a second language, dictionaries are permitted. Calculators are allowed.

DO NOT ASK QUESTIONS OF THE INVIGILATORS about interpretations of the exam questions. If you have a concern about a question, please summarize it at the top of this page. All concerns will be evaluated after the exam. GOOD LUCK!

**Question 1 8 points** (*Assume all transitions are possible*)

a. 6 points Calculate the **energy (in kJ/mol)** required to ionize (completely remove) the electron from the **$n=3$** level of **He^+ ion** (He: Helium). Show your work including equations. (3 sig. figs.)

b. 2 points

(I) Select the largest (longest) wavelength from the three choices below:

- i. Wavelength of photon to *completely remove* an electron from *1s orbital of Hydrogen atom*
- ii. Wavelength of photon to *completely remove* an electron from *1s orbital of Boron atom (B)*
- iii. Wavelength of photon to *completely remove* an electron from *1s orbital of Boron cation (B^+)*

i or ii or iii

(II) Select the smallest (shortest) wavelength from the three choices below (*for H atom*):

- i. Wavelength of photon *emitted* for an electron transitioning from *2p to 1s orbital*
- ii. Wavelength of photon *absorbed* for an electron transitioning from *2s to 4p orbital*
- iii. Wavelength of photon *absorbed* for an electron transitioning from *1s to 3p orbital*

i or ii or iii



Question 2 8 points



A Chem110 student plots the graph shown on the left, from data obtained for the kinetic energy (eV) of the ejected electron versus frequency (s^{-1}) of the incident light, for an unknown metal. ($1\text{eV} = 1.6022 \times 10^{-19} \text{ J}$)

Based on the plot, answer the questions **a to c** below:

a. 1 point On the plot above, label the **threshold frequency**.

b. 2 points What is the value of the work potential (in J) of the unknown metal?

c. 3 points What is the *longest* wavelength (in nm) required to eject an electron from this metal's surface? Show your work including equation(s) used. (3 sig figs)

d. 2 points In the space provided below, plot the following

(Assume frequency is *above* threshold frequency):

- i. *Kinetic energy* of an ejected electron versus the *intensity* of the incident light (left)
- ii. *Number of electrons* ejected versus the *frequency* of the incident light (right)



**Question 3** **6 points****a. 2 points**

In the ground state of Ge (Germanium; $Z = 32$) atom, **name** the following orbitals:

i. Orbital containing the electron that requires the maximum energy to be removed.

ii. Orbital containing the electron that experiences the minimum effective nuclear charge.

b. 2 points

An unknown anion (A^-) has the following *excited* electronic configuration: $[Ar] 4s^2 3d^3 4p^1$

(Ar: Argon; $Z = 18$)

Give the **condensed ground state electronic configuration** of the atom, **A** and the cation, **A^{2+}** .
Show your work. (Assume all electronic transitions are possible)

Electronic Configuration of A:

Electronic Configuration of A^{2+} :

c. 2 points

In a single atom, determine how many electrons can have the following set of quantum numbers?

(i) $n = 5; l = 3$

In a single atom, determine how many electrons can have the following set of quantum numbers?

(i) $n = 3; l = 2; m_s = +1/2$

**Question 4** **8 points**

Rank the following from 1 to 4 in **increasing** order. Provide the ranking in the boxes provided.

Note: Atomic number (Z) are provided for relevant atoms

a) Increasing (lowest to highest) order of *ionization energy*: (1= lowest; 4 = highest)

Mg

Mg: Z = 12

Rb

Rb: Z = 37

Cs

Cs: Z = 55

K

K : Z = 19

b) Increasing (lowest to highest) order of *bond angle*: (1= lowest; 4 = highest)

H₂OCCl₄XeF₄

HCN

c) Increasing (smallest to largest) order of *ionic size* (1= smallest; 4 = largest)

S²⁻

S: Z = 16

O²⁻

O: Z = 8

Be²⁺

Be: Z = 4

Na⁺

Na: Z = 11

d) Increasing (lowest to highest) order of *bond energy* (1= lowest; 4 = highest)

P-F

P-Cl

P-I

P-Br

**Question 5** **12 points**

a. 9 points For the molecules HClO_4 and TeF_6 , draw the most stable *Lewis structure* showing appropriate shape and *dashed/wedged bonds*, based on VSEPR. Determine the **molecular geometry** around the central atom. For HClO_4 , label the *most polar bond*.

HClO_4 (there are no O-O bonds)

Label the most polar bond

Molecular Geometry around Cl

TeF_6

Molecular Geometry around Te

b. 3 points (Note: Lone pairs are not shown in structures below)

(i) For the three *resonance* structures (1, 2 and 3) shown below, answer the following question:



(I) Circle the *most contributing* resonance structure.

Structure 1

Structure 2

Structure 3

(ii) For the three structures given (4, 5, and 6) below, answer the following questions:



(I) Circle all the structures that represent *resonance structures* of each other:

Structure 4

Structure 5

Structure 6

(II) Among the resonance structures, circle the *most contributing* resonance structure.

Structure 4

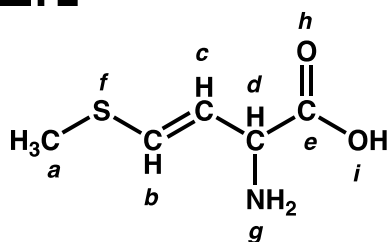
Structure 5

Structure 6

**Question 6** **6 points**

Using the data given below, calculate the **lattice energy** for SrO(s).

	ΔH° (kJ/mol)
$\text{Sr}(s) \longrightarrow \text{Sr}(g)$	+180
$\text{Sr}(g) \longrightarrow \text{Sr}^{2+}(g) + 2e^-$	+1468
$\text{O}_2(g) \longrightarrow 2\text{O}(g)$	+500
$\text{O}(g) + 2e^- \longrightarrow \text{O}^{2-}(g)$	+600
$\text{Sr}(s) + \frac{1}{2}\text{O}_2(g) \longrightarrow \text{SrO}(s)$	-550

**Question 7 10 points**

Consider the molecule given on the left, and answer questions (i) to (vi) using chemical bonding theories that we have learnt in class (valence bond theory, hybridization).

Note that the lone pairs are not shown in the structure:

i. Consider the carbon atoms marked in as ***a***, ***b***, ***c***, ***d***, and ***e*** in the molecule, and determine **hybridization** at each of the carbon centers. (2.5 points)

Atom	Hybridization
Carbon (<i>a</i>)	
Carbon (<i>b</i>)	
Carbon (<i>c</i>)	
Carbon (<i>d</i>)	
Carbon (<i>e</i>)	

ii. Now, consider the sulfur atom (S) marked as ***f***, nitrogen (N) atom (***g***), and oxygen(O) atoms (***h*** and ***i***), and determine their **hybridization**. (2 points)

Atom	Hybridization
Sulfur (<i>f</i>)	
Nitrogen (<i>g</i>)	
Oxygen (<i>h</i>)	
Oxygen (<i>i</i>)	

iii. **Describe** all the bonds formed by the carbon atoms ***a*** and ***b***, using sigma and pi designation as appropriate. (2 points)

Atom	Bonds formed (use sigma and pi as appropriate)
Carbon (<i>a</i>)	
Carbon (<i>b</i>)	

iv. How many total sp^3 -s sigma bonds are in the molecule? (1 point)

v. Determine **electronic** and **molecular geometry** at carbon (C) labeled ***e*** and sulfur (S) labeled ***f*** centres. (2 points)

Atom	Electronic Geometry	Molecular Geometry
Carbon (<i>e</i>)		
Sulfur (<i>f</i>)		

vi. Circle the value from the choices given below, that describes the **C-S-C bond angle at sulfur** (S) marked ***f***: (0.5 points)

>90° to <109.5°

109.5°

>109.5 to <120°

120°

>120° to <180°


Question 8 12 points

Using the **valence shell orbitals only** draw a **complete** molecular orbital (MO) diagram (with no s-p mixing) for the molecule **PS**, in the space provided below. **(6 points)**

Assume that 3s and 3p atomic orbitals will form molecular orbitals like those formed by the combination of 2s and 2p atomic orbitals that we learnt in class, and there is no s-p mixing.

Show **energy arrow**, **atomic orbitals** involved, **atom labels**, the **molecular orbitals** that are formed, **filled electrons** in the MO diagram etc. Make sure to label all the atomic and molecular orbitals.

Using the MO diagram above, answer the following questions:

- (i) Which one of the following will have the highest bond order? Circle the correct choice
 A) PS^{1-} B) PS C) PS^{1+}
- (ii) Which one of the following will have the longest bond length? Circle the correct choice
 A) PS^{1-} B) PS C) PS^{1+}
- (iii) Which of the following will be diamagnetic? Circle the correct choice.
 A) PS^{1-} B) PS C) PS^{1+}
- (iv) Draw Lewis dot structure of the molecule with the shortest bond length (based on the MO diagram) from among the three (PS^{1-} , PS, PS^{1+}), and indicate the bond order calculated from the Lewis dot structure. **(3 points)**

Lewis Dot Structure

Bond Order



CF877EE1-ACA0-45EF-ACB7-8AB500332959

midterm-exam-93388

#124 Page 10 of 10

SCRAP PAGE

**CHEMISTRY 110 –Midterm Exam****Thursday November 7, 2024, 6:30 – 8:30 pm**

Examiners: Pallavi Sirjoosingh and Ashok Kakkar

First name (please write as legibly as possible within the boxes)

--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--

Last name

--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--

Student ID number

--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--

This booklet has 8 questions on 10 pages (The last page can be used as a scrap page).

The staple must NOT be removed from the question booklet. Your intact question booklet must be returned at the end of the exam.

A separate booklet contains a periodic table, and datasheet.

Write your answers directly under the questions in the space provided. Do not write on the QR codes or near the top edges (as it may be cut off).

Please write **legibly and dark enough** for the scan to be readable. For students who use English as a second language, dictionaries are permitted. Calculators are allowed.

DO NOT ASK QUESTIONS OF THE INVIGILATORS about interpretations of the exam questions. If you have a concern about a question, please summarize it at the top of this page. All concerns will be evaluated after the exam. GOOD LUCK!

**Question 1 8 points** (*Assume all transitions are possible*)

a. 6 points Calculate the **energy (in kJ/mol)** required to ionize (completely remove) the electron from the **$n=3$** level of **He^+ ion** (He: Helium). Show your work including equations. (3 sig. figs.)

b. 2 points

(I) Select the largest (longest) wavelength from the three choices below:

- i. Wavelength of photon to *completely remove* an electron from *1s orbital of Hydrogen atom*
- ii. Wavelength of photon to *completely remove* an electron from *1s orbital of Boron atom (B)*
- iii. Wavelength of photon to *completely remove* an electron from *1s orbital of Boron cation (B^+)*

i or ii or iii

(II) Select the smallest (shortest) wavelength from the three choices below (*for H atom*):

- i. Wavelength of photon *emitted* for an electron transitioning from *2p to 1s orbital*
- ii. Wavelength of photon *absorbed* for an electron transitioning from *2s to 4p orbital*
- iii. Wavelength of photon *absorbed* for an electron transitioning from *1s to 3p orbital*

i or ii or iii



Question 2 8 points



A Chem110 student plots the graph shown on the left, from data obtained for the kinetic energy (eV) of the ejected electron versus frequency (s^{-1}) of the incident light, for an unknown metal. ($1\text{eV} = 1.6022 \times 10^{-19} \text{ J}$)

Based on the plot, answer the questions **a to c** below:

a. 1 point On the plot above, label the **threshold frequency**.

b. 2 points What is the value of the work potential (in J) of the unknown metal?

c. 3 points What is the *longest* wavelength (in nm) required to eject an electron from this metal's surface? Show your work including equation(s) used. (3 sig figs)

d. 2 points In the space provided below, plot the following

(Assume frequency is *above* threshold frequency):

- i. *Kinetic energy* of an ejected electron versus the *intensity* of the incident light (left)
- ii. *Number of electrons* ejected versus the *frequency* of the incident light (right)



**Question 3** **6 points****a. 2 points**

In the ground state of Ge (Germanium; $Z = 32$) atom, **name** the following orbitals:

i. Orbital containing the electron that requires the maximum energy to be removed.

ii. Orbital containing the electron that experiences the minimum effective nuclear charge.

b. 2 points

An unknown anion (A^-) has the following *excited* electronic configuration: $[Ar] 4s^2 3d^3 4p^1$

(Ar: Argon; $Z = 18$)

Give the **condensed ground state electronic configuration** of the atom, **A** and the cation, **A^{2+}** .
Show your work. (Assume all electronic transitions are possible)

Electronic Configuration of A:

Electronic Configuration of A^{2+} :

c. 2 points

In a single atom, determine how many electrons can have the following set of quantum numbers?

(i) $n = 5; l = 3$

In a single atom, determine how many electrons can have the following set of quantum numbers?

(i) $n = 3; l = 2; m_s = +1/2$

**Question 4** **8 points**

Rank the following from 1 to 4 in **increasing** order. Provide the ranking in the boxes provided.

Note: Atomic number (Z) are provided for relevant atoms

a) Increasing (lowest to highest) order of *ionization energy*: (1= lowest; 4 = highest)

Mg

Mg: Z = 12

Rb

Rb: Z = 37

Cs

Cs: Z = 55

K

K : Z = 19

b) Increasing (lowest to highest) order of *bond angle*: (1= lowest; 4 = highest)

H₂OCCl₄XeF₄

HCN

c) Increasing (smallest to largest) order of *ionic size* (1= smallest; 4 = largest)

S²⁻

S: Z = 16

O²⁻

O: Z = 8

Be²⁺

Be: Z = 4

Na⁺

Na: Z = 11

d) Increasing (lowest to highest) order of *bond energy* (1= lowest; 4 = highest)

P-F

P-Cl

P-I

P-Br

**Question 5** **12 points**

a. 9 points For the molecules HClO_4 and TeF_6 , draw the most stable *Lewis structure* showing appropriate shape and *dashed/wedged bonds*, based on VSEPR. Determine the **molecular geometry** around the central atom. For HClO_4 , label the *most polar bond*.



b. 3 points (Note: Lone pairs are not shown in structures below)

(i) For the three *resonance* structures (1, 2 and 3) shown below, answer the following question:



(I) Circle the *most contributing* resonance structure.

Structure 1

Structure 2

Structure 3

(ii) For the three structures given (4, 5, and 6) below, answer the following questions:



(I) Circle all the structures that represent *resonance structures* of each other:

Structure 4

Structure 5

Structure 6

(II) Among the resonance structures, circle the *most contributing* resonance structure.

Structure 4

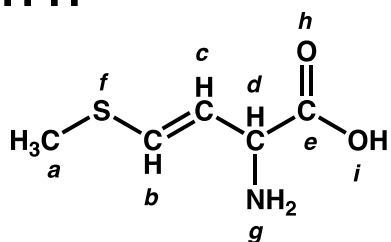
Structure 5

Structure 6

**Question 6** **6 points**

Using the data given below, calculate the **lattice energy** for SrO(s).

	ΔH° (kJ/mol)
$\text{Sr}(s) \longrightarrow \text{Sr}(g)$	+180
$\text{Sr}(g) \longrightarrow \text{Sr}^{2+}(g) + 2e^-$	+1468
$\text{O}_2(g) \longrightarrow 2\text{O}(g)$	+500
$\text{O}(g) + 2e^- \longrightarrow \text{O}^{2-}(g)$	+600
$\text{Sr}(s) + \frac{1}{2}\text{O}_2(g) \longrightarrow \text{SrO}(s)$	-550

**Question 7 10 points**

Consider the molecule given on the left, and answer questions (i) to (vi) using chemical bonding theories that we have learnt in class (valence bond theory, hybridization).

Note that the lone pairs are not shown in the structure:

i. Consider the carbon atoms marked in as ***a***, ***b***, ***c***, ***d***, and ***e*** in the molecule, and determine **hybridization** at each of the carbon centers. (2.5 points)

Atom	Hybridization
Carbon (<i>a</i>)	
Carbon (<i>b</i>)	
Carbon (<i>c</i>)	
Carbon (<i>d</i>)	
Carbon (<i>e</i>)	

ii. Now, consider the sulfur atom (S) marked as ***f***, nitrogen (N) atom (***g***), and oxygen(O) atoms (***h*** and ***i***), and determine their **hybridization**. (2 points)

Atom	Hybridization
Sulfur (<i>f</i>)	
Nitrogen (<i>g</i>)	
Oxygen (<i>h</i>)	
Oxygen (<i>i</i>)	

iii. **Describe** all the bonds formed by the carbon atoms ***a*** and ***b***, using sigma and pi designation as appropriate. (2 points)

Atom	Bonds formed (use sigma and pi as appropriate)
Carbon (<i>a</i>)	
Carbon (<i>b</i>)	

iv. How many total sp^3 -s sigma bonds are in the molecule? (1 point)

v. Determine **electronic** and **molecular geometry** at carbon (C) labeled ***e*** and sulfur (S) labeled ***f*** centres. (2 points)

Atom	Electronic Geometry	Molecular Geometry
Carbon (<i>e</i>)		
Sulfur (<i>f</i>)		

vi. Circle the value from the choices given below, that describes the **C-S-C bond angle at sulfur** (S) marked ***f***: (0.5 points)

$>90^\circ$ to $<109.5^\circ$ 109.5° >109.5 to $<120^\circ$ 120° $>120^\circ$ to $<180^\circ$


Question 8 12 points

Using the **valence shell orbitals only** draw a **complete** molecular orbital (MO) diagram (with no s-p mixing) for the molecule **PS**, in the space provided below. **(6 points)**

Assume that 3s and 3p atomic orbitals will form molecular orbitals like those formed by the combination of 2s and 2p atomic orbitals that we learnt in class, and there is no s-p mixing.

Show **energy arrow**, **atomic orbitals** involved, **atom labels**, the **molecular orbitals** that are formed, **filled electrons** in the MO diagram etc. Make sure to label all the atomic and molecular orbitals.

Using the MO diagram above, answer the following questions:

- (i) Which one of the following will have the highest bond order? Circle the correct choice
 A) PS^{1-} B) PS C) PS^{1+}
- (ii) Which one of the following will have the longest bond length? Circle the correct choice
 A) PS^{1-} B) PS C) PS^{1+}
- (iii) Which of the following will be diamagnetic? Circle the correct choice.
 A) PS^{1-} B) PS C) PS^{1+}
- (iv) Draw Lewis dot structure of the molecule with the shortest bond length (based on the MO diagram) from among the three (PS^{1-} , PS, PS^{1+}), and indicate the bond order calculated from the Lewis dot structure. **(3 points)**

Lewis Dot Structure

Bond Order



23747125-1E40-4938-B55C-CADA22DE9C06

midterm-exam-93388

#125 Page 10 of 10

SCRAP PAGE

**CHEMISTRY 110 –Midterm Exam****Thursday November 7, 2024, 6:30 – 8:30 pm****Examiners: Pallavi Sirjoosingh and Ashok Kakkar**

First name (please write as legibly as possible within the boxes)

--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--

Last name

--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--

Student ID number

--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--

This booklet has 8 questions on 10 pages (The last page can be used as a scrap page).

The staple must NOT be removed from the question booklet. Your intact question booklet must be returned at the end of the exam.

A separate booklet contains a periodic table, and datasheet.

Write your answers directly under the questions in the space provided. Do not write on the QR codes or near the top edges (as it may be cut off).

Please write **legibly and dark enough** for the scan to be readable. For students who use English as a second language, dictionaries are permitted. Calculators are allowed.

DO NOT ASK QUESTIONS OF THE INVIGILATORS about interpretations of the exam questions. If you have a concern about a question, please summarize it at the top of this page. All concerns will be evaluated after the exam. GOOD LUCK!

**Question 1 8 points** (*Assume all transitions are possible*)

a. 6 points Calculate the **energy (in kJ/mol)** required to ionize (completely remove) the electron from the **n=3** level of **He⁺ ion** (He: Helium). Show your work including equations. (3 sig. figs.)

b. 2 points

(I) Select the largest (longest) wavelength from the three choices below:

- i. Wavelength of photon to *completely remove* an electron from *1s orbital of Hydrogen atom*
- ii. Wavelength of photon to *completely remove* an electron from *1s orbital of Boron atom (B)*
- iii. Wavelength of photon to *completely remove* an electron from *1s orbital of Boron cation (B⁺)*

i or ii or iii

(II) Select the smallest (shortest) wavelength from the three choices below (*for H atom*):

- i. Wavelength of photon *emitted* for an electron transitioning from *2p to 1s orbital*
- ii. Wavelength of photon *absorbed* for an electron transitioning from *2s to 4p orbital*
- iii. Wavelength of photon *absorbed* for an electron transitioning from *1s to 3p orbital*

i or ii or iii



Question 2 8 points



A Chem110 student plots the graph shown on the left, from data obtained for the kinetic energy (eV) of the ejected electron versus frequency (s^{-1}) of the incident light, for an unknown metal. ($1\text{eV} = 1.6022 \times 10^{-19} \text{ J}$)

Based on the plot, answer the questions **a to c** below:

a. 1 point On the plot above, label the **threshold frequency**.

b. 2 points What is the value of the work potential (in J) of the unknown metal?

c. 3 points What is the *longest* wavelength (in nm) required to eject an electron from this metal's surface? Show your work including equation(s) used. (3 sig figs)

d. 2 points In the space provided below, plot the following

(Assume frequency is *above* threshold frequency):

- Kinetic energy* of an ejected electron versus the *intensity* of the incident light (left)
- Number of electrons* ejected versus the *frequency* of the incident light (right)



**Question 3** **6 points****a. 2 points**

In the ground state of Ge (Germanium; $Z = 32$) atom, **name** the following orbitals:

i. Orbital containing the electron that requires the maximum energy to be removed.

ii. Orbital containing the electron that experiences the minimum effective nuclear charge.

b. 2 points

An unknown anion (A^-) has the following *excited* electronic configuration: $[Ar] 4s^2 3d^3 4p^1$

(Ar: Argon; $Z = 18$)

Give the **condensed ground state electronic configuration** of the atom, **A** and the cation, **A^{2+}** .
Show your work. (Assume all electronic transitions are possible)

Electronic Configuration of A:

Electronic Configuration of A^{2+} :

c. 2 points

In a single atom, determine how many electrons can have the following set of quantum numbers?

(i) $n = 5; l = 3$

In a single atom, determine how many electrons can have the following set of quantum numbers?

(i) $n = 3; l = 2; m_s = +1/2$

**Question 4** **8 points**

Rank the following from 1 to 4 in **increasing** order. Provide the ranking in the boxes provided.

Note: Atomic number (Z) are provided for relevant atoms

a) Increasing (lowest to highest) order of *ionization energy*: (1= lowest; 4 = highest)

Mg

Mg: Z = 12

Rb

Rb: Z = 37

Cs

Cs: Z = 55

K

K : Z = 19

b) Increasing (lowest to highest) order of *bond angle*: (1= lowest; 4 = highest)

H₂OCCl₄XeF₄

HCN

c) Increasing (smallest to largest) order of *ionic size* (1= smallest; 4 = largest)

S²⁻

S: Z = 16

O²⁻

O: Z = 8

Be²⁺

Be: Z = 4

Na⁺

Na: Z = 11

d) Increasing (lowest to highest) order of *bond energy* (1= lowest; 4 = highest)

P-F

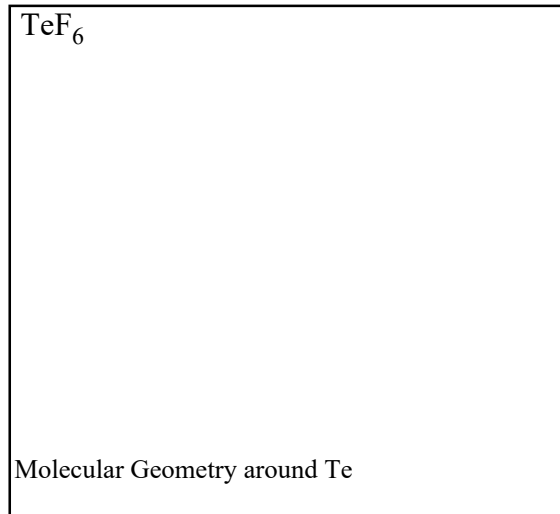
P-Cl

P-I

P-Br

**Question 5** **12 points**

a. 9 points For the molecules HClO_4 and TeF_6 , draw the most stable *Lewis structure* showing appropriate shape and *dashed/wedged bonds*, based on VSEPR. Determine the **molecular geometry** around the central atom. For HClO_4 , label the *most polar bond*.



b. 3 points (Note: Lone pairs are not shown in structures below)

(i) For the three *resonance* structures (1, 2 and 3) shown below, answer the following question:



(I) Circle the *most contributing* resonance structure.

Structure 1

Structure 2

Structure 3

(ii) For the three structures given (4, 5, and 6) below, answer the following questions:



(I) Circle all the structures that represent *resonance structures* of each other:

Structure 4

Structure 5

Structure 6

(II) Among the resonance structures, circle the *most contributing* resonance structure.

Structure 4

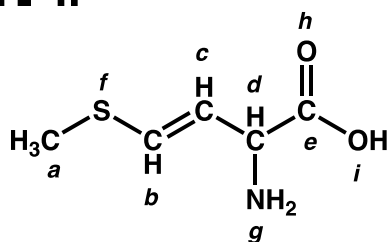
Structure 5

Structure 6

**Question 6** **6 points**

Using the data given below, calculate the **lattice energy** for SrO(s).

	ΔH° (kJ/mol)
$\text{Sr}(s) \longrightarrow \text{Sr}(g)$	+180
$\text{Sr}(g) \longrightarrow \text{Sr}^{2+}(g) + 2e^-$	+1468
$\text{O}_2(g) \longrightarrow 2\text{O}(g)$	+500
$\text{O}(g) + 2e^- \longrightarrow \text{O}^{2-}(g)$	+600
$\text{Sr}(s) + \frac{1}{2}\text{O}_2(g) \longrightarrow \text{SrO}(s)$	-550

**Question 7 10 points**

Consider the molecule given on the left, and answer questions (i) to (vi) using chemical bonding theories that we have learnt in class (valence bond theory, hybridization).

Note that the lone pairs are not shown in the structure:

i. Consider the carbon atoms marked in as ***a***, ***b***, ***c***, ***d***, and ***e*** in the molecule, and determine **hybridization** at each of the carbon centers. (2.5 points)

Atom	Hybridization
Carbon (<i>a</i>)	
Carbon (<i>b</i>)	
Carbon (<i>c</i>)	
Carbon (<i>d</i>)	
Carbon (<i>e</i>)	

ii. Now, consider the sulfur atom (S) marked as ***f***, nitrogen (N) atom (***g***), and oxygen(O) atoms (***h*** and ***i***), and determine their **hybridization**. (2 points)

Atom	Hybridization
Sulfur (<i>f</i>)	
Nitrogen (<i>g</i>)	
Oxygen (<i>h</i>)	
Oxygen (<i>i</i>)	

iii. **Describe** all the bonds formed by the carbon atoms ***a*** and ***b***, using sigma and pi designation as appropriate. (2 points)

Atom	Bonds formed (use sigma and pi as appropriate)
Carbon (<i>a</i>)	
Carbon (<i>b</i>)	

iv. How many total sp^3 -s sigma bonds are in the molecule? (1 point)

v. Determine **electronic** and **molecular geometry** at carbon (C) labeled ***e*** and sulfur (S) labeled ***f*** centres. (2 points)

Atom	Electronic Geometry	Molecular Geometry
Carbon (<i>e</i>)		
Sulfur (<i>f</i>)		

vi. Circle the value from the choices given below, that describes the **C-S-C bond angle at sulfur** (S) marked ***f***: (0.5 points)

>90° to <109.5° 109.5° >109.5 to <120° 120° >120° to <180°


Question 8 12 points

Using the **valence shell orbitals only** draw a **complete** molecular orbital (MO) diagram (with no s-p mixing) for the molecule **PS**, in the space provided below. **(6 points)**

Assume that 3s and 3p atomic orbitals will form molecular orbitals like those formed by the combination of 2s and 2p atomic orbitals that we learnt in class, and there is no s-p mixing.

Show **energy arrow**, **atomic orbitals** involved, **atom labels**, the **molecular orbitals** that are formed, **filled electrons** in the MO diagram etc. Make sure to **label all the atomic and molecular orbitals**.

Using the MO diagram above, answer the following questions:

- (i) Which one of the following will have the highest bond order? Circle the correct choice
 A) PS^{1-} B) PS C) PS^{1+}
- (ii) Which one of the following will have the longest bond length? Circle the correct choice
 A) PS^{1-} B) PS C) PS^{1+}
- (iii) Which of the following will be diamagnetic? Circle the correct choice.
 A) PS^{1-} B) PS C) PS^{1+}
- (iv) Draw Lewis dot structure of the molecule with the shortest bond length (based on the MO diagram) from among the three (PS^{1-} , PS, PS^{1+}), and indicate the bond order calculated from the Lewis dot structure. **(3 points)**

Lewis Dot Structure

Bond Order



54A15937-EAE3-415F-8E88-5FB8489C3196

midterm-exam-93388

#126 Page 10 of 10

SCRAP PAGE

**CHEMISTRY 110 –Midterm Exam****Thursday November 7, 2024, 6:30 – 8:30 pm****Examiners: Pallavi Sirjoosingh and Ashok Kakkar**

First name (please write as legibly as possible within the boxes)

--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--

Last name

--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--

Student ID number

--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--

This booklet has 8 questions on 10 pages (The last page can be used as a scrap page).

The staple must NOT be removed from the question booklet. Your intact question booklet must be returned at the end of the exam.

A separate booklet contains a periodic table, and datasheet.

Write your answers directly under the questions in the space provided. Do not write on the QR codes or near the top edges (as it may be cut off).

Please write **legibly and dark enough** for the scan to be readable. For students who use English as a second language, dictionaries are permitted. Calculators are allowed.

DO NOT ASK QUESTIONS OF THE INVIGILATORS about interpretations of the exam questions. If you have a concern about a question, please summarize it at the top of this page. All concerns will be evaluated after the exam. GOOD LUCK!

**Question 1 8 points** (*Assume all transitions are possible*)

a. 6 points Calculate the **energy (in kJ/mol)** required to ionize (completely remove) the electron from the **n=3** level of **He⁺ ion** (He: Helium). Show your work including equations. (3 sig. figs.)

b. 2 points

(I) Select the largest (longest) wavelength from the three choices below:

- i. Wavelength of photon to *completely remove* an electron from *1s orbital of Hydrogen atom*
- ii. Wavelength of photon to *completely remove* an electron from *1s orbital of Boron atom (B)*
- iii. Wavelength of photon to *completely remove* an electron from *1s orbital of Boron cation (B⁺)*

i or ii or iii

(II) Select the smallest (shortest) wavelength from the three choices below (*for H atom*):

- i. Wavelength of photon *emitted* for an electron transitioning from *2p to 1s orbital*
- ii. Wavelength of photon *absorbed* for an electron transitioning from *2s to 4p orbital*
- iii. Wavelength of photon *absorbed* for an electron transitioning from *1s to 3p orbital*

i or ii or iii



Question 2 8 points



A Chem110 student plots the graph shown on the left, from data obtained for the kinetic energy (eV) of the ejected electron versus frequency (s^{-1}) of the incident light, for an unknown metal. ($1\text{eV} = 1.6022 \times 10^{-19} \text{ J}$)

Based on the plot, answer the questions **a to c** below:

a. 1 point On the plot above, label the **threshold frequency**.

b. 2 points What is the value of the work potential (in J) of the unknown metal?

c. 3 points What is the *longest* wavelength (in nm) required to eject an electron from this metal's surface? Show your work including equation(s) used. (3 sig figs)

d. 2 points In the space provided below, plot the following

(Assume frequency is *above* threshold frequency):

- i. *Kinetic energy* of an ejected electron versus the *intensity* of the incident light (left)
- ii. *Number of electrons* ejected versus the *frequency* of the incident light (right)



**Question 3** **6 points****a. 2 points**

In the ground state of Ge (Germanium; $Z = 32$) atom, **name** the following orbitals:

i. Orbital containing the electron that requires the maximum energy to be removed.

ii. Orbital containing the electron that experiences the minimum effective nuclear charge.

b. 2 points

An unknown anion (A^-) has the following *excited* electronic configuration: $[Ar] 4s^2 3d^3 4p^1$

(Ar: Argon; $Z = 18$)

Give the **condensed ground state electronic configuration** of the atom, **A** and the cation, **A^{2+}** .
Show your work. (Assume all electronic transitions are possible)

Electronic Configuration of A:

Electronic Configuration of A^{2+} :

c. 2 points

In a single atom, determine how many electrons can have the following set of quantum numbers?

(i) $n = 5; l = 3$

In a single atom, determine how many electrons can have the following set of quantum numbers?

(i) $n = 3; l = 2; m_s = +1/2$

**Question 4** **8 points**

Rank the following from 1 to 4 in **increasing** order. Provide the ranking in the boxes provided.

Note: Atomic number (Z) are provided for relevant atoms

a) Increasing (lowest to highest) order of *ionization energy*: (1= lowest; 4 = highest)

Mg

Mg: Z = 12

Rb

Rb: Z = 37

Cs

Cs: Z = 55

K

K : Z = 19

b) Increasing (lowest to highest) order of *bond angle*: (1= lowest; 4 = highest)

H₂OCCl₄XeF₄

HCN

c) Increasing (smallest to largest) order of *ionic size* (1= smallest; 4 = largest)

S²⁻

S: Z = 16

O²⁻

O: Z = 8

Be²⁺

Be: Z = 4

Na⁺

Na: Z = 11

d) Increasing (lowest to highest) order of *bond energy* (1= lowest; 4 = highest)

P-F

P-Cl

P-I

P-Br

**Question 5** **12 points**

a. 9 points For the molecules HClO_4 and TeF_6 , draw the most stable *Lewis structure* showing appropriate shape and *dashed/wedged bonds*, based on VSEPR. Determine the **molecular geometry** around the central atom. For HClO_4 , label the *most polar bond*.



b. 3 points (Note: Lone pairs are not shown in structures below)

(i) For the three *resonance* structures (1, 2 and 3) shown below, answer the following question:



(I) Circle the *most contributing* resonance structure.

Structure 1

Structure 2

Structure 3

(ii) For the three structures given (4, 5, and 6) below, answer the following questions:



(I) Circle all the structures that represent *resonance structures* of each other:

Structure 4

Structure 5

Structure 6

(II) Among the resonance structures, circle the *most contributing* resonance structure.

Structure 4

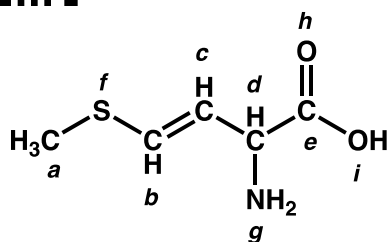
Structure 5

Structure 6

**Question 6** **6 points**

Using the data given below, calculate the **lattice energy** for SrO(s).

	ΔH° (kJ/mol)
$\text{Sr}(s) \longrightarrow \text{Sr}(g)$	+180
$\text{Sr}(g) \longrightarrow \text{Sr}^{2+}(g) + 2e^-$	+1468
$\text{O}_2(g) \longrightarrow 2\text{O}(g)$	+500
$\text{O}(g) + 2e^- \longrightarrow \text{O}^{2-}(g)$	+600
$\text{Sr}(s) + \frac{1}{2}\text{O}_2(g) \longrightarrow \text{SrO}(s)$	-550

**Question 7 10 points**

Consider the molecule given on the left, and answer questions (i) to (vi) using chemical bonding theories that we have learnt in class (valence bond theory, hybridization).

Note that the lone pairs are not shown in the structure:

i. Consider the carbon atoms marked in as ***a***, ***b***, ***c***, ***d***, and ***e*** in the molecule, and determine **hybridization** at each of the carbon centers. (2.5 points)

Atom	Hybridization
Carbon (<i>a</i>)	
Carbon (<i>b</i>)	
Carbon (<i>c</i>)	
Carbon (<i>d</i>)	
Carbon (<i>e</i>)	

ii. Now, consider the sulfur atom (S) marked as ***f***, nitrogen (N) atom (***g***), and oxygen(O) atoms (***h*** and ***i***), and determine their **hybridization**. (2 points)

Atom	Hybridization
Sulfur (<i>f</i>)	
Nitrogen (<i>g</i>)	
Oxygen (<i>h</i>)	
Oxygen (<i>i</i>)	

iii. **Describe** all the bonds formed by the carbon atoms ***a*** and ***b***, using sigma and pi designation as appropriate. (2 points)

Atom	Bonds formed (use sigma and pi as appropriate)
Carbon (<i>a</i>)	
Carbon (<i>b</i>)	

iv. How many total sp^3 -s sigma bonds are in the molecule? (1 point)

v. Determine **electronic** and **molecular geometry** at carbon (C) labeled ***e*** and sulfur (S) labeled ***f*** centres. (2 points)

Atom	Electronic Geometry	Molecular Geometry
Carbon (<i>e</i>)		
Sulfur (<i>f</i>)		

vi. Circle the value from the choices given below, that describes the **C-S-C bond angle at sulfur** (S) marked ***f***: (0.5 points)

$>90^\circ$ to $<109.5^\circ$ 109.5° >109.5 to $<120^\circ$ 120° $>120^\circ$ to $<180^\circ$


Question 8 12 points

Using the **valence shell orbitals only** draw a **complete** molecular orbital (MO) diagram (with no s-p mixing) for the molecule **PS**, in the space provided below. **(6 points)**

Assume that 3s and 3p atomic orbitals will form molecular orbitals like those formed by the combination of 2s and 2p atomic orbitals that we learnt in class, and there is no s-p mixing.

Show **energy arrow**, **atomic orbitals** involved, **atom labels**, the **molecular orbitals** that are formed, **filled electrons** in the MO diagram etc. Make sure to label all the atomic and molecular orbitals.

Using the MO diagram above, answer the following questions:

- (i) Which one of the following will have the highest bond order? Circle the correct choice
 A) PS^{1-} B) PS C) PS^{1+}
- (ii) Which one of the following will have the longest bond length? Circle the correct choice
 A) PS^{1-} B) PS C) PS^{1+}
- (iii) Which of the following will be diamagnetic? Circle the correct choice.
 A) PS^{1-} B) PS C) PS^{1+}
- (iv) Draw Lewis dot structure of the molecule with the shortest bond length (based on the MO diagram) from among the three (PS^{1-} , PS, PS^{1+}), and indicate the bond order calculated from the Lewis dot structure. **(3 points)**

Lewis Dot Structure

Bond Order



5151420F-296F-469C-A0A9-CD01E984BE34

midterm-exam-93388

#127 Page 10 of 10

SCRAP PAGE

**CHEMISTRY 110 –Midterm Exam****Thursday November 7, 2024, 6:30 – 8:30 pm****Examiners: Pallavi Sirjoosingh and Ashok Kakkar**

First name (please write as legibly as possible within the boxes)

--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--

Last name

--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--

Student ID number

--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--

This booklet has 8 questions on 10 pages (The last page can be used as a scrap page).

The staple must NOT be removed from the question booklet. Your intact question booklet must be returned at the end of the exam.

A separate booklet contains a periodic table, and datasheet.

Write your answers directly under the questions in the space provided. Do not write on the QR codes or near the top edges (as it may be cut off).

Please write **legibly and dark enough** for the scan to be readable. For students who use English as a second language, dictionaries are permitted. Calculators are allowed.

DO NOT ASK QUESTIONS OF THE INVIGILATORS about interpretations of the exam questions. If you have a concern about a question, please summarize it at the top of this page. All concerns will be evaluated after the exam. GOOD LUCK!

**Question 1 8 points** (*Assume all transitions are possible*)

a. 6 points Calculate the **energy (in kJ/mol)** required to ionize (completely remove) the electron from the **$n=3$** level of **He^+ ion** (He: Helium). Show your work including equations. (3 sig. figs.)

b. 2 points

(I) Select the largest (longest) wavelength from the three choices below:

- i. Wavelength of photon to *completely remove* an electron from *1s orbital of Hydrogen atom*
- ii. Wavelength of photon to *completely remove* an electron from *1s orbital of Boron atom (B)*
- iii. Wavelength of photon to *completely remove* an electron from *1s orbital of Boron cation (B^+)*

i or ii or iii

(II) Select the smallest (shortest) wavelength from the three choices below (*for H atom*):

- i. Wavelength of photon *emitted* for an electron transitioning from *2p to 1s orbital*
- ii. Wavelength of photon *absorbed* for an electron transitioning from *2s to 4p orbital*
- iii. Wavelength of photon *absorbed* for an electron transitioning from *1s to 3p orbital*

i or ii or iii



Question 2 8 points



A Chem110 student plots the graph shown on the left, from data obtained for the kinetic energy (eV) of the ejected electron versus frequency (s^{-1}) of the incident light, for an unknown metal. ($1\text{eV} = 1.6022 \times 10^{-19} \text{ J}$)

Based on the plot, answer the questions **a to c** below:

a. 1 point On the plot above, label the **threshold frequency**.

b. 2 points What is the value of the work potential (in J) of the unknown metal?

c. 3 points What is the *longest* wavelength (in nm) required to eject an electron from this metal's surface? Show your work including equation(s) used. (3 sig figs)

d. 2 points In the space provided below, plot the following

(Assume frequency is *above* threshold frequency):

- Kinetic energy* of an ejected electron versus the *intensity* of the incident light (left)
- Number of electrons* ejected versus the *frequency* of the incident light (right)



**Question 3** **6 points****a. 2 points**

In the ground state of Ge (Germanium; $Z = 32$) atom, **name** the following orbitals:

i. Orbital containing the electron that requires the maximum energy to be removed.

ii. Orbital containing the electron that experiences the minimum effective nuclear charge.

b. 2 points

An unknown anion (A^-) has the following *excited* electronic configuration: $[Ar] 4s^2 3d^3 4p^1$

(Ar: Argon; $Z = 18$)

Give the **condensed ground state electronic configuration** of the atom, **A** and the cation, **A^{2+}** .
Show your work. (Assume all electronic transitions are possible)

Electronic Configuration of A:

Electronic Configuration of A^{2+} :

c. 2 points

In a single atom, determine how many electrons can have the following set of quantum numbers?

(i) $n = 5; l = 3$

In a single atom, determine how many electrons can have the following set of quantum numbers?

(i) $n = 3; l = 2; m_s = +1/2$

**Question 4** **8 points**

Rank the following from 1 to 4 in **increasing** order. Provide the ranking in the boxes provided.

Note: Atomic number (Z) are provided for relevant atoms

a) Increasing (lowest to highest) order of *ionization energy*: (1= lowest; 4 = highest)

Mg

Mg: Z = 12

Rb

Rb: Z = 37

Cs

Cs: Z = 55

K

K : Z = 19

b) Increasing (lowest to highest) order of *bond angle*: (1= lowest; 4 = highest)

H₂OCCl₄XeF₄

HCN

c) Increasing (smallest to largest) order of *ionic size* (1= smallest; 4 = largest)

S²⁻

S: Z = 16

O²⁻

O: Z = 8

Be²⁺

Be: Z = 4

Na⁺

Na: Z = 11

d) Increasing (lowest to highest) order of *bond energy* (1= lowest; 4 = highest)

P-F

P-Cl

P-I

P-Br

**Question 5** **12 points**

a. 9 points For the molecules HClO_4 and TeF_6 , draw the most stable *Lewis structure* showing appropriate shape and *dashed/wedged bonds*, based on VSEPR. Determine the **molecular geometry** around the central atom. For HClO_4 , label the *most polar bond*.



b. 3 points (Note: Lone pairs are not shown in structures below)

(i) For the three *resonance* structures (1, 2 and 3) shown below, answer the following question:



(I) Circle the *most contributing* resonance structure.

Structure 1

Structure 2

Structure 3

(ii) For the three structures given (4, 5, and 6) below, answer the following questions:



(I) Circle all the structures that represent *resonance structures* of each other:

Structure 4

Structure 5

Structure 6

(II) Among the resonance structures, circle the *most contributing* resonance structure.

Structure 4

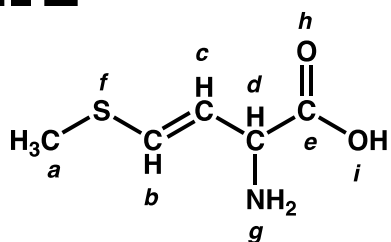
Structure 5

Structure 6

**Question 6** **6 points**

Using the data given below, calculate the **lattice energy** for SrO(s).

	ΔH° (kJ/mol)
$\text{Sr}(s) \longrightarrow \text{Sr}(g)$	+180
$\text{Sr}(g) \longrightarrow \text{Sr}^{2+}(g) + 2e^-$	+1468
$\text{O}_2(g) \longrightarrow 2\text{O}(g)$	+500
$\text{O}(g) + 2e^- \longrightarrow \text{O}^{2-}(g)$	+600
$\text{Sr}(s) + \frac{1}{2}\text{O}_2(g) \longrightarrow \text{SrO}(s)$	-550

**Question 7 10 points**

Consider the molecule given on the left, and answer questions (i) to (vi) using chemical bonding theories that we have learnt in class (valence bond theory, hybridization).

Note that the lone pairs are not shown in the structure:

i. Consider the carbon atoms marked in as ***a***, ***b***, ***c***, ***d***, and ***e*** in the molecule, and determine **hybridization** at each of the carbon centers. (2.5 points)

Atom	Hybridization
Carbon (<i>a</i>)	
Carbon (<i>b</i>)	
Carbon (<i>c</i>)	
Carbon (<i>d</i>)	
Carbon (<i>e</i>)	

ii. Now, consider the sulfur atom (S) marked as ***f***, nitrogen (N) atom (***g***), and oxygen(O) atoms (***h*** and ***i***), and determine their **hybridization**. (2 points)

Atom	Hybridization
Sulfur (<i>f</i>)	
Nitrogen (<i>g</i>)	
Oxygen (<i>h</i>)	
Oxygen (<i>i</i>)	

iii. **Describe** all the bonds formed by the carbon atoms ***a*** and ***b***, using sigma and pi designation as appropriate. (2 points)

Atom	Bonds formed (use sigma and pi as appropriate)
Carbon (<i>a</i>)	
Carbon (<i>b</i>)	

iv. How many total sp^3 -s sigma bonds are in the molecule? (1 point)

v. Determine **electronic** and **molecular geometry** at carbon (C) labeled ***e*** and sulfur (S) labeled ***f*** centres. (2 points)

Atom	Electronic Geometry	Molecular Geometry
Carbon (<i>e</i>)		
Sulfur (<i>f</i>)		

vi. Circle the value from the choices given below, that describes the **C-S-C bond angle at sulfur** (S) marked ***f***: (0.5 points)

>90° to <109.5° 109.5° >109.5 to <120° 120° >120° to <180°


Question 8 12 points

Using the **valence shell orbitals only** draw a **complete** molecular orbital (MO) diagram (with no s-p mixing) for the molecule **PS**, in the space provided below. **(6 points)**

Assume that 3s and 3p atomic orbitals will form molecular orbitals like those formed by the combination of 2s and 2p atomic orbitals that we learnt in class, and there is no s-p mixing.

Show **energy arrow**, **atomic orbitals** involved, **atom labels**, the **molecular orbitals** that are formed, **filled electrons** in the MO diagram etc. Make sure to label all the atomic and molecular orbitals.

Using the MO diagram above, answer the following questions:

- (i) Which one of the following will have the highest bond order? Circle the correct choice
 A) PS^{1-} B) PS C) PS^{1+}
- (ii) Which one of the following will have the longest bond length? Circle the correct choice
 A) PS^{1-} B) PS C) PS^{1+}
- (iii) Which of the following will be diamagnetic? Circle the correct choice.
 A) PS^{1-} B) PS C) PS^{1+}
- (iv) Draw Lewis dot structure of the molecule with the shortest bond length (based on the MO diagram) from among the three (PS^{1-} , PS, PS^{1+}), and indicate the bond order calculated from the Lewis dot structure. **(3 points)**

Lewis Dot Structure

Bond Order



CBF987FC-D7FD-47F7-993F-D803001D5DD1

midterm-exam-93388

#128 Page 10 of 10

SCRAP PAGE

**CHEMISTRY 110 –Midterm Exam****Thursday November 7, 2024, 6:30 – 8:30 pm****Examiners: Pallavi Sirjoosingh and Ashok Kakkar**

First name (please write as legibly as possible within the boxes)

--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--

Last name

--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--

Student ID number

--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--

This booklet has 8 questions on 10 pages (The last page can be used as a scrap page).

The staple must NOT be removed from the question booklet. Your intact question booklet must be returned at the end of the exam.

A separate booklet contains a periodic table, and datasheet.

Write your answers directly under the questions in the space provided. Do not write on the QR codes or near the top edges (as it may be cut off).

Please write **legibly and dark enough** for the scan to be readable. For students who use English as a second language, dictionaries are permitted. Calculators are allowed.

DO NOT ASK QUESTIONS OF THE INVIGILATORS about interpretations of the exam questions. If you have a concern about a question, please summarize it at the top of this page. All concerns will be evaluated after the exam. GOOD LUCK!

**Question 1 8 points** (*Assume all transitions are possible*)

a. 6 points Calculate the **energy (in kJ/mol)** required to ionize (completely remove) the electron from the **n=3** level of **He⁺ ion** (He: Helium). Show your work including equations. (3 sig. figs.)

b. 2 points

(I) Select the largest (longest) wavelength from the three choices below:

- i. Wavelength of photon to *completely remove* an electron from *1s orbital of Hydrogen atom*
- ii. Wavelength of photon to *completely remove* an electron from *1s orbital of Boron atom (B)*
- iii. Wavelength of photon to *completely remove* an electron from *1s orbital of Boron cation (B⁺)*

i or ii or iii

(II) Select the smallest (shortest) wavelength from the three choices below (*for H atom*):

- i. Wavelength of photon *emitted* for an electron transitioning from *2p to 1s orbital*
- ii. Wavelength of photon *absorbed* for an electron transitioning from *2s to 4p orbital*
- iii. Wavelength of photon *absorbed* for an electron transitioning from *1s to 3p orbital*

i or ii or iii



Question 2 8 points



A Chem110 student plots the graph shown on the left, from data obtained for the kinetic energy (eV) of the ejected electron versus frequency (s^{-1}) of the incident light, for an unknown metal. ($1\text{eV} = 1.6022 \times 10^{-19} \text{ J}$)

Based on the plot, answer the questions **a to c** below:

a. 1 point On the plot above, label the **threshold frequency**.

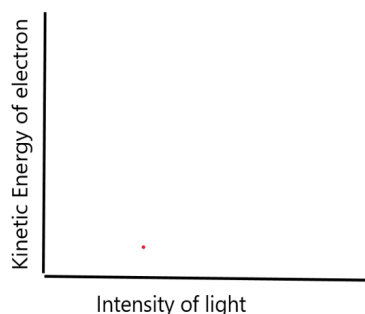
b. 2 points What is the value of the work potential (in J) of the unknown metal?

c. 3 points What is the *longest* wavelength (in nm) required to eject an electron from this metal's surface? Show your work including equation(s) used. (3 sig figs)

d. 2 points In the space provided below, plot the following

(Assume frequency is *above* threshold frequency):

- i. *Kinetic energy* of an ejected electron versus the *intensity* of the incident light (left)
- ii. *Number of electrons* ejected versus the *frequency* of the incident light (right)



**Question 3** **6 points****a. 2 points**

In the ground state of Ge (Germanium; $Z = 32$) atom, **name** the following orbitals:

i. Orbital containing the electron that requires the maximum energy to be removed.

ii. Orbital containing the electron that experiences the minimum effective nuclear charge.

b. 2 points

An unknown anion (A^-) has the following *excited* electronic configuration: $[Ar] 4s^2 3d^3 4p^1$

(Ar: Argon; $Z = 18$)

Give the **condensed ground state electronic configuration** of the atom, **A** and the cation, **A^{2+}** .
Show your work. (Assume all electronic transitions are possible)

Electronic Configuration of A:

Electronic Configuration of A^{2+} :

c. 2 points

In a single atom, determine how many electrons can have the following set of quantum numbers?

(i) $n = 5; l = 3$

In a single atom, determine how many electrons can have the following set of quantum numbers?

(i) $n = 3; l = 2; m_s = +1/2$

**Question 4** **8 points**

Rank the following from 1 to 4 in **increasing** order. Provide the ranking in the boxes provided.

Note: Atomic number (Z) are provided for relevant atoms

a) Increasing (lowest to highest) order of *ionization energy*: (1= lowest; 4 = highest)

Mg

Mg: Z = 12

Rb

Rb: Z = 37

Cs

Cs: Z = 55

K

K : Z = 19

b) Increasing (lowest to highest) order of *bond angle*: (1= lowest; 4 = highest)

H₂OCCl₄XeF₄

HCN

c) Increasing (smallest to largest) order of *ionic size* (1= smallest; 4 = largest)

S²⁻

S: Z = 16

O²⁻

O: Z = 8

Be²⁺

Be: Z = 4

Na⁺

Na: Z = 11

d) Increasing (lowest to highest) order of *bond energy* (1= lowest; 4 = highest)

P-F

P-Cl

P-I

P-Br

**Question 5** **12 points**

a. 9 points For the molecules HClO_4 and TeF_6 , draw the most stable *Lewis structure* showing appropriate shape and *dashed/wedged bonds*, based on VSEPR. Determine the **molecular geometry** around the central atom. For HClO_4 , label the *most polar bond*.



b. 3 points (Note: Lone pairs are not shown in structures below)

(i) For the three *resonance* structures (1, 2 and 3) shown below, answer the following question:



(I) Circle the *most contributing* resonance structure.

Structure 1

Structure 2

Structure 3

(ii) For the three structures given (4, 5, and 6) below, answer the following questions:



(I) Circle all the structures that represent *resonance structures* of each other:

Structure 4

Structure 5

Structure 6

(II) Among the resonance structures, circle the *most contributing* resonance structure.

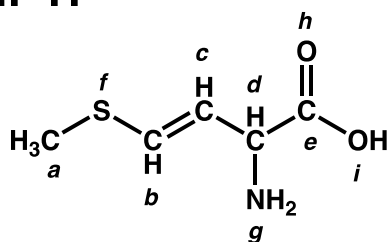
Structure 4

Structure 5

Structure 6

**Question 6** **6 points**Using the data given below, calculate the **lattice energy** for SrO(s).

	ΔH° (kJ/mol)
$\text{Sr}(s) \longrightarrow \text{Sr}(g)$	+180
$\text{Sr}(g) \longrightarrow \text{Sr}^{2+}(g) + 2e^-$	+1468
$\text{O}_2(g) \longrightarrow 2\text{O}(g)$	+500
$\text{O}(g) + 2e^- \longrightarrow \text{O}^{2-}(g)$	+600
$\text{Sr}(s) + \frac{1}{2}\text{O}_2(g) \longrightarrow \text{SrO}(s)$	-550

**Question 7 10 points**

Consider the molecule given on the left, and answer questions (i) to (vi) using chemical bonding theories that we have learnt in class (valence bond theory, hybridization).

Note that the lone pairs are not shown in the structure:

i. Consider the carbon atoms marked in as ***a***, ***b***, ***c***, ***d***, and ***e*** in the molecule, and determine **hybridization** at each of the carbon centers. (2.5 points)

Atom	Hybridization
Carbon (<i>a</i>)	
Carbon (<i>b</i>)	
Carbon (<i>c</i>)	
Carbon (<i>d</i>)	
Carbon (<i>e</i>)	

ii. Now, consider the sulfur atom (S) marked as ***f***, nitrogen (N) atom (***g***), and oxygen(O) atoms (***h*** and ***i***), and determine their **hybridization**. (2 points)

Atom	Hybridization
Sulfur (<i>f</i>)	
Nitrogen (<i>g</i>)	
Oxygen (<i>h</i>)	
Oxygen (<i>i</i>)	

iii. **Describe** all the bonds formed by the carbon atoms ***a*** and ***b***, using sigma and pi designation as appropriate. (2 points)

Atom	Bonds formed (use sigma and pi as appropriate)
Carbon (<i>a</i>)	
Carbon (<i>b</i>)	

iv. How many total sp^3 -s sigma bonds are in the molecule? (1 point)

v. Determine **electronic** and **molecular geometry** at carbon (C) labeled ***e*** and sulfur (S) labeled ***f*** centres. (2 points)

Atom	Electronic Geometry	Molecular Geometry
Carbon (<i>e</i>)		
Sulfur (<i>f</i>)		

vi. Circle the value from the choices given below, that describes the **C-S-C bond angle at sulfur** (S) marked ***f***: (0.5 points)

>90° to <109.5° 109.5° >109.5 to <120° 120° >120° to <180°


Question 8 12 points

Using the **valence shell orbitals only** draw a **complete** molecular orbital (MO) diagram (with no s-p mixing) for the molecule **PS**, in the space provided below. **(6 points)**

Assume that 3s and 3p atomic orbitals will form molecular orbitals like those formed by the combination of 2s and 2p atomic orbitals that we learnt in class, and there is no s-p mixing.

Show **energy arrow**, **atomic orbitals** involved, **atom labels**, the **molecular orbitals** that are formed, **filled electrons** in the MO diagram etc. Make sure to label all the atomic and molecular orbitals.

Using the MO diagram above, answer the following questions:

- (i) Which one of the following will have the highest bond order? Circle the correct choice
 A) PS^{1-} B) PS C) PS^{1+}
- (ii) Which one of the following will have the longest bond length? Circle the correct choice
 A) PS^{1-} B) PS C) PS^{1+}
- (iii) Which of the following will be diamagnetic? Circle the correct choice.
 A) PS^{1-} B) PS C) PS^{1+}
- (iv) Draw Lewis dot structure of the molecule with the shortest bond length (based on the MO diagram) from among the three (PS^{1-} , PS, PS^{1+}), and indicate the bond order calculated from the Lewis dot structure. **(3 points)**

Lewis Dot Structure

Bond Order



C3534C83-EAA3-4C81-937A-747434738B04

midterm-exam-93388

#129 Page 10 of 10

SCRAP PAGE

**CHEMISTRY 110 –Midterm Exam****Thursday November 7, 2024, 6:30 – 8:30 pm****Examiners: Pallavi Sirjoosingh and Ashok Kakkar**

First name (please write as legibly as possible within the boxes)

--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--

Last name

--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--

Student ID number

--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--

This booklet has 8 questions on 10 pages (The last page can be used as a scrap page).

The staple must NOT be removed from the question booklet. Your intact question booklet must be returned at the end of the exam.

A separate booklet contains a periodic table, and datasheet.

Write your answers directly under the questions in the space provided. Do not write on the QR codes or near the top edges (as it may be cut off).

Please write **legibly and dark enough** for the scan to be readable. For students who use English as a second language, dictionaries are permitted. Calculators are allowed.

DO NOT ASK QUESTIONS OF THE INVIGILATORS about interpretations of the exam questions. If you have a concern about a question, please summarize it at the top of this page. All concerns will be evaluated after the exam. GOOD LUCK!

**Question 1 8 points** (*Assume all transitions are possible*)

a. 6 points Calculate the **energy (in kJ/mol)** required to ionize (completely remove) the electron from the **n=3** level of **He⁺ ion** (He: Helium). Show your work including equations. (3 sig. figs.)

b. 2 points

(I) Select the largest (longest) wavelength from the three choices below:

- i. Wavelength of photon to *completely remove* an electron from *1s orbital of Hydrogen atom*
- ii. Wavelength of photon to *completely remove* an electron from *1s orbital of Boron atom (B)*
- iii. Wavelength of photon to *completely remove* an electron from *1s orbital of Boron cation (B⁺)*

i or ii or iii

(II) Select the smallest (shortest) wavelength from the three choices below (*for H atom*):

- i. Wavelength of photon *emitted* for an electron transitioning from *2p to 1s orbital*
- ii. Wavelength of photon *absorbed* for an electron transitioning from *2s to 4p orbital*
- iii. Wavelength of photon *absorbed* for an electron transitioning from *1s to 3p orbital*

i or ii or iii



Question 2 8 points



A Chem110 student plots the graph shown on the left, from data obtained for the kinetic energy (eV) of the ejected electron versus frequency (s^{-1}) of the incident light, for an unknown metal. ($1\text{eV} = 1.6022 \times 10^{-19} \text{ J}$)

Based on the plot, answer the questions **a to c** below:

a. 1 point On the plot above, label the **threshold frequency**.

b. 2 points What is the value of the work potential (in J) of the unknown metal?

c. 3 points What is the *longest* wavelength (in nm) required to eject an electron from this metal's surface? Show your work including equation(s) used. (3 sig figs)

d. 2 points In the space provided below, plot the following

(Assume frequency is *above* threshold frequency):

- Kinetic energy* of an ejected electron versus the *intensity* of the incident light (left)
- Number of electrons* ejected versus the *frequency* of the incident light (right)



**Question 3** **6 points****a. 2 points**

In the ground state of Ge (Germanium; $Z = 32$) atom, **name** the following orbitals:

i. Orbital containing the electron that requires the maximum energy to be removed.

ii. Orbital containing the electron that experiences the minimum effective nuclear charge.

b. 2 points

An unknown anion (A^-) has the following *excited* electronic configuration: $[Ar] 4s^2 3d^3 4p^1$

(Ar: Argon; $Z = 18$)

Give the **condensed ground state electronic configuration** of the atom, **A** and the cation, **A^{2+}** .
Show your work. (Assume all electronic transitions are possible)

Electronic Configuration of A:

Electronic Configuration of A^{2+} :

c. 2 points

In a single atom, determine how many electrons can have the following set of quantum numbers?

(i) $n = 5; l = 3$

In a single atom, determine how many electrons can have the following set of quantum numbers?

(i) $n = 3; l = 2; m_s = +1/2$

**Question 4** **8 points**

Rank the following from 1 to 4 in **increasing** order. Provide the ranking in the boxes provided.

Note: Atomic number (Z) are provided for relevant atoms

a) Increasing (lowest to highest) order of *ionization energy*: (1= lowest; 4 = highest)

Mg

Mg: Z = 12

Rb

Rb: Z = 37

Cs

Cs: Z = 55

K

K : Z = 19

b) Increasing (lowest to highest) order of *bond angle*: (1= lowest; 4 = highest)

H₂OCCl₄XeF₄

HCN

c) Increasing (smallest to largest) order of *ionic size* (1= smallest; 4 = largest)

S²⁻

S: Z = 16

O²⁻

O: Z = 8

Be²⁺

Be: Z = 4

Na⁺

Na: Z = 11

d) Increasing (lowest to highest) order of *bond energy* (1= lowest; 4 = highest)

P-F

P-Cl

P-I

P-Br

**Question 5** **12 points**

a. 9 points For the molecules HClO_4 and TeF_6 , draw the most stable *Lewis structure* showing appropriate shape and *dashed/wedged bonds*, based on VSEPR. Determine the **molecular geometry** around the central atom. For HClO_4 , label the *most polar bond*.



b. 3 points (Note: Lone pairs are not shown in structures below)

(i) For the three *resonance* structures (1, 2 and 3) shown below, answer the following question:



(I) Circle the *most contributing* resonance structure.

Structure 1

Structure 2

Structure 3

(ii) For the three structures given (4, 5, and 6) below, answer the following questions:



(I) Circle all the structures that represent *resonance structures* of each other:

Structure 4

Structure 5

Structure 6

(II) Among the resonance structures, circle the *most contributing* resonance structure.

Structure 4

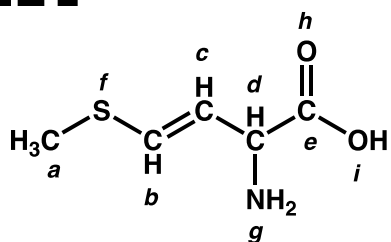
Structure 5

Structure 6

**Question 6** **6 points**

Using the data given below, calculate the **lattice energy** for SrO(s).

	ΔH° (kJ/mol)
$\text{Sr}(s) \longrightarrow \text{Sr}(g)$	+180
$\text{Sr}(g) \longrightarrow \text{Sr}^{2+}(g) + 2e^-$	+1468
$\text{O}_2(g) \longrightarrow 2\text{O}(g)$	+500
$\text{O}(g) + 2e^- \longrightarrow \text{O}^{2-}(g)$	+600
$\text{Sr}(s) + \frac{1}{2}\text{O}_2(g) \longrightarrow \text{SrO}(s)$	-550

**Question 7 10 points**

Consider the molecule given on the left, and answer questions (i) to (vi) using chemical bonding theories that we have learnt in class (valence bond theory, hybridization).

Note that the lone pairs are not shown in the structure:

i. Consider the carbon atoms marked in as ***a***, ***b***, ***c***, ***d***, and ***e*** in the molecule, and determine **hybridization** at each of the carbon centers. (2.5 points)

Atom	Hybridization
Carbon (<i>a</i>)	
Carbon (<i>b</i>)	
Carbon (<i>c</i>)	
Carbon (<i>d</i>)	
Carbon (<i>e</i>)	

ii. Now, consider the sulfur atom (S) marked as ***f***, nitrogen (N) atom (***g***), and oxygen(O) atoms (***h*** and ***i***), and determine their **hybridization**. (2 points)

Atom	Hybridization
Sulfur (<i>f</i>)	
Nitrogen (<i>g</i>)	
Oxygen (<i>h</i>)	
Oxygen (<i>i</i>)	

iii. **Describe** all the bonds formed by the carbon atoms ***a*** and ***b***, using sigma and pi designation as appropriate. (2 points)

Atom	Bonds formed (use sigma and pi as appropriate)
Carbon (<i>a</i>)	
Carbon (<i>b</i>)	

iv. How many total sp^3 -s sigma bonds are in the molecule? (1 point)

v. Determine **electronic** and **molecular geometry** at carbon (C) labeled ***e*** and sulfur (S) labeled ***f*** centres. (2 points)

Atom	Electronic Geometry	Molecular Geometry
Carbon (<i>e</i>)		
Sulfur (<i>f</i>)		

vi. Circle the value from the choices given below, that describes the **C-S-C bond angle at sulfur** (S) marked ***f***: (0.5 points)

>90° to <109.5° 109.5° >109.5 to <120° 120° >120° to <180°


Question 8 12 points

Using the **valence shell orbitals only** draw a **complete** molecular orbital (MO) diagram (with no s-p mixing) for the molecule **PS**, in the space provided below. **(6 points)**

Assume that 3s and 3p atomic orbitals will form molecular orbitals like those formed by the combination of 2s and 2p atomic orbitals that we learnt in class, and there is no s-p mixing.

Show **energy arrow**, **atomic orbitals** involved, **atom labels**, the **molecular orbitals** that are formed, **filled electrons** in the MO diagram etc. Make sure to label all the atomic and molecular orbitals.

Using the MO diagram above, answer the following questions:

- (i) Which one of the following will have the highest bond order? Circle the correct choice
 A) PS^{1-} B) PS C) PS^{1+}
- (ii) Which one of the following will have the longest bond length? Circle the correct choice
 A) PS^{1-} B) PS C) PS^{1+}
- (iii) Which of the following will be diamagnetic? Circle the correct choice.
 A) PS^{1-} B) PS C) PS^{1+}
- (iv) Draw Lewis dot structure of the molecule with the shortest bond length (based on the MO diagram) from among the three (PS^{1-} , PS, PS^{1+}), and indicate the bond order calculated from the Lewis dot structure. **(3 points)**

Lewis Dot Structure

Bond Order



4FD894DC-9CDB-4013-9C6C-519006FB78D2

midterm-exam-93388

#130 Page 10 of 10

SCRAP PAGE

**CHEMISTRY 110 –Midterm Exam****Thursday November 7, 2024, 6:30 – 8:30 pm****Examiners: Pallavi Sirjoosingh and Ashok Kakkar**

First name (please write as legibly as possible within the boxes)

--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--

Last name

--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--

Student ID number

--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--

This booklet has 8 questions on 10 pages (The last page can be used as a scrap page).

The staple must NOT be removed from the question booklet. Your intact question booklet must be returned at the end of the exam.

A separate booklet contains a periodic table, and datasheet.

Write your answers directly under the questions in the space provided. Do not write on the QR codes or near the top edges (as it may be cut off).

Please write **legibly and dark enough** for the scan to be readable. For students who use English as a second language, dictionaries are permitted. Calculators are allowed.

DO NOT ASK QUESTIONS OF THE INVIGILATORS about interpretations of the exam questions. If you have a concern about a question, please summarize it at the top of this page. All concerns will be evaluated after the exam. GOOD LUCK!

**Question 1 8 points** (*Assume all transitions are possible*)

a. 6 points Calculate the **energy (in kJ/mol)** required to ionize (completely remove) the electron from the **n=3** level of **He⁺ ion** (He: Helium). Show your work including equations. (3 sig. figs.)

b. 2 points

(I) Select the largest (longest) wavelength from the three choices below:

- i. Wavelength of photon to *completely remove* an electron from *1s orbital of Hydrogen atom*
- ii. Wavelength of photon to *completely remove* an electron from *1s orbital of Boron atom (B)*
- iii. Wavelength of photon to *completely remove* an electron from *1s orbital of Boron cation (B⁺)*

i or ii or iii

(II) Select the smallest (shortest) wavelength from the three choices below (*for H atom*):

- i. Wavelength of photon *emitted* for an electron transitioning from *2p to 1s orbital*
- ii. Wavelength of photon *absorbed* for an electron transitioning from *2s to 4p orbital*
- iii. Wavelength of photon *absorbed* for an electron transitioning from *1s to 3p orbital*

i or ii or iii



Question 2 8 points



A Chem110 student plots the graph shown on the left, from data obtained for the kinetic energy (eV) of the ejected electron versus frequency (s^{-1}) of the incident light, for an unknown metal. ($1\text{eV} = 1.6022 \times 10^{-19} \text{ J}$)

Based on the plot, answer the questions **a to c** below:

a. 1 point On the plot above, label the **threshold frequency**.

b. 2 points What is the value of the work potential (in J) of the unknown metal?

c. 3 points What is the *longest* wavelength (in nm) required to eject an electron from this metal's surface? Show your work including equation(s) used. (3 sig figs)

d. 2 points In the space provided below, plot the following

(Assume frequency is *above* threshold frequency):

- Kinetic energy* of an ejected electron versus the *intensity* of the incident light (left)
- Number of electrons* ejected versus the *frequency* of the incident light (right)



**Question 3** **6 points****a. 2 points**

In the ground state of Ge (Germanium; $Z = 32$) atom, **name** the following orbitals:

i. Orbital containing the electron that requires the maximum energy to be removed.

ii. Orbital containing the electron that experiences the minimum effective nuclear charge.

b. 2 points

An unknown anion (A^-) has the following *excited* electronic configuration: $[Ar] 4s^2 3d^3 4p^1$

(Ar: Argon; $Z = 18$)

Give the **condensed ground state electronic configuration** of the atom, **A** and the cation, **A^{2+}** .
Show your work. (Assume all electronic transitions are possible)

Electronic Configuration of A:

Electronic Configuration of A^{2+} :

c. 2 points

In a single atom, determine how many electrons can have the following set of quantum numbers?

(i) $n = 5; l = 3$

In a single atom, determine how many electrons can have the following set of quantum numbers?

(i) $n = 3; l = 2; m_s = +1/2$

**Question 4** **8 points**

Rank the following from 1 to 4 in **increasing** order. Provide the ranking in the boxes provided.

Note: Atomic number (Z) are provided for relevant atoms

a) Increasing (lowest to highest) order of *ionization energy*: (1= lowest; 4 = highest)

Mg

Mg: Z = 12

Rb

Rb: Z = 37

Cs

Cs: Z = 55

K

K : Z = 19

b) Increasing (lowest to highest) order of *bond angle*: (1= lowest; 4 = highest)

H₂OCCl₄XeF₄

HCN

c) Increasing (smallest to largest) order of *ionic size* (1= smallest; 4 = largest)

S²⁻

S: Z = 16

O²⁻

O: Z = 8

Be²⁺

Be: Z = 4

Na⁺

Na: Z = 11

d) Increasing (lowest to highest) order of *bond energy* (1= lowest; 4 = highest)

P-F

P-Cl

P-I

P-Br

**Question 5** **12 points**

a. 9 points For the molecules HClO_4 and TeF_6 , draw the most stable *Lewis structure* showing appropriate shape and *dashed/wedged bonds*, based on VSEPR. Determine the **molecular geometry** around the central atom. For HClO_4 , label the *most polar bond*.



b. 3 points (Note: Lone pairs are not shown in structures below)

(i) For the three *resonance* structures (1, 2 and 3) shown below, answer the following question:



(I) Circle the *most contributing* resonance structure.

Structure 1

Structure 2

Structure 3

(ii) For the three structures given (4, 5, and 6) below, answer the following questions:



(I) Circle all the structures that represent *resonance structures* of each other:

Structure 4

Structure 5

Structure 6

(II) Among the resonance structures, circle the *most contributing* resonance structure.

Structure 4

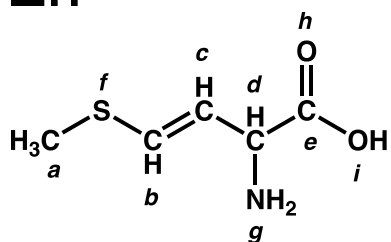
Structure 5

Structure 6

**Question 6** **6 points**

Using the data given below, calculate the **lattice energy** for SrO(s).

	ΔH° (kJ/mol)
$\text{Sr}(s) \longrightarrow \text{Sr}(g)$	+180
$\text{Sr}(g) \longrightarrow \text{Sr}^{2+}(g) + 2e^-$	+1468
$\text{O}_2(g) \longrightarrow 2\text{O}(g)$	+500
$\text{O}(g) + 2e^- \longrightarrow \text{O}^{2-}(g)$	+600
$\text{Sr}(s) + \frac{1}{2}\text{O}_2(g) \longrightarrow \text{SrO}(s)$	-550

**Question 7 10 points**

Consider the molecule given on the left, and answer questions (i) to (vi) using chemical bonding theories that we have learnt in class (valence bond theory, hybridization).

Note that the lone pairs are not shown in the structure:

i. Consider the carbon atoms marked in as ***a***, ***b***, ***c***, ***d***, and ***e*** in the molecule, and determine **hybridization** at each of the carbon centers. (2.5 points)

Atom	Hybridization
Carbon (<i>a</i>)	
Carbon (<i>b</i>)	
Carbon (<i>c</i>)	
Carbon (<i>d</i>)	
Carbon (<i>e</i>)	

ii. Now, consider the sulfur atom (S) marked as ***f***, nitrogen (N) atom (***g***), and oxygen(O) atoms (***h*** and ***i***), and determine their **hybridization**. (2 points)

Atom	Hybridization
Sulfur (<i>f</i>)	
Nitrogen (<i>g</i>)	
Oxygen (<i>h</i>)	
Oxygen (<i>i</i>)	

iii. **Describe** all the bonds formed by the carbon atoms ***a*** and ***b***, using sigma and pi designation as appropriate. (2 points)

Atom	Bonds formed (use sigma and pi as appropriate)
Carbon (<i>a</i>)	
Carbon (<i>b</i>)	

iv. How many total sp^3 -s sigma bonds are in the molecule? (1 point)

v. Determine **electronic** and **molecular geometry** at carbon (C) labeled ***e*** and sulfur (S) labeled ***f*** centres. (2 points)

Atom	Electronic Geometry	Molecular Geometry
Carbon (<i>e</i>)		
Sulfur (<i>f</i>)		

vi. Circle the value from the choices given below, that describes the **C-S-C bond angle at sulfur** (S) marked ***f***: (0.5 points)

$>90^\circ$ to $<109.5^\circ$ 109.5° >109.5 to $<120^\circ$ 120° $>120^\circ$ to $<180^\circ$


Question 8 12 points

Using the **valence shell orbitals only** draw a **complete** molecular orbital (MO) diagram (with no s-p mixing) for the molecule **PS**, in the space provided below. **(6 points)**

Assume that 3s and 3p atomic orbitals will form molecular orbitals like those formed by the combination of 2s and 2p atomic orbitals that we learnt in class, and there is no s-p mixing.

Show **energy arrow**, **atomic orbitals** involved, **atom labels**, the **molecular orbitals** that are formed, **filled electrons** in the MO diagram etc. Make sure to label all the atomic and molecular orbitals.

Using the MO diagram above, answer the following questions:

- (i) Which one of the following will have the highest bond order? Circle the correct choice
 A) PS^{1-} B) PS C) PS^{1+}
- (ii) Which one of the following will have the longest bond length? Circle the correct choice
 A) PS^{1-} B) PS C) PS^{1+}
- (iii) Which of the following will be diamagnetic? Circle the correct choice.
 A) PS^{1-} B) PS C) PS^{1+}
- (iv) Draw Lewis dot structure of the molecule with the shortest bond length (based on the MO diagram) from among the three (PS^{1-} , PS, PS^{1+}), and indicate the bond order calculated from the Lewis dot structure. **(3 points)**

Lewis Dot Structure

Bond Order



1C104B65-5638-48A6-8E56-15A6596F2529

midterm-exam-93388

#131 Page 10 of 10

SCRAP PAGE

**CHEMISTRY 110 –Midterm Exam****Thursday November 7, 2024, 6:30 – 8:30 pm**

Examiners: Pallavi Sirjoosingh and Ashok Kakkar

First name (please write as legibly as possible within the boxes)

--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--

Last name

--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--

Student ID number

--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--

This booklet has 8 questions on 10 pages (The last page can be used as a scrap page).

The staple must NOT be removed from the question booklet. Your intact question booklet must be returned at the end of the exam.

A separate booklet contains a periodic table, and datasheet.

Write your answers directly under the questions in the space provided. Do not write on the QR codes or near the top edges (as it may be cut off).

Please write **legibly and dark enough** for the scan to be readable. For students who use English as a second language, dictionaries are permitted. Calculators are allowed.

DO NOT ASK QUESTIONS OF THE INVIGILATORS about interpretations of the exam questions. If you have a concern about a question, please summarize it at the top of this page. All concerns will be evaluated after the exam. GOOD LUCK!

**Question 1 8 points** (*Assume all transitions are possible*)

a. 6 points Calculate the **energy (in kJ/mol)** required to ionize (completely remove) the electron from the **$n=3$** level of **He^+ ion** (He: Helium). Show your work including equations. (3 sig. figs.)

b. 2 points

(I) Select the largest (longest) wavelength from the three choices below:

- i. Wavelength of photon to *completely remove* an electron from *1s orbital of Hydrogen atom*
- ii. Wavelength of photon to *completely remove* an electron from *1s orbital of Boron atom (B)*
- iii. Wavelength of photon to *completely remove* an electron from *1s orbital of Boron cation (B^+)*

i or ii or iii

(II) Select the smallest (shortest) wavelength from the three choices below (*for H atom*):

- i. Wavelength of photon *emitted* for an electron transitioning from *2p to 1s orbital*
- ii. Wavelength of photon *absorbed* for an electron transitioning from *2s to 4p orbital*
- iii. Wavelength of photon *absorbed* for an electron transitioning from *1s to 3p orbital*

i or ii or iii



Question 2 8 points



A Chem110 student plots the graph shown on the left, from data obtained for the kinetic energy (eV) of the ejected electron versus frequency (s^{-1}) of the incident light, for an unknown metal. ($1\text{eV} = 1.6022 \times 10^{-19} \text{ J}$)

Based on the plot, answer the questions **a to c** below:

a. 1 point On the plot above, label the **threshold frequency**.

b. 2 points What is the value of the work potential (in J) of the unknown metal?

c. 3 points What is the *longest* wavelength (in nm) required to eject an electron from this metal's surface? Show your work including equation(s) used. (3 sig figs)

d. 2 points In the space provided below, plot the following

(Assume frequency is *above* threshold frequency):

- Kinetic energy* of an ejected electron versus the *intensity* of the incident light (left)
- Number of electrons* ejected versus the *frequency* of the incident light (right)



**Question 3** **6 points****a. 2 points**

In the ground state of Ge (Germanium; $Z = 32$) atom, **name** the following orbitals:

i. Orbital containing the electron that requires the maximum energy to be removed.

ii. Orbital containing the electron that experiences the minimum effective nuclear charge.

b. 2 points

An unknown anion (A^-) has the following *excited* electronic configuration: $[Ar] 4s^2 3d^3 4p^1$

(Ar: Argon; $Z = 18$)

Give the **condensed ground state electronic configuration** of the atom, **A** and the cation, **A^{2+}** .
Show your work. (Assume all electronic transitions are possible)

Electronic Configuration of A:

Electronic Configuration of A^{2+} :

c. 2 points

In a single atom, determine how many electrons can have the following set of quantum numbers?

(i) $n = 5; l = 3$

In a single atom, determine how many electrons can have the following set of quantum numbers?

(i) $n = 3; l = 2; m_s = +1/2$

**Question 4** **8 points**

Rank the following from 1 to 4 in **increasing** order. Provide the ranking in the boxes provided.

Note: Atomic number (Z) are provided for relevant atoms

a) Increasing (lowest to highest) order of *ionization energy*: (1= lowest; 4 = highest)

Mg

Mg: Z = 12

Rb

Rb: Z = 37

Cs

Cs: Z = 55

K

K : Z = 19

b) Increasing (lowest to highest) order of *bond angle*: (1= lowest; 4 = highest)

H₂OCCl₄XeF₄

HCN

c) Increasing (smallest to largest) order of *ionic size* (1= smallest; 4 = largest)

S²⁻

S: Z = 16

O²⁻

O: Z = 8

Be²⁺

Be: Z = 4

Na⁺

Na: Z = 11

d) Increasing (lowest to highest) order of *bond energy* (1= lowest; 4 = highest)

P-F

P-Cl

P-I

P-Br

**Question 5** **12 points**

a. 9 points For the molecules HClO_4 and TeF_6 , draw the most stable *Lewis structure* showing appropriate shape and *dashed/wedged bonds*, based on VSEPR. Determine the **molecular geometry** around the central atom. For HClO_4 , label the *most polar bond*.



b. 3 points (Note: Lone pairs are not shown in structures below)

(i) For the three *resonance* structures (1, 2 and 3) shown below, answer the following question:



(I) Circle the *most contributing* resonance structure.

Structure 1

Structure 2

Structure 3

(ii) For the three structures given (4, 5, and 6) below, answer the following questions:



(I) Circle all the structures that represent *resonance structures* of each other:

Structure 4

Structure 5

Structure 6

(II) Among the resonance structures, circle the *most contributing* resonance structure.

Structure 4

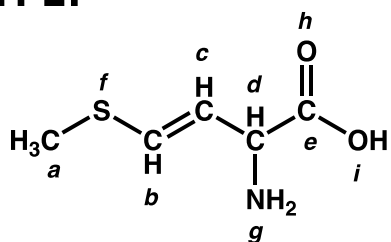
Structure 5

Structure 6

**Question 6** **6 points**

Using the data given below, calculate the **lattice energy** for SrO(s).

	ΔH° (kJ/mol)
$\text{Sr}(s) \longrightarrow \text{Sr}(g)$	+180
$\text{Sr}(g) \longrightarrow \text{Sr}^{2+}(g) + 2e^-$	+1468
$\text{O}_2(g) \longrightarrow 2\text{O}(g)$	+500
$\text{O}(g) + 2e^- \longrightarrow \text{O}^{2-}(g)$	+600
$\text{Sr}(s) + \frac{1}{2}\text{O}_2(g) \longrightarrow \text{SrO}(s)$	-550

**Question 7 10 points**

Consider the molecule given on the left, and answer questions (i) to (vi) using chemical bonding theories that we have learnt in class (valence bond theory, hybridization).

Note that the lone pairs are not shown in the structure:

i. Consider the carbon atoms marked in as ***a***, ***b***, ***c***, ***d***, and ***e*** in the molecule, and determine **hybridization** at each of the carbon centers. (2.5 points)

Atom	Hybridization
Carbon (<i>a</i>)	
Carbon (<i>b</i>)	
Carbon (<i>c</i>)	
Carbon (<i>d</i>)	
Carbon (<i>e</i>)	

ii. Now, consider the sulfur atom (S) marked as ***f***, nitrogen (N) atom (***g***), and oxygen(O) atoms (***h*** and ***i***), and determine their **hybridization**. (2 points)

Atom	Hybridization
Sulfur (<i>f</i>)	
Nitrogen (<i>g</i>)	
Oxygen (<i>h</i>)	
Oxygen (<i>i</i>)	

iii. **Describe** all the bonds formed by the carbon atoms ***a*** and ***b***, using sigma and pi designation as appropriate. (2 points)

Atom	Bonds formed (use sigma and pi as appropriate)
Carbon (<i>a</i>)	
Carbon (<i>b</i>)	

iv. How many total sp^3 -s sigma bonds are in the molecule? (1 point)

v. Determine **electronic** and **molecular geometry** at carbon (C) labeled ***e*** and sulfur (S) labeled ***f*** centres. (2 points)

Atom	Electronic Geometry	Molecular Geometry
Carbon (<i>e</i>)		
Sulfur (<i>f</i>)		

vi. Circle the value from the choices given below, that describes the **C-S-C bond angle at sulfur** (S) marked ***f***: (0.5 points)

$>90^\circ$ to $<109.5^\circ$ 109.5° >109.5 to $<120^\circ$ 120° $>120^\circ$ to $<180^\circ$

**Question 8 12 points**

Using the **valence shell orbitals only** draw a **complete** molecular orbital (MO) diagram (with no s-p mixing) for the molecule **PS**, in the space provided below. **(6 points)**

Assume that 3s and 3p atomic orbitals will form molecular orbitals like those formed by the combination of 2s and 2p atomic orbitals that we learnt in class, and there is no s-p mixing.

Show **energy arrow**, **atomic orbitals** involved, **atom labels**, the **molecular orbitals** that are formed, **filled electrons** in the MO diagram etc. Make sure to label all the atomic and molecular orbitals.

Using the MO diagram above, answer the following questions:

- (i) Which one of the following will have the highest bond order? Circle the correct choice
A) PS^{1-} B) PS C) PS^{1+}
- (ii) Which one of the following will have the longest bond length? Circle the correct choice
A) PS^{1-} B) PS C) PS^{1+}
- (iii) Which of the following will be diamagnetic? Circle the correct choice.
A) PS^{1-} B) PS C) PS^{1+}
- (iv) Draw Lewis dot structure of the molecule with the shortest bond length (based on the MO diagram) from among the three (PS^{1-} , PS, PS^{1+}), and indicate the bond order calculated from the Lewis dot structure. **(3 points)**

Lewis Dot Structure

Bond Order



E86FDDCC-D903-4EB2-9718-E8742DD22A93

midterm-exam-93388

#132 Page 10 of 10

SCRAP PAGE

**CHEMISTRY 110 –Midterm Exam****Thursday November 7, 2024, 6:30 – 8:30 pm**

Examiners: Pallavi Sirjoosingh and Ashok Kakkar

First name (please write as legibly as possible within the boxes)

--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--

Last name

--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--

Student ID number

--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--

This booklet has 8 questions on 10 pages (The last page can be used as a scrap page).

The staple must NOT be removed from the question booklet. Your intact question booklet must be returned at the end of the exam.

A separate booklet contains a periodic table, and datasheet.

Write your answers directly under the questions in the space provided. Do not write on the QR codes or near the top edges (as it may be cut off).

Please write **legibly and dark enough** for the scan to be readable. For students who use English as a second language, dictionaries are permitted. Calculators are allowed.

DO NOT ASK QUESTIONS OF THE INVIGILATORS about interpretations of the exam questions. If you have a concern about a question, please summarize it at the top of this page. All concerns will be evaluated after the exam. GOOD LUCK!

**Question 1 8 points** (*Assume all transitions are possible*)

a. 6 points Calculate the **energy (in kJ/mol)** required to ionize (completely remove) the electron from the **n=3** level of **He⁺ ion** (He: Helium). Show your work including equations. (3 sig. figs.)

b. 2 points

(I) Select the largest (longest) wavelength from the three choices below:

- i. Wavelength of photon to *completely remove* an electron from *1s orbital of Hydrogen atom*
- ii. Wavelength of photon to *completely remove* an electron from *1s orbital of Boron atom (B)*
- iii. Wavelength of photon to *completely remove* an electron from *1s orbital of Boron cation (B⁺)*

i or ii or iii

(II) Select the smallest (shortest) wavelength from the three choices below (*for H atom*):

- i. Wavelength of photon *emitted* for an electron transitioning from *2p to 1s orbital*
- ii. Wavelength of photon *absorbed* for an electron transitioning from *2s to 4p orbital*
- iii. Wavelength of photon *absorbed* for an electron transitioning from *1s to 3p orbital*

i or ii or iii



Question 2 8 points



A Chem110 student plots the graph shown on the left, from data obtained for the kinetic energy (eV) of the ejected electron versus frequency (s^{-1}) of the incident light, for an unknown metal. ($1\text{eV} = 1.6022 \times 10^{-19} \text{ J}$)

Based on the plot, answer the questions **a to c** below:

a. 1 point On the plot above, label the **threshold frequency**.

b. 2 points What is the value of the work potential (in J) of the unknown metal?

c. 3 points What is the *longest* wavelength (in nm) required to eject an electron from this metal's surface? Show your work including equation(s) used. (3 sig figs)

d. 2 points In the space provided below, plot the following

(Assume frequency is *above* threshold frequency):

- i. *Kinetic energy* of an ejected electron versus the *intensity* of the incident light (left)
- ii. *Number of electrons* ejected versus the *frequency* of the incident light (right)



**Question 3** **6 points****a. 2 points**

In the ground state of Ge (Germanium; $Z = 32$) atom, **name** the following orbitals:

i. Orbital containing the electron that requires the maximum energy to be removed.

ii. Orbital containing the electron that experiences the minimum effective nuclear charge.

b. 2 points

An unknown anion (A^-) has the following *excited* electronic configuration: $[Ar] 4s^2 3d^3 4p^1$

(Ar: Argon; $Z = 18$)

Give the **condensed ground state electronic configuration** of the atom, **A** and the cation, **A^{2+}** .
Show your work. (Assume all electronic transitions are possible)

Electronic Configuration of A:

Electronic Configuration of A^{2+} :

c. 2 points

In a single atom, determine how many electrons can have the following set of quantum numbers?

(i) $n = 5; l = 3$

In a single atom, determine how many electrons can have the following set of quantum numbers?

(i) $n = 3; l = 2; m_s = +1/2$

**Question 4** **8 points**

Rank the following from 1 to 4 in **increasing** order. Provide the ranking in the boxes provided.

Note: Atomic number (Z) are provided for relevant atoms

a) Increasing (lowest to highest) order of *ionization energy*: (1= lowest; 4 = highest)

Mg

Mg: Z = 12

Rb

Rb: Z = 37

Cs

Cs: Z = 55

K

K : Z = 19

b) Increasing (lowest to highest) order of *bond angle*: (1= lowest; 4 = highest)

H₂OCCl₄XeF₄

HCN

c) Increasing (smallest to largest) order of *ionic size* (1= smallest; 4 = largest)

S²⁻

S: Z = 16

O²⁻

O: Z = 8

Be²⁺

Be: Z = 4

Na⁺

Na: Z = 11

d) Increasing (lowest to highest) order of *bond energy* (1= lowest; 4 = highest)

P-F

P-Cl

P-I

P-Br

**Question 5** **12 points**

a. 9 points For the molecules HClO_4 and TeF_6 , draw the most stable *Lewis structure* showing appropriate shape and *dashed/wedged bonds*, based on VSEPR. Determine the **molecular geometry** around the central atom. For HClO_4 , label the *most polar bond*.



b. 3 points (Note: Lone pairs are not shown in structures below)

(i) For the three *resonance* structures (1, 2 and 3) shown below, answer the following question:



(I) Circle the *most contributing* resonance structure.

Structure 1

Structure 2

Structure 3

(ii) For the three structures given (4, 5, and 6) below, answer the following questions:



(I) Circle all the structures that represent *resonance structures* of each other:

Structure 4

Structure 5

Structure 6

(II) Among the resonance structures, circle the *most contributing* resonance structure.

Structure 4

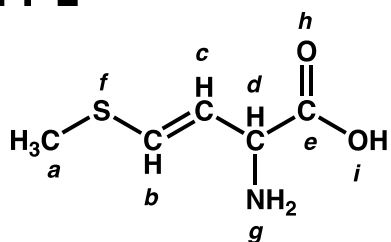
Structure 5

Structure 6

**Question 6** **6 points**

Using the data given below, calculate the **lattice energy** for SrO(s).

	ΔH° (kJ/mol)
$\text{Sr}(s) \longrightarrow \text{Sr}(g)$	+180
$\text{Sr}(g) \longrightarrow \text{Sr}^{2+}(g) + 2e^-$	+1468
$\text{O}_2(g) \longrightarrow 2\text{O}(g)$	+500
$\text{O}(g) + 2e^- \longrightarrow \text{O}^{2-}(g)$	+600
$\text{Sr}(s) + \frac{1}{2}\text{O}_2(g) \longrightarrow \text{SrO}(s)$	-550

**Question 7 10 points**

Consider the molecule given on the left, and answer questions (i) to (vi) using chemical bonding theories that we have learnt in class (valence bond theory, hybridization).

Note that the lone pairs are not shown in the structure:

i. Consider the carbon atoms marked in as ***a***, ***b***, ***c***, ***d***, and ***e*** in the molecule, and determine **hybridization** at each of the carbon centers. (2.5 points)

Atom	Hybridization
Carbon (<i>a</i>)	
Carbon (<i>b</i>)	
Carbon (<i>c</i>)	
Carbon (<i>d</i>)	
Carbon (<i>e</i>)	

ii. Now, consider the sulfur atom (S) marked as ***f***, nitrogen (N) atom (***g***), and oxygen(O) atoms (***h*** and ***i***), and determine their **hybridization**. (2 points)

Atom	Hybridization
Sulfur (<i>f</i>)	
Nitrogen (<i>g</i>)	
Oxygen (<i>h</i>)	
Oxygen (<i>i</i>)	

iii. **Describe** all the bonds formed by the carbon atoms ***a*** and ***b***, using sigma and pi designation as appropriate. (2 points)

Atom	Bonds formed (use sigma and pi as appropriate)
Carbon (<i>a</i>)	
Carbon (<i>b</i>)	

iv. How many total sp^3 -s sigma bonds are in the molecule? (1 point)

v. Determine **electronic** and **molecular geometry** at carbon (C) labeled ***e*** and sulfur (S) labeled ***f*** centres. (2 points)

Atom	Electronic Geometry	Molecular Geometry
Carbon (<i>e</i>)		
Sulfur (<i>f</i>)		

vi. Circle the value from the choices given below, that describes the **C-S-C bond angle at sulfur** (S) marked ***f***: (0.5 points)

$>90^\circ$ to $<109.5^\circ$ 109.5° >109.5 to $<120^\circ$ 120° $>120^\circ$ to $<180^\circ$


Question 8 12 points

Using the **valence shell orbitals only** draw a **complete** molecular orbital (MO) diagram (with no s-p mixing) for the molecule **PS**, in the space provided below. **(6 points)**

Assume that 3s and 3p atomic orbitals will form molecular orbitals like those formed by the combination of 2s and 2p atomic orbitals that we learnt in class, and there is no s-p mixing.

Show **energy arrow**, **atomic orbitals** involved, **atom labels**, the **molecular orbitals** that are formed, **filled electrons** in the MO diagram etc. Make sure to label all the atomic and molecular orbitals.

Using the MO diagram above, answer the following questions:

- (i) Which one of the following will have the highest bond order? Circle the correct choice
 A) PS^{1-} B) PS C) PS^{1+}
- (ii) Which one of the following will have the longest bond length? Circle the correct choice
 A) PS^{1-} B) PS C) PS^{1+}
- (iii) Which of the following will be diamagnetic? Circle the correct choice.
 A) PS^{1-} B) PS C) PS^{1+}
- (iv) Draw Lewis dot structure of the molecule with the shortest bond length (based on the MO diagram) from among the three (PS^{1-} , PS, PS^{1+}), and indicate the bond order calculated from the Lewis dot structure. **(3 points)**

Lewis Dot Structure

Bond Order



84DDFE12-B2B5-423B-9888-534D31886025

midterm-exam-93388

#133 Page 10 of 10

SCRAP PAGE

**CHEMISTRY 110 –Midterm Exam****Thursday November 7, 2024, 6:30 – 8:30 pm**

Examiners: Pallavi Sirjoosingh and Ashok Kakkar

First name (please write as legibly as possible within the boxes)

--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--

Last name

--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--

Student ID number

--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--

This booklet has 8 questions on 10 pages (The last page can be used as a scrap page).

The staple must NOT be removed from the question booklet. Your intact question booklet must be returned at the end of the exam.

A separate booklet contains a periodic table, and datasheet.

Write your answers directly under the questions in the space provided. Do not write on the QR codes or near the top edges (as it may be cut off).

Please write **legibly and dark enough** for the scan to be readable. For students who use English as a second language, dictionaries are permitted. Calculators are allowed.

DO NOT ASK QUESTIONS OF THE INVIGILATORS about interpretations of the exam questions. If you have a concern about a question, please summarize it at the top of this page. All concerns will be evaluated after the exam. GOOD LUCK!

**Question 1 8 points** (*Assume all transitions are possible*)

a. 6 points Calculate the **energy (in kJ/mol)** required to ionize (completely remove) the electron from the **n=3** level of **He⁺ ion** (He: Helium). Show your work including equations. (3 sig. figs.)

b. 2 points

(I) Select the largest (longest) wavelength from the three choices below:

- i. Wavelength of photon to *completely remove* an electron from *1s orbital of Hydrogen atom*
- ii. Wavelength of photon to *completely remove* an electron from *1s orbital of Boron atom (B)*
- iii. Wavelength of photon to *completely remove* an electron from *1s orbital of Boron cation (B⁺)*

i or ii or iii

(II) Select the smallest (shortest) wavelength from the three choices below (*for H atom*):

- i. Wavelength of photon *emitted* for an electron transitioning from *2p to 1s orbital*
- ii. Wavelength of photon *absorbed* for an electron transitioning from *2s to 4p orbital*
- iii. Wavelength of photon *absorbed* for an electron transitioning from *1s to 3p orbital*

i or ii or iii



Question 2 8 points



A Chem110 student plots the graph shown on the left, from data obtained for the kinetic energy (eV) of the ejected electron versus frequency (s^{-1}) of the incident light, for an unknown metal. ($1\text{eV} = 1.6022 \times 10^{-19} \text{ J}$)

Based on the plot, answer the questions **a to c** below:

a. 1 point On the plot above, label the **threshold frequency**.

b. 2 points What is the value of the work potential (in J) of the unknown metal?

c. 3 points What is the *longest* wavelength (in nm) required to eject an electron from this metal's surface? Show your work including equation(s) used. (3 sig figs)

d. 2 points In the space provided below, plot the following

(Assume frequency is *above* threshold frequency):

- i. *Kinetic energy* of an ejected electron versus the *intensity* of the incident light (left)
- ii. *Number of electrons* ejected versus the *frequency* of the incident light (right)



**Question 3** **6 points****a. 2 points**

In the ground state of Ge (Germanium; $Z = 32$) atom, **name** the following orbitals:

i. Orbital containing the electron that requires the maximum energy to be removed.

ii. Orbital containing the electron that experiences the minimum effective nuclear charge.

b. 2 points

An unknown anion (A^-) has the following *excited* electronic configuration: $[Ar] 4s^2 3d^3 4p^1$

(Ar: Argon; $Z = 18$)

Give the **condensed ground state electronic configuration** of the atom, **A** and the cation, **A^{2+}** .
Show your work. (Assume all electronic transitions are possible)

Electronic Configuration of A:

Electronic Configuration of A^{2+} :

c. 2 points

In a single atom, determine how many electrons can have the following set of quantum numbers?

(i) $n = 5; l = 3$

In a single atom, determine how many electrons can have the following set of quantum numbers?

(i) $n = 3; l = 2; m_s = +1/2$

**Question 4** **8 points**

Rank the following from 1 to 4 in **increasing** order. Provide the ranking in the boxes provided.

Note: Atomic number (Z) are provided for relevant atoms

a) Increasing (lowest to highest) order of *ionization energy*: (1= lowest; 4 = highest)

Mg

Mg: Z = 12

Rb

Rb: Z = 37

Cs

Cs: Z = 55

K

K : Z = 19

b) Increasing (lowest to highest) order of *bond angle*: (1= lowest; 4 = highest)

H₂OCCl₄XeF₄

HCN

c) Increasing (smallest to largest) order of *ionic size* (1= smallest; 4 = largest)

S²⁻

S: Z = 16

O²⁻

O: Z = 8

Be²⁺

Be: Z = 4

Na⁺

Na: Z = 11

d) Increasing (lowest to highest) order of *bond energy* (1= lowest; 4 = highest)

P-F

P-Cl

P-I

P-Br

**Question 5** **12 points**

a. 9 points For the molecules HClO_4 and TeF_6 , draw the most stable *Lewis structure* showing appropriate shape and *dashed/wedged bonds*, based on VSEPR. Determine the **molecular geometry** around the central atom. For HClO_4 , label the *most polar bond*.



b. 3 points (Note: Lone pairs are not shown in structures below)

(i) For the three *resonance* structures (1, 2 and 3) shown below, answer the following question:



(I) Circle the *most contributing* resonance structure.

Structure 1

Structure 2

Structure 3

(ii) For the three structures given (4, 5, and 6) below, answer the following questions:



(I) Circle all the structures that represent *resonance structures* of each other:

Structure 4

Structure 5

Structure 6

(II) Among the resonance structures, circle the *most contributing* resonance structure.

Structure 4

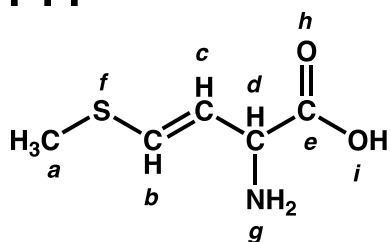
Structure 5

Structure 6

**Question 6** **6 points**

Using the data given below, calculate the **lattice energy** for SrO(s).

	ΔH° (kJ/mol)
$\text{Sr}(s) \longrightarrow \text{Sr}(g)$	+180
$\text{Sr}(g) \longrightarrow \text{Sr}^{2+}(g) + 2e^-$	+1468
$\text{O}_2(g) \longrightarrow 2\text{O}(g)$	+500
$\text{O}(g) + 2e^- \longrightarrow \text{O}^{2-}(g)$	+600
$\text{Sr}(s) + \frac{1}{2}\text{O}_2(g) \longrightarrow \text{SrO}(s)$	-550

**Question 7 10 points**

Consider the molecule given on the left, and answer questions (i) to (vi) using chemical bonding theories that we have learnt in class (valence bond theory, hybridization).

Note that the lone pairs are not shown in the structure:

i. Consider the carbon atoms marked in as ***a***, ***b***, ***c***, ***d***, and ***e*** in the molecule, and determine **hybridization** at each of the carbon centers. (2.5 points)

Atom	Hybridization
Carbon (<i>a</i>)	
Carbon (<i>b</i>)	
Carbon (<i>c</i>)	
Carbon (<i>d</i>)	
Carbon (<i>e</i>)	

ii. Now, consider the sulfur atom (S) marked as ***f***, nitrogen (N) atom (***g***), and oxygen(O) atoms (***h*** and ***i***), and determine their **hybridization**. (2 points)

Atom	Hybridization
Sulfur (<i>f</i>)	
Nitrogen (<i>g</i>)	
Oxygen (<i>h</i>)	
Oxygen (<i>i</i>)	

iii. **Describe** all the bonds formed by the carbon atoms ***a*** and ***b***, using sigma and pi designation as appropriate. (2 points)

Atom	Bonds formed (use sigma and pi as appropriate)
Carbon (<i>a</i>)	
Carbon (<i>b</i>)	

iv. How many total sp^3 -s sigma bonds are in the molecule? (1 point)

v. Determine **electronic** and **molecular geometry** at carbon (C) labeled ***e*** and sulfur (S) labeled ***f*** centres. (2 points)

Atom	Electronic Geometry	Molecular Geometry
Carbon (<i>e</i>)		
Sulfur (<i>f</i>)		

vi. Circle the value from the choices given below, that describes the **C-S-C bond angle at sulfur** (S) marked ***f***: (0.5 points)

>90° to <109.5° 109.5° >109.5 to <120° 120° >120° to <180°


Question 8 12 points

Using the **valence shell orbitals only** draw a **complete** molecular orbital (MO) diagram (with no s-p mixing) for the molecule **PS**, in the space provided below. **(6 points)**

Assume that 3s and 3p atomic orbitals will form molecular orbitals like those formed by the combination of 2s and 2p atomic orbitals that we learnt in class, and there is no s-p mixing.

Show **energy arrow**, **atomic orbitals** involved, **atom labels**, the **molecular orbitals** that are formed, **filled electrons** in the MO diagram etc. Make sure to label all the atomic and molecular orbitals.

Using the MO diagram above, answer the following questions:

- (i) Which one of the following will have the highest bond order? Circle the correct choice
 A) PS^{1-} B) PS C) PS^{1+}
- (ii) Which one of the following will have the longest bond length? Circle the correct choice
 A) PS^{1-} B) PS C) PS^{1+}
- (iii) Which of the following will be diamagnetic? Circle the correct choice.
 A) PS^{1-} B) PS C) PS^{1+}
- (iv) Draw Lewis dot structure of the molecule with the shortest bond length (based on the MO diagram) from among the three (PS^{1-} , PS, PS^{1+}), and indicate the bond order calculated from the Lewis dot structure. **(3 points)**

Lewis Dot Structure

Bond Order



E4840B88-7055-4884-B6BA-337900D47B32

midterm-exam-93388

#134 Page 10 of 10

SCRAP PAGE

**CHEMISTRY 110 –Midterm Exam****Thursday November 7, 2024, 6:30 – 8:30 pm****Examiners: Pallavi Sirjoosingh and Ashok Kakkar**

First name (please write as legibly as possible within the boxes)

--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--

Last name

--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--

Student ID number

--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--

This booklet has 8 questions on 10 pages (The last page can be used as a scrap page).

The staple must NOT be removed from the question booklet. Your intact question booklet must be returned at the end of the exam.

A separate booklet contains a periodic table, and datasheet.

Write your answers directly under the questions in the space provided. Do not write on the QR codes or near the top edges (as it may be cut off).

Please write **legibly and dark enough** for the scan to be readable. For students who use English as a second language, dictionaries are permitted. Calculators are allowed.

DO NOT ASK QUESTIONS OF THE INVIGILATORS about interpretations of the exam questions. If you have a concern about a question, please summarize it at the top of this page. All concerns will be evaluated after the exam. GOOD LUCK!

**Question 1 8 points** (*Assume all transitions are possible*)

a. 6 points Calculate the **energy (in kJ/mol)** required to ionize (completely remove) the electron from the **n=3** level of **He⁺ ion** (He: Helium). Show your work including equations. (3 sig. figs.)

b. 2 points

(I) Select the largest (longest) wavelength from the three choices below:

- i. Wavelength of photon to *completely remove* an electron from *1s orbital of Hydrogen atom*
- ii. Wavelength of photon to *completely remove* an electron from *1s orbital of Boron atom (B)*
- iii. Wavelength of photon to *completely remove* an electron from *1s orbital of Boron cation (B⁺)*

i or ii or iii

(II) Select the smallest (shortest) wavelength from the three choices below (*for H atom*):

- i. Wavelength of photon *emitted* for an electron transitioning from *2p to 1s orbital*
- ii. Wavelength of photon *absorbed* for an electron transitioning from *2s to 4p orbital*
- iii. Wavelength of photon *absorbed* for an electron transitioning from *1s to 3p orbital*

i or ii or iii



Question 2 8 points



A Chem110 student plots the graph shown on the left, from data obtained for the kinetic energy (eV) of the ejected electron versus frequency (s^{-1}) of the incident light, for an unknown metal. ($1\text{eV} = 1.6022 \times 10^{-19} \text{ J}$)

Based on the plot, answer the questions **a to c** below:

a. 1 point On the plot above, label the **threshold frequency**.

b. 2 points What is the value of the work potential (in J) of the unknown metal?

c. 3 points What is the *longest* wavelength (in nm) required to eject an electron from this metal's surface? Show your work including equation(s) used. (3 sig figs)

d. 2 points In the space provided below, plot the following

(Assume frequency is *above* threshold frequency):

- i. *Kinetic energy* of an ejected electron versus the *intensity* of the incident light (left)
- ii. *Number of electrons* ejected versus the *frequency* of the incident light (right)



**Question 3** **6 points****a. 2 points**

In the ground state of Ge (Germanium; $Z = 32$) atom, **name** the following orbitals:

i. Orbital containing the electron that requires the maximum energy to be removed.

ii. Orbital containing the electron that experiences the minimum effective nuclear charge.

b. 2 points

An unknown anion (A^-) has the following *excited* electronic configuration: $[Ar] 4s^2 3d^3 4p^1$

(Ar: Argon; $Z = 18$)

Give the **condensed ground state electronic configuration** of the atom, **A** and the cation, **A^{2+}** .
Show your work. (Assume all electronic transitions are possible)

Electronic Configuration of A:

Electronic Configuration of A^{2+} :

c. 2 points

In a single atom, determine how many electrons can have the following set of quantum numbers?

(i) $n = 5; l = 3$

In a single atom, determine how many electrons can have the following set of quantum numbers?

(i) $n = 3; l = 2; m_s = +1/2$

**Question 4** **8 points**

Rank the following from 1 to 4 in **increasing** order. Provide the ranking in the boxes provided.

Note: Atomic number (Z) are provided for relevant atoms

a) Increasing (lowest to highest) order of *ionization energy*: (1= lowest; 4 = highest)

Mg

Mg: Z = 12

Rb

Rb: Z = 37

Cs

Cs: Z = 55

K

K : Z = 19

b) Increasing (lowest to highest) order of *bond angle*: (1= lowest; 4 = highest)

H₂OCCl₄XeF₄

HCN

c) Increasing (smallest to largest) order of *ionic size* (1= smallest; 4 = largest)

S²⁻

S: Z = 16

O²⁻

O: Z = 8

Be²⁺

Be: Z = 4

Na⁺

Na: Z = 11

d) Increasing (lowest to highest) order of *bond energy* (1= lowest; 4 = highest)

P-F

P-Cl

P-I

P-Br

**Question 5** **12 points**

a. 9 points For the molecules HClO_4 and TeF_6 , draw the most stable *Lewis structure* showing appropriate shape and *dashed/wedged bonds*, based on VSEPR. Determine the **molecular geometry** around the central atom. For HClO_4 , label the *most polar bond*.



b. 3 points (Note: Lone pairs are not shown in structures below)

(i) For the three *resonance* structures (1, 2 and 3) shown below, answer the following question:



(I) Circle the *most contributing* resonance structure.

Structure 1

Structure 2

Structure 3

(ii) For the three structures given (4, 5, and 6) below, answer the following questions:



(I) Circle all the structures that represent *resonance structures* of each other:

Structure 4

Structure 5

Structure 6

(II) Among the resonance structures, circle the *most contributing* resonance structure.

Structure 4

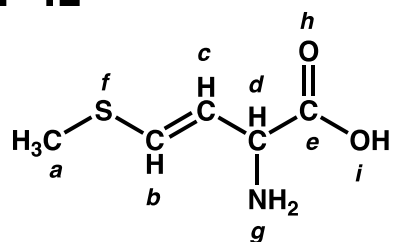
Structure 5

Structure 6

**Question 6** **6 points**

Using the data given below, calculate the **lattice energy** for SrO(s).

	ΔH° (kJ/mol)
$\text{Sr}(s) \longrightarrow \text{Sr}(g)$	+180
$\text{Sr}(g) \longrightarrow \text{Sr}^{2+}(g) + 2e^-$	+1468
$\text{O}_2(g) \longrightarrow 2\text{O}(g)$	+500
$\text{O}(g) + 2e^- \longrightarrow \text{O}^{2-}(g)$	+600
$\text{Sr}(s) + \frac{1}{2}\text{O}_2(g) \longrightarrow \text{SrO}(s)$	-550

**Question 7 10 points**

Consider the molecule given on the left, and answer questions (i) to (vi) using chemical bonding theories that we have learnt in class (valence bond theory, hybridization).

Note that the lone pairs are not shown in the structure:

i. Consider the carbon atoms marked in as ***a***, ***b***, ***c***, ***d***, and ***e*** in the molecule, and determine **hybridization** at each of the carbon centers. (2.5 points)

Atom	Hybridization
Carbon (<i>a</i>)	
Carbon (<i>b</i>)	
Carbon (<i>c</i>)	
Carbon (<i>d</i>)	
Carbon (<i>e</i>)	

ii. Now, consider the sulfur atom (S) marked as ***f***, nitrogen (N) atom (***g***), and oxygen(O) atoms (***h*** and ***i***), and determine their **hybridization**. (2 points)

Atom	Hybridization
Sulfur (<i>f</i>)	
Nitrogen (<i>g</i>)	
Oxygen (<i>h</i>)	
Oxygen (<i>i</i>)	

iii. **Describe** all the bonds formed by the carbon atoms ***a*** and ***b***, using sigma and pi designation as appropriate. (2 points)

Atom	Bonds formed (use sigma and pi as appropriate)
Carbon (<i>a</i>)	
Carbon (<i>b</i>)	

iv. How many total sp^3 -s sigma bonds are in the molecule? (1 point)

v. Determine **electronic** and **molecular geometry** at carbon (C) labeled ***e*** and sulfur (S) labeled ***f*** centres. (2 points)

Atom	Electronic Geometry	Molecular Geometry
Carbon (<i>e</i>)		
Sulfur (<i>f</i>)		

vi. Circle the value from the choices given below, that describes the **C-S-C bond angle at sulfur** (S) marked ***f***: (0.5 points)

>90° to <109.5° 109.5° >109.5 to <120° 120° >120° to <180°


Question 8 12 points

Using the **valence shell orbitals only** draw a **complete** molecular orbital (MO) diagram (with no s-p mixing) for the molecule **PS**, in the space provided below. **(6 points)**

Assume that 3s and 3p atomic orbitals will form molecular orbitals like those formed by the combination of 2s and 2p atomic orbitals that we learnt in class, and there is no s-p mixing.

Show **energy arrow**, **atomic orbitals** involved, **atom labels**, the **molecular orbitals** that are formed, **filled electrons** in the MO diagram etc. Make sure to **label all the atomic and molecular orbitals**.

Using the MO diagram above, answer the following questions:

- (i) Which one of the following will have the highest bond order? Circle the correct choice
 A) PS^{1-} B) PS C) PS^{1+}
- (ii) Which one of the following will have the longest bond length? Circle the correct choice
 A) PS^{1-} B) PS C) PS^{1+}
- (iii) Which of the following will be diamagnetic? Circle the correct choice.
 A) PS^{1-} B) PS C) PS^{1+}
- (iv) Draw Lewis dot structure of the molecule with the shortest bond length (based on the MO diagram) from among the three (PS^{1-} , PS, PS^{1+}), and indicate the bond order calculated from the Lewis dot structure. **(3 points)**

Lewis Dot Structure

Bond Order



4431B303-66B7-44D0-9589-5AE653F00B67

midterm-exam-93388

#135 Page 10 of 10

SCRAP PAGE

**CHEMISTRY 110 –Midterm Exam****Thursday November 7, 2024, 6:30 – 8:30 pm**

Examiners: Pallavi Sirjoosingh and Ashok Kakkar

First name (please write as legibly as possible within the boxes)

--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--

Last name

--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--

Student ID number

--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--

This booklet has 8 questions on 10 pages (The last page can be used as a scrap page).

The staple must NOT be removed from the question booklet. Your intact question booklet must be returned at the end of the exam.

A separate booklet contains a periodic table, and datasheet.

Write your answers directly under the questions in the space provided. Do not write on the QR codes or near the top edges (as it may be cut off).

Please write **legibly and dark enough** for the scan to be readable. For students who use English as a second language, dictionaries are permitted. Calculators are allowed.

DO NOT ASK QUESTIONS OF THE INVIGILATORS about interpretations of the exam questions. If you have a concern about a question, please summarize it at the top of this page. All concerns will be evaluated after the exam. GOOD LUCK!

**Question 1 8 points** (*Assume all transitions are possible*)

a. 6 points Calculate the **energy (in kJ/mol)** required to ionize (completely remove) the electron from the **$n=3$** level of **He^+ ion** (He: Helium). Show your work including equations. (3 sig. figs.)

b. 2 points

(I) Select the largest (longest) wavelength from the three choices below:

- i. Wavelength of photon to *completely remove* an electron from *1s orbital of Hydrogen atom*
- ii. Wavelength of photon to *completely remove* an electron from *1s orbital of Boron atom (B)*
- iii. Wavelength of photon to *completely remove* an electron from *1s orbital of Boron cation (B^+)*

i or ii or iii

(II) Select the smallest (shortest) wavelength from the three choices below (*for H atom*):

- i. Wavelength of photon *emitted* for an electron transitioning from *2p to 1s orbital*
- ii. Wavelength of photon *absorbed* for an electron transitioning from *2s to 4p orbital*
- iii. Wavelength of photon *absorbed* for an electron transitioning from *1s to 3p orbital*

i or ii or iii



Question 2 8 points



A Chem110 student plots the graph shown on the left, from data obtained for the kinetic energy (eV) of the ejected electron versus frequency (s^{-1}) of the incident light, for an unknown metal. ($1\text{eV} = 1.6022 \times 10^{-19} \text{ J}$)

Based on the plot, answer the questions **a to c** below:

a. 1 point On the plot above, label the **threshold frequency**.

b. 2 points What is the value of the work potential (in J) of the unknown metal?

c. 3 points What is the *longest* wavelength (in nm) required to eject an electron from this metal's surface? Show your work including equation(s) used. (3 sig figs)

d. 2 points In the space provided below, plot the following

(Assume frequency is *above* threshold frequency):

- i. *Kinetic energy* of an ejected electron versus the *intensity* of the incident light (left)
- ii. *Number of electrons* ejected versus the *frequency* of the incident light (right)



**Question 3** **6 points****a. 2 points**

In the ground state of Ge (Germanium; $Z = 32$) atom, **name** the following orbitals:

i. Orbital containing the electron that requires the maximum energy to be removed.

ii. Orbital containing the electron that experiences the minimum effective nuclear charge.

b. 2 points

An unknown anion (A^-) has the following *excited* electronic configuration: $[Ar] 4s^2 3d^3 4p^1$

(Ar: Argon; $Z = 18$)

Give the **condensed ground state electronic configuration** of the atom, **A** and the cation, **A^{2+}** .
Show your work. (Assume all electronic transitions are possible)

Electronic Configuration of A:

Electronic Configuration of A^{2+} :

c. 2 points

In a single atom, determine how many electrons can have the following set of quantum numbers?

(i) $n = 5; l = 3$

In a single atom, determine how many electrons can have the following set of quantum numbers?

(i) $n = 3; l = 2; m_s = +1/2$

**Question 4** **8 points**

Rank the following from 1 to 4 in **increasing** order. Provide the ranking in the boxes provided.

Note: Atomic number (Z) are provided for relevant atoms

a) Increasing (lowest to highest) order of *ionization energy*: (1= lowest; 4 = highest)

Mg

Mg: Z = 12

Rb

Rb: Z = 37

Cs

Cs: Z = 55

K

K : Z = 19

b) Increasing (lowest to highest) order of *bond angle*: (1= lowest; 4 = highest)

H₂OCCl₄XeF₄

HCN

c) Increasing (smallest to largest) order of *ionic size* (1= smallest; 4 = largest)

S²⁻

S: Z = 16

O²⁻

O: Z = 8

Be²⁺

Be: Z = 4

Na⁺

Na: Z = 11

d) Increasing (lowest to highest) order of *bond energy* (1= lowest; 4 = highest)

P-F

P-Cl

P-I

P-Br

**Question 5** **12 points**

a. 9 points For the molecules HClO_4 and TeF_6 , draw the most stable *Lewis structure* showing appropriate shape and *dashed/wedged bonds*, based on VSEPR. Determine the **molecular geometry** around the central atom. For HClO_4 , label the *most polar bond*.



b. 3 points (Note: Lone pairs are not shown in structures below)

(i) For the three *resonance* structures (1, 2 and 3) shown below, answer the following question:



(I) Circle the *most contributing* resonance structure.

Structure 1

Structure 2

Structure 3

(ii) For the three structures given (4, 5, and 6) below, answer the following questions:



(I) Circle all the structures that represent *resonance structures* of each other:

Structure 4

Structure 5

Structure 6

(II) Among the resonance structures, circle the *most contributing* resonance structure.

Structure 4

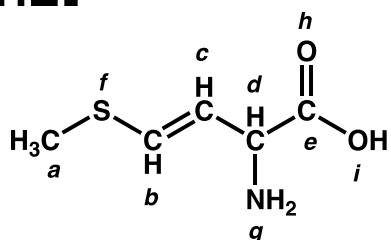
Structure 5

Structure 6

**Question 6** **6 points**

Using the data given below, calculate the **lattice energy** for SrO(s).

	ΔH° (kJ/mol)
$\text{Sr}(s) \longrightarrow \text{Sr}(g)$	+180
$\text{Sr}(g) \longrightarrow \text{Sr}^{2+}(g) + 2e^-$	+1468
$\text{O}_2(g) \longrightarrow 2\text{O}(g)$	+500
$\text{O}(g) + 2e^- \longrightarrow \text{O}^{2-}(g)$	+600
$\text{Sr}(s) + \frac{1}{2}\text{O}_2(g) \longrightarrow \text{SrO}(s)$	-550

**Question 7 10 points**

Consider the molecule given on the left, and answer questions (i) to (vi) using chemical bonding theories that we have learnt in class (valence bond theory, hybridization).

Note that the lone pairs are not shown in the structure:

i. Consider the carbon atoms marked in as ***a***, ***b***, ***c***, ***d***, and ***e*** in the molecule, and determine **hybridization** at each of the carbon centers. (2.5 points)

Atom	Hybridization
Carbon (<i>a</i>)	
Carbon (<i>b</i>)	
Carbon (<i>c</i>)	
Carbon (<i>d</i>)	
Carbon (<i>e</i>)	

ii. Now, consider the sulfur atom (S) marked as ***f***, nitrogen (N) atom (***g***), and oxygen(O) atoms (***h*** and ***i***), and determine their **hybridization**. (2 points)

Atom	Hybridization
Sulfur (<i>f</i>)	
Nitrogen (<i>g</i>)	
Oxygen (<i>h</i>)	
Oxygen (<i>i</i>)	

iii. **Describe** all the bonds formed by the carbon atoms ***a*** and ***b***, using sigma and pi designation as appropriate. (2 points)

Atom	Bonds formed (use sigma and pi as appropriate)
Carbon (<i>a</i>)	
Carbon (<i>b</i>)	

iv. How many total sp^3 -s sigma bonds are in the molecule? (1 point)

v. Determine **electronic** and **molecular geometry** at carbon (C) labeled ***e*** and sulfur (S) labeled ***f*** centres. (2 points)

Atom	Electronic Geometry	Molecular Geometry
Carbon (<i>e</i>)		
Sulfur (<i>f</i>)		

vi. Circle the value from the choices given below, that describes the **C-S-C bond angle at sulfur** (S) marked ***f***: (0.5 points)

$>90^\circ$ to $<109.5^\circ$ 109.5° >109.5 to $<120^\circ$ 120° $>120^\circ$ to $<180^\circ$


Question 8 12 points

Using the **valence shell orbitals only** draw a **complete** molecular orbital (MO) diagram (with no s-p mixing) for the molecule **PS**, in the space provided below. **(6 points)**

Assume that 3s and 3p atomic orbitals will form molecular orbitals like those formed by the combination of 2s and 2p atomic orbitals that we learnt in class, and there is no s-p mixing.

Show **energy arrow**, **atomic orbitals** involved, **atom labels**, the **molecular orbitals** that are formed, **filled electrons** in the MO diagram etc. Make sure to label all the atomic and molecular orbitals.

Using the MO diagram above, answer the following questions:

- (i) Which one of the following will have the highest bond order? Circle the correct choice
 A) PS^{1-} B) PS C) PS^{1+}
- (ii) Which one of the following will have the longest bond length? Circle the correct choice
 A) PS^{1-} B) PS C) PS^{1+}
- (iii) Which of the following will be diamagnetic? Circle the correct choice.
 A) PS^{1-} B) PS C) PS^{1+}
- (iv) Draw Lewis dot structure of the molecule with the shortest bond length (based on the MO diagram) from among the three (PS^{1-} , PS, PS^{1+}), and indicate the bond order calculated from the Lewis dot structure. **(3 points)**

Lewis Dot Structure

Bond Order



36782293-5ACF-4644-BE4A-D617987CE474

midterm-exam-93388

#136 Page 10 of 10

SCRAP PAGE

**CHEMISTRY 110 –Midterm Exam****Thursday November 7, 2024, 6:30 – 8:30 pm**

Examiners: Pallavi Sirjoosingh and Ashok Kakkar

First name (please write as legibly as possible within the boxes)

--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--

Last name

--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--

Student ID number

--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--

This booklet has 8 questions on 10 pages (The last page can be used as a scrap page).

The staple must NOT be removed from the question booklet. Your intact question booklet must be returned at the end of the exam.

A separate booklet contains a periodic table, and datasheet.

Write your answers directly under the questions in the space provided. Do not write on the QR codes or near the top edges (as it may be cut off).

Please write **legibly and dark enough** for the scan to be readable. For students who use English as a second language, dictionaries are permitted. Calculators are allowed.

DO NOT ASK QUESTIONS OF THE INVIGILATORS about interpretations of the exam questions. If you have a concern about a question, please summarize it at the top of this page. All concerns will be evaluated after the exam. GOOD LUCK!

**Question 1 8 points** (*Assume all transitions are possible*)

a. 6 points Calculate the **energy (in kJ/mol)** required to ionize (completely remove) the electron from the **n=3** level of **He⁺ ion** (He: Helium). Show your work including equations. (3 sig. figs.)

b. 2 points

(I) Select the largest (longest) wavelength from the three choices below:

- i. Wavelength of photon to *completely remove* an electron from *1s orbital of Hydrogen atom*
- ii. Wavelength of photon to *completely remove* an electron from *1s orbital of Boron atom (B)*
- iii. Wavelength of photon to *completely remove* an electron from *1s orbital of Boron cation (B⁺)*

i or ii or iii

(II) Select the smallest (shortest) wavelength from the three choices below (*for H atom*):

- i. Wavelength of photon *emitted* for an electron transitioning from *2p to 1s orbital*
- ii. Wavelength of photon *absorbed* for an electron transitioning from *2s to 4p orbital*
- iii. Wavelength of photon *absorbed* for an electron transitioning from *1s to 3p orbital*

i or ii or iii



Question 2 8 points



A Chem110 student plots the graph shown on the left, from data obtained for the kinetic energy (eV) of the ejected electron versus frequency (s^{-1}) of the incident light, for an unknown metal. ($1\text{eV} = 1.6022 \times 10^{-19} \text{ J}$)

Based on the plot, answer the questions **a to c** below:

a. 1 point On the plot above, label the **threshold frequency**.

b. 2 points What is the value of the work potential (in J) of the unknown metal?

c. 3 points What is the *longest* wavelength (in nm) required to eject an electron from this metal's surface? Show your work including equation(s) used. (3 sig figs)

d. 2 points In the space provided below, plot the following

(Assume frequency is *above* threshold frequency):

- i. *Kinetic energy* of an ejected electron versus the *intensity* of the incident light (left)
- ii. *Number of electrons* ejected versus the *frequency* of the incident light (right)



**Question 3** **6 points****a. 2 points**

In the ground state of Ge (Germanium; $Z = 32$) atom, **name** the following orbitals:

i. Orbital containing the electron that requires the maximum energy to be removed.

ii. Orbital containing the electron that experiences the minimum effective nuclear charge.

b. 2 points

An unknown anion (A^-) has the following *excited* electronic configuration: $[Ar] 4s^2 3d^3 4p^1$

(Ar: Argon; $Z = 18$)

Give the **condensed ground state electronic configuration** of the atom, **A** and the cation, **A^{2+}** .
Show your work. (Assume all electronic transitions are possible)

Electronic Configuration of A:

Electronic Configuration of A^{2+} :

c. 2 points

In a single atom, determine how many electrons can have the following set of quantum numbers?

(i) $n = 5; l = 3$

In a single atom, determine how many electrons can have the following set of quantum numbers?

(i) $n = 3; l = 2; m_s = +1/2$

**Question 4** **8 points**

Rank the following from 1 to 4 in **increasing** order. Provide the ranking in the boxes provided.

Note: Atomic number (Z) are provided for relevant atoms

a) Increasing (lowest to highest) order of *ionization energy*: (1= lowest; 4 = highest)

Mg

Mg: Z = 12

Rb

Rb: Z = 37

Cs

Cs: Z = 55

K

K : Z = 19

b) Increasing (lowest to highest) order of *bond angle*: (1= lowest; 4 = highest)

H₂OCCl₄XeF₄

HCN

c) Increasing (smallest to largest) order of *ionic size* (1= smallest; 4 = largest)

S²⁻

S: Z = 16

O²⁻

O: Z = 8

Be²⁺

Be: Z = 4

Na⁺

Na: Z = 11

d) Increasing (lowest to highest) order of *bond energy* (1= lowest; 4 = highest)

P-F

P-Cl

P-I

P-Br

**Question 5** **12 points**

a. 9 points For the molecules HClO_4 and TeF_6 , draw the most stable *Lewis structure* showing appropriate shape and *dashed/wedged bonds*, based on VSEPR. Determine the **molecular geometry** around the central atom. For HClO_4 , label the *most polar bond*.

HClO_4 (there are no O-O bonds)

Label the most polar bond

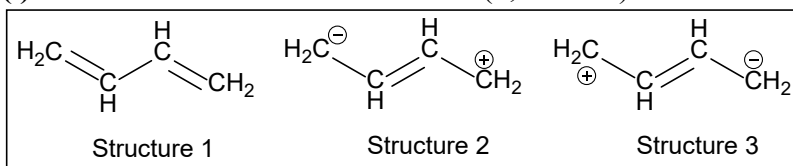
Molecular Geometry around Cl

TeF_6

Molecular Geometry around Te

b. 3 points (Note: Lone pairs are not shown in structures below)

(i) For the three *resonance* structures (1, 2 and 3) shown below, answer the following question:



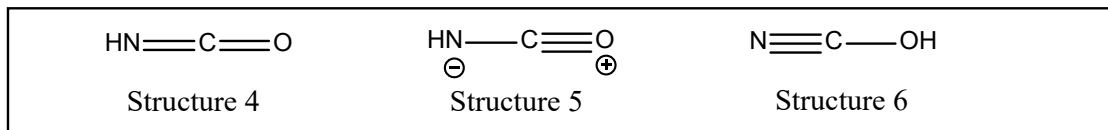
(I) Circle the *most contributing* resonance structure.

Structure 1

Structure 2

Structure 3

(ii) For the three structures given (4, 5, and 6) below, answer the following questions:



(I) Circle all the structures that represent *resonance structures* of each other:

Structure 4

Structure 5

Structure 6

(II) Among the resonance structures, circle the *most contributing* resonance structure.

Structure 4

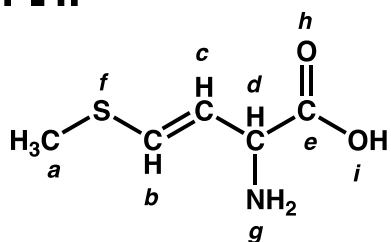
Structure 5

Structure 6

**Question 6** **6 points**

Using the data given below, calculate the **lattice energy** for SrO(s).

	ΔH° (kJ/mol)
$\text{Sr}(s) \longrightarrow \text{Sr}(g)$	+180
$\text{Sr}(g) \longrightarrow \text{Sr}^{2+}(g) + 2e^-$	+1468
$\text{O}_2(g) \longrightarrow 2\text{O}(g)$	+500
$\text{O}(g) + 2e^- \longrightarrow \text{O}^{2-}(g)$	+600
$\text{Sr}(s) + \frac{1}{2}\text{O}_2(g) \longrightarrow \text{SrO}(s)$	-550

**Question 7 10 points**

Consider the molecule given on the left, and answer questions (i) to (vi) using chemical bonding theories that we have learnt in class (valence bond theory, hybridization).

Note that the lone pairs are not shown in the structure:

i. Consider the carbon atoms marked in as ***a***, ***b***, ***c***, ***d***, and ***e*** in the molecule, and determine **hybridization** at each of the carbon centers. (2.5 points)

Atom	Hybridization
Carbon (<i>a</i>)	
Carbon (<i>b</i>)	
Carbon (<i>c</i>)	
Carbon (<i>d</i>)	
Carbon (<i>e</i>)	

ii. Now, consider the sulfur atom (S) marked as ***f***, nitrogen (N) atom (***g***), and oxygen(O) atoms (***h*** and ***i***), and determine their **hybridization**. (2 points)

Atom	Hybridization
Sulfur (<i>f</i>)	
Nitrogen (<i>g</i>)	
Oxygen (<i>h</i>)	
Oxygen (<i>i</i>)	

iii. **Describe** all the bonds formed by the carbon atoms ***a*** and ***b***, using sigma and pi designation as appropriate. (2 points)

Atom	Bonds formed (use sigma and pi as appropriate)
Carbon (<i>a</i>)	
Carbon (<i>b</i>)	

iv. How many total sp^3 -s sigma bonds are in the molecule? (1 point)

v. Determine **electronic** and **molecular geometry** at carbon (C) labeled ***e*** and sulfur (S) labeled ***f*** centres. (2 points)

Atom	Electronic Geometry	Molecular Geometry
Carbon (<i>e</i>)		
Sulfur (<i>f</i>)		

vi. Circle the value from the choices given below, that describes the **C-S-C bond angle at sulfur** (S) marked ***f***: (0.5 points)

>90° to <109.5° 109.5° >109.5 to <120° 120° >120° to <180°

**Question 8 12 points**

Using the **valence shell orbitals only** draw a **complete** molecular orbital (MO) diagram (with no s-p mixing) for the molecule **PS**, in the space provided below. **(6 points)**

Assume that 3s and 3p atomic orbitals will form molecular orbitals like those formed by the combination of 2s and 2p atomic orbitals that we learnt in class, and there is no s-p mixing.

Show **energy arrow**, **atomic orbitals** involved, **atom labels**, the **molecular orbitals** that are formed, **filled electrons** in the MO diagram etc. Make sure to label all the atomic and molecular orbitals.

Using the MO diagram above, answer the following questions:

- (i) Which one of the following will have the highest bond order? Circle the correct choice
A) PS^{1-} B) PS C) PS^{1+}
- (ii) Which one of the following will have the longest bond length? Circle the correct choice
A) PS^{1-} B) PS C) PS^{1+}
- (iii) Which of the following will be diamagnetic? Circle the correct choice.
A) PS^{1-} B) PS C) PS^{1+}
- (iv) Draw Lewis dot structure of the molecule with the shortest bond length (based on the MO diagram) from among the three (PS^{1-} , PS, PS^{1+}), and indicate the bond order calculated from the Lewis dot structure. **(3 points)**

Lewis Dot Structure

Bond Order



78B427CC-A0C1-4FB5-BB67-F457D75DA1F7

midterm-exam-93388

#137 Page 10 of 10

SCRAP PAGE

**CHEMISTRY 110 –Midterm Exam****Thursday November 7, 2024, 6:30 – 8:30 pm**

Examiners: Pallavi Sirjoosingh and Ashok Kakkar

First name (please write as legibly as possible within the boxes)

--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--

Last name

--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--

Student ID number

--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--

This booklet has 8 questions on 10 pages (The last page can be used as a scrap page).

The staple must NOT be removed from the question booklet. Your intact question booklet must be returned at the end of the exam.

A separate booklet contains a periodic table, and datasheet.

Write your answers directly under the questions in the space provided. Do not write on the QR codes or near the top edges (as it may be cut off).

Please write **legibly and dark enough** for the scan to be readable. For students who use English as a second language, dictionaries are permitted. Calculators are allowed.

DO NOT ASK QUESTIONS OF THE INVIGILATORS about interpretations of the exam questions. If you have a concern about a question, please summarize it at the top of this page. All concerns will be evaluated after the exam. GOOD LUCK!

**Question 1 8 points** (*Assume all transitions are possible*)

a. 6 points Calculate the **energy (in kJ/mol)** required to ionize (completely remove) the electron from the **n=3** level of **He⁺ ion** (He: Helium). Show your work including equations. (3 sig. figs.)

b. 2 points

(I) Select the largest (longest) wavelength from the three choices below:

- i. Wavelength of photon to *completely remove* an electron from *1s orbital of Hydrogen atom*
- ii. Wavelength of photon to *completely remove* an electron from *1s orbital of Boron atom (B)*
- iii. Wavelength of photon to *completely remove* an electron from *1s orbital of Boron cation (B⁺)*

i or ii or iii

(II) Select the smallest (shortest) wavelength from the three choices below (*for H atom*):

- i. Wavelength of photon *emitted* for an electron transitioning from *2p to 1s orbital*
- ii. Wavelength of photon *absorbed* for an electron transitioning from *2s to 4p orbital*
- iii. Wavelength of photon *absorbed* for an electron transitioning from *1s to 3p orbital*

i or ii or iii



Question 2 8 points



A Chem110 student plots the graph shown on the left, from data obtained for the kinetic energy (eV) of the ejected electron versus frequency (s^{-1}) of the incident light, for an unknown metal. ($1\text{eV} = 1.6022 \times 10^{-19} \text{ J}$)

Based on the plot, answer the questions **a to c** below:

a. 1 point On the plot above, label the **threshold frequency**.

b. 2 points What is the value of the work potential (in J) of the unknown metal?

c. 3 points What is the *longest* wavelength (in nm) required to eject an electron from this metal's surface? Show your work including equation(s) used. (3 sig figs)

d. 2 points In the space provided below, plot the following

(Assume frequency is *above* threshold frequency):

- Kinetic energy* of an ejected electron versus the *intensity* of the incident light (left)
- Number of electrons* ejected versus the *frequency* of the incident light (right)



**Question 3** **6 points****a. 2 points**

In the ground state of Ge (Germanium; $Z = 32$) atom, **name** the following orbitals:

i. Orbital containing the electron that requires the maximum energy to be removed.

ii. Orbital containing the electron that experiences the minimum effective nuclear charge.

b. 2 points

An unknown anion (A^-) has the following *excited* electronic configuration: $[Ar] 4s^2 3d^3 4p^1$

(Ar: Argon; $Z = 18$)

Give the **condensed ground state electronic configuration** of the atom, **A** and the cation, **A^{2+}** .
Show your work. (Assume all electronic transitions are possible)

Electronic Configuration of A:

Electronic Configuration of A^{2+} :

c. 2 points

In a single atom, determine how many electrons can have the following set of quantum numbers?

(i) $n = 5; l = 3$

In a single atom, determine how many electrons can have the following set of quantum numbers?

(i) $n = 3; l = 2; m_s = +1/2$

**Question 4** **8 points**

Rank the following from 1 to 4 in **increasing** order. Provide the ranking in the boxes provided.

Note: Atomic number (Z) are provided for relevant atoms

a) Increasing (lowest to highest) order of *ionization energy*: (1= lowest; 4 = highest)

Mg

Mg: Z = 12

Rb

Rb: Z = 37

Cs

Cs: Z = 55

K

K : Z = 19

b) Increasing (lowest to highest) order of *bond angle*: (1= lowest; 4 = highest)

H₂OCCl₄XeF₄

HCN

c) Increasing (smallest to largest) order of *ionic size* (1= smallest; 4 = largest)

S²⁻

S: Z = 16

O²⁻

O: Z = 8

Be²⁺

Be: Z = 4

Na⁺

Na: Z = 11

d) Increasing (lowest to highest) order of *bond energy* (1= lowest; 4 = highest)

P-F

P-Cl

P-I

P-Br

**Question 5** **12 points**

a. 9 points For the molecules HClO_4 and TeF_6 , draw the most stable *Lewis structure* showing appropriate shape and *dashed/wedged bonds*, based on VSEPR. Determine the **molecular geometry** around the central atom. For HClO_4 , label the *most polar bond*.



b. 3 points (Note: Lone pairs are not shown in structures below)

(i) For the three *resonance* structures (1, 2 and 3) shown below, answer the following question:



(I) Circle the *most contributing* resonance structure.

Structure 1

Structure 2

Structure 3

(ii) For the three structures given (4, 5, and 6) below, answer the following questions:



(I) Circle all the structures that represent *resonance structures* of each other:

Structure 4

Structure 5

Structure 6

(II) Among the resonance structures, circle the *most contributing* resonance structure.

Structure 4

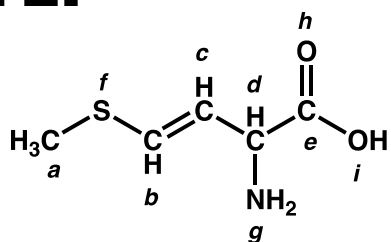
Structure 5

Structure 6

**Question 6** **6 points**

Using the data given below, calculate the **lattice energy** for SrO(s).

	ΔH° (kJ/mol)
$\text{Sr}(s) \longrightarrow \text{Sr}(g)$	+180
$\text{Sr}(g) \longrightarrow \text{Sr}^{2+}(g) + 2e^-$	+1468
$\text{O}_2(g) \longrightarrow 2\text{O}(g)$	+500
$\text{O}(g) + 2e^- \longrightarrow \text{O}^{2-}(g)$	+600
$\text{Sr}(s) + \frac{1}{2}\text{O}_2(g) \longrightarrow \text{SrO}(s)$	-550

**Question 7 10 points**

Consider the molecule given on the left, and answer questions (i) to (vi) using chemical bonding theories that we have learnt in class (valence bond theory, hybridization).

Note that the lone pairs are not shown in the structure:

i. Consider the carbon atoms marked in as ***a***, ***b***, ***c***, ***d***, and ***e*** in the molecule, and determine **hybridization** at each of the carbon centers. (2.5 points)

Atom	Hybridization
Carbon (<i>a</i>)	
Carbon (<i>b</i>)	
Carbon (<i>c</i>)	
Carbon (<i>d</i>)	
Carbon (<i>e</i>)	

ii. Now, consider the sulfur atom (S) marked as ***f***, nitrogen (N) atom (***g***), and oxygen(O) atoms (***h*** and ***i***), and determine their **hybridization**. (2 points)

Atom	Hybridization
Sulfur (<i>f</i>)	
Nitrogen (<i>g</i>)	
Oxygen (<i>h</i>)	
Oxygen (<i>i</i>)	

iii. **Describe** all the bonds formed by the carbon atoms ***a*** and ***b***, using sigma and pi designation as appropriate. (2 points)

Atom	Bonds formed (use sigma and pi as appropriate)
Carbon (<i>a</i>)	
Carbon (<i>b</i>)	

iv. How many total sp^3 -s sigma bonds are in the molecule? (1 point)

v. Determine **electronic** and **molecular geometry** at carbon (C) labeled ***e*** and sulfur (S) labeled ***f*** centres. (2 points)

Atom	Electronic Geometry	Molecular Geometry
Carbon (<i>e</i>)		
Sulfur (<i>f</i>)		

vi. Circle the value from the choices given below, that describes the **C-S-C bond angle at sulfur** (S) marked ***f***: (0.5 points)

>90° to <109.5° 109.5° >109.5 to <120° 120° >120° to <180°


Question 8 12 points

Using the **valence shell orbitals only** draw a **complete** molecular orbital (MO) diagram (with no s-p mixing) for the molecule **PS**, in the space provided below. **(6 points)**

Assume that 3s and 3p atomic orbitals will form molecular orbitals like those formed by the combination of 2s and 2p atomic orbitals that we learnt in class, and there is no s-p mixing.

Show **energy arrow**, **atomic orbitals** involved, **atom labels**, the **molecular orbitals** that are formed, **filled electrons** in the MO diagram etc. Make sure to label all the atomic and molecular orbitals.

Using the MO diagram above, answer the following questions:

- (i) Which one of the following will have the highest bond order? Circle the correct choice
 A) PS^{1-} B) PS C) PS^{1+}
- (ii) Which one of the following will have the longest bond length? Circle the correct choice
 A) PS^{1-} B) PS C) PS^{1+}
- (iii) Which of the following will be diamagnetic? Circle the correct choice.
 A) PS^{1-} B) PS C) PS^{1+}
- (iv) Draw Lewis dot structure of the molecule with the shortest bond length (based on the MO diagram) from among the three (PS^{1-} , PS, PS^{1+}), and indicate the bond order calculated from the Lewis dot structure. **(3 points)**

Lewis Dot Structure

Bond Order



5F5AE01D-E828-43F5-909D-DDAD5BB2B21E

midterm-exam-93388

#138 Page 10 of 10

SCRAP PAGE

**CHEMISTRY 110 –Midterm Exam****Thursday November 7, 2024, 6:30 – 8:30 pm****Examiners: Pallavi Sirjoosingh and Ashok Kakkar**

First name (please write as legibly as possible within the boxes)

--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--

Last name

--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--

Student ID number

--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--

This booklet has 8 questions on 10 pages (The last page can be used as a scrap page).

The staple must NOT be removed from the question booklet. Your intact question booklet must be returned at the end of the exam.

A separate booklet contains a periodic table, and datasheet.

Write your answers directly under the questions in the space provided. Do not write on the QR codes or near the top edges (as it may be cut off).

Please write **legibly and dark enough** for the scan to be readable. For students who use English as a second language, dictionaries are permitted. Calculators are allowed.

DO NOT ASK QUESTIONS OF THE INVIGILATORS about interpretations of the exam questions. If you have a concern about a question, please summarize it at the top of this page. All concerns will be evaluated after the exam. GOOD LUCK!

**Question 1 8 points** (*Assume all transitions are possible*)

a. 6 points Calculate the **energy (in kJ/mol)** required to ionize (completely remove) the electron from the **$n=3$** level of **He^+ ion** (He: Helium). Show your work including equations. (3 sig. figs.)

b. 2 points

(I) Select the largest (longest) wavelength from the three choices below:

- i. Wavelength of photon to *completely remove* an electron from *1s orbital of Hydrogen atom*
- ii. Wavelength of photon to *completely remove* an electron from *1s orbital of Boron atom (B)*
- iii. Wavelength of photon to *completely remove* an electron from *1s orbital of Boron cation (B^+)*

i or ii or iii

(II) Select the smallest (shortest) wavelength from the three choices below (*for H atom*):

- i. Wavelength of photon *emitted* for an electron transitioning from *2p to 1s orbital*
- ii. Wavelength of photon *absorbed* for an electron transitioning from *2s to 4p orbital*
- iii. Wavelength of photon *absorbed* for an electron transitioning from *1s to 3p orbital*

i or ii or iii



Question 2 8 points



A Chem110 student plots the graph shown on the left, from data obtained for the kinetic energy (eV) of the ejected electron versus frequency (s^{-1}) of the incident light, for an unknown metal. ($1\text{eV} = 1.6022 \times 10^{-19} \text{ J}$)

Based on the plot, answer the questions **a to c** below:

a. 1 point On the plot above, label the **threshold frequency**.

b. 2 points What is the value of the work potential (in J) of the unknown metal?

c. 3 points What is the *longest* wavelength (in nm) required to eject an electron from this metal's surface? Show your work including equation(s) used. (3 sig figs)

d. 2 points In the space provided below, plot the following

(Assume frequency is *above* threshold frequency):

- i. *Kinetic energy* of an ejected electron versus the *intensity* of the incident light (left)
- ii. *Number of electrons* ejected versus the *frequency* of the incident light (right)



**Question 3** **6 points****a. 2 points**

In the ground state of Ge (Germanium; $Z = 32$) atom, **name** the following orbitals:

i. Orbital containing the electron that requires the maximum energy to be removed.

ii. Orbital containing the electron that experiences the minimum effective nuclear charge.

b. 2 points

An unknown anion (A^-) has the following *excited* electronic configuration: $[Ar] 4s^2 3d^3 4p^1$

(Ar: Argon; $Z = 18$)

Give the **condensed ground state electronic configuration** of the atom, **A** and the cation, **A^{2+}** .
Show your work. (Assume all electronic transitions are possible)

Electronic Configuration of A:

Electronic Configuration of A^{2+} :

c. 2 points

In a single atom, determine how many electrons can have the following set of quantum numbers?

(i) $n = 5; l = 3$

In a single atom, determine how many electrons can have the following set of quantum numbers?

(i) $n = 3; l = 2; m_s = +1/2$

**Question 4** **8 points**

Rank the following from 1 to 4 in **increasing** order. Provide the ranking in the boxes provided.

Note: Atomic number (Z) are provided for relevant atoms

a) Increasing (lowest to highest) order of *ionization energy*: (1= lowest; 4 = highest)

Mg

Mg: Z = 12

Rb

Rb: Z = 37

Cs

Cs: Z = 55

K

K : Z = 19

b) Increasing (lowest to highest) order of *bond angle*: (1= lowest; 4 = highest)

H₂OCCl₄XeF₄

HCN

c) Increasing (smallest to largest) order of *ionic size* (1= smallest; 4 = largest)

S²⁻

S: Z = 16

O²⁻

O: Z = 8

Be²⁺

Be: Z = 4

Na⁺

Na: Z = 11

d) Increasing (lowest to highest) order of *bond energy* (1= lowest; 4 = highest)

P-F

P-Cl

P-I

P-Br

**Question 5** **12 points**

a. 9 points For the molecules HClO_4 and TeF_6 , draw the most stable *Lewis structure* showing appropriate shape and *dashed/wedged bonds*, based on VSEPR. Determine the **molecular geometry** around the central atom. For HClO_4 , label the *most polar bond*.

HClO_4 (there are no O-O bonds)

Label the most polar bond

Molecular Geometry around Cl

TeF_6

Molecular Geometry around Te

b. 3 points (Note: Lone pairs are not shown in structures below)

(i) For the three *resonance* structures (1, 2 and 3) shown below, answer the following question:



(I) Circle the *most contributing* resonance structure.

Structure 1

Structure 2

Structure 3

(ii) For the three structures given (4, 5, and 6) below, answer the following questions:



(I) Circle all the structures that represent *resonance structures* of each other:

Structure 4

Structure 5

Structure 6

(II) Among the resonance structures, circle the *most contributing* resonance structure.

Structure 4

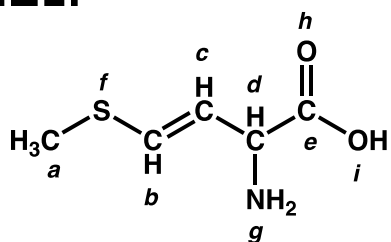
Structure 5

Structure 6

**Question 6** **6 points**

Using the data given below, calculate the **lattice energy** for SrO(s).

	ΔH° (kJ/mol)
$\text{Sr}(s) \longrightarrow \text{Sr}(g)$	+180
$\text{Sr}(g) \longrightarrow \text{Sr}^{2+}(g) + 2e^-$	+1468
$\text{O}_2(g) \longrightarrow 2\text{O}(g)$	+500
$\text{O}(g) + 2e^- \longrightarrow \text{O}^{2-}(g)$	+600
$\text{Sr}(s) + \frac{1}{2}\text{O}_2(g) \longrightarrow \text{SrO}(s)$	-550

**Question 7 10 points**

Consider the molecule given on the left, and answer questions (i) to (vi) using chemical bonding theories that we have learnt in class (valence bond theory, hybridization).

Note that the lone pairs are not shown in the structure:

i. Consider the carbon atoms marked in as ***a***, ***b***, ***c***, ***d***, and ***e*** in the molecule, and determine **hybridization** at each of the carbon centers. (2.5 points)

Atom	Hybridization
Carbon (<i>a</i>)	
Carbon (<i>b</i>)	
Carbon (<i>c</i>)	
Carbon (<i>d</i>)	
Carbon (<i>e</i>)	

ii. Now, consider the sulfur atom (S) marked as ***f***, nitrogen (N) atom (***g***), and oxygen(O) atoms (***h*** and ***i***), and determine their **hybridization**. (2 points)

Atom	Hybridization
Sulfur (<i>f</i>)	
Nitrogen (<i>g</i>)	
Oxygen (<i>h</i>)	
Oxygen (<i>i</i>)	

iii. **Describe** all the bonds formed by the carbon atoms ***a*** and ***b***, using sigma and pi designation as appropriate. (2 points)

Atom	Bonds formed (use sigma and pi as appropriate)
Carbon (<i>a</i>)	
Carbon (<i>b</i>)	

iv. How many total sp^3 -s sigma bonds are in the molecule? (1 point)

v. Determine **electronic** and **molecular geometry** at carbon (C) labeled ***e*** and sulfur (S) labeled ***f*** centres. (2 points)

Atom	Electronic Geometry	Molecular Geometry
Carbon (<i>e</i>)		
Sulfur (<i>f</i>)		

vi. Circle the value from the choices given below, that describes the **C-S-C bond angle at sulfur** (S) marked ***f***: (0.5 points)

>90° to <109.5° 109.5° >109.5 to <120° 120° >120° to <180°


Question 8 12 points

Using the **valence shell orbitals only** draw a **complete** molecular orbital (MO) diagram (with no s-p mixing) for the molecule **PS**, in the space provided below. **(6 points)**

Assume that 3s and 3p atomic orbitals will form molecular orbitals like those formed by the combination of 2s and 2p atomic orbitals that we learnt in class, and there is no s-p mixing.

Show **energy arrow**, **atomic orbitals** involved, **atom labels**, the **molecular orbitals** that are formed, **filled electrons** in the MO diagram etc. Make sure to label all the atomic and molecular orbitals.

Using the MO diagram above, answer the following questions:

- (i) Which one of the following will have the highest bond order? Circle the correct choice
 A) PS^{1-} B) PS C) PS^{1+}
- (ii) Which one of the following will have the longest bond length? Circle the correct choice
 A) PS^{1-} B) PS C) PS^{1+}
- (iii) Which of the following will be diamagnetic? Circle the correct choice.
 A) PS^{1-} B) PS C) PS^{1+}
- (iv) Draw Lewis dot structure of the molecule with the shortest bond length (based on the MO diagram) from among the three (PS^{1-} , PS, PS^{1+}), and indicate the bond order calculated from the Lewis dot structure. **(3 points)**

Lewis Dot Structure

Bond Order



562F173E-B7DE-4FD3-964F-DC6FCF0149BB

midterm-exam-93388

#139 Page 10 of 10

SCRAP PAGE

**CHEMISTRY 110 –Midterm Exam****Thursday November 7, 2024, 6:30 – 8:30 pm**

Examiners: Pallavi Sirjoosingh and Ashok Kakkar

First name (please write as legibly as possible within the boxes)

--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--

Last name

--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--

Student ID number

--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--

This booklet has 8 questions on 10 pages (The last page can be used as a scrap page).

The staple must NOT be removed from the question booklet. Your intact question booklet must be returned at the end of the exam.

A separate booklet contains a periodic table, and datasheet.

Write your answers directly under the questions in the space provided. Do not write on the QR codes or near the top edges (as it may be cut off).

Please write **legibly and dark enough** for the scan to be readable. For students who use English as a second language, dictionaries are permitted. Calculators are allowed.

DO NOT ASK QUESTIONS OF THE INVIGILATORS about interpretations of the exam questions. If you have a concern about a question, please summarize it at the top of this page. All concerns will be evaluated after the exam. GOOD LUCK!

**Question 1 8 points** (*Assume all transitions are possible*)

a. 6 points Calculate the **energy (in kJ/mol)** required to ionize (completely remove) the electron from the **n=3** level of **He⁺ ion** (He: Helium). Show your work including equations. (3 sig. figs.)

b. 2 points

(I) Select the largest (longest) wavelength from the three choices below:

- i. Wavelength of photon to *completely remove* an electron from *1s orbital of Hydrogen atom*
- ii. Wavelength of photon to *completely remove* an electron from *1s orbital of Boron atom (B)*
- iii. Wavelength of photon to *completely remove* an electron from *1s orbital of Boron cation (B⁺)*

i or ii or iii

(II) Select the smallest (shortest) wavelength from the three choices below (*for H atom*):

- i. Wavelength of photon *emitted* for an electron transitioning from *2p to 1s orbital*
- ii. Wavelength of photon *absorbed* for an electron transitioning from *2s to 4p orbital*
- iii. Wavelength of photon *absorbed* for an electron transitioning from *1s to 3p orbital*

i or ii or iii



Question 2 8 points



A Chem110 student plots the graph shown on the left, from data obtained for the kinetic energy (eV) of the ejected electron versus frequency (s^{-1}) of the incident light, for an unknown metal. ($1\text{eV} = 1.6022 \times 10^{-19} \text{ J}$)

Based on the plot, answer the questions **a to c** below:

a. 1 point On the plot above, label the **threshold frequency**.

b. 2 points What is the value of the work potential (in J) of the unknown metal?

c. 3 points What is the *longest* wavelength (in nm) required to eject an electron from this metal's surface? Show your work including equation(s) used. (3 sig figs)

d. 2 points In the space provided below, plot the following

(Assume frequency is *above* threshold frequency):

- Kinetic energy* of an ejected electron versus the *intensity* of the incident light (left)
- Number of electrons* ejected versus the *frequency* of the incident light (right)



**Question 3** **6 points****a. 2 points**

In the ground state of Ge (Germanium; $Z = 32$) atom, **name** the following orbitals:

i. Orbital containing the electron that requires the maximum energy to be removed.

ii. Orbital containing the electron that experiences the minimum effective nuclear charge.

b. 2 points

An unknown anion (A^-) has the following *excited* electronic configuration: $[Ar] 4s^2 3d^3 4p^1$

(Ar: Argon; $Z = 18$)

Give the **condensed ground state electronic configuration** of the atom, **A** and the cation, **A^{2+}** .
Show your work. (Assume all electronic transitions are possible)

Electronic Configuration of A:

Electronic Configuration of A^{2+} :

c. 2 points

In a single atom, determine how many electrons can have the following set of quantum numbers?

(i) $n = 5; l = 3$

In a single atom, determine how many electrons can have the following set of quantum numbers?

(i) $n = 3; l = 2; m_s = +1/2$

**Question 4** **8 points**

Rank the following from 1 to 4 in **increasing** order. Provide the ranking in the boxes provided.

Note: Atomic number (Z) are provided for relevant atoms

a) Increasing (lowest to highest) order of *ionization energy*: (1= lowest; 4 = highest)

Mg

Mg: Z = 12

Rb

Rb: Z = 37

Cs

Cs: Z = 55

K

K : Z = 19

b) Increasing (lowest to highest) order of *bond angle*: (1= lowest; 4 = highest)

H₂OCCl₄XeF₄

HCN

c) Increasing (smallest to largest) order of *ionic size* (1= smallest; 4 = largest)

S²⁻

S: Z = 16

O²⁻

O: Z = 8

Be²⁺

Be: Z = 4

Na⁺

Na: Z = 11

d) Increasing (lowest to highest) order of *bond energy* (1= lowest; 4 = highest)

P-F

P-Cl

P-I

P-Br

**Question 5** **12 points**

a. 9 points For the molecules HClO_4 and TeF_6 , draw the most stable *Lewis structure* showing appropriate shape and *dashed/wedged bonds*, based on VSEPR. Determine the **molecular geometry** around the central atom. For HClO_4 , label the *most polar bond*.

HClO_4 (there are no O-O bonds)

Label the most polar bond

Molecular Geometry around Cl

TeF_6

Molecular Geometry around Te

b. 3 points (Note: Lone pairs are not shown in structures below)

(i) For the three *resonance* structures (1, 2 and 3) shown below, answer the following question:



(I) Circle the *most contributing* resonance structure.

Structure 1

Structure 2

Structure 3

(ii) For the three structures given (4, 5, and 6) below, answer the following questions:



(I) Circle all the structures that represent *resonance structures* of each other:

Structure 4

Structure 5

Structure 6

(II) Among the resonance structures, circle the *most contributing* resonance structure.

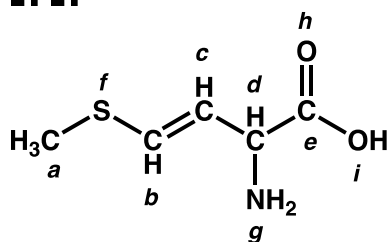
Structure 4

Structure 5

Structure 6

**Question 6** **6 points**Using the data given below, calculate the **lattice energy** for SrO(s).

	ΔH° (kJ/mol)
$\text{Sr}(s) \longrightarrow \text{Sr}(g)$	+180
$\text{Sr}(g) \longrightarrow \text{Sr}^{2+}(g) + 2e^-$	+1468
$\text{O}_2(g) \longrightarrow 2\text{O}(g)$	+500
$\text{O}(g) + 2e^- \longrightarrow \text{O}^{2-}(g)$	+600
$\text{Sr}(s) + \frac{1}{2}\text{O}_2(g) \longrightarrow \text{SrO}(s)$	-550

**Question 7 10 points**

Consider the molecule given on the left, and answer questions (i) to (vi) using chemical bonding theories that we have learnt in class (valence bond theory, hybridization).

Note that the lone pairs are not shown in the structure:

i. Consider the carbon atoms marked in as ***a***, ***b***, ***c***, ***d***, and ***e*** in the molecule, and determine **hybridization** at each of the carbon centers. (2.5 points)

Atom	Hybridization
Carbon (<i>a</i>)	
Carbon (<i>b</i>)	
Carbon (<i>c</i>)	
Carbon (<i>d</i>)	
Carbon (<i>e</i>)	

ii. Now, consider the sulfur atom (S) marked as ***f***, nitrogen (N) atom (***g***), and oxygen(O) atoms (***h*** and ***i***), and determine their **hybridization**. (2 points)

Atom	Hybridization
Sulfur (<i>f</i>)	
Nitrogen (<i>g</i>)	
Oxygen (<i>h</i>)	
Oxygen (<i>i</i>)	

iii. **Describe** all the bonds formed by the carbon atoms ***a*** and ***b***, using sigma and pi designation as appropriate. (2 points)

Atom	Bonds formed (use sigma and pi as appropriate)
Carbon (<i>a</i>)	
Carbon (<i>b</i>)	

iv. How many total sp^3 -s sigma bonds are in the molecule? (1 point)

v. Determine **electronic** and **molecular geometry** at carbon (C) labeled ***e*** and sulfur (S) labeled ***f*** centres. (2 points)

Atom	Electronic Geometry	Molecular Geometry
Carbon (<i>e</i>)		
Sulfur (<i>f</i>)		

vi. Circle the value from the choices given below, that describes the **C-S-C bond angle at sulfur** (S) marked ***f***: (0.5 points)

>90° to <109.5° 109.5° >109.5 to <120° 120° >120° to <180°


Question 8 12 points

Using the **valence shell orbitals only** draw a **complete** molecular orbital (MO) diagram (with no s-p mixing) for the molecule **PS**, in the space provided below. **(6 points)**

Assume that 3s and 3p atomic orbitals will form molecular orbitals like those formed by the combination of 2s and 2p atomic orbitals that we learnt in class, and there is no s-p mixing.

Show **energy arrow**, **atomic orbitals** involved, **atom labels**, the **molecular orbitals** that are formed, **filled electrons** in the MO diagram etc. Make sure to **label all the atomic and molecular orbitals**.

Using the MO diagram above, answer the following questions:

- (i) Which one of the following will have the highest bond order? Circle the correct choice
 A) PS^{1-} B) PS C) PS^{1+}
- (ii) Which one of the following will have the longest bond length? Circle the correct choice
 A) PS^{1-} B) PS C) PS^{1+}
- (iii) Which of the following will be diamagnetic? Circle the correct choice.
 A) PS^{1-} B) PS C) PS^{1+}
- (iv) Draw Lewis dot structure of the molecule with the shortest bond length (based on the MO diagram) from among the three (PS^{1-} , PS, PS^{1+}), and indicate the bond order calculated from the Lewis dot structure. **(3 points)**

Lewis Dot Structure

Bond Order



4FF20024-048F-4ECD-80FC-3BADAF06B344

midterm-exam-93388

#140 Page 10 of 10

SCRAP PAGE

**CHEMISTRY 110 –Midterm Exam****Thursday November 7, 2024, 6:30 – 8:30 pm****Examiners: Pallavi Sirjoosingh and Ashok Kakkar**

First name (please write as legibly as possible within the boxes)

--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--

Last name

--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--

Student ID number

--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--

This booklet has 8 questions on 10 pages (The last page can be used as a scrap page).

The staple must NOT be removed from the question booklet. Your intact question booklet must be returned at the end of the exam.

A separate booklet contains a periodic table, and datasheet.

Write your answers directly under the questions in the space provided. Do not write on the QR codes or near the top edges (as it may be cut off).

Please write **legibly and dark enough** for the scan to be readable. For students who use English as a second language, dictionaries are permitted. Calculators are allowed.

DO NOT ASK QUESTIONS OF THE INVIGILATORS about interpretations of the exam questions. If you have a concern about a question, please summarize it at the top of this page. All concerns will be evaluated after the exam. GOOD LUCK!

**Question 1 8 points** (*Assume all transitions are possible*)

a. 6 points Calculate the **energy (in kJ/mol)** required to ionize (completely remove) the electron from the **n=3** level of **He⁺ ion** (He: Helium). Show your work including equations. (3 sig. figs.)

b. 2 points

(I) Select the largest (longest) wavelength from the three choices below:

- i. Wavelength of photon to *completely remove* an electron from *1s orbital of Hydrogen atom*
- ii. Wavelength of photon to *completely remove* an electron from *1s orbital of Boron atom (B)*
- iii. Wavelength of photon to *completely remove* an electron from *1s orbital of Boron cation (B⁺)*

i or ii or iii

(II) Select the smallest (shortest) wavelength from the three choices below (*for H atom*):

- i. Wavelength of photon *emitted* for an electron transitioning from *2p to 1s orbital*
- ii. Wavelength of photon *absorbed* for an electron transitioning from *2s to 4p orbital*
- iii. Wavelength of photon *absorbed* for an electron transitioning from *1s to 3p orbital*

i or ii or iii



Question 2 8 points



A Chem110 student plots the graph shown on the left, from data obtained for the kinetic energy (eV) of the ejected electron versus frequency (s^{-1}) of the incident light, for an unknown metal. ($1\text{eV} = 1.6022 \times 10^{-19} \text{ J}$)

Based on the plot, answer the questions **a to c** below:

a. 1 point On the plot above, label the **threshold frequency**.

b. 2 points What is the value of the work potential (in J) of the unknown metal?

c. 3 points What is the *longest* wavelength (in nm) required to eject an electron from this metal's surface? Show your work including equation(s) used. (3 sig figs)

d. 2 points In the space provided below, plot the following

(Assume frequency is *above* threshold frequency):

- i. *Kinetic energy* of an ejected electron versus the *intensity* of the incident light (left)
- ii. *Number of electrons* ejected versus the *frequency* of the incident light (right)



**Question 3** **6 points****a. 2 points**

In the ground state of Ge (Germanium; $Z = 32$) atom, **name** the following orbitals:

i. Orbital containing the electron that requires the maximum energy to be removed.

ii. Orbital containing the electron that experiences the minimum effective nuclear charge.

b. 2 points

An unknown anion (A^-) has the following *excited* electronic configuration: $[Ar] 4s^2 3d^3 4p^1$

(Ar: Argon; $Z = 18$)

Give the **condensed ground state electronic configuration** of the atom, **A** and the cation, A^{2+} .
Show your work. (Assume all electronic transitions are possible)

Electronic Configuration of A:

Electronic Configuration of A^{2+} :

c. 2 points

In a single atom, determine how many electrons can have the following set of quantum numbers?

(i) $n = 5; l = 3$

In a single atom, determine how many electrons can have the following set of quantum numbers?

(i) $n = 3; l = 2; m_s = +1/2$

**Question 4** **8 points**

Rank the following from 1 to 4 in **increasing** order. Provide the ranking in the boxes provided.

Note: Atomic number (Z) are provided for relevant atoms

a) Increasing (lowest to highest) order of *ionization energy*: (1= lowest; 4 = highest)

Mg

Mg: Z = 12

Rb

Rb: Z = 37

Cs

Cs: Z = 55

K

K : Z = 19

b) Increasing (lowest to highest) order of *bond angle*: (1= lowest; 4 = highest)

H₂OCCl₄XeF₄

HCN

c) Increasing (smallest to largest) order of *ionic size* (1= smallest; 4 = largest)

S²⁻

S: Z = 16

O²⁻

O: Z = 8

Be²⁺

Be: Z = 4

Na⁺

Na: Z = 11

d) Increasing (lowest to highest) order of *bond energy* (1= lowest; 4 = highest)

P-F

P-Cl

P-I

P-Br

**Question 5** **12 points**

a. 9 points For the molecules HClO_4 and TeF_6 , draw the most stable *Lewis structure* showing appropriate shape and *dashed/wedged bonds*, based on VSEPR. Determine the **molecular geometry** around the central atom. For HClO_4 , label the *most polar bond*.

HClO_4 (there are no O-O bonds)

Label the most polar bond

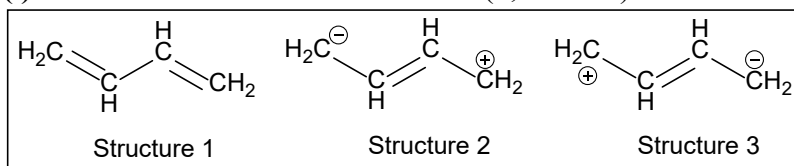
Molecular Geometry around Cl

TeF_6

Molecular Geometry around Te

b. 3 points (Note: Lone pairs are not shown in structures below)

(i) For the three *resonance* structures (1, 2 and 3) shown below, answer the following question:



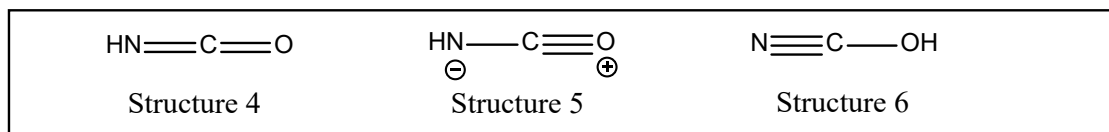
(I) Circle the *most contributing* resonance structure.

Structure 1

Structure 2

Structure 3

(ii) For the three structures given (4, 5, and 6) below, answer the following questions:



(I) Circle all the structures that represent *resonance structures* of each other:

Structure 4

Structure 5

Structure 6

(II) Among the resonance structures, circle the *most contributing* resonance structure.

Structure 4

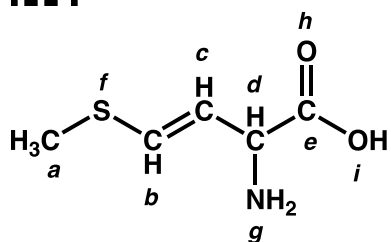
Structure 5

Structure 6

**Question 6** **6 points**

Using the data given below, calculate the **lattice energy** for SrO(s).

	ΔH° (kJ/mol)
$\text{Sr}(s) \longrightarrow \text{Sr}(g)$	+180
$\text{Sr}(g) \longrightarrow \text{Sr}^{2+}(g) + 2e^-$	+1468
$\text{O}_2(g) \longrightarrow 2\text{O}(g)$	+500
$\text{O}(g) + 2e^- \longrightarrow \text{O}^{2-}(g)$	+600
$\text{Sr}(s) + \frac{1}{2}\text{O}_2(g) \longrightarrow \text{SrO}(s)$	-550

**Question 7 10 points**

Consider the molecule given on the left, and answer questions (i) to (vi) using chemical bonding theories that we have learnt in class (valence bond theory, hybridization).

Note that the lone pairs are not shown in the structure:

i. Consider the carbon atoms marked in as ***a***, ***b***, ***c***, ***d***, and ***e*** in the molecule, and determine **hybridization** at each of the carbon centers. (2.5 points)

Atom	Hybridization
Carbon (<i>a</i>)	
Carbon (<i>b</i>)	
Carbon (<i>c</i>)	
Carbon (<i>d</i>)	
Carbon (<i>e</i>)	

ii. Now, consider the sulfur atom (S) marked as ***f***, nitrogen (N) atom (***g***), and oxygen(O) atoms (***h*** and ***i***), and determine their **hybridization**. (2 points)

Atom	Hybridization
Sulfur (<i>f</i>)	
Nitrogen (<i>g</i>)	
Oxygen (<i>h</i>)	
Oxygen (<i>i</i>)	

iii. **Describe** all the bonds formed by the carbon atoms ***a*** and ***b***, using sigma and pi designation as appropriate. (2 points)

Atom	Bonds formed (use sigma and pi as appropriate)
Carbon (<i>a</i>)	
Carbon (<i>b</i>)	

iv. How many total sp^3 -s sigma bonds are in the molecule? (1 point)

v. Determine **electronic** and **molecular geometry** at carbon (C) labeled ***e*** and sulfur (S) labeled ***f*** centres. (2 points)

Atom	Electronic Geometry	Molecular Geometry
Carbon (<i>e</i>)		
Sulfur (<i>f</i>)		

vi. Circle the value from the choices given below, that describes the **C-S-C bond angle at sulfur** (S) marked ***f***: (0.5 points)

>90° to <109.5° 109.5° >109.5 to <120° 120° >120° to <180°


Question 8 12 points

Using the **valence shell orbitals only** draw a **complete** molecular orbital (MO) diagram (with no s-p mixing) for the molecule **PS**, in the space provided below. **(6 points)**

Assume that 3s and 3p atomic orbitals will form molecular orbitals like those formed by the combination of 2s and 2p atomic orbitals that we learnt in class, and there is no s-p mixing.

Show **energy arrow**, **atomic orbitals** involved, **atom labels**, the **molecular orbitals** that are formed, **filled electrons** in the MO diagram etc. Make sure to label all the atomic and molecular orbitals.

Using the MO diagram above, answer the following questions:

- (i) Which one of the following will have the highest bond order? Circle the correct choice
 A) PS^{1-} B) PS C) PS^{1+}
- (ii) Which one of the following will have the longest bond length? Circle the correct choice
 A) PS^{1-} B) PS C) PS^{1+}
- (iii) Which of the following will be diamagnetic? Circle the correct choice.
 A) PS^{1-} B) PS C) PS^{1+}
- (iv) Draw Lewis dot structure of the molecule with the shortest bond length (based on the MO diagram) from among the three (PS^{1-} , PS, PS^{1+}), and indicate the bond order calculated from the Lewis dot structure. **(3 points)**

Lewis Dot Structure

Bond Order



E87E5AA8-8662-4577-9196-5383966DF724

midterm-exam-93388

#141 Page 10 of 10

SCRAP PAGE

**CHEMISTRY 110 –Midterm Exam****Thursday November 7, 2024, 6:30 – 8:30 pm****Examiners: Pallavi Sirjoosingh and Ashok Kakkar**

First name (please write as legibly as possible within the boxes)

--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--

Last name

--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--

Student ID number

--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--

This booklet has 8 questions on 10 pages (The last page can be used as a scrap page).

The staple must NOT be removed from the question booklet. Your intact question booklet must be returned at the end of the exam.

A separate booklet contains a periodic table, and datasheet.

Write your answers directly under the questions in the space provided. Do not write on the QR codes or near the top edges (as it may be cut off).

Please write **legibly and dark enough** for the scan to be readable. For students who use English as a second language, dictionaries are permitted. Calculators are allowed.

DO NOT ASK QUESTIONS OF THE INVIGILATORS about interpretations of the exam questions. If you have a concern about a question, please summarize it at the top of this page. All concerns will be evaluated after the exam. GOOD LUCK!

**Question 1 8 points** (*Assume all transitions are possible*)

a. 6 points Calculate the **energy (in kJ/mol)** required to ionize (completely remove) the electron from the **$n=3$** level of **He^+ ion** (He: Helium). Show your work including equations. (3 sig. figs.)

b. 2 points

(I) Select the largest (longest) wavelength from the three choices below:

- i. Wavelength of photon to *completely remove* an electron from *1s orbital of Hydrogen atom*
- ii. Wavelength of photon to *completely remove* an electron from *1s orbital of Boron atom (B)*
- iii. Wavelength of photon to *completely remove* an electron from *1s orbital of Boron cation (B^+)*

i or ii or iii

(II) Select the smallest (shortest) wavelength from the three choices below (*for H atom*):

- i. Wavelength of photon *emitted* for an electron transitioning from *2p to 1s orbital*
- ii. Wavelength of photon *absorbed* for an electron transitioning from *2s to 4p orbital*
- iii. Wavelength of photon *absorbed* for an electron transitioning from *1s to 3p orbital*

i or ii or iii



Question 2 8 points



A Chem110 student plots the graph shown on the left, from data obtained for the kinetic energy (eV) of the ejected electron versus frequency (s^{-1}) of the incident light, for an unknown metal. ($1\text{eV} = 1.6022 \times 10^{-19} \text{ J}$)

Based on the plot, answer the questions **a to c** below:

a. 1 point On the plot above, label the **threshold frequency**.

b. 2 points What is the value of the work potential (in J) of the unknown metal?

c. 3 points What is the *longest* wavelength (in nm) required to eject an electron from this metal's surface? Show your work including equation(s) used. (3 sig figs)

d. 2 points In the space provided below, plot the following

(Assume frequency is *above* threshold frequency):

- i. *Kinetic energy* of an ejected electron versus the *intensity* of the incident light (left)
- ii. *Number of electrons* ejected versus the *frequency* of the incident light (right)



**Question 3** **6 points****a. 2 points**

In the ground state of Ge (Germanium; $Z = 32$) atom, **name** the following orbitals:

i. Orbital containing the electron that requires the maximum energy to be removed.

ii. Orbital containing the electron that experiences the minimum effective nuclear charge.

b. 2 points

An unknown anion (A^-) has the following *excited* electronic configuration: $[Ar] 4s^2 3d^3 4p^1$

(Ar: Argon; $Z = 18$)

Give the **condensed ground state electronic configuration** of the atom, **A** and the cation, **A^{2+}** .
Show your work. (Assume all electronic transitions are possible)

Electronic Configuration of A:

Electronic Configuration of A^{2+} :

c. 2 points

In a single atom, determine how many electrons can have the following set of quantum numbers?

(i) $n = 5; l = 3$

In a single atom, determine how many electrons can have the following set of quantum numbers?

(i) $n = 3; l = 2; m_s = +1/2$

**Question 4** **8 points**

Rank the following from 1 to 4 in **increasing** order. Provide the ranking in the boxes provided.

Note: Atomic number (Z) are provided for relevant atoms

a) Increasing (lowest to highest) order of *ionization energy*: (1= lowest; 4 = highest)

Mg

Mg: Z = 12

Rb

Rb: Z = 37

Cs

Cs: Z = 55

K

K : Z = 19

b) Increasing (lowest to highest) order of *bond angle*: (1= lowest; 4 = highest)

H₂OCCl₄XeF₄

HCN

c) Increasing (smallest to largest) order of *ionic size* (1= smallest; 4 = largest)

S²⁻

S: Z = 16

O²⁻

O: Z = 8

Be²⁺

Be: Z = 4

Na⁺

Na: Z = 11

d) Increasing (lowest to highest) order of *bond energy* (1= lowest; 4 = highest)

P-F

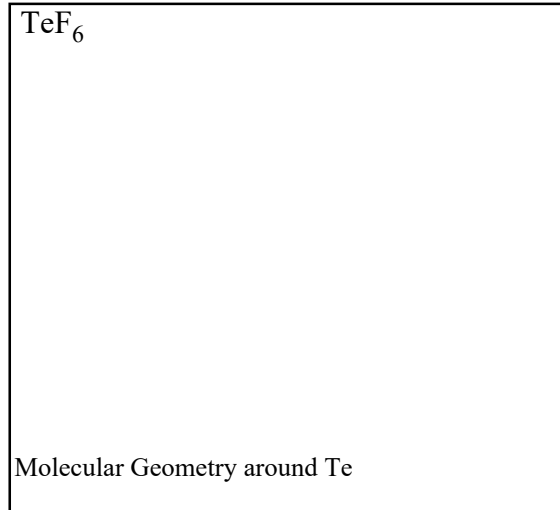
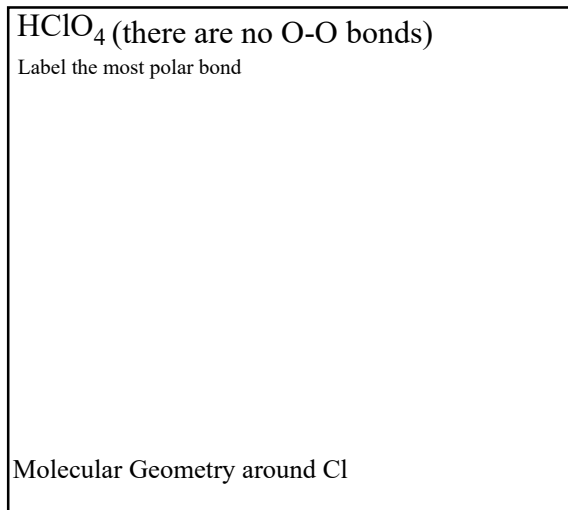
P-Cl

P-I

P-Br

**Question 5** **12 points**

a. 9 points For the molecules HClO_4 and TeF_6 , draw the most stable *Lewis structure* showing appropriate shape and *dashed/wedged bonds*, based on VSEPR. Determine the **molecular geometry** around the central atom. For HClO_4 , label the *most polar bond*.



b. 3 points (Note: Lone pairs are not shown in structures below)

(i) For the three *resonance* structures (1, 2 and 3) shown below, answer the following question:



(I) Circle the *most contributing* resonance structure.

Structure 1

Structure 2

Structure 3

(ii) For the three structures given (4, 5, and 6) below, answer the following questions:



(I) Circle all the structures that represent *resonance structures* of each other:

Structure 4

Structure 5

Structure 6

(II) Among the resonance structures, circle the *most contributing* resonance structure.

Structure 4

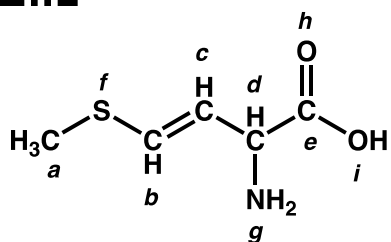
Structure 5

Structure 6

**Question 6** **6 points**

Using the data given below, calculate the **lattice energy** for SrO(s).

	ΔH° (kJ/mol)
$\text{Sr}(s) \longrightarrow \text{Sr}(g)$	+180
$\text{Sr}(g) \longrightarrow \text{Sr}^{2+}(g) + 2e^-$	+1468
$\text{O}_2(g) \longrightarrow 2\text{O}(g)$	+500
$\text{O}(g) + 2e^- \longrightarrow \text{O}^{2-}(g)$	+600
$\text{Sr}(s) + \frac{1}{2}\text{O}_2(g) \longrightarrow \text{SrO}(s)$	-550

**Question 7 10 points**

Consider the molecule given on the left, and answer questions (i) to (vi) using chemical bonding theories that we have learnt in class (valence bond theory, hybridization).

Note that the lone pairs are not shown in the structure:

i. Consider the carbon atoms marked in as ***a***, ***b***, ***c***, ***d***, and ***e*** in the molecule, and determine **hybridization** at each of the carbon centers. (2.5 points)

Atom	Hybridization
Carbon (<i>a</i>)	
Carbon (<i>b</i>)	
Carbon (<i>c</i>)	
Carbon (<i>d</i>)	
Carbon (<i>e</i>)	

ii. Now, consider the sulfur atom (S) marked as ***f***, nitrogen (N) atom (***g***), and oxygen(O) atoms (***h*** and ***i***), and determine their **hybridization**. (2 points)

Atom	Hybridization
Sulfur (<i>f</i>)	
Nitrogen (<i>g</i>)	
Oxygen (<i>h</i>)	
Oxygen (<i>i</i>)	

iii. **Describe** all the bonds formed by the carbon atoms ***a*** and ***b***, using sigma and pi designation as appropriate. (2 points)

Atom	Bonds formed (use sigma and pi as appropriate)
Carbon (<i>a</i>)	
Carbon (<i>b</i>)	

iv. How many total sp^3 -s sigma bonds are in the molecule? (1 point)

v. Determine **electronic** and **molecular geometry** at carbon (C) labeled ***e*** and sulfur (S) labeled ***f*** centres. (2 points)

Atom	Electronic Geometry	Molecular Geometry
Carbon (<i>e</i>)		
Sulfur (<i>f</i>)		

vi. Circle the value from the choices given below, that describes the **C-S-C bond angle at sulfur** (S) marked ***f***: (0.5 points)

>90° to <109.5°

109.5°

>109.5 to <120°

120°

>120° to <180°


Question 8 12 points

Using the **valence shell orbitals only** draw a **complete** molecular orbital (MO) diagram (with no s-p mixing) for the molecule **PS**, in the space provided below. **(6 points)**

Assume that 3s and 3p atomic orbitals will form molecular orbitals like those formed by the combination of 2s and 2p atomic orbitals that we learnt in class, and there is no s-p mixing.

Show **energy arrow**, **atomic orbitals** involved, **atom labels**, the **molecular orbitals** that are formed, **filled electrons** in the MO diagram etc. Make sure to label all the atomic and molecular orbitals.

Using the MO diagram above, answer the following questions:

- (i) Which one of the following will have the highest bond order? Circle the correct choice
 A) PS^{1-} B) PS C) PS^{1+}
- (ii) Which one of the following will have the longest bond length? Circle the correct choice
 A) PS^{1-} B) PS C) PS^{1+}
- (iii) Which of the following will be diamagnetic? Circle the correct choice.
 A) PS^{1-} B) PS C) PS^{1+}
- (iv) Draw Lewis dot structure of the molecule with the shortest bond length (based on the MO diagram) from among the three (PS^{1-} , PS, PS^{1+}), and indicate the bond order calculated from the Lewis dot structure. **(3 points)**

Lewis Dot Structure

Bond Order



830992B4-FB03-4DBD-BD41-4CF603E3DAD5

midterm-exam-93388

#142 Page 10 of 10

SCRAP PAGE

**CHEMISTRY 110 –Midterm Exam****Thursday November 7, 2024, 6:30 – 8:30 pm****Examiners: Pallavi Sirjoosingh and Ashok Kakkar**

First name (please write as legibly as possible within the boxes)

--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--

Last name

--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--

Student ID number

--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--

This booklet has 8 questions on 10 pages (The last page can be used as a scrap page).

The staple must NOT be removed from the question booklet. Your intact question booklet must be returned at the end of the exam.

A separate booklet contains a periodic table, and datasheet.

Write your answers directly under the questions in the space provided. Do not write on the QR codes or near the top edges (as it may be cut off).

Please write **legibly and dark enough** for the scan to be readable. For students who use English as a second language, dictionaries are permitted. Calculators are allowed.

DO NOT ASK QUESTIONS OF THE INVIGILATORS about interpretations of the exam questions. If you have a concern about a question, please summarize it at the top of this page. All concerns will be evaluated after the exam. GOOD LUCK!

**Question 1 8 points** (*Assume all transitions are possible*)

a. 6 points Calculate the **energy (in kJ/mol)** required to ionize (completely remove) the electron from the **$n=3$** level of **He^+ ion** (He: Helium). Show your work including equations. (3 sig. figs.)

b. 2 points

(I) Select the largest (longest) wavelength from the three choices below:

- i. Wavelength of photon to *completely remove* an electron from *1s orbital of Hydrogen atom*
- ii. Wavelength of photon to *completely remove* an electron from *1s orbital of Boron atom (B)*
- iii. Wavelength of photon to *completely remove* an electron from *1s orbital of Boron cation (B^+)*

i or ii or iii

(II) Select the smallest (shortest) wavelength from the three choices below (*for H atom*):

- i. Wavelength of photon *emitted* for an electron transitioning from *2p to 1s orbital*
- ii. Wavelength of photon *absorbed* for an electron transitioning from *2s to 4p orbital*
- iii. Wavelength of photon *absorbed* for an electron transitioning from *1s to 3p orbital*

i or ii or iii



Question 2 8 points



A Chem110 student plots the graph shown on the left, from data obtained for the kinetic energy (eV) of the ejected electron versus frequency (s^{-1}) of the incident light, for an unknown metal. ($1\text{eV} = 1.6022 \times 10^{-19} \text{ J}$)

Based on the plot, answer the questions **a to c** below:

a. 1 point On the plot above, label the **threshold frequency**.

b. 2 points What is the value of the work potential (in J) of the unknown metal?

c. 3 points What is the *longest* wavelength (in nm) required to eject an electron from this metal's surface? Show your work including equation(s) used. (3 sig figs)

d. 2 points In the space provided below, plot the following

(Assume frequency is *above* threshold frequency):

- Kinetic energy* of an ejected electron versus the *intensity* of the incident light (left)
- Number of electrons* ejected versus the *frequency* of the incident light (right)



**Question 3** **6 points****a. 2 points**

In the ground state of Ge (Germanium; $Z = 32$) atom, **name** the following orbitals:

i. Orbital containing the electron that requires the maximum energy to be removed.

ii. Orbital containing the electron that experiences the minimum effective nuclear charge.

b. 2 points

An unknown anion (A^-) has the following *excited* electronic configuration: $[Ar] 4s^2 3d^3 4p^1$

(Ar: Argon; $Z = 18$)

Give the **condensed ground state electronic configuration** of the atom, **A** and the cation, **A^{2+}** .
Show your work. (Assume all electronic transitions are possible)

Electronic Configuration of A:

Electronic Configuration of A^{2+} :

c. 2 points

In a single atom, determine how many electrons can have the following set of quantum numbers?

(i) $n = 5; l = 3$

In a single atom, determine how many electrons can have the following set of quantum numbers?

(i) $n = 3; l = 2; m_s = +1/2$

**Question 4** **8 points**

Rank the following from 1 to 4 in **increasing** order. Provide the ranking in the boxes provided.

Note: Atomic number (Z) are provided for relevant atoms

a) Increasing (lowest to highest) order of *ionization energy*: (1= lowest; 4 = highest)

Mg

Mg: Z = 12

Rb

Rb: Z = 37

Cs

Cs: Z = 55

K

K : Z = 19

b) Increasing (lowest to highest) order of *bond angle*: (1= lowest; 4 = highest)

H₂OCCl₄XeF₄

HCN

c) Increasing (smallest to largest) order of *ionic size* (1= smallest; 4 = largest)

S²⁻

S: Z = 16

O²⁻

O: Z = 8

Be²⁺

Be: Z = 4

Na⁺

Na: Z = 11

d) Increasing (lowest to highest) order of *bond energy* (1= lowest; 4 = highest)

P-F

P-Cl

P-I

P-Br

**Question 5** **12 points**

a. 9 points For the molecules HClO_4 and TeF_6 , draw the most stable *Lewis structure* showing appropriate shape and *dashed/wedged bonds*, based on VSEPR. Determine the **molecular geometry** around the central atom. For HClO_4 , label the *most polar bond*.



b. 3 points (Note: Lone pairs are not shown in structures below)

(i) For the three *resonance* structures (1, 2 and 3) shown below, answer the following question:



(I) Circle the *most contributing* resonance structure.

Structure 1

Structure 2

Structure 3

(ii) For the three structures given (4, 5, and 6) below, answer the following questions:



(I) Circle all the structures that represent *resonance structures* of each other:

Structure 4

Structure 5

Structure 6

(II) Among the resonance structures, circle the *most contributing* resonance structure.

Structure 4

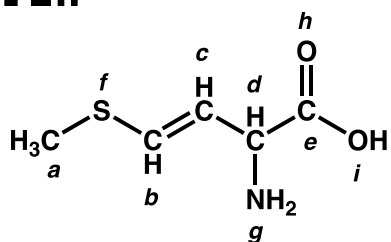
Structure 5

Structure 6

**Question 6** **6 points**

Using the data given below, calculate the **lattice energy** for SrO(s).

	ΔH° (kJ/mol)
$\text{Sr}(s) \longrightarrow \text{Sr}(g)$	+180
$\text{Sr}(g) \longrightarrow \text{Sr}^{2+}(g) + 2e^-$	+1468
$\text{O}_2(g) \longrightarrow 2\text{O}(g)$	+500
$\text{O}(g) + 2e^- \longrightarrow \text{O}^{2-}(g)$	+600
$\text{Sr}(s) + \frac{1}{2}\text{O}_2(g) \longrightarrow \text{SrO}(s)$	-550

**Question 7 10 points**

Consider the molecule given on the left, and answer questions (i) to (vi) using chemical bonding theories that we have learnt in class (valence bond theory, hybridization).

Note that the lone pairs are not shown in the structure:

i. Consider the carbon atoms marked in as ***a***, ***b***, ***c***, ***d***, and ***e*** in the molecule, and determine **hybridization** at each of the carbon centers. (2.5 points)

Atom	Hybridization
Carbon (<i>a</i>)	
Carbon (<i>b</i>)	
Carbon (<i>c</i>)	
Carbon (<i>d</i>)	
Carbon (<i>e</i>)	

ii. Now, consider the sulfur atom (S) marked as ***f***, nitrogen (N) atom (***g***), and oxygen(O) atoms (***h*** and ***i***), and determine their **hybridization**. (2 points)

Atom	Hybridization
Sulfur (<i>f</i>)	
Nitrogen (<i>g</i>)	
Oxygen (<i>h</i>)	
Oxygen (<i>i</i>)	

iii. **Describe** all the bonds formed by the carbon atoms ***a*** and ***b***, using sigma and pi designation as appropriate. (2 points)

Atom	Bonds formed (use sigma and pi as appropriate)
Carbon (<i>a</i>)	
Carbon (<i>b</i>)	

iv. How many total sp^3 -s sigma bonds are in the molecule? (1 point)

v. Determine **electronic** and **molecular geometry** at carbon (C) labeled ***e*** and sulfur (S) labeled ***f*** centres. (2 points)

Atom	Electronic Geometry	Molecular Geometry
Carbon (<i>e</i>)		
Sulfur (<i>f</i>)		

vi. Circle the value from the choices given below, that describes the **C-S-C bond angle at sulfur** (S) marked ***f***: (0.5 points)

>90° to <109.5° 109.5° >109.5 to <120° 120° >120° to <180°


Question 8 12 points

Using the **valence shell orbitals only** draw a **complete** molecular orbital (MO) diagram (with no s-p mixing) for the molecule **PS**, in the space provided below. **(6 points)**

Assume that 3s and 3p atomic orbitals will form molecular orbitals like those formed by the combination of 2s and 2p atomic orbitals that we learnt in class, and there is no s-p mixing.

Show **energy arrow**, **atomic orbitals** involved, **atom labels**, the **molecular orbitals** that are formed, **filled electrons** in the MO diagram etc. Make sure to label all the atomic and molecular orbitals.

Using the MO diagram above, answer the following questions:

- (i) Which one of the following will have the highest bond order? Circle the correct choice
 A) PS^{1-} B) PS C) PS^{1+}
- (ii) Which one of the following will have the longest bond length? Circle the correct choice
 A) PS^{1-} B) PS C) PS^{1+}
- (iii) Which of the following will be diamagnetic? Circle the correct choice.
 A) PS^{1-} B) PS C) PS^{1+}
- (iv) Draw Lewis dot structure of the molecule with the shortest bond length (based on the MO diagram) from among the three (PS^{1-} , PS, PS^{1+}), and indicate the bond order calculated from the Lewis dot structure. **(3 points)**

Lewis Dot Structure

Bond Order



1AC73D18-679A-4097-8936-71F36DFE22B3

midterm-exam-93388

#143 Page 10 of 10

SCRAP PAGE

**CHEMISTRY 110 –Midterm Exam****Thursday November 7, 2024, 6:30 – 8:30 pm**

Examiners: Pallavi Sirjoosingh and Ashok Kakkar

First name (please write as legibly as possible within the boxes)

--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--

Last name

--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--

Student ID number

--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--

This booklet has 8 questions on 10 pages (The last page can be used as a scrap page).

The staple must NOT be removed from the question booklet. Your intact question booklet must be returned at the end of the exam.

A separate booklet contains a periodic table, and datasheet.

Write your answers directly under the questions in the space provided. Do not write on the QR codes or near the top edges (as it may be cut off).

Please write **legibly and dark enough** for the scan to be readable. For students who use English as a second language, dictionaries are permitted. Calculators are allowed.

DO NOT ASK QUESTIONS OF THE INVIGILATORS about interpretations of the exam questions. If you have a concern about a question, please summarize it at the top of this page. All concerns will be evaluated after the exam. GOOD LUCK!

**Question 1 8 points** (*Assume all transitions are possible*)

a. 6 points Calculate the **energy (in kJ/mol)** required to ionize (completely remove) the electron from the **$n=3$** level of **He^+ ion** (He: Helium). Show your work including equations. (3 sig. figs.)

b. 2 points

(I) Select the largest (longest) wavelength from the three choices below:

- i. Wavelength of photon to *completely remove* an electron from *1s orbital of Hydrogen atom*
- ii. Wavelength of photon to *completely remove* an electron from *1s orbital of Boron atom (B)*
- iii. Wavelength of photon to *completely remove* an electron from *1s orbital of Boron cation (B^+)*

i or ii or iii

(II) Select the smallest (shortest) wavelength from the three choices below (*for H atom*):

- i. Wavelength of photon *emitted* for an electron transitioning from *2p to 1s orbital*
- ii. Wavelength of photon *absorbed* for an electron transitioning from *2s to 4p orbital*
- iii. Wavelength of photon *absorbed* for an electron transitioning from *1s to 3p orbital*

i or ii or iii



Question 2 8 points



A Chem110 student plots the graph shown on the left, from data obtained for the kinetic energy (eV) of the ejected electron versus frequency (s^{-1}) of the incident light, for an unknown metal. ($1\text{eV} = 1.6022 \times 10^{-19} \text{ J}$)

Based on the plot, answer the questions **a to c** below:

a. 1 point On the plot above, label the **threshold frequency**.

b. 2 points What is the value of the work potential (in J) of the unknown metal?

c. 3 points What is the *longest* wavelength (in nm) required to eject an electron from this metal's surface? Show your work including equation(s) used. (3 sig figs)

d. 2 points In the space provided below, plot the following

(Assume frequency is *above* threshold frequency):

- i. *Kinetic energy* of an ejected electron versus the *intensity* of the incident light (left)
- ii. *Number of electrons* ejected versus the *frequency* of the incident light (right)



**Question 3** **6 points****a. 2 points**

In the ground state of Ge (Germanium; $Z = 32$) atom, **name** the following orbitals:

i. Orbital containing the electron that requires the maximum energy to be removed.

ii. Orbital containing the electron that experiences the minimum effective nuclear charge.

b. 2 points

An unknown anion (A^-) has the following *excited* electronic configuration: $[Ar] 4s^2 3d^3 4p^1$

(Ar: Argon; $Z = 18$)

Give the **condensed ground state electronic configuration** of the atom, **A** and the cation, **A^{2+}** .
Show your work. (Assume all electronic transitions are possible)

Electronic Configuration of A:

Electronic Configuration of A^{2+} :

c. 2 points

In a single atom, determine how many electrons can have the following set of quantum numbers?

(i) $n = 5; l = 3$

In a single atom, determine how many electrons can have the following set of quantum numbers?

(i) $n = 3; l = 2; m_s = +1/2$

**Question 4** **8 points**

Rank the following from 1 to 4 in **increasing** order. Provide the ranking in the boxes provided.

Note: Atomic number (Z) are provided for relevant atoms

a) Increasing (lowest to highest) order of *ionization energy*: (1= lowest; 4 = highest)

Mg

Mg: Z = 12

Rb

Rb: Z = 37

Cs

Cs: Z = 55

K

K : Z = 19

b) Increasing (lowest to highest) order of *bond angle*: (1= lowest; 4 = highest)

H₂OCCl₄XeF₄

HCN

c) Increasing (smallest to largest) order of *ionic size* (1= smallest; 4 = largest)

S²⁻

S: Z = 16

O²⁻

O: Z = 8

Be²⁺

Be: Z = 4

Na⁺

Na: Z = 11

d) Increasing (lowest to highest) order of *bond energy* (1= lowest; 4 = highest)

P-F

P-Cl

P-I

P-Br

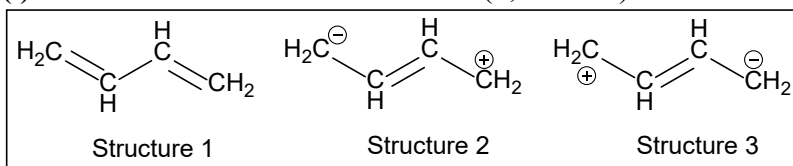
**Question 5** **12 points**

a. 9 points For the molecules HClO_4 and TeF_6 , draw the most stable *Lewis structure* showing appropriate shape and *dashed/wedged bonds*, based on VSEPR. Determine the **molecular geometry** around the central atom. For HClO_4 , label the *most polar bond*.



b. 3 points (Note: Lone pairs are not shown in structures below)

(i) For the three *resonance* structures (1, 2 and 3) shown below, answer the following question:



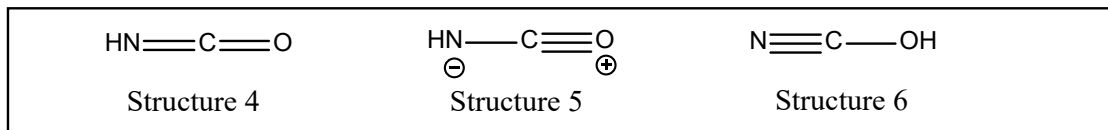
(I) Circle the *most contributing* resonance structure.

Structure 1

Structure 2

Structure 3

(ii) For the three structures given (4, 5, and 6) below, answer the following questions:



(I) Circle all the structures that represent *resonance structures* of each other:

Structure 4

Structure 5

Structure 6

(II) Among the resonance structures, circle the *most contributing* resonance structure.

Structure 4

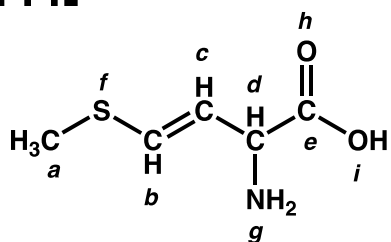
Structure 5

Structure 6

**Question 6** **6 points**

Using the data given below, calculate the **lattice energy** for SrO(s).

	ΔH° (kJ/mol)
$\text{Sr}(s) \longrightarrow \text{Sr}(g)$	+180
$\text{Sr}(g) \longrightarrow \text{Sr}^{2+}(g) + 2e^-$	+1468
$\text{O}_2(g) \longrightarrow 2\text{O}(g)$	+500
$\text{O}(g) + 2e^- \longrightarrow \text{O}^{2-}(g)$	+600
$\text{Sr}(s) + \frac{1}{2}\text{O}_2(g) \longrightarrow \text{SrO}(s)$	-550

**Question 7 10 points**

Consider the molecule given on the left, and answer questions (i) to (vi) using chemical bonding theories that we have learnt in class (valence bond theory, hybridization).

Note that the lone pairs are not shown in the structure:

i. Consider the carbon atoms marked in as ***a***, ***b***, ***c***, ***d***, and ***e*** in the molecule, and determine **hybridization** at each of the carbon centers. (2.5 points)

Atom	Hybridization
Carbon (<i>a</i>)	
Carbon (<i>b</i>)	
Carbon (<i>c</i>)	
Carbon (<i>d</i>)	
Carbon (<i>e</i>)	

ii. Now, consider the sulfur atom (S) marked as ***f***, nitrogen (N) atom (***g***), and oxygen(O) atoms (***h*** and ***i***), and determine their **hybridization**. (2 points)

Atom	Hybridization
Sulfur (<i>f</i>)	
Nitrogen (<i>g</i>)	
Oxygen (<i>h</i>)	
Oxygen (<i>i</i>)	

iii. **Describe** all the bonds formed by the carbon atoms ***a*** and ***b***, using sigma and pi designation as appropriate. (2 points)

Atom	Bonds formed (use sigma and pi as appropriate)
Carbon (<i>a</i>)	
Carbon (<i>b</i>)	

iv. How many total sp^3 -s sigma bonds are in the molecule? (1 point)

v. Determine **electronic** and **molecular geometry** at carbon (C) labeled ***e*** and sulfur (S) labeled ***f*** centres. (2 points)

Atom	Electronic Geometry	Molecular Geometry
Carbon (<i>e</i>)		
Sulfur (<i>f</i>)		

vi. Circle the value from the choices given below, that describes the **C-S-C bond angle at sulfur** (S) marked ***f***: (0.5 points)

>90° to <109.5° 109.5° >109.5 to <120° 120° >120° to <180°

**Question 8 12 points**

Using the **valence shell orbitals only** draw a **complete** molecular orbital (MO) diagram (with no s-p mixing) for the molecule **PS**, in the space provided below. **(6 points)**

Assume that 3s and 3p atomic orbitals will form molecular orbitals like those formed by the combination of 2s and 2p atomic orbitals that we learnt in class, and there is no s-p mixing.

Show **energy arrow**, **atomic orbitals** involved, **atom labels**, the **molecular orbitals** that are formed, **filled electrons** in the MO diagram etc. Make sure to label all the atomic and molecular orbitals.

Using the MO diagram above, answer the following questions:

- (i) Which one of the following will have the highest bond order? Circle the correct choice
A) PS^{1-} B) PS C) PS^{1+}
- (ii) Which one of the following will have the longest bond length? Circle the correct choice
A) PS^{1-} B) PS C) PS^{1+}
- (iii) Which of the following will be diamagnetic? Circle the correct choice.
A) PS^{1-} B) PS C) PS^{1+}
- (iv) Draw Lewis dot structure of the molecule with the shortest bond length (based on the MO diagram) from among the three (PS^{1-} , PS, PS^{1+}), and indicate the bond order calculated from the Lewis dot structure. **(3 points)**

Lewis Dot Structure

Bond Order



20984B0B-25A0-4E1D-866C-DBF34CE0DF71

midterm-exam-93388

#144 Page 10 of 10

SCRAP PAGE

**CHEMISTRY 110 –Midterm Exam****Thursday November 7, 2024, 6:30 – 8:30 pm**

Examiners: Pallavi Sirjoosingh and Ashok Kakkar

First name (please write as legibly as possible within the boxes)

--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--

Last name

--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--

Student ID number

--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--

This booklet has 8 questions on 10 pages (The last page can be used as a scrap page).

The staple must NOT be removed from the question booklet. Your intact question booklet must be returned at the end of the exam.

A separate booklet contains a periodic table, and datasheet.

Write your answers directly under the questions in the space provided. Do not write on the QR codes or near the top edges (as it may be cut off).

Please write **legibly and dark enough** for the scan to be readable. For students who use English as a second language, dictionaries are permitted. Calculators are allowed.

DO NOT ASK QUESTIONS OF THE INVIGILATORS about interpretations of the exam questions. If you have a concern about a question, please summarize it at the top of this page. All concerns will be evaluated after the exam. GOOD LUCK!

**Question 1 8 points** (*Assume all transitions are possible*)

a. 6 points Calculate the **energy (in kJ/mol)** required to ionize (completely remove) the electron from the **n=3** level of **He⁺ ion** (He: Helium). Show your work including equations. (3 sig. figs.)

b. 2 points

(I) Select the largest (longest) wavelength from the three choices below:

- i. Wavelength of photon to *completely remove* an electron from *1s orbital of Hydrogen atom*
- ii. Wavelength of photon to *completely remove* an electron from *1s orbital of Boron atom (B)*
- iii. Wavelength of photon to *completely remove* an electron from *1s orbital of Boron cation (B⁺)*

i or ii or iii

(II) Select the smallest (shortest) wavelength from the three choices below (*for H atom*):

- i. Wavelength of photon *emitted* for an electron transitioning from *2p to 1s orbital*
- ii. Wavelength of photon *absorbed* for an electron transitioning from *2s to 4p orbital*
- iii. Wavelength of photon *absorbed* for an electron transitioning from *1s to 3p orbital*

i or ii or iii



Question 2 8 points



A Chem110 student plots the graph shown on the left, from data obtained for the kinetic energy (eV) of the ejected electron versus frequency (s^{-1}) of the incident light, for an unknown metal. ($1\text{eV} = 1.6022 \times 10^{-19} \text{ J}$)

Based on the plot, answer the questions **a to c** below:

a. 1 point On the plot above, label the **threshold frequency**.

b. 2 points What is the value of the work potential (in J) of the unknown metal?

c. 3 points What is the *longest* wavelength (in nm) required to eject an electron from this metal's surface? Show your work including equation(s) used. (3 sig figs)

d. 2 points In the space provided below, plot the following

(Assume frequency is *above* threshold frequency):

- i. *Kinetic energy* of an ejected electron versus the *intensity* of the incident light (left)
- ii. *Number of electrons* ejected versus the *frequency* of the incident light (right)



**Question 3** **6 points****a. 2 points**

In the ground state of Ge (Germanium; $Z = 32$) atom, **name** the following orbitals:

i. Orbital containing the electron that requires the maximum energy to be removed.

ii. Orbital containing the electron that experiences the minimum effective nuclear charge.

b. 2 points

An unknown anion (A^-) has the following *excited* electronic configuration: $[Ar] 4s^2 3d^3 4p^1$

(Ar: Argon; $Z = 18$)

Give the **condensed ground state electronic configuration** of the atom, **A** and the cation, **A^{2+}** .
Show your work. (Assume all electronic transitions are possible)

Electronic Configuration of A:

Electronic Configuration of A^{2+} :

c. 2 points

In a single atom, determine how many electrons can have the following set of quantum numbers?

(i) $n = 5; l = 3$

In a single atom, determine how many electrons can have the following set of quantum numbers?

(i) $n = 3; l = 2; m_s = +1/2$

**Question 4** **8 points**

Rank the following from 1 to 4 in **increasing** order. Provide the ranking in the boxes provided.

Note: Atomic number (Z) are provided for relevant atoms

a) Increasing (lowest to highest) order of *ionization energy*: (1= lowest; 4 = highest)

Mg

Mg: Z = 12

Rb

Rb: Z = 37

Cs

Cs: Z = 55

K

K : Z = 19

b) Increasing (lowest to highest) order of *bond angle*: (1= lowest; 4 = highest)

H₂OCCl₄XeF₄

HCN

c) Increasing (smallest to largest) order of *ionic size* (1= smallest; 4 = largest)

S²⁻

S: Z = 16

O²⁻

O: Z = 8

Be²⁺

Be: Z = 4

Na⁺

Na: Z = 11

d) Increasing (lowest to highest) order of *bond energy* (1= lowest; 4 = highest)

P-F

P-Cl

P-I

P-Br

**Question 5** **12 points**

a. 9 points For the molecules HClO_4 and TeF_6 , draw the most stable *Lewis structure* showing appropriate shape and *dashed/wedged bonds*, based on VSEPR. Determine the **molecular geometry** around the central atom. For HClO_4 , label the *most polar bond*.



b. 3 points (Note: Lone pairs are not shown in structures below)

(i) For the three *resonance* structures (1, 2 and 3) shown below, answer the following question:



(I) Circle the *most contributing* resonance structure.

Structure 1

Structure 2

Structure 3

(ii) For the three structures given (4, 5, and 6) below, answer the following questions:



(I) Circle all the structures that represent *resonance structures* of each other:

Structure 4

Structure 5

Structure 6

(II) Among the resonance structures, circle the *most contributing* resonance structure.

Structure 4

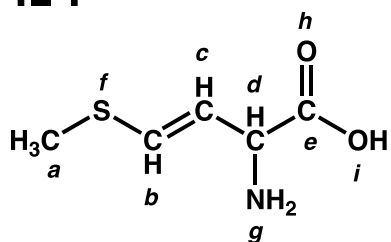
Structure 5

Structure 6

**Question 6** **6 points**

Using the data given below, calculate the **lattice energy** for SrO(s).

	ΔH° (kJ/mol)
$\text{Sr}(s) \longrightarrow \text{Sr}(g)$	+180
$\text{Sr}(g) \longrightarrow \text{Sr}^{2+}(g) + 2e^-$	+1468
$\text{O}_2(g) \longrightarrow 2\text{O}(g)$	+500
$\text{O}(g) + 2e^- \longrightarrow \text{O}^{2-}(g)$	+600
$\text{Sr}(s) + \frac{1}{2}\text{O}_2(g) \longrightarrow \text{SrO}(s)$	-550

**Question 7 10 points**

Consider the molecule given on the left, and answer questions (i) to (vi) using chemical bonding theories that we have learnt in class (valence bond theory, hybridization).

Note that the lone pairs are not shown in the structure:

i. Consider the carbon atoms marked in as ***a***, ***b***, ***c***, ***d***, and ***e*** in the molecule, and determine **hybridization** at each of the carbon centers. (2.5 points)

Atom	Hybridization
Carbon (<i>a</i>)	
Carbon (<i>b</i>)	
Carbon (<i>c</i>)	
Carbon (<i>d</i>)	
Carbon (<i>e</i>)	

ii. Now, consider the sulfur atom (S) marked as ***f***, nitrogen (N) atom (***g***), and oxygen(O) atoms (***h*** and ***i***), and determine their **hybridization**. (2 points)

Atom	Hybridization
Sulfur (<i>f</i>)	
Nitrogen (<i>g</i>)	
Oxygen (<i>h</i>)	
Oxygen (<i>i</i>)	

iii. **Describe** all the bonds formed by the carbon atoms ***a*** and ***b***, using sigma and pi designation as appropriate. (2 points)

Atom	Bonds formed (use sigma and pi as appropriate)
Carbon (<i>a</i>)	
Carbon (<i>b</i>)	

iv. How many total sp^3 -s sigma bonds are in the molecule? (1 point)

v. Determine **electronic** and **molecular geometry** at carbon (C) labeled ***e*** and sulfur (S) labeled ***f*** centres. (2 points)

Atom	Electronic Geometry	Molecular Geometry
Carbon (<i>e</i>)		
Sulfur (<i>f</i>)		

vi. Circle the value from the choices given below, that describes the **C-S-C bond angle at sulfur** (S) marked ***f***: (0.5 points)

>90° to <109.5° 109.5° >109.5 to <120° 120° >120° to <180°


Question 8 12 points

Using the **valence shell orbitals only** draw a **complete** molecular orbital (MO) diagram (with no s-p mixing) for the molecule **PS**, in the space provided below. **(6 points)**

Assume that 3s and 3p atomic orbitals will form molecular orbitals like those formed by the combination of 2s and 2p atomic orbitals that we learnt in class, and there is no s-p mixing.

Show **energy arrow**, **atomic orbitals** involved, **atom labels**, the **molecular orbitals** that are formed, **filled electrons** in the MO diagram etc. Make sure to label all the atomic and molecular orbitals.

Using the MO diagram above, answer the following questions:

- (i) Which one of the following will have the highest bond order? Circle the correct choice
 A) PS^{1-} B) PS C) PS^{1+}
- (ii) Which one of the following will have the longest bond length? Circle the correct choice
 A) PS^{1-} B) PS C) PS^{1+}
- (iii) Which of the following will be diamagnetic? Circle the correct choice.
 A) PS^{1-} B) PS C) PS^{1+}
- (iv) Draw Lewis dot structure of the molecule with the shortest bond length (based on the MO diagram) from among the three (PS^{1-} , PS, PS^{1+}), and indicate the bond order calculated from the Lewis dot structure. **(3 points)**

Lewis Dot Structure

Bond Order



BEEB29DA-52DE-4828-A164-54C6677D9C62

midterm-exam-93388

#145 Page 10 of 10

SCRAP PAGE

**CHEMISTRY 110 –Midterm Exam****Thursday November 7, 2024, 6:30 – 8:30 pm****Examiners: Pallavi Sirjoosingh and Ashok Kakkar**

First name (please write as legibly as possible within the boxes)

--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--

Last name

--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--

Student ID number

--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--

This booklet has 8 questions on 10 pages (The last page can be used as a scrap page).

The staple must NOT be removed from the question booklet. Your intact question booklet must be returned at the end of the exam.

A separate booklet contains a periodic table, and datasheet.

Write your answers directly under the questions in the space provided. Do not write on the QR codes or near the top edges (as it may be cut off).

Please write **legibly and dark enough** for the scan to be readable. For students who use English as a second language, dictionaries are permitted. Calculators are allowed.

DO NOT ASK QUESTIONS OF THE INVIGILATORS about interpretations of the exam questions. If you have a concern about a question, please summarize it at the top of this page. All concerns will be evaluated after the exam. GOOD LUCK!

**Question 1 8 points** (*Assume all transitions are possible*)

a. 6 points Calculate the **energy (in kJ/mol)** required to ionize (completely remove) the electron from the **$n=3$** level of **He^+ ion** (He: Helium). Show your work including equations. (3 sig. figs.)

b. 2 points

(I) Select the largest (longest) wavelength from the three choices below:

- i. Wavelength of photon to *completely remove* an electron from *1s orbital of Hydrogen atom*
- ii. Wavelength of photon to *completely remove* an electron from *1s orbital of Boron atom (B)*
- iii. Wavelength of photon to *completely remove* an electron from *1s orbital of Boron cation (B^+)*

i or ii or iii

(II) Select the smallest (shortest) wavelength from the three choices below (*for H atom*):

- i. Wavelength of photon *emitted* for an electron transitioning from *2p to 1s orbital*
- ii. Wavelength of photon *absorbed* for an electron transitioning from *2s to 4p orbital*
- iii. Wavelength of photon *absorbed* for an electron transitioning from *1s to 3p orbital*

i or ii or iii



Question 2 8 points



A Chem110 student plots the graph shown on the left, from data obtained for the kinetic energy (eV) of the ejected electron versus frequency (s^{-1}) of the incident light, for an unknown metal. ($1\text{eV} = 1.6022 \times 10^{-19} \text{ J}$)

Based on the plot, answer the questions **a to c** below:

a. 1 point On the plot above, label the **threshold frequency**.

b. 2 points What is the value of the work potential (in J) of the unknown metal?

c. 3 points What is the *longest* wavelength (in nm) required to eject an electron from this metal's surface? Show your work including equation(s) used. (3 sig figs)

d. 2 points In the space provided below, plot the following

(Assume frequency is *above* threshold frequency):

- i. *Kinetic energy* of an ejected electron versus the *intensity* of the incident light (left)
- ii. *Number of electrons* ejected versus the *frequency* of the incident light (right)



**Question 3** **6 points****a. 2 points**

In the ground state of Ge (Germanium; $Z = 32$) atom, **name** the following orbitals:

i. Orbital containing the electron that requires the maximum energy to be removed.

ii. Orbital containing the electron that experiences the minimum effective nuclear charge.

b. 2 points

An unknown anion (A^-) has the following *excited* electronic configuration: $[Ar] 4s^2 3d^3 4p^1$

(Ar: Argon; $Z = 18$)

Give the **condensed ground state electronic configuration** of the atom, **A** and the cation, **A^{2+}** .
Show your work. (Assume all electronic transitions are possible)

Electronic Configuration of A:

Electronic Configuration of A^{2+} :

c. 2 points

In a single atom, determine how many electrons can have the following set of quantum numbers?

(i) $n = 5; l = 3$

In a single atom, determine how many electrons can have the following set of quantum numbers?

(i) $n = 3; l = 2; m_s = +1/2$

**Question 4** **8 points**

Rank the following from 1 to 4 in **increasing** order. Provide the ranking in the boxes provided.

Note: Atomic number (Z) are provided for relevant atoms

a) Increasing (lowest to highest) order of *ionization energy*: (1= lowest; 4 = highest)

Mg

Mg: Z = 12

Rb

Rb: Z = 37

Cs

Cs: Z = 55

K

K : Z = 19

b) Increasing (lowest to highest) order of *bond angle*: (1= lowest; 4 = highest)

H₂OCCl₄XeF₄

HCN

c) Increasing (smallest to largest) order of *ionic size* (1= smallest; 4 = largest)

S²⁻

S: Z = 16

O²⁻

O: Z = 8

Be²⁺

Be: Z = 4

Na⁺

Na: Z = 11

d) Increasing (lowest to highest) order of *bond energy* (1= lowest; 4 = highest)

P-F

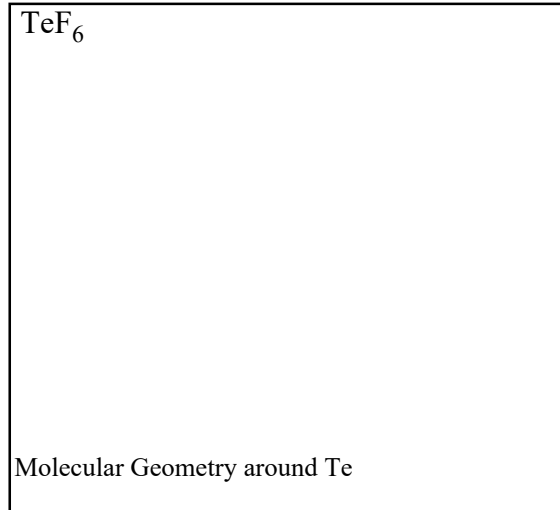
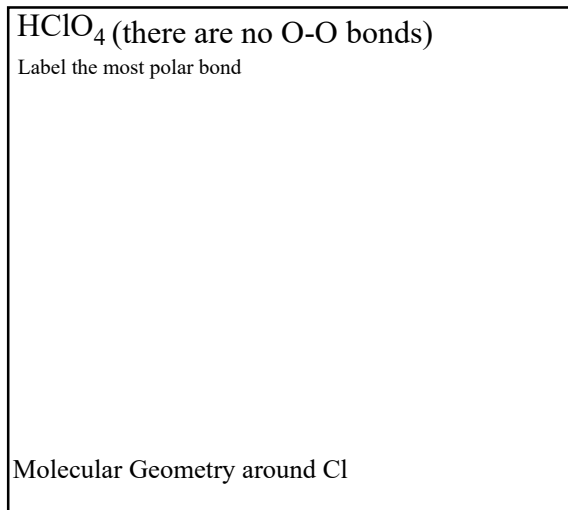
P-Cl

P-I

P-Br

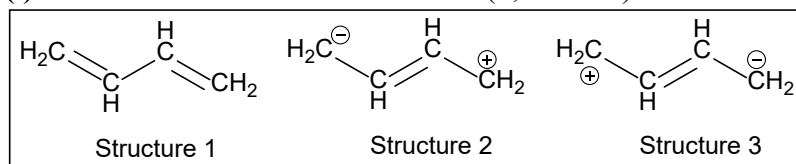
**Question 5** **12 points**

a. 9 points For the molecules HClO_4 and TeF_6 , draw the most stable *Lewis structure* showing appropriate shape and *dashed/wedged bonds*, based on VSEPR. Determine the **molecular geometry** around the central atom. For HClO_4 , label the *most polar bond*.



b. 3 points (Note: Lone pairs are not shown in structures below)

(i) For the three *resonance* structures (1, 2 and 3) shown below, answer the following question:



(I) Circle the *most contributing* resonance structure.

Structure 1

Structure 2

Structure 3

(ii) For the three structures given (4, 5, and 6) below, answer the following questions:



(I) Circle all the structures that represent *resonance structures* of each other:

Structure 4

Structure 5

Structure 6

(II) Among the resonance structures, circle the *most contributing* resonance structure.

Structure 4

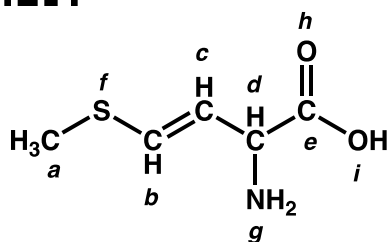
Structure 5

Structure 6

**Question 6** **6 points**

Using the data given below, calculate the **lattice energy** for SrO(s).

	ΔH° (kJ/mol)
$\text{Sr}(s) \longrightarrow \text{Sr}(g)$	+180
$\text{Sr}(g) \longrightarrow \text{Sr}^{2+}(g) + 2e^-$	+1468
$\text{O}_2(g) \longrightarrow 2\text{O}(g)$	+500
$\text{O}(g) + 2e^- \longrightarrow \text{O}^{2-}(g)$	+600
$\text{Sr}(s) + \frac{1}{2}\text{O}_2(g) \longrightarrow \text{SrO}(s)$	-550

**Question 7 10 points**

Consider the molecule given on the left, and answer questions (i) to (vi) using chemical bonding theories that we have learnt in class (valence bond theory, hybridization).

Note that the lone pairs are not shown in the structure:

i. Consider the carbon atoms marked in as ***a***, ***b***, ***c***, ***d***, and ***e*** in the molecule, and determine **hybridization** at each of the carbon centers. (2.5 points)

Atom	Hybridization
Carbon (<i>a</i>)	
Carbon (<i>b</i>)	
Carbon (<i>c</i>)	
Carbon (<i>d</i>)	
Carbon (<i>e</i>)	

ii. Now, consider the sulfur atom (S) marked as ***f***, nitrogen (N) atom (***g***), and oxygen(O) atoms (***h*** and ***i***), and determine their **hybridization**. (2 points)

Atom	Hybridization
Sulfur (<i>f</i>)	
Nitrogen (<i>g</i>)	
Oxygen (<i>h</i>)	
Oxygen (<i>i</i>)	

iii. **Describe** all the bonds formed by the carbon atoms ***a*** and ***b***, using sigma and pi designation as appropriate. (2 points)

Atom	Bonds formed (use sigma and pi as appropriate)
Carbon (<i>a</i>)	
Carbon (<i>b</i>)	

iv. How many total sp^3 -s sigma bonds are in the molecule? (1 point)

v. Determine **electronic** and **molecular geometry** at carbon (C) labeled ***e*** and sulfur (S) labeled ***f*** centres. (2 points)

Atom	Electronic Geometry	Molecular Geometry
Carbon (<i>e</i>)		
Sulfur (<i>f</i>)		

vi. Circle the value from the choices given below, that describes the **C-S-C bond angle at sulfur** (S) marked ***f***: (0.5 points)

$>90^\circ$ to $<109.5^\circ$ 109.5° >109.5 to $<120^\circ$ 120° $>120^\circ$ to $<180^\circ$



Question 8 12 points

Using the **valence shell orbitals only** draw a **complete** molecular orbital (MO) diagram (with no s-p mixing) for the molecule **PS**, in the space provided below. **(6 points)**

Assume that 3s and 3p atomic orbitals will form molecular orbitals like those formed by the combination of 2s and 2p atomic orbitals that we learnt in class, and there is no s-p mixing.

Show **energy arrow**, **atomic orbitals** involved, **atom labels**, the **molecular orbitals** that are formed, **filled electrons** in the MO diagram etc. Make sure to label all the atomic and molecular orbitals.

Using the MO diagram above, answer the following questions:

- (i) Which one of the following will have the highest bond order? Circle the correct choice
 A) PS^{1-} B) PS C) PS^{1+}
- (ii) Which one of the following will have the longest bond length? Circle the correct choice
 A) PS^{1-} B) PS C) PS^{1+}
- (iii) Which of the following will be diamagnetic? Circle the correct choice.
 A) PS^{1-} B) PS C) PS^{1+}
- (iv) Draw Lewis dot structure of the molecule with the shortest bond length (based on the MO diagram) from among the three (PS^{1-} , PS, PS^{1+}), and indicate the bond order calculated from the Lewis dot structure. **(3 points)**

Lewis Dot Structure

Bond Order



A346A08F-60D6-4BA3-860D-14702706A27B

midterm-exam-93388

#146 Page 10 of 10

SCRAP PAGE

**CHEMISTRY 110 –Midterm Exam****Thursday November 7, 2024, 6:30 – 8:30 pm****Examiners: Pallavi Sirjoosingh and Ashok Kakkar**

First name (please write as legibly as possible within the boxes)

--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--

Last name

--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--

Student ID number

--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--

This booklet has 8 questions on 10 pages (The last page can be used as a scrap page).

The staple must NOT be removed from the question booklet. Your intact question booklet must be returned at the end of the exam.

A separate booklet contains a periodic table, and datasheet.

Write your answers directly under the questions in the space provided. Do not write on the QR codes or near the top edges (as it may be cut off).

Please write **legibly and dark enough** for the scan to be readable. For students who use English as a second language, dictionaries are permitted. Calculators are allowed.

DO NOT ASK QUESTIONS OF THE INVIGILATORS about interpretations of the exam questions. If you have a concern about a question, please summarize it at the top of this page. All concerns will be evaluated after the exam. GOOD LUCK!

**Question 1 8 points** (*Assume all transitions are possible*)

a. 6 points Calculate the **energy (in kJ/mol)** required to ionize (completely remove) the electron from the **$n=3$** level of **He^+ ion** (He: Helium). Show your work including equations. (3 sig. figs.)

b. 2 points

(I) Select the largest (longest) wavelength from the three choices below:

- i. Wavelength of photon to *completely remove* an electron from *1s orbital of Hydrogen atom*
- ii. Wavelength of photon to *completely remove* an electron from *1s orbital of Boron atom (B)*
- iii. Wavelength of photon to *completely remove* an electron from *1s orbital of Boron cation (B^+)*

i or ii or iii

(II) Select the smallest (shortest) wavelength from the three choices below (*for H atom*):

- i. Wavelength of photon *emitted* for an electron transitioning from *2p to 1s orbital*
- ii. Wavelength of photon *absorbed* for an electron transitioning from *2s to 4p orbital*
- iii. Wavelength of photon *absorbed* for an electron transitioning from *1s to 3p orbital*

i or ii or iii



Question 2 8 points



A Chem110 student plots the graph shown on the left, from data obtained for the kinetic energy (eV) of the ejected electron versus frequency (s^{-1}) of the incident light, for an unknown metal. ($1\text{eV} = 1.6022 \times 10^{-19} \text{ J}$)

Based on the plot, answer the questions **a to c** below:

a. 1 point On the plot above, label the **threshold frequency**.

b. 2 points What is the value of the work potential (in J) of the unknown metal?

c. 3 points What is the *longest* wavelength (in nm) required to eject an electron from this metal's surface? Show your work including equation(s) used. (3 sig figs)

d. 2 points In the space provided below, plot the following

(Assume frequency is *above* threshold frequency):

- Kinetic energy* of an ejected electron versus the *intensity* of the incident light (left)
- Number of electrons* ejected versus the *frequency* of the incident light (right)



**Question 3** **6 points****a. 2 points**

In the ground state of Ge (Germanium; $Z = 32$) atom, **name** the following orbitals:

i. Orbital containing the electron that requires the maximum energy to be removed.

ii. Orbital containing the electron that experiences the minimum effective nuclear charge.

b. 2 points

An unknown anion (A^-) has the following *excited* electronic configuration: $[Ar] 4s^2 3d^3 4p^1$

(Ar: Argon; $Z = 18$)

Give the **condensed ground state electronic configuration** of the atom, **A** and the cation, **A^{2+}** .
Show your work. (Assume all electronic transitions are possible)

Electronic Configuration of A:

Electronic Configuration of A^{2+} :

c. 2 points

In a single atom, determine how many electrons can have the following set of quantum numbers?

(i) $n = 5; l = 3$

In a single atom, determine how many electrons can have the following set of quantum numbers?

(i) $n = 3; l = 2; m_s = +1/2$

**Question 4** **8 points**

Rank the following from 1 to 4 in **increasing** order. Provide the ranking in the boxes provided.

Note: Atomic number (Z) are provided for relevant atoms

a) Increasing (lowest to highest) order of *ionization energy*: (1= lowest; 4 = highest)

Mg

Mg: Z = 12

Rb

Rb: Z = 37

Cs

Cs: Z = 55

K

K : Z = 19

b) Increasing (lowest to highest) order of *bond angle*: (1= lowest; 4 = highest)

H₂OCCl₄XeF₄

HCN

c) Increasing (smallest to largest) order of *ionic size* (1= smallest; 4 = largest)

S²⁻

S: Z = 16

O²⁻

O: Z = 8

Be²⁺

Be: Z = 4

Na⁺

Na: Z = 11

d) Increasing (lowest to highest) order of *bond energy* (1= lowest; 4 = highest)

P-F

P-Cl

P-I

P-Br

**Question 5** **12 points**

a. 9 points For the molecules HClO_4 and TeF_6 , draw the most stable *Lewis structure* showing appropriate shape and *dashed/wedged bonds*, based on VSEPR. Determine the **molecular geometry** around the central atom. For HClO_4 , label the *most polar bond*.

HClO_4 (there are no O-O bonds)

Label the most polar bond

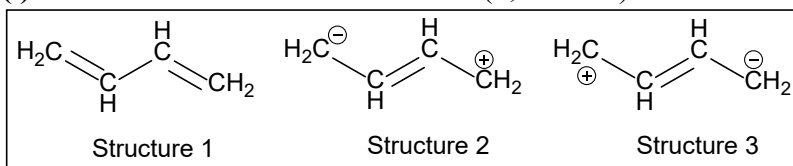
Molecular Geometry around Cl

TeF_6

Molecular Geometry around Te

b. 3 points (Note: Lone pairs are not shown in structures below)

(i) For the three *resonance* structures (1, 2 and 3) shown below, answer the following question:



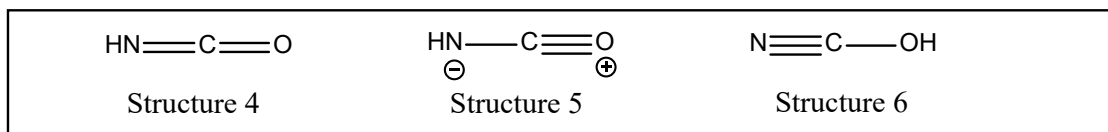
(I) Circle the *most contributing* resonance structure.

Structure 1

Structure 2

Structure 3

(ii) For the three structures given (4, 5, and 6) below, answer the following questions:



(I) Circle all the structures that represent *resonance structures* of each other:

Structure 4

Structure 5

Structure 6

(II) Among the resonance structures, circle the *most contributing* resonance structure.

Structure 4

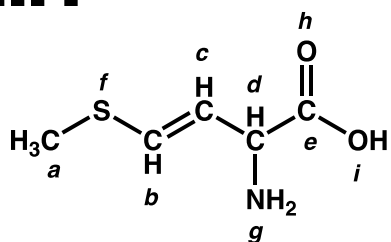
Structure 5

Structure 6

**Question 6** **6 points**

Using the data given below, calculate the **lattice energy** for SrO(s).

	ΔH° (kJ/mol)
$\text{Sr}(s) \longrightarrow \text{Sr}(g)$	+180
$\text{Sr}(g) \longrightarrow \text{Sr}^{2+}(g) + 2e^-$	+1468
$\text{O}_2(g) \longrightarrow 2\text{O}(g)$	+500
$\text{O}(g) + 2e^- \longrightarrow \text{O}^{2-}(g)$	+600
$\text{Sr}(s) + \frac{1}{2}\text{O}_2(g) \longrightarrow \text{SrO}(s)$	-550

**Question 7 10 points**

Consider the molecule given on the left, and answer questions (i) to (vi) using chemical bonding theories that we have learnt in class (valence bond theory, hybridization).

Note that the lone pairs are not shown in the structure:

i. Consider the carbon atoms marked in as ***a***, ***b***, ***c***, ***d***, and ***e*** in the molecule, and determine **hybridization** at each of the carbon centers. (2.5 points)

Atom	Hybridization
Carbon (<i>a</i>)	
Carbon (<i>b</i>)	
Carbon (<i>c</i>)	
Carbon (<i>d</i>)	
Carbon (<i>e</i>)	

ii. Now, consider the sulfur atom (S) marked as ***f***, nitrogen (N) atom (***g***), and oxygen(O) atoms (***h*** and ***i***), and determine their **hybridization**. (2 points)

Atom	Hybridization
Sulfur (<i>f</i>)	
Nitrogen (<i>g</i>)	
Oxygen (<i>h</i>)	
Oxygen (<i>i</i>)	

iii. **Describe** all the bonds formed by the carbon atoms ***a*** and ***b***, using sigma and pi designation as appropriate. (2 points)

Atom	Bonds formed (use sigma and pi as appropriate)
Carbon (<i>a</i>)	
Carbon (<i>b</i>)	

iv. How many total sp^3 -s sigma bonds are in the molecule? (1 point)

v. Determine **electronic** and **molecular geometry** at carbon (C) labeled ***e*** and sulfur (S) labeled ***f*** centres. (2 points)

Atom	Electronic Geometry	Molecular Geometry
Carbon (<i>e</i>)		
Sulfur (<i>f</i>)		

vi. Circle the value from the choices given below, that describes the **C-S-C bond angle at sulfur** (S) marked ***f***: (0.5 points)

>90° to <109.5° 109.5° >109.5 to <120° 120° >120° to <180°

**Question 8 12 points**

Using the **valence shell orbitals only** draw a **complete** molecular orbital (MO) diagram (with no s-p mixing) for the molecule **PS**, in the space provided below. **(6 points)**

Assume that 3s and 3p atomic orbitals will form molecular orbitals like those formed by the combination of 2s and 2p atomic orbitals that we learnt in class, and there is no s-p mixing.

Show **energy arrow**, **atomic orbitals** involved, **atom labels**, the **molecular orbitals** that are formed, **filled electrons** in the MO diagram etc. Make sure to label all the atomic and molecular orbitals.

Using the MO diagram above, answer the following questions:

- (i) Which one of the following will have the highest bond order? Circle the correct choice
A) PS^{1-} B) PS C) PS^{1+}
- (ii) Which one of the following will have the longest bond length? Circle the correct choice
A) PS^{1-} B) PS C) PS^{1+}
- (iii) Which of the following will be diamagnetic? Circle the correct choice.
A) PS^{1-} B) PS C) PS^{1+}
- (iv) Draw Lewis dot structure of the molecule with the shortest bond length (based on the MO diagram) from among the three (PS^{1-} , PS, PS^{1+}), and indicate the bond order calculated from the Lewis dot structure. **(3 points)**

Lewis Dot Structure

Bond Order



9C7F2D1C-464D-472D-A194-E298778ED5EE

midterm-exam-93388

#147 Page 10 of 10

SCRAP PAGE

**CHEMISTRY 110 –Midterm Exam****Thursday November 7, 2024, 6:30 – 8:30 pm**

Examiners: Pallavi Sirjoosingh and Ashok Kakkar

First name (please write as legibly as possible within the boxes)

--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--

Last name

--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--

Student ID number

--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--

This booklet has 8 questions on 10 pages (The last page can be used as a scrap page).

The staple must NOT be removed from the question booklet. Your intact question booklet must be returned at the end of the exam.

A separate booklet contains a periodic table, and datasheet.

Write your answers directly under the questions in the space provided. Do not write on the QR codes or near the top edges (as it may be cut off).

Please write **legibly and dark enough** for the scan to be readable. For students who use English as a second language, dictionaries are permitted. Calculators are allowed.

DO NOT ASK QUESTIONS OF THE INVIGILATORS about interpretations of the exam questions. If you have a concern about a question, please summarize it at the top of this page. All concerns will be evaluated after the exam. GOOD LUCK!

**Question 1 8 points** (*Assume all transitions are possible*)

a. 6 points Calculate the **energy (in kJ/mol)** required to ionize (completely remove) the electron from the **$n=3$** level of **He^+ ion** (He: Helium). Show your work including equations. (3 sig. figs.)

b. 2 points

(I) Select the largest (longest) wavelength from the three choices below:

- i. Wavelength of photon to *completely remove* an electron from *1s orbital of Hydrogen atom*
- ii. Wavelength of photon to *completely remove* an electron from *1s orbital of Boron atom (B)*
- iii. Wavelength of photon to *completely remove* an electron from *1s orbital of Boron cation (B^+)*

i or ii or iii

(II) Select the smallest (shortest) wavelength from the three choices below (*for H atom*):

- i. Wavelength of photon *emitted* for an electron transitioning from *2p to 1s orbital*
- ii. Wavelength of photon *absorbed* for an electron transitioning from *2s to 4p orbital*
- iii. Wavelength of photon *absorbed* for an electron transitioning from *1s to 3p orbital*

i or ii or iii



Question 2 8 points



A Chem110 student plots the graph shown on the left, from data obtained for the kinetic energy (eV) of the ejected electron versus frequency (s^{-1}) of the incident light, for an unknown metal. ($1\text{eV} = 1.6022 \times 10^{-19} \text{ J}$)

Based on the plot, answer the questions **a to c** below:

a. 1 point On the plot above, label the **threshold frequency**.

b. 2 points What is the value of the work potential (in J) of the unknown metal?

c. 3 points What is the *longest* wavelength (in nm) required to eject an electron from this metal's surface? Show your work including equation(s) used. (3 sig figs)

d. 2 points In the space provided below, plot the following

(Assume frequency is *above* threshold frequency):

- i. *Kinetic energy* of an ejected electron versus the *intensity* of the incident light (left)
- ii. *Number of electrons* ejected versus the *frequency* of the incident light (right)



**Question 3** **6 points****a. 2 points**

In the ground state of Ge (Germanium; $Z = 32$) atom, **name** the following orbitals:

i. Orbital containing the electron that requires the maximum energy to be removed.

ii. Orbital containing the electron that experiences the minimum effective nuclear charge.

b. 2 points

An unknown anion (A^-) has the following *excited* electronic configuration: $[Ar] 4s^2 3d^3 4p^1$

(Ar: Argon; $Z = 18$)

Give the **condensed ground state electronic configuration** of the atom, **A** and the cation, **A^{2+}** .
Show your work. (Assume all electronic transitions are possible)

Electronic Configuration of A:

Electronic Configuration of A^{2+} :

c. 2 points

In a single atom, determine how many electrons can have the following set of quantum numbers?

(i) $n = 5; l = 3$

In a single atom, determine how many electrons can have the following set of quantum numbers?

(i) $n = 3; l = 2; m_s = +1/2$

**Question 4** **8 points**

Rank the following from 1 to 4 in **increasing** order. Provide the ranking in the boxes provided.

Note: Atomic number (Z) are provided for relevant atoms

a) Increasing (lowest to highest) order of *ionization energy*: (1= lowest; 4 = highest)

Mg

Mg: Z = 12

Rb

Rb: Z = 37

Cs

Cs: Z = 55

K

K : Z = 19

b) Increasing (lowest to highest) order of *bond angle*: (1= lowest; 4 = highest)

H₂OCCl₄XeF₄

HCN

c) Increasing (smallest to largest) order of *ionic size* (1= smallest; 4 = largest)

S²⁻

S: Z = 16

O²⁻

O: Z = 8

Be²⁺

Be: Z = 4

Na⁺

Na: Z = 11

d) Increasing (lowest to highest) order of *bond energy* (1= lowest; 4 = highest)

P-F

P-Cl

P-I

P-Br

**Question 5** **12 points**

a. 9 points For the molecules HClO_4 and TeF_6 , draw the most stable *Lewis structure* showing appropriate shape and *dashed/wedged bonds*, based on VSEPR. Determine the **molecular geometry** around the central atom. For HClO_4 , label the *most polar bond*.



b. 3 points (Note: Lone pairs are not shown in structures below)

(i) For the three *resonance* structures (1, 2 and 3) shown below, answer the following question:



(I) Circle the *most contributing* resonance structure.

Structure 1

Structure 2

Structure 3

(ii) For the three structures given (4, 5, and 6) below, answer the following questions:



(I) Circle all the structures that represent *resonance structures* of each other:

Structure 4

Structure 5

Structure 6

(II) Among the resonance structures, circle the *most contributing* resonance structure.

Structure 4

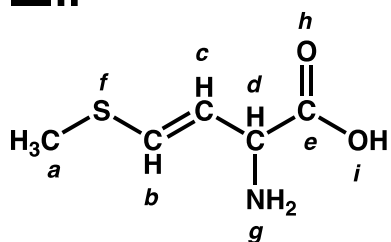
Structure 5

Structure 6

**Question 6** **6 points**

Using the data given below, calculate the **lattice energy** for SrO(s).

	ΔH° (kJ/mol)
$\text{Sr}(s) \longrightarrow \text{Sr}(g)$	+180
$\text{Sr}(g) \longrightarrow \text{Sr}^{2+}(g) + 2e^-$	+1468
$\text{O}_2(g) \longrightarrow 2\text{O}(g)$	+500
$\text{O}(g) + 2e^- \longrightarrow \text{O}^{2-}(g)$	+600
$\text{Sr}(s) + \frac{1}{2}\text{O}_2(g) \longrightarrow \text{SrO}(s)$	-550

**Question 7 10 points**

Consider the molecule given on the left, and answer questions (i) to (vi) using chemical bonding theories that we have learnt in class (valence bond theory, hybridization).

Note that the lone pairs are not shown in the structure:

i. Consider the carbon atoms marked in as ***a***, ***b***, ***c***, ***d***, and ***e*** in the molecule, and determine **hybridization** at each of the carbon centers. (2.5 points)

Atom	Hybridization
Carbon (<i>a</i>)	
Carbon (<i>b</i>)	
Carbon (<i>c</i>)	
Carbon (<i>d</i>)	
Carbon (<i>e</i>)	

ii. Now, consider the sulfur atom (S) marked as ***f***, nitrogen (N) atom (***g***), and oxygen(O) atoms (***h*** and ***i***), and determine their **hybridization**. (2 points)

Atom	Hybridization
Sulfur (<i>f</i>)	
Nitrogen (<i>g</i>)	
Oxygen (<i>h</i>)	
Oxygen (<i>i</i>)	

iii. **Describe** all the bonds formed by the carbon atoms ***a*** and ***b***, using sigma and pi designation as appropriate. (2 points)

Atom	Bonds formed (use sigma and pi as appropriate)
Carbon (<i>a</i>)	
Carbon (<i>b</i>)	

iv. How many total sp^3 -s sigma bonds are in the molecule? (1 point)

v. Determine **electronic** and **molecular geometry** at carbon (C) labeled ***e*** and sulfur (S) labeled ***f*** centres. (2 points)

Atom	Electronic Geometry	Molecular Geometry
Carbon (<i>e</i>)		
Sulfur (<i>f</i>)		

vi. Circle the value from the choices given below, that describes the **C-S-C bond angle at sulfur** (S) marked ***f***: (0.5 points)

>90° to <109.5° 109.5° >109.5 to <120° 120° >120° to <180°

**Question 8 12 points**

Using the **valence shell orbitals only** draw a **complete** molecular orbital (MO) diagram (with no s-p mixing) for the molecule **PS**, in the space provided below. **(6 points)**

Assume that 3s and 3p atomic orbitals will form molecular orbitals like those formed by the combination of 2s and 2p atomic orbitals that we learnt in class, and there is no s-p mixing.

Show **energy arrow**, **atomic orbitals** involved, **atom labels**, the **molecular orbitals** that are formed, **filled electrons** in the MO diagram etc. Make sure to label all the atomic and molecular orbitals.

Using the MO diagram above, answer the following questions:

- (i) Which one of the following will have the highest bond order? Circle the correct choice
A) PS^{1-} B) PS C) PS^{1+}
- (ii) Which one of the following will have the longest bond length? Circle the correct choice
A) PS^{1-} B) PS C) PS^{1+}
- (iii) Which of the following will be diamagnetic? Circle the correct choice.
A) PS^{1-} B) PS C) PS^{1+}
- (iv) Draw Lewis dot structure of the molecule with the shortest bond length (based on the MO diagram) from among the three (PS^{1-} , PS, PS^{1+}), and indicate the bond order calculated from the Lewis dot structure. **(3 points)**

Lewis Dot Structure

Bond Order



DFCEE535-57DF-4526-A98C-A09DF7C96C45

midterm-exam-93388

#148 Page 10 of 10

SCRAP PAGE

**CHEMISTRY 110 –Midterm Exam****Thursday November 7, 2024, 6:30 – 8:30 pm****Examiners: Pallavi Sirjoosingh and Ashok Kakkar**

First name (please write as legibly as possible within the boxes)

--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--

Last name

--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--

Student ID number

--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--

This booklet has 8 questions on 10 pages (The last page can be used as a scrap page).

The staple must NOT be removed from the question booklet. Your intact question booklet must be returned at the end of the exam.

A separate booklet contains a periodic table, and datasheet.

Write your answers directly under the questions in the space provided. Do not write on the QR codes or near the top edges (as it may be cut off).

Please write **legibly and dark enough** for the scan to be readable. For students who use English as a second language, dictionaries are permitted. Calculators are allowed.

DO NOT ASK QUESTIONS OF THE INVIGILATORS about interpretations of the exam questions. If you have a concern about a question, please summarize it at the top of this page. All concerns will be evaluated after the exam. GOOD LUCK!

**Question 1 8 points** (*Assume all transitions are possible*)

a. 6 points Calculate the **energy (in kJ/mol)** required to ionize (completely remove) the electron from the **n=3** level of **He⁺ ion** (He: Helium). Show your work including equations. (3 sig. figs.)

b. 2 points

(I) Select the largest (longest) wavelength from the three choices below:

- i. Wavelength of photon to *completely remove* an electron from *1s orbital of Hydrogen atom*
- ii. Wavelength of photon to *completely remove* an electron from *1s orbital of Boron atom (B)*
- iii. Wavelength of photon to *completely remove* an electron from *1s orbital of Boron cation (B⁺)*

i or ii or iii

(II) Select the smallest (shortest) wavelength from the three choices below (*for H atom*):

- i. Wavelength of photon *emitted* for an electron transitioning from *2p to 1s orbital*
- ii. Wavelength of photon *absorbed* for an electron transitioning from *2s to 4p orbital*
- iii. Wavelength of photon *absorbed* for an electron transitioning from *1s to 3p orbital*

i or ii or iii



Question 2 8 points



A Chem110 student plots the graph shown on the left, from data obtained for the kinetic energy (eV) of the ejected electron versus frequency (s^{-1}) of the incident light, for an unknown metal. ($1\text{eV} = 1.6022 \times 10^{-19} \text{ J}$)

Based on the plot, answer the questions **a to c** below:

a. 1 point On the plot above, label the **threshold frequency**.

b. 2 points What is the value of the work potential (in J) of the unknown metal?

c. 3 points What is the *longest* wavelength (in nm) required to eject an electron from this metal's surface? Show your work including equation(s) used. (3 sig figs)

d. 2 points In the space provided below, plot the following

(Assume frequency is *above* threshold frequency):

- i. *Kinetic energy* of an ejected electron versus the *intensity* of the incident light (left)
- ii. *Number of electrons* ejected versus the *frequency* of the incident light (right)



**Question 3** **6 points****a. 2 points**

In the ground state of Ge (Germanium; $Z = 32$) atom, **name** the following orbitals:

i. Orbital containing the electron that requires the maximum energy to be removed.

ii. Orbital containing the electron that experiences the minimum effective nuclear charge.

b. 2 points

An unknown anion (A^-) has the following *excited* electronic configuration: $[Ar] 4s^2 3d^3 4p^1$

(Ar: Argon; $Z = 18$)

Give the **condensed ground state electronic configuration** of the atom, **A** and the cation, **A^{2+}** .
Show your work. (Assume all electronic transitions are possible)

Electronic Configuration of A:

Electronic Configuration of A^{2+} :

c. 2 points

In a single atom, determine how many electrons can have the following set of quantum numbers?

(i) $n = 5; l = 3$

In a single atom, determine how many electrons can have the following set of quantum numbers?

(i) $n = 3; l = 2; m_s = +1/2$

**Question 4** **8 points**

Rank the following from 1 to 4 in **increasing** order. Provide the ranking in the boxes provided.

Note: Atomic number (Z) are provided for relevant atoms

a) Increasing (lowest to highest) order of *ionization energy*: (1= lowest; 4 = highest)

Mg

Mg: Z = 12

Rb

Rb: Z = 37

Cs

Cs: Z = 55

K

K : Z = 19

b) Increasing (lowest to highest) order of *bond angle*: (1= lowest; 4 = highest)

H₂OCCl₄XeF₄

HCN

c) Increasing (smallest to largest) order of *ionic size* (1= smallest; 4 = largest)

S²⁻

S: Z = 16

O²⁻

O: Z = 8

Be²⁺

Be: Z = 4

Na⁺

Na: Z = 11

d) Increasing (lowest to highest) order of *bond energy* (1= lowest; 4 = highest)

P-F

P-Cl

P-I

P-Br

**Question 5** **12 points**

a. 9 points For the molecules HClO_4 and TeF_6 , draw the most stable *Lewis structure* showing appropriate shape and *dashed/wedged bonds*, based on VSEPR. Determine the **molecular geometry** around the central atom. For HClO_4 , label the *most polar bond*.



b. 3 points (Note: Lone pairs are not shown in structures below)

(i) For the three *resonance* structures (1, 2 and 3) shown below, answer the following question:



(I) Circle the *most contributing* resonance structure.

Structure 1

Structure 2

Structure 3

(ii) For the three structures given (4, 5, and 6) below, answer the following questions:



(I) Circle all the structures that represent *resonance structures* of each other:

Structure 4

Structure 5

Structure 6

(II) Among the resonance structures, circle the *most contributing* resonance structure.

Structure 4

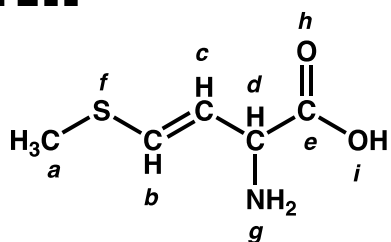
Structure 5

Structure 6

**Question 6** **6 points**

Using the data given below, calculate the **lattice energy** for SrO(s).

	ΔH° (kJ/mol)
$\text{Sr}(s) \longrightarrow \text{Sr}(g)$	+180
$\text{Sr}(g) \longrightarrow \text{Sr}^{2+}(g) + 2e^-$	+1468
$\text{O}_2(g) \longrightarrow 2\text{O}(g)$	+500
$\text{O}(g) + 2e^- \longrightarrow \text{O}^{2-}(g)$	+600
$\text{Sr}(s) + \frac{1}{2}\text{O}_2(g) \longrightarrow \text{SrO}(s)$	-550

**Question 7 10 points**

Consider the molecule given on the left, and answer questions (i) to (vi) using chemical bonding theories that we have learnt in class (valence bond theory, hybridization).

Note that the lone pairs are not shown in the structure:

i. Consider the carbon atoms marked in as ***a***, ***b***, ***c***, ***d***, and ***e*** in the molecule, and determine **hybridization** at each of the carbon centers. (2.5 points)

Atom	Hybridization
Carbon (<i>a</i>)	
Carbon (<i>b</i>)	
Carbon (<i>c</i>)	
Carbon (<i>d</i>)	
Carbon (<i>e</i>)	

ii. Now, consider the sulfur atom (S) marked as ***f***, nitrogen (N) atom (***g***), and oxygen(O) atoms (***h*** and ***i***), and determine their **hybridization**. (2 points)

Atom	Hybridization
Sulfur (<i>f</i>)	
Nitrogen (<i>g</i>)	
Oxygen (<i>h</i>)	
Oxygen (<i>i</i>)	

iii. **Describe** all the bonds formed by the carbon atoms ***a*** and ***b***, using sigma and pi designation as appropriate. (2 points)

Atom	Bonds formed (use sigma and pi as appropriate)
Carbon (<i>a</i>)	
Carbon (<i>b</i>)	

iv. How many total sp^3 -s sigma bonds are in the molecule? (1 point)

v. Determine **electronic** and **molecular geometry** at carbon (C) labeled ***e*** and sulfur (S) labeled ***f*** centres. (2 points)

Atom	Electronic Geometry	Molecular Geometry
Carbon (<i>e</i>)		
Sulfur (<i>f</i>)		

vi. Circle the value from the choices given below, that describes the **C-S-C bond angle at sulfur** (S) marked ***f***: (0.5 points)

>90° to <109.5° 109.5° >109.5 to <120° 120° >120° to <180°

**Question 8 12 points**

Using the **valence shell orbitals only** draw a **complete** molecular orbital (MO) diagram (with no s-p mixing) for the molecule **PS**, in the space provided below. **(6 points)**

Assume that 3s and 3p atomic orbitals will form molecular orbitals like those formed by the combination of 2s and 2p atomic orbitals that we learnt in class, and there is no s-p mixing.

Show **energy arrow**, **atomic orbitals** involved, **atom labels**, the **molecular orbitals** that are formed, **filled electrons** in the MO diagram etc. Make sure to **label all the atomic and molecular orbitals**.

Using the MO diagram above, answer the following questions:

- (i) Which one of the following will have the highest bond order? Circle the correct choice
A) PS^{1-} B) PS C) PS^{1+}
- (ii) Which one of the following will have the longest bond length? Circle the correct choice
A) PS^{1-} B) PS C) PS^{1+}
- (iii) Which of the following will be diamagnetic? Circle the correct choice.
A) PS^{1-} B) PS C) PS^{1+}
- (iv) Draw Lewis dot structure of the molecule with the shortest bond length (based on the MO diagram) from among the three (PS^{1-} , PS, PS^{1+}), and indicate the bond order calculated from the Lewis dot structure. **(3 points)**

Lewis Dot Structure

Bond Order



B9A231E8-2D4A-43BE-882E-0AC427652429

midterm-exam-93388

#149 Page 10 of 10

SCRAP PAGE

**CHEMISTRY 110 –Midterm Exam****Thursday November 7, 2024, 6:30 – 8:30 pm****Examiners: Pallavi Sirjoosingh and Ashok Kakkar**

First name (please write as legibly as possible within the boxes)

--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--

Last name

--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--

Student ID number

--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--

This booklet has 8 questions on 10 pages (The last page can be used as a scrap page).

The staple must NOT be removed from the question booklet. Your intact question booklet must be returned at the end of the exam.

A separate booklet contains a periodic table, and datasheet.

Write your answers directly under the questions in the space provided. Do not write on the QR codes or near the top edges (as it may be cut off).

Please write **legibly and dark enough** for the scan to be readable. For students who use English as a second language, dictionaries are permitted. Calculators are allowed.

DO NOT ASK QUESTIONS OF THE INVIGILATORS about interpretations of the exam questions. If you have a concern about a question, please summarize it at the top of this page. All concerns will be evaluated after the exam. GOOD LUCK!

**Question 1 8 points** (*Assume all transitions are possible*)

a. 6 points Calculate the **energy (in kJ/mol)** required to ionize (completely remove) the electron from the **n=3** level of **He⁺ ion** (He: Helium). Show your work including equations. (3 sig. figs.)

b. 2 points

(I) Select the largest (longest) wavelength from the three choices below:

- i. Wavelength of photon to *completely remove* an electron from *1s orbital of Hydrogen atom*
- ii. Wavelength of photon to *completely remove* an electron from *1s orbital of Boron atom (B)*
- iii. Wavelength of photon to *completely remove* an electron from *1s orbital of Boron cation (B⁺)*

i or ii or iii

(II) Select the smallest (shortest) wavelength from the three choices below (*for H atom*):

- i. Wavelength of photon *emitted* for an electron transitioning from *2p to 1s orbital*
- ii. Wavelength of photon *absorbed* for an electron transitioning from *2s to 4p orbital*
- iii. Wavelength of photon *absorbed* for an electron transitioning from *1s to 3p orbital*

i or ii or iii



Question 2 8 points



A Chem110 student plots the graph shown on the left, from data obtained for the kinetic energy (eV) of the ejected electron versus frequency (s^{-1}) of the incident light, for an unknown metal. ($1\text{eV} = 1.6022 \times 10^{-19} \text{ J}$)

Based on the plot, answer the questions **a to c** below:

a. 1 point On the plot above, label the **threshold frequency**.

b. 2 points What is the value of the work potential (in J) of the unknown metal?

c. 3 points What is the *longest* wavelength (in nm) required to eject an electron from this metal's surface? Show your work including equation(s) used. (3 sig figs)

d. 2 points In the space provided below, plot the following

(Assume frequency is *above* threshold frequency):

- Kinetic energy* of an ejected electron versus the *intensity* of the incident light (left)
- Number of electrons* ejected versus the *frequency* of the incident light (right)



**Question 3** **6 points****a. 2 points**

In the ground state of Ge (Germanium; $Z = 32$) atom, **name** the following orbitals:

i. Orbital containing the electron that requires the maximum energy to be removed.

ii. Orbital containing the electron that experiences the minimum effective nuclear charge.

b. 2 points

An unknown anion (A^-) has the following *excited* electronic configuration: $[Ar] 4s^2 3d^3 4p^1$

(Ar: Argon; $Z = 18$)

Give the **condensed ground state electronic configuration** of the atom, **A** and the cation, **A^{2+}** .
Show your work. (Assume all electronic transitions are possible)

Electronic Configuration of A:

Electronic Configuration of A^{2+} :

c. 2 points

In a single atom, determine how many electrons can have the following set of quantum numbers?

(i) $n = 5; l = 3$

In a single atom, determine how many electrons can have the following set of quantum numbers?

(i) $n = 3; l = 2; m_s = +1/2$

**Question 4** **8 points**

Rank the following from 1 to 4 in **increasing** order. Provide the ranking in the boxes provided.

Note: Atomic number (Z) are provided for relevant atoms

a) Increasing (lowest to highest) order of *ionization energy*: (1= lowest; 4 = highest)

Mg

Mg: Z = 12

Rb

Rb: Z = 37

Cs

Cs: Z = 55

K

K : Z = 19

b) Increasing (lowest to highest) order of *bond angle*: (1= lowest; 4 = highest)

H₂OCCl₄XeF₄

HCN

c) Increasing (smallest to largest) order of *ionic size* (1= smallest; 4 = largest)

S²⁻

S: Z = 16

O²⁻

O: Z = 8

Be²⁺

Be: Z = 4

Na⁺

Na: Z = 11

d) Increasing (lowest to highest) order of *bond energy* (1= lowest; 4 = highest)

P-F

P-Cl

P-I

P-Br

**Question 5** **12 points**

a. 9 points For the molecules HClO_4 and TeF_6 , draw the most stable *Lewis structure* showing appropriate shape and *dashed/wedged bonds*, based on VSEPR. Determine the **molecular geometry** around the central atom. For HClO_4 , label the *most polar bond*.



b. 3 points (Note: Lone pairs are not shown in structures below)

(i) For the three *resonance* structures (1, 2 and 3) shown below, answer the following question:



(I) Circle the *most contributing* resonance structure.

Structure 1

Structure 2

Structure 3

(ii) For the three structures given (4, 5, and 6) below, answer the following questions:



(I) Circle all the structures that represent *resonance structures* of each other:

Structure 4

Structure 5

Structure 6

(II) Among the resonance structures, circle the *most contributing* resonance structure.

Structure 4

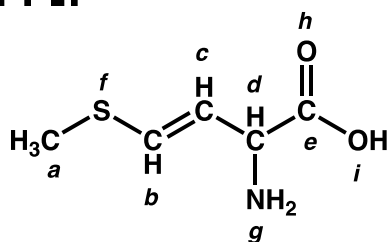
Structure 5

Structure 6

**Question 6** **6 points**

Using the data given below, calculate the **lattice energy** for SrO(s).

	ΔH° (kJ/mol)
$\text{Sr}(s) \longrightarrow \text{Sr}(g)$	+180
$\text{Sr}(g) \longrightarrow \text{Sr}^{2+}(g) + 2e^-$	+1468
$\text{O}_2(g) \longrightarrow 2\text{O}(g)$	+500
$\text{O}(g) + 2e^- \longrightarrow \text{O}^{2-}(g)$	+600
$\text{Sr}(s) + \frac{1}{2}\text{O}_2(g) \longrightarrow \text{SrO}(s)$	-550

**Question 7 10 points**

Consider the molecule given on the left, and answer questions (i) to (vi) using chemical bonding theories that we have learnt in class (valence bond theory, hybridization).

Note that the lone pairs are not shown in the structure:

i. Consider the carbon atoms marked in as ***a***, ***b***, ***c***, ***d***, and ***e*** in the molecule, and determine **hybridization** at each of the carbon centers. (2.5 points)

Atom	Hybridization
Carbon (<i>a</i>)	
Carbon (<i>b</i>)	
Carbon (<i>c</i>)	
Carbon (<i>d</i>)	
Carbon (<i>e</i>)	

ii. Now, consider the sulfur atom (S) marked as ***f***, nitrogen (N) atom (***g***), and oxygen(O) atoms (***h*** and ***i***), and determine their **hybridization**. (2 points)

Atom	Hybridization
Sulfur (<i>f</i>)	
Nitrogen (<i>g</i>)	
Oxygen (<i>h</i>)	
Oxygen (<i>i</i>)	

iii. **Describe** all the bonds formed by the carbon atoms ***a*** and ***b***, using sigma and pi designation as appropriate. (2 points)

Atom	Bonds formed (use sigma and pi as appropriate)
Carbon (<i>a</i>)	
Carbon (<i>b</i>)	

iv. How many total sp^3 -s sigma bonds are in the molecule? (1 point)

v. Determine **electronic** and **molecular geometry** at carbon (C) labeled ***e*** and sulfur (S) labeled ***f*** centres. (2 points)

Atom	Electronic Geometry	Molecular Geometry
Carbon (<i>e</i>)		
Sulfur (<i>f</i>)		

vi. Circle the value from the choices given below, that describes the **C-S-C bond angle at sulfur** (S) marked ***f***: (0.5 points)

>90° to <109.5° 109.5° >109.5 to <120° 120° >120° to <180°


Question 8 12 points

Using the **valence shell orbitals only** draw a **complete** molecular orbital (MO) diagram (with no s-p mixing) for the molecule **PS**, in the space provided below. **(6 points)**

Assume that 3s and 3p atomic orbitals will form molecular orbitals like those formed by the combination of 2s and 2p atomic orbitals that we learnt in class, and there is no s-p mixing.

Show **energy arrow**, **atomic orbitals** involved, **atom labels**, the **molecular orbitals** that are formed, **filled electrons** in the MO diagram etc. Make sure to label all the atomic and molecular orbitals.

Using the MO diagram above, answer the following questions:

- (i) Which one of the following will have the highest bond order? Circle the correct choice
 A) PS^{1-} B) PS C) PS^{1+}
- (ii) Which one of the following will have the longest bond length? Circle the correct choice
 A) PS^{1-} B) PS C) PS^{1+}
- (iii) Which of the following will be diamagnetic? Circle the correct choice.
 A) PS^{1-} B) PS C) PS^{1+}
- (iv) Draw Lewis dot structure of the molecule with the shortest bond length (based on the MO diagram) from among the three (PS^{1-} , PS, PS^{1+}), and indicate the bond order calculated from the Lewis dot structure. **(3 points)**

Lewis Dot Structure

Bond Order



A2419E8D-7BC5-47FC-A39D-9ACD252D866A

midterm-exam-93388

#150 Page 10 of 10

SCRAP PAGE

**CHEMISTRY 110 –Midterm Exam****Thursday November 7, 2024, 6:30 – 8:30 pm**

Examiners: Pallavi Sirjoosingh and Ashok Kakkar

First name (please write as legibly as possible within the boxes)

--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--

Last name

--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--

Student ID number

--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--

This booklet has 8 questions on 10 pages (The last page can be used as a scrap page).

The staple must NOT be removed from the question booklet. Your intact question booklet must be returned at the end of the exam.

A separate booklet contains a periodic table, and datasheet.

Write your answers directly under the questions in the space provided. Do not write on the QR codes or near the top edges (as it may be cut off).

Please write **legibly and dark enough** for the scan to be readable. For students who use English as a second language, dictionaries are permitted. Calculators are allowed.

DO NOT ASK QUESTIONS OF THE INVIGILATORS about interpretations of the exam questions. If you have a concern about a question, please summarize it at the top of this page. All concerns will be evaluated after the exam. GOOD LUCK!

**Question 1 8 points** (*Assume all transitions are possible*)

a. 6 points Calculate the **energy (in kJ/mol)** required to ionize (completely remove) the electron from the **n=3** level of **He⁺ ion** (He: Helium). Show your work including equations. (3 sig. figs.)

b. 2 points

(I) Select the largest (longest) wavelength from the three choices below:

- i. Wavelength of photon to *completely remove* an electron from *1s orbital of Hydrogen atom*
- ii. Wavelength of photon to *completely remove* an electron from *1s orbital of Boron atom (B)*
- iii. Wavelength of photon to *completely remove* an electron from *1s orbital of Boron cation (B⁺)*

i or ii or iii

(II) Select the smallest (shortest) wavelength from the three choices below (*for H atom*):

- i. Wavelength of photon *emitted* for an electron transitioning from *2p to 1s orbital*
- ii. Wavelength of photon *absorbed* for an electron transitioning from *2s to 4p orbital*
- iii. Wavelength of photon *absorbed* for an electron transitioning from *1s to 3p orbital*

i or ii or iii



Question 2 8 points



A Chem110 student plots the graph shown on the left, from data obtained for the kinetic energy (eV) of the ejected electron versus frequency (s^{-1}) of the incident light, for an unknown metal. ($1\text{eV} = 1.6022 \times 10^{-19} \text{ J}$)

Based on the plot, answer the questions **a to c** below:

a. 1 point On the plot above, label the **threshold frequency**.

b. 2 points What is the value of the work potential (in J) of the unknown metal?

c. 3 points What is the *longest* wavelength (in nm) required to eject an electron from this metal's surface? Show your work including equation(s) used. (3 sig figs)

d. 2 points In the space provided below, plot the following

(Assume frequency is *above* threshold frequency):

- i. *Kinetic energy* of an ejected electron versus the *intensity* of the incident light (left)
- ii. *Number of electrons* ejected versus the *frequency* of the incident light (right)



**Question 3** **6 points****a. 2 points**

In the ground state of Ge (Germanium; $Z = 32$) atom, **name** the following orbitals:

i. Orbital containing the electron that requires the maximum energy to be removed.

ii. Orbital containing the electron that experiences the minimum effective nuclear charge.

b. 2 points

An unknown anion (A^-) has the following *excited* electronic configuration: $[Ar] 4s^2 3d^3 4p^1$

(Ar: Argon; $Z = 18$)

Give the **condensed ground state electronic configuration** of the atom, **A** and the cation, **A^{2+}** .
Show your work. (Assume all electronic transitions are possible)

Electronic Configuration of A:

Electronic Configuration of A^{2+} :

c. 2 points

In a single atom, determine how many electrons can have the following set of quantum numbers?

(i) $n = 5; l = 3$

In a single atom, determine how many electrons can have the following set of quantum numbers?

(i) $n = 3; l = 2; m_s = +1/2$

**Question 4** **8 points**

Rank the following from 1 to 4 in **increasing** order. Provide the ranking in the boxes provided.

Note: Atomic number (Z) are provided for relevant atoms

a) Increasing (lowest to highest) order of *ionization energy*: (1= lowest; 4 = highest)

Mg

Mg: Z = 12

Rb

Rb: Z = 37

Cs

Cs: Z = 55

K

K : Z = 19

b) Increasing (lowest to highest) order of *bond angle*: (1= lowest; 4 = highest)

H₂OCCl₄XeF₄

HCN

c) Increasing (smallest to largest) order of *ionic size* (1= smallest; 4 = largest)

S²⁻

S: Z = 16

O²⁻

O: Z = 8

Be²⁺

Be: Z = 4

Na⁺

Na: Z = 11

d) Increasing (lowest to highest) order of *bond energy* (1= lowest; 4 = highest)

P-F

P-Cl

P-I

P-Br

**Question 5** **12 points**

a. 9 points For the molecules HClO_4 and TeF_6 , draw the most stable *Lewis structure* showing appropriate shape and *dashed/wedged bonds*, based on VSEPR. Determine the **molecular geometry** around the central atom. For HClO_4 , label the *most polar bond*.

HClO_4 (there are no O-O bonds)

Label the most polar bond

Molecular Geometry around Cl

TeF_6

Molecular Geometry around Te

b. 3 points (Note: Lone pairs are not shown in structures below)

(i) For the three *resonance* structures (1, 2 and 3) shown below, answer the following question:



(I) Circle the *most contributing* resonance structure.

Structure 1

Structure 2

Structure 3

(ii) For the three structures given (4, 5, and 6) below, answer the following questions:



(I) Circle all the structures that represent *resonance structures* of each other:

Structure 4

Structure 5

Structure 6

(II) Among the resonance structures, circle the *most contributing* resonance structure.

Structure 4

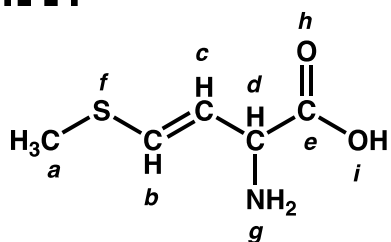
Structure 5

Structure 6

**Question 6** **6 points**

Using the data given below, calculate the **lattice energy** for SrO(s).

	ΔH° (kJ/mol)
$\text{Sr}(s) \longrightarrow \text{Sr}(g)$	+180
$\text{Sr}(g) \longrightarrow \text{Sr}^{2+}(g) + 2e^-$	+1468
$\text{O}_2(g) \longrightarrow 2\text{O}(g)$	+500
$\text{O}(g) + 2e^- \longrightarrow \text{O}^{2-}(g)$	+600
$\text{Sr}(s) + \frac{1}{2}\text{O}_2(g) \longrightarrow \text{SrO}(s)$	-550

**Question 7 10 points**

Consider the molecule given on the left, and answer questions (i) to (vi) using chemical bonding theories that we have learnt in class (valence bond theory, hybridization).

Note that the lone pairs are not shown in the structure:

i. Consider the carbon atoms marked in as ***a***, ***b***, ***c***, ***d***, and ***e*** in the molecule, and determine **hybridization** at each of the carbon centers. (2.5 points)

Atom	Hybridization
Carbon (<i>a</i>)	
Carbon (<i>b</i>)	
Carbon (<i>c</i>)	
Carbon (<i>d</i>)	
Carbon (<i>e</i>)	

ii. Now, consider the sulfur atom (S) marked as ***f***, nitrogen (N) atom (***g***), and oxygen(O) atoms (***h*** and ***i***), and determine their **hybridization**. (2 points)

Atom	Hybridization
Sulfur (<i>f</i>)	
Nitrogen (<i>g</i>)	
Oxygen (<i>h</i>)	
Oxygen (<i>i</i>)	

iii. **Describe** all the bonds formed by the carbon atoms ***a*** and ***b***, using sigma and pi designation as appropriate. (2 points)

Atom	Bonds formed (use sigma and pi as appropriate)
Carbon (<i>a</i>)	
Carbon (<i>b</i>)	

iv. How many total sp^3 -s sigma bonds are in the molecule? (1 point)

v. Determine **electronic** and **molecular geometry** at carbon (C) labeled ***e*** and sulfur (S) labeled ***f*** centres. (2 points)

Atom	Electronic Geometry	Molecular Geometry
Carbon (<i>e</i>)		
Sulfur (<i>f</i>)		

vi. Circle the value from the choices given below, that describes the **C-S-C bond angle at sulfur** (S) marked ***f***: (0.5 points)

>90° to <109.5°

109.5°

>109.5 to <120°

120°

>120° to <180°


Question 8 12 points

Using the **valence shell orbitals only** draw a **complete** molecular orbital (MO) diagram (with no s-p mixing) for the molecule **PS**, in the space provided below. **(6 points)**

Assume that 3s and 3p atomic orbitals will form molecular orbitals like those formed by the combination of 2s and 2p atomic orbitals that we learnt in class, and there is no s-p mixing.

Show **energy arrow**, **atomic orbitals** involved, **atom labels**, the **molecular orbitals** that are formed, **filled electrons** in the MO diagram etc. Make sure to label all the atomic and molecular orbitals.

Using the MO diagram above, answer the following questions:

- (i) Which one of the following will have the highest bond order? Circle the correct choice
 A) PS^{1-} B) PS C) PS^{1+}
- (ii) Which one of the following will have the longest bond length? Circle the correct choice
 A) PS^{1-} B) PS C) PS^{1+}
- (iii) Which of the following will be diamagnetic? Circle the correct choice.
 A) PS^{1-} B) PS C) PS^{1+}
- (iv) Draw Lewis dot structure of the molecule with the shortest bond length (based on the MO diagram) from among the three (PS^{1-} , PS, PS^{1+}), and indicate the bond order calculated from the Lewis dot structure. **(3 points)**

Lewis Dot Structure

Bond Order



24F4D7FC-4C74-4D63-A208-5963A292E99A

midterm-exam-93388

#151 Page 10 of 10

SCRAP PAGE

**CHEMISTRY 110 –Midterm Exam****Thursday November 7, 2024, 6:30 – 8:30 pm****Examiners: Pallavi Sirjoosingh and Ashok Kakkar**

First name (please write as legibly as possible within the boxes)

--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--

Last name

--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--

Student ID number

--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--

This booklet has 8 questions on 10 pages (The last page can be used as a scrap page).

The staple must NOT be removed from the question booklet. Your intact question booklet must be returned at the end of the exam.

A separate booklet contains a periodic table, and datasheet.

Write your answers directly under the questions in the space provided. Do not write on the QR codes or near the top edges (as it may be cut off).

Please write **legibly and dark enough** for the scan to be readable. For students who use English as a second language, dictionaries are permitted. Calculators are allowed.

DO NOT ASK QUESTIONS OF THE INVIGILATORS about interpretations of the exam questions. If you have a concern about a question, please summarize it at the top of this page. All concerns will be evaluated after the exam. GOOD LUCK!

**Question 1 8 points** (*Assume all transitions are possible*)

a. 6 points Calculate the **energy (in kJ/mol)** required to ionize (completely remove) the electron from the **n=3** level of **He⁺ ion** (He: Helium). Show your work including equations. (3 sig. figs.)

b. 2 points

(I) Select the largest (longest) wavelength from the three choices below:

- i. Wavelength of photon to *completely remove* an electron from *1s orbital of Hydrogen atom*
- ii. Wavelength of photon to *completely remove* an electron from *1s orbital of Boron atom (B)*
- iii. Wavelength of photon to *completely remove* an electron from *1s orbital of Boron cation (B⁺)*

i or ii or iii

(II) Select the smallest (shortest) wavelength from the three choices below (*for H atom*):

- i. Wavelength of photon *emitted* for an electron transitioning from *2p to 1s orbital*
- ii. Wavelength of photon *absorbed* for an electron transitioning from *2s to 4p orbital*
- iii. Wavelength of photon *absorbed* for an electron transitioning from *1s to 3p orbital*

i or ii or iii



Question 2 8 points



A Chem110 student plots the graph shown on the left, from data obtained for the kinetic energy (eV) of the ejected electron versus frequency (s^{-1}) of the incident light, for an unknown metal. ($1\text{eV} = 1.6022 \times 10^{-19} \text{ J}$)

Based on the plot, answer the questions **a to c** below:

a. 1 point On the plot above, label the **threshold frequency**.

b. 2 points What is the value of the work potential (in J) of the unknown metal?

c. 3 points What is the *longest* wavelength (in nm) required to eject an electron from this metal's surface? Show your work including equation(s) used. (3 sig figs)

d. 2 points In the space provided below, plot the following

(Assume frequency is *above* threshold frequency):

- i. *Kinetic energy* of an ejected electron versus the *intensity* of the incident light (left)
- ii. *Number of electrons* ejected versus the *frequency* of the incident light (right)



**Question 3** **6 points****a. 2 points**

In the ground state of Ge (Germanium; $Z = 32$) atom, **name** the following orbitals:

i. Orbital containing the electron that requires the maximum energy to be removed.

ii. Orbital containing the electron that experiences the minimum effective nuclear charge.

b. 2 points

An unknown anion (A^-) has the following *excited* electronic configuration: $[Ar] 4s^2 3d^3 4p^1$

(Ar: Argon; $Z = 18$)

Give the **condensed ground state electronic configuration** of the atom, **A** and the cation, **A^{2+}** .
Show your work. (Assume all electronic transitions are possible)

Electronic Configuration of A:

Electronic Configuration of A^{2+} :

c. 2 points

In a single atom, determine how many electrons can have the following set of quantum numbers?

(i) $n = 5; l = 3$

In a single atom, determine how many electrons can have the following set of quantum numbers?

(i) $n = 3; l = 2; m_s = +1/2$

**Question 4** **8 points**

Rank the following from 1 to 4 in **increasing** order. Provide the ranking in the boxes provided.

Note: Atomic number (Z) are provided for relevant atoms

a) Increasing (lowest to highest) order of *ionization energy*: (1= lowest; 4 = highest)

Mg

Mg: Z = 12

Rb

Rb: Z = 37

Cs

Cs: Z = 55

K

K : Z = 19

b) Increasing (lowest to highest) order of *bond angle*: (1= lowest; 4 = highest)

H₂OCCl₄XeF₄

HCN

c) Increasing (smallest to largest) order of *ionic size* (1= smallest; 4 = largest)

S²⁻

S: Z = 16

O²⁻

O: Z = 8

Be²⁺

Be: Z = 4

Na⁺

Na: Z = 11

d) Increasing (lowest to highest) order of *bond energy* (1= lowest; 4 = highest)

P-F

P-Cl

P-I

P-Br

**Question 5** **12 points**

a. 9 points For the molecules HClO_4 and TeF_6 , draw the most stable *Lewis structure* showing appropriate shape and *dashed/wedged bonds*, based on VSEPR. Determine the **molecular geometry** around the central atom. For HClO_4 , label the *most polar bond*.



b. 3 points (Note: Lone pairs are not shown in structures below)

(i) For the three *resonance* structures (1, 2 and 3) shown below, answer the following question:



(I) Circle the *most contributing* resonance structure.

Structure 1

Structure 2

Structure 3

(ii) For the three structures given (4, 5, and 6) below, answer the following questions:



(I) Circle all the structures that represent *resonance structures* of each other:

Structure 4

Structure 5

Structure 6

(II) Among the resonance structures, circle the *most contributing* resonance structure.

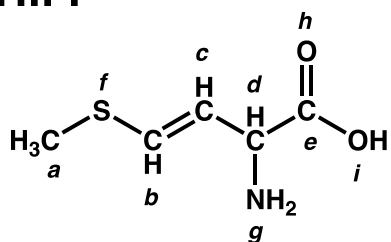
Structure 4

Structure 5

Structure 6

**Question 6** **6 points**Using the data given below, calculate the **lattice energy** for SrO(s).

	ΔH° (kJ/mol)
$\text{Sr}(s) \longrightarrow \text{Sr}(g)$	+180
$\text{Sr}(g) \longrightarrow \text{Sr}^{2+}(g) + 2e^-$	+1468
$\text{O}_2(g) \longrightarrow 2\text{O}(g)$	+500
$\text{O}(g) + 2e^- \longrightarrow \text{O}^{2-}(g)$	+600
$\text{Sr}(s) + \frac{1}{2}\text{O}_2(g) \longrightarrow \text{SrO}(s)$	-550

**Question 7 10 points**

Consider the molecule given on the left, and answer questions (i) to (vi) using chemical bonding theories that we have learnt in class (valence bond theory, hybridization).

Note that the lone pairs are not shown in the structure:

i. Consider the carbon atoms marked in as ***a***, ***b***, ***c***, ***d***, and ***e*** in the molecule, and determine **hybridization** at each of the carbon centers. (2.5 points)

Atom	Hybridization
Carbon (<i>a</i>)	
Carbon (<i>b</i>)	
Carbon (<i>c</i>)	
Carbon (<i>d</i>)	
Carbon (<i>e</i>)	

ii. Now, consider the sulfur atom (S) marked as ***f***, nitrogen (N) atom (***g***), and oxygen(O) atoms (***h*** and ***i***), and determine their **hybridization**. (2 points)

Atom	Hybridization
Sulfur (<i>f</i>)	
Nitrogen (<i>g</i>)	
Oxygen (<i>h</i>)	
Oxygen (<i>i</i>)	

iii. **Describe** all the bonds formed by the carbon atoms ***a*** and ***b***, using sigma and pi designation as appropriate. (2 points)

Atom	Bonds formed (use sigma and pi as appropriate)
Carbon (<i>a</i>)	
Carbon (<i>b</i>)	

iv. How many total sp^3 -s sigma bonds are in the molecule? (1 point)

v. Determine **electronic** and **molecular geometry** at carbon (C) labeled ***e*** and sulfur (S) labeled ***f*** centres. (2 points)

Atom	Electronic Geometry	Molecular Geometry
Carbon (<i>e</i>)		
Sulfur (<i>f</i>)		

vi. Circle the value from the choices given below, that describes the **C-S-C bond angle at sulfur** (S) marked ***f***: (0.5 points)

>90° to <109.5° 109.5° >109.5 to <120° 120° >120° to <180°


Question 8 12 points

Using the **valence shell orbitals only** draw a **complete** molecular orbital (MO) diagram (with no s-p mixing) for the molecule **PS**, in the space provided below. **(6 points)**

Assume that 3s and 3p atomic orbitals will form molecular orbitals like those formed by the combination of 2s and 2p atomic orbitals that we learnt in class, and there is no s-p mixing.

Show **energy arrow**, **atomic orbitals** involved, **atom labels**, the **molecular orbitals** that are formed, **filled electrons** in the MO diagram etc. Make sure to label all the atomic and molecular orbitals.

Using the MO diagram above, answer the following questions:

- (i) Which one of the following will have the highest bond order? Circle the correct choice
 A) PS^{1-} B) PS C) PS^{1+}
- (ii) Which one of the following will have the longest bond length? Circle the correct choice
 A) PS^{1-} B) PS C) PS^{1+}
- (iii) Which of the following will be diamagnetic? Circle the correct choice.
 A) PS^{1-} B) PS C) PS^{1+}
- (iv) Draw Lewis dot structure of the molecule with the shortest bond length (based on the MO diagram) from among the three (PS^{1-} , PS, PS^{1+}), and indicate the bond order calculated from the Lewis dot structure. **(3 points)**

Lewis Dot Structure

Bond Order



6F7C0630-C80A-41C3-A305-81EBC491840B

midterm-exam-93388

#152 Page 10 of 10

SCRAP PAGE

**CHEMISTRY 110 –Midterm Exam****Thursday November 7, 2024, 6:30 – 8:30 pm****Examiners: Pallavi Sirjoosingh and Ashok Kakkar**

First name (please write as legibly as possible within the boxes)

--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--

Last name

--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--

Student ID number

--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--

This booklet has 8 questions on 10 pages (The last page can be used as a scrap page).

The staple must NOT be removed from the question booklet. Your intact question booklet must be returned at the end of the exam.

A separate booklet contains a periodic table, and datasheet.

Write your answers directly under the questions in the space provided. Do not write on the QR codes or near the top edges (as it may be cut off).

Please write **legibly and dark enough** for the scan to be readable. For students who use English as a second language, dictionaries are permitted. Calculators are allowed.

DO NOT ASK QUESTIONS OF THE INVIGILATORS about interpretations of the exam questions. If you have a concern about a question, please summarize it at the top of this page. All concerns will be evaluated after the exam. GOOD LUCK!

**Question 1 8 points** (*Assume all transitions are possible*)

a. 6 points Calculate the **energy (in kJ/mol)** required to ionize (completely remove) the electron from the **$n=3$** level of **He^+ ion** (He: Helium). Show your work including equations. (3 sig. figs.)

b. 2 points

(I) Select the largest (longest) wavelength from the three choices below:

- i. Wavelength of photon to *completely remove* an electron from *1s orbital of Hydrogen atom*
- ii. Wavelength of photon to *completely remove* an electron from *1s orbital of Boron atom (B)*
- iii. Wavelength of photon to *completely remove* an electron from *1s orbital of Boron cation (B^+)*

i or ii or iii

(II) Select the smallest (shortest) wavelength from the three choices below (*for H atom*):

- i. Wavelength of photon *emitted* for an electron transitioning from *2p to 1s orbital*
- ii. Wavelength of photon *absorbed* for an electron transitioning from *2s to 4p orbital*
- iii. Wavelength of photon *absorbed* for an electron transitioning from *1s to 3p orbital*

i or ii or iii



Question 2 8 points



A Chem110 student plots the graph shown on the left, from data obtained for the kinetic energy (eV) of the ejected electron versus frequency (s^{-1}) of the incident light, for an unknown metal. ($1\text{eV} = 1.6022 \times 10^{-19} \text{ J}$)

Based on the plot, answer the questions **a to c** below:

a. 1 point On the plot above, label the **threshold frequency**.

b. 2 points What is the value of the work potential (in J) of the unknown metal?

c. 3 points What is the *longest* wavelength (in nm) required to eject an electron from this metal's surface? Show your work including equation(s) used. (3 sig figs)

d. 2 points In the space provided below, plot the following

(Assume frequency is *above* threshold frequency):

- Kinetic energy* of an ejected electron versus the *intensity* of the incident light (left)
- Number of electrons* ejected versus the *frequency* of the incident light (right)



**Question 3** **6 points****a. 2 points**

In the ground state of Ge (Germanium; $Z = 32$) atom, **name** the following orbitals:

i. Orbital containing the electron that requires the maximum energy to be removed.

ii. Orbital containing the electron that experiences the minimum effective nuclear charge.

b. 2 points

An unknown anion (A^-) has the following *excited* electronic configuration: $[Ar] 4s^2 3d^3 4p^1$

(Ar: Argon; $Z = 18$)

Give the **condensed ground state electronic configuration** of the atom, **A** and the cation, A^{2+} .
Show your work. (Assume all electronic transitions are possible)

Electronic Configuration of A:

Electronic Configuration of A^{2+} :

c. 2 points

In a single atom, determine how many electrons can have the following set of quantum numbers?

(i) $n = 5; l = 3$

In a single atom, determine how many electrons can have the following set of quantum numbers?

(i) $n = 3; l = 2; m_s = +1/2$

**Question 4** **8 points**

Rank the following from 1 to 4 in **increasing** order. Provide the ranking in the boxes provided.

Note: Atomic number (Z) are provided for relevant atoms

a) Increasing (lowest to highest) order of *ionization energy*: (1= lowest; 4 = highest)

Mg

Mg: Z = 12

Rb

Rb: Z = 37

Cs

Cs: Z = 55

K

K : Z = 19

b) Increasing (lowest to highest) order of *bond angle*: (1= lowest; 4 = highest)

H₂OCCl₄XeF₄

HCN

c) Increasing (smallest to largest) order of *ionic size* (1= smallest; 4 = largest)

S²⁻

S: Z = 16

O²⁻

O: Z = 8

Be²⁺

Be: Z = 4

Na⁺

Na: Z = 11

d) Increasing (lowest to highest) order of *bond energy* (1= lowest; 4 = highest)

P-F

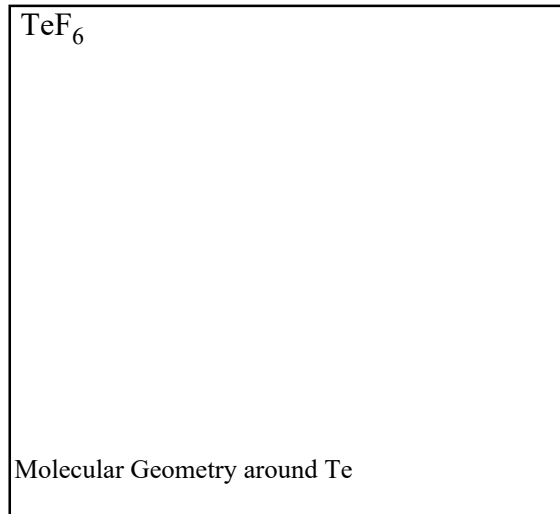
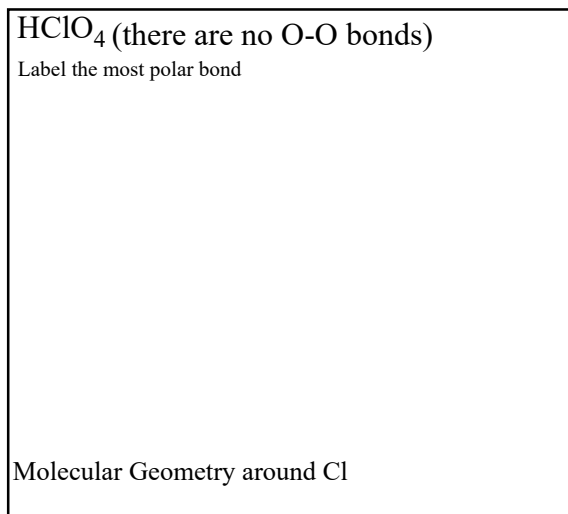
P-Cl

P-I

P-Br

**Question 5** **12 points**

a. 9 points For the molecules HClO_4 and TeF_6 , draw the most stable *Lewis structure* showing appropriate shape and *dashed/wedged bonds*, based on VSEPR. Determine the **molecular geometry** around the central atom. For HClO_4 , label the *most polar bond*.



b. 3 points (Note: Lone pairs are not shown in structures below)

(i) For the three *resonance* structures (1, 2 and 3) shown below, answer the following question:



(I) Circle the *most contributing* resonance structure.

Structure 1

Structure 2

Structure 3

(ii) For the three structures given (4, 5, and 6) below, answer the following questions:



(I) Circle all the structures that represent *resonance structures* of each other:

Structure 4

Structure 5

Structure 6

(II) Among the resonance structures, circle the *most contributing* resonance structure.

Structure 4

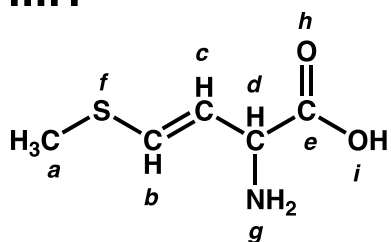
Structure 5

Structure 6

**Question 6** **6 points**

Using the data given below, calculate the **lattice energy** for SrO(s).

	ΔH° (kJ/mol)
$\text{Sr}(s) \longrightarrow \text{Sr}(g)$	+180
$\text{Sr}(g) \longrightarrow \text{Sr}^{2+}(g) + 2e^-$	+1468
$\text{O}_2(g) \longrightarrow 2\text{O}(g)$	+500
$\text{O}(g) + 2e^- \longrightarrow \text{O}^{2-}(g)$	+600
$\text{Sr}(s) + \frac{1}{2}\text{O}_2(g) \longrightarrow \text{SrO}(s)$	-550

**Question 7 10 points**

Consider the molecule given on the left, and answer questions (i) to (vi) using chemical bonding theories that we have learnt in class (valence bond theory, hybridization).

Note that the lone pairs are not shown in the structure:

i. Consider the carbon atoms marked in as ***a***, ***b***, ***c***, ***d***, and ***e*** in the molecule, and determine **hybridization** at each of the carbon centers. (2.5 points)

Atom	Hybridization
Carbon (<i>a</i>)	
Carbon (<i>b</i>)	
Carbon (<i>c</i>)	
Carbon (<i>d</i>)	
Carbon (<i>e</i>)	

ii. Now, consider the sulfur atom (S) marked as ***f***, nitrogen (N) atom (***g***), and oxygen(O) atoms (***h*** and ***i***), and determine their **hybridization**. (2 points)

Atom	Hybridization
Sulfur (<i>f</i>)	
Nitrogen (<i>g</i>)	
Oxygen (<i>h</i>)	
Oxygen (<i>i</i>)	

iii. **Describe** all the bonds formed by the carbon atoms ***a*** and ***b***, using sigma and pi designation as appropriate. (2 points)

Atom	Bonds formed (use sigma and pi as appropriate)
Carbon (<i>a</i>)	
Carbon (<i>b</i>)	

iv. How many total sp^3 -s sigma bonds are in the molecule? (1 point)

v. Determine **electronic** and **molecular geometry** at carbon (C) labeled ***e*** and sulfur (S) labeled ***f*** centres. (2 points)

Atom	Electronic Geometry	Molecular Geometry
Carbon (<i>e</i>)		
Sulfur (<i>f</i>)		

vi. Circle the value from the choices given below, that describes the **C-S-C bond angle at sulfur** (S) marked ***f***: (0.5 points)

$>90^\circ$ to $<109.5^\circ$ 109.5° >109.5 to $<120^\circ$ 120° $>120^\circ$ to $<180^\circ$


Question 8 12 points

Using the **valence shell orbitals only** draw a **complete** molecular orbital (MO) diagram (with no s-p mixing) for the molecule **PS**, in the space provided below. **(6 points)**

Assume that 3s and 3p atomic orbitals will form molecular orbitals like those formed by the combination of 2s and 2p atomic orbitals that we learnt in class, and there is no s-p mixing.

Show **energy arrow**, **atomic orbitals** involved, **atom labels**, the **molecular orbitals** that are formed, **filled electrons** in the MO diagram etc. Make sure to label all the atomic and molecular orbitals.

Using the MO diagram above, answer the following questions:

- (i) Which one of the following will have the highest bond order? Circle the correct choice
 A) PS^{1-} B) PS C) PS^{1+}
- (ii) Which one of the following will have the longest bond length? Circle the correct choice
 A) PS^{1-} B) PS C) PS^{1+}
- (iii) Which of the following will be diamagnetic? Circle the correct choice.
 A) PS^{1-} B) PS C) PS^{1+}
- (iv) Draw Lewis dot structure of the molecule with the shortest bond length (based on the MO diagram) from among the three (PS^{1-} , PS, PS^{1+}), and indicate the bond order calculated from the Lewis dot structure. **(3 points)**

Lewis Dot Structure

Bond Order



0BA74D83-40C0-4DE3-9C79-D6DB29565855

midterm-exam-93388

#153 Page 10 of 10

SCRAP PAGE

**CHEMISTRY 110 –Midterm Exam****Thursday November 7, 2024, 6:30 – 8:30 pm****Examiners: Pallavi Sirjoosingh and Ashok Kakkar**

First name (please write as legibly as possible within the boxes)

--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--

Last name

--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--

Student ID number

--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--

This booklet has 8 questions on 10 pages (The last page can be used as a scrap page).

The staple must NOT be removed from the question booklet. Your intact question booklet must be returned at the end of the exam.

A separate booklet contains a periodic table, and datasheet.

Write your answers directly under the questions in the space provided. Do not write on the QR codes or near the top edges (as it may be cut off).

Please write **legibly and dark enough** for the scan to be readable. For students who use English as a second language, dictionaries are permitted. Calculators are allowed.

DO NOT ASK QUESTIONS OF THE INVIGILATORS about interpretations of the exam questions. If you have a concern about a question, please summarize it at the top of this page. All concerns will be evaluated after the exam. GOOD LUCK!

**Question 1 8 points** (*Assume all transitions are possible*)

a. 6 points Calculate the **energy (in kJ/mol)** required to ionize (completely remove) the electron from the **$n=3$** level of **He^+ ion** (He: Helium). Show your work including equations. (3 sig. figs.)

b. 2 points

(I) Select the largest (longest) wavelength from the three choices below:

- i. Wavelength of photon to *completely remove* an electron from *1s orbital of Hydrogen atom*
- ii. Wavelength of photon to *completely remove* an electron from *1s orbital of Boron atom (B)*
- iii. Wavelength of photon to *completely remove* an electron from *1s orbital of Boron cation (B^+)*

i or ii or iii

(II) Select the smallest (shortest) wavelength from the three choices below (*for H atom*):

- i. Wavelength of photon *emitted* for an electron transitioning from *2p to 1s orbital*
- ii. Wavelength of photon *absorbed* for an electron transitioning from *2s to 4p orbital*
- iii. Wavelength of photon *absorbed* for an electron transitioning from *1s to 3p orbital*

i or ii or iii



Question 2 8 points



A Chem110 student plots the graph shown on the left, from data obtained for the kinetic energy (eV) of the ejected electron versus frequency (s^{-1}) of the incident light, for an unknown metal. ($1\text{eV} = 1.6022 \times 10^{-19} \text{ J}$)

Based on the plot, answer the questions **a to c** below:

a. 1 point On the plot above, label the **threshold frequency**.

b. 2 points What is the value of the work potential (in J) of the unknown metal?

c. 3 points What is the *longest* wavelength (in nm) required to eject an electron from this metal's surface? Show your work including equation(s) used. (3 sig figs)

d. 2 points In the space provided below, plot the following

(Assume frequency is *above* threshold frequency):

- Kinetic energy* of an ejected electron versus the *intensity* of the incident light (left)
- Number of electrons* ejected versus the *frequency* of the incident light (right)



**Question 3** **6 points****a. 2 points**

In the ground state of Ge (Germanium; $Z = 32$) atom, **name** the following orbitals:

i. Orbital containing the electron that requires the maximum energy to be removed.

ii. Orbital containing the electron that experiences the minimum effective nuclear charge.

b. 2 points

An unknown anion (A^-) has the following *excited* electronic configuration: $[Ar] 4s^2 3d^3 4p^1$

(Ar: Argon; $Z = 18$)

Give the **condensed ground state electronic configuration** of the atom, **A** and the cation, **A^{2+}** .
Show your work. (Assume all electronic transitions are possible)

Electronic Configuration of A:

Electronic Configuration of A^{2+} :

c. 2 points

In a single atom, determine how many electrons can have the following set of quantum numbers?

(i) $n = 5; l = 3$

In a single atom, determine how many electrons can have the following set of quantum numbers?

(i) $n = 3; l = 2; m_s = +1/2$

**Question 4** **8 points**

Rank the following from 1 to 4 in **increasing** order. Provide the ranking in the boxes provided.

Note: Atomic number (Z) are provided for relevant atoms

a) Increasing (lowest to highest) order of *ionization energy*: (1= lowest; 4 = highest)

Mg

Mg: Z = 12

Rb

Rb: Z = 37

Cs

Cs: Z = 55

K

K : Z = 19

b) Increasing (lowest to highest) order of *bond angle*: (1= lowest; 4 = highest)

H₂OCCl₄XeF₄

HCN

c) Increasing (smallest to largest) order of *ionic size* (1= smallest; 4 = largest)

S²⁻

S: Z = 16

O²⁻

O: Z = 8

Be²⁺

Be: Z = 4

Na⁺

Na: Z = 11

d) Increasing (lowest to highest) order of *bond energy* (1= lowest; 4 = highest)

P-F

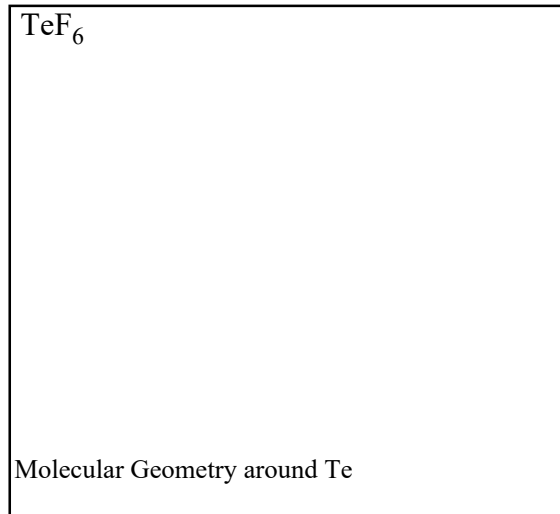
P-Cl

P-I

P-Br

**Question 5** **12 points**

a. 9 points For the molecules HClO_4 and TeF_6 , draw the most stable *Lewis structure* showing appropriate shape and *dashed/wedged bonds*, based on VSEPR. Determine the **molecular geometry** around the central atom. For HClO_4 , label the *most polar bond*.



b. 3 points (Note: Lone pairs are not shown in structures below)

(i) For the three *resonance* structures (1, 2 and 3) shown below, answer the following question:



(I) Circle the *most contributing* resonance structure.

Structure 1

Structure 2

Structure 3

(ii) For the three structures given (4, 5, and 6) below, answer the following questions:



(I) Circle all the structures that represent *resonance structures* of each other:

Structure 4

Structure 5

Structure 6

(II) Among the resonance structures, circle the *most contributing* resonance structure.

Structure 4

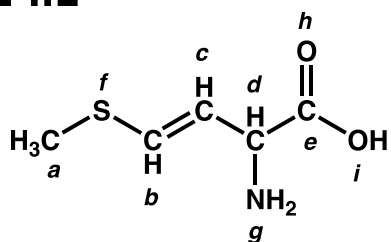
Structure 5

Structure 6

**Question 6** **6 points**

Using the data given below, calculate the **lattice energy** for SrO(s).

	ΔH° (kJ/mol)
$\text{Sr}(s) \longrightarrow \text{Sr}(g)$	+180
$\text{Sr}(g) \longrightarrow \text{Sr}^{2+}(g) + 2e^-$	+1468
$\text{O}_2(g) \longrightarrow 2\text{O}(g)$	+500
$\text{O}(g) + 2e^- \longrightarrow \text{O}^{2-}(g)$	+600
$\text{Sr}(s) + \frac{1}{2}\text{O}_2(g) \longrightarrow \text{SrO}(s)$	-550

**Question 7 10 points**

Consider the molecule given on the left, and answer questions (i) to (vi) using chemical bonding theories that we have learnt in class (valence bond theory, hybridization).

Note that the lone pairs are not shown in the structure:

i. Consider the carbon atoms marked in as ***a***, ***b***, ***c***, ***d***, and ***e*** in the molecule, and determine **hybridization** at each of the carbon centers. (2.5 points)

Atom	Hybridization
Carbon (<i>a</i>)	
Carbon (<i>b</i>)	
Carbon (<i>c</i>)	
Carbon (<i>d</i>)	
Carbon (<i>e</i>)	

ii. Now, consider the sulfur atom (S) marked as ***f***, nitrogen (N) atom (***g***), and oxygen(O) atoms (***h*** and ***i***), and determine their **hybridization**. (2 points)

Atom	Hybridization
Sulfur (<i>f</i>)	
Nitrogen (<i>g</i>)	
Oxygen (<i>h</i>)	
Oxygen (<i>i</i>)	

iii. **Describe** all the bonds formed by the carbon atoms ***a*** and ***b***, using sigma and pi designation as appropriate. (2 points)

Atom	Bonds formed (use sigma and pi as appropriate)
Carbon (<i>a</i>)	
Carbon (<i>b</i>)	

iv. How many total sp^3 -s sigma bonds are in the molecule? (1 point)

v. Determine **electronic** and **molecular geometry** at carbon (C) labeled ***e*** and sulfur (S) labeled ***f*** centres. (2 points)

Atom	Electronic Geometry	Molecular Geometry
Carbon (<i>e</i>)		
Sulfur (<i>f</i>)		

vi. Circle the value from the choices given below, that describes the **C-S-C bond angle at sulfur** (S) marked ***f***: (0.5 points)

>90° to <109.5° 109.5° >109.5 to <120° 120° >120° to <180°


Question 8 12 points

Using the **valence shell orbitals only** draw a **complete** molecular orbital (MO) diagram (with no s-p mixing) for the molecule **PS**, in the space provided below. **(6 points)**

Assume that 3s and 3p atomic orbitals will form molecular orbitals like those formed by the combination of 2s and 2p atomic orbitals that we learnt in class, and there is no s-p mixing.

Show **energy arrow**, **atomic orbitals** involved, **atom labels**, the **molecular orbitals** that are formed, **filled electrons** in the MO diagram etc. Make sure to **label all the atomic and molecular orbitals**.

Using the MO diagram above, answer the following questions:

- (i) Which one of the following will have the highest bond order? Circle the correct choice
 A) PS^{1-} B) PS C) PS^{1+}
- (ii) Which one of the following will have the longest bond length? Circle the correct choice
 A) PS^{1-} B) PS C) PS^{1+}
- (iii) Which of the following will be diamagnetic? Circle the correct choice.
 A) PS^{1-} B) PS C) PS^{1+}
- (iv) Draw Lewis dot structure of the molecule with the shortest bond length (based on the MO diagram) from among the three (PS^{1-} , PS, PS^{1+}), and indicate the bond order calculated from the Lewis dot structure. **(3 points)**

Lewis Dot Structure

Bond Order



0C0805B5-20A1-401B-AC71-CF0919E5008A

midterm-exam-93388

#154 Page 10 of 10

SCRAP PAGE

**CHEMISTRY 110 –Midterm Exam****Thursday November 7, 2024, 6:30 – 8:30 pm****Examiners: Pallavi Sirjoosingh and Ashok Kakkar**

First name (please write as legibly as possible within the boxes)

--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--

Last name

--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--

Student ID number

--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--

This booklet has 8 questions on 10 pages (The last page can be used as a scrap page).

The staple must NOT be removed from the question booklet. Your intact question booklet must be returned at the end of the exam.

A separate booklet contains a periodic table, and datasheet.

Write your answers directly under the questions in the space provided. Do not write on the QR codes or near the top edges (as it may be cut off).

Please write **legibly and dark enough** for the scan to be readable. For students who use English as a second language, dictionaries are permitted. Calculators are allowed.

DO NOT ASK QUESTIONS OF THE INVIGILATORS about interpretations of the exam questions. If you have a concern about a question, please summarize it at the top of this page. All concerns will be evaluated after the exam. GOOD LUCK!

**Question 1 8 points** (*Assume all transitions are possible*)

a. 6 points Calculate the **energy (in kJ/mol)** required to ionize (completely remove) the electron from the **n=3** level of **He⁺ ion** (He: Helium). Show your work including equations. (3 sig. figs.)

b. 2 points

(I) Select the largest (longest) wavelength from the three choices below:

- i. Wavelength of photon to *completely remove* an electron from *1s orbital of Hydrogen atom*
- ii. Wavelength of photon to *completely remove* an electron from *1s orbital of Boron atom (B)*
- iii. Wavelength of photon to *completely remove* an electron from *1s orbital of Boron cation (B⁺)*

i or ii or iii

(II) Select the smallest (shortest) wavelength from the three choices below (*for H atom*):

- i. Wavelength of photon *emitted* for an electron transitioning from *2p to 1s orbital*
- ii. Wavelength of photon *absorbed* for an electron transitioning from *2s to 4p orbital*
- iii. Wavelength of photon *absorbed* for an electron transitioning from *1s to 3p orbital*

i or ii or iii



Question 2 8 points



A Chem110 student plots the graph shown on the left, from data obtained for the kinetic energy (eV) of the ejected electron versus frequency (s^{-1}) of the incident light, for an unknown metal. ($1\text{eV} = 1.6022 \times 10^{-19} \text{ J}$)

Based on the plot, answer the questions **a to c** below:

a. 1 point On the plot above, label the **threshold frequency**.

b. 2 points What is the value of the work potential (in J) of the unknown metal?

c. 3 points What is the *longest* wavelength (in nm) required to eject an electron from this metal's surface? Show your work including equation(s) used. (3 sig figs)

d. 2 points In the space provided below, plot the following

(Assume frequency is *above* threshold frequency):

- i. *Kinetic energy* of an ejected electron versus the *intensity* of the incident light (left)
- ii. *Number of electrons* ejected versus the *frequency* of the incident light (right)



**Question 3** **6 points****a. 2 points**

In the ground state of Ge (Germanium; $Z = 32$) atom, **name** the following orbitals:

i. Orbital containing the electron that requires the maximum energy to be removed.

ii. Orbital containing the electron that experiences the minimum effective nuclear charge.

b. 2 points

An unknown anion (A^-) has the following *excited* electronic configuration: $[Ar] 4s^2 3d^3 4p^1$

(Ar: Argon; $Z = 18$)

Give the **condensed ground state electronic configuration** of the atom, **A** and the cation, **A^{2+}** .
Show your work. (Assume all electronic transitions are possible)

Electronic Configuration of A:

Electronic Configuration of A^{2+} :

c. 2 points

In a single atom, determine how many electrons can have the following set of quantum numbers?

(i) $n = 5; l = 3$

In a single atom, determine how many electrons can have the following set of quantum numbers?

(i) $n = 3; l = 2; m_s = +1/2$

**Question 4** **8 points**

Rank the following from 1 to 4 in **increasing** order. Provide the ranking in the boxes provided.

Note: Atomic number (Z) are provided for relevant atoms

a) Increasing (lowest to highest) order of *ionization energy*: (1= lowest; 4 = highest)

Mg

Mg: Z = 12

Rb

Rb: Z = 37

Cs

Cs: Z = 55

K

K : Z = 19

b) Increasing (lowest to highest) order of *bond angle*: (1= lowest; 4 = highest)

H₂OCCl₄XeF₄

HCN

c) Increasing (smallest to largest) order of *ionic size* (1= smallest; 4 = largest)

S²⁻

S: Z = 16

O²⁻

O: Z = 8

Be²⁺

Be: Z = 4

Na⁺

Na: Z = 11

d) Increasing (lowest to highest) order of *bond energy* (1= lowest; 4 = highest)

P-F

P-Cl

P-I

P-Br

**Question 5** **12 points**

a. 9 points For the molecules HClO_4 and TeF_6 , draw the most stable *Lewis structure* showing appropriate shape and *dashed/wedged bonds*, based on VSEPR. Determine the **molecular geometry** around the central atom. For HClO_4 , label the *most polar bond*.



b. 3 points (Note: Lone pairs are not shown in structures below)

(i) For the three *resonance* structures (1, 2 and 3) shown below, answer the following question:



(I) Circle the *most contributing* resonance structure.

Structure 1

Structure 2

Structure 3

(ii) For the three structures given (4, 5, and 6) below, answer the following questions:



(I) Circle all the structures that represent *resonance structures* of each other:

Structure 4

Structure 5

Structure 6

(II) Among the resonance structures, circle the *most contributing* resonance structure.

Structure 4

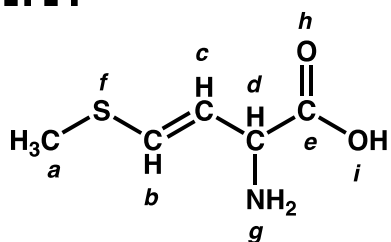
Structure 5

Structure 6

**Question 6** **6 points**

Using the data given below, calculate the **lattice energy** for SrO(s).

	ΔH° (kJ/mol)
$\text{Sr}(s) \longrightarrow \text{Sr}(g)$	+180
$\text{Sr}(g) \longrightarrow \text{Sr}^{2+}(g) + 2e^-$	+1468
$\text{O}_2(g) \longrightarrow 2\text{O}(g)$	+500
$\text{O}(g) + 2e^- \longrightarrow \text{O}^{2-}(g)$	+600
$\text{Sr}(s) + \frac{1}{2}\text{O}_2(g) \longrightarrow \text{SrO}(s)$	-550

**Question 7 10 points**

Consider the molecule given on the left, and answer questions (i) to (vi) using chemical bonding theories that we have learnt in class (valence bond theory, hybridization).

Note that the lone pairs are not shown in the structure:

i. Consider the carbon atoms marked in as ***a***, ***b***, ***c***, ***d***, and ***e*** in the molecule, and determine **hybridization** at each of the carbon centers. (2.5 points)

Atom	Hybridization
Carbon (<i>a</i>)	
Carbon (<i>b</i>)	
Carbon (<i>c</i>)	
Carbon (<i>d</i>)	
Carbon (<i>e</i>)	

ii. Now, consider the sulfur atom (S) marked as ***f***, nitrogen (N) atom (***g***), and oxygen(O) atoms (***h*** and ***i***), and determine their **hybridization**. (2 points)

Atom	Hybridization
Sulfur (<i>f</i>)	
Nitrogen (<i>g</i>)	
Oxygen (<i>h</i>)	
Oxygen (<i>i</i>)	

iii. **Describe** all the bonds formed by the carbon atoms ***a*** and ***b***, using sigma and pi designation as appropriate. (2 points)

Atom	Bonds formed (use sigma and pi as appropriate)
Carbon (<i>a</i>)	
Carbon (<i>b</i>)	

iv. How many total sp^3 -s sigma bonds are in the molecule? (1 point)

v. Determine **electronic** and **molecular geometry** at carbon (C) labeled ***e*** and sulfur (S) labeled ***f*** centres. (2 points)

Atom	Electronic Geometry	Molecular Geometry
Carbon (<i>e</i>)		
Sulfur (<i>f</i>)		

vi. Circle the value from the choices given below, that describes the **C-S-C bond angle at sulfur** (S) marked ***f***: (0.5 points)

>90° to <109.5° 109.5° >109.5 to <120° 120° >120° to <180°


Question 8 12 points

Using the **valence shell orbitals only** draw a **complete** molecular orbital (MO) diagram (with no s-p mixing) for the molecule **PS**, in the space provided below. **(6 points)**

Assume that 3s and 3p atomic orbitals will form molecular orbitals like those formed by the combination of 2s and 2p atomic orbitals that we learnt in class, and there is no s-p mixing.

Show **energy arrow**, **atomic orbitals** involved, **atom labels**, the **molecular orbitals** that are formed, **filled electrons** in the MO diagram etc. Make sure to label all the atomic and molecular orbitals.

Using the MO diagram above, answer the following questions:

- (i) Which one of the following will have the highest bond order? Circle the correct choice
 A) PS^{1-} B) PS C) PS^{1+}
- (ii) Which one of the following will have the longest bond length? Circle the correct choice
 A) PS^{1-} B) PS C) PS^{1+}
- (iii) Which of the following will be diamagnetic? Circle the correct choice.
 A) PS^{1-} B) PS C) PS^{1+}
- (iv) Draw Lewis dot structure of the molecule with the shortest bond length (based on the MO diagram) from among the three (PS^{1-} , PS, PS^{1+}), and indicate the bond order calculated from the Lewis dot structure. **(3 points)**

Lewis Dot Structure

Bond Order



4A974DD7-BEB7-483F-BDC3-E98BFCE1E95F

midterm-exam-93388

#155 Page 10 of 10

SCRAP PAGE

**CHEMISTRY 110 –Midterm Exam****Thursday November 7, 2024, 6:30 – 8:30 pm**

Examiners: Pallavi Sirjoosingh and Ashok Kakkar

First name (please write as legibly as possible within the boxes)

--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--

Last name

--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--

Student ID number

--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--

This booklet has 8 questions on 10 pages (The last page can be used as a scrap page).

The staple must NOT be removed from the question booklet. Your intact question booklet must be returned at the end of the exam.

A separate booklet contains a periodic table, and datasheet.

Write your answers directly under the questions in the space provided. Do not write on the QR codes or near the top edges (as it may be cut off).

Please write **legibly and dark enough** for the scan to be readable. For students who use English as a second language, dictionaries are permitted. Calculators are allowed.

DO NOT ASK QUESTIONS OF THE INVIGILATORS about interpretations of the exam questions. If you have a concern about a question, please summarize it at the top of this page. All concerns will be evaluated after the exam. GOOD LUCK!

**Question 1 8 points** (*Assume all transitions are possible*)

a. 6 points Calculate the **energy (in kJ/mol)** required to ionize (completely remove) the electron from the **n=3** level of **He⁺ ion** (He: Helium). Show your work including equations. (3 sig. figs.)

b. 2 points

(I) Select the largest (longest) wavelength from the three choices below:

- i. Wavelength of photon to *completely remove* an electron from *1s orbital of Hydrogen atom*
- ii. Wavelength of photon to *completely remove* an electron from *1s orbital of Boron atom (B)*
- iii. Wavelength of photon to *completely remove* an electron from *1s orbital of Boron cation (B⁺)*

i or ii or iii

(II) Select the smallest (shortest) wavelength from the three choices below (*for H atom*):

- i. Wavelength of photon *emitted* for an electron transitioning from *2p to 1s orbital*
- ii. Wavelength of photon *absorbed* for an electron transitioning from *2s to 4p orbital*
- iii. Wavelength of photon *absorbed* for an electron transitioning from *1s to 3p orbital*

i or ii or iii



Question 2 8 points



A Chem110 student plots the graph shown on the left, from data obtained for the kinetic energy (eV) of the ejected electron versus frequency (s^{-1}) of the incident light, for an unknown metal. ($1\text{eV} = 1.6022 \times 10^{-19} \text{ J}$)

Based on the plot, answer the questions **a to c** below:

a. 1 point On the plot above, label the **threshold frequency**.

b. 2 points What is the value of the work potential (in J) of the unknown metal?

c. 3 points What is the *longest* wavelength (in nm) required to eject an electron from this metal's surface? Show your work including equation(s) used. (3 sig figs)

d. 2 points In the space provided below, plot the following

(Assume frequency is *above* threshold frequency):

- Kinetic energy* of an ejected electron versus the *intensity* of the incident light (left)
- Number of electrons* ejected versus the *frequency* of the incident light (right)



**Question 3** **6 points****a. 2 points**

In the ground state of Ge (Germanium; $Z = 32$) atom, **name** the following orbitals:

i. Orbital containing the electron that requires the maximum energy to be removed.

ii. Orbital containing the electron that experiences the minimum effective nuclear charge.

b. 2 points

An unknown anion (A^-) has the following *excited* electronic configuration: $[Ar] 4s^2 3d^3 4p^1$

(Ar: Argon; $Z = 18$)

Give the **condensed ground state electronic configuration** of the atom, **A** and the cation, **A^{2+}** .
Show your work. (Assume all electronic transitions are possible)

Electronic Configuration of A:

Electronic Configuration of A^{2+} :

c. 2 points

In a single atom, determine how many electrons can have the following set of quantum numbers?

(i) $n = 5; l = 3$

In a single atom, determine how many electrons can have the following set of quantum numbers?

(i) $n = 3; l = 2; m_s = +1/2$

**Question 4** **8 points**

Rank the following from 1 to 4 in **increasing** order. Provide the ranking in the boxes provided.

Note: Atomic number (Z) are provided for relevant atoms

a) Increasing (lowest to highest) order of *ionization energy*: (1= lowest; 4 = highest)

Mg

Mg: Z = 12

Rb

Rb: Z = 37

Cs

Cs: Z = 55

K

K : Z = 19

b) Increasing (lowest to highest) order of *bond angle*: (1= lowest; 4 = highest)

H₂OCCl₄XeF₄

HCN

c) Increasing (smallest to largest) order of *ionic size* (1= smallest; 4 = largest)

S²⁻

S: Z = 16

O²⁻

O: Z = 8

Be²⁺

Be: Z = 4

Na⁺

Na: Z = 11

d) Increasing (lowest to highest) order of *bond energy* (1= lowest; 4 = highest)

P-F

P-Cl

P-I

P-Br

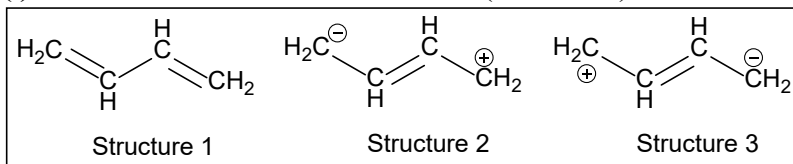
**Question 5** **12 points**

a. 9 points For the molecules HClO_4 and TeF_6 , draw the most stable *Lewis structure* showing appropriate shape and *dashed/wedged bonds*, based on VSEPR. Determine the **molecular geometry** around the central atom. For HClO_4 , label the *most polar bond*.



b. 3 points (Note: Lone pairs are not shown in structures below)

(i) For the three *resonance* structures (1, 2 and 3) shown below, answer the following question:



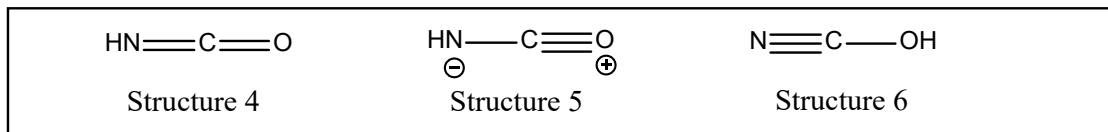
(I) Circle the *most contributing* resonance structure.

Structure 1

Structure 2

Structure 3

(ii) For the three structures given (4, 5, and 6) below, answer the following questions:



(I) Circle all the structures that represent *resonance structures* of each other:

Structure 4

Structure 5

Structure 6

(II) Among the resonance structures, circle the *most contributing* resonance structure.

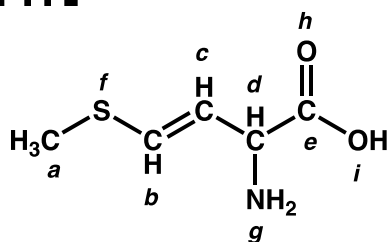
Structure 4

Structure 5

Structure 6

**Question 6** **6 points**Using the data given below, calculate the **lattice energy** for SrO(s).

	ΔH° (kJ/mol)
$\text{Sr}(s) \longrightarrow \text{Sr}(g)$	+180
$\text{Sr}(g) \longrightarrow \text{Sr}^{2+}(g) + 2e^-$	+1468
$\text{O}_2(g) \longrightarrow 2\text{O}(g)$	+500
$\text{O}(g) + 2e^- \longrightarrow \text{O}^{2-}(g)$	+600
$\text{Sr}(s) + \frac{1}{2}\text{O}_2(g) \longrightarrow \text{SrO}(s)$	-550

**Question 7 10 points**

Consider the molecule given on the left, and answer questions (i) to (vi) using chemical bonding theories that we have learnt in class (valence bond theory, hybridization).

Note that the lone pairs are not shown in the structure:

i. Consider the carbon atoms marked in as ***a***, ***b***, ***c***, ***d***, and ***e*** in the molecule, and determine **hybridization** at each of the carbon centers. (2.5 points)

Atom	Hybridization
Carbon (<i>a</i>)	
Carbon (<i>b</i>)	
Carbon (<i>c</i>)	
Carbon (<i>d</i>)	
Carbon (<i>e</i>)	

ii. Now, consider the sulfur atom (S) marked as ***f***, nitrogen (N) atom (***g***), and oxygen(O) atoms (***h*** and ***i***), and determine their **hybridization**. (2 points)

Atom	Hybridization
Sulfur (<i>f</i>)	
Nitrogen (<i>g</i>)	
Oxygen (<i>h</i>)	
Oxygen (<i>i</i>)	

iii. **Describe** all the bonds formed by the carbon atoms ***a*** and ***b***, using sigma and pi designation as appropriate. (2 points)

Atom	Bonds formed (use sigma and pi as appropriate)
Carbon (<i>a</i>)	
Carbon (<i>b</i>)	

iv. How many total sp^3 -s sigma bonds are in the molecule? (1 point)

v. Determine **electronic** and **molecular geometry** at carbon (C) labeled ***e*** and sulfur (S) labeled ***f*** centres. (2 points)

Atom	Electronic Geometry	Molecular Geometry
Carbon (<i>e</i>)		
Sulfur (<i>f</i>)		

vi. Circle the value from the choices given below, that describes the **C-S-C bond angle at sulfur** (S) marked ***f***: (0.5 points)

>90° to <109.5° 109.5° >109.5 to <120° 120° >120° to <180°


Question 8 12 points

Using the **valence shell orbitals only** draw a **complete** molecular orbital (MO) diagram (with no s-p mixing) for the molecule **PS**, in the space provided below. **(6 points)**

Assume that 3s and 3p atomic orbitals will form molecular orbitals like those formed by the combination of 2s and 2p atomic orbitals that we learnt in class, and there is no s-p mixing.

Show **energy arrow**, **atomic orbitals** involved, **atom labels**, the **molecular orbitals** that are formed, **filled electrons** in the MO diagram etc. Make sure to label all the atomic and molecular orbitals.

Using the MO diagram above, answer the following questions:

- (i) Which one of the following will have the highest bond order? Circle the correct choice
 A) PS^{1-} B) PS C) PS^{1+}
- (ii) Which one of the following will have the longest bond length? Circle the correct choice
 A) PS^{1-} B) PS C) PS^{1+}
- (iii) Which of the following will be diamagnetic? Circle the correct choice.
 A) PS^{1-} B) PS C) PS^{1+}
- (iv) Draw Lewis dot structure of the molecule with the shortest bond length (based on the MO diagram) from among the three (PS^{1-} , PS, PS^{1+}), and indicate the bond order calculated from the Lewis dot structure. **(3 points)**

Lewis Dot Structure

Bond Order



6B5B9196-D264-456D-BC34-EC351C85BE59

midterm-exam-93388

#156 Page 10 of 10

SCRAP PAGE

**CHEMISTRY 110 –Midterm Exam****Thursday November 7, 2024, 6:30 – 8:30 pm**

Examiners: Pallavi Sirjoosingh and Ashok Kakkar

First name (please write as legibly as possible within the boxes)

--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--

Last name

--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--

Student ID number

--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--

This booklet has 8 questions on 10 pages (The last page can be used as a scrap page).

The staple must NOT be removed from the question booklet. Your intact question booklet must be returned at the end of the exam.

A separate booklet contains a periodic table, and datasheet.

Write your answers directly under the questions in the space provided. Do not write on the QR codes or near the top edges (as it may be cut off).

Please write **legibly and dark enough** for the scan to be readable. For students who use English as a second language, dictionaries are permitted. Calculators are allowed.

DO NOT ASK QUESTIONS OF THE INVIGILATORS about interpretations of the exam questions. If you have a concern about a question, please summarize it at the top of this page. All concerns will be evaluated after the exam. GOOD LUCK!

**Question 1 8 points** (*Assume all transitions are possible*)

a. 6 points Calculate the **energy (in kJ/mol)** required to ionize (completely remove) the electron from the **n=3** level of **He⁺ ion** (He: Helium). Show your work including equations. (3 sig. figs.)

b. 2 points

(I) Select the largest (longest) wavelength from the three choices below:

- i. Wavelength of photon to *completely remove* an electron from *1s orbital of Hydrogen atom*
- ii. Wavelength of photon to *completely remove* an electron from *1s orbital of Boron atom (B)*
- iii. Wavelength of photon to *completely remove* an electron from *1s orbital of Boron cation (B⁺)*

i or ii or iii

(II) Select the smallest (shortest) wavelength from the three choices below (*for H atom*):

- i. Wavelength of photon *emitted* for an electron transitioning from *2p to 1s orbital*
- ii. Wavelength of photon *absorbed* for an electron transitioning from *2s to 4p orbital*
- iii. Wavelength of photon *absorbed* for an electron transitioning from *1s to 3p orbital*

i or ii or iii



Question 2 8 points



A Chem110 student plots the graph shown on the left, from data obtained for the kinetic energy (eV) of the ejected electron versus frequency (s^{-1}) of the incident light, for an unknown metal. ($1\text{eV} = 1.6022 \times 10^{-19} \text{ J}$)

Based on the plot, answer the questions **a to c** below:

a. 1 point On the plot above, label the **threshold frequency**.

b. 2 points What is the value of the work potential (in J) of the unknown metal?

c. 3 points What is the *longest* wavelength (in nm) required to eject an electron from this metal's surface? Show your work including equation(s) used. (3 sig figs)

d. 2 points In the space provided below, plot the following

(Assume frequency is *above* threshold frequency):

- i. *Kinetic energy* of an ejected electron versus the *intensity* of the incident light (left)
- ii. *Number of electrons* ejected versus the *frequency* of the incident light (right)



**Question 3** **6 points****a. 2 points**

In the ground state of Ge (Germanium; $Z = 32$) atom, **name** the following orbitals:

i. Orbital containing the electron that requires the maximum energy to be removed.

ii. Orbital containing the electron that experiences the minimum effective nuclear charge.

b. 2 points

An unknown anion (A^-) has the following *excited* electronic configuration: $[Ar] 4s^2 3d^3 4p^1$

(Ar: Argon; $Z = 18$)

Give the **condensed ground state electronic configuration** of the atom, **A** and the cation, **A^{2+}** .
Show your work. (Assume all electronic transitions are possible)

Electronic Configuration of A:

Electronic Configuration of A^{2+} :

c. 2 points

In a single atom, determine how many electrons can have the following set of quantum numbers?

(i) $n = 5; l = 3$

In a single atom, determine how many electrons can have the following set of quantum numbers?

(i) $n = 3; l = 2; m_s = +1/2$

**Question 4** **8 points**

Rank the following from 1 to 4 in **increasing** order. Provide the ranking in the boxes provided.

Note: Atomic number (Z) are provided for relevant atoms

a) Increasing (lowest to highest) order of *ionization energy*: (1= lowest; 4 = highest)

Mg

Mg: Z = 12

Rb

Rb: Z = 37

Cs

Cs: Z = 55

K

K : Z = 19

b) Increasing (lowest to highest) order of *bond angle*: (1= lowest; 4 = highest)

H₂OCCl₄XeF₄

HCN

c) Increasing (smallest to largest) order of *ionic size* (1= smallest; 4 = largest)

S²⁻

S: Z = 16

O²⁻

O: Z = 8

Be²⁺

Be: Z = 4

Na⁺

Na: Z = 11

d) Increasing (lowest to highest) order of *bond energy* (1= lowest; 4 = highest)

P-F

P-Cl

P-I

P-Br

**Question 5** **12 points**

a. 9 points For the molecules HClO_4 and TeF_6 , draw the most stable *Lewis structure* showing appropriate shape and *dashed/wedged bonds*, based on VSEPR. Determine the **molecular geometry** around the central atom. For HClO_4 , label the *most polar bond*.

HClO_4 (there are no O-O bonds)

Label the most polar bond

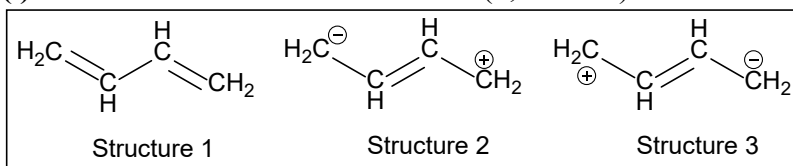
Molecular Geometry around Cl

TeF_6

Molecular Geometry around Te

b. 3 points (Note: Lone pairs are not shown in structures below)

(i) For the three *resonance* structures (1, 2 and 3) shown below, answer the following question:



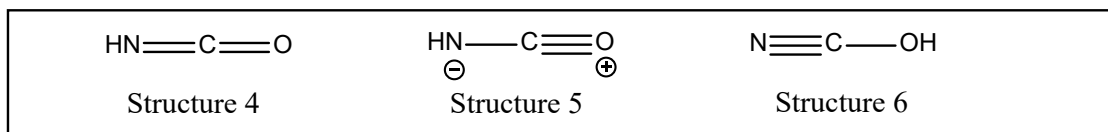
(I) Circle the *most contributing* resonance structure.

Structure 1

Structure 2

Structure 3

(ii) For the three structures given (4, 5, and 6) below, answer the following questions:



(I) Circle all the structures that represent *resonance structures* of each other:

Structure 4

Structure 5

Structure 6

(II) Among the resonance structures, circle the *most contributing* resonance structure.

Structure 4

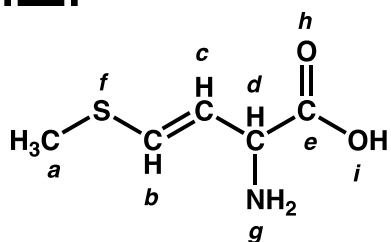
Structure 5

Structure 6

**Question 6** **6 points**

Using the data given below, calculate the **lattice energy** for SrO(s).

	ΔH° (kJ/mol)
$\text{Sr}(s) \longrightarrow \text{Sr}(g)$	+180
$\text{Sr}(g) \longrightarrow \text{Sr}^{2+}(g) + 2e^-$	+1468
$\text{O}_2(g) \longrightarrow 2\text{O}(g)$	+500
$\text{O}(g) + 2e^- \longrightarrow \text{O}^{2-}(g)$	+600
$\text{Sr}(s) + \frac{1}{2}\text{O}_2(g) \longrightarrow \text{SrO}(s)$	-550

**Question 7 10 points**

Consider the molecule given on the left, and answer questions (i) to (vi) using chemical bonding theories that we have learnt in class (valence bond theory, hybridization).

Note that the lone pairs are not shown in the structure:

i. Consider the carbon atoms marked in as ***a***, ***b***, ***c***, ***d***, and ***e*** in the molecule, and determine **hybridization** at each of the carbon centers. (2.5 points)

Atom	Hybridization
Carbon (<i>a</i>)	
Carbon (<i>b</i>)	
Carbon (<i>c</i>)	
Carbon (<i>d</i>)	
Carbon (<i>e</i>)	

ii. Now, consider the sulfur atom (S) marked as ***f***, nitrogen (N) atom (***g***), and oxygen(O) atoms (***h*** and ***i***), and determine their **hybridization**. (2 points)

Atom	Hybridization
Sulfur (<i>f</i>)	
Nitrogen (<i>g</i>)	
Oxygen (<i>h</i>)	
Oxygen (<i>i</i>)	

iii. **Describe** all the bonds formed by the carbon atoms ***a*** and ***b***, using sigma and pi designation as appropriate. (2 points)

Atom	Bonds formed (use sigma and pi as appropriate)
Carbon (<i>a</i>)	
Carbon (<i>b</i>)	

iv. How many total sp^3 -s sigma bonds are in the molecule? (1 point)

v. Determine **electronic** and **molecular geometry** at carbon (C) labeled ***e*** and sulfur (S) labeled ***f*** centres. (2 points)

Atom	Electronic Geometry	Molecular Geometry
Carbon (<i>e</i>)		
Sulfur (<i>f</i>)		

vi. Circle the value from the choices given below, that describes the **C-S-C bond angle at sulfur** (S) marked ***f***: (0.5 points)

$>90^\circ$ to $<109.5^\circ$ 109.5° >109.5 to $<120^\circ$ 120° $>120^\circ$ to $<180^\circ$


Question 8 12 points

Using the **valence shell orbitals only** draw a **complete** molecular orbital (MO) diagram (with no s-p mixing) for the molecule **PS**, in the space provided below. **(6 points)**

Assume that 3s and 3p atomic orbitals will form molecular orbitals like those formed by the combination of 2s and 2p atomic orbitals that we learnt in class, and there is no s-p mixing.

Show **energy arrow**, **atomic orbitals** involved, **atom labels**, the **molecular orbitals** that are formed, **filled electrons** in the MO diagram etc. Make sure to label all the atomic and molecular orbitals.

Using the MO diagram above, answer the following questions:

- (i) Which one of the following will have the highest bond order? Circle the correct choice
 A) PS^{1-} B) PS C) PS^{1+}
- (ii) Which one of the following will have the longest bond length? Circle the correct choice
 A) PS^{1-} B) PS C) PS^{1+}
- (iii) Which of the following will be diamagnetic? Circle the correct choice.
 A) PS^{1-} B) PS C) PS^{1+}
- (iv) Draw Lewis dot structure of the molecule with the shortest bond length (based on the MO diagram) from among the three (PS^{1-} , PS, PS^{1+}), and indicate the bond order calculated from the Lewis dot structure. **(3 points)**

Lewis Dot Structure

Bond Order



E639F7E3-1BD0-414C-96D8-50EAFE54446B

midterm-exam-93388

#157 Page 10 of 10

SCRAP PAGE

**CHEMISTRY 110 –Midterm Exam****Thursday November 7, 2024, 6:30 – 8:30 pm**

Examiners: Pallavi Sirjoosingh and Ashok Kakkar

First name (please write as legibly as possible within the boxes)

--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--

Last name

--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--

Student ID number

--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--

This booklet has 8 questions on 10 pages (The last page can be used as a scrap page).

The staple must NOT be removed from the question booklet. Your intact question booklet must be returned at the end of the exam.

A separate booklet contains a periodic table, and datasheet.

Write your answers directly under the questions in the space provided. Do not write on the QR codes or near the top edges (as it may be cut off).

Please write **legibly and dark enough** for the scan to be readable. For students who use English as a second language, dictionaries are permitted. Calculators are allowed.

DO NOT ASK QUESTIONS OF THE INVIGILATORS about interpretations of the exam questions. If you have a concern about a question, please summarize it at the top of this page. All concerns will be evaluated after the exam. GOOD LUCK!

**Question 1 8 points** (*Assume all transitions are possible*)

a. 6 points Calculate the **energy (in kJ/mol)** required to ionize (completely remove) the electron from the **n=3** level of **He⁺ ion** (He: Helium). Show your work including equations. (3 sig. figs.)

b. 2 points

(I) Select the largest (longest) wavelength from the three choices below:

- i. Wavelength of photon to *completely remove* an electron from *1s orbital of Hydrogen atom*
- ii. Wavelength of photon to *completely remove* an electron from *1s orbital of Boron atom (B)*
- iii. Wavelength of photon to *completely remove* an electron from *1s orbital of Boron cation (B⁺)*

i or ii or iii

(II) Select the smallest (shortest) wavelength from the three choices below (*for H atom*):

- i. Wavelength of photon *emitted* for an electron transitioning from *2p to 1s orbital*
- ii. Wavelength of photon *absorbed* for an electron transitioning from *2s to 4p orbital*
- iii. Wavelength of photon *absorbed* for an electron transitioning from *1s to 3p orbital*

i or ii or iii



Question 2 8 points



A Chem110 student plots the graph shown on the left, from data obtained for the kinetic energy (eV) of the ejected electron versus frequency (s^{-1}) of the incident light, for an unknown metal. ($1\text{eV} = 1.6022 \times 10^{-19} \text{ J}$)

Based on the plot, answer the questions **a to c** below:

a. 1 point On the plot above, label the **threshold frequency**.

b. 2 points What is the value of the work potential (in J) of the unknown metal?

c. 3 points What is the *longest* wavelength (in nm) required to eject an electron from this metal's surface? Show your work including equation(s) used. (3 sig figs)

d. 2 points In the space provided below, plot the following

(Assume frequency is *above* threshold frequency):

- Kinetic energy* of an ejected electron versus the *intensity* of the incident light (left)
- Number of electrons* ejected versus the *frequency* of the incident light (right)



**Question 3** **6 points****a. 2 points**

In the ground state of Ge (Germanium; $Z = 32$) atom, **name** the following orbitals:

i. Orbital containing the electron that requires the maximum energy to be removed.

ii. Orbital containing the electron that experiences the minimum effective nuclear charge.

b. 2 points

An unknown anion (A^-) has the following *excited* electronic configuration: $[Ar] 4s^2 3d^3 4p^1$

(Ar: Argon; $Z = 18$)

Give the **condensed ground state electronic configuration** of the atom, **A** and the cation, **A^{2+}** .
Show your work. (Assume all electronic transitions are possible)

Electronic Configuration of A:

Electronic Configuration of A^{2+} :

c. 2 points

In a single atom, determine how many electrons can have the following set of quantum numbers?

(i) $n = 5; l = 3$

In a single atom, determine how many electrons can have the following set of quantum numbers?

(i) $n = 3; l = 2; m_s = +1/2$

**Question 4** **8 points**

Rank the following from 1 to 4 in **increasing** order. Provide the ranking in the boxes provided.

Note: Atomic number (Z) are provided for relevant atoms

a) Increasing (lowest to highest) order of *ionization energy*: (1= lowest; 4 = highest)

Mg

Mg: Z = 12

Rb

Rb: Z = 37

Cs

Cs: Z = 55

K

K : Z = 19

b) Increasing (lowest to highest) order of *bond angle*: (1= lowest; 4 = highest)

H₂OCCl₄XeF₄

HCN

c) Increasing (smallest to largest) order of *ionic size* (1= smallest; 4 = largest)

S²⁻

S: Z = 16

O²⁻

O: Z = 8

Be²⁺

Be: Z = 4

Na⁺

Na: Z = 11

d) Increasing (lowest to highest) order of *bond energy* (1= lowest; 4 = highest)

P-F

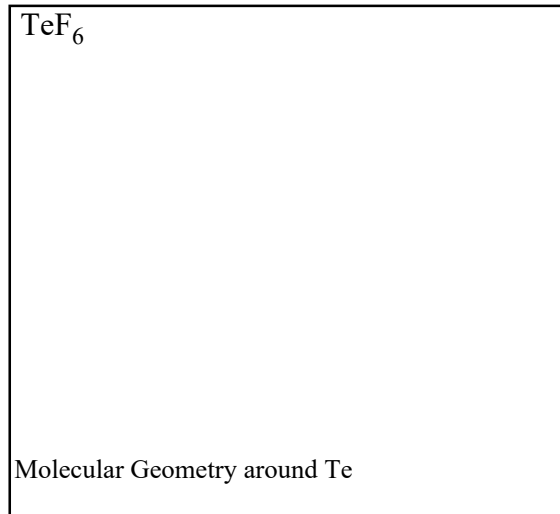
P-Cl

P-I

P-Br

**Question 5** **12 points**

a. 9 points For the molecules HClO_4 and TeF_6 , draw the most stable *Lewis structure* showing appropriate shape and *dashed/wedged bonds*, based on VSEPR. Determine the **molecular geometry** around the central atom. For HClO_4 , label the *most polar bond*.



b. 3 points (Note: Lone pairs are not shown in structures below)

(i) For the three *resonance* structures (1, 2 and 3) shown below, answer the following question:



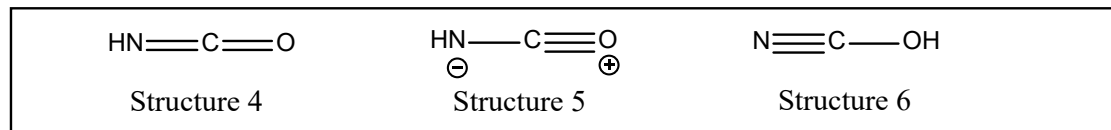
(I) Circle the *most contributing* resonance structure.

Structure 1

Structure 2

Structure 3

(ii) For the three structures given (4, 5, and 6) below, answer the following questions:



(I) Circle all the structures that represent *resonance structures* of each other:

Structure 4

Structure 5

Structure 6

(II) Among the resonance structures, circle the *most contributing* resonance structure.

Structure 4

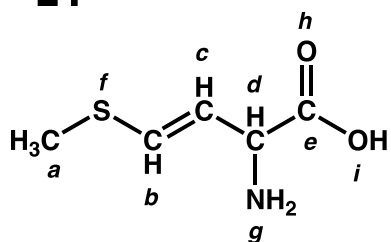
Structure 5

Structure 6

**Question 6** **6 points**

Using the data given below, calculate the **lattice energy** for SrO(s).

	ΔH° (kJ/mol)
$\text{Sr}(s) \longrightarrow \text{Sr}(g)$	+180
$\text{Sr}(g) \longrightarrow \text{Sr}^{2+}(g) + 2e^-$	+1468
$\text{O}_2(g) \longrightarrow 2\text{O}(g)$	+500
$\text{O}(g) + 2e^- \longrightarrow \text{O}^{2-}(g)$	+600
$\text{Sr}(s) + \frac{1}{2}\text{O}_2(g) \longrightarrow \text{SrO}(s)$	-550

**Question 7 10 points**

Consider the molecule given on the left, and answer questions (i) to (vi) using chemical bonding theories that we have learnt in class (valence bond theory, hybridization).

Note that the lone pairs are not shown in the structure:

i. Consider the carbon atoms marked in as ***a***, ***b***, ***c***, ***d***, and ***e*** in the molecule, and determine **hybridization** at each of the carbon centers. (2.5 points)

Atom	Hybridization
Carbon (<i>a</i>)	
Carbon (<i>b</i>)	
Carbon (<i>c</i>)	
Carbon (<i>d</i>)	
Carbon (<i>e</i>)	

ii. Now, consider the sulfur atom (S) marked as ***f***, nitrogen (N) atom (***g***), and oxygen(O) atoms (***h*** and ***i***), and determine their **hybridization**. (2 points)

Atom	Hybridization
Sulfur (<i>f</i>)	
Nitrogen (<i>g</i>)	
Oxygen (<i>h</i>)	
Oxygen (<i>i</i>)	

iii. **Describe** all the bonds formed by the carbon atoms ***a*** and ***b***, using sigma and pi designation as appropriate. (2 points)

Atom	Bonds formed (use sigma and pi as appropriate)
Carbon (<i>a</i>)	
Carbon (<i>b</i>)	

iv. How many total sp^3 -s sigma bonds are in the molecule? (1 point)

v. Determine **electronic** and **molecular geometry** at carbon (C) labeled ***e*** and sulfur (S) labeled ***f*** centres. (2 points)

Atom	Electronic Geometry	Molecular Geometry
Carbon (<i>e</i>)		
Sulfur (<i>f</i>)		

vi. Circle the value from the choices given below, that describes the **C-S-C bond angle at sulfur** (S) marked ***f***: (0.5 points)

>90° to <109.5° 109.5° >109.5 to <120° 120° >120° to <180°


Question 8 12 points

Using the **valence shell orbitals only** draw a **complete** molecular orbital (MO) diagram (with no s-p mixing) for the molecule **PS**, in the space provided below. **(6 points)**

Assume that 3s and 3p atomic orbitals will form molecular orbitals like those formed by the combination of 2s and 2p atomic orbitals that we learnt in class, and there is no s-p mixing.

Show **energy arrow**, **atomic orbitals** involved, **atom labels**, the **molecular orbitals** that are formed, **filled electrons** in the MO diagram etc. Make sure to label all the atomic and molecular orbitals.

Using the MO diagram above, answer the following questions:

- (i) Which one of the following will have the highest bond order? Circle the correct choice
 A) PS^{1-} B) PS C) PS^{1+}
- (ii) Which one of the following will have the longest bond length? Circle the correct choice
 A) PS^{1-} B) PS C) PS^{1+}
- (iii) Which of the following will be diamagnetic? Circle the correct choice.
 A) PS^{1-} B) PS C) PS^{1+}
- (iv) Draw Lewis dot structure of the molecule with the shortest bond length (based on the MO diagram) from among the three (PS^{1-} , PS, PS^{1+}), and indicate the bond order calculated from the Lewis dot structure. **(3 points)**

Lewis Dot Structure

Bond Order



392D9EFD-6AEE-409B-B6C2-C4ECC5C92272

midterm-exam-93388

#158 Page 10 of 10

SCRAP PAGE

**CHEMISTRY 110 –Midterm Exam****Thursday November 7, 2024, 6:30 – 8:30 pm**

Examiners: Pallavi Sirjoosingh and Ashok Kakkar

First name (please write as legibly as possible within the boxes)

--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--

Last name

--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--

Student ID number

--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--

This booklet has 8 questions on 10 pages (The last page can be used as a scrap page).

The staple must NOT be removed from the question booklet. Your intact question booklet must be returned at the end of the exam.

A separate booklet contains a periodic table, and datasheet.

Write your answers directly under the questions in the space provided. Do not write on the QR codes or near the top edges (as it may be cut off).

Please write **legibly and dark enough** for the scan to be readable. For students who use English as a second language, dictionaries are permitted. Calculators are allowed.

DO NOT ASK QUESTIONS OF THE INVIGILATORS about interpretations of the exam questions. If you have a concern about a question, please summarize it at the top of this page. All concerns will be evaluated after the exam. GOOD LUCK!

**Question 1 8 points** (*Assume all transitions are possible*)

a. 6 points Calculate the **energy (in kJ/mol)** required to ionize (completely remove) the electron from the **$n=3$** level of **He^+ ion** (He: Helium). Show your work including equations. (3 sig. figs.)

b. 2 points

(I) Select the largest (longest) wavelength from the three choices below:

- i. Wavelength of photon to *completely remove* an electron from *1s orbital of Hydrogen atom*
- ii. Wavelength of photon to *completely remove* an electron from *1s orbital of Boron atom (B)*
- iii. Wavelength of photon to *completely remove* an electron from *1s orbital of Boron cation (B^+)*

i or ii or iii

(II) Select the smallest (shortest) wavelength from the three choices below (*for H atom*):

- i. Wavelength of photon *emitted* for an electron transitioning from *2p to 1s orbital*
- ii. Wavelength of photon *absorbed* for an electron transitioning from *2s to 4p orbital*
- iii. Wavelength of photon *absorbed* for an electron transitioning from *1s to 3p orbital*

i or ii or iii



Question 2 8 points



A Chem110 student plots the graph shown on the left, from data obtained for the kinetic energy (eV) of the ejected electron versus frequency (s^{-1}) of the incident light, for an unknown metal. ($1\text{eV} = 1.6022 \times 10^{-19} \text{ J}$)

Based on the plot, answer the questions **a to c** below:

a. 1 point On the plot above, label the **threshold frequency**.

b. 2 points What is the value of the work potential (in J) of the unknown metal?

c. 3 points What is the *longest* wavelength (in nm) required to eject an electron from this metal's surface? Show your work including equation(s) used. (3 sig figs)

d. 2 points In the space provided below, plot the following

(Assume frequency is *above* threshold frequency):

- i. *Kinetic energy* of an ejected electron versus the *intensity* of the incident light (left)
- ii. *Number of electrons* ejected versus the *frequency* of the incident light (right)



**Question 3** **6 points****a. 2 points**

In the ground state of Ge (Germanium; $Z = 32$) atom, **name** the following orbitals:

i. Orbital containing the electron that requires the maximum energy to be removed.

ii. Orbital containing the electron that experiences the minimum effective nuclear charge.

b. 2 points

An unknown anion (A^-) has the following *excited* electronic configuration: $[Ar] 4s^2 3d^3 4p^1$

(Ar: Argon; $Z = 18$)

Give the **condensed ground state electronic configuration** of the atom, **A** and the cation, **A^{2+}** .
Show your work. (Assume all electronic transitions are possible)

Electronic Configuration of A:

Electronic Configuration of A^{2+} :

c. 2 points

In a single atom, determine how many electrons can have the following set of quantum numbers?

(i) $n = 5; l = 3$

In a single atom, determine how many electrons can have the following set of quantum numbers?

(i) $n = 3; l = 2; m_s = +1/2$

**Question 4** **8 points**

Rank the following from 1 to 4 in **increasing** order. Provide the ranking in the boxes provided.

Note: Atomic number (Z) are provided for relevant atoms

a) Increasing (lowest to highest) order of *ionization energy*: (1= lowest; 4 = highest)

Mg

Mg: Z = 12

Rb

Rb: Z = 37

Cs

Cs: Z = 55

K

K : Z = 19

b) Increasing (lowest to highest) order of *bond angle*: (1= lowest; 4 = highest)

H₂OCCl₄XeF₄

HCN

c) Increasing (smallest to largest) order of *ionic size* (1= smallest; 4 = largest)

S²⁻

S: Z = 16

O²⁻

O: Z = 8

Be²⁺

Be: Z = 4

Na⁺

Na: Z = 11

d) Increasing (lowest to highest) order of *bond energy* (1= lowest; 4 = highest)

P-F

P-Cl

P-I

P-Br

**Question 5** **12 points**

a. 9 points For the molecules HClO_4 and TeF_6 , draw the most stable *Lewis structure* showing appropriate shape and *dashed/wedged bonds*, based on VSEPR. Determine the **molecular geometry** around the central atom. For HClO_4 , label the *most polar bond*.



b. 3 points (Note: Lone pairs are not shown in structures below)

(i) For the three *resonance* structures (1, 2 and 3) shown below, answer the following question:



(I) Circle the *most contributing* resonance structure.

Structure 1

Structure 2

Structure 3

(ii) For the three structures given (4, 5, and 6) below, answer the following questions:



(I) Circle all the structures that represent *resonance structures* of each other:

Structure 4

Structure 5

Structure 6

(II) Among the resonance structures, circle the *most contributing* resonance structure.

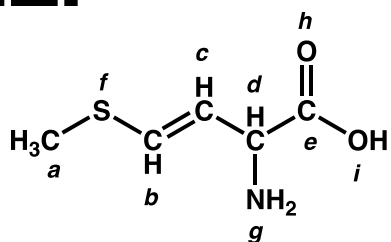
Structure 4

Structure 5

Structure 6

**Question 6** **6 points**Using the data given below, calculate the **lattice energy** for SrO(s).

	ΔH° (kJ/mol)
$\text{Sr}(s) \longrightarrow \text{Sr}(g)$	+180
$\text{Sr}(g) \longrightarrow \text{Sr}^{2+}(g) + 2e^-$	+1468
$\text{O}_2(g) \longrightarrow 2\text{O}(g)$	+500
$\text{O}(g) + 2e^- \longrightarrow \text{O}^{2-}(g)$	+600
$\text{Sr}(s) + \frac{1}{2}\text{O}_2(g) \longrightarrow \text{SrO}(s)$	-550

**Question 7 10 points**

Consider the molecule given on the left, and answer questions (i) to (vi) using chemical bonding theories that we have learnt in class (valence bond theory, hybridization).

Note that the lone pairs are not shown in the structure:

i. Consider the carbon atoms marked in as ***a***, ***b***, ***c***, ***d***, and ***e*** in the molecule, and determine **hybridization** at each of the carbon centers. (2.5 points)

Atom	Hybridization
Carbon (<i>a</i>)	
Carbon (<i>b</i>)	
Carbon (<i>c</i>)	
Carbon (<i>d</i>)	
Carbon (<i>e</i>)	

ii. Now, consider the sulfur atom (S) marked as ***f***, nitrogen (N) atom (***g***), and oxygen(O) atoms (***h*** and ***i***), and determine their **hybridization**. (2 points)

Atom	Hybridization
Sulfur (<i>f</i>)	
Nitrogen (<i>g</i>)	
Oxygen (<i>h</i>)	
Oxygen (<i>i</i>)	

iii. **Describe** all the bonds formed by the carbon atoms ***a*** and ***b***, using sigma and pi designation as appropriate. (2 points)

Atom	Bonds formed (use sigma and pi as appropriate)
Carbon (<i>a</i>)	
Carbon (<i>b</i>)	

iv. How many total sp^3 -s sigma bonds are in the molecule? (1 point)

v. Determine **electronic** and **molecular geometry** at carbon (C) labeled ***e*** and sulfur (S) labeled ***f*** centres. (2 points)

Atom	Electronic Geometry	Molecular Geometry
Carbon (<i>e</i>)		
Sulfur (<i>f</i>)		

vi. Circle the value from the choices given below, that describes the **C-S-C bond angle at sulfur** (S) marked ***f***: (0.5 points)

$>90^\circ$ to $<109.5^\circ$ 109.5° >109.5 to $<120^\circ$ 120° $>120^\circ$ to $<180^\circ$


Question 8 12 points

Using the **valence shell orbitals only** draw a **complete** molecular orbital (MO) diagram (with no s-p mixing) for the molecule **PS**, in the space provided below. **(6 points)**

Assume that 3s and 3p atomic orbitals will form molecular orbitals like those formed by the combination of 2s and 2p atomic orbitals that we learnt in class, and there is no s-p mixing.

Show **energy arrow**, **atomic orbitals** involved, **atom labels**, the **molecular orbitals** that are formed, **filled electrons** in the MO diagram etc. Make sure to label all the atomic and molecular orbitals.

Using the MO diagram above, answer the following questions:

- (i) Which one of the following will have the highest bond order? Circle the correct choice
 A) PS^{1-} B) PS C) PS^{1+}
- (ii) Which one of the following will have the longest bond length? Circle the correct choice
 A) PS^{1-} B) PS C) PS^{1+}
- (iii) Which of the following will be diamagnetic? Circle the correct choice.
 A) PS^{1-} B) PS C) PS^{1+}
- (iv) Draw Lewis dot structure of the molecule with the shortest bond length (based on the MO diagram) from among the three (PS^{1-} , PS, PS^{1+}), and indicate the bond order calculated from the Lewis dot structure. **(3 points)**

Lewis Dot Structure

Bond Order



7A95A8D2-83A5-48A4-AE5C-4A2004546C65

midterm-exam-93388

#159 Page 10 of 10

SCRAP PAGE

**CHEMISTRY 110 –Midterm Exam****Thursday November 7, 2024, 6:30 – 8:30 pm**

Examiners: Pallavi Sirjoosingh and Ashok Kakkar

First name (please write as legibly as possible within the boxes)

--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--

Last name

--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--

Student ID number

--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--

This booklet has 8 questions on 10 pages (The last page can be used as a scrap page).

The staple must NOT be removed from the question booklet. Your intact question booklet must be returned at the end of the exam.

A separate booklet contains a periodic table, and datasheet.

Write your answers directly under the questions in the space provided. Do not write on the QR codes or near the top edges (as it may be cut off).

Please write **legibly and dark enough** for the scan to be readable. For students who use English as a second language, dictionaries are permitted. Calculators are allowed.

DO NOT ASK QUESTIONS OF THE INVIGILATORS about interpretations of the exam questions. If you have a concern about a question, please summarize it at the top of this page. All concerns will be evaluated after the exam. GOOD LUCK!

**Question 1 8 points** (*Assume all transitions are possible*)

a. 6 points Calculate the **energy (in kJ/mol)** required to ionize (completely remove) the electron from the **n=3** level of **He⁺ ion** (He: Helium). Show your work including equations. (3 sig. figs.)

b. 2 points

(I) Select the largest (longest) wavelength from the three choices below:

- i. Wavelength of photon to *completely remove* an electron from *1s orbital of Hydrogen atom*
- ii. Wavelength of photon to *completely remove* an electron from *1s orbital of Boron atom (B)*
- iii. Wavelength of photon to *completely remove* an electron from *1s orbital of Boron cation (B⁺)*

i or ii or iii

(II) Select the smallest (shortest) wavelength from the three choices below (*for H atom*):

- i. Wavelength of photon *emitted* for an electron transitioning from *2p to 1s orbital*
- ii. Wavelength of photon *absorbed* for an electron transitioning from *2s to 4p orbital*
- iii. Wavelength of photon *absorbed* for an electron transitioning from *1s to 3p orbital*

i or ii or iii



Question 2 8 points



A Chem110 student plots the graph shown on the left, from data obtained for the kinetic energy (eV) of the ejected electron versus frequency (s^{-1}) of the incident light, for an unknown metal. ($1\text{eV} = 1.6022 \times 10^{-19} \text{ J}$)

Based on the plot, answer the questions **a to c** below:

a. 1 point On the plot above, label the **threshold frequency**.

b. 2 points What is the value of the work potential (in J) of the unknown metal?

c. 3 points What is the *longest* wavelength (in nm) required to eject an electron from this metal's surface? Show your work including equation(s) used. (3 sig figs)

d. 2 points In the space provided below, plot the following

(Assume frequency is *above* threshold frequency):

- Kinetic energy* of an ejected electron versus the *intensity* of the incident light (left)
- Number of electrons* ejected versus the *frequency* of the incident light (right)



**Question 3** **6 points****a. 2 points**

In the ground state of Ge (Germanium; $Z = 32$) atom, **name** the following orbitals:

i. Orbital containing the electron that requires the maximum energy to be removed.

ii. Orbital containing the electron that experiences the minimum effective nuclear charge.

b. 2 points

An unknown anion (A^-) has the following *excited* electronic configuration: $[Ar] 4s^2 3d^3 4p^1$

(Ar: Argon; $Z = 18$)

Give the **condensed ground state electronic configuration** of the atom, **A** and the cation, **A^{2+}** .
Show your work. (Assume all electronic transitions are possible)

Electronic Configuration of A:

Electronic Configuration of A^{2+} :

c. 2 points

In a single atom, determine how many electrons can have the following set of quantum numbers?

(i) $n = 5; l = 3$

In a single atom, determine how many electrons can have the following set of quantum numbers?

(i) $n = 3; l = 2; m_s = +1/2$

**Question 4** **8 points**

Rank the following from 1 to 4 in **increasing** order. Provide the ranking in the boxes provided.

Note: Atomic number (Z) are provided for relevant atoms

a) Increasing (lowest to highest) order of *ionization energy*: (1= lowest; 4 = highest)

Mg

Mg: Z = 12

Rb

Rb: Z = 37

Cs

Cs: Z = 55

K

K : Z = 19

b) Increasing (lowest to highest) order of *bond angle*: (1= lowest; 4 = highest)

H₂OCCl₄XeF₄

HCN

c) Increasing (smallest to largest) order of *ionic size* (1= smallest; 4 = largest)

S²⁻

S: Z = 16

O²⁻

O: Z = 8

Be²⁺

Be: Z = 4

Na⁺

Na: Z = 11

d) Increasing (lowest to highest) order of *bond energy* (1= lowest; 4 = highest)

P-F

P-Cl

P-I

P-Br

**Question 5** **12 points**

a. 9 points For the molecules HClO_4 and TeF_6 , draw the most stable *Lewis structure* showing appropriate shape and *dashed/wedged bonds*, based on VSEPR. Determine the **molecular geometry** around the central atom. For HClO_4 , label the *most polar bond*.



b. 3 points (Note: Lone pairs are not shown in structures below)

(i) For the three *resonance* structures (1, 2 and 3) shown below, answer the following question:



(I) Circle the *most contributing* resonance structure.

Structure 1

Structure 2

Structure 3

(ii) For the three structures given (4, 5, and 6) below, answer the following questions:



(I) Circle all the structures that represent *resonance structures* of each other:

Structure 4

Structure 5

Structure 6

(II) Among the resonance structures, circle the *most contributing* resonance structure.

Structure 4

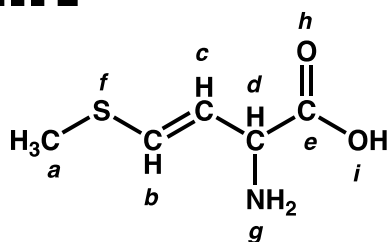
Structure 5

Structure 6

**Question 6** **6 points**

Using the data given below, calculate the **lattice energy** for SrO(s).

	ΔH° (kJ/mol)
$\text{Sr}(s) \longrightarrow \text{Sr}(g)$	+180
$\text{Sr}(g) \longrightarrow \text{Sr}^{2+}(g) + 2e^-$	+1468
$\text{O}_2(g) \longrightarrow 2\text{O}(g)$	+500
$\text{O}(g) + 2e^- \longrightarrow \text{O}^{2-}(g)$	+600
$\text{Sr}(s) + \frac{1}{2}\text{O}_2(g) \longrightarrow \text{SrO}(s)$	-550

**Question 7 10 points**

Consider the molecule given on the left, and answer questions (i) to (vi) using chemical bonding theories that we have learnt in class (valence bond theory, hybridization).

Note that the lone pairs are not shown in the structure:

i. Consider the carbon atoms marked in as ***a***, ***b***, ***c***, ***d***, and ***e*** in the molecule, and determine **hybridization** at each of the carbon centers. (2.5 points)

Atom	Hybridization
Carbon (<i>a</i>)	
Carbon (<i>b</i>)	
Carbon (<i>c</i>)	
Carbon (<i>d</i>)	
Carbon (<i>e</i>)	

ii. Now, consider the sulfur atom (S) marked as ***f***, nitrogen (N) atom (***g***), and oxygen(O) atoms (***h*** and ***i***), and determine their **hybridization**. (2 points)

Atom	Hybridization
Sulfur (<i>f</i>)	
Nitrogen (<i>g</i>)	
Oxygen (<i>h</i>)	
Oxygen (<i>i</i>)	

iii. **Describe** all the bonds formed by the carbon atoms ***a*** and ***b***, using sigma and pi designation as appropriate. (2 points)

Atom	Bonds formed (use sigma and pi as appropriate)
Carbon (<i>a</i>)	
Carbon (<i>b</i>)	

iv. How many total sp^3 -s sigma bonds are in the molecule? (1 point)

v. Determine **electronic** and **molecular geometry** at carbon (C) labeled ***e*** and sulfur (S) labeled ***f*** centres. (2 points)

Atom	Electronic Geometry	Molecular Geometry
Carbon (<i>e</i>)		
Sulfur (<i>f</i>)		

vi. Circle the value from the choices given below, that describes the **C-S-C bond angle at sulfur** (S) marked ***f***: (0.5 points)

>90° to <109.5° 109.5° >109.5 to <120° 120° >120° to <180°

**Question 8 12 points**

Using the **valence shell orbitals only** draw a **complete** molecular orbital (MO) diagram (with no s-p mixing) for the molecule **PS**, in the space provided below. **(6 points)**

Assume that 3s and 3p atomic orbitals will form molecular orbitals like those formed by the combination of 2s and 2p atomic orbitals that we learnt in class, and there is no s-p mixing.

Show **energy arrow**, **atomic orbitals** involved, **atom labels**, the **molecular orbitals** that are formed, **filled electrons** in the MO diagram etc. Make sure to label all the atomic and molecular orbitals.

Using the MO diagram above, answer the following questions:

- (i) Which one of the following will have the highest bond order? Circle the correct choice
A) PS^{1-} B) PS C) PS^{1+}
- (ii) Which one of the following will have the longest bond length? Circle the correct choice
A) PS^{1-} B) PS C) PS^{1+}
- (iii) Which of the following will be diamagnetic? Circle the correct choice.
A) PS^{1-} B) PS C) PS^{1+}
- (iv) Draw Lewis dot structure of the molecule with the shortest bond length (based on the MO diagram) from among the three (PS^{1-} , PS, PS^{1+}), and indicate the bond order calculated from the Lewis dot structure. **(3 points)**

Lewis Dot Structure

Bond Order



CB0A0966-06B1-4F41-9805-531FC1F18F53

midterm-exam-93388

#160 Page 10 of 10

SCRAP PAGE

**CHEMISTRY 110 –Midterm Exam****Thursday November 7, 2024, 6:30 – 8:30 pm****Examiners: Pallavi Sirjoosingh and Ashok Kakkar**

First name (please write as legibly as possible within the boxes)

--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--

Last name

--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--

Student ID number

--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--

This booklet has 8 questions on 10 pages (The last page can be used as a scrap page).

The staple must NOT be removed from the question booklet. Your intact question booklet must be returned at the end of the exam.

A separate booklet contains a periodic table, and datasheet.

Write your answers directly under the questions in the space provided. Do not write on the QR codes or near the top edges (as it may be cut off).

Please write **legibly and dark enough** for the scan to be readable. For students who use English as a second language, dictionaries are permitted. Calculators are allowed.

DO NOT ASK QUESTIONS OF THE INVIGILATORS about interpretations of the exam questions. If you have a concern about a question, please summarize it at the top of this page. All concerns will be evaluated after the exam. GOOD LUCK!

**Question 1 8 points** (*Assume all transitions are possible*)

a. 6 points Calculate the **energy (in kJ/mol)** required to ionize (completely remove) the electron from the **n=3** level of **He⁺ ion** (He: Helium). Show your work including equations. (3 sig. figs.)

b. 2 points

(I) Select the largest (longest) wavelength from the three choices below:

- i. Wavelength of photon to *completely remove* an electron from *1s orbital of Hydrogen atom*
- ii. Wavelength of photon to *completely remove* an electron from *1s orbital of Boron atom (B)*
- iii. Wavelength of photon to *completely remove* an electron from *1s orbital of Boron cation (B⁺)*

i or ii or iii

(II) Select the smallest (shortest) wavelength from the three choices below (*for H atom*):

- i. Wavelength of photon *emitted* for an electron transitioning from *2p to 1s orbital*
- ii. Wavelength of photon *absorbed* for an electron transitioning from *2s to 4p orbital*
- iii. Wavelength of photon *absorbed* for an electron transitioning from *1s to 3p orbital*

i or ii or iii



Question 2 8 points



A Chem110 student plots the graph shown on the left, from data obtained for the kinetic energy (eV) of the ejected electron versus frequency (s^{-1}) of the incident light, for an unknown metal. ($1\text{eV} = 1.6022 \times 10^{-19} \text{ J}$)

Based on the plot, answer the questions **a to c** below:

a. 1 point On the plot above, label the **threshold frequency**.

b. 2 points What is the value of the work potential (in J) of the unknown metal?

c. 3 points What is the *longest* wavelength (in nm) required to eject an electron from this metal's surface? Show your work including equation(s) used. (3 sig figs)

d. 2 points In the space provided below, plot the following

(Assume frequency is *above* threshold frequency):

- i. *Kinetic energy* of an ejected electron versus the *intensity* of the incident light (left)
- ii. *Number of electrons* ejected versus the *frequency* of the incident light (right)



**Question 3** **6 points****a. 2 points**

In the ground state of Ge (Germanium; $Z = 32$) atom, **name** the following orbitals:

i. Orbital containing the electron that requires the maximum energy to be removed.

ii. Orbital containing the electron that experiences the minimum effective nuclear charge.

b. 2 points

An unknown anion (A^-) has the following *excited* electronic configuration: $[Ar] 4s^2 3d^3 4p^1$

(Ar: Argon; $Z = 18$)

Give the **condensed ground state electronic configuration** of the atom, **A** and the cation, **A^{2+}** .
Show your work. (Assume all electronic transitions are possible)

Electronic Configuration of A:

Electronic Configuration of A^{2+} :

c. 2 points

In a single atom, determine how many electrons can have the following set of quantum numbers?

(i) $n = 5; l = 3$

In a single atom, determine how many electrons can have the following set of quantum numbers?

(i) $n = 3; l = 2; m_s = +1/2$

**Question 4** **8 points**

Rank the following from 1 to 4 in **increasing** order. Provide the ranking in the boxes provided.

Note: Atomic number (Z) are provided for relevant atoms

a) Increasing (lowest to highest) order of *ionization energy*: (1= lowest; 4 = highest)

Mg

Mg: Z = 12

Rb

Rb: Z = 37

Cs

Cs: Z = 55

K

K : Z = 19

b) Increasing (lowest to highest) order of *bond angle*: (1= lowest; 4 = highest)

H₂OCCl₄XeF₄

HCN

c) Increasing (smallest to largest) order of *ionic size* (1= smallest; 4 = largest)

S²⁻

S: Z = 16

O²⁻

O: Z = 8

Be²⁺

Be: Z = 4

Na⁺

Na: Z = 11

d) Increasing (lowest to highest) order of *bond energy* (1= lowest; 4 = highest)

P-F

P-Cl

P-I

P-Br

**Question 5** **12 points**

a. 9 points For the molecules HClO_4 and TeF_6 , draw the most stable *Lewis structure* showing appropriate shape and *dashed/wedged bonds*, based on VSEPR. Determine the **molecular geometry** around the central atom. For HClO_4 , label the *most polar bond*.



b. 3 points (Note: Lone pairs are not shown in structures below)

(i) For the three *resonance* structures (1, 2 and 3) shown below, answer the following question:



(I) Circle the *most contributing* resonance structure.

Structure 1

Structure 2

Structure 3

(ii) For the three structures given (4, 5, and 6) below, answer the following questions:



(I) Circle all the structures that represent *resonance structures* of each other:

Structure 4

Structure 5

Structure 6

(II) Among the resonance structures, circle the *most contributing* resonance structure.

Structure 4

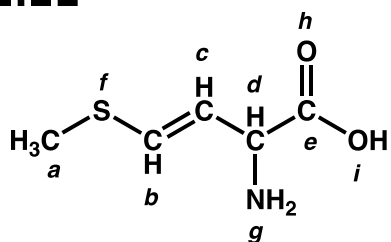
Structure 5

Structure 6

**Question 6** **6 points**

Using the data given below, calculate the **lattice energy** for SrO(s).

	ΔH° (kJ/mol)
$\text{Sr}(s) \longrightarrow \text{Sr}(g)$	+180
$\text{Sr}(g) \longrightarrow \text{Sr}^{2+}(g) + 2e^-$	+1468
$\text{O}_2(g) \longrightarrow 2\text{O}(g)$	+500
$\text{O}(g) + 2e^- \longrightarrow \text{O}^{2-}(g)$	+600
$\text{Sr}(s) + \frac{1}{2}\text{O}_2(g) \longrightarrow \text{SrO}(s)$	-550

**Question 7 10 points**

Consider the molecule given on the left, and answer questions (i) to (vi) using chemical bonding theories that we have learnt in class (valence bond theory, hybridization).

Note that the lone pairs are not shown in the structure:

i. Consider the carbon atoms marked in as ***a***, ***b***, ***c***, ***d***, and ***e*** in the molecule, and determine **hybridization** at each of the carbon centers. (2.5 points)

Atom	Hybridization
Carbon (<i>a</i>)	
Carbon (<i>b</i>)	
Carbon (<i>c</i>)	
Carbon (<i>d</i>)	
Carbon (<i>e</i>)	

ii. Now, consider the sulfur atom (S) marked as ***f***, nitrogen (N) atom (***g***), and oxygen(O) atoms (***h*** and ***i***), and determine their **hybridization**. (2 points)

Atom	Hybridization
Sulfur (<i>f</i>)	
Nitrogen (<i>g</i>)	
Oxygen (<i>h</i>)	
Oxygen (<i>i</i>)	

iii. **Describe** all the bonds formed by the carbon atoms ***a*** and ***b***, using sigma and pi designation as appropriate. (2 points)

Atom	Bonds formed (use sigma and pi as appropriate)
Carbon (<i>a</i>)	
Carbon (<i>b</i>)	

iv. How many total sp^3 -s sigma bonds are in the molecule? (1 point)

v. Determine **electronic** and **molecular geometry** at carbon (C) labeled ***e*** and sulfur (S) labeled ***f*** centres. (2 points)

Atom	Electronic Geometry	Molecular Geometry
Carbon (<i>e</i>)		
Sulfur (<i>f</i>)		

vi. Circle the value from the choices given below, that describes the **C-S-C bond angle at sulfur** (S) marked ***f***: (0.5 points)

>90° to <109.5° 109.5° >109.5 to <120° 120° >120° to <180°


Question 8 12 points

Using the **valence shell orbitals only** draw a **complete** molecular orbital (MO) diagram (with no s-p mixing) for the molecule **PS**, in the space provided below. **(6 points)**

Assume that 3s and 3p atomic orbitals will form molecular orbitals like those formed by the combination of 2s and 2p atomic orbitals that we learnt in class, and there is no s-p mixing.

Show **energy arrow**, **atomic orbitals** involved, **atom labels**, the **molecular orbitals** that are formed, **filled electrons** in the MO diagram etc. Make sure to **label all the atomic and molecular orbitals**.

Using the MO diagram above, answer the following questions:

- (i) Which one of the following will have the highest bond order? Circle the correct choice
 A) PS^{1-} B) PS C) PS^{1+}
- (ii) Which one of the following will have the longest bond length? Circle the correct choice
 A) PS^{1-} B) PS C) PS^{1+}
- (iii) Which of the following will be diamagnetic? Circle the correct choice.
 A) PS^{1-} B) PS C) PS^{1+}
- (iv) Draw Lewis dot structure of the molecule with the shortest bond length (based on the MO diagram) from among the three (PS^{1-} , PS, PS^{1+}), and indicate the bond order calculated from the Lewis dot structure. **(3 points)**

Lewis Dot Structure

Bond Order



613C08A5-2AAF-45B2-8216-D564BA5F7B12

midterm-exam-93388

#161 Page 10 of 10

SCRAP PAGE

**CHEMISTRY 110 –Midterm Exam****Thursday November 7, 2024, 6:30 – 8:30 pm****Examiners: Pallavi Sirjoosingh and Ashok Kakkar**

First name (please write as legibly as possible within the boxes)

--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--

Last name

--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--

Student ID number

--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--

This booklet has 8 questions on 10 pages (The last page can be used as a scrap page).

The staple must NOT be removed from the question booklet. Your intact question booklet must be returned at the end of the exam.

A separate booklet contains a periodic table, and datasheet.

Write your answers directly under the questions in the space provided. Do not write on the QR codes or near the top edges (as it may be cut off).

Please write **legibly and dark enough** for the scan to be readable. For students who use English as a second language, dictionaries are permitted. Calculators are allowed.

DO NOT ASK QUESTIONS OF THE INVIGILATORS about interpretations of the exam questions. If you have a concern about a question, please summarize it at the top of this page. All concerns will be evaluated after the exam. GOOD LUCK!

**Question 1 8 points** (*Assume all transitions are possible*)

a. 6 points Calculate the **energy (in kJ/mol)** required to ionize (completely remove) the electron from the **$n=3$** level of **He^+ ion** (He: Helium). Show your work including equations. (3 sig. figs.)

b. 2 points

(I) Select the largest (longest) wavelength from the three choices below:

- i. Wavelength of photon to *completely remove* an electron from *1s orbital of Hydrogen atom*
- ii. Wavelength of photon to *completely remove* an electron from *1s orbital of Boron atom (B)*
- iii. Wavelength of photon to *completely remove* an electron from *1s orbital of Boron cation (B^+)*

i or ii or iii

(II) Select the smallest (shortest) wavelength from the three choices below (*for H atom*):

- i. Wavelength of photon *emitted* for an electron transitioning from *2p to 1s orbital*
- ii. Wavelength of photon *absorbed* for an electron transitioning from *2s to 4p orbital*
- iii. Wavelength of photon *absorbed* for an electron transitioning from *1s to 3p orbital*

i or ii or iii



Question 2 8 points



A Chem110 student plots the graph shown on the left, from data obtained for the kinetic energy (eV) of the ejected electron versus frequency (s^{-1}) of the incident light, for an unknown metal. ($1\text{eV} = 1.6022 \times 10^{-19} \text{ J}$)

Based on the plot, answer the questions **a to c** below:

a. 1 point On the plot above, label the **threshold frequency**.

b. 2 points What is the value of the work potential (in J) of the unknown metal?

c. 3 points What is the *longest* wavelength (in nm) required to eject an electron from this metal's surface? Show your work including equation(s) used. (3 sig figs)

d. 2 points In the space provided below, plot the following

(Assume frequency is *above* threshold frequency):

- i. *Kinetic energy* of an ejected electron versus the *intensity* of the incident light (left)
- ii. *Number of electrons* ejected versus the *frequency* of the incident light (right)



**Question 3** **6 points****a. 2 points**

In the ground state of Ge (Germanium; $Z = 32$) atom, **name** the following orbitals:

i. Orbital containing the electron that requires the maximum energy to be removed.

ii. Orbital containing the electron that experiences the minimum effective nuclear charge.

b. 2 points

An unknown anion (A^-) has the following *excited* electronic configuration: $[Ar] 4s^2 3d^3 4p^1$

(Ar: Argon; $Z = 18$)

Give the **condensed ground state electronic configuration** of the atom, **A** and the cation, **A^{2+}** .
Show your work. (Assume all electronic transitions are possible)

Electronic Configuration of A:

Electronic Configuration of A^{2+} :

c. 2 points

In a single atom, determine how many electrons can have the following set of quantum numbers?

(i) $n = 5; l = 3$

In a single atom, determine how many electrons can have the following set of quantum numbers?

(i) $n = 3; l = 2; m_s = +1/2$

**Question 4** **8 points**

Rank the following from 1 to 4 in **increasing** order. Provide the ranking in the boxes provided.

Note: Atomic number (Z) are provided for relevant atoms

a) Increasing (lowest to highest) order of *ionization energy*: (1= lowest; 4 = highest)

Mg

Mg: Z = 12

Rb

Rb: Z = 37

Cs

Cs: Z = 55

K

K : Z = 19

b) Increasing (lowest to highest) order of *bond angle*: (1= lowest; 4 = highest)

H₂OCCl₄XeF₄

HCN

c) Increasing (smallest to largest) order of *ionic size* (1= smallest; 4 = largest)

S²⁻

S: Z = 16

O²⁻

O: Z = 8

Be²⁺

Be: Z = 4

Na⁺

Na: Z = 11

d) Increasing (lowest to highest) order of *bond energy* (1= lowest; 4 = highest)

P-F

P-Cl

P-I

P-Br

**Question 5** **12 points**

a. 9 points For the molecules HClO_4 and TeF_6 , draw the most stable *Lewis structure* showing appropriate shape and *dashed/wedged bonds*, based on VSEPR. Determine the **molecular geometry** around the central atom. For HClO_4 , label the *most polar bond*.



b. 3 points (Note: Lone pairs are not shown in structures below)

(i) For the three *resonance* structures (1, 2 and 3) shown below, answer the following question:



(I) Circle the *most contributing* resonance structure.

Structure 1

Structure 2

Structure 3

(ii) For the three structures given (4, 5, and 6) below, answer the following questions:



(I) Circle all the structures that represent *resonance structures* of each other:

Structure 4

Structure 5

Structure 6

(II) Among the resonance structures, circle the *most contributing* resonance structure.

Structure 4

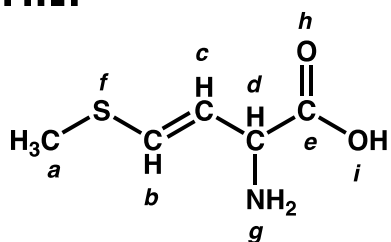
Structure 5

Structure 6

**Question 6** **6 points**

Using the data given below, calculate the **lattice energy** for SrO(s).

	ΔH° (kJ/mol)
$\text{Sr}(s) \longrightarrow \text{Sr}(g)$	+180
$\text{Sr}(g) \longrightarrow \text{Sr}^{2+}(g) + 2e^-$	+1468
$\text{O}_2(g) \longrightarrow 2\text{O}(g)$	+500
$\text{O}(g) + 2e^- \longrightarrow \text{O}^{2-}(g)$	+600
$\text{Sr}(s) + \frac{1}{2}\text{O}_2(g) \longrightarrow \text{SrO}(s)$	-550

**Question 7 10 points**

Consider the molecule given on the left, and answer questions (i) to (vi) using chemical bonding theories that we have learnt in class (valence bond theory, hybridization).

Note that the lone pairs are not shown in the structure:

i. Consider the carbon atoms marked in as ***a***, ***b***, ***c***, ***d***, and ***e*** in the molecule, and determine **hybridization** at each of the carbon centers. (2.5 points)

Atom	Hybridization
Carbon (<i>a</i>)	
Carbon (<i>b</i>)	
Carbon (<i>c</i>)	
Carbon (<i>d</i>)	
Carbon (<i>e</i>)	

ii. Now, consider the sulfur atom (S) marked as ***f***, nitrogen (N) atom (***g***), and oxygen(O) atoms (***h*** and ***i***), and determine their **hybridization**. (2 points)

Atom	Hybridization
Sulfur (<i>f</i>)	
Nitrogen (<i>g</i>)	
Oxygen (<i>h</i>)	
Oxygen (<i>i</i>)	

iii. **Describe** all the bonds formed by the carbon atoms ***a*** and ***b***, using sigma and pi designation as appropriate. (2 points)

Atom	Bonds formed (use sigma and pi as appropriate)
Carbon (<i>a</i>)	
Carbon (<i>b</i>)	

iv. How many total sp^3 -s sigma bonds are in the molecule? (1 point)

v. Determine **electronic** and **molecular geometry** at carbon (C) labeled ***e*** and sulfur (S) labeled ***f*** centres. (2 points)

Atom	Electronic Geometry	Molecular Geometry
Carbon (<i>e</i>)		
Sulfur (<i>f</i>)		

vi. Circle the value from the choices given below, that describes the **C-S-C bond angle at sulfur** (S) marked ***f***: (0.5 points)

>90° to <109.5°

109.5°

>109.5 to <120°

120°

>120° to <180°


Question 8 12 points

Using the **valence shell orbitals only** draw a **complete** molecular orbital (MO) diagram (with no s-p mixing) for the molecule **PS**, in the space provided below. **(6 points)**

Assume that 3s and 3p atomic orbitals will form molecular orbitals like those formed by the combination of 2s and 2p atomic orbitals that we learnt in class, and there is no s-p mixing.

Show **energy arrow**, **atomic orbitals** involved, **atom labels**, the **molecular orbitals** that are formed, **filled electrons** in the MO diagram etc. Make sure to label all the atomic and molecular orbitals.

Using the MO diagram above, answer the following questions:

- (i) Which one of the following will have the highest bond order? Circle the correct choice
 A) PS^{1-} B) PS C) PS^{1+}
- (ii) Which one of the following will have the longest bond length? Circle the correct choice
 A) PS^{1-} B) PS C) PS^{1+}
- (iii) Which of the following will be diamagnetic? Circle the correct choice.
 A) PS^{1-} B) PS C) PS^{1+}
- (iv) Draw Lewis dot structure of the molecule with the shortest bond length (based on the MO diagram) from among the three (PS^{1-} , PS, PS^{1+}), and indicate the bond order calculated from the Lewis dot structure. **(3 points)**

Lewis Dot Structure

Bond Order



19979B15-CE4A-4C00-9C55-63AFFCFF1CA2

midterm-exam-93388

#162 Page 10 of 10

SCRAP PAGE

**CHEMISTRY 110 –Midterm Exam****Thursday November 7, 2024, 6:30 – 8:30 pm**

Examiners: Pallavi Sirjoosingh and Ashok Kakkar

First name (please write as legibly as possible within the boxes)

--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--

Last name

--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--

Student ID number

--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--

This booklet has 8 questions on 10 pages (The last page can be used as a scrap page).

The staple must NOT be removed from the question booklet. Your intact question booklet must be returned at the end of the exam.

A separate booklet contains a periodic table, and datasheet.

Write your answers directly under the questions in the space provided. Do not write on the QR codes or near the top edges (as it may be cut off).

Please write **legibly and dark enough** for the scan to be readable. For students who use English as a second language, dictionaries are permitted. Calculators are allowed.

DO NOT ASK QUESTIONS OF THE INVIGILATORS about interpretations of the exam questions. If you have a concern about a question, please summarize it at the top of this page. All concerns will be evaluated after the exam. GOOD LUCK!

**Question 1 8 points** (*Assume all transitions are possible*)

a. 6 points Calculate the **energy (in kJ/mol)** required to ionize (completely remove) the electron from the **$n=3$** level of **He^+ ion** (He: Helium). Show your work including equations. (3 sig. figs.)

b. 2 points

(I) Select the largest (longest) wavelength from the three choices below:

- i. Wavelength of photon to *completely remove* an electron from *1s orbital of Hydrogen atom*
- ii. Wavelength of photon to *completely remove* an electron from *1s orbital of Boron atom (B)*
- iii. Wavelength of photon to *completely remove* an electron from *1s orbital of Boron cation (B^+)*

i or ii or iii

(II) Select the smallest (shortest) wavelength from the three choices below (*for H atom*):

- i. Wavelength of photon *emitted* for an electron transitioning from *2p to 1s orbital*
- ii. Wavelength of photon *absorbed* for an electron transitioning from *2s to 4p orbital*
- iii. Wavelength of photon *absorbed* for an electron transitioning from *1s to 3p orbital*

i or ii or iii



Question 2 8 points



A Chem110 student plots the graph shown on the left, from data obtained for the kinetic energy (eV) of the ejected electron versus frequency (s^{-1}) of the incident light, for an unknown metal. ($1\text{eV} = 1.6022 \times 10^{-19} \text{ J}$)

Based on the plot, answer the questions **a to c** below:

a. 1 point On the plot above, label the **threshold frequency**.

b. 2 points What is the value of the work potential (in J) of the unknown metal?

c. 3 points What is the *longest* wavelength (in nm) required to eject an electron from this metal's surface? Show your work including equation(s) used. (3 sig figs)

d. 2 points In the space provided below, plot the following

(Assume frequency is *above* threshold frequency):

- i. *Kinetic energy* of an ejected electron versus the *intensity* of the incident light (left)
- ii. *Number of electrons* ejected versus the *frequency* of the incident light (right)



**Question 3** **6 points****a. 2 points**

In the ground state of Ge (Germanium; $Z = 32$) atom, **name** the following orbitals:

i. Orbital containing the electron that requires the maximum energy to be removed.

ii. Orbital containing the electron that experiences the minimum effective nuclear charge.

b. 2 points

An unknown anion (A^-) has the following *excited* electronic configuration: $[Ar] 4s^2 3d^3 4p^1$

(Ar: Argon; $Z = 18$)

Give the **condensed ground state electronic configuration** of the atom, **A** and the cation, **A^{2+}** .
Show your work. (Assume all electronic transitions are possible)

Electronic Configuration of A:

Electronic Configuration of A^{2+} :

c. 2 points

In a single atom, determine how many electrons can have the following set of quantum numbers?

(i) $n = 5; l = 3$

In a single atom, determine how many electrons can have the following set of quantum numbers?

(i) $n = 3; l = 2; m_s = +1/2$

**Question 4** **8 points**

Rank the following from 1 to 4 in **increasing** order. Provide the ranking in the boxes provided.

Note: Atomic number (Z) are provided for relevant atoms

a) Increasing (lowest to highest) order of *ionization energy*: (1= lowest; 4 = highest)

Mg

Mg: Z = 12

Rb

Rb: Z = 37

Cs

Cs: Z = 55

K

K : Z = 19

b) Increasing (lowest to highest) order of *bond angle*: (1= lowest; 4 = highest)

H₂OCCl₄XeF₄

HCN

c) Increasing (smallest to largest) order of *ionic size* (1= smallest; 4 = largest)

S²⁻

S: Z = 16

O²⁻

O: Z = 8

Be²⁺

Be: Z = 4

Na⁺

Na: Z = 11

d) Increasing (lowest to highest) order of *bond energy* (1= lowest; 4 = highest)

P-F

P-Cl

P-I

P-Br

**Question 5** **12 points**

a. 9 points For the molecules HClO_4 and TeF_6 , draw the most stable *Lewis structure* showing appropriate shape and *dashed/wedged bonds*, based on VSEPR. Determine the **molecular geometry** around the central atom. For HClO_4 , label the *most polar bond*.



b. 3 points (Note: Lone pairs are not shown in structures below)

(i) For the three *resonance* structures (1, 2 and 3) shown below, answer the following question:



(I) Circle the *most contributing* resonance structure.

Structure 1

Structure 2

Structure 3

(ii) For the three structures given (4, 5, and 6) below, answer the following questions:



(I) Circle all the structures that represent *resonance structures* of each other:

Structure 4

Structure 5

Structure 6

(II) Among the resonance structures, circle the *most contributing* resonance structure.

Structure 4

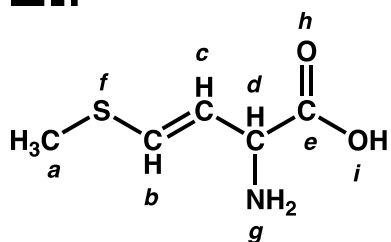
Structure 5

Structure 6

**Question 6** **6 points**

Using the data given below, calculate the **lattice energy** for SrO(s).

	ΔH° (kJ/mol)
$\text{Sr}(s) \longrightarrow \text{Sr}(g)$	+180
$\text{Sr}(g) \longrightarrow \text{Sr}^{2+}(g) + 2e^-$	+1468
$\text{O}_2(g) \longrightarrow 2\text{O}(g)$	+500
$\text{O}(g) + 2e^- \longrightarrow \text{O}^{2-}(g)$	+600
$\text{Sr}(s) + \frac{1}{2}\text{O}_2(g) \longrightarrow \text{SrO}(s)$	-550

**Question 7 10 points**

Consider the molecule given on the left, and answer questions (i) to (vi) using chemical bonding theories that we have learnt in class (valence bond theory, hybridization).

Note that the lone pairs are not shown in the structure:

i. Consider the carbon atoms marked in as ***a***, ***b***, ***c***, ***d***, and ***e*** in the molecule, and determine **hybridization** at each of the carbon centers. (2.5 points)

Atom	Hybridization
Carbon (<i>a</i>)	
Carbon (<i>b</i>)	
Carbon (<i>c</i>)	
Carbon (<i>d</i>)	
Carbon (<i>e</i>)	

ii. Now, consider the sulfur atom (S) marked as ***f***, nitrogen (N) atom (***g***), and oxygen(O) atoms (***h*** and ***i***), and determine their **hybridization**. (2 points)

Atom	Hybridization
Sulfur (<i>f</i>)	
Nitrogen (<i>g</i>)	
Oxygen (<i>h</i>)	
Oxygen (<i>i</i>)	

iii. **Describe** all the bonds formed by the carbon atoms ***a*** and ***b***, using sigma and pi designation as appropriate. (2 points)

Atom	Bonds formed (use sigma and pi as appropriate)
Carbon (<i>a</i>)	
Carbon (<i>b</i>)	

iv. How many total sp^3 -s sigma bonds are in the molecule? (1 point)

v. Determine **electronic** and **molecular geometry** at carbon (C) labeled ***e*** and sulfur (S) labeled ***f*** centres. (2 points)

Atom	Electronic Geometry	Molecular Geometry
Carbon (<i>e</i>)		
Sulfur (<i>f</i>)		

vi. Circle the value from the choices given below, that describes the **C-S-C bond angle at sulfur** (S) marked ***f***: (0.5 points)

>90° to <109.5° 109.5° >109.5 to <120° 120° >120° to <180°


Question 8 12 points

Using the **valence shell orbitals only** draw a **complete** molecular orbital (MO) diagram (with no s-p mixing) for the molecule **PS**, in the space provided below. **(6 points)**

Assume that 3s and 3p atomic orbitals will form molecular orbitals like those formed by the combination of 2s and 2p atomic orbitals that we learnt in class, and there is no s-p mixing.

Show **energy arrow**, **atomic orbitals** involved, **atom labels**, the **molecular orbitals** that are formed, **filled electrons** in the MO diagram etc. Make sure to label all the atomic and molecular orbitals.

Using the MO diagram above, answer the following questions:

- (i) Which one of the following will have the highest bond order? Circle the correct choice
 A) PS^{1-} B) PS C) PS^{1+}
- (ii) Which one of the following will have the longest bond length? Circle the correct choice
 A) PS^{1-} B) PS C) PS^{1+}
- (iii) Which of the following will be diamagnetic? Circle the correct choice.
 A) PS^{1-} B) PS C) PS^{1+}
- (iv) Draw Lewis dot structure of the molecule with the shortest bond length (based on the MO diagram) from among the three (PS^{1-} , PS, PS^{1+}), and indicate the bond order calculated from the Lewis dot structure. **(3 points)**

Lewis Dot Structure

Bond Order



8D586FE7-A652-4815-978B-146E08BE4E4D

midterm-exam-93388

#163 Page 10 of 10

SCRAP PAGE

**CHEMISTRY 110 –Midterm Exam****Thursday November 7, 2024, 6:30 – 8:30 pm**

Examiners: Pallavi Sirjoosingh and Ashok Kakkar

First name (please write as legibly as possible within the boxes)

--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--

Last name

--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--

Student ID number

--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--

This booklet has 8 questions on 10 pages (The last page can be used as a scrap page).

The staple must NOT be removed from the question booklet. Your intact question booklet must be returned at the end of the exam.

A separate booklet contains a periodic table, and datasheet.

Write your answers directly under the questions in the space provided. Do not write on the QR codes or near the top edges (as it may be cut off).

Please write **legibly and dark enough** for the scan to be readable. For students who use English as a second language, dictionaries are permitted. Calculators are allowed.

DO NOT ASK QUESTIONS OF THE INVIGILATORS about interpretations of the exam questions. If you have a concern about a question, please summarize it at the top of this page. All concerns will be evaluated after the exam. GOOD LUCK!

**Question 1 8 points** (*Assume all transitions are possible*)

a. 6 points Calculate the **energy (in kJ/mol)** required to ionize (completely remove) the electron from the **$n=3$** level of **He^+ ion** (He: Helium). Show your work including equations. (3 sig. figs.)

b. 2 points

(I) Select the largest (longest) wavelength from the three choices below:

- i. Wavelength of photon to *completely remove* an electron from *1s orbital of Hydrogen atom*
- ii. Wavelength of photon to *completely remove* an electron from *1s orbital of Boron atom (B)*
- iii. Wavelength of photon to *completely remove* an electron from *1s orbital of Boron cation (B^+)*

i or ii or iii

(II) Select the smallest (shortest) wavelength from the three choices below (*for H atom*):

- i. Wavelength of photon *emitted* for an electron transitioning from *2p to 1s orbital*
- ii. Wavelength of photon *absorbed* for an electron transitioning from *2s to 4p orbital*
- iii. Wavelength of photon *absorbed* for an electron transitioning from *1s to 3p orbital*

i or ii or iii



Question 2 8 points



A Chem110 student plots the graph shown on the left, from data obtained for the kinetic energy (eV) of the ejected electron versus frequency (s^{-1}) of the incident light, for an unknown metal. ($1\text{eV} = 1.6022 \times 10^{-19} \text{ J}$)

Based on the plot, answer the questions **a to c** below:

a. 1 point On the plot above, label the **threshold frequency**.

b. 2 points What is the value of the work potential (in J) of the unknown metal?

c. 3 points What is the *longest* wavelength (in nm) required to eject an electron from this metal's surface? Show your work including equation(s) used. (3 sig figs)

d. 2 points In the space provided below, plot the following

(Assume frequency is *above* threshold frequency):

- Kinetic energy* of an ejected electron versus the *intensity* of the incident light (left)
- Number of electrons* ejected versus the *frequency* of the incident light (right)



**Question 3** **6 points****a. 2 points**

In the ground state of Ge (Germanium; $Z = 32$) atom, **name** the following orbitals:

i. Orbital containing the electron that requires the maximum energy to be removed.

ii. Orbital containing the electron that experiences the minimum effective nuclear charge.

b. 2 points

An unknown anion (A^-) has the following *excited* electronic configuration: $[Ar] 4s^2 3d^3 4p^1$

(Ar: Argon; $Z = 18$)

Give the **condensed ground state electronic configuration** of the atom, **A** and the cation, **A^{2+}** .
Show your work. (Assume all electronic transitions are possible)

Electronic Configuration of A:

Electronic Configuration of A^{2+} :

c. 2 points

In a single atom, determine how many electrons can have the following set of quantum numbers?

(i) $n = 5; l = 3$

In a single atom, determine how many electrons can have the following set of quantum numbers?

(i) $n = 3; l = 2; m_s = +1/2$

**Question 4** **8 points**

Rank the following from 1 to 4 in **increasing** order. Provide the ranking in the boxes provided.

Note: Atomic number (Z) are provided for relevant atoms

a) Increasing (lowest to highest) order of *ionization energy*: (1= lowest; 4 = highest)

Mg

Mg: Z = 12

Rb

Rb: Z = 37

Cs

Cs: Z = 55

K

K : Z = 19

b) Increasing (lowest to highest) order of *bond angle*: (1= lowest; 4 = highest)

H₂OCCl₄XeF₄

HCN

c) Increasing (smallest to largest) order of *ionic size* (1= smallest; 4 = largest)

S²⁻

S: Z = 16

O²⁻

O: Z = 8

Be²⁺

Be: Z = 4

Na⁺

Na: Z = 11

d) Increasing (lowest to highest) order of *bond energy* (1= lowest; 4 = highest)

P-F

P-Cl

P-I

P-Br

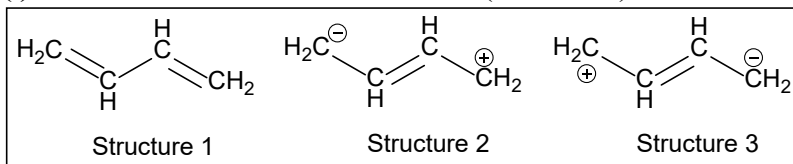
**Question 5** **12 points**

a. 9 points For the molecules HClO_4 and TeF_6 , draw the most stable *Lewis structure* showing appropriate shape and *dashed/wedged bonds*, based on VSEPR. Determine the **molecular geometry** around the central atom. For HClO_4 , label the *most polar bond*.



b. 3 points (Note: Lone pairs are not shown in structures below)

(i) For the three *resonance* structures (1, 2 and 3) shown below, answer the following question:



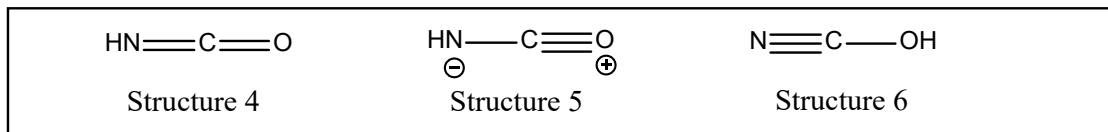
(I) Circle the *most contributing* resonance structure.

Structure 1

Structure 2

Structure 3

(ii) For the three structures given (4, 5, and 6) below, answer the following questions:



(I) Circle all the structures that represent *resonance structures* of each other:

Structure 4

Structure 5

Structure 6

(II) Among the resonance structures, circle the *most contributing* resonance structure.

Structure 4

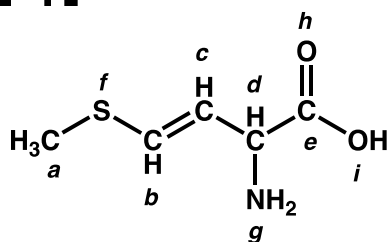
Structure 5

Structure 6

**Question 6** **6 points**

Using the data given below, calculate the **lattice energy** for SrO(s).

	ΔH° (kJ/mol)
$\text{Sr}(s) \longrightarrow \text{Sr}(g)$	+180
$\text{Sr}(g) \longrightarrow \text{Sr}^{2+}(g) + 2e^-$	+1468
$\text{O}_2(g) \longrightarrow 2\text{O}(g)$	+500
$\text{O}(g) + 2e^- \longrightarrow \text{O}^{2-}(g)$	+600
$\text{Sr}(s) + \frac{1}{2}\text{O}_2(g) \longrightarrow \text{SrO}(s)$	-550

**Question 7 10 points**

Consider the molecule given on the left, and answer questions (i) to (vi) using chemical bonding theories that we have learnt in class (valence bond theory, hybridization).

Note that the lone pairs are not shown in the structure:

i. Consider the carbon atoms marked in as ***a***, ***b***, ***c***, ***d***, and ***e*** in the molecule, and determine **hybridization** at each of the carbon centers. (2.5 points)

Atom	Hybridization
Carbon (<i>a</i>)	
Carbon (<i>b</i>)	
Carbon (<i>c</i>)	
Carbon (<i>d</i>)	
Carbon (<i>e</i>)	

ii. Now, consider the sulfur atom (S) marked as ***f***, nitrogen (N) atom (***g***), and oxygen(O) atoms (***h*** and ***i***), and determine their **hybridization**. (2 points)

Atom	Hybridization
Sulfur (<i>f</i>)	
Nitrogen (<i>g</i>)	
Oxygen (<i>h</i>)	
Oxygen (<i>i</i>)	

iii. **Describe** all the bonds formed by the carbon atoms ***a*** and ***b***, using sigma and pi designation as appropriate. (2 points)

Atom	Bonds formed (use sigma and pi as appropriate)
Carbon (<i>a</i>)	
Carbon (<i>b</i>)	

iv. How many total sp^3 -s sigma bonds are in the molecule? (1 point)

v. Determine **electronic** and **molecular geometry** at carbon (C) labeled ***e*** and sulfur (S) labeled ***f*** centres. (2 points)

Atom	Electronic Geometry	Molecular Geometry
Carbon (<i>e</i>)		
Sulfur (<i>f</i>)		

vi. Circle the value from the choices given below, that describes the **C-S-C bond angle at sulfur** (S) marked ***f***: (0.5 points)

$>90^\circ$ to $<109.5^\circ$

109.5°

>109.5 to $<120^\circ$

120°

$>120^\circ$ to $<180^\circ$


Question 8 12 points

Using the **valence shell orbitals only** draw a **complete** molecular orbital (MO) diagram (with no s-p mixing) for the molecule **PS**, in the space provided below. **(6 points)**

Assume that 3s and 3p atomic orbitals will form molecular orbitals like those formed by the combination of 2s and 2p atomic orbitals that we learnt in class, and there is no s-p mixing.

Show **energy arrow**, **atomic orbitals** involved, **atom labels**, the **molecular orbitals** that are formed, **filled electrons** in the MO diagram etc. Make sure to label all the atomic and molecular orbitals.

Using the MO diagram above, answer the following questions:

- (i) Which one of the following will have the highest bond order? Circle the correct choice
 A) PS^{1-} B) PS C) PS^{1+}
- (ii) Which one of the following will have the longest bond length? Circle the correct choice
 A) PS^{1-} B) PS C) PS^{1+}
- (iii) Which of the following will be diamagnetic? Circle the correct choice.
 A) PS^{1-} B) PS C) PS^{1+}
- (iv) Draw Lewis dot structure of the molecule with the shortest bond length (based on the MO diagram) from among the three (PS^{1-} , PS, PS^{1+}), and indicate the bond order calculated from the Lewis dot structure. **(3 points)**

Lewis Dot Structure

Bond Order



10960C94-B959-4F6B-BAE4-6E115A1C4689

midterm-exam-93388

#164 Page 10 of 10

SCRAP PAGE

**CHEMISTRY 110 –Midterm Exam****Thursday November 7, 2024, 6:30 – 8:30 pm**

Examiners: Pallavi Sirjoosingh and Ashok Kakkar

First name (please write as legibly as possible within the boxes)

--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--

Last name

--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--

Student ID number

--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--

This booklet has 8 questions on 10 pages (The last page can be used as a scrap page).

The staple must NOT be removed from the question booklet. Your intact question booklet must be returned at the end of the exam.

A separate booklet contains a periodic table, and datasheet.

Write your answers directly under the questions in the space provided. Do not write on the QR codes or near the top edges (as it may be cut off).

Please write **legibly and dark enough** for the scan to be readable. For students who use English as a second language, dictionaries are permitted. Calculators are allowed.

DO NOT ASK QUESTIONS OF THE INVIGILATORS about interpretations of the exam questions. If you have a concern about a question, please summarize it at the top of this page. All concerns will be evaluated after the exam. GOOD LUCK!

**Question 1 8 points** (*Assume all transitions are possible*)

a. 6 points Calculate the **energy (in kJ/mol)** required to ionize (completely remove) the electron from the **n=3** level of **He⁺ ion** (He: Helium). Show your work including equations. (3 sig. figs.)

b. 2 points

(I) Select the largest (longest) wavelength from the three choices below:

- i. Wavelength of photon to *completely remove* an electron from *1s orbital of Hydrogen atom*
- ii. Wavelength of photon to *completely remove* an electron from *1s orbital of Boron atom (B)*
- iii. Wavelength of photon to *completely remove* an electron from *1s orbital of Boron cation (B⁺)*

i or ii or iii

(II) Select the smallest (shortest) wavelength from the three choices below (*for H atom*):

- i. Wavelength of photon *emitted* for an electron transitioning from *2p to 1s orbital*
- ii. Wavelength of photon *absorbed* for an electron transitioning from *2s to 4p orbital*
- iii. Wavelength of photon *absorbed* for an electron transitioning from *1s to 3p orbital*

i or ii or iii



Question 2 8 points



A Chem110 student plots the graph shown on the left, from data obtained for the kinetic energy (eV) of the ejected electron versus frequency (s^{-1}) of the incident light, for an unknown metal. ($1\text{eV} = 1.6022 \times 10^{-19} \text{ J}$)

Based on the plot, answer the questions **a to c** below:

a. 1 point On the plot above, label the **threshold frequency**.

b. 2 points What is the value of the work potential (in J) of the unknown metal?

c. 3 points What is the *longest* wavelength (in nm) required to eject an electron from this metal's surface? Show your work including equation(s) used. (3 sig figs)

d. 2 points In the space provided below, plot the following

(Assume frequency is *above* threshold frequency):

- Kinetic energy* of an ejected electron versus the *intensity* of the incident light (left)
- Number of electrons* ejected versus the *frequency* of the incident light (right)



**Question 3** **6 points****a. 2 points**

In the ground state of Ge (Germanium; $Z = 32$) atom, **name** the following orbitals:

i. Orbital containing the electron that requires the maximum energy to be removed.

ii. Orbital containing the electron that experiences the minimum effective nuclear charge.

b. 2 points

An unknown anion (A^-) has the following *excited* electronic configuration: $[Ar] 4s^2 3d^3 4p^1$

(Ar: Argon; $Z = 18$)

Give the **condensed ground state electronic configuration** of the atom, **A** and the cation, **A^{2+}** .
Show your work. (Assume all electronic transitions are possible)

Electronic Configuration of A:

Electronic Configuration of A^{2+} :

c. 2 points

In a single atom, determine how many electrons can have the following set of quantum numbers?

(i) $n = 5; l = 3$

In a single atom, determine how many electrons can have the following set of quantum numbers?

(i) $n = 3; l = 2; m_s = +1/2$

**Question 4** **8 points**

Rank the following from 1 to 4 in **increasing** order. Provide the ranking in the boxes provided.

Note: Atomic number (Z) are provided for relevant atoms

a) Increasing (lowest to highest) order of *ionization energy*: (1= lowest; 4 = highest)

Mg

Mg: Z = 12

Rb

Rb: Z = 37

Cs

Cs: Z = 55

K

K : Z = 19

b) Increasing (lowest to highest) order of *bond angle*: (1= lowest; 4 = highest)

H₂OCCl₄XeF₄

HCN

c) Increasing (smallest to largest) order of *ionic size* (1= smallest; 4 = largest)

S²⁻

S: Z = 16

O²⁻

O: Z = 8

Be²⁺

Be: Z = 4

Na⁺

Na: Z = 11

d) Increasing (lowest to highest) order of *bond energy* (1= lowest; 4 = highest)

P-F

P-Cl

P-I

P-Br

**Question 5** **12 points**

a. 9 points For the molecules HClO_4 and TeF_6 , draw the most stable *Lewis structure* showing appropriate shape and *dashed/wedged bonds*, based on VSEPR. Determine the **molecular geometry** around the central atom. For HClO_4 , label the *most polar bond*.



b. 3 points (Note: Lone pairs are not shown in structures below)

(i) For the three *resonance* structures (1, 2 and 3) shown below, answer the following question:



(I) Circle the *most contributing* resonance structure.

Structure 1

Structure 2

Structure 3

(ii) For the three structures given (4, 5, and 6) below, answer the following questions:



(I) Circle all the structures that represent *resonance structures* of each other:

Structure 4

Structure 5

Structure 6

(II) Among the resonance structures, circle the *most contributing* resonance structure.

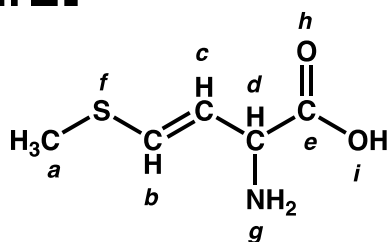
Structure 4

Structure 5

Structure 6

**Question 6** **6 points**Using the data given below, calculate the **lattice energy** for SrO(s).

	ΔH° (kJ/mol)
$\text{Sr}(s) \longrightarrow \text{Sr}(g)$	+180
$\text{Sr}(g) \longrightarrow \text{Sr}^{2+}(g) + 2e^-$	+1468
$\text{O}_2(g) \longrightarrow 2\text{O}(g)$	+500
$\text{O}(g) + 2e^- \longrightarrow \text{O}^{2-}(g)$	+600
$\text{Sr}(s) + \frac{1}{2}\text{O}_2(g) \longrightarrow \text{SrO}(s)$	-550

**Question 7 10 points**

Consider the molecule given on the left, and answer questions (i) to (vi) using chemical bonding theories that we have learnt in class (valence bond theory, hybridization).

Note that the lone pairs are not shown in the structure:

i. Consider the carbon atoms marked in as ***a***, ***b***, ***c***, ***d***, and ***e*** in the molecule, and determine **hybridization** at each of the carbon centers. (2.5 points)

Atom	Hybridization
Carbon (<i>a</i>)	
Carbon (<i>b</i>)	
Carbon (<i>c</i>)	
Carbon (<i>d</i>)	
Carbon (<i>e</i>)	

ii. Now, consider the sulfur atom (S) marked as ***f***, nitrogen (N) atom (***g***), and oxygen(O) atoms (***h*** and ***i***), and determine their **hybridization**. (2 points)

Atom	Hybridization
Sulfur (<i>f</i>)	
Nitrogen (<i>g</i>)	
Oxygen (<i>h</i>)	
Oxygen (<i>i</i>)	

iii. **Describe** all the bonds formed by the carbon atoms ***a*** and ***b***, using sigma and pi designation as appropriate. (2 points)

Atom	Bonds formed (use sigma and pi as appropriate)
Carbon (<i>a</i>)	
Carbon (<i>b</i>)	

iv. How many total sp^3 -s sigma bonds are in the molecule? (1 point)

v. Determine **electronic** and **molecular geometry** at carbon (C) labeled ***e*** and sulfur (S) labeled ***f*** centres. (2 points)

Atom	Electronic Geometry	Molecular Geometry
Carbon (<i>e</i>)		
Sulfur (<i>f</i>)		

vi. Circle the value from the choices given below, that describes the **C-S-C bond angle at sulfur** (S) marked ***f***: (0.5 points)

>90° to <109.5°

109.5°

>109.5 to <120°

120°

>120° to <180°


Question 8 12 points

Using the **valence shell orbitals only** draw a **complete** molecular orbital (MO) diagram (with no s-p mixing) for the molecule **PS**, in the space provided below. **(6 points)**

Assume that 3s and 3p atomic orbitals will form molecular orbitals like those formed by the combination of 2s and 2p atomic orbitals that we learnt in class, and there is no s-p mixing.

Show **energy arrow**, **atomic orbitals** involved, **atom labels**, the **molecular orbitals** that are formed, **filled electrons** in the MO diagram etc. Make sure to label all the atomic and molecular orbitals.

Using the MO diagram above, answer the following questions:

- (i) Which one of the following will have the highest bond order? Circle the correct choice
 A) PS^{1-} B) PS C) PS^{1+}
- (ii) Which one of the following will have the longest bond length? Circle the correct choice
 A) PS^{1-} B) PS C) PS^{1+}
- (iii) Which of the following will be diamagnetic? Circle the correct choice.
 A) PS^{1-} B) PS C) PS^{1+}
- (iv) Draw Lewis dot structure of the molecule with the shortest bond length (based on the MO diagram) from among the three (PS^{1-} , PS, PS^{1+}), and indicate the bond order calculated from the Lewis dot structure. **(3 points)**

Lewis Dot Structure

Bond Order



E616ADB3-09F1-45FD-AA73-7A9F4D11627D

midterm-exam-93388

#165 Page 10 of 10

SCRAP PAGE

**CHEMISTRY 110 –Midterm Exam****Thursday November 7, 2024, 6:30 – 8:30 pm**

Examiners: Pallavi Sirjoosingh and Ashok Kakkar

First name (please write as legibly as possible within the boxes)

--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--

Last name

--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--

Student ID number

--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--

This booklet has 8 questions on 10 pages (The last page can be used as a scrap page).

The staple must NOT be removed from the question booklet. Your intact question booklet must be returned at the end of the exam.

A separate booklet contains a periodic table, and datasheet.

Write your answers directly under the questions in the space provided. Do not write on the QR codes or near the top edges (as it may be cut off).

Please write **legibly and dark enough** for the scan to be readable. For students who use English as a second language, dictionaries are permitted. Calculators are allowed.

DO NOT ASK QUESTIONS OF THE INVIGILATORS about interpretations of the exam questions. If you have a concern about a question, please summarize it at the top of this page. All concerns will be evaluated after the exam. GOOD LUCK!

**Question 1 8 points** (*Assume all transitions are possible*)

a. 6 points Calculate the **energy (in kJ/mol)** required to ionize (completely remove) the electron from the **n=3** level of **He⁺ ion** (He: Helium). Show your work including equations. (3 sig. figs.)

b. 2 points

(I) Select the largest (longest) wavelength from the three choices below:

- i. Wavelength of photon to *completely remove* an electron from *1s orbital of Hydrogen atom*
- ii. Wavelength of photon to *completely remove* an electron from *1s orbital of Boron atom (B)*
- iii. Wavelength of photon to *completely remove* an electron from *1s orbital of Boron cation (B⁺)*

i or ii or iii

(II) Select the smallest (shortest) wavelength from the three choices below (*for H atom*):

- i. Wavelength of photon *emitted* for an electron transitioning from *2p to 1s orbital*
- ii. Wavelength of photon *absorbed* for an electron transitioning from *2s to 4p orbital*
- iii. Wavelength of photon *absorbed* for an electron transitioning from *1s to 3p orbital*

i or ii or iii



Question 2 8 points



A Chem110 student plots the graph shown on the left, from data obtained for the kinetic energy (eV) of the ejected electron versus frequency (s^{-1}) of the incident light, for an unknown metal. ($1\text{eV} = 1.6022 \times 10^{-19} \text{ J}$)

Based on the plot, answer the questions **a to c** below:

a. 1 point On the plot above, label the **threshold frequency**.

b. 2 points What is the value of the work potential (in J) of the unknown metal?

c. 3 points What is the *longest* wavelength (in nm) required to eject an electron from this metal's surface? Show your work including equation(s) used. (3 sig figs)

d. 2 points In the space provided below, plot the following

(Assume frequency is *above* threshold frequency):

- Kinetic energy* of an ejected electron versus the *intensity* of the incident light (left)
- Number of electrons* ejected versus the *frequency* of the incident light (right)



**Question 3** **6 points****a. 2 points**

In the ground state of Ge (Germanium; $Z = 32$) atom, **name** the following orbitals:

i. Orbital containing the electron that requires the maximum energy to be removed.

ii. Orbital containing the electron that experiences the minimum effective nuclear charge.

b. 2 points

An unknown anion (A^-) has the following *excited* electronic configuration: $[Ar] 4s^2 3d^3 4p^1$

(Ar: Argon; $Z = 18$)

Give the **condensed ground state electronic configuration** of the atom, **A** and the cation, **A^{2+}** .
Show your work. (Assume all electronic transitions are possible)

Electronic Configuration of A:

Electronic Configuration of A^{2+} :

c. 2 points

In a single atom, determine how many electrons can have the following set of quantum numbers?

(i) $n = 5; l = 3$

In a single atom, determine how many electrons can have the following set of quantum numbers?

(i) $n = 3; l = 2; m_s = +1/2$

**Question 4** **8 points**

Rank the following from 1 to 4 in **increasing** order. Provide the ranking in the boxes provided.

Note: Atomic number (Z) are provided for relevant atoms

a) Increasing (lowest to highest) order of *ionization energy*: (1= lowest; 4 = highest)

Mg

Mg: Z = 12

Rb

Rb: Z = 37

Cs

Cs: Z = 55

K

K : Z = 19

b) Increasing (lowest to highest) order of *bond angle*: (1= lowest; 4 = highest)

H₂OCCl₄XeF₄

HCN

c) Increasing (smallest to largest) order of *ionic size* (1= smallest; 4 = largest)

S²⁻

S: Z = 16

O²⁻

O: Z = 8

Be²⁺

Be: Z = 4

Na⁺

Na: Z = 11

d) Increasing (lowest to highest) order of *bond energy* (1= lowest; 4 = highest)

P-F

P-Cl

P-I

P-Br

**Question 5** **12 points**

a. 9 points For the molecules HClO_4 and TeF_6 , draw the most stable *Lewis structure* showing appropriate shape and *dashed/wedged bonds*, based on VSEPR. Determine the **molecular geometry** around the central atom. For HClO_4 , label the *most polar bond*.



b. 3 points (Note: Lone pairs are not shown in structures below)

(i) For the three *resonance* structures (1, 2 and 3) shown below, answer the following question:



(I) Circle the *most contributing* resonance structure.

Structure 1

Structure 2

Structure 3

(ii) For the three structures given (4, 5, and 6) below, answer the following questions:



(I) Circle all the structures that represent *resonance structures* of each other:

Structure 4

Structure 5

Structure 6

(II) Among the resonance structures, circle the *most contributing* resonance structure.

Structure 4

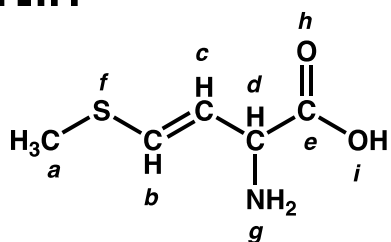
Structure 5

Structure 6

**Question 6** **6 points**

Using the data given below, calculate the **lattice energy** for SrO(s).

	ΔH° (kJ/mol)
$\text{Sr}(s) \longrightarrow \text{Sr}(g)$	+180
$\text{Sr}(g) \longrightarrow \text{Sr}^{2+}(g) + 2e^-$	+1468
$\text{O}_2(g) \longrightarrow 2\text{O}(g)$	+500
$\text{O}(g) + 2e^- \longrightarrow \text{O}^{2-}(g)$	+600
$\text{Sr}(s) + \frac{1}{2}\text{O}_2(g) \longrightarrow \text{SrO}(s)$	-550

**Question 7 10 points**

Consider the molecule given on the left, and answer questions (i) to (vi) using chemical bonding theories that we have learnt in class (valence bond theory, hybridization).

Note that the lone pairs are not shown in the structure:

i. Consider the carbon atoms marked in as ***a***, ***b***, ***c***, ***d***, and ***e*** in the molecule, and determine **hybridization** at each of the carbon centers. (2.5 points)

Atom	Hybridization
Carbon (<i>a</i>)	
Carbon (<i>b</i>)	
Carbon (<i>c</i>)	
Carbon (<i>d</i>)	
Carbon (<i>e</i>)	

ii. Now, consider the sulfur atom (S) marked as ***f***, nitrogen (N) atom (***g***), and oxygen(O) atoms (***h*** and ***i***), and determine their **hybridization**. (2 points)

Atom	Hybridization
Sulfur (<i>f</i>)	
Nitrogen (<i>g</i>)	
Oxygen (<i>h</i>)	
Oxygen (<i>i</i>)	

iii. **Describe** all the bonds formed by the carbon atoms ***a*** and ***b***, using sigma and pi designation as appropriate. (2 points)

Atom	Bonds formed (use sigma and pi as appropriate)
Carbon (<i>a</i>)	
Carbon (<i>b</i>)	

iv. How many total sp^3 -s sigma bonds are in the molecule? (1 point)

v. Determine **electronic** and **molecular geometry** at carbon (C) labeled ***e*** and sulfur (S) labeled ***f*** centres. (2 points)

Atom	Electronic Geometry	Molecular Geometry
Carbon (<i>e</i>)		
Sulfur (<i>f</i>)		

vi. Circle the value from the choices given below, that describes the **C-S-C bond angle at sulfur** (S) marked ***f***: (0.5 points)

>90° to <109.5°

109.5°

>109.5 to <120°

120°

>120° to <180°


Question 8 12 points

Using the **valence shell orbitals only** draw a **complete** molecular orbital (MO) diagram (with no s-p mixing) for the molecule **PS**, in the space provided below. **(6 points)**

Assume that 3s and 3p atomic orbitals will form molecular orbitals like those formed by the combination of 2s and 2p atomic orbitals that we learnt in class, and there is no s-p mixing.

Show **energy arrow**, **atomic orbitals** involved, **atom labels**, the **molecular orbitals** that are formed, **filled electrons** in the MO diagram etc. Make sure to label all the atomic and molecular orbitals.

Using the MO diagram above, answer the following questions:

- (i) Which one of the following will have the highest bond order? Circle the correct choice
 A) PS^{1-} B) PS C) PS^{1+}
- (ii) Which one of the following will have the longest bond length? Circle the correct choice
 A) PS^{1-} B) PS C) PS^{1+}
- (iii) Which of the following will be diamagnetic? Circle the correct choice.
 A) PS^{1-} B) PS C) PS^{1+}
- (iv) Draw Lewis dot structure of the molecule with the shortest bond length (based on the MO diagram) from among the three (PS^{1-} , PS, PS^{1+}), and indicate the bond order calculated from the Lewis dot structure. **(3 points)**

Lewis Dot Structure

Bond Order



0D8C24D7-A876-44DA-92FA-CDBD94469DBE

midterm-exam-93388

#166 Page 10 of 10

SCRAP PAGE

**CHEMISTRY 110 –Midterm Exam****Thursday November 7, 2024, 6:30 – 8:30 pm**

Examiners: Pallavi Sirjoosingh and Ashok Kakkar

First name (please write as legibly as possible within the boxes)

--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--

Last name

--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--

Student ID number

--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--

This booklet has 8 questions on 10 pages (The last page can be used as a scrap page).

The staple must NOT be removed from the question booklet. Your intact question booklet must be returned at the end of the exam.

A separate booklet contains a periodic table, and datasheet.

Write your answers directly under the questions in the space provided. Do not write on the QR codes or near the top edges (as it may be cut off).

Please write **legibly and dark enough** for the scan to be readable. For students who use English as a second language, dictionaries are permitted. Calculators are allowed.

DO NOT ASK QUESTIONS OF THE INVIGILATORS about interpretations of the exam questions. If you have a concern about a question, please summarize it at the top of this page. All concerns will be evaluated after the exam. GOOD LUCK!

**Question 1 8 points** (*Assume all transitions are possible*)

a. 6 points Calculate the **energy (in kJ/mol)** required to ionize (completely remove) the electron from the **$n=3$** level of **He^+ ion** (He: Helium). Show your work including equations. (3 sig. figs.)

b. 2 points

(I) Select the largest (longest) wavelength from the three choices below:

- i. Wavelength of photon to *completely remove* an electron from *1s orbital of Hydrogen atom*
- ii. Wavelength of photon to *completely remove* an electron from *1s orbital of Boron atom (B)*
- iii. Wavelength of photon to *completely remove* an electron from *1s orbital of Boron cation (B^+)*

i or ii or iii

(II) Select the smallest (shortest) wavelength from the three choices below (*for H atom*):

- i. Wavelength of photon *emitted* for an electron transitioning from *2p to 1s orbital*
- ii. Wavelength of photon *absorbed* for an electron transitioning from *2s to 4p orbital*
- iii. Wavelength of photon *absorbed* for an electron transitioning from *1s to 3p orbital*

i or ii or iii



Question 2 8 points



A Chem110 student plots the graph shown on the left, from data obtained for the kinetic energy (eV) of the ejected electron versus frequency (s^{-1}) of the incident light, for an unknown metal. ($1\text{eV} = 1.6022 \times 10^{-19} \text{ J}$)

Based on the plot, answer the questions **a to c** below:

a. 1 point On the plot above, label the **threshold frequency**.

b. 2 points What is the value of the work potential (in J) of the unknown metal?

c. 3 points What is the *longest* wavelength (in nm) required to eject an electron from this metal's surface? Show your work including equation(s) used. (3 sig figs)

d. 2 points In the space provided below, plot the following

(Assume frequency is *above* threshold frequency):

- Kinetic energy* of an ejected electron versus the *intensity* of the incident light (left)
- Number of electrons* ejected versus the *frequency* of the incident light (right)



**Question 3** **6 points****a. 2 points**

In the ground state of Ge (Germanium; $Z = 32$) atom, **name** the following orbitals:

i. Orbital containing the electron that requires the maximum energy to be removed.

ii. Orbital containing the electron that experiences the minimum effective nuclear charge.

b. 2 points

An unknown anion (A^-) has the following *excited* electronic configuration: $[Ar] 4s^2 3d^3 4p^1$

(Ar: Argon; $Z = 18$)

Give the **condensed ground state electronic configuration** of the atom, **A** and the cation, A^{2+} .
Show your work. (Assume all electronic transitions are possible)

Electronic Configuration of A:

Electronic Configuration of A^{2+} :

c. 2 points

In a single atom, determine how many electrons can have the following set of quantum numbers?

(i) $n = 5; l = 3$

In a single atom, determine how many electrons can have the following set of quantum numbers?

(i) $n = 3; l = 2; m_s = +1/2$

**Question 4** **8 points**

Rank the following from 1 to 4 in **increasing** order. Provide the ranking in the boxes provided.

Note: Atomic number (Z) are provided for relevant atoms

a) Increasing (lowest to highest) order of *ionization energy*: (1= lowest; 4 = highest)

Mg

Mg: Z = 12

Rb

Rb: Z = 37

Cs

Cs: Z = 55

K

K : Z = 19

b) Increasing (lowest to highest) order of *bond angle*: (1= lowest; 4 = highest)

H₂OCCl₄XeF₄

HCN

c) Increasing (smallest to largest) order of *ionic size* (1= smallest; 4 = largest)

S²⁻

S: Z = 16

O²⁻

O: Z = 8

Be²⁺

Be: Z = 4

Na⁺

Na: Z = 11

d) Increasing (lowest to highest) order of *bond energy* (1= lowest; 4 = highest)

P-F

P-Cl

P-I

P-Br

**Question 5** **12 points**

a. 9 points For the molecules HClO_4 and TeF_6 , draw the most stable *Lewis structure* showing appropriate shape and *dashed/wedged bonds*, based on VSEPR. Determine the **molecular geometry** around the central atom. For HClO_4 , label the *most polar bond*.



b. 3 points (Note: Lone pairs are not shown in structures below)

(i) For the three *resonance* structures (1, 2 and 3) shown below, answer the following question:



(I) Circle the *most contributing* resonance structure.

Structure 1

Structure 2

Structure 3

(ii) For the three structures given (4, 5, and 6) below, answer the following questions:



(I) Circle all the structures that represent *resonance structures* of each other:

Structure 4

Structure 5

Structure 6

(II) Among the resonance structures, circle the *most contributing* resonance structure.

Structure 4

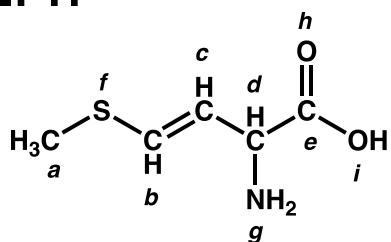
Structure 5

Structure 6

**Question 6** **6 points**

Using the data given below, calculate the **lattice energy** for SrO(s).

	ΔH° (kJ/mol)
$\text{Sr}(s) \longrightarrow \text{Sr}(g)$	+180
$\text{Sr}(g) \longrightarrow \text{Sr}^{2+}(g) + 2e^-$	+1468
$\text{O}_2(g) \longrightarrow 2\text{O}(g)$	+500
$\text{O}(g) + 2e^- \longrightarrow \text{O}^{2-}(g)$	+600
$\text{Sr}(s) + \frac{1}{2}\text{O}_2(g) \longrightarrow \text{SrO}(s)$	-550

**Question 7 10 points**

Consider the molecule given on the left, and answer questions (i) to (vi) using chemical bonding theories that we have learnt in class (valence bond theory, hybridization).

Note that the lone pairs are not shown in the structure:

i. Consider the carbon atoms marked in as ***a***, ***b***, ***c***, ***d***, and ***e*** in the molecule, and determine **hybridization** at each of the carbon centers. (2.5 points)

Atom	Hybridization
Carbon (<i>a</i>)	
Carbon (<i>b</i>)	
Carbon (<i>c</i>)	
Carbon (<i>d</i>)	
Carbon (<i>e</i>)	

ii. Now, consider the sulfur atom (S) marked as ***f***, nitrogen (N) atom (***g***), and oxygen(O) atoms (***h*** and ***i***), and determine their **hybridization**. (2 points)

Atom	Hybridization
Sulfur (<i>f</i>)	
Nitrogen (<i>g</i>)	
Oxygen (<i>h</i>)	
Oxygen (<i>i</i>)	

iii. **Describe** all the bonds formed by the carbon atoms ***a*** and ***b***, using sigma and pi designation as appropriate. (2 points)

Atom	Bonds formed (use sigma and pi as appropriate)
Carbon (<i>a</i>)	
Carbon (<i>b</i>)	

iv. How many total sp^3 -s sigma bonds are in the molecule? (1 point)

v. Determine **electronic** and **molecular geometry** at carbon (C) labeled ***e*** and sulfur (S) labeled ***f*** centres. (2 points)

Atom	Electronic Geometry	Molecular Geometry
Carbon (<i>e</i>)		
Sulfur (<i>f</i>)		

vi. Circle the value from the choices given below, that describes the **C-S-C bond angle at sulfur** (S) marked ***f***: (0.5 points)

$>90^\circ$ to $<109.5^\circ$

109.5°

>109.5 to $<120^\circ$

120°

$>120^\circ$ to $<180^\circ$

**Question 8 12 points**

Using the **valence shell orbitals only** draw a **complete** molecular orbital (MO) diagram (with no s-p mixing) for the molecule **PS**, in the space provided below. **(6 points)**

Assume that 3s and 3p atomic orbitals will form molecular orbitals like those formed by the combination of 2s and 2p atomic orbitals that we learnt in class, and there is no s-p mixing.

Show **energy arrow**, **atomic orbitals** involved, **atom labels**, the **molecular orbitals** that are formed, **filled electrons** in the MO diagram etc. Make sure to label all the atomic and molecular orbitals.

Using the MO diagram above, answer the following questions:

- (i) Which one of the following will have the highest bond order? Circle the correct choice
A) PS^{1-} B) PS C) PS^{1+}
- (ii) Which one of the following will have the longest bond length? Circle the correct choice
A) PS^{1-} B) PS C) PS^{1+}
- (iii) Which of the following will be diamagnetic? Circle the correct choice.
A) PS^{1-} B) PS C) PS^{1+}
- (iv) Draw Lewis dot structure of the molecule with the shortest bond length (based on the MO diagram) from among the three (PS^{1-} , PS, PS^{1+}), and indicate the bond order calculated from the Lewis dot structure. **(3 points)**

Lewis Dot Structure

Bond Order



806EB263-6965-472A-B0FD-E34F11D47A6E

midterm-exam-93388

#167 Page 10 of 10

SCRAP PAGE

**CHEMISTRY 110 –Midterm Exam****Thursday November 7, 2024, 6:30 – 8:30 pm****Examiners: Pallavi Sirjoosingh and Ashok Kakkar**

First name (please write as legibly as possible within the boxes)

--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--

Last name

--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--

Student ID number

--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--

This booklet has 8 questions on 10 pages (The last page can be used as a scrap page).

The staple must NOT be removed from the question booklet. Your intact question booklet must be returned at the end of the exam.

A separate booklet contains a periodic table, and datasheet.

Write your answers directly under the questions in the space provided. Do not write on the QR codes or near the top edges (as it may be cut off).

Please write **legibly and dark enough** for the scan to be readable. For students who use English as a second language, dictionaries are permitted. Calculators are allowed.

DO NOT ASK QUESTIONS OF THE INVIGILATORS about interpretations of the exam questions. If you have a concern about a question, please summarize it at the top of this page. All concerns will be evaluated after the exam. GOOD LUCK!

**Question 1 8 points** (*Assume all transitions are possible*)

a. 6 points Calculate the **energy (in kJ/mol)** required to ionize (completely remove) the electron from the **n=3** level of **He⁺ ion** (He: Helium). Show your work including equations. (3 sig. figs.)

b. 2 points

(I) Select the largest (longest) wavelength from the three choices below:

- i. Wavelength of photon to *completely remove* an electron from *1s orbital of Hydrogen atom*
- ii. Wavelength of photon to *completely remove* an electron from *1s orbital of Boron atom (B)*
- iii. Wavelength of photon to *completely remove* an electron from *1s orbital of Boron cation (B⁺)*

i or ii or iii

(II) Select the smallest (shortest) wavelength from the three choices below (*for H atom*):

- i. Wavelength of photon *emitted* for an electron transitioning from *2p to 1s orbital*
- ii. Wavelength of photon *absorbed* for an electron transitioning from *2s to 4p orbital*
- iii. Wavelength of photon *absorbed* for an electron transitioning from *1s to 3p orbital*

i or ii or iii



Question 2 8 points



A Chem110 student plots the graph shown on the left, from data obtained for the kinetic energy (eV) of the ejected electron versus frequency (s^{-1}) of the incident light, for an unknown metal. ($1\text{eV} = 1.6022 \times 10^{-19} \text{ J}$)

Based on the plot, answer the questions **a to c** below:

a. 1 point On the plot above, label the **threshold frequency**.

b. 2 points What is the value of the work potential (in J) of the unknown metal?

c. 3 points What is the *longest* wavelength (in nm) required to eject an electron from this metal's surface? Show your work including equation(s) used. (3 sig figs)

d. 2 points In the space provided below, plot the following

(Assume frequency is *above* threshold frequency):

- Kinetic energy* of an ejected electron versus the *intensity* of the incident light (left)
- Number of electrons* ejected versus the *frequency* of the incident light (right)



**Question 3** **6 points****a. 2 points**

In the ground state of Ge (Germanium; $Z = 32$) atom, **name** the following orbitals:

i. Orbital containing the electron that requires the maximum energy to be removed.

ii. Orbital containing the electron that experiences the minimum effective nuclear charge.

b. 2 points

An unknown anion (A^-) has the following *excited* electronic configuration: $[Ar] 4s^2 3d^3 4p^1$

(Ar: Argon; $Z = 18$)

Give the **condensed ground state electronic configuration** of the atom, **A** and the cation, **A^{2+}** .
Show your work. (Assume all electronic transitions are possible)

Electronic Configuration of A:

Electronic Configuration of A^{2+} :

c. 2 points

In a single atom, determine how many electrons can have the following set of quantum numbers?

(i) $n = 5; l = 3$

In a single atom, determine how many electrons can have the following set of quantum numbers?

(i) $n = 3; l = 2; m_s = +1/2$

**Question 4** **8 points**

Rank the following from 1 to 4 in **increasing** order. Provide the ranking in the boxes provided.

Note: Atomic number (Z) are provided for relevant atoms

a) Increasing (lowest to highest) order of *ionization energy*: (1= lowest; 4 = highest)

Mg

Mg: Z = 12

Rb

Rb: Z = 37

Cs

Cs: Z = 55

K

K : Z = 19

b) Increasing (lowest to highest) order of *bond angle*: (1= lowest; 4 = highest)

H₂OCCl₄XeF₄

HCN

c) Increasing (smallest to largest) order of *ionic size* (1= smallest; 4 = largest)

S²⁻

S: Z = 16

O²⁻

O: Z = 8

Be²⁺

Be: Z = 4

Na⁺

Na: Z = 11

d) Increasing (lowest to highest) order of *bond energy* (1= lowest; 4 = highest)

P-F

P-Cl

P-I

P-Br

**Question 5** **12 points**

a. 9 points For the molecules HClO_4 and TeF_6 , draw the most stable *Lewis structure* showing appropriate shape and *dashed/wedged bonds*, based on VSEPR. Determine the **molecular geometry** around the central atom. For HClO_4 , label the *most polar bond*.



b. 3 points (Note: Lone pairs are not shown in structures below)

(i) For the three *resonance* structures (1, 2 and 3) shown below, answer the following question:



(I) Circle the *most contributing* resonance structure.

Structure 1

Structure 2

Structure 3

(ii) For the three structures given (4, 5, and 6) below, answer the following questions:



(I) Circle all the structures that represent *resonance structures* of each other:

Structure 4

Structure 5

Structure 6

(II) Among the resonance structures, circle the *most contributing* resonance structure.

Structure 4

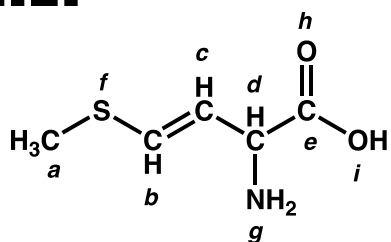
Structure 5

Structure 6

**Question 6** **6 points**

Using the data given below, calculate the **lattice energy** for SrO(s).

	ΔH° (kJ/mol)
$\text{Sr}(s) \longrightarrow \text{Sr}(g)$	+180
$\text{Sr}(g) \longrightarrow \text{Sr}^{2+}(g) + 2e^-$	+1468
$\text{O}_2(g) \longrightarrow 2\text{O}(g)$	+500
$\text{O}(g) + 2e^- \longrightarrow \text{O}^{2-}(g)$	+600
$\text{Sr}(s) + \frac{1}{2}\text{O}_2(g) \longrightarrow \text{SrO}(s)$	-550

**Question 7 10 points**

Consider the molecule given on the left, and answer questions (i) to (vi) using chemical bonding theories that we have learnt in class (valence bond theory, hybridization).

Note that the lone pairs are not shown in the structure:

i. Consider the carbon atoms marked in as ***a***, ***b***, ***c***, ***d***, and ***e*** in the molecule, and determine **hybridization** at each of the carbon centers. (2.5 points)

Atom	Hybridization
Carbon (<i>a</i>)	
Carbon (<i>b</i>)	
Carbon (<i>c</i>)	
Carbon (<i>d</i>)	
Carbon (<i>e</i>)	

ii. Now, consider the sulfur atom (S) marked as ***f***, nitrogen (N) atom (***g***), and oxygen(O) atoms (***h*** and ***i***), and determine their **hybridization**. (2 points)

Atom	Hybridization
Sulfur (<i>f</i>)	
Nitrogen (<i>g</i>)	
Oxygen (<i>h</i>)	
Oxygen (<i>i</i>)	

iii. **Describe** all the bonds formed by the carbon atoms ***a*** and ***b***, using sigma and pi designation as appropriate. (2 points)

Atom	Bonds formed (use sigma and pi as appropriate)
Carbon (<i>a</i>)	
Carbon (<i>b</i>)	

iv. How many total sp^3 -s sigma bonds are in the molecule? (1 point)

v. Determine **electronic** and **molecular geometry** at carbon (C) labeled ***e*** and sulfur (S) labeled ***f*** centres. (2 points)

Atom	Electronic Geometry	Molecular Geometry
Carbon (<i>e</i>)		
Sulfur (<i>f</i>)		

vi. Circle the value from the choices given below, that describes the **C-S-C bond angle at sulfur** (S) marked ***f***: (0.5 points)

>90° to <109.5° 109.5° >109.5 to <120° 120° >120° to <180°


Question 8 12 points

Using the **valence shell orbitals only** draw a **complete** molecular orbital (MO) diagram (with no s-p mixing) for the molecule **PS**, in the space provided below. **(6 points)**

Assume that 3s and 3p atomic orbitals will form molecular orbitals like those formed by the combination of 2s and 2p atomic orbitals that we learnt in class, and there is no s-p mixing.

Show **energy arrow**, **atomic orbitals** involved, **atom labels**, the **molecular orbitals** that are formed, **filled electrons** in the MO diagram etc. Make sure to label all the atomic and molecular orbitals.

Using the MO diagram above, answer the following questions:

- (i) Which one of the following will have the highest bond order? Circle the correct choice
 A) PS^{1-} B) PS C) PS^{1+}
- (ii) Which one of the following will have the longest bond length? Circle the correct choice
 A) PS^{1-} B) PS C) PS^{1+}
- (iii) Which of the following will be diamagnetic? Circle the correct choice.
 A) PS^{1-} B) PS C) PS^{1+}
- (iv) Draw Lewis dot structure of the molecule with the shortest bond length (based on the MO diagram) from among the three (PS^{1-} , PS, PS^{1+}), and indicate the bond order calculated from the Lewis dot structure. **(3 points)**

Lewis Dot Structure

Bond Order



4CF3694A-9DB3-4478-AD2D-BC1079DC6139

midterm-exam-93388

#168 Page 10 of 10

SCRAP PAGE

**CHEMISTRY 110 –Midterm Exam****Thursday November 7, 2024, 6:30 – 8:30 pm****Examiners: Pallavi Sirjoosingh and Ashok Kakkar**

First name (please write as legibly as possible within the boxes)

--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--

Last name

--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--

Student ID number

--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--

This booklet has 8 questions on 10 pages (The last page can be used as a scrap page).

The staple must NOT be removed from the question booklet. Your intact question booklet must be returned at the end of the exam.

A separate booklet contains a periodic table, and datasheet.

Write your answers directly under the questions in the space provided. Do not write on the QR codes or near the top edges (as it may be cut off).

Please write **legibly and dark enough** for the scan to be readable. For students who use English as a second language, dictionaries are permitted. Calculators are allowed.

DO NOT ASK QUESTIONS OF THE INVIGILATORS about interpretations of the exam questions. If you have a concern about a question, please summarize it at the top of this page. All concerns will be evaluated after the exam. GOOD LUCK!

**Question 1 8 points** (*Assume all transitions are possible*)

a. 6 points Calculate the **energy (in kJ/mol)** required to ionize (completely remove) the electron from the **$n=3$** level of **He^+ ion** (He: Helium). Show your work including equations. (3 sig. figs.)

b. 2 points

(I) Select the largest (longest) wavelength from the three choices below:

- i. Wavelength of photon to *completely remove* an electron from *1s orbital of Hydrogen atom*
- ii. Wavelength of photon to *completely remove* an electron from *1s orbital of Boron atom (B)*
- iii. Wavelength of photon to *completely remove* an electron from *1s orbital of Boron cation (B^+)*

i or ii or iii

(II) Select the smallest (shortest) wavelength from the three choices below (*for H atom*):

- i. Wavelength of photon *emitted* for an electron transitioning from *2p to 1s orbital*
- ii. Wavelength of photon *absorbed* for an electron transitioning from *2s to 4p orbital*
- iii. Wavelength of photon *absorbed* for an electron transitioning from *1s to 3p orbital*

i or ii or iii



Question 2 8 points



A Chem110 student plots the graph shown on the left, from data obtained for the kinetic energy (eV) of the ejected electron versus frequency (s^{-1}) of the incident light, for an unknown metal. ($1\text{eV} = 1.6022 \times 10^{-19} \text{ J}$)

Based on the plot, answer the questions **a to c** below:

a. 1 point On the plot above, label the **threshold frequency**.

b. 2 points What is the value of the work potential (in J) of the unknown metal?

c. 3 points What is the *longest* wavelength (in nm) required to eject an electron from this metal's surface? Show your work including equation(s) used. (3 sig figs)

d. 2 points In the space provided below, plot the following

(Assume frequency is *above* threshold frequency):

- Kinetic energy* of an ejected electron versus the *intensity* of the incident light (left)
- Number of electrons* ejected versus the *frequency* of the incident light (right)



**Question 3** **6 points****a. 2 points**

In the ground state of Ge (Germanium; $Z = 32$) atom, **name** the following orbitals:

i. Orbital containing the electron that requires the maximum energy to be removed.

ii. Orbital containing the electron that experiences the minimum effective nuclear charge.

b. 2 points

An unknown anion (A^-) has the following *excited* electronic configuration: $[Ar] 4s^2 3d^3 4p^1$

(Ar: Argon; $Z = 18$)

Give the **condensed ground state electronic configuration** of the atom, **A** and the cation, **A^{2+}** .
Show your work. (Assume all electronic transitions are possible)

Electronic Configuration of A:

Electronic Configuration of A^{2+} :

c. 2 points

In a single atom, determine how many electrons can have the following set of quantum numbers?

(i) $n = 5; l = 3$

In a single atom, determine how many electrons can have the following set of quantum numbers?

(i) $n = 3; l = 2; m_s = +1/2$

**Question 4** **8 points**

Rank the following from 1 to 4 in **increasing** order. Provide the ranking in the boxes provided.

Note: Atomic number (Z) are provided for relevant atoms

a) Increasing (lowest to highest) order of *ionization energy*: (1= lowest; 4 = highest)

Mg

Mg: Z = 12

Rb

Rb: Z = 37

Cs

Cs: Z = 55

K

K : Z = 19

b) Increasing (lowest to highest) order of *bond angle*: (1= lowest; 4 = highest)

H₂OCCl₄XeF₄

HCN

c) Increasing (smallest to largest) order of *ionic size* (1= smallest; 4 = largest)

S²⁻

S: Z = 16

O²⁻

O: Z = 8

Be²⁺

Be: Z = 4

Na⁺

Na: Z = 11

d) Increasing (lowest to highest) order of *bond energy* (1= lowest; 4 = highest)

P-F

P-Cl

P-I

P-Br

**Question 5** **12 points**

a. 9 points For the molecules HClO_4 and TeF_6 , draw the most stable *Lewis structure* showing appropriate shape and *dashed/wedged bonds*, based on VSEPR. Determine the **molecular geometry** around the central atom. For HClO_4 , label the *most polar bond*.



b. 3 points (Note: Lone pairs are not shown in structures below)

(i) For the three *resonance* structures (1, 2 and 3) shown below, answer the following question:



(I) Circle the *most contributing* resonance structure.

Structure 1

Structure 2

Structure 3

(ii) For the three structures given (4, 5, and 6) below, answer the following questions:



(I) Circle all the structures that represent *resonance structures* of each other:

Structure 4

Structure 5

Structure 6

(II) Among the resonance structures, circle the *most contributing* resonance structure.

Structure 4

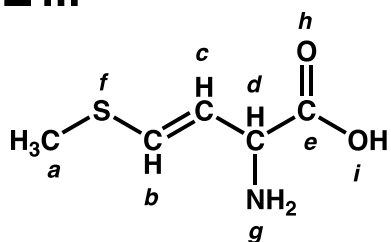
Structure 5

Structure 6

**Question 6** **6 points**

Using the data given below, calculate the **lattice energy** for SrO(s).

	ΔH° (kJ/mol)
$\text{Sr}(s) \longrightarrow \text{Sr}(g)$	+180
$\text{Sr}(g) \longrightarrow \text{Sr}^{2+}(g) + 2e^-$	+1468
$\text{O}_2(g) \longrightarrow 2\text{O}(g)$	+500
$\text{O}(g) + 2e^- \longrightarrow \text{O}^{2-}(g)$	+600
$\text{Sr}(s) + \frac{1}{2}\text{O}_2(g) \longrightarrow \text{SrO}(s)$	-550

**Question 7 10 points**

Consider the molecule given on the left, and answer questions (i) to (vi) using chemical bonding theories that we have learnt in class (valence bond theory, hybridization).

Note that the lone pairs are not shown in the structure:

i. Consider the carbon atoms marked in as ***a***, ***b***, ***c***, ***d***, and ***e*** in the molecule, and determine **hybridization** at each of the carbon centers. (2.5 points)

Atom	Hybridization
Carbon (<i>a</i>)	
Carbon (<i>b</i>)	
Carbon (<i>c</i>)	
Carbon (<i>d</i>)	
Carbon (<i>e</i>)	

ii. Now, consider the sulfur atom (S) marked as ***f***, nitrogen (N) atom (***g***), and oxygen(O) atoms (***h*** and ***i***), and determine their **hybridization**. (2 points)

Atom	Hybridization
Sulfur (<i>f</i>)	
Nitrogen (<i>g</i>)	
Oxygen (<i>h</i>)	
Oxygen (<i>i</i>)	

iii. **Describe** all the bonds formed by the carbon atoms ***a*** and ***b***, using sigma and pi designation as appropriate. (2 points)

Atom	Bonds formed (use sigma and pi as appropriate)
Carbon (<i>a</i>)	
Carbon (<i>b</i>)	

iv. How many total sp^3 -s sigma bonds are in the molecule? (1 point)

v. Determine **electronic** and **molecular geometry** at carbon (C) labeled ***e*** and sulfur (S) labeled ***f*** centres. (2 points)

Atom	Electronic Geometry	Molecular Geometry
Carbon (<i>e</i>)		
Sulfur (<i>f</i>)		

vi. Circle the value from the choices given below, that describes the **C-S-C bond angle at sulfur** (S) marked ***f***: (0.5 points)

>90° to <109.5°

109.5°

>109.5 to <120°

120°

>120° to <180°

**Question 8 12 points**

Using the **valence shell orbitals only** draw a **complete** molecular orbital (MO) diagram (with no s-p mixing) for the molecule **PS**, in the space provided below. **(6 points)**

Assume that 3s and 3p atomic orbitals will form molecular orbitals like those formed by the combination of 2s and 2p atomic orbitals that we learnt in class, and there is no s-p mixing.

Show **energy arrow**, **atomic orbitals** involved, **atom labels**, the **molecular orbitals** that are formed, **filled electrons** in the MO diagram etc. Make sure to label all the atomic and molecular orbitals.

Using the MO diagram above, answer the following questions:

- (i) Which one of the following will have the highest bond order? Circle the correct choice
A) PS^{1-} B) PS C) PS^{1+}
- (ii) Which one of the following will have the longest bond length? Circle the correct choice
A) PS^{1-} B) PS C) PS^{1+}
- (iii) Which of the following will be diamagnetic? Circle the correct choice.
A) PS^{1-} B) PS C) PS^{1+}
- (iv) Draw Lewis dot structure of the molecule with the shortest bond length (based on the MO diagram) from among the three (PS^{1-} , PS, PS^{1+}), and indicate the bond order calculated from the Lewis dot structure. **(3 points)**

Lewis Dot Structure

Bond Order



46E7FDB2-17AE-419D-8F9F-5194A7F0BBDF

midterm-exam-93388

#169 Page 10 of 10

SCRAP PAGE

**CHEMISTRY 110 –Midterm Exam****Thursday November 7, 2024, 6:30 – 8:30 pm**

Examiners: Pallavi Sirjoosingh and Ashok Kakkar

First name (please write as legibly as possible within the boxes)

--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--

Last name

--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--

Student ID number

--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--

This booklet has 8 questions on 10 pages (The last page can be used as a scrap page).

The staple must NOT be removed from the question booklet. Your intact question booklet must be returned at the end of the exam.

A separate booklet contains a periodic table, and datasheet.

Write your answers directly under the questions in the space provided. Do not write on the QR codes or near the top edges (as it may be cut off).

Please write **legibly and dark enough** for the scan to be readable. For students who use English as a second language, dictionaries are permitted. Calculators are allowed.

DO NOT ASK QUESTIONS OF THE INVIGILATORS about interpretations of the exam questions. If you have a concern about a question, please summarize it at the top of this page. All concerns will be evaluated after the exam. GOOD LUCK!

**Question 1 8 points** (*Assume all transitions are possible*)

a. 6 points Calculate the **energy (in kJ/mol)** required to ionize (completely remove) the electron from the **n=3** level of **He⁺ ion** (He: Helium). Show your work including equations. (3 sig. figs.)

b. 2 points

(I) Select the largest (longest) wavelength from the three choices below:

- i. Wavelength of photon to *completely remove* an electron from *1s orbital of Hydrogen atom*
- ii. Wavelength of photon to *completely remove* an electron from *1s orbital of Boron atom (B)*
- iii. Wavelength of photon to *completely remove* an electron from *1s orbital of Boron cation (B⁺)*

i or ii or iii

(II) Select the smallest (shortest) wavelength from the three choices below (*for H atom*):

- i. Wavelength of photon *emitted* for an electron transitioning from *2p to 1s orbital*
- ii. Wavelength of photon *absorbed* for an electron transitioning from *2s to 4p orbital*
- iii. Wavelength of photon *absorbed* for an electron transitioning from *1s to 3p orbital*

i or ii or iii



Question 2 8 points



A Chem110 student plots the graph shown on the left, from data obtained for the kinetic energy (eV) of the ejected electron versus frequency (s^{-1}) of the incident light, for an unknown metal. ($1\text{eV} = 1.6022 \times 10^{-19} \text{ J}$)

Based on the plot, answer the questions **a to c** below:

a. 1 point On the plot above, label the **threshold frequency**.

b. 2 points What is the value of the work potential (in J) of the unknown metal?

c. 3 points What is the *longest* wavelength (in nm) required to eject an electron from this metal's surface? Show your work including equation(s) used. (3 sig figs)

d. 2 points In the space provided below, plot the following

(Assume frequency is *above* threshold frequency):

- Kinetic energy* of an ejected electron versus the *intensity* of the incident light (left)
- Number of electrons* ejected versus the *frequency* of the incident light (right)



**Question 3** **6 points****a. 2 points**

In the ground state of Ge (Germanium; $Z = 32$) atom, **name** the following orbitals:

i. Orbital containing the electron that requires the maximum energy to be removed.

ii. Orbital containing the electron that experiences the minimum effective nuclear charge.

b. 2 points

An unknown anion (A^-) has the following *excited* electronic configuration: $[Ar] 4s^2 3d^3 4p^1$

(Ar: Argon; $Z = 18$)

Give the **condensed ground state electronic configuration** of the atom, **A** and the cation, **A^{2+}** .
Show your work. (Assume all electronic transitions are possible)

Electronic Configuration of A:

Electronic Configuration of A^{2+} :

c. 2 points

In a single atom, determine how many electrons can have the following set of quantum numbers?

(i) $n = 5; l = 3$

In a single atom, determine how many electrons can have the following set of quantum numbers?

(i) $n = 3; l = 2; m_s = +1/2$

**Question 4** **8 points**

Rank the following from 1 to 4 in **increasing** order. Provide the ranking in the boxes provided.

Note: Atomic number (Z) are provided for relevant atoms

a) Increasing (lowest to highest) order of *ionization energy*: (1= lowest; 4 = highest)

Mg

Mg: Z = 12

Rb

Rb: Z = 37

Cs

Cs: Z = 55

K

K : Z = 19

b) Increasing (lowest to highest) order of *bond angle*: (1= lowest; 4 = highest)

H₂OCCl₄XeF₄

HCN

c) Increasing (smallest to largest) order of *ionic size* (1= smallest; 4 = largest)

S²⁻

S: Z = 16

O²⁻

O: Z = 8

Be²⁺

Be: Z = 4

Na⁺

Na: Z = 11

d) Increasing (lowest to highest) order of *bond energy* (1= lowest; 4 = highest)

P-F

P-Cl

P-I

P-Br

**Question 5** **12 points**

a. 9 points For the molecules HClO_4 and TeF_6 , draw the most stable *Lewis structure* showing appropriate shape and *dashed/wedged bonds*, based on VSEPR. Determine the **molecular geometry** around the central atom. For HClO_4 , label the *most polar bond*.



b. 3 points (Note: Lone pairs are not shown in structures below)

(i) For the three *resonance* structures (1, 2 and 3) shown below, answer the following question:



(I) Circle the *most contributing* resonance structure.

Structure 1

Structure 2

Structure 3

(ii) For the three structures given (4, 5, and 6) below, answer the following questions:



(I) Circle all the structures that represent *resonance structures* of each other:

Structure 4

Structure 5

Structure 6

(II) Among the resonance structures, circle the *most contributing* resonance structure.

Structure 4

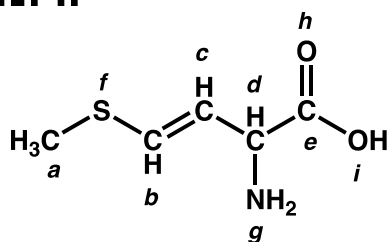
Structure 5

Structure 6

**Question 6** **6 points**

Using the data given below, calculate the **lattice energy** for SrO(s).

	ΔH° (kJ/mol)
$\text{Sr}(s) \longrightarrow \text{Sr}(g)$	+180
$\text{Sr}(g) \longrightarrow \text{Sr}^{2+}(g) + 2e^-$	+1468
$\text{O}_2(g) \longrightarrow 2\text{O}(g)$	+500
$\text{O}(g) + 2e^- \longrightarrow \text{O}^{2-}(g)$	+600
$\text{Sr}(s) + \frac{1}{2}\text{O}_2(g) \longrightarrow \text{SrO}(s)$	-550

**Question 7 10 points**

Consider the molecule given on the left, and answer questions (i) to (vi) using chemical bonding theories that we have learnt in class (valence bond theory, hybridization).

Note that the lone pairs are not shown in the structure:

i. Consider the carbon atoms marked in as ***a***, ***b***, ***c***, ***d***, and ***e*** in the molecule, and determine **hybridization** at each of the carbon centers. (2.5 points)

Atom	Hybridization
Carbon (<i>a</i>)	
Carbon (<i>b</i>)	
Carbon (<i>c</i>)	
Carbon (<i>d</i>)	
Carbon (<i>e</i>)	

ii. Now, consider the sulfur atom (S) marked as ***f***, nitrogen (N) atom (***g***), and oxygen(O) atoms (***h*** and ***i***), and determine their **hybridization**. (2 points)

Atom	Hybridization
Sulfur (<i>f</i>)	
Nitrogen (<i>g</i>)	
Oxygen (<i>h</i>)	
Oxygen (<i>i</i>)	

iii. **Describe** all the bonds formed by the carbon atoms ***a*** and ***b***, using sigma and pi designation as appropriate. (2 points)

Atom	Bonds formed (use sigma and pi as appropriate)
Carbon (<i>a</i>)	
Carbon (<i>b</i>)	

iv. How many total sp^3 -s sigma bonds are in the molecule? (1 point)

v. Determine **electronic** and **molecular geometry** at carbon (C) labeled ***e*** and sulfur (S) labeled ***f*** centres. (2 points)

Atom	Electronic Geometry	Molecular Geometry
Carbon (<i>e</i>)		
Sulfur (<i>f</i>)		

vi. Circle the value from the choices given below, that describes the **C-S-C bond angle at sulfur** (S) marked ***f***: (0.5 points)

>90° to <109.5° 109.5° >109.5 to <120° 120° >120° to <180°


Question 8 12 points

Using the **valence shell orbitals only** draw a **complete** molecular orbital (MO) diagram (with no s-p mixing) for the molecule **PS**, in the space provided below. **(6 points)**

Assume that 3s and 3p atomic orbitals will form molecular orbitals like those formed by the combination of 2s and 2p atomic orbitals that we learnt in class, and there is no s-p mixing.

Show **energy arrow**, **atomic orbitals** involved, **atom labels**, the **molecular orbitals** that are formed, **filled electrons** in the MO diagram etc. Make sure to label all the atomic and molecular orbitals.

Using the MO diagram above, answer the following questions:

- (i) Which one of the following will have the highest bond order? Circle the correct choice
 A) PS^{1-} B) PS C) PS^{1+}
- (ii) Which one of the following will have the longest bond length? Circle the correct choice
 A) PS^{1-} B) PS C) PS^{1+}
- (iii) Which of the following will be diamagnetic? Circle the correct choice.
 A) PS^{1-} B) PS C) PS^{1+}
- (iv) Draw Lewis dot structure of the molecule with the shortest bond length (based on the MO diagram) from among the three (PS^{1-} , PS, PS^{1+}), and indicate the bond order calculated from the Lewis dot structure. **(3 points)**

Lewis Dot Structure

Bond Order



7EBFFF6B-FB9D-40C7-A2D3-8E5AF21AC2BE

midterm-exam-93388

#170 Page 10 of 10

SCRAP PAGE

**CHEMISTRY 110 –Midterm Exam****Thursday November 7, 2024, 6:30 – 8:30 pm**

Examiners: Pallavi Sirjoosingh and Ashok Kakkar

First name (please write as legibly as possible within the boxes)

--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--

Last name

--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--

Student ID number

--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--

This booklet has 8 questions on 10 pages (The last page can be used as a scrap page).

The staple must NOT be removed from the question booklet. Your intact question booklet must be returned at the end of the exam.

A separate booklet contains a periodic table, and datasheet.

Write your answers directly under the questions in the space provided. Do not write on the QR codes or near the top edges (as it may be cut off).

Please write **legibly and dark enough** for the scan to be readable. For students who use English as a second language, dictionaries are permitted. Calculators are allowed.

DO NOT ASK QUESTIONS OF THE INVIGILATORS about interpretations of the exam questions. If you have a concern about a question, please summarize it at the top of this page. All concerns will be evaluated after the exam. GOOD LUCK!

**Question 1 8 points** (*Assume all transitions are possible*)

a. 6 points Calculate the **energy (in kJ/mol)** required to ionize (completely remove) the electron from the **n=3** level of **He⁺ ion** (He: Helium). Show your work including equations. (3 sig. figs.)

b. 2 points

(I) Select the largest (longest) wavelength from the three choices below:

- i. Wavelength of photon to *completely remove* an electron from *1s orbital of Hydrogen atom*
- ii. Wavelength of photon to *completely remove* an electron from *1s orbital of Boron atom (B)*
- iii. Wavelength of photon to *completely remove* an electron from *1s orbital of Boron cation (B⁺)*

i or ii or iii

(II) Select the smallest (shortest) wavelength from the three choices below (*for H atom*):

- i. Wavelength of photon *emitted* for an electron transitioning from *2p to 1s orbital*
- ii. Wavelength of photon *absorbed* for an electron transitioning from *2s to 4p orbital*
- iii. Wavelength of photon *absorbed* for an electron transitioning from *1s to 3p orbital*

i or ii or iii



Question 2 8 points



A Chem110 student plots the graph shown on the left, from data obtained for the kinetic energy (eV) of the ejected electron versus frequency (s^{-1}) of the incident light, for an unknown metal. ($1\text{eV} = 1.6022 \times 10^{-19} \text{ J}$)

Based on the plot, answer the questions **a to c** below:

a. 1 point On the plot above, label the **threshold frequency**.

b. 2 points What is the value of the work potential (in J) of the unknown metal?

c. 3 points What is the *longest* wavelength (in nm) required to eject an electron from this metal's surface? Show your work including equation(s) used. (3 sig figs)

d. 2 points In the space provided below, plot the following

(Assume frequency is *above* threshold frequency):

- i. *Kinetic energy* of an ejected electron versus the *intensity* of the incident light (left)
- ii. *Number of electrons* ejected versus the *frequency* of the incident light (right)



**Question 3** **6 points****a. 2 points**

In the ground state of Ge (Germanium; $Z = 32$) atom, **name** the following orbitals:

i. Orbital containing the electron that requires the maximum energy to be removed.

ii. Orbital containing the electron that experiences the minimum effective nuclear charge.

b. 2 points

An unknown anion (A^-) has the following *excited* electronic configuration: $[Ar] 4s^2 3d^3 4p^1$

(Ar: Argon; $Z = 18$)

Give the **condensed ground state electronic configuration** of the atom, **A** and the cation, **A^{2+}** .
Show your work. (Assume all electronic transitions are possible)

Electronic Configuration of A:

Electronic Configuration of A^{2+} :

c. 2 points

In a single atom, determine how many electrons can have the following set of quantum numbers?

(i) $n = 5; l = 3$

In a single atom, determine how many electrons can have the following set of quantum numbers?

(i) $n = 3; l = 2; m_s = +1/2$

**Question 4** **8 points**

Rank the following from 1 to 4 in **increasing** order. Provide the ranking in the boxes provided.

Note: Atomic number (Z) are provided for relevant atoms

a) Increasing (lowest to highest) order of *ionization energy*: (1= lowest; 4 = highest)

Mg

Mg: Z = 12

Rb

Rb: Z = 37

Cs

Cs: Z = 55

K

K : Z = 19

b) Increasing (lowest to highest) order of *bond angle*: (1= lowest; 4 = highest)

H₂OCCl₄XeF₄

HCN

c) Increasing (smallest to largest) order of *ionic size* (1= smallest; 4 = largest)

S²⁻

S: Z = 16

O²⁻

O: Z = 8

Be²⁺

Be: Z = 4

Na⁺

Na: Z = 11

d) Increasing (lowest to highest) order of *bond energy* (1= lowest; 4 = highest)

P-F

P-Cl

P-I

P-Br

**Question 5** **12 points**

a. 9 points For the molecules HClO_4 and TeF_6 , draw the most stable *Lewis structure* showing appropriate shape and *dashed/wedged bonds*, based on VSEPR. Determine the **molecular geometry** around the central atom. For HClO_4 , label the *most polar bond*.



b. 3 points (Note: Lone pairs are not shown in structures below)

(i) For the three *resonance* structures (1, 2 and 3) shown below, answer the following question:



(I) Circle the *most contributing* resonance structure.

Structure 1

Structure 2

Structure 3

(ii) For the three structures given (4, 5, and 6) below, answer the following questions:



(I) Circle all the structures that represent *resonance structures* of each other:

Structure 4

Structure 5

Structure 6

(II) Among the resonance structures, circle the *most contributing* resonance structure.

Structure 4

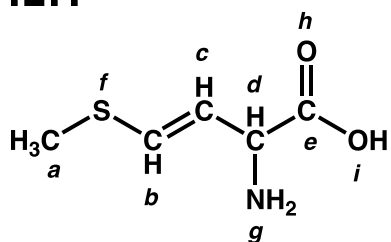
Structure 5

Structure 6

**Question 6** **6 points**

Using the data given below, calculate the **lattice energy** for SrO(s).

	ΔH° (kJ/mol)
$\text{Sr}(s) \longrightarrow \text{Sr}(g)$	+180
$\text{Sr}(g) \longrightarrow \text{Sr}^{2+}(g) + 2e^-$	+1468
$\text{O}_2(g) \longrightarrow 2\text{O}(g)$	+500
$\text{O}(g) + 2e^- \longrightarrow \text{O}^{2-}(g)$	+600
$\text{Sr}(s) + \frac{1}{2}\text{O}_2(g) \longrightarrow \text{SrO}(s)$	-550

**Question 7 10 points**

Consider the molecule given on the left, and answer questions (i) to (vi) using chemical bonding theories that we have learnt in class (valence bond theory, hybridization).

Note that the lone pairs are not shown in the structure:

i. Consider the carbon atoms marked in as ***a***, ***b***, ***c***, ***d***, and ***e*** in the molecule, and determine **hybridization** at each of the carbon centers. (2.5 points)

Atom	Hybridization
Carbon (<i>a</i>)	
Carbon (<i>b</i>)	
Carbon (<i>c</i>)	
Carbon (<i>d</i>)	
Carbon (<i>e</i>)	

ii. Now, consider the sulfur atom (S) marked as ***f***, nitrogen (N) atom (***g***), and oxygen(O) atoms (***h*** and ***i***), and determine their **hybridization**. (2 points)

Atom	Hybridization
Sulfur (<i>f</i>)	
Nitrogen (<i>g</i>)	
Oxygen (<i>h</i>)	
Oxygen (<i>i</i>)	

iii. **Describe** all the bonds formed by the carbon atoms ***a*** and ***b***, using sigma and pi designation as appropriate. (2 points)

Atom	Bonds formed (use sigma and pi as appropriate)
Carbon (<i>a</i>)	
Carbon (<i>b</i>)	

iv. How many total sp^3 -s sigma bonds are in the molecule? (1 point)

v. Determine **electronic** and **molecular geometry** at carbon (C) labeled ***e*** and sulfur (S) labeled ***f*** centres. (2 points)

Atom	Electronic Geometry	Molecular Geometry
Carbon (<i>e</i>)		
Sulfur (<i>f</i>)		

vi. Circle the value from the choices given below, that describes the **C-S-C bond angle at sulfur** (S) marked ***f***: (0.5 points)

>90° to <109.5° 109.5° >109.5 to <120° 120° >120° to <180°


Question 8 12 points

Using the **valence shell orbitals only** draw a **complete** molecular orbital (MO) diagram (with no s-p mixing) for the molecule **PS**, in the space provided below. **(6 points)**

Assume that 3s and 3p atomic orbitals will form molecular orbitals like those formed by the combination of 2s and 2p atomic orbitals that we learnt in class, and there is no s-p mixing.

Show **energy arrow**, **atomic orbitals** involved, **atom labels**, the **molecular orbitals** that are formed, **filled electrons** in the MO diagram etc. Make sure to label all the atomic and molecular orbitals.

Using the MO diagram above, answer the following questions:

- (i) Which one of the following will have the highest bond order? Circle the correct choice
 A) PS^{1-} B) PS C) PS^{1+}
- (ii) Which one of the following will have the longest bond length? Circle the correct choice
 A) PS^{1-} B) PS C) PS^{1+}
- (iii) Which of the following will be diamagnetic? Circle the correct choice.
 A) PS^{1-} B) PS C) PS^{1+}
- (iv) Draw Lewis dot structure of the molecule with the shortest bond length (based on the MO diagram) from among the three (PS^{1-} , PS, PS^{1+}), and indicate the bond order calculated from the Lewis dot structure. **(3 points)**

Lewis Dot Structure

Bond Order



43D3A3FE-B0C5-4DFA-94ED-D0A945AC3B1F

midterm-exam-93388

#171 Page 10 of 10

SCRAP PAGE

**CHEMISTRY 110 –Midterm Exam****Thursday November 7, 2024, 6:30 – 8:30 pm****Examiners: Pallavi Sirjoosingh and Ashok Kakkar**

First name (please write as legibly as possible within the boxes)

--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--

Last name

--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--

Student ID number

--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--

This booklet has 8 questions on 10 pages (The last page can be used as a scrap page).

The staple must NOT be removed from the question booklet. Your intact question booklet must be returned at the end of the exam.

A separate booklet contains a periodic table, and datasheet.

Write your answers directly under the questions in the space provided. Do not write on the QR codes or near the top edges (as it may be cut off).

Please write **legibly and dark enough** for the scan to be readable. For students who use English as a second language, dictionaries are permitted. Calculators are allowed.

DO NOT ASK QUESTIONS OF THE INVIGILATORS about interpretations of the exam questions. If you have a concern about a question, please summarize it at the top of this page. All concerns will be evaluated after the exam. GOOD LUCK!

**Question 1 8 points** (*Assume all transitions are possible*)

a. 6 points Calculate the **energy (in kJ/mol)** required to ionize (completely remove) the electron from the **n=3** level of **He⁺ ion** (He: Helium). Show your work including equations. (3 sig. figs.)

b. 2 points

(I) Select the largest (longest) wavelength from the three choices below:

- i. Wavelength of photon to *completely remove* an electron from *1s orbital of Hydrogen atom*
- ii. Wavelength of photon to *completely remove* an electron from *1s orbital of Boron atom (B)*
- iii. Wavelength of photon to *completely remove* an electron from *1s orbital of Boron cation (B⁺)*

i or ii or iii

(II) Select the smallest (shortest) wavelength from the three choices below (*for H atom*):

- i. Wavelength of photon *emitted* for an electron transitioning from *2p to 1s orbital*
- ii. Wavelength of photon *absorbed* for an electron transitioning from *2s to 4p orbital*
- iii. Wavelength of photon *absorbed* for an electron transitioning from *1s to 3p orbital*

i or ii or iii



Question 2 8 points



A Chem110 student plots the graph shown on the left, from data obtained for the kinetic energy (eV) of the ejected electron versus frequency (s^{-1}) of the incident light, for an unknown metal. ($1\text{eV} = 1.6022 \times 10^{-19} \text{ J}$)

Based on the plot, answer the questions **a to c** below:

a. 1 point On the plot above, label the **threshold frequency**.

b. 2 points What is the value of the work potential (in J) of the unknown metal?

c. 3 points What is the *longest* wavelength (in nm) required to eject an electron from this metal's surface? Show your work including equation(s) used. (3 sig figs)

d. 2 points In the space provided below, plot the following

(Assume frequency is *above* threshold frequency):

- Kinetic energy* of an ejected electron versus the *intensity* of the incident light (left)
- Number of electrons* ejected versus the *frequency* of the incident light (right)



**Question 3** **6 points****a. 2 points**

In the ground state of Ge (Germanium; $Z = 32$) atom, **name** the following orbitals:

i. Orbital containing the electron that requires the maximum energy to be removed.

ii. Orbital containing the electron that experiences the minimum effective nuclear charge.

b. 2 points

An unknown anion (A^-) has the following *excited* electronic configuration: $[Ar] 4s^2 3d^3 4p^1$

(Ar: Argon; $Z = 18$)

Give the **condensed ground state electronic configuration** of the atom, **A** and the cation, A^{2+} .
Show your work. (Assume all electronic transitions are possible)

Electronic Configuration of A:

Electronic Configuration of A^{2+} :

c. 2 points

In a single atom, determine how many electrons can have the following set of quantum numbers?

(i) $n = 5; l = 3$

In a single atom, determine how many electrons can have the following set of quantum numbers?

(i) $n = 3; l = 2; m_s = +1/2$

**Question 4** **8 points**

Rank the following from 1 to 4 in **increasing** order. Provide the ranking in the boxes provided.

Note: Atomic number (Z) are provided for relevant atoms

a) Increasing (lowest to highest) order of *ionization energy*: (1= lowest; 4 = highest)

Mg

Mg: Z = 12

Rb

Rb: Z = 37

Cs

Cs: Z = 55

K

K : Z = 19

b) Increasing (lowest to highest) order of *bond angle*: (1= lowest; 4 = highest)

H₂OCCl₄XeF₄

HCN

c) Increasing (smallest to largest) order of *ionic size* (1= smallest; 4 = largest)

S²⁻

S: Z = 16

O²⁻

O: Z = 8

Be²⁺

Be: Z = 4

Na⁺

Na: Z = 11

d) Increasing (lowest to highest) order of *bond energy* (1= lowest; 4 = highest)

P-F

P-Cl

P-I

P-Br

**Question 5** **12 points**

a. 9 points For the molecules HClO_4 and TeF_6 , draw the most stable *Lewis structure* showing appropriate shape and *dashed/wedged bonds*, based on VSEPR. Determine the **molecular geometry** around the central atom. For HClO_4 , label the *most polar bond*.



b. 3 points (Note: Lone pairs are not shown in structures below)

(i) For the three *resonance* structures (1, 2 and 3) shown below, answer the following question:



(I) Circle the *most contributing* resonance structure.

Structure 1

Structure 2

Structure 3

(ii) For the three structures given (4, 5, and 6) below, answer the following questions:



(I) Circle all the structures that represent *resonance structures* of each other:

Structure 4

Structure 5

Structure 6

(II) Among the resonance structures, circle the *most contributing* resonance structure.

Structure 4

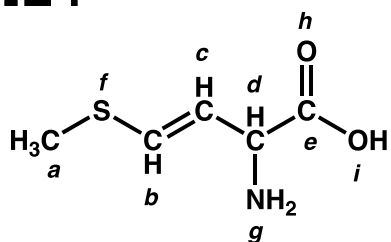
Structure 5

Structure 6

**Question 6** **6 points**

Using the data given below, calculate the **lattice energy** for SrO(s).

	ΔH° (kJ/mol)
$\text{Sr}(s) \longrightarrow \text{Sr}(g)$	+180
$\text{Sr}(g) \longrightarrow \text{Sr}^{2+}(g) + 2e^-$	+1468
$\text{O}_2(g) \longrightarrow 2\text{O}(g)$	+500
$\text{O}(g) + 2e^- \longrightarrow \text{O}^{2-}(g)$	+600
$\text{Sr}(s) + \frac{1}{2}\text{O}_2(g) \longrightarrow \text{SrO}(s)$	-550

**Question 7 10 points**

Consider the molecule given on the left, and answer questions (i) to (vi) using chemical bonding theories that we have learnt in class (valence bond theory, hybridization).

Note that the lone pairs are not shown in the structure:

i. Consider the carbon atoms marked in as ***a***, ***b***, ***c***, ***d***, and ***e*** in the molecule, and determine **hybridization** at each of the carbon centers. (2.5 points)

Atom	Hybridization
Carbon (<i>a</i>)	
Carbon (<i>b</i>)	
Carbon (<i>c</i>)	
Carbon (<i>d</i>)	
Carbon (<i>e</i>)	

ii. Now, consider the sulfur atom (S) marked as ***f***, nitrogen (N) atom (***g***), and oxygen(O) atoms (***h*** and ***i***), and determine their **hybridization**. (2 points)

Atom	Hybridization
Sulfur (<i>f</i>)	
Nitrogen (<i>g</i>)	
Oxygen (<i>h</i>)	
Oxygen (<i>i</i>)	

iii. **Describe** all the bonds formed by the carbon atoms ***a*** and ***b***, using sigma and pi designation as appropriate. (2 points)

Atom	Bonds formed (use sigma and pi as appropriate)
Carbon (<i>a</i>)	
Carbon (<i>b</i>)	

iv. How many total sp^3 -s sigma bonds are in the molecule? (1 point)

v. Determine **electronic** and **molecular geometry** at carbon (C) labeled ***e*** and sulfur (S) labeled ***f*** centres. (2 points)

Atom	Electronic Geometry	Molecular Geometry
Carbon (<i>e</i>)		
Sulfur (<i>f</i>)		

vi. Circle the value from the choices given below, that describes the **C-S-C bond angle at sulfur** (S) marked ***f***: (0.5 points)

>90° to <109.5° 109.5° >109.5 to <120° 120° >120° to <180°


Question 8 12 points

Using the **valence shell orbitals only** draw a **complete** molecular orbital (MO) diagram (with no s-p mixing) for the molecule **PS**, in the space provided below. **(6 points)**

Assume that 3s and 3p atomic orbitals will form molecular orbitals like those formed by the combination of 2s and 2p atomic orbitals that we learnt in class, and there is no s-p mixing.

Show **energy arrow**, **atomic orbitals** involved, **atom labels**, the **molecular orbitals** that are formed, **filled electrons** in the MO diagram etc. Make sure to label all the atomic and molecular orbitals.

Using the MO diagram above, answer the following questions:

- (i) Which one of the following will have the highest bond order? Circle the correct choice
 A) PS^{1-} B) PS C) PS^{1+}
- (ii) Which one of the following will have the longest bond length? Circle the correct choice
 A) PS^{1-} B) PS C) PS^{1+}
- (iii) Which of the following will be diamagnetic? Circle the correct choice.
 A) PS^{1-} B) PS C) PS^{1+}
- (iv) Draw Lewis dot structure of the molecule with the shortest bond length (based on the MO diagram) from among the three (PS^{1-} , PS, PS^{1+}), and indicate the bond order calculated from the Lewis dot structure. **(3 points)**

Lewis Dot Structure

Bond Order



121B9124-2612-4A6C-9F84-DEA03E33FD30

midterm-exam-93388

#172 Page 10 of 10

SCRAP PAGE

**CHEMISTRY 110 –Midterm Exam****Thursday November 7, 2024, 6:30 – 8:30 pm****Examiners: Pallavi Sirjoosingh and Ashok Kakkar**

First name (please write as legibly as possible within the boxes)

--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--

Last name

--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--

Student ID number

--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--

This booklet has 8 questions on 10 pages (The last page can be used as a scrap page).

The staple must NOT be removed from the question booklet. Your intact question booklet must be returned at the end of the exam.

A separate booklet contains a periodic table, and datasheet.

Write your answers directly under the questions in the space provided. Do not write on the QR codes or near the top edges (as it may be cut off).

Please write **legibly and dark enough** for the scan to be readable. For students who use English as a second language, dictionaries are permitted. Calculators are allowed.

DO NOT ASK QUESTIONS OF THE INVIGILATORS about interpretations of the exam questions. If you have a concern about a question, please summarize it at the top of this page. All concerns will be evaluated after the exam. GOOD LUCK!

**Question 1 8 points** (*Assume all transitions are possible*)

a. 6 points Calculate the **energy (in kJ/mol)** required to ionize (completely remove) the electron from the **n=3** level of **He⁺ ion** (He: Helium). Show your work including equations. (3 sig. figs.)

b. 2 points

(I) Select the largest (longest) wavelength from the three choices below:

- i. Wavelength of photon to *completely remove* an electron from *1s orbital of Hydrogen atom*
- ii. Wavelength of photon to *completely remove* an electron from *1s orbital of Boron atom (B)*
- iii. Wavelength of photon to *completely remove* an electron from *1s orbital of Boron cation (B⁺)*

i or ii or iii

(II) Select the smallest (shortest) wavelength from the three choices below (*for H atom*):

- i. Wavelength of photon *emitted* for an electron transitioning from *2p to 1s orbital*
- ii. Wavelength of photon *absorbed* for an electron transitioning from *2s to 4p orbital*
- iii. Wavelength of photon *absorbed* for an electron transitioning from *1s to 3p orbital*

i or ii or iii



Question 2 8 points



A Chem110 student plots the graph shown on the left, from data obtained for the kinetic energy (eV) of the ejected electron versus frequency (s^{-1}) of the incident light, for an unknown metal. ($1\text{eV} = 1.6022 \times 10^{-19} \text{ J}$)

Based on the plot, answer the questions **a to c** below:

a. 1 point On the plot above, label the **threshold frequency**.

b. 2 points What is the value of the work potential (in J) of the unknown metal?

c. 3 points What is the *longest* wavelength (in nm) required to eject an electron from this metal's surface? Show your work including equation(s) used. (3 sig figs)

d. 2 points In the space provided below, plot the following

(Assume frequency is *above* threshold frequency):

- Kinetic energy* of an ejected electron versus the *intensity* of the incident light (left)
- Number of electrons* ejected versus the *frequency* of the incident light (right)



**Question 3** **6 points****a. 2 points**

In the ground state of Ge (Germanium; $Z = 32$) atom, **name** the following orbitals:

i. Orbital containing the electron that requires the maximum energy to be removed.

ii. Orbital containing the electron that experiences the minimum effective nuclear charge.

b. 2 points

An unknown anion (A^-) has the following *excited* electronic configuration: $[Ar] 4s^2 3d^3 4p^1$

(Ar: Argon; $Z = 18$)

Give the **condensed ground state electronic configuration** of the atom, **A** and the cation, **A^{2+}** .
Show your work. (Assume all electronic transitions are possible)

Electronic Configuration of A:

Electronic Configuration of A^{2+} :

c. 2 points

In a single atom, determine how many electrons can have the following set of quantum numbers?

(i) $n = 5; l = 3$

In a single atom, determine how many electrons can have the following set of quantum numbers?

(i) $n = 3; l = 2; m_s = +1/2$

**Question 4** **8 points**

Rank the following from 1 to 4 in **increasing** order. Provide the ranking in the boxes provided.

Note: Atomic number (Z) are provided for relevant atoms

a) Increasing (lowest to highest) order of *ionization energy*: (1= lowest; 4 = highest)

Mg

Mg: Z = 12

Rb

Rb: Z = 37

Cs

Cs: Z = 55

K

K : Z = 19

b) Increasing (lowest to highest) order of *bond angle*: (1= lowest; 4 = highest)

H₂OCCl₄XeF₄

HCN

c) Increasing (smallest to largest) order of *ionic size* (1= smallest; 4 = largest)

S²⁻

S: Z = 16

O²⁻

O: Z = 8

Be²⁺

Be: Z = 4

Na⁺

Na: Z = 11

d) Increasing (lowest to highest) order of *bond energy* (1= lowest; 4 = highest)

P-F

P-Cl

P-I

P-Br

**Question 5** **12 points**

a. 9 points For the molecules HClO_4 and TeF_6 , draw the most stable *Lewis structure* showing appropriate shape and *dashed/wedged bonds*, based on VSEPR. Determine the **molecular geometry** around the central atom. For HClO_4 , label the *most polar bond*.



b. 3 points (Note: Lone pairs are not shown in structures below)

(i) For the three *resonance* structures (1, 2 and 3) shown below, answer the following question:



(I) Circle the *most contributing* resonance structure.

Structure 1

Structure 2

Structure 3

(ii) For the three structures given (4, 5, and 6) below, answer the following questions:



(I) Circle all the structures that represent *resonance structures* of each other:

Structure 4

Structure 5

Structure 6

(II) Among the resonance structures, circle the *most contributing* resonance structure.

Structure 4

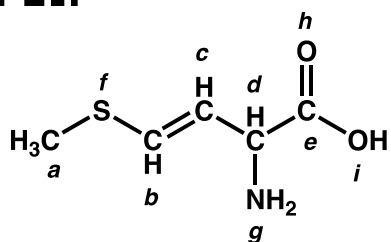
Structure 5

Structure 6

**Question 6** **6 points**

Using the data given below, calculate the **lattice energy** for SrO(s).

	ΔH° (kJ/mol)
$\text{Sr}(s) \longrightarrow \text{Sr}(g)$	+180
$\text{Sr}(g) \longrightarrow \text{Sr}^{2+}(g) + 2e^-$	+1468
$\text{O}_2(g) \longrightarrow 2\text{O}(g)$	+500
$\text{O}(g) + 2e^- \longrightarrow \text{O}^{2-}(g)$	+600
$\text{Sr}(s) + \frac{1}{2}\text{O}_2(g) \longrightarrow \text{SrO}(s)$	-550

**Question 7 10 points**

Consider the molecule given on the left, and answer questions (i) to (vi) using chemical bonding theories that we have learnt in class (valence bond theory, hybridization).

Note that the lone pairs are not shown in the structure:

i. Consider the carbon atoms marked in as ***a***, ***b***, ***c***, ***d***, and ***e*** in the molecule, and determine **hybridization** at each of the carbon centers. (2.5 points)

Atom	Hybridization
Carbon (<i>a</i>)	
Carbon (<i>b</i>)	
Carbon (<i>c</i>)	
Carbon (<i>d</i>)	
Carbon (<i>e</i>)	

ii. Now, consider the sulfur atom (S) marked as ***f***, nitrogen (N) atom (***g***), and oxygen(O) atoms (***h*** and ***i***), and determine their **hybridization**. (2 points)

Atom	Hybridization
Sulfur (<i>f</i>)	
Nitrogen (<i>g</i>)	
Oxygen (<i>h</i>)	
Oxygen (<i>i</i>)	

iii. **Describe** all the bonds formed by the carbon atoms ***a*** and ***b***, using sigma and pi designation as appropriate. (2 points)

Atom	Bonds formed (use sigma and pi as appropriate)
Carbon (<i>a</i>)	
Carbon (<i>b</i>)	

iv. How many total sp^3 -s sigma bonds are in the molecule? (1 point)

v. Determine **electronic** and **molecular geometry** at carbon (C) labeled ***e*** and sulfur (S) labeled ***f*** centres. (2 points)

Atom	Electronic Geometry	Molecular Geometry
Carbon (<i>e</i>)		
Sulfur (<i>f</i>)		

vi. Circle the value from the choices given below, that describes the **C-S-C bond angle at sulfur** (S) marked ***f***: (0.5 points)

>90° to <109.5°

109.5°

>109.5 to <120°

120°

>120° to <180°


Question 8 12 points

Using the **valence shell orbitals only** draw a **complete** molecular orbital (MO) diagram (with no s-p mixing) for the molecule **PS**, in the space provided below. **(6 points)**

Assume that 3s and 3p atomic orbitals will form molecular orbitals like those formed by the combination of 2s and 2p atomic orbitals that we learnt in class, and there is no s-p mixing.

Show **energy arrow**, **atomic orbitals** involved, **atom labels**, the **molecular orbitals** that are formed, **filled electrons** in the MO diagram etc. Make sure to label all the atomic and molecular orbitals.

Using the MO diagram above, answer the following questions:

- (i) Which one of the following will have the highest bond order? Circle the correct choice
 A) PS^{1-} B) PS C) PS^{1+}
- (ii) Which one of the following will have the longest bond length? Circle the correct choice
 A) PS^{1-} B) PS C) PS^{1+}
- (iii) Which of the following will be diamagnetic? Circle the correct choice.
 A) PS^{1-} B) PS C) PS^{1+}
- (iv) Draw Lewis dot structure of the molecule with the shortest bond length (based on the MO diagram) from among the three (PS^{1-} , PS, PS^{1+}), and indicate the bond order calculated from the Lewis dot structure. **(3 points)**

Lewis Dot Structure

Bond Order



E3A547DE-8CD4-4101-B271-E92B0995239C

midterm-exam-93388

#173 Page 10 of 10

SCRAP PAGE

**CHEMISTRY 110 –Midterm Exam****Thursday November 7, 2024, 6:30 – 8:30 pm**

Examiners: Pallavi Sirjoosingh and Ashok Kakkar

First name (please write as legibly as possible within the boxes)

--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--

Last name

--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--

Student ID number

--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--

This booklet has 8 questions on 10 pages (The last page can be used as a scrap page).

The staple must NOT be removed from the question booklet. Your intact question booklet must be returned at the end of the exam.

A separate booklet contains a periodic table, and datasheet.

Write your answers directly under the questions in the space provided. Do not write on the QR codes or near the top edges (as it may be cut off).

Please write **legibly and dark enough** for the scan to be readable. For students who use English as a second language, dictionaries are permitted. Calculators are allowed.

DO NOT ASK QUESTIONS OF THE INVIGILATORS about interpretations of the exam questions. If you have a concern about a question, please summarize it at the top of this page. All concerns will be evaluated after the exam. GOOD LUCK!

**Question 1 8 points** (*Assume all transitions are possible*)

a. 6 points Calculate the **energy (in kJ/mol)** required to ionize (completely remove) the electron from the **n=3** level of **He⁺ ion** (He: Helium). Show your work including equations. (3 sig. figs.)

b. 2 points

(I) Select the largest (longest) wavelength from the three choices below:

- i. Wavelength of photon to *completely remove* an electron from *1s orbital of Hydrogen atom*
- ii. Wavelength of photon to *completely remove* an electron from *1s orbital of Boron atom (B)*
- iii. Wavelength of photon to *completely remove* an electron from *1s orbital of Boron cation (B⁺)*

i or ii or iii

(II) Select the smallest (shortest) wavelength from the three choices below (*for H atom*):

- i. Wavelength of photon *emitted* for an electron transitioning from *2p to 1s orbital*
- ii. Wavelength of photon *absorbed* for an electron transitioning from *2s to 4p orbital*
- iii. Wavelength of photon *absorbed* for an electron transitioning from *1s to 3p orbital*

i or ii or iii



Question 2 8 points



A Chem110 student plots the graph shown on the left, from data obtained for the kinetic energy (eV) of the ejected electron versus frequency (s^{-1}) of the incident light, for an unknown metal. ($1\text{eV} = 1.6022 \times 10^{-19} \text{ J}$)

Based on the plot, answer the questions **a to c** below:

a. 1 point On the plot above, label the **threshold frequency**.

b. 2 points What is the value of the work potential (in J) of the unknown metal?

c. 3 points What is the *longest* wavelength (in nm) required to eject an electron from this metal's surface? Show your work including equation(s) used. (3 sig figs)

d. 2 points In the space provided below, plot the following

(Assume frequency is *above* threshold frequency):

- i. *Kinetic energy* of an ejected electron versus the *intensity* of the incident light (left)
- ii. *Number of electrons* ejected versus the *frequency* of the incident light (right)



**Question 3** **6 points****a. 2 points**

In the ground state of Ge (Germanium; $Z = 32$) atom, **name** the following orbitals:

i. Orbital containing the electron that requires the maximum energy to be removed.

ii. Orbital containing the electron that experiences the minimum effective nuclear charge.

b. 2 points

An unknown anion (A^-) has the following *excited* electronic configuration: $[Ar] 4s^2 3d^3 4p^1$

(Ar: Argon; $Z = 18$)

Give the **condensed ground state electronic configuration** of the atom, **A** and the cation, A^{2+} .
Show your work. (Assume all electronic transitions are possible)

Electronic Configuration of A:

Electronic Configuration of A^{2+} :

c. 2 points

In a single atom, determine how many electrons can have the following set of quantum numbers?

(i) $n = 5; l = 3$

In a single atom, determine how many electrons can have the following set of quantum numbers?

(i) $n = 3; l = 2; m_s = +1/2$

**Question 4** **8 points**

Rank the following from 1 to 4 in **increasing** order. Provide the ranking in the boxes provided.

Note: Atomic number (Z) are provided for relevant atoms

a) Increasing (lowest to highest) order of *ionization energy*: (1= lowest; 4 = highest)

Mg

Mg: Z = 12

Rb

Rb: Z = 37

Cs

Cs: Z = 55

K

K : Z = 19

b) Increasing (lowest to highest) order of *bond angle*: (1= lowest; 4 = highest)

H₂OCCl₄XeF₄

HCN

c) Increasing (smallest to largest) order of *ionic size* (1= smallest; 4 = largest)

S²⁻

S: Z = 16

O²⁻

O: Z = 8

Be²⁺

Be: Z = 4

Na⁺

Na: Z = 11

d) Increasing (lowest to highest) order of *bond energy* (1= lowest; 4 = highest)

P-F

P-Cl

P-I

P-Br

**Question 5** **12 points**

a. 9 points For the molecules HClO_4 and TeF_6 , draw the most stable *Lewis structure* showing appropriate shape and *dashed/wedged bonds*, based on VSEPR. Determine the **molecular geometry** around the central atom. For HClO_4 , label the *most polar bond*.



b. 3 points (Note: Lone pairs are not shown in structures below)

(i) For the three *resonance* structures (1, 2 and 3) shown below, answer the following question:



(I) Circle the *most contributing* resonance structure.

Structure 1

Structure 2

Structure 3

(ii) For the three structures given (4, 5, and 6) below, answer the following questions:



(I) Circle all the structures that represent *resonance structures* of each other:

Structure 4

Structure 5

Structure 6

(II) Among the resonance structures, circle the *most contributing* resonance structure.

Structure 4

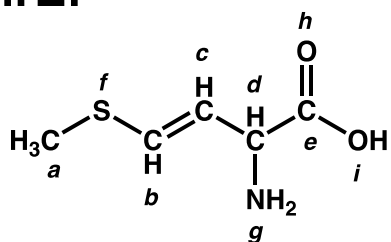
Structure 5

Structure 6

**Question 6** **6 points**

Using the data given below, calculate the **lattice energy** for SrO(s).

	ΔH° (kJ/mol)
$\text{Sr}(s) \longrightarrow \text{Sr}(g)$	+180
$\text{Sr}(g) \longrightarrow \text{Sr}^{2+}(g) + 2e^-$	+1468
$\text{O}_2(g) \longrightarrow 2\text{O}(g)$	+500
$\text{O}(g) + 2e^- \longrightarrow \text{O}^{2-}(g)$	+600
$\text{Sr}(s) + \frac{1}{2}\text{O}_2(g) \longrightarrow \text{SrO}(s)$	-550

**Question 7 10 points**

Consider the molecule given on the left, and answer questions (i) to (vi) using chemical bonding theories that we have learnt in class (valence bond theory, hybridization).

Note that the lone pairs are not shown in the structure:

i. Consider the carbon atoms marked in as ***a***, ***b***, ***c***, ***d***, and ***e*** in the molecule, and determine **hybridization** at each of the carbon centers. (2.5 points)

Atom	Hybridization
Carbon (<i>a</i>)	
Carbon (<i>b</i>)	
Carbon (<i>c</i>)	
Carbon (<i>d</i>)	
Carbon (<i>e</i>)	

ii. Now, consider the sulfur atom (S) marked as ***f***, nitrogen (N) atom (***g***), and oxygen(O) atoms (***h*** and ***i***), and determine their **hybridization**. (2 points)

Atom	Hybridization
Sulfur (<i>f</i>)	
Nitrogen (<i>g</i>)	
Oxygen (<i>h</i>)	
Oxygen (<i>i</i>)	

iii. **Describe** all the bonds formed by the carbon atoms ***a*** and ***b***, using sigma and pi designation as appropriate. (2 points)

Atom	Bonds formed (use sigma and pi as appropriate)
Carbon (<i>a</i>)	
Carbon (<i>b</i>)	

iv. How many total sp^3 -s sigma bonds are in the molecule? (1 point)

v. Determine **electronic** and **molecular geometry** at carbon (C) labeled ***e*** and sulfur (S) labeled ***f*** centres. (2 points)

Atom	Electronic Geometry	Molecular Geometry
Carbon (<i>e</i>)		
Sulfur (<i>f</i>)		

vi. Circle the value from the choices given below, that describes the **C-S-C bond angle at sulfur** (S) marked ***f***: (0.5 points)

>90° to <109.5°

109.5°

>109.5 to <120°

120°

>120° to <180°


Question 8 12 points

Using the **valence shell orbitals only** draw a **complete** molecular orbital (MO) diagram (with no s-p mixing) for the molecule **PS**, in the space provided below. **(6 points)**

Assume that 3s and 3p atomic orbitals will form molecular orbitals like those formed by the combination of 2s and 2p atomic orbitals that we learnt in class, and there is no s-p mixing.

Show **energy arrow**, **atomic orbitals** involved, **atom labels**, the **molecular orbitals** that are formed, **filled electrons** in the MO diagram etc. Make sure to label all the atomic and molecular orbitals.

Using the MO diagram above, answer the following questions:

- (i) Which one of the following will have the highest bond order? Circle the correct choice
 A) PS^{1-} B) PS C) PS^{1+}
- (ii) Which one of the following will have the longest bond length? Circle the correct choice
 A) PS^{1-} B) PS C) PS^{1+}
- (iii) Which of the following will be diamagnetic? Circle the correct choice.
 A) PS^{1-} B) PS C) PS^{1+}
- (iv) Draw Lewis dot structure of the molecule with the shortest bond length (based on the MO diagram) from among the three (PS^{1-} , PS, PS^{1+}), and indicate the bond order calculated from the Lewis dot structure. **(3 points)**

Lewis Dot Structure

Bond Order



82734B51-D0D2-4DF5-BE69-7CFBF9E74C6F

midterm-exam-93388

#174 Page 10 of 10

SCRAP PAGE

**CHEMISTRY 110 –Midterm Exam****Thursday November 7, 2024, 6:30 – 8:30 pm****Examiners: Pallavi Sirjoosingh and Ashok Kakkar**

First name (please write as legibly as possible within the boxes)

--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--

Last name

--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--

Student ID number

--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--

This booklet has 8 questions on 10 pages (The last page can be used as a scrap page).

The staple must NOT be removed from the question booklet. Your intact question booklet must be returned at the end of the exam.

A separate booklet contains a periodic table, and datasheet.

Write your answers directly under the questions in the space provided. Do not write on the QR codes or near the top edges (as it may be cut off).

Please write **legibly and dark enough** for the scan to be readable. For students who use English as a second language, dictionaries are permitted. Calculators are allowed.

DO NOT ASK QUESTIONS OF THE INVIGILATORS about interpretations of the exam questions. If you have a concern about a question, please summarize it at the top of this page. All concerns will be evaluated after the exam. GOOD LUCK!

**Question 1 8 points** (*Assume all transitions are possible*)

a. 6 points Calculate the **energy (in kJ/mol)** required to ionize (completely remove) the electron from the **n=3** level of **He⁺ ion** (He: Helium). Show your work including equations. (3 sig. figs.)

b. 2 points

(I) Select the largest (longest) wavelength from the three choices below:

- i. Wavelength of photon to *completely remove* an electron from *1s orbital of Hydrogen atom*
- ii. Wavelength of photon to *completely remove* an electron from *1s orbital of Boron atom (B)*
- iii. Wavelength of photon to *completely remove* an electron from *1s orbital of Boron cation (B⁺)*

i or ii or iii

(II) Select the smallest (shortest) wavelength from the three choices below (*for H atom*):

- i. Wavelength of photon *emitted* for an electron transitioning from *2p to 1s orbital*
- ii. Wavelength of photon *absorbed* for an electron transitioning from *2s to 4p orbital*
- iii. Wavelength of photon *absorbed* for an electron transitioning from *1s to 3p orbital*

i or ii or iii



Question 2 8 points



A Chem110 student plots the graph shown on the left, from data obtained for the kinetic energy (eV) of the ejected electron versus frequency (s^{-1}) of the incident light, for an unknown metal. ($1\text{eV} = 1.6022 \times 10^{-19} \text{ J}$)

Based on the plot, answer the questions **a to c** below:

a. 1 point On the plot above, label the **threshold frequency**.

b. 2 points What is the value of the work potential (in J) of the unknown metal?

c. 3 points What is the *longest* wavelength (in nm) required to eject an electron from this metal's surface? Show your work including equation(s) used. (3 sig figs)

d. 2 points In the space provided below, plot the following

(Assume frequency is *above* threshold frequency):

- Kinetic energy* of an ejected electron versus the *intensity* of the incident light (left)
- Number of electrons* ejected versus the *frequency* of the incident light (right)



**Question 3** **6 points****a. 2 points**

In the ground state of Ge (Germanium; $Z = 32$) atom, **name** the following orbitals:

i. Orbital containing the electron that requires the maximum energy to be removed.

ii. Orbital containing the electron that experiences the minimum effective nuclear charge.

b. 2 points

An unknown anion (A^-) has the following *excited* electronic configuration: $[Ar] 4s^2 3d^3 4p^1$

(Ar: Argon; $Z = 18$)

Give the **condensed ground state electronic configuration** of the atom, **A** and the cation, **A^{2+}** .
Show your work. (Assume all electronic transitions are possible)

Electronic Configuration of A:

Electronic Configuration of A^{2+} :

c. 2 points

In a single atom, determine how many electrons can have the following set of quantum numbers?

(i) $n = 5; l = 3$

In a single atom, determine how many electrons can have the following set of quantum numbers?

(i) $n = 3; l = 2; m_s = +1/2$

**Question 4** **8 points**

Rank the following from 1 to 4 in **increasing** order. Provide the ranking in the boxes provided.

Note: Atomic number (Z) are provided for relevant atoms

a) Increasing (lowest to highest) order of *ionization energy*: (1= lowest; 4 = highest)

Mg

Mg: Z = 12

Rb

Rb: Z = 37

Cs

Cs: Z = 55

K

K : Z = 19

b) Increasing (lowest to highest) order of *bond angle*: (1= lowest; 4 = highest)

H₂OCCl₄XeF₄

HCN

c) Increasing (smallest to largest) order of *ionic size* (1= smallest; 4 = largest)

S²⁻

S: Z = 16

O²⁻

O: Z = 8

Be²⁺

Be: Z = 4

Na⁺

Na: Z = 11

d) Increasing (lowest to highest) order of *bond energy* (1= lowest; 4 = highest)

P-F

P-Cl

P-I

P-Br

**Question 5** **12 points**

a. 9 points For the molecules HClO_4 and TeF_6 , draw the most stable *Lewis structure* showing appropriate shape and *dashed/wedged bonds*, based on VSEPR. Determine the **molecular geometry** around the central atom. For HClO_4 , label the *most polar bond*.



b. 3 points (Note: Lone pairs are not shown in structures below)

(i) For the three *resonance* structures (1, 2 and 3) shown below, answer the following question:



(I) Circle the *most contributing* resonance structure.

Structure 1

Structure 2

Structure 3

(ii) For the three structures given (4, 5, and 6) below, answer the following questions:



(I) Circle all the structures that represent *resonance structures* of each other:

Structure 4

Structure 5

Structure 6

(II) Among the resonance structures, circle the *most contributing* resonance structure.

Structure 4

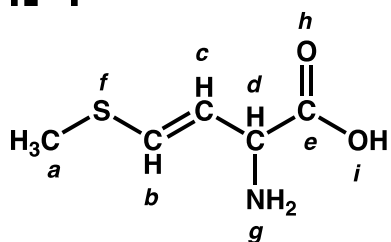
Structure 5

Structure 6

**Question 6** **6 points**

Using the data given below, calculate the **lattice energy** for SrO(s).

	ΔH° (kJ/mol)
$\text{Sr}(s) \longrightarrow \text{Sr}(g)$	+180
$\text{Sr}(g) \longrightarrow \text{Sr}^{2+}(g) + 2e^-$	+1468
$\text{O}_2(g) \longrightarrow 2\text{O}(g)$	+500
$\text{O}(g) + 2e^- \longrightarrow \text{O}^{2-}(g)$	+600
$\text{Sr}(s) + \frac{1}{2}\text{O}_2(g) \longrightarrow \text{SrO}(s)$	-550

**Question 7 10 points**

Consider the molecule given on the left, and answer questions (i) to (vi) using chemical bonding theories that we have learnt in class (valence bond theory, hybridization).

Note that the lone pairs are not shown in the structure:

i. Consider the carbon atoms marked in as ***a***, ***b***, ***c***, ***d***, and ***e*** in the molecule, and determine **hybridization** at each of the carbon centers. (2.5 points)

Atom	Hybridization
Carbon (<i>a</i>)	
Carbon (<i>b</i>)	
Carbon (<i>c</i>)	
Carbon (<i>d</i>)	
Carbon (<i>e</i>)	

ii. Now, consider the sulfur atom (S) marked as ***f***, nitrogen (N) atom (***g***), and oxygen(O) atoms (***h*** and ***i***), and determine their **hybridization**. (2 points)

Atom	Hybridization
Sulfur (<i>f</i>)	
Nitrogen (<i>g</i>)	
Oxygen (<i>h</i>)	
Oxygen (<i>i</i>)	

iii. **Describe** all the bonds formed by the carbon atoms ***a*** and ***b***, using sigma and pi designation as appropriate. (2 points)

Atom	Bonds formed (use sigma and pi as appropriate)
Carbon (<i>a</i>)	
Carbon (<i>b</i>)	

iv. How many total sp^3 -s sigma bonds are in the molecule? (1 point)

v. Determine **electronic** and **molecular geometry** at carbon (C) labeled ***e*** and sulfur (S) labeled ***f*** centres. (2 points)

Atom	Electronic Geometry	Molecular Geometry
Carbon (<i>e</i>)		
Sulfur (<i>f</i>)		

vi. Circle the value from the choices given below, that describes the **C-S-C bond angle at sulfur** (S) marked ***f***: (0.5 points)

>90° to <109.5° 109.5° >109.5 to <120° 120° >120° to <180°

**Question 8 12 points**

Using the **valence shell orbitals only** draw a **complete** molecular orbital (MO) diagram (with no s-p mixing) for the molecule **PS**, in the space provided below. **(6 points)**

Assume that 3s and 3p atomic orbitals will form molecular orbitals like those formed by the combination of 2s and 2p atomic orbitals that we learnt in class, and there is no s-p mixing.

Show **energy arrow**, **atomic orbitals** involved, **atom labels**, the **molecular orbitals** that are formed, **filled electrons** in the MO diagram etc. Make sure to label all the atomic and molecular orbitals.

Using the MO diagram above, answer the following questions:

- (i) Which one of the following will have the highest bond order? Circle the correct choice
A) PS^{1-} B) PS C) PS^{1+}
- (ii) Which one of the following will have the longest bond length? Circle the correct choice
A) PS^{1-} B) PS C) PS^{1+}
- (iii) Which of the following will be diamagnetic? Circle the correct choice.
A) PS^{1-} B) PS C) PS^{1+}
- (iv) Draw Lewis dot structure of the molecule with the shortest bond length (based on the MO diagram) from among the three (PS^{1-} , PS, PS^{1+}), and indicate the bond order calculated from the Lewis dot structure. **(3 points)**

Lewis Dot Structure

Bond Order



1FD3F0E8-A801-4B59-919C-24C660C689FC

midterm-exam-93388

#175 Page 10 of 10

SCRAP PAGE

**CHEMISTRY 110 –Midterm Exam****Thursday November 7, 2024, 6:30 – 8:30 pm**

Examiners: Pallavi Sirjoosingh and Ashok Kakkar

First name (please write as legibly as possible within the boxes)

--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--

Last name

--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--

Student ID number

--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--

This booklet has 8 questions on 10 pages (The last page can be used as a scrap page).

The staple must NOT be removed from the question booklet. Your intact question booklet must be returned at the end of the exam.

A separate booklet contains a periodic table, and datasheet.

Write your answers directly under the questions in the space provided. Do not write on the QR codes or near the top edges (as it may be cut off).

Please write **legibly and dark enough** for the scan to be readable. For students who use English as a second language, dictionaries are permitted. Calculators are allowed.

DO NOT ASK QUESTIONS OF THE INVIGILATORS about interpretations of the exam questions. If you have a concern about a question, please summarize it at the top of this page. All concerns will be evaluated after the exam. GOOD LUCK!

**Question 1 8 points** (*Assume all transitions are possible*)

a. 6 points Calculate the **energy (in kJ/mol)** required to ionize (completely remove) the electron from the **$n=3$** level of **He^+ ion** (He: Helium). Show your work including equations. (3 sig. figs.)

b. 2 points

(I) **Select** the **largest (longest)** wavelength from the three choices below:

- i. Wavelength of photon to *completely remove* an electron from *1s orbital of Hydrogen atom*
- ii. Wavelength of photon to *completely remove* an electron from *1s orbital of Boron atom (B)*
- iii. Wavelength of photon to *completely remove* an electron from *1s orbital of Boron cation (B^+)*

i **or** **ii** **or** **iii**

(II) **Select** the **smallest (shortest)** wavelength from the three choices below (*for H atom*):

- i. Wavelength of photon *emitted* for an electron transitioning from *2p to 1s orbital*
- ii. Wavelength of photon *absorbed* for an electron transitioning from *2s to 4p orbital*
- iii. Wavelength of photon *absorbed* for an electron transitioning from *1s to 3p orbital*

i **or** **ii** **or** **iii**



Question 2 8 points



A Chem110 student plots the graph shown on the left, from data obtained for the kinetic energy (eV) of the ejected electron versus frequency (s^{-1}) of the incident light, for an unknown metal. ($1\text{eV} = 1.6022 \times 10^{-19} \text{ J}$)

Based on the plot, answer the questions **a to c** below:

a. 1 point On the plot above, label the **threshold frequency**.

b. 2 points What is the value of the work potential (in J) of the unknown metal?

c. 3 points What is the *longest* wavelength (in nm) required to eject an electron from this metal's surface? Show your work including equation(s) used. (3 sig figs)

d. 2 points In the space provided below, plot the following

(Assume frequency is *above* threshold frequency):

- Kinetic energy* of an ejected electron versus the *intensity* of the incident light (left)
- Number of electrons* ejected versus the *frequency* of the incident light (right)



**Question 3** **6 points****a. 2 points**

In the ground state of Ge (Germanium; $Z = 32$) atom, **name** the following orbitals:

i. Orbital containing the electron that requires the maximum energy to be removed.

ii. Orbital containing the electron that experiences the minimum effective nuclear charge.

b. 2 points

An unknown anion (A^-) has the following *excited* electronic configuration: $[Ar] 4s^2 3d^3 4p^1$

(Ar: Argon; $Z = 18$)

Give the **condensed ground state electronic configuration** of the atom, **A** and the cation, **A^{2+}** .
Show your work. (Assume all electronic transitions are possible)

Electronic Configuration of A:

Electronic Configuration of A^{2+} :

c. 2 points

In a single atom, determine how many electrons can have the following set of quantum numbers?

(i) $n = 5; l = 3$

In a single atom, determine how many electrons can have the following set of quantum numbers?

(i) $n = 3; l = 2; m_s = +1/2$

**Question 4** **8 points**

Rank the following from 1 to 4 in **increasing** order. Provide the ranking in the boxes provided.

Note: Atomic number (Z) are provided for relevant atoms

a) Increasing (lowest to highest) order of *ionization energy*: (1= lowest; 4 = highest)

Mg

Mg: Z = 12

Rb

Rb: Z = 37

Cs

Cs: Z = 55

K

K : Z = 19

b) Increasing (lowest to highest) order of *bond angle*: (1= lowest; 4 = highest)

H₂OCCl₄XeF₄

HCN

c) Increasing (smallest to largest) order of *ionic size* (1= smallest; 4 = largest)

S²⁻

S: Z = 16

O²⁻

O: Z = 8

Be²⁺

Be: Z = 4

Na⁺

Na: Z = 11

d) Increasing (lowest to highest) order of *bond energy* (1= lowest; 4 = highest)

P-F

P-Cl

P-I

P-Br

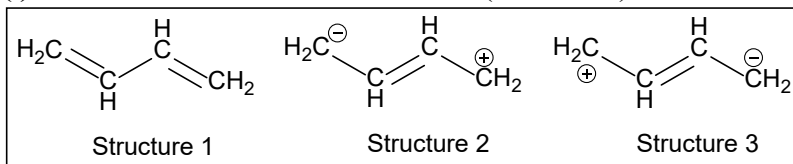
**Question 5** **12 points**

a. 9 points For the molecules HClO_4 and TeF_6 , draw the most stable *Lewis structure* showing appropriate shape and *dashed/wedged bonds*, based on VSEPR. Determine the **molecular geometry** around the central atom. For HClO_4 , label the *most polar bond*.



b. 3 points (Note: Lone pairs are not shown in structures below)

(i) For the three *resonance* structures (1, 2 and 3) shown below, answer the following question:



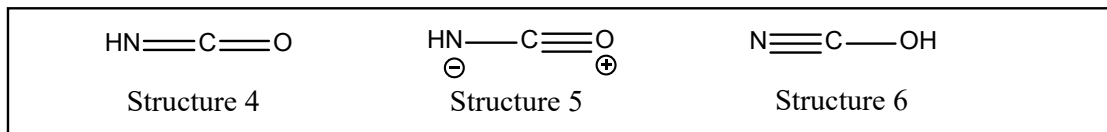
(I) Circle the *most contributing* resonance structure.

Structure 1

Structure 2

Structure 3

(ii) For the three structures given (4, 5, and 6) below, answer the following questions:



(I) Circle all the structures that represent *resonance structures* of each other:

Structure 4

Structure 5

Structure 6

(II) Among the resonance structures, circle the *most contributing* resonance structure.

Structure 4

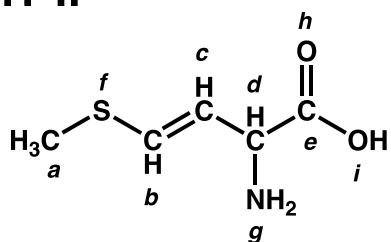
Structure 5

Structure 6

**Question 6** **6 points**

Using the data given below, calculate the **lattice energy** for SrO(s).

	ΔH° (kJ/mol)
$\text{Sr}(s) \longrightarrow \text{Sr}(g)$	+180
$\text{Sr}(g) \longrightarrow \text{Sr}^{2+}(g) + 2e^-$	+1468
$\text{O}_2(g) \longrightarrow 2\text{O}(g)$	+500
$\text{O}(g) + 2e^- \longrightarrow \text{O}^{2-}(g)$	+600
$\text{Sr}(s) + \frac{1}{2}\text{O}_2(g) \longrightarrow \text{SrO}(s)$	-550

**Question 7 10 points**

Consider the molecule given on the left, and answer questions (i) to (vi) using chemical bonding theories that we have learnt in class (valence bond theory, hybridization).

Note that the lone pairs are not shown in the structure:

i. Consider the carbon atoms marked in as ***a***, ***b***, ***c***, ***d***, and ***e*** in the molecule, and determine **hybridization** at each of the carbon centers. (2.5 points)

Atom	Hybridization
Carbon (<i>a</i>)	
Carbon (<i>b</i>)	
Carbon (<i>c</i>)	
Carbon (<i>d</i>)	
Carbon (<i>e</i>)	

ii. Now, consider the sulfur atom (S) marked as ***f***, nitrogen (N) atom (***g***), and oxygen(O) atoms (***h*** and ***i***), and determine their **hybridization**. (2 points)

Atom	Hybridization
Sulfur (<i>f</i>)	
Nitrogen (<i>g</i>)	
Oxygen (<i>h</i>)	
Oxygen (<i>i</i>)	

iii. **Describe** all the bonds formed by the carbon atoms ***a*** and ***b***, using sigma and pi designation as appropriate. (2 points)

Atom	Bonds formed (use sigma and pi as appropriate)
Carbon (<i>a</i>)	
Carbon (<i>b</i>)	

iv. How many total sp^3 -s sigma bonds are in the molecule? (1 point)

v. Determine **electronic** and **molecular geometry** at carbon (C) labeled ***e*** and sulfur (S) labeled ***f*** centres. (2 points)

Atom	Electronic Geometry	Molecular Geometry
Carbon (<i>e</i>)		
Sulfur (<i>f</i>)		

vi. Circle the value from the choices given below, that describes the **C-S-C bond angle at sulfur** (S) marked ***f***: (0.5 points)

>90° to <109.5° 109.5° >109.5 to <120° 120° >120° to <180°


Question 8 12 points

Using the **valence shell orbitals only** draw a **complete** molecular orbital (MO) diagram (with no s-p mixing) for the molecule **PS**, in the space provided below. **(6 points)**

Assume that 3s and 3p atomic orbitals will form molecular orbitals like those formed by the combination of 2s and 2p atomic orbitals that we learnt in class, and there is no s-p mixing.

Show **energy arrow**, **atomic orbitals** involved, **atom labels**, the **molecular orbitals** that are formed, **filled electrons** in the MO diagram etc. Make sure to label all the atomic and molecular orbitals.

Using the MO diagram above, answer the following questions:

- (i) Which one of the following will have the highest bond order? Circle the correct choice
 A) PS^{1-} B) PS C) PS^{1+}
- (ii) Which one of the following will have the longest bond length? Circle the correct choice
 A) PS^{1-} B) PS C) PS^{1+}
- (iii) Which of the following will be diamagnetic? Circle the correct choice.
 A) PS^{1-} B) PS C) PS^{1+}
- (iv) Draw Lewis dot structure of the molecule with the shortest bond length (based on the MO diagram) from among the three (PS^{1-} , PS, PS^{1+}), and indicate the bond order calculated from the Lewis dot structure. **(3 points)**

Lewis Dot Structure

Bond Order



F967DBEC-5431-4CC1-85A6-54AF85F60AC8

midterm-exam-93388

#176 Page 10 of 10

SCRAP PAGE

**CHEMISTRY 110 –Midterm Exam****Thursday November 7, 2024, 6:30 – 8:30 pm**

Examiners: Pallavi Sirjoosingh and Ashok Kakkar

First name (please write as legibly as possible within the boxes)

--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--

Last name

--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--

Student ID number

--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--

This booklet has 8 questions on 10 pages (The last page can be used as a scrap page).

The staple must NOT be removed from the question booklet. Your intact question booklet must be returned at the end of the exam.

A separate booklet contains a periodic table, and datasheet.

Write your answers directly under the questions in the space provided. Do not write on the QR codes or near the top edges (as it may be cut off).

Please write **legibly and dark enough** for the scan to be readable. For students who use English as a second language, dictionaries are permitted. Calculators are allowed.

DO NOT ASK QUESTIONS OF THE INVIGILATORS about interpretations of the exam questions. If you have a concern about a question, please summarize it at the top of this page. All concerns will be evaluated after the exam. GOOD LUCK!

**Question 1 8 points** (*Assume all transitions are possible*)

a. 6 points Calculate the **energy (in kJ/mol)** required to ionize (completely remove) the electron from the **$n=3$** level of **He^+ ion** (He: Helium). Show your work including equations. (3 sig. figs.)

b. 2 points

(I) Select the largest (longest) wavelength from the three choices below:

- i. Wavelength of photon to *completely remove* an electron from *1s orbital of Hydrogen atom*
- ii. Wavelength of photon to *completely remove* an electron from *1s orbital of Boron atom (B)*
- iii. Wavelength of photon to *completely remove* an electron from *1s orbital of Boron cation (B^+)*

i or ii or iii

(II) Select the smallest (shortest) wavelength from the three choices below (*for H atom*):

- i. Wavelength of photon *emitted* for an electron transitioning from *2p to 1s orbital*
- ii. Wavelength of photon *absorbed* for an electron transitioning from *2s to 4p orbital*
- iii. Wavelength of photon *absorbed* for an electron transitioning from *1s to 3p orbital*

i or ii or iii



Question 2 8 points



A Chem110 student plots the graph shown on the left, from data obtained for the kinetic energy (eV) of the ejected electron versus frequency (s^{-1}) of the incident light, for an unknown metal. ($1\text{eV} = 1.6022 \times 10^{-19} \text{ J}$)

Based on the plot, answer the questions **a to c** below:

a. 1 point On the plot above, label the **threshold frequency**.

b. 2 points What is the value of the work potential (in J) of the unknown metal?

c. 3 points What is the *longest* wavelength (in nm) required to eject an electron from this metal's surface? Show your work including equation(s) used. (3 sig figs)

d. 2 points In the space provided below, plot the following

(Assume frequency is *above* threshold frequency):

- i. *Kinetic energy* of an ejected electron versus the *intensity* of the incident light (left)
- ii. *Number of electrons* ejected versus the *frequency* of the incident light (right)



**Question 3** **6 points****a. 2 points**

In the ground state of Ge (Germanium; $Z = 32$) atom, **name** the following orbitals:

i. Orbital containing the electron that requires the maximum energy to be removed.

ii. Orbital containing the electron that experiences the minimum effective nuclear charge.

b. 2 points

An unknown anion (A^-) has the following *excited* electronic configuration: $[Ar] 4s^2 3d^3 4p^1$

(Ar: Argon; $Z = 18$)

Give the **condensed ground state electronic configuration** of the atom, **A** and the cation, **A^{2+}** .
Show your work. (Assume all electronic transitions are possible)

Electronic Configuration of A:

Electronic Configuration of A^{2+} :

c. 2 points

In a single atom, determine how many electrons can have the following set of quantum numbers?

(i) $n = 5; l = 3$

In a single atom, determine how many electrons can have the following set of quantum numbers?

(i) $n = 3; l = 2; m_s = +1/2$

**Question 4** **8 points**

Rank the following from 1 to 4 in **increasing** order. Provide the ranking in the boxes provided.

Note: Atomic number (Z) are provided for relevant atoms

a) Increasing (lowest to highest) order of *ionization energy*: (1= lowest; 4 = highest)

Mg

Mg: Z = 12

Rb

Rb: Z = 37

Cs

Cs: Z = 55

K

K : Z = 19

b) Increasing (lowest to highest) order of *bond angle*: (1= lowest; 4 = highest)

H₂OCCl₄XeF₄

HCN

c) Increasing (smallest to largest) order of *ionic size* (1= smallest; 4 = largest)

S²⁻

S: Z = 16

O²⁻

O: Z = 8

Be²⁺

Be: Z = 4

Na⁺

Na: Z = 11

d) Increasing (lowest to highest) order of *bond energy* (1= lowest; 4 = highest)

P-F

P-Cl

P-I

P-Br

**Question 5** **12 points**

a. 9 points For the molecules HClO_4 and TeF_6 , draw the most stable *Lewis structure* showing appropriate shape and *dashed/wedged bonds*, based on VSEPR. Determine the **molecular geometry** around the central atom. For HClO_4 , label the *most polar bond*.



b. 3 points (Note: Lone pairs are not shown in structures below)

(i) For the three *resonance* structures (1, 2 and 3) shown below, answer the following question:



(I) Circle the *most contributing* resonance structure.

Structure 1

Structure 2

Structure 3

(ii) For the three structures given (4, 5, and 6) below, answer the following questions:



(I) Circle all the structures that represent *resonance structures* of each other:

Structure 4

Structure 5

Structure 6

(II) Among the resonance structures, circle the *most contributing* resonance structure.

Structure 4

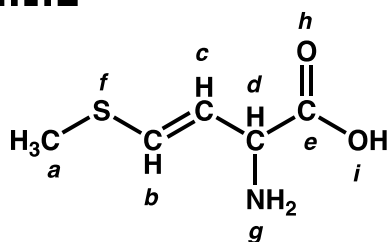
Structure 5

Structure 6

**Question 6** **6 points**

Using the data given below, calculate the **lattice energy** for SrO(s).

	ΔH° (kJ/mol)
$\text{Sr}(s) \longrightarrow \text{Sr}(g)$	+180
$\text{Sr}(g) \longrightarrow \text{Sr}^{2+}(g) + 2e^-$	+1468
$\text{O}_2(g) \longrightarrow 2\text{O}(g)$	+500
$\text{O}(g) + 2e^- \longrightarrow \text{O}^{2-}(g)$	+600
$\text{Sr}(s) + \frac{1}{2}\text{O}_2(g) \longrightarrow \text{SrO}(s)$	-550

**Question 7 10 points**

Consider the molecule given on the left, and answer questions (i) to (vi) using chemical bonding theories that we have learnt in class (valence bond theory, hybridization).

Note that the lone pairs are not shown in the structure:

i. Consider the carbon atoms marked in as ***a***, ***b***, ***c***, ***d***, and ***e*** in the molecule, and determine **hybridization** at each of the carbon centers. (2.5 points)

Atom	Hybridization
Carbon (<i>a</i>)	
Carbon (<i>b</i>)	
Carbon (<i>c</i>)	
Carbon (<i>d</i>)	
Carbon (<i>e</i>)	

ii. Now, consider the sulfur atom (S) marked as ***f***, nitrogen (N) atom (***g***), and oxygen(O) atoms (***h*** and ***i***), and determine their **hybridization**. (2 points)

Atom	Hybridization
Sulfur (<i>f</i>)	
Nitrogen (<i>g</i>)	
Oxygen (<i>h</i>)	
Oxygen (<i>i</i>)	

iii. **Describe** all the bonds formed by the carbon atoms ***a*** and ***b***, using sigma and pi designation as appropriate. (2 points)

Atom	Bonds formed (use sigma and pi as appropriate)
Carbon (<i>a</i>)	
Carbon (<i>b</i>)	

iv. How many total sp^3 -s sigma bonds are in the molecule? (1 point)

v. Determine **electronic** and **molecular geometry** at carbon (C) labeled ***e*** and sulfur (S) labeled ***f*** centres. (2 points)

Atom	Electronic Geometry	Molecular Geometry
Carbon (<i>e</i>)		
Sulfur (<i>f</i>)		

vi. Circle the value from the choices given below, that describes the **C-S-C bond angle at sulfur** (S) marked ***f***: (0.5 points)

>90° to <109.5° 109.5° >109.5 to <120° 120° >120° to <180°


Question 8 12 points

Using the **valence shell orbitals only** draw a **complete** molecular orbital (MO) diagram (with no s-p mixing) for the molecule **PS**, in the space provided below. **(6 points)**

Assume that 3s and 3p atomic orbitals will form molecular orbitals like those formed by the combination of 2s and 2p atomic orbitals that we learnt in class, and there is no s-p mixing.

Show **energy arrow**, **atomic orbitals** involved, **atom labels**, the **molecular orbitals** that are formed, **filled electrons** in the MO diagram etc. Make sure to **label all the atomic and molecular orbitals**.

Using the MO diagram above, answer the following questions:

- (i) Which one of the following will have the highest bond order? Circle the correct choice
 A) PS^{1-} B) PS C) PS^{1+}
- (ii) Which one of the following will have the longest bond length? Circle the correct choice
 A) PS^{1-} B) PS C) PS^{1+}
- (iii) Which of the following will be diamagnetic? Circle the correct choice.
 A) PS^{1-} B) PS C) PS^{1+}
- (iv) Draw Lewis dot structure of the molecule with the shortest bond length (based on the MO diagram) from among the three (PS^{1-} , PS, PS^{1+}), and indicate the bond order calculated from the Lewis dot structure. **(3 points)**

Lewis Dot Structure

Bond Order



2523AABE-1F78-4F8B-B382-A79C9F9EE36F

midterm-exam-93388

#177 Page 10 of 10

SCRAP PAGE

**CHEMISTRY 110 –Midterm Exam****Thursday November 7, 2024, 6:30 – 8:30 pm**

Examiners: Pallavi Sirjoosingh and Ashok Kakkar

First name (please write as legibly as possible within the boxes)

--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--

Last name

--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--

Student ID number

--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--

This booklet has 8 questions on 10 pages (The last page can be used as a scrap page).

The staple must NOT be removed from the question booklet. Your intact question booklet must be returned at the end of the exam.

A separate booklet contains a periodic table, and datasheet.

Write your answers directly under the questions in the space provided. Do not write on the QR codes or near the top edges (as it may be cut off).

Please write **legibly and dark enough** for the scan to be readable. For students who use English as a second language, dictionaries are permitted. Calculators are allowed.

DO NOT ASK QUESTIONS OF THE INVIGILATORS about interpretations of the exam questions. If you have a concern about a question, please summarize it at the top of this page. All concerns will be evaluated after the exam. GOOD LUCK!

**Question 1 8 points** (*Assume all transitions are possible*)

a. 6 points Calculate the **energy (in kJ/mol)** required to ionize (completely remove) the electron from the **n=3** level of **He⁺ ion** (He: Helium). Show your work including equations. (3 sig. figs.)

b. 2 points

(I) Select the largest (longest) wavelength from the three choices below:

- i. Wavelength of photon to *completely remove* an electron from *1s orbital of Hydrogen atom*
- ii. Wavelength of photon to *completely remove* an electron from *1s orbital of Boron atom (B)*
- iii. Wavelength of photon to *completely remove* an electron from *1s orbital of Boron cation (B⁺)*

i

or

ii

or

iii

(II) Select the smallest (shortest) wavelength from the three choices below (*for H atom*):

- i. Wavelength of photon *emitted* for an electron transitioning from *2p to 1s orbital*
- ii. Wavelength of photon *absorbed* for an electron transitioning from *2s to 4p orbital*
- iii. Wavelength of photon *absorbed* for an electron transitioning from *1s to 3p orbital*

i

or

ii

or

iii



Question 2 8 points



A Chem110 student plots the graph shown on the left, from data obtained for the kinetic energy (eV) of the ejected electron versus frequency (s^{-1}) of the incident light, for an unknown metal. ($1\text{eV} = 1.6022 \times 10^{-19} \text{ J}$)

Based on the plot, answer the questions **a to c** below:

a. 1 point On the plot above, label the **threshold frequency**.

b. 2 points What is the value of the work potential (in J) of the unknown metal?

c. 3 points What is the *longest* wavelength (in nm) required to eject an electron from this metal's surface? Show your work including equation(s) used. (3 sig figs)

d. 2 points In the space provided below, plot the following

(Assume frequency is *above* threshold frequency):

- Kinetic energy* of an ejected electron versus the *intensity* of the incident light (left)
- Number of electrons* ejected versus the *frequency* of the incident light (right)



**Question 3** **6 points****a. 2 points**

In the ground state of Ge (Germanium; $Z = 32$) atom, **name** the following orbitals:

i. Orbital containing the electron that requires the maximum energy to be removed.

ii. Orbital containing the electron that experiences the minimum effective nuclear charge.

b. 2 points

An unknown anion (A^-) has the following *excited* electronic configuration: $[Ar] 4s^2 3d^3 4p^1$

(Ar: Argon; $Z = 18$)

Give the **condensed ground state electronic configuration** of the atom, **A** and the cation, **A^{2+}** .
Show your work. (Assume all electronic transitions are possible)

Electronic Configuration of A:

Electronic Configuration of A^{2+} :

c. 2 points

In a single atom, determine how many electrons can have the following set of quantum numbers?

(i) $n = 5; l = 3$

In a single atom, determine how many electrons can have the following set of quantum numbers?

(i) $n = 3; l = 2; m_s = +1/2$

**Question 4** **8 points**

Rank the following from 1 to 4 in **increasing** order. Provide the ranking in the boxes provided.

Note: Atomic number (Z) are provided for relevant atoms

a) Increasing (lowest to highest) order of *ionization energy*: (1= lowest; 4 = highest)

Mg

Mg: Z = 12

Rb

Rb: Z = 37

Cs

Cs: Z = 55

K

K : Z = 19

b) Increasing (lowest to highest) order of *bond angle*: (1= lowest; 4 = highest)

H₂OCCl₄XeF₄

HCN

c) Increasing (smallest to largest) order of *ionic size* (1= smallest; 4 = largest)

S²⁻

S: Z = 16

O²⁻

O: Z = 8

Be²⁺

Be: Z = 4

Na⁺

Na: Z = 11

d) Increasing (lowest to highest) order of *bond energy* (1= lowest; 4 = highest)

P-F

P-Cl

P-I

P-Br

**Question 5** **12 points**

a. 9 points For the molecules HClO_4 and TeF_6 , draw the most stable *Lewis structure* showing appropriate shape and *dashed/wedged bonds*, based on VSEPR. Determine the **molecular geometry** around the central atom. For HClO_4 , label the *most polar bond*.



b. 3 points (Note: Lone pairs are not shown in structures below)

(i) For the three *resonance* structures (1, 2 and 3) shown below, answer the following question:



(I) Circle the *most contributing* resonance structure.

Structure 1

Structure 2

Structure 3

(ii) For the three structures given (4, 5, and 6) below, answer the following questions:



(I) Circle all the structures that represent *resonance structures* of each other:

Structure 4

Structure 5

Structure 6

(II) Among the resonance structures, circle the *most contributing* resonance structure.

Structure 4

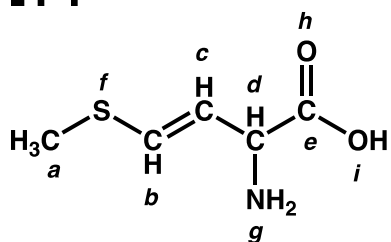
Structure 5

Structure 6

**Question 6** **6 points**

Using the data given below, calculate the **lattice energy** for SrO(s).

	ΔH° (kJ/mol)
$\text{Sr}(s) \longrightarrow \text{Sr}(g)$	+180
$\text{Sr}(g) \longrightarrow \text{Sr}^{2+}(g) + 2e^-$	+1468
$\text{O}_2(g) \longrightarrow 2\text{O}(g)$	+500
$\text{O}(g) + 2e^- \longrightarrow \text{O}^{2-}(g)$	+600
$\text{Sr}(s) + \frac{1}{2}\text{O}_2(g) \longrightarrow \text{SrO}(s)$	-550

**Question 7 10 points**

Consider the molecule given on the left, and answer questions (i) to (vi) using chemical bonding theories that we have learnt in class (valence bond theory, hybridization).

Note that the lone pairs are not shown in the structure:

i. Consider the carbon atoms marked in as ***a***, ***b***, ***c***, ***d***, and ***e*** in the molecule, and determine **hybridization** at each of the carbon centers. (2.5 points)

Atom	Hybridization
Carbon (<i>a</i>)	
Carbon (<i>b</i>)	
Carbon (<i>c</i>)	
Carbon (<i>d</i>)	
Carbon (<i>e</i>)	

ii. Now, consider the sulfur atom (S) marked as ***f***, nitrogen (N) atom (***g***), and oxygen(O) atoms (***h*** and ***i***), and determine their **hybridization**. (2 points)

Atom	Hybridization
Sulfur (<i>f</i>)	
Nitrogen (<i>g</i>)	
Oxygen (<i>h</i>)	
Oxygen (<i>i</i>)	

iii. **Describe** all the bonds formed by the carbon atoms ***a*** and ***b***, using sigma and pi designation as appropriate. (2 points)

Atom	Bonds formed (use sigma and pi as appropriate)
Carbon (<i>a</i>)	
Carbon (<i>b</i>)	

iv. How many total sp^3 -s sigma bonds are in the molecule? (1 point)

v. Determine **electronic** and **molecular geometry** at carbon (C) labeled ***e*** and sulfur (S) labeled ***f*** centres. (2 points)

Atom	Electronic Geometry	Molecular Geometry
Carbon (<i>e</i>)		
Sulfur (<i>f</i>)		

vi. Circle the value from the choices given below, that describes the **C-S-C bond angle at sulfur** (S) marked ***f***: (0.5 points)

>90° to <109.5° 109.5° >109.5 to <120° 120° >120° to <180°


Question 8 12 points

Using the **valence shell orbitals only** draw a **complete** molecular orbital (MO) diagram (with no s-p mixing) for the molecule **PS**, in the space provided below. **(6 points)**

Assume that 3s and 3p atomic orbitals will form molecular orbitals like those formed by the combination of 2s and 2p atomic orbitals that we learnt in class, and there is no s-p mixing.

Show **energy arrow**, **atomic orbitals** involved, **atom labels**, the **molecular orbitals** that are formed, **filled electrons** in the MO diagram etc. Make sure to label all the atomic and molecular orbitals.

Using the MO diagram above, answer the following questions:

- (i) Which one of the following will have the highest bond order? Circle the correct choice
 A) PS^{1-} B) PS C) PS^{1+}
- (ii) Which one of the following will have the longest bond length? Circle the correct choice
 A) PS^{1-} B) PS C) PS^{1+}
- (iii) Which of the following will be diamagnetic? Circle the correct choice.
 A) PS^{1-} B) PS C) PS^{1+}
- (iv) Draw Lewis dot structure of the molecule with the shortest bond length (based on the MO diagram) from among the three (PS^{1-} , PS, PS^{1+}), and indicate the bond order calculated from the Lewis dot structure. **(3 points)**

Lewis Dot Structure

Bond Order



AA468F3C-F1E9-4895-A5EA-C223231864DE

midterm-exam-93388

#178 Page 10 of 10

SCRAP PAGE

**CHEMISTRY 110 –Midterm Exam****Thursday November 7, 2024, 6:30 – 8:30 pm**

Examiners: Pallavi Sirjoosingh and Ashok Kakkar

First name (please write as legibly as possible within the boxes)

--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--

Last name

--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--

Student ID number

--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--

This booklet has 8 questions on 10 pages (The last page can be used as a scrap page).

The staple must NOT be removed from the question booklet. Your intact question booklet must be returned at the end of the exam.

A separate booklet contains a periodic table, and datasheet.

Write your answers directly under the questions in the space provided. Do not write on the QR codes or near the top edges (as it may be cut off).

Please write **legibly and dark enough** for the scan to be readable. For students who use English as a second language, dictionaries are permitted. Calculators are allowed.

DO NOT ASK QUESTIONS OF THE INVIGILATORS about interpretations of the exam questions. If you have a concern about a question, please summarize it at the top of this page. All concerns will be evaluated after the exam. GOOD LUCK!

**Question 1 8 points** (*Assume all transitions are possible*)

a. 6 points Calculate the **energy (in kJ/mol)** required to ionize (completely remove) the electron from the **$n=3$** level of **He^+ ion** (He: Helium). Show your work including equations. (3 sig. figs.)

b. 2 points

(I) Select the largest (longest) wavelength from the three choices below:

- i. Wavelength of photon to *completely remove* an electron from *1s orbital of Hydrogen atom*
- ii. Wavelength of photon to *completely remove* an electron from *1s orbital of Boron atom (B)*
- iii. Wavelength of photon to *completely remove* an electron from *1s orbital of Boron cation (B^+)*

i or ii or iii

(II) Select the smallest (shortest) wavelength from the three choices below (*for H atom*):

- i. Wavelength of photon *emitted* for an electron transitioning from *2p to 1s orbital*
- ii. Wavelength of photon *absorbed* for an electron transitioning from *2s to 4p orbital*
- iii. Wavelength of photon *absorbed* for an electron transitioning from *1s to 3p orbital*

i or ii or iii



Question 2 8 points



A Chem110 student plots the graph shown on the left, from data obtained for the kinetic energy (eV) of the ejected electron versus frequency (s^{-1}) of the incident light, for an unknown metal. ($1\text{eV} = 1.6022 \times 10^{-19} \text{ J}$)

Based on the plot, answer the questions **a to c** below:

a. 1 point On the plot above, label the **threshold frequency**.

b. 2 points What is the value of the work potential (in J) of the unknown metal?

c. 3 points What is the *longest* wavelength (in nm) required to eject an electron from this metal's surface? Show your work including equation(s) used. (3 sig figs)

d. 2 points In the space provided below, plot the following

(Assume frequency is *above* threshold frequency):

- Kinetic energy* of an ejected electron versus the *intensity* of the incident light (left)
- Number of electrons* ejected versus the *frequency* of the incident light (right)



**Question 3** **6 points****a. 2 points**

In the ground state of Ge (Germanium; $Z = 32$) atom, **name** the following orbitals:

i. Orbital containing the electron that requires the maximum energy to be removed.

ii. Orbital containing the electron that experiences the minimum effective nuclear charge.

b. 2 points

An unknown anion (A^-) has the following *excited* electronic configuration: $[Ar] 4s^2 3d^3 4p^1$

(Ar: Argon; $Z = 18$)

Give the **condensed ground state electronic configuration** of the atom, **A** and the cation, **A^{2+}** .
Show your work. (Assume all electronic transitions are possible)

Electronic Configuration of A:

Electronic Configuration of A^{2+} :

c. 2 points

In a single atom, determine how many electrons can have the following set of quantum numbers?

(i) $n = 5; l = 3$

In a single atom, determine how many electrons can have the following set of quantum numbers?

(i) $n = 3; l = 2; m_s = +1/2$

**Question 4** **8 points**

Rank the following from 1 to 4 in **increasing** order. Provide the ranking in the boxes provided.

Note: Atomic number (Z) are provided for relevant atoms

a) Increasing (lowest to highest) order of *ionization energy*: (1= lowest; 4 = highest)

Mg

Mg: Z = 12

Rb

Rb: Z = 37

Cs

Cs: Z = 55

K

K : Z = 19

b) Increasing (lowest to highest) order of *bond angle*: (1= lowest; 4 = highest)

H₂OCCl₄XeF₄

HCN

c) Increasing (smallest to largest) order of *ionic size* (1= smallest; 4 = largest)

S²⁻

S: Z = 16

O²⁻

O: Z = 8

Be²⁺

Be: Z = 4

Na⁺

Na: Z = 11

d) Increasing (lowest to highest) order of *bond energy* (1= lowest; 4 = highest)

P-F

P-Cl

P-I

P-Br

**Question 5** **12 points**

a. 9 points For the molecules HClO_4 and TeF_6 , draw the most stable *Lewis structure* showing appropriate shape and *dashed/wedged bonds*, based on VSEPR. Determine the **molecular geometry** around the central atom. For HClO_4 , label the *most polar bond*.

HClO_4 (there are no O-O bonds)

Label the most polar bond

Molecular Geometry around Cl

TeF_6

Molecular Geometry around Te

b. 3 points (Note: Lone pairs are not shown in structures below)

(i) For the three *resonance* structures (1, 2 and 3) shown below, answer the following question:



(I) Circle the *most contributing* resonance structure.

Structure 1

Structure 2

Structure 3

(ii) For the three structures given (4, 5, and 6) below, answer the following questions:



(I) Circle all the structures that represent *resonance structures* of each other:

Structure 4

Structure 5

Structure 6

(II) Among the resonance structures, circle the *most contributing* resonance structure.

Structure 4

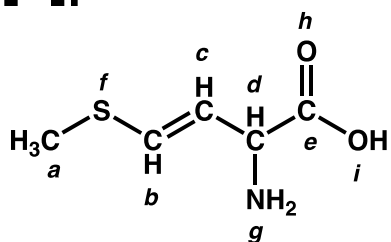
Structure 5

Structure 6

**Question 6** **6 points**

Using the data given below, calculate the **lattice energy** for SrO(s).

	ΔH° (kJ/mol)
$\text{Sr}(s) \longrightarrow \text{Sr}(g)$	+180
$\text{Sr}(g) \longrightarrow \text{Sr}^{2+}(g) + 2e^-$	+1468
$\text{O}_2(g) \longrightarrow 2\text{O}(g)$	+500
$\text{O}(g) + 2e^- \longrightarrow \text{O}^{2-}(g)$	+600
$\text{Sr}(s) + \frac{1}{2}\text{O}_2(g) \longrightarrow \text{SrO}(s)$	-550

**Question 7 10 points**

Consider the molecule given on the left, and answer questions (i) to (vi) using chemical bonding theories that we have learnt in class (valence bond theory, hybridization).

Note that the lone pairs are not shown in the structure:

i. Consider the carbon atoms marked in as ***a***, ***b***, ***c***, ***d***, and ***e*** in the molecule, and determine **hybridization** at each of the carbon centers. (2.5 points)

Atom	Hybridization
Carbon (<i>a</i>)	
Carbon (<i>b</i>)	
Carbon (<i>c</i>)	
Carbon (<i>d</i>)	
Carbon (<i>e</i>)	

ii. Now, consider the sulfur atom (S) marked as ***f***, nitrogen (N) atom (***g***), and oxygen(O) atoms (***h*** and ***i***), and determine their **hybridization**. (2 points)

Atom	Hybridization
Sulfur (<i>f</i>)	
Nitrogen (<i>g</i>)	
Oxygen (<i>h</i>)	
Oxygen (<i>i</i>)	

iii. **Describe** all the bonds formed by the carbon atoms ***a*** and ***b***, using sigma and pi designation as appropriate. (2 points)

Atom	Bonds formed (use sigma and pi as appropriate)
Carbon (<i>a</i>)	
Carbon (<i>b</i>)	

iv. How many total sp^3 -s sigma bonds are in the molecule? (1 point)

v. Determine **electronic** and **molecular geometry** at carbon (C) labeled ***e*** and sulfur (S) labeled ***f*** centres. (2 points)

Atom	Electronic Geometry	Molecular Geometry
Carbon (<i>e</i>)		
Sulfur (<i>f</i>)		

vi. Circle the value from the choices given below, that describes the **C-S-C bond angle at sulfur** (S) marked ***f***: (0.5 points)

>90° to <109.5° 109.5° >109.5 to <120° 120° >120° to <180°


Question 8 12 points

Using the **valence shell orbitals only** draw a **complete** molecular orbital (MO) diagram (with no s-p mixing) for the molecule **PS**, in the space provided below. **(6 points)**

Assume that 3s and 3p atomic orbitals will form molecular orbitals like those formed by the combination of 2s and 2p atomic orbitals that we learnt in class, and there is no s-p mixing.

Show **energy arrow**, **atomic orbitals** involved, **atom labels**, the **molecular orbitals** that are formed, **filled electrons** in the MO diagram etc. Make sure to label all the atomic and molecular orbitals.

Using the MO diagram above, answer the following questions:

- (i) Which one of the following will have the highest bond order? Circle the correct choice
 A) PS^{1-} B) PS C) PS^{1+}
- (ii) Which one of the following will have the longest bond length? Circle the correct choice
 A) PS^{1-} B) PS C) PS^{1+}
- (iii) Which of the following will be diamagnetic? Circle the correct choice.
 A) PS^{1-} B) PS C) PS^{1+}
- (iv) Draw Lewis dot structure of the molecule with the shortest bond length (based on the MO diagram) from among the three (PS^{1-} , PS, PS^{1+}), and indicate the bond order calculated from the Lewis dot structure. **(3 points)**

Lewis Dot Structure

Bond Order



67E8748F-C505-432B-868A-E2F9D121A1E1

midterm-exam-93388

#179 Page 10 of 10

SCRAP PAGE

**CHEMISTRY 110 –Midterm Exam****Thursday November 7, 2024, 6:30 – 8:30 pm**

Examiners: Pallavi Sirjoosingh and Ashok Kakkar

First name (please write as legibly as possible within the boxes)

--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--

Last name

--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--

Student ID number

--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--

This booklet has 8 questions on 10 pages (The last page can be used as a scrap page).

The staple must NOT be removed from the question booklet. Your intact question booklet must be returned at the end of the exam.

A separate booklet contains a periodic table, and datasheet.

Write your answers directly under the questions in the space provided. Do not write on the QR codes or near the top edges (as it may be cut off).

Please write **legibly and dark enough** for the scan to be readable. For students who use English as a second language, dictionaries are permitted. Calculators are allowed.

DO NOT ASK QUESTIONS OF THE INVIGILATORS about interpretations of the exam questions. If you have a concern about a question, please summarize it at the top of this page. All concerns will be evaluated after the exam. GOOD LUCK!

**Question 1 8 points** (*Assume all transitions are possible*)

a. 6 points Calculate the **energy (in kJ/mol)** required to ionize (completely remove) the electron from the **n=3** level of **He⁺ ion** (He: Helium). Show your work including equations. (3 sig. figs.)

b. 2 points

(I) Select the largest (longest) wavelength from the three choices below:

- i. Wavelength of photon to *completely remove* an electron from *1s orbital of Hydrogen atom*
- ii. Wavelength of photon to *completely remove* an electron from *1s orbital of Boron atom (B)*
- iii. Wavelength of photon to *completely remove* an electron from *1s orbital of Boron cation (B⁺)*

i or ii or iii

(II) Select the smallest (shortest) wavelength from the three choices below (*for H atom*):

- i. Wavelength of photon *emitted* for an electron transitioning from *2p to 1s orbital*
- ii. Wavelength of photon *absorbed* for an electron transitioning from *2s to 4p orbital*
- iii. Wavelength of photon *absorbed* for an electron transitioning from *1s to 3p orbital*

i or ii or iii



Question 2 8 points



A Chem110 student plots the graph shown on the left, from data obtained for the kinetic energy (eV) of the ejected electron versus frequency (s^{-1}) of the incident light, for an unknown metal. ($1\text{eV} = 1.6022 \times 10^{-19} \text{ J}$)

Based on the plot, answer the questions **a to c** below:

a. 1 point On the plot above, label the **threshold frequency**.

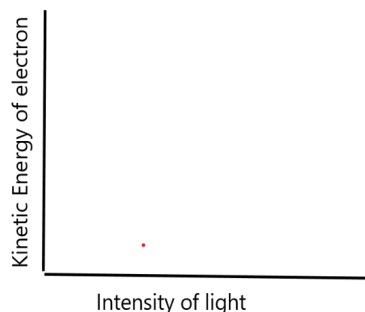
b. 2 points What is the value of the work potential (in J) of the unknown metal?

c. 3 points What is the *longest* wavelength (in nm) required to eject an electron from this metal's surface? Show your work including equation(s) used. (3 sig figs)

d. 2 points In the space provided below, plot the following

(Assume frequency is *above* threshold frequency):

- Kinetic energy* of an ejected electron versus the *intensity* of the incident light (left)
- Number of electrons* ejected versus the *frequency* of the incident light (right)



**Question 3** **6 points****a. 2 points**

In the ground state of Ge (Germanium; $Z = 32$) atom, **name** the following orbitals:

i. Orbital containing the electron that requires the maximum energy to be removed.

ii. Orbital containing the electron that experiences the minimum effective nuclear charge.

b. 2 points

An unknown anion (A^-) has the following *excited* electronic configuration: $[Ar] 4s^2 3d^3 4p^1$

(Ar: Argon; $Z = 18$)

Give the **condensed ground state electronic configuration** of the atom, **A** and the cation, **A^{2+}** .
Show your work. (Assume all electronic transitions are possible)

Electronic Configuration of A:

Electronic Configuration of A^{2+} :

c. 2 points

In a single atom, determine how many electrons can have the following set of quantum numbers?

(i) $n = 5; l = 3$

In a single atom, determine how many electrons can have the following set of quantum numbers?

(i) $n = 3; l = 2; m_s = +1/2$

**Question 4** **8 points**

Rank the following from 1 to 4 in **increasing** order. Provide the ranking in the boxes provided.

Note: Atomic number (Z) are provided for relevant atoms

a) Increasing (lowest to highest) order of *ionization energy*: (1= lowest; 4 = highest)

Mg

Mg: Z = 12

Rb

Rb: Z = 37

Cs

Cs: Z = 55

K

K : Z = 19

b) Increasing (lowest to highest) order of *bond angle*: (1= lowest; 4 = highest)

H₂OCCl₄XeF₄

HCN

c) Increasing (smallest to largest) order of *ionic size* (1= smallest; 4 = largest)

S²⁻

S: Z = 16

O²⁻

O: Z = 8

Be²⁺

Be: Z = 4

Na⁺

Na: Z = 11

d) Increasing (lowest to highest) order of *bond energy* (1= lowest; 4 = highest)

P-F

P-Cl

P-I

P-Br

**Question 5** **12 points**

a. 9 points For the molecules HClO_4 and TeF_6 , draw the most stable *Lewis structure* showing appropriate shape and *dashed/wedged bonds*, based on VSEPR. Determine the **molecular geometry** around the central atom. For HClO_4 , label the *most polar bond*.

HClO_4 (there are no O-O bonds)

Label the most polar bond

Molecular Geometry around Cl

TeF_6

Molecular Geometry around Te

b. 3 points (Note: Lone pairs are not shown in structures below)

(i) For the three *resonance* structures (1, 2 and 3) shown below, answer the following question:



(I) Circle the *most contributing* resonance structure.

Structure 1

Structure 2

Structure 3

(ii) For the three structures given (4, 5, and 6) below, answer the following questions:



(I) Circle all the structures that represent *resonance structures* of each other:

Structure 4

Structure 5

Structure 6

(II) Among the resonance structures, circle the *most contributing* resonance structure.

Structure 4

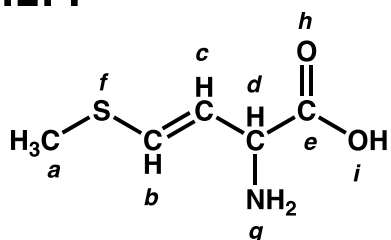
Structure 5

Structure 6

**Question 6** **6 points**

Using the data given below, calculate the **lattice energy** for SrO(s).

	ΔH° (kJ/mol)
$\text{Sr}(s) \longrightarrow \text{Sr}(g)$	+180
$\text{Sr}(g) \longrightarrow \text{Sr}^{2+}(g) + 2e^-$	+1468
$\text{O}_2(g) \longrightarrow 2\text{O}(g)$	+500
$\text{O}(g) + 2e^- \longrightarrow \text{O}^{2-}(g)$	+600
$\text{Sr}(s) + \frac{1}{2}\text{O}_2(g) \longrightarrow \text{SrO}(s)$	-550

**Question 7 10 points**

Consider the molecule given on the left, and answer questions (i) to (vi) using chemical bonding theories that we have learnt in class (valence bond theory, hybridization).

Note that the lone pairs are not shown in the structure:

i. Consider the carbon atoms marked in as ***a***, ***b***, ***c***, ***d***, and ***e*** in the molecule, and determine **hybridization** at each of the carbon centers. (2.5 points)

Atom	Hybridization
Carbon (<i>a</i>)	
Carbon (<i>b</i>)	
Carbon (<i>c</i>)	
Carbon (<i>d</i>)	
Carbon (<i>e</i>)	

ii. Now, consider the sulfur atom (S) marked as ***f***, nitrogen (N) atom (***g***), and oxygen(O) atoms (***h*** and ***i***), and determine their **hybridization**. (2 points)

Atom	Hybridization
Sulfur (<i>f</i>)	
Nitrogen (<i>g</i>)	
Oxygen (<i>h</i>)	
Oxygen (<i>i</i>)	

iii. **Describe** all the bonds formed by the carbon atoms ***a*** and ***b***, using sigma and pi designation as appropriate. (2 points)

Atom	Bonds formed (use sigma and pi as appropriate)
Carbon (<i>a</i>)	
Carbon (<i>b</i>)	

iv. How many total sp^3 -s sigma bonds are in the molecule? (1 point)

v. Determine **electronic** and **molecular geometry** at carbon (C) labeled ***e*** and sulfur (S) labeled ***f*** centres. (2 points)

Atom	Electronic Geometry	Molecular Geometry
Carbon (<i>e</i>)		
Sulfur (<i>f</i>)		

vi. Circle the value from the choices given below, that describes the **C-S-C bond angle at sulfur** (S) marked ***f***: (0.5 points)

>90° to <109.5° 109.5° >109.5 to <120° 120° >120° to <180°


Question 8 12 points

Using the **valence shell orbitals only** draw a **complete** molecular orbital (MO) diagram (with no s-p mixing) for the molecule **PS**, in the space provided below. **(6 points)**

Assume that 3s and 3p atomic orbitals will form molecular orbitals like those formed by the combination of 2s and 2p atomic orbitals that we learnt in class, and there is no s-p mixing.

Show **energy arrow**, **atomic orbitals** involved, **atom labels**, the **molecular orbitals** that are formed, **filled electrons** in the MO diagram etc. Make sure to label all the atomic and molecular orbitals.

Using the MO diagram above, answer the following questions:

- (i) Which one of the following will have the highest bond order? Circle the correct choice
 A) PS^{1-} B) PS C) PS^{1+}
- (ii) Which one of the following will have the longest bond length? Circle the correct choice
 A) PS^{1-} B) PS C) PS^{1+}
- (iii) Which of the following will be diamagnetic? Circle the correct choice.
 A) PS^{1-} B) PS C) PS^{1+}
- (iv) Draw Lewis dot structure of the molecule with the shortest bond length (based on the MO diagram) from among the three (PS^{1-} , PS, PS^{1+}), and indicate the bond order calculated from the Lewis dot structure. **(3 points)**

Lewis Dot Structure

Bond Order



50D52C05-FBA5-4656-A9D4-E30278FC5AA6

midterm-exam-93388

#180 Page 10 of 10

SCRAP PAGE

**CHEMISTRY 110 –Midterm Exam****Thursday November 7, 2024, 6:30 – 8:30 pm****Examiners: Pallavi Sirjoosingh and Ashok Kakkar**

First name (please write as legibly as possible within the boxes)

--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--

Last name

--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--

Student ID number

--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--

This booklet has 8 questions on 10 pages (The last page can be used as a scrap page).

The staple must NOT be removed from the question booklet. Your intact question booklet must be returned at the end of the exam.

A separate booklet contains a periodic table, and datasheet.

Write your answers directly under the questions in the space provided. Do not write on the QR codes or near the top edges (as it may be cut off).

Please write **legibly and dark enough** for the scan to be readable. For students who use English as a second language, dictionaries are permitted. Calculators are allowed.

DO NOT ASK QUESTIONS OF THE INVIGILATORS about interpretations of the exam questions. If you have a concern about a question, please summarize it at the top of this page. All concerns will be evaluated after the exam. GOOD LUCK!

**Question 1 8 points** (*Assume all transitions are possible*)

a. 6 points Calculate the **energy (in kJ/mol)** required to ionize (completely remove) the electron from the **n=3** level of **He⁺ ion** (He: Helium). Show your work including equations. (3 sig. figs.)

b. 2 points

(I) Select the largest (longest) wavelength from the three choices below:

- i. Wavelength of photon to *completely remove* an electron from *1s orbital of Hydrogen atom*
- ii. Wavelength of photon to *completely remove* an electron from *1s orbital of Boron atom (B)*
- iii. Wavelength of photon to *completely remove* an electron from *1s orbital of Boron cation (B⁺)*

i or ii or iii

(II) Select the smallest (shortest) wavelength from the three choices below (*for H atom*):

- i. Wavelength of photon *emitted* for an electron transitioning from *2p to 1s orbital*
- ii. Wavelength of photon *absorbed* for an electron transitioning from *2s to 4p orbital*
- iii. Wavelength of photon *absorbed* for an electron transitioning from *1s to 3p orbital*

i or ii or iii



Question 2 8 points



A Chem110 student plots the graph shown on the left, from data obtained for the kinetic energy (eV) of the ejected electron versus frequency (s^{-1}) of the incident light, for an unknown metal. ($1\text{eV} = 1.6022 \times 10^{-19} \text{ J}$)

Based on the plot, answer the questions **a to c** below:

a. 1 point On the plot above, label the **threshold frequency**.

b. 2 points What is the value of the work potential (in J) of the unknown metal?

c. 3 points What is the *longest* wavelength (in nm) required to eject an electron from this metal's surface? Show your work including equation(s) used. (3 sig figs)

d. 2 points In the space provided below, plot the following

(Assume frequency is *above* threshold frequency):

- Kinetic energy* of an ejected electron versus the *intensity* of the incident light (left)
- Number of electrons* ejected versus the *frequency* of the incident light (right)



**Question 3** **6 points****a. 2 points**

In the ground state of Ge (Germanium; $Z = 32$) atom, **name** the following orbitals:

i. Orbital containing the electron that requires the maximum energy to be removed.

ii. Orbital containing the electron that experiences the minimum effective nuclear charge.

b. 2 points

An unknown anion (A^-) has the following *excited* electronic configuration: $[Ar] 4s^2 3d^3 4p^1$

(Ar: Argon; $Z = 18$)

Give the **condensed ground state electronic configuration** of the atom, **A** and the cation, A^{2+} .
Show your work. (Assume all electronic transitions are possible)

Electronic Configuration of A:

Electronic Configuration of A^{2+} :

c. 2 points

In a single atom, determine how many electrons can have the following set of quantum numbers?

(i) $n = 5; l = 3$

In a single atom, determine how many electrons can have the following set of quantum numbers?

(i) $n = 3; l = 2; m_s = +1/2$

**Question 4** **8 points**

Rank the following from 1 to 4 in **increasing** order. Provide the ranking in the boxes provided.

Note: Atomic number (Z) are provided for relevant atoms

a) Increasing (lowest to highest) order of *ionization energy*: (1= lowest; 4 = highest)

Mg

Mg: Z = 12

Rb

Rb: Z = 37

Cs

Cs: Z = 55

K

K : Z = 19

b) Increasing (lowest to highest) order of *bond angle*: (1= lowest; 4 = highest)

H₂OCCl₄XeF₄

HCN

c) Increasing (smallest to largest) order of *ionic size* (1= smallest; 4 = largest)

S²⁻

S: Z = 16

O²⁻

O: Z = 8

Be²⁺

Be: Z = 4

Na⁺

Na: Z = 11

d) Increasing (lowest to highest) order of *bond energy* (1= lowest; 4 = highest)

P-F

P-Cl

P-I

P-Br

**Question 5** **12 points**

a. 9 points For the molecules HClO_4 and TeF_6 , draw the most stable *Lewis structure* showing appropriate shape and *dashed/wedged bonds*, based on VSEPR. Determine the **molecular geometry** around the central atom. For HClO_4 , label the *most polar bond*.

HClO_4 (there are no O-O bonds)

Label the most polar bond

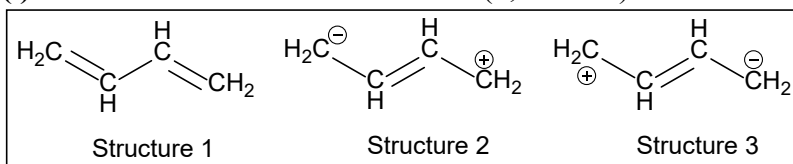
Molecular Geometry around Cl

TeF_6

Molecular Geometry around Te

b. 3 points (Note: Lone pairs are not shown in structures below)

(i) For the three *resonance* structures (1, 2 and 3) shown below, answer the following question:



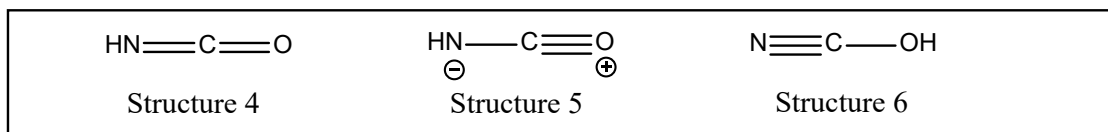
(I) Circle the *most contributing* resonance structure.

Structure 1

Structure 2

Structure 3

(ii) For the three structures given (4, 5, and 6) below, answer the following questions:



(I) Circle all the structures that represent *resonance structures* of each other:

Structure 4

Structure 5

Structure 6

(II) Among the resonance structures, circle the *most contributing* resonance structure.

Structure 4

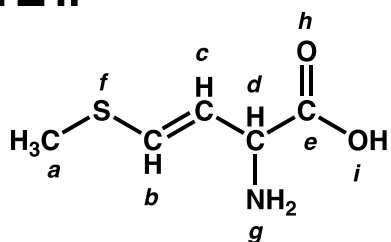
Structure 5

Structure 6

**Question 6** **6 points**

Using the data given below, calculate the **lattice energy** for SrO(s).

	ΔH° (kJ/mol)
$\text{Sr}(s) \longrightarrow \text{Sr}(g)$	+180
$\text{Sr}(g) \longrightarrow \text{Sr}^{2+}(g) + 2e^-$	+1468
$\text{O}_2(g) \longrightarrow 2\text{O}(g)$	+500
$\text{O}(g) + 2e^- \longrightarrow \text{O}^{2-}(g)$	+600
$\text{Sr}(s) + \frac{1}{2}\text{O}_2(g) \longrightarrow \text{SrO}(s)$	-550

**Question 7 10 points**

Consider the molecule given on the left, and answer questions (i) to (vi) using chemical bonding theories that we have learnt in class (valence bond theory, hybridization).

Note that the lone pairs are not shown in the structure:

i. Consider the carbon atoms marked in as ***a***, ***b***, ***c***, ***d***, and ***e*** in the molecule, and determine **hybridization** at each of the carbon centers. (2.5 points)

Atom	Hybridization
Carbon (<i>a</i>)	
Carbon (<i>b</i>)	
Carbon (<i>c</i>)	
Carbon (<i>d</i>)	
Carbon (<i>e</i>)	

ii. Now, consider the sulfur atom (S) marked as ***f***, nitrogen (N) atom (***g***), and oxygen(O) atoms (***h*** and ***i***), and determine their **hybridization**. (2 points)

Atom	Hybridization
Sulfur (<i>f</i>)	
Nitrogen (<i>g</i>)	
Oxygen (<i>h</i>)	
Oxygen (<i>i</i>)	

iii. **Describe** all the bonds formed by the carbon atoms ***a*** and ***b***, using sigma and pi designation as appropriate. (2 points)

Atom	Bonds formed (use sigma and pi as appropriate)
Carbon (<i>a</i>)	
Carbon (<i>b</i>)	

iv. How many total sp^3 -s sigma bonds are in the molecule? (1 point)

v. Determine **electronic** and **molecular geometry** at carbon (C) labeled ***e*** and sulfur (S) labeled ***f*** centres. (2 points)

Atom	Electronic Geometry	Molecular Geometry
Carbon (<i>e</i>)		
Sulfur (<i>f</i>)		

vi. Circle the value from the choices given below, that describes the **C-S-C bond angle at sulfur** (S) marked ***f***: (0.5 points)

>90° to <109.5° 109.5° >109.5 to <120° 120° >120° to <180°


Question 8 12 points

Using the **valence shell orbitals only** draw a **complete** molecular orbital (MO) diagram (with no s-p mixing) for the molecule **PS**, in the space provided below. **(6 points)**

Assume that 3s and 3p atomic orbitals will form molecular orbitals like those formed by the combination of 2s and 2p atomic orbitals that we learnt in class, and there is no s-p mixing.

Show **energy arrow**, **atomic orbitals** involved, **atom labels**, the **molecular orbitals** that are formed, **filled electrons** in the MO diagram etc. Make sure to **label all the atomic and molecular orbitals**.

Using the MO diagram above, answer the following questions:

- (i) Which one of the following will have the highest bond order? Circle the correct choice
 A) PS^{1-} B) PS C) PS^{1+}
- (ii) Which one of the following will have the longest bond length? Circle the correct choice
 A) PS^{1-} B) PS C) PS^{1+}
- (iii) Which of the following will be diamagnetic? Circle the correct choice.
 A) PS^{1-} B) PS C) PS^{1+}
- (iv) Draw Lewis dot structure of the molecule with the shortest bond length (based on the MO diagram) from among the three (PS^{1-} , PS, PS^{1+}), and indicate the bond order calculated from the Lewis dot structure. **(3 points)**

Lewis Dot Structure

Bond Order



449831B8-025A-4AC1-B155-000C67D63EB9

midterm-exam-93388

#181 Page 10 of 10

SCRAP PAGE

**CHEMISTRY 110 –Midterm Exam****Thursday November 7, 2024, 6:30 – 8:30 pm**

Examiners: Pallavi Sirjoosingh and Ashok Kakkar

First name (please write as legibly as possible within the boxes)

--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--

Last name

--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--

Student ID number

--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--

This booklet has 8 questions on 10 pages (The last page can be used as a scrap page).

The staple must NOT be removed from the question booklet. Your intact question booklet must be returned at the end of the exam.

A separate booklet contains a periodic table, and datasheet.

Write your answers directly under the questions in the space provided. Do not write on the QR codes or near the top edges (as it may be cut off).

Please write **legibly and dark enough** for the scan to be readable. For students who use English as a second language, dictionaries are permitted. Calculators are allowed.

DO NOT ASK QUESTIONS OF THE INVIGILATORS about interpretations of the exam questions. If you have a concern about a question, please summarize it at the top of this page. All concerns will be evaluated after the exam. GOOD LUCK!

**Question 1 8 points** (*Assume all transitions are possible*)

a. 6 points Calculate the **energy (in kJ/mol)** required to ionize (completely remove) the electron from the **$n=3$** level of **He^+ ion** (He: Helium). Show your work including equations. (3 sig. figs.)

b. 2 points

(I) Select the largest (longest) wavelength from the three choices below:

- i. Wavelength of photon to *completely remove* an electron from *1s orbital of Hydrogen atom*
- ii. Wavelength of photon to *completely remove* an electron from *1s orbital of Boron atom (B)*
- iii. Wavelength of photon to *completely remove* an electron from *1s orbital of Boron cation (B^+)*

i or ii or iii

(II) Select the smallest (shortest) wavelength from the three choices below (*for H atom*):

- i. Wavelength of photon *emitted* for an electron transitioning from *2p to 1s orbital*
- ii. Wavelength of photon *absorbed* for an electron transitioning from *2s to 4p orbital*
- iii. Wavelength of photon *absorbed* for an electron transitioning from *1s to 3p orbital*

i or ii or iii



Question 2 8 points



A Chem110 student plots the graph shown on the left, from data obtained for the kinetic energy (eV) of the ejected electron versus frequency (s^{-1}) of the incident light, for an unknown metal. ($1\text{eV} = 1.6022 \times 10^{-19} \text{ J}$)

Based on the plot, answer the questions **a to c** below:

a. 1 point On the plot above, label the **threshold frequency**.

b. 2 points What is the value of the work potential (in J) of the unknown metal?

c. 3 points What is the *longest* wavelength (in nm) required to eject an electron from this metal's surface? Show your work including equation(s) used. (3 sig figs)

d. 2 points In the space provided below, plot the following

(Assume frequency is *above* threshold frequency):

- Kinetic energy* of an ejected electron versus the *intensity* of the incident light (left)
- Number of electrons* ejected versus the *frequency* of the incident light (right)



**Question 3** **6 points****a. 2 points**

In the ground state of Ge (Germanium; $Z = 32$) atom, **name** the following orbitals:

i. Orbital containing the electron that requires the maximum energy to be removed.

ii. Orbital containing the electron that experiences the minimum effective nuclear charge.

b. 2 points

An unknown anion (A^-) has the following *excited* electronic configuration: $[Ar] 4s^2 3d^3 4p^1$

(Ar: Argon; $Z = 18$)

Give the **condensed ground state electronic configuration** of the atom, **A** and the cation, **A^{2+}** .
Show your work. (Assume all electronic transitions are possible)

Electronic Configuration of A:

Electronic Configuration of A^{2+} :

c. 2 points

In a single atom, determine how many electrons can have the following set of quantum numbers?

(i) $n = 5; l = 3$

In a single atom, determine how many electrons can have the following set of quantum numbers?

(i) $n = 3; l = 2; m_s = +1/2$

**Question 4** **8 points**

Rank the following from 1 to 4 in **increasing** order. Provide the ranking in the boxes provided.

Note: Atomic number (Z) are provided for relevant atoms

a) Increasing (lowest to highest) order of *ionization energy*: (1= lowest; 4 = highest)

Mg

Mg: Z = 12

Rb

Rb: Z = 37

Cs

Cs: Z = 55

K

K : Z = 19

b) Increasing (lowest to highest) order of *bond angle*: (1= lowest; 4 = highest)

H₂OCCl₄XeF₄

HCN

c) Increasing (smallest to largest) order of *ionic size* (1= smallest; 4 = largest)

S²⁻

S: Z = 16

O²⁻

O: Z = 8

Be²⁺

Be: Z = 4

Na⁺

Na: Z = 11

d) Increasing (lowest to highest) order of *bond energy* (1= lowest; 4 = highest)

P-F

P-Cl

P-I

P-Br

**Question 5** **12 points**

a. 9 points For the molecules HClO_4 and TeF_6 , draw the most stable *Lewis structure* showing appropriate shape and *dashed/wedged bonds*, based on VSEPR. Determine the **molecular geometry** around the central atom. For HClO_4 , label the *most polar bond*.



b. 3 points (Note: Lone pairs are not shown in structures below)

(i) For the three *resonance* structures (1, 2 and 3) shown below, answer the following question:



(I) Circle the *most contributing* resonance structure.

Structure 1

Structure 2

Structure 3

(ii) For the three structures given (4, 5, and 6) below, answer the following questions:



(I) Circle all the structures that represent *resonance structures* of each other:

Structure 4

Structure 5

Structure 6

(II) Among the resonance structures, circle the *most contributing* resonance structure.

Structure 4

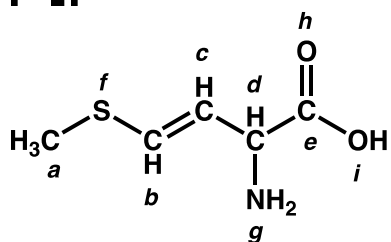
Structure 5

Structure 6

**Question 6** **6 points**

Using the data given below, calculate the **lattice energy** for SrO(s).

	ΔH° (kJ/mol)
$\text{Sr}(s) \longrightarrow \text{Sr}(g)$	+180
$\text{Sr}(g) \longrightarrow \text{Sr}^{2+}(g) + 2e^-$	+1468
$\text{O}_2(g) \longrightarrow 2\text{O}(g)$	+500
$\text{O}(g) + 2e^- \longrightarrow \text{O}^{2-}(g)$	+600
$\text{Sr}(s) + \frac{1}{2}\text{O}_2(g) \longrightarrow \text{SrO}(s)$	-550

**Question 7 10 points**

Consider the molecule given on the left, and answer questions (i) to (vi) using chemical bonding theories that we have learnt in class (valence bond theory, hybridization).

Note that the lone pairs are not shown in the structure:

i. Consider the carbon atoms marked in as ***a***, ***b***, ***c***, ***d***, and ***e*** in the molecule, and determine **hybridization** at each of the carbon centers. (2.5 points)

Atom	Hybridization
Carbon (<i>a</i>)	
Carbon (<i>b</i>)	
Carbon (<i>c</i>)	
Carbon (<i>d</i>)	
Carbon (<i>e</i>)	

ii. Now, consider the sulfur atom (S) marked as ***f***, nitrogen (N) atom (***g***), and oxygen(O) atoms (***h*** and ***i***), and determine their **hybridization**. (2 points)

Atom	Hybridization
Sulfur (<i>f</i>)	
Nitrogen (<i>g</i>)	
Oxygen (<i>h</i>)	
Oxygen (<i>i</i>)	

iii. **Describe** all the bonds formed by the carbon atoms ***a*** and ***b***, using sigma and pi designation as appropriate. (2 points)

Atom	Bonds formed (use sigma and pi as appropriate)
Carbon (<i>a</i>)	
Carbon (<i>b</i>)	

iv. How many total sp^3 -s sigma bonds are in the molecule? (1 point)

v. Determine **electronic** and **molecular geometry** at carbon (C) labeled ***e*** and sulfur (S) labeled ***f*** centres. (2 points)

Atom	Electronic Geometry	Molecular Geometry
Carbon (<i>e</i>)		
Sulfur (<i>f</i>)		

vi. Circle the value from the choices given below, that describes the **C-S-C bond angle at sulfur** (S) marked ***f***: (0.5 points)

>90° to <109.5°

109.5°

>109.5 to <120°

120°

>120° to <180°


Question 8 12 points

Using the **valence shell orbitals only** draw a **complete** molecular orbital (MO) diagram (with no s-p mixing) for the molecule **PS**, in the space provided below. **(6 points)**

Assume that 3s and 3p atomic orbitals will form molecular orbitals like those formed by the combination of 2s and 2p atomic orbitals that we learnt in class, and there is no s-p mixing.

Show **energy arrow**, **atomic orbitals** involved, **atom labels**, the **molecular orbitals** that are formed, **filled electrons** in the MO diagram etc. Make sure to label all the atomic and molecular orbitals.

Using the MO diagram above, answer the following questions:

- (i) Which one of the following will have the highest bond order? Circle the correct choice
 A) PS^{1-} B) PS C) PS^{1+}
- (ii) Which one of the following will have the longest bond length? Circle the correct choice
 A) PS^{1-} B) PS C) PS^{1+}
- (iii) Which of the following will be diamagnetic? Circle the correct choice.
 A) PS^{1-} B) PS C) PS^{1+}
- (iv) Draw Lewis dot structure of the molecule with the shortest bond length (based on the MO diagram) from among the three (PS^{1-} , PS, PS^{1+}), and indicate the bond order calculated from the Lewis dot structure. **(3 points)**

Lewis Dot Structure

Bond Order



C463076A-ED95-49EA-8CE3-B6A29F114AEF

midterm-exam-93388

#182 Page 10 of 10

SCRAP PAGE

**CHEMISTRY 110 –Midterm Exam****Thursday November 7, 2024, 6:30 – 8:30 pm****Examiners: Pallavi Sirjoosingh and Ashok Kakkar**

First name (please write as legibly as possible within the boxes)

--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--

Last name

--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--

Student ID number

--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--

This booklet has 8 questions on 10 pages (The last page can be used as a scrap page).

The staple must NOT be removed from the question booklet. Your intact question booklet must be returned at the end of the exam.

A separate booklet contains a periodic table, and datasheet.

Write your answers directly under the questions in the space provided. Do not write on the QR codes or near the top edges (as it may be cut off).

Please write **legibly and dark enough** for the scan to be readable. For students who use English as a second language, dictionaries are permitted. Calculators are allowed.

DO NOT ASK QUESTIONS OF THE INVIGILATORS about interpretations of the exam questions. If you have a concern about a question, please summarize it at the top of this page. All concerns will be evaluated after the exam. GOOD LUCK!

**Question 1 8 points** (*Assume all transitions are possible*)

a. 6 points Calculate the **energy (in kJ/mol)** required to ionize (completely remove) the electron from the **$n=3$** level of **He^+ ion** (He: Helium). Show your work including equations. (3 sig. figs.)

b. 2 points

(I) Select the largest (longest) wavelength from the three choices below:

- i. Wavelength of photon to *completely remove* an electron from *1s orbital of Hydrogen atom*
- ii. Wavelength of photon to *completely remove* an electron from *1s orbital of Boron atom (B)*
- iii. Wavelength of photon to *completely remove* an electron from *1s orbital of Boron cation (B^+)*

i or ii or iii

(II) Select the smallest (shortest) wavelength from the three choices below (*for H atom*):

- i. Wavelength of photon *emitted* for an electron transitioning from *2p to 1s orbital*
- ii. Wavelength of photon *absorbed* for an electron transitioning from *2s to 4p orbital*
- iii. Wavelength of photon *absorbed* for an electron transitioning from *1s to 3p orbital*

i or ii or iii



Question 2 8 points



A Chem110 student plots the graph shown on the left, from data obtained for the kinetic energy (eV) of the ejected electron versus frequency (s^{-1}) of the incident light, for an unknown metal. ($1\text{eV} = 1.6022 \times 10^{-19} \text{ J}$)

Based on the plot, answer the questions **a to c** below:

a. 1 point On the plot above, label the **threshold frequency**.

b. 2 points What is the value of the work potential (in J) of the unknown metal?

c. 3 points What is the *longest* wavelength (in nm) required to eject an electron from this metal's surface? Show your work including equation(s) used. (3 sig figs)

d. 2 points In the space provided below, plot the following

(Assume frequency is *above* threshold frequency):

- i. *Kinetic energy* of an ejected electron versus the *intensity* of the incident light (left)
- ii. *Number of electrons* ejected versus the *frequency* of the incident light (right)



**Question 3** **6 points****a. 2 points**

In the ground state of Ge (Germanium; $Z = 32$) atom, **name** the following orbitals:

i. Orbital containing the electron that requires the maximum energy to be removed.

ii. Orbital containing the electron that experiences the minimum effective nuclear charge.

b. 2 points

An unknown anion (A^-) has the following *excited* electronic configuration: $[Ar] 4s^2 3d^3 4p^1$

(Ar: Argon; $Z = 18$)

Give the **condensed ground state electronic configuration** of the atom, **A** and the cation, **A^{2+}** .
Show your work. (Assume all electronic transitions are possible)

Electronic Configuration of A:

Electronic Configuration of A^{2+} :

c. 2 points

In a single atom, determine how many electrons can have the following set of quantum numbers?

(i) $n = 5; l = 3$

In a single atom, determine how many electrons can have the following set of quantum numbers?

(i) $n = 3; l = 2; m_s = +1/2$

**Question 4** **8 points**

Rank the following from 1 to 4 in **increasing** order. Provide the ranking in the boxes provided.

Note: Atomic number (Z) are provided for relevant atoms

a) Increasing (lowest to highest) order of *ionization energy*: (1= lowest; 4 = highest)

Mg

Mg: Z = 12

Rb

Rb: Z = 37

Cs

Cs: Z = 55

K

K : Z = 19

b) Increasing (lowest to highest) order of *bond angle*: (1= lowest; 4 = highest)

H₂OCCl₄XeF₄

HCN

c) Increasing (smallest to largest) order of *ionic size* (1= smallest; 4 = largest)

S²⁻

S: Z = 16

O²⁻

O: Z = 8

Be²⁺

Be: Z = 4

Na⁺

Na: Z = 11

d) Increasing (lowest to highest) order of *bond energy* (1= lowest; 4 = highest)

P-F

P-Cl

P-I

P-Br

**Question 5** **12 points**

a. 9 points For the molecules HClO_4 and TeF_6 , draw the most stable *Lewis structure* showing appropriate shape and *dashed/wedged bonds*, based on VSEPR. Determine the **molecular geometry** around the central atom. For HClO_4 , label the *most polar bond*.

HClO_4 (there are no O-O bonds)

Label the most polar bond

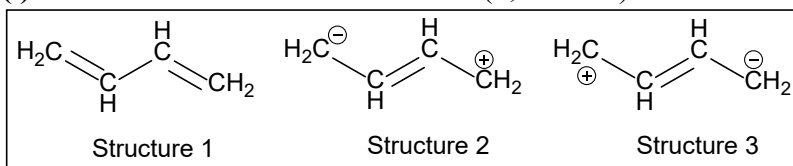
Molecular Geometry around Cl

TeF_6

Molecular Geometry around Te

b. 3 points (Note: Lone pairs are not shown in structures below)

(i) For the three *resonance* structures (1, 2 and 3) shown below, answer the following question:



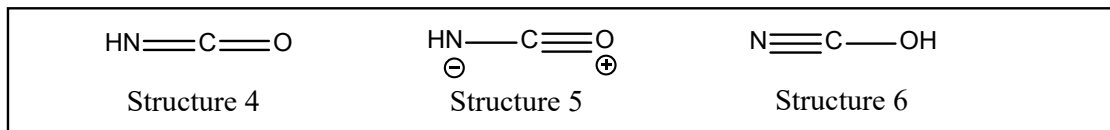
(I) Circle the *most contributing* resonance structure.

Structure 1

Structure 2

Structure 3

(ii) For the three structures given (4, 5, and 6) below, answer the following questions:



(I) Circle all the structures that represent *resonance structures* of each other:

Structure 4

Structure 5

Structure 6

(II) Among the resonance structures, circle the *most contributing* resonance structure.

Structure 4

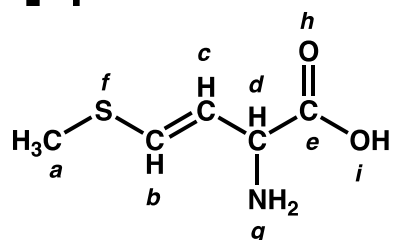
Structure 5

Structure 6

**Question 6** **6 points**

Using the data given below, calculate the **lattice energy** for SrO(s).

	ΔH° (kJ/mol)
$\text{Sr}(s) \longrightarrow \text{Sr}(g)$	+180
$\text{Sr}(g) \longrightarrow \text{Sr}^{2+}(g) + 2e^-$	+1468
$\text{O}_2(g) \longrightarrow 2\text{O}(g)$	+500
$\text{O}(g) + 2e^- \longrightarrow \text{O}^{2-}(g)$	+600
$\text{Sr}(s) + \frac{1}{2}\text{O}_2(g) \longrightarrow \text{SrO}(s)$	-550

**Question 7 10 points**

Consider the molecule given on the left, and answer questions (i) to (vi) using chemical bonding theories that we have learnt in class (valence bond theory, hybridization).

Note that the lone pairs are not shown in the structure:

i. Consider the carbon atoms marked in as ***a***, ***b***, ***c***, ***d***, and ***e*** in the molecule, and determine **hybridization** at each of the carbon centers. (2.5 points)

Atom	Hybridization
Carbon (<i>a</i>)	
Carbon (<i>b</i>)	
Carbon (<i>c</i>)	
Carbon (<i>d</i>)	
Carbon (<i>e</i>)	

ii. Now, consider the sulfur atom (S) marked as ***f***, nitrogen (N) atom (***g***), and oxygen(O) atoms (***h*** and ***i***), and determine their **hybridization**. (2 points)

Atom	Hybridization
Sulfur (<i>f</i>)	
Nitrogen (<i>g</i>)	
Oxygen (<i>h</i>)	
Oxygen (<i>i</i>)	

iii. **Describe** all the bonds formed by the carbon atoms ***a*** and ***b***, using sigma and pi designation as appropriate. (2 points)

Atom	Bonds formed (use sigma and pi as appropriate)
Carbon (<i>a</i>)	
Carbon (<i>b</i>)	

iv. How many total sp^3 -s sigma bonds are in the molecule? (1 point)

v. Determine **electronic** and **molecular geometry** at carbon (C) labeled ***e*** and sulfur (S) labeled ***f*** centres. (2 points)

Atom	Electronic Geometry	Molecular Geometry
Carbon (<i>e</i>)		
Sulfur (<i>f</i>)		

vi. Circle the value from the choices given below, that describes the **C-S-C bond angle at sulfur** (S) marked ***f***: (0.5 points)

$>90^\circ$ to $<109.5^\circ$ 109.5° >109.5 to $<120^\circ$ 120° $>120^\circ$ to $<180^\circ$


Question 8 12 points

Using the **valence shell orbitals only** draw a **complete** molecular orbital (MO) diagram (with no s-p mixing) for the molecule **PS**, in the space provided below. **(6 points)**

Assume that 3s and 3p atomic orbitals will form molecular orbitals like those formed by the combination of 2s and 2p atomic orbitals that we learnt in class, and there is no s-p mixing.

Show **energy arrow**, **atomic orbitals** involved, **atom labels**, the **molecular orbitals** that are formed, **filled electrons** in the MO diagram etc. Make sure to label all the atomic and molecular orbitals.

Using the MO diagram above, answer the following questions:

- (i) Which one of the following will have the highest bond order? Circle the correct choice
 A) PS^{1-} B) PS C) PS^{1+}
- (ii) Which one of the following will have the longest bond length? Circle the correct choice
 A) PS^{1-} B) PS C) PS^{1+}
- (iii) Which of the following will be diamagnetic? Circle the correct choice.
 A) PS^{1-} B) PS C) PS^{1+}
- (iv) Draw Lewis dot structure of the molecule with the shortest bond length (based on the MO diagram) from among the three (PS^{1-} , PS, PS^{1+}), and indicate the bond order calculated from the Lewis dot structure. **(3 points)**

Lewis Dot Structure

Bond Order



EED3B36C-FBA6-4F31-BA2C-13982BD4BE00

midterm-exam-93388

#183 Page 10 of 10

SCRAP PAGE

**CHEMISTRY 110 –Midterm Exam****Thursday November 7, 2024, 6:30 – 8:30 pm****Examiners: Pallavi Sirjoosingh and Ashok Kakkar**

First name (please write as legibly as possible within the boxes)

--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--

Last name

--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--

Student ID number

--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--

This booklet has 8 questions on 10 pages (The last page can be used as a scrap page).

The staple must NOT be removed from the question booklet. Your intact question booklet must be returned at the end of the exam.

A separate booklet contains a periodic table, and datasheet.

Write your answers directly under the questions in the space provided. Do not write on the QR codes or near the top edges (as it may be cut off).

Please write **legibly and dark enough** for the scan to be readable. For students who use English as a second language, dictionaries are permitted. Calculators are allowed.

DO NOT ASK QUESTIONS OF THE INVIGILATORS about interpretations of the exam questions. If you have a concern about a question, please summarize it at the top of this page. All concerns will be evaluated after the exam. GOOD LUCK!

**Question 1 8 points** (*Assume all transitions are possible*)

a. 6 points Calculate the **energy (in kJ/mol)** required to ionize (completely remove) the electron from the **$n=3$** level of **He^+ ion** (He: Helium). Show your work including equations. (3 sig. figs.)

b. 2 points

(I) Select the largest (longest) wavelength from the three choices below:

- i. Wavelength of photon to *completely remove* an electron from *1s orbital of Hydrogen atom*
- ii. Wavelength of photon to *completely remove* an electron from *1s orbital of Boron atom (B)*
- iii. Wavelength of photon to *completely remove* an electron from *1s orbital of Boron cation (B^+)*

i or ii or iii

(II) Select the smallest (shortest) wavelength from the three choices below (*for H atom*):

- i. Wavelength of photon *emitted* for an electron transitioning from *2p to 1s orbital*
- ii. Wavelength of photon *absorbed* for an electron transitioning from *2s to 4p orbital*
- iii. Wavelength of photon *absorbed* for an electron transitioning from *1s to 3p orbital*

i or ii or iii



Question 2 8 points



A Chem110 student plots the graph shown on the left, from data obtained for the kinetic energy (eV) of the ejected electron versus frequency (s^{-1}) of the incident light, for an unknown metal. ($1\text{eV} = 1.6022 \times 10^{-19} \text{ J}$)

Based on the plot, answer the questions **a to c** below:

a. 1 point On the plot above, label the **threshold frequency**.

b. 2 points What is the value of the work potential (in J) of the unknown metal?

c. 3 points What is the *longest* wavelength (in nm) required to eject an electron from this metal's surface? Show your work including equation(s) used. (3 sig figs)

d. 2 points In the space provided below, plot the following

(Assume frequency is *above* threshold frequency):

- i. *Kinetic energy* of an ejected electron versus the *intensity* of the incident light (left)
- ii. *Number of electrons* ejected versus the *frequency* of the incident light (right)



**Question 3** **6 points****a. 2 points**

In the ground state of Ge (Germanium; $Z = 32$) atom, **name** the following orbitals:

i. Orbital containing the electron that requires the maximum energy to be removed.

ii. Orbital containing the electron that experiences the minimum effective nuclear charge.

b. 2 points

An unknown anion (A^-) has the following *excited* electronic configuration: $[Ar] 4s^2 3d^3 4p^1$

(Ar: Argon; $Z = 18$)

Give the **condensed ground state electronic configuration** of the atom, **A** and the cation, **A^{2+}** .
Show your work. (Assume all electronic transitions are possible)

Electronic Configuration of A:

Electronic Configuration of A^{2+} :

c. 2 points

In a single atom, determine how many electrons can have the following set of quantum numbers?

(i) $n = 5; l = 3$

In a single atom, determine how many electrons can have the following set of quantum numbers?

(i) $n = 3; l = 2; m_s = +1/2$

**Question 4** **8 points**

Rank the following from 1 to 4 in **increasing** order. Provide the ranking in the boxes provided.

Note: Atomic number (Z) are provided for relevant atoms

a) Increasing (lowest to highest) order of *ionization energy*: (1= lowest; 4 = highest)

Mg

Mg: Z = 12

Rb

Rb: Z = 37

Cs

Cs: Z = 55

K

K : Z = 19

b) Increasing (lowest to highest) order of *bond angle*: (1= lowest; 4 = highest)

H₂OCCl₄XeF₄

HCN

c) Increasing (smallest to largest) order of *ionic size* (1= smallest; 4 = largest)

S²⁻

S: Z = 16

O²⁻

O: Z = 8

Be²⁺

Be: Z = 4

Na⁺

Na: Z = 11

d) Increasing (lowest to highest) order of *bond energy* (1= lowest; 4 = highest)

P-F

P-Cl

P-I

P-Br

**Question 5** **12 points**

a. 9 points For the molecules HClO_4 and TeF_6 , draw the most stable *Lewis structure* showing appropriate shape and *dashed/wedged bonds*, based on VSEPR. Determine the **molecular geometry** around the central atom. For HClO_4 , label the *most polar bond*.



b. 3 points (Note: Lone pairs are not shown in structures below)

(i) For the three *resonance* structures (1, 2 and 3) shown below, answer the following question:



(I) Circle the *most contributing* resonance structure.

Structure 1

Structure 2

Structure 3

(ii) For the three structures given (4, 5, and 6) below, answer the following questions:



(I) Circle all the structures that represent *resonance structures* of each other:

Structure 4

Structure 5

Structure 6

(II) Among the resonance structures, circle the *most contributing* resonance structure.

Structure 4

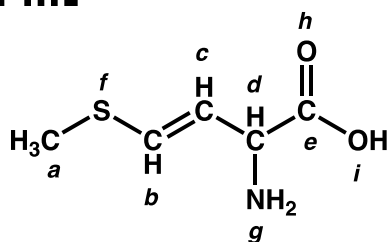
Structure 5

Structure 6

**Question 6** **6 points**

Using the data given below, calculate the **lattice energy** for SrO(s).

	ΔH° (kJ/mol)
$\text{Sr}(s) \longrightarrow \text{Sr}(g)$	+180
$\text{Sr}(g) \longrightarrow \text{Sr}^{2+}(g) + 2e^-$	+1468
$\text{O}_2(g) \longrightarrow 2\text{O}(g)$	+500
$\text{O}(g) + 2e^- \longrightarrow \text{O}^{2-}(g)$	+600
$\text{Sr}(s) + \frac{1}{2}\text{O}_2(g) \longrightarrow \text{SrO}(s)$	-550

**Question 7 10 points**

Consider the molecule given on the left, and answer questions (i) to (vi) using chemical bonding theories that we have learnt in class (valence bond theory, hybridization).

Note that the lone pairs are not shown in the structure:

i. Consider the carbon atoms marked in as ***a***, ***b***, ***c***, ***d***, and ***e*** in the molecule, and determine **hybridization** at each of the carbon centers. (2.5 points)

Atom	Hybridization
Carbon (<i>a</i>)	
Carbon (<i>b</i>)	
Carbon (<i>c</i>)	
Carbon (<i>d</i>)	
Carbon (<i>e</i>)	

ii. Now, consider the sulfur atom (S) marked as ***f***, nitrogen (N) atom (***g***), and oxygen(O) atoms (***h*** and ***i***), and determine their **hybridization**. (2 points)

Atom	Hybridization
Sulfur (<i>f</i>)	
Nitrogen (<i>g</i>)	
Oxygen (<i>h</i>)	
Oxygen (<i>i</i>)	

iii. **Describe** all the bonds formed by the carbon atoms ***a*** and ***b***, using sigma and pi designation as appropriate. (2 points)

Atom	Bonds formed (use sigma and pi as appropriate)
Carbon (<i>a</i>)	
Carbon (<i>b</i>)	

iv. How many total sp^3 -s sigma bonds are in the molecule? (1 point)

v. Determine **electronic** and **molecular geometry** at carbon (C) labeled ***e*** and sulfur (S) labeled ***f*** centres. (2 points)

Atom	Electronic Geometry	Molecular Geometry
Carbon (<i>e</i>)		
Sulfur (<i>f</i>)		

vi. Circle the value from the choices given below, that describes the **C-S-C bond angle at sulfur** (S) marked ***f***: (0.5 points)

>90° to <109.5° 109.5° >109.5 to <120° 120° >120° to <180°

**Question 8 12 points**

Using the **valence shell orbitals only** draw a **complete** molecular orbital (MO) diagram (with no s-p mixing) for the molecule **PS**, in the space provided below. **(6 points)**

Assume that 3s and 3p atomic orbitals will form molecular orbitals like those formed by the combination of 2s and 2p atomic orbitals that we learnt in class, and there is no s-p mixing.

Show **energy arrow**, **atomic orbitals** involved, **atom labels**, the **molecular orbitals** that are formed, **filled electrons** in the MO diagram etc. Make sure to label all the atomic and molecular orbitals.

Using the MO diagram above, answer the following questions:

- (i) Which one of the following will have the highest bond order? Circle the correct choice
A) PS^{1-} B) PS C) PS^{1+}
- (ii) Which one of the following will have the longest bond length? Circle the correct choice
A) PS^{1-} B) PS C) PS^{1+}
- (iii) Which of the following will be diamagnetic? Circle the correct choice.
A) PS^{1-} B) PS C) PS^{1+}
- (iv) Draw Lewis dot structure of the molecule with the shortest bond length (based on the MO diagram) from among the three (PS^{1-} , PS, PS^{1+}), and indicate the bond order calculated from the Lewis dot structure. **(3 points)**

Lewis Dot Structure

Bond Order



5DB1DA9D-BAC6-42DB-B060-3D5BA84BCD4F

midterm-exam-93388

#184 Page 10 of 10

SCRAP PAGE

**CHEMISTRY 110 –Midterm Exam****Thursday November 7, 2024, 6:30 – 8:30 pm**

Examiners: Pallavi Sirjoosingh and Ashok Kakkar

First name (please write as legibly as possible within the boxes)

--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--

Last name

--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--

Student ID number

--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--

This booklet has 8 questions on 10 pages (The last page can be used as a scrap page).

The staple must NOT be removed from the question booklet. Your intact question booklet must be returned at the end of the exam.

A separate booklet contains a periodic table, and datasheet.

Write your answers directly under the questions in the space provided. Do not write on the QR codes or near the top edges (as it may be cut off).

Please write **legibly and dark enough** for the scan to be readable. For students who use English as a second language, dictionaries are permitted. Calculators are allowed.

DO NOT ASK QUESTIONS OF THE INVIGILATORS about interpretations of the exam questions. If you have a concern about a question, please summarize it at the top of this page. All concerns will be evaluated after the exam. GOOD LUCK!

**Question 1 8 points** (*Assume all transitions are possible*)

a. 6 points Calculate the **energy (in kJ/mol)** required to ionize (completely remove) the electron from the **n=3** level of **He⁺ ion** (He: Helium). Show your work including equations. (3 sig. figs.)

b. 2 points

(I) Select the largest (longest) wavelength from the three choices below:

- i. Wavelength of photon to *completely remove* an electron from *1s orbital of Hydrogen atom*
- ii. Wavelength of photon to *completely remove* an electron from *1s orbital of Boron atom (B)*
- iii. Wavelength of photon to *completely remove* an electron from *1s orbital of Boron cation (B⁺)*

i or ii or iii

(II) Select the smallest (shortest) wavelength from the three choices below (*for H atom*):

- i. Wavelength of photon *emitted* for an electron transitioning from *2p to 1s orbital*
- ii. Wavelength of photon *absorbed* for an electron transitioning from *2s to 4p orbital*
- iii. Wavelength of photon *absorbed* for an electron transitioning from *1s to 3p orbital*

i or ii or iii



Question 2 8 points



A Chem110 student plots the graph shown on the left, from data obtained for the kinetic energy (eV) of the ejected electron versus frequency (s^{-1}) of the incident light, for an unknown metal. ($1\text{eV} = 1.6022 \times 10^{-19} \text{ J}$)

Based on the plot, answer the questions **a to c** below:

a. 1 point On the plot above, label the **threshold frequency**.

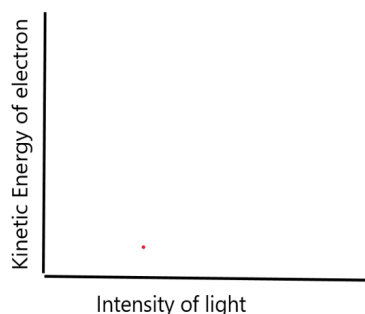
b. 2 points What is the value of the work potential (in J) of the unknown metal?

c. 3 points What is the *longest* wavelength (in nm) required to eject an electron from this metal's surface? Show your work including equation(s) used. (3 sig figs)

d. 2 points In the space provided below, plot the following

(Assume frequency is *above* threshold frequency):

- i. *Kinetic energy* of an ejected electron versus the *intensity* of the incident light (left)
- ii. *Number of electrons* ejected versus the *frequency* of the incident light (right)



**Question 3** **6 points****a. 2 points**

In the ground state of Ge (Germanium; $Z = 32$) atom, **name** the following orbitals:

i. Orbital containing the electron that requires the maximum energy to be removed.

ii. Orbital containing the electron that experiences the minimum effective nuclear charge.

b. 2 points

An unknown anion (A^-) has the following *excited* electronic configuration: $[Ar] 4s^2 3d^3 4p^1$

(Ar: Argon; $Z = 18$)

Give the **condensed ground state electronic configuration** of the atom, **A** and the cation, **A^{2+}** .
Show your work. (Assume all electronic transitions are possible)

Electronic Configuration of A:

Electronic Configuration of A^{2+} :

c. 2 points

In a single atom, determine how many electrons can have the following set of quantum numbers?

(i) $n = 5; l = 3$

In a single atom, determine how many electrons can have the following set of quantum numbers?

(i) $n = 3; l = 2; m_s = +1/2$

**Question 4** **8 points**

Rank the following from 1 to 4 in **increasing** order. Provide the ranking in the boxes provided.

Note: Atomic number (Z) are provided for relevant atoms

a) Increasing (lowest to highest) order of *ionization energy*: (1= lowest; 4 = highest)

Mg

Mg: Z = 12

Rb

Rb: Z = 37

Cs

Cs: Z = 55

K

K : Z = 19

b) Increasing (lowest to highest) order of *bond angle*: (1= lowest; 4 = highest)

H₂OCCl₄XeF₄

HCN

c) Increasing (smallest to largest) order of *ionic size* (1= smallest; 4 = largest)

S²⁻

S: Z = 16

O²⁻

O: Z = 8

Be²⁺

Be: Z = 4

Na⁺

Na: Z = 11

d) Increasing (lowest to highest) order of *bond energy* (1= lowest; 4 = highest)

P-F

P-Cl

P-I

P-Br

**Question 5** **12 points**

a. 9 points For the molecules HClO_4 and TeF_6 , draw the most stable *Lewis structure* showing appropriate shape and *dashed/wedged bonds*, based on VSEPR. Determine the **molecular geometry** around the central atom. For HClO_4 , label the *most polar bond*.



b. 3 points (Note: Lone pairs are not shown in structures below)

(i) For the three *resonance* structures (1, 2 and 3) shown below, answer the following question:



(I) Circle the *most contributing* resonance structure.

Structure 1

Structure 2

Structure 3

(ii) For the three structures given (4, 5, and 6) below, answer the following questions:



(I) Circle all the structures that represent *resonance structures* of each other:

Structure 4

Structure 5

Structure 6

(II) Among the resonance structures, circle the *most contributing* resonance structure.

Structure 4

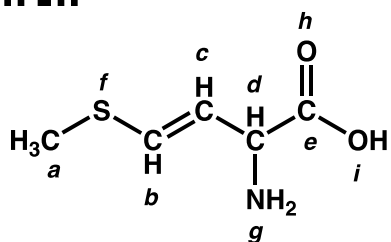
Structure 5

Structure 6

**Question 6** **6 points**

Using the data given below, calculate the **lattice energy** for SrO(s).

	ΔH° (kJ/mol)
$\text{Sr}(s) \longrightarrow \text{Sr}(g)$	+180
$\text{Sr}(g) \longrightarrow \text{Sr}^{2+}(g) + 2e^-$	+1468
$\text{O}_2(g) \longrightarrow 2\text{O}(g)$	+500
$\text{O}(g) + 2e^- \longrightarrow \text{O}^{2-}(g)$	+600
$\text{Sr}(s) + \frac{1}{2}\text{O}_2(g) \longrightarrow \text{SrO}(s)$	-550

**Question 7 10 points**

Consider the molecule given on the left, and answer questions (i) to (vi) using chemical bonding theories that we have learnt in class (valence bond theory, hybridization).

Note that the lone pairs are not shown in the structure:

i. Consider the carbon atoms marked in as ***a***, ***b***, ***c***, ***d***, and ***e*** in the molecule, and determine **hybridization** at each of the carbon centers. (2.5 points)

Atom	Hybridization
Carbon (<i>a</i>)	
Carbon (<i>b</i>)	
Carbon (<i>c</i>)	
Carbon (<i>d</i>)	
Carbon (<i>e</i>)	

ii. Now, consider the sulfur atom (S) marked as ***f***, nitrogen (N) atom (***g***), and oxygen(O) atoms (***h*** and ***i***), and determine their **hybridization**. (2 points)

Atom	Hybridization
Sulfur (<i>f</i>)	
Nitrogen (<i>g</i>)	
Oxygen (<i>h</i>)	
Oxygen (<i>i</i>)	

iii. **Describe** all the bonds formed by the carbon atoms ***a*** and ***b***, using sigma and pi designation as appropriate. (2 points)

Atom	Bonds formed (use sigma and pi as appropriate)
Carbon (<i>a</i>)	
Carbon (<i>b</i>)	

iv. How many total sp^3 -s sigma bonds are in the molecule? (1 point)

v. Determine **electronic** and **molecular geometry** at carbon (C) labeled ***e*** and sulfur (S) labeled ***f*** centres. (2 points)

Atom	Electronic Geometry	Molecular Geometry
Carbon (<i>e</i>)		
Sulfur (<i>f</i>)		

vi. Circle the value from the choices given below, that describes the **C-S-C bond angle at sulfur** (S) marked ***f***: (0.5 points)

$>90^\circ$ to $<109.5^\circ$ 109.5° >109.5 to $<120^\circ$ 120° $>120^\circ$ to $<180^\circ$

**Question 8 12 points**

Using the **valence shell orbitals only** draw a **complete** molecular orbital (MO) diagram (with no s-p mixing) for the molecule **PS**, in the space provided below. **(6 points)**

Assume that 3s and 3p atomic orbitals will form molecular orbitals like those formed by the combination of 2s and 2p atomic orbitals that we learnt in class, and there is no s-p mixing.

Show **energy arrow**, **atomic orbitals** involved, **atom labels**, the **molecular orbitals** that are formed, **filled electrons** in the MO diagram etc. Make sure to label all the atomic and molecular orbitals.

Using the MO diagram above, answer the following questions:

- (i) Which one of the following will have the highest bond order? Circle the correct choice
A) PS^{1-} B) PS C) PS^{1+}
- (ii) Which one of the following will have the longest bond length? Circle the correct choice
A) PS^{1-} B) PS C) PS^{1+}
- (iii) Which of the following will be diamagnetic? Circle the correct choice.
A) PS^{1-} B) PS C) PS^{1+}
- (iv) Draw Lewis dot structure of the molecule with the shortest bond length (based on the MO diagram) from among the three (PS^{1-} , PS, PS^{1+}), and indicate the bond order calculated from the Lewis dot structure. **(3 points)**

Lewis Dot Structure

Bond Order



E4AF62B3-D283-4693-9DF3-9E55444C7BBB

midterm-exam-93388

#185 Page 10 of 10

SCRAP PAGE

**CHEMISTRY 110 –Midterm Exam****Thursday November 7, 2024, 6:30 – 8:30 pm**

Examiners: Pallavi Sirjoosingh and Ashok Kakkar

First name (please write as legibly as possible within the boxes)

--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--

Last name

--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--

Student ID number

--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--

This booklet has 8 questions on 10 pages (The last page can be used as a scrap page).

The staple must NOT be removed from the question booklet. Your intact question booklet must be returned at the end of the exam.

A separate booklet contains a periodic table, and datasheet.

Write your answers directly under the questions in the space provided. Do not write on the QR codes or near the top edges (as it may be cut off).

Please write **legibly and dark enough** for the scan to be readable. For students who use English as a second language, dictionaries are permitted. Calculators are allowed.

DO NOT ASK QUESTIONS OF THE INVIGILATORS about interpretations of the exam questions. If you have a concern about a question, please summarize it at the top of this page. All concerns will be evaluated after the exam. GOOD LUCK!

**Question 1 8 points** (*Assume all transitions are possible*)

a. 6 points Calculate the **energy (in kJ/mol)** required to ionize (completely remove) the electron from the **$n=3$** level of **He^+ ion** (He: Helium). Show your work including equations. (3 sig. figs.)

b. 2 points

(I) Select the largest (longest) wavelength from the three choices below:

- i. Wavelength of photon to *completely remove* an electron from *1s orbital of Hydrogen atom*
- ii. Wavelength of photon to *completely remove* an electron from *1s orbital of Boron atom (B)*
- iii. Wavelength of photon to *completely remove* an electron from *1s orbital of Boron cation (B^+)*

i or ii or iii

(II) Select the smallest (shortest) wavelength from the three choices below (*for H atom*):

- i. Wavelength of photon *emitted* for an electron transitioning from *2p to 1s orbital*
- ii. Wavelength of photon *absorbed* for an electron transitioning from *2s to 4p orbital*
- iii. Wavelength of photon *absorbed* for an electron transitioning from *1s to 3p orbital*

i or ii or iii



Question 2 8 points



A Chem110 student plots the graph shown on the left, from data obtained for the kinetic energy (eV) of the ejected electron versus frequency (s^{-1}) of the incident light, for an unknown metal. ($1\text{eV} = 1.6022 \times 10^{-19} \text{ J}$)

Based on the plot, answer the questions **a to c** below:

a. 1 point On the plot above, label the **threshold frequency**.

b. 2 points What is the value of the work potential (in J) of the unknown metal?

c. 3 points What is the *longest* wavelength (in nm) required to eject an electron from this metal's surface? Show your work including equation(s) used. (3 sig figs)

d. 2 points In the space provided below, plot the following

(Assume frequency is *above* threshold frequency):

- Kinetic energy* of an ejected electron versus the *intensity* of the incident light (left)
- Number of electrons* ejected versus the *frequency* of the incident light (right)



**Question 3** **6 points****a. 2 points**

In the ground state of Ge (Germanium; $Z = 32$) atom, **name** the following orbitals:

i. Orbital containing the electron that requires the maximum energy to be removed.

ii. Orbital containing the electron that experiences the minimum effective nuclear charge.

b. 2 points

An unknown anion (A^-) has the following *excited* electronic configuration: $[Ar] 4s^2 3d^3 4p^1$

(Ar: Argon; $Z = 18$)

Give the **condensed ground state electronic configuration** of the atom, **A** and the cation, **A^{2+}** .
Show your work. (Assume all electronic transitions are possible)

Electronic Configuration of A:

Electronic Configuration of A^{2+} :

c. 2 points

In a single atom, determine how many electrons can have the following set of quantum numbers?

(i) $n = 5; l = 3$

In a single atom, determine how many electrons can have the following set of quantum numbers?

(i) $n = 3; l = 2; m_s = +1/2$

**Question 4** **8 points**

Rank the following from 1 to 4 in **increasing** order. Provide the ranking in the boxes provided.

Note: Atomic number (Z) are provided for relevant atoms

a) Increasing (lowest to highest) order of *ionization energy*: (1= lowest; 4 = highest)

Mg

Mg: Z = 12

Rb

Rb: Z = 37

Cs

Cs: Z = 55

K

K : Z = 19

b) Increasing (lowest to highest) order of *bond angle*: (1= lowest; 4 = highest)

H₂OCCl₄XeF₄

HCN

c) Increasing (smallest to largest) order of *ionic size* (1= smallest; 4 = largest)

S²⁻

S: Z = 16

O²⁻

O: Z = 8

Be²⁺

Be: Z = 4

Na⁺

Na: Z = 11

d) Increasing (lowest to highest) order of *bond energy* (1= lowest; 4 = highest)

P-F

P-Cl

P-I

P-Br

**Question 5** **12 points**

a. 9 points For the molecules HClO_4 and TeF_6 , draw the most stable *Lewis structure* showing appropriate shape and *dashed/wedged bonds*, based on VSEPR. Determine the **molecular geometry** around the central atom. For HClO_4 , label the *most polar bond*.



b. 3 points (Note: Lone pairs are not shown in structures below)

(i) For the three *resonance* structures (1, 2 and 3) shown below, answer the following question:



(I) Circle the *most contributing* resonance structure.

Structure 1

Structure 2

Structure 3

(ii) For the three structures given (4, 5, and 6) below, answer the following questions:



(I) Circle all the structures that represent *resonance structures* of each other:

Structure 4

Structure 5

Structure 6

(II) Among the resonance structures, circle the *most contributing* resonance structure.

Structure 4

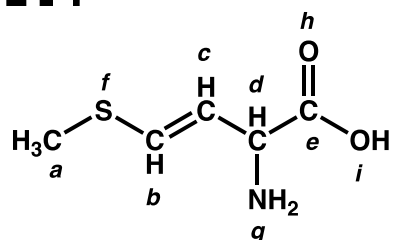
Structure 5

Structure 6

**Question 6** **6 points**

Using the data given below, calculate the **lattice energy** for SrO(s).

	ΔH° (kJ/mol)
$\text{Sr}(s) \longrightarrow \text{Sr}(g)$	+180
$\text{Sr}(g) \longrightarrow \text{Sr}^{2+}(g) + 2e^-$	+1468
$\text{O}_2(g) \longrightarrow 2\text{O}(g)$	+500
$\text{O}(g) + 2e^- \longrightarrow \text{O}^{2-}(g)$	+600
$\text{Sr}(s) + \frac{1}{2}\text{O}_2(g) \longrightarrow \text{SrO}(s)$	-550

**Question 7 10 points**

Consider the molecule given on the left, and answer questions (i) to (vi) using chemical bonding theories that we have learnt in class (valence bond theory, hybridization).

Note that the lone pairs are not shown in the structure:

i. Consider the carbon atoms marked in as ***a***, ***b***, ***c***, ***d***, and ***e*** in the molecule, and determine **hybridization** at each of the carbon centers. (2.5 points)

Atom	Hybridization
Carbon (<i>a</i>)	
Carbon (<i>b</i>)	
Carbon (<i>c</i>)	
Carbon (<i>d</i>)	
Carbon (<i>e</i>)	

ii. Now, consider the sulfur atom (S) marked as ***f***, nitrogen (N) atom (***g***), and oxygen(O) atoms (***h*** and ***i***), and determine their **hybridization**. (2 points)

Atom	Hybridization
Sulfur (<i>f</i>)	
Nitrogen (<i>g</i>)	
Oxygen (<i>h</i>)	
Oxygen (<i>i</i>)	

iii. **Describe** all the bonds formed by the carbon atoms ***a*** and ***b***, using sigma and pi designation as appropriate. (2 points)

Atom	Bonds formed (use sigma and pi as appropriate)
Carbon (<i>a</i>)	
Carbon (<i>b</i>)	

iv. How many total sp^3 -s sigma bonds are in the molecule? (1 point)

v. Determine **electronic** and **molecular geometry** at carbon (C) labeled ***e*** and sulfur (S) labeled ***f*** centres. (2 points)

Atom	Electronic Geometry	Molecular Geometry
Carbon (<i>e</i>)		
Sulfur (<i>f</i>)		

vi. Circle the value from the choices given below, that describes the **C-S-C bond angle at sulfur** (S) marked ***f***: (0.5 points)

>90° to <109.5°

109.5°

>109.5 to <120°

120°

>120° to <180°


Question 8 12 points

Using the **valence shell orbitals only** draw a **complete** molecular orbital (MO) diagram (with no s-p mixing) for the molecule **PS**, in the space provided below. **(6 points)**

Assume that 3s and 3p atomic orbitals will form molecular orbitals like those formed by the combination of 2s and 2p atomic orbitals that we learnt in class, and there is no s-p mixing.

Show **energy arrow**, **atomic orbitals** involved, **atom labels**, the **molecular orbitals** that are formed, **filled electrons** in the MO diagram etc. Make sure to label all the atomic and molecular orbitals.

Using the MO diagram above, answer the following questions:

- (i) Which one of the following will have the highest bond order? Circle the correct choice
 A) PS^{1-} B) PS C) PS^{1+}
- (ii) Which one of the following will have the longest bond length? Circle the correct choice
 A) PS^{1-} B) PS C) PS^{1+}
- (iii) Which of the following will be diamagnetic? Circle the correct choice.
 A) PS^{1-} B) PS C) PS^{1+}
- (iv) Draw Lewis dot structure of the molecule with the shortest bond length (based on the MO diagram) from among the three (PS^{1-} , PS, PS^{1+}), and indicate the bond order calculated from the Lewis dot structure. **(3 points)**

Lewis Dot Structure

Bond Order



AE1F1D38-04A7-4B39-88AC-A5BBBB9CFA94

midterm-exam-93388

#186 Page 10 of 10

SCRAP PAGE

**CHEMISTRY 110 –Midterm Exam****Thursday November 7, 2024, 6:30 – 8:30 pm****Examiners: Pallavi Sirjoosingh and Ashok Kakkar**

First name (please write as legibly as possible within the boxes)

--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--

Last name

--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--

Student ID number

--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--

This booklet has 8 questions on 10 pages (The last page can be used as a scrap page).

The staple must NOT be removed from the question booklet. Your intact question booklet must be returned at the end of the exam.

A separate booklet contains a periodic table, and datasheet.

Write your answers directly under the questions in the space provided. Do not write on the QR codes or near the top edges (as it may be cut off).

Please write **legibly and dark enough** for the scan to be readable. For students who use English as a second language, dictionaries are permitted. Calculators are allowed.

DO NOT ASK QUESTIONS OF THE INVIGILATORS about interpretations of the exam questions. If you have a concern about a question, please summarize it at the top of this page. All concerns will be evaluated after the exam. GOOD LUCK!

**Question 1 8 points** (*Assume all transitions are possible*)

a. 6 points Calculate the **energy (in kJ/mol)** required to ionize (completely remove) the electron from the **n=3** level of **He⁺ ion** (He: Helium). Show your work including equations. (3 sig. figs.)

b. 2 points

(I) Select the largest (longest) wavelength from the three choices below:

- i. Wavelength of photon to *completely remove* an electron from *1s orbital of Hydrogen atom*
- ii. Wavelength of photon to *completely remove* an electron from *1s orbital of Boron atom (B)*
- iii. Wavelength of photon to *completely remove* an electron from *1s orbital of Boron cation (B⁺)*

i or ii or iii

(II) Select the smallest (shortest) wavelength from the three choices below (*for H atom*):

- i. Wavelength of photon *emitted* for an electron transitioning from *2p to 1s orbital*
- ii. Wavelength of photon *absorbed* for an electron transitioning from *2s to 4p orbital*
- iii. Wavelength of photon *absorbed* for an electron transitioning from *1s to 3p orbital*

i or ii or iii



Question 2 8 points



A Chem110 student plots the graph shown on the left, from data obtained for the kinetic energy (eV) of the ejected electron versus frequency (s^{-1}) of the incident light, for an unknown metal. ($1\text{eV} = 1.6022 \times 10^{-19} \text{ J}$)

Based on the plot, answer the questions **a to c** below:

a. 1 point On the plot above, label the **threshold frequency**.

b. 2 points What is the value of the work potential (in J) of the unknown metal?

c. 3 points What is the *longest* wavelength (in nm) required to eject an electron from this metal's surface? Show your work including equation(s) used. (3 sig figs)

d. 2 points In the space provided below, plot the following

(Assume frequency is *above* threshold frequency):

- i. *Kinetic energy* of an ejected electron versus the *intensity* of the incident light (left)
- ii. *Number of electrons* ejected versus the *frequency* of the incident light (right)



**Question 3** **6 points****a. 2 points**

In the ground state of Ge (Germanium; $Z = 32$) atom, **name** the following orbitals:

i. Orbital containing the electron that requires the maximum energy to be removed.

ii. Orbital containing the electron that experiences the minimum effective nuclear charge.

b. 2 points

An unknown anion (A^-) has the following *excited* electronic configuration: $[Ar] 4s^2 3d^3 4p^1$

(Ar: Argon; $Z = 18$)

Give the **condensed ground state electronic configuration** of the atom, **A** and the cation, **A^{2+}** .
Show your work. (Assume all electronic transitions are possible)

Electronic Configuration of A:

Electronic Configuration of A^{2+} :

c. 2 points

In a single atom, determine how many electrons can have the following set of quantum numbers?

(i) $n = 5; l = 3$

In a single atom, determine how many electrons can have the following set of quantum numbers?

(i) $n = 3; l = 2; m_s = +1/2$

**Question 4** **8 points**

Rank the following from 1 to 4 in **increasing** order. Provide the ranking in the boxes provided.

Note: Atomic number (Z) are provided for relevant atoms

a) Increasing (lowest to highest) order of *ionization energy*: (1= lowest; 4 = highest)

Mg

Mg: Z = 12

Rb

Rb: Z = 37

Cs

Cs: Z = 55

K

K : Z = 19

b) Increasing (lowest to highest) order of *bond angle*: (1= lowest; 4 = highest)

H₂OCCl₄XeF₄

HCN

c) Increasing (smallest to largest) order of *ionic size* (1= smallest; 4 = largest)

S²⁻

S: Z = 16

O²⁻

O: Z = 8

Be²⁺

Be: Z = 4

Na⁺

Na: Z = 11

d) Increasing (lowest to highest) order of *bond energy* (1= lowest; 4 = highest)

P-F

P-Cl

P-I

P-Br

**Question 5** **12 points**

a. 9 points For the molecules HClO_4 and TeF_6 , draw the most stable *Lewis structure* showing appropriate shape and *dashed/wedged bonds*, based on VSEPR. Determine the **molecular geometry** around the central atom. For HClO_4 , label the *most polar bond*.



b. 3 points (Note: Lone pairs are not shown in structures below)

(i) For the three *resonance* structures (1, 2 and 3) shown below, answer the following question:



(I) Circle the *most contributing* resonance structure.

Structure 1

Structure 2

Structure 3

(ii) For the three structures given (4, 5, and 6) below, answer the following questions:



(I) Circle all the structures that represent *resonance structures* of each other:

Structure 4

Structure 5

Structure 6

(II) Among the resonance structures, circle the *most contributing* resonance structure.

Structure 4

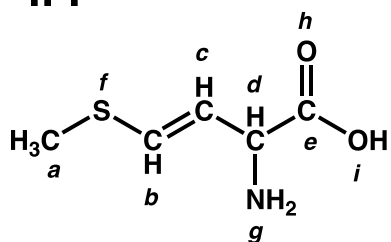
Structure 5

Structure 6

**Question 6** **6 points**

Using the data given below, calculate the **lattice energy** for SrO(s).

	ΔH° (kJ/mol)
$\text{Sr}(s) \longrightarrow \text{Sr}(g)$	+180
$\text{Sr}(g) \longrightarrow \text{Sr}^{2+}(g) + 2e^-$	+1468
$\text{O}_2(g) \longrightarrow 2\text{O}(g)$	+500
$\text{O}(g) + 2e^- \longrightarrow \text{O}^{2-}(g)$	+600
$\text{Sr}(s) + \frac{1}{2}\text{O}_2(g) \longrightarrow \text{SrO}(s)$	-550

**Question 7 10 points**

Consider the molecule given on the left, and answer questions (i) to (vi) using chemical bonding theories that we have learnt in class (valence bond theory, hybridization).

Note that the lone pairs are not shown in the structure:

i. Consider the carbon atoms marked in as ***a***, ***b***, ***c***, ***d***, and ***e*** in the molecule, and determine **hybridization** at each of the carbon centers. (2.5 points)

Atom	Hybridization
Carbon (<i>a</i>)	
Carbon (<i>b</i>)	
Carbon (<i>c</i>)	
Carbon (<i>d</i>)	
Carbon (<i>e</i>)	

ii. Now, consider the sulfur atom (S) marked as ***f***, nitrogen (N) atom (***g***), and oxygen(O) atoms (***h*** and ***i***), and determine their **hybridization**. (2 points)

Atom	Hybridization
Sulfur (<i>f</i>)	
Nitrogen (<i>g</i>)	
Oxygen (<i>h</i>)	
Oxygen (<i>i</i>)	

iii. **Describe** all the bonds formed by the carbon atoms ***a*** and ***b***, using sigma and pi designation as appropriate. (2 points)

Atom	Bonds formed (use sigma and pi as appropriate)
Carbon (<i>a</i>)	
Carbon (<i>b</i>)	

iv. How many total sp^3 -s sigma bonds are in the molecule? (1 point)

v. Determine **electronic** and **molecular geometry** at carbon (C) labeled ***e*** and sulfur (S) labeled ***f*** centres. (2 points)

Atom	Electronic Geometry	Molecular Geometry
Carbon (<i>e</i>)		
Sulfur (<i>f</i>)		

vi. Circle the value from the choices given below, that describes the **C-S-C bond angle at sulfur** (S) marked ***f***: (0.5 points)

>90° to <109.5°

109.5°

>109.5 to <120°

120°

>120° to <180°


Question 8 12 points

Using the **valence shell orbitals only** draw a **complete** molecular orbital (MO) diagram (with no s-p mixing) for the molecule **PS**, in the space provided below. **(6 points)**

Assume that 3s and 3p atomic orbitals will form molecular orbitals like those formed by the combination of 2s and 2p atomic orbitals that we learnt in class, and there is no s-p mixing.

Show **energy arrow**, **atomic orbitals** involved, **atom labels**, the **molecular orbitals** that are formed, **filled electrons** in the MO diagram etc. Make sure to label all the atomic and molecular orbitals.

Using the MO diagram above, answer the following questions:

- (i) Which one of the following will have the highest bond order? Circle the correct choice
 A) PS^{1-} B) PS C) PS^{1+}
- (ii) Which one of the following will have the longest bond length? Circle the correct choice
 A) PS^{1-} B) PS C) PS^{1+}
- (iii) Which of the following will be diamagnetic? Circle the correct choice.
 A) PS^{1-} B) PS C) PS^{1+}
- (iv) Draw Lewis dot structure of the molecule with the shortest bond length (based on the MO diagram) from among the three (PS^{1-} , PS, PS^{1+}), and indicate the bond order calculated from the Lewis dot structure. **(3 points)**

Lewis Dot Structure

Bond Order



CFDDC1A0-A0AD-48BD-BD12-FAA0EAAF19DD

midterm-exam-93388

#187 Page 10 of 10

SCRAP PAGE

**CHEMISTRY 110 –Midterm Exam****Thursday November 7, 2024, 6:30 – 8:30 pm**

Examiners: Pallavi Sirjoosingh and Ashok Kakkar

First name (please write as legibly as possible within the boxes)

--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--

Last name

--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--

Student ID number

--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--

This booklet has 8 questions on 10 pages (The last page can be used as a scrap page).

The staple must NOT be removed from the question booklet. Your intact question booklet must be returned at the end of the exam.

A separate booklet contains a periodic table, and datasheet.

Write your answers directly under the questions in the space provided. Do not write on the QR codes or near the top edges (as it may be cut off).

Please write **legibly and dark enough** for the scan to be readable. For students who use English as a second language, dictionaries are permitted. Calculators are allowed.

DO NOT ASK QUESTIONS OF THE INVIGILATORS about interpretations of the exam questions. If you have a concern about a question, please summarize it at the top of this page. All concerns will be evaluated after the exam. GOOD LUCK!

**Question 1 8 points** (*Assume all transitions are possible*)

a. 6 points Calculate the **energy (in kJ/mol)** required to ionize (completely remove) the electron from the **n=3** level of **He⁺ ion** (He: Helium). Show your work including equations. (3 sig. figs.)

b. 2 points

(I) Select the largest (longest) wavelength from the three choices below:

- i. Wavelength of photon to *completely remove* an electron from *1s orbital of Hydrogen atom*
- ii. Wavelength of photon to *completely remove* an electron from *1s orbital of Boron atom (B)*
- iii. Wavelength of photon to *completely remove* an electron from *1s orbital of Boron cation (B⁺)*

i or ii or iii

(II) Select the smallest (shortest) wavelength from the three choices below (*for H atom*):

- i. Wavelength of photon *emitted* for an electron transitioning from *2p to 1s orbital*
- ii. Wavelength of photon *absorbed* for an electron transitioning from *2s to 4p orbital*
- iii. Wavelength of photon *absorbed* for an electron transitioning from *1s to 3p orbital*

i or ii or iii



Question 2 8 points



A Chem110 student plots the graph shown on the left, from data obtained for the kinetic energy (eV) of the ejected electron versus frequency (s^{-1}) of the incident light, for an unknown metal. ($1\text{eV} = 1.6022 \times 10^{-19} \text{ J}$)

Based on the plot, answer the questions **a to c** below:

a. 1 point On the plot above, label the **threshold frequency**.

b. 2 points What is the value of the work potential (in J) of the unknown metal?

c. 3 points What is the *longest* wavelength (in nm) required to eject an electron from this metal's surface? Show your work including equation(s) used. (3 sig figs)

d. 2 points In the space provided below, plot the following

(Assume frequency is *above* threshold frequency):

- i. *Kinetic energy* of an ejected electron versus the *intensity* of the incident light (left)
- ii. *Number of electrons* ejected versus the *frequency* of the incident light (right)



**Question 3** **6 points****a. 2 points**

In the ground state of Ge (Germanium; $Z = 32$) atom, **name** the following orbitals:

i. Orbital containing the electron that requires the maximum energy to be removed.

ii. Orbital containing the electron that experiences the minimum effective nuclear charge.

b. 2 points

An unknown anion (A^-) has the following *excited* electronic configuration: $[Ar] 4s^2 3d^3 4p^1$

(Ar: Argon; $Z = 18$)

Give the **condensed ground state electronic configuration** of the atom, **A** and the cation, **A^{2+}** .
Show your work. (Assume all electronic transitions are possible)

Electronic Configuration of A:

Electronic Configuration of A^{2+} :

c. 2 points

In a single atom, determine how many electrons can have the following set of quantum numbers?

(i) $n = 5; l = 3$

In a single atom, determine how many electrons can have the following set of quantum numbers?

(i) $n = 3; l = 2; m_s = +1/2$

**Question 4** **8 points**

Rank the following from 1 to 4 in **increasing** order. Provide the ranking in the boxes provided.

Note: Atomic number (Z) are provided for relevant atoms

a) Increasing (lowest to highest) order of *ionization energy*: (1= lowest; 4 = highest)

Mg

Mg: Z = 12

Rb

Rb: Z = 37

Cs

Cs: Z = 55

K

K : Z = 19

b) Increasing (lowest to highest) order of *bond angle*: (1= lowest; 4 = highest)

H₂OCCl₄XeF₄

HCN

c) Increasing (smallest to largest) order of *ionic size* (1= smallest; 4 = largest)

S²⁻

S: Z = 16

O²⁻

O: Z = 8

Be²⁺

Be: Z = 4

Na⁺

Na: Z = 11

d) Increasing (lowest to highest) order of *bond energy* (1= lowest; 4 = highest)

P-F

P-Cl

P-I

P-Br

**Question 5** **12 points**

a. 9 points For the molecules HClO_4 and TeF_6 , draw the most stable *Lewis structure* showing appropriate shape and *dashed/wedged bonds*, based on VSEPR. Determine the **molecular geometry** around the central atom. For HClO_4 , label the *most polar bond*.



b. 3 points (Note: Lone pairs are not shown in structures below)

(i) For the three *resonance* structures (1, 2 and 3) shown below, answer the following question:



(I) Circle the *most contributing* resonance structure.

Structure 1

Structure 2

Structure 3

(ii) For the three structures given (4, 5, and 6) below, answer the following questions:



(I) Circle all the structures that represent *resonance structures* of each other:

Structure 4

Structure 5

Structure 6

(II) Among the resonance structures, circle the *most contributing* resonance structure.

Structure 4

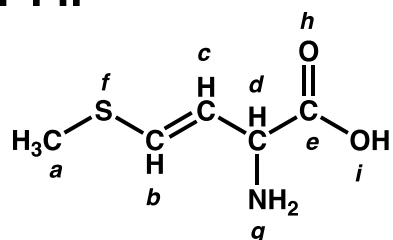
Structure 5

Structure 6

**Question 6** **6 points**

Using the data given below, calculate the **lattice energy** for SrO(s).

	ΔH° (kJ/mol)
$\text{Sr}(s) \longrightarrow \text{Sr}(g)$	+180
$\text{Sr}(g) \longrightarrow \text{Sr}^{2+}(g) + 2e^-$	+1468
$\text{O}_2(g) \longrightarrow 2\text{O}(g)$	+500
$\text{O}(g) + 2e^- \longrightarrow \text{O}^{2-}(g)$	+600
$\text{Sr}(s) + \frac{1}{2}\text{O}_2(g) \longrightarrow \text{SrO}(s)$	-550

**Question 7 10 points**

Consider the molecule given on the left, and answer questions (i) to (vi) using chemical bonding theories that we have learnt in class (valence bond theory, hybridization).

Note that the lone pairs are not shown in the structure:

i. Consider the carbon atoms marked in as ***a***, ***b***, ***c***, ***d***, and ***e*** in the molecule, and determine **hybridization** at each of the carbon centers. (2.5 points)

Atom	Hybridization
Carbon (<i>a</i>)	
Carbon (<i>b</i>)	
Carbon (<i>c</i>)	
Carbon (<i>d</i>)	
Carbon (<i>e</i>)	

ii. Now, consider the sulfur atom (S) marked as ***f***, nitrogen (N) atom (***g***), and oxygen(O) atoms (***h*** and ***i***), and determine their **hybridization**. (2 points)

Atom	Hybridization
Sulfur (<i>f</i>)	
Nitrogen (<i>g</i>)	
Oxygen (<i>h</i>)	
Oxygen (<i>i</i>)	

iii. **Describe** all the bonds formed by the carbon atoms ***a*** and ***b***, using sigma and pi designation as appropriate. (2 points)

Atom	Bonds formed (use sigma and pi as appropriate)
Carbon (<i>a</i>)	
Carbon (<i>b</i>)	

iv. How many total sp^3 -s sigma bonds are in the molecule? (1 point)

v. Determine **electronic** and **molecular geometry** at carbon (C) labeled ***e*** and sulfur (S) labeled ***f*** centres. (2 points)

Atom	Electronic Geometry	Molecular Geometry
Carbon (<i>e</i>)		
Sulfur (<i>f</i>)		

vi. Circle the value from the choices given below, that describes the **C-S-C bond angle at sulfur** (S) marked ***f***: (0.5 points)

>90° to <109.5° 109.5° >109.5 to <120° 120° >120° to <180°


Question 8 12 points

Using the **valence shell orbitals only** draw a **complete** molecular orbital (MO) diagram (with no s-p mixing) for the molecule **PS**, in the space provided below. **(6 points)**

Assume that 3s and 3p atomic orbitals will form molecular orbitals like those formed by the combination of 2s and 2p atomic orbitals that we learnt in class, and there is no s-p mixing.

Show **energy arrow**, **atomic orbitals** involved, **atom labels**, the **molecular orbitals** that are formed, **filled electrons** in the MO diagram etc. Make sure to label all the atomic and molecular orbitals.

Using the MO diagram above, answer the following questions:

- (i) Which one of the following will have the highest bond order? Circle the correct choice
 A) PS^{1-} B) PS C) PS^{1+}
- (ii) Which one of the following will have the longest bond length? Circle the correct choice
 A) PS^{1-} B) PS C) PS^{1+}
- (iii) Which of the following will be diamagnetic? Circle the correct choice.
 A) PS^{1-} B) PS C) PS^{1+}
- (iv) Draw Lewis dot structure of the molecule with the shortest bond length (based on the MO diagram) from among the three (PS^{1-} , PS, PS^{1+}), and indicate the bond order calculated from the Lewis dot structure. **(3 points)**

Lewis Dot Structure

Bond Order



81CCD49B-CEC1-436A-BAAF-1D5E72CB8D1E

midterm-exam-93388

#188 Page 10 of 10

SCRAP PAGE

**CHEMISTRY 110 –Midterm Exam****Thursday November 7, 2024, 6:30 – 8:30 pm****Examiners: Pallavi Sirjoosingh and Ashok Kakkar**

First name (please write as legibly as possible within the boxes)

--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--

Last name

--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--

Student ID number

--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--

This booklet has 8 questions on 10 pages (The last page can be used as a scrap page).

The staple must NOT be removed from the question booklet. Your intact question booklet must be returned at the end of the exam.

A separate booklet contains a periodic table, and datasheet.

Write your answers directly under the questions in the space provided. Do not write on the QR codes or near the top edges (as it may be cut off).

Please write **legibly and dark enough** for the scan to be readable. For students who use English as a second language, dictionaries are permitted. Calculators are allowed.

DO NOT ASK QUESTIONS OF THE INVIGILATORS about interpretations of the exam questions. If you have a concern about a question, please summarize it at the top of this page. All concerns will be evaluated after the exam. GOOD LUCK!

**Question 1 8 points** (*Assume all transitions are possible*)

a. 6 points Calculate the **energy (in kJ/mol)** required to ionize (completely remove) the electron from the **n=3** level of **He⁺ ion** (He: Helium). Show your work including equations. (3 sig. figs.)

b. 2 points

(I) Select the largest (longest) wavelength from the three choices below:

- i. Wavelength of photon to *completely remove* an electron from *1s orbital of Hydrogen atom*
- ii. Wavelength of photon to *completely remove* an electron from *1s orbital of Boron atom (B)*
- iii. Wavelength of photon to *completely remove* an electron from *1s orbital of Boron cation (B⁺)*

i or ii or iii

(II) Select the smallest (shortest) wavelength from the three choices below (*for H atom*):

- i. Wavelength of photon *emitted* for an electron transitioning from *2p to 1s orbital*
- ii. Wavelength of photon *absorbed* for an electron transitioning from *2s to 4p orbital*
- iii. Wavelength of photon *absorbed* for an electron transitioning from *1s to 3p orbital*

i or ii or iii



Question 2 8 points



A Chem110 student plots the graph shown on the left, from data obtained for the kinetic energy (eV) of the ejected electron versus frequency (s^{-1}) of the incident light, for an unknown metal. ($1\text{eV} = 1.6022 \times 10^{-19} \text{ J}$)

Based on the plot, answer the questions **a to c** below:

a. 1 point On the plot above, label the **threshold frequency**.

b. 2 points What is the value of the work potential (in J) of the unknown metal?

c. 3 points What is the *longest* wavelength (in nm) required to eject an electron from this metal's surface? Show your work including equation(s) used. (3 sig figs)

d. 2 points In the space provided below, plot the following

(Assume frequency is *above* threshold frequency):

- i. *Kinetic energy* of an ejected electron versus the *intensity* of the incident light (left)
- ii. *Number of electrons* ejected versus the *frequency* of the incident light (right)



**Question 3** **6 points****a. 2 points**

In the ground state of Ge (Germanium; $Z = 32$) atom, **name** the following orbitals:

i. Orbital containing the electron that requires the maximum energy to be removed.

ii. Orbital containing the electron that experiences the minimum effective nuclear charge.

b. 2 points

An unknown anion (A^-) has the following *excited* electronic configuration: $[Ar] 4s^2 3d^3 4p^1$

(Ar: Argon; $Z = 18$)

Give the **condensed ground state electronic configuration** of the atom, **A** and the cation, **A^{2+}** .
Show your work. (Assume all electronic transitions are possible)

Electronic Configuration of A:

Electronic Configuration of A^{2+} :

c. 2 points

In a single atom, determine how many electrons can have the following set of quantum numbers?

(i) $n = 5; l = 3$

In a single atom, determine how many electrons can have the following set of quantum numbers?

(i) $n = 3; l = 2; m_s = +1/2$

**Question 4** **8 points**

Rank the following from 1 to 4 in **increasing** order. Provide the ranking in the boxes provided.

Note: Atomic number (Z) are provided for relevant atoms

a) Increasing (lowest to highest) order of *ionization energy*: (1= lowest; 4 = highest)

Mg

Mg: Z = 12

Rb

Rb: Z = 37

Cs

Cs: Z = 55

K

K : Z = 19

b) Increasing (lowest to highest) order of *bond angle*: (1= lowest; 4 = highest)

H₂OCCl₄XeF₄

HCN

c) Increasing (smallest to largest) order of *ionic size* (1= smallest; 4 = largest)

S²⁻

S: Z = 16

O²⁻

O: Z = 8

Be²⁺

Be: Z = 4

Na⁺

Na: Z = 11

d) Increasing (lowest to highest) order of *bond energy* (1= lowest; 4 = highest)

P-F

P-Cl

P-I

P-Br

**Question 5** **12 points**

a. 9 points For the molecules HClO_4 and TeF_6 , draw the most stable *Lewis structure* showing appropriate shape and *dashed/wedged bonds*, based on VSEPR. Determine the **molecular geometry** around the central atom. For HClO_4 , label the *most polar bond*.



b. 3 points (Note: Lone pairs are not shown in structures below)

(i) For the three *resonance* structures (1, 2 and 3) shown below, answer the following question:



(I) Circle the *most contributing* resonance structure.

Structure 1

Structure 2

Structure 3

(ii) For the three structures given (4, 5, and 6) below, answer the following questions:



(I) Circle all the structures that represent *resonance structures* of each other:

Structure 4

Structure 5

Structure 6

(II) Among the resonance structures, circle the *most contributing* resonance structure.

Structure 4

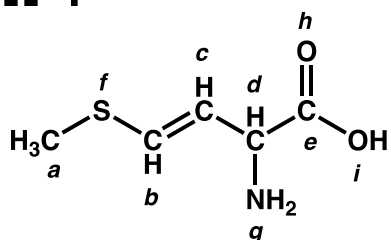
Structure 5

Structure 6

**Question 6** **6 points**

Using the data given below, calculate the **lattice energy** for SrO(s).

	ΔH° (kJ/mol)
$\text{Sr}(s) \longrightarrow \text{Sr}(g)$	+180
$\text{Sr}(g) \longrightarrow \text{Sr}^{2+}(g) + 2e^-$	+1468
$\text{O}_2(g) \longrightarrow 2\text{O}(g)$	+500
$\text{O}(g) + 2e^- \longrightarrow \text{O}^{2-}(g)$	+600
$\text{Sr}(s) + \frac{1}{2}\text{O}_2(g) \longrightarrow \text{SrO}(s)$	-550

**Question 7 10 points**

Consider the molecule given on the left, and answer questions (i) to (vi) using chemical bonding theories that we have learnt in class (valence bond theory, hybridization).

Note that the lone pairs are not shown in the structure:

i. Consider the carbon atoms marked in as ***a***, ***b***, ***c***, ***d***, and ***e*** in the molecule, and determine **hybridization** at each of the carbon centers. (2.5 points)

Atom	Hybridization
Carbon (<i>a</i>)	
Carbon (<i>b</i>)	
Carbon (<i>c</i>)	
Carbon (<i>d</i>)	
Carbon (<i>e</i>)	

ii. Now, consider the sulfur atom (S) marked as ***f***, nitrogen (N) atom (***g***), and oxygen(O) atoms (***h*** and ***i***), and determine their **hybridization**. (2 points)

Atom	Hybridization
Sulfur (<i>f</i>)	
Nitrogen (<i>g</i>)	
Oxygen (<i>h</i>)	
Oxygen (<i>i</i>)	

iii. **Describe** all the bonds formed by the carbon atoms ***a*** and ***b***, using sigma and pi designation as appropriate. (2 points)

Atom	Bonds formed (use sigma and pi as appropriate)
Carbon (<i>a</i>)	
Carbon (<i>b</i>)	

iv. How many total sp^3 -s sigma bonds are in the molecule? (1 point)

v. Determine **electronic** and **molecular geometry** at carbon (C) labeled ***e*** and sulfur (S) labeled ***f*** centres. (2 points)

Atom	Electronic Geometry	Molecular Geometry
Carbon (<i>e</i>)		
Sulfur (<i>f</i>)		

vi. Circle the value from the choices given below, that describes the **C-S-C bond angle at sulfur** (S) marked ***f***: (0.5 points)

>90° to <109.5°

109.5°

>109.5 to <120°

120°

>120° to <180°

**Question 8 12 points**

Using the **valence shell orbitals only** draw a **complete** molecular orbital (MO) diagram (with no s-p mixing) for the molecule **PS**, in the space provided below. **(6 points)**

Assume that 3s and 3p atomic orbitals will form molecular orbitals like those formed by the combination of 2s and 2p atomic orbitals that we learnt in class, and there is no s-p mixing.

Show **energy arrow**, **atomic orbitals** involved, **atom labels**, the **molecular orbitals** that are formed, **filled electrons** in the MO diagram etc. Make sure to label all the atomic and molecular orbitals.

Using the MO diagram above, answer the following questions:

- (i) Which one of the following will have the highest bond order? Circle the correct choice
A) PS^{1-} B) PS C) PS^{1+}
- (ii) Which one of the following will have the longest bond length? Circle the correct choice
A) PS^{1-} B) PS C) PS^{1+}
- (iii) Which of the following will be diamagnetic? Circle the correct choice.
A) PS^{1-} B) PS C) PS^{1+}
- (iv) Draw Lewis dot structure of the molecule with the shortest bond length (based on the MO diagram) from among the three (PS^{1-} , PS, PS^{1+}), and indicate the bond order calculated from the Lewis dot structure. **(3 points)**

Lewis Dot Structure

Bond Order



DEEAD447-C87A-4CF2-9F58-F11D4DBA1102

midterm-exam-93388

#189 Page 10 of 10

SCRAP PAGE

**CHEMISTRY 110 –Midterm Exam****Thursday November 7, 2024, 6:30 – 8:30 pm****Examiners: Pallavi Sirjoosingh and Ashok Kakkar**

First name (please write as legibly as possible within the boxes)

--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--

Last name

--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--

Student ID number

--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--

This booklet has 8 questions on 10 pages (The last page can be used as a scrap page).

The staple must NOT be removed from the question booklet. Your intact question booklet must be returned at the end of the exam.

A separate booklet contains a periodic table, and datasheet.

Write your answers directly under the questions in the space provided. Do not write on the QR codes or near the top edges (as it may be cut off).

Please write **legibly and dark enough** for the scan to be readable. For students who use English as a second language, dictionaries are permitted. Calculators are allowed.

DO NOT ASK QUESTIONS OF THE INVIGILATORS about interpretations of the exam questions. If you have a concern about a question, please summarize it at the top of this page. All concerns will be evaluated after the exam. GOOD LUCK!

**Question 1 8 points** (*Assume all transitions are possible*)

a. 6 points Calculate the **energy (in kJ/mol)** required to ionize (completely remove) the electron from the **n=3** level of **He⁺ ion** (He: Helium). Show your work including equations. (3 sig. figs.)

b. 2 points

(I) Select the largest (longest) wavelength from the three choices below:

- i. Wavelength of photon to *completely remove* an electron from *1s orbital of Hydrogen atom*
- ii. Wavelength of photon to *completely remove* an electron from *1s orbital of Boron atom (B)*
- iii. Wavelength of photon to *completely remove* an electron from *1s orbital of Boron cation (B⁺)*

i or ii or iii

(II) Select the smallest (shortest) wavelength from the three choices below (*for H atom*):

- i. Wavelength of photon *emitted* for an electron transitioning from *2p to 1s orbital*
- ii. Wavelength of photon *absorbed* for an electron transitioning from *2s to 4p orbital*
- iii. Wavelength of photon *absorbed* for an electron transitioning from *1s to 3p orbital*

i or ii or iii



Question 2 8 points



A Chem110 student plots the graph shown on the left, from data obtained for the kinetic energy (eV) of the ejected electron versus frequency (s^{-1}) of the incident light, for an unknown metal. ($1\text{eV} = 1.6022 \times 10^{-19} \text{ J}$)

Based on the plot, answer the questions **a to c** below:

a. 1 point On the plot above, label the **threshold frequency**.

b. 2 points What is the value of the work potential (in J) of the unknown metal?

c. 3 points What is the *longest* wavelength (in nm) required to eject an electron from this metal's surface? Show your work including equation(s) used. (3 sig figs)

d. 2 points In the space provided below, plot the following

(Assume frequency is *above* threshold frequency):

- i. *Kinetic energy* of an ejected electron versus the *intensity* of the incident light (left)
- ii. *Number of electrons* ejected versus the *frequency* of the incident light (right)



**Question 3** **6 points****a. 2 points**

In the ground state of Ge (Germanium; $Z = 32$) atom, **name** the following orbitals:

i. Orbital containing the electron that requires the maximum energy to be removed.

ii. Orbital containing the electron that experiences the minimum effective nuclear charge.

b. 2 points

An unknown anion (A^-) has the following *excited* electronic configuration: $[Ar] 4s^2 3d^3 4p^1$

(Ar: Argon; $Z = 18$)

Give the **condensed ground state electronic configuration** of the atom, **A** and the cation, **A^{2+}** .
Show your work. (Assume all electronic transitions are possible)

Electronic Configuration of A:

Electronic Configuration of A^{2+} :

c. 2 points

In a single atom, determine how many electrons can have the following set of quantum numbers?

(i) $n = 5; l = 3$

In a single atom, determine how many electrons can have the following set of quantum numbers?

(i) $n = 3; l = 2; m_s = +1/2$

**Question 4** **8 points**

Rank the following from 1 to 4 in **increasing** order. Provide the ranking in the boxes provided.

Note: Atomic number (Z) are provided for relevant atoms

a) Increasing (lowest to highest) order of *ionization energy*: (1= lowest; 4 = highest)

Mg

Mg: Z = 12

Rb

Rb: Z = 37

Cs

Cs: Z = 55

K

K : Z = 19

b) Increasing (lowest to highest) order of *bond angle*: (1= lowest; 4 = highest)

H₂OCCl₄XeF₄

HCN

c) Increasing (smallest to largest) order of *ionic size* (1= smallest; 4 = largest)

S²⁻

S: Z = 16

O²⁻

O: Z = 8

Be²⁺

Be: Z = 4

Na⁺

Na: Z = 11

d) Increasing (lowest to highest) order of *bond energy* (1= lowest; 4 = highest)

P-F

P-Cl

P-I

P-Br

**Question 5** **12 points**

a. 9 points For the molecules HClO_4 and TeF_6 , draw the most stable *Lewis structure* showing appropriate shape and *dashed/wedged bonds*, based on VSEPR. Determine the **molecular geometry** around the central atom. For HClO_4 , label the *most polar bond*.



b. 3 points (Note: Lone pairs are not shown in structures below)

(i) For the three *resonance* structures (1, 2 and 3) shown below, answer the following question:



(I) Circle the *most contributing* resonance structure.

Structure 1

Structure 2

Structure 3

(ii) For the three structures given (4, 5, and 6) below, answer the following questions:



(I) Circle all the structures that represent *resonance structures* of each other:

Structure 4

Structure 5

Structure 6

(II) Among the resonance structures, circle the *most contributing* resonance structure.

Structure 4

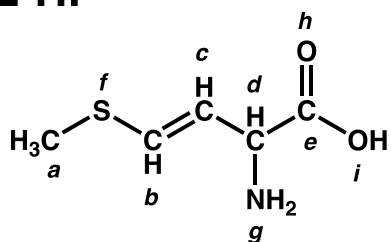
Structure 5

Structure 6

**Question 6** **6 points**

Using the data given below, calculate the **lattice energy** for SrO(s).

	ΔH° (kJ/mol)
$\text{Sr}(s) \longrightarrow \text{Sr}(g)$	+180
$\text{Sr}(g) \longrightarrow \text{Sr}^{2+}(g) + 2e^-$	+1468
$\text{O}_2(g) \longrightarrow 2\text{O}(g)$	+500
$\text{O}(g) + 2e^- \longrightarrow \text{O}^{2-}(g)$	+600
$\text{Sr}(s) + \frac{1}{2}\text{O}_2(g) \longrightarrow \text{SrO}(s)$	-550

**Question 7 10 points**

Consider the molecule given on the left, and answer questions (i) to (vi) using chemical bonding theories that we have learnt in class (valence bond theory, hybridization).

Note that the lone pairs are not shown in the structure:

i. Consider the carbon atoms marked in as ***a***, ***b***, ***c***, ***d***, and ***e*** in the molecule, and determine **hybridization** at each of the carbon centers. (2.5 points)

Atom	Hybridization
Carbon (<i>a</i>)	
Carbon (<i>b</i>)	
Carbon (<i>c</i>)	
Carbon (<i>d</i>)	
Carbon (<i>e</i>)	

ii. Now, consider the sulfur atom (S) marked as ***f***, nitrogen (N) atom (***g***), and oxygen(O) atoms (***h*** and ***i***), and determine their **hybridization**. (2 points)

Atom	Hybridization
Sulfur (<i>f</i>)	
Nitrogen (<i>g</i>)	
Oxygen (<i>h</i>)	
Oxygen (<i>i</i>)	

iii. **Describe** all the bonds formed by the carbon atoms ***a*** and ***b***, using sigma and pi designation as appropriate. (2 points)

Atom	Bonds formed (use sigma and pi as appropriate)
Carbon (<i>a</i>)	
Carbon (<i>b</i>)	

iv. How many total sp^3 -s sigma bonds are in the molecule? (1 point)

v. Determine **electronic** and **molecular geometry** at carbon (C) labeled ***e*** and sulfur (S) labeled ***f*** centres. (2 points)

Atom	Electronic Geometry	Molecular Geometry
Carbon (<i>e</i>)		
Sulfur (<i>f</i>)		

vi. Circle the value from the choices given below, that describes the **C-S-C bond angle at sulfur** (S) marked ***f***: (0.5 points)

>90° to <109.5° 109.5° >109.5 to <120° 120° >120° to <180°


Question 8 12 points

Using the **valence shell orbitals only** draw a **complete** molecular orbital (MO) diagram (with no s-p mixing) for the molecule **PS**, in the space provided below. **(6 points)**

Assume that 3s and 3p atomic orbitals will form molecular orbitals like those formed by the combination of 2s and 2p atomic orbitals that we learnt in class, and there is no s-p mixing.

Show **energy arrow**, **atomic orbitals** involved, **atom labels**, the **molecular orbitals** that are formed, **filled electrons** in the MO diagram etc. Make sure to **label all the atomic and molecular orbitals**.

Using the MO diagram above, answer the following questions:

- (i) Which one of the following will have the highest bond order? Circle the correct choice
 A) PS^{1-} B) PS C) PS^{1+}
- (ii) Which one of the following will have the longest bond length? Circle the correct choice
 A) PS^{1-} B) PS C) PS^{1+}
- (iii) Which of the following will be diamagnetic? Circle the correct choice.
 A) PS^{1-} B) PS C) PS^{1+}
- (iv) Draw Lewis dot structure of the molecule with the shortest bond length (based on the MO diagram) from among the three (PS^{1-} , PS, PS^{1+}), and indicate the bond order calculated from the Lewis dot structure. **(3 points)**

Lewis Dot Structure

Bond Order



6A6048E2-E602-4E37-8C60-F55EF3E6C28B

midterm-exam-93388

#190 Page 10 of 10

SCRAP PAGE

**CHEMISTRY 110 –Midterm Exam****Thursday November 7, 2024, 6:30 – 8:30 pm****Examiners: Pallavi Sirjoosingh and Ashok Kakkar**

First name (please write as legibly as possible within the boxes)

--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--

Last name

--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--

Student ID number

--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--

This booklet has 8 questions on 10 pages (The last page can be used as a scrap page).

The staple must NOT be removed from the question booklet. Your intact question booklet must be returned at the end of the exam.

A separate booklet contains a periodic table, and datasheet.

Write your answers directly under the questions in the space provided. Do not write on the QR codes or near the top edges (as it may be cut off).

Please write **legibly and dark enough** for the scan to be readable. For students who use English as a second language, dictionaries are permitted. Calculators are allowed.

DO NOT ASK QUESTIONS OF THE INVIGILATORS about interpretations of the exam questions. If you have a concern about a question, please summarize it at the top of this page. All concerns will be evaluated after the exam. GOOD LUCK!

**Question 1 8 points** (*Assume all transitions are possible*)

a. 6 points Calculate the **energy (in kJ/mol)** required to ionize (completely remove) the electron from the **$n=3$** level of **He^+ ion** (He: Helium). Show your work including equations. (3 sig. figs.)

b. 2 points

(I) Select the largest (longest) wavelength from the three choices below:

- i. Wavelength of photon to *completely remove* an electron from *1s orbital of Hydrogen atom*
- ii. Wavelength of photon to *completely remove* an electron from *1s orbital of Boron atom (B)*
- iii. Wavelength of photon to *completely remove* an electron from *1s orbital of Boron cation (B^+)*

i or ii or iii

(II) Select the smallest (shortest) wavelength from the three choices below (*for H atom*):

- i. Wavelength of photon *emitted* for an electron transitioning from *2p to 1s orbital*
- ii. Wavelength of photon *absorbed* for an electron transitioning from *2s to 4p orbital*
- iii. Wavelength of photon *absorbed* for an electron transitioning from *1s to 3p orbital*

i or ii or iii



Question 2 8 points



A Chem110 student plots the graph shown on the left, from data obtained for the kinetic energy (eV) of the ejected electron versus frequency (s^{-1}) of the incident light, for an unknown metal. ($1\text{eV} = 1.6022 \times 10^{-19} \text{ J}$)

Based on the plot, answer the questions **a to c** below:

a. 1 point On the plot above, label the **threshold frequency**.

b. 2 points What is the value of the work potential (in J) of the unknown metal?

c. 3 points What is the *longest* wavelength (in nm) required to eject an electron from this metal's surface? Show your work including equation(s) used. (3 sig figs)

d. 2 points In the space provided below, plot the following

(Assume frequency is *above* threshold frequency):

- Kinetic energy* of an ejected electron versus the *intensity* of the incident light (left)
- Number of electrons* ejected versus the *frequency* of the incident light (right)



**Question 3** **6 points****a. 2 points**

In the ground state of Ge (Germanium; $Z = 32$) atom, **name** the following orbitals:

i. Orbital containing the electron that requires the maximum energy to be removed.

ii. Orbital containing the electron that experiences the minimum effective nuclear charge.

b. 2 points

An unknown anion (A^-) has the following *excited* electronic configuration: $[Ar] 4s^2 3d^3 4p^1$

(Ar: Argon; $Z = 18$)

Give the **condensed ground state electronic configuration** of the atom, **A** and the cation, **A^{2+}** .
Show your work. (Assume all electronic transitions are possible)

Electronic Configuration of A:

Electronic Configuration of A^{2+} :

c. 2 points

In a single atom, determine how many electrons can have the following set of quantum numbers?

(i) $n = 5; l = 3$

In a single atom, determine how many electrons can have the following set of quantum numbers?

(i) $n = 3; l = 2; m_s = +1/2$

**Question 4** **8 points**

Rank the following from 1 to 4 in **increasing** order. Provide the ranking in the boxes provided.

Note: Atomic number (Z) are provided for relevant atoms

a) Increasing (lowest to highest) order of *ionization energy*: (1= lowest; 4 = highest)

Mg

Mg: Z = 12

Rb

Rb: Z = 37

Cs

Cs: Z = 55

K

K : Z = 19

b) Increasing (lowest to highest) order of *bond angle*: (1= lowest; 4 = highest)

H₂OCCl₄XeF₄

HCN

c) Increasing (smallest to largest) order of *ionic size* (1= smallest; 4 = largest)

S²⁻

S: Z = 16

O²⁻

O: Z = 8

Be²⁺

Be: Z = 4

Na⁺

Na: Z = 11

d) Increasing (lowest to highest) order of *bond energy* (1= lowest; 4 = highest)

P-F

P-Cl

P-I

P-Br

**Question 5** **12 points**

a. 9 points For the molecules HClO_4 and TeF_6 , draw the most stable *Lewis structure* showing appropriate shape and *dashed/wedged bonds*, based on VSEPR. Determine the **molecular geometry** around the central atom. For HClO_4 , label the *most polar bond*.

HClO_4 (there are no O-O bonds)

Label the most polar bond

Molecular Geometry around Cl

TeF_6

Molecular Geometry around Te

b. 3 points (Note: Lone pairs are not shown in structures below)

(i) For the three *resonance* structures (1, 2 and 3) shown below, answer the following question:



(I) Circle the *most contributing* resonance structure.

Structure 1

Structure 2

Structure 3

(ii) For the three structures given (4, 5, and 6) below, answer the following questions:



(I) Circle all the structures that represent *resonance structures* of each other:

Structure 4

Structure 5

Structure 6

(II) Among the resonance structures, circle the *most contributing* resonance structure.

Structure 4

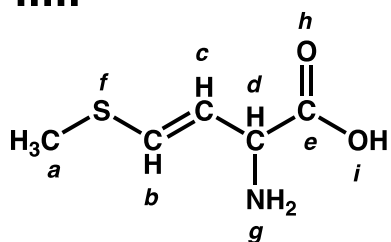
Structure 5

Structure 6

**Question 6** **6 points**

Using the data given below, calculate the **lattice energy** for SrO(s).

	ΔH° (kJ/mol)
$\text{Sr}(s) \longrightarrow \text{Sr}(g)$	+180
$\text{Sr}(g) \longrightarrow \text{Sr}^{2+}(g) + 2e^-$	+1468
$\text{O}_2(g) \longrightarrow 2\text{O}(g)$	+500
$\text{O}(g) + 2e^- \longrightarrow \text{O}^{2-}(g)$	+600
$\text{Sr}(s) + \frac{1}{2}\text{O}_2(g) \longrightarrow \text{SrO}(s)$	-550

**Question 7 10 points**

Consider the molecule given on the left, and answer questions (i) to (vi) using chemical bonding theories that we have learnt in class (valence bond theory, hybridization).

Note that the lone pairs are not shown in the structure:

i. Consider the carbon atoms marked in as ***a***, ***b***, ***c***, ***d***, and ***e*** in the molecule, and determine **hybridization** at each of the carbon centers. (2.5 points)

Atom	Hybridization
Carbon (<i>a</i>)	
Carbon (<i>b</i>)	
Carbon (<i>c</i>)	
Carbon (<i>d</i>)	
Carbon (<i>e</i>)	

ii. Now, consider the sulfur atom (S) marked as ***f***, nitrogen (N) atom (***g***), and oxygen(O) atoms (***h*** and ***i***), and determine their **hybridization**. (2 points)

Atom	Hybridization
Sulfur (<i>f</i>)	
Nitrogen (<i>g</i>)	
Oxygen (<i>h</i>)	
Oxygen (<i>i</i>)	

iii. **Describe** all the bonds formed by the carbon atoms ***a*** and ***b***, using sigma and pi designation as appropriate. (2 points)

Atom	Bonds formed (use sigma and pi as appropriate)
Carbon (<i>a</i>)	
Carbon (<i>b</i>)	

iv. How many total sp^3 -s sigma bonds are in the molecule? (1 point)

v. Determine **electronic** and **molecular geometry** at carbon (C) labeled ***e*** and sulfur (S) labeled ***f*** centres. (2 points)

Atom	Electronic Geometry	Molecular Geometry
Carbon (<i>e</i>)		
Sulfur (<i>f</i>)		

vi. Circle the value from the choices given below, that describes the **C-S-C bond angle at sulfur** (S) marked ***f***: (0.5 points)

>90° to <109.5°

109.5°

>109.5 to <120°

120°

>120° to <180°

**Question 8 12 points**

Using the **valence shell orbitals only** draw a **complete** molecular orbital (MO) diagram (with no s-p mixing) for the molecule **PS**, in the space provided below. **(6 points)**

Assume that 3s and 3p atomic orbitals will form molecular orbitals like those formed by the combination of 2s and 2p atomic orbitals that we learnt in class, and there is no s-p mixing.

Show **energy arrow**, **atomic orbitals** involved, **atom labels**, the **molecular orbitals** that are formed, **filled electrons** in the MO diagram etc. Make sure to label all the atomic and molecular orbitals.

Using the MO diagram above, answer the following questions:

- (i) Which one of the following will have the highest bond order? Circle the correct choice
A) PS^{1-} B) PS C) PS^{1+}
- (ii) Which one of the following will have the longest bond length? Circle the correct choice
A) PS^{1-} B) PS C) PS^{1+}
- (iii) Which of the following will be diamagnetic? Circle the correct choice.
A) PS^{1-} B) PS C) PS^{1+}
- (iv) Draw Lewis dot structure of the molecule with the shortest bond length (based on the MO diagram) from among the three (PS^{1-} , PS, PS^{1+}), and indicate the bond order calculated from the Lewis dot structure. **(3 points)**

Lewis Dot Structure

Bond Order



71CCD993-2662-451A-8422-75F3FFBE77DA

midterm-exam-93388

#191 Page 10 of 10

SCRAP PAGE

**CHEMISTRY 110 –Midterm Exam****Thursday November 7, 2024, 6:30 – 8:30 pm****Examiners: Pallavi Sirjoosingh and Ashok Kakkar**

First name (please write as legibly as possible within the boxes)

--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--

Last name

--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--

Student ID number

--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--

This booklet has 8 questions on 10 pages (The last page can be used as a scrap page).

The staple must NOT be removed from the question booklet. Your intact question booklet must be returned at the end of the exam.

A separate booklet contains a periodic table, and datasheet.

Write your answers directly under the questions in the space provided. Do not write on the QR codes or near the top edges (as it may be cut off).

Please write **legibly and dark enough** for the scan to be readable. For students who use English as a second language, dictionaries are permitted. Calculators are allowed.

DO NOT ASK QUESTIONS OF THE INVIGILATORS about interpretations of the exam questions. If you have a concern about a question, please summarize it at the top of this page. All concerns will be evaluated after the exam. GOOD LUCK!

**Question 1 8 points** (*Assume all transitions are possible*)

a. 6 points Calculate the **energy (in kJ/mol)** required to ionize (completely remove) the electron from the **n=3** level of **He⁺ ion** (He: Helium). Show your work including equations. (3 sig. figs.)

b. 2 points

(I) Select the largest (longest) wavelength from the three choices below:

- i. Wavelength of photon to *completely remove* an electron from *1s orbital of Hydrogen atom*
- ii. Wavelength of photon to *completely remove* an electron from *1s orbital of Boron atom (B)*
- iii. Wavelength of photon to *completely remove* an electron from *1s orbital of Boron cation (B⁺)*

i or ii or iii

(II) Select the smallest (shortest) wavelength from the three choices below (*for H atom*):

- i. Wavelength of photon *emitted* for an electron transitioning from *2p to 1s orbital*
- ii. Wavelength of photon *absorbed* for an electron transitioning from *2s to 4p orbital*
- iii. Wavelength of photon *absorbed* for an electron transitioning from *1s to 3p orbital*

i or ii or iii



Question 2 8 points



A Chem110 student plots the graph shown on the left, from data obtained for the kinetic energy (eV) of the ejected electron versus frequency (s^{-1}) of the incident light, for an unknown metal. ($1\text{eV} = 1.6022 \times 10^{-19} \text{ J}$)

Based on the plot, answer the questions **a to c** below:

a. 1 point On the plot above, label the **threshold frequency**.

b. 2 points What is the value of the work potential (in J) of the unknown metal?

c. 3 points What is the *longest* wavelength (in nm) required to eject an electron from this metal's surface? Show your work including equation(s) used. (3 sig figs)

d. 2 points In the space provided below, plot the following

(Assume frequency is *above* threshold frequency):

- Kinetic energy* of an ejected electron versus the *intensity* of the incident light (left)
- Number of electrons* ejected versus the *frequency* of the incident light (right)



**Question 3** **6 points****a. 2 points**

In the ground state of Ge (Germanium; $Z = 32$) atom, **name** the following orbitals:

i. Orbital containing the electron that requires the maximum energy to be removed.

ii. Orbital containing the electron that experiences the minimum effective nuclear charge.

b. 2 points

An unknown anion (A^-) has the following *excited* electronic configuration: $[Ar] 4s^2 3d^3 4p^1$

(Ar: Argon; $Z = 18$)

Give the **condensed ground state electronic configuration** of the atom, **A** and the cation, **A^{2+}** .
Show your work. (Assume all electronic transitions are possible)

Electronic Configuration of A:

Electronic Configuration of A^{2+} :

c. 2 points

In a single atom, determine how many electrons can have the following set of quantum numbers?

(i) $n = 5; l = 3$

In a single atom, determine how many electrons can have the following set of quantum numbers?

(i) $n = 3; l = 2; m_s = +1/2$

**Question 4** **8 points**

Rank the following from 1 to 4 in **increasing** order. Provide the ranking in the boxes provided.

Note: Atomic number (Z) are provided for relevant atoms

a) Increasing (lowest to highest) order of *ionization energy*: (1= lowest; 4 = highest)

Mg

Mg: Z = 12

Rb

Rb: Z = 37

Cs

Cs: Z = 55

K

K : Z = 19

b) Increasing (lowest to highest) order of *bond angle*: (1= lowest; 4 = highest)

H₂OCCl₄XeF₄

HCN

c) Increasing (smallest to largest) order of *ionic size* (1= smallest; 4 = largest)

S²⁻

S: Z = 16

O²⁻

O: Z = 8

Be²⁺

Be: Z = 4

Na⁺

Na: Z = 11

d) Increasing (lowest to highest) order of *bond energy* (1= lowest; 4 = highest)

P-F

P-Cl

P-I

P-Br

**Question 5** **12 points**

a. 9 points For the molecules HClO_4 and TeF_6 , draw the most stable *Lewis structure* showing appropriate shape and *dashed/wedged bonds*, based on VSEPR. Determine the **molecular geometry** around the central atom. For HClO_4 , label the *most polar bond*.



b. 3 points (Note: Lone pairs are not shown in structures below)

(i) For the three *resonance* structures (1, 2 and 3) shown below, answer the following question:



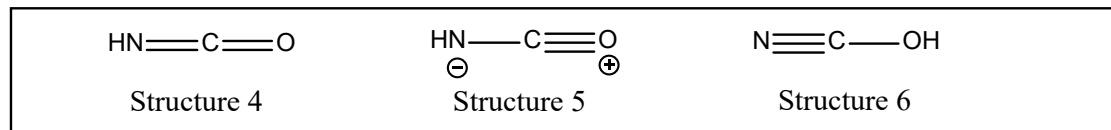
(I) Circle the *most contributing* resonance structure.

Structure 1

Structure 2

Structure 3

(ii) For the three structures given (4, 5, and 6) below, answer the following questions:



(I) Circle all the structures that represent *resonance structures* of each other:

Structure 4

Structure 5

Structure 6

(II) Among the resonance structures, circle the *most contributing* resonance structure.

Structure 4

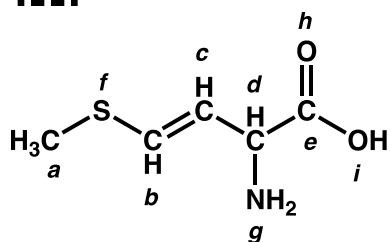
Structure 5

Structure 6

**Question 6** **6 points**

Using the data given below, calculate the **lattice energy** for SrO(s).

	ΔH° (kJ/mol)
$\text{Sr}(s) \longrightarrow \text{Sr}(g)$	+180
$\text{Sr}(g) \longrightarrow \text{Sr}^{2+}(g) + 2e^-$	+1468
$\text{O}_2(g) \longrightarrow 2\text{O}(g)$	+500
$\text{O}(g) + 2e^- \longrightarrow \text{O}^{2-}(g)$	+600
$\text{Sr}(s) + \frac{1}{2}\text{O}_2(g) \longrightarrow \text{SrO}(s)$	-550

**Question 7 10 points**

Consider the molecule given on the left, and answer questions (i) to (vi) using chemical bonding theories that we have learnt in class (valence bond theory, hybridization).

Note that the lone pairs are not shown in the structure:

i. Consider the carbon atoms marked in as ***a***, ***b***, ***c***, ***d***, and ***e*** in the molecule, and determine **hybridization** at each of the carbon centers. (2.5 points)

Atom	Hybridization
Carbon (<i>a</i>)	
Carbon (<i>b</i>)	
Carbon (<i>c</i>)	
Carbon (<i>d</i>)	
Carbon (<i>e</i>)	

ii. Now, consider the sulfur atom (S) marked as ***f***, nitrogen (N) atom (***g***), and oxygen(O) atoms (***h*** and ***i***), and determine their **hybridization**. (2 points)

Atom	Hybridization
Sulfur (<i>f</i>)	
Nitrogen (<i>g</i>)	
Oxygen (<i>h</i>)	
Oxygen (<i>i</i>)	

iii. **Describe** all the bonds formed by the carbon atoms ***a*** and ***b***, using sigma and pi designation as appropriate. (2 points)

Atom	Bonds formed (use sigma and pi as appropriate)
Carbon (<i>a</i>)	
Carbon (<i>b</i>)	

iv. How many total sp^3 -s sigma bonds are in the molecule? (1 point)

v. Determine **electronic** and **molecular geometry** at carbon (C) labeled ***e*** and sulfur (S) labeled ***f*** centres. (2 points)

Atom	Electronic Geometry	Molecular Geometry
Carbon (<i>e</i>)		
Sulfur (<i>f</i>)		

vi. Circle the value from the choices given below, that describes the **C-S-C bond angle at sulfur** (S) marked ***f***: (0.5 points)

>90° to <109.5° 109.5° >109.5 to <120° 120° >120° to <180°


Question 8 12 points

Using the **valence shell orbitals only** draw a **complete** molecular orbital (MO) diagram (with no s-p mixing) for the molecule **PS**, in the space provided below. **(6 points)**

Assume that 3s and 3p atomic orbitals will form molecular orbitals like those formed by the combination of 2s and 2p atomic orbitals that we learnt in class, and there is no s-p mixing.

Show **energy arrow**, **atomic orbitals** involved, **atom labels**, the **molecular orbitals** that are formed, **filled electrons** in the MO diagram etc. Make sure to label all the atomic and molecular orbitals.

Using the MO diagram above, answer the following questions:

- (i) Which one of the following will have the highest bond order? Circle the correct choice
 A) PS^{1-} B) PS C) PS^{1+}
- (ii) Which one of the following will have the longest bond length? Circle the correct choice
 A) PS^{1-} B) PS C) PS^{1+}
- (iii) Which of the following will be diamagnetic? Circle the correct choice.
 A) PS^{1-} B) PS C) PS^{1+}
- (iv) Draw Lewis dot structure of the molecule with the shortest bond length (based on the MO diagram) from among the three (PS^{1-} , PS, PS^{1+}), and indicate the bond order calculated from the Lewis dot structure. **(3 points)**

Lewis Dot Structure

Bond Order



F9E85199-857E-411D-96A9-A3252CF95BDC

midterm-exam-93388

#192 Page 10 of 10

SCRAP PAGE

**CHEMISTRY 110 –Midterm Exam****Thursday November 7, 2024, 6:30 – 8:30 pm**

Examiners: Pallavi Sirjoosingh and Ashok Kakkar

First name (please write as legibly as possible within the boxes)

--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--

Last name

--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--

Student ID number

--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--

This booklet has 8 questions on 10 pages (The last page can be used as a scrap page).

The staple must NOT be removed from the question booklet. Your intact question booklet must be returned at the end of the exam.

A separate booklet contains a periodic table, and datasheet.

Write your answers directly under the questions in the space provided. Do not write on the QR codes or near the top edges (as it may be cut off).

Please write **legibly and dark enough** for the scan to be readable. For students who use English as a second language, dictionaries are permitted. Calculators are allowed.

DO NOT ASK QUESTIONS OF THE INVIGILATORS about interpretations of the exam questions. If you have a concern about a question, please summarize it at the top of this page. All concerns will be evaluated after the exam. GOOD LUCK!

**Question 1 8 points** (*Assume all transitions are possible*)

a. 6 points Calculate the **energy (in kJ/mol)** required to ionize (completely remove) the electron from the **$n=3$** level of **He^+ ion** (He: Helium). Show your work including equations. (3 sig. figs.)

b. 2 points

(I) Select the largest (longest) wavelength from the three choices below:

- i. Wavelength of photon to *completely remove* an electron from *1s orbital of Hydrogen atom*
- ii. Wavelength of photon to *completely remove* an electron from *1s orbital of Boron atom (B)*
- iii. Wavelength of photon to *completely remove* an electron from *1s orbital of Boron cation (B^+)*

i or ii or iii

(II) Select the smallest (shortest) wavelength from the three choices below (*for H atom*):

- i. Wavelength of photon *emitted* for an electron transitioning from *2p to 1s orbital*
- ii. Wavelength of photon *absorbed* for an electron transitioning from *2s to 4p orbital*
- iii. Wavelength of photon *absorbed* for an electron transitioning from *1s to 3p orbital*

i or ii or iii



Question 2 8 points



A Chem110 student plots the graph shown on the left, from data obtained for the kinetic energy (eV) of the ejected electron versus frequency (s^{-1}) of the incident light, for an unknown metal. ($1\text{eV} = 1.6022 \times 10^{-19} \text{ J}$)

Based on the plot, answer the questions **a to c** below:

a. 1 point On the plot above, label the **threshold frequency**.

b. 2 points What is the value of the work potential (in J) of the unknown metal?

c. 3 points What is the *longest* wavelength (in nm) required to eject an electron from this metal's surface? Show your work including equation(s) used. (3 sig figs)

d. 2 points In the space provided below, plot the following

(Assume frequency is *above* threshold frequency):

- Kinetic energy* of an ejected electron versus the *intensity* of the incident light (left)
- Number of electrons* ejected versus the *frequency* of the incident light (right)



**Question 3** **6 points****a. 2 points**

In the ground state of Ge (Germanium; $Z = 32$) atom, **name** the following orbitals:

i. Orbital containing the electron that requires the maximum energy to be removed.

ii. Orbital containing the electron that experiences the minimum effective nuclear charge.

b. 2 points

An unknown anion (A^-) has the following *excited* electronic configuration: $[Ar] 4s^2 3d^3 4p^1$

(Ar: Argon; $Z = 18$)

Give the **condensed ground state electronic configuration** of the atom, **A** and the cation, **A^{2+}** .
Show your work. (Assume all electronic transitions are possible)

Electronic Configuration of A:

Electronic Configuration of A^{2+} :

c. 2 points

In a single atom, determine how many electrons can have the following set of quantum numbers?

(i) $n = 5; l = 3$

In a single atom, determine how many electrons can have the following set of quantum numbers?

(i) $n = 3; l = 2; m_s = +1/2$

**Question 4** **8 points**

Rank the following from 1 to 4 in **increasing** order. Provide the ranking in the boxes provided.

Note: Atomic number (Z) are provided for relevant atoms

a) Increasing (lowest to highest) order of *ionization energy*: (1= lowest; 4 = highest)

Mg

Mg: Z = 12

Rb

Rb: Z = 37

Cs

Cs: Z = 55

K

K : Z = 19

b) Increasing (lowest to highest) order of *bond angle*: (1= lowest; 4 = highest)

H₂OCCl₄XeF₄

HCN

c) Increasing (smallest to largest) order of *ionic size* (1= smallest; 4 = largest)

S²⁻

S: Z = 16

O²⁻

O: Z = 8

Be²⁺

Be: Z = 4

Na⁺

Na: Z = 11

d) Increasing (lowest to highest) order of *bond energy* (1= lowest; 4 = highest)

P-F

P-Cl

P-I

P-Br

**Question 5** **12 points**

a. 9 points For the molecules HClO_4 and TeF_6 , draw the most stable *Lewis structure* showing appropriate shape and *dashed/wedged bonds*, based on VSEPR. Determine the **molecular geometry** around the central atom. For HClO_4 , label the *most polar bond*.



b. 3 points (Note: Lone pairs are not shown in structures below)

(i) For the three *resonance* structures (1, 2 and 3) shown below, answer the following question:



(I) Circle the *most contributing* resonance structure.

Structure 1

Structure 2

Structure 3

(ii) For the three structures given (4, 5, and 6) below, answer the following questions:



(I) Circle all the structures that represent *resonance structures* of each other:

Structure 4

Structure 5

Structure 6

(II) Among the resonance structures, circle the *most contributing* resonance structure.

Structure 4

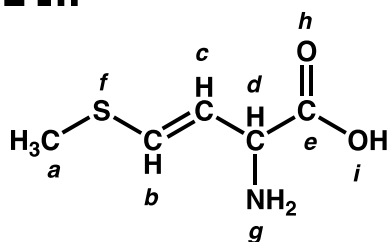
Structure 5

Structure 6

**Question 6** **6 points**

Using the data given below, calculate the **lattice energy** for SrO(s).

	ΔH° (kJ/mol)
$\text{Sr}(s) \longrightarrow \text{Sr}(g)$	+180
$\text{Sr}(g) \longrightarrow \text{Sr}^{2+}(g) + 2e^-$	+1468
$\text{O}_2(g) \longrightarrow 2\text{O}(g)$	+500
$\text{O}(g) + 2e^- \longrightarrow \text{O}^{2-}(g)$	+600
$\text{Sr}(s) + \frac{1}{2}\text{O}_2(g) \longrightarrow \text{SrO}(s)$	-550

**Question 7 10 points**

Consider the molecule given on the left, and answer questions (i) to (vi) using chemical bonding theories that we have learnt in class (valence bond theory, hybridization).

Note that the lone pairs are not shown in the structure:

i. Consider the carbon atoms marked in as ***a***, ***b***, ***c***, ***d***, and ***e*** in the molecule, and determine **hybridization** at each of the carbon centers. (2.5 points)

Atom	Hybridization
Carbon (<i>a</i>)	
Carbon (<i>b</i>)	
Carbon (<i>c</i>)	
Carbon (<i>d</i>)	
Carbon (<i>e</i>)	

ii. Now, consider the sulfur atom (S) marked as ***f***, nitrogen (N) atom (***g***), and oxygen(O) atoms (***h*** and ***i***), and determine their **hybridization**. (2 points)

Atom	Hybridization
Sulfur (<i>f</i>)	
Nitrogen (<i>g</i>)	
Oxygen (<i>h</i>)	
Oxygen (<i>i</i>)	

iii. **Describe** all the bonds formed by the carbon atoms ***a*** and ***b***, using sigma and pi designation as appropriate. (2 points)

Atom	Bonds formed (use sigma and pi as appropriate)
Carbon (<i>a</i>)	
Carbon (<i>b</i>)	

iv. How many total sp^3 -s sigma bonds are in the molecule? (1 point)

v. Determine **electronic** and **molecular geometry** at carbon (C) labeled ***e*** and sulfur (S) labeled ***f*** centres. (2 points)

Atom	Electronic Geometry	Molecular Geometry
Carbon (<i>e</i>)		
Sulfur (<i>f</i>)		

vi. Circle the value from the choices given below, that describes the **C-S-C bond angle at sulfur** (S) marked ***f***: (0.5 points)

>90° to <109.5° 109.5° >109.5 to <120° 120° >120° to <180°


Question 8 12 points

Using the **valence shell orbitals only** draw a **complete** molecular orbital (MO) diagram (with no s-p mixing) for the molecule **PS**, in the space provided below. **(6 points)**

Assume that 3s and 3p atomic orbitals will form molecular orbitals like those formed by the combination of 2s and 2p atomic orbitals that we learnt in class, and there is no s-p mixing.

Show **energy arrow**, **atomic orbitals** involved, **atom labels**, the **molecular orbitals** that are formed, **filled electrons** in the MO diagram etc. Make sure to **label all the atomic and molecular orbitals**.

Using the MO diagram above, answer the following questions:

- (i) Which one of the following will have the highest bond order? Circle the correct choice
 A) PS^{1-} B) PS C) PS^{1+}
- (ii) Which one of the following will have the longest bond length? Circle the correct choice
 A) PS^{1-} B) PS C) PS^{1+}
- (iii) Which of the following will be diamagnetic? Circle the correct choice.
 A) PS^{1-} B) PS C) PS^{1+}
- (iv) Draw Lewis dot structure of the molecule with the shortest bond length (based on the MO diagram) from among the three (PS^{1-} , PS, PS^{1+}), and indicate the bond order calculated from the Lewis dot structure. **(3 points)**

Lewis Dot Structure

Bond Order



25248006-DEDB-4F30-BD13-5138D9FFCA2C

midterm-exam-93388

#193 Page 10 of 10

SCRAP PAGE

**CHEMISTRY 110 –Midterm Exam****Thursday November 7, 2024, 6:30 – 8:30 pm**

Examiners: Pallavi Sirjoosingh and Ashok Kakkar

First name (please write as legibly as possible within the boxes)

--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--

Last name

--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--

Student ID number

--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--

This booklet has 8 questions on 10 pages (The last page can be used as a scrap page).

The staple must NOT be removed from the question booklet. Your intact question booklet must be returned at the end of the exam.

A separate booklet contains a periodic table, and datasheet.

Write your answers directly under the questions in the space provided. Do not write on the QR codes or near the top edges (as it may be cut off).

Please write **legibly and dark enough** for the scan to be readable. For students who use English as a second language, dictionaries are permitted. Calculators are allowed.

DO NOT ASK QUESTIONS OF THE INVIGILATORS about interpretations of the exam questions. If you have a concern about a question, please summarize it at the top of this page. All concerns will be evaluated after the exam. GOOD LUCK!

**Question 1 8 points** (*Assume all transitions are possible*)

a. 6 points Calculate the **energy (in kJ/mol)** required to ionize (completely remove) the electron from the **$n=3$** level of **He^+ ion** (He: Helium). Show your work including equations. (3 sig. figs.)

b. 2 points

(I) Select the largest (longest) wavelength from the three choices below:

- i. Wavelength of photon to *completely remove* an electron from *1s orbital of Hydrogen atom*
- ii. Wavelength of photon to *completely remove* an electron from *1s orbital of Boron atom (B)*
- iii. Wavelength of photon to *completely remove* an electron from *1s orbital of Boron cation (B^+)*

i or ii or iii

(II) Select the smallest (shortest) wavelength from the three choices below (*for H atom*):

- i. Wavelength of photon *emitted* for an electron transitioning from *2p to 1s orbital*
- ii. Wavelength of photon *absorbed* for an electron transitioning from *2s to 4p orbital*
- iii. Wavelength of photon *absorbed* for an electron transitioning from *1s to 3p orbital*

i or ii or iii



Question 2 8 points



A Chem110 student plots the graph shown on the left, from data obtained for the kinetic energy (eV) of the ejected electron versus frequency (s^{-1}) of the incident light, for an unknown metal. ($1\text{eV} = 1.6022 \times 10^{-19} \text{ J}$)

Based on the plot, answer the questions **a to c** below:

a. 1 point On the plot above, label the **threshold frequency**.

b. 2 points What is the value of the work potential (in J) of the unknown metal?

c. 3 points What is the *longest* wavelength (in nm) required to eject an electron from this metal's surface? Show your work including equation(s) used. (3 sig figs)

d. 2 points In the space provided below, plot the following

(Assume frequency is *above* threshold frequency):

- Kinetic energy* of an ejected electron versus the *intensity* of the incident light (left)
- Number of electrons* ejected versus the *frequency* of the incident light (right)



**Question 3** **6 points****a. 2 points**

In the ground state of Ge (Germanium; $Z = 32$) atom, **name** the following orbitals:

i. Orbital containing the electron that requires the maximum energy to be removed.

ii. Orbital containing the electron that experiences the minimum effective nuclear charge.

b. 2 points

An unknown anion (A^-) has the following *excited* electronic configuration: $[Ar] 4s^2 3d^3 4p^1$

(Ar: Argon; $Z = 18$)

Give the **condensed ground state electronic configuration** of the atom, **A** and the cation, **A^{2+}** .
Show your work. (Assume all electronic transitions are possible)

Electronic Configuration of A:

Electronic Configuration of A^{2+} :

c. 2 points

In a single atom, determine how many electrons can have the following set of quantum numbers?

(i) $n = 5; l = 3$

In a single atom, determine how many electrons can have the following set of quantum numbers?

(i) $n = 3; l = 2; m_s = +1/2$

**Question 4** **8 points**

Rank the following from 1 to 4 in **increasing** order. Provide the ranking in the boxes provided.

Note: Atomic number (Z) are provided for relevant atoms

a) Increasing (lowest to highest) order of *ionization energy*: (1= lowest; 4 = highest)

Mg

Mg: Z = 12

Rb

Rb: Z = 37

Cs

Cs: Z = 55

K

K : Z = 19

b) Increasing (lowest to highest) order of *bond angle*: (1= lowest; 4 = highest)

H₂OCCl₄XeF₄

HCN

c) Increasing (smallest to largest) order of *ionic size* (1= smallest; 4 = largest)

S²⁻

S: Z = 16

O²⁻

O: Z = 8

Be²⁺

Be: Z = 4

Na⁺

Na: Z = 11

d) Increasing (lowest to highest) order of *bond energy* (1= lowest; 4 = highest)

P-F

P-Cl

P-I

P-Br

**Question 5** **12 points**

a. 9 points For the molecules HClO_4 and TeF_6 , draw the most stable *Lewis structure* showing appropriate shape and *dashed/wedged bonds*, based on VSEPR. Determine the **molecular geometry** around the central atom. For HClO_4 , label the *most polar bond*.



b. 3 points (Note: Lone pairs are not shown in structures below)

(i) For the three *resonance* structures (1, 2 and 3) shown below, answer the following question:



(I) Circle the *most contributing* resonance structure.

Structure 1

Structure 2

Structure 3

(ii) For the three structures given (4, 5, and 6) below, answer the following questions:



(I) Circle all the structures that represent *resonance structures* of each other:

Structure 4

Structure 5

Structure 6

(II) Among the resonance structures, circle the *most contributing* resonance structure.

Structure 4

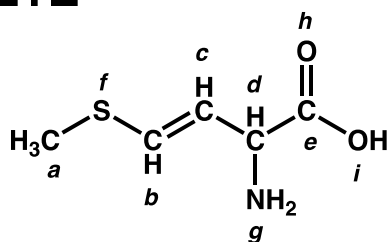
Structure 5

Structure 6

**Question 6** **6 points**

Using the data given below, calculate the **lattice energy** for SrO(s).

	ΔH° (kJ/mol)
$\text{Sr}(s) \longrightarrow \text{Sr}(g)$	+180
$\text{Sr}(g) \longrightarrow \text{Sr}^{2+}(g) + 2e^-$	+1468
$\text{O}_2(g) \longrightarrow 2\text{O}(g)$	+500
$\text{O}(g) + 2e^- \longrightarrow \text{O}^{2-}(g)$	+600
$\text{Sr}(s) + \frac{1}{2}\text{O}_2(g) \longrightarrow \text{SrO}(s)$	-550

**Question 7 10 points**

Consider the molecule given on the left, and answer questions (i) to (vi) using chemical bonding theories that we have learnt in class (valence bond theory, hybridization).

Note that the lone pairs are not shown in the structure:

i. Consider the carbon atoms marked in as ***a***, ***b***, ***c***, ***d***, and ***e*** in the molecule, and determine **hybridization** at each of the carbon centers. (2.5 points)

Atom	Hybridization
Carbon (<i>a</i>)	
Carbon (<i>b</i>)	
Carbon (<i>c</i>)	
Carbon (<i>d</i>)	
Carbon (<i>e</i>)	

ii. Now, consider the sulfur atom (S) marked as ***f***, nitrogen (N) atom (***g***), and oxygen(O) atoms (***h*** and ***i***), and determine their **hybridization**. (2 points)

Atom	Hybridization
Sulfur (<i>f</i>)	
Nitrogen (<i>g</i>)	
Oxygen (<i>h</i>)	
Oxygen (<i>i</i>)	

iii. **Describe** all the bonds formed by the carbon atoms ***a*** and ***b***, using sigma and pi designation as appropriate. (2 points)

Atom	Bonds formed (use sigma and pi as appropriate)
Carbon (<i>a</i>)	
Carbon (<i>b</i>)	

iv. How many total sp^3 -s sigma bonds are in the molecule? (1 point)

v. Determine **electronic** and **molecular geometry** at carbon (C) labeled ***e*** and sulfur (S) labeled ***f*** centres. (2 points)

Atom	Electronic Geometry	Molecular Geometry
Carbon (<i>e</i>)		
Sulfur (<i>f</i>)		

vi. Circle the value from the choices given below, that describes the **C-S-C bond angle at sulfur** (S) marked ***f***: (0.5 points)

>90° to <109.5°

109.5°

>109.5 to <120°

120°

>120° to <180°


Question 8 12 points

Using the **valence shell orbitals only** draw a **complete** molecular orbital (MO) diagram (with no s-p mixing) for the molecule **PS**, in the space provided below. **(6 points)**

Assume that 3s and 3p atomic orbitals will form molecular orbitals like those formed by the combination of 2s and 2p atomic orbitals that we learnt in class, and there is no s-p mixing.

Show **energy arrow**, **atomic orbitals** involved, **atom labels**, the **molecular orbitals** that are formed, **filled electrons** in the MO diagram etc. Make sure to label all the atomic and molecular orbitals.

Using the MO diagram above, answer the following questions:

- (i) Which one of the following will have the highest bond order? Circle the correct choice
 A) PS^{1-} B) PS C) PS^{1+}
- (ii) Which one of the following will have the longest bond length? Circle the correct choice
 A) PS^{1-} B) PS C) PS^{1+}
- (iii) Which of the following will be diamagnetic? Circle the correct choice.
 A) PS^{1-} B) PS C) PS^{1+}
- (iv) Draw Lewis dot structure of the molecule with the shortest bond length (based on the MO diagram) from among the three (PS^{1-} , PS, PS^{1+}), and indicate the bond order calculated from the Lewis dot structure. **(3 points)**

Lewis Dot Structure

Bond Order



74F53B8F-2EB8-4A66-9467-CBB5F92F34D8

midterm-exam-93388

#194 Page 10 of 10

SCRAP PAGE

**CHEMISTRY 110 –Midterm Exam****Thursday November 7, 2024, 6:30 – 8:30 pm****Examiners: Pallavi Sirjoosingh and Ashok Kakkar**

First name (please write as legibly as possible within the boxes)

--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--

Last name

--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--

Student ID number

--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--

This booklet has 8 questions on 10 pages (The last page can be used as a scrap page).

The staple must NOT be removed from the question booklet. Your intact question booklet must be returned at the end of the exam.

A separate booklet contains a periodic table, and datasheet.

Write your answers directly under the questions in the space provided. Do not write on the QR codes or near the top edges (as it may be cut off).

Please write **legibly and dark enough** for the scan to be readable. For students who use English as a second language, dictionaries are permitted. Calculators are allowed.

DO NOT ASK QUESTIONS OF THE INVIGILATORS about interpretations of the exam questions. If you have a concern about a question, please summarize it at the top of this page. All concerns will be evaluated after the exam. GOOD LUCK!

**Question 1 8 points** (*Assume all transitions are possible*)

a. 6 points Calculate the **energy (in kJ/mol)** required to ionize (completely remove) the electron from the **n=3** level of **He⁺ ion** (He: Helium). Show your work including equations. (3 sig. figs.)

b. 2 points

(I) Select the largest (longest) wavelength from the three choices below:

- i. Wavelength of photon to *completely remove* an electron from *1s orbital of Hydrogen atom*
- ii. Wavelength of photon to *completely remove* an electron from *1s orbital of Boron atom (B)*
- iii. Wavelength of photon to *completely remove* an electron from *1s orbital of Boron cation (B⁺)*

i or ii or iii

(II) Select the smallest (shortest) wavelength from the three choices below (*for H atom*):

- i. Wavelength of photon *emitted* for an electron transitioning from *2p to 1s orbital*
- ii. Wavelength of photon *absorbed* for an electron transitioning from *2s to 4p orbital*
- iii. Wavelength of photon *absorbed* for an electron transitioning from *1s to 3p orbital*

i or ii or iii



Question 2 8 points



A Chem110 student plots the graph shown on the left, from data obtained for the kinetic energy (eV) of the ejected electron versus frequency (s^{-1}) of the incident light, for an unknown metal. ($1\text{eV} = 1.6022 \times 10^{-19} \text{ J}$)

Based on the plot, answer the questions **a to c** below:

a. 1 point On the plot above, label the **threshold frequency**.

b. 2 points What is the value of the work potential (in J) of the unknown metal?

c. 3 points What is the *longest* wavelength (in nm) required to eject an electron from this metal's surface? Show your work including equation(s) used. (3 sig figs)

d. 2 points In the space provided below, plot the following

(Assume frequency is *above* threshold frequency):

- i. *Kinetic energy* of an ejected electron versus the *intensity* of the incident light (left)
- ii. *Number of electrons* ejected versus the *frequency* of the incident light (right)



**Question 3** **6 points****a. 2 points**

In the ground state of Ge (Germanium; $Z = 32$) atom, **name** the following orbitals:

i. Orbital containing the electron that requires the maximum energy to be removed.

ii. Orbital containing the electron that experiences the minimum effective nuclear charge.

b. 2 points

An unknown anion (A^-) has the following *excited* electronic configuration: $[Ar] 4s^2 3d^3 4p^1$

(Ar: Argon; $Z = 18$)

Give the **condensed ground state electronic configuration** of the atom, **A** and the cation, **A^{2+}** .
Show your work. (Assume all electronic transitions are possible)

Electronic Configuration of A:

Electronic Configuration of A^{2+} :

c. 2 points

In a single atom, determine how many electrons can have the following set of quantum numbers?

(i) $n = 5; l = 3$

In a single atom, determine how many electrons can have the following set of quantum numbers?

(i) $n = 3; l = 2; m_s = +1/2$

**Question 4** **8 points**

Rank the following from 1 to 4 in **increasing** order. Provide the ranking in the boxes provided.

Note: Atomic number (Z) are provided for relevant atoms

a) Increasing (lowest to highest) order of *ionization energy*: (1= lowest; 4 = highest)

Mg

Mg: Z = 12

Rb

Rb: Z = 37

Cs

Cs: Z = 55

K

K : Z = 19

b) Increasing (lowest to highest) order of *bond angle*: (1= lowest; 4 = highest)

H₂OCCl₄XeF₄

HCN

c) Increasing (smallest to largest) order of *ionic size* (1= smallest; 4 = largest)

S²⁻

S: Z = 16

O²⁻

O: Z = 8

Be²⁺

Be: Z = 4

Na⁺

Na: Z = 11

d) Increasing (lowest to highest) order of *bond energy* (1= lowest; 4 = highest)

P-F

P-Cl

P-I

P-Br

**Question 5** **12 points**

a. 9 points For the molecules HClO_4 and TeF_6 , draw the most stable *Lewis structure* showing appropriate shape and *dashed/wedged bonds*, based on VSEPR. Determine the **molecular geometry** around the central atom. For HClO_4 , label the *most polar bond*.



b. 3 points (Note: Lone pairs are not shown in structures below)

(i) For the three *resonance* structures (1, 2 and 3) shown below, answer the following question:



(I) Circle the *most contributing* resonance structure.

Structure 1

Structure 2

Structure 3

(ii) For the three structures given (4, 5, and 6) below, answer the following questions:



(I) Circle all the structures that represent *resonance structures* of each other:

Structure 4

Structure 5

Structure 6

(II) Among the resonance structures, circle the *most contributing* resonance structure.

Structure 4

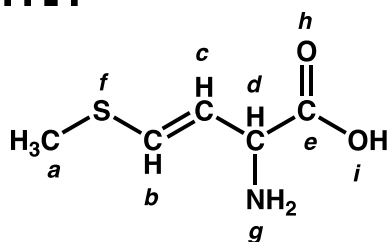
Structure 5

Structure 6

**Question 6** **6 points**

Using the data given below, calculate the **lattice energy** for SrO(s).

	ΔH° (kJ/mol)
$\text{Sr}(s) \longrightarrow \text{Sr}(g)$	+180
$\text{Sr}(g) \longrightarrow \text{Sr}^{2+}(g) + 2e^-$	+1468
$\text{O}_2(g) \longrightarrow 2\text{O}(g)$	+500
$\text{O}(g) + 2e^- \longrightarrow \text{O}^{2-}(g)$	+600
$\text{Sr}(s) + \frac{1}{2}\text{O}_2(g) \longrightarrow \text{SrO}(s)$	-550

**Question 7 10 points**

Consider the molecule given on the left, and answer questions (i) to (vi) using chemical bonding theories that we have learnt in class (valence bond theory, hybridization).

Note that the lone pairs are not shown in the structure:

i. Consider the carbon atoms marked in as ***a***, ***b***, ***c***, ***d***, and ***e*** in the molecule, and determine **hybridization** at each of the carbon centers. (2.5 points)

Atom	Hybridization
Carbon (<i>a</i>)	
Carbon (<i>b</i>)	
Carbon (<i>c</i>)	
Carbon (<i>d</i>)	
Carbon (<i>e</i>)	

ii. Now, consider the sulfur atom (S) marked as ***f***, nitrogen (N) atom (***g***), and oxygen(O) atoms (***h*** and ***i***), and determine their **hybridization**. (2 points)

Atom	Hybridization
Sulfur (<i>f</i>)	
Nitrogen (<i>g</i>)	
Oxygen (<i>h</i>)	
Oxygen (<i>i</i>)	

iii. **Describe** all the bonds formed by the carbon atoms ***a*** and ***b***, using sigma and pi designation as appropriate. (2 points)

Atom	Bonds formed (use sigma and pi as appropriate)
Carbon (<i>a</i>)	
Carbon (<i>b</i>)	

iv. How many total sp^3 -s sigma bonds are in the molecule? (1 point)

v. Determine **electronic** and **molecular geometry** at carbon (C) labeled ***e*** and sulfur (S) labeled ***f*** centres. (2 points)

Atom	Electronic Geometry	Molecular Geometry
Carbon (<i>e</i>)		
Sulfur (<i>f</i>)		

vi. Circle the value from the choices given below, that describes the **C-S-C bond angle at sulfur** (S) marked ***f***: (0.5 points)

>90° to <109.5° 109.5° >109.5 to <120° 120° >120° to <180°

**Question 8 12 points**

Using the **valence shell orbitals only** draw a **complete** molecular orbital (MO) diagram (with no s-p mixing) for the molecule **PS**, in the space provided below. **(6 points)**

Assume that 3s and 3p atomic orbitals will form molecular orbitals like those formed by the combination of 2s and 2p atomic orbitals that we learnt in class, and there is no s-p mixing.

Show **energy arrow**, **atomic orbitals** involved, **atom labels**, the **molecular orbitals** that are formed, **filled electrons** in the MO diagram etc. Make sure to label all the atomic and molecular orbitals.

Using the MO diagram above, answer the following questions:

- (i) Which one of the following will have the highest bond order? Circle the correct choice
A) PS^{1-} B) PS C) PS^{1+}
- (ii) Which one of the following will have the longest bond length? Circle the correct choice
A) PS^{1-} B) PS C) PS^{1+}
- (iii) Which of the following will be diamagnetic? Circle the correct choice.
A) PS^{1-} B) PS C) PS^{1+}
- (iv) Draw Lewis dot structure of the molecule with the shortest bond length (based on the MO diagram) from among the three (PS^{1-} , PS, PS^{1+}), and indicate the bond order calculated from the Lewis dot structure. **(3 points)**

Lewis Dot Structure

Bond Order



B5D36E03-C3A1-46C2-B011-5D47183FDDCF

midterm-exam-93388

#195 Page 10 of 10

SCRAP PAGE

**CHEMISTRY 110 –Midterm Exam****Thursday November 7, 2024, 6:30 – 8:30 pm**

Examiners: Pallavi Sirjoosingh and Ashok Kakkar

First name (please write as legibly as possible within the boxes)

--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--

Last name

--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--

Student ID number

--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--

This booklet has 8 questions on 10 pages (The last page can be used as a scrap page).

The staple must NOT be removed from the question booklet. Your intact question booklet must be returned at the end of the exam.

A separate booklet contains a periodic table, and datasheet.

Write your answers directly under the questions in the space provided. Do not write on the QR codes or near the top edges (as it may be cut off).

Please write **legibly and dark enough** for the scan to be readable. For students who use English as a second language, dictionaries are permitted. Calculators are allowed.

DO NOT ASK QUESTIONS OF THE INVIGILATORS about interpretations of the exam questions. If you have a concern about a question, please summarize it at the top of this page. All concerns will be evaluated after the exam. GOOD LUCK!

**Question 1 8 points** (*Assume all transitions are possible*)

a. 6 points Calculate the **energy (in kJ/mol)** required to ionize (completely remove) the electron from the **$n=3$** level of **He^+ ion** (He: Helium). Show your work including equations. (3 sig. figs.)

b. 2 points

(I) Select the largest (longest) wavelength from the three choices below:

- i. Wavelength of photon to *completely remove* an electron from *1s orbital of Hydrogen atom*
- ii. Wavelength of photon to *completely remove* an electron from *1s orbital of Boron atom (B)*
- iii. Wavelength of photon to *completely remove* an electron from *1s orbital of Boron cation (B^+)*

i or ii or iii

(II) Select the smallest (shortest) wavelength from the three choices below (*for H atom*):

- i. Wavelength of photon *emitted* for an electron transitioning from *2p to 1s orbital*
- ii. Wavelength of photon *absorbed* for an electron transitioning from *2s to 4p orbital*
- iii. Wavelength of photon *absorbed* for an electron transitioning from *1s to 3p orbital*

i or ii or iii



Question 2 8 points



A Chem110 student plots the graph shown on the left, from data obtained for the kinetic energy (eV) of the ejected electron versus frequency (s^{-1}) of the incident light, for an unknown metal. ($1\text{eV} = 1.6022 \times 10^{-19} \text{ J}$)

Based on the plot, answer the questions **a to c** below:

a. 1 point On the plot above, label the **threshold frequency**.

b. 2 points What is the value of the work potential (in J) of the unknown metal?

c. 3 points What is the *longest* wavelength (in nm) required to eject an electron from this metal's surface? Show your work including equation(s) used. (3 sig figs)

d. 2 points In the space provided below, plot the following

(Assume frequency is *above* threshold frequency):

- Kinetic energy* of an ejected electron versus the *intensity* of the incident light (left)
- Number of electrons* ejected versus the *frequency* of the incident light (right)



**Question 3** **6 points****a. 2 points**

In the ground state of Ge (Germanium; $Z = 32$) atom, **name** the following orbitals:

i. Orbital containing the electron that requires the maximum energy to be removed.

ii. Orbital containing the electron that experiences the minimum effective nuclear charge.

b. 2 points

An unknown anion (A^-) has the following *excited* electronic configuration: $[Ar] 4s^2 3d^3 4p^1$

(Ar: Argon; $Z = 18$)

Give the **condensed ground state electronic configuration** of the atom, **A** and the cation, **A^{2+}** .
Show your work. (Assume all electronic transitions are possible)

Electronic Configuration of A:

Electronic Configuration of A^{2+} :

c. 2 points

In a single atom, determine how many electrons can have the following set of quantum numbers?

(i) $n = 5; l = 3$

In a single atom, determine how many electrons can have the following set of quantum numbers?

(i) $n = 3; l = 2; m_s = +1/2$

**Question 4** **8 points**

Rank the following from 1 to 4 in **increasing** order. Provide the ranking in the boxes provided.

Note: Atomic number (Z) are provided for relevant atoms

a) Increasing (lowest to highest) order of *ionization energy*: (1= lowest; 4 = highest)

Mg

Mg: Z = 12

Rb

Rb: Z = 37

Cs

Cs: Z = 55

K

K : Z = 19

b) Increasing (lowest to highest) order of *bond angle*: (1= lowest; 4 = highest)

H₂OCCl₄XeF₄

HCN

c) Increasing (smallest to largest) order of *ionic size* (1= smallest; 4 = largest)

S²⁻

S: Z = 16

O²⁻

O: Z = 8

Be²⁺

Be: Z = 4

Na⁺

Na: Z = 11

d) Increasing (lowest to highest) order of *bond energy* (1= lowest; 4 = highest)

P-F

P-Cl

P-I

P-Br

**Question 5** **12 points**

a. 9 points For the molecules HClO_4 and TeF_6 , draw the most stable *Lewis structure* showing appropriate shape and *dashed/wedged bonds*, based on VSEPR. Determine the **molecular geometry** around the central atom. For HClO_4 , label the *most polar bond*.



b. 3 points (Note: Lone pairs are not shown in structures below)

(i) For the three *resonance* structures (1, 2 and 3) shown below, answer the following question:



(I) Circle the *most contributing* resonance structure.

Structure 1

Structure 2

Structure 3

(ii) For the three structures given (4, 5, and 6) below, answer the following questions:



(I) Circle all the structures that represent *resonance structures* of each other:

Structure 4

Structure 5

Structure 6

(II) Among the resonance structures, circle the *most contributing* resonance structure.

Structure 4

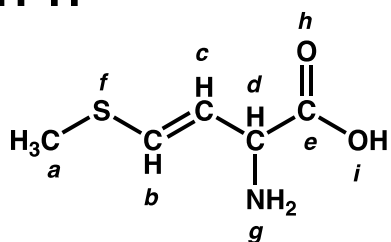
Structure 5

Structure 6

**Question 6** **6 points**

Using the data given below, calculate the **lattice energy** for SrO(s).

	ΔH° (kJ/mol)
$\text{Sr}(s) \longrightarrow \text{Sr}(g)$	+180
$\text{Sr}(g) \longrightarrow \text{Sr}^{2+}(g) + 2e^-$	+1468
$\text{O}_2(g) \longrightarrow 2\text{O}(g)$	+500
$\text{O}(g) + 2e^- \longrightarrow \text{O}^{2-}(g)$	+600
$\text{Sr}(s) + \frac{1}{2}\text{O}_2(g) \longrightarrow \text{SrO}(s)$	-550

**Question 7 10 points**

Consider the molecule given on the left, and answer questions (i) to (vi) using chemical bonding theories that we have learnt in class (valence bond theory, hybridization).

Note that the lone pairs are not shown in the structure:

i. Consider the carbon atoms marked in as ***a***, ***b***, ***c***, ***d***, and ***e*** in the molecule, and determine **hybridization** at each of the carbon centers. (2.5 points)

Atom	Hybridization
Carbon (<i>a</i>)	
Carbon (<i>b</i>)	
Carbon (<i>c</i>)	
Carbon (<i>d</i>)	
Carbon (<i>e</i>)	

ii. Now, consider the sulfur atom (S) marked as ***f***, nitrogen (N) atom (***g***), and oxygen(O) atoms (***h*** and ***i***), and determine their **hybridization**. (2 points)

Atom	Hybridization
Sulfur (<i>f</i>)	
Nitrogen (<i>g</i>)	
Oxygen (<i>h</i>)	
Oxygen (<i>i</i>)	

iii. **Describe** all the bonds formed by the carbon atoms ***a*** and ***b***, using sigma and pi designation as appropriate. (2 points)

Atom	Bonds formed (use sigma and pi as appropriate)
Carbon (<i>a</i>)	
Carbon (<i>b</i>)	

iv. How many total sp^3 -s sigma bonds are in the molecule? (1 point)

v. Determine **electronic** and **molecular geometry** at carbon (C) labeled ***e*** and sulfur (S) labeled ***f*** centres. (2 points)

Atom	Electronic Geometry	Molecular Geometry
Carbon (<i>e</i>)		
Sulfur (<i>f</i>)		

vi. Circle the value from the choices given below, that describes the **C-S-C bond angle at sulfur** (S) marked ***f***: (0.5 points)

>90° to <109.5°

109.5°

>109.5 to <120°

120°

>120° to <180°


Question 8 12 points

Using the **valence shell orbitals only** draw a **complete** molecular orbital (MO) diagram (with no s-p mixing) for the molecule **PS**, in the space provided below. **(6 points)**

Assume that 3s and 3p atomic orbitals will form molecular orbitals like those formed by the combination of 2s and 2p atomic orbitals that we learnt in class, and there is no s-p mixing.

Show **energy arrow**, **atomic orbitals** involved, **atom labels**, the **molecular orbitals** that are formed, **filled electrons** in the MO diagram etc. Make sure to label all the atomic and molecular orbitals.

Using the MO diagram above, answer the following questions:

- (i) Which one of the following will have the highest bond order? Circle the correct choice
 A) PS^{1-} B) PS C) PS^{1+}
- (ii) Which one of the following will have the longest bond length? Circle the correct choice
 A) PS^{1-} B) PS C) PS^{1+}
- (iii) Which of the following will be diamagnetic? Circle the correct choice.
 A) PS^{1-} B) PS C) PS^{1+}
- (iv) Draw Lewis dot structure of the molecule with the shortest bond length (based on the MO diagram) from among the three (PS^{1-} , PS, PS^{1+}), and indicate the bond order calculated from the Lewis dot structure. **(3 points)**

Lewis Dot Structure

Bond Order



95309053-8EBD-4FA7-AC69-AE7C8B55072B

midterm-exam-93388

#196 Page 10 of 10

SCRAP PAGE

**CHEMISTRY 110 –Midterm Exam****Thursday November 7, 2024, 6:30 – 8:30 pm**

Examiners: Pallavi Sirjoosingh and Ashok Kakkar

First name (please write as legibly as possible within the boxes)

--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--

Last name

--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--

Student ID number

--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--

This booklet has 8 questions on 10 pages (The last page can be used as a scrap page).

The staple must NOT be removed from the question booklet. Your intact question booklet must be returned at the end of the exam.

A separate booklet contains a periodic table, and datasheet.

Write your answers directly under the questions in the space provided. Do not write on the QR codes or near the top edges (as it may be cut off).

Please write **legibly and dark enough** for the scan to be readable. For students who use English as a second language, dictionaries are permitted. Calculators are allowed.

DO NOT ASK QUESTIONS OF THE INVIGILATORS about interpretations of the exam questions. If you have a concern about a question, please summarize it at the top of this page. All concerns will be evaluated after the exam. GOOD LUCK!

**Question 1 8 points** (*Assume all transitions are possible*)

a. 6 points Calculate the **energy (in kJ/mol)** required to ionize (completely remove) the electron from the **n=3** level of **He⁺ ion** (He: Helium). Show your work including equations. (3 sig. figs.)

b. 2 points

(I) Select the largest (longest) wavelength from the three choices below:

- i. Wavelength of photon to *completely remove* an electron from *1s orbital of Hydrogen atom*
- ii. Wavelength of photon to *completely remove* an electron from *1s orbital of Boron atom (B)*
- iii. Wavelength of photon to *completely remove* an electron from *1s orbital of Boron cation (B⁺)*

i or ii or iii

(II) Select the smallest (shortest) wavelength from the three choices below (*for H atom*):

- i. Wavelength of photon *emitted* for an electron transitioning from *2p to 1s orbital*
- ii. Wavelength of photon *absorbed* for an electron transitioning from *2s to 4p orbital*
- iii. Wavelength of photon *absorbed* for an electron transitioning from *1s to 3p orbital*

i or ii or iii



Question 2 8 points



A Chem110 student plots the graph shown on the left, from data obtained for the kinetic energy (eV) of the ejected electron versus frequency (s^{-1}) of the incident light, for an unknown metal. ($1\text{eV} = 1.6022 \times 10^{-19} \text{ J}$)

Based on the plot, answer the questions **a to c** below:

a. 1 point On the plot above, label the **threshold frequency**.

b. 2 points What is the value of the work potential (in J) of the unknown metal?

c. 3 points What is the *longest* wavelength (in nm) required to eject an electron from this metal's surface? Show your work including equation(s) used. (3 sig figs)

d. 2 points In the space provided below, plot the following

(Assume frequency is *above* threshold frequency):

- Kinetic energy* of an ejected electron versus the *intensity* of the incident light (left)
- Number of electrons* ejected versus the *frequency* of the incident light (right)



**Question 3** **6 points****a. 2 points**

In the ground state of Ge (Germanium; $Z = 32$) atom, **name** the following orbitals:

i. Orbital containing the electron that requires the maximum energy to be removed.

ii. Orbital containing the electron that experiences the minimum effective nuclear charge.

b. 2 points

An unknown anion (A^-) has the following *excited* electronic configuration: $[Ar] 4s^2 3d^3 4p^1$

(Ar: Argon; $Z = 18$)

Give the **condensed ground state electronic configuration** of the atom, **A** and the cation, **A^{2+}** .
Show your work. (Assume all electronic transitions are possible)

Electronic Configuration of A:

Electronic Configuration of A^{2+} :

c. 2 points

In a single atom, determine how many electrons can have the following set of quantum numbers?

(i) $n = 5; l = 3$

In a single atom, determine how many electrons can have the following set of quantum numbers?

(i) $n = 3; l = 2; m_s = +1/2$

**Question 4** **8 points**

Rank the following from 1 to 4 in **increasing** order. Provide the ranking in the boxes provided.

Note: Atomic number (Z) are provided for relevant atoms

a) Increasing (lowest to highest) order of *ionization energy*: (1= lowest; 4 = highest)

Mg

Mg: Z = 12

Rb

Rb: Z = 37

Cs

Cs: Z = 55

K

K : Z = 19

b) Increasing (lowest to highest) order of *bond angle*: (1= lowest; 4 = highest)

H₂OCCl₄XeF₄

HCN

c) Increasing (smallest to largest) order of *ionic size* (1= smallest; 4 = largest)

S²⁻

S: Z = 16

O²⁻

O: Z = 8

Be²⁺

Be: Z = 4

Na⁺

Na: Z = 11

d) Increasing (lowest to highest) order of *bond energy* (1= lowest; 4 = highest)

P-F

P-Cl

P-I

P-Br

**Question 5** **12 points**

a. 9 points For the molecules HClO_4 and TeF_6 , draw the most stable *Lewis structure* showing appropriate shape and *dashed/wedged bonds*, based on VSEPR. Determine the **molecular geometry** around the central atom. For HClO_4 , label the *most polar bond*.



b. 3 points (Note: Lone pairs are not shown in structures below)

(i) For the three *resonance* structures (1, 2 and 3) shown below, answer the following question:



(I) Circle the *most contributing* resonance structure.

Structure 1

Structure 2

Structure 3

(ii) For the three structures given (4, 5, and 6) below, answer the following questions:



(I) Circle all the structures that represent *resonance structures* of each other:

Structure 4

Structure 5

Structure 6

(II) Among the resonance structures, circle the *most contributing* resonance structure.

Structure 4

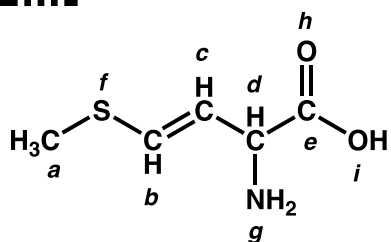
Structure 5

Structure 6

**Question 6** **6 points**

Using the data given below, calculate the **lattice energy** for SrO(s).

	ΔH° (kJ/mol)
$\text{Sr}(s) \longrightarrow \text{Sr}(g)$	+180
$\text{Sr}(g) \longrightarrow \text{Sr}^{2+}(g) + 2e^-$	+1468
$\text{O}_2(g) \longrightarrow 2\text{O}(g)$	+500
$\text{O}(g) + 2e^- \longrightarrow \text{O}^{2-}(g)$	+600
$\text{Sr}(s) + \frac{1}{2}\text{O}_2(g) \longrightarrow \text{SrO}(s)$	-550

**Question 7 10 points**

Consider the molecule given on the left, and answer questions (i) to (vi) using chemical bonding theories that we have learnt in class (valence bond theory, hybridization).

Note that the lone pairs are not shown in the structure:

i. Consider the carbon atoms marked in as ***a***, ***b***, ***c***, ***d***, and ***e*** in the molecule, and determine **hybridization** at each of the carbon centers. (2.5 points)

Atom	Hybridization
Carbon (<i>a</i>)	
Carbon (<i>b</i>)	
Carbon (<i>c</i>)	
Carbon (<i>d</i>)	
Carbon (<i>e</i>)	

ii. Now, consider the sulfur atom (S) marked as ***f***, nitrogen (N) atom (***g***), and oxygen(O) atoms (***h*** and ***i***), and determine their **hybridization**. (2 points)

Atom	Hybridization
Sulfur (<i>f</i>)	
Nitrogen (<i>g</i>)	
Oxygen (<i>h</i>)	
Oxygen (<i>i</i>)	

iii. **Describe** all the bonds formed by the carbon atoms ***a*** and ***b***, using sigma and pi designation as appropriate. (2 points)

Atom	Bonds formed (use sigma and pi as appropriate)
Carbon (<i>a</i>)	
Carbon (<i>b</i>)	

iv. How many total sp^3 -s sigma bonds are in the molecule? (1 point)

v. Determine **electronic** and **molecular geometry** at carbon (C) labeled ***e*** and sulfur (S) labeled ***f*** centres. (2 points)

Atom	Electronic Geometry	Molecular Geometry
Carbon (<i>e</i>)		
Sulfur (<i>f</i>)		

vi. Circle the value from the choices given below, that describes the **C-S-C bond angle at sulfur** (S) marked ***f***: (0.5 points)

>90° to <109.5° 109.5° >109.5 to <120° 120° >120° to <180°

**Question 8 12 points**

Using the **valence shell orbitals only** draw a **complete** molecular orbital (MO) diagram (with no s-p mixing) for the molecule **PS**, in the space provided below. **(6 points)**

Assume that 3s and 3p atomic orbitals will form molecular orbitals like those formed by the combination of 2s and 2p atomic orbitals that we learnt in class, and there is no s-p mixing.

Show **energy arrow**, **atomic orbitals** involved, **atom labels**, the **molecular orbitals** that are formed, **filled electrons** in the MO diagram etc. Make sure to label all the atomic and molecular orbitals.

Using the MO diagram above, answer the following questions:

- (i) Which one of the following will have the highest bond order? Circle the correct choice
A) PS^{1-} B) PS C) PS^{1+}
- (ii) Which one of the following will have the longest bond length? Circle the correct choice
A) PS^{1-} B) PS C) PS^{1+}
- (iii) Which of the following will be diamagnetic? Circle the correct choice.
A) PS^{1-} B) PS C) PS^{1+}
- (iv) Draw Lewis dot structure of the molecule with the shortest bond length (based on the MO diagram) from among the three (PS^{1-} , PS, PS^{1+}), and indicate the bond order calculated from the Lewis dot structure. **(3 points)**

Lewis Dot Structure

Bond Order



1FADF9EB-00EF-4337-AF82-EF4F45E5D072

midterm-exam-93388

#197 Page 10 of 10

SCRAP PAGE

**CHEMISTRY 110 –Midterm Exam****Thursday November 7, 2024, 6:30 – 8:30 pm****Examiners: Pallavi Sirjoosingh and Ashok Kakkar**

First name (please write as legibly as possible within the boxes)

--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--

Last name

--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--

Student ID number

--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--

This booklet has 8 questions on 10 pages (The last page can be used as a scrap page).

The staple must NOT be removed from the question booklet. Your intact question booklet must be returned at the end of the exam.

A separate booklet contains a periodic table, and datasheet.

Write your answers directly under the questions in the space provided. Do not write on the QR codes or near the top edges (as it may be cut off).

Please write **legibly and dark enough** for the scan to be readable. For students who use English as a second language, dictionaries are permitted. Calculators are allowed.

DO NOT ASK QUESTIONS OF THE INVIGILATORS about interpretations of the exam questions. If you have a concern about a question, please summarize it at the top of this page. All concerns will be evaluated after the exam. GOOD LUCK!

**Question 1 8 points** (*Assume all transitions are possible*)

a. 6 points Calculate the **energy (in kJ/mol)** required to ionize (completely remove) the electron from the **n=3** level of **He⁺ ion** (He: Helium). Show your work including equations. (3 sig. figs.)

b. 2 points

(I) Select the largest (longest) wavelength from the three choices below:

- i. Wavelength of photon to *completely remove* an electron from *1s orbital of Hydrogen atom*
- ii. Wavelength of photon to *completely remove* an electron from *1s orbital of Boron atom (B)*
- iii. Wavelength of photon to *completely remove* an electron from *1s orbital of Boron cation (B⁺)*

i or ii or iii

(II) Select the smallest (shortest) wavelength from the three choices below (*for H atom*):

- i. Wavelength of photon *emitted* for an electron transitioning from *2p to 1s orbital*
- ii. Wavelength of photon *absorbed* for an electron transitioning from *2s to 4p orbital*
- iii. Wavelength of photon *absorbed* for an electron transitioning from *1s to 3p orbital*

i or ii or iii



Question 2 8 points



A Chem110 student plots the graph shown on the left, from data obtained for the kinetic energy (eV) of the ejected electron versus frequency (s^{-1}) of the incident light, for an unknown metal. ($1\text{eV} = 1.6022 \times 10^{-19} \text{ J}$)

Based on the plot, answer the questions **a to c** below:

a. 1 point On the plot above, label the **threshold frequency**.

b. 2 points What is the value of the work potential (in J) of the unknown metal?

c. 3 points What is the *longest* wavelength (in nm) required to eject an electron from this metal's surface? Show your work including equation(s) used. (3 sig figs)

d. 2 points In the space provided below, plot the following

(Assume frequency is *above* threshold frequency):

- Kinetic energy* of an ejected electron versus the *intensity* of the incident light (left)
- Number of electrons* ejected versus the *frequency* of the incident light (right)



**Question 3** **6 points****a. 2 points**

In the ground state of Ge (Germanium; $Z = 32$) atom, **name** the following orbitals:

i. Orbital containing the electron that requires the maximum energy to be removed.

ii. Orbital containing the electron that experiences the minimum effective nuclear charge.

b. 2 points

An unknown anion (A^-) has the following *excited* electronic configuration: $[Ar] 4s^2 3d^3 4p^1$

(Ar: Argon; $Z = 18$)

Give the **condensed ground state electronic configuration** of the atom, **A** and the cation, **A^{2+}** .
Show your work. (Assume all electronic transitions are possible)

Electronic Configuration of A:

Electronic Configuration of A^{2+} :

c. 2 points

In a single atom, determine how many electrons can have the following set of quantum numbers?

(i) $n = 5; l = 3$

In a single atom, determine how many electrons can have the following set of quantum numbers?

(i) $n = 3; l = 2; m_s = +1/2$

**Question 4** **8 points**

Rank the following from 1 to 4 in **increasing** order. Provide the ranking in the boxes provided.

Note: Atomic number (Z) are provided for relevant atoms

a) Increasing (lowest to highest) order of *ionization energy*: (1= lowest; 4 = highest)

Mg

Mg: Z = 12

Rb

Rb: Z = 37

Cs

Cs: Z = 55

K

K : Z = 19

b) Increasing (lowest to highest) order of *bond angle*: (1= lowest; 4 = highest)

H₂OCCl₄XeF₄

HCN

c) Increasing (smallest to largest) order of *ionic size* (1= smallest; 4 = largest)

S²⁻

S: Z = 16

O²⁻

O: Z = 8

Be²⁺

Be: Z = 4

Na⁺

Na: Z = 11

d) Increasing (lowest to highest) order of *bond energy* (1= lowest; 4 = highest)

P-F

P-Cl

P-I

P-Br

**Question 5** **12 points**

a. 9 points For the molecules HClO_4 and TeF_6 , draw the most stable *Lewis structure* showing appropriate shape and *dashed/wedged bonds*, based on VSEPR. Determine the **molecular geometry** around the central atom. For HClO_4 , label the *most polar bond*.



b. 3 points (Note: Lone pairs are not shown in structures below)

(i) For the three *resonance* structures (1, 2 and 3) shown below, answer the following question:



(I) Circle the *most contributing* resonance structure.

Structure 1

Structure 2

Structure 3

(ii) For the three structures given (4, 5, and 6) below, answer the following questions:



(I) Circle all the structures that represent *resonance structures* of each other:

Structure 4

Structure 5

Structure 6

(II) Among the resonance structures, circle the *most contributing* resonance structure.

Structure 4

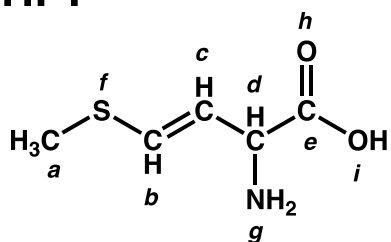
Structure 5

Structure 6

**Question 6** **6 points**

Using the data given below, calculate the **lattice energy** for SrO(s).

	ΔH° (kJ/mol)
$\text{Sr}(s) \longrightarrow \text{Sr}(g)$	+180
$\text{Sr}(g) \longrightarrow \text{Sr}^{2+}(g) + 2e^-$	+1468
$\text{O}_2(g) \longrightarrow 2\text{O}(g)$	+500
$\text{O}(g) + 2e^- \longrightarrow \text{O}^{2-}(g)$	+600
$\text{Sr}(s) + \frac{1}{2}\text{O}_2(g) \longrightarrow \text{SrO}(s)$	-550

**Question 7 10 points**

Consider the molecule given on the left, and answer questions (i) to (vi) using chemical bonding theories that we have learnt in class (valence bond theory, hybridization).

Note that the lone pairs are not shown in the structure:

i. Consider the carbon atoms marked in as ***a***, ***b***, ***c***, ***d***, and ***e*** in the molecule, and determine **hybridization** at each of the carbon centers. (2.5 points)

Atom	Hybridization
Carbon (<i>a</i>)	
Carbon (<i>b</i>)	
Carbon (<i>c</i>)	
Carbon (<i>d</i>)	
Carbon (<i>e</i>)	

ii. Now, consider the sulfur atom (S) marked as ***f***, nitrogen (N) atom (***g***), and oxygen(O) atoms (***h*** and ***i***), and determine their **hybridization**. (2 points)

Atom	Hybridization
Sulfur (<i>f</i>)	
Nitrogen (<i>g</i>)	
Oxygen (<i>h</i>)	
Oxygen (<i>i</i>)	

iii. **Describe** all the bonds formed by the carbon atoms ***a*** and ***b***, using sigma and pi designation as appropriate. (2 points)

Atom	Bonds formed (use sigma and pi as appropriate)
Carbon (<i>a</i>)	
Carbon (<i>b</i>)	

iv. How many total sp^3 -s sigma bonds are in the molecule? (1 point)

v. Determine **electronic** and **molecular geometry** at carbon (C) labeled ***e*** and sulfur (S) labeled ***f*** centres. (2 points)

Atom	Electronic Geometry	Molecular Geometry
Carbon (<i>e</i>)		
Sulfur (<i>f</i>)		

vi. Circle the value from the choices given below, that describes the **C-S-C bond angle at sulfur** (S) marked ***f***: (0.5 points)

>90° to <109.5°

109.5°

>109.5 to <120°

120°

>120° to <180°


Question 8 12 points

Using the **valence shell orbitals only** draw a **complete** molecular orbital (MO) diagram (with no s-p mixing) for the molecule **PS**, in the space provided below. **(6 points)**

Assume that 3s and 3p atomic orbitals will form molecular orbitals like those formed by the combination of 2s and 2p atomic orbitals that we learnt in class, and there is no s-p mixing.

Show **energy arrow**, **atomic orbitals** involved, **atom labels**, the **molecular orbitals** that are formed, **filled electrons** in the MO diagram etc. Make sure to **label all the atomic and molecular orbitals**.

Using the MO diagram above, answer the following questions:

- (i) Which one of the following will have the highest bond order? Circle the correct choice
 A) PS^{1-} B) PS C) PS^{1+}
- (ii) Which one of the following will have the longest bond length? Circle the correct choice
 A) PS^{1-} B) PS C) PS^{1+}
- (iii) Which of the following will be diamagnetic? Circle the correct choice.
 A) PS^{1-} B) PS C) PS^{1+}
- (iv) Draw Lewis dot structure of the molecule with the shortest bond length (based on the MO diagram) from among the three (PS^{1-} , PS, PS^{1+}), and indicate the bond order calculated from the Lewis dot structure. **(3 points)**

Lewis Dot Structure

Bond Order



D3D1B8E9-A512-4C03-9469-495602437AA9

midterm-exam-93388

#198 Page 10 of 10

SCRAP PAGE

**CHEMISTRY 110 –Midterm Exam****Thursday November 7, 2024, 6:30 – 8:30 pm**

Examiners: Pallavi Sirjoosingh and Ashok Kakkar

First name (please write as legibly as possible within the boxes)

--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--

Last name

--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--

Student ID number

--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--

This booklet has 8 questions on 10 pages (The last page can be used as a scrap page).

The staple must NOT be removed from the question booklet. Your intact question booklet must be returned at the end of the exam.

A separate booklet contains a periodic table, and datasheet.

Write your answers directly under the questions in the space provided. Do not write on the QR codes or near the top edges (as it may be cut off).

Please write **legibly and dark enough** for the scan to be readable. For students who use English as a second language, dictionaries are permitted. Calculators are allowed.

DO NOT ASK QUESTIONS OF THE INVIGILATORS about interpretations of the exam questions. If you have a concern about a question, please summarize it at the top of this page. All concerns will be evaluated after the exam. GOOD LUCK!

**Question 1 8 points** (*Assume all transitions are possible*)

a. 6 points Calculate the **energy (in kJ/mol)** required to ionize (completely remove) the electron from the **n=3** level of **He⁺ ion** (He: Helium). Show your work including equations. (3 sig. figs.)

b. 2 points

(I) Select the largest (longest) wavelength from the three choices below:

- i. Wavelength of photon to *completely remove* an electron from *1s orbital of Hydrogen atom*
- ii. Wavelength of photon to *completely remove* an electron from *1s orbital of Boron atom (B)*
- iii. Wavelength of photon to *completely remove* an electron from *1s orbital of Boron cation (B⁺)*

i or ii or iii

(II) Select the smallest (shortest) wavelength from the three choices below (*for H atom*):

- i. Wavelength of photon *emitted* for an electron transitioning from *2p to 1s orbital*
- ii. Wavelength of photon *absorbed* for an electron transitioning from *2s to 4p orbital*
- iii. Wavelength of photon *absorbed* for an electron transitioning from *1s to 3p orbital*

i or ii or iii



Question 2 8 points



A Chem110 student plots the graph shown on the left, from data obtained for the kinetic energy (eV) of the ejected electron versus frequency (s^{-1}) of the incident light, for an unknown metal. ($1\text{eV} = 1.6022 \times 10^{-19} \text{ J}$)

Based on the plot, answer the questions **a to c** below:

a. 1 point On the plot above, label the **threshold frequency**.

b. 2 points What is the value of the work potential (in J) of the unknown metal?

c. 3 points What is the *longest* wavelength (in nm) required to eject an electron from this metal's surface? Show your work including equation(s) used. (3 sig figs)

d. 2 points In the space provided below, plot the following

(Assume frequency is *above* threshold frequency):

- i. *Kinetic energy* of an ejected electron versus the *intensity* of the incident light (left)
- ii. *Number of electrons* ejected versus the *frequency* of the incident light (right)



**Question 3** **6 points****a. 2 points**

In the ground state of Ge (Germanium; $Z = 32$) atom, **name** the following orbitals:

i. Orbital containing the electron that requires the maximum energy to be removed.

ii. Orbital containing the electron that experiences the minimum effective nuclear charge.

b. 2 points

An unknown anion (A^-) has the following *excited* electronic configuration: $[Ar] 4s^2 3d^3 4p^1$

(Ar: Argon; $Z = 18$)

Give the **condensed ground state electronic configuration** of the atom, **A** and the cation, **A^{2+}** .
Show your work. (Assume all electronic transitions are possible)

Electronic Configuration of A:

Electronic Configuration of A^{2+} :

c. 2 points

In a single atom, determine how many electrons can have the following set of quantum numbers?

(i) $n = 5; l = 3$

In a single atom, determine how many electrons can have the following set of quantum numbers?

(i) $n = 3; l = 2; m_s = +1/2$

**Question 4** **8 points**

Rank the following from 1 to 4 in **increasing** order. Provide the ranking in the boxes provided.

Note: Atomic number (Z) are provided for relevant atoms

a) Increasing (lowest to highest) order of *ionization energy*: (1= lowest; 4 = highest)

Mg

Mg: Z = 12

Rb

Rb: Z = 37

Cs

Cs: Z = 55

K

K : Z = 19

b) Increasing (lowest to highest) order of *bond angle*: (1= lowest; 4 = highest)

H₂OCCl₄XeF₄

HCN

c) Increasing (smallest to largest) order of *ionic size* (1= smallest; 4 = largest)

S²⁻

S: Z = 16

O²⁻

O: Z = 8

Be²⁺

Be: Z = 4

Na⁺

Na: Z = 11

d) Increasing (lowest to highest) order of *bond energy* (1= lowest; 4 = highest)

P-F

P-Cl

P-I

P-Br

**Question 5** **12 points**

a. 9 points For the molecules HClO_4 and TeF_6 , draw the most stable *Lewis structure* showing appropriate shape and *dashed/wedged bonds*, based on VSEPR. Determine the **molecular geometry** around the central atom. For HClO_4 , label the *most polar bond*.



b. 3 points (Note: Lone pairs are not shown in structures below)

(i) For the three *resonance* structures (1, 2 and 3) shown below, answer the following question:



(I) Circle the *most contributing* resonance structure.

Structure 1

Structure 2

Structure 3

(ii) For the three structures given (4, 5, and 6) below, answer the following questions:



(I) Circle all the structures that represent *resonance structures* of each other:

Structure 4

Structure 5

Structure 6

(II) Among the resonance structures, circle the *most contributing* resonance structure.

Structure 4

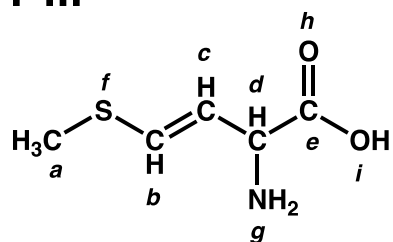
Structure 5

Structure 6

**Question 6** **6 points**

Using the data given below, calculate the **lattice energy** for SrO(s).

	ΔH° (kJ/mol)
$\text{Sr}(s) \longrightarrow \text{Sr}(g)$	+180
$\text{Sr}(g) \longrightarrow \text{Sr}^{2+}(g) + 2e^-$	+1468
$\text{O}_2(g) \longrightarrow 2\text{O}(g)$	+500
$\text{O}(g) + 2e^- \longrightarrow \text{O}^{2-}(g)$	+600
$\text{Sr}(s) + \frac{1}{2}\text{O}_2(g) \longrightarrow \text{SrO}(s)$	-550

**Question 7 10 points**

Consider the molecule given on the left, and answer questions (i) to (vi) using chemical bonding theories that we have learnt in class (valence bond theory, hybridization).

Note that the lone pairs are not shown in the structure:

i. Consider the carbon atoms marked in as ***a***, ***b***, ***c***, ***d***, and ***e*** in the molecule, and determine **hybridization** at each of the carbon centers. (2.5 points)

Atom	Hybridization
Carbon (<i>a</i>)	
Carbon (<i>b</i>)	
Carbon (<i>c</i>)	
Carbon (<i>d</i>)	
Carbon (<i>e</i>)	

ii. Now, consider the sulfur atom (S) marked as ***f***, nitrogen (N) atom (***g***), and oxygen(O) atoms (***h*** and ***i***), and determine their **hybridization**. (2 points)

Atom	Hybridization
Sulfur (<i>f</i>)	
Nitrogen (<i>g</i>)	
Oxygen (<i>h</i>)	
Oxygen (<i>i</i>)	

iii. **Describe** all the bonds formed by the carbon atoms ***a*** and ***b***, using sigma and pi designation as appropriate. (2 points)

Atom	Bonds formed (use sigma and pi as appropriate)
Carbon (<i>a</i>)	
Carbon (<i>b</i>)	

iv. How many total sp^3 -s sigma bonds are in the molecule? (1 point)

v. Determine **electronic** and **molecular geometry** at carbon (C) labeled ***e*** and sulfur (S) labeled ***f*** centres. (2 points)

Atom	Electronic Geometry	Molecular Geometry
Carbon (<i>e</i>)		
Sulfur (<i>f</i>)		

vi. Circle the value from the choices given below, that describes the **C-S-C bond angle at sulfur** (S) marked ***f***: (0.5 points)

$>90^\circ$ to $<109.5^\circ$ 109.5° >109.5 to $<120^\circ$ 120° $>120^\circ$ to $<180^\circ$


Question 8 12 points

Using the **valence shell orbitals only** draw a **complete** molecular orbital (MO) diagram (with no s-p mixing) for the molecule **PS**, in the space provided below. **(6 points)**

Assume that 3s and 3p atomic orbitals will form molecular orbitals like those formed by the combination of 2s and 2p atomic orbitals that we learnt in class, and there is no s-p mixing.

Show **energy arrow**, **atomic orbitals** involved, **atom labels**, the **molecular orbitals** that are formed, **filled electrons** in the MO diagram etc. Make sure to label all the atomic and molecular orbitals.

Using the MO diagram above, answer the following questions:

- (i) Which one of the following will have the highest bond order? Circle the correct choice
 A) PS^{1-} B) PS C) PS^{1+}
- (ii) Which one of the following will have the longest bond length? Circle the correct choice
 A) PS^{1-} B) PS C) PS^{1+}
- (iii) Which of the following will be diamagnetic? Circle the correct choice.
 A) PS^{1-} B) PS C) PS^{1+}
- (iv) Draw Lewis dot structure of the molecule with the shortest bond length (based on the MO diagram) from among the three (PS^{1-} , PS, PS^{1+}), and indicate the bond order calculated from the Lewis dot structure. **(3 points)**

Lewis Dot Structure

Bond Order



5B7E6F00-BDB9-41C4-BF2D-9CF4845B0284

midterm-exam-93388

#199 Page 10 of 10

SCRAP PAGE

**CHEMISTRY 110 –Midterm Exam****Thursday November 7, 2024, 6:30 – 8:30 pm**

Examiners: Pallavi Sirjoosingh and Ashok Kakkar

First name (please write as legibly as possible within the boxes)

--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--

Last name

--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--

Student ID number

--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--

This booklet has 8 questions on 10 pages (The last page can be used as a scrap page).

The staple must NOT be removed from the question booklet. Your intact question booklet must be returned at the end of the exam.

A separate booklet contains a periodic table, and datasheet.

Write your answers directly under the questions in the space provided. Do not write on the QR codes or near the top edges (as it may be cut off).

Please write **legibly and dark enough** for the scan to be readable. For students who use English as a second language, dictionaries are permitted. Calculators are allowed.

DO NOT ASK QUESTIONS OF THE INVIGILATORS about interpretations of the exam questions. If you have a concern about a question, please summarize it at the top of this page. All concerns will be evaluated after the exam. GOOD LUCK!

**Question 1 8 points** (*Assume all transitions are possible*)

a. 6 points Calculate the **energy (in kJ/mol)** required to ionize (completely remove) the electron from the **n=3** level of **He⁺ ion** (He: Helium). Show your work including equations. (3 sig. figs.)

b. 2 points

(I) Select the largest (longest) wavelength from the three choices below:

- i. Wavelength of photon to *completely remove* an electron from *1s orbital of Hydrogen atom*
- ii. Wavelength of photon to *completely remove* an electron from *1s orbital of Boron atom (B)*
- iii. Wavelength of photon to *completely remove* an electron from *1s orbital of Boron cation (B⁺)*

i or ii or iii

(II) Select the smallest (shortest) wavelength from the three choices below (*for H atom*):

- i. Wavelength of photon *emitted* for an electron transitioning from *2p to 1s orbital*
- ii. Wavelength of photon *absorbed* for an electron transitioning from *2s to 4p orbital*
- iii. Wavelength of photon *absorbed* for an electron transitioning from *1s to 3p orbital*

i or ii or iii



Question 2 8 points



A Chem110 student plots the graph shown on the left, from data obtained for the kinetic energy (eV) of the ejected electron versus frequency (s^{-1}) of the incident light, for an unknown metal. ($1\text{eV} = 1.6022 \times 10^{-19} \text{ J}$)

Based on the plot, answer the questions **a to c** below:

a. 1 point On the plot above, label the **threshold frequency**.

b. 2 points What is the value of the work potential (in J) of the unknown metal?

c. 3 points What is the *longest* wavelength (in nm) required to eject an electron from this metal's surface? Show your work including equation(s) used. (3 sig figs)

d. 2 points In the space provided below, plot the following

(Assume frequency is *above* threshold frequency):

- i. *Kinetic energy* of an ejected electron versus the *intensity* of the incident light (left)
- ii. *Number of electrons* ejected versus the *frequency* of the incident light (right)



**Question 3** **6 points****a. 2 points**

In the ground state of Ge (Germanium; $Z = 32$) atom, **name** the following orbitals:

i. Orbital containing the electron that requires the maximum energy to be removed.

ii. Orbital containing the electron that experiences the minimum effective nuclear charge.

b. 2 points

An unknown anion (A^-) has the following *excited* electronic configuration: $[Ar] 4s^2 3d^3 4p^1$

(Ar: Argon; $Z = 18$)

Give the **condensed ground state electronic configuration** of the atom, **A** and the cation, **A^{2+}** .
Show your work. (Assume all electronic transitions are possible)

Electronic Configuration of A:

Electronic Configuration of A^{2+} :

c. 2 points

In a single atom, determine how many electrons can have the following set of quantum numbers?

(i) $n = 5; l = 3$

In a single atom, determine how many electrons can have the following set of quantum numbers?

(i) $n = 3; l = 2; m_s = +1/2$

**Question 4** **8 points**

Rank the following from 1 to 4 in **increasing** order. Provide the ranking in the boxes provided.

Note: Atomic number (Z) are provided for relevant atoms

a) Increasing (lowest to highest) order of *ionization energy*: (1= lowest; 4 = highest)

Mg

Mg: Z = 12

Rb

Rb: Z = 37

Cs

Cs: Z = 55

K

K : Z = 19

b) Increasing (lowest to highest) order of *bond angle*: (1= lowest; 4 = highest)

H₂OCCl₄XeF₄

HCN

c) Increasing (smallest to largest) order of *ionic size* (1= smallest; 4 = largest)

S²⁻

S: Z = 16

O²⁻

O: Z = 8

Be²⁺

Be: Z = 4

Na⁺

Na: Z = 11

d) Increasing (lowest to highest) order of *bond energy* (1= lowest; 4 = highest)

P-F

P-Cl

P-I

P-Br

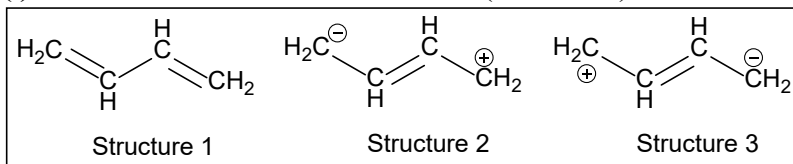
**Question 5** **12 points**

a. 9 points For the molecules HClO_4 and TeF_6 , draw the most stable *Lewis structure* showing appropriate shape and *dashed/wedged bonds*, based on VSEPR. Determine the **molecular geometry** around the central atom. For HClO_4 , label the *most polar bond*.



b. 3 points (Note: Lone pairs are not shown in structures below)

(i) For the three *resonance* structures (1, 2 and 3) shown below, answer the following question:



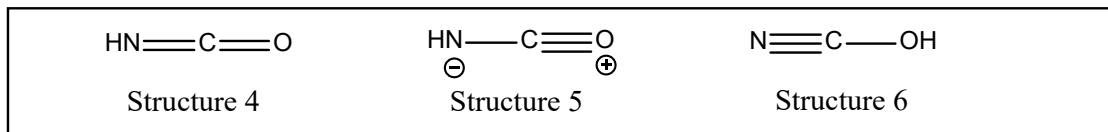
(I) Circle the *most contributing* resonance structure.

Structure 1

Structure 2

Structure 3

(ii) For the three structures given (4, 5, and 6) below, answer the following questions:



(I) Circle all the structures that represent *resonance structures* of each other:

Structure 4

Structure 5

Structure 6

(II) Among the resonance structures, circle the *most contributing* resonance structure.

Structure 4

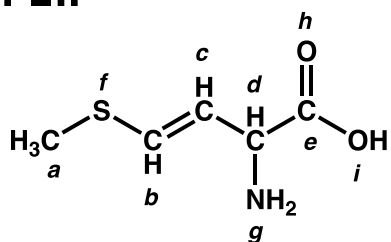
Structure 5

Structure 6

**Question 6** **6 points**

Using the data given below, calculate the **lattice energy** for SrO(s).

	ΔH° (kJ/mol)
$\text{Sr}(s) \longrightarrow \text{Sr}(g)$	+180
$\text{Sr}(g) \longrightarrow \text{Sr}^{2+}(g) + 2e^-$	+1468
$\text{O}_2(g) \longrightarrow 2\text{O}(g)$	+500
$\text{O}(g) + 2e^- \longrightarrow \text{O}^{2-}(g)$	+600
$\text{Sr}(s) + \frac{1}{2}\text{O}_2(g) \longrightarrow \text{SrO}(s)$	-550

**Question 7 10 points**

Consider the molecule given on the left, and answer questions (i) to (vi) using chemical bonding theories that we have learnt in class (valence bond theory, hybridization).

Note that the lone pairs are not shown in the structure:

i. Consider the carbon atoms marked in as ***a***, ***b***, ***c***, ***d***, and ***e*** in the molecule, and determine **hybridization** at each of the carbon centers. (2.5 points)

Atom	Hybridization
Carbon (<i>a</i>)	
Carbon (<i>b</i>)	
Carbon (<i>c</i>)	
Carbon (<i>d</i>)	
Carbon (<i>e</i>)	

ii. Now, consider the sulfur atom (S) marked as ***f***, nitrogen (N) atom (***g***), and oxygen(O) atoms (***h*** and ***i***), and determine their **hybridization**. (2 points)

Atom	Hybridization
Sulfur (<i>f</i>)	
Nitrogen (<i>g</i>)	
Oxygen (<i>h</i>)	
Oxygen (<i>i</i>)	

iii. **Describe** all the bonds formed by the carbon atoms ***a*** and ***b***, using sigma and pi designation as appropriate. (2 points)

Atom	Bonds formed (use sigma and pi as appropriate)
Carbon (<i>a</i>)	
Carbon (<i>b</i>)	

iv. How many total sp^3 -s sigma bonds are in the molecule? (1 point)

v. Determine **electronic** and **molecular geometry** at carbon (C) labeled ***e*** and sulfur (S) labeled ***f*** centres. (2 points)

Atom	Electronic Geometry	Molecular Geometry
Carbon (<i>e</i>)		
Sulfur (<i>f</i>)		

vi. Circle the value from the choices given below, that describes the **C-S-C bond angle at sulfur** (S) marked ***f***: (0.5 points)

>90° to <109.5° 109.5° >109.5 to <120° 120° >120° to <180°


Question 8 12 points

Using the **valence shell orbitals only** draw a **complete** molecular orbital (MO) diagram (with no s-p mixing) for the molecule **PS**, in the space provided below. **(6 points)**

Assume that 3s and 3p atomic orbitals will form molecular orbitals like those formed by the combination of 2s and 2p atomic orbitals that we learnt in class, and there is no s-p mixing.

Show **energy arrow**, **atomic orbitals** involved, **atom labels**, the **molecular orbitals** that are formed, **filled electrons** in the MO diagram etc. Make sure to label all the atomic and molecular orbitals.

Using the MO diagram above, answer the following questions:

- (i) Which one of the following will have the highest bond order? Circle the correct choice
 A) PS^{1-} B) PS C) PS^{1+}
- (ii) Which one of the following will have the longest bond length? Circle the correct choice
 A) PS^{1-} B) PS C) PS^{1+}
- (iii) Which of the following will be diamagnetic? Circle the correct choice.
 A) PS^{1-} B) PS C) PS^{1+}
- (iv) Draw Lewis dot structure of the molecule with the shortest bond length (based on the MO diagram) from among the three (PS^{1-} , PS, PS^{1+}), and indicate the bond order calculated from the Lewis dot structure. **(3 points)**

Lewis Dot Structure

Bond Order



FA8D7FFE-7D89-4370-88A6-F2E1F3678A2C

midterm-exam-93388

#200 Page 10 of 10

SCRAP PAGE

**CHEMISTRY 110 –Midterm Exam****Thursday November 7, 2024, 6:30 – 8:30 pm**

Examiners: Pallavi Sirjoosingh and Ashok Kakkar

First name (please write as legibly as possible within the boxes)

--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--

Last name

--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--

Student ID number

--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--

This booklet has 8 questions on 10 pages (The last page can be used as a scrap page).

The staple must NOT be removed from the question booklet. Your intact question booklet must be returned at the end of the exam.

A separate booklet contains a periodic table, and datasheet.

Write your answers directly under the questions in the space provided. Do not write on the QR codes or near the top edges (as it may be cut off).

Please write **legibly and dark enough** for the scan to be readable. For students who use English as a second language, dictionaries are permitted. Calculators are allowed.

DO NOT ASK QUESTIONS OF THE INVIGILATORS about interpretations of the exam questions. If you have a concern about a question, please summarize it at the top of this page. All concerns will be evaluated after the exam. GOOD LUCK!

**Question 1 8 points** (*Assume all transitions are possible*)

a. 6 points Calculate the **energy (in kJ/mol)** required to ionize (completely remove) the electron from the **$n=3$** level of **He^+ ion** (He: Helium). Show your work including equations. (3 sig. figs.)

b. 2 points

(I) Select the largest (longest) wavelength from the three choices below:

- i. Wavelength of photon to *completely remove* an electron from *1s orbital of Hydrogen atom*
- ii. Wavelength of photon to *completely remove* an electron from *1s orbital of Boron atom (B)*
- iii. Wavelength of photon to *completely remove* an electron from *1s orbital of Boron cation (B^+)*

i or ii or iii

(II) Select the smallest (shortest) wavelength from the three choices below (*for H atom*):

- i. Wavelength of photon *emitted* for an electron transitioning from *2p to 1s orbital*
- ii. Wavelength of photon *absorbed* for an electron transitioning from *2s to 4p orbital*
- iii. Wavelength of photon *absorbed* for an electron transitioning from *1s to 3p orbital*

i or ii or iii



Question 2 8 points



A Chem110 student plots the graph shown on the left, from data obtained for the kinetic energy (eV) of the ejected electron versus frequency (s^{-1}) of the incident light, for an unknown metal. ($1\text{eV} = 1.6022 \times 10^{-19} \text{ J}$)

Based on the plot, answer the questions **a to c** below:

a. 1 point On the plot above, label the **threshold frequency**.

b. 2 points What is the value of the work potential (in J) of the unknown metal?

c. 3 points What is the *longest* wavelength (in nm) required to eject an electron from this metal's surface? Show your work including equation(s) used. (3 sig figs)

d. 2 points In the space provided below, plot the following

(Assume frequency is *above* threshold frequency):

- Kinetic energy* of an ejected electron versus the *intensity* of the incident light (left)
- Number of electrons* ejected versus the *frequency* of the incident light (right)



**Question 3** **6 points****a. 2 points**

In the ground state of Ge (Germanium; $Z = 32$) atom, **name** the following orbitals:

i. Orbital containing the electron that requires the maximum energy to be removed.

ii. Orbital containing the electron that experiences the minimum effective nuclear charge.

b. 2 points

An unknown anion (A^-) has the following *excited* electronic configuration: $[Ar] 4s^2 3d^3 4p^1$

(Ar: Argon; $Z = 18$)

Give the **condensed ground state electronic configuration** of the atom, **A** and the cation, **A^{2+}** .
Show your work. (Assume all electronic transitions are possible)

Electronic Configuration of A:

Electronic Configuration of A^{2+} :

c. 2 points

In a single atom, determine how many electrons can have the following set of quantum numbers?

(i) $n = 5; l = 3$

In a single atom, determine how many electrons can have the following set of quantum numbers?

(i) $n = 3; l = 2; m_s = +1/2$

**Question 4** **8 points**

Rank the following from 1 to 4 in **increasing** order. Provide the ranking in the boxes provided.

Note: Atomic number (Z) are provided for relevant atoms

a) Increasing (lowest to highest) order of *ionization energy*: (1= lowest; 4 = highest)

Mg

Mg: Z = 12

Rb

Rb: Z = 37

Cs

Cs: Z = 55

K

K : Z = 19

b) Increasing (lowest to highest) order of *bond angle*: (1= lowest; 4 = highest)

H₂OCCl₄XeF₄

HCN

c) Increasing (smallest to largest) order of *ionic size* (1= smallest; 4 = largest)

S²⁻

S: Z = 16

O²⁻

O: Z = 8

Be²⁺

Be: Z = 4

Na⁺

Na: Z = 11

d) Increasing (lowest to highest) order of *bond energy* (1= lowest; 4 = highest)

P-F

P-Cl

P-I

P-Br

**Question 5** **12 points**

a. 9 points For the molecules HClO_4 and TeF_6 , draw the most stable *Lewis structure* showing appropriate shape and *dashed/wedged bonds*, based on VSEPR. Determine the **molecular geometry** around the central atom. For HClO_4 , label the *most polar bond*.



b. 3 points (Note: Lone pairs are not shown in structures below)

(i) For the three *resonance* structures (1, 2 and 3) shown below, answer the following question:



(I) Circle the *most contributing* resonance structure.

Structure 1

Structure 2

Structure 3

(ii) For the three structures given (4, 5, and 6) below, answer the following questions:



(I) Circle all the structures that represent *resonance structures* of each other:

Structure 4

Structure 5

Structure 6

(II) Among the resonance structures, circle the *most contributing* resonance structure.

Structure 4

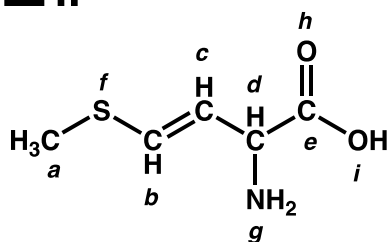
Structure 5

Structure 6

**Question 6** **6 points**

Using the data given below, calculate the **lattice energy** for SrO(s).

	ΔH° (kJ/mol)
$\text{Sr}(s) \longrightarrow \text{Sr}(g)$	+180
$\text{Sr}(g) \longrightarrow \text{Sr}^{2+}(g) + 2e^-$	+1468
$\text{O}_2(g) \longrightarrow 2\text{O}(g)$	+500
$\text{O}(g) + 2e^- \longrightarrow \text{O}^{2-}(g)$	+600
$\text{Sr}(s) + \frac{1}{2}\text{O}_2(g) \longrightarrow \text{SrO}(s)$	-550

**Question 7 10 points**

Consider the molecule given on the left, and answer questions (i) to (vi) using chemical bonding theories that we have learnt in class (valence bond theory, hybridization).

Note that the lone pairs are not shown in the structure:

i. Consider the carbon atoms marked in as ***a***, ***b***, ***c***, ***d***, and ***e*** in the molecule, and determine **hybridization** at each of the carbon centers. (2.5 points)

Atom	Hybridization
Carbon (<i>a</i>)	
Carbon (<i>b</i>)	
Carbon (<i>c</i>)	
Carbon (<i>d</i>)	
Carbon (<i>e</i>)	

ii. Now, consider the sulfur atom (S) marked as ***f***, nitrogen (N) atom (***g***), and oxygen(O) atoms (***h*** and ***i***), and determine their **hybridization**. (2 points)

Atom	Hybridization
Sulfur (<i>f</i>)	
Nitrogen (<i>g</i>)	
Oxygen (<i>h</i>)	
Oxygen (<i>i</i>)	

iii. **Describe** all the bonds formed by the carbon atoms ***a*** and ***b***, using sigma and pi designation as appropriate. (2 points)

Atom	Bonds formed (use sigma and pi as appropriate)
Carbon (<i>a</i>)	
Carbon (<i>b</i>)	

iv. How many total sp^3 -s sigma bonds are in the molecule? (1 point)

v. Determine **electronic** and **molecular geometry** at carbon (C) labeled ***e*** and sulfur (S) labeled ***f*** centres. (2 points)

Atom	Electronic Geometry	Molecular Geometry
Carbon (<i>e</i>)		
Sulfur (<i>f</i>)		

vi. Circle the value from the choices given below, that describes the **C-S-C bond angle at sulfur** (S) marked ***f***: (0.5 points)

>90° to <109.5° 109.5° >109.5 to <120° 120° >120° to <180°


Question 8 12 points

Using the **valence shell orbitals only** draw a **complete** molecular orbital (MO) diagram (with no s-p mixing) for the molecule **PS**, in the space provided below. **(6 points)**

Assume that 3s and 3p atomic orbitals will form molecular orbitals like those formed by the combination of 2s and 2p atomic orbitals that we learnt in class, and there is no s-p mixing.

Show **energy arrow**, **atomic orbitals** involved, **atom labels**, the **molecular orbitals** that are formed, **filled electrons** in the MO diagram etc. Make sure to label all the atomic and molecular orbitals.

Using the MO diagram above, answer the following questions:

- (i) Which one of the following will have the highest bond order? Circle the correct choice
 A) PS^{1-} B) PS C) PS^{1+}
- (ii) Which one of the following will have the longest bond length? Circle the correct choice
 A) PS^{1-} B) PS C) PS^{1+}
- (iii) Which of the following will be diamagnetic? Circle the correct choice.
 A) PS^{1-} B) PS C) PS^{1+}
- (iv) Draw Lewis dot structure of the molecule with the shortest bond length (based on the MO diagram) from among the three (PS^{1-} , PS, PS^{1+}), and indicate the bond order calculated from the Lewis dot structure. **(3 points)**

Lewis Dot Structure

Bond Order



88635252-2C96-4CC7-95F6-C4EF75D14E94

midterm-exam-93388

#201 Page 10 of 10

SCRAP PAGE

**CHEMISTRY 110 –Midterm Exam****Thursday November 7, 2024, 6:30 – 8:30 pm**

Examiners: Pallavi Sirjoosingh and Ashok Kakkar

First name (please write as legibly as possible within the boxes)

--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--

Last name

--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--

Student ID number

--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--

This booklet has 8 questions on 10 pages (The last page can be used as a scrap page).

The staple must NOT be removed from the question booklet. Your intact question booklet must be returned at the end of the exam.

A separate booklet contains a periodic table, and datasheet.

Write your answers directly under the questions in the space provided. Do not write on the QR codes or near the top edges (as it may be cut off).

Please write **legibly and dark enough** for the scan to be readable. For students who use English as a second language, dictionaries are permitted. Calculators are allowed.

DO NOT ASK QUESTIONS OF THE INVIGILATORS about interpretations of the exam questions. If you have a concern about a question, please summarize it at the top of this page. All concerns will be evaluated after the exam. GOOD LUCK!

**Question 1 8 points** (*Assume all transitions are possible*)

a. 6 points Calculate the **energy (in kJ/mol)** required to ionize (completely remove) the electron from the **$n=3$** level of **He^+ ion** (He: Helium). Show your work including equations. (3 sig. figs.)

b. 2 points

(I) Select the largest (longest) wavelength from the three choices below:

- i. Wavelength of photon to *completely remove* an electron from *1s orbital of Hydrogen atom*
- ii. Wavelength of photon to *completely remove* an electron from *1s orbital of Boron atom (B)*
- iii. Wavelength of photon to *completely remove* an electron from *1s orbital of Boron cation (B^+)*

i or ii or iii

(II) Select the smallest (shortest) wavelength from the three choices below (*for H atom*):

- i. Wavelength of photon *emitted* for an electron transitioning from *2p to 1s orbital*
- ii. Wavelength of photon *absorbed* for an electron transitioning from *2s to 4p orbital*
- iii. Wavelength of photon *absorbed* for an electron transitioning from *1s to 3p orbital*

i or ii or iii



Question 2 8 points



A Chem110 student plots the graph shown on the left, from data obtained for the kinetic energy (eV) of the ejected electron versus frequency (s^{-1}) of the incident light, for an unknown metal. ($1\text{eV} = 1.6022 \times 10^{-19} \text{ J}$)

Based on the plot, answer the questions **a to c** below:

a. 1 point On the plot above, label the **threshold frequency**.

b. 2 points What is the value of the work potential (in J) of the unknown metal?

c. 3 points What is the *longest* wavelength (in nm) required to eject an electron from this metal's surface? Show your work including equation(s) used. (3 sig figs)

d. 2 points In the space provided below, plot the following

(Assume frequency is *above* threshold frequency):

- Kinetic energy* of an ejected electron versus the *intensity* of the incident light (left)
- Number of electrons* ejected versus the *frequency* of the incident light (right)



**Question 3** **6 points****a. 2 points**

In the ground state of Ge (Germanium; $Z = 32$) atom, **name** the following orbitals:

i. Orbital containing the electron that requires the maximum energy to be removed.

ii. Orbital containing the electron that experiences the minimum effective nuclear charge.

b. 2 points

An unknown anion (A^-) has the following *excited* electronic configuration: $[Ar] 4s^2 3d^3 4p^1$

(Ar: Argon; $Z = 18$)

Give the **condensed ground state electronic configuration** of the atom, **A** and the cation, **A^{2+}** .
Show your work. (Assume all electronic transitions are possible)

Electronic Configuration of A:

Electronic Configuration of A^{2+} :

c. 2 points

In a single atom, determine how many electrons can have the following set of quantum numbers?

(i) $n = 5; l = 3$

In a single atom, determine how many electrons can have the following set of quantum numbers?

(i) $n = 3; l = 2; m_s = +1/2$

**Question 4** **8 points**

Rank the following from 1 to 4 in **increasing** order. Provide the ranking in the boxes provided.

Note: Atomic number (Z) are provided for relevant atoms

a) Increasing (lowest to highest) order of *ionization energy*: (1= lowest; 4 = highest)

Mg

Mg: Z = 12

Rb

Rb: Z = 37

Cs

Cs: Z = 55

K

K : Z = 19

b) Increasing (lowest to highest) order of *bond angle*: (1= lowest; 4 = highest)

H₂OCCl₄XeF₄

HCN

c) Increasing (smallest to largest) order of *ionic size* (1= smallest; 4 = largest)

S²⁻

S: Z = 16

O²⁻

O: Z = 8

Be²⁺

Be: Z = 4

Na⁺

Na: Z = 11

d) Increasing (lowest to highest) order of *bond energy* (1= lowest; 4 = highest)

P-F

P-Cl

P-I

P-Br

**Question 5** **12 points**

a. 9 points For the molecules HClO_4 and TeF_6 , draw the most stable *Lewis structure* showing appropriate shape and *dashed/wedged bonds*, based on VSEPR. Determine the **molecular geometry** around the central atom. For HClO_4 , label the *most polar bond*.



b. 3 points (Note: Lone pairs are not shown in structures below)

(i) For the three *resonance* structures (1, 2 and 3) shown below, answer the following question:



(I) Circle the *most contributing* resonance structure.

Structure 1

Structure 2

Structure 3

(ii) For the three structures given (4, 5, and 6) below, answer the following questions:



(I) Circle all the structures that represent *resonance structures* of each other:

Structure 4

Structure 5

Structure 6

(II) Among the resonance structures, circle the *most contributing* resonance structure.

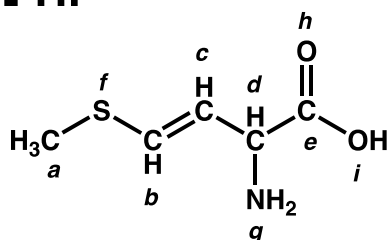
Structure 4

Structure 5

Structure 6

**Question 6** **6 points**Using the data given below, calculate the **lattice energy** for SrO(s).

	ΔH° (kJ/mol)
$\text{Sr}(s) \longrightarrow \text{Sr}(g)$	+180
$\text{Sr}(g) \longrightarrow \text{Sr}^{2+}(g) + 2e^-$	+1468
$\text{O}_2(g) \longrightarrow 2\text{O}(g)$	+500
$\text{O}(g) + 2e^- \longrightarrow \text{O}^{2-}(g)$	+600
$\text{Sr}(s) + \frac{1}{2}\text{O}_2(g) \longrightarrow \text{SrO}(s)$	-550

**Question 7 10 points**

Consider the molecule given on the left, and answer questions (i) to (vi) using chemical bonding theories that we have learnt in class (valence bond theory, hybridization).

Note that the lone pairs are not shown in the structure:

i. Consider the carbon atoms marked in as ***a***, ***b***, ***c***, ***d***, and ***e*** in the molecule, and determine **hybridization** at each of the carbon centers. (2.5 points)

Atom	Hybridization
Carbon (<i>a</i>)	
Carbon (<i>b</i>)	
Carbon (<i>c</i>)	
Carbon (<i>d</i>)	
Carbon (<i>e</i>)	

ii. Now, consider the sulfur atom (S) marked as ***f***, nitrogen (N) atom (***g***), and oxygen(O) atoms (***h*** and ***i***), and determine their **hybridization**. (2 points)

Atom	Hybridization
Sulfur (<i>f</i>)	
Nitrogen (<i>g</i>)	
Oxygen (<i>h</i>)	
Oxygen (<i>i</i>)	

iii. **Describe** all the bonds formed by the carbon atoms ***a*** and ***b***, using sigma and pi designation as appropriate. (2 points)

Atom	Bonds formed (use sigma and pi as appropriate)
Carbon (<i>a</i>)	
Carbon (<i>b</i>)	

iv. How many total sp^3 -s sigma bonds are in the molecule? (1 point)

v. Determine **electronic** and **molecular geometry** at carbon (C) labeled ***e*** and sulfur (S) labeled ***f*** centres. (2 points)

Atom	Electronic Geometry	Molecular Geometry
Carbon (<i>e</i>)		
Sulfur (<i>f</i>)		

vi. Circle the value from the choices given below, that describes the **C-S-C bond angle at sulfur** (S) marked ***f***: (0.5 points)

$>90^\circ$ to $<109.5^\circ$ 109.5° >109.5 to $<120^\circ$ 120° $>120^\circ$ to $<180^\circ$

**Question 8 12 points**

Using the **valence shell orbitals only** draw a **complete** molecular orbital (MO) diagram (with no s-p mixing) for the molecule **PS**, in the space provided below. **(6 points)**

Assume that 3s and 3p atomic orbitals will form molecular orbitals like those formed by the combination of 2s and 2p atomic orbitals that we learnt in class, and there is no s-p mixing.

Show **energy arrow**, **atomic orbitals** involved, **atom labels**, the **molecular orbitals** that are formed, **filled electrons** in the MO diagram etc. Make sure to label all the atomic and molecular orbitals.

Using the MO diagram above, answer the following questions:

- (i) Which one of the following will have the highest bond order? Circle the correct choice
A) PS^{1-} B) PS C) PS^{1+}
- (ii) Which one of the following will have the longest bond length? Circle the correct choice
A) PS^{1-} B) PS C) PS^{1+}
- (iii) Which of the following will be diamagnetic? Circle the correct choice.
A) PS^{1-} B) PS C) PS^{1+}
- (iv) Draw Lewis dot structure of the molecule with the shortest bond length (based on the MO diagram) from among the three (PS^{1-} , PS, PS^{1+}), and indicate the bond order calculated from the Lewis dot structure. **(3 points)**

Lewis Dot Structure

Bond Order



3D30F64B-7E7F-4149-B875-17030FD905A5

midterm-exam-93388

#202 Page 10 of 10

SCRAP PAGE

**CHEMISTRY 110 –Midterm Exam****Thursday November 7, 2024, 6:30 – 8:30 pm****Examiners: Pallavi Sirjoosingh and Ashok Kakkar**

First name (please write as legibly as possible within the boxes)

--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--

Last name

--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--

Student ID number

--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--

This booklet has 8 questions on 10 pages (The last page can be used as a scrap page).

The staple must NOT be removed from the question booklet. Your intact question booklet must be returned at the end of the exam.

A separate booklet contains a periodic table, and datasheet.

Write your answers directly under the questions in the space provided. Do not write on the QR codes or near the top edges (as it may be cut off).

Please write **legibly and dark enough** for the scan to be readable. For students who use English as a second language, dictionaries are permitted. Calculators are allowed.

DO NOT ASK QUESTIONS OF THE INVIGILATORS about interpretations of the exam questions. If you have a concern about a question, please summarize it at the top of this page. All concerns will be evaluated after the exam. GOOD LUCK!

**Question 1 8 points** (*Assume all transitions are possible*)

a. 6 points Calculate the **energy (in kJ/mol)** required to ionize (completely remove) the electron from the **$n=3$** level of **He^+ ion** (He: Helium). Show your work including equations. (3 sig. figs.)

b. 2 points

(I) Select the largest (longest) wavelength from the three choices below:

- i. Wavelength of photon to *completely remove* an electron from *1s orbital of Hydrogen atom*
- ii. Wavelength of photon to *completely remove* an electron from *1s orbital of Boron atom (B)*
- iii. Wavelength of photon to *completely remove* an electron from *1s orbital of Boron cation (B^+)*

i or ii or iii

(II) Select the smallest (shortest) wavelength from the three choices below (*for H atom*):

- i. Wavelength of photon *emitted* for an electron transitioning from *2p to 1s orbital*
- ii. Wavelength of photon *absorbed* for an electron transitioning from *2s to 4p orbital*
- iii. Wavelength of photon *absorbed* for an electron transitioning from *1s to 3p orbital*

i or ii or iii



Question 2 8 points



A Chem110 student plots the graph shown on the left, from data obtained for the kinetic energy (eV) of the ejected electron versus frequency (s^{-1}) of the incident light, for an unknown metal. ($1\text{eV} = 1.6022 \times 10^{-19} \text{ J}$)

Based on the plot, answer the questions **a to c** below:

a. 1 point On the plot above, label the **threshold frequency**.

b. 2 points What is the value of the work potential (in J) of the unknown metal?

c. 3 points What is the *longest* wavelength (in nm) required to eject an electron from this metal's surface? Show your work including equation(s) used. (3 sig figs)

d. 2 points In the space provided below, plot the following

(Assume frequency is *above* threshold frequency):

- i. *Kinetic energy* of an ejected electron versus the *intensity* of the incident light (left)
- ii. *Number of electrons* ejected versus the *frequency* of the incident light (right)



**Question 3** **6 points****a. 2 points**

In the ground state of Ge (Germanium; $Z = 32$) atom, **name** the following orbitals:

i. Orbital containing the electron that requires the maximum energy to be removed.

ii. Orbital containing the electron that experiences the minimum effective nuclear charge.

b. 2 points

An unknown anion (A^-) has the following *excited* electronic configuration: $[Ar] 4s^2 3d^3 4p^1$

(Ar: Argon; $Z = 18$)

Give the **condensed ground state electronic configuration** of the atom, **A** and the cation, **A^{2+}** .
Show your work. (Assume all electronic transitions are possible)

Electronic Configuration of A:

Electronic Configuration of A^{2+} :

c. 2 points

In a single atom, determine how many electrons can have the following set of quantum numbers?

(i) $n = 5; l = 3$

In a single atom, determine how many electrons can have the following set of quantum numbers?

(i) $n = 3; l = 2; m_s = +1/2$

**Question 4** **8 points**

Rank the following from 1 to 4 in **increasing** order. Provide the ranking in the boxes provided.

Note: Atomic number (Z) are provided for relevant atoms

a) Increasing (lowest to highest) order of *ionization energy*: (1= lowest; 4 = highest)

Mg

Mg: Z = 12

Rb

Rb: Z = 37

Cs

Cs: Z = 55

K

K : Z = 19

b) Increasing (lowest to highest) order of *bond angle*: (1= lowest; 4 = highest)

H₂OCCl₄XeF₄

HCN

c) Increasing (smallest to largest) order of *ionic size* (1= smallest; 4 = largest)

S²⁻

S: Z = 16

O²⁻

O: Z = 8

Be²⁺

Be: Z = 4

Na⁺

Na: Z = 11

d) Increasing (lowest to highest) order of *bond energy* (1= lowest; 4 = highest)

P-F

P-Cl

P-I

P-Br

**Question 5** **12 points**

a. 9 points For the molecules HClO_4 and TeF_6 , draw the most stable *Lewis structure* showing appropriate shape and *dashed/wedged bonds*, based on VSEPR. Determine the **molecular geometry** around the central atom. For HClO_4 , label the *most polar bond*.



b. 3 points (Note: Lone pairs are not shown in structures below)

(i) For the three *resonance* structures (1, 2 and 3) shown below, answer the following question:



(I) Circle the *most contributing* resonance structure.

Structure 1

Structure 2

Structure 3

(ii) For the three structures given (4, 5, and 6) below, answer the following questions:



(I) Circle all the structures that represent *resonance structures* of each other:

Structure 4

Structure 5

Structure 6

(II) Among the resonance structures, circle the *most contributing* resonance structure.

Structure 4

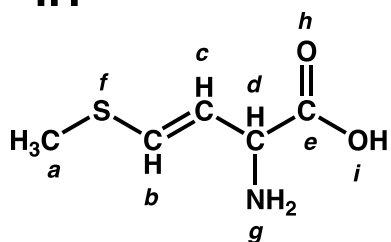
Structure 5

Structure 6

**Question 6** **6 points**

Using the data given below, calculate the **lattice energy** for SrO(s).

	ΔH° (kJ/mol)
$\text{Sr}(s) \longrightarrow \text{Sr}(g)$	+180
$\text{Sr}(g) \longrightarrow \text{Sr}^{2+}(g) + 2e^-$	+1468
$\text{O}_2(g) \longrightarrow 2\text{O}(g)$	+500
$\text{O}(g) + 2e^- \longrightarrow \text{O}^{2-}(g)$	+600
$\text{Sr}(s) + \frac{1}{2}\text{O}_2(g) \longrightarrow \text{SrO}(s)$	-550

**Question 7 10 points**

Consider the molecule given on the left, and answer questions (i) to (vi) using chemical bonding theories that we have learnt in class (valence bond theory, hybridization).

Note that the lone pairs are not shown in the structure:

i. Consider the carbon atoms marked in as ***a***, ***b***, ***c***, ***d***, and ***e*** in the molecule, and determine **hybridization** at each of the carbon centers. (2.5 points)

Atom	Hybridization
Carbon (<i>a</i>)	
Carbon (<i>b</i>)	
Carbon (<i>c</i>)	
Carbon (<i>d</i>)	
Carbon (<i>e</i>)	

ii. Now, consider the sulfur atom (S) marked as ***f***, nitrogen (N) atom (***g***), and oxygen(O) atoms (***h*** and ***i***), and determine their **hybridization**. (2 points)

Atom	Hybridization
Sulfur (<i>f</i>)	
Nitrogen (<i>g</i>)	
Oxygen (<i>h</i>)	
Oxygen (<i>i</i>)	

iii. **Describe** all the bonds formed by the carbon atoms ***a*** and ***b***, using sigma and pi designation as appropriate. (2 points)

Atom	Bonds formed (use sigma and pi as appropriate)
Carbon (<i>a</i>)	
Carbon (<i>b</i>)	

iv. How many total sp^3 -s sigma bonds are in the molecule? (1 point)

v. Determine **electronic** and **molecular geometry** at carbon (C) labeled ***e*** and sulfur (S) labeled ***f*** centres. (2 points)

Atom	Electronic Geometry	Molecular Geometry
Carbon (<i>e</i>)		
Sulfur (<i>f</i>)		

vi. Circle the value from the choices given below, that describes the **C-S-C bond angle at sulfur** (S) marked ***f***: (0.5 points)

>90° to <109.5°

109.5°

>109.5 to <120°

120°

>120° to <180°

**Question 8 12 points**

Using the **valence shell orbitals only** draw a **complete** molecular orbital (MO) diagram (with no s-p mixing) for the molecule **PS**, in the space provided below. **(6 points)**

Assume that 3s and 3p atomic orbitals will form molecular orbitals like those formed by the combination of 2s and 2p atomic orbitals that we learnt in class, and there is no s-p mixing.

Show **energy arrow**, **atomic orbitals** involved, **atom labels**, the **molecular orbitals** that are formed, **filled electrons** in the MO diagram etc. Make sure to label all the atomic and molecular orbitals.

Using the MO diagram above, answer the following questions:

- (i) Which one of the following will have the highest bond order? Circle the correct choice
A) PS^{1-} B) PS C) PS^{1+}
- (ii) Which one of the following will have the longest bond length? Circle the correct choice
A) PS^{1-} B) PS C) PS^{1+}
- (iii) Which of the following will be diamagnetic? Circle the correct choice.
A) PS^{1-} B) PS C) PS^{1+}
- (iv) Draw Lewis dot structure of the molecule with the shortest bond length (based on the MO diagram) from among the three (PS^{1-} , PS, PS^{1+}), and indicate the bond order calculated from the Lewis dot structure. **(3 points)**

Lewis Dot Structure

Bond Order



032EBD57-11B2-4D07-B6B9-E95FF2948BDA

midterm-exam-93388

#203 Page 10 of 10

SCRAP PAGE

**CHEMISTRY 110 –Midterm Exam****Thursday November 7, 2024, 6:30 – 8:30 pm**

Examiners: Pallavi Sirjoosingh and Ashok Kakkar

First name (please write as legibly as possible within the boxes)

--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--

Last name

--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--

Student ID number

--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--

This booklet has 8 questions on 10 pages (The last page can be used as a scrap page).

The staple must NOT be removed from the question booklet. Your intact question booklet must be returned at the end of the exam.

A separate booklet contains a periodic table, and datasheet.

Write your answers directly under the questions in the space provided. Do not write on the QR codes or near the top edges (as it may be cut off).

Please write **legibly and dark enough** for the scan to be readable. For students who use English as a second language, dictionaries are permitted. Calculators are allowed.

DO NOT ASK QUESTIONS OF THE INVIGILATORS about interpretations of the exam questions. If you have a concern about a question, please summarize it at the top of this page. All concerns will be evaluated after the exam. GOOD LUCK!

**Question 1 8 points** (*Assume all transitions are possible*)

a. 6 points Calculate the **energy (in kJ/mol)** required to ionize (completely remove) the electron from the **n=3** level of **He⁺ ion** (He: Helium). Show your work including equations. (3 sig. figs.)

b. 2 points

(I) Select the largest (longest) wavelength from the three choices below:

- i. Wavelength of photon to *completely remove* an electron from *1s orbital of Hydrogen atom*
- ii. Wavelength of photon to *completely remove* an electron from *1s orbital of Boron atom (B)*
- iii. Wavelength of photon to *completely remove* an electron from *1s orbital of Boron cation (B⁺)*

i or ii or iii

(II) Select the smallest (shortest) wavelength from the three choices below (*for H atom*):

- i. Wavelength of photon *emitted* for an electron transitioning from *2p to 1s orbital*
- ii. Wavelength of photon *absorbed* for an electron transitioning from *2s to 4p orbital*
- iii. Wavelength of photon *absorbed* for an electron transitioning from *1s to 3p orbital*

i or ii or iii



Question 2 8 points



A Chem110 student plots the graph shown on the left, from data obtained for the kinetic energy (eV) of the ejected electron versus frequency (s^{-1}) of the incident light, for an unknown metal. ($1\text{eV} = 1.6022 \times 10^{-19} \text{ J}$)

Based on the plot, answer the questions **a to c** below:

a. 1 point On the plot above, label the **threshold frequency**.

b. 2 points What is the value of the work potential (in J) of the unknown metal?

c. 3 points What is the *longest* wavelength (in nm) required to eject an electron from this metal's surface? Show your work including equation(s) used. (3 sig figs)

d. 2 points In the space provided below, plot the following

(Assume frequency is *above* threshold frequency):

- Kinetic energy* of an ejected electron versus the *intensity* of the incident light (left)
- Number of electrons* ejected versus the *frequency* of the incident light (right)



**Question 3** **6 points****a. 2 points**

In the ground state of Ge (Germanium; $Z = 32$) atom, **name** the following orbitals:

i. Orbital containing the electron that requires the maximum energy to be removed.

ii. Orbital containing the electron that experiences the minimum effective nuclear charge.

b. 2 points

An unknown anion (A^-) has the following *excited* electronic configuration: $[Ar] 4s^2 3d^3 4p^1$

(Ar: Argon; $Z = 18$)

Give the **condensed ground state electronic configuration** of the atom, **A** and the cation, **A^{2+}** .
Show your work. (Assume all electronic transitions are possible)

Electronic Configuration of A:

Electronic Configuration of A^{2+} :

c. 2 points

In a single atom, determine how many electrons can have the following set of quantum numbers?

(i) $n = 5; l = 3$

In a single atom, determine how many electrons can have the following set of quantum numbers?

(i) $n = 3; l = 2; m_s = +1/2$

**Question 4** **8 points**

Rank the following from 1 to 4 in **increasing** order. Provide the ranking in the boxes provided.

Note: Atomic number (Z) are provided for relevant atoms

a) Increasing (lowest to highest) order of *ionization energy*: (1= lowest; 4 = highest)

Mg

Mg: Z = 12

Rb

Rb: Z = 37

Cs

Cs: Z = 55

K

K : Z = 19

b) Increasing (lowest to highest) order of *bond angle*: (1= lowest; 4 = highest)

H₂OCCl₄XeF₄

HCN

c) Increasing (smallest to largest) order of *ionic size* (1= smallest; 4 = largest)

S²⁻

S: Z = 16

O²⁻

O: Z = 8

Be²⁺

Be: Z = 4

Na⁺

Na: Z = 11

d) Increasing (lowest to highest) order of *bond energy* (1= lowest; 4 = highest)

P-F

P-Cl

P-I

P-Br

**Question 5** **12 points**

a. 9 points For the molecules HClO_4 and TeF_6 , draw the most stable *Lewis structure* showing appropriate shape and *dashed/wedged bonds*, based on VSEPR. Determine the **molecular geometry** around the central atom. For HClO_4 , label the *most polar bond*.



b. 3 points (Note: Lone pairs are not shown in structures below)

(i) For the three *resonance* structures (1, 2 and 3) shown below, answer the following question:



(I) Circle the *most contributing* resonance structure.

Structure 1

Structure 2

Structure 3

(ii) For the three structures given (4, 5, and 6) below, answer the following questions:



(I) Circle all the structures that represent *resonance structures* of each other:

Structure 4

Structure 5

Structure 6

(II) Among the resonance structures, circle the *most contributing* resonance structure.

Structure 4

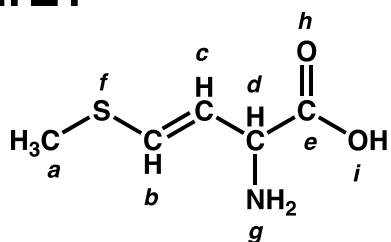
Structure 5

Structure 6

**Question 6** **6 points**

Using the data given below, calculate the **lattice energy** for SrO(s).

	ΔH° (kJ/mol)
$\text{Sr}(s) \longrightarrow \text{Sr}(g)$	+180
$\text{Sr}(g) \longrightarrow \text{Sr}^{2+}(g) + 2e^-$	+1468
$\text{O}_2(g) \longrightarrow 2\text{O}(g)$	+500
$\text{O}(g) + 2e^- \longrightarrow \text{O}^{2-}(g)$	+600
$\text{Sr}(s) + \frac{1}{2}\text{O}_2(g) \longrightarrow \text{SrO}(s)$	-550

**Question 7 10 points**

Consider the molecule given on the left, and answer questions (i) to (vi) using chemical bonding theories that we have learnt in class (valence bond theory, hybridization).

Note that the lone pairs are not shown in the structure:

i. Consider the carbon atoms marked in as ***a***, ***b***, ***c***, ***d***, and ***e*** in the molecule, and determine **hybridization** at each of the carbon centers. (2.5 points)

Atom	Hybridization
Carbon (<i>a</i>)	
Carbon (<i>b</i>)	
Carbon (<i>c</i>)	
Carbon (<i>d</i>)	
Carbon (<i>e</i>)	

ii. Now, consider the sulfur atom (S) marked as ***f***, nitrogen (N) atom (***g***), and oxygen(O) atoms (***h*** and ***i***), and determine their **hybridization**. (2 points)

Atom	Hybridization
Sulfur (<i>f</i>)	
Nitrogen (<i>g</i>)	
Oxygen (<i>h</i>)	
Oxygen (<i>i</i>)	

iii. **Describe** all the bonds formed by the carbon atoms ***a*** and ***b***, using sigma and pi designation as appropriate. (2 points)

Atom	Bonds formed (use sigma and pi as appropriate)
Carbon (<i>a</i>)	
Carbon (<i>b</i>)	

iv. How many total sp^3 -s sigma bonds are in the molecule? (1 point)

v. Determine **electronic** and **molecular geometry** at carbon (C) labeled ***e*** and sulfur (S) labeled ***f*** centres. (2 points)

Atom	Electronic Geometry	Molecular Geometry
Carbon (<i>e</i>)		
Sulfur (<i>f</i>)		

vi. Circle the value from the choices given below, that describes the **C-S-C bond angle at sulfur** (S) marked ***f***: (0.5 points)

>90° to <109.5° 109.5° >109.5 to <120° 120° >120° to <180°

**Question 8 12 points**

Using the **valence shell orbitals only** draw a **complete** molecular orbital (MO) diagram (with no s-p mixing) for the molecule **PS**, in the space provided below. **(6 points)**

Assume that 3s and 3p atomic orbitals will form molecular orbitals like those formed by the combination of 2s and 2p atomic orbitals that we learnt in class, and there is no s-p mixing.

Show **energy arrow**, **atomic orbitals** involved, **atom labels**, the **molecular orbitals** that are formed, **filled electrons** in the MO diagram etc. Make sure to label all the atomic and molecular orbitals.

Using the MO diagram above, answer the following questions:

- (i) Which one of the following will have the highest bond order? Circle the correct choice
A) PS^{1-} B) PS C) PS^{1+}
- (ii) Which one of the following will have the longest bond length? Circle the correct choice
A) PS^{1-} B) PS C) PS^{1+}
- (iii) Which of the following will be diamagnetic? Circle the correct choice.
A) PS^{1-} B) PS C) PS^{1+}
- (iv) Draw Lewis dot structure of the molecule with the shortest bond length (based on the MO diagram) from among the three (PS^{1-} , PS, PS^{1+}), and indicate the bond order calculated from the Lewis dot structure. **(3 points)**

Lewis Dot Structure

Bond Order



4017E2F2-5AB2-4720-A67F-2C02C5F5FDD2

midterm-exam-93388

#204 Page 10 of 10

SCRAP PAGE

**CHEMISTRY 110 –Midterm Exam****Thursday November 7, 2024, 6:30 – 8:30 pm**

Examiners: Pallavi Sirjoosingh and Ashok Kakkar

First name (please write as legibly as possible within the boxes)

--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--

Last name

--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--

Student ID number

--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--

This booklet has 8 questions on 10 pages (The last page can be used as a scrap page).

The staple must NOT be removed from the question booklet. Your intact question booklet must be returned at the end of the exam.

A separate booklet contains a periodic table, and datasheet.

Write your answers directly under the questions in the space provided. Do not write on the QR codes or near the top edges (as it may be cut off).

Please write **legibly and dark enough** for the scan to be readable. For students who use English as a second language, dictionaries are permitted. Calculators are allowed.

DO NOT ASK QUESTIONS OF THE INVIGILATORS about interpretations of the exam questions. If you have a concern about a question, please summarize it at the top of this page. All concerns will be evaluated after the exam. GOOD LUCK!

**Question 1 8 points** (*Assume all transitions are possible*)

a. 6 points Calculate the **energy (in kJ/mol)** required to ionize (completely remove) the electron from the **$n=3$** level of **He^+ ion** (He: Helium). Show your work including equations. (3 sig. figs.)

b. 2 points

(I) Select the largest (longest) wavelength from the three choices below:

- i. Wavelength of photon to *completely remove* an electron from *1s orbital of Hydrogen atom*
- ii. Wavelength of photon to *completely remove* an electron from *1s orbital of Boron atom (B)*
- iii. Wavelength of photon to *completely remove* an electron from *1s orbital of Boron cation (B^+)*

i or ii or iii

(II) Select the smallest (shortest) wavelength from the three choices below (*for H atom*):

- i. Wavelength of photon *emitted* for an electron transitioning from *2p to 1s orbital*
- ii. Wavelength of photon *absorbed* for an electron transitioning from *2s to 4p orbital*
- iii. Wavelength of photon *absorbed* for an electron transitioning from *1s to 3p orbital*

i or ii or iii



Question 2 8 points



A Chem110 student plots the graph shown on the left, from data obtained for the kinetic energy (eV) of the ejected electron versus frequency (s^{-1}) of the incident light, for an unknown metal. ($1\text{eV} = 1.6022 \times 10^{-19} \text{ J}$)

Based on the plot, answer the questions **a to c** below:

a. 1 point On the plot above, label the **threshold frequency**.

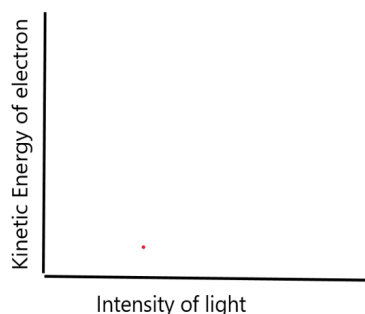
b. 2 points What is the value of the work potential (in J) of the unknown metal?

c. 3 points What is the *longest* wavelength (in nm) required to eject an electron from this metal's surface? Show your work including equation(s) used. (3 sig figs)

d. 2 points In the space provided below, plot the following

(Assume frequency is *above* threshold frequency):

- Kinetic energy* of an ejected electron versus the *intensity* of the incident light (left)
- Number of electrons* ejected versus the *frequency* of the incident light (right)



**Question 3** **6 points****a. 2 points**

In the ground state of Ge (Germanium; $Z = 32$) atom, **name** the following orbitals:

i. Orbital containing the electron that requires the maximum energy to be removed.

ii. Orbital containing the electron that experiences the minimum effective nuclear charge.

b. 2 points

An unknown anion (A^-) has the following *excited* electronic configuration: $[Ar] 4s^2 3d^3 4p^1$

(Ar: Argon; $Z = 18$)

Give the **condensed ground state electronic configuration** of the atom, **A** and the cation, **A^{2+}** .
Show your work. (Assume all electronic transitions are possible)

Electronic Configuration of A:

Electronic Configuration of A^{2+} :

c. 2 points

In a single atom, determine how many electrons can have the following set of quantum numbers?

(i) $n = 5; l = 3$

In a single atom, determine how many electrons can have the following set of quantum numbers?

(i) $n = 3; l = 2; m_s = +1/2$

**Question 4** **8 points**

Rank the following from 1 to 4 in **increasing** order. Provide the ranking in the boxes provided.

Note: Atomic number (Z) are provided for relevant atoms

a) Increasing (lowest to highest) order of *ionization energy*: (1= lowest; 4 = highest)

Mg

Mg: Z = 12

Rb

Rb: Z = 37

Cs

Cs: Z = 55

K

K : Z = 19

b) Increasing (lowest to highest) order of *bond angle*: (1= lowest; 4 = highest)

H₂OCCl₄XeF₄

HCN

c) Increasing (smallest to largest) order of *ionic size* (1= smallest; 4 = largest)

S²⁻

S: Z = 16

O²⁻

O: Z = 8

Be²⁺

Be: Z = 4

Na⁺

Na: Z = 11

d) Increasing (lowest to highest) order of *bond energy* (1= lowest; 4 = highest)

P-F

P-Cl

P-I

P-Br

**Question 5** **12 points**

a. 9 points For the molecules HClO_4 and TeF_6 , draw the most stable *Lewis structure* showing appropriate shape and *dashed/wedged bonds*, based on VSEPR. Determine the **molecular geometry** around the central atom. For HClO_4 , label the *most polar bond*.



b. 3 points (Note: Lone pairs are not shown in structures below)

(i) For the three *resonance* structures (1, 2 and 3) shown below, answer the following question:



(I) Circle the *most contributing* resonance structure.

Structure 1

Structure 2

Structure 3

(ii) For the three structures given (4, 5, and 6) below, answer the following questions:



(I) Circle all the structures that represent *resonance structures* of each other:

Structure 4

Structure 5

Structure 6

(II) Among the resonance structures, circle the *most contributing* resonance structure.

Structure 4

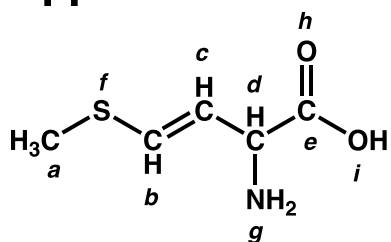
Structure 5

Structure 6

**Question 6** **6 points**

Using the data given below, calculate the **lattice energy** for SrO(s).

	ΔH° (kJ/mol)
$\text{Sr}(s) \longrightarrow \text{Sr}(g)$	+180
$\text{Sr}(g) \longrightarrow \text{Sr}^{2+}(g) + 2e^-$	+1468
$\text{O}_2(g) \longrightarrow 2\text{O}(g)$	+500
$\text{O}(g) + 2e^- \longrightarrow \text{O}^{2-}(g)$	+600
$\text{Sr}(s) + \frac{1}{2}\text{O}_2(g) \longrightarrow \text{SrO}(s)$	-550

**Question 7 10 points**

Consider the molecule given on the left, and answer questions (i) to (vi) using chemical bonding theories that we have learnt in class (valence bond theory, hybridization).

Note that the lone pairs are not shown in the structure:

i. Consider the carbon atoms marked in as ***a***, ***b***, ***c***, ***d***, and ***e*** in the molecule, and determine **hybridization** at each of the carbon centers. (2.5 points)

Atom	Hybridization
Carbon (<i>a</i>)	
Carbon (<i>b</i>)	
Carbon (<i>c</i>)	
Carbon (<i>d</i>)	
Carbon (<i>e</i>)	

ii. Now, consider the sulfur atom (S) marked as ***f***, nitrogen (N) atom (***g***), and oxygen(O) atoms (***h*** and ***i***), and determine their **hybridization**. (2 points)

Atom	Hybridization
Sulfur (<i>f</i>)	
Nitrogen (<i>g</i>)	
Oxygen (<i>h</i>)	
Oxygen (<i>i</i>)	

iii. **Describe** all the bonds formed by the carbon atoms ***a*** and ***b***, using sigma and pi designation as appropriate. (2 points)

Atom	Bonds formed (use sigma and pi as appropriate)
Carbon (<i>a</i>)	
Carbon (<i>b</i>)	

iv. How many total sp^3 -s sigma bonds are in the molecule? (1 point)

v. Determine **electronic** and **molecular geometry** at carbon (C) labeled ***e*** and sulfur (S) labeled ***f*** centres. (2 points)

Atom	Electronic Geometry	Molecular Geometry
Carbon (<i>e</i>)		
Sulfur (<i>f</i>)		

vi. Circle the value from the choices given below, that describes the **C-S-C bond angle at sulfur** (S) marked ***f***: (0.5 points)

$>90^\circ$ to $<109.5^\circ$ 109.5° >109.5 to $<120^\circ$ 120° $>120^\circ$ to $<180^\circ$


Question 8 12 points

Using the **valence shell orbitals only** draw a **complete** molecular orbital (MO) diagram (with no s-p mixing) for the molecule **PS**, in the space provided below. **(6 points)**

Assume that 3s and 3p atomic orbitals will form molecular orbitals like those formed by the combination of 2s and 2p atomic orbitals that we learnt in class, and there is no s-p mixing.

Show **energy arrow**, **atomic orbitals** involved, **atom labels**, the **molecular orbitals** that are formed, **filled electrons** in the MO diagram etc. Make sure to label all the atomic and molecular orbitals.

Using the MO diagram above, answer the following questions:

- (i) Which one of the following will have the highest bond order? Circle the correct choice
 A) PS^{1-} B) PS C) PS^{1+}
- (ii) Which one of the following will have the longest bond length? Circle the correct choice
 A) PS^{1-} B) PS C) PS^{1+}
- (iii) Which of the following will be diamagnetic? Circle the correct choice.
 A) PS^{1-} B) PS C) PS^{1+}
- (iv) Draw Lewis dot structure of the molecule with the shortest bond length (based on the MO diagram) from among the three (PS^{1-} , PS, PS^{1+}), and indicate the bond order calculated from the Lewis dot structure. **(3 points)**

Lewis Dot Structure

Bond Order



5FE80BC3-100D-45F3-829A-5B33E10BEE94

midterm-exam-93388

#205 Page 10 of 10

SCRAP PAGE

**CHEMISTRY 110 –Midterm Exam****Thursday November 7, 2024, 6:30 – 8:30 pm**

Examiners: Pallavi Sirjoosingh and Ashok Kakkar

First name (please write as legibly as possible within the boxes)

--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--

Last name

--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--

Student ID number

--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--

This booklet has 8 questions on 10 pages (The last page can be used as a scrap page).

The staple must NOT be removed from the question booklet. Your intact question booklet must be returned at the end of the exam.

A separate booklet contains a periodic table, and datasheet.

Write your answers directly under the questions in the space provided. Do not write on the QR codes or near the top edges (as it may be cut off).

Please write **legibly and dark enough** for the scan to be readable. For students who use English as a second language, dictionaries are permitted. Calculators are allowed.

DO NOT ASK QUESTIONS OF THE INVIGILATORS about interpretations of the exam questions. If you have a concern about a question, please summarize it at the top of this page. All concerns will be evaluated after the exam. GOOD LUCK!

**Question 1 8 points** (*Assume all transitions are possible*)

a. 6 points Calculate the **energy (in kJ/mol)** required to ionize (completely remove) the electron from the **$n=3$** level of **He^+ ion** (He: Helium). Show your work including equations. (3 sig. figs.)

b. 2 points

(I) Select the largest (longest) wavelength from the three choices below:

- i. Wavelength of photon to *completely remove* an electron from *1s orbital of Hydrogen atom*
- ii. Wavelength of photon to *completely remove* an electron from *1s orbital of Boron atom (B)*
- iii. Wavelength of photon to *completely remove* an electron from *1s orbital of Boron cation (B^+)*

i or ii or iii

(II) Select the smallest (shortest) wavelength from the three choices below (*for H atom*):

- i. Wavelength of photon *emitted* for an electron transitioning from *2p to 1s orbital*
- ii. Wavelength of photon *absorbed* for an electron transitioning from *2s to 4p orbital*
- iii. Wavelength of photon *absorbed* for an electron transitioning from *1s to 3p orbital*

i or ii or iii



Question 2 8 points



A Chem110 student plots the graph shown on the left, from data obtained for the kinetic energy (eV) of the ejected electron versus frequency (s^{-1}) of the incident light, for an unknown metal. ($1\text{eV} = 1.6022 \times 10^{-19} \text{ J}$)

Based on the plot, answer the questions **a to c** below:

a. 1 point On the plot above, label the **threshold frequency**.

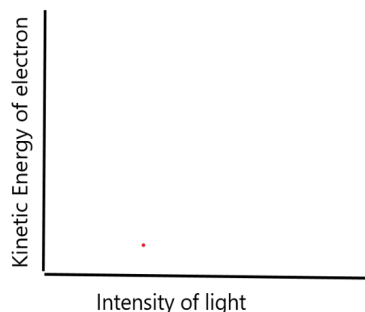
b. 2 points What is the value of the work potential (in J) of the unknown metal?

c. 3 points What is the *longest* wavelength (in nm) required to eject an electron from this metal's surface? Show your work including equation(s) used. (3 sig figs)

d. 2 points In the space provided below, plot the following

(Assume frequency is *above* threshold frequency):

- Kinetic energy* of an ejected electron versus the *intensity* of the incident light (left)
- Number of electrons* ejected versus the *frequency* of the incident light (right)



**Question 3** **6 points****a. 2 points**

In the ground state of Ge (Germanium; $Z = 32$) atom, **name** the following orbitals:

i. Orbital containing the electron that requires the maximum energy to be removed.

ii. Orbital containing the electron that experiences the minimum effective nuclear charge.

b. 2 points

An unknown anion (A^-) has the following *excited* electronic configuration: $[Ar] 4s^2 3d^3 4p^1$

(Ar: Argon; $Z = 18$)

Give the **condensed ground state electronic configuration** of the atom, **A** and the cation, **A^{2+}** .
Show your work. (Assume all electronic transitions are possible)

Electronic Configuration of A:

Electronic Configuration of A^{2+} :

c. 2 points

In a single atom, determine how many electrons can have the following set of quantum numbers?

(i) $n = 5; l = 3$

In a single atom, determine how many electrons can have the following set of quantum numbers?

(i) $n = 3; l = 2; m_s = +1/2$

**Question 4** **8 points**

Rank the following from 1 to 4 in **increasing** order. Provide the ranking in the boxes provided.

Note: Atomic number (Z) are provided for relevant atoms

a) Increasing (lowest to highest) order of *ionization energy*: (1= lowest; 4 = highest)

Mg

Mg: Z = 12

Rb

Rb: Z = 37

Cs

Cs: Z = 55

K

K : Z = 19

b) Increasing (lowest to highest) order of *bond angle*: (1= lowest; 4 = highest)

H₂OCCl₄XeF₄

HCN

c) Increasing (smallest to largest) order of *ionic size* (1= smallest; 4 = largest)

S²⁻

S: Z = 16

O²⁻

O: Z = 8

Be²⁺

Be: Z = 4

Na⁺

Na: Z = 11

d) Increasing (lowest to highest) order of *bond energy* (1= lowest; 4 = highest)

P-F

P-Cl

P-I

P-Br

**Question 5** **12 points**

a. 9 points For the molecules HClO_4 and TeF_6 , draw the most stable *Lewis structure* showing appropriate shape and *dashed/wedged bonds*, based on VSEPR. Determine the **molecular geometry** around the central atom. For HClO_4 , label the *most polar bond*.



b. 3 points (Note: Lone pairs are not shown in structures below)

(i) For the three *resonance* structures (1, 2 and 3) shown below, answer the following question:



(I) Circle the *most contributing* resonance structure.

Structure 1

Structure 2

Structure 3

(ii) For the three structures given (4, 5, and 6) below, answer the following questions:



(I) Circle all the structures that represent *resonance structures* of each other:

Structure 4

Structure 5

Structure 6

(II) Among the resonance structures, circle the *most contributing* resonance structure.

Structure 4

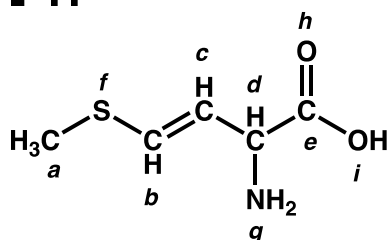
Structure 5

Structure 6

**Question 6** **6 points**

Using the data given below, calculate the **lattice energy** for SrO(s).

	ΔH° (kJ/mol)
$\text{Sr}(s) \longrightarrow \text{Sr}(g)$	+180
$\text{Sr}(g) \longrightarrow \text{Sr}^{2+}(g) + 2e^-$	+1468
$\text{O}_2(g) \longrightarrow 2\text{O}(g)$	+500
$\text{O}(g) + 2e^- \longrightarrow \text{O}^{2-}(g)$	+600
$\text{Sr}(s) + \frac{1}{2}\text{O}_2(g) \longrightarrow \text{SrO}(s)$	-550

**Question 7 10 points**

Consider the molecule given on the left, and answer questions (i) to (vi) using chemical bonding theories that we have learnt in class (valence bond theory, hybridization).

Note that the lone pairs are not shown in the structure:

i. Consider the carbon atoms marked in as ***a***, ***b***, ***c***, ***d***, and ***e*** in the molecule, and determine **hybridization** at each of the carbon centers. (2.5 points)

Atom	Hybridization
Carbon (<i>a</i>)	
Carbon (<i>b</i>)	
Carbon (<i>c</i>)	
Carbon (<i>d</i>)	
Carbon (<i>e</i>)	

ii. Now, consider the sulfur atom (S) marked as ***f***, nitrogen (N) atom (***g***), and oxygen(O) atoms (***h*** and ***i***), and determine their **hybridization**. (2 points)

Atom	Hybridization
Sulfur (<i>f</i>)	
Nitrogen (<i>g</i>)	
Oxygen (<i>h</i>)	
Oxygen (<i>i</i>)	

iii. **Describe** all the bonds formed by the carbon atoms ***a*** and ***b***, using sigma and pi designation as appropriate. (2 points)

Atom	Bonds formed (use sigma and pi as appropriate)
Carbon (<i>a</i>)	
Carbon (<i>b</i>)	

iv. How many total sp^3 -s sigma bonds are in the molecule? (1 point)

v. Determine **electronic** and **molecular geometry** at carbon (C) labeled ***e*** and sulfur (S) labeled ***f*** centres. (2 points)

Atom	Electronic Geometry	Molecular Geometry
Carbon (<i>e</i>)		
Sulfur (<i>f</i>)		

vi. Circle the value from the choices given below, that describes the **C-S-C bond angle at sulfur** (S) marked ***f***: (0.5 points)

>90° to <109.5° 109.5° >109.5 to <120° 120° >120° to <180°


Question 8 12 points

Using the **valence shell orbitals only** draw a **complete** molecular orbital (MO) diagram (with no s-p mixing) for the molecule **PS**, in the space provided below. **(6 points)**

Assume that 3s and 3p atomic orbitals will form molecular orbitals like those formed by the combination of 2s and 2p atomic orbitals that we learnt in class, and there is no s-p mixing.

Show **energy arrow**, **atomic orbitals** involved, **atom labels**, the **molecular orbitals** that are formed, **filled electrons** in the MO diagram etc. Make sure to label all the atomic and molecular orbitals.

Using the MO diagram above, answer the following questions:

- (i) Which one of the following will have the highest bond order? Circle the correct choice
 A) PS^{1-} B) PS C) PS^{1+}
- (ii) Which one of the following will have the longest bond length? Circle the correct choice
 A) PS^{1-} B) PS C) PS^{1+}
- (iii) Which of the following will be diamagnetic? Circle the correct choice.
 A) PS^{1-} B) PS C) PS^{1+}
- (iv) Draw Lewis dot structure of the molecule with the shortest bond length (based on the MO diagram) from among the three (PS^{1-} , PS, PS^{1+}), and indicate the bond order calculated from the Lewis dot structure. **(3 points)**

Lewis Dot Structure

Bond Order



5656DB4E-BC4B-4F80-BFB8-0C24F9A255DE

midterm-exam-93388

#206 Page 10 of 10

SCRAP PAGE

**CHEMISTRY 110 –Midterm Exam****Thursday November 7, 2024, 6:30 – 8:30 pm**

Examiners: Pallavi Sirjoosingh and Ashok Kakkar

First name (please write as legibly as possible within the boxes)

--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--

Last name

--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--

Student ID number

--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--

This booklet has 8 questions on 10 pages (The last page can be used as a scrap page).

The staple must NOT be removed from the question booklet. Your intact question booklet must be returned at the end of the exam.

A separate booklet contains a periodic table, and datasheet.

Write your answers directly under the questions in the space provided. Do not write on the QR codes or near the top edges (as it may be cut off).

Please write **legibly and dark enough** for the scan to be readable. For students who use English as a second language, dictionaries are permitted. Calculators are allowed.

DO NOT ASK QUESTIONS OF THE INVIGILATORS about interpretations of the exam questions. If you have a concern about a question, please summarize it at the top of this page. All concerns will be evaluated after the exam. GOOD LUCK!

**Question 1 8 points** (*Assume all transitions are possible*)

a. 6 points Calculate the **energy (in kJ/mol)** required to ionize (completely remove) the electron from the **n=3** level of **He⁺ ion** (He: Helium). Show your work including equations. (3 sig. figs.)

b. 2 points

(I) Select the largest (longest) wavelength from the three choices below:

- i. Wavelength of photon to *completely remove* an electron from *1s orbital of Hydrogen atom*
- ii. Wavelength of photon to *completely remove* an electron from *1s orbital of Boron atom (B)*
- iii. Wavelength of photon to *completely remove* an electron from *1s orbital of Boron cation (B⁺)*

i or ii or iii

(II) Select the smallest (shortest) wavelength from the three choices below (*for H atom*):

- i. Wavelength of photon *emitted* for an electron transitioning from *2p to 1s orbital*
- ii. Wavelength of photon *absorbed* for an electron transitioning from *2s to 4p orbital*
- iii. Wavelength of photon *absorbed* for an electron transitioning from *1s to 3p orbital*

i or ii or iii



Question 2 8 points



A Chem110 student plots the graph shown on the left, from data obtained for the kinetic energy (eV) of the ejected electron versus frequency (s^{-1}) of the incident light, for an unknown metal. ($1\text{eV} = 1.6022 \times 10^{-19} \text{ J}$)

Based on the plot, answer the questions **a to c** below:

a. 1 point On the plot above, label the **threshold frequency**.

b. 2 points What is the value of the work potential (in J) of the unknown metal?

c. 3 points What is the *longest* wavelength (in nm) required to eject an electron from this metal's surface? Show your work including equation(s) used. (3 sig figs)

d. 2 points In the space provided below, plot the following

(Assume frequency is *above* threshold frequency):

- i. *Kinetic energy* of an ejected electron versus the *intensity* of the incident light (left)
- ii. *Number of electrons* ejected versus the *frequency* of the incident light (right)



**Question 3** **6 points****a. 2 points**

In the ground state of Ge (Germanium; $Z = 32$) atom, **name** the following orbitals:

i. Orbital containing the electron that requires the maximum energy to be removed.

ii. Orbital containing the electron that experiences the minimum effective nuclear charge.

b. 2 points

An unknown anion (A^-) has the following *excited* electronic configuration: $[Ar] 4s^2 3d^3 4p^1$

(Ar: Argon; $Z = 18$)

Give the **condensed ground state electronic configuration** of the atom, **A** and the cation, **A^{2+}** .
Show your work. (Assume all electronic transitions are possible)

Electronic Configuration of A:

Electronic Configuration of A^{2+} :

c. 2 points

In a single atom, determine how many electrons can have the following set of quantum numbers?

(i) $n = 5; l = 3$

In a single atom, determine how many electrons can have the following set of quantum numbers?

(i) $n = 3; l = 2; m_s = +1/2$

**Question 4** **8 points**

Rank the following from 1 to 4 in **increasing** order. Provide the ranking in the boxes provided.

Note: Atomic number (Z) are provided for relevant atoms

a) Increasing (lowest to highest) order of *ionization energy*: (1= lowest; 4 = highest)

Mg

Mg: Z = 12

Rb

Rb: Z = 37

Cs

Cs: Z = 55

K

K : Z = 19

b) Increasing (lowest to highest) order of *bond angle*: (1= lowest; 4 = highest)

H₂OCCl₄XeF₄

HCN

c) Increasing (smallest to largest) order of *ionic size* (1= smallest; 4 = largest)

S²⁻

S: Z = 16

O²⁻

O: Z = 8

Be²⁺

Be: Z = 4

Na⁺

Na: Z = 11

d) Increasing (lowest to highest) order of *bond energy* (1= lowest; 4 = highest)

P-F

P-Cl

P-I

P-Br

**Question 5** **12 points**

a. 9 points For the molecules HClO_4 and TeF_6 , draw the most stable *Lewis structure* showing appropriate shape and *dashed/wedged bonds*, based on VSEPR. Determine the **molecular geometry** around the central atom. For HClO_4 , label the *most polar bond*.

HClO_4 (there are no O-O bonds)

Label the most polar bond

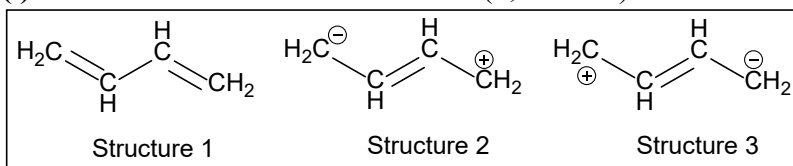
Molecular Geometry around Cl

TeF_6

Molecular Geometry around Te

b. 3 points (Note: Lone pairs are not shown in structures below)

(i) For the three *resonance* structures (1, 2 and 3) shown below, answer the following question:



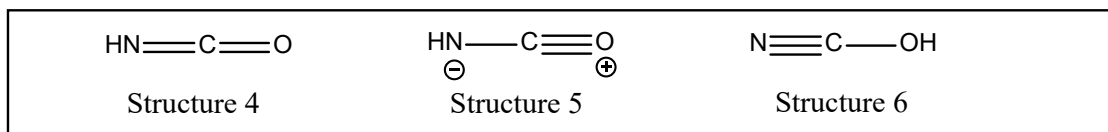
(I) Circle the *most contributing* resonance structure.

Structure 1

Structure 2

Structure 3

(ii) For the three structures given (4, 5, and 6) below, answer the following questions:



(I) Circle all the structures that represent *resonance structures* of each other:

Structure 4

Structure 5

Structure 6

(II) Among the resonance structures, circle the *most contributing* resonance structure.

Structure 4

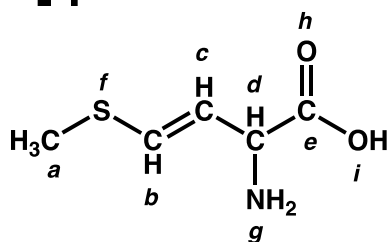
Structure 5

Structure 6

**Question 6** **6 points**

Using the data given below, calculate the **lattice energy** for SrO(s).

	ΔH° (kJ/mol)
$\text{Sr}(s) \longrightarrow \text{Sr}(g)$	+180
$\text{Sr}(g) \longrightarrow \text{Sr}^{2+}(g) + 2e^-$	+1468
$\text{O}_2(g) \longrightarrow 2\text{O}(g)$	+500
$\text{O}(g) + 2e^- \longrightarrow \text{O}^{2-}(g)$	+600
$\text{Sr}(s) + \frac{1}{2}\text{O}_2(g) \longrightarrow \text{SrO}(s)$	-550

**Question 7 10 points**

Consider the molecule given on the left, and answer questions (i) to (vi) using chemical bonding theories that we have learnt in class (valence bond theory, hybridization).

Note that the lone pairs are not shown in the structure:

i. Consider the carbon atoms marked in as ***a***, ***b***, ***c***, ***d***, and ***e*** in the molecule, and determine **hybridization** at each of the carbon centers. (2.5 points)

Atom	Hybridization
Carbon (<i>a</i>)	
Carbon (<i>b</i>)	
Carbon (<i>c</i>)	
Carbon (<i>d</i>)	
Carbon (<i>e</i>)	

ii. Now, consider the sulfur atom (S) marked as ***f***, nitrogen (N) atom (***g***), and oxygen(O) atoms (***h*** and ***i***), and determine their **hybridization**. (2 points)

Atom	Hybridization
Sulfur (<i>f</i>)	
Nitrogen (<i>g</i>)	
Oxygen (<i>h</i>)	
Oxygen (<i>i</i>)	

iii. **Describe** all the bonds formed by the carbon atoms ***a*** and ***b***, using sigma and pi designation as appropriate. (2 points)

Atom	Bonds formed (use sigma and pi as appropriate)
Carbon (<i>a</i>)	
Carbon (<i>b</i>)	

iv. How many total sp^3 -s sigma bonds are in the molecule? (1 point)

v. Determine **electronic** and **molecular geometry** at carbon (C) labeled ***e*** and sulfur (S) labeled ***f*** centres. (2 points)

Atom	Electronic Geometry	Molecular Geometry
Carbon (<i>e</i>)		
Sulfur (<i>f</i>)		

vi. Circle the value from the choices given below, that describes the **C-S-C bond angle at sulfur** (S) marked ***f***: (0.5 points)

>90° to <109.5°

109.5°

>109.5 to <120°

120°

>120° to <180°



Question 8 12 points

Using the **valence shell orbitals only** draw a **complete** molecular orbital (MO) diagram (with no s-p mixing) for the molecule **PS**, in the space provided below. **(6 points)**

Assume that 3s and 3p atomic orbitals will form molecular orbitals like those formed by the combination of 2s and 2p atomic orbitals that we learnt in class, and there is no s-p mixing.

Show **energy arrow**, **atomic orbitals** involved, **atom labels**, the **molecular orbitals** that are formed, **filled electrons** in the MO diagram etc. Make sure to label all the atomic and molecular orbitals.

Using the MO diagram above, answer the following questions:

- (i) Which one of the following will have the highest bond order? Circle the correct choice
 A) PS^{1-} B) PS C) PS^{1+}
- (ii) Which one of the following will have the longest bond length? Circle the correct choice
 A) PS^{1-} B) PS C) PS^{1+}
- (iii) Which of the following will be diamagnetic? Circle the correct choice.
 A) PS^{1-} B) PS C) PS^{1+}
- (iv) Draw Lewis dot structure of the molecule with the shortest bond length (based on the MO diagram) from among the three (PS^{1-} , PS, PS^{1+}), and indicate the bond order calculated from the Lewis dot structure. **(3 points)**

Lewis Dot Structure

Bond Order



06543B71-AA07-4FF3-8CE6-7A97D7F5C40C

midterm-exam-93388

#207 Page 10 of 10

SCRAP PAGE

**CHEMISTRY 110 –Midterm Exam****Thursday November 7, 2024, 6:30 – 8:30 pm****Examiners: Pallavi Sirjoosingh and Ashok Kakkar**

First name (please write as legibly as possible within the boxes)

--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--

Last name

--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--

Student ID number

--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--

This booklet has 8 questions on 10 pages (The last page can be used as a scrap page).

The staple must NOT be removed from the question booklet. Your intact question booklet must be returned at the end of the exam.

A separate booklet contains a periodic table, and datasheet.

Write your answers directly under the questions in the space provided. Do not write on the QR codes or near the top edges (as it may be cut off).

Please write **legibly and dark enough** for the scan to be readable. For students who use English as a second language, dictionaries are permitted. Calculators are allowed.

DO NOT ASK QUESTIONS OF THE INVIGILATORS about interpretations of the exam questions. If you have a concern about a question, please summarize it at the top of this page. All concerns will be evaluated after the exam. GOOD LUCK!

**Question 1 8 points** (*Assume all transitions are possible*)

a. 6 points Calculate the **energy (in kJ/mol)** required to ionize (completely remove) the electron from the **$n=3$** level of **He^+ ion** (He: Helium). Show your work including equations. (3 sig. figs.)

b. 2 points

(I) Select the largest (longest) wavelength from the three choices below:

- i. Wavelength of photon to *completely remove* an electron from *1s orbital of Hydrogen atom*
- ii. Wavelength of photon to *completely remove* an electron from *1s orbital of Boron atom (B)*
- iii. Wavelength of photon to *completely remove* an electron from *1s orbital of Boron cation (B^+)*

i or ii or iii

(II) Select the smallest (shortest) wavelength from the three choices below (*for H atom*):

- i. Wavelength of photon *emitted* for an electron transitioning from *2p to 1s orbital*
- ii. Wavelength of photon *absorbed* for an electron transitioning from *2s to 4p orbital*
- iii. Wavelength of photon *absorbed* for an electron transitioning from *1s to 3p orbital*

i or ii or iii



Question 2 8 points



A Chem110 student plots the graph shown on the left, from data obtained for the kinetic energy (eV) of the ejected electron versus frequency (s^{-1}) of the incident light, for an unknown metal. ($1\text{eV} = 1.6022 \times 10^{-19} \text{ J}$)

Based on the plot, answer the questions **a to c** below:

a. 1 point On the plot above, label the **threshold frequency**.

b. 2 points What is the value of the work potential (in J) of the unknown metal?

c. 3 points What is the *longest* wavelength (in nm) required to eject an electron from this metal's surface? Show your work including equation(s) used. (3 sig figs)

d. 2 points In the space provided below, plot the following

(Assume frequency is *above* threshold frequency):

- i. *Kinetic energy* of an ejected electron versus the *intensity* of the incident light (left)
- ii. *Number of electrons* ejected versus the *frequency* of the incident light (right)



**Question 3** **6 points****a. 2 points**

In the ground state of Ge (Germanium; $Z = 32$) atom, **name** the following orbitals:

i. Orbital containing the electron that requires the maximum energy to be removed.

ii. Orbital containing the electron that experiences the minimum effective nuclear charge.

b. 2 points

An unknown anion (A^-) has the following *excited* electronic configuration: $[Ar] 4s^2 3d^3 4p^1$

(Ar: Argon; $Z = 18$)

Give the **condensed ground state electronic configuration** of the atom, **A** and the cation, **A^{2+}** .
Show your work. (Assume all electronic transitions are possible)

Electronic Configuration of A:

Electronic Configuration of A^{2+} :

c. 2 points

In a single atom, determine how many electrons can have the following set of quantum numbers?

(i) $n = 5; l = 3$

In a single atom, determine how many electrons can have the following set of quantum numbers?

(i) $n = 3; l = 2; m_s = +1/2$

**Question 4** **8 points**

Rank the following from 1 to 4 in **increasing** order. Provide the ranking in the boxes provided.

Note: Atomic number (Z) are provided for relevant atoms

a) Increasing (lowest to highest) order of *ionization energy*: (1= lowest; 4 = highest)

Mg

Mg: Z = 12

Rb

Rb: Z = 37

Cs

Cs: Z = 55

K

K : Z = 19

b) Increasing (lowest to highest) order of *bond angle*: (1= lowest; 4 = highest)

H₂OCCl₄XeF₄

HCN

c) Increasing (smallest to largest) order of *ionic size* (1= smallest; 4 = largest)

S²⁻

S: Z = 16

O²⁻

O: Z = 8

Be²⁺

Be: Z = 4

Na⁺

Na: Z = 11

d) Increasing (lowest to highest) order of *bond energy* (1= lowest; 4 = highest)

P-F

P-Cl

P-I

P-Br

**Question 5** **12 points**

a. 9 points For the molecules HClO_4 and TeF_6 , draw the most stable *Lewis structure* showing appropriate shape and *dashed/wedged bonds*, based on VSEPR. Determine the **molecular geometry** around the central atom. For HClO_4 , label the *most polar bond*.



b. 3 points (Note: Lone pairs are not shown in structures below)

(i) For the three *resonance* structures (1, 2 and 3) shown below, answer the following question:



(I) Circle the *most contributing* resonance structure.

Structure 1

Structure 2

Structure 3

(ii) For the three structures given (4, 5, and 6) below, answer the following questions:



(I) Circle all the structures that represent *resonance structures* of each other:

Structure 4

Structure 5

Structure 6

(II) Among the resonance structures, circle the *most contributing* resonance structure.

Structure 4

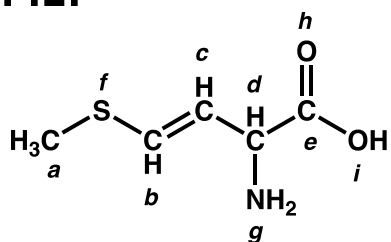
Structure 5

Structure 6

**Question 6** **6 points**

Using the data given below, calculate the **lattice energy** for SrO(s).

	ΔH° (kJ/mol)
$\text{Sr}(s) \longrightarrow \text{Sr}(g)$	+180
$\text{Sr}(g) \longrightarrow \text{Sr}^{2+}(g) + 2e^-$	+1468
$\text{O}_2(g) \longrightarrow 2\text{O}(g)$	+500
$\text{O}(g) + 2e^- \longrightarrow \text{O}^{2-}(g)$	+600
$\text{Sr}(s) + \frac{1}{2}\text{O}_2(g) \longrightarrow \text{SrO}(s)$	-550

**Question 7 10 points**

Consider the molecule given on the left, and answer questions (i) to (vi) using chemical bonding theories that we have learnt in class (valence bond theory, hybridization).

Note that the lone pairs are not shown in the structure:

i. Consider the carbon atoms marked in as ***a***, ***b***, ***c***, ***d***, and ***e*** in the molecule, and determine **hybridization** at each of the carbon centers. (2.5 points)

Atom	Hybridization
Carbon (<i>a</i>)	
Carbon (<i>b</i>)	
Carbon (<i>c</i>)	
Carbon (<i>d</i>)	
Carbon (<i>e</i>)	

ii. Now, consider the sulfur atom (S) marked as ***f***, nitrogen (N) atom (***g***), and oxygen(O) atoms (***h*** and ***i***), and determine their **hybridization**. (2 points)

Atom	Hybridization
Sulfur (<i>f</i>)	
Nitrogen (<i>g</i>)	
Oxygen (<i>h</i>)	
Oxygen (<i>i</i>)	

iii. **Describe** all the bonds formed by the carbon atoms ***a*** and ***b***, using sigma and pi designation as appropriate. (2 points)

Atom	Bonds formed (use sigma and pi as appropriate)
Carbon (<i>a</i>)	
Carbon (<i>b</i>)	

iv. How many total sp^3 -s sigma bonds are in the molecule? (1 point)

v. Determine **electronic** and **molecular geometry** at carbon (C) labeled ***e*** and sulfur (S) labeled ***f*** centres. (2 points)

Atom	Electronic Geometry	Molecular Geometry
Carbon (<i>e</i>)		
Sulfur (<i>f</i>)		

vi. Circle the value from the choices given below, that describes the **C-S-C bond angle at sulfur** (S) marked ***f***: (0.5 points)

$>90^\circ$ to $<109.5^\circ$ 109.5° >109.5 to $<120^\circ$ 120° $>120^\circ$ to $<180^\circ$


Question 8 12 points

Using the **valence shell orbitals only** draw a **complete** molecular orbital (MO) diagram (with no s-p mixing) for the molecule **PS**, in the space provided below. **(6 points)**

Assume that 3s and 3p atomic orbitals will form molecular orbitals like those formed by the combination of 2s and 2p atomic orbitals that we learnt in class, and there is no s-p mixing.

Show **energy arrow**, **atomic orbitals** involved, **atom labels**, the **molecular orbitals** that are formed, **filled electrons** in the MO diagram etc. Make sure to label all the atomic and molecular orbitals.

Using the MO diagram above, answer the following questions:

- (i) Which one of the following will have the highest bond order? Circle the correct choice
 A) PS^{1-} B) PS C) PS^{1+}
- (ii) Which one of the following will have the longest bond length? Circle the correct choice
 A) PS^{1-} B) PS C) PS^{1+}
- (iii) Which of the following will be diamagnetic? Circle the correct choice.
 A) PS^{1-} B) PS C) PS^{1+}
- (iv) Draw Lewis dot structure of the molecule with the shortest bond length (based on the MO diagram) from among the three (PS^{1-} , PS, PS^{1+}), and indicate the bond order calculated from the Lewis dot structure. **(3 points)**

Lewis Dot Structure

Bond Order



A7399EFB-900B-49AC-B3FC-34CD9D6F1E5A

midterm-exam-93388

#208 Page 10 of 10

SCRAP PAGE

**CHEMISTRY 110 –Midterm Exam****Thursday November 7, 2024, 6:30 – 8:30 pm**

Examiners: Pallavi Sirjoosingh and Ashok Kakkar

First name (please write as legibly as possible within the boxes)

--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--

Last name

--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--

Student ID number

--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--

This booklet has 8 questions on 10 pages (The last page can be used as a scrap page).

The staple must NOT be removed from the question booklet. Your intact question booklet must be returned at the end of the exam.

A separate booklet contains a periodic table, and datasheet.

Write your answers directly under the questions in the space provided. Do not write on the QR codes or near the top edges (as it may be cut off).

Please write **legibly and dark enough** for the scan to be readable. For students who use English as a second language, dictionaries are permitted. Calculators are allowed.

DO NOT ASK QUESTIONS OF THE INVIGILATORS about interpretations of the exam questions. If you have a concern about a question, please summarize it at the top of this page. All concerns will be evaluated after the exam. GOOD LUCK!

**Question 1 8 points** (*Assume all transitions are possible*)

a. 6 points Calculate the **energy (in kJ/mol)** required to ionize (completely remove) the electron from the **n=3** level of **He⁺ ion** (He: Helium). Show your work including equations. (3 sig. figs.)

b. 2 points

(I) Select the largest (longest) wavelength from the three choices below:

- i. Wavelength of photon to *completely remove* an electron from *1s orbital of Hydrogen atom*
- ii. Wavelength of photon to *completely remove* an electron from *1s orbital of Boron atom (B)*
- iii. Wavelength of photon to *completely remove* an electron from *1s orbital of Boron cation (B⁺)*

i or ii or iii

(II) Select the smallest (shortest) wavelength from the three choices below (*for H atom*):

- i. Wavelength of photon *emitted* for an electron transitioning from *2p to 1s orbital*
- ii. Wavelength of photon *absorbed* for an electron transitioning from *2s to 4p orbital*
- iii. Wavelength of photon *absorbed* for an electron transitioning from *1s to 3p orbital*

i or ii or iii



Question 2 8 points



A Chem110 student plots the graph shown on the left, from data obtained for the kinetic energy (eV) of the ejected electron versus frequency (s^{-1}) of the incident light, for an unknown metal. ($1\text{eV} = 1.6022 \times 10^{-19} \text{ J}$)

Based on the plot, answer the questions **a to c** below:

a. 1 point On the plot above, label the **threshold frequency**.

b. 2 points What is the value of the work potential (in J) of the unknown metal?

c. 3 points What is the *longest* wavelength (in nm) required to eject an electron from this metal's surface? Show your work including equation(s) used. (3 sig figs)

d. 2 points In the space provided below, plot the following

(Assume frequency is *above* threshold frequency):

- i. *Kinetic energy* of an ejected electron versus the *intensity* of the incident light (left)
- ii. *Number of electrons* ejected versus the *frequency* of the incident light (right)



**Question 3** **6 points****a. 2 points**

In the ground state of Ge (Germanium; $Z = 32$) atom, **name** the following orbitals:

i. Orbital containing the electron that requires the maximum energy to be removed.

ii. Orbital containing the electron that experiences the minimum effective nuclear charge.

b. 2 points

An unknown anion (A^-) has the following *excited* electronic configuration: $[Ar] 4s^2 3d^3 4p^1$

(Ar: Argon; $Z = 18$)

Give the **condensed ground state electronic configuration** of the atom, **A** and the cation, **A^{2+}** .
Show your work. (Assume all electronic transitions are possible)

Electronic Configuration of A:

Electronic Configuration of A^{2+} :

c. 2 points

In a single atom, determine how many electrons can have the following set of quantum numbers?

(i) $n = 5; l = 3$

In a single atom, determine how many electrons can have the following set of quantum numbers?

(i) $n = 3; l = 2; m_s = +1/2$

**Question 4** **8 points**

Rank the following from 1 to 4 in **increasing** order. Provide the ranking in the boxes provided.

Note: Atomic number (Z) are provided for relevant atoms

a) Increasing (lowest to highest) order of *ionization energy*: (1= lowest; 4 = highest)

Mg

Mg: Z = 12

Rb

Rb: Z = 37

Cs

Cs: Z = 55

K

K : Z = 19

b) Increasing (lowest to highest) order of *bond angle*: (1= lowest; 4 = highest)

H₂OCCl₄XeF₄

HCN

c) Increasing (smallest to largest) order of *ionic size* (1= smallest; 4 = largest)

S²⁻

S: Z = 16

O²⁻

O: Z = 8

Be²⁺

Be: Z = 4

Na⁺

Na: Z = 11

d) Increasing (lowest to highest) order of *bond energy* (1= lowest; 4 = highest)

P-F

P-Cl

P-I

P-Br

**Question 5** **12 points**

a. 9 points For the molecules HClO_4 and TeF_6 , draw the most stable *Lewis structure* showing appropriate shape and *dashed/wedged bonds*, based on VSEPR. Determine the **molecular geometry** around the central atom. For HClO_4 , label the *most polar bond*.



b. 3 points (Note: Lone pairs are not shown in structures below)

(i) For the three *resonance* structures (1, 2 and 3) shown below, answer the following question:



(I) Circle the *most contributing* resonance structure.

Structure 1

Structure 2

Structure 3

(ii) For the three structures given (4, 5, and 6) below, answer the following questions:



(I) Circle all the structures that represent *resonance structures* of each other:

Structure 4

Structure 5

Structure 6

(II) Among the resonance structures, circle the *most contributing* resonance structure.

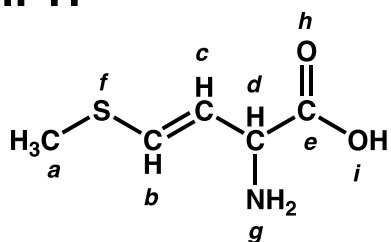
Structure 4

Structure 5

Structure 6

**Question 6** **6 points**Using the data given below, calculate the **lattice energy** for SrO(s).

	ΔH° (kJ/mol)
$\text{Sr}(s) \longrightarrow \text{Sr}(g)$	+180
$\text{Sr}(g) \longrightarrow \text{Sr}^{2+}(g) + 2e^-$	+1468
$\text{O}_2(g) \longrightarrow 2\text{O}(g)$	+500
$\text{O}(g) + 2e^- \longrightarrow \text{O}^{2-}(g)$	+600
$\text{Sr}(s) + \frac{1}{2} \text{O}_2(g) \longrightarrow \text{SrO}(s)$	-550

**Question 7 10 points**

Consider the molecule given on the left, and answer questions (i) to (vi) using chemical bonding theories that we have learnt in class (valence bond theory, hybridization).

Note that the lone pairs are not shown in the structure:

i. Consider the carbon atoms marked in as ***a***, ***b***, ***c***, ***d***, and ***e*** in the molecule, and determine **hybridization** at each of the carbon centers. (2.5 points)

Atom	Hybridization
Carbon (<i>a</i>)	
Carbon (<i>b</i>)	
Carbon (<i>c</i>)	
Carbon (<i>d</i>)	
Carbon (<i>e</i>)	

ii. Now, consider the sulfur atom (S) marked as ***f***, nitrogen (N) atom (***g***), and oxygen(O) atoms (***h*** and ***i***), and determine their **hybridization**. (2 points)

Atom	Hybridization
Sulfur (<i>f</i>)	
Nitrogen (<i>g</i>)	
Oxygen (<i>h</i>)	
Oxygen (<i>i</i>)	

iii. **Describe** all the bonds formed by the carbon atoms ***a*** and ***b***, using sigma and pi designation as appropriate. (2 points)

Atom	Bonds formed (use sigma and pi as appropriate)
Carbon (<i>a</i>)	
Carbon (<i>b</i>)	

iv. How many total sp^3 -s sigma bonds are in the molecule? (1 point)

v. Determine **electronic** and **molecular geometry** at carbon (C) labeled ***e*** and sulfur (S) labeled ***f*** centres. (2 points)

Atom	Electronic Geometry	Molecular Geometry
Carbon (<i>e</i>)		
Sulfur (<i>f</i>)		

vi. Circle the value from the choices given below, that describes the **C-S-C bond angle at sulfur** (S) marked ***f***: (0.5 points)

>90° to <109.5° 109.5° >109.5 to <120° 120° >120° to <180°


Question 8 12 points

Using the **valence shell orbitals only** draw a **complete** molecular orbital (MO) diagram (with no s-p mixing) for the molecule **PS**, in the space provided below. **(6 points)**

Assume that 3s and 3p atomic orbitals will form molecular orbitals like those formed by the combination of 2s and 2p atomic orbitals that we learnt in class, and there is no s-p mixing.

Show **energy arrow**, **atomic orbitals** involved, **atom labels**, the **molecular orbitals** that are formed, **filled electrons** in the MO diagram etc. Make sure to **label all the atomic and molecular orbitals**.

Using the MO diagram above, answer the following questions:

- (i) Which one of the following will have the highest bond order? Circle the correct choice
 A) PS^{1-} B) PS C) PS^{1+}
- (ii) Which one of the following will have the longest bond length? Circle the correct choice
 A) PS^{1-} B) PS C) PS^{1+}
- (iii) Which of the following will be diamagnetic? Circle the correct choice.
 A) PS^{1-} B) PS C) PS^{1+}
- (iv) Draw Lewis dot structure of the molecule with the shortest bond length (based on the MO diagram) from among the three (PS^{1-} , PS, PS^{1+}), and indicate the bond order calculated from the Lewis dot structure. **(3 points)**

Lewis Dot Structure

Bond Order



0A982D36-F627-4962-9E77-4D343603C74A

midterm-exam-93388

#209 Page 10 of 10

SCRAP PAGE

**CHEMISTRY 110 –Midterm Exam****Thursday November 7, 2024, 6:30 – 8:30 pm****Examiners: Pallavi Sirjoosingh and Ashok Kakkar**

First name (please write as legibly as possible within the boxes)

--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--

Last name

--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--

Student ID number

--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--

This booklet has 8 questions on 10 pages (The last page can be used as a scrap page).

The staple must NOT be removed from the question booklet. Your intact question booklet must be returned at the end of the exam.

A separate booklet contains a periodic table, and datasheet.

Write your answers directly under the questions in the space provided. Do not write on the QR codes or near the top edges (as it may be cut off).

Please write **legibly and dark enough** for the scan to be readable. For students who use English as a second language, dictionaries are permitted. Calculators are allowed.

DO NOT ASK QUESTIONS OF THE INVIGILATORS about interpretations of the exam questions. If you have a concern about a question, please summarize it at the top of this page. All concerns will be evaluated after the exam. GOOD LUCK!

**Question 1 8 points** (*Assume all transitions are possible*)

a. 6 points Calculate the **energy (in kJ/mol)** required to ionize (completely remove) the electron from the **$n=3$** level of **He^+ ion** (He: Helium). Show your work including equations. (3 sig. figs.)

b. 2 points

(I) Select the largest (longest) wavelength from the three choices below:

- i. Wavelength of photon to *completely remove* an electron from *1s orbital of Hydrogen atom*
- ii. Wavelength of photon to *completely remove* an electron from *1s orbital of Boron atom (B)*
- iii. Wavelength of photon to *completely remove* an electron from *1s orbital of Boron cation (B^+)*

i or ii or iii

(II) Select the smallest (shortest) wavelength from the three choices below (*for H atom*):

- i. Wavelength of photon *emitted* for an electron transitioning from *2p to 1s orbital*
- ii. Wavelength of photon *absorbed* for an electron transitioning from *2s to 4p orbital*
- iii. Wavelength of photon *absorbed* for an electron transitioning from *1s to 3p orbital*

i or ii or iii



Question 2 8 points



A Chem110 student plots the graph shown on the left, from data obtained for the kinetic energy (eV) of the ejected electron versus frequency (s^{-1}) of the incident light, for an unknown metal. ($1\text{eV} = 1.6022 \times 10^{-19} \text{ J}$)

Based on the plot, answer the questions **a to c** below:

a. 1 point On the plot above, label the **threshold frequency**.

b. 2 points What is the value of the work potential (in J) of the unknown metal?

c. 3 points What is the *longest* wavelength (in nm) required to eject an electron from this metal's surface? Show your work including equation(s) used. (3 sig figs)

d. 2 points In the space provided below, plot the following

(Assume frequency is *above* threshold frequency):

- i. *Kinetic energy* of an ejected electron versus the *intensity* of the incident light (left)
- ii. *Number of electrons* ejected versus the *frequency* of the incident light (right)



**Question 3** **6 points****a. 2 points**

In the ground state of Ge (Germanium; $Z = 32$) atom, **name** the following orbitals:

i. Orbital containing the electron that requires the maximum energy to be removed.

ii. Orbital containing the electron that experiences the minimum effective nuclear charge.

b. 2 points

An unknown anion (A^-) has the following *excited* electronic configuration: $[Ar] 4s^2 3d^3 4p^1$

(Ar: Argon; $Z = 18$)

Give the **condensed ground state electronic configuration** of the atom, **A** and the cation, **A^{2+}** .
Show your work. (Assume all electronic transitions are possible)

Electronic Configuration of A:

Electronic Configuration of A^{2+} :

c. 2 points

In a single atom, determine how many electrons can have the following set of quantum numbers?

(i) $n = 5; l = 3$

In a single atom, determine how many electrons can have the following set of quantum numbers?

(i) $n = 3; l = 2; m_s = +1/2$

**Question 4** **8 points**

Rank the following from 1 to 4 in **increasing** order. Provide the ranking in the boxes provided.

Note: Atomic number (Z) are provided for relevant atoms

a) Increasing (lowest to highest) order of *ionization energy*: (1= lowest; 4 = highest)

Mg

Mg: Z = 12

Rb

Rb: Z = 37

Cs

Cs: Z = 55

K

K : Z = 19

b) Increasing (lowest to highest) order of *bond angle*: (1= lowest; 4 = highest)

H₂OCCl₄XeF₄

HCN

c) Increasing (smallest to largest) order of *ionic size* (1= smallest; 4 = largest)

S²⁻

S: Z = 16

O²⁻

O: Z = 8

Be²⁺

Be: Z = 4

Na⁺

Na: Z = 11

d) Increasing (lowest to highest) order of *bond energy* (1= lowest; 4 = highest)

P-F

P-Cl

P-I

P-Br

**Question 5** **12 points**

a. 9 points For the molecules HClO_4 and TeF_6 , draw the most stable *Lewis structure* showing appropriate shape and *dashed/wedged bonds*, based on VSEPR. Determine the **molecular geometry** around the central atom. For HClO_4 , label the *most polar bond*.



b. 3 points (Note: Lone pairs are not shown in structures below)

(i) For the three *resonance* structures (1, 2 and 3) shown below, answer the following question:



(I) Circle the *most contributing* resonance structure.

Structure 1

Structure 2

Structure 3

(ii) For the three structures given (4, 5, and 6) below, answer the following questions:



(I) Circle all the structures that represent *resonance structures* of each other:

Structure 4

Structure 5

Structure 6

(II) Among the resonance structures, circle the *most contributing* resonance structure.

Structure 4

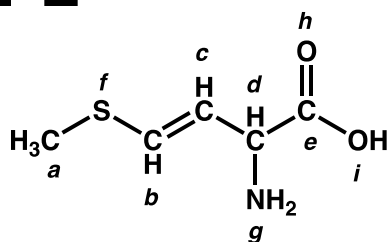
Structure 5

Structure 6

**Question 6** **6 points**

Using the data given below, calculate the **lattice energy** for SrO(s).

	ΔH° (kJ/mol)
$\text{Sr}(s) \longrightarrow \text{Sr}(g)$	+180
$\text{Sr}(g) \longrightarrow \text{Sr}^{2+}(g) + 2e^-$	+1468
$\text{O}_2(g) \longrightarrow 2\text{O}(g)$	+500
$\text{O}(g) + 2e^- \longrightarrow \text{O}^{2-}(g)$	+600
$\text{Sr}(s) + \frac{1}{2}\text{O}_2(g) \longrightarrow \text{SrO}(s)$	-550

**Question 7 10 points**

Consider the molecule given on the left, and answer questions (i) to (vi) using chemical bonding theories that we have learnt in class (valence bond theory, hybridization).

Note that the lone pairs are not shown in the structure:

i. Consider the carbon atoms marked in as ***a***, ***b***, ***c***, ***d***, and ***e*** in the molecule, and determine **hybridization** at each of the carbon centers. (2.5 points)

Atom	Hybridization
Carbon (<i>a</i>)	
Carbon (<i>b</i>)	
Carbon (<i>c</i>)	
Carbon (<i>d</i>)	
Carbon (<i>e</i>)	

ii. Now, consider the sulfur atom (S) marked as ***f***, nitrogen (N) atom (***g***), and oxygen(O) atoms (***h*** and ***i***), and determine their **hybridization**. (2 points)

Atom	Hybridization
Sulfur (<i>f</i>)	
Nitrogen (<i>g</i>)	
Oxygen (<i>h</i>)	
Oxygen (<i>i</i>)	

iii. **Describe** all the bonds formed by the carbon atoms ***a*** and ***b***, using sigma and pi designation as appropriate. (2 points)

Atom	Bonds formed (use sigma and pi as appropriate)
Carbon (<i>a</i>)	
Carbon (<i>b</i>)	

iv. How many total sp^3 -s sigma bonds are in the molecule? (1 point)

v. Determine **electronic** and **molecular geometry** at carbon (C) labeled ***e*** and sulfur (S) labeled ***f*** centres. (2 points)

Atom	Electronic Geometry	Molecular Geometry
Carbon (<i>e</i>)		
Sulfur (<i>f</i>)		

vi. Circle the value from the choices given below, that describes the **C-S-C bond angle at sulfur** (S) marked ***f***: (0.5 points)

>90° to <109.5° 109.5° >109.5 to <120° 120° >120° to <180°


Question 8 12 points

Using the **valence shell orbitals only** draw a **complete** molecular orbital (MO) diagram (with no s-p mixing) for the molecule **PS**, in the space provided below. **(6 points)**

Assume that 3s and 3p atomic orbitals will form molecular orbitals like those formed by the combination of 2s and 2p atomic orbitals that we learnt in class, and there is no s-p mixing.

Show **energy arrow**, **atomic orbitals** involved, **atom labels**, the **molecular orbitals** that are formed, **filled electrons** in the MO diagram etc. Make sure to label all the atomic and molecular orbitals.

Using the MO diagram above, answer the following questions:

- (i) Which one of the following will have the highest bond order? Circle the correct choice
 A) PS^{1-} B) PS C) PS^{1+}
- (ii) Which one of the following will have the longest bond length? Circle the correct choice
 A) PS^{1-} B) PS C) PS^{1+}
- (iii) Which of the following will be diamagnetic? Circle the correct choice.
 A) PS^{1-} B) PS C) PS^{1+}
- (iv) Draw Lewis dot structure of the molecule with the shortest bond length (based on the MO diagram) from among the three (PS^{1-} , PS, PS^{1+}), and indicate the bond order calculated from the Lewis dot structure. **(3 points)**

Lewis Dot Structure

Bond Order



E118CC26-E001-4569-A168-18F774F5425F

midterm-exam-93388

#210 Page 10 of 10

SCRAP PAGE

**CHEMISTRY 110 –Midterm Exam****Thursday November 7, 2024, 6:30 – 8:30 pm****Examiners: Pallavi Sirjoosingh and Ashok Kakkar**

First name (please write as legibly as possible within the boxes)

--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--

Last name

--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--

Student ID number

--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--

This booklet has 8 questions on 10 pages (The last page can be used as a scrap page).

The staple must NOT be removed from the question booklet. Your intact question booklet must be returned at the end of the exam.

A separate booklet contains a periodic table, and datasheet.

Write your answers directly under the questions in the space provided. Do not write on the QR codes or near the top edges (as it may be cut off).

Please write **legibly and dark enough** for the scan to be readable. For students who use English as a second language, dictionaries are permitted. Calculators are allowed.

DO NOT ASK QUESTIONS OF THE INVIGILATORS about interpretations of the exam questions. If you have a concern about a question, please summarize it at the top of this page. All concerns will be evaluated after the exam. GOOD LUCK!

**Question 1 8 points** (*Assume all transitions are possible*)

a. 6 points Calculate the **energy (in kJ/mol)** required to ionize (completely remove) the electron from the **n=3** level of **He⁺ ion** (He: Helium). Show your work including equations. (3 sig. figs.)

b. 2 points

(I) Select the largest (longest) wavelength from the three choices below:

- i. Wavelength of photon to *completely remove* an electron from *1s orbital of Hydrogen atom*
- ii. Wavelength of photon to *completely remove* an electron from *1s orbital of Boron atom (B)*
- iii. Wavelength of photon to *completely remove* an electron from *1s orbital of Boron cation (B⁺)*

i or ii or iii

(II) Select the smallest (shortest) wavelength from the three choices below (*for H atom*):

- i. Wavelength of photon *emitted* for an electron transitioning from *2p to 1s orbital*
- ii. Wavelength of photon *absorbed* for an electron transitioning from *2s to 4p orbital*
- iii. Wavelength of photon *absorbed* for an electron transitioning from *1s to 3p orbital*

i or ii or iii



Question 2 8 points



A Chem110 student plots the graph shown on the left, from data obtained for the kinetic energy (eV) of the ejected electron versus frequency (s^{-1}) of the incident light, for an unknown metal. ($1\text{eV} = 1.6022 \times 10^{-19} \text{ J}$)

Based on the plot, answer the questions **a to c** below:

a. 1 point On the plot above, label the **threshold frequency**.

b. 2 points What is the value of the work potential (in J) of the unknown metal?

c. 3 points What is the *longest* wavelength (in nm) required to eject an electron from this metal's surface? Show your work including equation(s) used. (3 sig figs)

d. 2 points In the space provided below, plot the following

(Assume frequency is *above* threshold frequency):

- Kinetic energy* of an ejected electron versus the *intensity* of the incident light (left)
- Number of electrons* ejected versus the *frequency* of the incident light (right)



**Question 3** **6 points****a. 2 points**

In the ground state of Ge (Germanium; $Z = 32$) atom, **name** the following orbitals:

i. Orbital containing the electron that requires the maximum energy to be removed.

ii. Orbital containing the electron that experiences the minimum effective nuclear charge.

b. 2 points

An unknown anion (A^-) has the following *excited* electronic configuration: $[Ar] 4s^2 3d^3 4p^1$

(Ar: Argon; $Z = 18$)

Give the **condensed ground state electronic configuration** of the atom, **A** and the cation, **A^{2+}** .
Show your work. (Assume all electronic transitions are possible)

Electronic Configuration of A:

Electronic Configuration of A^{2+} :

c. 2 points

In a single atom, determine how many electrons can have the following set of quantum numbers?

(i) $n = 5; l = 3$

In a single atom, determine how many electrons can have the following set of quantum numbers?

(i) $n = 3; l = 2; m_s = +1/2$

**Question 4** **8 points**

Rank the following from 1 to 4 in **increasing** order. Provide the ranking in the boxes provided.

Note: Atomic number (Z) are provided for relevant atoms

a) Increasing (lowest to highest) order of *ionization energy*: (1= lowest; 4 = highest)

Mg

Mg: Z = 12

Rb

Rb: Z = 37

Cs

Cs: Z = 55

K

K : Z = 19

b) Increasing (lowest to highest) order of *bond angle*: (1= lowest; 4 = highest)

H₂OCCl₄XeF₄

HCN

c) Increasing (smallest to largest) order of *ionic size* (1= smallest; 4 = largest)

S²⁻

S: Z = 16

O²⁻

O: Z = 8

Be²⁺

Be: Z = 4

Na⁺

Na: Z = 11

d) Increasing (lowest to highest) order of *bond energy* (1= lowest; 4 = highest)

P-F

P-Cl

P-I

P-Br

**Question 5** **12 points**

a. 9 points For the molecules HClO_4 and TeF_6 , draw the most stable *Lewis structure* showing appropriate shape and *dashed/wedged bonds*, based on VSEPR. Determine the **molecular geometry** around the central atom. For HClO_4 , label the *most polar bond*.



b. 3 points (Note: Lone pairs are not shown in structures below)

(i) For the three *resonance* structures (1, 2 and 3) shown below, answer the following question:



(I) Circle the *most contributing* resonance structure.

Structure 1

Structure 2

Structure 3

(ii) For the three structures given (4, 5, and 6) below, answer the following questions:



(I) Circle all the structures that represent *resonance structures* of each other:

Structure 4

Structure 5

Structure 6

(II) Among the resonance structures, circle the *most contributing* resonance structure.

Structure 4

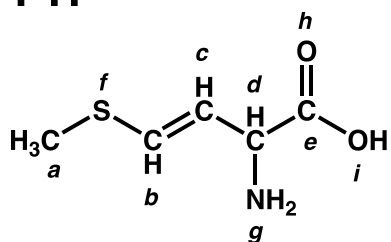
Structure 5

Structure 6

**Question 6** **6 points**

Using the data given below, calculate the **lattice energy** for SrO(s).

	ΔH° (kJ/mol)
$\text{Sr}(s) \longrightarrow \text{Sr}(g)$	+180
$\text{Sr}(g) \longrightarrow \text{Sr}^{2+}(g) + 2e^-$	+1468
$\text{O}_2(g) \longrightarrow 2\text{O}(g)$	+500
$\text{O}(g) + 2e^- \longrightarrow \text{O}^{2-}(g)$	+600
$\text{Sr}(s) + \frac{1}{2}\text{O}_2(g) \longrightarrow \text{SrO}(s)$	-550

**Question 7 10 points**

Consider the molecule given on the left, and answer questions (i) to (vi) using chemical bonding theories that we have learnt in class (valence bond theory, hybridization).

Note that the lone pairs are not shown in the structure:

i. Consider the carbon atoms marked in as ***a***, ***b***, ***c***, ***d***, and ***e*** in the molecule, and determine **hybridization** at each of the carbon centers. (2.5 points)

Atom	Hybridization
Carbon (<i>a</i>)	
Carbon (<i>b</i>)	
Carbon (<i>c</i>)	
Carbon (<i>d</i>)	
Carbon (<i>e</i>)	

ii. Now, consider the sulfur atom (S) marked as ***f***, nitrogen (N) atom (***g***), and oxygen(O) atoms (***h*** and ***i***), and determine their **hybridization**. (2 points)

Atom	Hybridization
Sulfur (<i>f</i>)	
Nitrogen (<i>g</i>)	
Oxygen (<i>h</i>)	
Oxygen (<i>i</i>)	

iii. **Describe** all the bonds formed by the carbon atoms ***a*** and ***b***, using sigma and pi designation as appropriate. (2 points)

Atom	Bonds formed (use sigma and pi as appropriate)
Carbon (<i>a</i>)	
Carbon (<i>b</i>)	

iv. How many total sp^3 -s sigma bonds are in the molecule? (1 point)

v. Determine **electronic** and **molecular geometry** at carbon (C) labeled ***e*** and sulfur (S) labeled ***f*** centres. (2 points)

Atom	Electronic Geometry	Molecular Geometry
Carbon (<i>e</i>)		
Sulfur (<i>f</i>)		

vi. Circle the value from the choices given below, that describes the **C-S-C bond angle at sulfur** (S) marked ***f***: (0.5 points)

>90° to <109.5° 109.5° >109.5 to <120° 120° >120° to <180°


Question 8 12 points

Using the **valence shell orbitals only** draw a **complete** molecular orbital (MO) diagram (with no s-p mixing) for the molecule **PS**, in the space provided below. **(6 points)**

Assume that 3s and 3p atomic orbitals will form molecular orbitals like those formed by the combination of 2s and 2p atomic orbitals that we learnt in class, and there is no s-p mixing.

Show **energy arrow**, **atomic orbitals** involved, **atom labels**, the **molecular orbitals** that are formed, **filled electrons** in the MO diagram etc. Make sure to label all the atomic and molecular orbitals.

Using the MO diagram above, answer the following questions:

- (i) Which one of the following will have the highest bond order? Circle the correct choice
 A) PS^{1-} B) PS C) PS^{1+}
- (ii) Which one of the following will have the longest bond length? Circle the correct choice
 A) PS^{1-} B) PS C) PS^{1+}
- (iii) Which of the following will be diamagnetic? Circle the correct choice.
 A) PS^{1-} B) PS C) PS^{1+}
- (iv) Draw Lewis dot structure of the molecule with the shortest bond length (based on the MO diagram) from among the three (PS^{1-} , PS, PS^{1+}), and indicate the bond order calculated from the Lewis dot structure. **(3 points)**

Lewis Dot Structure

Bond Order



F1D2CC5C-3B15-4AB8-96F2-17BDA2FF81BB

midterm-exam-93388

#211 Page 10 of 10

SCRAP PAGE

**CHEMISTRY 110 –Midterm Exam****Thursday November 7, 2024, 6:30 – 8:30 pm**

Examiners: Pallavi Sirjoosingh and Ashok Kakkar

First name (please write as legibly as possible within the boxes)

--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--

Last name

--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--

Student ID number

--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--

This booklet has 8 questions on 10 pages (The last page can be used as a scrap page).

The staple must NOT be removed from the question booklet. Your intact question booklet must be returned at the end of the exam.

A separate booklet contains a periodic table, and datasheet.

Write your answers directly under the questions in the space provided. Do not write on the QR codes or near the top edges (as it may be cut off).

Please write **legibly and dark enough** for the scan to be readable. For students who use English as a second language, dictionaries are permitted. Calculators are allowed.

DO NOT ASK QUESTIONS OF THE INVIGILATORS about interpretations of the exam questions. If you have a concern about a question, please summarize it at the top of this page. All concerns will be evaluated after the exam. GOOD LUCK!

**Question 1 8 points** (*Assume all transitions are possible*)

a. 6 points Calculate the **energy (in kJ/mol)** required to ionize (completely remove) the electron from the **n=3** level of **He⁺ ion** (He: Helium). Show your work including equations. (3 sig. figs.)

b. 2 points

(I) Select the largest (longest) wavelength from the three choices below:

- i. Wavelength of photon to *completely remove* an electron from *1s orbital of Hydrogen atom*
- ii. Wavelength of photon to *completely remove* an electron from *1s orbital of Boron atom (B)*
- iii. Wavelength of photon to *completely remove* an electron from *1s orbital of Boron cation (B⁺)*

i or ii or iii

(II) Select the smallest (shortest) wavelength from the three choices below (*for H atom*):

- i. Wavelength of photon *emitted* for an electron transitioning from *2p to 1s orbital*
- ii. Wavelength of photon *absorbed* for an electron transitioning from *2s to 4p orbital*
- iii. Wavelength of photon *absorbed* for an electron transitioning from *1s to 3p orbital*

i or ii or iii



Question 2 8 points



A Chem110 student plots the graph shown on the left, from data obtained for the kinetic energy (eV) of the ejected electron versus frequency (s^{-1}) of the incident light, for an unknown metal. ($1\text{eV} = 1.6022 \times 10^{-19} \text{ J}$)

Based on the plot, answer the questions **a to c** below:

a. 1 point On the plot above, label the **threshold frequency**.

b. 2 points What is the value of the work potential (in J) of the unknown metal?

c. 3 points What is the *longest* wavelength (in nm) required to eject an electron from this metal's surface? Show your work including equation(s) used. (3 sig figs)

d. 2 points In the space provided below, plot the following

(Assume frequency is *above* threshold frequency):

- i. *Kinetic energy* of an ejected electron versus the *intensity* of the incident light (left)
- ii. *Number of electrons* ejected versus the *frequency* of the incident light (right)



**Question 3** **6 points****a. 2 points**

In the ground state of Ge (Germanium; $Z = 32$) atom, **name** the following orbitals:

i. Orbital containing the electron that requires the maximum energy to be removed.

ii. Orbital containing the electron that experiences the minimum effective nuclear charge.

b. 2 points

An unknown anion (A^-) has the following *excited* electronic configuration: $[Ar] 4s^2 3d^3 4p^1$

(Ar: Argon; $Z = 18$)

Give the **condensed ground state electronic configuration** of the atom, **A** and the cation, **A^{2+}** .
Show your work. (Assume all electronic transitions are possible)

Electronic Configuration of A:

Electronic Configuration of A^{2+} :

c. 2 points

In a single atom, determine how many electrons can have the following set of quantum numbers?

(i) $n = 5; l = 3$

In a single atom, determine how many electrons can have the following set of quantum numbers?

(i) $n = 3; l = 2; m_s = +1/2$

**Question 4** **8 points**

Rank the following from 1 to 4 in **increasing** order. Provide the ranking in the boxes provided.

Note: Atomic number (Z) are provided for relevant atoms

a) Increasing (lowest to highest) order of *ionization energy*: (1= lowest; 4 = highest)

Mg

Mg: Z = 12

Rb

Rb: Z = 37

Cs

Cs: Z = 55

K

K : Z = 19

b) Increasing (lowest to highest) order of *bond angle*: (1= lowest; 4 = highest)

H₂OCCl₄XeF₄

HCN

c) Increasing (smallest to largest) order of *ionic size* (1= smallest; 4 = largest)

S²⁻

S: Z = 16

O²⁻

O: Z = 8

Be²⁺

Be: Z = 4

Na⁺

Na: Z = 11

d) Increasing (lowest to highest) order of *bond energy* (1= lowest; 4 = highest)

P-F

P-Cl

P-I

P-Br

**Question 5** **12 points**

a. 9 points For the molecules HClO_4 and TeF_6 , draw the most stable *Lewis structure* showing appropriate shape and *dashed/wedged bonds*, based on VSEPR. Determine the **molecular geometry** around the central atom. For HClO_4 , label the *most polar bond*.



b. 3 points (Note: Lone pairs are not shown in structures below)

(i) For the three *resonance* structures (1, 2 and 3) shown below, answer the following question:



(I) Circle the *most contributing* resonance structure.

Structure 1

Structure 2

Structure 3

(ii) For the three structures given (4, 5, and 6) below, answer the following questions:



(I) Circle all the structures that represent *resonance structures* of each other:

Structure 4

Structure 5

Structure 6

(II) Among the resonance structures, circle the *most contributing* resonance structure.

Structure 4

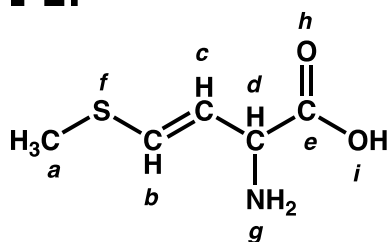
Structure 5

Structure 6

**Question 6** **6 points**

Using the data given below, calculate the **lattice energy** for SrO(s).

	ΔH° (kJ/mol)
$\text{Sr}(s) \longrightarrow \text{Sr}(g)$	+180
$\text{Sr}(g) \longrightarrow \text{Sr}^{2+}(g) + 2e^-$	+1468
$\text{O}_2(g) \longrightarrow 2\text{O}(g)$	+500
$\text{O}(g) + 2e^- \longrightarrow \text{O}^{2-}(g)$	+600
$\text{Sr}(s) + \frac{1}{2}\text{O}_2(g) \longrightarrow \text{SrO}(s)$	-550

**Question 7 10 points**

Consider the molecule given on the left, and answer questions (i) to (vi) using chemical bonding theories that we have learnt in class (valence bond theory, hybridization).

Note that the lone pairs are not shown in the structure:

i. Consider the carbon atoms marked in as ***a***, ***b***, ***c***, ***d***, and ***e*** in the molecule, and determine **hybridization** at each of the carbon centers. (2.5 points)

Atom	Hybridization
Carbon (<i>a</i>)	
Carbon (<i>b</i>)	
Carbon (<i>c</i>)	
Carbon (<i>d</i>)	
Carbon (<i>e</i>)	

ii. Now, consider the sulfur atom (S) marked as ***f***, nitrogen (N) atom (***g***), and oxygen(O) atoms (***h*** and ***i***), and determine their **hybridization**. (2 points)

Atom	Hybridization
Sulfur (<i>f</i>)	
Nitrogen (<i>g</i>)	
Oxygen (<i>h</i>)	
Oxygen (<i>i</i>)	

iii. **Describe** all the bonds formed by the carbon atoms ***a*** and ***b***, using sigma and pi designation as appropriate. (2 points)

Atom	Bonds formed (use sigma and pi as appropriate)
Carbon (<i>a</i>)	
Carbon (<i>b</i>)	

iv. How many total sp^3 -s sigma bonds are in the molecule? (1 point)

v. Determine **electronic** and **molecular geometry** at carbon (C) labeled ***e*** and sulfur (S) labeled ***f*** centres. (2 points)

Atom	Electronic Geometry	Molecular Geometry
Carbon (<i>e</i>)		
Sulfur (<i>f</i>)		

vi. Circle the value from the choices given below, that describes the **C-S-C bond angle at sulfur** (S) marked ***f***: (0.5 points)

>90° to <109.5° 109.5° >109.5 to <120° 120° >120° to <180°

**Question 8 12 points**

Using the **valence shell orbitals only** draw a **complete** molecular orbital (MO) diagram (with no s-p mixing) for the molecule **PS**, in the space provided below. **(6 points)**

Assume that 3s and 3p atomic orbitals will form molecular orbitals like those formed by the combination of 2s and 2p atomic orbitals that we learnt in class, and there is no s-p mixing.

Show **energy arrow**, **atomic orbitals** involved, **atom labels**, the **molecular orbitals** that are formed, **filled electrons** in the MO diagram etc. Make sure to label all the atomic and molecular orbitals.

Using the MO diagram above, answer the following questions:

- (i) Which one of the following will have the highest bond order? Circle the correct choice
A) PS^{1-} B) PS C) PS^{1+}
- (ii) Which one of the following will have the longest bond length? Circle the correct choice
A) PS^{1-} B) PS C) PS^{1+}
- (iii) Which of the following will be diamagnetic? Circle the correct choice.
A) PS^{1-} B) PS C) PS^{1+}
- (iv) Draw Lewis dot structure of the molecule with the shortest bond length (based on the MO diagram) from among the three (PS^{1-} , PS, PS^{1+}), and indicate the bond order calculated from the Lewis dot structure. **(3 points)**

Lewis Dot Structure

Bond Order



A42C0743-B4D3-4CB2-BFB7-CD0DDEDA6719

midterm-exam-93388

#212 Page 10 of 10

SCRAP PAGE

**CHEMISTRY 110 –Midterm Exam****Thursday November 7, 2024, 6:30 – 8:30 pm**

Examiners: Pallavi Sirjoosingh and Ashok Kakkar

First name (please write as legibly as possible within the boxes)

--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--

Last name

--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--

Student ID number

--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--

This booklet has 8 questions on 10 pages (The last page can be used as a scrap page).

The staple must NOT be removed from the question booklet. Your intact question booklet must be returned at the end of the exam.

A separate booklet contains a periodic table, and datasheet.

Write your answers directly under the questions in the space provided. Do not write on the QR codes or near the top edges (as it may be cut off).

Please write **legibly and dark enough** for the scan to be readable. For students who use English as a second language, dictionaries are permitted. Calculators are allowed.

DO NOT ASK QUESTIONS OF THE INVIGILATORS about interpretations of the exam questions. If you have a concern about a question, please summarize it at the top of this page. All concerns will be evaluated after the exam. GOOD LUCK!

**Question 1 8 points** (*Assume all transitions are possible*)

a. 6 points Calculate the **energy (in kJ/mol)** required to ionize (completely remove) the electron from the **$n=3$** level of **He^+ ion** (He: Helium). Show your work including equations. (3 sig. figs.)

b. 2 points

(I) Select the largest (longest) wavelength from the three choices below:

- i. Wavelength of photon to *completely remove* an electron from *1s orbital of Hydrogen atom*
- ii. Wavelength of photon to *completely remove* an electron from *1s orbital of Boron atom (B)*
- iii. Wavelength of photon to *completely remove* an electron from *1s orbital of Boron cation (B^+)*

i or ii or iii

(II) Select the smallest (shortest) wavelength from the three choices below (*for H atom*):

- i. Wavelength of photon *emitted* for an electron transitioning from *2p to 1s orbital*
- ii. Wavelength of photon *absorbed* for an electron transitioning from *2s to 4p orbital*
- iii. Wavelength of photon *absorbed* for an electron transitioning from *1s to 3p orbital*

i or ii or iii



Question 2 8 points



A Chem110 student plots the graph shown on the left, from data obtained for the kinetic energy (eV) of the ejected electron versus frequency (s^{-1}) of the incident light, for an unknown metal. ($1\text{eV} = 1.6022 \times 10^{-19} \text{ J}$)

Based on the plot, answer the questions **a to c** below:

a. 1 point On the plot above, label the **threshold frequency**.

b. 2 points What is the value of the work potential (in J) of the unknown metal?

c. 3 points What is the *longest* wavelength (in nm) required to eject an electron from this metal's surface? Show your work including equation(s) used. (3 sig figs)

d. 2 points In the space provided below, plot the following

(Assume frequency is *above* threshold frequency):

- Kinetic energy* of an ejected electron versus the *intensity* of the incident light (left)
- Number of electrons* ejected versus the *frequency* of the incident light (right)



**Question 3** **6 points****a. 2 points**

In the ground state of Ge (Germanium; $Z = 32$) atom, **name** the following orbitals:

i. Orbital containing the electron that requires the maximum energy to be removed.

ii. Orbital containing the electron that experiences the minimum effective nuclear charge.

b. 2 points

An unknown anion (A^-) has the following *excited* electronic configuration: $[Ar] 4s^2 3d^3 4p^1$

(Ar: Argon; $Z = 18$)

Give the **condensed ground state electronic configuration** of the atom, **A** and the cation, **A^{2+}** .
Show your work. (Assume all electronic transitions are possible)

Electronic Configuration of A:

Electronic Configuration of A^{2+} :

c. 2 points

In a single atom, determine how many electrons can have the following set of quantum numbers?

(i) $n = 5; l = 3$

In a single atom, determine how many electrons can have the following set of quantum numbers?

(i) $n = 3; l = 2; m_s = +1/2$

**Question 4** **8 points**

Rank the following from 1 to 4 in **increasing** order. Provide the ranking in the boxes provided.

Note: Atomic number (Z) are provided for relevant atoms

a) Increasing (lowest to highest) order of *ionization energy*: (1= lowest; 4 = highest)

Mg

Mg: Z = 12

Rb

Rb: Z = 37

Cs

Cs: Z = 55

K

K : Z = 19

b) Increasing (lowest to highest) order of *bond angle*: (1= lowest; 4 = highest)

H₂OCCl₄XeF₄

HCN

c) Increasing (smallest to largest) order of *ionic size* (1= smallest; 4 = largest)

S²⁻

S: Z = 16

O²⁻

O: Z = 8

Be²⁺

Be: Z = 4

Na⁺

Na: Z = 11

d) Increasing (lowest to highest) order of *bond energy* (1= lowest; 4 = highest)

P-F

P-Cl

P-I

P-Br

**Question 5** **12 points**

a. 9 points For the molecules HClO_4 and TeF_6 , draw the most stable *Lewis structure* showing appropriate shape and *dashed/wedged bonds*, based on VSEPR. Determine the **molecular geometry** around the central atom. For HClO_4 , label the *most polar bond*.



b. 3 points (Note: Lone pairs are not shown in structures below)

(i) For the three *resonance* structures (1, 2 and 3) shown below, answer the following question:



(I) Circle the *most contributing* resonance structure.

Structure 1

Structure 2

Structure 3

(ii) For the three structures given (4, 5, and 6) below, answer the following questions:



(I) Circle all the structures that represent *resonance structures* of each other:

Structure 4

Structure 5

Structure 6

(II) Among the resonance structures, circle the *most contributing* resonance structure.

Structure 4

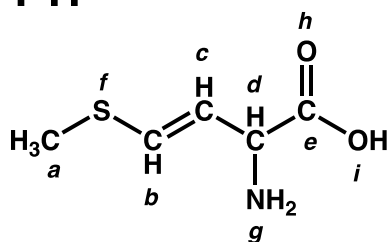
Structure 5

Structure 6

**Question 6** **6 points**

Using the data given below, calculate the **lattice energy** for SrO(s).

	ΔH° (kJ/mol)
$\text{Sr}(s) \longrightarrow \text{Sr}(g)$	+180
$\text{Sr}(g) \longrightarrow \text{Sr}^{2+}(g) + 2e^-$	+1468
$\text{O}_2(g) \longrightarrow 2\text{O}(g)$	+500
$\text{O}(g) + 2e^- \longrightarrow \text{O}^{2-}(g)$	+600
$\text{Sr}(s) + \frac{1}{2}\text{O}_2(g) \longrightarrow \text{SrO}(s)$	-550

**Question 7 10 points**

Consider the molecule given on the left, and answer questions (i) to (vi) using chemical bonding theories that we have learnt in class (valence bond theory, hybridization).

Note that the lone pairs are not shown in the structure:

i. Consider the carbon atoms marked in as ***a***, ***b***, ***c***, ***d***, and ***e*** in the molecule, and determine **hybridization** at each of the carbon centers. (2.5 points)

Atom	Hybridization
Carbon (<i>a</i>)	
Carbon (<i>b</i>)	
Carbon (<i>c</i>)	
Carbon (<i>d</i>)	
Carbon (<i>e</i>)	

ii. Now, consider the sulfur atom (S) marked as ***f***, nitrogen (N) atom (***g***), and oxygen(O) atoms (***h*** and ***i***), and determine their **hybridization**. (2 points)

Atom	Hybridization
Sulfur (<i>f</i>)	
Nitrogen (<i>g</i>)	
Oxygen (<i>h</i>)	
Oxygen (<i>i</i>)	

iii. **Describe** all the bonds formed by the carbon atoms ***a*** and ***b***, using sigma and pi designation as appropriate. (2 points)

Atom	Bonds formed (use sigma and pi as appropriate)
Carbon (<i>a</i>)	
Carbon (<i>b</i>)	

iv. How many total sp^3 -s sigma bonds are in the molecule? (1 point)

v. Determine **electronic** and **molecular geometry** at carbon (C) labeled ***e*** and sulfur (S) labeled ***f*** centres. (2 points)

Atom	Electronic Geometry	Molecular Geometry
Carbon (<i>e</i>)		
Sulfur (<i>f</i>)		

vi. Circle the value from the choices given below, that describes the **C-S-C bond angle at sulfur** (S) marked ***f***: (0.5 points)

>90° to <109.5° 109.5° >109.5 to <120° 120° >120° to <180°

**Question 8 12 points**

Using the **valence shell orbitals only** draw a **complete** molecular orbital (MO) diagram (with no s-p mixing) for the molecule **PS**, in the space provided below. **(6 points)**

Assume that 3s and 3p atomic orbitals will form molecular orbitals like those formed by the combination of 2s and 2p atomic orbitals that we learnt in class, and there is no s-p mixing.

Show **energy arrow**, **atomic orbitals** involved, **atom labels**, the **molecular orbitals** that are formed, **filled electrons** in the MO diagram etc. Make sure to label all the atomic and molecular orbitals.

Using the MO diagram above, answer the following questions:

- (i) Which one of the following will have the highest bond order? Circle the correct choice
A) PS^{1-} B) PS C) PS^{1+}
- (ii) Which one of the following will have the longest bond length? Circle the correct choice
A) PS^{1-} B) PS C) PS^{1+}
- (iii) Which of the following will be diamagnetic? Circle the correct choice.
A) PS^{1-} B) PS C) PS^{1+}
- (iv) Draw Lewis dot structure of the molecule with the shortest bond length (based on the MO diagram) from among the three (PS^{1-} , PS, PS^{1+}), and indicate the bond order calculated from the Lewis dot structure. **(3 points)**

Lewis Dot Structure

Bond Order



64962638-49BB-49A8-A55B-FE05D9B3B411

midterm-exam-93388

#213 Page 10 of 10

SCRAP PAGE

**CHEMISTRY 110 –Midterm Exam****Thursday November 7, 2024, 6:30 – 8:30 pm**

Examiners: Pallavi Sirjoosingh and Ashok Kakkar

First name (please write as legibly as possible within the boxes)

--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--

Last name

--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--

Student ID number

--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--

This booklet has 8 questions on 10 pages (The last page can be used as a scrap page).

The staple must NOT be removed from the question booklet. Your intact question booklet must be returned at the end of the exam.

A separate booklet contains a periodic table, and datasheet.

Write your answers directly under the questions in the space provided. Do not write on the QR codes or near the top edges (as it may be cut off).

Please write **legibly and dark enough** for the scan to be readable. For students who use English as a second language, dictionaries are permitted. Calculators are allowed.

DO NOT ASK QUESTIONS OF THE INVIGILATORS about interpretations of the exam questions. If you have a concern about a question, please summarize it at the top of this page. All concerns will be evaluated after the exam. GOOD LUCK!

**Question 1 8 points** (*Assume all transitions are possible*)

a. 6 points Calculate the **energy (in kJ/mol)** required to ionize (completely remove) the electron from the **n=3** level of **He⁺ ion** (He: Helium). Show your work including equations. (3 sig. figs.)

b. 2 points

(I) Select the largest (longest) wavelength from the three choices below:

- i. Wavelength of photon to *completely remove* an electron from *1s orbital of Hydrogen atom*
- ii. Wavelength of photon to *completely remove* an electron from *1s orbital of Boron atom (B)*
- iii. Wavelength of photon to *completely remove* an electron from *1s orbital of Boron cation (B⁺)*

i or ii or iii

(II) Select the smallest (shortest) wavelength from the three choices below (*for H atom*):

- i. Wavelength of photon *emitted* for an electron transitioning from *2p to 1s orbital*
- ii. Wavelength of photon *absorbed* for an electron transitioning from *2s to 4p orbital*
- iii. Wavelength of photon *absorbed* for an electron transitioning from *1s to 3p orbital*

i or ii or iii



Question 2 8 points



A Chem110 student plots the graph shown on the left, from data obtained for the kinetic energy (eV) of the ejected electron versus frequency (s^{-1}) of the incident light, for an unknown metal. ($1\text{eV} = 1.6022 \times 10^{-19} \text{ J}$)

Based on the plot, answer the questions **a to c** below:

a. 1 point On the plot above, label the **threshold frequency**.

b. 2 points What is the value of the work potential (in J) of the unknown metal?

c. 3 points What is the *longest* wavelength (in nm) required to eject an electron from this metal's surface? Show your work including equation(s) used. (3 sig figs)

d. 2 points In the space provided below, plot the following

(Assume frequency is *above* threshold frequency):

- Kinetic energy* of an ejected electron versus the *intensity* of the incident light (left)
- Number of electrons* ejected versus the *frequency* of the incident light (right)



**Question 3** **6 points****a. 2 points**

In the ground state of Ge (Germanium; $Z = 32$) atom, **name** the following orbitals:

i. Orbital containing the electron that requires the maximum energy to be removed.

ii. Orbital containing the electron that experiences the minimum effective nuclear charge.

b. 2 points

An unknown anion (A^-) has the following *excited* electronic configuration: $[Ar] 4s^2 3d^3 4p^1$

(Ar: Argon; $Z = 18$)

Give the **condensed ground state electronic configuration** of the atom, **A** and the cation, **A^{2+}** .
Show your work. (Assume all electronic transitions are possible)

Electronic Configuration of A:

Electronic Configuration of A^{2+} :

c. 2 points

In a single atom, determine how many electrons can have the following set of quantum numbers?

(i) $n = 5; l = 3$

In a single atom, determine how many electrons can have the following set of quantum numbers?

(i) $n = 3; l = 2; m_s = +1/2$

**Question 4** **8 points**

Rank the following from 1 to 4 in **increasing** order. Provide the ranking in the boxes provided.

Note: Atomic number (Z) are provided for relevant atoms

a) Increasing (lowest to highest) order of *ionization energy*: (1= lowest; 4 = highest)

Mg

Mg: Z = 12

Rb

Rb: Z = 37

Cs

Cs: Z = 55

K

K : Z = 19

b) Increasing (lowest to highest) order of *bond angle*: (1= lowest; 4 = highest)

H₂OCCl₄XeF₄

HCN

c) Increasing (smallest to largest) order of *ionic size* (1= smallest; 4 = largest)

S²⁻

S: Z = 16

O²⁻

O: Z = 8

Be²⁺

Be: Z = 4

Na⁺

Na: Z = 11

d) Increasing (lowest to highest) order of *bond energy* (1= lowest; 4 = highest)

P-F

P-Cl

P-I

P-Br

**Question 5** **12 points**

a. 9 points For the molecules HClO_4 and TeF_6 , draw the most stable *Lewis structure* showing appropriate shape and *dashed/wedged bonds*, based on VSEPR. Determine the **molecular geometry** around the central atom. For HClO_4 , label the *most polar bond*.

HClO_4 (there are no O-O bonds)

Label the most polar bond

Molecular Geometry around Cl

TeF_6

Molecular Geometry around Te

b. 3 points (Note: Lone pairs are not shown in structures below)

(i) For the three *resonance* structures (1, 2 and 3) shown below, answer the following question:



(I) Circle the *most contributing* resonance structure.

Structure 1

Structure 2

Structure 3

(ii) For the three structures given (4, 5, and 6) below, answer the following questions:



(I) Circle all the structures that represent *resonance structures* of each other:

Structure 4

Structure 5

Structure 6

(II) Among the resonance structures, circle the *most contributing* resonance structure.

Structure 4

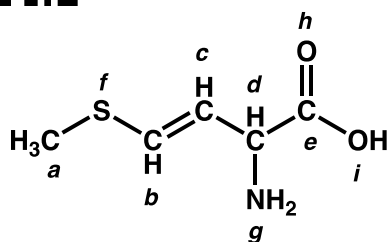
Structure 5

Structure 6

**Question 6** **6 points**

Using the data given below, calculate the **lattice energy** for SrO(s).

	ΔH° (kJ/mol)
$\text{Sr}(s) \longrightarrow \text{Sr}(g)$	+180
$\text{Sr}(g) \longrightarrow \text{Sr}^{2+}(g) + 2e^-$	+1468
$\text{O}_2(g) \longrightarrow 2\text{O}(g)$	+500
$\text{O}(g) + 2e^- \longrightarrow \text{O}^{2-}(g)$	+600
$\text{Sr}(s) + \frac{1}{2}\text{O}_2(g) \longrightarrow \text{SrO}(s)$	-550

**Question 7 10 points**

Consider the molecule given on the left, and answer questions (i) to (vi) using chemical bonding theories that we have learnt in class (valence bond theory, hybridization).

Note that the lone pairs are not shown in the structure:

i. Consider the carbon atoms marked in as ***a***, ***b***, ***c***, ***d***, and ***e*** in the molecule, and determine **hybridization** at each of the carbon centers. (2.5 points)

Atom	Hybridization
Carbon (<i>a</i>)	
Carbon (<i>b</i>)	
Carbon (<i>c</i>)	
Carbon (<i>d</i>)	
Carbon (<i>e</i>)	

ii. Now, consider the sulfur atom (S) marked as ***f***, nitrogen (N) atom (***g***), and oxygen(O) atoms (***h*** and ***i***), and determine their **hybridization**. (2 points)

Atom	Hybridization
Sulfur (<i>f</i>)	
Nitrogen (<i>g</i>)	
Oxygen (<i>h</i>)	
Oxygen (<i>i</i>)	

iii. **Describe** all the bonds formed by the carbon atoms ***a*** and ***b***, using sigma and pi designation as appropriate. (2 points)

Atom	Bonds formed (use sigma and pi as appropriate)
Carbon (<i>a</i>)	
Carbon (<i>b</i>)	

iv. How many total sp^3 -s sigma bonds are in the molecule? (1 point)

v. Determine **electronic** and **molecular geometry** at carbon (C) labeled ***e*** and sulfur (S) labeled ***f*** centres. (2 points)

Atom	Electronic Geometry	Molecular Geometry
Carbon (<i>e</i>)		
Sulfur (<i>f</i>)		

vi. Circle the value from the choices given below, that describes the **C-S-C bond angle at sulfur** (S) marked ***f***: (0.5 points)

>90° to <109.5° 109.5° >109.5 to <120° 120° >120° to <180°



Question 8 12 points

Using the **valence shell orbitals only** draw a **complete** molecular orbital (MO) diagram (with no s-p mixing) for the molecule **PS**, in the space provided below. **(6 points)**

Assume that 3s and 3p atomic orbitals will form molecular orbitals like those formed by the combination of 2s and 2p atomic orbitals that we learnt in class, and there is no s-p mixing.

Show **energy arrow**, **atomic orbitals** involved, **atom labels**, the **molecular orbitals** that are formed, **filled electrons** in the MO diagram etc. Make sure to label all the atomic and molecular orbitals.

Using the MO diagram above, answer the following questions:

- (i) Which one of the following will have the highest bond order? Circle the correct choice
 A) PS^{1-} B) PS C) PS^{1+}
- (ii) Which one of the following will have the longest bond length? Circle the correct choice
 A) PS^{1-} B) PS C) PS^{1+}
- (iii) Which of the following will be diamagnetic? Circle the correct choice.
 A) PS^{1-} B) PS C) PS^{1+}
- (iv) Draw Lewis dot structure of the molecule with the shortest bond length (based on the MO diagram) from among the three (PS^{1-} , PS, PS^{1+}), and indicate the bond order calculated from the Lewis dot structure. **(3 points)**

Lewis Dot Structure

Bond Order



98EA3040-4C28-4FE2-AD8B-11DDE1E6F1D1

midterm-exam-93388

#214 Page 10 of 10

SCRAP PAGE

**CHEMISTRY 110 –Midterm Exam****Thursday November 7, 2024, 6:30 – 8:30 pm****Examiners: Pallavi Sirjoosingh and Ashok Kakkar**

First name (please write as legibly as possible within the boxes)

--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--

Last name

--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--

Student ID number

--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--

This booklet has 8 questions on 10 pages (The last page can be used as a scrap page).

The staple must NOT be removed from the question booklet. Your intact question booklet must be returned at the end of the exam.

A separate booklet contains a periodic table, and datasheet.

Write your answers directly under the questions in the space provided. Do not write on the QR codes or near the top edges (as it may be cut off).

Please write **legibly and dark enough** for the scan to be readable. For students who use English as a second language, dictionaries are permitted. Calculators are allowed.

DO NOT ASK QUESTIONS OF THE INVIGILATORS about interpretations of the exam questions. If you have a concern about a question, please summarize it at the top of this page. All concerns will be evaluated after the exam. GOOD LUCK!

**Question 1 8 points** (*Assume all transitions are possible*)

a. 6 points Calculate the **energy (in kJ/mol)** required to ionize (completely remove) the electron from the **n=3** level of **He⁺ ion** (He: Helium). Show your work including equations. (3 sig. figs.)

b. 2 points

(I) Select the largest (longest) wavelength from the three choices below:

- i. Wavelength of photon to *completely remove* an electron from *1s orbital of Hydrogen atom*
- ii. Wavelength of photon to *completely remove* an electron from *1s orbital of Boron atom (B)*
- iii. Wavelength of photon to *completely remove* an electron from *1s orbital of Boron cation (B⁺)*

i or ii or iii

(II) Select the smallest (shortest) wavelength from the three choices below (*for H atom*):

- i. Wavelength of photon *emitted* for an electron transitioning from *2p to 1s orbital*
- ii. Wavelength of photon *absorbed* for an electron transitioning from *2s to 4p orbital*
- iii. Wavelength of photon *absorbed* for an electron transitioning from *1s to 3p orbital*

i or ii or iii



Question 2 8 points



A Chem110 student plots the graph shown on the left, from data obtained for the kinetic energy (eV) of the ejected electron versus frequency (s^{-1}) of the incident light, for an unknown metal. ($1\text{eV} = 1.6022 \times 10^{-19} \text{ J}$)

Based on the plot, answer the questions **a to c** below:

a. 1 point On the plot above, label the **threshold frequency**.

b. 2 points What is the value of the work potential (in J) of the unknown metal?

c. 3 points What is the *longest* wavelength (in nm) required to eject an electron from this metal's surface? Show your work including equation(s) used. (3 sig figs)

d. 2 points In the space provided below, plot the following

(Assume frequency is *above* threshold frequency):

- Kinetic energy* of an ejected electron versus the *intensity* of the incident light (left)
- Number of electrons* ejected versus the *frequency* of the incident light (right)



**Question 3** **6 points****a. 2 points**

In the ground state of Ge (Germanium; $Z = 32$) atom, **name** the following orbitals:

i. Orbital containing the electron that requires the maximum energy to be removed.

ii. Orbital containing the electron that experiences the minimum effective nuclear charge.

b. 2 points

An unknown anion (A^-) has the following *excited* electronic configuration: $[Ar] 4s^2 3d^3 4p^1$

(Ar: Argon; $Z = 18$)

Give the **condensed ground state electronic configuration** of the atom, **A** and the cation, **A^{2+}** .
Show your work. (Assume all electronic transitions are possible)

Electronic Configuration of A:

Electronic Configuration of A^{2+} :

c. 2 points

In a single atom, determine how many electrons can have the following set of quantum numbers?

(i) $n = 5; l = 3$

In a single atom, determine how many electrons can have the following set of quantum numbers?

(i) $n = 3; l = 2; m_s = +1/2$

**Question 4** **8 points**

Rank the following from 1 to 4 in **increasing** order. Provide the ranking in the boxes provided.

Note: Atomic number (Z) are provided for relevant atoms

a) Increasing (lowest to highest) order of *ionization energy*: (1= lowest; 4 = highest)

Mg

Mg: Z = 12

Rb

Rb: Z = 37

Cs

Cs: Z = 55

K

K : Z = 19

b) Increasing (lowest to highest) order of *bond angle*: (1= lowest; 4 = highest)

H₂OCCl₄XeF₄

HCN

c) Increasing (smallest to largest) order of *ionic size* (1= smallest; 4 = largest)

S²⁻

S: Z = 16

O²⁻

O: Z = 8

Be²⁺

Be: Z = 4

Na⁺

Na: Z = 11

d) Increasing (lowest to highest) order of *bond energy* (1= lowest; 4 = highest)

P-F

P-Cl

P-I

P-Br

**Question 5** **12 points**

a. 9 points For the molecules HClO_4 and TeF_6 , draw the most stable *Lewis structure* showing appropriate shape and *dashed/wedged bonds*, based on VSEPR. Determine the **molecular geometry** around the central atom. For HClO_4 , label the *most polar bond*.



b. 3 points (Note: Lone pairs are not shown in structures below)

(i) For the three *resonance* structures (1, 2 and 3) shown below, answer the following question:



(I) Circle the *most contributing* resonance structure.

Structure 1

Structure 2

Structure 3

(ii) For the three structures given (4, 5, and 6) below, answer the following questions:



(I) Circle all the structures that represent *resonance structures* of each other:

Structure 4

Structure 5

Structure 6

(II) Among the resonance structures, circle the *most contributing* resonance structure.

Structure 4

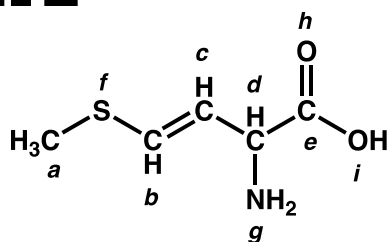
Structure 5

Structure 6

**Question 6** **6 points**

Using the data given below, calculate the **lattice energy** for SrO(s).

	ΔH° (kJ/mol)
$\text{Sr}(s) \longrightarrow \text{Sr}(g)$	+180
$\text{Sr}(g) \longrightarrow \text{Sr}^{2+}(g) + 2e^-$	+1468
$\text{O}_2(g) \longrightarrow 2\text{O}(g)$	+500
$\text{O}(g) + 2e^- \longrightarrow \text{O}^{2-}(g)$	+600
$\text{Sr}(s) + \frac{1}{2}\text{O}_2(g) \longrightarrow \text{SrO}(s)$	-550

**Question 7 10 points**

Consider the molecule given on the left, and answer questions (i) to (vi) using chemical bonding theories that we have learnt in class (valence bond theory, hybridization).

Note that the lone pairs are not shown in the structure:

i. Consider the carbon atoms marked in as ***a***, ***b***, ***c***, ***d***, and ***e*** in the molecule, and determine **hybridization** at each of the carbon centers. (2.5 points)

Atom	Hybridization
Carbon (<i>a</i>)	
Carbon (<i>b</i>)	
Carbon (<i>c</i>)	
Carbon (<i>d</i>)	
Carbon (<i>e</i>)	

ii. Now, consider the sulfur atom (S) marked as ***f***, nitrogen (N) atom (***g***), and oxygen(O) atoms (***h*** and ***i***), and determine their **hybridization**. (2 points)

Atom	Hybridization
Sulfur (<i>f</i>)	
Nitrogen (<i>g</i>)	
Oxygen (<i>h</i>)	
Oxygen (<i>i</i>)	

iii. **Describe** all the bonds formed by the carbon atoms ***a*** and ***b***, using sigma and pi designation as appropriate. (2 points)

Atom	Bonds formed (use sigma and pi as appropriate)
Carbon (<i>a</i>)	
Carbon (<i>b</i>)	

iv. How many total sp^3 -s sigma bonds are in the molecule? (1 point)

v. Determine **electronic** and **molecular geometry** at carbon (C) labeled ***e*** and sulfur (S) labeled ***f*** centres. (2 points)

Atom	Electronic Geometry	Molecular Geometry
Carbon (<i>e</i>)		
Sulfur (<i>f</i>)		

vi. Circle the value from the choices given below, that describes the **C-S-C bond angle at sulfur** (S) marked ***f***: (0.5 points)

$>90^\circ$ to $<109.5^\circ$ 109.5° >109.5 to $<120^\circ$ 120° $>120^\circ$ to $<180^\circ$


Question 8 12 points

Using the **valence shell orbitals only** draw a **complete** molecular orbital (MO) diagram (with no s-p mixing) for the molecule **PS**, in the space provided below. **(6 points)**

Assume that 3s and 3p atomic orbitals will form molecular orbitals like those formed by the combination of 2s and 2p atomic orbitals that we learnt in class, and there is no s-p mixing.

Show **energy arrow**, **atomic orbitals** involved, **atom labels**, the **molecular orbitals** that are formed, **filled electrons** in the MO diagram etc. Make sure to label all the atomic and molecular orbitals.

Using the MO diagram above, answer the following questions:

- (i) Which one of the following will have the highest bond order? Circle the correct choice
 A) PS^{1-} B) PS C) PS^{1+}
- (ii) Which one of the following will have the longest bond length? Circle the correct choice
 A) PS^{1-} B) PS C) PS^{1+}
- (iii) Which of the following will be diamagnetic? Circle the correct choice.
 A) PS^{1-} B) PS C) PS^{1+}
- (iv) Draw Lewis dot structure of the molecule with the shortest bond length (based on the MO diagram) from among the three (PS^{1-} , PS, PS^{1+}), and indicate the bond order calculated from the Lewis dot structure. **(3 points)**

Lewis Dot Structure

Bond Order



13EE72B2-789E-4A1F-9249-65CCA5ABB8E8

midterm-exam-93388

#215 Page 10 of 10

SCRAP PAGE

**CHEMISTRY 110 –Midterm Exam****Thursday November 7, 2024, 6:30 – 8:30 pm**

Examiners: Pallavi Sirjoosingh and Ashok Kakkar

First name (please write as legibly as possible within the boxes)

--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--

Last name

--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--

Student ID number

--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--

This booklet has 8 questions on 10 pages (The last page can be used as a scrap page).

The staple must NOT be removed from the question booklet. Your intact question booklet must be returned at the end of the exam.

A separate booklet contains a periodic table, and datasheet.

Write your answers directly under the questions in the space provided. Do not write on the QR codes or near the top edges (as it may be cut off).

Please write **legibly and dark enough** for the scan to be readable. For students who use English as a second language, dictionaries are permitted. Calculators are allowed.

DO NOT ASK QUESTIONS OF THE INVIGILATORS about interpretations of the exam questions. If you have a concern about a question, please summarize it at the top of this page. All concerns will be evaluated after the exam. GOOD LUCK!

**Question 1 8 points** (*Assume all transitions are possible*)

a. 6 points Calculate the **energy (in kJ/mol)** required to ionize (completely remove) the electron from the **n=3** level of **He⁺ ion** (He: Helium). Show your work including equations. (3 sig. figs.)

b. 2 points

(I) Select the largest (longest) wavelength from the three choices below:

- i. Wavelength of photon to *completely remove* an electron from *1s orbital of Hydrogen atom*
- ii. Wavelength of photon to *completely remove* an electron from *1s orbital of Boron atom (B)*
- iii. Wavelength of photon to *completely remove* an electron from *1s orbital of Boron cation (B⁺)*

i or ii or iii

(II) Select the smallest (shortest) wavelength from the three choices below (*for H atom*):

- i. Wavelength of photon *emitted* for an electron transitioning from *2p to 1s orbital*
- ii. Wavelength of photon *absorbed* for an electron transitioning from *2s to 4p orbital*
- iii. Wavelength of photon *absorbed* for an electron transitioning from *1s to 3p orbital*

i or ii or iii



Question 2 8 points



A Chem110 student plots the graph shown on the left, from data obtained for the kinetic energy (eV) of the ejected electron versus frequency (s^{-1}) of the incident light, for an unknown metal. ($1\text{eV} = 1.6022 \times 10^{-19} \text{ J}$)

Based on the plot, answer the questions **a to c** below:

a. 1 point On the plot above, label the **threshold frequency**.

b. 2 points What is the value of the work potential (in J) of the unknown metal?

c. 3 points What is the *longest* wavelength (in nm) required to eject an electron from this metal's surface? Show your work including equation(s) used. (3 sig figs)

d. 2 points In the space provided below, plot the following

(Assume frequency is *above* threshold frequency):

- i. *Kinetic energy* of an ejected electron versus the *intensity* of the incident light (left)
- ii. *Number of electrons* ejected versus the *frequency* of the incident light (right)



**Question 3** **6 points****a. 2 points**

In the ground state of Ge (Germanium; $Z = 32$) atom, **name** the following orbitals:

i. Orbital containing the electron that requires the maximum energy to be removed.

ii. Orbital containing the electron that experiences the minimum effective nuclear charge.

b. 2 points

An unknown anion (A^-) has the following *excited* electronic configuration: $[Ar] 4s^2 3d^3 4p^1$

(Ar: Argon; $Z = 18$)

Give the **condensed ground state electronic configuration** of the atom, **A** and the cation, **A^{2+}** .
Show your work. (Assume all electronic transitions are possible)

Electronic Configuration of A:

Electronic Configuration of A^{2+} :

c. 2 points

In a single atom, determine how many electrons can have the following set of quantum numbers?

(i) $n = 5; l = 3$

In a single atom, determine how many electrons can have the following set of quantum numbers?

(i) $n = 3; l = 2; m_s = +1/2$

**Question 4** **8 points**

Rank the following from 1 to 4 in **increasing** order. Provide the ranking in the boxes provided.

Note: Atomic number (Z) are provided for relevant atoms

a) Increasing (lowest to highest) order of *ionization energy*: (1= lowest; 4 = highest)

Mg

Mg: Z = 12

Rb

Rb: Z = 37

Cs

Cs: Z = 55

K

K : Z = 19

b) Increasing (lowest to highest) order of *bond angle*: (1= lowest; 4 = highest)

H₂OCCl₄XeF₄

HCN

c) Increasing (smallest to largest) order of *ionic size* (1= smallest; 4 = largest)

S²⁻

S: Z = 16

O²⁻

O: Z = 8

Be²⁺

Be: Z = 4

Na⁺

Na: Z = 11

d) Increasing (lowest to highest) order of *bond energy* (1= lowest; 4 = highest)

P-F

P-Cl

P-I

P-Br

**Question 5** **12 points**

a. 9 points For the molecules HClO_4 and TeF_6 , draw the most stable *Lewis structure* showing appropriate shape and *dashed/wedged bonds*, based on VSEPR. Determine the **molecular geometry** around the central atom. For HClO_4 , label the *most polar bond*.



b. 3 points (Note: Lone pairs are not shown in structures below)

(i) For the three *resonance* structures (1, 2 and 3) shown below, answer the following question:



(I) Circle the *most contributing* resonance structure.

Structure 1

Structure 2

Structure 3

(ii) For the three structures given (4, 5, and 6) below, answer the following questions:



(I) Circle all the structures that represent *resonance structures* of each other:

Structure 4

Structure 5

Structure 6

(II) Among the resonance structures, circle the *most contributing* resonance structure.

Structure 4

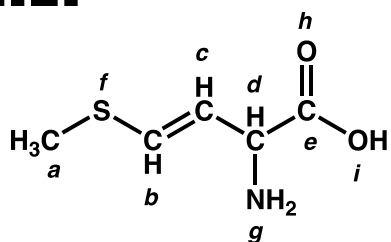
Structure 5

Structure 6

**Question 6** **6 points**

Using the data given below, calculate the **lattice energy** for SrO(s).

	ΔH° (kJ/mol)
$\text{Sr}(s) \longrightarrow \text{Sr}(g)$	+180
$\text{Sr}(g) \longrightarrow \text{Sr}^{2+}(g) + 2e^-$	+1468
$\text{O}_2(g) \longrightarrow 2\text{O}(g)$	+500
$\text{O}(g) + 2e^- \longrightarrow \text{O}^{2-}(g)$	+600
$\text{Sr}(s) + \frac{1}{2}\text{O}_2(g) \longrightarrow \text{SrO}(s)$	-550

**Question 7 10 points**

Consider the molecule given on the left, and answer questions (i) to (vi) using chemical bonding theories that we have learnt in class (valence bond theory, hybridization).

Note that the lone pairs are not shown in the structure:

i. Consider the carbon atoms marked in as ***a***, ***b***, ***c***, ***d***, and ***e*** in the molecule, and determine **hybridization** at each of the carbon centers. (2.5 points)

Atom	Hybridization
Carbon (<i>a</i>)	
Carbon (<i>b</i>)	
Carbon (<i>c</i>)	
Carbon (<i>d</i>)	
Carbon (<i>e</i>)	

ii. Now, consider the sulfur atom (S) marked as ***f***, nitrogen (N) atom (***g***), and oxygen(O) atoms (***h*** and ***i***), and determine their **hybridization**. (2 points)

Atom	Hybridization
Sulfur (<i>f</i>)	
Nitrogen (<i>g</i>)	
Oxygen (<i>h</i>)	
Oxygen (<i>i</i>)	

iii. **Describe** all the bonds formed by the carbon atoms ***a*** and ***b***, using sigma and pi designation as appropriate. (2 points)

Atom	Bonds formed (use sigma and pi as appropriate)
Carbon (<i>a</i>)	
Carbon (<i>b</i>)	

iv. How many total sp^3 -s sigma bonds are in the molecule? (1 point)

v. Determine **electronic** and **molecular geometry** at carbon (C) labeled ***e*** and sulfur (S) labeled ***f*** centres. (2 points)

Atom	Electronic Geometry	Molecular Geometry
Carbon (<i>e</i>)		
Sulfur (<i>f</i>)		

vi. Circle the value from the choices given below, that describes the **C-S-C bond angle at sulfur** (S) marked ***f***: (0.5 points)

>90° to <109.5° 109.5° >109.5 to <120° 120° >120° to <180°


Question 8 12 points

Using the **valence shell orbitals only** draw a **complete** molecular orbital (MO) diagram (with no s-p mixing) for the molecule **PS**, in the space provided below. **(6 points)**

Assume that 3s and 3p atomic orbitals will form molecular orbitals like those formed by the combination of 2s and 2p atomic orbitals that we learnt in class, and there is no s-p mixing.

Show **energy arrow**, **atomic orbitals** involved, **atom labels**, the **molecular orbitals** that are formed, **filled electrons** in the MO diagram etc. Make sure to label all the atomic and molecular orbitals.

Using the MO diagram above, answer the following questions:

- (i) Which one of the following will have the highest bond order? Circle the correct choice
 A) PS^{1-} B) PS C) PS^{1+}
- (ii) Which one of the following will have the longest bond length? Circle the correct choice
 A) PS^{1-} B) PS C) PS^{1+}
- (iii) Which of the following will be diamagnetic? Circle the correct choice.
 A) PS^{1-} B) PS C) PS^{1+}
- (iv) Draw Lewis dot structure of the molecule with the shortest bond length (based on the MO diagram) from among the three (PS^{1-} , PS, PS^{1+}), and indicate the bond order calculated from the Lewis dot structure. **(3 points)**

Lewis Dot Structure

Bond Order



B84E79D5-C064-4A84-9DF7-6AA7D2EE6DF3

midterm-exam-93388

#216 Page 10 of 10

SCRAP PAGE

**CHEMISTRY 110 –Midterm Exam****Thursday November 7, 2024, 6:30 – 8:30 pm**

Examiners: Pallavi Sirjoosingh and Ashok Kakkar

First name (please write as legibly as possible within the boxes)

--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--

Last name

--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--

Student ID number

--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--

This booklet has 8 questions on 10 pages (The last page can be used as a scrap page).

The staple must NOT be removed from the question booklet. Your intact question booklet must be returned at the end of the exam.

A separate booklet contains a periodic table, and datasheet.

Write your answers directly under the questions in the space provided. Do not write on the QR codes or near the top edges (as it may be cut off).

Please write **legibly and dark enough** for the scan to be readable. For students who use English as a second language, dictionaries are permitted. Calculators are allowed.

DO NOT ASK QUESTIONS OF THE INVIGILATORS about interpretations of the exam questions. If you have a concern about a question, please summarize it at the top of this page. All concerns will be evaluated after the exam. GOOD LUCK!

**Question 1 8 points** (*Assume all transitions are possible*)

a. 6 points Calculate the **energy (in kJ/mol)** required to ionize (completely remove) the electron from the **$n=3$** level of **He^+ ion** (He: Helium). Show your work including equations. (3 sig. figs.)

b. 2 points

(I) Select the largest (longest) wavelength from the three choices below:

- i. Wavelength of photon to *completely remove* an electron from *1s orbital of Hydrogen atom*
- ii. Wavelength of photon to *completely remove* an electron from *1s orbital of Boron atom (B)*
- iii. Wavelength of photon to *completely remove* an electron from *1s orbital of Boron cation (B^+)*

i or ii or iii

(II) Select the smallest (shortest) wavelength from the three choices below (*for H atom*):

- i. Wavelength of photon *emitted* for an electron transitioning from *2p to 1s orbital*
- ii. Wavelength of photon *absorbed* for an electron transitioning from *2s to 4p orbital*
- iii. Wavelength of photon *absorbed* for an electron transitioning from *1s to 3p orbital*

i or ii or iii



Question 2 8 points



A Chem110 student plots the graph shown on the left, from data obtained for the kinetic energy (eV) of the ejected electron versus frequency (s^{-1}) of the incident light, for an unknown metal. ($1\text{eV} = 1.6022 \times 10^{-19} \text{ J}$)

Based on the plot, answer the questions **a to c** below:

a. 1 point On the plot above, label the **threshold frequency**.

b. 2 points What is the value of the work potential (in J) of the unknown metal?

c. 3 points What is the *longest* wavelength (in nm) required to eject an electron from this metal's surface? Show your work including equation(s) used. (3 sig figs)

d. 2 points In the space provided below, plot the following

(Assume frequency is *above* threshold frequency):

- Kinetic energy* of an ejected electron versus the *intensity* of the incident light (left)
- Number of electrons* ejected versus the *frequency* of the incident light (right)



**Question 3** **6 points****a. 2 points**

In the ground state of Ge (Germanium; $Z = 32$) atom, **name** the following orbitals:

i. Orbital containing the electron that requires the maximum energy to be removed.

ii. Orbital containing the electron that experiences the minimum effective nuclear charge.

b. 2 points

An unknown anion (A^-) has the following *excited* electronic configuration: $[Ar] 4s^2 3d^3 4p^1$

(Ar: Argon; $Z = 18$)

Give the **condensed ground state electronic configuration** of the atom, **A** and the cation, **A^{2+}** .
Show your work. (Assume all electronic transitions are possible)

Electronic Configuration of A:

Electronic Configuration of A^{2+} :

c. 2 points

In a single atom, determine how many electrons can have the following set of quantum numbers?

(i) $n = 5; l = 3$

In a single atom, determine how many electrons can have the following set of quantum numbers?

(i) $n = 3; l = 2; m_s = +1/2$

**Question 4** **8 points**

Rank the following from 1 to 4 in **increasing** order. Provide the ranking in the boxes provided.

Note: Atomic number (Z) are provided for relevant atoms

a) Increasing (lowest to highest) order of *ionization energy*: (1= lowest; 4 = highest)

Mg

Mg: Z = 12

Rb

Rb: Z = 37

Cs

Cs: Z = 55

K

K : Z = 19

b) Increasing (lowest to highest) order of *bond angle*: (1= lowest; 4 = highest)

H₂OCCl₄XeF₄

HCN

c) Increasing (smallest to largest) order of *ionic size* (1= smallest; 4 = largest)

S²⁻

S: Z = 16

O²⁻

O: Z = 8

Be²⁺

Be: Z = 4

Na⁺

Na: Z = 11

d) Increasing (lowest to highest) order of *bond energy* (1= lowest; 4 = highest)

P-F

P-Cl

P-I

P-Br

**Question 5** **12 points**

a. 9 points For the molecules HClO_4 and TeF_6 , draw the most stable *Lewis structure* showing appropriate shape and *dashed/wedged bonds*, based on VSEPR. Determine the **molecular geometry** around the central atom. For HClO_4 , label the *most polar bond*.



b. 3 points (Note: Lone pairs are not shown in structures below)

(i) For the three *resonance* structures (1, 2 and 3) shown below, answer the following question:



(I) Circle the *most contributing* resonance structure.

Structure 1

Structure 2

Structure 3

(ii) For the three structures given (4, 5, and 6) below, answer the following questions:



(I) Circle all the structures that represent *resonance structures* of each other:

Structure 4

Structure 5

Structure 6

(II) Among the resonance structures, circle the *most contributing* resonance structure.

Structure 4

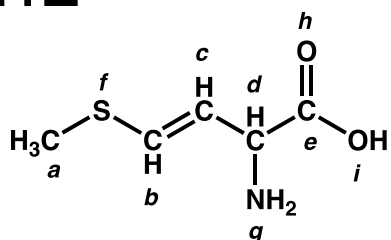
Structure 5

Structure 6

**Question 6** **6 points**

Using the data given below, calculate the **lattice energy** for SrO(s).

	ΔH° (kJ/mol)
$\text{Sr}(s) \longrightarrow \text{Sr}(g)$	+180
$\text{Sr}(g) \longrightarrow \text{Sr}^{2+}(g) + 2e^-$	+1468
$\text{O}_2(g) \longrightarrow 2\text{O}(g)$	+500
$\text{O}(g) + 2e^- \longrightarrow \text{O}^{2-}(g)$	+600
$\text{Sr}(s) + \frac{1}{2}\text{O}_2(g) \longrightarrow \text{SrO}(s)$	-550

**Question 7 10 points**

Consider the molecule given on the left, and answer questions (i) to (vi) using chemical bonding theories that we have learnt in class (valence bond theory, hybridization).

Note that the lone pairs are not shown in the structure:

i. Consider the carbon atoms marked in as ***a***, ***b***, ***c***, ***d***, and ***e*** in the molecule, and determine **hybridization** at each of the carbon centers. (2.5 points)

Atom	Hybridization
Carbon (<i>a</i>)	
Carbon (<i>b</i>)	
Carbon (<i>c</i>)	
Carbon (<i>d</i>)	
Carbon (<i>e</i>)	

ii. Now, consider the sulfur atom (S) marked as ***f***, nitrogen (N) atom (***g***), and oxygen(O) atoms (***h*** and ***i***), and determine their **hybridization**. (2 points)

Atom	Hybridization
Sulfur (<i>f</i>)	
Nitrogen (<i>g</i>)	
Oxygen (<i>h</i>)	
Oxygen (<i>i</i>)	

iii. **Describe** all the bonds formed by the carbon atoms ***a*** and ***b***, using sigma and pi designation as appropriate. (2 points)

Atom	Bonds formed (use sigma and pi as appropriate)
Carbon (<i>a</i>)	
Carbon (<i>b</i>)	

iv. How many total sp^3 -s sigma bonds are in the molecule? (1 point)

v. Determine **electronic** and **molecular geometry** at carbon (C) labeled ***e*** and sulfur (S) labeled ***f*** centres. (2 points)

Atom	Electronic Geometry	Molecular Geometry
Carbon (<i>e</i>)		
Sulfur (<i>f</i>)		

vi. Circle the value from the choices given below, that describes the **C-S-C bond angle at sulfur** (S) marked ***f***: (0.5 points)

>90° to <109.5° 109.5° >109.5 to <120° 120° >120° to <180°


Question 8 12 points

Using the **valence shell orbitals only** draw a **complete** molecular orbital (MO) diagram (with no s-p mixing) for the molecule **PS**, in the space provided below. **(6 points)**

Assume that 3s and 3p atomic orbitals will form molecular orbitals like those formed by the combination of 2s and 2p atomic orbitals that we learnt in class, and there is no s-p mixing.

Show **energy arrow**, **atomic orbitals** involved, **atom labels**, the **molecular orbitals** that are formed, **filled electrons** in the MO diagram etc. Make sure to label all the atomic and molecular orbitals.

Using the MO diagram above, answer the following questions:

- (i) Which one of the following will have the highest bond order? Circle the correct choice
 A) PS^{1-} B) PS C) PS^{1+}
- (ii) Which one of the following will have the longest bond length? Circle the correct choice
 A) PS^{1-} B) PS C) PS^{1+}
- (iii) Which of the following will be diamagnetic? Circle the correct choice.
 A) PS^{1-} B) PS C) PS^{1+}
- (iv) Draw Lewis dot structure of the molecule with the shortest bond length (based on the MO diagram) from among the three (PS^{1-} , PS, PS^{1+}), and indicate the bond order calculated from the Lewis dot structure. **(3 points)**

Lewis Dot Structure

Bond Order



87363A91-50DD-41D4-B576-B894FDE85C41

midterm-exam-93388

#217 Page 10 of 10

SCRAP PAGE

**CHEMISTRY 110 –Midterm Exam****Thursday November 7, 2024, 6:30 – 8:30 pm**

Examiners: Pallavi Sirjoosingh and Ashok Kakkar

First name (please write as legibly as possible within the boxes)

--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--

Last name

--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--

Student ID number

--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--

This booklet has 8 questions on 10 pages (The last page can be used as a scrap page).

The staple must NOT be removed from the question booklet. Your intact question booklet must be returned at the end of the exam.

A separate booklet contains a periodic table, and datasheet.

Write your answers directly under the questions in the space provided. Do not write on the QR codes or near the top edges (as it may be cut off).

Please write **legibly and dark enough** for the scan to be readable. For students who use English as a second language, dictionaries are permitted. Calculators are allowed.

DO NOT ASK QUESTIONS OF THE INVIGILATORS about interpretations of the exam questions. If you have a concern about a question, please summarize it at the top of this page. All concerns will be evaluated after the exam. GOOD LUCK!

**Question 1 8 points** (*Assume all transitions are possible*)

a. 6 points Calculate the **energy (in kJ/mol)** required to ionize (completely remove) the electron from the **n=3** level of **He⁺ ion** (He: Helium). Show your work including equations. (3 sig. figs.)

b. 2 points

(I) Select the largest (longest) wavelength from the three choices below:

- i. Wavelength of photon to *completely remove* an electron from *1s orbital of Hydrogen atom*
- ii. Wavelength of photon to *completely remove* an electron from *1s orbital of Boron atom (B)*
- iii. Wavelength of photon to *completely remove* an electron from *1s orbital of Boron cation (B⁺)*

i or ii or iii

(II) Select the smallest (shortest) wavelength from the three choices below (*for H atom*):

- i. Wavelength of photon *emitted* for an electron transitioning from *2p to 1s orbital*
- ii. Wavelength of photon *absorbed* for an electron transitioning from *2s to 4p orbital*
- iii. Wavelength of photon *absorbed* for an electron transitioning from *1s to 3p orbital*

i or ii or iii



Question 2 8 points



A Chem110 student plots the graph shown on the left, from data obtained for the kinetic energy (eV) of the ejected electron versus frequency (s^{-1}) of the incident light, for an unknown metal. ($1\text{eV} = 1.6022 \times 10^{-19} \text{ J}$)

Based on the plot, answer the questions **a to c** below:

a. 1 point On the plot above, label the **threshold frequency**.

b. 2 points What is the value of the work potential (in J) of the unknown metal?

c. 3 points What is the *longest* wavelength (in nm) required to eject an electron from this metal's surface? Show your work including equation(s) used. (3 sig figs)

d. 2 points In the space provided below, plot the following

(Assume frequency is *above* threshold frequency):

- Kinetic energy* of an ejected electron versus the *intensity* of the incident light (left)
- Number of electrons* ejected versus the *frequency* of the incident light (right)



**Question 3** **6 points****a. 2 points**

In the ground state of Ge (Germanium; $Z = 32$) atom, **name** the following orbitals:

i. Orbital containing the electron that requires the maximum energy to be removed.

ii. Orbital containing the electron that experiences the minimum effective nuclear charge.

b. 2 points

An unknown anion (A^-) has the following *excited* electronic configuration: $[Ar] 4s^2 3d^3 4p^1$

(Ar: Argon; $Z = 18$)

Give the **condensed ground state electronic configuration** of the atom, **A** and the cation, **A^{2+}** .
Show your work. (Assume all electronic transitions are possible)

Electronic Configuration of A:

Electronic Configuration of A^{2+} :

c. 2 points

In a single atom, determine how many electrons can have the following set of quantum numbers?

(i) $n = 5; l = 3$

In a single atom, determine how many electrons can have the following set of quantum numbers?

(i) $n = 3; l = 2; m_s = +1/2$

**Question 4** **8 points**

Rank the following from 1 to 4 in **increasing** order. Provide the ranking in the boxes provided.

Note: Atomic number (Z) are provided for relevant atoms

a) Increasing (lowest to highest) order of *ionization energy*: (1= lowest; 4 = highest)

Mg

Mg: Z = 12

Rb

Rb: Z = 37

Cs

Cs: Z = 55

K

K : Z = 19

b) Increasing (lowest to highest) order of *bond angle*: (1= lowest; 4 = highest)

H₂OCCl₄XeF₄

HCN

c) Increasing (smallest to largest) order of *ionic size* (1= smallest; 4 = largest)

S²⁻

S: Z = 16

O²⁻

O: Z = 8

Be²⁺

Be: Z = 4

Na⁺

Na: Z = 11

d) Increasing (lowest to highest) order of *bond energy* (1= lowest; 4 = highest)

P-F

P-Cl

P-I

P-Br

**Question 5** **12 points**

a. 9 points For the molecules HClO_4 and TeF_6 , draw the most stable *Lewis structure* showing appropriate shape and *dashed/wedged bonds*, based on VSEPR. Determine the **molecular geometry** around the central atom. For HClO_4 , label the *most polar bond*.



b. 3 points (Note: Lone pairs are not shown in structures below)

(i) For the three *resonance* structures (1, 2 and 3) shown below, answer the following question:



(I) Circle the *most contributing* resonance structure.

Structure 1

Structure 2

Structure 3

(ii) For the three structures given (4, 5, and 6) below, answer the following questions:



(I) Circle all the structures that represent *resonance structures* of each other:

Structure 4

Structure 5

Structure 6

(II) Among the resonance structures, circle the *most contributing* resonance structure.

Structure 4

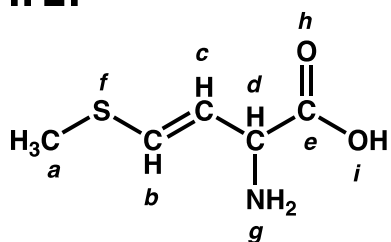
Structure 5

Structure 6

**Question 6** **6 points**

Using the data given below, calculate the **lattice energy** for SrO(s).

	ΔH° (kJ/mol)
$\text{Sr}(s) \longrightarrow \text{Sr}(g)$	+180
$\text{Sr}(g) \longrightarrow \text{Sr}^{2+}(g) + 2e^-$	+1468
$\text{O}_2(g) \longrightarrow 2\text{O}(g)$	+500
$\text{O}(g) + 2e^- \longrightarrow \text{O}^{2-}(g)$	+600
$\text{Sr}(s) + \frac{1}{2}\text{O}_2(g) \longrightarrow \text{SrO}(s)$	-550

**Question 7 10 points**

Consider the molecule given on the left, and answer questions (i) to (vi) using chemical bonding theories that we have learnt in class (valence bond theory, hybridization).

Note that the lone pairs are not shown in the structure:

i. Consider the carbon atoms marked in as ***a***, ***b***, ***c***, ***d***, and ***e*** in the molecule, and determine **hybridization** at each of the carbon centers. (2.5 points)

Atom	Hybridization
Carbon (<i>a</i>)	
Carbon (<i>b</i>)	
Carbon (<i>c</i>)	
Carbon (<i>d</i>)	
Carbon (<i>e</i>)	

ii. Now, consider the sulfur atom (S) marked as ***f***, nitrogen (N) atom (***g***), and oxygen(O) atoms (***h*** and ***i***), and determine their **hybridization**. (2 points)

Atom	Hybridization
Sulfur (<i>f</i>)	
Nitrogen (<i>g</i>)	
Oxygen (<i>h</i>)	
Oxygen (<i>i</i>)	

iii. **Describe** all the bonds formed by the carbon atoms ***a*** and ***b***, using sigma and pi designation as appropriate. (2 points)

Atom	Bonds formed (use sigma and pi as appropriate)
Carbon (<i>a</i>)	
Carbon (<i>b</i>)	

iv. How many total sp^3 -s sigma bonds are in the molecule? (1 point)

v. Determine **electronic** and **molecular geometry** at carbon (C) labeled ***e*** and sulfur (S) labeled ***f*** centres. (2 points)

Atom	Electronic Geometry	Molecular Geometry
Carbon (<i>e</i>)		
Sulfur (<i>f</i>)		

vi. Circle the value from the choices given below, that describes the **C-S-C bond angle at sulfur** (S) marked ***f***: (0.5 points)

$>90^\circ$ to $<109.5^\circ$ 109.5° >109.5 to $<120^\circ$ 120° $>120^\circ$ to $<180^\circ$


Question 8 12 points

Using the **valence shell orbitals only** draw a **complete** molecular orbital (MO) diagram (with no s-p mixing) for the molecule **PS**, in the space provided below. **(6 points)**

Assume that 3s and 3p atomic orbitals will form molecular orbitals like those formed by the combination of 2s and 2p atomic orbitals that we learnt in class, and there is no s-p mixing.

Show **energy arrow**, **atomic orbitals** involved, **atom labels**, the **molecular orbitals** that are formed, **filled electrons** in the MO diagram etc. Make sure to label all the atomic and molecular orbitals.

Using the MO diagram above, answer the following questions:

- (i) Which one of the following will have the highest bond order? Circle the correct choice
 A) PS^{1-} B) PS C) PS^{1+}
- (ii) Which one of the following will have the longest bond length? Circle the correct choice
 A) PS^{1-} B) PS C) PS^{1+}
- (iii) Which of the following will be diamagnetic? Circle the correct choice.
 A) PS^{1-} B) PS C) PS^{1+}
- (iv) Draw Lewis dot structure of the molecule with the shortest bond length (based on the MO diagram) from among the three (PS^{1-} , PS, PS^{1+}), and indicate the bond order calculated from the Lewis dot structure. **(3 points)**

Lewis Dot Structure

Bond Order



E94D591D-6F59-48F8-AC57-69F7FF98DE9E

midterm-exam-93388

#218 Page 10 of 10

SCRAP PAGE

**CHEMISTRY 110 –Midterm Exam****Thursday November 7, 2024, 6:30 – 8:30 pm****Examiners: Pallavi Sirjoosingh and Ashok Kakkar**

First name (please write as legibly as possible within the boxes)

--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--

Last name

--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--

Student ID number

--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--

This booklet has 8 questions on 10 pages (The last page can be used as a scrap page).

The staple must NOT be removed from the question booklet. Your intact question booklet must be returned at the end of the exam.

A separate booklet contains a periodic table, and datasheet.

Write your answers directly under the questions in the space provided. Do not write on the QR codes or near the top edges (as it may be cut off).

Please write **legibly and dark enough** for the scan to be readable. For students who use English as a second language, dictionaries are permitted. Calculators are allowed.

DO NOT ASK QUESTIONS OF THE INVIGILATORS about interpretations of the exam questions. If you have a concern about a question, please summarize it at the top of this page. All concerns will be evaluated after the exam. GOOD LUCK!

**Question 1 8 points** (*Assume all transitions are possible*)

a. 6 points Calculate the **energy (in kJ/mol)** required to ionize (completely remove) the electron from the **$n=3$** level of **He^+ ion** (He: Helium). Show your work including equations. (3 sig. figs.)

b. 2 points

(I) Select the largest (longest) wavelength from the three choices below:

- i. Wavelength of photon to *completely remove* an electron from *1s orbital of Hydrogen atom*
- ii. Wavelength of photon to *completely remove* an electron from *1s orbital of Boron atom (B)*
- iii. Wavelength of photon to *completely remove* an electron from *1s orbital of Boron cation (B^+)*

i or ii or iii

(II) Select the smallest (shortest) wavelength from the three choices below (*for H atom*):

- i. Wavelength of photon *emitted* for an electron transitioning from *2p to 1s orbital*
- ii. Wavelength of photon *absorbed* for an electron transitioning from *2s to 4p orbital*
- iii. Wavelength of photon *absorbed* for an electron transitioning from *1s to 3p orbital*

i or ii or iii



Question 2 8 points



A Chem110 student plots the graph shown on the left, from data obtained for the kinetic energy (eV) of the ejected electron versus frequency (s^{-1}) of the incident light, for an unknown metal. ($1\text{eV} = 1.6022 \times 10^{-19} \text{ J}$)

Based on the plot, answer the questions **a to c** below:

a. 1 point On the plot above, label the **threshold frequency**.

b. 2 points What is the value of the work potential (in J) of the unknown metal?

c. 3 points What is the *longest* wavelength (in nm) required to eject an electron from this metal's surface? Show your work including equation(s) used. (3 sig figs)

d. 2 points In the space provided below, plot the following

(Assume frequency is *above* threshold frequency):

- Kinetic energy* of an ejected electron versus the *intensity* of the incident light (left)
- Number of electrons* ejected versus the *frequency* of the incident light (right)



**Question 3** **6 points****a. 2 points**

In the ground state of Ge (Germanium; $Z = 32$) atom, **name** the following orbitals:

i. Orbital containing the electron that requires the maximum energy to be removed.

ii. Orbital containing the electron that experiences the minimum effective nuclear charge.

b. 2 points

An unknown anion (A^-) has the following *excited* electronic configuration: $[Ar] 4s^2 3d^3 4p^1$

(Ar: Argon; $Z = 18$)

Give the **condensed ground state electronic configuration** of the atom, **A** and the cation, **A^{2+}** .
Show your work. (Assume all electronic transitions are possible)

Electronic Configuration of A:

Electronic Configuration of A^{2+} :

c. 2 points

In a single atom, determine how many electrons can have the following set of quantum numbers?

(i) $n = 5; l = 3$

In a single atom, determine how many electrons can have the following set of quantum numbers?

(i) $n = 3; l = 2; m_s = +1/2$

**Question 4** **8 points**

Rank the following from 1 to 4 in **increasing** order. Provide the ranking in the boxes provided.

Note: Atomic number (Z) are provided for relevant atoms

a) Increasing (lowest to highest) order of *ionization energy*: (1= lowest; 4 = highest)

Mg

Mg: Z = 12

Rb

Rb: Z = 37

Cs

Cs: Z = 55

K

K : Z = 19

b) Increasing (lowest to highest) order of *bond angle*: (1= lowest; 4 = highest)

H₂OCCl₄XeF₄

HCN

c) Increasing (smallest to largest) order of *ionic size* (1= smallest; 4 = largest)

S²⁻

S: Z = 16

O²⁻

O: Z = 8

Be²⁺

Be: Z = 4

Na⁺

Na: Z = 11

d) Increasing (lowest to highest) order of *bond energy* (1= lowest; 4 = highest)

P-F

P-Cl

P-I

P-Br

**Question 5** **12 points**

a. 9 points For the molecules HClO_4 and TeF_6 , draw the most stable *Lewis structure* showing appropriate shape and *dashed/wedged bonds*, based on VSEPR. Determine the **molecular geometry** around the central atom. For HClO_4 , label the *most polar bond*.



b. 3 points (Note: Lone pairs are not shown in structures below)

(i) For the three *resonance* structures (1, 2 and 3) shown below, answer the following question:



(I) Circle the *most contributing* resonance structure.

Structure 1

Structure 2

Structure 3

(ii) For the three structures given (4, 5, and 6) below, answer the following questions:



(I) Circle all the structures that represent *resonance structures* of each other:

Structure 4

Structure 5

Structure 6

(II) Among the resonance structures, circle the *most contributing* resonance structure.

Structure 4

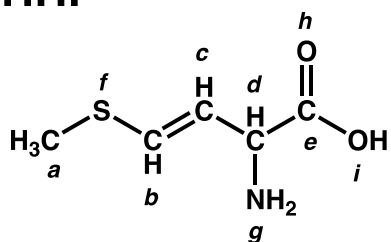
Structure 5

Structure 6

**Question 6** **6 points**

Using the data given below, calculate the **lattice energy** for SrO(s).

	ΔH° (kJ/mol)
$\text{Sr}(s) \longrightarrow \text{Sr}(g)$	+180
$\text{Sr}(g) \longrightarrow \text{Sr}^{2+}(g) + 2e^-$	+1468
$\text{O}_2(g) \longrightarrow 2\text{O}(g)$	+500
$\text{O}(g) + 2e^- \longrightarrow \text{O}^{2-}(g)$	+600
$\text{Sr}(s) + \frac{1}{2}\text{O}_2(g) \longrightarrow \text{SrO}(s)$	-550

**Question 7 10 points**

Consider the molecule given on the left, and answer questions (i) to (vi) using chemical bonding theories that we have learnt in class (valence bond theory, hybridization).

Note that the lone pairs are not shown in the structure:

i. Consider the carbon atoms marked in as ***a***, ***b***, ***c***, ***d***, and ***e*** in the molecule, and determine **hybridization** at each of the carbon centers. (2.5 points)

Atom	Hybridization
Carbon (<i>a</i>)	
Carbon (<i>b</i>)	
Carbon (<i>c</i>)	
Carbon (<i>d</i>)	
Carbon (<i>e</i>)	

ii. Now, consider the sulfur atom (S) marked as ***f***, nitrogen (N) atom (***g***), and oxygen(O) atoms (***h*** and ***i***), and determine their **hybridization**. (2 points)

Atom	Hybridization
Sulfur (<i>f</i>)	
Nitrogen (<i>g</i>)	
Oxygen (<i>h</i>)	
Oxygen (<i>i</i>)	

iii. **Describe** all the bonds formed by the carbon atoms ***a*** and ***b***, using sigma and pi designation as appropriate. (2 points)

Atom	Bonds formed (use sigma and pi as appropriate)
Carbon (<i>a</i>)	
Carbon (<i>b</i>)	

iv. How many total sp^3 -s sigma bonds are in the molecule? (1 point)

v. Determine **electronic** and **molecular geometry** at carbon (C) labeled ***e*** and sulfur (S) labeled ***f*** centres. (2 points)

Atom	Electronic Geometry	Molecular Geometry
Carbon (<i>e</i>)		
Sulfur (<i>f</i>)		

vi. Circle the value from the choices given below, that describes the **C-S-C bond angle at sulfur** (S) marked ***f***: (0.5 points)

>90° to <109.5°

109.5°

>109.5 to <120°

120°

>120° to <180°

**Question 8 12 points**

Using the **valence shell orbitals only** draw a **complete** molecular orbital (MO) diagram (with no s-p mixing) for the molecule **PS**, in the space provided below. **(6 points)**

Assume that 3s and 3p atomic orbitals will form molecular orbitals like those formed by the combination of 2s and 2p atomic orbitals that we learnt in class, and there is no s-p mixing.

Show **energy arrow**, **atomic orbitals** involved, **atom labels**, the **molecular orbitals** that are formed, **filled electrons** in the MO diagram etc. Make sure to label all the atomic and molecular orbitals.

Using the MO diagram above, answer the following questions:

- (i) Which one of the following will have the highest bond order? Circle the correct choice
A) PS^{1-} B) PS C) PS^{1+}
- (ii) Which one of the following will have the longest bond length? Circle the correct choice
A) PS^{1-} B) PS C) PS^{1+}
- (iii) Which of the following will be diamagnetic? Circle the correct choice.
A) PS^{1-} B) PS C) PS^{1+}
- (iv) Draw Lewis dot structure of the molecule with the shortest bond length (based on the MO diagram) from among the three (PS^{1-} , PS, PS^{1+}), and indicate the bond order calculated from the Lewis dot structure. **(3 points)**

Lewis Dot Structure

Bond Order



A6F883F6-5001-4D9A-8FED-BD3292E60923

midterm-exam-93388

#219 Page 10 of 10

SCRAP PAGE

**CHEMISTRY 110 –Midterm Exam****Thursday November 7, 2024, 6:30 – 8:30 pm**

Examiners: Pallavi Sirjoosingh and Ashok Kakkar

First name (please write as legibly as possible within the boxes)

--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--

Last name

--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--

Student ID number

--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--

This booklet has 8 questions on 10 pages (The last page can be used as a scrap page).

The staple must NOT be removed from the question booklet. Your intact question booklet must be returned at the end of the exam.

A separate booklet contains a periodic table, and datasheet.

Write your answers directly under the questions in the space provided. Do not write on the QR codes or near the top edges (as it may be cut off).

Please write **legibly and dark enough** for the scan to be readable. For students who use English as a second language, dictionaries are permitted. Calculators are allowed.

DO NOT ASK QUESTIONS OF THE INVIGILATORS about interpretations of the exam questions. If you have a concern about a question, please summarize it at the top of this page. All concerns will be evaluated after the exam. GOOD LUCK!

**Question 1 8 points** (*Assume all transitions are possible*)

a. 6 points Calculate the **energy (in kJ/mol)** required to ionize (completely remove) the electron from the **n=3** level of **He⁺ ion** (He: Helium). Show your work including equations. (3 sig. figs.)

b. 2 points

(I) Select the largest (longest) wavelength from the three choices below:

- i. Wavelength of photon to *completely remove* an electron from *1s orbital of Hydrogen atom*
- ii. Wavelength of photon to *completely remove* an electron from *1s orbital of Boron atom (B)*
- iii. Wavelength of photon to *completely remove* an electron from *1s orbital of Boron cation (B⁺)*

i or ii or iii

(II) Select the smallest (shortest) wavelength from the three choices below (*for H atom*):

- i. Wavelength of photon *emitted* for an electron transitioning from *2p to 1s orbital*
- ii. Wavelength of photon *absorbed* for an electron transitioning from *2s to 4p orbital*
- iii. Wavelength of photon *absorbed* for an electron transitioning from *1s to 3p orbital*

i or ii or iii



Question 2 8 points



A Chem110 student plots the graph shown on the left, from data obtained for the kinetic energy (eV) of the ejected electron versus frequency (s^{-1}) of the incident light, for an unknown metal. ($1\text{eV} = 1.6022 \times 10^{-19} \text{ J}$)

Based on the plot, answer the questions **a to c** below:

a. 1 point On the plot above, label the **threshold frequency**.

b. 2 points What is the value of the work potential (in J) of the unknown metal?

c. 3 points What is the *longest* wavelength (in nm) required to eject an electron from this metal's surface? Show your work including equation(s) used. (3 sig figs)

d. 2 points In the space provided below, plot the following

(Assume frequency is *above* threshold frequency):

- Kinetic energy* of an ejected electron versus the *intensity* of the incident light (left)
- Number of electrons* ejected versus the *frequency* of the incident light (right)



**Question 3** **6 points****a. 2 points**

In the ground state of Ge (Germanium; $Z = 32$) atom, **name** the following orbitals:

i. Orbital containing the electron that requires the maximum energy to be removed.

ii. Orbital containing the electron that experiences the minimum effective nuclear charge.

b. 2 points

An unknown anion (A^-) has the following *excited* electronic configuration: $[Ar] 4s^2 3d^3 4p^1$

(Ar: Argon; $Z = 18$)

Give the **condensed ground state electronic configuration** of the atom, **A** and the cation, **A^{2+}** .
Show your work. (Assume all electronic transitions are possible)

Electronic Configuration of A:

Electronic Configuration of A^{2+} :

c. 2 points

In a single atom, determine how many electrons can have the following set of quantum numbers?

(i) $n = 5; l = 3$

In a single atom, determine how many electrons can have the following set of quantum numbers?

(i) $n = 3; l = 2; m_s = +1/2$

**Question 4** **8 points**

Rank the following from 1 to 4 in **increasing** order. Provide the ranking in the boxes provided.

Note: Atomic number (Z) are provided for relevant atoms

a) Increasing (lowest to highest) order of *ionization energy*: (1= lowest; 4 = highest)

Mg

Mg: Z = 12

Rb

Rb: Z = 37

Cs

Cs: Z = 55

K

K : Z = 19

b) Increasing (lowest to highest) order of *bond angle*: (1= lowest; 4 = highest)

H₂OCCl₄XeF₄

HCN

c) Increasing (smallest to largest) order of *ionic size* (1= smallest; 4 = largest)

S²⁻

S: Z = 16

O²⁻

O: Z = 8

Be²⁺

Be: Z = 4

Na⁺

Na: Z = 11

d) Increasing (lowest to highest) order of *bond energy* (1= lowest; 4 = highest)

P-F

P-Cl

P-I

P-Br

**Question 5** **12 points**

a. 9 points For the molecules HClO_4 and TeF_6 , draw the most stable *Lewis structure* showing appropriate shape and *dashed/wedged bonds*, based on VSEPR. Determine the **molecular geometry** around the central atom. For HClO_4 , label the *most polar bond*.



b. 3 points (Note: Lone pairs are not shown in structures below)

(i) For the three *resonance* structures (1, 2 and 3) shown below, answer the following question:



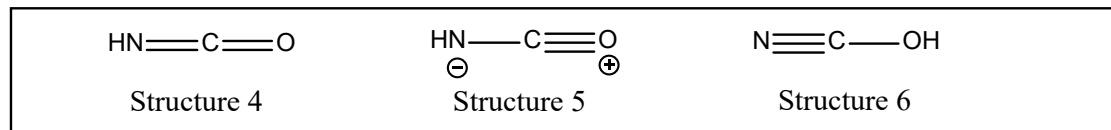
(I) Circle the *most contributing* resonance structure.

Structure 1

Structure 2

Structure 3

(ii) For the three structures given (4, 5, and 6) below, answer the following questions:



(I) Circle all the structures that represent *resonance structures* of each other:

Structure 4

Structure 5

Structure 6

(II) Among the resonance structures, circle the *most contributing* resonance structure.

Structure 4

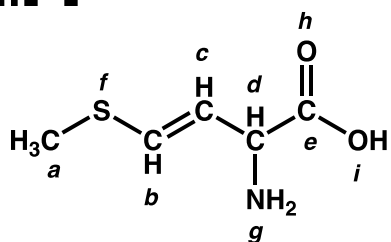
Structure 5

Structure 6

**Question 6** **6 points**

Using the data given below, calculate the **lattice energy** for SrO(s).

	ΔH° (kJ/mol)
$\text{Sr}(s) \longrightarrow \text{Sr}(g)$	+180
$\text{Sr}(g) \longrightarrow \text{Sr}^{2+}(g) + 2e^-$	+1468
$\text{O}_2(g) \longrightarrow 2\text{O}(g)$	+500
$\text{O}(g) + 2e^- \longrightarrow \text{O}^{2-}(g)$	+600
$\text{Sr}(s) + \frac{1}{2}\text{O}_2(g) \longrightarrow \text{SrO}(s)$	-550

**Question 7 10 points**

Consider the molecule given on the left, and answer questions (i) to (vi) using chemical bonding theories that we have learnt in class (valence bond theory, hybridization).

Note that the lone pairs are not shown in the structure:

i. Consider the carbon atoms marked in as ***a***, ***b***, ***c***, ***d***, and ***e*** in the molecule, and determine **hybridization** at each of the carbon centers. (2.5 points)

Atom	Hybridization
Carbon (<i>a</i>)	
Carbon (<i>b</i>)	
Carbon (<i>c</i>)	
Carbon (<i>d</i>)	
Carbon (<i>e</i>)	

ii. Now, consider the sulfur atom (S) marked as ***f***, nitrogen (N) atom (***g***), and oxygen(O) atoms (***h*** and ***i***), and determine their **hybridization**. (2 points)

Atom	Hybridization
Sulfur (<i>f</i>)	
Nitrogen (<i>g</i>)	
Oxygen (<i>h</i>)	
Oxygen (<i>i</i>)	

iii. **Describe** all the bonds formed by the carbon atoms ***a*** and ***b***, using sigma and pi designation as appropriate. (2 points)

Atom	Bonds formed (use sigma and pi as appropriate)
Carbon (<i>a</i>)	
Carbon (<i>b</i>)	

iv. How many total sp^3 -s sigma bonds are in the molecule? (1 point)

v. Determine **electronic** and **molecular geometry** at carbon (C) labeled ***e*** and sulfur (S) labeled ***f*** centres. (2 points)

Atom	Electronic Geometry	Molecular Geometry
Carbon (<i>e</i>)		
Sulfur (<i>f</i>)		

vi. Circle the value from the choices given below, that describes the **C-S-C bond angle at sulfur** (S) marked ***f***: (0.5 points)

>90° to <109.5° 109.5° >109.5 to <120° 120° >120° to <180°

**Question 8 12 points**

Using the **valence shell orbitals only** draw a **complete** molecular orbital (MO) diagram (with no s-p mixing) for the molecule **PS**, in the space provided below. **(6 points)**

Assume that 3s and 3p atomic orbitals will form molecular orbitals like those formed by the combination of 2s and 2p atomic orbitals that we learnt in class, and there is no s-p mixing.

Show **energy arrow**, **atomic orbitals** involved, **atom labels**, the **molecular orbitals** that are formed, **filled electrons** in the MO diagram etc. Make sure to label all the atomic and molecular orbitals.

Using the MO diagram above, answer the following questions:

- (i) Which one of the following will have the highest bond order? Circle the correct choice
A) PS^{1-} B) PS C) PS^{1+}
- (ii) Which one of the following will have the longest bond length? Circle the correct choice
A) PS^{1-} B) PS C) PS^{1+}
- (iii) Which of the following will be diamagnetic? Circle the correct choice.
A) PS^{1-} B) PS C) PS^{1+}
- (iv) Draw Lewis dot structure of the molecule with the shortest bond length (based on the MO diagram) from among the three (PS^{1-} , PS, PS^{1+}), and indicate the bond order calculated from the Lewis dot structure. **(3 points)**

Lewis Dot Structure

Bond Order



F3D3ABC2-5207-4A66-A8CF-649B45F90C4F

midterm-exam-93388

#220 Page 10 of 10

SCRAP PAGE

**CHEMISTRY 110 –Midterm Exam****Thursday November 7, 2024, 6:30 – 8:30 pm****Examiners: Pallavi Sirjoosingh and Ashok Kakkar**

First name (please write as legibly as possible within the boxes)

--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--

Last name

--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--

Student ID number

--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--

This booklet has 8 questions on 10 pages (The last page can be used as a scrap page).

The staple must NOT be removed from the question booklet. Your intact question booklet must be returned at the end of the exam.

A separate booklet contains a periodic table, and datasheet.

Write your answers directly under the questions in the space provided. Do not write on the QR codes or near the top edges (as it may be cut off).

Please write **legibly and dark enough** for the scan to be readable. For students who use English as a second language, dictionaries are permitted. Calculators are allowed.

DO NOT ASK QUESTIONS OF THE INVIGILATORS about interpretations of the exam questions. If you have a concern about a question, please summarize it at the top of this page. All concerns will be evaluated after the exam. GOOD LUCK!

**Question 1 8 points** (*Assume all transitions are possible*)

a. 6 points Calculate the **energy (in kJ/mol)** required to ionize (completely remove) the electron from the **$n=3$** level of **He^+ ion** (He: Helium). Show your work including equations. (3 sig. figs.)

b. 2 points

(I) Select the largest (longest) wavelength from the three choices below:

- i. Wavelength of photon to *completely remove* an electron from *1s orbital of Hydrogen atom*
- ii. Wavelength of photon to *completely remove* an electron from *1s orbital of Boron atom (B)*
- iii. Wavelength of photon to *completely remove* an electron from *1s orbital of Boron cation (B^+)*

i or ii or iii

(II) Select the smallest (shortest) wavelength from the three choices below (*for H atom*):

- i. Wavelength of photon *emitted* for an electron transitioning from *2p to 1s orbital*
- ii. Wavelength of photon *absorbed* for an electron transitioning from *2s to 4p orbital*
- iii. Wavelength of photon *absorbed* for an electron transitioning from *1s to 3p orbital*

i or ii or iii



Question 2 8 points



A Chem110 student plots the graph shown on the left, from data obtained for the kinetic energy (eV) of the ejected electron versus frequency (s^{-1}) of the incident light, for an unknown metal. ($1\text{eV} = 1.6022 \times 10^{-19} \text{ J}$)

Based on the plot, answer the questions **a to c** below:

a. 1 point On the plot above, label the **threshold frequency**.

b. 2 points What is the value of the work potential (in J) of the unknown metal?

c. 3 points What is the *longest* wavelength (in nm) required to eject an electron from this metal's surface? Show your work including equation(s) used. (3 sig figs)

d. 2 points In the space provided below, plot the following

(Assume frequency is *above* threshold frequency):

- i. *Kinetic energy* of an ejected electron versus the *intensity* of the incident light (left)
- ii. *Number of electrons* ejected versus the *frequency* of the incident light (right)



**Question 3** **6 points****a. 2 points**

In the ground state of Ge (Germanium; $Z = 32$) atom, **name** the following orbitals:

i. Orbital containing the electron that requires the maximum energy to be removed.

ii. Orbital containing the electron that experiences the minimum effective nuclear charge.

b. 2 points

An unknown anion (A^-) has the following *excited* electronic configuration: $[Ar] 4s^2 3d^3 4p^1$

(Ar: Argon; $Z = 18$)

Give the **condensed ground state electronic configuration** of the atom, **A** and the cation, **A^{2+}** .
Show your work. (Assume all electronic transitions are possible)

Electronic Configuration of A:

Electronic Configuration of A^{2+} :

c. 2 points

In a single atom, determine how many electrons can have the following set of quantum numbers?

(i) $n = 5; l = 3$

In a single atom, determine how many electrons can have the following set of quantum numbers?

(i) $n = 3; l = 2; m_s = +1/2$

**Question 4** **8 points**

Rank the following from 1 to 4 in **increasing** order. Provide the ranking in the boxes provided.

Note: Atomic number (Z) are provided for relevant atoms

a) Increasing (lowest to highest) order of *ionization energy*: (1= lowest; 4 = highest)

Mg

Mg: Z = 12

Rb

Rb: Z = 37

Cs

Cs: Z = 55

K

K : Z = 19

b) Increasing (lowest to highest) order of *bond angle*: (1= lowest; 4 = highest)

H₂OCCl₄XeF₄

HCN

c) Increasing (smallest to largest) order of *ionic size* (1= smallest; 4 = largest)

S²⁻

S: Z = 16

O²⁻

O: Z = 8

Be²⁺

Be: Z = 4

Na⁺

Na: Z = 11

d) Increasing (lowest to highest) order of *bond energy* (1= lowest; 4 = highest)

P-F

P-Cl

P-I

P-Br

**Question 5** **12 points**

a. 9 points For the molecules HClO_4 and TeF_6 , draw the most stable *Lewis structure* showing appropriate shape and *dashed/wedged bonds*, based on VSEPR. Determine the **molecular geometry** around the central atom. For HClO_4 , label the *most polar bond*.



b. 3 points (Note: Lone pairs are not shown in structures below)

(i) For the three *resonance* structures (1, 2 and 3) shown below, answer the following question:



(I) Circle the *most contributing* resonance structure.

Structure 1

Structure 2

Structure 3

(ii) For the three structures given (4, 5, and 6) below, answer the following questions:



(I) Circle all the structures that represent *resonance structures* of each other:

Structure 4

Structure 5

Structure 6

(II) Among the resonance structures, circle the *most contributing* resonance structure.

Structure 4

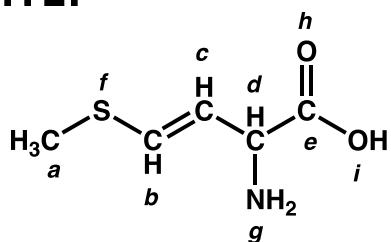
Structure 5

Structure 6

**Question 6** **6 points**

Using the data given below, calculate the **lattice energy** for SrO(s).

	ΔH° (kJ/mol)
$\text{Sr}(s) \longrightarrow \text{Sr}(g)$	+180
$\text{Sr}(g) \longrightarrow \text{Sr}^{2+}(g) + 2e^-$	+1468
$\text{O}_2(g) \longrightarrow 2\text{O}(g)$	+500
$\text{O}(g) + 2e^- \longrightarrow \text{O}^{2-}(g)$	+600
$\text{Sr}(s) + \frac{1}{2}\text{O}_2(g) \longrightarrow \text{SrO}(s)$	-550

**Question 7 10 points**

Consider the molecule given on the left, and answer questions (i) to (vi) using chemical bonding theories that we have learnt in class (valence bond theory, hybridization).

Note that the lone pairs are not shown in the structure:

i. Consider the carbon atoms marked in as ***a***, ***b***, ***c***, ***d***, and ***e*** in the molecule, and determine **hybridization** at each of the carbon centers. (2.5 points)

Atom	Hybridization
Carbon (<i>a</i>)	
Carbon (<i>b</i>)	
Carbon (<i>c</i>)	
Carbon (<i>d</i>)	
Carbon (<i>e</i>)	

ii. Now, consider the sulfur atom (S) marked as ***f***, nitrogen (N) atom (***g***), and oxygen(O) atoms (***h*** and ***i***), and determine their **hybridization**. (2 points)

Atom	Hybridization
Sulfur (<i>f</i>)	
Nitrogen (<i>g</i>)	
Oxygen (<i>h</i>)	
Oxygen (<i>i</i>)	

iii. **Describe** all the bonds formed by the carbon atoms ***a*** and ***b***, using sigma and pi designation as appropriate. (2 points)

Atom	Bonds formed (use sigma and pi as appropriate)
Carbon (<i>a</i>)	
Carbon (<i>b</i>)	

iv. How many total sp^3 -s sigma bonds are in the molecule? (1 point)

v. Determine **electronic** and **molecular geometry** at carbon (C) labeled ***e*** and sulfur (S) labeled ***f*** centres. (2 points)

Atom	Electronic Geometry	Molecular Geometry
Carbon (<i>e</i>)		
Sulfur (<i>f</i>)		

vi. Circle the value from the choices given below, that describes the **C-S-C bond angle at sulfur** (S) marked ***f***: (0.5 points)

>90° to <109.5° 109.5° >109.5 to <120° 120° >120° to <180°


Question 8 12 points

Using the **valence shell orbitals only** draw a **complete** molecular orbital (MO) diagram (with no s-p mixing) for the molecule **PS**, in the space provided below. **(6 points)**

Assume that 3s and 3p atomic orbitals will form molecular orbitals like those formed by the combination of 2s and 2p atomic orbitals that we learnt in class, and there is no s-p mixing.

Show **energy arrow**, **atomic orbitals** involved, **atom labels**, the **molecular orbitals** that are formed, **filled electrons** in the MO diagram etc. Make sure to label all the atomic and molecular orbitals.

Using the MO diagram above, answer the following questions:

- (i) Which one of the following will have the highest bond order? Circle the correct choice
 A) PS^{1-} B) PS C) PS^{1+}
- (ii) Which one of the following will have the longest bond length? Circle the correct choice
 A) PS^{1-} B) PS C) PS^{1+}
- (iii) Which of the following will be diamagnetic? Circle the correct choice.
 A) PS^{1-} B) PS C) PS^{1+}
- (iv) Draw Lewis dot structure of the molecule with the shortest bond length (based on the MO diagram) from among the three (PS^{1-} , PS, PS^{1+}), and indicate the bond order calculated from the Lewis dot structure. **(3 points)**

Lewis Dot Structure

Bond Order



AD640F11-0752-4542-B1C4-6020004B1239

midterm-exam-93388

#221 Page 10 of 10

SCRAP PAGE

**CHEMISTRY 110 –Midterm Exam****Thursday November 7, 2024, 6:30 – 8:30 pm****Examiners: Pallavi Sirjoosingh and Ashok Kakkar**

First name (please write as legibly as possible within the boxes)

--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--

Last name

--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--

Student ID number

--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--

This booklet has 8 questions on 10 pages (The last page can be used as a scrap page).

The staple must NOT be removed from the question booklet. Your intact question booklet must be returned at the end of the exam.

A separate booklet contains a periodic table, and datasheet.

Write your answers directly under the questions in the space provided. Do not write on the QR codes or near the top edges (as it may be cut off).

Please write **legibly and dark enough** for the scan to be readable. For students who use English as a second language, dictionaries are permitted. Calculators are allowed.

DO NOT ASK QUESTIONS OF THE INVIGILATORS about interpretations of the exam questions. If you have a concern about a question, please summarize it at the top of this page. All concerns will be evaluated after the exam. GOOD LUCK!

**Question 1 8 points** (*Assume all transitions are possible*)

a. 6 points Calculate the **energy (in kJ/mol)** required to ionize (completely remove) the electron from the **$n=3$** level of **He^+ ion** (He: Helium). Show your work including equations. (3 sig. figs.)

b. 2 points

(I) Select the largest (longest) wavelength from the three choices below:

- i. Wavelength of photon to *completely remove* an electron from *1s orbital of Hydrogen atom*
- ii. Wavelength of photon to *completely remove* an electron from *1s orbital of Boron atom (B)*
- iii. Wavelength of photon to *completely remove* an electron from *1s orbital of Boron cation (B^+)*

i or ii or iii

(II) Select the smallest (shortest) wavelength from the three choices below (*for H atom*):

- i. Wavelength of photon *emitted* for an electron transitioning from *2p to 1s orbital*
- ii. Wavelength of photon *absorbed* for an electron transitioning from *2s to 4p orbital*
- iii. Wavelength of photon *absorbed* for an electron transitioning from *1s to 3p orbital*

i or ii or iii



Question 2 8 points



A Chem110 student plots the graph shown on the left, from data obtained for the kinetic energy (eV) of the ejected electron versus frequency (s^{-1}) of the incident light, for an unknown metal. ($1\text{eV} = 1.6022 \times 10^{-19} \text{ J}$)

Based on the plot, answer the questions **a to c** below:

a. 1 point On the plot above, label the **threshold frequency**.

b. 2 points What is the value of the work potential (in J) of the unknown metal?

c. 3 points What is the *longest* wavelength (in nm) required to eject an electron from this metal's surface? Show your work including equation(s) used. (3 sig figs)

d. 2 points In the space provided below, plot the following

(Assume frequency is *above* threshold frequency):

- i. *Kinetic energy* of an ejected electron versus the *intensity* of the incident light (left)
- ii. *Number of electrons* ejected versus the *frequency* of the incident light (right)



**Question 3** **6 points****a. 2 points**

In the ground state of Ge (Germanium; $Z = 32$) atom, **name** the following orbitals:

i. Orbital containing the electron that requires the maximum energy to be removed.

ii. Orbital containing the electron that experiences the minimum effective nuclear charge.

b. 2 points

An unknown anion (A^-) has the following *excited* electronic configuration: $[Ar] 4s^2 3d^3 4p^1$

(Ar: Argon; $Z = 18$)

Give the **condensed ground state electronic configuration** of the atom, **A** and the cation, **A^{2+}** .
Show your work. (Assume all electronic transitions are possible)

Electronic Configuration of A:

Electronic Configuration of A^{2+} :

c. 2 points

In a single atom, determine how many electrons can have the following set of quantum numbers?

(i) $n = 5; l = 3$

In a single atom, determine how many electrons can have the following set of quantum numbers?

(i) $n = 3; l = 2; m_s = +1/2$

**Question 4** **8 points**

Rank the following from 1 to 4 in **increasing** order. Provide the ranking in the boxes provided.

Note: Atomic number (Z) are provided for relevant atoms

a) Increasing (lowest to highest) order of *ionization energy*: (1= lowest; 4 = highest)

Mg

Mg: Z = 12

Rb

Rb: Z = 37

Cs

Cs: Z = 55

K

K : Z = 19

b) Increasing (lowest to highest) order of *bond angle*: (1= lowest; 4 = highest)

H₂OCCl₄XeF₄

HCN

c) Increasing (smallest to largest) order of *ionic size* (1= smallest; 4 = largest)

S²⁻

S: Z = 16

O²⁻

O: Z = 8

Be²⁺

Be: Z = 4

Na⁺

Na: Z = 11

d) Increasing (lowest to highest) order of *bond energy* (1= lowest; 4 = highest)

P-F

P-Cl

P-I

P-Br

**Question 5** **12 points**

a. 9 points For the molecules HClO_4 and TeF_6 , draw the most stable *Lewis structure* showing appropriate shape and *dashed/wedged bonds*, based on VSEPR. Determine the **molecular geometry** around the central atom. For HClO_4 , label the *most polar bond*.



b. 3 points (Note: Lone pairs are not shown in structures below)

(i) For the three *resonance* structures (1, 2 and 3) shown below, answer the following question:



(I) Circle the *most contributing* resonance structure.

Structure 1

Structure 2

Structure 3

(ii) For the three structures given (4, 5, and 6) below, answer the following questions:



(I) Circle all the structures that represent *resonance structures* of each other:

Structure 4

Structure 5

Structure 6

(II) Among the resonance structures, circle the *most contributing* resonance structure.

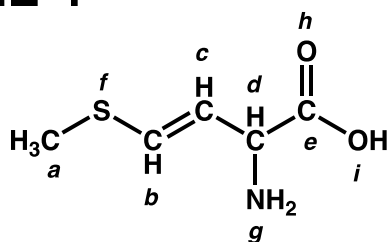
Structure 4

Structure 5

Structure 6

**Question 6** **6 points**Using the data given below, calculate the **lattice energy** for SrO(s).

	ΔH° (kJ/mol)
$\text{Sr}(s) \longrightarrow \text{Sr}(g)$	+180
$\text{Sr}(g) \longrightarrow \text{Sr}^{2+}(g) + 2e^-$	+1468
$\text{O}_2(g) \longrightarrow 2\text{O}(g)$	+500
$\text{O}(g) + 2e^- \longrightarrow \text{O}^{2-}(g)$	+600
$\text{Sr}(s) + \frac{1}{2}\text{O}_2(g) \longrightarrow \text{SrO}(s)$	-550

**Question 7 10 points**

Consider the molecule given on the left, and answer questions (i) to (vi) using chemical bonding theories that we have learnt in class (valence bond theory, hybridization).

Note that the lone pairs are not shown in the structure:

i. Consider the carbon atoms marked in as ***a***, ***b***, ***c***, ***d***, and ***e*** in the molecule, and determine **hybridization** at each of the carbon centers. (2.5 points)

Atom	Hybridization
Carbon (<i>a</i>)	
Carbon (<i>b</i>)	
Carbon (<i>c</i>)	
Carbon (<i>d</i>)	
Carbon (<i>e</i>)	

ii. Now, consider the sulfur atom (S) marked as ***f***, nitrogen (N) atom (***g***), and oxygen(O) atoms (***h*** and ***i***), and determine their **hybridization**. (2 points)

Atom	Hybridization
Sulfur (<i>f</i>)	
Nitrogen (<i>g</i>)	
Oxygen (<i>h</i>)	
Oxygen (<i>i</i>)	

iii. **Describe** all the bonds formed by the carbon atoms ***a*** and ***b***, using sigma and pi designation as appropriate. (2 points)

Atom	Bonds formed (use sigma and pi as appropriate)
Carbon (<i>a</i>)	
Carbon (<i>b</i>)	

iv. How many total sp^3 -s sigma bonds are in the molecule? (1 point)

v. Determine **electronic** and **molecular geometry** at carbon (C) labeled ***e*** and sulfur (S) labeled ***f*** centres. (2 points)

Atom	Electronic Geometry	Molecular Geometry
Carbon (<i>e</i>)		
Sulfur (<i>f</i>)		

vi. Circle the value from the choices given below, that describes the **C-S-C bond angle at sulfur** (S) marked ***f***: (0.5 points)

>90° to <109.5° 109.5° >109.5 to <120° 120° >120° to <180°


Question 8 12 points

Using the **valence shell orbitals only** draw a **complete** molecular orbital (MO) diagram (with no s-p mixing) for the molecule **PS**, in the space provided below. **(6 points)**

Assume that 3s and 3p atomic orbitals will form molecular orbitals like those formed by the combination of 2s and 2p atomic orbitals that we learnt in class, and there is no s-p mixing.

Show **energy arrow**, **atomic orbitals** involved, **atom labels**, the **molecular orbitals** that are formed, **filled electrons** in the MO diagram etc. Make sure to label all the atomic and molecular orbitals.

Using the MO diagram above, answer the following questions:

- (i) Which one of the following will have the highest bond order? Circle the correct choice
 A) PS^{1-} B) PS C) PS^{1+}
- (ii) Which one of the following will have the longest bond length? Circle the correct choice
 A) PS^{1-} B) PS C) PS^{1+}
- (iii) Which of the following will be diamagnetic? Circle the correct choice.
 A) PS^{1-} B) PS C) PS^{1+}
- (iv) Draw Lewis dot structure of the molecule with the shortest bond length (based on the MO diagram) from among the three (PS^{1-} , PS, PS^{1+}), and indicate the bond order calculated from the Lewis dot structure. **(3 points)**

Lewis Dot Structure

Bond Order



5937D031-A19C-40D2-AA5E-CEEB27108ECD

midterm-exam-93388

#222 Page 10 of 10

SCRAP PAGE

**CHEMISTRY 110 –Midterm Exam****Thursday November 7, 2024, 6:30 – 8:30 pm****Examiners: Pallavi Sirjoosingh and Ashok Kakkar**

First name (please write as legibly as possible within the boxes)

--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--

Last name

--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--

Student ID number

--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--

This booklet has 8 questions on 10 pages (The last page can be used as a scrap page).

The staple must NOT be removed from the question booklet. Your intact question booklet must be returned at the end of the exam.

A separate booklet contains a periodic table, and datasheet.

Write your answers directly under the questions in the space provided. Do not write on the QR codes or near the top edges (as it may be cut off).

Please write **legibly and dark enough** for the scan to be readable. For students who use English as a second language, dictionaries are permitted. Calculators are allowed.

DO NOT ASK QUESTIONS OF THE INVIGILATORS about interpretations of the exam questions. If you have a concern about a question, please summarize it at the top of this page. All concerns will be evaluated after the exam. GOOD LUCK!

**Question 1 8 points** (*Assume all transitions are possible*)

a. 6 points Calculate the **energy (in kJ/mol)** required to ionize (completely remove) the electron from the **n=3** level of **He⁺ ion** (He: Helium). Show your work including equations. (3 sig. figs.)

b. 2 points

(I) Select the largest (longest) wavelength from the three choices below:

- i. Wavelength of photon to *completely remove* an electron from *1s orbital of Hydrogen atom*
- ii. Wavelength of photon to *completely remove* an electron from *1s orbital of Boron atom (B)*
- iii. Wavelength of photon to *completely remove* an electron from *1s orbital of Boron cation (B⁺)*

i or ii or iii

(II) Select the smallest (shortest) wavelength from the three choices below (*for H atom*):

- i. Wavelength of photon *emitted* for an electron transitioning from *2p to 1s orbital*
- ii. Wavelength of photon *absorbed* for an electron transitioning from *2s to 4p orbital*
- iii. Wavelength of photon *absorbed* for an electron transitioning from *1s to 3p orbital*

i or ii or iii



Question 2 8 points



A Chem110 student plots the graph shown on the left, from data obtained for the kinetic energy (eV) of the ejected electron versus frequency (s^{-1}) of the incident light, for an unknown metal. ($1\text{eV} = 1.6022 \times 10^{-19} \text{ J}$)

Based on the plot, answer the questions **a to c** below:

a. 1 point On the plot above, label the **threshold frequency**.

b. 2 points What is the value of the work potential (in J) of the unknown metal?

c. 3 points What is the *longest* wavelength (in nm) required to eject an electron from this metal's surface? Show your work including equation(s) used. (3 sig figs)

d. 2 points In the space provided below, plot the following

(Assume frequency is *above* threshold frequency):

- Kinetic energy* of an ejected electron versus the *intensity* of the incident light (left)
- Number of electrons* ejected versus the *frequency* of the incident light (right)



**Question 3** **6 points****a. 2 points**

In the ground state of Ge (Germanium; $Z = 32$) atom, **name** the following orbitals:

i. Orbital containing the electron that requires the maximum energy to be removed.

ii. Orbital containing the electron that experiences the minimum effective nuclear charge.

b. 2 points

An unknown anion (A^-) has the following *excited* electronic configuration: $[Ar] 4s^2 3d^3 4p^1$

(Ar: Argon; $Z = 18$)

Give the **condensed ground state electronic configuration** of the atom, **A** and the cation, **A^{2+}** .
Show your work. (Assume all electronic transitions are possible)

Electronic Configuration of A:

Electronic Configuration of A^{2+} :

c. 2 points

In a single atom, determine how many electrons can have the following set of quantum numbers?

(i) $n = 5; l = 3$

In a single atom, determine how many electrons can have the following set of quantum numbers?

(i) $n = 3; l = 2; m_s = +1/2$

**Question 4** **8 points**

Rank the following from 1 to 4 in **increasing** order. Provide the ranking in the boxes provided.

Note: Atomic number (Z) are provided for relevant atoms

a) Increasing (lowest to highest) order of *ionization energy*: (1= lowest; 4 = highest)

Mg

Mg: Z = 12

Rb

Rb: Z = 37

Cs

Cs: Z = 55

K

K : Z = 19

b) Increasing (lowest to highest) order of *bond angle*: (1= lowest; 4 = highest)

H₂OCCl₄XeF₄

HCN

c) Increasing (smallest to largest) order of *ionic size* (1= smallest; 4 = largest)

S²⁻

S: Z = 16

O²⁻

O: Z = 8

Be²⁺

Be: Z = 4

Na⁺

Na: Z = 11

d) Increasing (lowest to highest) order of *bond energy* (1= lowest; 4 = highest)

P-F

P-Cl

P-I

P-Br

**Question 5** **12 points**

a. 9 points For the molecules HClO_4 and TeF_6 , draw the most stable *Lewis structure* showing appropriate shape and *dashed/wedged bonds*, based on VSEPR. Determine the **molecular geometry** around the central atom. For HClO_4 , label the *most polar bond*.



b. 3 points (Note: Lone pairs are not shown in structures below)

(i) For the three *resonance* structures (1, 2 and 3) shown below, answer the following question:



(I) Circle the *most contributing* resonance structure.

Structure 1

Structure 2

Structure 3

(ii) For the three structures given (4, 5, and 6) below, answer the following questions:



(I) Circle all the structures that represent *resonance structures* of each other:

Structure 4

Structure 5

Structure 6

(II) Among the resonance structures, circle the *most contributing* resonance structure.

Structure 4

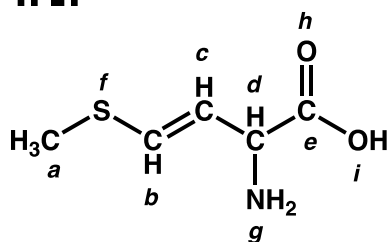
Structure 5

Structure 6

**Question 6** **6 points**

Using the data given below, calculate the **lattice energy** for SrO(s).

	ΔH° (kJ/mol)
$\text{Sr}(s) \longrightarrow \text{Sr}(g)$	+180
$\text{Sr}(g) \longrightarrow \text{Sr}^{2+}(g) + 2e^-$	+1468
$\text{O}_2(g) \longrightarrow 2\text{O}(g)$	+500
$\text{O}(g) + 2e^- \longrightarrow \text{O}^{2-}(g)$	+600
$\text{Sr}(s) + \frac{1}{2}\text{O}_2(g) \longrightarrow \text{SrO}(s)$	-550

**Question 7 10 points**

Consider the molecule given on the left, and answer questions (i) to (vi) using chemical bonding theories that we have learnt in class (valence bond theory, hybridization).

Note that the lone pairs are not shown in the structure:

i. Consider the carbon atoms marked in as ***a***, ***b***, ***c***, ***d***, and ***e*** in the molecule, and determine **hybridization** at each of the carbon centers. (2.5 points)

Atom	Hybridization
Carbon (<i>a</i>)	
Carbon (<i>b</i>)	
Carbon (<i>c</i>)	
Carbon (<i>d</i>)	
Carbon (<i>e</i>)	

ii. Now, consider the sulfur atom (S) marked as ***f***, nitrogen (N) atom (***g***), and oxygen(O) atoms (***h*** and ***i***), and determine their **hybridization**. (2 points)

Atom	Hybridization
Sulfur (<i>f</i>)	
Nitrogen (<i>g</i>)	
Oxygen (<i>h</i>)	
Oxygen (<i>i</i>)	

iii. **Describe** all the bonds formed by the carbon atoms ***a*** and ***b***, using sigma and pi designation as appropriate. (2 points)

Atom	Bonds formed (use sigma and pi as appropriate)
Carbon (<i>a</i>)	
Carbon (<i>b</i>)	

iv. How many total sp^3 -s sigma bonds are in the molecule? (1 point)

v. Determine **electronic** and **molecular geometry** at carbon (C) labeled ***e*** and sulfur (S) labeled ***f*** centres. (2 points)

Atom	Electronic Geometry	Molecular Geometry
Carbon (<i>e</i>)		
Sulfur (<i>f</i>)		

vi. Circle the value from the choices given below, that describes the **C-S-C bond angle at sulfur** (S) marked ***f***: (0.5 points)

>90° to <109.5° 109.5° >109.5 to <120° 120° >120° to <180°


Question 8 12 points

Using the **valence shell orbitals only** draw a **complete** molecular orbital (MO) diagram (with no s-p mixing) for the molecule **PS**, in the space provided below. **(6 points)**

Assume that 3s and 3p atomic orbitals will form molecular orbitals like those formed by the combination of 2s and 2p atomic orbitals that we learnt in class, and there is no s-p mixing.

Show **energy arrow**, **atomic orbitals** involved, **atom labels**, the **molecular orbitals** that are formed, **filled electrons** in the MO diagram etc. Make sure to label all the atomic and molecular orbitals.

Using the MO diagram above, answer the following questions:

- (i) Which one of the following will have the highest bond order? Circle the correct choice
 A) PS^{1-} B) PS C) PS^{1+}
- (ii) Which one of the following will have the longest bond length? Circle the correct choice
 A) PS^{1-} B) PS C) PS^{1+}
- (iii) Which of the following will be diamagnetic? Circle the correct choice.
 A) PS^{1-} B) PS C) PS^{1+}
- (iv) Draw Lewis dot structure of the molecule with the shortest bond length (based on the MO diagram) from among the three (PS^{1-} , PS, PS^{1+}), and indicate the bond order calculated from the Lewis dot structure. **(3 points)**

Lewis Dot Structure

Bond Order



335B0263-3585-4284-8BD4-34496464E9D6

midterm-exam-93388

#223 Page 10 of 10

SCRAP PAGE

**CHEMISTRY 110 –Midterm Exam****Thursday November 7, 2024, 6:30 – 8:30 pm**

Examiners: Pallavi Sirjoosingh and Ashok Kakkar

First name (please write as legibly as possible within the boxes)

--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--

Last name

--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--

Student ID number

--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--

This booklet has 8 questions on 10 pages (The last page can be used as a scrap page).

The staple must NOT be removed from the question booklet. Your intact question booklet must be returned at the end of the exam.

A separate booklet contains a periodic table, and datasheet.

Write your answers directly under the questions in the space provided. Do not write on the QR codes or near the top edges (as it may be cut off).

Please write **legibly and dark enough** for the scan to be readable. For students who use English as a second language, dictionaries are permitted. Calculators are allowed.

DO NOT ASK QUESTIONS OF THE INVIGILATORS about interpretations of the exam questions. If you have a concern about a question, please summarize it at the top of this page. All concerns will be evaluated after the exam. GOOD LUCK!

**Question 1 8 points** (*Assume all transitions are possible*)

a. 6 points Calculate the **energy (in kJ/mol)** required to ionize (completely remove) the electron from the **n=3** level of **He⁺ ion** (He: Helium). Show your work including equations. (3 sig. figs.)

b. 2 points

(I) Select the largest (longest) wavelength from the three choices below:

- i. Wavelength of photon to *completely remove* an electron from *1s orbital of Hydrogen atom*
- ii. Wavelength of photon to *completely remove* an electron from *1s orbital of Boron atom (B)*
- iii. Wavelength of photon to *completely remove* an electron from *1s orbital of Boron cation (B⁺)*

i or ii or iii

(II) Select the smallest (shortest) wavelength from the three choices below (*for H atom*):

- i. Wavelength of photon *emitted* for an electron transitioning from *2p to 1s orbital*
- ii. Wavelength of photon *absorbed* for an electron transitioning from *2s to 4p orbital*
- iii. Wavelength of photon *absorbed* for an electron transitioning from *1s to 3p orbital*

i or ii or iii



Question 2 8 points



A Chem110 student plots the graph shown on the left, from data obtained for the kinetic energy (eV) of the ejected electron versus frequency (s^{-1}) of the incident light, for an unknown metal. ($1\text{eV} = 1.6022 \times 10^{-19} \text{ J}$)

Based on the plot, answer the questions **a to c** below:

a. 1 point On the plot above, label the **threshold frequency**.

b. 2 points What is the value of the work potential (in J) of the unknown metal?

c. 3 points What is the *longest* wavelength (in nm) required to eject an electron from this metal's surface? Show your work including equation(s) used. (3 sig figs)

d. 2 points In the space provided below, plot the following

(Assume frequency is *above* threshold frequency):

- Kinetic energy* of an ejected electron versus the *intensity* of the incident light (left)
- Number of electrons* ejected versus the *frequency* of the incident light (right)



**Question 3** **6 points****a. 2 points**

In the ground state of Ge (Germanium; $Z = 32$) atom, **name** the following orbitals:

i. Orbital containing the electron that requires the maximum energy to be removed.

ii. Orbital containing the electron that experiences the minimum effective nuclear charge.

b. 2 points

An unknown anion (A^-) has the following *excited* electronic configuration: $[Ar] 4s^2 3d^3 4p^1$

(Ar: Argon; $Z = 18$)

Give the **condensed ground state electronic configuration** of the atom, **A** and the cation, **A^{2+}** .
Show your work. (Assume all electronic transitions are possible)

Electronic Configuration of A:

Electronic Configuration of A^{2+} :

c. 2 points

In a single atom, determine how many electrons can have the following set of quantum numbers?

(i) $n = 5; l = 3$

In a single atom, determine how many electrons can have the following set of quantum numbers?

(i) $n = 3; l = 2; m_s = +1/2$

**Question 4** **8 points**

Rank the following from 1 to 4 in **increasing** order. Provide the ranking in the boxes provided.

Note: Atomic number (Z) are provided for relevant atoms

a) Increasing (lowest to highest) order of *ionization energy*: (1= lowest; 4 = highest)

Mg

Mg: Z = 12

Rb

Rb: Z = 37

Cs

Cs: Z = 55

K

K : Z = 19

b) Increasing (lowest to highest) order of *bond angle*: (1= lowest; 4 = highest)

H₂OCCl₄XeF₄

HCN

c) Increasing (smallest to largest) order of *ionic size* (1= smallest; 4 = largest)

S²⁻

S: Z = 16

O²⁻

O: Z = 8

Be²⁺

Be: Z = 4

Na⁺

Na: Z = 11

d) Increasing (lowest to highest) order of *bond energy* (1= lowest; 4 = highest)

P-F

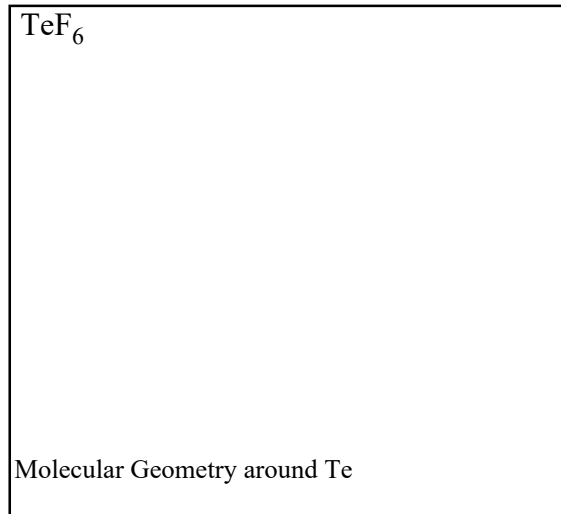
P-Cl

P-I

P-Br

**Question 5** **12 points**

a. 9 points For the molecules HClO_4 and TeF_6 , draw the most stable *Lewis structure* showing appropriate shape and *dashed/wedged bonds*, based on VSEPR. Determine the **molecular geometry** around the central atom. For HClO_4 , label the *most polar bond*.



b. 3 points (Note: Lone pairs are not shown in structures below)

(i) For the three *resonance* structures (1, 2 and 3) shown below, answer the following question:



(I) Circle the *most contributing* resonance structure.

Structure 1

Structure 2

Structure 3

(ii) For the three structures given (4, 5, and 6) below, answer the following questions:



(I) Circle all the structures that represent *resonance structures* of each other:

Structure 4

Structure 5

Structure 6

(II) Among the resonance structures, circle the *most contributing* resonance structure.

Structure 4

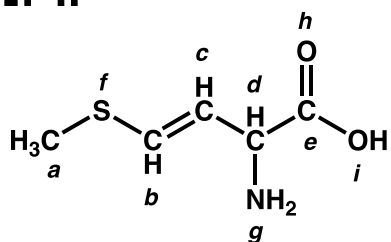
Structure 5

Structure 6

**Question 6** **6 points**

Using the data given below, calculate the **lattice energy** for SrO(s).

	ΔH° (kJ/mol)
$\text{Sr}(s) \longrightarrow \text{Sr}(g)$	+180
$\text{Sr}(g) \longrightarrow \text{Sr}^{2+}(g) + 2e^-$	+1468
$\text{O}_2(g) \longrightarrow 2\text{O}(g)$	+500
$\text{O}(g) + 2e^- \longrightarrow \text{O}^{2-}(g)$	+600
$\text{Sr}(s) + \frac{1}{2}\text{O}_2(g) \longrightarrow \text{SrO}(s)$	-550

**Question 7 10 points**

Consider the molecule given on the left, and answer questions (i) to (vi) using chemical bonding theories that we have learnt in class (valence bond theory, hybridization).

Note that the lone pairs are not shown in the structure:

i. Consider the carbon atoms marked in as ***a***, ***b***, ***c***, ***d***, and ***e*** in the molecule, and determine **hybridization** at each of the carbon centers. (2.5 points)

Atom	Hybridization
Carbon (<i>a</i>)	
Carbon (<i>b</i>)	
Carbon (<i>c</i>)	
Carbon (<i>d</i>)	
Carbon (<i>e</i>)	

ii. Now, consider the sulfur atom (S) marked as ***f***, nitrogen (N) atom (***g***), and oxygen(O) atoms (***h*** and ***i***), and determine their **hybridization**. (2 points)

Atom	Hybridization
Sulfur (<i>f</i>)	
Nitrogen (<i>g</i>)	
Oxygen (<i>h</i>)	
Oxygen (<i>i</i>)	

iii. **Describe** all the bonds formed by the carbon atoms ***a*** and ***b***, using sigma and pi designation as appropriate. (2 points)

Atom	Bonds formed (use sigma and pi as appropriate)
Carbon (<i>a</i>)	
Carbon (<i>b</i>)	

iv. How many total sp^3 -s sigma bonds are in the molecule? (1 point)

v. Determine **electronic** and **molecular geometry** at carbon (C) labeled ***e*** and sulfur (S) labeled ***f*** centres. (2 points)

Atom	Electronic Geometry	Molecular Geometry
Carbon (<i>e</i>)		
Sulfur (<i>f</i>)		

vi. Circle the value from the choices given below, that describes the **C-S-C bond angle at sulfur** (S) marked ***f***: (0.5 points)

>90° to <109.5° 109.5° >109.5 to <120° 120° >120° to <180°

**Question 8 12 points**

Using the **valence shell orbitals only** draw a **complete** molecular orbital (MO) diagram (with no s-p mixing) for the molecule **PS**, in the space provided below. **(6 points)**

Assume that 3s and 3p atomic orbitals will form molecular orbitals like those formed by the combination of 2s and 2p atomic orbitals that we learnt in class, and there is no s-p mixing.

Show **energy arrow**, **atomic orbitals** involved, **atom labels**, the **molecular orbitals** that are formed, **filled electrons** in the MO diagram etc. Make sure to label all the atomic and molecular orbitals.

Using the MO diagram above, answer the following questions:

- (i) Which one of the following will have the highest bond order? Circle the correct choice
A) PS^{1-} B) PS C) PS^{1+}
- (ii) Which one of the following will have the longest bond length? Circle the correct choice
A) PS^{1-} B) PS C) PS^{1+}
- (iii) Which of the following will be diamagnetic? Circle the correct choice.
A) PS^{1-} B) PS C) PS^{1+}
- (iv) Draw Lewis dot structure of the molecule with the shortest bond length (based on the MO diagram) from among the three (PS^{1-} , PS, PS^{1+}), and indicate the bond order calculated from the Lewis dot structure. **(3 points)**

Lewis Dot Structure

Bond Order



50637793-7C63-4932-A736-1D98EADF42FB

midterm-exam-93388

#224 Page 10 of 10

SCRAP PAGE

**CHEMISTRY 110 –Midterm Exam****Thursday November 7, 2024, 6:30 – 8:30 pm****Examiners: Pallavi Sirjoosingh and Ashok Kakkar**

First name (please write as legibly as possible within the boxes)

--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--

Last name

--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--

Student ID number

--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--

This booklet has 8 questions on 10 pages (The last page can be used as a scrap page).

The staple must NOT be removed from the question booklet. Your intact question booklet must be returned at the end of the exam.

A separate booklet contains a periodic table, and datasheet.

Write your answers directly under the questions in the space provided. Do not write on the QR codes or near the top edges (as it may be cut off).

Please write **legibly and dark enough** for the scan to be readable. For students who use English as a second language, dictionaries are permitted. Calculators are allowed.

DO NOT ASK QUESTIONS OF THE INVIGILATORS about interpretations of the exam questions. If you have a concern about a question, please summarize it at the top of this page. All concerns will be evaluated after the exam. GOOD LUCK!

**Question 1 8 points** (*Assume all transitions are possible*)

a. 6 points Calculate the **energy (in kJ/mol)** required to ionize (completely remove) the electron from the **n=3** level of **He⁺ ion** (He: Helium). Show your work including equations. (3 sig. figs.)

b. 2 points

(I) Select the largest (longest) wavelength from the three choices below:

- i. Wavelength of photon to *completely remove* an electron from *1s orbital of Hydrogen atom*
- ii. Wavelength of photon to *completely remove* an electron from *1s orbital of Boron atom (B)*
- iii. Wavelength of photon to *completely remove* an electron from *1s orbital of Boron cation (B⁺)*

i or ii or iii

(II) Select the smallest (shortest) wavelength from the three choices below (*for H atom*):

- i. Wavelength of photon *emitted* for an electron transitioning from *2p to 1s orbital*
- ii. Wavelength of photon *absorbed* for an electron transitioning from *2s to 4p orbital*
- iii. Wavelength of photon *absorbed* for an electron transitioning from *1s to 3p orbital*

i or ii or iii



Question 2 8 points



A Chem110 student plots the graph shown on the left, from data obtained for the kinetic energy (eV) of the ejected electron versus frequency (s^{-1}) of the incident light, for an unknown metal. ($1\text{eV} = 1.6022 \times 10^{-19} \text{ J}$)

Based on the plot, answer the questions **a to c** below:

a. 1 point On the plot above, label the **threshold frequency**.

b. 2 points What is the value of the work potential (in J) of the unknown metal?

c. 3 points What is the *longest* wavelength (in nm) required to eject an electron from this metal's surface? Show your work including equation(s) used. (3 sig figs)

d. 2 points In the space provided below, plot the following

(Assume frequency is *above* threshold frequency):

- i. *Kinetic energy* of an ejected electron versus the *intensity* of the incident light (left)
- ii. *Number of electrons* ejected versus the *frequency* of the incident light (right)



**Question 3** **6 points****a. 2 points**

In the ground state of Ge (Germanium; $Z = 32$) atom, **name** the following orbitals:

i. Orbital containing the electron that requires the maximum energy to be removed.

ii. Orbital containing the electron that experiences the minimum effective nuclear charge.

b. 2 points

An unknown anion (A^-) has the following *excited* electronic configuration: $[Ar] 4s^2 3d^3 4p^1$

(Ar: Argon; $Z = 18$)

Give the **condensed ground state electronic configuration** of the atom, **A** and the cation, **A^{2+}** .
Show your work. (Assume all electronic transitions are possible)

Electronic Configuration of A:

Electronic Configuration of A^{2+} :

c. 2 points

In a single atom, determine how many electrons can have the following set of quantum numbers?

(i) $n = 5; l = 3$

In a single atom, determine how many electrons can have the following set of quantum numbers?

(i) $n = 3; l = 2; m_s = +1/2$

**Question 4** **8 points**

Rank the following from 1 to 4 in **increasing** order. Provide the ranking in the boxes provided.

Note: Atomic number (Z) are provided for relevant atoms

a) Increasing (lowest to highest) order of *ionization energy*: (1= lowest; 4 = highest)

Mg

Mg: Z = 12

Rb

Rb: Z = 37

Cs

Cs: Z = 55

K

K : Z = 19

b) Increasing (lowest to highest) order of *bond angle*: (1= lowest; 4 = highest)

H₂OCCl₄XeF₄

HCN

c) Increasing (smallest to largest) order of *ionic size* (1= smallest; 4 = largest)

S²⁻

S: Z = 16

O²⁻

O: Z = 8

Be²⁺

Be: Z = 4

Na⁺

Na: Z = 11

d) Increasing (lowest to highest) order of *bond energy* (1= lowest; 4 = highest)

P-F

P-Cl

P-I

P-Br

**Question 5** **12 points**

a. 9 points For the molecules HClO_4 and TeF_6 , draw the most stable *Lewis structure* showing appropriate shape and *dashed/wedged bonds*, based on VSEPR. Determine the **molecular geometry** around the central atom. For HClO_4 , label the *most polar bond*.



b. 3 points (Note: Lone pairs are not shown in structures below)

(i) For the three *resonance* structures (1, 2 and 3) shown below, answer the following question:



(I) Circle the *most contributing* resonance structure.

Structure 1

Structure 2

Structure 3

(ii) For the three structures given (4, 5, and 6) below, answer the following questions:



(I) Circle all the structures that represent *resonance structures* of each other:

Structure 4

Structure 5

Structure 6

(II) Among the resonance structures, circle the *most contributing* resonance structure.

Structure 4

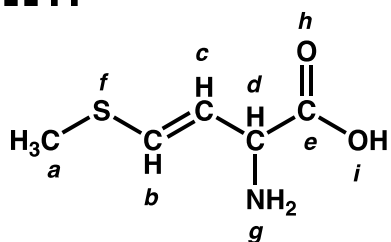
Structure 5

Structure 6

**Question 6** **6 points**

Using the data given below, calculate the **lattice energy** for SrO(s).

	ΔH° (kJ/mol)
$\text{Sr}(s) \longrightarrow \text{Sr}(g)$	+180
$\text{Sr}(g) \longrightarrow \text{Sr}^{2+}(g) + 2e^-$	+1468
$\text{O}_2(g) \longrightarrow 2\text{O}(g)$	+500
$\text{O}(g) + 2e^- \longrightarrow \text{O}^{2-}(g)$	+600
$\text{Sr}(s) + \frac{1}{2}\text{O}_2(g) \longrightarrow \text{SrO}(s)$	-550

**Question 7 10 points**

Consider the molecule given on the left, and answer questions (i) to (vi) using chemical bonding theories that we have learnt in class (valence bond theory, hybridization).

Note that the lone pairs are not shown in the structure:

i. Consider the carbon atoms marked in as ***a***, ***b***, ***c***, ***d***, and ***e*** in the molecule, and determine **hybridization** at each of the carbon centers. (2.5 points)

Atom	Hybridization
Carbon (<i>a</i>)	
Carbon (<i>b</i>)	
Carbon (<i>c</i>)	
Carbon (<i>d</i>)	
Carbon (<i>e</i>)	

ii. Now, consider the sulfur atom (S) marked as ***f***, nitrogen (N) atom (***g***), and oxygen(O) atoms (***h*** and ***i***), and determine their **hybridization**. (2 points)

Atom	Hybridization
Sulfur (<i>f</i>)	
Nitrogen (<i>g</i>)	
Oxygen (<i>h</i>)	
Oxygen (<i>i</i>)	

iii. **Describe** all the bonds formed by the carbon atoms ***a*** and ***b***, using sigma and pi designation as appropriate. (2 points)

Atom	Bonds formed (use sigma and pi as appropriate)
Carbon (<i>a</i>)	
Carbon (<i>b</i>)	

iv. How many total sp^3 -s sigma bonds are in the molecule? (1 point)

v. Determine **electronic** and **molecular geometry** at carbon (C) labeled ***e*** and sulfur (S) labeled ***f*** centres. (2 points)

Atom	Electronic Geometry	Molecular Geometry
Carbon (<i>e</i>)		
Sulfur (<i>f</i>)		

vi. Circle the value from the choices given below, that describes the **C-S-C bond angle at sulfur** (S) marked ***f***: (0.5 points)

>90° to <109.5° 109.5° >109.5 to <120° 120° >120° to <180°

**Question 8 12 points**

Using the **valence shell orbitals only** draw a **complete** molecular orbital (MO) diagram (with no s-p mixing) for the molecule **PS**, in the space provided below. **(6 points)**

Assume that 3s and 3p atomic orbitals will form molecular orbitals like those formed by the combination of 2s and 2p atomic orbitals that we learnt in class, and there is no s-p mixing.

Show **energy arrow**, **atomic orbitals** involved, **atom labels**, the **molecular orbitals** that are formed, **filled electrons** in the MO diagram etc. Make sure to label all the atomic and molecular orbitals.

Using the MO diagram above, answer the following questions:

- (i) Which one of the following will have the highest bond order? Circle the correct choice
A) PS^{1-} B) PS C) PS^{1+}
- (ii) Which one of the following will have the longest bond length? Circle the correct choice
A) PS^{1-} B) PS C) PS^{1+}
- (iii) Which of the following will be diamagnetic? Circle the correct choice.
A) PS^{1-} B) PS C) PS^{1+}
- (iv) Draw Lewis dot structure of the molecule with the shortest bond length (based on the MO diagram) from among the three (PS^{1-} , PS, PS^{1+}), and indicate the bond order calculated from the Lewis dot structure. **(3 points)**

Lewis Dot Structure

Bond Order



5BE95F75-D72D-4BC0-BE37-651185227C7C

midterm-exam-93388

#225 Page 10 of 10

SCRAP PAGE

**CHEMISTRY 110 –Midterm Exam****Thursday November 7, 2024, 6:30 – 8:30 pm****Examiners: Pallavi Sirjoosingh and Ashok Kakkar**

First name (please write as legibly as possible within the boxes)

--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--

Last name

--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--

Student ID number

--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--

This booklet has 8 questions on 10 pages (The last page can be used as a scrap page).

The staple must NOT be removed from the question booklet. Your intact question booklet must be returned at the end of the exam.

A separate booklet contains a periodic table, and datasheet.

Write your answers directly under the questions in the space provided. Do not write on the QR codes or near the top edges (as it may be cut off).

Please write **legibly and dark enough** for the scan to be readable. For students who use English as a second language, dictionaries are permitted. Calculators are allowed.

DO NOT ASK QUESTIONS OF THE INVIGILATORS about interpretations of the exam questions. If you have a concern about a question, please summarize it at the top of this page. All concerns will be evaluated after the exam. GOOD LUCK!

**Question 1 8 points** (*Assume all transitions are possible*)

a. 6 points Calculate the **energy (in kJ/mol)** required to ionize (completely remove) the electron from the **$n=3$** level of **He^+ ion** (He: Helium). Show your work including equations. (3 sig. figs.)

b. 2 points

(I) Select the largest (longest) wavelength from the three choices below:

- i. Wavelength of photon to *completely remove* an electron from *1s orbital of Hydrogen atom*
- ii. Wavelength of photon to *completely remove* an electron from *1s orbital of Boron atom (B)*
- iii. Wavelength of photon to *completely remove* an electron from *1s orbital of Boron cation (B^+)*

i or ii or iii

(II) Select the smallest (shortest) wavelength from the three choices below (*for H atom*):

- i. Wavelength of photon *emitted* for an electron transitioning from *2p to 1s orbital*
- ii. Wavelength of photon *absorbed* for an electron transitioning from *2s to 4p orbital*
- iii. Wavelength of photon *absorbed* for an electron transitioning from *1s to 3p orbital*

i or ii or iii



Question 2 8 points



A Chem110 student plots the graph shown on the left, from data obtained for the kinetic energy (eV) of the ejected electron versus frequency (s^{-1}) of the incident light, for an unknown metal. ($1\text{eV} = 1.6022 \times 10^{-19} \text{ J}$)

Based on the plot, answer the questions **a to c** below:

a. 1 point On the plot above, label the **threshold frequency**.

b. 2 points What is the value of the work potential (in J) of the unknown metal?

c. 3 points What is the *longest* wavelength (in nm) required to eject an electron from this metal's surface? Show your work including equation(s) used. (3 sig figs)

d. 2 points In the space provided below, plot the following

(Assume frequency is *above* threshold frequency):

- Kinetic energy* of an ejected electron versus the *intensity* of the incident light (left)
- Number of electrons* ejected versus the *frequency* of the incident light (right)



**Question 3** **6 points****a. 2 points**

In the ground state of Ge (Germanium; $Z = 32$) atom, **name** the following orbitals:

i. Orbital containing the electron that requires the maximum energy to be removed.

ii. Orbital containing the electron that experiences the minimum effective nuclear charge.

b. 2 points

An unknown anion (A^-) has the following *excited* electronic configuration: $[Ar] 4s^2 3d^3 4p^1$

(Ar: Argon; $Z = 18$)

Give the **condensed ground state electronic configuration** of the atom, **A** and the cation, **A^{2+}** .
Show your work. (Assume all electronic transitions are possible)

Electronic Configuration of A:

Electronic Configuration of A^{2+} :

c. 2 points

In a single atom, determine how many electrons can have the following set of quantum numbers?

(i) $n = 5; l = 3$

In a single atom, determine how many electrons can have the following set of quantum numbers?

(i) $n = 3; l = 2; m_s = +1/2$

**Question 4** **8 points**

Rank the following from 1 to 4 in **increasing** order. Provide the ranking in the boxes provided.

Note: Atomic number (Z) are provided for relevant atoms

a) Increasing (lowest to highest) order of *ionization energy*: (1= lowest; 4 = highest)

Mg

Mg: Z = 12

Rb

Rb: Z = 37

Cs

Cs: Z = 55

K

K : Z = 19

b) Increasing (lowest to highest) order of *bond angle*: (1= lowest; 4 = highest)

H₂OCCl₄XeF₄

HCN

c) Increasing (smallest to largest) order of *ionic size* (1= smallest; 4 = largest)

S²⁻

S: Z = 16

O²⁻

O: Z = 8

Be²⁺

Be: Z = 4

Na⁺

Na: Z = 11

d) Increasing (lowest to highest) order of *bond energy* (1= lowest; 4 = highest)

P-F

P-Cl

P-I

P-Br

**Question 5** **12 points**

a. 9 points For the molecules HClO_4 and TeF_6 , draw the most stable *Lewis structure* showing appropriate shape and *dashed/wedged bonds*, based on VSEPR. Determine the **molecular geometry** around the central atom. For HClO_4 , label the *most polar bond*.



b. 3 points (Note: Lone pairs are not shown in structures below)

(i) For the three *resonance* structures (1, 2 and 3) shown below, answer the following question:



(I) Circle the *most contributing* resonance structure.

Structure 1

Structure 2

Structure 3

(ii) For the three structures given (4, 5, and 6) below, answer the following questions:



(I) Circle all the structures that represent *resonance structures* of each other:

Structure 4

Structure 5

Structure 6

(II) Among the resonance structures, circle the *most contributing* resonance structure.

Structure 4

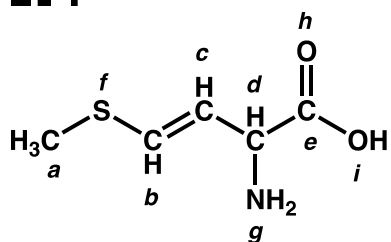
Structure 5

Structure 6

**Question 6** **6 points**

Using the data given below, calculate the **lattice energy** for SrO(s).

	ΔH° (kJ/mol)
$\text{Sr}(s) \longrightarrow \text{Sr}(g)$	+180
$\text{Sr}(g) \longrightarrow \text{Sr}^{2+}(g) + 2e^-$	+1468
$\text{O}_2(g) \longrightarrow 2\text{O}(g)$	+500
$\text{O}(g) + 2e^- \longrightarrow \text{O}^{2-}(g)$	+600
$\text{Sr}(s) + \frac{1}{2}\text{O}_2(g) \longrightarrow \text{SrO}(s)$	-550

**Question 7 10 points**

Consider the molecule given on the left, and answer questions (i) to (vi) using chemical bonding theories that we have learnt in class (valence bond theory, hybridization).

Note that the lone pairs are not shown in the structure:

i. Consider the carbon atoms marked in as ***a***, ***b***, ***c***, ***d***, and ***e*** in the molecule, and determine **hybridization** at each of the carbon centers. (2.5 points)

Atom	Hybridization
Carbon (<i>a</i>)	
Carbon (<i>b</i>)	
Carbon (<i>c</i>)	
Carbon (<i>d</i>)	
Carbon (<i>e</i>)	

ii. Now, consider the sulfur atom (S) marked as ***f***, nitrogen (N) atom (***g***), and oxygen(O) atoms (***h*** and ***i***), and determine their **hybridization**. (2 points)

Atom	Hybridization
Sulfur (<i>f</i>)	
Nitrogen (<i>g</i>)	
Oxygen (<i>h</i>)	
Oxygen (<i>i</i>)	

iii. **Describe** all the bonds formed by the carbon atoms ***a*** and ***b***, using sigma and pi designation as appropriate. (2 points)

Atom	Bonds formed (use sigma and pi as appropriate)
Carbon (<i>a</i>)	
Carbon (<i>b</i>)	

iv. How many total sp^3 -s sigma bonds are in the molecule? (1 point)

v. Determine **electronic** and **molecular geometry** at carbon (C) labeled ***e*** and sulfur (S) labeled ***f*** centres. (2 points)

Atom	Electronic Geometry	Molecular Geometry
Carbon (<i>e</i>)		
Sulfur (<i>f</i>)		

vi. Circle the value from the choices given below, that describes the **C-S-C bond angle at sulfur** (S) marked ***f***: (0.5 points)

>90° to <109.5° 109.5° >109.5 to <120° 120° >120° to <180°

**Question 8 12 points**

Using the **valence shell orbitals only** draw a **complete** molecular orbital (MO) diagram (with no s-p mixing) for the molecule **PS**, in the space provided below. **(6 points)**

Assume that 3s and 3p atomic orbitals will form molecular orbitals like those formed by the combination of 2s and 2p atomic orbitals that we learnt in class, and there is no s-p mixing.

Show **energy arrow**, **atomic orbitals** involved, **atom labels**, the **molecular orbitals** that are formed, **filled electrons** in the MO diagram etc. Make sure to label all the atomic and molecular orbitals.

Using the MO diagram above, answer the following questions:

- (i) Which one of the following will have the highest bond order? Circle the correct choice
A) PS^{1-} B) PS C) PS^{1+}
- (ii) Which one of the following will have the longest bond length? Circle the correct choice
A) PS^{1-} B) PS C) PS^{1+}
- (iii) Which of the following will be diamagnetic? Circle the correct choice.
A) PS^{1-} B) PS C) PS^{1+}
- (iv) Draw Lewis dot structure of the molecule with the shortest bond length (based on the MO diagram) from among the three (PS^{1-} , PS, PS^{1+}), and indicate the bond order calculated from the Lewis dot structure. **(3 points)**

Lewis Dot Structure

Bond Order



B6088302-71DC-41E4-9177-ED1F4AF998ED

midterm-exam-93388

#226 Page 10 of 10

SCRAP PAGE

**CHEMISTRY 110 –Midterm Exam****Thursday November 7, 2024, 6:30 – 8:30 pm****Examiners: Pallavi Sirjoosingh and Ashok Kakkar**

First name (please write as legibly as possible within the boxes)

--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--

Last name

--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--

Student ID number

--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--

This booklet has 8 questions on 10 pages (The last page can be used as a scrap page).

The staple must NOT be removed from the question booklet. Your intact question booklet must be returned at the end of the exam.

A separate booklet contains a periodic table, and datasheet.

Write your answers directly under the questions in the space provided. Do not write on the QR codes or near the top edges (as it may be cut off).

Please write **legibly and dark enough** for the scan to be readable. For students who use English as a second language, dictionaries are permitted. Calculators are allowed.

DO NOT ASK QUESTIONS OF THE INVIGILATORS about interpretations of the exam questions. If you have a concern about a question, please summarize it at the top of this page. All concerns will be evaluated after the exam. GOOD LUCK!

**Question 1 8 points** (*Assume all transitions are possible*)

a. 6 points Calculate the **energy (in kJ/mol)** required to ionize (completely remove) the electron from the **$n=3$** level of **He^+ ion** (He: Helium). Show your work including equations. (3 sig. figs.)

b. 2 points

(I) Select the largest (longest) wavelength from the three choices below:

- i. Wavelength of photon to *completely remove* an electron from *1s orbital of Hydrogen atom*
- ii. Wavelength of photon to *completely remove* an electron from *1s orbital of Boron atom (B)*
- iii. Wavelength of photon to *completely remove* an electron from *1s orbital of Boron cation (B^+)*

i or ii or iii

(II) Select the smallest (shortest) wavelength from the three choices below (*for H atom*):

- i. Wavelength of photon *emitted* for an electron transitioning from *2p to 1s orbital*
- ii. Wavelength of photon *absorbed* for an electron transitioning from *2s to 4p orbital*
- iii. Wavelength of photon *absorbed* for an electron transitioning from *1s to 3p orbital*

i or ii or iii



Question 2 8 points



A Chem110 student plots the graph shown on the left, from data obtained for the kinetic energy (eV) of the ejected electron versus frequency (s^{-1}) of the incident light, for an unknown metal. ($1\text{eV} = 1.6022 \times 10^{-19} \text{ J}$)

Based on the plot, answer the questions **a to c** below:

a. 1 point On the plot above, label the **threshold frequency**.

b. 2 points What is the value of the work potential (in J) of the unknown metal?

c. 3 points What is the *longest* wavelength (in nm) required to eject an electron from this metal's surface? Show your work including equation(s) used. (3 sig figs)

d. 2 points In the space provided below, plot the following

(Assume frequency is *above* threshold frequency):

- Kinetic energy* of an ejected electron versus the *intensity* of the incident light (left)
- Number of electrons* ejected versus the *frequency* of the incident light (right)



**Question 3** **6 points****a. 2 points**

In the ground state of Ge (Germanium; $Z = 32$) atom, **name** the following orbitals:

i. Orbital containing the electron that requires the maximum energy to be removed.

ii. Orbital containing the electron that experiences the minimum effective nuclear charge.

b. 2 points

An unknown anion (A^-) has the following *excited* electronic configuration: $[Ar] 4s^2 3d^3 4p^1$

(Ar: Argon; $Z = 18$)

Give the **condensed ground state electronic configuration** of the atom, **A** and the cation, **A^{2+}** .
Show your work. (Assume all electronic transitions are possible)

Electronic Configuration of A:

Electronic Configuration of A^{2+} :

c. 2 points

In a single atom, determine how many electrons can have the following set of quantum numbers?

(i) $n = 5; l = 3$

In a single atom, determine how many electrons can have the following set of quantum numbers?

(i) $n = 3; l = 2; m_s = +1/2$

**Question 4** **8 points**

Rank the following from 1 to 4 in **increasing** order. Provide the ranking in the boxes provided.

Note: Atomic number (Z) are provided for relevant atoms

a) Increasing (lowest to highest) order of *ionization energy*: (1= lowest; 4 = highest)

Mg

Mg: Z = 12

Rb

Rb: Z = 37

Cs

Cs: Z = 55

K

K : Z = 19

b) Increasing (lowest to highest) order of *bond angle*: (1= lowest; 4 = highest)

H₂OCCl₄XeF₄

HCN

c) Increasing (smallest to largest) order of *ionic size* (1= smallest; 4 = largest)

S²⁻

S: Z = 16

O²⁻

O: Z = 8

Be²⁺

Be: Z = 4

Na⁺

Na: Z = 11

d) Increasing (lowest to highest) order of *bond energy* (1= lowest; 4 = highest)

P-F

P-Cl

P-I

P-Br

**Question 5** **12 points**

a. 9 points For the molecules HClO_4 and TeF_6 , draw the most stable *Lewis structure* showing appropriate shape and *dashed/wedged bonds*, based on VSEPR. Determine the **molecular geometry** around the central atom. For HClO_4 , label the *most polar bond*.

HClO_4 (there are no O-O bonds)

Label the most polar bond

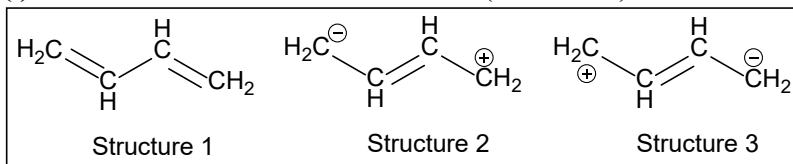
Molecular Geometry around Cl

TeF_6

Molecular Geometry around Te

b. 3 points (Note: Lone pairs are not shown in structures below)

(i) For the three *resonance* structures (1, 2 and 3) shown below, answer the following question:



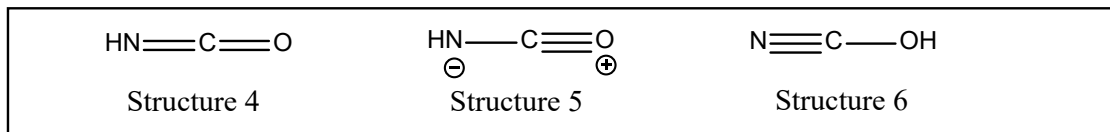
(I) Circle the *most contributing* resonance structure.

Structure 1

Structure 2

Structure 3

(ii) For the three structures given (4, 5, and 6) below, answer the following questions:



(I) Circle all the structures that represent *resonance structures* of each other:

Structure 4

Structure 5

Structure 6

(II) Among the resonance structures, circle the *most contributing* resonance structure.

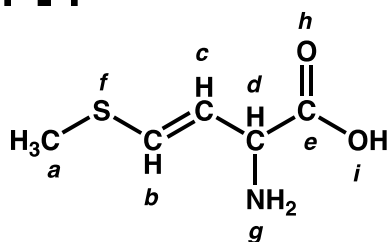
Structure 4

Structure 5

Structure 6

**Question 6** **6 points**Using the data given below, calculate the **lattice energy** for SrO(s).

	ΔH° (kJ/mol)
$\text{Sr}(s) \longrightarrow \text{Sr}(g)$	+180
$\text{Sr}(g) \longrightarrow \text{Sr}^{2+}(g) + 2e^-$	+1468
$\text{O}_2(g) \longrightarrow 2\text{O}(g)$	+500
$\text{O}(g) + 2e^- \longrightarrow \text{O}^{2-}(g)$	+600
$\text{Sr}(s) + \frac{1}{2}\text{O}_2(g) \longrightarrow \text{SrO}(s)$	-550

**Question 7 10 points**

Consider the molecule given on the left, and answer questions (i) to (vi) using chemical bonding theories that we have learnt in class (valence bond theory, hybridization).

Note that the lone pairs are not shown in the structure:

i. Consider the carbon atoms marked in as ***a***, ***b***, ***c***, ***d***, and ***e*** in the molecule, and determine **hybridization** at each of the carbon centers. (2.5 points)

Atom	Hybridization
Carbon (<i>a</i>)	
Carbon (<i>b</i>)	
Carbon (<i>c</i>)	
Carbon (<i>d</i>)	
Carbon (<i>e</i>)	

ii. Now, consider the sulfur atom (S) marked as ***f***, nitrogen (N) atom (***g***), and oxygen(O) atoms (***h*** and ***i***), and determine their **hybridization**. (2 points)

Atom	Hybridization
Sulfur (<i>f</i>)	
Nitrogen (<i>g</i>)	
Oxygen (<i>h</i>)	
Oxygen (<i>i</i>)	

iii. **Describe** all the bonds formed by the carbon atoms ***a*** and ***b***, using sigma and pi designation as appropriate. (2 points)

Atom	Bonds formed (use sigma and pi as appropriate)
Carbon (<i>a</i>)	
Carbon (<i>b</i>)	

iv. How many total sp^3 -s sigma bonds are in the molecule? (1 point)

v. Determine **electronic** and **molecular geometry** at carbon (C) labeled ***e*** and sulfur (S) labeled ***f*** centres. (2 points)

Atom	Electronic Geometry	Molecular Geometry
Carbon (<i>e</i>)		
Sulfur (<i>f</i>)		

vi. Circle the value from the choices given below, that describes the **C-S-C bond angle at sulfur** (S) marked ***f***: (0.5 points)

>90° to <109.5° 109.5° >109.5 to <120° 120° >120° to <180°


Question 8 12 points

Using the **valence shell orbitals only** draw a **complete** molecular orbital (MO) diagram (with no s-p mixing) for the molecule **PS**, in the space provided below. **(6 points)**

Assume that 3s and 3p atomic orbitals will form molecular orbitals like those formed by the combination of 2s and 2p atomic orbitals that we learnt in class, and there is no s-p mixing.

Show **energy arrow**, **atomic orbitals** involved, **atom labels**, the **molecular orbitals** that are formed, **filled electrons** in the MO diagram etc. Make sure to label all the atomic and molecular orbitals.

Using the MO diagram above, answer the following questions:

- (i) Which one of the following will have the highest bond order? Circle the correct choice
 A) PS^{1-} B) PS C) PS^{1+}
- (ii) Which one of the following will have the longest bond length? Circle the correct choice
 A) PS^{1-} B) PS C) PS^{1+}
- (iii) Which of the following will be diamagnetic? Circle the correct choice.
 A) PS^{1-} B) PS C) PS^{1+}
- (iv) Draw Lewis dot structure of the molecule with the shortest bond length (based on the MO diagram) from among the three (PS^{1-} , PS, PS^{1+}), and indicate the bond order calculated from the Lewis dot structure. **(3 points)**

Lewis Dot Structure

Bond Order



DD3CAA19-41BE-44A4-935E-432DAE1B3B92

midterm-exam-93388

#227 Page 10 of 10

SCRAP PAGE

**CHEMISTRY 110 –Midterm Exam****Thursday November 7, 2024, 6:30 – 8:30 pm**

Examiners: Pallavi Sirjoosingh and Ashok Kakkar

First name (please write as legibly as possible within the boxes)

--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--

Last name

--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--

Student ID number

--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--

This booklet has 8 questions on 10 pages (The last page can be used as a scrap page).

The staple must NOT be removed from the question booklet. Your intact question booklet must be returned at the end of the exam.

A separate booklet contains a periodic table, and datasheet.

Write your answers directly under the questions in the space provided. Do not write on the QR codes or near the top edges (as it may be cut off).

Please write **legibly and dark enough** for the scan to be readable. For students who use English as a second language, dictionaries are permitted. Calculators are allowed.

DO NOT ASK QUESTIONS OF THE INVIGILATORS about interpretations of the exam questions. If you have a concern about a question, please summarize it at the top of this page. All concerns will be evaluated after the exam. GOOD LUCK!

**Question 1 8 points** (*Assume all transitions are possible*)

a. 6 points Calculate the **energy (in kJ/mol)** required to ionize (completely remove) the electron from the **$n=3$** level of **He^+ ion** (He: Helium). Show your work including equations. (3 sig. figs.)

b. 2 points

(I) Select the largest (longest) wavelength from the three choices below:

- i. Wavelength of photon to *completely remove* an electron from *1s orbital of Hydrogen atom*
- ii. Wavelength of photon to *completely remove* an electron from *1s orbital of Boron atom (B)*
- iii. Wavelength of photon to *completely remove* an electron from *1s orbital of Boron cation (B^+)*

i or ii or iii

(II) Select the smallest (shortest) wavelength from the three choices below (*for H atom*):

- i. Wavelength of photon *emitted* for an electron transitioning from *2p to 1s orbital*
- ii. Wavelength of photon *absorbed* for an electron transitioning from *2s to 4p orbital*
- iii. Wavelength of photon *absorbed* for an electron transitioning from *1s to 3p orbital*

i or ii or iii



Question 2 8 points



A Chem110 student plots the graph shown on the left, from data obtained for the kinetic energy (eV) of the ejected electron versus frequency (s^{-1}) of the incident light, for an unknown metal. ($1\text{eV} = 1.6022 \times 10^{-19} \text{ J}$)

Based on the plot, answer the questions **a to c** below:

a. 1 point On the plot above, label the **threshold frequency**.

b. 2 points What is the value of the work potential (in J) of the unknown metal?

c. 3 points What is the *longest* wavelength (in nm) required to eject an electron from this metal's surface? Show your work including equation(s) used. (3 sig figs)

d. 2 points In the space provided below, plot the following

(Assume frequency is *above* threshold frequency):

- Kinetic energy* of an ejected electron versus the *intensity* of the incident light (left)
- Number of electrons* ejected versus the *frequency* of the incident light (right)



**Question 3** **6 points****a. 2 points**

In the ground state of Ge (Germanium; $Z = 32$) atom, **name** the following orbitals:

i. Orbital containing the electron that requires the maximum energy to be removed.

ii. Orbital containing the electron that experiences the minimum effective nuclear charge.

b. 2 points

An unknown anion (A^-) has the following *excited* electronic configuration: $[Ar] 4s^2 3d^3 4p^1$

(Ar: Argon; $Z = 18$)

Give the **condensed ground state electronic configuration** of the atom, **A** and the cation, A^{2+} .
Show your work. (Assume all electronic transitions are possible)

Electronic Configuration of A:

Electronic Configuration of A^{2+} :

c. 2 points

In a single atom, determine how many electrons can have the following set of quantum numbers?

(i) $n = 5; l = 3$

In a single atom, determine how many electrons can have the following set of quantum numbers?

(i) $n = 3; l = 2; m_s = +1/2$

**Question 4** **8 points**

Rank the following from 1 to 4 in **increasing** order. Provide the ranking in the boxes provided.

Note: Atomic number (Z) are provided for relevant atoms

a) Increasing (lowest to highest) order of *ionization energy*: (1= lowest; 4 = highest)

Mg

Mg: Z = 12

Rb

Rb: Z = 37

Cs

Cs: Z = 55

K

K : Z = 19

b) Increasing (lowest to highest) order of *bond angle*: (1= lowest; 4 = highest)

H₂OCCl₄XeF₄

HCN

c) Increasing (smallest to largest) order of *ionic size* (1= smallest; 4 = largest)

S²⁻

S: Z = 16

O²⁻

O: Z = 8

Be²⁺

Be: Z = 4

Na⁺

Na: Z = 11

d) Increasing (lowest to highest) order of *bond energy* (1= lowest; 4 = highest)

P-F

P-Cl

P-I

P-Br

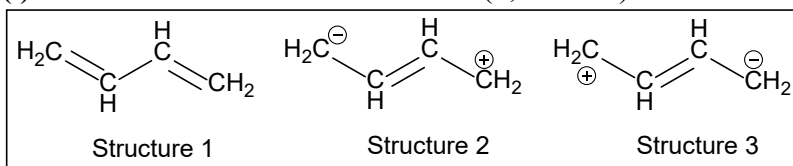
**Question 5** **12 points**

a. 9 points For the molecules HClO_4 and TeF_6 , draw the most stable *Lewis structure* showing appropriate shape and *dashed/wedged bonds*, based on VSEPR. Determine the **molecular geometry** around the central atom. For HClO_4 , label the *most polar bond*.



b. 3 points (Note: Lone pairs are not shown in structures below)

(i) For the three *resonance* structures (1, 2 and 3) shown below, answer the following question:



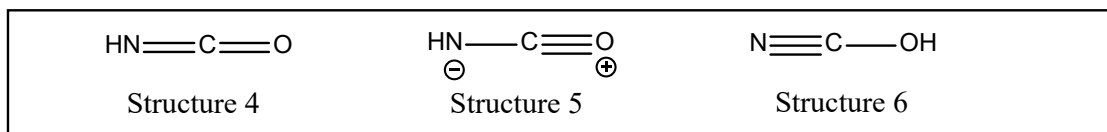
(I) Circle the *most contributing* resonance structure.

Structure 1

Structure 2

Structure 3

(ii) For the three structures given (4, 5, and 6) below, answer the following questions:



(I) Circle all the structures that represent *resonance structures* of each other:

Structure 4

Structure 5

Structure 6

(II) Among the resonance structures, circle the *most contributing* resonance structure.

Structure 4

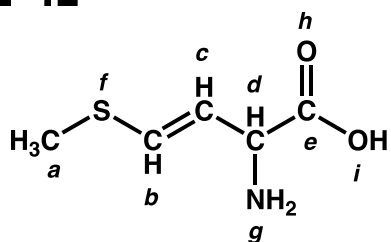
Structure 5

Structure 6

**Question 6** **6 points**

Using the data given below, calculate the **lattice energy** for SrO(s).

	ΔH° (kJ/mol)
$\text{Sr}(s) \longrightarrow \text{Sr}(g)$	+180
$\text{Sr}(g) \longrightarrow \text{Sr}^{2+}(g) + 2e^-$	+1468
$\text{O}_2(g) \longrightarrow 2\text{O}(g)$	+500
$\text{O}(g) + 2e^- \longrightarrow \text{O}^{2-}(g)$	+600
$\text{Sr}(s) + \frac{1}{2}\text{O}_2(g) \longrightarrow \text{SrO}(s)$	-550

**Question 7 10 points**

Consider the molecule given on the left, and answer questions (i) to (vi) using chemical bonding theories that we have learnt in class (valence bond theory, hybridization).

Note that the lone pairs are not shown in the structure:

i. Consider the carbon atoms marked in as ***a***, ***b***, ***c***, ***d***, and ***e*** in the molecule, and determine **hybridization** at each of the carbon centers. (2.5 points)

Atom	Hybridization
Carbon (<i>a</i>)	
Carbon (<i>b</i>)	
Carbon (<i>c</i>)	
Carbon (<i>d</i>)	
Carbon (<i>e</i>)	

ii. Now, consider the sulfur atom (S) marked as ***f***, nitrogen (N) atom (***g***), and oxygen(O) atoms (***h*** and ***i***), and determine their **hybridization**. (2 points)

Atom	Hybridization
Sulfur (<i>f</i>)	
Nitrogen (<i>g</i>)	
Oxygen (<i>h</i>)	
Oxygen (<i>i</i>)	

iii. **Describe** all the bonds formed by the carbon atoms ***a*** and ***b***, using sigma and pi designation as appropriate. (2 points)

Atom	Bonds formed (use sigma and pi as appropriate)
Carbon (<i>a</i>)	
Carbon (<i>b</i>)	

iv. How many total sp^3 -s sigma bonds are in the molecule? (1 point)

v. Determine **electronic** and **molecular geometry** at carbon (C) labeled ***e*** and sulfur (S) labeled ***f*** centres. (2 points)

Atom	Electronic Geometry	Molecular Geometry
Carbon (<i>e</i>)		
Sulfur (<i>f</i>)		

vi. Circle the value from the choices given below, that describes the **C-S-C bond angle at sulfur** (S) marked ***f***: (0.5 points)

>90° to <109.5°

109.5°

>109.5 to <120°

120°

>120° to <180°


Question 8 12 points

Using the **valence shell orbitals only** draw a **complete** molecular orbital (MO) diagram (with no s-p mixing) for the molecule **PS**, in the space provided below. **(6 points)**

Assume that 3s and 3p atomic orbitals will form molecular orbitals like those formed by the combination of 2s and 2p atomic orbitals that we learnt in class, and there is no s-p mixing.

Show **energy arrow**, **atomic orbitals** involved, **atom labels**, the **molecular orbitals** that are formed, **filled electrons** in the MO diagram etc. Make sure to label all the atomic and molecular orbitals.

Using the MO diagram above, answer the following questions:

- (i) Which one of the following will have the highest bond order? Circle the correct choice
 A) PS^{1-} B) PS C) PS^{1+}
- (ii) Which one of the following will have the longest bond length? Circle the correct choice
 A) PS^{1-} B) PS C) PS^{1+}
- (iii) Which of the following will be diamagnetic? Circle the correct choice.
 A) PS^{1-} B) PS C) PS^{1+}
- (iv) Draw Lewis dot structure of the molecule with the shortest bond length (based on the MO diagram) from among the three (PS^{1-} , PS, PS^{1+}), and indicate the bond order calculated from the Lewis dot structure. **(3 points)**

Lewis Dot Structure

Bond Order



609901DE-6F19-4C97-96F7-DAE086DE1D2E

midterm-exam-93388

#228 Page 10 of 10

SCRAP PAGE

**CHEMISTRY 110 –Midterm Exam****Thursday November 7, 2024, 6:30 – 8:30 pm**

Examiners: Pallavi Sirjoosingh and Ashok Kakkar

First name (please write as legibly as possible within the boxes)

--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--

Last name

--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--

Student ID number

--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--

This booklet has 8 questions on 10 pages (The last page can be used as a scrap page).

The staple must NOT be removed from the question booklet. Your intact question booklet must be returned at the end of the exam.

A separate booklet contains a periodic table, and datasheet.

Write your answers directly under the questions in the space provided. Do not write on the QR codes or near the top edges (as it may be cut off).

Please write **legibly and dark enough** for the scan to be readable. For students who use English as a second language, dictionaries are permitted. Calculators are allowed.

DO NOT ASK QUESTIONS OF THE INVIGILATORS about interpretations of the exam questions. If you have a concern about a question, please summarize it at the top of this page. All concerns will be evaluated after the exam. GOOD LUCK!

**Question 1 8 points** (*Assume all transitions are possible*)

a. 6 points Calculate the **energy (in kJ/mol)** required to ionize (completely remove) the electron from the **n=3** level of **He⁺ ion** (He: Helium). Show your work including equations. (3 sig. figs.)

b. 2 points

(I) Select the largest (longest) wavelength from the three choices below:

- i. Wavelength of photon to *completely remove* an electron from *1s orbital of Hydrogen atom*
- ii. Wavelength of photon to *completely remove* an electron from *1s orbital of Boron atom (B)*
- iii. Wavelength of photon to *completely remove* an electron from *1s orbital of Boron cation (B⁺)*

i or ii or iii

(II) Select the smallest (shortest) wavelength from the three choices below (*for H atom*):

- i. Wavelength of photon *emitted* for an electron transitioning from *2p to 1s orbital*
- ii. Wavelength of photon *absorbed* for an electron transitioning from *2s to 4p orbital*
- iii. Wavelength of photon *absorbed* for an electron transitioning from *1s to 3p orbital*

i or ii or iii



Question 2 8 points



A Chem110 student plots the graph shown on the left, from data obtained for the kinetic energy (eV) of the ejected electron versus frequency (s^{-1}) of the incident light, for an unknown metal. ($1\text{eV} = 1.6022 \times 10^{-19} \text{ J}$)

Based on the plot, answer the questions **a to c** below:

a. 1 point On the plot above, label the **threshold frequency**.

b. 2 points What is the value of the work potential (in J) of the unknown metal?

c. 3 points What is the *longest* wavelength (in nm) required to eject an electron from this metal's surface? Show your work including equation(s) used. (3 sig figs)

d. 2 points In the space provided below, plot the following

(Assume frequency is *above* threshold frequency):

- Kinetic energy* of an ejected electron versus the *intensity* of the incident light (left)
- Number of electrons* ejected versus the *frequency* of the incident light (right)



**Question 3** **6 points****a. 2 points**

In the ground state of Ge (Germanium; $Z = 32$) atom, **name** the following orbitals:

i. Orbital containing the electron that requires the maximum energy to be removed.

ii. Orbital containing the electron that experiences the minimum effective nuclear charge.

b. 2 points

An unknown anion (A^-) has the following *excited* electronic configuration: $[Ar] 4s^2 3d^3 4p^1$

(Ar: Argon; $Z = 18$)

Give the **condensed ground state electronic configuration** of the atom, **A** and the cation, **A^{2+}** .
Show your work. (Assume all electronic transitions are possible)

Electronic Configuration of A:

Electronic Configuration of A^{2+} :

c. 2 points

In a single atom, determine how many electrons can have the following set of quantum numbers?

(i) $n = 5; l = 3$

In a single atom, determine how many electrons can have the following set of quantum numbers?

(i) $n = 3; l = 2; m_s = +1/2$

**Question 4** **8 points**

Rank the following from 1 to 4 in **increasing** order. Provide the ranking in the boxes provided.

Note: Atomic number (Z) are provided for relevant atoms

a) Increasing (lowest to highest) order of *ionization energy*: (1= lowest; 4 = highest)

Mg

Mg: Z = 12

Rb

Rb: Z = 37

Cs

Cs: Z = 55

K

K : Z = 19

b) Increasing (lowest to highest) order of *bond angle*: (1= lowest; 4 = highest)

H₂OCCl₄XeF₄

HCN

c) Increasing (smallest to largest) order of *ionic size* (1= smallest; 4 = largest)

S²⁻

S: Z = 16

O²⁻

O: Z = 8

Be²⁺

Be: Z = 4

Na⁺

Na: Z = 11

d) Increasing (lowest to highest) order of *bond energy* (1= lowest; 4 = highest)

P-F

P-Cl

P-I

P-Br

**Question 5** **12 points**

a. 9 points For the molecules HClO_4 and TeF_6 , draw the most stable *Lewis structure* showing appropriate shape and *dashed/wedged bonds*, based on VSEPR. Determine the **molecular geometry** around the central atom. For HClO_4 , label the *most polar bond*.

HClO_4 (there are no O-O bonds)

Label the most polar bond

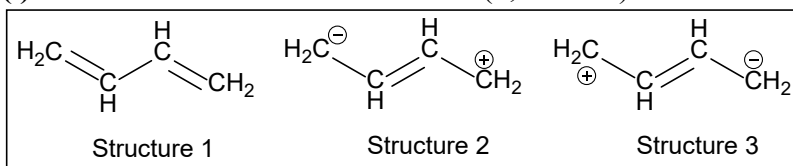
Molecular Geometry around Cl

TeF_6

Molecular Geometry around Te

b. 3 points (Note: Lone pairs are not shown in structures below)

(i) For the three *resonance* structures (1, 2 and 3) shown below, answer the following question:



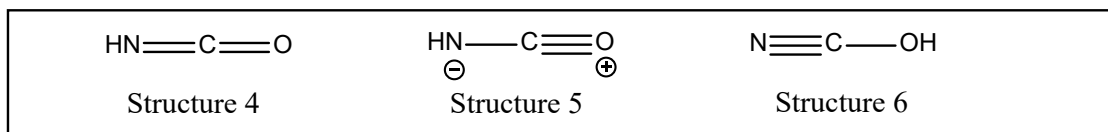
(I) Circle the *most contributing* resonance structure.

Structure 1

Structure 2

Structure 3

(ii) For the three structures given (4, 5, and 6) below, answer the following questions:



(I) Circle all the structures that represent *resonance structures* of each other:

Structure 4

Structure 5

Structure 6

(II) Among the resonance structures, circle the *most contributing* resonance structure.

Structure 4

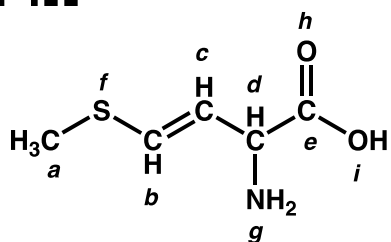
Structure 5

Structure 6

**Question 6** **6 points**

Using the data given below, calculate the **lattice energy** for SrO(s).

	ΔH° (kJ/mol)
$\text{Sr}(s) \longrightarrow \text{Sr}(g)$	+180
$\text{Sr}(g) \longrightarrow \text{Sr}^{2+}(g) + 2e^-$	+1468
$\text{O}_2(g) \longrightarrow 2\text{O}(g)$	+500
$\text{O}(g) + 2e^- \longrightarrow \text{O}^{2-}(g)$	+600
$\text{Sr}(s) + \frac{1}{2}\text{O}_2(g) \longrightarrow \text{SrO}(s)$	-550

**Question 7 10 points**

Consider the molecule given on the left, and answer questions (i) to (vi) using chemical bonding theories that we have learnt in class (valence bond theory, hybridization).

Note that the lone pairs are not shown in the structure:

i. Consider the carbon atoms marked in as ***a***, ***b***, ***c***, ***d***, and ***e*** in the molecule, and determine **hybridization** at each of the carbon centers. (2.5 points)

Atom	Hybridization
Carbon (<i>a</i>)	
Carbon (<i>b</i>)	
Carbon (<i>c</i>)	
Carbon (<i>d</i>)	
Carbon (<i>e</i>)	

ii. Now, consider the sulfur atom (S) marked as ***f***, nitrogen (N) atom (***g***), and oxygen(O) atoms (***h*** and ***i***), and determine their **hybridization**. (2 points)

Atom	Hybridization
Sulfur (<i>f</i>)	
Nitrogen (<i>g</i>)	
Oxygen (<i>h</i>)	
Oxygen (<i>i</i>)	

iii. **Describe** all the bonds formed by the carbon atoms ***a*** and ***b***, using sigma and pi designation as appropriate. (2 points)

Atom	Bonds formed (use sigma and pi as appropriate)
Carbon (<i>a</i>)	
Carbon (<i>b</i>)	

iv. How many total sp^3 -s sigma bonds are in the molecule? (1 point)

v. Determine **electronic** and **molecular geometry** at carbon (C) labeled ***e*** and sulfur (S) labeled ***f*** centres. (2 points)

Atom	Electronic Geometry	Molecular Geometry
Carbon (<i>e</i>)		
Sulfur (<i>f</i>)		

vi. Circle the value from the choices given below, that describes the **C-S-C bond angle at sulfur** (S) marked ***f***: (0.5 points)

>90° to <109.5° 109.5° >109.5 to <120° 120° >120° to <180°


Question 8 12 points

Using the **valence shell orbitals only** draw a **complete** molecular orbital (MO) diagram (with no s-p mixing) for the molecule **PS**, in the space provided below. **(6 points)**

Assume that 3s and 3p atomic orbitals will form molecular orbitals like those formed by the combination of 2s and 2p atomic orbitals that we learnt in class, and there is no s-p mixing.

Show **energy arrow**, **atomic orbitals** involved, **atom labels**, the **molecular orbitals** that are formed, **filled electrons** in the MO diagram etc. Make sure to label all the atomic and molecular orbitals.

Using the MO diagram above, answer the following questions:

- (i) Which one of the following will have the highest bond order? Circle the correct choice
 A) PS^{1-} B) PS C) PS^{1+}
- (ii) Which one of the following will have the longest bond length? Circle the correct choice
 A) PS^{1-} B) PS C) PS^{1+}
- (iii) Which of the following will be diamagnetic? Circle the correct choice.
 A) PS^{1-} B) PS C) PS^{1+}
- (iv) Draw Lewis dot structure of the molecule with the shortest bond length (based on the MO diagram) from among the three (PS^{1-} , PS, PS^{1+}), and indicate the bond order calculated from the Lewis dot structure. **(3 points)**

Lewis Dot Structure

Bond Order



49BCA370-21A0-429D-B78A-2CB899852198

midterm-exam-93388

#229 Page 10 of 10

SCRAP PAGE

**CHEMISTRY 110 –Midterm Exam****Thursday November 7, 2024, 6:30 – 8:30 pm**

Examiners: Pallavi Sirjoosingh and Ashok Kakkar

First name (please write as legibly as possible within the boxes)

--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--

Last name

--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--

Student ID number

--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--

This booklet has 8 questions on 10 pages (The last page can be used as a scrap page).

The staple must NOT be removed from the question booklet. Your intact question booklet must be returned at the end of the exam.

A separate booklet contains a periodic table, and datasheet.

Write your answers directly under the questions in the space provided. Do not write on the QR codes or near the top edges (as it may be cut off).

Please write **legibly and dark enough** for the scan to be readable. For students who use English as a second language, dictionaries are permitted. Calculators are allowed.

DO NOT ASK QUESTIONS OF THE INVIGILATORS about interpretations of the exam questions. If you have a concern about a question, please summarize it at the top of this page. All concerns will be evaluated after the exam. GOOD LUCK!

**Question 1 8 points** (*Assume all transitions are possible*)

a. 6 points Calculate the **energy (in kJ/mol)** required to ionize (completely remove) the electron from the **n=3** level of **He⁺ ion** (He: Helium). Show your work including equations. (3 sig. figs.)

b. 2 points

(I) Select the largest (longest) wavelength from the three choices below:

- i. Wavelength of photon to *completely remove* an electron from *1s orbital of Hydrogen atom*
- ii. Wavelength of photon to *completely remove* an electron from *1s orbital of Boron atom (B)*
- iii. Wavelength of photon to *completely remove* an electron from *1s orbital of Boron cation (B⁺)*

i or ii or iii

(II) Select the smallest (shortest) wavelength from the three choices below (*for H atom*):

- i. Wavelength of photon *emitted* for an electron transitioning from *2p to 1s orbital*
- ii. Wavelength of photon *absorbed* for an electron transitioning from *2s to 4p orbital*
- iii. Wavelength of photon *absorbed* for an electron transitioning from *1s to 3p orbital*

i or ii or iii



Question 2 8 points



A Chem110 student plots the graph shown on the left, from data obtained for the kinetic energy (eV) of the ejected electron versus frequency (s^{-1}) of the incident light, for an unknown metal. ($1\text{eV} = 1.6022 \times 10^{-19} \text{ J}$)

Based on the plot, answer the questions **a to c** below:

a. 1 point On the plot above, label the **threshold frequency**.

b. 2 points What is the value of the work potential (in J) of the unknown metal?

c. 3 points What is the *longest* wavelength (in nm) required to eject an electron from this metal's surface? Show your work including equation(s) used. (3 sig figs)

d. 2 points In the space provided below, plot the following

(Assume frequency is *above* threshold frequency):

- Kinetic energy* of an ejected electron versus the *intensity* of the incident light (left)
- Number of electrons* ejected versus the *frequency* of the incident light (right)



**Question 3** **6 points****a. 2 points**

In the ground state of Ge (Germanium; $Z = 32$) atom, **name** the following orbitals:

i. Orbital containing the electron that requires the maximum energy to be removed.

ii. Orbital containing the electron that experiences the minimum effective nuclear charge.

b. 2 points

An unknown anion (A^-) has the following *excited* electronic configuration: $[Ar] 4s^2 3d^3 4p^1$

(Ar: Argon; $Z = 18$)

Give the **condensed ground state electronic configuration** of the atom, **A** and the cation, **A^{2+}** .
Show your work. (Assume all electronic transitions are possible)

Electronic Configuration of A:

Electronic Configuration of A^{2+} :

c. 2 points

In a single atom, determine how many electrons can have the following set of quantum numbers?

(i) $n = 5; l = 3$

In a single atom, determine how many electrons can have the following set of quantum numbers?

(i) $n = 3; l = 2; m_s = +1/2$

**Question 4** **8 points**

Rank the following from 1 to 4 in **increasing** order. Provide the ranking in the boxes provided.

Note: Atomic number (Z) are provided for relevant atoms

a) Increasing (lowest to highest) order of *ionization energy*: (1= lowest; 4 = highest)

Mg

Mg: Z = 12

Rb

Rb: Z = 37

Cs

Cs: Z = 55

K

K : Z = 19

b) Increasing (lowest to highest) order of *bond angle*: (1= lowest; 4 = highest)

H₂OCCl₄XeF₄

HCN

c) Increasing (smallest to largest) order of *ionic size* (1= smallest; 4 = largest)

S²⁻

S: Z = 16

O²⁻

O: Z = 8

Be²⁺

Be: Z = 4

Na⁺

Na: Z = 11

d) Increasing (lowest to highest) order of *bond energy* (1= lowest; 4 = highest)

P-F

P-Cl

P-I

P-Br

**Question 5** **12 points**

a. 9 points For the molecules HClO_4 and TeF_6 , draw the most stable *Lewis structure* showing appropriate shape and *dashed/wedged bonds*, based on VSEPR. Determine the **molecular geometry** around the central atom. For HClO_4 , label the *most polar bond*.



b. 3 points (Note: Lone pairs are not shown in structures below)

(i) For the three *resonance* structures (1, 2 and 3) shown below, answer the following question:



(I) Circle the *most contributing* resonance structure.

Structure 1

Structure 2

Structure 3

(ii) For the three structures given (4, 5, and 6) below, answer the following questions:



(I) Circle all the structures that represent *resonance structures* of each other:

Structure 4

Structure 5

Structure 6

(II) Among the resonance structures, circle the *most contributing* resonance structure.

Structure 4

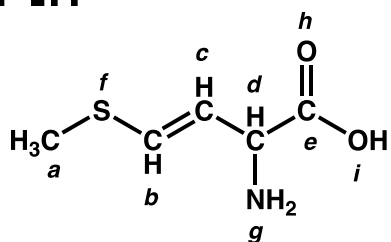
Structure 5

Structure 6

**Question 6** **6 points**

Using the data given below, calculate the **lattice energy** for SrO(s).

	ΔH° (kJ/mol)
$\text{Sr}(s) \longrightarrow \text{Sr}(g)$	+180
$\text{Sr}(g) \longrightarrow \text{Sr}^{2+}(g) + 2e^-$	+1468
$\text{O}_2(g) \longrightarrow 2\text{O}(g)$	+500
$\text{O}(g) + 2e^- \longrightarrow \text{O}^{2-}(g)$	+600
$\text{Sr}(s) + \frac{1}{2}\text{O}_2(g) \longrightarrow \text{SrO}(s)$	-550

**Question 7 10 points**

Consider the molecule given on the left, and answer questions (i) to (vi) using chemical bonding theories that we have learnt in class (valence bond theory, hybridization).

Note that the lone pairs are not shown in the structure:

i. Consider the carbon atoms marked in as ***a***, ***b***, ***c***, ***d***, and ***e*** in the molecule, and determine **hybridization** at each of the carbon centers. (2.5 points)

Atom	Hybridization
Carbon (<i>a</i>)	
Carbon (<i>b</i>)	
Carbon (<i>c</i>)	
Carbon (<i>d</i>)	
Carbon (<i>e</i>)	

ii. Now, consider the sulfur atom (S) marked as ***f***, nitrogen (N) atom (***g***), and oxygen(O) atoms (***h*** and ***i***), and determine their **hybridization**. (2 points)

Atom	Hybridization
Sulfur (<i>f</i>)	
Nitrogen (<i>g</i>)	
Oxygen (<i>h</i>)	
Oxygen (<i>i</i>)	

iii. **Describe** all the bonds formed by the carbon atoms ***a*** and ***b***, using sigma and pi designation as appropriate. (2 points)

Atom	Bonds formed (use sigma and pi as appropriate)
Carbon (<i>a</i>)	
Carbon (<i>b</i>)	

iv. How many total sp^3 -s sigma bonds are in the molecule? (1 point)

v. Determine **electronic** and **molecular geometry** at carbon (C) labeled ***e*** and sulfur (S) labeled ***f*** centres. (2 points)

Atom	Electronic Geometry	Molecular Geometry
Carbon (<i>e</i>)		
Sulfur (<i>f</i>)		

vi. Circle the value from the choices given below, that describes the **C-S-C bond angle at sulfur** (S) marked ***f***: (0.5 points)

>90° to <109.5° 109.5° >109.5 to <120° 120° >120° to <180°

**Question 8 12 points**

Using the **valence shell orbitals only** draw a **complete** molecular orbital (MO) diagram (with no s-p mixing) for the molecule **PS**, in the space provided below. **(6 points)**

Assume that 3s and 3p atomic orbitals will form molecular orbitals like those formed by the combination of 2s and 2p atomic orbitals that we learnt in class, and there is no s-p mixing.

Show **energy arrow**, **atomic orbitals** involved, **atom labels**, the **molecular orbitals** that are formed, **filled electrons** in the MO diagram etc. Make sure to label all the atomic and molecular orbitals.

Using the MO diagram above, answer the following questions:

- (i) Which one of the following will have the highest bond order? Circle the correct choice
A) PS^{1-} B) PS C) PS^{1+}
- (ii) Which one of the following will have the longest bond length? Circle the correct choice
A) PS^{1-} B) PS C) PS^{1+}
- (iii) Which of the following will be diamagnetic? Circle the correct choice.
A) PS^{1-} B) PS C) PS^{1+}
- (iv) Draw Lewis dot structure of the molecule with the shortest bond length (based on the MO diagram) from among the three (PS^{1-} , PS, PS^{1+}), and indicate the bond order calculated from the Lewis dot structure. **(3 points)**

Lewis Dot Structure

Bond Order



1D1B2807-09B4-406F-B0DC-A9476E9FBA7B

midterm-exam-93388

#230 Page 10 of 10

SCRAP PAGE

**CHEMISTRY 110 –Midterm Exam****Thursday November 7, 2024, 6:30 – 8:30 pm**

Examiners: Pallavi Sirjoosingh and Ashok Kakkar

First name (please write as legibly as possible within the boxes)

--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--

Last name

--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--

Student ID number

--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--

This booklet has 8 questions on 10 pages (The last page can be used as a scrap page).

The staple must NOT be removed from the question booklet. Your intact question booklet must be returned at the end of the exam.

A separate booklet contains a periodic table, and datasheet.

Write your answers directly under the questions in the space provided. Do not write on the QR codes or near the top edges (as it may be cut off).

Please write **legibly and dark enough** for the scan to be readable. For students who use English as a second language, dictionaries are permitted. Calculators are allowed.

DO NOT ASK QUESTIONS OF THE INVIGILATORS about interpretations of the exam questions. If you have a concern about a question, please summarize it at the top of this page. All concerns will be evaluated after the exam. GOOD LUCK!

**Question 1 8 points** (*Assume all transitions are possible*)

a. 6 points Calculate the **energy (in kJ/mol)** required to ionize (completely remove) the electron from the **$n=3$** level of **He^+ ion** (He: Helium). Show your work including equations. (3 sig. figs.)

b. 2 points

(I) Select the largest (longest) wavelength from the three choices below:

- i. Wavelength of photon to *completely remove* an electron from *1s orbital of Hydrogen atom*
- ii. Wavelength of photon to *completely remove* an electron from *1s orbital of Boron atom (B)*
- iii. Wavelength of photon to *completely remove* an electron from *1s orbital of Boron cation (B^+)*

i or ii or iii

(II) Select the smallest (shortest) wavelength from the three choices below (*for H atom*):

- i. Wavelength of photon *emitted* for an electron transitioning from *2p to 1s orbital*
- ii. Wavelength of photon *absorbed* for an electron transitioning from *2s to 4p orbital*
- iii. Wavelength of photon *absorbed* for an electron transitioning from *1s to 3p orbital*

i or ii or iii



Question 2 8 points



A Chem110 student plots the graph shown on the left, from data obtained for the kinetic energy (eV) of the ejected electron versus frequency (s^{-1}) of the incident light, for an unknown metal. ($1\text{eV} = 1.6022 \times 10^{-19} \text{ J}$)

Based on the plot, answer the questions **a to c** below:

a. 1 point On the plot above, label the **threshold frequency**.

b. 2 points What is the value of the work potential (in J) of the unknown metal?

c. 3 points What is the *longest* wavelength (in nm) required to eject an electron from this metal's surface? Show your work including equation(s) used. (3 sig figs)

d. 2 points In the space provided below, plot the following

(Assume frequency is *above* threshold frequency):

- Kinetic energy* of an ejected electron versus the *intensity* of the incident light (left)
- Number of electrons* ejected versus the *frequency* of the incident light (right)



**Question 3** **6 points****a. 2 points**

In the ground state of Ge (Germanium; $Z = 32$) atom, **name** the following orbitals:

i. Orbital containing the electron that requires the maximum energy to be removed.

ii. Orbital containing the electron that experiences the minimum effective nuclear charge.

b. 2 points

An unknown anion (A^-) has the following *excited* electronic configuration: $[Ar] 4s^2 3d^3 4p^1$

(Ar: Argon; $Z = 18$)

Give the **condensed ground state electronic configuration** of the atom, **A** and the cation, **A^{2+}** .
Show your work. (Assume all electronic transitions are possible)

Electronic Configuration of A:

Electronic Configuration of A^{2+} :

c. 2 points

In a single atom, determine how many electrons can have the following set of quantum numbers?

(i) $n = 5; l = 3$

In a single atom, determine how many electrons can have the following set of quantum numbers?

(i) $n = 3; l = 2; m_s = +1/2$

**Question 4** **8 points**

Rank the following from 1 to 4 in **increasing** order. Provide the ranking in the boxes provided.

Note: Atomic number (Z) are provided for relevant atoms

a) Increasing (lowest to highest) order of *ionization energy*: (1= lowest; 4 = highest)

Mg

Mg: Z = 12

Rb

Rb: Z = 37

Cs

Cs: Z = 55

K

K : Z = 19

b) Increasing (lowest to highest) order of *bond angle*: (1= lowest; 4 = highest)

H₂OCCl₄XeF₄

HCN

c) Increasing (smallest to largest) order of *ionic size* (1= smallest; 4 = largest)

S²⁻

S: Z = 16

O²⁻

O: Z = 8

Be²⁺

Be: Z = 4

Na⁺

Na: Z = 11

d) Increasing (lowest to highest) order of *bond energy* (1= lowest; 4 = highest)

P-F

P-Cl

P-I

P-Br

**Question 5** **12 points**

a. 9 points For the molecules HClO_4 and TeF_6 , draw the most stable *Lewis structure* showing appropriate shape and *dashed/wedged bonds*, based on VSEPR. Determine the **molecular geometry** around the central atom. For HClO_4 , label the *most polar bond*.



b. 3 points (Note: Lone pairs are not shown in structures below)

(i) For the three *resonance* structures (1, 2 and 3) shown below, answer the following question:



(I) Circle the *most contributing* resonance structure.

Structure 1

Structure 2

Structure 3

(ii) For the three structures given (4, 5, and 6) below, answer the following questions:



(I) Circle all the structures that represent *resonance structures* of each other:

Structure 4

Structure 5

Structure 6

(II) Among the resonance structures, circle the *most contributing* resonance structure.

Structure 4

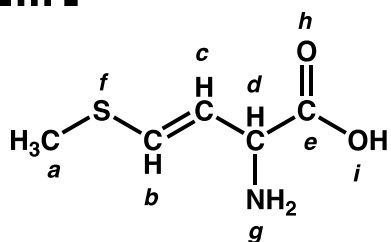
Structure 5

Structure 6

**Question 6** **6 points**

Using the data given below, calculate the **lattice energy** for SrO(s).

	ΔH° (kJ/mol)
$\text{Sr}(s) \longrightarrow \text{Sr}(g)$	+180
$\text{Sr}(g) \longrightarrow \text{Sr}^{2+}(g) + 2e^-$	+1468
$\text{O}_2(g) \longrightarrow 2\text{O}(g)$	+500
$\text{O}(g) + 2e^- \longrightarrow \text{O}^{2-}(g)$	+600
$\text{Sr}(s) + \frac{1}{2}\text{O}_2(g) \longrightarrow \text{SrO}(s)$	-550

**Question 7 10 points**

Consider the molecule given on the left, and answer questions (i) to (vi) using chemical bonding theories that we have learnt in class (valence bond theory, hybridization).

Note that the lone pairs are not shown in the structure:

i. Consider the carbon atoms marked in as ***a***, ***b***, ***c***, ***d***, and ***e*** in the molecule, and determine **hybridization** at each of the carbon centers. (2.5 points)

Atom	Hybridization
Carbon (<i>a</i>)	
Carbon (<i>b</i>)	
Carbon (<i>c</i>)	
Carbon (<i>d</i>)	
Carbon (<i>e</i>)	

ii. Now, consider the sulfur atom (S) marked as ***f***, nitrogen (N) atom (***g***), and oxygen(O) atoms (***h*** and ***i***), and determine their **hybridization**. (2 points)

Atom	Hybridization
Sulfur (<i>f</i>)	
Nitrogen (<i>g</i>)	
Oxygen (<i>h</i>)	
Oxygen (<i>i</i>)	

iii. **Describe** all the bonds formed by the carbon atoms ***a*** and ***b***, using sigma and pi designation as appropriate. (2 points)

Atom	Bonds formed (use sigma and pi as appropriate)
Carbon (<i>a</i>)	
Carbon (<i>b</i>)	

iv. How many total sp^3 -s sigma bonds are in the molecule? (1 point)

v. Determine **electronic** and **molecular geometry** at carbon (C) labeled ***e*** and sulfur (S) labeled ***f*** centres. (2 points)

Atom	Electronic Geometry	Molecular Geometry
Carbon (<i>e</i>)		
Sulfur (<i>f</i>)		

vi. Circle the value from the choices given below, that describes the **C-S-C bond angle at sulfur** (S) marked ***f***: (0.5 points)

>90° to <109.5°

109.5°

>109.5 to <120°

120°

>120° to <180°


Question 8 12 points

Using the **valence shell orbitals only** draw a **complete** molecular orbital (MO) diagram (with no s-p mixing) for the molecule **PS**, in the space provided below. **(6 points)**

Assume that 3s and 3p atomic orbitals will form molecular orbitals like those formed by the combination of 2s and 2p atomic orbitals that we learnt in class, and there is no s-p mixing.

Show **energy arrow**, **atomic orbitals** involved, **atom labels**, the **molecular orbitals** that are formed, **filled electrons** in the MO diagram etc. Make sure to label all the atomic and molecular orbitals.

Using the MO diagram above, answer the following questions:

- (i) Which one of the following will have the highest bond order? Circle the correct choice
 A) PS^{1-} B) PS C) PS^{1+}
- (ii) Which one of the following will have the longest bond length? Circle the correct choice
 A) PS^{1-} B) PS C) PS^{1+}
- (iii) Which of the following will be diamagnetic? Circle the correct choice.
 A) PS^{1-} B) PS C) PS^{1+}
- (iv) Draw Lewis dot structure of the molecule with the shortest bond length (based on the MO diagram) from among the three (PS^{1-} , PS, PS^{1+}), and indicate the bond order calculated from the Lewis dot structure. **(3 points)**

Lewis Dot Structure

Bond Order



B7802DAA-3D74-48CB-BCE4-77862339A78E

midterm-exam-93388

#231 Page 10 of 10

SCRAP PAGE

**CHEMISTRY 110 –Midterm Exam****Thursday November 7, 2024, 6:30 – 8:30 pm**

Examiners: Pallavi Sirjoosingh and Ashok Kakkar

First name (please write as legibly as possible within the boxes)

--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--

Last name

--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--

Student ID number

--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--

This booklet has 8 questions on 10 pages (The last page can be used as a scrap page).

The staple must NOT be removed from the question booklet. Your intact question booklet must be returned at the end of the exam.

A separate booklet contains a periodic table, and datasheet.

Write your answers directly under the questions in the space provided. Do not write on the QR codes or near the top edges (as it may be cut off).

Please write **legibly and dark enough** for the scan to be readable. For students who use English as a second language, dictionaries are permitted. Calculators are allowed.

DO NOT ASK QUESTIONS OF THE INVIGILATORS about interpretations of the exam questions. If you have a concern about a question, please summarize it at the top of this page. All concerns will be evaluated after the exam. GOOD LUCK!

**Question 1 8 points** (*Assume all transitions are possible*)

a. 6 points Calculate the **energy (in kJ/mol)** required to ionize (completely remove) the electron from the **n=3** level of **He⁺ ion** (He: Helium). Show your work including equations. (3 sig. figs.)

b. 2 points

(I) Select the largest (longest) wavelength from the three choices below:

- i. Wavelength of photon to *completely remove* an electron from *1s orbital of Hydrogen atom*
- ii. Wavelength of photon to *completely remove* an electron from *1s orbital of Boron atom (B)*
- iii. Wavelength of photon to *completely remove* an electron from *1s orbital of Boron cation (B⁺)*

i or ii or iii

(II) Select the smallest (shortest) wavelength from the three choices below (*for H atom*):

- i. Wavelength of photon *emitted* for an electron transitioning from *2p to 1s orbital*
- ii. Wavelength of photon *absorbed* for an electron transitioning from *2s to 4p orbital*
- iii. Wavelength of photon *absorbed* for an electron transitioning from *1s to 3p orbital*

i or ii or iii



Question 2 8 points



A Chem110 student plots the graph shown on the left, from data obtained for the kinetic energy (eV) of the ejected electron versus frequency (s^{-1}) of the incident light, for an unknown metal. ($1\text{eV} = 1.6022 \times 10^{-19} \text{ J}$)

Based on the plot, answer the questions **a to c** below:

a. 1 point On the plot above, label the **threshold frequency**.

b. 2 points What is the value of the work potential (in J) of the unknown metal?

c. 3 points What is the *longest* wavelength (in nm) required to eject an electron from this metal's surface? Show your work including equation(s) used. (3 sig figs)

d. 2 points In the space provided below, plot the following

(Assume frequency is *above* threshold frequency):

- i. *Kinetic energy* of an ejected electron versus the *intensity* of the incident light (left)
- ii. *Number of electrons* ejected versus the *frequency* of the incident light (right)



**Question 3** **6 points****a. 2 points**

In the ground state of Ge (Germanium; $Z = 32$) atom, **name** the following orbitals:

i. Orbital containing the electron that requires the maximum energy to be removed.

ii. Orbital containing the electron that experiences the minimum effective nuclear charge.

b. 2 points

An unknown anion (A^-) has the following *excited* electronic configuration: $[Ar] 4s^2 3d^3 4p^1$

(Ar: Argon; $Z = 18$)

Give the **condensed ground state electronic configuration** of the atom, **A** and the cation, **A^{2+}** .
Show your work. (Assume all electronic transitions are possible)

Electronic Configuration of A:

Electronic Configuration of A^{2+} :

c. 2 points

In a single atom, determine how many electrons can have the following set of quantum numbers?

(i) $n = 5; l = 3$

In a single atom, determine how many electrons can have the following set of quantum numbers?

(i) $n = 3; l = 2; m_s = +1/2$

**Question 4** **8 points**

Rank the following from 1 to 4 in **increasing** order. Provide the ranking in the boxes provided.

Note: Atomic number (Z) are provided for relevant atoms

a) Increasing (lowest to highest) order of *ionization energy*: (1= lowest; 4 = highest)

Mg

Mg: Z = 12

Rb

Rb: Z = 37

Cs

Cs: Z = 55

K

K : Z = 19

b) Increasing (lowest to highest) order of *bond angle*: (1= lowest; 4 = highest)

H₂OCCl₄XeF₄

HCN

c) Increasing (smallest to largest) order of *ionic size* (1= smallest; 4 = largest)

S²⁻

S: Z = 16

O²⁻

O: Z = 8

Be²⁺

Be: Z = 4

Na⁺

Na: Z = 11

d) Increasing (lowest to highest) order of *bond energy* (1= lowest; 4 = highest)

P-F

P-Cl

P-I

P-Br

**Question 5** **12 points**

a. 9 points For the molecules HClO_4 and TeF_6 , draw the most stable *Lewis structure* showing appropriate shape and *dashed/wedged bonds*, based on VSEPR. Determine the **molecular geometry** around the central atom. For HClO_4 , label the *most polar bond*.



b. 3 points (Note: Lone pairs are not shown in structures below)

(i) For the three *resonance* structures (1, 2 and 3) shown below, answer the following question:



(I) Circle the *most contributing* resonance structure.

Structure 1

Structure 2

Structure 3

(ii) For the three structures given (4, 5, and 6) below, answer the following questions:



(I) Circle all the structures that represent *resonance structures* of each other:

Structure 4

Structure 5

Structure 6

(II) Among the resonance structures, circle the *most contributing* resonance structure.

Structure 4

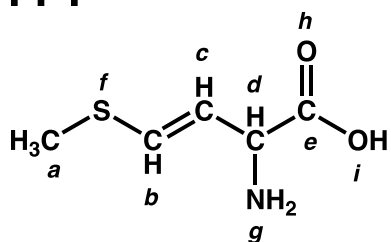
Structure 5

Structure 6

**Question 6** **6 points**

Using the data given below, calculate the **lattice energy** for SrO(s).

	ΔH° (kJ/mol)
$\text{Sr}(s) \longrightarrow \text{Sr}(g)$	+180
$\text{Sr}(g) \longrightarrow \text{Sr}^{2+}(g) + 2e^-$	+1468
$\text{O}_2(g) \longrightarrow 2\text{O}(g)$	+500
$\text{O}(g) + 2e^- \longrightarrow \text{O}^{2-}(g)$	+600
$\text{Sr}(s) + \frac{1}{2}\text{O}_2(g) \longrightarrow \text{SrO}(s)$	-550

**Question 7 10 points**

Consider the molecule given on the left, and answer questions (i) to (vi) using chemical bonding theories that we have learnt in class (valence bond theory, hybridization).

Note that the lone pairs are not shown in the structure:

i. Consider the carbon atoms marked in as ***a***, ***b***, ***c***, ***d***, and ***e*** in the molecule, and determine **hybridization** at each of the carbon centers. (2.5 points)

Atom	Hybridization
Carbon (<i>a</i>)	
Carbon (<i>b</i>)	
Carbon (<i>c</i>)	
Carbon (<i>d</i>)	
Carbon (<i>e</i>)	

ii. Now, consider the sulfur atom (S) marked as ***f***, nitrogen (N) atom (***g***), and oxygen(O) atoms (***h*** and ***i***), and determine their **hybridization**. (2 points)

Atom	Hybridization
Sulfur (<i>f</i>)	
Nitrogen (<i>g</i>)	
Oxygen (<i>h</i>)	
Oxygen (<i>i</i>)	

iii. **Describe** all the bonds formed by the carbon atoms ***a*** and ***b***, using sigma and pi designation as appropriate. (2 points)

Atom	Bonds formed (use sigma and pi as appropriate)
Carbon (<i>a</i>)	
Carbon (<i>b</i>)	

iv. How many total sp^3 -s sigma bonds are in the molecule? (1 point)

v. Determine **electronic** and **molecular geometry** at carbon (C) labeled ***e*** and sulfur (S) labeled ***f*** centres. (2 points)

Atom	Electronic Geometry	Molecular Geometry
Carbon (<i>e</i>)		
Sulfur (<i>f</i>)		

vi. Circle the value from the choices given below, that describes the **C-S-C bond angle at sulfur** (S) marked ***f***: (0.5 points)

>90° to <109.5°

109.5°

>109.5 to <120°

120°

>120° to <180°


Question 8 12 points

Using the **valence shell orbitals only** draw a **complete** molecular orbital (MO) diagram (with no s-p mixing) for the molecule **PS**, in the space provided below. **(6 points)**

Assume that 3s and 3p atomic orbitals will form molecular orbitals like those formed by the combination of 2s and 2p atomic orbitals that we learnt in class, and there is no s-p mixing.

Show **energy arrow**, **atomic orbitals** involved, **atom labels**, the **molecular orbitals** that are formed, **filled electrons** in the MO diagram etc. Make sure to label all the atomic and molecular orbitals.

Using the MO diagram above, answer the following questions:

- (i) Which one of the following will have the highest bond order? Circle the correct choice
 A) PS^{1-} B) PS C) PS^{1+}
- (ii) Which one of the following will have the longest bond length? Circle the correct choice
 A) PS^{1-} B) PS C) PS^{1+}
- (iii) Which of the following will be diamagnetic? Circle the correct choice.
 A) PS^{1-} B) PS C) PS^{1+}
- (iv) Draw Lewis dot structure of the molecule with the shortest bond length (based on the MO diagram) from among the three (PS^{1-} , PS, PS^{1+}), and indicate the bond order calculated from the Lewis dot structure. **(3 points)**

Lewis Dot Structure

Bond Order



E133DA21-62C0-4ECE-B412-51F0A03B1092

midterm-exam-93388

#232 Page 10 of 10

SCRAP PAGE

**CHEMISTRY 110 –Midterm Exam****Thursday November 7, 2024, 6:30 – 8:30 pm****Examiners: Pallavi Sirjoosingh and Ashok Kakkar**

First name (please write as legibly as possible within the boxes)

--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--

Last name

--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--

Student ID number

--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--

This booklet has 8 questions on 10 pages (The last page can be used as a scrap page).

The staple must NOT be removed from the question booklet. Your intact question booklet must be returned at the end of the exam.

A separate booklet contains a periodic table, and datasheet.

Write your answers directly under the questions in the space provided. Do not write on the QR codes or near the top edges (as it may be cut off).

Please write **legibly and dark enough** for the scan to be readable. For students who use English as a second language, dictionaries are permitted. Calculators are allowed.

DO NOT ASK QUESTIONS OF THE INVIGILATORS about interpretations of the exam questions. If you have a concern about a question, please summarize it at the top of this page. All concerns will be evaluated after the exam. GOOD LUCK!

**Question 1 8 points** (*Assume all transitions are possible*)

a. 6 points Calculate the **energy (in kJ/mol)** required to ionize (completely remove) the electron from the **$n=3$** level of **He^+ ion** (He: Helium). Show your work including equations. (3 sig. figs.)

b. 2 points

(I) Select the largest (longest) wavelength from the three choices below:

- i. Wavelength of photon to *completely remove* an electron from *1s orbital of Hydrogen atom*
- ii. Wavelength of photon to *completely remove* an electron from *1s orbital of Boron atom (B)*
- iii. Wavelength of photon to *completely remove* an electron from *1s orbital of Boron cation (B^+)*

i or ii or iii

(II) Select the smallest (shortest) wavelength from the three choices below (*for H atom*):

- i. Wavelength of photon *emitted* for an electron transitioning from *2p to 1s orbital*
- ii. Wavelength of photon *absorbed* for an electron transitioning from *2s to 4p orbital*
- iii. Wavelength of photon *absorbed* for an electron transitioning from *1s to 3p orbital*

i or ii or iii



Question 2 8 points



A Chem110 student plots the graph shown on the left, from data obtained for the kinetic energy (eV) of the ejected electron versus frequency (s^{-1}) of the incident light, for an unknown metal. ($1\text{eV} = 1.6022 \times 10^{-19} \text{ J}$)

Based on the plot, answer the questions **a to c** below:

a. 1 point On the plot above, label the **threshold frequency**.

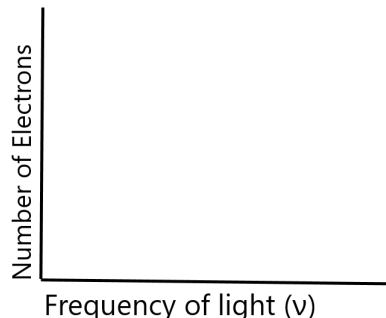
b. 2 points What is the value of the work potential (in J) of the unknown metal?

c. 3 points What is the *longest* wavelength (in nm) required to eject an electron from this metal's surface? Show your work including equation(s) used. (3 sig figs)

d. 2 points In the space provided below, plot the following

(Assume frequency is *above* threshold frequency):

- i. *Kinetic energy* of an ejected electron versus the *intensity* of the incident light (left)
- ii. *Number of electrons* ejected versus the *frequency* of the incident light (right)



**Question 3** **6 points****a. 2 points**

In the ground state of Ge (Germanium; $Z = 32$) atom, **name** the following orbitals:

i. Orbital containing the electron that requires the maximum energy to be removed.

ii. Orbital containing the electron that experiences the minimum effective nuclear charge.

b. 2 points

An unknown anion (A^-) has the following *excited* electronic configuration: $[Ar] 4s^2 3d^3 4p^1$

(Ar: Argon; $Z = 18$)

Give the **condensed ground state electronic configuration** of the atom, **A** and the cation, **A^{2+}** .
Show your work. (Assume all electronic transitions are possible)

Electronic Configuration of A:

Electronic Configuration of A^{2+} :

c. 2 points

In a single atom, determine how many electrons can have the following set of quantum numbers?

(i) $n = 5; l = 3$

In a single atom, determine how many electrons can have the following set of quantum numbers?

(i) $n = 3; l = 2; m_s = +1/2$

**Question 4** **8 points**

Rank the following from 1 to 4 in **increasing** order. Provide the ranking in the boxes provided.

Note: Atomic number (Z) are provided for relevant atoms

a) Increasing (lowest to highest) order of *ionization energy*: (1= lowest; 4 = highest)

Mg

Mg: Z = 12

Rb

Rb: Z = 37

Cs

Cs: Z = 55

K

K : Z = 19

b) Increasing (lowest to highest) order of *bond angle*: (1= lowest; 4 = highest)

H₂OCCl₄XeF₄

HCN

c) Increasing (smallest to largest) order of *ionic size* (1= smallest; 4 = largest)

S²⁻

S: Z = 16

O²⁻

O: Z = 8

Be²⁺

Be: Z = 4

Na⁺

Na: Z = 11

d) Increasing (lowest to highest) order of *bond energy* (1= lowest; 4 = highest)

P-F

P-Cl

P-I

P-Br

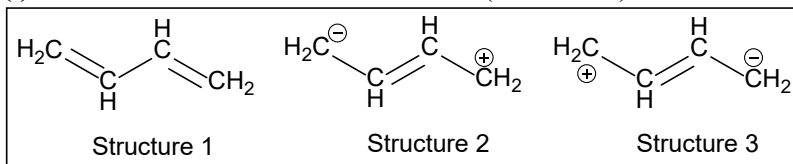
**Question 5** **12 points**

a. 9 points For the molecules HClO_4 and TeF_6 , draw the most stable *Lewis structure* showing appropriate shape and *dashed/wedged bonds*, based on VSEPR. Determine the **molecular geometry** around the central atom. For HClO_4 , label the *most polar bond*.



b. 3 points (Note: Lone pairs are not shown in structures below)

(i) For the three *resonance* structures (1, 2 and 3) shown below, answer the following question:



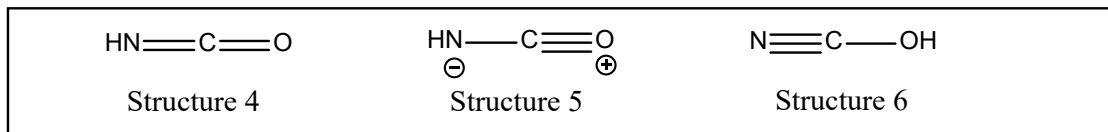
(I) Circle the *most contributing* resonance structure.

Structure 1

Structure 2

Structure 3

(ii) For the three structures given (4, 5, and 6) below, answer the following questions:



(I) Circle all the structures that represent *resonance structures* of each other:

Structure 4

Structure 5

Structure 6

(II) Among the resonance structures, circle the *most contributing* resonance structure.

Structure 4

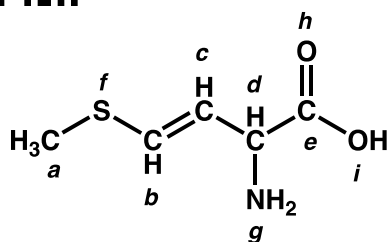
Structure 5

Structure 6

**Question 6** **6 points**

Using the data given below, calculate the **lattice energy** for SrO(s).

	ΔH° (kJ/mol)
$\text{Sr}(s) \longrightarrow \text{Sr}(g)$	+180
$\text{Sr}(g) \longrightarrow \text{Sr}^{2+}(g) + 2e^-$	+1468
$\text{O}_2(g) \longrightarrow 2\text{O}(g)$	+500
$\text{O}(g) + 2e^- \longrightarrow \text{O}^{2-}(g)$	+600
$\text{Sr}(s) + \frac{1}{2}\text{O}_2(g) \longrightarrow \text{SrO}(s)$	-550

**Question 7 10 points**

Consider the molecule given on the left, and answer questions (i) to (vi) using chemical bonding theories that we have learnt in class (valence bond theory, hybridization).

Note that the lone pairs are not shown in the structure:

i. Consider the carbon atoms marked in as ***a***, ***b***, ***c***, ***d***, and ***e*** in the molecule, and determine **hybridization** at each of the carbon centers. (2.5 points)

Atom	Hybridization
Carbon (<i>a</i>)	
Carbon (<i>b</i>)	
Carbon (<i>c</i>)	
Carbon (<i>d</i>)	
Carbon (<i>e</i>)	

ii. Now, consider the sulfur atom (S) marked as ***f***, nitrogen (N) atom (***g***), and oxygen(O) atoms (***h*** and ***i***), and determine their **hybridization**. (2 points)

Atom	Hybridization
Sulfur (<i>f</i>)	
Nitrogen (<i>g</i>)	
Oxygen (<i>h</i>)	
Oxygen (<i>i</i>)	

iii. **Describe** all the bonds formed by the carbon atoms ***a*** and ***b***, using sigma and pi designation as appropriate. (2 points)

Atom	Bonds formed (use sigma and pi as appropriate)
Carbon (<i>a</i>)	
Carbon (<i>b</i>)	

iv. How many total sp^3 -s sigma bonds are in the molecule? (1 point)

v. Determine **electronic** and **molecular geometry** at carbon (C) labeled ***e*** and sulfur (S) labeled ***f*** centres. (2 points)

Atom	Electronic Geometry	Molecular Geometry
Carbon (<i>e</i>)		
Sulfur (<i>f</i>)		

vi. Circle the value from the choices given below, that describes the **C-S-C bond angle at sulfur** (S) marked ***f***: (0.5 points)

>90° to <109.5° 109.5° >109.5 to <120° 120° >120° to <180°


Question 8 12 points

Using the **valence shell orbitals only** draw a **complete** molecular orbital (MO) diagram (with no s-p mixing) for the molecule **PS**, in the space provided below. **(6 points)**

Assume that 3s and 3p atomic orbitals will form molecular orbitals like those formed by the combination of 2s and 2p atomic orbitals that we learnt in class, and there is no s-p mixing.

Show **energy arrow**, **atomic orbitals** involved, **atom labels**, the **molecular orbitals** that are formed, **filled electrons** in the MO diagram etc. Make sure to label all the atomic and molecular orbitals.

Using the MO diagram above, answer the following questions:

- (i) Which one of the following will have the highest bond order? Circle the correct choice
 A) PS^{1-} B) PS C) PS^{1+}
- (ii) Which one of the following will have the longest bond length? Circle the correct choice
 A) PS^{1-} B) PS C) PS^{1+}
- (iii) Which of the following will be diamagnetic? Circle the correct choice.
 A) PS^{1-} B) PS C) PS^{1+}
- (iv) Draw Lewis dot structure of the molecule with the shortest bond length (based on the MO diagram) from among the three (PS^{1-} , PS, PS^{1+}), and indicate the bond order calculated from the Lewis dot structure. **(3 points)**

Lewis Dot Structure

Bond Order



6BA428BE-73D9-4045-B256-D647CBEA59ED

midterm-exam-93388

#233 Page 10 of 10

SCRAP PAGE

**CHEMISTRY 110 –Midterm Exam****Thursday November 7, 2024, 6:30 – 8:30 pm****Examiners: Pallavi Sirjoosingh and Ashok Kakkar**

First name (please write as legibly as possible within the boxes)

--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--

Last name

--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--

Student ID number

--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--

This booklet has 8 questions on 10 pages (The last page can be used as a scrap page).

The staple must NOT be removed from the question booklet. Your intact question booklet must be returned at the end of the exam.

A separate booklet contains a periodic table, and datasheet.

Write your answers directly under the questions in the space provided. Do not write on the QR codes or near the top edges (as it may be cut off).

Please write **legibly and dark enough** for the scan to be readable. For students who use English as a second language, dictionaries are permitted. Calculators are allowed.

DO NOT ASK QUESTIONS OF THE INVIGILATORS about interpretations of the exam questions. If you have a concern about a question, please summarize it at the top of this page. All concerns will be evaluated after the exam. GOOD LUCK!

**Question 1 8 points** (*Assume all transitions are possible*)

a. 6 points Calculate the **energy (in kJ/mol)** required to ionize (completely remove) the electron from the **$n=3$** level of **He^+ ion** (He: Helium). Show your work including equations. (3 sig. figs.)

b. 2 points

(I) Select the largest (longest) wavelength from the three choices below:

- i. Wavelength of photon to *completely remove* an electron from *1s orbital of Hydrogen atom*
- ii. Wavelength of photon to *completely remove* an electron from *1s orbital of Boron atom (B)*
- iii. Wavelength of photon to *completely remove* an electron from *1s orbital of Boron cation (B^+)*

i or ii or iii

(II) Select the smallest (shortest) wavelength from the three choices below (*for H atom*):

- i. Wavelength of photon *emitted* for an electron transitioning from *2p to 1s orbital*
- ii. Wavelength of photon *absorbed* for an electron transitioning from *2s to 4p orbital*
- iii. Wavelength of photon *absorbed* for an electron transitioning from *1s to 3p orbital*

i or ii or iii



Question 2 8 points



A Chem110 student plots the graph shown on the left, from data obtained for the kinetic energy (eV) of the ejected electron versus frequency (s^{-1}) of the incident light, for an unknown metal. ($1\text{eV} = 1.6022 \times 10^{-19} \text{ J}$)

Based on the plot, answer the questions **a to c** below:

a. 1 point On the plot above, label the **threshold frequency**.

b. 2 points What is the value of the work potential (in J) of the unknown metal?

c. 3 points What is the *longest* wavelength (in nm) required to eject an electron from this metal's surface? Show your work including equation(s) used. (3 sig figs)

d. 2 points In the space provided below, plot the following

(Assume frequency is *above* threshold frequency):

- Kinetic energy* of an ejected electron versus the *intensity* of the incident light (left)
- Number of electrons* ejected versus the *frequency* of the incident light (right)



**Question 3** **6 points****a. 2 points**

In the ground state of Ge (Germanium; $Z = 32$) atom, **name** the following orbitals:

i. Orbital containing the electron that requires the maximum energy to be removed.

ii. Orbital containing the electron that experiences the minimum effective nuclear charge.

b. 2 points

An unknown anion (A^-) has the following *excited* electronic configuration: $[Ar] 4s^2 3d^3 4p^1$

(Ar: Argon; $Z = 18$)

Give the **condensed ground state electronic configuration** of the atom, **A** and the cation, **A^{2+}** .
Show your work. (Assume all electronic transitions are possible)

Electronic Configuration of A:

Electronic Configuration of A^{2+} :

c. 2 points

In a single atom, determine how many electrons can have the following set of quantum numbers?

(i) $n = 5; l = 3$

In a single atom, determine how many electrons can have the following set of quantum numbers?

(i) $n = 3; l = 2; m_s = +1/2$

**Question 4** **8 points**

Rank the following from 1 to 4 in **increasing** order. Provide the ranking in the boxes provided.

Note: Atomic number (Z) are provided for relevant atoms

a) Increasing (lowest to highest) order of *ionization energy*: (1= lowest; 4 = highest)

Mg

Mg: Z = 12

Rb

Rb: Z = 37

Cs

Cs: Z = 55

K

K : Z = 19

b) Increasing (lowest to highest) order of *bond angle*: (1= lowest; 4 = highest)

H₂OCCl₄XeF₄

HCN

c) Increasing (smallest to largest) order of *ionic size* (1= smallest; 4 = largest)

S²⁻

S: Z = 16

O²⁻

O: Z = 8

Be²⁺

Be: Z = 4

Na⁺

Na: Z = 11

d) Increasing (lowest to highest) order of *bond energy* (1= lowest; 4 = highest)

P-F

P-Cl

P-I

P-Br

**Question 5** **12 points**

a. 9 points For the molecules HClO_4 and TeF_6 , draw the most stable *Lewis structure* showing appropriate shape and *dashed/wedged bonds*, based on VSEPR. Determine the **molecular geometry** around the central atom. For HClO_4 , label the *most polar bond*.



b. 3 points (Note: Lone pairs are not shown in structures below)

(i) For the three *resonance* structures (1, 2 and 3) shown below, answer the following question:



(I) Circle the *most contributing* resonance structure.

Structure 1

Structure 2

Structure 3

(ii) For the three structures given (4, 5, and 6) below, answer the following questions:



(I) Circle all the structures that represent *resonance structures* of each other:

Structure 4

Structure 5

Structure 6

(II) Among the resonance structures, circle the *most contributing* resonance structure.

Structure 4

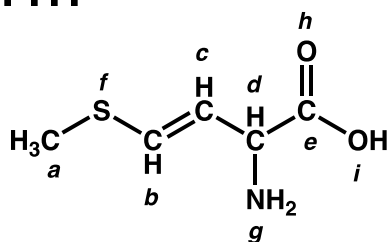
Structure 5

Structure 6

**Question 6** **6 points**

Using the data given below, calculate the **lattice energy** for SrO(s).

	ΔH° (kJ/mol)
$\text{Sr}(s) \longrightarrow \text{Sr}(g)$	+180
$\text{Sr}(g) \longrightarrow \text{Sr}^{2+}(g) + 2e^-$	+1468
$\text{O}_2(g) \longrightarrow 2\text{O}(g)$	+500
$\text{O}(g) + 2e^- \longrightarrow \text{O}^{2-}(g)$	+600
$\text{Sr}(s) + \frac{1}{2}\text{O}_2(g) \longrightarrow \text{SrO}(s)$	-550

**Question 7 10 points**

Consider the molecule given on the left, and answer questions (i) to (vi) using chemical bonding theories that we have learnt in class (valence bond theory, hybridization).

Note that the lone pairs are not shown in the structure:

i. Consider the carbon atoms marked in as ***a***, ***b***, ***c***, ***d***, and ***e*** in the molecule, and determine **hybridization** at each of the carbon centers. (2.5 points)

Atom	Hybridization
Carbon (<i>a</i>)	
Carbon (<i>b</i>)	
Carbon (<i>c</i>)	
Carbon (<i>d</i>)	
Carbon (<i>e</i>)	

ii. Now, consider the sulfur atom (S) marked as ***f***, nitrogen (N) atom (***g***), and oxygen(O) atoms (***h*** and ***i***), and determine their **hybridization**. (2 points)

Atom	Hybridization
Sulfur (<i>f</i>)	
Nitrogen (<i>g</i>)	
Oxygen (<i>h</i>)	
Oxygen (<i>i</i>)	

iii. **Describe** all the bonds formed by the carbon atoms ***a*** and ***b***, using sigma and pi designation as appropriate. (2 points)

Atom	Bonds formed (use sigma and pi as appropriate)
Carbon (<i>a</i>)	
Carbon (<i>b</i>)	

iv. How many total sp^3 -s sigma bonds are in the molecule? (1 point)

v. Determine **electronic** and **molecular geometry** at carbon (C) labeled ***e*** and sulfur (S) labeled ***f*** centres. (2 points)

Atom	Electronic Geometry	Molecular Geometry
Carbon (<i>e</i>)		
Sulfur (<i>f</i>)		

vi. Circle the value from the choices given below, that describes the **C-S-C bond angle at sulfur** (S) marked ***f***: (0.5 points)

>90° to <109.5°

109.5°

>109.5 to <120°

120°

>120° to <180°



Question 8 12 points

Using the **valence shell orbitals only** draw a **complete** molecular orbital (MO) diagram (with no s-p mixing) for the molecule **PS**, in the space provided below. **(6 points)**

Assume that 3s and 3p atomic orbitals will form molecular orbitals like those formed by the combination of 2s and 2p atomic orbitals that we learnt in class, and there is no s-p mixing.

Show **energy arrow**, **atomic orbitals** involved, **atom labels**, the **molecular orbitals** that are formed, **filled electrons** in the MO diagram etc. Make sure to label all the atomic and molecular orbitals.

Using the MO diagram above, answer the following questions:

- (i) Which one of the following will have the highest bond order? Circle the correct choice
 A) PS^{1-} B) PS C) PS^{1+}
- (ii) Which one of the following will have the longest bond length? Circle the correct choice
 A) PS^{1-} B) PS C) PS^{1+}
- (iii) Which of the following will be diamagnetic? Circle the correct choice.
 A) PS^{1-} B) PS C) PS^{1+}
- (iv) Draw Lewis dot structure of the molecule with the shortest bond length (based on the MO diagram) from among the three (PS^{1-} , PS, PS^{1+}), and indicate the bond order calculated from the Lewis dot structure. **(3 points)**

Lewis Dot Structure

Bond Order



90653702-4DC7-4BC5-9E86-225AFB7C6695

midterm-exam-93388

#234 Page 10 of 10

SCRAP PAGE

**CHEMISTRY 110 –Midterm Exam****Thursday November 7, 2024, 6:30 – 8:30 pm****Examiners: Pallavi Sirjoosingh and Ashok Kakkar**

First name (please write as legibly as possible within the boxes)

--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--

Last name

--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--

Student ID number

--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--

This booklet has 8 questions on 10 pages (The last page can be used as a scrap page).

The staple must NOT be removed from the question booklet. Your intact question booklet must be returned at the end of the exam.

A separate booklet contains a periodic table, and datasheet.

Write your answers directly under the questions in the space provided. Do not write on the QR codes or near the top edges (as it may be cut off).

Please write **legibly and dark enough** for the scan to be readable. For students who use English as a second language, dictionaries are permitted. Calculators are allowed.

DO NOT ASK QUESTIONS OF THE INVIGILATORS about interpretations of the exam questions. If you have a concern about a question, please summarize it at the top of this page. All concerns will be evaluated after the exam. GOOD LUCK!

**Question 1 8 points** (*Assume all transitions are possible*)

a. 6 points Calculate the **energy (in kJ/mol)** required to ionize (completely remove) the electron from the **n=3** level of **He⁺ ion** (He: Helium). Show your work including equations. (3 sig. figs.)

b. 2 points

(I) Select the largest (longest) wavelength from the three choices below:

- i. Wavelength of photon to *completely remove* an electron from *1s orbital of Hydrogen atom*
- ii. Wavelength of photon to *completely remove* an electron from *1s orbital of Boron atom (B)*
- iii. Wavelength of photon to *completely remove* an electron from *1s orbital of Boron cation (B⁺)*

i or ii or iii

(II) Select the smallest (shortest) wavelength from the three choices below (*for H atom*):

- i. Wavelength of photon *emitted* for an electron transitioning from *2p to 1s orbital*
- ii. Wavelength of photon *absorbed* for an electron transitioning from *2s to 4p orbital*
- iii. Wavelength of photon *absorbed* for an electron transitioning from *1s to 3p orbital*

i or ii or iii



Question 2 8 points



A Chem110 student plots the graph shown on the left, from data obtained for the kinetic energy (eV) of the ejected electron versus frequency (s^{-1}) of the incident light, for an unknown metal. ($1\text{eV} = 1.6022 \times 10^{-19} \text{ J}$)

Based on the plot, answer the questions **a to c** below:

a. 1 point On the plot above, label the **threshold frequency**.

b. 2 points What is the value of the work potential (in J) of the unknown metal?

c. 3 points What is the *longest* wavelength (in nm) required to eject an electron from this metal's surface? Show your work including equation(s) used. (3 sig figs)

d. 2 points In the space provided below, plot the following

(Assume frequency is *above* threshold frequency):

- Kinetic energy* of an ejected electron versus the *intensity* of the incident light (left)
- Number of electrons* ejected versus the *frequency* of the incident light (right)



**Question 3** **6 points****a. 2 points**

In the ground state of Ge (Germanium; $Z = 32$) atom, **name** the following orbitals:

i. Orbital containing the electron that requires the maximum energy to be removed.

ii. Orbital containing the electron that experiences the minimum effective nuclear charge.

b. 2 points

An unknown anion (A^-) has the following *excited* electronic configuration: $[Ar] 4s^2 3d^3 4p^1$

(Ar: Argon; $Z = 18$)

Give the **condensed ground state electronic configuration** of the atom, **A** and the cation, **A^{2+}** .
Show your work. (Assume all electronic transitions are possible)

Electronic Configuration of A:

Electronic Configuration of A^{2+} :

c. 2 points

In a single atom, determine how many electrons can have the following set of quantum numbers?

(i) $n = 5; l = 3$

In a single atom, determine how many electrons can have the following set of quantum numbers?

(i) $n = 3; l = 2; m_s = +1/2$

**Question 4** **8 points**

Rank the following from 1 to 4 in **increasing** order. Provide the ranking in the boxes provided.

Note: Atomic number (Z) are provided for relevant atoms

a) Increasing (lowest to highest) order of *ionization energy*: (1= lowest; 4 = highest)

Mg

Mg: Z = 12

Rb

Rb: Z = 37

Cs

Cs: Z = 55

K

K : Z = 19

b) Increasing (lowest to highest) order of *bond angle*: (1= lowest; 4 = highest)

H₂OCCl₄XeF₄

HCN

c) Increasing (smallest to largest) order of *ionic size* (1= smallest; 4 = largest)

S²⁻

S: Z = 16

O²⁻

O: Z = 8

Be²⁺

Be: Z = 4

Na⁺

Na: Z = 11

d) Increasing (lowest to highest) order of *bond energy* (1= lowest; 4 = highest)

P-F

P-Cl

P-I

P-Br

**Question 5** **12 points**

a. 9 points For the molecules HClO_4 and TeF_6 , draw the most stable *Lewis structure* showing appropriate shape and *dashed/wedged bonds*, based on VSEPR. Determine the **molecular geometry** around the central atom. For HClO_4 , label the *most polar bond*.



b. 3 points (Note: Lone pairs are not shown in structures below)

(i) For the three *resonance* structures (1, 2 and 3) shown below, answer the following question:



(I) Circle the *most contributing* resonance structure.

Structure 1

Structure 2

Structure 3

(ii) For the three structures given (4, 5, and 6) below, answer the following questions:



(I) Circle all the structures that represent *resonance structures* of each other:

Structure 4

Structure 5

Structure 6

(II) Among the resonance structures, circle the *most contributing* resonance structure.

Structure 4

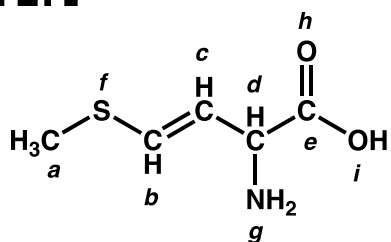
Structure 5

Structure 6

**Question 6** **6 points**

Using the data given below, calculate the **lattice energy** for SrO(s).

	ΔH° (kJ/mol)
$\text{Sr}(s) \longrightarrow \text{Sr}(g)$	+180
$\text{Sr}(g) \longrightarrow \text{Sr}^{2+}(g) + 2e^-$	+1468
$\text{O}_2(g) \longrightarrow 2\text{O}(g)$	+500
$\text{O}(g) + 2e^- \longrightarrow \text{O}^{2-}(g)$	+600
$\text{Sr}(s) + \frac{1}{2}\text{O}_2(g) \longrightarrow \text{SrO}(s)$	-550

**Question 7 10 points**

Consider the molecule given on the left, and answer questions (i) to (vi) using chemical bonding theories that we have learnt in class (valence bond theory, hybridization).

Note that the lone pairs are not shown in the structure:

i. Consider the carbon atoms marked in as ***a***, ***b***, ***c***, ***d***, and ***e*** in the molecule, and determine **hybridization** at each of the carbon centers. (2.5 points)

Atom	Hybridization
Carbon (<i>a</i>)	
Carbon (<i>b</i>)	
Carbon (<i>c</i>)	
Carbon (<i>d</i>)	
Carbon (<i>e</i>)	

ii. Now, consider the sulfur atom (S) marked as ***f***, nitrogen (N) atom (***g***), and oxygen(O) atoms (***h*** and ***i***), and determine their **hybridization**. (2 points)

Atom	Hybridization
Sulfur (<i>f</i>)	
Nitrogen (<i>g</i>)	
Oxygen (<i>h</i>)	
Oxygen (<i>i</i>)	

iii. **Describe** all the bonds formed by the carbon atoms ***a*** and ***b***, using sigma and pi designation as appropriate. (2 points)

Atom	Bonds formed (use sigma and pi as appropriate)
Carbon (<i>a</i>)	
Carbon (<i>b</i>)	

iv. How many total sp^3 -s sigma bonds are in the molecule? (1 point)

v. Determine **electronic** and **molecular geometry** at carbon (C) labeled ***e*** and sulfur (S) labeled ***f*** centres. (2 points)

Atom	Electronic Geometry	Molecular Geometry
Carbon (<i>e</i>)		
Sulfur (<i>f</i>)		

vi. Circle the value from the choices given below, that describes the **C-S-C bond angle at sulfur** (S) marked ***f***: (0.5 points)

>90° to <109.5° 109.5° >109.5 to <120° 120° >120° to <180°


Question 8 12 points

Using the **valence shell orbitals only** draw a **complete** molecular orbital (MO) diagram (with no s-p mixing) for the molecule **PS**, in the space provided below. **(6 points)**

Assume that 3s and 3p atomic orbitals will form molecular orbitals like those formed by the combination of 2s and 2p atomic orbitals that we learnt in class, and there is no s-p mixing.

Show **energy arrow**, **atomic orbitals** involved, **atom labels**, the **molecular orbitals** that are formed, **filled electrons** in the MO diagram etc. Make sure to label all the atomic and molecular orbitals.

Using the MO diagram above, answer the following questions:

- (i) Which one of the following will have the highest bond order? Circle the correct choice
 A) PS^{1-} B) PS C) PS^{1+}
- (ii) Which one of the following will have the longest bond length? Circle the correct choice
 A) PS^{1-} B) PS C) PS^{1+}
- (iii) Which of the following will be diamagnetic? Circle the correct choice.
 A) PS^{1-} B) PS C) PS^{1+}
- (iv) Draw Lewis dot structure of the molecule with the shortest bond length (based on the MO diagram) from among the three (PS^{1-} , PS, PS^{1+}), and indicate the bond order calculated from the Lewis dot structure. **(3 points)**

Lewis Dot Structure

Bond Order



21AB9318-0BA3-49EB-AD54-013AF0E5CDDF

midterm-exam-93388

#235 Page 10 of 10

SCRAP PAGE

**CHEMISTRY 110 –Midterm Exam****Thursday November 7, 2024, 6:30 – 8:30 pm****Examiners: Pallavi Sirjoosingh and Ashok Kakkar**

First name (please write as legibly as possible within the boxes)

--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--

Last name

--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--

Student ID number

--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--

This booklet has 8 questions on 10 pages (The last page can be used as a scrap page).

The staple must NOT be removed from the question booklet. Your intact question booklet must be returned at the end of the exam.

A separate booklet contains a periodic table, and datasheet.

Write your answers directly under the questions in the space provided. Do not write on the QR codes or near the top edges (as it may be cut off).

Please write **legibly and dark enough** for the scan to be readable. For students who use English as a second language, dictionaries are permitted. Calculators are allowed.

DO NOT ASK QUESTIONS OF THE INVIGILATORS about interpretations of the exam questions. If you have a concern about a question, please summarize it at the top of this page. All concerns will be evaluated after the exam. GOOD LUCK!

**Question 1 8 points** (*Assume all transitions are possible*)

a. 6 points Calculate the **energy (in kJ/mol)** required to ionize (completely remove) the electron from the **n=3** level of **He⁺ ion** (He: Helium). Show your work including equations. (3 sig. figs.)

b. 2 points

(I) Select the largest (longest) wavelength from the three choices below:

- i. Wavelength of photon to *completely remove* an electron from *1s orbital of Hydrogen atom*
- ii. Wavelength of photon to *completely remove* an electron from *1s orbital of Boron atom (B)*
- iii. Wavelength of photon to *completely remove* an electron from *1s orbital of Boron cation (B⁺)*

i or ii or iii

(II) Select the smallest (shortest) wavelength from the three choices below (*for H atom*):

- i. Wavelength of photon *emitted* for an electron transitioning from *2p to 1s orbital*
- ii. Wavelength of photon *absorbed* for an electron transitioning from *2s to 4p orbital*
- iii. Wavelength of photon *absorbed* for an electron transitioning from *1s to 3p orbital*

i or ii or iii



Question 2 8 points



A Chem110 student plots the graph shown on the left, from data obtained for the kinetic energy (eV) of the ejected electron versus frequency (s^{-1}) of the incident light, for an unknown metal. ($1\text{eV} = 1.6022 \times 10^{-19} \text{ J}$)

Based on the plot, answer the questions **a to c** below:

a. 1 point On the plot above, label the **threshold frequency**.

b. 2 points What is the value of the work potential (in J) of the unknown metal?

c. 3 points What is the *longest* wavelength (in nm) required to eject an electron from this metal's surface? Show your work including equation(s) used. (3 sig figs)

d. 2 points In the space provided below, plot the following

(Assume frequency is *above* threshold frequency):

- i. *Kinetic energy* of an ejected electron versus the *intensity* of the incident light (left)
- ii. *Number of electrons* ejected versus the *frequency* of the incident light (right)



**Question 3** **6 points****a. 2 points**

In the ground state of Ge (Germanium; $Z = 32$) atom, **name** the following orbitals:

i. Orbital containing the electron that requires the maximum energy to be removed.

ii. Orbital containing the electron that experiences the minimum effective nuclear charge.

b. 2 points

An unknown anion (A^-) has the following *excited* electronic configuration: $[Ar] 4s^2 3d^3 4p^1$

(Ar: Argon; $Z = 18$)

Give the **condensed ground state electronic configuration** of the atom, **A** and the cation, **A^{2+}** .
Show your work. (Assume all electronic transitions are possible)

Electronic Configuration of A:

Electronic Configuration of A^{2+} :

c. 2 points

In a single atom, determine how many electrons can have the following set of quantum numbers?

(i) $n = 5; l = 3$

In a single atom, determine how many electrons can have the following set of quantum numbers?

(i) $n = 3; l = 2; m_s = +1/2$

**Question 4** **8 points**

Rank the following from 1 to 4 in **increasing** order. Provide the ranking in the boxes provided.

Note: Atomic number (Z) are provided for relevant atoms

a) Increasing (lowest to highest) order of *ionization energy*: (1= lowest; 4 = highest)

Mg

Mg: Z = 12

Rb

Rb: Z = 37

Cs

Cs: Z = 55

K

K : Z = 19

b) Increasing (lowest to highest) order of *bond angle*: (1= lowest; 4 = highest)

H₂OCCl₄XeF₄

HCN

c) Increasing (smallest to largest) order of *ionic size* (1= smallest; 4 = largest)

S²⁻

S: Z = 16

O²⁻

O: Z = 8

Be²⁺

Be: Z = 4

Na⁺

Na: Z = 11

d) Increasing (lowest to highest) order of *bond energy* (1= lowest; 4 = highest)

P-F

P-Cl

P-I

P-Br

**Question 5** **12 points**

a. 9 points For the molecules HClO_4 and TeF_6 , draw the most stable *Lewis structure* showing appropriate shape and *dashed/wedged bonds*, based on VSEPR. Determine the **molecular geometry** around the central atom. For HClO_4 , label the *most polar bond*.



b. 3 points (Note: Lone pairs are not shown in structures below)

(i) For the three *resonance* structures (1, 2 and 3) shown below, answer the following question:



(I) Circle the *most contributing* resonance structure.

Structure 1

Structure 2

Structure 3

(ii) For the three structures given (4, 5, and 6) below, answer the following questions:



(I) Circle all the structures that represent *resonance structures* of each other:

Structure 4

Structure 5

Structure 6

(II) Among the resonance structures, circle the *most contributing* resonance structure.

Structure 4

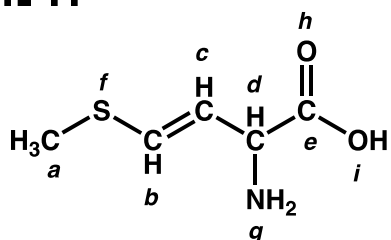
Structure 5

Structure 6

**Question 6** **6 points**

Using the data given below, calculate the **lattice energy** for SrO(s).

	ΔH° (kJ/mol)
$\text{Sr}(s) \longrightarrow \text{Sr}(g)$	+180
$\text{Sr}(g) \longrightarrow \text{Sr}^{2+}(g) + 2e^-$	+1468
$\text{O}_2(g) \longrightarrow 2\text{O}(g)$	+500
$\text{O}(g) + 2e^- \longrightarrow \text{O}^{2-}(g)$	+600
$\text{Sr}(s) + \frac{1}{2}\text{O}_2(g) \longrightarrow \text{SrO}(s)$	-550

**Question 7 10 points**

Consider the molecule given on the left, and answer questions (i) to (vi) using chemical bonding theories that we have learnt in class (valence bond theory, hybridization).

Note that the lone pairs are not shown in the structure:

i. Consider the carbon atoms marked in as ***a***, ***b***, ***c***, ***d***, and ***e*** in the molecule, and determine **hybridization** at each of the carbon centers. (2.5 points)

Atom	Hybridization
Carbon (<i>a</i>)	
Carbon (<i>b</i>)	
Carbon (<i>c</i>)	
Carbon (<i>d</i>)	
Carbon (<i>e</i>)	

ii. Now, consider the sulfur atom (S) marked as ***f***, nitrogen (N) atom (***g***), and oxygen(O) atoms (***h*** and ***i***), and determine their **hybridization**. (2 points)

Atom	Hybridization
Sulfur (<i>f</i>)	
Nitrogen (<i>g</i>)	
Oxygen (<i>h</i>)	
Oxygen (<i>i</i>)	

iii. **Describe** all the bonds formed by the carbon atoms ***a*** and ***b***, using sigma and pi designation as appropriate. (2 points)

Atom	Bonds formed (use sigma and pi as appropriate)
Carbon (<i>a</i>)	
Carbon (<i>b</i>)	

iv. How many total sp^3 -s sigma bonds are in the molecule? (1 point)

v. Determine **electronic** and **molecular geometry** at carbon (C) labeled ***e*** and sulfur (S) labeled ***f*** centres. (2 points)

Atom	Electronic Geometry	Molecular Geometry
Carbon (<i>e</i>)		
Sulfur (<i>f</i>)		

vi. Circle the value from the choices given below, that describes the **C-S-C bond angle at sulfur** (S) marked ***f***: (0.5 points)

>90° to <109.5°

109.5°

>109.5 to <120°

120°

>120° to <180°


Question 8 12 points

Using the **valence shell orbitals only** draw a **complete** molecular orbital (MO) diagram (with no s-p mixing) for the molecule **PS**, in the space provided below. **(6 points)**

Assume that 3s and 3p atomic orbitals will form molecular orbitals like those formed by the combination of 2s and 2p atomic orbitals that we learnt in class, and there is no s-p mixing.

Show **energy arrow**, **atomic orbitals** involved, **atom labels**, the **molecular orbitals** that are formed, **filled electrons** in the MO diagram etc. Make sure to label all the atomic and molecular orbitals.

Using the MO diagram above, answer the following questions:

- (i) Which one of the following will have the highest bond order? Circle the correct choice
 A) PS^{1-} B) PS C) PS^{1+}
- (ii) Which one of the following will have the longest bond length? Circle the correct choice
 A) PS^{1-} B) PS C) PS^{1+}
- (iii) Which of the following will be diamagnetic? Circle the correct choice.
 A) PS^{1-} B) PS C) PS^{1+}
- (iv) Draw Lewis dot structure of the molecule with the shortest bond length (based on the MO diagram) from among the three (PS^{1-} , PS, PS^{1+}), and indicate the bond order calculated from the Lewis dot structure. **(3 points)**

Lewis Dot Structure

Bond Order



A04DE0D1-7346-4F38-B3B6-9C5D0B88C063

midterm-exam-93388

#236 Page 10 of 10

SCRAP PAGE

**CHEMISTRY 110 –Midterm Exam****Thursday November 7, 2024, 6:30 – 8:30 pm****Examiners: Pallavi Sirjoosingh and Ashok Kakkar**

First name (please write as legibly as possible within the boxes)

--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--

Last name

--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--

Student ID number

--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--

This booklet has 8 questions on 10 pages (The last page can be used as a scrap page).

The staple must NOT be removed from the question booklet. Your intact question booklet must be returned at the end of the exam.

A separate booklet contains a periodic table, and datasheet.

Write your answers directly under the questions in the space provided. Do not write on the QR codes or near the top edges (as it may be cut off).

Please write **legibly and dark enough** for the scan to be readable. For students who use English as a second language, dictionaries are permitted. Calculators are allowed.

DO NOT ASK QUESTIONS OF THE INVIGILATORS about interpretations of the exam questions. If you have a concern about a question, please summarize it at the top of this page. All concerns will be evaluated after the exam. GOOD LUCK!

**Question 1 8 points** (*Assume all transitions are possible*)

a. 6 points Calculate the **energy (in kJ/mol)** required to ionize (completely remove) the electron from the **n=3** level of **He⁺ ion** (He: Helium). Show your work including equations. (3 sig. figs.)

b. 2 points

(I) Select the largest (longest) wavelength from the three choices below:

- i. Wavelength of photon to *completely remove* an electron from *1s orbital of Hydrogen atom*
- ii. Wavelength of photon to *completely remove* an electron from *1s orbital of Boron atom (B)*
- iii. Wavelength of photon to *completely remove* an electron from *1s orbital of Boron cation (B⁺)*

i or ii or iii

(II) Select the smallest (shortest) wavelength from the three choices below (*for H atom*):

- i. Wavelength of photon *emitted* for an electron transitioning from *2p to 1s orbital*
- ii. Wavelength of photon *absorbed* for an electron transitioning from *2s to 4p orbital*
- iii. Wavelength of photon *absorbed* for an electron transitioning from *1s to 3p orbital*

i or ii or iii



Question 2 8 points



A Chem110 student plots the graph shown on the left, from data obtained for the kinetic energy (eV) of the ejected electron versus frequency (s^{-1}) of the incident light, for an unknown metal. ($1\text{eV} = 1.6022 \times 10^{-19} \text{ J}$)

Based on the plot, answer the questions **a to c** below:

a. 1 point On the plot above, label the **threshold frequency**.

b. 2 points What is the value of the work potential (in J) of the unknown metal?

c. 3 points What is the *longest* wavelength (in nm) required to eject an electron from this metal's surface? Show your work including equation(s) used. (3 sig figs)

d. 2 points In the space provided below, plot the following

(Assume frequency is *above* threshold frequency):

- i. *Kinetic energy* of an ejected electron versus the *intensity* of the incident light (left)
- ii. *Number of electrons* ejected versus the *frequency* of the incident light (right)



**Question 3** **6 points****a. 2 points**

In the ground state of Ge (Germanium; $Z = 32$) atom, **name** the following orbitals:

i. Orbital containing the electron that requires the maximum energy to be removed.

ii. Orbital containing the electron that experiences the minimum effective nuclear charge.

b. 2 points

An unknown anion (A^-) has the following *excited* electronic configuration: $[Ar] 4s^2 3d^3 4p^1$

(Ar: Argon; $Z = 18$)

Give the **condensed ground state electronic configuration** of the atom, **A** and the cation, **A^{2+}** .
Show your work. (Assume all electronic transitions are possible)

Electronic Configuration of A:

Electronic Configuration of A^{2+} :

c. 2 points

In a single atom, determine how many electrons can have the following set of quantum numbers?

(i) $n = 5; l = 3$

In a single atom, determine how many electrons can have the following set of quantum numbers?

(i) $n = 3; l = 2; m_s = +1/2$

**Question 4** **8 points**

Rank the following from 1 to 4 in **increasing** order. Provide the ranking in the boxes provided.

Note: Atomic number (Z) are provided for relevant atoms

a) Increasing (lowest to highest) order of *ionization energy*: (1= lowest; 4 = highest)

Mg

Mg: Z = 12

Rb

Rb: Z = 37

Cs

Cs: Z = 55

K

K : Z = 19

b) Increasing (lowest to highest) order of *bond angle*: (1= lowest; 4 = highest)

H₂OCCl₄XeF₄

HCN

c) Increasing (smallest to largest) order of *ionic size* (1= smallest; 4 = largest)

S²⁻

S: Z = 16

O²⁻

O: Z = 8

Be²⁺

Be: Z = 4

Na⁺

Na: Z = 11

d) Increasing (lowest to highest) order of *bond energy* (1= lowest; 4 = highest)

P-F

P-Cl

P-I

P-Br

**Question 5** **12 points**

a. 9 points For the molecules HClO_4 and TeF_6 , draw the most stable *Lewis structure* showing appropriate shape and *dashed/wedged bonds*, based on VSEPR. Determine the **molecular geometry** around the central atom. For HClO_4 , label the *most polar bond*.



b. 3 points (Note: Lone pairs are not shown in structures below)

(i) For the three *resonance* structures (1, 2 and 3) shown below, answer the following question:



(I) Circle the *most contributing* resonance structure.

Structure 1

Structure 2

Structure 3

(ii) For the three structures given (4, 5, and 6) below, answer the following questions:



(I) Circle all the structures that represent *resonance structures* of each other:

Structure 4

Structure 5

Structure 6

(II) Among the resonance structures, circle the *most contributing* resonance structure.

Structure 4

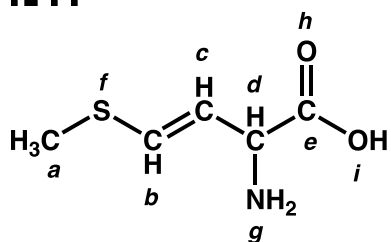
Structure 5

Structure 6

**Question 6** **6 points**

Using the data given below, calculate the **lattice energy** for SrO(s).

	ΔH° (kJ/mol)
$\text{Sr}(s) \longrightarrow \text{Sr}(g)$	+180
$\text{Sr}(g) \longrightarrow \text{Sr}^{2+}(g) + 2e^-$	+1468
$\text{O}_2(g) \longrightarrow 2\text{O}(g)$	+500
$\text{O}(g) + 2e^- \longrightarrow \text{O}^{2-}(g)$	+600
$\text{Sr}(s) + \frac{1}{2}\text{O}_2(g) \longrightarrow \text{SrO}(s)$	-550

**Question 7 10 points**

Consider the molecule given on the left, and answer questions (i) to (vi) using chemical bonding theories that we have learnt in class (valence bond theory, hybridization).

Note that the lone pairs are not shown in the structure:

i. Consider the carbon atoms marked in as ***a***, ***b***, ***c***, ***d***, and ***e*** in the molecule, and determine **hybridization** at each of the carbon centers. (2.5 points)

Atom	Hybridization
Carbon (<i>a</i>)	
Carbon (<i>b</i>)	
Carbon (<i>c</i>)	
Carbon (<i>d</i>)	
Carbon (<i>e</i>)	

ii. Now, consider the sulfur atom (S) marked as ***f***, nitrogen (N) atom (***g***), and oxygen(O) atoms (***h*** and ***i***), and determine their **hybridization**. (2 points)

Atom	Hybridization
Sulfur (<i>f</i>)	
Nitrogen (<i>g</i>)	
Oxygen (<i>h</i>)	
Oxygen (<i>i</i>)	

iii. **Describe** all the bonds formed by the carbon atoms ***a*** and ***b***, using sigma and pi designation as appropriate. (2 points)

Atom	Bonds formed (use sigma and pi as appropriate)
Carbon (<i>a</i>)	
Carbon (<i>b</i>)	

iv. How many total sp^3 -s sigma bonds are in the molecule? (1 point)

v. Determine **electronic** and **molecular geometry** at carbon (C) labeled ***e*** and sulfur (S) labeled ***f*** centres. (2 points)

Atom	Electronic Geometry	Molecular Geometry
Carbon (<i>e</i>)		
Sulfur (<i>f</i>)		

vi. Circle the value from the choices given below, that describes the **C-S-C bond angle at sulfur** (S) marked ***f***: (0.5 points)

>90° to <109.5° 109.5° >109.5 to <120° 120° >120° to <180°


Question 8 12 points

Using the **valence shell orbitals only** draw a **complete** molecular orbital (MO) diagram (with no s-p mixing) for the molecule **PS**, in the space provided below. **(6 points)**

Assume that 3s and 3p atomic orbitals will form molecular orbitals like those formed by the combination of 2s and 2p atomic orbitals that we learnt in class, and there is no s-p mixing.

Show **energy arrow**, **atomic orbitals** involved, **atom labels**, the **molecular orbitals** that are formed, **filled electrons** in the MO diagram etc. Make sure to label all the atomic and molecular orbitals.

Using the MO diagram above, answer the following questions:

- (i) Which one of the following will have the highest bond order? Circle the correct choice
 A) PS^{1-} B) PS C) PS^{1+}
- (ii) Which one of the following will have the longest bond length? Circle the correct choice
 A) PS^{1-} B) PS C) PS^{1+}
- (iii) Which of the following will be diamagnetic? Circle the correct choice.
 A) PS^{1-} B) PS C) PS^{1+}
- (iv) Draw Lewis dot structure of the molecule with the shortest bond length (based on the MO diagram) from among the three (PS^{1-} , PS, PS^{1+}), and indicate the bond order calculated from the Lewis dot structure. **(3 points)**

Lewis Dot Structure

Bond Order



2F4BC6C8-5BC7-4888-B648-ACF321C4ECD8

midterm-exam-93388

#237 Page 10 of 10

SCRAP PAGE

**CHEMISTRY 110 –Midterm Exam****Thursday November 7, 2024, 6:30 – 8:30 pm**

Examiners: Pallavi Sirjoosingh and Ashok Kakkar

First name (please write as legibly as possible within the boxes)

--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--

Last name

--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--

Student ID number

--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--

This booklet has 8 questions on 10 pages (The last page can be used as a scrap page).

The staple must NOT be removed from the question booklet. Your intact question booklet must be returned at the end of the exam.

A separate booklet contains a periodic table, and datasheet.

Write your answers directly under the questions in the space provided. Do not write on the QR codes or near the top edges (as it may be cut off).

Please write **legibly and dark enough** for the scan to be readable. For students who use English as a second language, dictionaries are permitted. Calculators are allowed.

DO NOT ASK QUESTIONS OF THE INVIGILATORS about interpretations of the exam questions. If you have a concern about a question, please summarize it at the top of this page. All concerns will be evaluated after the exam. GOOD LUCK!

**Question 1 8 points** (*Assume all transitions are possible*)

a. 6 points Calculate the **energy (in kJ/mol)** required to ionize (completely remove) the electron from the **$n=3$** level of **He^+ ion** (He: Helium). Show your work including equations. (3 sig. figs.)

b. 2 points

(I) Select the largest (longest) wavelength from the three choices below:

- i. Wavelength of photon to *completely remove* an electron from *1s orbital of Hydrogen atom*
- ii. Wavelength of photon to *completely remove* an electron from *1s orbital of Boron atom (B)*
- iii. Wavelength of photon to *completely remove* an electron from *1s orbital of Boron cation (B^+)*

i or ii or iii

(II) Select the smallest (shortest) wavelength from the three choices below (*for H atom*):

- i. Wavelength of photon *emitted* for an electron transitioning from *2p to 1s orbital*
- ii. Wavelength of photon *absorbed* for an electron transitioning from *2s to 4p orbital*
- iii. Wavelength of photon *absorbed* for an electron transitioning from *1s to 3p orbital*

i or ii or iii



Question 2 8 points



A Chem110 student plots the graph shown on the left, from data obtained for the kinetic energy (eV) of the ejected electron versus frequency (s^{-1}) of the incident light, for an unknown metal. ($1\text{eV} = 1.6022 \times 10^{-19} \text{ J}$)

Based on the plot, answer the questions **a to c** below:

a. 1 point On the plot above, label the **threshold frequency**.

b. 2 points What is the value of the work potential (in J) of the unknown metal?

c. 3 points What is the *longest* wavelength (in nm) required to eject an electron from this metal's surface? Show your work including equation(s) used. (3 sig figs)

d. 2 points In the space provided below, plot the following

(Assume frequency is *above* threshold frequency):

- Kinetic energy* of an ejected electron versus the *intensity* of the incident light (left)
- Number of electrons* ejected versus the *frequency* of the incident light (right)



**Question 3** **6 points****a. 2 points**

In the ground state of Ge (Germanium; $Z = 32$) atom, **name** the following orbitals:

i. Orbital containing the electron that requires the maximum energy to be removed.

ii. Orbital containing the electron that experiences the minimum effective nuclear charge.

b. 2 points

An unknown anion (A^-) has the following *excited* electronic configuration: $[Ar] 4s^2 3d^3 4p^1$

(Ar: Argon; $Z = 18$)

Give the **condensed ground state electronic configuration** of the atom, **A** and the cation, **A^{2+}** .
Show your work. (Assume all electronic transitions are possible)

Electronic Configuration of A:

Electronic Configuration of A^{2+} :

c. 2 points

In a single atom, determine how many electrons can have the following set of quantum numbers?

(i) $n = 5; l = 3$

In a single atom, determine how many electrons can have the following set of quantum numbers?

(i) $n = 3; l = 2; m_s = +1/2$

**Question 4** **8 points**

Rank the following from 1 to 4 in **increasing** order. Provide the ranking in the boxes provided.

Note: Atomic number (Z) are provided for relevant atoms

a) Increasing (lowest to highest) order of *ionization energy*: (1= lowest; 4 = highest)

Mg

Mg: Z = 12

Rb

Rb: Z = 37

Cs

Cs: Z = 55

K

K : Z = 19

b) Increasing (lowest to highest) order of *bond angle*: (1= lowest; 4 = highest)

H₂OCCl₄XeF₄

HCN

c) Increasing (smallest to largest) order of *ionic size* (1= smallest; 4 = largest)

S²⁻

S: Z = 16

O²⁻

O: Z = 8

Be²⁺

Be: Z = 4

Na⁺

Na: Z = 11

d) Increasing (lowest to highest) order of *bond energy* (1= lowest; 4 = highest)

P-F

P-Cl

P-I

P-Br

**Question 5** **12 points**

a. 9 points For the molecules HClO_4 and TeF_6 , draw the most stable *Lewis structure* showing appropriate shape and *dashed/wedged bonds*, based on VSEPR. Determine the **molecular geometry** around the central atom. For HClO_4 , label the *most polar bond*.



b. 3 points (Note: Lone pairs are not shown in structures below)

(i) For the three *resonance* structures (1, 2 and 3) shown below, answer the following question:



(I) Circle the *most contributing* resonance structure.

Structure 1

Structure 2

Structure 3

(ii) For the three structures given (4, 5, and 6) below, answer the following questions:



(I) Circle all the structures that represent *resonance structures* of each other:

Structure 4

Structure 5

Structure 6

(II) Among the resonance structures, circle the *most contributing* resonance structure.

Structure 4

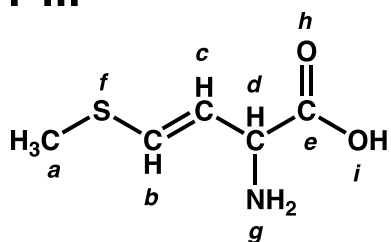
Structure 5

Structure 6

**Question 6** **6 points**

Using the data given below, calculate the **lattice energy** for SrO(s).

	ΔH° (kJ/mol)
$\text{Sr}(s) \longrightarrow \text{Sr}(g)$	+180
$\text{Sr}(g) \longrightarrow \text{Sr}^{2+}(g) + 2e^-$	+1468
$\text{O}_2(g) \longrightarrow 2\text{O}(g)$	+500
$\text{O}(g) + 2e^- \longrightarrow \text{O}^{2-}(g)$	+600
$\text{Sr}(s) + \frac{1}{2}\text{O}_2(g) \longrightarrow \text{SrO}(s)$	-550

**Question 7 10 points**

Consider the molecule given on the left, and answer questions (i) to (vi) using chemical bonding theories that we have learnt in class (valence bond theory, hybridization).

Note that the lone pairs are not shown in the structure:

i. Consider the carbon atoms marked in as ***a***, ***b***, ***c***, ***d***, and ***e*** in the molecule, and determine **hybridization** at each of the carbon centers. (2.5 points)

Atom	Hybridization
Carbon (<i>a</i>)	
Carbon (<i>b</i>)	
Carbon (<i>c</i>)	
Carbon (<i>d</i>)	
Carbon (<i>e</i>)	

ii. Now, consider the sulfur atom (S) marked as ***f***, nitrogen (N) atom (***g***), and oxygen(O) atoms (***h*** and ***i***), and determine their **hybridization**. (2 points)

Atom	Hybridization
Sulfur (<i>f</i>)	
Nitrogen (<i>g</i>)	
Oxygen (<i>h</i>)	
Oxygen (<i>i</i>)	

iii. **Describe** all the bonds formed by the carbon atoms ***a*** and ***b***, using sigma and pi designation as appropriate. (2 points)

Atom	Bonds formed (use sigma and pi as appropriate)
Carbon (<i>a</i>)	
Carbon (<i>b</i>)	

iv. How many total sp^3 -s sigma bonds are in the molecule? (1 point)

v. Determine **electronic** and **molecular geometry** at carbon (C) labeled ***e*** and sulfur (S) labeled ***f*** centres. (2 points)

Atom	Electronic Geometry	Molecular Geometry
Carbon (<i>e</i>)		
Sulfur (<i>f</i>)		

vi. Circle the value from the choices given below, that describes the **C-S-C bond angle at sulfur** (S) marked ***f***: (0.5 points)

>90° to <109.5° 109.5° >109.5 to <120° 120° >120° to <180°


Question 8 12 points

Using the **valence shell orbitals only** draw a **complete** molecular orbital (MO) diagram (with no s-p mixing) for the molecule **PS**, in the space provided below. **(6 points)**

Assume that 3s and 3p atomic orbitals will form molecular orbitals like those formed by the combination of 2s and 2p atomic orbitals that we learnt in class, and there is no s-p mixing.

Show **energy arrow**, **atomic orbitals** involved, **atom labels**, the **molecular orbitals** that are formed, **filled electrons** in the MO diagram etc. Make sure to label all the atomic and molecular orbitals.

Using the MO diagram above, answer the following questions:

- (i) Which one of the following will have the highest bond order? Circle the correct choice
 A) PS^{1-} B) PS C) PS^{1+}
- (ii) Which one of the following will have the longest bond length? Circle the correct choice
 A) PS^{1-} B) PS C) PS^{1+}
- (iii) Which of the following will be diamagnetic? Circle the correct choice.
 A) PS^{1-} B) PS C) PS^{1+}
- (iv) Draw Lewis dot structure of the molecule with the shortest bond length (based on the MO diagram) from among the three (PS^{1-} , PS, PS^{1+}), and indicate the bond order calculated from the Lewis dot structure. **(3 points)**

Lewis Dot Structure

Bond Order



AFDB5E9A-9B08-4A19-A5C7-C43352A61451

midterm-exam-93388

#238 Page 10 of 10

SCRAP PAGE

**CHEMISTRY 110 –Midterm Exam****Thursday November 7, 2024, 6:30 – 8:30 pm**

Examiners: Pallavi Sirjoosingh and Ashok Kakkar

First name (please write as legibly as possible within the boxes)

--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--

Last name

--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--

Student ID number

--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--

This booklet has 8 questions on 10 pages (The last page can be used as a scrap page).

The staple must NOT be removed from the question booklet. Your intact question booklet must be returned at the end of the exam.

A separate booklet contains a periodic table, and datasheet.

Write your answers directly under the questions in the space provided. Do not write on the QR codes or near the top edges (as it may be cut off).

Please write **legibly and dark enough** for the scan to be readable. For students who use English as a second language, dictionaries are permitted. Calculators are allowed.

DO NOT ASK QUESTIONS OF THE INVIGILATORS about interpretations of the exam questions. If you have a concern about a question, please summarize it at the top of this page. All concerns will be evaluated after the exam. GOOD LUCK!

**Question 1 8 points** (*Assume all transitions are possible*)

a. 6 points Calculate the **energy (in kJ/mol)** required to ionize (completely remove) the electron from the **$n=3$** level of **He^+ ion** (He: Helium). Show your work including equations. (3 sig. figs.)

b. 2 points

(I) Select the largest (longest) wavelength from the three choices below:

- i. Wavelength of photon to *completely remove* an electron from *1s orbital of Hydrogen atom*
- ii. Wavelength of photon to *completely remove* an electron from *1s orbital of Boron atom (B)*
- iii. Wavelength of photon to *completely remove* an electron from *1s orbital of Boron cation (B^+)*

i or ii or iii

(II) Select the smallest (shortest) wavelength from the three choices below (*for H atom*):

- i. Wavelength of photon *emitted* for an electron transitioning from *2p to 1s orbital*
- ii. Wavelength of photon *absorbed* for an electron transitioning from *2s to 4p orbital*
- iii. Wavelength of photon *absorbed* for an electron transitioning from *1s to 3p orbital*

i or ii or iii



Question 2 8 points



A Chem110 student plots the graph shown on the left, from data obtained for the kinetic energy (eV) of the ejected electron versus frequency (s^{-1}) of the incident light, for an unknown metal. ($1\text{eV} = 1.6022 \times 10^{-19} \text{ J}$)

Based on the plot, answer the questions **a to c** below:

a. 1 point On the plot above, label the **threshold frequency**.

b. 2 points What is the value of the work potential (in J) of the unknown metal?

c. 3 points What is the *longest* wavelength (in nm) required to eject an electron from this metal's surface? Show your work including equation(s) used. (3 sig figs)

d. 2 points In the space provided below, plot the following

(Assume frequency is *above* threshold frequency):

- Kinetic energy* of an ejected electron versus the *intensity* of the incident light (left)
- Number of electrons* ejected versus the *frequency* of the incident light (right)



**Question 3** **6 points****a. 2 points**

In the ground state of Ge (Germanium; $Z = 32$) atom, **name** the following orbitals:

i. Orbital containing the electron that requires the maximum energy to be removed.

ii. Orbital containing the electron that experiences the minimum effective nuclear charge.

b. 2 points

An unknown anion (A^-) has the following *excited* electronic configuration: $[Ar] 4s^2 3d^3 4p^1$

(Ar: Argon; $Z = 18$)

Give the **condensed ground state electronic configuration** of the atom, **A** and the cation, **A^{2+}** .
Show your work. (Assume all electronic transitions are possible)

Electronic Configuration of A:

Electronic Configuration of A^{2+} :

c. 2 points

In a single atom, determine how many electrons can have the following set of quantum numbers?

(i) $n = 5; l = 3$

In a single atom, determine how many electrons can have the following set of quantum numbers?

(i) $n = 3; l = 2; m_s = +1/2$

**Question 4** **8 points**

Rank the following from 1 to 4 in **increasing** order. Provide the ranking in the boxes provided.

Note: Atomic number (Z) are provided for relevant atoms

a) Increasing (lowest to highest) order of *ionization energy*: (1= lowest; 4 = highest)

Mg

Mg: Z = 12

Rb

Rb: Z = 37

Cs

Cs: Z = 55

K

K : Z = 19

b) Increasing (lowest to highest) order of *bond angle*: (1= lowest; 4 = highest)

H₂OCCl₄XeF₄

HCN

c) Increasing (smallest to largest) order of *ionic size* (1= smallest; 4 = largest)

S²⁻

S: Z = 16

O²⁻

O: Z = 8

Be²⁺

Be: Z = 4

Na⁺

Na: Z = 11

d) Increasing (lowest to highest) order of *bond energy* (1= lowest; 4 = highest)

P-F

P-Cl

P-I

P-Br

**Question 5** **12 points**

a. 9 points For the molecules HClO_4 and TeF_6 , draw the most stable *Lewis structure* showing appropriate shape and *dashed/wedged bonds*, based on VSEPR. Determine the **molecular geometry** around the central atom. For HClO_4 , label the *most polar bond*.



b. 3 points (Note: Lone pairs are not shown in structures below)

(i) For the three *resonance* structures (1, 2 and 3) shown below, answer the following question:



(I) Circle the *most contributing* resonance structure.

Structure 1

Structure 2

Structure 3

(ii) For the three structures given (4, 5, and 6) below, answer the following questions:



(I) Circle all the structures that represent *resonance structures* of each other:

Structure 4

Structure 5

Structure 6

(II) Among the resonance structures, circle the *most contributing* resonance structure.

Structure 4

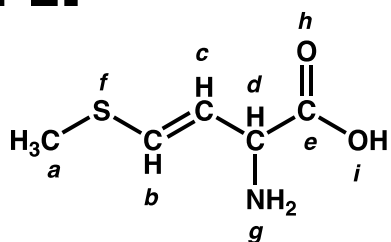
Structure 5

Structure 6

**Question 6** **6 points**

Using the data given below, calculate the **lattice energy** for SrO(s).

	ΔH° (kJ/mol)
$\text{Sr}(s) \longrightarrow \text{Sr}(g)$	+180
$\text{Sr}(g) \longrightarrow \text{Sr}^{2+}(g) + 2e^-$	+1468
$\text{O}_2(g) \longrightarrow 2\text{O}(g)$	+500
$\text{O}(g) + 2e^- \longrightarrow \text{O}^{2-}(g)$	+600
$\text{Sr}(s) + \frac{1}{2}\text{O}_2(g) \longrightarrow \text{SrO}(s)$	-550

**Question 7 10 points**

Consider the molecule given on the left, and answer questions (i) to (vi) using chemical bonding theories that we have learnt in class (valence bond theory, hybridization).

Note that the lone pairs are not shown in the structure:

i. Consider the carbon atoms marked in as ***a***, ***b***, ***c***, ***d***, and ***e*** in the molecule, and determine **hybridization** at each of the carbon centers. (2.5 points)

Atom	Hybridization
Carbon (<i>a</i>)	
Carbon (<i>b</i>)	
Carbon (<i>c</i>)	
Carbon (<i>d</i>)	
Carbon (<i>e</i>)	

ii. Now, consider the sulfur atom (S) marked as ***f***, nitrogen (N) atom (***g***), and oxygen(O) atoms (***h*** and ***i***), and determine their **hybridization**. (2 points)

Atom	Hybridization
Sulfur (<i>f</i>)	
Nitrogen (<i>g</i>)	
Oxygen (<i>h</i>)	
Oxygen (<i>i</i>)	

iii. **Describe** all the bonds formed by the carbon atoms ***a*** and ***b***, using sigma and pi designation as appropriate. (2 points)

Atom	Bonds formed (use sigma and pi as appropriate)
Carbon (<i>a</i>)	
Carbon (<i>b</i>)	

iv. How many total sp^3 -s sigma bonds are in the molecule? (1 point)

v. Determine **electronic** and **molecular geometry** at carbon (C) labeled ***e*** and sulfur (S) labeled ***f*** centres. (2 points)

Atom	Electronic Geometry	Molecular Geometry
Carbon (<i>e</i>)		
Sulfur (<i>f</i>)		

vi. Circle the value from the choices given below, that describes the **C-S-C bond angle at sulfur** (S) marked ***f***: (0.5 points)

>90° to <109.5° 109.5° >109.5 to <120° 120° >120° to <180°

**Question 8 12 points**

Using the **valence shell orbitals only** draw a **complete** molecular orbital (MO) diagram (with no s-p mixing) for the molecule **PS**, in the space provided below. **(6 points)**

Assume that 3s and 3p atomic orbitals will form molecular orbitals like those formed by the combination of 2s and 2p atomic orbitals that we learnt in class, and there is no s-p mixing.

Show **energy arrow**, **atomic orbitals** involved, **atom labels**, the **molecular orbitals** that are formed, **filled electrons** in the MO diagram etc. Make sure to label all the atomic and molecular orbitals.

Using the MO diagram above, answer the following questions:

- (i) Which one of the following will have the highest bond order? Circle the correct choice
A) PS^{1-} B) PS C) PS^{1+}
- (ii) Which one of the following will have the longest bond length? Circle the correct choice
A) PS^{1-} B) PS C) PS^{1+}
- (iii) Which of the following will be diamagnetic? Circle the correct choice.
A) PS^{1-} B) PS C) PS^{1+}
- (iv) Draw Lewis dot structure of the molecule with the shortest bond length (based on the MO diagram) from among the three (PS^{1-} , PS, PS^{1+}), and indicate the bond order calculated from the Lewis dot structure. **(3 points)**

Lewis Dot Structure

Bond Order



6CEE473C-A783-42A7-BC38-879C2BCBB383

midterm-exam-93388

#239 Page 10 of 10

SCRAP PAGE

**CHEMISTRY 110 –Midterm Exam****Thursday November 7, 2024, 6:30 – 8:30 pm****Examiners: Pallavi Sirjoosingh and Ashok Kakkar**

First name (please write as legibly as possible within the boxes)

--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--

Last name

--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--

Student ID number

--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--

This booklet has 8 questions on 10 pages (The last page can be used as a scrap page).

The staple must NOT be removed from the question booklet. Your intact question booklet must be returned at the end of the exam.

A separate booklet contains a periodic table, and datasheet.

Write your answers directly under the questions in the space provided. Do not write on the QR codes or near the top edges (as it may be cut off).

Please write **legibly and dark enough** for the scan to be readable. For students who use English as a second language, dictionaries are permitted. Calculators are allowed.

DO NOT ASK QUESTIONS OF THE INVIGILATORS about interpretations of the exam questions. If you have a concern about a question, please summarize it at the top of this page. All concerns will be evaluated after the exam. GOOD LUCK!

**Question 1 8 points** (*Assume all transitions are possible*)

a. 6 points Calculate the **energy (in kJ/mol)** required to ionize (completely remove) the electron from the **n=3** level of **He⁺ ion** (He: Helium). Show your work including equations. (3 sig. figs.)

b. 2 points

(I) Select the largest (longest) wavelength from the three choices below:

- i. Wavelength of photon to *completely remove* an electron from *1s orbital of Hydrogen atom*
- ii. Wavelength of photon to *completely remove* an electron from *1s orbital of Boron atom (B)*
- iii. Wavelength of photon to *completely remove* an electron from *1s orbital of Boron cation (B⁺)*

i or ii or iii

(II) Select the smallest (shortest) wavelength from the three choices below (*for H atom*):

- i. Wavelength of photon *emitted* for an electron transitioning from *2p to 1s orbital*
- ii. Wavelength of photon *absorbed* for an electron transitioning from *2s to 4p orbital*
- iii. Wavelength of photon *absorbed* for an electron transitioning from *1s to 3p orbital*

i or ii or iii



Question 2 8 points



A Chem110 student plots the graph shown on the left, from data obtained for the kinetic energy (eV) of the ejected electron versus frequency (s^{-1}) of the incident light, for an unknown metal. ($1\text{eV} = 1.6022 \times 10^{-19} \text{ J}$)

Based on the plot, answer the questions **a to c** below:

a. 1 point On the plot above, label the **threshold frequency**.

b. 2 points What is the value of the work potential (in J) of the unknown metal?

c. 3 points What is the *longest* wavelength (in nm) required to eject an electron from this metal's surface? Show your work including equation(s) used. (3 sig figs)

d. 2 points In the space provided below, plot the following

(Assume frequency is *above* threshold frequency):

- Kinetic energy* of an ejected electron versus the *intensity* of the incident light (left)
- Number of electrons* ejected versus the *frequency* of the incident light (right)



**Question 3** **6 points****a. 2 points**

In the ground state of Ge (Germanium; $Z = 32$) atom, **name** the following orbitals:

i. Orbital containing the electron that requires the maximum energy to be removed.

ii. Orbital containing the electron that experiences the minimum effective nuclear charge.

b. 2 points

An unknown anion (A^-) has the following *excited* electronic configuration: $[Ar] 4s^2 3d^3 4p^1$

(Ar: Argon; $Z = 18$)

Give the **condensed ground state electronic configuration** of the atom, **A** and the cation, **A^{2+}** .
Show your work. (Assume all electronic transitions are possible)

Electronic Configuration of A:

Electronic Configuration of A^{2+} :

c. 2 points

In a single atom, determine how many electrons can have the following set of quantum numbers?

(i) $n = 5; l = 3$

In a single atom, determine how many electrons can have the following set of quantum numbers?

(i) $n = 3; l = 2; m_s = +1/2$

**Question 4** **8 points**

Rank the following from 1 to 4 in **increasing** order. Provide the ranking in the boxes provided.

Note: Atomic number (Z) are provided for relevant atoms

a) Increasing (lowest to highest) order of *ionization energy*: (1= lowest; 4 = highest)

Mg

Mg: Z = 12

Rb

Rb: Z = 37

Cs

Cs: Z = 55

K

K : Z = 19

b) Increasing (lowest to highest) order of *bond angle*: (1= lowest; 4 = highest)

H₂OCCl₄XeF₄

HCN

c) Increasing (smallest to largest) order of *ionic size* (1= smallest; 4 = largest)

S²⁻

S: Z = 16

O²⁻

O: Z = 8

Be²⁺

Be: Z = 4

Na⁺

Na: Z = 11

d) Increasing (lowest to highest) order of *bond energy* (1= lowest; 4 = highest)

P-F

P-Cl

P-I

P-Br

**Question 5** **12 points**

a. 9 points For the molecules HClO_4 and TeF_6 , draw the most stable *Lewis structure* showing appropriate shape and *dashed/wedged bonds*, based on VSEPR. Determine the **molecular geometry** around the central atom. For HClO_4 , label the *most polar bond*.



b. 3 points (Note: Lone pairs are not shown in structures below)

(i) For the three *resonance* structures (1, 2 and 3) shown below, answer the following question:



(I) Circle the *most contributing* resonance structure.

Structure 1

Structure 2

Structure 3

(ii) For the three structures given (4, 5, and 6) below, answer the following questions:



(I) Circle all the structures that represent *resonance structures* of each other:

Structure 4

Structure 5

Structure 6

(II) Among the resonance structures, circle the *most contributing* resonance structure.

Structure 4

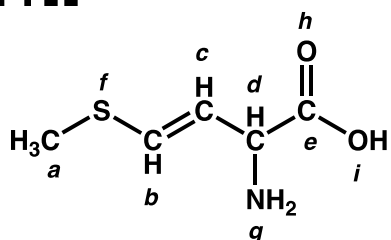
Structure 5

Structure 6

**Question 6** **6 points**

Using the data given below, calculate the **lattice energy** for SrO(s).

	ΔH° (kJ/mol)
$\text{Sr}(s) \longrightarrow \text{Sr}(g)$	+180
$\text{Sr}(g) \longrightarrow \text{Sr}^{2+}(g) + 2e^-$	+1468
$\text{O}_2(g) \longrightarrow 2\text{O}(g)$	+500
$\text{O}(g) + 2e^- \longrightarrow \text{O}^{2-}(g)$	+600
$\text{Sr}(s) + \frac{1}{2}\text{O}_2(g) \longrightarrow \text{SrO}(s)$	-550

**Question 7 10 points**

Consider the molecule given on the left, and answer questions (i) to (vi) using chemical bonding theories that we have learnt in class (valence bond theory, hybridization).

Note that the lone pairs are not shown in the structure:

i. Consider the carbon atoms marked in as ***a***, ***b***, ***c***, ***d***, and ***e*** in the molecule, and determine **hybridization** at each of the carbon centers. (2.5 points)

Atom	Hybridization
Carbon (<i>a</i>)	
Carbon (<i>b</i>)	
Carbon (<i>c</i>)	
Carbon (<i>d</i>)	
Carbon (<i>e</i>)	

ii. Now, consider the sulfur atom (S) marked as ***f***, nitrogen (N) atom (***g***), and oxygen(O) atoms (***h*** and ***i***), and determine their **hybridization**. (2 points)

Atom	Hybridization
Sulfur (<i>f</i>)	
Nitrogen (<i>g</i>)	
Oxygen (<i>h</i>)	
Oxygen (<i>i</i>)	

iii. **Describe** all the bonds formed by the carbon atoms ***a*** and ***b***, using sigma and pi designation as appropriate. (2 points)

Atom	Bonds formed (use sigma and pi as appropriate)
Carbon (<i>a</i>)	
Carbon (<i>b</i>)	

iv. How many total sp^3 -s sigma bonds are in the molecule? (1 point)

v. Determine **electronic** and **molecular geometry** at carbon (C) labeled ***e*** and sulfur (S) labeled ***f*** centres. (2 points)

Atom	Electronic Geometry	Molecular Geometry
Carbon (<i>e</i>)		
Sulfur (<i>f</i>)		

vi. Circle the value from the choices given below, that describes the **C-S-C bond angle at sulfur** (S) marked ***f***: (0.5 points)

>90° to <109.5° 109.5° >109.5 to <120° 120° >120° to <180°


Question 8 12 points

Using the **valence shell orbitals only** draw a **complete** molecular orbital (MO) diagram (with no s-p mixing) for the molecule **PS**, in the space provided below. **(6 points)**

Assume that 3s and 3p atomic orbitals will form molecular orbitals like those formed by the combination of 2s and 2p atomic orbitals that we learnt in class, and there is no s-p mixing.

Show **energy arrow**, **atomic orbitals** involved, **atom labels**, the **molecular orbitals** that are formed, **filled electrons** in the MO diagram etc. Make sure to label all the atomic and molecular orbitals.

Using the MO diagram above, answer the following questions:

- (i) Which one of the following will have the highest bond order? Circle the correct choice
 A) PS^{1-} B) PS C) PS^{1+}
- (ii) Which one of the following will have the longest bond length? Circle the correct choice
 A) PS^{1-} B) PS C) PS^{1+}
- (iii) Which of the following will be diamagnetic? Circle the correct choice.
 A) PS^{1-} B) PS C) PS^{1+}
- (iv) Draw Lewis dot structure of the molecule with the shortest bond length (based on the MO diagram) from among the three (PS^{1-} , PS, PS^{1+}), and indicate the bond order calculated from the Lewis dot structure. **(3 points)**

Lewis Dot Structure

Bond Order



66D985B8-53D0-4E8A-9A32-9253A3D2E348

midterm-exam-93388

#240 Page 10 of 10

SCRAP PAGE

**CHEMISTRY 110 –Midterm Exam****Thursday November 7, 2024, 6:30 – 8:30 pm**

Examiners: Pallavi Sirjoosingh and Ashok Kakkar

First name (please write as legibly as possible within the boxes)

--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--

Last name

--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--

Student ID number

--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--

This booklet has 8 questions on 10 pages (The last page can be used as a scrap page).

The staple must NOT be removed from the question booklet. Your intact question booklet must be returned at the end of the exam.

A separate booklet contains a periodic table, and datasheet.

Write your answers directly under the questions in the space provided. Do not write on the QR codes or near the top edges (as it may be cut off).

Please write **legibly and dark enough** for the scan to be readable. For students who use English as a second language, dictionaries are permitted. Calculators are allowed.

DO NOT ASK QUESTIONS OF THE INVIGILATORS about interpretations of the exam questions. If you have a concern about a question, please summarize it at the top of this page. All concerns will be evaluated after the exam. GOOD LUCK!

**Question 1 8 points** (*Assume all transitions are possible*)

a. 6 points Calculate the **energy (in kJ/mol)** required to ionize (completely remove) the electron from the **n=3** level of **He⁺ ion** (He: Helium). Show your work including equations. (3 sig. figs.)

b. 2 points

(I) Select the largest (longest) wavelength from the three choices below:

- i. Wavelength of photon to *completely remove* an electron from *1s orbital of Hydrogen atom*
- ii. Wavelength of photon to *completely remove* an electron from *1s orbital of Boron atom (B)*
- iii. Wavelength of photon to *completely remove* an electron from *1s orbital of Boron cation (B⁺)*

i or ii or iii

(II) Select the smallest (shortest) wavelength from the three choices below (*for H atom*):

- i. Wavelength of photon *emitted* for an electron transitioning from *2p to 1s orbital*
- ii. Wavelength of photon *absorbed* for an electron transitioning from *2s to 4p orbital*
- iii. Wavelength of photon *absorbed* for an electron transitioning from *1s to 3p orbital*

i or ii or iii



Question 2 8 points



A Chem110 student plots the graph shown on the left, from data obtained for the kinetic energy (eV) of the ejected electron versus frequency (s^{-1}) of the incident light, for an unknown metal. ($1\text{eV} = 1.6022 \times 10^{-19} \text{ J}$)

Based on the plot, answer the questions **a to c** below:

a. 1 point On the plot above, label the **threshold frequency**.

b. 2 points What is the value of the work potential (in J) of the unknown metal?

c. 3 points What is the *longest* wavelength (in nm) required to eject an electron from this metal's surface? Show your work including equation(s) used. (3 sig figs)

d. 2 points In the space provided below, plot the following

(Assume frequency is *above* threshold frequency):

- i. *Kinetic energy* of an ejected electron versus the *intensity* of the incident light (left)
- ii. *Number of electrons* ejected versus the *frequency* of the incident light (right)



**Question 3** **6 points****a. 2 points**

In the ground state of Ge (Germanium; $Z = 32$) atom, **name** the following orbitals:

i. Orbital containing the electron that requires the maximum energy to be removed.

ii. Orbital containing the electron that experiences the minimum effective nuclear charge.

b. 2 points

An unknown anion (A^-) has the following *excited* electronic configuration: $[Ar] 4s^2 3d^3 4p^1$

(Ar: Argon; $Z = 18$)

Give the **condensed ground state electronic configuration** of the atom, **A** and the cation, **A^{2+}** .
Show your work. (Assume all electronic transitions are possible)

Electronic Configuration of A:

Electronic Configuration of A^{2+} :

c. 2 points

In a single atom, determine how many electrons can have the following set of quantum numbers?

(i) $n = 5; l = 3$

In a single atom, determine how many electrons can have the following set of quantum numbers?

(i) $n = 3; l = 2; m_s = +1/2$

**Question 4** **8 points**

Rank the following from 1 to 4 in **increasing** order. Provide the ranking in the boxes provided.

Note: Atomic number (Z) are provided for relevant atoms

a) Increasing (lowest to highest) order of *ionization energy*: (1= lowest; 4 = highest)

Mg

Mg: Z = 12

Rb

Rb: Z = 37

Cs

Cs: Z = 55

K

K : Z = 19

b) Increasing (lowest to highest) order of *bond angle*: (1= lowest; 4 = highest)

H₂OCCl₄XeF₄

HCN

c) Increasing (smallest to largest) order of *ionic size* (1= smallest; 4 = largest)

S²⁻

S: Z = 16

O²⁻

O: Z = 8

Be²⁺

Be: Z = 4

Na⁺

Na: Z = 11

d) Increasing (lowest to highest) order of *bond energy* (1= lowest; 4 = highest)

P-F

P-Cl

P-I

P-Br

**Question 5** **12 points**

a. 9 points For the molecules HClO_4 and TeF_6 , draw the most stable *Lewis structure* showing appropriate shape and *dashed/wedged bonds*, based on VSEPR. Determine the **molecular geometry** around the central atom. For HClO_4 , label the *most polar bond*.



b. 3 points (Note: Lone pairs are not shown in structures below)

(i) For the three *resonance* structures (1, 2 and 3) shown below, answer the following question:



(I) Circle the *most contributing* resonance structure.

Structure 1

Structure 2

Structure 3

(ii) For the three structures given (4, 5, and 6) below, answer the following questions:



(I) Circle all the structures that represent *resonance structures* of each other:

Structure 4

Structure 5

Structure 6

(II) Among the resonance structures, circle the *most contributing* resonance structure.

Structure 4

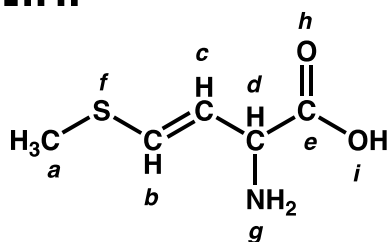
Structure 5

Structure 6

**Question 6** **6 points**

Using the data given below, calculate the **lattice energy** for SrO(s).

	ΔH° (kJ/mol)
$\text{Sr}(s) \longrightarrow \text{Sr}(g)$	+180
$\text{Sr}(g) \longrightarrow \text{Sr}^{2+}(g) + 2e^-$	+1468
$\text{O}_2(g) \longrightarrow 2\text{O}(g)$	+500
$\text{O}(g) + 2e^- \longrightarrow \text{O}^{2-}(g)$	+600
$\text{Sr}(s) + \frac{1}{2}\text{O}_2(g) \longrightarrow \text{SrO}(s)$	-550

**Question 7 10 points**

Consider the molecule given on the left, and answer questions (i) to (vi) using chemical bonding theories that we have learnt in class (valence bond theory, hybridization).

Note that the lone pairs are not shown in the structure:

i. Consider the carbon atoms marked in as ***a***, ***b***, ***c***, ***d***, and ***e*** in the molecule, and determine **hybridization** at each of the carbon centers. (2.5 points)

Atom	Hybridization
Carbon (<i>a</i>)	
Carbon (<i>b</i>)	
Carbon (<i>c</i>)	
Carbon (<i>d</i>)	
Carbon (<i>e</i>)	

ii. Now, consider the sulfur atom (S) marked as ***f***, nitrogen (N) atom (***g***), and oxygen(O) atoms (***h*** and ***i***), and determine their **hybridization**. (2 points)

Atom	Hybridization
Sulfur (<i>f</i>)	
Nitrogen (<i>g</i>)	
Oxygen (<i>h</i>)	
Oxygen (<i>i</i>)	

iii. **Describe** all the bonds formed by the carbon atoms ***a*** and ***b***, using sigma and pi designation as appropriate. (2 points)

Atom	Bonds formed (use sigma and pi as appropriate)
Carbon (<i>a</i>)	
Carbon (<i>b</i>)	

iv. How many total sp^3 -s sigma bonds are in the molecule? (1 point)

v. Determine **electronic** and **molecular geometry** at carbon (C) labeled ***e*** and sulfur (S) labeled ***f*** centres. (2 points)

Atom	Electronic Geometry	Molecular Geometry
Carbon (<i>e</i>)		
Sulfur (<i>f</i>)		

vi. Circle the value from the choices given below, that describes the **C-S-C bond angle at sulfur** (S) marked ***f***: (0.5 points)

>90° to <109.5° 109.5° >109.5 to <120° 120° >120° to <180°


Question 8 12 points

Using the **valence shell orbitals only** draw a **complete** molecular orbital (MO) diagram (with no s-p mixing) for the molecule **PS**, in the space provided below. **(6 points)**

Assume that 3s and 3p atomic orbitals will form molecular orbitals like those formed by the combination of 2s and 2p atomic orbitals that we learnt in class, and there is no s-p mixing.

Show **energy arrow**, **atomic orbitals** involved, **atom labels**, the **molecular orbitals** that are formed, **filled electrons** in the MO diagram etc. Make sure to label all the atomic and molecular orbitals.

Using the MO diagram above, answer the following questions:

- (i) Which one of the following will have the highest bond order? Circle the correct choice
 A) PS^{1-} B) PS C) PS^{1+}
- (ii) Which one of the following will have the longest bond length? Circle the correct choice
 A) PS^{1-} B) PS C) PS^{1+}
- (iii) Which of the following will be diamagnetic? Circle the correct choice.
 A) PS^{1-} B) PS C) PS^{1+}
- (iv) Draw Lewis dot structure of the molecule with the shortest bond length (based on the MO diagram) from among the three (PS^{1-} , PS, PS^{1+}), and indicate the bond order calculated from the Lewis dot structure. **(3 points)**

Lewis Dot Structure

Bond Order



9E6127C3-2495-445D-A742-87009047D779

midterm-exam-93388

#241 Page 10 of 10

SCRAP PAGE

**CHEMISTRY 110 –Midterm Exam****Thursday November 7, 2024, 6:30 – 8:30 pm**

Examiners: Pallavi Sirjoosingh and Ashok Kakkar

First name (please write as legibly as possible within the boxes)

--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--

Last name

--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--

Student ID number

--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--

This booklet has 8 questions on 10 pages (The last page can be used as a scrap page).

The staple must NOT be removed from the question booklet. Your intact question booklet must be returned at the end of the exam.

A separate booklet contains a periodic table, and datasheet.

Write your answers directly under the questions in the space provided. Do not write on the QR codes or near the top edges (as it may be cut off).

Please write **legibly and dark enough** for the scan to be readable. For students who use English as a second language, dictionaries are permitted. Calculators are allowed.

DO NOT ASK QUESTIONS OF THE INVIGILATORS about interpretations of the exam questions. If you have a concern about a question, please summarize it at the top of this page. All concerns will be evaluated after the exam. GOOD LUCK!

**Question 1 8 points** (*Assume all transitions are possible*)

a. 6 points Calculate the **energy (in kJ/mol)** required to ionize (completely remove) the electron from the **n=3** level of **He⁺ ion** (He: Helium). Show your work including equations. (3 sig. figs.)

b. 2 points

(I) Select the largest (longest) wavelength from the three choices below:

- i. Wavelength of photon to *completely remove* an electron from *1s orbital of Hydrogen atom*
- ii. Wavelength of photon to *completely remove* an electron from *1s orbital of Boron atom (B)*
- iii. Wavelength of photon to *completely remove* an electron from *1s orbital of Boron cation (B⁺)*

i or ii or iii

(II) Select the smallest (shortest) wavelength from the three choices below (*for H atom*):

- i. Wavelength of photon *emitted* for an electron transitioning from *2p to 1s orbital*
- ii. Wavelength of photon *absorbed* for an electron transitioning from *2s to 4p orbital*
- iii. Wavelength of photon *absorbed* for an electron transitioning from *1s to 3p orbital*

i or ii or iii



Question 2 8 points



A Chem110 student plots the graph shown on the left, from data obtained for the kinetic energy (eV) of the ejected electron versus frequency (s^{-1}) of the incident light, for an unknown metal. ($1\text{eV} = 1.6022 \times 10^{-19} \text{ J}$)

Based on the plot, answer the questions **a to c** below:

a. 1 point On the plot above, label the **threshold frequency**.

b. 2 points What is the value of the work potential (in J) of the unknown metal?

c. 3 points What is the *longest* wavelength (in nm) required to eject an electron from this metal's surface? Show your work including equation(s) used. (3 sig figs)

d. 2 points In the space provided below, plot the following

(Assume frequency is *above* threshold frequency):

- Kinetic energy* of an ejected electron versus the *intensity* of the incident light (left)
- Number of electrons* ejected versus the *frequency* of the incident light (right)



**Question 3** **6 points****a. 2 points**

In the ground state of Ge (Germanium; $Z = 32$) atom, **name** the following orbitals:

i. Orbital containing the electron that requires the maximum energy to be removed.

ii. Orbital containing the electron that experiences the minimum effective nuclear charge.

b. 2 points

An unknown anion (A^-) has the following *excited* electronic configuration: $[Ar] 4s^2 3d^3 4p^1$

(Ar: Argon; $Z = 18$)

Give the **condensed ground state electronic configuration** of the atom, **A** and the cation, **A^{2+}** .
Show your work. (Assume all electronic transitions are possible)

Electronic Configuration of A:

Electronic Configuration of A^{2+} :

c. 2 points

In a single atom, determine how many electrons can have the following set of quantum numbers?

(i) $n = 5; l = 3$

In a single atom, determine how many electrons can have the following set of quantum numbers?

(i) $n = 3; l = 2; m_s = +1/2$

**Question 4** **8 points**

Rank the following from 1 to 4 in **increasing** order. Provide the ranking in the boxes provided.

Note: Atomic number (Z) are provided for relevant atoms

a) Increasing (lowest to highest) order of *ionization energy*: (1= lowest; 4 = highest)

Mg

Mg: Z = 12

Rb

Rb: Z = 37

Cs

Cs: Z = 55

K

K : Z = 19

b) Increasing (lowest to highest) order of *bond angle*: (1= lowest; 4 = highest)

H₂OCCl₄XeF₄

HCN

c) Increasing (smallest to largest) order of *ionic size* (1= smallest; 4 = largest)

S²⁻

S: Z = 16

O²⁻

O: Z = 8

Be²⁺

Be: Z = 4

Na⁺

Na: Z = 11

d) Increasing (lowest to highest) order of *bond energy* (1= lowest; 4 = highest)

P-F

P-Cl

P-I

P-Br

**Question 5** **12 points**

a. 9 points For the molecules HClO_4 and TeF_6 , draw the most stable *Lewis structure* showing appropriate shape and *dashed/wedged bonds*, based on VSEPR. Determine the **molecular geometry** around the central atom. For HClO_4 , label the *most polar bond*.

HClO_4 (there are no O-O bonds)

Label the most polar bond

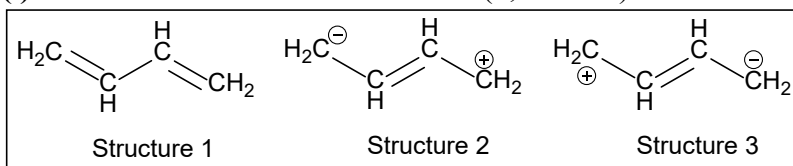
Molecular Geometry around Cl

TeF_6

Molecular Geometry around Te

b. 3 points (Note: Lone pairs are not shown in structures below)

(i) For the three *resonance* structures (1, 2 and 3) shown below, answer the following question:



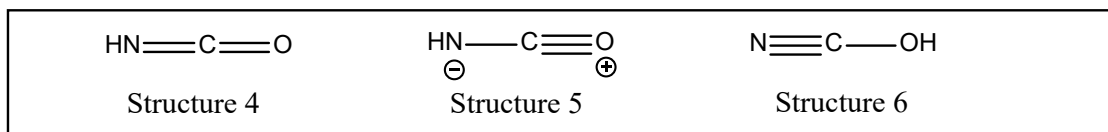
(I) Circle the *most contributing* resonance structure.

Structure 1

Structure 2

Structure 3

(ii) For the three structures given (4, 5, and 6) below, answer the following questions:



(I) Circle all the structures that represent *resonance structures* of each other:

Structure 4

Structure 5

Structure 6

(II) Among the resonance structures, circle the *most contributing* resonance structure.

Structure 4

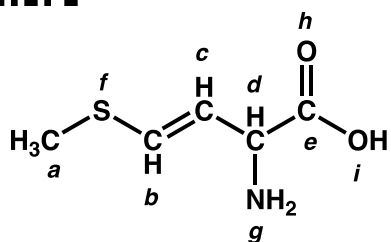
Structure 5

Structure 6

**Question 6** **6 points**

Using the data given below, calculate the **lattice energy** for SrO(s).

	ΔH° (kJ/mol)
$\text{Sr}(s) \longrightarrow \text{Sr}(g)$	+180
$\text{Sr}(g) \longrightarrow \text{Sr}^{2+}(g) + 2e^-$	+1468
$\text{O}_2(g) \longrightarrow 2\text{O}(g)$	+500
$\text{O}(g) + 2e^- \longrightarrow \text{O}^{2-}(g)$	+600
$\text{Sr}(s) + \frac{1}{2}\text{O}_2(g) \longrightarrow \text{SrO}(s)$	-550

**Question 7 10 points**

Consider the molecule given on the left, and answer questions (i) to (vi) using chemical bonding theories that we have learnt in class (valence bond theory, hybridization).

Note that the lone pairs are not shown in the structure:

i. Consider the carbon atoms marked in as ***a***, ***b***, ***c***, ***d***, and ***e*** in the molecule, and determine **hybridization** at each of the carbon centers. (2.5 points)

Atom	Hybridization
Carbon (<i>a</i>)	
Carbon (<i>b</i>)	
Carbon (<i>c</i>)	
Carbon (<i>d</i>)	
Carbon (<i>e</i>)	

ii. Now, consider the sulfur atom (S) marked as ***f***, nitrogen (N) atom (***g***), and oxygen(O) atoms (***h*** and ***i***), and determine their **hybridization**. (2 points)

Atom	Hybridization
Sulfur (<i>f</i>)	
Nitrogen (<i>g</i>)	
Oxygen (<i>h</i>)	
Oxygen (<i>i</i>)	

iii. **Describe** all the bonds formed by the carbon atoms ***a*** and ***b***, using sigma and pi designation as appropriate. (2 points)

Atom	Bonds formed (use sigma and pi as appropriate)
Carbon (<i>a</i>)	
Carbon (<i>b</i>)	

iv. How many total sp^3 -s sigma bonds are in the molecule? (1 point)

v. Determine **electronic** and **molecular geometry** at carbon (C) labeled ***e*** and sulfur (S) labeled ***f*** centres. (2 points)

Atom	Electronic Geometry	Molecular Geometry
Carbon (<i>e</i>)		
Sulfur (<i>f</i>)		

vi. Circle the value from the choices given below, that describes the **C-S-C bond angle at sulfur** (S) marked ***f***: (0.5 points)

>90° to <109.5° 109.5° >109.5 to <120° 120° >120° to <180°

**Question 8 12 points**

Using the **valence shell orbitals only** draw a **complete** molecular orbital (MO) diagram (with no s-p mixing) for the molecule **PS**, in the space provided below. **(6 points)**

Assume that 3s and 3p atomic orbitals will form molecular orbitals like those formed by the combination of 2s and 2p atomic orbitals that we learnt in class, and there is no s-p mixing.

Show **energy arrow**, **atomic orbitals** involved, **atom labels**, the **molecular orbitals** that are formed, **filled electrons** in the MO diagram etc. Make sure to label all the atomic and molecular orbitals.

Using the MO diagram above, answer the following questions:

- (i) Which one of the following will have the highest bond order? Circle the correct choice
A) PS^{1-} B) PS C) PS^{1+}
- (ii) Which one of the following will have the longest bond length? Circle the correct choice
A) PS^{1-} B) PS C) PS^{1+}
- (iii) Which of the following will be diamagnetic? Circle the correct choice.
A) PS^{1-} B) PS C) PS^{1+}
- (iv) Draw Lewis dot structure of the molecule with the shortest bond length (based on the MO diagram) from among the three (PS^{1-} , PS, PS^{1+}), and indicate the bond order calculated from the Lewis dot structure. **(3 points)**

Lewis Dot Structure

Bond Order



1C079108-6A5F-4AD7-9075-5BFD719685EF

midterm-exam-93388

#242 Page 10 of 10

SCRAP PAGE

**CHEMISTRY 110 –Midterm Exam****Thursday November 7, 2024, 6:30 – 8:30 pm**

Examiners: Pallavi Sirjoosingh and Ashok Kakkar

First name (please write as legibly as possible within the boxes)

--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--

Last name

--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--

Student ID number

--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--

This booklet has 8 questions on 10 pages (The last page can be used as a scrap page).

The staple must NOT be removed from the question booklet. Your intact question booklet must be returned at the end of the exam.

A separate booklet contains a periodic table, and datasheet.

Write your answers directly under the questions in the space provided. Do not write on the QR codes or near the top edges (as it may be cut off).

Please write **legibly and dark enough** for the scan to be readable. For students who use English as a second language, dictionaries are permitted. Calculators are allowed.

DO NOT ASK QUESTIONS OF THE INVIGILATORS about interpretations of the exam questions. If you have a concern about a question, please summarize it at the top of this page. All concerns will be evaluated after the exam. GOOD LUCK!

**Question 1 8 points** (*Assume all transitions are possible*)

a. 6 points Calculate the **energy (in kJ/mol)** required to ionize (completely remove) the electron from the **$n=3$** level of **He^+ ion** (He: Helium). Show your work including equations. (3 sig. figs.)

b. 2 points

(I) Select the largest (longest) wavelength from the three choices below:

- i. Wavelength of photon to *completely remove* an electron from *1s orbital of Hydrogen atom*
- ii. Wavelength of photon to *completely remove* an electron from *1s orbital of Boron atom (B)*
- iii. Wavelength of photon to *completely remove* an electron from *1s orbital of Boron cation (B^+)*

i or ii or iii

(II) Select the smallest (shortest) wavelength from the three choices below (*for H atom*):

- i. Wavelength of photon *emitted* for an electron transitioning from *2p to 1s orbital*
- ii. Wavelength of photon *absorbed* for an electron transitioning from *2s to 4p orbital*
- iii. Wavelength of photon *absorbed* for an electron transitioning from *1s to 3p orbital*

i or ii or iii



Question 2 8 points



A Chem110 student plots the graph shown on the left, from data obtained for the kinetic energy (eV) of the ejected electron versus frequency (s^{-1}) of the incident light, for an unknown metal. ($1\text{eV} = 1.6022 \times 10^{-19} \text{ J}$)

Based on the plot, answer the questions **a to c** below:

a. 1 point On the plot above, label the **threshold frequency**.

b. 2 points What is the value of the work potential (in J) of the unknown metal?

c. 3 points What is the *longest* wavelength (in nm) required to eject an electron from this metal's surface? Show your work including equation(s) used. (3 sig figs)

d. 2 points In the space provided below, plot the following

(Assume frequency is *above* threshold frequency):

- Kinetic energy* of an ejected electron versus the *intensity* of the incident light (left)
- Number of electrons* ejected versus the *frequency* of the incident light (right)



**Question 3** **6 points****a. 2 points**

In the ground state of Ge (Germanium; $Z = 32$) atom, **name** the following orbitals:

i. Orbital containing the electron that requires the maximum energy to be removed.

ii. Orbital containing the electron that experiences the minimum effective nuclear charge.

b. 2 points

An unknown anion (A^-) has the following *excited* electronic configuration: $[Ar] 4s^2 3d^3 4p^1$

(Ar: Argon; $Z = 18$)

Give the **condensed ground state electronic configuration** of the atom, **A** and the cation, **A^{2+}** .
Show your work. (Assume all electronic transitions are possible)

Electronic Configuration of A:

Electronic Configuration of A^{2+} :

c. 2 points

In a single atom, determine how many electrons can have the following set of quantum numbers?

(i) $n = 5; l = 3$

In a single atom, determine how many electrons can have the following set of quantum numbers?

(i) $n = 3; l = 2; m_s = +1/2$

**Question 4** **8 points**

Rank the following from 1 to 4 in **increasing** order. Provide the ranking in the boxes provided.

Note: Atomic number (Z) are provided for relevant atoms

a) Increasing (lowest to highest) order of *ionization energy*: (1= lowest; 4 = highest)

Mg

Mg: Z = 12

Rb

Rb: Z = 37

Cs

Cs: Z = 55

K

K : Z = 19

b) Increasing (lowest to highest) order of *bond angle*: (1= lowest; 4 = highest)

H₂OCCl₄XeF₄

HCN

c) Increasing (smallest to largest) order of *ionic size* (1= smallest; 4 = largest)

S²⁻

S: Z = 16

O²⁻

O: Z = 8

Be²⁺

Be: Z = 4

Na⁺

Na: Z = 11

d) Increasing (lowest to highest) order of *bond energy* (1= lowest; 4 = highest)

P-F

P-Cl

P-I

P-Br

**Question 5** **12 points**

a. 9 points For the molecules HClO_4 and TeF_6 , draw the most stable *Lewis structure* showing appropriate shape and *dashed/wedged bonds*, based on VSEPR. Determine the **molecular geometry** around the central atom. For HClO_4 , label the *most polar bond*.



b. 3 points (Note: Lone pairs are not shown in structures below)

(i) For the three *resonance* structures (1, 2 and 3) shown below, answer the following question:



(I) Circle the *most contributing* resonance structure.

Structure 1

Structure 2

Structure 3

(ii) For the three structures given (4, 5, and 6) below, answer the following questions:



(I) Circle all the structures that represent *resonance structures* of each other:

Structure 4

Structure 5

Structure 6

(II) Among the resonance structures, circle the *most contributing* resonance structure.

Structure 4

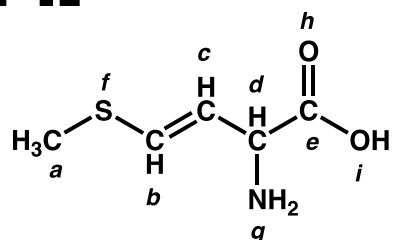
Structure 5

Structure 6

**Question 6** **6 points**

Using the data given below, calculate the **lattice energy** for SrO(s).

	ΔH° (kJ/mol)
$\text{Sr}(s) \longrightarrow \text{Sr}(g)$	+180
$\text{Sr}(g) \longrightarrow \text{Sr}^{2+}(g) + 2e^-$	+1468
$\text{O}_2(g) \longrightarrow 2\text{O}(g)$	+500
$\text{O}(g) + 2e^- \longrightarrow \text{O}^{2-}(g)$	+600
$\text{Sr}(s) + \frac{1}{2}\text{O}_2(g) \longrightarrow \text{SrO}(s)$	-550

**Question 7 10 points**

Consider the molecule given on the left, and answer questions (i) to (vi) using chemical bonding theories that we have learnt in class (valence bond theory, hybridization).

Note that the lone pairs are not shown in the structure:

i. Consider the carbon atoms marked in as ***a***, ***b***, ***c***, ***d***, and ***e*** in the molecule, and determine **hybridization** at each of the carbon centers. (2.5 points)

Atom	Hybridization
Carbon (<i>a</i>)	
Carbon (<i>b</i>)	
Carbon (<i>c</i>)	
Carbon (<i>d</i>)	
Carbon (<i>e</i>)	

ii. Now, consider the sulfur atom (S) marked as ***f***, nitrogen (N) atom (***g***), and oxygen(O) atoms (***h*** and ***i***), and determine their **hybridization**. (2 points)

Atom	Hybridization
Sulfur (<i>f</i>)	
Nitrogen (<i>g</i>)	
Oxygen (<i>h</i>)	
Oxygen (<i>i</i>)	

iii. **Describe** all the bonds formed by the carbon atoms ***a*** and ***b***, using sigma and pi designation as appropriate. (2 points)

Atom	Bonds formed (use sigma and pi as appropriate)
Carbon (<i>a</i>)	
Carbon (<i>b</i>)	

iv. How many total sp^3 -s sigma bonds are in the molecule? (1 point)

v. Determine **electronic** and **molecular geometry** at carbon (C) labeled ***e*** and sulfur (S) labeled ***f*** centres. (2 points)

Atom	Electronic Geometry	Molecular Geometry
Carbon (<i>e</i>)		
Sulfur (<i>f</i>)		

vi. Circle the value from the choices given below, that describes the **C-S-C bond angle at sulfur** (S) marked ***f***: (0.5 points)

>90° to <109.5° 109.5° >109.5 to <120° 120° >120° to <180°

**Question 8 12 points**

Using the **valence shell orbitals only** draw a **complete** molecular orbital (MO) diagram (with no s-p mixing) for the molecule **PS**, in the space provided below. **(6 points)**

Assume that 3s and 3p atomic orbitals will form molecular orbitals like those formed by the combination of 2s and 2p atomic orbitals that we learnt in class, and there is no s-p mixing.

Show **energy arrow**, **atomic orbitals** involved, **atom labels**, the **molecular orbitals** that are formed, **filled electrons** in the MO diagram etc. Make sure to label all the atomic and molecular orbitals.

Using the MO diagram above, answer the following questions:

- (i) Which one of the following will have the highest bond order? Circle the correct choice
A) PS^{1-} B) PS C) PS^{1+}
- (ii) Which one of the following will have the longest bond length? Circle the correct choice
A) PS^{1-} B) PS C) PS^{1+}
- (iii) Which of the following will be diamagnetic? Circle the correct choice.
A) PS^{1-} B) PS C) PS^{1+}
- (iv) Draw Lewis dot structure of the molecule with the shortest bond length (based on the MO diagram) from among the three (PS^{1-} , PS, PS^{1+}), and indicate the bond order calculated from the Lewis dot structure. **(3 points)**

Lewis Dot Structure

Bond Order



64FE187F-959F-45E6-9522-CAF0C901B2DF

midterm-exam-93388

#243 Page 10 of 10

SCRAP PAGE

**CHEMISTRY 110 –Midterm Exam****Thursday November 7, 2024, 6:30 – 8:30 pm****Examiners: Pallavi Sirjoosingh and Ashok Kakkar**

First name (please write as legibly as possible within the boxes)

--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--

Last name

--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--

Student ID number

--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--

This booklet has 8 questions on 10 pages (The last page can be used as a scrap page).

The staple must NOT be removed from the question booklet. Your intact question booklet must be returned at the end of the exam.

A separate booklet contains a periodic table, and datasheet.

Write your answers directly under the questions in the space provided. Do not write on the QR codes or near the top edges (as it may be cut off).

Please write **legibly and dark enough** for the scan to be readable. For students who use English as a second language, dictionaries are permitted. Calculators are allowed.

DO NOT ASK QUESTIONS OF THE INVIGILATORS about interpretations of the exam questions. If you have a concern about a question, please summarize it at the top of this page. All concerns will be evaluated after the exam. GOOD LUCK!

**Question 1 8 points** (*Assume all transitions are possible*)

a. 6 points Calculate the **energy (in kJ/mol)** required to ionize (completely remove) the electron from the **n=3** level of **He⁺ ion** (He: Helium). Show your work including equations. (3 sig. figs.)

b. 2 points

(I) Select the largest (longest) wavelength from the three choices below:

- i. Wavelength of photon to *completely remove* an electron from *1s orbital of Hydrogen atom*
- ii. Wavelength of photon to *completely remove* an electron from *1s orbital of Boron atom (B)*
- iii. Wavelength of photon to *completely remove* an electron from *1s orbital of Boron cation (B⁺)*

i or ii or iii

(II) Select the smallest (shortest) wavelength from the three choices below (*for H atom*):

- i. Wavelength of photon *emitted* for an electron transitioning from *2p to 1s orbital*
- ii. Wavelength of photon *absorbed* for an electron transitioning from *2s to 4p orbital*
- iii. Wavelength of photon *absorbed* for an electron transitioning from *1s to 3p orbital*

i or ii or iii



Question 2 8 points



A Chem110 student plots the graph shown on the left, from data obtained for the kinetic energy (eV) of the ejected electron versus frequency (s^{-1}) of the incident light, for an unknown metal. ($1\text{eV} = 1.6022 \times 10^{-19} \text{ J}$)

Based on the plot, answer the questions **a to c** below:

a. 1 point On the plot above, label the **threshold frequency**.

b. 2 points What is the value of the work potential (in J) of the unknown metal?

c. 3 points What is the *longest* wavelength (in nm) required to eject an electron from this metal's surface? Show your work including equation(s) used. (3 sig figs)

d. 2 points In the space provided below, plot the following

(Assume frequency is *above* threshold frequency):

- Kinetic energy* of an ejected electron versus the *intensity* of the incident light (left)
- Number of electrons* ejected versus the *frequency* of the incident light (right)



**Question 3** **6 points****a. 2 points**

In the ground state of Ge (Germanium; $Z = 32$) atom, **name** the following orbitals:

i. Orbital containing the electron that requires the maximum energy to be removed.

ii. Orbital containing the electron that experiences the minimum effective nuclear charge.

b. 2 points

An unknown anion (A^-) has the following *excited* electronic configuration: $[Ar] 4s^2 3d^3 4p^1$

(Ar: Argon; $Z = 18$)

Give the **condensed ground state electronic configuration** of the atom, **A** and the cation, **A^{2+}** .
Show your work. (Assume all electronic transitions are possible)

Electronic Configuration of A:

Electronic Configuration of A^{2+} :

c. 2 points

In a single atom, determine how many electrons can have the following set of quantum numbers?

(i) $n = 5; l = 3$

In a single atom, determine how many electrons can have the following set of quantum numbers?

(i) $n = 3; l = 2; m_s = +1/2$

**Question 4** **8 points**

Rank the following from 1 to 4 in **increasing** order. Provide the ranking in the boxes provided.

Note: Atomic number (Z) are provided for relevant atoms

a) Increasing (lowest to highest) order of *ionization energy*: (1= lowest; 4 = highest)

Mg

Mg: Z = 12

Rb

Rb: Z = 37

Cs

Cs: Z = 55

K

K : Z = 19

b) Increasing (lowest to highest) order of *bond angle*: (1= lowest; 4 = highest)

H₂OCCl₄XeF₄

HCN

c) Increasing (smallest to largest) order of *ionic size* (1= smallest; 4 = largest)

S²⁻

S: Z = 16

O²⁻

O: Z = 8

Be²⁺

Be: Z = 4

Na⁺

Na: Z = 11

d) Increasing (lowest to highest) order of *bond energy* (1= lowest; 4 = highest)

P-F

P-Cl

P-I

P-Br

**Question 5** **12 points**

a. 9 points For the molecules HClO_4 and TeF_6 , draw the most stable *Lewis structure* showing appropriate shape and *dashed/wedged bonds*, based on VSEPR. Determine the **molecular geometry** around the central atom. For HClO_4 , label the *most polar bond*.



b. 3 points (Note: Lone pairs are not shown in structures below)

(i) For the three *resonance* structures (1, 2 and 3) shown below, answer the following question:



(I) Circle the *most contributing* resonance structure.

Structure 1

Structure 2

Structure 3

(ii) For the three structures given (4, 5, and 6) below, answer the following questions:



(I) Circle all the structures that represent *resonance structures* of each other:

Structure 4

Structure 5

Structure 6

(II) Among the resonance structures, circle the *most contributing* resonance structure.

Structure 4

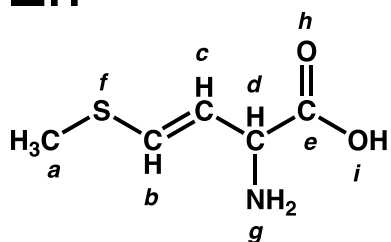
Structure 5

Structure 6

**Question 6** **6 points**

Using the data given below, calculate the **lattice energy** for SrO(s).

	ΔH° (kJ/mol)
$\text{Sr}(s) \longrightarrow \text{Sr}(g)$	+180
$\text{Sr}(g) \longrightarrow \text{Sr}^{2+}(g) + 2e^-$	+1468
$\text{O}_2(g) \longrightarrow 2\text{O}(g)$	+500
$\text{O}(g) + 2e^- \longrightarrow \text{O}^{2-}(g)$	+600
$\text{Sr}(s) + \frac{1}{2}\text{O}_2(g) \longrightarrow \text{SrO}(s)$	-550

**Question 7 10 points**

Consider the molecule given on the left, and answer questions (i) to (vi) using chemical bonding theories that we have learnt in class (valence bond theory, hybridization).

Note that the lone pairs are not shown in the structure:

i. Consider the carbon atoms marked in as ***a***, ***b***, ***c***, ***d***, and ***e*** in the molecule, and determine **hybridization** at each of the carbon centers. (2.5 points)

Atom	Hybridization
Carbon (<i>a</i>)	
Carbon (<i>b</i>)	
Carbon (<i>c</i>)	
Carbon (<i>d</i>)	
Carbon (<i>e</i>)	

ii. Now, consider the sulfur atom (S) marked as ***f***, nitrogen (N) atom (***g***), and oxygen(O) atoms (***h*** and ***i***), and determine their **hybridization**. (2 points)

Atom	Hybridization
Sulfur (<i>f</i>)	
Nitrogen (<i>g</i>)	
Oxygen (<i>h</i>)	
Oxygen (<i>i</i>)	

iii. **Describe** all the bonds formed by the carbon atoms ***a*** and ***b***, using sigma and pi designation as appropriate. (2 points)

Atom	Bonds formed (use sigma and pi as appropriate)
Carbon (<i>a</i>)	
Carbon (<i>b</i>)	

iv. How many total sp^3 -s sigma bonds are in the molecule? (1 point)

v. Determine **electronic** and **molecular geometry** at carbon (C) labeled ***e*** and sulfur (S) labeled ***f*** centres. (2 points)

Atom	Electronic Geometry	Molecular Geometry
Carbon (<i>e</i>)		
Sulfur (<i>f</i>)		

vi. Circle the value from the choices given below, that describes the **C-S-C bond angle at sulfur** (S) marked ***f***: (0.5 points)

>90° to <109.5° 109.5° >109.5 to <120° 120° >120° to <180°

**Question 8 12 points**

Using the **valence shell orbitals only** draw a **complete** molecular orbital (MO) diagram (with no s-p mixing) for the molecule **PS**, in the space provided below. **(6 points)**

Assume that 3s and 3p atomic orbitals will form molecular orbitals like those formed by the combination of 2s and 2p atomic orbitals that we learnt in class, and there is no s-p mixing.

Show **energy arrow**, **atomic orbitals** involved, **atom labels**, the **molecular orbitals** that are formed, **filled electrons** in the MO diagram etc. Make sure to label all the atomic and molecular orbitals.

Using the MO diagram above, answer the following questions:

- (i) Which one of the following will have the highest bond order? Circle the correct choice
A) PS^{1-} B) PS C) PS^{1+}
- (ii) Which one of the following will have the longest bond length? Circle the correct choice
A) PS^{1-} B) PS C) PS^{1+}
- (iii) Which of the following will be diamagnetic? Circle the correct choice.
A) PS^{1-} B) PS C) PS^{1+}
- (iv) Draw Lewis dot structure of the molecule with the shortest bond length (based on the MO diagram) from among the three (PS^{1-} , PS, PS^{1+}), and indicate the bond order calculated from the Lewis dot structure. **(3 points)**

Lewis Dot Structure

Bond Order



3CA4C34B-AFD4-4108-916F-6E7F8468BA22

midterm-exam-93388

#244 Page 10 of 10

SCRAP PAGE

**CHEMISTRY 110 –Midterm Exam****Thursday November 7, 2024, 6:30 – 8:30 pm**

Examiners: Pallavi Sirjoosingh and Ashok Kakkar

First name (please write as legibly as possible within the boxes)

--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--

Last name

--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--

Student ID number

--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--

This booklet has 8 questions on 10 pages (The last page can be used as a scrap page).

The staple must NOT be removed from the question booklet. Your intact question booklet must be returned at the end of the exam.

A separate booklet contains a periodic table, and datasheet.

Write your answers directly under the questions in the space provided. Do not write on the QR codes or near the top edges (as it may be cut off).

Please write **legibly and dark enough** for the scan to be readable. For students who use English as a second language, dictionaries are permitted. Calculators are allowed.

DO NOT ASK QUESTIONS OF THE INVIGILATORS about interpretations of the exam questions. If you have a concern about a question, please summarize it at the top of this page. All concerns will be evaluated after the exam. GOOD LUCK!

**Question 1 8 points** (*Assume all transitions are possible*)

a. 6 points Calculate the **energy (in kJ/mol)** required to ionize (completely remove) the electron from the **n=3** level of **He⁺ ion** (He: Helium). Show your work including equations. (3 sig. figs.)

b. 2 points

(I) Select the largest (longest) wavelength from the three choices below:

- i. Wavelength of photon to *completely remove* an electron from *1s orbital of Hydrogen atom*
- ii. Wavelength of photon to *completely remove* an electron from *1s orbital of Boron atom (B)*
- iii. Wavelength of photon to *completely remove* an electron from *1s orbital of Boron cation (B⁺)*

i or ii or iii

(II) Select the smallest (shortest) wavelength from the three choices below (*for H atom*):

- i. Wavelength of photon *emitted* for an electron transitioning from *2p to 1s orbital*
- ii. Wavelength of photon *absorbed* for an electron transitioning from *2s to 4p orbital*
- iii. Wavelength of photon *absorbed* for an electron transitioning from *1s to 3p orbital*

i or ii or iii



Question 2 8 points



A Chem110 student plots the graph shown on the left, from data obtained for the kinetic energy (eV) of the ejected electron versus frequency (s^{-1}) of the incident light, for an unknown metal. ($1\text{eV} = 1.6022 \times 10^{-19} \text{ J}$)

Based on the plot, answer the questions **a to c** below:

a. 1 point On the plot above, label the **threshold frequency**.

b. 2 points What is the value of the work potential (in J) of the unknown metal?

c. 3 points What is the *longest* wavelength (in nm) required to eject an electron from this metal's surface? Show your work including equation(s) used. (3 sig figs)

d. 2 points In the space provided below, plot the following

(Assume frequency is *above* threshold frequency):

- Kinetic energy* of an ejected electron versus the *intensity* of the incident light (left)
- Number of electrons* ejected versus the *frequency* of the incident light (right)



**Question 3** **6 points****a. 2 points**

In the ground state of Ge (Germanium; $Z = 32$) atom, **name** the following orbitals:

i. Orbital containing the electron that requires the maximum energy to be removed.

ii. Orbital containing the electron that experiences the minimum effective nuclear charge.

b. 2 points

An unknown anion (A^-) has the following *excited* electronic configuration: $[Ar] 4s^2 3d^3 4p^1$

(Ar: Argon; $Z = 18$)

Give the **condensed ground state electronic configuration** of the atom, **A** and the cation, **A^{2+}** .
Show your work. (Assume all electronic transitions are possible)

Electronic Configuration of A:

Electronic Configuration of A^{2+} :

c. 2 points

In a single atom, determine how many electrons can have the following set of quantum numbers?

(i) $n = 5; l = 3$

In a single atom, determine how many electrons can have the following set of quantum numbers?

(i) $n = 3; l = 2; m_s = +1/2$

**Question 4** **8 points**

Rank the following from 1 to 4 in **increasing** order. Provide the ranking in the boxes provided.

Note: Atomic number (Z) are provided for relevant atoms

a) Increasing (lowest to highest) order of *ionization energy*: (1= lowest; 4 = highest)

Mg

Mg: Z = 12

Rb

Rb: Z = 37

Cs

Cs: Z = 55

K

K : Z = 19

b) Increasing (lowest to highest) order of *bond angle*: (1= lowest; 4 = highest)

H₂OCCl₄XeF₄

HCN

c) Increasing (smallest to largest) order of *ionic size* (1= smallest; 4 = largest)

S²⁻

S: Z = 16

O²⁻

O: Z = 8

Be²⁺

Be: Z = 4

Na⁺

Na: Z = 11

d) Increasing (lowest to highest) order of *bond energy* (1= lowest; 4 = highest)

P-F

P-Cl

P-I

P-Br

**Question 5** **12 points**

a. 9 points For the molecules HClO_4 and TeF_6 , draw the most stable *Lewis structure* showing appropriate shape and *dashed/wedged bonds*, based on VSEPR. Determine the **molecular geometry** around the central atom. For HClO_4 , label the *most polar bond*.



b. 3 points (Note: Lone pairs are not shown in structures below)

(i) For the three *resonance* structures (1, 2 and 3) shown below, answer the following question:



(I) Circle the *most contributing* resonance structure.

Structure 1

Structure 2

Structure 3

(ii) For the three structures given (4, 5, and 6) below, answer the following questions:



(I) Circle all the structures that represent *resonance structures* of each other:

Structure 4

Structure 5

Structure 6

(II) Among the resonance structures, circle the *most contributing* resonance structure.

Structure 4

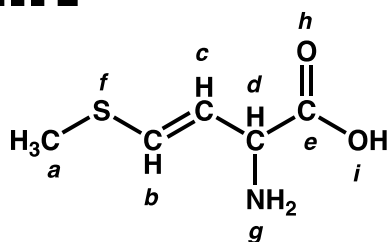
Structure 5

Structure 6

**Question 6** **6 points**

Using the data given below, calculate the **lattice energy** for SrO(s).

	ΔH° (kJ/mol)
$\text{Sr}(s) \longrightarrow \text{Sr}(g)$	+180
$\text{Sr}(g) \longrightarrow \text{Sr}^{2+}(g) + 2e^-$	+1468
$\text{O}_2(g) \longrightarrow 2\text{O}(g)$	+500
$\text{O}(g) + 2e^- \longrightarrow \text{O}^{2-}(g)$	+600
$\text{Sr}(s) + \frac{1}{2}\text{O}_2(g) \longrightarrow \text{SrO}(s)$	-550

**Question 7 10 points**

Consider the molecule given on the left, and answer questions (i) to (vi) using chemical bonding theories that we have learnt in class (valence bond theory, hybridization).

Note that the lone pairs are not shown in the structure:

i. Consider the carbon atoms marked in as ***a***, ***b***, ***c***, ***d***, and ***e*** in the molecule, and determine **hybridization** at each of the carbon centers. (2.5 points)

Atom	Hybridization
Carbon (<i>a</i>)	
Carbon (<i>b</i>)	
Carbon (<i>c</i>)	
Carbon (<i>d</i>)	
Carbon (<i>e</i>)	

ii. Now, consider the sulfur atom (S) marked as ***f***, nitrogen (N) atom (***g***), and oxygen(O) atoms (***h*** and ***i***), and determine their **hybridization**. (2 points)

Atom	Hybridization
Sulfur (<i>f</i>)	
Nitrogen (<i>g</i>)	
Oxygen (<i>h</i>)	
Oxygen (<i>i</i>)	

iii. **Describe** all the bonds formed by the carbon atoms ***a*** and ***b***, using sigma and pi designation as appropriate. (2 points)

Atom	Bonds formed (use sigma and pi as appropriate)
Carbon (<i>a</i>)	
Carbon (<i>b</i>)	

iv. How many total sp^3 -s sigma bonds are in the molecule? (1 point)

v. Determine **electronic** and **molecular geometry** at carbon (C) labeled ***e*** and sulfur (S) labeled ***f*** centres. (2 points)

Atom	Electronic Geometry	Molecular Geometry
Carbon (<i>e</i>)		
Sulfur (<i>f</i>)		

vi. Circle the value from the choices given below, that describes the **C-S-C bond angle at sulfur** (S) marked ***f***: (0.5 points)

$>90^\circ$ to $<109.5^\circ$ 109.5° >109.5 to $<120^\circ$ 120° $>120^\circ$ to $<180^\circ$

**Question 8 12 points**

Using the **valence shell orbitals only** draw a **complete** molecular orbital (MO) diagram (with no s-p mixing) for the molecule **PS**, in the space provided below. **(6 points)**

Assume that 3s and 3p atomic orbitals will form molecular orbitals like those formed by the combination of 2s and 2p atomic orbitals that we learnt in class, and there is no s-p mixing.

Show **energy arrow**, **atomic orbitals** involved, **atom labels**, the **molecular orbitals** that are formed, **filled electrons** in the MO diagram etc. Make sure to label all the atomic and molecular orbitals.

Using the MO diagram above, answer the following questions:

- (i) Which one of the following will have the highest bond order? Circle the correct choice
A) PS^{1-} B) PS C) PS^{1+}
- (ii) Which one of the following will have the longest bond length? Circle the correct choice
A) PS^{1-} B) PS C) PS^{1+}
- (iii) Which of the following will be diamagnetic? Circle the correct choice.
A) PS^{1-} B) PS C) PS^{1+}
- (iv) Draw Lewis dot structure of the molecule with the shortest bond length (based on the MO diagram) from among the three (PS^{1-} , PS, PS^{1+}), and indicate the bond order calculated from the Lewis dot structure. **(3 points)**

Lewis Dot Structure

Bond Order



F29FCEBA-659D-44BC-9310-B31D78522397

midterm-exam-93388

#245 Page 10 of 10

SCRAP PAGE

**CHEMISTRY 110 –Midterm Exam****Thursday November 7, 2024, 6:30 – 8:30 pm**

Examiners: Pallavi Sirjoosingh and Ashok Kakkar

First name (please write as legibly as possible within the boxes)

--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--

Last name

--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--

Student ID number

--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--

This booklet has 8 questions on 10 pages (The last page can be used as a scrap page).

The staple must NOT be removed from the question booklet. Your intact question booklet must be returned at the end of the exam.

A separate booklet contains a periodic table, and datasheet.

Write your answers directly under the questions in the space provided. Do not write on the QR codes or near the top edges (as it may be cut off).

Please write **legibly and dark enough** for the scan to be readable. For students who use English as a second language, dictionaries are permitted. Calculators are allowed.

DO NOT ASK QUESTIONS OF THE INVIGILATORS about interpretations of the exam questions. If you have a concern about a question, please summarize it at the top of this page. All concerns will be evaluated after the exam. GOOD LUCK!

**Question 1 8 points** (*Assume all transitions are possible*)

a. 6 points Calculate the **energy (in kJ/mol)** required to ionize (completely remove) the electron from the **$n=3$** level of **He^+ ion** (He: Helium). Show your work including equations. (3 sig. figs.)

b. 2 points

(I) Select the largest (longest) wavelength from the three choices below:

- i. Wavelength of photon to *completely remove* an electron from *1s orbital of Hydrogen atom*
- ii. Wavelength of photon to *completely remove* an electron from *1s orbital of Boron atom (B)*
- iii. Wavelength of photon to *completely remove* an electron from *1s orbital of Boron cation (B^+)*

i or ii or iii

(II) Select the smallest (shortest) wavelength from the three choices below (*for H atom*):

- i. Wavelength of photon *emitted* for an electron transitioning from *2p to 1s orbital*
- ii. Wavelength of photon *absorbed* for an electron transitioning from *2s to 4p orbital*
- iii. Wavelength of photon *absorbed* for an electron transitioning from *1s to 3p orbital*

i or ii or iii



Question 2 8 points



A Chem110 student plots the graph shown on the left, from data obtained for the kinetic energy (eV) of the ejected electron versus frequency (s^{-1}) of the incident light, for an unknown metal. ($1\text{eV} = 1.6022 \times 10^{-19} \text{ J}$)

Based on the plot, answer the questions **a to c** below:

a. 1 point On the plot above, label the **threshold frequency**.

b. 2 points What is the value of the work potential (in J) of the unknown metal?

c. 3 points What is the *longest* wavelength (in nm) required to eject an electron from this metal's surface? Show your work including equation(s) used. (3 sig figs)

d. 2 points In the space provided below, plot the following

(Assume frequency is *above* threshold frequency):

- Kinetic energy* of an ejected electron versus the *intensity* of the incident light (left)
- Number of electrons* ejected versus the *frequency* of the incident light (right)



**Question 3** **6 points****a. 2 points**

In the ground state of Ge (Germanium; $Z = 32$) atom, **name** the following orbitals:

i. Orbital containing the electron that requires the maximum energy to be removed.

ii. Orbital containing the electron that experiences the minimum effective nuclear charge.

b. 2 points

An unknown anion (A^-) has the following *excited* electronic configuration: $[Ar] 4s^2 3d^3 4p^1$

(Ar: Argon; $Z = 18$)

Give the **condensed ground state electronic configuration** of the atom, **A** and the cation, **A^{2+}** .
Show your work. (Assume all electronic transitions are possible)

Electronic Configuration of A:

Electronic Configuration of A^{2+} :

c. 2 points

In a single atom, determine how many electrons can have the following set of quantum numbers?

(i) $n = 5; l = 3$

In a single atom, determine how many electrons can have the following set of quantum numbers?

(i) $n = 3; l = 2; m_s = +1/2$

**Question 4** **8 points**

Rank the following from 1 to 4 in **increasing** order. Provide the ranking in the boxes provided.

Note: Atomic number (Z) are provided for relevant atoms

a) Increasing (lowest to highest) order of *ionization energy*: (1= lowest; 4 = highest)

Mg

Mg: Z = 12

Rb

Rb: Z = 37

Cs

Cs: Z = 55

K

K : Z = 19

b) Increasing (lowest to highest) order of *bond angle*: (1= lowest; 4 = highest)

H₂OCCl₄XeF₄

HCN

c) Increasing (smallest to largest) order of *ionic size* (1= smallest; 4 = largest)

S²⁻

S: Z = 16

O²⁻

O: Z = 8

Be²⁺

Be: Z = 4

Na⁺

Na: Z = 11

d) Increasing (lowest to highest) order of *bond energy* (1= lowest; 4 = highest)

P-F

P-Cl

P-I

P-Br

**Question 5** **12 points**

a. 9 points For the molecules HClO_4 and TeF_6 , draw the most stable *Lewis structure* showing appropriate shape and *dashed/wedged bonds*, based on VSEPR. Determine the **molecular geometry** around the central atom. For HClO_4 , label the *most polar bond*.



b. 3 points (Note: Lone pairs are not shown in structures below)

(i) For the three *resonance* structures (1, 2 and 3) shown below, answer the following question:



(I) Circle the *most contributing* resonance structure.

Structure 1

Structure 2

Structure 3

(ii) For the three structures given (4, 5, and 6) below, answer the following questions:



(I) Circle all the structures that represent *resonance structures* of each other:

Structure 4

Structure 5

Structure 6

(II) Among the resonance structures, circle the *most contributing* resonance structure.

Structure 4

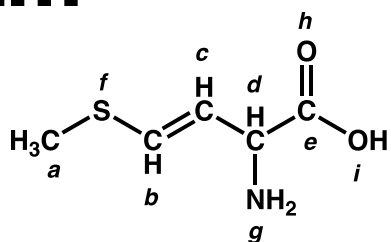
Structure 5

Structure 6

**Question 6** **6 points**

Using the data given below, calculate the **lattice energy** for SrO(s).

	ΔH° (kJ/mol)
$\text{Sr}(s) \longrightarrow \text{Sr}(g)$	+180
$\text{Sr}(g) \longrightarrow \text{Sr}^{2+}(g) + 2e^-$	+1468
$\text{O}_2(g) \longrightarrow 2\text{O}(g)$	+500
$\text{O}(g) + 2e^- \longrightarrow \text{O}^{2-}(g)$	+600
$\text{Sr}(s) + \frac{1}{2}\text{O}_2(g) \longrightarrow \text{SrO}(s)$	-550

**Question 7 10 points**

Consider the molecule given on the left, and answer questions (i) to (vi) using chemical bonding theories that we have learnt in class (valence bond theory, hybridization).

Note that the lone pairs are not shown in the structure:

i. Consider the carbon atoms marked in as ***a***, ***b***, ***c***, ***d***, and ***e*** in the molecule, and determine **hybridization** at each of the carbon centers. (2.5 points)

Atom	Hybridization
Carbon (<i>a</i>)	
Carbon (<i>b</i>)	
Carbon (<i>c</i>)	
Carbon (<i>d</i>)	
Carbon (<i>e</i>)	

ii. Now, consider the sulfur atom (S) marked as ***f***, nitrogen (N) atom (***g***), and oxygen(O) atoms (***h*** and ***i***), and determine their **hybridization**. (2 points)

Atom	Hybridization
Sulfur (<i>f</i>)	
Nitrogen (<i>g</i>)	
Oxygen (<i>h</i>)	
Oxygen (<i>i</i>)	

iii. **Describe** all the bonds formed by the carbon atoms ***a*** and ***b***, using sigma and pi designation as appropriate. (2 points)

Atom	Bonds formed (use sigma and pi as appropriate)
Carbon (<i>a</i>)	
Carbon (<i>b</i>)	

iv. How many total sp^3 -s sigma bonds are in the molecule? (1 point)

v. Determine **electronic** and **molecular geometry** at carbon (C) labeled ***e*** and sulfur (S) labeled ***f*** centres. (2 points)

Atom	Electronic Geometry	Molecular Geometry
Carbon (<i>e</i>)		
Sulfur (<i>f</i>)		

vi. Circle the value from the choices given below, that describes the **C-S-C bond angle at sulfur** (S) marked ***f***: (0.5 points)

>90° to <109.5° 109.5° >109.5 to <120° 120° >120° to <180°


Question 8 12 points

Using the **valence shell orbitals only** draw a **complete** molecular orbital (MO) diagram (with no s-p mixing) for the molecule **PS**, in the space provided below. **(6 points)**

Assume that 3s and 3p atomic orbitals will form molecular orbitals like those formed by the combination of 2s and 2p atomic orbitals that we learnt in class, and there is no s-p mixing.

Show **energy arrow**, **atomic orbitals** involved, **atom labels**, the **molecular orbitals** that are formed, **filled electrons** in the MO diagram etc. Make sure to label all the atomic and molecular orbitals.

Using the MO diagram above, answer the following questions:

- (i) Which one of the following will have the highest bond order? Circle the correct choice
 A) PS^{1-} B) PS C) PS^{1+}
- (ii) Which one of the following will have the longest bond length? Circle the correct choice
 A) PS^{1-} B) PS C) PS^{1+}
- (iii) Which of the following will be diamagnetic? Circle the correct choice.
 A) PS^{1-} B) PS C) PS^{1+}
- (iv) Draw Lewis dot structure of the molecule with the shortest bond length (based on the MO diagram) from among the three (PS^{1-} , PS, PS^{1+}), and indicate the bond order calculated from the Lewis dot structure. **(3 points)**

Lewis Dot Structure

Bond Order



FD9D2EAA-0CD2-44E3-BA15-85B9D8607A3D

midterm-exam-93388

#246 Page 10 of 10

SCRAP PAGE

**CHEMISTRY 110 –Midterm Exam****Thursday November 7, 2024, 6:30 – 8:30 pm****Examiners: Pallavi Sirjoosingh and Ashok Kakkar**

First name (please write as legibly as possible within the boxes)

--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--

Last name

--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--

Student ID number

--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--

This booklet has 8 questions on 10 pages (The last page can be used as a scrap page).

The staple must NOT be removed from the question booklet. Your intact question booklet must be returned at the end of the exam.

A separate booklet contains a periodic table, and datasheet.

Write your answers directly under the questions in the space provided. Do not write on the QR codes or near the top edges (as it may be cut off).

Please write **legibly and dark enough** for the scan to be readable. For students who use English as a second language, dictionaries are permitted. Calculators are allowed.

DO NOT ASK QUESTIONS OF THE INVIGILATORS about interpretations of the exam questions. If you have a concern about a question, please summarize it at the top of this page. All concerns will be evaluated after the exam. GOOD LUCK!

**Question 1 8 points** (*Assume all transitions are possible*)

a. 6 points Calculate the **energy (in kJ/mol)** required to ionize (completely remove) the electron from the **n=3** level of **He⁺ ion** (He: Helium). Show your work including equations. (3 sig. figs.)

b. 2 points

(I) Select the largest (longest) wavelength from the three choices below:

- i. Wavelength of photon to *completely remove* an electron from *1s orbital of Hydrogen atom*
- ii. Wavelength of photon to *completely remove* an electron from *1s orbital of Boron atom (B)*
- iii. Wavelength of photon to *completely remove* an electron from *1s orbital of Boron cation (B⁺)*

i or ii or iii

(II) Select the smallest (shortest) wavelength from the three choices below (*for H atom*):

- i. Wavelength of photon *emitted* for an electron transitioning from *2p to 1s orbital*
- ii. Wavelength of photon *absorbed* for an electron transitioning from *2s to 4p orbital*
- iii. Wavelength of photon *absorbed* for an electron transitioning from *1s to 3p orbital*

i or ii or iii



Question 2 8 points



A Chem110 student plots the graph shown on the left, from data obtained for the kinetic energy (eV) of the ejected electron versus frequency (s^{-1}) of the incident light, for an unknown metal. ($1\text{eV} = 1.6022 \times 10^{-19} \text{ J}$)

Based on the plot, answer the questions **a to c** below:

a. 1 point On the plot above, label the **threshold frequency**.

b. 2 points What is the value of the work potential (in J) of the unknown metal?

c. 3 points What is the *longest* wavelength (in nm) required to eject an electron from this metal's surface? Show your work including equation(s) used. (3 sig figs)

d. 2 points In the space provided below, plot the following

(Assume frequency is *above* threshold frequency):

- Kinetic energy* of an ejected electron versus the *intensity* of the incident light (left)
- Number of electrons* ejected versus the *frequency* of the incident light (right)



**Question 3** **6 points****a. 2 points**

In the ground state of Ge (Germanium; $Z = 32$) atom, **name** the following orbitals:

i. Orbital containing the electron that requires the maximum energy to be removed.

ii. Orbital containing the electron that experiences the minimum effective nuclear charge.

b. 2 points

An unknown anion (A^-) has the following *excited* electronic configuration: $[Ar] 4s^2 3d^3 4p^1$

(Ar: Argon; $Z = 18$)

Give the **condensed ground state electronic configuration** of the atom, **A** and the cation, **A^{2+}** .
Show your work. (Assume all electronic transitions are possible)

Electronic Configuration of A:

Electronic Configuration of A^{2+} :

c. 2 points

In a single atom, determine how many electrons can have the following set of quantum numbers?

(i) $n = 5; l = 3$

In a single atom, determine how many electrons can have the following set of quantum numbers?

(i) $n = 3; l = 2; m_s = +1/2$

**Question 4** **8 points**

Rank the following from 1 to 4 in **increasing** order. Provide the ranking in the boxes provided.

Note: Atomic number (Z) are provided for relevant atoms

a) Increasing (lowest to highest) order of *ionization energy*: (1= lowest; 4 = highest)

Mg

Mg: Z = 12

Rb

Rb: Z = 37

Cs

Cs: Z = 55

K

K : Z = 19

b) Increasing (lowest to highest) order of *bond angle*: (1= lowest; 4 = highest)

H₂OCCl₄XeF₄

HCN

c) Increasing (smallest to largest) order of *ionic size* (1= smallest; 4 = largest)

S²⁻

S: Z = 16

O²⁻

O: Z = 8

Be²⁺

Be: Z = 4

Na⁺

Na: Z = 11

d) Increasing (lowest to highest) order of *bond energy* (1= lowest; 4 = highest)

P-F

P-Cl

P-I

P-Br

**Question 5** **12 points**

a. 9 points For the molecules HClO_4 and TeF_6 , draw the most stable *Lewis structure* showing appropriate shape and *dashed/wedged bonds*, based on VSEPR. Determine the **molecular geometry** around the central atom. For HClO_4 , label the *most polar bond*.



b. 3 points (Note: Lone pairs are not shown in structures below)

(i) For the three *resonance* structures (1, 2 and 3) shown below, answer the following question:



(I) Circle the *most contributing* resonance structure.

Structure 1

Structure 2

Structure 3

(ii) For the three structures given (4, 5, and 6) below, answer the following questions:



(I) Circle all the structures that represent *resonance structures* of each other:

Structure 4

Structure 5

Structure 6

(II) Among the resonance structures, circle the *most contributing* resonance structure.

Structure 4

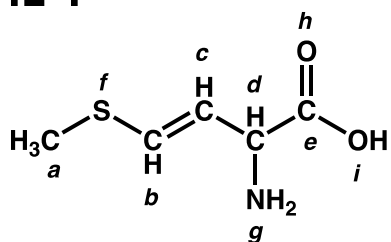
Structure 5

Structure 6

**Question 6** **6 points**

Using the data given below, calculate the **lattice energy** for SrO(s).

	ΔH° (kJ/mol)
$\text{Sr}(s) \longrightarrow \text{Sr}(g)$	+180
$\text{Sr}(g) \longrightarrow \text{Sr}^{2+}(g) + 2e^-$	+1468
$\text{O}_2(g) \longrightarrow 2\text{O}(g)$	+500
$\text{O}(g) + 2e^- \longrightarrow \text{O}^{2-}(g)$	+600
$\text{Sr}(s) + \frac{1}{2}\text{O}_2(g) \longrightarrow \text{SrO}(s)$	-550

**Question 7 10 points**

Consider the molecule given on the left, and answer questions (i) to (vi) using chemical bonding theories that we have learnt in class (valence bond theory, hybridization).

Note that the lone pairs are not shown in the structure:

i. Consider the carbon atoms marked in as ***a***, ***b***, ***c***, ***d***, and ***e*** in the molecule, and determine **hybridization** at each of the carbon centers. (2.5 points)

Atom	Hybridization
Carbon (<i>a</i>)	
Carbon (<i>b</i>)	
Carbon (<i>c</i>)	
Carbon (<i>d</i>)	
Carbon (<i>e</i>)	

ii. Now, consider the sulfur atom (S) marked as ***f***, nitrogen (N) atom (***g***), and oxygen(O) atoms (***h*** and ***i***), and determine their **hybridization**. (2 points)

Atom	Hybridization
Sulfur (<i>f</i>)	
Nitrogen (<i>g</i>)	
Oxygen (<i>h</i>)	
Oxygen (<i>i</i>)	

iii. **Describe** all the bonds formed by the carbon atoms ***a*** and ***b***, using sigma and pi designation as appropriate. (2 points)

Atom	Bonds formed (use sigma and pi as appropriate)
Carbon (<i>a</i>)	
Carbon (<i>b</i>)	

iv. How many total sp^3 -s sigma bonds are in the molecule? (1 point)

v. Determine **electronic** and **molecular geometry** at carbon (C) labeled ***e*** and sulfur (S) labeled ***f*** centres. (2 points)

Atom	Electronic Geometry	Molecular Geometry
Carbon (<i>e</i>)		
Sulfur (<i>f</i>)		

vi. Circle the value from the choices given below, that describes the **C-S-C bond angle at sulfur** (S) marked ***f***: (0.5 points)

>90° to <109.5° 109.5° >109.5 to <120° 120° >120° to <180°


Question 8 12 points

Using the **valence shell orbitals only** draw a **complete** molecular orbital (MO) diagram (with no s-p mixing) for the molecule **PS**, in the space provided below. **(6 points)**

Assume that 3s and 3p atomic orbitals will form molecular orbitals like those formed by the combination of 2s and 2p atomic orbitals that we learnt in class, and there is no s-p mixing.

Show **energy arrow**, **atomic orbitals** involved, **atom labels**, the **molecular orbitals** that are formed, **filled electrons** in the MO diagram etc. Make sure to label all the atomic and molecular orbitals.

Using the MO diagram above, answer the following questions:

- (i) Which one of the following will have the highest bond order? Circle the correct choice
 A) PS^{1-} B) PS C) PS^{1+}
- (ii) Which one of the following will have the longest bond length? Circle the correct choice
 A) PS^{1-} B) PS C) PS^{1+}
- (iii) Which of the following will be diamagnetic? Circle the correct choice.
 A) PS^{1-} B) PS C) PS^{1+}
- (iv) Draw Lewis dot structure of the molecule with the shortest bond length (based on the MO diagram) from among the three (PS^{1-} , PS, PS^{1+}), and indicate the bond order calculated from the Lewis dot structure. **(3 points)**

Lewis Dot Structure

Bond Order



99E238E4-2111-4E16-8D95-E3EDBF22C49D

midterm-exam-93388

#247 Page 10 of 10

SCRAP PAGE

**CHEMISTRY 110 –Midterm Exam****Thursday November 7, 2024, 6:30 – 8:30 pm****Examiners: Pallavi Sirjoosingh and Ashok Kakkar**

First name (please write as legibly as possible within the boxes)

--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--

Last name

--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--

Student ID number

--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--

This booklet has 8 questions on 10 pages (The last page can be used as a scrap page).

The staple must NOT be removed from the question booklet. Your intact question booklet must be returned at the end of the exam.

A separate booklet contains a periodic table, and datasheet.

Write your answers directly under the questions in the space provided. Do not write on the QR codes or near the top edges (as it may be cut off).

Please write **legibly and dark enough** for the scan to be readable. For students who use English as a second language, dictionaries are permitted. Calculators are allowed.

DO NOT ASK QUESTIONS OF THE INVIGILATORS about interpretations of the exam questions. If you have a concern about a question, please summarize it at the top of this page. All concerns will be evaluated after the exam. GOOD LUCK!

**Question 1 8 points** (*Assume all transitions are possible*)

a. 6 points Calculate the **energy (in kJ/mol)** required to ionize (completely remove) the electron from the **n=3** level of **He⁺ ion** (He: Helium). Show your work including equations. (3 sig. figs.)

b. 2 points

(I) Select the largest (longest) wavelength from the three choices below:

- i. Wavelength of photon to *completely remove* an electron from *1s orbital of Hydrogen atom*
- ii. Wavelength of photon to *completely remove* an electron from *1s orbital of Boron atom (B)*
- iii. Wavelength of photon to *completely remove* an electron from *1s orbital of Boron cation (B⁺)*

i or ii or iii

(II) Select the smallest (shortest) wavelength from the three choices below (*for H atom*):

- i. Wavelength of photon *emitted* for an electron transitioning from *2p to 1s orbital*
- ii. Wavelength of photon *absorbed* for an electron transitioning from *2s to 4p orbital*
- iii. Wavelength of photon *absorbed* for an electron transitioning from *1s to 3p orbital*

i or ii or iii



Question 2 8 points



A Chem110 student plots the graph shown on the left, from data obtained for the kinetic energy (eV) of the ejected electron versus frequency (s^{-1}) of the incident light, for an unknown metal. ($1\text{eV} = 1.6022 \times 10^{-19} \text{ J}$)

Based on the plot, answer the questions **a to c** below:

a. 1 point On the plot above, label the **threshold frequency**.

b. 2 points What is the value of the work potential (in J) of the unknown metal?

c. 3 points What is the *longest* wavelength (in nm) required to eject an electron from this metal's surface? Show your work including equation(s) used. (3 sig figs)

d. 2 points In the space provided below, plot the following

(Assume frequency is *above* threshold frequency):

- Kinetic energy* of an ejected electron versus the *intensity* of the incident light (left)
- Number of electrons* ejected versus the *frequency* of the incident light (right)



**Question 3** **6 points****a. 2 points**

In the ground state of Ge (Germanium; $Z = 32$) atom, **name** the following orbitals:

i. Orbital containing the electron that requires the maximum energy to be removed.

ii. Orbital containing the electron that experiences the minimum effective nuclear charge.

b. 2 points

An unknown anion (A^-) has the following *excited* electronic configuration: $[Ar] 4s^2 3d^3 4p^1$

(Ar: Argon; $Z = 18$)

Give the **condensed ground state electronic configuration** of the atom, **A** and the cation, **A^{2+}** .
Show your work. (Assume all electronic transitions are possible)

Electronic Configuration of A:

Electronic Configuration of A^{2+} :

c. 2 points

In a single atom, determine how many electrons can have the following set of quantum numbers?

(i) $n = 5; l = 3$

In a single atom, determine how many electrons can have the following set of quantum numbers?

(i) $n = 3; l = 2; m_s = +1/2$

**Question 4** **8 points**

Rank the following from 1 to 4 in **increasing** order. Provide the ranking in the boxes provided.

Note: Atomic number (Z) are provided for relevant atoms

a) Increasing (lowest to highest) order of *ionization energy*: (1= lowest; 4 = highest)

Mg

Mg: Z = 12

Rb

Rb: Z = 37

Cs

Cs: Z = 55

K

K : Z = 19

b) Increasing (lowest to highest) order of *bond angle*: (1= lowest; 4 = highest)

H₂OCCl₄XeF₄

HCN

c) Increasing (smallest to largest) order of *ionic size* (1= smallest; 4 = largest)

S²⁻

S: Z = 16

O²⁻

O: Z = 8

Be²⁺

Be: Z = 4

Na⁺

Na: Z = 11

d) Increasing (lowest to highest) order of *bond energy* (1= lowest; 4 = highest)

P-F

P-Cl

P-I

P-Br

**Question 5** **12 points**

a. 9 points For the molecules HClO_4 and TeF_6 , draw the most stable *Lewis structure* showing appropriate shape and *dashed/wedged bonds*, based on VSEPR. Determine the **molecular geometry** around the central atom. For HClO_4 , label the *most polar bond*.



b. 3 points (Note: Lone pairs are not shown in structures below)

(i) For the three *resonance* structures (1, 2 and 3) shown below, answer the following question:



(I) Circle the *most contributing* resonance structure.

Structure 1

Structure 2

Structure 3

(ii) For the three structures given (4, 5, and 6) below, answer the following questions:



(I) Circle all the structures that represent *resonance structures* of each other:

Structure 4

Structure 5

Structure 6

(II) Among the resonance structures, circle the *most contributing* resonance structure.

Structure 4

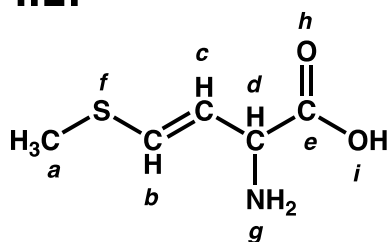
Structure 5

Structure 6

**Question 6** **6 points**

Using the data given below, calculate the **lattice energy** for SrO(s).

	ΔH° (kJ/mol)
$\text{Sr}(s) \longrightarrow \text{Sr}(g)$	+180
$\text{Sr}(g) \longrightarrow \text{Sr}^{2+}(g) + 2e^-$	+1468
$\text{O}_2(g) \longrightarrow 2\text{O}(g)$	+500
$\text{O}(g) + 2e^- \longrightarrow \text{O}^{2-}(g)$	+600
$\text{Sr}(s) + \frac{1}{2}\text{O}_2(g) \longrightarrow \text{SrO}(s)$	-550

**Question 7 10 points**

Consider the molecule given on the left, and answer questions (i) to (vi) using chemical bonding theories that we have learnt in class (valence bond theory, hybridization).

Note that the lone pairs are not shown in the structure:

i. Consider the carbon atoms marked in as ***a***, ***b***, ***c***, ***d***, and ***e*** in the molecule, and determine **hybridization** at each of the carbon centers. (2.5 points)

Atom	Hybridization
Carbon (<i>a</i>)	
Carbon (<i>b</i>)	
Carbon (<i>c</i>)	
Carbon (<i>d</i>)	
Carbon (<i>e</i>)	

ii. Now, consider the sulfur atom (S) marked as ***f***, nitrogen (N) atom (***g***), and oxygen(O) atoms (***h*** and ***i***), and determine their **hybridization**. (2 points)

Atom	Hybridization
Sulfur (<i>f</i>)	
Nitrogen (<i>g</i>)	
Oxygen (<i>h</i>)	
Oxygen (<i>i</i>)	

iii. **Describe** all the bonds formed by the carbon atoms ***a*** and ***b***, using sigma and pi designation as appropriate. (2 points)

Atom	Bonds formed (use sigma and pi as appropriate)
Carbon (<i>a</i>)	
Carbon (<i>b</i>)	

iv. How many total sp^3 -s sigma bonds are in the molecule? (1 point)

v. Determine **electronic** and **molecular geometry** at carbon (C) labeled ***e*** and sulfur (S) labeled ***f*** centres. (2 points)

Atom	Electronic Geometry	Molecular Geometry
Carbon (<i>e</i>)		
Sulfur (<i>f</i>)		

vi. Circle the value from the choices given below, that describes the **C-S-C bond angle at sulfur** (S) marked ***f***: (0.5 points)

>90° to <109.5° 109.5° >109.5 to <120° 120° >120° to <180°


Question 8 12 points

Using the **valence shell orbitals only** draw a **complete** molecular orbital (MO) diagram (with no s-p mixing) for the molecule **PS**, in the space provided below. **(6 points)**

Assume that 3s and 3p atomic orbitals will form molecular orbitals like those formed by the combination of 2s and 2p atomic orbitals that we learnt in class, and there is no s-p mixing.

Show **energy arrow**, **atomic orbitals** involved, **atom labels**, the **molecular orbitals** that are formed, **filled electrons** in the MO diagram etc. Make sure to label all the atomic and molecular orbitals.

Using the MO diagram above, answer the following questions:

- (i) Which one of the following will have the highest bond order? Circle the correct choice
 A) PS^{1-} B) PS C) PS^{1+}
- (ii) Which one of the following will have the longest bond length? Circle the correct choice
 A) PS^{1-} B) PS C) PS^{1+}
- (iii) Which of the following will be diamagnetic? Circle the correct choice.
 A) PS^{1-} B) PS C) PS^{1+}
- (iv) Draw Lewis dot structure of the molecule with the shortest bond length (based on the MO diagram) from among the three (PS^{1-} , PS, PS^{1+}), and indicate the bond order calculated from the Lewis dot structure. **(3 points)**

Lewis Dot Structure

Bond Order



2DE50B7A-348D-4C69-8F63-44EE7E2C94CC

midterm-exam-93388

#248 Page 10 of 10

SCRAP PAGE

**CHEMISTRY 110 –Midterm Exam****Thursday November 7, 2024, 6:30 – 8:30 pm****Examiners: Pallavi Sirjoosingh and Ashok Kakkar**

First name (please write as legibly as possible within the boxes)

--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--

Last name

--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--

Student ID number

--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--

This booklet has 8 questions on 10 pages (The last page can be used as a scrap page).

The staple must NOT be removed from the question booklet. Your intact question booklet must be returned at the end of the exam.

A separate booklet contains a periodic table, and datasheet.

Write your answers directly under the questions in the space provided. Do not write on the QR codes or near the top edges (as it may be cut off).

Please write **legibly and dark enough** for the scan to be readable. For students who use English as a second language, dictionaries are permitted. Calculators are allowed.

DO NOT ASK QUESTIONS OF THE INVIGILATORS about interpretations of the exam questions. If you have a concern about a question, please summarize it at the top of this page. All concerns will be evaluated after the exam. GOOD LUCK!

**Question 1 8 points** (*Assume all transitions are possible*)

a. 6 points Calculate the **energy (in kJ/mol)** required to ionize (completely remove) the electron from the **$n=3$** level of **He^+ ion** (He: Helium). Show your work including equations. (3 sig. figs.)

b. 2 points

(I) **Select** the **largest (longest)** wavelength from the three choices below:

- i. Wavelength of photon to *completely remove* an electron from *1s orbital of Hydrogen atom*
- ii. Wavelength of photon to *completely remove* an electron from *1s orbital of Boron atom (B)*
- iii. Wavelength of photon to *completely remove* an electron from *1s orbital of Boron cation (B^+)*

i **or** **ii** **or** **iii**

(II) **Select** the **smallest (shortest)** wavelength from the three choices below (*for H atom*):

- i. Wavelength of photon *emitted* for an electron transitioning from *2p to 1s orbital*
- ii. Wavelength of photon *absorbed* for an electron transitioning from *2s to 4p orbital*
- iii. Wavelength of photon *absorbed* for an electron transitioning from *1s to 3p orbital*

i **or** **ii** **or** **iii**



Question 2 8 points



A Chem110 student plots the graph shown on the left, from data obtained for the kinetic energy (eV) of the ejected electron versus frequency (s^{-1}) of the incident light, for an unknown metal. ($1\text{eV} = 1.6022 \times 10^{-19} \text{ J}$)

Based on the plot, answer the questions **a to c** below:

a. 1 point On the plot above, label the **threshold frequency**.

b. 2 points What is the value of the work potential (in J) of the unknown metal?

c. 3 points What is the *longest* wavelength (in nm) required to eject an electron from this metal's surface? Show your work including equation(s) used. (3 sig figs)

d. 2 points In the space provided below, plot the following

(Assume frequency is *above* threshold frequency):

- i. *Kinetic energy* of an ejected electron versus the *intensity* of the incident light (left)
- ii. *Number of electrons* ejected versus the *frequency* of the incident light (right)



**Question 3** **6 points****a. 2 points**

In the ground state of Ge (Germanium; $Z = 32$) atom, **name** the following orbitals:

i. Orbital containing the electron that requires the maximum energy to be removed.

ii. Orbital containing the electron that experiences the minimum effective nuclear charge.

b. 2 points

An unknown anion (A^-) has the following *excited* electronic configuration: $[Ar] 4s^2 3d^3 4p^1$

(Ar: Argon; $Z = 18$)

Give the **condensed ground state electronic configuration** of the atom, **A** and the cation, **A^{2+}** .
Show your work. (Assume all electronic transitions are possible)

Electronic Configuration of A:

Electronic Configuration of A^{2+} :

c. 2 points

In a single atom, determine how many electrons can have the following set of quantum numbers?

(i) $n = 5; l = 3$

In a single atom, determine how many electrons can have the following set of quantum numbers?

(i) $n = 3; l = 2; m_s = +1/2$

**Question 4** **8 points**

Rank the following from 1 to 4 in **increasing** order. Provide the ranking in the boxes provided.

Note: Atomic number (Z) are provided for relevant atoms

a) Increasing (lowest to highest) order of *ionization energy*: (1= lowest; 4 = highest)

Mg

Mg: Z = 12

Rb

Rb: Z = 37

Cs

Cs: Z = 55

K

K : Z = 19

b) Increasing (lowest to highest) order of *bond angle*: (1= lowest; 4 = highest)

H₂OCCl₄XeF₄

HCN

c) Increasing (smallest to largest) order of *ionic size* (1= smallest; 4 = largest)

S²⁻

S: Z = 16

O²⁻

O: Z = 8

Be²⁺

Be: Z = 4

Na⁺

Na: Z = 11

d) Increasing (lowest to highest) order of *bond energy* (1= lowest; 4 = highest)

P-F

P-Cl

P-I

P-Br

**Question 5** **12 points**

a. 9 points For the molecules HClO_4 and TeF_6 , draw the most stable *Lewis structure* showing appropriate shape and *dashed/wedged bonds*, based on VSEPR. Determine the **molecular geometry** around the central atom. For HClO_4 , label the *most polar bond*.

HClO_4 (there are no O-O bonds)

Label the most polar bond

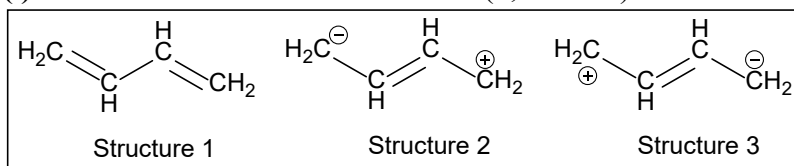
Molecular Geometry around Cl

TeF_6

Molecular Geometry around Te

b. 3 points (Note: Lone pairs are not shown in structures below)

(i) For the three *resonance* structures (1, 2 and 3) shown below, answer the following question:



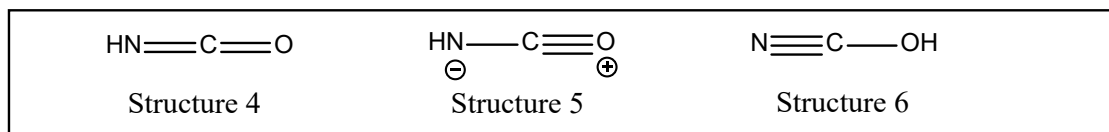
(I) Circle the *most contributing* resonance structure.

Structure 1

Structure 2

Structure 3

(ii) For the three structures given (4, 5, and 6) below, answer the following questions:



(I) Circle all the structures that represent *resonance structures* of each other:

Structure 4

Structure 5

Structure 6

(II) Among the resonance structures, circle the *most contributing* resonance structure.

Structure 4

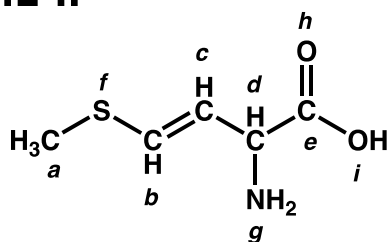
Structure 5

Structure 6

**Question 6** **6 points**

Using the data given below, calculate the **lattice energy** for SrO(s).

	ΔH° (kJ/mol)
$\text{Sr}(s) \longrightarrow \text{Sr}(g)$	+180
$\text{Sr}(g) \longrightarrow \text{Sr}^{2+}(g) + 2e^-$	+1468
$\text{O}_2(g) \longrightarrow 2\text{O}(g)$	+500
$\text{O}(g) + 2e^- \longrightarrow \text{O}^{2-}(g)$	+600
$\text{Sr}(s) + \frac{1}{2}\text{O}_2(g) \longrightarrow \text{SrO}(s)$	-550

**Question 7 10 points**

Consider the molecule given on the left, and answer questions (i) to (vi) using chemical bonding theories that we have learnt in class (valence bond theory, hybridization).

Note that the lone pairs are not shown in the structure:

i. Consider the carbon atoms marked in as ***a***, ***b***, ***c***, ***d***, and ***e*** in the molecule, and determine **hybridization** at each of the carbon centers. (2.5 points)

Atom	Hybridization
Carbon (<i>a</i>)	
Carbon (<i>b</i>)	
Carbon (<i>c</i>)	
Carbon (<i>d</i>)	
Carbon (<i>e</i>)	

ii. Now, consider the sulfur atom (S) marked as ***f***, nitrogen (N) atom (***g***), and oxygen(O) atoms (***h*** and ***i***), and determine their **hybridization**. (2 points)

Atom	Hybridization
Sulfur (<i>f</i>)	
Nitrogen (<i>g</i>)	
Oxygen (<i>h</i>)	
Oxygen (<i>i</i>)	

iii. **Describe** all the bonds formed by the carbon atoms ***a*** and ***b***, using sigma and pi designation as appropriate. (2 points)

Atom	Bonds formed (use sigma and pi as appropriate)
Carbon (<i>a</i>)	
Carbon (<i>b</i>)	

iv. How many total sp^3 -s sigma bonds are in the molecule? (1 point)

v. Determine **electronic** and **molecular geometry** at carbon (C) labeled ***e*** and sulfur (S) labeled ***f*** centres. (2 points)

Atom	Electronic Geometry	Molecular Geometry
Carbon (<i>e</i>)		
Sulfur (<i>f</i>)		

vi. Circle the value from the choices given below, that describes the **C-S-C bond angle at sulfur** (S) marked ***f***: (0.5 points)

>90° to <109.5° 109.5° >109.5 to <120° 120° >120° to <180°


Question 8 12 points

Using the **valence shell orbitals only** draw a **complete** molecular orbital (MO) diagram (with no s-p mixing) for the molecule **PS**, in the space provided below. **(6 points)**

Assume that 3s and 3p atomic orbitals will form molecular orbitals like those formed by the combination of 2s and 2p atomic orbitals that we learnt in class, and there is no s-p mixing.

Show **energy arrow**, **atomic orbitals** involved, **atom labels**, the **molecular orbitals** that are formed, **filled electrons** in the MO diagram etc. Make sure to label all the atomic and molecular orbitals.

Using the MO diagram above, answer the following questions:

- (i) Which one of the following will have the highest bond order? Circle the correct choice
 A) PS^{1-} B) PS C) PS^{1+}
- (ii) Which one of the following will have the longest bond length? Circle the correct choice
 A) PS^{1-} B) PS C) PS^{1+}
- (iii) Which of the following will be diamagnetic? Circle the correct choice.
 A) PS^{1-} B) PS C) PS^{1+}
- (iv) Draw Lewis dot structure of the molecule with the shortest bond length (based on the MO diagram) from among the three (PS^{1-} , PS, PS^{1+}), and indicate the bond order calculated from the Lewis dot structure. **(3 points)**

Lewis Dot Structure

Bond Order



1745B48B-906F-4CF3-AE58-DC0E93CB9937

midterm-exam-93388

#249 Page 10 of 10

SCRAP PAGE

**CHEMISTRY 110 –Midterm Exam****Thursday November 7, 2024, 6:30 – 8:30 pm****Examiners: Pallavi Sirjoosingh and Ashok Kakkar**

First name (please write as legibly as possible within the boxes)

--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--

Last name

--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--

Student ID number

--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--

This booklet has 8 questions on 10 pages (The last page can be used as a scrap page).

The staple must NOT be removed from the question booklet. Your intact question booklet must be returned at the end of the exam.

A separate booklet contains a periodic table, and datasheet.

Write your answers directly under the questions in the space provided. Do not write on the QR codes or near the top edges (as it may be cut off).

Please write **legibly and dark enough** for the scan to be readable. For students who use English as a second language, dictionaries are permitted. Calculators are allowed.

DO NOT ASK QUESTIONS OF THE INVIGILATORS about interpretations of the exam questions. If you have a concern about a question, please summarize it at the top of this page. All concerns will be evaluated after the exam. GOOD LUCK!

**Question 1 8 points** (*Assume all transitions are possible*)

a. 6 points Calculate the **energy (in kJ/mol)** required to ionize (completely remove) the electron from the **$n=3$** level of **He^+ ion** (He: Helium). Show your work including equations. (3 sig. figs.)

b. 2 points

(I) Select the largest (longest) wavelength from the three choices below:

- i. Wavelength of photon to *completely remove* an electron from *1s orbital of Hydrogen atom*
- ii. Wavelength of photon to *completely remove* an electron from *1s orbital of Boron atom (B)*
- iii. Wavelength of photon to *completely remove* an electron from *1s orbital of Boron cation (B^+)*

i or ii or iii

(II) Select the smallest (shortest) wavelength from the three choices below (*for H atom*):

- i. Wavelength of photon *emitted* for an electron transitioning from *2p to 1s orbital*
- ii. Wavelength of photon *absorbed* for an electron transitioning from *2s to 4p orbital*
- iii. Wavelength of photon *absorbed* for an electron transitioning from *1s to 3p orbital*

i or ii or iii



Question 2 8 points



A Chem110 student plots the graph shown on the left, from data obtained for the kinetic energy (eV) of the ejected electron versus frequency (s^{-1}) of the incident light, for an unknown metal. ($1\text{eV} = 1.6022 \times 10^{-19} \text{ J}$)

Based on the plot, answer the questions **a to c** below:

a. 1 point On the plot above, label the **threshold frequency**.

b. 2 points What is the value of the work potential (in J) of the unknown metal?

c. 3 points What is the *longest* wavelength (in nm) required to eject an electron from this metal's surface? Show your work including equation(s) used. (3 sig figs)

d. 2 points In the space provided below, plot the following

(Assume frequency is *above* threshold frequency):

- i. *Kinetic energy* of an ejected electron versus the *intensity* of the incident light (left)
- ii. *Number of electrons* ejected versus the *frequency* of the incident light (right)



**Question 3** **6 points****a. 2 points**

In the ground state of Ge (Germanium; $Z = 32$) atom, **name** the following orbitals:

i. Orbital containing the electron that requires the maximum energy to be removed.

ii. Orbital containing the electron that experiences the minimum effective nuclear charge.

b. 2 points

An unknown anion (A^-) has the following *excited* electronic configuration: $[Ar] 4s^2 3d^3 4p^1$

(Ar: Argon; $Z = 18$)

Give the **condensed ground state electronic configuration** of the atom, **A** and the cation, **A^{2+}** .
Show your work. (Assume all electronic transitions are possible)

Electronic Configuration of A:

Electronic Configuration of A^{2+} :

c. 2 points

In a single atom, determine how many electrons can have the following set of quantum numbers?

(i) $n = 5; l = 3$

In a single atom, determine how many electrons can have the following set of quantum numbers?

(i) $n = 3; l = 2; m_s = +1/2$

**Question 4** **8 points**

Rank the following from 1 to 4 in **increasing** order. Provide the ranking in the boxes provided.

Note: Atomic number (Z) are provided for relevant atoms

a) Increasing (lowest to highest) order of *ionization energy*: (1= lowest; 4 = highest)

Mg

Mg: Z = 12

Rb

Rb: Z = 37

Cs

Cs: Z = 55

K

K : Z = 19

b) Increasing (lowest to highest) order of *bond angle*: (1= lowest; 4 = highest)

H₂OCCl₄XeF₄

HCN

c) Increasing (smallest to largest) order of *ionic size* (1= smallest; 4 = largest)

S²⁻

S: Z = 16

O²⁻

O: Z = 8

Be²⁺

Be: Z = 4

Na⁺

Na: Z = 11

d) Increasing (lowest to highest) order of *bond energy* (1= lowest; 4 = highest)

P-F

P-Cl

P-I

P-Br

**Question 5** **12 points**

a. 9 points For the molecules HClO_4 and TeF_6 , draw the most stable *Lewis structure* showing appropriate shape and *dashed/wedged bonds*, based on VSEPR. Determine the **molecular geometry** around the central atom. For HClO_4 , label the *most polar bond*.



b. 3 points (Note: Lone pairs are not shown in structures below)

(i) For the three *resonance* structures (1, 2 and 3) shown below, answer the following question:



(I) Circle the *most contributing* resonance structure.

Structure 1

Structure 2

Structure 3

(ii) For the three structures given (4, 5, and 6) below, answer the following questions:



(I) Circle all the structures that represent *resonance structures* of each other:

Structure 4

Structure 5

Structure 6

(II) Among the resonance structures, circle the *most contributing* resonance structure.

Structure 4

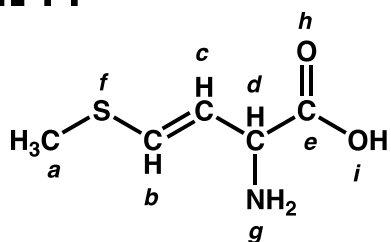
Structure 5

Structure 6

**Question 6** **6 points**

Using the data given below, calculate the **lattice energy** for SrO(s).

	ΔH° (kJ/mol)
$\text{Sr}(s) \longrightarrow \text{Sr}(g)$	+180
$\text{Sr}(g) \longrightarrow \text{Sr}^{2+}(g) + 2e^-$	+1468
$\text{O}_2(g) \longrightarrow 2\text{O}(g)$	+500
$\text{O}(g) + 2e^- \longrightarrow \text{O}^{2-}(g)$	+600
$\text{Sr}(s) + \frac{1}{2}\text{O}_2(g) \longrightarrow \text{SrO}(s)$	-550

**Question 7 10 points**

Consider the molecule given on the left, and answer questions (i) to (vi) using chemical bonding theories that we have learnt in class (valence bond theory, hybridization).

Note that the lone pairs are not shown in the structure:

i. Consider the carbon atoms marked in as ***a***, ***b***, ***c***, ***d***, and ***e*** in the molecule, and determine **hybridization** at each of the carbon centers. (2.5 points)

Atom	Hybridization
Carbon (<i>a</i>)	
Carbon (<i>b</i>)	
Carbon (<i>c</i>)	
Carbon (<i>d</i>)	
Carbon (<i>e</i>)	

ii. Now, consider the sulfur atom (S) marked as ***f***, nitrogen (N) atom (***g***), and oxygen(O) atoms (***h*** and ***i***), and determine their **hybridization**. (2 points)

Atom	Hybridization
Sulfur (<i>f</i>)	
Nitrogen (<i>g</i>)	
Oxygen (<i>h</i>)	
Oxygen (<i>i</i>)	

iii. **Describe** all the bonds formed by the carbon atoms ***a*** and ***b***, using sigma and pi designation as appropriate. (2 points)

Atom	Bonds formed (use sigma and pi as appropriate)
Carbon (<i>a</i>)	
Carbon (<i>b</i>)	

iv. How many total sp^3 -s sigma bonds are in the molecule? (1 point)

v. Determine **electronic** and **molecular geometry** at carbon (C) labeled ***e*** and sulfur (S) labeled ***f*** centres. (2 points)

Atom	Electronic Geometry	Molecular Geometry
Carbon (<i>e</i>)		
Sulfur (<i>f</i>)		

vi. Circle the value from the choices given below, that describes the **C-S-C bond angle at sulfur** (S) marked ***f***: (0.5 points)

$>90^\circ$ to $<109.5^\circ$ 109.5° >109.5 to $<120^\circ$ 120° $>120^\circ$ to $<180^\circ$

**Question 8 12 points**

Using the **valence shell orbitals only** draw a **complete** molecular orbital (MO) diagram (with no s-p mixing) for the molecule **PS**, in the space provided below. **(6 points)**

Assume that 3s and 3p atomic orbitals will form molecular orbitals like those formed by the combination of 2s and 2p atomic orbitals that we learnt in class, and there is no s-p mixing.

Show **energy arrow**, **atomic orbitals** involved, **atom labels**, the **molecular orbitals** that are formed, **filled electrons** in the MO diagram etc. Make sure to label all the atomic and molecular orbitals.

Using the MO diagram above, answer the following questions:

- (i) Which one of the following will have the highest bond order? Circle the correct choice
A) PS^{1-} B) PS C) PS^{1+}
- (ii) Which one of the following will have the longest bond length? Circle the correct choice
A) PS^{1-} B) PS C) PS^{1+}
- (iii) Which of the following will be diamagnetic? Circle the correct choice.
A) PS^{1-} B) PS C) PS^{1+}
- (iv) Draw Lewis dot structure of the molecule with the shortest bond length (based on the MO diagram) from among the three (PS^{1-} , PS, PS^{1+}), and indicate the bond order calculated from the Lewis dot structure. **(3 points)**

Lewis Dot Structure

Bond Order



ADDB577A-8107-42A8-B778-CC9A1A555135

midterm-exam-93388

#250 Page 10 of 10

SCRAP PAGE

**CHEMISTRY 110 –Midterm Exam****Thursday November 7, 2024, 6:30 – 8:30 pm**

Examiners: Pallavi Sirjoosingh and Ashok Kakkar

First name (please write as legibly as possible within the boxes)

--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--

Last name

--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--

Student ID number

--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--

This booklet has 8 questions on 10 pages (The last page can be used as a scrap page).

The staple must NOT be removed from the question booklet. Your intact question booklet must be returned at the end of the exam.

A separate booklet contains a periodic table, and datasheet.

Write your answers directly under the questions in the space provided. Do not write on the QR codes or near the top edges (as it may be cut off).

Please write **legibly and dark enough** for the scan to be readable. For students who use English as a second language, dictionaries are permitted. Calculators are allowed.

DO NOT ASK QUESTIONS OF THE INVIGILATORS about interpretations of the exam questions. If you have a concern about a question, please summarize it at the top of this page. All concerns will be evaluated after the exam. GOOD LUCK!

**Question 1 8 points** (*Assume all transitions are possible*)

a. 6 points Calculate the **energy (in kJ/mol)** required to ionize (completely remove) the electron from the **n=3** level of **He⁺ ion** (He: Helium). Show your work including equations. (3 sig. figs.)

b. 2 points

(I) Select the largest (longest) wavelength from the three choices below:

- i. Wavelength of photon to *completely remove* an electron from *1s orbital of Hydrogen atom*
- ii. Wavelength of photon to *completely remove* an electron from *1s orbital of Boron atom (B)*
- iii. Wavelength of photon to *completely remove* an electron from *1s orbital of Boron cation (B⁺)*

i or ii or iii

(II) Select the smallest (shortest) wavelength from the three choices below (*for H atom*):

- i. Wavelength of photon *emitted* for an electron transitioning from *2p to 1s orbital*
- ii. Wavelength of photon *absorbed* for an electron transitioning from *2s to 4p orbital*
- iii. Wavelength of photon *absorbed* for an electron transitioning from *1s to 3p orbital*

i or ii or iii



Question 2 8 points



A Chem110 student plots the graph shown on the left, from data obtained for the kinetic energy (eV) of the ejected electron versus frequency (s^{-1}) of the incident light, for an unknown metal. ($1\text{eV} = 1.6022 \times 10^{-19} \text{ J}$)

Based on the plot, answer the questions **a to c** below:

a. 1 point On the plot above, label the **threshold frequency**.

b. 2 points What is the value of the work potential (in J) of the unknown metal?

c. 3 points What is the *longest* wavelength (in nm) required to eject an electron from this metal's surface? Show your work including equation(s) used. (3 sig figs)

d. 2 points In the space provided below, plot the following

(Assume frequency is *above* threshold frequency):

- i. *Kinetic energy* of an ejected electron versus the *intensity* of the incident light (left)
- ii. *Number of electrons* ejected versus the *frequency* of the incident light (right)



**Question 3** **6 points****a. 2 points**

In the ground state of Ge (Germanium; $Z = 32$) atom, **name** the following orbitals:

i. Orbital containing the electron that requires the maximum energy to be removed.

ii. Orbital containing the electron that experiences the minimum effective nuclear charge.

b. 2 points

An unknown anion (A^-) has the following *excited* electronic configuration: $[Ar] 4s^2 3d^3 4p^1$

(Ar: Argon; $Z = 18$)

Give the **condensed ground state electronic configuration** of the atom, **A** and the cation, **A^{2+}** .
Show your work. (Assume all electronic transitions are possible)

Electronic Configuration of A:

Electronic Configuration of A^{2+} :

c. 2 points

In a single atom, determine how many electrons can have the following set of quantum numbers?

(i) $n = 5; l = 3$

In a single atom, determine how many electrons can have the following set of quantum numbers?

(i) $n = 3; l = 2; m_s = +1/2$

**Question 4** **8 points**

Rank the following from 1 to 4 in **increasing** order. Provide the ranking in the boxes provided.

Note: Atomic number (Z) are provided for relevant atoms

a) Increasing (lowest to highest) order of *ionization energy*: (1= lowest; 4 = highest)

Mg

Mg: Z = 12

Rb

Rb: Z = 37

Cs

Cs: Z = 55

K

K : Z = 19

b) Increasing (lowest to highest) order of *bond angle*: (1= lowest; 4 = highest)

H₂OCCl₄XeF₄

HCN

c) Increasing (smallest to largest) order of *ionic size* (1= smallest; 4 = largest)

S²⁻

S: Z = 16

O²⁻

O: Z = 8

Be²⁺

Be: Z = 4

Na⁺

Na: Z = 11

d) Increasing (lowest to highest) order of *bond energy* (1= lowest; 4 = highest)

P-F

P-Cl

P-I

P-Br

**Question 5** **12 points**

a. 9 points For the molecules HClO_4 and TeF_6 , draw the most stable *Lewis structure* showing appropriate shape and *dashed/wedged bonds*, based on VSEPR. Determine the **molecular geometry** around the central atom. For HClO_4 , label the *most polar bond*.



b. 3 points (Note: Lone pairs are not shown in structures below)

(i) For the three *resonance* structures (1, 2 and 3) shown below, answer the following question:



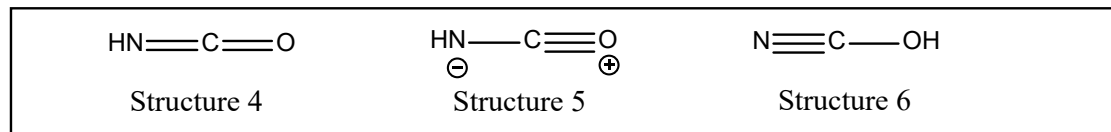
(I) Circle the *most contributing* resonance structure.

Structure 1

Structure 2

Structure 3

(ii) For the three structures given (4, 5, and 6) below, answer the following questions:



(I) Circle all the structures that represent *resonance structures* of each other:

Structure 4

Structure 5

Structure 6

(II) Among the resonance structures, circle the *most contributing* resonance structure.

Structure 4

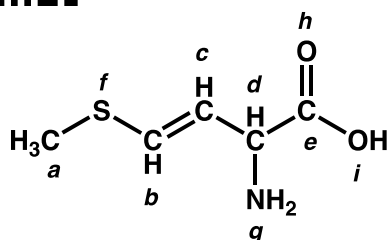
Structure 5

Structure 6

**Question 6** **6 points**

Using the data given below, calculate the **lattice energy** for SrO(s).

	ΔH° (kJ/mol)
$\text{Sr}(s) \longrightarrow \text{Sr}(g)$	+180
$\text{Sr}(g) \longrightarrow \text{Sr}^{2+}(g) + 2e^-$	+1468
$\text{O}_2(g) \longrightarrow 2\text{O}(g)$	+500
$\text{O}(g) + 2e^- \longrightarrow \text{O}^{2-}(g)$	+600
$\text{Sr}(s) + \frac{1}{2}\text{O}_2(g) \longrightarrow \text{SrO}(s)$	-550

**Question 7 10 points**

Consider the molecule given on the left, and answer questions (i) to (vi) using chemical bonding theories that we have learnt in class (valence bond theory, hybridization).

Note that the lone pairs are not shown in the structure:

i. Consider the carbon atoms marked in as ***a***, ***b***, ***c***, ***d***, and ***e*** in the molecule, and determine **hybridization** at each of the carbon centers. (2.5 points)

Atom	Hybridization
Carbon (<i>a</i>)	
Carbon (<i>b</i>)	
Carbon (<i>c</i>)	
Carbon (<i>d</i>)	
Carbon (<i>e</i>)	

ii. Now, consider the sulfur atom (S) marked as ***f***, nitrogen (N) atom (***g***), and oxygen(O) atoms (***h*** and ***i***), and determine their **hybridization**. (2 points)

Atom	Hybridization
Sulfur (<i>f</i>)	
Nitrogen (<i>g</i>)	
Oxygen (<i>h</i>)	
Oxygen (<i>i</i>)	

iii. **Describe** all the bonds formed by the carbon atoms ***a*** and ***b***, using sigma and pi designation as appropriate. (2 points)

Atom	Bonds formed (use sigma and pi as appropriate)
Carbon (<i>a</i>)	
Carbon (<i>b</i>)	

iv. How many total sp^3 -s sigma bonds are in the molecule? (1 point)

v. Determine **electronic** and **molecular geometry** at carbon (C) labeled ***e*** and sulfur (S) labeled ***f*** centres. (2 points)

Atom	Electronic Geometry	Molecular Geometry
Carbon (<i>e</i>)		
Sulfur (<i>f</i>)		

vi. Circle the value from the choices given below, that describes the **C-S-C bond angle at sulfur** (S) marked ***f***: (0.5 points)

>90° to <109.5° 109.5° >109.5 to <120° 120° >120° to <180°


Question 8 12 points

Using the **valence shell orbitals only** draw a **complete** molecular orbital (MO) diagram (with no s-p mixing) for the molecule **PS**, in the space provided below. **(6 points)**

Assume that 3s and 3p atomic orbitals will form molecular orbitals like those formed by the combination of 2s and 2p atomic orbitals that we learnt in class, and there is no s-p mixing.

Show **energy arrow**, **atomic orbitals** involved, **atom labels**, the **molecular orbitals** that are formed, **filled electrons** in the MO diagram etc. Make sure to label all the atomic and molecular orbitals.

Using the MO diagram above, answer the following questions:

- (i) Which one of the following will have the highest bond order? Circle the correct choice
 A) PS^{1-} B) PS C) PS^{1+}
- (ii) Which one of the following will have the longest bond length? Circle the correct choice
 A) PS^{1-} B) PS C) PS^{1+}
- (iii) Which of the following will be diamagnetic? Circle the correct choice.
 A) PS^{1-} B) PS C) PS^{1+}
- (iv) Draw Lewis dot structure of the molecule with the shortest bond length (based on the MO diagram) from among the three (PS^{1-} , PS, PS^{1+}), and indicate the bond order calculated from the Lewis dot structure. **(3 points)**

Lewis Dot Structure

Bond Order



6E5BBE32-F927-4647-AA46-A3A262D83ED5

midterm-exam-93388

#251 Page 10 of 10

SCRAP PAGE

**CHEMISTRY 110 –Midterm Exam****Thursday November 7, 2024, 6:30 – 8:30 pm****Examiners: Pallavi Sirjoosingh and Ashok Kakkar**

First name (please write as legibly as possible within the boxes)

--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--

Last name

--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--

Student ID number

--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--

This booklet has 8 questions on 10 pages (The last page can be used as a scrap page).

The staple must NOT be removed from the question booklet. Your intact question booklet must be returned at the end of the exam.

A separate booklet contains a periodic table, and datasheet.

Write your answers directly under the questions in the space provided. Do not write on the QR codes or near the top edges (as it may be cut off).

Please write **legibly and dark enough** for the scan to be readable. For students who use English as a second language, dictionaries are permitted. Calculators are allowed.

DO NOT ASK QUESTIONS OF THE INVIGILATORS about interpretations of the exam questions. If you have a concern about a question, please summarize it at the top of this page. All concerns will be evaluated after the exam. GOOD LUCK!

**Question 1 8 points** (*Assume all transitions are possible*)

a. 6 points Calculate the **energy (in kJ/mol)** required to ionize (completely remove) the electron from the **$n=3$** level of **He^+ ion** (He: Helium). Show your work including equations. (3 sig. figs.)

b. 2 points

(I) Select the largest (longest) wavelength from the three choices below:

- i. Wavelength of photon to *completely remove* an electron from *1s orbital of Hydrogen atom*
- ii. Wavelength of photon to *completely remove* an electron from *1s orbital of Boron atom (B)*
- iii. Wavelength of photon to *completely remove* an electron from *1s orbital of Boron cation (B^+)*

i or ii or iii

(II) Select the smallest (shortest) wavelength from the three choices below (*for H atom*):

- i. Wavelength of photon *emitted* for an electron transitioning from *2p to 1s orbital*
- ii. Wavelength of photon *absorbed* for an electron transitioning from *2s to 4p orbital*
- iii. Wavelength of photon *absorbed* for an electron transitioning from *1s to 3p orbital*

i or ii or iii



Question 2 8 points



A Chem110 student plots the graph shown on the left, from data obtained for the kinetic energy (eV) of the ejected electron versus frequency (s^{-1}) of the incident light, for an unknown metal. ($1\text{eV} = 1.6022 \times 10^{-19} \text{ J}$)

Based on the plot, answer the questions **a to c** below:

a. 1 point On the plot above, label the **threshold frequency**.

b. 2 points What is the value of the work potential (in J) of the unknown metal?

c. 3 points What is the *longest* wavelength (in nm) required to eject an electron from this metal's surface? Show your work including equation(s) used. (3 sig figs)

d. 2 points In the space provided below, plot the following

(Assume frequency is *above* threshold frequency):

- Kinetic energy* of an ejected electron versus the *intensity* of the incident light (left)
- Number of electrons* ejected versus the *frequency* of the incident light (right)



**Question 3** **6 points****a. 2 points**

In the ground state of Ge (Germanium; $Z = 32$) atom, **name** the following orbitals:

i. Orbital containing the electron that requires the maximum energy to be removed.

ii. Orbital containing the electron that experiences the minimum effective nuclear charge.

b. 2 points

An unknown anion (A^-) has the following *excited* electronic configuration: $[Ar] 4s^2 3d^3 4p^1$

(Ar: Argon; $Z = 18$)

Give the **condensed ground state electronic configuration** of the atom, **A** and the cation, **A^{2+}** .
Show your work. (Assume all electronic transitions are possible)

Electronic Configuration of A:

Electronic Configuration of A^{2+} :

c. 2 points

In a single atom, determine how many electrons can have the following set of quantum numbers?

(i) $n = 5; l = 3$

In a single atom, determine how many electrons can have the following set of quantum numbers?

(i) $n = 3; l = 2; m_s = +1/2$

**Question 4** **8 points**

Rank the following from 1 to 4 in **increasing** order. Provide the ranking in the boxes provided.

Note: Atomic number (Z) are provided for relevant atoms

a) Increasing (lowest to highest) order of *ionization energy*: (1= lowest; 4 = highest)

Mg

Mg: Z = 12

Rb

Rb: Z = 37

Cs

Cs: Z = 55

K

K : Z = 19

b) Increasing (lowest to highest) order of *bond angle*: (1= lowest; 4 = highest)

H₂OCCl₄XeF₄

HCN

c) Increasing (smallest to largest) order of *ionic size* (1= smallest; 4 = largest)

S²⁻

S: Z = 16

O²⁻

O: Z = 8

Be²⁺

Be: Z = 4

Na⁺

Na: Z = 11

d) Increasing (lowest to highest) order of *bond energy* (1= lowest; 4 = highest)

P-F

P-Cl

P-I

P-Br

**Question 5** **12 points**

a. 9 points For the molecules HClO_4 and TeF_6 , draw the most stable *Lewis structure* showing appropriate shape and *dashed/wedged bonds*, based on VSEPR. Determine the **molecular geometry** around the central atom. For HClO_4 , label the *most polar bond*.



b. 3 points (Note: Lone pairs are not shown in structures below)

(i) For the three *resonance* structures (1, 2 and 3) shown below, answer the following question:



(I) Circle the *most contributing* resonance structure.

Structure 1

Structure 2

Structure 3

(ii) For the three structures given (4, 5, and 6) below, answer the following questions:



(I) Circle all the structures that represent *resonance structures* of each other:

Structure 4

Structure 5

Structure 6

(II) Among the resonance structures, circle the *most contributing* resonance structure.

Structure 4

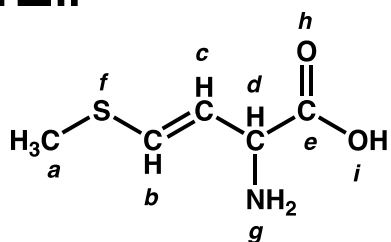
Structure 5

Structure 6

**Question 6** **6 points**

Using the data given below, calculate the **lattice energy** for SrO(s).

	ΔH° (kJ/mol)
$\text{Sr}(s) \longrightarrow \text{Sr}(g)$	+180
$\text{Sr}(g) \longrightarrow \text{Sr}^{2+}(g) + 2e^-$	+1468
$\text{O}_2(g) \longrightarrow 2\text{O}(g)$	+500
$\text{O}(g) + 2e^- \longrightarrow \text{O}^{2-}(g)$	+600
$\text{Sr}(s) + \frac{1}{2}\text{O}_2(g) \longrightarrow \text{SrO}(s)$	-550

**Question 7 10 points**

Consider the molecule given on the left, and answer questions (i) to (vi) using chemical bonding theories that we have learnt in class (valence bond theory, hybridization).

Note that the lone pairs are not shown in the structure:

i. Consider the carbon atoms marked in as ***a***, ***b***, ***c***, ***d***, and ***e*** in the molecule, and determine **hybridization** at each of the carbon centers. (2.5 points)

Atom	Hybridization
Carbon (<i>a</i>)	
Carbon (<i>b</i>)	
Carbon (<i>c</i>)	
Carbon (<i>d</i>)	
Carbon (<i>e</i>)	

ii. Now, consider the sulfur atom (S) marked as ***f***, nitrogen (N) atom (***g***), and oxygen(O) atoms (***h*** and ***i***), and determine their **hybridization**. (2 points)

Atom	Hybridization
Sulfur (<i>f</i>)	
Nitrogen (<i>g</i>)	
Oxygen (<i>h</i>)	
Oxygen (<i>i</i>)	

iii. **Describe** all the bonds formed by the carbon atoms ***a*** and ***b***, using sigma and pi designation as appropriate. (2 points)

Atom	Bonds formed (use sigma and pi as appropriate)
Carbon (<i>a</i>)	
Carbon (<i>b</i>)	

iv. How many total sp^3 -s sigma bonds are in the molecule? (1 point)

v. Determine **electronic** and **molecular geometry** at carbon (C) labeled ***e*** and sulfur (S) labeled ***f*** centres. (2 points)

Atom	Electronic Geometry	Molecular Geometry
Carbon (<i>e</i>)		
Sulfur (<i>f</i>)		

vi. Circle the value from the choices given below, that describes the **C-S-C bond angle at sulfur** (S) marked ***f***: (0.5 points)

>90° to <109.5°

109.5°

>109.5 to <120°

120°

>120° to <180°


Question 8 12 points

Using the **valence shell orbitals only** draw a **complete** molecular orbital (MO) diagram (with no s-p mixing) for the molecule **PS**, in the space provided below. **(6 points)**

Assume that 3s and 3p atomic orbitals will form molecular orbitals like those formed by the combination of 2s and 2p atomic orbitals that we learnt in class, and there is no s-p mixing.

Show **energy arrow**, **atomic orbitals** involved, **atom labels**, the **molecular orbitals** that are formed, **filled electrons** in the MO diagram etc. Make sure to label all the atomic and molecular orbitals.

Using the MO diagram above, answer the following questions:

- (i) Which one of the following will have the highest bond order? Circle the correct choice
 A) PS^{1-} B) PS C) PS^{1+}
- (ii) Which one of the following will have the longest bond length? Circle the correct choice
 A) PS^{1-} B) PS C) PS^{1+}
- (iii) Which of the following will be diamagnetic? Circle the correct choice.
 A) PS^{1-} B) PS C) PS^{1+}
- (iv) Draw Lewis dot structure of the molecule with the shortest bond length (based on the MO diagram) from among the three (PS^{1-} , PS, PS^{1+}), and indicate the bond order calculated from the Lewis dot structure. **(3 points)**

Lewis Dot Structure

Bond Order



3B045CE6-AD42-4263-AEE6-BE2171C18A5C

midterm-exam-93388

#252 Page 10 of 10

SCRAP PAGE

**CHEMISTRY 110 –Midterm Exam****Thursday November 7, 2024, 6:30 – 8:30 pm**

Examiners: Pallavi Sirjoosingh and Ashok Kakkar

First name (please write as legibly as possible within the boxes)

--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--

Last name

--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--

Student ID number

--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--

This booklet has 8 questions on 10 pages (The last page can be used as a scrap page).

The staple must NOT be removed from the question booklet. Your intact question booklet must be returned at the end of the exam.

A separate booklet contains a periodic table, and datasheet.

Write your answers directly under the questions in the space provided. Do not write on the QR codes or near the top edges (as it may be cut off).

Please write **legibly and dark enough** for the scan to be readable. For students who use English as a second language, dictionaries are permitted. Calculators are allowed.

DO NOT ASK QUESTIONS OF THE INVIGILATORS about interpretations of the exam questions. If you have a concern about a question, please summarize it at the top of this page. All concerns will be evaluated after the exam. GOOD LUCK!

**Question 1 8 points** (*Assume all transitions are possible*)

a. 6 points Calculate the **energy (in kJ/mol)** required to ionize (completely remove) the electron from the **n=3** level of **He⁺ ion** (He: Helium). Show your work including equations. (3 sig. figs.)

b. 2 points

(I) Select the largest (longest) wavelength from the three choices below:

- i. Wavelength of photon to *completely remove* an electron from *1s orbital of Hydrogen atom*
- ii. Wavelength of photon to *completely remove* an electron from *1s orbital of Boron atom (B)*
- iii. Wavelength of photon to *completely remove* an electron from *1s orbital of Boron cation (B⁺)*

i or ii or iii

(II) Select the smallest (shortest) wavelength from the three choices below (*for H atom*):

- i. Wavelength of photon *emitted* for an electron transitioning from *2p to 1s orbital*
- ii. Wavelength of photon *absorbed* for an electron transitioning from *2s to 4p orbital*
- iii. Wavelength of photon *absorbed* for an electron transitioning from *1s to 3p orbital*

i or ii or iii



Question 2 8 points



A Chem110 student plots the graph shown on the left, from data obtained for the kinetic energy (eV) of the ejected electron versus frequency (s^{-1}) of the incident light, for an unknown metal. ($1\text{eV} = 1.6022 \times 10^{-19} \text{ J}$)

Based on the plot, answer the questions **a to c** below:

a. 1 point On the plot above, label the **threshold frequency**.

b. 2 points What is the value of the work potential (in J) of the unknown metal?

c. 3 points What is the *longest* wavelength (in nm) required to eject an electron from this metal's surface? Show your work including equation(s) used. (3 sig figs)

d. 2 points In the space provided below, plot the following

(Assume frequency is *above* threshold frequency):

- i. *Kinetic energy* of an ejected electron versus the *intensity* of the incident light (left)
- ii. *Number of electrons* ejected versus the *frequency* of the incident light (right)



**Question 3** **6 points****a. 2 points**

In the ground state of Ge (Germanium; $Z = 32$) atom, **name** the following orbitals:

i. Orbital containing the electron that requires the maximum energy to be removed.

ii. Orbital containing the electron that experiences the minimum effective nuclear charge.

b. 2 points

An unknown anion (A^-) has the following *excited* electronic configuration: $[Ar] 4s^2 3d^3 4p^1$

(Ar: Argon; $Z = 18$)

Give the **condensed ground state electronic configuration** of the atom, **A** and the cation, **A^{2+}** .
Show your work. (Assume all electronic transitions are possible)

Electronic Configuration of A:

Electronic Configuration of A^{2+} :

c. 2 points

In a single atom, determine how many electrons can have the following set of quantum numbers?

(i) $n = 5; l = 3$

In a single atom, determine how many electrons can have the following set of quantum numbers?

(i) $n = 3; l = 2; m_s = +1/2$

**Question 4** **8 points**

Rank the following from 1 to 4 in **increasing** order. Provide the ranking in the boxes provided.

Note: Atomic number (Z) are provided for relevant atoms

a) Increasing (lowest to highest) order of *ionization energy*: (1= lowest; 4 = highest)

Mg

Mg: Z = 12

Rb

Rb: Z = 37

Cs

Cs: Z = 55

K

K : Z = 19

b) Increasing (lowest to highest) order of *bond angle*: (1= lowest; 4 = highest)

H₂OCCl₄XeF₄

HCN

c) Increasing (smallest to largest) order of *ionic size* (1= smallest; 4 = largest)

S²⁻

S: Z = 16

O²⁻

O: Z = 8

Be²⁺

Be: Z = 4

Na⁺

Na: Z = 11

d) Increasing (lowest to highest) order of *bond energy* (1= lowest; 4 = highest)

P-F

P-Cl

P-I

P-Br

**Question 5** **12 points**

a. 9 points For the molecules HClO_4 and TeF_6 , draw the most stable *Lewis structure* showing appropriate shape and *dashed/wedged bonds*, based on VSEPR. Determine the **molecular geometry** around the central atom. For HClO_4 , label the *most polar bond*.



b. 3 points (Note: Lone pairs are not shown in structures below)

(i) For the three *resonance* structures (1, 2 and 3) shown below, answer the following question:



(I) Circle the *most contributing* resonance structure.

Structure 1

Structure 2

Structure 3

(ii) For the three structures given (4, 5, and 6) below, answer the following questions:



(I) Circle all the structures that represent *resonance structures* of each other:

Structure 4

Structure 5

Structure 6

(II) Among the resonance structures, circle the *most contributing* resonance structure.

Structure 4

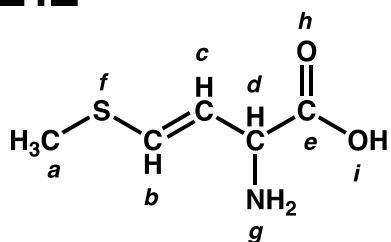
Structure 5

Structure 6

**Question 6** **6 points**

Using the data given below, calculate the **lattice energy** for SrO(s).

	ΔH° (kJ/mol)
$\text{Sr}(s) \longrightarrow \text{Sr}(g)$	+180
$\text{Sr}(g) \longrightarrow \text{Sr}^{2+}(g) + 2e^-$	+1468
$\text{O}_2(g) \longrightarrow 2\text{O}(g)$	+500
$\text{O}(g) + 2e^- \longrightarrow \text{O}^{2-}(g)$	+600
$\text{Sr}(s) + \frac{1}{2}\text{O}_2(g) \longrightarrow \text{SrO}(s)$	-550

**Question 7 10 points**

Consider the molecule given on the left, and answer questions (i) to (vi) using chemical bonding theories that we have learnt in class (valence bond theory, hybridization).

Note that the lone pairs are not shown in the structure:

i. Consider the carbon atoms marked in as ***a***, ***b***, ***c***, ***d***, and ***e*** in the molecule, and determine **hybridization** at each of the carbon centers. (2.5 points)

Atom	Hybridization
Carbon (<i>a</i>)	
Carbon (<i>b</i>)	
Carbon (<i>c</i>)	
Carbon (<i>d</i>)	
Carbon (<i>e</i>)	

ii. Now, consider the sulfur atom (S) marked as ***f***, nitrogen (N) atom (***g***), and oxygen(O) atoms (***h*** and ***i***), and determine their **hybridization**. (2 points)

Atom	Hybridization
Sulfur (<i>f</i>)	
Nitrogen (<i>g</i>)	
Oxygen (<i>h</i>)	
Oxygen (<i>i</i>)	

iii. **Describe** all the bonds formed by the carbon atoms ***a*** and ***b***, using sigma and pi designation as appropriate. (2 points)

Atom	Bonds formed (use sigma and pi as appropriate)
Carbon (<i>a</i>)	
Carbon (<i>b</i>)	

iv. How many total sp^3 -s sigma bonds are in the molecule? (1 point)

v. Determine **electronic** and **molecular geometry** at carbon (C) labeled ***e*** and sulfur (S) labeled ***f*** centres. (2 points)

Atom	Electronic Geometry	Molecular Geometry
Carbon (<i>e</i>)		
Sulfur (<i>f</i>)		

vi. Circle the value from the choices given below, that describes the **C-S-C bond angle at sulfur** (S) marked ***f***: (0.5 points)

>90° to <109.5° 109.5° >109.5 to <120° 120° >120° to <180°

**Question 8 12 points**

Using the **valence shell orbitals only** draw a **complete** molecular orbital (MO) diagram (with no s-p mixing) for the molecule **PS**, in the space provided below. **(6 points)**

Assume that 3s and 3p atomic orbitals will form molecular orbitals like those formed by the combination of 2s and 2p atomic orbitals that we learnt in class, and there is no s-p mixing.

Show **energy arrow**, **atomic orbitals** involved, **atom labels**, the **molecular orbitals** that are formed, **filled electrons** in the MO diagram etc. Make sure to label all the atomic and molecular orbitals.

Using the MO diagram above, answer the following questions:

- (i) Which one of the following will have the highest bond order? Circle the correct choice
A) PS^{1-} B) PS C) PS^{1+}
- (ii) Which one of the following will have the longest bond length? Circle the correct choice
A) PS^{1-} B) PS C) PS^{1+}
- (iii) Which of the following will be diamagnetic? Circle the correct choice.
A) PS^{1-} B) PS C) PS^{1+}
- (iv) Draw Lewis dot structure of the molecule with the shortest bond length (based on the MO diagram) from among the three (PS^{1-} , PS, PS^{1+}), and indicate the bond order calculated from the Lewis dot structure. **(3 points)**

Lewis Dot Structure

Bond Order



2E079D26-559E-4CCF-9832-F344B2449AF5

midterm-exam-93388

#253 Page 10 of 10

SCRAP PAGE

**CHEMISTRY 110 –Midterm Exam****Thursday November 7, 2024, 6:30 – 8:30 pm**

Examiners: Pallavi Sirjoosingh and Ashok Kakkar

First name (please write as legibly as possible within the boxes)

--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--

Last name

--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--

Student ID number

--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--

This booklet has 8 questions on 10 pages (The last page can be used as a scrap page).

The staple must NOT be removed from the question booklet. Your intact question booklet must be returned at the end of the exam.

A separate booklet contains a periodic table, and datasheet.

Write your answers directly under the questions in the space provided. Do not write on the QR codes or near the top edges (as it may be cut off).

Please write **legibly and dark enough** for the scan to be readable. For students who use English as a second language, dictionaries are permitted. Calculators are allowed.

DO NOT ASK QUESTIONS OF THE INVIGILATORS about interpretations of the exam questions. If you have a concern about a question, please summarize it at the top of this page. All concerns will be evaluated after the exam. GOOD LUCK!

**Question 1 8 points** (*Assume all transitions are possible*)

a. 6 points Calculate the **energy (in kJ/mol)** required to ionize (completely remove) the electron from the **$n=3$** level of **He^+ ion** (He: Helium). Show your work including equations. (3 sig. figs.)

b. 2 points

(I) Select the largest (longest) wavelength from the three choices below:

- i. Wavelength of photon to *completely remove* an electron from *1s orbital of Hydrogen atom*
- ii. Wavelength of photon to *completely remove* an electron from *1s orbital of Boron atom (B)*
- iii. Wavelength of photon to *completely remove* an electron from *1s orbital of Boron cation (B^+)*

i or ii or iii

(II) Select the smallest (shortest) wavelength from the three choices below (*for H atom*):

- i. Wavelength of photon *emitted* for an electron transitioning from *2p to 1s orbital*
- ii. Wavelength of photon *absorbed* for an electron transitioning from *2s to 4p orbital*
- iii. Wavelength of photon *absorbed* for an electron transitioning from *1s to 3p orbital*

i or ii or iii



Question 2 8 points



A Chem110 student plots the graph shown on the left, from data obtained for the kinetic energy (eV) of the ejected electron versus frequency (s^{-1}) of the incident light, for an unknown metal. ($1\text{eV} = 1.6022 \times 10^{-19} \text{ J}$)

Based on the plot, answer the questions **a to c** below:

a. 1 point On the plot above, label the **threshold frequency**.

b. 2 points What is the value of the work potential (in J) of the unknown metal?

c. 3 points What is the *longest* wavelength (in nm) required to eject an electron from this metal's surface? Show your work including equation(s) used. (3 sig figs)

d. 2 points In the space provided below, plot the following

(Assume frequency is *above* threshold frequency):

- i. *Kinetic energy* of an ejected electron versus the *intensity* of the incident light (left)
- ii. *Number of electrons* ejected versus the *frequency* of the incident light (right)



**Question 3** **6 points****a. 2 points**

In the ground state of Ge (Germanium; $Z = 32$) atom, **name** the following orbitals:

i. Orbital containing the electron that requires the maximum energy to be removed.

ii. Orbital containing the electron that experiences the minimum effective nuclear charge.

b. 2 points

An unknown anion (A^-) has the following *excited* electronic configuration: $[Ar] 4s^2 3d^3 4p^1$

(Ar: Argon; $Z = 18$)

Give the **condensed ground state electronic configuration** of the atom, **A** and the cation, **A^{2+}** .
Show your work. (Assume all electronic transitions are possible)

Electronic Configuration of A:

Electronic Configuration of A^{2+} :

c. 2 points

In a single atom, determine how many electrons can have the following set of quantum numbers?

(i) $n = 5; l = 3$

In a single atom, determine how many electrons can have the following set of quantum numbers?

(i) $n = 3; l = 2; m_s = +1/2$

**Question 4** **8 points**

Rank the following from 1 to 4 in **increasing** order. Provide the ranking in the boxes provided.

Note: Atomic number (Z) are provided for relevant atoms

a) Increasing (lowest to highest) order of *ionization energy*: (1= lowest; 4 = highest)

Mg

Mg: Z = 12

Rb

Rb: Z = 37

Cs

Cs: Z = 55

K

K : Z = 19

b) Increasing (lowest to highest) order of *bond angle*: (1= lowest; 4 = highest)

H₂OCCl₄XeF₄

HCN

c) Increasing (smallest to largest) order of *ionic size* (1= smallest; 4 = largest)

S²⁻

S: Z = 16

O²⁻

O: Z = 8

Be²⁺

Be: Z = 4

Na⁺

Na: Z = 11

d) Increasing (lowest to highest) order of *bond energy* (1= lowest; 4 = highest)

P-F

P-Cl

P-I

P-Br

**Question 5** **12 points**

a. 9 points For the molecules HClO_4 and TeF_6 , draw the most stable *Lewis structure* showing appropriate shape and *dashed/wedged bonds*, based on VSEPR. Determine the **molecular geometry** around the central atom. For HClO_4 , label the *most polar bond*.

HClO_4 (there are no O-O bonds)

Label the most polar bond

Molecular Geometry around Cl

TeF_6

Molecular Geometry around Te

b. 3 points (Note: Lone pairs are not shown in structures below)

(i) For the three *resonance* structures (1, 2 and 3) shown below, answer the following question:



(I) Circle the *most contributing* resonance structure.

Structure 1

Structure 2

Structure 3

(ii) For the three structures given (4, 5, and 6) below, answer the following questions:



(I) Circle all the structures that represent *resonance structures* of each other:

Structure 4

Structure 5

Structure 6

(II) Among the resonance structures, circle the *most contributing* resonance structure.

Structure 4

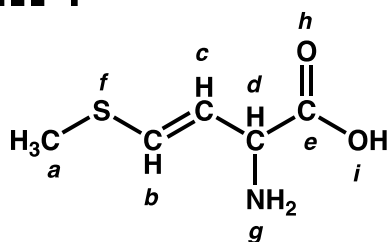
Structure 5

Structure 6

**Question 6** **6 points**

Using the data given below, calculate the **lattice energy** for SrO(s).

	ΔH° (kJ/mol)
$\text{Sr}(s) \longrightarrow \text{Sr}(g)$	+180
$\text{Sr}(g) \longrightarrow \text{Sr}^{2+}(g) + 2e^-$	+1468
$\text{O}_2(g) \longrightarrow 2\text{O}(g)$	+500
$\text{O}(g) + 2e^- \longrightarrow \text{O}^{2-}(g)$	+600
$\text{Sr}(s) + \frac{1}{2}\text{O}_2(g) \longrightarrow \text{SrO}(s)$	-550

**Question 7 10 points**

Consider the molecule given on the left, and answer questions (i) to (vi) using chemical bonding theories that we have learnt in class (valence bond theory, hybridization).

Note that the lone pairs are not shown in the structure:

i. Consider the carbon atoms marked in as ***a***, ***b***, ***c***, ***d***, and ***e*** in the molecule, and determine **hybridization** at each of the carbon centers. (2.5 points)

Atom	Hybridization
Carbon (<i>a</i>)	
Carbon (<i>b</i>)	
Carbon (<i>c</i>)	
Carbon (<i>d</i>)	
Carbon (<i>e</i>)	

ii. Now, consider the sulfur atom (S) marked as ***f***, nitrogen (N) atom (***g***), and oxygen(O) atoms (***h*** and ***i***), and determine their **hybridization**. (2 points)

Atom	Hybridization
Sulfur (<i>f</i>)	
Nitrogen (<i>g</i>)	
Oxygen (<i>h</i>)	
Oxygen (<i>i</i>)	

iii. **Describe** all the bonds formed by the carbon atoms ***a*** and ***b***, using sigma and pi designation as appropriate. (2 points)

Atom	Bonds formed (use sigma and pi as appropriate)
Carbon (<i>a</i>)	
Carbon (<i>b</i>)	

iv. How many total sp^3 -s sigma bonds are in the molecule? (1 point)

v. Determine **electronic** and **molecular geometry** at carbon (C) labeled ***e*** and sulfur (S) labeled ***f*** centres. (2 points)

Atom	Electronic Geometry	Molecular Geometry
Carbon (<i>e</i>)		
Sulfur (<i>f</i>)		

vi. Circle the value from the choices given below, that describes the **C-S-C bond angle at sulfur** (S) marked ***f***: (0.5 points)

>90° to <109.5° 109.5° >109.5 to <120° 120° >120° to <180°



Question 8 12 points

Using the **valence shell orbitals only** draw a **complete** molecular orbital (MO) diagram (with no s-p mixing) for the molecule **PS**, in the space provided below. **(6 points)**

Assume that 3s and 3p atomic orbitals will form molecular orbitals like those formed by the combination of 2s and 2p atomic orbitals that we learnt in class, and there is no s-p mixing.

Show **energy arrow**, **atomic orbitals** involved, **atom labels**, the **molecular orbitals** that are formed, **filled electrons** in the MO diagram etc. Make sure to label all the atomic and molecular orbitals.

Using the MO diagram above, answer the following questions:

- (i) Which one of the following will have the highest bond order? Circle the correct choice
 A) PS^{1-} B) PS C) PS^{1+}
- (ii) Which one of the following will have the longest bond length? Circle the correct choice
 A) PS^{1-} B) PS C) PS^{1+}
- (iii) Which of the following will be diamagnetic? Circle the correct choice.
 A) PS^{1-} B) PS C) PS^{1+}
- (iv) Draw Lewis dot structure of the molecule with the shortest bond length (based on the MO diagram) from among the three (PS^{1-} , PS, PS^{1+}), and indicate the bond order calculated from the Lewis dot structure. **(3 points)**

Lewis Dot Structure

Bond Order



E81EB6F9-4096-4E29-A8CF-92E6589544E3

midterm-exam-93388

#254 Page 10 of 10

SCRAP PAGE

**CHEMISTRY 110 –Midterm Exam****Thursday November 7, 2024, 6:30 – 8:30 pm**

Examiners: Pallavi Sirjoosingh and Ashok Kakkar

First name (please write as legibly as possible within the boxes)

--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--

Last name

--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--

Student ID number

--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--

This booklet has 8 questions on 10 pages (The last page can be used as a scrap page).

The staple must NOT be removed from the question booklet. Your intact question booklet must be returned at the end of the exam.

A separate booklet contains a periodic table, and datasheet.

Write your answers directly under the questions in the space provided. Do not write on the QR codes or near the top edges (as it may be cut off).

Please write **legibly and dark enough** for the scan to be readable. For students who use English as a second language, dictionaries are permitted. Calculators are allowed.

DO NOT ASK QUESTIONS OF THE INVIGILATORS about interpretations of the exam questions. If you have a concern about a question, please summarize it at the top of this page. All concerns will be evaluated after the exam. GOOD LUCK!

**Question 1 8 points** (*Assume all transitions are possible*)

a. 6 points Calculate the **energy (in kJ/mol)** required to ionize (completely remove) the electron from the **n=3** level of **He⁺ ion** (He: Helium). Show your work including equations. (3 sig. figs.)

b. 2 points

(I) Select the largest (longest) wavelength from the three choices below:

- i. Wavelength of photon to *completely remove* an electron from *1s orbital of Hydrogen atom*
- ii. Wavelength of photon to *completely remove* an electron from *1s orbital of Boron atom (B)*
- iii. Wavelength of photon to *completely remove* an electron from *1s orbital of Boron cation (B⁺)*

i or ii or iii

(II) Select the smallest (shortest) wavelength from the three choices below (*for H atom*):

- i. Wavelength of photon *emitted* for an electron transitioning from *2p to 1s orbital*
- ii. Wavelength of photon *absorbed* for an electron transitioning from *2s to 4p orbital*
- iii. Wavelength of photon *absorbed* for an electron transitioning from *1s to 3p orbital*

i or ii or iii



Question 2 8 points



A Chem110 student plots the graph shown on the left, from data obtained for the kinetic energy (eV) of the ejected electron versus frequency (s^{-1}) of the incident light, for an unknown metal. ($1\text{eV} = 1.6022 \times 10^{-19} \text{ J}$)

Based on the plot, answer the questions **a to c** below:

a. 1 point On the plot above, label the **threshold frequency**.

b. 2 points What is the value of the work potential (in J) of the unknown metal?

c. 3 points What is the *longest* wavelength (in nm) required to eject an electron from this metal's surface? Show your work including equation(s) used. (3 sig figs)

d. 2 points In the space provided below, plot the following

(Assume frequency is *above* threshold frequency):

- i. *Kinetic energy* of an ejected electron versus the *intensity* of the incident light (left)
- ii. *Number of electrons* ejected versus the *frequency* of the incident light (right)



**Question 3** **6 points****a. 2 points**

In the ground state of Ge (Germanium; $Z = 32$) atom, **name** the following orbitals:

i. Orbital containing the electron that requires the maximum energy to be removed.

ii. Orbital containing the electron that experiences the minimum effective nuclear charge.

b. 2 points

An unknown anion (A^-) has the following *excited* electronic configuration: $[Ar] 4s^2 3d^3 4p^1$

(Ar: Argon; $Z = 18$)

Give the **condensed ground state electronic configuration** of the atom, **A** and the cation, **A^{2+}** .
Show your work. (Assume all electronic transitions are possible)

Electronic Configuration of A:

Electronic Configuration of A^{2+} :

c. 2 points

In a single atom, determine how many electrons can have the following set of quantum numbers?

(i) $n = 5; l = 3$

In a single atom, determine how many electrons can have the following set of quantum numbers?

(i) $n = 3; l = 2; m_s = +1/2$

**Question 4** **8 points**

Rank the following from 1 to 4 in **increasing** order. Provide the ranking in the boxes provided.

Note: Atomic number (Z) are provided for relevant atoms

a) Increasing (lowest to highest) order of *ionization energy*: (1= lowest; 4 = highest)

Mg

Mg: Z = 12

Rb

Rb: Z = 37

Cs

Cs: Z = 55

K

K : Z = 19

b) Increasing (lowest to highest) order of *bond angle*: (1= lowest; 4 = highest)

H₂OCCl₄XeF₄

HCN

c) Increasing (smallest to largest) order of *ionic size* (1= smallest; 4 = largest)

S²⁻

S: Z = 16

O²⁻

O: Z = 8

Be²⁺

Be: Z = 4

Na⁺

Na: Z = 11

d) Increasing (lowest to highest) order of *bond energy* (1= lowest; 4 = highest)

P-F

P-Cl

P-I

P-Br

**Question 5** **12 points**

a. 9 points For the molecules HClO_4 and TeF_6 , draw the most stable *Lewis structure* showing appropriate shape and *dashed/wedged bonds*, based on VSEPR. Determine the **molecular geometry** around the central atom. For HClO_4 , label the *most polar bond*.

HClO_4 (there are no O-O bonds)

Label the most polar bond

Molecular Geometry around Cl

TeF_6

Molecular Geometry around Te

b. 3 points (Note: Lone pairs are not shown in structures below)

(i) For the three *resonance* structures (1, 2 and 3) shown below, answer the following question:



(I) Circle the *most contributing* resonance structure.

Structure 1

Structure 2

Structure 3

(ii) For the three structures given (4, 5, and 6) below, answer the following questions:



(I) Circle all the structures that represent *resonance structures* of each other:

Structure 4

Structure 5

Structure 6

(II) Among the resonance structures, circle the *most contributing* resonance structure.

Structure 4

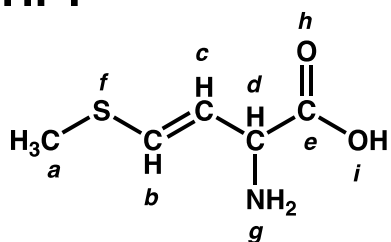
Structure 5

Structure 6

**Question 6** **6 points**

Using the data given below, calculate the **lattice energy** for SrO(s).

	ΔH° (kJ/mol)
$\text{Sr}(s) \longrightarrow \text{Sr}(g)$	+180
$\text{Sr}(g) \longrightarrow \text{Sr}^{2+}(g) + 2e^-$	+1468
$\text{O}_2(g) \longrightarrow 2\text{O}(g)$	+500
$\text{O}(g) + 2e^- \longrightarrow \text{O}^{2-}(g)$	+600
$\text{Sr}(s) + \frac{1}{2}\text{O}_2(g) \longrightarrow \text{SrO}(s)$	-550

**Question 7 10 points**

Consider the molecule given on the left, and answer questions (i) to (vi) using chemical bonding theories that we have learnt in class (valence bond theory, hybridization).

Note that the lone pairs are not shown in the structure:

i. Consider the carbon atoms marked in as ***a***, ***b***, ***c***, ***d***, and ***e*** in the molecule, and determine **hybridization** at each of the carbon centers. (2.5 points)

Atom	Hybridization
Carbon (<i>a</i>)	
Carbon (<i>b</i>)	
Carbon (<i>c</i>)	
Carbon (<i>d</i>)	
Carbon (<i>e</i>)	

ii. Now, consider the sulfur atom (S) marked as ***f***, nitrogen (N) atom (***g***), and oxygen(O) atoms (***h*** and ***i***), and determine their **hybridization**. (2 points)

Atom	Hybridization
Sulfur (<i>f</i>)	
Nitrogen (<i>g</i>)	
Oxygen (<i>h</i>)	
Oxygen (<i>i</i>)	

iii. **Describe** all the bonds formed by the carbon atoms ***a*** and ***b***, using sigma and pi designation as appropriate. (2 points)

Atom	Bonds formed (use sigma and pi as appropriate)
Carbon (<i>a</i>)	
Carbon (<i>b</i>)	

iv. How many total sp^3 -s sigma bonds are in the molecule? (1 point)

v. Determine **electronic** and **molecular geometry** at carbon (C) labeled ***e*** and sulfur (S) labeled ***f*** centres. (2 points)

Atom	Electronic Geometry	Molecular Geometry
Carbon (<i>e</i>)		
Sulfur (<i>f</i>)		

vi. Circle the value from the choices given below, that describes the **C-S-C bond angle at sulfur** (S) marked ***f***: (0.5 points)

>90° to <109.5° 109.5° >109.5 to <120° 120° >120° to <180°

**Question 8 12 points**

Using the **valence shell orbitals only** draw a **complete** molecular orbital (MO) diagram (with no s-p mixing) for the molecule **PS**, in the space provided below. **(6 points)**

Assume that 3s and 3p atomic orbitals will form molecular orbitals like those formed by the combination of 2s and 2p atomic orbitals that we learnt in class, and there is no s-p mixing.

Show **energy arrow**, **atomic orbitals** involved, **atom labels**, the **molecular orbitals** that are formed, **filled electrons** in the MO diagram etc. Make sure to label all the atomic and molecular orbitals.

Using the MO diagram above, answer the following questions:

- (i) Which one of the following will have the highest bond order? Circle the correct choice
A) PS^{1-} B) PS C) PS^{1+}
- (ii) Which one of the following will have the longest bond length? Circle the correct choice
A) PS^{1-} B) PS C) PS^{1+}
- (iii) Which of the following will be diamagnetic? Circle the correct choice.
A) PS^{1-} B) PS C) PS^{1+}
- (iv) Draw Lewis dot structure of the molecule with the shortest bond length (based on the MO diagram) from among the three (PS^{1-} , PS, PS^{1+}), and indicate the bond order calculated from the Lewis dot structure. **(3 points)**

Lewis Dot Structure

Bond Order



CBB0CBFE-C0AF-40CA-B894-850B7FACA5E9

midterm-exam-93388

#255 Page 10 of 10

SCRAP PAGE

**CHEMISTRY 110 –Midterm Exam****Thursday November 7, 2024, 6:30 – 8:30 pm**

Examiners: Pallavi Sirjoosingh and Ashok Kakkar

First name (please write as legibly as possible within the boxes)

--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--

Last name

--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--

Student ID number

--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--

This booklet has 8 questions on 10 pages (The last page can be used as a scrap page).

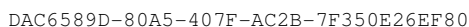
The staple must NOT be removed from the question booklet. Your intact question booklet must be returned at the end of the exam.

A separate booklet contains a periodic table, and datasheet.

Write your answers directly under the questions in the space provided. Do not write on the QR codes or near the top edges (as it may be cut off).

Please write **legibly and dark enough** for the scan to be readable. For students who use English as a second language, dictionaries are permitted. Calculators are allowed.

DO NOT ASK QUESTIONS OF THE INVIGILATORS about interpretations of the exam questions. If you have a concern about a question, please summarize it at the top of this page. All concerns will be evaluated after the exam. GOOD LUCK!



midterm-exam-93388

#256 Page 2 of 10

a. 6 points Calculate the **energy (in kJ/mol)** required to ionize (completely remove) the electron from the **n=3** level of **He⁺ ion** (He: Helium). Show your work including equations. (3 sig. figs.)

(I) Select the largest (longest) wavelength from the three choices below:

- Wavelength of photon to *completely remove* an electron from *1s orbital of Hydrogen atom*
- Wavelength of photon to *completely remove* an electron from *1s orbital of Boron atom (B)*
- Wavelength of photon to *completely remove* an electron from *1s orbital of Boron cation (B⁺)*

i
or
ii
or
iii

(II) Select the smallest (shortest) wavelength from the three choices below (*for H atom*):

- Wavelength of photon *emitted* for an electron transitioning from $2p$ to $1s$ orbital
- Wavelength of photon *absorbed* for an electron transitioning from $2s$ to $4p$ orbital
- Wavelength of photon *absorbed* for an electron transitioning from $1s$ to $3p$ orbital

i or ii or iii



Question 2 8 points



A Chem110 student plots the graph shown on the left, from data obtained for the kinetic energy (eV) of the ejected electron versus frequency (s^{-1}) of the incident light, for an unknown metal. ($1\text{eV} = 1.6022 \times 10^{-19} \text{ J}$)

Based on the plot, answer the questions **a to c** below:

a. 1 point On the plot above, label the **threshold frequency**.

b. 2 points What is the value of the work potential (in J) of the unknown metal?

c. 3 points What is the *longest* wavelength (in nm) required to eject an electron from this metal's surface? Show your work including equation(s) used. (3 sig figs)

d. 2 points In the space provided below, plot the following

(Assume frequency is *above* threshold frequency):

- i. *Kinetic energy* of an ejected electron versus the *intensity* of the incident light (left)
- ii. *Number of electrons* ejected versus the *frequency* of the incident light (right)



**Question 3** **6 points****a. 2 points**

In the ground state of Ge (Germanium; $Z = 32$) atom, **name** the following orbitals:

i. Orbital containing the electron that requires the maximum energy to be removed.

ii. Orbital containing the electron that experiences the minimum effective nuclear charge.

b. 2 points

An unknown anion (A^-) has the following *excited* electronic configuration: $[Ar] 4s^2 3d^3 4p^1$

(Ar: Argon; $Z = 18$)

Give the **condensed ground state electronic configuration** of the atom, **A** and the cation, **A^{2+}** .
Show your work. (Assume all electronic transitions are possible)

Electronic Configuration of A:

Electronic Configuration of A^{2+} :

c. 2 points

In a single atom, determine how many electrons can have the following set of quantum numbers?

(i) $n = 5; l = 3$

In a single atom, determine how many electrons can have the following set of quantum numbers?

(i) $n = 3; l = 2; m_s = +1/2$

**Question 4** **8 points**

Rank the following from 1 to 4 in **increasing** order. Provide the ranking in the boxes provided.

Note: Atomic number (Z) are provided for relevant atoms

a) Increasing (lowest to highest) order of *ionization energy*: (1= lowest; 4 = highest)

Mg

Mg: Z = 12

Rb

Rb: Z = 37

Cs

Cs: Z = 55

K

K : Z = 19

b) Increasing (lowest to highest) order of *bond angle*: (1= lowest; 4 = highest)

H₂OCCl₄XeF₄

HCN

c) Increasing (smallest to largest) order of *ionic size* (1= smallest; 4 = largest)

S²⁻

S: Z = 16

O²⁻

O: Z = 8

Be²⁺

Be: Z = 4

Na⁺

Na: Z = 11

d) Increasing (lowest to highest) order of *bond energy* (1= lowest; 4 = highest)

P-F

P-Cl

P-I

P-Br

**Question 5** **12 points**

a. 9 points For the molecules HClO_4 and TeF_6 , draw the most stable *Lewis structure* showing appropriate shape and *dashed/wedged bonds*, based on VSEPR. Determine the **molecular geometry** around the central atom. For HClO_4 , label the *most polar bond*.



b. 3 points (Note: Lone pairs are not shown in structures below)

(i) For the three *resonance* structures (1, 2 and 3) shown below, answer the following question:



(I) Circle the *most contributing* resonance structure.

Structure 1

Structure 2

Structure 3

(ii) For the three structures given (4, 5, and 6) below, answer the following questions:



(I) Circle all the structures that represent *resonance structures* of each other:

Structure 4

Structure 5

Structure 6

(II) Among the resonance structures, circle the *most contributing* resonance structure.

Structure 4

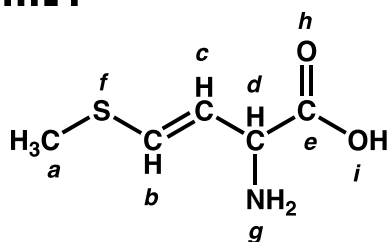
Structure 5

Structure 6

**Question 6** **6 points**

Using the data given below, calculate the **lattice energy** for SrO(s).

	ΔH° (kJ/mol)
$\text{Sr}(s) \longrightarrow \text{Sr}(g)$	+180
$\text{Sr}(g) \longrightarrow \text{Sr}^{2+}(g) + 2e^-$	+1468
$\text{O}_2(g) \longrightarrow 2\text{O}(g)$	+500
$\text{O}(g) + 2e^- \longrightarrow \text{O}^{2-}(g)$	+600
$\text{Sr}(s) + \frac{1}{2}\text{O}_2(g) \longrightarrow \text{SrO}(s)$	-550

**Question 7 10 points**

Consider the molecule given on the left, and answer questions (i) to (vi) using chemical bonding theories that we have learnt in class (valence bond theory, hybridization).

Note that the lone pairs are not shown in the structure:

i. Consider the carbon atoms marked in as ***a***, ***b***, ***c***, ***d***, and ***e*** in the molecule, and determine **hybridization** at each of the carbon centers. (2.5 points)

Atom	Hybridization
Carbon (<i>a</i>)	
Carbon (<i>b</i>)	
Carbon (<i>c</i>)	
Carbon (<i>d</i>)	
Carbon (<i>e</i>)	

ii. Now, consider the sulfur atom (S) marked as ***f***, nitrogen (N) atom (***g***), and oxygen(O) atoms (***h*** and ***i***), and determine their **hybridization**. (2 points)

Atom	Hybridization
Sulfur (<i>f</i>)	
Nitrogen (<i>g</i>)	
Oxygen (<i>h</i>)	
Oxygen (<i>i</i>)	

iii. **Describe** all the bonds formed by the carbon atoms ***a*** and ***b***, using sigma and pi designation as appropriate. (2 points)

Atom	Bonds formed (use sigma and pi as appropriate)
Carbon (<i>a</i>)	
Carbon (<i>b</i>)	

iv. How many total sp^3 -s sigma bonds are in the molecule? (1 point)

v. Determine **electronic** and **molecular geometry** at carbon (C) labeled ***e*** and sulfur (S) labeled ***f*** centres. (2 points)

Atom	Electronic Geometry	Molecular Geometry
Carbon (<i>e</i>)		
Sulfur (<i>f</i>)		

vi. Circle the value from the choices given below, that describes the **C-S-C bond angle at sulfur** (S) marked ***f***: (0.5 points)

>90° to <109.5° 109.5° >109.5 to <120° 120° >120° to <180°


Question 8 12 points

Using the **valence shell orbitals only** draw a **complete** molecular orbital (MO) diagram (with no s-p mixing) for the molecule **PS**, in the space provided below. **(6 points)**

Assume that 3s and 3p atomic orbitals will form molecular orbitals like those formed by the combination of 2s and 2p atomic orbitals that we learnt in class, and there is no s-p mixing.

Show **energy arrow**, **atomic orbitals** involved, **atom labels**, the **molecular orbitals** that are formed, **filled electrons** in the MO diagram etc. Make sure to label all the atomic and molecular orbitals.

Using the MO diagram above, answer the following questions:

- (i) Which one of the following will have the highest bond order? Circle the correct choice
 A) PS^{1-} B) PS C) PS^{1+}
- (ii) Which one of the following will have the longest bond length? Circle the correct choice
 A) PS^{1-} B) PS C) PS^{1+}
- (iii) Which of the following will be diamagnetic? Circle the correct choice.
 A) PS^{1-} B) PS C) PS^{1+}
- (iv) Draw Lewis dot structure of the molecule with the shortest bond length (based on the MO diagram) from among the three (PS^{1-} , PS, PS^{1+}), and indicate the bond order calculated from the Lewis dot structure. **(3 points)**

Lewis Dot Structure

Bond Order



A195BA42-05B1-4D8A-9FBD-D790BE46FBCE

midterm-exam-93388

#256 Page 10 of 10

SCRAP PAGE

**CHEMISTRY 110 –Midterm Exam****Thursday November 7, 2024, 6:30 – 8:30 pm****Examiners: Pallavi Sirjoosingh and Ashok Kakkar**

First name (please write as legibly as possible within the boxes)

--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--

Last name

--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--

Student ID number

--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--

This booklet has 8 questions on 10 pages (The last page can be used as a scrap page).

The staple must NOT be removed from the question booklet. Your intact question booklet must be returned at the end of the exam.

A separate booklet contains a periodic table, and datasheet.

Write your answers directly under the questions in the space provided. Do not write on the QR codes or near the top edges (as it may be cut off).

Please write **legibly and dark enough** for the scan to be readable. For students who use English as a second language, dictionaries are permitted. Calculators are allowed.

DO NOT ASK QUESTIONS OF THE INVIGILATORS about interpretations of the exam questions. If you have a concern about a question, please summarize it at the top of this page. All concerns will be evaluated after the exam. GOOD LUCK!

**Question 1 8 points** (*Assume all transitions are possible*)

a. 6 points Calculate the **energy (in kJ/mol)** required to ionize (completely remove) the electron from the **n=3** level of **He⁺ ion** (He: Helium). Show your work including equations. (3 sig. figs.)

b. 2 points

(I) Select the largest (longest) wavelength from the three choices below:

- i. Wavelength of photon to *completely remove* an electron from *1s orbital of Hydrogen atom*
- ii. Wavelength of photon to *completely remove* an electron from *1s orbital of Boron atom (B)*
- iii. Wavelength of photon to *completely remove* an electron from *1s orbital of Boron cation (B⁺)*

i or ii or iii

(II) Select the smallest (shortest) wavelength from the three choices below (*for H atom*):

- i. Wavelength of photon *emitted* for an electron transitioning from *2p to 1s orbital*
- ii. Wavelength of photon *absorbed* for an electron transitioning from *2s to 4p orbital*
- iii. Wavelength of photon *absorbed* for an electron transitioning from *1s to 3p orbital*

i or ii or iii



Question 2 8 points



A Chem110 student plots the graph shown on the left, from data obtained for the kinetic energy (eV) of the ejected electron versus frequency (s^{-1}) of the incident light, for an unknown metal. ($1\text{eV} = 1.6022 \times 10^{-19} \text{ J}$)

Based on the plot, answer the questions **a to c** below:

a. 1 point On the plot above, label the **threshold frequency**.

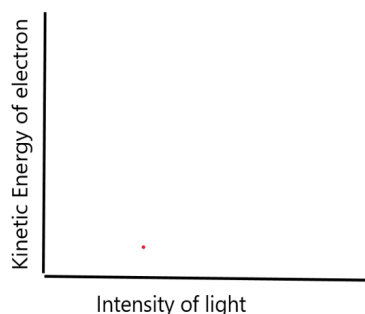
b. 2 points What is the value of the work potential (in J) of the unknown metal?

c. 3 points What is the *longest* wavelength (in nm) required to eject an electron from this metal's surface? Show your work including equation(s) used. (3 sig figs)

d. 2 points In the space provided below, plot the following

(Assume frequency is *above* threshold frequency):

- Kinetic energy* of an ejected electron versus the *intensity* of the incident light (left)
- Number of electrons* ejected versus the *frequency* of the incident light (right)



**Question 3** **6 points****a. 2 points**

In the ground state of Ge (Germanium; $Z = 32$) atom, **name** the following orbitals:

i. Orbital containing the electron that requires the maximum energy to be removed.

ii. Orbital containing the electron that experiences the minimum effective nuclear charge.

b. 2 points

An unknown anion (A^-) has the following *excited* electronic configuration: $[Ar] 4s^2 3d^3 4p^1$

(Ar: Argon; $Z = 18$)

Give the **condensed ground state electronic configuration** of the atom, **A** and the cation, **A^{2+}** .
Show your work. (Assume all electronic transitions are possible)

Electronic Configuration of A:

Electronic Configuration of A^{2+} :

c. 2 points

In a single atom, determine how many electrons can have the following set of quantum numbers?

(i) $n = 5; l = 3$

In a single atom, determine how many electrons can have the following set of quantum numbers?

(i) $n = 3; l = 2; m_s = +1/2$

**Question 4** **8 points**

Rank the following from 1 to 4 in **increasing** order. Provide the ranking in the boxes provided.

Note: Atomic number (Z) are provided for relevant atoms

a) Increasing (lowest to highest) order of *ionization energy*: (1= lowest; 4 = highest)

Mg

Mg: Z = 12

Rb

Rb: Z = 37

Cs

Cs: Z = 55

K

K : Z = 19

b) Increasing (lowest to highest) order of *bond angle*: (1= lowest; 4 = highest)

H₂OCCl₄XeF₄

HCN

c) Increasing (smallest to largest) order of *ionic size* (1= smallest; 4 = largest)

S²⁻

S: Z = 16

O²⁻

O: Z = 8

Be²⁺

Be: Z = 4

Na⁺

Na: Z = 11

d) Increasing (lowest to highest) order of *bond energy* (1= lowest; 4 = highest)

P-F

P-Cl

P-I

P-Br

**Question 5** **12 points**

a. 9 points For the molecules HClO_4 and TeF_6 , draw the most stable *Lewis structure* showing appropriate shape and *dashed/wedged bonds*, based on VSEPR. Determine the **molecular geometry** around the central atom. For HClO_4 , label the *most polar bond*.

HClO_4 (there are no O-O bonds)

Label the most polar bond

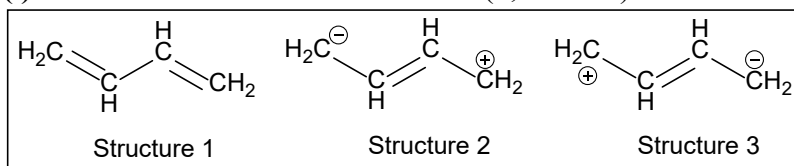
Molecular Geometry around Cl

TeF_6

Molecular Geometry around Te

b. 3 points (Note: Lone pairs are not shown in structures below)

(i) For the three *resonance* structures (1, 2 and 3) shown below, answer the following question:



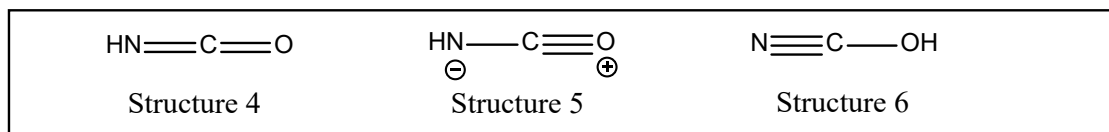
(I) Circle the *most contributing* resonance structure.

Structure 1

Structure 2

Structure 3

(ii) For the three structures given (4, 5, and 6) below, answer the following questions:



(I) Circle all the structures that represent *resonance structures* of each other:

Structure 4

Structure 5

Structure 6

(II) Among the resonance structures, circle the *most contributing* resonance structure.

Structure 4

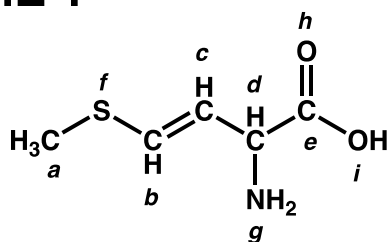
Structure 5

Structure 6

**Question 6** **6 points**

Using the data given below, calculate the **lattice energy** for SrO(s).

	ΔH° (kJ/mol)
$\text{Sr}(s) \longrightarrow \text{Sr}(g)$	+180
$\text{Sr}(g) \longrightarrow \text{Sr}^{2+}(g) + 2e^-$	+1468
$\text{O}_2(g) \longrightarrow 2\text{O}(g)$	+500
$\text{O}(g) + 2e^- \longrightarrow \text{O}^{2-}(g)$	+600
$\text{Sr}(s) + \frac{1}{2}\text{O}_2(g) \longrightarrow \text{SrO}(s)$	-550

**Question 7 10 points**

Consider the molecule given on the left, and answer questions (i) to (vi) using chemical bonding theories that we have learnt in class (valence bond theory, hybridization).

Note that the lone pairs are not shown in the structure:

i. Consider the carbon atoms marked in as ***a***, ***b***, ***c***, ***d***, and ***e*** in the molecule, and determine **hybridization** at each of the carbon centers. (2.5 points)

Atom	Hybridization
Carbon (<i>a</i>)	
Carbon (<i>b</i>)	
Carbon (<i>c</i>)	
Carbon (<i>d</i>)	
Carbon (<i>e</i>)	

ii. Now, consider the sulfur atom (S) marked as ***f***, nitrogen (N) atom (***g***), and oxygen(O) atoms (***h*** and ***i***), and determine their **hybridization**. (2 points)

Atom	Hybridization
Sulfur (<i>f</i>)	
Nitrogen (<i>g</i>)	
Oxygen (<i>h</i>)	
Oxygen (<i>i</i>)	

iii. **Describe** all the bonds formed by the carbon atoms ***a*** and ***b***, using sigma and pi designation as appropriate. (2 points)

Atom	Bonds formed (use sigma and pi as appropriate)
Carbon (<i>a</i>)	
Carbon (<i>b</i>)	

iv. How many total sp^3 -s sigma bonds are in the molecule? (1 point)

v. Determine **electronic** and **molecular geometry** at carbon (C) labeled ***e*** and sulfur (S) labeled ***f*** centres. (2 points)

Atom	Electronic Geometry	Molecular Geometry
Carbon (<i>e</i>)		
Sulfur (<i>f</i>)		

vi. Circle the value from the choices given below, that describes the **C-S-C bond angle at sulfur** (S) marked ***f***: (0.5 points)

>90° to <109.5° 109.5° >109.5 to <120° 120° >120° to <180°


Question 8 12 points

Using the **valence shell orbitals only** draw a **complete** molecular orbital (MO) diagram (with no s-p mixing) for the molecule **PS**, in the space provided below. **(6 points)**

Assume that 3s and 3p atomic orbitals will form molecular orbitals like those formed by the combination of 2s and 2p atomic orbitals that we learnt in class, and there is no s-p mixing.

Show **energy arrow**, **atomic orbitals** involved, **atom labels**, the **molecular orbitals** that are formed, **filled electrons** in the MO diagram etc. Make sure to **label all the atomic and molecular orbitals**.

Using the MO diagram above, answer the following questions:

- (i) Which one of the following will have the highest bond order? Circle the correct choice
 A) PS^{1-} B) PS C) PS^{1+}
- (ii) Which one of the following will have the longest bond length? Circle the correct choice
 A) PS^{1-} B) PS C) PS^{1+}
- (iii) Which of the following will be diamagnetic? Circle the correct choice.
 A) PS^{1-} B) PS C) PS^{1+}
- (iv) Draw Lewis dot structure of the molecule with the shortest bond length (based on the MO diagram) from among the three (PS^{1-} , PS, PS^{1+}), and indicate the bond order calculated from the Lewis dot structure. **(3 points)**

Lewis Dot Structure

Bond Order



F904FBFD-6778-44F0-8682-8719AAAE2AC2

midterm-exam-93388

#257 Page 10 of 10

SCRAP PAGE

**CHEMISTRY 110 –Midterm Exam****Thursday November 7, 2024, 6:30 – 8:30 pm**

Examiners: Pallavi Sirjoosingh and Ashok Kakkar

First name (please write as legibly as possible within the boxes)

--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--

Last name

--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--

Student ID number

--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--

This booklet has 8 questions on 10 pages (The last page can be used as a scrap page).

The staple must NOT be removed from the question booklet. Your intact question booklet must be returned at the end of the exam.

A separate booklet contains a periodic table, and datasheet.

Write your answers directly under the questions in the space provided. Do not write on the QR codes or near the top edges (as it may be cut off).

Please write **legibly and dark enough** for the scan to be readable. For students who use English as a second language, dictionaries are permitted. Calculators are allowed.

DO NOT ASK QUESTIONS OF THE INVIGILATORS about interpretations of the exam questions. If you have a concern about a question, please summarize it at the top of this page. All concerns will be evaluated after the exam. GOOD LUCK!

**Question 1 8 points** (*Assume all transitions are possible*)

a. 6 points Calculate the **energy (in kJ/mol)** required to ionize (completely remove) the electron from the **n=3** level of **He⁺ ion** (He: Helium). Show your work including equations. (3 sig. figs.)

b. 2 points

(I) Select the largest (longest) wavelength from the three choices below:

- i. Wavelength of photon to *completely remove* an electron from *1s orbital of Hydrogen atom*
- ii. Wavelength of photon to *completely remove* an electron from *1s orbital of Boron atom (B)*
- iii. Wavelength of photon to *completely remove* an electron from *1s orbital of Boron cation (B⁺)*

i or ii or iii

(II) Select the smallest (shortest) wavelength from the three choices below (*for H atom*):

- i. Wavelength of photon *emitted* for an electron transitioning from *2p to 1s orbital*
- ii. Wavelength of photon *absorbed* for an electron transitioning from *2s to 4p orbital*
- iii. Wavelength of photon *absorbed* for an electron transitioning from *1s to 3p orbital*

i or ii or iii



Question 2 8 points



A Chem110 student plots the graph shown on the left, from data obtained for the kinetic energy (eV) of the ejected electron versus frequency (s^{-1}) of the incident light, for an unknown metal. ($1\text{eV} = 1.6022 \times 10^{-19} \text{ J}$)

Based on the plot, answer the questions **a to c** below:

a. 1 point On the plot above, label the **threshold frequency**.

b. 2 points What is the value of the work potential (in J) of the unknown metal?

c. 3 points What is the *longest* wavelength (in nm) required to eject an electron from this metal's surface? Show your work including equation(s) used. (3 sig figs)

d. 2 points In the space provided below, plot the following

(Assume frequency is *above* threshold frequency):

- Kinetic energy* of an ejected electron versus the *intensity* of the incident light (left)
- Number of electrons* ejected versus the *frequency* of the incident light (right)



**Question 3** **6 points****a. 2 points**

In the ground state of Ge (Germanium; $Z = 32$) atom, **name** the following orbitals:

i. Orbital containing the electron that requires the maximum energy to be removed.

ii. Orbital containing the electron that experiences the minimum effective nuclear charge.

b. 2 points

An unknown anion (A^-) has the following *excited* electronic configuration: $[Ar] 4s^2 3d^3 4p^1$

(Ar: Argon; $Z = 18$)

Give the **condensed ground state electronic configuration** of the atom, **A** and the cation, A^{2+} .
Show your work. (Assume all electronic transitions are possible)

Electronic Configuration of A:

Electronic Configuration of A^{2+} :

c. 2 points

In a single atom, determine how many electrons can have the following set of quantum numbers?

(i) $n = 5; l = 3$

In a single atom, determine how many electrons can have the following set of quantum numbers?

(i) $n = 3; l = 2; m_s = +1/2$

**Question 4** **8 points**

Rank the following from 1 to 4 in **increasing** order. Provide the ranking in the boxes provided.

Note: Atomic number (Z) are provided for relevant atoms

a) Increasing (lowest to highest) order of *ionization energy*: (1= lowest; 4 = highest)

Mg

Mg: Z = 12

Rb

Rb: Z = 37

Cs

Cs: Z = 55

K

K : Z = 19

b) Increasing (lowest to highest) order of *bond angle*: (1= lowest; 4 = highest)

H₂OCCl₄XeF₄

HCN

c) Increasing (smallest to largest) order of *ionic size* (1= smallest; 4 = largest)

S²⁻

S: Z = 16

O²⁻

O: Z = 8

Be²⁺

Be: Z = 4

Na⁺

Na: Z = 11

d) Increasing (lowest to highest) order of *bond energy* (1= lowest; 4 = highest)

P-F

P-Cl

P-I

P-Br

**Question 5** **12 points**

a. 9 points For the molecules HClO_4 and TeF_6 , draw the most stable *Lewis structure* showing appropriate shape and *dashed/wedged bonds*, based on VSEPR. Determine the **molecular geometry** around the central atom. For HClO_4 , label the *most polar bond*.



b. 3 points (Note: Lone pairs are not shown in structures below)

(i) For the three *resonance* structures (1, 2 and 3) shown below, answer the following question:



(I) Circle the *most contributing* resonance structure.

Structure 1

Structure 2

Structure 3

(ii) For the three structures given (4, 5, and 6) below, answer the following questions:



(I) Circle all the structures that represent *resonance structures* of each other:

Structure 4

Structure 5

Structure 6

(II) Among the resonance structures, circle the *most contributing* resonance structure.

Structure 4

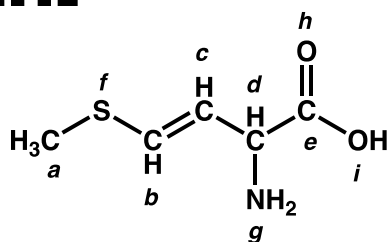
Structure 5

Structure 6

**Question 6** **6 points**

Using the data given below, calculate the **lattice energy** for SrO(s).

	ΔH° (kJ/mol)
$\text{Sr}(s) \longrightarrow \text{Sr}(g)$	+180
$\text{Sr}(g) \longrightarrow \text{Sr}^{2+}(g) + 2e^-$	+1468
$\text{O}_2(g) \longrightarrow 2\text{O}(g)$	+500
$\text{O}(g) + 2e^- \longrightarrow \text{O}^{2-}(g)$	+600
$\text{Sr}(s) + \frac{1}{2}\text{O}_2(g) \longrightarrow \text{SrO}(s)$	-550

**Question 7 10 points**

Consider the molecule given on the left, and answer questions (i) to (vi) using chemical bonding theories that we have learnt in class (valence bond theory, hybridization).

Note that the lone pairs are not shown in the structure:

i. Consider the carbon atoms marked in as ***a***, ***b***, ***c***, ***d***, and ***e*** in the molecule, and determine **hybridization** at each of the carbon centers. (2.5 points)

Atom	Hybridization
Carbon (<i>a</i>)	
Carbon (<i>b</i>)	
Carbon (<i>c</i>)	
Carbon (<i>d</i>)	
Carbon (<i>e</i>)	

ii. Now, consider the sulfur atom (S) marked as ***f***, nitrogen (N) atom (***g***), and oxygen(O) atoms (***h*** and ***i***), and determine their **hybridization**. (2 points)

Atom	Hybridization
Sulfur (<i>f</i>)	
Nitrogen (<i>g</i>)	
Oxygen (<i>h</i>)	
Oxygen (<i>i</i>)	

iii. **Describe** all the bonds formed by the carbon atoms ***a*** and ***b***, using sigma and pi designation as appropriate. (2 points)

Atom	Bonds formed (use sigma and pi as appropriate)
Carbon (<i>a</i>)	
Carbon (<i>b</i>)	

iv. How many total sp^3 -s sigma bonds are in the molecule? (1 point)

v. Determine **electronic** and **molecular geometry** at carbon (C) labeled ***e*** and sulfur (S) labeled ***f*** centres. (2 points)

Atom	Electronic Geometry	Molecular Geometry
Carbon (<i>e</i>)		
Sulfur (<i>f</i>)		

vi. Circle the value from the choices given below, that describes the **C-S-C bond angle at sulfur** (S) marked ***f***: (0.5 points)

>90° to <109.5° 109.5° >109.5 to <120° 120° >120° to <180°


Question 8 12 points

Using the **valence shell orbitals only** draw a **complete** molecular orbital (MO) diagram (with no s-p mixing) for the molecule **PS**, in the space provided below. **(6 points)**

Assume that 3s and 3p atomic orbitals will form molecular orbitals like those formed by the combination of 2s and 2p atomic orbitals that we learnt in class, and there is no s-p mixing.

Show **energy arrow**, **atomic orbitals** involved, **atom labels**, the **molecular orbitals** that are formed, **filled electrons** in the MO diagram etc. Make sure to label all the atomic and molecular orbitals.

Using the MO diagram above, answer the following questions:

- (i) Which one of the following will have the highest bond order? Circle the correct choice
 A) PS^{1-} B) PS C) PS^{1+}
- (ii) Which one of the following will have the longest bond length? Circle the correct choice
 A) PS^{1-} B) PS C) PS^{1+}
- (iii) Which of the following will be diamagnetic? Circle the correct choice.
 A) PS^{1-} B) PS C) PS^{1+}
- (iv) Draw Lewis dot structure of the molecule with the shortest bond length (based on the MO diagram) from among the three (PS^{1-} , PS, PS^{1+}), and indicate the bond order calculated from the Lewis dot structure. **(3 points)**

Lewis Dot Structure

Bond Order



8A3E2ED2-3EDC-435E-9CB6-66C4EDA9A49B

midterm-exam-93388

#258 Page 10 of 10

SCRAP PAGE

**CHEMISTRY 110 –Midterm Exam****Thursday November 7, 2024, 6:30 – 8:30 pm****Examiners: Pallavi Sirjoosingh and Ashok Kakkar**

First name (please write as legibly as possible within the boxes)

--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--

Last name

--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--

Student ID number

--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--

This booklet has 8 questions on 10 pages (The last page can be used as a scrap page).

The staple must NOT be removed from the question booklet. Your intact question booklet must be returned at the end of the exam.

A separate booklet contains a periodic table, and datasheet.

Write your answers directly under the questions in the space provided. Do not write on the QR codes or near the top edges (as it may be cut off).

Please write **legibly and dark enough** for the scan to be readable. For students who use English as a second language, dictionaries are permitted. Calculators are allowed.

DO NOT ASK QUESTIONS OF THE INVIGILATORS about interpretations of the exam questions. If you have a concern about a question, please summarize it at the top of this page. All concerns will be evaluated after the exam. GOOD LUCK!

**Question 1 8 points** (*Assume all transitions are possible*)

a. 6 points Calculate the **energy (in kJ/mol)** required to ionize (completely remove) the electron from the **n=3** level of **He⁺ ion** (He: Helium). Show your work including equations. (3 sig. figs.)

b. 2 points

(I) Select the largest (longest) wavelength from the three choices below:

- i. Wavelength of photon to *completely remove* an electron from *1s orbital of Hydrogen atom*
- ii. Wavelength of photon to *completely remove* an electron from *1s orbital of Boron atom (B)*
- iii. Wavelength of photon to *completely remove* an electron from *1s orbital of Boron cation (B⁺)*

i or ii or iii

(II) Select the smallest (shortest) wavelength from the three choices below (*for H atom*):

- i. Wavelength of photon *emitted* for an electron transitioning from *2p to 1s orbital*
- ii. Wavelength of photon *absorbed* for an electron transitioning from *2s to 4p orbital*
- iii. Wavelength of photon *absorbed* for an electron transitioning from *1s to 3p orbital*

i or ii or iii



Question 2 8 points



A Chem110 student plots the graph shown on the left, from data obtained for the kinetic energy (eV) of the ejected electron versus frequency (s^{-1}) of the incident light, for an unknown metal. ($1\text{eV} = 1.6022 \times 10^{-19} \text{ J}$)

Based on the plot, answer the questions **a to c** below:

a. 1 point On the plot above, label the **threshold frequency**.

b. 2 points What is the value of the work potential (in J) of the unknown metal?

c. 3 points What is the *longest* wavelength (in nm) required to eject an electron from this metal's surface? Show your work including equation(s) used. (3 sig figs)

d. 2 points In the space provided below, plot the following

(Assume frequency is *above* threshold frequency):

- i. *Kinetic energy* of an ejected electron versus the *intensity* of the incident light (left)
- ii. *Number of electrons* ejected versus the *frequency* of the incident light (right)



**Question 3** **6 points****a. 2 points**

In the ground state of Ge (Germanium; $Z = 32$) atom, **name** the following orbitals:

i. Orbital containing the electron that requires the maximum energy to be removed.

ii. Orbital containing the electron that experiences the minimum effective nuclear charge.

b. 2 points

An unknown anion (A^-) has the following *excited* electronic configuration: $[Ar] 4s^2 3d^3 4p^1$

(Ar: Argon; $Z = 18$)

Give the **condensed ground state electronic configuration** of the atom, **A** and the cation, **A^{2+}** .
Show your work. (Assume all electronic transitions are possible)

Electronic Configuration of A:

Electronic Configuration of A^{2+} :

c. 2 points

In a single atom, determine how many electrons can have the following set of quantum numbers?

(i) $n = 5; l = 3$

In a single atom, determine how many electrons can have the following set of quantum numbers?

(i) $n = 3; l = 2; m_s = +1/2$

**Question 4** **8 points**

Rank the following from 1 to 4 in **increasing** order. Provide the ranking in the boxes provided.

Note: Atomic number (Z) are provided for relevant atoms

a) Increasing (lowest to highest) order of *ionization energy*: (1= lowest; 4 = highest)

Mg

Mg: Z = 12

Rb

Rb: Z = 37

Cs

Cs: Z = 55

K

K : Z = 19

b) Increasing (lowest to highest) order of *bond angle*: (1= lowest; 4 = highest)

H₂OCCl₄XeF₄

HCN

c) Increasing (smallest to largest) order of *ionic size* (1= smallest; 4 = largest)

S²⁻

S: Z = 16

O²⁻

O: Z = 8

Be²⁺

Be: Z = 4

Na⁺

Na: Z = 11

d) Increasing (lowest to highest) order of *bond energy* (1= lowest; 4 = highest)

P-F

P-Cl

P-I

P-Br

**Question 5** **12 points**

a. 9 points For the molecules HClO_4 and TeF_6 , draw the most stable *Lewis structure* showing appropriate shape and *dashed/wedged bonds*, based on VSEPR. Determine the **molecular geometry** around the central atom. For HClO_4 , label the *most polar bond*.

HClO_4 (there are no O-O bonds)

Label the most polar bond

Molecular Geometry around Cl

TeF_6

Molecular Geometry around Te

b. 3 points (Note: Lone pairs are not shown in structures below)

(i) For the three *resonance* structures (1, 2 and 3) shown below, answer the following question:



(I) Circle the *most contributing* resonance structure.

Structure 1

Structure 2

Structure 3

(ii) For the three structures given (4, 5, and 6) below, answer the following questions:



(I) Circle all the structures that represent *resonance structures* of each other:

Structure 4

Structure 5

Structure 6

(II) Among the resonance structures, circle the *most contributing* resonance structure.

Structure 4

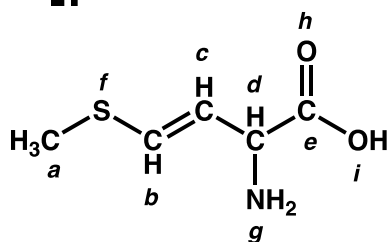
Structure 5

Structure 6

**Question 6** **6 points**

Using the data given below, calculate the **lattice energy** for SrO(s).

	ΔH° (kJ/mol)
$\text{Sr}(s) \longrightarrow \text{Sr}(g)$	+180
$\text{Sr}(g) \longrightarrow \text{Sr}^{2+}(g) + 2e^-$	+1468
$\text{O}_2(g) \longrightarrow 2\text{O}(g)$	+500
$\text{O}(g) + 2e^- \longrightarrow \text{O}^{2-}(g)$	+600
$\text{Sr}(s) + \frac{1}{2}\text{O}_2(g) \longrightarrow \text{SrO}(s)$	-550

**Question 7 10 points**

Consider the molecule given on the left, and answer questions (i) to (vi) using chemical bonding theories that we have learnt in class (valence bond theory, hybridization).

Note that the lone pairs are not shown in the structure:

i. Consider the carbon atoms marked in as ***a***, ***b***, ***c***, ***d***, and ***e*** in the molecule, and determine **hybridization** at each of the carbon centers. (2.5 points)

Atom	Hybridization
Carbon (<i>a</i>)	
Carbon (<i>b</i>)	
Carbon (<i>c</i>)	
Carbon (<i>d</i>)	
Carbon (<i>e</i>)	

ii. Now, consider the sulfur atom (S) marked as ***f***, nitrogen (N) atom (***g***), and oxygen(O) atoms (***h*** and ***i***), and determine their **hybridization**. (2 points)

Atom	Hybridization
Sulfur (<i>f</i>)	
Nitrogen (<i>g</i>)	
Oxygen (<i>h</i>)	
Oxygen (<i>i</i>)	

iii. **Describe** all the bonds formed by the carbon atoms ***a*** and ***b***, using sigma and pi designation as appropriate. (2 points)

Atom	Bonds formed (use sigma and pi as appropriate)
Carbon (<i>a</i>)	
Carbon (<i>b</i>)	

iv. How many total sp^3 -s sigma bonds are in the molecule? (1 point)

v. Determine **electronic** and **molecular geometry** at carbon (C) labeled ***e*** and sulfur (S) labeled ***f*** centres. (2 points)

Atom	Electronic Geometry	Molecular Geometry
Carbon (<i>e</i>)		
Sulfur (<i>f</i>)		

vi. Circle the value from the choices given below, that describes the **C-S-C bond angle at sulfur** (S) marked ***f***: (0.5 points)

>90° to <109.5° 109.5° >109.5 to <120° 120° >120° to <180°

**Question 8 12 points**

Using the **valence shell orbitals only** draw a **complete** molecular orbital (MO) diagram (with no s-p mixing) for the molecule **PS**, in the space provided below. **(6 points)**

Assume that 3s and 3p atomic orbitals will form molecular orbitals like those formed by the combination of 2s and 2p atomic orbitals that we learnt in class, and there is no s-p mixing.

Show **energy arrow**, **atomic orbitals** involved, **atom labels**, the **molecular orbitals** that are formed, **filled electrons** in the MO diagram etc. Make sure to label all the atomic and molecular orbitals.

Using the MO diagram above, answer the following questions:

- (i) Which one of the following will have the highest bond order? Circle the correct choice
A) PS^{1-} B) PS C) PS^{1+}
- (ii) Which one of the following will have the longest bond length? Circle the correct choice
A) PS^{1-} B) PS C) PS^{1+}
- (iii) Which of the following will be diamagnetic? Circle the correct choice.
A) PS^{1-} B) PS C) PS^{1+}
- (iv) Draw Lewis dot structure of the molecule with the shortest bond length (based on the MO diagram) from among the three (PS^{1-} , PS, PS^{1+}), and indicate the bond order calculated from the Lewis dot structure. **(3 points)**

Lewis Dot Structure

Bond Order



CD4FBFA0-F673-49E3-9AF6-60FBD1FB787C

midterm-exam-93388

#259 Page 10 of 10

SCRAP PAGE

**CHEMISTRY 110 –Midterm Exam****Thursday November 7, 2024, 6:30 – 8:30 pm**

Examiners: Pallavi Sirjoosingh and Ashok Kakkar

First name (please write as legibly as possible within the boxes)

--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--

Last name

--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--

Student ID number

--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--

This booklet has 8 questions on 10 pages (The last page can be used as a scrap page).

The staple must NOT be removed from the question booklet. Your intact question booklet must be returned at the end of the exam.

A separate booklet contains a periodic table, and datasheet.

Write your answers directly under the questions in the space provided. Do not write on the QR codes or near the top edges (as it may be cut off).

Please write **legibly and dark enough** for the scan to be readable. For students who use English as a second language, dictionaries are permitted. Calculators are allowed.

DO NOT ASK QUESTIONS OF THE INVIGILATORS about interpretations of the exam questions. If you have a concern about a question, please summarize it at the top of this page. All concerns will be evaluated after the exam. GOOD LUCK!

**Question 1 8 points** (*Assume all transitions are possible*)

a. 6 points Calculate the **energy (in kJ/mol)** required to ionize (completely remove) the electron from the **n=3** level of **He⁺ ion** (He: Helium). Show your work including equations. (3 sig. figs.)

b. 2 points

(I) Select the largest (longest) wavelength from the three choices below:

- i. Wavelength of photon to *completely remove* an electron from *1s orbital of Hydrogen atom*
- ii. Wavelength of photon to *completely remove* an electron from *1s orbital of Boron atom (B)*
- iii. Wavelength of photon to *completely remove* an electron from *1s orbital of Boron cation (B⁺)*

i or ii or iii

(II) Select the smallest (shortest) wavelength from the three choices below (*for H atom*):

- i. Wavelength of photon *emitted* for an electron transitioning from *2p to 1s orbital*
- ii. Wavelength of photon *absorbed* for an electron transitioning from *2s to 4p orbital*
- iii. Wavelength of photon *absorbed* for an electron transitioning from *1s to 3p orbital*

i or ii or iii



Question 2 8 points



A Chem110 student plots the graph shown on the left, from data obtained for the kinetic energy (eV) of the ejected electron versus frequency (s^{-1}) of the incident light, for an unknown metal. ($1\text{eV} = 1.6022 \times 10^{-19} \text{ J}$)

Based on the plot, answer the questions **a to c** below:

a. 1 point On the plot above, label the **threshold frequency**.

b. 2 points What is the value of the work potential (in J) of the unknown metal?

c. 3 points What is the *longest* wavelength (in nm) required to eject an electron from this metal's surface? Show your work including equation(s) used. (3 sig figs)

d. 2 points In the space provided below, plot the following

(Assume frequency is *above* threshold frequency):

- Kinetic energy* of an ejected electron versus the *intensity* of the incident light (left)
- Number of electrons* ejected versus the *frequency* of the incident light (right)



**Question 3** **6 points****a. 2 points**

In the ground state of Ge (Germanium; $Z = 32$) atom, **name** the following orbitals:

i. Orbital containing the electron that requires the maximum energy to be removed.

ii. Orbital containing the electron that experiences the minimum effective nuclear charge.

b. 2 points

An unknown anion (A^-) has the following *excited* electronic configuration: $[Ar] 4s^2 3d^3 4p^1$

(Ar: Argon; $Z = 18$)

Give the **condensed ground state electronic configuration** of the atom, **A** and the cation, **A^{2+}** .
Show your work. (Assume all electronic transitions are possible)

Electronic Configuration of A:

Electronic Configuration of A^{2+} :

c. 2 points

In a single atom, determine how many electrons can have the following set of quantum numbers?

(i) $n = 5; l = 3$

In a single atom, determine how many electrons can have the following set of quantum numbers?

(i) $n = 3; l = 2; m_s = +1/2$

**Question 4** **8 points**

Rank the following from 1 to 4 in **increasing** order. Provide the ranking in the boxes provided.

Note: Atomic number (Z) are provided for relevant atoms

a) Increasing (lowest to highest) order of *ionization energy*: (1= lowest; 4 = highest)

Mg

Mg: Z = 12

Rb

Rb: Z = 37

Cs

Cs: Z = 55

K

K : Z = 19

b) Increasing (lowest to highest) order of *bond angle*: (1= lowest; 4 = highest)

H₂OCCl₄XeF₄

HCN

c) Increasing (smallest to largest) order of *ionic size* (1= smallest; 4 = largest)

S²⁻

S: Z = 16

O²⁻

O: Z = 8

Be²⁺

Be: Z = 4

Na⁺

Na: Z = 11

d) Increasing (lowest to highest) order of *bond energy* (1= lowest; 4 = highest)

P-F

P-Cl

P-I

P-Br

**Question 5** **12 points**

a. 9 points For the molecules HClO_4 and TeF_6 , draw the most stable *Lewis structure* showing appropriate shape and *dashed/wedged bonds*, based on VSEPR. Determine the **molecular geometry** around the central atom. For HClO_4 , label the *most polar bond*.



b. 3 points (Note: Lone pairs are not shown in structures below)

(i) For the three *resonance* structures (1, 2 and 3) shown below, answer the following question:



(I) Circle the *most contributing* resonance structure.

Structure 1

Structure 2

Structure 3

(ii) For the three structures given (4, 5, and 6) below, answer the following questions:



(I) Circle all the structures that represent *resonance structures* of each other:

Structure 4

Structure 5

Structure 6

(II) Among the resonance structures, circle the *most contributing* resonance structure.

Structure 4

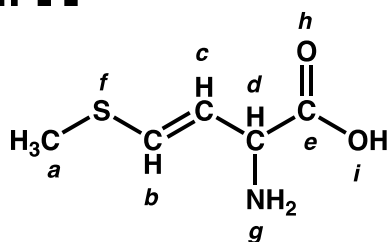
Structure 5

Structure 6

**Question 6** **6 points**

Using the data given below, calculate the **lattice energy** for SrO(s).

	ΔH° (kJ/mol)
$\text{Sr}(s) \longrightarrow \text{Sr}(g)$	+180
$\text{Sr}(g) \longrightarrow \text{Sr}^{2+}(g) + 2e^-$	+1468
$\text{O}_2(g) \longrightarrow 2\text{O}(g)$	+500
$\text{O}(g) + 2e^- \longrightarrow \text{O}^{2-}(g)$	+600
$\text{Sr}(s) + \frac{1}{2}\text{O}_2(g) \longrightarrow \text{SrO}(s)$	-550

**Question 7 10 points**

Consider the molecule given on the left, and answer questions (i) to (vi) using chemical bonding theories that we have learnt in class (valence bond theory, hybridization).

Note that the lone pairs are not shown in the structure:

i. Consider the carbon atoms marked in as ***a***, ***b***, ***c***, ***d***, and ***e*** in the molecule, and determine **hybridization** at each of the carbon centers. (2.5 points)

Atom	Hybridization
Carbon (<i>a</i>)	
Carbon (<i>b</i>)	
Carbon (<i>c</i>)	
Carbon (<i>d</i>)	
Carbon (<i>e</i>)	

ii. Now, consider the sulfur atom (S) marked as ***f***, nitrogen (N) atom (***g***), and oxygen(O) atoms (***h*** and ***i***), and determine their **hybridization**. (2 points)

Atom	Hybridization
Sulfur (<i>f</i>)	
Nitrogen (<i>g</i>)	
Oxygen (<i>h</i>)	
Oxygen (<i>i</i>)	

iii. **Describe** all the bonds formed by the carbon atoms ***a*** and ***b***, using sigma and pi designation as appropriate. (2 points)

Atom	Bonds formed (use sigma and pi as appropriate)
Carbon (<i>a</i>)	
Carbon (<i>b</i>)	

iv. How many total sp^3 -s sigma bonds are in the molecule? (1 point)

v. Determine **electronic** and **molecular geometry** at carbon (C) labeled ***e*** and sulfur (S) labeled ***f*** centres. (2 points)

Atom	Electronic Geometry	Molecular Geometry
Carbon (<i>e</i>)		
Sulfur (<i>f</i>)		

vi. Circle the value from the choices given below, that describes the **C-S-C bond angle at sulfur** (S) marked ***f***: (0.5 points)

>90° to <109.5° 109.5° >109.5 to <120° 120° >120° to <180°


Question 8 12 points

Using the **valence shell orbitals only** draw a **complete** molecular orbital (MO) diagram (with no s-p mixing) for the molecule **PS**, in the space provided below. **(6 points)**

Assume that 3s and 3p atomic orbitals will form molecular orbitals like those formed by the combination of 2s and 2p atomic orbitals that we learnt in class, and there is no s-p mixing.

Show **energy arrow**, **atomic orbitals** involved, **atom labels**, the **molecular orbitals** that are formed, **filled electrons** in the MO diagram etc. Make sure to label all the atomic and molecular orbitals.

Using the MO diagram above, answer the following questions:

- (i) Which one of the following will have the highest bond order? Circle the correct choice
 A) PS^{1-} B) PS C) PS^{1+}
- (ii) Which one of the following will have the longest bond length? Circle the correct choice
 A) PS^{1-} B) PS C) PS^{1+}
- (iii) Which of the following will be diamagnetic? Circle the correct choice.
 A) PS^{1-} B) PS C) PS^{1+}
- (iv) Draw Lewis dot structure of the molecule with the shortest bond length (based on the MO diagram) from among the three (PS^{1-} , PS, PS^{1+}), and indicate the bond order calculated from the Lewis dot structure. **(3 points)**

Lewis Dot Structure

Bond Order



2E874C1D-8F85-45E9-A174-1575D3C154A6

midterm-exam-93388

#260 Page 10 of 10

SCRAP PAGE

**CHEMISTRY 110 –Midterm Exam****Thursday November 7, 2024, 6:30 – 8:30 pm****Examiners: Pallavi Sirjoosingh and Ashok Kakkar**

First name (please write as legibly as possible within the boxes)

--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--

Last name

--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--

Student ID number

--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--

This booklet has 8 questions on 10 pages (The last page can be used as a scrap page).

The staple must NOT be removed from the question booklet. Your intact question booklet must be returned at the end of the exam.

A separate booklet contains a periodic table, and datasheet.

Write your answers directly under the questions in the space provided. Do not write on the QR codes or near the top edges (as it may be cut off).

Please write **legibly and dark enough** for the scan to be readable. For students who use English as a second language, dictionaries are permitted. Calculators are allowed.

DO NOT ASK QUESTIONS OF THE INVIGILATORS about interpretations of the exam questions. If you have a concern about a question, please summarize it at the top of this page. All concerns will be evaluated after the exam. GOOD LUCK!

**Question 1 8 points** (*Assume all transitions are possible*)

a. 6 points Calculate the **energy (in kJ/mol)** required to ionize (completely remove) the electron from the **n=3** level of **He⁺ ion** (He: Helium). Show your work including equations. (3 sig. figs.)

b. 2 points

(I) Select the largest (longest) wavelength from the three choices below:

- i. Wavelength of photon to *completely remove* an electron from *1s orbital of Hydrogen atom*
- ii. Wavelength of photon to *completely remove* an electron from *1s orbital of Boron atom (B)*
- iii. Wavelength of photon to *completely remove* an electron from *1s orbital of Boron cation (B⁺)*

i or ii or iii

(II) Select the smallest (shortest) wavelength from the three choices below (*for H atom*):

- i. Wavelength of photon *emitted* for an electron transitioning from *2p to 1s orbital*
- ii. Wavelength of photon *absorbed* for an electron transitioning from *2s to 4p orbital*
- iii. Wavelength of photon *absorbed* for an electron transitioning from *1s to 3p orbital*

i or ii or iii



Question 2 8 points



A Chem110 student plots the graph shown on the left, from data obtained for the kinetic energy (eV) of the ejected electron versus frequency (s^{-1}) of the incident light, for an unknown metal. ($1\text{eV} = 1.6022 \times 10^{-19} \text{ J}$)

Based on the plot, answer the questions **a to c** below:

a. 1 point On the plot above, label the **threshold frequency**.

b. 2 points What is the value of the work potential (in J) of the unknown metal?

c. 3 points What is the *longest* wavelength (in nm) required to eject an electron from this metal's surface? Show your work including equation(s) used. (3 sig figs)

d. 2 points In the space provided below, plot the following

(Assume frequency is *above* threshold frequency):

- i. *Kinetic energy* of an ejected electron versus the *intensity* of the incident light (left)
- ii. *Number of electrons* ejected versus the *frequency* of the incident light (right)



**Question 3** **6 points****a. 2 points**

In the ground state of Ge (Germanium; $Z = 32$) atom, **name** the following orbitals:

i. Orbital containing the electron that requires the maximum energy to be removed.

ii. Orbital containing the electron that experiences the minimum effective nuclear charge.

b. 2 points

An unknown anion (A^-) has the following *excited* electronic configuration: $[Ar] 4s^2 3d^3 4p^1$

(Ar: Argon; $Z = 18$)

Give the **condensed ground state electronic configuration** of the atom, **A** and the cation, A^{2+} .
Show your work. (Assume all electronic transitions are possible)

Electronic Configuration of A:

Electronic Configuration of A^{2+} :

c. 2 points

In a single atom, determine how many electrons can have the following set of quantum numbers?

(i) $n = 5; l = 3$

In a single atom, determine how many electrons can have the following set of quantum numbers?

(i) $n = 3; l = 2; m_s = +1/2$

**Question 4** **8 points**

Rank the following from 1 to 4 in **increasing** order. Provide the ranking in the boxes provided.

Note: Atomic number (Z) are provided for relevant atoms

a) Increasing (lowest to highest) order of *ionization energy*: (1= lowest; 4 = highest)

Mg

Mg: Z = 12

Rb

Rb: Z = 37

Cs

Cs: Z = 55

K

K : Z = 19

b) Increasing (lowest to highest) order of *bond angle*: (1= lowest; 4 = highest)

H₂OCCl₄XeF₄

HCN

c) Increasing (smallest to largest) order of *ionic size* (1= smallest; 4 = largest)

S²⁻

S: Z = 16

O²⁻

O: Z = 8

Be²⁺

Be: Z = 4

Na⁺

Na: Z = 11

d) Increasing (lowest to highest) order of *bond energy* (1= lowest; 4 = highest)

P-F

P-Cl

P-I

P-Br

**Question 5** **12 points**

a. 9 points For the molecules HClO_4 and TeF_6 , draw the most stable *Lewis structure* showing appropriate shape and *dashed/wedged bonds*, based on VSEPR. Determine the **molecular geometry** around the central atom. For HClO_4 , label the *most polar bond*.



b. 3 points (Note: Lone pairs are not shown in structures below)

(i) For the three *resonance* structures (1, 2 and 3) shown below, answer the following question:



(I) Circle the *most contributing* resonance structure.

Structure 1

Structure 2

Structure 3

(ii) For the three structures given (4, 5, and 6) below, answer the following questions:



(I) Circle all the structures that represent *resonance structures* of each other:

Structure 4

Structure 5

Structure 6

(II) Among the resonance structures, circle the *most contributing* resonance structure.

Structure 4

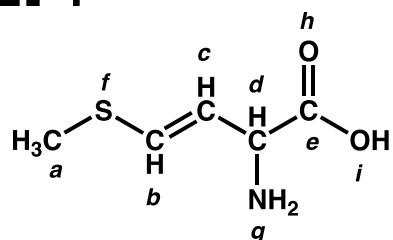
Structure 5

Structure 6

**Question 6** **6 points**

Using the data given below, calculate the **lattice energy** for SrO(s).

	ΔH° (kJ/mol)
$\text{Sr}(s) \longrightarrow \text{Sr}(g)$	+180
$\text{Sr}(g) \longrightarrow \text{Sr}^{2+}(g) + 2e^-$	+1468
$\text{O}_2(g) \longrightarrow 2\text{O}(g)$	+500
$\text{O}(g) + 2e^- \longrightarrow \text{O}^{2-}(g)$	+600
$\text{Sr}(s) + \frac{1}{2}\text{O}_2(g) \longrightarrow \text{SrO}(s)$	-550

**Question 7 10 points**

Consider the molecule given on the left, and answer questions (i) to (vi) using chemical bonding theories that we have learnt in class (valence bond theory, hybridization).

Note that the lone pairs are not shown in the structure:

i. Consider the carbon atoms marked in as ***a***, ***b***, ***c***, ***d***, and ***e*** in the molecule, and determine **hybridization** at each of the carbon centers. (2.5 points)

Atom	Hybridization
Carbon (<i>a</i>)	
Carbon (<i>b</i>)	
Carbon (<i>c</i>)	
Carbon (<i>d</i>)	
Carbon (<i>e</i>)	

ii. Now, consider the sulfur atom (S) marked as ***f***, nitrogen (N) atom (***g***), and oxygen(O) atoms (***h*** and ***i***), and determine their **hybridization**. (2 points)

Atom	Hybridization
Sulfur (<i>f</i>)	
Nitrogen (<i>g</i>)	
Oxygen (<i>h</i>)	
Oxygen (<i>i</i>)	

iii. **Describe** all the bonds formed by the carbon atoms ***a*** and ***b***, using sigma and pi designation as appropriate. (2 points)

Atom	Bonds formed (use sigma and pi as appropriate)
Carbon (<i>a</i>)	
Carbon (<i>b</i>)	

iv. How many total sp^3 -s sigma bonds are in the molecule? (1 point)

v. Determine **electronic** and **molecular geometry** at carbon (C) labeled ***e*** and sulfur (S) labeled ***f*** centres. (2 points)

Atom	Electronic Geometry	Molecular Geometry
Carbon (<i>e</i>)		
Sulfur (<i>f</i>)		

vi. Circle the value from the choices given below, that describes the **C-S-C bond angle at sulfur** (S) marked ***f***: (0.5 points)

>90° to <109.5° 109.5° >109.5 to <120° 120° >120° to <180°


Question 8 12 points

Using the **valence shell orbitals only** draw a **complete** molecular orbital (MO) diagram (with no s-p mixing) for the molecule **PS**, in the space provided below. **(6 points)**

Assume that 3s and 3p atomic orbitals will form molecular orbitals like those formed by the combination of 2s and 2p atomic orbitals that we learnt in class, and there is no s-p mixing.

Show **energy arrow**, **atomic orbitals** involved, **atom labels**, the **molecular orbitals** that are formed, **filled electrons** in the MO diagram etc. Make sure to **label all the atomic and molecular orbitals**.

Using the MO diagram above, answer the following questions:

- (i) Which one of the following will have the highest bond order? Circle the correct choice
 A) PS^{1-} B) PS C) PS^{1+}
- (ii) Which one of the following will have the longest bond length? Circle the correct choice
 A) PS^{1-} B) PS C) PS^{1+}
- (iii) Which of the following will be diamagnetic? Circle the correct choice.
 A) PS^{1-} B) PS C) PS^{1+}
- (iv) Draw Lewis dot structure of the molecule with the shortest bond length (based on the MO diagram) from among the three (PS^{1-} , PS, PS^{1+}), and indicate the bond order calculated from the Lewis dot structure. **(3 points)**

Lewis Dot Structure

Bond Order



CC96F313-0F63-42D6-B59B-DB4B0295AB25

midterm-exam-93388

#261 Page 10 of 10

SCRAP PAGE

**CHEMISTRY 110 –Midterm Exam****Thursday November 7, 2024, 6:30 – 8:30 pm****Examiners: Pallavi Sirjoosingh and Ashok Kakkar**

First name (please write as legibly as possible within the boxes)

--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--

Last name

--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--

Student ID number

--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--

This booklet has 8 questions on 10 pages (The last page can be used as a scrap page).

The staple must NOT be removed from the question booklet. Your intact question booklet must be returned at the end of the exam.

A separate booklet contains a periodic table, and datasheet.

Write your answers directly under the questions in the space provided. Do not write on the QR codes or near the top edges (as it may be cut off).

Please write **legibly and dark enough** for the scan to be readable. For students who use English as a second language, dictionaries are permitted. Calculators are allowed.

DO NOT ASK QUESTIONS OF THE INVIGILATORS about interpretations of the exam questions. If you have a concern about a question, please summarize it at the top of this page. All concerns will be evaluated after the exam. GOOD LUCK!

**Question 1 8 points** (*Assume all transitions are possible*)

a. 6 points Calculate the **energy (in kJ/mol)** required to ionize (completely remove) the electron from the **n=3** level of **He⁺ ion** (He: Helium). Show your work including equations. (3 sig. figs.)

b. 2 points

(I) Select the largest (longest) wavelength from the three choices below:

- i. Wavelength of photon to *completely remove* an electron from *1s orbital of Hydrogen atom*
- ii. Wavelength of photon to *completely remove* an electron from *1s orbital of Boron atom (B)*
- iii. Wavelength of photon to *completely remove* an electron from *1s orbital of Boron cation (B⁺)*

i or ii or iii

(II) Select the smallest (shortest) wavelength from the three choices below (*for H atom*):

- i. Wavelength of photon *emitted* for an electron transitioning from *2p to 1s orbital*
- ii. Wavelength of photon *absorbed* for an electron transitioning from *2s to 4p orbital*
- iii. Wavelength of photon *absorbed* for an electron transitioning from *1s to 3p orbital*

i or ii or iii



Question 2 8 points



A Chem110 student plots the graph shown on the left, from data obtained for the kinetic energy (eV) of the ejected electron versus frequency (s^{-1}) of the incident light, for an unknown metal. ($1\text{eV} = 1.6022 \times 10^{-19} \text{ J}$)

Based on the plot, answer the questions **a to c** below:

a. 1 point On the plot above, label the **threshold frequency**.

b. 2 points What is the value of the work potential (in J) of the unknown metal?

c. 3 points What is the *longest* wavelength (in nm) required to eject an electron from this metal's surface? Show your work including equation(s) used. (3 sig figs)

d. 2 points In the space provided below, plot the following

(Assume frequency is *above* threshold frequency):

- Kinetic energy* of an ejected electron versus the *intensity* of the incident light (left)
- Number of electrons* ejected versus the *frequency* of the incident light (right)



**Question 3** **6 points****a. 2 points**

In the ground state of Ge (Germanium; $Z = 32$) atom, **name** the following orbitals:

i. Orbital containing the electron that requires the maximum energy to be removed.

ii. Orbital containing the electron that experiences the minimum effective nuclear charge.

b. 2 points

An unknown anion (A^-) has the following *excited* electronic configuration: $[Ar] 4s^2 3d^3 4p^1$

(Ar: Argon; $Z = 18$)

Give the **condensed ground state electronic configuration** of the atom, **A** and the cation, **A^{2+}** .
Show your work. (Assume all electronic transitions are possible)

Electronic Configuration of A:

Electronic Configuration of A^{2+} :

c. 2 points

In a single atom, determine how many electrons can have the following set of quantum numbers?

(i) $n = 5; l = 3$

In a single atom, determine how many electrons can have the following set of quantum numbers?

(i) $n = 3; l = 2; m_s = +1/2$

**Question 4** **8 points**

Rank the following from 1 to 4 in **increasing** order. Provide the ranking in the boxes provided.

Note: Atomic number (Z) are provided for relevant atoms

a) Increasing (lowest to highest) order of *ionization energy*: (1= lowest; 4 = highest)

Mg

Mg: Z = 12

Rb

Rb: Z = 37

Cs

Cs: Z = 55

K

K : Z = 19

b) Increasing (lowest to highest) order of *bond angle*: (1= lowest; 4 = highest)

H₂OCCl₄XeF₄

HCN

c) Increasing (smallest to largest) order of *ionic size* (1= smallest; 4 = largest)

S²⁻

S: Z = 16

O²⁻

O: Z = 8

Be²⁺

Be: Z = 4

Na⁺

Na: Z = 11

d) Increasing (lowest to highest) order of *bond energy* (1= lowest; 4 = highest)

P-F

P-Cl

P-I

P-Br

**Question 5** **12 points**

a. 9 points For the molecules HClO_4 and TeF_6 , draw the most stable *Lewis structure* showing appropriate shape and *dashed/wedged bonds*, based on VSEPR. Determine the **molecular geometry** around the central atom. For HClO_4 , label the *most polar bond*.



b. 3 points (Note: Lone pairs are not shown in structures below)

(i) For the three *resonance* structures (1, 2 and 3) shown below, answer the following question:



(I) Circle the *most contributing* resonance structure.

Structure 1

Structure 2

Structure 3

(ii) For the three structures given (4, 5, and 6) below, answer the following questions:



(I) Circle all the structures that represent *resonance structures* of each other:

Structure 4

Structure 5

Structure 6

(II) Among the resonance structures, circle the *most contributing* resonance structure.

Structure 4

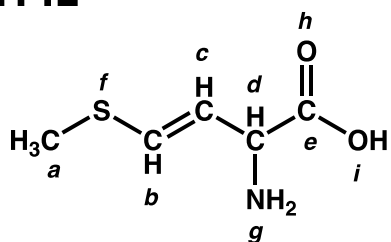
Structure 5

Structure 6

**Question 6** **6 points**

Using the data given below, calculate the **lattice energy** for SrO(s).

	ΔH° (kJ/mol)
$\text{Sr}(s) \longrightarrow \text{Sr}(g)$	+180
$\text{Sr}(g) \longrightarrow \text{Sr}^{2+}(g) + 2e^-$	+1468
$\text{O}_2(g) \longrightarrow 2\text{O}(g)$	+500
$\text{O}(g) + 2e^- \longrightarrow \text{O}^{2-}(g)$	+600
$\text{Sr}(s) + \frac{1}{2}\text{O}_2(g) \longrightarrow \text{SrO}(s)$	-550

**Question 7 10 points**

Consider the molecule given on the left, and answer questions (i) to (vi) using chemical bonding theories that we have learnt in class (valence bond theory, hybridization).

Note that the lone pairs are not shown in the structure:

i. Consider the carbon atoms marked in as ***a***, ***b***, ***c***, ***d***, and ***e*** in the molecule, and determine **hybridization** at each of the carbon centers. (2.5 points)

Atom	Hybridization
Carbon (<i>a</i>)	
Carbon (<i>b</i>)	
Carbon (<i>c</i>)	
Carbon (<i>d</i>)	
Carbon (<i>e</i>)	

ii. Now, consider the sulfur atom (S) marked as ***f***, nitrogen (N) atom (***g***), and oxygen(O) atoms (***h*** and ***i***), and determine their **hybridization**. (2 points)

Atom	Hybridization
Sulfur (<i>f</i>)	
Nitrogen (<i>g</i>)	
Oxygen (<i>h</i>)	
Oxygen (<i>i</i>)	

iii. **Describe** all the bonds formed by the carbon atoms ***a*** and ***b***, using sigma and pi designation as appropriate. (2 points)

Atom	Bonds formed (use sigma and pi as appropriate)
Carbon (<i>a</i>)	
Carbon (<i>b</i>)	

iv. How many total sp^3 -s sigma bonds are in the molecule? (1 point)

v. Determine **electronic** and **molecular geometry** at carbon (C) labeled ***e*** and sulfur (S) labeled ***f*** centres. (2 points)

Atom	Electronic Geometry	Molecular Geometry
Carbon (<i>e</i>)		
Sulfur (<i>f</i>)		

vi. Circle the value from the choices given below, that describes the **C-S-C bond angle at sulfur** (S) marked ***f***: (0.5 points)

>90° to <109.5°

109.5°

>109.5 to <120°

120°

>120° to <180°


Question 8 12 points

Using the **valence shell orbitals only** draw a **complete** molecular orbital (MO) diagram (with no s-p mixing) for the molecule **PS**, in the space provided below. **(6 points)**

Assume that 3s and 3p atomic orbitals will form molecular orbitals like those formed by the combination of 2s and 2p atomic orbitals that we learnt in class, and there is no s-p mixing.

Show **energy arrow**, **atomic orbitals** involved, **atom labels**, the **molecular orbitals** that are formed, **filled electrons** in the MO diagram etc. Make sure to label all the atomic and molecular orbitals.

Using the MO diagram above, answer the following questions:

- (i) Which one of the following will have the highest bond order? Circle the correct choice
 A) PS^{1-} B) PS C) PS^{1+}
- (ii) Which one of the following will have the longest bond length? Circle the correct choice
 A) PS^{1-} B) PS C) PS^{1+}
- (iii) Which of the following will be diamagnetic? Circle the correct choice.
 A) PS^{1-} B) PS C) PS^{1+}
- (iv) Draw Lewis dot structure of the molecule with the shortest bond length (based on the MO diagram) from among the three (PS^{1-} , PS, PS^{1+}), and indicate the bond order calculated from the Lewis dot structure. **(3 points)**

Lewis Dot Structure

Bond Order



0A06487B-49BB-45C8-BED5-5A78F8F0A8F2

midterm-exam-93388

#262 Page 10 of 10

SCRAP PAGE

**CHEMISTRY 110 –Midterm Exam****Thursday November 7, 2024, 6:30 – 8:30 pm**

Examiners: Pallavi Sirjoosingh and Ashok Kakkar

First name (please write as legibly as possible within the boxes)

--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--

Last name

--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--

Student ID number

--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--

This booklet has 8 questions on 10 pages (The last page can be used as a scrap page).

The staple must NOT be removed from the question booklet. Your intact question booklet must be returned at the end of the exam.

A separate booklet contains a periodic table, and datasheet.

Write your answers directly under the questions in the space provided. Do not write on the QR codes or near the top edges (as it may be cut off).

Please write **legibly and dark enough** for the scan to be readable. For students who use English as a second language, dictionaries are permitted. Calculators are allowed.

DO NOT ASK QUESTIONS OF THE INVIGILATORS about interpretations of the exam questions. If you have a concern about a question, please summarize it at the top of this page. All concerns will be evaluated after the exam. GOOD LUCK!

**Question 1 8 points** (*Assume all transitions are possible*)

a. 6 points Calculate the **energy (in kJ/mol)** required to ionize (completely remove) the electron from the **n=3** level of **He⁺ ion** (He: Helium). Show your work including equations. (3 sig. figs.)

b. 2 points

(I) Select the largest (longest) wavelength from the three choices below:

- i. Wavelength of photon to *completely remove* an electron from *1s orbital of Hydrogen atom*
- ii. Wavelength of photon to *completely remove* an electron from *1s orbital of Boron atom (B)*
- iii. Wavelength of photon to *completely remove* an electron from *1s orbital of Boron cation (B⁺)*

i or ii or iii

(II) Select the smallest (shortest) wavelength from the three choices below (*for H atom*):

- i. Wavelength of photon *emitted* for an electron transitioning from *2p to 1s orbital*
- ii. Wavelength of photon *absorbed* for an electron transitioning from *2s to 4p orbital*
- iii. Wavelength of photon *absorbed* for an electron transitioning from *1s to 3p orbital*

i or ii or iii



Question 2 8 points



A Chem110 student plots the graph shown on the left, from data obtained for the kinetic energy (eV) of the ejected electron versus frequency (s^{-1}) of the incident light, for an unknown metal. ($1\text{eV} = 1.6022 \times 10^{-19} \text{ J}$)

Based on the plot, answer the questions **a to c** below:

a. 1 point On the plot above, label the **threshold frequency**.

b. 2 points What is the value of the work potential (in J) of the unknown metal?

c. 3 points What is the *longest* wavelength (in nm) required to eject an electron from this metal's surface? Show your work including equation(s) used. (3 sig figs)

d. 2 points In the space provided below, plot the following

(Assume frequency is *above* threshold frequency):

- i. *Kinetic energy* of an ejected electron versus the *intensity* of the incident light (left)
- ii. *Number of electrons* ejected versus the *frequency* of the incident light (right)



**Question 3** **6 points****a. 2 points**

In the ground state of Ge (Germanium; $Z = 32$) atom, **name** the following orbitals:

i. Orbital containing the electron that requires the maximum energy to be removed.

ii. Orbital containing the electron that experiences the minimum effective nuclear charge.

b. 2 points

An unknown anion (A^-) has the following *excited* electronic configuration: $[Ar] 4s^2 3d^3 4p^1$

(Ar: Argon; $Z = 18$)

Give the **condensed ground state electronic configuration** of the atom, **A** and the cation, **A^{2+}** .
Show your work. (Assume all electronic transitions are possible)

Electronic Configuration of A:

Electronic Configuration of A^{2+} :

c. 2 points

In a single atom, determine how many electrons can have the following set of quantum numbers?

(i) $n = 5; l = 3$

In a single atom, determine how many electrons can have the following set of quantum numbers?

(i) $n = 3; l = 2; m_s = +1/2$

**Question 4** **8 points**

Rank the following from 1 to 4 in **increasing** order. Provide the ranking in the boxes provided.

Note: Atomic number (Z) are provided for relevant atoms

a) Increasing (lowest to highest) order of *ionization energy*: (1= lowest; 4 = highest)

Mg

Mg: Z = 12

Rb

Rb: Z = 37

Cs

Cs: Z = 55

K

K : Z = 19

b) Increasing (lowest to highest) order of *bond angle*: (1= lowest; 4 = highest)

H₂OCCl₄XeF₄

HCN

c) Increasing (smallest to largest) order of *ionic size* (1= smallest; 4 = largest)

S²⁻

S: Z = 16

O²⁻

O: Z = 8

Be²⁺

Be: Z = 4

Na⁺

Na: Z = 11

d) Increasing (lowest to highest) order of *bond energy* (1= lowest; 4 = highest)

P-F

P-Cl

P-I

P-Br

**Question 5** **12 points**

a. 9 points For the molecules HClO_4 and TeF_6 , draw the most stable *Lewis structure* showing appropriate shape and *dashed/wedged bonds*, based on VSEPR. Determine the **molecular geometry** around the central atom. For HClO_4 , label the *most polar bond*.



b. 3 points (Note: Lone pairs are not shown in structures below)

(i) For the three *resonance* structures (1, 2 and 3) shown below, answer the following question:



(I) Circle the *most contributing* resonance structure.

Structure 1

Structure 2

Structure 3

(ii) For the three structures given (4, 5, and 6) below, answer the following questions:



(I) Circle all the structures that represent *resonance structures* of each other:

Structure 4

Structure 5

Structure 6

(II) Among the resonance structures, circle the *most contributing* resonance structure.

Structure 4

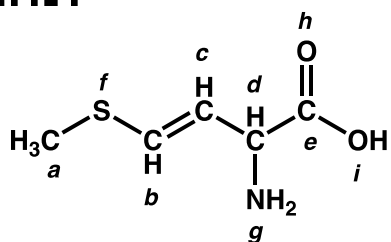
Structure 5

Structure 6

**Question 6** **6 points**

Using the data given below, calculate the **lattice energy** for SrO(s).

	ΔH° (kJ/mol)
$\text{Sr}(s) \longrightarrow \text{Sr}(g)$	+180
$\text{Sr}(g) \longrightarrow \text{Sr}^{2+}(g) + 2e^-$	+1468
$\text{O}_2(g) \longrightarrow 2\text{O}(g)$	+500
$\text{O}(g) + 2e^- \longrightarrow \text{O}^{2-}(g)$	+600
$\text{Sr}(s) + \frac{1}{2}\text{O}_2(g) \longrightarrow \text{SrO}(s)$	-550

**Question 7 10 points**

Consider the molecule given on the left, and answer questions (i) to (vi) using chemical bonding theories that we have learnt in class (valence bond theory, hybridization).

Note that the lone pairs are not shown in the structure:

i. Consider the carbon atoms marked in as ***a***, ***b***, ***c***, ***d***, and ***e*** in the molecule, and determine **hybridization** at each of the carbon centers. (2.5 points)

Atom	Hybridization
Carbon (<i>a</i>)	
Carbon (<i>b</i>)	
Carbon (<i>c</i>)	
Carbon (<i>d</i>)	
Carbon (<i>e</i>)	

ii. Now, consider the sulfur atom (S) marked as ***f***, nitrogen (N) atom (***g***), and oxygen(O) atoms (***h*** and ***i***), and determine their **hybridization**. (2 points)

Atom	Hybridization
Sulfur (<i>f</i>)	
Nitrogen (<i>g</i>)	
Oxygen (<i>h</i>)	
Oxygen (<i>i</i>)	

iii. **Describe** all the bonds formed by the carbon atoms ***a*** and ***b***, using sigma and pi designation as appropriate. (2 points)

Atom	Bonds formed (use sigma and pi as appropriate)
Carbon (<i>a</i>)	
Carbon (<i>b</i>)	

iv. How many total sp^3 -s sigma bonds are in the molecule? (1 point)

v. Determine **electronic** and **molecular geometry** at carbon (C) labeled ***e*** and sulfur (S) labeled ***f*** centres. (2 points)

Atom	Electronic Geometry	Molecular Geometry
Carbon (<i>e</i>)		
Sulfur (<i>f</i>)		

vi. Circle the value from the choices given below, that describes the **C-S-C bond angle at sulfur** (S) marked ***f***: (0.5 points)

>90° to <109.5° 109.5° >109.5 to <120° 120° >120° to <180°


Question 8 12 points

Using the **valence shell orbitals only** draw a **complete** molecular orbital (MO) diagram (with no s-p mixing) for the molecule **PS**, in the space provided below. **(6 points)**

Assume that 3s and 3p atomic orbitals will form molecular orbitals like those formed by the combination of 2s and 2p atomic orbitals that we learnt in class, and there is no s-p mixing.

Show **energy arrow**, **atomic orbitals** involved, **atom labels**, the **molecular orbitals** that are formed, **filled electrons** in the MO diagram etc. Make sure to label all the atomic and molecular orbitals.

Using the MO diagram above, answer the following questions:

- (i) Which one of the following will have the highest bond order? Circle the correct choice
 A) PS^{1-} B) PS C) PS^{1+}
- (ii) Which one of the following will have the longest bond length? Circle the correct choice
 A) PS^{1-} B) PS C) PS^{1+}
- (iii) Which of the following will be diamagnetic? Circle the correct choice.
 A) PS^{1-} B) PS C) PS^{1+}
- (iv) Draw Lewis dot structure of the molecule with the shortest bond length (based on the MO diagram) from among the three (PS^{1-} , PS, PS^{1+}), and indicate the bond order calculated from the Lewis dot structure. **(3 points)**

Lewis Dot Structure

Bond Order



66999688-1F0B-4976-8ADB-AEB5077B1807

midterm-exam-93388

#263 Page 10 of 10

SCRAP PAGE

**CHEMISTRY 110 –Midterm Exam****Thursday November 7, 2024, 6:30 – 8:30 pm****Examiners: Pallavi Sirjoosingh and Ashok Kakkar**

First name (please write as legibly as possible within the boxes)

--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--

Last name

--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--

Student ID number

--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--

This booklet has 8 questions on 10 pages (The last page can be used as a scrap page).

The staple must NOT be removed from the question booklet. Your intact question booklet must be returned at the end of the exam.

A separate booklet contains a periodic table, and datasheet.

Write your answers directly under the questions in the space provided. Do not write on the QR codes or near the top edges (as it may be cut off).

Please write **legibly and dark enough** for the scan to be readable. For students who use English as a second language, dictionaries are permitted. Calculators are allowed.

DO NOT ASK QUESTIONS OF THE INVIGILATORS about interpretations of the exam questions. If you have a concern about a question, please summarize it at the top of this page. All concerns will be evaluated after the exam. GOOD LUCK!

**Question 1 8 points** (*Assume all transitions are possible*)

a. 6 points Calculate the **energy (in kJ/mol)** required to ionize (completely remove) the electron from the **n=3** level of **He⁺ ion** (He: Helium). Show your work including equations. (3 sig. figs.)

b. 2 points

(I) Select the largest (longest) wavelength from the three choices below:

- i. Wavelength of photon to *completely remove* an electron from *1s orbital of Hydrogen atom*
- ii. Wavelength of photon to *completely remove* an electron from *1s orbital of Boron atom (B)*
- iii. Wavelength of photon to *completely remove* an electron from *1s orbital of Boron cation (B⁺)*

i or ii or iii

(II) Select the smallest (shortest) wavelength from the three choices below (*for H atom*):

- i. Wavelength of photon *emitted* for an electron transitioning from *2p to 1s orbital*
- ii. Wavelength of photon *absorbed* for an electron transitioning from *2s to 4p orbital*
- iii. Wavelength of photon *absorbed* for an electron transitioning from *1s to 3p orbital*

i or ii or iii



Question 2 8 points



A Chem110 student plots the graph shown on the left, from data obtained for the kinetic energy (eV) of the ejected electron versus frequency (s^{-1}) of the incident light, for an unknown metal. ($1\text{eV} = 1.6022 \times 10^{-19} \text{ J}$)

Based on the plot, answer the questions **a to c** below:

a. 1 point On the plot above, label the **threshold frequency**.

b. 2 points What is the value of the work potential (in J) of the unknown metal?

c. 3 points What is the *longest* wavelength (in nm) required to eject an electron from this metal's surface? Show your work including equation(s) used. (3 sig figs)

d. 2 points In the space provided below, plot the following

(Assume frequency is *above* threshold frequency):

- Kinetic energy* of an ejected electron versus the *intensity* of the incident light (left)
- Number of electrons* ejected versus the *frequency* of the incident light (right)



**Question 3** **6 points****a. 2 points**

In the ground state of Ge (Germanium; $Z = 32$) atom, **name** the following orbitals:

i. Orbital containing the electron that requires the maximum energy to be removed.

ii. Orbital containing the electron that experiences the minimum effective nuclear charge.

b. 2 points

An unknown anion (A^-) has the following *excited* electronic configuration: $[Ar] 4s^2 3d^3 4p^1$

(Ar: Argon; $Z = 18$)

Give the **condensed ground state electronic configuration** of the atom, **A** and the cation, **A^{2+}** .
Show your work. (Assume all electronic transitions are possible)

Electronic Configuration of A:

Electronic Configuration of A^{2+} :

c. 2 points

In a single atom, determine how many electrons can have the following set of quantum numbers?

(i) $n = 5; l = 3$

In a single atom, determine how many electrons can have the following set of quantum numbers?

(i) $n = 3; l = 2; m_s = +1/2$

**Question 4** **8 points**

Rank the following from 1 to 4 in **increasing** order. Provide the ranking in the boxes provided.

Note: Atomic number (Z) are provided for relevant atoms

a) Increasing (lowest to highest) order of *ionization energy*: (1= lowest; 4 = highest)

Mg

Mg: Z = 12

Rb

Rb: Z = 37

Cs

Cs: Z = 55

K

K : Z = 19

b) Increasing (lowest to highest) order of *bond angle*: (1= lowest; 4 = highest)

H₂OCCl₄XeF₄

HCN

c) Increasing (smallest to largest) order of *ionic size* (1= smallest; 4 = largest)

S²⁻

S: Z = 16

O²⁻

O: Z = 8

Be²⁺

Be: Z = 4

Na⁺

Na: Z = 11

d) Increasing (lowest to highest) order of *bond energy* (1= lowest; 4 = highest)

P-F

P-Cl

P-I

P-Br

**Question 5** **12 points**

a. 9 points For the molecules HClO_4 and TeF_6 , draw the most stable *Lewis structure* showing appropriate shape and *dashed/wedged bonds*, based on VSEPR. Determine the **molecular geometry** around the central atom. For HClO_4 , label the *most polar bond*.



b. 3 points (Note: Lone pairs are not shown in structures below)

(i) For the three *resonance* structures (1, 2 and 3) shown below, answer the following question:



(I) Circle the *most contributing* resonance structure.

Structure 1

Structure 2

Structure 3

(ii) For the three structures given (4, 5, and 6) below, answer the following questions:



(I) Circle all the structures that represent *resonance structures* of each other:

Structure 4

Structure 5

Structure 6

(II) Among the resonance structures, circle the *most contributing* resonance structure.

Structure 4

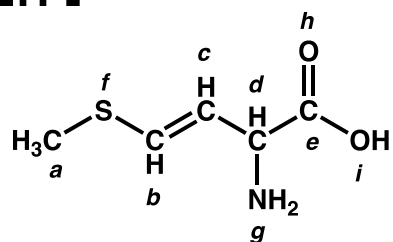
Structure 5

Structure 6

**Question 6** **6 points**

Using the data given below, calculate the **lattice energy** for SrO(s).

	ΔH° (kJ/mol)
$\text{Sr}(s) \longrightarrow \text{Sr}(g)$	+180
$\text{Sr}(g) \longrightarrow \text{Sr}^{2+}(g) + 2e^-$	+1468
$\text{O}_2(g) \longrightarrow 2\text{O}(g)$	+500
$\text{O}(g) + 2e^- \longrightarrow \text{O}^{2-}(g)$	+600
$\text{Sr}(s) + \frac{1}{2}\text{O}_2(g) \longrightarrow \text{SrO}(s)$	-550

**Question 7 10 points**

Consider the molecule given on the left, and answer questions (i) to (vi) using chemical bonding theories that we have learnt in class (valence bond theory, hybridization).

Note that the lone pairs are not shown in the structure:

i. Consider the carbon atoms marked in as ***a***, ***b***, ***c***, ***d***, and ***e*** in the molecule, and determine **hybridization** at each of the carbon centers. (2.5 points)

Atom	Hybridization
Carbon (<i>a</i>)	
Carbon (<i>b</i>)	
Carbon (<i>c</i>)	
Carbon (<i>d</i>)	
Carbon (<i>e</i>)	

ii. Now, consider the sulfur atom (S) marked as ***f***, nitrogen (N) atom (***g***), and oxygen(O) atoms (***h*** and ***i***), and determine their **hybridization**. (2 points)

Atom	Hybridization
Sulfur (<i>f</i>)	
Nitrogen (<i>g</i>)	
Oxygen (<i>h</i>)	
Oxygen (<i>i</i>)	

iii. **Describe** all the bonds formed by the carbon atoms ***a*** and ***b***, using sigma and pi designation as appropriate. (2 points)

Atom	Bonds formed (use sigma and pi as appropriate)
Carbon (<i>a</i>)	
Carbon (<i>b</i>)	

iv. How many total sp^3 -s sigma bonds are in the molecule? (1 point)

v. Determine **electronic** and **molecular geometry** at carbon (C) labeled ***e*** and sulfur (S) labeled ***f*** centres. (2 points)

Atom	Electronic Geometry	Molecular Geometry
Carbon (<i>e</i>)		
Sulfur (<i>f</i>)		

vi. Circle the value from the choices given below, that describes the **C-S-C bond angle at sulfur** (S) marked ***f***: (0.5 points)

>90° to <109.5° 109.5° >109.5 to <120° 120° >120° to <180°

**Question 8 12 points**

Using the **valence shell orbitals only** draw a **complete** molecular orbital (MO) diagram (with no s-p mixing) for the molecule **PS**, in the space provided below. **(6 points)**

Assume that 3s and 3p atomic orbitals will form molecular orbitals like those formed by the combination of 2s and 2p atomic orbitals that we learnt in class, and there is no s-p mixing.

Show **energy arrow**, **atomic orbitals** involved, **atom labels**, the **molecular orbitals** that are formed, **filled electrons** in the MO diagram etc. Make sure to label all the atomic and molecular orbitals.

Using the MO diagram above, answer the following questions:

- (i) Which one of the following will have the highest bond order? Circle the correct choice
A) PS^{1-} B) PS C) PS^{1+}
- (ii) Which one of the following will have the longest bond length? Circle the correct choice
A) PS^{1-} B) PS C) PS^{1+}
- (iii) Which of the following will be diamagnetic? Circle the correct choice.
A) PS^{1-} B) PS C) PS^{1+}
- (iv) Draw Lewis dot structure of the molecule with the shortest bond length (based on the MO diagram) from among the three (PS^{1-} , PS, PS^{1+}), and indicate the bond order calculated from the Lewis dot structure. **(3 points)**

Lewis Dot Structure

Bond Order



03F84471-BA3E-48D2-8D37-A468FCF0DED8

midterm-exam-93388

#264 Page 10 of 10

SCRAP PAGE

**CHEMISTRY 110 –Midterm Exam****Thursday November 7, 2024, 6:30 – 8:30 pm**

Examiners: Pallavi Sirjoosingh and Ashok Kakkar

First name (please write as legibly as possible within the boxes)

--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--

Last name

--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--

Student ID number

--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--

This booklet has 8 questions on 10 pages (The last page can be used as a scrap page).

The staple must NOT be removed from the question booklet. Your intact question booklet must be returned at the end of the exam.

A separate booklet contains a periodic table, and datasheet.

Write your answers directly under the questions in the space provided. Do not write on the QR codes or near the top edges (as it may be cut off).

Please write **legibly and dark enough** for the scan to be readable. For students who use English as a second language, dictionaries are permitted. Calculators are allowed.

DO NOT ASK QUESTIONS OF THE INVIGILATORS about interpretations of the exam questions. If you have a concern about a question, please summarize it at the top of this page. All concerns will be evaluated after the exam. GOOD LUCK!

**Question 1 8 points** (*Assume all transitions are possible*)

a. 6 points Calculate the **energy (in kJ/mol)** required to ionize (completely remove) the electron from the **n=3** level of **He⁺ ion** (He: Helium). Show your work including equations. (3 sig. figs.)

b. 2 points

(I) Select the largest (longest) wavelength from the three choices below:

- i. Wavelength of photon to *completely remove* an electron from *1s orbital of Hydrogen atom*
- ii. Wavelength of photon to *completely remove* an electron from *1s orbital of Boron atom (B)*
- iii. Wavelength of photon to *completely remove* an electron from *1s orbital of Boron cation (B⁺)*

i or ii or iii

(II) Select the smallest (shortest) wavelength from the three choices below (*for H atom*):

- i. Wavelength of photon *emitted* for an electron transitioning from *2p to 1s orbital*
- ii. Wavelength of photon *absorbed* for an electron transitioning from *2s to 4p orbital*
- iii. Wavelength of photon *absorbed* for an electron transitioning from *1s to 3p orbital*

i or ii or iii



Question 2 8 points



A Chem110 student plots the graph shown on the left, from data obtained for the kinetic energy (eV) of the ejected electron versus frequency (s^{-1}) of the incident light, for an unknown metal. ($1\text{eV} = 1.6022 \times 10^{-19} \text{ J}$)

Based on the plot, answer the questions **a to c** below:

a. 1 point On the plot above, label the **threshold frequency**.

b. 2 points What is the value of the work potential (in J) of the unknown metal?

c. 3 points What is the *longest* wavelength (in nm) required to eject an electron from this metal's surface? Show your work including equation(s) used. (3 sig figs)

d. 2 points In the space provided below, plot the following

(Assume frequency is *above* threshold frequency):

- i. *Kinetic energy* of an ejected electron versus the *intensity* of the incident light (left)
- ii. *Number of electrons* ejected versus the *frequency* of the incident light (right)



**Question 3** **6 points****a. 2 points**

In the ground state of Ge (Germanium; $Z = 32$) atom, **name** the following orbitals:

i. Orbital containing the electron that requires the maximum energy to be removed.

ii. Orbital containing the electron that experiences the minimum effective nuclear charge.

b. 2 points

An unknown anion (A^-) has the following *excited* electronic configuration: $[Ar] 4s^2 3d^3 4p^1$

(Ar: Argon; $Z = 18$)

Give the **condensed ground state electronic configuration** of the atom, **A** and the cation, **A^{2+}** .
Show your work. (Assume all electronic transitions are possible)

Electronic Configuration of A:

Electronic Configuration of A^{2+} :

c. 2 points

In a single atom, determine how many electrons can have the following set of quantum numbers?

(i) $n = 5; l = 3$

In a single atom, determine how many electrons can have the following set of quantum numbers?

(i) $n = 3; l = 2; m_s = +1/2$

**Question 4** **8 points**

Rank the following from 1 to 4 in **increasing** order. Provide the ranking in the boxes provided.

Note: Atomic number (Z) are provided for relevant atoms

a) Increasing (lowest to highest) order of *ionization energy*: (1= lowest; 4 = highest)

Mg

Mg: Z = 12

Rb

Rb: Z = 37

Cs

Cs: Z = 55

K

K : Z = 19

b) Increasing (lowest to highest) order of *bond angle*: (1= lowest; 4 = highest)

H₂OCCl₄XeF₄

HCN

c) Increasing (smallest to largest) order of *ionic size* (1= smallest; 4 = largest)

S²⁻

S: Z = 16

O²⁻

O: Z = 8

Be²⁺

Be: Z = 4

Na⁺

Na: Z = 11

d) Increasing (lowest to highest) order of *bond energy* (1= lowest; 4 = highest)

P-F

P-Cl

P-I

P-Br

**Question 5** **12 points**

a. 9 points For the molecules HClO_4 and TeF_6 , draw the most stable *Lewis structure* showing appropriate shape and *dashed/wedged bonds*, based on VSEPR. Determine the **molecular geometry** around the central atom. For HClO_4 , label the *most polar bond*.

HClO_4 (there are no O-O bonds)

Label the most polar bond

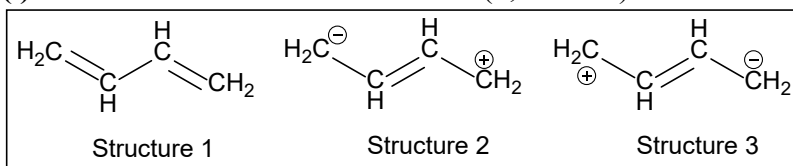
Molecular Geometry around Cl

TeF_6

Molecular Geometry around Te

b. 3 points (Note: Lone pairs are not shown in structures below)

(i) For the three *resonance* structures (1, 2 and 3) shown below, answer the following question:



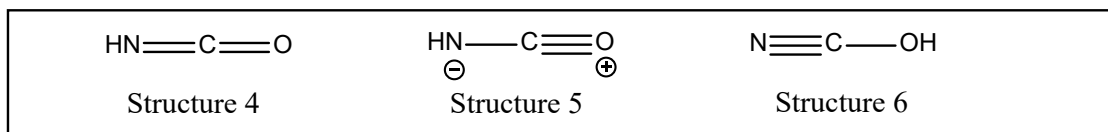
(I) Circle the *most contributing* resonance structure.

Structure 1

Structure 2

Structure 3

(ii) For the three structures given (4, 5, and 6) below, answer the following questions:



(I) Circle all the structures that represent *resonance structures* of each other:

Structure 4

Structure 5

Structure 6

(II) Among the resonance structures, circle the *most contributing* resonance structure.

Structure 4

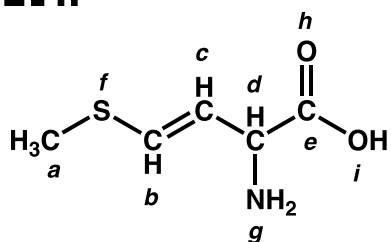
Structure 5

Structure 6

**Question 6** **6 points**

Using the data given below, calculate the **lattice energy** for SrO(s).

	ΔH° (kJ/mol)
$\text{Sr}(s) \longrightarrow \text{Sr}(g)$	+180
$\text{Sr}(g) \longrightarrow \text{Sr}^{2+}(g) + 2e^-$	+1468
$\text{O}_2(g) \longrightarrow 2\text{O}(g)$	+500
$\text{O}(g) + 2e^- \longrightarrow \text{O}^{2-}(g)$	+600
$\text{Sr}(s) + \frac{1}{2}\text{O}_2(g) \longrightarrow \text{SrO}(s)$	-550

**Question 7 10 points**

Consider the molecule given on the left, and answer questions (i) to (vi) using chemical bonding theories that we have learnt in class (valence bond theory, hybridization).

Note that the lone pairs are not shown in the structure:

i. Consider the carbon atoms marked in as ***a***, ***b***, ***c***, ***d***, and ***e*** in the molecule, and determine **hybridization** at each of the carbon centers. (2.5 points)

Atom	Hybridization
Carbon (<i>a</i>)	
Carbon (<i>b</i>)	
Carbon (<i>c</i>)	
Carbon (<i>d</i>)	
Carbon (<i>e</i>)	

ii. Now, consider the sulfur atom (S) marked as ***f***, nitrogen (N) atom (***g***), and oxygen(O) atoms (***h*** and ***i***), and determine their **hybridization**. (2 points)

Atom	Hybridization
Sulfur (<i>f</i>)	
Nitrogen (<i>g</i>)	
Oxygen (<i>h</i>)	
Oxygen (<i>i</i>)	

iii. **Describe** all the bonds formed by the carbon atoms ***a*** and ***b***, using sigma and pi designation as appropriate. (2 points)

Atom	Bonds formed (use sigma and pi as appropriate)
Carbon (<i>a</i>)	
Carbon (<i>b</i>)	

iv. How many total sp^3 -s sigma bonds are in the molecule? (1 point)

v. Determine **electronic** and **molecular geometry** at carbon (C) labeled ***e*** and sulfur (S) labeled ***f*** centres. (2 points)

Atom	Electronic Geometry	Molecular Geometry
Carbon (<i>e</i>)		
Sulfur (<i>f</i>)		

vi. Circle the value from the choices given below, that describes the **C-S-C bond angle at sulfur** (S) marked ***f***: (0.5 points)

>90° to <109.5° 109.5° >109.5 to <120° 120° >120° to <180°


Question 8 12 points

Using the **valence shell orbitals only** draw a **complete** molecular orbital (MO) diagram (with no s-p mixing) for the molecule **PS**, in the space provided below. **(6 points)**

Assume that 3s and 3p atomic orbitals will form molecular orbitals like those formed by the combination of 2s and 2p atomic orbitals that we learnt in class, and there is no s-p mixing.

Show **energy arrow**, **atomic orbitals** involved, **atom labels**, the **molecular orbitals** that are formed, **filled electrons** in the MO diagram etc. Make sure to label all the atomic and molecular orbitals.

Using the MO diagram above, answer the following questions:

- (i) Which one of the following will have the highest bond order? Circle the correct choice
 A) PS^{1-} B) PS C) PS^{1+}
- (ii) Which one of the following will have the longest bond length? Circle the correct choice
 A) PS^{1-} B) PS C) PS^{1+}
- (iii) Which of the following will be diamagnetic? Circle the correct choice.
 A) PS^{1-} B) PS C) PS^{1+}
- (iv) Draw Lewis dot structure of the molecule with the shortest bond length (based on the MO diagram) from among the three (PS^{1-} , PS, PS^{1+}), and indicate the bond order calculated from the Lewis dot structure. **(3 points)**

Lewis Dot Structure

Bond Order



0B43BA75-291B-4431-B891-5DE50AFC2D8D

midterm-exam-93388

#265 Page 10 of 10

SCRAP PAGE

**CHEMISTRY 110 –Midterm Exam****Thursday November 7, 2024, 6:30 – 8:30 pm**

Examiners: Pallavi Sirjoosingh and Ashok Kakkar

First name (please write as legibly as possible within the boxes)

--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--

Last name

--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--

Student ID number

--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--

This booklet has 8 questions on 10 pages (The last page can be used as a scrap page).

The staple must NOT be removed from the question booklet. Your intact question booklet must be returned at the end of the exam.

A separate booklet contains a periodic table, and datasheet.

Write your answers directly under the questions in the space provided. Do not write on the QR codes or near the top edges (as it may be cut off).

Please write **legibly and dark enough** for the scan to be readable. For students who use English as a second language, dictionaries are permitted. Calculators are allowed.

DO NOT ASK QUESTIONS OF THE INVIGILATORS about interpretations of the exam questions. If you have a concern about a question, please summarize it at the top of this page. All concerns will be evaluated after the exam. GOOD LUCK!

**Question 1 8 points** (*Assume all transitions are possible*)

a. 6 points Calculate the **energy (in kJ/mol)** required to ionize (completely remove) the electron from the **$n=3$** level of **He^+ ion** (He: Helium). Show your work including equations. (3 sig. figs.)

b. 2 points

(I) Select the largest (longest) wavelength from the three choices below:

- i. Wavelength of photon to *completely remove* an electron from *1s orbital of Hydrogen atom*
- ii. Wavelength of photon to *completely remove* an electron from *1s orbital of Boron atom (B)*
- iii. Wavelength of photon to *completely remove* an electron from *1s orbital of Boron cation (B^+)*

i or ii or iii

(II) Select the smallest (shortest) wavelength from the three choices below (*for H atom*):

- i. Wavelength of photon *emitted* for an electron transitioning from *2p to 1s orbital*
- ii. Wavelength of photon *absorbed* for an electron transitioning from *2s to 4p orbital*
- iii. Wavelength of photon *absorbed* for an electron transitioning from *1s to 3p orbital*

i or ii or iii



Question 2 8 points



A Chem110 student plots the graph shown on the left, from data obtained for the kinetic energy (eV) of the ejected electron versus frequency (s^{-1}) of the incident light, for an unknown metal. ($1\text{eV} = 1.6022 \times 10^{-19} \text{ J}$)

Based on the plot, answer the questions **a to c** below:

a. 1 point On the plot above, label the **threshold frequency**.

b. 2 points What is the value of the work potential (in J) of the unknown metal?

c. 3 points What is the *longest* wavelength (in nm) required to eject an electron from this metal's surface? Show your work including equation(s) used. (3 sig figs)

d. 2 points In the space provided below, plot the following

(Assume frequency is *above* threshold frequency):

- i. *Kinetic energy* of an ejected electron versus the *intensity* of the incident light (left)
- ii. *Number of electrons* ejected versus the *frequency* of the incident light (right)



**Question 3** **6 points****a. 2 points**

In the ground state of Ge (Germanium; $Z = 32$) atom, **name** the following orbitals:

i. Orbital containing the electron that requires the maximum energy to be removed.

ii. Orbital containing the electron that experiences the minimum effective nuclear charge.

b. 2 points

An unknown anion (A^-) has the following *excited* electronic configuration: $[Ar] 4s^2 3d^3 4p^1$

(Ar: Argon; $Z = 18$)

Give the **condensed ground state electronic configuration** of the atom, **A** and the cation, **A^{2+}** .
Show your work. (Assume all electronic transitions are possible)

Electronic Configuration of A:

Electronic Configuration of A^{2+} :

c. 2 points

In a single atom, determine how many electrons can have the following set of quantum numbers?

(i) $n = 5; l = 3$

In a single atom, determine how many electrons can have the following set of quantum numbers?

(i) $n = 3; l = 2; m_s = +1/2$

**Question 4** **8 points**

Rank the following from 1 to 4 in **increasing** order. Provide the ranking in the boxes provided.

Note: Atomic number (Z) are provided for relevant atoms

a) Increasing (lowest to highest) order of *ionization energy*: (1= lowest; 4 = highest)

Mg

Mg: Z = 12

Rb

Rb: Z = 37

Cs

Cs: Z = 55

K

K : Z = 19

b) Increasing (lowest to highest) order of *bond angle*: (1= lowest; 4 = highest)

H₂OCCl₄XeF₄

HCN

c) Increasing (smallest to largest) order of *ionic size* (1= smallest; 4 = largest)

S²⁻

S: Z = 16

O²⁻

O: Z = 8

Be²⁺

Be: Z = 4

Na⁺

Na: Z = 11

d) Increasing (lowest to highest) order of *bond energy* (1= lowest; 4 = highest)

P-F

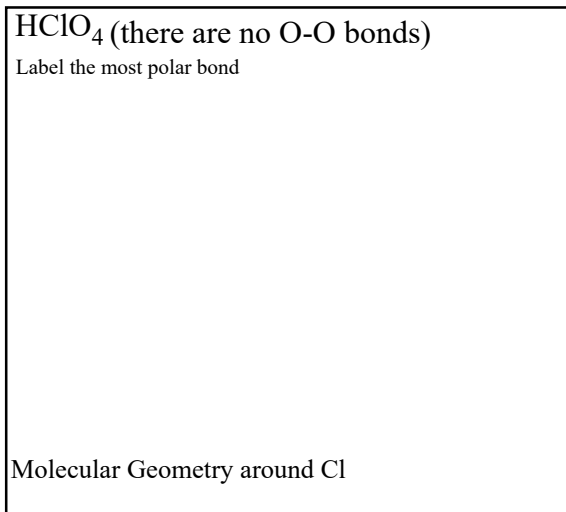
P-Cl

P-I

P-Br

**Question 5** **12 points**

a. 9 points For the molecules HClO_4 and TeF_6 , draw the most stable *Lewis structure* showing appropriate shape and *dashed/wedged bonds*, based on VSEPR. Determine the **molecular geometry** around the central atom. For HClO_4 , label the *most polar bond*.



b. 3 points (Note: Lone pairs are not shown in structures below)

(i) For the three *resonance* structures (1, 2 and 3) shown below, answer the following question:



(I) Circle the *most contributing* resonance structure.

Structure 1

Structure 2

Structure 3

(ii) For the three structures given (4, 5, and 6) below, answer the following questions:



(I) Circle all the structures that represent *resonance structures* of each other:

Structure 4

Structure 5

Structure 6

(II) Among the resonance structures, circle the *most contributing* resonance structure.

Structure 4

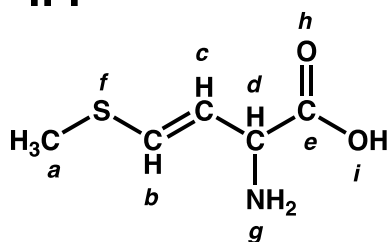
Structure 5

Structure 6

**Question 6** **6 points**

Using the data given below, calculate the **lattice energy** for SrO(s).

	ΔH° (kJ/mol)
$\text{Sr}(s) \longrightarrow \text{Sr}(g)$	+180
$\text{Sr}(g) \longrightarrow \text{Sr}^{2+}(g) + 2e^-$	+1468
$\text{O}_2(g) \longrightarrow 2\text{O}(g)$	+500
$\text{O}(g) + 2e^- \longrightarrow \text{O}^{2-}(g)$	+600
$\text{Sr}(s) + \frac{1}{2}\text{O}_2(g) \longrightarrow \text{SrO}(s)$	-550

**Question 7 10 points**

Consider the molecule given on the left, and answer questions (i) to (vi) using chemical bonding theories that we have learnt in class (valence bond theory, hybridization).

Note that the lone pairs are not shown in the structure:

i. Consider the carbon atoms marked in as ***a***, ***b***, ***c***, ***d***, and ***e*** in the molecule, and determine **hybridization** at each of the carbon centers. (2.5 points)

Atom	Hybridization
Carbon (<i>a</i>)	
Carbon (<i>b</i>)	
Carbon (<i>c</i>)	
Carbon (<i>d</i>)	
Carbon (<i>e</i>)	

ii. Now, consider the sulfur atom (S) marked as ***f***, nitrogen (N) atom (***g***), and oxygen(O) atoms (***h*** and ***i***), and determine their **hybridization**. (2 points)

Atom	Hybridization
Sulfur (<i>f</i>)	
Nitrogen (<i>g</i>)	
Oxygen (<i>h</i>)	
Oxygen (<i>i</i>)	

iii. **Describe** all the bonds formed by the carbon atoms ***a*** and ***b***, using sigma and pi designation as appropriate. (2 points)

Atom	Bonds formed (use sigma and pi as appropriate)
Carbon (<i>a</i>)	
Carbon (<i>b</i>)	

iv. How many total sp^3 -s sigma bonds are in the molecule? (1 point)

v. Determine **electronic** and **molecular geometry** at carbon (C) labeled ***e*** and sulfur (S) labeled ***f*** centres. (2 points)

Atom	Electronic Geometry	Molecular Geometry
Carbon (<i>e</i>)		
Sulfur (<i>f</i>)		

vi. Circle the value from the choices given below, that describes the **C-S-C bond angle at sulfur** (S) marked ***f***: (0.5 points)

>90° to <109.5° 109.5° >109.5 to <120° 120° >120° to <180°

**Question 8 12 points**

Using the **valence shell orbitals only** draw a **complete** molecular orbital (MO) diagram (with no s-p mixing) for the molecule **PS**, in the space provided below. **(6 points)**

Assume that 3s and 3p atomic orbitals will form molecular orbitals like those formed by the combination of 2s and 2p atomic orbitals that we learnt in class, and there is no s-p mixing.

Show **energy arrow**, **atomic orbitals** involved, **atom labels**, the **molecular orbitals** that are formed, **filled electrons** in the MO diagram etc. Make sure to label all the atomic and molecular orbitals.

Using the MO diagram above, answer the following questions:

- (i) Which one of the following will have the highest bond order? Circle the correct choice
A) PS^{1-} B) PS C) PS^{1+}
- (ii) Which one of the following will have the longest bond length? Circle the correct choice
A) PS^{1-} B) PS C) PS^{1+}
- (iii) Which of the following will be diamagnetic? Circle the correct choice.
A) PS^{1-} B) PS C) PS^{1+}
- (iv) Draw Lewis dot structure of the molecule with the shortest bond length (based on the MO diagram) from among the three (PS^{1-} , PS, PS^{1+}), and indicate the bond order calculated from the Lewis dot structure. **(3 points)**

Lewis Dot Structure

Bond Order



20DE722B-32E1-4C2A-8435-5E4D8DA47564

midterm-exam-93388

#266 Page 10 of 10

SCRAP PAGE

**CHEMISTRY 110 –Midterm Exam****Thursday November 7, 2024, 6:30 – 8:30 pm****Examiners: Pallavi Sirjoosingh and Ashok Kakkar**

First name (please write as legibly as possible within the boxes)

--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--

Last name

--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--

Student ID number

--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--

This booklet has 8 questions on 10 pages (The last page can be used as a scrap page).

The staple must NOT be removed from the question booklet. Your intact question booklet must be returned at the end of the exam.

A separate booklet contains a periodic table, and datasheet.

Write your answers directly under the questions in the space provided. Do not write on the QR codes or near the top edges (as it may be cut off).

Please write **legibly and dark enough** for the scan to be readable. For students who use English as a second language, dictionaries are permitted. Calculators are allowed.

DO NOT ASK QUESTIONS OF THE INVIGILATORS about interpretations of the exam questions. If you have a concern about a question, please summarize it at the top of this page. All concerns will be evaluated after the exam. GOOD LUCK!

**Question 1 8 points** (*Assume all transitions are possible*)

a. 6 points Calculate the **energy (in kJ/mol)** required to ionize (completely remove) the electron from the **n=3** level of **He⁺ ion** (He: Helium). Show your work including equations. (3 sig. figs.)

b. 2 points

(I) Select the largest (longest) wavelength from the three choices below:

- i. Wavelength of photon to *completely remove* an electron from *1s orbital of Hydrogen atom*
- ii. Wavelength of photon to *completely remove* an electron from *1s orbital of Boron atom (B)*
- iii. Wavelength of photon to *completely remove* an electron from *1s orbital of Boron cation (B⁺)*

i or ii or iii

(II) Select the smallest (shortest) wavelength from the three choices below (*for H atom*):

- i. Wavelength of photon *emitted* for an electron transitioning from *2p to 1s orbital*
- ii. Wavelength of photon *absorbed* for an electron transitioning from *2s to 4p orbital*
- iii. Wavelength of photon *absorbed* for an electron transitioning from *1s to 3p orbital*

i or ii or iii



Question 2 8 points



A Chem110 student plots the graph shown on the left, from data obtained for the kinetic energy (eV) of the ejected electron versus frequency (s^{-1}) of the incident light, for an unknown metal. ($1\text{eV} = 1.6022 \times 10^{-19} \text{ J}$)

Based on the plot, answer the questions **a to c** below:

a. 1 point On the plot above, label the **threshold frequency**.

b. 2 points What is the value of the work potential (in J) of the unknown metal?

c. 3 points What is the *longest* wavelength (in nm) required to eject an electron from this metal's surface? Show your work including equation(s) used. (3 sig figs)

d. 2 points In the space provided below, plot the following

(Assume frequency is *above* threshold frequency):

- Kinetic energy* of an ejected electron versus the *intensity* of the incident light (left)
- Number of electrons* ejected versus the *frequency* of the incident light (right)



**Question 3** **6 points****a. 2 points**

In the ground state of Ge (Germanium; $Z = 32$) atom, **name** the following orbitals:

i. Orbital containing the electron that requires the maximum energy to be removed.

ii. Orbital containing the electron that experiences the minimum effective nuclear charge.

b. 2 points

An unknown anion (A^-) has the following *excited* electronic configuration: $[Ar] 4s^2 3d^3 4p^1$

(Ar: Argon; $Z = 18$)

Give the **condensed ground state electronic configuration** of the atom, **A** and the cation, **A^{2+}** .
Show your work. (Assume all electronic transitions are possible)

Electronic Configuration of A:

Electronic Configuration of A^{2+} :

c. 2 points

In a single atom, determine how many electrons can have the following set of quantum numbers?

(i) $n = 5; l = 3$

In a single atom, determine how many electrons can have the following set of quantum numbers?

(i) $n = 3; l = 2; m_s = +1/2$

**Question 4** **8 points**

Rank the following from 1 to 4 in **increasing** order. Provide the ranking in the boxes provided.

Note: Atomic number (Z) are provided for relevant atoms

a) Increasing (lowest to highest) order of *ionization energy*: (1= lowest; 4 = highest)

Mg

Mg: Z = 12

Rb

Rb: Z = 37

Cs

Cs: Z = 55

K

K : Z = 19

b) Increasing (lowest to highest) order of *bond angle*: (1= lowest; 4 = highest)

H₂OCCl₄XeF₄

HCN

c) Increasing (smallest to largest) order of *ionic size* (1= smallest; 4 = largest)

S²⁻

S: Z = 16

O²⁻

O: Z = 8

Be²⁺

Be: Z = 4

Na⁺

Na: Z = 11

d) Increasing (lowest to highest) order of *bond energy* (1= lowest; 4 = highest)

P-F

P-Cl

P-I

P-Br

**Question 5** **12 points**

a. 9 points For the molecules HClO_4 and TeF_6 , draw the most stable *Lewis structure* showing appropriate shape and *dashed/wedged bonds*, based on VSEPR. Determine the **molecular geometry** around the central atom. For HClO_4 , label the *most polar bond*.

HClO_4 (there are no O-O bonds)

Label the most polar bond

Molecular Geometry around Cl

TeF_6

Molecular Geometry around Te

b. 3 points (Note: Lone pairs are not shown in structures below)

(i) For the three *resonance* structures (1, 2 and 3) shown below, answer the following question:



(I) Circle the *most contributing* resonance structure.

Structure 1

Structure 2

Structure 3

(ii) For the three structures given (4, 5, and 6) below, answer the following questions:



(I) Circle all the structures that represent *resonance structures* of each other:

Structure 4

Structure 5

Structure 6

(II) Among the resonance structures, circle the *most contributing* resonance structure.

Structure 4

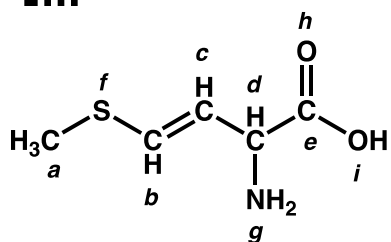
Structure 5

Structure 6

**Question 6** **6 points**

Using the data given below, calculate the **lattice energy** for SrO(s).

	ΔH° (kJ/mol)
$\text{Sr}(s) \longrightarrow \text{Sr}(g)$	+180
$\text{Sr}(g) \longrightarrow \text{Sr}^{2+}(g) + 2e^-$	+1468
$\text{O}_2(g) \longrightarrow 2\text{O}(g)$	+500
$\text{O}(g) + 2e^- \longrightarrow \text{O}^{2-}(g)$	+600
$\text{Sr}(s) + \frac{1}{2}\text{O}_2(g) \longrightarrow \text{SrO}(s)$	-550

**Question 7 10 points**

Consider the molecule given on the left, and answer questions (i) to (vi) using chemical bonding theories that we have learnt in class (valence bond theory, hybridization).

Note that the lone pairs are not shown in the structure:

i. Consider the carbon atoms marked in as ***a***, ***b***, ***c***, ***d***, and ***e*** in the molecule, and determine **hybridization** at each of the carbon centers. (2.5 points)

Atom	Hybridization
Carbon (<i>a</i>)	
Carbon (<i>b</i>)	
Carbon (<i>c</i>)	
Carbon (<i>d</i>)	
Carbon (<i>e</i>)	

ii. Now, consider the sulfur atom (S) marked as ***f***, nitrogen (N) atom (***g***), and oxygen(O) atoms (***h*** and ***i***), and determine their **hybridization**. (2 points)

Atom	Hybridization
Sulfur (<i>f</i>)	
Nitrogen (<i>g</i>)	
Oxygen (<i>h</i>)	
Oxygen (<i>i</i>)	

iii. **Describe** all the bonds formed by the carbon atoms ***a*** and ***b***, using sigma and pi designation as appropriate. (2 points)

Atom	Bonds formed (use sigma and pi as appropriate)
Carbon (<i>a</i>)	
Carbon (<i>b</i>)	

iv. How many total sp^3 -s sigma bonds are in the molecule? (1 point)

v. Determine **electronic** and **molecular geometry** at carbon (C) labeled ***e*** and sulfur (S) labeled ***f*** centres. (2 points)

Atom	Electronic Geometry	Molecular Geometry
Carbon (<i>e</i>)		
Sulfur (<i>f</i>)		

vi. Circle the value from the choices given below, that describes the **C-S-C bond angle at sulfur** (S) marked ***f***: (0.5 points)

>90° to <109.5°

109.5°

>109.5 to <120°

120°

>120° to <180°


Question 8 12 points

Using the **valence shell orbitals only** draw a **complete** molecular orbital (MO) diagram (with no s-p mixing) for the molecule **PS**, in the space provided below. **(6 points)**

Assume that 3s and 3p atomic orbitals will form molecular orbitals like those formed by the combination of 2s and 2p atomic orbitals that we learnt in class, and there is no s-p mixing.

Show **energy arrow**, **atomic orbitals** involved, **atom labels**, the **molecular orbitals** that are formed, **filled electrons** in the MO diagram etc. Make sure to **label all the atomic and molecular orbitals**.

Using the MO diagram above, answer the following questions:

- (i) Which one of the following will have the highest bond order? Circle the correct choice
 A) PS^{1-} B) PS C) PS^{1+}
- (ii) Which one of the following will have the longest bond length? Circle the correct choice
 A) PS^{1-} B) PS C) PS^{1+}
- (iii) Which of the following will be diamagnetic? Circle the correct choice.
 A) PS^{1-} B) PS C) PS^{1+}
- (iv) Draw Lewis dot structure of the molecule with the shortest bond length (based on the MO diagram) from among the three (PS^{1-} , PS, PS^{1+}), and indicate the bond order calculated from the Lewis dot structure. **(3 points)**

Lewis Dot Structure

Bond Order



387A723F-42FE-4C20-A6E6-8BAE496B8FC1

midterm-exam-93388

#267 Page 10 of 10

SCRAP PAGE

**CHEMISTRY 110 –Midterm Exam****Thursday November 7, 2024, 6:30 – 8:30 pm****Examiners: Pallavi Sirjoosingh and Ashok Kakkar**

First name (please write as legibly as possible within the boxes)

--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--

Last name

--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--

Student ID number

--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--

This booklet has 8 questions on 10 pages (The last page can be used as a scrap page).

The staple must NOT be removed from the question booklet. Your intact question booklet must be returned at the end of the exam.

A separate booklet contains a periodic table, and datasheet.

Write your answers directly under the questions in the space provided. Do not write on the QR codes or near the top edges (as it may be cut off).

Please write **legibly and dark enough** for the scan to be readable. For students who use English as a second language, dictionaries are permitted. Calculators are allowed.

DO NOT ASK QUESTIONS OF THE INVIGILATORS about interpretations of the exam questions. If you have a concern about a question, please summarize it at the top of this page. All concerns will be evaluated after the exam. GOOD LUCK!

**Question 1 8 points** (*Assume all transitions are possible*)

a. 6 points Calculate the **energy (in kJ/mol)** required to ionize (completely remove) the electron from the **n=3** level of **He⁺ ion** (He: Helium). Show your work including equations. (3 sig. figs.)

b. 2 points

(I) Select the largest (longest) wavelength from the three choices below:

- i. Wavelength of photon to *completely remove* an electron from *1s orbital of Hydrogen atom*
- ii. Wavelength of photon to *completely remove* an electron from *1s orbital of Boron atom (B)*
- iii. Wavelength of photon to *completely remove* an electron from *1s orbital of Boron cation (B⁺)*

i or ii or iii

(II) Select the smallest (shortest) wavelength from the three choices below (*for H atom*):

- i. Wavelength of photon *emitted* for an electron transitioning from *2p to 1s orbital*
- ii. Wavelength of photon *absorbed* for an electron transitioning from *2s to 4p orbital*
- iii. Wavelength of photon *absorbed* for an electron transitioning from *1s to 3p orbital*

i or ii or iii



Question 2 8 points



A Chem110 student plots the graph shown on the left, from data obtained for the kinetic energy (eV) of the ejected electron versus frequency (s^{-1}) of the incident light, for an unknown metal. ($1\text{eV} = 1.6022 \times 10^{-19} \text{ J}$)

Based on the plot, answer the questions **a to c** below:

a. 1 point On the plot above, label the **threshold frequency**.

b. 2 points What is the value of the work potential (in J) of the unknown metal?

c. 3 points What is the *longest* wavelength (in nm) required to eject an electron from this metal's surface? Show your work including equation(s) used. (3 sig figs)

d. 2 points In the space provided below, plot the following

(Assume frequency is *above* threshold frequency):

- i. *Kinetic energy* of an ejected electron versus the *intensity* of the incident light (left)
- ii. *Number of electrons* ejected versus the *frequency* of the incident light (right)



**Question 3** **6 points****a. 2 points**

In the ground state of Ge (Germanium; $Z = 32$) atom, **name** the following orbitals:

i. Orbital containing the electron that requires the maximum energy to be removed.

ii. Orbital containing the electron that experiences the minimum effective nuclear charge.

b. 2 points

An unknown anion (A^-) has the following *excited* electronic configuration: $[Ar] 4s^2 3d^3 4p^1$

(Ar: Argon; $Z = 18$)

Give the **condensed ground state electronic configuration** of the atom, **A** and the cation, **A^{2+}** .
Show your work. (Assume all electronic transitions are possible)

Electronic Configuration of A:

Electronic Configuration of A^{2+} :

c. 2 points

In a single atom, determine how many electrons can have the following set of quantum numbers?

(i) $n = 5; l = 3$

In a single atom, determine how many electrons can have the following set of quantum numbers?

(i) $n = 3; l = 2; m_s = +1/2$

**Question 4** **8 points**

Rank the following from 1 to 4 in **increasing** order. Provide the ranking in the boxes provided.

Note: Atomic number (Z) are provided for relevant atoms

a) Increasing (lowest to highest) order of *ionization energy*: (1= lowest; 4 = highest)

Mg

Mg: Z = 12

Rb

Rb: Z = 37

Cs

Cs: Z = 55

K

K : Z = 19

b) Increasing (lowest to highest) order of *bond angle*: (1= lowest; 4 = highest)

H₂OCCl₄XeF₄

HCN

c) Increasing (smallest to largest) order of *ionic size* (1= smallest; 4 = largest)

S²⁻

S: Z = 16

O²⁻

O: Z = 8

Be²⁺

Be: Z = 4

Na⁺

Na: Z = 11

d) Increasing (lowest to highest) order of *bond energy* (1= lowest; 4 = highest)

P-F

P-Cl

P-I

P-Br

**Question 5** **12 points**

a. 9 points For the molecules HClO_4 and TeF_6 , draw the most stable *Lewis structure* showing appropriate shape and *dashed/wedged bonds*, based on VSEPR. Determine the **molecular geometry** around the central atom. For HClO_4 , label the *most polar bond*.



b. 3 points (Note: Lone pairs are not shown in structures below)

(i) For the three *resonance* structures (1, 2 and 3) shown below, answer the following question:



(I) Circle the *most contributing* resonance structure.

Structure 1

Structure 2

Structure 3

(ii) For the three structures given (4, 5, and 6) below, answer the following questions:



(I) Circle all the structures that represent *resonance structures* of each other:

Structure 4

Structure 5

Structure 6

(II) Among the resonance structures, circle the *most contributing* resonance structure.

Structure 4

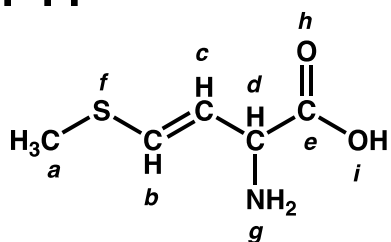
Structure 5

Structure 6

**Question 6** **6 points**

Using the data given below, calculate the **lattice energy** for SrO(s).

	ΔH° (kJ/mol)
$\text{Sr}(s) \longrightarrow \text{Sr}(g)$	+180
$\text{Sr}(g) \longrightarrow \text{Sr}^{2+}(g) + 2e^-$	+1468
$\text{O}_2(g) \longrightarrow 2\text{O}(g)$	+500
$\text{O}(g) + 2e^- \longrightarrow \text{O}^{2-}(g)$	+600
$\text{Sr}(s) + \frac{1}{2}\text{O}_2(g) \longrightarrow \text{SrO}(s)$	-550

**Question 7 10 points**

Consider the molecule given on the left, and answer questions (i) to (vi) using chemical bonding theories that we have learnt in class (valence bond theory, hybridization).

Note that the lone pairs are not shown in the structure:

i. Consider the carbon atoms marked in as ***a***, ***b***, ***c***, ***d***, and ***e*** in the molecule, and determine **hybridization** at each of the carbon centers. (2.5 points)

Atom	Hybridization
Carbon (<i>a</i>)	
Carbon (<i>b</i>)	
Carbon (<i>c</i>)	
Carbon (<i>d</i>)	
Carbon (<i>e</i>)	

ii. Now, consider the sulfur atom (S) marked as ***f***, nitrogen (N) atom (***g***), and oxygen(O) atoms (***h*** and ***i***), and determine their **hybridization**. (2 points)

Atom	Hybridization
Sulfur (<i>f</i>)	
Nitrogen (<i>g</i>)	
Oxygen (<i>h</i>)	
Oxygen (<i>i</i>)	

iii. **Describe** all the bonds formed by the carbon atoms ***a*** and ***b***, using sigma and pi designation as appropriate. (2 points)

Atom	Bonds formed (use sigma and pi as appropriate)
Carbon (<i>a</i>)	
Carbon (<i>b</i>)	

iv. How many total sp^3 -s sigma bonds are in the molecule? (1 point)

v. Determine **electronic** and **molecular geometry** at carbon (C) labeled ***e*** and sulfur (S) labeled ***f*** centres. (2 points)

Atom	Electronic Geometry	Molecular Geometry
Carbon (<i>e</i>)		
Sulfur (<i>f</i>)		

vi. Circle the value from the choices given below, that describes the **C-S-C bond angle at sulfur** (S) marked ***f***: (0.5 points)

>90° to <109.5° 109.5° >109.5 to <120° 120° >120° to <180°


Question 8 12 points

Using the **valence shell orbitals only** draw a **complete** molecular orbital (MO) diagram (with no s-p mixing) for the molecule **PS**, in the space provided below. **(6 points)**

Assume that 3s and 3p atomic orbitals will form molecular orbitals like those formed by the combination of 2s and 2p atomic orbitals that we learnt in class, and there is no s-p mixing.

Show **energy arrow**, **atomic orbitals** involved, **atom labels**, the **molecular orbitals** that are formed, **filled electrons** in the MO diagram etc. Make sure to label all the atomic and molecular orbitals.

Using the MO diagram above, answer the following questions:

- (i) Which one of the following will have the highest bond order? Circle the correct choice
 A) PS^{1-} B) PS C) PS^{1+}
- (ii) Which one of the following will have the longest bond length? Circle the correct choice
 A) PS^{1-} B) PS C) PS^{1+}
- (iii) Which of the following will be diamagnetic? Circle the correct choice.
 A) PS^{1-} B) PS C) PS^{1+}
- (iv) Draw Lewis dot structure of the molecule with the shortest bond length (based on the MO diagram) from among the three (PS^{1-} , PS, PS^{1+}), and indicate the bond order calculated from the Lewis dot structure. **(3 points)**

Lewis Dot Structure

Bond Order



0C399AE8-2C64-42A4-B6FF-A940D42BF205

midterm-exam-93388

#268 Page 10 of 10

SCRAP PAGE

**CHEMISTRY 110 –Midterm Exam****Thursday November 7, 2024, 6:30 – 8:30 pm**

Examiners: Pallavi Sirjoosingh and Ashok Kakkar

First name (please write as legibly as possible within the boxes)

--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--

Last name

--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--

Student ID number

--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--

This booklet has 8 questions on 10 pages (The last page can be used as a scrap page).

The staple must NOT be removed from the question booklet. Your intact question booklet must be returned at the end of the exam.

A separate booklet contains a periodic table, and datasheet.

Write your answers directly under the questions in the space provided. Do not write on the QR codes or near the top edges (as it may be cut off).

Please write **legibly and dark enough** for the scan to be readable. For students who use English as a second language, dictionaries are permitted. Calculators are allowed.

DO NOT ASK QUESTIONS OF THE INVIGILATORS about interpretations of the exam questions. If you have a concern about a question, please summarize it at the top of this page. All concerns will be evaluated after the exam. GOOD LUCK!

**Question 1 8 points** (*Assume all transitions are possible*)

a. 6 points Calculate the **energy (in kJ/mol)** required to ionize (completely remove) the electron from the **n=3** level of **He⁺ ion** (He: Helium). Show your work including equations. (3 sig. figs.)

b. 2 points

(I) Select the largest (longest) wavelength from the three choices below:

- i. Wavelength of photon to *completely remove* an electron from *1s orbital of Hydrogen atom*
- ii. Wavelength of photon to *completely remove* an electron from *1s orbital of Boron atom (B)*
- iii. Wavelength of photon to *completely remove* an electron from *1s orbital of Boron cation (B⁺)*

i or ii or iii

(II) Select the smallest (shortest) wavelength from the three choices below (*for H atom*):

- i. Wavelength of photon *emitted* for an electron transitioning from *2p to 1s orbital*
- ii. Wavelength of photon *absorbed* for an electron transitioning from *2s to 4p orbital*
- iii. Wavelength of photon *absorbed* for an electron transitioning from *1s to 3p orbital*

i or ii or iii



Question 2 8 points



A Chem110 student plots the graph shown on the left, from data obtained for the kinetic energy (eV) of the ejected electron versus frequency (s^{-1}) of the incident light, for an unknown metal. ($1\text{eV} = 1.6022 \times 10^{-19} \text{ J}$)

Based on the plot, answer the questions **a to c** below:

a. 1 point On the plot above, label the **threshold frequency**.

b. 2 points What is the value of the work potential (in J) of the unknown metal?

c. 3 points What is the *longest* wavelength (in nm) required to eject an electron from this metal's surface? Show your work including equation(s) used. (3 sig figs)

d. 2 points In the space provided below, plot the following

(Assume frequency is *above* threshold frequency):

- i. *Kinetic energy* of an ejected electron versus the *intensity* of the incident light (left)
- ii. *Number of electrons* ejected versus the *frequency* of the incident light (right)



**Question 3** **6 points****a. 2 points**

In the ground state of Ge (Germanium; $Z = 32$) atom, **name** the following orbitals:

i. Orbital containing the electron that requires the maximum energy to be removed.

ii. Orbital containing the electron that experiences the minimum effective nuclear charge.

b. 2 points

An unknown anion (A^-) has the following *excited* electronic configuration: $[Ar] 4s^2 3d^3 4p^1$

(Ar: Argon; $Z = 18$)

Give the **condensed ground state electronic configuration** of the atom, **A** and the cation, **A^{2+}** .
Show your work. (Assume all electronic transitions are possible)

Electronic Configuration of A:

Electronic Configuration of A^{2+} :

c. 2 points

In a single atom, determine how many electrons can have the following set of quantum numbers?

(i) $n = 5; l = 3$

In a single atom, determine how many electrons can have the following set of quantum numbers?

(i) $n = 3; l = 2; m_s = +1/2$

**Question 4** **8 points**

Rank the following from 1 to 4 in **increasing** order. Provide the ranking in the boxes provided.

Note: Atomic number (Z) are provided for relevant atoms

a) Increasing (lowest to highest) order of *ionization energy*: (1= lowest; 4 = highest)

Mg

Mg: Z = 12

Rb

Rb: Z = 37

Cs

Cs: Z = 55

K

K : Z = 19

b) Increasing (lowest to highest) order of *bond angle*: (1= lowest; 4 = highest)

H₂OCCl₄XeF₄

HCN

c) Increasing (smallest to largest) order of *ionic size* (1= smallest; 4 = largest)

S²⁻

S: Z = 16

O²⁻

O: Z = 8

Be²⁺

Be: Z = 4

Na⁺

Na: Z = 11

d) Increasing (lowest to highest) order of *bond energy* (1= lowest; 4 = highest)

P-F

P-Cl

P-I

P-Br

**Question 5** **12 points**

a. 9 points For the molecules HClO_4 and TeF_6 , draw the most stable *Lewis structure* showing appropriate shape and *dashed/wedged bonds*, based on VSEPR. Determine the **molecular geometry** around the central atom. For HClO_4 , label the *most polar bond*.



b. 3 points (Note: Lone pairs are not shown in structures below)

(i) For the three *resonance* structures (1, 2 and 3) shown below, answer the following question:



(I) Circle the *most contributing* resonance structure.

Structure 1

Structure 2

Structure 3

(ii) For the three structures given (4, 5, and 6) below, answer the following questions:



(I) Circle all the structures that represent *resonance structures* of each other:

Structure 4

Structure 5

Structure 6

(II) Among the resonance structures, circle the *most contributing* resonance structure.

Structure 4

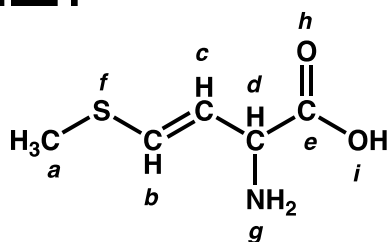
Structure 5

Structure 6

**Question 6** **6 points**

Using the data given below, calculate the **lattice energy** for SrO(s).

	ΔH° (kJ/mol)
$\text{Sr}(s) \longrightarrow \text{Sr}(g)$	+180
$\text{Sr}(g) \longrightarrow \text{Sr}^{2+}(g) + 2e^-$	+1468
$\text{O}_2(g) \longrightarrow 2\text{O}(g)$	+500
$\text{O}(g) + 2e^- \longrightarrow \text{O}^{2-}(g)$	+600
$\text{Sr}(s) + \frac{1}{2}\text{O}_2(g) \longrightarrow \text{SrO}(s)$	-550

**Question 7 10 points**

Consider the molecule given on the left, and answer questions (i) to (vi) using chemical bonding theories that we have learnt in class (valence bond theory, hybridization).

Note that the lone pairs are not shown in the structure:

i. Consider the carbon atoms marked in as ***a***, ***b***, ***c***, ***d***, and ***e*** in the molecule, and determine **hybridization** at each of the carbon centers. (2.5 points)

Atom	Hybridization
Carbon (<i>a</i>)	
Carbon (<i>b</i>)	
Carbon (<i>c</i>)	
Carbon (<i>d</i>)	
Carbon (<i>e</i>)	

ii. Now, consider the sulfur atom (S) marked as ***f***, nitrogen (N) atom (***g***), and oxygen(O) atoms (***h*** and ***i***), and determine their **hybridization**. (2 points)

Atom	Hybridization
Sulfur (<i>f</i>)	
Nitrogen (<i>g</i>)	
Oxygen (<i>h</i>)	
Oxygen (<i>i</i>)	

iii. **Describe** all the bonds formed by the carbon atoms ***a*** and ***b***, using sigma and pi designation as appropriate. (2 points)

Atom	Bonds formed (use sigma and pi as appropriate)
Carbon (<i>a</i>)	
Carbon (<i>b</i>)	

iv. How many total sp^3 -s sigma bonds are in the molecule? (1 point)

v. Determine **electronic** and **molecular geometry** at carbon (C) labeled ***e*** and sulfur (S) labeled ***f*** centres. (2 points)

Atom	Electronic Geometry	Molecular Geometry
Carbon (<i>e</i>)		
Sulfur (<i>f</i>)		

vi. Circle the value from the choices given below, that describes the **C-S-C bond angle at sulfur** (S) marked ***f***: (0.5 points)

>90° to <109.5°

109.5°

>109.5 to <120°

120°

>120° to <180°

**Question 8 12 points**

Using the **valence shell orbitals only** draw a **complete** molecular orbital (MO) diagram (with no s-p mixing) for the molecule **PS**, in the space provided below. **(6 points)**

Assume that 3s and 3p atomic orbitals will form molecular orbitals like those formed by the combination of 2s and 2p atomic orbitals that we learnt in class, and there is no s-p mixing.

Show **energy arrow**, **atomic orbitals** involved, **atom labels**, the **molecular orbitals** that are formed, **filled electrons** in the MO diagram etc. Make sure to label all the atomic and molecular orbitals.

Using the MO diagram above, answer the following questions:

- (i) Which one of the following will have the highest bond order? Circle the correct choice
A) PS^{1-} B) PS C) PS^{1+}
- (ii) Which one of the following will have the longest bond length? Circle the correct choice
A) PS^{1-} B) PS C) PS^{1+}
- (iii) Which of the following will be diamagnetic? Circle the correct choice.
A) PS^{1-} B) PS C) PS^{1+}
- (iv) Draw Lewis dot structure of the molecule with the shortest bond length (based on the MO diagram) from among the three (PS^{1-} , PS, PS^{1+}), and indicate the bond order calculated from the Lewis dot structure. **(3 points)**

Lewis Dot Structure

Bond Order



7E1F7868-7572-439D-97AB-05C25AFA4C9A

midterm-exam-93388

#269 Page 10 of 10

SCRAP PAGE

**CHEMISTRY 110 –Midterm Exam****Thursday November 7, 2024, 6:30 – 8:30 pm**

Examiners: Pallavi Sirjoosingh and Ashok Kakkar

First name (please write as legibly as possible within the boxes)

--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--

Last name

--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--

Student ID number

--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--

This booklet has 8 questions on 10 pages (The last page can be used as a scrap page).

The staple must NOT be removed from the question booklet. Your intact question booklet must be returned at the end of the exam.

A separate booklet contains a periodic table, and datasheet.

Write your answers directly under the questions in the space provided. Do not write on the QR codes or near the top edges (as it may be cut off).

Please write **legibly and dark enough** for the scan to be readable. For students who use English as a second language, dictionaries are permitted. Calculators are allowed.

DO NOT ASK QUESTIONS OF THE INVIGILATORS about interpretations of the exam questions. If you have a concern about a question, please summarize it at the top of this page. All concerns will be evaluated after the exam. GOOD LUCK!

**Question 1 8 points** (*Assume all transitions are possible*)

a. 6 points Calculate the **energy (in kJ/mol)** required to ionize (completely remove) the electron from the **n=3** level of **He⁺ ion** (He: Helium). Show your work including equations. (3 sig. figs.)

b. 2 points

(I) Select the largest (longest) wavelength from the three choices below:

- i. Wavelength of photon to *completely remove* an electron from *1s orbital of Hydrogen atom*
- ii. Wavelength of photon to *completely remove* an electron from *1s orbital of Boron atom (B)*
- iii. Wavelength of photon to *completely remove* an electron from *1s orbital of Boron cation (B⁺)*

i or ii or iii

(II) Select the smallest (shortest) wavelength from the three choices below (*for H atom*):

- i. Wavelength of photon *emitted* for an electron transitioning from *2p to 1s orbital*
- ii. Wavelength of photon *absorbed* for an electron transitioning from *2s to 4p orbital*
- iii. Wavelength of photon *absorbed* for an electron transitioning from *1s to 3p orbital*

i or ii or iii



Question 2 8 points



A Chem110 student plots the graph shown on the left, from data obtained for the kinetic energy (eV) of the ejected electron versus frequency (s^{-1}) of the incident light, for an unknown metal. ($1\text{eV} = 1.6022 \times 10^{-19} \text{ J}$)

Based on the plot, answer the questions **a to c** below:

a. 1 point On the plot above, label the **threshold frequency**.

b. 2 points What is the value of the work potential (in J) of the unknown metal?

c. 3 points What is the *longest* wavelength (in nm) required to eject an electron from this metal's surface? Show your work including equation(s) used. (3 sig figs)

d. 2 points In the space provided below, plot the following

(Assume frequency is *above* threshold frequency):

- i. *Kinetic energy* of an ejected electron versus the *intensity* of the incident light (left)
- ii. *Number of electrons* ejected versus the *frequency* of the incident light (right)



**Question 3** **6 points****a. 2 points**

In the ground state of Ge (Germanium; $Z = 32$) atom, **name** the following orbitals:

i. Orbital containing the electron that requires the maximum energy to be removed.

ii. Orbital containing the electron that experiences the minimum effective nuclear charge.

b. 2 points

An unknown anion (A^-) has the following *excited* electronic configuration: $[Ar] 4s^2 3d^3 4p^1$

(Ar: Argon; $Z = 18$)

Give the **condensed ground state electronic configuration** of the atom, **A** and the cation, **A^{2+}** .
Show your work. (Assume all electronic transitions are possible)

Electronic Configuration of A:

Electronic Configuration of A^{2+} :

c. 2 points

In a single atom, determine how many electrons can have the following set of quantum numbers?

(i) $n = 5; l = 3$

In a single atom, determine how many electrons can have the following set of quantum numbers?

(i) $n = 3; l = 2; m_s = +1/2$

**Question 4** **8 points**

Rank the following from 1 to 4 in **increasing** order. Provide the ranking in the boxes provided.

Note: Atomic number (Z) are provided for relevant atoms

a) Increasing (lowest to highest) order of *ionization energy*: (1= lowest; 4 = highest)

Mg

Mg: Z = 12

Rb

Rb: Z = 37

Cs

Cs: Z = 55

K

K : Z = 19

b) Increasing (lowest to highest) order of *bond angle*: (1= lowest; 4 = highest)

H₂OCCl₄XeF₄

HCN

c) Increasing (smallest to largest) order of *ionic size* (1= smallest; 4 = largest)

S²⁻

S: Z = 16

O²⁻

O: Z = 8

Be²⁺

Be: Z = 4

Na⁺

Na: Z = 11

d) Increasing (lowest to highest) order of *bond energy* (1= lowest; 4 = highest)

P-F

P-Cl

P-I

P-Br

**Question 5** **12 points**

a. 9 points For the molecules HClO_4 and TeF_6 , draw the most stable *Lewis structure* showing appropriate shape and *dashed/wedged bonds*, based on VSEPR. Determine the **molecular geometry** around the central atom. For HClO_4 , label the *most polar bond*.

HClO_4 (there are no O-O bonds)

Label the most polar bond

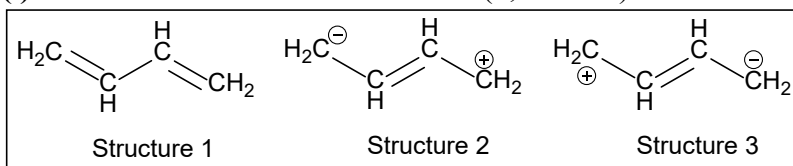
Molecular Geometry around Cl

TeF_6

Molecular Geometry around Te

b. 3 points (Note: Lone pairs are not shown in structures below)

(i) For the three *resonance* structures (1, 2 and 3) shown below, answer the following question:



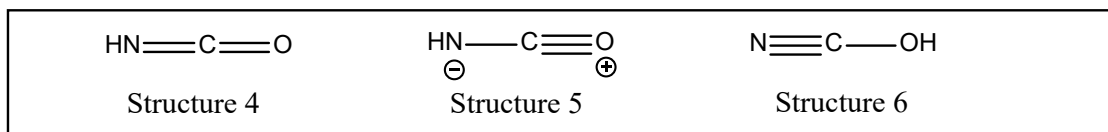
(I) Circle the *most contributing* resonance structure.

Structure 1

Structure 2

Structure 3

(ii) For the three structures given (4, 5, and 6) below, answer the following questions:



(I) Circle all the structures that represent *resonance structures* of each other:

Structure 4

Structure 5

Structure 6

(II) Among the resonance structures, circle the *most contributing* resonance structure.

Structure 4

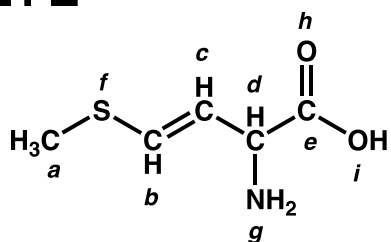
Structure 5

Structure 6

**Question 6** **6 points**

Using the data given below, calculate the **lattice energy** for SrO(s).

	ΔH° (kJ/mol)
$\text{Sr}(s) \longrightarrow \text{Sr}(g)$	+180
$\text{Sr}(g) \longrightarrow \text{Sr}^{2+}(g) + 2e^-$	+1468
$\text{O}_2(g) \longrightarrow 2\text{O}(g)$	+500
$\text{O}(g) + 2e^- \longrightarrow \text{O}^{2-}(g)$	+600
$\text{Sr}(s) + \frac{1}{2}\text{O}_2(g) \longrightarrow \text{SrO}(s)$	-550

**Question 7 10 points**

Consider the molecule given on the left, and answer questions (i) to (vi) using chemical bonding theories that we have learnt in class (valence bond theory, hybridization).

Note that the lone pairs are not shown in the structure:

i. Consider the carbon atoms marked in as ***a***, ***b***, ***c***, ***d***, and ***e*** in the molecule, and determine **hybridization** at each of the carbon centers. (2.5 points)

Atom	Hybridization
Carbon (<i>a</i>)	
Carbon (<i>b</i>)	
Carbon (<i>c</i>)	
Carbon (<i>d</i>)	
Carbon (<i>e</i>)	

ii. Now, consider the sulfur atom (S) marked as ***f***, nitrogen (N) atom (***g***), and oxygen(O) atoms (***h*** and ***i***), and determine their **hybridization**. (2 points)

Atom	Hybridization
Sulfur (<i>f</i>)	
Nitrogen (<i>g</i>)	
Oxygen (<i>h</i>)	
Oxygen (<i>i</i>)	

iii. **Describe** all the bonds formed by the carbon atoms ***a*** and ***b***, using sigma and pi designation as appropriate. (2 points)

Atom	Bonds formed (use sigma and pi as appropriate)
Carbon (<i>a</i>)	
Carbon (<i>b</i>)	

iv. How many total sp^3 -s sigma bonds are in the molecule? (1 point)

v. Determine **electronic** and **molecular geometry** at carbon (C) labeled ***e*** and sulfur (S) labeled ***f*** centres. (2 points)

Atom	Electronic Geometry	Molecular Geometry
Carbon (<i>e</i>)		
Sulfur (<i>f</i>)		

vi. Circle the value from the choices given below, that describes the **C-S-C bond angle at sulfur** (S) marked ***f***: (0.5 points)

$>90^\circ$ to $<109.5^\circ$ 109.5° >109.5 to $<120^\circ$ 120° $>120^\circ$ to $<180^\circ$

**Question 8 12 points**

Using the **valence shell orbitals only** draw a **complete** molecular orbital (MO) diagram (with no s-p mixing) for the molecule **PS**, in the space provided below. **(6 points)**

Assume that 3s and 3p atomic orbitals will form molecular orbitals like those formed by the combination of 2s and 2p atomic orbitals that we learnt in class, and there is no s-p mixing.

Show **energy arrow**, **atomic orbitals** involved, **atom labels**, the **molecular orbitals** that are formed, **filled electrons** in the MO diagram etc. Make sure to label all the atomic and molecular orbitals.

Using the MO diagram above, answer the following questions:

- (i) Which one of the following will have the highest bond order? Circle the correct choice
A) PS^{1-} B) PS C) PS^{1+}
- (ii) Which one of the following will have the longest bond length? Circle the correct choice
A) PS^{1-} B) PS C) PS^{1+}
- (iii) Which of the following will be diamagnetic? Circle the correct choice.
A) PS^{1-} B) PS C) PS^{1+}
- (iv) Draw Lewis dot structure of the molecule with the shortest bond length (based on the MO diagram) from among the three (PS^{1-} , PS, PS^{1+}), and indicate the bond order calculated from the Lewis dot structure. **(3 points)**

Lewis Dot Structure

Bond Order



DB0DA7A7-9095-4538-A997-34A634BF940C

midterm-exam-93388

#270 Page 10 of 10

SCRAP PAGE

**CHEMISTRY 110 –Midterm Exam****Thursday November 7, 2024, 6:30 – 8:30 pm****Examiners: Pallavi Sirjoosingh and Ashok Kakkar**

First name (please write as legibly as possible within the boxes)

--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--

Last name

--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--

Student ID number

--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--

This booklet has 8 questions on 10 pages (The last page can be used as a scrap page).

The staple must NOT be removed from the question booklet. Your intact question booklet must be returned at the end of the exam.

A separate booklet contains a periodic table, and datasheet.

Write your answers directly under the questions in the space provided. Do not write on the QR codes or near the top edges (as it may be cut off).

Please write **legibly and dark enough** for the scan to be readable. For students who use English as a second language, dictionaries are permitted. Calculators are allowed.

DO NOT ASK QUESTIONS OF THE INVIGILATORS about interpretations of the exam questions. If you have a concern about a question, please summarize it at the top of this page. All concerns will be evaluated after the exam. GOOD LUCK!

**Question 1 8 points** (*Assume all transitions are possible*)

a. 6 points Calculate the **energy (in kJ/mol)** required to ionize (completely remove) the electron from the **$n=3$** level of **He^+ ion** (He: Helium). Show your work including equations. (3 sig. figs.)

b. 2 points

(I) Select the largest (longest) wavelength from the three choices below:

- i. Wavelength of photon to *completely remove* an electron from *1s orbital of Hydrogen atom*
- ii. Wavelength of photon to *completely remove* an electron from *1s orbital of Boron atom (B)*
- iii. Wavelength of photon to *completely remove* an electron from *1s orbital of Boron cation (B^+)*

i or ii or iii

(II) Select the smallest (shortest) wavelength from the three choices below (*for H atom*):

- i. Wavelength of photon *emitted* for an electron transitioning from *2p to 1s orbital*
- ii. Wavelength of photon *absorbed* for an electron transitioning from *2s to 4p orbital*
- iii. Wavelength of photon *absorbed* for an electron transitioning from *1s to 3p orbital*

i or ii or iii



Question 2 8 points



A Chem110 student plots the graph shown on the left, from data obtained for the kinetic energy (eV) of the ejected electron versus frequency (s^{-1}) of the incident light, for an unknown metal. ($1\text{eV} = 1.6022 \times 10^{-19} \text{ J}$)

Based on the plot, answer the questions **a to c** below:

a. 1 point On the plot above, label the **threshold frequency**.

b. 2 points What is the value of the work potential (in J) of the unknown metal?

c. 3 points What is the *longest* wavelength (in nm) required to eject an electron from this metal's surface? Show your work including equation(s) used. (3 sig figs)

d. 2 points In the space provided below, plot the following

(Assume frequency is *above* threshold frequency):

- i. *Kinetic energy* of an ejected electron versus the *intensity* of the incident light (left)
- ii. *Number of electrons* ejected versus the *frequency* of the incident light (right)



**Question 3** **6 points****a. 2 points**

In the ground state of Ge (Germanium; $Z = 32$) atom, **name** the following orbitals:

i. Orbital containing the electron that requires the maximum energy to be removed.

ii. Orbital containing the electron that experiences the minimum effective nuclear charge.

b. 2 points

An unknown anion (A^-) has the following *excited* electronic configuration: $[Ar] 4s^2 3d^3 4p^1$

(Ar: Argon; $Z = 18$)

Give the **condensed ground state electronic configuration** of the atom, **A** and the cation, **A^{2+}** .
Show your work. (Assume all electronic transitions are possible)

Electronic Configuration of A:

Electronic Configuration of A^{2+} :

c. 2 points

In a single atom, determine how many electrons can have the following set of quantum numbers?

(i) $n = 5; l = 3$

In a single atom, determine how many electrons can have the following set of quantum numbers?

(i) $n = 3; l = 2; m_s = +1/2$

**Question 4** **8 points**

Rank the following from 1 to 4 in **increasing** order. Provide the ranking in the boxes provided.

Note: Atomic number (Z) are provided for relevant atoms

a) Increasing (lowest to highest) order of *ionization energy*: (1= lowest; 4 = highest)

Mg

Mg: Z = 12

Rb

Rb: Z = 37

Cs

Cs: Z = 55

K

K : Z = 19

b) Increasing (lowest to highest) order of *bond angle*: (1= lowest; 4 = highest)

H₂OCCl₄XeF₄

HCN

c) Increasing (smallest to largest) order of *ionic size* (1= smallest; 4 = largest)

S²⁻

S: Z = 16

O²⁻

O: Z = 8

Be²⁺

Be: Z = 4

Na⁺

Na: Z = 11

d) Increasing (lowest to highest) order of *bond energy* (1= lowest; 4 = highest)

P-F

P-Cl

P-I

P-Br

**Question 5** **12 points**

a. 9 points For the molecules HClO_4 and TeF_6 , draw the most stable *Lewis structure* showing appropriate shape and *dashed/wedged bonds*, based on VSEPR. Determine the **molecular geometry** around the central atom. For HClO_4 , label the *most polar bond*.



b. 3 points (Note: Lone pairs are not shown in structures below)

(i) For the three *resonance* structures (1, 2 and 3) shown below, answer the following question:



(I) Circle the *most contributing* resonance structure.

Structure 1

Structure 2

Structure 3

(ii) For the three structures given (4, 5, and 6) below, answer the following questions:



(I) Circle all the structures that represent *resonance structures* of each other:

Structure 4

Structure 5

Structure 6

(II) Among the resonance structures, circle the *most contributing* resonance structure.

Structure 4

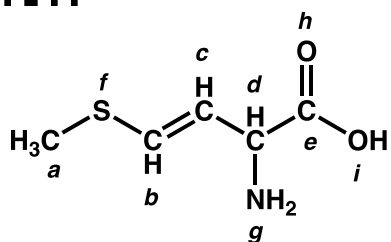
Structure 5

Structure 6

**Question 6** **6 points**

Using the data given below, calculate the **lattice energy** for SrO(s).

	ΔH° (kJ/mol)
$\text{Sr}(s) \longrightarrow \text{Sr}(g)$	+180
$\text{Sr}(g) \longrightarrow \text{Sr}^{2+}(g) + 2e^-$	+1468
$\text{O}_2(g) \longrightarrow 2\text{O}(g)$	+500
$\text{O}(g) + 2e^- \longrightarrow \text{O}^{2-}(g)$	+600
$\text{Sr}(s) + \frac{1}{2}\text{O}_2(g) \longrightarrow \text{SrO}(s)$	-550

**Question 7 10 points**

Consider the molecule given on the left, and answer questions (i) to (vi) using chemical bonding theories that we have learnt in class (valence bond theory, hybridization).

Note that the lone pairs are not shown in the structure:

i. Consider the carbon atoms marked in as ***a***, ***b***, ***c***, ***d***, and ***e*** in the molecule, and determine **hybridization** at each of the carbon centers. (2.5 points)

Atom	Hybridization
Carbon (<i>a</i>)	
Carbon (<i>b</i>)	
Carbon (<i>c</i>)	
Carbon (<i>d</i>)	
Carbon (<i>e</i>)	

ii. Now, consider the sulfur atom (S) marked as ***f***, nitrogen (N) atom (***g***), and oxygen(O) atoms (***h*** and ***i***), and determine their **hybridization**. (2 points)

Atom	Hybridization
Sulfur (<i>f</i>)	
Nitrogen (<i>g</i>)	
Oxygen (<i>h</i>)	
Oxygen (<i>i</i>)	

iii. **Describe** all the bonds formed by the carbon atoms ***a*** and ***b***, using sigma and pi designation as appropriate. (2 points)

Atom	Bonds formed (use sigma and pi as appropriate)
Carbon (<i>a</i>)	
Carbon (<i>b</i>)	

iv. How many total sp^3 -s sigma bonds are in the molecule? (1 point)

v. Determine **electronic** and **molecular geometry** at carbon (C) labeled ***e*** and sulfur (S) labeled ***f*** centres. (2 points)

Atom	Electronic Geometry	Molecular Geometry
Carbon (<i>e</i>)		
Sulfur (<i>f</i>)		

vi. Circle the value from the choices given below, that describes the **C-S-C bond angle at sulfur** (S) marked ***f***: (0.5 points)

>90° to <109.5° 109.5° >109.5 to <120° 120° >120° to <180°


Question 8 12 points

Using the **valence shell orbitals only** draw a **complete** molecular orbital (MO) diagram (with no s-p mixing) for the molecule **PS**, in the space provided below. **(6 points)**

Assume that 3s and 3p atomic orbitals will form molecular orbitals like those formed by the combination of 2s and 2p atomic orbitals that we learnt in class, and there is no s-p mixing.

Show **energy arrow**, **atomic orbitals** involved, **atom labels**, the **molecular orbitals** that are formed, **filled electrons** in the MO diagram etc. Make sure to **label all the atomic and molecular orbitals**.

Using the MO diagram above, answer the following questions:

- (i) Which one of the following will have the highest bond order? Circle the correct choice
 A) PS^{1-} B) PS C) PS^{1+}
- (ii) Which one of the following will have the longest bond length? Circle the correct choice
 A) PS^{1-} B) PS C) PS^{1+}
- (iii) Which of the following will be diamagnetic? Circle the correct choice.
 A) PS^{1-} B) PS C) PS^{1+}
- (iv) Draw Lewis dot structure of the molecule with the shortest bond length (based on the MO diagram) from among the three (PS^{1-} , PS, PS^{1+}), and indicate the bond order calculated from the Lewis dot structure. **(3 points)**

Lewis Dot Structure

Bond Order



EF008EC0-1AC9-4EA4-90D4-4E2CB41E1696

midterm-exam-93388

#271 Page 10 of 10

SCRAP PAGE

**CHEMISTRY 110 –Midterm Exam****Thursday November 7, 2024, 6:30 – 8:30 pm**

Examiners: Pallavi Sirjoosingh and Ashok Kakkar

First name (please write as legibly as possible within the boxes)

--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--

Last name

--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--

Student ID number

--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--

This booklet has 8 questions on 10 pages (The last page can be used as a scrap page).

The staple must NOT be removed from the question booklet. Your intact question booklet must be returned at the end of the exam.

A separate booklet contains a periodic table, and datasheet.

Write your answers directly under the questions in the space provided. Do not write on the QR codes or near the top edges (as it may be cut off).

Please write **legibly and dark enough** for the scan to be readable. For students who use English as a second language, dictionaries are permitted. Calculators are allowed.

DO NOT ASK QUESTIONS OF THE INVIGILATORS about interpretations of the exam questions. If you have a concern about a question, please summarize it at the top of this page. All concerns will be evaluated after the exam. GOOD LUCK!

**Question 1 8 points** (*Assume all transitions are possible*)

a. 6 points Calculate the **energy (in kJ/mol)** required to ionize (completely remove) the electron from the **n=3** level of **He⁺ ion** (He: Helium). Show your work including equations. (3 sig. figs.)

b. 2 points

(I) Select the largest (longest) wavelength from the three choices below:

- i. Wavelength of photon to *completely remove* an electron from *1s orbital of Hydrogen atom*
- ii. Wavelength of photon to *completely remove* an electron from *1s orbital of Boron atom (B)*
- iii. Wavelength of photon to *completely remove* an electron from *1s orbital of Boron cation (B⁺)*

i or ii or iii

(II) Select the smallest (shortest) wavelength from the three choices below (*for H atom*):

- i. Wavelength of photon *emitted* for an electron transitioning from *2p to 1s orbital*
- ii. Wavelength of photon *absorbed* for an electron transitioning from *2s to 4p orbital*
- iii. Wavelength of photon *absorbed* for an electron transitioning from *1s to 3p orbital*

i or ii or iii



Question 2 8 points



A Chem110 student plots the graph shown on the left, from data obtained for the kinetic energy (eV) of the ejected electron versus frequency (s^{-1}) of the incident light, for an unknown metal. ($1\text{eV} = 1.6022 \times 10^{-19} \text{ J}$)

Based on the plot, answer the questions **a to c** below:

a. 1 point On the plot above, label the **threshold frequency**.

b. 2 points What is the value of the work potential (in J) of the unknown metal?

c. 3 points What is the *longest* wavelength (in nm) required to eject an electron from this metal's surface? Show your work including equation(s) used. (3 sig figs)

d. 2 points In the space provided below, plot the following

(Assume frequency is *above* threshold frequency):

- Kinetic energy* of an ejected electron versus the *intensity* of the incident light (left)
- Number of electrons* ejected versus the *frequency* of the incident light (right)



**Question 3** **6 points****a. 2 points**

In the ground state of Ge (Germanium; $Z = 32$) atom, **name** the following orbitals:

i. Orbital containing the electron that requires the maximum energy to be removed.

ii. Orbital containing the electron that experiences the minimum effective nuclear charge.

b. 2 points

An unknown anion (A^-) has the following *excited* electronic configuration: $[Ar] 4s^2 3d^3 4p^1$

(Ar: Argon; $Z = 18$)

Give the **condensed ground state electronic configuration** of the atom, **A** and the cation, **A^{2+}** .
Show your work. (Assume all electronic transitions are possible)

Electronic Configuration of A:

Electronic Configuration of A^{2+} :

c. 2 points

In a single atom, determine how many electrons can have the following set of quantum numbers?

(i) $n = 5; l = 3$

In a single atom, determine how many electrons can have the following set of quantum numbers?

(i) $n = 3; l = 2; m_s = +1/2$

**Question 4** **8 points**

Rank the following from 1 to 4 in **increasing** order. Provide the ranking in the boxes provided.

Note: Atomic number (Z) are provided for relevant atoms

a) Increasing (lowest to highest) order of *ionization energy*: (1= lowest; 4 = highest)

Mg

Mg: Z = 12

Rb

Rb: Z = 37

Cs

Cs: Z = 55

K

K : Z = 19

b) Increasing (lowest to highest) order of *bond angle*: (1= lowest; 4 = highest)

H₂OCCl₄XeF₄

HCN

c) Increasing (smallest to largest) order of *ionic size* (1= smallest; 4 = largest)

S²⁻

S: Z = 16

O²⁻

O: Z = 8

Be²⁺

Be: Z = 4

Na⁺

Na: Z = 11

d) Increasing (lowest to highest) order of *bond energy* (1= lowest; 4 = highest)

P-F

P-Cl

P-I

P-Br

**Question 5** **12 points**

a. 9 points For the molecules HClO_4 and TeF_6 , draw the most stable *Lewis structure* showing appropriate shape and *dashed/wedged bonds*, based on VSEPR. Determine the **molecular geometry** around the central atom. For HClO_4 , label the *most polar bond*.



b. 3 points (Note: Lone pairs are not shown in structures below)

(i) For the three *resonance* structures (1, 2 and 3) shown below, answer the following question:



(I) Circle the *most contributing* resonance structure.

Structure 1

Structure 2

Structure 3

(ii) For the three structures given (4, 5, and 6) below, answer the following questions:



(I) Circle all the structures that represent *resonance structures* of each other:

Structure 4

Structure 5

Structure 6

(II) Among the resonance structures, circle the *most contributing* resonance structure.

Structure 4

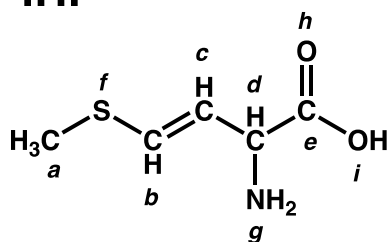
Structure 5

Structure 6

**Question 6** **6 points**

Using the data given below, calculate the **lattice energy** for SrO(s).

	ΔH° (kJ/mol)
$\text{Sr}(s) \longrightarrow \text{Sr}(g)$	+180
$\text{Sr}(g) \longrightarrow \text{Sr}^{2+}(g) + 2e^-$	+1468
$\text{O}_2(g) \longrightarrow 2\text{O}(g)$	+500
$\text{O}(g) + 2e^- \longrightarrow \text{O}^{2-}(g)$	+600
$\text{Sr}(s) + \frac{1}{2}\text{O}_2(g) \longrightarrow \text{SrO}(s)$	-550

**Question 7 10 points**

Consider the molecule given on the left, and answer questions (i) to (vi) using chemical bonding theories that we have learnt in class (valence bond theory, hybridization).

Note that the lone pairs are not shown in the structure:

i. Consider the carbon atoms marked in as ***a***, ***b***, ***c***, ***d***, and ***e*** in the molecule, and determine **hybridization** at each of the carbon centers. (2.5 points)

Atom	Hybridization
Carbon (<i>a</i>)	
Carbon (<i>b</i>)	
Carbon (<i>c</i>)	
Carbon (<i>d</i>)	
Carbon (<i>e</i>)	

ii. Now, consider the sulfur atom (S) marked as ***f***, nitrogen (N) atom (***g***), and oxygen(O) atoms (***h*** and ***i***), and determine their **hybridization**. (2 points)

Atom	Hybridization
Sulfur (<i>f</i>)	
Nitrogen (<i>g</i>)	
Oxygen (<i>h</i>)	
Oxygen (<i>i</i>)	

iii. **Describe** all the bonds formed by the carbon atoms ***a*** and ***b***, using sigma and pi designation as appropriate. (2 points)

Atom	Bonds formed (use sigma and pi as appropriate)
Carbon (<i>a</i>)	
Carbon (<i>b</i>)	

iv. How many total sp^3 -s sigma bonds are in the molecule? (1 point)

v. Determine **electronic** and **molecular geometry** at carbon (C) labeled ***e*** and sulfur (S) labeled ***f*** centres. (2 points)

Atom	Electronic Geometry	Molecular Geometry
Carbon (<i>e</i>)		
Sulfur (<i>f</i>)		

vi. Circle the value from the choices given below, that describes the **C-S-C bond angle at sulfur** (S) marked ***f***: (0.5 points)

>90° to <109.5° 109.5° >109.5 to <120° 120° >120° to <180°


Question 8 12 points

Using the **valence shell orbitals only** draw a **complete** molecular orbital (MO) diagram (with no s-p mixing) for the molecule **PS**, in the space provided below. **(6 points)**

Assume that 3s and 3p atomic orbitals will form molecular orbitals like those formed by the combination of 2s and 2p atomic orbitals that we learnt in class, and there is no s-p mixing.

Show **energy arrow**, **atomic orbitals** involved, **atom labels**, the **molecular orbitals** that are formed, **filled electrons** in the MO diagram etc. Make sure to label all the atomic and molecular orbitals.

Using the MO diagram above, answer the following questions:

- (i) Which one of the following will have the highest bond order? Circle the correct choice
 A) PS^{1-} B) PS C) PS^{1+}
- (ii) Which one of the following will have the longest bond length? Circle the correct choice
 A) PS^{1-} B) PS C) PS^{1+}
- (iii) Which of the following will be diamagnetic? Circle the correct choice.
 A) PS^{1-} B) PS C) PS^{1+}
- (iv) Draw Lewis dot structure of the molecule with the shortest bond length (based on the MO diagram) from among the three (PS^{1-} , PS, PS^{1+}), and indicate the bond order calculated from the Lewis dot structure. **(3 points)**

Lewis Dot Structure

Bond Order



024E02E9-D174-4827-8CC9-2342CE01E5DA

midterm-exam-93388

#272 Page 10 of 10

SCRAP PAGE

**CHEMISTRY 110 –Midterm Exam****Thursday November 7, 2024, 6:30 – 8:30 pm**

Examiners: Pallavi Sirjoosingh and Ashok Kakkar

First name (please write as legibly as possible within the boxes)

--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--

Last name

--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--

Student ID number

--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--

This booklet has 8 questions on 10 pages (The last page can be used as a scrap page).

The staple must NOT be removed from the question booklet. Your intact question booklet must be returned at the end of the exam.

A separate booklet contains a periodic table, and datasheet.

Write your answers directly under the questions in the space provided. Do not write on the QR codes or near the top edges (as it may be cut off).

Please write **legibly and dark enough** for the scan to be readable. For students who use English as a second language, dictionaries are permitted. Calculators are allowed.

DO NOT ASK QUESTIONS OF THE INVIGILATORS about interpretations of the exam questions. If you have a concern about a question, please summarize it at the top of this page. All concerns will be evaluated after the exam. GOOD LUCK!

**Question 1 8 points** (*Assume all transitions are possible*)

a. 6 points Calculate the **energy (in kJ/mol)** required to ionize (completely remove) the electron from the **n=3** level of **He⁺ ion** (He: Helium). Show your work including equations. (3 sig. figs.)

b. 2 points

(I) Select the largest (longest) wavelength from the three choices below:

- i. Wavelength of photon to *completely remove* an electron from *1s orbital of Hydrogen atom*
- ii. Wavelength of photon to *completely remove* an electron from *1s orbital of Boron atom (B)*
- iii. Wavelength of photon to *completely remove* an electron from *1s orbital of Boron cation (B⁺)*

i or ii or iii

(II) Select the smallest (shortest) wavelength from the three choices below (*for H atom*):

- i. Wavelength of photon *emitted* for an electron transitioning from *2p to 1s orbital*
- ii. Wavelength of photon *absorbed* for an electron transitioning from *2s to 4p orbital*
- iii. Wavelength of photon *absorbed* for an electron transitioning from *1s to 3p orbital*

i or ii or iii



Question 2 8 points



A Chem110 student plots the graph shown on the left, from data obtained for the kinetic energy (eV) of the ejected electron versus frequency (s^{-1}) of the incident light, for an unknown metal. ($1\text{eV} = 1.6022 \times 10^{-19} \text{ J}$)

Based on the plot, answer the questions **a to c** below:

a. 1 point On the plot above, label the **threshold frequency**.

b. 2 points What is the value of the work potential (in J) of the unknown metal?

c. 3 points What is the *longest* wavelength (in nm) required to eject an electron from this metal's surface? Show your work including equation(s) used. (3 sig figs)

d. 2 points In the space provided below, plot the following

(Assume frequency is *above* threshold frequency):

- Kinetic energy* of an ejected electron versus the *intensity* of the incident light (left)
- Number of electrons* ejected versus the *frequency* of the incident light (right)



**Question 3** **6 points****a. 2 points**

In the ground state of Ge (Germanium; $Z = 32$) atom, **name** the following orbitals:

i. Orbital containing the electron that requires the maximum energy to be removed.

ii. Orbital containing the electron that experiences the minimum effective nuclear charge.

b. 2 points

An unknown anion (A^-) has the following *excited* electronic configuration: $[Ar] 4s^2 3d^3 4p^1$

(Ar: Argon; $Z = 18$)

Give the **condensed ground state electronic configuration** of the atom, **A** and the cation, **A^{2+}** .
Show your work. (Assume all electronic transitions are possible)

Electronic Configuration of A:

Electronic Configuration of A^{2+} :

c. 2 points

In a single atom, determine how many electrons can have the following set of quantum numbers?

(i) $n = 5; l = 3$

In a single atom, determine how many electrons can have the following set of quantum numbers?

(i) $n = 3; l = 2; m_s = +1/2$

**Question 4** **8 points**

Rank the following from 1 to 4 in **increasing** order. Provide the ranking in the boxes provided.

Note: Atomic number (Z) are provided for relevant atoms

a) Increasing (lowest to highest) order of *ionization energy*: (1= lowest; 4 = highest)

Mg

Mg: Z = 12

Rb

Rb: Z = 37

Cs

Cs: Z = 55

K

K : Z = 19

b) Increasing (lowest to highest) order of *bond angle*: (1= lowest; 4 = highest)

H₂OCCl₄XeF₄

HCN

c) Increasing (smallest to largest) order of *ionic size* (1= smallest; 4 = largest)

S²⁻

S: Z = 16

O²⁻

O: Z = 8

Be²⁺

Be: Z = 4

Na⁺

Na: Z = 11

d) Increasing (lowest to highest) order of *bond energy* (1= lowest; 4 = highest)

P-F

P-Cl

P-I

P-Br

**Question 5** **12 points**

a. 9 points For the molecules HClO_4 and TeF_6 , draw the most stable *Lewis structure* showing appropriate shape and *dashed/wedged bonds*, based on VSEPR. Determine the **molecular geometry** around the central atom. For HClO_4 , label the *most polar bond*.



b. 3 points (Note: Lone pairs are not shown in structures below)

(i) For the three *resonance* structures (1, 2 and 3) shown below, answer the following question:



(I) Circle the *most contributing* resonance structure.

Structure 1

Structure 2

Structure 3

(ii) For the three structures given (4, 5, and 6) below, answer the following questions:



(I) Circle all the structures that represent *resonance structures* of each other:

Structure 4

Structure 5

Structure 6

(II) Among the resonance structures, circle the *most contributing* resonance structure.

Structure 4

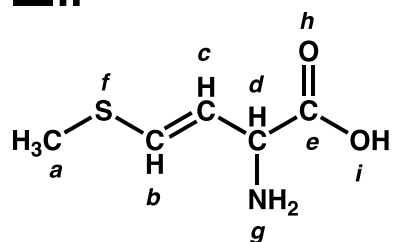
Structure 5

Structure 6

**Question 6** **6 points**

Using the data given below, calculate the **lattice energy** for SrO(s).

	ΔH° (kJ/mol)
$\text{Sr}(s) \longrightarrow \text{Sr}(g)$	+180
$\text{Sr}(g) \longrightarrow \text{Sr}^{2+}(g) + 2e^-$	+1468
$\text{O}_2(g) \longrightarrow 2\text{O}(g)$	+500
$\text{O}(g) + 2e^- \longrightarrow \text{O}^{2-}(g)$	+600
$\text{Sr}(s) + \frac{1}{2}\text{O}_2(g) \longrightarrow \text{SrO}(s)$	-550

**Question 7 10 points**

Consider the molecule given on the left, and answer questions (i) to (vi) using chemical bonding theories that we have learnt in class (valence bond theory, hybridization).

Note that the lone pairs are not shown in the structure:

i. Consider the carbon atoms marked in as ***a***, ***b***, ***c***, ***d***, and ***e*** in the molecule, and determine **hybridization** at each of the carbon centers. (2.5 points)

Atom	Hybridization
Carbon (<i>a</i>)	
Carbon (<i>b</i>)	
Carbon (<i>c</i>)	
Carbon (<i>d</i>)	
Carbon (<i>e</i>)	

ii. Now, consider the sulfur atom (S) marked as ***f***, nitrogen (N) atom (***g***), and oxygen(O) atoms (***h*** and ***i***), and determine their **hybridization**. (2 points)

Atom	Hybridization
Sulfur (<i>f</i>)	
Nitrogen (<i>g</i>)	
Oxygen (<i>h</i>)	
Oxygen (<i>i</i>)	

iii. **Describe** all the bonds formed by the carbon atoms ***a*** and ***b***, using sigma and pi designation as appropriate. (2 points)

Atom	Bonds formed (use sigma and pi as appropriate)
Carbon (<i>a</i>)	
Carbon (<i>b</i>)	

iv. How many total sp^3 -s sigma bonds are in the molecule? (1 point)

v. Determine **electronic** and **molecular geometry** at carbon (C) labeled ***e*** and sulfur (S) labeled ***f*** centres. (2 points)

Atom	Electronic Geometry	Molecular Geometry
Carbon (<i>e</i>)		
Sulfur (<i>f</i>)		

vi. Circle the value from the choices given below, that describes the **C-S-C bond angle at sulfur** (S) marked ***f***: (0.5 points)

>90° to <109.5° 109.5° >109.5 to <120° 120° >120° to <180°

**Question 8 12 points**

Using the **valence shell orbitals only** draw a **complete** molecular orbital (MO) diagram (with no s-p mixing) for the molecule **PS**, in the space provided below. **(6 points)**

Assume that 3s and 3p atomic orbitals will form molecular orbitals like those formed by the combination of 2s and 2p atomic orbitals that we learnt in class, and there is no s-p mixing.

Show **energy arrow**, **atomic orbitals** involved, **atom labels**, the **molecular orbitals** that are formed, **filled electrons** in the MO diagram etc. Make sure to label all the atomic and molecular orbitals.

Using the MO diagram above, answer the following questions:

- (i) Which one of the following will have the highest bond order? Circle the correct choice
A) PS^{1-} B) PS C) PS^{1+}
- (ii) Which one of the following will have the longest bond length? Circle the correct choice
A) PS^{1-} B) PS C) PS^{1+}
- (iii) Which of the following will be diamagnetic? Circle the correct choice.
A) PS^{1-} B) PS C) PS^{1+}
- (iv) Draw Lewis dot structure of the molecule with the shortest bond length (based on the MO diagram) from among the three (PS^{1-} , PS, PS^{1+}), and indicate the bond order calculated from the Lewis dot structure. **(3 points)**

Lewis Dot Structure

Bond Order



43D6EDB9-01A8-404F-ACE8-9F16D67D1B57

midterm-exam-93388

#273 Page 10 of 10

SCRAP PAGE

**CHEMISTRY 110 –Midterm Exam****Thursday November 7, 2024, 6:30 – 8:30 pm**

Examiners: Pallavi Sirjoosingh and Ashok Kakkar

First name (please write as legibly as possible within the boxes)

--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--

Last name

--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--

Student ID number

--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--

This booklet has 8 questions on 10 pages (The last page can be used as a scrap page).

The staple must NOT be removed from the question booklet. Your intact question booklet must be returned at the end of the exam.

A separate booklet contains a periodic table, and datasheet.

Write your answers directly under the questions in the space provided. Do not write on the QR codes or near the top edges (as it may be cut off).

Please write **legibly and dark enough** for the scan to be readable. For students who use English as a second language, dictionaries are permitted. Calculators are allowed.

DO NOT ASK QUESTIONS OF THE INVIGILATORS about interpretations of the exam questions. If you have a concern about a question, please summarize it at the top of this page. All concerns will be evaluated after the exam. GOOD LUCK!

**Question 1 8 points** (*Assume all transitions are possible*)

a. 6 points Calculate the **energy (in kJ/mol)** required to ionize (completely remove) the electron from the **$n=3$** level of **He^+ ion** (He: Helium). Show your work including equations. (3 sig. figs.)

b. 2 points

(I) Select the largest (longest) wavelength from the three choices below:

- i. Wavelength of photon to *completely remove* an electron from *1s orbital of Hydrogen atom*
- ii. Wavelength of photon to *completely remove* an electron from *1s orbital of Boron atom (B)*
- iii. Wavelength of photon to *completely remove* an electron from *1s orbital of Boron cation (B^+)*

i or ii or iii

(II) Select the smallest (shortest) wavelength from the three choices below (*for H atom*):

- i. Wavelength of photon *emitted* for an electron transitioning from *2p to 1s orbital*
- ii. Wavelength of photon *absorbed* for an electron transitioning from *2s to 4p orbital*
- iii. Wavelength of photon *absorbed* for an electron transitioning from *1s to 3p orbital*

i or ii or iii



Question 2 8 points



A Chem110 student plots the graph shown on the left, from data obtained for the kinetic energy (eV) of the ejected electron versus frequency (s^{-1}) of the incident light, for an unknown metal. ($1\text{eV} = 1.6022 \times 10^{-19} \text{ J}$)

Based on the plot, answer the questions **a to c** below:

a. 1 point On the plot above, label the **threshold frequency**.

b. 2 points What is the value of the work potential (in J) of the unknown metal?

c. 3 points What is the *longest* wavelength (in nm) required to eject an electron from this metal's surface? Show your work including equation(s) used. (3 sig figs)

d. 2 points In the space provided below, plot the following

(Assume frequency is *above* threshold frequency):

- i. *Kinetic energy* of an ejected electron versus the *intensity* of the incident light (left)
- ii. *Number of electrons* ejected versus the *frequency* of the incident light (right)



**Question 3** **6 points****a. 2 points**

In the ground state of Ge (Germanium; $Z = 32$) atom, **name** the following orbitals:

i. Orbital containing the electron that requires the maximum energy to be removed.

ii. Orbital containing the electron that experiences the minimum effective nuclear charge.

b. 2 points

An unknown anion (A^-) has the following *excited* electronic configuration: $[Ar] 4s^2 3d^3 4p^1$

(Ar: Argon; $Z = 18$)

Give the **condensed ground state electronic configuration** of the atom, **A** and the cation, **A^{2+}** .
Show your work. (Assume all electronic transitions are possible)

Electronic Configuration of A:

Electronic Configuration of A^{2+} :

c. 2 points

In a single atom, determine how many electrons can have the following set of quantum numbers?

(i) $n = 5; l = 3$

In a single atom, determine how many electrons can have the following set of quantum numbers?

(i) $n = 3; l = 2; m_s = +1/2$

**Question 4** **8 points**

Rank the following from 1 to 4 in **increasing** order. Provide the ranking in the boxes provided.

Note: Atomic number (Z) are provided for relevant atoms

a) Increasing (lowest to highest) order of *ionization energy*: (1= lowest; 4 = highest)

Mg

Mg: Z = 12

Rb

Rb: Z = 37

Cs

Cs: Z = 55

K

K : Z = 19

b) Increasing (lowest to highest) order of *bond angle*: (1= lowest; 4 = highest)

H₂OCCl₄XeF₄

HCN

c) Increasing (smallest to largest) order of *ionic size* (1= smallest; 4 = largest)

S²⁻

S: Z = 16

O²⁻

O: Z = 8

Be²⁺

Be: Z = 4

Na⁺

Na: Z = 11

d) Increasing (lowest to highest) order of *bond energy* (1= lowest; 4 = highest)

P-F

P-Cl

P-I

P-Br

**Question 5** **12 points**

a. 9 points For the molecules HClO_4 and TeF_6 , draw the most stable *Lewis structure* showing appropriate shape and *dashed/wedged bonds*, based on VSEPR. Determine the **molecular geometry** around the central atom. For HClO_4 , label the *most polar bond*.



b. 3 points (Note: Lone pairs are not shown in structures below)

(i) For the three *resonance* structures (1, 2 and 3) shown below, answer the following question:



(I) Circle the *most contributing* resonance structure.

Structure 1

Structure 2

Structure 3

(ii) For the three structures given (4, 5, and 6) below, answer the following questions:



(I) Circle all the structures that represent *resonance structures* of each other:

Structure 4

Structure 5

Structure 6

(II) Among the resonance structures, circle the *most contributing* resonance structure.

Structure 4

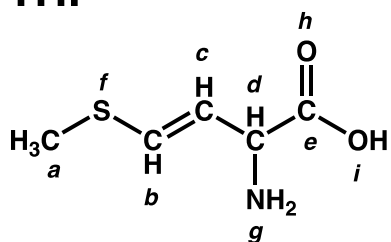
Structure 5

Structure 6

**Question 6** **6 points**

Using the data given below, calculate the **lattice energy** for SrO(s).

	ΔH° (kJ/mol)
$\text{Sr}(s) \longrightarrow \text{Sr}(g)$	+180
$\text{Sr}(g) \longrightarrow \text{Sr}^{2+}(g) + 2e^-$	+1468
$\text{O}_2(g) \longrightarrow 2\text{O}(g)$	+500
$\text{O}(g) + 2e^- \longrightarrow \text{O}^{2-}(g)$	+600
$\text{Sr}(s) + \frac{1}{2}\text{O}_2(g) \longrightarrow \text{SrO}(s)$	-550

**Question 7 10 points**

Consider the molecule given on the left, and answer questions (i) to (vi) using chemical bonding theories that we have learnt in class (valence bond theory, hybridization).

Note that the lone pairs are not shown in the structure:

i. Consider the carbon atoms marked in as ***a***, ***b***, ***c***, ***d***, and ***e*** in the molecule, and determine **hybridization** at each of the carbon centers. (2.5 points)

Atom	Hybridization
Carbon (<i>a</i>)	
Carbon (<i>b</i>)	
Carbon (<i>c</i>)	
Carbon (<i>d</i>)	
Carbon (<i>e</i>)	

ii. Now, consider the sulfur atom (S) marked as ***f***, nitrogen (N) atom (***g***), and oxygen(O) atoms (***h*** and ***i***), and determine their **hybridization**. (2 points)

Atom	Hybridization
Sulfur (<i>f</i>)	
Nitrogen (<i>g</i>)	
Oxygen (<i>h</i>)	
Oxygen (<i>i</i>)	

iii. **Describe** all the bonds formed by the carbon atoms ***a*** and ***b***, using sigma and pi designation as appropriate. (2 points)

Atom	Bonds formed (use sigma and pi as appropriate)
Carbon (<i>a</i>)	
Carbon (<i>b</i>)	

iv. How many total sp^3 -s sigma bonds are in the molecule? (1 point)

v. Determine **electronic** and **molecular geometry** at carbon (C) labeled ***e*** and sulfur (S) labeled ***f*** centres. (2 points)

Atom	Electronic Geometry	Molecular Geometry
Carbon (<i>e</i>)		
Sulfur (<i>f</i>)		

vi. Circle the value from the choices given below, that describes the **C-S-C bond angle at sulfur** (S) marked ***f***: (0.5 points)

>90° to <109.5°

109.5°

>109.5 to <120°

120°

>120° to <180°


Question 8 12 points

Using the **valence shell orbitals only** draw a **complete** molecular orbital (MO) diagram (with no s-p mixing) for the molecule **PS**, in the space provided below. **(6 points)**

Assume that 3s and 3p atomic orbitals will form molecular orbitals like those formed by the combination of 2s and 2p atomic orbitals that we learnt in class, and there is no s-p mixing.

Show **energy arrow**, **atomic orbitals** involved, **atom labels**, the **molecular orbitals** that are formed, **filled electrons** in the MO diagram etc. Make sure to label all the atomic and molecular orbitals.

Using the MO diagram above, answer the following questions:

- (i) Which one of the following will have the highest bond order? Circle the correct choice
 A) PS^{1-} B) PS C) PS^{1+}
- (ii) Which one of the following will have the longest bond length? Circle the correct choice
 A) PS^{1-} B) PS C) PS^{1+}
- (iii) Which of the following will be diamagnetic? Circle the correct choice.
 A) PS^{1-} B) PS C) PS^{1+}
- (iv) Draw Lewis dot structure of the molecule with the shortest bond length (based on the MO diagram) from among the three (PS^{1-} , PS, PS^{1+}), and indicate the bond order calculated from the Lewis dot structure. **(3 points)**

Lewis Dot Structure

Bond Order



B2240AF6-DAE6-4CED-A51E-0C64655F4ABE

midterm-exam-93388

#274 Page 10 of 10

SCRAP PAGE

**CHEMISTRY 110 –Midterm Exam****Thursday November 7, 2024, 6:30 – 8:30 pm**

Examiners: Pallavi Sirjoosingh and Ashok Kakkar

First name (please write as legibly as possible within the boxes)

--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--

Last name

--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--

Student ID number

--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--

This booklet has 8 questions on 10 pages (The last page can be used as a scrap page).

The staple must NOT be removed from the question booklet. Your intact question booklet must be returned at the end of the exam.

A separate booklet contains a periodic table, and datasheet.

Write your answers directly under the questions in the space provided. Do not write on the QR codes or near the top edges (as it may be cut off).

Please write **legibly and dark enough** for the scan to be readable. For students who use English as a second language, dictionaries are permitted. Calculators are allowed.

DO NOT ASK QUESTIONS OF THE INVIGILATORS about interpretations of the exam questions. If you have a concern about a question, please summarize it at the top of this page. All concerns will be evaluated after the exam. GOOD LUCK!

**Question 1 8 points** (*Assume all transitions are possible*)

a. 6 points Calculate the **energy (in kJ/mol)** required to ionize (completely remove) the electron from the **n=3** level of **He⁺ ion** (He: Helium). Show your work including equations. (3 sig. figs.)

b. 2 points

(I) Select the largest (longest) wavelength from the three choices below:

- i. Wavelength of photon to *completely remove* an electron from *1s orbital of Hydrogen atom*
- ii. Wavelength of photon to *completely remove* an electron from *1s orbital of Boron atom (B)*
- iii. Wavelength of photon to *completely remove* an electron from *1s orbital of Boron cation (B⁺)*

i

or

ii

or

iii

(II) Select the smallest (shortest) wavelength from the three choices below (*for H atom*):

- i. Wavelength of photon *emitted* for an electron transitioning from *2p to 1s orbital*
- ii. Wavelength of photon *absorbed* for an electron transitioning from *2s to 4p orbital*
- iii. Wavelength of photon *absorbed* for an electron transitioning from *1s to 3p orbital*

i

or

ii

or

iii



Question 2 8 points



A Chem110 student plots the graph shown on the left, from data obtained for the kinetic energy (eV) of the ejected electron versus frequency (s^{-1}) of the incident light, for an unknown metal. ($1\text{eV} = 1.6022 \times 10^{-19} \text{ J}$)

Based on the plot, answer the questions **a to c** below:

a. 1 point On the plot above, label the **threshold frequency**.

b. 2 points What is the value of the work potential (in J) of the unknown metal?

c. 3 points What is the *longest* wavelength (in nm) required to eject an electron from this metal's surface? Show your work including equation(s) used. (3 sig figs)

d. 2 points In the space provided below, plot the following

(Assume frequency is *above* threshold frequency):

- Kinetic energy* of an ejected electron versus the *intensity* of the incident light (left)
- Number of electrons* ejected versus the *frequency* of the incident light (right)



**Question 3** **6 points****a. 2 points**

In the ground state of Ge (Germanium; $Z = 32$) atom, **name** the following orbitals:

i. Orbital containing the electron that requires the maximum energy to be removed.

ii. Orbital containing the electron that experiences the minimum effective nuclear charge.

b. 2 points

An unknown anion (A^-) has the following *excited* electronic configuration: $[Ar] 4s^2 3d^3 4p^1$

(Ar: Argon; $Z = 18$)

Give the **condensed ground state electronic configuration** of the atom, **A** and the cation, **A^{2+}** .
Show your work. (Assume all electronic transitions are possible)

Electronic Configuration of A:

Electronic Configuration of A^{2+} :

c. 2 points

In a single atom, determine how many electrons can have the following set of quantum numbers?

(i) $n = 5; l = 3$

In a single atom, determine how many electrons can have the following set of quantum numbers?

(i) $n = 3; l = 2; m_s = +1/2$

**Question 4** **8 points**

Rank the following from 1 to 4 in **increasing** order. Provide the ranking in the boxes provided.

Note: Atomic number (Z) are provided for relevant atoms

a) Increasing (lowest to highest) order of *ionization energy*: (1= lowest; 4 = highest)

Mg

Mg: Z = 12

Rb

Rb: Z = 37

Cs

Cs: Z = 55

K

K : Z = 19

b) Increasing (lowest to highest) order of *bond angle*: (1= lowest; 4 = highest)

H₂OCCl₄XeF₄

HCN

c) Increasing (smallest to largest) order of *ionic size* (1= smallest; 4 = largest)

S²⁻

S: Z = 16

O²⁻

O: Z = 8

Be²⁺

Be: Z = 4

Na⁺

Na: Z = 11

d) Increasing (lowest to highest) order of *bond energy* (1= lowest; 4 = highest)

P-F

P-Cl

P-I

P-Br

**Question 5** **12 points**

a. 9 points For the molecules HClO_4 and TeF_6 , draw the most stable *Lewis structure* showing appropriate shape and *dashed/wedged bonds*, based on VSEPR. Determine the **molecular geometry** around the central atom. For HClO_4 , label the *most polar bond*.

HClO_4 (there are no O-O bonds)

Label the most polar bond

Molecular Geometry around Cl

TeF_6

Molecular Geometry around Te

b. 3 points (Note: Lone pairs are not shown in structures below)

(i) For the three *resonance* structures (1, 2 and 3) shown below, answer the following question:



(I) Circle the *most contributing* resonance structure.

Structure 1

Structure 2

Structure 3

(ii) For the three structures given (4, 5, and 6) below, answer the following questions:



(I) Circle all the structures that represent *resonance structures* of each other:

Structure 4

Structure 5

Structure 6

(II) Among the resonance structures, circle the *most contributing* resonance structure.

Structure 4

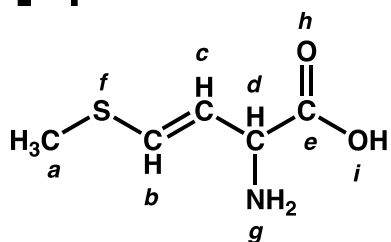
Structure 5

Structure 6

**Question 6** **6 points**

Using the data given below, calculate the **lattice energy** for SrO(s).

	ΔH° (kJ/mol)
$\text{Sr}(s) \longrightarrow \text{Sr}(g)$	+180
$\text{Sr}(g) \longrightarrow \text{Sr}^{2+}(g) + 2e^-$	+1468
$\text{O}_2(g) \longrightarrow 2\text{O}(g)$	+500
$\text{O}(g) + 2e^- \longrightarrow \text{O}^{2-}(g)$	+600
$\text{Sr}(s) + \frac{1}{2}\text{O}_2(g) \longrightarrow \text{SrO}(s)$	-550

**Question 7 10 points**

Consider the molecule given on the left, and answer questions (i) to (vi) using chemical bonding theories that we have learnt in class (valence bond theory, hybridization).

Note that the lone pairs are not shown in the structure:

i. Consider the carbon atoms marked in as ***a***, ***b***, ***c***, ***d***, and ***e*** in the molecule, and determine **hybridization** at each of the carbon centers. (2.5 points)

Atom	Hybridization
Carbon (<i>a</i>)	
Carbon (<i>b</i>)	
Carbon (<i>c</i>)	
Carbon (<i>d</i>)	
Carbon (<i>e</i>)	

ii. Now, consider the sulfur atom (S) marked as ***f***, nitrogen (N) atom (***g***), and oxygen(O) atoms (***h*** and ***i***), and determine their **hybridization**. (2 points)

Atom	Hybridization
Sulfur (<i>f</i>)	
Nitrogen (<i>g</i>)	
Oxygen (<i>h</i>)	
Oxygen (<i>i</i>)	

iii. **Describe** all the bonds formed by the carbon atoms ***a*** and ***b***, using sigma and pi designation as appropriate. (2 points)

Atom	Bonds formed (use sigma and pi as appropriate)
Carbon (<i>a</i>)	
Carbon (<i>b</i>)	

iv. How many total sp^3 -s sigma bonds are in the molecule? (1 point)

v. Determine **electronic** and **molecular geometry** at carbon (C) labeled ***e*** and sulfur (S) labeled ***f*** centres. (2 points)

Atom	Electronic Geometry	Molecular Geometry
Carbon (<i>e</i>)		
Sulfur (<i>f</i>)		

vi. Circle the value from the choices given below, that describes the **C-S-C bond angle at sulfur** (S) marked ***f***: (0.5 points)

>90° to <109.5° 109.5° >109.5 to <120° 120° >120° to <180°


Question 8 12 points

Using the **valence shell orbitals only** draw a **complete** molecular orbital (MO) diagram (with no s-p mixing) for the molecule **PS**, in the space provided below. **(6 points)**

Assume that 3s and 3p atomic orbitals will form molecular orbitals like those formed by the combination of 2s and 2p atomic orbitals that we learnt in class, and there is no s-p mixing.

Show **energy arrow**, **atomic orbitals** involved, **atom labels**, the **molecular orbitals** that are formed, **filled electrons** in the MO diagram etc. Make sure to label all the atomic and molecular orbitals.

Using the MO diagram above, answer the following questions:

- (i) Which one of the following will have the highest bond order? Circle the correct choice
 A) PS^{1-} B) PS C) PS^{1+}
- (ii) Which one of the following will have the longest bond length? Circle the correct choice
 A) PS^{1-} B) PS C) PS^{1+}
- (iii) Which of the following will be diamagnetic? Circle the correct choice.
 A) PS^{1-} B) PS C) PS^{1+}
- (iv) Draw Lewis dot structure of the molecule with the shortest bond length (based on the MO diagram) from among the three (PS^{1-} , PS, PS^{1+}), and indicate the bond order calculated from the Lewis dot structure. **(3 points)**

Lewis Dot Structure

Bond Order



CBBC6A83-13CF-479A-B8EA-BAE740B4046E

midterm-exam-93388

#275 Page 10 of 10

SCRAP PAGE

**CHEMISTRY 110 –Midterm Exam****Thursday November 7, 2024, 6:30 – 8:30 pm****Examiners: Pallavi Sirjoosingh and Ashok Kakkar**

First name (please write as legibly as possible within the boxes)

--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--

Last name

--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--

Student ID number

--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--

This booklet has 8 questions on 10 pages (The last page can be used as a scrap page).

The staple must NOT be removed from the question booklet. Your intact question booklet must be returned at the end of the exam.

A separate booklet contains a periodic table, and datasheet.

Write your answers directly under the questions in the space provided. Do not write on the QR codes or near the top edges (as it may be cut off).

Please write **legibly and dark enough** for the scan to be readable. For students who use English as a second language, dictionaries are permitted. Calculators are allowed.

DO NOT ASK QUESTIONS OF THE INVIGILATORS about interpretations of the exam questions. If you have a concern about a question, please summarize it at the top of this page. All concerns will be evaluated after the exam. GOOD LUCK!

**Question 1 8 points** (*Assume all transitions are possible*)

a. 6 points Calculate the **energy (in kJ/mol)** required to ionize (completely remove) the electron from the **n=3** level of **He⁺ ion** (He: Helium). Show your work including equations. (3 sig. figs.)

b. 2 points

(I) Select the largest (longest) wavelength from the three choices below:

- i. Wavelength of photon to *completely remove* an electron from *1s orbital of Hydrogen atom*
- ii. Wavelength of photon to *completely remove* an electron from *1s orbital of Boron atom (B)*
- iii. Wavelength of photon to *completely remove* an electron from *1s orbital of Boron cation (B⁺)*

i or ii or iii

(II) Select the smallest (shortest) wavelength from the three choices below (*for H atom*):

- i. Wavelength of photon *emitted* for an electron transitioning from *2p to 1s orbital*
- ii. Wavelength of photon *absorbed* for an electron transitioning from *2s to 4p orbital*
- iii. Wavelength of photon *absorbed* for an electron transitioning from *1s to 3p orbital*

i or ii or iii



Question 2 8 points



A Chem110 student plots the graph shown on the left, from data obtained for the kinetic energy (eV) of the ejected electron versus frequency (s^{-1}) of the incident light, for an unknown metal. ($1\text{eV} = 1.6022 \times 10^{-19} \text{ J}$)

Based on the plot, answer the questions **a to c** below:

a. 1 point On the plot above, label the **threshold frequency**.

b. 2 points What is the value of the work potential (in J) of the unknown metal?

c. 3 points What is the *longest* wavelength (in nm) required to eject an electron from this metal's surface? Show your work including equation(s) used. (3 sig figs)

d. 2 points In the space provided below, plot the following

(Assume frequency is *above* threshold frequency):

- Kinetic energy* of an ejected electron versus the *intensity* of the incident light (left)
- Number of electrons* ejected versus the *frequency* of the incident light (right)



**Question 3** **6 points****a. 2 points**

In the ground state of Ge (Germanium; $Z = 32$) atom, **name** the following orbitals:

i. Orbital containing the electron that requires the maximum energy to be removed.

ii. Orbital containing the electron that experiences the minimum effective nuclear charge.

b. 2 points

An unknown anion (A^-) has the following *excited* electronic configuration: $[Ar] 4s^2 3d^3 4p^1$

(Ar: Argon; $Z = 18$)

Give the **condensed ground state electronic configuration** of the atom, **A** and the cation, **A^{2+}** .
Show your work. (Assume all electronic transitions are possible)

Electronic Configuration of A:

Electronic Configuration of A^{2+} :

c. 2 points

In a single atom, determine how many electrons can have the following set of quantum numbers?

(i) $n = 5; l = 3$

In a single atom, determine how many electrons can have the following set of quantum numbers?

(i) $n = 3; l = 2; m_s = +1/2$

**Question 4** **8 points**

Rank the following from 1 to 4 in **increasing** order. Provide the ranking in the boxes provided.

Note: Atomic number (Z) are provided for relevant atoms

a) Increasing (lowest to highest) order of *ionization energy*: (1= lowest; 4 = highest)

Mg

Mg: Z = 12

Rb

Rb: Z = 37

Cs

Cs: Z = 55

K

K : Z = 19

b) Increasing (lowest to highest) order of *bond angle*: (1= lowest; 4 = highest)

H₂OCCl₄XeF₄

HCN

c) Increasing (smallest to largest) order of *ionic size* (1= smallest; 4 = largest)

S²⁻

S: Z = 16

O²⁻

O: Z = 8

Be²⁺

Be: Z = 4

Na⁺

Na: Z = 11

d) Increasing (lowest to highest) order of *bond energy* (1= lowest; 4 = highest)

P-F

P-Cl

P-I

P-Br

**Question 5** **12 points**

a. 9 points For the molecules HClO_4 and TeF_6 , draw the most stable *Lewis structure* showing appropriate shape and *dashed/wedged bonds*, based on VSEPR. Determine the **molecular geometry** around the central atom. For HClO_4 , label the *most polar bond*.



b. 3 points (Note: Lone pairs are not shown in structures below)

(i) For the three *resonance* structures (1, 2 and 3) shown below, answer the following question:



(I) Circle the *most contributing* resonance structure.

Structure 1

Structure 2

Structure 3

(ii) For the three structures given (4, 5, and 6) below, answer the following questions:



(I) Circle all the structures that represent *resonance structures* of each other:

Structure 4

Structure 5

Structure 6

(II) Among the resonance structures, circle the *most contributing* resonance structure.

Structure 4

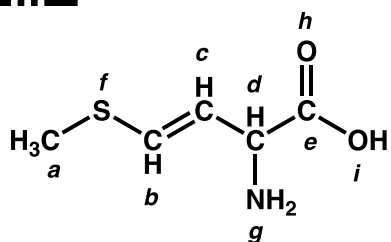
Structure 5

Structure 6

**Question 6** **6 points**

Using the data given below, calculate the **lattice energy** for SrO(s).

	ΔH° (kJ/mol)
$\text{Sr}(s) \longrightarrow \text{Sr}(g)$	+180
$\text{Sr}(g) \longrightarrow \text{Sr}^{2+}(g) + 2e^-$	+1468
$\text{O}_2(g) \longrightarrow 2\text{O}(g)$	+500
$\text{O}(g) + 2e^- \longrightarrow \text{O}^{2-}(g)$	+600
$\text{Sr}(s) + \frac{1}{2}\text{O}_2(g) \longrightarrow \text{SrO}(s)$	-550

**Question 7 10 points**

Consider the molecule given on the left, and answer questions (i) to (vi) using chemical bonding theories that we have learnt in class (valence bond theory, hybridization).

Note that the lone pairs are not shown in the structure:

i. Consider the carbon atoms marked in as ***a***, ***b***, ***c***, ***d***, and ***e*** in the molecule, and determine **hybridization** at each of the carbon centers. (2.5 points)

Atom	Hybridization
Carbon (<i>a</i>)	
Carbon (<i>b</i>)	
Carbon (<i>c</i>)	
Carbon (<i>d</i>)	
Carbon (<i>e</i>)	

ii. Now, consider the sulfur atom (S) marked as ***f***, nitrogen (N) atom (***g***), and oxygen(O) atoms (***h*** and ***i***), and determine their **hybridization**. (2 points)

Atom	Hybridization
Sulfur (<i>f</i>)	
Nitrogen (<i>g</i>)	
Oxygen (<i>h</i>)	
Oxygen (<i>i</i>)	

iii. **Describe** all the bonds formed by the carbon atoms ***a*** and ***b***, using sigma and pi designation as appropriate. (2 points)

Atom	Bonds formed (use sigma and pi as appropriate)
Carbon (<i>a</i>)	
Carbon (<i>b</i>)	

iv. How many total sp^3 -s sigma bonds are in the molecule? (1 point)

v. Determine **electronic** and **molecular geometry** at carbon (C) labeled ***e*** and sulfur (S) labeled ***f*** centres. (2 points)

Atom	Electronic Geometry	Molecular Geometry
Carbon (<i>e</i>)		
Sulfur (<i>f</i>)		

vi. Circle the value from the choices given below, that describes the **C-S-C bond angle at sulfur** (S) marked ***f***: (0.5 points)

>90° to <109.5° 109.5° >109.5 to <120° 120° >120° to <180°


Question 8 12 points

Using the **valence shell orbitals only** draw a **complete** molecular orbital (MO) diagram (with no s-p mixing) for the molecule **PS**, in the space provided below. **(6 points)**

Assume that 3s and 3p atomic orbitals will form molecular orbitals like those formed by the combination of 2s and 2p atomic orbitals that we learnt in class, and there is no s-p mixing.

Show **energy arrow**, **atomic orbitals** involved, **atom labels**, the **molecular orbitals** that are formed, **filled electrons** in the MO diagram etc. Make sure to label all the atomic and molecular orbitals.

Using the MO diagram above, answer the following questions:

- (i) Which one of the following will have the highest bond order? Circle the correct choice
 A) PS^{1-} B) PS C) PS^{1+}
- (ii) Which one of the following will have the longest bond length? Circle the correct choice
 A) PS^{1-} B) PS C) PS^{1+}
- (iii) Which of the following will be diamagnetic? Circle the correct choice.
 A) PS^{1-} B) PS C) PS^{1+}
- (iv) Draw Lewis dot structure of the molecule with the shortest bond length (based on the MO diagram) from among the three (PS^{1-} , PS, PS^{1+}), and indicate the bond order calculated from the Lewis dot structure. **(3 points)**

Lewis Dot Structure

Bond Order



50C809B4-3E08-405E-B894-B94C4412A356

midterm-exam-93388

#276 Page 10 of 10

SCRAP PAGE

**CHEMISTRY 110 –Midterm Exam****Thursday November 7, 2024, 6:30 – 8:30 pm**

Examiners: Pallavi Sirjoosingh and Ashok Kakkar

First name (please write as legibly as possible within the boxes)

--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--

Last name

--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--

Student ID number

--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--

This booklet has 8 questions on 10 pages (The last page can be used as a scrap page).

The staple must NOT be removed from the question booklet. Your intact question booklet must be returned at the end of the exam.

A separate booklet contains a periodic table, and datasheet.

Write your answers directly under the questions in the space provided. Do not write on the QR codes or near the top edges (as it may be cut off).

Please write **legibly and dark enough** for the scan to be readable. For students who use English as a second language, dictionaries are permitted. Calculators are allowed.

DO NOT ASK QUESTIONS OF THE INVIGILATORS about interpretations of the exam questions. If you have a concern about a question, please summarize it at the top of this page. All concerns will be evaluated after the exam. GOOD LUCK!

**Question 1 8 points** (*Assume all transitions are possible*)

a. 6 points Calculate the **energy (in kJ/mol)** required to ionize (completely remove) the electron from the **$n=3$** level of **He^+ ion** (He: Helium). Show your work including equations. (3 sig. figs.)

b. 2 points

(I) Select the largest (longest) wavelength from the three choices below:

- i. Wavelength of photon to *completely remove* an electron from *1s orbital of Hydrogen atom*
- ii. Wavelength of photon to *completely remove* an electron from *1s orbital of Boron atom (B)*
- iii. Wavelength of photon to *completely remove* an electron from *1s orbital of Boron cation (B^+)*

i or ii or iii

(II) Select the smallest (shortest) wavelength from the three choices below (*for H atom*):

- i. Wavelength of photon *emitted* for an electron transitioning from *2p to 1s orbital*
- ii. Wavelength of photon *absorbed* for an electron transitioning from *2s to 4p orbital*
- iii. Wavelength of photon *absorbed* for an electron transitioning from *1s to 3p orbital*

i or ii or iii



Question 2 8 points



A Chem110 student plots the graph shown on the left, from data obtained for the kinetic energy (eV) of the ejected electron versus frequency (s^{-1}) of the incident light, for an unknown metal. ($1\text{eV} = 1.6022 \times 10^{-19} \text{ J}$)

Based on the plot, answer the questions **a to c** below:

a. 1 point On the plot above, label the **threshold frequency**.

b. 2 points What is the value of the work potential (in J) of the unknown metal?

c. 3 points What is the *longest* wavelength (in nm) required to eject an electron from this metal's surface? Show your work including equation(s) used. (3 sig figs)

d. 2 points In the space provided below, plot the following

(Assume frequency is *above* threshold frequency):

- Kinetic energy* of an ejected electron versus the *intensity* of the incident light (left)
- Number of electrons* ejected versus the *frequency* of the incident light (right)



**Question 3** **6 points****a. 2 points**

In the ground state of Ge (Germanium; $Z = 32$) atom, **name** the following orbitals:

i. Orbital containing the electron that requires the maximum energy to be removed.

ii. Orbital containing the electron that experiences the minimum effective nuclear charge.

b. 2 points

An unknown anion (A^-) has the following *excited* electronic configuration: $[Ar] 4s^2 3d^3 4p^1$

(Ar: Argon; $Z = 18$)

Give the **condensed ground state electronic configuration** of the atom, **A** and the cation, **A^{2+}** .
Show your work. (Assume all electronic transitions are possible)

Electronic Configuration of A:

Electronic Configuration of A^{2+} :

c. 2 points

In a single atom, determine how many electrons can have the following set of quantum numbers?

(i) $n = 5; l = 3$

In a single atom, determine how many electrons can have the following set of quantum numbers?

(i) $n = 3; l = 2; m_s = +1/2$

**Question 4** **8 points**

Rank the following from 1 to 4 in **increasing** order. Provide the ranking in the boxes provided.

Note: Atomic number (Z) are provided for relevant atoms

a) Increasing (lowest to highest) order of *ionization energy*: (1= lowest; 4 = highest)

Mg

Mg: Z = 12

Rb

Rb: Z = 37

Cs

Cs: Z = 55

K

K : Z = 19

b) Increasing (lowest to highest) order of *bond angle*: (1= lowest; 4 = highest)

H₂OCCl₄XeF₄

HCN

c) Increasing (smallest to largest) order of *ionic size* (1= smallest; 4 = largest)

S²⁻

S: Z = 16

O²⁻

O: Z = 8

Be²⁺

Be: Z = 4

Na⁺

Na: Z = 11

d) Increasing (lowest to highest) order of *bond energy* (1= lowest; 4 = highest)

P-F

P-Cl

P-I

P-Br

**Question 5** **12 points**

a. 9 points For the molecules HClO_4 and TeF_6 , draw the most stable *Lewis structure* showing appropriate shape and *dashed/wedged bonds*, based on VSEPR. Determine the **molecular geometry** around the central atom. For HClO_4 , label the *most polar bond*.



b. 3 points (Note: Lone pairs are not shown in structures below)

(i) For the three *resonance* structures (1, 2 and 3) shown below, answer the following question:



(I) Circle the *most contributing* resonance structure.

Structure 1

Structure 2

Structure 3

(ii) For the three structures given (4, 5, and 6) below, answer the following questions:



(I) Circle all the structures that represent *resonance structures* of each other:

Structure 4

Structure 5

Structure 6

(II) Among the resonance structures, circle the *most contributing* resonance structure.

Structure 4

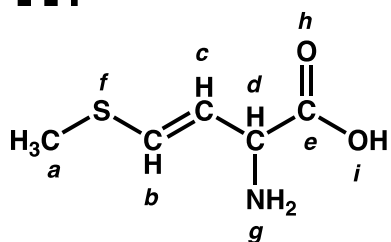
Structure 5

Structure 6

**Question 6** **6 points**

Using the data given below, calculate the **lattice energy** for SrO(s).

	ΔH° (kJ/mol)
$\text{Sr}(s) \longrightarrow \text{Sr}(g)$	+180
$\text{Sr}(g) \longrightarrow \text{Sr}^{2+}(g) + 2e^-$	+1468
$\text{O}_2(g) \longrightarrow 2\text{O}(g)$	+500
$\text{O}(g) + 2e^- \longrightarrow \text{O}^{2-}(g)$	+600
$\text{Sr}(s) + \frac{1}{2}\text{O}_2(g) \longrightarrow \text{SrO}(s)$	-550

**Question 7 10 points**

Consider the molecule given on the left, and answer questions (i) to (vi) using chemical bonding theories that we have learnt in class (valence bond theory, hybridization).

Note that the lone pairs are not shown in the structure:

i. Consider the carbon atoms marked in as ***a***, ***b***, ***c***, ***d***, and ***e*** in the molecule, and determine **hybridization** at each of the carbon centers. (2.5 points)

Atom	Hybridization
Carbon (<i>a</i>)	
Carbon (<i>b</i>)	
Carbon (<i>c</i>)	
Carbon (<i>d</i>)	
Carbon (<i>e</i>)	

ii. Now, consider the sulfur atom (S) marked as ***f***, nitrogen (N) atom (***g***), and oxygen(O) atoms (***h*** and ***i***), and determine their **hybridization**. (2 points)

Atom	Hybridization
Sulfur (<i>f</i>)	
Nitrogen (<i>g</i>)	
Oxygen (<i>h</i>)	
Oxygen (<i>i</i>)	

iii. **Describe** all the bonds formed by the carbon atoms ***a*** and ***b***, using sigma and pi designation as appropriate. (2 points)

Atom	Bonds formed (use sigma and pi as appropriate)
Carbon (<i>a</i>)	
Carbon (<i>b</i>)	

iv. How many total sp^3 -s sigma bonds are in the molecule? (1 point)

v. Determine **electronic** and **molecular geometry** at carbon (C) labeled ***e*** and sulfur (S) labeled ***f*** centres. (2 points)

Atom	Electronic Geometry	Molecular Geometry
Carbon (<i>e</i>)		
Sulfur (<i>f</i>)		

vi. Circle the value from the choices given below, that describes the **C-S-C bond angle at sulfur** (S) marked ***f***: (0.5 points)

$>90^\circ$ to $<109.5^\circ$ 109.5° >109.5 to $<120^\circ$ 120° $>120^\circ$ to $<180^\circ$


Question 8 12 points

Using the **valence shell orbitals only** draw a **complete** molecular orbital (MO) diagram (with no s-p mixing) for the molecule **PS**, in the space provided below. **(6 points)**

Assume that 3s and 3p atomic orbitals will form molecular orbitals like those formed by the combination of 2s and 2p atomic orbitals that we learnt in class, and there is no s-p mixing.

Show **energy arrow**, **atomic orbitals** involved, **atom labels**, the **molecular orbitals** that are formed, **filled electrons** in the MO diagram etc. Make sure to label all the atomic and molecular orbitals.

Using the MO diagram above, answer the following questions:

- (i) Which one of the following will have the highest bond order? Circle the correct choice
 A) PS^{1-} B) PS C) PS^{1+}
- (ii) Which one of the following will have the longest bond length? Circle the correct choice
 A) PS^{1-} B) PS C) PS^{1+}
- (iii) Which of the following will be diamagnetic? Circle the correct choice.
 A) PS^{1-} B) PS C) PS^{1+}
- (iv) Draw Lewis dot structure of the molecule with the shortest bond length (based on the MO diagram) from among the three (PS^{1-} , PS, PS^{1+}), and indicate the bond order calculated from the Lewis dot structure. **(3 points)**

Lewis Dot Structure

Bond Order



6BAE965E-77BB-4E3D-982B-5F3E1E5D9A47

midterm-exam-93388

#277 Page 10 of 10

SCRAP PAGE

**CHEMISTRY 110 –Midterm Exam****Thursday November 7, 2024, 6:30 – 8:30 pm**

Examiners: Pallavi Sirjoosingh and Ashok Kakkar

First name (please write as legibly as possible within the boxes)

--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--

Last name

--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--

Student ID number

--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--

This booklet has 8 questions on 10 pages (The last page can be used as a scrap page).

The staple must NOT be removed from the question booklet. Your intact question booklet must be returned at the end of the exam.

A separate booklet contains a periodic table, and datasheet.

Write your answers directly under the questions in the space provided. Do not write on the QR codes or near the top edges (as it may be cut off).

Please write **legibly and dark enough** for the scan to be readable. For students who use English as a second language, dictionaries are permitted. Calculators are allowed.

DO NOT ASK QUESTIONS OF THE INVIGILATORS about interpretations of the exam questions. If you have a concern about a question, please summarize it at the top of this page. All concerns will be evaluated after the exam. GOOD LUCK!

**Question 1 8 points** (*Assume all transitions are possible*)

a. 6 points Calculate the **energy (in kJ/mol)** required to ionize (completely remove) the electron from the **n=3** level of **He⁺ ion** (He: Helium). Show your work including equations. (3 sig. figs.)

b. 2 points

(I) Select the largest (longest) wavelength from the three choices below:

- i. Wavelength of photon to *completely remove* an electron from *1s orbital of Hydrogen atom*
- ii. Wavelength of photon to *completely remove* an electron from *1s orbital of Boron atom (B)*
- iii. Wavelength of photon to *completely remove* an electron from *1s orbital of Boron cation (B⁺)*

i or ii or iii

(II) Select the smallest (shortest) wavelength from the three choices below (*for H atom*):

- i. Wavelength of photon *emitted* for an electron transitioning from *2p to 1s orbital*
- ii. Wavelength of photon *absorbed* for an electron transitioning from *2s to 4p orbital*
- iii. Wavelength of photon *absorbed* for an electron transitioning from *1s to 3p orbital*

i or ii or iii



Question 2 8 points



A Chem110 student plots the graph shown on the left, from data obtained for the kinetic energy (eV) of the ejected electron versus frequency (s^{-1}) of the incident light, for an unknown metal. ($1\text{eV} = 1.6022 \times 10^{-19} \text{ J}$)

Based on the plot, answer the questions **a to c** below:

a. 1 point On the plot above, label the **threshold frequency**.

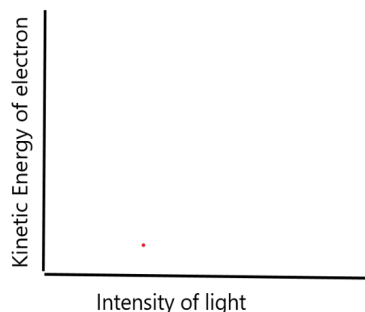
b. 2 points What is the value of the work potential (in J) of the unknown metal?

c. 3 points What is the *longest* wavelength (in nm) required to eject an electron from this metal's surface? Show your work including equation(s) used. (3 sig figs)

d. 2 points In the space provided below, plot the following

(Assume frequency is *above* threshold frequency):

- i. *Kinetic energy* of an ejected electron versus the *intensity* of the incident light (left)
- ii. *Number of electrons* ejected versus the *frequency* of the incident light (right)



**Question 3** **6 points****a. 2 points**

In the ground state of Ge (Germanium; $Z = 32$) atom, **name** the following orbitals:

i. Orbital containing the electron that requires the maximum energy to be removed.

ii. Orbital containing the electron that experiences the minimum effective nuclear charge.

b. 2 points

An unknown anion (A^-) has the following *excited* electronic configuration: $[Ar] 4s^2 3d^3 4p^1$

(Ar: Argon; $Z = 18$)

Give the **condensed ground state electronic configuration** of the atom, **A** and the cation, **A^{2+}** .
Show your work. (Assume all electronic transitions are possible)

Electronic Configuration of A:

Electronic Configuration of A^{2+} :

c. 2 points

In a single atom, determine how many electrons can have the following set of quantum numbers?

(i) $n = 5; l = 3$

In a single atom, determine how many electrons can have the following set of quantum numbers?

(i) $n = 3; l = 2; m_s = +1/2$

**Question 4** **8 points**

Rank the following from 1 to 4 in **increasing** order. Provide the ranking in the boxes provided.

Note: Atomic number (Z) are provided for relevant atoms

a) Increasing (lowest to highest) order of *ionization energy*: (1= lowest; 4 = highest)

Mg

Mg: Z = 12

Rb

Rb: Z = 37

Cs

Cs: Z = 55

K

K : Z = 19

b) Increasing (lowest to highest) order of *bond angle*: (1= lowest; 4 = highest)

H₂OCCl₄XeF₄

HCN

c) Increasing (smallest to largest) order of *ionic size* (1= smallest; 4 = largest)

S²⁻

S: Z = 16

O²⁻

O: Z = 8

Be²⁺

Be: Z = 4

Na⁺

Na: Z = 11

d) Increasing (lowest to highest) order of *bond energy* (1= lowest; 4 = highest)

P-F

P-Cl

P-I

P-Br

**Question 5** **12 points**

a. 9 points For the molecules HClO_4 and TeF_6 , draw the most stable *Lewis structure* showing appropriate shape and *dashed/wedged bonds*, based on VSEPR. Determine the **molecular geometry** around the central atom. For HClO_4 , label the *most polar bond*.

HClO_4 (there are no O-O bonds)

Label the most polar bond

Molecular Geometry around Cl

TeF_6

Molecular Geometry around Te

b. 3 points (Note: Lone pairs are not shown in structures below)

(i) For the three *resonance* structures (1, 2 and 3) shown below, answer the following question:



(I) Circle the *most contributing* resonance structure.

Structure 1

Structure 2

Structure 3

(ii) For the three structures given (4, 5, and 6) below, answer the following questions:



(I) Circle all the structures that represent *resonance structures* of each other:

Structure 4

Structure 5

Structure 6

(II) Among the resonance structures, circle the *most contributing* resonance structure.

Structure 4

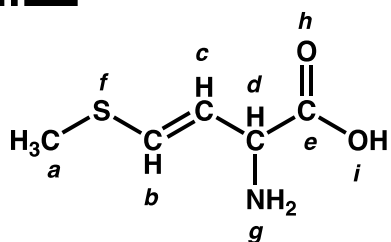
Structure 5

Structure 6

**Question 6** **6 points**

Using the data given below, calculate the **lattice energy** for SrO(s).

	ΔH° (kJ/mol)
$\text{Sr}(s) \longrightarrow \text{Sr}(g)$	+180
$\text{Sr}(g) \longrightarrow \text{Sr}^{2+}(g) + 2e^-$	+1468
$\text{O}_2(g) \longrightarrow 2\text{O}(g)$	+500
$\text{O}(g) + 2e^- \longrightarrow \text{O}^{2-}(g)$	+600
$\text{Sr}(s) + \frac{1}{2}\text{O}_2(g) \longrightarrow \text{SrO}(s)$	-550

**Question 7 10 points**

Consider the molecule given on the left, and answer questions (i) to (vi) using chemical bonding theories that we have learnt in class (valence bond theory, hybridization).

Note that the lone pairs are not shown in the structure:

i. Consider the carbon atoms marked in as ***a***, ***b***, ***c***, ***d***, and ***e*** in the molecule, and determine **hybridization** at each of the carbon centers. (2.5 points)

Atom	Hybridization
Carbon (<i>a</i>)	
Carbon (<i>b</i>)	
Carbon (<i>c</i>)	
Carbon (<i>d</i>)	
Carbon (<i>e</i>)	

ii. Now, consider the sulfur atom (S) marked as ***f***, nitrogen (N) atom (***g***), and oxygen(O) atoms (***h*** and ***i***), and determine their **hybridization**. (2 points)

Atom	Hybridization
Sulfur (<i>f</i>)	
Nitrogen (<i>g</i>)	
Oxygen (<i>h</i>)	
Oxygen (<i>i</i>)	

iii. **Describe** all the bonds formed by the carbon atoms ***a*** and ***b***, using sigma and pi designation as appropriate. (2 points)

Atom	Bonds formed (use sigma and pi as appropriate)
Carbon (<i>a</i>)	
Carbon (<i>b</i>)	

iv. How many total sp^3 -s sigma bonds are in the molecule? (1 point)

v. Determine **electronic** and **molecular geometry** at carbon (C) labeled ***e*** and sulfur (S) labeled ***f*** centres. (2 points)

Atom	Electronic Geometry	Molecular Geometry
Carbon (<i>e</i>)		
Sulfur (<i>f</i>)		

vi. Circle the value from the choices given below, that describes the **C-S-C bond angle at sulfur** (S) marked ***f***: (0.5 points)

$>90^\circ$ to $<109.5^\circ$ 109.5° >109.5 to $<120^\circ$ 120° $>120^\circ$ to $<180^\circ$


Question 8 12 points

Using the **valence shell orbitals only** draw a **complete** molecular orbital (MO) diagram (with no s-p mixing) for the molecule **PS**, in the space provided below. **(6 points)**

Assume that 3s and 3p atomic orbitals will form molecular orbitals like those formed by the combination of 2s and 2p atomic orbitals that we learnt in class, and there is no s-p mixing.

Show **energy arrow**, **atomic orbitals** involved, **atom labels**, the **molecular orbitals** that are formed, **filled electrons** in the MO diagram etc. Make sure to label all the atomic and molecular orbitals.

Using the MO diagram above, answer the following questions:

- (i) Which one of the following will have the highest bond order? Circle the correct choice
 A) PS^{1-} B) PS C) PS^{1+}
- (ii) Which one of the following will have the longest bond length? Circle the correct choice
 A) PS^{1-} B) PS C) PS^{1+}
- (iii) Which of the following will be diamagnetic? Circle the correct choice.
 A) PS^{1-} B) PS C) PS^{1+}
- (iv) Draw Lewis dot structure of the molecule with the shortest bond length (based on the MO diagram) from among the three (PS^{1-} , PS, PS^{1+}), and indicate the bond order calculated from the Lewis dot structure. **(3 points)**

Lewis Dot Structure

Bond Order



D6B94264-8CA2-4BC6-8207-65ED6EB3B783

midterm-exam-93388

#278 Page 10 of 10

SCRAP PAGE

**CHEMISTRY 110 –Midterm Exam****Thursday November 7, 2024, 6:30 – 8:30 pm****Examiners: Pallavi Sirjoosingh and Ashok Kakkar**

First name (please write as legibly as possible within the boxes)

--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--

Last name

--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--

Student ID number

--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--

This booklet has 8 questions on 10 pages (The last page can be used as a scrap page).

The staple must NOT be removed from the question booklet. Your intact question booklet must be returned at the end of the exam.

A separate booklet contains a periodic table, and datasheet.

Write your answers directly under the questions in the space provided. Do not write on the QR codes or near the top edges (as it may be cut off).

Please write **legibly and dark enough** for the scan to be readable. For students who use English as a second language, dictionaries are permitted. Calculators are allowed.

DO NOT ASK QUESTIONS OF THE INVIGILATORS about interpretations of the exam questions. If you have a concern about a question, please summarize it at the top of this page. All concerns will be evaluated after the exam. GOOD LUCK!

**Question 1 8 points** (*Assume all transitions are possible*)

a. 6 points Calculate the **energy (in kJ/mol)** required to ionize (completely remove) the electron from the **$n=3$** level of **He^+ ion** (He: Helium). Show your work including equations. (3 sig. figs.)

b. 2 points

(I) Select the largest (longest) wavelength from the three choices below:

- i. Wavelength of photon to *completely remove* an electron from *1s orbital of Hydrogen atom*
- ii. Wavelength of photon to *completely remove* an electron from *1s orbital of Boron atom (B)*
- iii. Wavelength of photon to *completely remove* an electron from *1s orbital of Boron cation (B^+)*

i or ii or iii

(II) Select the smallest (shortest) wavelength from the three choices below (*for H atom*):

- i. Wavelength of photon *emitted* for an electron transitioning from *2p to 1s orbital*
- ii. Wavelength of photon *absorbed* for an electron transitioning from *2s to 4p orbital*
- iii. Wavelength of photon *absorbed* for an electron transitioning from *1s to 3p orbital*

i or ii or iii



Question 2 8 points



A Chem110 student plots the graph shown on the left, from data obtained for the kinetic energy (eV) of the ejected electron versus frequency (s^{-1}) of the incident light, for an unknown metal. ($1\text{eV} = 1.6022 \times 10^{-19} \text{ J}$)

Based on the plot, answer the questions **a to c** below:

a. 1 point On the plot above, label the **threshold frequency**.

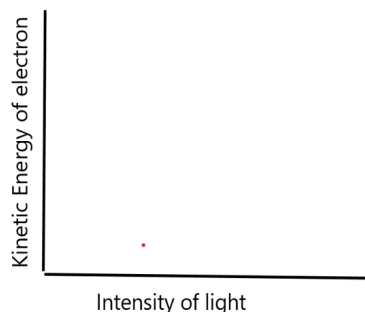
b. 2 points What is the value of the work potential (in J) of the unknown metal?

c. 3 points What is the *longest* wavelength (in nm) required to eject an electron from this metal's surface? Show your work including equation(s) used. (3 sig figs)

d. 2 points In the space provided below, plot the following

(Assume frequency is *above* threshold frequency):

- Kinetic energy* of an ejected electron versus the *intensity* of the incident light (left)
- Number of electrons* ejected versus the *frequency* of the incident light (right)



**Question 3** **6 points****a. 2 points**

In the ground state of Ge (Germanium; $Z = 32$) atom, **name** the following orbitals:

i. Orbital containing the electron that requires the maximum energy to be removed.

ii. Orbital containing the electron that experiences the minimum effective nuclear charge.

b. 2 points

An unknown anion (A^-) has the following *excited* electronic configuration: $[Ar] 4s^2 3d^3 4p^1$

(Ar: Argon; $Z = 18$)

Give the **condensed ground state electronic configuration** of the atom, **A** and the cation, **A^{2+}** .
Show your work. (Assume all electronic transitions are possible)

Electronic Configuration of A:

Electronic Configuration of A^{2+} :

c. 2 points

In a single atom, determine how many electrons can have the following set of quantum numbers?

(i) $n = 5; l = 3$

In a single atom, determine how many electrons can have the following set of quantum numbers?

(i) $n = 3; l = 2; m_s = +1/2$

**Question 4** **8 points**

Rank the following from 1 to 4 in **increasing** order. Provide the ranking in the boxes provided.

Note: Atomic number (Z) are provided for relevant atoms

a) Increasing (lowest to highest) order of *ionization energy*: (1= lowest; 4 = highest)

Mg

Mg: Z = 12

Rb

Rb: Z = 37

Cs

Cs: Z = 55

K

K : Z = 19

b) Increasing (lowest to highest) order of *bond angle*: (1= lowest; 4 = highest)

H₂OCCl₄XeF₄

HCN

c) Increasing (smallest to largest) order of *ionic size* (1= smallest; 4 = largest)

S²⁻

S: Z = 16

O²⁻

O: Z = 8

Be²⁺

Be: Z = 4

Na⁺

Na: Z = 11

d) Increasing (lowest to highest) order of *bond energy* (1= lowest; 4 = highest)

P-F

P-Cl

P-I

P-Br

**Question 5** **12 points**

a. 9 points For the molecules HClO_4 and TeF_6 , draw the most stable *Lewis structure* showing appropriate shape and *dashed/wedged bonds*, based on VSEPR. Determine the **molecular geometry** around the central atom. For HClO_4 , label the *most polar bond*.



b. 3 points (Note: Lone pairs are not shown in structures below)

(i) For the three *resonance* structures (1, 2 and 3) shown below, answer the following question:



(I) Circle the *most contributing* resonance structure.

Structure 1

Structure 2

Structure 3

(ii) For the three structures given (4, 5, and 6) below, answer the following questions:



(I) Circle all the structures that represent *resonance structures* of each other:

Structure 4

Structure 5

Structure 6

(II) Among the resonance structures, circle the *most contributing* resonance structure.

Structure 4

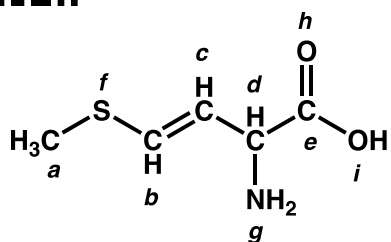
Structure 5

Structure 6

**Question 6** **6 points**

Using the data given below, calculate the **lattice energy** for SrO(s).

	ΔH° (kJ/mol)
$\text{Sr}(s) \longrightarrow \text{Sr}(g)$	+180
$\text{Sr}(g) \longrightarrow \text{Sr}^{2+}(g) + 2e^-$	+1468
$\text{O}_2(g) \longrightarrow 2\text{O}(g)$	+500
$\text{O}(g) + 2e^- \longrightarrow \text{O}^{2-}(g)$	+600
$\text{Sr}(s) + \frac{1}{2}\text{O}_2(g) \longrightarrow \text{SrO}(s)$	-550

**Question 7 10 points**

Consider the molecule given on the left, and answer questions (i) to (vi) using chemical bonding theories that we have learnt in class (valence bond theory, hybridization).

Note that the lone pairs are not shown in the structure:

i. Consider the carbon atoms marked in as ***a***, ***b***, ***c***, ***d***, and ***e*** in the molecule, and determine **hybridization** at each of the carbon centers. (2.5 points)

Atom	Hybridization
Carbon (<i>a</i>)	
Carbon (<i>b</i>)	
Carbon (<i>c</i>)	
Carbon (<i>d</i>)	
Carbon (<i>e</i>)	

ii. Now, consider the sulfur atom (S) marked as ***f***, nitrogen (N) atom (***g***), and oxygen(O) atoms (***h*** and ***i***), and determine their **hybridization**. (2 points)

Atom	Hybridization
Sulfur (<i>f</i>)	
Nitrogen (<i>g</i>)	
Oxygen (<i>h</i>)	
Oxygen (<i>i</i>)	

iii. **Describe** all the bonds formed by the carbon atoms ***a*** and ***b***, using sigma and pi designation as appropriate. (2 points)

Atom	Bonds formed (use sigma and pi as appropriate)
Carbon (<i>a</i>)	
Carbon (<i>b</i>)	

iv. How many total sp^3 -s sigma bonds are in the molecule? (1 point)

v. Determine **electronic** and **molecular geometry** at carbon (C) labeled ***e*** and sulfur (S) labeled ***f*** centres. (2 points)

Atom	Electronic Geometry	Molecular Geometry
Carbon (<i>e</i>)		
Sulfur (<i>f</i>)		

vi. Circle the value from the choices given below, that describes the **C-S-C bond angle at sulfur** (S) marked ***f***: (0.5 points)

>90° to <109.5° 109.5° >109.5 to <120° 120° >120° to <180°

**Question 8 12 points**

Using the **valence shell orbitals only** draw a **complete** molecular orbital (MO) diagram (with no s-p mixing) for the molecule **PS**, in the space provided below. **(6 points)**

Assume that 3s and 3p atomic orbitals will form molecular orbitals like those formed by the combination of 2s and 2p atomic orbitals that we learnt in class, and there is no s-p mixing.

Show **energy arrow**, **atomic orbitals** involved, **atom labels**, the **molecular orbitals** that are formed, **filled electrons** in the MO diagram etc. Make sure to label all the atomic and molecular orbitals.

Using the MO diagram above, answer the following questions:

- (i) Which one of the following will have the highest bond order? Circle the correct choice
A) PS^{1-} B) PS C) PS^{1+}
- (ii) Which one of the following will have the longest bond length? Circle the correct choice
A) PS^{1-} B) PS C) PS^{1+}
- (iii) Which of the following will be diamagnetic? Circle the correct choice.
A) PS^{1-} B) PS C) PS^{1+}
- (iv) Draw Lewis dot structure of the molecule with the shortest bond length (based on the MO diagram) from among the three (PS^{1-} , PS, PS^{1+}), and indicate the bond order calculated from the Lewis dot structure. **(3 points)**

Lewis Dot Structure

Bond Order



88DF0DCF-B87D-459B-B9D8-DBD3D77E14F9

midterm-exam-93388

#279 Page 10 of 10

SCRAP PAGE

**CHEMISTRY 110 –Midterm Exam****Thursday November 7, 2024, 6:30 – 8:30 pm****Examiners: Pallavi Sirjoosingh and Ashok Kakkar**

First name (please write as legibly as possible within the boxes)

--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--

Last name

--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--

Student ID number

--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--

This booklet has 8 questions on 10 pages (The last page can be used as a scrap page).

The staple must NOT be removed from the question booklet. Your intact question booklet must be returned at the end of the exam.

A separate booklet contains a periodic table, and datasheet.

Write your answers directly under the questions in the space provided. Do not write on the QR codes or near the top edges (as it may be cut off).

Please write **legibly and dark enough** for the scan to be readable. For students who use English as a second language, dictionaries are permitted. Calculators are allowed.

DO NOT ASK QUESTIONS OF THE INVIGILATORS about interpretations of the exam questions. If you have a concern about a question, please summarize it at the top of this page. All concerns will be evaluated after the exam. GOOD LUCK!

**Question 1 8 points** (*Assume all transitions are possible*)

a. 6 points Calculate the **energy (in kJ/mol)** required to ionize (completely remove) the electron from the **n=3** level of **He⁺ ion** (He: Helium). Show your work including equations. (3 sig. figs.)

b. 2 points

(I) Select the largest (longest) wavelength from the three choices below:

- i. Wavelength of photon to *completely remove* an electron from *1s orbital of Hydrogen atom*
- ii. Wavelength of photon to *completely remove* an electron from *1s orbital of Boron atom (B)*
- iii. Wavelength of photon to *completely remove* an electron from *1s orbital of Boron cation (B⁺)*

i or ii or iii

(II) Select the smallest (shortest) wavelength from the three choices below (*for H atom*):

- i. Wavelength of photon *emitted* for an electron transitioning from *2p to 1s orbital*
- ii. Wavelength of photon *absorbed* for an electron transitioning from *2s to 4p orbital*
- iii. Wavelength of photon *absorbed* for an electron transitioning from *1s to 3p orbital*

i or ii or iii



Question 2 8 points



A Chem110 student plots the graph shown on the left, from data obtained for the kinetic energy (eV) of the ejected electron versus frequency (s^{-1}) of the incident light, for an unknown metal. ($1\text{eV} = 1.6022 \times 10^{-19} \text{ J}$)

Based on the plot, answer the questions **a to c** below:

a. 1 point On the plot above, label the **threshold frequency**.

b. 2 points What is the value of the work potential (in J) of the unknown metal?

c. 3 points What is the *longest* wavelength (in nm) required to eject an electron from this metal's surface? Show your work including equation(s) used. (3 sig figs)

d. 2 points In the space provided below, plot the following

(Assume frequency is *above* threshold frequency):

- Kinetic energy* of an ejected electron versus the *intensity* of the incident light (left)
- Number of electrons* ejected versus the *frequency* of the incident light (right)



**Question 3** **6 points****a. 2 points**

In the ground state of Ge (Germanium; $Z = 32$) atom, **name** the following orbitals:

i. Orbital containing the electron that requires the maximum energy to be removed.

ii. Orbital containing the electron that experiences the minimum effective nuclear charge.

b. 2 points

An unknown anion (A^-) has the following *excited* electronic configuration: $[Ar] 4s^2 3d^3 4p^1$

(Ar: Argon; $Z = 18$)

Give the **condensed ground state electronic configuration** of the atom, **A** and the cation, **A^{2+}** .
Show your work. (Assume all electronic transitions are possible)

Electronic Configuration of A:

Electronic Configuration of A^{2+} :

c. 2 points

In a single atom, determine how many electrons can have the following set of quantum numbers?

(i) $n = 5; l = 3$

In a single atom, determine how many electrons can have the following set of quantum numbers?

(i) $n = 3; l = 2; m_s = +1/2$

**Question 4** **8 points**

Rank the following from 1 to 4 in **increasing** order. Provide the ranking in the boxes provided.

Note: Atomic number (Z) are provided for relevant atoms

a) Increasing (lowest to highest) order of *ionization energy*: (1= lowest; 4 = highest)

Mg

Mg: Z = 12

Rb

Rb: Z = 37

Cs

Cs: Z = 55

K

K : Z = 19

b) Increasing (lowest to highest) order of *bond angle*: (1= lowest; 4 = highest)

H₂OCCl₄XeF₄

HCN

c) Increasing (smallest to largest) order of *ionic size* (1= smallest; 4 = largest)

S²⁻

S: Z = 16

O²⁻

O: Z = 8

Be²⁺

Be: Z = 4

Na⁺

Na: Z = 11

d) Increasing (lowest to highest) order of *bond energy* (1= lowest; 4 = highest)

P-F

P-Cl

P-I

P-Br

**Question 5** **12 points**

a. 9 points For the molecules HClO_4 and TeF_6 , draw the most stable *Lewis structure* showing appropriate shape and *dashed/wedged bonds*, based on VSEPR. Determine the **molecular geometry** around the central atom. For HClO_4 , label the *most polar bond*.



b. 3 points (Note: Lone pairs are not shown in structures below)

(i) For the three *resonance* structures (1, 2 and 3) shown below, answer the following question:



(I) Circle the *most contributing* resonance structure.

Structure 1

Structure 2

Structure 3

(ii) For the three structures given (4, 5, and 6) below, answer the following questions:



(I) Circle all the structures that represent *resonance structures* of each other:

Structure 4

Structure 5

Structure 6

(II) Among the resonance structures, circle the *most contributing* resonance structure.

Structure 4

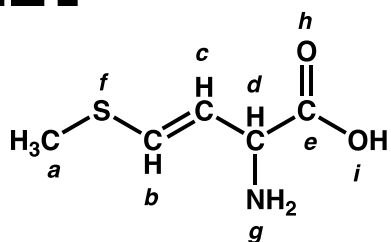
Structure 5

Structure 6

**Question 6** **6 points**

Using the data given below, calculate the **lattice energy** for SrO(s).

	ΔH° (kJ/mol)
$\text{Sr}(s) \longrightarrow \text{Sr}(g)$	+180
$\text{Sr}(g) \longrightarrow \text{Sr}^{2+}(g) + 2e^-$	+1468
$\text{O}_2(g) \longrightarrow 2\text{O}(g)$	+500
$\text{O}(g) + 2e^- \longrightarrow \text{O}^{2-}(g)$	+600
$\text{Sr}(s) + \frac{1}{2}\text{O}_2(g) \longrightarrow \text{SrO}(s)$	-550

**Question 7 10 points**

Consider the molecule given on the left, and answer questions (i) to (vi) using chemical bonding theories that we have learnt in class (valence bond theory, hybridization).

Note that the lone pairs are not shown in the structure:

i. Consider the carbon atoms marked in as ***a***, ***b***, ***c***, ***d***, and ***e*** in the molecule, and determine **hybridization** at each of the carbon centers. (2.5 points)

Atom	Hybridization
Carbon (<i>a</i>)	
Carbon (<i>b</i>)	
Carbon (<i>c</i>)	
Carbon (<i>d</i>)	
Carbon (<i>e</i>)	

ii. Now, consider the sulfur atom (S) marked as ***f***, nitrogen (N) atom (***g***), and oxygen(O) atoms (***h*** and ***i***), and determine their **hybridization**. (2 points)

Atom	Hybridization
Sulfur (<i>f</i>)	
Nitrogen (<i>g</i>)	
Oxygen (<i>h</i>)	
Oxygen (<i>i</i>)	

iii. **Describe** all the bonds formed by the carbon atoms ***a*** and ***b***, using sigma and pi designation as appropriate. (2 points)

Atom	Bonds formed (use sigma and pi as appropriate)
Carbon (<i>a</i>)	
Carbon (<i>b</i>)	

iv. How many total sp^3 -s sigma bonds are in the molecule? (1 point)

v. Determine **electronic** and **molecular geometry** at carbon (C) labeled ***e*** and sulfur (S) labeled ***f*** centres. (2 points)

Atom	Electronic Geometry	Molecular Geometry
Carbon (<i>e</i>)		
Sulfur (<i>f</i>)		

vi. Circle the value from the choices given below, that describes the **C-S-C bond angle at sulfur** (S) marked ***f***: (0.5 points)

>90° to <109.5° 109.5° >109.5 to <120° 120° >120° to <180°


Question 8 12 points

Using the **valence shell orbitals only** draw a **complete** molecular orbital (MO) diagram (with no s-p mixing) for the molecule **PS**, in the space provided below. **(6 points)**

Assume that 3s and 3p atomic orbitals will form molecular orbitals like those formed by the combination of 2s and 2p atomic orbitals that we learnt in class, and there is no s-p mixing.

Show **energy arrow**, **atomic orbitals** involved, **atom labels**, the **molecular orbitals** that are formed, **filled electrons** in the MO diagram etc. Make sure to label all the atomic and molecular orbitals.

Using the MO diagram above, answer the following questions:

- (i) Which one of the following will have the highest bond order? Circle the correct choice
 A) PS^{1-} B) PS C) PS^{1+}
- (ii) Which one of the following will have the longest bond length? Circle the correct choice
 A) PS^{1-} B) PS C) PS^{1+}
- (iii) Which of the following will be diamagnetic? Circle the correct choice.
 A) PS^{1-} B) PS C) PS^{1+}
- (iv) Draw Lewis dot structure of the molecule with the shortest bond length (based on the MO diagram) from among the three (PS^{1-} , PS, PS^{1+}), and indicate the bond order calculated from the Lewis dot structure. **(3 points)**

Lewis Dot Structure

Bond Order



217D5863-F2EC-40E7-9B84-17C47D3BEC12

midterm-exam-93388

#280 Page 10 of 10

SCRAP PAGE

**CHEMISTRY 110 –Midterm Exam****Thursday November 7, 2024, 6:30 – 8:30 pm****Examiners: Pallavi Sirjoosingh and Ashok Kakkar**

First name (please write as legibly as possible within the boxes)

--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--

Last name

--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--

Student ID number

--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--

This booklet has 8 questions on 10 pages (The last page can be used as a scrap page).

The staple must NOT be removed from the question booklet. Your intact question booklet must be returned at the end of the exam.

A separate booklet contains a periodic table, and datasheet.

Write your answers directly under the questions in the space provided. Do not write on the QR codes or near the top edges (as it may be cut off).

Please write **legibly and dark enough** for the scan to be readable. For students who use English as a second language, dictionaries are permitted. Calculators are allowed.

DO NOT ASK QUESTIONS OF THE INVIGILATORS about interpretations of the exam questions. If you have a concern about a question, please summarize it at the top of this page. All concerns will be evaluated after the exam. GOOD LUCK!

**Question 1 8 points** (*Assume all transitions are possible*)

a. 6 points Calculate the **energy (in kJ/mol)** required to ionize (completely remove) the electron from the **$n=3$** level of **He^+ ion** (He: Helium). Show your work including equations. (3 sig. figs.)

b. 2 points

(I) Select the largest (longest) wavelength from the three choices below:

- i. Wavelength of photon to *completely remove* an electron from *1s orbital of Hydrogen atom*
- ii. Wavelength of photon to *completely remove* an electron from *1s orbital of Boron atom (B)*
- iii. Wavelength of photon to *completely remove* an electron from *1s orbital of Boron cation (B^+)*

i or ii or iii

(II) Select the smallest (shortest) wavelength from the three choices below (*for H atom*):

- i. Wavelength of photon *emitted* for an electron transitioning from *2p to 1s orbital*
- ii. Wavelength of photon *absorbed* for an electron transitioning from *2s to 4p orbital*
- iii. Wavelength of photon *absorbed* for an electron transitioning from *1s to 3p orbital*

i or ii or iii



Question 2 8 points



A Chem110 student plots the graph shown on the left, from data obtained for the kinetic energy (eV) of the ejected electron versus frequency (s^{-1}) of the incident light, for an unknown metal. ($1\text{eV} = 1.6022 \times 10^{-19} \text{ J}$)

Based on the plot, answer the questions **a to c** below:

a. 1 point On the plot above, label the **threshold frequency**.

b. 2 points What is the value of the work potential (in J) of the unknown metal?

c. 3 points What is the *longest* wavelength (in nm) required to eject an electron from this metal's surface? Show your work including equation(s) used. (3 sig figs)

d. 2 points In the space provided below, plot the following

(Assume frequency is *above* threshold frequency):

- i. *Kinetic energy* of an ejected electron versus the *intensity* of the incident light (left)
- ii. *Number of electrons* ejected versus the *frequency* of the incident light (right)



**Question 3** **6 points****a. 2 points**

In the ground state of Ge (Germanium; $Z = 32$) atom, **name** the following orbitals:

i. Orbital containing the electron that requires the maximum energy to be removed.

ii. Orbital containing the electron that experiences the minimum effective nuclear charge.

b. 2 points

An unknown anion (A^-) has the following *excited* electronic configuration: $[Ar] 4s^2 3d^3 4p^1$

(Ar: Argon; $Z = 18$)

Give the **condensed ground state electronic configuration** of the atom, **A** and the cation, **A^{2+}** .
Show your work. (Assume all electronic transitions are possible)

Electronic Configuration of A:

Electronic Configuration of A^{2+} :

c. 2 points

In a single atom, determine how many electrons can have the following set of quantum numbers?

(i) $n = 5; l = 3$

In a single atom, determine how many electrons can have the following set of quantum numbers?

(i) $n = 3; l = 2; m_s = +1/2$

**Question 4** **8 points**

Rank the following from 1 to 4 in **increasing** order. Provide the ranking in the boxes provided.

Note: Atomic number (Z) are provided for relevant atoms

a) Increasing (lowest to highest) order of *ionization energy*: (1= lowest; 4 = highest)

Mg

Mg: Z = 12

Rb

Rb: Z = 37

Cs

Cs: Z = 55

K

K : Z = 19

b) Increasing (lowest to highest) order of *bond angle*: (1= lowest; 4 = highest)

H₂OCCl₄XeF₄

HCN

c) Increasing (smallest to largest) order of *ionic size* (1= smallest; 4 = largest)

S²⁻

S: Z = 16

O²⁻

O: Z = 8

Be²⁺

Be: Z = 4

Na⁺

Na: Z = 11

d) Increasing (lowest to highest) order of *bond energy* (1= lowest; 4 = highest)

P-F

P-Cl

P-I

P-Br

**Question 5** **12 points**

a. 9 points For the molecules HClO_4 and TeF_6 , draw the most stable *Lewis structure* showing appropriate shape and *dashed/wedged bonds*, based on VSEPR. Determine the **molecular geometry** around the central atom. For HClO_4 , label the *most polar bond*.

HClO_4 (there are no O-O bonds)

Label the most polar bond

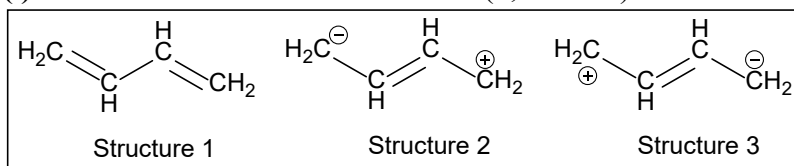
Molecular Geometry around Cl

TeF_6

Molecular Geometry around Te

b. 3 points (Note: Lone pairs are not shown in structures below)

(i) For the three *resonance* structures (1, 2 and 3) shown below, answer the following question:



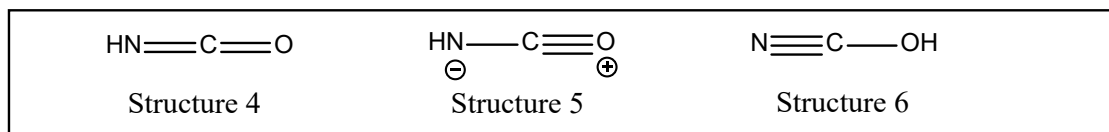
(I) Circle the *most contributing* resonance structure.

Structure 1

Structure 2

Structure 3

(ii) For the three structures given (4, 5, and 6) below, answer the following questions:



(I) Circle all the structures that represent *resonance structures* of each other:

Structure 4

Structure 5

Structure 6

(II) Among the resonance structures, circle the *most contributing* resonance structure.

Structure 4

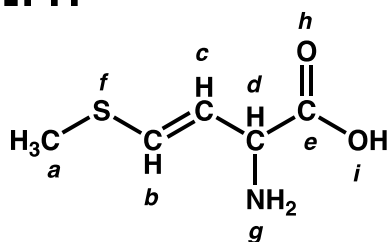
Structure 5

Structure 6

**Question 6** **6 points**

Using the data given below, calculate the **lattice energy** for SrO(s).

	ΔH° (kJ/mol)
$\text{Sr}(s) \longrightarrow \text{Sr}(g)$	+180
$\text{Sr}(g) \longrightarrow \text{Sr}^{2+}(g) + 2e^-$	+1468
$\text{O}_2(g) \longrightarrow 2\text{O}(g)$	+500
$\text{O}(g) + 2e^- \longrightarrow \text{O}^{2-}(g)$	+600
$\text{Sr}(s) + \frac{1}{2}\text{O}_2(g) \longrightarrow \text{SrO}(s)$	-550

**Question 7 10 points**

Consider the molecule given on the left, and answer questions (i) to (vi) using chemical bonding theories that we have learnt in class (valence bond theory, hybridization).

Note that the lone pairs are not shown in the structure:

i. Consider the carbon atoms marked in as ***a***, ***b***, ***c***, ***d***, and ***e*** in the molecule, and determine **hybridization** at each of the carbon centers. (2.5 points)

Atom	Hybridization
Carbon (<i>a</i>)	
Carbon (<i>b</i>)	
Carbon (<i>c</i>)	
Carbon (<i>d</i>)	
Carbon (<i>e</i>)	

ii. Now, consider the sulfur atom (S) marked as ***f***, nitrogen (N) atom (***g***), and oxygen(O) atoms (***h*** and ***i***), and determine their **hybridization**. (2 points)

Atom	Hybridization
Sulfur (<i>f</i>)	
Nitrogen (<i>g</i>)	
Oxygen (<i>h</i>)	
Oxygen (<i>i</i>)	

iii. **Describe** all the bonds formed by the carbon atoms ***a*** and ***b***, using sigma and pi designation as appropriate. (2 points)

Atom	Bonds formed (use sigma and pi as appropriate)
Carbon (<i>a</i>)	
Carbon (<i>b</i>)	

iv. How many total sp^3 -s sigma bonds are in the molecule? (1 point)

v. Determine **electronic** and **molecular geometry** at carbon (C) labeled ***e*** and sulfur (S) labeled ***f*** centres. (2 points)

Atom	Electronic Geometry	Molecular Geometry
Carbon (<i>e</i>)		
Sulfur (<i>f</i>)		

vi. Circle the value from the choices given below, that describes the **C-S-C bond angle at sulfur** (S) marked ***f***: (0.5 points)

>90° to <109.5° 109.5° >109.5 to <120° 120° >120° to <180°

**Question 8 12 points**

Using the **valence shell orbitals only** draw a **complete** molecular orbital (MO) diagram (with no s-p mixing) for the molecule **PS**, in the space provided below. **(6 points)**

Assume that 3s and 3p atomic orbitals will form molecular orbitals like those formed by the combination of 2s and 2p atomic orbitals that we learnt in class, and there is no s-p mixing.

Show **energy arrow**, **atomic orbitals** involved, **atom labels**, the **molecular orbitals** that are formed, **filled electrons** in the MO diagram etc. Make sure to label all the atomic and molecular orbitals.

Using the MO diagram above, answer the following questions:

- (i) Which one of the following will have the highest bond order? Circle the correct choice
A) PS^{1-} B) PS C) PS^{1+}
- (ii) Which one of the following will have the longest bond length? Circle the correct choice
A) PS^{1-} B) PS C) PS^{1+}
- (iii) Which of the following will be diamagnetic? Circle the correct choice.
A) PS^{1-} B) PS C) PS^{1+}
- (iv) Draw Lewis dot structure of the molecule with the shortest bond length (based on the MO diagram) from among the three (PS^{1-} , PS, PS^{1+}), and indicate the bond order calculated from the Lewis dot structure. **(3 points)**

Lewis Dot Structure

Bond Order



53CF6072-7188-4B07-B525-11049D90F94E

midterm-exam-93388

#281 Page 10 of 10

SCRAP PAGE

**CHEMISTRY 110 –Midterm Exam****Thursday November 7, 2024, 6:30 – 8:30 pm****Examiners: Pallavi Sirjoosingh and Ashok Kakkar**

First name (please write as legibly as possible within the boxes)

--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--

Last name

--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--

Student ID number

--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--

This booklet has 8 questions on 10 pages (The last page can be used as a scrap page).

The staple must NOT be removed from the question booklet. Your intact question booklet must be returned at the end of the exam.

A separate booklet contains a periodic table, and datasheet.

Write your answers directly under the questions in the space provided. Do not write on the QR codes or near the top edges (as it may be cut off).

Please write **legibly and dark enough** for the scan to be readable. For students who use English as a second language, dictionaries are permitted. Calculators are allowed.

DO NOT ASK QUESTIONS OF THE INVIGILATORS about interpretations of the exam questions. If you have a concern about a question, please summarize it at the top of this page. All concerns will be evaluated after the exam. GOOD LUCK!

**Question 1 8 points** (*Assume all transitions are possible*)

a. 6 points Calculate the **energy (in kJ/mol)** required to ionize (completely remove) the electron from the **$n=3$** level of **He^+ ion** (He: Helium). Show your work including equations. (3 sig. figs.)

b. 2 points

(I) Select the largest (longest) wavelength from the three choices below:

- i. Wavelength of photon to *completely remove* an electron from *1s orbital of Hydrogen atom*
- ii. Wavelength of photon to *completely remove* an electron from *1s orbital of Boron atom (B)*
- iii. Wavelength of photon to *completely remove* an electron from *1s orbital of Boron cation (B^+)*

i or ii or iii

(II) Select the smallest (shortest) wavelength from the three choices below (*for H atom*):

- i. Wavelength of photon *emitted* for an electron transitioning from *2p to 1s orbital*
- ii. Wavelength of photon *absorbed* for an electron transitioning from *2s to 4p orbital*
- iii. Wavelength of photon *absorbed* for an electron transitioning from *1s to 3p orbital*

i or ii or iii



Question 2 8 points



A Chem110 student plots the graph shown on the left, from data obtained for the kinetic energy (eV) of the ejected electron versus frequency (s^{-1}) of the incident light, for an unknown metal. ($1\text{eV} = 1.6022 \times 10^{-19} \text{ J}$)

Based on the plot, answer the questions **a to c** below:

a. 1 point On the plot above, label the **threshold frequency**.

b. 2 points What is the value of the work potential (in J) of the unknown metal?

c. 3 points What is the *longest* wavelength (in nm) required to eject an electron from this metal's surface? Show your work including equation(s) used. (3 sig figs)

d. 2 points In the space provided below, plot the following

(Assume frequency is *above* threshold frequency):

- i. *Kinetic energy* of an ejected electron versus the *intensity* of the incident light (left)
- ii. *Number of electrons* ejected versus the *frequency* of the incident light (right)



**Question 3** **6 points****a. 2 points**

In the ground state of Ge (Germanium; $Z = 32$) atom, **name** the following orbitals:

i. Orbital containing the electron that requires the maximum energy to be removed.

ii. Orbital containing the electron that experiences the minimum effective nuclear charge.

b. 2 points

An unknown anion (A^-) has the following *excited* electronic configuration: $[Ar] 4s^2 3d^3 4p^1$

(Ar: Argon; $Z = 18$)

Give the **condensed ground state electronic configuration** of the atom, **A** and the cation, **A^{2+}** .
Show your work. (Assume all electronic transitions are possible)

Electronic Configuration of A:

Electronic Configuration of A^{2+} :

c. 2 points

In a single atom, determine how many electrons can have the following set of quantum numbers?

(i) $n = 5; l = 3$

In a single atom, determine how many electrons can have the following set of quantum numbers?

(i) $n = 3; l = 2; m_s = +1/2$

**Question 4** **8 points**

Rank the following from 1 to 4 in **increasing** order. Provide the ranking in the boxes provided.

Note: Atomic number (Z) are provided for relevant atoms

a) Increasing (lowest to highest) order of *ionization energy*: (1= lowest; 4 = highest)

Mg

Mg: Z = 12

Rb

Rb: Z = 37

Cs

Cs: Z = 55

K

K : Z = 19

b) Increasing (lowest to highest) order of *bond angle*: (1= lowest; 4 = highest)

H₂OCCl₄XeF₄

HCN

c) Increasing (smallest to largest) order of *ionic size* (1= smallest; 4 = largest)

S²⁻

S: Z = 16

O²⁻

O: Z = 8

Be²⁺

Be: Z = 4

Na⁺

Na: Z = 11

d) Increasing (lowest to highest) order of *bond energy* (1= lowest; 4 = highest)

P-F

P-Cl

P-I

P-Br

**Question 5** **12 points**

a. 9 points For the molecules HClO_4 and TeF_6 , draw the most stable *Lewis structure* showing appropriate shape and *dashed/wedged bonds*, based on VSEPR. Determine the **molecular geometry** around the central atom. For HClO_4 , label the *most polar bond*.



b. 3 points (Note: Lone pairs are not shown in structures below)

(i) For the three *resonance* structures (1, 2 and 3) shown below, answer the following question:



(I) Circle the *most contributing* resonance structure.

Structure 1

Structure 2

Structure 3

(ii) For the three structures given (4, 5, and 6) below, answer the following questions:



(I) Circle all the structures that represent *resonance structures* of each other:

Structure 4

Structure 5

Structure 6

(II) Among the resonance structures, circle the *most contributing* resonance structure.

Structure 4

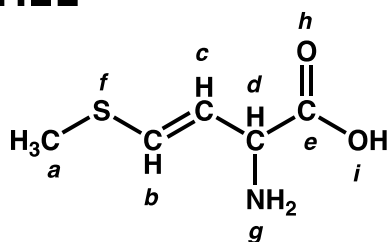
Structure 5

Structure 6

**Question 6** **6 points**

Using the data given below, calculate the **lattice energy** for SrO(s).

	ΔH° (kJ/mol)
$\text{Sr}(s) \longrightarrow \text{Sr}(g)$	+180
$\text{Sr}(g) \longrightarrow \text{Sr}^{2+}(g) + 2e^-$	+1468
$\text{O}_2(g) \longrightarrow 2\text{O}(g)$	+500
$\text{O}(g) + 2e^- \longrightarrow \text{O}^{2-}(g)$	+600
$\text{Sr}(s) + \frac{1}{2}\text{O}_2(g) \longrightarrow \text{SrO}(s)$	-550

**Question 7 10 points**

Consider the molecule given on the left, and answer questions (i) to (vi) using chemical bonding theories that we have learnt in class (valence bond theory, hybridization).

Note that the lone pairs are not shown in the structure:

i. Consider the carbon atoms marked in as ***a***, ***b***, ***c***, ***d***, and ***e*** in the molecule, and determine **hybridization** at each of the carbon centers. (2.5 points)

Atom	Hybridization
Carbon (<i>a</i>)	
Carbon (<i>b</i>)	
Carbon (<i>c</i>)	
Carbon (<i>d</i>)	
Carbon (<i>e</i>)	

ii. Now, consider the sulfur atom (S) marked as ***f***, nitrogen (N) atom (***g***), and oxygen(O) atoms (***h*** and ***i***), and determine their **hybridization**. (2 points)

Atom	Hybridization
Sulfur (<i>f</i>)	
Nitrogen (<i>g</i>)	
Oxygen (<i>h</i>)	
Oxygen (<i>i</i>)	

iii. **Describe** all the bonds formed by the carbon atoms ***a*** and ***b***, using sigma and pi designation as appropriate. (2 points)

Atom	Bonds formed (use sigma and pi as appropriate)
Carbon (<i>a</i>)	
Carbon (<i>b</i>)	

iv. How many total sp^3 -s sigma bonds are in the molecule? (1 point)

v. Determine **electronic** and **molecular geometry** at carbon (C) labeled ***e*** and sulfur (S) labeled ***f*** centres. (2 points)

Atom	Electronic Geometry	Molecular Geometry
Carbon (<i>e</i>)		
Sulfur (<i>f</i>)		

vi. Circle the value from the choices given below, that describes the **C-S-C bond angle at sulfur** (S) marked ***f***: (0.5 points)

$>90^\circ$ to $<109.5^\circ$ 109.5° >109.5 to $<120^\circ$ 120° $>120^\circ$ to $<180^\circ$


Question 8 12 points

Using the **valence shell orbitals only** draw a **complete** molecular orbital (MO) diagram (with no s-p mixing) for the molecule **PS**, in the space provided below. **(6 points)**

Assume that 3s and 3p atomic orbitals will form molecular orbitals like those formed by the combination of 2s and 2p atomic orbitals that we learnt in class, and there is no s-p mixing.

Show **energy arrow**, **atomic orbitals** involved, **atom labels**, the **molecular orbitals** that are formed, **filled electrons** in the MO diagram etc. Make sure to label all the atomic and molecular orbitals.

Using the MO diagram above, answer the following questions:

- (i) Which one of the following will have the highest bond order? Circle the correct choice
 A) PS^{1-} B) PS C) PS^{1+}
- (ii) Which one of the following will have the longest bond length? Circle the correct choice
 A) PS^{1-} B) PS C) PS^{1+}
- (iii) Which of the following will be diamagnetic? Circle the correct choice.
 A) PS^{1-} B) PS C) PS^{1+}
- (iv) Draw Lewis dot structure of the molecule with the shortest bond length (based on the MO diagram) from among the three (PS^{1-} , PS, PS^{1+}), and indicate the bond order calculated from the Lewis dot structure. **(3 points)**

Lewis Dot Structure

Bond Order



794BAD69-3626-4619-B747-90E7F7EF1E3F

midterm-exam-93388

#282 Page 10 of 10

SCRAP PAGE

**CHEMISTRY 110 –Midterm Exam****Thursday November 7, 2024, 6:30 – 8:30 pm****Examiners: Pallavi Sirjoosingh and Ashok Kakkar**

First name (please write as legibly as possible within the boxes)

--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--

Last name

--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--

Student ID number

--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--

This booklet has 8 questions on 10 pages (The last page can be used as a scrap page).

The staple must NOT be removed from the question booklet. Your intact question booklet must be returned at the end of the exam.

A separate booklet contains a periodic table, and datasheet.

Write your answers directly under the questions in the space provided. Do not write on the QR codes or near the top edges (as it may be cut off).

Please write **legibly and dark enough** for the scan to be readable. For students who use English as a second language, dictionaries are permitted. Calculators are allowed.

DO NOT ASK QUESTIONS OF THE INVIGILATORS about interpretations of the exam questions. If you have a concern about a question, please summarize it at the top of this page. All concerns will be evaluated after the exam. GOOD LUCK!

**Question 1 8 points** (*Assume all transitions are possible*)

a. 6 points Calculate the **energy (in kJ/mol)** required to ionize (completely remove) the electron from the **$n=3$** level of **He^+ ion** (He: Helium). Show your work including equations. (3 sig. figs.)

b. 2 points

(I) Select the largest (longest) wavelength from the three choices below:

- i. Wavelength of photon to *completely remove* an electron from *1s orbital of Hydrogen atom*
- ii. Wavelength of photon to *completely remove* an electron from *1s orbital of Boron atom (B)*
- iii. Wavelength of photon to *completely remove* an electron from *1s orbital of Boron cation (B^+)*

i or ii or iii

(II) Select the smallest (shortest) wavelength from the three choices below (*for H atom*):

- i. Wavelength of photon *emitted* for an electron transitioning from *2p to 1s orbital*
- ii. Wavelength of photon *absorbed* for an electron transitioning from *2s to 4p orbital*
- iii. Wavelength of photon *absorbed* for an electron transitioning from *1s to 3p orbital*

i or ii or iii



Question 2 8 points



A Chem110 student plots the graph shown on the left, from data obtained for the kinetic energy (eV) of the ejected electron versus frequency (s^{-1}) of the incident light, for an unknown metal. ($1\text{eV} = 1.6022 \times 10^{-19} \text{ J}$)

Based on the plot, answer the questions **a to c** below:

a. 1 point On the plot above, label the **threshold frequency**.

b. 2 points What is the value of the work potential (in J) of the unknown metal?

c. 3 points What is the *longest* wavelength (in nm) required to eject an electron from this metal's surface? Show your work including equation(s) used. (3 sig figs)

d. 2 points In the space provided below, plot the following

(Assume frequency is *above* threshold frequency):

- i. *Kinetic energy* of an ejected electron versus the *intensity* of the incident light (left)
- ii. *Number of electrons* ejected versus the *frequency* of the incident light (right)



**Question 3** **6 points****a. 2 points**

In the ground state of Ge (Germanium; $Z = 32$) atom, **name** the following orbitals:

i. Orbital containing the electron that requires the maximum energy to be removed.

ii. Orbital containing the electron that experiences the minimum effective nuclear charge.

b. 2 points

An unknown anion (A^-) has the following *excited* electronic configuration: $[Ar] 4s^2 3d^3 4p^1$

(Ar: Argon; $Z = 18$)

Give the **condensed ground state electronic configuration** of the atom, **A** and the cation, **A^{2+}** .
Show your work. (Assume all electronic transitions are possible)

Electronic Configuration of A:

Electronic Configuration of A^{2+} :

c. 2 points

In a single atom, determine how many electrons can have the following set of quantum numbers?

(i) $n = 5; l = 3$

In a single atom, determine how many electrons can have the following set of quantum numbers?

(i) $n = 3; l = 2; m_s = +1/2$

**Question 4** **8 points**

Rank the following from 1 to 4 in **increasing** order. Provide the ranking in the boxes provided.

Note: Atomic number (Z) are provided for relevant atoms

a) Increasing (lowest to highest) order of *ionization energy*: (1= lowest; 4 = highest)

Mg

Mg: Z = 12

Rb

Rb: Z = 37

Cs

Cs: Z = 55

K

K : Z = 19

b) Increasing (lowest to highest) order of *bond angle*: (1= lowest; 4 = highest)

H₂OCCl₄XeF₄

HCN

c) Increasing (smallest to largest) order of *ionic size* (1= smallest; 4 = largest)

S²⁻

S: Z = 16

O²⁻

O: Z = 8

Be²⁺

Be: Z = 4

Na⁺

Na: Z = 11

d) Increasing (lowest to highest) order of *bond energy* (1= lowest; 4 = highest)

P-F

P-Cl

P-I

P-Br

**Question 5** **12 points**

a. 9 points For the molecules HClO_4 and TeF_6 , draw the most stable *Lewis structure* showing appropriate shape and *dashed/wedged bonds*, based on VSEPR. Determine the **molecular geometry** around the central atom. For HClO_4 , label the *most polar bond*.

HClO_4 (there are no O-O bonds)

Label the most polar bond

Molecular Geometry around Cl

TeF_6

Molecular Geometry around Te

b. 3 points (Note: Lone pairs are not shown in structures below)

(i) For the three *resonance* structures (1, 2 and 3) shown below, answer the following question:



(I) Circle the *most contributing* resonance structure.

Structure 1

Structure 2

Structure 3

(ii) For the three structures given (4, 5, and 6) below, answer the following questions:



(I) Circle all the structures that represent *resonance structures* of each other:

Structure 4

Structure 5

Structure 6

(II) Among the resonance structures, circle the *most contributing* resonance structure.

Structure 4

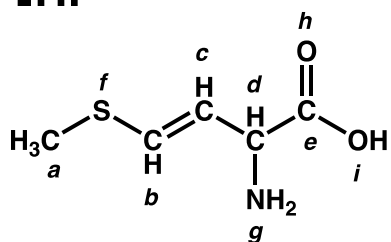
Structure 5

Structure 6

**Question 6** **6 points**

Using the data given below, calculate the **lattice energy** for SrO(s).

	ΔH° (kJ/mol)
$\text{Sr}(s) \longrightarrow \text{Sr}(g)$	+180
$\text{Sr}(g) \longrightarrow \text{Sr}^{2+}(g) + 2e^-$	+1468
$\text{O}_2(g) \longrightarrow 2\text{O}(g)$	+500
$\text{O}(g) + 2e^- \longrightarrow \text{O}^{2-}(g)$	+600
$\text{Sr}(s) + \frac{1}{2}\text{O}_2(g) \longrightarrow \text{SrO}(s)$	-550

**Question 7 10 points**

Consider the molecule given on the left, and answer questions (i) to (vi) using chemical bonding theories that we have learnt in class (valence bond theory, hybridization).

Note that the lone pairs are not shown in the structure:

i. Consider the carbon atoms marked in as ***a***, ***b***, ***c***, ***d***, and ***e*** in the molecule, and determine **hybridization** at each of the carbon centers. (2.5 points)

Atom	Hybridization
Carbon (<i>a</i>)	
Carbon (<i>b</i>)	
Carbon (<i>c</i>)	
Carbon (<i>d</i>)	
Carbon (<i>e</i>)	

ii. Now, consider the sulfur atom (S) marked as ***f***, nitrogen (N) atom (***g***), and oxygen(O) atoms (***h*** and ***i***), and determine their **hybridization**. (2 points)

Atom	Hybridization
Sulfur (<i>f</i>)	
Nitrogen (<i>g</i>)	
Oxygen (<i>h</i>)	
Oxygen (<i>i</i>)	

iii. **Describe** all the bonds formed by the carbon atoms ***a*** and ***b***, using sigma and pi designation as appropriate. (2 points)

Atom	Bonds formed (use sigma and pi as appropriate)
Carbon (<i>a</i>)	
Carbon (<i>b</i>)	

iv. How many total sp^3 -s sigma bonds are in the molecule? (1 point)

v. Determine **electronic** and **molecular geometry** at carbon (C) labeled ***e*** and sulfur (S) labeled ***f*** centres. (2 points)

Atom	Electronic Geometry	Molecular Geometry
Carbon (<i>e</i>)		
Sulfur (<i>f</i>)		

vi. Circle the value from the choices given below, that describes the **C-S-C bond angle at sulfur** (S) marked ***f***: (0.5 points)

>90° to <109.5° 109.5° >109.5 to <120° 120° >120° to <180°


Question 8 12 points

Using the **valence shell orbitals only** draw a **complete** molecular orbital (MO) diagram (with no s-p mixing) for the molecule **PS**, in the space provided below. **(6 points)**

Assume that 3s and 3p atomic orbitals will form molecular orbitals like those formed by the combination of 2s and 2p atomic orbitals that we learnt in class, and there is no s-p mixing.

Show **energy arrow**, **atomic orbitals** involved, **atom labels**, the **molecular orbitals** that are formed, **filled electrons** in the MO diagram etc. Make sure to label all the atomic and molecular orbitals.

Using the MO diagram above, answer the following questions:

- (i) Which one of the following will have the highest bond order? Circle the correct choice
 A) PS^{1-} B) PS C) PS^{1+}
- (ii) Which one of the following will have the longest bond length? Circle the correct choice
 A) PS^{1-} B) PS C) PS^{1+}
- (iii) Which of the following will be diamagnetic? Circle the correct choice.
 A) PS^{1-} B) PS C) PS^{1+}
- (iv) Draw Lewis dot structure of the molecule with the shortest bond length (based on the MO diagram) from among the three (PS^{1-} , PS, PS^{1+}), and indicate the bond order calculated from the Lewis dot structure. **(3 points)**

Lewis Dot Structure

Bond Order



E4BD4EFF-A343-40A6-9687-D8BB16D8AE94

midterm-exam-93388

#283 Page 10 of 10

SCRAP PAGE

**CHEMISTRY 110 –Midterm Exam****Thursday November 7, 2024, 6:30 – 8:30 pm****Examiners: Pallavi Sirjoosingh and Ashok Kakkar**

First name (please write as legibly as possible within the boxes)

--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--

Last name

--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--

Student ID number

--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--

This booklet has 8 questions on 10 pages (The last page can be used as a scrap page).

The staple must NOT be removed from the question booklet. Your intact question booklet must be returned at the end of the exam.

A separate booklet contains a periodic table, and datasheet.

Write your answers directly under the questions in the space provided. Do not write on the QR codes or near the top edges (as it may be cut off).

Please write **legibly and dark enough** for the scan to be readable. For students who use English as a second language, dictionaries are permitted. Calculators are allowed.

DO NOT ASK QUESTIONS OF THE INVIGILATORS about interpretations of the exam questions. If you have a concern about a question, please summarize it at the top of this page. All concerns will be evaluated after the exam. GOOD LUCK!

**Question 1 8 points** (*Assume all transitions are possible*)

a. 6 points Calculate the **energy (in kJ/mol)** required to ionize (completely remove) the electron from the **n=3** level of **He⁺ ion** (He: Helium). Show your work including equations. (3 sig. figs.)

b. 2 points

(I) Select the largest (longest) wavelength from the three choices below:

- i. Wavelength of photon to *completely remove* an electron from *1s orbital of Hydrogen atom*
- ii. Wavelength of photon to *completely remove* an electron from *1s orbital of Boron atom (B)*
- iii. Wavelength of photon to *completely remove* an electron from *1s orbital of Boron cation (B⁺)*

i or ii or iii

(II) Select the smallest (shortest) wavelength from the three choices below (*for H atom*):

- i. Wavelength of photon *emitted* for an electron transitioning from *2p to 1s orbital*
- ii. Wavelength of photon *absorbed* for an electron transitioning from *2s to 4p orbital*
- iii. Wavelength of photon *absorbed* for an electron transitioning from *1s to 3p orbital*

i or ii or iii



Question 2 8 points



A Chem110 student plots the graph shown on the left, from data obtained for the kinetic energy (eV) of the ejected electron versus frequency (s^{-1}) of the incident light, for an unknown metal. ($1\text{eV} = 1.6022 \times 10^{-19} \text{ J}$)

Based on the plot, answer the questions **a to c** below:

a. 1 point On the plot above, label the **threshold frequency**.

b. 2 points What is the value of the work potential (in J) of the unknown metal?

c. 3 points What is the *longest* wavelength (in nm) required to eject an electron from this metal's surface? Show your work including equation(s) used. (3 sig figs)

d. 2 points In the space provided below, plot the following

(Assume frequency is *above* threshold frequency):

- Kinetic energy* of an ejected electron versus the *intensity* of the incident light (left)
- Number of electrons* ejected versus the *frequency* of the incident light (right)



**Question 3** **6 points****a. 2 points**

In the ground state of Ge (Germanium; $Z = 32$) atom, **name** the following orbitals:

i. Orbital containing the electron that requires the maximum energy to be removed.

ii. Orbital containing the electron that experiences the minimum effective nuclear charge.

b. 2 points

An unknown anion (A^-) has the following *excited* electronic configuration: $[Ar] 4s^2 3d^3 4p^1$

(Ar: Argon; $Z = 18$)

Give the **condensed ground state electronic configuration** of the atom, **A** and the cation, **A^{2+}** .
Show your work. (Assume all electronic transitions are possible)

Electronic Configuration of A:

Electronic Configuration of A^{2+} :

c. 2 points

In a single atom, determine how many electrons can have the following set of quantum numbers?

(i) $n = 5; l = 3$

In a single atom, determine how many electrons can have the following set of quantum numbers?

(i) $n = 3; l = 2; m_s = +1/2$

**Question 4** **8 points**

Rank the following from 1 to 4 in **increasing** order. Provide the ranking in the boxes provided.

Note: Atomic number (Z) are provided for relevant atoms

a) Increasing (lowest to highest) order of *ionization energy*: (1= lowest; 4 = highest)

Mg

Mg: Z = 12

Rb

Rb: Z = 37

Cs

Cs: Z = 55

K

K : Z = 19

b) Increasing (lowest to highest) order of *bond angle*: (1= lowest; 4 = highest)

H₂OCCl₄XeF₄

HCN

c) Increasing (smallest to largest) order of *ionic size* (1= smallest; 4 = largest)

S²⁻

S: Z = 16

O²⁻

O: Z = 8

Be²⁺

Be: Z = 4

Na⁺

Na: Z = 11

d) Increasing (lowest to highest) order of *bond energy* (1= lowest; 4 = highest)

P-F

P-Cl

P-I

P-Br

**Question 5** **12 points**

a. 9 points For the molecules HClO_4 and TeF_6 , draw the most stable *Lewis structure* showing appropriate shape and *dashed/wedged bonds*, based on VSEPR. Determine the **molecular geometry** around the central atom. For HClO_4 , label the *most polar bond*.

HClO_4 (there are no O-O bonds)

Label the most polar bond

Molecular Geometry around Cl

TeF_6

Molecular Geometry around Te

b. 3 points (Note: Lone pairs are not shown in structures below)

(i) For the three *resonance* structures (1, 2 and 3) shown below, answer the following question:



(I) Circle the *most contributing* resonance structure.

Structure 1

Structure 2

Structure 3

(ii) For the three structures given (4, 5, and 6) below, answer the following questions:



(I) Circle all the structures that represent *resonance structures* of each other:

Structure 4

Structure 5

Structure 6

(II) Among the resonance structures, circle the *most contributing* resonance structure.

Structure 4

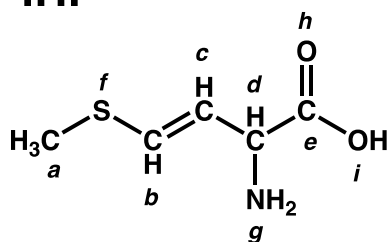
Structure 5

Structure 6

**Question 6** **6 points**

Using the data given below, calculate the **lattice energy** for SrO(s).

	ΔH° (kJ/mol)
$\text{Sr}(s) \longrightarrow \text{Sr}(g)$	+180
$\text{Sr}(g) \longrightarrow \text{Sr}^{2+}(g) + 2e^-$	+1468
$\text{O}_2(g) \longrightarrow 2\text{O}(g)$	+500
$\text{O}(g) + 2e^- \longrightarrow \text{O}^{2-}(g)$	+600
$\text{Sr}(s) + \frac{1}{2}\text{O}_2(g) \longrightarrow \text{SrO}(s)$	-550

**Question 7 10 points**

Consider the molecule given on the left, and answer questions (i) to (vi) using chemical bonding theories that we have learnt in class (valence bond theory, hybridization).

Note that the lone pairs are not shown in the structure:

i. Consider the carbon atoms marked in as ***a***, ***b***, ***c***, ***d***, and ***e*** in the molecule, and determine **hybridization** at each of the carbon centers. (2.5 points)

Atom	Hybridization
Carbon (<i>a</i>)	
Carbon (<i>b</i>)	
Carbon (<i>c</i>)	
Carbon (<i>d</i>)	
Carbon (<i>e</i>)	

ii. Now, consider the sulfur atom (S) marked as ***f***, nitrogen (N) atom (***g***), and oxygen(O) atoms (***h*** and ***i***), and determine their **hybridization**. (2 points)

Atom	Hybridization
Sulfur (<i>f</i>)	
Nitrogen (<i>g</i>)	
Oxygen (<i>h</i>)	
Oxygen (<i>i</i>)	

iii. **Describe** all the bonds formed by the carbon atoms ***a*** and ***b***, using sigma and pi designation as appropriate. (2 points)

Atom	Bonds formed (use sigma and pi as appropriate)
Carbon (<i>a</i>)	
Carbon (<i>b</i>)	

iv. How many total sp^3 -s sigma bonds are in the molecule? (1 point)

v. Determine **electronic** and **molecular geometry** at carbon (C) labeled ***e*** and sulfur (S) labeled ***f*** centres. (2 points)

Atom	Electronic Geometry	Molecular Geometry
Carbon (<i>e</i>)		
Sulfur (<i>f</i>)		

vi. Circle the value from the choices given below, that describes the **C-S-C bond angle at sulfur** (S) marked ***f***: (0.5 points)

>90° to <109.5° 109.5° >109.5 to <120° 120° >120° to <180°


Question 8 12 points

Using the **valence shell orbitals only** draw a **complete** molecular orbital (MO) diagram (with no s-p mixing) for the molecule **PS**, in the space provided below. **(6 points)**

Assume that 3s and 3p atomic orbitals will form molecular orbitals like those formed by the combination of 2s and 2p atomic orbitals that we learnt in class, and there is no s-p mixing.

Show **energy arrow**, **atomic orbitals** involved, **atom labels**, the **molecular orbitals** that are formed, **filled electrons** in the MO diagram etc. Make sure to label all the atomic and molecular orbitals.

Using the MO diagram above, answer the following questions:

- (i) Which one of the following will have the highest bond order? Circle the correct choice
 A) PS^{1-} B) PS C) PS^{1+}
- (ii) Which one of the following will have the longest bond length? Circle the correct choice
 A) PS^{1-} B) PS C) PS^{1+}
- (iii) Which of the following will be diamagnetic? Circle the correct choice.
 A) PS^{1-} B) PS C) PS^{1+}
- (iv) Draw Lewis dot structure of the molecule with the shortest bond length (based on the MO diagram) from among the three (PS^{1-} , PS, PS^{1+}), and indicate the bond order calculated from the Lewis dot structure. **(3 points)**

Lewis Dot Structure

Bond Order



A777263B-6D6D-4810-BAC6-F6C9D9EBB6C1

midterm-exam-93388

#284 Page 10 of 10

SCRAP PAGE

**CHEMISTRY 110 –Midterm Exam****Thursday November 7, 2024, 6:30 – 8:30 pm**

Examiners: Pallavi Sirjoosingh and Ashok Kakkar

First name (please write as legibly as possible within the boxes)

--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--

Last name

--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--

Student ID number

--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--

This booklet has 8 questions on 10 pages (The last page can be used as a scrap page).

The staple must NOT be removed from the question booklet. Your intact question booklet must be returned at the end of the exam.

A separate booklet contains a periodic table, and datasheet.

Write your answers directly under the questions in the space provided. Do not write on the QR codes or near the top edges (as it may be cut off).

Please write **legibly and dark enough** for the scan to be readable. For students who use English as a second language, dictionaries are permitted. Calculators are allowed.

DO NOT ASK QUESTIONS OF THE INVIGILATORS about interpretations of the exam questions. If you have a concern about a question, please summarize it at the top of this page. All concerns will be evaluated after the exam. GOOD LUCK!

**Question 1 8 points** (*Assume all transitions are possible*)

a. 6 points Calculate the **energy (in kJ/mol)** required to ionize (completely remove) the electron from the **n=3** level of **He⁺ ion** (He: Helium). Show your work including equations. (3 sig. figs.)

b. 2 points

(I) Select the largest (longest) wavelength from the three choices below:

- i. Wavelength of photon to *completely remove* an electron from *1s orbital of Hydrogen atom*
- ii. Wavelength of photon to *completely remove* an electron from *1s orbital of Boron atom (B)*
- iii. Wavelength of photon to *completely remove* an electron from *1s orbital of Boron cation (B⁺)*

i or ii or iii

(II) Select the smallest (shortest) wavelength from the three choices below (*for H atom*):

- i. Wavelength of photon *emitted* for an electron transitioning from *2p to 1s orbital*
- ii. Wavelength of photon *absorbed* for an electron transitioning from *2s to 4p orbital*
- iii. Wavelength of photon *absorbed* for an electron transitioning from *1s to 3p orbital*

i or ii or iii



Question 2 8 points



A Chem110 student plots the graph shown on the left, from data obtained for the kinetic energy (eV) of the ejected electron versus frequency (s^{-1}) of the incident light, for an unknown metal. ($1\text{eV} = 1.6022 \times 10^{-19} \text{ J}$)

Based on the plot, answer the questions **a to c** below:

a. 1 point On the plot above, label the **threshold frequency**.

b. 2 points What is the value of the work potential (in J) of the unknown metal?

c. 3 points What is the *longest* wavelength (in nm) required to eject an electron from this metal's surface? Show your work including equation(s) used. (3 sig figs)

d. 2 points In the space provided below, plot the following

(Assume frequency is *above* threshold frequency):

- i. *Kinetic energy* of an ejected electron versus the *intensity* of the incident light (left)
- ii. *Number of electrons* ejected versus the *frequency* of the incident light (right)



**Question 3** **6 points****a. 2 points**

In the ground state of Ge (Germanium; $Z = 32$) atom, **name** the following orbitals:

i. Orbital containing the electron that requires the maximum energy to be removed.

ii. Orbital containing the electron that experiences the minimum effective nuclear charge.

b. 2 points

An unknown anion (A^-) has the following *excited* electronic configuration: $[Ar] 4s^2 3d^3 4p^1$

(Ar: Argon; $Z = 18$)

Give the **condensed ground state electronic configuration** of the atom, **A** and the cation, **A^{2+}** .
Show your work. (Assume all electronic transitions are possible)

Electronic Configuration of A:

Electronic Configuration of A^{2+} :

c. 2 points

In a single atom, determine how many electrons can have the following set of quantum numbers?

(i) $n = 5; l = 3$

In a single atom, determine how many electrons can have the following set of quantum numbers?

(i) $n = 3; l = 2; m_s = +1/2$

**Question 4** **8 points**

Rank the following from 1 to 4 in **increasing** order. Provide the ranking in the boxes provided.

Note: Atomic number (Z) are provided for relevant atoms

a) Increasing (lowest to highest) order of *ionization energy*: (1= lowest; 4 = highest)

Mg

Mg: Z = 12

Rb

Rb: Z = 37

Cs

Cs: Z = 55

K

K : Z = 19

b) Increasing (lowest to highest) order of *bond angle*: (1= lowest; 4 = highest)

H₂OCCl₄XeF₄

HCN

c) Increasing (smallest to largest) order of *ionic size* (1= smallest; 4 = largest)

S²⁻

S: Z = 16

O²⁻

O: Z = 8

Be²⁺

Be: Z = 4

Na⁺

Na: Z = 11

d) Increasing (lowest to highest) order of *bond energy* (1= lowest; 4 = highest)

P-F

P-Cl

P-I

P-Br

**Question 5** **12 points**

a. 9 points For the molecules HClO_4 and TeF_6 , draw the most stable *Lewis structure* showing appropriate shape and *dashed/wedged bonds*, based on VSEPR. Determine the **molecular geometry** around the central atom. For HClO_4 , label the *most polar bond*.



b. 3 points (Note: Lone pairs are not shown in structures below)

(i) For the three *resonance* structures (1, 2 and 3) shown below, answer the following question:



(I) Circle the *most contributing* resonance structure.

Structure 1

Structure 2

Structure 3

(ii) For the three structures given (4, 5, and 6) below, answer the following questions:



(I) Circle all the structures that represent *resonance structures* of each other:

Structure 4

Structure 5

Structure 6

(II) Among the resonance structures, circle the *most contributing* resonance structure.

Structure 4

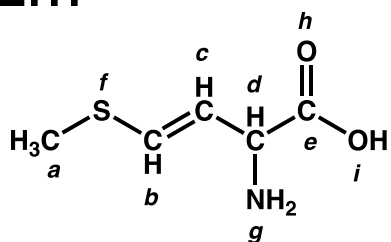
Structure 5

Structure 6

**Question 6** **6 points**

Using the data given below, calculate the **lattice energy** for SrO(s).

	ΔH° (kJ/mol)
$\text{Sr}(s) \longrightarrow \text{Sr}(g)$	+180
$\text{Sr}(g) \longrightarrow \text{Sr}^{2+}(g) + 2e^-$	+1468
$\text{O}_2(g) \longrightarrow 2\text{O}(g)$	+500
$\text{O}(g) + 2e^- \longrightarrow \text{O}^{2-}(g)$	+600
$\text{Sr}(s) + \frac{1}{2}\text{O}_2(g) \longrightarrow \text{SrO}(s)$	-550

**Question 7 10 points**

Consider the molecule given on the left, and answer questions (i) to (vi) using chemical bonding theories that we have learnt in class (valence bond theory, hybridization).

Note that the lone pairs are not shown in the structure:

i. Consider the carbon atoms marked in as ***a***, ***b***, ***c***, ***d***, and ***e*** in the molecule, and determine **hybridization** at each of the carbon centers. (2.5 points)

Atom	Hybridization
Carbon (<i>a</i>)	
Carbon (<i>b</i>)	
Carbon (<i>c</i>)	
Carbon (<i>d</i>)	
Carbon (<i>e</i>)	

ii. Now, consider the sulfur atom (S) marked as ***f***, nitrogen (N) atom (***g***), and oxygen(O) atoms (***h*** and ***i***), and determine their **hybridization**. (2 points)

Atom	Hybridization
Sulfur (<i>f</i>)	
Nitrogen (<i>g</i>)	
Oxygen (<i>h</i>)	
Oxygen (<i>i</i>)	

iii. **Describe** all the bonds formed by the carbon atoms ***a*** and ***b***, using sigma and pi designation as appropriate. (2 points)

Atom	Bonds formed (use sigma and pi as appropriate)
Carbon (<i>a</i>)	
Carbon (<i>b</i>)	

iv. How many total sp^3 -s sigma bonds are in the molecule? (1 point)

v. Determine **electronic** and **molecular geometry** at carbon (C) labeled ***e*** and sulfur (S) labeled ***f*** centres. (2 points)

Atom	Electronic Geometry	Molecular Geometry
Carbon (<i>e</i>)		
Sulfur (<i>f</i>)		

vi. Circle the value from the choices given below, that describes the **C-S-C bond angle at sulfur** (S) marked ***f***: (0.5 points)

>90° to <109.5° 109.5° >109.5 to <120° 120° >120° to <180°


Question 8 12 points

Using the **valence shell orbitals only** draw a **complete** molecular orbital (MO) diagram (with no s-p mixing) for the molecule **PS**, in the space provided below. **(6 points)**

Assume that 3s and 3p atomic orbitals will form molecular orbitals like those formed by the combination of 2s and 2p atomic orbitals that we learnt in class, and there is no s-p mixing.

Show **energy arrow**, **atomic orbitals** involved, **atom labels**, the **molecular orbitals** that are formed, **filled electrons** in the MO diagram etc. Make sure to label all the atomic and molecular orbitals.

Using the MO diagram above, answer the following questions:

- (i) Which one of the following will have the highest bond order? Circle the correct choice
 A) PS^{1-} B) PS C) PS^{1+}
- (ii) Which one of the following will have the longest bond length? Circle the correct choice
 A) PS^{1-} B) PS C) PS^{1+}
- (iii) Which of the following will be diamagnetic? Circle the correct choice.
 A) PS^{1-} B) PS C) PS^{1+}
- (iv) Draw Lewis dot structure of the molecule with the shortest bond length (based on the MO diagram) from among the three (PS^{1-} , PS, PS^{1+}), and indicate the bond order calculated from the Lewis dot structure. **(3 points)**

Lewis Dot Structure

Bond Order



0747858B-3E30-436B-A62A-56579BF26598

midterm-exam-93388

#285 Page 10 of 10

SCRAP PAGE

**CHEMISTRY 110 –Midterm Exam****Thursday November 7, 2024, 6:30 – 8:30 pm**

Examiners: Pallavi Sirjoosingh and Ashok Kakkar

First name (please write as legibly as possible within the boxes)

--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--

Last name

--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--

Student ID number

--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--

This booklet has 8 questions on 10 pages (The last page can be used as a scrap page).

The staple must NOT be removed from the question booklet. Your intact question booklet must be returned at the end of the exam.

A separate booklet contains a periodic table, and datasheet.

Write your answers directly under the questions in the space provided. Do not write on the QR codes or near the top edges (as it may be cut off).

Please write **legibly and dark enough** for the scan to be readable. For students who use English as a second language, dictionaries are permitted. Calculators are allowed.

DO NOT ASK QUESTIONS OF THE INVIGILATORS about interpretations of the exam questions. If you have a concern about a question, please summarize it at the top of this page. All concerns will be evaluated after the exam. GOOD LUCK!

**Question 1 8 points** (*Assume all transitions are possible*)

a. 6 points Calculate the **energy (in kJ/mol)** required to ionize (completely remove) the electron from the **$n=3$** level of **He^+ ion** (He: Helium). Show your work including equations. (3 sig. figs.)

b. 2 points

(I) Select the largest (longest) wavelength from the three choices below:

- i. Wavelength of photon to *completely remove* an electron from *1s orbital of Hydrogen atom*
- ii. Wavelength of photon to *completely remove* an electron from *1s orbital of Boron atom (B)*
- iii. Wavelength of photon to *completely remove* an electron from *1s orbital of Boron cation (B^+)*

i or ii or iii

(II) Select the smallest (shortest) wavelength from the three choices below (*for H atom*):

- i. Wavelength of photon *emitted* for an electron transitioning from *2p to 1s orbital*
- ii. Wavelength of photon *absorbed* for an electron transitioning from *2s to 4p orbital*
- iii. Wavelength of photon *absorbed* for an electron transitioning from *1s to 3p orbital*

i or ii or iii



Question 2 8 points



A Chem110 student plots the graph shown on the left, from data obtained for the kinetic energy (eV) of the ejected electron versus frequency (s^{-1}) of the incident light, for an unknown metal. ($1\text{eV} = 1.6022 \times 10^{-19} \text{ J}$)

Based on the plot, answer the questions **a to c** below:

a. 1 point On the plot above, label the **threshold frequency**.

b. 2 points What is the value of the work potential (in J) of the unknown metal?

c. 3 points What is the *longest* wavelength (in nm) required to eject an electron from this metal's surface? Show your work including equation(s) used. (3 sig figs)

d. 2 points In the space provided below, plot the following

(Assume frequency is *above* threshold frequency):

- Kinetic energy* of an ejected electron versus the *intensity* of the incident light (left)
- Number of electrons* ejected versus the *frequency* of the incident light (right)



**Question 3** **6 points****a. 2 points**

In the ground state of Ge (Germanium; $Z = 32$) atom, **name** the following orbitals:

i. Orbital containing the electron that requires the maximum energy to be removed.

ii. Orbital containing the electron that experiences the minimum effective nuclear charge.

b. 2 points

An unknown anion (A^-) has the following *excited* electronic configuration: $[Ar] 4s^2 3d^3 4p^1$

(Ar: Argon; $Z = 18$)

Give the **condensed ground state electronic configuration** of the atom, **A** and the cation, **A^{2+}** .
Show your work. (Assume all electronic transitions are possible)

Electronic Configuration of A:

Electronic Configuration of A^{2+} :

c. 2 points

In a single atom, determine how many electrons can have the following set of quantum numbers?

(i) $n = 5; l = 3$

In a single atom, determine how many electrons can have the following set of quantum numbers?

(i) $n = 3; l = 2; m_s = +1/2$

**Question 4** **8 points**

Rank the following from 1 to 4 in **increasing** order. Provide the ranking in the boxes provided.

Note: Atomic number (Z) are provided for relevant atoms

a) Increasing (lowest to highest) order of *ionization energy*: (1= lowest; 4 = highest)

Mg

Mg: Z = 12

Rb

Rb: Z = 37

Cs

Cs: Z = 55

K

K : Z = 19

b) Increasing (lowest to highest) order of *bond angle*: (1= lowest; 4 = highest)

H₂OCCl₄XeF₄

HCN

c) Increasing (smallest to largest) order of *ionic size* (1= smallest; 4 = largest)

S²⁻

S: Z = 16

O²⁻

O: Z = 8

Be²⁺

Be: Z = 4

Na⁺

Na: Z = 11

d) Increasing (lowest to highest) order of *bond energy* (1= lowest; 4 = highest)

P-F

P-Cl

P-I

P-Br

**Question 5** **12 points**

a. 9 points For the molecules HClO_4 and TeF_6 , draw the most stable *Lewis structure* showing appropriate shape and *dashed/wedged bonds*, based on VSEPR. Determine the **molecular geometry** around the central atom. For HClO_4 , label the *most polar bond*.



b. 3 points (Note: Lone pairs are not shown in structures below)

(i) For the three *resonance* structures (1, 2 and 3) shown below, answer the following question:



(I) Circle the *most contributing* resonance structure.

Structure 1

Structure 2

Structure 3

(ii) For the three structures given (4, 5, and 6) below, answer the following questions:



(I) Circle all the structures that represent *resonance structures* of each other:

Structure 4

Structure 5

Structure 6

(II) Among the resonance structures, circle the *most contributing* resonance structure.

Structure 4

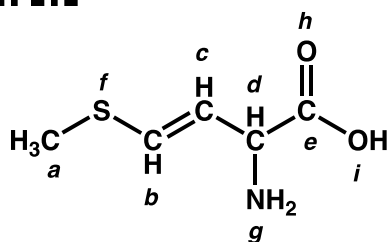
Structure 5

Structure 6

**Question 6** **6 points**

Using the data given below, calculate the **lattice energy** for SrO(s).

	ΔH° (kJ/mol)
$\text{Sr}(s) \longrightarrow \text{Sr}(g)$	+180
$\text{Sr}(g) \longrightarrow \text{Sr}^{2+}(g) + 2e^-$	+1468
$\text{O}_2(g) \longrightarrow 2\text{O}(g)$	+500
$\text{O}(g) + 2e^- \longrightarrow \text{O}^{2-}(g)$	+600
$\text{Sr}(s) + \frac{1}{2}\text{O}_2(g) \longrightarrow \text{SrO}(s)$	-550

**Question 7 10 points**

Consider the molecule given on the left, and answer questions (i) to (vi) using chemical bonding theories that we have learnt in class (valence bond theory, hybridization).

Note that the lone pairs are not shown in the structure:

i. Consider the carbon atoms marked in as ***a***, ***b***, ***c***, ***d***, and ***e*** in the molecule, and determine **hybridization** at each of the carbon centers. (2.5 points)

Atom	Hybridization
Carbon (<i>a</i>)	
Carbon (<i>b</i>)	
Carbon (<i>c</i>)	
Carbon (<i>d</i>)	
Carbon (<i>e</i>)	

ii. Now, consider the sulfur atom (S) marked as ***f***, nitrogen (N) atom (***g***), and oxygen(O) atoms (***h*** and ***i***), and determine their **hybridization**. (2 points)

Atom	Hybridization
Sulfur (<i>f</i>)	
Nitrogen (<i>g</i>)	
Oxygen (<i>h</i>)	
Oxygen (<i>i</i>)	

iii. **Describe** all the bonds formed by the carbon atoms ***a*** and ***b***, using sigma and pi designation as appropriate. (2 points)

Atom	Bonds formed (use sigma and pi as appropriate)
Carbon (<i>a</i>)	
Carbon (<i>b</i>)	

iv. How many total sp^3 -s sigma bonds are in the molecule? (1 point)

v. Determine **electronic** and **molecular geometry** at carbon (C) labeled ***e*** and sulfur (S) labeled ***f*** centres. (2 points)

Atom	Electronic Geometry	Molecular Geometry
Carbon (<i>e</i>)		
Sulfur (<i>f</i>)		

vi. Circle the value from the choices given below, that describes the **C-S-C bond angle at sulfur** (S) marked ***f***: (0.5 points)

$>90^\circ$ to $<109.5^\circ$ 109.5° >109.5 to $<120^\circ$ 120° $>120^\circ$ to $<180^\circ$


Question 8 12 points

Using the **valence shell orbitals only** draw a **complete** molecular orbital (MO) diagram (with no s-p mixing) for the molecule **PS**, in the space provided below. **(6 points)**

Assume that 3s and 3p atomic orbitals will form molecular orbitals like those formed by the combination of 2s and 2p atomic orbitals that we learnt in class, and there is no s-p mixing.

Show **energy arrow**, **atomic orbitals** involved, **atom labels**, the **molecular orbitals** that are formed, **filled electrons** in the MO diagram etc. Make sure to label all the atomic and molecular orbitals.

Using the MO diagram above, answer the following questions:

- (i) Which one of the following will have the highest bond order? Circle the correct choice
 A) PS^{1-} B) PS C) PS^{1+}
- (ii) Which one of the following will have the longest bond length? Circle the correct choice
 A) PS^{1-} B) PS C) PS^{1+}
- (iii) Which of the following will be diamagnetic? Circle the correct choice.
 A) PS^{1-} B) PS C) PS^{1+}
- (iv) Draw Lewis dot structure of the molecule with the shortest bond length (based on the MO diagram) from among the three (PS^{1-} , PS, PS^{1+}), and indicate the bond order calculated from the Lewis dot structure. **(3 points)**

Lewis Dot Structure

Bond Order



82E15216-F6DE-4AE7-AD8A-E7549A528548

midterm-exam-93388

#286 Page 10 of 10

SCRAP PAGE

**CHEMISTRY 110 –Midterm Exam****Thursday November 7, 2024, 6:30 – 8:30 pm****Examiners: Pallavi Sirjoosingh and Ashok Kakkar**

First name (please write as legibly as possible within the boxes)

--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--

Last name

--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--

Student ID number

--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--

This booklet has 8 questions on 10 pages (The last page can be used as a scrap page).

The staple must NOT be removed from the question booklet. Your intact question booklet must be returned at the end of the exam.

A separate booklet contains a periodic table, and datasheet.

Write your answers directly under the questions in the space provided. Do not write on the QR codes or near the top edges (as it may be cut off).

Please write **legibly and dark enough** for the scan to be readable. For students who use English as a second language, dictionaries are permitted. Calculators are allowed.

DO NOT ASK QUESTIONS OF THE INVIGILATORS about interpretations of the exam questions. If you have a concern about a question, please summarize it at the top of this page. All concerns will be evaluated after the exam. GOOD LUCK!

**Question 1 8 points** (*Assume all transitions are possible*)

a. 6 points Calculate the **energy (in kJ/mol)** required to ionize (completely remove) the electron from the **$n=3$** level of **He^+ ion** (He: Helium). Show your work including equations. (3 sig. figs.)

b. 2 points

(I) Select the largest (longest) wavelength from the three choices below:

- i. Wavelength of photon to *completely remove* an electron from *1s orbital of Hydrogen atom*
- ii. Wavelength of photon to *completely remove* an electron from *1s orbital of Boron atom (B)*
- iii. Wavelength of photon to *completely remove* an electron from *1s orbital of Boron cation (B^+)*

i or ii or iii

(II) Select the smallest (shortest) wavelength from the three choices below (*for H atom*):

- i. Wavelength of photon *emitted* for an electron transitioning from *2p to 1s orbital*
- ii. Wavelength of photon *absorbed* for an electron transitioning from *2s to 4p orbital*
- iii. Wavelength of photon *absorbed* for an electron transitioning from *1s to 3p orbital*

i or ii or iii



Question 2 8 points



A Chem110 student plots the graph shown on the left, from data obtained for the kinetic energy (eV) of the ejected electron versus frequency (s^{-1}) of the incident light, for an unknown metal. ($1\text{eV} = 1.6022 \times 10^{-19} \text{ J}$)

Based on the plot, answer the questions **a to c** below:

a. 1 point On the plot above, label the **threshold frequency**.

b. 2 points What is the value of the work potential (in J) of the unknown metal?

c. 3 points What is the *longest* wavelength (in nm) required to eject an electron from this metal's surface? Show your work including equation(s) used. (3 sig figs)

d. 2 points In the space provided below, plot the following

(Assume frequency is *above* threshold frequency):

- Kinetic energy* of an ejected electron versus the *intensity* of the incident light (left)
- Number of electrons* ejected versus the *frequency* of the incident light (right)



**Question 3** **6 points****a. 2 points**

In the ground state of Ge (Germanium; $Z = 32$) atom, **name** the following orbitals:

i. Orbital containing the electron that requires the maximum energy to be removed.

ii. Orbital containing the electron that experiences the minimum effective nuclear charge.

b. 2 points

An unknown anion (A^-) has the following *excited* electronic configuration: $[Ar] 4s^2 3d^3 4p^1$

(Ar: Argon; $Z = 18$)

Give the **condensed ground state electronic configuration** of the atom, **A** and the cation, **A^{2+}** .
Show your work. (Assume all electronic transitions are possible)

Electronic Configuration of A:

Electronic Configuration of A^{2+} :

c. 2 points

In a single atom, determine how many electrons can have the following set of quantum numbers?

(i) $n = 5; l = 3$

In a single atom, determine how many electrons can have the following set of quantum numbers?

(i) $n = 3; l = 2; m_s = +1/2$

**Question 4** **8 points**

Rank the following from 1 to 4 in **increasing** order. Provide the ranking in the boxes provided.

Note: Atomic number (Z) are provided for relevant atoms

a) Increasing (lowest to highest) order of *ionization energy*: (1= lowest; 4 = highest)

Mg

Mg: Z = 12

Rb

Rb: Z = 37

Cs

Cs: Z = 55

K

K : Z = 19

b) Increasing (lowest to highest) order of *bond angle*: (1= lowest; 4 = highest)

H₂OCCl₄XeF₄

HCN

c) Increasing (smallest to largest) order of *ionic size* (1= smallest; 4 = largest)

S²⁻

S: Z = 16

O²⁻

O: Z = 8

Be²⁺

Be: Z = 4

Na⁺

Na: Z = 11

d) Increasing (lowest to highest) order of *bond energy* (1= lowest; 4 = highest)

P-F

P-Cl

P-I

P-Br

**Question 5** **12 points**

a. 9 points For the molecules HClO_4 and TeF_6 , draw the most stable *Lewis structure* showing appropriate shape and *dashed/wedged bonds*, based on VSEPR. Determine the **molecular geometry** around the central atom. For HClO_4 , label the *most polar bond*.

HClO_4 (there are no O-O bonds)

Label the most polar bond

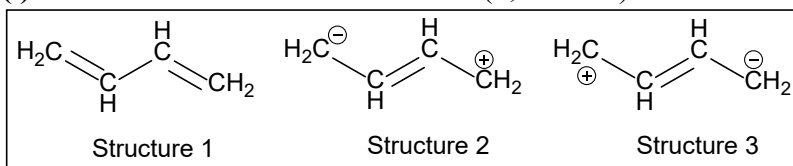
Molecular Geometry around Cl

TeF_6

Molecular Geometry around Te

b. 3 points (Note: Lone pairs are not shown in structures below)

(i) For the three *resonance* structures (1, 2 and 3) shown below, answer the following question:



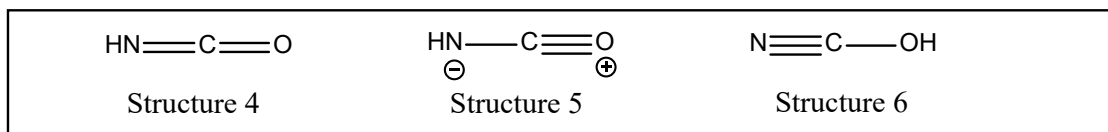
(I) Circle the *most contributing* resonance structure.

Structure 1

Structure 2

Structure 3

(ii) For the three structures given (4, 5, and 6) below, answer the following questions:



(I) Circle all the structures that represent *resonance structures* of each other:

Structure 4

Structure 5

Structure 6

(II) Among the resonance structures, circle the *most contributing* resonance structure.

Structure 4

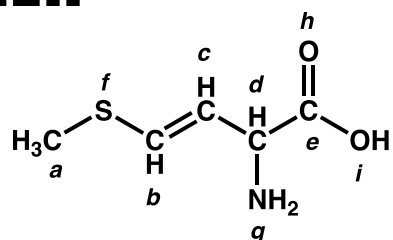
Structure 5

Structure 6

**Question 6** **6 points**

Using the data given below, calculate the **lattice energy** for SrO(s).

	ΔH° (kJ/mol)
$\text{Sr}(s) \longrightarrow \text{Sr}(g)$	+180
$\text{Sr}(g) \longrightarrow \text{Sr}^{2+}(g) + 2e^-$	+1468
$\text{O}_2(g) \longrightarrow 2\text{O}(g)$	+500
$\text{O}(g) + 2e^- \longrightarrow \text{O}^{2-}(g)$	+600
$\text{Sr}(s) + \frac{1}{2}\text{O}_2(g) \longrightarrow \text{SrO}(s)$	-550

**Question 7 10 points**

Consider the molecule given on the left, and answer questions (i) to (vi) using chemical bonding theories that we have learnt in class (valence bond theory, hybridization).

Note that the lone pairs are not shown in the structure:

i. Consider the carbon atoms marked in as ***a***, ***b***, ***c***, ***d***, and ***e*** in the molecule, and determine **hybridization** at each of the carbon centers. (2.5 points)

Atom	Hybridization
Carbon (<i>a</i>)	
Carbon (<i>b</i>)	
Carbon (<i>c</i>)	
Carbon (<i>d</i>)	
Carbon (<i>e</i>)	

ii. Now, consider the sulfur atom (S) marked as ***f***, nitrogen (N) atom (***g***), and oxygen(O) atoms (***h*** and ***i***), and determine their **hybridization**. (2 points)

Atom	Hybridization
Sulfur (<i>f</i>)	
Nitrogen (<i>g</i>)	
Oxygen (<i>h</i>)	
Oxygen (<i>i</i>)	

iii. **Describe** all the bonds formed by the carbon atoms ***a*** and ***b***, using sigma and pi designation as appropriate. (2 points)

Atom	Bonds formed (use sigma and pi as appropriate)
Carbon (<i>a</i>)	
Carbon (<i>b</i>)	

iv. How many total sp^3 -s sigma bonds are in the molecule? (1 point)

v. Determine **electronic** and **molecular geometry** at carbon (C) labeled ***e*** and sulfur (S) labeled ***f*** centres. (2 points)

Atom	Electronic Geometry	Molecular Geometry
Carbon (<i>e</i>)		
Sulfur (<i>f</i>)		

vi. Circle the value from the choices given below, that describes the **C-S-C bond angle at sulfur** (S) marked ***f***: (0.5 points)

>90° to <109.5° 109.5° >109.5 to <120° 120° >120° to <180°


Question 8 12 points

Using the **valence shell orbitals only** draw a **complete** molecular orbital (MO) diagram (with no s-p mixing) for the molecule **PS**, in the space provided below. **(6 points)**

Assume that 3s and 3p atomic orbitals will form molecular orbitals like those formed by the combination of 2s and 2p atomic orbitals that we learnt in class, and there is no s-p mixing.

Show **energy arrow**, **atomic orbitals** involved, **atom labels**, the **molecular orbitals** that are formed, **filled electrons** in the MO diagram etc. Make sure to label all the atomic and molecular orbitals.

Using the MO diagram above, answer the following questions:

- (i) Which one of the following will have the highest bond order? Circle the correct choice
 A) PS^{1-} B) PS C) PS^{1+}
- (ii) Which one of the following will have the longest bond length? Circle the correct choice
 A) PS^{1-} B) PS C) PS^{1+}
- (iii) Which of the following will be diamagnetic? Circle the correct choice.
 A) PS^{1-} B) PS C) PS^{1+}
- (iv) Draw Lewis dot structure of the molecule with the shortest bond length (based on the MO diagram) from among the three (PS^{1-} , PS, PS^{1+}), and indicate the bond order calculated from the Lewis dot structure. **(3 points)**

Lewis Dot Structure

Bond Order



0CD34983-A8F7-49D9-8F89-B3D5B6242AF4

midterm-exam-93388

#287 Page 10 of 10

SCRAP PAGE

**CHEMISTRY 110 –Midterm Exam****Thursday November 7, 2024, 6:30 – 8:30 pm****Examiners: Pallavi Sirjoosingh and Ashok Kakkar**

First name (please write as legibly as possible within the boxes)

--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--

Last name

--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--

Student ID number

--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--

This booklet has 8 questions on 10 pages (The last page can be used as a scrap page).

The staple must NOT be removed from the question booklet. Your intact question booklet must be returned at the end of the exam.

A separate booklet contains a periodic table, and datasheet.

Write your answers directly under the questions in the space provided. Do not write on the QR codes or near the top edges (as it may be cut off).

Please write **legibly and dark enough** for the scan to be readable. For students who use English as a second language, dictionaries are permitted. Calculators are allowed.

DO NOT ASK QUESTIONS OF THE INVIGILATORS about interpretations of the exam questions. If you have a concern about a question, please summarize it at the top of this page. All concerns will be evaluated after the exam. GOOD LUCK!

**Question 1 8 points** (*Assume all transitions are possible*)

a. 6 points Calculate the **energy (in kJ/mol)** required to ionize (completely remove) the electron from the **n=3** level of **He⁺ ion** (He: Helium). Show your work including equations. (3 sig. figs.)

b. 2 points

(I) Select the largest (longest) wavelength from the three choices below:

- i. Wavelength of photon to *completely remove* an electron from *1s orbital of Hydrogen atom*
- ii. Wavelength of photon to *completely remove* an electron from *1s orbital of Boron atom (B)*
- iii. Wavelength of photon to *completely remove* an electron from *1s orbital of Boron cation (B⁺)*

i or ii or iii

(II) Select the smallest (shortest) wavelength from the three choices below (*for H atom*):

- i. Wavelength of photon *emitted* for an electron transitioning from *2p to 1s orbital*
- ii. Wavelength of photon *absorbed* for an electron transitioning from *2s to 4p orbital*
- iii. Wavelength of photon *absorbed* for an electron transitioning from *1s to 3p orbital*

i or ii or iii



Question 2 8 points



A Chem110 student plots the graph shown on the left, from data obtained for the kinetic energy (eV) of the ejected electron versus frequency (s^{-1}) of the incident light, for an unknown metal. ($1\text{eV} = 1.6022 \times 10^{-19} \text{ J}$)

Based on the plot, answer the questions **a to c** below:

a. 1 point On the plot above, label the **threshold frequency**.

b. 2 points What is the value of the work potential (in J) of the unknown metal?

c. 3 points What is the *longest* wavelength (in nm) required to eject an electron from this metal's surface? Show your work including equation(s) used. (3 sig figs)

d. 2 points In the space provided below, plot the following

(Assume frequency is *above* threshold frequency):

- i. *Kinetic energy* of an ejected electron versus the *intensity* of the incident light (left)
- ii. *Number of electrons* ejected versus the *frequency* of the incident light (right)



**Question 3** **6 points****a. 2 points**

In the ground state of Ge (Germanium; $Z = 32$) atom, **name** the following orbitals:

i. Orbital containing the electron that requires the maximum energy to be removed.

ii. Orbital containing the electron that experiences the minimum effective nuclear charge.

b. 2 points

An unknown anion (A^-) has the following *excited* electronic configuration: $[Ar] 4s^2 3d^3 4p^1$

(Ar: Argon; $Z = 18$)

Give the **condensed ground state electronic configuration** of the atom, **A** and the cation, **A^{2+}** .
Show your work. (Assume all electronic transitions are possible)

Electronic Configuration of A:

Electronic Configuration of A^{2+} :

c. 2 points

In a single atom, determine how many electrons can have the following set of quantum numbers?

(i) $n = 5; l = 3$

In a single atom, determine how many electrons can have the following set of quantum numbers?

(i) $n = 3; l = 2; m_s = +1/2$

**Question 4** **8 points**

Rank the following from 1 to 4 in **increasing** order. Provide the ranking in the boxes provided.

Note: Atomic number (Z) are provided for relevant atoms

a) Increasing (lowest to highest) order of *ionization energy*: (1= lowest; 4 = highest)

Mg

Mg: Z = 12

Rb

Rb: Z = 37

Cs

Cs: Z = 55

K

K : Z = 19

b) Increasing (lowest to highest) order of *bond angle*: (1= lowest; 4 = highest)

H₂OCCl₄XeF₄

HCN

c) Increasing (smallest to largest) order of *ionic size* (1= smallest; 4 = largest)

S²⁻

S: Z = 16

O²⁻

O: Z = 8

Be²⁺

Be: Z = 4

Na⁺

Na: Z = 11

d) Increasing (lowest to highest) order of *bond energy* (1= lowest; 4 = highest)

P-F

P-Cl

P-I

P-Br

**Question 5** **12 points**

a. 9 points For the molecules HClO_4 and TeF_6 , draw the most stable *Lewis structure* showing appropriate shape and *dashed/wedged bonds*, based on VSEPR. Determine the **molecular geometry** around the central atom. For HClO_4 , label the *most polar bond*.



b. 3 points (Note: Lone pairs are not shown in structures below)

(i) For the three *resonance* structures (1, 2 and 3) shown below, answer the following question:



(I) Circle the *most contributing* resonance structure.

Structure 1

Structure 2

Structure 3

(ii) For the three structures given (4, 5, and 6) below, answer the following questions:



(I) Circle all the structures that represent *resonance structures* of each other:

Structure 4

Structure 5

Structure 6

(II) Among the resonance structures, circle the *most contributing* resonance structure.

Structure 4

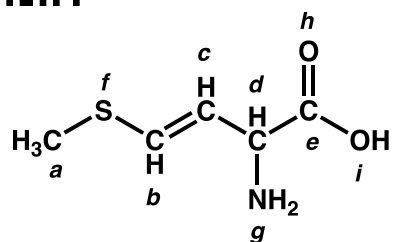
Structure 5

Structure 6

**Question 6** **6 points**

Using the data given below, calculate the **lattice energy** for SrO(s).

	ΔH° (kJ/mol)
$\text{Sr}(s) \longrightarrow \text{Sr}(g)$	+180
$\text{Sr}(g) \longrightarrow \text{Sr}^{2+}(g) + 2e^-$	+1468
$\text{O}_2(g) \longrightarrow 2\text{O}(g)$	+500
$\text{O}(g) + 2e^- \longrightarrow \text{O}^{2-}(g)$	+600
$\text{Sr}(s) + \frac{1}{2}\text{O}_2(g) \longrightarrow \text{SrO}(s)$	-550

**Question 7 10 points**

Consider the molecule given on the left, and answer questions (i) to (vi) using chemical bonding theories that we have learnt in class (valence bond theory, hybridization).

Note that the lone pairs are not shown in the structure:

i. Consider the carbon atoms marked in as ***a***, ***b***, ***c***, ***d***, and ***e*** in the molecule, and determine **hybridization** at each of the carbon centers. (2.5 points)

Atom	Hybridization
Carbon (<i>a</i>)	
Carbon (<i>b</i>)	
Carbon (<i>c</i>)	
Carbon (<i>d</i>)	
Carbon (<i>e</i>)	

ii. Now, consider the sulfur atom (S) marked as ***f***, nitrogen (N) atom (***g***), and oxygen(O) atoms (***h*** and ***i***), and determine their **hybridization**. (2 points)

Atom	Hybridization
Sulfur (<i>f</i>)	
Nitrogen (<i>g</i>)	
Oxygen (<i>h</i>)	
Oxygen (<i>i</i>)	

iii. **Describe** all the bonds formed by the carbon atoms ***a*** and ***b***, using sigma and pi designation as appropriate. (2 points)

Atom	Bonds formed (use sigma and pi as appropriate)
Carbon (<i>a</i>)	
Carbon (<i>b</i>)	

iv. How many total sp^3 -s sigma bonds are in the molecule? (1 point)

v. Determine **electronic** and **molecular geometry** at carbon (C) labeled ***e*** and sulfur (S) labeled ***f*** centres. (2 points)

Atom	Electronic Geometry	Molecular Geometry
Carbon (<i>e</i>)		
Sulfur (<i>f</i>)		

vi. Circle the value from the choices given below, that describes the **C-S-C bond angle at sulfur** (S) marked ***f***: (0.5 points)

>90° to <109.5° 109.5° >109.5 to <120° 120° >120° to <180°

**Question 8 12 points**

Using the **valence shell orbitals only** draw a **complete** molecular orbital (MO) diagram (with no s-p mixing) for the molecule **PS**, in the space provided below. **(6 points)**

Assume that 3s and 3p atomic orbitals will form molecular orbitals like those formed by the combination of 2s and 2p atomic orbitals that we learnt in class, and there is no s-p mixing.

Show **energy arrow**, **atomic orbitals** involved, **atom labels**, the **molecular orbitals** that are formed, **filled electrons** in the MO diagram etc. Make sure to label all the atomic and molecular orbitals.

Using the MO diagram above, answer the following questions:

- (i) Which one of the following will have the highest bond order? Circle the correct choice
A) PS^{1-} B) PS C) PS^{1+}
- (ii) Which one of the following will have the longest bond length? Circle the correct choice
A) PS^{1-} B) PS C) PS^{1+}
- (iii) Which of the following will be diamagnetic? Circle the correct choice.
A) PS^{1-} B) PS C) PS^{1+}
- (iv) Draw Lewis dot structure of the molecule with the shortest bond length (based on the MO diagram) from among the three (PS^{1-} , PS, PS^{1+}), and indicate the bond order calculated from the Lewis dot structure. **(3 points)**

Lewis Dot Structure

Bond Order



5F645946-916C-4154-8F3F-7F429289F00F

midterm-exam-93388

#288 Page 10 of 10

SCRAP PAGE

**CHEMISTRY 110 –Midterm Exam****Thursday November 7, 2024, 6:30 – 8:30 pm****Examiners: Pallavi Sirjoosingh and Ashok Kakkar**

First name (please write as legibly as possible within the boxes)

--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--

Last name

--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--

Student ID number

--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--

This booklet has 8 questions on 10 pages (The last page can be used as a scrap page).

The staple must NOT be removed from the question booklet. Your intact question booklet must be returned at the end of the exam.

A separate booklet contains a periodic table, and datasheet.

Write your answers directly under the questions in the space provided. Do not write on the QR codes or near the top edges (as it may be cut off).

Please write **legibly and dark enough** for the scan to be readable. For students who use English as a second language, dictionaries are permitted. Calculators are allowed.

DO NOT ASK QUESTIONS OF THE INVIGILATORS about interpretations of the exam questions. If you have a concern about a question, please summarize it at the top of this page. All concerns will be evaluated after the exam. GOOD LUCK!

**Question 1 8 points** (*Assume all transitions are possible*)

a. 6 points Calculate the **energy (in kJ/mol)** required to ionize (completely remove) the electron from the **$n=3$** level of **He^+ ion** (He: Helium). Show your work including equations. (3 sig. figs.)

b. 2 points

(I) Select the largest (longest) wavelength from the three choices below:

- i. Wavelength of photon to *completely remove* an electron from *1s orbital of Hydrogen atom*
- ii. Wavelength of photon to *completely remove* an electron from *1s orbital of Boron atom (B)*
- iii. Wavelength of photon to *completely remove* an electron from *1s orbital of Boron cation (B^+)*

i or ii or iii

(II) Select the smallest (shortest) wavelength from the three choices below (*for H atom*):

- i. Wavelength of photon *emitted* for an electron transitioning from *2p to 1s orbital*
- ii. Wavelength of photon *absorbed* for an electron transitioning from *2s to 4p orbital*
- iii. Wavelength of photon *absorbed* for an electron transitioning from *1s to 3p orbital*

i or ii or iii



Question 2 8 points



A Chem110 student plots the graph shown on the left, from data obtained for the kinetic energy (eV) of the ejected electron versus frequency (s^{-1}) of the incident light, for an unknown metal. ($1\text{eV} = 1.6022 \times 10^{-19} \text{ J}$)

Based on the plot, answer the questions **a to c** below:

a. 1 point On the plot above, label the **threshold frequency**.

b. 2 points What is the value of the work potential (in J) of the unknown metal?

c. 3 points What is the *longest* wavelength (in nm) required to eject an electron from this metal's surface? Show your work including equation(s) used. (3 sig figs)

d. 2 points In the space provided below, plot the following

(Assume frequency is *above* threshold frequency):

- Kinetic energy* of an ejected electron versus the *intensity* of the incident light (left)
- Number of electrons* ejected versus the *frequency* of the incident light (right)



**Question 3** **6 points****a. 2 points**

In the ground state of Ge (Germanium; $Z = 32$) atom, **name** the following orbitals:

i. Orbital containing the electron that requires the maximum energy to be removed.

ii. Orbital containing the electron that experiences the minimum effective nuclear charge.

b. 2 points

An unknown anion (A^-) has the following *excited* electronic configuration: $[Ar] 4s^2 3d^3 4p^1$

(Ar: Argon; $Z = 18$)

Give the **condensed ground state electronic configuration** of the atom, **A** and the cation, **A^{2+}** .
Show your work. (Assume all electronic transitions are possible)

Electronic Configuration of A:

Electronic Configuration of A^{2+} :

c. 2 points

In a single atom, determine how many electrons can have the following set of quantum numbers?

(i) $n = 5; l = 3$

In a single atom, determine how many electrons can have the following set of quantum numbers?

(i) $n = 3; l = 2; m_s = +1/2$

**Question 4** **8 points**

Rank the following from 1 to 4 in **increasing** order. Provide the ranking in the boxes provided.

Note: Atomic number (Z) are provided for relevant atoms

a) Increasing (lowest to highest) order of *ionization energy*: (1= lowest; 4 = highest)

Mg

Mg: Z = 12

Rb

Rb: Z = 37

Cs

Cs: Z = 55

K

K : Z = 19

b) Increasing (lowest to highest) order of *bond angle*: (1= lowest; 4 = highest)

H₂OCCl₄XeF₄

HCN

c) Increasing (smallest to largest) order of *ionic size* (1= smallest; 4 = largest)

S²⁻

S: Z = 16

O²⁻

O: Z = 8

Be²⁺

Be: Z = 4

Na⁺

Na: Z = 11

d) Increasing (lowest to highest) order of *bond energy* (1= lowest; 4 = highest)

P-F

P-Cl

P-I

P-Br

**Question 5** **12 points**

a. 9 points For the molecules HClO_4 and TeF_6 , draw the most stable *Lewis structure* showing appropriate shape and *dashed/wedged bonds*, based on VSEPR. Determine the **molecular geometry** around the central atom. For HClO_4 , label the *most polar bond*.



b. 3 points (Note: Lone pairs are not shown in structures below)

(i) For the three *resonance* structures (1, 2 and 3) shown below, answer the following question:



(I) Circle the *most contributing* resonance structure.

Structure 1

Structure 2

Structure 3

(ii) For the three structures given (4, 5, and 6) below, answer the following questions:



(I) Circle all the structures that represent *resonance structures* of each other:

Structure 4

Structure 5

Structure 6

(II) Among the resonance structures, circle the *most contributing* resonance structure.

Structure 4

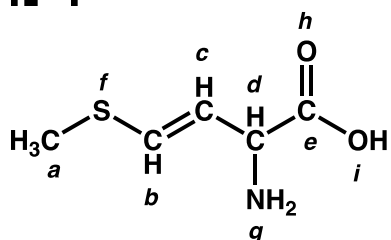
Structure 5

Structure 6

**Question 6** **6 points**

Using the data given below, calculate the **lattice energy** for SrO(s).

	ΔH° (kJ/mol)
$\text{Sr}(s) \longrightarrow \text{Sr}(g)$	+180
$\text{Sr}(g) \longrightarrow \text{Sr}^{2+}(g) + 2e^-$	+1468
$\text{O}_2(g) \longrightarrow 2\text{O}(g)$	+500
$\text{O}(g) + 2e^- \longrightarrow \text{O}^{2-}(g)$	+600
$\text{Sr}(s) + \frac{1}{2}\text{O}_2(g) \longrightarrow \text{SrO}(s)$	-550

**Question 7 10 points**

Consider the molecule given on the left, and answer questions (i) to (vi) using chemical bonding theories that we have learnt in class (valence bond theory, hybridization).

Note that the lone pairs are not shown in the structure:

i. Consider the carbon atoms marked in as ***a***, ***b***, ***c***, ***d***, and ***e*** in the molecule, and determine **hybridization** at each of the carbon centers. (2.5 points)

Atom	Hybridization
Carbon (<i>a</i>)	
Carbon (<i>b</i>)	
Carbon (<i>c</i>)	
Carbon (<i>d</i>)	
Carbon (<i>e</i>)	

ii. Now, consider the sulfur atom (S) marked as ***f***, nitrogen (N) atom (***g***), and oxygen(O) atoms (***h*** and ***i***), and determine their **hybridization**. (2 points)

Atom	Hybridization
Sulfur (<i>f</i>)	
Nitrogen (<i>g</i>)	
Oxygen (<i>h</i>)	
Oxygen (<i>i</i>)	

iii. **Describe** all the bonds formed by the carbon atoms ***a*** and ***b***, using sigma and pi designation as appropriate. (2 points)

Atom	Bonds formed (use sigma and pi as appropriate)
Carbon (<i>a</i>)	
Carbon (<i>b</i>)	

iv. How many total sp^3 -s sigma bonds are in the molecule? (1 point)

v. Determine **electronic** and **molecular geometry** at carbon (C) labeled ***e*** and sulfur (S) labeled ***f*** centres. (2 points)

Atom	Electronic Geometry	Molecular Geometry
Carbon (<i>e</i>)		
Sulfur (<i>f</i>)		

vi. Circle the value from the choices given below, that describes the **C-S-C bond angle at sulfur** (S) marked ***f***: (0.5 points)

>90° to <109.5° 109.5° >109.5 to <120° 120° >120° to <180°

**Question 8 12 points**

Using the **valence shell orbitals only** draw a **complete** molecular orbital (MO) diagram (with no s-p mixing) for the molecule **PS**, in the space provided below. **(6 points)**

Assume that 3s and 3p atomic orbitals will form molecular orbitals like those formed by the combination of 2s and 2p atomic orbitals that we learnt in class, and there is no s-p mixing.

Show **energy arrow**, **atomic orbitals** involved, **atom labels**, the **molecular orbitals** that are formed, **filled electrons** in the MO diagram etc. Make sure to label all the atomic and molecular orbitals.

Using the MO diagram above, answer the following questions:

- (i) Which one of the following will have the highest bond order? Circle the correct choice
A) PS^{1-} B) PS C) PS^{1+}
- (ii) Which one of the following will have the longest bond length? Circle the correct choice
A) PS^{1-} B) PS C) PS^{1+}
- (iii) Which of the following will be diamagnetic? Circle the correct choice.
A) PS^{1-} B) PS C) PS^{1+}
- (iv) Draw Lewis dot structure of the molecule with the shortest bond length (based on the MO diagram) from among the three (PS^{1-} , PS, PS^{1+}), and indicate the bond order calculated from the Lewis dot structure. **(3 points)**

Lewis Dot Structure

Bond Order



34C11E9B-FA76-4B0F-89C1-0C805E50CB49

midterm-exam-93388

#289 Page 10 of 10

SCRAP PAGE

**CHEMISTRY 110 –Midterm Exam****Thursday November 7, 2024, 6:30 – 8:30 pm****Examiners: Pallavi Sirjoosingh and Ashok Kakkar**

First name (please write as legibly as possible within the boxes)

--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--

Last name

--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--

Student ID number

--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--

This booklet has 8 questions on 10 pages (The last page can be used as a scrap page).

The staple must NOT be removed from the question booklet. Your intact question booklet must be returned at the end of the exam.

A separate booklet contains a periodic table, and datasheet.

Write your answers directly under the questions in the space provided. Do not write on the QR codes or near the top edges (as it may be cut off).

Please write **legibly and dark enough** for the scan to be readable. For students who use English as a second language, dictionaries are permitted. Calculators are allowed.

DO NOT ASK QUESTIONS OF THE INVIGILATORS about interpretations of the exam questions. If you have a concern about a question, please summarize it at the top of this page. All concerns will be evaluated after the exam. GOOD LUCK!

**Question 1 8 points** (*Assume all transitions are possible*)

a. 6 points Calculate the **energy (in kJ/mol)** required to ionize (completely remove) the electron from the **n=3** level of **He⁺ ion** (He: Helium). Show your work including equations. (3 sig. figs.)

b. 2 points

(I) Select the largest (longest) wavelength from the three choices below:

- i. Wavelength of photon to *completely remove* an electron from *1s orbital of Hydrogen atom*
- ii. Wavelength of photon to *completely remove* an electron from *1s orbital of Boron atom (B)*
- iii. Wavelength of photon to *completely remove* an electron from *1s orbital of Boron cation (B⁺)*

i or ii or iii

(II) Select the smallest (shortest) wavelength from the three choices below (*for H atom*):

- i. Wavelength of photon *emitted* for an electron transitioning from *2p to 1s orbital*
- ii. Wavelength of photon *absorbed* for an electron transitioning from *2s to 4p orbital*
- iii. Wavelength of photon *absorbed* for an electron transitioning from *1s to 3p orbital*

i or ii or iii



Question 2 8 points



A Chem110 student plots the graph shown on the left, from data obtained for the kinetic energy (eV) of the ejected electron versus frequency (s^{-1}) of the incident light, for an unknown metal. ($1\text{eV} = 1.6022 \times 10^{-19} \text{ J}$)

Based on the plot, answer the questions **a to c** below:

a. 1 point On the plot above, label the **threshold frequency**.

b. 2 points What is the value of the work potential (in J) of the unknown metal?

c. 3 points What is the *longest* wavelength (in nm) required to eject an electron from this metal's surface? Show your work including equation(s) used. (3 sig figs)

d. 2 points In the space provided below, plot the following

(Assume frequency is *above* threshold frequency):

- i. *Kinetic energy* of an ejected electron versus the *intensity* of the incident light (left)
- ii. *Number of electrons* ejected versus the *frequency* of the incident light (right)



**Question 3** **6 points****a. 2 points**

In the ground state of Ge (Germanium; $Z = 32$) atom, **name** the following orbitals:

i. Orbital containing the electron that requires the maximum energy to be removed.

ii. Orbital containing the electron that experiences the minimum effective nuclear charge.

b. 2 points

An unknown anion (A^-) has the following *excited* electronic configuration: $[Ar] 4s^2 3d^3 4p^1$

(Ar: Argon; $Z = 18$)

Give the **condensed ground state electronic configuration** of the atom, **A** and the cation, **A^{2+}** .
Show your work. (Assume all electronic transitions are possible)

Electronic Configuration of A:

Electronic Configuration of A^{2+} :

c. 2 points

In a single atom, determine how many electrons can have the following set of quantum numbers?

(i) $n = 5; l = 3$

In a single atom, determine how many electrons can have the following set of quantum numbers?

(i) $n = 3; l = 2; m_s = +1/2$

**Question 4** **8 points**

Rank the following from 1 to 4 in **increasing** order. Provide the ranking in the boxes provided.

Note: Atomic number (Z) are provided for relevant atoms

a) Increasing (lowest to highest) order of *ionization energy*: (1= lowest; 4 = highest)

Mg

Mg: Z = 12

Rb

Rb: Z = 37

Cs

Cs: Z = 55

K

K : Z = 19

b) Increasing (lowest to highest) order of *bond angle*: (1= lowest; 4 = highest)

H₂OCCl₄XeF₄

HCN

c) Increasing (smallest to largest) order of *ionic size* (1= smallest; 4 = largest)

S²⁻

S: Z = 16

O²⁻

O: Z = 8

Be²⁺

Be: Z = 4

Na⁺

Na: Z = 11

d) Increasing (lowest to highest) order of *bond energy* (1= lowest; 4 = highest)

P-F

P-Cl

P-I

P-Br

**Question 5** **12 points**

a. 9 points For the molecules HClO_4 and TeF_6 , draw the most stable *Lewis structure* showing appropriate shape and *dashed/wedged bonds*, based on VSEPR. Determine the **molecular geometry** around the central atom. For HClO_4 , label the *most polar bond*.



b. 3 points (Note: Lone pairs are not shown in structures below)

(i) For the three *resonance* structures (1, 2 and 3) shown below, answer the following question:



(I) Circle the *most contributing* resonance structure.

Structure 1

Structure 2

Structure 3

(ii) For the three structures given (4, 5, and 6) below, answer the following questions:



(I) Circle all the structures that represent *resonance structures* of each other:

Structure 4

Structure 5

Structure 6

(II) Among the resonance structures, circle the *most contributing* resonance structure.

Structure 4

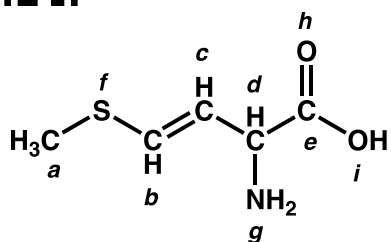
Structure 5

Structure 6

**Question 6** **6 points**

Using the data given below, calculate the **lattice energy** for SrO(s).

	ΔH° (kJ/mol)
$\text{Sr}(s) \longrightarrow \text{Sr}(g)$	+180
$\text{Sr}(g) \longrightarrow \text{Sr}^{2+}(g) + 2e^-$	+1468
$\text{O}_2(g) \longrightarrow 2\text{O}(g)$	+500
$\text{O}(g) + 2e^- \longrightarrow \text{O}^{2-}(g)$	+600
$\text{Sr}(s) + \frac{1}{2}\text{O}_2(g) \longrightarrow \text{SrO}(s)$	-550

**Question 7 10 points**

Consider the molecule given on the left, and answer questions (i) to (vi) using chemical bonding theories that we have learnt in class (valence bond theory, hybridization).

Note that the lone pairs are not shown in the structure:

i. Consider the carbon atoms marked in as ***a***, ***b***, ***c***, ***d***, and ***e*** in the molecule, and determine **hybridization** at each of the carbon centers. (2.5 points)

Atom	Hybridization
Carbon (<i>a</i>)	
Carbon (<i>b</i>)	
Carbon (<i>c</i>)	
Carbon (<i>d</i>)	
Carbon (<i>e</i>)	

ii. Now, consider the sulfur atom (S) marked as ***f***, nitrogen (N) atom (***g***), and oxygen(O) atoms (***h*** and ***i***), and determine their **hybridization**. (2 points)

Atom	Hybridization
Sulfur (<i>f</i>)	
Nitrogen (<i>g</i>)	
Oxygen (<i>h</i>)	
Oxygen (<i>i</i>)	

iii. **Describe** all the bonds formed by the carbon atoms ***a*** and ***b***, using sigma and pi designation as appropriate. (2 points)

Atom	Bonds formed (use sigma and pi as appropriate)
Carbon (<i>a</i>)	
Carbon (<i>b</i>)	

iv. How many total sp^3 -s sigma bonds are in the molecule? (1 point)

v. Determine **electronic** and **molecular geometry** at carbon (C) labeled ***e*** and sulfur (S) labeled ***f*** centres. (2 points)

Atom	Electronic Geometry	Molecular Geometry
Carbon (<i>e</i>)		
Sulfur (<i>f</i>)		

vi. Circle the value from the choices given below, that describes the **C-S-C bond angle at sulfur** (S) marked ***f***: (0.5 points)

>90° to <109.5° 109.5° >109.5 to <120° 120° >120° to <180°

**Question 8 12 points**

Using the **valence shell orbitals only** draw a **complete** molecular orbital (MO) diagram (with no s-p mixing) for the molecule **PS**, in the space provided below. **(6 points)**

Assume that 3s and 3p atomic orbitals will form molecular orbitals like those formed by the combination of 2s and 2p atomic orbitals that we learnt in class, and there is no s-p mixing.

Show **energy arrow**, **atomic orbitals** involved, **atom labels**, the **molecular orbitals** that are formed, **filled electrons** in the MO diagram etc. Make sure to label all the atomic and molecular orbitals.

Using the MO diagram above, answer the following questions:

- (i) Which one of the following will have the highest bond order? Circle the correct choice
A) PS^{1-} B) PS C) PS^{1+}
- (ii) Which one of the following will have the longest bond length? Circle the correct choice
A) PS^{1-} B) PS C) PS^{1+}
- (iii) Which of the following will be diamagnetic? Circle the correct choice.
A) PS^{1-} B) PS C) PS^{1+}
- (iv) Draw Lewis dot structure of the molecule with the shortest bond length (based on the MO diagram) from among the three (PS^{1-} , PS, PS^{1+}), and indicate the bond order calculated from the Lewis dot structure. **(3 points)**

Lewis Dot Structure

Bond Order



91C06623-D842-4D7A-A1B4-A2B871D2543A

midterm-exam-93388

#290 Page 10 of 10

SCRAP PAGE

**CHEMISTRY 110 –Midterm Exam****Thursday November 7, 2024, 6:30 – 8:30 pm****Examiners: Pallavi Sirjoosingh and Ashok Kakkar**

First name (please write as legibly as possible within the boxes)

--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--

Last name

--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--

Student ID number

--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--

This booklet has 8 questions on 10 pages (The last page can be used as a scrap page).

The staple must NOT be removed from the question booklet. Your intact question booklet must be returned at the end of the exam.

A separate booklet contains a periodic table, and datasheet.

Write your answers directly under the questions in the space provided. Do not write on the QR codes or near the top edges (as it may be cut off).

Please write **legibly and dark enough** for the scan to be readable. For students who use English as a second language, dictionaries are permitted. Calculators are allowed.

DO NOT ASK QUESTIONS OF THE INVIGILATORS about interpretations of the exam questions. If you have a concern about a question, please summarize it at the top of this page. All concerns will be evaluated after the exam. GOOD LUCK!

**Question 1 8 points** (*Assume all transitions are possible*)

a. 6 points Calculate the **energy (in kJ/mol)** required to ionize (completely remove) the electron from the **n=3** level of **He⁺ ion** (He: Helium). Show your work including equations. (3 sig. figs.)

b. 2 points

(I) Select the largest (longest) wavelength from the three choices below:

- i. Wavelength of photon to *completely remove* an electron from *1s orbital of Hydrogen atom*
- ii. Wavelength of photon to *completely remove* an electron from *1s orbital of Boron atom (B)*
- iii. Wavelength of photon to *completely remove* an electron from *1s orbital of Boron cation (B⁺)*

i or ii or iii

(II) Select the smallest (shortest) wavelength from the three choices below (*for H atom*):

- i. Wavelength of photon *emitted* for an electron transitioning from *2p to 1s orbital*
- ii. Wavelength of photon *absorbed* for an electron transitioning from *2s to 4p orbital*
- iii. Wavelength of photon *absorbed* for an electron transitioning from *1s to 3p orbital*

i or ii or iii



Question 2 8 points



A Chem110 student plots the graph shown on the left, from data obtained for the kinetic energy (eV) of the ejected electron versus frequency (s^{-1}) of the incident light, for an unknown metal. ($1\text{eV} = 1.6022 \times 10^{-19} \text{ J}$)

Based on the plot, answer the questions **a to c** below:

a. 1 point On the plot above, label the **threshold frequency**.

b. 2 points What is the value of the work potential (in J) of the unknown metal?

c. 3 points What is the *longest* wavelength (in nm) required to eject an electron from this metal's surface? Show your work including equation(s) used. (3 sig figs)

d. 2 points In the space provided below, plot the following

(Assume frequency is *above* threshold frequency):

- i. *Kinetic energy* of an ejected electron versus the *intensity* of the incident light (left)
- ii. *Number of electrons* ejected versus the *frequency* of the incident light (right)



**Question 3** **6 points****a. 2 points**

In the ground state of Ge (Germanium; $Z = 32$) atom, **name** the following orbitals:

i. Orbital containing the electron that requires the maximum energy to be removed.

ii. Orbital containing the electron that experiences the minimum effective nuclear charge.

b. 2 points

An unknown anion (A^-) has the following *excited* electronic configuration: $[Ar] 4s^2 3d^3 4p^1$

(Ar: Argon; $Z = 18$)

Give the **condensed ground state electronic configuration** of the atom, **A** and the cation, **A^{2+}** .
Show your work. (Assume all electronic transitions are possible)

Electronic Configuration of A:

Electronic Configuration of A^{2+} :

c. 2 points

In a single atom, determine how many electrons can have the following set of quantum numbers?

(i) $n = 5; l = 3$

In a single atom, determine how many electrons can have the following set of quantum numbers?

(i) $n = 3; l = 2; m_s = +1/2$

**Question 4** **8 points**

Rank the following from 1 to 4 in **increasing** order. Provide the ranking in the boxes provided.

Note: Atomic number (Z) are provided for relevant atoms

a) Increasing (lowest to highest) order of *ionization energy*: (1= lowest; 4 = highest)

Mg

Mg: Z = 12

Rb

Rb: Z = 37

Cs

Cs: Z = 55

K

K : Z = 19

b) Increasing (lowest to highest) order of *bond angle*: (1= lowest; 4 = highest)

H₂OCCl₄XeF₄

HCN

c) Increasing (smallest to largest) order of *ionic size* (1= smallest; 4 = largest)

S²⁻

S: Z = 16

O²⁻

O: Z = 8

Be²⁺

Be: Z = 4

Na⁺

Na: Z = 11

d) Increasing (lowest to highest) order of *bond energy* (1= lowest; 4 = highest)

P-F

P-Cl

P-I

P-Br

**Question 5** **12 points**

a. 9 points For the molecules HClO_4 and TeF_6 , draw the most stable *Lewis structure* showing appropriate shape and *dashed/wedged bonds*, based on VSEPR. Determine the **molecular geometry** around the central atom. For HClO_4 , label the *most polar bond*.



b. 3 points (Note: Lone pairs are not shown in structures below)

(i) For the three *resonance* structures (1, 2 and 3) shown below, answer the following question:



(I) Circle the *most contributing* resonance structure.

Structure 1

Structure 2

Structure 3

(ii) For the three structures given (4, 5, and 6) below, answer the following questions:



(I) Circle all the structures that represent *resonance structures* of each other:

Structure 4

Structure 5

Structure 6

(II) Among the resonance structures, circle the *most contributing* resonance structure.

Structure 4

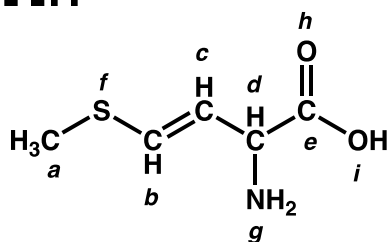
Structure 5

Structure 6

**Question 6** **6 points**

Using the data given below, calculate the **lattice energy** for SrO(s).

	ΔH° (kJ/mol)
$\text{Sr}(s) \longrightarrow \text{Sr}(g)$	+180
$\text{Sr}(g) \longrightarrow \text{Sr}^{2+}(g) + 2e^-$	+1468
$\text{O}_2(g) \longrightarrow 2\text{O}(g)$	+500
$\text{O}(g) + 2e^- \longrightarrow \text{O}^{2-}(g)$	+600
$\text{Sr}(s) + \frac{1}{2}\text{O}_2(g) \longrightarrow \text{SrO}(s)$	-550

**Question 7 10 points**

Consider the molecule given on the left, and answer questions (i) to (vi) using chemical bonding theories that we have learnt in class (valence bond theory, hybridization).

Note that the lone pairs are not shown in the structure:

i. Consider the carbon atoms marked in as ***a***, ***b***, ***c***, ***d***, and ***e*** in the molecule, and determine **hybridization** at each of the carbon centers. (2.5 points)

Atom	Hybridization
Carbon (<i>a</i>)	
Carbon (<i>b</i>)	
Carbon (<i>c</i>)	
Carbon (<i>d</i>)	
Carbon (<i>e</i>)	

ii. Now, consider the sulfur atom (S) marked as ***f***, nitrogen (N) atom (***g***), and oxygen(O) atoms (***h*** and ***i***), and determine their **hybridization**. (2 points)

Atom	Hybridization
Sulfur (<i>f</i>)	
Nitrogen (<i>g</i>)	
Oxygen (<i>h</i>)	
Oxygen (<i>i</i>)	

iii. **Describe** all the bonds formed by the carbon atoms ***a*** and ***b***, using sigma and pi designation as appropriate. (2 points)

Atom	Bonds formed (use sigma and pi as appropriate)
Carbon (<i>a</i>)	
Carbon (<i>b</i>)	

iv. How many total sp^3 -s sigma bonds are in the molecule? (1 point)

v. Determine **electronic** and **molecular geometry** at carbon (C) labeled ***e*** and sulfur (S) labeled ***f*** centres. (2 points)

Atom	Electronic Geometry	Molecular Geometry
Carbon (<i>e</i>)		
Sulfur (<i>f</i>)		

vi. Circle the value from the choices given below, that describes the **C-S-C bond angle at sulfur** (S) marked ***f***: (0.5 points)

$>90^\circ$ to $<109.5^\circ$ 109.5° >109.5 to $<120^\circ$ 120° $>120^\circ$ to $<180^\circ$

**Question 8 12 points**

Using the **valence shell orbitals only** draw a **complete** molecular orbital (MO) diagram (with no s-p mixing) for the molecule **PS**, in the space provided below. **(6 points)**

Assume that 3s and 3p atomic orbitals will form molecular orbitals like those formed by the combination of 2s and 2p atomic orbitals that we learnt in class, and there is no s-p mixing.

Show **energy arrow**, **atomic orbitals** involved, **atom labels**, the **molecular orbitals** that are formed, **filled electrons** in the MO diagram etc. Make sure to label all the atomic and molecular orbitals.

Using the MO diagram above, answer the following questions:

- (i) Which one of the following will have the highest bond order? Circle the correct choice
A) PS^{1-} B) PS C) PS^{1+}
- (ii) Which one of the following will have the longest bond length? Circle the correct choice
A) PS^{1-} B) PS C) PS^{1+}
- (iii) Which of the following will be diamagnetic? Circle the correct choice.
A) PS^{1-} B) PS C) PS^{1+}
- (iv) Draw Lewis dot structure of the molecule with the shortest bond length (based on the MO diagram) from among the three (PS^{1-} , PS, PS^{1+}), and indicate the bond order calculated from the Lewis dot structure. **(3 points)**

Lewis Dot Structure

Bond Order



2443B8FE-9B38-4119-804A-50A358544B7B

midterm-exam-93388

#291 Page 10 of 10

SCRAP PAGE

**CHEMISTRY 110 –Midterm Exam****Thursday November 7, 2024, 6:30 – 8:30 pm****Examiners: Pallavi Sirjoosingh and Ashok Kakkar**

First name (please write as legibly as possible within the boxes)

--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--

Last name

--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--

Student ID number

--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--

This booklet has 8 questions on 10 pages (The last page can be used as a scrap page).

The staple must NOT be removed from the question booklet. Your intact question booklet must be returned at the end of the exam.

A separate booklet contains a periodic table, and datasheet.

Write your answers directly under the questions in the space provided. Do not write on the QR codes or near the top edges (as it may be cut off).

Please write **legibly and dark enough** for the scan to be readable. For students who use English as a second language, dictionaries are permitted. Calculators are allowed.

DO NOT ASK QUESTIONS OF THE INVIGILATORS about interpretations of the exam questions. If you have a concern about a question, please summarize it at the top of this page. All concerns will be evaluated after the exam. GOOD LUCK!

**Question 1 8 points** (*Assume all transitions are possible*)

a. 6 points Calculate the **energy (in kJ/mol)** required to ionize (completely remove) the electron from the **n=3** level of **He⁺ ion** (He: Helium). Show your work including equations. (3 sig. figs.)

b. 2 points

(I) Select the largest (longest) wavelength from the three choices below:

- i. Wavelength of photon to *completely remove* an electron from *1s orbital of Hydrogen atom*
- ii. Wavelength of photon to *completely remove* an electron from *1s orbital of Boron atom (B)*
- iii. Wavelength of photon to *completely remove* an electron from *1s orbital of Boron cation (B⁺)*

i or ii or iii

(II) Select the smallest (shortest) wavelength from the three choices below (*for H atom*):

- i. Wavelength of photon *emitted* for an electron transitioning from *2p to 1s orbital*
- ii. Wavelength of photon *absorbed* for an electron transitioning from *2s to 4p orbital*
- iii. Wavelength of photon *absorbed* for an electron transitioning from *1s to 3p orbital*

i or ii or iii



Question 2 8 points



A Chem110 student plots the graph shown on the left, from data obtained for the kinetic energy (eV) of the ejected electron versus frequency (s^{-1}) of the incident light, for an unknown metal. ($1\text{eV} = 1.6022 \times 10^{-19} \text{ J}$)

Based on the plot, answer the questions **a to c** below:

a. 1 point On the plot above, label the **threshold frequency**.

b. 2 points What is the value of the work potential (in J) of the unknown metal?

c. 3 points What is the *longest* wavelength (in nm) required to eject an electron from this metal's surface? Show your work including equation(s) used. (3 sig figs)

d. 2 points In the space provided below, plot the following

(Assume frequency is *above* threshold frequency):

- Kinetic energy* of an ejected electron versus the *intensity* of the incident light (left)
- Number of electrons* ejected versus the *frequency* of the incident light (right)



**Question 3** **6 points****a. 2 points**

In the ground state of Ge (Germanium; $Z = 32$) atom, **name** the following orbitals:

i. Orbital containing the electron that requires the maximum energy to be removed.

ii. Orbital containing the electron that experiences the minimum effective nuclear charge.

b. 2 points

An unknown anion (A^-) has the following *excited* electronic configuration: $[Ar] 4s^2 3d^3 4p^1$

(Ar: Argon; $Z = 18$)

Give the **condensed ground state electronic configuration** of the atom, **A** and the cation, **A^{2+}** .
Show your work. (Assume all electronic transitions are possible)

Electronic Configuration of A:

Electronic Configuration of A^{2+} :

c. 2 points

In a single atom, determine how many electrons can have the following set of quantum numbers?

(i) $n = 5; l = 3$

In a single atom, determine how many electrons can have the following set of quantum numbers?

(i) $n = 3; l = 2; m_s = +1/2$

**Question 4** **8 points**

Rank the following from 1 to 4 in **increasing** order. Provide the ranking in the boxes provided.

Note: Atomic number (Z) are provided for relevant atoms

a) Increasing (lowest to highest) order of *ionization energy*: (1= lowest; 4 = highest)

Mg

Mg: Z = 12

Rb

Rb: Z = 37

Cs

Cs: Z = 55

K

K : Z = 19

b) Increasing (lowest to highest) order of *bond angle*: (1= lowest; 4 = highest)

H₂OCCl₄XeF₄

HCN

c) Increasing (smallest to largest) order of *ionic size* (1= smallest; 4 = largest)

S²⁻

S: Z = 16

O²⁻

O: Z = 8

Be²⁺

Be: Z = 4

Na⁺

Na: Z = 11

d) Increasing (lowest to highest) order of *bond energy* (1= lowest; 4 = highest)

P-F

P-Cl

P-I

P-Br

**Question 5** **12 points**

a. 9 points For the molecules HClO_4 and TeF_6 , draw the most stable *Lewis structure* showing appropriate shape and *dashed/wedged bonds*, based on VSEPR. Determine the **molecular geometry** around the central atom. For HClO_4 , label the *most polar bond*.



b. 3 points (Note: Lone pairs are not shown in structures below)

(i) For the three *resonance* structures (1, 2 and 3) shown below, answer the following question:



(I) Circle the *most contributing* resonance structure.

Structure 1

Structure 2

Structure 3

(ii) For the three structures given (4, 5, and 6) below, answer the following questions:



(I) Circle all the structures that represent *resonance structures* of each other:

Structure 4

Structure 5

Structure 6

(II) Among the resonance structures, circle the *most contributing* resonance structure.

Structure 4

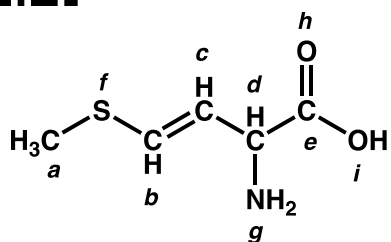
Structure 5

Structure 6

**Question 6** **6 points**

Using the data given below, calculate the **lattice energy** for SrO(s).

	ΔH° (kJ/mol)
$\text{Sr}(s) \longrightarrow \text{Sr}(g)$	+180
$\text{Sr}(g) \longrightarrow \text{Sr}^{2+}(g) + 2e^-$	+1468
$\text{O}_2(g) \longrightarrow 2\text{O}(g)$	+500
$\text{O}(g) + 2e^- \longrightarrow \text{O}^{2-}(g)$	+600
$\text{Sr}(s) + \frac{1}{2}\text{O}_2(g) \longrightarrow \text{SrO}(s)$	-550

**Question 7 10 points**

Consider the molecule given on the left, and answer questions (i) to (vi) using chemical bonding theories that we have learnt in class (valence bond theory, hybridization).

Note that the lone pairs are not shown in the structure:

i. Consider the carbon atoms marked in as ***a***, ***b***, ***c***, ***d***, and ***e*** in the molecule, and determine **hybridization** at each of the carbon centers. (2.5 points)

Atom	Hybridization
Carbon (<i>a</i>)	
Carbon (<i>b</i>)	
Carbon (<i>c</i>)	
Carbon (<i>d</i>)	
Carbon (<i>e</i>)	

ii. Now, consider the sulfur atom (S) marked as ***f***, nitrogen (N) atom (***g***), and oxygen(O) atoms (***h*** and ***i***), and determine their **hybridization**. (2 points)

Atom	Hybridization
Sulfur (<i>f</i>)	
Nitrogen (<i>g</i>)	
Oxygen (<i>h</i>)	
Oxygen (<i>i</i>)	

iii. **Describe** all the bonds formed by the carbon atoms ***a*** and ***b***, using sigma and pi designation as appropriate. (2 points)

Atom	Bonds formed (use sigma and pi as appropriate)
Carbon (<i>a</i>)	
Carbon (<i>b</i>)	

iv. How many total sp^3 -s sigma bonds are in the molecule? (1 point)

v. Determine **electronic** and **molecular geometry** at carbon (C) labeled ***e*** and sulfur (S) labeled ***f*** centres. (2 points)

Atom	Electronic Geometry	Molecular Geometry
Carbon (<i>e</i>)		
Sulfur (<i>f</i>)		

vi. Circle the value from the choices given below, that describes the **C-S-C bond angle at sulfur** (S) marked ***f***: (0.5 points)

>90° to <109.5°

109.5°

>109.5 to <120°

120°

>120° to <180°


Question 8 12 points

Using the **valence shell orbitals only** draw a **complete** molecular orbital (MO) diagram (with no s-p mixing) for the molecule **PS**, in the space provided below. **(6 points)**

Assume that 3s and 3p atomic orbitals will form molecular orbitals like those formed by the combination of 2s and 2p atomic orbitals that we learnt in class, and there is no s-p mixing.

Show **energy arrow**, **atomic orbitals** involved, **atom labels**, the **molecular orbitals** that are formed, **filled electrons** in the MO diagram etc. Make sure to label all the atomic and molecular orbitals.

Using the MO diagram above, answer the following questions:

- (i) Which one of the following will have the highest bond order? Circle the correct choice
 A) PS^{1-} B) PS C) PS^{1+}
- (ii) Which one of the following will have the longest bond length? Circle the correct choice
 A) PS^{1-} B) PS C) PS^{1+}
- (iii) Which of the following will be diamagnetic? Circle the correct choice.
 A) PS^{1-} B) PS C) PS^{1+}
- (iv) Draw Lewis dot structure of the molecule with the shortest bond length (based on the MO diagram) from among the three (PS^{1-} , PS, PS^{1+}), and indicate the bond order calculated from the Lewis dot structure. **(3 points)**

Lewis Dot Structure

Bond Order



765F55F2-64E7-4D20-8A10-3F5F1C2492D9

midterm-exam-93388

#292 Page 10 of 10

SCRAP PAGE

**CHEMISTRY 110 –Midterm Exam****Thursday November 7, 2024, 6:30 – 8:30 pm**

Examiners: Pallavi Sirjoosingh and Ashok Kakkar

First name (please write as legibly as possible within the boxes)

--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--

Last name

--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--

Student ID number

--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--

This booklet has 8 questions on 10 pages (The last page can be used as a scrap page).

The staple must NOT be removed from the question booklet. Your intact question booklet must be returned at the end of the exam.

A separate booklet contains a periodic table, and datasheet.

Write your answers directly under the questions in the space provided. Do not write on the QR codes or near the top edges (as it may be cut off).

Please write **legibly and dark enough** for the scan to be readable. For students who use English as a second language, dictionaries are permitted. Calculators are allowed.

DO NOT ASK QUESTIONS OF THE INVIGILATORS about interpretations of the exam questions. If you have a concern about a question, please summarize it at the top of this page. All concerns will be evaluated after the exam. GOOD LUCK!

**Question 1 8 points** (*Assume all transitions are possible*)

a. 6 points Calculate the **energy (in kJ/mol)** required to ionize (completely remove) the electron from the **n=3** level of **He⁺ ion** (He: Helium). Show your work including equations. (3 sig. figs.)

b. 2 points

(I) Select the largest (longest) wavelength from the three choices below:

- i. Wavelength of photon to *completely remove* an electron from *1s orbital of Hydrogen atom*
- ii. Wavelength of photon to *completely remove* an electron from *1s orbital of Boron atom (B)*
- iii. Wavelength of photon to *completely remove* an electron from *1s orbital of Boron cation (B⁺)*

i or ii or iii

(II) Select the smallest (shortest) wavelength from the three choices below (*for H atom*):

- i. Wavelength of photon *emitted* for an electron transitioning from *2p to 1s orbital*
- ii. Wavelength of photon *absorbed* for an electron transitioning from *2s to 4p orbital*
- iii. Wavelength of photon *absorbed* for an electron transitioning from *1s to 3p orbital*

i or ii or iii



Question 2 8 points



A Chem110 student plots the graph shown on the left, from data obtained for the kinetic energy (eV) of the ejected electron versus frequency (s^{-1}) of the incident light, for an unknown metal. ($1\text{eV} = 1.6022 \times 10^{-19} \text{ J}$)

Based on the plot, answer the questions **a to c** below:

a. 1 point On the plot above, label the **threshold frequency**.

b. 2 points What is the value of the work potential (in J) of the unknown metal?

c. 3 points What is the *longest* wavelength (in nm) required to eject an electron from this metal's surface? Show your work including equation(s) used. (3 sig figs)

d. 2 points In the space provided below, plot the following

(Assume frequency is *above* threshold frequency):

- Kinetic energy* of an ejected electron versus the *intensity* of the incident light (left)
- Number of electrons* ejected versus the *frequency* of the incident light (right)



**Question 3** **6 points****a. 2 points**

In the ground state of Ge (Germanium; $Z = 32$) atom, **name** the following orbitals:

i. Orbital containing the electron that requires the maximum energy to be removed.

ii. Orbital containing the electron that experiences the minimum effective nuclear charge.

b. 2 points

An unknown anion (A^-) has the following *excited* electronic configuration: $[Ar] 4s^2 3d^3 4p^1$

(Ar: Argon; $Z = 18$)

Give the **condensed ground state electronic configuration** of the atom, **A** and the cation, **A^{2+}** .
Show your work. (Assume all electronic transitions are possible)

Electronic Configuration of A:

Electronic Configuration of A^{2+} :

c. 2 points

In a single atom, determine how many electrons can have the following set of quantum numbers?

(i) $n = 5; l = 3$

In a single atom, determine how many electrons can have the following set of quantum numbers?

(i) $n = 3; l = 2; m_s = +1/2$

**Question 4** **8 points**

Rank the following from 1 to 4 in **increasing** order. Provide the ranking in the boxes provided.

Note: Atomic number (Z) are provided for relevant atoms

a) Increasing (lowest to highest) order of *ionization energy*: (1= lowest; 4 = highest)

Mg

Mg: Z = 12

Rb

Rb: Z = 37

Cs

Cs: Z = 55

K

K : Z = 19

b) Increasing (lowest to highest) order of *bond angle*: (1= lowest; 4 = highest)

H₂OCCl₄XeF₄

HCN

c) Increasing (smallest to largest) order of *ionic size* (1= smallest; 4 = largest)

S²⁻

S: Z = 16

O²⁻

O: Z = 8

Be²⁺

Be: Z = 4

Na⁺

Na: Z = 11

d) Increasing (lowest to highest) order of *bond energy* (1= lowest; 4 = highest)

P-F

P-Cl

P-I

P-Br

**Question 5** **12 points**

a. 9 points For the molecules HClO_4 and TeF_6 , draw the most stable *Lewis structure* showing appropriate shape and *dashed/wedged bonds*, based on VSEPR. Determine the **molecular geometry** around the central atom. For HClO_4 , label the *most polar bond*.



b. 3 points (Note: Lone pairs are not shown in structures below)

(i) For the three *resonance* structures (1, 2 and 3) shown below, answer the following question:



(I) Circle the *most contributing* resonance structure.

Structure 1

Structure 2

Structure 3

(ii) For the three structures given (4, 5, and 6) below, answer the following questions:



(I) Circle all the structures that represent *resonance structures* of each other:

Structure 4

Structure 5

Structure 6

(II) Among the resonance structures, circle the *most contributing* resonance structure.

Structure 4

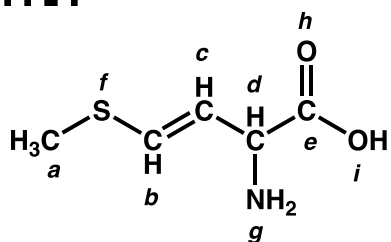
Structure 5

Structure 6

**Question 6** **6 points**

Using the data given below, calculate the **lattice energy** for SrO(s).

	ΔH° (kJ/mol)
$\text{Sr}(s) \longrightarrow \text{Sr}(g)$	+180
$\text{Sr}(g) \longrightarrow \text{Sr}^{2+}(g) + 2e^-$	+1468
$\text{O}_2(g) \longrightarrow 2\text{O}(g)$	+500
$\text{O}(g) + 2e^- \longrightarrow \text{O}^{2-}(g)$	+600
$\text{Sr}(s) + \frac{1}{2}\text{O}_2(g) \longrightarrow \text{SrO}(s)$	-550

**Question 7 10 points**

Consider the molecule given on the left, and answer questions (i) to (vi) using chemical bonding theories that we have learnt in class (valence bond theory, hybridization).

Note that the lone pairs are not shown in the structure:

i. Consider the carbon atoms marked in as ***a***, ***b***, ***c***, ***d***, and ***e*** in the molecule, and determine **hybridization** at each of the carbon centers. (2.5 points)

Atom	Hybridization
Carbon (<i>a</i>)	
Carbon (<i>b</i>)	
Carbon (<i>c</i>)	
Carbon (<i>d</i>)	
Carbon (<i>e</i>)	

ii. Now, consider the sulfur atom (S) marked as ***f***, nitrogen (N) atom (***g***), and oxygen(O) atoms (***h*** and ***i***), and determine their **hybridization**. (2 points)

Atom	Hybridization
Sulfur (<i>f</i>)	
Nitrogen (<i>g</i>)	
Oxygen (<i>h</i>)	
Oxygen (<i>i</i>)	

iii. **Describe** all the bonds formed by the carbon atoms ***a*** and ***b***, using sigma and pi designation as appropriate. (2 points)

Atom	Bonds formed (use sigma and pi as appropriate)
Carbon (<i>a</i>)	
Carbon (<i>b</i>)	

iv. How many total sp^3 -s sigma bonds are in the molecule? (1 point)

v. Determine **electronic** and **molecular geometry** at carbon (C) labeled ***e*** and sulfur (S) labeled ***f*** centres. (2 points)

Atom	Electronic Geometry	Molecular Geometry
Carbon (<i>e</i>)		
Sulfur (<i>f</i>)		

vi. Circle the value from the choices given below, that describes the **C-S-C bond angle at sulfur** (S) marked ***f***: (0.5 points)

>90° to <109.5°

109.5°

>109.5 to <120°

120°

>120° to <180°

**Question 8 12 points**

Using the **valence shell orbitals only** draw a **complete** molecular orbital (MO) diagram (with no s-p mixing) for the molecule **PS**, in the space provided below. **(6 points)**

Assume that 3s and 3p atomic orbitals will form molecular orbitals like those formed by the combination of 2s and 2p atomic orbitals that we learnt in class, and there is no s-p mixing.

Show **energy arrow**, **atomic orbitals** involved, **atom labels**, the **molecular orbitals** that are formed, **filled electrons** in the MO diagram etc. Make sure to label all the atomic and molecular orbitals.

Using the MO diagram above, answer the following questions:

- (i) Which one of the following will have the highest bond order? Circle the correct choice
A) PS^{1-} B) PS C) PS^{1+}
- (ii) Which one of the following will have the longest bond length? Circle the correct choice
A) PS^{1-} B) PS C) PS^{1+}
- (iii) Which of the following will be diamagnetic? Circle the correct choice.
A) PS^{1-} B) PS C) PS^{1+}
- (iv) Draw Lewis dot structure of the molecule with the shortest bond length (based on the MO diagram) from among the three (PS^{1-} , PS, PS^{1+}), and indicate the bond order calculated from the Lewis dot structure. **(3 points)**

Lewis Dot Structure

Bond Order



01355131-41DB-473E-8E57-1D9A8D741479

midterm-exam-93388

#293 Page 10 of 10

SCRAP PAGE

**CHEMISTRY 110 –Midterm Exam****Thursday November 7, 2024, 6:30 – 8:30 pm**

Examiners: Pallavi Sirjoosingh and Ashok Kakkar

First name (please write as legibly as possible within the boxes)

--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--

Last name

--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--

Student ID number

--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--

This booklet has 8 questions on 10 pages (The last page can be used as a scrap page).

The staple must NOT be removed from the question booklet. Your intact question booklet must be returned at the end of the exam.

A separate booklet contains a periodic table, and datasheet.

Write your answers directly under the questions in the space provided. Do not write on the QR codes or near the top edges (as it may be cut off).

Please write **legibly and dark enough** for the scan to be readable. For students who use English as a second language, dictionaries are permitted. Calculators are allowed.

DO NOT ASK QUESTIONS OF THE INVIGILATORS about interpretations of the exam questions. If you have a concern about a question, please summarize it at the top of this page. All concerns will be evaluated after the exam. GOOD LUCK!

**Question 1 8 points** (*Assume all transitions are possible*)

a. 6 points Calculate the **energy (in kJ/mol)** required to ionize (completely remove) the electron from the **n=3** level of **He⁺ ion** (He: Helium). Show your work including equations. (3 sig. figs.)

b. 2 points

(I) Select the largest (longest) wavelength from the three choices below:

- i. Wavelength of photon to *completely remove* an electron from *1s orbital of Hydrogen atom*
- ii. Wavelength of photon to *completely remove* an electron from *1s orbital of Boron atom (B)*
- iii. Wavelength of photon to *completely remove* an electron from *1s orbital of Boron cation (B⁺)*

i or ii or iii

(II) Select the smallest (shortest) wavelength from the three choices below (*for H atom*):

- i. Wavelength of photon *emitted* for an electron transitioning from *2p to 1s orbital*
- ii. Wavelength of photon *absorbed* for an electron transitioning from *2s to 4p orbital*
- iii. Wavelength of photon *absorbed* for an electron transitioning from *1s to 3p orbital*

i or ii or iii



Question 2 8 points



A Chem110 student plots the graph shown on the left, from data obtained for the kinetic energy (eV) of the ejected electron versus frequency (s^{-1}) of the incident light, for an unknown metal. ($1\text{eV} = 1.6022 \times 10^{-19} \text{ J}$)

Based on the plot, answer the questions **a to c** below:

a. 1 point On the plot above, label the **threshold frequency**.

b. 2 points What is the value of the work potential (in J) of the unknown metal?

c. 3 points What is the *longest* wavelength (in nm) required to eject an electron from this metal's surface? Show your work including equation(s) used. (3 sig figs)

d. 2 points In the space provided below, plot the following

(Assume frequency is *above* threshold frequency):

- Kinetic energy* of an ejected electron versus the *intensity* of the incident light (left)
- Number of electrons* ejected versus the *frequency* of the incident light (right)



**Question 3** **6 points****a. 2 points**

In the ground state of Ge (Germanium; $Z = 32$) atom, **name** the following orbitals:

i. Orbital containing the electron that requires the maximum energy to be removed.

ii. Orbital containing the electron that experiences the minimum effective nuclear charge.

b. 2 points

An unknown anion (A^-) has the following *excited* electronic configuration: $[Ar] 4s^2 3d^3 4p^1$

(Ar: Argon; $Z = 18$)

Give the **condensed ground state electronic configuration** of the atom, **A** and the cation, **A^{2+}** .
Show your work. (Assume all electronic transitions are possible)

Electronic Configuration of A:

Electronic Configuration of A^{2+} :

c. 2 points

In a single atom, determine how many electrons can have the following set of quantum numbers?

(i) $n = 5; l = 3$

In a single atom, determine how many electrons can have the following set of quantum numbers?

(i) $n = 3; l = 2; m_s = +1/2$

**Question 4** **8 points**

Rank the following from 1 to 4 in **increasing** order. Provide the ranking in the boxes provided.

Note: Atomic number (Z) are provided for relevant atoms

a) Increasing (lowest to highest) order of *ionization energy*: (1= lowest; 4 = highest)

Mg

Mg: Z = 12

Rb

Rb: Z = 37

Cs

Cs: Z = 55

K

K : Z = 19

b) Increasing (lowest to highest) order of *bond angle*: (1= lowest; 4 = highest)

H₂OCCl₄XeF₄

HCN

c) Increasing (smallest to largest) order of *ionic size* (1= smallest; 4 = largest)

S²⁻

S: Z = 16

O²⁻

O: Z = 8

Be²⁺

Be: Z = 4

Na⁺

Na: Z = 11

d) Increasing (lowest to highest) order of *bond energy* (1= lowest; 4 = highest)

P-F

P-Cl

P-I

P-Br

**Question 5** **12 points**

a. 9 points For the molecules HClO_4 and TeF_6 , draw the most stable *Lewis structure* showing appropriate shape and *dashed/wedged bonds*, based on VSEPR. Determine the **molecular geometry** around the central atom. For HClO_4 , label the *most polar bond*.



b. 3 points (Note: Lone pairs are not shown in structures below)

(i) For the three *resonance* structures (1, 2 and 3) shown below, answer the following question:



(I) Circle the *most contributing* resonance structure.

Structure 1

Structure 2

Structure 3

(ii) For the three structures given (4, 5, and 6) below, answer the following questions:



(I) Circle all the structures that represent *resonance structures* of each other:

Structure 4

Structure 5

Structure 6

(II) Among the resonance structures, circle the *most contributing* resonance structure.

Structure 4

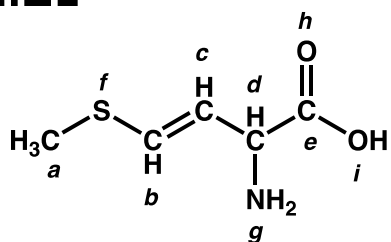
Structure 5

Structure 6

**Question 6** **6 points**

Using the data given below, calculate the **lattice energy** for SrO(s).

	ΔH° (kJ/mol)
$\text{Sr}(s) \longrightarrow \text{Sr}(g)$	+180
$\text{Sr}(g) \longrightarrow \text{Sr}^{2+}(g) + 2e^-$	+1468
$\text{O}_2(g) \longrightarrow 2\text{O}(g)$	+500
$\text{O}(g) + 2e^- \longrightarrow \text{O}^{2-}(g)$	+600
$\text{Sr}(s) + \frac{1}{2}\text{O}_2(g) \longrightarrow \text{SrO}(s)$	-550

**Question 7 10 points**

Consider the molecule given on the left, and answer questions (i) to (vi) using chemical bonding theories that we have learnt in class (valence bond theory, hybridization).

Note that the lone pairs are not shown in the structure:

i. Consider the carbon atoms marked in as ***a***, ***b***, ***c***, ***d***, and ***e*** in the molecule, and determine **hybridization** at each of the carbon centers. (2.5 points)

Atom	Hybridization
Carbon (<i>a</i>)	
Carbon (<i>b</i>)	
Carbon (<i>c</i>)	
Carbon (<i>d</i>)	
Carbon (<i>e</i>)	

ii. Now, consider the sulfur atom (S) marked as ***f***, nitrogen (N) atom (***g***), and oxygen(O) atoms (***h*** and ***i***), and determine their **hybridization**. (2 points)

Atom	Hybridization
Sulfur (<i>f</i>)	
Nitrogen (<i>g</i>)	
Oxygen (<i>h</i>)	
Oxygen (<i>i</i>)	

iii. **Describe** all the bonds formed by the carbon atoms ***a*** and ***b***, using sigma and pi designation as appropriate. (2 points)

Atom	Bonds formed (use sigma and pi as appropriate)
Carbon (<i>a</i>)	
Carbon (<i>b</i>)	

iv. How many total sp^3 -s sigma bonds are in the molecule? (1 point)

v. Determine **electronic** and **molecular geometry** at carbon (C) labeled ***e*** and sulfur (S) labeled ***f*** centres. (2 points)

Atom	Electronic Geometry	Molecular Geometry
Carbon (<i>e</i>)		
Sulfur (<i>f</i>)		

vi. Circle the value from the choices given below, that describes the **C-S-C bond angle at sulfur** (S) marked ***f***: (0.5 points)

>90° to <109.5° 109.5° >109.5 to <120° 120° >120° to <180°



Question 8 12 points

Using the **valence shell orbitals only** draw a **complete** molecular orbital (MO) diagram (with no s-p mixing) for the molecule **PS**, in the space provided below. **(6 points)**

Assume that 3s and 3p atomic orbitals will form molecular orbitals like those formed by the combination of 2s and 2p atomic orbitals that we learnt in class, and there is no s-p mixing.

Show **energy arrow**, **atomic orbitals** involved, **atom labels**, the **molecular orbitals** that are formed, **filled electrons** in the MO diagram etc. Make sure to label all the atomic and molecular orbitals.

Using the MO diagram above, answer the following questions:

- (i) Which one of the following will have the highest bond order? Circle the correct choice
 A) PS^{1-} B) PS C) PS^{1+}
- (ii) Which one of the following will have the longest bond length? Circle the correct choice
 A) PS^{1-} B) PS C) PS^{1+}
- (iii) Which of the following will be diamagnetic? Circle the correct choice.
 A) PS^{1-} B) PS C) PS^{1+}
- (iv) Draw Lewis dot structure of the molecule with the shortest bond length (based on the MO diagram) from among the three (PS^{1-} , PS, PS^{1+}), and indicate the bond order calculated from the Lewis dot structure. **(3 points)**

Lewis Dot Structure

Bond Order



07F69D98-B1BB-4DE8-BCB2-18EAADA0F5B1

midterm-exam-93388

#294 Page 10 of 10

SCRAP PAGE

**CHEMISTRY 110 –Midterm Exam****Thursday November 7, 2024, 6:30 – 8:30 pm**

Examiners: Pallavi Sirjoosingh and Ashok Kakkar

First name (please write as legibly as possible within the boxes)

--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--

Last name

--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--

Student ID number

--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--

This booklet has 8 questions on 10 pages (The last page can be used as a scrap page).

The staple must NOT be removed from the question booklet. Your intact question booklet must be returned at the end of the exam.

A separate booklet contains a periodic table, and datasheet.

Write your answers directly under the questions in the space provided. Do not write on the QR codes or near the top edges (as it may be cut off).

Please write **legibly and dark enough** for the scan to be readable. For students who use English as a second language, dictionaries are permitted. Calculators are allowed.

DO NOT ASK QUESTIONS OF THE INVIGILATORS about interpretations of the exam questions. If you have a concern about a question, please summarize it at the top of this page. All concerns will be evaluated after the exam. GOOD LUCK!

**Question 1 8 points** (*Assume all transitions are possible*)

a. 6 points Calculate the **energy (in kJ/mol)** required to ionize (completely remove) the electron from the **n=3** level of **He⁺ ion** (He: Helium). Show your work including equations. (3 sig. figs.)

b. 2 points

(I) Select the largest (longest) wavelength from the three choices below:

- i. Wavelength of photon to *completely remove* an electron from *1s orbital of Hydrogen atom*
- ii. Wavelength of photon to *completely remove* an electron from *1s orbital of Boron atom (B)*
- iii. Wavelength of photon to *completely remove* an electron from *1s orbital of Boron cation (B⁺)*

i or ii or iii

(II) Select the smallest (shortest) wavelength from the three choices below (*for H atom*):

- i. Wavelength of photon *emitted* for an electron transitioning from *2p to 1s orbital*
- ii. Wavelength of photon *absorbed* for an electron transitioning from *2s to 4p orbital*
- iii. Wavelength of photon *absorbed* for an electron transitioning from *1s to 3p orbital*

i or ii or iii



Question 2 8 points



A Chem110 student plots the graph shown on the left, from data obtained for the kinetic energy (eV) of the ejected electron versus frequency (s^{-1}) of the incident light, for an unknown metal. ($1\text{eV} = 1.6022 \times 10^{-19} \text{ J}$)

Based on the plot, answer the questions **a to c** below:

a. 1 point On the plot above, label the **threshold frequency**.

b. 2 points What is the value of the work potential (in J) of the unknown metal?

c. 3 points What is the *longest* wavelength (in nm) required to eject an electron from this metal's surface? Show your work including equation(s) used. (3 sig figs)

d. 2 points In the space provided below, plot the following

(Assume frequency is *above* threshold frequency):

- i. *Kinetic energy* of an ejected electron versus the *intensity* of the incident light (left)
- ii. *Number of electrons* ejected versus the *frequency* of the incident light (right)



**Question 3** **6 points****a. 2 points**

In the ground state of Ge (Germanium; $Z = 32$) atom, **name** the following orbitals:

i. Orbital containing the electron that requires the maximum energy to be removed.

ii. Orbital containing the electron that experiences the minimum effective nuclear charge.

b. 2 points

An unknown anion (A^-) has the following *excited* electronic configuration: $[Ar] 4s^2 3d^3 4p^1$

(Ar: Argon; $Z = 18$)

Give the **condensed ground state electronic configuration** of the atom, **A** and the cation, **A^{2+}** .
Show your work. (Assume all electronic transitions are possible)

Electronic Configuration of A:

Electronic Configuration of A^{2+} :

c. 2 points

In a single atom, determine how many electrons can have the following set of quantum numbers?

(i) $n = 5; l = 3$

In a single atom, determine how many electrons can have the following set of quantum numbers?

(i) $n = 3; l = 2; m_s = +1/2$

**Question 4** **8 points**

Rank the following from 1 to 4 in **increasing** order. Provide the ranking in the boxes provided.

Note: Atomic number (Z) are provided for relevant atoms

a) Increasing (lowest to highest) order of *ionization energy*: (1= lowest; 4 = highest)

Mg

Mg: Z = 12

Rb

Rb: Z = 37

Cs

Cs: Z = 55

K

K : Z = 19

b) Increasing (lowest to highest) order of *bond angle*: (1= lowest; 4 = highest)

H₂OCCl₄XeF₄

HCN

c) Increasing (smallest to largest) order of *ionic size* (1= smallest; 4 = largest)

S²⁻

S: Z = 16

O²⁻

O: Z = 8

Be²⁺

Be: Z = 4

Na⁺

Na: Z = 11

d) Increasing (lowest to highest) order of *bond energy* (1= lowest; 4 = highest)

P-F

P-Cl

P-I

P-Br

**Question 5** **12 points**

a. 9 points For the molecules HClO_4 and TeF_6 , draw the most stable *Lewis structure* showing appropriate shape and *dashed/wedged bonds*, based on VSEPR. Determine the **molecular geometry** around the central atom. For HClO_4 , label the *most polar bond*.

HClO_4 (there are no O-O bonds)

Label the most polar bond

Molecular Geometry around Cl

TeF_6

Molecular Geometry around Te

b. 3 points (Note: Lone pairs are not shown in structures below)

(i) For the three *resonance* structures (1, 2 and 3) shown below, answer the following question:



(I) Circle the *most contributing* resonance structure.

Structure 1

Structure 2

Structure 3

(ii) For the three structures given (4, 5, and 6) below, answer the following questions:



(I) Circle all the structures that represent *resonance structures* of each other:

Structure 4

Structure 5

Structure 6

(II) Among the resonance structures, circle the *most contributing* resonance structure.

Structure 4

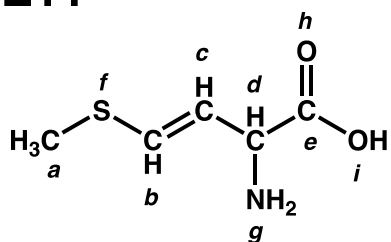
Structure 5

Structure 6

**Question 6** **6 points**

Using the data given below, calculate the **lattice energy** for SrO(s).

	ΔH° (kJ/mol)
$\text{Sr}(s) \longrightarrow \text{Sr}(g)$	+180
$\text{Sr}(g) \longrightarrow \text{Sr}^{2+}(g) + 2e^-$	+1468
$\text{O}_2(g) \longrightarrow 2\text{O}(g)$	+500
$\text{O}(g) + 2e^- \longrightarrow \text{O}^{2-}(g)$	+600
$\text{Sr}(s) + \frac{1}{2}\text{O}_2(g) \longrightarrow \text{SrO}(s)$	-550

**Question 7 10 points**

Consider the molecule given on the left, and answer questions (i) to (vi) using chemical bonding theories that we have learnt in class (valence bond theory, hybridization).

Note that the lone pairs are not shown in the structure:

i. Consider the carbon atoms marked in as ***a***, ***b***, ***c***, ***d***, and ***e*** in the molecule, and determine **hybridization** at each of the carbon centers. (2.5 points)

Atom	Hybridization
Carbon (<i>a</i>)	
Carbon (<i>b</i>)	
Carbon (<i>c</i>)	
Carbon (<i>d</i>)	
Carbon (<i>e</i>)	

ii. Now, consider the sulfur atom (S) marked as ***f***, nitrogen (N) atom (***g***), and oxygen(O) atoms (***h*** and ***i***), and determine their **hybridization**. (2 points)

Atom	Hybridization
Sulfur (<i>f</i>)	
Nitrogen (<i>g</i>)	
Oxygen (<i>h</i>)	
Oxygen (<i>i</i>)	

iii. **Describe** all the bonds formed by the carbon atoms ***a*** and ***b***, using sigma and pi designation as appropriate. (2 points)

Atom	Bonds formed (use sigma and pi as appropriate)
Carbon (<i>a</i>)	
Carbon (<i>b</i>)	

iv. How many total sp^3 -s sigma bonds are in the molecule? (1 point)

v. Determine **electronic** and **molecular geometry** at carbon (C) labeled ***e*** and sulfur (S) labeled ***f*** centres. (2 points)

Atom	Electronic Geometry	Molecular Geometry
Carbon (<i>e</i>)		
Sulfur (<i>f</i>)		

vi. Circle the value from the choices given below, that describes the **C-S-C bond angle at sulfur** (S) marked ***f***: (0.5 points)

>90° to <109.5° 109.5° >109.5 to <120° 120° >120° to <180°



Question 8 12 points

Using the **valence shell orbitals only** draw a **complete** molecular orbital (MO) diagram (with no s-p mixing) for the molecule **PS**, in the space provided below. **(6 points)**

Assume that 3s and 3p atomic orbitals will form molecular orbitals like those formed by the combination of 2s and 2p atomic orbitals that we learnt in class, and there is no s-p mixing.

Show **energy arrow**, **atomic orbitals** involved, **atom labels**, the **molecular orbitals** that are formed, **filled electrons** in the MO diagram etc. Make sure to label all the atomic and molecular orbitals.

Using the MO diagram above, answer the following questions:

- (i) Which one of the following will have the highest bond order? Circle the correct choice
 A) PS^{1-} B) PS C) PS^{1+}
- (ii) Which one of the following will have the longest bond length? Circle the correct choice
 A) PS^{1-} B) PS C) PS^{1+}
- (iii) Which of the following will be diamagnetic? Circle the correct choice.
 A) PS^{1-} B) PS C) PS^{1+}
- (iv) Draw Lewis dot structure of the molecule with the shortest bond length (based on the MO diagram) from among the three (PS^{1-} , PS, PS^{1+}), and indicate the bond order calculated from the Lewis dot structure. **(3 points)**

Lewis Dot Structure

Bond Order



E942D117-47A5-417C-AF10-0ECCE0A12D84

midterm-exam-93388

#295 Page 10 of 10

SCRAP PAGE

**CHEMISTRY 110 –Midterm Exam****Thursday November 7, 2024, 6:30 – 8:30 pm**

Examiners: Pallavi Sirjoosingh and Ashok Kakkar

First name (please write as legibly as possible within the boxes)

--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--

Last name

--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--

Student ID number

--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--

This booklet has 8 questions on 10 pages (The last page can be used as a scrap page).

The staple must NOT be removed from the question booklet. Your intact question booklet must be returned at the end of the exam.

A separate booklet contains a periodic table, and datasheet.

Write your answers directly under the questions in the space provided. Do not write on the QR codes or near the top edges (as it may be cut off).

Please write **legibly and dark enough** for the scan to be readable. For students who use English as a second language, dictionaries are permitted. Calculators are allowed.

DO NOT ASK QUESTIONS OF THE INVIGILATORS about interpretations of the exam questions. If you have a concern about a question, please summarize it at the top of this page. All concerns will be evaluated after the exam. GOOD LUCK!

**Question 1 8 points** (*Assume all transitions are possible*)

a. 6 points Calculate the **energy (in kJ/mol)** required to ionize (completely remove) the electron from the **n=3** level of **He⁺ ion** (He: Helium). Show your work including equations. (3 sig. figs.)

b. 2 points

(I) Select the largest (longest) wavelength from the three choices below:

- i. Wavelength of photon to *completely remove* an electron from *1s orbital of Hydrogen atom*
- ii. Wavelength of photon to *completely remove* an electron from *1s orbital of Boron atom (B)*
- iii. Wavelength of photon to *completely remove* an electron from *1s orbital of Boron cation (B⁺)*

i or ii or iii

(II) Select the smallest (shortest) wavelength from the three choices below (*for H atom*):

- i. Wavelength of photon *emitted* for an electron transitioning from *2p to 1s orbital*
- ii. Wavelength of photon *absorbed* for an electron transitioning from *2s to 4p orbital*
- iii. Wavelength of photon *absorbed* for an electron transitioning from *1s to 3p orbital*

i or ii or iii



Question 2 8 points



A Chem110 student plots the graph shown on the left, from data obtained for the kinetic energy (eV) of the ejected electron versus frequency (s^{-1}) of the incident light, for an unknown metal. ($1\text{eV} = 1.6022 \times 10^{-19} \text{ J}$)

Based on the plot, answer the questions **a to c** below:

a. 1 point On the plot above, label the **threshold frequency**.

b. 2 points What is the value of the work potential (in J) of the unknown metal?

c. 3 points What is the *longest* wavelength (in nm) required to eject an electron from this metal's surface? Show your work including equation(s) used. (3 sig figs)

d. 2 points In the space provided below, plot the following

(Assume frequency is *above* threshold frequency):

- Kinetic energy* of an ejected electron versus the *intensity* of the incident light (left)
- Number of electrons* ejected versus the *frequency* of the incident light (right)



**Question 3** **6 points****a. 2 points**

In the ground state of Ge (Germanium; $Z = 32$) atom, **name** the following orbitals:

i. Orbital containing the electron that requires the maximum energy to be removed.

ii. Orbital containing the electron that experiences the minimum effective nuclear charge.

b. 2 points

An unknown anion (A^-) has the following *excited* electronic configuration: $[Ar] 4s^2 3d^3 4p^1$

(Ar: Argon; $Z = 18$)

Give the **condensed ground state electronic configuration** of the atom, **A** and the cation, A^{2+} .
Show your work. (Assume all electronic transitions are possible)

Electronic Configuration of A:

Electronic Configuration of A^{2+} :

c. 2 points

In a single atom, determine how many electrons can have the following set of quantum numbers?

(i) $n = 5; l = 3$

In a single atom, determine how many electrons can have the following set of quantum numbers?

(i) $n = 3; l = 2; m_s = +1/2$

**Question 4** **8 points**

Rank the following from 1 to 4 in **increasing** order. Provide the ranking in the boxes provided.

Note: Atomic number (Z) are provided for relevant atoms

a) Increasing (lowest to highest) order of *ionization energy*: (1= lowest; 4 = highest)

Mg

Mg: Z = 12

Rb

Rb: Z = 37

Cs

Cs: Z = 55

K

K : Z = 19

b) Increasing (lowest to highest) order of *bond angle*: (1= lowest; 4 = highest)

H₂OCCl₄XeF₄

HCN

c) Increasing (smallest to largest) order of *ionic size* (1= smallest; 4 = largest)

S²⁻

S: Z = 16

O²⁻

O: Z = 8

Be²⁺

Be: Z = 4

Na⁺

Na: Z = 11

d) Increasing (lowest to highest) order of *bond energy* (1= lowest; 4 = highest)

P-F

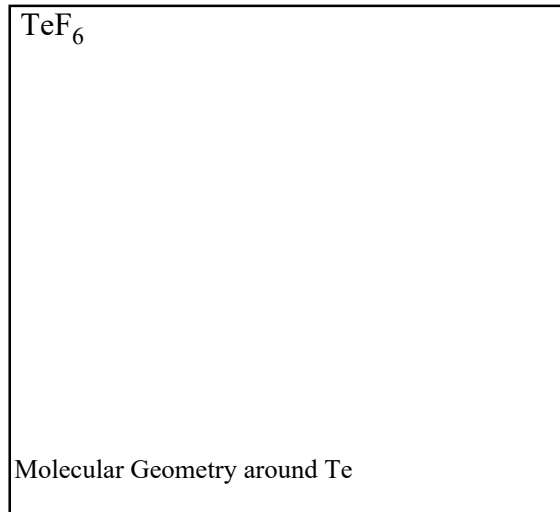
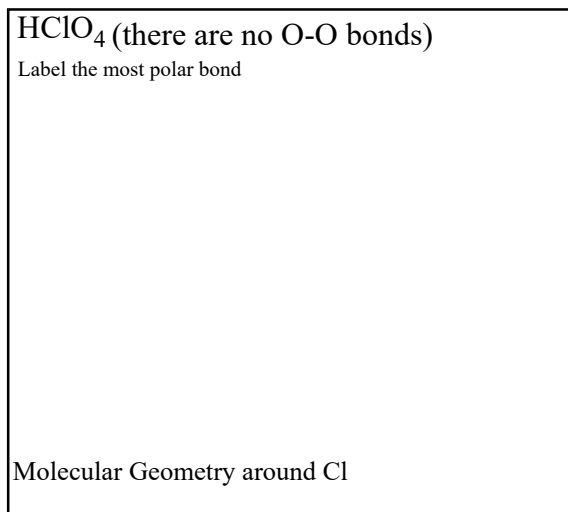
P-Cl

P-I

P-Br

**Question 5** **12 points**

a. 9 points For the molecules HClO_4 and TeF_6 , draw the most stable *Lewis structure* showing appropriate shape and *dashed/wedged bonds*, based on VSEPR. Determine the **molecular geometry** around the central atom. For HClO_4 , label the *most polar bond*.



b. 3 points (Note: Lone pairs are not shown in structures below)

(i) For the three *resonance* structures (1, 2 and 3) shown below, answer the following question:



(I) Circle the *most contributing* resonance structure.

Structure 1

Structure 2

Structure 3

(ii) For the three structures given (4, 5, and 6) below, answer the following questions:



(I) Circle all the structures that represent *resonance structures* of each other:

Structure 4

Structure 5

Structure 6

(II) Among the resonance structures, circle the *most contributing* resonance structure.

Structure 4

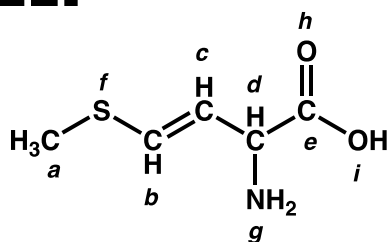
Structure 5

Structure 6

**Question 6** **6 points**

Using the data given below, calculate the **lattice energy** for SrO(s).

	ΔH° (kJ/mol)
$\text{Sr}(s) \longrightarrow \text{Sr}(g)$	+180
$\text{Sr}(g) \longrightarrow \text{Sr}^{2+}(g) + 2e^-$	+1468
$\text{O}_2(g) \longrightarrow 2\text{O}(g)$	+500
$\text{O}(g) + 2e^- \longrightarrow \text{O}^{2-}(g)$	+600
$\text{Sr}(s) + \frac{1}{2}\text{O}_2(g) \longrightarrow \text{SrO}(s)$	-550

**Question 7 10 points**

Consider the molecule given on the left, and answer questions (i) to (vi) using chemical bonding theories that we have learnt in class (valence bond theory, hybridization).

Note that the lone pairs are not shown in the structure:

i. Consider the carbon atoms marked in as ***a***, ***b***, ***c***, ***d***, and ***e*** in the molecule, and determine **hybridization** at each of the carbon centers. (2.5 points)

Atom	Hybridization
Carbon (<i>a</i>)	
Carbon (<i>b</i>)	
Carbon (<i>c</i>)	
Carbon (<i>d</i>)	
Carbon (<i>e</i>)	

ii. Now, consider the sulfur atom (S) marked as ***f***, nitrogen (N) atom (***g***), and oxygen(O) atoms (***h*** and ***i***), and determine their **hybridization**. (2 points)

Atom	Hybridization
Sulfur (<i>f</i>)	
Nitrogen (<i>g</i>)	
Oxygen (<i>h</i>)	
Oxygen (<i>i</i>)	

iii. **Describe** all the bonds formed by the carbon atoms ***a*** and ***b***, using sigma and pi designation as appropriate. (2 points)

Atom	Bonds formed (use sigma and pi as appropriate)
Carbon (<i>a</i>)	
Carbon (<i>b</i>)	

iv. How many total sp^3 -s sigma bonds are in the molecule? (1 point)

v. Determine **electronic** and **molecular geometry** at carbon (C) labeled ***e*** and sulfur (S) labeled ***f*** centres. (2 points)

Atom	Electronic Geometry	Molecular Geometry
Carbon (<i>e</i>)		
Sulfur (<i>f</i>)		

vi. Circle the value from the choices given below, that describes the **C-S-C bond angle at sulfur** (S) marked ***f***: (0.5 points)

>90° to <109.5° 109.5° >109.5 to <120° 120° >120° to <180°

**Question 8 12 points**

Using the **valence shell orbitals only** draw a **complete** molecular orbital (MO) diagram (with no s-p mixing) for the molecule **PS**, in the space provided below. **(6 points)**

Assume that 3s and 3p atomic orbitals will form molecular orbitals like those formed by the combination of 2s and 2p atomic orbitals that we learnt in class, and there is no s-p mixing.

Show **energy arrow**, **atomic orbitals** involved, **atom labels**, the **molecular orbitals** that are formed, **filled electrons** in the MO diagram etc. Make sure to label all the atomic and molecular orbitals.

Using the MO diagram above, answer the following questions:

- (i) Which one of the following will have the highest bond order? Circle the correct choice
A) PS^{1-} B) PS C) PS^{1+}
- (ii) Which one of the following will have the longest bond length? Circle the correct choice
A) PS^{1-} B) PS C) PS^{1+}
- (iii) Which of the following will be diamagnetic? Circle the correct choice.
A) PS^{1-} B) PS C) PS^{1+}
- (iv) Draw Lewis dot structure of the molecule with the shortest bond length (based on the MO diagram) from among the three (PS^{1-} , PS, PS^{1+}), and indicate the bond order calculated from the Lewis dot structure. **(3 points)**

Lewis Dot Structure

Bond Order



AF6AF8EC-4EFF-4466-834A-C22A00EBC432

midterm-exam-93388

#296 Page 10 of 10

SCRAP PAGE

**CHEMISTRY 110 –Midterm Exam****Thursday November 7, 2024, 6:30 – 8:30 pm****Examiners: Pallavi Sirjoosingh and Ashok Kakkar**

First name (please write as legibly as possible within the boxes)

--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--

Last name

--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--

Student ID number

--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--

This booklet has 8 questions on 10 pages (The last page can be used as a scrap page).

The staple must NOT be removed from the question booklet. Your intact question booklet must be returned at the end of the exam.

A separate booklet contains a periodic table, and datasheet.

Write your answers directly under the questions in the space provided. Do not write on the QR codes or near the top edges (as it may be cut off).

Please write **legibly and dark enough** for the scan to be readable. For students who use English as a second language, dictionaries are permitted. Calculators are allowed.

DO NOT ASK QUESTIONS OF THE INVIGILATORS about interpretations of the exam questions. If you have a concern about a question, please summarize it at the top of this page. All concerns will be evaluated after the exam. GOOD LUCK!

**Question 1 8 points** (*Assume all transitions are possible*)

a. 6 points Calculate the **energy (in kJ/mol)** required to ionize (completely remove) the electron from the **n=3** level of **He⁺ ion** (He: Helium). Show your work including equations. (3 sig. figs.)

b. 2 points

(I) Select the largest (longest) wavelength from the three choices below:

- i. Wavelength of photon to *completely remove* an electron from *1s orbital of Hydrogen atom*
- ii. Wavelength of photon to *completely remove* an electron from *1s orbital of Boron atom (B)*
- iii. Wavelength of photon to *completely remove* an electron from *1s orbital of Boron cation (B⁺)*

i or ii or iii

(II) Select the smallest (shortest) wavelength from the three choices below (*for H atom*):

- i. Wavelength of photon *emitted* for an electron transitioning from *2p to 1s orbital*
- ii. Wavelength of photon *absorbed* for an electron transitioning from *2s to 4p orbital*
- iii. Wavelength of photon *absorbed* for an electron transitioning from *1s to 3p orbital*

i or ii or iii



Question 2 8 points



A Chem110 student plots the graph shown on the left, from data obtained for the kinetic energy (eV) of the ejected electron versus frequency (s^{-1}) of the incident light, for an unknown metal. ($1\text{eV} = 1.6022 \times 10^{-19} \text{ J}$)

Based on the plot, answer the questions **a to c** below:

a. 1 point On the plot above, label the **threshold frequency**.

b. 2 points What is the value of the work potential (in J) of the unknown metal?

c. 3 points What is the *longest* wavelength (in nm) required to eject an electron from this metal's surface? Show your work including equation(s) used. (3 sig figs)

d. 2 points In the space provided below, plot the following

(Assume frequency is *above* threshold frequency):

- i. *Kinetic energy* of an ejected electron versus the *intensity* of the incident light (left)
- ii. *Number of electrons* ejected versus the *frequency* of the incident light (right)



**Question 3** **6 points****a. 2 points**

In the ground state of Ge (Germanium; $Z = 32$) atom, **name** the following orbitals:

i. Orbital containing the electron that requires the maximum energy to be removed.

ii. Orbital containing the electron that experiences the minimum effective nuclear charge.

b. 2 points

An unknown anion (A^-) has the following *excited* electronic configuration: $[Ar] 4s^2 3d^3 4p^1$

(Ar: Argon; $Z = 18$)

Give the **condensed ground state electronic configuration** of the atom, **A** and the cation, **A^{2+}** .
Show your work. (Assume all electronic transitions are possible)

Electronic Configuration of A:

Electronic Configuration of A^{2+} :

c. 2 points

In a single atom, determine how many electrons can have the following set of quantum numbers?

(i) $n = 5; l = 3$

In a single atom, determine how many electrons can have the following set of quantum numbers?

(i) $n = 3; l = 2; m_s = +1/2$

**Question 4** **8 points**

Rank the following from 1 to 4 in **increasing** order. Provide the ranking in the boxes provided.

Note: Atomic number (Z) are provided for relevant atoms

a) Increasing (lowest to highest) order of *ionization energy*: (1= lowest; 4 = highest)

Mg

Mg: Z = 12

Rb

Rb: Z = 37

Cs

Cs: Z = 55

K

K : Z = 19

b) Increasing (lowest to highest) order of *bond angle*: (1= lowest; 4 = highest)

H₂OCCl₄XeF₄

HCN

c) Increasing (smallest to largest) order of *ionic size* (1= smallest; 4 = largest)

S²⁻

S: Z = 16

O²⁻

O: Z = 8

Be²⁺

Be: Z = 4

Na⁺

Na: Z = 11

d) Increasing (lowest to highest) order of *bond energy* (1= lowest; 4 = highest)

P-F

P-Cl

P-I

P-Br

**Question 5** **12 points**

a. 9 points For the molecules HClO_4 and TeF_6 , draw the most stable *Lewis structure* showing appropriate shape and *dashed/wedged bonds*, based on VSEPR. Determine the **molecular geometry** around the central atom. For HClO_4 , label the *most polar bond*.



b. 3 points (Note: Lone pairs are not shown in structures below)

(i) For the three *resonance* structures (1, 2 and 3) shown below, answer the following question:



(I) Circle the *most contributing* resonance structure.

Structure 1

Structure 2

Structure 3

(ii) For the three structures given (4, 5, and 6) below, answer the following questions:



(I) Circle all the structures that represent *resonance structures* of each other:

Structure 4

Structure 5

Structure 6

(II) Among the resonance structures, circle the *most contributing* resonance structure.

Structure 4

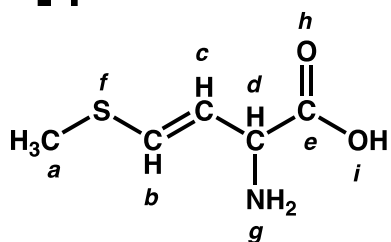
Structure 5

Structure 6

**Question 6** **6 points**

Using the data given below, calculate the **lattice energy** for SrO(s).

	ΔH° (kJ/mol)
$\text{Sr}(s) \longrightarrow \text{Sr}(g)$	+180
$\text{Sr}(g) \longrightarrow \text{Sr}^{2+}(g) + 2e^-$	+1468
$\text{O}_2(g) \longrightarrow 2\text{O}(g)$	+500
$\text{O}(g) + 2e^- \longrightarrow \text{O}^{2-}(g)$	+600
$\text{Sr}(s) + \frac{1}{2}\text{O}_2(g) \longrightarrow \text{SrO}(s)$	-550

**Question 7 10 points**

Consider the molecule given on the left, and answer questions (i) to (vi) using chemical bonding theories that we have learnt in class (valence bond theory, hybridization).

Note that the lone pairs are not shown in the structure:

i. Consider the carbon atoms marked in as ***a***, ***b***, ***c***, ***d***, and ***e*** in the molecule, and determine **hybridization** at each of the carbon centers. (2.5 points)

Atom	Hybridization
Carbon (<i>a</i>)	
Carbon (<i>b</i>)	
Carbon (<i>c</i>)	
Carbon (<i>d</i>)	
Carbon (<i>e</i>)	

ii. Now, consider the sulfur atom (S) marked as ***f***, nitrogen (N) atom (***g***), and oxygen(O) atoms (***h*** and ***i***), and determine their **hybridization**. (2 points)

Atom	Hybridization
Sulfur (<i>f</i>)	
Nitrogen (<i>g</i>)	
Oxygen (<i>h</i>)	
Oxygen (<i>i</i>)	

iii. **Describe** all the bonds formed by the carbon atoms ***a*** and ***b***, using sigma and pi designation as appropriate. (2 points)

Atom	Bonds formed (use sigma and pi as appropriate)
Carbon (<i>a</i>)	
Carbon (<i>b</i>)	

iv. How many total sp^3 -s sigma bonds are in the molecule? (1 point)

v. Determine **electronic** and **molecular geometry** at carbon (C) labeled ***e*** and sulfur (S) labeled ***f*** centres. (2 points)

Atom	Electronic Geometry	Molecular Geometry
Carbon (<i>e</i>)		
Sulfur (<i>f</i>)		

vi. Circle the value from the choices given below, that describes the **C-S-C bond angle at sulfur** (S) marked ***f***: (0.5 points)

>90° to <109.5° 109.5° >109.5 to <120° 120° >120° to <180°


Question 8 12 points

Using the **valence shell orbitals only** draw a **complete** molecular orbital (MO) diagram (with no s-p mixing) for the molecule **PS**, in the space provided below. **(6 points)**

Assume that 3s and 3p atomic orbitals will form molecular orbitals like those formed by the combination of 2s and 2p atomic orbitals that we learnt in class, and there is no s-p mixing.

Show **energy arrow**, **atomic orbitals** involved, **atom labels**, the **molecular orbitals** that are formed, **filled electrons** in the MO diagram etc. Make sure to label all the atomic and molecular orbitals.

Using the MO diagram above, answer the following questions:

- (i) Which one of the following will have the highest bond order? Circle the correct choice
 A) PS^{1-} B) PS C) PS^{1+}
- (ii) Which one of the following will have the longest bond length? Circle the correct choice
 A) PS^{1-} B) PS C) PS^{1+}
- (iii) Which of the following will be diamagnetic? Circle the correct choice.
 A) PS^{1-} B) PS C) PS^{1+}
- (iv) Draw Lewis dot structure of the molecule with the shortest bond length (based on the MO diagram) from among the three (PS^{1-} , PS, PS^{1+}), and indicate the bond order calculated from the Lewis dot structure. **(3 points)**

Lewis Dot Structure

Bond Order



36A59C16-B588-4CA3-8751-A26BC108BF4B

midterm-exam-93388

#297 Page 10 of 10

SCRAP PAGE

**CHEMISTRY 110 –Midterm Exam****Thursday November 7, 2024, 6:30 – 8:30 pm****Examiners: Pallavi Sirjoosingh and Ashok Kakkar**

First name (please write as legibly as possible within the boxes)

--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--

Last name

--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--

Student ID number

--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--

This booklet has 8 questions on 10 pages (The last page can be used as a scrap page).

The staple must NOT be removed from the question booklet. Your intact question booklet must be returned at the end of the exam.

A separate booklet contains a periodic table, and datasheet.

Write your answers directly under the questions in the space provided. Do not write on the QR codes or near the top edges (as it may be cut off).

Please write **legibly and dark enough** for the scan to be readable. For students who use English as a second language, dictionaries are permitted. Calculators are allowed.

DO NOT ASK QUESTIONS OF THE INVIGILATORS about interpretations of the exam questions. If you have a concern about a question, please summarize it at the top of this page. All concerns will be evaluated after the exam. GOOD LUCK!

**Question 1 8 points** (*Assume all transitions are possible*)

a. 6 points Calculate the **energy (in kJ/mol)** required to ionize (completely remove) the electron from the **n=3** level of **He⁺ ion** (He: Helium). Show your work including equations. (3 sig. figs.)

b. 2 points

(I) Select the largest (longest) wavelength from the three choices below:

- i. Wavelength of photon to *completely remove* an electron from *1s orbital of Hydrogen atom*
- ii. Wavelength of photon to *completely remove* an electron from *1s orbital of Boron atom (B)*
- iii. Wavelength of photon to *completely remove* an electron from *1s orbital of Boron cation (B⁺)*

i or ii or iii

(II) Select the smallest (shortest) wavelength from the three choices below (*for H atom*):

- i. Wavelength of photon *emitted* for an electron transitioning from *2p to 1s orbital*
- ii. Wavelength of photon *absorbed* for an electron transitioning from *2s to 4p orbital*
- iii. Wavelength of photon *absorbed* for an electron transitioning from *1s to 3p orbital*

i or ii or iii



Question 2 8 points



A Chem110 student plots the graph shown on the left, from data obtained for the kinetic energy (eV) of the ejected electron versus frequency (s^{-1}) of the incident light, for an unknown metal. ($1\text{eV} = 1.6022 \times 10^{-19} \text{ J}$)

Based on the plot, answer the questions **a to c** below:

a. 1 point On the plot above, label the **threshold frequency**.

b. 2 points What is the value of the work potential (in J) of the unknown metal?

c. 3 points What is the *longest* wavelength (in nm) required to eject an electron from this metal's surface? Show your work including equation(s) used. (3 sig figs)

d. 2 points In the space provided below, plot the following

(Assume frequency is *above* threshold frequency):

- Kinetic energy* of an ejected electron versus the *intensity* of the incident light (left)
- Number of electrons* ejected versus the *frequency* of the incident light (right)



**Question 3** **6 points****a. 2 points**

In the ground state of Ge (Germanium; $Z = 32$) atom, **name** the following orbitals:

i. Orbital containing the electron that requires the maximum energy to be removed.

ii. Orbital containing the electron that experiences the minimum effective nuclear charge.

b. 2 points

An unknown anion (A^-) has the following *excited* electronic configuration: $[Ar] 4s^2 3d^3 4p^1$

(Ar: Argon; $Z = 18$)

Give the **condensed ground state electronic configuration** of the atom, **A** and the cation, **A^{2+}** .
Show your work. (Assume all electronic transitions are possible)

Electronic Configuration of A:

Electronic Configuration of A^{2+} :

c. 2 points

In a single atom, determine how many electrons can have the following set of quantum numbers?

(i) $n = 5; l = 3$

In a single atom, determine how many electrons can have the following set of quantum numbers?

(i) $n = 3; l = 2; m_s = +1/2$

**Question 4** **8 points**

Rank the following from 1 to 4 in **increasing** order. Provide the ranking in the boxes provided.

Note: Atomic number (Z) are provided for relevant atoms

a) Increasing (lowest to highest) order of *ionization energy*: (1= lowest; 4 = highest)

Mg

Mg: Z = 12

Rb

Rb: Z = 37

Cs

Cs: Z = 55

K

K : Z = 19

b) Increasing (lowest to highest) order of *bond angle*: (1= lowest; 4 = highest)

H₂OCCl₄XeF₄

HCN

c) Increasing (smallest to largest) order of *ionic size* (1= smallest; 4 = largest)

S²⁻

S: Z = 16

O²⁻

O: Z = 8

Be²⁺

Be: Z = 4

Na⁺

Na: Z = 11

d) Increasing (lowest to highest) order of *bond energy* (1= lowest; 4 = highest)

P-F

P-Cl

P-I

P-Br

**Question 5** **12 points**

a. 9 points For the molecules HClO_4 and TeF_6 , draw the most stable *Lewis structure* showing appropriate shape and *dashed/wedged bonds*, based on VSEPR. Determine the **molecular geometry** around the central atom. For HClO_4 , label the *most polar bond*.

HClO_4 (there are no O-O bonds)

Label the most polar bond

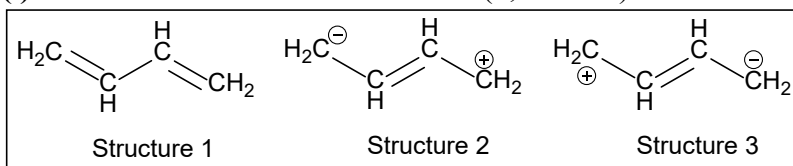
Molecular Geometry around Cl

TeF_6

Molecular Geometry around Te

b. 3 points (Note: Lone pairs are not shown in structures below)

(i) For the three *resonance* structures (1, 2 and 3) shown below, answer the following question:



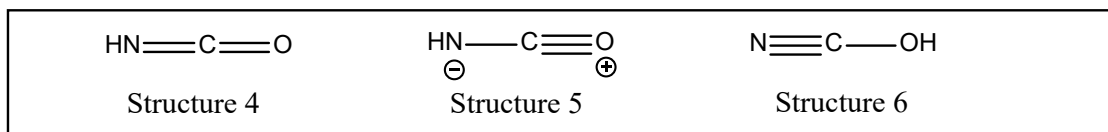
(I) Circle the *most contributing* resonance structure.

Structure 1

Structure 2

Structure 3

(ii) For the three structures given (4, 5, and 6) below, answer the following questions:



(I) Circle all the structures that represent *resonance structures* of each other:

Structure 4

Structure 5

Structure 6

(II) Among the resonance structures, circle the *most contributing* resonance structure.

Structure 4

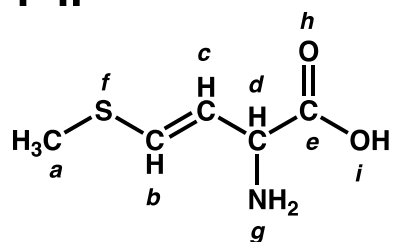
Structure 5

Structure 6

**Question 6** **6 points**

Using the data given below, calculate the **lattice energy** for SrO(s).

	ΔH° (kJ/mol)
$\text{Sr}(s) \longrightarrow \text{Sr}(g)$	+180
$\text{Sr}(g) \longrightarrow \text{Sr}^{2+}(g) + 2e^-$	+1468
$\text{O}_2(g) \longrightarrow 2\text{O}(g)$	+500
$\text{O}(g) + 2e^- \longrightarrow \text{O}^{2-}(g)$	+600
$\text{Sr}(s) + \frac{1}{2}\text{O}_2(g) \longrightarrow \text{SrO}(s)$	-550

**Question 7 10 points**

Consider the molecule given on the left, and answer questions (i) to (vi) using chemical bonding theories that we have learnt in class (valence bond theory, hybridization).

Note that the lone pairs are not shown in the structure:

i. Consider the carbon atoms marked in as ***a***, ***b***, ***c***, ***d***, and ***e*** in the molecule, and determine **hybridization** at each of the carbon centers. (2.5 points)

Atom	Hybridization
Carbon (<i>a</i>)	
Carbon (<i>b</i>)	
Carbon (<i>c</i>)	
Carbon (<i>d</i>)	
Carbon (<i>e</i>)	

ii. Now, consider the sulfur atom (S) marked as ***f***, nitrogen (N) atom (***g***), and oxygen(O) atoms (***h*** and ***i***), and determine their **hybridization**. (2 points)

Atom	Hybridization
Sulfur (<i>f</i>)	
Nitrogen (<i>g</i>)	
Oxygen (<i>h</i>)	
Oxygen (<i>i</i>)	

iii. **Describe** all the bonds formed by the carbon atoms ***a*** and ***b***, using sigma and pi designation as appropriate. (2 points)

Atom	Bonds formed (use sigma and pi as appropriate)
Carbon (<i>a</i>)	
Carbon (<i>b</i>)	

iv. How many total sp^3 -s sigma bonds are in the molecule? (1 point)

v. Determine **electronic** and **molecular geometry** at carbon (C) labeled ***e*** and sulfur (S) labeled ***f*** centres. (2 points)

Atom	Electronic Geometry	Molecular Geometry
Carbon (<i>e</i>)		
Sulfur (<i>f</i>)		

vi. Circle the value from the choices given below, that describes the **C-S-C bond angle at sulfur** (S) marked ***f***: (0.5 points)

>90° to <109.5° 109.5° >109.5 to <120° 120° >120° to <180°

**Question 8 12 points**

Using the **valence shell orbitals only** draw a **complete** molecular orbital (MO) diagram (with no s-p mixing) for the molecule **PS**, in the space provided below. **(6 points)**

Assume that 3s and 3p atomic orbitals will form molecular orbitals like those formed by the combination of 2s and 2p atomic orbitals that we learnt in class, and there is no s-p mixing.

Show **energy arrow**, **atomic orbitals** involved, **atom labels**, the **molecular orbitals** that are formed, **filled electrons** in the MO diagram etc. Make sure to label all the atomic and molecular orbitals.

Using the MO diagram above, answer the following questions:

- (i) Which one of the following will have the highest bond order? Circle the correct choice
A) PS^{1-} B) PS C) PS^{1+}
- (ii) Which one of the following will have the longest bond length? Circle the correct choice
A) PS^{1-} B) PS C) PS^{1+}
- (iii) Which of the following will be diamagnetic? Circle the correct choice.
A) PS^{1-} B) PS C) PS^{1+}
- (iv) Draw Lewis dot structure of the molecule with the shortest bond length (based on the MO diagram) from among the three (PS^{1-} , PS, PS^{1+}), and indicate the bond order calculated from the Lewis dot structure. **(3 points)**

Lewis Dot Structure

Bond Order



C00E34B8-8246-4CCD-928A-116EA6FC065C

midterm-exam-93388

#298 Page 10 of 10

SCRAP PAGE

**CHEMISTRY 110 –Midterm Exam****Thursday November 7, 2024, 6:30 – 8:30 pm**

Examiners: Pallavi Sirjoosingh and Ashok Kakkar

First name (please write as legibly as possible within the boxes)

--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--

Last name

--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--

Student ID number

--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--

This booklet has 8 questions on 10 pages (The last page can be used as a scrap page).

The staple must NOT be removed from the question booklet. Your intact question booklet must be returned at the end of the exam.

A separate booklet contains a periodic table, and datasheet.

Write your answers directly under the questions in the space provided. Do not write on the QR codes or near the top edges (as it may be cut off).

Please write **legibly and dark enough** for the scan to be readable. For students who use English as a second language, dictionaries are permitted. Calculators are allowed.

DO NOT ASK QUESTIONS OF THE INVIGILATORS about interpretations of the exam questions. If you have a concern about a question, please summarize it at the top of this page. All concerns will be evaluated after the exam. GOOD LUCK!

**Question 1 8 points** (*Assume all transitions are possible*)

a. 6 points Calculate the **energy (in kJ/mol)** required to ionize (completely remove) the electron from the **n=3** level of **He⁺ ion** (He: Helium). Show your work including equations. (3 sig. figs.)

b. 2 points

(I) Select the largest (longest) wavelength from the three choices below:

- i. Wavelength of photon to *completely remove* an electron from *1s orbital of Hydrogen atom*
- ii. Wavelength of photon to *completely remove* an electron from *1s orbital of Boron atom (B)*
- iii. Wavelength of photon to *completely remove* an electron from *1s orbital of Boron cation (B⁺)*

i or ii or iii

(II) Select the smallest (shortest) wavelength from the three choices below (*for H atom*):

- i. Wavelength of photon *emitted* for an electron transitioning from *2p to 1s orbital*
- ii. Wavelength of photon *absorbed* for an electron transitioning from *2s to 4p orbital*
- iii. Wavelength of photon *absorbed* for an electron transitioning from *1s to 3p orbital*

i or ii or iii



Question 2 8 points



A Chem110 student plots the graph shown on the left, from data obtained for the kinetic energy (eV) of the ejected electron versus frequency (s^{-1}) of the incident light, for an unknown metal. ($1\text{eV} = 1.6022 \times 10^{-19} \text{ J}$)

Based on the plot, answer the questions **a to c** below:

a. 1 point On the plot above, label the **threshold frequency**.

b. 2 points What is the value of the work potential (in J) of the unknown metal?

c. 3 points What is the *longest* wavelength (in nm) required to eject an electron from this metal's surface? Show your work including equation(s) used. (3 sig figs)

d. 2 points In the space provided below, plot the following

(Assume frequency is *above* threshold frequency):

- Kinetic energy* of an ejected electron versus the *intensity* of the incident light (left)
- Number of electrons* ejected versus the *frequency* of the incident light (right)



**Question 3** **6 points****a. 2 points**

In the ground state of Ge (Germanium; $Z = 32$) atom, **name** the following orbitals:

i. Orbital containing the electron that requires the maximum energy to be removed.

ii. Orbital containing the electron that experiences the minimum effective nuclear charge.

b. 2 points

An unknown anion (A^-) has the following *excited* electronic configuration: $[Ar] 4s^2 3d^3 4p^1$

(Ar: Argon; $Z = 18$)

Give the **condensed ground state electronic configuration** of the atom, **A** and the cation, **A^{2+}** .
Show your work. (Assume all electronic transitions are possible)

Electronic Configuration of A:

Electronic Configuration of A^{2+} :

c. 2 points

In a single atom, determine how many electrons can have the following set of quantum numbers?

(i) $n = 5; l = 3$

In a single atom, determine how many electrons can have the following set of quantum numbers?

(i) $n = 3; l = 2; m_s = +1/2$

**Question 4** **8 points**

Rank the following from 1 to 4 in **increasing** order. Provide the ranking in the boxes provided.

Note: Atomic number (Z) are provided for relevant atoms

a) Increasing (lowest to highest) order of *ionization energy*: (1= lowest; 4 = highest)

Mg

Mg: Z = 12

Rb

Rb: Z = 37

Cs

Cs: Z = 55

K

K : Z = 19

b) Increasing (lowest to highest) order of *bond angle*: (1= lowest; 4 = highest)

H₂OCCl₄XeF₄

HCN

c) Increasing (smallest to largest) order of *ionic size* (1= smallest; 4 = largest)

S²⁻

S: Z = 16

O²⁻

O: Z = 8

Be²⁺

Be: Z = 4

Na⁺

Na: Z = 11

d) Increasing (lowest to highest) order of *bond energy* (1= lowest; 4 = highest)

P-F

P-Cl

P-I

P-Br

**Question 5** **12 points**

a. 9 points For the molecules HClO_4 and TeF_6 , draw the most stable *Lewis structure* showing appropriate shape and *dashed/wedged bonds*, based on VSEPR. Determine the **molecular geometry** around the central atom. For HClO_4 , label the *most polar bond*.



b. 3 points (Note: Lone pairs are not shown in structures below)

(i) For the three *resonance* structures (1, 2 and 3) shown below, answer the following question:



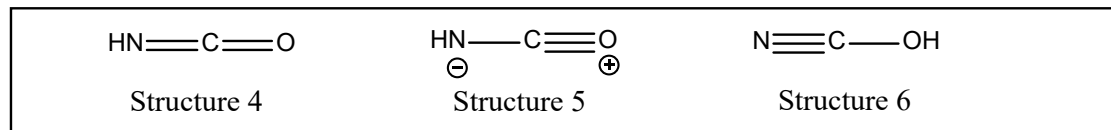
(I) Circle the *most contributing* resonance structure.

Structure 1

Structure 2

Structure 3

(ii) For the three structures given (4, 5, and 6) below, answer the following questions:



(I) Circle all the structures that represent *resonance structures* of each other:

Structure 4

Structure 5

Structure 6

(II) Among the resonance structures, circle the *most contributing* resonance structure.

Structure 4

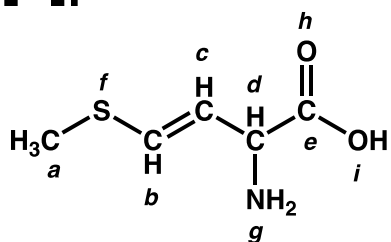
Structure 5

Structure 6

**Question 6** **6 points**

Using the data given below, calculate the **lattice energy** for SrO(s).

	ΔH° (kJ/mol)
$\text{Sr}(s) \longrightarrow \text{Sr}(g)$	+180
$\text{Sr}(g) \longrightarrow \text{Sr}^{2+}(g) + 2e^-$	+1468
$\text{O}_2(g) \longrightarrow 2\text{O}(g)$	+500
$\text{O}(g) + 2e^- \longrightarrow \text{O}^{2-}(g)$	+600
$\text{Sr}(s) + \frac{1}{2}\text{O}_2(g) \longrightarrow \text{SrO}(s)$	-550

**Question 7 10 points**

Consider the molecule given on the left, and answer questions (i) to (vi) using chemical bonding theories that we have learnt in class (valence bond theory, hybridization).

Note that the lone pairs are not shown in the structure:

i. Consider the carbon atoms marked in as ***a***, ***b***, ***c***, ***d***, and ***e*** in the molecule, and determine **hybridization** at each of the carbon centers. (2.5 points)

Atom	Hybridization
Carbon (<i>a</i>)	
Carbon (<i>b</i>)	
Carbon (<i>c</i>)	
Carbon (<i>d</i>)	
Carbon (<i>e</i>)	

ii. Now, consider the sulfur atom (S) marked as ***f***, nitrogen (N) atom (***g***), and oxygen(O) atoms (***h*** and ***i***), and determine their **hybridization**. (2 points)

Atom	Hybridization
Sulfur (<i>f</i>)	
Nitrogen (<i>g</i>)	
Oxygen (<i>h</i>)	
Oxygen (<i>i</i>)	

iii. **Describe** all the bonds formed by the carbon atoms ***a*** and ***b***, using sigma and pi designation as appropriate. (2 points)

Atom	Bonds formed (use sigma and pi as appropriate)
Carbon (<i>a</i>)	
Carbon (<i>b</i>)	

iv. How many total sp^3 -s sigma bonds are in the molecule? (1 point)

v. Determine **electronic** and **molecular geometry** at carbon (C) labeled ***e*** and sulfur (S) labeled ***f*** centres. (2 points)

Atom	Electronic Geometry	Molecular Geometry
Carbon (<i>e</i>)		
Sulfur (<i>f</i>)		

vi. Circle the value from the choices given below, that describes the **C-S-C bond angle at sulfur** (S) marked ***f***: (0.5 points)

$>90^\circ$ to $<109.5^\circ$ 109.5° >109.5 to $<120^\circ$ 120° $>120^\circ$ to $<180^\circ$


Question 8 12 points

Using the **valence shell orbitals only** draw a **complete** molecular orbital (MO) diagram (with no s-p mixing) for the molecule **PS**, in the space provided below. **(6 points)**

Assume that 3s and 3p atomic orbitals will form molecular orbitals like those formed by the combination of 2s and 2p atomic orbitals that we learnt in class, and there is no s-p mixing.

Show **energy arrow**, **atomic orbitals** involved, **atom labels**, the **molecular orbitals** that are formed, **filled electrons** in the MO diagram etc. Make sure to label all the atomic and molecular orbitals.

Using the MO diagram above, answer the following questions:

- (i) Which one of the following will have the highest bond order? Circle the correct choice
 A) PS^{1-} B) PS C) PS^{1+}
- (ii) Which one of the following will have the longest bond length? Circle the correct choice
 A) PS^{1-} B) PS C) PS^{1+}
- (iii) Which of the following will be diamagnetic? Circle the correct choice.
 A) PS^{1-} B) PS C) PS^{1+}
- (iv) Draw Lewis dot structure of the molecule with the shortest bond length (based on the MO diagram) from among the three (PS^{1-} , PS, PS^{1+}), and indicate the bond order calculated from the Lewis dot structure. **(3 points)**

Lewis Dot Structure

Bond Order



D84CB5F8-74EC-4FEF-8F46-4C65C4AA105C

midterm-exam-93388

#299 Page 10 of 10

SCRAP PAGE

**CHEMISTRY 110 –Midterm Exam****Thursday November 7, 2024, 6:30 – 8:30 pm**

Examiners: Pallavi Sirjoosingh and Ashok Kakkar

First name (please write as legibly as possible within the boxes)

--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--

Last name

--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--

Student ID number

--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--

This booklet has 8 questions on 10 pages (The last page can be used as a scrap page).

The staple must NOT be removed from the question booklet. Your intact question booklet must be returned at the end of the exam.

A separate booklet contains a periodic table, and datasheet.

Write your answers directly under the questions in the space provided. Do not write on the QR codes or near the top edges (as it may be cut off).

Please write **legibly and dark enough** for the scan to be readable. For students who use English as a second language, dictionaries are permitted. Calculators are allowed.

DO NOT ASK QUESTIONS OF THE INVIGILATORS about interpretations of the exam questions. If you have a concern about a question, please summarize it at the top of this page. All concerns will be evaluated after the exam. GOOD LUCK!

**Question 1 8 points** (*Assume all transitions are possible*)

a. 6 points Calculate the **energy (in kJ/mol)** required to ionize (completely remove) the electron from the **$n=3$** level of **He^+ ion** (He: Helium). Show your work including equations. (3 sig. figs.)

b. 2 points

(I) Select the largest (longest) wavelength from the three choices below:

- i. Wavelength of photon to *completely remove* an electron from *1s orbital of Hydrogen atom*
- ii. Wavelength of photon to *completely remove* an electron from *1s orbital of Boron atom (B)*
- iii. Wavelength of photon to *completely remove* an electron from *1s orbital of Boron cation (B^+)*

i or ii or iii

(II) Select the smallest (shortest) wavelength from the three choices below (*for H atom*):

- i. Wavelength of photon *emitted* for an electron transitioning from *2p to 1s orbital*
- ii. Wavelength of photon *absorbed* for an electron transitioning from *2s to 4p orbital*
- iii. Wavelength of photon *absorbed* for an electron transitioning from *1s to 3p orbital*

i or ii or iii



Question 2 8 points



A Chem110 student plots the graph shown on the left, from data obtained for the kinetic energy (eV) of the ejected electron versus frequency (s^{-1}) of the incident light, for an unknown metal. ($1\text{eV} = 1.6022 \times 10^{-19} \text{ J}$)

Based on the plot, answer the questions **a to c** below:

a. 1 point On the plot above, label the **threshold frequency**.

b. 2 points What is the value of the work potential (in J) of the unknown metal?

c. 3 points What is the *longest* wavelength (in nm) required to eject an electron from this metal's surface? Show your work including equation(s) used. (3 sig figs)

d. 2 points In the space provided below, plot the following

(Assume frequency is *above* threshold frequency):

- Kinetic energy* of an ejected electron versus the *intensity* of the incident light (left)
- Number of electrons* ejected versus the *frequency* of the incident light (right)



**Question 3** **6 points****a. 2 points**

In the ground state of Ge (Germanium; $Z = 32$) atom, **name** the following orbitals:

i. Orbital containing the electron that requires the maximum energy to be removed.

ii. Orbital containing the electron that experiences the minimum effective nuclear charge.

b. 2 points

An unknown anion (A^-) has the following *excited* electronic configuration: $[Ar] 4s^2 3d^3 4p^1$

(Ar: Argon; $Z = 18$)

Give the **condensed ground state electronic configuration** of the atom, **A** and the cation, **A^{2+}** .
Show your work. (Assume all electronic transitions are possible)

Electronic Configuration of A:

Electronic Configuration of A^{2+} :

c. 2 points

In a single atom, determine how many electrons can have the following set of quantum numbers?

(i) $n = 5; l = 3$

In a single atom, determine how many electrons can have the following set of quantum numbers?

(i) $n = 3; l = 2; m_s = +1/2$

**Question 4** **8 points**

Rank the following from 1 to 4 in **increasing** order. Provide the ranking in the boxes provided.

Note: Atomic number (Z) are provided for relevant atoms

a) Increasing (lowest to highest) order of *ionization energy*: (1= lowest; 4 = highest)

Mg

Mg: Z = 12

Rb

Rb: Z = 37

Cs

Cs: Z = 55

K

K : Z = 19

b) Increasing (lowest to highest) order of *bond angle*: (1= lowest; 4 = highest)

H₂OCCl₄XeF₄

HCN

c) Increasing (smallest to largest) order of *ionic size* (1= smallest; 4 = largest)

S²⁻

S: Z = 16

O²⁻

O: Z = 8

Be²⁺

Be: Z = 4

Na⁺

Na: Z = 11

d) Increasing (lowest to highest) order of *bond energy* (1= lowest; 4 = highest)

P-F

P-Cl

P-I

P-Br

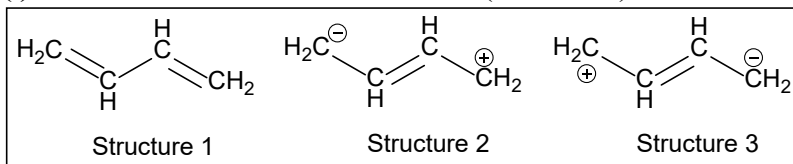
**Question 5** **12 points**

a. 9 points For the molecules HClO_4 and TeF_6 , draw the most stable *Lewis structure* showing appropriate shape and *dashed/wedged bonds*, based on VSEPR. Determine the **molecular geometry** around the central atom. For HClO_4 , label the *most polar bond*.



b. 3 points (Note: Lone pairs are not shown in structures below)

(i) For the three *resonance* structures (1, 2 and 3) shown below, answer the following question:



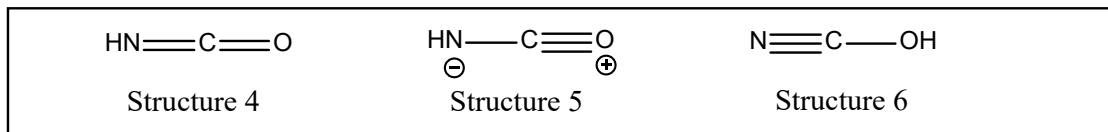
(I) Circle the *most contributing* resonance structure.

Structure 1

Structure 2

Structure 3

(ii) For the three structures given (4, 5, and 6) below, answer the following questions:



(I) Circle all the structures that represent *resonance structures* of each other:

Structure 4

Structure 5

Structure 6

(II) Among the resonance structures, circle the *most contributing* resonance structure.

Structure 4

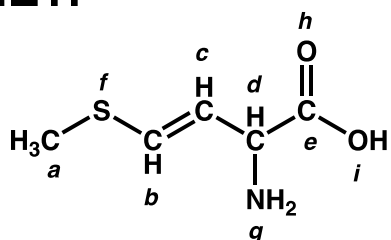
Structure 5

Structure 6

**Question 6** **6 points**

Using the data given below, calculate the **lattice energy** for SrO(s).

	ΔH° (kJ/mol)
$\text{Sr}(s) \longrightarrow \text{Sr}(g)$	+180
$\text{Sr}(g) \longrightarrow \text{Sr}^{2+}(g) + 2e^-$	+1468
$\text{O}_2(g) \longrightarrow 2\text{O}(g)$	+500
$\text{O}(g) + 2e^- \longrightarrow \text{O}^{2-}(g)$	+600
$\text{Sr}(s) + \frac{1}{2}\text{O}_2(g) \longrightarrow \text{SrO}(s)$	-550

**Question 7 10 points**

Consider the molecule given on the left, and answer questions (i) to (vi) using chemical bonding theories that we have learnt in class (valence bond theory, hybridization).

Note that the lone pairs are not shown in the structure:

i. Consider the carbon atoms marked in as ***a***, ***b***, ***c***, ***d***, and ***e*** in the molecule, and determine **hybridization** at each of the carbon centers. (2.5 points)

Atom	Hybridization
Carbon (<i>a</i>)	
Carbon (<i>b</i>)	
Carbon (<i>c</i>)	
Carbon (<i>d</i>)	
Carbon (<i>e</i>)	

ii. Now, consider the sulfur atom (S) marked as ***f***, nitrogen (N) atom (***g***), and oxygen(O) atoms (***h*** and ***i***), and determine their **hybridization**. (2 points)

Atom	Hybridization
Sulfur (<i>f</i>)	
Nitrogen (<i>g</i>)	
Oxygen (<i>h</i>)	
Oxygen (<i>i</i>)	

iii. **Describe** all the bonds formed by the carbon atoms ***a*** and ***b***, using sigma and pi designation as appropriate. (2 points)

Atom	Bonds formed (use sigma and pi as appropriate)
Carbon (<i>a</i>)	
Carbon (<i>b</i>)	

iv. How many total sp^3 -s sigma bonds are in the molecule? (1 point)

v. Determine **electronic** and **molecular geometry** at carbon (C) labeled ***e*** and sulfur (S) labeled ***f*** centres. (2 points)

Atom	Electronic Geometry	Molecular Geometry
Carbon (<i>e</i>)		
Sulfur (<i>f</i>)		

vi. Circle the value from the choices given below, that describes the **C-S-C bond angle at sulfur** (S) marked ***f***: (0.5 points)

>90° to <109.5° 109.5° >109.5 to <120° 120° >120° to <180°


Question 8 12 points

Using the **valence shell orbitals only** draw a **complete** molecular orbital (MO) diagram (with no s-p mixing) for the molecule **PS**, in the space provided below. **(6 points)**

Assume that 3s and 3p atomic orbitals will form molecular orbitals like those formed by the combination of 2s and 2p atomic orbitals that we learnt in class, and there is no s-p mixing.

Show **energy arrow**, **atomic orbitals** involved, **atom labels**, the **molecular orbitals** that are formed, **filled electrons** in the MO diagram etc. Make sure to label all the atomic and molecular orbitals.

Using the MO diagram above, answer the following questions:

- (i) Which one of the following will have the highest bond order? Circle the correct choice
 A) PS^{1-} B) PS C) PS^{1+}
- (ii) Which one of the following will have the longest bond length? Circle the correct choice
 A) PS^{1-} B) PS C) PS^{1+}
- (iii) Which of the following will be diamagnetic? Circle the correct choice.
 A) PS^{1-} B) PS C) PS^{1+}
- (iv) Draw Lewis dot structure of the molecule with the shortest bond length (based on the MO diagram) from among the three (PS^{1-} , PS, PS^{1+}), and indicate the bond order calculated from the Lewis dot structure. **(3 points)**

Lewis Dot Structure

Bond Order



EC174123-DA78-4CFC-B84F-6048A4A0FE45

midterm-exam-93388

#300 Page 10 of 10

SCRAP PAGE

**CHEMISTRY 110 –Midterm Exam****Thursday November 7, 2024, 6:30 – 8:30 pm****Examiners: Pallavi Sirjoosingh and Ashok Kakkar**

First name (please write as legibly as possible within the boxes)

--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--

Last name

--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--

Student ID number

--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--

This booklet has 8 questions on 10 pages (The last page can be used as a scrap page).

The staple must NOT be removed from the question booklet. Your intact question booklet must be returned at the end of the exam.

A separate booklet contains a periodic table, and datasheet.

Write your answers directly under the questions in the space provided. Do not write on the QR codes or near the top edges (as it may be cut off).

Please write **legibly and dark enough** for the scan to be readable. For students who use English as a second language, dictionaries are permitted. Calculators are allowed.

DO NOT ASK QUESTIONS OF THE INVIGILATORS about interpretations of the exam questions. If you have a concern about a question, please summarize it at the top of this page. All concerns will be evaluated after the exam. GOOD LUCK!

**Question 1 8 points** (*Assume all transitions are possible*)

a. 6 points Calculate the **energy (in kJ/mol)** required to ionize (completely remove) the electron from the **n=3** level of **He⁺ ion** (He: Helium). Show your work including equations. (3 sig. figs.)

b. 2 points

(I) Select the largest (longest) wavelength from the three choices below:

- i. Wavelength of photon to *completely remove* an electron from *1s orbital of Hydrogen atom*
- ii. Wavelength of photon to *completely remove* an electron from *1s orbital of Boron atom (B)*
- iii. Wavelength of photon to *completely remove* an electron from *1s orbital of Boron cation (B⁺)*

i or ii or iii

(II) Select the smallest (shortest) wavelength from the three choices below (*for H atom*):

- i. Wavelength of photon *emitted* for an electron transitioning from *2p to 1s orbital*
- ii. Wavelength of photon *absorbed* for an electron transitioning from *2s to 4p orbital*
- iii. Wavelength of photon *absorbed* for an electron transitioning from *1s to 3p orbital*

i or ii or iii



Question 2 8 points



A Chem110 student plots the graph shown on the left, from data obtained for the kinetic energy (eV) of the ejected electron versus frequency (s^{-1}) of the incident light, for an unknown metal. ($1\text{eV} = 1.6022 \times 10^{-19} \text{ J}$)

Based on the plot, answer the questions **a to c** below:

a. 1 point On the plot above, label the **threshold frequency**.

b. 2 points What is the value of the work potential (in J) of the unknown metal?

c. 3 points What is the *longest* wavelength (in nm) required to eject an electron from this metal's surface? Show your work including equation(s) used. (3 sig figs)

d. 2 points In the space provided below, plot the following

(Assume frequency is *above* threshold frequency):

- Kinetic energy* of an ejected electron versus the *intensity* of the incident light (left)
- Number of electrons* ejected versus the *frequency* of the incident light (right)



**Question 3** **6 points****a. 2 points**

In the ground state of Ge (Germanium; $Z = 32$) atom, **name** the following orbitals:

i. Orbital containing the electron that requires the maximum energy to be removed.

ii. Orbital containing the electron that experiences the minimum effective nuclear charge.

b. 2 points

An unknown anion (A^-) has the following *excited* electronic configuration: $[Ar] 4s^2 3d^3 4p^1$

(Ar: Argon; $Z = 18$)

Give the **condensed ground state electronic configuration** of the atom, **A** and the cation, **A^{2+}** .
Show your work. (Assume all electronic transitions are possible)

Electronic Configuration of A:

Electronic Configuration of A^{2+} :

c. 2 points

In a single atom, determine how many electrons can have the following set of quantum numbers?

(i) $n = 5; l = 3$

In a single atom, determine how many electrons can have the following set of quantum numbers?

(i) $n = 3; l = 2; m_s = +1/2$

**Question 4** **8 points**

Rank the following from 1 to 4 in **increasing** order. Provide the ranking in the boxes provided.

Note: Atomic number (Z) are provided for relevant atoms

a) Increasing (lowest to highest) order of *ionization energy*: (1= lowest; 4 = highest)

Mg

Mg: Z = 12

Rb

Rb: Z = 37

Cs

Cs: Z = 55

K

K : Z = 19

b) Increasing (lowest to highest) order of *bond angle*: (1= lowest; 4 = highest)

H₂OCCl₄XeF₄

HCN

c) Increasing (smallest to largest) order of *ionic size* (1= smallest; 4 = largest)

S²⁻

S: Z = 16

O²⁻

O: Z = 8

Be²⁺

Be: Z = 4

Na⁺

Na: Z = 11

d) Increasing (lowest to highest) order of *bond energy* (1= lowest; 4 = highest)

P-F

P-Cl

P-I

P-Br

**Question 5** **12 points**

a. 9 points For the molecules HClO_4 and TeF_6 , draw the most stable *Lewis structure* showing appropriate shape and *dashed/wedged bonds*, based on VSEPR. Determine the **molecular geometry** around the central atom. For HClO_4 , label the *most polar bond*.



b. 3 points (Note: Lone pairs are not shown in structures below)

(i) For the three *resonance* structures (1, 2 and 3) shown below, answer the following question:



(I) Circle the *most contributing* resonance structure.

Structure 1

Structure 2

Structure 3

(ii) For the three structures given (4, 5, and 6) below, answer the following questions:



(I) Circle all the structures that represent *resonance structures* of each other:

Structure 4

Structure 5

Structure 6

(II) Among the resonance structures, circle the *most contributing* resonance structure.

Structure 4

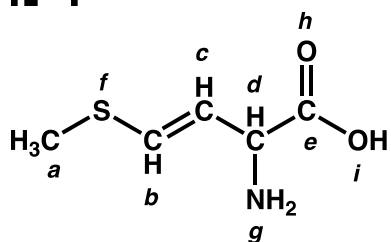
Structure 5

Structure 6

**Question 6** **6 points**

Using the data given below, calculate the **lattice energy** for SrO(s).

	ΔH° (kJ/mol)
$\text{Sr}(s) \longrightarrow \text{Sr}(g)$	+180
$\text{Sr}(g) \longrightarrow \text{Sr}^{2+}(g) + 2e^-$	+1468
$\text{O}_2(g) \longrightarrow 2\text{O}(g)$	+500
$\text{O}(g) + 2e^- \longrightarrow \text{O}^{2-}(g)$	+600
$\text{Sr}(s) + \frac{1}{2}\text{O}_2(g) \longrightarrow \text{SrO}(s)$	-550

**Question 7 10 points**

Consider the molecule given on the left, and answer questions (i) to (vi) using chemical bonding theories that we have learnt in class (valence bond theory, hybridization).

Note that the lone pairs are not shown in the structure:

i. Consider the carbon atoms marked in as ***a***, ***b***, ***c***, ***d***, and ***e*** in the molecule, and determine **hybridization** at each of the carbon centers. (2.5 points)

Atom	Hybridization
Carbon (<i>a</i>)	
Carbon (<i>b</i>)	
Carbon (<i>c</i>)	
Carbon (<i>d</i>)	
Carbon (<i>e</i>)	

ii. Now, consider the sulfur atom (S) marked as ***f***, nitrogen (N) atom (***g***), and oxygen(O) atoms (***h*** and ***i***), and determine their **hybridization**. (2 points)

Atom	Hybridization
Sulfur (<i>f</i>)	
Nitrogen (<i>g</i>)	
Oxygen (<i>h</i>)	
Oxygen (<i>i</i>)	

iii. **Describe** all the bonds formed by the carbon atoms ***a*** and ***b***, using sigma and pi designation as appropriate. (2 points)

Atom	Bonds formed (use sigma and pi as appropriate)
Carbon (<i>a</i>)	
Carbon (<i>b</i>)	

iv. How many total sp^3 -s sigma bonds are in the molecule? (1 point)

v. Determine **electronic** and **molecular geometry** at carbon (C) labeled ***e*** and sulfur (S) labeled ***f*** centres. (2 points)

Atom	Electronic Geometry	Molecular Geometry
Carbon (<i>e</i>)		
Sulfur (<i>f</i>)		

vi. Circle the value from the choices given below, that describes the **C-S-C bond angle at sulfur** (S) marked ***f***: (0.5 points)

>90° to <109.5°

109.5°

>109.5 to <120°

120°

>120° to <180°

**Question 8 12 points**

Using the **valence shell orbitals only** draw a **complete** molecular orbital (MO) diagram (with no s-p mixing) for the molecule **PS**, in the space provided below. **(6 points)**

Assume that 3s and 3p atomic orbitals will form molecular orbitals like those formed by the combination of 2s and 2p atomic orbitals that we learnt in class, and there is no s-p mixing.

Show **energy arrow**, **atomic orbitals** involved, **atom labels**, the **molecular orbitals** that are formed, **filled electrons** in the MO diagram etc. Make sure to label all the atomic and molecular orbitals.

Using the MO diagram above, answer the following questions:

- (i) Which one of the following will have the highest bond order? Circle the correct choice
A) PS^{1-} B) PS C) PS^{1+}
- (ii) Which one of the following will have the longest bond length? Circle the correct choice
A) PS^{1-} B) PS C) PS^{1+}
- (iii) Which of the following will be diamagnetic? Circle the correct choice.
A) PS^{1-} B) PS C) PS^{1+}
- (iv) Draw Lewis dot structure of the molecule with the shortest bond length (based on the MO diagram) from among the three (PS^{1-} , PS, PS^{1+}), and indicate the bond order calculated from the Lewis dot structure. **(3 points)**

Lewis Dot Structure

Bond Order



20DE3EDE-E852-4A08-8DBA-E98461ACB8C5

midterm-exam-93388

#301 Page 10 of 10

SCRAP PAGE

**CHEMISTRY 110 –Midterm Exam****Thursday November 7, 2024, 6:30 – 8:30 pm**

Examiners: Pallavi Sirjoosingh and Ashok Kakkar

First name (please write as legibly as possible within the boxes)

--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--

Last name

--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--

Student ID number

--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--

This booklet has 8 questions on 10 pages (The last page can be used as a scrap page).

The staple must NOT be removed from the question booklet. Your intact question booklet must be returned at the end of the exam.

A separate booklet contains a periodic table, and datasheet.

Write your answers directly under the questions in the space provided. Do not write on the QR codes or near the top edges (as it may be cut off).

Please write **legibly and dark enough** for the scan to be readable. For students who use English as a second language, dictionaries are permitted. Calculators are allowed.

DO NOT ASK QUESTIONS OF THE INVIGILATORS about interpretations of the exam questions. If you have a concern about a question, please summarize it at the top of this page. All concerns will be evaluated after the exam. GOOD LUCK!

**Question 1 8 points** (*Assume all transitions are possible*)

a. 6 points Calculate the **energy (in kJ/mol)** required to ionize (completely remove) the electron from the **$n=3$** level of **He^+ ion** (He: Helium). Show your work including equations. (3 sig. figs.)

b. 2 points

(I) Select the largest (longest) wavelength from the three choices below:

- i. Wavelength of photon to *completely remove* an electron from *1s orbital of Hydrogen atom*
- ii. Wavelength of photon to *completely remove* an electron from *1s orbital of Boron atom (B)*
- iii. Wavelength of photon to *completely remove* an electron from *1s orbital of Boron cation (B^+)*

i or ii or iii

(II) Select the smallest (shortest) wavelength from the three choices below (*for H atom*):

- i. Wavelength of photon *emitted* for an electron transitioning from *2p to 1s orbital*
- ii. Wavelength of photon *absorbed* for an electron transitioning from *2s to 4p orbital*
- iii. Wavelength of photon *absorbed* for an electron transitioning from *1s to 3p orbital*

i or ii or iii



Question 2 8 points



A Chem110 student plots the graph shown on the left, from data obtained for the kinetic energy (eV) of the ejected electron versus frequency (s^{-1}) of the incident light, for an unknown metal. ($1\text{eV} = 1.6022 \times 10^{-19} \text{ J}$)

Based on the plot, answer the questions **a to c** below:

a. 1 point On the plot above, label the **threshold frequency**.

b. 2 points What is the value of the work potential (in J) of the unknown metal?

c. 3 points What is the *longest* wavelength (in nm) required to eject an electron from this metal's surface? Show your work including equation(s) used. (3 sig figs)

d. 2 points In the space provided below, plot the following

(Assume frequency is *above* threshold frequency):

- i. *Kinetic energy* of an ejected electron versus the *intensity* of the incident light (left)
- ii. *Number of electrons* ejected versus the *frequency* of the incident light (right)



**Question 3** **6 points****a. 2 points**

In the ground state of Ge (Germanium; $Z = 32$) atom, **name** the following orbitals:

i. Orbital containing the electron that requires the maximum energy to be removed.

ii. Orbital containing the electron that experiences the minimum effective nuclear charge.

b. 2 points

An unknown anion (A^-) has the following *excited* electronic configuration: $[Ar] 4s^2 3d^3 4p^1$

(Ar: Argon; $Z = 18$)

Give the **condensed ground state electronic configuration** of the atom, **A** and the cation, **A^{2+}** .
Show your work. (Assume all electronic transitions are possible)

Electronic Configuration of A:

Electronic Configuration of A^{2+} :

c. 2 points

In a single atom, determine how many electrons can have the following set of quantum numbers?

(i) $n = 5; l = 3$

In a single atom, determine how many electrons can have the following set of quantum numbers?

(i) $n = 3; l = 2; m_s = +1/2$

**Question 4** **8 points**

Rank the following from 1 to 4 in **increasing** order. Provide the ranking in the boxes provided.

Note: Atomic number (Z) are provided for relevant atoms

a) Increasing (lowest to highest) order of *ionization energy*: (1= lowest; 4 = highest)

Mg

Mg: Z = 12

Rb

Rb: Z = 37

Cs

Cs: Z = 55

K

K : Z = 19

b) Increasing (lowest to highest) order of *bond angle*: (1= lowest; 4 = highest)

H₂OCCl₄XeF₄

HCN

c) Increasing (smallest to largest) order of *ionic size* (1= smallest; 4 = largest)

S²⁻

S: Z = 16

O²⁻

O: Z = 8

Be²⁺

Be: Z = 4

Na⁺

Na: Z = 11

d) Increasing (lowest to highest) order of *bond energy* (1= lowest; 4 = highest)

P-F

P-Cl

P-I

P-Br

**Question 5** **12 points**

a. 9 points For the molecules HClO_4 and TeF_6 , draw the most stable *Lewis structure* showing appropriate shape and *dashed/wedged bonds*, based on VSEPR. Determine the **molecular geometry** around the central atom. For HClO_4 , label the *most polar bond*.



b. 3 points (Note: Lone pairs are not shown in structures below)

(i) For the three *resonance* structures (1, 2 and 3) shown below, answer the following question:



(I) Circle the *most contributing* resonance structure.

Structure 1

Structure 2

Structure 3

(ii) For the three structures given (4, 5, and 6) below, answer the following questions:



(I) Circle all the structures that represent *resonance structures* of each other:

Structure 4

Structure 5

Structure 6

(II) Among the resonance structures, circle the *most contributing* resonance structure.

Structure 4

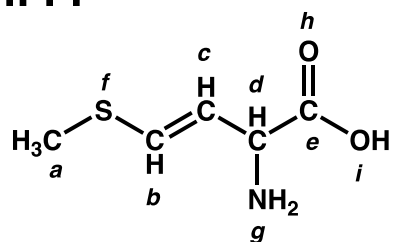
Structure 5

Structure 6

**Question 6** **6 points**

Using the data given below, calculate the **lattice energy** for SrO(s).

	ΔH° (kJ/mol)
$\text{Sr}(s) \longrightarrow \text{Sr}(g)$	+180
$\text{Sr}(g) \longrightarrow \text{Sr}^{2+}(g) + 2e^-$	+1468
$\text{O}_2(g) \longrightarrow 2\text{O}(g)$	+500
$\text{O}(g) + 2e^- \longrightarrow \text{O}^{2-}(g)$	+600
$\text{Sr}(s) + \frac{1}{2}\text{O}_2(g) \longrightarrow \text{SrO}(s)$	-550

**Question 7 10 points**

Consider the molecule given on the left, and answer questions (i) to (vi) using chemical bonding theories that we have learnt in class (valence bond theory, hybridization).

Note that the lone pairs are not shown in the structure:

i. Consider the carbon atoms marked in as ***a***, ***b***, ***c***, ***d***, and ***e*** in the molecule, and determine **hybridization** at each of the carbon centers. (2.5 points)

Atom	Hybridization
Carbon (<i>a</i>)	
Carbon (<i>b</i>)	
Carbon (<i>c</i>)	
Carbon (<i>d</i>)	
Carbon (<i>e</i>)	

ii. Now, consider the sulfur atom (S) marked as ***f***, nitrogen (N) atom (***g***), and oxygen(O) atoms (***h*** and ***i***), and determine their **hybridization**. (2 points)

Atom	Hybridization
Sulfur (<i>f</i>)	
Nitrogen (<i>g</i>)	
Oxygen (<i>h</i>)	
Oxygen (<i>i</i>)	

iii. **Describe** all the bonds formed by the carbon atoms ***a*** and ***b***, using sigma and pi designation as appropriate. (2 points)

Atom	Bonds formed (use sigma and pi as appropriate)
Carbon (<i>a</i>)	
Carbon (<i>b</i>)	

iv. How many total sp^3 -s sigma bonds are in the molecule? (1 point)

v. Determine **electronic** and **molecular geometry** at carbon (C) labeled ***e*** and sulfur (S) labeled ***f*** centres. (2 points)

Atom	Electronic Geometry	Molecular Geometry
Carbon (<i>e</i>)		
Sulfur (<i>f</i>)		

vi. Circle the value from the choices given below, that describes the **C-S-C bond angle at sulfur** (S) marked ***f***: (0.5 points)

>90° to <109.5° 109.5° >109.5 to <120° 120° >120° to <180°


Question 8 12 points

Using the **valence shell orbitals only** draw a **complete** molecular orbital (MO) diagram (with no s-p mixing) for the molecule **PS**, in the space provided below. **(6 points)**

Assume that 3s and 3p atomic orbitals will form molecular orbitals like those formed by the combination of 2s and 2p atomic orbitals that we learnt in class, and there is no s-p mixing.

Show **energy arrow**, **atomic orbitals** involved, **atom labels**, the **molecular orbitals** that are formed, **filled electrons** in the MO diagram etc. Make sure to label all the atomic and molecular orbitals.

Using the MO diagram above, answer the following questions:

- (i) Which one of the following will have the highest bond order? Circle the correct choice
 A) PS^{1-} B) PS C) PS^{1+}
- (ii) Which one of the following will have the longest bond length? Circle the correct choice
 A) PS^{1-} B) PS C) PS^{1+}
- (iii) Which of the following will be diamagnetic? Circle the correct choice.
 A) PS^{1-} B) PS C) PS^{1+}
- (iv) Draw Lewis dot structure of the molecule with the shortest bond length (based on the MO diagram) from among the three (PS^{1-} , PS, PS^{1+}), and indicate the bond order calculated from the Lewis dot structure. **(3 points)**

Lewis Dot Structure

Bond Order



620E7C35-677A-4926-9F72-CDF006C57C98

midterm-exam-93388

#302 Page 10 of 10

SCRAP PAGE

**CHEMISTRY 110 –Midterm Exam****Thursday November 7, 2024, 6:30 – 8:30 pm**

Examiners: Pallavi Sirjoosingh and Ashok Kakkar

First name (please write as legibly as possible within the boxes)

--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--

Last name

--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--

Student ID number

--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--

This booklet has 8 questions on 10 pages (The last page can be used as a scrap page).

The staple must NOT be removed from the question booklet. Your intact question booklet must be returned at the end of the exam.

A separate booklet contains a periodic table, and datasheet.

Write your answers directly under the questions in the space provided. Do not write on the QR codes or near the top edges (as it may be cut off).

Please write **legibly and dark enough** for the scan to be readable. For students who use English as a second language, dictionaries are permitted. Calculators are allowed.

DO NOT ASK QUESTIONS OF THE INVIGILATORS about interpretations of the exam questions. If you have a concern about a question, please summarize it at the top of this page. All concerns will be evaluated after the exam. GOOD LUCK!

**Question 1 8 points** (*Assume all transitions are possible*)

a. 6 points Calculate the **energy (in kJ/mol)** required to ionize (completely remove) the electron from the **n=3** level of **He⁺ ion** (He: Helium). Show your work including equations. (3 sig. figs.)

b. 2 points

(I) Select the largest (longest) wavelength from the three choices below:

- i. Wavelength of photon to *completely remove* an electron from *1s orbital of Hydrogen atom*
- ii. Wavelength of photon to *completely remove* an electron from *1s orbital of Boron atom (B)*
- iii. Wavelength of photon to *completely remove* an electron from *1s orbital of Boron cation (B⁺)*

i or ii or iii

(II) Select the smallest (shortest) wavelength from the three choices below (*for H atom*):

- i. Wavelength of photon *emitted* for an electron transitioning from *2p to 1s orbital*
- ii. Wavelength of photon *absorbed* for an electron transitioning from *2s to 4p orbital*
- iii. Wavelength of photon *absorbed* for an electron transitioning from *1s to 3p orbital*

i or ii or iii



Question 2 8 points



A Chem110 student plots the graph shown on the left, from data obtained for the kinetic energy (eV) of the ejected electron versus frequency (s^{-1}) of the incident light, for an unknown metal. ($1\text{eV} = 1.6022 \times 10^{-19} \text{ J}$)

Based on the plot, answer the questions **a to c** below:

a. 1 point On the plot above, label the **threshold frequency**.

b. 2 points What is the value of the work potential (in J) of the unknown metal?

c. 3 points What is the *longest* wavelength (in nm) required to eject an electron from this metal's surface? Show your work including equation(s) used. (3 sig figs)

d. 2 points In the space provided below, plot the following

(Assume frequency is *above* threshold frequency):

- Kinetic energy* of an ejected electron versus the *intensity* of the incident light (left)
- Number of electrons* ejected versus the *frequency* of the incident light (right)



**Question 3** **6 points****a. 2 points**

In the ground state of Ge (Germanium; $Z = 32$) atom, **name** the following orbitals:

i. Orbital containing the electron that requires the maximum energy to be removed.

ii. Orbital containing the electron that experiences the minimum effective nuclear charge.

b. 2 points

An unknown anion (A^-) has the following *excited* electronic configuration: $[Ar] 4s^2 3d^3 4p^1$

(Ar: Argon; $Z = 18$)

Give the **condensed ground state electronic configuration** of the atom, **A** and the cation, **A^{2+}** .
Show your work. (Assume all electronic transitions are possible)

Electronic Configuration of A:

Electronic Configuration of A^{2+} :

c. 2 points

In a single atom, determine how many electrons can have the following set of quantum numbers?

(i) $n = 5; l = 3$

In a single atom, determine how many electrons can have the following set of quantum numbers?

(i) $n = 3; l = 2; m_s = +1/2$

**Question 4** **8 points**

Rank the following from 1 to 4 in **increasing** order. Provide the ranking in the boxes provided.

Note: Atomic number (Z) are provided for relevant atoms

a) Increasing (lowest to highest) order of *ionization energy*: (1= lowest; 4 = highest)

Mg

Mg: Z = 12

Rb

Rb: Z = 37

Cs

Cs: Z = 55

K

K : Z = 19

b) Increasing (lowest to highest) order of *bond angle*: (1= lowest; 4 = highest)

H₂OCCl₄XeF₄

HCN

c) Increasing (smallest to largest) order of *ionic size* (1= smallest; 4 = largest)

S²⁻

S: Z = 16

O²⁻

O: Z = 8

Be²⁺

Be: Z = 4

Na⁺

Na: Z = 11

d) Increasing (lowest to highest) order of *bond energy* (1= lowest; 4 = highest)

P-F

P-Cl

P-I

P-Br

**Question 5** **12 points**

a. 9 points For the molecules HClO_4 and TeF_6 , draw the most stable *Lewis structure* showing appropriate shape and *dashed/wedged bonds*, based on VSEPR. Determine the **molecular geometry** around the central atom. For HClO_4 , label the *most polar bond*.

HClO_4 (there are no O-O bonds)

Label the most polar bond

Molecular Geometry around Cl

TeF_6

Molecular Geometry around Te

b. 3 points (Note: Lone pairs are not shown in structures below)

(i) For the three *resonance* structures (1, 2 and 3) shown below, answer the following question:



(I) Circle the *most contributing* resonance structure.

Structure 1

Structure 2

Structure 3

(ii) For the three structures given (4, 5, and 6) below, answer the following questions:



(I) Circle all the structures that represent *resonance structures* of each other:

Structure 4

Structure 5

Structure 6

(II) Among the resonance structures, circle the *most contributing* resonance structure.

Structure 4

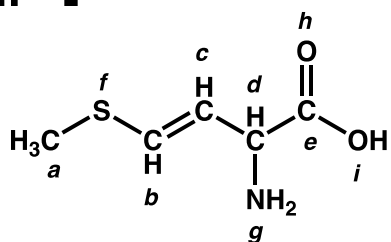
Structure 5

Structure 6

**Question 6** **6 points**

Using the data given below, calculate the **lattice energy** for SrO(s).

	ΔH° (kJ/mol)
$\text{Sr}(s) \longrightarrow \text{Sr}(g)$	+180
$\text{Sr}(g) \longrightarrow \text{Sr}^{2+}(g) + 2e^-$	+1468
$\text{O}_2(g) \longrightarrow 2\text{O}(g)$	+500
$\text{O}(g) + 2e^- \longrightarrow \text{O}^{2-}(g)$	+600
$\text{Sr}(s) + \frac{1}{2}\text{O}_2(g) \longrightarrow \text{SrO}(s)$	-550

**Question 7 10 points**

Consider the molecule given on the left, and answer questions (i) to (vi) using chemical bonding theories that we have learnt in class (valence bond theory, hybridization).

Note that the lone pairs are not shown in the structure:

i. Consider the carbon atoms marked in as ***a***, ***b***, ***c***, ***d***, and ***e*** in the molecule, and determine **hybridization** at each of the carbon centers. (2.5 points)

Atom	Hybridization
Carbon (<i>a</i>)	
Carbon (<i>b</i>)	
Carbon (<i>c</i>)	
Carbon (<i>d</i>)	
Carbon (<i>e</i>)	

ii. Now, consider the sulfur atom (S) marked as ***f***, nitrogen (N) atom (***g***), and oxygen(O) atoms (***h*** and ***i***), and determine their **hybridization**. (2 points)

Atom	Hybridization
Sulfur (<i>f</i>)	
Nitrogen (<i>g</i>)	
Oxygen (<i>h</i>)	
Oxygen (<i>i</i>)	

iii. **Describe** all the bonds formed by the carbon atoms ***a*** and ***b***, using sigma and pi designation as appropriate. (2 points)

Atom	Bonds formed (use sigma and pi as appropriate)
Carbon (<i>a</i>)	
Carbon (<i>b</i>)	

iv. How many total sp^3 -s sigma bonds are in the molecule? (1 point)

v. Determine **electronic** and **molecular geometry** at carbon (C) labeled ***e*** and sulfur (S) labeled ***f*** centres. (2 points)

Atom	Electronic Geometry	Molecular Geometry
Carbon (<i>e</i>)		
Sulfur (<i>f</i>)		

vi. Circle the value from the choices given below, that describes the **C-S-C bond angle at sulfur** (S) marked ***f***: (0.5 points)

$>90^\circ$ to $<109.5^\circ$ 109.5° >109.5 to $<120^\circ$ 120° $>120^\circ$ to $<180^\circ$


Question 8 12 points

Using the **valence shell orbitals only** draw a **complete** molecular orbital (MO) diagram (with no s-p mixing) for the molecule **PS**, in the space provided below. **(6 points)**

Assume that 3s and 3p atomic orbitals will form molecular orbitals like those formed by the combination of 2s and 2p atomic orbitals that we learnt in class, and there is no s-p mixing.

Show **energy arrow**, **atomic orbitals** involved, **atom labels**, the **molecular orbitals** that are formed, **filled electrons** in the MO diagram etc. Make sure to label all the atomic and molecular orbitals.

Using the MO diagram above, answer the following questions:

- (i) Which one of the following will have the highest bond order? Circle the correct choice
 A) PS^{1-} B) PS C) PS^{1+}
- (ii) Which one of the following will have the longest bond length? Circle the correct choice
 A) PS^{1-} B) PS C) PS^{1+}
- (iii) Which of the following will be diamagnetic? Circle the correct choice.
 A) PS^{1-} B) PS C) PS^{1+}
- (iv) Draw Lewis dot structure of the molecule with the shortest bond length (based on the MO diagram) from among the three (PS^{1-} , PS, PS^{1+}), and indicate the bond order calculated from the Lewis dot structure. **(3 points)**

Lewis Dot Structure

Bond Order



D9A6882E-94AD-4DB1-A6D1-525FC7EAA6FA

midterm-exam-93388

#303 Page 10 of 10

SCRAP PAGE

**CHEMISTRY 110 –Midterm Exam****Thursday November 7, 2024, 6:30 – 8:30 pm**

Examiners: Pallavi Sirjoosingh and Ashok Kakkar

First name (please write as legibly as possible within the boxes)

--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--

Last name

--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--

Student ID number

--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--

This booklet has 8 questions on 10 pages (The last page can be used as a scrap page).

The staple must NOT be removed from the question booklet. Your intact question booklet must be returned at the end of the exam.

A separate booklet contains a periodic table, and datasheet.

Write your answers directly under the questions in the space provided. Do not write on the QR codes or near the top edges (as it may be cut off).

Please write **legibly and dark enough** for the scan to be readable. For students who use English as a second language, dictionaries are permitted. Calculators are allowed.

DO NOT ASK QUESTIONS OF THE INVIGILATORS about interpretations of the exam questions. If you have a concern about a question, please summarize it at the top of this page. All concerns will be evaluated after the exam. GOOD LUCK!

**Question 1 8 points** (*Assume all transitions are possible*)

a. 6 points Calculate the **energy (in kJ/mol)** required to ionize (completely remove) the electron from the **$n=3$** level of **He^+ ion** (He: Helium). Show your work including equations. (3 sig. figs.)

b. 2 points

(I) Select the largest (longest) wavelength from the three choices below:

- i. Wavelength of photon to *completely remove* an electron from *1s orbital of Hydrogen atom*
- ii. Wavelength of photon to *completely remove* an electron from *1s orbital of Boron atom (B)*
- iii. Wavelength of photon to *completely remove* an electron from *1s orbital of Boron cation (B^+)*

i or ii or iii

(II) Select the smallest (shortest) wavelength from the three choices below (*for H atom*):

- i. Wavelength of photon *emitted* for an electron transitioning from *2p to 1s orbital*
- ii. Wavelength of photon *absorbed* for an electron transitioning from *2s to 4p orbital*
- iii. Wavelength of photon *absorbed* for an electron transitioning from *1s to 3p orbital*

i or ii or iii



Question 2 8 points



A Chem110 student plots the graph shown on the left, from data obtained for the kinetic energy (eV) of the ejected electron versus frequency (s^{-1}) of the incident light, for an unknown metal. ($1\text{eV} = 1.6022 \times 10^{-19} \text{ J}$)

Based on the plot, answer the questions **a to c** below:

a. 1 point On the plot above, label the **threshold frequency**.

b. 2 points What is the value of the work potential (in J) of the unknown metal?

c. 3 points What is the *longest* wavelength (in nm) required to eject an electron from this metal's surface? Show your work including equation(s) used. (3 sig figs)

d. 2 points In the space provided below, plot the following

(Assume frequency is *above* threshold frequency):

- Kinetic energy* of an ejected electron versus the *intensity* of the incident light (left)
- Number of electrons* ejected versus the *frequency* of the incident light (right)



**Question 3** **6 points****a. 2 points**

In the ground state of Ge (Germanium; $Z = 32$) atom, **name** the following orbitals:

i. Orbital containing the electron that requires the maximum energy to be removed.

ii. Orbital containing the electron that experiences the minimum effective nuclear charge.

b. 2 points

An unknown anion (A^-) has the following *excited* electronic configuration: $[Ar] 4s^2 3d^3 4p^1$

(Ar: Argon; $Z = 18$)

Give the **condensed ground state electronic configuration** of the atom, **A** and the cation, **A^{2+}** .
Show your work. (Assume all electronic transitions are possible)

Electronic Configuration of A:

Electronic Configuration of A^{2+} :

c. 2 points

In a single atom, determine how many electrons can have the following set of quantum numbers?

(i) $n = 5; l = 3$

In a single atom, determine how many electrons can have the following set of quantum numbers?

(i) $n = 3; l = 2; m_s = +1/2$

**Question 4** **8 points**

Rank the following from 1 to 4 in **increasing** order. Provide the ranking in the boxes provided.

Note: Atomic number (Z) are provided for relevant atoms

a) Increasing (lowest to highest) order of *ionization energy*: (1= lowest; 4 = highest)

Mg

Mg: Z = 12

Rb

Rb: Z = 37

Cs

Cs: Z = 55

K

K : Z = 19

b) Increasing (lowest to highest) order of *bond angle*: (1= lowest; 4 = highest)

H₂OCCl₄XeF₄

HCN

c) Increasing (smallest to largest) order of *ionic size* (1= smallest; 4 = largest)

S²⁻

S: Z = 16

O²⁻

O: Z = 8

Be²⁺

Be: Z = 4

Na⁺

Na: Z = 11

d) Increasing (lowest to highest) order of *bond energy* (1= lowest; 4 = highest)

P-F

P-Cl

P-I

P-Br

**Question 5** **12 points**

a. 9 points For the molecules HClO_4 and TeF_6 , draw the most stable *Lewis structure* showing appropriate shape and *dashed/wedged bonds*, based on VSEPR. Determine the **molecular geometry** around the central atom. For HClO_4 , label the *most polar bond*.



b. 3 points (Note: Lone pairs are not shown in structures below)

(i) For the three *resonance* structures (1, 2 and 3) shown below, answer the following question:



(I) Circle the *most contributing* resonance structure.

Structure 1

Structure 2

Structure 3

(ii) For the three structures given (4, 5, and 6) below, answer the following questions:



(I) Circle all the structures that represent *resonance structures* of each other:

Structure 4

Structure 5

Structure 6

(II) Among the resonance structures, circle the *most contributing* resonance structure.

Structure 4

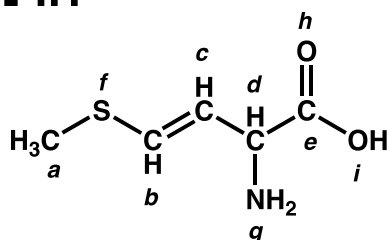
Structure 5

Structure 6

**Question 6** **6 points**

Using the data given below, calculate the **lattice energy** for SrO(s).

	ΔH° (kJ/mol)
$\text{Sr}(s) \longrightarrow \text{Sr}(g)$	+180
$\text{Sr}(g) \longrightarrow \text{Sr}^{2+}(g) + 2e^-$	+1468
$\text{O}_2(g) \longrightarrow 2\text{O}(g)$	+500
$\text{O}(g) + 2e^- \longrightarrow \text{O}^{2-}(g)$	+600
$\text{Sr}(s) + \frac{1}{2}\text{O}_2(g) \longrightarrow \text{SrO}(s)$	-550

**Question 7 10 points**

Consider the molecule given on the left, and answer questions (i) to (vi) using chemical bonding theories that we have learnt in class (valence bond theory, hybridization).

Note that the lone pairs are not shown in the structure:

i. Consider the carbon atoms marked in as ***a***, ***b***, ***c***, ***d***, and ***e*** in the molecule, and determine **hybridization** at each of the carbon centers. (2.5 points)

Atom	Hybridization
Carbon (<i>a</i>)	
Carbon (<i>b</i>)	
Carbon (<i>c</i>)	
Carbon (<i>d</i>)	
Carbon (<i>e</i>)	

ii. Now, consider the sulfur atom (S) marked as ***f***, nitrogen (N) atom (***g***), and oxygen(O) atoms (***h*** and ***i***), and determine their **hybridization**. (2 points)

Atom	Hybridization
Sulfur (<i>f</i>)	
Nitrogen (<i>g</i>)	
Oxygen (<i>h</i>)	
Oxygen (<i>i</i>)	

iii. **Describe** all the bonds formed by the carbon atoms ***a*** and ***b***, using sigma and pi designation as appropriate. (2 points)

Atom	Bonds formed (use sigma and pi as appropriate)
Carbon (<i>a</i>)	
Carbon (<i>b</i>)	

iv. How many total sp^3 -s sigma bonds are in the molecule? (1 point)

v. Determine **electronic** and **molecular geometry** at carbon (C) labeled ***e*** and sulfur (S) labeled ***f*** centres. (2 points)

Atom	Electronic Geometry	Molecular Geometry
Carbon (<i>e</i>)		
Sulfur (<i>f</i>)		

vi. Circle the value from the choices given below, that describes the **C-S-C bond angle at sulfur** (S) marked ***f***: (0.5 points)

$>90^\circ$ to $<109.5^\circ$ 109.5° >109.5 to $<120^\circ$ 120° $>120^\circ$ to $<180^\circ$



Question 8 12 points

Using the **valence shell orbitals only** draw a **complete** molecular orbital (MO) diagram (with no s-p mixing) for the molecule **PS**, in the space provided below. **(6 points)**

Assume that 3s and 3p atomic orbitals will form molecular orbitals like those formed by the combination of 2s and 2p atomic orbitals that we learnt in class, and there is no s-p mixing.

Show **energy arrow**, **atomic orbitals** involved, **atom labels**, the **molecular orbitals** that are formed, **filled electrons** in the MO diagram etc. Make sure to label all the atomic and molecular orbitals.

Using the MO diagram above, answer the following questions:

- (i) Which one of the following will have the highest bond order? Circle the correct choice
 A) PS^{1-} B) PS C) PS^{1+}
- (ii) Which one of the following will have the longest bond length? Circle the correct choice
 A) PS^{1-} B) PS C) PS^{1+}
- (iii) Which of the following will be diamagnetic? Circle the correct choice.
 A) PS^{1-} B) PS C) PS^{1+}
- (iv) Draw Lewis dot structure of the molecule with the shortest bond length (based on the MO diagram) from among the three (PS^{1-} , PS, PS^{1+}), and indicate the bond order calculated from the Lewis dot structure. **(3 points)**

Lewis Dot Structure

Bond Order



57F89D7A-A3B2-411B-B388-2CBA57FD4D56

midterm-exam-93388

#304 Page 10 of 10

SCRAP PAGE

**CHEMISTRY 110 –Midterm Exam****Thursday November 7, 2024, 6:30 – 8:30 pm**

Examiners: Pallavi Sirjoosingh and Ashok Kakkar

First name (please write as legibly as possible within the boxes)

--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--

Last name

--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--

Student ID number

--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--

This booklet has 8 questions on 10 pages (The last page can be used as a scrap page).

The staple must NOT be removed from the question booklet. Your intact question booklet must be returned at the end of the exam.

A separate booklet contains a periodic table, and datasheet.

Write your answers directly under the questions in the space provided. Do not write on the QR codes or near the top edges (as it may be cut off).

Please write **legibly and dark enough** for the scan to be readable. For students who use English as a second language, dictionaries are permitted. Calculators are allowed.

DO NOT ASK QUESTIONS OF THE INVIGILATORS about interpretations of the exam questions. If you have a concern about a question, please summarize it at the top of this page. All concerns will be evaluated after the exam. GOOD LUCK!

**Question 1 8 points** (*Assume all transitions are possible*)

a. 6 points Calculate the **energy (in kJ/mol)** required to ionize (completely remove) the electron from the **$n=3$** level of **He^+ ion** (He: Helium). Show your work including equations. (3 sig. figs.)

b. 2 points

(I) Select the largest (longest) wavelength from the three choices below:

- i. Wavelength of photon to *completely remove* an electron from *1s orbital of Hydrogen atom*
- ii. Wavelength of photon to *completely remove* an electron from *1s orbital of Boron atom (B)*
- iii. Wavelength of photon to *completely remove* an electron from *1s orbital of Boron cation (B^+)*

i or ii or iii

(II) Select the smallest (shortest) wavelength from the three choices below (*for H atom*):

- i. Wavelength of photon *emitted* for an electron transitioning from *2p to 1s orbital*
- ii. Wavelength of photon *absorbed* for an electron transitioning from *2s to 4p orbital*
- iii. Wavelength of photon *absorbed* for an electron transitioning from *1s to 3p orbital*

i or ii or iii



Question 2 8 points



A Chem110 student plots the graph shown on the left, from data obtained for the kinetic energy (eV) of the ejected electron versus frequency (s^{-1}) of the incident light, for an unknown metal. ($1\text{eV} = 1.6022 \times 10^{-19} \text{ J}$)

Based on the plot, answer the questions **a to c** below:

a. 1 point On the plot above, label the **threshold frequency**.

b. 2 points What is the value of the work potential (in J) of the unknown metal?

c. 3 points What is the *longest* wavelength (in nm) required to eject an electron from this metal's surface? Show your work including equation(s) used. (3 sig figs)

d. 2 points In the space provided below, plot the following

(Assume frequency is *above* threshold frequency):

- i. *Kinetic energy* of an ejected electron versus the *intensity* of the incident light (left)
- ii. *Number of electrons* ejected versus the *frequency* of the incident light (right)



**Question 3** **6 points****a. 2 points**

In the ground state of Ge (Germanium; $Z = 32$) atom, **name** the following orbitals:

i. Orbital containing the electron that requires the maximum energy to be removed.

ii. Orbital containing the electron that experiences the minimum effective nuclear charge.

b. 2 points

An unknown anion (A^-) has the following *excited* electronic configuration: $[Ar] 4s^2 3d^3 4p^1$

(Ar: Argon; $Z = 18$)

Give the **condensed ground state electronic configuration** of the atom, **A** and the cation, **A^{2+}** .
Show your work. (Assume all electronic transitions are possible)

Electronic Configuration of A:

Electronic Configuration of A^{2+} :

c. 2 points

In a single atom, determine how many electrons can have the following set of quantum numbers?

(i) $n = 5; l = 3$

In a single atom, determine how many electrons can have the following set of quantum numbers?

(i) $n = 3; l = 2; m_s = +1/2$

**Question 4** **8 points**

Rank the following from 1 to 4 in **increasing** order. Provide the ranking in the boxes provided.

Note: Atomic number (Z) are provided for relevant atoms

a) Increasing (lowest to highest) order of *ionization energy*: (1= lowest; 4 = highest)

Mg

Mg: Z = 12

Rb

Rb: Z = 37

Cs

Cs: Z = 55

K

K : Z = 19

b) Increasing (lowest to highest) order of *bond angle*: (1= lowest; 4 = highest)

H₂OCCl₄XeF₄

HCN

c) Increasing (smallest to largest) order of *ionic size* (1= smallest; 4 = largest)

S²⁻

S: Z = 16

O²⁻

O: Z = 8

Be²⁺

Be: Z = 4

Na⁺

Na: Z = 11

d) Increasing (lowest to highest) order of *bond energy* (1= lowest; 4 = highest)

P-F

P-Cl

P-I

P-Br

**Question 5** **12 points**

a. 9 points For the molecules HClO_4 and TeF_6 , draw the most stable *Lewis structure* showing appropriate shape and *dashed/wedged bonds*, based on VSEPR. Determine the **molecular geometry** around the central atom. For HClO_4 , label the *most polar bond*.



b. 3 points (Note: Lone pairs are not shown in structures below)

(i) For the three *resonance* structures (1, 2 and 3) shown below, answer the following question:



(I) Circle the *most contributing* resonance structure.

Structure 1

Structure 2

Structure 3

(ii) For the three structures given (4, 5, and 6) below, answer the following questions:



(I) Circle all the structures that represent *resonance structures* of each other:

Structure 4

Structure 5

Structure 6

(II) Among the resonance structures, circle the *most contributing* resonance structure.

Structure 4

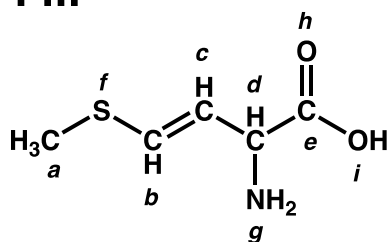
Structure 5

Structure 6

**Question 6** **6 points**

Using the data given below, calculate the **lattice energy** for SrO(s).

	ΔH° (kJ/mol)
$\text{Sr}(s) \longrightarrow \text{Sr}(g)$	+180
$\text{Sr}(g) \longrightarrow \text{Sr}^{2+}(g) + 2e^-$	+1468
$\text{O}_2(g) \longrightarrow 2\text{O}(g)$	+500
$\text{O}(g) + 2e^- \longrightarrow \text{O}^{2-}(g)$	+600
$\text{Sr}(s) + \frac{1}{2}\text{O}_2(g) \longrightarrow \text{SrO}(s)$	-550

**Question 7 10 points**

Consider the molecule given on the left, and answer questions (i) to (vi) using chemical bonding theories that we have learnt in class (valence bond theory, hybridization).

Note that the lone pairs are not shown in the structure:

i. Consider the carbon atoms marked in as ***a***, ***b***, ***c***, ***d***, and ***e*** in the molecule, and determine **hybridization** at each of the carbon centers. (2.5 points)

Atom	Hybridization
Carbon (<i>a</i>)	
Carbon (<i>b</i>)	
Carbon (<i>c</i>)	
Carbon (<i>d</i>)	
Carbon (<i>e</i>)	

ii. Now, consider the sulfur atom (S) marked as ***f***, nitrogen (N) atom (***g***), and oxygen(O) atoms (***h*** and ***i***), and determine their **hybridization**. (2 points)

Atom	Hybridization
Sulfur (<i>f</i>)	
Nitrogen (<i>g</i>)	
Oxygen (<i>h</i>)	
Oxygen (<i>i</i>)	

iii. **Describe** all the bonds formed by the carbon atoms ***a*** and ***b***, using sigma and pi designation as appropriate. (2 points)

Atom	Bonds formed (use sigma and pi as appropriate)
Carbon (<i>a</i>)	
Carbon (<i>b</i>)	

iv. How many total sp^3 -s sigma bonds are in the molecule? (1 point)

v. Determine **electronic** and **molecular geometry** at carbon (C) labeled ***e*** and sulfur (S) labeled ***f*** centres. (2 points)

Atom	Electronic Geometry	Molecular Geometry
Carbon (<i>e</i>)		
Sulfur (<i>f</i>)		

vi. Circle the value from the choices given below, that describes the **C-S-C bond angle at sulfur** (S) marked ***f***: (0.5 points)

>90° to <109.5° 109.5° >109.5 to <120° 120° >120° to <180°


Question 8 12 points

Using the **valence shell orbitals only** draw a **complete** molecular orbital (MO) diagram (with no s-p mixing) for the molecule **PS**, in the space provided below. **(6 points)**

Assume that 3s and 3p atomic orbitals will form molecular orbitals like those formed by the combination of 2s and 2p atomic orbitals that we learnt in class, and there is no s-p mixing.

Show **energy arrow**, **atomic orbitals** involved, **atom labels**, the **molecular orbitals** that are formed, **filled electrons** in the MO diagram etc. Make sure to label all the atomic and molecular orbitals.

Using the MO diagram above, answer the following questions:

- (i) Which one of the following will have the highest bond order? Circle the correct choice
 A) PS^{1-} B) PS C) PS^{1+}
- (ii) Which one of the following will have the longest bond length? Circle the correct choice
 A) PS^{1-} B) PS C) PS^{1+}
- (iii) Which of the following will be diamagnetic? Circle the correct choice.
 A) PS^{1-} B) PS C) PS^{1+}
- (iv) Draw Lewis dot structure of the molecule with the shortest bond length (based on the MO diagram) from among the three (PS^{1-} , PS, PS^{1+}), and indicate the bond order calculated from the Lewis dot structure. **(3 points)**

Lewis Dot Structure

Bond Order



C3A3F707-02E2-48D5-B7C7-7B0E56B76A3F

midterm-exam-93388

#305 Page 10 of 10

SCRAP PAGE

**CHEMISTRY 110 –Midterm Exam****Thursday November 7, 2024, 6:30 – 8:30 pm****Examiners: Pallavi Sirjoosingh and Ashok Kakkar**

First name (please write as legibly as possible within the boxes)

--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--

Last name

--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--

Student ID number

--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--

This booklet has 8 questions on 10 pages (The last page can be used as a scrap page).

The staple must NOT be removed from the question booklet. Your intact question booklet must be returned at the end of the exam.

A separate booklet contains a periodic table, and datasheet.

Write your answers directly under the questions in the space provided. Do not write on the QR codes or near the top edges (as it may be cut off).

Please write **legibly and dark enough** for the scan to be readable. For students who use English as a second language, dictionaries are permitted. Calculators are allowed.

DO NOT ASK QUESTIONS OF THE INVIGILATORS about interpretations of the exam questions. If you have a concern about a question, please summarize it at the top of this page. All concerns will be evaluated after the exam. GOOD LUCK!

**Question 1 8 points** (*Assume all transitions are possible*)

a. 6 points Calculate the **energy (in kJ/mol)** required to ionize (completely remove) the electron from the **n=3** level of **He⁺ ion** (He: Helium). Show your work including equations. (3 sig. figs.)

b. 2 points

(I) Select the largest (longest) wavelength from the three choices below:

- i. Wavelength of photon to *completely remove* an electron from *1s orbital of Hydrogen atom*
- ii. Wavelength of photon to *completely remove* an electron from *1s orbital of Boron atom (B)*
- iii. Wavelength of photon to *completely remove* an electron from *1s orbital of Boron cation (B⁺)*

i or ii or iii

(II) Select the smallest (shortest) wavelength from the three choices below (*for H atom*):

- i. Wavelength of photon *emitted* for an electron transitioning from *2p to 1s orbital*
- ii. Wavelength of photon *absorbed* for an electron transitioning from *2s to 4p orbital*
- iii. Wavelength of photon *absorbed* for an electron transitioning from *1s to 3p orbital*

i or ii or iii



Question 2 8 points



A Chem110 student plots the graph shown on the left, from data obtained for the kinetic energy (eV) of the ejected electron versus frequency (s^{-1}) of the incident light, for an unknown metal. ($1\text{eV} = 1.6022 \times 10^{-19} \text{ J}$)

Based on the plot, answer the questions **a to c** below:

a. 1 point On the plot above, label the **threshold frequency**.

b. 2 points What is the value of the work potential (in J) of the unknown metal?

c. 3 points What is the *longest* wavelength (in nm) required to eject an electron from this metal's surface? Show your work including equation(s) used. (3 sig figs)

d. 2 points In the space provided below, plot the following

(Assume frequency is *above* threshold frequency):

- i. *Kinetic energy* of an ejected electron versus the *intensity* of the incident light (left)
- ii. *Number of electrons* ejected versus the *frequency* of the incident light (right)



**Question 3** **6 points****a. 2 points**

In the ground state of Ge (Germanium; $Z = 32$) atom, **name** the following orbitals:

i. Orbital containing the electron that requires the maximum energy to be removed.

ii. Orbital containing the electron that experiences the minimum effective nuclear charge.

b. 2 points

An unknown anion (A^-) has the following *excited* electronic configuration: $[Ar] 4s^2 3d^3 4p^1$

(Ar: Argon; $Z = 18$)

Give the **condensed ground state electronic configuration** of the atom, **A** and the cation, **A^{2+}** .
Show your work. (Assume all electronic transitions are possible)

Electronic Configuration of A:

Electronic Configuration of A^{2+} :

c. 2 points

In a single atom, determine how many electrons can have the following set of quantum numbers?

(i) $n = 5; l = 3$

In a single atom, determine how many electrons can have the following set of quantum numbers?

(i) $n = 3; l = 2; m_s = +1/2$

**Question 4** **8 points**

Rank the following from 1 to 4 in **increasing** order. Provide the ranking in the boxes provided.

Note: Atomic number (Z) are provided for relevant atoms

a) Increasing (lowest to highest) order of *ionization energy*: (1= lowest; 4 = highest)

Mg

Mg: Z = 12

Rb

Rb: Z = 37

Cs

Cs: Z = 55

K

K : Z = 19

b) Increasing (lowest to highest) order of *bond angle*: (1= lowest; 4 = highest)

H₂OCCl₄XeF₄

HCN

c) Increasing (smallest to largest) order of *ionic size* (1= smallest; 4 = largest)

S²⁻

S: Z = 16

O²⁻

O: Z = 8

Be²⁺

Be: Z = 4

Na⁺

Na: Z = 11

d) Increasing (lowest to highest) order of *bond energy* (1= lowest; 4 = highest)

P-F

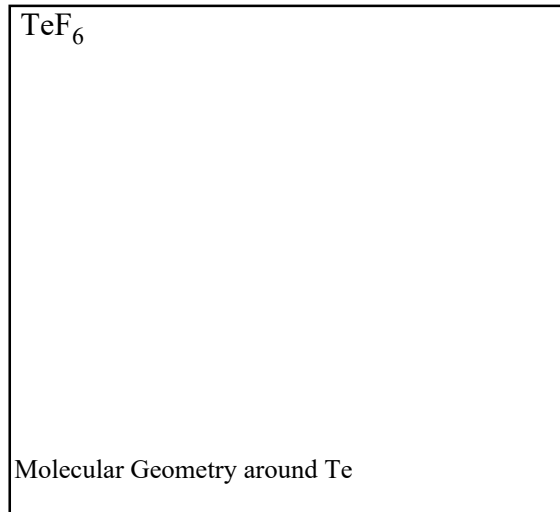
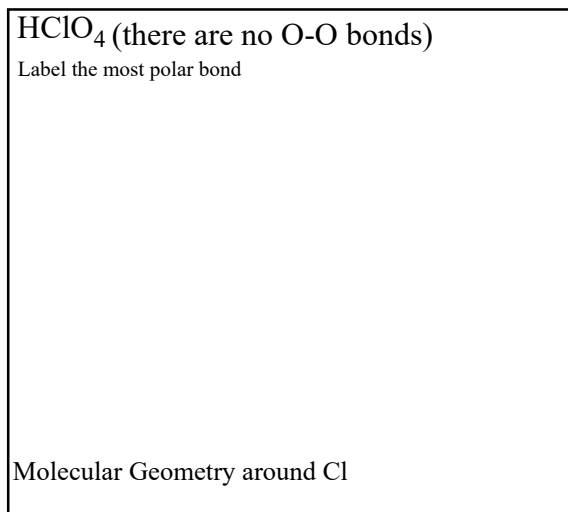
P-Cl

P-I

P-Br

**Question 5** **12 points**

a. 9 points For the molecules HClO_4 and TeF_6 , draw the most stable *Lewis structure* showing appropriate shape and *dashed/wedged bonds*, based on VSEPR. Determine the **molecular geometry** around the central atom. For HClO_4 , label the *most polar bond*.



b. 3 points (Note: Lone pairs are not shown in structures below)

(i) For the three *resonance* structures (1, 2 and 3) shown below, answer the following question:



(I) Circle the *most contributing* resonance structure.

Structure 1

Structure 2

Structure 3

(ii) For the three structures given (4, 5, and 6) below, answer the following questions:



(I) Circle all the structures that represent *resonance structures* of each other:

Structure 4

Structure 5

Structure 6

(II) Among the resonance structures, circle the *most contributing* resonance structure.

Structure 4

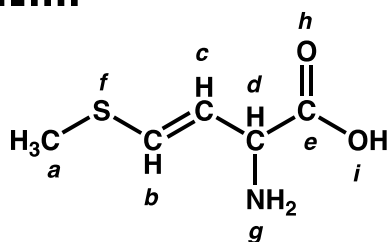
Structure 5

Structure 6

**Question 6** **6 points**

Using the data given below, calculate the **lattice energy** for SrO(s).

	ΔH° (kJ/mol)
$\text{Sr}(s) \longrightarrow \text{Sr}(g)$	+180
$\text{Sr}(g) \longrightarrow \text{Sr}^{2+}(g) + 2e^-$	+1468
$\text{O}_2(g) \longrightarrow 2\text{O}(g)$	+500
$\text{O}(g) + 2e^- \longrightarrow \text{O}^{2-}(g)$	+600
$\text{Sr}(s) + \frac{1}{2}\text{O}_2(g) \longrightarrow \text{SrO}(s)$	-550

**Question 7 10 points**

Consider the molecule given on the left, and answer questions (i) to (vi) using chemical bonding theories that we have learnt in class (valence bond theory, hybridization).

Note that the lone pairs are not shown in the structure:

i. Consider the carbon atoms marked in as ***a***, ***b***, ***c***, ***d***, and ***e*** in the molecule, and determine **hybridization** at each of the carbon centers. (2.5 points)

Atom	Hybridization
Carbon (<i>a</i>)	
Carbon (<i>b</i>)	
Carbon (<i>c</i>)	
Carbon (<i>d</i>)	
Carbon (<i>e</i>)	

ii. Now, consider the sulfur atom (S) marked as ***f***, nitrogen (N) atom (***g***), and oxygen(O) atoms (***h*** and ***i***), and determine their **hybridization**. (2 points)

Atom	Hybridization
Sulfur (<i>f</i>)	
Nitrogen (<i>g</i>)	
Oxygen (<i>h</i>)	
Oxygen (<i>i</i>)	

iii. **Describe** all the bonds formed by the carbon atoms ***a*** and ***b***, using sigma and pi designation as appropriate. (2 points)

Atom	Bonds formed (use sigma and pi as appropriate)
Carbon (<i>a</i>)	
Carbon (<i>b</i>)	

iv. How many total sp^3 -s sigma bonds are in the molecule? (1 point)

v. Determine **electronic** and **molecular geometry** at carbon (C) labeled ***e*** and sulfur (S) labeled ***f*** centres. (2 points)

Atom	Electronic Geometry	Molecular Geometry
Carbon (<i>e</i>)		
Sulfur (<i>f</i>)		

vi. Circle the value from the choices given below, that describes the **C-S-C bond angle at sulfur** (S) marked ***f***: (0.5 points)

$>90^\circ$ to $<109.5^\circ$ 109.5° >109.5 to $<120^\circ$ 120° $>120^\circ$ to $<180^\circ$


Question 8 12 points

Using the **valence shell orbitals only** draw a **complete** molecular orbital (MO) diagram (with no s-p mixing) for the molecule **PS**, in the space provided below. **(6 points)**

Assume that 3s and 3p atomic orbitals will form molecular orbitals like those formed by the combination of 2s and 2p atomic orbitals that we learnt in class, and there is no s-p mixing.

Show **energy arrow**, **atomic orbitals** involved, **atom labels**, the **molecular orbitals** that are formed, **filled electrons** in the MO diagram etc. Make sure to label all the atomic and molecular orbitals.

Using the MO diagram above, answer the following questions:

- (i) Which one of the following will have the highest bond order? Circle the correct choice
 A) PS^{1-} B) PS C) PS^{1+}
- (ii) Which one of the following will have the longest bond length? Circle the correct choice
 A) PS^{1-} B) PS C) PS^{1+}
- (iii) Which of the following will be diamagnetic? Circle the correct choice.
 A) PS^{1-} B) PS C) PS^{1+}
- (iv) Draw Lewis dot structure of the molecule with the shortest bond length (based on the MO diagram) from among the three (PS^{1-} , PS, PS^{1+}), and indicate the bond order calculated from the Lewis dot structure. **(3 points)**

Lewis Dot Structure

Bond Order



58B1EBDC-61A0-40AB-AD82-89EE233C9878

midterm-exam-93388

#306 Page 10 of 10

SCRAP PAGE

**CHEMISTRY 110 –Midterm Exam****Thursday November 7, 2024, 6:30 – 8:30 pm**

Examiners: Pallavi Sirjoosingh and Ashok Kakkar

First name (please write as legibly as possible within the boxes)

--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--

Last name

--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--

Student ID number

--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--

This booklet has 8 questions on 10 pages (The last page can be used as a scrap page).

The staple must NOT be removed from the question booklet. Your intact question booklet must be returned at the end of the exam.

A separate booklet contains a periodic table, and datasheet.

Write your answers directly under the questions in the space provided. Do not write on the QR codes or near the top edges (as it may be cut off).

Please write **legibly and dark enough** for the scan to be readable. For students who use English as a second language, dictionaries are permitted. Calculators are allowed.

DO NOT ASK QUESTIONS OF THE INVIGILATORS about interpretations of the exam questions. If you have a concern about a question, please summarize it at the top of this page. All concerns will be evaluated after the exam. GOOD LUCK!

**Question 1 8 points** (*Assume all transitions are possible*)

a. 6 points Calculate the **energy (in kJ/mol)** required to ionize (completely remove) the electron from the **n=3** level of **He⁺ ion** (He: Helium). Show your work including equations. (3 sig. figs.)

b. 2 points

(I) Select the largest (longest) wavelength from the three choices below:

- i. Wavelength of photon to *completely remove* an electron from *1s orbital of Hydrogen atom*
- ii. Wavelength of photon to *completely remove* an electron from *1s orbital of Boron atom (B)*
- iii. Wavelength of photon to *completely remove* an electron from *1s orbital of Boron cation (B⁺)*

i or ii or iii

(II) Select the smallest (shortest) wavelength from the three choices below (*for H atom*):

- i. Wavelength of photon *emitted* for an electron transitioning from *2p to 1s orbital*
- ii. Wavelength of photon *absorbed* for an electron transitioning from *2s to 4p orbital*
- iii. Wavelength of photon *absorbed* for an electron transitioning from *1s to 3p orbital*

i or ii or iii



Question 2 8 points



A Chem110 student plots the graph shown on the left, from data obtained for the kinetic energy (eV) of the ejected electron versus frequency (s^{-1}) of the incident light, for an unknown metal. ($1\text{eV} = 1.6022 \times 10^{-19} \text{ J}$)

Based on the plot, answer the questions **a to c** below:

a. 1 point On the plot above, label the **threshold frequency**.

b. 2 points What is the value of the work potential (in J) of the unknown metal?

c. 3 points What is the *longest* wavelength (in nm) required to eject an electron from this metal's surface? Show your work including equation(s) used. (3 sig figs)

d. 2 points In the space provided below, plot the following

(Assume frequency is *above* threshold frequency):

- Kinetic energy* of an ejected electron versus the *intensity* of the incident light (left)
- Number of electrons* ejected versus the *frequency* of the incident light (right)



**Question 3** **6 points****a. 2 points**

In the ground state of Ge (Germanium; $Z = 32$) atom, **name** the following orbitals:

i. Orbital containing the electron that requires the maximum energy to be removed.

ii. Orbital containing the electron that experiences the minimum effective nuclear charge.

b. 2 points

An unknown anion (A^-) has the following *excited* electronic configuration: $[Ar] 4s^2 3d^3 4p^1$

(Ar: Argon; $Z = 18$)

Give the **condensed ground state electronic configuration** of the atom, **A** and the cation, **A^{2+}** .
Show your work. (Assume all electronic transitions are possible)

Electronic Configuration of A:

Electronic Configuration of A^{2+} :

c. 2 points

In a single atom, determine how many electrons can have the following set of quantum numbers?

(i) $n = 5; l = 3$

In a single atom, determine how many electrons can have the following set of quantum numbers?

(i) $n = 3; l = 2; m_s = +1/2$

**Question 4** **8 points**

Rank the following from 1 to 4 in **increasing** order. Provide the ranking in the boxes provided.

Note: Atomic number (Z) are provided for relevant atoms

a) Increasing (lowest to highest) order of *ionization energy*: (1= lowest; 4 = highest)

Mg

Mg: Z = 12

Rb

Rb: Z = 37

Cs

Cs: Z = 55

K

K : Z = 19

b) Increasing (lowest to highest) order of *bond angle*: (1= lowest; 4 = highest)

H₂OCCl₄XeF₄

HCN

c) Increasing (smallest to largest) order of *ionic size* (1= smallest; 4 = largest)

S²⁻

S: Z = 16

O²⁻

O: Z = 8

Be²⁺

Be: Z = 4

Na⁺

Na: Z = 11

d) Increasing (lowest to highest) order of *bond energy* (1= lowest; 4 = highest)

P-F

P-Cl

P-I

P-Br

**Question 5** **12 points**

a. 9 points For the molecules HClO_4 and TeF_6 , draw the most stable *Lewis structure* showing appropriate shape and *dashed/wedged bonds*, based on VSEPR. Determine the **molecular geometry** around the central atom. For HClO_4 , label the *most polar bond*.

HClO_4 (there are no O-O bonds)

Label the most polar bond

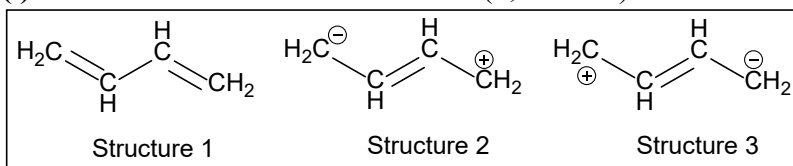
Molecular Geometry around Cl

TeF_6

Molecular Geometry around Te

b. 3 points (Note: Lone pairs are not shown in structures below)

(i) For the three *resonance* structures (1, 2 and 3) shown below, answer the following question:



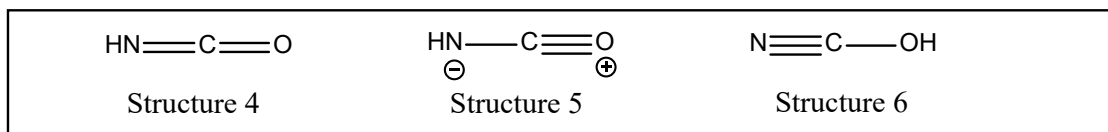
(I) Circle the *most contributing* resonance structure.

Structure 1

Structure 2

Structure 3

(ii) For the three structures given (4, 5, and 6) below, answer the following questions:



(I) Circle all the structures that represent *resonance structures* of each other:

Structure 4

Structure 5

Structure 6

(II) Among the resonance structures, circle the *most contributing* resonance structure.

Structure 4

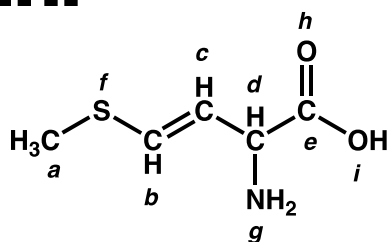
Structure 5

Structure 6

**Question 6** **6 points**

Using the data given below, calculate the **lattice energy** for SrO(s).

	ΔH° (kJ/mol)
$\text{Sr}(s) \longrightarrow \text{Sr}(g)$	+180
$\text{Sr}(g) \longrightarrow \text{Sr}^{2+}(g) + 2e^-$	+1468
$\text{O}_2(g) \longrightarrow 2\text{O}(g)$	+500
$\text{O}(g) + 2e^- \longrightarrow \text{O}^{2-}(g)$	+600
$\text{Sr}(s) + \frac{1}{2}\text{O}_2(g) \longrightarrow \text{SrO}(s)$	-550

**Question 7 10 points**

Consider the molecule given on the left, and answer questions (i) to (vi) using chemical bonding theories that we have learnt in class (valence bond theory, hybridization).

Note that the lone pairs are not shown in the structure:

i. Consider the carbon atoms marked in as ***a***, ***b***, ***c***, ***d***, and ***e*** in the molecule, and determine **hybridization** at each of the carbon centers. (2.5 points)

Atom	Hybridization
Carbon (<i>a</i>)	
Carbon (<i>b</i>)	
Carbon (<i>c</i>)	
Carbon (<i>d</i>)	
Carbon (<i>e</i>)	

ii. Now, consider the sulfur atom (S) marked as ***f***, nitrogen (N) atom (***g***), and oxygen(O) atoms (***h*** and ***i***), and determine their **hybridization**. (2 points)

Atom	Hybridization
Sulfur (<i>f</i>)	
Nitrogen (<i>g</i>)	
Oxygen (<i>h</i>)	
Oxygen (<i>i</i>)	

iii. **Describe** all the bonds formed by the carbon atoms ***a*** and ***b***, using sigma and pi designation as appropriate. (2 points)

Atom	Bonds formed (use sigma and pi as appropriate)
Carbon (<i>a</i>)	
Carbon (<i>b</i>)	

iv. How many total sp^3 -s sigma bonds are in the molecule? (1 point)

v. Determine **electronic** and **molecular geometry** at carbon (C) labeled ***e*** and sulfur (S) labeled ***f*** centres. (2 points)

Atom	Electronic Geometry	Molecular Geometry
Carbon (<i>e</i>)		
Sulfur (<i>f</i>)		

vi. Circle the value from the choices given below, that describes the **C-S-C bond angle at sulfur** (S) marked ***f***: (0.5 points)

>90° to <109.5° 109.5° >109.5 to <120° 120° >120° to <180°


Question 8 12 points

Using the **valence shell orbitals only** draw a **complete** molecular orbital (MO) diagram (with no s-p mixing) for the molecule **PS**, in the space provided below. **(6 points)**

Assume that 3s and 3p atomic orbitals will form molecular orbitals like those formed by the combination of 2s and 2p atomic orbitals that we learnt in class, and there is no s-p mixing.

Show **energy arrow**, **atomic orbitals** involved, **atom labels**, the **molecular orbitals** that are formed, **filled electrons** in the MO diagram etc. Make sure to label all the atomic and molecular orbitals.

Using the MO diagram above, answer the following questions:

- (i) Which one of the following will have the highest bond order? Circle the correct choice
 A) PS^{1-} B) PS C) PS^{1+}
- (ii) Which one of the following will have the longest bond length? Circle the correct choice
 A) PS^{1-} B) PS C) PS^{1+}
- (iii) Which of the following will be diamagnetic? Circle the correct choice.
 A) PS^{1-} B) PS C) PS^{1+}
- (iv) Draw Lewis dot structure of the molecule with the shortest bond length (based on the MO diagram) from among the three (PS^{1-} , PS, PS^{1+}), and indicate the bond order calculated from the Lewis dot structure. **(3 points)**

Lewis Dot Structure

Bond Order



431A3578-9FFA-46EA-BCC0-FF521E9E75B0

midterm-exam-93388

#307 Page 10 of 10

SCRAP PAGE

**CHEMISTRY 110 –Midterm Exam****Thursday November 7, 2024, 6:30 – 8:30 pm**

Examiners: Pallavi Sirjoosingh and Ashok Kakkar

First name (please write as legibly as possible within the boxes)

--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--

Last name

--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--

Student ID number

--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--

This booklet has 8 questions on 10 pages (The last page can be used as a scrap page).

The staple must NOT be removed from the question booklet. Your intact question booklet must be returned at the end of the exam.

A separate booklet contains a periodic table, and datasheet.

Write your answers directly under the questions in the space provided. Do not write on the QR codes or near the top edges (as it may be cut off).

Please write **legibly and dark enough** for the scan to be readable. For students who use English as a second language, dictionaries are permitted. Calculators are allowed.

DO NOT ASK QUESTIONS OF THE INVIGILATORS about interpretations of the exam questions. If you have a concern about a question, please summarize it at the top of this page. All concerns will be evaluated after the exam. GOOD LUCK!

**Question 1 8 points** (*Assume all transitions are possible*)

a. 6 points Calculate the **energy (in kJ/mol)** required to ionize (completely remove) the electron from the **n=3** level of **He⁺ ion** (He: Helium). Show your work including equations. (3 sig. figs.)

b. 2 points

(I) Select the largest (longest) wavelength from the three choices below:

- i. Wavelength of photon to *completely remove* an electron from *1s orbital of Hydrogen atom*
- ii. Wavelength of photon to *completely remove* an electron from *1s orbital of Boron atom (B)*
- iii. Wavelength of photon to *completely remove* an electron from *1s orbital of Boron cation (B⁺)*

i or ii or iii

(II) Select the smallest (shortest) wavelength from the three choices below (*for H atom*):

- i. Wavelength of photon *emitted* for an electron transitioning from *2p to 1s orbital*
- ii. Wavelength of photon *absorbed* for an electron transitioning from *2s to 4p orbital*
- iii. Wavelength of photon *absorbed* for an electron transitioning from *1s to 3p orbital*

i or ii or iii



Question 2 8 points



A Chem110 student plots the graph shown on the left, from data obtained for the kinetic energy (eV) of the ejected electron versus frequency (s^{-1}) of the incident light, for an unknown metal. ($1\text{eV} = 1.6022 \times 10^{-19} \text{ J}$)

Based on the plot, answer the questions **a to c** below:

a. 1 point On the plot above, label the **threshold frequency**.

b. 2 points What is the value of the work potential (in J) of the unknown metal?

c. 3 points What is the *longest* wavelength (in nm) required to eject an electron from this metal's surface? Show your work including equation(s) used. (3 sig figs)

d. 2 points In the space provided below, plot the following

(Assume frequency is *above* threshold frequency):

- Kinetic energy* of an ejected electron versus the *intensity* of the incident light (left)
- Number of electrons* ejected versus the *frequency* of the incident light (right)



**Question 3** **6 points****a. 2 points**

In the ground state of Ge (Germanium; $Z = 32$) atom, **name** the following orbitals:

i. Orbital containing the electron that requires the maximum energy to be removed.

ii. Orbital containing the electron that experiences the minimum effective nuclear charge.

b. 2 points

An unknown anion (A^-) has the following *excited* electronic configuration: $[Ar] 4s^2 3d^3 4p^1$

(Ar: Argon; $Z = 18$)

Give the **condensed ground state electronic configuration** of the atom, **A** and the cation, **A^{2+}** .
Show your work. (Assume all electronic transitions are possible)

Electronic Configuration of A:

Electronic Configuration of A^{2+} :

c. 2 points

In a single atom, determine how many electrons can have the following set of quantum numbers?

(i) $n = 5; l = 3$

In a single atom, determine how many electrons can have the following set of quantum numbers?

(i) $n = 3; l = 2; m_s = +1/2$

**Question 4** **8 points**

Rank the following from 1 to 4 in **increasing** order. Provide the ranking in the boxes provided.

Note: Atomic number (Z) are provided for relevant atoms

a) Increasing (lowest to highest) order of *ionization energy*: (1= lowest; 4 = highest)

Mg

Mg: Z = 12

Rb

Rb: Z = 37

Cs

Cs: Z = 55

K

K : Z = 19

b) Increasing (lowest to highest) order of *bond angle*: (1= lowest; 4 = highest)

H₂OCCl₄XeF₄

HCN

c) Increasing (smallest to largest) order of *ionic size* (1= smallest; 4 = largest)

S²⁻

S: Z = 16

O²⁻

O: Z = 8

Be²⁺

Be: Z = 4

Na⁺

Na: Z = 11

d) Increasing (lowest to highest) order of *bond energy* (1= lowest; 4 = highest)

P-F

P-Cl

P-I

P-Br

**Question 5** **12 points**

a. 9 points For the molecules HClO_4 and TeF_6 , draw the most stable *Lewis structure* showing appropriate shape and *dashed/wedged bonds*, based on VSEPR. Determine the **molecular geometry** around the central atom. For HClO_4 , label the *most polar bond*.



b. 3 points (Note: Lone pairs are not shown in structures below)

(i) For the three *resonance* structures (1, 2 and 3) shown below, answer the following question:



(I) Circle the *most contributing* resonance structure.

Structure 1

Structure 2

Structure 3

(ii) For the three structures given (4, 5, and 6) below, answer the following questions:



(I) Circle all the structures that represent *resonance structures* of each other:

Structure 4

Structure 5

Structure 6

(II) Among the resonance structures, circle the *most contributing* resonance structure.

Structure 4

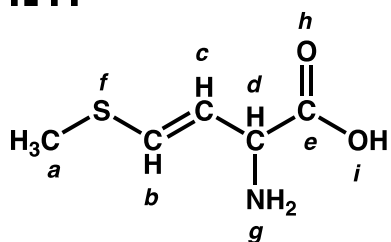
Structure 5

Structure 6

**Question 6** **6 points**

Using the data given below, calculate the **lattice energy** for SrO(s).

	ΔH° (kJ/mol)
$\text{Sr}(s) \longrightarrow \text{Sr}(g)$	+180
$\text{Sr}(g) \longrightarrow \text{Sr}^{2+}(g) + 2e^-$	+1468
$\text{O}_2(g) \longrightarrow 2\text{O}(g)$	+500
$\text{O}(g) + 2e^- \longrightarrow \text{O}^{2-}(g)$	+600
$\text{Sr}(s) + \frac{1}{2}\text{O}_2(g) \longrightarrow \text{SrO}(s)$	-550

**Question 7 10 points**

Consider the molecule given on the left, and answer questions (i) to (vi) using chemical bonding theories that we have learnt in class (valence bond theory, hybridization).

Note that the lone pairs are not shown in the structure:

i. Consider the carbon atoms marked in as ***a***, ***b***, ***c***, ***d***, and ***e*** in the molecule, and determine **hybridization** at each of the carbon centers. (2.5 points)

Atom	Hybridization
Carbon (<i>a</i>)	
Carbon (<i>b</i>)	
Carbon (<i>c</i>)	
Carbon (<i>d</i>)	
Carbon (<i>e</i>)	

ii. Now, consider the sulfur atom (S) marked as ***f***, nitrogen (N) atom (***g***), and oxygen(O) atoms (***h*** and ***i***), and determine their **hybridization**. (2 points)

Atom	Hybridization
Sulfur (<i>f</i>)	
Nitrogen (<i>g</i>)	
Oxygen (<i>h</i>)	
Oxygen (<i>i</i>)	

iii. **Describe** all the bonds formed by the carbon atoms ***a*** and ***b***, using sigma and pi designation as appropriate. (2 points)

Atom	Bonds formed (use sigma and pi as appropriate)
Carbon (<i>a</i>)	
Carbon (<i>b</i>)	

iv. How many total sp^3 -s sigma bonds are in the molecule? (1 point)

v. Determine **electronic** and **molecular geometry** at carbon (C) labeled ***e*** and sulfur (S) labeled ***f*** centres. (2 points)

Atom	Electronic Geometry	Molecular Geometry
Carbon (<i>e</i>)		
Sulfur (<i>f</i>)		

vi. Circle the value from the choices given below, that describes the **C-S-C bond angle at sulfur** (S) marked ***f***: (0.5 points)

>90° to <109.5°

109.5°

>109.5 to <120°

120°

>120° to <180°

**Question 8 12 points**

Using the **valence shell orbitals only** draw a **complete** molecular orbital (MO) diagram (with no s-p mixing) for the molecule **PS**, in the space provided below. **(6 points)**

Assume that 3s and 3p atomic orbitals will form molecular orbitals like those formed by the combination of 2s and 2p atomic orbitals that we learnt in class, and there is no s-p mixing.

Show **energy arrow**, **atomic orbitals** involved, **atom labels**, the **molecular orbitals** that are formed, **filled electrons** in the MO diagram etc. Make sure to label all the atomic and molecular orbitals.

Using the MO diagram above, answer the following questions:

- (i) Which one of the following will have the highest bond order? Circle the correct choice
A) PS^{1-} B) PS C) PS^{1+}
- (ii) Which one of the following will have the longest bond length? Circle the correct choice
A) PS^{1-} B) PS C) PS^{1+}
- (iii) Which of the following will be diamagnetic? Circle the correct choice.
A) PS^{1-} B) PS C) PS^{1+}
- (iv) Draw Lewis dot structure of the molecule with the shortest bond length (based on the MO diagram) from among the three (PS^{1-} , PS, PS^{1+}), and indicate the bond order calculated from the Lewis dot structure. **(3 points)**

Lewis Dot Structure

Bond Order



A66CB97F-135F-468D-A160-8242FBB63FE1

midterm-exam-93388

#308 Page 10 of 10

SCRAP PAGE

**CHEMISTRY 110 –Midterm Exam****Thursday November 7, 2024, 6:30 – 8:30 pm**

Examiners: Pallavi Sirjoosingh and Ashok Kakkar

First name (please write as legibly as possible within the boxes)

--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--

Last name

--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--

Student ID number

--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--

This booklet has 8 questions on 10 pages (The last page can be used as a scrap page).

The staple must NOT be removed from the question booklet. Your intact question booklet must be returned at the end of the exam.

A separate booklet contains a periodic table, and datasheet.

Write your answers directly under the questions in the space provided. Do not write on the QR codes or near the top edges (as it may be cut off).

Please write **legibly and dark enough** for the scan to be readable. For students who use English as a second language, dictionaries are permitted. Calculators are allowed.

DO NOT ASK QUESTIONS OF THE INVIGILATORS about interpretations of the exam questions. If you have a concern about a question, please summarize it at the top of this page. All concerns will be evaluated after the exam. GOOD LUCK!

**Question 1 8 points** (*Assume all transitions are possible*)

a. 6 points Calculate the **energy (in kJ/mol)** required to ionize (completely remove) the electron from the **$n=3$** level of **He^+ ion** (He: Helium). Show your work including equations. (3 sig. figs.)

b. 2 points

(I) Select the largest (longest) wavelength from the three choices below:

- i. Wavelength of photon to *completely remove* an electron from *1s orbital of Hydrogen atom*
- ii. Wavelength of photon to *completely remove* an electron from *1s orbital of Boron atom (B)*
- iii. Wavelength of photon to *completely remove* an electron from *1s orbital of Boron cation (B^+)*

i or ii or iii

(II) Select the smallest (shortest) wavelength from the three choices below (*for H atom*):

- i. Wavelength of photon *emitted* for an electron transitioning from *2p to 1s orbital*
- ii. Wavelength of photon *absorbed* for an electron transitioning from *2s to 4p orbital*
- iii. Wavelength of photon *absorbed* for an electron transitioning from *1s to 3p orbital*

i or ii or iii



Question 2 8 points



A Chem110 student plots the graph shown on the left, from data obtained for the kinetic energy (eV) of the ejected electron versus frequency (s^{-1}) of the incident light, for an unknown metal. ($1\text{eV} = 1.6022 \times 10^{-19} \text{ J}$)

Based on the plot, answer the questions **a to c** below:

a. 1 point On the plot above, label the **threshold frequency**.

b. 2 points What is the value of the work potential (in J) of the unknown metal?

c. 3 points What is the *longest* wavelength (in nm) required to eject an electron from this metal's surface? Show your work including equation(s) used. (3 sig figs)

d. 2 points In the space provided below, plot the following

(Assume frequency is *above* threshold frequency):

- i. *Kinetic energy* of an ejected electron versus the *intensity* of the incident light (left)
- ii. *Number of electrons* ejected versus the *frequency* of the incident light (right)



**Question 3** **6 points****a. 2 points**

In the ground state of Ge (Germanium; $Z = 32$) atom, **name** the following orbitals:

i. Orbital containing the electron that requires the maximum energy to be removed.

ii. Orbital containing the electron that experiences the minimum effective nuclear charge.

b. 2 points

An unknown anion (A^-) has the following *excited* electronic configuration: $[Ar] 4s^2 3d^3 4p^1$

(Ar: Argon; $Z = 18$)

Give the **condensed ground state electronic configuration** of the atom, **A** and the cation, **A^{2+}** .
Show your work. (Assume all electronic transitions are possible)

Electronic Configuration of A:

Electronic Configuration of A^{2+} :

c. 2 points

In a single atom, determine how many electrons can have the following set of quantum numbers?

(i) $n = 5; l = 3$

In a single atom, determine how many electrons can have the following set of quantum numbers?

(i) $n = 3; l = 2; m_s = +1/2$

**Question 4** **8 points**

Rank the following from 1 to 4 in **increasing** order. Provide the ranking in the boxes provided.

Note: Atomic number (Z) are provided for relevant atoms

a) Increasing (lowest to highest) order of *ionization energy*: (1= lowest; 4 = highest)

Mg

Mg: Z = 12

Rb

Rb: Z = 37

Cs

Cs: Z = 55

K

K : Z = 19

b) Increasing (lowest to highest) order of *bond angle*: (1= lowest; 4 = highest)

H₂OCCl₄XeF₄

HCN

c) Increasing (smallest to largest) order of *ionic size* (1= smallest; 4 = largest)

S²⁻

S: Z = 16

O²⁻

O: Z = 8

Be²⁺

Be: Z = 4

Na⁺

Na: Z = 11

d) Increasing (lowest to highest) order of *bond energy* (1= lowest; 4 = highest)

P-F

P-Cl

P-I

P-Br

**Question 5** **12 points**

a. 9 points For the molecules HClO_4 and TeF_6 , draw the most stable *Lewis structure* showing appropriate shape and *dashed/wedged bonds*, based on VSEPR. Determine the **molecular geometry** around the central atom. For HClO_4 , label the *most polar bond*.



b. 3 points (Note: Lone pairs are not shown in structures below)

(i) For the three *resonance* structures (1, 2 and 3) shown below, answer the following question:



(I) Circle the *most contributing* resonance structure.

Structure 1

Structure 2

Structure 3

(ii) For the three structures given (4, 5, and 6) below, answer the following questions:



(I) Circle all the structures that represent *resonance structures* of each other:

Structure 4

Structure 5

Structure 6

(II) Among the resonance structures, circle the *most contributing* resonance structure.

Structure 4

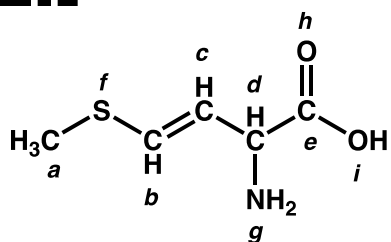
Structure 5

Structure 6

**Question 6** **6 points**

Using the data given below, calculate the **lattice energy** for SrO(s).

	ΔH° (kJ/mol)
$\text{Sr}(s) \longrightarrow \text{Sr}(g)$	+180
$\text{Sr}(g) \longrightarrow \text{Sr}^{2+}(g) + 2e^-$	+1468
$\text{O}_2(g) \longrightarrow 2\text{O}(g)$	+500
$\text{O}(g) + 2e^- \longrightarrow \text{O}^{2-}(g)$	+600
$\text{Sr}(s) + \frac{1}{2}\text{O}_2(g) \longrightarrow \text{SrO}(s)$	-550

**Question 7 10 points**

Consider the molecule given on the left, and answer questions (i) to (vi) using chemical bonding theories that we have learnt in class (valence bond theory, hybridization).

Note that the lone pairs are not shown in the structure:

i. Consider the carbon atoms marked in as ***a***, ***b***, ***c***, ***d***, and ***e*** in the molecule, and determine **hybridization** at each of the carbon centers. (2.5 points)

Atom	Hybridization
Carbon (<i>a</i>)	
Carbon (<i>b</i>)	
Carbon (<i>c</i>)	
Carbon (<i>d</i>)	
Carbon (<i>e</i>)	

ii. Now, consider the sulfur atom (S) marked as ***f***, nitrogen (N) atom (***g***), and oxygen(O) atoms (***h*** and ***i***), and determine their **hybridization**. (2 points)

Atom	Hybridization
Sulfur (<i>f</i>)	
Nitrogen (<i>g</i>)	
Oxygen (<i>h</i>)	
Oxygen (<i>i</i>)	

iii. **Describe** all the bonds formed by the carbon atoms ***a*** and ***b***, using sigma and pi designation as appropriate. (2 points)

Atom	Bonds formed (use sigma and pi as appropriate)
Carbon (<i>a</i>)	
Carbon (<i>b</i>)	

iv. How many total sp^3 -s sigma bonds are in the molecule? (1 point)

v. Determine **electronic** and **molecular geometry** at carbon (C) labeled ***e*** and sulfur (S) labeled ***f*** centres. (2 points)

Atom	Electronic Geometry	Molecular Geometry
Carbon (<i>e</i>)		
Sulfur (<i>f</i>)		

vi. Circle the value from the choices given below, that describes the **C-S-C bond angle at sulfur** (S) marked ***f***: (0.5 points)

>90° to <109.5° 109.5° >109.5 to <120° 120° >120° to <180°


Question 8 12 points

Using the **valence shell orbitals only** draw a **complete** molecular orbital (MO) diagram (with no s-p mixing) for the molecule **PS**, in the space provided below. **(6 points)**

Assume that 3s and 3p atomic orbitals will form molecular orbitals like those formed by the combination of 2s and 2p atomic orbitals that we learnt in class, and there is no s-p mixing.

Show **energy arrow**, **atomic orbitals** involved, **atom labels**, the **molecular orbitals** that are formed, **filled electrons** in the MO diagram etc. Make sure to label all the atomic and molecular orbitals.

Using the MO diagram above, answer the following questions:

- (i) Which one of the following will have the highest bond order? Circle the correct choice
 A) PS^{1-} B) PS C) PS^{1+}
- (ii) Which one of the following will have the longest bond length? Circle the correct choice
 A) PS^{1-} B) PS C) PS^{1+}
- (iii) Which of the following will be diamagnetic? Circle the correct choice.
 A) PS^{1-} B) PS C) PS^{1+}
- (iv) Draw Lewis dot structure of the molecule with the shortest bond length (based on the MO diagram) from among the three (PS^{1-} , PS, PS^{1+}), and indicate the bond order calculated from the Lewis dot structure. **(3 points)**

Lewis Dot Structure

Bond Order



B7CB3AC1-E446-4B7E-8FC8-1E28F64DDC36

midterm-exam-93388

#309 Page 10 of 10

SCRAP PAGE

**CHEMISTRY 110 –Midterm Exam****Thursday November 7, 2024, 6:30 – 8:30 pm****Examiners: Pallavi Sirjoosingh and Ashok Kakkar**

First name (please write as legibly as possible within the boxes)

--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--

Last name

--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--

Student ID number

--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--

This booklet has 8 questions on 10 pages (The last page can be used as a scrap page).

The staple must NOT be removed from the question booklet. Your intact question booklet must be returned at the end of the exam.

A separate booklet contains a periodic table, and datasheet.

Write your answers directly under the questions in the space provided. Do not write on the QR codes or near the top edges (as it may be cut off).

Please write **legibly and dark enough** for the scan to be readable. For students who use English as a second language, dictionaries are permitted. Calculators are allowed.

DO NOT ASK QUESTIONS OF THE INVIGILATORS about interpretations of the exam questions. If you have a concern about a question, please summarize it at the top of this page. All concerns will be evaluated after the exam. GOOD LUCK!

**Question 1 8 points** (*Assume all transitions are possible*)

a. 6 points Calculate the **energy (in kJ/mol)** required to ionize (completely remove) the electron from the **n=3** level of **He⁺ ion** (He: Helium). Show your work including equations. (3 sig. figs.)

b. 2 points

(I) Select the largest (longest) wavelength from the three choices below:

- i. Wavelength of photon to *completely remove* an electron from *1s orbital of Hydrogen atom*
- ii. Wavelength of photon to *completely remove* an electron from *1s orbital of Boron atom (B)*
- iii. Wavelength of photon to *completely remove* an electron from *1s orbital of Boron cation (B⁺)*

i or ii or iii

(II) Select the smallest (shortest) wavelength from the three choices below (*for H atom*):

- i. Wavelength of photon *emitted* for an electron transitioning from *2p to 1s orbital*
- ii. Wavelength of photon *absorbed* for an electron transitioning from *2s to 4p orbital*
- iii. Wavelength of photon *absorbed* for an electron transitioning from *1s to 3p orbital*

i or ii or iii



Question 2 8 points



A Chem110 student plots the graph shown on the left, from data obtained for the kinetic energy (eV) of the ejected electron versus frequency (s^{-1}) of the incident light, for an unknown metal. ($1\text{eV} = 1.6022 \times 10^{-19} \text{ J}$)

Based on the plot, answer the questions **a to c** below:

a. 1 point On the plot above, label the **threshold frequency**.

b. 2 points What is the value of the work potential (in J) of the unknown metal?

c. 3 points What is the *longest* wavelength (in nm) required to eject an electron from this metal's surface? Show your work including equation(s) used. (3 sig figs)

d. 2 points In the space provided below, plot the following

(Assume frequency is *above* threshold frequency):

- Kinetic energy* of an ejected electron versus the *intensity* of the incident light (left)
- Number of electrons* ejected versus the *frequency* of the incident light (right)



**Question 3** **6 points****a. 2 points**

In the ground state of Ge (Germanium; $Z = 32$) atom, **name** the following orbitals:

i. Orbital containing the electron that requires the maximum energy to be removed.

ii. Orbital containing the electron that experiences the minimum effective nuclear charge.

b. 2 points

An unknown anion (A^-) has the following *excited* electronic configuration: $[Ar] 4s^2 3d^3 4p^1$

(Ar: Argon; $Z = 18$)

Give the **condensed ground state electronic configuration** of the atom, **A** and the cation, **A^{2+}** .
Show your work. (Assume all electronic transitions are possible)

Electronic Configuration of A:

Electronic Configuration of A^{2+} :

c. 2 points

In a single atom, determine how many electrons can have the following set of quantum numbers?

(i) $n = 5; l = 3$

In a single atom, determine how many electrons can have the following set of quantum numbers?

(i) $n = 3; l = 2; m_s = +1/2$

**Question 4** **8 points**

Rank the following from 1 to 4 in **increasing** order. Provide the ranking in the boxes provided.

Note: Atomic number (Z) are provided for relevant atoms

a) Increasing (lowest to highest) order of *ionization energy*: (1= lowest; 4 = highest)

Mg

Mg: Z = 12

Rb

Rb: Z = 37

Cs

Cs: Z = 55

K

K : Z = 19

b) Increasing (lowest to highest) order of *bond angle*: (1= lowest; 4 = highest)

H₂OCCl₄XeF₄

HCN

c) Increasing (smallest to largest) order of *ionic size* (1= smallest; 4 = largest)

S²⁻

S: Z = 16

O²⁻

O: Z = 8

Be²⁺

Be: Z = 4

Na⁺

Na: Z = 11

d) Increasing (lowest to highest) order of *bond energy* (1= lowest; 4 = highest)

P-F

P-Cl

P-I

P-Br

**Question 5** **12 points**

a. 9 points For the molecules HClO_4 and TeF_6 , draw the most stable *Lewis structure* showing appropriate shape and *dashed/wedged bonds*, based on VSEPR. Determine the **molecular geometry** around the central atom. For HClO_4 , label the *most polar bond*.



b. 3 points (Note: Lone pairs are not shown in structures below)

(i) For the three *resonance* structures (1, 2 and 3) shown below, answer the following question:



(I) Circle the *most contributing* resonance structure.

Structure 1

Structure 2

Structure 3

(ii) For the three structures given (4, 5, and 6) below, answer the following questions:



(I) Circle all the structures that represent *resonance structures* of each other:

Structure 4

Structure 5

Structure 6

(II) Among the resonance structures, circle the *most contributing* resonance structure.

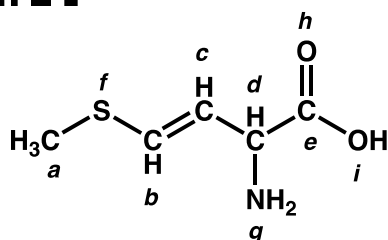
Structure 4

Structure 5

Structure 6

**Question 6** **6 points**Using the data given below, calculate the **lattice energy** for SrO(s).

	ΔH° (kJ/mol)
$\text{Sr}(s) \longrightarrow \text{Sr}(g)$	+180
$\text{Sr}(g) \longrightarrow \text{Sr}^{2+}(g) + 2e^-$	+1468
$\text{O}_2(g) \longrightarrow 2\text{O}(g)$	+500
$\text{O}(g) + 2e^- \longrightarrow \text{O}^{2-}(g)$	+600
$\text{Sr}(s) + \frac{1}{2}\text{O}_2(g) \longrightarrow \text{SrO}(s)$	-550

**Question 7 10 points**

Consider the molecule given on the left, and answer questions (i) to (vi) using chemical bonding theories that we have learnt in class (valence bond theory, hybridization).

Note that the lone pairs are not shown in the structure:

i. Consider the carbon atoms marked in as ***a***, ***b***, ***c***, ***d***, and ***e*** in the molecule, and determine **hybridization** at each of the carbon centers. (2.5 points)

Atom	Hybridization
Carbon (<i>a</i>)	
Carbon (<i>b</i>)	
Carbon (<i>c</i>)	
Carbon (<i>d</i>)	
Carbon (<i>e</i>)	

ii. Now, consider the sulfur atom (S) marked as ***f***, nitrogen (N) atom (***g***), and oxygen(O) atoms (***h*** and ***i***), and determine their **hybridization**. (2 points)

Atom	Hybridization
Sulfur (<i>f</i>)	
Nitrogen (<i>g</i>)	
Oxygen (<i>h</i>)	
Oxygen (<i>i</i>)	

iii. **Describe** all the bonds formed by the carbon atoms ***a*** and ***b***, using sigma and pi designation as appropriate. (2 points)

Atom	Bonds formed (use sigma and pi as appropriate)
Carbon (<i>a</i>)	
Carbon (<i>b</i>)	

iv. How many total sp^3 -s sigma bonds are in the molecule? (1 point)

v. Determine **electronic** and **molecular geometry** at carbon (C) labeled ***e*** and sulfur (S) labeled ***f*** centres. (2 points)

Atom	Electronic Geometry	Molecular Geometry
Carbon (<i>e</i>)		
Sulfur (<i>f</i>)		

vi. Circle the value from the choices given below, that describes the **C-S-C bond angle at sulfur** (S) marked ***f***: (0.5 points)

>90° to <109.5° 109.5° >109.5 to <120° 120° >120° to <180°

**Question 8 12 points**

Using the **valence shell orbitals only** draw a **complete** molecular orbital (MO) diagram (with no s-p mixing) for the molecule **PS**, in the space provided below. **(6 points)**

Assume that 3s and 3p atomic orbitals will form molecular orbitals like those formed by the combination of 2s and 2p atomic orbitals that we learnt in class, and there is no s-p mixing.

Show **energy arrow**, **atomic orbitals** involved, **atom labels**, the **molecular orbitals** that are formed, **filled electrons** in the MO diagram etc. Make sure to label all the atomic and molecular orbitals.

Using the MO diagram above, answer the following questions:

- (i) Which one of the following will have the highest bond order? Circle the correct choice
A) PS^{1-} B) PS C) PS^{1+}
- (ii) Which one of the following will have the longest bond length? Circle the correct choice
A) PS^{1-} B) PS C) PS^{1+}
- (iii) Which of the following will be diamagnetic? Circle the correct choice.
A) PS^{1-} B) PS C) PS^{1+}
- (iv) Draw Lewis dot structure of the molecule with the shortest bond length (based on the MO diagram) from among the three (PS^{1-} , PS, PS^{1+}), and indicate the bond order calculated from the Lewis dot structure. **(3 points)**

Lewis Dot Structure

Bond Order



934660CB-E380-41F1-9EEF-86576A288D8B

midterm-exam-93388

#310 Page 10 of 10

SCRAP PAGE

**CHEMISTRY 110 –Midterm Exam****Thursday November 7, 2024, 6:30 – 8:30 pm****Examiners: Pallavi Sirjoosingh and Ashok Kakkar**

First name (please write as legibly as possible within the boxes)

--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--

Last name

--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--

Student ID number

--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--

This booklet has 8 questions on 10 pages (The last page can be used as a scrap page).

The staple must NOT be removed from the question booklet. Your intact question booklet must be returned at the end of the exam.

A separate booklet contains a periodic table, and datasheet.

Write your answers directly under the questions in the space provided. Do not write on the QR codes or near the top edges (as it may be cut off).

Please write **legibly and dark enough** for the scan to be readable. For students who use English as a second language, dictionaries are permitted. Calculators are allowed.

DO NOT ASK QUESTIONS OF THE INVIGILATORS about interpretations of the exam questions. If you have a concern about a question, please summarize it at the top of this page. All concerns will be evaluated after the exam. GOOD LUCK!

**Question 1 8 points** (*Assume all transitions are possible*)

a. 6 points Calculate the **energy (in kJ/mol)** required to ionize (completely remove) the electron from the **$n=3$** level of **He^+ ion** (He: Helium). Show your work including equations. (3 sig. figs.)

b. 2 points

(I) Select the largest (longest) wavelength from the three choices below:

- i. Wavelength of photon to *completely remove* an electron from *1s orbital of Hydrogen atom*
- ii. Wavelength of photon to *completely remove* an electron from *1s orbital of Boron atom (B)*
- iii. Wavelength of photon to *completely remove* an electron from *1s orbital of Boron cation (B^+)*

i or ii or iii

(II) Select the smallest (shortest) wavelength from the three choices below (*for H atom*):

- i. Wavelength of photon *emitted* for an electron transitioning from *2p to 1s orbital*
- ii. Wavelength of photon *absorbed* for an electron transitioning from *2s to 4p orbital*
- iii. Wavelength of photon *absorbed* for an electron transitioning from *1s to 3p orbital*

i or ii or iii



Question 2 8 points



A Chem110 student plots the graph shown on the left, from data obtained for the kinetic energy (eV) of the ejected electron versus frequency (s^{-1}) of the incident light, for an unknown metal. ($1\text{eV} = 1.6022 \times 10^{-19} \text{ J}$)

Based on the plot, answer the questions **a to c** below:

a. 1 point On the plot above, label the **threshold frequency**.

b. 2 points What is the value of the work potential (in J) of the unknown metal?

c. 3 points What is the *longest* wavelength (in nm) required to eject an electron from this metal's surface? Show your work including equation(s) used. (3 sig figs)

d. 2 points In the space provided below, plot the following

(Assume frequency is *above* threshold frequency):

- i. *Kinetic energy* of an ejected electron versus the *intensity* of the incident light (left)
- ii. *Number of electrons* ejected versus the *frequency* of the incident light (right)



**Question 3** **6 points****a. 2 points**

In the ground state of Ge (Germanium; $Z = 32$) atom, **name** the following orbitals:

i. Orbital containing the electron that requires the maximum energy to be removed.

ii. Orbital containing the electron that experiences the minimum effective nuclear charge.

b. 2 points

An unknown anion (A^-) has the following *excited* electronic configuration: $[Ar] 4s^2 3d^3 4p^1$

(Ar: Argon; $Z = 18$)

Give the **condensed ground state electronic configuration** of the atom, **A** and the cation, **A^{2+}** .
Show your work. (Assume all electronic transitions are possible)

Electronic Configuration of A:

Electronic Configuration of A^{2+} :

c. 2 points

In a single atom, determine how many electrons can have the following set of quantum numbers?

(i) $n = 5; l = 3$

In a single atom, determine how many electrons can have the following set of quantum numbers?

(i) $n = 3; l = 2; m_s = +1/2$

**Question 4** **8 points**

Rank the following from 1 to 4 in **increasing** order. Provide the ranking in the boxes provided.

Note: Atomic number (Z) are provided for relevant atoms

a) Increasing (lowest to highest) order of *ionization energy*: (1= lowest; 4 = highest)

Mg

Mg: Z = 12

Rb

Rb: Z = 37

Cs

Cs: Z = 55

K

K : Z = 19

b) Increasing (lowest to highest) order of *bond angle*: (1= lowest; 4 = highest)

H₂OCCl₄XeF₄

HCN

c) Increasing (smallest to largest) order of *ionic size* (1= smallest; 4 = largest)

S²⁻

S: Z = 16

O²⁻

O: Z = 8

Be²⁺

Be: Z = 4

Na⁺

Na: Z = 11

d) Increasing (lowest to highest) order of *bond energy* (1= lowest; 4 = highest)

P-F

P-Cl

P-I

P-Br

**Question 5** **12 points**

a. 9 points For the molecules HClO_4 and TeF_6 , draw the most stable *Lewis structure* showing appropriate shape and *dashed/wedged bonds*, based on VSEPR. Determine the **molecular geometry** around the central atom. For HClO_4 , label the *most polar bond*.

HClO_4 (there are no O-O bonds)

Label the most polar bond

Molecular Geometry around Cl

TeF_6

Molecular Geometry around Te

b. 3 points (Note: Lone pairs are not shown in structures below)

(i) For the three *resonance* structures (1, 2 and 3) shown below, answer the following question:



(I) Circle the *most contributing* resonance structure.

Structure 1

Structure 2

Structure 3

(ii) For the three structures given (4, 5, and 6) below, answer the following questions:



(I) Circle all the structures that represent *resonance structures* of each other:

Structure 4

Structure 5

Structure 6

(II) Among the resonance structures, circle the *most contributing* resonance structure.

Structure 4

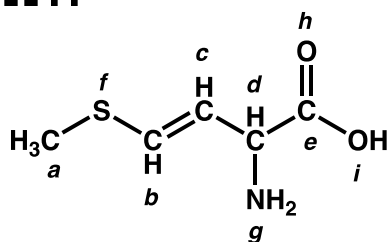
Structure 5

Structure 6

**Question 6** **6 points**

Using the data given below, calculate the **lattice energy** for SrO(s).

	ΔH° (kJ/mol)
$\text{Sr}(s) \longrightarrow \text{Sr}(g)$	+180
$\text{Sr}(g) \longrightarrow \text{Sr}^{2+}(g) + 2e^-$	+1468
$\text{O}_2(g) \longrightarrow 2\text{O}(g)$	+500
$\text{O}(g) + 2e^- \longrightarrow \text{O}^{2-}(g)$	+600
$\text{Sr}(s) + \frac{1}{2}\text{O}_2(g) \longrightarrow \text{SrO}(s)$	-550

**Question 7 10 points**

Consider the molecule given on the left, and answer questions (i) to (vi) using chemical bonding theories that we have learnt in class (valence bond theory, hybridization).

Note that the lone pairs are not shown in the structure:

i. Consider the carbon atoms marked in as ***a***, ***b***, ***c***, ***d***, and ***e*** in the molecule, and determine **hybridization** at each of the carbon centers. (2.5 points)

Atom	Hybridization
Carbon (<i>a</i>)	
Carbon (<i>b</i>)	
Carbon (<i>c</i>)	
Carbon (<i>d</i>)	
Carbon (<i>e</i>)	

ii. Now, consider the sulfur atom (S) marked as ***f***, nitrogen (N) atom (***g***), and oxygen(O) atoms (***h*** and ***i***), and determine their **hybridization**. (2 points)

Atom	Hybridization
Sulfur (<i>f</i>)	
Nitrogen (<i>g</i>)	
Oxygen (<i>h</i>)	
Oxygen (<i>i</i>)	

iii. **Describe** all the bonds formed by the carbon atoms ***a*** and ***b***, using sigma and pi designation as appropriate. (2 points)

Atom	Bonds formed (use sigma and pi as appropriate)
Carbon (<i>a</i>)	
Carbon (<i>b</i>)	

iv. How many total sp^3 -s sigma bonds are in the molecule? (1 point)

v. Determine **electronic** and **molecular geometry** at carbon (C) labeled ***e*** and sulfur (S) labeled ***f*** centres. (2 points)

Atom	Electronic Geometry	Molecular Geometry
Carbon (<i>e</i>)		
Sulfur (<i>f</i>)		

vi. Circle the value from the choices given below, that describes the **C-S-C bond angle at sulfur** (S) marked ***f***: (0.5 points)

>90° to <109.5° 109.5° >109.5 to <120° 120° >120° to <180°

**Question 8 12 points**

Using the **valence shell orbitals only** draw a **complete** molecular orbital (MO) diagram (with no s-p mixing) for the molecule **PS**, in the space provided below. **(6 points)**

Assume that 3s and 3p atomic orbitals will form molecular orbitals like those formed by the combination of 2s and 2p atomic orbitals that we learnt in class, and there is no s-p mixing.

Show **energy arrow**, **atomic orbitals** involved, **atom labels**, the **molecular orbitals** that are formed, **filled electrons** in the MO diagram etc. Make sure to label all the atomic and molecular orbitals.

Using the MO diagram above, answer the following questions:

- (i) Which one of the following will have the highest bond order? Circle the correct choice
A) PS^{1-} B) PS C) PS^{1+}
- (ii) Which one of the following will have the longest bond length? Circle the correct choice
A) PS^{1-} B) PS C) PS^{1+}
- (iii) Which of the following will be diamagnetic? Circle the correct choice.
A) PS^{1-} B) PS C) PS^{1+}
- (iv) Draw Lewis dot structure of the molecule with the shortest bond length (based on the MO diagram) from among the three (PS^{1-} , PS, PS^{1+}), and indicate the bond order calculated from the Lewis dot structure. **(3 points)**

Lewis Dot Structure

Bond Order



72A11D20-5AC6-4024-BE24-CDE4EB93F7C3

midterm-exam-93388

#311 Page 10 of 10

SCRAP PAGE

**CHEMISTRY 110 –Midterm Exam****Thursday November 7, 2024, 6:30 – 8:30 pm**

Examiners: Pallavi Sirjoosingh and Ashok Kakkar

First name (please write as legibly as possible within the boxes)

--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--

Last name

--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--

Student ID number

--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--

This booklet has 8 questions on 10 pages (The last page can be used as a scrap page).

The staple must NOT be removed from the question booklet. Your intact question booklet must be returned at the end of the exam.

A separate booklet contains a periodic table, and datasheet.

Write your answers directly under the questions in the space provided. Do not write on the QR codes or near the top edges (as it may be cut off).

Please write **legibly and dark enough** for the scan to be readable. For students who use English as a second language, dictionaries are permitted. Calculators are allowed.

DO NOT ASK QUESTIONS OF THE INVIGILATORS about interpretations of the exam questions. If you have a concern about a question, please summarize it at the top of this page. All concerns will be evaluated after the exam. GOOD LUCK!

**Question 1 8 points** (*Assume all transitions are possible*)

a. 6 points Calculate the **energy (in kJ/mol)** required to ionize (completely remove) the electron from the **n=3** level of **He⁺ ion** (He: Helium). Show your work including equations. (3 sig. figs.)

b. 2 points

(I) Select the largest (longest) wavelength from the three choices below:

- i. Wavelength of photon to *completely remove* an electron from *1s orbital of Hydrogen atom*
- ii. Wavelength of photon to *completely remove* an electron from *1s orbital of Boron atom (B)*
- iii. Wavelength of photon to *completely remove* an electron from *1s orbital of Boron cation (B⁺)*

i or ii or iii

(II) Select the smallest (shortest) wavelength from the three choices below (*for H atom*):

- i. Wavelength of photon *emitted* for an electron transitioning from *2p to 1s orbital*
- ii. Wavelength of photon *absorbed* for an electron transitioning from *2s to 4p orbital*
- iii. Wavelength of photon *absorbed* for an electron transitioning from *1s to 3p orbital*

i or ii or iii



Question 2 8 points



A Chem110 student plots the graph shown on the left, from data obtained for the kinetic energy (eV) of the ejected electron versus frequency (s^{-1}) of the incident light, for an unknown metal. ($1\text{eV} = 1.6022 \times 10^{-19} \text{ J}$)

Based on the plot, answer the questions **a to c** below:

a. 1 point On the plot above, label the **threshold frequency**.

b. 2 points What is the value of the work potential (in J) of the unknown metal?

c. 3 points What is the *longest* wavelength (in nm) required to eject an electron from this metal's surface? Show your work including equation(s) used. (3 sig figs)

d. 2 points In the space provided below, plot the following

(Assume frequency is *above* threshold frequency):

- i. *Kinetic energy* of an ejected electron versus the *intensity* of the incident light (left)
- ii. *Number of electrons* ejected versus the *frequency* of the incident light (right)



**Question 3** **6 points****a. 2 points**

In the ground state of Ge (Germanium; $Z = 32$) atom, **name** the following orbitals:

i. Orbital containing the electron that requires the maximum energy to be removed.

ii. Orbital containing the electron that experiences the minimum effective nuclear charge.

b. 2 points

An unknown anion (A^-) has the following *excited* electronic configuration: $[Ar] 4s^2 3d^3 4p^1$

(Ar: Argon; $Z = 18$)

Give the **condensed ground state electronic configuration** of the atom, **A** and the cation, **A^{2+}** .
Show your work. (Assume all electronic transitions are possible)

Electronic Configuration of A:

Electronic Configuration of A^{2+} :

c. 2 points

In a single atom, determine how many electrons can have the following set of quantum numbers?

(i) $n = 5; l = 3$

In a single atom, determine how many electrons can have the following set of quantum numbers?

(i) $n = 3; l = 2; m_s = +1/2$

**Question 4** **8 points**

Rank the following from 1 to 4 in **increasing** order. Provide the ranking in the boxes provided.

Note: Atomic number (Z) are provided for relevant atoms

a) Increasing (lowest to highest) order of *ionization energy*: (1= lowest; 4 = highest)

Mg

Mg: Z = 12

Rb

Rb: Z = 37

Cs

Cs: Z = 55

K

K : Z = 19

b) Increasing (lowest to highest) order of *bond angle*: (1= lowest; 4 = highest)

H₂OCCl₄XeF₄

HCN

c) Increasing (smallest to largest) order of *ionic size* (1= smallest; 4 = largest)

S²⁻

S: Z = 16

O²⁻

O: Z = 8

Be²⁺

Be: Z = 4

Na⁺

Na: Z = 11

d) Increasing (lowest to highest) order of *bond energy* (1= lowest; 4 = highest)

P-F

P-Cl

P-I

P-Br

**Question 5** **12 points**

a. 9 points For the molecules HClO_4 and TeF_6 , draw the most stable *Lewis structure* showing appropriate shape and *dashed/wedged bonds*, based on VSEPR. Determine the **molecular geometry** around the central atom. For HClO_4 , label the *most polar bond*.



b. 3 points (Note: Lone pairs are not shown in structures below)

(i) For the three *resonance* structures (1, 2 and 3) shown below, answer the following question:



(I) Circle the *most contributing* resonance structure.

Structure 1

Structure 2

Structure 3

(ii) For the three structures given (4, 5, and 6) below, answer the following questions:



(I) Circle all the structures that represent *resonance structures* of each other:

Structure 4

Structure 5

Structure 6

(II) Among the resonance structures, circle the *most contributing* resonance structure.

Structure 4

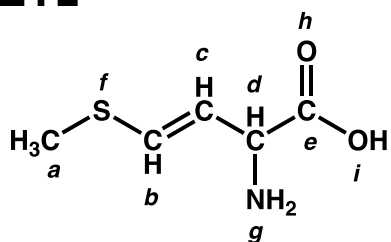
Structure 5

Structure 6

**Question 6** **6 points**

Using the data given below, calculate the **lattice energy** for SrO(s).

	ΔH° (kJ/mol)
$\text{Sr}(s) \longrightarrow \text{Sr}(g)$	+180
$\text{Sr}(g) \longrightarrow \text{Sr}^{2+}(g) + 2e^-$	+1468
$\text{O}_2(g) \longrightarrow 2\text{O}(g)$	+500
$\text{O}(g) + 2e^- \longrightarrow \text{O}^{2-}(g)$	+600
$\text{Sr}(s) + \frac{1}{2}\text{O}_2(g) \longrightarrow \text{SrO}(s)$	-550

**Question 7 10 points**

Consider the molecule given on the left, and answer questions (i) to (vi) using chemical bonding theories that we have learnt in class (valence bond theory, hybridization).

Note that the lone pairs are not shown in the structure:

i. Consider the carbon atoms marked in as ***a***, ***b***, ***c***, ***d***, and ***e*** in the molecule, and determine **hybridization** at each of the carbon centers. (2.5 points)

Atom	Hybridization
Carbon (<i>a</i>)	
Carbon (<i>b</i>)	
Carbon (<i>c</i>)	
Carbon (<i>d</i>)	
Carbon (<i>e</i>)	

ii. Now, consider the sulfur atom (S) marked as ***f***, nitrogen (N) atom (***g***), and oxygen(O) atoms (***h*** and ***i***), and determine their **hybridization**. (2 points)

Atom	Hybridization
Sulfur (<i>f</i>)	
Nitrogen (<i>g</i>)	
Oxygen (<i>h</i>)	
Oxygen (<i>i</i>)	

iii. **Describe** all the bonds formed by the carbon atoms ***a*** and ***b***, using sigma and pi designation as appropriate. (2 points)

Atom	Bonds formed (use sigma and pi as appropriate)
Carbon (<i>a</i>)	
Carbon (<i>b</i>)	

iv. How many total sp^3 -s sigma bonds are in the molecule? (1 point)

v. Determine **electronic** and **molecular geometry** at carbon (C) labeled ***e*** and sulfur (S) labeled ***f*** centres. (2 points)

Atom	Electronic Geometry	Molecular Geometry
Carbon (<i>e</i>)		
Sulfur (<i>f</i>)		

vi. Circle the value from the choices given below, that describes the **C-S-C bond angle at sulfur** (S) marked ***f***: (0.5 points)

>90° to <109.5°

109.5°

>109.5 to <120°

120°

>120° to <180°


Question 8 12 points

Using the **valence shell orbitals only** draw a **complete** molecular orbital (MO) diagram (with no s-p mixing) for the molecule **PS**, in the space provided below. **(6 points)**

Assume that 3s and 3p atomic orbitals will form molecular orbitals like those formed by the combination of 2s and 2p atomic orbitals that we learnt in class, and there is no s-p mixing.

Show **energy arrow**, **atomic orbitals** involved, **atom labels**, the **molecular orbitals** that are formed, **filled electrons** in the MO diagram etc. Make sure to label all the atomic and molecular orbitals.

Using the MO diagram above, answer the following questions:

- (i) Which one of the following will have the highest bond order? Circle the correct choice
 A) PS^{1-} B) PS C) PS^{1+}
- (ii) Which one of the following will have the longest bond length? Circle the correct choice
 A) PS^{1-} B) PS C) PS^{1+}
- (iii) Which of the following will be diamagnetic? Circle the correct choice.
 A) PS^{1-} B) PS C) PS^{1+}
- (iv) Draw Lewis dot structure of the molecule with the shortest bond length (based on the MO diagram) from among the three (PS^{1-} , PS, PS^{1+}), and indicate the bond order calculated from the Lewis dot structure. **(3 points)**

Lewis Dot Structure

Bond Order



6541916E-F82F-4E61-BC65-513277A43657

midterm-exam-93388

#312 Page 10 of 10

SCRAP PAGE

**CHEMISTRY 110 –Midterm Exam****Thursday November 7, 2024, 6:30 – 8:30 pm**

Examiners: Pallavi Sirjoosingh and Ashok Kakkar

First name (please write as legibly as possible within the boxes)

--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--

Last name

--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--

Student ID number

--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--

This booklet has 8 questions on 10 pages (The last page can be used as a scrap page).

The staple must NOT be removed from the question booklet. Your intact question booklet must be returned at the end of the exam.

A separate booklet contains a periodic table, and datasheet.

Write your answers directly under the questions in the space provided. Do not write on the QR codes or near the top edges (as it may be cut off).

Please write **legibly and dark enough** for the scan to be readable. For students who use English as a second language, dictionaries are permitted. Calculators are allowed.

DO NOT ASK QUESTIONS OF THE INVIGILATORS about interpretations of the exam questions. If you have a concern about a question, please summarize it at the top of this page. All concerns will be evaluated after the exam. GOOD LUCK!

**Question 1 8 points** (*Assume all transitions are possible*)

a. 6 points Calculate the **energy (in kJ/mol)** required to ionize (completely remove) the electron from the **n=3** level of **He⁺ ion** (He: Helium). Show your work including equations. (3 sig. figs.)

b. 2 points

(I) Select the largest (longest) wavelength from the three choices below:

- i. Wavelength of photon to *completely remove* an electron from *1s orbital of Hydrogen atom*
- ii. Wavelength of photon to *completely remove* an electron from *1s orbital of Boron atom (B)*
- iii. Wavelength of photon to *completely remove* an electron from *1s orbital of Boron cation (B⁺)*

i or ii or iii

(II) Select the smallest (shortest) wavelength from the three choices below (*for H atom*):

- i. Wavelength of photon *emitted* for an electron transitioning from *2p to 1s orbital*
- ii. Wavelength of photon *absorbed* for an electron transitioning from *2s to 4p orbital*
- iii. Wavelength of photon *absorbed* for an electron transitioning from *1s to 3p orbital*

i or ii or iii



Question 2 8 points



A Chem110 student plots the graph shown on the left, from data obtained for the kinetic energy (eV) of the ejected electron versus frequency (s^{-1}) of the incident light, for an unknown metal. ($1\text{eV} = 1.6022 \times 10^{-19} \text{ J}$)

Based on the plot, answer the questions **a to c** below:

a. 1 point On the plot above, label the **threshold frequency**.

b. 2 points What is the value of the work potential (in J) of the unknown metal?

c. 3 points What is the *longest* wavelength (in nm) required to eject an electron from this metal's surface? Show your work including equation(s) used. (3 sig figs)

d. 2 points In the space provided below, plot the following

(Assume frequency is *above* threshold frequency):

- Kinetic energy* of an ejected electron versus the *intensity* of the incident light (left)
- Number of electrons* ejected versus the *frequency* of the incident light (right)



**Question 3** **6 points****a. 2 points**

In the ground state of Ge (Germanium; $Z = 32$) atom, **name** the following orbitals:

i. Orbital containing the electron that requires the maximum energy to be removed.

ii. Orbital containing the electron that experiences the minimum effective nuclear charge.

b. 2 points

An unknown anion (A^-) has the following *excited* electronic configuration: $[Ar] 4s^2 3d^3 4p^1$

(Ar: Argon; $Z = 18$)

Give the **condensed ground state electronic configuration** of the atom, **A** and the cation, **A^{2+}** .
Show your work. (Assume all electronic transitions are possible)

Electronic Configuration of A:

Electronic Configuration of A^{2+} :

c. 2 points

In a single atom, determine how many electrons can have the following set of quantum numbers?

(i) $n = 5; l = 3$

In a single atom, determine how many electrons can have the following set of quantum numbers?

(i) $n = 3; l = 2; m_s = +1/2$

**Question 4** **8 points**

Rank the following from 1 to 4 in **increasing** order. Provide the ranking in the boxes provided.

Note: Atomic number (Z) are provided for relevant atoms

a) Increasing (lowest to highest) order of *ionization energy*: (1= lowest; 4 = highest)

Mg

Mg: Z = 12

Rb

Rb: Z = 37

Cs

Cs: Z = 55

K

K : Z = 19

b) Increasing (lowest to highest) order of *bond angle*: (1= lowest; 4 = highest)

H₂OCCl₄XeF₄

HCN

c) Increasing (smallest to largest) order of *ionic size* (1= smallest; 4 = largest)

S²⁻

S: Z = 16

O²⁻

O: Z = 8

Be²⁺

Be: Z = 4

Na⁺

Na: Z = 11

d) Increasing (lowest to highest) order of *bond energy* (1= lowest; 4 = highest)

P-F

P-Cl

P-I

P-Br

**Question 5** **12 points**

a. 9 points For the molecules HClO_4 and TeF_6 , draw the most stable *Lewis structure* showing appropriate shape and *dashed/wedged bonds*, based on VSEPR. Determine the **molecular geometry** around the central atom. For HClO_4 , label the *most polar bond*.

HClO_4 (there are no O-O bonds)

Label the most polar bond

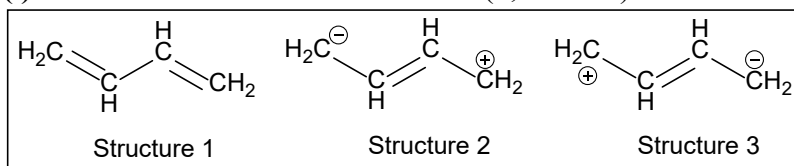
Molecular Geometry around Cl

TeF_6

Molecular Geometry around Te

b. 3 points (Note: Lone pairs are not shown in structures below)

(i) For the three *resonance* structures (1, 2 and 3) shown below, answer the following question:



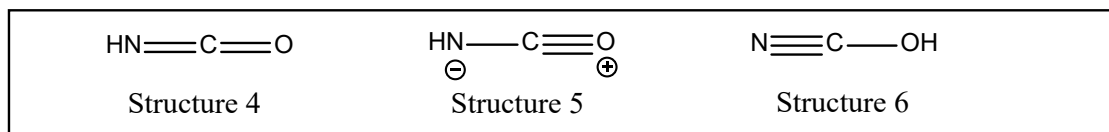
(I) Circle the *most contributing* resonance structure.

Structure 1

Structure 2

Structure 3

(ii) For the three structures given (4, 5, and 6) below, answer the following questions:



(I) Circle all the structures that represent *resonance structures* of each other:

Structure 4

Structure 5

Structure 6

(II) Among the resonance structures, circle the *most contributing* resonance structure.

Structure 4

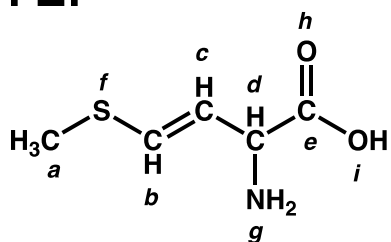
Structure 5

Structure 6

**Question 6** **6 points**

Using the data given below, calculate the **lattice energy** for SrO(s).

	ΔH° (kJ/mol)
$\text{Sr}(s) \longrightarrow \text{Sr}(g)$	+180
$\text{Sr}(g) \longrightarrow \text{Sr}^{2+}(g) + 2e^-$	+1468
$\text{O}_2(g) \longrightarrow 2\text{O}(g)$	+500
$\text{O}(g) + 2e^- \longrightarrow \text{O}^{2-}(g)$	+600
$\text{Sr}(s) + \frac{1}{2}\text{O}_2(g) \longrightarrow \text{SrO}(s)$	-550

**Question 7 10 points**

Consider the molecule given on the left, and answer questions (i) to (vi) using chemical bonding theories that we have learnt in class (valence bond theory, hybridization).

Note that the lone pairs are not shown in the structure:

i. Consider the carbon atoms marked in as ***a***, ***b***, ***c***, ***d***, and ***e*** in the molecule, and determine **hybridization** at each of the carbon centers. (2.5 points)

Atom	Hybridization
Carbon (<i>a</i>)	
Carbon (<i>b</i>)	
Carbon (<i>c</i>)	
Carbon (<i>d</i>)	
Carbon (<i>e</i>)	

ii. Now, consider the sulfur atom (S) marked as ***f***, nitrogen (N) atom (***g***), and oxygen(O) atoms (***h*** and ***i***), and determine their **hybridization**. (2 points)

Atom	Hybridization
Sulfur (<i>f</i>)	
Nitrogen (<i>g</i>)	
Oxygen (<i>h</i>)	
Oxygen (<i>i</i>)	

iii. **Describe** all the bonds formed by the carbon atoms ***a*** and ***b***, using sigma and pi designation as appropriate. (2 points)

Atom	Bonds formed (use sigma and pi as appropriate)
Carbon (<i>a</i>)	
Carbon (<i>b</i>)	

iv. How many total sp^3 -s sigma bonds are in the molecule? (1 point)

v. Determine **electronic** and **molecular geometry** at carbon (C) labeled ***e*** and sulfur (S) labeled ***f*** centres. (2 points)

Atom	Electronic Geometry	Molecular Geometry
Carbon (<i>e</i>)		
Sulfur (<i>f</i>)		

vi. Circle the value from the choices given below, that describes the **C-S-C bond angle at sulfur** (S) marked ***f***: (0.5 points)

>90° to <109.5° 109.5° >109.5 to <120° 120° >120° to <180°

**Question 8 12 points**

Using the **valence shell orbitals only** draw a **complete** molecular orbital (MO) diagram (with no s-p mixing) for the molecule **PS**, in the space provided below. **(6 points)**

Assume that 3s and 3p atomic orbitals will form molecular orbitals like those formed by the combination of 2s and 2p atomic orbitals that we learnt in class, and there is no s-p mixing.

Show **energy arrow**, **atomic orbitals** involved, **atom labels**, the **molecular orbitals** that are formed, **filled electrons** in the MO diagram etc. Make sure to label all the atomic and molecular orbitals.

Using the MO diagram above, answer the following questions:

- (i) Which one of the following will have the highest bond order? Circle the correct choice
A) PS^{1-} B) PS C) PS^{1+}
- (ii) Which one of the following will have the longest bond length? Circle the correct choice
A) PS^{1-} B) PS C) PS^{1+}
- (iii) Which of the following will be diamagnetic? Circle the correct choice.
A) PS^{1-} B) PS C) PS^{1+}
- (iv) Draw Lewis dot structure of the molecule with the shortest bond length (based on the MO diagram) from among the three (PS^{1-} , PS, PS^{1+}), and indicate the bond order calculated from the Lewis dot structure. **(3 points)**

Lewis Dot Structure

Bond Order



AB1FF8E2-3594-4FCA-99E9-D77799586342

midterm-exam-93388

#313 Page 10 of 10

SCRAP PAGE

**CHEMISTRY 110 –Midterm Exam****Thursday November 7, 2024, 6:30 – 8:30 pm**

Examiners: Pallavi Sirjoosingh and Ashok Kakkar

First name (please write as legibly as possible within the boxes)

--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--

Last name

--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--

Student ID number

--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--

This booklet has 8 questions on 10 pages (The last page can be used as a scrap page).

The staple must NOT be removed from the question booklet. Your intact question booklet must be returned at the end of the exam.

A separate booklet contains a periodic table, and datasheet.

Write your answers directly under the questions in the space provided. Do not write on the QR codes or near the top edges (as it may be cut off).

Please write **legibly and dark enough** for the scan to be readable. For students who use English as a second language, dictionaries are permitted. Calculators are allowed.

DO NOT ASK QUESTIONS OF THE INVIGILATORS about interpretations of the exam questions. If you have a concern about a question, please summarize it at the top of this page. All concerns will be evaluated after the exam. GOOD LUCK!

**Question 1 8 points** (*Assume all transitions are possible*)

a. 6 points Calculate the **energy (in kJ/mol)** required to ionize (completely remove) the electron from the **n=3** level of **He⁺ ion** (He: Helium). Show your work including equations. (3 sig. figs.)

b. 2 points

(I) Select the largest (longest) wavelength from the three choices below:

- i. Wavelength of photon to *completely remove* an electron from *1s orbital of Hydrogen atom*
- ii. Wavelength of photon to *completely remove* an electron from *1s orbital of Boron atom (B)*
- iii. Wavelength of photon to *completely remove* an electron from *1s orbital of Boron cation (B⁺)*

i or ii or iii

(II) Select the smallest (shortest) wavelength from the three choices below (*for H atom*):

- i. Wavelength of photon *emitted* for an electron transitioning from *2p to 1s orbital*
- ii. Wavelength of photon *absorbed* for an electron transitioning from *2s to 4p orbital*
- iii. Wavelength of photon *absorbed* for an electron transitioning from *1s to 3p orbital*

i or ii or iii



Question 2 8 points



A Chem110 student plots the graph shown on the left, from data obtained for the kinetic energy (eV) of the ejected electron versus frequency (s^{-1}) of the incident light, for an unknown metal. ($1\text{eV} = 1.6022 \times 10^{-19} \text{ J}$)

Based on the plot, answer the questions **a to c** below:

a. 1 point On the plot above, label the **threshold frequency**.

b. 2 points What is the value of the work potential (in J) of the unknown metal?

c. 3 points What is the *longest* wavelength (in nm) required to eject an electron from this metal's surface? Show your work including equation(s) used. (3 sig figs)

d. 2 points In the space provided below, plot the following

(Assume frequency is *above* threshold frequency):

- Kinetic energy* of an ejected electron versus the *intensity* of the incident light (left)
- Number of electrons* ejected versus the *frequency* of the incident light (right)



**Question 3** **6 points****a. 2 points**

In the ground state of Ge (Germanium; $Z = 32$) atom, **name** the following orbitals:

i. Orbital containing the electron that requires the maximum energy to be removed.

ii. Orbital containing the electron that experiences the minimum effective nuclear charge.

b. 2 points

An unknown anion (A^-) has the following *excited* electronic configuration: $[Ar] 4s^2 3d^3 4p^1$

(Ar: Argon; $Z = 18$)

Give the **condensed ground state electronic configuration** of the atom, **A** and the cation, **A^{2+}** .
Show your work. (Assume all electronic transitions are possible)

Electronic Configuration of A:

Electronic Configuration of A^{2+} :

c. 2 points

In a single atom, determine how many electrons can have the following set of quantum numbers?

(i) $n = 5; l = 3$

In a single atom, determine how many electrons can have the following set of quantum numbers?

(i) $n = 3; l = 2; m_s = +1/2$

**Question 4** **8 points**

Rank the following from 1 to 4 in **increasing** order. Provide the ranking in the boxes provided.

Note: Atomic number (Z) are provided for relevant atoms

a) Increasing (lowest to highest) order of *ionization energy*: (1= lowest; 4 = highest)

Mg

Mg: Z = 12

Rb

Rb: Z = 37

Cs

Cs: Z = 55

K

K : Z = 19

b) Increasing (lowest to highest) order of *bond angle*: (1= lowest; 4 = highest)

H₂OCCl₄XeF₄

HCN

c) Increasing (smallest to largest) order of *ionic size* (1= smallest; 4 = largest)

S²⁻

S: Z = 16

O²⁻

O: Z = 8

Be²⁺

Be: Z = 4

Na⁺

Na: Z = 11

d) Increasing (lowest to highest) order of *bond energy* (1= lowest; 4 = highest)

P-F

P-Cl

P-I

P-Br

**Question 5** **12 points**

a. 9 points For the molecules HClO_4 and TeF_6 , draw the most stable *Lewis structure* showing appropriate shape and *dashed/wedged bonds*, based on VSEPR. Determine the **molecular geometry** around the central atom. For HClO_4 , label the *most polar bond*.



b. 3 points (Note: Lone pairs are not shown in structures below)

(i) For the three *resonance* structures (1, 2 and 3) shown below, answer the following question:



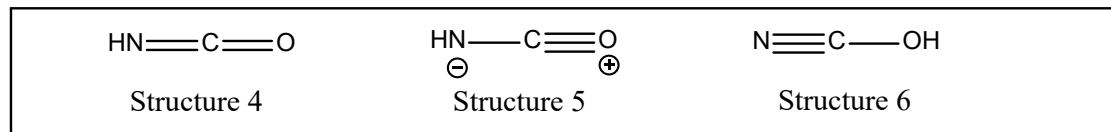
(I) Circle the *most contributing* resonance structure.

Structure 1

Structure 2

Structure 3

(ii) For the three structures given (4, 5, and 6) below, answer the following questions:



(I) Circle all the structures that represent *resonance structures* of each other:

Structure 4

Structure 5

Structure 6

(II) Among the resonance structures, circle the *most contributing* resonance structure.

Structure 4

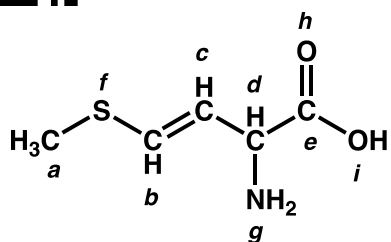
Structure 5

Structure 6

**Question 6** **6 points**

Using the data given below, calculate the **lattice energy** for SrO(s).

	ΔH° (kJ/mol)
$\text{Sr}(s) \longrightarrow \text{Sr}(g)$	+180
$\text{Sr}(g) \longrightarrow \text{Sr}^{2+}(g) + 2e^-$	+1468
$\text{O}_2(g) \longrightarrow 2\text{O}(g)$	+500
$\text{O}(g) + 2e^- \longrightarrow \text{O}^{2-}(g)$	+600
$\text{Sr}(s) + \frac{1}{2}\text{O}_2(g) \longrightarrow \text{SrO}(s)$	-550

**Question 7 10 points**

Consider the molecule given on the left, and answer questions (i) to (vi) using chemical bonding theories that we have learnt in class (valence bond theory, hybridization).

Note that the lone pairs are not shown in the structure:

i. Consider the carbon atoms marked in as ***a***, ***b***, ***c***, ***d***, and ***e*** in the molecule, and determine **hybridization** at each of the carbon centers. (2.5 points)

Atom	Hybridization
Carbon (<i>a</i>)	
Carbon (<i>b</i>)	
Carbon (<i>c</i>)	
Carbon (<i>d</i>)	
Carbon (<i>e</i>)	

ii. Now, consider the sulfur atom (S) marked as ***f***, nitrogen (N) atom (***g***), and oxygen(O) atoms (***h*** and ***i***), and determine their **hybridization**. (2 points)

Atom	Hybridization
Sulfur (<i>f</i>)	
Nitrogen (<i>g</i>)	
Oxygen (<i>h</i>)	
Oxygen (<i>i</i>)	

iii. **Describe** all the bonds formed by the carbon atoms ***a*** and ***b***, using sigma and pi designation as appropriate. (2 points)

Atom	Bonds formed (use sigma and pi as appropriate)
Carbon (<i>a</i>)	
Carbon (<i>b</i>)	

iv. How many total sp^3 -s sigma bonds are in the molecule? (1 point)

v. Determine **electronic** and **molecular geometry** at carbon (C) labeled ***e*** and sulfur (S) labeled ***f*** centres. (2 points)

Atom	Electronic Geometry	Molecular Geometry
Carbon (<i>e</i>)		
Sulfur (<i>f</i>)		

vi. Circle the value from the choices given below, that describes the **C-S-C bond angle at sulfur** (S) marked ***f***: (0.5 points)

>90° to <109.5° 109.5° >109.5 to <120° 120° >120° to <180°


Question 8 12 points

Using the **valence shell orbitals only** draw a **complete** molecular orbital (MO) diagram (with no s-p mixing) for the molecule **PS**, in the space provided below. **(6 points)**

Assume that 3s and 3p atomic orbitals will form molecular orbitals like those formed by the combination of 2s and 2p atomic orbitals that we learnt in class, and there is no s-p mixing.

Show **energy arrow**, **atomic orbitals** involved, **atom labels**, the **molecular orbitals** that are formed, **filled electrons** in the MO diagram etc. Make sure to label all the atomic and molecular orbitals.

Using the MO diagram above, answer the following questions:

- (i) Which one of the following will have the highest bond order? Circle the correct choice
 A) PS^{1-} B) PS C) PS^{1+}
- (ii) Which one of the following will have the longest bond length? Circle the correct choice
 A) PS^{1-} B) PS C) PS^{1+}
- (iii) Which of the following will be diamagnetic? Circle the correct choice.
 A) PS^{1-} B) PS C) PS^{1+}
- (iv) Draw Lewis dot structure of the molecule with the shortest bond length (based on the MO diagram) from among the three (PS^{1-} , PS, PS^{1+}), and indicate the bond order calculated from the Lewis dot structure. **(3 points)**

Lewis Dot Structure

Bond Order



7349A575-E92E-42CA-9E87-16D387A97435

midterm-exam-93388

#314 Page 10 of 10

SCRAP PAGE

**CHEMISTRY 110 –Midterm Exam****Thursday November 7, 2024, 6:30 – 8:30 pm**

Examiners: Pallavi Sirjoosingh and Ashok Kakkar

First name (please write as legibly as possible within the boxes)

--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--

Last name

--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--

Student ID number

--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--

This booklet has 8 questions on 10 pages (The last page can be used as a scrap page).

The staple must NOT be removed from the question booklet. Your intact question booklet must be returned at the end of the exam.

A separate booklet contains a periodic table, and datasheet.

Write your answers directly under the questions in the space provided. Do not write on the QR codes or near the top edges (as it may be cut off).

Please write **legibly and dark enough** for the scan to be readable. For students who use English as a second language, dictionaries are permitted. Calculators are allowed.

DO NOT ASK QUESTIONS OF THE INVIGILATORS about interpretations of the exam questions. If you have a concern about a question, please summarize it at the top of this page. All concerns will be evaluated after the exam. GOOD LUCK!

**Question 1 8 points** (*Assume all transitions are possible*)

a. 6 points Calculate the **energy (in kJ/mol)** required to ionize (completely remove) the electron from the **$n=3$** level of **He^+ ion** (He: Helium). Show your work including equations. (3 sig. figs.)

b. 2 points

(I) Select the largest (longest) wavelength from the three choices below:

- i. Wavelength of photon to *completely remove* an electron from *1s orbital of Hydrogen atom*
- ii. Wavelength of photon to *completely remove* an electron from *1s orbital of Boron atom (B)*
- iii. Wavelength of photon to *completely remove* an electron from *1s orbital of Boron cation (B^+)*

i or ii or iii

(II) Select the smallest (shortest) wavelength from the three choices below (*for H atom*):

- i. Wavelength of photon *emitted* for an electron transitioning from *2p to 1s orbital*
- ii. Wavelength of photon *absorbed* for an electron transitioning from *2s to 4p orbital*
- iii. Wavelength of photon *absorbed* for an electron transitioning from *1s to 3p orbital*

i or ii or iii



Question 2 8 points



A Chem110 student plots the graph shown on the left, from data obtained for the kinetic energy (eV) of the ejected electron versus frequency (s^{-1}) of the incident light, for an unknown metal. ($1\text{eV} = 1.6022 \times 10^{-19} \text{ J}$)

Based on the plot, answer the questions **a to c** below:

a. 1 point On the plot above, label the **threshold frequency**.

b. 2 points What is the value of the work potential (in J) of the unknown metal?

c. 3 points What is the *longest* wavelength (in nm) required to eject an electron from this metal's surface? Show your work including equation(s) used. (3 sig figs)

d. 2 points In the space provided below, plot the following

(Assume frequency is *above* threshold frequency):

- i. *Kinetic energy* of an ejected electron versus the *intensity* of the incident light (left)
- ii. *Number of electrons* ejected versus the *frequency* of the incident light (right)



**Question 3** **6 points****a. 2 points**

In the ground state of Ge (Germanium; $Z = 32$) atom, **name** the following orbitals:

i. Orbital containing the electron that requires the maximum energy to be removed.

ii. Orbital containing the electron that experiences the minimum effective nuclear charge.

b. 2 points

An unknown anion (A^-) has the following *excited* electronic configuration: $[Ar] 4s^2 3d^3 4p^1$

(Ar: Argon; $Z = 18$)

Give the **condensed ground state electronic configuration** of the atom, **A** and the cation, **A^{2+}** .
Show your work. (Assume all electronic transitions are possible)

Electronic Configuration of A:

Electronic Configuration of A^{2+} :

c. 2 points

In a single atom, determine how many electrons can have the following set of quantum numbers?

(i) $n = 5; l = 3$

In a single atom, determine how many electrons can have the following set of quantum numbers?

(i) $n = 3; l = 2; m_s = +1/2$

**Question 4** **8 points**

Rank the following from 1 to 4 in **increasing** order. Provide the ranking in the boxes provided.

Note: Atomic number (Z) are provided for relevant atoms

a) Increasing (lowest to highest) order of *ionization energy*: (1= lowest; 4 = highest)

Mg

Mg: Z = 12

Rb

Rb: Z = 37

Cs

Cs: Z = 55

K

K : Z = 19

b) Increasing (lowest to highest) order of *bond angle*: (1= lowest; 4 = highest)

H₂OCCl₄XeF₄

HCN

c) Increasing (smallest to largest) order of *ionic size* (1= smallest; 4 = largest)

S²⁻

S: Z = 16

O²⁻

O: Z = 8

Be²⁺

Be: Z = 4

Na⁺

Na: Z = 11

d) Increasing (lowest to highest) order of *bond energy* (1= lowest; 4 = highest)

P-F

P-Cl

P-I

P-Br

**Question 5** **12 points**

a. 9 points For the molecules HClO_4 and TeF_6 , draw the most stable *Lewis structure* showing appropriate shape and *dashed/wedged bonds*, based on VSEPR. Determine the **molecular geometry** around the central atom. For HClO_4 , label the *most polar bond*.

HClO_4 (there are no O-O bonds)

Label the most polar bond

Molecular Geometry around Cl

TeF_6

Molecular Geometry around Te

b. 3 points (Note: Lone pairs are not shown in structures below)

(i) For the three *resonance* structures (1, 2 and 3) shown below, answer the following question:



(I) Circle the *most contributing* resonance structure.

Structure 1

Structure 2

Structure 3

(ii) For the three structures given (4, 5, and 6) below, answer the following questions:



(I) Circle all the structures that represent *resonance structures* of each other:

Structure 4

Structure 5

Structure 6

(II) Among the resonance structures, circle the *most contributing* resonance structure.

Structure 4

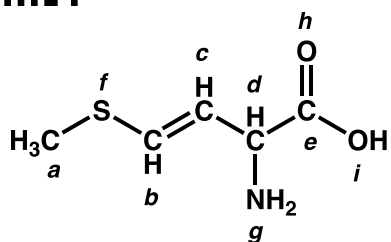
Structure 5

Structure 6

**Question 6** **6 points**

Using the data given below, calculate the **lattice energy** for SrO(s).

	ΔH° (kJ/mol)
$\text{Sr}(s) \longrightarrow \text{Sr}(g)$	+180
$\text{Sr}(g) \longrightarrow \text{Sr}^{2+}(g) + 2e^-$	+1468
$\text{O}_2(g) \longrightarrow 2\text{O}(g)$	+500
$\text{O}(g) + 2e^- \longrightarrow \text{O}^{2-}(g)$	+600
$\text{Sr}(s) + \frac{1}{2}\text{O}_2(g) \longrightarrow \text{SrO}(s)$	-550

**Question 7 10 points**

Consider the molecule given on the left, and answer questions (i) to (vi) using chemical bonding theories that we have learnt in class (valence bond theory, hybridization).

Note that the lone pairs are not shown in the structure:

i. Consider the carbon atoms marked in as ***a***, ***b***, ***c***, ***d***, and ***e*** in the molecule, and determine **hybridization** at each of the carbon centers. (2.5 points)

Atom	Hybridization
Carbon (<i>a</i>)	
Carbon (<i>b</i>)	
Carbon (<i>c</i>)	
Carbon (<i>d</i>)	
Carbon (<i>e</i>)	

ii. Now, consider the sulfur atom (S) marked as ***f***, nitrogen (N) atom (***g***), and oxygen(O) atoms (***h*** and ***i***), and determine their **hybridization**. (2 points)

Atom	Hybridization
Sulfur (<i>f</i>)	
Nitrogen (<i>g</i>)	
Oxygen (<i>h</i>)	
Oxygen (<i>i</i>)	

iii. **Describe** all the bonds formed by the carbon atoms ***a*** and ***b***, using sigma and pi designation as appropriate. (2 points)

Atom	Bonds formed (use sigma and pi as appropriate)
Carbon (<i>a</i>)	
Carbon (<i>b</i>)	

iv. How many total sp^3 -s sigma bonds are in the molecule? (1 point)

v. Determine **electronic** and **molecular geometry** at carbon (C) labeled ***e*** and sulfur (S) labeled ***f*** centres. (2 points)

Atom	Electronic Geometry	Molecular Geometry
Carbon (<i>e</i>)		
Sulfur (<i>f</i>)		

vi. Circle the value from the choices given below, that describes the **C-S-C bond angle at sulfur** (S) marked ***f***: (0.5 points)

>90° to <109.5° 109.5° >109.5 to <120° 120° >120° to <180°



Question 8 12 points

Using the **valence shell orbitals only** draw a **complete** molecular orbital (MO) diagram (with no s-p mixing) for the molecule **PS**, in the space provided below. **(6 points)**

Assume that 3s and 3p atomic orbitals will form molecular orbitals like those formed by the combination of 2s and 2p atomic orbitals that we learnt in class, and there is no s-p mixing.

Show **energy arrow**, **atomic orbitals** involved, **atom labels**, the **molecular orbitals** that are formed, **filled electrons** in the MO diagram etc. Make sure to label all the atomic and molecular orbitals.

Using the MO diagram above, answer the following questions:

- (i) Which one of the following will have the highest bond order? Circle the correct choice
 A) PS^{1-} B) PS C) PS^{1+}
- (ii) Which one of the following will have the longest bond length? Circle the correct choice
 A) PS^{1-} B) PS C) PS^{1+}
- (iii) Which of the following will be diamagnetic? Circle the correct choice.
 A) PS^{1-} B) PS C) PS^{1+}
- (iv) Draw Lewis dot structure of the molecule with the shortest bond length (based on the MO diagram) from among the three (PS^{1-} , PS, PS^{1+}), and indicate the bond order calculated from the Lewis dot structure. **(3 points)**

Lewis Dot Structure

Bond Order



1A605F05-CC20-4D42-B2D7-ADFF87C542AB

midterm-exam-93388

#315 Page 10 of 10

SCRAP PAGE

**CHEMISTRY 110 –Midterm Exam****Thursday November 7, 2024, 6:30 – 8:30 pm**

Examiners: Pallavi Sirjoosingh and Ashok Kakkar

First name (please write as legibly as possible within the boxes)

--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--

Last name

--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--

Student ID number

--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--

This booklet has 8 questions on 10 pages (The last page can be used as a scrap page).

The staple must NOT be removed from the question booklet. Your intact question booklet must be returned at the end of the exam.

A separate booklet contains a periodic table, and datasheet.

Write your answers directly under the questions in the space provided. Do not write on the QR codes or near the top edges (as it may be cut off).

Please write **legibly and dark enough** for the scan to be readable. For students who use English as a second language, dictionaries are permitted. Calculators are allowed.

DO NOT ASK QUESTIONS OF THE INVIGILATORS about interpretations of the exam questions. If you have a concern about a question, please summarize it at the top of this page. All concerns will be evaluated after the exam. GOOD LUCK!

**Question 1 8 points** (*Assume all transitions are possible*)

a. 6 points Calculate the **energy (in kJ/mol)** required to ionize (completely remove) the electron from the **n=3** level of **He⁺ ion** (He: Helium). Show your work including equations. (3 sig. figs.)

b. 2 points

(I) Select the largest (longest) wavelength from the three choices below:

- i. Wavelength of photon to *completely remove* an electron from *1s orbital of Hydrogen atom*
- ii. Wavelength of photon to *completely remove* an electron from *1s orbital of Boron atom (B)*
- iii. Wavelength of photon to *completely remove* an electron from *1s orbital of Boron cation (B⁺)*

i or ii or iii

(II) Select the smallest (shortest) wavelength from the three choices below (*for H atom*):

- i. Wavelength of photon *emitted* for an electron transitioning from *2p to 1s orbital*
- ii. Wavelength of photon *absorbed* for an electron transitioning from *2s to 4p orbital*
- iii. Wavelength of photon *absorbed* for an electron transitioning from *1s to 3p orbital*

i or ii or iii



Question 2 8 points



A Chem110 student plots the graph shown on the left, from data obtained for the kinetic energy (eV) of the ejected electron versus frequency (s^{-1}) of the incident light, for an unknown metal. ($1\text{eV} = 1.6022 \times 10^{-19} \text{ J}$)

Based on the plot, answer the questions **a to c** below:

a. 1 point On the plot above, label the **threshold frequency**.

b. 2 points What is the value of the work potential (in J) of the unknown metal?

c. 3 points What is the *longest* wavelength (in nm) required to eject an electron from this metal's surface? Show your work including equation(s) used. (3 sig figs)

d. 2 points In the space provided below, plot the following

(Assume frequency is *above* threshold frequency):

- Kinetic energy* of an ejected electron versus the *intensity* of the incident light (left)
- Number of electrons* ejected versus the *frequency* of the incident light (right)



**Question 3** **6 points****a. 2 points**

In the ground state of Ge (Germanium; $Z = 32$) atom, **name** the following orbitals:

i. Orbital containing the electron that requires the maximum energy to be removed.

ii. Orbital containing the electron that experiences the minimum effective nuclear charge.

b. 2 points

An unknown anion (A^-) has the following *excited* electronic configuration: $[Ar] 4s^2 3d^3 4p^1$

(Ar: Argon; $Z = 18$)

Give the **condensed ground state electronic configuration** of the atom, **A** and the cation, **A^{2+}** .
Show your work. (Assume all electronic transitions are possible)

Electronic Configuration of A:

Electronic Configuration of A^{2+} :

c. 2 points

In a single atom, determine how many electrons can have the following set of quantum numbers?

(i) $n = 5; l = 3$

In a single atom, determine how many electrons can have the following set of quantum numbers?

(i) $n = 3; l = 2; m_s = +1/2$

**Question 4** **8 points**

Rank the following from 1 to 4 in **increasing** order. Provide the ranking in the boxes provided.

Note: Atomic number (Z) are provided for relevant atoms

a) Increasing (lowest to highest) order of *ionization energy*: (1= lowest; 4 = highest)

Mg

Mg: Z = 12

Rb

Rb: Z = 37

Cs

Cs: Z = 55

K

K : Z = 19

b) Increasing (lowest to highest) order of *bond angle*: (1= lowest; 4 = highest)

H₂OCCl₄XeF₄

HCN

c) Increasing (smallest to largest) order of *ionic size* (1= smallest; 4 = largest)

S²⁻

S: Z = 16

O²⁻

O: Z = 8

Be²⁺

Be: Z = 4

Na⁺

Na: Z = 11

d) Increasing (lowest to highest) order of *bond energy* (1= lowest; 4 = highest)

P-F

P-Cl

P-I

P-Br

**Question 5** **12 points**

a. 9 points For the molecules HClO_4 and TeF_6 , draw the most stable *Lewis structure* showing appropriate shape and *dashed/wedged bonds*, based on VSEPR. Determine the **molecular geometry** around the central atom. For HClO_4 , label the *most polar bond*.



b. 3 points (Note: Lone pairs are not shown in structures below)

(i) For the three *resonance* structures (1, 2 and 3) shown below, answer the following question:



(I) Circle the *most contributing* resonance structure.

Structure 1

Structure 2

Structure 3

(ii) For the three structures given (4, 5, and 6) below, answer the following questions:



(I) Circle all the structures that represent *resonance structures* of each other:

Structure 4

Structure 5

Structure 6

(II) Among the resonance structures, circle the *most contributing* resonance structure.

Structure 4

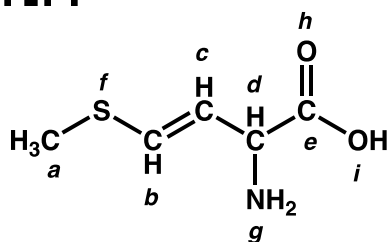
Structure 5

Structure 6

**Question 6** **6 points**

Using the data given below, calculate the **lattice energy** for SrO(s).

	ΔH° (kJ/mol)
$\text{Sr}(s) \longrightarrow \text{Sr}(g)$	+180
$\text{Sr}(g) \longrightarrow \text{Sr}^{2+}(g) + 2e^-$	+1468
$\text{O}_2(g) \longrightarrow 2\text{O}(g)$	+500
$\text{O}(g) + 2e^- \longrightarrow \text{O}^{2-}(g)$	+600
$\text{Sr}(s) + \frac{1}{2}\text{O}_2(g) \longrightarrow \text{SrO}(s)$	-550

**Question 7 10 points**

Consider the molecule given on the left, and answer questions (i) to (vi) using chemical bonding theories that we have learnt in class (valence bond theory, hybridization).

Note that the lone pairs are not shown in the structure:

i. Consider the carbon atoms marked in as ***a***, ***b***, ***c***, ***d***, and ***e*** in the molecule, and determine **hybridization** at each of the carbon centers. (2.5 points)

Atom	Hybridization
Carbon (<i>a</i>)	
Carbon (<i>b</i>)	
Carbon (<i>c</i>)	
Carbon (<i>d</i>)	
Carbon (<i>e</i>)	

ii. Now, consider the sulfur atom (S) marked as ***f***, nitrogen (N) atom (***g***), and oxygen(O) atoms (***h*** and ***i***), and determine their **hybridization**. (2 points)

Atom	Hybridization
Sulfur (<i>f</i>)	
Nitrogen (<i>g</i>)	
Oxygen (<i>h</i>)	
Oxygen (<i>i</i>)	

iii. **Describe** all the bonds formed by the carbon atoms ***a*** and ***b***, using sigma and pi designation as appropriate. (2 points)

Atom	Bonds formed (use sigma and pi as appropriate)
Carbon (<i>a</i>)	
Carbon (<i>b</i>)	

iv. How many total sp^3 -s sigma bonds are in the molecule? (1 point)

v. Determine **electronic** and **molecular geometry** at carbon (C) labeled ***e*** and sulfur (S) labeled ***f*** centres. (2 points)

Atom	Electronic Geometry	Molecular Geometry
Carbon (<i>e</i>)		
Sulfur (<i>f</i>)		

vi. Circle the value from the choices given below, that describes the **C-S-C bond angle at sulfur** (S) marked ***f***: (0.5 points)

>90° to <109.5° 109.5° >109.5 to <120° 120° >120° to <180°


Question 8 12 points

Using the **valence shell orbitals only** draw a **complete** molecular orbital (MO) diagram (with no s-p mixing) for the molecule **PS**, in the space provided below. **(6 points)**

Assume that 3s and 3p atomic orbitals will form molecular orbitals like those formed by the combination of 2s and 2p atomic orbitals that we learnt in class, and there is no s-p mixing.

Show **energy arrow**, **atomic orbitals** involved, **atom labels**, the **molecular orbitals** that are formed, **filled electrons** in the MO diagram etc. Make sure to label all the atomic and molecular orbitals.

Using the MO diagram above, answer the following questions:

- (i) Which one of the following will have the highest bond order? Circle the correct choice
 A) PS^{1-} B) PS C) PS^{1+}
- (ii) Which one of the following will have the longest bond length? Circle the correct choice
 A) PS^{1-} B) PS C) PS^{1+}
- (iii) Which of the following will be diamagnetic? Circle the correct choice.
 A) PS^{1-} B) PS C) PS^{1+}
- (iv) Draw Lewis dot structure of the molecule with the shortest bond length (based on the MO diagram) from among the three (PS^{1-} , PS, PS^{1+}), and indicate the bond order calculated from the Lewis dot structure. **(3 points)**

Lewis Dot Structure

Bond Order



0D0271C5-F740-4EA2-B0E0-8488BAF295A8

midterm-exam-93388

#316 Page 10 of 10

SCRAP PAGE

**CHEMISTRY 110 –Midterm Exam****Thursday November 7, 2024, 6:30 – 8:30 pm****Examiners: Pallavi Sirjoosingh and Ashok Kakkar**

First name (please write as legibly as possible within the boxes)

--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--

Last name

--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--

Student ID number

--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--

This booklet has 8 questions on 10 pages (The last page can be used as a scrap page).

The staple must NOT be removed from the question booklet. Your intact question booklet must be returned at the end of the exam.

A separate booklet contains a periodic table, and datasheet.

Write your answers directly under the questions in the space provided. Do not write on the QR codes or near the top edges (as it may be cut off).

Please write **legibly and dark enough** for the scan to be readable. For students who use English as a second language, dictionaries are permitted. Calculators are allowed.

DO NOT ASK QUESTIONS OF THE INVIGILATORS about interpretations of the exam questions. If you have a concern about a question, please summarize it at the top of this page. All concerns will be evaluated after the exam. GOOD LUCK!

**Question 1 8 points** (*Assume all transitions are possible*)

a. 6 points Calculate the **energy (in kJ/mol)** required to ionize (completely remove) the electron from the **n=3** level of **He⁺ ion** (He: Helium). Show your work including equations. (3 sig. figs.)

b. 2 points

(I) Select the largest (longest) wavelength from the three choices below:

- i. Wavelength of photon to *completely remove* an electron from *1s orbital of Hydrogen atom*
- ii. Wavelength of photon to *completely remove* an electron from *1s orbital of Boron atom (B)*
- iii. Wavelength of photon to *completely remove* an electron from *1s orbital of Boron cation (B⁺)*

i or ii or iii

(II) Select the smallest (shortest) wavelength from the three choices below (*for H atom*):

- i. Wavelength of photon *emitted* for an electron transitioning from *2p to 1s orbital*
- ii. Wavelength of photon *absorbed* for an electron transitioning from *2s to 4p orbital*
- iii. Wavelength of photon *absorbed* for an electron transitioning from *1s to 3p orbital*

i or ii or iii



Question 2 8 points



A Chem110 student plots the graph shown on the left, from data obtained for the kinetic energy (eV) of the ejected electron versus frequency (s^{-1}) of the incident light, for an unknown metal. ($1\text{eV} = 1.6022 \times 10^{-19} \text{ J}$)

Based on the plot, answer the questions **a to c** below:

a. 1 point On the plot above, label the **threshold frequency**.

b. 2 points What is the value of the work potential (in J) of the unknown metal?

c. 3 points What is the *longest* wavelength (in nm) required to eject an electron from this metal's surface? Show your work including equation(s) used. (3 sig figs)

d. 2 points In the space provided below, plot the following

(Assume frequency is *above* threshold frequency):

- Kinetic energy* of an ejected electron versus the *intensity* of the incident light (left)
- Number of electrons* ejected versus the *frequency* of the incident light (right)



**Question 3** **6 points****a. 2 points**

In the ground state of Ge (Germanium; $Z = 32$) atom, **name** the following orbitals:

i. Orbital containing the electron that requires the maximum energy to be removed.

ii. Orbital containing the electron that experiences the minimum effective nuclear charge.

b. 2 points

An unknown anion (A^-) has the following *excited* electronic configuration: $[Ar] 4s^2 3d^3 4p^1$

(Ar: Argon; $Z = 18$)

Give the **condensed ground state electronic configuration** of the atom, **A** and the cation, **A^{2+}** .
Show your work. (Assume all electronic transitions are possible)

Electronic Configuration of A:

Electronic Configuration of A^{2+} :

c. 2 points

In a single atom, determine how many electrons can have the following set of quantum numbers?

(i) $n = 5; l = 3$

In a single atom, determine how many electrons can have the following set of quantum numbers?

(i) $n = 3; l = 2; m_s = +1/2$

**Question 4** **8 points**

Rank the following from 1 to 4 in **increasing** order. Provide the ranking in the boxes provided.

Note: Atomic number (Z) are provided for relevant atoms

a) Increasing (lowest to highest) order of *ionization energy*: (1= lowest; 4 = highest)

Mg

Mg: Z = 12

Rb

Rb: Z = 37

Cs

Cs: Z = 55

K

K : Z = 19

b) Increasing (lowest to highest) order of *bond angle*: (1= lowest; 4 = highest)

H₂OCCl₄XeF₄

HCN

c) Increasing (smallest to largest) order of *ionic size* (1= smallest; 4 = largest)

S²⁻

S: Z = 16

O²⁻

O: Z = 8

Be²⁺

Be: Z = 4

Na⁺

Na: Z = 11

d) Increasing (lowest to highest) order of *bond energy* (1= lowest; 4 = highest)

P-F

P-Cl

P-I

P-Br

**Question 5** **12 points**

a. 9 points For the molecules HClO_4 and TeF_6 , draw the most stable *Lewis structure* showing appropriate shape and *dashed/wedged bonds*, based on VSEPR. Determine the **molecular geometry** around the central atom. For HClO_4 , label the *most polar bond*.



b. 3 points (Note: Lone pairs are not shown in structures below)

(i) For the three *resonance* structures (1, 2 and 3) shown below, answer the following question:



(I) Circle the *most contributing* resonance structure.

Structure 1

Structure 2

Structure 3

(ii) For the three structures given (4, 5, and 6) below, answer the following questions:



(I) Circle all the structures that represent *resonance structures* of each other:

Structure 4

Structure 5

Structure 6

(II) Among the resonance structures, circle the *most contributing* resonance structure.

Structure 4

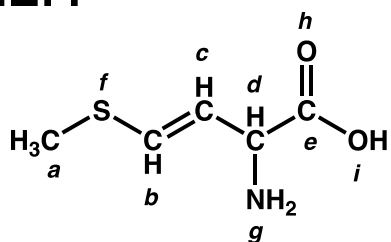
Structure 5

Structure 6

**Question 6** **6 points**

Using the data given below, calculate the **lattice energy** for SrO(s).

	ΔH° (kJ/mol)
$\text{Sr}(s) \longrightarrow \text{Sr}(g)$	+180
$\text{Sr}(g) \longrightarrow \text{Sr}^{2+}(g) + 2e^-$	+1468
$\text{O}_2(g) \longrightarrow 2\text{O}(g)$	+500
$\text{O}(g) + 2e^- \longrightarrow \text{O}^{2-}(g)$	+600
$\text{Sr}(s) + \frac{1}{2}\text{O}_2(g) \longrightarrow \text{SrO}(s)$	-550

**Question 7 10 points**

Consider the molecule given on the left, and answer questions (i) to (vi) using chemical bonding theories that we have learnt in class (valence bond theory, hybridization).

Note that the lone pairs are not shown in the structure:

i. Consider the carbon atoms marked in as ***a***, ***b***, ***c***, ***d***, and ***e*** in the molecule, and determine **hybridization** at each of the carbon centers. (2.5 points)

Atom	Hybridization
Carbon (<i>a</i>)	
Carbon (<i>b</i>)	
Carbon (<i>c</i>)	
Carbon (<i>d</i>)	
Carbon (<i>e</i>)	

ii. Now, consider the sulfur atom (S) marked as ***f***, nitrogen (N) atom (***g***), and oxygen(O) atoms (***h*** and ***i***), and determine their **hybridization**. (2 points)

Atom	Hybridization
Sulfur (<i>f</i>)	
Nitrogen (<i>g</i>)	
Oxygen (<i>h</i>)	
Oxygen (<i>i</i>)	

iii. **Describe** all the bonds formed by the carbon atoms ***a*** and ***b***, using sigma and pi designation as appropriate. (2 points)

Atom	Bonds formed (use sigma and pi as appropriate)
Carbon (<i>a</i>)	
Carbon (<i>b</i>)	

iv. How many total sp^3 -s sigma bonds are in the molecule? (1 point)

v. Determine **electronic** and **molecular geometry** at carbon (C) labeled ***e*** and sulfur (S) labeled ***f*** centres. (2 points)

Atom	Electronic Geometry	Molecular Geometry
Carbon (<i>e</i>)		
Sulfur (<i>f</i>)		

vi. Circle the value from the choices given below, that describes the **C-S-C bond angle at sulfur** (S) marked ***f***: (0.5 points)

>90° to <109.5° 109.5° >109.5 to <120° 120° >120° to <180°

**Question 8 12 points**

Using the **valence shell orbitals only** draw a **complete** molecular orbital (MO) diagram (with no s-p mixing) for the molecule **PS**, in the space provided below. **(6 points)**

Assume that 3s and 3p atomic orbitals will form molecular orbitals like those formed by the combination of 2s and 2p atomic orbitals that we learnt in class, and there is no s-p mixing.

Show **energy arrow**, **atomic orbitals** involved, **atom labels**, the **molecular orbitals** that are formed, **filled electrons** in the MO diagram etc. Make sure to label all the atomic and molecular orbitals.

Using the MO diagram above, answer the following questions:

- (i) Which one of the following will have the highest bond order? Circle the correct choice
A) PS^{1-} B) PS C) PS^{1+}
- (ii) Which one of the following will have the longest bond length? Circle the correct choice
A) PS^{1-} B) PS C) PS^{1+}
- (iii) Which of the following will be diamagnetic? Circle the correct choice.
A) PS^{1-} B) PS C) PS^{1+}
- (iv) Draw Lewis dot structure of the molecule with the shortest bond length (based on the MO diagram) from among the three (PS^{1-} , PS, PS^{1+}), and indicate the bond order calculated from the Lewis dot structure. **(3 points)**

Lewis Dot Structure

Bond Order



03E883C9-62FC-41BB-82EF-8418061656F6

midterm-exam-93388

#317 Page 10 of 10

SCRAP PAGE

**CHEMISTRY 110 –Midterm Exam****Thursday November 7, 2024, 6:30 – 8:30 pm****Examiners: Pallavi Sirjoosingh and Ashok Kakkar**

First name (please write as legibly as possible within the boxes)

--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--

Last name

--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--

Student ID number

--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--

This booklet has 8 questions on 10 pages (The last page can be used as a scrap page).

The staple must NOT be removed from the question booklet. Your intact question booklet must be returned at the end of the exam.

A separate booklet contains a periodic table, and datasheet.

Write your answers directly under the questions in the space provided. Do not write on the QR codes or near the top edges (as it may be cut off).

Please write **legibly and dark enough** for the scan to be readable. For students who use English as a second language, dictionaries are permitted. Calculators are allowed.

DO NOT ASK QUESTIONS OF THE INVIGILATORS about interpretations of the exam questions. If you have a concern about a question, please summarize it at the top of this page. All concerns will be evaluated after the exam. GOOD LUCK!

**Question 1 8 points** (*Assume all transitions are possible*)

a. 6 points Calculate the **energy (in kJ/mol)** required to ionize (completely remove) the electron from the **n=3** level of **He⁺ ion** (He: Helium). Show your work including equations. (3 sig. figs.)

b. 2 points

(I) Select the largest (longest) wavelength from the three choices below:

- i. Wavelength of photon to *completely remove* an electron from *1s orbital of Hydrogen atom*
- ii. Wavelength of photon to *completely remove* an electron from *1s orbital of Boron atom (B)*
- iii. Wavelength of photon to *completely remove* an electron from *1s orbital of Boron cation (B⁺)*

i or ii or iii

(II) Select the smallest (shortest) wavelength from the three choices below (*for H atom*):

- i. Wavelength of photon *emitted* for an electron transitioning from *2p to 1s orbital*
- ii. Wavelength of photon *absorbed* for an electron transitioning from *2s to 4p orbital*
- iii. Wavelength of photon *absorbed* for an electron transitioning from *1s to 3p orbital*

i or ii or iii



Question 2 8 points



A Chem110 student plots the graph shown on the left, from data obtained for the kinetic energy (eV) of the ejected electron versus frequency (s^{-1}) of the incident light, for an unknown metal. ($1\text{eV} = 1.6022 \times 10^{-19} \text{ J}$)

Based on the plot, answer the questions **a to c** below:

a. 1 point On the plot above, label the **threshold frequency**.

b. 2 points What is the value of the work potential (in J) of the unknown metal?

c. 3 points What is the *longest* wavelength (in nm) required to eject an electron from this metal's surface? Show your work including equation(s) used. (3 sig figs)

d. 2 points In the space provided below, plot the following

(Assume frequency is *above* threshold frequency):

- Kinetic energy* of an ejected electron versus the *intensity* of the incident light (left)
- Number of electrons* ejected versus the *frequency* of the incident light (right)



**Question 3** **6 points****a. 2 points**

In the ground state of Ge (Germanium; $Z = 32$) atom, **name** the following orbitals:

i. Orbital containing the electron that requires the maximum energy to be removed.

ii. Orbital containing the electron that experiences the minimum effective nuclear charge.

b. 2 points

An unknown anion (A^-) has the following *excited* electronic configuration: $[Ar] 4s^2 3d^3 4p^1$

(Ar: Argon; $Z = 18$)

Give the **condensed ground state electronic configuration** of the atom, **A** and the cation, **A^{2+}** .
Show your work. (Assume all electronic transitions are possible)

Electronic Configuration of A:

Electronic Configuration of A^{2+} :

c. 2 points

In a single atom, determine how many electrons can have the following set of quantum numbers?

(i) $n = 5; l = 3$

In a single atom, determine how many electrons can have the following set of quantum numbers?

(i) $n = 3; l = 2; m_s = +1/2$

**Question 4** **8 points**

Rank the following from 1 to 4 in **increasing** order. Provide the ranking in the boxes provided.

Note: Atomic number (Z) are provided for relevant atoms

a) Increasing (lowest to highest) order of *ionization energy*: (1= lowest; 4 = highest)

Mg

Mg: Z = 12

Rb

Rb: Z = 37

Cs

Cs: Z = 55

K

K : Z = 19

b) Increasing (lowest to highest) order of *bond angle*: (1= lowest; 4 = highest)

H₂OCCl₄XeF₄

HCN

c) Increasing (smallest to largest) order of *ionic size* (1= smallest; 4 = largest)

S²⁻

S: Z = 16

O²⁻

O: Z = 8

Be²⁺

Be: Z = 4

Na⁺

Na: Z = 11

d) Increasing (lowest to highest) order of *bond energy* (1= lowest; 4 = highest)

P-F

P-Cl

P-I

P-Br

**Question 5** **12 points**

a. 9 points For the molecules HClO_4 and TeF_6 , draw the most stable *Lewis structure* showing appropriate shape and *dashed/wedged bonds*, based on VSEPR. Determine the **molecular geometry** around the central atom. For HClO_4 , label the *most polar bond*.



b. 3 points (Note: Lone pairs are not shown in structures below)

(i) For the three *resonance* structures (1, 2 and 3) shown below, answer the following question:



(I) Circle the *most contributing* resonance structure.

Structure 1

Structure 2

Structure 3

(ii) For the three structures given (4, 5, and 6) below, answer the following questions:



(I) Circle all the structures that represent *resonance structures* of each other:

Structure 4

Structure 5

Structure 6

(II) Among the resonance structures, circle the *most contributing* resonance structure.

Structure 4

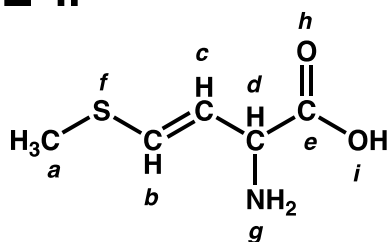
Structure 5

Structure 6

**Question 6** **6 points**

Using the data given below, calculate the **lattice energy** for SrO(s).

	ΔH° (kJ/mol)
$\text{Sr}(s) \longrightarrow \text{Sr}(g)$	+180
$\text{Sr}(g) \longrightarrow \text{Sr}^{2+}(g) + 2e^-$	+1468
$\text{O}_2(g) \longrightarrow 2\text{O}(g)$	+500
$\text{O}(g) + 2e^- \longrightarrow \text{O}^{2-}(g)$	+600
$\text{Sr}(s) + \frac{1}{2}\text{O}_2(g) \longrightarrow \text{SrO}(s)$	-550

**Question 7 10 points**

Consider the molecule given on the left, and answer questions (i) to (vi) using chemical bonding theories that we have learnt in class (valence bond theory, hybridization).

Note that the lone pairs are not shown in the structure:

i. Consider the carbon atoms marked in as ***a***, ***b***, ***c***, ***d***, and ***e*** in the molecule, and determine **hybridization** at each of the carbon centers. (2.5 points)

Atom	Hybridization
Carbon (<i>a</i>)	
Carbon (<i>b</i>)	
Carbon (<i>c</i>)	
Carbon (<i>d</i>)	
Carbon (<i>e</i>)	

ii. Now, consider the sulfur atom (S) marked as ***f***, nitrogen (N) atom (***g***), and oxygen(O) atoms (***h*** and ***i***), and determine their **hybridization**. (2 points)

Atom	Hybridization
Sulfur (<i>f</i>)	
Nitrogen (<i>g</i>)	
Oxygen (<i>h</i>)	
Oxygen (<i>i</i>)	

iii. **Describe** all the bonds formed by the carbon atoms ***a*** and ***b***, using sigma and pi designation as appropriate. (2 points)

Atom	Bonds formed (use sigma and pi as appropriate)
Carbon (<i>a</i>)	
Carbon (<i>b</i>)	

iv. How many total sp^3 -s sigma bonds are in the molecule? (1 point)

v. Determine **electronic** and **molecular geometry** at carbon (C) labeled ***e*** and sulfur (S) labeled ***f*** centres. (2 points)

Atom	Electronic Geometry	Molecular Geometry
Carbon (<i>e</i>)		
Sulfur (<i>f</i>)		

vi. Circle the value from the choices given below, that describes the **C-S-C bond angle at sulfur** (S) marked ***f***: (0.5 points)

>90° to <109.5° 109.5° >109.5 to <120° 120° >120° to <180°


Question 8 12 points

Using the **valence shell orbitals only** draw a **complete** molecular orbital (MO) diagram (with no s-p mixing) for the molecule **PS**, in the space provided below. **(6 points)**

Assume that 3s and 3p atomic orbitals will form molecular orbitals like those formed by the combination of 2s and 2p atomic orbitals that we learnt in class, and there is no s-p mixing.

Show **energy arrow**, **atomic orbitals** involved, **atom labels**, the **molecular orbitals** that are formed, **filled electrons** in the MO diagram etc. Make sure to label all the atomic and molecular orbitals.

Using the MO diagram above, answer the following questions:

- (i) Which one of the following will have the highest bond order? Circle the correct choice
 A) PS^{1-} B) PS C) PS^{1+}
- (ii) Which one of the following will have the longest bond length? Circle the correct choice
 A) PS^{1-} B) PS C) PS^{1+}
- (iii) Which of the following will be diamagnetic? Circle the correct choice.
 A) PS^{1-} B) PS C) PS^{1+}
- (iv) Draw Lewis dot structure of the molecule with the shortest bond length (based on the MO diagram) from among the three (PS^{1-} , PS, PS^{1+}), and indicate the bond order calculated from the Lewis dot structure. **(3 points)**

Lewis Dot Structure

Bond Order



9DD099D6-D077-4EED-8668-1D7F10445D7F

midterm-exam-93388

#318 Page 10 of 10

SCRAP PAGE

**CHEMISTRY 110 –Midterm Exam****Thursday November 7, 2024, 6:30 – 8:30 pm****Examiners: Pallavi Sirjoosingh and Ashok Kakkar**

First name (please write as legibly as possible within the boxes)

--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--

Last name

--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--

Student ID number

--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--

This booklet has 8 questions on 10 pages (The last page can be used as a scrap page).

The staple must NOT be removed from the question booklet. Your intact question booklet must be returned at the end of the exam.

A separate booklet contains a periodic table, and datasheet.

Write your answers directly under the questions in the space provided. Do not write on the QR codes or near the top edges (as it may be cut off).

Please write **legibly and dark enough** for the scan to be readable. For students who use English as a second language, dictionaries are permitted. Calculators are allowed.

DO NOT ASK QUESTIONS OF THE INVIGILATORS about interpretations of the exam questions. If you have a concern about a question, please summarize it at the top of this page. All concerns will be evaluated after the exam. GOOD LUCK!

**Question 1 8 points** (*Assume all transitions are possible*)

a. 6 points Calculate the **energy (in kJ/mol)** required to ionize (completely remove) the electron from the **n=3** level of **He⁺ ion** (He: Helium). Show your work including equations. (3 sig. figs.)

b. 2 points

(I) Select the largest (longest) wavelength from the three choices below:

- i. Wavelength of photon to *completely remove* an electron from *1s orbital of Hydrogen atom*
- ii. Wavelength of photon to *completely remove* an electron from *1s orbital of Boron atom (B)*
- iii. Wavelength of photon to *completely remove* an electron from *1s orbital of Boron cation (B⁺)*

i or ii or iii

(II) Select the smallest (shortest) wavelength from the three choices below (*for H atom*):

- i. Wavelength of photon *emitted* for an electron transitioning from *2p to 1s orbital*
- ii. Wavelength of photon *absorbed* for an electron transitioning from *2s to 4p orbital*
- iii. Wavelength of photon *absorbed* for an electron transitioning from *1s to 3p orbital*

i or ii or iii



Question 2 8 points



A Chem110 student plots the graph shown on the left, from data obtained for the kinetic energy (eV) of the ejected electron versus frequency (s^{-1}) of the incident light, for an unknown metal. ($1\text{eV} = 1.6022 \times 10^{-19} \text{ J}$)

Based on the plot, answer the questions **a to c** below:

a. 1 point On the plot above, label the **threshold frequency**.

b. 2 points What is the value of the work potential (in J) of the unknown metal?

c. 3 points What is the *longest* wavelength (in nm) required to eject an electron from this metal's surface? Show your work including equation(s) used. (3 sig figs)

d. 2 points In the space provided below, plot the following

(Assume frequency is *above* threshold frequency):

- Kinetic energy* of an ejected electron versus the *intensity* of the incident light (left)
- Number of electrons* ejected versus the *frequency* of the incident light (right)



**Question 3** **6 points****a. 2 points**

In the ground state of Ge (Germanium; $Z = 32$) atom, **name** the following orbitals:

i. Orbital containing the electron that requires the maximum energy to be removed.

ii. Orbital containing the electron that experiences the minimum effective nuclear charge.

b. 2 points

An unknown anion (A^-) has the following *excited* electronic configuration: $[Ar] 4s^2 3d^3 4p^1$

(Ar: Argon; $Z = 18$)

Give the **condensed ground state electronic configuration** of the atom, **A** and the cation, A^{2+} .
Show your work. (Assume all electronic transitions are possible)

Electronic Configuration of A:

Electronic Configuration of A^{2+} :

c. 2 points

In a single atom, determine how many electrons can have the following set of quantum numbers?

(i) $n = 5; l = 3$

In a single atom, determine how many electrons can have the following set of quantum numbers?

(i) $n = 3; l = 2; m_s = +1/2$

**Question 4** **8 points**

Rank the following from 1 to 4 in **increasing** order. Provide the ranking in the boxes provided.

Note: Atomic number (Z) are provided for relevant atoms

a) Increasing (lowest to highest) order of *ionization energy*: (1= lowest; 4 = highest)

Mg

Mg: Z = 12

Rb

Rb: Z = 37

Cs

Cs: Z = 55

K

K : Z = 19

b) Increasing (lowest to highest) order of *bond angle*: (1= lowest; 4 = highest)

H₂OCCl₄XeF₄

HCN

c) Increasing (smallest to largest) order of *ionic size* (1= smallest; 4 = largest)

S²⁻

S: Z = 16

O²⁻

O: Z = 8

Be²⁺

Be: Z = 4

Na⁺

Na: Z = 11

d) Increasing (lowest to highest) order of *bond energy* (1= lowest; 4 = highest)

P-F

P-Cl

P-I

P-Br

**Question 5** **12 points**

a. 9 points For the molecules HClO_4 and TeF_6 , draw the most stable *Lewis structure* showing appropriate shape and *dashed/wedged bonds*, based on VSEPR. Determine the **molecular geometry** around the central atom. For HClO_4 , label the *most polar bond*.



b. 3 points (Note: Lone pairs are not shown in structures below)

(i) For the three *resonance* structures (1, 2 and 3) shown below, answer the following question:



(I) Circle the *most contributing* resonance structure.

Structure 1

Structure 2

Structure 3

(ii) For the three structures given (4, 5, and 6) below, answer the following questions:



(I) Circle all the structures that represent *resonance structures* of each other:

Structure 4

Structure 5

Structure 6

(II) Among the resonance structures, circle the *most contributing* resonance structure.

Structure 4

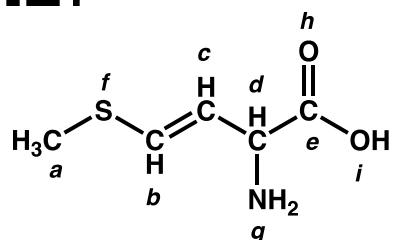
Structure 5

Structure 6

**Question 6** **6 points**

Using the data given below, calculate the **lattice energy** for SrO(s).

	ΔH° (kJ/mol)
$\text{Sr}(s) \longrightarrow \text{Sr}(g)$	+180
$\text{Sr}(g) \longrightarrow \text{Sr}^{2+}(g) + 2e^-$	+1468
$\text{O}_2(g) \longrightarrow 2\text{O}(g)$	+500
$\text{O}(g) + 2e^- \longrightarrow \text{O}^{2-}(g)$	+600
$\text{Sr}(s) + \frac{1}{2}\text{O}_2(g) \longrightarrow \text{SrO}(s)$	-550

**Question 7 10 points**

Consider the molecule given on the left, and answer questions (i) to (vi) using chemical bonding theories that we have learnt in class (valence bond theory, hybridization).

Note that the lone pairs are not shown in the structure:

i. Consider the carbon atoms marked in as ***a***, ***b***, ***c***, ***d***, and ***e*** in the molecule, and determine **hybridization** at each of the carbon centers. (2.5 points)

Atom	Hybridization
Carbon (<i>a</i>)	
Carbon (<i>b</i>)	
Carbon (<i>c</i>)	
Carbon (<i>d</i>)	
Carbon (<i>e</i>)	

ii. Now, consider the sulfur atom (S) marked as ***f***, nitrogen (N) atom (***g***), and oxygen(O) atoms (***h*** and ***i***), and determine their **hybridization**. (2 points)

Atom	Hybridization
Sulfur (<i>f</i>)	
Nitrogen (<i>g</i>)	
Oxygen (<i>h</i>)	
Oxygen (<i>i</i>)	

iii. **Describe** all the bonds formed by the carbon atoms ***a*** and ***b***, using sigma and pi designation as appropriate. (2 points)

Atom	Bonds formed (use sigma and pi as appropriate)
Carbon (<i>a</i>)	
Carbon (<i>b</i>)	

iv. How many total sp^3 -s sigma bonds are in the molecule? (1 point)

v. Determine **electronic** and **molecular geometry** at carbon (C) labeled ***e*** and sulfur (S) labeled ***f*** centres. (2 points)

Atom	Electronic Geometry	Molecular Geometry
Carbon (<i>e</i>)		
Sulfur (<i>f</i>)		

vi. Circle the value from the choices given below, that describes the **C-S-C bond angle at sulfur** (S) marked ***f***: (0.5 points)

>90° to <109.5° 109.5° >109.5 to <120° 120° >120° to <180°

**Question 8 12 points**

Using the **valence shell orbitals only** draw a **complete** molecular orbital (MO) diagram (with no s-p mixing) for the molecule **PS**, in the space provided below. **(6 points)**

Assume that 3s and 3p atomic orbitals will form molecular orbitals like those formed by the combination of 2s and 2p atomic orbitals that we learnt in class, and there is no s-p mixing.

Show **energy arrow**, **atomic orbitals** involved, **atom labels**, the **molecular orbitals** that are formed, **filled electrons** in the MO diagram etc. Make sure to label all the atomic and molecular orbitals.

Using the MO diagram above, answer the following questions:

- (i) Which one of the following will have the highest bond order? Circle the correct choice
A) PS^{1-} B) PS C) PS^{1+}
- (ii) Which one of the following will have the longest bond length? Circle the correct choice
A) PS^{1-} B) PS C) PS^{1+}
- (iii) Which of the following will be diamagnetic? Circle the correct choice.
A) PS^{1-} B) PS C) PS^{1+}
- (iv) Draw Lewis dot structure of the molecule with the shortest bond length (based on the MO diagram) from among the three (PS^{1-} , PS, PS^{1+}), and indicate the bond order calculated from the Lewis dot structure. **(3 points)**

Lewis Dot Structure

Bond Order



701EF982-5235-4BFF-B229-75C8644BCE09

midterm-exam-93388

#319 Page 10 of 10

SCRAP PAGE

**CHEMISTRY 110 –Midterm Exam****Thursday November 7, 2024, 6:30 – 8:30 pm**

Examiners: Pallavi Sirjoosingh and Ashok Kakkar

First name (please write as legibly as possible within the boxes)

--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--

Last name

--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--

Student ID number

--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--

This booklet has 8 questions on 10 pages (The last page can be used as a scrap page).

The staple must NOT be removed from the question booklet. Your intact question booklet must be returned at the end of the exam.

A separate booklet contains a periodic table, and datasheet.

Write your answers directly under the questions in the space provided. Do not write on the QR codes or near the top edges (as it may be cut off).

Please write **legibly and dark enough** for the scan to be readable. For students who use English as a second language, dictionaries are permitted. Calculators are allowed.

DO NOT ASK QUESTIONS OF THE INVIGILATORS about interpretations of the exam questions. If you have a concern about a question, please summarize it at the top of this page. All concerns will be evaluated after the exam. GOOD LUCK!

**Question 1 8 points** (*Assume all transitions are possible*)

a. 6 points Calculate the **energy (in kJ/mol)** required to ionize (completely remove) the electron from the **n=3** level of **He⁺ ion** (He: Helium). Show your work including equations. (3 sig. figs.)

b. 2 points

(I) Select the largest (longest) wavelength from the three choices below:

- i. Wavelength of photon to *completely remove* an electron from *1s orbital of Hydrogen atom*
- ii. Wavelength of photon to *completely remove* an electron from *1s orbital of Boron atom (B)*
- iii. Wavelength of photon to *completely remove* an electron from *1s orbital of Boron cation (B⁺)*

i or ii or iii

(II) Select the smallest (shortest) wavelength from the three choices below (*for H atom*):

- i. Wavelength of photon *emitted* for an electron transitioning from *2p to 1s orbital*
- ii. Wavelength of photon *absorbed* for an electron transitioning from *2s to 4p orbital*
- iii. Wavelength of photon *absorbed* for an electron transitioning from *1s to 3p orbital*

i or ii or iii



Question 2 8 points



A Chem110 student plots the graph shown on the left, from data obtained for the kinetic energy (eV) of the ejected electron versus frequency (s^{-1}) of the incident light, for an unknown metal. ($1\text{eV} = 1.6022 \times 10^{-19} \text{ J}$)

Based on the plot, answer the questions **a to c** below:

a. 1 point On the plot above, label the **threshold frequency**.

b. 2 points What is the value of the work potential (in J) of the unknown metal?

c. 3 points What is the *longest* wavelength (in nm) required to eject an electron from this metal's surface? Show your work including equation(s) used. (3 sig figs)

d. 2 points In the space provided below, plot the following

(Assume frequency is *above* threshold frequency):

- i. *Kinetic energy* of an ejected electron versus the *intensity* of the incident light (left)
- ii. *Number of electrons* ejected versus the *frequency* of the incident light (right)



**Question 3** **6 points****a. 2 points**

In the ground state of Ge (Germanium; $Z = 32$) atom, **name** the following orbitals:

i. Orbital containing the electron that requires the maximum energy to be removed.

ii. Orbital containing the electron that experiences the minimum effective nuclear charge.

b. 2 points

An unknown anion (A^-) has the following *excited* electronic configuration: $[Ar] 4s^2 3d^3 4p^1$

(Ar: Argon; $Z = 18$)

Give the **condensed ground state electronic configuration** of the atom, **A** and the cation, **A^{2+}** .
Show your work. (Assume all electronic transitions are possible)

Electronic Configuration of A:

Electronic Configuration of A^{2+} :

c. 2 points

In a single atom, determine how many electrons can have the following set of quantum numbers?

(i) $n = 5; l = 3$

In a single atom, determine how many electrons can have the following set of quantum numbers?

(i) $n = 3; l = 2; m_s = +1/2$

**Question 4** **8 points**

Rank the following from 1 to 4 in **increasing** order. Provide the ranking in the boxes provided.

Note: Atomic number (Z) are provided for relevant atoms

a) Increasing (lowest to highest) order of *ionization energy*: (1= lowest; 4 = highest)

Mg

Mg: Z = 12

Rb

Rb: Z = 37

Cs

Cs: Z = 55

K

K : Z = 19

b) Increasing (lowest to highest) order of *bond angle*: (1= lowest; 4 = highest)

H₂OCCl₄XeF₄

HCN

c) Increasing (smallest to largest) order of *ionic size* (1= smallest; 4 = largest)

S²⁻

S: Z = 16

O²⁻

O: Z = 8

Be²⁺

Be: Z = 4

Na⁺

Na: Z = 11

d) Increasing (lowest to highest) order of *bond energy* (1= lowest; 4 = highest)

P-F

P-Cl

P-I

P-Br

**Question 5** **12 points**

a. 9 points For the molecules HClO_4 and TeF_6 , draw the most stable *Lewis structure* showing appropriate shape and *dashed/wedged bonds*, based on VSEPR. Determine the **molecular geometry** around the central atom. For HClO_4 , label the *most polar bond*.

HClO_4 (there are no O-O bonds)

Label the most polar bond

Molecular Geometry around Cl

TeF_6

Molecular Geometry around Te

b. 3 points (Note: Lone pairs are not shown in structures below)

(i) For the three *resonance* structures (1, 2 and 3) shown below, answer the following question:



(I) Circle the *most contributing* resonance structure.

Structure 1

Structure 2

Structure 3

(ii) For the three structures given (4, 5, and 6) below, answer the following questions:



(I) Circle all the structures that represent *resonance structures* of each other:

Structure 4

Structure 5

Structure 6

(II) Among the resonance structures, circle the *most contributing* resonance structure.

Structure 4

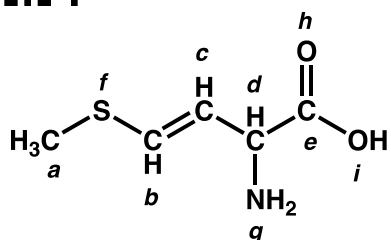
Structure 5

Structure 6

**Question 6** **6 points**

Using the data given below, calculate the **lattice energy** for SrO(s).

	ΔH° (kJ/mol)
$\text{Sr}(s) \longrightarrow \text{Sr}(g)$	+180
$\text{Sr}(g) \longrightarrow \text{Sr}^{2+}(g) + 2e^-$	+1468
$\text{O}_2(g) \longrightarrow 2\text{O}(g)$	+500
$\text{O}(g) + 2e^- \longrightarrow \text{O}^{2-}(g)$	+600
$\text{Sr}(s) + \frac{1}{2}\text{O}_2(g) \longrightarrow \text{SrO}(s)$	-550

**Question 7 10 points**

Consider the molecule given on the left, and answer questions (i) to (vi) using chemical bonding theories that we have learnt in class (valence bond theory, hybridization).

Note that the lone pairs are not shown in the structure:

i. Consider the carbon atoms marked in as ***a***, ***b***, ***c***, ***d***, and ***e*** in the molecule, and determine **hybridization** at each of the carbon centers. (2.5 points)

Atom	Hybridization
Carbon (<i>a</i>)	
Carbon (<i>b</i>)	
Carbon (<i>c</i>)	
Carbon (<i>d</i>)	
Carbon (<i>e</i>)	

ii. Now, consider the sulfur atom (S) marked as ***f***, nitrogen (N) atom (***g***), and oxygen(O) atoms (***h*** and ***i***), and determine their **hybridization**. (2 points)

Atom	Hybridization
Sulfur (<i>f</i>)	
Nitrogen (<i>g</i>)	
Oxygen (<i>h</i>)	
Oxygen (<i>i</i>)	

iii. **Describe** all the bonds formed by the carbon atoms ***a*** and ***b***, using sigma and pi designation as appropriate. (2 points)

Atom	Bonds formed (use sigma and pi as appropriate)
Carbon (<i>a</i>)	
Carbon (<i>b</i>)	

iv. How many total sp^3 -s sigma bonds are in the molecule? (1 point)

v. Determine **electronic** and **molecular geometry** at carbon (C) labeled ***e*** and sulfur (S) labeled ***f*** centres. (2 points)

Atom	Electronic Geometry	Molecular Geometry
Carbon (<i>e</i>)		
Sulfur (<i>f</i>)		

vi. Circle the value from the choices given below, that describes the **C-S-C bond angle at sulfur** (S) marked ***f***: (0.5 points)

>90° to <109.5° 109.5° >109.5 to <120° 120° >120° to <180°


Question 8 12 points

Using the **valence shell orbitals only** draw a **complete** molecular orbital (MO) diagram (with no s-p mixing) for the molecule **PS**, in the space provided below. **(6 points)**

Assume that 3s and 3p atomic orbitals will form molecular orbitals like those formed by the combination of 2s and 2p atomic orbitals that we learnt in class, and there is no s-p mixing.

Show **energy arrow**, **atomic orbitals** involved, **atom labels**, the **molecular orbitals** that are formed, **filled electrons** in the MO diagram etc. Make sure to **label all the atomic and molecular orbitals**.

Using the MO diagram above, answer the following questions:

- (i) Which one of the following will have the highest bond order? Circle the correct choice
 A) PS^{1-} B) PS C) PS^{1+}
- (ii) Which one of the following will have the longest bond length? Circle the correct choice
 A) PS^{1-} B) PS C) PS^{1+}
- (iii) Which of the following will be diamagnetic? Circle the correct choice.
 A) PS^{1-} B) PS C) PS^{1+}
- (iv) Draw Lewis dot structure of the molecule with the shortest bond length (based on the MO diagram) from among the three (PS^{1-} , PS, PS^{1+}), and indicate the bond order calculated from the Lewis dot structure. **(3 points)**

Lewis Dot Structure

Bond Order



3736C23E-6714-4F52-A047-099F79C3EB8D

midterm-exam-93388

#320 Page 10 of 10

SCRAP PAGE

**CHEMISTRY 110 –Midterm Exam****Thursday November 7, 2024, 6:30 – 8:30 pm****Examiners: Pallavi Sirjoosingh and Ashok Kakkar**

First name (please write as legibly as possible within the boxes)

--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--

Last name

--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--

Student ID number

--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--

This booklet has 8 questions on 10 pages (The last page can be used as a scrap page).

The staple must NOT be removed from the question booklet. Your intact question booklet must be returned at the end of the exam.

A separate booklet contains a periodic table, and datasheet.

Write your answers directly under the questions in the space provided. Do not write on the QR codes or near the top edges (as it may be cut off).

Please write **legibly and dark enough** for the scan to be readable. For students who use English as a second language, dictionaries are permitted. Calculators are allowed.

DO NOT ASK QUESTIONS OF THE INVIGILATORS about interpretations of the exam questions. If you have a concern about a question, please summarize it at the top of this page. All concerns will be evaluated after the exam. GOOD LUCK!

**Question 1 8 points** (*Assume all transitions are possible*)

a. 6 points Calculate the **energy (in kJ/mol)** required to ionize (completely remove) the electron from the **n=3** level of **He⁺ ion** (He: Helium). Show your work including equations. (3 sig. figs.)

b. 2 points

(I) Select the largest (longest) wavelength from the three choices below:

- i. Wavelength of photon to *completely remove* an electron from *1s orbital of Hydrogen atom*
- ii. Wavelength of photon to *completely remove* an electron from *1s orbital of Boron atom (B)*
- iii. Wavelength of photon to *completely remove* an electron from *1s orbital of Boron cation (B⁺)*

i or ii or iii

(II) Select the smallest (shortest) wavelength from the three choices below (*for H atom*):

- i. Wavelength of photon *emitted* for an electron transitioning from *2p to 1s orbital*
- ii. Wavelength of photon *absorbed* for an electron transitioning from *2s to 4p orbital*
- iii. Wavelength of photon *absorbed* for an electron transitioning from *1s to 3p orbital*

i or ii or iii



Question 2 8 points



A Chem110 student plots the graph shown on the left, from data obtained for the kinetic energy (eV) of the ejected electron versus frequency (s^{-1}) of the incident light, for an unknown metal. ($1\text{eV} = 1.6022 \times 10^{-19} \text{ J}$)

Based on the plot, answer the questions **a to c** below:

a. 1 point On the plot above, label the **threshold frequency**.

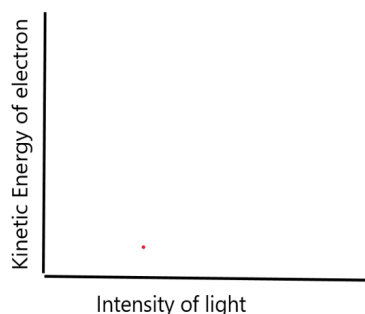
b. 2 points What is the value of the work potential (in J) of the unknown metal?

c. 3 points What is the *longest* wavelength (in nm) required to eject an electron from this metal's surface? Show your work including equation(s) used. (3 sig figs)

d. 2 points In the space provided below, plot the following

(Assume frequency is *above* threshold frequency):

- i. *Kinetic energy* of an ejected electron versus the *intensity* of the incident light (left)
- ii. *Number of electrons* ejected versus the *frequency* of the incident light (right)



**Question 3** **6 points****a. 2 points**

In the ground state of Ge (Germanium; $Z = 32$) atom, **name** the following orbitals:

i. Orbital containing the electron that requires the maximum energy to be removed.

ii. Orbital containing the electron that experiences the minimum effective nuclear charge.

b. 2 points

An unknown anion (A^-) has the following *excited* electronic configuration: $[Ar] 4s^2 3d^3 4p^1$

(Ar: Argon; $Z = 18$)

Give the **condensed ground state electronic configuration** of the atom, **A** and the cation, **A^{2+}** .
Show your work. (Assume all electronic transitions are possible)

Electronic Configuration of A:

Electronic Configuration of A^{2+} :

c. 2 points

In a single atom, determine how many electrons can have the following set of quantum numbers?

(i) $n = 5; l = 3$

In a single atom, determine how many electrons can have the following set of quantum numbers?

(i) $n = 3; l = 2; m_s = +1/2$

**Question 4** **8 points**

Rank the following from 1 to 4 in **increasing** order. Provide the ranking in the boxes provided.

Note: Atomic number (Z) are provided for relevant atoms

a) Increasing (lowest to highest) order of *ionization energy*: (1= lowest; 4 = highest)

Mg

Mg: Z = 12

Rb

Rb: Z = 37

Cs

Cs: Z = 55

K

K : Z = 19

b) Increasing (lowest to highest) order of *bond angle*: (1= lowest; 4 = highest)

H₂OCCl₄XeF₄

HCN

c) Increasing (smallest to largest) order of *ionic size* (1= smallest; 4 = largest)

S²⁻

S: Z = 16

O²⁻

O: Z = 8

Be²⁺

Be: Z = 4

Na⁺

Na: Z = 11

d) Increasing (lowest to highest) order of *bond energy* (1= lowest; 4 = highest)

P-F

P-Cl

P-I

P-Br

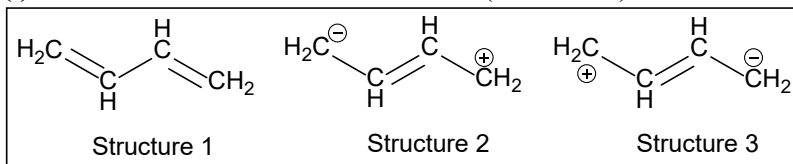
**Question 5** **12 points**

a. 9 points For the molecules HClO_4 and TeF_6 , draw the most stable *Lewis structure* showing appropriate shape and *dashed/wedged bonds*, based on VSEPR. Determine the **molecular geometry** around the central atom. For HClO_4 , label the *most polar bond*.



b. 3 points (Note: Lone pairs are not shown in structures below)

(i) For the three *resonance* structures (1, 2 and 3) shown below, answer the following question:



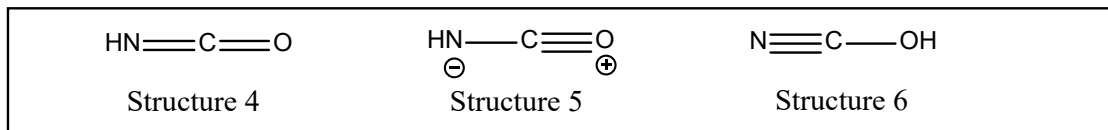
(I) Circle the *most contributing* resonance structure.

Structure 1

Structure 2

Structure 3

(ii) For the three structures given (4, 5, and 6) below, answer the following questions:



(I) Circle all the structures that represent *resonance structures* of each other:

Structure 4

Structure 5

Structure 6

(II) Among the resonance structures, circle the *most contributing* resonance structure.

Structure 4

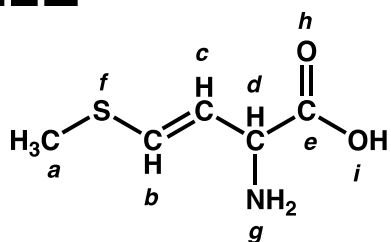
Structure 5

Structure 6

**Question 6** **6 points**

Using the data given below, calculate the **lattice energy** for SrO(s).

	ΔH° (kJ/mol)
$\text{Sr}(s) \longrightarrow \text{Sr}(g)$	+180
$\text{Sr}(g) \longrightarrow \text{Sr}^{2+}(g) + 2e^-$	+1468
$\text{O}_2(g) \longrightarrow 2\text{O}(g)$	+500
$\text{O}(g) + 2e^- \longrightarrow \text{O}^{2-}(g)$	+600
$\text{Sr}(s) + \frac{1}{2}\text{O}_2(g) \longrightarrow \text{SrO}(s)$	-550

**Question 7 10 points**

Consider the molecule given on the left, and answer questions (i) to (vi) using chemical bonding theories that we have learnt in class (valence bond theory, hybridization).

Note that the lone pairs are not shown in the structure:

i. Consider the carbon atoms marked in as ***a***, ***b***, ***c***, ***d***, and ***e*** in the molecule, and determine **hybridization** at each of the carbon centers. (2.5 points)

Atom	Hybridization
Carbon (<i>a</i>)	
Carbon (<i>b</i>)	
Carbon (<i>c</i>)	
Carbon (<i>d</i>)	
Carbon (<i>e</i>)	

ii. Now, consider the sulfur atom (S) marked as ***f***, nitrogen (N) atom (***g***), and oxygen(O) atoms (***h*** and ***i***), and determine their **hybridization**. (2 points)

Atom	Hybridization
Sulfur (<i>f</i>)	
Nitrogen (<i>g</i>)	
Oxygen (<i>h</i>)	
Oxygen (<i>i</i>)	

iii. **Describe** all the bonds formed by the carbon atoms ***a*** and ***b***, using sigma and pi designation as appropriate. (2 points)

Atom	Bonds formed (use sigma and pi as appropriate)
Carbon (<i>a</i>)	
Carbon (<i>b</i>)	

iv. How many total sp^3 -s sigma bonds are in the molecule? (1 point)

v. Determine **electronic** and **molecular geometry** at carbon (C) labeled ***e*** and sulfur (S) labeled ***f*** centres. (2 points)

Atom	Electronic Geometry	Molecular Geometry
Carbon (<i>e</i>)		
Sulfur (<i>f</i>)		

vi. Circle the value from the choices given below, that describes the **C-S-C bond angle at sulfur** (S) marked ***f***: (0.5 points)

>90° to <109.5° 109.5° >109.5 to <120° 120° >120° to <180°


Question 8 12 points

Using the **valence shell orbitals only** draw a **complete** molecular orbital (MO) diagram (with no s-p mixing) for the molecule **PS**, in the space provided below. **(6 points)**

Assume that 3s and 3p atomic orbitals will form molecular orbitals like those formed by the combination of 2s and 2p atomic orbitals that we learnt in class, and there is no s-p mixing.

Show **energy arrow**, **atomic orbitals** involved, **atom labels**, the **molecular orbitals** that are formed, **filled electrons** in the MO diagram etc. Make sure to label all the atomic and molecular orbitals.

Using the MO diagram above, answer the following questions:

- (i) Which one of the following will have the highest bond order? Circle the correct choice
 A) PS^{1-} B) PS C) PS^{1+}
- (ii) Which one of the following will have the longest bond length? Circle the correct choice
 A) PS^{1-} B) PS C) PS^{1+}
- (iii) Which of the following will be diamagnetic? Circle the correct choice.
 A) PS^{1-} B) PS C) PS^{1+}
- (iv) Draw Lewis dot structure of the molecule with the shortest bond length (based on the MO diagram) from among the three (PS^{1-} , PS, PS^{1+}), and indicate the bond order calculated from the Lewis dot structure. **(3 points)**

Lewis Dot Structure

Bond Order



3F135F8E-5095-4E30-8FC2-177F20F3B0F0

midterm-exam-93388

#321 Page 10 of 10

SCRAP PAGE

**CHEMISTRY 110 –Midterm Exam****Thursday November 7, 2024, 6:30 – 8:30 pm**

Examiners: Pallavi Sirjoosingh and Ashok Kakkar

First name (please write as legibly as possible within the boxes)

--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--

Last name

--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--

Student ID number

--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--

This booklet has 8 questions on 10 pages (The last page can be used as a scrap page).

The staple must NOT be removed from the question booklet. Your intact question booklet must be returned at the end of the exam.

A separate booklet contains a periodic table, and datasheet.

Write your answers directly under the questions in the space provided. Do not write on the QR codes or near the top edges (as it may be cut off).

Please write **legibly and dark enough** for the scan to be readable. For students who use English as a second language, dictionaries are permitted. Calculators are allowed.

DO NOT ASK QUESTIONS OF THE INVIGILATORS about interpretations of the exam questions. If you have a concern about a question, please summarize it at the top of this page. All concerns will be evaluated after the exam. GOOD LUCK!

**Question 1 8 points** (*Assume all transitions are possible*)

a. 6 points Calculate the **energy (in kJ/mol)** required to ionize (completely remove) the electron from the **n=3** level of **He⁺ ion** (He: Helium). Show your work including equations. (3 sig. figs.)

b. 2 points

(I) Select the largest (longest) wavelength from the three choices below:

- i. Wavelength of photon to *completely remove* an electron from *1s orbital of Hydrogen atom*
- ii. Wavelength of photon to *completely remove* an electron from *1s orbital of Boron atom (B)*
- iii. Wavelength of photon to *completely remove* an electron from *1s orbital of Boron cation (B⁺)*

i or ii or iii

(II) Select the smallest (shortest) wavelength from the three choices below (*for H atom*):

- i. Wavelength of photon *emitted* for an electron transitioning from *2p to 1s orbital*
- ii. Wavelength of photon *absorbed* for an electron transitioning from *2s to 4p orbital*
- iii. Wavelength of photon *absorbed* for an electron transitioning from *1s to 3p orbital*

i or ii or iii



Question 2 8 points



A Chem110 student plots the graph shown on the left, from data obtained for the kinetic energy (eV) of the ejected electron versus frequency (s^{-1}) of the incident light, for an unknown metal. ($1\text{eV} = 1.6022 \times 10^{-19} \text{ J}$)

Based on the plot, answer the questions **a to c** below:

a. 1 point On the plot above, label the **threshold frequency**.

b. 2 points What is the value of the work potential (in J) of the unknown metal?

c. 3 points What is the *longest* wavelength (in nm) required to eject an electron from this metal's surface? Show your work including equation(s) used. (3 sig figs)

d. 2 points In the space provided below, plot the following

(Assume frequency is *above* threshold frequency):

- i. *Kinetic energy* of an ejected electron versus the *intensity* of the incident light (left)
- ii. *Number of electrons* ejected versus the *frequency* of the incident light (right)



**Question 3** **6 points****a. 2 points**

In the ground state of Ge (Germanium; $Z = 32$) atom, **name** the following orbitals:

i. Orbital containing the electron that requires the maximum energy to be removed.

ii. Orbital containing the electron that experiences the minimum effective nuclear charge.

b. 2 points

An unknown anion (A^-) has the following *excited* electronic configuration: $[Ar] 4s^2 3d^3 4p^1$

(Ar: Argon; $Z = 18$)

Give the **condensed ground state electronic configuration** of the atom, **A** and the cation, **A^{2+}** .
Show your work. (Assume all electronic transitions are possible)

Electronic Configuration of A:

Electronic Configuration of A^{2+} :

c. 2 points

In a single atom, determine how many electrons can have the following set of quantum numbers?

(i) $n = 5; l = 3$

In a single atom, determine how many electrons can have the following set of quantum numbers?

(i) $n = 3; l = 2; m_s = +1/2$

**Question 4** **8 points**

Rank the following from 1 to 4 in **increasing** order. Provide the ranking in the boxes provided.

Note: Atomic number (Z) are provided for relevant atoms

a) Increasing (lowest to highest) order of *ionization energy*: (1= lowest; 4 = highest)

Mg

Mg: Z = 12

Rb

Rb: Z = 37

Cs

Cs: Z = 55

K

K : Z = 19

b) Increasing (lowest to highest) order of *bond angle*: (1= lowest; 4 = highest)

H₂OCCl₄XeF₄

HCN

c) Increasing (smallest to largest) order of *ionic size* (1= smallest; 4 = largest)

S²⁻

S: Z = 16

O²⁻

O: Z = 8

Be²⁺

Be: Z = 4

Na⁺

Na: Z = 11

d) Increasing (lowest to highest) order of *bond energy* (1= lowest; 4 = highest)

P-F

P-Cl

P-I

P-Br

**Question 5** **12 points**

a. 9 points For the molecules HClO_4 and TeF_6 , draw the most stable *Lewis structure* showing appropriate shape and *dashed/wedged bonds*, based on VSEPR. Determine the **molecular geometry** around the central atom. For HClO_4 , label the *most polar bond*.



b. 3 points (Note: Lone pairs are not shown in structures below)

(i) For the three *resonance* structures (1, 2 and 3) shown below, answer the following question:



(I) Circle the *most contributing* resonance structure.

Structure 1

Structure 2

Structure 3

(ii) For the three structures given (4, 5, and 6) below, answer the following questions:



(I) Circle all the structures that represent *resonance structures* of each other:

Structure 4

Structure 5

Structure 6

(II) Among the resonance structures, circle the *most contributing* resonance structure.

Structure 4

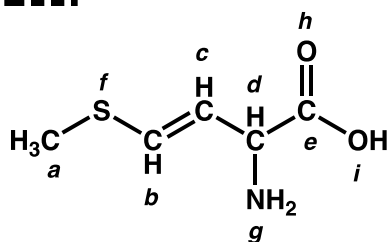
Structure 5

Structure 6

**Question 6** **6 points**

Using the data given below, calculate the **lattice energy** for SrO(s).

	ΔH° (kJ/mol)
$\text{Sr}(s) \longrightarrow \text{Sr}(g)$	+180
$\text{Sr}(g) \longrightarrow \text{Sr}^{2+}(g) + 2e^-$	+1468
$\text{O}_2(g) \longrightarrow 2\text{O}(g)$	+500
$\text{O}(g) + 2e^- \longrightarrow \text{O}^{2-}(g)$	+600
$\text{Sr}(s) + \frac{1}{2}\text{O}_2(g) \longrightarrow \text{SrO}(s)$	-550

**Question 7 10 points**

Consider the molecule given on the left, and answer questions (i) to (vi) using chemical bonding theories that we have learnt in class (valence bond theory, hybridization).

Note that the lone pairs are not shown in the structure:

i. Consider the carbon atoms marked in as ***a***, ***b***, ***c***, ***d***, and ***e*** in the molecule, and determine **hybridization** at each of the carbon centers. (2.5 points)

Atom	Hybridization
Carbon (<i>a</i>)	
Carbon (<i>b</i>)	
Carbon (<i>c</i>)	
Carbon (<i>d</i>)	
Carbon (<i>e</i>)	

ii. Now, consider the sulfur atom (S) marked as ***f***, nitrogen (N) atom (***g***), and oxygen(O) atoms (***h*** and ***i***), and determine their **hybridization**. (2 points)

Atom	Hybridization
Sulfur (<i>f</i>)	
Nitrogen (<i>g</i>)	
Oxygen (<i>h</i>)	
Oxygen (<i>i</i>)	

iii. **Describe** all the bonds formed by the carbon atoms ***a*** and ***b***, using sigma and pi designation as appropriate. (2 points)

Atom	Bonds formed (use sigma and pi as appropriate)
Carbon (<i>a</i>)	
Carbon (<i>b</i>)	

iv. How many total sp^3 -s sigma bonds are in the molecule? (1 point)

v. Determine **electronic** and **molecular geometry** at carbon (C) labeled ***e*** and sulfur (S) labeled ***f*** centres. (2 points)

Atom	Electronic Geometry	Molecular Geometry
Carbon (<i>e</i>)		
Sulfur (<i>f</i>)		

vi. Circle the value from the choices given below, that describes the **C-S-C bond angle at sulfur** (S) marked ***f***: (0.5 points)

>90° to <109.5° 109.5° >109.5 to <120° 120° >120° to <180°


Question 8 12 points

Using the **valence shell orbitals only** draw a **complete** molecular orbital (MO) diagram (with no s-p mixing) for the molecule **PS**, in the space provided below. **(6 points)**

Assume that 3s and 3p atomic orbitals will form molecular orbitals like those formed by the combination of 2s and 2p atomic orbitals that we learnt in class, and there is no s-p mixing.

Show **energy arrow**, **atomic orbitals** involved, **atom labels**, the **molecular orbitals** that are formed, **filled electrons** in the MO diagram etc. Make sure to label all the atomic and molecular orbitals.

Using the MO diagram above, answer the following questions:

- (i) Which one of the following will have the highest bond order? Circle the correct choice
 A) PS^{1-} B) PS C) PS^{1+}
- (ii) Which one of the following will have the longest bond length? Circle the correct choice
 A) PS^{1-} B) PS C) PS^{1+}
- (iii) Which of the following will be diamagnetic? Circle the correct choice.
 A) PS^{1-} B) PS C) PS^{1+}
- (iv) Draw Lewis dot structure of the molecule with the shortest bond length (based on the MO diagram) from among the three (PS^{1-} , PS, PS^{1+}), and indicate the bond order calculated from the Lewis dot structure. **(3 points)**

Lewis Dot Structure

Bond Order



E47B2C30-C2AE-4D35-81D2-971A6A86748D

midterm-exam-93388

#322 Page 10 of 10

SCRAP PAGE

**CHEMISTRY 110 –Midterm Exam****Thursday November 7, 2024, 6:30 – 8:30 pm****Examiners: Pallavi Sirjoosingh and Ashok Kakkar**

First name (please write as legibly as possible within the boxes)

--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--

Last name

--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--

Student ID number

--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--

This booklet has 8 questions on 10 pages (The last page can be used as a scrap page).

The staple must NOT be removed from the question booklet. Your intact question booklet must be returned at the end of the exam.

A separate booklet contains a periodic table, and datasheet.

Write your answers directly under the questions in the space provided. Do not write on the QR codes or near the top edges (as it may be cut off).

Please write **legibly and dark enough** for the scan to be readable. For students who use English as a second language, dictionaries are permitted. Calculators are allowed.

DO NOT ASK QUESTIONS OF THE INVIGILATORS about interpretations of the exam questions. If you have a concern about a question, please summarize it at the top of this page. All concerns will be evaluated after the exam. GOOD LUCK!

**Question 1 8 points** (*Assume all transitions are possible*)

a. 6 points Calculate the **energy (in kJ/mol)** required to ionize (completely remove) the electron from the **n=3** level of **He⁺ ion** (He: Helium). Show your work including equations. (3 sig. figs.)

b. 2 points

(I) Select the largest (longest) wavelength from the three choices below:

- i. Wavelength of photon to *completely remove* an electron from *1s orbital of Hydrogen atom*
- ii. Wavelength of photon to *completely remove* an electron from *1s orbital of Boron atom (B)*
- iii. Wavelength of photon to *completely remove* an electron from *1s orbital of Boron cation (B⁺)*

i or ii or iii

(II) Select the smallest (shortest) wavelength from the three choices below (*for H atom*):

- i. Wavelength of photon *emitted* for an electron transitioning from *2p to 1s orbital*
- ii. Wavelength of photon *absorbed* for an electron transitioning from *2s to 4p orbital*
- iii. Wavelength of photon *absorbed* for an electron transitioning from *1s to 3p orbital*

i or ii or iii



Question 2 8 points



A Chem110 student plots the graph shown on the left, from data obtained for the kinetic energy (eV) of the ejected electron versus frequency (s^{-1}) of the incident light, for an unknown metal. ($1\text{eV} = 1.6022 \times 10^{-19} \text{ J}$)

Based on the plot, answer the questions **a to c** below:

a. 1 point On the plot above, label the **threshold frequency**.

b. 2 points What is the value of the work potential (in J) of the unknown metal?

c. 3 points What is the *longest* wavelength (in nm) required to eject an electron from this metal's surface? Show your work including equation(s) used. (3 sig figs)

d. 2 points In the space provided below, plot the following

(Assume frequency is *above* threshold frequency):

- i. *Kinetic energy* of an ejected electron versus the *intensity* of the incident light (left)
- ii. *Number of electrons* ejected versus the *frequency* of the incident light (right)



**Question 3** **6 points****a. 2 points**

In the ground state of Ge (Germanium; $Z = 32$) atom, **name** the following orbitals:

i. Orbital containing the electron that requires the maximum energy to be removed.

ii. Orbital containing the electron that experiences the minimum effective nuclear charge.

b. 2 points

An unknown anion (A^-) has the following *excited* electronic configuration: $[Ar] 4s^2 3d^3 4p^1$

(Ar: Argon; $Z = 18$)

Give the **condensed ground state electronic configuration** of the atom, **A** and the cation, **A^{2+}** .
Show your work. (Assume all electronic transitions are possible)

Electronic Configuration of A:

Electronic Configuration of A^{2+} :

c. 2 points

In a single atom, determine how many electrons can have the following set of quantum numbers?

(i) $n = 5; l = 3$

In a single atom, determine how many electrons can have the following set of quantum numbers?

(i) $n = 3; l = 2; m_s = +1/2$

**Question 4** **8 points**

Rank the following from 1 to 4 in **increasing** order. Provide the ranking in the boxes provided.

Note: Atomic number (Z) are provided for relevant atoms

a) Increasing (lowest to highest) order of *ionization energy*: (1= lowest; 4 = highest)

Mg

Mg: Z = 12

Rb

Rb: Z = 37

Cs

Cs: Z = 55

K

K : Z = 19

b) Increasing (lowest to highest) order of *bond angle*: (1= lowest; 4 = highest)

H₂OCCl₄XeF₄

HCN

c) Increasing (smallest to largest) order of *ionic size* (1= smallest; 4 = largest)

S²⁻

S: Z = 16

O²⁻

O: Z = 8

Be²⁺

Be: Z = 4

Na⁺

Na: Z = 11

d) Increasing (lowest to highest) order of *bond energy* (1= lowest; 4 = highest)

P-F

P-Cl

P-I

P-Br

**Question 5** **12 points**

a. 9 points For the molecules HClO_4 and TeF_6 , draw the most stable *Lewis structure* showing appropriate shape and *dashed/wedged bonds*, based on VSEPR. Determine the **molecular geometry** around the central atom. For HClO_4 , label the *most polar bond*.



b. 3 points (Note: Lone pairs are not shown in structures below)

(i) For the three *resonance* structures (1, 2 and 3) shown below, answer the following question:



(I) Circle the *most contributing* resonance structure.

Structure 1

Structure 2

Structure 3

(ii) For the three structures given (4, 5, and 6) below, answer the following questions:



(I) Circle all the structures that represent *resonance structures* of each other:

Structure 4

Structure 5

Structure 6

(II) Among the resonance structures, circle the *most contributing* resonance structure.

Structure 4

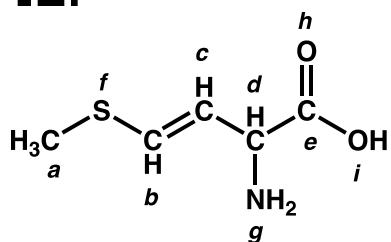
Structure 5

Structure 6

**Question 6** **6 points**

Using the data given below, calculate the **lattice energy** for SrO(s).

	ΔH° (kJ/mol)
$\text{Sr}(s) \longrightarrow \text{Sr}(g)$	+180
$\text{Sr}(g) \longrightarrow \text{Sr}^{2+}(g) + 2e^-$	+1468
$\text{O}_2(g) \longrightarrow 2\text{O}(g)$	+500
$\text{O}(g) + 2e^- \longrightarrow \text{O}^{2-}(g)$	+600
$\text{Sr}(s) + \frac{1}{2}\text{O}_2(g) \longrightarrow \text{SrO}(s)$	-550

**Question 7 10 points**

Consider the molecule given on the left, and answer questions (i) to (vi) using chemical bonding theories that we have learnt in class (valence bond theory, hybridization).

Note that the lone pairs are not shown in the structure:

i. Consider the carbon atoms marked in as ***a***, ***b***, ***c***, ***d***, and ***e*** in the molecule, and determine **hybridization** at each of the carbon centers. (2.5 points)

Atom	Hybridization
Carbon (<i>a</i>)	
Carbon (<i>b</i>)	
Carbon (<i>c</i>)	
Carbon (<i>d</i>)	
Carbon (<i>e</i>)	

ii. Now, consider the sulfur atom (S) marked as ***f***, nitrogen (N) atom (***g***), and oxygen(O) atoms (***h*** and ***i***), and determine their **hybridization**. (2 points)

Atom	Hybridization
Sulfur (<i>f</i>)	
Nitrogen (<i>g</i>)	
Oxygen (<i>h</i>)	
Oxygen (<i>i</i>)	

iii. **Describe** all the bonds formed by the carbon atoms ***a*** and ***b***, using sigma and pi designation as appropriate. (2 points)

Atom	Bonds formed (use sigma and pi as appropriate)
Carbon (<i>a</i>)	
Carbon (<i>b</i>)	

iv. How many total sp^3 -s sigma bonds are in the molecule? (1 point)

v. Determine **electronic** and **molecular geometry** at carbon (C) labeled ***e*** and sulfur (S) labeled ***f*** centres. (2 points)

Atom	Electronic Geometry	Molecular Geometry
Carbon (<i>e</i>)		
Sulfur (<i>f</i>)		

vi. Circle the value from the choices given below, that describes the **C-S-C bond angle at sulfur** (S) marked ***f***: (0.5 points)

>90° to <109.5° 109.5° >109.5 to <120° 120° >120° to <180°


Question 8 12 points

Using the **valence shell orbitals only** draw a **complete** molecular orbital (MO) diagram (with no s-p mixing) for the molecule **PS**, in the space provided below. **(6 points)**

Assume that 3s and 3p atomic orbitals will form molecular orbitals like those formed by the combination of 2s and 2p atomic orbitals that we learnt in class, and there is no s-p mixing.

Show **energy arrow**, **atomic orbitals** involved, **atom labels**, the **molecular orbitals** that are formed, **filled electrons** in the MO diagram etc. Make sure to label all the atomic and molecular orbitals.

Using the MO diagram above, answer the following questions:

- (i) Which one of the following will have the highest bond order? Circle the correct choice
 A) PS^{1-} B) PS C) PS^{1+}
- (ii) Which one of the following will have the longest bond length? Circle the correct choice
 A) PS^{1-} B) PS C) PS^{1+}
- (iii) Which of the following will be diamagnetic? Circle the correct choice.
 A) PS^{1-} B) PS C) PS^{1+}
- (iv) Draw Lewis dot structure of the molecule with the shortest bond length (based on the MO diagram) from among the three (PS^{1-} , PS, PS^{1+}), and indicate the bond order calculated from the Lewis dot structure. **(3 points)**

Lewis Dot Structure

Bond Order



133F6985-18FF-43BC-8F5C-BCD59B721C08

midterm-exam-93388

#323 Page 10 of 10

SCRAP PAGE

**CHEMISTRY 110 –Midterm Exam****Thursday November 7, 2024, 6:30 – 8:30 pm****Examiners: Pallavi Sirjoosingh and Ashok Kakkar**

First name (please write as legibly as possible within the boxes)

--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--

Last name

--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--

Student ID number

--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--

This booklet has 8 questions on 10 pages (The last page can be used as a scrap page).

The staple must NOT be removed from the question booklet. Your intact question booklet must be returned at the end of the exam.

A separate booklet contains a periodic table, and datasheet.

Write your answers directly under the questions in the space provided. Do not write on the QR codes or near the top edges (as it may be cut off).

Please write **legibly and dark enough** for the scan to be readable. For students who use English as a second language, dictionaries are permitted. Calculators are allowed.

DO NOT ASK QUESTIONS OF THE INVIGILATORS about interpretations of the exam questions. If you have a concern about a question, please summarize it at the top of this page. All concerns will be evaluated after the exam. GOOD LUCK!

**Question 1 8 points** (*Assume all transitions are possible*)

a. 6 points Calculate the **energy (in kJ/mol)** required to ionize (completely remove) the electron from the **n=3** level of **He⁺ ion** (He: Helium). Show your work including equations. (3 sig. figs.)

b. 2 points

(I) Select the largest (longest) wavelength from the three choices below:

- i. Wavelength of photon to *completely remove* an electron from *1s orbital of Hydrogen atom*
- ii. Wavelength of photon to *completely remove* an electron from *1s orbital of Boron atom (B)*
- iii. Wavelength of photon to *completely remove* an electron from *1s orbital of Boron cation (B⁺)*

i or ii or iii

(II) Select the smallest (shortest) wavelength from the three choices below (*for H atom*):

- i. Wavelength of photon *emitted* for an electron transitioning from *2p to 1s orbital*
- ii. Wavelength of photon *absorbed* for an electron transitioning from *2s to 4p orbital*
- iii. Wavelength of photon *absorbed* for an electron transitioning from *1s to 3p orbital*

i or ii or iii



Question 2 8 points



A Chem110 student plots the graph shown on the left, from data obtained for the kinetic energy (eV) of the ejected electron versus frequency (s^{-1}) of the incident light, for an unknown metal. ($1\text{eV} = 1.6022 \times 10^{-19} \text{ J}$)

Based on the plot, answer the questions **a to c** below:

a. 1 point On the plot above, label the **threshold frequency**.

b. 2 points What is the value of the work potential (in J) of the unknown metal?

c. 3 points What is the *longest* wavelength (in nm) required to eject an electron from this metal's surface? Show your work including equation(s) used. (3 sig figs)

d. 2 points In the space provided below, plot the following

(Assume frequency is *above* threshold frequency):

- Kinetic energy* of an ejected electron versus the *intensity* of the incident light (left)
- Number of electrons* ejected versus the *frequency* of the incident light (right)



**Question 3** **6 points****a. 2 points**

In the ground state of Ge (Germanium; $Z = 32$) atom, **name** the following orbitals:

i. Orbital containing the electron that requires the maximum energy to be removed.

ii. Orbital containing the electron that experiences the minimum effective nuclear charge.

b. 2 points

An unknown anion (A^-) has the following *excited* electronic configuration: $[Ar] 4s^2 3d^3 4p^1$

(Ar: Argon; $Z = 18$)

Give the **condensed ground state electronic configuration** of the atom, **A** and the cation, **A^{2+}** .
Show your work. (Assume all electronic transitions are possible)

Electronic Configuration of A:

Electronic Configuration of A^{2+} :

c. 2 points

In a single atom, determine how many electrons can have the following set of quantum numbers?

(i) $n = 5; l = 3$

In a single atom, determine how many electrons can have the following set of quantum numbers?

(i) $n = 3; l = 2; m_s = +1/2$

**Question 4** **8 points**

Rank the following from 1 to 4 in **increasing** order. Provide the ranking in the boxes provided.

Note: Atomic number (Z) are provided for relevant atoms

a) Increasing (lowest to highest) order of *ionization energy*: (1= lowest; 4 = highest)

Mg

Mg: Z = 12

Rb

Rb: Z = 37

Cs

Cs: Z = 55

K

K : Z = 19

b) Increasing (lowest to highest) order of *bond angle*: (1= lowest; 4 = highest)

H₂OCCl₄XeF₄

HCN

c) Increasing (smallest to largest) order of *ionic size* (1= smallest; 4 = largest)

S²⁻

S: Z = 16

O²⁻

O: Z = 8

Be²⁺

Be: Z = 4

Na⁺

Na: Z = 11

d) Increasing (lowest to highest) order of *bond energy* (1= lowest; 4 = highest)

P-F

P-Cl

P-I

P-Br

**Question 5** **12 points**

a. 9 points For the molecules HClO_4 and TeF_6 , draw the most stable *Lewis structure* showing appropriate shape and *dashed/wedged bonds*, based on VSEPR. Determine the **molecular geometry** around the central atom. For HClO_4 , label the *most polar bond*.



b. 3 points (Note: Lone pairs are not shown in structures below)

(i) For the three *resonance* structures (1, 2 and 3) shown below, answer the following question:



(I) Circle the *most contributing* resonance structure.

Structure 1

Structure 2

Structure 3

(ii) For the three structures given (4, 5, and 6) below, answer the following questions:



(I) Circle all the structures that represent *resonance structures* of each other:

Structure 4

Structure 5

Structure 6

(II) Among the resonance structures, circle the *most contributing* resonance structure.

Structure 4

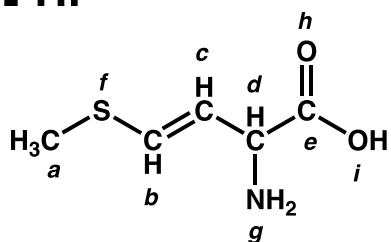
Structure 5

Structure 6

**Question 6** **6 points**

Using the data given below, calculate the **lattice energy** for SrO(s).

	ΔH° (kJ/mol)
$\text{Sr}(s) \longrightarrow \text{Sr}(g)$	+180
$\text{Sr}(g) \longrightarrow \text{Sr}^{2+}(g) + 2e^-$	+1468
$\text{O}_2(g) \longrightarrow 2\text{O}(g)$	+500
$\text{O}(g) + 2e^- \longrightarrow \text{O}^{2-}(g)$	+600
$\text{Sr}(s) + \frac{1}{2}\text{O}_2(g) \longrightarrow \text{SrO}(s)$	-550

**Question 7 10 points**

Consider the molecule given on the left, and answer questions (i) to (vi) using chemical bonding theories that we have learnt in class (valence bond theory, hybridization).

Note that the lone pairs are not shown in the structure:

i. Consider the carbon atoms marked in as ***a***, ***b***, ***c***, ***d***, and ***e*** in the molecule, and determine **hybridization** at each of the carbon centers. (2.5 points)

Atom	Hybridization
Carbon (<i>a</i>)	
Carbon (<i>b</i>)	
Carbon (<i>c</i>)	
Carbon (<i>d</i>)	
Carbon (<i>e</i>)	

ii. Now, consider the sulfur atom (S) marked as ***f***, nitrogen (N) atom (***g***), and oxygen(O) atoms (***h*** and ***i***), and determine their **hybridization**. (2 points)

Atom	Hybridization
Sulfur (<i>f</i>)	
Nitrogen (<i>g</i>)	
Oxygen (<i>h</i>)	
Oxygen (<i>i</i>)	

iii. **Describe** all the bonds formed by the carbon atoms ***a*** and ***b***, using sigma and pi designation as appropriate. (2 points)

Atom	Bonds formed (use sigma and pi as appropriate)
Carbon (<i>a</i>)	
Carbon (<i>b</i>)	

iv. How many total sp^3 -s sigma bonds are in the molecule? (1 point)

v. Determine **electronic** and **molecular geometry** at carbon (C) labeled ***e*** and sulfur (S) labeled ***f*** centres. (2 points)

Atom	Electronic Geometry	Molecular Geometry
Carbon (<i>e</i>)		
Sulfur (<i>f</i>)		

vi. Circle the value from the choices given below, that describes the **C-S-C bond angle at sulfur** (S) marked ***f***: (0.5 points)

>90° to <109.5° 109.5° >109.5 to <120° 120° >120° to <180°

**Question 8 12 points**

Using the **valence shell orbitals only** draw a **complete** molecular orbital (MO) diagram (with no s-p mixing) for the molecule **PS**, in the space provided below. **(6 points)**

Assume that 3s and 3p atomic orbitals will form molecular orbitals like those formed by the combination of 2s and 2p atomic orbitals that we learnt in class, and there is no s-p mixing.

Show **energy arrow**, **atomic orbitals** involved, **atom labels**, the **molecular orbitals** that are formed, **filled electrons** in the MO diagram etc. Make sure to label all the atomic and molecular orbitals.

Using the MO diagram above, answer the following questions:

- (i) Which one of the following will have the highest bond order? Circle the correct choice
A) PS^{1-} B) PS C) PS^{1+}
- (ii) Which one of the following will have the longest bond length? Circle the correct choice
A) PS^{1-} B) PS C) PS^{1+}
- (iii) Which of the following will be diamagnetic? Circle the correct choice.
A) PS^{1-} B) PS C) PS^{1+}
- (iv) Draw Lewis dot structure of the molecule with the shortest bond length (based on the MO diagram) from among the three (PS^{1-} , PS, PS^{1+}), and indicate the bond order calculated from the Lewis dot structure. **(3 points)**

Lewis Dot Structure

Bond Order



EA415FB4-6D77-450A-AF90-4C4D20354F39

midterm-exam-93388

#324 Page 10 of 10

SCRAP PAGE

**CHEMISTRY 110 –Midterm Exam****Thursday November 7, 2024, 6:30 – 8:30 pm****Examiners: Pallavi Sirjoosingh and Ashok Kakkar**

First name (please write as legibly as possible within the boxes)

--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--

Last name

--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--

Student ID number

--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--

This booklet has 8 questions on 10 pages (The last page can be used as a scrap page).

The staple must NOT be removed from the question booklet. Your intact question booklet must be returned at the end of the exam.

A separate booklet contains a periodic table, and datasheet.

Write your answers directly under the questions in the space provided. Do not write on the QR codes or near the top edges (as it may be cut off).

Please write **legibly and dark enough** for the scan to be readable. For students who use English as a second language, dictionaries are permitted. Calculators are allowed.

DO NOT ASK QUESTIONS OF THE INVIGILATORS about interpretations of the exam questions. If you have a concern about a question, please summarize it at the top of this page. All concerns will be evaluated after the exam. GOOD LUCK!

**Question 1 8 points** (*Assume all transitions are possible*)

a. 6 points Calculate the **energy (in kJ/mol)** required to ionize (completely remove) the electron from the **n=3** level of **He⁺ ion** (He: Helium). Show your work including equations. (3 sig. figs.)

b. 2 points

(I) Select the largest (longest) wavelength from the three choices below:

- i. Wavelength of photon to *completely remove* an electron from *1s orbital of Hydrogen atom*
- ii. Wavelength of photon to *completely remove* an electron from *1s orbital of Boron atom (B)*
- iii. Wavelength of photon to *completely remove* an electron from *1s orbital of Boron cation (B⁺)*

i or ii or iii

(II) Select the smallest (shortest) wavelength from the three choices below (*for H atom*):

- i. Wavelength of photon *emitted* for an electron transitioning from *2p to 1s orbital*
- ii. Wavelength of photon *absorbed* for an electron transitioning from *2s to 4p orbital*
- iii. Wavelength of photon *absorbed* for an electron transitioning from *1s to 3p orbital*

i or ii or iii



Question 2 8 points



A Chem110 student plots the graph shown on the left, from data obtained for the kinetic energy (eV) of the ejected electron versus frequency (s^{-1}) of the incident light, for an unknown metal. ($1\text{eV} = 1.6022 \times 10^{-19} \text{ J}$)

Based on the plot, answer the questions **a to c** below:

a. 1 point On the plot above, label the **threshold frequency**.

b. 2 points What is the value of the work potential (in J) of the unknown metal?

c. 3 points What is the *longest* wavelength (in nm) required to eject an electron from this metal's surface? Show your work including equation(s) used. (3 sig figs)

d. 2 points In the space provided below, plot the following

(Assume frequency is *above* threshold frequency):

- i. *Kinetic energy* of an ejected electron versus the *intensity* of the incident light (left)
- ii. *Number of electrons* ejected versus the *frequency* of the incident light (right)



**Question 3** **6 points****a. 2 points**

In the ground state of Ge (Germanium; $Z = 32$) atom, **name** the following orbitals:

i. Orbital containing the electron that requires the maximum energy to be removed.

ii. Orbital containing the electron that experiences the minimum effective nuclear charge.

b. 2 points

An unknown anion (A^-) has the following *excited* electronic configuration: $[Ar] 4s^2 3d^3 4p^1$

(Ar: Argon; $Z = 18$)

Give the **condensed ground state electronic configuration** of the atom, **A** and the cation, **A^{2+}** .
Show your work. (Assume all electronic transitions are possible)

Electronic Configuration of A:

Electronic Configuration of A^{2+} :

c. 2 points

In a single atom, determine how many electrons can have the following set of quantum numbers?

(i) $n = 5; l = 3$

In a single atom, determine how many electrons can have the following set of quantum numbers?

(i) $n = 3; l = 2; m_s = +1/2$

**Question 4** **8 points**

Rank the following from 1 to 4 in **increasing** order. Provide the ranking in the boxes provided.

Note: Atomic number (Z) are provided for relevant atoms

a) Increasing (lowest to highest) order of *ionization energy*: (1= lowest; 4 = highest)

Mg

Mg: Z = 12

Rb

Rb: Z = 37

Cs

Cs: Z = 55

K

K : Z = 19

b) Increasing (lowest to highest) order of *bond angle*: (1= lowest; 4 = highest)

H₂OCCl₄XeF₄

HCN

c) Increasing (smallest to largest) order of *ionic size* (1= smallest; 4 = largest)

S²⁻

S: Z = 16

O²⁻

O: Z = 8

Be²⁺

Be: Z = 4

Na⁺

Na: Z = 11

d) Increasing (lowest to highest) order of *bond energy* (1= lowest; 4 = highest)

P-F

P-Cl

P-I

P-Br

**Question 5** **12 points**

a. 9 points For the molecules HClO_4 and TeF_6 , draw the most stable *Lewis structure* showing appropriate shape and *dashed/wedged bonds*, based on VSEPR. Determine the **molecular geometry** around the central atom. For HClO_4 , label the *most polar bond*.



b. 3 points (Note: Lone pairs are not shown in structures below)

(i) For the three *resonance* structures (1, 2 and 3) shown below, answer the following question:



(I) Circle the *most contributing* resonance structure.

Structure 1

Structure 2

Structure 3

(ii) For the three structures given (4, 5, and 6) below, answer the following questions:



(I) Circle all the structures that represent *resonance structures* of each other:

Structure 4

Structure 5

Structure 6

(II) Among the resonance structures, circle the *most contributing* resonance structure.

Structure 4

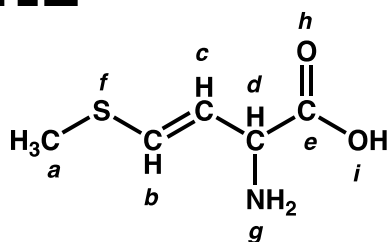
Structure 5

Structure 6

**Question 6** **6 points**

Using the data given below, calculate the **lattice energy** for SrO(s).

	ΔH° (kJ/mol)
$\text{Sr}(s) \longrightarrow \text{Sr}(g)$	+180
$\text{Sr}(g) \longrightarrow \text{Sr}^{2+}(g) + 2e^-$	+1468
$\text{O}_2(g) \longrightarrow 2\text{O}(g)$	+500
$\text{O}(g) + 2e^- \longrightarrow \text{O}^{2-}(g)$	+600
$\text{Sr}(s) + \frac{1}{2}\text{O}_2(g) \longrightarrow \text{SrO}(s)$	-550

**Question 7 10 points**

Consider the molecule given on the left, and answer questions (i) to (vi) using chemical bonding theories that we have learnt in class (valence bond theory, hybridization).

Note that the lone pairs are not shown in the structure:

i. Consider the carbon atoms marked in as ***a***, ***b***, ***c***, ***d***, and ***e*** in the molecule, and determine **hybridization** at each of the carbon centers. (2.5 points)

Atom	Hybridization
Carbon (<i>a</i>)	
Carbon (<i>b</i>)	
Carbon (<i>c</i>)	
Carbon (<i>d</i>)	
Carbon (<i>e</i>)	

ii. Now, consider the sulfur atom (S) marked as ***f***, nitrogen (N) atom (***g***), and oxygen(O) atoms (***h*** and ***i***), and determine their **hybridization**. (2 points)

Atom	Hybridization
Sulfur (<i>f</i>)	
Nitrogen (<i>g</i>)	
Oxygen (<i>h</i>)	
Oxygen (<i>i</i>)	

iii. **Describe** all the bonds formed by the carbon atoms ***a*** and ***b***, using sigma and pi designation as appropriate. (2 points)

Atom	Bonds formed (use sigma and pi as appropriate)
Carbon (<i>a</i>)	
Carbon (<i>b</i>)	

iv. How many total sp^3 -s sigma bonds are in the molecule? (1 point)

v. Determine **electronic** and **molecular geometry** at carbon (C) labeled ***e*** and sulfur (S) labeled ***f*** centres. (2 points)

Atom	Electronic Geometry	Molecular Geometry
Carbon (<i>e</i>)		
Sulfur (<i>f</i>)		

vi. Circle the value from the choices given below, that describes the **C-S-C bond angle at sulfur** (S) marked ***f***: (0.5 points)

>90° to <109.5° 109.5° >109.5 to <120° 120° >120° to <180°

**Question 8 12 points**

Using the **valence shell orbitals only** draw a **complete** molecular orbital (MO) diagram (with no s-p mixing) for the molecule **PS**, in the space provided below. **(6 points)**

Assume that 3s and 3p atomic orbitals will form molecular orbitals like those formed by the combination of 2s and 2p atomic orbitals that we learnt in class, and there is no s-p mixing.

Show **energy arrow**, **atomic orbitals** involved, **atom labels**, the **molecular orbitals** that are formed, **filled electrons** in the MO diagram etc. Make sure to label all the atomic and molecular orbitals.

Using the MO diagram above, answer the following questions:

- (i) Which one of the following will have the highest bond order? Circle the correct choice
A) PS^{1-} B) PS C) PS^{1+}
- (ii) Which one of the following will have the longest bond length? Circle the correct choice
A) PS^{1-} B) PS C) PS^{1+}
- (iii) Which of the following will be diamagnetic? Circle the correct choice.
A) PS^{1-} B) PS C) PS^{1+}
- (iv) Draw Lewis dot structure of the molecule with the shortest bond length (based on the MO diagram) from among the three (PS^{1-} , PS, PS^{1+}), and indicate the bond order calculated from the Lewis dot structure. **(3 points)**

Lewis Dot Structure

Bond Order



3BDA02D1-3A9D-4E2D-85A5-C69A26429742

midterm-exam-93388

#325 Page 10 of 10

SCRAP PAGE

**CHEMISTRY 110 –Midterm Exam****Thursday November 7, 2024, 6:30 – 8:30 pm**

Examiners: Pallavi Sirjoosingh and Ashok Kakkar

First name (please write as legibly as possible within the boxes)

--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--

Last name

--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--

Student ID number

--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--

This booklet has 8 questions on 10 pages (The last page can be used as a scrap page).

The staple must NOT be removed from the question booklet. Your intact question booklet must be returned at the end of the exam.

A separate booklet contains a periodic table, and datasheet.

Write your answers directly under the questions in the space provided. Do not write on the QR codes or near the top edges (as it may be cut off).

Please write **legibly and dark enough** for the scan to be readable. For students who use English as a second language, dictionaries are permitted. Calculators are allowed.

DO NOT ASK QUESTIONS OF THE INVIGILATORS about interpretations of the exam questions. If you have a concern about a question, please summarize it at the top of this page. All concerns will be evaluated after the exam. GOOD LUCK!

**Question 1 8 points** (*Assume all transitions are possible*)

a. 6 points Calculate the **energy (in kJ/mol)** required to ionize (completely remove) the electron from the **$n=3$** level of **He^+ ion** (He: Helium). Show your work including equations. (3 sig. figs.)

b. 2 points

(I) Select the largest (longest) wavelength from the three choices below:

- i. Wavelength of photon to *completely remove* an electron from *1s orbital of Hydrogen atom*
- ii. Wavelength of photon to *completely remove* an electron from *1s orbital of Boron atom (B)*
- iii. Wavelength of photon to *completely remove* an electron from *1s orbital of Boron cation (B^+)*

i or ii or iii

(II) Select the smallest (shortest) wavelength from the three choices below (*for H atom*):

- i. Wavelength of photon *emitted* for an electron transitioning from *2p to 1s orbital*
- ii. Wavelength of photon *absorbed* for an electron transitioning from *2s to 4p orbital*
- iii. Wavelength of photon *absorbed* for an electron transitioning from *1s to 3p orbital*

i or ii or iii



Question 2 8 points



A Chem110 student plots the graph shown on the left, from data obtained for the kinetic energy (eV) of the ejected electron versus frequency (s^{-1}) of the incident light, for an unknown metal. ($1\text{eV} = 1.6022 \times 10^{-19} \text{ J}$)

Based on the plot, answer the questions **a to c** below:

a. 1 point On the plot above, label the **threshold frequency**.

b. 2 points What is the value of the work potential (in J) of the unknown metal?

c. 3 points What is the *longest* wavelength (in nm) required to eject an electron from this metal's surface? Show your work including equation(s) used. (3 sig figs)

d. 2 points In the space provided below, plot the following

(Assume frequency is *above* threshold frequency):

- i. *Kinetic energy* of an ejected electron versus the *intensity* of the incident light (left)
- ii. *Number of electrons* ejected versus the *frequency* of the incident light (right)



**Question 3** **6 points****a. 2 points**

In the ground state of Ge (Germanium; $Z = 32$) atom, **name** the following orbitals:

i. Orbital containing the electron that requires the maximum energy to be removed.

ii. Orbital containing the electron that experiences the minimum effective nuclear charge.

b. 2 points

An unknown anion (A^-) has the following *excited* electronic configuration: $[Ar] 4s^2 3d^3 4p^1$

(Ar: Argon; $Z = 18$)

Give the **condensed ground state electronic configuration** of the atom, **A** and the cation, **A^{2+}** .
Show your work. (Assume all electronic transitions are possible)

Electronic Configuration of A:

Electronic Configuration of A^{2+} :

c. 2 points

In a single atom, determine how many electrons can have the following set of quantum numbers?

(i) $n = 5; l = 3$

In a single atom, determine how many electrons can have the following set of quantum numbers?

(i) $n = 3; l = 2; m_s = +1/2$

**Question 4** **8 points**

Rank the following from 1 to 4 in **increasing** order. Provide the ranking in the boxes provided.

Note: Atomic number (Z) are provided for relevant atoms

a) Increasing (lowest to highest) order of *ionization energy*: (1= lowest; 4 = highest)

Mg

Mg: Z = 12

Rb

Rb: Z = 37

Cs

Cs: Z = 55

K

K : Z = 19

b) Increasing (lowest to highest) order of *bond angle*: (1= lowest; 4 = highest)

H₂OCCl₄XeF₄

HCN

c) Increasing (smallest to largest) order of *ionic size* (1= smallest; 4 = largest)

S²⁻

S: Z = 16

O²⁻

O: Z = 8

Be²⁺

Be: Z = 4

Na⁺

Na: Z = 11

d) Increasing (lowest to highest) order of *bond energy* (1= lowest; 4 = highest)

P-F

P-Cl

P-I

P-Br

**Question 5** **12 points**

a. 9 points For the molecules HClO_4 and TeF_6 , draw the most stable *Lewis structure* showing appropriate shape and *dashed/wedged bonds*, based on VSEPR. Determine the **molecular geometry** around the central atom. For HClO_4 , label the *most polar bond*.

HClO_4 (there are no O-O bonds)

Label the most polar bond

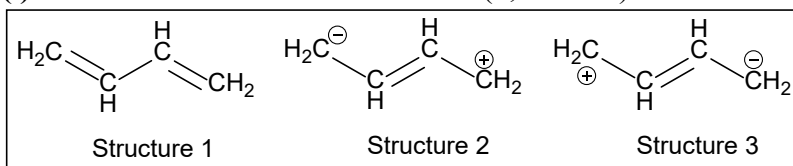
Molecular Geometry around Cl

TeF_6

Molecular Geometry around Te

b. 3 points (Note: Lone pairs are not shown in structures below)

(i) For the three *resonance* structures (1, 2 and 3) shown below, answer the following question:



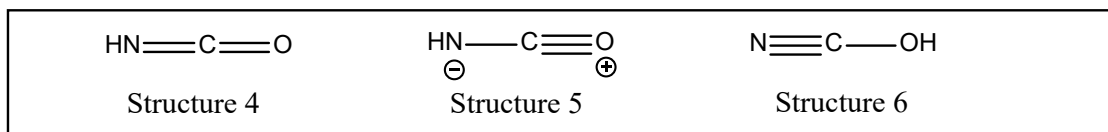
(I) Circle the *most contributing* resonance structure.

Structure 1

Structure 2

Structure 3

(ii) For the three structures given (4, 5, and 6) below, answer the following questions:



(I) Circle all the structures that represent *resonance structures* of each other:

Structure 4

Structure 5

Structure 6

(II) Among the resonance structures, circle the *most contributing* resonance structure.

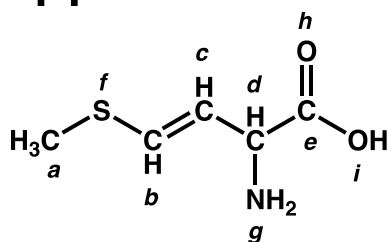
Structure 4

Structure 5

Structure 6

**Question 6** **6 points**Using the data given below, calculate the **lattice energy** for SrO(s).

	ΔH° (kJ/mol)
$\text{Sr}(s) \longrightarrow \text{Sr}(g)$	+180
$\text{Sr}(g) \longrightarrow \text{Sr}^{2+}(g) + 2e^-$	+1468
$\text{O}_2(g) \longrightarrow 2\text{O}(g)$	+500
$\text{O}(g) + 2e^- \longrightarrow \text{O}^{2-}(g)$	+600
$\text{Sr}(s) + \frac{1}{2}\text{O}_2(g) \longrightarrow \text{SrO}(s)$	-550

**Question 7 10 points**

Consider the molecule given on the left, and answer questions (i) to (vi) using chemical bonding theories that we have learnt in class (valence bond theory, hybridization).

Note that the lone pairs are not shown in the structure:

i. Consider the carbon atoms marked in as ***a***, ***b***, ***c***, ***d***, and ***e*** in the molecule, and determine **hybridization** at each of the carbon centers. (2.5 points)

Atom	Hybridization
Carbon (<i>a</i>)	
Carbon (<i>b</i>)	
Carbon (<i>c</i>)	
Carbon (<i>d</i>)	
Carbon (<i>e</i>)	

ii. Now, consider the sulfur atom (S) marked as ***f***, nitrogen (N) atom (***g***), and oxygen(O) atoms (***h*** and ***i***), and determine their **hybridization**. (2 points)

Atom	Hybridization
Sulfur (<i>f</i>)	
Nitrogen (<i>g</i>)	
Oxygen (<i>h</i>)	
Oxygen (<i>i</i>)	

iii. **Describe** all the bonds formed by the carbon atoms ***a*** and ***b***, using sigma and pi designation as appropriate. (2 points)

Atom	Bonds formed (use sigma and pi as appropriate)
Carbon (<i>a</i>)	
Carbon (<i>b</i>)	

iv. How many total sp^3 -s sigma bonds are in the molecule? (1 point)

v. Determine **electronic** and **molecular geometry** at carbon (C) labeled ***e*** and sulfur (S) labeled ***f*** centres. (2 points)

Atom	Electronic Geometry	Molecular Geometry
Carbon (<i>e</i>)		
Sulfur (<i>f</i>)		

vi. Circle the value from the choices given below, that describes the **C-S-C bond angle at sulfur** (S) marked ***f***: (0.5 points)

>90° to <109.5° 109.5° >109.5 to <120° 120° >120° to <180°


Question 8 12 points

Using the **valence shell orbitals only** draw a **complete** molecular orbital (MO) diagram (with no s-p mixing) for the molecule **PS**, in the space provided below. **(6 points)**

Assume that 3s and 3p atomic orbitals will form molecular orbitals like those formed by the combination of 2s and 2p atomic orbitals that we learnt in class, and there is no s-p mixing.

Show **energy arrow**, **atomic orbitals** involved, **atom labels**, the **molecular orbitals** that are formed, **filled electrons** in the MO diagram etc. Make sure to **label all the atomic and molecular orbitals**.

Using the MO diagram above, answer the following questions:

- (i) Which one of the following will have the highest bond order? Circle the correct choice
 A) PS^{1-} B) PS C) PS^{1+}
- (ii) Which one of the following will have the longest bond length? Circle the correct choice
 A) PS^{1-} B) PS C) PS^{1+}
- (iii) Which of the following will be diamagnetic? Circle the correct choice.
 A) PS^{1-} B) PS C) PS^{1+}
- (iv) Draw Lewis dot structure of the molecule with the shortest bond length (based on the MO diagram) from among the three (PS^{1-} , PS, PS^{1+}), and indicate the bond order calculated from the Lewis dot structure. **(3 points)**

Lewis Dot Structure

Bond Order



B87A3911-8157-4505-9796-6D49653451F6

midterm-exam-93388

#326 Page 10 of 10

SCRAP PAGE

**CHEMISTRY 110 –Midterm Exam****Thursday November 7, 2024, 6:30 – 8:30 pm**

Examiners: Pallavi Sirjoosingh and Ashok Kakkar

First name (please write as legibly as possible within the boxes)

--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--

Last name

--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--

Student ID number

--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--

This booklet has 8 questions on 10 pages (The last page can be used as a scrap page).

The staple must NOT be removed from the question booklet. Your intact question booklet must be returned at the end of the exam.

A separate booklet contains a periodic table, and datasheet.

Write your answers directly under the questions in the space provided. Do not write on the QR codes or near the top edges (as it may be cut off).

Please write **legibly and dark enough** for the scan to be readable. For students who use English as a second language, dictionaries are permitted. Calculators are allowed.

DO NOT ASK QUESTIONS OF THE INVIGILATORS about interpretations of the exam questions. If you have a concern about a question, please summarize it at the top of this page. All concerns will be evaluated after the exam. GOOD LUCK!

**Question 1 8 points** (*Assume all transitions are possible*)

a. 6 points Calculate the **energy (in kJ/mol)** required to ionize (completely remove) the electron from the **n=3** level of **He⁺ ion** (He: Helium). Show your work including equations. (3 sig. figs.)

b. 2 points

(I) Select the largest (longest) wavelength from the three choices below:

- i. Wavelength of photon to *completely remove* an electron from *1s orbital of Hydrogen atom*
- ii. Wavelength of photon to *completely remove* an electron from *1s orbital of Boron atom (B)*
- iii. Wavelength of photon to *completely remove* an electron from *1s orbital of Boron cation (B⁺)*

i or ii or iii

(II) Select the smallest (shortest) wavelength from the three choices below (*for H atom*):

- i. Wavelength of photon *emitted* for an electron transitioning from *2p to 1s orbital*
- ii. Wavelength of photon *absorbed* for an electron transitioning from *2s to 4p orbital*
- iii. Wavelength of photon *absorbed* for an electron transitioning from *1s to 3p orbital*

i or ii or iii



Question 2 8 points



A Chem110 student plots the graph shown on the left, from data obtained for the kinetic energy (eV) of the ejected electron versus frequency (s^{-1}) of the incident light, for an unknown metal. ($1\text{eV} = 1.6022 \times 10^{-19} \text{ J}$)

Based on the plot, answer the questions **a to c** below:

a. 1 point On the plot above, label the **threshold frequency**.

b. 2 points What is the value of the work potential (in J) of the unknown metal?

c. 3 points What is the *longest* wavelength (in nm) required to eject an electron from this metal's surface? Show your work including equation(s) used. (3 sig figs)

d. 2 points In the space provided below, plot the following

(Assume frequency is *above* threshold frequency):

- Kinetic energy* of an ejected electron versus the *intensity* of the incident light (left)
- Number of electrons* ejected versus the *frequency* of the incident light (right)



**Question 3** **6 points****a. 2 points**

In the ground state of Ge (Germanium; $Z = 32$) atom, **name** the following orbitals:

i. Orbital containing the electron that requires the maximum energy to be removed.

ii. Orbital containing the electron that experiences the minimum effective nuclear charge.

b. 2 points

An unknown anion (A^-) has the following *excited* electronic configuration: $[Ar] 4s^2 3d^3 4p^1$

(Ar: Argon; $Z = 18$)

Give the **condensed ground state electronic configuration** of the atom, **A** and the cation, A^{2+} .
Show your work. (Assume all electronic transitions are possible)

Electronic Configuration of A:

Electronic Configuration of A^{2+} :

c. 2 points

In a single atom, determine how many electrons can have the following set of quantum numbers?

(i) $n = 5; l = 3$

In a single atom, determine how many electrons can have the following set of quantum numbers?

(i) $n = 3; l = 2; m_s = +1/2$

**Question 4** **8 points**

Rank the following from 1 to 4 in **increasing** order. Provide the ranking in the boxes provided.

Note: Atomic number (Z) are provided for relevant atoms

a) Increasing (lowest to highest) order of *ionization energy*: (1= lowest; 4 = highest)

Mg

Mg: Z = 12

Rb

Rb: Z = 37

Cs

Cs: Z = 55

K

K : Z = 19

b) Increasing (lowest to highest) order of *bond angle*: (1= lowest; 4 = highest)

H₂OCCl₄XeF₄

HCN

c) Increasing (smallest to largest) order of *ionic size* (1= smallest; 4 = largest)

S²⁻

S: Z = 16

O²⁻

O: Z = 8

Be²⁺

Be: Z = 4

Na⁺

Na: Z = 11

d) Increasing (lowest to highest) order of *bond energy* (1= lowest; 4 = highest)

P-F

P-Cl

P-I

P-Br

**Question 5** **12 points**

a. 9 points For the molecules HClO_4 and TeF_6 , draw the most stable *Lewis structure* showing appropriate shape and *dashed/wedged bonds*, based on VSEPR. Determine the **molecular geometry** around the central atom. For HClO_4 , label the *most polar bond*.



b. 3 points (Note: Lone pairs are not shown in structures below)

(i) For the three *resonance* structures (1, 2 and 3) shown below, answer the following question:



(I) Circle the *most contributing* resonance structure.

Structure 1

Structure 2

Structure 3

(ii) For the three structures given (4, 5, and 6) below, answer the following questions:



(I) Circle all the structures that represent *resonance structures* of each other:

Structure 4

Structure 5

Structure 6

(II) Among the resonance structures, circle the *most contributing* resonance structure.

Structure 4

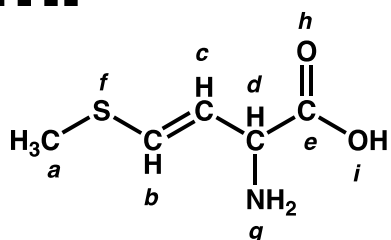
Structure 5

Structure 6

**Question 6** **6 points**

Using the data given below, calculate the **lattice energy** for SrO(s).

	ΔH° (kJ/mol)
$\text{Sr}(s) \longrightarrow \text{Sr}(g)$	+180
$\text{Sr}(g) \longrightarrow \text{Sr}^{2+}(g) + 2e^-$	+1468
$\text{O}_2(g) \longrightarrow 2\text{O}(g)$	+500
$\text{O}(g) + 2e^- \longrightarrow \text{O}^{2-}(g)$	+600
$\text{Sr}(s) + \frac{1}{2}\text{O}_2(g) \longrightarrow \text{SrO}(s)$	-550

**Question 7 10 points**

Consider the molecule given on the left, and answer questions (i) to (vi) using chemical bonding theories that we have learnt in class (valence bond theory, hybridization).

Note that the lone pairs are not shown in the structure:

i. Consider the carbon atoms marked in as ***a***, ***b***, ***c***, ***d***, and ***e*** in the molecule, and determine **hybridization** at each of the carbon centers. (2.5 points)

Atom	Hybridization
Carbon (<i>a</i>)	
Carbon (<i>b</i>)	
Carbon (<i>c</i>)	
Carbon (<i>d</i>)	
Carbon (<i>e</i>)	

ii. Now, consider the sulfur atom (S) marked as ***f***, nitrogen (N) atom (***g***), and oxygen(O) atoms (***h*** and ***i***), and determine their **hybridization**. (2 points)

Atom	Hybridization
Sulfur (<i>f</i>)	
Nitrogen (<i>g</i>)	
Oxygen (<i>h</i>)	
Oxygen (<i>i</i>)	

iii. **Describe** all the bonds formed by the carbon atoms ***a*** and ***b***, using sigma and pi designation as appropriate. (2 points)

Atom	Bonds formed (use sigma and pi as appropriate)
Carbon (<i>a</i>)	
Carbon (<i>b</i>)	

iv. How many total sp^3 -s sigma bonds are in the molecule? (1 point)

v. Determine **electronic** and **molecular geometry** at carbon (C) labeled ***e*** and sulfur (S) labeled ***f*** centres. (2 points)

Atom	Electronic Geometry	Molecular Geometry
Carbon (<i>e</i>)		
Sulfur (<i>f</i>)		

vi. Circle the value from the choices given below, that describes the **C-S-C bond angle at sulfur** (S) marked ***f***: (0.5 points)

>90° to <109.5° 109.5° >109.5 to <120° 120° >120° to <180°

**Question 8 12 points**

Using the **valence shell orbitals only** draw a **complete** molecular orbital (MO) diagram (with no s-p mixing) for the molecule **PS**, in the space provided below. **(6 points)**

Assume that 3s and 3p atomic orbitals will form molecular orbitals like those formed by the combination of 2s and 2p atomic orbitals that we learnt in class, and there is no s-p mixing.

Show **energy arrow**, **atomic orbitals** involved, **atom labels**, the **molecular orbitals** that are formed, **filled electrons** in the MO diagram etc. Make sure to label all the atomic and molecular orbitals.

Using the MO diagram above, answer the following questions:

- (i) Which one of the following will have the highest bond order? Circle the correct choice
A) PS^{1-} B) PS C) PS^{1+}
- (ii) Which one of the following will have the longest bond length? Circle the correct choice
A) PS^{1-} B) PS C) PS^{1+}
- (iii) Which of the following will be diamagnetic? Circle the correct choice.
A) PS^{1-} B) PS C) PS^{1+}
- (iv) Draw Lewis dot structure of the molecule with the shortest bond length (based on the MO diagram) from among the three (PS^{1-} , PS, PS^{1+}), and indicate the bond order calculated from the Lewis dot structure. **(3 points)**

Lewis Dot Structure

Bond Order



00CD075B-7E53-41B4-AA9C-1F5BA51B3144

midterm-exam-93388

#327 Page 10 of 10

SCRAP PAGE

**CHEMISTRY 110 –Midterm Exam****Thursday November 7, 2024, 6:30 – 8:30 pm**

Examiners: Pallavi Sirjoosingh and Ashok Kakkar

First name (please write as legibly as possible within the boxes)

--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--

Last name

--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--

Student ID number

--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--

This booklet has 8 questions on 10 pages (The last page can be used as a scrap page).

The staple must NOT be removed from the question booklet. Your intact question booklet must be returned at the end of the exam.

A separate booklet contains a periodic table, and datasheet.

Write your answers directly under the questions in the space provided. Do not write on the QR codes or near the top edges (as it may be cut off).

Please write **legibly and dark enough** for the scan to be readable. For students who use English as a second language, dictionaries are permitted. Calculators are allowed.

DO NOT ASK QUESTIONS OF THE INVIGILATORS about interpretations of the exam questions. If you have a concern about a question, please summarize it at the top of this page. All concerns will be evaluated after the exam. GOOD LUCK!

**Question 1 8 points** (*Assume all transitions are possible*)

a. 6 points Calculate the **energy (in kJ/mol)** required to ionize (completely remove) the electron from the **n=3** level of **He⁺ ion** (He: Helium). Show your work including equations. (3 sig. figs.)

b. 2 points

(I) Select the largest (longest) wavelength from the three choices below:

- i. Wavelength of photon to *completely remove* an electron from *1s orbital of Hydrogen atom*
- ii. Wavelength of photon to *completely remove* an electron from *1s orbital of Boron atom (B)*
- iii. Wavelength of photon to *completely remove* an electron from *1s orbital of Boron cation (B⁺)*

i or ii or iii

(II) Select the smallest (shortest) wavelength from the three choices below (*for H atom*):

- i. Wavelength of photon *emitted* for an electron transitioning from *2p to 1s orbital*
- ii. Wavelength of photon *absorbed* for an electron transitioning from *2s to 4p orbital*
- iii. Wavelength of photon *absorbed* for an electron transitioning from *1s to 3p orbital*

i or ii or iii



Question 2 8 points



A Chem110 student plots the graph shown on the left, from data obtained for the kinetic energy (eV) of the ejected electron versus frequency (s^{-1}) of the incident light, for an unknown metal. ($1\text{eV} = 1.6022 \times 10^{-19} \text{ J}$)

Based on the plot, answer the questions **a to c** below:

a. 1 point On the plot above, label the **threshold frequency**.

b. 2 points What is the value of the work potential (in J) of the unknown metal?

c. 3 points What is the *longest* wavelength (in nm) required to eject an electron from this metal's surface? Show your work including equation(s) used. (3 sig figs)

d. 2 points In the space provided below, plot the following

(Assume frequency is *above* threshold frequency):

- Kinetic energy* of an ejected electron versus the *intensity* of the incident light (left)
- Number of electrons* ejected versus the *frequency* of the incident light (right)



**Question 3** **6 points****a. 2 points**

In the ground state of Ge (Germanium; $Z = 32$) atom, **name** the following orbitals:

i. Orbital containing the electron that requires the maximum energy to be removed.

ii. Orbital containing the electron that experiences the minimum effective nuclear charge.

b. 2 points

An unknown anion (A^-) has the following *excited* electronic configuration: $[Ar] 4s^2 3d^3 4p^1$

(Ar: Argon; $Z = 18$)

Give the **condensed ground state electronic configuration** of the atom, **A** and the cation, **A^{2+}** .
Show your work. (Assume all electronic transitions are possible)

Electronic Configuration of A:

Electronic Configuration of A^{2+} :

c. 2 points

In a single atom, determine how many electrons can have the following set of quantum numbers?

(i) $n = 5; l = 3$

In a single atom, determine how many electrons can have the following set of quantum numbers?

(i) $n = 3; l = 2; m_s = +1/2$

**Question 4** **8 points**

Rank the following from 1 to 4 in **increasing** order. Provide the ranking in the boxes provided.

Note: Atomic number (Z) are provided for relevant atoms

a) Increasing (lowest to highest) order of *ionization energy*: (1= lowest; 4 = highest)

Mg

Mg: Z = 12

Rb

Rb: Z = 37

Cs

Cs: Z = 55

K

K : Z = 19

b) Increasing (lowest to highest) order of *bond angle*: (1= lowest; 4 = highest)

H₂OCCl₄XeF₄

HCN

c) Increasing (smallest to largest) order of *ionic size* (1= smallest; 4 = largest)

S²⁻

S: Z = 16

O²⁻

O: Z = 8

Be²⁺

Be: Z = 4

Na⁺

Na: Z = 11

d) Increasing (lowest to highest) order of *bond energy* (1= lowest; 4 = highest)

P-F

P-Cl

P-I

P-Br

**Question 5** **12 points**

a. 9 points For the molecules HClO_4 and TeF_6 , draw the most stable *Lewis structure* showing appropriate shape and *dashed/wedged bonds*, based on VSEPR. Determine the **molecular geometry** around the central atom. For HClO_4 , label the *most polar bond*.



b. 3 points (Note: Lone pairs are not shown in structures below)

(i) For the three *resonance* structures (1, 2 and 3) shown below, answer the following question:



(I) Circle the *most contributing* resonance structure.

Structure 1

Structure 2

Structure 3

(ii) For the three structures given (4, 5, and 6) below, answer the following questions:



(I) Circle all the structures that represent *resonance structures* of each other:

Structure 4

Structure 5

Structure 6

(II) Among the resonance structures, circle the *most contributing* resonance structure.

Structure 4

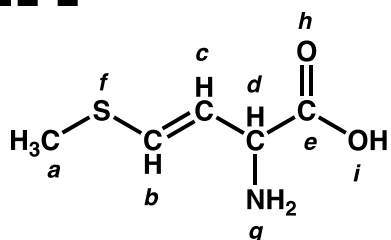
Structure 5

Structure 6

**Question 6** **6 points**

Using the data given below, calculate the **lattice energy** for SrO(s).

	ΔH° (kJ/mol)
$\text{Sr}(s) \longrightarrow \text{Sr}(g)$	+180
$\text{Sr}(g) \longrightarrow \text{Sr}^{2+}(g) + 2e^-$	+1468
$\text{O}_2(g) \longrightarrow 2\text{O}(g)$	+500
$\text{O}(g) + 2e^- \longrightarrow \text{O}^{2-}(g)$	+600
$\text{Sr}(s) + \frac{1}{2}\text{O}_2(g) \longrightarrow \text{SrO}(s)$	-550

**Question 7 10 points**

Consider the molecule given on the left, and answer questions (i) to (vi) using chemical bonding theories that we have learnt in class (valence bond theory, hybridization).

Note that the lone pairs are not shown in the structure:

i. Consider the carbon atoms marked in as ***a***, ***b***, ***c***, ***d***, and ***e*** in the molecule, and determine **hybridization** at each of the carbon centers. (2.5 points)

Atom	Hybridization
Carbon (<i>a</i>)	
Carbon (<i>b</i>)	
Carbon (<i>c</i>)	
Carbon (<i>d</i>)	
Carbon (<i>e</i>)	

ii. Now, consider the sulfur atom (S) marked as ***f***, nitrogen (N) atom (***g***), and oxygen(O) atoms (***h*** and ***i***), and determine their **hybridization**. (2 points)

Atom	Hybridization
Sulfur (<i>f</i>)	
Nitrogen (<i>g</i>)	
Oxygen (<i>h</i>)	
Oxygen (<i>i</i>)	

iii. **Describe** all the bonds formed by the carbon atoms ***a*** and ***b***, using sigma and pi designation as appropriate. (2 points)

Atom	Bonds formed (use sigma and pi as appropriate)
Carbon (<i>a</i>)	
Carbon (<i>b</i>)	

iv. How many total sp^3 -s sigma bonds are in the molecule? (1 point)

v. Determine **electronic** and **molecular geometry** at carbon (C) labeled ***e*** and sulfur (S) labeled ***f*** centres. (2 points)

Atom	Electronic Geometry	Molecular Geometry
Carbon (<i>e</i>)		
Sulfur (<i>f</i>)		

vi. Circle the value from the choices given below, that describes the **C-S-C bond angle at sulfur** (S) marked ***f***: (0.5 points)

>90° to <109.5° 109.5° >109.5 to <120° 120° >120° to <180°

**Question 8 12 points**

Using the **valence shell orbitals only** draw a **complete** molecular orbital (MO) diagram (with no s-p mixing) for the molecule **PS**, in the space provided below. **(6 points)**

Assume that 3s and 3p atomic orbitals will form molecular orbitals like those formed by the combination of 2s and 2p atomic orbitals that we learnt in class, and there is no s-p mixing.

Show **energy arrow**, **atomic orbitals** involved, **atom labels**, the **molecular orbitals** that are formed, **filled electrons** in the MO diagram etc. Make sure to label all the atomic and molecular orbitals.

Using the MO diagram above, answer the following questions:

- (i) Which one of the following will have the highest bond order? Circle the correct choice
A) PS^{1-} B) PS C) PS^{1+}
- (ii) Which one of the following will have the longest bond length? Circle the correct choice
A) PS^{1-} B) PS C) PS^{1+}
- (iii) Which of the following will be diamagnetic? Circle the correct choice.
A) PS^{1-} B) PS C) PS^{1+}
- (iv) Draw Lewis dot structure of the molecule with the shortest bond length (based on the MO diagram) from among the three (PS^{1-} , PS, PS^{1+}), and indicate the bond order calculated from the Lewis dot structure. **(3 points)**

Lewis Dot Structure

Bond Order



3D282176-B2FB-4DAA-94A8-906B8E5AED77

midterm-exam-93388

#328 Page 10 of 10

SCRAP PAGE

**CHEMISTRY 110 –Midterm Exam****Thursday November 7, 2024, 6:30 – 8:30 pm****Examiners: Pallavi Sirjoosingh and Ashok Kakkar**

First name (please write as legibly as possible within the boxes)

--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--

Last name

--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--

Student ID number

--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--

This booklet has 8 questions on 10 pages (The last page can be used as a scrap page).

The staple must NOT be removed from the question booklet. Your intact question booklet must be returned at the end of the exam.

A separate booklet contains a periodic table, and datasheet.

Write your answers directly under the questions in the space provided. Do not write on the QR codes or near the top edges (as it may be cut off).

Please write **legibly and dark enough** for the scan to be readable. For students who use English as a second language, dictionaries are permitted. Calculators are allowed.

DO NOT ASK QUESTIONS OF THE INVIGILATORS about interpretations of the exam questions. If you have a concern about a question, please summarize it at the top of this page. All concerns will be evaluated after the exam. GOOD LUCK!

**Question 1 8 points** (*Assume all transitions are possible*)

a. 6 points Calculate the **energy (in kJ/mol)** required to ionize (completely remove) the electron from the **n=3** level of **He⁺ ion** (He: Helium). Show your work including equations. (3 sig. figs.)

b. 2 points

(I) Select the largest (longest) wavelength from the three choices below:

- i. Wavelength of photon to *completely remove* an electron from *1s orbital of Hydrogen atom*
- ii. Wavelength of photon to *completely remove* an electron from *1s orbital of Boron atom (B)*
- iii. Wavelength of photon to *completely remove* an electron from *1s orbital of Boron cation (B⁺)*

i or ii or iii

(II) Select the smallest (shortest) wavelength from the three choices below (*for H atom*):

- i. Wavelength of photon *emitted* for an electron transitioning from *2p to 1s orbital*
- ii. Wavelength of photon *absorbed* for an electron transitioning from *2s to 4p orbital*
- iii. Wavelength of photon *absorbed* for an electron transitioning from *1s to 3p orbital*

i or ii or iii



Question 2 8 points



A Chem110 student plots the graph shown on the left, from data obtained for the kinetic energy (eV) of the ejected electron versus frequency (s^{-1}) of the incident light, for an unknown metal. ($1\text{eV} = 1.6022 \times 10^{-19} \text{ J}$)

Based on the plot, answer the questions **a to c** below:

a. 1 point On the plot above, label the **threshold frequency**.

b. 2 points What is the value of the work potential (in J) of the unknown metal?

c. 3 points What is the *longest* wavelength (in nm) required to eject an electron from this metal's surface? Show your work including equation(s) used. (3 sig figs)

d. 2 points In the space provided below, plot the following

(Assume frequency is *above* threshold frequency):

- Kinetic energy* of an ejected electron versus the *intensity* of the incident light (left)
- Number of electrons* ejected versus the *frequency* of the incident light (right)



**Question 3** **6 points****a. 2 points**

In the ground state of Ge (Germanium; $Z = 32$) atom, **name** the following orbitals:

i. Orbital containing the electron that requires the maximum energy to be removed.

ii. Orbital containing the electron that experiences the minimum effective nuclear charge.

b. 2 points

An unknown anion (A^-) has the following *excited* electronic configuration: $[Ar] 4s^2 3d^3 4p^1$

(Ar: Argon; $Z = 18$)

Give the **condensed ground state electronic configuration** of the atom, **A** and the cation, **A^{2+}** .
Show your work. (Assume all electronic transitions are possible)

Electronic Configuration of A:

Electronic Configuration of A^{2+} :

c. 2 points

In a single atom, determine how many electrons can have the following set of quantum numbers?

(i) $n = 5; l = 3$

In a single atom, determine how many electrons can have the following set of quantum numbers?

(i) $n = 3; l = 2; m_s = +1/2$

**Question 4** **8 points**

Rank the following from 1 to 4 in **increasing** order. Provide the ranking in the boxes provided.

Note: Atomic number (Z) are provided for relevant atoms

a) Increasing (lowest to highest) order of *ionization energy*: (1= lowest; 4 = highest)

Mg

Mg: Z = 12

Rb

Rb: Z = 37

Cs

Cs: Z = 55

K

K : Z = 19

b) Increasing (lowest to highest) order of *bond angle*: (1= lowest; 4 = highest)

H₂OCCl₄XeF₄

HCN

c) Increasing (smallest to largest) order of *ionic size* (1= smallest; 4 = largest)

S²⁻

S: Z = 16

O²⁻

O: Z = 8

Be²⁺

Be: Z = 4

Na⁺

Na: Z = 11

d) Increasing (lowest to highest) order of *bond energy* (1= lowest; 4 = highest)

P-F

P-Cl

P-I

P-Br

**Question 5** **12 points**

a. 9 points For the molecules HClO_4 and TeF_6 , draw the most stable *Lewis structure* showing appropriate shape and *dashed/wedged bonds*, based on VSEPR. Determine the **molecular geometry** around the central atom. For HClO_4 , label the *most polar bond*.



b. 3 points (Note: Lone pairs are not shown in structures below)

(i) For the three *resonance* structures (1, 2 and 3) shown below, answer the following question:



(I) Circle the *most contributing* resonance structure.

Structure 1

Structure 2

Structure 3

(ii) For the three structures given (4, 5, and 6) below, answer the following questions:



(I) Circle all the structures that represent *resonance structures* of each other:

Structure 4

Structure 5

Structure 6

(II) Among the resonance structures, circle the *most contributing* resonance structure.

Structure 4

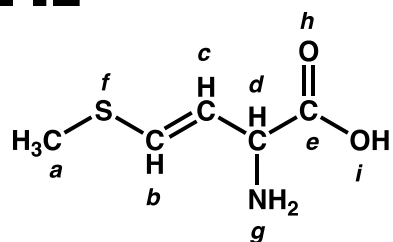
Structure 5

Structure 6

**Question 6** **6 points**

Using the data given below, calculate the **lattice energy** for SrO(s).

	ΔH° (kJ/mol)
$\text{Sr}(s) \longrightarrow \text{Sr}(g)$	+180
$\text{Sr}(g) \longrightarrow \text{Sr}^{2+}(g) + 2e^-$	+1468
$\text{O}_2(g) \longrightarrow 2\text{O}(g)$	+500
$\text{O}(g) + 2e^- \longrightarrow \text{O}^{2-}(g)$	+600
$\text{Sr}(s) + \frac{1}{2}\text{O}_2(g) \longrightarrow \text{SrO}(s)$	-550

**Question 7 10 points**

Consider the molecule given on the left, and answer questions (i) to (vi) using chemical bonding theories that we have learnt in class (valence bond theory, hybridization).

Note that the lone pairs are not shown in the structure:

i. Consider the carbon atoms marked in as ***a***, ***b***, ***c***, ***d***, and ***e*** in the molecule, and determine **hybridization** at each of the carbon centers. (2.5 points)

Atom	Hybridization
Carbon (<i>a</i>)	
Carbon (<i>b</i>)	
Carbon (<i>c</i>)	
Carbon (<i>d</i>)	
Carbon (<i>e</i>)	

ii. Now, consider the sulfur atom (S) marked as ***f***, nitrogen (N) atom (***g***), and oxygen(O) atoms (***h*** and ***i***), and determine their **hybridization**. (2 points)

Atom	Hybridization
Sulfur (<i>f</i>)	
Nitrogen (<i>g</i>)	
Oxygen (<i>h</i>)	
Oxygen (<i>i</i>)	

iii. **Describe** all the bonds formed by the carbon atoms ***a*** and ***b***, using sigma and pi designation as appropriate. (2 points)

Atom	Bonds formed (use sigma and pi as appropriate)
Carbon (<i>a</i>)	
Carbon (<i>b</i>)	

iv. How many total sp^3 -s sigma bonds are in the molecule? (1 point)

v. Determine **electronic** and **molecular geometry** at carbon (C) labeled ***e*** and sulfur (S) labeled ***f*** centres. (2 points)

Atom	Electronic Geometry	Molecular Geometry
Carbon (<i>e</i>)		
Sulfur (<i>f</i>)		

vi. Circle the value from the choices given below, that describes the **C-S-C bond angle at sulfur** (S) marked ***f***: (0.5 points)

$>90^\circ$ to $<109.5^\circ$ 109.5° >109.5 to $<120^\circ$ 120° $>120^\circ$ to $<180^\circ$


Question 8 12 points

Using the **valence shell orbitals only** draw a **complete** molecular orbital (MO) diagram (with no s-p mixing) for the molecule **PS**, in the space provided below. **(6 points)**

Assume that 3s and 3p atomic orbitals will form molecular orbitals like those formed by the combination of 2s and 2p atomic orbitals that we learnt in class, and there is no s-p mixing.

Show **energy arrow**, **atomic orbitals** involved, **atom labels**, the **molecular orbitals** that are formed, **filled electrons** in the MO diagram etc. Make sure to label all the atomic and molecular orbitals.

Using the MO diagram above, answer the following questions:

- (i) Which one of the following will have the highest bond order? Circle the correct choice
 A) PS^{1-} B) PS C) PS^{1+}
- (ii) Which one of the following will have the longest bond length? Circle the correct choice
 A) PS^{1-} B) PS C) PS^{1+}
- (iii) Which of the following will be diamagnetic? Circle the correct choice.
 A) PS^{1-} B) PS C) PS^{1+}
- (iv) Draw Lewis dot structure of the molecule with the shortest bond length (based on the MO diagram) from among the three (PS^{1-} , PS, PS^{1+}), and indicate the bond order calculated from the Lewis dot structure. **(3 points)**

Lewis Dot Structure

Bond Order



F1C26EC2-2DB4-4704-B216-23018E019102

midterm-exam-93388

#329 Page 10 of 10

SCRAP PAGE

**CHEMISTRY 110 –Midterm Exam****Thursday November 7, 2024, 6:30 – 8:30 pm****Examiners: Pallavi Sirjoosingh and Ashok Kakkar**

First name (please write as legibly as possible within the boxes)

--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--

Last name

--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--

Student ID number

--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--

This booklet has 8 questions on 10 pages (The last page can be used as a scrap page).

The staple must NOT be removed from the question booklet. Your intact question booklet must be returned at the end of the exam.

A separate booklet contains a periodic table, and datasheet.

Write your answers directly under the questions in the space provided. Do not write on the QR codes or near the top edges (as it may be cut off).

Please write **legibly and dark enough** for the scan to be readable. For students who use English as a second language, dictionaries are permitted. Calculators are allowed.

DO NOT ASK QUESTIONS OF THE INVIGILATORS about interpretations of the exam questions. If you have a concern about a question, please summarize it at the top of this page. All concerns will be evaluated after the exam. GOOD LUCK!

**Question 1 8 points** (*Assume all transitions are possible*)

a. 6 points Calculate the **energy (in kJ/mol)** required to ionize (completely remove) the electron from the **$n=3$** level of **He^+ ion** (He: Helium). Show your work including equations. (3 sig. figs.)

b. 2 points

(I) Select the largest (longest) wavelength from the three choices below:

- i. Wavelength of photon to *completely remove* an electron from *1s orbital of Hydrogen atom*
- ii. Wavelength of photon to *completely remove* an electron from *1s orbital of Boron atom (B)*
- iii. Wavelength of photon to *completely remove* an electron from *1s orbital of Boron cation (B^+)*

i or ii or iii

(II) Select the smallest (shortest) wavelength from the three choices below (*for H atom*):

- i. Wavelength of photon *emitted* for an electron transitioning from *2p to 1s orbital*
- ii. Wavelength of photon *absorbed* for an electron transitioning from *2s to 4p orbital*
- iii. Wavelength of photon *absorbed* for an electron transitioning from *1s to 3p orbital*

i or ii or iii



Question 2 8 points



A Chem110 student plots the graph shown on the left, from data obtained for the kinetic energy (eV) of the ejected electron versus frequency (s^{-1}) of the incident light, for an unknown metal. ($1\text{eV} = 1.6022 \times 10^{-19} \text{ J}$)

Based on the plot, answer the questions **a to c** below:

a. 1 point On the plot above, label the **threshold frequency**.

b. 2 points What is the value of the work potential (in J) of the unknown metal?

c. 3 points What is the *longest* wavelength (in nm) required to eject an electron from this metal's surface? Show your work including equation(s) used. (3 sig figs)

d. 2 points In the space provided below, plot the following

(Assume frequency is *above* threshold frequency):

- Kinetic energy* of an ejected electron versus the *intensity* of the incident light (left)
- Number of electrons* ejected versus the *frequency* of the incident light (right)



**Question 3** **6 points****a. 2 points**

In the ground state of Ge (Germanium; $Z = 32$) atom, **name** the following orbitals:

i. Orbital containing the electron that requires the maximum energy to be removed.

ii. Orbital containing the electron that experiences the minimum effective nuclear charge.

b. 2 points

An unknown anion (A^-) has the following *excited* electronic configuration: $[Ar] 4s^2 3d^3 4p^1$

(Ar: Argon; $Z = 18$)

Give the **condensed ground state electronic configuration** of the atom, **A** and the cation, **A^{2+}** .
Show your work. (Assume all electronic transitions are possible)

Electronic Configuration of A:

Electronic Configuration of A^{2+} :

c. 2 points

In a single atom, determine how many electrons can have the following set of quantum numbers?

(i) $n = 5; l = 3$

In a single atom, determine how many electrons can have the following set of quantum numbers?

(i) $n = 3; l = 2; m_s = +1/2$

**Question 4** **8 points**

Rank the following from 1 to 4 in **increasing** order. Provide the ranking in the boxes provided.

Note: Atomic number (Z) are provided for relevant atoms

a) Increasing (lowest to highest) order of *ionization energy*: (1= lowest; 4 = highest)

Mg

Mg: Z = 12

Rb

Rb: Z = 37

Cs

Cs: Z = 55

K

K : Z = 19

b) Increasing (lowest to highest) order of *bond angle*: (1= lowest; 4 = highest)

H₂OCCl₄XeF₄

HCN

c) Increasing (smallest to largest) order of *ionic size* (1= smallest; 4 = largest)

S²⁻

S: Z = 16

O²⁻

O: Z = 8

Be²⁺

Be: Z = 4

Na⁺

Na: Z = 11

d) Increasing (lowest to highest) order of *bond energy* (1= lowest; 4 = highest)

P-F

P-Cl

P-I

P-Br

**Question 5** **12 points**

a. 9 points For the molecules HClO_4 and TeF_6 , draw the most stable *Lewis structure* showing appropriate shape and *dashed/wedged bonds*, based on VSEPR. Determine the **molecular geometry** around the central atom. For HClO_4 , label the *most polar bond*.

HClO_4 (there are no O-O bonds)

Label the most polar bond

Molecular Geometry around Cl

TeF_6

Molecular Geometry around Te

b. 3 points (Note: Lone pairs are not shown in structures below)

(i) For the three *resonance* structures (1, 2 and 3) shown below, answer the following question:



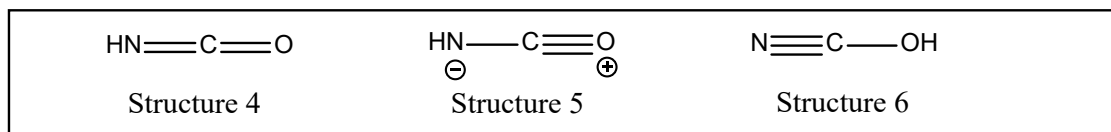
(I) Circle the *most contributing* resonance structure.

Structure 1

Structure 2

Structure 3

(ii) For the three structures given (4, 5, and 6) below, answer the following questions:



(I) Circle all the structures that represent *resonance structures* of each other:

Structure 4

Structure 5

Structure 6

(II) Among the resonance structures, circle the *most contributing* resonance structure.

Structure 4

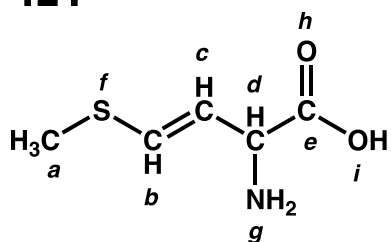
Structure 5

Structure 6

**Question 6** **6 points**

Using the data given below, calculate the **lattice energy** for SrO(s).

	ΔH° (kJ/mol)
$\text{Sr}(s) \longrightarrow \text{Sr}(g)$	+180
$\text{Sr}(g) \longrightarrow \text{Sr}^{2+}(g) + 2e^-$	+1468
$\text{O}_2(g) \longrightarrow 2\text{O}(g)$	+500
$\text{O}(g) + 2e^- \longrightarrow \text{O}^{2-}(g)$	+600
$\text{Sr}(s) + \frac{1}{2}\text{O}_2(g) \longrightarrow \text{SrO}(s)$	-550

**Question 7 10 points**

Consider the molecule given on the left, and answer questions (i) to (vi) using chemical bonding theories that we have learnt in class (valence bond theory, hybridization).

Note that the lone pairs are not shown in the structure:

i. Consider the carbon atoms marked in as ***a***, ***b***, ***c***, ***d***, and ***e*** in the molecule, and determine **hybridization** at each of the carbon centers. (2.5 points)

Atom	Hybridization
Carbon (<i>a</i>)	
Carbon (<i>b</i>)	
Carbon (<i>c</i>)	
Carbon (<i>d</i>)	
Carbon (<i>e</i>)	

ii. Now, consider the sulfur atom (S) marked as ***f***, nitrogen (N) atom (***g***), and oxygen(O) atoms (***h*** and ***i***), and determine their **hybridization**. (2 points)

Atom	Hybridization
Sulfur (<i>f</i>)	
Nitrogen (<i>g</i>)	
Oxygen (<i>h</i>)	
Oxygen (<i>i</i>)	

iii. **Describe** all the bonds formed by the carbon atoms ***a*** and ***b***, using sigma and pi designation as appropriate. (2 points)

Atom	Bonds formed (use sigma and pi as appropriate)
Carbon (<i>a</i>)	
Carbon (<i>b</i>)	

iv. How many total sp^3 -s sigma bonds are in the molecule? (1 point)

v. Determine **electronic** and **molecular geometry** at carbon (C) labeled ***e*** and sulfur (S) labeled ***f*** centres. (2 points)

Atom	Electronic Geometry	Molecular Geometry
Carbon (<i>e</i>)		
Sulfur (<i>f</i>)		

vi. Circle the value from the choices given below, that describes the **C-S-C bond angle at sulfur** (S) marked ***f***: (0.5 points)

>90° to <109.5° 109.5° >109.5 to <120° 120° >120° to <180°


Question 8 12 points

Using the **valence shell orbitals only** draw a **complete** molecular orbital (MO) diagram (with no s-p mixing) for the molecule **PS**, in the space provided below. **(6 points)**

Assume that 3s and 3p atomic orbitals will form molecular orbitals like those formed by the combination of 2s and 2p atomic orbitals that we learnt in class, and there is no s-p mixing.

Show **energy arrow**, **atomic orbitals** involved, **atom labels**, the **molecular orbitals** that are formed, **filled electrons** in the MO diagram etc. Make sure to label all the atomic and molecular orbitals.

Using the MO diagram above, answer the following questions:

- (i) Which one of the following will have the highest bond order? Circle the correct choice
 A) PS^{1-} B) PS C) PS^{1+}
- (ii) Which one of the following will have the longest bond length? Circle the correct choice
 A) PS^{1-} B) PS C) PS^{1+}
- (iii) Which of the following will be diamagnetic? Circle the correct choice.
 A) PS^{1-} B) PS C) PS^{1+}
- (iv) Draw Lewis dot structure of the molecule with the shortest bond length (based on the MO diagram) from among the three (PS^{1-} , PS, PS^{1+}), and indicate the bond order calculated from the Lewis dot structure. **(3 points)**

Lewis Dot Structure

Bond Order



3D791276-77E4-4B18-BA1D-1768C1B733A4

midterm-exam-93388

#330 Page 10 of 10

SCRAP PAGE

**CHEMISTRY 110 –Midterm Exam****Thursday November 7, 2024, 6:30 – 8:30 pm****Examiners: Pallavi Sirjoosingh and Ashok Kakkar**

First name (please write as legibly as possible within the boxes)

--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--

Last name

--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--

Student ID number

--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--

This booklet has 8 questions on 10 pages (The last page can be used as a scrap page).

The staple must NOT be removed from the question booklet. Your intact question booklet must be returned at the end of the exam.

A separate booklet contains a periodic table, and datasheet.

Write your answers directly under the questions in the space provided. Do not write on the QR codes or near the top edges (as it may be cut off).

Please write **legibly and dark enough** for the scan to be readable. For students who use English as a second language, dictionaries are permitted. Calculators are allowed.

DO NOT ASK QUESTIONS OF THE INVIGILATORS about interpretations of the exam questions. If you have a concern about a question, please summarize it at the top of this page. All concerns will be evaluated after the exam. GOOD LUCK!

**Question 1 8 points** (*Assume all transitions are possible*)

a. 6 points Calculate the **energy (in kJ/mol)** required to ionize (completely remove) the electron from the **$n=3$** level of **He^+ ion** (He: Helium). Show your work including equations. (3 sig. figs.)

b. 2 points

(I) Select the largest (longest) wavelength from the three choices below:

- i. Wavelength of photon to *completely remove* an electron from *1s orbital of Hydrogen atom*
- ii. Wavelength of photon to *completely remove* an electron from *1s orbital of Boron atom (B)*
- iii. Wavelength of photon to *completely remove* an electron from *1s orbital of Boron cation (B^+)*

i or ii or iii

(II) Select the smallest (shortest) wavelength from the three choices below (*for H atom*):

- i. Wavelength of photon *emitted* for an electron transitioning from *2p to 1s orbital*
- ii. Wavelength of photon *absorbed* for an electron transitioning from *2s to 4p orbital*
- iii. Wavelength of photon *absorbed* for an electron transitioning from *1s to 3p orbital*

i or ii or iii



Question 2 8 points



A Chem110 student plots the graph shown on the left, from data obtained for the kinetic energy (eV) of the ejected electron versus frequency (s^{-1}) of the incident light, for an unknown metal. ($1\text{eV} = 1.6022 \times 10^{-19} \text{ J}$)

Based on the plot, answer the questions **a to c** below:

a. 1 point On the plot above, label the **threshold frequency**.

b. 2 points What is the value of the work potential (in J) of the unknown metal?

c. 3 points What is the *longest* wavelength (in nm) required to eject an electron from this metal's surface? Show your work including equation(s) used. (3 sig figs)

d. 2 points In the space provided below, plot the following

(Assume frequency is *above* threshold frequency):

- Kinetic energy* of an ejected electron versus the *intensity* of the incident light (left)
- Number of electrons* ejected versus the *frequency* of the incident light (right)



**Question 3** **6 points****a. 2 points**

In the ground state of Ge (Germanium; $Z = 32$) atom, **name** the following orbitals:

i. Orbital containing the electron that requires the maximum energy to be removed.

ii. Orbital containing the electron that experiences the minimum effective nuclear charge.

b. 2 points

An unknown anion (A^-) has the following *excited* electronic configuration: $[Ar] 4s^2 3d^3 4p^1$

(Ar: Argon; $Z = 18$)

Give the **condensed ground state electronic configuration** of the atom, **A** and the cation, **A^{2+}** .
Show your work. (Assume all electronic transitions are possible)

Electronic Configuration of A:

Electronic Configuration of A^{2+} :

c. 2 points

In a single atom, determine how many electrons can have the following set of quantum numbers?

(i) $n = 5; l = 3$

In a single atom, determine how many electrons can have the following set of quantum numbers?

(i) $n = 3; l = 2; m_s = +1/2$

**Question 4** **8 points**

Rank the following from 1 to 4 in **increasing** order. Provide the ranking in the boxes provided.

Note: Atomic number (Z) are provided for relevant atoms

a) Increasing (lowest to highest) order of *ionization energy*: (1= lowest; 4 = highest)

Mg

Mg: Z = 12

Rb

Rb: Z = 37

Cs

Cs: Z = 55

K

K : Z = 19

b) Increasing (lowest to highest) order of *bond angle*: (1= lowest; 4 = highest)

H₂OCCl₄XeF₄

HCN

c) Increasing (smallest to largest) order of *ionic size* (1= smallest; 4 = largest)

S²⁻

S: Z = 16

O²⁻

O: Z = 8

Be²⁺

Be: Z = 4

Na⁺

Na: Z = 11

d) Increasing (lowest to highest) order of *bond energy* (1= lowest; 4 = highest)

P-F

P-Cl

P-I

P-Br

**Question 5** **12 points**

a. 9 points For the molecules HClO_4 and TeF_6 , draw the most stable *Lewis structure* showing appropriate shape and *dashed/wedged bonds*, based on VSEPR. Determine the **molecular geometry** around the central atom. For HClO_4 , label the *most polar bond*.



b. 3 points (Note: Lone pairs are not shown in structures below)

(i) For the three *resonance* structures (1, 2 and 3) shown below, answer the following question:



(I) Circle the *most contributing* resonance structure.

Structure 1

Structure 2

Structure 3

(ii) For the three structures given (4, 5, and 6) below, answer the following questions:



(I) Circle all the structures that represent *resonance structures* of each other:

Structure 4

Structure 5

Structure 6

(II) Among the resonance structures, circle the *most contributing* resonance structure.

Structure 4

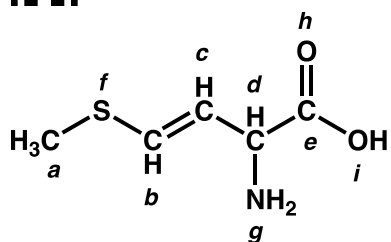
Structure 5

Structure 6

**Question 6** **6 points**

Using the data given below, calculate the **lattice energy** for SrO(s).

	ΔH° (kJ/mol)
$\text{Sr}(s) \longrightarrow \text{Sr}(g)$	+180
$\text{Sr}(g) \longrightarrow \text{Sr}^{2+}(g) + 2e^-$	+1468
$\text{O}_2(g) \longrightarrow 2\text{O}(g)$	+500
$\text{O}(g) + 2e^- \longrightarrow \text{O}^{2-}(g)$	+600
$\text{Sr}(s) + \frac{1}{2}\text{O}_2(g) \longrightarrow \text{SrO}(s)$	-550

**Question 7 10 points**

Consider the molecule given on the left, and answer questions (i) to (vi) using chemical bonding theories that we have learnt in class (valence bond theory, hybridization).

Note that the lone pairs are not shown in the structure:

i. Consider the carbon atoms marked in as ***a***, ***b***, ***c***, ***d***, and ***e*** in the molecule, and determine **hybridization** at each of the carbon centers. (2.5 points)

Atom	Hybridization
Carbon (<i>a</i>)	
Carbon (<i>b</i>)	
Carbon (<i>c</i>)	
Carbon (<i>d</i>)	
Carbon (<i>e</i>)	

ii. Now, consider the sulfur atom (S) marked as ***f***, nitrogen (N) atom (***g***), and oxygen(O) atoms (***h*** and ***i***), and determine their **hybridization**. (2 points)

Atom	Hybridization
Sulfur (<i>f</i>)	
Nitrogen (<i>g</i>)	
Oxygen (<i>h</i>)	
Oxygen (<i>i</i>)	

iii. **Describe** all the bonds formed by the carbon atoms ***a*** and ***b***, using sigma and pi designation as appropriate. (2 points)

Atom	Bonds formed (use sigma and pi as appropriate)
Carbon (<i>a</i>)	
Carbon (<i>b</i>)	

iv. How many total sp^3 -s sigma bonds are in the molecule? (1 point)

v. Determine **electronic** and **molecular geometry** at carbon (C) labeled ***e*** and sulfur (S) labeled ***f*** centres. (2 points)

Atom	Electronic Geometry	Molecular Geometry
Carbon (<i>e</i>)		
Sulfur (<i>f</i>)		

vi. Circle the value from the choices given below, that describes the **C-S-C bond angle at sulfur** (S) marked ***f***: (0.5 points)

$>90^\circ$ to $<109.5^\circ$ 109.5° >109.5 to $<120^\circ$ 120° $>120^\circ$ to $<180^\circ$

**Question 8 12 points**

Using the **valence shell orbitals only** draw a **complete** molecular orbital (MO) diagram (with no s-p mixing) for the molecule **PS**, in the space provided below. **(6 points)**

Assume that 3s and 3p atomic orbitals will form molecular orbitals like those formed by the combination of 2s and 2p atomic orbitals that we learnt in class, and there is no s-p mixing.

Show **energy arrow**, **atomic orbitals** involved, **atom labels**, the **molecular orbitals** that are formed, **filled electrons** in the MO diagram etc. Make sure to label all the atomic and molecular orbitals.

Using the MO diagram above, answer the following questions:

- (i) Which one of the following will have the highest bond order? Circle the correct choice
A) PS^{1-} B) PS C) PS^{1+}
- (ii) Which one of the following will have the longest bond length? Circle the correct choice
A) PS^{1-} B) PS C) PS^{1+}
- (iii) Which of the following will be diamagnetic? Circle the correct choice.
A) PS^{1-} B) PS C) PS^{1+}
- (iv) Draw Lewis dot structure of the molecule with the shortest bond length (based on the MO diagram) from among the three (PS^{1-} , PS, PS^{1+}), and indicate the bond order calculated from the Lewis dot structure. **(3 points)**

Lewis Dot Structure

Bond Order



00DAEFFE-12C2-43D1-B758-5DE548CE1567

midterm-exam-93388

#331 Page 10 of 10

SCRAP PAGE

**CHEMISTRY 110 –Midterm Exam****Thursday November 7, 2024, 6:30 – 8:30 pm**

Examiners: Pallavi Sirjoosingh and Ashok Kakkar

First name (please write as legibly as possible within the boxes)

--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--

Last name

--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--

Student ID number

--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--

This booklet has 8 questions on 10 pages (The last page can be used as a scrap page).

The staple must NOT be removed from the question booklet. Your intact question booklet must be returned at the end of the exam.

A separate booklet contains a periodic table, and datasheet.

Write your answers directly under the questions in the space provided. Do not write on the QR codes or near the top edges (as it may be cut off).

Please write **legibly and dark enough** for the scan to be readable. For students who use English as a second language, dictionaries are permitted. Calculators are allowed.

DO NOT ASK QUESTIONS OF THE INVIGILATORS about interpretations of the exam questions. If you have a concern about a question, please summarize it at the top of this page. All concerns will be evaluated after the exam. GOOD LUCK!

**Question 1 8 points** (*Assume all transitions are possible*)

a. 6 points Calculate the **energy (in kJ/mol)** required to ionize (completely remove) the electron from the **n=3** level of **He⁺ ion** (He: Helium). Show your work including equations. (3 sig. figs.)

b. 2 points

(I) Select the largest (longest) wavelength from the three choices below:

- i. Wavelength of photon to *completely remove* an electron from *1s orbital of Hydrogen atom*
- ii. Wavelength of photon to *completely remove* an electron from *1s orbital of Boron atom (B)*
- iii. Wavelength of photon to *completely remove* an electron from *1s orbital of Boron cation (B⁺)*

i or ii or iii

(II) Select the smallest (shortest) wavelength from the three choices below (*for H atom*):

- i. Wavelength of photon *emitted* for an electron transitioning from *2p to 1s orbital*
- ii. Wavelength of photon *absorbed* for an electron transitioning from *2s to 4p orbital*
- iii. Wavelength of photon *absorbed* for an electron transitioning from *1s to 3p orbital*

i or ii or iii



Question 2 8 points



A Chem110 student plots the graph shown on the left, from data obtained for the kinetic energy (eV) of the ejected electron versus frequency (s^{-1}) of the incident light, for an unknown metal. ($1\text{eV} = 1.6022 \times 10^{-19} \text{ J}$)

Based on the plot, answer the questions **a to c** below:

a. 1 point On the plot above, label the **threshold frequency**.

b. 2 points What is the value of the work potential (in J) of the unknown metal?

c. 3 points What is the *longest* wavelength (in nm) required to eject an electron from this metal's surface? Show your work including equation(s) used. (3 sig figs)

d. 2 points In the space provided below, plot the following

(Assume frequency is *above* threshold frequency):

- i. *Kinetic energy* of an ejected electron versus the *intensity* of the incident light (left)
- ii. *Number of electrons* ejected versus the *frequency* of the incident light (right)



**Question 3** **6 points****a. 2 points**

In the ground state of Ge (Germanium; $Z = 32$) atom, **name** the following orbitals:

i. Orbital containing the electron that requires the maximum energy to be removed.

ii. Orbital containing the electron that experiences the minimum effective nuclear charge.

b. 2 points

An unknown anion (A^-) has the following *excited* electronic configuration: $[Ar] 4s^2 3d^3 4p^1$

(Ar: Argon; $Z = 18$)

Give the **condensed ground state electronic configuration** of the atom, **A** and the cation, **A^{2+}** .
Show your work. (Assume all electronic transitions are possible)

Electronic Configuration of A:

Electronic Configuration of A^{2+} :

c. 2 points

In a single atom, determine how many electrons can have the following set of quantum numbers?

(i) $n = 5; l = 3$

In a single atom, determine how many electrons can have the following set of quantum numbers?

(i) $n = 3; l = 2; m_s = +1/2$

**Question 4** **8 points**

Rank the following from 1 to 4 in **increasing** order. Provide the ranking in the boxes provided.

Note: Atomic number (Z) are provided for relevant atoms

a) Increasing (lowest to highest) order of *ionization energy*: (1= lowest; 4 = highest)

Mg

Mg: Z = 12

Rb

Rb: Z = 37

Cs

Cs: Z = 55

K

K : Z = 19

b) Increasing (lowest to highest) order of *bond angle*: (1= lowest; 4 = highest)

H₂OCCl₄XeF₄

HCN

c) Increasing (smallest to largest) order of *ionic size* (1= smallest; 4 = largest)

S²⁻

S: Z = 16

O²⁻

O: Z = 8

Be²⁺

Be: Z = 4

Na⁺

Na: Z = 11

d) Increasing (lowest to highest) order of *bond energy* (1= lowest; 4 = highest)

P-F

P-Cl

P-I

P-Br

**Question 5** **12 points**

a. 9 points For the molecules HClO_4 and TeF_6 , draw the most stable *Lewis structure* showing appropriate shape and *dashed/wedged bonds*, based on VSEPR. Determine the **molecular geometry** around the central atom. For HClO_4 , label the *most polar bond*.

HClO_4 (there are no O-O bonds)

Label the most polar bond

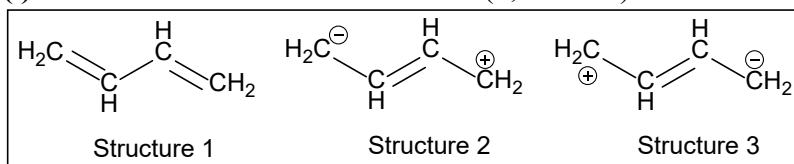
Molecular Geometry around Cl

TeF_6

Molecular Geometry around Te

b. 3 points (Note: Lone pairs are not shown in structures below)

(i) For the three *resonance* structures (1, 2 and 3) shown below, answer the following question:



(I) Circle the *most contributing* resonance structure.

Structure 1

Structure 2

Structure 3

(ii) For the three structures given (4, 5, and 6) below, answer the following questions:



(I) Circle all the structures that represent *resonance structures* of each other:

Structure 4

Structure 5

Structure 6

(II) Among the resonance structures, circle the *most contributing* resonance structure.

Structure 4

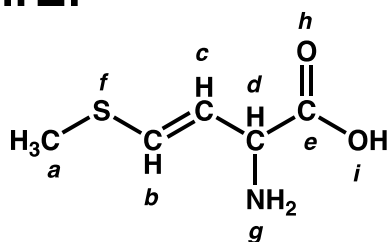
Structure 5

Structure 6

**Question 6** **6 points**

Using the data given below, calculate the **lattice energy** for SrO(s).

	ΔH° (kJ/mol)
$\text{Sr}(s) \longrightarrow \text{Sr}(g)$	+180
$\text{Sr}(g) \longrightarrow \text{Sr}^{2+}(g) + 2e^-$	+1468
$\text{O}_2(g) \longrightarrow 2\text{O}(g)$	+500
$\text{O}(g) + 2e^- \longrightarrow \text{O}^{2-}(g)$	+600
$\text{Sr}(s) + \frac{1}{2}\text{O}_2(g) \longrightarrow \text{SrO}(s)$	-550

**Question 7 10 points**

Consider the molecule given on the left, and answer questions (i) to (vi) using chemical bonding theories that we have learnt in class (valence bond theory, hybridization).

Note that the lone pairs are not shown in the structure:

i. Consider the carbon atoms marked in as ***a***, ***b***, ***c***, ***d***, and ***e*** in the molecule, and determine **hybridization** at each of the carbon centers. (2.5 points)

Atom	Hybridization
Carbon (<i>a</i>)	
Carbon (<i>b</i>)	
Carbon (<i>c</i>)	
Carbon (<i>d</i>)	
Carbon (<i>e</i>)	

ii. Now, consider the sulfur atom (S) marked as ***f***, nitrogen (N) atom (***g***), and oxygen(O) atoms (***h*** and ***i***), and determine their **hybridization**. (2 points)

Atom	Hybridization
Sulfur (<i>f</i>)	
Nitrogen (<i>g</i>)	
Oxygen (<i>h</i>)	
Oxygen (<i>i</i>)	

iii. **Describe** all the bonds formed by the carbon atoms ***a*** and ***b***, using sigma and pi designation as appropriate. (2 points)

Atom	Bonds formed (use sigma and pi as appropriate)
Carbon (<i>a</i>)	
Carbon (<i>b</i>)	

iv. How many total sp^3 -s sigma bonds are in the molecule? (1 point)

v. Determine **electronic** and **molecular geometry** at carbon (C) labeled ***e*** and sulfur (S) labeled ***f*** centres. (2 points)

Atom	Electronic Geometry	Molecular Geometry
Carbon (<i>e</i>)		
Sulfur (<i>f</i>)		

vi. Circle the value from the choices given below, that describes the **C-S-C bond angle at sulfur** (S) marked ***f***: (0.5 points)

>90° to <109.5° 109.5° >109.5 to <120° 120° >120° to <180°


Question 8 12 points

Using the **valence shell orbitals only** draw a **complete** molecular orbital (MO) diagram (with no s-p mixing) for the molecule **PS**, in the space provided below. **(6 points)**

Assume that 3s and 3p atomic orbitals will form molecular orbitals like those formed by the combination of 2s and 2p atomic orbitals that we learnt in class, and there is no s-p mixing.

Show **energy arrow**, **atomic orbitals** involved, **atom labels**, the **molecular orbitals** that are formed, **filled electrons** in the MO diagram etc. Make sure to label all the atomic and molecular orbitals.

Using the MO diagram above, answer the following questions:

- (i) Which one of the following will have the highest bond order? Circle the correct choice
 A) PS^{1-} B) PS C) PS^{1+}
- (ii) Which one of the following will have the longest bond length? Circle the correct choice
 A) PS^{1-} B) PS C) PS^{1+}
- (iii) Which of the following will be diamagnetic? Circle the correct choice.
 A) PS^{1-} B) PS C) PS^{1+}
- (iv) Draw Lewis dot structure of the molecule with the shortest bond length (based on the MO diagram) from among the three (PS^{1-} , PS, PS^{1+}), and indicate the bond order calculated from the Lewis dot structure. **(3 points)**

Lewis Dot Structure

Bond Order



4D3EFEC2-1445-4517-8CDB-1283C7A1F3B7

midterm-exam-93388

#332 Page 10 of 10

SCRAP PAGE

**CHEMISTRY 110 –Midterm Exam****Thursday November 7, 2024, 6:30 – 8:30 pm**

Examiners: Pallavi Sirjoosingh and Ashok Kakkar

First name (please write as legibly as possible within the boxes)

--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--

Last name

--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--

Student ID number

--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--

This booklet has 8 questions on 10 pages (The last page can be used as a scrap page).

The staple must NOT be removed from the question booklet. Your intact question booklet must be returned at the end of the exam.

A separate booklet contains a periodic table, and datasheet.

Write your answers directly under the questions in the space provided. Do not write on the QR codes or near the top edges (as it may be cut off).

Please write **legibly and dark enough** for the scan to be readable. For students who use English as a second language, dictionaries are permitted. Calculators are allowed.

DO NOT ASK QUESTIONS OF THE INVIGILATORS about interpretations of the exam questions. If you have a concern about a question, please summarize it at the top of this page. All concerns will be evaluated after the exam. GOOD LUCK!

**Question 1 8 points** (*Assume all transitions are possible*)

a. 6 points Calculate the **energy (in kJ/mol)** required to ionize (completely remove) the electron from the **n=3** level of **He⁺ ion** (He: Helium). Show your work including equations. (3 sig. figs.)

b. 2 points

(I) Select the largest (longest) wavelength from the three choices below:

- i. Wavelength of photon to *completely remove* an electron from *1s orbital of Hydrogen atom*
- ii. Wavelength of photon to *completely remove* an electron from *1s orbital of Boron atom (B)*
- iii. Wavelength of photon to *completely remove* an electron from *1s orbital of Boron cation (B⁺)*

i or ii or iii

(II) Select the smallest (shortest) wavelength from the three choices below (*for H atom*):

- i. Wavelength of photon *emitted* for an electron transitioning from *2p to 1s orbital*
- ii. Wavelength of photon *absorbed* for an electron transitioning from *2s to 4p orbital*
- iii. Wavelength of photon *absorbed* for an electron transitioning from *1s to 3p orbital*

i or ii or iii



Question 2 8 points



A Chem110 student plots the graph shown on the left, from data obtained for the kinetic energy (eV) of the ejected electron versus frequency (s^{-1}) of the incident light, for an unknown metal. ($1\text{eV} = 1.6022 \times 10^{-19} \text{ J}$)

Based on the plot, answer the questions **a to c** below:

a. 1 point On the plot above, label the **threshold frequency**.

b. 2 points What is the value of the work potential (in J) of the unknown metal?

c. 3 points What is the *longest* wavelength (in nm) required to eject an electron from this metal's surface? Show your work including equation(s) used. (3 sig figs)

d. 2 points In the space provided below, plot the following

(Assume frequency is *above* threshold frequency):

- i. *Kinetic energy* of an ejected electron versus the *intensity* of the incident light (left)
- ii. *Number of electrons* ejected versus the *frequency* of the incident light (right)



**Question 3** **6 points****a. 2 points**

In the ground state of Ge (Germanium; $Z = 32$) atom, **name** the following orbitals:

i. Orbital containing the electron that requires the maximum energy to be removed.

ii. Orbital containing the electron that experiences the minimum effective nuclear charge.

b. 2 points

An unknown anion (A^-) has the following *excited* electronic configuration: $[Ar] 4s^2 3d^3 4p^1$

(Ar: Argon; $Z = 18$)

Give the **condensed ground state electronic configuration** of the atom, **A** and the cation, **A^{2+}** .
Show your work. (Assume all electronic transitions are possible)

Electronic Configuration of A:

Electronic Configuration of A^{2+} :

c. 2 points

In a single atom, determine how many electrons can have the following set of quantum numbers?

(i) $n = 5; l = 3$

In a single atom, determine how many electrons can have the following set of quantum numbers?

(i) $n = 3; l = 2; m_s = +1/2$

**Question 4** **8 points**

Rank the following from 1 to 4 in **increasing** order. Provide the ranking in the boxes provided.

Note: Atomic number (Z) are provided for relevant atoms

a) Increasing (lowest to highest) order of *ionization energy*: (1= lowest; 4 = highest)

Mg

Mg: Z = 12

Rb

Rb: Z = 37

Cs

Cs: Z = 55

K

K : Z = 19

b) Increasing (lowest to highest) order of *bond angle*: (1= lowest; 4 = highest)

H₂OCCl₄XeF₄

HCN

c) Increasing (smallest to largest) order of *ionic size* (1= smallest; 4 = largest)

S²⁻

S: Z = 16

O²⁻

O: Z = 8

Be²⁺

Be: Z = 4

Na⁺

Na: Z = 11

d) Increasing (lowest to highest) order of *bond energy* (1= lowest; 4 = highest)

P-F

P-Cl

P-I

P-Br

**Question 5** **12 points**

a. 9 points For the molecules HClO_4 and TeF_6 , draw the most stable *Lewis structure* showing appropriate shape and *dashed/wedged bonds*, based on VSEPR. Determine the **molecular geometry** around the central atom. For HClO_4 , label the *most polar bond*.



b. 3 points (Note: Lone pairs are not shown in structures below)

(i) For the three *resonance* structures (1, 2 and 3) shown below, answer the following question:



(I) Circle the *most contributing* resonance structure.

Structure 1

Structure 2

Structure 3

(ii) For the three structures given (4, 5, and 6) below, answer the following questions:



(I) Circle all the structures that represent *resonance structures* of each other:

Structure 4

Structure 5

Structure 6

(II) Among the resonance structures, circle the *most contributing* resonance structure.

Structure 4

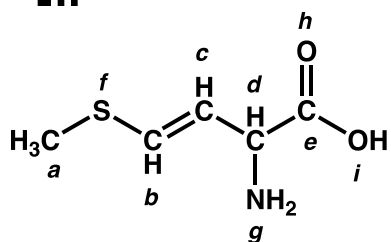
Structure 5

Structure 6

**Question 6** **6 points**

Using the data given below, calculate the **lattice energy** for SrO(s).

	ΔH° (kJ/mol)
$\text{Sr}(s) \longrightarrow \text{Sr}(g)$	+180
$\text{Sr}(g) \longrightarrow \text{Sr}^{2+}(g) + 2e^-$	+1468
$\text{O}_2(g) \longrightarrow 2\text{O}(g)$	+500
$\text{O}(g) + 2e^- \longrightarrow \text{O}^{2-}(g)$	+600
$\text{Sr}(s) + \frac{1}{2}\text{O}_2(g) \longrightarrow \text{SrO}(s)$	-550

**Question 7 10 points**

Consider the molecule given on the left, and answer questions (i) to (vi) using chemical bonding theories that we have learnt in class (valence bond theory, hybridization).

Note that the lone pairs are not shown in the structure:

i. Consider the carbon atoms marked in as ***a***, ***b***, ***c***, ***d***, and ***e*** in the molecule, and determine **hybridization** at each of the carbon centers. (2.5 points)

Atom	Hybridization
Carbon (<i>a</i>)	
Carbon (<i>b</i>)	
Carbon (<i>c</i>)	
Carbon (<i>d</i>)	
Carbon (<i>e</i>)	

ii. Now, consider the sulfur atom (S) marked as ***f***, nitrogen (N) atom (***g***), and oxygen(O) atoms (***h*** and ***i***), and determine their **hybridization**. (2 points)

Atom	Hybridization
Sulfur (<i>f</i>)	
Nitrogen (<i>g</i>)	
Oxygen (<i>h</i>)	
Oxygen (<i>i</i>)	

iii. **Describe** all the bonds formed by the carbon atoms ***a*** and ***b***, using sigma and pi designation as appropriate. (2 points)

Atom	Bonds formed (use sigma and pi as appropriate)
Carbon (<i>a</i>)	
Carbon (<i>b</i>)	

iv. How many total sp^3 -s sigma bonds are in the molecule? (1 point)

v. Determine **electronic** and **molecular geometry** at carbon (C) labeled ***e*** and sulfur (S) labeled ***f*** centres. (2 points)

Atom	Electronic Geometry	Molecular Geometry
Carbon (<i>e</i>)		
Sulfur (<i>f</i>)		

vi. Circle the value from the choices given below, that describes the **C-S-C bond angle at sulfur** (S) marked ***f***: (0.5 points)

>90° to <109.5° 109.5° >109.5 to <120° 120° >120° to <180°


Question 8 12 points

Using the **valence shell orbitals only** draw a **complete** molecular orbital (MO) diagram (with no s-p mixing) for the molecule **PS**, in the space provided below. **(6 points)**

Assume that 3s and 3p atomic orbitals will form molecular orbitals like those formed by the combination of 2s and 2p atomic orbitals that we learnt in class, and there is no s-p mixing.

Show **energy arrow**, **atomic orbitals** involved, **atom labels**, the **molecular orbitals** that are formed, **filled electrons** in the MO diagram etc. Make sure to label all the atomic and molecular orbitals.

Using the MO diagram above, answer the following questions:

- (i) Which one of the following will have the highest bond order? Circle the correct choice
 A) PS^{1-} B) PS C) PS^{1+}
- (ii) Which one of the following will have the longest bond length? Circle the correct choice
 A) PS^{1-} B) PS C) PS^{1+}
- (iii) Which of the following will be diamagnetic? Circle the correct choice.
 A) PS^{1-} B) PS C) PS^{1+}
- (iv) Draw Lewis dot structure of the molecule with the shortest bond length (based on the MO diagram) from among the three (PS^{1-} , PS, PS^{1+}), and indicate the bond order calculated from the Lewis dot structure. **(3 points)**

Lewis Dot Structure

Bond Order



77853C7B-7EF4-42DF-805C-512C4348BD9E

midterm-exam-93388

#333 Page 10 of 10

SCRAP PAGE

**CHEMISTRY 110 –Midterm Exam****Thursday November 7, 2024, 6:30 – 8:30 pm**

Examiners: Pallavi Sirjoosingh and Ashok Kakkar

First name (please write as legibly as possible within the boxes)

--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--

Last name

--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--

Student ID number

--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--

This booklet has 8 questions on 10 pages (The last page can be used as a scrap page).

The staple must NOT be removed from the question booklet. Your intact question booklet must be returned at the end of the exam.

A separate booklet contains a periodic table, and datasheet.

Write your answers directly under the questions in the space provided. Do not write on the QR codes or near the top edges (as it may be cut off).

Please write **legibly and dark enough** for the scan to be readable. For students who use English as a second language, dictionaries are permitted. Calculators are allowed.

DO NOT ASK QUESTIONS OF THE INVIGILATORS about interpretations of the exam questions. If you have a concern about a question, please summarize it at the top of this page. All concerns will be evaluated after the exam. GOOD LUCK!

**Question 1 8 points** (*Assume all transitions are possible*)

a. 6 points Calculate the **energy (in kJ/mol)** required to ionize (completely remove) the electron from the **$n=3$** level of **He^+ ion** (He: Helium). Show your work including equations. (3 sig. figs.)

b. 2 points

(I) Select the largest (longest) wavelength from the three choices below:

- i. Wavelength of photon to *completely remove* an electron from *1s orbital of Hydrogen atom*
- ii. Wavelength of photon to *completely remove* an electron from *1s orbital of Boron atom (B)*
- iii. Wavelength of photon to *completely remove* an electron from *1s orbital of Boron cation (B^+)*

i or ii or iii

(II) Select the smallest (shortest) wavelength from the three choices below (*for H atom*):

- i. Wavelength of photon *emitted* for an electron transitioning from *2p to 1s orbital*
- ii. Wavelength of photon *absorbed* for an electron transitioning from *2s to 4p orbital*
- iii. Wavelength of photon *absorbed* for an electron transitioning from *1s to 3p orbital*

i or ii or iii



Question 2 8 points



A Chem110 student plots the graph shown on the left, from data obtained for the kinetic energy (eV) of the ejected electron versus frequency (s^{-1}) of the incident light, for an unknown metal. ($1\text{eV} = 1.6022 \times 10^{-19} \text{ J}$)

Based on the plot, answer the questions **a to c** below:

a. 1 point On the plot above, label the **threshold frequency**.

b. 2 points What is the value of the work potential (in J) of the unknown metal?

c. 3 points What is the *longest* wavelength (in nm) required to eject an electron from this metal's surface? Show your work including equation(s) used. (3 sig figs)

d. 2 points In the space provided below, plot the following

(Assume frequency is *above* threshold frequency):

- i. *Kinetic energy* of an ejected electron versus the *intensity* of the incident light (left)
- ii. *Number of electrons* ejected versus the *frequency* of the incident light (right)



**Question 3** **6 points****a. 2 points**

In the ground state of Ge (Germanium; $Z = 32$) atom, **name** the following orbitals:

i. Orbital containing the electron that requires the maximum energy to be removed.

ii. Orbital containing the electron that experiences the minimum effective nuclear charge.

b. 2 points

An unknown anion (A^-) has the following *excited* electronic configuration: $[Ar] 4s^2 3d^3 4p^1$

(Ar: Argon; $Z = 18$)

Give the **condensed ground state electronic configuration** of the atom, **A** and the cation, **A^{2+}** .
Show your work. (Assume all electronic transitions are possible)

Electronic Configuration of A:

Electronic Configuration of A^{2+} :

c. 2 points

In a single atom, determine how many electrons can have the following set of quantum numbers?

(i) $n = 5; l = 3$

In a single atom, determine how many electrons can have the following set of quantum numbers?

(i) $n = 3; l = 2; m_s = +1/2$

**Question 4** **8 points**

Rank the following from 1 to 4 in **increasing** order. Provide the ranking in the boxes provided.

Note: Atomic number (Z) are provided for relevant atoms

a) Increasing (lowest to highest) order of *ionization energy*: (1= lowest; 4 = highest)

Mg

Mg: Z = 12

Rb

Rb: Z = 37

Cs

Cs: Z = 55

K

K : Z = 19

b) Increasing (lowest to highest) order of *bond angle*: (1= lowest; 4 = highest)

H₂OCCl₄XeF₄

HCN

c) Increasing (smallest to largest) order of *ionic size* (1= smallest; 4 = largest)

S²⁻

S: Z = 16

O²⁻

O: Z = 8

Be²⁺

Be: Z = 4

Na⁺

Na: Z = 11

d) Increasing (lowest to highest) order of *bond energy* (1= lowest; 4 = highest)

P-F

P-Cl

P-I

P-Br

**Question 5** **12 points**

a. 9 points For the molecules HClO_4 and TeF_6 , draw the most stable *Lewis structure* showing appropriate shape and *dashed/wedged bonds*, based on VSEPR. Determine the **molecular geometry** around the central atom. For HClO_4 , label the *most polar bond*.



b. 3 points (Note: Lone pairs are not shown in structures below)

(i) For the three *resonance* structures (1, 2 and 3) shown below, answer the following question:



(I) Circle the *most contributing* resonance structure.

Structure 1

Structure 2

Structure 3

(ii) For the three structures given (4, 5, and 6) below, answer the following questions:



(I) Circle all the structures that represent *resonance structures* of each other:

Structure 4

Structure 5

Structure 6

(II) Among the resonance structures, circle the *most contributing* resonance structure.

Structure 4

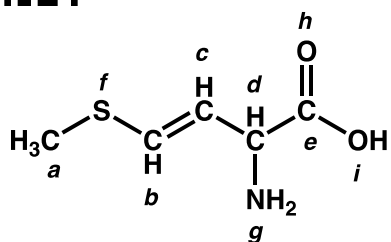
Structure 5

Structure 6

**Question 6** **6 points**

Using the data given below, calculate the **lattice energy** for SrO(s).

	ΔH° (kJ/mol)
$\text{Sr}(s) \longrightarrow \text{Sr}(g)$	+180
$\text{Sr}(g) \longrightarrow \text{Sr}^{2+}(g) + 2e^-$	+1468
$\text{O}_2(g) \longrightarrow 2\text{O}(g)$	+500
$\text{O}(g) + 2e^- \longrightarrow \text{O}^{2-}(g)$	+600
$\text{Sr}(s) + \frac{1}{2}\text{O}_2(g) \longrightarrow \text{SrO}(s)$	-550

**Question 7 10 points**

Consider the molecule given on the left, and answer questions (i) to (vi) using chemical bonding theories that we have learnt in class (valence bond theory, hybridization).

Note that the lone pairs are not shown in the structure:

i. Consider the carbon atoms marked in as ***a***, ***b***, ***c***, ***d***, and ***e*** in the molecule, and determine **hybridization** at each of the carbon centers. (2.5 points)

Atom	Hybridization
Carbon (<i>a</i>)	
Carbon (<i>b</i>)	
Carbon (<i>c</i>)	
Carbon (<i>d</i>)	
Carbon (<i>e</i>)	

ii. Now, consider the sulfur atom (S) marked as ***f***, nitrogen (N) atom (***g***), and oxygen(O) atoms (***h*** and ***i***), and determine their **hybridization**. (2 points)

Atom	Hybridization
Sulfur (<i>f</i>)	
Nitrogen (<i>g</i>)	
Oxygen (<i>h</i>)	
Oxygen (<i>i</i>)	

iii. **Describe** all the bonds formed by the carbon atoms ***a*** and ***b***, using sigma and pi designation as appropriate. (2 points)

Atom	Bonds formed (use sigma and pi as appropriate)
Carbon (<i>a</i>)	
Carbon (<i>b</i>)	

iv. How many total sp^3 -s sigma bonds are in the molecule? (1 point)

v. Determine **electronic** and **molecular geometry** at carbon (C) labeled ***e*** and sulfur (S) labeled ***f*** centres. (2 points)

Atom	Electronic Geometry	Molecular Geometry
Carbon (<i>e</i>)		
Sulfur (<i>f</i>)		

vi. Circle the value from the choices given below, that describes the **C-S-C bond angle at sulfur** (S) marked ***f***: (0.5 points)

>90° to <109.5° 109.5° >109.5 to <120° 120° >120° to <180°


Question 8 12 points

Using the **valence shell orbitals only** draw a **complete** molecular orbital (MO) diagram (with no s-p mixing) for the molecule **PS**, in the space provided below. **(6 points)**

Assume that 3s and 3p atomic orbitals will form molecular orbitals like those formed by the combination of 2s and 2p atomic orbitals that we learnt in class, and there is no s-p mixing.

Show **energy arrow**, **atomic orbitals** involved, **atom labels**, the **molecular orbitals** that are formed, **filled electrons** in the MO diagram etc. Make sure to label all the atomic and molecular orbitals.

Using the MO diagram above, answer the following questions:

- (i) Which one of the following will have the highest bond order? Circle the correct choice
 A) PS^{1-} B) PS C) PS^{1+}
- (ii) Which one of the following will have the longest bond length? Circle the correct choice
 A) PS^{1-} B) PS C) PS^{1+}
- (iii) Which of the following will be diamagnetic? Circle the correct choice.
 A) PS^{1-} B) PS C) PS^{1+}
- (iv) Draw Lewis dot structure of the molecule with the shortest bond length (based on the MO diagram) from among the three (PS^{1-} , PS, PS^{1+}), and indicate the bond order calculated from the Lewis dot structure. **(3 points)**

Lewis Dot Structure

Bond Order



838B54C6-DE5F-4086-B00A-493EE98CB95B

midterm-exam-93388

#334 Page 10 of 10

SCRAP PAGE

**CHEMISTRY 110 –Midterm Exam****Thursday November 7, 2024, 6:30 – 8:30 pm**

Examiners: Pallavi Sirjoosingh and Ashok Kakkar

First name (please write as legibly as possible within the boxes)

--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--

Last name

--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--

Student ID number

--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--

This booklet has 8 questions on 10 pages (The last page can be used as a scrap page).

The staple must NOT be removed from the question booklet. Your intact question booklet must be returned at the end of the exam.

A separate booklet contains a periodic table, and datasheet.

Write your answers directly under the questions in the space provided. Do not write on the QR codes or near the top edges (as it may be cut off).

Please write **legibly and dark enough** for the scan to be readable. For students who use English as a second language, dictionaries are permitted. Calculators are allowed.

DO NOT ASK QUESTIONS OF THE INVIGILATORS about interpretations of the exam questions. If you have a concern about a question, please summarize it at the top of this page. All concerns will be evaluated after the exam. GOOD LUCK!

**Question 1 8 points** (*Assume all transitions are possible*)

a. 6 points Calculate the **energy (in kJ/mol)** required to ionize (completely remove) the electron from the **n=3** level of **He⁺ ion** (He: Helium). Show your work including equations. (3 sig. figs.)

b. 2 points

(I) Select the largest (longest) wavelength from the three choices below:

- i. Wavelength of photon to *completely remove* an electron from *1s orbital of Hydrogen atom*
- ii. Wavelength of photon to *completely remove* an electron from *1s orbital of Boron atom (B)*
- iii. Wavelength of photon to *completely remove* an electron from *1s orbital of Boron cation (B⁺)*

i or ii or iii

(II) Select the smallest (shortest) wavelength from the three choices below (*for H atom*):

- i. Wavelength of photon *emitted* for an electron transitioning from *2p to 1s orbital*
- ii. Wavelength of photon *absorbed* for an electron transitioning from *2s to 4p orbital*
- iii. Wavelength of photon *absorbed* for an electron transitioning from *1s to 3p orbital*

i or ii or iii



Question 2 8 points



A Chem110 student plots the graph shown on the left, from data obtained for the kinetic energy (eV) of the ejected electron versus frequency (s^{-1}) of the incident light, for an unknown metal. ($1\text{eV} = 1.6022 \times 10^{-19} \text{ J}$)

Based on the plot, answer the questions **a to c** below:

a. 1 point On the plot above, label the **threshold frequency**.

b. 2 points What is the value of the work potential (in J) of the unknown metal?

c. 3 points What is the *longest* wavelength (in nm) required to eject an electron from this metal's surface? Show your work including equation(s) used. (3 sig figs)

d. 2 points In the space provided below, plot the following

(Assume frequency is *above* threshold frequency):

- Kinetic energy* of an ejected electron versus the *intensity* of the incident light (left)
- Number of electrons* ejected versus the *frequency* of the incident light (right)



**Question 3** **6 points****a. 2 points**

In the ground state of Ge (Germanium; $Z = 32$) atom, **name** the following orbitals:

i. Orbital containing the electron that requires the maximum energy to be removed.

ii. Orbital containing the electron that experiences the minimum effective nuclear charge.

b. 2 points

An unknown anion (A^-) has the following *excited* electronic configuration: $[Ar] 4s^2 3d^3 4p^1$

(Ar: Argon; $Z = 18$)

Give the **condensed ground state electronic configuration** of the atom, **A** and the cation, **A^{2+}** .
Show your work. (Assume all electronic transitions are possible)

Electronic Configuration of A:

Electronic Configuration of A^{2+} :

c. 2 points

In a single atom, determine how many electrons can have the following set of quantum numbers?

(i) $n = 5; l = 3$

In a single atom, determine how many electrons can have the following set of quantum numbers?

(i) $n = 3; l = 2; m_s = +1/2$

**Question 4** **8 points**

Rank the following from 1 to 4 in **increasing** order. Provide the ranking in the boxes provided.

Note: Atomic number (Z) are provided for relevant atoms

a) Increasing (lowest to highest) order of *ionization energy*: (1= lowest; 4 = highest)

Mg

Mg: Z = 12

Rb

Rb: Z = 37

Cs

Cs: Z = 55

K

K : Z = 19

b) Increasing (lowest to highest) order of *bond angle*: (1= lowest; 4 = highest)

H₂OCCl₄XeF₄

HCN

c) Increasing (smallest to largest) order of *ionic size* (1= smallest; 4 = largest)

S²⁻

S: Z = 16

O²⁻

O: Z = 8

Be²⁺

Be: Z = 4

Na⁺

Na: Z = 11

d) Increasing (lowest to highest) order of *bond energy* (1= lowest; 4 = highest)

P-F

P-Cl

P-I

P-Br

**Question 5** **12 points**

a. 9 points For the molecules HClO_4 and TeF_6 , draw the most stable *Lewis structure* showing appropriate shape and *dashed/wedged bonds*, based on VSEPR. Determine the **molecular geometry** around the central atom. For HClO_4 , label the *most polar bond*.



b. 3 points (Note: Lone pairs are not shown in structures below)

(i) For the three *resonance* structures (1, 2 and 3) shown below, answer the following question:



(I) Circle the *most contributing* resonance structure.

Structure 1

Structure 2

Structure 3

(ii) For the three structures given (4, 5, and 6) below, answer the following questions:



(I) Circle all the structures that represent *resonance structures* of each other:

Structure 4

Structure 5

Structure 6

(II) Among the resonance structures, circle the *most contributing* resonance structure.

Structure 4

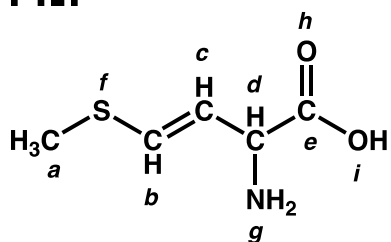
Structure 5

Structure 6

**Question 6** **6 points**

Using the data given below, calculate the **lattice energy** for SrO(s).

	ΔH° (kJ/mol)
$\text{Sr}(s) \longrightarrow \text{Sr}(g)$	+180
$\text{Sr}(g) \longrightarrow \text{Sr}^{2+}(g) + 2e^-$	+1468
$\text{O}_2(g) \longrightarrow 2\text{O}(g)$	+500
$\text{O}(g) + 2e^- \longrightarrow \text{O}^{2-}(g)$	+600
$\text{Sr}(s) + \frac{1}{2}\text{O}_2(g) \longrightarrow \text{SrO}(s)$	-550

**Question 7 10 points**

Consider the molecule given on the left, and answer questions (i) to (vi) using chemical bonding theories that we have learnt in class (valence bond theory, hybridization).

Note that the lone pairs are not shown in the structure:

i. Consider the carbon atoms marked in as ***a***, ***b***, ***c***, ***d***, and ***e*** in the molecule, and determine **hybridization** at each of the carbon centers. (2.5 points)

Atom	Hybridization
Carbon (<i>a</i>)	
Carbon (<i>b</i>)	
Carbon (<i>c</i>)	
Carbon (<i>d</i>)	
Carbon (<i>e</i>)	

ii. Now, consider the sulfur atom (S) marked as ***f***, nitrogen (N) atom (***g***), and oxygen(O) atoms (***h*** and ***i***), and determine their **hybridization**. (2 points)

Atom	Hybridization
Sulfur (<i>f</i>)	
Nitrogen (<i>g</i>)	
Oxygen (<i>h</i>)	
Oxygen (<i>i</i>)	

iii. **Describe** all the bonds formed by the carbon atoms ***a*** and ***b***, using sigma and pi designation as appropriate. (2 points)

Atom	Bonds formed (use sigma and pi as appropriate)
Carbon (<i>a</i>)	
Carbon (<i>b</i>)	

iv. How many total sp^3 -s sigma bonds are in the molecule? (1 point)

v. Determine **electronic** and **molecular geometry** at carbon (C) labeled ***e*** and sulfur (S) labeled ***f*** centres. (2 points)

Atom	Electronic Geometry	Molecular Geometry
Carbon (<i>e</i>)		
Sulfur (<i>f</i>)		

vi. Circle the value from the choices given below, that describes the **C-S-C bond angle at sulfur** (S) marked ***f***: (0.5 points)

>90° to <109.5°

109.5°

>109.5 to <120°

120°

>120° to <180°


Question 8 12 points

Using the **valence shell orbitals only** draw a **complete** molecular orbital (MO) diagram (with no s-p mixing) for the molecule **PS**, in the space provided below. **(6 points)**

Assume that 3s and 3p atomic orbitals will form molecular orbitals like those formed by the combination of 2s and 2p atomic orbitals that we learnt in class, and there is no s-p mixing.

Show **energy arrow**, **atomic orbitals** involved, **atom labels**, the **molecular orbitals** that are formed, **filled electrons** in the MO diagram etc. Make sure to label all the atomic and molecular orbitals.

Using the MO diagram above, answer the following questions:

- (i) Which one of the following will have the highest bond order? Circle the correct choice
 A) PS^{1-} B) PS C) PS^{1+}
- (ii) Which one of the following will have the longest bond length? Circle the correct choice
 A) PS^{1-} B) PS C) PS^{1+}
- (iii) Which of the following will be diamagnetic? Circle the correct choice.
 A) PS^{1-} B) PS C) PS^{1+}
- (iv) Draw Lewis dot structure of the molecule with the shortest bond length (based on the MO diagram) from among the three (PS^{1-} , PS, PS^{1+}), and indicate the bond order calculated from the Lewis dot structure. **(3 points)**

Lewis Dot Structure

Bond Order



24FE717F-0D1A-41B0-A1A5-1F1FE4FB41C8

midterm-exam-93388

#335 Page 10 of 10

SCRAP PAGE

**CHEMISTRY 110 –Midterm Exam****Thursday November 7, 2024, 6:30 – 8:30 pm**

Examiners: Pallavi Sirjoosingh and Ashok Kakkar

First name (please write as legibly as possible within the boxes)

--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--

Last name

--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--

Student ID number

--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--

This booklet has 8 questions on 10 pages (The last page can be used as a scrap page).

The staple must NOT be removed from the question booklet. Your intact question booklet must be returned at the end of the exam.

A separate booklet contains a periodic table, and datasheet.

Write your answers directly under the questions in the space provided. Do not write on the QR codes or near the top edges (as it may be cut off).

Please write **legibly and dark enough** for the scan to be readable. For students who use English as a second language, dictionaries are permitted. Calculators are allowed.

DO NOT ASK QUESTIONS OF THE INVIGILATORS about interpretations of the exam questions. If you have a concern about a question, please summarize it at the top of this page. All concerns will be evaluated after the exam. GOOD LUCK!

**Question 1 8 points** (*Assume all transitions are possible*)

a. 6 points Calculate the **energy (in kJ/mol)** required to ionize (completely remove) the electron from the **$n=3$** level of **He^+ ion** (He: Helium). Show your work including equations. (3 sig. figs.)

b. 2 points

(I) Select the largest (longest) wavelength from the three choices below:

- i. Wavelength of photon to *completely remove* an electron from *1s orbital of Hydrogen atom*
- ii. Wavelength of photon to *completely remove* an electron from *1s orbital of Boron atom (B)*
- iii. Wavelength of photon to *completely remove* an electron from *1s orbital of Boron cation (B^+)*

i or ii or iii

(II) Select the smallest (shortest) wavelength from the three choices below (*for H atom*):

- i. Wavelength of photon *emitted* for an electron transitioning from *2p to 1s orbital*
- ii. Wavelength of photon *absorbed* for an electron transitioning from *2s to 4p orbital*
- iii. Wavelength of photon *absorbed* for an electron transitioning from *1s to 3p orbital*

i or ii or iii



Question 2 8 points



A Chem110 student plots the graph shown on the left, from data obtained for the kinetic energy (eV) of the ejected electron versus frequency (s^{-1}) of the incident light, for an unknown metal. ($1\text{eV} = 1.6022 \times 10^{-19} \text{ J}$)

Based on the plot, answer the questions **a to c** below:

a. 1 point On the plot above, label the **threshold frequency**.

b. 2 points What is the value of the work potential (in J) of the unknown metal?

c. 3 points What is the *longest* wavelength (in nm) required to eject an electron from this metal's surface? Show your work including equation(s) used. (3 sig figs)

d. 2 points In the space provided below, plot the following

(Assume frequency is *above* threshold frequency):

- i. *Kinetic energy* of an ejected electron versus the *intensity* of the incident light (left)
- ii. *Number of electrons* ejected versus the *frequency* of the incident light (right)



**Question 3** **6 points****a. 2 points**

In the ground state of Ge (Germanium; $Z = 32$) atom, **name** the following orbitals:

i. Orbital containing the electron that requires the maximum energy to be removed.

ii. Orbital containing the electron that experiences the minimum effective nuclear charge.

b. 2 points

An unknown anion (A^-) has the following *excited* electronic configuration: $[Ar] 4s^2 3d^3 4p^1$

(Ar: Argon; $Z = 18$)

Give the **condensed ground state electronic configuration** of the atom, **A** and the cation, **A^{2+}** .
Show your work. (Assume all electronic transitions are possible)

Electronic Configuration of A:

Electronic Configuration of A^{2+} :

c. 2 points

In a single atom, determine how many electrons can have the following set of quantum numbers?

(i) $n = 5; l = 3$

In a single atom, determine how many electrons can have the following set of quantum numbers?

(i) $n = 3; l = 2; m_s = +1/2$

**Question 4** **8 points**

Rank the following from 1 to 4 in **increasing** order. Provide the ranking in the boxes provided.

Note: Atomic number (Z) are provided for relevant atoms

a) Increasing (lowest to highest) order of *ionization energy*: (1= lowest; 4 = highest)

Mg

Mg: Z = 12

Rb

Rb: Z = 37

Cs

Cs: Z = 55

K

K : Z = 19

b) Increasing (lowest to highest) order of *bond angle*: (1= lowest; 4 = highest)

H₂OCCl₄XeF₄

HCN

c) Increasing (smallest to largest) order of *ionic size* (1= smallest; 4 = largest)

S²⁻

S: Z = 16

O²⁻

O: Z = 8

Be²⁺

Be: Z = 4

Na⁺

Na: Z = 11

d) Increasing (lowest to highest) order of *bond energy* (1= lowest; 4 = highest)

P-F

P-Cl

P-I

P-Br

**Question 5** **12 points**

a. 9 points For the molecules HClO_4 and TeF_6 , draw the most stable *Lewis structure* showing appropriate shape and *dashed/wedged bonds*, based on VSEPR. Determine the **molecular geometry** around the central atom. For HClO_4 , label the *most polar bond*.



b. 3 points (Note: Lone pairs are not shown in structures below)

(i) For the three *resonance* structures (1, 2 and 3) shown below, answer the following question:



(I) Circle the *most contributing* resonance structure.

Structure 1

Structure 2

Structure 3

(ii) For the three structures given (4, 5, and 6) below, answer the following questions:



(I) Circle all the structures that represent *resonance structures* of each other:

Structure 4

Structure 5

Structure 6

(II) Among the resonance structures, circle the *most contributing* resonance structure.

Structure 4

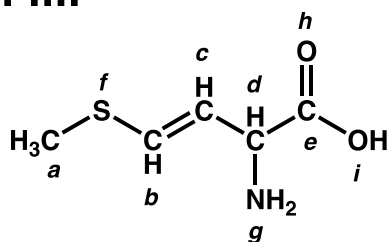
Structure 5

Structure 6

**Question 6** **6 points**

Using the data given below, calculate the **lattice energy** for SrO(s).

	ΔH° (kJ/mol)
$\text{Sr}(s) \longrightarrow \text{Sr}(g)$	+180
$\text{Sr}(g) \longrightarrow \text{Sr}^{2+}(g) + 2e^-$	+1468
$\text{O}_2(g) \longrightarrow 2\text{O}(g)$	+500
$\text{O}(g) + 2e^- \longrightarrow \text{O}^{2-}(g)$	+600
$\text{Sr}(s) + \frac{1}{2}\text{O}_2(g) \longrightarrow \text{SrO}(s)$	-550

**Question 7 10 points**

Consider the molecule given on the left, and answer questions (i) to (vi) using chemical bonding theories that we have learnt in class (valence bond theory, hybridization).

Note that the lone pairs are not shown in the structure:

i. Consider the carbon atoms marked in as ***a***, ***b***, ***c***, ***d***, and ***e*** in the molecule, and determine **hybridization** at each of the carbon centers. (2.5 points)

Atom	Hybridization
Carbon (<i>a</i>)	
Carbon (<i>b</i>)	
Carbon (<i>c</i>)	
Carbon (<i>d</i>)	
Carbon (<i>e</i>)	

ii. Now, consider the sulfur atom (S) marked as ***f***, nitrogen (N) atom (***g***), and oxygen(O) atoms (***h*** and ***i***), and determine their **hybridization**. (2 points)

Atom	Hybridization
Sulfur (<i>f</i>)	
Nitrogen (<i>g</i>)	
Oxygen (<i>h</i>)	
Oxygen (<i>i</i>)	

iii. **Describe** all the bonds formed by the carbon atoms ***a*** and ***b***, using sigma and pi designation as appropriate. (2 points)

Atom	Bonds formed (use sigma and pi as appropriate)
Carbon (<i>a</i>)	
Carbon (<i>b</i>)	

iv. How many total sp^3 -s sigma bonds are in the molecule? (1 point)

v. Determine **electronic** and **molecular geometry** at carbon (C) labeled ***e*** and sulfur (S) labeled ***f*** centres. (2 points)

Atom	Electronic Geometry	Molecular Geometry
Carbon (<i>e</i>)		
Sulfur (<i>f</i>)		

vi. Circle the value from the choices given below, that describes the **C-S-C bond angle at sulfur** (S) marked ***f***: (0.5 points)

>90° to <109.5° 109.5° >109.5 to <120° 120° >120° to <180°


Question 8 12 points

Using the **valence shell orbitals only** draw a **complete** molecular orbital (MO) diagram (with no s-p mixing) for the molecule **PS**, in the space provided below. **(6 points)**

Assume that 3s and 3p atomic orbitals will form molecular orbitals like those formed by the combination of 2s and 2p atomic orbitals that we learnt in class, and there is no s-p mixing.

Show **energy arrow**, **atomic orbitals** involved, **atom labels**, the **molecular orbitals** that are formed, **filled electrons** in the MO diagram etc. Make sure to **label all the atomic and molecular orbitals**.

Using the MO diagram above, answer the following questions:

- (i) Which one of the following will have the highest bond order? Circle the correct choice
 A) PS^{1-} B) PS C) PS^{1+}
- (ii) Which one of the following will have the longest bond length? Circle the correct choice
 A) PS^{1-} B) PS C) PS^{1+}
- (iii) Which of the following will be diamagnetic? Circle the correct choice.
 A) PS^{1-} B) PS C) PS^{1+}
- (iv) Draw Lewis dot structure of the molecule with the shortest bond length (based on the MO diagram) from among the three (PS^{1-} , PS, PS^{1+}), and indicate the bond order calculated from the Lewis dot structure. **(3 points)**

Lewis Dot Structure

Bond Order



9A738685-ACB6-4168-B65A-E40ECC79746F

midterm-exam-93388

#336 Page 10 of 10

SCRAP PAGE

**CHEMISTRY 110 –Midterm Exam****Thursday November 7, 2024, 6:30 – 8:30 pm**

Examiners: Pallavi Sirjoosingh and Ashok Kakkar

First name (please write as legibly as possible within the boxes)

--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--

Last name

--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--

Student ID number

--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--

This booklet has 8 questions on 10 pages (The last page can be used as a scrap page).

The staple must NOT be removed from the question booklet. Your intact question booklet must be returned at the end of the exam.

A separate booklet contains a periodic table, and datasheet.

Write your answers directly under the questions in the space provided. Do not write on the QR codes or near the top edges (as it may be cut off).

Please write **legibly and dark enough** for the scan to be readable. For students who use English as a second language, dictionaries are permitted. Calculators are allowed.

DO NOT ASK QUESTIONS OF THE INVIGILATORS about interpretations of the exam questions. If you have a concern about a question, please summarize it at the top of this page. All concerns will be evaluated after the exam. GOOD LUCK!

**Question 1 8 points** (*Assume all transitions are possible*)

a. 6 points Calculate the **energy (in kJ/mol)** required to ionize (completely remove) the electron from the **$n=3$** level of **He^+ ion** (He: Helium). Show your work including equations. (3 sig. figs.)

b. 2 points

(I) Select the largest (longest) wavelength from the three choices below:

- i. Wavelength of photon to *completely remove* an electron from *1s orbital of Hydrogen atom*
- ii. Wavelength of photon to *completely remove* an electron from *1s orbital of Boron atom (B)*
- iii. Wavelength of photon to *completely remove* an electron from *1s orbital of Boron cation (B^+)*

i or ii or iii

(II) Select the smallest (shortest) wavelength from the three choices below (*for H atom*):

- i. Wavelength of photon *emitted* for an electron transitioning from *2p to 1s orbital*
- ii. Wavelength of photon *absorbed* for an electron transitioning from *2s to 4p orbital*
- iii. Wavelength of photon *absorbed* for an electron transitioning from *1s to 3p orbital*

i or ii or iii



Question 2 8 points



A Chem110 student plots the graph shown on the left, from data obtained for the kinetic energy (eV) of the ejected electron versus frequency (s^{-1}) of the incident light, for an unknown metal. ($1\text{eV} = 1.6022 \times 10^{-19} \text{ J}$)

Based on the plot, answer the questions **a to c** below:

a. 1 point On the plot above, label the **threshold frequency**.

b. 2 points What is the value of the work potential (in J) of the unknown metal?

c. 3 points What is the *longest* wavelength (in nm) required to eject an electron from this metal's surface? Show your work including equation(s) used. (3 sig figs)

d. 2 points In the space provided below, plot the following

(Assume frequency is *above* threshold frequency):

- Kinetic energy* of an ejected electron versus the *intensity* of the incident light (left)
- Number of electrons* ejected versus the *frequency* of the incident light (right)



**Question 3** **6 points****a. 2 points**

In the ground state of Ge (Germanium; $Z = 32$) atom, **name** the following orbitals:

i. Orbital containing the electron that requires the maximum energy to be removed.

ii. Orbital containing the electron that experiences the minimum effective nuclear charge.

b. 2 points

An unknown anion (A^-) has the following *excited* electronic configuration: $[Ar] 4s^2 3d^3 4p^1$

(Ar: Argon; $Z = 18$)

Give the **condensed ground state electronic configuration** of the atom, **A** and the cation, **A^{2+}** .
Show your work. (Assume all electronic transitions are possible)

Electronic Configuration of A:

Electronic Configuration of A^{2+} :

c. 2 points

In a single atom, determine how many electrons can have the following set of quantum numbers?

(i) $n = 5; l = 3$

In a single atom, determine how many electrons can have the following set of quantum numbers?

(i) $n = 3; l = 2; m_s = +1/2$

**Question 4** **8 points**

Rank the following from 1 to 4 in **increasing** order. Provide the ranking in the boxes provided.

Note: Atomic number (Z) are provided for relevant atoms

a) Increasing (lowest to highest) order of *ionization energy*: (1= lowest; 4 = highest)

Mg

Mg: Z = 12

Rb

Rb: Z = 37

Cs

Cs: Z = 55

K

K : Z = 19

b) Increasing (lowest to highest) order of *bond angle*: (1= lowest; 4 = highest)

H₂OCCl₄XeF₄

HCN

c) Increasing (smallest to largest) order of *ionic size* (1= smallest; 4 = largest)

S²⁻

S: Z = 16

O²⁻

O: Z = 8

Be²⁺

Be: Z = 4

Na⁺

Na: Z = 11

d) Increasing (lowest to highest) order of *bond energy* (1= lowest; 4 = highest)

P-F

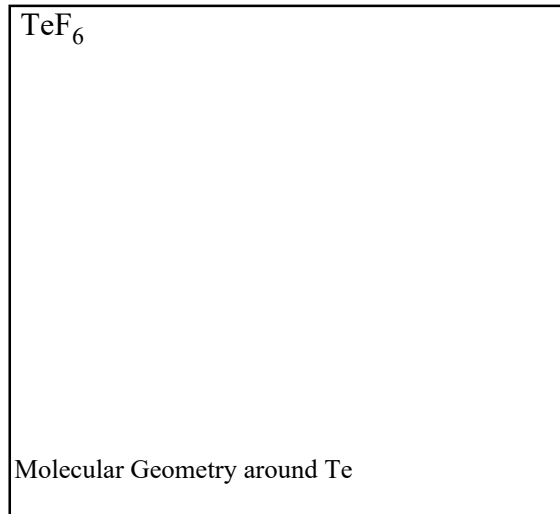
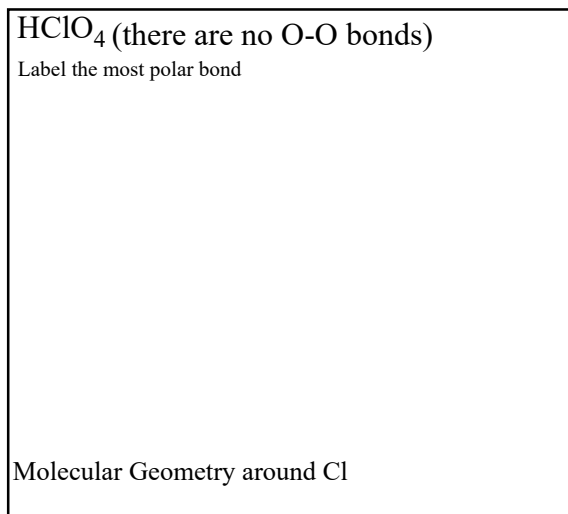
P-Cl

P-I

P-Br

**Question 5** **12 points**

a. 9 points For the molecules HClO_4 and TeF_6 , draw the most stable *Lewis structure* showing appropriate shape and *dashed/wedged bonds*, based on VSEPR. Determine the **molecular geometry** around the central atom. For HClO_4 , label the *most polar bond*.



b. 3 points (Note: Lone pairs are not shown in structures below)

(i) For the three *resonance* structures (1, 2 and 3) shown below, answer the following question:



(I) Circle the *most contributing* resonance structure.

Structure 1

Structure 2

Structure 3

(ii) For the three structures given (4, 5, and 6) below, answer the following questions:



(I) Circle all the structures that represent *resonance structures* of each other:

Structure 4

Structure 5

Structure 6

(II) Among the resonance structures, circle the *most contributing* resonance structure.

Structure 4

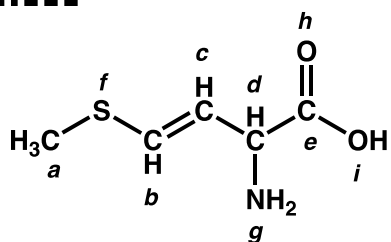
Structure 5

Structure 6

**Question 6** **6 points**

Using the data given below, calculate the **lattice energy** for SrO(s).

	ΔH° (kJ/mol)
$\text{Sr}(s) \longrightarrow \text{Sr}(g)$	+180
$\text{Sr}(g) \longrightarrow \text{Sr}^{2+}(g) + 2e^-$	+1468
$\text{O}_2(g) \longrightarrow 2\text{O}(g)$	+500
$\text{O}(g) + 2e^- \longrightarrow \text{O}^{2-}(g)$	+600
$\text{Sr}(s) + \frac{1}{2}\text{O}_2(g) \longrightarrow \text{SrO}(s)$	-550

**Question 7 10 points**

Consider the molecule given on the left, and answer questions (i) to (vi) using chemical bonding theories that we have learnt in class (valence bond theory, hybridization).

Note that the lone pairs are not shown in the structure:

i. Consider the carbon atoms marked in as ***a***, ***b***, ***c***, ***d***, and ***e*** in the molecule, and determine **hybridization** at each of the carbon centers. (2.5 points)

Atom	Hybridization
Carbon (<i>a</i>)	
Carbon (<i>b</i>)	
Carbon (<i>c</i>)	
Carbon (<i>d</i>)	
Carbon (<i>e</i>)	

ii. Now, consider the sulfur atom (S) marked as ***f***, nitrogen (N) atom (***g***), and oxygen(O) atoms (***h*** and ***i***), and determine their **hybridization**. (2 points)

Atom	Hybridization
Sulfur (<i>f</i>)	
Nitrogen (<i>g</i>)	
Oxygen (<i>h</i>)	
Oxygen (<i>i</i>)	

iii. **Describe** all the bonds formed by the carbon atoms ***a*** and ***b***, using sigma and pi designation as appropriate. (2 points)

Atom	Bonds formed (use sigma and pi as appropriate)
Carbon (<i>a</i>)	
Carbon (<i>b</i>)	

iv. How many total sp^3 -s sigma bonds are in the molecule? (1 point)

v. Determine **electronic** and **molecular geometry** at carbon (C) labeled ***e*** and sulfur (S) labeled ***f*** centres. (2 points)

Atom	Electronic Geometry	Molecular Geometry
Carbon (<i>e</i>)		
Sulfur (<i>f</i>)		

vi. Circle the value from the choices given below, that describes the **C-S-C bond angle at sulfur** (S) marked ***f***: (0.5 points)

>90° to <109.5° 109.5° >109.5 to <120° 120° >120° to <180°


Question 8 12 points

Using the **valence shell orbitals only** draw a **complete** molecular orbital (MO) diagram (with no s-p mixing) for the molecule **PS**, in the space provided below. **(6 points)**

Assume that 3s and 3p atomic orbitals will form molecular orbitals like those formed by the combination of 2s and 2p atomic orbitals that we learnt in class, and there is no s-p mixing.

Show **energy arrow**, **atomic orbitals** involved, **atom labels**, the **molecular orbitals** that are formed, **filled electrons** in the MO diagram etc. Make sure to label all the atomic and molecular orbitals.

Using the MO diagram above, answer the following questions:

- (i) Which one of the following will have the highest bond order? Circle the correct choice
 A) PS^{1-} B) PS C) PS^{1+}
- (ii) Which one of the following will have the longest bond length? Circle the correct choice
 A) PS^{1-} B) PS C) PS^{1+}
- (iii) Which of the following will be diamagnetic? Circle the correct choice.
 A) PS^{1-} B) PS C) PS^{1+}
- (iv) Draw Lewis dot structure of the molecule with the shortest bond length (based on the MO diagram) from among the three (PS^{1-} , PS, PS^{1+}), and indicate the bond order calculated from the Lewis dot structure. **(3 points)**

Lewis Dot Structure

Bond Order



DDC0B3C8-F064-4005-9C9B-406490A35781

midterm-exam-93388

#337 Page 10 of 10

SCRAP PAGE

**CHEMISTRY 110 –Midterm Exam****Thursday November 7, 2024, 6:30 – 8:30 pm**

Examiners: Pallavi Sirjoosingh and Ashok Kakkar

First name (please write as legibly as possible within the boxes)

--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--

Last name

--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--

Student ID number

--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--

This booklet has 8 questions on 10 pages (The last page can be used as a scrap page).

The staple must NOT be removed from the question booklet. Your intact question booklet must be returned at the end of the exam.

A separate booklet contains a periodic table, and datasheet.

Write your answers directly under the questions in the space provided. Do not write on the QR codes or near the top edges (as it may be cut off).

Please write **legibly and dark enough** for the scan to be readable. For students who use English as a second language, dictionaries are permitted. Calculators are allowed.

DO NOT ASK QUESTIONS OF THE INVIGILATORS about interpretations of the exam questions. If you have a concern about a question, please summarize it at the top of this page. All concerns will be evaluated after the exam. GOOD LUCK!

**Question 1 8 points** (*Assume all transitions are possible*)

a. 6 points Calculate the **energy (in kJ/mol)** required to ionize (completely remove) the electron from the **n=3** level of **He⁺ ion** (He: Helium). Show your work including equations. (3 sig. figs.)

b. 2 points

(I) Select the largest (longest) wavelength from the three choices below:

- i. Wavelength of photon to *completely remove* an electron from *1s orbital of Hydrogen atom*
- ii. Wavelength of photon to *completely remove* an electron from *1s orbital of Boron atom (B)*
- iii. Wavelength of photon to *completely remove* an electron from *1s orbital of Boron cation (B⁺)*

i or ii or iii

(II) Select the smallest (shortest) wavelength from the three choices below (*for H atom*):

- i. Wavelength of photon *emitted* for an electron transitioning from *2p to 1s orbital*
- ii. Wavelength of photon *absorbed* for an electron transitioning from *2s to 4p orbital*
- iii. Wavelength of photon *absorbed* for an electron transitioning from *1s to 3p orbital*

i or ii or iii



Question 2 8 points



A Chem110 student plots the graph shown on the left, from data obtained for the kinetic energy (eV) of the ejected electron versus frequency (s^{-1}) of the incident light, for an unknown metal. ($1\text{eV} = 1.6022 \times 10^{-19} \text{ J}$)

Based on the plot, answer the questions **a to c** below:

a. 1 point On the plot above, label the **threshold frequency**.

b. 2 points What is the value of the work potential (in J) of the unknown metal?

c. 3 points What is the *longest* wavelength (in nm) required to eject an electron from this metal's surface? Show your work including equation(s) used. (3 sig figs)

d. 2 points In the space provided below, plot the following

(Assume frequency is *above* threshold frequency):

- Kinetic energy* of an ejected electron versus the *intensity* of the incident light (left)
- Number of electrons* ejected versus the *frequency* of the incident light (right)



**Question 3** **6 points****a. 2 points**

In the ground state of Ge (Germanium; $Z = 32$) atom, **name** the following orbitals:

i. Orbital containing the electron that requires the maximum energy to be removed.

ii. Orbital containing the electron that experiences the minimum effective nuclear charge.

b. 2 points

An unknown anion (A^-) has the following *excited* electronic configuration: $[Ar] 4s^2 3d^3 4p^1$

(Ar: Argon; $Z = 18$)

Give the **condensed ground state electronic configuration** of the atom, **A** and the cation, **A^{2+}** .
Show your work. (Assume all electronic transitions are possible)

Electronic Configuration of A:

Electronic Configuration of A^{2+} :

c. 2 points

In a single atom, determine how many electrons can have the following set of quantum numbers?

(i) $n = 5; l = 3$

In a single atom, determine how many electrons can have the following set of quantum numbers?

(i) $n = 3; l = 2; m_s = +1/2$

**Question 4** **8 points**

Rank the following from 1 to 4 in **increasing** order. Provide the ranking in the boxes provided.

Note: Atomic number (Z) are provided for relevant atoms

a) Increasing (lowest to highest) order of *ionization energy*: (1= lowest; 4 = highest)

Mg

Mg: Z = 12

Rb

Rb: Z = 37

Cs

Cs: Z = 55

K

K : Z = 19

b) Increasing (lowest to highest) order of *bond angle*: (1= lowest; 4 = highest)

H₂OCCl₄XeF₄

HCN

c) Increasing (smallest to largest) order of *ionic size* (1= smallest; 4 = largest)

S²⁻

S: Z = 16

O²⁻

O: Z = 8

Be²⁺

Be: Z = 4

Na⁺

Na: Z = 11

d) Increasing (lowest to highest) order of *bond energy* (1= lowest; 4 = highest)

P-F

P-Cl

P-I

P-Br

**Question 5** **12 points**

a. 9 points For the molecules HClO_4 and TeF_6 , draw the most stable *Lewis structure* showing appropriate shape and *dashed/wedged bonds*, based on VSEPR. Determine the **molecular geometry** around the central atom. For HClO_4 , label the *most polar bond*.



b. 3 points (Note: Lone pairs are not shown in structures below)

(i) For the three *resonance* structures (1, 2 and 3) shown below, answer the following question:



(I) Circle the *most contributing* resonance structure.

Structure 1

Structure 2

Structure 3

(ii) For the three structures given (4, 5, and 6) below, answer the following questions:



(I) Circle all the structures that represent *resonance structures* of each other:

Structure 4

Structure 5

Structure 6

(II) Among the resonance structures, circle the *most contributing* resonance structure.

Structure 4

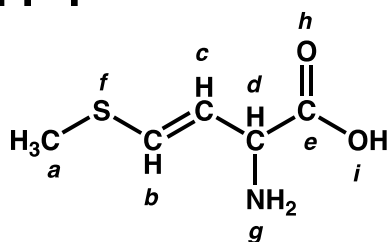
Structure 5

Structure 6

**Question 6** **6 points**

Using the data given below, calculate the **lattice energy** for SrO(s).

	ΔH° (kJ/mol)
$\text{Sr}(s) \longrightarrow \text{Sr}(g)$	+180
$\text{Sr}(g) \longrightarrow \text{Sr}^{2+}(g) + 2e^-$	+1468
$\text{O}_2(g) \longrightarrow 2\text{O}(g)$	+500
$\text{O}(g) + 2e^- \longrightarrow \text{O}^{2-}(g)$	+600
$\text{Sr}(s) + \frac{1}{2}\text{O}_2(g) \longrightarrow \text{SrO}(s)$	-550

**Question 7 10 points**

Consider the molecule given on the left, and answer questions (i) to (vi) using chemical bonding theories that we have learnt in class (valence bond theory, hybridization).

Note that the lone pairs are not shown in the structure:

i. Consider the carbon atoms marked in as ***a***, ***b***, ***c***, ***d***, and ***e*** in the molecule, and determine **hybridization** at each of the carbon centers. (2.5 points)

Atom	Hybridization
Carbon (<i>a</i>)	
Carbon (<i>b</i>)	
Carbon (<i>c</i>)	
Carbon (<i>d</i>)	
Carbon (<i>e</i>)	

ii. Now, consider the sulfur atom (S) marked as ***f***, nitrogen (N) atom (***g***), and oxygen(O) atoms (***h*** and ***i***), and determine their **hybridization**. (2 points)

Atom	Hybridization
Sulfur (<i>f</i>)	
Nitrogen (<i>g</i>)	
Oxygen (<i>h</i>)	
Oxygen (<i>i</i>)	

iii. **Describe** all the bonds formed by the carbon atoms ***a*** and ***b***, using sigma and pi designation as appropriate. (2 points)

Atom	Bonds formed (use sigma and pi as appropriate)
Carbon (<i>a</i>)	
Carbon (<i>b</i>)	

iv. How many total sp^3 -s sigma bonds are in the molecule? (1 point)

v. Determine **electronic** and **molecular geometry** at carbon (C) labeled ***e*** and sulfur (S) labeled ***f*** centres. (2 points)

Atom	Electronic Geometry	Molecular Geometry
Carbon (<i>e</i>)		
Sulfur (<i>f</i>)		

vi. Circle the value from the choices given below, that describes the **C-S-C bond angle at sulfur** (S) marked ***f***: (0.5 points)

>90° to <109.5°

109.5°

>109.5 to <120°

120°

>120° to <180°


Question 8 12 points

Using the **valence shell orbitals only** draw a **complete** molecular orbital (MO) diagram (with no s-p mixing) for the molecule **PS**, in the space provided below. **(6 points)**

Assume that 3s and 3p atomic orbitals will form molecular orbitals like those formed by the combination of 2s and 2p atomic orbitals that we learnt in class, and there is no s-p mixing.

Show **energy arrow**, **atomic orbitals** involved, **atom labels**, the **molecular orbitals** that are formed, **filled electrons** in the MO diagram etc. Make sure to label all the atomic and molecular orbitals.

Using the MO diagram above, answer the following questions:

- (i) Which one of the following will have the highest bond order? Circle the correct choice
 A) PS^{1-} B) PS C) PS^{1+}
- (ii) Which one of the following will have the longest bond length? Circle the correct choice
 A) PS^{1-} B) PS C) PS^{1+}
- (iii) Which of the following will be diamagnetic? Circle the correct choice.
 A) PS^{1-} B) PS C) PS^{1+}
- (iv) Draw Lewis dot structure of the molecule with the shortest bond length (based on the MO diagram) from among the three (PS^{1-} , PS, PS^{1+}), and indicate the bond order calculated from the Lewis dot structure. **(3 points)**

Lewis Dot Structure

Bond Order



E59C805A-0C76-4EE4-BBDC-0AA8156C9C3B

midterm-exam-93388

#338 Page 10 of 10

SCRAP PAGE

**CHEMISTRY 110 –Midterm Exam****Thursday November 7, 2024, 6:30 – 8:30 pm****Examiners: Pallavi Sirjoosingh and Ashok Kakkar**

First name (please write as legibly as possible within the boxes)

--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--

Last name

--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--

Student ID number

--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--

This booklet has 8 questions on 10 pages (The last page can be used as a scrap page).

The staple must NOT be removed from the question booklet. Your intact question booklet must be returned at the end of the exam.

A separate booklet contains a periodic table, and datasheet.

Write your answers directly under the questions in the space provided. Do not write on the QR codes or near the top edges (as it may be cut off).

Please write **legibly and dark enough** for the scan to be readable. For students who use English as a second language, dictionaries are permitted. Calculators are allowed.

DO NOT ASK QUESTIONS OF THE INVIGILATORS about interpretations of the exam questions. If you have a concern about a question, please summarize it at the top of this page. All concerns will be evaluated after the exam. GOOD LUCK!

**Question 1 8 points** (*Assume all transitions are possible*)

a. 6 points Calculate the **energy (in kJ/mol)** required to ionize (completely remove) the electron from the **n=3** level of **He⁺ ion** (He: Helium). Show your work including equations. (3 sig. figs.)

b. 2 points

(I) Select the largest (longest) wavelength from the three choices below:

- i. Wavelength of photon to *completely remove* an electron from *1s orbital of Hydrogen atom*
- ii. Wavelength of photon to *completely remove* an electron from *1s orbital of Boron atom (B)*
- iii. Wavelength of photon to *completely remove* an electron from *1s orbital of Boron cation (B⁺)*

i or ii or iii

(II) Select the smallest (shortest) wavelength from the three choices below (*for H atom*):

- i. Wavelength of photon *emitted* for an electron transitioning from *2p to 1s orbital*
- ii. Wavelength of photon *absorbed* for an electron transitioning from *2s to 4p orbital*
- iii. Wavelength of photon *absorbed* for an electron transitioning from *1s to 3p orbital*

i or ii or iii



Question 2 8 points



A Chem110 student plots the graph shown on the left, from data obtained for the kinetic energy (eV) of the ejected electron versus frequency (s^{-1}) of the incident light, for an unknown metal. ($1\text{eV} = 1.6022 \times 10^{-19} \text{ J}$)

Based on the plot, answer the questions **a to c** below:

a. 1 point On the plot above, label the **threshold frequency**.

b. 2 points What is the value of the work potential (in J) of the unknown metal?

c. 3 points What is the *longest* wavelength (in nm) required to eject an electron from this metal's surface? Show your work including equation(s) used. (3 sig figs)

d. 2 points In the space provided below, plot the following

(Assume frequency is *above* threshold frequency):

- Kinetic energy* of an ejected electron versus the *intensity* of the incident light (left)
- Number of electrons* ejected versus the *frequency* of the incident light (right)



**Question 3** **6 points****a. 2 points**

In the ground state of Ge (Germanium; $Z = 32$) atom, **name** the following orbitals:

i. Orbital containing the electron that requires the maximum energy to be removed.

ii. Orbital containing the electron that experiences the minimum effective nuclear charge.

b. 2 points

An unknown anion (A^-) has the following *excited* electronic configuration: $[Ar] 4s^2 3d^3 4p^1$

(Ar: Argon; $Z = 18$)

Give the **condensed ground state electronic configuration** of the atom, **A** and the cation, **A^{2+}** .
Show your work. (Assume all electronic transitions are possible)

Electronic Configuration of A:

Electronic Configuration of A^{2+} :

c. 2 points

In a single atom, determine how many electrons can have the following set of quantum numbers?

(i) $n = 5; l = 3$

In a single atom, determine how many electrons can have the following set of quantum numbers?

(i) $n = 3; l = 2; m_s = +1/2$

**Question 4** **8 points**

Rank the following from 1 to 4 in **increasing** order. Provide the ranking in the boxes provided.

Note: Atomic number (Z) are provided for relevant atoms

a) Increasing (lowest to highest) order of *ionization energy*: (1= lowest; 4 = highest)

Mg

Mg: Z = 12

Rb

Rb: Z = 37

Cs

Cs: Z = 55

K

K : Z = 19

b) Increasing (lowest to highest) order of *bond angle*: (1= lowest; 4 = highest)

H₂OCCl₄XeF₄

HCN

c) Increasing (smallest to largest) order of *ionic size* (1= smallest; 4 = largest)

S²⁻

S: Z = 16

O²⁻

O: Z = 8

Be²⁺

Be: Z = 4

Na⁺

Na: Z = 11

d) Increasing (lowest to highest) order of *bond energy* (1= lowest; 4 = highest)

P-F

P-Cl

P-I

P-Br

**Question 5** **12 points**

a. 9 points For the molecules HClO_4 and TeF_6 , draw the most stable *Lewis structure* showing appropriate shape and *dashed/wedged bonds*, based on VSEPR. Determine the **molecular geometry** around the central atom. For HClO_4 , label the *most polar bond*.



b. 3 points (Note: Lone pairs are not shown in structures below)

(i) For the three *resonance* structures (1, 2 and 3) shown below, answer the following question:



(I) Circle the *most contributing* resonance structure.

Structure 1

Structure 2

Structure 3

(ii) For the three structures given (4, 5, and 6) below, answer the following questions:



(I) Circle all the structures that represent *resonance structures* of each other:

Structure 4

Structure 5

Structure 6

(II) Among the resonance structures, circle the *most contributing* resonance structure.

Structure 4

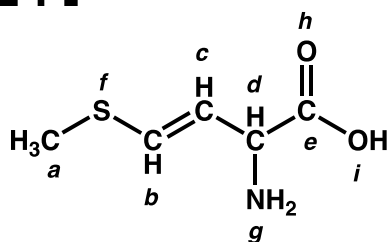
Structure 5

Structure 6

**Question 6** **6 points**

Using the data given below, calculate the **lattice energy** for SrO(s).

	ΔH° (kJ/mol)
$\text{Sr}(s) \longrightarrow \text{Sr}(g)$	+180
$\text{Sr}(g) \longrightarrow \text{Sr}^{2+}(g) + 2e^-$	+1468
$\text{O}_2(g) \longrightarrow 2\text{O}(g)$	+500
$\text{O}(g) + 2e^- \longrightarrow \text{O}^{2-}(g)$	+600
$\text{Sr}(s) + \frac{1}{2}\text{O}_2(g) \longrightarrow \text{SrO}(s)$	-550

**Question 7 10 points**

Consider the molecule given on the left, and answer questions (i) to (vi) using chemical bonding theories that we have learnt in class (valence bond theory, hybridization).

Note that the lone pairs are not shown in the structure:

i. Consider the carbon atoms marked in as ***a***, ***b***, ***c***, ***d***, and ***e*** in the molecule, and determine **hybridization** at each of the carbon centers. (2.5 points)

Atom	Hybridization
Carbon (<i>a</i>)	
Carbon (<i>b</i>)	
Carbon (<i>c</i>)	
Carbon (<i>d</i>)	
Carbon (<i>e</i>)	

ii. Now, consider the sulfur atom (S) marked as ***f***, nitrogen (N) atom (***g***), and oxygen(O) atoms (***h*** and ***i***), and determine their **hybridization**. (2 points)

Atom	Hybridization
Sulfur (<i>f</i>)	
Nitrogen (<i>g</i>)	
Oxygen (<i>h</i>)	
Oxygen (<i>i</i>)	

iii. **Describe** all the bonds formed by the carbon atoms ***a*** and ***b***, using sigma and pi designation as appropriate. (2 points)

Atom	Bonds formed (use sigma and pi as appropriate)
Carbon (<i>a</i>)	
Carbon (<i>b</i>)	

iv. How many total sp^3 -s sigma bonds are in the molecule? (1 point)

v. Determine **electronic** and **molecular geometry** at carbon (C) labeled ***e*** and sulfur (S) labeled ***f*** centres. (2 points)

Atom	Electronic Geometry	Molecular Geometry
Carbon (<i>e</i>)		
Sulfur (<i>f</i>)		

vi. Circle the value from the choices given below, that describes the **C-S-C bond angle at sulfur** (S) marked ***f***: (0.5 points)

$>90^\circ$ to $<109.5^\circ$ 109.5° >109.5 to $<120^\circ$ 120° $>120^\circ$ to $<180^\circ$


Question 8 12 points

Using the **valence shell orbitals only** draw a **complete** molecular orbital (MO) diagram (with no s-p mixing) for the molecule **PS**, in the space provided below. **(6 points)**

Assume that 3s and 3p atomic orbitals will form molecular orbitals like those formed by the combination of 2s and 2p atomic orbitals that we learnt in class, and there is no s-p mixing.

Show **energy arrow**, **atomic orbitals** involved, **atom labels**, the **molecular orbitals** that are formed, **filled electrons** in the MO diagram etc. Make sure to label all the atomic and molecular orbitals.

Using the MO diagram above, answer the following questions:

- (i) Which one of the following will have the highest bond order? Circle the correct choice
 A) PS^{1-} B) PS C) PS^{1+}
- (ii) Which one of the following will have the longest bond length? Circle the correct choice
 A) PS^{1-} B) PS C) PS^{1+}
- (iii) Which of the following will be diamagnetic? Circle the correct choice.
 A) PS^{1-} B) PS C) PS^{1+}
- (iv) Draw Lewis dot structure of the molecule with the shortest bond length (based on the MO diagram) from among the three (PS^{1-} , PS, PS^{1+}), and indicate the bond order calculated from the Lewis dot structure. **(3 points)**

Lewis Dot Structure

Bond Order



8D18EBE8-1970-4800-81C7-60D58FAA022F

midterm-exam-93388

#339 Page 10 of 10

SCRAP PAGE

**CHEMISTRY 110 –Midterm Exam****Thursday November 7, 2024, 6:30 – 8:30 pm**

Examiners: Pallavi Sirjoosingh and Ashok Kakkar

First name (please write as legibly as possible within the boxes)

--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--

Last name

--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--

Student ID number

--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--

This booklet has 8 questions on 10 pages (The last page can be used as a scrap page).

The staple must NOT be removed from the question booklet. Your intact question booklet must be returned at the end of the exam.

A separate booklet contains a periodic table, and datasheet.

Write your answers directly under the questions in the space provided. Do not write on the QR codes or near the top edges (as it may be cut off).

Please write **legibly and dark enough** for the scan to be readable. For students who use English as a second language, dictionaries are permitted. Calculators are allowed.

DO NOT ASK QUESTIONS OF THE INVIGILATORS about interpretations of the exam questions. If you have a concern about a question, please summarize it at the top of this page. All concerns will be evaluated after the exam. GOOD LUCK!

**Question 1 8 points** (*Assume all transitions are possible*)

a. 6 points Calculate the **energy (in kJ/mol)** required to ionize (completely remove) the electron from the **n=3** level of **He⁺ ion** (He: Helium). Show your work including equations. (3 sig. figs.)

b. 2 points

(I) Select the largest (longest) wavelength from the three choices below:

- i. Wavelength of photon to *completely remove* an electron from *1s orbital of Hydrogen atom*
- ii. Wavelength of photon to *completely remove* an electron from *1s orbital of Boron atom (B)*
- iii. Wavelength of photon to *completely remove* an electron from *1s orbital of Boron cation (B⁺)*

i or ii or iii

(II) Select the smallest (shortest) wavelength from the three choices below (*for H atom*):

- i. Wavelength of photon *emitted* for an electron transitioning from *2p to 1s orbital*
- ii. Wavelength of photon *absorbed* for an electron transitioning from *2s to 4p orbital*
- iii. Wavelength of photon *absorbed* for an electron transitioning from *1s to 3p orbital*

i or ii or iii



Question 2 8 points



A Chem110 student plots the graph shown on the left, from data obtained for the kinetic energy (eV) of the ejected electron versus frequency (s^{-1}) of the incident light, for an unknown metal. ($1\text{eV} = 1.6022 \times 10^{-19} \text{ J}$)

Based on the plot, answer the questions **a to c** below:

a. 1 point On the plot above, label the **threshold frequency**.

b. 2 points What is the value of the work potential (in J) of the unknown metal?

c. 3 points What is the *longest* wavelength (in nm) required to eject an electron from this metal's surface? Show your work including equation(s) used. (3 sig figs)

d. 2 points In the space provided below, plot the following

(Assume frequency is *above* threshold frequency):

- Kinetic energy* of an ejected electron versus the *intensity* of the incident light (left)
- Number of electrons* ejected versus the *frequency* of the incident light (right)



**Question 3** **6 points****a. 2 points**

In the ground state of Ge (Germanium; $Z = 32$) atom, **name** the following orbitals:

i. Orbital containing the electron that requires the maximum energy to be removed.

ii. Orbital containing the electron that experiences the minimum effective nuclear charge.

b. 2 points

An unknown anion (A^-) has the following *excited* electronic configuration: $[Ar] 4s^2 3d^3 4p^1$

(Ar: Argon; $Z = 18$)

Give the **condensed ground state electronic configuration** of the atom, **A** and the cation, **A^{2+}** .
Show your work. (Assume all electronic transitions are possible)

Electronic Configuration of A:

Electronic Configuration of A^{2+} :

c. 2 points

In a single atom, determine how many electrons can have the following set of quantum numbers?

(i) $n = 5; l = 3$

In a single atom, determine how many electrons can have the following set of quantum numbers?

(i) $n = 3; l = 2; m_s = +1/2$

**Question 4** **8 points**

Rank the following from 1 to 4 in **increasing** order. Provide the ranking in the boxes provided.

Note: Atomic number (Z) are provided for relevant atoms

a) Increasing (lowest to highest) order of *ionization energy*: (1= lowest; 4 = highest)

Mg

Mg: Z = 12

Rb

Rb: Z = 37

Cs

Cs: Z = 55

K

K : Z = 19

b) Increasing (lowest to highest) order of *bond angle*: (1= lowest; 4 = highest)

H₂OCCl₄XeF₄

HCN

c) Increasing (smallest to largest) order of *ionic size* (1= smallest; 4 = largest)

S²⁻

S: Z = 16

O²⁻

O: Z = 8

Be²⁺

Be: Z = 4

Na⁺

Na: Z = 11

d) Increasing (lowest to highest) order of *bond energy* (1= lowest; 4 = highest)

P-F

P-Cl

P-I

P-Br

**Question 5** **12 points**

a. 9 points For the molecules HClO_4 and TeF_6 , draw the most stable *Lewis structure* showing appropriate shape and *dashed/wedged bonds*, based on VSEPR. Determine the **molecular geometry** around the central atom. For HClO_4 , label the *most polar bond*.



b. 3 points (Note: Lone pairs are not shown in structures below)

(i) For the three *resonance* structures (1, 2 and 3) shown below, answer the following question:



(I) Circle the *most contributing* resonance structure.

Structure 1

Structure 2

Structure 3

(ii) For the three structures given (4, 5, and 6) below, answer the following questions:



(I) Circle all the structures that represent *resonance structures* of each other:

Structure 4

Structure 5

Structure 6

(II) Among the resonance structures, circle the *most contributing* resonance structure.

Structure 4

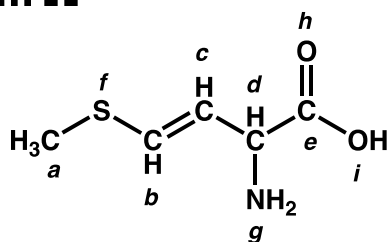
Structure 5

Structure 6

**Question 6** **6 points**

Using the data given below, calculate the **lattice energy** for SrO(s).

	ΔH° (kJ/mol)
$\text{Sr}(s) \longrightarrow \text{Sr}(g)$	+180
$\text{Sr}(g) \longrightarrow \text{Sr}^{2+}(g) + 2e^-$	+1468
$\text{O}_2(g) \longrightarrow 2\text{O}(g)$	+500
$\text{O}(g) + 2e^- \longrightarrow \text{O}^{2-}(g)$	+600
$\text{Sr}(s) + \frac{1}{2}\text{O}_2(g) \longrightarrow \text{SrO}(s)$	-550

**Question 7 10 points**

Consider the molecule given on the left, and answer questions (i) to (vi) using chemical bonding theories that we have learnt in class (valence bond theory, hybridization).

Note that the lone pairs are not shown in the structure:

i. Consider the carbon atoms marked in as ***a***, ***b***, ***c***, ***d***, and ***e*** in the molecule, and determine **hybridization** at each of the carbon centers. (2.5 points)

Atom	Hybridization
Carbon (<i>a</i>)	
Carbon (<i>b</i>)	
Carbon (<i>c</i>)	
Carbon (<i>d</i>)	
Carbon (<i>e</i>)	

ii. Now, consider the sulfur atom (S) marked as ***f***, nitrogen (N) atom (***g***), and oxygen(O) atoms (***h*** and ***i***), and determine their **hybridization**. (2 points)

Atom	Hybridization
Sulfur (<i>f</i>)	
Nitrogen (<i>g</i>)	
Oxygen (<i>h</i>)	
Oxygen (<i>i</i>)	

iii. **Describe** all the bonds formed by the carbon atoms ***a*** and ***b***, using sigma and pi designation as appropriate. (2 points)

Atom	Bonds formed (use sigma and pi as appropriate)
Carbon (<i>a</i>)	
Carbon (<i>b</i>)	

iv. How many total sp^3 -s sigma bonds are in the molecule? (1 point)

v. Determine **electronic** and **molecular geometry** at carbon (C) labeled ***e*** and sulfur (S) labeled ***f*** centres. (2 points)

Atom	Electronic Geometry	Molecular Geometry
Carbon (<i>e</i>)		
Sulfur (<i>f</i>)		

vi. Circle the value from the choices given below, that describes the **C-S-C bond angle at sulfur** (S) marked ***f***: (0.5 points)

>90° to <109.5° 109.5° >109.5 to <120° 120° >120° to <180°

**Question 8 12 points**

Using the **valence shell orbitals only** draw a **complete** molecular orbital (MO) diagram (with no s-p mixing) for the molecule **PS**, in the space provided below. **(6 points)**

Assume that 3s and 3p atomic orbitals will form molecular orbitals like those formed by the combination of 2s and 2p atomic orbitals that we learnt in class, and there is no s-p mixing.

Show **energy arrow**, **atomic orbitals** involved, **atom labels**, the **molecular orbitals** that are formed, **filled electrons** in the MO diagram etc. Make sure to label all the atomic and molecular orbitals.

Using the MO diagram above, answer the following questions:

- (i) Which one of the following will have the highest bond order? Circle the correct choice
A) PS^{1-} B) PS C) PS^{1+}
- (ii) Which one of the following will have the longest bond length? Circle the correct choice
A) PS^{1-} B) PS C) PS^{1+}
- (iii) Which of the following will be diamagnetic? Circle the correct choice.
A) PS^{1-} B) PS C) PS^{1+}
- (iv) Draw Lewis dot structure of the molecule with the shortest bond length (based on the MO diagram) from among the three (PS^{1-} , PS, PS^{1+}), and indicate the bond order calculated from the Lewis dot structure. **(3 points)**

Lewis Dot Structure

Bond Order



FC934CC2-5E22-4D38-B1EB-E3D8A5683670

midterm-exam-93388

#340 Page 10 of 10

SCRAP PAGE

**CHEMISTRY 110 –Midterm Exam****Thursday November 7, 2024, 6:30 – 8:30 pm****Examiners: Pallavi Sirjoosingh and Ashok Kakkar**

First name (please write as legibly as possible within the boxes)

--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--

Last name

--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--

Student ID number

--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--

This booklet has 8 questions on 10 pages (The last page can be used as a scrap page).

The staple must NOT be removed from the question booklet. Your intact question booklet must be returned at the end of the exam.

A separate booklet contains a periodic table, and datasheet.

Write your answers directly under the questions in the space provided. Do not write on the QR codes or near the top edges (as it may be cut off).

Please write **legibly and dark enough** for the scan to be readable. For students who use English as a second language, dictionaries are permitted. Calculators are allowed.

DO NOT ASK QUESTIONS OF THE INVIGILATORS about interpretations of the exam questions. If you have a concern about a question, please summarize it at the top of this page. All concerns will be evaluated after the exam. GOOD LUCK!

**Question 1 8 points** (*Assume all transitions are possible*)

a. 6 points Calculate the **energy (in kJ/mol)** required to ionize (completely remove) the electron from the **$n=3$** level of **He^+ ion** (He: Helium). Show your work including equations. (3 sig. figs.)

b. 2 points

(I) Select the largest (longest) wavelength from the three choices below:

- i. Wavelength of photon to *completely remove* an electron from *1s orbital of Hydrogen atom*
- ii. Wavelength of photon to *completely remove* an electron from *1s orbital of Boron atom (B)*
- iii. Wavelength of photon to *completely remove* an electron from *1s orbital of Boron cation (B^+)*

i or ii or iii

(II) Select the smallest (shortest) wavelength from the three choices below (*for H atom*):

- i. Wavelength of photon *emitted* for an electron transitioning from *2p to 1s orbital*
- ii. Wavelength of photon *absorbed* for an electron transitioning from *2s to 4p orbital*
- iii. Wavelength of photon *absorbed* for an electron transitioning from *1s to 3p orbital*

i or ii or iii



Question 2 8 points



A Chem110 student plots the graph shown on the left, from data obtained for the kinetic energy (eV) of the ejected electron versus frequency (s^{-1}) of the incident light, for an unknown metal. ($1\text{eV} = 1.6022 \times 10^{-19} \text{ J}$)

Based on the plot, answer the questions **a to c** below:

a. 1 point On the plot above, label the **threshold frequency**.

b. 2 points What is the value of the work potential (in J) of the unknown metal?

c. 3 points What is the *longest* wavelength (in nm) required to eject an electron from this metal's surface? Show your work including equation(s) used. (3 sig figs)

d. 2 points In the space provided below, plot the following

(Assume frequency is *above* threshold frequency):

- Kinetic energy* of an ejected electron versus the *intensity* of the incident light (left)
- Number of electrons* ejected versus the *frequency* of the incident light (right)



**Question 3** **6 points****a. 2 points**

In the ground state of Ge (Germanium; $Z = 32$) atom, **name** the following orbitals:

i. Orbital containing the electron that requires the maximum energy to be removed.

ii. Orbital containing the electron that experiences the minimum effective nuclear charge.

b. 2 points

An unknown anion (A^-) has the following *excited* electronic configuration: $[Ar] 4s^2 3d^3 4p^1$

(Ar: Argon; $Z = 18$)

Give the **condensed ground state electronic configuration** of the atom, **A** and the cation, **A^{2+}** .
Show your work. (Assume all electronic transitions are possible)

Electronic Configuration of A:

Electronic Configuration of A^{2+} :

c. 2 points

In a single atom, determine how many electrons can have the following set of quantum numbers?

(i) $n = 5; l = 3$

In a single atom, determine how many electrons can have the following set of quantum numbers?

(i) $n = 3; l = 2; m_s = +1/2$

**Question 4** **8 points**

Rank the following from 1 to 4 in **increasing** order. Provide the ranking in the boxes provided.

Note: Atomic number (Z) are provided for relevant atoms

a) Increasing (lowest to highest) order of *ionization energy*: (1= lowest; 4 = highest)

Mg

Mg: Z = 12

Rb

Rb: Z = 37

Cs

Cs: Z = 55

K

K : Z = 19

b) Increasing (lowest to highest) order of *bond angle*: (1= lowest; 4 = highest)

H₂OCCl₄XeF₄

HCN

c) Increasing (smallest to largest) order of *ionic size* (1= smallest; 4 = largest)

S²⁻

S: Z = 16

O²⁻

O: Z = 8

Be²⁺

Be: Z = 4

Na⁺

Na: Z = 11

d) Increasing (lowest to highest) order of *bond energy* (1= lowest; 4 = highest)

P-F

P-Cl

P-I

P-Br

**Question 5** **12 points**

a. 9 points For the molecules HClO_4 and TeF_6 , draw the most stable *Lewis structure* showing appropriate shape and *dashed/wedged bonds*, based on VSEPR. Determine the **molecular geometry** around the central atom. For HClO_4 , label the *most polar bond*.



b. 3 points (Note: Lone pairs are not shown in structures below)

(i) For the three *resonance* structures (1, 2 and 3) shown below, answer the following question:



(I) Circle the *most contributing* resonance structure.

Structure 1

Structure 2

Structure 3

(ii) For the three structures given (4, 5, and 6) below, answer the following questions:



(I) Circle all the structures that represent *resonance structures* of each other:

Structure 4

Structure 5

Structure 6

(II) Among the resonance structures, circle the *most contributing* resonance structure.

Structure 4

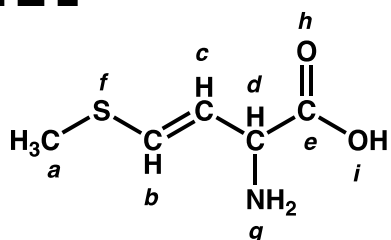
Structure 5

Structure 6

**Question 6** **6 points**

Using the data given below, calculate the **lattice energy** for SrO(s).

	ΔH° (kJ/mol)
$\text{Sr}(s) \longrightarrow \text{Sr}(g)$	+180
$\text{Sr}(g) \longrightarrow \text{Sr}^{2+}(g) + 2e^-$	+1468
$\text{O}_2(g) \longrightarrow 2\text{O}(g)$	+500
$\text{O}(g) + 2e^- \longrightarrow \text{O}^{2-}(g)$	+600
$\text{Sr}(s) + \frac{1}{2}\text{O}_2(g) \longrightarrow \text{SrO}(s)$	-550

**Question 7 10 points**

Consider the molecule given on the left, and answer questions (i) to (vi) using chemical bonding theories that we have learnt in class (valence bond theory, hybridization).

Note that the lone pairs are not shown in the structure:

i. Consider the carbon atoms marked in as ***a***, ***b***, ***c***, ***d***, and ***e*** in the molecule, and determine **hybridization** at each of the carbon centers. (2.5 points)

Atom	Hybridization
Carbon (<i>a</i>)	
Carbon (<i>b</i>)	
Carbon (<i>c</i>)	
Carbon (<i>d</i>)	
Carbon (<i>e</i>)	

ii. Now, consider the sulfur atom (S) marked as ***f***, nitrogen (N) atom (***g***), and oxygen(O) atoms (***h*** and ***i***), and determine their **hybridization**. (2 points)

Atom	Hybridization
Sulfur (<i>f</i>)	
Nitrogen (<i>g</i>)	
Oxygen (<i>h</i>)	
Oxygen (<i>i</i>)	

iii. **Describe** all the bonds formed by the carbon atoms ***a*** and ***b***, using sigma and pi designation as appropriate. (2 points)

Atom	Bonds formed (use sigma and pi as appropriate)
Carbon (<i>a</i>)	
Carbon (<i>b</i>)	

iv. How many total sp^3 -s sigma bonds are in the molecule? (1 point)

v. Determine **electronic** and **molecular geometry** at carbon (C) labeled ***e*** and sulfur (S) labeled ***f*** centres. (2 points)

Atom	Electronic Geometry	Molecular Geometry
Carbon (<i>e</i>)		
Sulfur (<i>f</i>)		

vi. Circle the value from the choices given below, that describes the **C-S-C bond angle at sulfur** (S) marked ***f***: (0.5 points)

>90° to <109.5°

109.5°

>109.5 to <120°

120°

>120° to <180°


Question 8 12 points

Using the **valence shell orbitals only** draw a **complete** molecular orbital (MO) diagram (with no s-p mixing) for the molecule **PS**, in the space provided below. **(6 points)**

Assume that 3s and 3p atomic orbitals will form molecular orbitals like those formed by the combination of 2s and 2p atomic orbitals that we learnt in class, and there is no s-p mixing.

Show **energy arrow**, **atomic orbitals** involved, **atom labels**, the **molecular orbitals** that are formed, **filled electrons** in the MO diagram etc. Make sure to label all the atomic and molecular orbitals.

Using the MO diagram above, answer the following questions:

- (i) Which one of the following will have the highest bond order? Circle the correct choice
 A) PS^{1-} B) PS C) PS^{1+}
- (ii) Which one of the following will have the longest bond length? Circle the correct choice
 A) PS^{1-} B) PS C) PS^{1+}
- (iii) Which of the following will be diamagnetic? Circle the correct choice.
 A) PS^{1-} B) PS C) PS^{1+}
- (iv) Draw Lewis dot structure of the molecule with the shortest bond length (based on the MO diagram) from among the three (PS^{1-} , PS, PS^{1+}), and indicate the bond order calculated from the Lewis dot structure. **(3 points)**

Lewis Dot Structure

Bond Order



C34B896A-D3B5-4B76-BF97-3B4E7222AA79

midterm-exam-93388

#341 Page 10 of 10

SCRAP PAGE

**CHEMISTRY 110 –Midterm Exam****Thursday November 7, 2024, 6:30 – 8:30 pm****Examiners: Pallavi Sirjoosingh and Ashok Kakkar**

First name (please write as legibly as possible within the boxes)

--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--

Last name

--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--

Student ID number

--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--

This booklet has 8 questions on 10 pages (The last page can be used as a scrap page).

The staple must NOT be removed from the question booklet. Your intact question booklet must be returned at the end of the exam.

A separate booklet contains a periodic table, and datasheet.

Write your answers directly under the questions in the space provided. Do not write on the QR codes or near the top edges (as it may be cut off).

Please write **legibly and dark enough** for the scan to be readable. For students who use English as a second language, dictionaries are permitted. Calculators are allowed.

DO NOT ASK QUESTIONS OF THE INVIGILATORS about interpretations of the exam questions. If you have a concern about a question, please summarize it at the top of this page. All concerns will be evaluated after the exam. GOOD LUCK!

**Question 1 8 points** (*Assume all transitions are possible*)

a. 6 points Calculate the **energy (in kJ/mol)** required to ionize (completely remove) the electron from the **n=3** level of **He⁺ ion** (He: Helium). Show your work including equations. (3 sig. figs.)

b. 2 points

(I) Select the largest (longest) wavelength from the three choices below:

- i. Wavelength of photon to *completely remove* an electron from *1s orbital of Hydrogen atom*
- ii. Wavelength of photon to *completely remove* an electron from *1s orbital of Boron atom (B)*
- iii. Wavelength of photon to *completely remove* an electron from *1s orbital of Boron cation (B⁺)*

i or ii or iii

(II) Select the smallest (shortest) wavelength from the three choices below (*for H atom*):

- i. Wavelength of photon *emitted* for an electron transitioning from *2p to 1s orbital*
- ii. Wavelength of photon *absorbed* for an electron transitioning from *2s to 4p orbital*
- iii. Wavelength of photon *absorbed* for an electron transitioning from *1s to 3p orbital*

i or ii or iii



Question 2 8 points



A Chem110 student plots the graph shown on the left, from data obtained for the kinetic energy (eV) of the ejected electron versus frequency (s^{-1}) of the incident light, for an unknown metal. ($1\text{eV} = 1.6022 \times 10^{-19} \text{ J}$)

Based on the plot, answer the questions **a to c** below:

a. 1 point On the plot above, label the **threshold frequency**.

b. 2 points What is the value of the work potential (in J) of the unknown metal?

c. 3 points What is the *longest* wavelength (in nm) required to eject an electron from this metal's surface? Show your work including equation(s) used. (3 sig figs)

d. 2 points In the space provided below, plot the following

(Assume frequency is *above* threshold frequency):

- Kinetic energy* of an ejected electron versus the *intensity* of the incident light (left)
- Number of electrons* ejected versus the *frequency* of the incident light (right)



**Question 3** **6 points****a. 2 points**

In the ground state of Ge (Germanium; $Z = 32$) atom, **name** the following orbitals:

i. Orbital containing the electron that requires the maximum energy to be removed.

ii. Orbital containing the electron that experiences the minimum effective nuclear charge.

b. 2 points

An unknown anion (A^-) has the following *excited* electronic configuration: $[Ar] 4s^2 3d^3 4p^1$

(Ar: Argon; $Z = 18$)

Give the **condensed ground state electronic configuration** of the atom, **A** and the cation, A^{2+} .
Show your work. (Assume all electronic transitions are possible)

Electronic Configuration of A:

Electronic Configuration of A^{2+} :

c. 2 points

In a single atom, determine how many electrons can have the following set of quantum numbers?

(i) $n = 5; l = 3$

In a single atom, determine how many electrons can have the following set of quantum numbers?

(i) $n = 3; l = 2; m_s = +1/2$

**Question 4** **8 points**

Rank the following from 1 to 4 in **increasing** order. Provide the ranking in the boxes provided.

Note: Atomic number (Z) are provided for relevant atoms

a) Increasing (lowest to highest) order of *ionization energy*: (1= lowest; 4 = highest)

Mg

Mg: Z = 12

Rb

Rb: Z = 37

Cs

Cs: Z = 55

K

K : Z = 19

b) Increasing (lowest to highest) order of *bond angle*: (1= lowest; 4 = highest)

H₂OCCl₄XeF₄

HCN

c) Increasing (smallest to largest) order of *ionic size* (1= smallest; 4 = largest)

S²⁻

S: Z = 16

O²⁻

O: Z = 8

Be²⁺

Be: Z = 4

Na⁺

Na: Z = 11

d) Increasing (lowest to highest) order of *bond energy* (1= lowest; 4 = highest)

P-F

P-Cl

P-I

P-Br

**Question 5** **12 points**

a. 9 points For the molecules HClO_4 and TeF_6 , draw the most stable *Lewis structure* showing appropriate shape and *dashed/wedged bonds*, based on VSEPR. Determine the **molecular geometry** around the central atom. For HClO_4 , label the *most polar bond*.

HClO_4 (there are no O-O bonds)

Label the most polar bond

Molecular Geometry around Cl

TeF_6

Molecular Geometry around Te

b. 3 points (Note: Lone pairs are not shown in structures below)

(i) For the three *resonance* structures (1, 2 and 3) shown below, answer the following question:



(I) Circle the *most contributing* resonance structure.

Structure 1

Structure 2

Structure 3

(ii) For the three structures given (4, 5, and 6) below, answer the following questions:



(I) Circle all the structures that represent *resonance structures* of each other:

Structure 4

Structure 5

Structure 6

(II) Among the resonance structures, circle the *most contributing* resonance structure.

Structure 4

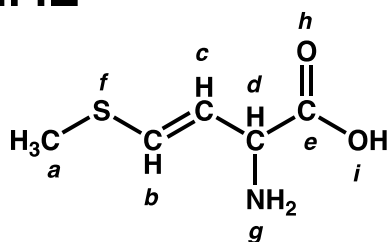
Structure 5

Structure 6

**Question 6** **6 points**

Using the data given below, calculate the **lattice energy** for SrO(s).

	ΔH° (kJ/mol)
$\text{Sr}(s) \longrightarrow \text{Sr}(g)$	+180
$\text{Sr}(g) \longrightarrow \text{Sr}^{2+}(g) + 2e^-$	+1468
$\text{O}_2(g) \longrightarrow 2\text{O}(g)$	+500
$\text{O}(g) + 2e^- \longrightarrow \text{O}^{2-}(g)$	+600
$\text{Sr}(s) + \frac{1}{2}\text{O}_2(g) \longrightarrow \text{SrO}(s)$	-550

**Question 7 10 points**

Consider the molecule given on the left, and answer questions (i) to (vi) using chemical bonding theories that we have learnt in class (valence bond theory, hybridization).

Note that the lone pairs are not shown in the structure:

i. Consider the carbon atoms marked in as ***a***, ***b***, ***c***, ***d***, and ***e*** in the molecule, and determine **hybridization** at each of the carbon centers. (2.5 points)

Atom	Hybridization
Carbon (<i>a</i>)	
Carbon (<i>b</i>)	
Carbon (<i>c</i>)	
Carbon (<i>d</i>)	
Carbon (<i>e</i>)	

ii. Now, consider the sulfur atom (S) marked as ***f***, nitrogen (N) atom (***g***), and oxygen(O) atoms (***h*** and ***i***), and determine their **hybridization**. (2 points)

Atom	Hybridization
Sulfur (<i>f</i>)	
Nitrogen (<i>g</i>)	
Oxygen (<i>h</i>)	
Oxygen (<i>i</i>)	

iii. **Describe** all the bonds formed by the carbon atoms ***a*** and ***b***, using sigma and pi designation as appropriate. (2 points)

Atom	Bonds formed (use sigma and pi as appropriate)
Carbon (<i>a</i>)	
Carbon (<i>b</i>)	

iv. How many total sp^3 -s sigma bonds are in the molecule? (1 point)

v. Determine **electronic** and **molecular geometry** at carbon (C) labeled ***e*** and sulfur (S) labeled ***f*** centres. (2 points)

Atom	Electronic Geometry	Molecular Geometry
Carbon (<i>e</i>)		
Sulfur (<i>f</i>)		

vi. Circle the value from the choices given below, that describes the **C-S-C bond angle at sulfur** (S) marked ***f***: (0.5 points)

>90° to <109.5° 109.5° >109.5 to <120° 120° >120° to <180°


Question 8 12 points

Using the **valence shell orbitals only** draw a **complete** molecular orbital (MO) diagram (with no s-p mixing) for the molecule **PS**, in the space provided below. **(6 points)**

Assume that 3s and 3p atomic orbitals will form molecular orbitals like those formed by the combination of 2s and 2p atomic orbitals that we learnt in class, and there is no s-p mixing.

Show **energy arrow**, **atomic orbitals** involved, **atom labels**, the **molecular orbitals** that are formed, **filled electrons** in the MO diagram etc. Make sure to label all the atomic and molecular orbitals.

Using the MO diagram above, answer the following questions:

- (i) Which one of the following will have the highest bond order? Circle the correct choice
 A) PS^{1-} B) PS C) PS^{1+}
- (ii) Which one of the following will have the longest bond length? Circle the correct choice
 A) PS^{1-} B) PS C) PS^{1+}
- (iii) Which of the following will be diamagnetic? Circle the correct choice.
 A) PS^{1-} B) PS C) PS^{1+}
- (iv) Draw Lewis dot structure of the molecule with the shortest bond length (based on the MO diagram) from among the three (PS^{1-} , PS, PS^{1+}), and indicate the bond order calculated from the Lewis dot structure. **(3 points)**

Lewis Dot Structure

Bond Order



477CDA50-CE10-41B0-9D29-C0841E5057F7

midterm-exam-93388

#342 Page 10 of 10

SCRAP PAGE

**CHEMISTRY 110 –Midterm Exam****Thursday November 7, 2024, 6:30 – 8:30 pm**

Examiners: Pallavi Sirjoosingh and Ashok Kakkar

First name (please write as legibly as possible within the boxes)

--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--

Last name

--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--

Student ID number

--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--

This booklet has 8 questions on 10 pages (The last page can be used as a scrap page).

The staple must NOT be removed from the question booklet. Your intact question booklet must be returned at the end of the exam.

A separate booklet contains a periodic table, and datasheet.

Write your answers directly under the questions in the space provided. Do not write on the QR codes or near the top edges (as it may be cut off).

Please write **legibly and dark enough** for the scan to be readable. For students who use English as a second language, dictionaries are permitted. Calculators are allowed.

DO NOT ASK QUESTIONS OF THE INVIGILATORS about interpretations of the exam questions. If you have a concern about a question, please summarize it at the top of this page. All concerns will be evaluated after the exam. GOOD LUCK!

**Question 1 8 points** (*Assume all transitions are possible*)

a. 6 points Calculate the **energy (in kJ/mol)** required to ionize (completely remove) the electron from the **$n=3$** level of **He^+ ion** (He: Helium). Show your work including equations. (3 sig. figs.)

b. 2 points

(I) Select the largest (longest) wavelength from the three choices below:

- i. Wavelength of photon to *completely remove* an electron from *1s orbital of Hydrogen atom*
- ii. Wavelength of photon to *completely remove* an electron from *1s orbital of Boron atom (B)*
- iii. Wavelength of photon to *completely remove* an electron from *1s orbital of Boron cation (B^+)*

i or ii or iii

(II) Select the smallest (shortest) wavelength from the three choices below (*for H atom*):

- i. Wavelength of photon *emitted* for an electron transitioning from *2p to 1s orbital*
- ii. Wavelength of photon *absorbed* for an electron transitioning from *2s to 4p orbital*
- iii. Wavelength of photon *absorbed* for an electron transitioning from *1s to 3p orbital*

i or ii or iii



Question 2 8 points



A Chem110 student plots the graph shown on the left, from data obtained for the kinetic energy (eV) of the ejected electron versus frequency (s^{-1}) of the incident light, for an unknown metal. ($1\text{eV} = 1.6022 \times 10^{-19} \text{ J}$)

Based on the plot, answer the questions **a to c** below:

a. 1 point On the plot above, label the **threshold frequency**.

b. 2 points What is the value of the work potential (in J) of the unknown metal?

c. 3 points What is the *longest* wavelength (in nm) required to eject an electron from this metal's surface? Show your work including equation(s) used. (3 sig figs)

d. 2 points In the space provided below, plot the following

(Assume frequency is *above* threshold frequency):

- i. *Kinetic energy* of an ejected electron versus the *intensity* of the incident light (left)
- ii. *Number of electrons* ejected versus the *frequency* of the incident light (right)



**Question 3** **6 points****a. 2 points**

In the ground state of Ge (Germanium; $Z = 32$) atom, **name** the following orbitals:

i. Orbital containing the electron that requires the maximum energy to be removed.

ii. Orbital containing the electron that experiences the minimum effective nuclear charge.

b. 2 points

An unknown anion (A^-) has the following *excited* electronic configuration: $[Ar] 4s^2 3d^3 4p^1$

(Ar: Argon; $Z = 18$)

Give the **condensed ground state electronic configuration** of the atom, **A** and the cation, **A^{2+}** .
Show your work. (Assume all electronic transitions are possible)

Electronic Configuration of A:

Electronic Configuration of A^{2+} :

c. 2 points

In a single atom, determine how many electrons can have the following set of quantum numbers?

(i) $n = 5; l = 3$

In a single atom, determine how many electrons can have the following set of quantum numbers?

(i) $n = 3; l = 2; m_s = +1/2$

**Question 4** **8 points**

Rank the following from 1 to 4 in **increasing** order. Provide the ranking in the boxes provided.

Note: Atomic number (Z) are provided for relevant atoms

a) Increasing (lowest to highest) order of *ionization energy*: (1= lowest; 4 = highest)

Mg

Mg: Z = 12

Rb

Rb: Z = 37

Cs

Cs: Z = 55

K

K : Z = 19

b) Increasing (lowest to highest) order of *bond angle*: (1= lowest; 4 = highest)

H₂OCCl₄XeF₄

HCN

c) Increasing (smallest to largest) order of *ionic size* (1= smallest; 4 = largest)

S²⁻

S: Z = 16

O²⁻

O: Z = 8

Be²⁺

Be: Z = 4

Na⁺

Na: Z = 11

d) Increasing (lowest to highest) order of *bond energy* (1= lowest; 4 = highest)

P-F

P-Cl

P-I

P-Br

**Question 5** **12 points**

a. 9 points For the molecules HClO_4 and TeF_6 , draw the most stable *Lewis structure* showing appropriate shape and *dashed/wedged bonds*, based on VSEPR. Determine the **molecular geometry** around the central atom. For HClO_4 , label the *most polar bond*.



b. 3 points (Note: Lone pairs are not shown in structures below)

(i) For the three *resonance* structures (1, 2 and 3) shown below, answer the following question:



(I) Circle the *most contributing* resonance structure.

Structure 1

Structure 2

Structure 3

(ii) For the three structures given (4, 5, and 6) below, answer the following questions:



(I) Circle all the structures that represent *resonance structures* of each other:

Structure 4

Structure 5

Structure 6

(II) Among the resonance structures, circle the *most contributing* resonance structure.

Structure 4

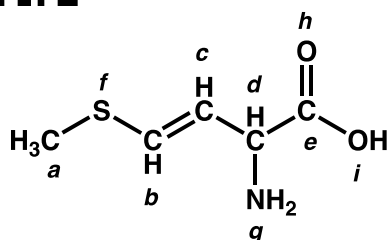
Structure 5

Structure 6

**Question 6** **6 points**

Using the data given below, calculate the **lattice energy** for SrO(s).

	ΔH° (kJ/mol)
$\text{Sr}(s) \longrightarrow \text{Sr}(g)$	+180
$\text{Sr}(g) \longrightarrow \text{Sr}^{2+}(g) + 2e^-$	+1468
$\text{O}_2(g) \longrightarrow 2\text{O}(g)$	+500
$\text{O}(g) + 2e^- \longrightarrow \text{O}^{2-}(g)$	+600
$\text{Sr}(s) + \frac{1}{2}\text{O}_2(g) \longrightarrow \text{SrO}(s)$	-550

**Question 7 10 points**

Consider the molecule given on the left, and answer questions (i) to (vi) using chemical bonding theories that we have learnt in class (valence bond theory, hybridization).

Note that the lone pairs are not shown in the structure:

i. Consider the carbon atoms marked in as ***a***, ***b***, ***c***, ***d***, and ***e*** in the molecule, and determine **hybridization** at each of the carbon centers. (2.5 points)

Atom	Hybridization
Carbon (<i>a</i>)	
Carbon (<i>b</i>)	
Carbon (<i>c</i>)	
Carbon (<i>d</i>)	
Carbon (<i>e</i>)	

ii. Now, consider the sulfur atom (S) marked as ***f***, nitrogen (N) atom (***g***), and oxygen(O) atoms (***h*** and ***i***), and determine their **hybridization**. (2 points)

Atom	Hybridization
Sulfur (<i>f</i>)	
Nitrogen (<i>g</i>)	
Oxygen (<i>h</i>)	
Oxygen (<i>i</i>)	

iii. **Describe** all the bonds formed by the carbon atoms ***a*** and ***b***, using sigma and pi designation as appropriate. (2 points)

Atom	Bonds formed (use sigma and pi as appropriate)
Carbon (<i>a</i>)	
Carbon (<i>b</i>)	

iv. How many total sp^3 -s sigma bonds are in the molecule? (1 point)

v. Determine **electronic** and **molecular geometry** at carbon (C) labeled ***e*** and sulfur (S) labeled ***f*** centres. (2 points)

Atom	Electronic Geometry	Molecular Geometry
Carbon (<i>e</i>)		
Sulfur (<i>f</i>)		

vi. Circle the value from the choices given below, that describes the **C-S-C bond angle at sulfur** (S) marked ***f***: (0.5 points)

>90° to <109.5° 109.5° >109.5 to <120° 120° >120° to <180°

**Question 8 12 points**

Using the **valence shell orbitals only** draw a **complete** molecular orbital (MO) diagram (with no s-p mixing) for the molecule **PS**, in the space provided below. **(6 points)**

Assume that 3s and 3p atomic orbitals will form molecular orbitals like those formed by the combination of 2s and 2p atomic orbitals that we learnt in class, and there is no s-p mixing.

Show **energy arrow**, **atomic orbitals** involved, **atom labels**, the **molecular orbitals** that are formed, **filled electrons** in the MO diagram etc. Make sure to label all the atomic and molecular orbitals.

Using the MO diagram above, answer the following questions:

- (i) Which one of the following will have the highest bond order? Circle the correct choice
A) PS^{1-} B) PS C) PS^{1+}
- (ii) Which one of the following will have the longest bond length? Circle the correct choice
A) PS^{1-} B) PS C) PS^{1+}
- (iii) Which of the following will be diamagnetic? Circle the correct choice.
A) PS^{1-} B) PS C) PS^{1+}
- (iv) Draw Lewis dot structure of the molecule with the shortest bond length (based on the MO diagram) from among the three (PS^{1-} , PS, PS^{1+}), and indicate the bond order calculated from the Lewis dot structure. **(3 points)**

Lewis Dot Structure

Bond Order



A4D09FC6-74F6-4B66-BBB7-F4321ED90C01

midterm-exam-93388

#343 Page 10 of 10

SCRAP PAGE

**CHEMISTRY 110 –Midterm Exam****Thursday November 7, 2024, 6:30 – 8:30 pm**

Examiners: Pallavi Sirjoosingh and Ashok Kakkar

First name (please write as legibly as possible within the boxes)

--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--

Last name

--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--

Student ID number

--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--

This booklet has 8 questions on 10 pages (The last page can be used as a scrap page).

The staple must NOT be removed from the question booklet. Your intact question booklet must be returned at the end of the exam.

A separate booklet contains a periodic table, and datasheet.

Write your answers directly under the questions in the space provided. Do not write on the QR codes or near the top edges (as it may be cut off).

Please write **legibly and dark enough** for the scan to be readable. For students who use English as a second language, dictionaries are permitted. Calculators are allowed.

DO NOT ASK QUESTIONS OF THE INVIGILATORS about interpretations of the exam questions. If you have a concern about a question, please summarize it at the top of this page. All concerns will be evaluated after the exam. GOOD LUCK!

**Question 1 8 points** (*Assume all transitions are possible*)

a. 6 points Calculate the **energy (in kJ/mol)** required to ionize (completely remove) the electron from the **n=3** level of **He⁺ ion** (He: Helium). Show your work including equations. (3 sig. figs.)

b. 2 points

(I) Select the largest (longest) wavelength from the three choices below:

- i. Wavelength of photon to *completely remove* an electron from *1s orbital of Hydrogen atom*
- ii. Wavelength of photon to *completely remove* an electron from *1s orbital of Boron atom (B)*
- iii. Wavelength of photon to *completely remove* an electron from *1s orbital of Boron cation (B⁺)*

i or ii or iii

(II) Select the smallest (shortest) wavelength from the three choices below (*for H atom*):

- i. Wavelength of photon *emitted* for an electron transitioning from *2p to 1s orbital*
- ii. Wavelength of photon *absorbed* for an electron transitioning from *2s to 4p orbital*
- iii. Wavelength of photon *absorbed* for an electron transitioning from *1s to 3p orbital*

i or ii or iii



Question 2 8 points



A Chem110 student plots the graph shown on the left, from data obtained for the kinetic energy (eV) of the ejected electron versus frequency (s^{-1}) of the incident light, for an unknown metal. ($1\text{eV} = 1.6022 \times 10^{-19} \text{ J}$)

Based on the plot, answer the questions **a to c** below:

a. 1 point On the plot above, label the **threshold frequency**.

b. 2 points What is the value of the work potential (in J) of the unknown metal?

c. 3 points What is the *longest* wavelength (in nm) required to eject an electron from this metal's surface? Show your work including equation(s) used. (3 sig figs)

d. 2 points In the space provided below, plot the following

(Assume frequency is *above* threshold frequency):

- i. *Kinetic energy* of an ejected electron versus the *intensity* of the incident light (left)
- ii. *Number of electrons* ejected versus the *frequency* of the incident light (right)



**Question 3** **6 points****a. 2 points**

In the ground state of Ge (Germanium; $Z = 32$) atom, **name** the following orbitals:

i. Orbital containing the electron that requires the maximum energy to be removed.

ii. Orbital containing the electron that experiences the minimum effective nuclear charge.

b. 2 points

An unknown anion (A^-) has the following *excited* electronic configuration: $[Ar] 4s^2 3d^3 4p^1$

(Ar: Argon; $Z = 18$)

Give the **condensed ground state electronic configuration** of the atom, **A** and the cation, **A^{2+}** .
Show your work. (Assume all electronic transitions are possible)

Electronic Configuration of A:

Electronic Configuration of A^{2+} :

c. 2 points

In a single atom, determine how many electrons can have the following set of quantum numbers?

(i) $n = 5; l = 3$

In a single atom, determine how many electrons can have the following set of quantum numbers?

(i) $n = 3; l = 2; m_s = +1/2$

**Question 4** **8 points**

Rank the following from 1 to 4 in **increasing** order. Provide the ranking in the boxes provided.

Note: Atomic number (Z) are provided for relevant atoms

a) Increasing (lowest to highest) order of *ionization energy*: (1= lowest; 4 = highest)

Mg

Mg: Z = 12

Rb

Rb: Z = 37

Cs

Cs: Z = 55

K

K : Z = 19

b) Increasing (lowest to highest) order of *bond angle*: (1= lowest; 4 = highest)

H₂OCCl₄XeF₄

HCN

c) Increasing (smallest to largest) order of *ionic size* (1= smallest; 4 = largest)

S²⁻

S: Z = 16

O²⁻

O: Z = 8

Be²⁺

Be: Z = 4

Na⁺

Na: Z = 11

d) Increasing (lowest to highest) order of *bond energy* (1= lowest; 4 = highest)

P-F

P-Cl

P-I

P-Br

**Question 5** **12 points**

a. 9 points For the molecules HClO_4 and TeF_6 , draw the most stable *Lewis structure* showing appropriate shape and *dashed/wedged bonds*, based on VSEPR. Determine the **molecular geometry** around the central atom. For HClO_4 , label the *most polar bond*.



b. 3 points (Note: Lone pairs are not shown in structures below)

(i) For the three *resonance* structures (1, 2 and 3) shown below, answer the following question:



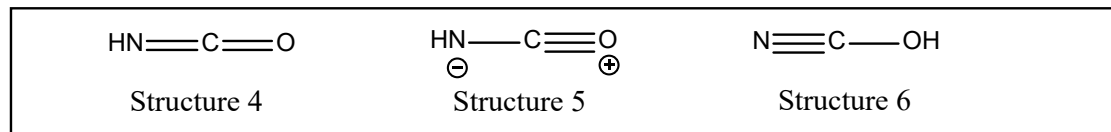
(I) Circle the *most contributing* resonance structure.

Structure 1

Structure 2

Structure 3

(ii) For the three structures given (4, 5, and 6) below, answer the following questions:



(I) Circle all the structures that represent *resonance structures* of each other:

Structure 4

Structure 5

Structure 6

(II) Among the resonance structures, circle the *most contributing* resonance structure.

Structure 4

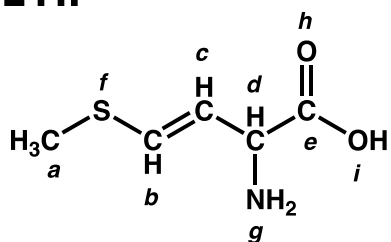
Structure 5

Structure 6

**Question 6** **6 points**

Using the data given below, calculate the **lattice energy** for SrO(s).

	ΔH° (kJ/mol)
$\text{Sr}(s) \longrightarrow \text{Sr}(g)$	+180
$\text{Sr}(g) \longrightarrow \text{Sr}^{2+}(g) + 2e^-$	+1468
$\text{O}_2(g) \longrightarrow 2\text{O}(g)$	+500
$\text{O}(g) + 2e^- \longrightarrow \text{O}^{2-}(g)$	+600
$\text{Sr}(s) + \frac{1}{2}\text{O}_2(g) \longrightarrow \text{SrO}(s)$	-550

**Question 7 10 points**

Consider the molecule given on the left, and answer questions (i) to (vi) using chemical bonding theories that we have learnt in class (valence bond theory, hybridization).

Note that the lone pairs are not shown in the structure:

i. Consider the carbon atoms marked in as ***a***, ***b***, ***c***, ***d***, and ***e*** in the molecule, and determine **hybridization** at each of the carbon centers. (2.5 points)

Atom	Hybridization
Carbon (<i>a</i>)	
Carbon (<i>b</i>)	
Carbon (<i>c</i>)	
Carbon (<i>d</i>)	
Carbon (<i>e</i>)	

ii. Now, consider the sulfur atom (S) marked as ***f***, nitrogen (N) atom (***g***), and oxygen(O) atoms (***h*** and ***i***), and determine their **hybridization**. (2 points)

Atom	Hybridization
Sulfur (<i>f</i>)	
Nitrogen (<i>g</i>)	
Oxygen (<i>h</i>)	
Oxygen (<i>i</i>)	

iii. **Describe** all the bonds formed by the carbon atoms ***a*** and ***b***, using sigma and pi designation as appropriate. (2 points)

Atom	Bonds formed (use sigma and pi as appropriate)
Carbon (<i>a</i>)	
Carbon (<i>b</i>)	

iv. How many total sp^3 -s sigma bonds are in the molecule? (1 point)

v. Determine **electronic** and **molecular geometry** at carbon (C) labeled ***e*** and sulfur (S) labeled ***f*** centres. (2 points)

Atom	Electronic Geometry	Molecular Geometry
Carbon (<i>e</i>)		
Sulfur (<i>f</i>)		

vi. Circle the value from the choices given below, that describes the **C-S-C bond angle at sulfur** (S) marked ***f***: (0.5 points)

>90° to <109.5° 109.5° >109.5 to <120° 120° >120° to <180°


Question 8 12 points

Using the **valence shell orbitals only** draw a **complete** molecular orbital (MO) diagram (with no s-p mixing) for the molecule **PS**, in the space provided below. **(6 points)**

Assume that 3s and 3p atomic orbitals will form molecular orbitals like those formed by the combination of 2s and 2p atomic orbitals that we learnt in class, and there is no s-p mixing.

Show **energy arrow**, **atomic orbitals** involved, **atom labels**, the **molecular orbitals** that are formed, **filled electrons** in the MO diagram etc. Make sure to label all the atomic and molecular orbitals.

Using the MO diagram above, answer the following questions:

- (i) Which one of the following will have the highest bond order? Circle the correct choice
 A) PS^{1-} B) PS C) PS^{1+}
- (ii) Which one of the following will have the longest bond length? Circle the correct choice
 A) PS^{1-} B) PS C) PS^{1+}
- (iii) Which of the following will be diamagnetic? Circle the correct choice.
 A) PS^{1-} B) PS C) PS^{1+}
- (iv) Draw Lewis dot structure of the molecule with the shortest bond length (based on the MO diagram) from among the three (PS^{1-} , PS, PS^{1+}), and indicate the bond order calculated from the Lewis dot structure. **(3 points)**

Lewis Dot Structure

Bond Order



5B8F1188-F50F-43F9-B992-2E2323EEAA40

midterm-exam-93388

#344 Page 10 of 10

SCRAP PAGE

**CHEMISTRY 110 –Midterm Exam****Thursday November 7, 2024, 6:30 – 8:30 pm**

Examiners: Pallavi Sirjoosingh and Ashok Kakkar

First name (please write as legibly as possible within the boxes)

--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--

Last name

--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--

Student ID number

--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--

This booklet has 8 questions on 10 pages (The last page can be used as a scrap page).

The staple must NOT be removed from the question booklet. Your intact question booklet must be returned at the end of the exam.

A separate booklet contains a periodic table, and datasheet.

Write your answers directly under the questions in the space provided. Do not write on the QR codes or near the top edges (as it may be cut off).

Please write **legibly and dark enough** for the scan to be readable. For students who use English as a second language, dictionaries are permitted. Calculators are allowed.

DO NOT ASK QUESTIONS OF THE INVIGILATORS about interpretations of the exam questions. If you have a concern about a question, please summarize it at the top of this page. All concerns will be evaluated after the exam. GOOD LUCK!

**Question 1 8 points** (*Assume all transitions are possible*)

a. 6 points Calculate the **energy (in kJ/mol)** required to ionize (completely remove) the electron from the **n=3** level of **He⁺ ion** (He: Helium). Show your work including equations. (3 sig. figs.)

b. 2 points

(I) Select the largest (longest) wavelength from the three choices below:

- i. Wavelength of photon to *completely remove* an electron from *1s orbital of Hydrogen atom*
- ii. Wavelength of photon to *completely remove* an electron from *1s orbital of Boron atom (B)*
- iii. Wavelength of photon to *completely remove* an electron from *1s orbital of Boron cation (B⁺)*

i or ii or iii

(II) Select the smallest (shortest) wavelength from the three choices below (*for H atom*):

- i. Wavelength of photon *emitted* for an electron transitioning from *2p to 1s orbital*
- ii. Wavelength of photon *absorbed* for an electron transitioning from *2s to 4p orbital*
- iii. Wavelength of photon *absorbed* for an electron transitioning from *1s to 3p orbital*

i or ii or iii



Question 2 8 points



A Chem110 student plots the graph shown on the left, from data obtained for the kinetic energy (eV) of the ejected electron versus frequency (s^{-1}) of the incident light, for an unknown metal. ($1\text{eV} = 1.6022 \times 10^{-19} \text{ J}$)

Based on the plot, answer the questions **a to c** below:

a. 1 point On the plot above, label the **threshold frequency**.

b. 2 points What is the value of the work potential (in J) of the unknown metal?

c. 3 points What is the *longest* wavelength (in nm) required to eject an electron from this metal's surface? Show your work including equation(s) used. (3 sig figs)

d. 2 points In the space provided below, plot the following

(Assume frequency is *above* threshold frequency):

- i. *Kinetic energy* of an ejected electron versus the *intensity* of the incident light (left)
- ii. *Number of electrons* ejected versus the *frequency* of the incident light (right)



**Question 3** **6 points****a. 2 points**

In the ground state of Ge (Germanium; $Z = 32$) atom, **name** the following orbitals:

i. Orbital containing the electron that requires the maximum energy to be removed.

ii. Orbital containing the electron that experiences the minimum effective nuclear charge.

b. 2 points

An unknown anion (A^-) has the following *excited* electronic configuration: $[Ar] 4s^2 3d^3 4p^1$

(Ar: Argon; $Z = 18$)

Give the **condensed ground state electronic configuration** of the atom, **A** and the cation, **A^{2+}** .
Show your work. (Assume all electronic transitions are possible)

Electronic Configuration of A:

Electronic Configuration of A^{2+} :

c. 2 points

In a single atom, determine how many electrons can have the following set of quantum numbers?

(i) $n = 5; l = 3$

In a single atom, determine how many electrons can have the following set of quantum numbers?

(i) $n = 3; l = 2; m_s = +1/2$

**Question 4** **8 points**

Rank the following from 1 to 4 in **increasing** order. Provide the ranking in the boxes provided.

Note: Atomic number (Z) are provided for relevant atoms

a) Increasing (lowest to highest) order of *ionization energy*: (1= lowest; 4 = highest)

Mg

Mg: Z = 12

Rb

Rb: Z = 37

Cs

Cs: Z = 55

K

K : Z = 19

b) Increasing (lowest to highest) order of *bond angle*: (1= lowest; 4 = highest)

H₂OCCl₄XeF₄

HCN

c) Increasing (smallest to largest) order of *ionic size* (1= smallest; 4 = largest)

S²⁻

S: Z = 16

O²⁻

O: Z = 8

Be²⁺

Be: Z = 4

Na⁺

Na: Z = 11

d) Increasing (lowest to highest) order of *bond energy* (1= lowest; 4 = highest)

P-F

P-Cl

P-I

P-Br

**Question 5** **12 points**

a. 9 points For the molecules HClO_4 and TeF_6 , draw the most stable *Lewis structure* showing appropriate shape and *dashed/wedged bonds*, based on VSEPR. Determine the **molecular geometry** around the central atom. For HClO_4 , label the *most polar bond*.



b. 3 points (Note: Lone pairs are not shown in structures below)

(i) For the three *resonance* structures (1, 2 and 3) shown below, answer the following question:



(I) Circle the *most contributing* resonance structure.

Structure 1

Structure 2

Structure 3

(ii) For the three structures given (4, 5, and 6) below, answer the following questions:



(I) Circle all the structures that represent *resonance structures* of each other:

Structure 4

Structure 5

Structure 6

(II) Among the resonance structures, circle the *most contributing* resonance structure.

Structure 4

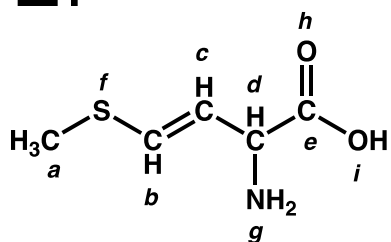
Structure 5

Structure 6

**Question 6** **6 points**

Using the data given below, calculate the **lattice energy** for SrO(s).

	ΔH° (kJ/mol)
$\text{Sr}(s) \longrightarrow \text{Sr}(g)$	+180
$\text{Sr}(g) \longrightarrow \text{Sr}^{2+}(g) + 2e^-$	+1468
$\text{O}_2(g) \longrightarrow 2\text{O}(g)$	+500
$\text{O}(g) + 2e^- \longrightarrow \text{O}^{2-}(g)$	+600
$\text{Sr}(s) + \frac{1}{2}\text{O}_2(g) \longrightarrow \text{SrO}(s)$	-550

**Question 7 10 points**

Consider the molecule given on the left, and answer questions (i) to (vi) using chemical bonding theories that we have learnt in class (valence bond theory, hybridization).

Note that the lone pairs are not shown in the structure:

i. Consider the carbon atoms marked in as ***a***, ***b***, ***c***, ***d***, and ***e*** in the molecule, and determine **hybridization** at each of the carbon centers. (2.5 points)

Atom	Hybridization
Carbon (<i>a</i>)	
Carbon (<i>b</i>)	
Carbon (<i>c</i>)	
Carbon (<i>d</i>)	
Carbon (<i>e</i>)	

ii. Now, consider the sulfur atom (S) marked as ***f***, nitrogen (N) atom (***g***), and oxygen(O) atoms (***h*** and ***i***), and determine their **hybridization**. (2 points)

Atom	Hybridization
Sulfur (<i>f</i>)	
Nitrogen (<i>g</i>)	
Oxygen (<i>h</i>)	
Oxygen (<i>i</i>)	

iii. **Describe** all the bonds formed by the carbon atoms ***a*** and ***b***, using sigma and pi designation as appropriate. (2 points)

Atom	Bonds formed (use sigma and pi as appropriate)
Carbon (<i>a</i>)	
Carbon (<i>b</i>)	

iv. How many total sp^3 -s sigma bonds are in the molecule? (1 point)

v. Determine **electronic** and **molecular geometry** at carbon (C) labeled ***e*** and sulfur (S) labeled ***f*** centres. (2 points)

Atom	Electronic Geometry	Molecular Geometry
Carbon (<i>e</i>)		
Sulfur (<i>f</i>)		

vi. Circle the value from the choices given below, that describes the **C-S-C bond angle at sulfur** (S) marked ***f***: (0.5 points)

$>90^\circ$ to $<109.5^\circ$ 109.5° >109.5 to $<120^\circ$ 120° $>120^\circ$ to $<180^\circ$


Question 8 12 points

Using the **valence shell orbitals only** draw a **complete** molecular orbital (MO) diagram (with no s-p mixing) for the molecule **PS**, in the space provided below. **(6 points)**

Assume that 3s and 3p atomic orbitals will form molecular orbitals like those formed by the combination of 2s and 2p atomic orbitals that we learnt in class, and there is no s-p mixing.

Show **energy arrow**, **atomic orbitals** involved, **atom labels**, the **molecular orbitals** that are formed, **filled electrons** in the MO diagram etc. Make sure to label all the atomic and molecular orbitals.

Using the MO diagram above, answer the following questions:

- (i) Which one of the following will have the highest bond order? Circle the correct choice
 A) PS^{1-} B) PS C) PS^{1+}
- (ii) Which one of the following will have the longest bond length? Circle the correct choice
 A) PS^{1-} B) PS C) PS^{1+}
- (iii) Which of the following will be diamagnetic? Circle the correct choice.
 A) PS^{1-} B) PS C) PS^{1+}
- (iv) Draw Lewis dot structure of the molecule with the shortest bond length (based on the MO diagram) from among the three (PS^{1-} , PS, PS^{1+}), and indicate the bond order calculated from the Lewis dot structure. **(3 points)**

Lewis Dot Structure

Bond Order



8DE6630F-F82D-4771-91CD-C51128DE4BC9

midterm-exam-93388

#345 Page 10 of 10

SCRAP PAGE

**CHEMISTRY 110 –Midterm Exam****Thursday November 7, 2024, 6:30 – 8:30 pm**

Examiners: Pallavi Sirjoosingh and Ashok Kakkar

First name (please write as legibly as possible within the boxes)

--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--

Last name

--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--

Student ID number

--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--

This booklet has 8 questions on 10 pages (The last page can be used as a scrap page).

The staple must NOT be removed from the question booklet. Your intact question booklet must be returned at the end of the exam.

A separate booklet contains a periodic table, and datasheet.

Write your answers directly under the questions in the space provided. Do not write on the QR codes or near the top edges (as it may be cut off).

Please write **legibly and dark enough** for the scan to be readable. For students who use English as a second language, dictionaries are permitted. Calculators are allowed.

DO NOT ASK QUESTIONS OF THE INVIGILATORS about interpretations of the exam questions. If you have a concern about a question, please summarize it at the top of this page. All concerns will be evaluated after the exam. GOOD LUCK!

**Question 1 8 points** (*Assume all transitions are possible*)

a. 6 points Calculate the **energy (in kJ/mol)** required to ionize (completely remove) the electron from the **n=3** level of **He⁺ ion** (He: Helium). Show your work including equations. (3 sig. figs.)

b. 2 points

(I) Select the largest (longest) wavelength from the three choices below:

- i. Wavelength of photon to *completely remove* an electron from *1s orbital of Hydrogen atom*
- ii. Wavelength of photon to *completely remove* an electron from *1s orbital of Boron atom (B)*
- iii. Wavelength of photon to *completely remove* an electron from *1s orbital of Boron cation (B⁺)*

i or ii or iii

(II) Select the smallest (shortest) wavelength from the three choices below (*for H atom*):

- i. Wavelength of photon *emitted* for an electron transitioning from *2p to 1s orbital*
- ii. Wavelength of photon *absorbed* for an electron transitioning from *2s to 4p orbital*
- iii. Wavelength of photon *absorbed* for an electron transitioning from *1s to 3p orbital*

i or ii or iii



Question 2 8 points



A Chem110 student plots the graph shown on the left, from data obtained for the kinetic energy (eV) of the ejected electron versus frequency (s^{-1}) of the incident light, for an unknown metal. ($1\text{eV} = 1.6022 \times 10^{-19} \text{ J}$)

Based on the plot, answer the questions **a to c** below:

a. 1 point On the plot above, label the **threshold frequency**.

b. 2 points What is the value of the work potential (in J) of the unknown metal?

c. 3 points What is the *longest* wavelength (in nm) required to eject an electron from this metal's surface? Show your work including equation(s) used. (3 sig figs)

d. 2 points In the space provided below, plot the following

(Assume frequency is *above* threshold frequency):

- i. *Kinetic energy* of an ejected electron versus the *intensity* of the incident light (left)
- ii. *Number of electrons* ejected versus the *frequency* of the incident light (right)



**Question 3** **6 points****a. 2 points**

In the ground state of Ge (Germanium; $Z = 32$) atom, **name** the following orbitals:

i. Orbital containing the electron that requires the maximum energy to be removed.

ii. Orbital containing the electron that experiences the minimum effective nuclear charge.

b. 2 points

An unknown anion (A^-) has the following *excited* electronic configuration: $[Ar] 4s^2 3d^3 4p^1$

(Ar: Argon; $Z = 18$)

Give the **condensed ground state electronic configuration** of the atom, **A** and the cation, **A^{2+}** .
Show your work. (Assume all electronic transitions are possible)

Electronic Configuration of A:

Electronic Configuration of A^{2+} :

c. 2 points

In a single atom, determine how many electrons can have the following set of quantum numbers?

(i) $n = 5; l = 3$

In a single atom, determine how many electrons can have the following set of quantum numbers?

(i) $n = 3; l = 2; m_s = +1/2$

**Question 4** **8 points**

Rank the following from 1 to 4 in **increasing** order. Provide the ranking in the boxes provided.

Note: Atomic number (Z) are provided for relevant atoms

a) Increasing (lowest to highest) order of *ionization energy*: (1= lowest; 4 = highest)

Mg

Mg: Z = 12

Rb

Rb: Z = 37

Cs

Cs: Z = 55

K

K : Z = 19

b) Increasing (lowest to highest) order of *bond angle*: (1= lowest; 4 = highest)

H₂OCCl₄XeF₄

HCN

c) Increasing (smallest to largest) order of *ionic size* (1= smallest; 4 = largest)

S²⁻

S: Z = 16

O²⁻

O: Z = 8

Be²⁺

Be: Z = 4

Na⁺

Na: Z = 11

d) Increasing (lowest to highest) order of *bond energy* (1= lowest; 4 = highest)

P-F

P-Cl

P-I

P-Br

**Question 5** **12 points**

a. 9 points For the molecules HClO_4 and TeF_6 , draw the most stable *Lewis structure* showing appropriate shape and *dashed/wedged bonds*, based on VSEPR. Determine the **molecular geometry** around the central atom. For HClO_4 , label the *most polar bond*.

HClO_4 (there are no O-O bonds)

Label the most polar bond

Molecular Geometry around Cl

TeF_6

Molecular Geometry around Te

b. 3 points (Note: Lone pairs are not shown in structures below)

(i) For the three *resonance* structures (1, 2 and 3) shown below, answer the following question:



(I) Circle the *most contributing* resonance structure.

Structure 1

Structure 2

Structure 3

(ii) For the three structures given (4, 5, and 6) below, answer the following questions:



(I) Circle all the structures that represent *resonance structures* of each other:

Structure 4

Structure 5

Structure 6

(II) Among the resonance structures, circle the *most contributing* resonance structure.

Structure 4

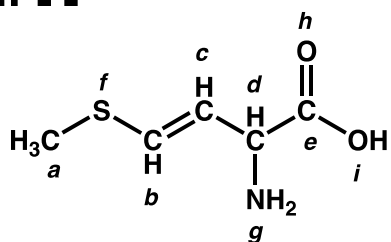
Structure 5

Structure 6

**Question 6** **6 points**

Using the data given below, calculate the **lattice energy** for SrO(s).

	ΔH° (kJ/mol)
$\text{Sr}(s) \longrightarrow \text{Sr}(g)$	+180
$\text{Sr}(g) \longrightarrow \text{Sr}^{2+}(g) + 2e^-$	+1468
$\text{O}_2(g) \longrightarrow 2\text{O}(g)$	+500
$\text{O}(g) + 2e^- \longrightarrow \text{O}^{2-}(g)$	+600
$\text{Sr}(s) + \frac{1}{2}\text{O}_2(g) \longrightarrow \text{SrO}(s)$	-550

**Question 7 10 points**

Consider the molecule given on the left, and answer questions (i) to (vi) using chemical bonding theories that we have learnt in class (valence bond theory, hybridization).

Note that the lone pairs are not shown in the structure:

i. Consider the carbon atoms marked in as ***a***, ***b***, ***c***, ***d***, and ***e*** in the molecule, and determine **hybridization** at each of the carbon centers. (2.5 points)

Atom	Hybridization
Carbon (<i>a</i>)	
Carbon (<i>b</i>)	
Carbon (<i>c</i>)	
Carbon (<i>d</i>)	
Carbon (<i>e</i>)	

ii. Now, consider the sulfur atom (S) marked as ***f***, nitrogen (N) atom (***g***), and oxygen(O) atoms (***h*** and ***i***), and determine their **hybridization**. (2 points)

Atom	Hybridization
Sulfur (<i>f</i>)	
Nitrogen (<i>g</i>)	
Oxygen (<i>h</i>)	
Oxygen (<i>i</i>)	

iii. **Describe** all the bonds formed by the carbon atoms ***a*** and ***b***, using sigma and pi designation as appropriate. (2 points)

Atom	Bonds formed (use sigma and pi as appropriate)
Carbon (<i>a</i>)	
Carbon (<i>b</i>)	

iv. How many total sp^3 -s sigma bonds are in the molecule? (1 point)

v. Determine **electronic** and **molecular geometry** at carbon (C) labeled ***e*** and sulfur (S) labeled ***f*** centres. (2 points)

Atom	Electronic Geometry	Molecular Geometry
Carbon (<i>e</i>)		
Sulfur (<i>f</i>)		

vi. Circle the value from the choices given below, that describes the **C-S-C bond angle at sulfur** (S) marked ***f***: (0.5 points)

>90° to <109.5° 109.5° >109.5 to <120° 120° >120° to <180°


Question 8 12 points

Using the **valence shell orbitals only** draw a **complete** molecular orbital (MO) diagram (with no s-p mixing) for the molecule **PS**, in the space provided below. **(6 points)**

Assume that 3s and 3p atomic orbitals will form molecular orbitals like those formed by the combination of 2s and 2p atomic orbitals that we learnt in class, and there is no s-p mixing.

Show **energy arrow**, **atomic orbitals** involved, **atom labels**, the **molecular orbitals** that are formed, **filled electrons** in the MO diagram etc. Make sure to label all the atomic and molecular orbitals.

Using the MO diagram above, answer the following questions:

- (i) Which one of the following will have the highest bond order? Circle the correct choice
 A) PS^{1-} B) PS C) PS^{1+}
- (ii) Which one of the following will have the longest bond length? Circle the correct choice
 A) PS^{1-} B) PS C) PS^{1+}
- (iii) Which of the following will be diamagnetic? Circle the correct choice.
 A) PS^{1-} B) PS C) PS^{1+}
- (iv) Draw Lewis dot structure of the molecule with the shortest bond length (based on the MO diagram) from among the three (PS^{1-} , PS, PS^{1+}), and indicate the bond order calculated from the Lewis dot structure. **(3 points)**

Lewis Dot Structure

Bond Order



2F58CFC7-967E-4C6F-8523-910C9FCCBF85

midterm-exam-93388

#346 Page 10 of 10

SCRAP PAGE

**CHEMISTRY 110 –Midterm Exam****Thursday November 7, 2024, 6:30 – 8:30 pm**

Examiners: Pallavi Sirjoosingh and Ashok Kakkar

First name (please write as legibly as possible within the boxes)

--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--

Last name

--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--

Student ID number

--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--

This booklet has 8 questions on 10 pages (The last page can be used as a scrap page).

The staple must NOT be removed from the question booklet. Your intact question booklet must be returned at the end of the exam.

A separate booklet contains a periodic table, and datasheet.

Write your answers directly under the questions in the space provided. Do not write on the QR codes or near the top edges (as it may be cut off).

Please write **legibly and dark enough** for the scan to be readable. For students who use English as a second language, dictionaries are permitted. Calculators are allowed.

DO NOT ASK QUESTIONS OF THE INVIGILATORS about interpretations of the exam questions. If you have a concern about a question, please summarize it at the top of this page. All concerns will be evaluated after the exam. GOOD LUCK!

**Question 1 8 points** (*Assume all transitions are possible*)

a. 6 points Calculate the **energy (in kJ/mol)** required to ionize (completely remove) the electron from the **n=3** level of **He⁺ ion** (He: Helium). Show your work including equations. (3 sig. figs.)

b. 2 points

(I) Select the largest (longest) wavelength from the three choices below:

- i. Wavelength of photon to *completely remove* an electron from *1s orbital of Hydrogen atom*
- ii. Wavelength of photon to *completely remove* an electron from *1s orbital of Boron atom (B)*
- iii. Wavelength of photon to *completely remove* an electron from *1s orbital of Boron cation (B⁺)*

i or ii or iii

(II) Select the smallest (shortest) wavelength from the three choices below (*for H atom*):

- i. Wavelength of photon *emitted* for an electron transitioning from *2p to 1s orbital*
- ii. Wavelength of photon *absorbed* for an electron transitioning from *2s to 4p orbital*
- iii. Wavelength of photon *absorbed* for an electron transitioning from *1s to 3p orbital*

i or ii or iii



Question 2 8 points



A Chem110 student plots the graph shown on the left, from data obtained for the kinetic energy (eV) of the ejected electron versus frequency (s^{-1}) of the incident light, for an unknown metal. ($1\text{eV} = 1.6022 \times 10^{-19} \text{ J}$)

Based on the plot, answer the questions **a to c** below:

a. 1 point On the plot above, label the **threshold frequency**.

b. 2 points What is the value of the work potential (in J) of the unknown metal?

c. 3 points What is the *longest* wavelength (in nm) required to eject an electron from this metal's surface? Show your work including equation(s) used. (3 sig figs)

d. 2 points In the space provided below, plot the following

(Assume frequency is *above* threshold frequency):

- i. *Kinetic energy* of an ejected electron versus the *intensity* of the incident light (left)
- ii. *Number of electrons* ejected versus the *frequency* of the incident light (right)



**Question 3** **6 points****a. 2 points**

In the ground state of Ge (Germanium; $Z = 32$) atom, **name** the following orbitals:

i. Orbital containing the electron that requires the maximum energy to be removed.

ii. Orbital containing the electron that experiences the minimum effective nuclear charge.

b. 2 points

An unknown anion (A^-) has the following *excited* electronic configuration: $[Ar] 4s^2 3d^3 4p^1$

(Ar: Argon; $Z = 18$)

Give the **condensed ground state electronic configuration** of the atom, **A** and the cation, **A^{2+}** .
Show your work. (Assume all electronic transitions are possible)

Electronic Configuration of A:

Electronic Configuration of A^{2+} :

c. 2 points

In a single atom, determine how many electrons can have the following set of quantum numbers?

(i) $n = 5; l = 3$

In a single atom, determine how many electrons can have the following set of quantum numbers?

(i) $n = 3; l = 2; m_s = +1/2$

**Question 4** **8 points**

Rank the following from 1 to 4 in **increasing** order. Provide the ranking in the boxes provided.

Note: Atomic number (Z) are provided for relevant atoms

a) Increasing (lowest to highest) order of *ionization energy*: (1= lowest; 4 = highest)

Mg

Mg: Z = 12

Rb

Rb: Z = 37

Cs

Cs: Z = 55

K

K : Z = 19

b) Increasing (lowest to highest) order of *bond angle*: (1= lowest; 4 = highest)

H₂OCCl₄XeF₄

HCN

c) Increasing (smallest to largest) order of *ionic size* (1= smallest; 4 = largest)

S²⁻

S: Z = 16

O²⁻

O: Z = 8

Be²⁺

Be: Z = 4

Na⁺

Na: Z = 11

d) Increasing (lowest to highest) order of *bond energy* (1= lowest; 4 = highest)

P-F

P-Cl

P-I

P-Br

**Question 5** **12 points**

a. 9 points For the molecules HClO_4 and TeF_6 , draw the most stable *Lewis structure* showing appropriate shape and *dashed/wedged bonds*, based on VSEPR. Determine the **molecular geometry** around the central atom. For HClO_4 , label the *most polar bond*.

HClO_4 (there are no O-O bonds)

Label the most polar bond

Molecular Geometry around Cl

TeF_6

Molecular Geometry around Te

b. 3 points (Note: Lone pairs are not shown in structures below)

(i) For the three *resonance* structures (1, 2 and 3) shown below, answer the following question:



(I) Circle the *most contributing* resonance structure.

Structure 1

Structure 2

Structure 3

(ii) For the three structures given (4, 5, and 6) below, answer the following questions:



(I) Circle all the structures that represent *resonance structures* of each other:

Structure 4

Structure 5

Structure 6

(II) Among the resonance structures, circle the *most contributing* resonance structure.

Structure 4

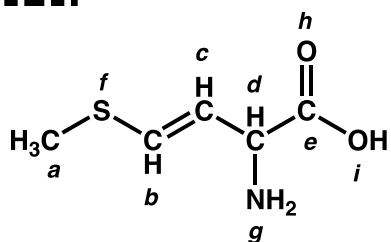
Structure 5

Structure 6

**Question 6** **6 points**

Using the data given below, calculate the **lattice energy** for SrO(s).

	ΔH° (kJ/mol)
$\text{Sr}(s) \longrightarrow \text{Sr}(g)$	+180
$\text{Sr}(g) \longrightarrow \text{Sr}^{2+}(g) + 2e^-$	+1468
$\text{O}_2(g) \longrightarrow 2\text{O}(g)$	+500
$\text{O}(g) + 2e^- \longrightarrow \text{O}^{2-}(g)$	+600
$\text{Sr}(s) + \frac{1}{2}\text{O}_2(g) \longrightarrow \text{SrO}(s)$	-550

**Question 7 10 points**

Consider the molecule given on the left, and answer questions (i) to (vi) using chemical bonding theories that we have learnt in class (valence bond theory, hybridization).

Note that the lone pairs are not shown in the structure:

i. Consider the carbon atoms marked in as ***a***, ***b***, ***c***, ***d***, and ***e*** in the molecule, and determine **hybridization** at each of the carbon centers. (2.5 points)

Atom	Hybridization
Carbon (<i>a</i>)	
Carbon (<i>b</i>)	
Carbon (<i>c</i>)	
Carbon (<i>d</i>)	
Carbon (<i>e</i>)	

ii. Now, consider the sulfur atom (S) marked as ***f***, nitrogen (N) atom (***g***), and oxygen(O) atoms (***h*** and ***i***), and determine their **hybridization**. (2 points)

Atom	Hybridization
Sulfur (<i>f</i>)	
Nitrogen (<i>g</i>)	
Oxygen (<i>h</i>)	
Oxygen (<i>i</i>)	

iii. **Describe** all the bonds formed by the carbon atoms ***a*** and ***b***, using sigma and pi designation as appropriate. (2 points)

Atom	Bonds formed (use sigma and pi as appropriate)
Carbon (<i>a</i>)	
Carbon (<i>b</i>)	

iv. How many total sp^3 -s sigma bonds are in the molecule? (1 point)

v. Determine **electronic** and **molecular geometry** at carbon (C) labeled ***e*** and sulfur (S) labeled ***f*** centres. (2 points)

Atom	Electronic Geometry	Molecular Geometry
Carbon (<i>e</i>)		
Sulfur (<i>f</i>)		

vi. Circle the value from the choices given below, that describes the **C-S-C bond angle at sulfur** (S) marked ***f***: (0.5 points)

>90° to <109.5° 109.5° >109.5 to <120° 120° >120° to <180°


Question 8 12 points

Using the **valence shell orbitals only** draw a **complete** molecular orbital (MO) diagram (with no s-p mixing) for the molecule **PS**, in the space provided below. **(6 points)**

Assume that 3s and 3p atomic orbitals will form molecular orbitals like those formed by the combination of 2s and 2p atomic orbitals that we learnt in class, and there is no s-p mixing.

Show **energy arrow**, **atomic orbitals** involved, **atom labels**, the **molecular orbitals** that are formed, **filled electrons** in the MO diagram etc. Make sure to label all the atomic and molecular orbitals.

Using the MO diagram above, answer the following questions:

- (i) Which one of the following will have the highest bond order? Circle the correct choice
 A) PS^{1-} B) PS C) PS^{1+}
- (ii) Which one of the following will have the longest bond length? Circle the correct choice
 A) PS^{1-} B) PS C) PS^{1+}
- (iii) Which of the following will be diamagnetic? Circle the correct choice.
 A) PS^{1-} B) PS C) PS^{1+}
- (iv) Draw Lewis dot structure of the molecule with the shortest bond length (based on the MO diagram) from among the three (PS^{1-} , PS, PS^{1+}), and indicate the bond order calculated from the Lewis dot structure. **(3 points)**

Lewis Dot Structure

Bond Order



5CEAD235-DD16-442F-9063-01670F08E8D8

midterm-exam-93388

#347 Page 10 of 10

SCRAP PAGE

**CHEMISTRY 110 –Midterm Exam****Thursday November 7, 2024, 6:30 – 8:30 pm**

Examiners: Pallavi Sirjoosingh and Ashok Kakkar

First name (please write as legibly as possible within the boxes)

--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--

Last name

--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--

Student ID number

--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--

This booklet has 8 questions on 10 pages (The last page can be used as a scrap page).

The staple must NOT be removed from the question booklet. Your intact question booklet must be returned at the end of the exam.

A separate booklet contains a periodic table, and datasheet.

Write your answers directly under the questions in the space provided. Do not write on the QR codes or near the top edges (as it may be cut off).

Please write **legibly and dark enough** for the scan to be readable. For students who use English as a second language, dictionaries are permitted. Calculators are allowed.

DO NOT ASK QUESTIONS OF THE INVIGILATORS about interpretations of the exam questions. If you have a concern about a question, please summarize it at the top of this page. All concerns will be evaluated after the exam. GOOD LUCK!

**Question 1 8 points** (*Assume all transitions are possible*)

a. 6 points Calculate the **energy (in kJ/mol)** required to ionize (completely remove) the electron from the **$n=3$** level of **He^+ ion** (He: Helium). Show your work including equations. (3 sig. figs.)

b. 2 points

(I) Select the largest (longest) wavelength from the three choices below:

- i. Wavelength of photon to *completely remove* an electron from *1s orbital of Hydrogen atom*
- ii. Wavelength of photon to *completely remove* an electron from *1s orbital of Boron atom (B)*
- iii. Wavelength of photon to *completely remove* an electron from *1s orbital of Boron cation (B^+)*

i or ii or iii

(II) Select the smallest (shortest) wavelength from the three choices below (*for H atom*):

- i. Wavelength of photon *emitted* for an electron transitioning from *2p to 1s orbital*
- ii. Wavelength of photon *absorbed* for an electron transitioning from *2s to 4p orbital*
- iii. Wavelength of photon *absorbed* for an electron transitioning from *1s to 3p orbital*

i or ii or iii



Question 2 8 points



A Chem110 student plots the graph shown on the left, from data obtained for the kinetic energy (eV) of the ejected electron versus frequency (s^{-1}) of the incident light, for an unknown metal. ($1\text{eV} = 1.6022 \times 10^{-19} \text{ J}$)

Based on the plot, answer the questions **a to c** below:

a. 1 point On the plot above, label the **threshold frequency**.

b. 2 points What is the value of the work potential (in J) of the unknown metal?

c. 3 points What is the *longest* wavelength (in nm) required to eject an electron from this metal's surface? Show your work including equation(s) used. (3 sig figs)

d. 2 points In the space provided below, plot the following

(Assume frequency is *above* threshold frequency):

- i. *Kinetic energy* of an ejected electron versus the *intensity* of the incident light (left)
- ii. *Number of electrons* ejected versus the *frequency* of the incident light (right)



**Question 3** **6 points****a. 2 points**

In the ground state of Ge (Germanium; $Z = 32$) atom, **name** the following orbitals:

i. Orbital containing the electron that requires the maximum energy to be removed.

ii. Orbital containing the electron that experiences the minimum effective nuclear charge.

b. 2 points

An unknown anion (A^-) has the following *excited* electronic configuration: $[Ar] 4s^2 3d^3 4p^1$

(Ar: Argon; $Z = 18$)

Give the **condensed ground state electronic configuration** of the atom, **A** and the cation, **A^{2+}** .
Show your work. (Assume all electronic transitions are possible)

Electronic Configuration of A:

Electronic Configuration of A^{2+} :

c. 2 points

In a single atom, determine how many electrons can have the following set of quantum numbers?

(i) $n = 5; l = 3$

In a single atom, determine how many electrons can have the following set of quantum numbers?

(i) $n = 3; l = 2; m_s = +1/2$

**Question 4** **8 points**

Rank the following from 1 to 4 in **increasing** order. Provide the ranking in the boxes provided.

Note: Atomic number (Z) are provided for relevant atoms

a) Increasing (lowest to highest) order of *ionization energy*: (1= lowest; 4 = highest)

Mg

Mg: Z = 12

Rb

Rb: Z = 37

Cs

Cs: Z = 55

K

K : Z = 19

b) Increasing (lowest to highest) order of *bond angle*: (1= lowest; 4 = highest)

H₂OCCl₄XeF₄

HCN

c) Increasing (smallest to largest) order of *ionic size* (1= smallest; 4 = largest)

S²⁻

S: Z = 16

O²⁻

O: Z = 8

Be²⁺

Be: Z = 4

Na⁺

Na: Z = 11

d) Increasing (lowest to highest) order of *bond energy* (1= lowest; 4 = highest)

P-F

P-Cl

P-I

P-Br

**Question 5** **12 points**

a. 9 points For the molecules HClO_4 and TeF_6 , draw the most stable *Lewis structure* showing appropriate shape and *dashed/wedged bonds*, based on VSEPR. Determine the **molecular geometry** around the central atom. For HClO_4 , label the *most polar bond*.



b. 3 points (Note: Lone pairs are not shown in structures below)

(i) For the three *resonance* structures (1, 2 and 3) shown below, answer the following question:



(I) Circle the *most contributing* resonance structure.

Structure 1

Structure 2

Structure 3

(ii) For the three structures given (4, 5, and 6) below, answer the following questions:



(I) Circle all the structures that represent *resonance structures* of each other:

Structure 4

Structure 5

Structure 6

(II) Among the resonance structures, circle the *most contributing* resonance structure.

Structure 4

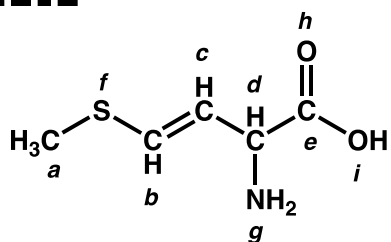
Structure 5

Structure 6

**Question 6** **6 points**

Using the data given below, calculate the **lattice energy** for SrO(s).

	ΔH° (kJ/mol)
$\text{Sr}(s) \longrightarrow \text{Sr}(g)$	+180
$\text{Sr}(g) \longrightarrow \text{Sr}^{2+}(g) + 2e^-$	+1468
$\text{O}_2(g) \longrightarrow 2\text{O}(g)$	+500
$\text{O}(g) + 2e^- \longrightarrow \text{O}^{2-}(g)$	+600
$\text{Sr}(s) + \frac{1}{2}\text{O}_2(g) \longrightarrow \text{SrO}(s)$	-550

**Question 7 10 points**

Consider the molecule given on the left, and answer questions (i) to (vi) using chemical bonding theories that we have learnt in class (valence bond theory, hybridization).

Note that the lone pairs are not shown in the structure:

i. Consider the carbon atoms marked in as ***a***, ***b***, ***c***, ***d***, and ***e*** in the molecule, and determine **hybridization** at each of the carbon centers. (2.5 points)

Atom	Hybridization
Carbon (<i>a</i>)	
Carbon (<i>b</i>)	
Carbon (<i>c</i>)	
Carbon (<i>d</i>)	
Carbon (<i>e</i>)	

ii. Now, consider the sulfur atom (S) marked as ***f***, nitrogen (N) atom (***g***), and oxygen(O) atoms (***h*** and ***i***), and determine their **hybridization**. (2 points)

Atom	Hybridization
Sulfur (<i>f</i>)	
Nitrogen (<i>g</i>)	
Oxygen (<i>h</i>)	
Oxygen (<i>i</i>)	

iii. **Describe** all the bonds formed by the carbon atoms ***a*** and ***b***, using sigma and pi designation as appropriate. (2 points)

Atom	Bonds formed (use sigma and pi as appropriate)
Carbon (<i>a</i>)	
Carbon (<i>b</i>)	

iv. How many total sp^3 -s sigma bonds are in the molecule? (1 point)

v. Determine **electronic** and **molecular geometry** at carbon (C) labeled ***e*** and sulfur (S) labeled ***f*** centres. (2 points)

Atom	Electronic Geometry	Molecular Geometry
Carbon (<i>e</i>)		
Sulfur (<i>f</i>)		

vi. Circle the value from the choices given below, that describes the **C-S-C bond angle at sulfur** (S) marked ***f***: (0.5 points)

>90° to <109.5° 109.5° >109.5 to <120° 120° >120° to <180°

**Question 8 12 points**

Using the **valence shell orbitals only** draw a **complete** molecular orbital (MO) diagram (with no s-p mixing) for the molecule **PS**, in the space provided below. **(6 points)**

Assume that 3s and 3p atomic orbitals will form molecular orbitals like those formed by the combination of 2s and 2p atomic orbitals that we learnt in class, and there is no s-p mixing.

Show **energy arrow**, **atomic orbitals** involved, **atom labels**, the **molecular orbitals** that are formed, **filled electrons** in the MO diagram etc. Make sure to label all the atomic and molecular orbitals.

Using the MO diagram above, answer the following questions:

- (i) Which one of the following will have the highest bond order? Circle the correct choice
A) PS^{1-} B) PS C) PS^{1+}
- (ii) Which one of the following will have the longest bond length? Circle the correct choice
A) PS^{1-} B) PS C) PS^{1+}
- (iii) Which of the following will be diamagnetic? Circle the correct choice.
A) PS^{1-} B) PS C) PS^{1+}
- (iv) Draw Lewis dot structure of the molecule with the shortest bond length (based on the MO diagram) from among the three (PS^{1-} , PS, PS^{1+}), and indicate the bond order calculated from the Lewis dot structure. **(3 points)**

Lewis Dot Structure

Bond Order



22E09219-9B18-4168-A2C3-CE09AEDC776A

midterm-exam-93388

#348 Page 10 of 10

SCRAP PAGE

**CHEMISTRY 110 –Midterm Exam****Thursday November 7, 2024, 6:30 – 8:30 pm**

Examiners: Pallavi Sirjoosingh and Ashok Kakkar

First name (please write as legibly as possible within the boxes)

--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--

Last name

--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--

Student ID number

--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--

This booklet has 8 questions on 10 pages (The last page can be used as a scrap page).

The staple must NOT be removed from the question booklet. Your intact question booklet must be returned at the end of the exam.

A separate booklet contains a periodic table, and datasheet.

Write your answers directly under the questions in the space provided. Do not write on the QR codes or near the top edges (as it may be cut off).

Please write **legibly and dark enough** for the scan to be readable. For students who use English as a second language, dictionaries are permitted. Calculators are allowed.

DO NOT ASK QUESTIONS OF THE INVIGILATORS about interpretations of the exam questions. If you have a concern about a question, please summarize it at the top of this page. All concerns will be evaluated after the exam. GOOD LUCK!

**Question 1 8 points** (*Assume all transitions are possible*)

a. 6 points Calculate the **energy (in kJ/mol)** required to ionize (completely remove) the electron from the **$n=3$** level of **He^+ ion** (He: Helium). Show your work including equations. (3 sig. figs.)

b. 2 points

(I) Select the largest (longest) wavelength from the three choices below:

- i. Wavelength of photon to *completely remove* an electron from *1s orbital of Hydrogen atom*
- ii. Wavelength of photon to *completely remove* an electron from *1s orbital of Boron atom (B)*
- iii. Wavelength of photon to *completely remove* an electron from *1s orbital of Boron cation (B^+)*

i or ii or iii

(II) Select the smallest (shortest) wavelength from the three choices below (*for H atom*):

- i. Wavelength of photon *emitted* for an electron transitioning from *2p to 1s orbital*
- ii. Wavelength of photon *absorbed* for an electron transitioning from *2s to 4p orbital*
- iii. Wavelength of photon *absorbed* for an electron transitioning from *1s to 3p orbital*

i or ii or iii



Question 2 8 points



A Chem110 student plots the graph shown on the left, from data obtained for the kinetic energy (eV) of the ejected electron versus frequency (s^{-1}) of the incident light, for an unknown metal. ($1\text{eV} = 1.6022 \times 10^{-19} \text{ J}$)

Based on the plot, answer the questions **a to c** below:

a. 1 point On the plot above, label the **threshold frequency**.

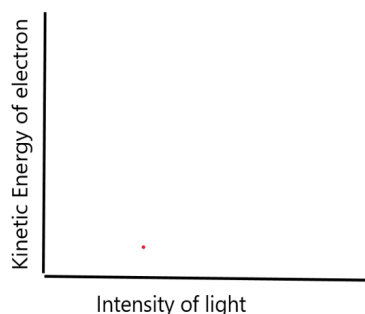
b. 2 points What is the value of the work potential (in J) of the unknown metal?

c. 3 points What is the *longest* wavelength (in nm) required to eject an electron from this metal's surface? Show your work including equation(s) used. (3 sig figs)

d. 2 points In the space provided below, plot the following

(Assume frequency is *above* threshold frequency):

- Kinetic energy* of an ejected electron versus the *intensity* of the incident light (left)
- Number of electrons* ejected versus the *frequency* of the incident light (right)



**Question 3** **6 points****a. 2 points**

In the ground state of Ge (Germanium; $Z = 32$) atom, **name** the following orbitals:

i. Orbital containing the electron that requires the maximum energy to be removed.

ii. Orbital containing the electron that experiences the minimum effective nuclear charge.

b. 2 points

An unknown anion (A^-) has the following *excited* electronic configuration: $[Ar] 4s^2 3d^3 4p^1$

(Ar: Argon; $Z = 18$)

Give the **condensed ground state electronic configuration** of the atom, **A** and the cation, **A^{2+}** .
Show your work. (Assume all electronic transitions are possible)

Electronic Configuration of A:

Electronic Configuration of A^{2+} :

c. 2 points

In a single atom, determine how many electrons can have the following set of quantum numbers?

(i) $n = 5; l = 3$

In a single atom, determine how many electrons can have the following set of quantum numbers?

(i) $n = 3; l = 2; m_s = +1/2$

**Question 4** **8 points**

Rank the following from 1 to 4 in **increasing** order. Provide the ranking in the boxes provided.

Note: Atomic number (Z) are provided for relevant atoms

a) Increasing (lowest to highest) order of *ionization energy*: (1= lowest; 4 = highest)

Mg

Mg: Z = 12

Rb

Rb: Z = 37

Cs

Cs: Z = 55

K

K : Z = 19

b) Increasing (lowest to highest) order of *bond angle*: (1= lowest; 4 = highest)

H₂OCCl₄XeF₄

HCN

c) Increasing (smallest to largest) order of *ionic size* (1= smallest; 4 = largest)

S²⁻

S: Z = 16

O²⁻

O: Z = 8

Be²⁺

Be: Z = 4

Na⁺

Na: Z = 11

d) Increasing (lowest to highest) order of *bond energy* (1= lowest; 4 = highest)

P-F

P-Cl

P-I

P-Br

**Question 5** **12 points**

a. 9 points For the molecules HClO_4 and TeF_6 , draw the most stable *Lewis structure* showing appropriate shape and *dashed/wedged bonds*, based on VSEPR. Determine the **molecular geometry** around the central atom. For HClO_4 , label the *most polar bond*.



b. 3 points (Note: Lone pairs are not shown in structures below)

(i) For the three *resonance* structures (1, 2 and 3) shown below, answer the following question:



(I) Circle the *most contributing* resonance structure.

Structure 1

Structure 2

Structure 3

(ii) For the three structures given (4, 5, and 6) below, answer the following questions:



(I) Circle all the structures that represent *resonance structures* of each other:

Structure 4

Structure 5

Structure 6

(II) Among the resonance structures, circle the *most contributing* resonance structure.

Structure 4

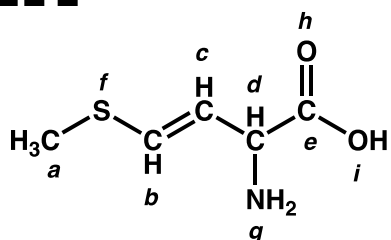
Structure 5

Structure 6

**Question 6** **6 points**

Using the data given below, calculate the **lattice energy** for SrO(s).

	ΔH° (kJ/mol)
$\text{Sr}(s) \longrightarrow \text{Sr}(g)$	+180
$\text{Sr}(g) \longrightarrow \text{Sr}^{2+}(g) + 2e^-$	+1468
$\text{O}_2(g) \longrightarrow 2\text{O}(g)$	+500
$\text{O}(g) + 2e^- \longrightarrow \text{O}^{2-}(g)$	+600
$\text{Sr}(s) + \frac{1}{2}\text{O}_2(g) \longrightarrow \text{SrO}(s)$	-550

**Question 7 10 points**

Consider the molecule given on the left, and answer questions (i) to (vi) using chemical bonding theories that we have learnt in class (valence bond theory, hybridization).

Note that the lone pairs are not shown in the structure:

i. Consider the carbon atoms marked in as ***a***, ***b***, ***c***, ***d***, and ***e*** in the molecule, and determine **hybridization** at each of the carbon centers. (2.5 points)

Atom	Hybridization
Carbon (<i>a</i>)	
Carbon (<i>b</i>)	
Carbon (<i>c</i>)	
Carbon (<i>d</i>)	
Carbon (<i>e</i>)	

ii. Now, consider the sulfur atom (S) marked as ***f***, nitrogen (N) atom (***g***), and oxygen(O) atoms (***h*** and ***i***), and determine their **hybridization**. (2 points)

Atom	Hybridization
Sulfur (<i>f</i>)	
Nitrogen (<i>g</i>)	
Oxygen (<i>h</i>)	
Oxygen (<i>i</i>)	

iii. **Describe** all the bonds formed by the carbon atoms ***a*** and ***b***, using sigma and pi designation as appropriate. (2 points)

Atom	Bonds formed (use sigma and pi as appropriate)
Carbon (<i>a</i>)	
Carbon (<i>b</i>)	

iv. How many total sp^3 -s sigma bonds are in the molecule? (1 point)

v. Determine **electronic** and **molecular geometry** at carbon (C) labeled ***e*** and sulfur (S) labeled ***f*** centres. (2 points)

Atom	Electronic Geometry	Molecular Geometry
Carbon (<i>e</i>)		
Sulfur (<i>f</i>)		

vi. Circle the value from the choices given below, that describes the **C-S-C bond angle at sulfur** (S) marked ***f***: (0.5 points)

$>90^\circ$ to $<109.5^\circ$ 109.5° >109.5 to $<120^\circ$ 120° $>120^\circ$ to $<180^\circ$


Question 8 12 points

Using the **valence shell orbitals only** draw a **complete** molecular orbital (MO) diagram (with no s-p mixing) for the molecule **PS**, in the space provided below. **(6 points)**

Assume that 3s and 3p atomic orbitals will form molecular orbitals like those formed by the combination of 2s and 2p atomic orbitals that we learnt in class, and there is no s-p mixing.

Show **energy arrow**, **atomic orbitals** involved, **atom labels**, the **molecular orbitals** that are formed, **filled electrons** in the MO diagram etc. Make sure to label all the atomic and molecular orbitals.

Using the MO diagram above, answer the following questions:

- (i) Which one of the following will have the highest bond order? Circle the correct choice
 A) PS^{1-} B) PS C) PS^{1+}
- (ii) Which one of the following will have the longest bond length? Circle the correct choice
 A) PS^{1-} B) PS C) PS^{1+}
- (iii) Which of the following will be diamagnetic? Circle the correct choice.
 A) PS^{1-} B) PS C) PS^{1+}
- (iv) Draw Lewis dot structure of the molecule with the shortest bond length (based on the MO diagram) from among the three (PS^{1-} , PS, PS^{1+}), and indicate the bond order calculated from the Lewis dot structure. **(3 points)**

Lewis Dot Structure

Bond Order



489A0136-5BB8-403F-814C-F0F54699A6A2

midterm-exam-93388

#349 Page 10 of 10

SCRAP PAGE

**CHEMISTRY 110 –Midterm Exam****Thursday November 7, 2024, 6:30 – 8:30 pm**

Examiners: Pallavi Sirjoosingh and Ashok Kakkar

First name (please write as legibly as possible within the boxes)

--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--

Last name

--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--

Student ID number

--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--

This booklet has 8 questions on 10 pages (The last page can be used as a scrap page).

The staple must NOT be removed from the question booklet. Your intact question booklet must be returned at the end of the exam.

A separate booklet contains a periodic table, and datasheet.

Write your answers directly under the questions in the space provided. Do not write on the QR codes or near the top edges (as it may be cut off).

Please write **legibly and dark enough** for the scan to be readable. For students who use English as a second language, dictionaries are permitted. Calculators are allowed.

DO NOT ASK QUESTIONS OF THE INVIGILATORS about interpretations of the exam questions. If you have a concern about a question, please summarize it at the top of this page. All concerns will be evaluated after the exam. GOOD LUCK!

**Question 1 8 points** (*Assume all transitions are possible*)

a. 6 points Calculate the **energy (in kJ/mol)** required to ionize (completely remove) the electron from the **n=3** level of **He⁺ ion** (He: Helium). Show your work including equations. (3 sig. figs.)

b. 2 points

(I) Select the largest (longest) wavelength from the three choices below:

- i. Wavelength of photon to *completely remove* an electron from *1s orbital of Hydrogen atom*
- ii. Wavelength of photon to *completely remove* an electron from *1s orbital of Boron atom (B)*
- iii. Wavelength of photon to *completely remove* an electron from *1s orbital of Boron cation (B⁺)*

i or ii or iii

(II) Select the smallest (shortest) wavelength from the three choices below (*for H atom*):

- i. Wavelength of photon *emitted* for an electron transitioning from *2p to 1s orbital*
- ii. Wavelength of photon *absorbed* for an electron transitioning from *2s to 4p orbital*
- iii. Wavelength of photon *absorbed* for an electron transitioning from *1s to 3p orbital*

i or ii or iii



Question 2 8 points



A Chem110 student plots the graph shown on the left, from data obtained for the kinetic energy (eV) of the ejected electron versus frequency (s^{-1}) of the incident light, for an unknown metal. ($1\text{eV} = 1.6022 \times 10^{-19} \text{ J}$)

Based on the plot, answer the questions **a to c** below:

a. 1 point On the plot above, label the **threshold frequency**.

b. 2 points What is the value of the work potential (in J) of the unknown metal?

c. 3 points What is the *longest* wavelength (in nm) required to eject an electron from this metal's surface? Show your work including equation(s) used. (3 sig figs)

d. 2 points In the space provided below, plot the following

(Assume frequency is *above* threshold frequency):

- Kinetic energy* of an ejected electron versus the *intensity* of the incident light (left)
- Number of electrons* ejected versus the *frequency* of the incident light (right)



**Question 3** **6 points****a. 2 points**

In the ground state of Ge (Germanium; $Z = 32$) atom, **name** the following orbitals:

i. Orbital containing the electron that requires the maximum energy to be removed.

ii. Orbital containing the electron that experiences the minimum effective nuclear charge.

b. 2 points

An unknown anion (A^-) has the following *excited* electronic configuration: $[Ar] 4s^2 3d^3 4p^1$

(Ar: Argon; $Z = 18$)

Give the **condensed ground state electronic configuration** of the atom, **A** and the cation, **A^{2+}** .
Show your work. (Assume all electronic transitions are possible)

Electronic Configuration of A:

Electronic Configuration of A^{2+} :

c. 2 points

In a single atom, determine how many electrons can have the following set of quantum numbers?

(i) $n = 5; l = 3$

In a single atom, determine how many electrons can have the following set of quantum numbers?

(i) $n = 3; l = 2; m_s = +1/2$

**Question 4** **8 points**

Rank the following from 1 to 4 in **increasing** order. Provide the ranking in the boxes provided.

Note: Atomic number (Z) are provided for relevant atoms

a) Increasing (lowest to highest) order of *ionization energy*: (1= lowest; 4 = highest)

Mg

Mg: Z = 12

Rb

Rb: Z = 37

Cs

Cs: Z = 55

K

K : Z = 19

b) Increasing (lowest to highest) order of *bond angle*: (1= lowest; 4 = highest)

H₂OCCl₄XeF₄

HCN

c) Increasing (smallest to largest) order of *ionic size* (1= smallest; 4 = largest)

S²⁻

S: Z = 16

O²⁻

O: Z = 8

Be²⁺

Be: Z = 4

Na⁺

Na: Z = 11

d) Increasing (lowest to highest) order of *bond energy* (1= lowest; 4 = highest)

P-F

P-Cl

P-I

P-Br

**Question 5** **12 points**

a. 9 points For the molecules HClO_4 and TeF_6 , draw the most stable *Lewis structure* showing appropriate shape and *dashed/wedged bonds*, based on VSEPR. Determine the **molecular geometry** around the central atom. For HClO_4 , label the *most polar bond*.



b. 3 points (Note: Lone pairs are not shown in structures below)

(i) For the three *resonance* structures (1, 2 and 3) shown below, answer the following question:



(I) Circle the *most contributing* resonance structure.

Structure 1

Structure 2

Structure 3

(ii) For the three structures given (4, 5, and 6) below, answer the following questions:



(I) Circle all the structures that represent *resonance structures* of each other:

Structure 4

Structure 5

Structure 6

(II) Among the resonance structures, circle the *most contributing* resonance structure.

Structure 4

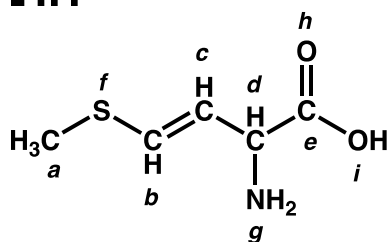
Structure 5

Structure 6

**Question 6** **6 points**

Using the data given below, calculate the **lattice energy** for SrO(s).

	ΔH° (kJ/mol)
$\text{Sr}(s) \longrightarrow \text{Sr}(g)$	+180
$\text{Sr}(g) \longrightarrow \text{Sr}^{2+}(g) + 2e^-$	+1468
$\text{O}_2(g) \longrightarrow 2\text{O}(g)$	+500
$\text{O}(g) + 2e^- \longrightarrow \text{O}^{2-}(g)$	+600
$\text{Sr}(s) + \frac{1}{2}\text{O}_2(g) \longrightarrow \text{SrO}(s)$	-550

**Question 7 10 points**

Consider the molecule given on the left, and answer questions (i) to (vi) using chemical bonding theories that we have learnt in class (valence bond theory, hybridization).

Note that the lone pairs are not shown in the structure:

i. Consider the carbon atoms marked in as ***a***, ***b***, ***c***, ***d***, and ***e*** in the molecule, and determine **hybridization** at each of the carbon centers. (2.5 points)

Atom	Hybridization
Carbon (<i>a</i>)	
Carbon (<i>b</i>)	
Carbon (<i>c</i>)	
Carbon (<i>d</i>)	
Carbon (<i>e</i>)	

ii. Now, consider the sulfur atom (S) marked as ***f***, nitrogen (N) atom (***g***), and oxygen(O) atoms (***h*** and ***i***), and determine their **hybridization**. (2 points)

Atom	Hybridization
Sulfur (<i>f</i>)	
Nitrogen (<i>g</i>)	
Oxygen (<i>h</i>)	
Oxygen (<i>i</i>)	

iii. **Describe** all the bonds formed by the carbon atoms ***a*** and ***b***, using sigma and pi designation as appropriate. (2 points)

Atom	Bonds formed (use sigma and pi as appropriate)
Carbon (<i>a</i>)	
Carbon (<i>b</i>)	

iv. How many total sp^3 -s sigma bonds are in the molecule? (1 point)

v. Determine **electronic** and **molecular geometry** at carbon (C) labeled ***e*** and sulfur (S) labeled ***f*** centres. (2 points)

Atom	Electronic Geometry	Molecular Geometry
Carbon (<i>e</i>)		
Sulfur (<i>f</i>)		

vi. Circle the value from the choices given below, that describes the **C-S-C bond angle at sulfur** (S) marked ***f***: (0.5 points)

>90° to <109.5° 109.5° >109.5 to <120° 120° >120° to <180°


Question 8 12 points

Using the **valence shell orbitals only** draw a **complete** molecular orbital (MO) diagram (with no s-p mixing) for the molecule **PS**, in the space provided below. **(6 points)**

Assume that 3s and 3p atomic orbitals will form molecular orbitals like those formed by the combination of 2s and 2p atomic orbitals that we learnt in class, and there is no s-p mixing.

Show **energy arrow**, **atomic orbitals** involved, **atom labels**, the **molecular orbitals** that are formed, **filled electrons** in the MO diagram etc. Make sure to label all the atomic and molecular orbitals.

Using the MO diagram above, answer the following questions:

- (i) Which one of the following will have the highest bond order? Circle the correct choice
 A) PS^{1-} B) PS C) PS^{1+}
- (ii) Which one of the following will have the longest bond length? Circle the correct choice
 A) PS^{1-} B) PS C) PS^{1+}
- (iii) Which of the following will be diamagnetic? Circle the correct choice.
 A) PS^{1-} B) PS C) PS^{1+}
- (iv) Draw Lewis dot structure of the molecule with the shortest bond length (based on the MO diagram) from among the three (PS^{1-} , PS, PS^{1+}), and indicate the bond order calculated from the Lewis dot structure. **(3 points)**

Lewis Dot Structure

Bond Order



6CF09985-AEB6-4E9D-BF4C-1588878A56DF

midterm-exam-93388

#350 Page 10 of 10

SCRAP PAGE

**CHEMISTRY 110 –Midterm Exam****Thursday November 7, 2024, 6:30 – 8:30 pm**

Examiners: Pallavi Sirjoosingh and Ashok Kakkar

First name (please write as legibly as possible within the boxes)

--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--

Last name

--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--

Student ID number

--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--

This booklet has 8 questions on 10 pages (The last page can be used as a scrap page).

The staple must NOT be removed from the question booklet. Your intact question booklet must be returned at the end of the exam.

A separate booklet contains a periodic table, and datasheet.

Write your answers directly under the questions in the space provided. Do not write on the QR codes or near the top edges (as it may be cut off).

Please write **legibly and dark enough** for the scan to be readable. For students who use English as a second language, dictionaries are permitted. Calculators are allowed.

DO NOT ASK QUESTIONS OF THE INVIGILATORS about interpretations of the exam questions. If you have a concern about a question, please summarize it at the top of this page. All concerns will be evaluated after the exam. GOOD LUCK!

**Question 1 8 points** (*Assume all transitions are possible*)

a. 6 points Calculate the **energy (in kJ/mol)** required to ionize (completely remove) the electron from the **$n=3$** level of **He^+ ion** (He: Helium). Show your work including equations. (3 sig. figs.)

b. 2 points

(I) Select the largest (longest) wavelength from the three choices below:

- i. Wavelength of photon to *completely remove* an electron from *1s orbital of Hydrogen atom*
- ii. Wavelength of photon to *completely remove* an electron from *1s orbital of Boron atom (B)*
- iii. Wavelength of photon to *completely remove* an electron from *1s orbital of Boron cation (B^+)*

i or ii or iii

(II) Select the smallest (shortest) wavelength from the three choices below (*for H atom*):

- i. Wavelength of photon *emitted* for an electron transitioning from *2p to 1s orbital*
- ii. Wavelength of photon *absorbed* for an electron transitioning from *2s to 4p orbital*
- iii. Wavelength of photon *absorbed* for an electron transitioning from *1s to 3p orbital*

i or ii or iii



Question 2 8 points



A Chem110 student plots the graph shown on the left, from data obtained for the kinetic energy (eV) of the ejected electron versus frequency (s^{-1}) of the incident light, for an unknown metal. ($1\text{eV} = 1.6022 \times 10^{-19} \text{ J}$)

Based on the plot, answer the questions **a to c** below:

a. 1 point On the plot above, label the **threshold frequency**.

b. 2 points What is the value of the work potential (in J) of the unknown metal?

c. 3 points What is the *longest* wavelength (in nm) required to eject an electron from this metal's surface? Show your work including equation(s) used. (3 sig figs)

d. 2 points In the space provided below, plot the following

(Assume frequency is *above* threshold frequency):

- Kinetic energy* of an ejected electron versus the *intensity* of the incident light (left)
- Number of electrons* ejected versus the *frequency* of the incident light (right)



**Question 3** **6 points****a. 2 points**

In the ground state of Ge (Germanium; $Z = 32$) atom, **name** the following orbitals:

i. Orbital containing the electron that requires the maximum energy to be removed.

ii. Orbital containing the electron that experiences the minimum effective nuclear charge.

b. 2 points

An unknown anion (A^-) has the following *excited* electronic configuration: $[Ar] 4s^2 3d^3 4p^1$

(Ar: Argon; $Z = 18$)

Give the **condensed ground state electronic configuration** of the atom, **A** and the cation, A^{2+} .
Show your work. (Assume all electronic transitions are possible)

Electronic Configuration of A:

Electronic Configuration of A^{2+} :

c. 2 points

In a single atom, determine how many electrons can have the following set of quantum numbers?

(i) $n = 5; l = 3$

In a single atom, determine how many electrons can have the following set of quantum numbers?

(i) $n = 3; l = 2; m_s = +1/2$

**Question 4** **8 points**

Rank the following from 1 to 4 in **increasing** order. Provide the ranking in the boxes provided.

Note: Atomic number (Z) are provided for relevant atoms

a) Increasing (lowest to highest) order of *ionization energy*: (1= lowest; 4 = highest)

Mg

Mg: Z = 12

Rb

Rb: Z = 37

Cs

Cs: Z = 55

K

K : Z = 19

b) Increasing (lowest to highest) order of *bond angle*: (1= lowest; 4 = highest)

H₂OCCl₄XeF₄

HCN

c) Increasing (smallest to largest) order of *ionic size* (1= smallest; 4 = largest)

S²⁻

S: Z = 16

O²⁻

O: Z = 8

Be²⁺

Be: Z = 4

Na⁺

Na: Z = 11

d) Increasing (lowest to highest) order of *bond energy* (1= lowest; 4 = highest)

P-F

P-Cl

P-I

P-Br

**Question 5** **12 points**

a. 9 points For the molecules HClO_4 and TeF_6 , draw the most stable *Lewis structure* showing appropriate shape and *dashed/wedged bonds*, based on VSEPR. Determine the **molecular geometry** around the central atom. For HClO_4 , label the *most polar bond*.



b. 3 points (Note: Lone pairs are not shown in structures below)

(i) For the three *resonance* structures (1, 2 and 3) shown below, answer the following question:



(I) Circle the *most contributing* resonance structure.

Structure 1

Structure 2

Structure 3

(ii) For the three structures given (4, 5, and 6) below, answer the following questions:



(I) Circle all the structures that represent *resonance structures* of each other:

Structure 4

Structure 5

Structure 6

(II) Among the resonance structures, circle the *most contributing* resonance structure.

Structure 4

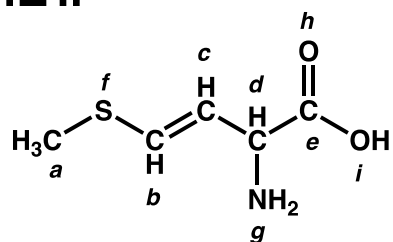
Structure 5

Structure 6

**Question 6** **6 points**

Using the data given below, calculate the **lattice energy** for SrO(s).

	ΔH° (kJ/mol)
$\text{Sr}(s) \longrightarrow \text{Sr}(g)$	+180
$\text{Sr}(g) \longrightarrow \text{Sr}^{2+}(g) + 2e^-$	+1468
$\text{O}_2(g) \longrightarrow 2\text{O}(g)$	+500
$\text{O}(g) + 2e^- \longrightarrow \text{O}^{2-}(g)$	+600
$\text{Sr}(s) + \frac{1}{2}\text{O}_2(g) \longrightarrow \text{SrO}(s)$	-550

**Question 7 10 points**

Consider the molecule given on the left, and answer questions (i) to (vi) using chemical bonding theories that we have learnt in class (valence bond theory, hybridization).

Note that the lone pairs are not shown in the structure:

i. Consider the carbon atoms marked in as ***a***, ***b***, ***c***, ***d***, and ***e*** in the molecule, and determine **hybridization** at each of the carbon centers. (2.5 points)

Atom	Hybridization
Carbon (<i>a</i>)	
Carbon (<i>b</i>)	
Carbon (<i>c</i>)	
Carbon (<i>d</i>)	
Carbon (<i>e</i>)	

ii. Now, consider the sulfur atom (S) marked as ***f***, nitrogen (N) atom (***g***), and oxygen(O) atoms (***h*** and ***i***), and determine their **hybridization**. (2 points)

Atom	Hybridization
Sulfur (<i>f</i>)	
Nitrogen (<i>g</i>)	
Oxygen (<i>h</i>)	
Oxygen (<i>i</i>)	

iii. **Describe** all the bonds formed by the carbon atoms ***a*** and ***b***, using sigma and pi designation as appropriate. (2 points)

Atom	Bonds formed (use sigma and pi as appropriate)
Carbon (<i>a</i>)	
Carbon (<i>b</i>)	

iv. How many total sp^3 -s sigma bonds are in the molecule? (1 point)

v. Determine **electronic** and **molecular geometry** at carbon (C) labeled ***e*** and sulfur (S) labeled ***f*** centres. (2 points)

Atom	Electronic Geometry	Molecular Geometry
Carbon (<i>e</i>)		
Sulfur (<i>f</i>)		

vi. Circle the value from the choices given below, that describes the **C-S-C bond angle at sulfur** (S) marked ***f***: (0.5 points)

>90° to <109.5° 109.5° >109.5 to <120° 120° >120° to <180°



Question 8 12 points

Using the **valence shell orbitals only** draw a **complete** molecular orbital (MO) diagram (with no s-p mixing) for the molecule **PS**, in the space provided below. **(6 points)**

Assume that 3s and 3p atomic orbitals will form molecular orbitals like those formed by the combination of 2s and 2p atomic orbitals that we learnt in class, and there is no s-p mixing.

Show **energy arrow**, **atomic orbitals** involved, **atom labels**, the **molecular orbitals** that are formed, **filled electrons** in the MO diagram etc. Make sure to label all the atomic and molecular orbitals.

Using the MO diagram above, answer the following questions:

- (i) Which one of the following will have the highest bond order? Circle the correct choice
 A) PS^{1-} B) PS C) PS^{1+}
- (ii) Which one of the following will have the longest bond length? Circle the correct choice
 A) PS^{1-} B) PS C) PS^{1+}
- (iii) Which of the following will be diamagnetic? Circle the correct choice.
 A) PS^{1-} B) PS C) PS^{1+}
- (iv) Draw Lewis dot structure of the molecule with the shortest bond length (based on the MO diagram) from among the three (PS^{1-} , PS, PS^{1+}), and indicate the bond order calculated from the Lewis dot structure. **(3 points)**

Lewis Dot Structure

Bond Order



C8D09B35-83C0-4B72-AB14-2ACAE443025

midterm-exam-93388

#351 Page 10 of 10

SCRAP PAGE

**CHEMISTRY 110 –Midterm Exam****Thursday November 7, 2024, 6:30 – 8:30 pm****Examiners: Pallavi Sirjoosingh and Ashok Kakkar**

First name (please write as legibly as possible within the boxes)

--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--

Last name

--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--

Student ID number

--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--

This booklet has 8 questions on 10 pages (The last page can be used as a scrap page).

The staple must NOT be removed from the question booklet. Your intact question booklet must be returned at the end of the exam.

A separate booklet contains a periodic table, and datasheet.

Write your answers directly under the questions in the space provided. Do not write on the QR codes or near the top edges (as it may be cut off).

Please write **legibly and dark enough** for the scan to be readable. For students who use English as a second language, dictionaries are permitted. Calculators are allowed.

DO NOT ASK QUESTIONS OF THE INVIGILATORS about interpretations of the exam questions. If you have a concern about a question, please summarize it at the top of this page. All concerns will be evaluated after the exam. GOOD LUCK!

**Question 1 8 points** (*Assume all transitions are possible*)

a. 6 points Calculate the **energy (in kJ/mol)** required to ionize (completely remove) the electron from the **n=3** level of **He⁺ ion** (He: Helium). Show your work including equations. (3 sig. figs.)

b. 2 points

(I) Select the largest (longest) wavelength from the three choices below:

- i. Wavelength of photon to *completely remove* an electron from *1s orbital of Hydrogen atom*
- ii. Wavelength of photon to *completely remove* an electron from *1s orbital of Boron atom (B)*
- iii. Wavelength of photon to *completely remove* an electron from *1s orbital of Boron cation (B⁺)*

i or ii or iii

(II) Select the smallest (shortest) wavelength from the three choices below (*for H atom*):

- i. Wavelength of photon *emitted* for an electron transitioning from *2p to 1s orbital*
- ii. Wavelength of photon *absorbed* for an electron transitioning from *2s to 4p orbital*
- iii. Wavelength of photon *absorbed* for an electron transitioning from *1s to 3p orbital*

i or ii or iii



Question 2 8 points



A Chem110 student plots the graph shown on the left, from data obtained for the kinetic energy (eV) of the ejected electron versus frequency (s^{-1}) of the incident light, for an unknown metal. ($1\text{eV} = 1.6022 \times 10^{-19} \text{ J}$)

Based on the plot, answer the questions **a to c** below:

a. 1 point On the plot above, label the **threshold frequency**.

b. 2 points What is the value of the work potential (in J) of the unknown metal?

c. 3 points What is the *longest* wavelength (in nm) required to eject an electron from this metal's surface? Show your work including equation(s) used. (3 sig figs)

d. 2 points In the space provided below, plot the following

(Assume frequency is *above* threshold frequency):

- i. *Kinetic energy* of an ejected electron versus the *intensity* of the incident light (left)
- ii. *Number of electrons* ejected versus the *frequency* of the incident light (right)



**Question 3** **6 points****a. 2 points**

In the ground state of Ge (Germanium; $Z = 32$) atom, **name** the following orbitals:

i. Orbital containing the electron that requires the maximum energy to be removed.

ii. Orbital containing the electron that experiences the minimum effective nuclear charge.

b. 2 points

An unknown anion (A^-) has the following *excited* electronic configuration: $[Ar] 4s^2 3d^3 4p^1$

(Ar: Argon; $Z = 18$)

Give the **condensed ground state electronic configuration** of the atom, **A** and the cation, **A^{2+}** .
Show your work. (Assume all electronic transitions are possible)

Electronic Configuration of A:

Electronic Configuration of A^{2+} :

c. 2 points

In a single atom, determine how many electrons can have the following set of quantum numbers?

(i) $n = 5; l = 3$

In a single atom, determine how many electrons can have the following set of quantum numbers?

(i) $n = 3; l = 2; m_s = +1/2$

**Question 4** **8 points**

Rank the following from 1 to 4 in **increasing** order. Provide the ranking in the boxes provided.

Note: Atomic number (Z) are provided for relevant atoms

a) Increasing (lowest to highest) order of *ionization energy*: (1= lowest; 4 = highest)

Mg

Mg: Z = 12

Rb

Rb: Z = 37

Cs

Cs: Z = 55

K

K : Z = 19

b) Increasing (lowest to highest) order of *bond angle*: (1= lowest; 4 = highest)

H₂OCCl₄XeF₄

HCN

c) Increasing (smallest to largest) order of *ionic size* (1= smallest; 4 = largest)

S²⁻

S: Z = 16

O²⁻

O: Z = 8

Be²⁺

Be: Z = 4

Na⁺

Na: Z = 11

d) Increasing (lowest to highest) order of *bond energy* (1= lowest; 4 = highest)

P-F

P-Cl

P-I

P-Br

**Question 5** **12 points**

a. 9 points For the molecules HClO_4 and TeF_6 , draw the most stable *Lewis structure* showing appropriate shape and *dashed/wedged bonds*, based on VSEPR. Determine the **molecular geometry** around the central atom. For HClO_4 , label the *most polar bond*.



b. 3 points (Note: Lone pairs are not shown in structures below)

(i) For the three *resonance* structures (1, 2 and 3) shown below, answer the following question:



(I) Circle the *most contributing* resonance structure.

Structure 1

Structure 2

Structure 3

(ii) For the three structures given (4, 5, and 6) below, answer the following questions:



(I) Circle all the structures that represent *resonance structures* of each other:

Structure 4

Structure 5

Structure 6

(II) Among the resonance structures, circle the *most contributing* resonance structure.

Structure 4

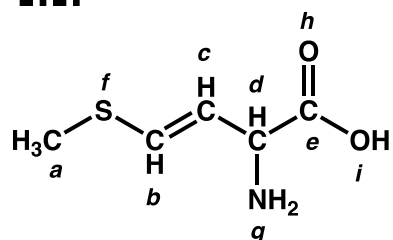
Structure 5

Structure 6

**Question 6** **6 points**

Using the data given below, calculate the **lattice energy** for SrO(s).

	ΔH° (kJ/mol)
$\text{Sr}(s) \longrightarrow \text{Sr}(g)$	+180
$\text{Sr}(g) \longrightarrow \text{Sr}^{2+}(g) + 2e^-$	+1468
$\text{O}_2(g) \longrightarrow 2\text{O}(g)$	+500
$\text{O}(g) + 2e^- \longrightarrow \text{O}^{2-}(g)$	+600
$\text{Sr}(s) + \frac{1}{2}\text{O}_2(g) \longrightarrow \text{SrO}(s)$	-550

**Question 7 10 points**

Consider the molecule given on the left, and answer questions (i) to (vi) using chemical bonding theories that we have learnt in class (valence bond theory, hybridization).

Note that the lone pairs are not shown in the structure:

i. Consider the carbon atoms marked in as ***a***, ***b***, ***c***, ***d***, and ***e*** in the molecule, and determine **hybridization** at each of the carbon centers. (2.5 points)

Atom	Hybridization
Carbon (<i>a</i>)	
Carbon (<i>b</i>)	
Carbon (<i>c</i>)	
Carbon (<i>d</i>)	
Carbon (<i>e</i>)	

ii. Now, consider the sulfur atom (S) marked as ***f***, nitrogen (N) atom (***g***), and oxygen(O) atoms (***h*** and ***i***), and determine their **hybridization**. (2 points)

Atom	Hybridization
Sulfur (<i>f</i>)	
Nitrogen (<i>g</i>)	
Oxygen (<i>h</i>)	
Oxygen (<i>i</i>)	

iii. **Describe** all the bonds formed by the carbon atoms ***a*** and ***b***, using sigma and pi designation as appropriate. (2 points)

Atom	Bonds formed (use sigma and pi as appropriate)
Carbon (<i>a</i>)	
Carbon (<i>b</i>)	

iv. How many total sp^3 -s sigma bonds are in the molecule? (1 point)

v. Determine **electronic** and **molecular geometry** at carbon (C) labeled ***e*** and sulfur (S) labeled ***f*** centres. (2 points)

Atom	Electronic Geometry	Molecular Geometry
Carbon (<i>e</i>)		
Sulfur (<i>f</i>)		

vi. Circle the value from the choices given below, that describes the **C-S-C bond angle at sulfur** (S) marked ***f***: (0.5 points)

$>90^\circ$ to $<109.5^\circ$ 109.5° >109.5 to $<120^\circ$ 120° $>120^\circ$ to $<180^\circ$

**Question 8 12 points**

Using the **valence shell orbitals only** draw a **complete** molecular orbital (MO) diagram (with no s-p mixing) for the molecule **PS**, in the space provided below. **(6 points)**

Assume that 3s and 3p atomic orbitals will form molecular orbitals like those formed by the combination of 2s and 2p atomic orbitals that we learnt in class, and there is no s-p mixing.

Show **energy arrow**, **atomic orbitals** involved, **atom labels**, the **molecular orbitals** that are formed, **filled electrons** in the MO diagram etc. Make sure to label all the atomic and molecular orbitals.

Using the MO diagram above, answer the following questions:

- (i) Which one of the following will have the highest bond order? Circle the correct choice
A) PS^{1-} B) PS C) PS^{1+}
- (ii) Which one of the following will have the longest bond length? Circle the correct choice
A) PS^{1-} B) PS C) PS^{1+}
- (iii) Which of the following will be diamagnetic? Circle the correct choice.
A) PS^{1-} B) PS C) PS^{1+}
- (iv) Draw Lewis dot structure of the molecule with the shortest bond length (based on the MO diagram) from among the three (PS^{1-} , PS, PS^{1+}), and indicate the bond order calculated from the Lewis dot structure. **(3 points)**

Lewis Dot Structure

Bond Order



3334B41A-215B-4625-A9A4-75E7ACEE2427

midterm-exam-93388

#352 Page 10 of 10

SCRAP PAGE

**CHEMISTRY 110 –Midterm Exam****Thursday November 7, 2024, 6:30 – 8:30 pm****Examiners: Pallavi Sirjoosingh and Ashok Kakkar**

First name (please write as legibly as possible within the boxes)

--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--

Last name

--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--

Student ID number

--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--

This booklet has 8 questions on 10 pages (The last page can be used as a scrap page).

The staple must NOT be removed from the question booklet. Your intact question booklet must be returned at the end of the exam.

A separate booklet contains a periodic table, and datasheet.

Write your answers directly under the questions in the space provided. Do not write on the QR codes or near the top edges (as it may be cut off).

Please write **legibly and dark enough** for the scan to be readable. For students who use English as a second language, dictionaries are permitted. Calculators are allowed.

DO NOT ASK QUESTIONS OF THE INVIGILATORS about interpretations of the exam questions. If you have a concern about a question, please summarize it at the top of this page. All concerns will be evaluated after the exam. GOOD LUCK!

**Question 1 8 points** (*Assume all transitions are possible*)

a. 6 points Calculate the **energy (in kJ/mol)** required to ionize (completely remove) the electron from the **n=3** level of **He⁺ ion** (He: Helium). Show your work including equations. (3 sig. figs.)

b. 2 points

(I) Select the largest (longest) wavelength from the three choices below:

- i. Wavelength of photon to *completely remove* an electron from *1s orbital of Hydrogen atom*
- ii. Wavelength of photon to *completely remove* an electron from *1s orbital of Boron atom (B)*
- iii. Wavelength of photon to *completely remove* an electron from *1s orbital of Boron cation (B⁺)*

i or ii or iii

(II) Select the smallest (shortest) wavelength from the three choices below (*for H atom*):

- i. Wavelength of photon *emitted* for an electron transitioning from *2p to 1s orbital*
- ii. Wavelength of photon *absorbed* for an electron transitioning from *2s to 4p orbital*
- iii. Wavelength of photon *absorbed* for an electron transitioning from *1s to 3p orbital*

i or ii or iii



Question 2 8 points



A Chem110 student plots the graph shown on the left, from data obtained for the kinetic energy (eV) of the ejected electron versus frequency (s^{-1}) of the incident light, for an unknown metal. ($1\text{eV} = 1.6022 \times 10^{-19} \text{ J}$)

Based on the plot, answer the questions **a to c** below:

a. 1 point On the plot above, label the **threshold frequency**.

b. 2 points What is the value of the work potential (in J) of the unknown metal?

c. 3 points What is the *longest* wavelength (in nm) required to eject an electron from this metal's surface? Show your work including equation(s) used. (3 sig figs)

d. 2 points In the space provided below, plot the following

(Assume frequency is *above* threshold frequency):

- Kinetic energy* of an ejected electron versus the *intensity* of the incident light (left)
- Number of electrons* ejected versus the *frequency* of the incident light (right)



**Question 3** **6 points****a. 2 points**

In the ground state of Ge (Germanium; $Z = 32$) atom, **name** the following orbitals:

i. Orbital containing the electron that requires the maximum energy to be removed.

ii. Orbital containing the electron that experiences the minimum effective nuclear charge.

b. 2 points

An unknown anion (A^-) has the following *excited* electronic configuration: $[Ar] 4s^2 3d^3 4p^1$

(Ar: Argon; $Z = 18$)

Give the **condensed ground state electronic configuration** of the atom, **A** and the cation, **A^{2+}** .
Show your work. (Assume all electronic transitions are possible)

Electronic Configuration of A:

Electronic Configuration of A^{2+} :

c. 2 points

In a single atom, determine how many electrons can have the following set of quantum numbers?

(i) $n = 5; l = 3$

In a single atom, determine how many electrons can have the following set of quantum numbers?

(i) $n = 3; l = 2; m_s = +1/2$

**Question 4** **8 points**

Rank the following from 1 to 4 in **increasing** order. Provide the ranking in the boxes provided.

Note: Atomic number (Z) are provided for relevant atoms

a) Increasing (lowest to highest) order of *ionization energy*: (1= lowest; 4 = highest)

Mg

Mg: Z = 12

Rb

Rb: Z = 37

Cs

Cs: Z = 55

K

K : Z = 19

b) Increasing (lowest to highest) order of *bond angle*: (1= lowest; 4 = highest)

H₂OCCl₄XeF₄

HCN

c) Increasing (smallest to largest) order of *ionic size* (1= smallest; 4 = largest)

S²⁻

S: Z = 16

O²⁻

O: Z = 8

Be²⁺

Be: Z = 4

Na⁺

Na: Z = 11

d) Increasing (lowest to highest) order of *bond energy* (1= lowest; 4 = highest)

P-F

P-Cl

P-I

P-Br

**Question 5** **12 points**

a. 9 points For the molecules HClO_4 and TeF_6 , draw the most stable *Lewis structure* showing appropriate shape and *dashed/wedged bonds*, based on VSEPR. Determine the **molecular geometry** around the central atom. For HClO_4 , label the *most polar bond*.



b. 3 points (Note: Lone pairs are not shown in structures below)

(i) For the three *resonance* structures (1, 2 and 3) shown below, answer the following question:



(I) Circle the *most contributing* resonance structure.

Structure 1

Structure 2

Structure 3

(ii) For the three structures given (4, 5, and 6) below, answer the following questions:



(I) Circle all the structures that represent *resonance structures* of each other:

Structure 4

Structure 5

Structure 6

(II) Among the resonance structures, circle the *most contributing* resonance structure.

Structure 4

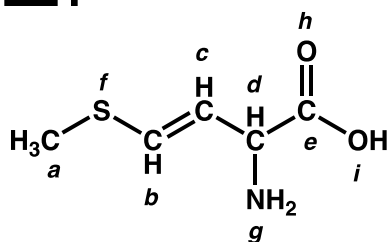
Structure 5

Structure 6

**Question 6** **6 points**

Using the data given below, calculate the **lattice energy** for SrO(s).

	ΔH° (kJ/mol)
$\text{Sr}(s) \longrightarrow \text{Sr}(g)$	+180
$\text{Sr}(g) \longrightarrow \text{Sr}^{2+}(g) + 2e^-$	+1468
$\text{O}_2(g) \longrightarrow 2\text{O}(g)$	+500
$\text{O}(g) + 2e^- \longrightarrow \text{O}^{2-}(g)$	+600
$\text{Sr}(s) + \frac{1}{2}\text{O}_2(g) \longrightarrow \text{SrO}(s)$	-550

**Question 7 10 points**

Consider the molecule given on the left, and answer questions (i) to (vi) using chemical bonding theories that we have learnt in class (valence bond theory, hybridization).

Note that the lone pairs are not shown in the structure:

i. Consider the carbon atoms marked in as ***a***, ***b***, ***c***, ***d***, and ***e*** in the molecule, and determine **hybridization** at each of the carbon centers. (2.5 points)

Atom	Hybridization
Carbon (<i>a</i>)	
Carbon (<i>b</i>)	
Carbon (<i>c</i>)	
Carbon (<i>d</i>)	
Carbon (<i>e</i>)	

ii. Now, consider the sulfur atom (S) marked as ***f***, nitrogen (N) atom (***g***), and oxygen(O) atoms (***h*** and ***i***), and determine their **hybridization**. (2 points)

Atom	Hybridization
Sulfur (<i>f</i>)	
Nitrogen (<i>g</i>)	
Oxygen (<i>h</i>)	
Oxygen (<i>i</i>)	

iii. **Describe** all the bonds formed by the carbon atoms ***a*** and ***b***, using sigma and pi designation as appropriate. (2 points)

Atom	Bonds formed (use sigma and pi as appropriate)
Carbon (<i>a</i>)	
Carbon (<i>b</i>)	

iv. How many total sp^3 -s sigma bonds are in the molecule? (1 point)

v. Determine **electronic** and **molecular geometry** at carbon (C) labeled ***e*** and sulfur (S) labeled ***f*** centres. (2 points)

Atom	Electronic Geometry	Molecular Geometry
Carbon (<i>e</i>)		
Sulfur (<i>f</i>)		

vi. Circle the value from the choices given below, that describes the **C-S-C bond angle at sulfur** (S) marked ***f***: (0.5 points)

>90° to <109.5° 109.5° >109.5 to <120° 120° >120° to <180°

**Question 8 12 points**

Using the **valence shell orbitals only** draw a **complete** molecular orbital (MO) diagram (with no s-p mixing) for the molecule **PS**, in the space provided below. **(6 points)**

Assume that 3s and 3p atomic orbitals will form molecular orbitals like those formed by the combination of 2s and 2p atomic orbitals that we learnt in class, and there is no s-p mixing.

Show **energy arrow**, **atomic orbitals** involved, **atom labels**, the **molecular orbitals** that are formed, **filled electrons** in the MO diagram etc. Make sure to **label all the atomic and molecular orbitals**.

Using the MO diagram above, answer the following questions:

- (i) Which one of the following will have the highest bond order? Circle the correct choice
A) PS^{1-} B) PS C) PS^{1+}
- (ii) Which one of the following will have the longest bond length? Circle the correct choice
A) PS^{1-} B) PS C) PS^{1+}
- (iii) Which of the following will be diamagnetic? Circle the correct choice.
A) PS^{1-} B) PS C) PS^{1+}
- (iv) Draw Lewis dot structure of the molecule with the shortest bond length (based on the MO diagram) from among the three (PS^{1-} , PS, PS^{1+}), and indicate the bond order calculated from the Lewis dot structure. **(3 points)**

Lewis Dot Structure

Bond Order



E03F8555-E840-4A6B-94EE-582F538EAD36

midterm-exam-93388

#353 Page 10 of 10

SCRAP PAGE

**CHEMISTRY 110 –Midterm Exam****Thursday November 7, 2024, 6:30 – 8:30 pm**

Examiners: Pallavi Sirjoosingh and Ashok Kakkar

First name (please write as legibly as possible within the boxes)

--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--

Last name

--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--

Student ID number

--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--

This booklet has 8 questions on 10 pages (The last page can be used as a scrap page).

The staple must NOT be removed from the question booklet. Your intact question booklet must be returned at the end of the exam.

A separate booklet contains a periodic table, and datasheet.

Write your answers directly under the questions in the space provided. Do not write on the QR codes or near the top edges (as it may be cut off).

Please write **legibly and dark enough** for the scan to be readable. For students who use English as a second language, dictionaries are permitted. Calculators are allowed.

DO NOT ASK QUESTIONS OF THE INVIGILATORS about interpretations of the exam questions. If you have a concern about a question, please summarize it at the top of this page. All concerns will be evaluated after the exam. GOOD LUCK!

**Question 1 8 points** (*Assume all transitions are possible*)

a. 6 points Calculate the **energy (in kJ/mol)** required to ionize (completely remove) the electron from the **n=3** level of **He⁺ ion** (He: Helium). Show your work including equations. (3 sig. figs.)

b. 2 points

(I) Select the largest (longest) wavelength from the three choices below:

- i. Wavelength of photon to *completely remove* an electron from *1s orbital of Hydrogen atom*
- ii. Wavelength of photon to *completely remove* an electron from *1s orbital of Boron atom (B)*
- iii. Wavelength of photon to *completely remove* an electron from *1s orbital of Boron cation (B⁺)*

i or ii or iii

(II) Select the smallest (shortest) wavelength from the three choices below (*for H atom*):

- i. Wavelength of photon *emitted* for an electron transitioning from *2p to 1s orbital*
- ii. Wavelength of photon *absorbed* for an electron transitioning from *2s to 4p orbital*
- iii. Wavelength of photon *absorbed* for an electron transitioning from *1s to 3p orbital*

i or ii or iii



Question 2 8 points



A Chem110 student plots the graph shown on the left, from data obtained for the kinetic energy (eV) of the ejected electron versus frequency (s^{-1}) of the incident light, for an unknown metal. ($1\text{eV} = 1.6022 \times 10^{-19} \text{ J}$)

Based on the plot, answer the questions **a to c** below:

a. 1 point On the plot above, label the **threshold frequency**.

b. 2 points What is the value of the work potential (in J) of the unknown metal?

c. 3 points What is the *longest* wavelength (in nm) required to eject an electron from this metal's surface? Show your work including equation(s) used. (3 sig figs)

d. 2 points In the space provided below, plot the following

(Assume frequency is *above* threshold frequency):

- Kinetic energy* of an ejected electron versus the *intensity* of the incident light (left)
- Number of electrons* ejected versus the *frequency* of the incident light (right)



**Question 3** **6 points****a. 2 points**

In the ground state of Ge (Germanium; $Z = 32$) atom, **name** the following orbitals:

i. Orbital containing the electron that requires the maximum energy to be removed.

ii. Orbital containing the electron that experiences the minimum effective nuclear charge.

b. 2 points

An unknown anion (A^-) has the following *excited* electronic configuration: $[Ar] 4s^2 3d^3 4p^1$

(Ar: Argon; $Z = 18$)

Give the **condensed ground state electronic configuration** of the atom, **A** and the cation, **A^{2+}** .
Show your work. (Assume all electronic transitions are possible)

Electronic Configuration of A:

Electronic Configuration of A^{2+} :

c. 2 points

In a single atom, determine how many electrons can have the following set of quantum numbers?

(i) $n = 5; l = 3$

In a single atom, determine how many electrons can have the following set of quantum numbers?

(i) $n = 3; l = 2; m_s = +1/2$

**Question 4** **8 points**

Rank the following from 1 to 4 in **increasing** order. Provide the ranking in the boxes provided.

Note: Atomic number (Z) are provided for relevant atoms

a) Increasing (lowest to highest) order of *ionization energy*: (1= lowest; 4 = highest)

Mg

Mg: Z = 12

Rb

Rb: Z = 37

Cs

Cs: Z = 55

K

K : Z = 19

b) Increasing (lowest to highest) order of *bond angle*: (1= lowest; 4 = highest)

H₂OCCl₄XeF₄

HCN

c) Increasing (smallest to largest) order of *ionic size* (1= smallest; 4 = largest)

S²⁻

S: Z = 16

O²⁻

O: Z = 8

Be²⁺

Be: Z = 4

Na⁺

Na: Z = 11

d) Increasing (lowest to highest) order of *bond energy* (1= lowest; 4 = highest)

P-F

P-Cl

P-I

P-Br

**Question 5** **12 points**

a. 9 points For the molecules HClO_4 and TeF_6 , draw the most stable *Lewis structure* showing appropriate shape and *dashed/wedged bonds*, based on VSEPR. Determine the **molecular geometry** around the central atom. For HClO_4 , label the *most polar bond*.



b. 3 points (Note: Lone pairs are not shown in structures below)

(i) For the three *resonance* structures (1, 2 and 3) shown below, answer the following question:



(I) Circle the *most contributing* resonance structure.

Structure 1

Structure 2

Structure 3

(ii) For the three structures given (4, 5, and 6) below, answer the following questions:



(I) Circle all the structures that represent *resonance structures* of each other:

Structure 4

Structure 5

Structure 6

(II) Among the resonance structures, circle the *most contributing* resonance structure.

Structure 4

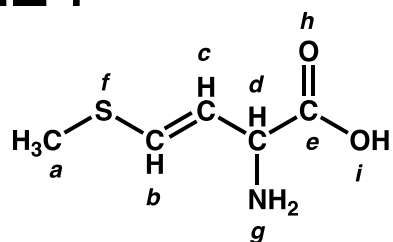
Structure 5

Structure 6

**Question 6** **6 points**

Using the data given below, calculate the **lattice energy** for SrO(s).

	ΔH° (kJ/mol)
$\text{Sr}(s) \longrightarrow \text{Sr}(g)$	+180
$\text{Sr}(g) \longrightarrow \text{Sr}^{2+}(g) + 2e^-$	+1468
$\text{O}_2(g) \longrightarrow 2\text{O}(g)$	+500
$\text{O}(g) + 2e^- \longrightarrow \text{O}^{2-}(g)$	+600
$\text{Sr}(s) + \frac{1}{2}\text{O}_2(g) \longrightarrow \text{SrO}(s)$	-550

**Question 7 10 points**

Consider the molecule given on the left, and answer questions (i) to (vi) using chemical bonding theories that we have learnt in class (valence bond theory, hybridization).

Note that the lone pairs are not shown in the structure:

i. Consider the carbon atoms marked in as ***a***, ***b***, ***c***, ***d***, and ***e*** in the molecule, and determine **hybridization** at each of the carbon centers. (2.5 points)

Atom	Hybridization
Carbon (<i>a</i>)	
Carbon (<i>b</i>)	
Carbon (<i>c</i>)	
Carbon (<i>d</i>)	
Carbon (<i>e</i>)	

ii. Now, consider the sulfur atom (S) marked as ***f***, nitrogen (N) atom (***g***), and oxygen(O) atoms (***h*** and ***i***), and determine their **hybridization**. (2 points)

Atom	Hybridization
Sulfur (<i>f</i>)	
Nitrogen (<i>g</i>)	
Oxygen (<i>h</i>)	
Oxygen (<i>i</i>)	

iii. **Describe** all the bonds formed by the carbon atoms ***a*** and ***b***, using sigma and pi designation as appropriate. (2 points)

Atom	Bonds formed (use sigma and pi as appropriate)
Carbon (<i>a</i>)	
Carbon (<i>b</i>)	

iv. How many total sp^3 -s sigma bonds are in the molecule? (1 point)

v. Determine **electronic** and **molecular geometry** at carbon (C) labeled ***e*** and sulfur (S) labeled ***f*** centres. (2 points)

Atom	Electronic Geometry	Molecular Geometry
Carbon (<i>e</i>)		
Sulfur (<i>f</i>)		

vi. Circle the value from the choices given below, that describes the **C-S-C bond angle at sulfur** (S) marked ***f***: (0.5 points)

>90° to <109.5° 109.5° >109.5 to <120° 120° >120° to <180°


Question 8 12 points

Using the **valence shell orbitals only** draw a **complete** molecular orbital (MO) diagram (with no s-p mixing) for the molecule **PS**, in the space provided below. **(6 points)**

Assume that 3s and 3p atomic orbitals will form molecular orbitals like those formed by the combination of 2s and 2p atomic orbitals that we learnt in class, and there is no s-p mixing.

Show **energy arrow**, **atomic orbitals** involved, **atom labels**, the **molecular orbitals** that are formed, **filled electrons** in the MO diagram etc. Make sure to label all the atomic and molecular orbitals.

Using the MO diagram above, answer the following questions:

- (i) Which one of the following will have the highest bond order? Circle the correct choice
 A) PS^{1-} B) PS C) PS^{1+}
- (ii) Which one of the following will have the longest bond length? Circle the correct choice
 A) PS^{1-} B) PS C) PS^{1+}
- (iii) Which of the following will be diamagnetic? Circle the correct choice.
 A) PS^{1-} B) PS C) PS^{1+}
- (iv) Draw Lewis dot structure of the molecule with the shortest bond length (based on the MO diagram) from among the three (PS^{1-} , PS, PS^{1+}), and indicate the bond order calculated from the Lewis dot structure. **(3 points)**

Lewis Dot Structure

Bond Order



62EB476E-4EBF-4BDE-999E-197F209F6558

midterm-exam-93388

#354 Page 10 of 10

SCRAP PAGE

**CHEMISTRY 110 –Midterm Exam****Thursday November 7, 2024, 6:30 – 8:30 pm****Examiners: Pallavi Sirjoosingh and Ashok Kakkar**

First name (please write as legibly as possible within the boxes)

--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--

Last name

--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--

Student ID number

--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--

This booklet has 8 questions on 10 pages (The last page can be used as a scrap page).

The staple must NOT be removed from the question booklet. Your intact question booklet must be returned at the end of the exam.

A separate booklet contains a periodic table, and datasheet.

Write your answers directly under the questions in the space provided. Do not write on the QR codes or near the top edges (as it may be cut off).

Please write **legibly and dark enough** for the scan to be readable. For students who use English as a second language, dictionaries are permitted. Calculators are allowed.

DO NOT ASK QUESTIONS OF THE INVIGILATORS about interpretations of the exam questions. If you have a concern about a question, please summarize it at the top of this page. All concerns will be evaluated after the exam. GOOD LUCK!

**Question 1 8 points** (*Assume all transitions are possible*)

a. 6 points Calculate the **energy (in kJ/mol)** required to ionize (completely remove) the electron from the **$n=3$** level of **He^+ ion** (He: Helium). Show your work including equations. (3 sig. figs.)

b. 2 points

(I) Select the largest (longest) wavelength from the three choices below:

- i. Wavelength of photon to *completely remove* an electron from *1s orbital of Hydrogen atom*
- ii. Wavelength of photon to *completely remove* an electron from *1s orbital of Boron atom (B)*
- iii. Wavelength of photon to *completely remove* an electron from *1s orbital of Boron cation (B^+)*

i or ii or iii

(II) Select the smallest (shortest) wavelength from the three choices below (*for H atom*):

- i. Wavelength of photon *emitted* for an electron transitioning from *2p to 1s orbital*
- ii. Wavelength of photon *absorbed* for an electron transitioning from *2s to 4p orbital*
- iii. Wavelength of photon *absorbed* for an electron transitioning from *1s to 3p orbital*

i or ii or iii



Question 2 8 points



A Chem110 student plots the graph shown on the left, from data obtained for the kinetic energy (eV) of the ejected electron versus frequency (s^{-1}) of the incident light, for an unknown metal. ($1\text{eV} = 1.6022 \times 10^{-19} \text{ J}$)

Based on the plot, answer the questions **a to c** below:

a. 1 point On the plot above, label the **threshold frequency**.

b. 2 points What is the value of the work potential (in J) of the unknown metal?

c. 3 points What is the *longest* wavelength (in nm) required to eject an electron from this metal's surface? Show your work including equation(s) used. (3 sig figs)

d. 2 points In the space provided below, plot the following

(Assume frequency is *above* threshold frequency):

- Kinetic energy* of an ejected electron versus the *intensity* of the incident light (left)
- Number of electrons* ejected versus the *frequency* of the incident light (right)



**Question 3** **6 points****a. 2 points**

In the ground state of Ge (Germanium; $Z = 32$) atom, **name** the following orbitals:

i. Orbital containing the electron that requires the maximum energy to be removed.

ii. Orbital containing the electron that experiences the minimum effective nuclear charge.

b. 2 points

An unknown anion (A^-) has the following *excited* electronic configuration: $[Ar] 4s^2 3d^3 4p^1$

(Ar: Argon; $Z = 18$)

Give the **condensed ground state electronic configuration** of the atom, **A** and the cation, **A^{2+}** .
Show your work. (Assume all electronic transitions are possible)

Electronic Configuration of A:

Electronic Configuration of A^{2+} :

c. 2 points

In a single atom, determine how many electrons can have the following set of quantum numbers?

(i) $n = 5; l = 3$

In a single atom, determine how many electrons can have the following set of quantum numbers?

(i) $n = 3; l = 2; m_s = +1/2$

**Question 4** **8 points**

Rank the following from 1 to 4 in **increasing** order. Provide the ranking in the boxes provided.

Note: Atomic number (Z) are provided for relevant atoms

a) Increasing (lowest to highest) order of *ionization energy*: (1= lowest; 4 = highest)

Mg

Mg: Z = 12

Rb

Rb: Z = 37

Cs

Cs: Z = 55

K

K : Z = 19

b) Increasing (lowest to highest) order of *bond angle*: (1= lowest; 4 = highest)

H₂OCCl₄XeF₄

HCN

c) Increasing (smallest to largest) order of *ionic size* (1= smallest; 4 = largest)

S²⁻

S: Z = 16

O²⁻

O: Z = 8

Be²⁺

Be: Z = 4

Na⁺

Na: Z = 11

d) Increasing (lowest to highest) order of *bond energy* (1= lowest; 4 = highest)

P-F

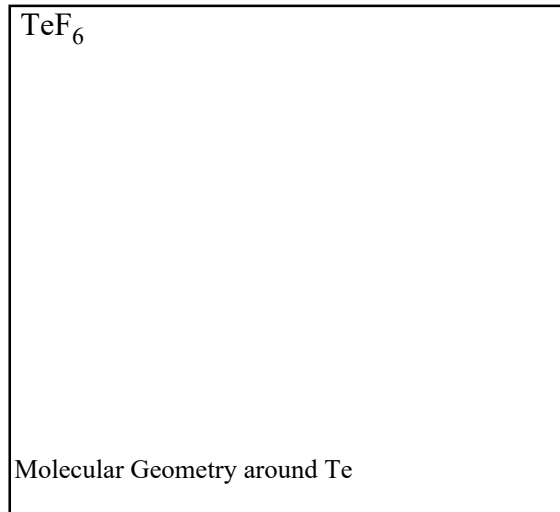
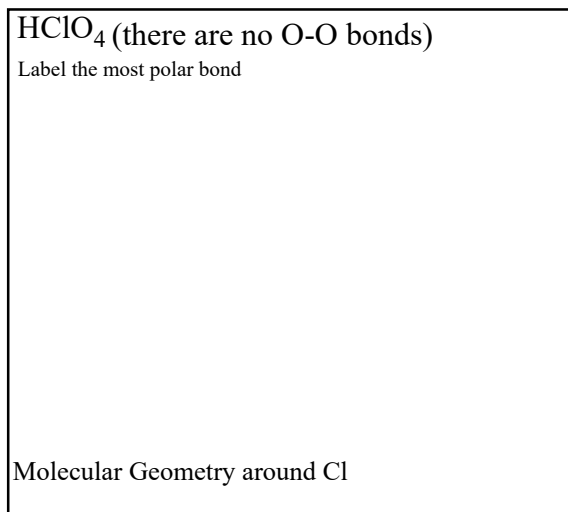
P-Cl

P-I

P-Br

**Question 5** **12 points**

a. 9 points For the molecules HClO_4 and TeF_6 , draw the most stable *Lewis structure* showing appropriate shape and *dashed/wedged bonds*, based on VSEPR. Determine the **molecular geometry** around the central atom. For HClO_4 , label the *most polar bond*.



b. 3 points (Note: Lone pairs are not shown in structures below)

(i) For the three *resonance* structures (1, 2 and 3) shown below, answer the following question:



(I) Circle the *most contributing* resonance structure.

Structure 1

Structure 2

Structure 3

(ii) For the three structures given (4, 5, and 6) below, answer the following questions:



(I) Circle all the structures that represent *resonance structures* of each other:

Structure 4

Structure 5

Structure 6

(II) Among the resonance structures, circle the *most contributing* resonance structure.

Structure 4

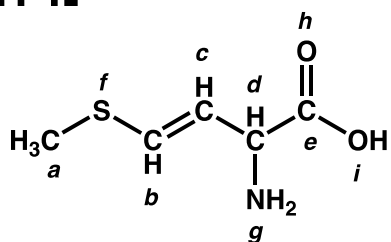
Structure 5

Structure 6

**Question 6** **6 points**

Using the data given below, calculate the **lattice energy** for SrO(s).

	ΔH° (kJ/mol)
$\text{Sr}(s) \longrightarrow \text{Sr}(g)$	+180
$\text{Sr}(g) \longrightarrow \text{Sr}^{2+}(g) + 2e^-$	+1468
$\text{O}_2(g) \longrightarrow 2\text{O}(g)$	+500
$\text{O}(g) + 2e^- \longrightarrow \text{O}^{2-}(g)$	+600
$\text{Sr}(s) + \frac{1}{2}\text{O}_2(g) \longrightarrow \text{SrO}(s)$	-550

**Question 7 10 points**

Consider the molecule given on the left, and answer questions (i) to (vi) using chemical bonding theories that we have learnt in class (valence bond theory, hybridization).

Note that the lone pairs are not shown in the structure:

i. Consider the carbon atoms marked in as ***a***, ***b***, ***c***, ***d***, and ***e*** in the molecule, and determine **hybridization** at each of the carbon centers. (2.5 points)

Atom	Hybridization
Carbon (<i>a</i>)	
Carbon (<i>b</i>)	
Carbon (<i>c</i>)	
Carbon (<i>d</i>)	
Carbon (<i>e</i>)	

ii. Now, consider the sulfur atom (S) marked as ***f***, nitrogen (N) atom (***g***), and oxygen(O) atoms (***h*** and ***i***), and determine their **hybridization**. (2 points)

Atom	Hybridization
Sulfur (<i>f</i>)	
Nitrogen (<i>g</i>)	
Oxygen (<i>h</i>)	
Oxygen (<i>i</i>)	

iii. **Describe** all the bonds formed by the carbon atoms ***a*** and ***b***, using sigma and pi designation as appropriate. (2 points)

Atom	Bonds formed (use sigma and pi as appropriate)
Carbon (<i>a</i>)	
Carbon (<i>b</i>)	

iv. How many total sp^3 -s sigma bonds are in the molecule? (1 point)

v. Determine **electronic** and **molecular geometry** at carbon (C) labeled ***e*** and sulfur (S) labeled ***f*** centres. (2 points)

Atom	Electronic Geometry	Molecular Geometry
Carbon (<i>e</i>)		
Sulfur (<i>f</i>)		

vi. Circle the value from the choices given below, that describes the **C-S-C bond angle at sulfur** (S) marked ***f***: (0.5 points)

>90° to <109.5° 109.5° >109.5 to <120° 120° >120° to <180°


Question 8 12 points

Using the **valence shell orbitals only** draw a **complete** molecular orbital (MO) diagram (with no s-p mixing) for the molecule **PS**, in the space provided below. **(6 points)**

Assume that 3s and 3p atomic orbitals will form molecular orbitals like those formed by the combination of 2s and 2p atomic orbitals that we learnt in class, and there is no s-p mixing.

Show **energy arrow**, **atomic orbitals** involved, **atom labels**, the **molecular orbitals** that are formed, **filled electrons** in the MO diagram etc. Make sure to label all the atomic and molecular orbitals.

Using the MO diagram above, answer the following questions:

- (i) Which one of the following will have the highest bond order? Circle the correct choice
 A) PS^{1-} B) PS C) PS^{1+}
- (ii) Which one of the following will have the longest bond length? Circle the correct choice
 A) PS^{1-} B) PS C) PS^{1+}
- (iii) Which of the following will be diamagnetic? Circle the correct choice.
 A) PS^{1-} B) PS C) PS^{1+}
- (iv) Draw Lewis dot structure of the molecule with the shortest bond length (based on the MO diagram) from among the three (PS^{1-} , PS, PS^{1+}), and indicate the bond order calculated from the Lewis dot structure. **(3 points)**

Lewis Dot Structure

Bond Order



12CA4AA5-FFED-4595-B94B-D4F33FC61AE4

midterm-exam-93388

#355 Page 10 of 10

SCRAP PAGE

**CHEMISTRY 110 –Midterm Exam****Thursday November 7, 2024, 6:30 – 8:30 pm**

Examiners: Pallavi Sirjoosingh and Ashok Kakkar

First name (please write as legibly as possible within the boxes)

--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--

Last name

--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--

Student ID number

--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--

This booklet has 8 questions on 10 pages (The last page can be used as a scrap page).

The staple must NOT be removed from the question booklet. Your intact question booklet must be returned at the end of the exam.

A separate booklet contains a periodic table, and datasheet.

Write your answers directly under the questions in the space provided. Do not write on the QR codes or near the top edges (as it may be cut off).

Please write **legibly and dark enough** for the scan to be readable. For students who use English as a second language, dictionaries are permitted. Calculators are allowed.

DO NOT ASK QUESTIONS OF THE INVIGILATORS about interpretations of the exam questions. If you have a concern about a question, please summarize it at the top of this page. All concerns will be evaluated after the exam. GOOD LUCK!

**Question 1 8 points** (*Assume all transitions are possible*)

a. 6 points Calculate the **energy (in kJ/mol)** required to ionize (completely remove) the electron from the **$n=3$** level of **He^+ ion** (He: Helium). Show your work including equations. (3 sig. figs.)

b. 2 points

(I) Select the largest (longest) wavelength from the three choices below:

- i. Wavelength of photon to *completely remove* an electron from *1s orbital of Hydrogen atom*
- ii. Wavelength of photon to *completely remove* an electron from *1s orbital of Boron atom (B)*
- iii. Wavelength of photon to *completely remove* an electron from *1s orbital of Boron cation (B^+)*

i or ii or iii

(II) Select the smallest (shortest) wavelength from the three choices below (*for H atom*):

- i. Wavelength of photon *emitted* for an electron transitioning from *2p to 1s orbital*
- ii. Wavelength of photon *absorbed* for an electron transitioning from *2s to 4p orbital*
- iii. Wavelength of photon *absorbed* for an electron transitioning from *1s to 3p orbital*

i or ii or iii



Question 2 8 points



A Chem110 student plots the graph shown on the left, from data obtained for the kinetic energy (eV) of the ejected electron versus frequency (s^{-1}) of the incident light, for an unknown metal. ($1\text{eV} = 1.6022 \times 10^{-19} \text{ J}$)

Based on the plot, answer the questions **a to c** below:

a. 1 point On the plot above, label the **threshold frequency**.

b. 2 points What is the value of the work potential (in J) of the unknown metal?

c. 3 points What is the *longest* wavelength (in nm) required to eject an electron from this metal's surface? Show your work including equation(s) used. (3 sig figs)

d. 2 points In the space provided below, plot the following

(Assume frequency is *above* threshold frequency):

- Kinetic energy* of an ejected electron versus the *intensity* of the incident light (left)
- Number of electrons* ejected versus the *frequency* of the incident light (right)



**Question 3** **6 points****a. 2 points**

In the ground state of Ge (Germanium; $Z = 32$) atom, **name** the following orbitals:

i. Orbital containing the electron that requires the maximum energy to be removed.

ii. Orbital containing the electron that experiences the minimum effective nuclear charge.

b. 2 points

An unknown anion (A^-) has the following *excited* electronic configuration: $[Ar] 4s^2 3d^3 4p^1$

(Ar: Argon; $Z = 18$)

Give the **condensed ground state electronic configuration** of the atom, **A** and the cation, **A^{2+}** .
Show your work. (Assume all electronic transitions are possible)

Electronic Configuration of A:

Electronic Configuration of A^{2+} :

c. 2 points

In a single atom, determine how many electrons can have the following set of quantum numbers?

(i) $n = 5; l = 3$

In a single atom, determine how many electrons can have the following set of quantum numbers?

(i) $n = 3; l = 2; m_s = +1/2$

**Question 4** **8 points**

Rank the following from 1 to 4 in **increasing** order. Provide the ranking in the boxes provided.

Note: Atomic number (Z) are provided for relevant atoms

a) Increasing (lowest to highest) order of *ionization energy*: (1= lowest; 4 = highest)

Mg

Mg: Z = 12

Rb

Rb: Z = 37

Cs

Cs: Z = 55

K

K : Z = 19

b) Increasing (lowest to highest) order of *bond angle*: (1= lowest; 4 = highest)

H₂OCCl₄XeF₄

HCN

c) Increasing (smallest to largest) order of *ionic size* (1= smallest; 4 = largest)

S²⁻

S: Z = 16

O²⁻

O: Z = 8

Be²⁺

Be: Z = 4

Na⁺

Na: Z = 11

d) Increasing (lowest to highest) order of *bond energy* (1= lowest; 4 = highest)

P-F

P-Cl

P-I

P-Br

**Question 5** **12 points**

a. 9 points For the molecules HClO_4 and TeF_6 , draw the most stable *Lewis structure* showing appropriate shape and *dashed/wedged bonds*, based on VSEPR. Determine the **molecular geometry** around the central atom. For HClO_4 , label the *most polar bond*.



b. 3 points (Note: Lone pairs are not shown in structures below)

(i) For the three *resonance* structures (1, 2 and 3) shown below, answer the following question:



(I) Circle the *most contributing* resonance structure.

Structure 1

Structure 2

Structure 3

(ii) For the three structures given (4, 5, and 6) below, answer the following questions:



(I) Circle all the structures that represent *resonance structures* of each other:

Structure 4

Structure 5

Structure 6

(II) Among the resonance structures, circle the *most contributing* resonance structure.

Structure 4

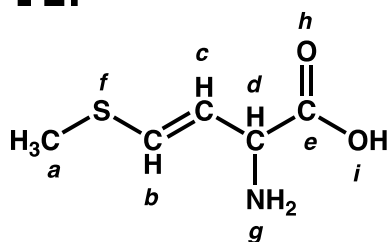
Structure 5

Structure 6

**Question 6** **6 points**

Using the data given below, calculate the **lattice energy** for SrO(s).

	ΔH° (kJ/mol)
$\text{Sr}(s) \longrightarrow \text{Sr}(g)$	+180
$\text{Sr}(g) \longrightarrow \text{Sr}^{2+}(g) + 2e^-$	+1468
$\text{O}_2(g) \longrightarrow 2\text{O}(g)$	+500
$\text{O}(g) + 2e^- \longrightarrow \text{O}^{2-}(g)$	+600
$\text{Sr}(s) + \frac{1}{2}\text{O}_2(g) \longrightarrow \text{SrO}(s)$	-550

**Question 7 10 points**

Consider the molecule given on the left, and answer questions (i) to (vi) using chemical bonding theories that we have learnt in class (valence bond theory, hybridization).

Note that the lone pairs are not shown in the structure:

i. Consider the carbon atoms marked in as ***a***, ***b***, ***c***, ***d***, and ***e*** in the molecule, and determine **hybridization** at each of the carbon centers. (2.5 points)

Atom	Hybridization
Carbon (<i>a</i>)	
Carbon (<i>b</i>)	
Carbon (<i>c</i>)	
Carbon (<i>d</i>)	
Carbon (<i>e</i>)	

ii. Now, consider the sulfur atom (S) marked as ***f***, nitrogen (N) atom (***g***), and oxygen(O) atoms (***h*** and ***i***), and determine their **hybridization**. (2 points)

Atom	Hybridization
Sulfur (<i>f</i>)	
Nitrogen (<i>g</i>)	
Oxygen (<i>h</i>)	
Oxygen (<i>i</i>)	

iii. **Describe** all the bonds formed by the carbon atoms ***a*** and ***b***, using sigma and pi designation as appropriate. (2 points)

Atom	Bonds formed (use sigma and pi as appropriate)
Carbon (<i>a</i>)	
Carbon (<i>b</i>)	

iv. How many total sp^3 -s sigma bonds are in the molecule? (1 point)

v. Determine **electronic** and **molecular geometry** at carbon (C) labeled ***e*** and sulfur (S) labeled ***f*** centres. (2 points)

Atom	Electronic Geometry	Molecular Geometry
Carbon (<i>e</i>)		
Sulfur (<i>f</i>)		

vi. Circle the value from the choices given below, that describes the **C-S-C bond angle at sulfur** (S) marked ***f***: (0.5 points)

>90° to <109.5°

109.5°

>109.5 to <120°

120°

>120° to <180°

**Question 8 12 points**

Using the **valence shell orbitals only** draw a **complete** molecular orbital (MO) diagram (with no s-p mixing) for the molecule **PS**, in the space provided below. **(6 points)**

Assume that 3s and 3p atomic orbitals will form molecular orbitals like those formed by the combination of 2s and 2p atomic orbitals that we learnt in class, and there is no s-p mixing.

Show **energy arrow**, **atomic orbitals** involved, **atom labels**, the **molecular orbitals** that are formed, **filled electrons** in the MO diagram etc. Make sure to label all the atomic and molecular orbitals.

Using the MO diagram above, answer the following questions:

- (i) Which one of the following will have the highest bond order? Circle the correct choice
A) PS^{1-} B) PS C) PS^{1+}
- (ii) Which one of the following will have the longest bond length? Circle the correct choice
A) PS^{1-} B) PS C) PS^{1+}
- (iii) Which of the following will be diamagnetic? Circle the correct choice.
A) PS^{1-} B) PS C) PS^{1+}
- (iv) Draw Lewis dot structure of the molecule with the shortest bond length (based on the MO diagram) from among the three (PS^{1-} , PS, PS^{1+}), and indicate the bond order calculated from the Lewis dot structure. **(3 points)**

Lewis Dot Structure

Bond Order



5BA8AB74-9236-457F-912D-0899DC2E0996

midterm-exam-93388

#356 Page 10 of 10

SCRAP PAGE

**CHEMISTRY 110 –Midterm Exam****Thursday November 7, 2024, 6:30 – 8:30 pm**

Examiners: Pallavi Sirjoosingh and Ashok Kakkar

First name (please write as legibly as possible within the boxes)

--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--

Last name

--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--

Student ID number

--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--

This booklet has 8 questions on 10 pages (The last page can be used as a scrap page).

The staple must NOT be removed from the question booklet. Your intact question booklet must be returned at the end of the exam.

A separate booklet contains a periodic table, and datasheet.

Write your answers directly under the questions in the space provided. Do not write on the QR codes or near the top edges (as it may be cut off).

Please write **legibly and dark enough** for the scan to be readable. For students who use English as a second language, dictionaries are permitted. Calculators are allowed.

DO NOT ASK QUESTIONS OF THE INVIGILATORS about interpretations of the exam questions. If you have a concern about a question, please summarize it at the top of this page. All concerns will be evaluated after the exam. GOOD LUCK!

**Question 1 8 points** (*Assume all transitions are possible*)

a. 6 points Calculate the **energy (in kJ/mol)** required to ionize (completely remove) the electron from the **$n=3$** level of **He^+ ion** (He: Helium). Show your work including equations. (3 sig. figs.)

b. 2 points

(I) Select the largest (longest) wavelength from the three choices below:

- i. Wavelength of photon to *completely remove* an electron from *1s orbital of Hydrogen atom*
- ii. Wavelength of photon to *completely remove* an electron from *1s orbital of Boron atom (B)*
- iii. Wavelength of photon to *completely remove* an electron from *1s orbital of Boron cation (B^+)*

i or ii or iii

(II) Select the smallest (shortest) wavelength from the three choices below (*for H atom*):

- i. Wavelength of photon *emitted* for an electron transitioning from *2p to 1s orbital*
- ii. Wavelength of photon *absorbed* for an electron transitioning from *2s to 4p orbital*
- iii. Wavelength of photon *absorbed* for an electron transitioning from *1s to 3p orbital*

i or ii or iii



Question 2 8 points



A Chem110 student plots the graph shown on the left, from data obtained for the kinetic energy (eV) of the ejected electron versus frequency (s^{-1}) of the incident light, for an unknown metal. ($1\text{eV} = 1.6022 \times 10^{-19} \text{ J}$)

Based on the plot, answer the questions **a to c** below:

a. 1 point On the plot above, label the **threshold frequency**.

b. 2 points What is the value of the work potential (in J) of the unknown metal?

c. 3 points What is the *longest* wavelength (in nm) required to eject an electron from this metal's surface? Show your work including equation(s) used. (3 sig figs)

d. 2 points In the space provided below, plot the following

(Assume frequency is *above* threshold frequency):

- i. *Kinetic energy* of an ejected electron versus the *intensity* of the incident light (left)
- ii. *Number of electrons* ejected versus the *frequency* of the incident light (right)



**Question 3** **6 points****a. 2 points**

In the ground state of Ge (Germanium; $Z = 32$) atom, **name** the following orbitals:

i. Orbital containing the electron that requires the maximum energy to be removed.

ii. Orbital containing the electron that experiences the minimum effective nuclear charge.

b. 2 points

An unknown anion (A^-) has the following *excited* electronic configuration: $[Ar] 4s^2 3d^3 4p^1$

(Ar: Argon; $Z = 18$)

Give the **condensed ground state electronic configuration** of the atom, **A** and the cation, **A^{2+}** .
Show your work. (Assume all electronic transitions are possible)

Electronic Configuration of A:

Electronic Configuration of A^{2+} :

c. 2 points

In a single atom, determine how many electrons can have the following set of quantum numbers?

(i) $n = 5; l = 3$

In a single atom, determine how many electrons can have the following set of quantum numbers?

(i) $n = 3; l = 2; m_s = +1/2$

**Question 4** **8 points**

Rank the following from 1 to 4 in **increasing** order. Provide the ranking in the boxes provided.

Note: Atomic number (Z) are provided for relevant atoms

a) Increasing (lowest to highest) order of *ionization energy*: (1= lowest; 4 = highest)

Mg

Mg: Z = 12

Rb

Rb: Z = 37

Cs

Cs: Z = 55

K

K : Z = 19

b) Increasing (lowest to highest) order of *bond angle*: (1= lowest; 4 = highest)

H₂OCCl₄XeF₄

HCN

c) Increasing (smallest to largest) order of *ionic size* (1= smallest; 4 = largest)

S²⁻

S: Z = 16

O²⁻

O: Z = 8

Be²⁺

Be: Z = 4

Na⁺

Na: Z = 11

d) Increasing (lowest to highest) order of *bond energy* (1= lowest; 4 = highest)

P-F

P-Cl

P-I

P-Br

**Question 5** **12 points**

a. 9 points For the molecules HClO_4 and TeF_6 , draw the most stable *Lewis structure* showing appropriate shape and *dashed/wedged bonds*, based on VSEPR. Determine the **molecular geometry** around the central atom. For HClO_4 , label the *most polar bond*.



b. 3 points (Note: Lone pairs are not shown in structures below)

(i) For the three *resonance* structures (1, 2 and 3) shown below, answer the following question:



(I) Circle the *most contributing* resonance structure.

Structure 1

Structure 2

Structure 3

(ii) For the three structures given (4, 5, and 6) below, answer the following questions:



(I) Circle all the structures that represent *resonance structures* of each other:

Structure 4

Structure 5

Structure 6

(II) Among the resonance structures, circle the *most contributing* resonance structure.

Structure 4

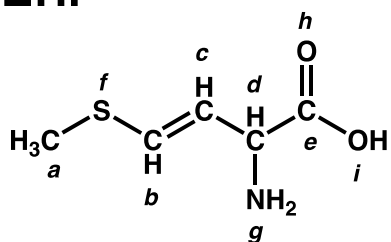
Structure 5

Structure 6

**Question 6** **6 points**

Using the data given below, calculate the **lattice energy** for SrO(s).

	ΔH° (kJ/mol)
$\text{Sr}(s) \longrightarrow \text{Sr}(g)$	+180
$\text{Sr}(g) \longrightarrow \text{Sr}^{2+}(g) + 2e^-$	+1468
$\text{O}_2(g) \longrightarrow 2\text{O}(g)$	+500
$\text{O}(g) + 2e^- \longrightarrow \text{O}^{2-}(g)$	+600
$\text{Sr}(s) + \frac{1}{2}\text{O}_2(g) \longrightarrow \text{SrO}(s)$	-550

**Question 7 10 points**

Consider the molecule given on the left, and answer questions (i) to (vi) using chemical bonding theories that we have learnt in class (valence bond theory, hybridization).

Note that the lone pairs are not shown in the structure:

i. Consider the carbon atoms marked in as ***a***, ***b***, ***c***, ***d***, and ***e*** in the molecule, and determine **hybridization** at each of the carbon centers. (2.5 points)

Atom	Hybridization
Carbon (<i>a</i>)	
Carbon (<i>b</i>)	
Carbon (<i>c</i>)	
Carbon (<i>d</i>)	
Carbon (<i>e</i>)	

ii. Now, consider the sulfur atom (S) marked as ***f***, nitrogen (N) atom (***g***), and oxygen(O) atoms (***h*** and ***i***), and determine their **hybridization**. (2 points)

Atom	Hybridization
Sulfur (<i>f</i>)	
Nitrogen (<i>g</i>)	
Oxygen (<i>h</i>)	
Oxygen (<i>i</i>)	

iii. **Describe** all the bonds formed by the carbon atoms ***a*** and ***b***, using sigma and pi designation as appropriate. (2 points)

Atom	Bonds formed (use sigma and pi as appropriate)
Carbon (<i>a</i>)	
Carbon (<i>b</i>)	

iv. How many total sp^3 -s sigma bonds are in the molecule? (1 point)

v. Determine **electronic** and **molecular geometry** at carbon (C) labeled ***e*** and sulfur (S) labeled ***f*** centres. (2 points)

Atom	Electronic Geometry	Molecular Geometry
Carbon (<i>e</i>)		
Sulfur (<i>f</i>)		

vi. Circle the value from the choices given below, that describes the **C-S-C bond angle at sulfur** (S) marked ***f***: (0.5 points)

>90° to <109.5° 109.5° >109.5 to <120° 120° >120° to <180°


Question 8 12 points

Using the **valence shell orbitals only** draw a **complete** molecular orbital (MO) diagram (with no s-p mixing) for the molecule **PS**, in the space provided below. **(6 points)**

Assume that 3s and 3p atomic orbitals will form molecular orbitals like those formed by the combination of 2s and 2p atomic orbitals that we learnt in class, and there is no s-p mixing.

Show **energy arrow**, **atomic orbitals** involved, **atom labels**, the **molecular orbitals** that are formed, **filled electrons** in the MO diagram etc. Make sure to label all the atomic and molecular orbitals.

Using the MO diagram above, answer the following questions:

- (i) Which one of the following will have the highest bond order? Circle the correct choice
 A) PS^{1-} B) PS C) PS^{1+}
- (ii) Which one of the following will have the longest bond length? Circle the correct choice
 A) PS^{1-} B) PS C) PS^{1+}
- (iii) Which of the following will be diamagnetic? Circle the correct choice.
 A) PS^{1-} B) PS C) PS^{1+}
- (iv) Draw Lewis dot structure of the molecule with the shortest bond length (based on the MO diagram) from among the three (PS^{1-} , PS, PS^{1+}), and indicate the bond order calculated from the Lewis dot structure. **(3 points)**

Lewis Dot Structure

Bond Order



7F4BC445-16F0-4EDF-860B-E63CC7639E33

midterm-exam-93388

#357 Page 10 of 10

SCRAP PAGE

**CHEMISTRY 110 –Midterm Exam****Thursday November 7, 2024, 6:30 – 8:30 pm**

Examiners: Pallavi Sirjoosingh and Ashok Kakkar

First name (please write as legibly as possible within the boxes)

--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--

Last name

--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--

Student ID number

--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--

This booklet has 8 questions on 10 pages (The last page can be used as a scrap page).

The staple must NOT be removed from the question booklet. Your intact question booklet must be returned at the end of the exam.

A separate booklet contains a periodic table, and datasheet.

Write your answers directly under the questions in the space provided. Do not write on the QR codes or near the top edges (as it may be cut off).

Please write **legibly and dark enough** for the scan to be readable. For students who use English as a second language, dictionaries are permitted. Calculators are allowed.

DO NOT ASK QUESTIONS OF THE INVIGILATORS about interpretations of the exam questions. If you have a concern about a question, please summarize it at the top of this page. All concerns will be evaluated after the exam. GOOD LUCK!

**Question 1 8 points** (*Assume all transitions are possible*)

a. 6 points Calculate the **energy (in kJ/mol)** required to ionize (completely remove) the electron from the **n=3** level of **He⁺ ion** (He: Helium). Show your work including equations. (3 sig. figs.)

b. 2 points

(I) Select the largest (longest) wavelength from the three choices below:

- i. Wavelength of photon to *completely remove* an electron from *1s orbital of Hydrogen atom*
- ii. Wavelength of photon to *completely remove* an electron from *1s orbital of Boron atom (B)*
- iii. Wavelength of photon to *completely remove* an electron from *1s orbital of Boron cation (B⁺)*

i or ii or iii

(II) Select the smallest (shortest) wavelength from the three choices below (*for H atom*):

- i. Wavelength of photon *emitted* for an electron transitioning from *2p to 1s orbital*
- ii. Wavelength of photon *absorbed* for an electron transitioning from *2s to 4p orbital*
- iii. Wavelength of photon *absorbed* for an electron transitioning from *1s to 3p orbital*

i or ii or iii



Question 2 8 points



A Chem110 student plots the graph shown on the left, from data obtained for the kinetic energy (eV) of the ejected electron versus frequency (s^{-1}) of the incident light, for an unknown metal. ($1\text{eV} = 1.6022 \times 10^{-19} \text{ J}$)

Based on the plot, answer the questions **a to c** below:

a. 1 point On the plot above, label the **threshold frequency**.

b. 2 points What is the value of the work potential (in J) of the unknown metal?

c. 3 points What is the *longest* wavelength (in nm) required to eject an electron from this metal's surface? Show your work including equation(s) used. (3 sig figs)

d. 2 points In the space provided below, plot the following

(Assume frequency is *above* threshold frequency):

- i. *Kinetic energy* of an ejected electron versus the *intensity* of the incident light (left)
- ii. *Number of electrons* ejected versus the *frequency* of the incident light (right)



**Question 3** **6 points****a. 2 points**

In the ground state of Ge (Germanium; $Z = 32$) atom, **name** the following orbitals:

i. Orbital containing the electron that requires the maximum energy to be removed.

ii. Orbital containing the electron that experiences the minimum effective nuclear charge.

b. 2 points

An unknown anion (A^-) has the following *excited* electronic configuration: $[Ar] 4s^2 3d^3 4p^1$

(Ar: Argon; $Z = 18$)

Give the **condensed ground state electronic configuration** of the atom, **A** and the cation, **A^{2+}** .
Show your work. (Assume all electronic transitions are possible)

Electronic Configuration of A:

Electronic Configuration of A^{2+} :

c. 2 points

In a single atom, determine how many electrons can have the following set of quantum numbers?

(i) $n = 5; l = 3$

In a single atom, determine how many electrons can have the following set of quantum numbers?

(i) $n = 3; l = 2; m_s = +1/2$

**Question 4** **8 points**

Rank the following from 1 to 4 in **increasing** order. Provide the ranking in the boxes provided.

Note: Atomic number (Z) are provided for relevant atoms

a) Increasing (lowest to highest) order of *ionization energy*: (1= lowest; 4 = highest)

Mg

Mg: Z = 12

Rb

Rb: Z = 37

Cs

Cs: Z = 55

K

K : Z = 19

b) Increasing (lowest to highest) order of *bond angle*: (1= lowest; 4 = highest)

H₂OCCl₄XeF₄

HCN

c) Increasing (smallest to largest) order of *ionic size* (1= smallest; 4 = largest)

S²⁻

S: Z = 16

O²⁻

O: Z = 8

Be²⁺

Be: Z = 4

Na⁺

Na: Z = 11

d) Increasing (lowest to highest) order of *bond energy* (1= lowest; 4 = highest)

P-F

P-Cl

P-I

P-Br

**Question 5** **12 points**

a. 9 points For the molecules HClO_4 and TeF_6 , draw the most stable *Lewis structure* showing appropriate shape and *dashed/wedged bonds*, based on VSEPR. Determine the **molecular geometry** around the central atom. For HClO_4 , label the *most polar bond*.



b. 3 points (Note: Lone pairs are not shown in structures below)

(i) For the three *resonance* structures (1, 2 and 3) shown below, answer the following question:



(I) Circle the *most contributing* resonance structure.

Structure 1

Structure 2

Structure 3

(ii) For the three structures given (4, 5, and 6) below, answer the following questions:



(I) Circle all the structures that represent *resonance structures* of each other:

Structure 4

Structure 5

Structure 6

(II) Among the resonance structures, circle the *most contributing* resonance structure.

Structure 4

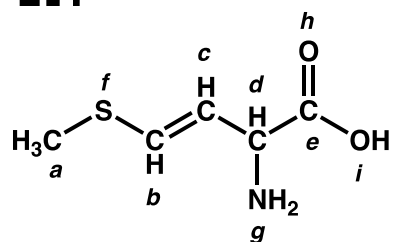
Structure 5

Structure 6

**Question 6** **6 points**

Using the data given below, calculate the **lattice energy** for SrO(s).

	ΔH° (kJ/mol)
$\text{Sr}(s) \longrightarrow \text{Sr}(g)$	+180
$\text{Sr}(g) \longrightarrow \text{Sr}^{2+}(g) + 2e^-$	+1468
$\text{O}_2(g) \longrightarrow 2\text{O}(g)$	+500
$\text{O}(g) + 2e^- \longrightarrow \text{O}^{2-}(g)$	+600
$\text{Sr}(s) + \frac{1}{2}\text{O}_2(g) \longrightarrow \text{SrO}(s)$	-550

**Question 7 10 points**

Consider the molecule given on the left, and answer questions (i) to (vi) using chemical bonding theories that we have learnt in class (valence bond theory, hybridization).

Note that the lone pairs are not shown in the structure:

i. Consider the carbon atoms marked in as ***a***, ***b***, ***c***, ***d***, and ***e*** in the molecule, and determine **hybridization** at each of the carbon centers. (2.5 points)

Atom	Hybridization
Carbon (<i>a</i>)	
Carbon (<i>b</i>)	
Carbon (<i>c</i>)	
Carbon (<i>d</i>)	
Carbon (<i>e</i>)	

ii. Now, consider the sulfur atom (S) marked as ***f***, nitrogen (N) atom (***g***), and oxygen(O) atoms (***h*** and ***i***), and determine their **hybridization**. (2 points)

Atom	Hybridization
Sulfur (<i>f</i>)	
Nitrogen (<i>g</i>)	
Oxygen (<i>h</i>)	
Oxygen (<i>i</i>)	

iii. **Describe** all the bonds formed by the carbon atoms ***a*** and ***b***, using sigma and pi designation as appropriate. (2 points)

Atom	Bonds formed (use sigma and pi as appropriate)
Carbon (<i>a</i>)	
Carbon (<i>b</i>)	

iv. How many total sp^3 -s sigma bonds are in the molecule? (1 point)

v. Determine **electronic** and **molecular geometry** at carbon (C) labeled ***e*** and sulfur (S) labeled ***f*** centres. (2 points)

Atom	Electronic Geometry	Molecular Geometry
Carbon (<i>e</i>)		
Sulfur (<i>f</i>)		

vi. Circle the value from the choices given below, that describes the **C-S-C bond angle at sulfur** (S) marked ***f***: (0.5 points)

>90° to <109.5° 109.5° >109.5 to <120° 120° >120° to <180°



Question 8 12 points

Using the **valence shell orbitals only** draw a **complete** molecular orbital (MO) diagram (with no s-p mixing) for the molecule **PS**, in the space provided below. **(6 points)**

Assume that 3s and 3p atomic orbitals will form molecular orbitals like those formed by the combination of 2s and 2p atomic orbitals that we learnt in class, and there is no s-p mixing.

Show **energy arrow**, **atomic orbitals** involved, **atom labels**, the **molecular orbitals** that are formed, **filled electrons** in the MO diagram etc. Make sure to label all the atomic and molecular orbitals.

Using the MO diagram above, answer the following questions:

- (i) Which one of the following will have the highest bond order? Circle the correct choice
 A) PS^{1-} B) PS C) PS^{1+}
- (ii) Which one of the following will have the longest bond length? Circle the correct choice
 A) PS^{1-} B) PS C) PS^{1+}
- (iii) Which of the following will be diamagnetic? Circle the correct choice.
 A) PS^{1-} B) PS C) PS^{1+}
- (iv) Draw Lewis dot structure of the molecule with the shortest bond length (based on the MO diagram) from among the three (PS^{1-} , PS, PS^{1+}), and indicate the bond order calculated from the Lewis dot structure. **(3 points)**

Lewis Dot Structure

Bond Order



2D89A950-70AE-4091-872F-32B6D22742E5

midterm-exam-93388

#358 Page 10 of 10

SCRAP PAGE

**CHEMISTRY 110 –Midterm Exam****Thursday November 7, 2024, 6:30 – 8:30 pm**

Examiners: Pallavi Sirjoosingh and Ashok Kakkar

First name (please write as legibly as possible within the boxes)

--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--

Last name

--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--

Student ID number

--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--

This booklet has 8 questions on 10 pages (The last page can be used as a scrap page).

The staple must NOT be removed from the question booklet. Your intact question booklet must be returned at the end of the exam.

A separate booklet contains a periodic table, and datasheet.

Write your answers directly under the questions in the space provided. Do not write on the QR codes or near the top edges (as it may be cut off).

Please write **legibly and dark enough** for the scan to be readable. For students who use English as a second language, dictionaries are permitted. Calculators are allowed.

DO NOT ASK QUESTIONS OF THE INVIGILATORS about interpretations of the exam questions. If you have a concern about a question, please summarize it at the top of this page. All concerns will be evaluated after the exam. GOOD LUCK!

**Question 1 8 points** (*Assume all transitions are possible*)

a. 6 points Calculate the **energy (in kJ/mol)** required to ionize (completely remove) the electron from the **n=3** level of **He⁺ ion** (He: Helium). Show your work including equations. (3 sig. figs.)

b. 2 points

(I) Select the largest (longest) wavelength from the three choices below:

- i. Wavelength of photon to *completely remove* an electron from *1s orbital of Hydrogen atom*
- ii. Wavelength of photon to *completely remove* an electron from *1s orbital of Boron atom (B)*
- iii. Wavelength of photon to *completely remove* an electron from *1s orbital of Boron cation (B⁺)*

i or ii or iii

(II) Select the smallest (shortest) wavelength from the three choices below (*for H atom*):

- i. Wavelength of photon *emitted* for an electron transitioning from *2p to 1s orbital*
- ii. Wavelength of photon *absorbed* for an electron transitioning from *2s to 4p orbital*
- iii. Wavelength of photon *absorbed* for an electron transitioning from *1s to 3p orbital*

i or ii or iii



Question 2 8 points



A Chem110 student plots the graph shown on the left, from data obtained for the kinetic energy (eV) of the ejected electron versus frequency (s^{-1}) of the incident light, for an unknown metal. ($1\text{eV} = 1.6022 \times 10^{-19} \text{ J}$)

Based on the plot, answer the questions **a to c** below:

a. 1 point On the plot above, label the **threshold frequency**.

b. 2 points What is the value of the work potential (in J) of the unknown metal?

c. 3 points What is the *longest* wavelength (in nm) required to eject an electron from this metal's surface? Show your work including equation(s) used. (3 sig figs)

d. 2 points In the space provided below, plot the following

(Assume frequency is *above* threshold frequency):

- i. *Kinetic energy* of an ejected electron versus the *intensity* of the incident light (left)
- ii. *Number of electrons* ejected versus the *frequency* of the incident light (right)



**Question 3** **6 points****a. 2 points**

In the ground state of Ge (Germanium; $Z = 32$) atom, **name** the following orbitals:

i. Orbital containing the electron that requires the maximum energy to be removed.

ii. Orbital containing the electron that experiences the minimum effective nuclear charge.

b. 2 points

An unknown anion (A^-) has the following *excited* electronic configuration: $[Ar] 4s^2 3d^3 4p^1$

(Ar: Argon; $Z = 18$)

Give the **condensed ground state electronic configuration** of the atom, **A** and the cation, **A^{2+}** .
Show your work. (Assume all electronic transitions are possible)

Electronic Configuration of A:

Electronic Configuration of A^{2+} :

c. 2 points

In a single atom, determine how many electrons can have the following set of quantum numbers?

(i) $n = 5; l = 3$

In a single atom, determine how many electrons can have the following set of quantum numbers?

(i) $n = 3; l = 2; m_s = +1/2$

**Question 4** **8 points**

Rank the following from 1 to 4 in **increasing** order. Provide the ranking in the boxes provided.

Note: Atomic number (Z) are provided for relevant atoms

a) Increasing (lowest to highest) order of *ionization energy*: (1= lowest; 4 = highest)

Mg

Mg: Z = 12

Rb

Rb: Z = 37

Cs

Cs: Z = 55

K

K : Z = 19

b) Increasing (lowest to highest) order of *bond angle*: (1= lowest; 4 = highest)

H₂OCCl₄XeF₄

HCN

c) Increasing (smallest to largest) order of *ionic size* (1= smallest; 4 = largest)

S²⁻

S: Z = 16

O²⁻

O: Z = 8

Be²⁺

Be: Z = 4

Na⁺

Na: Z = 11

d) Increasing (lowest to highest) order of *bond energy* (1= lowest; 4 = highest)

P-F

P-Cl

P-I

P-Br

**Question 5** **12 points**

a. 9 points For the molecules HClO_4 and TeF_6 , draw the most stable *Lewis structure* showing appropriate shape and *dashed/wedged bonds*, based on VSEPR. Determine the **molecular geometry** around the central atom. For HClO_4 , label the *most polar bond*.



b. 3 points (Note: Lone pairs are not shown in structures below)

(i) For the three *resonance* structures (1, 2 and 3) shown below, answer the following question:



(I) Circle the *most contributing* resonance structure.

Structure 1

Structure 2

Structure 3

(ii) For the three structures given (4, 5, and 6) below, answer the following questions:



(I) Circle all the structures that represent *resonance structures* of each other:

Structure 4

Structure 5

Structure 6

(II) Among the resonance structures, circle the *most contributing* resonance structure.

Structure 4

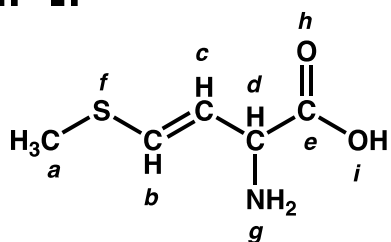
Structure 5

Structure 6

**Question 6** **6 points**

Using the data given below, calculate the **lattice energy** for SrO(s).

	ΔH° (kJ/mol)
$\text{Sr}(s) \longrightarrow \text{Sr}(g)$	+180
$\text{Sr}(g) \longrightarrow \text{Sr}^{2+}(g) + 2e^-$	+1468
$\text{O}_2(g) \longrightarrow 2\text{O}(g)$	+500
$\text{O}(g) + 2e^- \longrightarrow \text{O}^{2-}(g)$	+600
$\text{Sr}(s) + \frac{1}{2}\text{O}_2(g) \longrightarrow \text{SrO}(s)$	-550

**Question 7 10 points**

Consider the molecule given on the left, and answer questions (i) to (vi) using chemical bonding theories that we have learnt in class (valence bond theory, hybridization).

Note that the lone pairs are not shown in the structure:

i. Consider the carbon atoms marked in as ***a***, ***b***, ***c***, ***d***, and ***e*** in the molecule, and determine **hybridization** at each of the carbon centers. (2.5 points)

Atom	Hybridization
Carbon (<i>a</i>)	
Carbon (<i>b</i>)	
Carbon (<i>c</i>)	
Carbon (<i>d</i>)	
Carbon (<i>e</i>)	

ii. Now, consider the sulfur atom (S) marked as ***f***, nitrogen (N) atom (***g***), and oxygen(O) atoms (***h*** and ***i***), and determine their **hybridization**. (2 points)

Atom	Hybridization
Sulfur (<i>f</i>)	
Nitrogen (<i>g</i>)	
Oxygen (<i>h</i>)	
Oxygen (<i>i</i>)	

iii. **Describe** all the bonds formed by the carbon atoms ***a*** and ***b***, using sigma and pi designation as appropriate. (2 points)

Atom	Bonds formed (use sigma and pi as appropriate)
Carbon (<i>a</i>)	
Carbon (<i>b</i>)	

iv. How many total sp^3 -s sigma bonds are in the molecule? (1 point)

v. Determine **electronic** and **molecular geometry** at carbon (C) labeled ***e*** and sulfur (S) labeled ***f*** centres. (2 points)

Atom	Electronic Geometry	Molecular Geometry
Carbon (<i>e</i>)		
Sulfur (<i>f</i>)		

vi. Circle the value from the choices given below, that describes the **C-S-C bond angle at sulfur** (S) marked ***f***: (0.5 points)

>90° to <109.5°

109.5°

>109.5 to <120°

120°

>120° to <180°


Question 8 12 points

Using the **valence shell orbitals only** draw a **complete** molecular orbital (MO) diagram (with no s-p mixing) for the molecule **PS**, in the space provided below. **(6 points)**

Assume that 3s and 3p atomic orbitals will form molecular orbitals like those formed by the combination of 2s and 2p atomic orbitals that we learnt in class, and there is no s-p mixing.

Show **energy arrow**, **atomic orbitals** involved, **atom labels**, the **molecular orbitals** that are formed, **filled electrons** in the MO diagram etc. Make sure to label all the atomic and molecular orbitals.

Using the MO diagram above, answer the following questions:

- (i) Which one of the following will have the highest bond order? Circle the correct choice
 A) PS^{1-} B) PS C) PS^{1+}
- (ii) Which one of the following will have the longest bond length? Circle the correct choice
 A) PS^{1-} B) PS C) PS^{1+}
- (iii) Which of the following will be diamagnetic? Circle the correct choice.
 A) PS^{1-} B) PS C) PS^{1+}
- (iv) Draw Lewis dot structure of the molecule with the shortest bond length (based on the MO diagram) from among the three (PS^{1-} , PS, PS^{1+}), and indicate the bond order calculated from the Lewis dot structure. **(3 points)**

Lewis Dot Structure

Bond Order



0FCE5AED-DF86-49A0-82E8-B33FAC302B42

midterm-exam-93388

#359 Page 10 of 10

SCRAP PAGE

**CHEMISTRY 110 –Midterm Exam****Thursday November 7, 2024, 6:30 – 8:30 pm**

Examiners: Pallavi Sirjoosingh and Ashok Kakkar

First name (please write as legibly as possible within the boxes)

--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--

Last name

--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--

Student ID number

--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--

This booklet has 8 questions on 10 pages (The last page can be used as a scrap page).

The staple must NOT be removed from the question booklet. Your intact question booklet must be returned at the end of the exam.

A separate booklet contains a periodic table, and datasheet.

Write your answers directly under the questions in the space provided. Do not write on the QR codes or near the top edges (as it may be cut off).

Please write **legibly and dark enough** for the scan to be readable. For students who use English as a second language, dictionaries are permitted. Calculators are allowed.

DO NOT ASK QUESTIONS OF THE INVIGILATORS about interpretations of the exam questions. If you have a concern about a question, please summarize it at the top of this page. All concerns will be evaluated after the exam. GOOD LUCK!

**Question 1 8 points** (*Assume all transitions are possible*)

a. 6 points Calculate the **energy (in kJ/mol)** required to ionize (completely remove) the electron from the **$n=3$** level of **He^+ ion** (He: Helium). Show your work including equations. (3 sig. figs.)

b. 2 points

(I) Select the largest (longest) wavelength from the three choices below:

- i. Wavelength of photon to *completely remove* an electron from *1s orbital of Hydrogen atom*
- ii. Wavelength of photon to *completely remove* an electron from *1s orbital of Boron atom (B)*
- iii. Wavelength of photon to *completely remove* an electron from *1s orbital of Boron cation (B^+)*

i or ii or iii

(II) Select the smallest (shortest) wavelength from the three choices below (*for H atom*):

- i. Wavelength of photon *emitted* for an electron transitioning from *2p to 1s orbital*
- ii. Wavelength of photon *absorbed* for an electron transitioning from *2s to 4p orbital*
- iii. Wavelength of photon *absorbed* for an electron transitioning from *1s to 3p orbital*

i or ii or iii



Question 2 8 points



A Chem110 student plots the graph shown on the left, from data obtained for the kinetic energy (eV) of the ejected electron versus frequency (s^{-1}) of the incident light, for an unknown metal. ($1\text{eV} = 1.6022 \times 10^{-19} \text{ J}$)

Based on the plot, answer the questions **a to c** below:

a. 1 point On the plot above, label the **threshold frequency**.

b. 2 points What is the value of the work potential (in J) of the unknown metal?

c. 3 points What is the *longest* wavelength (in nm) required to eject an electron from this metal's surface? Show your work including equation(s) used. (3 sig figs)

d. 2 points In the space provided below, plot the following

(Assume frequency is *above* threshold frequency):

- Kinetic energy* of an ejected electron versus the *intensity* of the incident light (left)
- Number of electrons* ejected versus the *frequency* of the incident light (right)



**Question 3** **6 points****a. 2 points**

In the ground state of Ge (Germanium; $Z = 32$) atom, **name** the following orbitals:

i. Orbital containing the electron that requires the maximum energy to be removed.

ii. Orbital containing the electron that experiences the minimum effective nuclear charge.

b. 2 points

An unknown anion (A^-) has the following *excited* electronic configuration: $[Ar] 4s^2 3d^3 4p^1$

(Ar: Argon; $Z = 18$)

Give the **condensed ground state electronic configuration** of the atom, **A** and the cation, **A^{2+}** .
Show your work. (Assume all electronic transitions are possible)

Electronic Configuration of A:

Electronic Configuration of A^{2+} :

c. 2 points

In a single atom, determine how many electrons can have the following set of quantum numbers?

(i) $n = 5; l = 3$

In a single atom, determine how many electrons can have the following set of quantum numbers?

(i) $n = 3; l = 2; m_s = +1/2$

**Question 4** **8 points**

Rank the following from 1 to 4 in **increasing** order. Provide the ranking in the boxes provided.

Note: Atomic number (Z) are provided for relevant atoms

a) Increasing (lowest to highest) order of *ionization energy*: (1= lowest; 4 = highest)

Mg

Mg: Z = 12

Rb

Rb: Z = 37

Cs

Cs: Z = 55

K

K : Z = 19

b) Increasing (lowest to highest) order of *bond angle*: (1= lowest; 4 = highest)

H₂OCCl₄XeF₄

HCN

c) Increasing (smallest to largest) order of *ionic size* (1= smallest; 4 = largest)

S²⁻

S: Z = 16

O²⁻

O: Z = 8

Be²⁺

Be: Z = 4

Na⁺

Na: Z = 11

d) Increasing (lowest to highest) order of *bond energy* (1= lowest; 4 = highest)

P-F

P-Cl

P-I

P-Br

**Question 5** **12 points**

a. 9 points For the molecules HClO_4 and TeF_6 , draw the most stable *Lewis structure* showing appropriate shape and *dashed/wedged bonds*, based on VSEPR. Determine the **molecular geometry** around the central atom. For HClO_4 , label the *most polar bond*.

HClO_4 (there are no O-O bonds)

Label the most polar bond

Molecular Geometry around Cl

TeF_6

Molecular Geometry around Te

b. 3 points (Note: Lone pairs are not shown in structures below)

(i) For the three *resonance* structures (1, 2 and 3) shown below, answer the following question:



(I) Circle the *most contributing* resonance structure.

Structure 1

Structure 2

Structure 3

(ii) For the three structures given (4, 5, and 6) below, answer the following questions:



(I) Circle all the structures that represent *resonance structures* of each other:

Structure 4

Structure 5

Structure 6

(II) Among the resonance structures, circle the *most contributing* resonance structure.

Structure 4

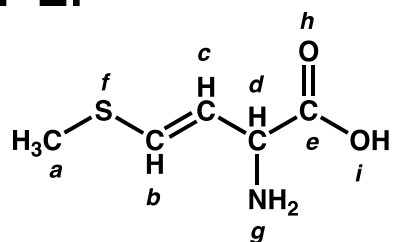
Structure 5

Structure 6

**Question 6** **6 points**

Using the data given below, calculate the **lattice energy** for SrO(s).

	ΔH° (kJ/mol)
$\text{Sr}(s) \longrightarrow \text{Sr}(g)$	+180
$\text{Sr}(g) \longrightarrow \text{Sr}^{2+}(g) + 2e^-$	+1468
$\text{O}_2(g) \longrightarrow 2\text{O}(g)$	+500
$\text{O}(g) + 2e^- \longrightarrow \text{O}^{2-}(g)$	+600
$\text{Sr}(s) + \frac{1}{2}\text{O}_2(g) \longrightarrow \text{SrO}(s)$	-550

**Question 7 10 points**

Consider the molecule given on the left, and answer questions (i) to (vi) using chemical bonding theories that we have learnt in class (valence bond theory, hybridization).

Note that the lone pairs are not shown in the structure:

i. Consider the carbon atoms marked in as ***a***, ***b***, ***c***, ***d***, and ***e*** in the molecule, and determine **hybridization** at each of the carbon centers. (2.5 points)

Atom	Hybridization
Carbon (<i>a</i>)	
Carbon (<i>b</i>)	
Carbon (<i>c</i>)	
Carbon (<i>d</i>)	
Carbon (<i>e</i>)	

ii. Now, consider the sulfur atom (S) marked as ***f***, nitrogen (N) atom (***g***), and oxygen(O) atoms (***h*** and ***i***), and determine their **hybridization**. (2 points)

Atom	Hybridization
Sulfur (<i>f</i>)	
Nitrogen (<i>g</i>)	
Oxygen (<i>h</i>)	
Oxygen (<i>i</i>)	

iii. **Describe** all the bonds formed by the carbon atoms ***a*** and ***b***, using sigma and pi designation as appropriate. (2 points)

Atom	Bonds formed (use sigma and pi as appropriate)
Carbon (<i>a</i>)	
Carbon (<i>b</i>)	

iv. How many total sp^3 -s sigma bonds are in the molecule? (1 point)

v. Determine **electronic** and **molecular geometry** at carbon (C) labeled ***e*** and sulfur (S) labeled ***f*** centres. (2 points)

Atom	Electronic Geometry	Molecular Geometry
Carbon (<i>e</i>)		
Sulfur (<i>f</i>)		

vi. Circle the value from the choices given below, that describes the **C-S-C bond angle at sulfur** (S) marked ***f***: (0.5 points)

$>90^\circ$ to $<109.5^\circ$ 109.5° >109.5 to $<120^\circ$ 120° $>120^\circ$ to $<180^\circ$



Question 8 12 points

Using the **valence shell orbitals only** draw a **complete** molecular orbital (MO) diagram (with no s-p mixing) for the molecule **PS**, in the space provided below. **(6 points)**

Assume that 3s and 3p atomic orbitals will form molecular orbitals like those formed by the combination of 2s and 2p atomic orbitals that we learnt in class, and there is no s-p mixing.

Show **energy arrow**, **atomic orbitals** involved, **atom labels**, the **molecular orbitals** that are formed, **filled electrons** in the MO diagram etc. Make sure to label all the atomic and molecular orbitals.

Using the MO diagram above, answer the following questions:

- (i) Which one of the following will have the highest bond order? Circle the correct choice
 A) PS^{1-} B) PS C) PS^{1+}
- (ii) Which one of the following will have the longest bond length? Circle the correct choice
 A) PS^{1-} B) PS C) PS^{1+}
- (iii) Which of the following will be diamagnetic? Circle the correct choice.
 A) PS^{1-} B) PS C) PS^{1+}
- (iv) Draw Lewis dot structure of the molecule with the shortest bond length (based on the MO diagram) from among the three (PS^{1-} , PS, PS^{1+}), and indicate the bond order calculated from the Lewis dot structure. **(3 points)**

Lewis Dot Structure

Bond Order



A383E072-E191-45F9-BE4B-F20559C91555

midterm-exam-93388

#360 Page 10 of 10

SCRAP PAGE

**CHEMISTRY 110 –Midterm Exam****Thursday November 7, 2024, 6:30 – 8:30 pm****Examiners: Pallavi Sirjoosingh and Ashok Kakkar**

First name (please write as legibly as possible within the boxes)

--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--

Last name

--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--

Student ID number

--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--

This booklet has 8 questions on 10 pages (The last page can be used as a scrap page).

The staple must NOT be removed from the question booklet. Your intact question booklet must be returned at the end of the exam.

A separate booklet contains a periodic table, and datasheet.

Write your answers directly under the questions in the space provided. Do not write on the QR codes or near the top edges (as it may be cut off).

Please write **legibly and dark enough** for the scan to be readable. For students who use English as a second language, dictionaries are permitted. Calculators are allowed.

DO NOT ASK QUESTIONS OF THE INVIGILATORS about interpretations of the exam questions. If you have a concern about a question, please summarize it at the top of this page. All concerns will be evaluated after the exam. GOOD LUCK!

**Question 1 8 points** (*Assume all transitions are possible*)

a. 6 points Calculate the **energy (in kJ/mol)** required to ionize (completely remove) the electron from the **n=3** level of **He⁺ ion** (He: Helium). Show your work including equations. (3 sig. figs.)

b. 2 points

(I) Select the largest (longest) wavelength from the three choices below:

- i. Wavelength of photon to *completely remove* an electron from *1s orbital of Hydrogen atom*
- ii. Wavelength of photon to *completely remove* an electron from *1s orbital of Boron atom (B)*
- iii. Wavelength of photon to *completely remove* an electron from *1s orbital of Boron cation (B⁺)*

i or ii or iii

(II) Select the smallest (shortest) wavelength from the three choices below (*for H atom*):

- i. Wavelength of photon *emitted* for an electron transitioning from *2p to 1s orbital*
- ii. Wavelength of photon *absorbed* for an electron transitioning from *2s to 4p orbital*
- iii. Wavelength of photon *absorbed* for an electron transitioning from *1s to 3p orbital*

i or ii or iii



Question 2 8 points



A Chem110 student plots the graph shown on the left, from data obtained for the kinetic energy (eV) of the ejected electron versus frequency (s^{-1}) of the incident light, for an unknown metal. ($1\text{eV} = 1.6022 \times 10^{-19} \text{ J}$)

Based on the plot, answer the questions **a to c** below:

a. 1 point On the plot above, label the **threshold frequency**.

b. 2 points What is the value of the work potential (in J) of the unknown metal?

c. 3 points What is the *longest* wavelength (in nm) required to eject an electron from this metal's surface? Show your work including equation(s) used. (3 sig figs)

d. 2 points In the space provided below, plot the following

(Assume frequency is *above* threshold frequency):

- Kinetic energy* of an ejected electron versus the *intensity* of the incident light (left)
- Number of electrons* ejected versus the *frequency* of the incident light (right)



**Question 3** **6 points****a. 2 points**

In the ground state of Ge (Germanium; $Z = 32$) atom, **name** the following orbitals:

i. Orbital containing the electron that requires the maximum energy to be removed.

ii. Orbital containing the electron that experiences the minimum effective nuclear charge.

b. 2 points

An unknown anion (A^-) has the following *excited* electronic configuration: $[Ar] 4s^2 3d^3 4p^1$

(Ar: Argon; $Z = 18$)

Give the **condensed ground state electronic configuration** of the atom, **A** and the cation, **A^{2+}** .
Show your work. (Assume all electronic transitions are possible)

Electronic Configuration of A:

Electronic Configuration of A^{2+} :

c. 2 points

In a single atom, determine how many electrons can have the following set of quantum numbers?

(i) $n = 5; l = 3$

In a single atom, determine how many electrons can have the following set of quantum numbers?

(i) $n = 3; l = 2; m_s = +1/2$

**Question 4** **8 points**

Rank the following from 1 to 4 in **increasing** order. Provide the ranking in the boxes provided.

Note: Atomic number (Z) are provided for relevant atoms

a) Increasing (lowest to highest) order of *ionization energy*: (1= lowest; 4 = highest)

Mg

Mg: Z = 12

Rb

Rb: Z = 37

Cs

Cs: Z = 55

K

K : Z = 19

b) Increasing (lowest to highest) order of *bond angle*: (1= lowest; 4 = highest)

H₂OCCl₄XeF₄

HCN

c) Increasing (smallest to largest) order of *ionic size* (1= smallest; 4 = largest)

S²⁻

S: Z = 16

O²⁻

O: Z = 8

Be²⁺

Be: Z = 4

Na⁺

Na: Z = 11

d) Increasing (lowest to highest) order of *bond energy* (1= lowest; 4 = highest)

P-F

P-Cl

P-I

P-Br

**Question 5** **12 points**

a. 9 points For the molecules HClO_4 and TeF_6 , draw the most stable *Lewis structure* showing appropriate shape and *dashed/wedged bonds*, based on VSEPR. Determine the **molecular geometry** around the central atom. For HClO_4 , label the *most polar bond*.



b. 3 points (Note: Lone pairs are not shown in structures below)

(i) For the three *resonance* structures (1, 2 and 3) shown below, answer the following question:



(I) Circle the *most contributing* resonance structure.

Structure 1

Structure 2

Structure 3

(ii) For the three structures given (4, 5, and 6) below, answer the following questions:



(I) Circle all the structures that represent *resonance structures* of each other:

Structure 4

Structure 5

Structure 6

(II) Among the resonance structures, circle the *most contributing* resonance structure.

Structure 4

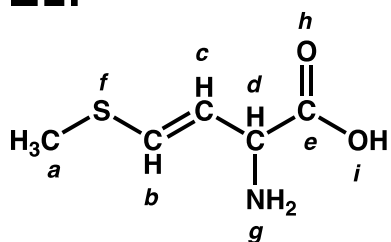
Structure 5

Structure 6

**Question 6** **6 points**

Using the data given below, calculate the **lattice energy** for SrO(s).

	ΔH° (kJ/mol)
$\text{Sr}(s) \longrightarrow \text{Sr}(g)$	+180
$\text{Sr}(g) \longrightarrow \text{Sr}^{2+}(g) + 2e^-$	+1468
$\text{O}_2(g) \longrightarrow 2\text{O}(g)$	+500
$\text{O}(g) + 2e^- \longrightarrow \text{O}^{2-}(g)$	+600
$\text{Sr}(s) + \frac{1}{2}\text{O}_2(g) \longrightarrow \text{SrO}(s)$	-550

**Question 7 10 points**

Consider the molecule given on the left, and answer questions (i) to (vi) using chemical bonding theories that we have learnt in class (valence bond theory, hybridization).

Note that the lone pairs are not shown in the structure:

i. Consider the carbon atoms marked in as ***a***, ***b***, ***c***, ***d***, and ***e*** in the molecule, and determine **hybridization** at each of the carbon centers. (2.5 points)

Atom	Hybridization
Carbon (<i>a</i>)	
Carbon (<i>b</i>)	
Carbon (<i>c</i>)	
Carbon (<i>d</i>)	
Carbon (<i>e</i>)	

ii. Now, consider the sulfur atom (S) marked as ***f***, nitrogen (N) atom (***g***), and oxygen(O) atoms (***h*** and ***i***), and determine their **hybridization**. (2 points)

Atom	Hybridization
Sulfur (<i>f</i>)	
Nitrogen (<i>g</i>)	
Oxygen (<i>h</i>)	
Oxygen (<i>i</i>)	

iii. **Describe** all the bonds formed by the carbon atoms ***a*** and ***b***, using sigma and pi designation as appropriate. (2 points)

Atom	Bonds formed (use sigma and pi as appropriate)
Carbon (<i>a</i>)	
Carbon (<i>b</i>)	

iv. How many total sp^3 -s sigma bonds are in the molecule? (1 point)

v. Determine **electronic** and **molecular geometry** at carbon (C) labeled ***e*** and sulfur (S) labeled ***f*** centres. (2 points)

Atom	Electronic Geometry	Molecular Geometry
Carbon (<i>e</i>)		
Sulfur (<i>f</i>)		

vi. Circle the value from the choices given below, that describes the **C-S-C bond angle at sulfur** (S) marked ***f***: (0.5 points)

>90° to <109.5° 109.5° >109.5 to <120° 120° >120° to <180°

**Question 8 12 points**

Using the **valence shell orbitals only** draw a **complete** molecular orbital (MO) diagram (with no s-p mixing) for the molecule **PS**, in the space provided below. **(6 points)**

Assume that 3s and 3p atomic orbitals will form molecular orbitals like those formed by the combination of 2s and 2p atomic orbitals that we learnt in class, and there is no s-p mixing.

Show **energy arrow**, **atomic orbitals** involved, **atom labels**, the **molecular orbitals** that are formed, **filled electrons** in the MO diagram etc. Make sure to label all the atomic and molecular orbitals.

Using the MO diagram above, answer the following questions:

- (i) Which one of the following will have the highest bond order? Circle the correct choice
A) PS^{1-} B) PS C) PS^{1+}
- (ii) Which one of the following will have the longest bond length? Circle the correct choice
A) PS^{1-} B) PS C) PS^{1+}
- (iii) Which of the following will be diamagnetic? Circle the correct choice.
A) PS^{1-} B) PS C) PS^{1+}
- (iv) Draw Lewis dot structure of the molecule with the shortest bond length (based on the MO diagram) from among the three (PS^{1-} , PS, PS^{1+}), and indicate the bond order calculated from the Lewis dot structure. **(3 points)**

Lewis Dot Structure

Bond Order



EDEDA8F5-A592-4F7D-9C5A-04DA73AA541F

midterm-exam-93388

#361 Page 10 of 10

SCRAP PAGE

**CHEMISTRY 110 –Midterm Exam****Thursday November 7, 2024, 6:30 – 8:30 pm****Examiners: Pallavi Sirjoosingh and Ashok Kakkar**

First name (please write as legibly as possible within the boxes)

--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--

Last name

--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--

Student ID number

--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--

This booklet has 8 questions on 10 pages (The last page can be used as a scrap page).

The staple must NOT be removed from the question booklet. Your intact question booklet must be returned at the end of the exam.

A separate booklet contains a periodic table, and datasheet.

Write your answers directly under the questions in the space provided. Do not write on the QR codes or near the top edges (as it may be cut off).

Please write **legibly and dark enough** for the scan to be readable. For students who use English as a second language, dictionaries are permitted. Calculators are allowed.

DO NOT ASK QUESTIONS OF THE INVIGILATORS about interpretations of the exam questions. If you have a concern about a question, please summarize it at the top of this page. All concerns will be evaluated after the exam. GOOD LUCK!

**Question 1 8 points** (*Assume all transitions are possible*)

a. 6 points Calculate the **energy (in kJ/mol)** required to ionize (completely remove) the electron from the **n=3** level of **He⁺ ion** (He: Helium). Show your work including equations. (3 sig. figs.)

b. 2 points

(I) Select the largest (longest) wavelength from the three choices below:

- i. Wavelength of photon to *completely remove* an electron from *1s orbital of Hydrogen atom*
- ii. Wavelength of photon to *completely remove* an electron from *1s orbital of Boron atom (B)*
- iii. Wavelength of photon to *completely remove* an electron from *1s orbital of Boron cation (B⁺)*

i or ii or iii

(II) Select the smallest (shortest) wavelength from the three choices below (*for H atom*):

- i. Wavelength of photon *emitted* for an electron transitioning from *2p to 1s orbital*
- ii. Wavelength of photon *absorbed* for an electron transitioning from *2s to 4p orbital*
- iii. Wavelength of photon *absorbed* for an electron transitioning from *1s to 3p orbital*

i or ii or iii



Question 2 8 points



A Chem110 student plots the graph shown on the left, from data obtained for the kinetic energy (eV) of the ejected electron versus frequency (s^{-1}) of the incident light, for an unknown metal. ($1\text{eV} = 1.6022 \times 10^{-19} \text{ J}$)

Based on the plot, answer the questions **a to c** below:

a. 1 point On the plot above, label the **threshold frequency**.

b. 2 points What is the value of the work potential (in J) of the unknown metal?

c. 3 points What is the *longest* wavelength (in nm) required to eject an electron from this metal's surface? Show your work including equation(s) used. (3 sig figs)

d. 2 points In the space provided below, plot the following

(Assume frequency is *above* threshold frequency):

- Kinetic energy* of an ejected electron versus the *intensity* of the incident light (left)
- Number of electrons* ejected versus the *frequency* of the incident light (right)



**Question 3** **6 points****a. 2 points**

In the ground state of Ge (Germanium; $Z = 32$) atom, **name** the following orbitals:

i. Orbital containing the electron that requires the maximum energy to be removed.

ii. Orbital containing the electron that experiences the minimum effective nuclear charge.

b. 2 points

An unknown anion (A^-) has the following *excited* electronic configuration: $[Ar] 4s^2 3d^3 4p^1$

(Ar: Argon; $Z = 18$)

Give the **condensed ground state electronic configuration** of the atom, **A** and the cation, **A^{2+}** .
Show your work. (Assume all electronic transitions are possible)

Electronic Configuration of A:

Electronic Configuration of A^{2+} :

c. 2 points

In a single atom, determine how many electrons can have the following set of quantum numbers?

(i) $n = 5; l = 3$

In a single atom, determine how many electrons can have the following set of quantum numbers?

(i) $n = 3; l = 2; m_s = +1/2$

**Question 4** **8 points**

Rank the following from 1 to 4 in **increasing** order. Provide the ranking in the boxes provided.

Note: Atomic number (Z) are provided for relevant atoms

a) Increasing (lowest to highest) order of *ionization energy*: (1= lowest; 4 = highest)

Mg

Mg: Z = 12

Rb

Rb: Z = 37

Cs

Cs: Z = 55

K

K : Z = 19

b) Increasing (lowest to highest) order of *bond angle*: (1= lowest; 4 = highest)

H₂OCCl₄XeF₄

HCN

c) Increasing (smallest to largest) order of *ionic size* (1= smallest; 4 = largest)

S²⁻

S: Z = 16

O²⁻

O: Z = 8

Be²⁺

Be: Z = 4

Na⁺

Na: Z = 11

d) Increasing (lowest to highest) order of *bond energy* (1= lowest; 4 = highest)

P-F

P-Cl

P-I

P-Br

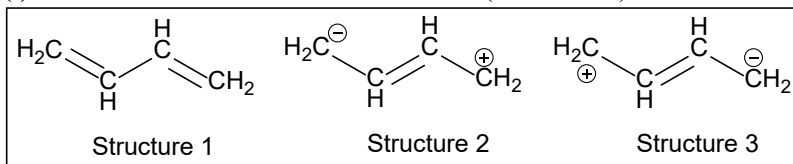
**Question 5** **12 points**

a. 9 points For the molecules HClO_4 and TeF_6 , draw the most stable *Lewis structure* showing appropriate shape and *dashed/wedged bonds*, based on VSEPR. Determine the **molecular geometry** around the central atom. For HClO_4 , label the *most polar bond*.



b. 3 points (Note: Lone pairs are not shown in structures below)

(i) For the three *resonance* structures (1, 2 and 3) shown below, answer the following question:



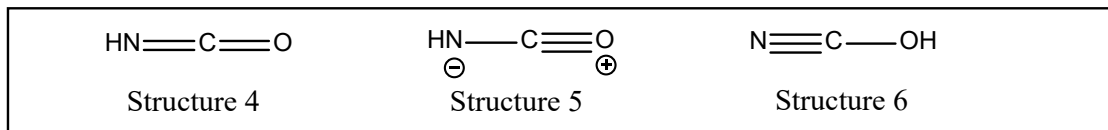
(I) Circle the *most contributing* resonance structure.

Structure 1

Structure 2

Structure 3

(ii) For the three structures given (4, 5, and 6) below, answer the following questions:



(I) Circle all the structures that represent *resonance structures* of each other:

Structure 4

Structure 5

Structure 6

(II) Among the resonance structures, circle the *most contributing* resonance structure.

Structure 4

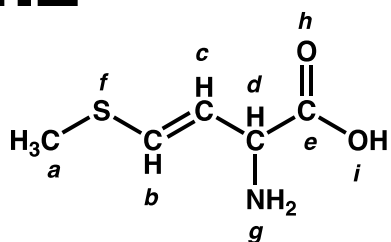
Structure 5

Structure 6

**Question 6** **6 points**

Using the data given below, calculate the **lattice energy** for SrO(s).

	ΔH° (kJ/mol)
$\text{Sr}(s) \longrightarrow \text{Sr}(g)$	+180
$\text{Sr}(g) \longrightarrow \text{Sr}^{2+}(g) + 2e^-$	+1468
$\text{O}_2(g) \longrightarrow 2\text{O}(g)$	+500
$\text{O}(g) + 2e^- \longrightarrow \text{O}^{2-}(g)$	+600
$\text{Sr}(s) + \frac{1}{2}\text{O}_2(g) \longrightarrow \text{SrO}(s)$	-550

**Question 7 10 points**

Consider the molecule given on the left, and answer questions (i) to (vi) using chemical bonding theories that we have learnt in class (valence bond theory, hybridization).

Note that the lone pairs are not shown in the structure:

i. Consider the carbon atoms marked in as ***a***, ***b***, ***c***, ***d***, and ***e*** in the molecule, and determine **hybridization** at each of the carbon centers. (2.5 points)

Atom	Hybridization
Carbon (<i>a</i>)	
Carbon (<i>b</i>)	
Carbon (<i>c</i>)	
Carbon (<i>d</i>)	
Carbon (<i>e</i>)	

ii. Now, consider the sulfur atom (S) marked as ***f***, nitrogen (N) atom (***g***), and oxygen(O) atoms (***h*** and ***i***), and determine their **hybridization**. (2 points)

Atom	Hybridization
Sulfur (<i>f</i>)	
Nitrogen (<i>g</i>)	
Oxygen (<i>h</i>)	
Oxygen (<i>i</i>)	

iii. **Describe** all the bonds formed by the carbon atoms ***a*** and ***b***, using sigma and pi designation as appropriate. (2 points)

Atom	Bonds formed (use sigma and pi as appropriate)
Carbon (<i>a</i>)	
Carbon (<i>b</i>)	

iv. How many total sp^3 -s sigma bonds are in the molecule? (1 point)

v. Determine **electronic** and **molecular geometry** at carbon (C) labeled ***e*** and sulfur (S) labeled ***f*** centres. (2 points)

Atom	Electronic Geometry	Molecular Geometry
Carbon (<i>e</i>)		
Sulfur (<i>f</i>)		

vi. Circle the value from the choices given below, that describes the **C-S-C bond angle at sulfur** (S) marked ***f***: (0.5 points)

>90° to <109.5° 109.5° >109.5 to <120° 120° >120° to <180°


Question 8 12 points

Using the **valence shell orbitals only** draw a **complete** molecular orbital (MO) diagram (with no s-p mixing) for the molecule **PS**, in the space provided below. **(6 points)**

Assume that 3s and 3p atomic orbitals will form molecular orbitals like those formed by the combination of 2s and 2p atomic orbitals that we learnt in class, and there is no s-p mixing.

Show **energy arrow**, **atomic orbitals** involved, **atom labels**, the **molecular orbitals** that are formed, **filled electrons** in the MO diagram etc. Make sure to label all the atomic and molecular orbitals.

Using the MO diagram above, answer the following questions:

- (i) Which one of the following will have the highest bond order? Circle the correct choice
 A) PS^{1-} B) PS C) PS^{1+}
- (ii) Which one of the following will have the longest bond length? Circle the correct choice
 A) PS^{1-} B) PS C) PS^{1+}
- (iii) Which of the following will be diamagnetic? Circle the correct choice.
 A) PS^{1-} B) PS C) PS^{1+}
- (iv) Draw Lewis dot structure of the molecule with the shortest bond length (based on the MO diagram) from among the three (PS^{1-} , PS, PS^{1+}), and indicate the bond order calculated from the Lewis dot structure. **(3 points)**

Lewis Dot Structure

Bond Order



8F7FA9F1-02E4-46C5-901A-AAC3EFAD76A2

midterm-exam-93388

#362 Page 10 of 10

SCRAP PAGE

**CHEMISTRY 110 –Midterm Exam****Thursday November 7, 2024, 6:30 – 8:30 pm**

Examiners: Pallavi Sirjoosingh and Ashok Kakkar

First name (please write as legibly as possible within the boxes)

--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--

Last name

--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--

Student ID number

--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--

This booklet has 8 questions on 10 pages (The last page can be used as a scrap page).

The staple must NOT be removed from the question booklet. Your intact question booklet must be returned at the end of the exam.

A separate booklet contains a periodic table, and datasheet.

Write your answers directly under the questions in the space provided. Do not write on the QR codes or near the top edges (as it may be cut off).

Please write **legibly and dark enough** for the scan to be readable. For students who use English as a second language, dictionaries are permitted. Calculators are allowed.

DO NOT ASK QUESTIONS OF THE INVIGILATORS about interpretations of the exam questions. If you have a concern about a question, please summarize it at the top of this page. All concerns will be evaluated after the exam. GOOD LUCK!

**Question 1 8 points** (*Assume all transitions are possible*)

a. 6 points Calculate the **energy (in kJ/mol)** required to ionize (completely remove) the electron from the **$n=3$** level of **He^+ ion** (He: Helium). Show your work including equations. (3 sig. figs.)

b. 2 points

(I) Select the largest (longest) wavelength from the three choices below:

- i. Wavelength of photon to *completely remove* an electron from *1s orbital of Hydrogen atom*
- ii. Wavelength of photon to *completely remove* an electron from *1s orbital of Boron atom (B)*
- iii. Wavelength of photon to *completely remove* an electron from *1s orbital of Boron cation (B^+)*

i or ii or iii

(II) Select the smallest (shortest) wavelength from the three choices below (*for H atom*):

- i. Wavelength of photon *emitted* for an electron transitioning from *2p to 1s orbital*
- ii. Wavelength of photon *absorbed* for an electron transitioning from *2s to 4p orbital*
- iii. Wavelength of photon *absorbed* for an electron transitioning from *1s to 3p orbital*

i or ii or iii



Question 2 8 points



A Chem110 student plots the graph shown on the left, from data obtained for the kinetic energy (eV) of the ejected electron versus frequency (s^{-1}) of the incident light, for an unknown metal. ($1\text{eV} = 1.6022 \times 10^{-19} \text{ J}$)

Based on the plot, answer the questions **a to c** below:

a. 1 point On the plot above, label the **threshold frequency**.

b. 2 points What is the value of the work potential (in J) of the unknown metal?

c. 3 points What is the *longest* wavelength (in nm) required to eject an electron from this metal's surface? Show your work including equation(s) used. (3 sig figs)

d. 2 points In the space provided below, plot the following

(Assume frequency is *above* threshold frequency):

- Kinetic energy* of an ejected electron versus the *intensity* of the incident light (left)
- Number of electrons* ejected versus the *frequency* of the incident light (right)



**Question 3** **6 points****a. 2 points**

In the ground state of Ge (Germanium; $Z = 32$) atom, **name** the following orbitals:

i. Orbital containing the electron that requires the maximum energy to be removed.

ii. Orbital containing the electron that experiences the minimum effective nuclear charge.

b. 2 points

An unknown anion (A^-) has the following *excited* electronic configuration: $[Ar] 4s^2 3d^3 4p^1$

(Ar: Argon; $Z = 18$)

Give the **condensed ground state electronic configuration** of the atom, **A** and the cation, **A^{2+}** .
Show your work. (Assume all electronic transitions are possible)

Electronic Configuration of A:

Electronic Configuration of A^{2+} :

c. 2 points

In a single atom, determine how many electrons can have the following set of quantum numbers?

(i) $n = 5; l = 3$

In a single atom, determine how many electrons can have the following set of quantum numbers?

(i) $n = 3; l = 2; m_s = +1/2$

**Question 4** **8 points**

Rank the following from 1 to 4 in **increasing** order. Provide the ranking in the boxes provided.

Note: Atomic number (Z) are provided for relevant atoms

a) Increasing (lowest to highest) order of *ionization energy*: (1= lowest; 4 = highest)

Mg

Mg: Z = 12

Rb

Rb: Z = 37

Cs

Cs: Z = 55

K

K : Z = 19

b) Increasing (lowest to highest) order of *bond angle*: (1= lowest; 4 = highest)

H₂OCCl₄XeF₄

HCN

c) Increasing (smallest to largest) order of *ionic size* (1= smallest; 4 = largest)

S²⁻

S: Z = 16

O²⁻

O: Z = 8

Be²⁺

Be: Z = 4

Na⁺

Na: Z = 11

d) Increasing (lowest to highest) order of *bond energy* (1= lowest; 4 = highest)

P-F

P-Cl

P-I

P-Br

**Question 5** **12 points**

a. 9 points For the molecules HClO_4 and TeF_6 , draw the most stable *Lewis structure* showing appropriate shape and *dashed/wedged bonds*, based on VSEPR. Determine the **molecular geometry** around the central atom. For HClO_4 , label the *most polar bond*.



b. 3 points (Note: Lone pairs are not shown in structures below)

(i) For the three *resonance* structures (1, 2 and 3) shown below, answer the following question:



(I) Circle the *most contributing* resonance structure.

Structure 1

Structure 2

Structure 3

(ii) For the three structures given (4, 5, and 6) below, answer the following questions:



(I) Circle all the structures that represent *resonance structures* of each other:

Structure 4

Structure 5

Structure 6

(II) Among the resonance structures, circle the *most contributing* resonance structure.

Structure 4

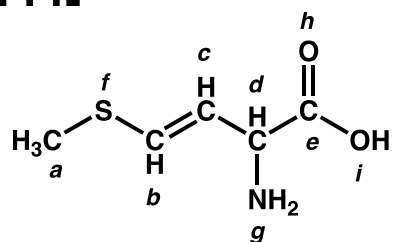
Structure 5

Structure 6

**Question 6** **6 points**

Using the data given below, calculate the **lattice energy** for SrO(s).

	ΔH° (kJ/mol)
$\text{Sr}(s) \longrightarrow \text{Sr}(g)$	+180
$\text{Sr}(g) \longrightarrow \text{Sr}^{2+}(g) + 2e^-$	+1468
$\text{O}_2(g) \longrightarrow 2\text{O}(g)$	+500
$\text{O}(g) + 2e^- \longrightarrow \text{O}^{2-}(g)$	+600
$\text{Sr}(s) + \frac{1}{2}\text{O}_2(g) \longrightarrow \text{SrO}(s)$	-550

**Question 7 10 points**

Consider the molecule given on the left, and answer questions (i) to (vi) using chemical bonding theories that we have learnt in class (valence bond theory, hybridization).

Note that the lone pairs are not shown in the structure:

i. Consider the carbon atoms marked in as ***a***, ***b***, ***c***, ***d***, and ***e*** in the molecule, and determine **hybridization** at each of the carbon centers. (2.5 points)

Atom	Hybridization
Carbon (<i>a</i>)	
Carbon (<i>b</i>)	
Carbon (<i>c</i>)	
Carbon (<i>d</i>)	
Carbon (<i>e</i>)	

ii. Now, consider the sulfur atom (S) marked as ***f***, nitrogen (N) atom (***g***), and oxygen(O) atoms (***h*** and ***i***), and determine their **hybridization**. (2 points)

Atom	Hybridization
Sulfur (<i>f</i>)	
Nitrogen (<i>g</i>)	
Oxygen (<i>h</i>)	
Oxygen (<i>i</i>)	

iii. **Describe** all the bonds formed by the carbon atoms ***a*** and ***b***, using sigma and pi designation as appropriate. (2 points)

Atom	Bonds formed (use sigma and pi as appropriate)
Carbon (<i>a</i>)	
Carbon (<i>b</i>)	

iv. How many total sp^3 -s sigma bonds are in the molecule? (1 point)

v. Determine **electronic** and **molecular geometry** at carbon (C) labeled ***e*** and sulfur (S) labeled ***f*** centres. (2 points)

Atom	Electronic Geometry	Molecular Geometry
Carbon (<i>e</i>)		
Sulfur (<i>f</i>)		

vi. Circle the value from the choices given below, that describes the **C-S-C bond angle at sulfur** (S) marked ***f***: (0.5 points)

$>90^\circ$ to $<109.5^\circ$ 109.5° >109.5 to $<120^\circ$ 120° $>120^\circ$ to $<180^\circ$

**Question 8 12 points**

Using the **valence shell orbitals only** draw a **complete** molecular orbital (MO) diagram (with no s-p mixing) for the molecule **PS**, in the space provided below. **(6 points)**

Assume that 3s and 3p atomic orbitals will form molecular orbitals like those formed by the combination of 2s and 2p atomic orbitals that we learnt in class, and there is no s-p mixing.

Show **energy arrow**, **atomic orbitals** involved, **atom labels**, the **molecular orbitals** that are formed, **filled electrons** in the MO diagram etc. Make sure to label all the atomic and molecular orbitals.

Using the MO diagram above, answer the following questions:

- (i) Which one of the following will have the highest bond order? Circle the correct choice
A) PS^{1-} B) PS C) PS^{1+}
- (ii) Which one of the following will have the longest bond length? Circle the correct choice
A) PS^{1-} B) PS C) PS^{1+}
- (iii) Which of the following will be diamagnetic? Circle the correct choice.
A) PS^{1-} B) PS C) PS^{1+}
- (iv) Draw Lewis dot structure of the molecule with the shortest bond length (based on the MO diagram) from among the three (PS^{1-} , PS, PS^{1+}), and indicate the bond order calculated from the Lewis dot structure. **(3 points)**

Lewis Dot Structure

Bond Order



8285EFC9-BFC2-4337-80B4-1268A52A403B

midterm-exam-93388

#363 Page 10 of 10

SCRAP PAGE

**CHEMISTRY 110 –Midterm Exam****Thursday November 7, 2024, 6:30 – 8:30 pm**

Examiners: Pallavi Sirjoosingh and Ashok Kakkar

First name (please write as legibly as possible within the boxes)

--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--

Last name

--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--

Student ID number

--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--

This booklet has 8 questions on 10 pages (The last page can be used as a scrap page).

The staple must NOT be removed from the question booklet. Your intact question booklet must be returned at the end of the exam.

A separate booklet contains a periodic table, and datasheet.

Write your answers directly under the questions in the space provided. Do not write on the QR codes or near the top edges (as it may be cut off).

Please write **legibly and dark enough** for the scan to be readable. For students who use English as a second language, dictionaries are permitted. Calculators are allowed.

DO NOT ASK QUESTIONS OF THE INVIGILATORS about interpretations of the exam questions. If you have a concern about a question, please summarize it at the top of this page. All concerns will be evaluated after the exam. GOOD LUCK!

**Question 1 8 points** (*Assume all transitions are possible*)

a. 6 points Calculate the **energy (in kJ/mol)** required to ionize (completely remove) the electron from the **$n=3$** level of **He^+ ion** (He: Helium). Show your work including equations. (3 sig. figs.)

b. 2 points

(I) Select the largest (longest) wavelength from the three choices below:

- i. Wavelength of photon to *completely remove* an electron from *1s orbital of Hydrogen atom*
- ii. Wavelength of photon to *completely remove* an electron from *1s orbital of Boron atom (B)*
- iii. Wavelength of photon to *completely remove* an electron from *1s orbital of Boron cation (B^+)*

i or ii or iii

(II) Select the smallest (shortest) wavelength from the three choices below (*for H atom*):

- i. Wavelength of photon *emitted* for an electron transitioning from *2p to 1s orbital*
- ii. Wavelength of photon *absorbed* for an electron transitioning from *2s to 4p orbital*
- iii. Wavelength of photon *absorbed* for an electron transitioning from *1s to 3p orbital*

i or ii or iii



Question 2 8 points



A Chem110 student plots the graph shown on the left, from data obtained for the kinetic energy (eV) of the ejected electron versus frequency (s^{-1}) of the incident light, for an unknown metal. ($1\text{eV} = 1.6022 \times 10^{-19} \text{ J}$)

Based on the plot, answer the questions **a to c** below:

a. 1 point On the plot above, label the **threshold frequency**.

b. 2 points What is the value of the work potential (in J) of the unknown metal?

c. 3 points What is the *longest* wavelength (in nm) required to eject an electron from this metal's surface? Show your work including equation(s) used. (3 sig figs)

d. 2 points In the space provided below, plot the following

(Assume frequency is *above* threshold frequency):

- Kinetic energy* of an ejected electron versus the *intensity* of the incident light (left)
- Number of electrons* ejected versus the *frequency* of the incident light (right)



**Question 3** **6 points****a. 2 points**

In the ground state of Ge (Germanium; $Z = 32$) atom, **name** the following orbitals:

i. Orbital containing the electron that requires the maximum energy to be removed.

ii. Orbital containing the electron that experiences the minimum effective nuclear charge.

b. 2 points

An unknown anion (A^-) has the following *excited* electronic configuration: $[Ar] 4s^2 3d^3 4p^1$

(Ar: Argon; $Z = 18$)

Give the **condensed ground state electronic configuration** of the atom, **A** and the cation, **A^{2+}** .
Show your work. (Assume all electronic transitions are possible)

Electronic Configuration of A:

Electronic Configuration of A^{2+} :

c. 2 points

In a single atom, determine how many electrons can have the following set of quantum numbers?

(i) $n = 5; l = 3$

In a single atom, determine how many electrons can have the following set of quantum numbers?

(i) $n = 3; l = 2; m_s = +1/2$

**Question 4** **8 points**

Rank the following from 1 to 4 in **increasing** order. Provide the ranking in the boxes provided.

Note: Atomic number (Z) are provided for relevant atoms

a) Increasing (lowest to highest) order of *ionization energy*: (1= lowest; 4 = highest)

Mg

Mg: Z = 12

Rb

Rb: Z = 37

Cs

Cs: Z = 55

K

K : Z = 19

b) Increasing (lowest to highest) order of *bond angle*: (1= lowest; 4 = highest)

H₂OCCl₄XeF₄

HCN

c) Increasing (smallest to largest) order of *ionic size* (1= smallest; 4 = largest)

S²⁻

S: Z = 16

O²⁻

O: Z = 8

Be²⁺

Be: Z = 4

Na⁺

Na: Z = 11

d) Increasing (lowest to highest) order of *bond energy* (1= lowest; 4 = highest)

P-F

P-Cl

P-I

P-Br

**Question 5** **12 points**

a. 9 points For the molecules HClO_4 and TeF_6 , draw the most stable *Lewis structure* showing appropriate shape and *dashed/wedged bonds*, based on VSEPR. Determine the **molecular geometry** around the central atom. For HClO_4 , label the *most polar bond*.

HClO_4 (there are no O-O bonds)

Label the most polar bond

Molecular Geometry around Cl

TeF_6

Molecular Geometry around Te

b. 3 points (Note: Lone pairs are not shown in structures below)

(i) For the three *resonance* structures (1, 2 and 3) shown below, answer the following question:



(I) Circle the *most contributing* resonance structure.

Structure 1

Structure 2

Structure 3

(ii) For the three structures given (4, 5, and 6) below, answer the following questions:



(I) Circle all the structures that represent *resonance structures* of each other:

Structure 4

Structure 5

Structure 6

(II) Among the resonance structures, circle the *most contributing* resonance structure.

Structure 4

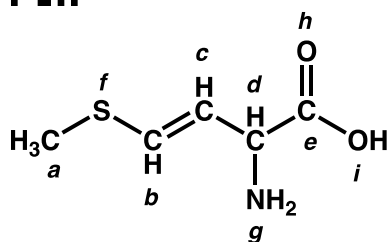
Structure 5

Structure 6

**Question 6** **6 points**

Using the data given below, calculate the **lattice energy** for SrO(s).

	ΔH° (kJ/mol)
$\text{Sr}(s) \longrightarrow \text{Sr}(g)$	+180
$\text{Sr}(g) \longrightarrow \text{Sr}^{2+}(g) + 2e^-$	+1468
$\text{O}_2(g) \longrightarrow 2\text{O}(g)$	+500
$\text{O}(g) + 2e^- \longrightarrow \text{O}^{2-}(g)$	+600
$\text{Sr}(s) + \frac{1}{2}\text{O}_2(g) \longrightarrow \text{SrO}(s)$	-550

**Question 7 10 points**

Consider the molecule given on the left, and answer questions (i) to (vi) using chemical bonding theories that we have learnt in class (valence bond theory, hybridization).

Note that the lone pairs are not shown in the structure:

i. Consider the carbon atoms marked in as ***a***, ***b***, ***c***, ***d***, and ***e*** in the molecule, and determine **hybridization** at each of the carbon centers. (2.5 points)

Atom	Hybridization
Carbon (<i>a</i>)	
Carbon (<i>b</i>)	
Carbon (<i>c</i>)	
Carbon (<i>d</i>)	
Carbon (<i>e</i>)	

ii. Now, consider the sulfur atom (S) marked as ***f***, nitrogen (N) atom (***g***), and oxygen(O) atoms (***h*** and ***i***), and determine their **hybridization**. (2 points)

Atom	Hybridization
Sulfur (<i>f</i>)	
Nitrogen (<i>g</i>)	
Oxygen (<i>h</i>)	
Oxygen (<i>i</i>)	

iii. **Describe** all the bonds formed by the carbon atoms ***a*** and ***b***, using sigma and pi designation as appropriate. (2 points)

Atom	Bonds formed (use sigma and pi as appropriate)
Carbon (<i>a</i>)	
Carbon (<i>b</i>)	

iv. How many total sp^3 -s sigma bonds are in the molecule? (1 point)

v. Determine **electronic** and **molecular geometry** at carbon (C) labeled ***e*** and sulfur (S) labeled ***f*** centres. (2 points)

Atom	Electronic Geometry	Molecular Geometry
Carbon (<i>e</i>)		
Sulfur (<i>f</i>)		

vi. Circle the value from the choices given below, that describes the **C-S-C bond angle at sulfur** (S) marked ***f***: (0.5 points)

>90° to <109.5° 109.5° >109.5 to <120° 120° >120° to <180°


Question 8 12 points

Using the **valence shell orbitals only** draw a **complete** molecular orbital (MO) diagram (with no s-p mixing) for the molecule **PS**, in the space provided below. **(6 points)**

Assume that 3s and 3p atomic orbitals will form molecular orbitals like those formed by the combination of 2s and 2p atomic orbitals that we learnt in class, and there is no s-p mixing.

Show **energy arrow**, **atomic orbitals** involved, **atom labels**, the **molecular orbitals** that are formed, **filled electrons** in the MO diagram etc. Make sure to label all the atomic and molecular orbitals.

Using the MO diagram above, answer the following questions:

- (i) Which one of the following will have the highest bond order? Circle the correct choice
 A) PS^{1-} B) PS C) PS^{1+}
- (ii) Which one of the following will have the longest bond length? Circle the correct choice
 A) PS^{1-} B) PS C) PS^{1+}
- (iii) Which of the following will be diamagnetic? Circle the correct choice.
 A) PS^{1-} B) PS C) PS^{1+}
- (iv) Draw Lewis dot structure of the molecule with the shortest bond length (based on the MO diagram) from among the three (PS^{1-} , PS, PS^{1+}), and indicate the bond order calculated from the Lewis dot structure. **(3 points)**

Lewis Dot Structure

Bond Order



086F4459-C6C2-4E53-901F-7E914F6152DE

midterm-exam-93388

#364 Page 10 of 10

SCRAP PAGE

**CHEMISTRY 110 –Midterm Exam****Thursday November 7, 2024, 6:30 – 8:30 pm****Examiners: Pallavi Sirjoosingh and Ashok Kakkar**

First name (please write as legibly as possible within the boxes)

--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--

Last name

--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--

Student ID number

--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--

This booklet has 8 questions on 10 pages (The last page can be used as a scrap page).

The staple must NOT be removed from the question booklet. Your intact question booklet must be returned at the end of the exam.

A separate booklet contains a periodic table, and datasheet.

Write your answers directly under the questions in the space provided. Do not write on the QR codes or near the top edges (as it may be cut off).

Please write **legibly and dark enough** for the scan to be readable. For students who use English as a second language, dictionaries are permitted. Calculators are allowed.

DO NOT ASK QUESTIONS OF THE INVIGILATORS about interpretations of the exam questions. If you have a concern about a question, please summarize it at the top of this page. All concerns will be evaluated after the exam. GOOD LUCK!

**Question 1 8 points** (*Assume all transitions are possible*)

a. 6 points Calculate the **energy (in kJ/mol)** required to ionize (completely remove) the electron from the **$n=3$** level of **He^+ ion** (He: Helium). Show your work including equations. (3 sig. figs.)

b. 2 points

(I) Select the largest (longest) wavelength from the three choices below:

- i. Wavelength of photon to *completely remove* an electron from *1s orbital of Hydrogen atom*
- ii. Wavelength of photon to *completely remove* an electron from *1s orbital of Boron atom (B)*
- iii. Wavelength of photon to *completely remove* an electron from *1s orbital of Boron cation (B^+)*

i or ii or iii

(II) Select the smallest (shortest) wavelength from the three choices below (*for H atom*):

- i. Wavelength of photon *emitted* for an electron transitioning from *2p to 1s orbital*
- ii. Wavelength of photon *absorbed* for an electron transitioning from *2s to 4p orbital*
- iii. Wavelength of photon *absorbed* for an electron transitioning from *1s to 3p orbital*

i or ii or iii



Question 2 8 points



A Chem110 student plots the graph shown on the left, from data obtained for the kinetic energy (eV) of the ejected electron versus frequency (s^{-1}) of the incident light, for an unknown metal. ($1\text{eV} = 1.6022 \times 10^{-19} \text{ J}$)

Based on the plot, answer the questions **a to c** below:

a. 1 point On the plot above, label the **threshold frequency**.

b. 2 points What is the value of the work potential (in J) of the unknown metal?

c. 3 points What is the *longest* wavelength (in nm) required to eject an electron from this metal's surface? Show your work including equation(s) used. (3 sig figs)

d. 2 points In the space provided below, plot the following

(Assume frequency is *above* threshold frequency):

- i. *Kinetic energy* of an ejected electron versus the *intensity* of the incident light (left)
- ii. *Number of electrons* ejected versus the *frequency* of the incident light (right)



**Question 3** **6 points****a. 2 points**

In the ground state of Ge (Germanium; $Z = 32$) atom, **name** the following orbitals:

i. Orbital containing the electron that requires the maximum energy to be removed.

ii. Orbital containing the electron that experiences the minimum effective nuclear charge.

b. 2 points

An unknown anion (A^-) has the following *excited* electronic configuration: $[Ar] 4s^2 3d^3 4p^1$

(Ar: Argon; $Z = 18$)

Give the **condensed ground state electronic configuration** of the atom, **A** and the cation, **A^{2+}** .
Show your work. (Assume all electronic transitions are possible)

Electronic Configuration of A:

Electronic Configuration of A^{2+} :

c. 2 points

In a single atom, determine how many electrons can have the following set of quantum numbers?

(i) $n = 5; l = 3$

In a single atom, determine how many electrons can have the following set of quantum numbers?

(i) $n = 3; l = 2; m_s = +1/2$

**Question 4** **8 points**

Rank the following from 1 to 4 in **increasing** order. Provide the ranking in the boxes provided.

Note: Atomic number (Z) are provided for relevant atoms

a) Increasing (lowest to highest) order of *ionization energy*: (1= lowest; 4 = highest)

Mg

Mg: Z = 12

Rb

Rb: Z = 37

Cs

Cs: Z = 55

K

K : Z = 19

b) Increasing (lowest to highest) order of *bond angle*: (1= lowest; 4 = highest)

H₂OCCl₄XeF₄

HCN

c) Increasing (smallest to largest) order of *ionic size* (1= smallest; 4 = largest)

S²⁻

S: Z = 16

O²⁻

O: Z = 8

Be²⁺

Be: Z = 4

Na⁺

Na: Z = 11

d) Increasing (lowest to highest) order of *bond energy* (1= lowest; 4 = highest)

P-F

P-Cl

P-I

P-Br

**Question 5** **12 points**

a. 9 points For the molecules HClO_4 and TeF_6 , draw the most stable *Lewis structure* showing appropriate shape and *dashed/wedged bonds*, based on VSEPR. Determine the **molecular geometry** around the central atom. For HClO_4 , label the *most polar bond*.



b. 3 points (Note: Lone pairs are not shown in structures below)

(i) For the three *resonance* structures (1, 2 and 3) shown below, answer the following question:



(I) Circle the *most contributing* resonance structure.

Structure 1

Structure 2

Structure 3

(ii) For the three structures given (4, 5, and 6) below, answer the following questions:



(I) Circle all the structures that represent *resonance structures* of each other:

Structure 4

Structure 5

Structure 6

(II) Among the resonance structures, circle the *most contributing* resonance structure.

Structure 4

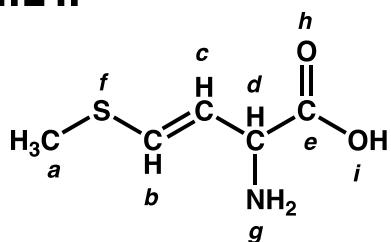
Structure 5

Structure 6

**Question 6** **6 points**

Using the data given below, calculate the **lattice energy** for SrO(s).

	ΔH° (kJ/mol)
$\text{Sr}(s) \longrightarrow \text{Sr}(g)$	+180
$\text{Sr}(g) \longrightarrow \text{Sr}^{2+}(g) + 2e^-$	+1468
$\text{O}_2(g) \longrightarrow 2\text{O}(g)$	+500
$\text{O}(g) + 2e^- \longrightarrow \text{O}^{2-}(g)$	+600
$\text{Sr}(s) + \frac{1}{2}\text{O}_2(g) \longrightarrow \text{SrO}(s)$	-550

**Question 7 10 points**

Consider the molecule given on the left, and answer questions (i) to (vi) using chemical bonding theories that we have learnt in class (valence bond theory, hybridization).

Note that the lone pairs are not shown in the structure:

i. Consider the carbon atoms marked in as ***a***, ***b***, ***c***, ***d***, and ***e*** in the molecule, and determine **hybridization** at each of the carbon centers. (2.5 points)

Atom	Hybridization
Carbon (<i>a</i>)	
Carbon (<i>b</i>)	
Carbon (<i>c</i>)	
Carbon (<i>d</i>)	
Carbon (<i>e</i>)	

ii. Now, consider the sulfur atom (S) marked as ***f***, nitrogen (N) atom (***g***), and oxygen(O) atoms (***h*** and ***i***), and determine their **hybridization**. (2 points)

Atom	Hybridization
Sulfur (<i>f</i>)	
Nitrogen (<i>g</i>)	
Oxygen (<i>h</i>)	
Oxygen (<i>i</i>)	

iii. **Describe** all the bonds formed by the carbon atoms ***a*** and ***b***, using sigma and pi designation as appropriate. (2 points)

Atom	Bonds formed (use sigma and pi as appropriate)
Carbon (<i>a</i>)	
Carbon (<i>b</i>)	

iv. How many total sp^3 -s sigma bonds are in the molecule? (1 point)

v. Determine **electronic** and **molecular geometry** at carbon (C) labeled ***e*** and sulfur (S) labeled ***f*** centres. (2 points)

Atom	Electronic Geometry	Molecular Geometry
Carbon (<i>e</i>)		
Sulfur (<i>f</i>)		

vi. Circle the value from the choices given below, that describes the **C-S-C bond angle at sulfur** (S) marked ***f***: (0.5 points)

>90° to <109.5° 109.5° >109.5 to <120° 120° >120° to <180°

**Question 8 12 points**

Using the **valence shell orbitals only** draw a **complete** molecular orbital (MO) diagram (with no s-p mixing) for the molecule **PS**, in the space provided below. **(6 points)**

Assume that 3s and 3p atomic orbitals will form molecular orbitals like those formed by the combination of 2s and 2p atomic orbitals that we learnt in class, and there is no s-p mixing.

Show **energy arrow**, **atomic orbitals** involved, **atom labels**, the **molecular orbitals** that are formed, **filled electrons** in the MO diagram etc. Make sure to label all the atomic and molecular orbitals.

Using the MO diagram above, answer the following questions:

- (i) Which one of the following will have the highest bond order? Circle the correct choice
A) PS^{1-} B) PS C) PS^{1+}
- (ii) Which one of the following will have the longest bond length? Circle the correct choice
A) PS^{1-} B) PS C) PS^{1+}
- (iii) Which of the following will be diamagnetic? Circle the correct choice.
A) PS^{1-} B) PS C) PS^{1+}
- (iv) Draw Lewis dot structure of the molecule with the shortest bond length (based on the MO diagram) from among the three (PS^{1-} , PS, PS^{1+}), and indicate the bond order calculated from the Lewis dot structure. **(3 points)**

Lewis Dot Structure

Bond Order



E46021F4-35FB-4C5D-A9DD-B8390CBE0D9A

midterm-exam-93388

#365 Page 10 of 10

SCRAP PAGE

**CHEMISTRY 110 –Midterm Exam****Thursday November 7, 2024, 6:30 – 8:30 pm**

Examiners: Pallavi Sirjoosingh and Ashok Kakkar

First name (please write as legibly as possible within the boxes)

--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--

Last name

--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--

Student ID number

--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--

This booklet has 8 questions on 10 pages (The last page can be used as a scrap page).

The staple must NOT be removed from the question booklet. Your intact question booklet must be returned at the end of the exam.

A separate booklet contains a periodic table, and datasheet.

Write your answers directly under the questions in the space provided. Do not write on the QR codes or near the top edges (as it may be cut off).

Please write **legibly and dark enough** for the scan to be readable. For students who use English as a second language, dictionaries are permitted. Calculators are allowed.

DO NOT ASK QUESTIONS OF THE INVIGILATORS about interpretations of the exam questions. If you have a concern about a question, please summarize it at the top of this page. All concerns will be evaluated after the exam. GOOD LUCK!

**Question 1 8 points** (*Assume all transitions are possible*)

a. 6 points Calculate the **energy (in kJ/mol)** required to ionize (completely remove) the electron from the **n=3** level of **He⁺ ion** (He: Helium). Show your work including equations. (3 sig. figs.)

b. 2 points

(I) Select the largest (longest) wavelength from the three choices below:

- i. Wavelength of photon to *completely remove* an electron from *1s orbital of Hydrogen atom*
- ii. Wavelength of photon to *completely remove* an electron from *1s orbital of Boron atom (B)*
- iii. Wavelength of photon to *completely remove* an electron from *1s orbital of Boron cation (B⁺)*

i or ii or iii

(II) Select the smallest (shortest) wavelength from the three choices below (*for H atom*):

- i. Wavelength of photon *emitted* for an electron transitioning from *2p to 1s orbital*
- ii. Wavelength of photon *absorbed* for an electron transitioning from *2s to 4p orbital*
- iii. Wavelength of photon *absorbed* for an electron transitioning from *1s to 3p orbital*

i or ii or iii



Question 2 8 points



A Chem110 student plots the graph shown on the left, from data obtained for the kinetic energy (eV) of the ejected electron versus frequency (s^{-1}) of the incident light, for an unknown metal. ($1\text{eV} = 1.6022 \times 10^{-19} \text{ J}$)

Based on the plot, answer the questions **a to c** below:

a. 1 point On the plot above, label the **threshold frequency**.

b. 2 points What is the value of the work potential (in J) of the unknown metal?

c. 3 points What is the *longest* wavelength (in nm) required to eject an electron from this metal's surface? Show your work including equation(s) used. (3 sig figs)

d. 2 points In the space provided below, plot the following

(Assume frequency is *above* threshold frequency):

- Kinetic energy* of an ejected electron versus the *intensity* of the incident light (left)
- Number of electrons* ejected versus the *frequency* of the incident light (right)



**Question 3** **6 points****a. 2 points**

In the ground state of Ge (Germanium; $Z = 32$) atom, **name** the following orbitals:

i. Orbital containing the electron that requires the maximum energy to be removed.

ii. Orbital containing the electron that experiences the minimum effective nuclear charge.

b. 2 points

An unknown anion (A^-) has the following *excited* electronic configuration: $[Ar] 4s^2 3d^3 4p^1$

(Ar: Argon; $Z = 18$)

Give the **condensed ground state electronic configuration** of the atom, **A** and the cation, **A^{2+}** .
Show your work. (Assume all electronic transitions are possible)

Electronic Configuration of A:

Electronic Configuration of A^{2+} :

c. 2 points

In a single atom, determine how many electrons can have the following set of quantum numbers?

(i) $n = 5; l = 3$

In a single atom, determine how many electrons can have the following set of quantum numbers?

(i) $n = 3; l = 2; m_s = +1/2$

**Question 4** **8 points**

Rank the following from 1 to 4 in **increasing** order. Provide the ranking in the boxes provided.

Note: Atomic number (Z) are provided for relevant atoms

a) Increasing (lowest to highest) order of *ionization energy*: (1= lowest; 4 = highest)

Mg

Mg: Z = 12

Rb

Rb: Z = 37

Cs

Cs: Z = 55

K

K : Z = 19

b) Increasing (lowest to highest) order of *bond angle*: (1= lowest; 4 = highest)

H₂OCCl₄XeF₄

HCN

c) Increasing (smallest to largest) order of *ionic size* (1= smallest; 4 = largest)

S²⁻

S: Z = 16

O²⁻

O: Z = 8

Be²⁺

Be: Z = 4

Na⁺

Na: Z = 11

d) Increasing (lowest to highest) order of *bond energy* (1= lowest; 4 = highest)

P-F

P-Cl

P-I

P-Br

**Question 5** **12 points**

a. 9 points For the molecules HClO_4 and TeF_6 , draw the most stable *Lewis structure* showing appropriate shape and *dashed/wedged bonds*, based on VSEPR. Determine the **molecular geometry** around the central atom. For HClO_4 , label the *most polar bond*.



b. 3 points (Note: Lone pairs are not shown in structures below)

(i) For the three *resonance* structures (1, 2 and 3) shown below, answer the following question:



(I) Circle the *most contributing* resonance structure.

Structure 1

Structure 2

Structure 3

(ii) For the three structures given (4, 5, and 6) below, answer the following questions:



(I) Circle all the structures that represent *resonance structures* of each other:

Structure 4

Structure 5

Structure 6

(II) Among the resonance structures, circle the *most contributing* resonance structure.

Structure 4

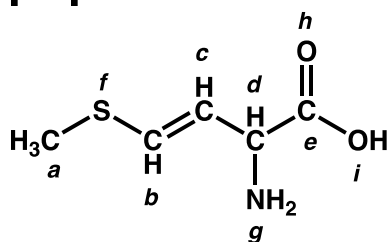
Structure 5

Structure 6

**Question 6** **6 points**

Using the data given below, calculate the **lattice energy** for SrO(s).

	ΔH° (kJ/mol)
$\text{Sr}(s) \longrightarrow \text{Sr}(g)$	+180
$\text{Sr}(g) \longrightarrow \text{Sr}^{2+}(g) + 2e^-$	+1468
$\text{O}_2(g) \longrightarrow 2\text{O}(g)$	+500
$\text{O}(g) + 2e^- \longrightarrow \text{O}^{2-}(g)$	+600
$\text{Sr}(s) + \frac{1}{2}\text{O}_2(g) \longrightarrow \text{SrO}(s)$	-550

**Question 7 10 points**

Consider the molecule given on the left, and answer questions (i) to (vi) using chemical bonding theories that we have learnt in class (valence bond theory, hybridization).

Note that the lone pairs are not shown in the structure:

i. Consider the carbon atoms marked in as ***a***, ***b***, ***c***, ***d***, and ***e*** in the molecule, and determine **hybridization** at each of the carbon centers. (2.5 points)

Atom	Hybridization
Carbon (<i>a</i>)	
Carbon (<i>b</i>)	
Carbon (<i>c</i>)	
Carbon (<i>d</i>)	
Carbon (<i>e</i>)	

ii. Now, consider the sulfur atom (S) marked as ***f***, nitrogen (N) atom (***g***), and oxygen(O) atoms (***h*** and ***i***), and determine their **hybridization**. (2 points)

Atom	Hybridization
Sulfur (<i>f</i>)	
Nitrogen (<i>g</i>)	
Oxygen (<i>h</i>)	
Oxygen (<i>i</i>)	

iii. **Describe** all the bonds formed by the carbon atoms ***a*** and ***b***, using sigma and pi designation as appropriate. (2 points)

Atom	Bonds formed (use sigma and pi as appropriate)
Carbon (<i>a</i>)	
Carbon (<i>b</i>)	

iv. How many total sp^3 -s sigma bonds are in the molecule? (1 point)

v. Determine **electronic** and **molecular geometry** at carbon (C) labeled ***e*** and sulfur (S) labeled ***f*** centres. (2 points)

Atom	Electronic Geometry	Molecular Geometry
Carbon (<i>e</i>)		
Sulfur (<i>f</i>)		

vi. Circle the value from the choices given below, that describes the **C-S-C bond angle at sulfur** (S) marked ***f***: (0.5 points)

>90° to <109.5° 109.5° >109.5 to <120° 120° >120° to <180°


Question 8 12 points

Using the **valence shell orbitals only** draw a **complete** molecular orbital (MO) diagram (with no s-p mixing) for the molecule **PS**, in the space provided below. **(6 points)**

Assume that 3s and 3p atomic orbitals will form molecular orbitals like those formed by the combination of 2s and 2p atomic orbitals that we learnt in class, and there is no s-p mixing.

Show **energy arrow**, **atomic orbitals** involved, **atom labels**, the **molecular orbitals** that are formed, **filled electrons** in the MO diagram etc. Make sure to label all the atomic and molecular orbitals.

Using the MO diagram above, answer the following questions:

- (i) Which one of the following will have the highest bond order? Circle the correct choice
 A) PS^{1-} B) PS C) PS^{1+}
- (ii) Which one of the following will have the longest bond length? Circle the correct choice
 A) PS^{1-} B) PS C) PS^{1+}
- (iii) Which of the following will be diamagnetic? Circle the correct choice.
 A) PS^{1-} B) PS C) PS^{1+}
- (iv) Draw Lewis dot structure of the molecule with the shortest bond length (based on the MO diagram) from among the three (PS^{1-} , PS, PS^{1+}), and indicate the bond order calculated from the Lewis dot structure. **(3 points)**

Lewis Dot Structure

Bond Order



E5E243D0-11BF-41F6-BDD3-AC196BA8CCB5

midterm-exam-93388

#366 Page 10 of 10

SCRAP PAGE

**CHEMISTRY 110 –Midterm Exam****Thursday November 7, 2024, 6:30 – 8:30 pm**

Examiners: Pallavi Sirjoosingh and Ashok Kakkar

First name (please write as legibly as possible within the boxes)

--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--

Last name

--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--

Student ID number

--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--

This booklet has 8 questions on 10 pages (The last page can be used as a scrap page).

The staple must NOT be removed from the question booklet. Your intact question booklet must be returned at the end of the exam.

A separate booklet contains a periodic table, and datasheet.

Write your answers directly under the questions in the space provided. Do not write on the QR codes or near the top edges (as it may be cut off).

Please write **legibly and dark enough** for the scan to be readable. For students who use English as a second language, dictionaries are permitted. Calculators are allowed.

DO NOT ASK QUESTIONS OF THE INVIGILATORS about interpretations of the exam questions. If you have a concern about a question, please summarize it at the top of this page. All concerns will be evaluated after the exam. GOOD LUCK!

**Question 1 8 points** (*Assume all transitions are possible*)

a. 6 points Calculate the **energy (in kJ/mol)** required to ionize (completely remove) the electron from the **n=3** level of **He⁺ ion** (He: Helium). Show your work including equations. (3 sig. figs.)

b. 2 points

(I) Select the largest (longest) wavelength from the three choices below:

- i. Wavelength of photon to *completely remove* an electron from *1s orbital of Hydrogen atom*
- ii. Wavelength of photon to *completely remove* an electron from *1s orbital of Boron atom (B)*
- iii. Wavelength of photon to *completely remove* an electron from *1s orbital of Boron cation (B⁺)*

i or ii or iii

(II) Select the smallest (shortest) wavelength from the three choices below (*for H atom*):

- i. Wavelength of photon *emitted* for an electron transitioning from *2p to 1s orbital*
- ii. Wavelength of photon *absorbed* for an electron transitioning from *2s to 4p orbital*
- iii. Wavelength of photon *absorbed* for an electron transitioning from *1s to 3p orbital*

i or ii or iii



Question 2 8 points



A Chem110 student plots the graph shown on the left, from data obtained for the kinetic energy (eV) of the ejected electron versus frequency (s^{-1}) of the incident light, for an unknown metal. ($1\text{eV} = 1.6022 \times 10^{-19} \text{ J}$)

Based on the plot, answer the questions **a to c** below:

a. 1 point On the plot above, label the **threshold frequency**.

b. 2 points What is the value of the work potential (in J) of the unknown metal?

c. 3 points What is the *longest* wavelength (in nm) required to eject an electron from this metal's surface? Show your work including equation(s) used. (3 sig figs)

d. 2 points In the space provided below, plot the following

(Assume frequency is *above* threshold frequency):

- i. *Kinetic energy* of an ejected electron versus the *intensity* of the incident light (left)
- ii. *Number of electrons* ejected versus the *frequency* of the incident light (right)



**Question 3** **6 points****a. 2 points**

In the ground state of Ge (Germanium; $Z = 32$) atom, **name** the following orbitals:

i. Orbital containing the electron that requires the maximum energy to be removed.

ii. Orbital containing the electron that experiences the minimum effective nuclear charge.

b. 2 points

An unknown anion (A^-) has the following *excited* electronic configuration: $[Ar] 4s^2 3d^3 4p^1$

(Ar: Argon; $Z = 18$)

Give the **condensed ground state electronic configuration** of the atom, **A** and the cation, **A^{2+}** .
Show your work. (Assume all electronic transitions are possible)

Electronic Configuration of A:

Electronic Configuration of A^{2+} :

c. 2 points

In a single atom, determine how many electrons can have the following set of quantum numbers?

(i) $n = 5; l = 3$

In a single atom, determine how many electrons can have the following set of quantum numbers?

(i) $n = 3; l = 2; m_s = +1/2$

**Question 4** **8 points**

Rank the following from 1 to 4 in **increasing** order. Provide the ranking in the boxes provided.

Note: Atomic number (Z) are provided for relevant atoms

a) Increasing (lowest to highest) order of *ionization energy*: (1= lowest; 4 = highest)

Mg

Mg: Z = 12

Rb

Rb: Z = 37

Cs

Cs: Z = 55

K

K : Z = 19

b) Increasing (lowest to highest) order of *bond angle*: (1= lowest; 4 = highest)

H₂OCCl₄XeF₄

HCN

c) Increasing (smallest to largest) order of *ionic size* (1= smallest; 4 = largest)

S²⁻

S: Z = 16

O²⁻

O: Z = 8

Be²⁺

Be: Z = 4

Na⁺

Na: Z = 11

d) Increasing (lowest to highest) order of *bond energy* (1= lowest; 4 = highest)

P-F

P-Cl

P-I

P-Br

**Question 5** **12 points**

a. 9 points For the molecules HClO_4 and TeF_6 , draw the most stable *Lewis structure* showing appropriate shape and *dashed/wedged bonds*, based on VSEPR. Determine the **molecular geometry** around the central atom. For HClO_4 , label the *most polar bond*.



b. 3 points (Note: Lone pairs are not shown in structures below)

(i) For the three *resonance* structures (1, 2 and 3) shown below, answer the following question:



(I) Circle the *most contributing* resonance structure.

Structure 1

Structure 2

Structure 3

(ii) For the three structures given (4, 5, and 6) below, answer the following questions:



(I) Circle all the structures that represent *resonance structures* of each other:

Structure 4

Structure 5

Structure 6

(II) Among the resonance structures, circle the *most contributing* resonance structure.

Structure 4

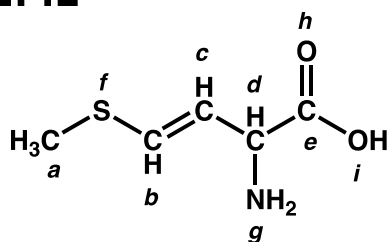
Structure 5

Structure 6

**Question 6** **6 points**

Using the data given below, calculate the **lattice energy** for SrO(s).

	ΔH° (kJ/mol)
$\text{Sr}(s) \longrightarrow \text{Sr}(g)$	+180
$\text{Sr}(g) \longrightarrow \text{Sr}^{2+}(g) + 2e^-$	+1468
$\text{O}_2(g) \longrightarrow 2\text{O}(g)$	+500
$\text{O}(g) + 2e^- \longrightarrow \text{O}^{2-}(g)$	+600
$\text{Sr}(s) + \frac{1}{2}\text{O}_2(g) \longrightarrow \text{SrO}(s)$	-550

**Question 7 10 points**

Consider the molecule given on the left, and answer questions (i) to (vi) using chemical bonding theories that we have learnt in class (valence bond theory, hybridization).

Note that the lone pairs are not shown in the structure:

i. Consider the carbon atoms marked in as ***a***, ***b***, ***c***, ***d***, and ***e*** in the molecule, and determine **hybridization** at each of the carbon centers. (2.5 points)

Atom	Hybridization
Carbon (<i>a</i>)	
Carbon (<i>b</i>)	
Carbon (<i>c</i>)	
Carbon (<i>d</i>)	
Carbon (<i>e</i>)	

ii. Now, consider the sulfur atom (S) marked as ***f***, nitrogen (N) atom (***g***), and oxygen(O) atoms (***h*** and ***i***), and determine their **hybridization**. (2 points)

Atom	Hybridization
Sulfur (<i>f</i>)	
Nitrogen (<i>g</i>)	
Oxygen (<i>h</i>)	
Oxygen (<i>i</i>)	

iii. **Describe** all the bonds formed by the carbon atoms ***a*** and ***b***, using sigma and pi designation as appropriate. (2 points)

Atom	Bonds formed (use sigma and pi as appropriate)
Carbon (<i>a</i>)	
Carbon (<i>b</i>)	

iv. How many total sp^3 -s sigma bonds are in the molecule? (1 point)

v. Determine **electronic** and **molecular geometry** at carbon (C) labeled ***e*** and sulfur (S) labeled ***f*** centres. (2 points)

Atom	Electronic Geometry	Molecular Geometry
Carbon (<i>e</i>)		
Sulfur (<i>f</i>)		

vi. Circle the value from the choices given below, that describes the **C-S-C bond angle at sulfur** (S) marked ***f***: (0.5 points)

>90° to <109.5° 109.5° >109.5 to <120° 120° >120° to <180°


Question 8 12 points

Using the **valence shell orbitals only** draw a **complete** molecular orbital (MO) diagram (with no s-p mixing) for the molecule **PS**, in the space provided below. **(6 points)**

Assume that 3s and 3p atomic orbitals will form molecular orbitals like those formed by the combination of 2s and 2p atomic orbitals that we learnt in class, and there is no s-p mixing.

Show **energy arrow**, **atomic orbitals** involved, **atom labels**, the **molecular orbitals** that are formed, **filled electrons** in the MO diagram etc. Make sure to label all the atomic and molecular orbitals.

Using the MO diagram above, answer the following questions:

- (i) Which one of the following will have the highest bond order? Circle the correct choice
 A) PS^{1-} B) PS C) PS^{1+}
- (ii) Which one of the following will have the longest bond length? Circle the correct choice
 A) PS^{1-} B) PS C) PS^{1+}
- (iii) Which of the following will be diamagnetic? Circle the correct choice.
 A) PS^{1-} B) PS C) PS^{1+}
- (iv) Draw Lewis dot structure of the molecule with the shortest bond length (based on the MO diagram) from among the three (PS^{1-} , PS, PS^{1+}), and indicate the bond order calculated from the Lewis dot structure. **(3 points)**

Lewis Dot Structure

Bond Order



0A94A6E7-3D78-4D90-AB59-585468C8A958

midterm-exam-93388

#367 Page 10 of 10

SCRAP PAGE

**CHEMISTRY 110 –Midterm Exam****Thursday November 7, 2024, 6:30 – 8:30 pm**

Examiners: Pallavi Sirjoosingh and Ashok Kakkar

First name (please write as legibly as possible within the boxes)

--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--

Last name

--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--

Student ID number

--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--

This booklet has 8 questions on 10 pages (The last page can be used as a scrap page).

The staple must NOT be removed from the question booklet. Your intact question booklet must be returned at the end of the exam.

A separate booklet contains a periodic table, and datasheet.

Write your answers directly under the questions in the space provided. Do not write on the QR codes or near the top edges (as it may be cut off).

Please write **legibly and dark enough** for the scan to be readable. For students who use English as a second language, dictionaries are permitted. Calculators are allowed.

DO NOT ASK QUESTIONS OF THE INVIGILATORS about interpretations of the exam questions. If you have a concern about a question, please summarize it at the top of this page. All concerns will be evaluated after the exam. GOOD LUCK!

**Question 1 8 points** (*Assume all transitions are possible*)

a. 6 points Calculate the **energy (in kJ/mol)** required to ionize (completely remove) the electron from the **n=3** level of **He⁺ ion** (He: Helium). Show your work including equations. (3 sig. figs.)

b. 2 points

(I) Select the largest (longest) wavelength from the three choices below:

- i. Wavelength of photon to *completely remove* an electron from *1s orbital of Hydrogen atom*
- ii. Wavelength of photon to *completely remove* an electron from *1s orbital of Boron atom (B)*
- iii. Wavelength of photon to *completely remove* an electron from *1s orbital of Boron cation (B⁺)*

i or ii or iii

(II) Select the smallest (shortest) wavelength from the three choices below (*for H atom*):

- i. Wavelength of photon *emitted* for an electron transitioning from *2p to 1s orbital*
- ii. Wavelength of photon *absorbed* for an electron transitioning from *2s to 4p orbital*
- iii. Wavelength of photon *absorbed* for an electron transitioning from *1s to 3p orbital*

i or ii or iii



Question 2 8 points



A Chem110 student plots the graph shown on the left, from data obtained for the kinetic energy (eV) of the ejected electron versus frequency (s^{-1}) of the incident light, for an unknown metal. ($1\text{eV} = 1.6022 \times 10^{-19} \text{ J}$)

Based on the plot, answer the questions **a to c** below:

a. 1 point On the plot above, label the **threshold frequency**.

b. 2 points What is the value of the work potential (in J) of the unknown metal?

c. 3 points What is the *longest* wavelength (in nm) required to eject an electron from this metal's surface? Show your work including equation(s) used. (3 sig figs)

d. 2 points In the space provided below, plot the following

(Assume frequency is *above* threshold frequency):

- i. *Kinetic energy* of an ejected electron versus the *intensity* of the incident light (left)
- ii. *Number of electrons* ejected versus the *frequency* of the incident light (right)



**Question 3** **6 points****a. 2 points**

In the ground state of Ge (Germanium; $Z = 32$) atom, **name** the following orbitals:

i. Orbital containing the electron that requires the maximum energy to be removed.

ii. Orbital containing the electron that experiences the minimum effective nuclear charge.

b. 2 points

An unknown anion (A^-) has the following *excited* electronic configuration: $[Ar] 4s^2 3d^3 4p^1$

(Ar: Argon; $Z = 18$)

Give the **condensed ground state electronic configuration** of the atom, **A** and the cation, A^{2+} .
Show your work. (Assume all electronic transitions are possible)

Electronic Configuration of A:

Electronic Configuration of A^{2+} :

c. 2 points

In a single atom, determine how many electrons can have the following set of quantum numbers?

(i) $n = 5; l = 3$

In a single atom, determine how many electrons can have the following set of quantum numbers?

(i) $n = 3; l = 2; m_s = +1/2$

**Question 4** **8 points**

Rank the following from 1 to 4 in **increasing** order. Provide the ranking in the boxes provided.

Note: Atomic number (Z) are provided for relevant atoms

a) Increasing (lowest to highest) order of *ionization energy*: (1= lowest; 4 = highest)

Mg

Mg: Z = 12

Rb

Rb: Z = 37

Cs

Cs: Z = 55

K

K : Z = 19

b) Increasing (lowest to highest) order of *bond angle*: (1= lowest; 4 = highest)

H₂OCCl₄XeF₄

HCN

c) Increasing (smallest to largest) order of *ionic size* (1= smallest; 4 = largest)

S²⁻

S: Z = 16

O²⁻

O: Z = 8

Be²⁺

Be: Z = 4

Na⁺

Na: Z = 11

d) Increasing (lowest to highest) order of *bond energy* (1= lowest; 4 = highest)

P-F

P-Cl

P-I

P-Br

**Question 5** **12 points**

a. 9 points For the molecules HClO_4 and TeF_6 , draw the most stable *Lewis structure* showing appropriate shape and *dashed/wedged bonds*, based on VSEPR. Determine the **molecular geometry** around the central atom. For HClO_4 , label the *most polar bond*.



b. 3 points (Note: Lone pairs are not shown in structures below)

(i) For the three *resonance* structures (1, 2 and 3) shown below, answer the following question:



(I) Circle the *most contributing* resonance structure.

Structure 1

Structure 2

Structure 3

(ii) For the three structures given (4, 5, and 6) below, answer the following questions:



(I) Circle all the structures that represent *resonance structures* of each other:

Structure 4

Structure 5

Structure 6

(II) Among the resonance structures, circle the *most contributing* resonance structure.

Structure 4

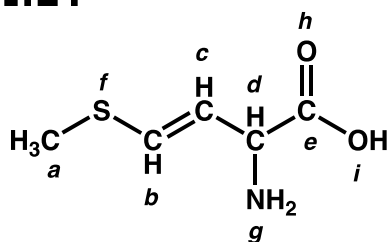
Structure 5

Structure 6

**Question 6** **6 points**

Using the data given below, calculate the **lattice energy** for SrO(s).

	ΔH° (kJ/mol)
$\text{Sr}(s) \longrightarrow \text{Sr}(g)$	+180
$\text{Sr}(g) \longrightarrow \text{Sr}^{2+}(g) + 2e^-$	+1468
$\text{O}_2(g) \longrightarrow 2\text{O}(g)$	+500
$\text{O}(g) + 2e^- \longrightarrow \text{O}^{2-}(g)$	+600
$\text{Sr}(s) + \frac{1}{2}\text{O}_2(g) \longrightarrow \text{SrO}(s)$	-550

**Question 7 10 points**

Consider the molecule given on the left, and answer questions (i) to (vi) using chemical bonding theories that we have learnt in class (valence bond theory, hybridization).

Note that the lone pairs are not shown in the structure:

i. Consider the carbon atoms marked in as ***a***, ***b***, ***c***, ***d***, and ***e*** in the molecule, and determine **hybridization** at each of the carbon centers. (2.5 points)

Atom	Hybridization
Carbon (<i>a</i>)	
Carbon (<i>b</i>)	
Carbon (<i>c</i>)	
Carbon (<i>d</i>)	
Carbon (<i>e</i>)	

ii. Now, consider the sulfur atom (S) marked as ***f***, nitrogen (N) atom (***g***), and oxygen(O) atoms (***h*** and ***i***), and determine their **hybridization**. (2 points)

Atom	Hybridization
Sulfur (<i>f</i>)	
Nitrogen (<i>g</i>)	
Oxygen (<i>h</i>)	
Oxygen (<i>i</i>)	

iii. **Describe** all the bonds formed by the carbon atoms ***a*** and ***b***, using sigma and pi designation as appropriate. (2 points)

Atom	Bonds formed (use sigma and pi as appropriate)
Carbon (<i>a</i>)	
Carbon (<i>b</i>)	

iv. How many total sp^3 -s sigma bonds are in the molecule? (1 point)

v. Determine **electronic** and **molecular geometry** at carbon (C) labeled ***e*** and sulfur (S) labeled ***f*** centres. (2 points)

Atom	Electronic Geometry	Molecular Geometry
Carbon (<i>e</i>)		
Sulfur (<i>f</i>)		

vi. Circle the value from the choices given below, that describes the **C-S-C bond angle at sulfur** (S) marked ***f***: (0.5 points)

$>90^\circ$ to $<109.5^\circ$ 109.5° >109.5 to $<120^\circ$ 120° $>120^\circ$ to $<180^\circ$


Question 8 12 points

Using the **valence shell orbitals only** draw a **complete** molecular orbital (MO) diagram (with no s-p mixing) for the molecule **PS**, in the space provided below. **(6 points)**

Assume that 3s and 3p atomic orbitals will form molecular orbitals like those formed by the combination of 2s and 2p atomic orbitals that we learnt in class, and there is no s-p mixing.

Show **energy arrow**, **atomic orbitals** involved, **atom labels**, the **molecular orbitals** that are formed, **filled electrons** in the MO diagram etc. Make sure to **label all the atomic and molecular orbitals**.

Using the MO diagram above, answer the following questions:

- (i) Which one of the following will have the highest bond order? Circle the correct choice
 A) PS^{1-} B) PS C) PS^{1+}
- (ii) Which one of the following will have the longest bond length? Circle the correct choice
 A) PS^{1-} B) PS C) PS^{1+}
- (iii) Which of the following will be diamagnetic? Circle the correct choice.
 A) PS^{1-} B) PS C) PS^{1+}
- (iv) Draw Lewis dot structure of the molecule with the shortest bond length (based on the MO diagram) from among the three (PS^{1-} , PS, PS^{1+}), and indicate the bond order calculated from the Lewis dot structure. **(3 points)**

Lewis Dot Structure

Bond Order



AB8BB486-E816-4F5C-997E-B5EEEBD61E5E

midterm-exam-93388

#368 Page 10 of 10

SCRAP PAGE

**CHEMISTRY 110 –Midterm Exam****Thursday November 7, 2024, 6:30 – 8:30 pm**

Examiners: Pallavi Sirjoosingh and Ashok Kakkar

First name (please write as legibly as possible within the boxes)

--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--

Last name

--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--

Student ID number

--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--

This booklet has 8 questions on 10 pages (The last page can be used as a scrap page).

The staple must NOT be removed from the question booklet. Your intact question booklet must be returned at the end of the exam.

A separate booklet contains a periodic table, and datasheet.

Write your answers directly under the questions in the space provided. Do not write on the QR codes or near the top edges (as it may be cut off).

Please write **legibly and dark enough** for the scan to be readable. For students who use English as a second language, dictionaries are permitted. Calculators are allowed.

DO NOT ASK QUESTIONS OF THE INVIGILATORS about interpretations of the exam questions. If you have a concern about a question, please summarize it at the top of this page. All concerns will be evaluated after the exam. GOOD LUCK!

**Question 1 8 points** (*Assume all transitions are possible*)

a. 6 points Calculate the **energy (in kJ/mol)** required to ionize (completely remove) the electron from the **n=3** level of **He⁺ ion** (He: Helium). Show your work including equations. (3 sig. figs.)

b. 2 points

(I) Select the largest (longest) wavelength from the three choices below:

- i. Wavelength of photon to *completely remove* an electron from *1s orbital of Hydrogen atom*
- ii. Wavelength of photon to *completely remove* an electron from *1s orbital of Boron atom (B)*
- iii. Wavelength of photon to *completely remove* an electron from *1s orbital of Boron cation (B⁺)*

i or ii or iii

(II) Select the smallest (shortest) wavelength from the three choices below (*for H atom*):

- i. Wavelength of photon *emitted* for an electron transitioning from *2p to 1s orbital*
- ii. Wavelength of photon *absorbed* for an electron transitioning from *2s to 4p orbital*
- iii. Wavelength of photon *absorbed* for an electron transitioning from *1s to 3p orbital*

i or ii or iii



Question 2 8 points



A Chem110 student plots the graph shown on the left, from data obtained for the kinetic energy (eV) of the ejected electron versus frequency (s^{-1}) of the incident light, for an unknown metal. ($1\text{eV} = 1.6022 \times 10^{-19} \text{ J}$)

Based on the plot, answer the questions **a to c** below:

a. 1 point On the plot above, label the **threshold frequency**.

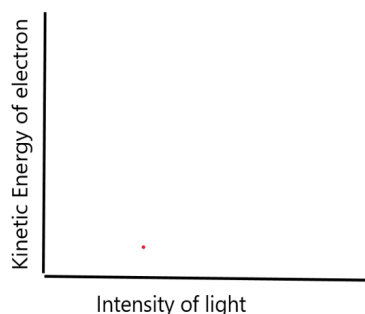
b. 2 points What is the value of the work potential (in J) of the unknown metal?

c. 3 points What is the *longest* wavelength (in nm) required to eject an electron from this metal's surface? Show your work including equation(s) used. (3 sig figs)

d. 2 points In the space provided below, plot the following

(Assume frequency is *above* threshold frequency):

- Kinetic energy* of an ejected electron versus the *intensity* of the incident light (left)
- Number of electrons* ejected versus the *frequency* of the incident light (right)



**Question 3** **6 points****a. 2 points**

In the ground state of Ge (Germanium; $Z = 32$) atom, **name** the following orbitals:

i. Orbital containing the electron that requires the maximum energy to be removed.

ii. Orbital containing the electron that experiences the minimum effective nuclear charge.

b. 2 points

An unknown anion (A^-) has the following *excited* electronic configuration: $[Ar] 4s^2 3d^3 4p^1$

(Ar: Argon; $Z = 18$)

Give the **condensed ground state electronic configuration** of the atom, **A** and the cation, **A^{2+}** .
Show your work. (Assume all electronic transitions are possible)

Electronic Configuration of A:

Electronic Configuration of A^{2+} :

c. 2 points

In a single atom, determine how many electrons can have the following set of quantum numbers?

(i) $n = 5; l = 3$

In a single atom, determine how many electrons can have the following set of quantum numbers?

(i) $n = 3; l = 2; m_s = +1/2$

**Question 4** **8 points**

Rank the following from 1 to 4 in **increasing** order. Provide the ranking in the boxes provided.

Note: Atomic number (Z) are provided for relevant atoms

a) Increasing (lowest to highest) order of *ionization energy*: (1= lowest; 4 = highest)

Mg

Mg: Z = 12

Rb

Rb: Z = 37

Cs

Cs: Z = 55

K

K : Z = 19

b) Increasing (lowest to highest) order of *bond angle*: (1= lowest; 4 = highest)

H₂OCCl₄XeF₄

HCN

c) Increasing (smallest to largest) order of *ionic size* (1= smallest; 4 = largest)

S²⁻

S: Z = 16

O²⁻

O: Z = 8

Be²⁺

Be: Z = 4

Na⁺

Na: Z = 11

d) Increasing (lowest to highest) order of *bond energy* (1= lowest; 4 = highest)

P-F

P-Cl

P-I

P-Br

**Question 5** **12 points**

a. 9 points For the molecules HClO_4 and TeF_6 , draw the most stable *Lewis structure* showing appropriate shape and *dashed/wedged bonds*, based on VSEPR. Determine the **molecular geometry** around the central atom. For HClO_4 , label the *most polar bond*.

HClO_4 (there are no O-O bonds)

Label the most polar bond

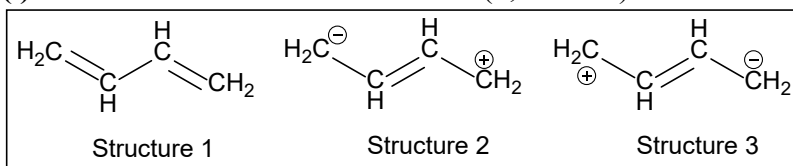
Molecular Geometry around Cl

TeF_6

Molecular Geometry around Te

b. 3 points (Note: Lone pairs are not shown in structures below)

(i) For the three *resonance* structures (1, 2 and 3) shown below, answer the following question:



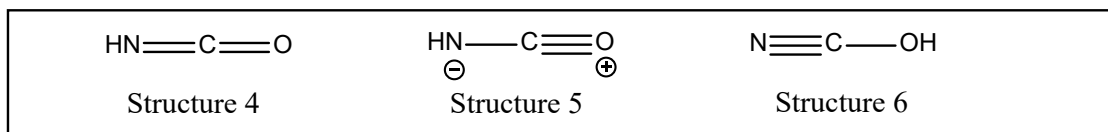
(I) Circle the *most contributing* resonance structure.

Structure 1

Structure 2

Structure 3

(ii) For the three structures given (4, 5, and 6) below, answer the following questions:



(I) Circle all the structures that represent *resonance structures* of each other:

Structure 4

Structure 5

Structure 6

(II) Among the resonance structures, circle the *most contributing* resonance structure.

Structure 4

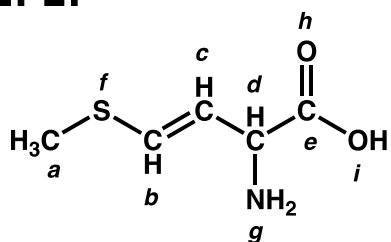
Structure 5

Structure 6

**Question 6** **6 points**

Using the data given below, calculate the **lattice energy** for SrO(s).

	ΔH° (kJ/mol)
$\text{Sr}(s) \longrightarrow \text{Sr}(g)$	+180
$\text{Sr}(g) \longrightarrow \text{Sr}^{2+}(g) + 2e^-$	+1468
$\text{O}_2(g) \longrightarrow 2\text{O}(g)$	+500
$\text{O}(g) + 2e^- \longrightarrow \text{O}^{2-}(g)$	+600
$\text{Sr}(s) + \frac{1}{2}\text{O}_2(g) \longrightarrow \text{SrO}(s)$	-550

**Question 7 10 points**

Consider the molecule given on the left, and answer questions (i) to (vi) using chemical bonding theories that we have learnt in class (valence bond theory, hybridization).

Note that the lone pairs are not shown in the structure:

i. Consider the carbon atoms marked in as ***a***, ***b***, ***c***, ***d***, and ***e*** in the molecule, and determine **hybridization** at each of the carbon centers. (2.5 points)

Atom	Hybridization
Carbon (<i>a</i>)	
Carbon (<i>b</i>)	
Carbon (<i>c</i>)	
Carbon (<i>d</i>)	
Carbon (<i>e</i>)	

ii. Now, consider the sulfur atom (S) marked as ***f***, nitrogen (N) atom (***g***), and oxygen(O) atoms (***h*** and ***i***), and determine their **hybridization**. (2 points)

Atom	Hybridization
Sulfur (<i>f</i>)	
Nitrogen (<i>g</i>)	
Oxygen (<i>h</i>)	
Oxygen (<i>i</i>)	

iii. **Describe** all the bonds formed by the carbon atoms ***a*** and ***b***, using sigma and pi designation as appropriate. (2 points)

Atom	Bonds formed (use sigma and pi as appropriate)
Carbon (<i>a</i>)	
Carbon (<i>b</i>)	

iv. How many total sp^3 -s sigma bonds are in the molecule? (1 point)

v. Determine **electronic** and **molecular geometry** at carbon (C) labeled ***e*** and sulfur (S) labeled ***f*** centres. (2 points)

Atom	Electronic Geometry	Molecular Geometry
Carbon (<i>e</i>)		
Sulfur (<i>f</i>)		

vi. Circle the value from the choices given below, that describes the **C-S-C bond angle at sulfur** (S) marked ***f***: (0.5 points)

>90° to <109.5° 109.5° >109.5 to <120° 120° >120° to <180°

**Question 8 12 points**

Using the **valence shell orbitals only** draw a **complete** molecular orbital (MO) diagram (with no s-p mixing) for the molecule **PS**, in the space provided below. **(6 points)**

Assume that 3s and 3p atomic orbitals will form molecular orbitals like those formed by the combination of 2s and 2p atomic orbitals that we learnt in class, and there is no s-p mixing.

Show **energy arrow**, **atomic orbitals** involved, **atom labels**, the **molecular orbitals** that are formed, **filled electrons** in the MO diagram etc. Make sure to label all the atomic and molecular orbitals.

Using the MO diagram above, answer the following questions:

- (i) Which one of the following will have the highest bond order? Circle the correct choice
A) PS^{1-} B) PS C) PS^{1+}
- (ii) Which one of the following will have the longest bond length? Circle the correct choice
A) PS^{1-} B) PS C) PS^{1+}
- (iii) Which of the following will be diamagnetic? Circle the correct choice.
A) PS^{1-} B) PS C) PS^{1+}
- (iv) Draw Lewis dot structure of the molecule with the shortest bond length (based on the MO diagram) from among the three (PS^{1-} , PS, PS^{1+}), and indicate the bond order calculated from the Lewis dot structure. **(3 points)**

Lewis Dot Structure

Bond Order



A541D1C2-0A85-4895-912B-2DAECB0C633D

midterm-exam-93388

#369 Page 10 of 10

SCRAP PAGE

**CHEMISTRY 110 –Midterm Exam****Thursday November 7, 2024, 6:30 – 8:30 pm****Examiners: Pallavi Sirjoosingh and Ashok Kakkar**

First name (please write as legibly as possible within the boxes)

--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--

Last name

--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--

Student ID number

--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--

This booklet has 8 questions on 10 pages (The last page can be used as a scrap page).

The staple must NOT be removed from the question booklet. Your intact question booklet must be returned at the end of the exam.

A separate booklet contains a periodic table, and datasheet.

Write your answers directly under the questions in the space provided. Do not write on the QR codes or near the top edges (as it may be cut off).

Please write **legibly and dark enough** for the scan to be readable. For students who use English as a second language, dictionaries are permitted. Calculators are allowed.

DO NOT ASK QUESTIONS OF THE INVIGILATORS about interpretations of the exam questions. If you have a concern about a question, please summarize it at the top of this page. All concerns will be evaluated after the exam. GOOD LUCK!

**Question 1 8 points** (*Assume all transitions are possible*)

a. 6 points Calculate the **energy (in kJ/mol)** required to ionize (completely remove) the electron from the **n=3** level of **He⁺ ion** (He: Helium). Show your work including equations. (3 sig. figs.)

b. 2 points

(I) Select the largest (longest) wavelength from the three choices below:

- i. Wavelength of photon to *completely remove* an electron from *1s orbital of Hydrogen atom*
- ii. Wavelength of photon to *completely remove* an electron from *1s orbital of Boron atom (B)*
- iii. Wavelength of photon to *completely remove* an electron from *1s orbital of Boron cation (B⁺)*

i or ii or iii

(II) Select the smallest (shortest) wavelength from the three choices below (*for H atom*):

- i. Wavelength of photon *emitted* for an electron transitioning from *2p to 1s orbital*
- ii. Wavelength of photon *absorbed* for an electron transitioning from *2s to 4p orbital*
- iii. Wavelength of photon *absorbed* for an electron transitioning from *1s to 3p orbital*

i or ii or iii



Question 2 8 points



A Chem110 student plots the graph shown on the left, from data obtained for the kinetic energy (eV) of the ejected electron versus frequency (s^{-1}) of the incident light, for an unknown metal. ($1\text{eV} = 1.6022 \times 10^{-19} \text{ J}$)

Based on the plot, answer the questions **a to c** below:

a. 1 point On the plot above, label the **threshold frequency**.

b. 2 points What is the value of the work potential (in J) of the unknown metal?

c. 3 points What is the *longest* wavelength (in nm) required to eject an electron from this metal's surface? Show your work including equation(s) used. (3 sig figs)

d. 2 points In the space provided below, plot the following

(Assume frequency is *above* threshold frequency):

- i. *Kinetic energy* of an ejected electron versus the *intensity* of the incident light (left)
- ii. *Number of electrons* ejected versus the *frequency* of the incident light (right)



**Question 3** **6 points****a. 2 points**

In the ground state of Ge (Germanium; $Z = 32$) atom, **name** the following orbitals:

i. Orbital containing the electron that requires the maximum energy to be removed.

ii. Orbital containing the electron that experiences the minimum effective nuclear charge.

b. 2 points

An unknown anion (A^-) has the following *excited* electronic configuration: $[Ar] 4s^2 3d^3 4p^1$

(Ar: Argon; $Z = 18$)

Give the **condensed ground state electronic configuration** of the atom, **A** and the cation, **A^{2+}** .
Show your work. (Assume all electronic transitions are possible)

Electronic Configuration of A:

Electronic Configuration of A^{2+} :

c. 2 points

In a single atom, determine how many electrons can have the following set of quantum numbers?

(i) $n = 5; l = 3$

In a single atom, determine how many electrons can have the following set of quantum numbers?

(i) $n = 3; l = 2; m_s = +1/2$

**Question 4** **8 points**

Rank the following from 1 to 4 in **increasing** order. Provide the ranking in the boxes provided.

Note: Atomic number (Z) are provided for relevant atoms

a) Increasing (lowest to highest) order of *ionization energy*: (1= lowest; 4 = highest)

Mg

Mg: Z = 12

Rb

Rb: Z = 37

Cs

Cs: Z = 55

K

K : Z = 19

b) Increasing (lowest to highest) order of *bond angle*: (1= lowest; 4 = highest)

H₂OCCl₄XeF₄

HCN

c) Increasing (smallest to largest) order of *ionic size* (1= smallest; 4 = largest)

S²⁻

S: Z = 16

O²⁻

O: Z = 8

Be²⁺

Be: Z = 4

Na⁺

Na: Z = 11

d) Increasing (lowest to highest) order of *bond energy* (1= lowest; 4 = highest)

P-F

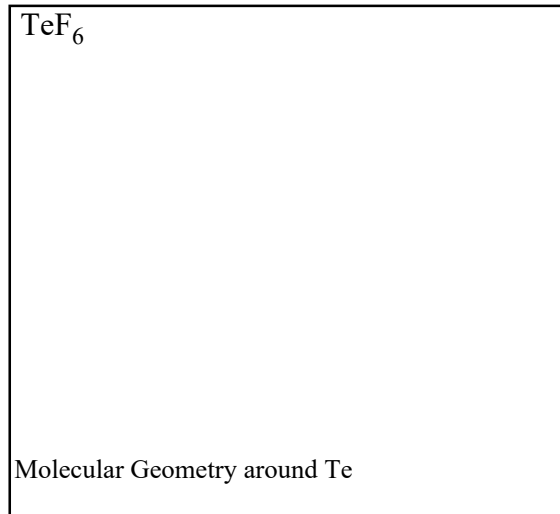
P-Cl

P-I

P-Br

**Question 5** **12 points**

a. 9 points For the molecules HClO_4 and TeF_6 , draw the most stable *Lewis structure* showing appropriate shape and *dashed/wedged bonds*, based on VSEPR. Determine the **molecular geometry** around the central atom. For HClO_4 , label the *most polar bond*.



b. 3 points (Note: Lone pairs are not shown in structures below)

(i) For the three *resonance* structures (1, 2 and 3) shown below, answer the following question:



(I) Circle the *most contributing* resonance structure.

Structure 1

Structure 2

Structure 3

(ii) For the three structures given (4, 5, and 6) below, answer the following questions:



(I) Circle all the structures that represent *resonance structures* of each other:

Structure 4

Structure 5

Structure 6

(II) Among the resonance structures, circle the *most contributing* resonance structure.

Structure 4

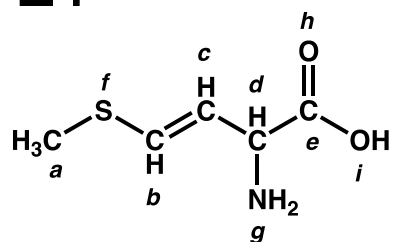
Structure 5

Structure 6

**Question 6** **6 points**

Using the data given below, calculate the **lattice energy** for SrO(s).

	ΔH° (kJ/mol)
$\text{Sr}(s) \longrightarrow \text{Sr}(g)$	+180
$\text{Sr}(g) \longrightarrow \text{Sr}^{2+}(g) + 2e^-$	+1468
$\text{O}_2(g) \longrightarrow 2\text{O}(g)$	+500
$\text{O}(g) + 2e^- \longrightarrow \text{O}^{2-}(g)$	+600
$\text{Sr}(s) + \frac{1}{2}\text{O}_2(g) \longrightarrow \text{SrO}(s)$	-550

**Question 7 10 points**

Consider the molecule given on the left, and answer questions (i) to (vi) using chemical bonding theories that we have learnt in class (valence bond theory, hybridization).

Note that the lone pairs are not shown in the structure:

i. Consider the carbon atoms marked in as ***a***, ***b***, ***c***, ***d***, and ***e*** in the molecule, and determine **hybridization** at each of the carbon centers. (2.5 points)

Atom	Hybridization
Carbon (<i>a</i>)	
Carbon (<i>b</i>)	
Carbon (<i>c</i>)	
Carbon (<i>d</i>)	
Carbon (<i>e</i>)	

ii. Now, consider the sulfur atom (S) marked as ***f***, nitrogen (N) atom (***g***), and oxygen(O) atoms (***h*** and ***i***), and determine their **hybridization**. (2 points)

Atom	Hybridization
Sulfur (<i>f</i>)	
Nitrogen (<i>g</i>)	
Oxygen (<i>h</i>)	
Oxygen (<i>i</i>)	

iii. **Describe** all the bonds formed by the carbon atoms ***a*** and ***b***, using sigma and pi designation as appropriate. (2 points)

Atom	Bonds formed (use sigma and pi as appropriate)
Carbon (<i>a</i>)	
Carbon (<i>b</i>)	

iv. How many total sp^3 -s sigma bonds are in the molecule? (1 point)

v. Determine **electronic** and **molecular geometry** at carbon (C) labeled ***e*** and sulfur (S) labeled ***f*** centres. (2 points)

Atom	Electronic Geometry	Molecular Geometry
Carbon (<i>e</i>)		
Sulfur (<i>f</i>)		

vi. Circle the value from the choices given below, that describes the **C-S-C bond angle at sulfur** (S) marked ***f***: (0.5 points)

>90° to <109.5° 109.5° >109.5 to <120° 120° >120° to <180°


Question 8 12 points

Using the **valence shell orbitals only** draw a **complete** molecular orbital (MO) diagram (with no s-p mixing) for the molecule **PS**, in the space provided below. **(6 points)**

Assume that 3s and 3p atomic orbitals will form molecular orbitals like those formed by the combination of 2s and 2p atomic orbitals that we learnt in class, and there is no s-p mixing.

Show **energy arrow**, **atomic orbitals** involved, **atom labels**, the **molecular orbitals** that are formed, **filled electrons** in the MO diagram etc. Make sure to label all the atomic and molecular orbitals.

Using the MO diagram above, answer the following questions:

- (i) Which one of the following will have the highest bond order? Circle the correct choice
 A) PS^{1-} B) PS C) PS^{1+}
- (ii) Which one of the following will have the longest bond length? Circle the correct choice
 A) PS^{1-} B) PS C) PS^{1+}
- (iii) Which of the following will be diamagnetic? Circle the correct choice.
 A) PS^{1-} B) PS C) PS^{1+}
- (iv) Draw Lewis dot structure of the molecule with the shortest bond length (based on the MO diagram) from among the three (PS^{1-} , PS, PS^{1+}), and indicate the bond order calculated from the Lewis dot structure. **(3 points)**

Lewis Dot Structure

Bond Order



F066C164-D1AD-46E6-94CD-52F334DFAB78

midterm-exam-93388

#370 Page 10 of 10

SCRAP PAGE

**CHEMISTRY 110 –Midterm Exam****Thursday November 7, 2024, 6:30 – 8:30 pm****Examiners: Pallavi Sirjoosingh and Ashok Kakkar**

First name (please write as legibly as possible within the boxes)

--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--

Last name

--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--

Student ID number

--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--

This booklet has 8 questions on 10 pages (The last page can be used as a scrap page).

The staple must NOT be removed from the question booklet. Your intact question booklet must be returned at the end of the exam.

A separate booklet contains a periodic table, and datasheet.

Write your answers directly under the questions in the space provided. Do not write on the QR codes or near the top edges (as it may be cut off).

Please write **legibly and dark enough** for the scan to be readable. For students who use English as a second language, dictionaries are permitted. Calculators are allowed.

DO NOT ASK QUESTIONS OF THE INVIGILATORS about interpretations of the exam questions. If you have a concern about a question, please summarize it at the top of this page. All concerns will be evaluated after the exam. GOOD LUCK!

**Question 1 8 points** (*Assume all transitions are possible*)

a. 6 points Calculate the **energy (in kJ/mol)** required to ionize (completely remove) the electron from the **n=3** level of **He⁺ ion** (He: Helium). Show your work including equations. (3 sig. figs.)

b. 2 points

(I) Select the largest (longest) wavelength from the three choices below:

- i. Wavelength of photon to *completely remove* an electron from *1s orbital of Hydrogen atom*
- ii. Wavelength of photon to *completely remove* an electron from *1s orbital of Boron atom (B)*
- iii. Wavelength of photon to *completely remove* an electron from *1s orbital of Boron cation (B⁺)*

i or ii or iii

(II) Select the smallest (shortest) wavelength from the three choices below (*for H atom*):

- i. Wavelength of photon *emitted* for an electron transitioning from *2p to 1s orbital*
- ii. Wavelength of photon *absorbed* for an electron transitioning from *2s to 4p orbital*
- iii. Wavelength of photon *absorbed* for an electron transitioning from *1s to 3p orbital*

i or ii or iii



Question 2 8 points



A Chem110 student plots the graph shown on the left, from data obtained for the kinetic energy (eV) of the ejected electron versus frequency (s^{-1}) of the incident light, for an unknown metal. ($1\text{eV} = 1.6022 \times 10^{-19} \text{ J}$)

Based on the plot, answer the questions **a to c** below:

a. 1 point On the plot above, label the **threshold frequency**.

b. 2 points What is the value of the work potential (in J) of the unknown metal?

c. 3 points What is the *longest* wavelength (in nm) required to eject an electron from this metal's surface? Show your work including equation(s) used. (3 sig figs)

d. 2 points In the space provided below, plot the following

(Assume frequency is *above* threshold frequency):

- i. *Kinetic energy* of an ejected electron versus the *intensity* of the incident light (left)
- ii. *Number of electrons* ejected versus the *frequency* of the incident light (right)



**Question 3** **6 points****a. 2 points**

In the ground state of Ge (Germanium; $Z = 32$) atom, **name** the following orbitals:

i. Orbital containing the electron that requires the maximum energy to be removed.

ii. Orbital containing the electron that experiences the minimum effective nuclear charge.

b. 2 points

An unknown anion (A^-) has the following *excited* electronic configuration: $[Ar] 4s^2 3d^3 4p^1$

(Ar: Argon; $Z = 18$)

Give the **condensed ground state electronic configuration** of the atom, **A** and the cation, **A^{2+}** .
Show your work. (Assume all electronic transitions are possible)

Electronic Configuration of A:

Electronic Configuration of A^{2+} :

c. 2 points

In a single atom, determine how many electrons can have the following set of quantum numbers?

(i) $n = 5; l = 3$

In a single atom, determine how many electrons can have the following set of quantum numbers?

(i) $n = 3; l = 2; m_s = +1/2$

**Question 4** **8 points**

Rank the following from 1 to 4 in **increasing** order. Provide the ranking in the boxes provided.

Note: Atomic number (Z) are provided for relevant atoms

a) Increasing (lowest to highest) order of *ionization energy*: (1= lowest; 4 = highest)

Mg

Mg: Z = 12

Rb

Rb: Z = 37

Cs

Cs: Z = 55

K

K : Z = 19

b) Increasing (lowest to highest) order of *bond angle*: (1= lowest; 4 = highest)

H₂OCCl₄XeF₄

HCN

c) Increasing (smallest to largest) order of *ionic size* (1= smallest; 4 = largest)

S²⁻

S: Z = 16

O²⁻

O: Z = 8

Be²⁺

Be: Z = 4

Na⁺

Na: Z = 11

d) Increasing (lowest to highest) order of *bond energy* (1= lowest; 4 = highest)

P-F

P-Cl

P-I

P-Br

**Question 5** **12 points**

a. 9 points For the molecules HClO_4 and TeF_6 , draw the most stable *Lewis structure* showing appropriate shape and *dashed/wedged bonds*, based on VSEPR. Determine the **molecular geometry** around the central atom. For HClO_4 , label the *most polar bond*.



b. 3 points (Note: Lone pairs are not shown in structures below)

(i) For the three *resonance* structures (1, 2 and 3) shown below, answer the following question:



(I) Circle the *most contributing* resonance structure.

Structure 1

Structure 2

Structure 3

(ii) For the three structures given (4, 5, and 6) below, answer the following questions:



(I) Circle all the structures that represent *resonance structures* of each other:

Structure 4

Structure 5

Structure 6

(II) Among the resonance structures, circle the *most contributing* resonance structure.

Structure 4

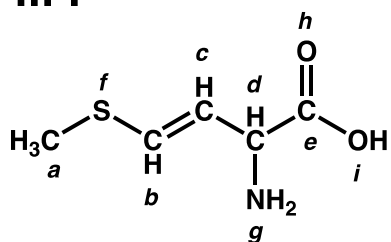
Structure 5

Structure 6

**Question 6** **6 points**

Using the data given below, calculate the **lattice energy** for SrO(s).

	ΔH° (kJ/mol)
$\text{Sr}(s) \longrightarrow \text{Sr}(g)$	+180
$\text{Sr}(g) \longrightarrow \text{Sr}^{2+}(g) + 2e^-$	+1468
$\text{O}_2(g) \longrightarrow 2\text{O}(g)$	+500
$\text{O}(g) + 2e^- \longrightarrow \text{O}^{2-}(g)$	+600
$\text{Sr}(s) + \frac{1}{2}\text{O}_2(g) \longrightarrow \text{SrO}(s)$	-550

**Question 7 10 points**

Consider the molecule given on the left, and answer questions (i) to (vi) using chemical bonding theories that we have learnt in class (valence bond theory, hybridization).

Note that the lone pairs are not shown in the structure:

i. Consider the carbon atoms marked in as ***a***, ***b***, ***c***, ***d***, and ***e*** in the molecule, and determine **hybridization** at each of the carbon centers. (2.5 points)

Atom	Hybridization
Carbon (<i>a</i>)	
Carbon (<i>b</i>)	
Carbon (<i>c</i>)	
Carbon (<i>d</i>)	
Carbon (<i>e</i>)	

ii. Now, consider the sulfur atom (S) marked as ***f***, nitrogen (N) atom (***g***), and oxygen(O) atoms (***h*** and ***i***), and determine their **hybridization**. (2 points)

Atom	Hybridization
Sulfur (<i>f</i>)	
Nitrogen (<i>g</i>)	
Oxygen (<i>h</i>)	
Oxygen (<i>i</i>)	

iii. **Describe** all the bonds formed by the carbon atoms ***a*** and ***b***, using sigma and pi designation as appropriate. (2 points)

Atom	Bonds formed (use sigma and pi as appropriate)
Carbon (<i>a</i>)	
Carbon (<i>b</i>)	

iv. How many total sp^3 -s sigma bonds are in the molecule? (1 point)

v. Determine **electronic** and **molecular geometry** at carbon (C) labeled ***e*** and sulfur (S) labeled ***f*** centres. (2 points)

Atom	Electronic Geometry	Molecular Geometry
Carbon (<i>e</i>)		
Sulfur (<i>f</i>)		

vi. Circle the value from the choices given below, that describes the **C-S-C bond angle at sulfur** (S) marked ***f***: (0.5 points)

>90° to <109.5° 109.5° >109.5 to <120° 120° >120° to <180°


Question 8 12 points

Using the **valence shell orbitals only** draw a **complete** molecular orbital (MO) diagram (with no s-p mixing) for the molecule **PS**, in the space provided below. **(6 points)**

Assume that 3s and 3p atomic orbitals will form molecular orbitals like those formed by the combination of 2s and 2p atomic orbitals that we learnt in class, and there is no s-p mixing.

Show **energy arrow**, **atomic orbitals** involved, **atom labels**, the **molecular orbitals** that are formed, **filled electrons** in the MO diagram etc. Make sure to label all the atomic and molecular orbitals.

Using the MO diagram above, answer the following questions:

- (i) Which one of the following will have the highest bond order? Circle the correct choice
 A) PS^{1-} B) PS C) PS^{1+}
- (ii) Which one of the following will have the longest bond length? Circle the correct choice
 A) PS^{1-} B) PS C) PS^{1+}
- (iii) Which of the following will be diamagnetic? Circle the correct choice.
 A) PS^{1-} B) PS C) PS^{1+}
- (iv) Draw Lewis dot structure of the molecule with the shortest bond length (based on the MO diagram) from among the three (PS^{1-} , PS, PS^{1+}), and indicate the bond order calculated from the Lewis dot structure. **(3 points)**

Lewis Dot Structure

Bond Order



BD57478D-1041-4D54-A2CD-4CB634E922F4

midterm-exam-93388

#371 Page 10 of 10

SCRAP PAGE

**CHEMISTRY 110 –Midterm Exam****Thursday November 7, 2024, 6:30 – 8:30 pm**

Examiners: Pallavi Sirjoosingh and Ashok Kakkar

First name (please write as legibly as possible within the boxes)

--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--

Last name

--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--

Student ID number

--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--

This booklet has 8 questions on 10 pages (The last page can be used as a scrap page).

The staple must NOT be removed from the question booklet. Your intact question booklet must be returned at the end of the exam.

A separate booklet contains a periodic table, and datasheet.

Write your answers directly under the questions in the space provided. Do not write on the QR codes or near the top edges (as it may be cut off).

Please write **legibly and dark enough** for the scan to be readable. For students who use English as a second language, dictionaries are permitted. Calculators are allowed.

DO NOT ASK QUESTIONS OF THE INVIGILATORS about interpretations of the exam questions. If you have a concern about a question, please summarize it at the top of this page. All concerns will be evaluated after the exam. GOOD LUCK!

**Question 1 8 points** (*Assume all transitions are possible*)

a. 6 points Calculate the **energy (in kJ/mol)** required to ionize (completely remove) the electron from the **n=3** level of **He⁺ ion** (He: Helium). Show your work including equations. (3 sig. figs.)

b. 2 points

(I) Select the largest (longest) wavelength from the three choices below:

- i. Wavelength of photon to *completely remove* an electron from *1s orbital of Hydrogen atom*
- ii. Wavelength of photon to *completely remove* an electron from *1s orbital of Boron atom (B)*
- iii. Wavelength of photon to *completely remove* an electron from *1s orbital of Boron cation (B⁺)*

i or ii or iii

(II) Select the smallest (shortest) wavelength from the three choices below (*for H atom*):

- i. Wavelength of photon *emitted* for an electron transitioning from *2p to 1s orbital*
- ii. Wavelength of photon *absorbed* for an electron transitioning from *2s to 4p orbital*
- iii. Wavelength of photon *absorbed* for an electron transitioning from *1s to 3p orbital*

i or ii or iii



Question 2 8 points



A Chem110 student plots the graph shown on the left, from data obtained for the kinetic energy (eV) of the ejected electron versus frequency (s^{-1}) of the incident light, for an unknown metal. ($1\text{eV} = 1.6022 \times 10^{-19} \text{ J}$)

Based on the plot, answer the questions **a to c** below:

a. 1 point On the plot above, label the **threshold frequency**.

b. 2 points What is the value of the work potential (in J) of the unknown metal?

c. 3 points What is the *longest* wavelength (in nm) required to eject an electron from this metal's surface? Show your work including equation(s) used. (3 sig figs)

d. 2 points In the space provided below, plot the following

(Assume frequency is *above* threshold frequency):

- i. *Kinetic energy* of an ejected electron versus the *intensity* of the incident light (left)
- ii. *Number of electrons* ejected versus the *frequency* of the incident light (right)



**Question 3** **6 points****a. 2 points**

In the ground state of Ge (Germanium; $Z = 32$) atom, **name** the following orbitals:

i. Orbital containing the electron that requires the maximum energy to be removed.

ii. Orbital containing the electron that experiences the minimum effective nuclear charge.

b. 2 points

An unknown anion (A^-) has the following *excited* electronic configuration: $[Ar] 4s^2 3d^3 4p^1$

(Ar: Argon; $Z = 18$)

Give the **condensed ground state electronic configuration** of the atom, **A** and the cation, **A^{2+}** .
Show your work. (Assume all electronic transitions are possible)

Electronic Configuration of A:

Electronic Configuration of A^{2+} :

c. 2 points

In a single atom, determine how many electrons can have the following set of quantum numbers?

(i) $n = 5; l = 3$

In a single atom, determine how many electrons can have the following set of quantum numbers?

(i) $n = 3; l = 2; m_s = +1/2$

**Question 4** **8 points**

Rank the following from 1 to 4 in **increasing** order. Provide the ranking in the boxes provided.

Note: Atomic number (Z) are provided for relevant atoms

a) Increasing (lowest to highest) order of *ionization energy*: (1= lowest; 4 = highest)

Mg

Mg: Z = 12

Rb

Rb: Z = 37

Cs

Cs: Z = 55

K

K : Z = 19

b) Increasing (lowest to highest) order of *bond angle*: (1= lowest; 4 = highest)

H₂OCCl₄XeF₄

HCN

c) Increasing (smallest to largest) order of *ionic size* (1= smallest; 4 = largest)

S²⁻

S: Z = 16

O²⁻

O: Z = 8

Be²⁺

Be: Z = 4

Na⁺

Na: Z = 11

d) Increasing (lowest to highest) order of *bond energy* (1= lowest; 4 = highest)

P-F

P-Cl

P-I

P-Br

**Question 5** **12 points**

a. 9 points For the molecules HClO_4 and TeF_6 , draw the most stable *Lewis structure* showing appropriate shape and *dashed/wedged bonds*, based on VSEPR. Determine the **molecular geometry** around the central atom. For HClO_4 , label the *most polar bond*.

HClO_4 (there are no O-O bonds)

Label the most polar bond

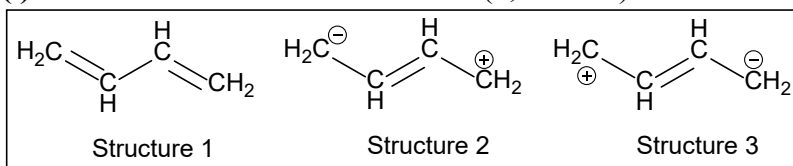
Molecular Geometry around Cl

TeF_6

Molecular Geometry around Te

b. 3 points (Note: Lone pairs are not shown in structures below)

(i) For the three *resonance* structures (1, 2 and 3) shown below, answer the following question:



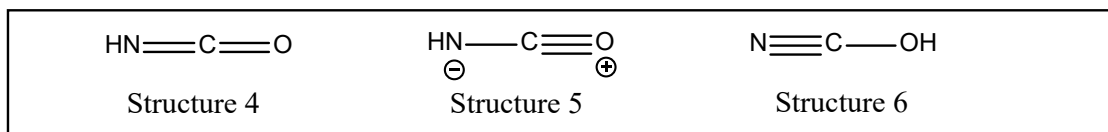
(I) Circle the *most contributing* resonance structure.

Structure 1

Structure 2

Structure 3

(ii) For the three structures given (4, 5, and 6) below, answer the following questions:



(I) Circle all the structures that represent *resonance structures* of each other:

Structure 4

Structure 5

Structure 6

(II) Among the resonance structures, circle the *most contributing* resonance structure.

Structure 4

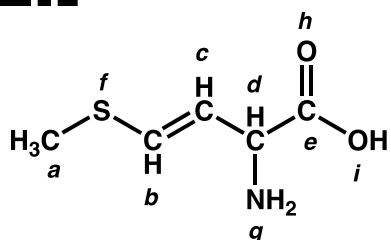
Structure 5

Structure 6

**Question 6** **6 points**

Using the data given below, calculate the **lattice energy** for SrO(s).

	ΔH° (kJ/mol)
$\text{Sr}(s) \longrightarrow \text{Sr}(g)$	+180
$\text{Sr}(g) \longrightarrow \text{Sr}^{2+}(g) + 2e^-$	+1468
$\text{O}_2(g) \longrightarrow 2\text{O}(g)$	+500
$\text{O}(g) + 2e^- \longrightarrow \text{O}^{2-}(g)$	+600
$\text{Sr}(s) + \frac{1}{2}\text{O}_2(g) \longrightarrow \text{SrO}(s)$	-550

**Question 7 10 points**

Consider the molecule given on the left, and answer questions (i) to (vi) using chemical bonding theories that we have learnt in class (valence bond theory, hybridization).

Note that the lone pairs are not shown in the structure:

i. Consider the carbon atoms marked in as ***a***, ***b***, ***c***, ***d***, and ***e*** in the molecule, and determine **hybridization** at each of the carbon centers. (2.5 points)

Atom	Hybridization
Carbon (<i>a</i>)	
Carbon (<i>b</i>)	
Carbon (<i>c</i>)	
Carbon (<i>d</i>)	
Carbon (<i>e</i>)	

ii. Now, consider the sulfur atom (S) marked as ***f***, nitrogen (N) atom (***g***), and oxygen(O) atoms (***h*** and ***i***), and determine their **hybridization**. (2 points)

Atom	Hybridization
Sulfur (<i>f</i>)	
Nitrogen (<i>g</i>)	
Oxygen (<i>h</i>)	
Oxygen (<i>i</i>)	

iii. **Describe** all the bonds formed by the carbon atoms ***a*** and ***b***, using sigma and pi designation as appropriate. (2 points)

Atom	Bonds formed (use sigma and pi as appropriate)
Carbon (<i>a</i>)	
Carbon (<i>b</i>)	

iv. How many total sp^3 -s sigma bonds are in the molecule? (1 point)

v. Determine **electronic** and **molecular geometry** at carbon (C) labeled ***e*** and sulfur (S) labeled ***f*** centres. (2 points)

Atom	Electronic Geometry	Molecular Geometry
Carbon (<i>e</i>)		
Sulfur (<i>f</i>)		

vi. Circle the value from the choices given below, that describes the **C-S-C bond angle at sulfur** (S) marked ***f***: (0.5 points)

$>90^\circ$ to $<109.5^\circ$ 109.5° >109.5 to $<120^\circ$ 120° $>120^\circ$ to $<180^\circ$


Question 8 12 points

Using the **valence shell orbitals only** draw a **complete** molecular orbital (MO) diagram (with no s-p mixing) for the molecule **PS**, in the space provided below. **(6 points)**

Assume that 3s and 3p atomic orbitals will form molecular orbitals like those formed by the combination of 2s and 2p atomic orbitals that we learnt in class, and there is no s-p mixing.

Show **energy arrow**, **atomic orbitals** involved, **atom labels**, the **molecular orbitals** that are formed, **filled electrons** in the MO diagram etc. Make sure to label all the atomic and molecular orbitals.

Using the MO diagram above, answer the following questions:

- (i) Which one of the following will have the highest bond order? Circle the correct choice
 A) PS^{1-} B) PS C) PS^{1+}
- (ii) Which one of the following will have the longest bond length? Circle the correct choice
 A) PS^{1-} B) PS C) PS^{1+}
- (iii) Which of the following will be diamagnetic? Circle the correct choice.
 A) PS^{1-} B) PS C) PS^{1+}
- (iv) Draw Lewis dot structure of the molecule with the shortest bond length (based on the MO diagram) from among the three (PS^{1-} , PS, PS^{1+}), and indicate the bond order calculated from the Lewis dot structure. **(3 points)**

Lewis Dot Structure

Bond Order



797BD3DE-586C-4A51-9279-9675CCA67D22

midterm-exam-93388

#372 Page 10 of 10

SCRAP PAGE

**CHEMISTRY 110 –Midterm Exam****Thursday November 7, 2024, 6:30 – 8:30 pm****Examiners: Pallavi Sirjoosingh and Ashok Kakkar**

First name (please write as legibly as possible within the boxes)

--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--

Last name

--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--

Student ID number

--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--

This booklet has 8 questions on 10 pages (The last page can be used as a scrap page).

The staple must NOT be removed from the question booklet. Your intact question booklet must be returned at the end of the exam.

A separate booklet contains a periodic table, and datasheet.

Write your answers directly under the questions in the space provided. Do not write on the QR codes or near the top edges (as it may be cut off).

Please write **legibly and dark enough** for the scan to be readable. For students who use English as a second language, dictionaries are permitted. Calculators are allowed.

DO NOT ASK QUESTIONS OF THE INVIGILATORS about interpretations of the exam questions. If you have a concern about a question, please summarize it at the top of this page. All concerns will be evaluated after the exam. GOOD LUCK!

**Question 1 8 points** (*Assume all transitions are possible*)

a. 6 points Calculate the **energy (in kJ/mol)** required to ionize (completely remove) the electron from the **n=3** level of **He⁺ ion** (He: Helium). Show your work including equations. (3 sig. figs.)

b. 2 points

(I) Select the largest (longest) wavelength from the three choices below:

- i. Wavelength of photon to *completely remove* an electron from *1s orbital of Hydrogen atom*
- ii. Wavelength of photon to *completely remove* an electron from *1s orbital of Boron atom (B)*
- iii. Wavelength of photon to *completely remove* an electron from *1s orbital of Boron cation (B⁺)*

i or ii or iii

(II) Select the smallest (shortest) wavelength from the three choices below (*for H atom*):

- i. Wavelength of photon *emitted* for an electron transitioning from *2p to 1s orbital*
- ii. Wavelength of photon *absorbed* for an electron transitioning from *2s to 4p orbital*
- iii. Wavelength of photon *absorbed* for an electron transitioning from *1s to 3p orbital*

i or ii or iii



Question 2 8 points



A Chem110 student plots the graph shown on the left, from data obtained for the kinetic energy (eV) of the ejected electron versus frequency (s^{-1}) of the incident light, for an unknown metal. ($1\text{eV} = 1.6022 \times 10^{-19} \text{ J}$)

Based on the plot, answer the questions **a to c** below:

a. 1 point On the plot above, label the **threshold frequency**.

b. 2 points What is the value of the work potential (in J) of the unknown metal?

c. 3 points What is the *longest* wavelength (in nm) required to eject an electron from this metal's surface? Show your work including equation(s) used. (3 sig figs)

d. 2 points In the space provided below, plot the following

(Assume frequency is *above* threshold frequency):

- i. *Kinetic energy* of an ejected electron versus the *intensity* of the incident light (left)
- ii. *Number of electrons* ejected versus the *frequency* of the incident light (right)



**Question 3** **6 points****a. 2 points**

In the ground state of Ge (Germanium; $Z = 32$) atom, **name** the following orbitals:

i. Orbital containing the electron that requires the maximum energy to be removed.

ii. Orbital containing the electron that experiences the minimum effective nuclear charge.

b. 2 points

An unknown anion (A^-) has the following *excited* electronic configuration: $[Ar] 4s^2 3d^3 4p^1$

(Ar: Argon; $Z = 18$)

Give the **condensed ground state electronic configuration** of the atom, **A** and the cation, **A^{2+}** .
Show your work. (Assume all electronic transitions are possible)

Electronic Configuration of A:

Electronic Configuration of A^{2+} :

c. 2 points

In a single atom, determine how many electrons can have the following set of quantum numbers?

(i) $n = 5; l = 3$

In a single atom, determine how many electrons can have the following set of quantum numbers?

(i) $n = 3; l = 2; m_s = +1/2$

**Question 4** **8 points**

Rank the following from 1 to 4 in **increasing** order. Provide the ranking in the boxes provided.

Note: Atomic number (Z) are provided for relevant atoms

a) Increasing (lowest to highest) order of *ionization energy*: (1= lowest; 4 = highest)

Mg

Mg: Z = 12

Rb

Rb: Z = 37

Cs

Cs: Z = 55

K

K : Z = 19

b) Increasing (lowest to highest) order of *bond angle*: (1= lowest; 4 = highest)

H₂OCCl₄XeF₄

HCN

c) Increasing (smallest to largest) order of *ionic size* (1= smallest; 4 = largest)

S²⁻

S: Z = 16

O²⁻

O: Z = 8

Be²⁺

Be: Z = 4

Na⁺

Na: Z = 11

d) Increasing (lowest to highest) order of *bond energy* (1= lowest; 4 = highest)

P-F

P-Cl

P-I

P-Br

**Question 5** **12 points**

a. 9 points For the molecules HClO_4 and TeF_6 , draw the most stable *Lewis structure* showing appropriate shape and *dashed/wedged bonds*, based on VSEPR. Determine the **molecular geometry** around the central atom. For HClO_4 , label the *most polar bond*.



b. 3 points (Note: Lone pairs are not shown in structures below)

(i) For the three *resonance* structures (1, 2 and 3) shown below, answer the following question:



(I) Circle the *most contributing* resonance structure.

Structure 1

Structure 2

Structure 3

(ii) For the three structures given (4, 5, and 6) below, answer the following questions:



(I) Circle all the structures that represent *resonance structures* of each other:

Structure 4

Structure 5

Structure 6

(II) Among the resonance structures, circle the *most contributing* resonance structure.

Structure 4

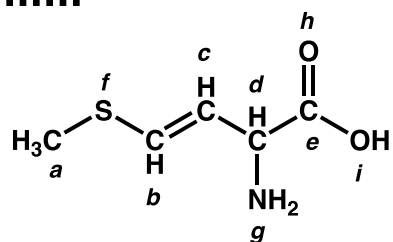
Structure 5

Structure 6

**Question 6** **6 points**

Using the data given below, calculate the **lattice energy** for SrO(s).

	ΔH° (kJ/mol)
$\text{Sr}(s) \longrightarrow \text{Sr}(g)$	+180
$\text{Sr}(g) \longrightarrow \text{Sr}^{2+}(g) + 2e^-$	+1468
$\text{O}_2(g) \longrightarrow 2\text{O}(g)$	+500
$\text{O}(g) + 2e^- \longrightarrow \text{O}^{2-}(g)$	+600
$\text{Sr}(s) + \frac{1}{2}\text{O}_2(g) \longrightarrow \text{SrO}(s)$	-550

**Question 7 10 points**

Consider the molecule given on the left, and answer questions (i) to (vi) using chemical bonding theories that we have learnt in class (valence bond theory, hybridization).

Note that the lone pairs are not shown in the structure:

i. Consider the carbon atoms marked in as ***a***, ***b***, ***c***, ***d***, and ***e*** in the molecule, and determine **hybridization** at each of the carbon centers. (2.5 points)

Atom	Hybridization
Carbon (<i>a</i>)	
Carbon (<i>b</i>)	
Carbon (<i>c</i>)	
Carbon (<i>d</i>)	
Carbon (<i>e</i>)	

ii. Now, consider the sulfur atom (S) marked as ***f***, nitrogen (N) atom (***g***), and oxygen(O) atoms (***h*** and ***i***), and determine their **hybridization**. (2 points)

Atom	Hybridization
Sulfur (<i>f</i>)	
Nitrogen (<i>g</i>)	
Oxygen (<i>h</i>)	
Oxygen (<i>i</i>)	

iii. **Describe** all the bonds formed by the carbon atoms ***a*** and ***b***, using sigma and pi designation as appropriate. (2 points)

Atom	Bonds formed (use sigma and pi as appropriate)
Carbon (<i>a</i>)	
Carbon (<i>b</i>)	

iv. How many total sp^3 -s sigma bonds are in the molecule? (1 point)

v. Determine **electronic** and **molecular geometry** at carbon (C) labeled ***e*** and sulfur (S) labeled ***f*** centres. (2 points)

Atom	Electronic Geometry	Molecular Geometry
Carbon (<i>e</i>)		
Sulfur (<i>f</i>)		

vi. Circle the value from the choices given below, that describes the **C-S-C bond angle at sulfur** (S) marked ***f***: (0.5 points)

>90° to <109.5° 109.5° >109.5 to <120° 120° >120° to <180°


Question 8 12 points

Using the **valence shell orbitals only** draw a **complete** molecular orbital (MO) diagram (with no s-p mixing) for the molecule **PS**, in the space provided below. **(6 points)**

Assume that 3s and 3p atomic orbitals will form molecular orbitals like those formed by the combination of 2s and 2p atomic orbitals that we learnt in class, and there is no s-p mixing.

Show **energy arrow**, **atomic orbitals** involved, **atom labels**, the **molecular orbitals** that are formed, **filled electrons** in the MO diagram etc. Make sure to label all the atomic and molecular orbitals.

Using the MO diagram above, answer the following questions:

- (i) Which one of the following will have the highest bond order? Circle the correct choice
 A) PS^{1-} B) PS C) PS^{1+}
- (ii) Which one of the following will have the longest bond length? Circle the correct choice
 A) PS^{1-} B) PS C) PS^{1+}
- (iii) Which of the following will be diamagnetic? Circle the correct choice.
 A) PS^{1-} B) PS C) PS^{1+}
- (iv) Draw Lewis dot structure of the molecule with the shortest bond length (based on the MO diagram) from among the three (PS^{1-} , PS, PS^{1+}), and indicate the bond order calculated from the Lewis dot structure. **(3 points)**

Lewis Dot Structure

Bond Order



D5043B91-17B1-4CC4-B563-1EFC01A51F06

midterm-exam-93388

#373 Page 10 of 10

SCRAP PAGE

**CHEMISTRY 110 –Midterm Exam****Thursday November 7, 2024, 6:30 – 8:30 pm**

Examiners: Pallavi Sirjoosingh and Ashok Kakkar

First name (please write as legibly as possible within the boxes)

--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--

Last name

--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--

Student ID number

--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--

This booklet has 8 questions on 10 pages (The last page can be used as a scrap page).

The staple must NOT be removed from the question booklet. Your intact question booklet must be returned at the end of the exam.

A separate booklet contains a periodic table, and datasheet.

Write your answers directly under the questions in the space provided. Do not write on the QR codes or near the top edges (as it may be cut off).

Please write **legibly and dark enough** for the scan to be readable. For students who use English as a second language, dictionaries are permitted. Calculators are allowed.

DO NOT ASK QUESTIONS OF THE INVIGILATORS about interpretations of the exam questions. If you have a concern about a question, please summarize it at the top of this page. All concerns will be evaluated after the exam. GOOD LUCK!

**Question 1 8 points** (*Assume all transitions are possible*)

a. 6 points Calculate the **energy (in kJ/mol)** required to ionize (completely remove) the electron from the **n=3** level of **He⁺ ion** (He: Helium). Show your work including equations. (3 sig. figs.)

b. 2 points

(I) Select the largest (longest) wavelength from the three choices below:

- i. Wavelength of photon to *completely remove* an electron from *1s orbital of Hydrogen atom*
- ii. Wavelength of photon to *completely remove* an electron from *1s orbital of Boron atom (B)*
- iii. Wavelength of photon to *completely remove* an electron from *1s orbital of Boron cation (B⁺)*

i or ii or iii

(II) Select the smallest (shortest) wavelength from the three choices below (*for H atom*):

- i. Wavelength of photon *emitted* for an electron transitioning from *2p to 1s orbital*
- ii. Wavelength of photon *absorbed* for an electron transitioning from *2s to 4p orbital*
- iii. Wavelength of photon *absorbed* for an electron transitioning from *1s to 3p orbital*

i or ii or iii



Question 2 8 points



A Chem110 student plots the graph shown on the left, from data obtained for the kinetic energy (eV) of the ejected electron versus frequency (s^{-1}) of the incident light, for an unknown metal. ($1\text{eV} = 1.6022 \times 10^{-19} \text{ J}$)

Based on the plot, answer the questions **a to c** below:

a. 1 point On the plot above, label the **threshold frequency**.

b. 2 points What is the value of the work potential (in J) of the unknown metal?

c. 3 points What is the *longest* wavelength (in nm) required to eject an electron from this metal's surface? Show your work including equation(s) used. (3 sig figs)

d. 2 points In the space provided below, plot the following

(Assume frequency is *above* threshold frequency):

- i. *Kinetic energy* of an ejected electron versus the *intensity* of the incident light (left)
- ii. *Number of electrons* ejected versus the *frequency* of the incident light (right)



**Question 3** **6 points****a. 2 points**

In the ground state of Ge (Germanium; $Z = 32$) atom, **name** the following orbitals:

i. Orbital containing the electron that requires the maximum energy to be removed.

ii. Orbital containing the electron that experiences the minimum effective nuclear charge.

b. 2 points

An unknown anion (A^-) has the following *excited* electronic configuration: $[Ar] 4s^2 3d^3 4p^1$

(Ar: Argon; $Z = 18$)

Give the **condensed ground state electronic configuration** of the atom, **A** and the cation, **A^{2+}** .
Show your work. (Assume all electronic transitions are possible)

Electronic Configuration of A:

Electronic Configuration of A^{2+} :

c. 2 points

In a single atom, determine how many electrons can have the following set of quantum numbers?

(i) $n = 5; l = 3$

In a single atom, determine how many electrons can have the following set of quantum numbers?

(i) $n = 3; l = 2; m_s = +1/2$

**Question 4** **8 points**

Rank the following from 1 to 4 in **increasing** order. Provide the ranking in the boxes provided.

Note: Atomic number (Z) are provided for relevant atoms

a) Increasing (lowest to highest) order of *ionization energy*: (1= lowest; 4 = highest)

Mg

Mg: Z = 12

Rb

Rb: Z = 37

Cs

Cs: Z = 55

K

K : Z = 19

b) Increasing (lowest to highest) order of *bond angle*: (1= lowest; 4 = highest)

H₂OCCl₄XeF₄

HCN

c) Increasing (smallest to largest) order of *ionic size* (1= smallest; 4 = largest)

S²⁻

S: Z = 16

O²⁻

O: Z = 8

Be²⁺

Be: Z = 4

Na⁺

Na: Z = 11

d) Increasing (lowest to highest) order of *bond energy* (1= lowest; 4 = highest)

P-F

P-Cl

P-I

P-Br

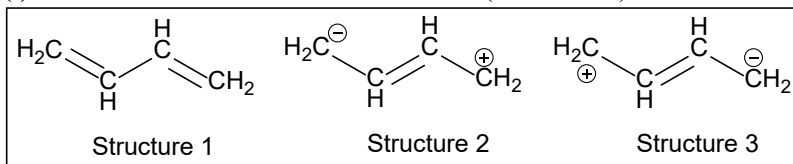
**Question 5** **12 points**

a. 9 points For the molecules HClO_4 and TeF_6 , draw the most stable *Lewis structure* showing appropriate shape and *dashed/wedged bonds*, based on VSEPR. Determine the **molecular geometry** around the central atom. For HClO_4 , label the *most polar bond*.



b. 3 points (Note: Lone pairs are not shown in structures below)

(i) For the three *resonance* structures (1, 2 and 3) shown below, answer the following question:



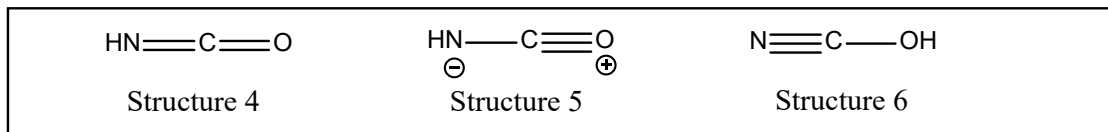
(I) Circle the *most contributing* resonance structure.

Structure 1

Structure 2

Structure 3

(ii) For the three structures given (4, 5, and 6) below, answer the following questions:



(I) Circle all the structures that represent *resonance structures* of each other:

Structure 4

Structure 5

Structure 6

(II) Among the resonance structures, circle the *most contributing* resonance structure.

Structure 4

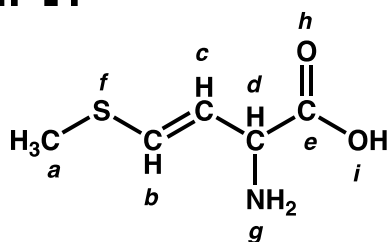
Structure 5

Structure 6

**Question 6** **6 points**

Using the data given below, calculate the **lattice energy** for SrO(s).

	ΔH° (kJ/mol)
$\text{Sr}(s) \longrightarrow \text{Sr}(g)$	+180
$\text{Sr}(g) \longrightarrow \text{Sr}^{2+}(g) + 2e^-$	+1468
$\text{O}_2(g) \longrightarrow 2\text{O}(g)$	+500
$\text{O}(g) + 2e^- \longrightarrow \text{O}^{2-}(g)$	+600
$\text{Sr}(s) + \frac{1}{2}\text{O}_2(g) \longrightarrow \text{SrO}(s)$	-550

**Question 7 10 points**

Consider the molecule given on the left, and answer questions (i) to (vi) using chemical bonding theories that we have learnt in class (valence bond theory, hybridization).

Note that the lone pairs are not shown in the structure:

i. Consider the carbon atoms marked in as ***a***, ***b***, ***c***, ***d***, and ***e*** in the molecule, and determine **hybridization** at each of the carbon centers. (2.5 points)

Atom	Hybridization
Carbon (<i>a</i>)	
Carbon (<i>b</i>)	
Carbon (<i>c</i>)	
Carbon (<i>d</i>)	
Carbon (<i>e</i>)	

ii. Now, consider the sulfur atom (S) marked as ***f***, nitrogen (N) atom (***g***), and oxygen(O) atoms (***h*** and ***i***), and determine their **hybridization**. (2 points)

Atom	Hybridization
Sulfur (<i>f</i>)	
Nitrogen (<i>g</i>)	
Oxygen (<i>h</i>)	
Oxygen (<i>i</i>)	

iii. **Describe** all the bonds formed by the carbon atoms ***a*** and ***b***, using sigma and pi designation as appropriate. (2 points)

Atom	Bonds formed (use sigma and pi as appropriate)
Carbon (<i>a</i>)	
Carbon (<i>b</i>)	

iv. How many total sp^3 -s sigma bonds are in the molecule? (1 point)

v. Determine **electronic** and **molecular geometry** at carbon (C) labeled ***e*** and sulfur (S) labeled ***f*** centres. (2 points)

Atom	Electronic Geometry	Molecular Geometry
Carbon (<i>e</i>)		
Sulfur (<i>f</i>)		

vi. Circle the value from the choices given below, that describes the **C-S-C bond angle at sulfur** (S) marked ***f***: (0.5 points)

>90° to <109.5° 109.5° >109.5 to <120° 120° >120° to <180°


Question 8 12 points

Using the **valence shell orbitals only** draw a **complete** molecular orbital (MO) diagram (with no s-p mixing) for the molecule **PS**, in the space provided below. **(6 points)**

Assume that 3s and 3p atomic orbitals will form molecular orbitals like those formed by the combination of 2s and 2p atomic orbitals that we learnt in class, and there is no s-p mixing.

Show **energy arrow**, **atomic orbitals** involved, **atom labels**, the **molecular orbitals** that are formed, **filled electrons** in the MO diagram etc. Make sure to label all the atomic and molecular orbitals.

Using the MO diagram above, answer the following questions:

- (i) Which one of the following will have the highest bond order? Circle the correct choice
 A) PS^{1-} B) PS C) PS^{1+}
- (ii) Which one of the following will have the longest bond length? Circle the correct choice
 A) PS^{1-} B) PS C) PS^{1+}
- (iii) Which of the following will be diamagnetic? Circle the correct choice.
 A) PS^{1-} B) PS C) PS^{1+}
- (iv) Draw Lewis dot structure of the molecule with the shortest bond length (based on the MO diagram) from among the three (PS^{1-} , PS, PS^{1+}), and indicate the bond order calculated from the Lewis dot structure. **(3 points)**

Lewis Dot Structure

Bond Order



3AE3F74F-E65B-4C52-B4EA-03F18824C0E4

midterm-exam-93388

#374 Page 10 of 10

SCRAP PAGE

**CHEMISTRY 110 –Midterm Exam****Thursday November 7, 2024, 6:30 – 8:30 pm****Examiners: Pallavi Sirjoosingh and Ashok Kakkar**

First name (please write as legibly as possible within the boxes)

--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--

Last name

--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--

Student ID number

--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--

This booklet has 8 questions on 10 pages (The last page can be used as a scrap page).

The staple must NOT be removed from the question booklet. Your intact question booklet must be returned at the end of the exam.

A separate booklet contains a periodic table, and datasheet.

Write your answers directly under the questions in the space provided. Do not write on the QR codes or near the top edges (as it may be cut off).

Please write **legibly and dark enough** for the scan to be readable. For students who use English as a second language, dictionaries are permitted. Calculators are allowed.

DO NOT ASK QUESTIONS OF THE INVIGILATORS about interpretations of the exam questions. If you have a concern about a question, please summarize it at the top of this page. All concerns will be evaluated after the exam. GOOD LUCK!

**Question 1 8 points** (*Assume all transitions are possible*)

a. 6 points Calculate the **energy (in kJ/mol)** required to ionize (completely remove) the electron from the **n=3** level of **He⁺ ion** (He: Helium). Show your work including equations. (3 sig. figs.)

b. 2 points

(I) Select the largest (longest) wavelength from the three choices below:

- i. Wavelength of photon to *completely remove* an electron from *1s orbital of Hydrogen atom*
- ii. Wavelength of photon to *completely remove* an electron from *1s orbital of Boron atom (B)*
- iii. Wavelength of photon to *completely remove* an electron from *1s orbital of Boron cation (B⁺)*

i or ii or iii

(II) Select the smallest (shortest) wavelength from the three choices below (*for H atom*):

- i. Wavelength of photon *emitted* for an electron transitioning from *2p to 1s orbital*
- ii. Wavelength of photon *absorbed* for an electron transitioning from *2s to 4p orbital*
- iii. Wavelength of photon *absorbed* for an electron transitioning from *1s to 3p orbital*

i or ii or iii



Question 2 8 points



A Chem110 student plots the graph shown on the left, from data obtained for the kinetic energy (eV) of the ejected electron versus frequency (s^{-1}) of the incident light, for an unknown metal. ($1\text{eV} = 1.6022 \times 10^{-19} \text{ J}$)

Based on the plot, answer the questions **a to c** below:

a. 1 point On the plot above, label the **threshold frequency**.

b. 2 points What is the value of the work potential (in J) of the unknown metal?

c. 3 points What is the *longest* wavelength (in nm) required to eject an electron from this metal's surface? Show your work including equation(s) used. (3 sig figs)

d. 2 points In the space provided below, plot the following

(Assume frequency is *above* threshold frequency):

- Kinetic energy* of an ejected electron versus the *intensity* of the incident light (left)
- Number of electrons* ejected versus the *frequency* of the incident light (right)



**Question 3** **6 points****a. 2 points**

In the ground state of Ge (Germanium; $Z = 32$) atom, **name** the following orbitals:

i. Orbital containing the electron that requires the maximum energy to be removed.

ii. Orbital containing the electron that experiences the minimum effective nuclear charge.

b. 2 points

An unknown anion (A^-) has the following *excited* electronic configuration: $[Ar] 4s^2 3d^3 4p^1$

(Ar: Argon; $Z = 18$)

Give the **condensed ground state electronic configuration** of the atom, **A** and the cation, **A^{2+}** .
Show your work. (Assume all electronic transitions are possible)

Electronic Configuration of A:

Electronic Configuration of A^{2+} :

c. 2 points

In a single atom, determine how many electrons can have the following set of quantum numbers?

(i) $n = 5; l = 3$

In a single atom, determine how many electrons can have the following set of quantum numbers?

(i) $n = 3; l = 2; m_s = +1/2$

**Question 4** **8 points**

Rank the following from 1 to 4 in **increasing** order. Provide the ranking in the boxes provided.

Note: Atomic number (Z) are provided for relevant atoms

a) Increasing (lowest to highest) order of *ionization energy*: (1= lowest; 4 = highest)

Mg

Mg: Z = 12

Rb

Rb: Z = 37

Cs

Cs: Z = 55

K

K : Z = 19

b) Increasing (lowest to highest) order of *bond angle*: (1= lowest; 4 = highest)

H₂OCCl₄XeF₄

HCN

c) Increasing (smallest to largest) order of *ionic size* (1= smallest; 4 = largest)

S²⁻

S: Z = 16

O²⁻

O: Z = 8

Be²⁺

Be: Z = 4

Na⁺

Na: Z = 11

d) Increasing (lowest to highest) order of *bond energy* (1= lowest; 4 = highest)

P-F

P-Cl

P-I

P-Br

**Question 5** **12 points**

a. 9 points For the molecules HClO_4 and TeF_6 , draw the most stable *Lewis structure* showing appropriate shape and *dashed/wedged bonds*, based on VSEPR. Determine the **molecular geometry** around the central atom. For HClO_4 , label the *most polar bond*.



b. 3 points (Note: Lone pairs are not shown in structures below)

(i) For the three *resonance* structures (1, 2 and 3) shown below, answer the following question:



(I) Circle the *most contributing* resonance structure.

Structure 1

Structure 2

Structure 3

(ii) For the three structures given (4, 5, and 6) below, answer the following questions:



(I) Circle all the structures that represent *resonance structures* of each other:

Structure 4

Structure 5

Structure 6

(II) Among the resonance structures, circle the *most contributing* resonance structure.

Structure 4

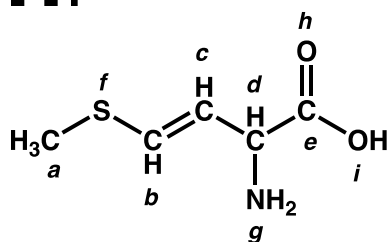
Structure 5

Structure 6

**Question 6** **6 points**

Using the data given below, calculate the **lattice energy** for SrO(s).

	ΔH° (kJ/mol)
$\text{Sr}(s) \longrightarrow \text{Sr}(g)$	+180
$\text{Sr}(g) \longrightarrow \text{Sr}^{2+}(g) + 2e^-$	+1468
$\text{O}_2(g) \longrightarrow 2\text{O}(g)$	+500
$\text{O}(g) + 2e^- \longrightarrow \text{O}^{2-}(g)$	+600
$\text{Sr}(s) + \frac{1}{2}\text{O}_2(g) \longrightarrow \text{SrO}(s)$	-550

**Question 7 10 points**

Consider the molecule given on the left, and answer questions (i) to (vi) using chemical bonding theories that we have learnt in class (valence bond theory, hybridization).

Note that the lone pairs are not shown in the structure:

i. Consider the carbon atoms marked in as ***a***, ***b***, ***c***, ***d***, and ***e*** in the molecule, and determine **hybridization** at each of the carbon centers. (2.5 points)

Atom	Hybridization
Carbon (<i>a</i>)	
Carbon (<i>b</i>)	
Carbon (<i>c</i>)	
Carbon (<i>d</i>)	
Carbon (<i>e</i>)	

ii. Now, consider the sulfur atom (S) marked as ***f***, nitrogen (N) atom (***g***), and oxygen(O) atoms (***h*** and ***i***), and determine their **hybridization**. (2 points)

Atom	Hybridization
Sulfur (<i>f</i>)	
Nitrogen (<i>g</i>)	
Oxygen (<i>h</i>)	
Oxygen (<i>i</i>)	

iii. **Describe** all the bonds formed by the carbon atoms ***a*** and ***b***, using sigma and pi designation as appropriate. (2 points)

Atom	Bonds formed (use sigma and pi as appropriate)
Carbon (<i>a</i>)	
Carbon (<i>b</i>)	

iv. How many total sp^3 -s sigma bonds are in the molecule? (1 point)

v. Determine **electronic** and **molecular geometry** at carbon (C) labeled ***e*** and sulfur (S) labeled ***f*** centres. (2 points)

Atom	Electronic Geometry	Molecular Geometry
Carbon (<i>e</i>)		
Sulfur (<i>f</i>)		

vi. Circle the value from the choices given below, that describes the **C-S-C bond angle at sulfur** (S) marked ***f***: (0.5 points)

>90° to <109.5° 109.5° >109.5 to <120° 120° >120° to <180°


Question 8 12 points

Using the **valence shell orbitals only** draw a **complete** molecular orbital (MO) diagram (with no s-p mixing) for the molecule **PS**, in the space provided below. **(6 points)**

Assume that 3s and 3p atomic orbitals will form molecular orbitals like those formed by the combination of 2s and 2p atomic orbitals that we learnt in class, and there is no s-p mixing.

Show **energy arrow**, **atomic orbitals** involved, **atom labels**, the **molecular orbitals** that are formed, **filled electrons** in the MO diagram etc. Make sure to label all the atomic and molecular orbitals.

Using the MO diagram above, answer the following questions:

- (i) Which one of the following will have the highest bond order? Circle the correct choice
 A) PS^{1-} B) PS C) PS^{1+}
- (ii) Which one of the following will have the longest bond length? Circle the correct choice
 A) PS^{1-} B) PS C) PS^{1+}
- (iii) Which of the following will be diamagnetic? Circle the correct choice.
 A) PS^{1-} B) PS C) PS^{1+}
- (iv) Draw Lewis dot structure of the molecule with the shortest bond length (based on the MO diagram) from among the three (PS^{1-} , PS, PS^{1+}), and indicate the bond order calculated from the Lewis dot structure. **(3 points)**

Lewis Dot Structure

Bond Order



6F2B8CA5-3AB3-46B6-9BB8-95D03E73164F

midterm-exam-93388

#375 Page 10 of 10

SCRAP PAGE

**CHEMISTRY 110 –Midterm Exam****Thursday November 7, 2024, 6:30 – 8:30 pm****Examiners: Pallavi Sirjoosingh and Ashok Kakkar**

First name (please write as legibly as possible within the boxes)

--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--

Last name

--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--

Student ID number

--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--

This booklet has 8 questions on 10 pages (The last page can be used as a scrap page).

The staple must NOT be removed from the question booklet. Your intact question booklet must be returned at the end of the exam.

A separate booklet contains a periodic table, and datasheet.

Write your answers directly under the questions in the space provided. Do not write on the QR codes or near the top edges (as it may be cut off).

Please write **legibly and dark enough** for the scan to be readable. For students who use English as a second language, dictionaries are permitted. Calculators are allowed.

DO NOT ASK QUESTIONS OF THE INVIGILATORS about interpretations of the exam questions. If you have a concern about a question, please summarize it at the top of this page. All concerns will be evaluated after the exam. GOOD LUCK!

**Question 1 8 points** (*Assume all transitions are possible*)

a. 6 points Calculate the **energy (in kJ/mol)** required to ionize (completely remove) the electron from the **n=3** level of **He⁺ ion** (He: Helium). Show your work including equations. (3 sig. figs.)

b. 2 points

(I) Select the largest (longest) wavelength from the three choices below:

- i. Wavelength of photon to *completely remove* an electron from *1s orbital of Hydrogen atom*
- ii. Wavelength of photon to *completely remove* an electron from *1s orbital of Boron atom (B)*
- iii. Wavelength of photon to *completely remove* an electron from *1s orbital of Boron cation (B⁺)*

i or ii or iii

(II) Select the smallest (shortest) wavelength from the three choices below (*for H atom*):

- i. Wavelength of photon *emitted* for an electron transitioning from *2p to 1s orbital*
- ii. Wavelength of photon *absorbed* for an electron transitioning from *2s to 4p orbital*
- iii. Wavelength of photon *absorbed* for an electron transitioning from *1s to 3p orbital*

i or ii or iii



Question 2 8 points



A Chem110 student plots the graph shown on the left, from data obtained for the kinetic energy (eV) of the ejected electron versus frequency (s^{-1}) of the incident light, for an unknown metal. ($1\text{eV} = 1.6022 \times 10^{-19} \text{ J}$)

Based on the plot, answer the questions **a to c** below:

a. 1 point On the plot above, label the **threshold frequency**.

b. 2 points What is the value of the work potential (in J) of the unknown metal?

c. 3 points What is the *longest* wavelength (in nm) required to eject an electron from this metal's surface? Show your work including equation(s) used. (3 sig figs)

d. 2 points In the space provided below, plot the following

(Assume frequency is *above* threshold frequency):

- Kinetic energy* of an ejected electron versus the *intensity* of the incident light (left)
- Number of electrons* ejected versus the *frequency* of the incident light (right)



**Question 3** **6 points****a. 2 points**

In the ground state of Ge (Germanium; $Z = 32$) atom, **name** the following orbitals:

i. Orbital containing the electron that requires the maximum energy to be removed.

ii. Orbital containing the electron that experiences the minimum effective nuclear charge.

b. 2 points

An unknown anion (A^-) has the following *excited* electronic configuration: $[Ar] 4s^2 3d^3 4p^1$

(Ar: Argon; $Z = 18$)

Give the **condensed ground state electronic configuration** of the atom, **A** and the cation, **A^{2+}** .
Show your work. (Assume all electronic transitions are possible)

Electronic Configuration of A:

Electronic Configuration of A^{2+} :

c. 2 points

In a single atom, determine how many electrons can have the following set of quantum numbers?

(i) $n = 5; l = 3$

In a single atom, determine how many electrons can have the following set of quantum numbers?

(i) $n = 3; l = 2; m_s = +1/2$

**Question 4** **8 points**

Rank the following from 1 to 4 in **increasing** order. Provide the ranking in the boxes provided.

Note: Atomic number (Z) are provided for relevant atoms

a) Increasing (lowest to highest) order of *ionization energy*: (1= lowest; 4 = highest)

Mg

Mg: Z = 12

Rb

Rb: Z = 37

Cs

Cs: Z = 55

K

K : Z = 19

b) Increasing (lowest to highest) order of *bond angle*: (1= lowest; 4 = highest)

H₂OCCl₄XeF₄

HCN

c) Increasing (smallest to largest) order of *ionic size* (1= smallest; 4 = largest)

S²⁻

S: Z = 16

O²⁻

O: Z = 8

Be²⁺

Be: Z = 4

Na⁺

Na: Z = 11

d) Increasing (lowest to highest) order of *bond energy* (1= lowest; 4 = highest)

P-F

P-Cl

P-I

P-Br

**Question 5** **12 points**

a. 9 points For the molecules HClO_4 and TeF_6 , draw the most stable *Lewis structure* showing appropriate shape and *dashed/wedged bonds*, based on VSEPR. Determine the **molecular geometry** around the central atom. For HClO_4 , label the *most polar bond*.



b. 3 points (Note: Lone pairs are not shown in structures below)

(i) For the three *resonance* structures (1, 2 and 3) shown below, answer the following question:



(I) Circle the *most contributing* resonance structure.

Structure 1

Structure 2

Structure 3

(ii) For the three structures given (4, 5, and 6) below, answer the following questions:



(I) Circle all the structures that represent *resonance structures* of each other:

Structure 4

Structure 5

Structure 6

(II) Among the resonance structures, circle the *most contributing* resonance structure.

Structure 4

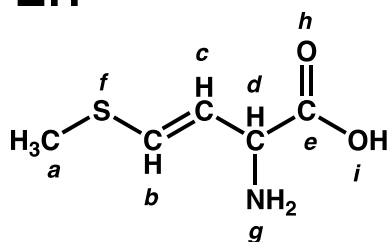
Structure 5

Structure 6

**Question 6** **6 points**

Using the data given below, calculate the **lattice energy** for SrO(s).

	ΔH° (kJ/mol)
$\text{Sr}(s) \longrightarrow \text{Sr}(g)$	+180
$\text{Sr}(g) \longrightarrow \text{Sr}^{2+}(g) + 2e^-$	+1468
$\text{O}_2(g) \longrightarrow 2\text{O}(g)$	+500
$\text{O}(g) + 2e^- \longrightarrow \text{O}^{2-}(g)$	+600
$\text{Sr}(s) + \frac{1}{2}\text{O}_2(g) \longrightarrow \text{SrO}(s)$	-550

**Question 7 10 points**

Consider the molecule given on the left, and answer questions (i) to (vi) using chemical bonding theories that we have learnt in class (valence bond theory, hybridization).

Note that the lone pairs are not shown in the structure:

i. Consider the carbon atoms marked in as ***a***, ***b***, ***c***, ***d***, and ***e*** in the molecule, and determine **hybridization** at each of the carbon centers. (2.5 points)

Atom	Hybridization
Carbon (<i>a</i>)	
Carbon (<i>b</i>)	
Carbon (<i>c</i>)	
Carbon (<i>d</i>)	
Carbon (<i>e</i>)	

ii. Now, consider the sulfur atom (S) marked as ***f***, nitrogen (N) atom (***g***), and oxygen(O) atoms (***h*** and ***i***), and determine their **hybridization**. (2 points)

Atom	Hybridization
Sulfur (<i>f</i>)	
Nitrogen (<i>g</i>)	
Oxygen (<i>h</i>)	
Oxygen (<i>i</i>)	

iii. **Describe** all the bonds formed by the carbon atoms ***a*** and ***b***, using sigma and pi designation as appropriate. (2 points)

Atom	Bonds formed (use sigma and pi as appropriate)
Carbon (<i>a</i>)	
Carbon (<i>b</i>)	

iv. How many total sp^3 -s sigma bonds are in the molecule? (1 point)

v. Determine **electronic** and **molecular geometry** at carbon (C) labeled ***e*** and sulfur (S) labeled ***f*** centres. (2 points)

Atom	Electronic Geometry	Molecular Geometry
Carbon (<i>e</i>)		
Sulfur (<i>f</i>)		

vi. Circle the value from the choices given below, that describes the **C-S-C bond angle at sulfur** (S) marked ***f***: (0.5 points)

>90° to <109.5° 109.5° >109.5 to <120° 120° >120° to <180°


Question 8 12 points

Using the **valence shell orbitals only** draw a **complete** molecular orbital (MO) diagram (with no s-p mixing) for the molecule **PS**, in the space provided below. **(6 points)**

Assume that 3s and 3p atomic orbitals will form molecular orbitals like those formed by the combination of 2s and 2p atomic orbitals that we learnt in class, and there is no s-p mixing.

Show **energy arrow**, **atomic orbitals** involved, **atom labels**, the **molecular orbitals** that are formed, **filled electrons** in the MO diagram etc. Make sure to **label all the atomic and molecular orbitals**.

Using the MO diagram above, answer the following questions:

- (i) Which one of the following will have the highest bond order? Circle the correct choice
 A) PS^{1-} B) PS C) PS^{1+}
- (ii) Which one of the following will have the longest bond length? Circle the correct choice
 A) PS^{1-} B) PS C) PS^{1+}
- (iii) Which of the following will be diamagnetic? Circle the correct choice.
 A) PS^{1-} B) PS C) PS^{1+}
- (iv) Draw Lewis dot structure of the molecule with the shortest bond length (based on the MO diagram) from among the three (PS^{1-} , PS, PS^{1+}), and indicate the bond order calculated from the Lewis dot structure. **(3 points)**

Lewis Dot Structure

Bond Order



97429EF0-2A2B-4AF4-B076-77DB306B6912

midterm-exam-93388

#376 Page 10 of 10

SCRAP PAGE

**CHEMISTRY 110 –Midterm Exam****Thursday November 7, 2024, 6:30 – 8:30 pm**

Examiners: Pallavi Sirjoosingh and Ashok Kakkar

First name (please write as legibly as possible within the boxes)

--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--

Last name

--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--

Student ID number

--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--

This booklet has 8 questions on 10 pages (The last page can be used as a scrap page).

The staple must NOT be removed from the question booklet. Your intact question booklet must be returned at the end of the exam.

A separate booklet contains a periodic table, and datasheet.

Write your answers directly under the questions in the space provided. Do not write on the QR codes or near the top edges (as it may be cut off).

Please write **legibly and dark enough** for the scan to be readable. For students who use English as a second language, dictionaries are permitted. Calculators are allowed.

DO NOT ASK QUESTIONS OF THE INVIGILATORS about interpretations of the exam questions. If you have a concern about a question, please summarize it at the top of this page. All concerns will be evaluated after the exam. GOOD LUCK!

**Question 1 8 points** (*Assume all transitions are possible*)

a. 6 points Calculate the **energy (in kJ/mol)** required to ionize (completely remove) the electron from the **n=3** level of **He⁺ ion** (He: Helium). Show your work including equations. (3 sig. figs.)

b. 2 points

(I) Select the largest (longest) wavelength from the three choices below:

- i. Wavelength of photon to *completely remove* an electron from *1s orbital of Hydrogen atom*
- ii. Wavelength of photon to *completely remove* an electron from *1s orbital of Boron atom (B)*
- iii. Wavelength of photon to *completely remove* an electron from *1s orbital of Boron cation (B⁺)*

i or ii or iii

(II) Select the smallest (shortest) wavelength from the three choices below (*for H atom*):

- i. Wavelength of photon *emitted* for an electron transitioning from *2p to 1s orbital*
- ii. Wavelength of photon *absorbed* for an electron transitioning from *2s to 4p orbital*
- iii. Wavelength of photon *absorbed* for an electron transitioning from *1s to 3p orbital*

i or ii or iii



Question 2 8 points



A Chem110 student plots the graph shown on the left, from data obtained for the kinetic energy (eV) of the ejected electron versus frequency (s^{-1}) of the incident light, for an unknown metal. ($1\text{eV} = 1.6022 \times 10^{-19} \text{ J}$)

Based on the plot, answer the questions **a to c** below:

a. 1 point On the plot above, label the **threshold frequency**.

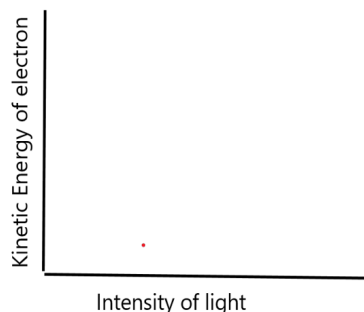
b. 2 points What is the value of the work potential (in J) of the unknown metal?

c. 3 points What is the *longest* wavelength (in nm) required to eject an electron from this metal's surface? Show your work including equation(s) used. (3 sig figs)

d. 2 points In the space provided below, plot the following

(Assume frequency is *above* threshold frequency):

- Kinetic energy* of an ejected electron versus the *intensity* of the incident light (left)
- Number of electrons* ejected versus the *frequency* of the incident light (right)



**Question 3** **6 points****a. 2 points**

In the ground state of Ge (Germanium; $Z = 32$) atom, **name** the following orbitals:

i. Orbital containing the electron that requires the maximum energy to be removed.

ii. Orbital containing the electron that experiences the minimum effective nuclear charge.

b. 2 points

An unknown anion (A^-) has the following *excited* electronic configuration: $[Ar] 4s^2 3d^3 4p^1$

(Ar: Argon; $Z = 18$)

Give the **condensed ground state electronic configuration** of the atom, **A** and the cation, **A^{2+}** .
Show your work. (Assume all electronic transitions are possible)

Electronic Configuration of A:

Electronic Configuration of A^{2+} :

c. 2 points

In a single atom, determine how many electrons can have the following set of quantum numbers?

(i) $n = 5; l = 3$

In a single atom, determine how many electrons can have the following set of quantum numbers?

(i) $n = 3; l = 2; m_s = +1/2$

**Question 4** **8 points**

Rank the following from 1 to 4 in **increasing** order. Provide the ranking in the boxes provided.

Note: Atomic number (Z) are provided for relevant atoms

a) Increasing (lowest to highest) order of *ionization energy*: (1= lowest; 4 = highest)

Mg

Mg: Z = 12

Rb

Rb: Z = 37

Cs

Cs: Z = 55

K

K : Z = 19

b) Increasing (lowest to highest) order of *bond angle*: (1= lowest; 4 = highest)

H₂OCCl₄XeF₄

HCN

c) Increasing (smallest to largest) order of *ionic size* (1= smallest; 4 = largest)

S²⁻

S: Z = 16

O²⁻

O: Z = 8

Be²⁺

Be: Z = 4

Na⁺

Na: Z = 11

d) Increasing (lowest to highest) order of *bond energy* (1= lowest; 4 = highest)

P-F

P-Cl

P-I

P-Br

**Question 5** **12 points**

a. 9 points For the molecules HClO_4 and TeF_6 , draw the most stable *Lewis structure* showing appropriate shape and *dashed/wedged bonds*, based on VSEPR. Determine the **molecular geometry** around the central atom. For HClO_4 , label the *most polar bond*.



b. 3 points (Note: Lone pairs are not shown in structures below)

(i) For the three *resonance* structures (1, 2 and 3) shown below, answer the following question:



(I) Circle the *most contributing* resonance structure.

Structure 1

Structure 2

Structure 3

(ii) For the three structures given (4, 5, and 6) below, answer the following questions:



(I) Circle all the structures that represent *resonance structures* of each other:

Structure 4

Structure 5

Structure 6

(II) Among the resonance structures, circle the *most contributing* resonance structure.

Structure 4

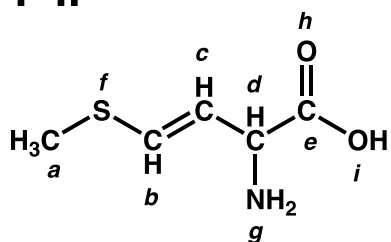
Structure 5

Structure 6

**Question 6** **6 points**

Using the data given below, calculate the **lattice energy** for SrO(s).

	ΔH° (kJ/mol)
$\text{Sr}(s) \longrightarrow \text{Sr}(g)$	+180
$\text{Sr}(g) \longrightarrow \text{Sr}^{2+}(g) + 2e^-$	+1468
$\text{O}_2(g) \longrightarrow 2\text{O}(g)$	+500
$\text{O}(g) + 2e^- \longrightarrow \text{O}^{2-}(g)$	+600
$\text{Sr}(s) + \frac{1}{2}\text{O}_2(g) \longrightarrow \text{SrO}(s)$	-550

**Question 7 10 points**

Consider the molecule given on the left, and answer questions (i) to (vi) using chemical bonding theories that we have learnt in class (valence bond theory, hybridization).

Note that the lone pairs are not shown in the structure:

i. Consider the carbon atoms marked in as ***a***, ***b***, ***c***, ***d***, and ***e*** in the molecule, and determine **hybridization** at each of the carbon centers. (2.5 points)

Atom	Hybridization
Carbon (<i>a</i>)	
Carbon (<i>b</i>)	
Carbon (<i>c</i>)	
Carbon (<i>d</i>)	
Carbon (<i>e</i>)	

ii. Now, consider the sulfur atom (S) marked as ***f***, nitrogen (N) atom (***g***), and oxygen(O) atoms (***h*** and ***i***), and determine their **hybridization**. (2 points)

Atom	Hybridization
Sulfur (<i>f</i>)	
Nitrogen (<i>g</i>)	
Oxygen (<i>h</i>)	
Oxygen (<i>i</i>)	

iii. **Describe** all the bonds formed by the carbon atoms ***a*** and ***b***, using sigma and pi designation as appropriate. (2 points)

Atom	Bonds formed (use sigma and pi as appropriate)
Carbon (<i>a</i>)	
Carbon (<i>b</i>)	

iv. How many total sp^3 -s sigma bonds are in the molecule? (1 point)

v. Determine **electronic** and **molecular geometry** at carbon (C) labeled ***e*** and sulfur (S) labeled ***f*** centres. (2 points)

Atom	Electronic Geometry	Molecular Geometry
Carbon (<i>e</i>)		
Sulfur (<i>f</i>)		

vi. Circle the value from the choices given below, that describes the **C-S-C bond angle at sulfur** (S) marked ***f***: (0.5 points)

>90° to <109.5°

109.5°

>109.5 to <120°

120°

>120° to <180°


Question 8 12 points

Using the **valence shell orbitals only** draw a **complete** molecular orbital (MO) diagram (with no s-p mixing) for the molecule **PS**, in the space provided below. **(6 points)**

Assume that 3s and 3p atomic orbitals will form molecular orbitals like those formed by the combination of 2s and 2p atomic orbitals that we learnt in class, and there is no s-p mixing.

Show **energy arrow**, **atomic orbitals** involved, **atom labels**, the **molecular orbitals** that are formed, **filled electrons** in the MO diagram etc. Make sure to label all the atomic and molecular orbitals.

Using the MO diagram above, answer the following questions:

- (i) Which one of the following will have the highest bond order? Circle the correct choice
 A) PS^{1-} B) PS C) PS^{1+}
- (ii) Which one of the following will have the longest bond length? Circle the correct choice
 A) PS^{1-} B) PS C) PS^{1+}
- (iii) Which of the following will be diamagnetic? Circle the correct choice.
 A) PS^{1-} B) PS C) PS^{1+}
- (iv) Draw Lewis dot structure of the molecule with the shortest bond length (based on the MO diagram) from among the three (PS^{1-} , PS, PS^{1+}), and indicate the bond order calculated from the Lewis dot structure. **(3 points)**

Lewis Dot Structure

Bond Order



8C181571-5436-42EF-87EF-5A51B180C891

midterm-exam-93388

#377 Page 10 of 10

SCRAP PAGE

**CHEMISTRY 110 –Midterm Exam****Thursday November 7, 2024, 6:30 – 8:30 pm**

Examiners: Pallavi Sirjoosingh and Ashok Kakkar

First name (please write as legibly as possible within the boxes)

--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--

Last name

--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--

Student ID number

--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--

This booklet has 8 questions on 10 pages (The last page can be used as a scrap page).

The staple must NOT be removed from the question booklet. Your intact question booklet must be returned at the end of the exam.

A separate booklet contains a periodic table, and datasheet.

Write your answers directly under the questions in the space provided. Do not write on the QR codes or near the top edges (as it may be cut off).

Please write **legibly and dark enough** for the scan to be readable. For students who use English as a second language, dictionaries are permitted. Calculators are allowed.

DO NOT ASK QUESTIONS OF THE INVIGILATORS about interpretations of the exam questions. If you have a concern about a question, please summarize it at the top of this page. All concerns will be evaluated after the exam. GOOD LUCK!

**Question 1 8 points** (*Assume all transitions are possible*)

a. 6 points Calculate the **energy (in kJ/mol)** required to ionize (completely remove) the electron from the **n=3** level of **He⁺ ion** (He: Helium). Show your work including equations. (3 sig. figs.)

b. 2 points

(I) Select the largest (longest) wavelength from the three choices below:

- i. Wavelength of photon to *completely remove* an electron from *1s orbital of Hydrogen atom*
- ii. Wavelength of photon to *completely remove* an electron from *1s orbital of Boron atom (B)*
- iii. Wavelength of photon to *completely remove* an electron from *1s orbital of Boron cation (B⁺)*

i or ii or iii

(II) Select the smallest (shortest) wavelength from the three choices below (*for H atom*):

- i. Wavelength of photon *emitted* for an electron transitioning from *2p to 1s orbital*
- ii. Wavelength of photon *absorbed* for an electron transitioning from *2s to 4p orbital*
- iii. Wavelength of photon *absorbed* for an electron transitioning from *1s to 3p orbital*

i or ii or iii



Question 2 8 points



A Chem110 student plots the graph shown on the left, from data obtained for the kinetic energy (eV) of the ejected electron versus frequency (s^{-1}) of the incident light, for an unknown metal. ($1\text{eV} = 1.6022 \times 10^{-19} \text{ J}$)

Based on the plot, answer the questions **a to c** below:

a. 1 point On the plot above, label the **threshold frequency**.

b. 2 points What is the value of the work potential (in J) of the unknown metal?

c. 3 points What is the *longest* wavelength (in nm) required to eject an electron from this metal's surface? Show your work including equation(s) used. (3 sig figs)

d. 2 points In the space provided below, plot the following

(Assume frequency is *above* threshold frequency):

- i. *Kinetic energy* of an ejected electron versus the *intensity* of the incident light (left)
- ii. *Number of electrons* ejected versus the *frequency* of the incident light (right)



**Question 3** **6 points****a. 2 points**

In the ground state of Ge (Germanium; $Z = 32$) atom, **name** the following orbitals:

i. Orbital containing the electron that requires the maximum energy to be removed.

ii. Orbital containing the electron that experiences the minimum effective nuclear charge.

b. 2 points

An unknown anion (A^-) has the following *excited* electronic configuration: $[Ar] 4s^2 3d^3 4p^1$

(Ar: Argon; $Z = 18$)

Give the **condensed ground state electronic configuration** of the atom, **A** and the cation, **A^{2+}** .
Show your work. (Assume all electronic transitions are possible)

Electronic Configuration of A:

Electronic Configuration of A^{2+} :

c. 2 points

In a single atom, determine how many electrons can have the following set of quantum numbers?

(i) $n = 5; l = 3$

In a single atom, determine how many electrons can have the following set of quantum numbers?

(i) $n = 3; l = 2; m_s = +1/2$

**Question 4** **8 points**

Rank the following from 1 to 4 in **increasing** order. Provide the ranking in the boxes provided.

Note: Atomic number (Z) are provided for relevant atoms

a) Increasing (lowest to highest) order of *ionization energy*: (1= lowest; 4 = highest)

Mg

Mg: Z = 12

Rb

Rb: Z = 37

Cs

Cs: Z = 55

K

K : Z = 19

b) Increasing (lowest to highest) order of *bond angle*: (1= lowest; 4 = highest)

H₂OCCl₄XeF₄

HCN

c) Increasing (smallest to largest) order of *ionic size* (1= smallest; 4 = largest)

S²⁻

S: Z = 16

O²⁻

O: Z = 8

Be²⁺

Be: Z = 4

Na⁺

Na: Z = 11

d) Increasing (lowest to highest) order of *bond energy* (1= lowest; 4 = highest)

P-F

P-Cl

P-I

P-Br

**Question 5** **12 points**

a. 9 points For the molecules HClO_4 and TeF_6 , draw the most stable *Lewis structure* showing appropriate shape and *dashed/wedged bonds*, based on VSEPR. Determine the **molecular geometry** around the central atom. For HClO_4 , label the *most polar bond*.



b. 3 points (Note: Lone pairs are not shown in structures below)

(i) For the three *resonance* structures (1, 2 and 3) shown below, answer the following question:



(I) Circle the *most contributing* resonance structure.

Structure 1

Structure 2

Structure 3

(ii) For the three structures given (4, 5, and 6) below, answer the following questions:



(I) Circle all the structures that represent *resonance structures* of each other:

Structure 4

Structure 5

Structure 6

(II) Among the resonance structures, circle the *most contributing* resonance structure.

Structure 4

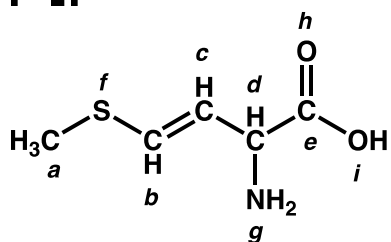
Structure 5

Structure 6

**Question 6** **6 points**

Using the data given below, calculate the **lattice energy** for SrO(s).

	ΔH° (kJ/mol)
$\text{Sr}(s) \longrightarrow \text{Sr}(g)$	+180
$\text{Sr}(g) \longrightarrow \text{Sr}^{2+}(g) + 2e^-$	+1468
$\text{O}_2(g) \longrightarrow 2\text{O}(g)$	+500
$\text{O}(g) + 2e^- \longrightarrow \text{O}^{2-}(g)$	+600
$\text{Sr}(s) + \frac{1}{2}\text{O}_2(g) \longrightarrow \text{SrO}(s)$	-550

**Question 7 10 points**

Consider the molecule given on the left, and answer questions (i) to (vi) using chemical bonding theories that we have learnt in class (valence bond theory, hybridization).

Note that the lone pairs are not shown in the structure:

i. Consider the carbon atoms marked in as ***a***, ***b***, ***c***, ***d***, and ***e*** in the molecule, and determine **hybridization** at each of the carbon centers. (2.5 points)

Atom	Hybridization
Carbon (<i>a</i>)	
Carbon (<i>b</i>)	
Carbon (<i>c</i>)	
Carbon (<i>d</i>)	
Carbon (<i>e</i>)	

ii. Now, consider the sulfur atom (S) marked as ***f***, nitrogen (N) atom (***g***), and oxygen(O) atoms (***h*** and ***i***), and determine their **hybridization**. (2 points)

Atom	Hybridization
Sulfur (<i>f</i>)	
Nitrogen (<i>g</i>)	
Oxygen (<i>h</i>)	
Oxygen (<i>i</i>)	

iii. **Describe** all the bonds formed by the carbon atoms ***a*** and ***b***, using sigma and pi designation as appropriate. (2 points)

Atom	Bonds formed (use sigma and pi as appropriate)
Carbon (<i>a</i>)	
Carbon (<i>b</i>)	

iv. How many total sp^3 -s sigma bonds are in the molecule? (1 point)

v. Determine **electronic** and **molecular geometry** at carbon (C) labeled ***e*** and sulfur (S) labeled ***f*** centres. (2 points)

Atom	Electronic Geometry	Molecular Geometry
Carbon (<i>e</i>)		
Sulfur (<i>f</i>)		

vi. Circle the value from the choices given below, that describes the **C-S-C bond angle at sulfur** (S) marked ***f***: (0.5 points)

>90° to <109.5°

109.5°

>109.5 to <120°

120°

>120° to <180°


Question 8 12 points

Using the **valence shell orbitals only** draw a **complete** molecular orbital (MO) diagram (with no s-p mixing) for the molecule **PS**, in the space provided below. **(6 points)**

Assume that 3s and 3p atomic orbitals will form molecular orbitals like those formed by the combination of 2s and 2p atomic orbitals that we learnt in class, and there is no s-p mixing.

Show **energy arrow**, **atomic orbitals** involved, **atom labels**, the **molecular orbitals** that are formed, **filled electrons** in the MO diagram etc. Make sure to label all the atomic and molecular orbitals.

Using the MO diagram above, answer the following questions:

- (i) Which one of the following will have the highest bond order? Circle the correct choice
 A) PS^{1-} B) PS C) PS^{1+}
- (ii) Which one of the following will have the longest bond length? Circle the correct choice
 A) PS^{1-} B) PS C) PS^{1+}
- (iii) Which of the following will be diamagnetic? Circle the correct choice.
 A) PS^{1-} B) PS C) PS^{1+}
- (iv) Draw Lewis dot structure of the molecule with the shortest bond length (based on the MO diagram) from among the three (PS^{1-} , PS, PS^{1+}), and indicate the bond order calculated from the Lewis dot structure. **(3 points)**

Lewis Dot Structure

Bond Order



3C0C33E8-E90E-4A07-AD90-145CBC89DF76

midterm-exam-93388

#378 Page 10 of 10

SCRAP PAGE

**CHEMISTRY 110 –Midterm Exam****Thursday November 7, 2024, 6:30 – 8:30 pm**

Examiners: Pallavi Sirjoosingh and Ashok Kakkar

First name (please write as legibly as possible within the boxes)

--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--

Last name

--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--

Student ID number

--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--

This booklet has 8 questions on 10 pages (The last page can be used as a scrap page).

The staple must NOT be removed from the question booklet. Your intact question booklet must be returned at the end of the exam.

A separate booklet contains a periodic table, and datasheet.

Write your answers directly under the questions in the space provided. Do not write on the QR codes or near the top edges (as it may be cut off).

Please write **legibly and dark enough** for the scan to be readable. For students who use English as a second language, dictionaries are permitted. Calculators are allowed.

DO NOT ASK QUESTIONS OF THE INVIGILATORS about interpretations of the exam questions. If you have a concern about a question, please summarize it at the top of this page. All concerns will be evaluated after the exam. GOOD LUCK!

**Question 1 8 points** (*Assume all transitions are possible*)

a. 6 points Calculate the **energy (in kJ/mol)** required to ionize (completely remove) the electron from the **n=3** level of **He⁺ ion** (He: Helium). Show your work including equations. (3 sig. figs.)

b. 2 points

(I) Select the largest (longest) wavelength from the three choices below:

- i. Wavelength of photon to *completely remove* an electron from *1s orbital of Hydrogen atom*
- ii. Wavelength of photon to *completely remove* an electron from *1s orbital of Boron atom (B)*
- iii. Wavelength of photon to *completely remove* an electron from *1s orbital of Boron cation (B⁺)*

i or ii or iii

(II) Select the smallest (shortest) wavelength from the three choices below (*for H atom*):

- i. Wavelength of photon *emitted* for an electron transitioning from *2p to 1s orbital*
- ii. Wavelength of photon *absorbed* for an electron transitioning from *2s to 4p orbital*
- iii. Wavelength of photon *absorbed* for an electron transitioning from *1s to 3p orbital*

i or ii or iii



Question 2 8 points



A Chem110 student plots the graph shown on the left, from data obtained for the kinetic energy (eV) of the ejected electron versus frequency (s^{-1}) of the incident light, for an unknown metal. ($1\text{eV} = 1.6022 \times 10^{-19} \text{ J}$)

Based on the plot, answer the questions **a to c** below:

a. 1 point On the plot above, label the **threshold frequency**.

b. 2 points What is the value of the work potential (in J) of the unknown metal?

c. 3 points What is the *longest* wavelength (in nm) required to eject an electron from this metal's surface? Show your work including equation(s) used. (3 sig figs)

d. 2 points In the space provided below, plot the following

(Assume frequency is *above* threshold frequency):

- i. *Kinetic energy* of an ejected electron versus the *intensity* of the incident light (left)
- ii. *Number of electrons* ejected versus the *frequency* of the incident light (right)



**Question 3** **6 points****a. 2 points**

In the ground state of Ge (Germanium; $Z = 32$) atom, **name** the following orbitals:

i. Orbital containing the electron that requires the maximum energy to be removed.

ii. Orbital containing the electron that experiences the minimum effective nuclear charge.

b. 2 points

An unknown anion (A^-) has the following *excited* electronic configuration: $[Ar] 4s^2 3d^3 4p^1$

(Ar: Argon; $Z = 18$)

Give the **condensed ground state electronic configuration** of the atom, **A** and the cation, **A^{2+}** .
Show your work. (Assume all electronic transitions are possible)

Electronic Configuration of A:

Electronic Configuration of A^{2+} :

c. 2 points

In a single atom, determine how many electrons can have the following set of quantum numbers?

(i) $n = 5; l = 3$

In a single atom, determine how many electrons can have the following set of quantum numbers?

(i) $n = 3; l = 2; m_s = +1/2$

**Question 4** **8 points**

Rank the following from 1 to 4 in **increasing** order. Provide the ranking in the boxes provided.

Note: Atomic number (Z) are provided for relevant atoms

a) Increasing (lowest to highest) order of *ionization energy*: (1= lowest; 4 = highest)

Mg

Mg: Z = 12

Rb

Rb: Z = 37

Cs

Cs: Z = 55

K

K : Z = 19

b) Increasing (lowest to highest) order of *bond angle*: (1= lowest; 4 = highest)

H₂OCCl₄XeF₄

HCN

c) Increasing (smallest to largest) order of *ionic size* (1= smallest; 4 = largest)

S²⁻

S: Z = 16

O²⁻

O: Z = 8

Be²⁺

Be: Z = 4

Na⁺

Na: Z = 11

d) Increasing (lowest to highest) order of *bond energy* (1= lowest; 4 = highest)

P-F

P-Cl

P-I

P-Br

**Question 5** **12 points**

a. 9 points For the molecules HClO_4 and TeF_6 , draw the most stable *Lewis structure* showing appropriate shape and *dashed/wedged bonds*, based on VSEPR. Determine the **molecular geometry** around the central atom. For HClO_4 , label the *most polar bond*.



b. 3 points (Note: Lone pairs are not shown in structures below)

(i) For the three *resonance* structures (1, 2 and 3) shown below, answer the following question:



(I) Circle the *most contributing* resonance structure.

Structure 1

Structure 2

Structure 3

(ii) For the three structures given (4, 5, and 6) below, answer the following questions:



(I) Circle all the structures that represent *resonance structures* of each other:

Structure 4

Structure 5

Structure 6

(II) Among the resonance structures, circle the *most contributing* resonance structure.

Structure 4

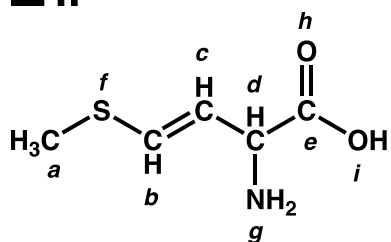
Structure 5

Structure 6

**Question 6** **6 points**

Using the data given below, calculate the **lattice energy** for SrO(s).

	ΔH° (kJ/mol)
$\text{Sr}(s) \longrightarrow \text{Sr}(g)$	+180
$\text{Sr}(g) \longrightarrow \text{Sr}^{2+}(g) + 2e^-$	+1468
$\text{O}_2(g) \longrightarrow 2\text{O}(g)$	+500
$\text{O}(g) + 2e^- \longrightarrow \text{O}^{2-}(g)$	+600
$\text{Sr}(s) + \frac{1}{2}\text{O}_2(g) \longrightarrow \text{SrO}(s)$	-550

**Question 7 10 points**

Consider the molecule given on the left, and answer questions (i) to (vi) using chemical bonding theories that we have learnt in class (valence bond theory, hybridization).

Note that the lone pairs are not shown in the structure:

i. Consider the carbon atoms marked in as ***a***, ***b***, ***c***, ***d***, and ***e*** in the molecule, and determine **hybridization** at each of the carbon centers. (2.5 points)

Atom	Hybridization
Carbon (<i>a</i>)	
Carbon (<i>b</i>)	
Carbon (<i>c</i>)	
Carbon (<i>d</i>)	
Carbon (<i>e</i>)	

ii. Now, consider the sulfur atom (S) marked as ***f***, nitrogen (N) atom (***g***), and oxygen(O) atoms (***h*** and ***i***), and determine their **hybridization**. (2 points)

Atom	Hybridization
Sulfur (<i>f</i>)	
Nitrogen (<i>g</i>)	
Oxygen (<i>h</i>)	
Oxygen (<i>i</i>)	

iii. **Describe** all the bonds formed by the carbon atoms ***a*** and ***b***, using sigma and pi designation as appropriate. (2 points)

Atom	Bonds formed (use sigma and pi as appropriate)
Carbon (<i>a</i>)	
Carbon (<i>b</i>)	

iv. How many total sp^3 -s sigma bonds are in the molecule? (1 point)

v. Determine **electronic** and **molecular geometry** at carbon (C) labeled ***e*** and sulfur (S) labeled ***f*** centres. (2 points)

Atom	Electronic Geometry	Molecular Geometry
Carbon (<i>e</i>)		
Sulfur (<i>f</i>)		

vi. Circle the value from the choices given below, that describes the **C-S-C bond angle at sulfur** (S) marked ***f***: (0.5 points)

>90° to <109.5°

109.5°

>109.5 to <120°

120°

>120° to <180°


Question 8 12 points

Using the **valence shell orbitals only** draw a **complete** molecular orbital (MO) diagram (with no s-p mixing) for the molecule **PS**, in the space provided below. **(6 points)**

Assume that 3s and 3p atomic orbitals will form molecular orbitals like those formed by the combination of 2s and 2p atomic orbitals that we learnt in class, and there is no s-p mixing.

Show **energy arrow**, **atomic orbitals** involved, **atom labels**, the **molecular orbitals** that are formed, **filled electrons** in the MO diagram etc. Make sure to label all the atomic and molecular orbitals.

Using the MO diagram above, answer the following questions:

- (i) Which one of the following will have the highest bond order? Circle the correct choice
 A) PS^{1-} B) PS C) PS^{1+}
- (ii) Which one of the following will have the longest bond length? Circle the correct choice
 A) PS^{1-} B) PS C) PS^{1+}
- (iii) Which of the following will be diamagnetic? Circle the correct choice.
 A) PS^{1-} B) PS C) PS^{1+}
- (iv) Draw Lewis dot structure of the molecule with the shortest bond length (based on the MO diagram) from among the three (PS^{1-} , PS, PS^{1+}), and indicate the bond order calculated from the Lewis dot structure. **(3 points)**

Lewis Dot Structure

Bond Order



B559B12D-A95F-422F-B6F2-53EF3BDC3FD8

midterm-exam-93388

#379 Page 10 of 10

SCRAP PAGE

**CHEMISTRY 110 –Midterm Exam****Thursday November 7, 2024, 6:30 – 8:30 pm**

Examiners: Pallavi Sirjoosingh and Ashok Kakkar

First name (please write as legibly as possible within the boxes)

--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--

Last name

--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--

Student ID number

--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--

This booklet has 8 questions on 10 pages (The last page can be used as a scrap page).

The staple must NOT be removed from the question booklet. Your intact question booklet must be returned at the end of the exam.

A separate booklet contains a periodic table, and datasheet.

Write your answers directly under the questions in the space provided. Do not write on the QR codes or near the top edges (as it may be cut off).

Please write **legibly and dark enough** for the scan to be readable. For students who use English as a second language, dictionaries are permitted. Calculators are allowed.

DO NOT ASK QUESTIONS OF THE INVIGILATORS about interpretations of the exam questions. If you have a concern about a question, please summarize it at the top of this page. All concerns will be evaluated after the exam. GOOD LUCK!

**Question 1 8 points** (*Assume all transitions are possible*)

a. 6 points Calculate the **energy (in kJ/mol)** required to ionize (completely remove) the electron from the **n=3** level of **He⁺ ion** (He: Helium). Show your work including equations. (3 sig. figs.)

b. 2 points

(I) Select the largest (longest) wavelength from the three choices below:

- i. Wavelength of photon to *completely remove* an electron from *1s orbital of Hydrogen atom*
- ii. Wavelength of photon to *completely remove* an electron from *1s orbital of Boron atom (B)*
- iii. Wavelength of photon to *completely remove* an electron from *1s orbital of Boron cation (B⁺)*

i or ii or iii

(II) Select the smallest (shortest) wavelength from the three choices below (*for H atom*):

- i. Wavelength of photon *emitted* for an electron transitioning from *2p to 1s orbital*
- ii. Wavelength of photon *absorbed* for an electron transitioning from *2s to 4p orbital*
- iii. Wavelength of photon *absorbed* for an electron transitioning from *1s to 3p orbital*

i or ii or iii



Question 2 8 points



A Chem110 student plots the graph shown on the left, from data obtained for the kinetic energy (eV) of the ejected electron versus frequency (s^{-1}) of the incident light, for an unknown metal. ($1\text{eV} = 1.6022 \times 10^{-19} \text{ J}$)

Based on the plot, answer the questions **a to c** below:

a. 1 point On the plot above, label the **threshold frequency**.

b. 2 points What is the value of the work potential (in J) of the unknown metal?

c. 3 points What is the *longest* wavelength (in nm) required to eject an electron from this metal's surface? Show your work including equation(s) used. (3 sig figs)

d. 2 points In the space provided below, plot the following

(Assume frequency is *above* threshold frequency):

- Kinetic energy* of an ejected electron versus the *intensity* of the incident light (left)
- Number of electrons* ejected versus the *frequency* of the incident light (right)



**Question 3** **6 points****a. 2 points**

In the ground state of Ge (Germanium; $Z = 32$) atom, **name** the following orbitals:

i. Orbital containing the electron that requires the maximum energy to be removed.

ii. Orbital containing the electron that experiences the minimum effective nuclear charge.

b. 2 points

An unknown anion (A^-) has the following *excited* electronic configuration: $[Ar] 4s^2 3d^3 4p^1$

(Ar: Argon; $Z = 18$)

Give the **condensed ground state electronic configuration** of the atom, **A** and the cation, A^{2+} .
Show your work. (Assume all electronic transitions are possible)

Electronic Configuration of A:

Electronic Configuration of A^{2+} :

c. 2 points

In a single atom, determine how many electrons can have the following set of quantum numbers?

(i) $n = 5; l = 3$

In a single atom, determine how many electrons can have the following set of quantum numbers?

(i) $n = 3; l = 2; m_s = +1/2$

**Question 4** **8 points**

Rank the following from 1 to 4 in **increasing** order. Provide the ranking in the boxes provided.

Note: Atomic number (Z) are provided for relevant atoms

a) Increasing (lowest to highest) order of *ionization energy*: (1= lowest; 4 = highest)

Mg

Mg: Z = 12

Rb

Rb: Z = 37

Cs

Cs: Z = 55

K

K : Z = 19

b) Increasing (lowest to highest) order of *bond angle*: (1= lowest; 4 = highest)

H₂OCCl₄XeF₄

HCN

c) Increasing (smallest to largest) order of *ionic size* (1= smallest; 4 = largest)

S²⁻

S: Z = 16

O²⁻

O: Z = 8

Be²⁺

Be: Z = 4

Na⁺

Na: Z = 11

d) Increasing (lowest to highest) order of *bond energy* (1= lowest; 4 = highest)

P-F

P-Cl

P-I

P-Br

**Question 5** **12 points**

a. 9 points For the molecules HClO_4 and TeF_6 , draw the most stable *Lewis structure* showing appropriate shape and *dashed/wedged bonds*, based on VSEPR. Determine the **molecular geometry** around the central atom. For HClO_4 , label the *most polar bond*.

HClO_4 (there are no O-O bonds)

Label the most polar bond

Molecular Geometry around Cl

TeF_6

Molecular Geometry around Te

b. 3 points (Note: Lone pairs are not shown in structures below)

(i) For the three *resonance* structures (1, 2 and 3) shown below, answer the following question:



(I) Circle the *most contributing* resonance structure.

Structure 1

Structure 2

Structure 3

(ii) For the three structures given (4, 5, and 6) below, answer the following questions:



(I) Circle all the structures that represent *resonance structures* of each other:

Structure 4

Structure 5

Structure 6

(II) Among the resonance structures, circle the *most contributing* resonance structure.

Structure 4

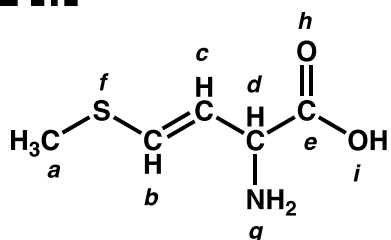
Structure 5

Structure 6

**Question 6** **6 points**

Using the data given below, calculate the **lattice energy** for SrO(s).

	ΔH° (kJ/mol)
$\text{Sr}(s) \longrightarrow \text{Sr}(g)$	+180
$\text{Sr}(g) \longrightarrow \text{Sr}^{2+}(g) + 2e^-$	+1468
$\text{O}_2(g) \longrightarrow 2\text{O}(g)$	+500
$\text{O}(g) + 2e^- \longrightarrow \text{O}^{2-}(g)$	+600
$\text{Sr}(s) + \frac{1}{2}\text{O}_2(g) \longrightarrow \text{SrO}(s)$	-550

**Question 7 10 points**

Consider the molecule given on the left, and answer questions (i) to (vi) using chemical bonding theories that we have learnt in class (valence bond theory, hybridization).

Note that the lone pairs are not shown in the structure:

i. Consider the carbon atoms marked in as ***a***, ***b***, ***c***, ***d***, and ***e*** in the molecule, and determine **hybridization** at each of the carbon centers. (2.5 points)

Atom	Hybridization
Carbon (<i>a</i>)	
Carbon (<i>b</i>)	
Carbon (<i>c</i>)	
Carbon (<i>d</i>)	
Carbon (<i>e</i>)	

ii. Now, consider the sulfur atom (S) marked as ***f***, nitrogen (N) atom (***g***), and oxygen(O) atoms (***h*** and ***i***), and determine their **hybridization**. (2 points)

Atom	Hybridization
Sulfur (<i>f</i>)	
Nitrogen (<i>g</i>)	
Oxygen (<i>h</i>)	
Oxygen (<i>i</i>)	

iii. **Describe** all the bonds formed by the carbon atoms ***a*** and ***b***, using sigma and pi designation as appropriate. (2 points)

Atom	Bonds formed (use sigma and pi as appropriate)
Carbon (<i>a</i>)	
Carbon (<i>b</i>)	

iv. How many total sp^3 -s sigma bonds are in the molecule? (1 point)

v. Determine **electronic** and **molecular geometry** at carbon (C) labeled ***e*** and sulfur (S) labeled ***f*** centres. (2 points)

Atom	Electronic Geometry	Molecular Geometry
Carbon (<i>e</i>)		
Sulfur (<i>f</i>)		

vi. Circle the value from the choices given below, that describes the **C-S-C bond angle at sulfur** (S) marked ***f***: (0.5 points)

>90° to <109.5°

109.5°

>109.5 to <120°

120°

>120° to <180°


Question 8 12 points

Using the **valence shell orbitals only** draw a **complete** molecular orbital (MO) diagram (with no s-p mixing) for the molecule **PS**, in the space provided below. **(6 points)**

Assume that 3s and 3p atomic orbitals will form molecular orbitals like those formed by the combination of 2s and 2p atomic orbitals that we learnt in class, and there is no s-p mixing.

Show **energy arrow**, **atomic orbitals** involved, **atom labels**, the **molecular orbitals** that are formed, **filled electrons** in the MO diagram etc. Make sure to label all the atomic and molecular orbitals.

Using the MO diagram above, answer the following questions:

- (i) Which one of the following will have the highest bond order? Circle the correct choice
 A) PS^{1-} B) PS C) PS^{1+}
- (ii) Which one of the following will have the longest bond length? Circle the correct choice
 A) PS^{1-} B) PS C) PS^{1+}
- (iii) Which of the following will be diamagnetic? Circle the correct choice.
 A) PS^{1-} B) PS C) PS^{1+}
- (iv) Draw Lewis dot structure of the molecule with the shortest bond length (based on the MO diagram) from among the three (PS^{1-} , PS, PS^{1+}), and indicate the bond order calculated from the Lewis dot structure. **(3 points)**

Lewis Dot Structure

Bond Order



AC5E61C7-4D49-48BC-BD57-978DF91EAB31

midterm-exam-93388

#380 Page 10 of 10

SCRAP PAGE

**CHEMISTRY 110 –Midterm Exam****Thursday November 7, 2024, 6:30 – 8:30 pm**

Examiners: Pallavi Sirjoosingh and Ashok Kakkar

First name (please write as legibly as possible within the boxes)

--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--

Last name

--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--

Student ID number

--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--

This booklet has 8 questions on 10 pages (The last page can be used as a scrap page).

The staple must NOT be removed from the question booklet. Your intact question booklet must be returned at the end of the exam.

A separate booklet contains a periodic table, and datasheet.

Write your answers directly under the questions in the space provided. Do not write on the QR codes or near the top edges (as it may be cut off).

Please write **legibly and dark enough** for the scan to be readable. For students who use English as a second language, dictionaries are permitted. Calculators are allowed.

DO NOT ASK QUESTIONS OF THE INVIGILATORS about interpretations of the exam questions. If you have a concern about a question, please summarize it at the top of this page. All concerns will be evaluated after the exam. GOOD LUCK!

**Question 1 8 points** (*Assume all transitions are possible*)

a. 6 points Calculate the **energy (in kJ/mol)** required to ionize (completely remove) the electron from the **$n=3$** level of **He^+ ion** (He: Helium). Show your work including equations. (3 sig. figs.)

b. 2 points

(I) Select the largest (longest) wavelength from the three choices below:

- i. Wavelength of photon to *completely remove* an electron from *1s orbital of Hydrogen atom*
- ii. Wavelength of photon to *completely remove* an electron from *1s orbital of Boron atom (B)*
- iii. Wavelength of photon to *completely remove* an electron from *1s orbital of Boron cation (B^+)*

i or ii or iii

(II) Select the smallest (shortest) wavelength from the three choices below (*for H atom*):

- i. Wavelength of photon *emitted* for an electron transitioning from *2p to 1s orbital*
- ii. Wavelength of photon *absorbed* for an electron transitioning from *2s to 4p orbital*
- iii. Wavelength of photon *absorbed* for an electron transitioning from *1s to 3p orbital*

i or ii or iii



Question 2 8 points



A Chem110 student plots the graph shown on the left, from data obtained for the kinetic energy (eV) of the ejected electron versus frequency (s^{-1}) of the incident light, for an unknown metal. ($1\text{eV} = 1.6022 \times 10^{-19} \text{ J}$)

Based on the plot, answer the questions **a to c** below:

a. 1 point On the plot above, label the **threshold frequency**.

b. 2 points What is the value of the work potential (in J) of the unknown metal?

c. 3 points What is the *longest* wavelength (in nm) required to eject an electron from this metal's surface? Show your work including equation(s) used. (3 sig figs)

d. 2 points In the space provided below, plot the following

(Assume frequency is *above* threshold frequency):

- i. *Kinetic energy* of an ejected electron versus the *intensity* of the incident light (left)
- ii. *Number of electrons* ejected versus the *frequency* of the incident light (right)



**Question 3** **6 points****a. 2 points**

In the ground state of Ge (Germanium; $Z = 32$) atom, **name** the following orbitals:

i. Orbital containing the electron that requires the maximum energy to be removed.

ii. Orbital containing the electron that experiences the minimum effective nuclear charge.

b. 2 points

An unknown anion (A^-) has the following *excited* electronic configuration: $[Ar] 4s^2 3d^3 4p^1$

(Ar: Argon; $Z = 18$)

Give the **condensed ground state electronic configuration** of the atom, **A** and the cation, **A^{2+}** .
Show your work. (Assume all electronic transitions are possible)

Electronic Configuration of A:

Electronic Configuration of A^{2+} :

c. 2 points

In a single atom, determine how many electrons can have the following set of quantum numbers?

(i) $n = 5; l = 3$

In a single atom, determine how many electrons can have the following set of quantum numbers?

(i) $n = 3; l = 2; m_s = +1/2$

**Question 4** **8 points**

Rank the following from 1 to 4 in **increasing** order. Provide the ranking in the boxes provided.

Note: Atomic number (Z) are provided for relevant atoms

a) Increasing (lowest to highest) order of *ionization energy*: (1= lowest; 4 = highest)

Mg

Mg: Z = 12

Rb

Rb: Z = 37

Cs

Cs: Z = 55

K

K : Z = 19

b) Increasing (lowest to highest) order of *bond angle*: (1= lowest; 4 = highest)

H₂OCCl₄XeF₄

HCN

c) Increasing (smallest to largest) order of *ionic size* (1= smallest; 4 = largest)

S²⁻

S: Z = 16

O²⁻

O: Z = 8

Be²⁺

Be: Z = 4

Na⁺

Na: Z = 11

d) Increasing (lowest to highest) order of *bond energy* (1= lowest; 4 = highest)

P-F

P-Cl

P-I

P-Br

**Question 5** **12 points**

a. 9 points For the molecules HClO_4 and TeF_6 , draw the most stable *Lewis structure* showing appropriate shape and *dashed/wedged bonds*, based on VSEPR. Determine the **molecular geometry** around the central atom. For HClO_4 , label the *most polar bond*.



b. 3 points (Note: Lone pairs are not shown in structures below)

(i) For the three *resonance* structures (1, 2 and 3) shown below, answer the following question:



(I) Circle the *most contributing* resonance structure.

Structure 1

Structure 2

Structure 3

(ii) For the three structures given (4, 5, and 6) below, answer the following questions:



(I) Circle all the structures that represent *resonance structures* of each other:

Structure 4

Structure 5

Structure 6

(II) Among the resonance structures, circle the *most contributing* resonance structure.

Structure 4

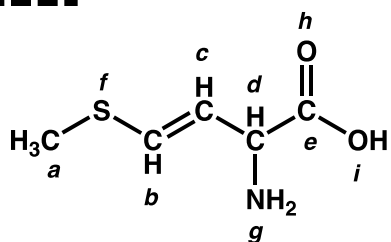
Structure 5

Structure 6

**Question 6** **6 points**

Using the data given below, calculate the **lattice energy** for SrO(s).

	ΔH° (kJ/mol)
$\text{Sr}(s) \longrightarrow \text{Sr}(g)$	+180
$\text{Sr}(g) \longrightarrow \text{Sr}^{2+}(g) + 2e^-$	+1468
$\text{O}_2(g) \longrightarrow 2\text{O}(g)$	+500
$\text{O}(g) + 2e^- \longrightarrow \text{O}^{2-}(g)$	+600
$\text{Sr}(s) + \frac{1}{2}\text{O}_2(g) \longrightarrow \text{SrO}(s)$	-550

**Question 7 10 points**

Consider the molecule given on the left, and answer questions (i) to (vi) using chemical bonding theories that we have learnt in class (valence bond theory, hybridization).

Note that the lone pairs are not shown in the structure:

i. Consider the carbon atoms marked in as ***a***, ***b***, ***c***, ***d***, and ***e*** in the molecule, and determine **hybridization** at each of the carbon centers. (2.5 points)

Atom	Hybridization
Carbon (<i>a</i>)	
Carbon (<i>b</i>)	
Carbon (<i>c</i>)	
Carbon (<i>d</i>)	
Carbon (<i>e</i>)	

ii. Now, consider the sulfur atom (S) marked as ***f***, nitrogen (N) atom (***g***), and oxygen(O) atoms (***h*** and ***i***), and determine their **hybridization**. (2 points)

Atom	Hybridization
Sulfur (<i>f</i>)	
Nitrogen (<i>g</i>)	
Oxygen (<i>h</i>)	
Oxygen (<i>i</i>)	

iii. **Describe** all the bonds formed by the carbon atoms ***a*** and ***b***, using sigma and pi designation as appropriate. (2 points)

Atom	Bonds formed (use sigma and pi as appropriate)
Carbon (<i>a</i>)	
Carbon (<i>b</i>)	

iv. How many total sp^3 -s sigma bonds are in the molecule? (1 point)

v. Determine **electronic** and **molecular geometry** at carbon (C) labeled ***e*** and sulfur (S) labeled ***f*** centres. (2 points)

Atom	Electronic Geometry	Molecular Geometry
Carbon (<i>e</i>)		
Sulfur (<i>f</i>)		

vi. Circle the value from the choices given below, that describes the **C-S-C bond angle at sulfur** (S) marked ***f***: (0.5 points)

>90° to <109.5° 109.5° >109.5 to <120° 120° >120° to <180°

**Question 8 12 points**

Using the **valence shell orbitals only** draw a **complete** molecular orbital (MO) diagram (with no s-p mixing) for the molecule **PS**, in the space provided below. **(6 points)**

Assume that 3s and 3p atomic orbitals will form molecular orbitals like those formed by the combination of 2s and 2p atomic orbitals that we learnt in class, and there is no s-p mixing.

Show **energy arrow**, **atomic orbitals** involved, **atom labels**, the **molecular orbitals** that are formed, **filled electrons** in the MO diagram etc. Make sure to **label all the atomic and molecular orbitals**.

Using the MO diagram above, answer the following questions:

- (i) Which one of the following will have the highest bond order? Circle the correct choice
A) PS^{1-} B) PS C) PS^{1+}
- (ii) Which one of the following will have the longest bond length? Circle the correct choice
A) PS^{1-} B) PS C) PS^{1+}
- (iii) Which of the following will be diamagnetic? Circle the correct choice.
A) PS^{1-} B) PS C) PS^{1+}
- (iv) Draw Lewis dot structure of the molecule with the shortest bond length (based on the MO diagram) from among the three (PS^{1-} , PS, PS^{1+}), and indicate the bond order calculated from the Lewis dot structure. **(3 points)**

Lewis Dot Structure

Bond Order



08C99556-80CA-4191-AF35-C1CACFBAE88F

midterm-exam-93388

#381 Page 10 of 10

SCRAP PAGE

**CHEMISTRY 110 –Midterm Exam****Thursday November 7, 2024, 6:30 – 8:30 pm****Examiners: Pallavi Sirjoosingh and Ashok Kakkar**

First name (please write as legibly as possible within the boxes)

--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--

Last name

--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--

Student ID number

--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--

This booklet has 8 questions on 10 pages (The last page can be used as a scrap page).

The staple must NOT be removed from the question booklet. Your intact question booklet must be returned at the end of the exam.

A separate booklet contains a periodic table, and datasheet.

Write your answers directly under the questions in the space provided. Do not write on the QR codes or near the top edges (as it may be cut off).

Please write **legibly and dark enough** for the scan to be readable. For students who use English as a second language, dictionaries are permitted. Calculators are allowed.

DO NOT ASK QUESTIONS OF THE INVIGILATORS about interpretations of the exam questions. If you have a concern about a question, please summarize it at the top of this page. All concerns will be evaluated after the exam. GOOD LUCK!

**Question 1 8 points** (*Assume all transitions are possible*)

a. 6 points Calculate the **energy (in kJ/mol)** required to ionize (completely remove) the electron from the **n=3** level of **He⁺ ion** (He: Helium). Show your work including equations. (3 sig. figs.)

b. 2 points

(I) Select the largest (longest) wavelength from the three choices below:

- i. Wavelength of photon to *completely remove* an electron from *1s orbital of Hydrogen atom*
- ii. Wavelength of photon to *completely remove* an electron from *1s orbital of Boron atom (B)*
- iii. Wavelength of photon to *completely remove* an electron from *1s orbital of Boron cation (B⁺)*

i or ii or iii

(II) Select the smallest (shortest) wavelength from the three choices below (*for H atom*):

- i. Wavelength of photon *emitted* for an electron transitioning from *2p to 1s orbital*
- ii. Wavelength of photon *absorbed* for an electron transitioning from *2s to 4p orbital*
- iii. Wavelength of photon *absorbed* for an electron transitioning from *1s to 3p orbital*

i or ii or iii



Question 2 8 points



A Chem110 student plots the graph shown on the left, from data obtained for the kinetic energy (eV) of the ejected electron versus frequency (s^{-1}) of the incident light, for an unknown metal. ($1\text{eV} = 1.6022 \times 10^{-19} \text{ J}$)

Based on the plot, answer the questions **a to c** below:

a. 1 point On the plot above, label the **threshold frequency**.

b. 2 points What is the value of the work potential (in J) of the unknown metal?

c. 3 points What is the *longest* wavelength (in nm) required to eject an electron from this metal's surface? Show your work including equation(s) used. (3 sig figs)

d. 2 points In the space provided below, plot the following

(Assume frequency is *above* threshold frequency):

- i. *Kinetic energy* of an ejected electron versus the *intensity* of the incident light (left)
- ii. *Number of electrons* ejected versus the *frequency* of the incident light (right)



**Question 3** **6 points****a. 2 points**

In the ground state of Ge (Germanium; $Z = 32$) atom, **name** the following orbitals:

i. Orbital containing the electron that requires the maximum energy to be removed.

ii. Orbital containing the electron that experiences the minimum effective nuclear charge.

b. 2 points

An unknown anion (A^-) has the following *excited* electronic configuration: $[Ar] 4s^2 3d^3 4p^1$

(Ar: Argon; $Z = 18$)

Give the **condensed ground state electronic configuration** of the atom, **A** and the cation, **A^{2+}** .
Show your work. (Assume all electronic transitions are possible)

Electronic Configuration of A:

Electronic Configuration of A^{2+} :

c. 2 points

In a single atom, determine how many electrons can have the following set of quantum numbers?

(i) $n = 5; l = 3$

In a single atom, determine how many electrons can have the following set of quantum numbers?

(i) $n = 3; l = 2; m_s = +1/2$

**Question 4** **8 points**

Rank the following from 1 to 4 in **increasing** order. Provide the ranking in the boxes provided.

Note: Atomic number (Z) are provided for relevant atoms

a) Increasing (lowest to highest) order of *ionization energy*: (1= lowest; 4 = highest)

Mg

Mg: Z = 12

Rb

Rb: Z = 37

Cs

Cs: Z = 55

K

K : Z = 19

b) Increasing (lowest to highest) order of *bond angle*: (1= lowest; 4 = highest)

H₂OCCl₄XeF₄

HCN

c) Increasing (smallest to largest) order of *ionic size* (1= smallest; 4 = largest)

S²⁻

S: Z = 16

O²⁻

O: Z = 8

Be²⁺

Be: Z = 4

Na⁺

Na: Z = 11

d) Increasing (lowest to highest) order of *bond energy* (1= lowest; 4 = highest)

P-F

P-Cl

P-I

P-Br

**Question 5** **12 points**

a. 9 points For the molecules HClO_4 and TeF_6 , draw the most stable *Lewis structure* showing appropriate shape and *dashed/wedged bonds*, based on VSEPR. Determine the **molecular geometry** around the central atom. For HClO_4 , label the *most polar bond*.



b. 3 points (Note: Lone pairs are not shown in structures below)

(i) For the three *resonance* structures (1, 2 and 3) shown below, answer the following question:



(I) Circle the *most contributing* resonance structure.

Structure 1

Structure 2

Structure 3

(ii) For the three structures given (4, 5, and 6) below, answer the following questions:



(I) Circle all the structures that represent *resonance structures* of each other:

Structure 4

Structure 5

Structure 6

(II) Among the resonance structures, circle the *most contributing* resonance structure.

Structure 4

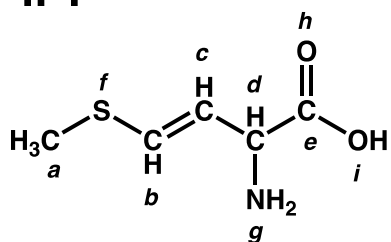
Structure 5

Structure 6

**Question 6** **6 points**

Using the data given below, calculate the **lattice energy** for SrO(s).

	ΔH° (kJ/mol)
$\text{Sr}(s) \longrightarrow \text{Sr}(g)$	+180
$\text{Sr}(g) \longrightarrow \text{Sr}^{2+}(g) + 2e^-$	+1468
$\text{O}_2(g) \longrightarrow 2\text{O}(g)$	+500
$\text{O}(g) + 2e^- \longrightarrow \text{O}^{2-}(g)$	+600
$\text{Sr}(s) + \frac{1}{2}\text{O}_2(g) \longrightarrow \text{SrO}(s)$	-550

**Question 7 10 points**

Consider the molecule given on the left, and answer questions (i) to (vi) using chemical bonding theories that we have learnt in class (valence bond theory, hybridization).

Note that the lone pairs are not shown in the structure:

i. Consider the carbon atoms marked in as ***a***, ***b***, ***c***, ***d***, and ***e*** in the molecule, and determine **hybridization** at each of the carbon centers. (2.5 points)

Atom	Hybridization
Carbon (<i>a</i>)	
Carbon (<i>b</i>)	
Carbon (<i>c</i>)	
Carbon (<i>d</i>)	
Carbon (<i>e</i>)	

ii. Now, consider the sulfur atom (S) marked as ***f***, nitrogen (N) atom (***g***), and oxygen(O) atoms (***h*** and ***i***), and determine their **hybridization**. (2 points)

Atom	Hybridization
Sulfur (<i>f</i>)	
Nitrogen (<i>g</i>)	
Oxygen (<i>h</i>)	
Oxygen (<i>i</i>)	

iii. **Describe** all the bonds formed by the carbon atoms ***a*** and ***b***, using sigma and pi designation as appropriate. (2 points)

Atom	Bonds formed (use sigma and pi as appropriate)
Carbon (<i>a</i>)	
Carbon (<i>b</i>)	

iv. How many total sp^3 -s sigma bonds are in the molecule? (1 point)

v. Determine **electronic** and **molecular geometry** at carbon (C) labeled ***e*** and sulfur (S) labeled ***f*** centres. (2 points)

Atom	Electronic Geometry	Molecular Geometry
Carbon (<i>e</i>)		
Sulfur (<i>f</i>)		

vi. Circle the value from the choices given below, that describes the **C-S-C bond angle at sulfur** (S) marked ***f***: (0.5 points)

>90° to <109.5° 109.5° >109.5 to <120° 120° >120° to <180°


Question 8 12 points

Using the **valence shell orbitals only** draw a **complete** molecular orbital (MO) diagram (with no s-p mixing) for the molecule **PS**, in the space provided below. **(6 points)**

Assume that 3s and 3p atomic orbitals will form molecular orbitals like those formed by the combination of 2s and 2p atomic orbitals that we learnt in class, and there is no s-p mixing.

Show **energy arrow**, **atomic orbitals** involved, **atom labels**, the **molecular orbitals** that are formed, **filled electrons** in the MO diagram etc. Make sure to label all the atomic and molecular orbitals.

Using the MO diagram above, answer the following questions:

- (i) Which one of the following will have the highest bond order? Circle the correct choice
 A) PS^{1-} B) PS C) PS^{1+}
- (ii) Which one of the following will have the longest bond length? Circle the correct choice
 A) PS^{1-} B) PS C) PS^{1+}
- (iii) Which of the following will be diamagnetic? Circle the correct choice.
 A) PS^{1-} B) PS C) PS^{1+}
- (iv) Draw Lewis dot structure of the molecule with the shortest bond length (based on the MO diagram) from among the three (PS^{1-} , PS, PS^{1+}), and indicate the bond order calculated from the Lewis dot structure. **(3 points)**

Lewis Dot Structure

Bond Order



5019D2B7-7FA2-4970-A137-4E8868AFCCD6

midterm-exam-93388

#382 Page 10 of 10

SCRAP PAGE

**CHEMISTRY 110 –Midterm Exam****Thursday November 7, 2024, 6:30 – 8:30 pm****Examiners: Pallavi Sirjoosingh and Ashok Kakkar**

First name (please write as legibly as possible within the boxes)

--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--

Last name

--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--

Student ID number

--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--

This booklet has 8 questions on 10 pages (The last page can be used as a scrap page).

The staple must NOT be removed from the question booklet. Your intact question booklet must be returned at the end of the exam.

A separate booklet contains a periodic table, and datasheet.

Write your answers directly under the questions in the space provided. Do not write on the QR codes or near the top edges (as it may be cut off).

Please write **legibly and dark enough** for the scan to be readable. For students who use English as a second language, dictionaries are permitted. Calculators are allowed.

DO NOT ASK QUESTIONS OF THE INVIGILATORS about interpretations of the exam questions. If you have a concern about a question, please summarize it at the top of this page. All concerns will be evaluated after the exam. GOOD LUCK!

**Question 1 8 points** (*Assume all transitions are possible*)

a. 6 points Calculate the **energy (in kJ/mol)** required to ionize (completely remove) the electron from the **n=3** level of **He⁺ ion** (He: Helium). Show your work including equations. (3 sig. figs.)

b. 2 points

(I) Select the largest (longest) wavelength from the three choices below:

- i. Wavelength of photon to *completely remove* an electron from *1s orbital of Hydrogen atom*
- ii. Wavelength of photon to *completely remove* an electron from *1s orbital of Boron atom (B)*
- iii. Wavelength of photon to *completely remove* an electron from *1s orbital of Boron cation (B⁺)*

i or ii or iii

(II) Select the smallest (shortest) wavelength from the three choices below (*for H atom*):

- i. Wavelength of photon *emitted* for an electron transitioning from *2p to 1s orbital*
- ii. Wavelength of photon *absorbed* for an electron transitioning from *2s to 4p orbital*
- iii. Wavelength of photon *absorbed* for an electron transitioning from *1s to 3p orbital*

i or ii or iii



Question 2 8 points



A Chem110 student plots the graph shown on the left, from data obtained for the kinetic energy (eV) of the ejected electron versus frequency (s^{-1}) of the incident light, for an unknown metal. ($1\text{eV} = 1.6022 \times 10^{-19} \text{ J}$)

Based on the plot, answer the questions **a to c** below:

a. 1 point On the plot above, label the **threshold frequency**.

b. 2 points What is the value of the work potential (in J) of the unknown metal?

c. 3 points What is the *longest* wavelength (in nm) required to eject an electron from this metal's surface? Show your work including equation(s) used. (3 sig figs)

d. 2 points In the space provided below, plot the following

(Assume frequency is *above* threshold frequency):

- i. *Kinetic energy* of an ejected electron versus the *intensity* of the incident light (left)
- ii. *Number of electrons* ejected versus the *frequency* of the incident light (right)



**Question 3** **6 points****a. 2 points**

In the ground state of Ge (Germanium; $Z = 32$) atom, **name** the following orbitals:

i. Orbital containing the electron that requires the maximum energy to be removed.

ii. Orbital containing the electron that experiences the minimum effective nuclear charge.

b. 2 points

An unknown anion (A^-) has the following *excited* electronic configuration: $[Ar] 4s^2 3d^3 4p^1$

(Ar: Argon; $Z = 18$)

Give the **condensed ground state electronic configuration** of the atom, **A** and the cation, **A^{2+}** .
Show your work. (Assume all electronic transitions are possible)

Electronic Configuration of A:

Electronic Configuration of A^{2+} :

c. 2 points

In a single atom, determine how many electrons can have the following set of quantum numbers?

(i) $n = 5; l = 3$

In a single atom, determine how many electrons can have the following set of quantum numbers?

(i) $n = 3; l = 2; m_s = +1/2$

**Question 4** **8 points**

Rank the following from 1 to 4 in **increasing** order. Provide the ranking in the boxes provided.

Note: Atomic number (Z) are provided for relevant atoms

a) Increasing (lowest to highest) order of *ionization energy*: (1= lowest; 4 = highest)

Mg

Mg: Z = 12

Rb

Rb: Z = 37

Cs

Cs: Z = 55

K

K : Z = 19

b) Increasing (lowest to highest) order of *bond angle*: (1= lowest; 4 = highest)

H₂OCCl₄XeF₄

HCN

c) Increasing (smallest to largest) order of *ionic size* (1= smallest; 4 = largest)

S²⁻

S: Z = 16

O²⁻

O: Z = 8

Be²⁺

Be: Z = 4

Na⁺

Na: Z = 11

d) Increasing (lowest to highest) order of *bond energy* (1= lowest; 4 = highest)

P-F

P-Cl

P-I

P-Br

**Question 5** **12 points**

a. 9 points For the molecules HClO_4 and TeF_6 , draw the most stable *Lewis structure* showing appropriate shape and *dashed/wedged bonds*, based on VSEPR. Determine the **molecular geometry** around the central atom. For HClO_4 , label the *most polar bond*.

HClO_4 (there are no O-O bonds)

Label the most polar bond

Molecular Geometry around Cl

TeF_6

Molecular Geometry around Te

b. 3 points (Note: Lone pairs are not shown in structures below)

(i) For the three *resonance* structures (1, 2 and 3) shown below, answer the following question:



(I) Circle the *most contributing* resonance structure.

Structure 1

Structure 2

Structure 3

(ii) For the three structures given (4, 5, and 6) below, answer the following questions:



(I) Circle all the structures that represent *resonance structures* of each other:

Structure 4

Structure 5

Structure 6

(II) Among the resonance structures, circle the *most contributing* resonance structure.

Structure 4

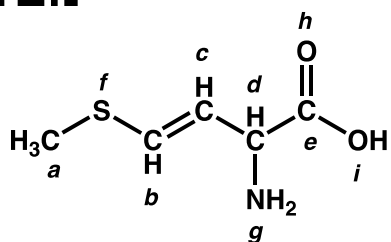
Structure 5

Structure 6

**Question 6** **6 points**

Using the data given below, calculate the **lattice energy** for SrO(s).

	ΔH° (kJ/mol)
$\text{Sr}(s) \longrightarrow \text{Sr}(g)$	+180
$\text{Sr}(g) \longrightarrow \text{Sr}^{2+}(g) + 2e^-$	+1468
$\text{O}_2(g) \longrightarrow 2\text{O}(g)$	+500
$\text{O}(g) + 2e^- \longrightarrow \text{O}^{2-}(g)$	+600
$\text{Sr}(s) + \frac{1}{2}\text{O}_2(g) \longrightarrow \text{SrO}(s)$	-550

**Question 7 10 points**

Consider the molecule given on the left, and answer questions (i) to (vi) using chemical bonding theories that we have learnt in class (valence bond theory, hybridization).

Note that the lone pairs are not shown in the structure:

i. Consider the carbon atoms marked in as ***a***, ***b***, ***c***, ***d***, and ***e*** in the molecule, and determine **hybridization** at each of the carbon centers. (2.5 points)

Atom	Hybridization
Carbon (<i>a</i>)	
Carbon (<i>b</i>)	
Carbon (<i>c</i>)	
Carbon (<i>d</i>)	
Carbon (<i>e</i>)	

ii. Now, consider the sulfur atom (S) marked as ***f***, nitrogen (N) atom (***g***), and oxygen(O) atoms (***h*** and ***i***), and determine their **hybridization**. (2 points)

Atom	Hybridization
Sulfur (<i>f</i>)	
Nitrogen (<i>g</i>)	
Oxygen (<i>h</i>)	
Oxygen (<i>i</i>)	

iii. **Describe** all the bonds formed by the carbon atoms ***a*** and ***b***, using sigma and pi designation as appropriate. (2 points)

Atom	Bonds formed (use sigma and pi as appropriate)
Carbon (<i>a</i>)	
Carbon (<i>b</i>)	

iv. How many total sp^3 -s sigma bonds are in the molecule? (1 point)

v. Determine **electronic** and **molecular geometry** at carbon (C) labeled ***e*** and sulfur (S) labeled ***f*** centres. (2 points)

Atom	Electronic Geometry	Molecular Geometry
Carbon (<i>e</i>)		
Sulfur (<i>f</i>)		

vi. Circle the value from the choices given below, that describes the **C-S-C bond angle at sulfur** (S) marked ***f***: (0.5 points)

>90° to <109.5° 109.5° >109.5 to <120° 120° >120° to <180°


Question 8 12 points

Using the **valence shell orbitals only** draw a **complete** molecular orbital (MO) diagram (with no s-p mixing) for the molecule **PS**, in the space provided below. **(6 points)**

Assume that 3s and 3p atomic orbitals will form molecular orbitals like those formed by the combination of 2s and 2p atomic orbitals that we learnt in class, and there is no s-p mixing.

Show **energy arrow**, **atomic orbitals** involved, **atom labels**, the **molecular orbitals** that are formed, **filled electrons** in the MO diagram etc. Make sure to label all the atomic and molecular orbitals.

Using the MO diagram above, answer the following questions:

- (i) Which one of the following will have the highest bond order? Circle the correct choice
 A) PS^{1-} B) PS C) PS^{1+}
- (ii) Which one of the following will have the longest bond length? Circle the correct choice
 A) PS^{1-} B) PS C) PS^{1+}
- (iii) Which of the following will be diamagnetic? Circle the correct choice.
 A) PS^{1-} B) PS C) PS^{1+}
- (iv) Draw Lewis dot structure of the molecule with the shortest bond length (based on the MO diagram) from among the three (PS^{1-} , PS, PS^{1+}), and indicate the bond order calculated from the Lewis dot structure. **(3 points)**

Lewis Dot Structure

Bond Order



D05E2A44-541B-49FF-9537-66FB7F77D49E

midterm-exam-93388

#383 Page 10 of 10

SCRAP PAGE

**CHEMISTRY 110 –Midterm Exam****Thursday November 7, 2024, 6:30 – 8:30 pm**

Examiners: Pallavi Sirjoosingh and Ashok Kakkar

First name (please write as legibly as possible within the boxes)

--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--

Last name

--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--

Student ID number

--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--

This booklet has 8 questions on 10 pages (The last page can be used as a scrap page).

The staple must NOT be removed from the question booklet. Your intact question booklet must be returned at the end of the exam.

A separate booklet contains a periodic table, and datasheet.

Write your answers directly under the questions in the space provided. Do not write on the QR codes or near the top edges (as it may be cut off).

Please write **legibly and dark enough** for the scan to be readable. For students who use English as a second language, dictionaries are permitted. Calculators are allowed.

DO NOT ASK QUESTIONS OF THE INVIGILATORS about interpretations of the exam questions. If you have a concern about a question, please summarize it at the top of this page. All concerns will be evaluated after the exam. GOOD LUCK!

**Question 1 8 points** (*Assume all transitions are possible*)

a. 6 points Calculate the **energy (in kJ/mol)** required to ionize (completely remove) the electron from the **n=3** level of **He⁺ ion** (He: Helium). Show your work including equations. (3 sig. figs.)

b. 2 points

(I) Select the largest (longest) wavelength from the three choices below:

- i. Wavelength of photon to *completely remove* an electron from *1s orbital of Hydrogen atom*
- ii. Wavelength of photon to *completely remove* an electron from *1s orbital of Boron atom (B)*
- iii. Wavelength of photon to *completely remove* an electron from *1s orbital of Boron cation (B⁺)*

i or ii or iii

(II) Select the smallest (shortest) wavelength from the three choices below (*for H atom*):

- i. Wavelength of photon *emitted* for an electron transitioning from *2p to 1s orbital*
- ii. Wavelength of photon *absorbed* for an electron transitioning from *2s to 4p orbital*
- iii. Wavelength of photon *absorbed* for an electron transitioning from *1s to 3p orbital*

i or ii or iii



Question 2 8 points



A Chem110 student plots the graph shown on the left, from data obtained for the kinetic energy (eV) of the ejected electron versus frequency (s^{-1}) of the incident light, for an unknown metal. ($1\text{eV} = 1.6022 \times 10^{-19} \text{ J}$)

Based on the plot, answer the questions **a to c** below:

a. 1 point On the plot above, label the **threshold frequency**.

b. 2 points What is the value of the work potential (in J) of the unknown metal?

c. 3 points What is the *longest* wavelength (in nm) required to eject an electron from this metal's surface? Show your work including equation(s) used. (3 sig figs)

d. 2 points In the space provided below, plot the following

(Assume frequency is *above* threshold frequency):

- i. *Kinetic energy* of an ejected electron versus the *intensity* of the incident light (left)
- ii. *Number of electrons* ejected versus the *frequency* of the incident light (right)



**Question 3** **6 points****a. 2 points**

In the ground state of Ge (Germanium; $Z = 32$) atom, **name** the following orbitals:

i. Orbital containing the electron that requires the maximum energy to be removed.

ii. Orbital containing the electron that experiences the minimum effective nuclear charge.

b. 2 points

An unknown anion (A^-) has the following *excited* electronic configuration: $[Ar] 4s^2 3d^3 4p^1$

(Ar: Argon; $Z = 18$)

Give the **condensed ground state electronic configuration** of the atom, **A** and the cation, **A^{2+}** .
Show your work. (Assume all electronic transitions are possible)

Electronic Configuration of A:

Electronic Configuration of A^{2+} :

c. 2 points

In a single atom, determine how many electrons can have the following set of quantum numbers?

(i) $n = 5; l = 3$

In a single atom, determine how many electrons can have the following set of quantum numbers?

(i) $n = 3; l = 2; m_s = +1/2$

**Question 4** **8 points**

Rank the following from 1 to 4 in **increasing** order. Provide the ranking in the boxes provided.

Note: Atomic number (Z) are provided for relevant atoms

a) Increasing (lowest to highest) order of *ionization energy*: (1= lowest; 4 = highest)

Mg

Mg: Z = 12

Rb

Rb: Z = 37

Cs

Cs: Z = 55

K

K : Z = 19

b) Increasing (lowest to highest) order of *bond angle*: (1= lowest; 4 = highest)

H₂OCCl₄XeF₄

HCN

c) Increasing (smallest to largest) order of *ionic size* (1= smallest; 4 = largest)

S²⁻

S: Z = 16

O²⁻

O: Z = 8

Be²⁺

Be: Z = 4

Na⁺

Na: Z = 11

d) Increasing (lowest to highest) order of *bond energy* (1= lowest; 4 = highest)

P-F

P-Cl

P-I

P-Br

**Question 5** **12 points**

a. 9 points For the molecules HClO_4 and TeF_6 , draw the most stable *Lewis structure* showing appropriate shape and *dashed/wedged bonds*, based on VSEPR. Determine the **molecular geometry** around the central atom. For HClO_4 , label the *most polar bond*.



b. 3 points (Note: Lone pairs are not shown in structures below)

(i) For the three *resonance* structures (1, 2 and 3) shown below, answer the following question:



(I) Circle the *most contributing* resonance structure.

Structure 1

Structure 2

Structure 3

(ii) For the three structures given (4, 5, and 6) below, answer the following questions:



(I) Circle all the structures that represent *resonance structures* of each other:

Structure 4

Structure 5

Structure 6

(II) Among the resonance structures, circle the *most contributing* resonance structure.

Structure 4

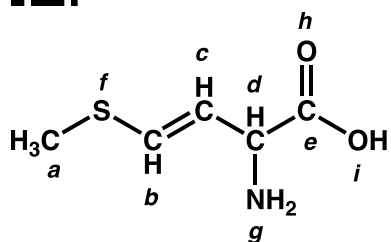
Structure 5

Structure 6

**Question 6** **6 points**

Using the data given below, calculate the **lattice energy** for SrO(s).

	ΔH° (kJ/mol)
$\text{Sr}(s) \longrightarrow \text{Sr}(g)$	+180
$\text{Sr}(g) \longrightarrow \text{Sr}^{2+}(g) + 2e^-$	+1468
$\text{O}_2(g) \longrightarrow 2\text{O}(g)$	+500
$\text{O}(g) + 2e^- \longrightarrow \text{O}^{2-}(g)$	+600
$\text{Sr}(s) + \frac{1}{2}\text{O}_2(g) \longrightarrow \text{SrO}(s)$	-550

**Question 7 10 points**

Consider the molecule given on the left, and answer questions (i) to (vi) using chemical bonding theories that we have learnt in class (valence bond theory, hybridization).

Note that the lone pairs are not shown in the structure:

i. Consider the carbon atoms marked in as ***a***, ***b***, ***c***, ***d***, and ***e*** in the molecule, and determine **hybridization** at each of the carbon centers. (2.5 points)

Atom	Hybridization
Carbon (<i>a</i>)	
Carbon (<i>b</i>)	
Carbon (<i>c</i>)	
Carbon (<i>d</i>)	
Carbon (<i>e</i>)	

ii. Now, consider the sulfur atom (S) marked as ***f***, nitrogen (N) atom (***g***), and oxygen(O) atoms (***h*** and ***i***), and determine their **hybridization**. (2 points)

Atom	Hybridization
Sulfur (<i>f</i>)	
Nitrogen (<i>g</i>)	
Oxygen (<i>h</i>)	
Oxygen (<i>i</i>)	

iii. **Describe** all the bonds formed by the carbon atoms ***a*** and ***b***, using sigma and pi designation as appropriate. (2 points)

Atom	Bonds formed (use sigma and pi as appropriate)
Carbon (<i>a</i>)	
Carbon (<i>b</i>)	

iv. How many total sp^3 -s sigma bonds are in the molecule? (1 point)

v. Determine **electronic** and **molecular geometry** at carbon (C) labeled ***e*** and sulfur (S) labeled ***f*** centres. (2 points)

Atom	Electronic Geometry	Molecular Geometry
Carbon (<i>e</i>)		
Sulfur (<i>f</i>)		

vi. Circle the value from the choices given below, that describes the **C-S-C bond angle at sulfur** (S) marked ***f***: (0.5 points)

>90° to <109.5°

109.5°

>109.5 to <120°

120°

>120° to <180°


Question 8 12 points

Using the **valence shell orbitals only** draw a **complete** molecular orbital (MO) diagram (with no s-p mixing) for the molecule **PS**, in the space provided below. **(6 points)**

Assume that 3s and 3p atomic orbitals will form molecular orbitals like those formed by the combination of 2s and 2p atomic orbitals that we learnt in class, and there is no s-p mixing.

Show **energy arrow**, **atomic orbitals** involved, **atom labels**, the **molecular orbitals** that are formed, **filled electrons** in the MO diagram etc. Make sure to label all the atomic and molecular orbitals.

Using the MO diagram above, answer the following questions:

- (i) Which one of the following will have the highest bond order? Circle the correct choice
 A) PS^{1-} B) PS C) PS^{1+}
- (ii) Which one of the following will have the longest bond length? Circle the correct choice
 A) PS^{1-} B) PS C) PS^{1+}
- (iii) Which of the following will be diamagnetic? Circle the correct choice.
 A) PS^{1-} B) PS C) PS^{1+}
- (iv) Draw Lewis dot structure of the molecule with the shortest bond length (based on the MO diagram) from among the three (PS^{1-} , PS, PS^{1+}), and indicate the bond order calculated from the Lewis dot structure. **(3 points)**

Lewis Dot Structure

Bond Order



D49CBD5B-0C64-4427-8DFD-F3BB0B328B9D

midterm-exam-93388

#384 Page 10 of 10

SCRAP PAGE

**CHEMISTRY 110 –Midterm Exam****Thursday November 7, 2024, 6:30 – 8:30 pm**

Examiners: Pallavi Sirjoosingh and Ashok Kakkar

First name (please write as legibly as possible within the boxes)

--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--

Last name

--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--

Student ID number

--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--

This booklet has 8 questions on 10 pages (The last page can be used as a scrap page).

The staple must NOT be removed from the question booklet. Your intact question booklet must be returned at the end of the exam.

A separate booklet contains a periodic table, and datasheet.

Write your answers directly under the questions in the space provided. Do not write on the QR codes or near the top edges (as it may be cut off).

Please write **legibly and dark enough** for the scan to be readable. For students who use English as a second language, dictionaries are permitted. Calculators are allowed.

DO NOT ASK QUESTIONS OF THE INVIGILATORS about interpretations of the exam questions. If you have a concern about a question, please summarize it at the top of this page. All concerns will be evaluated after the exam. GOOD LUCK!

**Question 1 8 points** (*Assume all transitions are possible*)

a. 6 points Calculate the **energy (in kJ/mol)** required to ionize (completely remove) the electron from the **n=3** level of **He⁺ ion** (He: Helium). Show your work including equations. (3 sig. figs.)

b. 2 points

(I) Select the largest (longest) wavelength from the three choices below:

- i. Wavelength of photon to *completely remove* an electron from *1s orbital of Hydrogen atom*
- ii. Wavelength of photon to *completely remove* an electron from *1s orbital of Boron atom (B)*
- iii. Wavelength of photon to *completely remove* an electron from *1s orbital of Boron cation (B⁺)*

i or ii or iii

(II) Select the smallest (shortest) wavelength from the three choices below (*for H atom*):

- i. Wavelength of photon *emitted* for an electron transitioning from *2p to 1s orbital*
- ii. Wavelength of photon *absorbed* for an electron transitioning from *2s to 4p orbital*
- iii. Wavelength of photon *absorbed* for an electron transitioning from *1s to 3p orbital*

i or ii or iii



Question 2 8 points



A Chem110 student plots the graph shown on the left, from data obtained for the kinetic energy (eV) of the ejected electron versus frequency (s^{-1}) of the incident light, for an unknown metal. ($1\text{eV} = 1.6022 \times 10^{-19} \text{ J}$)

Based on the plot, answer the questions **a to c** below:

a. 1 point On the plot above, label the **threshold frequency**.

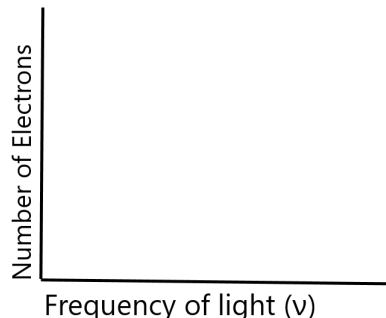
b. 2 points What is the value of the work potential (in J) of the unknown metal?

c. 3 points What is the *longest* wavelength (in nm) required to eject an electron from this metal's surface? Show your work including equation(s) used. (3 sig figs)

d. 2 points In the space provided below, plot the following

(Assume frequency is *above* threshold frequency):

- Kinetic energy* of an ejected electron versus the *intensity* of the incident light (left)
- Number of electrons* ejected versus the *frequency* of the incident light (right)



**Question 3** **6 points****a. 2 points**

In the ground state of Ge (Germanium; $Z = 32$) atom, **name** the following orbitals:

i. Orbital containing the electron that requires the maximum energy to be removed.

ii. Orbital containing the electron that experiences the minimum effective nuclear charge.

b. 2 points

An unknown anion (A^-) has the following *excited* electronic configuration: $[Ar] 4s^2 3d^3 4p^1$

(Ar: Argon; $Z = 18$)

Give the **condensed ground state electronic configuration** of the atom, **A** and the cation, **A^{2+}** .
Show your work. (Assume all electronic transitions are possible)

Electronic Configuration of A:

Electronic Configuration of A^{2+} :

c. 2 points

In a single atom, determine how many electrons can have the following set of quantum numbers?

(i) $n = 5; l = 3$

In a single atom, determine how many electrons can have the following set of quantum numbers?

(i) $n = 3; l = 2; m_s = +1/2$

**Question 4** **8 points**

Rank the following from 1 to 4 in **increasing** order. Provide the ranking in the boxes provided.

Note: Atomic number (Z) are provided for relevant atoms

a) Increasing (lowest to highest) order of *ionization energy*: (1= lowest; 4 = highest)

Mg

Mg: Z = 12

Rb

Rb: Z = 37

Cs

Cs: Z = 55

K

K : Z = 19

b) Increasing (lowest to highest) order of *bond angle*: (1= lowest; 4 = highest)

H₂OCCl₄XeF₄

HCN

c) Increasing (smallest to largest) order of *ionic size* (1= smallest; 4 = largest)

S²⁻

S: Z = 16

O²⁻

O: Z = 8

Be²⁺

Be: Z = 4

Na⁺

Na: Z = 11

d) Increasing (lowest to highest) order of *bond energy* (1= lowest; 4 = highest)

P-F

P-Cl

P-I

P-Br

**Question 5** **12 points**

a. 9 points For the molecules HClO_4 and TeF_6 , draw the most stable *Lewis structure* showing appropriate shape and *dashed/wedged bonds*, based on VSEPR. Determine the **molecular geometry** around the central atom. For HClO_4 , label the *most polar bond*.



b. 3 points (Note: Lone pairs are not shown in structures below)

(i) For the three *resonance* structures (1, 2 and 3) shown below, answer the following question:



(I) Circle the *most contributing* resonance structure.

Structure 1

Structure 2

Structure 3

(ii) For the three structures given (4, 5, and 6) below, answer the following questions:



(I) Circle all the structures that represent *resonance structures* of each other:

Structure 4

Structure 5

Structure 6

(II) Among the resonance structures, circle the *most contributing* resonance structure.

Structure 4

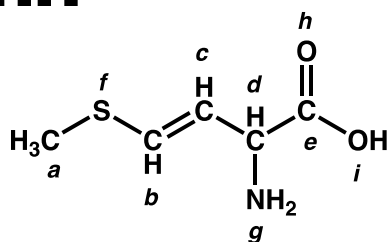
Structure 5

Structure 6

**Question 6** **6 points**

Using the data given below, calculate the **lattice energy** for SrO(s).

	ΔH° (kJ/mol)
$\text{Sr}(s) \longrightarrow \text{Sr}(g)$	+180
$\text{Sr}(g) \longrightarrow \text{Sr}^{2+}(g) + 2e^-$	+1468
$\text{O}_2(g) \longrightarrow 2\text{O}(g)$	+500
$\text{O}(g) + 2e^- \longrightarrow \text{O}^{2-}(g)$	+600
$\text{Sr}(s) + \frac{1}{2}\text{O}_2(g) \longrightarrow \text{SrO}(s)$	-550

**Question 7 10 points**

Consider the molecule given on the left, and answer questions (i) to (vi) using chemical bonding theories that we have learnt in class (valence bond theory, hybridization).

Note that the lone pairs are not shown in the structure:

i. Consider the carbon atoms marked in as ***a***, ***b***, ***c***, ***d***, and ***e*** in the molecule, and determine **hybridization** at each of the carbon centers. (2.5 points)

Atom	Hybridization
Carbon (<i>a</i>)	
Carbon (<i>b</i>)	
Carbon (<i>c</i>)	
Carbon (<i>d</i>)	
Carbon (<i>e</i>)	

ii. Now, consider the sulfur atom (S) marked as ***f***, nitrogen (N) atom (***g***), and oxygen(O) atoms (***h*** and ***i***), and determine their **hybridization**. (2 points)

Atom	Hybridization
Sulfur (<i>f</i>)	
Nitrogen (<i>g</i>)	
Oxygen (<i>h</i>)	
Oxygen (<i>i</i>)	

iii. **Describe** all the bonds formed by the carbon atoms ***a*** and ***b***, using sigma and pi designation as appropriate. (2 points)

Atom	Bonds formed (use sigma and pi as appropriate)
Carbon (<i>a</i>)	
Carbon (<i>b</i>)	

iv. How many total sp^3 -s sigma bonds are in the molecule? (1 point)

v. Determine **electronic** and **molecular geometry** at carbon (C) labeled ***e*** and sulfur (S) labeled ***f*** centres. (2 points)

Atom	Electronic Geometry	Molecular Geometry
Carbon (<i>e</i>)		
Sulfur (<i>f</i>)		

vi. Circle the value from the choices given below, that describes the **C-S-C bond angle at sulfur** (S) marked ***f***: (0.5 points)

>90° to <109.5° 109.5° >109.5 to <120° 120° >120° to <180°


Question 8 12 points

Using the **valence shell orbitals only** draw a **complete** molecular orbital (MO) diagram (with no s-p mixing) for the molecule **PS**, in the space provided below. **(6 points)**

Assume that 3s and 3p atomic orbitals will form molecular orbitals like those formed by the combination of 2s and 2p atomic orbitals that we learnt in class, and there is no s-p mixing.

Show **energy arrow**, **atomic orbitals** involved, **atom labels**, the **molecular orbitals** that are formed, **filled electrons** in the MO diagram etc. Make sure to label all the atomic and molecular orbitals.

Using the MO diagram above, answer the following questions:

- (i) Which one of the following will have the highest bond order? Circle the correct choice
 A) PS^{1-} B) PS C) PS^{1+}
- (ii) Which one of the following will have the longest bond length? Circle the correct choice
 A) PS^{1-} B) PS C) PS^{1+}
- (iii) Which of the following will be diamagnetic? Circle the correct choice.
 A) PS^{1-} B) PS C) PS^{1+}
- (iv) Draw Lewis dot structure of the molecule with the shortest bond length (based on the MO diagram) from among the three (PS^{1-} , PS, PS^{1+}), and indicate the bond order calculated from the Lewis dot structure. **(3 points)**

Lewis Dot Structure

Bond Order



E228FE0A-5A7B-4EDE-BDAA-D393A22681F7

midterm-exam-93388

#385 Page 10 of 10

SCRAP PAGE

**CHEMISTRY 110 –Midterm Exam****Thursday November 7, 2024, 6:30 – 8:30 pm****Examiners: Pallavi Sirjoosingh and Ashok Kakkar**

First name (please write as legibly as possible within the boxes)

--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--

Last name

--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--

Student ID number

--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--

This booklet has 8 questions on 10 pages (The last page can be used as a scrap page).

The staple must NOT be removed from the question booklet. Your intact question booklet must be returned at the end of the exam.

A separate booklet contains a periodic table, and datasheet.

Write your answers directly under the questions in the space provided. Do not write on the QR codes or near the top edges (as it may be cut off).

Please write **legibly and dark enough** for the scan to be readable. For students who use English as a second language, dictionaries are permitted. Calculators are allowed.

DO NOT ASK QUESTIONS OF THE INVIGILATORS about interpretations of the exam questions. If you have a concern about a question, please summarize it at the top of this page. All concerns will be evaluated after the exam. GOOD LUCK!

**Question 1 8 points** (*Assume all transitions are possible*)

a. 6 points Calculate the **energy (in kJ/mol)** required to ionize (completely remove) the electron from the **n=3** level of **He⁺ ion** (He: Helium). Show your work including equations. (3 sig. figs.)

b. 2 points

(I) Select the largest (longest) wavelength from the three choices below:

- i. Wavelength of photon to *completely remove* an electron from *1s orbital of Hydrogen atom*
- ii. Wavelength of photon to *completely remove* an electron from *1s orbital of Boron atom (B)*
- iii. Wavelength of photon to *completely remove* an electron from *1s orbital of Boron cation (B⁺)*

i or ii or iii

(II) Select the smallest (shortest) wavelength from the three choices below (*for H atom*):

- i. Wavelength of photon *emitted* for an electron transitioning from *2p to 1s orbital*
- ii. Wavelength of photon *absorbed* for an electron transitioning from *2s to 4p orbital*
- iii. Wavelength of photon *absorbed* for an electron transitioning from *1s to 3p orbital*

i or ii or iii



Question 2 8 points



A Chem110 student plots the graph shown on the left, from data obtained for the kinetic energy (eV) of the ejected electron versus frequency (s^{-1}) of the incident light, for an unknown metal. ($1\text{eV} = 1.6022 \times 10^{-19} \text{ J}$)

Based on the plot, answer the questions **a to c** below:

a. 1 point On the plot above, label the **threshold frequency**.

b. 2 points What is the value of the work potential (in J) of the unknown metal?

c. 3 points What is the *longest* wavelength (in nm) required to eject an electron from this metal's surface? Show your work including equation(s) used. (3 sig figs)

d. 2 points In the space provided below, plot the following

(Assume frequency is *above* threshold frequency):

- Kinetic energy* of an ejected electron versus the *intensity* of the incident light (left)
- Number of electrons* ejected versus the *frequency* of the incident light (right)



**Question 3** **6 points****a. 2 points**

In the ground state of Ge (Germanium; $Z = 32$) atom, **name** the following orbitals:

i. Orbital containing the electron that requires the maximum energy to be removed.

ii. Orbital containing the electron that experiences the minimum effective nuclear charge.

b. 2 points

An unknown anion (A^-) has the following *excited* electronic configuration: $[Ar] 4s^2 3d^3 4p^1$

(Ar: Argon; $Z = 18$)

Give the **condensed ground state electronic configuration** of the atom, **A** and the cation, A^{2+} .
Show your work. (Assume all electronic transitions are possible)

Electronic Configuration of A:

Electronic Configuration of A^{2+} :

c. 2 points

In a single atom, determine how many electrons can have the following set of quantum numbers?

(i) $n = 5; l = 3$

In a single atom, determine how many electrons can have the following set of quantum numbers?

(i) $n = 3; l = 2; m_s = +1/2$

**Question 4** **8 points**

Rank the following from 1 to 4 in **increasing** order. Provide the ranking in the boxes provided.

Note: Atomic number (Z) are provided for relevant atoms

a) Increasing (lowest to highest) order of *ionization energy*: (1= lowest; 4 = highest)

Mg

Mg: Z = 12

Rb

Rb: Z = 37

Cs

Cs: Z = 55

K

K : Z = 19

b) Increasing (lowest to highest) order of *bond angle*: (1= lowest; 4 = highest)

H₂OCCl₄XeF₄

HCN

c) Increasing (smallest to largest) order of *ionic size* (1= smallest; 4 = largest)

S²⁻

S: Z = 16

O²⁻

O: Z = 8

Be²⁺

Be: Z = 4

Na⁺

Na: Z = 11

d) Increasing (lowest to highest) order of *bond energy* (1= lowest; 4 = highest)

P-F

P-Cl

P-I

P-Br

**Question 5** **12 points**

a. 9 points For the molecules HClO_4 and TeF_6 , draw the most stable *Lewis structure* showing appropriate shape and *dashed/wedged bonds*, based on VSEPR. Determine the **molecular geometry** around the central atom. For HClO_4 , label the *most polar bond*.



b. 3 points (Note: Lone pairs are not shown in structures below)

(i) For the three *resonance* structures (1, 2 and 3) shown below, answer the following question:



(I) Circle the *most contributing* resonance structure.

Structure 1

Structure 2

Structure 3

(ii) For the three structures given (4, 5, and 6) below, answer the following questions:



(I) Circle all the structures that represent *resonance structures* of each other:

Structure 4

Structure 5

Structure 6

(II) Among the resonance structures, circle the *most contributing* resonance structure.

Structure 4

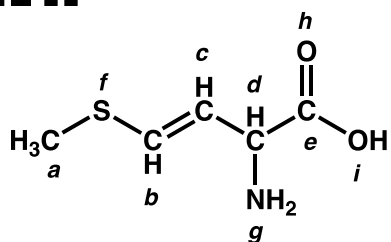
Structure 5

Structure 6

**Question 6** **6 points**

Using the data given below, calculate the **lattice energy** for SrO(s).

	ΔH° (kJ/mol)
$\text{Sr}(s) \longrightarrow \text{Sr}(g)$	+180
$\text{Sr}(g) \longrightarrow \text{Sr}^{2+}(g) + 2e^-$	+1468
$\text{O}_2(g) \longrightarrow 2\text{O}(g)$	+500
$\text{O}(g) + 2e^- \longrightarrow \text{O}^{2-}(g)$	+600
$\text{Sr}(s) + \frac{1}{2}\text{O}_2(g) \longrightarrow \text{SrO}(s)$	-550

**Question 7 10 points**

Consider the molecule given on the left, and answer questions (i) to (vi) using chemical bonding theories that we have learnt in class (valence bond theory, hybridization).

Note that the lone pairs are not shown in the structure:

i. Consider the carbon atoms marked in as ***a***, ***b***, ***c***, ***d***, and ***e*** in the molecule, and determine **hybridization** at each of the carbon centers. (2.5 points)

Atom	Hybridization
Carbon (<i>a</i>)	
Carbon (<i>b</i>)	
Carbon (<i>c</i>)	
Carbon (<i>d</i>)	
Carbon (<i>e</i>)	

ii. Now, consider the sulfur atom (S) marked as ***f***, nitrogen (N) atom (***g***), and oxygen(O) atoms (***h*** and ***i***), and determine their **hybridization**. (2 points)

Atom	Hybridization
Sulfur (<i>f</i>)	
Nitrogen (<i>g</i>)	
Oxygen (<i>h</i>)	
Oxygen (<i>i</i>)	

iii. **Describe** all the bonds formed by the carbon atoms ***a*** and ***b***, using sigma and pi designation as appropriate. (2 points)

Atom	Bonds formed (use sigma and pi as appropriate)
Carbon (<i>a</i>)	
Carbon (<i>b</i>)	

iv. How many total sp^3 -s sigma bonds are in the molecule? (1 point)

v. Determine **electronic** and **molecular geometry** at carbon (C) labeled ***e*** and sulfur (S) labeled ***f*** centres. (2 points)

Atom	Electronic Geometry	Molecular Geometry
Carbon (<i>e</i>)		
Sulfur (<i>f</i>)		

vi. Circle the value from the choices given below, that describes the **C-S-C bond angle at sulfur** (S) marked ***f***: (0.5 points)

>90° to <109.5° 109.5° >109.5 to <120° 120° >120° to <180°

**Question 8 12 points**

Using the **valence shell orbitals only** draw a **complete** molecular orbital (MO) diagram (with no s-p mixing) for the molecule **PS**, in the space provided below. **(6 points)**

Assume that 3s and 3p atomic orbitals will form molecular orbitals like those formed by the combination of 2s and 2p atomic orbitals that we learnt in class, and there is no s-p mixing.

Show **energy arrow**, **atomic orbitals** involved, **atom labels**, the **molecular orbitals** that are formed, **filled electrons** in the MO diagram etc. Make sure to label all the atomic and molecular orbitals.

Using the MO diagram above, answer the following questions:

- (i) Which one of the following will have the highest bond order? Circle the correct choice
A) PS^{1-} B) PS C) PS^{1+}
- (ii) Which one of the following will have the longest bond length? Circle the correct choice
A) PS^{1-} B) PS C) PS^{1+}
- (iii) Which of the following will be diamagnetic? Circle the correct choice.
A) PS^{1-} B) PS C) PS^{1+}
- (iv) Draw Lewis dot structure of the molecule with the shortest bond length (based on the MO diagram) from among the three (PS^{1-} , PS, PS^{1+}), and indicate the bond order calculated from the Lewis dot structure. **(3 points)**

Lewis Dot Structure

Bond Order



C8404398-11FA-465D-AB20-FBE5B3D1471C

midterm-exam-93388

#386 Page 10 of 10

SCRAP PAGE

**CHEMISTRY 110 –Midterm Exam****Thursday November 7, 2024, 6:30 – 8:30 pm****Examiners: Pallavi Sirjoosingh and Ashok Kakkar**

First name (please write as legibly as possible within the boxes)

--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--

Last name

--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--

Student ID number

--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--

This booklet has 8 questions on 10 pages (The last page can be used as a scrap page).

The staple must NOT be removed from the question booklet. Your intact question booklet must be returned at the end of the exam.

A separate booklet contains a periodic table, and datasheet.

Write your answers directly under the questions in the space provided. Do not write on the QR codes or near the top edges (as it may be cut off).

Please write **legibly and dark enough** for the scan to be readable. For students who use English as a second language, dictionaries are permitted. Calculators are allowed.

DO NOT ASK QUESTIONS OF THE INVIGILATORS about interpretations of the exam questions. If you have a concern about a question, please summarize it at the top of this page. All concerns will be evaluated after the exam. GOOD LUCK!

**Question 1 8 points** (*Assume all transitions are possible*)

a. 6 points Calculate the **energy (in kJ/mol)** required to ionize (completely remove) the electron from the **n=3** level of **He⁺ ion** (He: Helium). Show your work including equations. (3 sig. figs.)

b. 2 points

(I) Select the largest (longest) wavelength from the three choices below:

- i. Wavelength of photon to *completely remove* an electron from *1s orbital of Hydrogen atom*
- ii. Wavelength of photon to *completely remove* an electron from *1s orbital of Boron atom (B)*
- iii. Wavelength of photon to *completely remove* an electron from *1s orbital of Boron cation (B⁺)*

i or ii or iii

(II) Select the smallest (shortest) wavelength from the three choices below (*for H atom*):

- i. Wavelength of photon *emitted* for an electron transitioning from *2p to 1s orbital*
- ii. Wavelength of photon *absorbed* for an electron transitioning from *2s to 4p orbital*
- iii. Wavelength of photon *absorbed* for an electron transitioning from *1s to 3p orbital*

i or ii or iii



Question 2 8 points



A Chem110 student plots the graph shown on the left, from data obtained for the kinetic energy (eV) of the ejected electron versus frequency (s^{-1}) of the incident light, for an unknown metal. ($1\text{eV} = 1.6022 \times 10^{-19} \text{ J}$)

Based on the plot, answer the questions **a to c** below:

a. 1 point On the plot above, label the **threshold frequency**.

b. 2 points What is the value of the work potential (in J) of the unknown metal?

c. 3 points What is the *longest* wavelength (in nm) required to eject an electron from this metal's surface? Show your work including equation(s) used. (3 sig figs)

d. 2 points In the space provided below, plot the following

(Assume frequency is *above* threshold frequency):

- i. *Kinetic energy* of an ejected electron versus the *intensity* of the incident light (left)
- ii. *Number of electrons* ejected versus the *frequency* of the incident light (right)



**Question 3** **6 points****a. 2 points**

In the ground state of Ge (Germanium; $Z = 32$) atom, **name** the following orbitals:

i. Orbital containing the electron that requires the maximum energy to be removed.

ii. Orbital containing the electron that experiences the minimum effective nuclear charge.

b. 2 points

An unknown anion (A^-) has the following *excited* electronic configuration: $[Ar] 4s^2 3d^3 4p^1$

(Ar: Argon; $Z = 18$)

Give the **condensed ground state electronic configuration** of the atom, **A** and the cation, **A^{2+}** .
Show your work. (Assume all electronic transitions are possible)

Electronic Configuration of A:

Electronic Configuration of A^{2+} :

c. 2 points

In a single atom, determine how many electrons can have the following set of quantum numbers?

(i) $n = 5; l = 3$

In a single atom, determine how many electrons can have the following set of quantum numbers?

(i) $n = 3; l = 2; m_s = +1/2$

**Question 4** **8 points**

Rank the following from 1 to 4 in **increasing** order. Provide the ranking in the boxes provided.

Note: Atomic number (Z) are provided for relevant atoms

a) Increasing (lowest to highest) order of *ionization energy*: (1= lowest; 4 = highest)

Mg

Mg: Z = 12

Rb

Rb: Z = 37

Cs

Cs: Z = 55

K

K : Z = 19

b) Increasing (lowest to highest) order of *bond angle*: (1= lowest; 4 = highest)

H₂OCCl₄XeF₄

HCN

c) Increasing (smallest to largest) order of *ionic size* (1= smallest; 4 = largest)

S²⁻

S: Z = 16

O²⁻

O: Z = 8

Be²⁺

Be: Z = 4

Na⁺

Na: Z = 11

d) Increasing (lowest to highest) order of *bond energy* (1= lowest; 4 = highest)

P-F

P-Cl

P-I

P-Br

**Question 5** **12 points**

a. 9 points For the molecules HClO_4 and TeF_6 , draw the most stable *Lewis structure* showing appropriate shape and *dashed/wedged bonds*, based on VSEPR. Determine the **molecular geometry** around the central atom. For HClO_4 , label the *most polar bond*.



b. 3 points (Note: Lone pairs are not shown in structures below)

(i) For the three *resonance* structures (1, 2 and 3) shown below, answer the following question:



(I) Circle the *most contributing* resonance structure.

Structure 1

Structure 2

Structure 3

(ii) For the three structures given (4, 5, and 6) below, answer the following questions:



(I) Circle all the structures that represent *resonance structures* of each other:

Structure 4

Structure 5

Structure 6

(II) Among the resonance structures, circle the *most contributing* resonance structure.

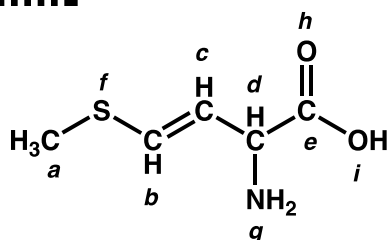
Structure 4

Structure 5

Structure 6

**Question 6** **6 points**Using the data given below, calculate the **lattice energy** for SrO(s).

	ΔH° (kJ/mol)
$\text{Sr}(s) \longrightarrow \text{Sr}(g)$	+180
$\text{Sr}(g) \longrightarrow \text{Sr}^{2+}(g) + 2e^-$	+1468
$\text{O}_2(g) \longrightarrow 2\text{O}(g)$	+500
$\text{O}(g) + 2e^- \longrightarrow \text{O}^{2-}(g)$	+600
$\text{Sr}(s) + \frac{1}{2}\text{O}_2(g) \longrightarrow \text{SrO}(s)$	-550

**Question 7 10 points**

Consider the molecule given on the left, and answer questions (i) to (vi) using chemical bonding theories that we have learnt in class (valence bond theory, hybridization).

Note that the lone pairs are not shown in the structure:

i. Consider the carbon atoms marked in as ***a***, ***b***, ***c***, ***d***, and ***e*** in the molecule, and determine **hybridization** at each of the carbon centers. (2.5 points)

Atom	Hybridization
Carbon (<i>a</i>)	
Carbon (<i>b</i>)	
Carbon (<i>c</i>)	
Carbon (<i>d</i>)	
Carbon (<i>e</i>)	

ii. Now, consider the sulfur atom (S) marked as ***f***, nitrogen (N) atom (***g***), and oxygen(O) atoms (***h*** and ***i***), and determine their **hybridization**. (2 points)

Atom	Hybridization
Sulfur (<i>f</i>)	
Nitrogen (<i>g</i>)	
Oxygen (<i>h</i>)	
Oxygen (<i>i</i>)	

iii. **Describe** all the bonds formed by the carbon atoms ***a*** and ***b***, using sigma and pi designation as appropriate. (2 points)

Atom	Bonds formed (use sigma and pi as appropriate)
Carbon (<i>a</i>)	
Carbon (<i>b</i>)	

iv. How many total sp^3 -s sigma bonds are in the molecule? (1 point)

v. Determine **electronic** and **molecular geometry** at carbon (C) labeled ***e*** and sulfur (S) labeled ***f*** centres. (2 points)

Atom	Electronic Geometry	Molecular Geometry
Carbon (<i>e</i>)		
Sulfur (<i>f</i>)		

vi. Circle the value from the choices given below, that describes the **C-S-C bond angle at sulfur** (S) marked ***f***: (0.5 points)

>90° to <109.5° 109.5° >109.5 to <120° 120° >120° to <180°


Question 8 12 points

Using the **valence shell orbitals only** draw a **complete** molecular orbital (MO) diagram (with no s-p mixing) for the molecule **PS**, in the space provided below. **(6 points)**

Assume that 3s and 3p atomic orbitals will form molecular orbitals like those formed by the combination of 2s and 2p atomic orbitals that we learnt in class, and there is no s-p mixing.

Show **energy arrow**, **atomic orbitals** involved, **atom labels**, the **molecular orbitals** that are formed, **filled electrons** in the MO diagram etc. Make sure to label all the atomic and molecular orbitals.

Using the MO diagram above, answer the following questions:

- (i) Which one of the following will have the highest bond order? Circle the correct choice
 A) PS^{1-} B) PS C) PS^{1+}
- (ii) Which one of the following will have the longest bond length? Circle the correct choice
 A) PS^{1-} B) PS C) PS^{1+}
- (iii) Which of the following will be diamagnetic? Circle the correct choice.
 A) PS^{1-} B) PS C) PS^{1+}
- (iv) Draw Lewis dot structure of the molecule with the shortest bond length (based on the MO diagram) from among the three (PS^{1-} , PS, PS^{1+}), and indicate the bond order calculated from the Lewis dot structure. **(3 points)**

Lewis Dot Structure

Bond Order



AC1124E3-5405-495C-AD2D-D09DD639EC37

midterm-exam-93388

#387 Page 10 of 10

SCRAP PAGE

**CHEMISTRY 110 –Midterm Exam****Thursday November 7, 2024, 6:30 – 8:30 pm****Examiners: Pallavi Sirjoosingh and Ashok Kakkar**

First name (please write as legibly as possible within the boxes)

--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--

Last name

--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--

Student ID number

--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--

This booklet has 8 questions on 10 pages (The last page can be used as a scrap page).

The staple must NOT be removed from the question booklet. Your intact question booklet must be returned at the end of the exam.

A separate booklet contains a periodic table, and datasheet.

Write your answers directly under the questions in the space provided. Do not write on the QR codes or near the top edges (as it may be cut off).

Please write **legibly and dark enough** for the scan to be readable. For students who use English as a second language, dictionaries are permitted. Calculators are allowed.

DO NOT ASK QUESTIONS OF THE INVIGILATORS about interpretations of the exam questions. If you have a concern about a question, please summarize it at the top of this page. All concerns will be evaluated after the exam. GOOD LUCK!

**Question 1 8 points** (*Assume all transitions are possible*)

a. 6 points Calculate the **energy (in kJ/mol)** required to ionize (completely remove) the electron from the **n=3** level of **He⁺ ion** (He: Helium). Show your work including equations. (3 sig. figs.)

b. 2 points

(I) Select the largest (longest) wavelength from the three choices below:

- i. Wavelength of photon to *completely remove* an electron from *1s orbital of Hydrogen atom*
- ii. Wavelength of photon to *completely remove* an electron from *1s orbital of Boron atom (B)*
- iii. Wavelength of photon to *completely remove* an electron from *1s orbital of Boron cation (B⁺)*

i or ii or iii

(II) Select the smallest (shortest) wavelength from the three choices below (*for H atom*):

- i. Wavelength of photon *emitted* for an electron transitioning from *2p to 1s orbital*
- ii. Wavelength of photon *absorbed* for an electron transitioning from *2s to 4p orbital*
- iii. Wavelength of photon *absorbed* for an electron transitioning from *1s to 3p orbital*

i or ii or iii



Question 2 8 points



A Chem110 student plots the graph shown on the left, from data obtained for the kinetic energy (eV) of the ejected electron versus frequency (s^{-1}) of the incident light, for an unknown metal. ($1\text{eV} = 1.6022 \times 10^{-19} \text{ J}$)

Based on the plot, answer the questions **a to c** below:

a. 1 point On the plot above, label the **threshold frequency**.

b. 2 points What is the value of the work potential (in J) of the unknown metal?

c. 3 points What is the *longest* wavelength (in nm) required to eject an electron from this metal's surface? Show your work including equation(s) used. (3 sig figs)

d. 2 points In the space provided below, plot the following

(Assume frequency is *above* threshold frequency):

- i. *Kinetic energy* of an ejected electron versus the *intensity* of the incident light (left)
- ii. *Number of electrons* ejected versus the *frequency* of the incident light (right)



**Question 3** **6 points****a. 2 points**

In the ground state of Ge (Germanium; $Z = 32$) atom, **name** the following orbitals:

i. Orbital containing the electron that requires the maximum energy to be removed.

ii. Orbital containing the electron that experiences the minimum effective nuclear charge.

b. 2 points

An unknown anion (A^-) has the following *excited* electronic configuration: $[Ar] 4s^2 3d^3 4p^1$

(Ar: Argon; $Z = 18$)

Give the **condensed ground state electronic configuration** of the atom, **A** and the cation, **A^{2+}** .
Show your work. (Assume all electronic transitions are possible)

Electronic Configuration of A:

Electronic Configuration of A^{2+} :

c. 2 points

In a single atom, determine how many electrons can have the following set of quantum numbers?

(i) $n = 5; l = 3$

In a single atom, determine how many electrons can have the following set of quantum numbers?

(i) $n = 3; l = 2; m_s = +1/2$

**Question 4** **8 points**

Rank the following from 1 to 4 in **increasing** order. Provide the ranking in the boxes provided.

Note: Atomic number (Z) are provided for relevant atoms

a) Increasing (lowest to highest) order of *ionization energy*: (1= lowest; 4 = highest)

Mg

Mg: Z = 12

Rb

Rb: Z = 37

Cs

Cs: Z = 55

K

K : Z = 19

b) Increasing (lowest to highest) order of *bond angle*: (1= lowest; 4 = highest)

H₂OCCl₄XeF₄

HCN

c) Increasing (smallest to largest) order of *ionic size* (1= smallest; 4 = largest)

S²⁻

S: Z = 16

O²⁻

O: Z = 8

Be²⁺

Be: Z = 4

Na⁺

Na: Z = 11

d) Increasing (lowest to highest) order of *bond energy* (1= lowest; 4 = highest)

P-F

P-Cl

P-I

P-Br

**Question 5** **12 points**

a. 9 points For the molecules HClO_4 and TeF_6 , draw the most stable *Lewis structure* showing appropriate shape and *dashed/wedged bonds*, based on VSEPR. Determine the **molecular geometry** around the central atom. For HClO_4 , label the *most polar bond*.



b. 3 points (Note: Lone pairs are not shown in structures below)

(i) For the three *resonance* structures (1, 2 and 3) shown below, answer the following question:



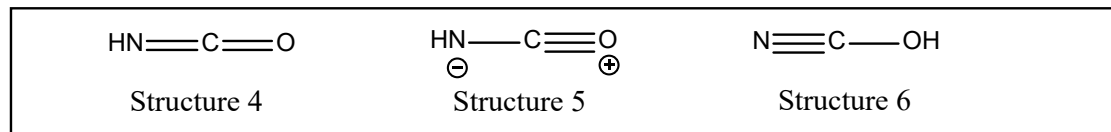
(I) Circle the *most contributing* resonance structure.

Structure 1

Structure 2

Structure 3

(ii) For the three structures given (4, 5, and 6) below, answer the following questions:



(I) Circle all the structures that represent *resonance structures* of each other:

Structure 4

Structure 5

Structure 6

(II) Among the resonance structures, circle the *most contributing* resonance structure.

Structure 4

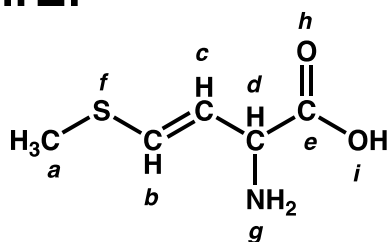
Structure 5

Structure 6

**Question 6** **6 points**

Using the data given below, calculate the **lattice energy** for SrO(s).

	ΔH° (kJ/mol)
$\text{Sr}(s) \longrightarrow \text{Sr}(g)$	+180
$\text{Sr}(g) \longrightarrow \text{Sr}^{2+}(g) + 2e^-$	+1468
$\text{O}_2(g) \longrightarrow 2\text{O}(g)$	+500
$\text{O}(g) + 2e^- \longrightarrow \text{O}^{2-}(g)$	+600
$\text{Sr}(s) + \frac{1}{2}\text{O}_2(g) \longrightarrow \text{SrO}(s)$	-550

**Question 7 10 points**

Consider the molecule given on the left, and answer questions (i) to (vi) using chemical bonding theories that we have learnt in class (valence bond theory, hybridization).

Note that the lone pairs are not shown in the structure:

i. Consider the carbon atoms marked in as ***a***, ***b***, ***c***, ***d***, and ***e*** in the molecule, and determine **hybridization** at each of the carbon centers. (2.5 points)

Atom	Hybridization
Carbon (<i>a</i>)	
Carbon (<i>b</i>)	
Carbon (<i>c</i>)	
Carbon (<i>d</i>)	
Carbon (<i>e</i>)	

ii. Now, consider the sulfur atom (S) marked as ***f***, nitrogen (N) atom (***g***), and oxygen(O) atoms (***h*** and ***i***), and determine their **hybridization**. (2 points)

Atom	Hybridization
Sulfur (<i>f</i>)	
Nitrogen (<i>g</i>)	
Oxygen (<i>h</i>)	
Oxygen (<i>i</i>)	

iii. **Describe** all the bonds formed by the carbon atoms ***a*** and ***b***, using sigma and pi designation as appropriate. (2 points)

Atom	Bonds formed (use sigma and pi as appropriate)
Carbon (<i>a</i>)	
Carbon (<i>b</i>)	

iv. How many total sp^3 -s sigma bonds are in the molecule? (1 point)

v. Determine **electronic** and **molecular geometry** at carbon (C) labeled ***e*** and sulfur (S) labeled ***f*** centres. (2 points)

Atom	Electronic Geometry	Molecular Geometry
Carbon (<i>e</i>)		
Sulfur (<i>f</i>)		

vi. Circle the value from the choices given below, that describes the **C-S-C bond angle at sulfur** (S) marked ***f***: (0.5 points)

$>90^\circ$ to $<109.5^\circ$ 109.5° >109.5 to $<120^\circ$ 120° $>120^\circ$ to $<180^\circ$


Question 8 12 points

Using the **valence shell orbitals only** draw a **complete** molecular orbital (MO) diagram (with no s-p mixing) for the molecule **PS**, in the space provided below. **(6 points)**

Assume that 3s and 3p atomic orbitals will form molecular orbitals like those formed by the combination of 2s and 2p atomic orbitals that we learnt in class, and there is no s-p mixing.

Show **energy arrow**, **atomic orbitals** involved, **atom labels**, the **molecular orbitals** that are formed, **filled electrons** in the MO diagram etc. Make sure to **label all the atomic and molecular orbitals**.

Using the MO diagram above, answer the following questions:

- (i) Which one of the following will have the highest bond order? Circle the correct choice
 A) PS^{1-} B) PS C) PS^{1+}
- (ii) Which one of the following will have the longest bond length? Circle the correct choice
 A) PS^{1-} B) PS C) PS^{1+}
- (iii) Which of the following will be diamagnetic? Circle the correct choice.
 A) PS^{1-} B) PS C) PS^{1+}
- (iv) Draw Lewis dot structure of the molecule with the shortest bond length (based on the MO diagram) from among the three (PS^{1-} , PS, PS^{1+}), and indicate the bond order calculated from the Lewis dot structure. **(3 points)**

Lewis Dot Structure

Bond Order



41FE4424-A642-463F-ADCB-8FD74081629E

midterm-exam-93388

#388 Page 10 of 10

SCRAP PAGE

**CHEMISTRY 110 –Midterm Exam****Thursday November 7, 2024, 6:30 – 8:30 pm**

Examiners: Pallavi Sirjoosingh and Ashok Kakkar

First name (please write as legibly as possible within the boxes)

--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--

Last name

--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--

Student ID number

--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--

This booklet has 8 questions on 10 pages (The last page can be used as a scrap page).

The staple must NOT be removed from the question booklet. Your intact question booklet must be returned at the end of the exam.

A separate booklet contains a periodic table, and datasheet.

Write your answers directly under the questions in the space provided. Do not write on the QR codes or near the top edges (as it may be cut off).

Please write **legibly and dark enough** for the scan to be readable. For students who use English as a second language, dictionaries are permitted. Calculators are allowed.

DO NOT ASK QUESTIONS OF THE INVIGILATORS about interpretations of the exam questions. If you have a concern about a question, please summarize it at the top of this page. All concerns will be evaluated after the exam. GOOD LUCK!

**Question 1 8 points** (*Assume all transitions are possible*)

a. 6 points Calculate the **energy (in kJ/mol)** required to ionize (completely remove) the electron from the **$n=3$** level of **He^+ ion** (He: Helium). Show your work including equations. (3 sig. figs.)

b. 2 points

(I) Select the largest (longest) wavelength from the three choices below:

- i. Wavelength of photon to *completely remove* an electron from *1s orbital of Hydrogen atom*
- ii. Wavelength of photon to *completely remove* an electron from *1s orbital of Boron atom (B)*
- iii. Wavelength of photon to *completely remove* an electron from *1s orbital of Boron cation (B^+)*

i or ii or iii

(II) Select the smallest (shortest) wavelength from the three choices below (*for H atom*):

- i. Wavelength of photon *emitted* for an electron transitioning from *2p to 1s orbital*
- ii. Wavelength of photon *absorbed* for an electron transitioning from *2s to 4p orbital*
- iii. Wavelength of photon *absorbed* for an electron transitioning from *1s to 3p orbital*

i or ii or iii



Question 2 8 points



A Chem110 student plots the graph shown on the left, from data obtained for the kinetic energy (eV) of the ejected electron versus frequency (s^{-1}) of the incident light, for an unknown metal. ($1\text{eV} = 1.6022 \times 10^{-19} \text{ J}$)

Based on the plot, answer the questions **a to c** below:

a. 1 point On the plot above, label the **threshold frequency**.

b. 2 points What is the value of the work potential (in J) of the unknown metal?

c. 3 points What is the *longest* wavelength (in nm) required to eject an electron from this metal's surface? Show your work including equation(s) used. (3 sig figs)

d. 2 points In the space provided below, plot the following

(Assume frequency is *above* threshold frequency):

- Kinetic energy* of an ejected electron versus the *intensity* of the incident light (left)
- Number of electrons* ejected versus the *frequency* of the incident light (right)



**Question 3** **6 points****a. 2 points**

In the ground state of Ge (Germanium; $Z = 32$) atom, **name** the following orbitals:

i. Orbital containing the electron that requires the maximum energy to be removed.

ii. Orbital containing the electron that experiences the minimum effective nuclear charge.

b. 2 points

An unknown anion (A^-) has the following *excited* electronic configuration: $[Ar] 4s^2 3d^3 4p^1$

(Ar: Argon; $Z = 18$)

Give the **condensed ground state electronic configuration** of the atom, **A** and the cation, **A^{2+}** .
Show your work. (Assume all electronic transitions are possible)

Electronic Configuration of A:

Electronic Configuration of A^{2+} :

c. 2 points

In a single atom, determine how many electrons can have the following set of quantum numbers?

(i) $n = 5; l = 3$

In a single atom, determine how many electrons can have the following set of quantum numbers?

(i) $n = 3; l = 2; m_s = +1/2$

**Question 4** **8 points**

Rank the following from 1 to 4 in **increasing** order. Provide the ranking in the boxes provided.

Note: Atomic number (Z) are provided for relevant atoms

a) Increasing (lowest to highest) order of *ionization energy*: (1= lowest; 4 = highest)

Mg

Mg: Z = 12

Rb

Rb: Z = 37

Cs

Cs: Z = 55

K

K : Z = 19

b) Increasing (lowest to highest) order of *bond angle*: (1= lowest; 4 = highest)

H₂OCCl₄XeF₄

HCN

c) Increasing (smallest to largest) order of *ionic size* (1= smallest; 4 = largest)

S²⁻

S: Z = 16

O²⁻

O: Z = 8

Be²⁺

Be: Z = 4

Na⁺

Na: Z = 11

d) Increasing (lowest to highest) order of *bond energy* (1= lowest; 4 = highest)

P-F

P-Cl

P-I

P-Br

**Question 5** **12 points**

a. 9 points For the molecules HClO_4 and TeF_6 , draw the most stable *Lewis structure* showing appropriate shape and *dashed/wedged bonds*, based on VSEPR. Determine the **molecular geometry** around the central atom. For HClO_4 , label the *most polar bond*.



b. 3 points (Note: Lone pairs are not shown in structures below)

(i) For the three *resonance* structures (1, 2 and 3) shown below, answer the following question:



(I) Circle the *most contributing* resonance structure.

Structure 1

Structure 2

Structure 3

(ii) For the three structures given (4, 5, and 6) below, answer the following questions:



(I) Circle all the structures that represent *resonance structures* of each other:

Structure 4

Structure 5

Structure 6

(II) Among the resonance structures, circle the *most contributing* resonance structure.

Structure 4

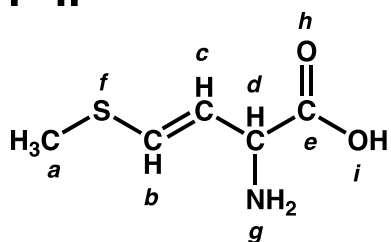
Structure 5

Structure 6

**Question 6** **6 points**

Using the data given below, calculate the **lattice energy** for SrO(s).

	ΔH° (kJ/mol)
$\text{Sr}(s) \longrightarrow \text{Sr}(g)$	+180
$\text{Sr}(g) \longrightarrow \text{Sr}^{2+}(g) + 2e^-$	+1468
$\text{O}_2(g) \longrightarrow 2\text{O}(g)$	+500
$\text{O}(g) + 2e^- \longrightarrow \text{O}^{2-}(g)$	+600
$\text{Sr}(s) + \frac{1}{2}\text{O}_2(g) \longrightarrow \text{SrO}(s)$	-550

**Question 7 10 points**

Consider the molecule given on the left, and answer questions (i) to (vi) using chemical bonding theories that we have learnt in class (valence bond theory, hybridization).

Note that the lone pairs are not shown in the structure:

i. Consider the carbon atoms marked in as ***a***, ***b***, ***c***, ***d***, and ***e*** in the molecule, and determine **hybridization** at each of the carbon centers. (2.5 points)

Atom	Hybridization
Carbon (<i>a</i>)	
Carbon (<i>b</i>)	
Carbon (<i>c</i>)	
Carbon (<i>d</i>)	
Carbon (<i>e</i>)	

ii. Now, consider the sulfur atom (S) marked as ***f***, nitrogen (N) atom (***g***), and oxygen(O) atoms (***h*** and ***i***), and determine their **hybridization**. (2 points)

Atom	Hybridization
Sulfur (<i>f</i>)	
Nitrogen (<i>g</i>)	
Oxygen (<i>h</i>)	
Oxygen (<i>i</i>)	

iii. **Describe** all the bonds formed by the carbon atoms ***a*** and ***b***, using sigma and pi designation as appropriate. (2 points)

Atom	Bonds formed (use sigma and pi as appropriate)
Carbon (<i>a</i>)	
Carbon (<i>b</i>)	

iv. How many total sp^3 -s sigma bonds are in the molecule? (1 point)

v. Determine **electronic** and **molecular geometry** at carbon (C) labeled ***e*** and sulfur (S) labeled ***f*** centres. (2 points)

Atom	Electronic Geometry	Molecular Geometry
Carbon (<i>e</i>)		
Sulfur (<i>f</i>)		

vi. Circle the value from the choices given below, that describes the **C-S-C bond angle at sulfur** (S) marked ***f***: (0.5 points)

$>90^\circ$ to $<109.5^\circ$ 109.5° >109.5 to $<120^\circ$ 120° $>120^\circ$ to $<180^\circ$



Question 8 12 points

Using the **valence shell orbitals only** draw a **complete** molecular orbital (MO) diagram (with no s-p mixing) for the molecule **PS**, in the space provided below. **(6 points)**

Assume that 3s and 3p atomic orbitals will form molecular orbitals like those formed by the combination of 2s and 2p atomic orbitals that we learnt in class, and there is no s-p mixing.

Show **energy arrow**, **atomic orbitals** involved, **atom labels**, the **molecular orbitals** that are formed, **filled electrons** in the MO diagram etc. Make sure to label all the atomic and molecular orbitals.

Using the MO diagram above, answer the following questions:

- (i) Which one of the following will have the highest bond order? Circle the correct choice
 A) PS^{1-} B) PS C) PS^{1+}
- (ii) Which one of the following will have the longest bond length? Circle the correct choice
 A) PS^{1-} B) PS C) PS^{1+}
- (iii) Which of the following will be diamagnetic? Circle the correct choice.
 A) PS^{1-} B) PS C) PS^{1+}
- (iv) Draw Lewis dot structure of the molecule with the shortest bond length (based on the MO diagram) from among the three (PS^{1-} , PS, PS^{1+}), and indicate the bond order calculated from the Lewis dot structure. **(3 points)**

Lewis Dot Structure

Bond Order



4D1A19DC-C3B6-47B7-9C64-AEC818A3E542

midterm-exam-93388

#389 Page 10 of 10

SCRAP PAGE

**CHEMISTRY 110 –Midterm Exam****Thursday November 7, 2024, 6:30 – 8:30 pm**

Examiners: Pallavi Sirjoosingh and Ashok Kakkar

First name (please write as legibly as possible within the boxes)

--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--

Last name

--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--

Student ID number

--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--

This booklet has 8 questions on 10 pages (The last page can be used as a scrap page).

The staple must NOT be removed from the question booklet. Your intact question booklet must be returned at the end of the exam.

A separate booklet contains a periodic table, and datasheet.

Write your answers directly under the questions in the space provided. Do not write on the QR codes or near the top edges (as it may be cut off).

Please write **legibly and dark enough** for the scan to be readable. For students who use English as a second language, dictionaries are permitted. Calculators are allowed.

DO NOT ASK QUESTIONS OF THE INVIGILATORS about interpretations of the exam questions. If you have a concern about a question, please summarize it at the top of this page. All concerns will be evaluated after the exam. GOOD LUCK!

**Question 1 8 points** (*Assume all transitions are possible*)

a. 6 points Calculate the **energy (in kJ/mol)** required to ionize (completely remove) the electron from the **n=3** level of **He⁺ ion** (He: Helium). Show your work including equations. (3 sig. figs.)

b. 2 points

(I) Select the largest (longest) wavelength from the three choices below:

- i. Wavelength of photon to *completely remove* an electron from *1s orbital of Hydrogen atom*
- ii. Wavelength of photon to *completely remove* an electron from *1s orbital of Boron atom (B)*
- iii. Wavelength of photon to *completely remove* an electron from *1s orbital of Boron cation (B⁺)*

i or ii or iii

(II) Select the smallest (shortest) wavelength from the three choices below (*for H atom*):

- i. Wavelength of photon *emitted* for an electron transitioning from *2p to 1s orbital*
- ii. Wavelength of photon *absorbed* for an electron transitioning from *2s to 4p orbital*
- iii. Wavelength of photon *absorbed* for an electron transitioning from *1s to 3p orbital*

i or ii or iii



Question 2 8 points



A Chem110 student plots the graph shown on the left, from data obtained for the kinetic energy (eV) of the ejected electron versus frequency (s^{-1}) of the incident light, for an unknown metal. ($1\text{eV} = 1.6022 \times 10^{-19} \text{ J}$)

Based on the plot, answer the questions **a to c** below:

a. 1 point On the plot above, label the **threshold frequency**.

b. 2 points What is the value of the work potential (in J) of the unknown metal?

c. 3 points What is the *longest* wavelength (in nm) required to eject an electron from this metal's surface? Show your work including equation(s) used. (3 sig figs)

d. 2 points In the space provided below, plot the following

(Assume frequency is *above* threshold frequency):

- i. *Kinetic energy* of an ejected electron versus the *intensity* of the incident light (left)
- ii. *Number of electrons* ejected versus the *frequency* of the incident light (right)



**Question 3** **6 points****a. 2 points**

In the ground state of Ge (Germanium; $Z = 32$) atom, **name** the following orbitals:

i. Orbital containing the electron that requires the maximum energy to be removed.

ii. Orbital containing the electron that experiences the minimum effective nuclear charge.

b. 2 points

An unknown anion (A^-) has the following *excited* electronic configuration: $[Ar] 4s^2 3d^3 4p^1$

(Ar: Argon; $Z = 18$)

Give the **condensed ground state electronic configuration** of the atom, **A** and the cation, **A^{2+}** .
Show your work. (Assume all electronic transitions are possible)

Electronic Configuration of A:

Electronic Configuration of A^{2+} :

c. 2 points

In a single atom, determine how many electrons can have the following set of quantum numbers?

(i) $n = 5; l = 3$

In a single atom, determine how many electrons can have the following set of quantum numbers?

(i) $n = 3; l = 2; m_s = +1/2$

**Question 4** **8 points**

Rank the following from 1 to 4 in **increasing** order. Provide the ranking in the boxes provided.

Note: Atomic number (Z) are provided for relevant atoms

a) Increasing (lowest to highest) order of *ionization energy*: (1= lowest; 4 = highest)

Mg

Mg: Z = 12

Rb

Rb: Z = 37

Cs

Cs: Z = 55

K

K : Z = 19

b) Increasing (lowest to highest) order of *bond angle*: (1= lowest; 4 = highest)

H₂OCCl₄XeF₄

HCN

c) Increasing (smallest to largest) order of *ionic size* (1= smallest; 4 = largest)

S²⁻

S: Z = 16

O²⁻

O: Z = 8

Be²⁺

Be: Z = 4

Na⁺

Na: Z = 11

d) Increasing (lowest to highest) order of *bond energy* (1= lowest; 4 = highest)

P-F

P-Cl

P-I

P-Br

**Question 5** **12 points**

a. 9 points For the molecules HClO_4 and TeF_6 , draw the most stable *Lewis structure* showing appropriate shape and *dashed/wedged bonds*, based on VSEPR. Determine the **molecular geometry** around the central atom. For HClO_4 , label the *most polar bond*.

HClO_4 (there are no O-O bonds)

Label the most polar bond

Molecular Geometry around Cl

TeF_6

Molecular Geometry around Te

b. 3 points (Note: Lone pairs are not shown in structures below)

(i) For the three *resonance* structures (1, 2 and 3) shown below, answer the following question:



(I) Circle the *most contributing* resonance structure.

Structure 1

Structure 2

Structure 3

(ii) For the three structures given (4, 5, and 6) below, answer the following questions:



(I) Circle all the structures that represent *resonance structures* of each other:

Structure 4

Structure 5

Structure 6

(II) Among the resonance structures, circle the *most contributing* resonance structure.

Structure 4

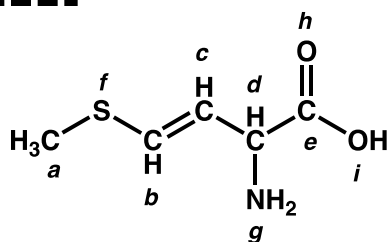
Structure 5

Structure 6

**Question 6** **6 points**

Using the data given below, calculate the **lattice energy** for SrO(s).

	ΔH° (kJ/mol)
$\text{Sr}(s) \longrightarrow \text{Sr}(g)$	+180
$\text{Sr}(g) \longrightarrow \text{Sr}^{2+}(g) + 2e^-$	+1468
$\text{O}_2(g) \longrightarrow 2\text{O}(g)$	+500
$\text{O}(g) + 2e^- \longrightarrow \text{O}^{2-}(g)$	+600
$\text{Sr}(s) + \frac{1}{2}\text{O}_2(g) \longrightarrow \text{SrO}(s)$	-550

**Question 7 10 points**

Consider the molecule given on the left, and answer questions (i) to (vi) using chemical bonding theories that we have learnt in class (valence bond theory, hybridization).

Note that the lone pairs are not shown in the structure:

i. Consider the carbon atoms marked in as ***a***, ***b***, ***c***, ***d***, and ***e*** in the molecule, and determine **hybridization** at each of the carbon centers. (2.5 points)

Atom	Hybridization
Carbon (<i>a</i>)	
Carbon (<i>b</i>)	
Carbon (<i>c</i>)	
Carbon (<i>d</i>)	
Carbon (<i>e</i>)	

ii. Now, consider the sulfur atom (S) marked as ***f***, nitrogen (N) atom (***g***), and oxygen(O) atoms (***h*** and ***i***), and determine their **hybridization**. (2 points)

Atom	Hybridization
Sulfur (<i>f</i>)	
Nitrogen (<i>g</i>)	
Oxygen (<i>h</i>)	
Oxygen (<i>i</i>)	

iii. **Describe** all the bonds formed by the carbon atoms ***a*** and ***b***, using sigma and pi designation as appropriate. (2 points)

Atom	Bonds formed (use sigma and pi as appropriate)
Carbon (<i>a</i>)	
Carbon (<i>b</i>)	

iv. How many total sp^3 -s sigma bonds are in the molecule? (1 point)

v. Determine **electronic** and **molecular geometry** at carbon (C) labeled ***e*** and sulfur (S) labeled ***f*** centres. (2 points)

Atom	Electronic Geometry	Molecular Geometry
Carbon (<i>e</i>)		
Sulfur (<i>f</i>)		

vi. Circle the value from the choices given below, that describes the **C-S-C bond angle at sulfur** (S) marked ***f***: (0.5 points)

>90° to <109.5° 109.5° >109.5 to <120° 120° >120° to <180°

**Question 8 12 points**

Using the **valence shell orbitals only** draw a **complete** molecular orbital (MO) diagram (with no s-p mixing) for the molecule **PS**, in the space provided below. **(6 points)**

Assume that 3s and 3p atomic orbitals will form molecular orbitals like those formed by the combination of 2s and 2p atomic orbitals that we learnt in class, and there is no s-p mixing.

Show **energy arrow**, **atomic orbitals** involved, **atom labels**, the **molecular orbitals** that are formed, **filled electrons** in the MO diagram etc. Make sure to label all the atomic and molecular orbitals.

Using the MO diagram above, answer the following questions:

- (i) Which one of the following will have the highest bond order? Circle the correct choice
A) PS^{1-} B) PS C) PS^{1+}
- (ii) Which one of the following will have the longest bond length? Circle the correct choice
A) PS^{1-} B) PS C) PS^{1+}
- (iii) Which of the following will be diamagnetic? Circle the correct choice.
A) PS^{1-} B) PS C) PS^{1+}
- (iv) Draw Lewis dot structure of the molecule with the shortest bond length (based on the MO diagram) from among the three (PS^{1-} , PS, PS^{1+}), and indicate the bond order calculated from the Lewis dot structure. **(3 points)**

Lewis Dot Structure

Bond Order



B4B49515-D2ED-4661-9EC3-C30EF398FCCB

midterm-exam-93388

#390 Page 10 of 10

SCRAP PAGE

**CHEMISTRY 110 –Midterm Exam****Thursday November 7, 2024, 6:30 – 8:30 pm****Examiners: Pallavi Sirjoosingh and Ashok Kakkar**

First name (please write as legibly as possible within the boxes)

--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--

Last name

--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--

Student ID number

--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--

This booklet has 8 questions on 10 pages (The last page can be used as a scrap page).

The staple must NOT be removed from the question booklet. Your intact question booklet must be returned at the end of the exam.

A separate booklet contains a periodic table, and datasheet.

Write your answers directly under the questions in the space provided. Do not write on the QR codes or near the top edges (as it may be cut off).

Please write **legibly and dark enough** for the scan to be readable. For students who use English as a second language, dictionaries are permitted. Calculators are allowed.

DO NOT ASK QUESTIONS OF THE INVIGILATORS about interpretations of the exam questions. If you have a concern about a question, please summarize it at the top of this page. All concerns will be evaluated after the exam. GOOD LUCK!

**Question 1 8 points** (*Assume all transitions are possible*)

a. 6 points Calculate the **energy (in kJ/mol)** required to ionize (completely remove) the electron from the **$n=3$** level of **He^+ ion** (He: Helium). Show your work including equations. (3 sig. figs.)

b. 2 points

(I) Select the largest (longest) wavelength from the three choices below:

- i. Wavelength of photon to *completely remove* an electron from *1s orbital of Hydrogen atom*
- ii. Wavelength of photon to *completely remove* an electron from *1s orbital of Boron atom (B)*
- iii. Wavelength of photon to *completely remove* an electron from *1s orbital of Boron cation (B^+)*

i or ii or iii

(II) Select the smallest (shortest) wavelength from the three choices below (*for H atom*):

- i. Wavelength of photon *emitted* for an electron transitioning from *2p to 1s orbital*
- ii. Wavelength of photon *absorbed* for an electron transitioning from *2s to 4p orbital*
- iii. Wavelength of photon *absorbed* for an electron transitioning from *1s to 3p orbital*

i or ii or iii



Question 2 8 points



A Chem110 student plots the graph shown on the left, from data obtained for the kinetic energy (eV) of the ejected electron versus frequency (s^{-1}) of the incident light, for an unknown metal. ($1\text{eV} = 1.6022 \times 10^{-19} \text{ J}$)

Based on the plot, answer the questions **a to c** below:

a. 1 point On the plot above, label the **threshold frequency**.

b. 2 points What is the value of the work potential (in J) of the unknown metal?

c. 3 points What is the *longest* wavelength (in nm) required to eject an electron from this metal's surface? Show your work including equation(s) used. (3 sig figs)

d. 2 points In the space provided below, plot the following

(Assume frequency is *above* threshold frequency):

- i. *Kinetic energy* of an ejected electron versus the *intensity* of the incident light (left)
- ii. *Number of electrons* ejected versus the *frequency* of the incident light (right)



**Question 3** **6 points****a. 2 points**

In the ground state of Ge (Germanium; $Z = 32$) atom, **name** the following orbitals:

i. Orbital containing the electron that requires the maximum energy to be removed.

ii. Orbital containing the electron that experiences the minimum effective nuclear charge.

b. 2 points

An unknown anion (A^-) has the following *excited* electronic configuration: $[Ar] 4s^2 3d^3 4p^1$

(Ar: Argon; $Z = 18$)

Give the **condensed ground state electronic configuration** of the atom, **A** and the cation, **A^{2+}** .
Show your work. (Assume all electronic transitions are possible)

Electronic Configuration of A:

Electronic Configuration of A^{2+} :

c. 2 points

In a single atom, determine how many electrons can have the following set of quantum numbers?

(i) $n = 5; l = 3$

In a single atom, determine how many electrons can have the following set of quantum numbers?

(i) $n = 3; l = 2; m_s = +1/2$

**Question 4** **8 points**

Rank the following from 1 to 4 in **increasing** order. Provide the ranking in the boxes provided.

Note: Atomic number (Z) are provided for relevant atoms

a) Increasing (lowest to highest) order of *ionization energy*: (1= lowest; 4 = highest)

Mg

Mg: Z = 12

Rb

Rb: Z = 37

Cs

Cs: Z = 55

K

K : Z = 19

b) Increasing (lowest to highest) order of *bond angle*: (1= lowest; 4 = highest)

H₂OCCl₄XeF₄

HCN

c) Increasing (smallest to largest) order of *ionic size* (1= smallest; 4 = largest)

S²⁻

S: Z = 16

O²⁻

O: Z = 8

Be²⁺

Be: Z = 4

Na⁺

Na: Z = 11

d) Increasing (lowest to highest) order of *bond energy* (1= lowest; 4 = highest)

P-F

P-Cl

P-I

P-Br

**Question 5** **12 points**

a. 9 points For the molecules HClO_4 and TeF_6 , draw the most stable *Lewis structure* showing appropriate shape and *dashed/wedged bonds*, based on VSEPR. Determine the **molecular geometry** around the central atom. For HClO_4 , label the *most polar bond*.



b. 3 points (Note: Lone pairs are not shown in structures below)

(i) For the three *resonance* structures (1, 2 and 3) shown below, answer the following question:



(I) Circle the *most contributing* resonance structure.

Structure 1

Structure 2

Structure 3

(ii) For the three structures given (4, 5, and 6) below, answer the following questions:



(I) Circle all the structures that represent *resonance structures* of each other:

Structure 4

Structure 5

Structure 6

(II) Among the resonance structures, circle the *most contributing* resonance structure.

Structure 4

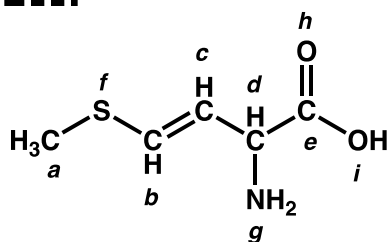
Structure 5

Structure 6

**Question 6** **6 points**

Using the data given below, calculate the **lattice energy** for SrO(s).

	ΔH° (kJ/mol)
$\text{Sr}(s) \longrightarrow \text{Sr}(g)$	+180
$\text{Sr}(g) \longrightarrow \text{Sr}^{2+}(g) + 2e^-$	+1468
$\text{O}_2(g) \longrightarrow 2\text{O}(g)$	+500
$\text{O}(g) + 2e^- \longrightarrow \text{O}^{2-}(g)$	+600
$\text{Sr}(s) + \frac{1}{2}\text{O}_2(g) \longrightarrow \text{SrO}(s)$	-550

**Question 7 10 points**

Consider the molecule given on the left, and answer questions (i) to (vi) using chemical bonding theories that we have learnt in class (valence bond theory, hybridization).

Note that the lone pairs are not shown in the structure:

i. Consider the carbon atoms marked in as ***a***, ***b***, ***c***, ***d***, and ***e*** in the molecule, and determine **hybridization** at each of the carbon centers. (2.5 points)

Atom	Hybridization
Carbon (<i>a</i>)	
Carbon (<i>b</i>)	
Carbon (<i>c</i>)	
Carbon (<i>d</i>)	
Carbon (<i>e</i>)	

ii. Now, consider the sulfur atom (S) marked as ***f***, nitrogen (N) atom (***g***), and oxygen(O) atoms (***h*** and ***i***), and determine their **hybridization**. (2 points)

Atom	Hybridization
Sulfur (<i>f</i>)	
Nitrogen (<i>g</i>)	
Oxygen (<i>h</i>)	
Oxygen (<i>i</i>)	

iii. **Describe** all the bonds formed by the carbon atoms ***a*** and ***b***, using sigma and pi designation as appropriate. (2 points)

Atom	Bonds formed (use sigma and pi as appropriate)
Carbon (<i>a</i>)	
Carbon (<i>b</i>)	

iv. How many total sp^3 -s sigma bonds are in the molecule? (1 point)

v. Determine **electronic** and **molecular geometry** at carbon (C) labeled ***e*** and sulfur (S) labeled ***f*** centres. (2 points)

Atom	Electronic Geometry	Molecular Geometry
Carbon (<i>e</i>)		
Sulfur (<i>f</i>)		

vi. Circle the value from the choices given below, that describes the **C-S-C bond angle at sulfur** (S) marked ***f***: (0.5 points)

>90° to <109.5° 109.5° >109.5 to <120° 120° >120° to <180°


Question 8 12 points

Using the **valence shell orbitals only** draw a **complete** molecular orbital (MO) diagram (with no s-p mixing) for the molecule **PS**, in the space provided below. **(6 points)**

Assume that 3s and 3p atomic orbitals will form molecular orbitals like those formed by the combination of 2s and 2p atomic orbitals that we learnt in class, and there is no s-p mixing.

Show **energy arrow**, **atomic orbitals** involved, **atom labels**, the **molecular orbitals** that are formed, **filled electrons** in the MO diagram etc. Make sure to label all the atomic and molecular orbitals.

Using the MO diagram above, answer the following questions:

- (i) Which one of the following will have the highest bond order? Circle the correct choice
 A) PS^{1-} B) PS C) PS^{1+}
- (ii) Which one of the following will have the longest bond length? Circle the correct choice
 A) PS^{1-} B) PS C) PS^{1+}
- (iii) Which of the following will be diamagnetic? Circle the correct choice.
 A) PS^{1-} B) PS C) PS^{1+}
- (iv) Draw Lewis dot structure of the molecule with the shortest bond length (based on the MO diagram) from among the three (PS^{1-} , PS, PS^{1+}), and indicate the bond order calculated from the Lewis dot structure. **(3 points)**

Lewis Dot Structure

Bond Order



8F3E1F5C-1A0B-4AC5-BF0B-3F3B63BBB73C

midterm-exam-93388

#391 Page 10 of 10

SCRAP PAGE

**CHEMISTRY 110 –Midterm Exam****Thursday November 7, 2024, 6:30 – 8:30 pm****Examiners: Pallavi Sirjoosingh and Ashok Kakkar**

First name (please write as legibly as possible within the boxes)

--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--

Last name

--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--

Student ID number

--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--

This booklet has 8 questions on 10 pages (The last page can be used as a scrap page).

The staple must NOT be removed from the question booklet. Your intact question booklet must be returned at the end of the exam.

A separate booklet contains a periodic table, and datasheet.

Write your answers directly under the questions in the space provided. Do not write on the QR codes or near the top edges (as it may be cut off).

Please write **legibly and dark enough** for the scan to be readable. For students who use English as a second language, dictionaries are permitted. Calculators are allowed.

DO NOT ASK QUESTIONS OF THE INVIGILATORS about interpretations of the exam questions. If you have a concern about a question, please summarize it at the top of this page. All concerns will be evaluated after the exam. GOOD LUCK!

**Question 1 8 points** (*Assume all transitions are possible*)

a. 6 points Calculate the **energy (in kJ/mol)** required to ionize (completely remove) the electron from the **$n=3$** level of **He^+ ion** (He: Helium). Show your work including equations. (3 sig. figs.)

b. 2 points

(I) Select the largest (longest) wavelength from the three choices below:

- i. Wavelength of photon to *completely remove* an electron from *1s orbital of Hydrogen atom*
- ii. Wavelength of photon to *completely remove* an electron from *1s orbital of Boron atom (B)*
- iii. Wavelength of photon to *completely remove* an electron from *1s orbital of Boron cation (B^+)*

i or ii or iii

(II) Select the smallest (shortest) wavelength from the three choices below (*for H atom*):

- i. Wavelength of photon *emitted* for an electron transitioning from *2p to 1s orbital*
- ii. Wavelength of photon *absorbed* for an electron transitioning from *2s to 4p orbital*
- iii. Wavelength of photon *absorbed* for an electron transitioning from *1s to 3p orbital*

i or ii or iii



Question 2 8 points



A Chem110 student plots the graph shown on the left, from data obtained for the kinetic energy (eV) of the ejected electron versus frequency (s^{-1}) of the incident light, for an unknown metal. ($1\text{eV} = 1.6022 \times 10^{-19} \text{ J}$)

Based on the plot, answer the questions **a to c** below:

a. 1 point On the plot above, label the **threshold frequency**.

b. 2 points What is the value of the work potential (in J) of the unknown metal?

c. 3 points What is the *longest* wavelength (in nm) required to eject an electron from this metal's surface? Show your work including equation(s) used. (3 sig figs)

d. 2 points In the space provided below, plot the following

(Assume frequency is *above* threshold frequency):

- i. *Kinetic energy* of an ejected electron versus the *intensity* of the incident light (left)
- ii. *Number of electrons* ejected versus the *frequency* of the incident light (right)



**Question 3** **6 points****a. 2 points**

In the ground state of Ge (Germanium; $Z = 32$) atom, **name** the following orbitals:

i. Orbital containing the electron that requires the maximum energy to be removed.

ii. Orbital containing the electron that experiences the minimum effective nuclear charge.

b. 2 points

An unknown anion (A^-) has the following *excited* electronic configuration: $[Ar] 4s^2 3d^3 4p^1$

(Ar: Argon; $Z = 18$)

Give the **condensed ground state electronic configuration** of the atom, **A** and the cation, **A^{2+}** .
Show your work. (Assume all electronic transitions are possible)

Electronic Configuration of A:

Electronic Configuration of A^{2+} :

c. 2 points

In a single atom, determine how many electrons can have the following set of quantum numbers?

(i) $n = 5; l = 3$

In a single atom, determine how many electrons can have the following set of quantum numbers?

(i) $n = 3; l = 2; m_s = +1/2$

**Question 4** **8 points**

Rank the following from 1 to 4 in **increasing** order. Provide the ranking in the boxes provided.

Note: Atomic number (Z) are provided for relevant atoms

a) Increasing (lowest to highest) order of *ionization energy*: (1= lowest; 4 = highest)

Mg

Mg: Z = 12

Rb

Rb: Z = 37

Cs

Cs: Z = 55

K

K : Z = 19

b) Increasing (lowest to highest) order of *bond angle*: (1= lowest; 4 = highest)

H₂OCCl₄XeF₄

HCN

c) Increasing (smallest to largest) order of *ionic size* (1= smallest; 4 = largest)

S²⁻

S: Z = 16

O²⁻

O: Z = 8

Be²⁺

Be: Z = 4

Na⁺

Na: Z = 11

d) Increasing (lowest to highest) order of *bond energy* (1= lowest; 4 = highest)

P-F

P-Cl

P-I

P-Br

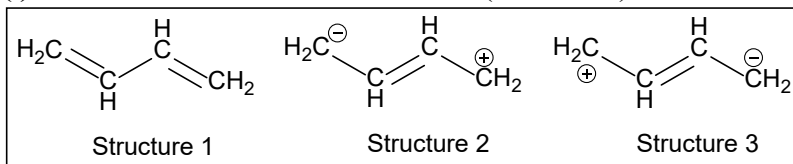
**Question 5** **12 points**

a. 9 points For the molecules HClO_4 and TeF_6 , draw the most stable *Lewis structure* showing appropriate shape and *dashed/wedged bonds*, based on VSEPR. Determine the **molecular geometry** around the central atom. For HClO_4 , label the *most polar bond*.



b. 3 points (Note: Lone pairs are not shown in structures below)

(i) For the three *resonance* structures (1, 2 and 3) shown below, answer the following question:



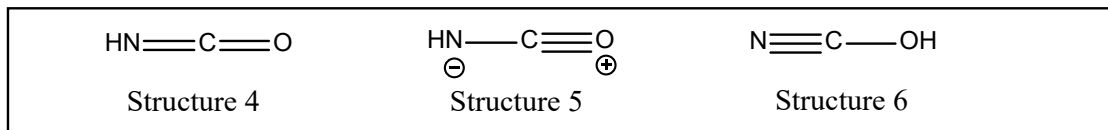
(I) Circle the *most contributing* resonance structure.

Structure 1

Structure 2

Structure 3

(ii) For the three structures given (4, 5, and 6) below, answer the following questions:



(I) Circle all the structures that represent *resonance structures* of each other:

Structure 4

Structure 5

Structure 6

(II) Among the resonance structures, circle the *most contributing* resonance structure.

Structure 4

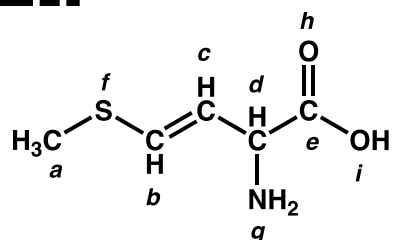
Structure 5

Structure 6

**Question 6** **6 points**

Using the data given below, calculate the **lattice energy** for SrO(s).

	ΔH° (kJ/mol)
$\text{Sr}(s) \longrightarrow \text{Sr}(g)$	+180
$\text{Sr}(g) \longrightarrow \text{Sr}^{2+}(g) + 2e^-$	+1468
$\text{O}_2(g) \longrightarrow 2\text{O}(g)$	+500
$\text{O}(g) + 2e^- \longrightarrow \text{O}^{2-}(g)$	+600
$\text{Sr}(s) + \frac{1}{2}\text{O}_2(g) \longrightarrow \text{SrO}(s)$	-550

**Question 7 10 points**

Consider the molecule given on the left, and answer questions (i) to (vi) using chemical bonding theories that we have learnt in class (valence bond theory, hybridization).

Note that the lone pairs are not shown in the structure:

i. Consider the carbon atoms marked in as ***a***, ***b***, ***c***, ***d***, and ***e*** in the molecule, and determine **hybridization** at each of the carbon centers. (2.5 points)

Atom	Hybridization
Carbon (<i>a</i>)	
Carbon (<i>b</i>)	
Carbon (<i>c</i>)	
Carbon (<i>d</i>)	
Carbon (<i>e</i>)	

ii. Now, consider the sulfur atom (S) marked as ***f***, nitrogen (N) atom (***g***), and oxygen(O) atoms (***h*** and ***i***), and determine their **hybridization**. (2 points)

Atom	Hybridization
Sulfur (<i>f</i>)	
Nitrogen (<i>g</i>)	
Oxygen (<i>h</i>)	
Oxygen (<i>i</i>)	

iii. **Describe** all the bonds formed by the carbon atoms ***a*** and ***b***, using sigma and pi designation as appropriate. (2 points)

Atom	Bonds formed (use sigma and pi as appropriate)
Carbon (<i>a</i>)	
Carbon (<i>b</i>)	

iv. How many total sp^3 -s sigma bonds are in the molecule? (1 point)

v. Determine **electronic** and **molecular geometry** at carbon (C) labeled ***e*** and sulfur (S) labeled ***f*** centres. (2 points)

Atom	Electronic Geometry	Molecular Geometry
Carbon (<i>e</i>)		
Sulfur (<i>f</i>)		

vi. Circle the value from the choices given below, that describes the **C-S-C bond angle at sulfur** (S) marked ***f***: (0.5 points)

$>90^\circ$ to $<109.5^\circ$ 109.5° >109.5 to $<120^\circ$ 120° $>120^\circ$ to $<180^\circ$


Question 8 12 points

Using the **valence shell orbitals only** draw a **complete** molecular orbital (MO) diagram (with no s-p mixing) for the molecule **PS**, in the space provided below. **(6 points)**

Assume that 3s and 3p atomic orbitals will form molecular orbitals like those formed by the combination of 2s and 2p atomic orbitals that we learnt in class, and there is no s-p mixing.

Show **energy arrow**, **atomic orbitals** involved, **atom labels**, the **molecular orbitals** that are formed, **filled electrons** in the MO diagram etc. Make sure to label all the atomic and molecular orbitals.

Using the MO diagram above, answer the following questions:

- (i) Which one of the following will have the highest bond order? Circle the correct choice
 A) PS^{1-} B) PS C) PS^{1+}
- (ii) Which one of the following will have the longest bond length? Circle the correct choice
 A) PS^{1-} B) PS C) PS^{1+}
- (iii) Which of the following will be diamagnetic? Circle the correct choice.
 A) PS^{1-} B) PS C) PS^{1+}
- (iv) Draw Lewis dot structure of the molecule with the shortest bond length (based on the MO diagram) from among the three (PS^{1-} , PS, PS^{1+}), and indicate the bond order calculated from the Lewis dot structure. **(3 points)**

Lewis Dot Structure

Bond Order



DBB80DD8-89F2-42DF-8429-E0B46A312892

midterm-exam-93388

#392 Page 10 of 10

SCRAP PAGE

**CHEMISTRY 110 –Midterm Exam****Thursday November 7, 2024, 6:30 – 8:30 pm**

Examiners: Pallavi Sirjoosingh and Ashok Kakkar

First name (please write as legibly as possible within the boxes)

--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--

Last name

--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--

Student ID number

--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--

This booklet has 8 questions on 10 pages (The last page can be used as a scrap page).

The staple must NOT be removed from the question booklet. Your intact question booklet must be returned at the end of the exam.

A separate booklet contains a periodic table, and datasheet.

Write your answers directly under the questions in the space provided. Do not write on the QR codes or near the top edges (as it may be cut off).

Please write **legibly and dark enough** for the scan to be readable. For students who use English as a second language, dictionaries are permitted. Calculators are allowed.

DO NOT ASK QUESTIONS OF THE INVIGILATORS about interpretations of the exam questions. If you have a concern about a question, please summarize it at the top of this page. All concerns will be evaluated after the exam. GOOD LUCK!

**Question 1 8 points** (*Assume all transitions are possible*)

a. 6 points Calculate the **energy (in kJ/mol)** required to ionize (completely remove) the electron from the **n=3** level of **He⁺ ion** (He: Helium). Show your work including equations. (3 sig. figs.)

b. 2 points

(I) Select the largest (longest) wavelength from the three choices below:

- i. Wavelength of photon to *completely remove* an electron from *1s orbital of Hydrogen atom*
- ii. Wavelength of photon to *completely remove* an electron from *1s orbital of Boron atom (B)*
- iii. Wavelength of photon to *completely remove* an electron from *1s orbital of Boron cation (B⁺)*

i or ii or iii

(II) Select the smallest (shortest) wavelength from the three choices below (*for H atom*):

- i. Wavelength of photon *emitted* for an electron transitioning from *2p to 1s orbital*
- ii. Wavelength of photon *absorbed* for an electron transitioning from *2s to 4p orbital*
- iii. Wavelength of photon *absorbed* for an electron transitioning from *1s to 3p orbital*

i or ii or iii



Question 2 8 points



A Chem110 student plots the graph shown on the left, from data obtained for the kinetic energy (eV) of the ejected electron versus frequency (s^{-1}) of the incident light, for an unknown metal. ($1\text{eV} = 1.6022 \times 10^{-19} \text{ J}$)

Based on the plot, answer the questions **a to c** below:

a. 1 point On the plot above, label the **threshold frequency**.

b. 2 points What is the value of the work potential (in J) of the unknown metal?

c. 3 points What is the *longest* wavelength (in nm) required to eject an electron from this metal's surface? Show your work including equation(s) used. (3 sig figs)

d. 2 points In the space provided below, plot the following

(Assume frequency is *above* threshold frequency):

- Kinetic energy* of an ejected electron versus the *intensity* of the incident light (left)
- Number of electrons* ejected versus the *frequency* of the incident light (right)



**Question 3** **6 points****a. 2 points**

In the ground state of Ge (Germanium; $Z = 32$) atom, **name** the following orbitals:

i. Orbital containing the electron that requires the maximum energy to be removed.

ii. Orbital containing the electron that experiences the minimum effective nuclear charge.

b. 2 points

An unknown anion (A^-) has the following *excited* electronic configuration: $[Ar] 4s^2 3d^3 4p^1$

(Ar: Argon; $Z = 18$)

Give the **condensed ground state electronic configuration** of the atom, **A** and the cation, **A^{2+}** .
Show your work. (Assume all electronic transitions are possible)

Electronic Configuration of A:

Electronic Configuration of A^{2+} :

c. 2 points

In a single atom, determine how many electrons can have the following set of quantum numbers?

(i) $n = 5; l = 3$

In a single atom, determine how many electrons can have the following set of quantum numbers?

(i) $n = 3; l = 2; m_s = +1/2$

**Question 4** **8 points**

Rank the following from 1 to 4 in **increasing** order. Provide the ranking in the boxes provided.

Note: Atomic number (Z) are provided for relevant atoms

a) Increasing (lowest to highest) order of *ionization energy*: (1= lowest; 4 = highest)

Mg

Mg: Z = 12

Rb

Rb: Z = 37

Cs

Cs: Z = 55

K

K : Z = 19

b) Increasing (lowest to highest) order of *bond angle*: (1= lowest; 4 = highest)

H₂OCCl₄XeF₄

HCN

c) Increasing (smallest to largest) order of *ionic size* (1= smallest; 4 = largest)

S²⁻

S: Z = 16

O²⁻

O: Z = 8

Be²⁺

Be: Z = 4

Na⁺

Na: Z = 11

d) Increasing (lowest to highest) order of *bond energy* (1= lowest; 4 = highest)

P-F

P-Cl

P-I

P-Br

**Question 5** **12 points**

a. 9 points For the molecules HClO_4 and TeF_6 , draw the most stable *Lewis structure* showing appropriate shape and *dashed/wedged bonds*, based on VSEPR. Determine the **molecular geometry** around the central atom. For HClO_4 , label the *most polar bond*.



b. 3 points (Note: Lone pairs are not shown in structures below)

(i) For the three *resonance* structures (1, 2 and 3) shown below, answer the following question:



(I) Circle the *most contributing* resonance structure.

Structure 1

Structure 2

Structure 3

(ii) For the three structures given (4, 5, and 6) below, answer the following questions:



(I) Circle all the structures that represent *resonance structures* of each other:

Structure 4

Structure 5

Structure 6

(II) Among the resonance structures, circle the *most contributing* resonance structure.

Structure 4

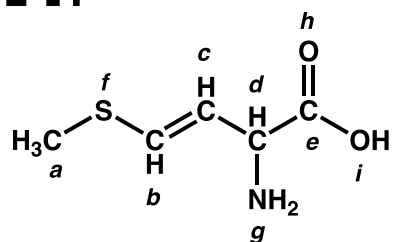
Structure 5

Structure 6

**Question 6** **6 points**

Using the data given below, calculate the **lattice energy** for SrO(s).

	ΔH° (kJ/mol)
$\text{Sr}(s) \longrightarrow \text{Sr}(g)$	+180
$\text{Sr}(g) \longrightarrow \text{Sr}^{2+}(g) + 2e^-$	+1468
$\text{O}_2(g) \longrightarrow 2\text{O}(g)$	+500
$\text{O}(g) + 2e^- \longrightarrow \text{O}^{2-}(g)$	+600
$\text{Sr}(s) + \frac{1}{2}\text{O}_2(g) \longrightarrow \text{SrO}(s)$	-550

**Question 7 10 points**

Consider the molecule given on the left, and answer questions (i) to (vi) using chemical bonding theories that we have learnt in class (valence bond theory, hybridization).

Note that the lone pairs are not shown in the structure:

i. Consider the carbon atoms marked in as ***a***, ***b***, ***c***, ***d***, and ***e*** in the molecule, and determine **hybridization** at each of the carbon centers. (2.5 points)

Atom	Hybridization
Carbon (<i>a</i>)	
Carbon (<i>b</i>)	
Carbon (<i>c</i>)	
Carbon (<i>d</i>)	
Carbon (<i>e</i>)	

ii. Now, consider the sulfur atom (S) marked as ***f***, nitrogen (N) atom (***g***), and oxygen(O) atoms (***h*** and ***i***), and determine their **hybridization**. (2 points)

Atom	Hybridization
Sulfur (<i>f</i>)	
Nitrogen (<i>g</i>)	
Oxygen (<i>h</i>)	
Oxygen (<i>i</i>)	

iii. **Describe** all the bonds formed by the carbon atoms ***a*** and ***b***, using sigma and pi designation as appropriate. (2 points)

Atom	Bonds formed (use sigma and pi as appropriate)
Carbon (<i>a</i>)	
Carbon (<i>b</i>)	

iv. How many total sp^3 -s sigma bonds are in the molecule? (1 point)

v. Determine **electronic** and **molecular geometry** at carbon (C) labeled ***e*** and sulfur (S) labeled ***f*** centres. (2 points)

Atom	Electronic Geometry	Molecular Geometry
Carbon (<i>e</i>)		
Sulfur (<i>f</i>)		

vi. Circle the value from the choices given below, that describes the **C-S-C bond angle at sulfur** (S) marked ***f***: (0.5 points)

>90° to <109.5° 109.5° >109.5 to <120° 120° >120° to <180°


Question 8 12 points

Using the **valence shell orbitals only** draw a **complete** molecular orbital (MO) diagram (with no s-p mixing) for the molecule **PS**, in the space provided below. **(6 points)**

Assume that 3s and 3p atomic orbitals will form molecular orbitals like those formed by the combination of 2s and 2p atomic orbitals that we learnt in class, and there is no s-p mixing.

Show **energy arrow**, **atomic orbitals** involved, **atom labels**, the **molecular orbitals** that are formed, **filled electrons** in the MO diagram etc. Make sure to label all the atomic and molecular orbitals.

Using the MO diagram above, answer the following questions:

- (i) Which one of the following will have the highest bond order? Circle the correct choice
 A) PS^{1-} B) PS C) PS^{1+}
- (ii) Which one of the following will have the longest bond length? Circle the correct choice
 A) PS^{1-} B) PS C) PS^{1+}
- (iii) Which of the following will be diamagnetic? Circle the correct choice.
 A) PS^{1-} B) PS C) PS^{1+}
- (iv) Draw Lewis dot structure of the molecule with the shortest bond length (based on the MO diagram) from among the three (PS^{1-} , PS, PS^{1+}), and indicate the bond order calculated from the Lewis dot structure. **(3 points)**

Lewis Dot Structure

Bond Order



77B22641-6471-4001-A01E-E98E3A56C646

midterm-exam-93388

#393 Page 10 of 10

SCRAP PAGE

**CHEMISTRY 110 –Midterm Exam****Thursday November 7, 2024, 6:30 – 8:30 pm**

Examiners: Pallavi Sirjoosingh and Ashok Kakkar

First name (please write as legibly as possible within the boxes)

--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--

Last name

--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--

Student ID number

--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--

This booklet has 8 questions on 10 pages (The last page can be used as a scrap page).

The staple must NOT be removed from the question booklet. Your intact question booklet must be returned at the end of the exam.

A separate booklet contains a periodic table, and datasheet.

Write your answers directly under the questions in the space provided. Do not write on the QR codes or near the top edges (as it may be cut off).

Please write **legibly and dark enough** for the scan to be readable. For students who use English as a second language, dictionaries are permitted. Calculators are allowed.

DO NOT ASK QUESTIONS OF THE INVIGILATORS about interpretations of the exam questions. If you have a concern about a question, please summarize it at the top of this page. All concerns will be evaluated after the exam. GOOD LUCK!

**Question 1 8 points** (*Assume all transitions are possible*)

a. 6 points Calculate the **energy (in kJ/mol)** required to ionize (completely remove) the electron from the **n=3** level of **He⁺ ion** (He: Helium). Show your work including equations. (3 sig. figs.)

b. 2 points

(I) Select the largest (longest) wavelength from the three choices below:

- i. Wavelength of photon to *completely remove* an electron from *1s orbital of Hydrogen atom*
- ii. Wavelength of photon to *completely remove* an electron from *1s orbital of Boron atom (B)*
- iii. Wavelength of photon to *completely remove* an electron from *1s orbital of Boron cation (B⁺)*

i or ii or iii

(II) Select the smallest (shortest) wavelength from the three choices below (*for H atom*):

- i. Wavelength of photon *emitted* for an electron transitioning from *2p to 1s orbital*
- ii. Wavelength of photon *absorbed* for an electron transitioning from *2s to 4p orbital*
- iii. Wavelength of photon *absorbed* for an electron transitioning from *1s to 3p orbital*

i or ii or iii



Question 2 8 points



A Chem110 student plots the graph shown on the left, from data obtained for the kinetic energy (eV) of the ejected electron versus frequency (s^{-1}) of the incident light, for an unknown metal. ($1\text{eV} = 1.6022 \times 10^{-19} \text{ J}$)

Based on the plot, answer the questions **a to c** below:

a. 1 point On the plot above, label the **threshold frequency**.

b. 2 points What is the value of the work potential (in J) of the unknown metal?

c. 3 points What is the *longest* wavelength (in nm) required to eject an electron from this metal's surface? Show your work including equation(s) used. (3 sig figs)

d. 2 points In the space provided below, plot the following

(Assume frequency is *above* threshold frequency):

- Kinetic energy* of an ejected electron versus the *intensity* of the incident light (left)
- Number of electrons* ejected versus the *frequency* of the incident light (right)



**Question 3** **6 points****a. 2 points**

In the ground state of Ge (Germanium; $Z = 32$) atom, **name** the following orbitals:

i. Orbital containing the electron that requires the maximum energy to be removed.

ii. Orbital containing the electron that experiences the minimum effective nuclear charge.

b. 2 points

An unknown anion (A^-) has the following *excited* electronic configuration: $[Ar] 4s^2 3d^3 4p^1$

(Ar: Argon; $Z = 18$)

Give the **condensed ground state electronic configuration** of the atom, **A** and the cation, **A^{2+}** .
Show your work. (Assume all electronic transitions are possible)

Electronic Configuration of A:

Electronic Configuration of A^{2+} :

c. 2 points

In a single atom, determine how many electrons can have the following set of quantum numbers?

(i) $n = 5; l = 3$

In a single atom, determine how many electrons can have the following set of quantum numbers?

(i) $n = 3; l = 2; m_s = +1/2$

**Question 4** **8 points**

Rank the following from 1 to 4 in **increasing** order. Provide the ranking in the boxes provided.

Note: Atomic number (Z) are provided for relevant atoms

a) Increasing (lowest to highest) order of *ionization energy*: (1= lowest; 4 = highest)

Mg

Mg: Z = 12

Rb

Rb: Z = 37

Cs

Cs: Z = 55

K

K : Z = 19

b) Increasing (lowest to highest) order of *bond angle*: (1= lowest; 4 = highest)

H₂OCCl₄XeF₄

HCN

c) Increasing (smallest to largest) order of *ionic size* (1= smallest; 4 = largest)

S²⁻

S: Z = 16

O²⁻

O: Z = 8

Be²⁺

Be: Z = 4

Na⁺

Na: Z = 11

d) Increasing (lowest to highest) order of *bond energy* (1= lowest; 4 = highest)

P-F

P-Cl

P-I

P-Br

**Question 5** **12 points**

a. 9 points For the molecules HClO_4 and TeF_6 , draw the most stable *Lewis structure* showing appropriate shape and *dashed/wedged bonds*, based on VSEPR. Determine the **molecular geometry** around the central atom. For HClO_4 , label the *most polar bond*.

HClO_4 (there are no O-O bonds)

Label the most polar bond

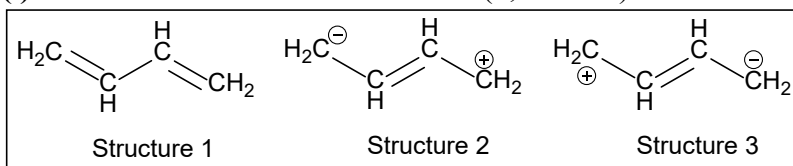
Molecular Geometry around Cl

TeF_6

Molecular Geometry around Te

b. 3 points (Note: Lone pairs are not shown in structures below)

(i) For the three *resonance* structures (1, 2 and 3) shown below, answer the following question:



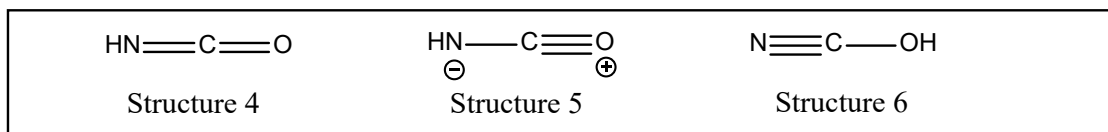
(I) Circle the *most contributing* resonance structure.

Structure 1

Structure 2

Structure 3

(ii) For the three structures given (4, 5, and 6) below, answer the following questions:



(I) Circle all the structures that represent *resonance structures* of each other:

Structure 4

Structure 5

Structure 6

(II) Among the resonance structures, circle the *most contributing* resonance structure.

Structure 4

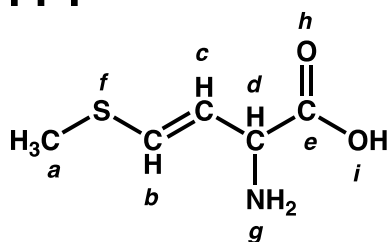
Structure 5

Structure 6

**Question 6** **6 points**

Using the data given below, calculate the **lattice energy** for SrO(s).

	ΔH° (kJ/mol)
$\text{Sr}(s) \longrightarrow \text{Sr}(g)$	+180
$\text{Sr}(g) \longrightarrow \text{Sr}^{2+}(g) + 2e^-$	+1468
$\text{O}_2(g) \longrightarrow 2\text{O}(g)$	+500
$\text{O}(g) + 2e^- \longrightarrow \text{O}^{2-}(g)$	+600
$\text{Sr}(s) + \frac{1}{2}\text{O}_2(g) \longrightarrow \text{SrO}(s)$	-550

**Question 7 10 points**

Consider the molecule given on the left, and answer questions (i) to (vi) using chemical bonding theories that we have learnt in class (valence bond theory, hybridization).

Note that the lone pairs are not shown in the structure:

i. Consider the carbon atoms marked in as ***a***, ***b***, ***c***, ***d***, and ***e*** in the molecule, and determine **hybridization** at each of the carbon centers. (2.5 points)

Atom	Hybridization
Carbon (<i>a</i>)	
Carbon (<i>b</i>)	
Carbon (<i>c</i>)	
Carbon (<i>d</i>)	
Carbon (<i>e</i>)	

ii. Now, consider the sulfur atom (S) marked as ***f***, nitrogen (N) atom (***g***), and oxygen(O) atoms (***h*** and ***i***), and determine their **hybridization**. (2 points)

Atom	Hybridization
Sulfur (<i>f</i>)	
Nitrogen (<i>g</i>)	
Oxygen (<i>h</i>)	
Oxygen (<i>i</i>)	

iii. **Describe** all the bonds formed by the carbon atoms ***a*** and ***b***, using sigma and pi designation as appropriate. (2 points)

Atom	Bonds formed (use sigma and pi as appropriate)
Carbon (<i>a</i>)	
Carbon (<i>b</i>)	

iv. How many total sp^3 -s sigma bonds are in the molecule? (1 point)

v. Determine **electronic** and **molecular geometry** at carbon (C) labeled ***e*** and sulfur (S) labeled ***f*** centres. (2 points)

Atom	Electronic Geometry	Molecular Geometry
Carbon (<i>e</i>)		
Sulfur (<i>f</i>)		

vi. Circle the value from the choices given below, that describes the **C-S-C bond angle at sulfur** (S) marked ***f***: (0.5 points)

$>90^\circ$ to $<109.5^\circ$ 109.5° >109.5 to $<120^\circ$ 120° $>120^\circ$ to $<180^\circ$

**Question 8 12 points**

Using the **valence shell orbitals only** draw a **complete** molecular orbital (MO) diagram (with no s-p mixing) for the molecule **PS**, in the space provided below. **(6 points)**

Assume that 3s and 3p atomic orbitals will form molecular orbitals like those formed by the combination of 2s and 2p atomic orbitals that we learnt in class, and there is no s-p mixing.

Show **energy arrow**, **atomic orbitals** involved, **atom labels**, the **molecular orbitals** that are formed, **filled electrons** in the MO diagram etc. Make sure to label all the atomic and molecular orbitals.

Using the MO diagram above, answer the following questions:

- (i) Which one of the following will have the highest bond order? Circle the correct choice
A) PS^{1-} B) PS C) PS^{1+}
- (ii) Which one of the following will have the longest bond length? Circle the correct choice
A) PS^{1-} B) PS C) PS^{1+}
- (iii) Which of the following will be diamagnetic? Circle the correct choice.
A) PS^{1-} B) PS C) PS^{1+}
- (iv) Draw Lewis dot structure of the molecule with the shortest bond length (based on the MO diagram) from among the three (PS^{1-} , PS, PS^{1+}), and indicate the bond order calculated from the Lewis dot structure. **(3 points)**

Lewis Dot Structure

Bond Order



7A3149CF-F74C-480A-9938-245BFA0166DC

midterm-exam-93388

#394 Page 10 of 10

SCRAP PAGE

**CHEMISTRY 110 –Midterm Exam****Thursday November 7, 2024, 6:30 – 8:30 pm**

Examiners: Pallavi Sirjoosingh and Ashok Kakkar

First name (please write as legibly as possible within the boxes)

--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--

Last name

--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--

Student ID number

--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--

This booklet has 8 questions on 10 pages (The last page can be used as a scrap page).

The staple must NOT be removed from the question booklet. Your intact question booklet must be returned at the end of the exam.

A separate booklet contains a periodic table, and datasheet.

Write your answers directly under the questions in the space provided. Do not write on the QR codes or near the top edges (as it may be cut off).

Please write **legibly and dark enough** for the scan to be readable. For students who use English as a second language, dictionaries are permitted. Calculators are allowed.

DO NOT ASK QUESTIONS OF THE INVIGILATORS about interpretations of the exam questions. If you have a concern about a question, please summarize it at the top of this page. All concerns will be evaluated after the exam. GOOD LUCK!

**Question 1 8 points** (*Assume all transitions are possible*)

a. 6 points Calculate the **energy (in kJ/mol)** required to ionize (completely remove) the electron from the **n=3** level of **He⁺ ion** (He: Helium). Show your work including equations. (3 sig. figs.)

b. 2 points

(I) Select the largest (longest) wavelength from the three choices below:

- i. Wavelength of photon to *completely remove* an electron from *1s orbital of Hydrogen atom*
- ii. Wavelength of photon to *completely remove* an electron from *1s orbital of Boron atom (B)*
- iii. Wavelength of photon to *completely remove* an electron from *1s orbital of Boron cation (B⁺)*

i or ii or iii

(II) Select the smallest (shortest) wavelength from the three choices below (*for H atom*):

- i. Wavelength of photon *emitted* for an electron transitioning from *2p to 1s orbital*
- ii. Wavelength of photon *absorbed* for an electron transitioning from *2s to 4p orbital*
- iii. Wavelength of photon *absorbed* for an electron transitioning from *1s to 3p orbital*

i or ii or iii



Question 2 8 points



A Chem110 student plots the graph shown on the left, from data obtained for the kinetic energy (eV) of the ejected electron versus frequency (s^{-1}) of the incident light, for an unknown metal. ($1\text{eV} = 1.6022 \times 10^{-19} \text{ J}$)

Based on the plot, answer the questions **a to c** below:

a. 1 point On the plot above, label the **threshold frequency**.

b. 2 points What is the value of the work potential (in J) of the unknown metal?

c. 3 points What is the *longest* wavelength (in nm) required to eject an electron from this metal's surface? Show your work including equation(s) used. (3 sig figs)

d. 2 points In the space provided below, plot the following

(Assume frequency is *above* threshold frequency):

- Kinetic energy* of an ejected electron versus the *intensity* of the incident light (left)
- Number of electrons* ejected versus the *frequency* of the incident light (right)



**Question 3** **6 points****a. 2 points**

In the ground state of Ge (Germanium; $Z = 32$) atom, **name** the following orbitals:

i. Orbital containing the electron that requires the maximum energy to be removed.

ii. Orbital containing the electron that experiences the minimum effective nuclear charge.

b. 2 points

An unknown anion (A^-) has the following *excited* electronic configuration: $[Ar] 4s^2 3d^3 4p^1$

(Ar: Argon; $Z = 18$)

Give the **condensed ground state electronic configuration** of the atom, **A** and the cation, **A^{2+}** .
Show your work. (Assume all electronic transitions are possible)

Electronic Configuration of A:

Electronic Configuration of A^{2+} :

c. 2 points

In a single atom, determine how many electrons can have the following set of quantum numbers?

(i) $n = 5; l = 3$

In a single atom, determine how many electrons can have the following set of quantum numbers?

(i) $n = 3; l = 2; m_s = +1/2$

**Question 4** **8 points**

Rank the following from 1 to 4 in **increasing** order. Provide the ranking in the boxes provided.

Note: Atomic number (Z) are provided for relevant atoms

a) Increasing (lowest to highest) order of *ionization energy*: (1= lowest; 4 = highest)

Mg

Mg: Z = 12

Rb

Rb: Z = 37

Cs

Cs: Z = 55

K

K : Z = 19

b) Increasing (lowest to highest) order of *bond angle*: (1= lowest; 4 = highest)

H₂OCCl₄XeF₄

HCN

c) Increasing (smallest to largest) order of *ionic size* (1= smallest; 4 = largest)

S²⁻

S: Z = 16

O²⁻

O: Z = 8

Be²⁺

Be: Z = 4

Na⁺

Na: Z = 11

d) Increasing (lowest to highest) order of *bond energy* (1= lowest; 4 = highest)

P-F

P-Cl

P-I

P-Br

**Question 5** **12 points**

a. 9 points For the molecules HClO_4 and TeF_6 , draw the most stable *Lewis structure* showing appropriate shape and *dashed/wedged bonds*, based on VSEPR. Determine the **molecular geometry** around the central atom. For HClO_4 , label the *most polar bond*.



b. 3 points (Note: Lone pairs are not shown in structures below)

(i) For the three *resonance* structures (1, 2 and 3) shown below, answer the following question:



(I) Circle the *most contributing* resonance structure.

Structure 1

Structure 2

Structure 3

(ii) For the three structures given (4, 5, and 6) below, answer the following questions:



(I) Circle all the structures that represent *resonance structures* of each other:

Structure 4

Structure 5

Structure 6

(II) Among the resonance structures, circle the *most contributing* resonance structure.

Structure 4

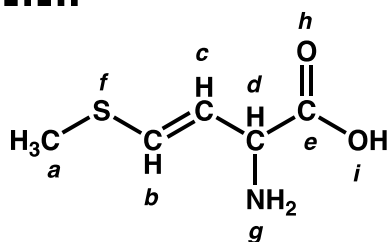
Structure 5

Structure 6

**Question 6** **6 points**

Using the data given below, calculate the **lattice energy** for SrO(s).

	ΔH° (kJ/mol)
$\text{Sr}(s) \longrightarrow \text{Sr}(g)$	+180
$\text{Sr}(g) \longrightarrow \text{Sr}^{2+}(g) + 2e^-$	+1468
$\text{O}_2(g) \longrightarrow 2\text{O}(g)$	+500
$\text{O}(g) + 2e^- \longrightarrow \text{O}^{2-}(g)$	+600
$\text{Sr}(s) + \frac{1}{2}\text{O}_2(g) \longrightarrow \text{SrO}(s)$	-550

**Question 7 10 points**

Consider the molecule given on the left, and answer questions (i) to (vi) using chemical bonding theories that we have learnt in class (valence bond theory, hybridization).

Note that the lone pairs are not shown in the structure:

i. Consider the carbon atoms marked in as ***a***, ***b***, ***c***, ***d***, and ***e*** in the molecule, and determine **hybridization** at each of the carbon centers. (2.5 points)

Atom	Hybridization
Carbon (<i>a</i>)	
Carbon (<i>b</i>)	
Carbon (<i>c</i>)	
Carbon (<i>d</i>)	
Carbon (<i>e</i>)	

ii. Now, consider the sulfur atom (S) marked as ***f***, nitrogen (N) atom (***g***), and oxygen(O) atoms (***h*** and ***i***), and determine their **hybridization**. (2 points)

Atom	Hybridization
Sulfur (<i>f</i>)	
Nitrogen (<i>g</i>)	
Oxygen (<i>h</i>)	
Oxygen (<i>i</i>)	

iii. **Describe** all the bonds formed by the carbon atoms ***a*** and ***b***, using sigma and pi designation as appropriate. (2 points)

Atom	Bonds formed (use sigma and pi as appropriate)
Carbon (<i>a</i>)	
Carbon (<i>b</i>)	

iv. How many total sp^3 -s sigma bonds are in the molecule? (1 point)

v. Determine **electronic** and **molecular geometry** at carbon (C) labeled ***e*** and sulfur (S) labeled ***f*** centres. (2 points)

Atom	Electronic Geometry	Molecular Geometry
Carbon (<i>e</i>)		
Sulfur (<i>f</i>)		

vi. Circle the value from the choices given below, that describes the **C-S-C bond angle at sulfur** (S) marked ***f***: (0.5 points)

>90° to <109.5° 109.5° >109.5 to <120° 120° >120° to <180°


Question 8 12 points

Using the **valence shell orbitals only** draw a **complete** molecular orbital (MO) diagram (with no s-p mixing) for the molecule **PS**, in the space provided below. **(6 points)**

Assume that 3s and 3p atomic orbitals will form molecular orbitals like those formed by the combination of 2s and 2p atomic orbitals that we learnt in class, and there is no s-p mixing.

Show **energy arrow**, **atomic orbitals** involved, **atom labels**, the **molecular orbitals** that are formed, **filled electrons** in the MO diagram etc. Make sure to label all the atomic and molecular orbitals.

Using the MO diagram above, answer the following questions:

- (i) Which one of the following will have the highest bond order? Circle the correct choice
 A) PS^{1-} B) PS C) PS^{1+}
- (ii) Which one of the following will have the longest bond length? Circle the correct choice
 A) PS^{1-} B) PS C) PS^{1+}
- (iii) Which of the following will be diamagnetic? Circle the correct choice.
 A) PS^{1-} B) PS C) PS^{1+}
- (iv) Draw Lewis dot structure of the molecule with the shortest bond length (based on the MO diagram) from among the three (PS^{1-} , PS, PS^{1+}), and indicate the bond order calculated from the Lewis dot structure. **(3 points)**

Lewis Dot Structure

Bond Order



E7B00F69-4658-407D-9BFE-D1D7F66974CA

midterm-exam-93388

#395 Page 10 of 10

SCRAP PAGE

**CHEMISTRY 110 –Midterm Exam****Thursday November 7, 2024, 6:30 – 8:30 pm**

Examiners: Pallavi Sirjoosingh and Ashok Kakkar

First name (please write as legibly as possible within the boxes)

--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--

Last name

--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--

Student ID number

--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--

This booklet has 8 questions on 10 pages (The last page can be used as a scrap page).

The staple must NOT be removed from the question booklet. Your intact question booklet must be returned at the end of the exam.

A separate booklet contains a periodic table, and datasheet.

Write your answers directly under the questions in the space provided. Do not write on the QR codes or near the top edges (as it may be cut off).

Please write **legibly and dark enough** for the scan to be readable. For students who use English as a second language, dictionaries are permitted. Calculators are allowed.

DO NOT ASK QUESTIONS OF THE INVIGILATORS about interpretations of the exam questions. If you have a concern about a question, please summarize it at the top of this page. All concerns will be evaluated after the exam. GOOD LUCK!

**Question 1 8 points** (*Assume all transitions are possible*)

a. 6 points Calculate the **energy (in kJ/mol)** required to ionize (completely remove) the electron from the **n=3** level of **He⁺ ion** (He: Helium). Show your work including equations. (3 sig. figs.)

b. 2 points

(I) Select the largest (longest) wavelength from the three choices below:

- i. Wavelength of photon to *completely remove* an electron from *1s orbital of Hydrogen atom*
- ii. Wavelength of photon to *completely remove* an electron from *1s orbital of Boron atom (B)*
- iii. Wavelength of photon to *completely remove* an electron from *1s orbital of Boron cation (B⁺)*

i or ii or iii

(II) Select the smallest (shortest) wavelength from the three choices below (*for H atom*):

- i. Wavelength of photon *emitted* for an electron transitioning from *2p to 1s orbital*
- ii. Wavelength of photon *absorbed* for an electron transitioning from *2s to 4p orbital*
- iii. Wavelength of photon *absorbed* for an electron transitioning from *1s to 3p orbital*

i or ii or iii



Question 2 8 points



A Chem110 student plots the graph shown on the left, from data obtained for the kinetic energy (eV) of the ejected electron versus frequency (s^{-1}) of the incident light, for an unknown metal. ($1\text{eV} = 1.6022 \times 10^{-19} \text{ J}$)

Based on the plot, answer the questions **a to c** below:

a. 1 point On the plot above, label the **threshold frequency**.

b. 2 points What is the value of the work potential (in J) of the unknown metal?

c. 3 points What is the *longest* wavelength (in nm) required to eject an electron from this metal's surface? Show your work including equation(s) used. (3 sig figs)

d. 2 points In the space provided below, plot the following

(Assume frequency is *above* threshold frequency):

- Kinetic energy* of an ejected electron versus the *intensity* of the incident light (left)
- Number of electrons* ejected versus the *frequency* of the incident light (right)



**Question 3** **6 points****a. 2 points**

In the ground state of Ge (Germanium; $Z = 32$) atom, **name** the following orbitals:

i. Orbital containing the electron that requires the maximum energy to be removed.

ii. Orbital containing the electron that experiences the minimum effective nuclear charge.

b. 2 points

An unknown anion (A^-) has the following *excited* electronic configuration: $[Ar] 4s^2 3d^3 4p^1$

(Ar: Argon; $Z = 18$)

Give the **condensed ground state electronic configuration** of the atom, **A** and the cation, **A^{2+}** .
Show your work. (Assume all electronic transitions are possible)

Electronic Configuration of A:

Electronic Configuration of A^{2+} :

c. 2 points

In a single atom, determine how many electrons can have the following set of quantum numbers?

(i) $n = 5; l = 3$

In a single atom, determine how many electrons can have the following set of quantum numbers?

(i) $n = 3; l = 2; m_s = +1/2$

**Question 4** **8 points**

Rank the following from 1 to 4 in **increasing** order. Provide the ranking in the boxes provided.

Note: Atomic number (Z) are provided for relevant atoms

a) Increasing (lowest to highest) order of *ionization energy*: (1= lowest; 4 = highest)

Mg

Mg: Z = 12

Rb

Rb: Z = 37

Cs

Cs: Z = 55

K

K : Z = 19

b) Increasing (lowest to highest) order of *bond angle*: (1= lowest; 4 = highest)

H₂OCCl₄XeF₄

HCN

c) Increasing (smallest to largest) order of *ionic size* (1= smallest; 4 = largest)

S²⁻

S: Z = 16

O²⁻

O: Z = 8

Be²⁺

Be: Z = 4

Na⁺

Na: Z = 11

d) Increasing (lowest to highest) order of *bond energy* (1= lowest; 4 = highest)

P-F

P-Cl

P-I

P-Br

**Question 5** **12 points**

a. 9 points For the molecules HClO_4 and TeF_6 , draw the most stable *Lewis structure* showing appropriate shape and *dashed/wedged bonds*, based on VSEPR. Determine the **molecular geometry** around the central atom. For HClO_4 , label the *most polar bond*.



b. 3 points (Note: Lone pairs are not shown in structures below)

(i) For the three *resonance* structures (1, 2 and 3) shown below, answer the following question:



(I) Circle the *most contributing* resonance structure.

Structure 1

Structure 2

Structure 3

(ii) For the three structures given (4, 5, and 6) below, answer the following questions:



(I) Circle all the structures that represent *resonance structures* of each other:

Structure 4

Structure 5

Structure 6

(II) Among the resonance structures, circle the *most contributing* resonance structure.

Structure 4

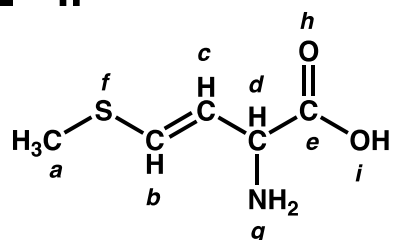
Structure 5

Structure 6

**Question 6** **6 points**

Using the data given below, calculate the **lattice energy** for SrO(s).

	ΔH° (kJ/mol)
$\text{Sr}(s) \longrightarrow \text{Sr}(g)$	+180
$\text{Sr}(g) \longrightarrow \text{Sr}^{2+}(g) + 2e^-$	+1468
$\text{O}_2(g) \longrightarrow 2\text{O}(g)$	+500
$\text{O}(g) + 2e^- \longrightarrow \text{O}^{2-}(g)$	+600
$\text{Sr}(s) + \frac{1}{2}\text{O}_2(g) \longrightarrow \text{SrO}(s)$	-550

**Question 7 10 points**

Consider the molecule given on the left, and answer questions (i) to (vi) using chemical bonding theories that we have learnt in class (valence bond theory, hybridization).

Note that the lone pairs are not shown in the structure:

i. Consider the carbon atoms marked in as ***a***, ***b***, ***c***, ***d***, and ***e*** in the molecule, and determine **hybridization** at each of the carbon centers. (2.5 points)

Atom	Hybridization
Carbon (<i>a</i>)	
Carbon (<i>b</i>)	
Carbon (<i>c</i>)	
Carbon (<i>d</i>)	
Carbon (<i>e</i>)	

ii. Now, consider the sulfur atom (S) marked as ***f***, nitrogen (N) atom (***g***), and oxygen(O) atoms (***h*** and ***i***), and determine their **hybridization**. (2 points)

Atom	Hybridization
Sulfur (<i>f</i>)	
Nitrogen (<i>g</i>)	
Oxygen (<i>h</i>)	
Oxygen (<i>i</i>)	

iii. **Describe** all the bonds formed by the carbon atoms ***a*** and ***b***, using sigma and pi designation as appropriate. (2 points)

Atom	Bonds formed (use sigma and pi as appropriate)
Carbon (<i>a</i>)	
Carbon (<i>b</i>)	

iv. How many total sp^3 -s sigma bonds are in the molecule? (1 point)

v. Determine **electronic** and **molecular geometry** at carbon (C) labeled ***e*** and sulfur (S) labeled ***f*** centres. (2 points)

Atom	Electronic Geometry	Molecular Geometry
Carbon (<i>e</i>)		
Sulfur (<i>f</i>)		

vi. Circle the value from the choices given below, that describes the **C-S-C bond angle at sulfur** (S) marked ***f***: (0.5 points)

$>90^\circ$ to $<109.5^\circ$ 109.5° >109.5 to $<120^\circ$ 120° $>120^\circ$ to $<180^\circ$


Question 8 12 points

Using the **valence shell orbitals only** draw a **complete** molecular orbital (MO) diagram (with no s-p mixing) for the molecule **PS**, in the space provided below. **(6 points)**

Assume that 3s and 3p atomic orbitals will form molecular orbitals like those formed by the combination of 2s and 2p atomic orbitals that we learnt in class, and there is no s-p mixing.

Show **energy arrow**, **atomic orbitals** involved, **atom labels**, the **molecular orbitals** that are formed, **filled electrons** in the MO diagram etc. Make sure to label all the atomic and molecular orbitals.

Using the MO diagram above, answer the following questions:

- (i) Which one of the following will have the highest bond order? Circle the correct choice
 A) PS^{1-} B) PS C) PS^{1+}
- (ii) Which one of the following will have the longest bond length? Circle the correct choice
 A) PS^{1-} B) PS C) PS^{1+}
- (iii) Which of the following will be diamagnetic? Circle the correct choice.
 A) PS^{1-} B) PS C) PS^{1+}
- (iv) Draw Lewis dot structure of the molecule with the shortest bond length (based on the MO diagram) from among the three (PS^{1-} , PS, PS^{1+}), and indicate the bond order calculated from the Lewis dot structure. **(3 points)**

Lewis Dot Structure

Bond Order



C1A893AE-4FAC-47A7-AF81-851F63CDC260

midterm-exam-93388

#396 Page 10 of 10

SCRAP PAGE

**CHEMISTRY 110 –Midterm Exam****Thursday November 7, 2024, 6:30 – 8:30 pm**

Examiners: Pallavi Sirjoosingh and Ashok Kakkar

First name (please write as legibly as possible within the boxes)

--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--

Last name

--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--

Student ID number

--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--

This booklet has 8 questions on 10 pages (The last page can be used as a scrap page).

The staple must NOT be removed from the question booklet. Your intact question booklet must be returned at the end of the exam.

A separate booklet contains a periodic table, and datasheet.

Write your answers directly under the questions in the space provided. Do not write on the QR codes or near the top edges (as it may be cut off).

Please write **legibly and dark enough** for the scan to be readable. For students who use English as a second language, dictionaries are permitted. Calculators are allowed.

DO NOT ASK QUESTIONS OF THE INVIGILATORS about interpretations of the exam questions. If you have a concern about a question, please summarize it at the top of this page. All concerns will be evaluated after the exam. GOOD LUCK!

**Question 1 8 points** (*Assume all transitions are possible*)

a. 6 points Calculate the **energy (in kJ/mol)** required to ionize (completely remove) the electron from the **n=3** level of **He⁺ ion** (He: Helium). Show your work including equations. (3 sig. figs.)

b. 2 points

(I) Select the largest (longest) wavelength from the three choices below:

- i. Wavelength of photon to *completely remove* an electron from *1s orbital of Hydrogen atom*
- ii. Wavelength of photon to *completely remove* an electron from *1s orbital of Boron atom (B)*
- iii. Wavelength of photon to *completely remove* an electron from *1s orbital of Boron cation (B⁺)*

i or ii or iii

(II) Select the smallest (shortest) wavelength from the three choices below (*for H atom*):

- i. Wavelength of photon *emitted* for an electron transitioning from *2p to 1s orbital*
- ii. Wavelength of photon *absorbed* for an electron transitioning from *2s to 4p orbital*
- iii. Wavelength of photon *absorbed* for an electron transitioning from *1s to 3p orbital*

i or ii or iii



Question 2 8 points



A Chem110 student plots the graph shown on the left, from data obtained for the kinetic energy (eV) of the ejected electron versus frequency (s^{-1}) of the incident light, for an unknown metal. ($1\text{eV} = 1.6022 \times 10^{-19} \text{ J}$)

Based on the plot, answer the questions **a to c** below:

a. 1 point On the plot above, label the **threshold frequency**.

b. 2 points What is the value of the work potential (in J) of the unknown metal?

c. 3 points What is the *longest* wavelength (in nm) required to eject an electron from this metal's surface? Show your work including equation(s) used. (3 sig figs)

d. 2 points In the space provided below, plot the following

(Assume frequency is *above* threshold frequency):

- i. *Kinetic energy* of an ejected electron versus the *intensity* of the incident light (left)
- ii. *Number of electrons* ejected versus the *frequency* of the incident light (right)



**Question 3** **6 points****a. 2 points**

In the ground state of Ge (Germanium; $Z = 32$) atom, **name** the following orbitals:

i. Orbital containing the electron that requires the maximum energy to be removed.

ii. Orbital containing the electron that experiences the minimum effective nuclear charge.

b. 2 points

An unknown anion (A^-) has the following *excited* electronic configuration: $[Ar] 4s^2 3d^3 4p^1$

(Ar: Argon; $Z = 18$)

Give the **condensed ground state electronic configuration** of the atom, **A** and the cation, **A^{2+}** .
Show your work. (Assume all electronic transitions are possible)

Electronic Configuration of A:

Electronic Configuration of A^{2+} :

c. 2 points

In a single atom, determine how many electrons can have the following set of quantum numbers?

(i) $n = 5; l = 3$

In a single atom, determine how many electrons can have the following set of quantum numbers?

(i) $n = 3; l = 2; m_s = +1/2$

**Question 4** **8 points**

Rank the following from 1 to 4 in **increasing** order. Provide the ranking in the boxes provided.

Note: Atomic number (Z) are provided for relevant atoms

a) Increasing (lowest to highest) order of *ionization energy*: (1= lowest; 4 = highest)

Mg

Mg: Z = 12

Rb

Rb: Z = 37

Cs

Cs: Z = 55

K

K : Z = 19

b) Increasing (lowest to highest) order of *bond angle*: (1= lowest; 4 = highest)

H₂OCCl₄XeF₄

HCN

c) Increasing (smallest to largest) order of *ionic size* (1= smallest; 4 = largest)

S²⁻

S: Z = 16

O²⁻

O: Z = 8

Be²⁺

Be: Z = 4

Na⁺

Na: Z = 11

d) Increasing (lowest to highest) order of *bond energy* (1= lowest; 4 = highest)

P-F

P-Cl

P-I

P-Br

**Question 5** **12 points**

a. 9 points For the molecules HClO_4 and TeF_6 , draw the most stable *Lewis structure* showing appropriate shape and *dashed/wedged bonds*, based on VSEPR. Determine the **molecular geometry** around the central atom. For HClO_4 , label the *most polar bond*.



b. 3 points (Note: Lone pairs are not shown in structures below)

(i) For the three *resonance* structures (1, 2 and 3) shown below, answer the following question:



(I) Circle the *most contributing* resonance structure.

Structure 1

Structure 2

Structure 3

(ii) For the three structures given (4, 5, and 6) below, answer the following questions:



(I) Circle all the structures that represent *resonance structures* of each other:

Structure 4

Structure 5

Structure 6

(II) Among the resonance structures, circle the *most contributing* resonance structure.

Structure 4

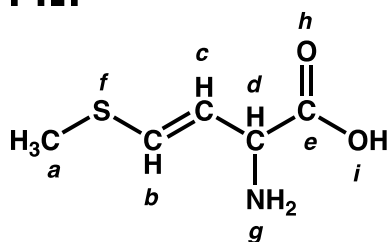
Structure 5

Structure 6

**Question 6** **6 points**

Using the data given below, calculate the **lattice energy** for SrO(s).

	ΔH° (kJ/mol)
$\text{Sr}(s) \longrightarrow \text{Sr}(g)$	+180
$\text{Sr}(g) \longrightarrow \text{Sr}^{2+}(g) + 2e^-$	+1468
$\text{O}_2(g) \longrightarrow 2\text{O}(g)$	+500
$\text{O}(g) + 2e^- \longrightarrow \text{O}^{2-}(g)$	+600
$\text{Sr}(s) + \frac{1}{2}\text{O}_2(g) \longrightarrow \text{SrO}(s)$	-550

**Question 7 10 points**

Consider the molecule given on the left, and answer questions (i) to (vi) using chemical bonding theories that we have learnt in class (valence bond theory, hybridization).

Note that the lone pairs are not shown in the structure:

i. Consider the carbon atoms marked in as ***a***, ***b***, ***c***, ***d***, and ***e*** in the molecule, and determine **hybridization** at each of the carbon centers. (2.5 points)

Atom	Hybridization
Carbon (<i>a</i>)	
Carbon (<i>b</i>)	
Carbon (<i>c</i>)	
Carbon (<i>d</i>)	
Carbon (<i>e</i>)	

ii. Now, consider the sulfur atom (S) marked as ***f***, nitrogen (N) atom (***g***), and oxygen(O) atoms (***h*** and ***i***), and determine their **hybridization**. (2 points)

Atom	Hybridization
Sulfur (<i>f</i>)	
Nitrogen (<i>g</i>)	
Oxygen (<i>h</i>)	
Oxygen (<i>i</i>)	

iii. **Describe** all the bonds formed by the carbon atoms ***a*** and ***b***, using sigma and pi designation as appropriate. (2 points)

Atom	Bonds formed (use sigma and pi as appropriate)
Carbon (<i>a</i>)	
Carbon (<i>b</i>)	

iv. How many total sp^3 -s sigma bonds are in the molecule? (1 point)

v. Determine **electronic** and **molecular geometry** at carbon (C) labeled ***e*** and sulfur (S) labeled ***f*** centres. (2 points)

Atom	Electronic Geometry	Molecular Geometry
Carbon (<i>e</i>)		
Sulfur (<i>f</i>)		

vi. Circle the value from the choices given below, that describes the **C-S-C bond angle at sulfur** (S) marked ***f***: (0.5 points)

>90° to <109.5°

109.5°

>109.5 to <120°

120°

>120° to <180°


Question 8 12 points

Using the **valence shell orbitals only** draw a **complete** molecular orbital (MO) diagram (with no s-p mixing) for the molecule **PS**, in the space provided below. **(6 points)**

Assume that 3s and 3p atomic orbitals will form molecular orbitals like those formed by the combination of 2s and 2p atomic orbitals that we learnt in class, and there is no s-p mixing.

Show **energy arrow**, **atomic orbitals** involved, **atom labels**, the **molecular orbitals** that are formed, **filled electrons** in the MO diagram etc. Make sure to label all the atomic and molecular orbitals.

Using the MO diagram above, answer the following questions:

- (i) Which one of the following will have the highest bond order? Circle the correct choice
 A) PS^{1-} B) PS C) PS^{1+}
- (ii) Which one of the following will have the longest bond length? Circle the correct choice
 A) PS^{1-} B) PS C) PS^{1+}
- (iii) Which of the following will be diamagnetic? Circle the correct choice.
 A) PS^{1-} B) PS C) PS^{1+}
- (iv) Draw Lewis dot structure of the molecule with the shortest bond length (based on the MO diagram) from among the three (PS^{1-} , PS, PS^{1+}), and indicate the bond order calculated from the Lewis dot structure. **(3 points)**

Lewis Dot Structure

Bond Order



BD6FF632-9E5F-4DBE-97D9-617981FA33DD

midterm-exam-93388

#397 Page 10 of 10

SCRAP PAGE

**CHEMISTRY 110 –Midterm Exam****Thursday November 7, 2024, 6:30 – 8:30 pm****Examiners: Pallavi Sirjoosingh and Ashok Kakkar**

First name (please write as legibly as possible within the boxes)

--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--

Last name

--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--

Student ID number

--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--

This booklet has 8 questions on 10 pages (The last page can be used as a scrap page).

The staple must NOT be removed from the question booklet. Your intact question booklet must be returned at the end of the exam.

A separate booklet contains a periodic table, and datasheet.

Write your answers directly under the questions in the space provided. Do not write on the QR codes or near the top edges (as it may be cut off).

Please write **legibly and dark enough** for the scan to be readable. For students who use English as a second language, dictionaries are permitted. Calculators are allowed.

DO NOT ASK QUESTIONS OF THE INVIGILATORS about interpretations of the exam questions. If you have a concern about a question, please summarize it at the top of this page. All concerns will be evaluated after the exam. GOOD LUCK!

**Question 1 8 points** (*Assume all transitions are possible*)

a. 6 points Calculate the **energy (in kJ/mol)** required to ionize (completely remove) the electron from the **$n=3$** level of **He^+ ion** (He: Helium). Show your work including equations. (3 sig. figs.)

b. 2 points

(I) Select the largest (longest) wavelength from the three choices below:

- i. Wavelength of photon to *completely remove* an electron from *1s orbital of Hydrogen atom*
- ii. Wavelength of photon to *completely remove* an electron from *1s orbital of Boron atom (B)*
- iii. Wavelength of photon to *completely remove* an electron from *1s orbital of Boron cation (B^+)*

i or ii or iii

(II) Select the smallest (shortest) wavelength from the three choices below (*for H atom*):

- i. Wavelength of photon *emitted* for an electron transitioning from *2p to 1s orbital*
- ii. Wavelength of photon *absorbed* for an electron transitioning from *2s to 4p orbital*
- iii. Wavelength of photon *absorbed* for an electron transitioning from *1s to 3p orbital*

i or ii or iii



Question 2 8 points



A Chem110 student plots the graph shown on the left, from data obtained for the kinetic energy (eV) of the ejected electron versus frequency (s^{-1}) of the incident light, for an unknown metal. ($1\text{eV} = 1.6022 \times 10^{-19} \text{ J}$)

Based on the plot, answer the questions **a to c** below:

a. 1 point On the plot above, label the **threshold frequency**.

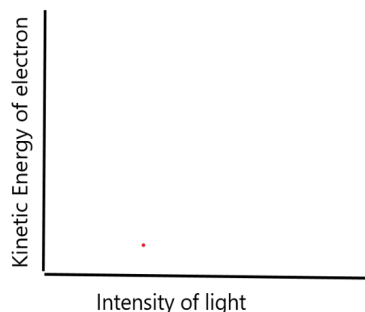
b. 2 points What is the value of the work potential (in J) of the unknown metal?

c. 3 points What is the *longest* wavelength (in nm) required to eject an electron from this metal's surface? Show your work including equation(s) used. (3 sig figs)

d. 2 points In the space provided below, plot the following

(Assume frequency is *above* threshold frequency):

- Kinetic energy* of an ejected electron versus the *intensity* of the incident light (left)
- Number of electrons* ejected versus the *frequency* of the incident light (right)



**Question 3** **6 points****a. 2 points**

In the ground state of Ge (Germanium; $Z = 32$) atom, **name** the following orbitals:

i. Orbital containing the electron that requires the maximum energy to be removed.

ii. Orbital containing the electron that experiences the minimum effective nuclear charge.

b. 2 points

An unknown anion (A^-) has the following *excited* electronic configuration: $[Ar] 4s^2 3d^3 4p^1$

(Ar: Argon; $Z = 18$)

Give the **condensed ground state electronic configuration** of the atom, **A** and the cation, **A^{2+}** .
Show your work. (Assume all electronic transitions are possible)

Electronic Configuration of A:

Electronic Configuration of A^{2+} :

c. 2 points

In a single atom, determine how many electrons can have the following set of quantum numbers?

(i) $n = 5; l = 3$

In a single atom, determine how many electrons can have the following set of quantum numbers?

(i) $n = 3; l = 2; m_s = +1/2$

**Question 4** **8 points**

Rank the following from 1 to 4 in **increasing** order. Provide the ranking in the boxes provided.

Note: Atomic number (Z) are provided for relevant atoms

a) Increasing (lowest to highest) order of *ionization energy*: (1= lowest; 4 = highest)

Mg

Mg: Z = 12

Rb

Rb: Z = 37

Cs

Cs: Z = 55

K

K : Z = 19

b) Increasing (lowest to highest) order of *bond angle*: (1= lowest; 4 = highest)

H₂OCCl₄XeF₄

HCN

c) Increasing (smallest to largest) order of *ionic size* (1= smallest; 4 = largest)

S²⁻

S: Z = 16

O²⁻

O: Z = 8

Be²⁺

Be: Z = 4

Na⁺

Na: Z = 11

d) Increasing (lowest to highest) order of *bond energy* (1= lowest; 4 = highest)

P-F

P-Cl

P-I

P-Br

**Question 5** **12 points**

a. 9 points For the molecules HClO_4 and TeF_6 , draw the most stable *Lewis structure* showing appropriate shape and *dashed/wedged bonds*, based on VSEPR. Determine the **molecular geometry** around the central atom. For HClO_4 , label the *most polar bond*.



b. 3 points (Note: Lone pairs are not shown in structures below)

(i) For the three *resonance* structures (1, 2 and 3) shown below, answer the following question:



(I) Circle the *most contributing* resonance structure.

Structure 1

Structure 2

Structure 3

(ii) For the three structures given (4, 5, and 6) below, answer the following questions:



(I) Circle all the structures that represent *resonance structures* of each other:

Structure 4

Structure 5

Structure 6

(II) Among the resonance structures, circle the *most contributing* resonance structure.

Structure 4

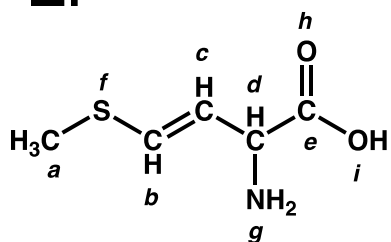
Structure 5

Structure 6

**Question 6** **6 points**

Using the data given below, calculate the **lattice energy** for SrO(s).

	ΔH° (kJ/mol)
$\text{Sr}(s) \longrightarrow \text{Sr}(g)$	+180
$\text{Sr}(g) \longrightarrow \text{Sr}^{2+}(g) + 2e^-$	+1468
$\text{O}_2(g) \longrightarrow 2\text{O}(g)$	+500
$\text{O}(g) + 2e^- \longrightarrow \text{O}^{2-}(g)$	+600
$\text{Sr}(s) + \frac{1}{2}\text{O}_2(g) \longrightarrow \text{SrO}(s)$	-550

**Question 7 10 points**

Consider the molecule given on the left, and answer questions (i) to (vi) using chemical bonding theories that we have learnt in class (valence bond theory, hybridization).

Note that the lone pairs are not shown in the structure:

i. Consider the carbon atoms marked in as ***a***, ***b***, ***c***, ***d***, and ***e*** in the molecule, and determine **hybridization** at each of the carbon centers. (2.5 points)

Atom	Hybridization
Carbon (<i>a</i>)	
Carbon (<i>b</i>)	
Carbon (<i>c</i>)	
Carbon (<i>d</i>)	
Carbon (<i>e</i>)	

ii. Now, consider the sulfur atom (S) marked as ***f***, nitrogen (N) atom (***g***), and oxygen(O) atoms (***h*** and ***i***), and determine their **hybridization**. (2 points)

Atom	Hybridization
Sulfur (<i>f</i>)	
Nitrogen (<i>g</i>)	
Oxygen (<i>h</i>)	
Oxygen (<i>i</i>)	

iii. **Describe** all the bonds formed by the carbon atoms ***a*** and ***b***, using sigma and pi designation as appropriate. (2 points)

Atom	Bonds formed (use sigma and pi as appropriate)
Carbon (<i>a</i>)	
Carbon (<i>b</i>)	

iv. How many total sp^3 -s sigma bonds are in the molecule? (1 point)

v. Determine **electronic** and **molecular geometry** at carbon (C) labeled ***e*** and sulfur (S) labeled ***f*** centres. (2 points)

Atom	Electronic Geometry	Molecular Geometry
Carbon (<i>e</i>)		
Sulfur (<i>f</i>)		

vi. Circle the value from the choices given below, that describes the **C-S-C bond angle at sulfur** (S) marked ***f***: (0.5 points)

>90° to <109.5° 109.5° >109.5 to <120° 120° >120° to <180°


Question 8 12 points

Using the **valence shell orbitals only** draw a **complete** molecular orbital (MO) diagram (with no s-p mixing) for the molecule **PS**, in the space provided below. **(6 points)**

Assume that 3s and 3p atomic orbitals will form molecular orbitals like those formed by the combination of 2s and 2p atomic orbitals that we learnt in class, and there is no s-p mixing.

Show **energy arrow**, **atomic orbitals** involved, **atom labels**, the **molecular orbitals** that are formed, **filled electrons** in the MO diagram etc. Make sure to label all the atomic and molecular orbitals.

Using the MO diagram above, answer the following questions:

- (i) Which one of the following will have the highest bond order? Circle the correct choice
 A) PS^{1-} B) PS C) PS^{1+}
- (ii) Which one of the following will have the longest bond length? Circle the correct choice
 A) PS^{1-} B) PS C) PS^{1+}
- (iii) Which of the following will be diamagnetic? Circle the correct choice.
 A) PS^{1-} B) PS C) PS^{1+}
- (iv) Draw Lewis dot structure of the molecule with the shortest bond length (based on the MO diagram) from among the three (PS^{1-} , PS, PS^{1+}), and indicate the bond order calculated from the Lewis dot structure. **(3 points)**

Lewis Dot Structure

Bond Order



994213AA-3D36-40E0-92B4-4D1A07032A92

midterm-exam-93388

#398 Page 10 of 10

SCRAP PAGE

**CHEMISTRY 110 –Midterm Exam****Thursday November 7, 2024, 6:30 – 8:30 pm****Examiners: Pallavi Sirjoosingh and Ashok Kakkar**

First name (please write as legibly as possible within the boxes)

--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--

Last name

--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--

Student ID number

--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--

This booklet has 8 questions on 10 pages (The last page can be used as a scrap page).

The staple must NOT be removed from the question booklet. Your intact question booklet must be returned at the end of the exam.

A separate booklet contains a periodic table, and datasheet.

Write your answers directly under the questions in the space provided. Do not write on the QR codes or near the top edges (as it may be cut off).

Please write **legibly and dark enough** for the scan to be readable. For students who use English as a second language, dictionaries are permitted. Calculators are allowed.

DO NOT ASK QUESTIONS OF THE INVIGILATORS about interpretations of the exam questions. If you have a concern about a question, please summarize it at the top of this page. All concerns will be evaluated after the exam. GOOD LUCK!

**Question 1 8 points** (*Assume all transitions are possible*)

a. 6 points Calculate the **energy (in kJ/mol)** required to ionize (completely remove) the electron from the **n=3** level of **He⁺ ion** (He: Helium). Show your work including equations. (3 sig. figs.)

b. 2 points

(I) Select the largest (longest) wavelength from the three choices below:

- i. Wavelength of photon to *completely remove* an electron from *1s orbital of Hydrogen atom*
- ii. Wavelength of photon to *completely remove* an electron from *1s orbital of Boron atom (B)*
- iii. Wavelength of photon to *completely remove* an electron from *1s orbital of Boron cation (B⁺)*

i or ii or iii

(II) Select the smallest (shortest) wavelength from the three choices below (*for H atom*):

- i. Wavelength of photon *emitted* for an electron transitioning from *2p to 1s orbital*
- ii. Wavelength of photon *absorbed* for an electron transitioning from *2s to 4p orbital*
- iii. Wavelength of photon *absorbed* for an electron transitioning from *1s to 3p orbital*

i or ii or iii



Question 2 8 points



A Chem110 student plots the graph shown on the left, from data obtained for the kinetic energy (eV) of the ejected electron versus frequency (s^{-1}) of the incident light, for an unknown metal. ($1\text{eV} = 1.6022 \times 10^{-19} \text{ J}$)

Based on the plot, answer the questions **a to c** below:

a. 1 point On the plot above, label the **threshold frequency**.

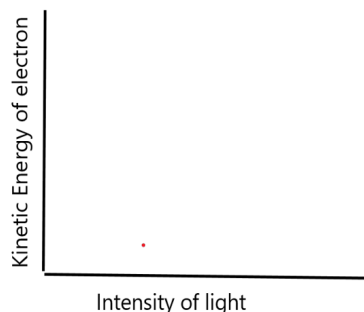
b. 2 points What is the value of the work potential (in J) of the unknown metal?

c. 3 points What is the *longest* wavelength (in nm) required to eject an electron from this metal's surface? Show your work including equation(s) used. (3 sig figs)

d. 2 points In the space provided below, plot the following

(Assume frequency is *above* threshold frequency):

- Kinetic energy* of an ejected electron versus the *intensity* of the incident light (left)
- Number of electrons* ejected versus the *frequency* of the incident light (right)



**Question 3** **6 points****a. 2 points**

In the ground state of Ge (Germanium; $Z = 32$) atom, **name** the following orbitals:

i. Orbital containing the electron that requires the maximum energy to be removed.

ii. Orbital containing the electron that experiences the minimum effective nuclear charge.

b. 2 points

An unknown anion (A^-) has the following *excited* electronic configuration: $[Ar] 4s^2 3d^3 4p^1$

(Ar: Argon; $Z = 18$)

Give the **condensed ground state electronic configuration** of the atom, **A** and the cation, **A^{2+}** .
Show your work. (Assume all electronic transitions are possible)

Electronic Configuration of A:

Electronic Configuration of A^{2+} :

c. 2 points

In a single atom, determine how many electrons can have the following set of quantum numbers?

(i) $n = 5; l = 3$

In a single atom, determine how many electrons can have the following set of quantum numbers?

(i) $n = 3; l = 2; m_s = +1/2$

**Question 4** **8 points**

Rank the following from 1 to 4 in **increasing** order. Provide the ranking in the boxes provided.

Note: Atomic number (Z) are provided for relevant atoms

a) Increasing (lowest to highest) order of *ionization energy*: (1= lowest; 4 = highest)

Mg

Mg: Z = 12

Rb

Rb: Z = 37

Cs

Cs: Z = 55

K

K : Z = 19

b) Increasing (lowest to highest) order of *bond angle*: (1= lowest; 4 = highest)

H₂OCCl₄XeF₄

HCN

c) Increasing (smallest to largest) order of *ionic size* (1= smallest; 4 = largest)

S²⁻

S: Z = 16

O²⁻

O: Z = 8

Be²⁺

Be: Z = 4

Na⁺

Na: Z = 11

d) Increasing (lowest to highest) order of *bond energy* (1= lowest; 4 = highest)

P-F

P-Cl

P-I

P-Br

**Question 5** **12 points**

a. 9 points For the molecules HClO_4 and TeF_6 , draw the most stable *Lewis structure* showing appropriate shape and *dashed/wedged bonds*, based on VSEPR. Determine the **molecular geometry** around the central atom. For HClO_4 , label the *most polar bond*.



b. 3 points (Note: Lone pairs are not shown in structures below)

(i) For the three *resonance* structures (1, 2 and 3) shown below, answer the following question:



(I) Circle the *most contributing* resonance structure.

Structure 1

Structure 2

Structure 3

(ii) For the three structures given (4, 5, and 6) below, answer the following questions:



(I) Circle all the structures that represent *resonance structures* of each other:

Structure 4

Structure 5

Structure 6

(II) Among the resonance structures, circle the *most contributing* resonance structure.

Structure 4

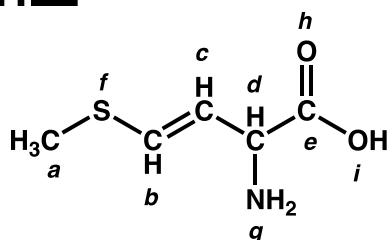
Structure 5

Structure 6

**Question 6** **6 points**

Using the data given below, calculate the **lattice energy** for SrO(s).

	ΔH° (kJ/mol)
$\text{Sr}(s) \longrightarrow \text{Sr}(g)$	+180
$\text{Sr}(g) \longrightarrow \text{Sr}^{2+}(g) + 2e^-$	+1468
$\text{O}_2(g) \longrightarrow 2\text{O}(g)$	+500
$\text{O}(g) + 2e^- \longrightarrow \text{O}^{2-}(g)$	+600
$\text{Sr}(s) + \frac{1}{2}\text{O}_2(g) \longrightarrow \text{SrO}(s)$	-550

**Question 7 10 points**

Consider the molecule given on the left, and answer questions (i) to (vi) using chemical bonding theories that we have learnt in class (valence bond theory, hybridization).

Note that the lone pairs are not shown in the structure:

i. Consider the carbon atoms marked in as ***a***, ***b***, ***c***, ***d***, and ***e*** in the molecule, and determine **hybridization** at each of the carbon centers. (2.5 points)

Atom	Hybridization
Carbon (<i>a</i>)	
Carbon (<i>b</i>)	
Carbon (<i>c</i>)	
Carbon (<i>d</i>)	
Carbon (<i>e</i>)	

ii. Now, consider the sulfur atom (S) marked as ***f***, nitrogen (N) atom (***g***), and oxygen(O) atoms (***h*** and ***i***), and determine their **hybridization**. (2 points)

Atom	Hybridization
Sulfur (<i>f</i>)	
Nitrogen (<i>g</i>)	
Oxygen (<i>h</i>)	
Oxygen (<i>i</i>)	

iii. **Describe** all the bonds formed by the carbon atoms ***a*** and ***b***, using sigma and pi designation as appropriate. (2 points)

Atom	Bonds formed (use sigma and pi as appropriate)
Carbon (<i>a</i>)	
Carbon (<i>b</i>)	

iv. How many total sp^3 -s sigma bonds are in the molecule? (1 point)

v. Determine **electronic** and **molecular geometry** at carbon (C) labeled ***e*** and sulfur (S) labeled ***f*** centres. (2 points)

Atom	Electronic Geometry	Molecular Geometry
Carbon (<i>e</i>)		
Sulfur (<i>f</i>)		

vi. Circle the value from the choices given below, that describes the **C-S-C bond angle at sulfur** (S) marked ***f***: (0.5 points)

$>90^\circ$ to $<109.5^\circ$ 109.5° >109.5 to $<120^\circ$ 120° $>120^\circ$ to $<180^\circ$


Question 8 12 points

Using the **valence shell orbitals only** draw a **complete** molecular orbital (MO) diagram (with no s-p mixing) for the molecule **PS**, in the space provided below. **(6 points)**

Assume that 3s and 3p atomic orbitals will form molecular orbitals like those formed by the combination of 2s and 2p atomic orbitals that we learnt in class, and there is no s-p mixing.

Show **energy arrow**, **atomic orbitals** involved, **atom labels**, the **molecular orbitals** that are formed, **filled electrons** in the MO diagram etc. Make sure to label all the atomic and molecular orbitals.

Using the MO diagram above, answer the following questions:

- (i) Which one of the following will have the highest bond order? Circle the correct choice
 A) PS^{1-} B) PS C) PS^{1+}
- (ii) Which one of the following will have the longest bond length? Circle the correct choice
 A) PS^{1-} B) PS C) PS^{1+}
- (iii) Which of the following will be diamagnetic? Circle the correct choice.
 A) PS^{1-} B) PS C) PS^{1+}
- (iv) Draw Lewis dot structure of the molecule with the shortest bond length (based on the MO diagram) from among the three (PS^{1-} , PS, PS^{1+}), and indicate the bond order calculated from the Lewis dot structure. **(3 points)**

Lewis Dot Structure

Bond Order



577E538E-5EBA-425E-A615-1396D4E2A514

midterm-exam-93388

#399 Page 10 of 10

SCRAP PAGE

**CHEMISTRY 110 –Midterm Exam****Thursday November 7, 2024, 6:30 – 8:30 pm**

Examiners: Pallavi Sirjoosingh and Ashok Kakkar

First name (please write as legibly as possible within the boxes)

--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--

Last name

--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--

Student ID number

--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--

This booklet has 8 questions on 10 pages (The last page can be used as a scrap page).

The staple must NOT be removed from the question booklet. Your intact question booklet must be returned at the end of the exam.

A separate booklet contains a periodic table, and datasheet.

Write your answers directly under the questions in the space provided. Do not write on the QR codes or near the top edges (as it may be cut off).

Please write **legibly and dark enough** for the scan to be readable. For students who use English as a second language, dictionaries are permitted. Calculators are allowed.

DO NOT ASK QUESTIONS OF THE INVIGILATORS about interpretations of the exam questions. If you have a concern about a question, please summarize it at the top of this page. All concerns will be evaluated after the exam. GOOD LUCK!

**Question 1 8 points** (*Assume all transitions are possible*)

a. 6 points Calculate the **energy (in kJ/mol)** required to ionize (completely remove) the electron from the **$n=3$** level of **He^+ ion** (He: Helium). Show your work including equations. (3 sig. figs.)

b. 2 points

(I) Select the largest (longest) wavelength from the three choices below:

- i. Wavelength of photon to *completely remove* an electron from *1s orbital of Hydrogen atom*
- ii. Wavelength of photon to *completely remove* an electron from *1s orbital of Boron atom (B)*
- iii. Wavelength of photon to *completely remove* an electron from *1s orbital of Boron cation (B^+)*

i or ii or iii

(II) Select the smallest (shortest) wavelength from the three choices below (*for H atom*):

- i. Wavelength of photon *emitted* for an electron transitioning from *2p to 1s orbital*
- ii. Wavelength of photon *absorbed* for an electron transitioning from *2s to 4p orbital*
- iii. Wavelength of photon *absorbed* for an electron transitioning from *1s to 3p orbital*

i or ii or iii



Question 2 8 points



A Chem110 student plots the graph shown on the left, from data obtained for the kinetic energy (eV) of the ejected electron versus frequency (s^{-1}) of the incident light, for an unknown metal. ($1\text{eV} = 1.6022 \times 10^{-19} \text{ J}$)

Based on the plot, answer the questions **a to c** below:

a. 1 point On the plot above, label the **threshold frequency**.

b. 2 points What is the value of the work potential (in J) of the unknown metal?

c. 3 points What is the *longest* wavelength (in nm) required to eject an electron from this metal's surface? Show your work including equation(s) used. (3 sig figs)

d. 2 points In the space provided below, plot the following

(Assume frequency is *above* threshold frequency):

- i. *Kinetic energy* of an ejected electron versus the *intensity* of the incident light (left)
- ii. *Number of electrons* ejected versus the *frequency* of the incident light (right)



**Question 3** **6 points****a. 2 points**

In the ground state of Ge (Germanium; $Z = 32$) atom, **name** the following orbitals:

i. Orbital containing the electron that requires the maximum energy to be removed.

ii. Orbital containing the electron that experiences the minimum effective nuclear charge.

b. 2 points

An unknown anion (A^-) has the following *excited* electronic configuration: $[Ar] 4s^2 3d^3 4p^1$

(Ar: Argon; $Z = 18$)

Give the **condensed ground state electronic configuration** of the atom, **A** and the cation, **A^{2+}** .
Show your work. (Assume all electronic transitions are possible)

Electronic Configuration of A:

Electronic Configuration of A^{2+} :

c. 2 points

In a single atom, determine how many electrons can have the following set of quantum numbers?

(i) $n = 5; l = 3$

In a single atom, determine how many electrons can have the following set of quantum numbers?

(i) $n = 3; l = 2; m_s = +1/2$

**Question 4** **8 points**

Rank the following from 1 to 4 in **increasing** order. Provide the ranking in the boxes provided.

Note: Atomic number (Z) are provided for relevant atoms

a) Increasing (lowest to highest) order of *ionization energy*: (1= lowest; 4 = highest)

Mg

Mg: Z = 12

Rb

Rb: Z = 37

Cs

Cs: Z = 55

K

K : Z = 19

b) Increasing (lowest to highest) order of *bond angle*: (1= lowest; 4 = highest)

H₂OCCl₄XeF₄

HCN

c) Increasing (smallest to largest) order of *ionic size* (1= smallest; 4 = largest)

S²⁻

S: Z = 16

O²⁻

O: Z = 8

Be²⁺

Be: Z = 4

Na⁺

Na: Z = 11

d) Increasing (lowest to highest) order of *bond energy* (1= lowest; 4 = highest)

P-F

P-Cl

P-I

P-Br

**Question 5** **12 points**

a. 9 points For the molecules HClO_4 and TeF_6 , draw the most stable *Lewis structure* showing appropriate shape and *dashed/wedged bonds*, based on VSEPR. Determine the **molecular geometry** around the central atom. For HClO_4 , label the *most polar bond*.

HClO_4 (there are no O-O bonds)

Label the most polar bond

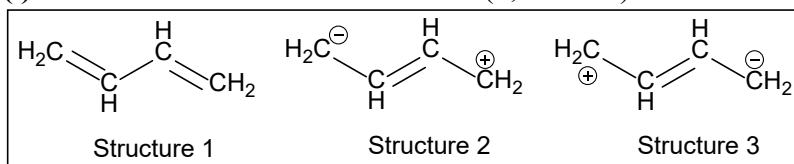
Molecular Geometry around Cl

TeF_6

Molecular Geometry around Te

b. 3 points (Note: Lone pairs are not shown in structures below)

(i) For the three *resonance* structures (1, 2 and 3) shown below, answer the following question:



(I) Circle the *most contributing* resonance structure.

Structure 1

Structure 2

Structure 3

(ii) For the three structures given (4, 5, and 6) below, answer the following questions:



(I) Circle all the structures that represent *resonance structures* of each other:

Structure 4

Structure 5

Structure 6

(II) Among the resonance structures, circle the *most contributing* resonance structure.

Structure 4

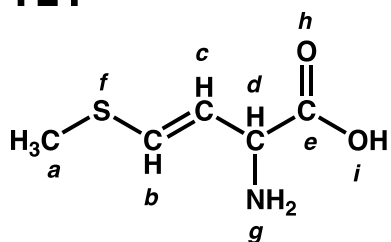
Structure 5

Structure 6

**Question 6** **6 points**

Using the data given below, calculate the **lattice energy** for SrO(s).

	ΔH° (kJ/mol)
$\text{Sr}(s) \longrightarrow \text{Sr}(g)$	+180
$\text{Sr}(g) \longrightarrow \text{Sr}^{2+}(g) + 2e^-$	+1468
$\text{O}_2(g) \longrightarrow 2\text{O}(g)$	+500
$\text{O}(g) + 2e^- \longrightarrow \text{O}^{2-}(g)$	+600
$\text{Sr}(s) + \frac{1}{2}\text{O}_2(g) \longrightarrow \text{SrO}(s)$	-550

**Question 7 10 points**

Consider the molecule given on the left, and answer questions (i) to (vi) using chemical bonding theories that we have learnt in class (valence bond theory, hybridization).

Note that the lone pairs are not shown in the structure:

i. Consider the carbon atoms marked in as ***a***, ***b***, ***c***, ***d***, and ***e*** in the molecule, and determine **hybridization** at each of the carbon centers. (2.5 points)

Atom	Hybridization
Carbon (<i>a</i>)	
Carbon (<i>b</i>)	
Carbon (<i>c</i>)	
Carbon (<i>d</i>)	
Carbon (<i>e</i>)	

ii. Now, consider the sulfur atom (S) marked as ***f***, nitrogen (N) atom (***g***), and oxygen(O) atoms (***h*** and ***i***), and determine their **hybridization**. (2 points)

Atom	Hybridization
Sulfur (<i>f</i>)	
Nitrogen (<i>g</i>)	
Oxygen (<i>h</i>)	
Oxygen (<i>i</i>)	

iii. **Describe** all the bonds formed by the carbon atoms ***a*** and ***b***, using sigma and pi designation as appropriate. (2 points)

Atom	Bonds formed (use sigma and pi as appropriate)
Carbon (<i>a</i>)	
Carbon (<i>b</i>)	

iv. How many total sp^3 -s sigma bonds are in the molecule? (1 point)

v. Determine **electronic** and **molecular geometry** at carbon (C) labeled ***e*** and sulfur (S) labeled ***f*** centres. (2 points)

Atom	Electronic Geometry	Molecular Geometry
Carbon (<i>e</i>)		
Sulfur (<i>f</i>)		

vi. Circle the value from the choices given below, that describes the **C-S-C bond angle at sulfur** (S) marked ***f***: (0.5 points)

>90° to <109.5° 109.5° >109.5 to <120° 120° >120° to <180°


Question 8 12 points

Using the **valence shell orbitals only** draw a **complete** molecular orbital (MO) diagram (with no s-p mixing) for the molecule **PS**, in the space provided below. **(6 points)**

Assume that 3s and 3p atomic orbitals will form molecular orbitals like those formed by the combination of 2s and 2p atomic orbitals that we learnt in class, and there is no s-p mixing.

Show **energy arrow**, **atomic orbitals** involved, **atom labels**, the **molecular orbitals** that are formed, **filled electrons** in the MO diagram etc. Make sure to label all the atomic and molecular orbitals.

Using the MO diagram above, answer the following questions:

- (i) Which one of the following will have the highest bond order? Circle the correct choice
 A) PS^{1-} B) PS C) PS^{1+}
- (ii) Which one of the following will have the longest bond length? Circle the correct choice
 A) PS^{1-} B) PS C) PS^{1+}
- (iii) Which of the following will be diamagnetic? Circle the correct choice.
 A) PS^{1-} B) PS C) PS^{1+}
- (iv) Draw Lewis dot structure of the molecule with the shortest bond length (based on the MO diagram) from among the three (PS^{1-} , PS, PS^{1+}), and indicate the bond order calculated from the Lewis dot structure. **(3 points)**

Lewis Dot Structure

Bond Order



49CCC248-BCA2-41AD-8B3D-48616A7C45C3

midterm-exam-93388

#400 Page 10 of 10

SCRAP PAGE

**CHEMISTRY 110 –Midterm Exam****Thursday November 7, 2024, 6:30 – 8:30 pm**

Examiners: Pallavi Sirjoosingh and Ashok Kakkar

First name (please write as legibly as possible within the boxes)

--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--

Last name

--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--

Student ID number

--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--

This booklet has 8 questions on 10 pages (The last page can be used as a scrap page).

The staple must NOT be removed from the question booklet. Your intact question booklet must be returned at the end of the exam.

A separate booklet contains a periodic table, and datasheet.

Write your answers directly under the questions in the space provided. Do not write on the QR codes or near the top edges (as it may be cut off).

Please write **legibly and dark enough** for the scan to be readable. For students who use English as a second language, dictionaries are permitted. Calculators are allowed.

DO NOT ASK QUESTIONS OF THE INVIGILATORS about interpretations of the exam questions. If you have a concern about a question, please summarize it at the top of this page. All concerns will be evaluated after the exam. GOOD LUCK!

**Question 1 8 points** (*Assume all transitions are possible*)

a. 6 points Calculate the **energy (in kJ/mol)** required to ionize (completely remove) the electron from the **n=3** level of **He⁺ ion** (He: Helium). Show your work including equations. (3 sig. figs.)

b. 2 points

(I) Select the largest (longest) wavelength from the three choices below:

- i. Wavelength of photon to *completely remove* an electron from *1s orbital of Hydrogen atom*
- ii. Wavelength of photon to *completely remove* an electron from *1s orbital of Boron atom (B)*
- iii. Wavelength of photon to *completely remove* an electron from *1s orbital of Boron cation (B⁺)*

i or ii or iii

(II) Select the smallest (shortest) wavelength from the three choices below (*for H atom*):

- i. Wavelength of photon *emitted* for an electron transitioning from *2p to 1s orbital*
- ii. Wavelength of photon *absorbed* for an electron transitioning from *2s to 4p orbital*
- iii. Wavelength of photon *absorbed* for an electron transitioning from *1s to 3p orbital*

i or ii or iii



Question 2 8 points



A Chem110 student plots the graph shown on the left, from data obtained for the kinetic energy (eV) of the ejected electron versus frequency (s^{-1}) of the incident light, for an unknown metal. ($1\text{eV} = 1.6022 \times 10^{-19} \text{ J}$)

Based on the plot, answer the questions **a to c** below:

a. 1 point On the plot above, label the **threshold frequency**.

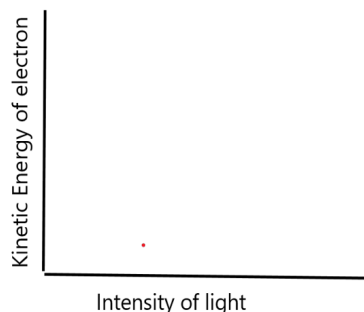
b. 2 points What is the value of the work potential (in J) of the unknown metal?

c. 3 points What is the *longest* wavelength (in nm) required to eject an electron from this metal's surface? Show your work including equation(s) used. (3 sig figs)

d. 2 points In the space provided below, plot the following

(Assume frequency is *above* threshold frequency):

- i. *Kinetic energy* of an ejected electron versus the *intensity* of the incident light (left)
- ii. *Number of electrons* ejected versus the *frequency* of the incident light (right)



**Question 3** **6 points****a. 2 points**

In the ground state of Ge (Germanium; $Z = 32$) atom, **name** the following orbitals:

i. Orbital containing the electron that requires the maximum energy to be removed.

ii. Orbital containing the electron that experiences the minimum effective nuclear charge.

b. 2 points

An unknown anion (A^-) has the following *excited* electronic configuration: $[Ar] 4s^2 3d^3 4p^1$

(Ar: Argon; $Z = 18$)

Give the **condensed ground state electronic configuration** of the atom, **A** and the cation, **A^{2+}** .
Show your work. (Assume all electronic transitions are possible)

Electronic Configuration of A:

Electronic Configuration of A^{2+} :

c. 2 points

In a single atom, determine how many electrons can have the following set of quantum numbers?

(i) $n = 5; l = 3$

In a single atom, determine how many electrons can have the following set of quantum numbers?

(i) $n = 3; l = 2; m_s = +1/2$

**Question 4** **8 points**

Rank the following from 1 to 4 in **increasing** order. Provide the ranking in the boxes provided.

Note: Atomic number (Z) are provided for relevant atoms

a) Increasing (lowest to highest) order of *ionization energy*: (1= lowest; 4 = highest)

Mg

Mg: Z = 12

Rb

Rb: Z = 37

Cs

Cs: Z = 55

K

K : Z = 19

b) Increasing (lowest to highest) order of *bond angle*: (1= lowest; 4 = highest)

H₂OCCl₄XeF₄

HCN

c) Increasing (smallest to largest) order of *ionic size* (1= smallest; 4 = largest)

S²⁻

S: Z = 16

O²⁻

O: Z = 8

Be²⁺

Be: Z = 4

Na⁺

Na: Z = 11

d) Increasing (lowest to highest) order of *bond energy* (1= lowest; 4 = highest)

P-F

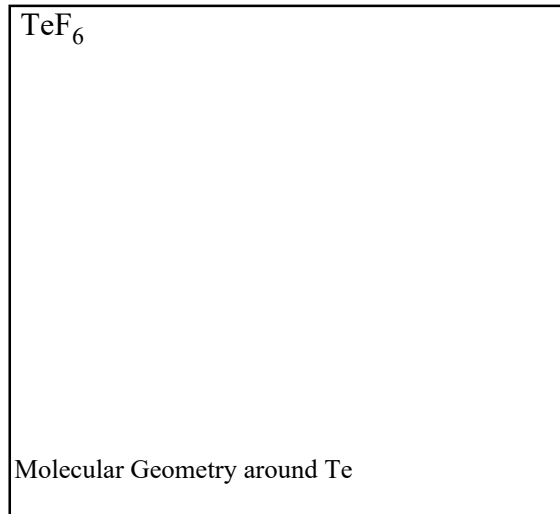
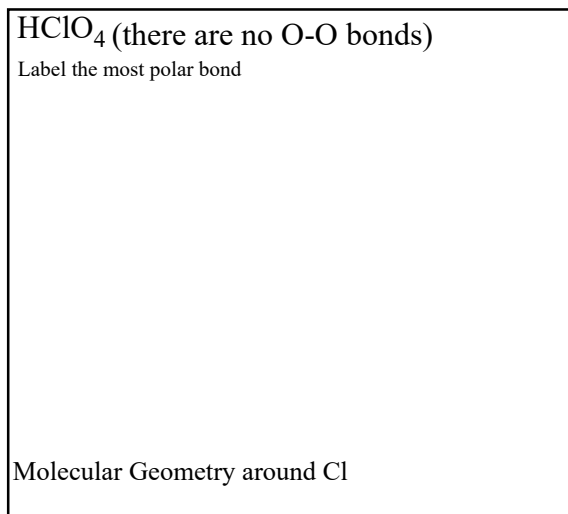
P-Cl

P-I

P-Br

**Question 5** **12 points**

a. 9 points For the molecules HClO_4 and TeF_6 , draw the most stable *Lewis structure* showing appropriate shape and *dashed/wedged bonds*, based on VSEPR. Determine the **molecular geometry** around the central atom. For HClO_4 , label the *most polar bond*.



b. 3 points (Note: Lone pairs are not shown in structures below)

(i) For the three *resonance* structures (1, 2 and 3) shown below, answer the following question:



(I) Circle the *most contributing* resonance structure.

Structure 1

Structure 2

Structure 3

(ii) For the three structures given (4, 5, and 6) below, answer the following questions:



(I) Circle all the structures that represent *resonance structures* of each other:

Structure 4

Structure 5

Structure 6

(II) Among the resonance structures, circle the *most contributing* resonance structure.

Structure 4

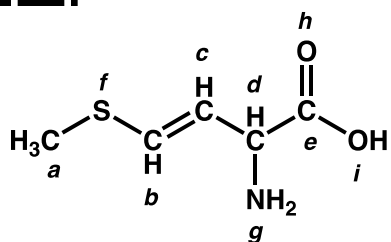
Structure 5

Structure 6

**Question 6** **6 points**

Using the data given below, calculate the **lattice energy** for SrO(s).

	ΔH° (kJ/mol)
$\text{Sr}(s) \longrightarrow \text{Sr}(g)$	+180
$\text{Sr}(g) \longrightarrow \text{Sr}^{2+}(g) + 2e^-$	+1468
$\text{O}_2(g) \longrightarrow 2\text{O}(g)$	+500
$\text{O}(g) + 2e^- \longrightarrow \text{O}^{2-}(g)$	+600
$\text{Sr}(s) + \frac{1}{2}\text{O}_2(g) \longrightarrow \text{SrO}(s)$	-550

**Question 7 10 points**

Consider the molecule given on the left, and answer questions (i) to (vi) using chemical bonding theories that we have learnt in class (valence bond theory, hybridization).

Note that the lone pairs are not shown in the structure:

i. Consider the carbon atoms marked in as ***a***, ***b***, ***c***, ***d***, and ***e*** in the molecule, and determine **hybridization** at each of the carbon centers. (2.5 points)

Atom	Hybridization
Carbon (<i>a</i>)	
Carbon (<i>b</i>)	
Carbon (<i>c</i>)	
Carbon (<i>d</i>)	
Carbon (<i>e</i>)	

ii. Now, consider the sulfur atom (S) marked as ***f***, nitrogen (N) atom (***g***), and oxygen(O) atoms (***h*** and ***i***), and determine their **hybridization**. (2 points)

Atom	Hybridization
Sulfur (<i>f</i>)	
Nitrogen (<i>g</i>)	
Oxygen (<i>h</i>)	
Oxygen (<i>i</i>)	

iii. **Describe** all the bonds formed by the carbon atoms ***a*** and ***b***, using sigma and pi designation as appropriate. (2 points)

Atom	Bonds formed (use sigma and pi as appropriate)
Carbon (<i>a</i>)	
Carbon (<i>b</i>)	

iv. How many total sp^3 -s sigma bonds are in the molecule? (1 point)

v. Determine **electronic** and **molecular geometry** at carbon (C) labeled ***e*** and sulfur (S) labeled ***f*** centres. (2 points)

Atom	Electronic Geometry	Molecular Geometry
Carbon (<i>e</i>)		
Sulfur (<i>f</i>)		

vi. Circle the value from the choices given below, that describes the **C-S-C bond angle at sulfur** (S) marked ***f***: (0.5 points)

>90° to <109.5°

109.5°

>109.5 to <120°

120°

>120° to <180°


Question 8 12 points

Using the **valence shell orbitals only** draw a **complete** molecular orbital (MO) diagram (with no s-p mixing) for the molecule **PS**, in the space provided below. **(6 points)**

Assume that 3s and 3p atomic orbitals will form molecular orbitals like those formed by the combination of 2s and 2p atomic orbitals that we learnt in class, and there is no s-p mixing.

Show **energy arrow**, **atomic orbitals** involved, **atom labels**, the **molecular orbitals** that are formed, **filled electrons** in the MO diagram etc. Make sure to label all the atomic and molecular orbitals.

Using the MO diagram above, answer the following questions:

- (i) Which one of the following will have the highest bond order? Circle the correct choice
 A) PS^{1-} B) PS C) PS^{1+}
- (ii) Which one of the following will have the longest bond length? Circle the correct choice
 A) PS^{1-} B) PS C) PS^{1+}
- (iii) Which of the following will be diamagnetic? Circle the correct choice.
 A) PS^{1-} B) PS C) PS^{1+}
- (iv) Draw Lewis dot structure of the molecule with the shortest bond length (based on the MO diagram) from among the three (PS^{1-} , PS, PS^{1+}), and indicate the bond order calculated from the Lewis dot structure. **(3 points)**

Lewis Dot Structure

Bond Order



D4C5A675-BBC3-49AC-8569-ECA06C59577E

midterm-exam-93388

#401 Page 10 of 10

SCRAP PAGE

**CHEMISTRY 110 –Midterm Exam****Thursday November 7, 2024, 6:30 – 8:30 pm**

Examiners: Pallavi Sirjoosingh and Ashok Kakkar

First name (please write as legibly as possible within the boxes)

--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--

Last name

--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--

Student ID number

--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--

This booklet has 8 questions on 10 pages (The last page can be used as a scrap page).

The staple must NOT be removed from the question booklet. Your intact question booklet must be returned at the end of the exam.

A separate booklet contains a periodic table, and datasheet.

Write your answers directly under the questions in the space provided. Do not write on the QR codes or near the top edges (as it may be cut off).

Please write **legibly and dark enough** for the scan to be readable. For students who use English as a second language, dictionaries are permitted. Calculators are allowed.

DO NOT ASK QUESTIONS OF THE INVIGILATORS about interpretations of the exam questions. If you have a concern about a question, please summarize it at the top of this page. All concerns will be evaluated after the exam. GOOD LUCK!

**Question 1 8 points** (*Assume all transitions are possible*)

a. 6 points Calculate the **energy (in kJ/mol)** required to ionize (completely remove) the electron from the **$n=3$** level of **He^+ ion** (He: Helium). Show your work including equations. (3 sig. figs.)

b. 2 points

(I) Select the largest (longest) wavelength from the three choices below:

- i. Wavelength of photon to *completely remove* an electron from *1s orbital of Hydrogen atom*
- ii. Wavelength of photon to *completely remove* an electron from *1s orbital of Boron atom (B)*
- iii. Wavelength of photon to *completely remove* an electron from *1s orbital of Boron cation (B^+)*

i or ii or iii

(II) Select the smallest (shortest) wavelength from the three choices below (*for H atom*):

- i. Wavelength of photon *emitted* for an electron transitioning from *2p to 1s orbital*
- ii. Wavelength of photon *absorbed* for an electron transitioning from *2s to 4p orbital*
- iii. Wavelength of photon *absorbed* for an electron transitioning from *1s to 3p orbital*

i or ii or iii



Question 2 8 points



A Chem110 student plots the graph shown on the left, from data obtained for the kinetic energy (eV) of the ejected electron versus frequency (s^{-1}) of the incident light, for an unknown metal. ($1\text{eV} = 1.6022 \times 10^{-19} \text{ J}$)

Based on the plot, answer the questions **a to c** below:

a. 1 point On the plot above, label the **threshold frequency**.

b. 2 points What is the value of the work potential (in J) of the unknown metal?

c. 3 points What is the *longest* wavelength (in nm) required to eject an electron from this metal's surface? Show your work including equation(s) used. (3 sig figs)

d. 2 points In the space provided below, plot the following

(Assume frequency is *above* threshold frequency):

- i. *Kinetic energy* of an ejected electron versus the *intensity* of the incident light (left)
- ii. *Number of electrons* ejected versus the *frequency* of the incident light (right)



**Question 3** **6 points****a. 2 points**

In the ground state of Ge (Germanium; $Z = 32$) atom, **name** the following orbitals:

i. Orbital containing the electron that requires the maximum energy to be removed.

ii. Orbital containing the electron that experiences the minimum effective nuclear charge.

b. 2 points

An unknown anion (A^-) has the following *excited* electronic configuration: $[Ar] 4s^2 3d^3 4p^1$

(Ar: Argon; $Z = 18$)

Give the **condensed ground state electronic configuration** of the atom, **A** and the cation, **A^{2+}** .
Show your work. (Assume all electronic transitions are possible)

Electronic Configuration of A:

Electronic Configuration of A^{2+} :

c. 2 points

In a single atom, determine how many electrons can have the following set of quantum numbers?

(i) $n = 5; l = 3$

In a single atom, determine how many electrons can have the following set of quantum numbers?

(i) $n = 3; l = 2; m_s = +1/2$

**Question 4** **8 points**

Rank the following from 1 to 4 in **increasing** order. Provide the ranking in the boxes provided.

Note: Atomic number (Z) are provided for relevant atoms

a) Increasing (lowest to highest) order of *ionization energy*: (1= lowest; 4 = highest)

Mg

Mg: Z = 12

Rb

Rb: Z = 37

Cs

Cs: Z = 55

K

K : Z = 19

b) Increasing (lowest to highest) order of *bond angle*: (1= lowest; 4 = highest)

H₂OCCl₄XeF₄

HCN

c) Increasing (smallest to largest) order of *ionic size* (1= smallest; 4 = largest)

S²⁻

S: Z = 16

O²⁻

O: Z = 8

Be²⁺

Be: Z = 4

Na⁺

Na: Z = 11

d) Increasing (lowest to highest) order of *bond energy* (1= lowest; 4 = highest)

P-F

P-Cl

P-I

P-Br

**Question 5** **12 points**

a. 9 points For the molecules HClO_4 and TeF_6 , draw the most stable *Lewis structure* showing appropriate shape and *dashed/wedged bonds*, based on VSEPR. Determine the **molecular geometry** around the central atom. For HClO_4 , label the *most polar bond*.



b. 3 points (Note: Lone pairs are not shown in structures below)

(i) For the three *resonance* structures (1, 2 and 3) shown below, answer the following question:



(I) Circle the *most contributing* resonance structure.

Structure 1

Structure 2

Structure 3

(ii) For the three structures given (4, 5, and 6) below, answer the following questions:



(I) Circle all the structures that represent *resonance structures* of each other:

Structure 4

Structure 5

Structure 6

(II) Among the resonance structures, circle the *most contributing* resonance structure.

Structure 4

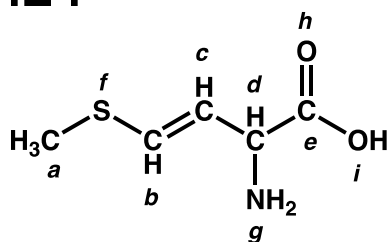
Structure 5

Structure 6

**Question 6** **6 points**

Using the data given below, calculate the **lattice energy** for SrO(s).

	ΔH° (kJ/mol)
$\text{Sr}(s) \longrightarrow \text{Sr}(g)$	+180
$\text{Sr}(g) \longrightarrow \text{Sr}^{2+}(g) + 2e^-$	+1468
$\text{O}_2(g) \longrightarrow 2\text{O}(g)$	+500
$\text{O}(g) + 2e^- \longrightarrow \text{O}^{2-}(g)$	+600
$\text{Sr}(s) + \frac{1}{2}\text{O}_2(g) \longrightarrow \text{SrO}(s)$	-550

**Question 7 10 points**

Consider the molecule given on the left, and answer questions (i) to (vi) using chemical bonding theories that we have learnt in class (valence bond theory, hybridization).

Note that the lone pairs are not shown in the structure:

i. Consider the carbon atoms marked in as ***a***, ***b***, ***c***, ***d***, and ***e*** in the molecule, and determine **hybridization** at each of the carbon centers. (2.5 points)

Atom	Hybridization
Carbon (<i>a</i>)	
Carbon (<i>b</i>)	
Carbon (<i>c</i>)	
Carbon (<i>d</i>)	
Carbon (<i>e</i>)	

ii. Now, consider the sulfur atom (S) marked as ***f***, nitrogen (N) atom (***g***), and oxygen(O) atoms (***h*** and ***i***), and determine their **hybridization**. (2 points)

Atom	Hybridization
Sulfur (<i>f</i>)	
Nitrogen (<i>g</i>)	
Oxygen (<i>h</i>)	
Oxygen (<i>i</i>)	

iii. **Describe** all the bonds formed by the carbon atoms ***a*** and ***b***, using sigma and pi designation as appropriate. (2 points)

Atom	Bonds formed (use sigma and pi as appropriate)
Carbon (<i>a</i>)	
Carbon (<i>b</i>)	

iv. How many total sp^3 -s sigma bonds are in the molecule? (1 point)

v. Determine **electronic** and **molecular geometry** at carbon (C) labeled ***e*** and sulfur (S) labeled ***f*** centres. (2 points)

Atom	Electronic Geometry	Molecular Geometry
Carbon (<i>e</i>)		
Sulfur (<i>f</i>)		

vi. Circle the value from the choices given below, that describes the **C-S-C bond angle at sulfur** (S) marked ***f***: (0.5 points)

>90° to <109.5°

109.5°

>109.5 to <120°

120°

>120° to <180°


Question 8 12 points

Using the **valence shell orbitals only** draw a **complete** molecular orbital (MO) diagram (with no s-p mixing) for the molecule **PS**, in the space provided below. **(6 points)**

Assume that 3s and 3p atomic orbitals will form molecular orbitals like those formed by the combination of 2s and 2p atomic orbitals that we learnt in class, and there is no s-p mixing.

Show **energy arrow**, **atomic orbitals** involved, **atom labels**, the **molecular orbitals** that are formed, **filled electrons** in the MO diagram etc. Make sure to **label all the atomic and molecular orbitals**.

Using the MO diagram above, answer the following questions:

- (i) Which one of the following will have the highest bond order? Circle the correct choice
 A) PS^{1-} B) PS C) PS^{1+}
- (ii) Which one of the following will have the longest bond length? Circle the correct choice
 A) PS^{1-} B) PS C) PS^{1+}
- (iii) Which of the following will be diamagnetic? Circle the correct choice.
 A) PS^{1-} B) PS C) PS^{1+}
- (iv) Draw Lewis dot structure of the molecule with the shortest bond length (based on the MO diagram) from among the three (PS^{1-} , PS, PS^{1+}), and indicate the bond order calculated from the Lewis dot structure. **(3 points)**

Lewis Dot Structure

Bond Order



11868962-4AE5-4765-8255-2F1BD6074306

midterm-exam-93388

#402 Page 10 of 10

SCRAP PAGE

**CHEMISTRY 110 –Midterm Exam****Thursday November 7, 2024, 6:30 – 8:30 pm**

Examiners: Pallavi Sirjoosingh and Ashok Kakkar

First name (please write as legibly as possible within the boxes)

--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--

Last name

--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--

Student ID number

--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--

This booklet has 8 questions on 10 pages (The last page can be used as a scrap page).

The staple must NOT be removed from the question booklet. Your intact question booklet must be returned at the end of the exam.

A separate booklet contains a periodic table, and datasheet.

Write your answers directly under the questions in the space provided. Do not write on the QR codes or near the top edges (as it may be cut off).

Please write **legibly and dark enough** for the scan to be readable. For students who use English as a second language, dictionaries are permitted. Calculators are allowed.

DO NOT ASK QUESTIONS OF THE INVIGILATORS about interpretations of the exam questions. If you have a concern about a question, please summarize it at the top of this page. All concerns will be evaluated after the exam. GOOD LUCK!

**Question 1 8 points** (*Assume all transitions are possible*)

a. 6 points Calculate the **energy (in kJ/mol)** required to ionize (completely remove) the electron from the **n=3** level of **He⁺ ion** (He: Helium). Show your work including equations. (3 sig. figs.)

b. 2 points

(I) Select the largest (longest) wavelength from the three choices below:

- i. Wavelength of photon to *completely remove* an electron from *1s orbital of Hydrogen atom*
- ii. Wavelength of photon to *completely remove* an electron from *1s orbital of Boron atom (B)*
- iii. Wavelength of photon to *completely remove* an electron from *1s orbital of Boron cation (B⁺)*

i or ii or iii

(II) Select the smallest (shortest) wavelength from the three choices below (*for H atom*):

- i. Wavelength of photon *emitted* for an electron transitioning from *2p to 1s orbital*
- ii. Wavelength of photon *absorbed* for an electron transitioning from *2s to 4p orbital*
- iii. Wavelength of photon *absorbed* for an electron transitioning from *1s to 3p orbital*

i or ii or iii



Question 2 8 points



A Chem110 student plots the graph shown on the left, from data obtained for the kinetic energy (eV) of the ejected electron versus frequency (s^{-1}) of the incident light, for an unknown metal. ($1\text{eV} = 1.6022 \times 10^{-19} \text{ J}$)

Based on the plot, answer the questions **a to c** below:

a. 1 point On the plot above, label the **threshold frequency**.

b. 2 points What is the value of the work potential (in J) of the unknown metal?

c. 3 points What is the *longest* wavelength (in nm) required to eject an electron from this metal's surface? Show your work including equation(s) used. (3 sig figs)

d. 2 points In the space provided below, plot the following

(Assume frequency is *above* threshold frequency):

- i. *Kinetic energy* of an ejected electron versus the *intensity* of the incident light (left)
- ii. *Number of electrons* ejected versus the *frequency* of the incident light (right)



**Question 3** **6 points****a. 2 points**

In the ground state of Ge (Germanium; $Z = 32$) atom, **name** the following orbitals:

i. Orbital containing the electron that requires the maximum energy to be removed.

ii. Orbital containing the electron that experiences the minimum effective nuclear charge.

b. 2 points

An unknown anion (A^-) has the following *excited* electronic configuration: $[Ar] 4s^2 3d^3 4p^1$

(Ar: Argon; $Z = 18$)

Give the **condensed ground state electronic configuration** of the atom, **A** and the cation, **A^{2+}** .
Show your work. (Assume all electronic transitions are possible)

Electronic Configuration of A:

Electronic Configuration of A^{2+} :

c. 2 points

In a single atom, determine how many electrons can have the following set of quantum numbers?

(i) $n = 5; l = 3$

In a single atom, determine how many electrons can have the following set of quantum numbers?

(i) $n = 3; l = 2; m_s = +1/2$

**Question 4** **8 points**

Rank the following from 1 to 4 in **increasing** order. Provide the ranking in the boxes provided.

Note: Atomic number (Z) are provided for relevant atoms

a) Increasing (lowest to highest) order of *ionization energy*: (1= lowest; 4 = highest)

Mg

Mg: Z = 12

Rb

Rb: Z = 37

Cs

Cs: Z = 55

K

K : Z = 19

b) Increasing (lowest to highest) order of *bond angle*: (1= lowest; 4 = highest)

H₂OCCl₄XeF₄

HCN

c) Increasing (smallest to largest) order of *ionic size* (1= smallest; 4 = largest)

S²⁻

S: Z = 16

O²⁻

O: Z = 8

Be²⁺

Be: Z = 4

Na⁺

Na: Z = 11

d) Increasing (lowest to highest) order of *bond energy* (1= lowest; 4 = highest)

P-F

P-Cl

P-I

P-Br

**Question 5** **12 points**

a. 9 points For the molecules HClO_4 and TeF_6 , draw the most stable *Lewis structure* showing appropriate shape and *dashed/wedged bonds*, based on VSEPR. Determine the **molecular geometry** around the central atom. For HClO_4 , label the *most polar bond*.



b. 3 points (Note: Lone pairs are not shown in structures below)

(i) For the three *resonance* structures (1, 2 and 3) shown below, answer the following question:



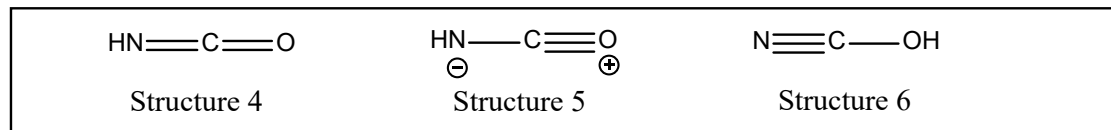
(I) Circle the *most contributing* resonance structure.

Structure 1

Structure 2

Structure 3

(ii) For the three structures given (4, 5, and 6) below, answer the following questions:



(I) Circle all the structures that represent *resonance structures* of each other:

Structure 4

Structure 5

Structure 6

(II) Among the resonance structures, circle the *most contributing* resonance structure.

Structure 4

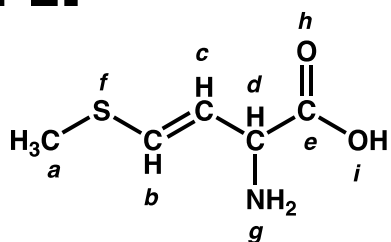
Structure 5

Structure 6

**Question 6** **6 points**

Using the data given below, calculate the **lattice energy** for SrO(s).

	ΔH° (kJ/mol)
$\text{Sr}(s) \longrightarrow \text{Sr}(g)$	+180
$\text{Sr}(g) \longrightarrow \text{Sr}^{2+}(g) + 2e^-$	+1468
$\text{O}_2(g) \longrightarrow 2\text{O}(g)$	+500
$\text{O}(g) + 2e^- \longrightarrow \text{O}^{2-}(g)$	+600
$\text{Sr}(s) + \frac{1}{2}\text{O}_2(g) \longrightarrow \text{SrO}(s)$	-550

**Question 7 10 points**

Consider the molecule given on the left, and answer questions (i) to (vi) using chemical bonding theories that we have learnt in class (valence bond theory, hybridization).

Note that the lone pairs are not shown in the structure:

i. Consider the carbon atoms marked in as ***a***, ***b***, ***c***, ***d***, and ***e*** in the molecule, and determine **hybridization** at each of the carbon centers. (2.5 points)

Atom	Hybridization
Carbon (<i>a</i>)	
Carbon (<i>b</i>)	
Carbon (<i>c</i>)	
Carbon (<i>d</i>)	
Carbon (<i>e</i>)	

ii. Now, consider the sulfur atom (S) marked as ***f***, nitrogen (N) atom (***g***), and oxygen(O) atoms (***h*** and ***i***), and determine their **hybridization**. (2 points)

Atom	Hybridization
Sulfur (<i>f</i>)	
Nitrogen (<i>g</i>)	
Oxygen (<i>h</i>)	
Oxygen (<i>i</i>)	

iii. **Describe** all the bonds formed by the carbon atoms ***a*** and ***b***, using sigma and pi designation as appropriate. (2 points)

Atom	Bonds formed (use sigma and pi as appropriate)
Carbon (<i>a</i>)	
Carbon (<i>b</i>)	

iv. How many total sp^3 -s sigma bonds are in the molecule? (1 point)

v. Determine **electronic** and **molecular geometry** at carbon (C) labeled ***e*** and sulfur (S) labeled ***f*** centres. (2 points)

Atom	Electronic Geometry	Molecular Geometry
Carbon (<i>e</i>)		
Sulfur (<i>f</i>)		

vi. Circle the value from the choices given below, that describes the **C-S-C bond angle at sulfur** (S) marked ***f***: (0.5 points)

>90° to <109.5° 109.5° >109.5 to <120° 120° >120° to <180°


Question 8 12 points

Using the **valence shell orbitals only** draw a **complete** molecular orbital (MO) diagram (with no s-p mixing) for the molecule **PS**, in the space provided below. **(6 points)**

Assume that 3s and 3p atomic orbitals will form molecular orbitals like those formed by the combination of 2s and 2p atomic orbitals that we learnt in class, and there is no s-p mixing.

Show **energy arrow**, **atomic orbitals** involved, **atom labels**, the **molecular orbitals** that are formed, **filled electrons** in the MO diagram etc. Make sure to label all the atomic and molecular orbitals.

Using the MO diagram above, answer the following questions:

- (i) Which one of the following will have the highest bond order? Circle the correct choice
 A) PS^{1-} B) PS C) PS^{1+}
- (ii) Which one of the following will have the longest bond length? Circle the correct choice
 A) PS^{1-} B) PS C) PS^{1+}
- (iii) Which of the following will be diamagnetic? Circle the correct choice.
 A) PS^{1-} B) PS C) PS^{1+}
- (iv) Draw Lewis dot structure of the molecule with the shortest bond length (based on the MO diagram) from among the three (PS^{1-} , PS, PS^{1+}), and indicate the bond order calculated from the Lewis dot structure. **(3 points)**

Lewis Dot Structure

Bond Order



2BF95D8B-0A63-4871-BA4E-5DD102C25A9E

midterm-exam-93388

#403 Page 10 of 10

SCRAP PAGE

**CHEMISTRY 110 –Midterm Exam****Thursday November 7, 2024, 6:30 – 8:30 pm****Examiners: Pallavi Sirjoosingh and Ashok Kakkar**

First name (please write as legibly as possible within the boxes)

--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--

Last name

--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--

Student ID number

--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--

This booklet has 8 questions on 10 pages (The last page can be used as a scrap page).

The staple must NOT be removed from the question booklet. Your intact question booklet must be returned at the end of the exam.

A separate booklet contains a periodic table, and datasheet.

Write your answers directly under the questions in the space provided. Do not write on the QR codes or near the top edges (as it may be cut off).

Please write **legibly and dark enough** for the scan to be readable. For students who use English as a second language, dictionaries are permitted. Calculators are allowed.

DO NOT ASK QUESTIONS OF THE INVIGILATORS about interpretations of the exam questions. If you have a concern about a question, please summarize it at the top of this page. All concerns will be evaluated after the exam. GOOD LUCK!

**Question 1 8 points** (*Assume all transitions are possible*)

a. 6 points Calculate the **energy (in kJ/mol)** required to ionize (completely remove) the electron from the **n=3** level of **He⁺ ion** (He: Helium). Show your work including equations. (3 sig. figs.)

b. 2 points

(I) Select the largest (longest) wavelength from the three choices below:

- i. Wavelength of photon to *completely remove* an electron from *1s orbital of Hydrogen atom*
- ii. Wavelength of photon to *completely remove* an electron from *1s orbital of Boron atom (B)*
- iii. Wavelength of photon to *completely remove* an electron from *1s orbital of Boron cation (B⁺)*

i or ii or iii

(II) Select the smallest (shortest) wavelength from the three choices below (*for H atom*):

- i. Wavelength of photon *emitted* for an electron transitioning from *2p to 1s orbital*
- ii. Wavelength of photon *absorbed* for an electron transitioning from *2s to 4p orbital*
- iii. Wavelength of photon *absorbed* for an electron transitioning from *1s to 3p orbital*

i or ii or iii



Question 2 8 points



A Chem110 student plots the graph shown on the left, from data obtained for the kinetic energy (eV) of the ejected electron versus frequency (s^{-1}) of the incident light, for an unknown metal. ($1\text{eV} = 1.6022 \times 10^{-19} \text{ J}$)

Based on the plot, answer the questions **a to c** below:

a. 1 point On the plot above, label the **threshold frequency**.

b. 2 points What is the value of the work potential (in J) of the unknown metal?

c. 3 points What is the *longest* wavelength (in nm) required to eject an electron from this metal's surface? Show your work including equation(s) used. (3 sig figs)

d. 2 points In the space provided below, plot the following

(Assume frequency is *above* threshold frequency):

- i. *Kinetic energy* of an ejected electron versus the *intensity* of the incident light (left)
- ii. *Number of electrons* ejected versus the *frequency* of the incident light (right)



**Question 3** **6 points****a. 2 points**

In the ground state of Ge (Germanium; $Z = 32$) atom, **name** the following orbitals:

i. Orbital containing the electron that requires the maximum energy to be removed.

ii. Orbital containing the electron that experiences the minimum effective nuclear charge.

b. 2 points

An unknown anion (A^-) has the following *excited* electronic configuration: $[Ar] 4s^2 3d^3 4p^1$

(Ar: Argon; $Z = 18$)

Give the **condensed ground state electronic configuration** of the atom, **A** and the cation, **A^{2+}** .
Show your work. (Assume all electronic transitions are possible)

Electronic Configuration of A:

Electronic Configuration of A^{2+} :

c. 2 points

In a single atom, determine how many electrons can have the following set of quantum numbers?

(i) $n = 5; l = 3$

In a single atom, determine how many electrons can have the following set of quantum numbers?

(i) $n = 3; l = 2; m_s = +1/2$

**Question 4** **8 points**

Rank the following from 1 to 4 in **increasing** order. Provide the ranking in the boxes provided.

Note: Atomic number (Z) are provided for relevant atoms

a) Increasing (lowest to highest) order of *ionization energy*: (1= lowest; 4 = highest)

Mg

Mg: Z = 12

Rb

Rb: Z = 37

Cs

Cs: Z = 55

K

K : Z = 19

b) Increasing (lowest to highest) order of *bond angle*: (1= lowest; 4 = highest)

H₂OCCl₄XeF₄

HCN

c) Increasing (smallest to largest) order of *ionic size* (1= smallest; 4 = largest)

S²⁻

S: Z = 16

O²⁻

O: Z = 8

Be²⁺

Be: Z = 4

Na⁺

Na: Z = 11

d) Increasing (lowest to highest) order of *bond energy* (1= lowest; 4 = highest)

P-F

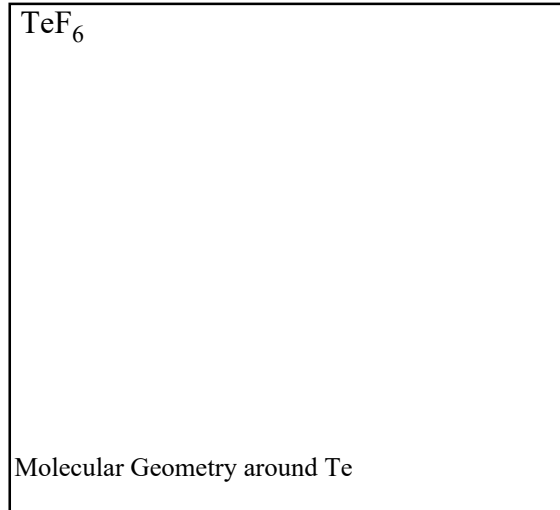
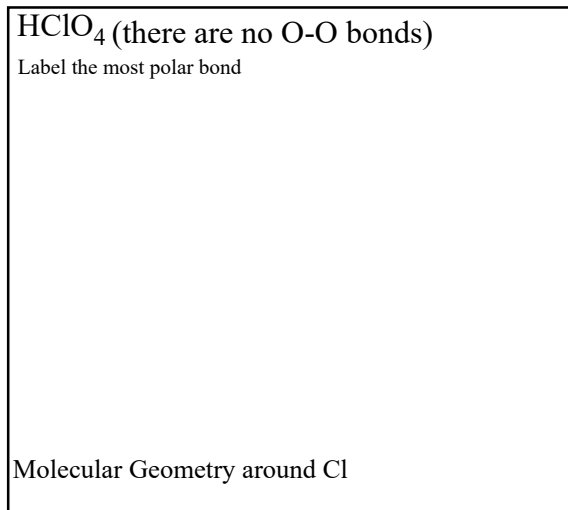
P-Cl

P-I

P-Br

**Question 5** **12 points**

a. 9 points For the molecules HClO_4 and TeF_6 , draw the most stable *Lewis structure* showing appropriate shape and *dashed/wedged bonds*, based on VSEPR. Determine the **molecular geometry** around the central atom. For HClO_4 , label the *most polar bond*.



b. 3 points (Note: Lone pairs are not shown in structures below)

(i) For the three *resonance* structures (1, 2 and 3) shown below, answer the following question:



(I) Circle the *most contributing* resonance structure.

Structure 1

Structure 2

Structure 3

(ii) For the three structures given (4, 5, and 6) below, answer the following questions:



(I) Circle all the structures that represent *resonance structures* of each other:

Structure 4

Structure 5

Structure 6

(II) Among the resonance structures, circle the *most contributing* resonance structure.

Structure 4

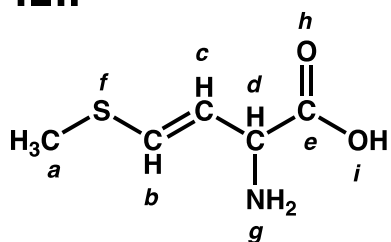
Structure 5

Structure 6

**Question 6** **6 points**

Using the data given below, calculate the **lattice energy** for SrO(s).

	ΔH° (kJ/mol)
$\text{Sr}(s) \longrightarrow \text{Sr}(g)$	+180
$\text{Sr}(g) \longrightarrow \text{Sr}^{2+}(g) + 2e^-$	+1468
$\text{O}_2(g) \longrightarrow 2\text{O}(g)$	+500
$\text{O}(g) + 2e^- \longrightarrow \text{O}^{2-}(g)$	+600
$\text{Sr}(s) + \frac{1}{2}\text{O}_2(g) \longrightarrow \text{SrO}(s)$	-550

**Question 7 10 points**

Consider the molecule given on the left, and answer questions (i) to (vi) using chemical bonding theories that we have learnt in class (valence bond theory, hybridization).

Note that the lone pairs are not shown in the structure:

i. Consider the carbon atoms marked in as ***a***, ***b***, ***c***, ***d***, and ***e*** in the molecule, and determine **hybridization** at each of the carbon centers. (2.5 points)

Atom	Hybridization
Carbon (<i>a</i>)	
Carbon (<i>b</i>)	
Carbon (<i>c</i>)	
Carbon (<i>d</i>)	
Carbon (<i>e</i>)	

ii. Now, consider the sulfur atom (S) marked as ***f***, nitrogen (N) atom (***g***), and oxygen(O) atoms (***h*** and ***i***), and determine their **hybridization**. (2 points)

Atom	Hybridization
Sulfur (<i>f</i>)	
Nitrogen (<i>g</i>)	
Oxygen (<i>h</i>)	
Oxygen (<i>i</i>)	

iii. **Describe** all the bonds formed by the carbon atoms ***a*** and ***b***, using sigma and pi designation as appropriate. (2 points)

Atom	Bonds formed (use sigma and pi as appropriate)
Carbon (<i>a</i>)	
Carbon (<i>b</i>)	

iv. How many total sp^3 -s sigma bonds are in the molecule? (1 point)

v. Determine **electronic** and **molecular geometry** at carbon (C) labeled ***e*** and sulfur (S) labeled ***f*** centres. (2 points)

Atom	Electronic Geometry	Molecular Geometry
Carbon (<i>e</i>)		
Sulfur (<i>f</i>)		

vi. Circle the value from the choices given below, that describes the **C-S-C bond angle at sulfur** (S) marked ***f***: (0.5 points)

>90° to <109.5° 109.5° >109.5 to <120° 120° >120° to <180°


Question 8 12 points

Using the **valence shell orbitals only** draw a **complete** molecular orbital (MO) diagram (with no s-p mixing) for the molecule **PS**, in the space provided below. **(6 points)**

Assume that 3s and 3p atomic orbitals will form molecular orbitals like those formed by the combination of 2s and 2p atomic orbitals that we learnt in class, and there is no s-p mixing.

Show **energy arrow**, **atomic orbitals** involved, **atom labels**, the **molecular orbitals** that are formed, **filled electrons** in the MO diagram etc. Make sure to label all the atomic and molecular orbitals.

Using the MO diagram above, answer the following questions:

- (i) Which one of the following will have the highest bond order? Circle the correct choice
 A) PS^{1-} B) PS C) PS^{1+}
- (ii) Which one of the following will have the longest bond length? Circle the correct choice
 A) PS^{1-} B) PS C) PS^{1+}
- (iii) Which of the following will be diamagnetic? Circle the correct choice.
 A) PS^{1-} B) PS C) PS^{1+}
- (iv) Draw Lewis dot structure of the molecule with the shortest bond length (based on the MO diagram) from among the three (PS^{1-} , PS, PS^{1+}), and indicate the bond order calculated from the Lewis dot structure. **(3 points)**

Lewis Dot Structure

Bond Order



6BD4DB53-19AC-4745-A32A-E63856D8D664

midterm-exam-93388

#404 Page 10 of 10

SCRAP PAGE

**CHEMISTRY 110 –Midterm Exam****Thursday November 7, 2024, 6:30 – 8:30 pm**

Examiners: Pallavi Sirjoosingh and Ashok Kakkar

First name (please write as legibly as possible within the boxes)

--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--

Last name

--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--

Student ID number

--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--

This booklet has 8 questions on 10 pages (The last page can be used as a scrap page).

The staple must NOT be removed from the question booklet. Your intact question booklet must be returned at the end of the exam.

A separate booklet contains a periodic table, and datasheet.

Write your answers directly under the questions in the space provided. Do not write on the QR codes or near the top edges (as it may be cut off).

Please write **legibly and dark enough** for the scan to be readable. For students who use English as a second language, dictionaries are permitted. Calculators are allowed.

DO NOT ASK QUESTIONS OF THE INVIGILATORS about interpretations of the exam questions. If you have a concern about a question, please summarize it at the top of this page. All concerns will be evaluated after the exam. GOOD LUCK!

**Question 1 8 points** (*Assume all transitions are possible*)

a. 6 points Calculate the **energy (in kJ/mol)** required to ionize (completely remove) the electron from the **$n=3$** level of **He^+ ion** (He: Helium). Show your work including equations. (3 sig. figs.)

b. 2 points

(I) Select the largest (longest) wavelength from the three choices below:

- i. Wavelength of photon to *completely remove* an electron from *1s orbital of Hydrogen atom*
- ii. Wavelength of photon to *completely remove* an electron from *1s orbital of Boron atom (B)*
- iii. Wavelength of photon to *completely remove* an electron from *1s orbital of Boron cation (B^+)*

i or ii or iii

(II) Select the smallest (shortest) wavelength from the three choices below (*for H atom*):

- i. Wavelength of photon *emitted* for an electron transitioning from *2p to 1s orbital*
- ii. Wavelength of photon *absorbed* for an electron transitioning from *2s to 4p orbital*
- iii. Wavelength of photon *absorbed* for an electron transitioning from *1s to 3p orbital*

i or ii or iii



Question 2 8 points



A Chem110 student plots the graph shown on the left, from data obtained for the kinetic energy (eV) of the ejected electron versus frequency (s^{-1}) of the incident light, for an unknown metal. ($1\text{eV} = 1.6022 \times 10^{-19} \text{ J}$)

Based on the plot, answer the questions **a to c** below:

a. 1 point On the plot above, label the **threshold frequency**.

b. 2 points What is the value of the work potential (in J) of the unknown metal?

c. 3 points What is the *longest* wavelength (in nm) required to eject an electron from this metal's surface? Show your work including equation(s) used. (3 sig figs)

d. 2 points In the space provided below, plot the following

(Assume frequency is *above* threshold frequency):

- Kinetic energy* of an ejected electron versus the *intensity* of the incident light (left)
- Number of electrons* ejected versus the *frequency* of the incident light (right)



**Question 3** **6 points****a. 2 points**

In the ground state of Ge (Germanium; $Z = 32$) atom, **name** the following orbitals:

i. Orbital containing the electron that requires the maximum energy to be removed.

ii. Orbital containing the electron that experiences the minimum effective nuclear charge.

b. 2 points

An unknown anion (A^-) has the following *excited* electronic configuration: $[Ar] 4s^2 3d^3 4p^1$

(Ar: Argon; $Z = 18$)

Give the **condensed ground state electronic configuration** of the atom, **A** and the cation, **A^{2+}** .
Show your work. (Assume all electronic transitions are possible)

Electronic Configuration of A:

Electronic Configuration of A^{2+} :

c. 2 points

In a single atom, determine how many electrons can have the following set of quantum numbers?

(i) $n = 5; l = 3$

In a single atom, determine how many electrons can have the following set of quantum numbers?

(i) $n = 3; l = 2; m_s = +1/2$

**Question 4** **8 points**

Rank the following from 1 to 4 in **increasing** order. Provide the ranking in the boxes provided.

Note: Atomic number (Z) are provided for relevant atoms

a) Increasing (lowest to highest) order of *ionization energy*: (1= lowest; 4 = highest)

Mg

Mg: Z = 12

Rb

Rb: Z = 37

Cs

Cs: Z = 55

K

K : Z = 19

b) Increasing (lowest to highest) order of *bond angle*: (1= lowest; 4 = highest)

H₂OCCl₄XeF₄

HCN

c) Increasing (smallest to largest) order of *ionic size* (1= smallest; 4 = largest)

S²⁻

S: Z = 16

O²⁻

O: Z = 8

Be²⁺

Be: Z = 4

Na⁺

Na: Z = 11

d) Increasing (lowest to highest) order of *bond energy* (1= lowest; 4 = highest)

P-F

P-Cl

P-I

P-Br

**Question 5** **12 points**

a. 9 points For the molecules HClO_4 and TeF_6 , draw the most stable *Lewis structure* showing appropriate shape and *dashed/wedged bonds*, based on VSEPR. Determine the **molecular geometry** around the central atom. For HClO_4 , label the *most polar bond*.



b. 3 points (Note: Lone pairs are not shown in structures below)

(i) For the three *resonance* structures (1, 2 and 3) shown below, answer the following question:



(I) Circle the *most contributing* resonance structure.

Structure 1

Structure 2

Structure 3

(ii) For the three structures given (4, 5, and 6) below, answer the following questions:



(I) Circle all the structures that represent *resonance structures* of each other:

Structure 4

Structure 5

Structure 6

(II) Among the resonance structures, circle the *most contributing* resonance structure.

Structure 4

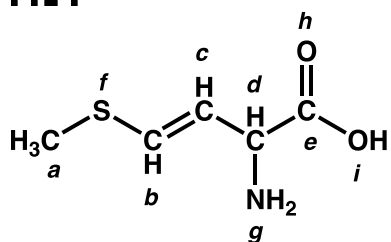
Structure 5

Structure 6

**Question 6** **6 points**

Using the data given below, calculate the **lattice energy** for SrO(s).

	ΔH° (kJ/mol)
$\text{Sr}(s) \longrightarrow \text{Sr}(g)$	+180
$\text{Sr}(g) \longrightarrow \text{Sr}^{2+}(g) + 2e^-$	+1468
$\text{O}_2(g) \longrightarrow 2\text{O}(g)$	+500
$\text{O}(g) + 2e^- \longrightarrow \text{O}^{2-}(g)$	+600
$\text{Sr}(s) + \frac{1}{2}\text{O}_2(g) \longrightarrow \text{SrO}(s)$	-550

**Question 7 10 points**

Consider the molecule given on the left, and answer questions (i) to (vi) using chemical bonding theories that we have learnt in class (valence bond theory, hybridization).

Note that the lone pairs are not shown in the structure:

i. Consider the carbon atoms marked in as ***a***, ***b***, ***c***, ***d***, and ***e*** in the molecule, and determine **hybridization** at each of the carbon centers. (2.5 points)

Atom	Hybridization
Carbon (<i>a</i>)	
Carbon (<i>b</i>)	
Carbon (<i>c</i>)	
Carbon (<i>d</i>)	
Carbon (<i>e</i>)	

ii. Now, consider the sulfur atom (S) marked as ***f***, nitrogen (N) atom (***g***), and oxygen(O) atoms (***h*** and ***i***), and determine their **hybridization**. (2 points)

Atom	Hybridization
Sulfur (<i>f</i>)	
Nitrogen (<i>g</i>)	
Oxygen (<i>h</i>)	
Oxygen (<i>i</i>)	

iii. **Describe** all the bonds formed by the carbon atoms ***a*** and ***b***, using sigma and pi designation as appropriate. (2 points)

Atom	Bonds formed (use sigma and pi as appropriate)
Carbon (<i>a</i>)	
Carbon (<i>b</i>)	

iv. How many total sp^3 -s sigma bonds are in the molecule? (1 point)

v. Determine **electronic** and **molecular geometry** at carbon (C) labeled ***e*** and sulfur (S) labeled ***f*** centres. (2 points)

Atom	Electronic Geometry	Molecular Geometry
Carbon (<i>e</i>)		
Sulfur (<i>f</i>)		

vi. Circle the value from the choices given below, that describes the **C-S-C bond angle at sulfur** (S) marked ***f***: (0.5 points)

$>90^\circ$ to $<109.5^\circ$ 109.5° >109.5 to $<120^\circ$ 120° $>120^\circ$ to $<180^\circ$


Question 8 12 points

Using the **valence shell orbitals only** draw a **complete** molecular orbital (MO) diagram (with no s-p mixing) for the molecule **PS**, in the space provided below. **(6 points)**

Assume that 3s and 3p atomic orbitals will form molecular orbitals like those formed by the combination of 2s and 2p atomic orbitals that we learnt in class, and there is no s-p mixing.

Show **energy arrow**, **atomic orbitals** involved, **atom labels**, the **molecular orbitals** that are formed, **filled electrons** in the MO diagram etc. Make sure to label all the atomic and molecular orbitals.

Using the MO diagram above, answer the following questions:

- (i) Which one of the following will have the highest bond order? Circle the correct choice
 A) PS^{1-} B) PS C) PS^{1+}
- (ii) Which one of the following will have the longest bond length? Circle the correct choice
 A) PS^{1-} B) PS C) PS^{1+}
- (iii) Which of the following will be diamagnetic? Circle the correct choice.
 A) PS^{1-} B) PS C) PS^{1+}
- (iv) Draw Lewis dot structure of the molecule with the shortest bond length (based on the MO diagram) from among the three (PS^{1-} , PS, PS^{1+}), and indicate the bond order calculated from the Lewis dot structure. **(3 points)**

Lewis Dot Structure

Bond Order



4DA7D7F1-5427-451E-9F98-A49E87A59500

midterm-exam-93388

#405 Page 10 of 10

SCRAP PAGE

**CHEMISTRY 110 –Midterm Exam****Thursday November 7, 2024, 6:30 – 8:30 pm**

Examiners: Pallavi Sirjoosingh and Ashok Kakkar

First name (please write as legibly as possible within the boxes)

--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--

Last name

--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--

Student ID number

--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--

This booklet has 8 questions on 10 pages (The last page can be used as a scrap page).

The staple must NOT be removed from the question booklet. Your intact question booklet must be returned at the end of the exam.

A separate booklet contains a periodic table, and datasheet.

Write your answers directly under the questions in the space provided. Do not write on the QR codes or near the top edges (as it may be cut off).

Please write **legibly and dark enough** for the scan to be readable. For students who use English as a second language, dictionaries are permitted. Calculators are allowed.

DO NOT ASK QUESTIONS OF THE INVIGILATORS about interpretations of the exam questions. If you have a concern about a question, please summarize it at the top of this page. All concerns will be evaluated after the exam. GOOD LUCK!

**Question 1 8 points** (*Assume all transitions are possible*)

a. 6 points Calculate the **energy (in kJ/mol)** required to ionize (completely remove) the electron from the **$n=3$** level of **He^+ ion** (He: Helium). Show your work including equations. (3 sig. figs.)

b. 2 points

(I) Select the largest (longest) wavelength from the three choices below:

- i. Wavelength of photon to *completely remove* an electron from *1s orbital of Hydrogen atom*
- ii. Wavelength of photon to *completely remove* an electron from *1s orbital of Boron atom (B)*
- iii. Wavelength of photon to *completely remove* an electron from *1s orbital of Boron cation (B^+)*

i or ii or iii

(II) Select the smallest (shortest) wavelength from the three choices below (*for H atom*):

- i. Wavelength of photon *emitted* for an electron transitioning from *2p to 1s orbital*
- ii. Wavelength of photon *absorbed* for an electron transitioning from *2s to 4p orbital*
- iii. Wavelength of photon *absorbed* for an electron transitioning from *1s to 3p orbital*

i or ii or iii



Question 2 8 points



A Chem110 student plots the graph shown on the left, from data obtained for the kinetic energy (eV) of the ejected electron versus frequency (s^{-1}) of the incident light, for an unknown metal. ($1\text{eV} = 1.6022 \times 10^{-19} \text{ J}$)

Based on the plot, answer the questions **a to c** below:

a. 1 point On the plot above, label the **threshold frequency**.

b. 2 points What is the value of the work potential (in J) of the unknown metal?

c. 3 points What is the *longest* wavelength (in nm) required to eject an electron from this metal's surface? Show your work including equation(s) used. (3 sig figs)

d. 2 points In the space provided below, plot the following

(Assume frequency is *above* threshold frequency):

- Kinetic energy* of an ejected electron versus the *intensity* of the incident light (left)
- Number of electrons* ejected versus the *frequency* of the incident light (right)



**Question 3** **6 points****a. 2 points**

In the ground state of Ge (Germanium; $Z = 32$) atom, **name** the following orbitals:

i. Orbital containing the electron that requires the maximum energy to be removed.

ii. Orbital containing the electron that experiences the minimum effective nuclear charge.

b. 2 points

An unknown anion (A^-) has the following *excited* electronic configuration: $[Ar] 4s^2 3d^3 4p^1$

(Ar: Argon; $Z = 18$)

Give the **condensed ground state electronic configuration** of the atom, **A** and the cation, **A^{2+}** .
Show your work. (Assume all electronic transitions are possible)

Electronic Configuration of A:

Electronic Configuration of A^{2+} :

c. 2 points

In a single atom, determine how many electrons can have the following set of quantum numbers?

(i) $n = 5; l = 3$

In a single atom, determine how many electrons can have the following set of quantum numbers?

(i) $n = 3; l = 2; m_s = +1/2$

**Question 4** **8 points**

Rank the following from 1 to 4 in **increasing** order. Provide the ranking in the boxes provided.

Note: Atomic number (Z) are provided for relevant atoms

a) Increasing (lowest to highest) order of *ionization energy*: (1= lowest; 4 = highest)

Mg

Mg: Z = 12

Rb

Rb: Z = 37

Cs

Cs: Z = 55

K

K : Z = 19

b) Increasing (lowest to highest) order of *bond angle*: (1= lowest; 4 = highest)

H₂OCCl₄XeF₄

HCN

c) Increasing (smallest to largest) order of *ionic size* (1= smallest; 4 = largest)

S²⁻

S: Z = 16

O²⁻

O: Z = 8

Be²⁺

Be: Z = 4

Na⁺

Na: Z = 11

d) Increasing (lowest to highest) order of *bond energy* (1= lowest; 4 = highest)

P-F

P-Cl

P-I

P-Br

**Question 5** **12 points**

a. 9 points For the molecules HClO_4 and TeF_6 , draw the most stable *Lewis structure* showing appropriate shape and *dashed/wedged bonds*, based on VSEPR. Determine the **molecular geometry** around the central atom. For HClO_4 , label the *most polar bond*.



b. 3 points (Note: Lone pairs are not shown in structures below)

(i) For the three *resonance* structures (1, 2 and 3) shown below, answer the following question:



(I) Circle the *most contributing* resonance structure.

Structure 1

Structure 2

Structure 3

(ii) For the three structures given (4, 5, and 6) below, answer the following questions:



(I) Circle all the structures that represent *resonance structures* of each other:

Structure 4

Structure 5

Structure 6

(II) Among the resonance structures, circle the *most contributing* resonance structure.

Structure 4

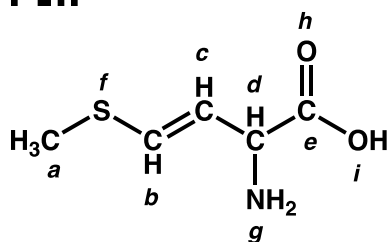
Structure 5

Structure 6

**Question 6** **6 points**

Using the data given below, calculate the **lattice energy** for SrO(s).

	ΔH° (kJ/mol)
$\text{Sr}(s) \longrightarrow \text{Sr}(g)$	+180
$\text{Sr}(g) \longrightarrow \text{Sr}^{2+}(g) + 2e^-$	+1468
$\text{O}_2(g) \longrightarrow 2\text{O}(g)$	+500
$\text{O}(g) + 2e^- \longrightarrow \text{O}^{2-}(g)$	+600
$\text{Sr}(s) + \frac{1}{2}\text{O}_2(g) \longrightarrow \text{SrO}(s)$	-550

**Question 7 10 points**

Consider the molecule given on the left, and answer questions (i) to (vi) using chemical bonding theories that we have learnt in class (valence bond theory, hybridization).

Note that the lone pairs are not shown in the structure:

i. Consider the carbon atoms marked in as ***a***, ***b***, ***c***, ***d***, and ***e*** in the molecule, and determine **hybridization** at each of the carbon centers. (2.5 points)

Atom	Hybridization
Carbon (<i>a</i>)	
Carbon (<i>b</i>)	
Carbon (<i>c</i>)	
Carbon (<i>d</i>)	
Carbon (<i>e</i>)	

ii. Now, consider the sulfur atom (S) marked as ***f***, nitrogen (N) atom (***g***), and oxygen(O) atoms (***h*** and ***i***), and determine their **hybridization**. (2 points)

Atom	Hybridization
Sulfur (<i>f</i>)	
Nitrogen (<i>g</i>)	
Oxygen (<i>h</i>)	
Oxygen (<i>i</i>)	

iii. **Describe** all the bonds formed by the carbon atoms ***a*** and ***b***, using sigma and pi designation as appropriate. (2 points)

Atom	Bonds formed (use sigma and pi as appropriate)
Carbon (<i>a</i>)	
Carbon (<i>b</i>)	

iv. How many total sp^3 -s sigma bonds are in the molecule? (1 point)

v. Determine **electronic** and **molecular geometry** at carbon (C) labeled ***e*** and sulfur (S) labeled ***f*** centres. (2 points)

Atom	Electronic Geometry	Molecular Geometry
Carbon (<i>e</i>)		
Sulfur (<i>f</i>)		

vi. Circle the value from the choices given below, that describes the **C-S-C bond angle at sulfur** (S) marked ***f***: (0.5 points)

>90° to <109.5° 109.5° >109.5 to <120° 120° >120° to <180°


Question 8 12 points

Using the **valence shell orbitals only** draw a **complete** molecular orbital (MO) diagram (with no s-p mixing) for the molecule **PS**, in the space provided below. **(6 points)**

Assume that 3s and 3p atomic orbitals will form molecular orbitals like those formed by the combination of 2s and 2p atomic orbitals that we learnt in class, and there is no s-p mixing.

Show **energy arrow**, **atomic orbitals** involved, **atom labels**, the **molecular orbitals** that are formed, **filled electrons** in the MO diagram etc. Make sure to label all the atomic and molecular orbitals.

Using the MO diagram above, answer the following questions:

- (i) Which one of the following will have the highest bond order? Circle the correct choice
 A) PS^{1-} B) PS C) PS^{1+}
- (ii) Which one of the following will have the longest bond length? Circle the correct choice
 A) PS^{1-} B) PS C) PS^{1+}
- (iii) Which of the following will be diamagnetic? Circle the correct choice.
 A) PS^{1-} B) PS C) PS^{1+}
- (iv) Draw Lewis dot structure of the molecule with the shortest bond length (based on the MO diagram) from among the three (PS^{1-} , PS, PS^{1+}), and indicate the bond order calculated from the Lewis dot structure. **(3 points)**

Lewis Dot Structure

Bond Order



D15C7D32-A9AA-4C7A-BD09-50F19AD5C889

midterm-exam-93388

#406 Page 10 of 10

SCRAP PAGE

**CHEMISTRY 110 –Midterm Exam****Thursday November 7, 2024, 6:30 – 8:30 pm**

Examiners: Pallavi Sirjoosingh and Ashok Kakkar

First name (please write as legibly as possible within the boxes)

--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--

Last name

--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--

Student ID number

--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--

This booklet has 8 questions on 10 pages (The last page can be used as a scrap page).

The staple must NOT be removed from the question booklet. Your intact question booklet must be returned at the end of the exam.

A separate booklet contains a periodic table, and datasheet.

Write your answers directly under the questions in the space provided. Do not write on the QR codes or near the top edges (as it may be cut off).

Please write **legibly and dark enough** for the scan to be readable. For students who use English as a second language, dictionaries are permitted. Calculators are allowed.

DO NOT ASK QUESTIONS OF THE INVIGILATORS about interpretations of the exam questions. If you have a concern about a question, please summarize it at the top of this page. All concerns will be evaluated after the exam. GOOD LUCK!

**Question 1 8 points** (*Assume all transitions are possible*)

a. 6 points Calculate the **energy (in kJ/mol)** required to ionize (completely remove) the electron from the **n=3** level of **He⁺ ion** (He: Helium). Show your work including equations. (3 sig. figs.)

b. 2 points

(I) Select the largest (longest) wavelength from the three choices below:

- i. Wavelength of photon to *completely remove* an electron from *1s orbital of Hydrogen atom*
- ii. Wavelength of photon to *completely remove* an electron from *1s orbital of Boron atom (B)*
- iii. Wavelength of photon to *completely remove* an electron from *1s orbital of Boron cation (B⁺)*

i or ii or iii

(II) Select the smallest (shortest) wavelength from the three choices below (*for H atom*):

- i. Wavelength of photon *emitted* for an electron transitioning from *2p to 1s orbital*
- ii. Wavelength of photon *absorbed* for an electron transitioning from *2s to 4p orbital*
- iii. Wavelength of photon *absorbed* for an electron transitioning from *1s to 3p orbital*

i or ii or iii



Question 2 8 points



A Chem110 student plots the graph shown on the left, from data obtained for the kinetic energy (eV) of the ejected electron versus frequency (s^{-1}) of the incident light, for an unknown metal. ($1\text{eV} = 1.6022 \times 10^{-19} \text{ J}$)

Based on the plot, answer the questions **a to c** below:

a. 1 point On the plot above, label the **threshold frequency**.

b. 2 points What is the value of the work potential (in J) of the unknown metal?

c. 3 points What is the *longest* wavelength (in nm) required to eject an electron from this metal's surface? Show your work including equation(s) used. (3 sig figs)

d. 2 points In the space provided below, plot the following

(Assume frequency is *above* threshold frequency):

- i. *Kinetic energy* of an ejected electron versus the *intensity* of the incident light (left)
- ii. *Number of electrons* ejected versus the *frequency* of the incident light (right)



**Question 3** **6 points****a. 2 points**

In the ground state of Ge (Germanium; $Z = 32$) atom, **name** the following orbitals:

i. Orbital containing the electron that requires the maximum energy to be removed.

ii. Orbital containing the electron that experiences the minimum effective nuclear charge.

b. 2 points

An unknown anion (A^-) has the following *excited* electronic configuration: $[Ar] 4s^2 3d^3 4p^1$

(Ar: Argon; $Z = 18$)

Give the **condensed ground state electronic configuration** of the atom, **A** and the cation, **A^{2+}** .
Show your work. (Assume all electronic transitions are possible)

Electronic Configuration of A:

Electronic Configuration of A^{2+} :

c. 2 points

In a single atom, determine how many electrons can have the following set of quantum numbers?

(i) $n = 5; l = 3$

In a single atom, determine how many electrons can have the following set of quantum numbers?

(i) $n = 3; l = 2; m_s = +1/2$

**Question 4** **8 points**

Rank the following from 1 to 4 in **increasing** order. Provide the ranking in the boxes provided.

Note: Atomic number (Z) are provided for relevant atoms

a) Increasing (lowest to highest) order of *ionization energy*: (1= lowest; 4 = highest)

Mg

Mg: Z = 12

Rb

Rb: Z = 37

Cs

Cs: Z = 55

K

K : Z = 19

b) Increasing (lowest to highest) order of *bond angle*: (1= lowest; 4 = highest)

H₂OCCl₄XeF₄

HCN

c) Increasing (smallest to largest) order of *ionic size* (1= smallest; 4 = largest)

S²⁻

S: Z = 16

O²⁻

O: Z = 8

Be²⁺

Be: Z = 4

Na⁺

Na: Z = 11

d) Increasing (lowest to highest) order of *bond energy* (1= lowest; 4 = highest)

P-F

P-Cl

P-I

P-Br

**Question 5** **12 points**

a. 9 points For the molecules HClO_4 and TeF_6 , draw the most stable *Lewis structure* showing appropriate shape and *dashed/wedged bonds*, based on VSEPR. Determine the **molecular geometry** around the central atom. For HClO_4 , label the *most polar bond*.



b. 3 points (Note: Lone pairs are not shown in structures below)

(i) For the three *resonance* structures (1, 2 and 3) shown below, answer the following question:



(I) Circle the *most contributing* resonance structure.

Structure 1

Structure 2

Structure 3

(ii) For the three structures given (4, 5, and 6) below, answer the following questions:



(I) Circle all the structures that represent *resonance structures* of each other:

Structure 4

Structure 5

Structure 6

(II) Among the resonance structures, circle the *most contributing* resonance structure.

Structure 4

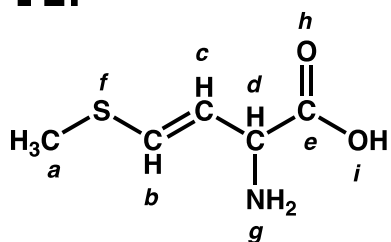
Structure 5

Structure 6

**Question 6** **6 points**

Using the data given below, calculate the **lattice energy** for SrO(s).

	ΔH° (kJ/mol)
$\text{Sr}(s) \longrightarrow \text{Sr}(g)$	+180
$\text{Sr}(g) \longrightarrow \text{Sr}^{2+}(g) + 2e^-$	+1468
$\text{O}_2(g) \longrightarrow 2\text{O}(g)$	+500
$\text{O}(g) + 2e^- \longrightarrow \text{O}^{2-}(g)$	+600
$\text{Sr}(s) + \frac{1}{2}\text{O}_2(g) \longrightarrow \text{SrO}(s)$	-550

**Question 7 10 points**

Consider the molecule given on the left, and answer questions (i) to (vi) using chemical bonding theories that we have learnt in class (valence bond theory, hybridization).

Note that the lone pairs are not shown in the structure:

i. Consider the carbon atoms marked in as ***a***, ***b***, ***c***, ***d***, and ***e*** in the molecule, and determine **hybridization** at each of the carbon centers. (2.5 points)

Atom	Hybridization
Carbon (<i>a</i>)	
Carbon (<i>b</i>)	
Carbon (<i>c</i>)	
Carbon (<i>d</i>)	
Carbon (<i>e</i>)	

ii. Now, consider the sulfur atom (S) marked as ***f***, nitrogen (N) atom (***g***), and oxygen(O) atoms (***h*** and ***i***), and determine their **hybridization**. (2 points)

Atom	Hybridization
Sulfur (<i>f</i>)	
Nitrogen (<i>g</i>)	
Oxygen (<i>h</i>)	
Oxygen (<i>i</i>)	

iii. **Describe** all the bonds formed by the carbon atoms ***a*** and ***b***, using sigma and pi designation as appropriate. (2 points)

Atom	Bonds formed (use sigma and pi as appropriate)
Carbon (<i>a</i>)	
Carbon (<i>b</i>)	

iv. How many total sp^3 -s sigma bonds are in the molecule? (1 point)

v. Determine **electronic** and **molecular geometry** at carbon (C) labeled ***e*** and sulfur (S) labeled ***f*** centres. (2 points)

Atom	Electronic Geometry	Molecular Geometry
Carbon (<i>e</i>)		
Sulfur (<i>f</i>)		

vi. Circle the value from the choices given below, that describes the **C-S-C bond angle at sulfur** (S) marked ***f***: (0.5 points)

>90° to <109.5° 109.5° >109.5 to <120° 120° >120° to <180°


Question 8 12 points

Using the **valence shell orbitals only** draw a **complete** molecular orbital (MO) diagram (with no s-p mixing) for the molecule **PS**, in the space provided below. **(6 points)**

Assume that 3s and 3p atomic orbitals will form molecular orbitals like those formed by the combination of 2s and 2p atomic orbitals that we learnt in class, and there is no s-p mixing.

Show **energy arrow**, **atomic orbitals** involved, **atom labels**, the **molecular orbitals** that are formed, **filled electrons** in the MO diagram etc. Make sure to label all the atomic and molecular orbitals.

Using the MO diagram above, answer the following questions:

- (i) Which one of the following will have the highest bond order? Circle the correct choice
 A) PS^{1-} B) PS C) PS^{1+}
- (ii) Which one of the following will have the longest bond length? Circle the correct choice
 A) PS^{1-} B) PS C) PS^{1+}
- (iii) Which of the following will be diamagnetic? Circle the correct choice.
 A) PS^{1-} B) PS C) PS^{1+}
- (iv) Draw Lewis dot structure of the molecule with the shortest bond length (based on the MO diagram) from among the three (PS^{1-} , PS, PS^{1+}), and indicate the bond order calculated from the Lewis dot structure. **(3 points)**

Lewis Dot Structure

Bond Order



C675870A-6035-44F6-AC1E-32A29EFB181F

midterm-exam-93388

#407 Page 10 of 10

SCRAP PAGE

**CHEMISTRY 110 –Midterm Exam****Thursday November 7, 2024, 6:30 – 8:30 pm****Examiners: Pallavi Sirjoosingh and Ashok Kakkar**

First name (please write as legibly as possible within the boxes)

--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--

Last name

--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--

Student ID number

--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--

This booklet has 8 questions on 10 pages (The last page can be used as a scrap page).

The staple must NOT be removed from the question booklet. Your intact question booklet must be returned at the end of the exam.

A separate booklet contains a periodic table, and datasheet.

Write your answers directly under the questions in the space provided. Do not write on the QR codes or near the top edges (as it may be cut off).

Please write **legibly and dark enough** for the scan to be readable. For students who use English as a second language, dictionaries are permitted. Calculators are allowed.

DO NOT ASK QUESTIONS OF THE INVIGILATORS about interpretations of the exam questions. If you have a concern about a question, please summarize it at the top of this page. All concerns will be evaluated after the exam. GOOD LUCK!

**Question 1 8 points** (*Assume all transitions are possible*)

a. 6 points Calculate the **energy (in kJ/mol)** required to ionize (completely remove) the electron from the **$n=3$** level of **He^+ ion** (He: Helium). Show your work including equations. (3 sig. figs.)

b. 2 points

(I) Select the largest (longest) wavelength from the three choices below:

- i. Wavelength of photon to *completely remove* an electron from *1s orbital of Hydrogen atom*
- ii. Wavelength of photon to *completely remove* an electron from *1s orbital of Boron atom (B)*
- iii. Wavelength of photon to *completely remove* an electron from *1s orbital of Boron cation (B^+)*

i or ii or iii

(II) Select the smallest (shortest) wavelength from the three choices below (*for H atom*):

- i. Wavelength of photon *emitted* for an electron transitioning from *2p to 1s orbital*
- ii. Wavelength of photon *absorbed* for an electron transitioning from *2s to 4p orbital*
- iii. Wavelength of photon *absorbed* for an electron transitioning from *1s to 3p orbital*

i or ii or iii



Question 2 8 points



A Chem110 student plots the graph shown on the left, from data obtained for the kinetic energy (eV) of the ejected electron versus frequency (s^{-1}) of the incident light, for an unknown metal. ($1\text{eV} = 1.6022 \times 10^{-19} \text{ J}$)

Based on the plot, answer the questions **a to c** below:

a. 1 point On the plot above, label the **threshold frequency**.

b. 2 points What is the value of the work potential (in J) of the unknown metal?

c. 3 points What is the *longest* wavelength (in nm) required to eject an electron from this metal's surface? Show your work including equation(s) used. (3 sig figs)

d. 2 points In the space provided below, plot the following

(Assume frequency is *above* threshold frequency):

- i. *Kinetic energy* of an ejected electron versus the *intensity* of the incident light (left)
- ii. *Number of electrons* ejected versus the *frequency* of the incident light (right)



**Question 3** **6 points****a. 2 points**

In the ground state of Ge (Germanium; $Z = 32$) atom, **name** the following orbitals:

i. Orbital containing the electron that requires the maximum energy to be removed.

ii. Orbital containing the electron that experiences the minimum effective nuclear charge.

b. 2 points

An unknown anion (A^-) has the following *excited* electronic configuration: $[Ar] 4s^2 3d^3 4p^1$

(Ar: Argon; $Z = 18$)

Give the **condensed ground state electronic configuration** of the atom, **A** and the cation, **A^{2+}** .
Show your work. (Assume all electronic transitions are possible)

Electronic Configuration of A:

Electronic Configuration of A^{2+} :

c. 2 points

In a single atom, determine how many electrons can have the following set of quantum numbers?

(i) $n = 5; l = 3$

In a single atom, determine how many electrons can have the following set of quantum numbers?

(i) $n = 3; l = 2; m_s = +1/2$

**Question 4** **8 points**

Rank the following from 1 to 4 in **increasing** order. Provide the ranking in the boxes provided.

Note: Atomic number (Z) are provided for relevant atoms

a) Increasing (lowest to highest) order of *ionization energy*: (1= lowest; 4 = highest)

Mg

Mg: Z = 12

Rb

Rb: Z = 37

Cs

Cs: Z = 55

K

K : Z = 19

b) Increasing (lowest to highest) order of *bond angle*: (1= lowest; 4 = highest)

H₂OCCl₄XeF₄

HCN

c) Increasing (smallest to largest) order of *ionic size* (1= smallest; 4 = largest)

S²⁻

S: Z = 16

O²⁻

O: Z = 8

Be²⁺

Be: Z = 4

Na⁺

Na: Z = 11

d) Increasing (lowest to highest) order of *bond energy* (1= lowest; 4 = highest)

P-F

P-Cl

P-I

P-Br

**Question 5** **12 points**

a. 9 points For the molecules HClO_4 and TeF_6 , draw the most stable *Lewis structure* showing appropriate shape and *dashed/wedged bonds*, based on VSEPR. Determine the **molecular geometry** around the central atom. For HClO_4 , label the *most polar bond*.



b. 3 points (Note: Lone pairs are not shown in structures below)

(i) For the three *resonance* structures (1, 2 and 3) shown below, answer the following question:



(I) Circle the *most contributing* resonance structure.

Structure 1

Structure 2

Structure 3

(ii) For the three structures given (4, 5, and 6) below, answer the following questions:



(I) Circle all the structures that represent *resonance structures* of each other:

Structure 4

Structure 5

Structure 6

(II) Among the resonance structures, circle the *most contributing* resonance structure.

Structure 4

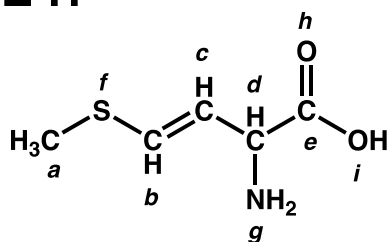
Structure 5

Structure 6

**Question 6** **6 points**

Using the data given below, calculate the **lattice energy** for SrO(s).

	ΔH° (kJ/mol)
$\text{Sr}(s) \longrightarrow \text{Sr}(g)$	+180
$\text{Sr}(g) \longrightarrow \text{Sr}^{2+}(g) + 2e^-$	+1468
$\text{O}_2(g) \longrightarrow 2\text{O}(g)$	+500
$\text{O}(g) + 2e^- \longrightarrow \text{O}^{2-}(g)$	+600
$\text{Sr}(s) + \frac{1}{2}\text{O}_2(g) \longrightarrow \text{SrO}(s)$	-550

**Question 7 10 points**

Consider the molecule given on the left, and answer questions (i) to (vi) using chemical bonding theories that we have learnt in class (valence bond theory, hybridization).

Note that the lone pairs are not shown in the structure:

i. Consider the carbon atoms marked in as ***a***, ***b***, ***c***, ***d***, and ***e*** in the molecule, and determine **hybridization** at each of the carbon centers. (2.5 points)

Atom	Hybridization
Carbon (<i>a</i>)	
Carbon (<i>b</i>)	
Carbon (<i>c</i>)	
Carbon (<i>d</i>)	
Carbon (<i>e</i>)	

ii. Now, consider the sulfur atom (S) marked as ***f***, nitrogen (N) atom (***g***), and oxygen(O) atoms (***h*** and ***i***), and determine their **hybridization**. (2 points)

Atom	Hybridization
Sulfur (<i>f</i>)	
Nitrogen (<i>g</i>)	
Oxygen (<i>h</i>)	
Oxygen (<i>i</i>)	

iii. **Describe** all the bonds formed by the carbon atoms ***a*** and ***b***, using sigma and pi designation as appropriate. (2 points)

Atom	Bonds formed (use sigma and pi as appropriate)
Carbon (<i>a</i>)	
Carbon (<i>b</i>)	

iv. How many total sp^3 -s sigma bonds are in the molecule? (1 point)

v. Determine **electronic** and **molecular geometry** at carbon (C) labeled ***e*** and sulfur (S) labeled ***f*** centres. (2 points)

Atom	Electronic Geometry	Molecular Geometry
Carbon (<i>e</i>)		
Sulfur (<i>f</i>)		

vi. Circle the value from the choices given below, that describes the **C-S-C bond angle at sulfur** (S) marked ***f***: (0.5 points)

>90° to <109.5°

109.5°

>109.5 to <120°

120°

>120° to <180°


Question 8 12 points

Using the **valence shell orbitals only** draw a **complete** molecular orbital (MO) diagram (with no s-p mixing) for the molecule **PS**, in the space provided below. **(6 points)**

Assume that 3s and 3p atomic orbitals will form molecular orbitals like those formed by the combination of 2s and 2p atomic orbitals that we learnt in class, and there is no s-p mixing.

Show **energy arrow**, **atomic orbitals** involved, **atom labels**, the **molecular orbitals** that are formed, **filled electrons** in the MO diagram etc. Make sure to label all the atomic and molecular orbitals.

Using the MO diagram above, answer the following questions:

- (i) Which one of the following will have the highest bond order? Circle the correct choice
 A) PS^{1-} B) PS C) PS^{1+}
- (ii) Which one of the following will have the longest bond length? Circle the correct choice
 A) PS^{1-} B) PS C) PS^{1+}
- (iii) Which of the following will be diamagnetic? Circle the correct choice.
 A) PS^{1-} B) PS C) PS^{1+}
- (iv) Draw Lewis dot structure of the molecule with the shortest bond length (based on the MO diagram) from among the three (PS^{1-} , PS, PS^{1+}), and indicate the bond order calculated from the Lewis dot structure. **(3 points)**

Lewis Dot Structure

Bond Order



579B5F9C-8F44-4FA3-9851-C6E2DAEA08BD

midterm-exam-93388

#408 Page 10 of 10

SCRAP PAGE

**CHEMISTRY 110 –Midterm Exam****Thursday November 7, 2024, 6:30 – 8:30 pm**

Examiners: Pallavi Sirjoosingh and Ashok Kakkar

First name (please write as legibly as possible within the boxes)

--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--

Last name

--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--

Student ID number

--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--

This booklet has 8 questions on 10 pages (The last page can be used as a scrap page).

The staple must NOT be removed from the question booklet. Your intact question booklet must be returned at the end of the exam.

A separate booklet contains a periodic table, and datasheet.

Write your answers directly under the questions in the space provided. Do not write on the QR codes or near the top edges (as it may be cut off).

Please write **legibly and dark enough** for the scan to be readable. For students who use English as a second language, dictionaries are permitted. Calculators are allowed.

DO NOT ASK QUESTIONS OF THE INVIGILATORS about interpretations of the exam questions. If you have a concern about a question, please summarize it at the top of this page. All concerns will be evaluated after the exam. GOOD LUCK!

**Question 1 8 points** (*Assume all transitions are possible*)

a. 6 points Calculate the **energy (in kJ/mol)** required to ionize (completely remove) the electron from the **n=3** level of **He⁺ ion** (He: Helium). Show your work including equations. (3 sig. figs.)

b. 2 points

(I) Select the largest (longest) wavelength from the three choices below:

- i. Wavelength of photon to *completely remove* an electron from *1s orbital of Hydrogen atom*
- ii. Wavelength of photon to *completely remove* an electron from *1s orbital of Boron atom (B)*
- iii. Wavelength of photon to *completely remove* an electron from *1s orbital of Boron cation (B⁺)*

i or ii or iii

(II) Select the smallest (shortest) wavelength from the three choices below (*for H atom*):

- i. Wavelength of photon *emitted* for an electron transitioning from *2p to 1s orbital*
- ii. Wavelength of photon *absorbed* for an electron transitioning from *2s to 4p orbital*
- iii. Wavelength of photon *absorbed* for an electron transitioning from *1s to 3p orbital*

i or ii or iii



Question 2 8 points



A Chem110 student plots the graph shown on the left, from data obtained for the kinetic energy (eV) of the ejected electron versus frequency (s^{-1}) of the incident light, for an unknown metal. ($1\text{eV} = 1.6022 \times 10^{-19} \text{ J}$)

Based on the plot, answer the questions **a to c** below:

a. 1 point On the plot above, label the **threshold frequency**.

b. 2 points What is the value of the work potential (in J) of the unknown metal?

c. 3 points What is the *longest* wavelength (in nm) required to eject an electron from this metal's surface? Show your work including equation(s) used. (3 sig figs)

d. 2 points In the space provided below, plot the following

(Assume frequency is *above* threshold frequency):

- Kinetic energy* of an ejected electron versus the *intensity* of the incident light (left)
- Number of electrons* ejected versus the *frequency* of the incident light (right)



**Question 3** **6 points****a. 2 points**

In the ground state of Ge (Germanium; $Z = 32$) atom, **name** the following orbitals:

i. Orbital containing the electron that requires the maximum energy to be removed.

ii. Orbital containing the electron that experiences the minimum effective nuclear charge.

b. 2 points

An unknown anion (A^-) has the following *excited* electronic configuration: $[Ar] 4s^2 3d^3 4p^1$

(Ar: Argon; $Z = 18$)

Give the **condensed ground state electronic configuration** of the atom, **A** and the cation, **A^{2+}** .
Show your work. (Assume all electronic transitions are possible)

Electronic Configuration of A:

Electronic Configuration of A^{2+} :

c. 2 points

In a single atom, determine how many electrons can have the following set of quantum numbers?

(i) $n = 5; l = 3$

In a single atom, determine how many electrons can have the following set of quantum numbers?

(i) $n = 3; l = 2; m_s = +1/2$

**Question 4** **8 points**

Rank the following from 1 to 4 in **increasing** order. Provide the ranking in the boxes provided.

Note: Atomic number (Z) are provided for relevant atoms

a) Increasing (lowest to highest) order of *ionization energy*: (1= lowest; 4 = highest)

Mg

Mg: Z = 12

Rb

Rb: Z = 37

Cs

Cs: Z = 55

K

K : Z = 19

b) Increasing (lowest to highest) order of *bond angle*: (1= lowest; 4 = highest)

H₂OCCl₄XeF₄

HCN

c) Increasing (smallest to largest) order of *ionic size* (1= smallest; 4 = largest)

S²⁻

S: Z = 16

O²⁻

O: Z = 8

Be²⁺

Be: Z = 4

Na⁺

Na: Z = 11

d) Increasing (lowest to highest) order of *bond energy* (1= lowest; 4 = highest)

P-F

P-Cl

P-I

P-Br

**Question 5** **12 points**

a. 9 points For the molecules HClO_4 and TeF_6 , draw the most stable *Lewis structure* showing appropriate shape and *dashed/wedged bonds*, based on VSEPR. Determine the **molecular geometry** around the central atom. For HClO_4 , label the *most polar bond*.

HClO_4 (there are no O-O bonds)

Label the most polar bond

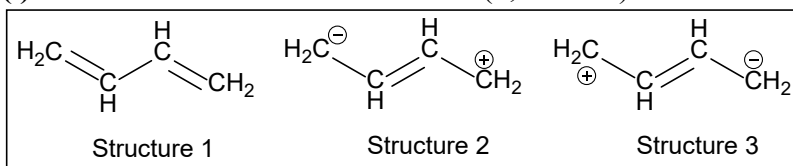
Molecular Geometry around Cl

TeF_6

Molecular Geometry around Te

b. 3 points (Note: Lone pairs are not shown in structures below)

(i) For the three *resonance* structures (1, 2 and 3) shown below, answer the following question:



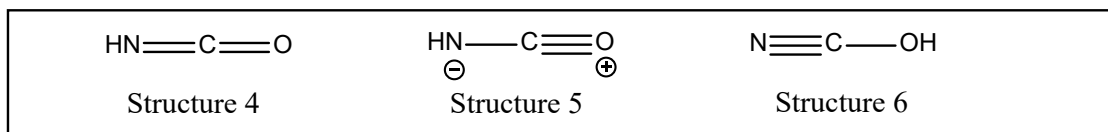
(I) Circle the *most contributing* resonance structure.

Structure 1

Structure 2

Structure 3

(ii) For the three structures given (4, 5, and 6) below, answer the following questions:



(I) Circle all the structures that represent *resonance structures* of each other:

Structure 4

Structure 5

Structure 6

(II) Among the resonance structures, circle the *most contributing* resonance structure.

Structure 4

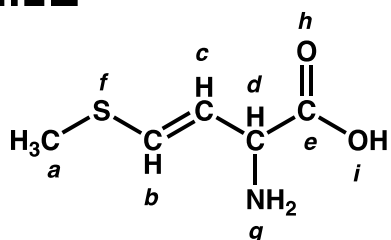
Structure 5

Structure 6

**Question 6** **6 points**

Using the data given below, calculate the **lattice energy** for SrO(s).

	ΔH° (kJ/mol)
$\text{Sr}(s) \longrightarrow \text{Sr}(g)$	+180
$\text{Sr}(g) \longrightarrow \text{Sr}^{2+}(g) + 2e^-$	+1468
$\text{O}_2(g) \longrightarrow 2\text{O}(g)$	+500
$\text{O}(g) + 2e^- \longrightarrow \text{O}^{2-}(g)$	+600
$\text{Sr}(s) + \frac{1}{2}\text{O}_2(g) \longrightarrow \text{SrO}(s)$	-550

**Question 7 10 points**

Consider the molecule given on the left, and answer questions (i) to (vi) using chemical bonding theories that we have learnt in class (valence bond theory, hybridization).

Note that the lone pairs are not shown in the structure:

i. Consider the carbon atoms marked in as ***a***, ***b***, ***c***, ***d***, and ***e*** in the molecule, and determine **hybridization** at each of the carbon centers. (2.5 points)

Atom	Hybridization
Carbon (<i>a</i>)	
Carbon (<i>b</i>)	
Carbon (<i>c</i>)	
Carbon (<i>d</i>)	
Carbon (<i>e</i>)	

ii. Now, consider the sulfur atom (S) marked as ***f***, nitrogen (N) atom (***g***), and oxygen(O) atoms (***h*** and ***i***), and determine their **hybridization**. (2 points)

Atom	Hybridization
Sulfur (<i>f</i>)	
Nitrogen (<i>g</i>)	
Oxygen (<i>h</i>)	
Oxygen (<i>i</i>)	

iii. **Describe** all the bonds formed by the carbon atoms ***a*** and ***b***, using sigma and pi designation as appropriate. (2 points)

Atom	Bonds formed (use sigma and pi as appropriate)
Carbon (<i>a</i>)	
Carbon (<i>b</i>)	

iv. How many total sp^3 -s sigma bonds are in the molecule? (1 point)

v. Determine **electronic** and **molecular geometry** at carbon (C) labeled ***e*** and sulfur (S) labeled ***f*** centres. (2 points)

Atom	Electronic Geometry	Molecular Geometry
Carbon (<i>e</i>)		
Sulfur (<i>f</i>)		

vi. Circle the value from the choices given below, that describes the **C-S-C bond angle at sulfur** (S) marked ***f***: (0.5 points)

>90° to <109.5° 109.5° >109.5 to <120° 120° >120° to <180°


Question 8 12 points

Using the **valence shell orbitals only** draw a **complete** molecular orbital (MO) diagram (with no s-p mixing) for the molecule **PS**, in the space provided below. **(6 points)**

Assume that 3s and 3p atomic orbitals will form molecular orbitals like those formed by the combination of 2s and 2p atomic orbitals that we learnt in class, and there is no s-p mixing.

Show **energy arrow**, **atomic orbitals** involved, **atom labels**, the **molecular orbitals** that are formed, **filled electrons** in the MO diagram etc. Make sure to label all the atomic and molecular orbitals.

Using the MO diagram above, answer the following questions:

- (i) Which one of the following will have the highest bond order? Circle the correct choice
 A) PS^{1-} B) PS C) PS^{1+}
- (ii) Which one of the following will have the longest bond length? Circle the correct choice
 A) PS^{1-} B) PS C) PS^{1+}
- (iii) Which of the following will be diamagnetic? Circle the correct choice.
 A) PS^{1-} B) PS C) PS^{1+}
- (iv) Draw Lewis dot structure of the molecule with the shortest bond length (based on the MO diagram) from among the three (PS^{1-} , PS, PS^{1+}), and indicate the bond order calculated from the Lewis dot structure. **(3 points)**

Lewis Dot Structure

Bond Order



12CCF460-0DF5-4A0E-9864-DCD91F2747B2

midterm-exam-93388

#409 Page 10 of 10

SCRAP PAGE

**CHEMISTRY 110 –Midterm Exam****Thursday November 7, 2024, 6:30 – 8:30 pm**

Examiners: Pallavi Sirjoosingh and Ashok Kakkar

First name (please write as legibly as possible within the boxes)

--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--

Last name

--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--

Student ID number

--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--

This booklet has 8 questions on 10 pages (The last page can be used as a scrap page).

The staple must NOT be removed from the question booklet. Your intact question booklet must be returned at the end of the exam.

A separate booklet contains a periodic table, and datasheet.

Write your answers directly under the questions in the space provided. Do not write on the QR codes or near the top edges (as it may be cut off).

Please write **legibly and dark enough** for the scan to be readable. For students who use English as a second language, dictionaries are permitted. Calculators are allowed.

DO NOT ASK QUESTIONS OF THE INVIGILATORS about interpretations of the exam questions. If you have a concern about a question, please summarize it at the top of this page. All concerns will be evaluated after the exam. GOOD LUCK!

**Question 1 8 points** (*Assume all transitions are possible*)

a. 6 points Calculate the **energy (in kJ/mol)** required to ionize (completely remove) the electron from the **$n=3$** level of **He^+ ion** (He: Helium). Show your work including equations. (3 sig. figs.)

b. 2 points

(I) Select the largest (longest) wavelength from the three choices below:

- i. Wavelength of photon to *completely remove* an electron from *1s orbital of Hydrogen atom*
- ii. Wavelength of photon to *completely remove* an electron from *1s orbital of Boron atom (B)*
- iii. Wavelength of photon to *completely remove* an electron from *1s orbital of Boron cation (B^+)*

i or ii or iii

(II) Select the smallest (shortest) wavelength from the three choices below (*for H atom*):

- i. Wavelength of photon *emitted* for an electron transitioning from *2p to 1s orbital*
- ii. Wavelength of photon *absorbed* for an electron transitioning from *2s to 4p orbital*
- iii. Wavelength of photon *absorbed* for an electron transitioning from *1s to 3p orbital*

i or ii or iii



Question 2 8 points



A Chem110 student plots the graph shown on the left, from data obtained for the kinetic energy (eV) of the ejected electron versus frequency (s^{-1}) of the incident light, for an unknown metal. ($1\text{eV} = 1.6022 \times 10^{-19} \text{ J}$)

Based on the plot, answer the questions **a to c** below:

a. 1 point On the plot above, label the **threshold frequency**.

b. 2 points What is the value of the work potential (in J) of the unknown metal?

c. 3 points What is the *longest* wavelength (in nm) required to eject an electron from this metal's surface? Show your work including equation(s) used. (3 sig figs)

d. 2 points In the space provided below, plot the following

(Assume frequency is *above* threshold frequency):

- i. *Kinetic energy* of an ejected electron versus the *intensity* of the incident light (left)
- ii. *Number of electrons* ejected versus the *frequency* of the incident light (right)



**Question 3** **6 points****a. 2 points**

In the ground state of Ge (Germanium; $Z = 32$) atom, **name** the following orbitals:

i. Orbital containing the electron that requires the maximum energy to be removed.

ii. Orbital containing the electron that experiences the minimum effective nuclear charge.

b. 2 points

An unknown anion (A^-) has the following *excited* electronic configuration: $[Ar] 4s^2 3d^3 4p^1$

(Ar: Argon; $Z = 18$)

Give the **condensed ground state electronic configuration** of the atom, **A** and the cation, **A^{2+}** .
Show your work. (Assume all electronic transitions are possible)

Electronic Configuration of A:

Electronic Configuration of A^{2+} :

c. 2 points

In a single atom, determine how many electrons can have the following set of quantum numbers?

(i) $n = 5; l = 3$

In a single atom, determine how many electrons can have the following set of quantum numbers?

(i) $n = 3; l = 2; m_s = +1/2$

**Question 4** **8 points**

Rank the following from 1 to 4 in **increasing** order. Provide the ranking in the boxes provided.

Note: Atomic number (Z) are provided for relevant atoms

a) Increasing (lowest to highest) order of *ionization energy*: (1= lowest; 4 = highest)

Mg

Mg: Z = 12

Rb

Rb: Z = 37

Cs

Cs: Z = 55

K

K : Z = 19

b) Increasing (lowest to highest) order of *bond angle*: (1= lowest; 4 = highest)

H₂OCCl₄XeF₄

HCN

c) Increasing (smallest to largest) order of *ionic size* (1= smallest; 4 = largest)

S²⁻

S: Z = 16

O²⁻

O: Z = 8

Be²⁺

Be: Z = 4

Na⁺

Na: Z = 11

d) Increasing (lowest to highest) order of *bond energy* (1= lowest; 4 = highest)

P-F

P-Cl

P-I

P-Br

**Question 5** **12 points**

a. 9 points For the molecules HClO_4 and TeF_6 , draw the most stable *Lewis structure* showing appropriate shape and *dashed/wedged bonds*, based on VSEPR. Determine the **molecular geometry** around the central atom. For HClO_4 , label the *most polar bond*.



b. 3 points (Note: Lone pairs are not shown in structures below)

(i) For the three *resonance* structures (1, 2 and 3) shown below, answer the following question:



(I) Circle the *most contributing* resonance structure.

Structure 1

Structure 2

Structure 3

(ii) For the three structures given (4, 5, and 6) below, answer the following questions:



(I) Circle all the structures that represent *resonance structures* of each other:

Structure 4

Structure 5

Structure 6

(II) Among the resonance structures, circle the *most contributing* resonance structure.

Structure 4

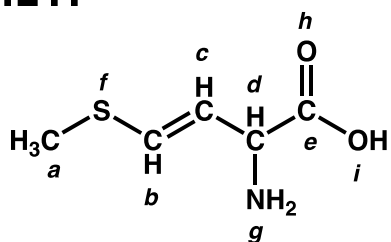
Structure 5

Structure 6

**Question 6** **6 points**

Using the data given below, calculate the **lattice energy** for SrO(s).

	ΔH° (kJ/mol)
$\text{Sr}(s) \longrightarrow \text{Sr}(g)$	+180
$\text{Sr}(g) \longrightarrow \text{Sr}^{2+}(g) + 2e^-$	+1468
$\text{O}_2(g) \longrightarrow 2\text{O}(g)$	+500
$\text{O}(g) + 2e^- \longrightarrow \text{O}^{2-}(g)$	+600
$\text{Sr}(s) + \frac{1}{2}\text{O}_2(g) \longrightarrow \text{SrO}(s)$	-550

**Question 7 10 points**

Consider the molecule given on the left, and answer questions (i) to (vi) using chemical bonding theories that we have learnt in class (valence bond theory, hybridization).

Note that the lone pairs are not shown in the structure:

i. Consider the carbon atoms marked in as ***a***, ***b***, ***c***, ***d***, and ***e*** in the molecule, and determine **hybridization** at each of the carbon centers. (2.5 points)

Atom	Hybridization
Carbon (<i>a</i>)	
Carbon (<i>b</i>)	
Carbon (<i>c</i>)	
Carbon (<i>d</i>)	
Carbon (<i>e</i>)	

ii. Now, consider the sulfur atom (S) marked as ***f***, nitrogen (N) atom (***g***), and oxygen(O) atoms (***h*** and ***i***), and determine their **hybridization**. (2 points)

Atom	Hybridization
Sulfur (<i>f</i>)	
Nitrogen (<i>g</i>)	
Oxygen (<i>h</i>)	
Oxygen (<i>i</i>)	

iii. **Describe** all the bonds formed by the carbon atoms ***a*** and ***b***, using sigma and pi designation as appropriate. (2 points)

Atom	Bonds formed (use sigma and pi as appropriate)
Carbon (<i>a</i>)	
Carbon (<i>b</i>)	

iv. How many total sp^3 -s sigma bonds are in the molecule? (1 point)

v. Determine **electronic** and **molecular geometry** at carbon (C) labeled ***e*** and sulfur (S) labeled ***f*** centres. (2 points)

Atom	Electronic Geometry	Molecular Geometry
Carbon (<i>e</i>)		
Sulfur (<i>f</i>)		

vi. Circle the value from the choices given below, that describes the **C-S-C bond angle at sulfur** (S) marked ***f***: (0.5 points)

>90° to <109.5° 109.5° >109.5 to <120° 120° >120° to <180°


Question 8 12 points

Using the **valence shell orbitals only** draw a **complete** molecular orbital (MO) diagram (with no s-p mixing) for the molecule **PS**, in the space provided below. **(6 points)**

Assume that 3s and 3p atomic orbitals will form molecular orbitals like those formed by the combination of 2s and 2p atomic orbitals that we learnt in class, and there is no s-p mixing.

Show **energy arrow**, **atomic orbitals** involved, **atom labels**, the **molecular orbitals** that are formed, **filled electrons** in the MO diagram etc. Make sure to label all the atomic and molecular orbitals.

Using the MO diagram above, answer the following questions:

- (i) Which one of the following will have the highest bond order? Circle the correct choice
 A) PS^{1-} B) PS C) PS^{1+}
- (ii) Which one of the following will have the longest bond length? Circle the correct choice
 A) PS^{1-} B) PS C) PS^{1+}
- (iii) Which of the following will be diamagnetic? Circle the correct choice.
 A) PS^{1-} B) PS C) PS^{1+}
- (iv) Draw Lewis dot structure of the molecule with the shortest bond length (based on the MO diagram) from among the three (PS^{1-} , PS, PS^{1+}), and indicate the bond order calculated from the Lewis dot structure. **(3 points)**

Lewis Dot Structure

Bond Order



D9B8A91F-7AA4-4400-92CA-44220568DB9F

midterm-exam-93388

#410 Page 10 of 10

SCRAP PAGE

**CHEMISTRY 110 –Midterm Exam****Thursday November 7, 2024, 6:30 – 8:30 pm**

Examiners: Pallavi Sirjoosingh and Ashok Kakkar

First name (please write as legibly as possible within the boxes)

--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--

Last name

--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--

Student ID number

--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--

This booklet has 8 questions on 10 pages (The last page can be used as a scrap page).

The staple must NOT be removed from the question booklet. Your intact question booklet must be returned at the end of the exam.

A separate booklet contains a periodic table, and datasheet.

Write your answers directly under the questions in the space provided. Do not write on the QR codes or near the top edges (as it may be cut off).

Please write **legibly and dark enough** for the scan to be readable. For students who use English as a second language, dictionaries are permitted. Calculators are allowed.

DO NOT ASK QUESTIONS OF THE INVIGILATORS about interpretations of the exam questions. If you have a concern about a question, please summarize it at the top of this page. All concerns will be evaluated after the exam. GOOD LUCK!

**Question 1 8 points** (*Assume all transitions are possible*)

a. 6 points Calculate the **energy (in kJ/mol)** required to ionize (completely remove) the electron from the **n=3** level of **He⁺ ion** (He: Helium). Show your work including equations. (3 sig. figs.)

b. 2 points

(I) Select the largest (longest) wavelength from the three choices below:

- i. Wavelength of photon to *completely remove* an electron from *1s orbital of Hydrogen atom*
- ii. Wavelength of photon to *completely remove* an electron from *1s orbital of Boron atom (B)*
- iii. Wavelength of photon to *completely remove* an electron from *1s orbital of Boron cation (B⁺)*

i or ii or iii

(II) Select the smallest (shortest) wavelength from the three choices below (*for H atom*):

- i. Wavelength of photon *emitted* for an electron transitioning from *2p to 1s orbital*
- ii. Wavelength of photon *absorbed* for an electron transitioning from *2s to 4p orbital*
- iii. Wavelength of photon *absorbed* for an electron transitioning from *1s to 3p orbital*

i or ii or iii



Question 2 8 points



A Chem110 student plots the graph shown on the left, from data obtained for the kinetic energy (eV) of the ejected electron versus frequency (s^{-1}) of the incident light, for an unknown metal. ($1\text{eV} = 1.6022 \times 10^{-19} \text{ J}$)

Based on the plot, answer the questions **a to c** below:

a. 1 point On the plot above, label the **threshold frequency**.

b. 2 points What is the value of the work potential (in J) of the unknown metal?

c. 3 points What is the *longest* wavelength (in nm) required to eject an electron from this metal's surface? Show your work including equation(s) used. (3 sig figs)

d. 2 points In the space provided below, plot the following

(Assume frequency is *above* threshold frequency):

- Kinetic energy* of an ejected electron versus the *intensity* of the incident light (left)
- Number of electrons* ejected versus the *frequency* of the incident light (right)



**Question 3** **6 points****a. 2 points**

In the ground state of Ge (Germanium; $Z = 32$) atom, **name** the following orbitals:

i. Orbital containing the electron that requires the maximum energy to be removed.

ii. Orbital containing the electron that experiences the minimum effective nuclear charge.

b. 2 points

An unknown anion (A^-) has the following *excited* electronic configuration: $[Ar] 4s^2 3d^3 4p^1$

(Ar: Argon; $Z = 18$)

Give the **condensed ground state electronic configuration** of the atom, **A** and the cation, **A^{2+}** .
Show your work. (Assume all electronic transitions are possible)

Electronic Configuration of A:

Electronic Configuration of A^{2+} :

c. 2 points

In a single atom, determine how many electrons can have the following set of quantum numbers?

(i) $n = 5; l = 3$

In a single atom, determine how many electrons can have the following set of quantum numbers?

(i) $n = 3; l = 2; m_s = +1/2$

**Question 4** **8 points**

Rank the following from 1 to 4 in **increasing** order. Provide the ranking in the boxes provided.

Note: Atomic number (Z) are provided for relevant atoms

a) Increasing (lowest to highest) order of *ionization energy*: (1= lowest; 4 = highest)

Mg

Mg: Z = 12

Rb

Rb: Z = 37

Cs

Cs: Z = 55

K

K : Z = 19

b) Increasing (lowest to highest) order of *bond angle*: (1= lowest; 4 = highest)

H₂OCCl₄XeF₄

HCN

c) Increasing (smallest to largest) order of *ionic size* (1= smallest; 4 = largest)

S²⁻

S: Z = 16

O²⁻

O: Z = 8

Be²⁺

Be: Z = 4

Na⁺

Na: Z = 11

d) Increasing (lowest to highest) order of *bond energy* (1= lowest; 4 = highest)

P-F

P-Cl

P-I

P-Br

**Question 5** **12 points**

a. 9 points For the molecules HClO_4 and TeF_6 , draw the most stable *Lewis structure* showing appropriate shape and *dashed/wedged bonds*, based on VSEPR. Determine the **molecular geometry** around the central atom. For HClO_4 , label the *most polar bond*.

HClO_4 (there are no O-O bonds)

Label the most polar bond

Molecular Geometry around Cl

TeF_6

Molecular Geometry around Te

b. 3 points (Note: Lone pairs are not shown in structures below)

(i) For the three *resonance* structures (1, 2 and 3) shown below, answer the following question:



(I) Circle the *most contributing* resonance structure.

Structure 1

Structure 2

Structure 3

(ii) For the three structures given (4, 5, and 6) below, answer the following questions:



(I) Circle all the structures that represent *resonance structures* of each other:

Structure 4

Structure 5

Structure 6

(II) Among the resonance structures, circle the *most contributing* resonance structure.

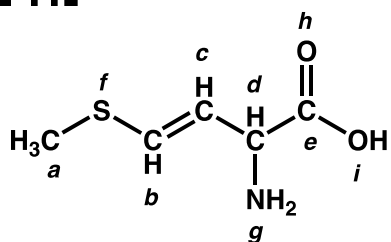
Structure 4

Structure 5

Structure 6

**Question 6** **6 points**Using the data given below, calculate the **lattice energy** for SrO(s).

	ΔH° (kJ/mol)
$\text{Sr}(s) \longrightarrow \text{Sr}(g)$	+180
$\text{Sr}(g) \longrightarrow \text{Sr}^{2+}(g) + 2e^-$	+1468
$\text{O}_2(g) \longrightarrow 2\text{O}(g)$	+500
$\text{O}(g) + 2e^- \longrightarrow \text{O}^{2-}(g)$	+600
$\text{Sr}(s) + \frac{1}{2}\text{O}_2(g) \longrightarrow \text{SrO}(s)$	-550

**Question 7 10 points**

Consider the molecule given on the left, and answer questions (i) to (vi) using chemical bonding theories that we have learnt in class (valence bond theory, hybridization).

Note that the lone pairs are not shown in the structure:

i. Consider the carbon atoms marked in as ***a***, ***b***, ***c***, ***d***, and ***e*** in the molecule, and determine **hybridization** at each of the carbon centers. (2.5 points)

Atom	Hybridization
Carbon (<i>a</i>)	
Carbon (<i>b</i>)	
Carbon (<i>c</i>)	
Carbon (<i>d</i>)	
Carbon (<i>e</i>)	

ii. Now, consider the sulfur atom (S) marked as ***f***, nitrogen (N) atom (***g***), and oxygen(O) atoms (***h*** and ***i***), and determine their **hybridization**. (2 points)

Atom	Hybridization
Sulfur (<i>f</i>)	
Nitrogen (<i>g</i>)	
Oxygen (<i>h</i>)	
Oxygen (<i>i</i>)	

iii. **Describe** all the bonds formed by the carbon atoms ***a*** and ***b***, using sigma and pi designation as appropriate. (2 points)

Atom	Bonds formed (use sigma and pi as appropriate)
Carbon (<i>a</i>)	
Carbon (<i>b</i>)	

iv. How many total sp^3 -s sigma bonds are in the molecule? (1 point)

v. Determine **electronic** and **molecular geometry** at carbon (C) labeled ***e*** and sulfur (S) labeled ***f*** centres. (2 points)

Atom	Electronic Geometry	Molecular Geometry
Carbon (<i>e</i>)		
Sulfur (<i>f</i>)		

vi. Circle the value from the choices given below, that describes the **C-S-C bond angle at sulfur** (S) marked ***f***: (0.5 points)

>90° to <109.5° 109.5° >109.5 to <120° 120° >120° to <180°


Question 8 12 points

Using the **valence shell orbitals only** draw a **complete** molecular orbital (MO) diagram (with no s-p mixing) for the molecule **PS**, in the space provided below. **(6 points)**

Assume that 3s and 3p atomic orbitals will form molecular orbitals like those formed by the combination of 2s and 2p atomic orbitals that we learnt in class, and there is no s-p mixing.

Show **energy arrow**, **atomic orbitals** involved, **atom labels**, the **molecular orbitals** that are formed, **filled electrons** in the MO diagram etc. Make sure to label all the atomic and molecular orbitals.

Using the MO diagram above, answer the following questions:

- (i) Which one of the following will have the highest bond order? Circle the correct choice
 A) PS^{1-} B) PS C) PS^{1+}
- (ii) Which one of the following will have the longest bond length? Circle the correct choice
 A) PS^{1-} B) PS C) PS^{1+}
- (iii) Which of the following will be diamagnetic? Circle the correct choice.
 A) PS^{1-} B) PS C) PS^{1+}
- (iv) Draw Lewis dot structure of the molecule with the shortest bond length (based on the MO diagram) from among the three (PS^{1-} , PS, PS^{1+}), and indicate the bond order calculated from the Lewis dot structure. **(3 points)**

Lewis Dot Structure

Bond Order



276021EA-08F5-4A4A-85EC-ACCF1B06EA10

midterm-exam-93388

#411 Page 10 of 10

SCRAP PAGE

**CHEMISTRY 110 –Midterm Exam****Thursday November 7, 2024, 6:30 – 8:30 pm**

Examiners: Pallavi Sirjoosingh and Ashok Kakkar

First name (please write as legibly as possible within the boxes)

--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--

Last name

--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--

Student ID number

--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--

This booklet has 8 questions on 10 pages (The last page can be used as a scrap page).

The staple must NOT be removed from the question booklet. Your intact question booklet must be returned at the end of the exam.

A separate booklet contains a periodic table, and datasheet.

Write your answers directly under the questions in the space provided. Do not write on the QR codes or near the top edges (as it may be cut off).

Please write **legibly and dark enough** for the scan to be readable. For students who use English as a second language, dictionaries are permitted. Calculators are allowed.

DO NOT ASK QUESTIONS OF THE INVIGILATORS about interpretations of the exam questions. If you have a concern about a question, please summarize it at the top of this page. All concerns will be evaluated after the exam. GOOD LUCK!

**Question 1 8 points** (*Assume all transitions are possible*)

a. 6 points Calculate the **energy (in kJ/mol)** required to ionize (completely remove) the electron from the **n=3** level of **He⁺ ion** (He: Helium). Show your work including equations. (3 sig. figs.)

b. 2 points

(I) Select the largest (longest) wavelength from the three choices below:

- i. Wavelength of photon to *completely remove* an electron from *1s orbital of Hydrogen atom*
- ii. Wavelength of photon to *completely remove* an electron from *1s orbital of Boron atom (B)*
- iii. Wavelength of photon to *completely remove* an electron from *1s orbital of Boron cation (B⁺)*

i or ii or iii

(II) Select the smallest (shortest) wavelength from the three choices below (*for H atom*):

- i. Wavelength of photon *emitted* for an electron transitioning from *2p to 1s orbital*
- ii. Wavelength of photon *absorbed* for an electron transitioning from *2s to 4p orbital*
- iii. Wavelength of photon *absorbed* for an electron transitioning from *1s to 3p orbital*

i or ii or iii



Question 2 8 points



A Chem110 student plots the graph shown on the left, from data obtained for the kinetic energy (eV) of the ejected electron versus frequency (s^{-1}) of the incident light, for an unknown metal. ($1\text{eV} = 1.6022 \times 10^{-19} \text{ J}$)

Based on the plot, answer the questions **a to c** below:

a. 1 point On the plot above, label the **threshold frequency**.

b. 2 points What is the value of the work potential (in J) of the unknown metal?

c. 3 points What is the *longest* wavelength (in nm) required to eject an electron from this metal's surface? Show your work including equation(s) used. (3 sig figs)

d. 2 points In the space provided below, plot the following

(Assume frequency is *above* threshold frequency):

- Kinetic energy* of an ejected electron versus the *intensity* of the incident light (left)
- Number of electrons* ejected versus the *frequency* of the incident light (right)



**Question 3** **6 points****a. 2 points**

In the ground state of Ge (Germanium; $Z = 32$) atom, **name** the following orbitals:

i. Orbital containing the electron that requires the maximum energy to be removed.

ii. Orbital containing the electron that experiences the minimum effective nuclear charge.

b. 2 points

An unknown anion (A^-) has the following *excited* electronic configuration: $[Ar] 4s^2 3d^3 4p^1$

(Ar: Argon; $Z = 18$)

Give the **condensed ground state electronic configuration** of the atom, **A** and the cation, **A^{2+}** .
Show your work. (Assume all electronic transitions are possible)

Electronic Configuration of A:

Electronic Configuration of A^{2+} :

c. 2 points

In a single atom, determine how many electrons can have the following set of quantum numbers?

(i) $n = 5; l = 3$

In a single atom, determine how many electrons can have the following set of quantum numbers?

(i) $n = 3; l = 2; m_s = +1/2$

**Question 4** **8 points**

Rank the following from 1 to 4 in **increasing** order. Provide the ranking in the boxes provided.

Note: Atomic number (Z) are provided for relevant atoms

a) Increasing (lowest to highest) order of *ionization energy*: (1= lowest; 4 = highest)

Mg

Mg: Z = 12

Rb

Rb: Z = 37

Cs

Cs: Z = 55

K

K : Z = 19

b) Increasing (lowest to highest) order of *bond angle*: (1= lowest; 4 = highest)

H₂OCCl₄XeF₄

HCN

c) Increasing (smallest to largest) order of *ionic size* (1= smallest; 4 = largest)

S²⁻

S: Z = 16

O²⁻

O: Z = 8

Be²⁺

Be: Z = 4

Na⁺

Na: Z = 11

d) Increasing (lowest to highest) order of *bond energy* (1= lowest; 4 = highest)

P-F

P-Cl

P-I

P-Br

**Question 5** **12 points**

a. 9 points For the molecules HClO_4 and TeF_6 , draw the most stable *Lewis structure* showing appropriate shape and *dashed/wedged bonds*, based on VSEPR. Determine the **molecular geometry** around the central atom. For HClO_4 , label the *most polar bond*.



b. 3 points (Note: Lone pairs are not shown in structures below)

(i) For the three *resonance* structures (1, 2 and 3) shown below, answer the following question:



(I) Circle the *most contributing* resonance structure.

Structure 1

Structure 2

Structure 3

(ii) For the three structures given (4, 5, and 6) below, answer the following questions:



(I) Circle all the structures that represent *resonance structures* of each other:

Structure 4

Structure 5

Structure 6

(II) Among the resonance structures, circle the *most contributing* resonance structure.

Structure 4

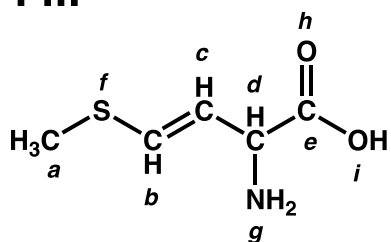
Structure 5

Structure 6

**Question 6** **6 points**

Using the data given below, calculate the **lattice energy** for SrO(s).

	ΔH° (kJ/mol)
$\text{Sr}(s) \longrightarrow \text{Sr}(g)$	+180
$\text{Sr}(g) \longrightarrow \text{Sr}^{2+}(g) + 2e^-$	+1468
$\text{O}_2(g) \longrightarrow 2\text{O}(g)$	+500
$\text{O}(g) + 2e^- \longrightarrow \text{O}^{2-}(g)$	+600
$\text{Sr}(s) + \frac{1}{2}\text{O}_2(g) \longrightarrow \text{SrO}(s)$	-550

**Question 7 10 points**

Consider the molecule given on the left, and answer questions (i) to (vi) using chemical bonding theories that we have learnt in class (valence bond theory, hybridization).

Note that the lone pairs are not shown in the structure:

i. Consider the carbon atoms marked in as ***a***, ***b***, ***c***, ***d***, and ***e*** in the molecule, and determine **hybridization** at each of the carbon centers. (2.5 points)

Atom	Hybridization
Carbon (<i>a</i>)	
Carbon (<i>b</i>)	
Carbon (<i>c</i>)	
Carbon (<i>d</i>)	
Carbon (<i>e</i>)	

ii. Now, consider the sulfur atom (S) marked as ***f***, nitrogen (N) atom (***g***), and oxygen(O) atoms (***h*** and ***i***), and determine their **hybridization**. (2 points)

Atom	Hybridization
Sulfur (<i>f</i>)	
Nitrogen (<i>g</i>)	
Oxygen (<i>h</i>)	
Oxygen (<i>i</i>)	

iii. **Describe** all the bonds formed by the carbon atoms ***a*** and ***b***, using sigma and pi designation as appropriate. (2 points)

Atom	Bonds formed (use sigma and pi as appropriate)
Carbon (<i>a</i>)	
Carbon (<i>b</i>)	

iv. How many total sp^3 -s sigma bonds are in the molecule? (1 point)

v. Determine **electronic** and **molecular geometry** at carbon (C) labeled ***e*** and sulfur (S) labeled ***f*** centres. (2 points)

Atom	Electronic Geometry	Molecular Geometry
Carbon (<i>e</i>)		
Sulfur (<i>f</i>)		

vi. Circle the value from the choices given below, that describes the **C-S-C bond angle at sulfur** (S) marked ***f***: (0.5 points)

>90° to <109.5°

109.5°

>109.5 to <120°

120°

>120° to <180°


Question 8 12 points

Using the **valence shell orbitals only** draw a **complete** molecular orbital (MO) diagram (with no s-p mixing) for the molecule **PS**, in the space provided below. **(6 points)**

Assume that 3s and 3p atomic orbitals will form molecular orbitals like those formed by the combination of 2s and 2p atomic orbitals that we learnt in class, and there is no s-p mixing.

Show **energy arrow**, **atomic orbitals** involved, **atom labels**, the **molecular orbitals** that are formed, **filled electrons** in the MO diagram etc. Make sure to label all the atomic and molecular orbitals.

Using the MO diagram above, answer the following questions:

- (i) Which one of the following will have the highest bond order? Circle the correct choice
 A) PS^{1-} B) PS C) PS^{1+}
- (ii) Which one of the following will have the longest bond length? Circle the correct choice
 A) PS^{1-} B) PS C) PS^{1+}
- (iii) Which of the following will be diamagnetic? Circle the correct choice.
 A) PS^{1-} B) PS C) PS^{1+}
- (iv) Draw Lewis dot structure of the molecule with the shortest bond length (based on the MO diagram) from among the three (PS^{1-} , PS, PS^{1+}), and indicate the bond order calculated from the Lewis dot structure. **(3 points)**

Lewis Dot Structure

Bond Order



D2F612C5-D851-4EFD-A4F6-61FACF15D2BF

midterm-exam-93388

#412 Page 10 of 10

SCRAP PAGE

**CHEMISTRY 110 –Midterm Exam****Thursday November 7, 2024, 6:30 – 8:30 pm**

Examiners: Pallavi Sirjoosingh and Ashok Kakkar

First name (please write as legibly as possible within the boxes)

--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--

Last name

--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--

Student ID number

--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--

This booklet has 8 questions on 10 pages (The last page can be used as a scrap page).

The staple must NOT be removed from the question booklet. Your intact question booklet must be returned at the end of the exam.

A separate booklet contains a periodic table, and datasheet.

Write your answers directly under the questions in the space provided. Do not write on the QR codes or near the top edges (as it may be cut off).

Please write **legibly and dark enough** for the scan to be readable. For students who use English as a second language, dictionaries are permitted. Calculators are allowed.

DO NOT ASK QUESTIONS OF THE INVIGILATORS about interpretations of the exam questions. If you have a concern about a question, please summarize it at the top of this page. All concerns will be evaluated after the exam. GOOD LUCK!

**Question 1 8 points** (*Assume all transitions are possible*)

a. 6 points Calculate the **energy (in kJ/mol)** required to ionize (completely remove) the electron from the **n=3** level of **He⁺ ion** (He: Helium). Show your work including equations. (3 sig. figs.)

b. 2 points

(I) Select the largest (longest) wavelength from the three choices below:

- i. Wavelength of photon to *completely remove* an electron from *1s orbital of Hydrogen atom*
- ii. Wavelength of photon to *completely remove* an electron from *1s orbital of Boron atom (B)*
- iii. Wavelength of photon to *completely remove* an electron from *1s orbital of Boron cation (B⁺)*

i or ii or iii

(II) Select the smallest (shortest) wavelength from the three choices below (*for H atom*):

- i. Wavelength of photon *emitted* for an electron transitioning from *2p to 1s orbital*
- ii. Wavelength of photon *absorbed* for an electron transitioning from *2s to 4p orbital*
- iii. Wavelength of photon *absorbed* for an electron transitioning from *1s to 3p orbital*

i or ii or iii



Question 2 8 points



A Chem110 student plots the graph shown on the left, from data obtained for the kinetic energy (eV) of the ejected electron versus frequency (s^{-1}) of the incident light, for an unknown metal. ($1\text{eV} = 1.6022 \times 10^{-19} \text{ J}$)

Based on the plot, answer the questions **a to c** below:

a. 1 point On the plot above, label the **threshold frequency**.

b. 2 points What is the value of the work potential (in J) of the unknown metal?

c. 3 points What is the *longest* wavelength (in nm) required to eject an electron from this metal's surface? Show your work including equation(s) used. (3 sig figs)

d. 2 points In the space provided below, plot the following

(Assume frequency is *above* threshold frequency):

- Kinetic energy* of an ejected electron versus the *intensity* of the incident light (left)
- Number of electrons* ejected versus the *frequency* of the incident light (right)



**Question 3** **6 points****a. 2 points**

In the ground state of Ge (Germanium; $Z = 32$) atom, **name** the following orbitals:

i. Orbital containing the electron that requires the maximum energy to be removed.

ii. Orbital containing the electron that experiences the minimum effective nuclear charge.

b. 2 points

An unknown anion (A^-) has the following *excited* electronic configuration: $[Ar] 4s^2 3d^3 4p^1$

(Ar: Argon; $Z = 18$)

Give the **condensed ground state electronic configuration** of the atom, **A** and the cation, **A^{2+}** .
Show your work. (Assume all electronic transitions are possible)

Electronic Configuration of A:

Electronic Configuration of A^{2+} :

c. 2 points

In a single atom, determine how many electrons can have the following set of quantum numbers?

(i) $n = 5; l = 3$

In a single atom, determine how many electrons can have the following set of quantum numbers?

(i) $n = 3; l = 2; m_s = +1/2$

**Question 4** **8 points**

Rank the following from 1 to 4 in **increasing** order. Provide the ranking in the boxes provided.

Note: Atomic number (Z) are provided for relevant atoms

a) Increasing (lowest to highest) order of *ionization energy*: (1= lowest; 4 = highest)

Mg

Mg: Z = 12

Rb

Rb: Z = 37

Cs

Cs: Z = 55

K

K : Z = 19

b) Increasing (lowest to highest) order of *bond angle*: (1= lowest; 4 = highest)

H₂OCCl₄XeF₄

HCN

c) Increasing (smallest to largest) order of *ionic size* (1= smallest; 4 = largest)

S²⁻

S: Z = 16

O²⁻

O: Z = 8

Be²⁺

Be: Z = 4

Na⁺

Na: Z = 11

d) Increasing (lowest to highest) order of *bond energy* (1= lowest; 4 = highest)

P-F

P-Cl

P-I

P-Br

**Question 5** **12 points**

a. 9 points For the molecules HClO_4 and TeF_6 , draw the most stable *Lewis structure* showing appropriate shape and *dashed/wedged bonds*, based on VSEPR. Determine the **molecular geometry** around the central atom. For HClO_4 , label the *most polar bond*.

HClO_4 (there are no O-O bonds)

Label the most polar bond

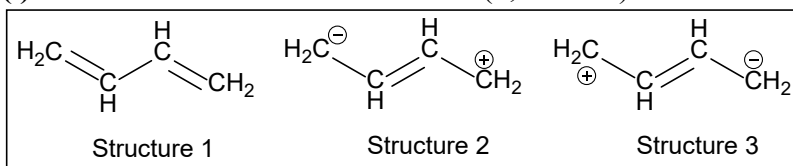
Molecular Geometry around Cl

TeF_6

Molecular Geometry around Te

b. 3 points (Note: Lone pairs are not shown in structures below)

(i) For the three *resonance* structures (1, 2 and 3) shown below, answer the following question:



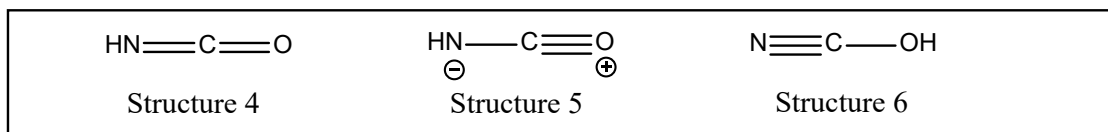
(I) Circle the *most contributing* resonance structure.

Structure 1

Structure 2

Structure 3

(ii) For the three structures given (4, 5, and 6) below, answer the following questions:



(I) Circle all the structures that represent *resonance structures* of each other:

Structure 4

Structure 5

Structure 6

(II) Among the resonance structures, circle the *most contributing* resonance structure.

Structure 4

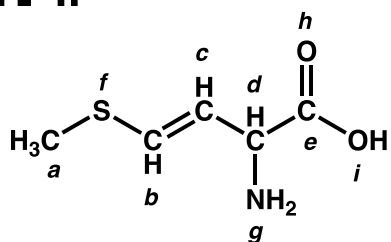
Structure 5

Structure 6

**Question 6** **6 points**

Using the data given below, calculate the **lattice energy** for SrO(s).

	ΔH° (kJ/mol)
$\text{Sr}(s) \longrightarrow \text{Sr}(g)$	+180
$\text{Sr}(g) \longrightarrow \text{Sr}^{2+}(g) + 2e^-$	+1468
$\text{O}_2(g) \longrightarrow 2\text{O}(g)$	+500
$\text{O}(g) + 2e^- \longrightarrow \text{O}^{2-}(g)$	+600
$\text{Sr}(s) + \frac{1}{2}\text{O}_2(g) \longrightarrow \text{SrO}(s)$	-550

**Question 7 10 points**

Consider the molecule given on the left, and answer questions (i) to (vi) using chemical bonding theories that we have learnt in class (valence bond theory, hybridization).

Note that the lone pairs are not shown in the structure:

i. Consider the carbon atoms marked in as ***a***, ***b***, ***c***, ***d***, and ***e*** in the molecule, and determine **hybridization** at each of the carbon centers. (2.5 points)

Atom	Hybridization
Carbon (<i>a</i>)	
Carbon (<i>b</i>)	
Carbon (<i>c</i>)	
Carbon (<i>d</i>)	
Carbon (<i>e</i>)	

ii. Now, consider the sulfur atom (S) marked as ***f***, nitrogen (N) atom (***g***), and oxygen(O) atoms (***h*** and ***i***), and determine their **hybridization**. (2 points)

Atom	Hybridization
Sulfur (<i>f</i>)	
Nitrogen (<i>g</i>)	
Oxygen (<i>h</i>)	
Oxygen (<i>i</i>)	

iii. **Describe** all the bonds formed by the carbon atoms ***a*** and ***b***, using sigma and pi designation as appropriate. (2 points)

Atom	Bonds formed (use sigma and pi as appropriate)
Carbon (<i>a</i>)	
Carbon (<i>b</i>)	

iv. How many total sp^3 -s sigma bonds are in the molecule? (1 point)

v. Determine **electronic** and **molecular geometry** at carbon (C) labeled ***e*** and sulfur (S) labeled ***f*** centres. (2 points)

Atom	Electronic Geometry	Molecular Geometry
Carbon (<i>e</i>)		
Sulfur (<i>f</i>)		

vi. Circle the value from the choices given below, that describes the **C-S-C bond angle at sulfur** (S) marked ***f***: (0.5 points)

>90° to <109.5° 109.5° >109.5 to <120° 120° >120° to <180°


Question 8 12 points

Using the **valence shell orbitals only** draw a **complete** molecular orbital (MO) diagram (with no s-p mixing) for the molecule **PS**, in the space provided below. **(6 points)**

Assume that 3s and 3p atomic orbitals will form molecular orbitals like those formed by the combination of 2s and 2p atomic orbitals that we learnt in class, and there is no s-p mixing.

Show **energy arrow**, **atomic orbitals** involved, **atom labels**, the **molecular orbitals** that are formed, **filled electrons** in the MO diagram etc. Make sure to label all the atomic and molecular orbitals.

Using the MO diagram above, answer the following questions:

- (i) Which one of the following will have the highest bond order? Circle the correct choice
 A) PS^{1-} B) PS C) PS^{1+}
- (ii) Which one of the following will have the longest bond length? Circle the correct choice
 A) PS^{1-} B) PS C) PS^{1+}
- (iii) Which of the following will be diamagnetic? Circle the correct choice.
 A) PS^{1-} B) PS C) PS^{1+}
- (iv) Draw Lewis dot structure of the molecule with the shortest bond length (based on the MO diagram) from among the three (PS^{1-} , PS, PS^{1+}), and indicate the bond order calculated from the Lewis dot structure. **(3 points)**

Lewis Dot Structure

Bond Order



E9AE3B6A-1954-4A2B-B4A9-4F8900F0A1E6

midterm-exam-93388

#413 Page 10 of 10

SCRAP PAGE

**CHEMISTRY 110 –Midterm Exam****Thursday November 7, 2024, 6:30 – 8:30 pm**

Examiners: Pallavi Sirjoosingh and Ashok Kakkar

First name (please write as legibly as possible within the boxes)

--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--

Last name

--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--

Student ID number

--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--

This booklet has 8 questions on 10 pages (The last page can be used as a scrap page).

The staple must NOT be removed from the question booklet. Your intact question booklet must be returned at the end of the exam.

A separate booklet contains a periodic table, and datasheet.

Write your answers directly under the questions in the space provided. Do not write on the QR codes or near the top edges (as it may be cut off).

Please write **legibly and dark enough** for the scan to be readable. For students who use English as a second language, dictionaries are permitted. Calculators are allowed.

DO NOT ASK QUESTIONS OF THE INVIGILATORS about interpretations of the exam questions. If you have a concern about a question, please summarize it at the top of this page. All concerns will be evaluated after the exam. GOOD LUCK!

**Question 1 8 points** (*Assume all transitions are possible*)

a. 6 points Calculate the **energy (in kJ/mol)** required to ionize (completely remove) the electron from the **n=3** level of **He⁺ ion** (He: Helium). Show your work including equations. (3 sig. figs.)

b. 2 points

(I) Select the largest (longest) wavelength from the three choices below:

- i. Wavelength of photon to *completely remove* an electron from *1s orbital of Hydrogen atom*
- ii. Wavelength of photon to *completely remove* an electron from *1s orbital of Boron atom (B)*
- iii. Wavelength of photon to *completely remove* an electron from *1s orbital of Boron cation (B⁺)*

i or ii or iii

(II) Select the smallest (shortest) wavelength from the three choices below (*for H atom*):

- i. Wavelength of photon *emitted* for an electron transitioning from *2p to 1s orbital*
- ii. Wavelength of photon *absorbed* for an electron transitioning from *2s to 4p orbital*
- iii. Wavelength of photon *absorbed* for an electron transitioning from *1s to 3p orbital*

i or ii or iii



Question 2 8 points



A Chem110 student plots the graph shown on the left, from data obtained for the kinetic energy (eV) of the ejected electron versus frequency (s^{-1}) of the incident light, for an unknown metal. ($1\text{eV} = 1.6022 \times 10^{-19} \text{ J}$)

Based on the plot, answer the questions **a to c** below:

a. 1 point On the plot above, label the **threshold frequency**.

b. 2 points What is the value of the work potential (in J) of the unknown metal?

c. 3 points What is the *longest* wavelength (in nm) required to eject an electron from this metal's surface? Show your work including equation(s) used. (3 sig figs)

d. 2 points In the space provided below, plot the following

(Assume frequency is *above* threshold frequency):

- i. *Kinetic energy* of an ejected electron versus the *intensity* of the incident light (left)
- ii. *Number of electrons* ejected versus the *frequency* of the incident light (right)



**Question 3** **6 points****a. 2 points**

In the ground state of Ge (Germanium; $Z = 32$) atom, **name** the following orbitals:

i. Orbital containing the electron that requires the maximum energy to be removed.

ii. Orbital containing the electron that experiences the minimum effective nuclear charge.

b. 2 points

An unknown anion (A^-) has the following *excited* electronic configuration: $[Ar] 4s^2 3d^3 4p^1$

(Ar: Argon; $Z = 18$)

Give the **condensed ground state electronic configuration** of the atom, **A** and the cation, **A^{2+}** .
Show your work. (Assume all electronic transitions are possible)

Electronic Configuration of A:

Electronic Configuration of A^{2+} :

c. 2 points

In a single atom, determine how many electrons can have the following set of quantum numbers?

(i) $n = 5; l = 3$

In a single atom, determine how many electrons can have the following set of quantum numbers?

(i) $n = 3; l = 2; m_s = +1/2$

**Question 4** **8 points**

Rank the following from 1 to 4 in **increasing** order. Provide the ranking in the boxes provided.

Note: Atomic number (Z) are provided for relevant atoms

a) Increasing (lowest to highest) order of *ionization energy*: (1= lowest; 4 = highest)

Mg

Mg: Z = 12

Rb

Rb: Z = 37

Cs

Cs: Z = 55

K

K : Z = 19

b) Increasing (lowest to highest) order of *bond angle*: (1= lowest; 4 = highest)

H₂OCCl₄XeF₄

HCN

c) Increasing (smallest to largest) order of *ionic size* (1= smallest; 4 = largest)

S²⁻

S: Z = 16

O²⁻

O: Z = 8

Be²⁺

Be: Z = 4

Na⁺

Na: Z = 11

d) Increasing (lowest to highest) order of *bond energy* (1= lowest; 4 = highest)

P-F

P-Cl

P-I

P-Br

**Question 5** **12 points**

a. 9 points For the molecules HClO_4 and TeF_6 , draw the most stable *Lewis structure* showing appropriate shape and *dashed/wedged bonds*, based on VSEPR. Determine the **molecular geometry** around the central atom. For HClO_4 , label the *most polar bond*.



b. 3 points (Note: Lone pairs are not shown in structures below)

(i) For the three *resonance* structures (1, 2 and 3) shown below, answer the following question:



(I) Circle the *most contributing* resonance structure.

Structure 1

Structure 2

Structure 3

(ii) For the three structures given (4, 5, and 6) below, answer the following questions:



(I) Circle all the structures that represent *resonance structures* of each other:

Structure 4

Structure 5

Structure 6

(II) Among the resonance structures, circle the *most contributing* resonance structure.

Structure 4

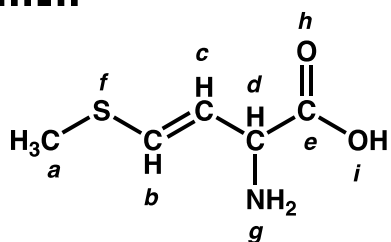
Structure 5

Structure 6

**Question 6** **6 points**

Using the data given below, calculate the **lattice energy** for SrO(s).

	ΔH° (kJ/mol)
$\text{Sr}(s) \longrightarrow \text{Sr}(g)$	+180
$\text{Sr}(g) \longrightarrow \text{Sr}^{2+}(g) + 2e^-$	+1468
$\text{O}_2(g) \longrightarrow 2\text{O}(g)$	+500
$\text{O}(g) + 2e^- \longrightarrow \text{O}^{2-}(g)$	+600
$\text{Sr}(s) + \frac{1}{2}\text{O}_2(g) \longrightarrow \text{SrO}(s)$	-550

**Question 7 10 points**

Consider the molecule given on the left, and answer questions (i) to (vi) using chemical bonding theories that we have learnt in class (valence bond theory, hybridization).

Note that the lone pairs are not shown in the structure:

i. Consider the carbon atoms marked in as ***a***, ***b***, ***c***, ***d***, and ***e*** in the molecule, and determine **hybridization** at each of the carbon centers. (2.5 points)

Atom	Hybridization
Carbon (<i>a</i>)	
Carbon (<i>b</i>)	
Carbon (<i>c</i>)	
Carbon (<i>d</i>)	
Carbon (<i>e</i>)	

ii. Now, consider the sulfur atom (S) marked as ***f***, nitrogen (N) atom (***g***), and oxygen(O) atoms (***h*** and ***i***), and determine their **hybridization**. (2 points)

Atom	Hybridization
Sulfur (<i>f</i>)	
Nitrogen (<i>g</i>)	
Oxygen (<i>h</i>)	
Oxygen (<i>i</i>)	

iii. **Describe** all the bonds formed by the carbon atoms ***a*** and ***b***, using sigma and pi designation as appropriate. (2 points)

Atom	Bonds formed (use sigma and pi as appropriate)
Carbon (<i>a</i>)	
Carbon (<i>b</i>)	

iv. How many total sp^3 -s sigma bonds are in the molecule? (1 point)

v. Determine **electronic** and **molecular geometry** at carbon (C) labeled ***e*** and sulfur (S) labeled ***f*** centres. (2 points)

Atom	Electronic Geometry	Molecular Geometry
Carbon (<i>e</i>)		
Sulfur (<i>f</i>)		

vi. Circle the value from the choices given below, that describes the **C-S-C bond angle at sulfur** (S) marked ***f***: (0.5 points)

>90° to <109.5° 109.5° >109.5 to <120° 120° >120° to <180°


Question 8 12 points

Using the **valence shell orbitals only** draw a **complete** molecular orbital (MO) diagram (with no s-p mixing) for the molecule **PS**, in the space provided below. **(6 points)**

Assume that 3s and 3p atomic orbitals will form molecular orbitals like those formed by the combination of 2s and 2p atomic orbitals that we learnt in class, and there is no s-p mixing.

Show **energy arrow**, **atomic orbitals** involved, **atom labels**, the **molecular orbitals** that are formed, **filled electrons** in the MO diagram etc. Make sure to label all the atomic and molecular orbitals.

Using the MO diagram above, answer the following questions:

- (i) Which one of the following will have the highest bond order? Circle the correct choice
 A) PS^{1-} B) PS C) PS^{1+}
- (ii) Which one of the following will have the longest bond length? Circle the correct choice
 A) PS^{1-} B) PS C) PS^{1+}
- (iii) Which of the following will be diamagnetic? Circle the correct choice.
 A) PS^{1-} B) PS C) PS^{1+}
- (iv) Draw Lewis dot structure of the molecule with the shortest bond length (based on the MO diagram) from among the three (PS^{1-} , PS, PS^{1+}), and indicate the bond order calculated from the Lewis dot structure. **(3 points)**

Lewis Dot Structure

Bond Order



51AC5530-DA1F-4D4F-9A47-5F42AC0968AD

midterm-exam-93388

#414 Page 10 of 10

SCRAP PAGE

**CHEMISTRY 110 –Midterm Exam****Thursday November 7, 2024, 6:30 – 8:30 pm**

Examiners: Pallavi Sirjoosingh and Ashok Kakkar

First name (please write as legibly as possible within the boxes)

--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--

Last name

--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--

Student ID number

--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--

This booklet has 8 questions on 10 pages (The last page can be used as a scrap page).

The staple must NOT be removed from the question booklet. Your intact question booklet must be returned at the end of the exam.

A separate booklet contains a periodic table, and datasheet.

Write your answers directly under the questions in the space provided. Do not write on the QR codes or near the top edges (as it may be cut off).

Please write **legibly and dark enough** for the scan to be readable. For students who use English as a second language, dictionaries are permitted. Calculators are allowed.

DO NOT ASK QUESTIONS OF THE INVIGILATORS about interpretations of the exam questions. If you have a concern about a question, please summarize it at the top of this page. All concerns will be evaluated after the exam. GOOD LUCK!

**Question 1 8 points** (*Assume all transitions are possible*)

a. 6 points Calculate the **energy (in kJ/mol)** required to ionize (completely remove) the electron from the **$n=3$** level of **He^+ ion** (He: Helium). Show your work including equations. (3 sig. figs.)

b. 2 points

(I) Select the largest (longest) wavelength from the three choices below:

- i. Wavelength of photon to *completely remove* an electron from *1s orbital of Hydrogen atom*
- ii. Wavelength of photon to *completely remove* an electron from *1s orbital of Boron atom (B)*
- iii. Wavelength of photon to *completely remove* an electron from *1s orbital of Boron cation (B^+)*

i or ii or iii

(II) Select the smallest (shortest) wavelength from the three choices below (*for H atom*):

- i. Wavelength of photon *emitted* for an electron transitioning from *2p to 1s orbital*
- ii. Wavelength of photon *absorbed* for an electron transitioning from *2s to 4p orbital*
- iii. Wavelength of photon *absorbed* for an electron transitioning from *1s to 3p orbital*

i or ii or iii



Question 2 8 points



A Chem110 student plots the graph shown on the left, from data obtained for the kinetic energy (eV) of the ejected electron versus frequency (s^{-1}) of the incident light, for an unknown metal. ($1\text{eV} = 1.6022 \times 10^{-19} \text{ J}$)

Based on the plot, answer the questions **a to c** below:

a. 1 point On the plot above, label the **threshold frequency**.

b. 2 points What is the value of the work potential (in J) of the unknown metal?

c. 3 points What is the *longest* wavelength (in nm) required to eject an electron from this metal's surface? Show your work including equation(s) used. (3 sig figs)

d. 2 points In the space provided below, plot the following

(Assume frequency is *above* threshold frequency):

- i. *Kinetic energy* of an ejected electron versus the *intensity* of the incident light (left)
- ii. *Number of electrons* ejected versus the *frequency* of the incident light (right)



**Question 3** **6 points****a. 2 points**

In the ground state of Ge (Germanium; $Z = 32$) atom, **name** the following orbitals:

i. Orbital containing the electron that requires the maximum energy to be removed.

ii. Orbital containing the electron that experiences the minimum effective nuclear charge.

b. 2 points

An unknown anion (A^-) has the following *excited* electronic configuration: $[Ar] 4s^2 3d^3 4p^1$

(Ar: Argon; $Z = 18$)

Give the **condensed ground state electronic configuration** of the atom, **A** and the cation, **A^{2+}** .
Show your work. (Assume all electronic transitions are possible)

Electronic Configuration of A:

Electronic Configuration of A^{2+} :

c. 2 points

In a single atom, determine how many electrons can have the following set of quantum numbers?

(i) $n = 5; l = 3$

In a single atom, determine how many electrons can have the following set of quantum numbers?

(i) $n = 3; l = 2; m_s = +1/2$

**Question 4** **8 points**

Rank the following from 1 to 4 in **increasing** order. Provide the ranking in the boxes provided.

Note: Atomic number (Z) are provided for relevant atoms

a) Increasing (lowest to highest) order of *ionization energy*: (1= lowest; 4 = highest)

Mg

Mg: Z = 12

Rb

Rb: Z = 37

Cs

Cs: Z = 55

K

K : Z = 19

b) Increasing (lowest to highest) order of *bond angle*: (1= lowest; 4 = highest)

H₂OCCl₄XeF₄

HCN

c) Increasing (smallest to largest) order of *ionic size* (1= smallest; 4 = largest)

S²⁻

S: Z = 16

O²⁻

O: Z = 8

Be²⁺

Be: Z = 4

Na⁺

Na: Z = 11

d) Increasing (lowest to highest) order of *bond energy* (1= lowest; 4 = highest)

P-F

P-Cl

P-I

P-Br

**Question 5** **12 points**

a. 9 points For the molecules HClO_4 and TeF_6 , draw the most stable *Lewis structure* showing appropriate shape and *dashed/wedged bonds*, based on VSEPR. Determine the **molecular geometry** around the central atom. For HClO_4 , label the *most polar bond*.



b. 3 points (Note: Lone pairs are not shown in structures below)

(i) For the three *resonance* structures (1, 2 and 3) shown below, answer the following question:



(I) Circle the *most contributing* resonance structure.

Structure 1

Structure 2

Structure 3

(ii) For the three structures given (4, 5, and 6) below, answer the following questions:



(I) Circle all the structures that represent *resonance structures* of each other:

Structure 4

Structure 5

Structure 6

(II) Among the resonance structures, circle the *most contributing* resonance structure.

Structure 4

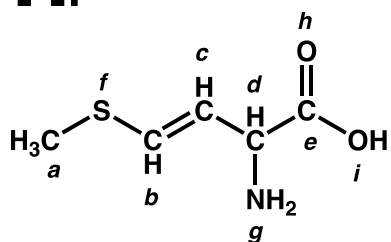
Structure 5

Structure 6

**Question 6** **6 points**

Using the data given below, calculate the **lattice energy** for SrO(s).

	ΔH° (kJ/mol)
$\text{Sr}(s) \longrightarrow \text{Sr}(g)$	+180
$\text{Sr}(g) \longrightarrow \text{Sr}^{2+}(g) + 2e^-$	+1468
$\text{O}_2(g) \longrightarrow 2\text{O}(g)$	+500
$\text{O}(g) + 2e^- \longrightarrow \text{O}^{2-}(g)$	+600
$\text{Sr}(s) + \frac{1}{2}\text{O}_2(g) \longrightarrow \text{SrO}(s)$	-550

**Question 7 10 points**

Consider the molecule given on the left, and answer questions (i) to (vi) using chemical bonding theories that we have learnt in class (valence bond theory, hybridization).

Note that the lone pairs are not shown in the structure:

i. Consider the carbon atoms marked in as ***a***, ***b***, ***c***, ***d***, and ***e*** in the molecule, and determine **hybridization** at each of the carbon centers. (2.5 points)

Atom	Hybridization
Carbon (<i>a</i>)	
Carbon (<i>b</i>)	
Carbon (<i>c</i>)	
Carbon (<i>d</i>)	
Carbon (<i>e</i>)	

ii. Now, consider the sulfur atom (S) marked as ***f***, nitrogen (N) atom (***g***), and oxygen(O) atoms (***h*** and ***i***), and determine their **hybridization**. (2 points)

Atom	Hybridization
Sulfur (<i>f</i>)	
Nitrogen (<i>g</i>)	
Oxygen (<i>h</i>)	
Oxygen (<i>i</i>)	

iii. **Describe** all the bonds formed by the carbon atoms ***a*** and ***b***, using sigma and pi designation as appropriate. (2 points)

Atom	Bonds formed (use sigma and pi as appropriate)
Carbon (<i>a</i>)	
Carbon (<i>b</i>)	

iv. How many total sp^3 -s sigma bonds are in the molecule? (1 point)

v. Determine **electronic** and **molecular geometry** at carbon (C) labeled ***e*** and sulfur (S) labeled ***f*** centres. (2 points)

Atom	Electronic Geometry	Molecular Geometry
Carbon (<i>e</i>)		
Sulfur (<i>f</i>)		

vi. Circle the value from the choices given below, that describes the **C-S-C bond angle at sulfur** (S) marked ***f***: (0.5 points)

>90° to <109.5° 109.5° >109.5 to <120° 120° >120° to <180°



Question 8 12 points

Using the **valence shell orbitals only** draw a **complete** molecular orbital (MO) diagram (with no s-p mixing) for the molecule **PS**, in the space provided below. **(6 points)**

Assume that 3s and 3p atomic orbitals will form molecular orbitals like those formed by the combination of 2s and 2p atomic orbitals that we learnt in class, and there is no s-p mixing.

Show **energy arrow**, **atomic orbitals** involved, **atom labels**, the **molecular orbitals** that are formed, **filled electrons** in the MO diagram etc. Make sure to label all the atomic and molecular orbitals.

Using the MO diagram above, answer the following questions:

- (i) Which one of the following will have the highest bond order? Circle the correct choice
 A) PS^{1-} B) PS C) PS^{1+}
- (ii) Which one of the following will have the longest bond length? Circle the correct choice
 A) PS^{1-} B) PS C) PS^{1+}
- (iii) Which of the following will be diamagnetic? Circle the correct choice.
 A) PS^{1-} B) PS C) PS^{1+}
- (iv) Draw Lewis dot structure of the molecule with the shortest bond length (based on the MO diagram) from among the three (PS^{1-} , PS, PS^{1+}), and indicate the bond order calculated from the Lewis dot structure. **(3 points)**

Lewis Dot Structure

Bond Order



43E2A336-CFFE-4925-9444-7E9D8164F8B3

midterm-exam-93388

#415 Page 10 of 10

SCRAP PAGE

**CHEMISTRY 110 –Midterm Exam****Thursday November 7, 2024, 6:30 – 8:30 pm**

Examiners: Pallavi Sirjoosingh and Ashok Kakkar

First name (please write as legibly as possible within the boxes)

--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--

Last name

--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--

Student ID number

--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--

This booklet has 8 questions on 10 pages (The last page can be used as a scrap page).

The staple must NOT be removed from the question booklet. Your intact question booklet must be returned at the end of the exam.

A separate booklet contains a periodic table, and datasheet.

Write your answers directly under the questions in the space provided. Do not write on the QR codes or near the top edges (as it may be cut off).

Please write **legibly and dark enough** for the scan to be readable. For students who use English as a second language, dictionaries are permitted. Calculators are allowed.

DO NOT ASK QUESTIONS OF THE INVIGILATORS about interpretations of the exam questions. If you have a concern about a question, please summarize it at the top of this page. All concerns will be evaluated after the exam. GOOD LUCK!

**Question 1 8 points** (*Assume all transitions are possible*)

a. 6 points Calculate the **energy (in kJ/mol)** required to ionize (completely remove) the electron from the **n=3** level of **He⁺ ion** (He: Helium). Show your work including equations. (3 sig. figs.)

b. 2 points

(I) Select the largest (longest) wavelength from the three choices below:

- i. Wavelength of photon to *completely remove* an electron from *1s orbital of Hydrogen atom*
- ii. Wavelength of photon to *completely remove* an electron from *1s orbital of Boron atom (B)*
- iii. Wavelength of photon to *completely remove* an electron from *1s orbital of Boron cation (B⁺)*

i or ii or iii

(II) Select the smallest (shortest) wavelength from the three choices below (*for H atom*):

- i. Wavelength of photon *emitted* for an electron transitioning from *2p to 1s orbital*
- ii. Wavelength of photon *absorbed* for an electron transitioning from *2s to 4p orbital*
- iii. Wavelength of photon *absorbed* for an electron transitioning from *1s to 3p orbital*

i or ii or iii



Question 2 8 points



A Chem110 student plots the graph shown on the left, from data obtained for the kinetic energy (eV) of the ejected electron versus frequency (s^{-1}) of the incident light, for an unknown metal. ($1\text{eV} = 1.6022 \times 10^{-19} \text{ J}$)

Based on the plot, answer the questions **a to c** below:

a. 1 point On the plot above, label the **threshold frequency**.

b. 2 points What is the value of the work potential (in J) of the unknown metal?

c. 3 points What is the *longest* wavelength (in nm) required to eject an electron from this metal's surface? Show your work including equation(s) used. (3 sig figs)

d. 2 points In the space provided below, plot the following

(Assume frequency is *above* threshold frequency):

- i. *Kinetic energy* of an ejected electron versus the *intensity* of the incident light (left)
- ii. *Number of electrons* ejected versus the *frequency* of the incident light (right)



**Question 3** **6 points****a. 2 points**

In the ground state of Ge (Germanium; $Z = 32$) atom, **name** the following orbitals:

i. Orbital containing the electron that requires the maximum energy to be removed.

ii. Orbital containing the electron that experiences the minimum effective nuclear charge.

b. 2 points

An unknown anion (A^-) has the following *excited* electronic configuration: $[Ar] 4s^2 3d^3 4p^1$

(Ar: Argon; $Z = 18$)

Give the **condensed ground state electronic configuration** of the atom, **A** and the cation, A^{2+} .
Show your work. (Assume all electronic transitions are possible)

Electronic Configuration of A:

Electronic Configuration of A^{2+} :

c. 2 points

In a single atom, determine how many electrons can have the following set of quantum numbers?

(i) $n = 5; l = 3$

In a single atom, determine how many electrons can have the following set of quantum numbers?

(i) $n = 3; l = 2; m_s = +1/2$

**Question 4** **8 points**

Rank the following from 1 to 4 in **increasing** order. Provide the ranking in the boxes provided.

Note: Atomic number (Z) are provided for relevant atoms

a) Increasing (lowest to highest) order of *ionization energy*: (1= lowest; 4 = highest)

Mg

Mg: Z = 12

Rb

Rb: Z = 37

Cs

Cs: Z = 55

K

K : Z = 19

b) Increasing (lowest to highest) order of *bond angle*: (1= lowest; 4 = highest)

H₂OCCl₄XeF₄

HCN

c) Increasing (smallest to largest) order of *ionic size* (1= smallest; 4 = largest)

S²⁻

S: Z = 16

O²⁻

O: Z = 8

Be²⁺

Be: Z = 4

Na⁺

Na: Z = 11

d) Increasing (lowest to highest) order of *bond energy* (1= lowest; 4 = highest)

P-F

P-Cl

P-I

P-Br

**Question 5** **12 points**

a. 9 points For the molecules HClO_4 and TeF_6 , draw the most stable *Lewis structure* showing appropriate shape and *dashed/wedged bonds*, based on VSEPR. Determine the **molecular geometry** around the central atom. For HClO_4 , label the *most polar bond*.

HClO_4 (there are no O-O bonds)

Label the most polar bond

Molecular Geometry around Cl

TeF_6

Molecular Geometry around Te

b. 3 points (Note: Lone pairs are not shown in structures below)

(i) For the three *resonance* structures (1, 2 and 3) shown below, answer the following question:



(I) Circle the *most contributing* resonance structure.

Structure 1

Structure 2

Structure 3

(ii) For the three structures given (4, 5, and 6) below, answer the following questions:



(I) Circle all the structures that represent *resonance structures* of each other:

Structure 4

Structure 5

Structure 6

(II) Among the resonance structures, circle the *most contributing* resonance structure.

Structure 4

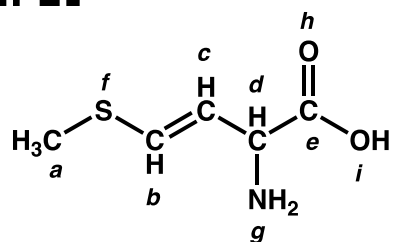
Structure 5

Structure 6

**Question 6** **6 points**

Using the data given below, calculate the **lattice energy** for SrO(s).

	ΔH° (kJ/mol)
$\text{Sr}(s) \longrightarrow \text{Sr}(g)$	+180
$\text{Sr}(g) \longrightarrow \text{Sr}^{2+}(g) + 2e^-$	+1468
$\text{O}_2(g) \longrightarrow 2\text{O}(g)$	+500
$\text{O}(g) + 2e^- \longrightarrow \text{O}^{2-}(g)$	+600
$\text{Sr}(s) + \frac{1}{2}\text{O}_2(g) \longrightarrow \text{SrO}(s)$	-550

**Question 7 10 points**

Consider the molecule given on the left, and answer questions (i) to (vi) using chemical bonding theories that we have learnt in class (valence bond theory, hybridization).

Note that the lone pairs are not shown in the structure:

i. Consider the carbon atoms marked in as ***a***, ***b***, ***c***, ***d***, and ***e*** in the molecule, and determine **hybridization** at each of the carbon centers. (2.5 points)

Atom	Hybridization
Carbon (<i>a</i>)	
Carbon (<i>b</i>)	
Carbon (<i>c</i>)	
Carbon (<i>d</i>)	
Carbon (<i>e</i>)	

ii. Now, consider the sulfur atom (S) marked as ***f***, nitrogen (N) atom (***g***), and oxygen(O) atoms (***h*** and ***i***), and determine their **hybridization**. (2 points)

Atom	Hybridization
Sulfur (<i>f</i>)	
Nitrogen (<i>g</i>)	
Oxygen (<i>h</i>)	
Oxygen (<i>i</i>)	

iii. **Describe** all the bonds formed by the carbon atoms ***a*** and ***b***, using sigma and pi designation as appropriate. (2 points)

Atom	Bonds formed (use sigma and pi as appropriate)
Carbon (<i>a</i>)	
Carbon (<i>b</i>)	

iv. How many total sp^3 -s sigma bonds are in the molecule? (1 point)

v. Determine **electronic** and **molecular geometry** at carbon (C) labeled ***e*** and sulfur (S) labeled ***f*** centres. (2 points)

Atom	Electronic Geometry	Molecular Geometry
Carbon (<i>e</i>)		
Sulfur (<i>f</i>)		

vi. Circle the value from the choices given below, that describes the **C-S-C bond angle at sulfur** (S) marked ***f***: (0.5 points)

>90° to <109.5° 109.5° >109.5 to <120° 120° >120° to <180°


Question 8 12 points

Using the **valence shell orbitals only** draw a **complete** molecular orbital (MO) diagram (with no s-p mixing) for the molecule **PS**, in the space provided below. **(6 points)**

Assume that 3s and 3p atomic orbitals will form molecular orbitals like those formed by the combination of 2s and 2p atomic orbitals that we learnt in class, and there is no s-p mixing.

Show **energy arrow**, **atomic orbitals** involved, **atom labels**, the **molecular orbitals** that are formed, **filled electrons** in the MO diagram etc. Make sure to label all the atomic and molecular orbitals.

Using the MO diagram above, answer the following questions:

- (i) Which one of the following will have the highest bond order? Circle the correct choice
 A) PS^{1-} B) PS C) PS^{1+}
- (ii) Which one of the following will have the longest bond length? Circle the correct choice
 A) PS^{1-} B) PS C) PS^{1+}
- (iii) Which of the following will be diamagnetic? Circle the correct choice.
 A) PS^{1-} B) PS C) PS^{1+}
- (iv) Draw Lewis dot structure of the molecule with the shortest bond length (based on the MO diagram) from among the three (PS^{1-} , PS, PS^{1+}), and indicate the bond order calculated from the Lewis dot structure. **(3 points)**

Lewis Dot Structure

Bond Order



F64C57A8-3335-4736-A25D-E9C6AB3BDBF0

midterm-exam-93388

#416 Page 10 of 10

SCRAP PAGE

**CHEMISTRY 110 –Midterm Exam****Thursday November 7, 2024, 6:30 – 8:30 pm**

Examiners: Pallavi Sirjoosingh and Ashok Kakkar

First name (please write as legibly as possible within the boxes)

--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--

Last name

--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--

Student ID number

--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--

This booklet has 8 questions on 10 pages (The last page can be used as a scrap page).

The staple must NOT be removed from the question booklet. Your intact question booklet must be returned at the end of the exam.

A separate booklet contains a periodic table, and datasheet.

Write your answers directly under the questions in the space provided. Do not write on the QR codes or near the top edges (as it may be cut off).

Please write **legibly and dark enough** for the scan to be readable. For students who use English as a second language, dictionaries are permitted. Calculators are allowed.

DO NOT ASK QUESTIONS OF THE INVIGILATORS about interpretations of the exam questions. If you have a concern about a question, please summarize it at the top of this page. All concerns will be evaluated after the exam. GOOD LUCK!

**Question 1 8 points** (*Assume all transitions are possible*)

a. 6 points Calculate the **energy (in kJ/mol)** required to ionize (completely remove) the electron from the **$n=3$** level of **He^+ ion** (He: Helium). Show your work including equations. (3 sig. figs.)

b. 2 points

(I) Select the largest (longest) wavelength from the three choices below:

- i. Wavelength of photon to *completely remove* an electron from *1s orbital of Hydrogen atom*
- ii. Wavelength of photon to *completely remove* an electron from *1s orbital of Boron atom (B)*
- iii. Wavelength of photon to *completely remove* an electron from *1s orbital of Boron cation (B^+)*

i or ii or iii

(II) Select the smallest (shortest) wavelength from the three choices below (*for H atom*):

- i. Wavelength of photon *emitted* for an electron transitioning from *2p to 1s orbital*
- ii. Wavelength of photon *absorbed* for an electron transitioning from *2s to 4p orbital*
- iii. Wavelength of photon *absorbed* for an electron transitioning from *1s to 3p orbital*

i or ii or iii



Question 2 8 points



A Chem110 student plots the graph shown on the left, from data obtained for the kinetic energy (eV) of the ejected electron versus frequency (s^{-1}) of the incident light, for an unknown metal. ($1\text{eV} = 1.6022 \times 10^{-19} \text{ J}$)

Based on the plot, answer the questions **a to c** below:

a. 1 point On the plot above, label the **threshold frequency**.

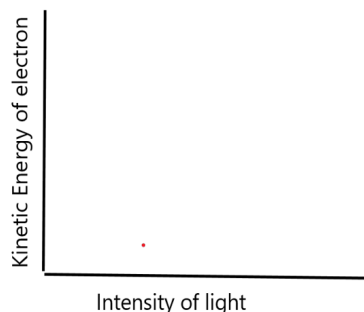
b. 2 points What is the value of the work potential (in J) of the unknown metal?

c. 3 points What is the *longest* wavelength (in nm) required to eject an electron from this metal's surface? Show your work including equation(s) used. (3 sig figs)

d. 2 points In the space provided below, plot the following

(Assume frequency is *above* threshold frequency):

- i. *Kinetic energy* of an ejected electron versus the *intensity* of the incident light (left)
- ii. *Number of electrons* ejected versus the *frequency* of the incident light (right)



**Question 3** **6 points****a. 2 points**

In the ground state of Ge (Germanium; $Z = 32$) atom, **name** the following orbitals:

i. Orbital containing the electron that requires the maximum energy to be removed.

ii. Orbital containing the electron that experiences the minimum effective nuclear charge.

b. 2 points

An unknown anion (A^-) has the following *excited* electronic configuration: $[Ar] 4s^2 3d^3 4p^1$

(Ar: Argon; $Z = 18$)

Give the **condensed ground state electronic configuration** of the atom, **A** and the cation, **A^{2+}** .
Show your work. (Assume all electronic transitions are possible)

Electronic Configuration of A:

Electronic Configuration of A^{2+} :

c. 2 points

In a single atom, determine how many electrons can have the following set of quantum numbers?

(i) $n = 5; l = 3$

In a single atom, determine how many electrons can have the following set of quantum numbers?

(i) $n = 3; l = 2; m_s = +1/2$

**Question 4** **8 points**

Rank the following from 1 to 4 in **increasing** order. Provide the ranking in the boxes provided.

Note: Atomic number (Z) are provided for relevant atoms

a) Increasing (lowest to highest) order of *ionization energy*: (1= lowest; 4 = highest)

Mg

Mg: Z = 12

Rb

Rb: Z = 37

Cs

Cs: Z = 55

K

K : Z = 19

b) Increasing (lowest to highest) order of *bond angle*: (1= lowest; 4 = highest)

H₂OCCl₄XeF₄

HCN

c) Increasing (smallest to largest) order of *ionic size* (1= smallest; 4 = largest)

S²⁻

S: Z = 16

O²⁻

O: Z = 8

Be²⁺

Be: Z = 4

Na⁺

Na: Z = 11

d) Increasing (lowest to highest) order of *bond energy* (1= lowest; 4 = highest)

P-F

P-Cl

P-I

P-Br

**Question 5** **12 points**

a. 9 points For the molecules HClO_4 and TeF_6 , draw the most stable *Lewis structure* showing appropriate shape and *dashed/wedged bonds*, based on VSEPR. Determine the **molecular geometry** around the central atom. For HClO_4 , label the *most polar bond*.



b. 3 points (Note: Lone pairs are not shown in structures below)

(i) For the three *resonance* structures (1, 2 and 3) shown below, answer the following question:



(I) Circle the *most contributing* resonance structure.

Structure 1

Structure 2

Structure 3

(ii) For the three structures given (4, 5, and 6) below, answer the following questions:



(I) Circle all the structures that represent *resonance structures* of each other:

Structure 4

Structure 5

Structure 6

(II) Among the resonance structures, circle the *most contributing* resonance structure.

Structure 4

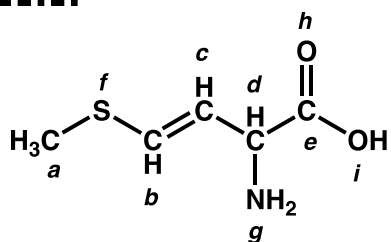
Structure 5

Structure 6

**Question 6** **6 points**

Using the data given below, calculate the **lattice energy** for SrO(s).

	ΔH° (kJ/mol)
$\text{Sr}(s) \longrightarrow \text{Sr}(g)$	+180
$\text{Sr}(g) \longrightarrow \text{Sr}^{2+}(g) + 2e^-$	+1468
$\text{O}_2(g) \longrightarrow 2\text{O}(g)$	+500
$\text{O}(g) + 2e^- \longrightarrow \text{O}^{2-}(g)$	+600
$\text{Sr}(s) + \frac{1}{2}\text{O}_2(g) \longrightarrow \text{SrO}(s)$	-550

**Question 7 10 points**

Consider the molecule given on the left, and answer questions (i) to (vi) using chemical bonding theories that we have learnt in class (valence bond theory, hybridization).

Note that the lone pairs are not shown in the structure:

i. Consider the carbon atoms marked in as ***a***, ***b***, ***c***, ***d***, and ***e*** in the molecule, and determine **hybridization** at each of the carbon centers. (2.5 points)

Atom	Hybridization
Carbon (<i>a</i>)	
Carbon (<i>b</i>)	
Carbon (<i>c</i>)	
Carbon (<i>d</i>)	
Carbon (<i>e</i>)	

ii. Now, consider the sulfur atom (S) marked as ***f***, nitrogen (N) atom (***g***), and oxygen(O) atoms (***h*** and ***i***), and determine their **hybridization**. (2 points)

Atom	Hybridization
Sulfur (<i>f</i>)	
Nitrogen (<i>g</i>)	
Oxygen (<i>h</i>)	
Oxygen (<i>i</i>)	

iii. **Describe** all the bonds formed by the carbon atoms ***a*** and ***b***, using sigma and pi designation as appropriate. (2 points)

Atom	Bonds formed (use sigma and pi as appropriate)
Carbon (<i>a</i>)	
Carbon (<i>b</i>)	

iv. How many total sp^3 -s sigma bonds are in the molecule? (1 point)

v. Determine **electronic** and **molecular geometry** at carbon (C) labeled ***e*** and sulfur (S) labeled ***f*** centres. (2 points)

Atom	Electronic Geometry	Molecular Geometry
Carbon (<i>e</i>)		
Sulfur (<i>f</i>)		

vi. Circle the value from the choices given below, that describes the **C-S-C bond angle at sulfur** (S) marked ***f***: (0.5 points)

>90° to <109.5° 109.5° >109.5 to <120° 120° >120° to <180°


Question 8 12 points

Using the **valence shell orbitals only** draw a **complete** molecular orbital (MO) diagram (with no s-p mixing) for the molecule **PS**, in the space provided below. **(6 points)**

Assume that 3s and 3p atomic orbitals will form molecular orbitals like those formed by the combination of 2s and 2p atomic orbitals that we learnt in class, and there is no s-p mixing.

Show **energy arrow**, **atomic orbitals** involved, **atom labels**, the **molecular orbitals** that are formed, **filled electrons** in the MO diagram etc. Make sure to label all the atomic and molecular orbitals.

Using the MO diagram above, answer the following questions:

- (i) Which one of the following will have the highest bond order? Circle the correct choice
 A) PS^{1-} B) PS C) PS^{1+}
- (ii) Which one of the following will have the longest bond length? Circle the correct choice
 A) PS^{1-} B) PS C) PS^{1+}
- (iii) Which of the following will be diamagnetic? Circle the correct choice.
 A) PS^{1-} B) PS C) PS^{1+}
- (iv) Draw Lewis dot structure of the molecule with the shortest bond length (based on the MO diagram) from among the three (PS^{1-} , PS, PS^{1+}), and indicate the bond order calculated from the Lewis dot structure. **(3 points)**

Lewis Dot Structure

Bond Order



BFB22990-EEE1-4244-B892-80B3B8737482

midterm-exam-93388

#417 Page 10 of 10

SCRAP PAGE

**CHEMISTRY 110 –Midterm Exam****Thursday November 7, 2024, 6:30 – 8:30 pm****Examiners: Pallavi Sirjoosingh and Ashok Kakkar**

First name (please write as legibly as possible within the boxes)

--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--

Last name

--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--

Student ID number

--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--

This booklet has 8 questions on 10 pages (The last page can be used as a scrap page).

The staple must NOT be removed from the question booklet. Your intact question booklet must be returned at the end of the exam.

A separate booklet contains a periodic table, and datasheet.

Write your answers directly under the questions in the space provided. Do not write on the QR codes or near the top edges (as it may be cut off).

Please write **legibly and dark enough** for the scan to be readable. For students who use English as a second language, dictionaries are permitted. Calculators are allowed.

DO NOT ASK QUESTIONS OF THE INVIGILATORS about interpretations of the exam questions. If you have a concern about a question, please summarize it at the top of this page. All concerns will be evaluated after the exam. GOOD LUCK!

**Question 1 8 points** (*Assume all transitions are possible*)

a. 6 points Calculate the **energy (in kJ/mol)** required to ionize (completely remove) the electron from the **n=3** level of **He⁺ ion** (He: Helium). Show your work including equations. (3 sig. figs.)

b. 2 points

(I) Select the largest (longest) wavelength from the three choices below:

- i. Wavelength of photon to *completely remove* an electron from *1s orbital of Hydrogen atom*
- ii. Wavelength of photon to *completely remove* an electron from *1s orbital of Boron atom (B)*
- iii. Wavelength of photon to *completely remove* an electron from *1s orbital of Boron cation (B⁺)*

i or ii or iii

(II) Select the smallest (shortest) wavelength from the three choices below (*for H atom*):

- i. Wavelength of photon *emitted* for an electron transitioning from *2p to 1s orbital*
- ii. Wavelength of photon *absorbed* for an electron transitioning from *2s to 4p orbital*
- iii. Wavelength of photon *absorbed* for an electron transitioning from *1s to 3p orbital*

i or ii or iii



Question 2 8 points



A Chem110 student plots the graph shown on the left, from data obtained for the kinetic energy (eV) of the ejected electron versus frequency (s^{-1}) of the incident light, for an unknown metal. ($1\text{eV} = 1.6022 \times 10^{-19} \text{ J}$)

Based on the plot, answer the questions **a to c** below:

a. 1 point On the plot above, label the **threshold frequency**.

b. 2 points What is the value of the work potential (in J) of the unknown metal?

c. 3 points What is the *longest* wavelength (in nm) required to eject an electron from this metal's surface? Show your work including equation(s) used. (3 sig figs)

d. 2 points In the space provided below, plot the following

(Assume frequency is *above* threshold frequency):

- Kinetic energy* of an ejected electron versus the *intensity* of the incident light (left)
- Number of electrons* ejected versus the *frequency* of the incident light (right)



**Question 3** **6 points****a. 2 points**

In the ground state of Ge (Germanium; $Z = 32$) atom, **name** the following orbitals:

i. Orbital containing the electron that requires the maximum energy to be removed.

ii. Orbital containing the electron that experiences the minimum effective nuclear charge.

b. 2 points

An unknown anion (A^-) has the following *excited* electronic configuration: $[Ar] 4s^2 3d^3 4p^1$

(Ar: Argon; $Z = 18$)

Give the **condensed ground state electronic configuration** of the atom, **A** and the cation, **A^{2+}** .
Show your work. (Assume all electronic transitions are possible)

Electronic Configuration of A:

Electronic Configuration of A^{2+} :

c. 2 points

In a single atom, determine how many electrons can have the following set of quantum numbers?

(i) $n = 5; l = 3$

In a single atom, determine how many electrons can have the following set of quantum numbers?

(i) $n = 3; l = 2; m_s = +1/2$

**Question 4** **8 points**

Rank the following from 1 to 4 in **increasing** order. Provide the ranking in the boxes provided.

Note: Atomic number (Z) are provided for relevant atoms

a) Increasing (lowest to highest) order of *ionization energy*: (1= lowest; 4 = highest)

Mg

Mg: Z = 12

Rb

Rb: Z = 37

Cs

Cs: Z = 55

K

K : Z = 19

b) Increasing (lowest to highest) order of *bond angle*: (1= lowest; 4 = highest)

H₂OCCl₄XeF₄

HCN

c) Increasing (smallest to largest) order of *ionic size* (1= smallest; 4 = largest)

S²⁻

S: Z = 16

O²⁻

O: Z = 8

Be²⁺

Be: Z = 4

Na⁺

Na: Z = 11

d) Increasing (lowest to highest) order of *bond energy* (1= lowest; 4 = highest)

P-F

P-Cl

P-I

P-Br

**Question 5** **12 points**

a. 9 points For the molecules HClO_4 and TeF_6 , draw the most stable *Lewis structure* showing appropriate shape and *dashed/wedged bonds*, based on VSEPR. Determine the **molecular geometry** around the central atom. For HClO_4 , label the *most polar bond*.



b. 3 points (Note: Lone pairs are not shown in structures below)

(i) For the three *resonance* structures (1, 2 and 3) shown below, answer the following question:



(I) Circle the *most contributing* resonance structure.

Structure 1

Structure 2

Structure 3

(ii) For the three structures given (4, 5, and 6) below, answer the following questions:



(I) Circle all the structures that represent *resonance structures* of each other:

Structure 4

Structure 5

Structure 6

(II) Among the resonance structures, circle the *most contributing* resonance structure.

Structure 4

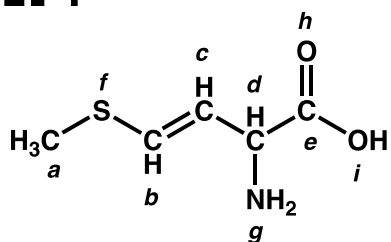
Structure 5

Structure 6

**Question 6** **6 points**

Using the data given below, calculate the **lattice energy** for SrO(s).

	ΔH° (kJ/mol)
$\text{Sr}(s) \longrightarrow \text{Sr}(g)$	+180
$\text{Sr}(g) \longrightarrow \text{Sr}^{2+}(g) + 2e^-$	+1468
$\text{O}_2(g) \longrightarrow 2\text{O}(g)$	+500
$\text{O}(g) + 2e^- \longrightarrow \text{O}^{2-}(g)$	+600
$\text{Sr}(s) + \frac{1}{2}\text{O}_2(g) \longrightarrow \text{SrO}(s)$	-550

**Question 7 10 points**

Consider the molecule given on the left, and answer questions (i) to (vi) using chemical bonding theories that we have learnt in class (valence bond theory, hybridization).

Note that the lone pairs are not shown in the structure:

i. Consider the carbon atoms marked in as ***a***, ***b***, ***c***, ***d***, and ***e*** in the molecule, and determine **hybridization** at each of the carbon centers. (2.5 points)

Atom	Hybridization
Carbon (<i>a</i>)	
Carbon (<i>b</i>)	
Carbon (<i>c</i>)	
Carbon (<i>d</i>)	
Carbon (<i>e</i>)	

ii. Now, consider the sulfur atom (S) marked as ***f***, nitrogen (N) atom (***g***), and oxygen(O) atoms (***h*** and ***i***), and determine their **hybridization**. (2 points)

Atom	Hybridization
Sulfur (<i>f</i>)	
Nitrogen (<i>g</i>)	
Oxygen (<i>h</i>)	
Oxygen (<i>i</i>)	

iii. **Describe** all the bonds formed by the carbon atoms ***a*** and ***b***, using sigma and pi designation as appropriate. (2 points)

Atom	Bonds formed (use sigma and pi as appropriate)
Carbon (<i>a</i>)	
Carbon (<i>b</i>)	

iv. How many total sp^3 -s sigma bonds are in the molecule? (1 point)

v. Determine **electronic** and **molecular geometry** at carbon (C) labeled ***e*** and sulfur (S) labeled ***f*** centres. (2 points)

Atom	Electronic Geometry	Molecular Geometry
Carbon (<i>e</i>)		
Sulfur (<i>f</i>)		

vi. Circle the value from the choices given below, that describes the **C-S-C bond angle at sulfur** (S) marked ***f***: (0.5 points)

>90° to <109.5° 109.5° >109.5 to <120° 120° >120° to <180°


Question 8 12 points

Using the **valence shell orbitals only** draw a **complete** molecular orbital (MO) diagram (with no s-p mixing) for the molecule **PS**, in the space provided below. **(6 points)**

Assume that 3s and 3p atomic orbitals will form molecular orbitals like those formed by the combination of 2s and 2p atomic orbitals that we learnt in class, and there is no s-p mixing.

Show **energy arrow**, **atomic orbitals** involved, **atom labels**, the **molecular orbitals** that are formed, **filled electrons** in the MO diagram etc. Make sure to label all the atomic and molecular orbitals.

Using the MO diagram above, answer the following questions:

- (i) Which one of the following will have the highest bond order? Circle the correct choice
 A) PS^{1-} B) PS C) PS^{1+}
- (ii) Which one of the following will have the longest bond length? Circle the correct choice
 A) PS^{1-} B) PS C) PS^{1+}
- (iii) Which of the following will be diamagnetic? Circle the correct choice.
 A) PS^{1-} B) PS C) PS^{1+}
- (iv) Draw Lewis dot structure of the molecule with the shortest bond length (based on the MO diagram) from among the three (PS^{1-} , PS, PS^{1+}), and indicate the bond order calculated from the Lewis dot structure. **(3 points)**

Lewis Dot Structure

Bond Order



CCDEDA13-D13C-41C7-8D62-A6EB235099B2

midterm-exam-93388

#418 Page 10 of 10

SCRAP PAGE

**CHEMISTRY 110 –Midterm Exam****Thursday November 7, 2024, 6:30 – 8:30 pm**

Examiners: Pallavi Sirjoosingh and Ashok Kakkar

First name (please write as legibly as possible within the boxes)

--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--

Last name

--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--

Student ID number

--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--

This booklet has 8 questions on 10 pages (The last page can be used as a scrap page).

The staple must NOT be removed from the question booklet. Your intact question booklet must be returned at the end of the exam.

A separate booklet contains a periodic table, and datasheet.

Write your answers directly under the questions in the space provided. Do not write on the QR codes or near the top edges (as it may be cut off).

Please write **legibly and dark enough** for the scan to be readable. For students who use English as a second language, dictionaries are permitted. Calculators are allowed.

DO NOT ASK QUESTIONS OF THE INVIGILATORS about interpretations of the exam questions. If you have a concern about a question, please summarize it at the top of this page. All concerns will be evaluated after the exam. GOOD LUCK!

**Question 1 8 points** (*Assume all transitions are possible*)

a. 6 points Calculate the **energy (in kJ/mol)** required to ionize (completely remove) the electron from the **n=3** level of **He⁺ ion** (He: Helium). Show your work including equations. (3 sig. figs.)

b. 2 points

(I) Select the largest (longest) wavelength from the three choices below:

- i. Wavelength of photon to *completely remove* an electron from *1s orbital of Hydrogen atom*
- ii. Wavelength of photon to *completely remove* an electron from *1s orbital of Boron atom (B)*
- iii. Wavelength of photon to *completely remove* an electron from *1s orbital of Boron cation (B⁺)*

i or ii or iii

(II) Select the smallest (shortest) wavelength from the three choices below (*for H atom*):

- i. Wavelength of photon *emitted* for an electron transitioning from *2p to 1s orbital*
- ii. Wavelength of photon *absorbed* for an electron transitioning from *2s to 4p orbital*
- iii. Wavelength of photon *absorbed* for an electron transitioning from *1s to 3p orbital*

i or ii or iii



Question 2 8 points



A Chem110 student plots the graph shown on the left, from data obtained for the kinetic energy (eV) of the ejected electron versus frequency (s^{-1}) of the incident light, for an unknown metal. ($1\text{eV} = 1.6022 \times 10^{-19} \text{ J}$)

Based on the plot, answer the questions **a to c** below:

a. 1 point On the plot above, label the **threshold frequency**.

b. 2 points What is the value of the work potential (in J) of the unknown metal?

c. 3 points What is the *longest* wavelength (in nm) required to eject an electron from this metal's surface? Show your work including equation(s) used. (3 sig figs)

d. 2 points In the space provided below, plot the following

(Assume frequency is *above* threshold frequency):

- i. *Kinetic energy* of an ejected electron versus the *intensity* of the incident light (left)
- ii. *Number of electrons* ejected versus the *frequency* of the incident light (right)



**Question 3** **6 points****a. 2 points**

In the ground state of Ge (Germanium; $Z = 32$) atom, **name** the following orbitals:

i. Orbital containing the electron that requires the maximum energy to be removed.

ii. Orbital containing the electron that experiences the minimum effective nuclear charge.

b. 2 points

An unknown anion (A^-) has the following *excited* electronic configuration: $[Ar] 4s^2 3d^3 4p^1$

(Ar: Argon; $Z = 18$)

Give the **condensed ground state electronic configuration** of the atom, **A** and the cation, **A^{2+}** .
Show your work. (Assume all electronic transitions are possible)

Electronic Configuration of A:

Electronic Configuration of A^{2+} :

c. 2 points

In a single atom, determine how many electrons can have the following set of quantum numbers?

(i) $n = 5; l = 3$

In a single atom, determine how many electrons can have the following set of quantum numbers?

(i) $n = 3; l = 2; m_s = +1/2$

**Question 4** **8 points**

Rank the following from 1 to 4 in **increasing** order. Provide the ranking in the boxes provided.

Note: Atomic number (Z) are provided for relevant atoms

a) Increasing (lowest to highest) order of *ionization energy*: (1= lowest; 4 = highest)

Mg

Mg: Z = 12

Rb

Rb: Z = 37

Cs

Cs: Z = 55

K

K : Z = 19

b) Increasing (lowest to highest) order of *bond angle*: (1= lowest; 4 = highest)

H₂OCCl₄XeF₄

HCN

c) Increasing (smallest to largest) order of *ionic size* (1= smallest; 4 = largest)

S²⁻

S: Z = 16

O²⁻

O: Z = 8

Be²⁺

Be: Z = 4

Na⁺

Na: Z = 11

d) Increasing (lowest to highest) order of *bond energy* (1= lowest; 4 = highest)

P-F

P-Cl

P-I

P-Br

**Question 5** **12 points**

a. 9 points For the molecules HClO_4 and TeF_6 , draw the most stable *Lewis structure* showing appropriate shape and *dashed/wedged bonds*, based on VSEPR. Determine the **molecular geometry** around the central atom. For HClO_4 , label the *most polar bond*.



b. 3 points (Note: Lone pairs are not shown in structures below)

(i) For the three *resonance* structures (1, 2 and 3) shown below, answer the following question:



(I) Circle the *most contributing* resonance structure.

Structure 1

Structure 2

Structure 3

(ii) For the three structures given (4, 5, and 6) below, answer the following questions:



(I) Circle all the structures that represent *resonance structures* of each other:

Structure 4

Structure 5

Structure 6

(II) Among the resonance structures, circle the *most contributing* resonance structure.

Structure 4

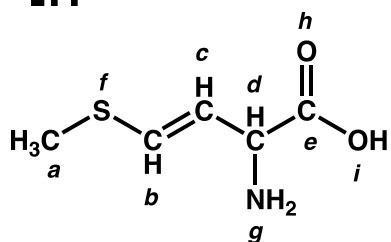
Structure 5

Structure 6

**Question 6** **6 points**

Using the data given below, calculate the **lattice energy** for SrO(s).

	ΔH° (kJ/mol)
$\text{Sr}(s) \longrightarrow \text{Sr}(g)$	+180
$\text{Sr}(g) \longrightarrow \text{Sr}^{2+}(g) + 2e^-$	+1468
$\text{O}_2(g) \longrightarrow 2\text{O}(g)$	+500
$\text{O}(g) + 2e^- \longrightarrow \text{O}^{2-}(g)$	+600
$\text{Sr}(s) + \frac{1}{2}\text{O}_2(g) \longrightarrow \text{SrO}(s)$	-550

**Question 7 10 points**

Consider the molecule given on the left, and answer questions (i) to (vi) using chemical bonding theories that we have learnt in class (valence bond theory, hybridization).

Note that the lone pairs are not shown in the structure:

i. Consider the carbon atoms marked in as ***a***, ***b***, ***c***, ***d***, and ***e*** in the molecule, and determine **hybridization** at each of the carbon centers. (2.5 points)

Atom	Hybridization
Carbon (<i>a</i>)	
Carbon (<i>b</i>)	
Carbon (<i>c</i>)	
Carbon (<i>d</i>)	
Carbon (<i>e</i>)	

ii. Now, consider the sulfur atom (S) marked as ***f***, nitrogen (N) atom (***g***), and oxygen(O) atoms (***h*** and ***i***), and determine their **hybridization**. (2 points)

Atom	Hybridization
Sulfur (<i>f</i>)	
Nitrogen (<i>g</i>)	
Oxygen (<i>h</i>)	
Oxygen (<i>i</i>)	

iii. **Describe** all the bonds formed by the carbon atoms ***a*** and ***b***, using sigma and pi designation as appropriate. (2 points)

Atom	Bonds formed (use sigma and pi as appropriate)
Carbon (<i>a</i>)	
Carbon (<i>b</i>)	

iv. How many total sp^3 -s sigma bonds are in the molecule? (1 point)

v. Determine **electronic** and **molecular geometry** at carbon (C) labeled ***e*** and sulfur (S) labeled ***f*** centres. (2 points)

Atom	Electronic Geometry	Molecular Geometry
Carbon (<i>e</i>)		
Sulfur (<i>f</i>)		

vi. Circle the value from the choices given below, that describes the **C-S-C bond angle at sulfur** (S) marked ***f***: (0.5 points)

>90° to <109.5° 109.5° >109.5 to <120° 120° >120° to <180°



Question 8 12 points

Using the **valence shell orbitals only** draw a **complete** molecular orbital (MO) diagram (with no s-p mixing) for the molecule **PS**, in the space provided below. **(6 points)**

Assume that 3s and 3p atomic orbitals will form molecular orbitals like those formed by the combination of 2s and 2p atomic orbitals that we learnt in class, and there is no s-p mixing.

Show **energy arrow**, **atomic orbitals** involved, **atom labels**, the **molecular orbitals** that are formed, **filled electrons** in the MO diagram etc. Make sure to label all the atomic and molecular orbitals.

Using the MO diagram above, answer the following questions:

- (i) Which one of the following will have the highest bond order? Circle the correct choice
 A) PS^{1-} B) PS C) PS^{1+}
- (ii) Which one of the following will have the longest bond length? Circle the correct choice
 A) PS^{1-} B) PS C) PS^{1+}
- (iii) Which of the following will be diamagnetic? Circle the correct choice.
 A) PS^{1-} B) PS C) PS^{1+}
- (iv) Draw Lewis dot structure of the molecule with the shortest bond length (based on the MO diagram) from among the three (PS^{1-} , PS, PS^{1+}), and indicate the bond order calculated from the Lewis dot structure. **(3 points)**

Lewis Dot Structure

Bond Order



63306526-7716-4DF7-82AC-4A9B085A534A

midterm-exam-93388

#419 Page 10 of 10

SCRAP PAGE

**CHEMISTRY 110 –Midterm Exam****Thursday November 7, 2024, 6:30 – 8:30 pm****Examiners: Pallavi Sirjoosingh and Ashok Kakkar**

First name (please write as legibly as possible within the boxes)

--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--

Last name

--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--

Student ID number

--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--

This booklet has 8 questions on 10 pages (The last page can be used as a scrap page).

The staple must NOT be removed from the question booklet. Your intact question booklet must be returned at the end of the exam.

A separate booklet contains a periodic table, and datasheet.

Write your answers directly under the questions in the space provided. Do not write on the QR codes or near the top edges (as it may be cut off).

Please write **legibly and dark enough** for the scan to be readable. For students who use English as a second language, dictionaries are permitted. Calculators are allowed.

DO NOT ASK QUESTIONS OF THE INVIGILATORS about interpretations of the exam questions. If you have a concern about a question, please summarize it at the top of this page. All concerns will be evaluated after the exam. GOOD LUCK!

**Question 1 8 points** (*Assume all transitions are possible*)

a. 6 points Calculate the **energy (in kJ/mol)** required to ionize (completely remove) the electron from the **n=3** level of **He⁺ ion** (He: Helium). Show your work including equations. (3 sig. figs.)

b. 2 points

(I) Select the largest (longest) wavelength from the three choices below:

- i. Wavelength of photon to *completely remove* an electron from *1s orbital of Hydrogen atom*
- ii. Wavelength of photon to *completely remove* an electron from *1s orbital of Boron atom (B)*
- iii. Wavelength of photon to *completely remove* an electron from *1s orbital of Boron cation (B⁺)*

i or ii or iii

(II) Select the smallest (shortest) wavelength from the three choices below (*for H atom*):

- i. Wavelength of photon *emitted* for an electron transitioning from *2p to 1s orbital*
- ii. Wavelength of photon *absorbed* for an electron transitioning from *2s to 4p orbital*
- iii. Wavelength of photon *absorbed* for an electron transitioning from *1s to 3p orbital*

i or ii or iii



Question 2 8 points



A Chem110 student plots the graph shown on the left, from data obtained for the kinetic energy (eV) of the ejected electron versus frequency (s^{-1}) of the incident light, for an unknown metal. ($1\text{eV} = 1.6022 \times 10^{-19} \text{ J}$)

Based on the plot, answer the questions **a to c** below:

a. 1 point On the plot above, label the **threshold frequency**.

b. 2 points What is the value of the work potential (in J) of the unknown metal?

c. 3 points What is the *longest* wavelength (in nm) required to eject an electron from this metal's surface? Show your work including equation(s) used. (3 sig figs)

d. 2 points In the space provided below, plot the following

(Assume frequency is *above* threshold frequency):

- i. *Kinetic energy* of an ejected electron versus the *intensity* of the incident light (left)
- ii. *Number of electrons* ejected versus the *frequency* of the incident light (right)



**Question 3** **6 points****a. 2 points**

In the ground state of Ge (Germanium; $Z = 32$) atom, **name** the following orbitals:

i. Orbital containing the electron that requires the maximum energy to be removed.

ii. Orbital containing the electron that experiences the minimum effective nuclear charge.

b. 2 points

An unknown anion (A^-) has the following *excited* electronic configuration: $[Ar] 4s^2 3d^3 4p^1$

(Ar: Argon; $Z = 18$)

Give the **condensed ground state electronic configuration** of the atom, **A** and the cation, **A^{2+}** .
Show your work. (Assume all electronic transitions are possible)

Electronic Configuration of A:

Electronic Configuration of A^{2+} :

c. 2 points

In a single atom, determine how many electrons can have the following set of quantum numbers?

(i) $n = 5; l = 3$

In a single atom, determine how many electrons can have the following set of quantum numbers?

(i) $n = 3; l = 2; m_s = +1/2$

**Question 4** **8 points**

Rank the following from 1 to 4 in **increasing** order. Provide the ranking in the boxes provided.

Note: Atomic number (Z) are provided for relevant atoms

a) Increasing (lowest to highest) order of *ionization energy*: (1= lowest; 4 = highest)

Mg

Mg: Z = 12

Rb

Rb: Z = 37

Cs

Cs: Z = 55

K

K : Z = 19

b) Increasing (lowest to highest) order of *bond angle*: (1= lowest; 4 = highest)

H₂OCCl₄XeF₄

HCN

c) Increasing (smallest to largest) order of *ionic size* (1= smallest; 4 = largest)

S²⁻

S: Z = 16

O²⁻

O: Z = 8

Be²⁺

Be: Z = 4

Na⁺

Na: Z = 11

d) Increasing (lowest to highest) order of *bond energy* (1= lowest; 4 = highest)

P-F

P-Cl

P-I

P-Br

**Question 5** **12 points**

a. 9 points For the molecules HClO_4 and TeF_6 , draw the most stable *Lewis structure* showing appropriate shape and *dashed/wedged bonds*, based on VSEPR. Determine the **molecular geometry** around the central atom. For HClO_4 , label the *most polar bond*.



b. 3 points (Note: Lone pairs are not shown in structures below)

(i) For the three *resonance* structures (1, 2 and 3) shown below, answer the following question:



(I) Circle the *most contributing* resonance structure.

Structure 1

Structure 2

Structure 3

(ii) For the three structures given (4, 5, and 6) below, answer the following questions:



(I) Circle all the structures that represent *resonance structures* of each other:

Structure 4

Structure 5

Structure 6

(II) Among the resonance structures, circle the *most contributing* resonance structure.

Structure 4

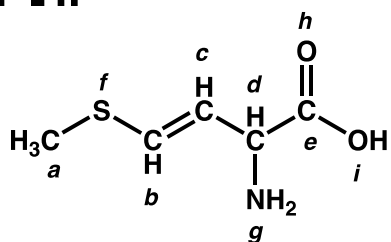
Structure 5

Structure 6

**Question 6** **6 points**

Using the data given below, calculate the **lattice energy** for SrO(s).

	ΔH° (kJ/mol)
$\text{Sr}(s) \longrightarrow \text{Sr}(g)$	+180
$\text{Sr}(g) \longrightarrow \text{Sr}^{2+}(g) + 2e^-$	+1468
$\text{O}_2(g) \longrightarrow 2\text{O}(g)$	+500
$\text{O}(g) + 2e^- \longrightarrow \text{O}^{2-}(g)$	+600
$\text{Sr}(s) + \frac{1}{2}\text{O}_2(g) \longrightarrow \text{SrO}(s)$	-550

**Question 7 10 points**

Consider the molecule given on the left, and answer questions (i) to (vi) using chemical bonding theories that we have learnt in class (valence bond theory, hybridization).

Note that the lone pairs are not shown in the structure:

i. Consider the carbon atoms marked in as ***a***, ***b***, ***c***, ***d***, and ***e*** in the molecule, and determine **hybridization** at each of the carbon centers. (2.5 points)

Atom	Hybridization
Carbon (<i>a</i>)	
Carbon (<i>b</i>)	
Carbon (<i>c</i>)	
Carbon (<i>d</i>)	
Carbon (<i>e</i>)	

ii. Now, consider the sulfur atom (S) marked as ***f***, nitrogen (N) atom (***g***), and oxygen(O) atoms (***h*** and ***i***), and determine their **hybridization**. (2 points)

Atom	Hybridization
Sulfur (<i>f</i>)	
Nitrogen (<i>g</i>)	
Oxygen (<i>h</i>)	
Oxygen (<i>i</i>)	

iii. **Describe** all the bonds formed by the carbon atoms ***a*** and ***b***, using sigma and pi designation as appropriate. (2 points)

Atom	Bonds formed (use sigma and pi as appropriate)
Carbon (<i>a</i>)	
Carbon (<i>b</i>)	

iv. How many total sp^3 -s sigma bonds are in the molecule? (1 point)

v. Determine **electronic** and **molecular geometry** at carbon (C) labeled ***e*** and sulfur (S) labeled ***f*** centres. (2 points)

Atom	Electronic Geometry	Molecular Geometry
Carbon (<i>e</i>)		
Sulfur (<i>f</i>)		

vi. Circle the value from the choices given below, that describes the **C-S-C bond angle at sulfur** (S) marked ***f***: (0.5 points)

>90° to <109.5° 109.5° >109.5 to <120° 120° >120° to <180°


Question 8 12 points

Using the **valence shell orbitals only** draw a **complete** molecular orbital (MO) diagram (with no s-p mixing) for the molecule **PS**, in the space provided below. **(6 points)**

Assume that 3s and 3p atomic orbitals will form molecular orbitals like those formed by the combination of 2s and 2p atomic orbitals that we learnt in class, and there is no s-p mixing.

Show **energy arrow**, **atomic orbitals** involved, **atom labels**, the **molecular orbitals** that are formed, **filled electrons** in the MO diagram etc. Make sure to label all the atomic and molecular orbitals.

Using the MO diagram above, answer the following questions:

- (i) Which one of the following will have the highest bond order? Circle the correct choice
 A) PS^{1-} B) PS C) PS^{1+}
- (ii) Which one of the following will have the longest bond length? Circle the correct choice
 A) PS^{1-} B) PS C) PS^{1+}
- (iii) Which of the following will be diamagnetic? Circle the correct choice.
 A) PS^{1-} B) PS C) PS^{1+}
- (iv) Draw Lewis dot structure of the molecule with the shortest bond length (based on the MO diagram) from among the three (PS^{1-} , PS, PS^{1+}), and indicate the bond order calculated from the Lewis dot structure. **(3 points)**

Lewis Dot Structure

Bond Order



E2C4310F-4F5B-4A92-9E0B-034F4A2F380F

midterm-exam-93388

#420 Page 10 of 10

SCRAP PAGE

**CHEMISTRY 110 –Midterm Exam****Thursday November 7, 2024, 6:30 – 8:30 pm**

Examiners: Pallavi Sirjoosingh and Ashok Kakkar

First name (please write as legibly as possible within the boxes)

--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--

Last name

--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--

Student ID number

--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--

This booklet has 8 questions on 10 pages (The last page can be used as a scrap page).

The staple must NOT be removed from the question booklet. Your intact question booklet must be returned at the end of the exam.

A separate booklet contains a periodic table, and datasheet.

Write your answers directly under the questions in the space provided. Do not write on the QR codes or near the top edges (as it may be cut off).

Please write **legibly and dark enough** for the scan to be readable. For students who use English as a second language, dictionaries are permitted. Calculators are allowed.

DO NOT ASK QUESTIONS OF THE INVIGILATORS about interpretations of the exam questions. If you have a concern about a question, please summarize it at the top of this page. All concerns will be evaluated after the exam. GOOD LUCK!

**Question 1 8 points** (*Assume all transitions are possible*)

a. 6 points Calculate the **energy (in kJ/mol)** required to ionize (completely remove) the electron from the **$n=3$** level of **He^+ ion** (He: Helium). Show your work including equations. (3 sig. figs.)

b. 2 points

(I) Select the largest (longest) wavelength from the three choices below:

- i. Wavelength of photon to *completely remove* an electron from *1s orbital of Hydrogen atom*
- ii. Wavelength of photon to *completely remove* an electron from *1s orbital of Boron atom (B)*
- iii. Wavelength of photon to *completely remove* an electron from *1s orbital of Boron cation (B^+)*

i or ii or iii

(II) Select the smallest (shortest) wavelength from the three choices below (*for H atom*):

- i. Wavelength of photon *emitted* for an electron transitioning from *2p to 1s orbital*
- ii. Wavelength of photon *absorbed* for an electron transitioning from *2s to 4p orbital*
- iii. Wavelength of photon *absorbed* for an electron transitioning from *1s to 3p orbital*

i or ii or iii



Question 2 8 points



A Chem110 student plots the graph shown on the left, from data obtained for the kinetic energy (eV) of the ejected electron versus frequency (s^{-1}) of the incident light, for an unknown metal. ($1\text{eV} = 1.6022 \times 10^{-19} \text{ J}$)

Based on the plot, answer the questions **a to c** below:

a. 1 point On the plot above, label the **threshold frequency**.

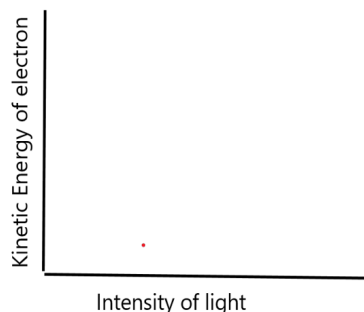
b. 2 points What is the value of the work potential (in J) of the unknown metal?

c. 3 points What is the *longest* wavelength (in nm) required to eject an electron from this metal's surface? Show your work including equation(s) used. (3 sig figs)

d. 2 points In the space provided below, plot the following

(Assume frequency is *above* threshold frequency):

- Kinetic energy* of an ejected electron versus the *intensity* of the incident light (left)
- Number of electrons* ejected versus the *frequency* of the incident light (right)



**Question 3** **6 points****a. 2 points**

In the ground state of Ge (Germanium; $Z = 32$) atom, **name** the following orbitals:

i. Orbital containing the electron that requires the maximum energy to be removed.

ii. Orbital containing the electron that experiences the minimum effective nuclear charge.

b. 2 points

An unknown anion (A^-) has the following *excited* electronic configuration: $[Ar] 4s^2 3d^3 4p^1$

(Ar: Argon; $Z = 18$)

Give the **condensed ground state electronic configuration** of the atom, **A** and the cation, **A^{2+}** .
Show your work. (Assume all electronic transitions are possible)

Electronic Configuration of A:

Electronic Configuration of A^{2+} :

c. 2 points

In a single atom, determine how many electrons can have the following set of quantum numbers?

(i) $n = 5; l = 3$

In a single atom, determine how many electrons can have the following set of quantum numbers?

(i) $n = 3; l = 2; m_s = +1/2$

**Question 4** **8 points**

Rank the following from 1 to 4 in **increasing** order. Provide the ranking in the boxes provided.

Note: Atomic number (Z) are provided for relevant atoms

a) Increasing (lowest to highest) order of *ionization energy*: (1= lowest; 4 = highest)

Mg

Mg: Z = 12

Rb

Rb: Z = 37

Cs

Cs: Z = 55

K

K : Z = 19

b) Increasing (lowest to highest) order of *bond angle*: (1= lowest; 4 = highest)

H₂OCCl₄XeF₄

HCN

c) Increasing (smallest to largest) order of *ionic size* (1= smallest; 4 = largest)

S²⁻

S: Z = 16

O²⁻

O: Z = 8

Be²⁺

Be: Z = 4

Na⁺

Na: Z = 11

d) Increasing (lowest to highest) order of *bond energy* (1= lowest; 4 = highest)

P-F

P-Cl

P-I

P-Br

**Question 5** **12 points**

a. 9 points For the molecules HClO_4 and TeF_6 , draw the most stable *Lewis structure* showing appropriate shape and *dashed/wedged bonds*, based on VSEPR. Determine the **molecular geometry** around the central atom. For HClO_4 , label the *most polar bond*.



b. 3 points (Note: Lone pairs are not shown in structures below)

(i) For the three *resonance* structures (1, 2 and 3) shown below, answer the following question:



(I) Circle the *most contributing* resonance structure.

Structure 1

Structure 2

Structure 3

(ii) For the three structures given (4, 5, and 6) below, answer the following questions:



(I) Circle all the structures that represent *resonance structures* of each other:

Structure 4

Structure 5

Structure 6

(II) Among the resonance structures, circle the *most contributing* resonance structure.

Structure 4

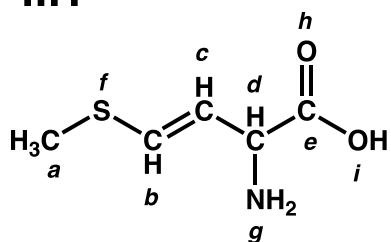
Structure 5

Structure 6

**Question 6** **6 points**

Using the data given below, calculate the **lattice energy** for SrO(s).

	ΔH° (kJ/mol)
$\text{Sr}(s) \longrightarrow \text{Sr}(g)$	+180
$\text{Sr}(g) \longrightarrow \text{Sr}^{2+}(g) + 2e^-$	+1468
$\text{O}_2(g) \longrightarrow 2\text{O}(g)$	+500
$\text{O}(g) + 2e^- \longrightarrow \text{O}^{2-}(g)$	+600
$\text{Sr}(s) + \frac{1}{2}\text{O}_2(g) \longrightarrow \text{SrO}(s)$	-550

**Question 7 10 points**

Consider the molecule given on the left, and answer questions (i) to (vi) using chemical bonding theories that we have learnt in class (valence bond theory, hybridization).

Note that the lone pairs are not shown in the structure:

i. Consider the carbon atoms marked in as ***a***, ***b***, ***c***, ***d***, and ***e*** in the molecule, and determine **hybridization** at each of the carbon centers. (2.5 points)

Atom	Hybridization
Carbon (<i>a</i>)	
Carbon (<i>b</i>)	
Carbon (<i>c</i>)	
Carbon (<i>d</i>)	
Carbon (<i>e</i>)	

ii. Now, consider the sulfur atom (S) marked as ***f***, nitrogen (N) atom (***g***), and oxygen(O) atoms (***h*** and ***i***), and determine their **hybridization**. (2 points)

Atom	Hybridization
Sulfur (<i>f</i>)	
Nitrogen (<i>g</i>)	
Oxygen (<i>h</i>)	
Oxygen (<i>i</i>)	

iii. **Describe** all the bonds formed by the carbon atoms ***a*** and ***b***, using sigma and pi designation as appropriate. (2 points)

Atom	Bonds formed (use sigma and pi as appropriate)
Carbon (<i>a</i>)	
Carbon (<i>b</i>)	

iv. How many total sp^3 -s sigma bonds are in the molecule? (1 point)

v. Determine **electronic** and **molecular geometry** at carbon (C) labeled ***e*** and sulfur (S) labeled ***f*** centres. (2 points)

Atom	Electronic Geometry	Molecular Geometry
Carbon (<i>e</i>)		
Sulfur (<i>f</i>)		

vi. Circle the value from the choices given below, that describes the **C-S-C bond angle at sulfur** (S) marked ***f***: (0.5 points)

>90° to <109.5° 109.5° >109.5 to <120° 120° >120° to <180°

**Question 8 12 points**

Using the **valence shell orbitals only** draw a **complete** molecular orbital (MO) diagram (with no s-p mixing) for the molecule **PS**, in the space provided below. **(6 points)**

Assume that 3s and 3p atomic orbitals will form molecular orbitals like those formed by the combination of 2s and 2p atomic orbitals that we learnt in class, and there is no s-p mixing.

Show **energy arrow**, **atomic orbitals** involved, **atom labels**, the **molecular orbitals** that are formed, **filled electrons** in the MO diagram etc. Make sure to label all the atomic and molecular orbitals.

Using the MO diagram above, answer the following questions:

- (i) Which one of the following will have the highest bond order? Circle the correct choice
A) PS^{1-} B) PS C) PS^{1+}
- (ii) Which one of the following will have the longest bond length? Circle the correct choice
A) PS^{1-} B) PS C) PS^{1+}
- (iii) Which of the following will be diamagnetic? Circle the correct choice.
A) PS^{1-} B) PS C) PS^{1+}
- (iv) Draw Lewis dot structure of the molecule with the shortest bond length (based on the MO diagram) from among the three (PS^{1-} , PS, PS^{1+}), and indicate the bond order calculated from the Lewis dot structure. **(3 points)**

Lewis Dot Structure

Bond Order



6CCE0BAF-9E7B-407B-9E01-361A92FD3FA9

midterm-exam-93388

#421 Page 10 of 10

SCRAP PAGE

**CHEMISTRY 110 –Midterm Exam****Thursday November 7, 2024, 6:30 – 8:30 pm****Examiners: Pallavi Sirjoosingh and Ashok Kakkar**

First name (please write as legibly as possible within the boxes)

--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--

Last name

--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--

Student ID number

--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--

This booklet has 8 questions on 10 pages (The last page can be used as a scrap page).

The staple must NOT be removed from the question booklet. Your intact question booklet must be returned at the end of the exam.

A separate booklet contains a periodic table, and datasheet.

Write your answers directly under the questions in the space provided. Do not write on the QR codes or near the top edges (as it may be cut off).

Please write **legibly and dark enough** for the scan to be readable. For students who use English as a second language, dictionaries are permitted. Calculators are allowed.

DO NOT ASK QUESTIONS OF THE INVIGILATORS about interpretations of the exam questions. If you have a concern about a question, please summarize it at the top of this page. All concerns will be evaluated after the exam. GOOD LUCK!

**Question 1 8 points** (*Assume all transitions are possible*)

a. 6 points Calculate the **energy (in kJ/mol)** required to ionize (completely remove) the electron from the **n=3** level of **He⁺ ion** (He: Helium). Show your work including equations. (3 sig. figs.)

b. 2 points

(I) Select the largest (longest) wavelength from the three choices below:

- i. Wavelength of photon to *completely remove* an electron from *1s orbital of Hydrogen atom*
- ii. Wavelength of photon to *completely remove* an electron from *1s orbital of Boron atom (B)*
- iii. Wavelength of photon to *completely remove* an electron from *1s orbital of Boron cation (B⁺)*

i or ii or iii

(II) Select the smallest (shortest) wavelength from the three choices below (*for H atom*):

- i. Wavelength of photon *emitted* for an electron transitioning from *2p to 1s orbital*
- ii. Wavelength of photon *absorbed* for an electron transitioning from *2s to 4p orbital*
- iii. Wavelength of photon *absorbed* for an electron transitioning from *1s to 3p orbital*

i or ii or iii



Question 2 8 points



A Chem110 student plots the graph shown on the left, from data obtained for the kinetic energy (eV) of the ejected electron versus frequency (s^{-1}) of the incident light, for an unknown metal. ($1\text{eV} = 1.6022 \times 10^{-19} \text{ J}$)

Based on the plot, answer the questions **a to c** below:

a. 1 point On the plot above, label the **threshold frequency**.

b. 2 points What is the value of the work potential (in J) of the unknown metal?

c. 3 points What is the *longest* wavelength (in nm) required to eject an electron from this metal's surface? Show your work including equation(s) used. (3 sig figs)

d. 2 points In the space provided below, plot the following

(Assume frequency is *above* threshold frequency):

- Kinetic energy* of an ejected electron versus the *intensity* of the incident light (left)
- Number of electrons* ejected versus the *frequency* of the incident light (right)



**Question 3** **6 points****a. 2 points**

In the ground state of Ge (Germanium; $Z = 32$) atom, **name** the following orbitals:

i. Orbital containing the electron that requires the maximum energy to be removed.

ii. Orbital containing the electron that experiences the minimum effective nuclear charge.

b. 2 points

An unknown anion (A^-) has the following *excited* electronic configuration: $[Ar] 4s^2 3d^3 4p^1$

(Ar: Argon; $Z = 18$)

Give the **condensed ground state electronic configuration** of the atom, **A** and the cation, **A^{2+}** .
Show your work. (Assume all electronic transitions are possible)

Electronic Configuration of A:

Electronic Configuration of A^{2+} :

c. 2 points

In a single atom, determine how many electrons can have the following set of quantum numbers?

(i) $n = 5; l = 3$

In a single atom, determine how many electrons can have the following set of quantum numbers?

(i) $n = 3; l = 2; m_s = +1/2$

**Question 4** **8 points**

Rank the following from 1 to 4 in **increasing** order. Provide the ranking in the boxes provided.

Note: Atomic number (Z) are provided for relevant atoms

a) Increasing (lowest to highest) order of *ionization energy*: (1= lowest; 4 = highest)

Mg

Mg: Z = 12

Rb

Rb: Z = 37

Cs

Cs: Z = 55

K

K : Z = 19

b) Increasing (lowest to highest) order of *bond angle*: (1= lowest; 4 = highest)

H₂OCCl₄XeF₄

HCN

c) Increasing (smallest to largest) order of *ionic size* (1= smallest; 4 = largest)

S²⁻

S: Z = 16

O²⁻

O: Z = 8

Be²⁺

Be: Z = 4

Na⁺

Na: Z = 11

d) Increasing (lowest to highest) order of *bond energy* (1= lowest; 4 = highest)

P-F

P-Cl

P-I

P-Br

**Question 5** **12 points**

a. 9 points For the molecules HClO_4 and TeF_6 , draw the most stable *Lewis structure* showing appropriate shape and *dashed/wedged bonds*, based on VSEPR. Determine the **molecular geometry** around the central atom. For HClO_4 , label the *most polar bond*.



b. 3 points (Note: Lone pairs are not shown in structures below)

(i) For the three *resonance* structures (1, 2 and 3) shown below, answer the following question:



(I) Circle the *most contributing* resonance structure.

Structure 1

Structure 2

Structure 3

(ii) For the three structures given (4, 5, and 6) below, answer the following questions:



(I) Circle all the structures that represent *resonance structures* of each other:

Structure 4

Structure 5

Structure 6

(II) Among the resonance structures, circle the *most contributing* resonance structure.

Structure 4

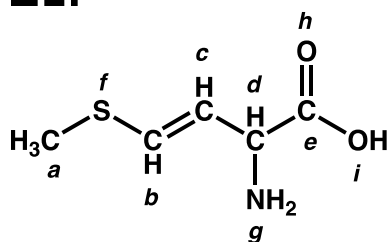
Structure 5

Structure 6

**Question 6** **6 points**

Using the data given below, calculate the **lattice energy** for SrO(s).

	ΔH° (kJ/mol)
$\text{Sr}(s) \longrightarrow \text{Sr}(g)$	+180
$\text{Sr}(g) \longrightarrow \text{Sr}^{2+}(g) + 2e^-$	+1468
$\text{O}_2(g) \longrightarrow 2\text{O}(g)$	+500
$\text{O}(g) + 2e^- \longrightarrow \text{O}^{2-}(g)$	+600
$\text{Sr}(s) + \frac{1}{2}\text{O}_2(g) \longrightarrow \text{SrO}(s)$	-550

**Question 7 10 points**

Consider the molecule given on the left, and answer questions (i) to (vi) using chemical bonding theories that we have learnt in class (valence bond theory, hybridization).

Note that the lone pairs are not shown in the structure:

i. Consider the carbon atoms marked in as ***a***, ***b***, ***c***, ***d***, and ***e*** in the molecule, and determine **hybridization** at each of the carbon centers. (2.5 points)

Atom	Hybridization
Carbon (<i>a</i>)	
Carbon (<i>b</i>)	
Carbon (<i>c</i>)	
Carbon (<i>d</i>)	
Carbon (<i>e</i>)	

ii. Now, consider the sulfur atom (S) marked as ***f***, nitrogen (N) atom (***g***), and oxygen(O) atoms (***h*** and ***i***), and determine their **hybridization**. (2 points)

Atom	Hybridization
Sulfur (<i>f</i>)	
Nitrogen (<i>g</i>)	
Oxygen (<i>h</i>)	
Oxygen (<i>i</i>)	

iii. **Describe** all the bonds formed by the carbon atoms ***a*** and ***b***, using sigma and pi designation as appropriate. (2 points)

Atom	Bonds formed (use sigma and pi as appropriate)
Carbon (<i>a</i>)	
Carbon (<i>b</i>)	

iv. How many total sp^3 -s sigma bonds are in the molecule? (1 point)

v. Determine **electronic** and **molecular geometry** at carbon (C) labeled ***e*** and sulfur (S) labeled ***f*** centres. (2 points)

Atom	Electronic Geometry	Molecular Geometry
Carbon (<i>e</i>)		
Sulfur (<i>f</i>)		

vi. Circle the value from the choices given below, that describes the **C-S-C bond angle at sulfur** (S) marked ***f***: (0.5 points)

>90° to <109.5°

109.5°

>109.5 to <120°

120°

>120° to <180°



Question 8 12 points

Using the **valence shell orbitals only** draw a **complete** molecular orbital (MO) diagram (with no s-p mixing) for the molecule **PS**, in the space provided below. **(6 points)**

Assume that 3s and 3p atomic orbitals will form molecular orbitals like those formed by the combination of 2s and 2p atomic orbitals that we learnt in class, and there is no s-p mixing.

Show **energy arrow**, **atomic orbitals** involved, **atom labels**, the **molecular orbitals** that are formed, **filled electrons** in the MO diagram etc. Make sure to label all the atomic and molecular orbitals.

Using the MO diagram above, answer the following questions:

- (i) Which one of the following will have the highest bond order? Circle the correct choice
 A) PS^{1-} B) PS C) PS^{1+}
- (ii) Which one of the following will have the longest bond length? Circle the correct choice
 A) PS^{1-} B) PS C) PS^{1+}
- (iii) Which of the following will be diamagnetic? Circle the correct choice.
 A) PS^{1-} B) PS C) PS^{1+}
- (iv) Draw Lewis dot structure of the molecule with the shortest bond length (based on the MO diagram) from among the three (PS^{1-} , PS, PS^{1+}), and indicate the bond order calculated from the Lewis dot structure. **(3 points)**

Lewis Dot Structure

Bond Order



8691A35D-D1D2-4B10-AD3C-B63D0BD6A2FA

midterm-exam-93388

#422 Page 10 of 10

SCRAP PAGE

**CHEMISTRY 110 –Midterm Exam****Thursday November 7, 2024, 6:30 – 8:30 pm****Examiners: Pallavi Sirjoosingh and Ashok Kakkar**

First name (please write as legibly as possible within the boxes)

--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--

Last name

--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--

Student ID number

--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--

This booklet has 8 questions on 10 pages (The last page can be used as a scrap page).

The staple must NOT be removed from the question booklet. Your intact question booklet must be returned at the end of the exam.

A separate booklet contains a periodic table, and datasheet.

Write your answers directly under the questions in the space provided. Do not write on the QR codes or near the top edges (as it may be cut off).

Please write **legibly and dark enough** for the scan to be readable. For students who use English as a second language, dictionaries are permitted. Calculators are allowed.

DO NOT ASK QUESTIONS OF THE INVIGILATORS about interpretations of the exam questions. If you have a concern about a question, please summarize it at the top of this page. All concerns will be evaluated after the exam. GOOD LUCK!

**Question 1 8 points** (*Assume all transitions are possible*)

a. 6 points Calculate the **energy (in kJ/mol)** required to ionize (completely remove) the electron from the **n=3** level of **He⁺ ion** (He: Helium). Show your work including equations. (3 sig. figs.)

b. 2 points

(I) Select the largest (longest) wavelength from the three choices below:

- i. Wavelength of photon to *completely remove* an electron from *1s orbital of Hydrogen atom*
- ii. Wavelength of photon to *completely remove* an electron from *1s orbital of Boron atom (B)*
- iii. Wavelength of photon to *completely remove* an electron from *1s orbital of Boron cation (B⁺)*

i or ii or iii

(II) Select the smallest (shortest) wavelength from the three choices below (*for H atom*):

- i. Wavelength of photon *emitted* for an electron transitioning from *2p to 1s orbital*
- ii. Wavelength of photon *absorbed* for an electron transitioning from *2s to 4p orbital*
- iii. Wavelength of photon *absorbed* for an electron transitioning from *1s to 3p orbital*

i or ii or iii



Question 2 8 points



A Chem110 student plots the graph shown on the left, from data obtained for the kinetic energy (eV) of the ejected electron versus frequency (s^{-1}) of the incident light, for an unknown metal. ($1\text{eV} = 1.6022 \times 10^{-19} \text{ J}$)

Based on the plot, answer the questions **a to c** below:

a. 1 point On the plot above, label the **threshold frequency**.

b. 2 points What is the value of the work potential (in J) of the unknown metal?

c. 3 points What is the *longest* wavelength (in nm) required to eject an electron from this metal's surface? Show your work including equation(s) used. (3 sig figs)

d. 2 points In the space provided below, plot the following

(Assume frequency is *above* threshold frequency):

- i. *Kinetic energy* of an ejected electron versus the *intensity* of the incident light (left)
- ii. *Number of electrons* ejected versus the *frequency* of the incident light (right)



**Question 3** **6 points****a. 2 points**

In the ground state of Ge (Germanium; $Z = 32$) atom, **name** the following orbitals:

i. Orbital containing the electron that requires the maximum energy to be removed.

ii. Orbital containing the electron that experiences the minimum effective nuclear charge.

b. 2 points

An unknown anion (A^-) has the following *excited* electronic configuration: $[Ar] 4s^2 3d^3 4p^1$

(Ar: Argon; $Z = 18$)

Give the **condensed ground state electronic configuration** of the atom, **A** and the cation, **A^{2+}** .
Show your work. (Assume all electronic transitions are possible)

Electronic Configuration of A:

Electronic Configuration of A^{2+} :

c. 2 points

In a single atom, determine how many electrons can have the following set of quantum numbers?

(i) $n = 5; l = 3$

In a single atom, determine how many electrons can have the following set of quantum numbers?

(i) $n = 3; l = 2; m_s = +1/2$

**Question 4** **8 points**

Rank the following from 1 to 4 in **increasing** order. Provide the ranking in the boxes provided.

Note: Atomic number (Z) are provided for relevant atoms

a) Increasing (lowest to highest) order of *ionization energy*: (1= lowest; 4 = highest)

Mg

Mg: Z = 12

Rb

Rb: Z = 37

Cs

Cs: Z = 55

K

K : Z = 19

b) Increasing (lowest to highest) order of *bond angle*: (1= lowest; 4 = highest)

H₂OCCl₄XeF₄

HCN

c) Increasing (smallest to largest) order of *ionic size* (1= smallest; 4 = largest)

S²⁻

S: Z = 16

O²⁻

O: Z = 8

Be²⁺

Be: Z = 4

Na⁺

Na: Z = 11

d) Increasing (lowest to highest) order of *bond energy* (1= lowest; 4 = highest)

P-F

P-Cl

P-I

P-Br

**Question 5** **12 points**

a. 9 points For the molecules HClO_4 and TeF_6 , draw the most stable *Lewis structure* showing appropriate shape and *dashed/wedged bonds*, based on VSEPR. Determine the **molecular geometry** around the central atom. For HClO_4 , label the *most polar bond*.



b. 3 points (Note: Lone pairs are not shown in structures below)

(i) For the three *resonance* structures (1, 2 and 3) shown below, answer the following question:



(I) Circle the *most contributing* resonance structure.

Structure 1

Structure 2

Structure 3

(ii) For the three structures given (4, 5, and 6) below, answer the following questions:



(I) Circle all the structures that represent *resonance structures* of each other:

Structure 4

Structure 5

Structure 6

(II) Among the resonance structures, circle the *most contributing* resonance structure.

Structure 4

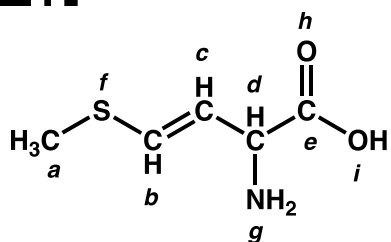
Structure 5

Structure 6

**Question 6** **6 points**

Using the data given below, calculate the **lattice energy** for SrO(s).

	ΔH° (kJ/mol)
$\text{Sr}(s) \longrightarrow \text{Sr}(g)$	+180
$\text{Sr}(g) \longrightarrow \text{Sr}^{2+}(g) + 2e^-$	+1468
$\text{O}_2(g) \longrightarrow 2\text{O}(g)$	+500
$\text{O}(g) + 2e^- \longrightarrow \text{O}^{2-}(g)$	+600
$\text{Sr}(s) + \frac{1}{2}\text{O}_2(g) \longrightarrow \text{SrO}(s)$	-550

**Question 7 10 points**

Consider the molecule given on the left, and answer questions (i) to (vi) using chemical bonding theories that we have learnt in class (valence bond theory, hybridization).

Note that the lone pairs are not shown in the structure:

i. Consider the carbon atoms marked in as ***a***, ***b***, ***c***, ***d***, and ***e*** in the molecule, and determine **hybridization** at each of the carbon centers. (2.5 points)

Atom	Hybridization
Carbon (<i>a</i>)	
Carbon (<i>b</i>)	
Carbon (<i>c</i>)	
Carbon (<i>d</i>)	
Carbon (<i>e</i>)	

ii. Now, consider the sulfur atom (S) marked as ***f***, nitrogen (N) atom (***g***), and oxygen(O) atoms (***h*** and ***i***), and determine their **hybridization**. (2 points)

Atom	Hybridization
Sulfur (<i>f</i>)	
Nitrogen (<i>g</i>)	
Oxygen (<i>h</i>)	
Oxygen (<i>i</i>)	

iii. **Describe** all the bonds formed by the carbon atoms ***a*** and ***b***, using sigma and pi designation as appropriate. (2 points)

Atom	Bonds formed (use sigma and pi as appropriate)
Carbon (<i>a</i>)	
Carbon (<i>b</i>)	

iv. How many total sp^3 -s sigma bonds are in the molecule? (1 point)

v. Determine **electronic** and **molecular geometry** at carbon (C) labeled ***e*** and sulfur (S) labeled ***f*** centres. (2 points)

Atom	Electronic Geometry	Molecular Geometry
Carbon (<i>e</i>)		
Sulfur (<i>f</i>)		

vi. Circle the value from the choices given below, that describes the **C-S-C bond angle at sulfur** (S) marked ***f***: (0.5 points)

>90° to <109.5°

109.5°

>109.5 to <120°

120°

>120° to <180°

**Question 8 12 points**

Using the **valence shell orbitals only** draw a **complete** molecular orbital (MO) diagram (with no s-p mixing) for the molecule **PS**, in the space provided below. **(6 points)**

Assume that 3s and 3p atomic orbitals will form molecular orbitals like those formed by the combination of 2s and 2p atomic orbitals that we learnt in class, and there is no s-p mixing.

Show **energy arrow**, **atomic orbitals** involved, **atom labels**, the **molecular orbitals** that are formed, **filled electrons** in the MO diagram etc. Make sure to label all the atomic and molecular orbitals.

Using the MO diagram above, answer the following questions:

- (i) Which one of the following will have the highest bond order? Circle the correct choice
A) PS^{1-} B) PS C) PS^{1+}
- (ii) Which one of the following will have the longest bond length? Circle the correct choice
A) PS^{1-} B) PS C) PS^{1+}
- (iii) Which of the following will be diamagnetic? Circle the correct choice.
A) PS^{1-} B) PS C) PS^{1+}
- (iv) Draw Lewis dot structure of the molecule with the shortest bond length (based on the MO diagram) from among the three (PS^{1-} , PS, PS^{1+}), and indicate the bond order calculated from the Lewis dot structure. **(3 points)**

Lewis Dot Structure

Bond Order



8101B8CA-9B89-45DC-8BAB-04C20CFC0C25

midterm-exam-93388

#423 Page 10 of 10

SCRAP PAGE

**CHEMISTRY 110 –Midterm Exam****Thursday November 7, 2024, 6:30 – 8:30 pm**

Examiners: Pallavi Sirjoosingh and Ashok Kakkar

First name (please write as legibly as possible within the boxes)

--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--

Last name

--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--

Student ID number

--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--

This booklet has 8 questions on 10 pages (The last page can be used as a scrap page).

The staple must NOT be removed from the question booklet. Your intact question booklet must be returned at the end of the exam.

A separate booklet contains a periodic table, and datasheet.

Write your answers directly under the questions in the space provided. Do not write on the QR codes or near the top edges (as it may be cut off).

Please write **legibly and dark enough** for the scan to be readable. For students who use English as a second language, dictionaries are permitted. Calculators are allowed.

DO NOT ASK QUESTIONS OF THE INVIGILATORS about interpretations of the exam questions. If you have a concern about a question, please summarize it at the top of this page. All concerns will be evaluated after the exam. GOOD LUCK!

**Question 1 8 points** (*Assume all transitions are possible*)

a. 6 points Calculate the **energy (in kJ/mol)** required to ionize (completely remove) the electron from the **$n=3$** level of **He^+ ion** (He: Helium). Show your work including equations. (3 sig. figs.)

b. 2 points

(I) Select the largest (longest) wavelength from the three choices below:

- i. Wavelength of photon to *completely remove* an electron from *1s orbital of Hydrogen atom*
- ii. Wavelength of photon to *completely remove* an electron from *1s orbital of Boron atom (B)*
- iii. Wavelength of photon to *completely remove* an electron from *1s orbital of Boron cation (B^+)*

i or ii or iii

(II) Select the smallest (shortest) wavelength from the three choices below (*for H atom*):

- i. Wavelength of photon *emitted* for an electron transitioning from *2p to 1s orbital*
- ii. Wavelength of photon *absorbed* for an electron transitioning from *2s to 4p orbital*
- iii. Wavelength of photon *absorbed* for an electron transitioning from *1s to 3p orbital*

i or ii or iii



Question 2 8 points



A Chem110 student plots the graph shown on the left, from data obtained for the kinetic energy (eV) of the ejected electron versus frequency (s^{-1}) of the incident light, for an unknown metal. ($1\text{eV} = 1.6022 \times 10^{-19} \text{ J}$)

Based on the plot, answer the questions **a to c** below:

a. 1 point On the plot above, label the **threshold frequency**.

b. 2 points What is the value of the work potential (in J) of the unknown metal?

c. 3 points What is the *longest* wavelength (in nm) required to eject an electron from this metal's surface? Show your work including equation(s) used. (3 sig figs)

d. 2 points In the space provided below, plot the following

(Assume frequency is *above* threshold frequency):

- Kinetic energy* of an ejected electron versus the *intensity* of the incident light (left)
- Number of electrons* ejected versus the *frequency* of the incident light (right)



**Question 3** **6 points****a. 2 points**

In the ground state of Ge (Germanium; $Z = 32$) atom, **name** the following orbitals:

i. Orbital containing the electron that requires the maximum energy to be removed.

ii. Orbital containing the electron that experiences the minimum effective nuclear charge.

b. 2 points

An unknown anion (A^-) has the following *excited* electronic configuration: $[Ar] 4s^2 3d^3 4p^1$

(Ar: Argon; $Z = 18$)

Give the **condensed ground state electronic configuration** of the atom, **A** and the cation, **A^{2+}** .
Show your work. (Assume all electronic transitions are possible)

Electronic Configuration of A:

Electronic Configuration of A^{2+} :

c. 2 points

In a single atom, determine how many electrons can have the following set of quantum numbers?

(i) $n = 5; l = 3$

In a single atom, determine how many electrons can have the following set of quantum numbers?

(i) $n = 3; l = 2; m_s = +1/2$

**Question 4** **8 points**

Rank the following from 1 to 4 in **increasing** order. Provide the ranking in the boxes provided.

Note: Atomic number (Z) are provided for relevant atoms

a) Increasing (lowest to highest) order of *ionization energy*: (1= lowest; 4 = highest)

Mg

Mg: Z = 12

Rb

Rb: Z = 37

Cs

Cs: Z = 55

K

K : Z = 19

b) Increasing (lowest to highest) order of *bond angle*: (1= lowest; 4 = highest)

H₂OCCl₄XeF₄

HCN

c) Increasing (smallest to largest) order of *ionic size* (1= smallest; 4 = largest)

S²⁻

S: Z = 16

O²⁻

O: Z = 8

Be²⁺

Be: Z = 4

Na⁺

Na: Z = 11

d) Increasing (lowest to highest) order of *bond energy* (1= lowest; 4 = highest)

P-F

P-Cl

P-I

P-Br

**Question 5** **12 points**

a. 9 points For the molecules HClO_4 and TeF_6 , draw the most stable *Lewis structure* showing appropriate shape and *dashed/wedged bonds*, based on VSEPR. Determine the **molecular geometry** around the central atom. For HClO_4 , label the *most polar bond*.



b. 3 points (Note: Lone pairs are not shown in structures below)

(i) For the three *resonance* structures (1, 2 and 3) shown below, answer the following question:



(I) Circle the *most contributing* resonance structure.

Structure 1

Structure 2

Structure 3

(ii) For the three structures given (4, 5, and 6) below, answer the following questions:



(I) Circle all the structures that represent *resonance structures* of each other:

Structure 4

Structure 5

Structure 6

(II) Among the resonance structures, circle the *most contributing* resonance structure.

Structure 4

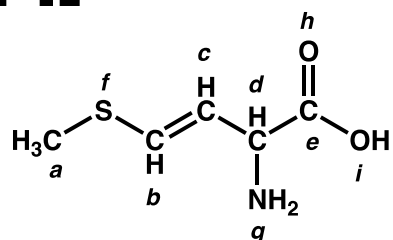
Structure 5

Structure 6

**Question 6** **6 points**

Using the data given below, calculate the **lattice energy** for SrO(s).

	ΔH° (kJ/mol)
$\text{Sr}(s) \longrightarrow \text{Sr}(g)$	+180
$\text{Sr}(g) \longrightarrow \text{Sr}^{2+}(g) + 2e^-$	+1468
$\text{O}_2(g) \longrightarrow 2\text{O}(g)$	+500
$\text{O}(g) + 2e^- \longrightarrow \text{O}^{2-}(g)$	+600
$\text{Sr}(s) + \frac{1}{2}\text{O}_2(g) \longrightarrow \text{SrO}(s)$	-550

**Question 7 10 points**

Consider the molecule given on the left, and answer questions (i) to (vi) using chemical bonding theories that we have learnt in class (valence bond theory, hybridization).

Note that the lone pairs are not shown in the structure:

i. Consider the carbon atoms marked in as ***a***, ***b***, ***c***, ***d***, and ***e*** in the molecule, and determine **hybridization** at each of the carbon centers. (2.5 points)

Atom	Hybridization
Carbon (<i>a</i>)	
Carbon (<i>b</i>)	
Carbon (<i>c</i>)	
Carbon (<i>d</i>)	
Carbon (<i>e</i>)	

ii. Now, consider the sulfur atom (S) marked as ***f***, nitrogen (N) atom (***g***), and oxygen(O) atoms (***h*** and ***i***), and determine their **hybridization**. (2 points)

Atom	Hybridization
Sulfur (<i>f</i>)	
Nitrogen (<i>g</i>)	
Oxygen (<i>h</i>)	
Oxygen (<i>i</i>)	

iii. **Describe** all the bonds formed by the carbon atoms ***a*** and ***b***, using sigma and pi designation as appropriate. (2 points)

Atom	Bonds formed (use sigma and pi as appropriate)
Carbon (<i>a</i>)	
Carbon (<i>b</i>)	

iv. How many total sp^3 -s sigma bonds are in the molecule? (1 point)

v. Determine **electronic** and **molecular geometry** at carbon (C) labeled ***e*** and sulfur (S) labeled ***f*** centres. (2 points)

Atom	Electronic Geometry	Molecular Geometry
Carbon (<i>e</i>)		
Sulfur (<i>f</i>)		

vi. Circle the value from the choices given below, that describes the **C-S-C bond angle at sulfur** (S) marked ***f***: (0.5 points)

>90° to <109.5° 109.5° >109.5 to <120° 120° >120° to <180°


Question 8 12 points

Using the **valence shell orbitals only** draw a **complete** molecular orbital (MO) diagram (with no s-p mixing) for the molecule **PS**, in the space provided below. **(6 points)**

Assume that 3s and 3p atomic orbitals will form molecular orbitals like those formed by the combination of 2s and 2p atomic orbitals that we learnt in class, and there is no s-p mixing.

Show **energy arrow**, **atomic orbitals** involved, **atom labels**, the **molecular orbitals** that are formed, **filled electrons** in the MO diagram etc. Make sure to label all the atomic and molecular orbitals.

Using the MO diagram above, answer the following questions:

- (i) Which one of the following will have the highest bond order? Circle the correct choice
 A) PS^{1-} B) PS C) PS^{1+}
- (ii) Which one of the following will have the longest bond length? Circle the correct choice
 A) PS^{1-} B) PS C) PS^{1+}
- (iii) Which of the following will be diamagnetic? Circle the correct choice.
 A) PS^{1-} B) PS C) PS^{1+}
- (iv) Draw Lewis dot structure of the molecule with the shortest bond length (based on the MO diagram) from among the three (PS^{1-} , PS, PS^{1+}), and indicate the bond order calculated from the Lewis dot structure. **(3 points)**

Lewis Dot Structure

Bond Order



C82F4B9F-417E-49D4-ADD0-2A5F1E957B6C

midterm-exam-93388

#424 Page 10 of 10

SCRAP PAGE

**CHEMISTRY 110 –Midterm Exam****Thursday November 7, 2024, 6:30 – 8:30 pm**

Examiners: Pallavi Sirjoosingh and Ashok Kakkar

First name (please write as legibly as possible within the boxes)

--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--

Last name

--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--

Student ID number

--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--

This booklet has 8 questions on 10 pages (The last page can be used as a scrap page).

The staple must NOT be removed from the question booklet. Your intact question booklet must be returned at the end of the exam.

A separate booklet contains a periodic table, and datasheet.

Write your answers directly under the questions in the space provided. Do not write on the QR codes or near the top edges (as it may be cut off).

Please write **legibly and dark enough** for the scan to be readable. For students who use English as a second language, dictionaries are permitted. Calculators are allowed.

DO NOT ASK QUESTIONS OF THE INVIGILATORS about interpretations of the exam questions. If you have a concern about a question, please summarize it at the top of this page. All concerns will be evaluated after the exam. GOOD LUCK!

**Question 1 8 points** (*Assume all transitions are possible*)

a. 6 points Calculate the **energy (in kJ/mol)** required to ionize (completely remove) the electron from the **n=3** level of **He⁺ ion** (He: Helium). Show your work including equations. (3 sig. figs.)

b. 2 points

(I) Select the largest (longest) wavelength from the three choices below:

- i. Wavelength of photon to *completely remove* an electron from *1s orbital of Hydrogen atom*
- ii. Wavelength of photon to *completely remove* an electron from *1s orbital of Boron atom (B)*
- iii. Wavelength of photon to *completely remove* an electron from *1s orbital of Boron cation (B⁺)*

i or ii or iii

(II) Select the smallest (shortest) wavelength from the three choices below (*for H atom*):

- i. Wavelength of photon *emitted* for an electron transitioning from *2p to 1s orbital*
- ii. Wavelength of photon *absorbed* for an electron transitioning from *2s to 4p orbital*
- iii. Wavelength of photon *absorbed* for an electron transitioning from *1s to 3p orbital*

i or ii or iii



Question 2 8 points



A Chem110 student plots the graph shown on the left, from data obtained for the kinetic energy (eV) of the ejected electron versus frequency (s^{-1}) of the incident light, for an unknown metal. ($1\text{eV} = 1.6022 \times 10^{-19} \text{ J}$)

Based on the plot, answer the questions **a to c** below:

a. 1 point On the plot above, label the **threshold frequency**.

b. 2 points What is the value of the work potential (in J) of the unknown metal?

c. 3 points What is the *longest* wavelength (in nm) required to eject an electron from this metal's surface? Show your work including equation(s) used. (3 sig figs)

d. 2 points In the space provided below, plot the following

(Assume frequency is *above* threshold frequency):

- i. *Kinetic energy* of an ejected electron versus the *intensity* of the incident light (left)
- ii. *Number of electrons* ejected versus the *frequency* of the incident light (right)



**Question 3** **6 points****a. 2 points**

In the ground state of Ge (Germanium; $Z = 32$) atom, **name** the following orbitals:

i. Orbital containing the electron that requires the maximum energy to be removed.

ii. Orbital containing the electron that experiences the minimum effective nuclear charge.

b. 2 points

An unknown anion (A^-) has the following *excited* electronic configuration: $[Ar] 4s^2 3d^3 4p^1$

(Ar: Argon; $Z = 18$)

Give the **condensed ground state electronic configuration** of the atom, **A** and the cation, **A^{2+}** .
Show your work. (Assume all electronic transitions are possible)

Electronic Configuration of A:

Electronic Configuration of A^{2+} :

c. 2 points

In a single atom, determine how many electrons can have the following set of quantum numbers?

(i) $n = 5; l = 3$

In a single atom, determine how many electrons can have the following set of quantum numbers?

(i) $n = 3; l = 2; m_s = +1/2$

**Question 4** **8 points**

Rank the following from 1 to 4 in **increasing** order. Provide the ranking in the boxes provided.

Note: Atomic number (Z) are provided for relevant atoms

a) Increasing (lowest to highest) order of *ionization energy*: (1= lowest; 4 = highest)

Mg

Mg: Z = 12

Rb

Rb: Z = 37

Cs

Cs: Z = 55

K

K : Z = 19

b) Increasing (lowest to highest) order of *bond angle*: (1= lowest; 4 = highest)

H₂OCCl₄XeF₄

HCN

c) Increasing (smallest to largest) order of *ionic size* (1= smallest; 4 = largest)

S²⁻

S: Z = 16

O²⁻

O: Z = 8

Be²⁺

Be: Z = 4

Na⁺

Na: Z = 11

d) Increasing (lowest to highest) order of *bond energy* (1= lowest; 4 = highest)

P-F

P-Cl

P-I

P-Br

**Question 5** **12 points**

a. 9 points For the molecules HClO_4 and TeF_6 , draw the most stable *Lewis structure* showing appropriate shape and *dashed/wedged bonds*, based on VSEPR. Determine the **molecular geometry** around the central atom. For HClO_4 , label the *most polar bond*.

HClO_4 (there are no O-O bonds)

Label the most polar bond

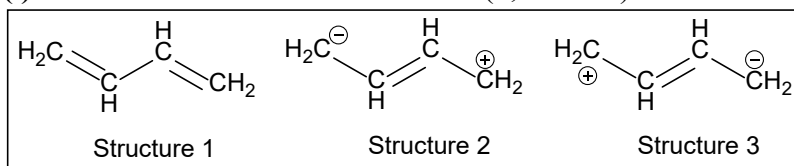
Molecular Geometry around Cl

TeF_6

Molecular Geometry around Te

b. 3 points (Note: Lone pairs are not shown in structures below)

(i) For the three *resonance* structures (1, 2 and 3) shown below, answer the following question:



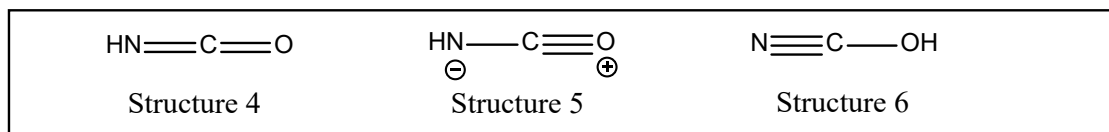
(I) Circle the *most contributing* resonance structure.

Structure 1

Structure 2

Structure 3

(ii) For the three structures given (4, 5, and 6) below, answer the following questions:



(I) Circle all the structures that represent *resonance structures* of each other:

Structure 4

Structure 5

Structure 6

(II) Among the resonance structures, circle the *most contributing* resonance structure.

Structure 4

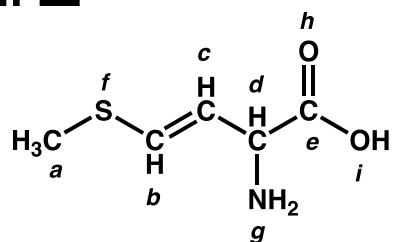
Structure 5

Structure 6

**Question 6** **6 points**

Using the data given below, calculate the **lattice energy** for SrO(s).

	ΔH° (kJ/mol)
$\text{Sr}(s) \longrightarrow \text{Sr}(g)$	+180
$\text{Sr}(g) \longrightarrow \text{Sr}^{2+}(g) + 2e^-$	+1468
$\text{O}_2(g) \longrightarrow 2\text{O}(g)$	+500
$\text{O}(g) + 2e^- \longrightarrow \text{O}^{2-}(g)$	+600
$\text{Sr}(s) + \frac{1}{2}\text{O}_2(g) \longrightarrow \text{SrO}(s)$	-550

**Question 7 10 points**

Consider the molecule given on the left, and answer questions (i) to (vi) using chemical bonding theories that we have learnt in class (valence bond theory, hybridization).

Note that the lone pairs are not shown in the structure:

i. Consider the carbon atoms marked in as ***a***, ***b***, ***c***, ***d***, and ***e*** in the molecule, and determine **hybridization** at each of the carbon centers. (2.5 points)

Atom	Hybridization
Carbon (<i>a</i>)	
Carbon (<i>b</i>)	
Carbon (<i>c</i>)	
Carbon (<i>d</i>)	
Carbon (<i>e</i>)	

ii. Now, consider the sulfur atom (S) marked as ***f***, nitrogen (N) atom (***g***), and oxygen(O) atoms (***h*** and ***i***), and determine their **hybridization**. (2 points)

Atom	Hybridization
Sulfur (<i>f</i>)	
Nitrogen (<i>g</i>)	
Oxygen (<i>h</i>)	
Oxygen (<i>i</i>)	

iii. **Describe** all the bonds formed by the carbon atoms ***a*** and ***b***, using sigma and pi designation as appropriate. (2 points)

Atom	Bonds formed (use sigma and pi as appropriate)
Carbon (<i>a</i>)	
Carbon (<i>b</i>)	

iv. How many total sp^3 -s sigma bonds are in the molecule? (1 point)

v. Determine **electronic** and **molecular geometry** at carbon (C) labeled ***e*** and sulfur (S) labeled ***f*** centres. (2 points)

Atom	Electronic Geometry	Molecular Geometry
Carbon (<i>e</i>)		
Sulfur (<i>f</i>)		

vi. Circle the value from the choices given below, that describes the **C-S-C bond angle at sulfur** (S) marked ***f***: (0.5 points)

>90° to <109.5°

109.5°

>109.5 to <120°

120°

>120° to <180°

**Question 8 12 points**

Using the **valence shell orbitals only** draw a **complete** molecular orbital (MO) diagram (with no s-p mixing) for the molecule **PS**, in the space provided below. **(6 points)**

Assume that 3s and 3p atomic orbitals will form molecular orbitals like those formed by the combination of 2s and 2p atomic orbitals that we learnt in class, and there is no s-p mixing.

Show **energy arrow**, **atomic orbitals** involved, **atom labels**, the **molecular orbitals** that are formed, **filled electrons** in the MO diagram etc. Make sure to label all the atomic and molecular orbitals.

Using the MO diagram above, answer the following questions:

- (i) Which one of the following will have the highest bond order? Circle the correct choice
A) PS^{1-} B) PS C) PS^{1+}
- (ii) Which one of the following will have the longest bond length? Circle the correct choice
A) PS^{1-} B) PS C) PS^{1+}
- (iii) Which of the following will be diamagnetic? Circle the correct choice.
A) PS^{1-} B) PS C) PS^{1+}
- (iv) Draw Lewis dot structure of the molecule with the shortest bond length (based on the MO diagram) from among the three (PS^{1-} , PS, PS^{1+}), and indicate the bond order calculated from the Lewis dot structure. **(3 points)**

Lewis Dot Structure

Bond Order



8CA8CA5C-E764-44B2-A73F-A716BFE6A712

midterm-exam-93388

#425 Page 10 of 10

SCRAP PAGE

**CHEMISTRY 110 –Midterm Exam****Thursday November 7, 2024, 6:30 – 8:30 pm****Examiners: Pallavi Sirjoosingh and Ashok Kakkar**

First name (please write as legibly as possible within the boxes)

--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--

Last name

--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--

Student ID number

--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--

This booklet has 8 questions on 10 pages (The last page can be used as a scrap page).

The staple must NOT be removed from the question booklet. Your intact question booklet must be returned at the end of the exam.

A separate booklet contains a periodic table, and datasheet.

Write your answers directly under the questions in the space provided. Do not write on the QR codes or near the top edges (as it may be cut off).

Please write **legibly and dark enough** for the scan to be readable. For students who use English as a second language, dictionaries are permitted. Calculators are allowed.

DO NOT ASK QUESTIONS OF THE INVIGILATORS about interpretations of the exam questions. If you have a concern about a question, please summarize it at the top of this page. All concerns will be evaluated after the exam. GOOD LUCK!

**Question 1 8 points** (*Assume all transitions are possible*)

a. 6 points Calculate the **energy (in kJ/mol)** required to ionize (completely remove) the electron from the **n=3** level of **He⁺ ion** (He: Helium). Show your work including equations. (3 sig. figs.)

b. 2 points

(I) Select the largest (longest) wavelength from the three choices below:

- i. Wavelength of photon to *completely remove* an electron from *1s orbital of Hydrogen atom*
- ii. Wavelength of photon to *completely remove* an electron from *1s orbital of Boron atom (B)*
- iii. Wavelength of photon to *completely remove* an electron from *1s orbital of Boron cation (B⁺)*

i or ii or iii

(II) Select the smallest (shortest) wavelength from the three choices below (*for H atom*):

- i. Wavelength of photon *emitted* for an electron transitioning from *2p to 1s orbital*
- ii. Wavelength of photon *absorbed* for an electron transitioning from *2s to 4p orbital*
- iii. Wavelength of photon *absorbed* for an electron transitioning from *1s to 3p orbital*

i or ii or iii



Question 2 8 points



A Chem110 student plots the graph shown on the left, from data obtained for the kinetic energy (eV) of the ejected electron versus frequency (s^{-1}) of the incident light, for an unknown metal. ($1\text{eV} = 1.6022 \times 10^{-19} \text{ J}$)

Based on the plot, answer the questions **a to c** below:

a. 1 point On the plot above, label the **threshold frequency**.

b. 2 points What is the value of the work potential (in J) of the unknown metal?

c. 3 points What is the *longest* wavelength (in nm) required to eject an electron from this metal's surface? Show your work including equation(s) used. (3 sig figs)

d. 2 points In the space provided below, plot the following

(Assume frequency is *above* threshold frequency):

- Kinetic energy* of an ejected electron versus the *intensity* of the incident light (left)
- Number of electrons* ejected versus the *frequency* of the incident light (right)



**Question 3** **6 points****a. 2 points**

In the ground state of Ge (Germanium; $Z = 32$) atom, **name** the following orbitals:

i. Orbital containing the electron that requires the maximum energy to be removed.

ii. Orbital containing the electron that experiences the minimum effective nuclear charge.

b. 2 points

An unknown anion (A^-) has the following *excited* electronic configuration: $[Ar] 4s^2 3d^3 4p^1$

(Ar: Argon; $Z = 18$)

Give the **condensed ground state electronic configuration** of the atom, **A** and the cation, **A^{2+}** .
Show your work. (Assume all electronic transitions are possible)

Electronic Configuration of A:

Electronic Configuration of A^{2+} :

c. 2 points

In a single atom, determine how many electrons can have the following set of quantum numbers?

(i) $n = 5; l = 3$

In a single atom, determine how many electrons can have the following set of quantum numbers?

(i) $n = 3; l = 2; m_s = +1/2$

**Question 4** **8 points**

Rank the following from 1 to 4 in **increasing** order. Provide the ranking in the boxes provided.

Note: Atomic number (Z) are provided for relevant atoms

a) Increasing (lowest to highest) order of *ionization energy*: (1= lowest; 4 = highest)

Mg

Mg: Z = 12

Rb

Rb: Z = 37

Cs

Cs: Z = 55

K

K : Z = 19

b) Increasing (lowest to highest) order of *bond angle*: (1= lowest; 4 = highest)

H₂OCCl₄XeF₄

HCN

c) Increasing (smallest to largest) order of *ionic size* (1= smallest; 4 = largest)

S²⁻

S: Z = 16

O²⁻

O: Z = 8

Be²⁺

Be: Z = 4

Na⁺

Na: Z = 11

d) Increasing (lowest to highest) order of *bond energy* (1= lowest; 4 = highest)

P-F

P-Cl

P-I

P-Br

**Question 5** **12 points**

a. 9 points For the molecules HClO_4 and TeF_6 , draw the most stable *Lewis structure* showing appropriate shape and *dashed/wedged bonds*, based on VSEPR. Determine the **molecular geometry** around the central atom. For HClO_4 , label the *most polar bond*.



b. 3 points (Note: Lone pairs are not shown in structures below)

(i) For the three *resonance* structures (1, 2 and 3) shown below, answer the following question:



(I) Circle the *most contributing* resonance structure.

Structure 1

Structure 2

Structure 3

(ii) For the three structures given (4, 5, and 6) below, answer the following questions:



(I) Circle all the structures that represent *resonance structures* of each other:

Structure 4

Structure 5

Structure 6

(II) Among the resonance structures, circle the *most contributing* resonance structure.

Structure 4

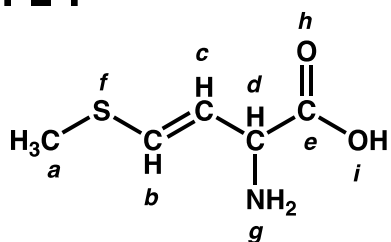
Structure 5

Structure 6

**Question 6** **6 points**

Using the data given below, calculate the **lattice energy** for SrO(s).

	ΔH° (kJ/mol)
$\text{Sr}(s) \longrightarrow \text{Sr}(g)$	+180
$\text{Sr}(g) \longrightarrow \text{Sr}^{2+}(g) + 2e^-$	+1468
$\text{O}_2(g) \longrightarrow 2\text{O}(g)$	+500
$\text{O}(g) + 2e^- \longrightarrow \text{O}^{2-}(g)$	+600
$\text{Sr}(s) + \frac{1}{2}\text{O}_2(g) \longrightarrow \text{SrO}(s)$	-550

**Question 7 10 points**

Consider the molecule given on the left, and answer questions (i) to (vi) using chemical bonding theories that we have learnt in class (valence bond theory, hybridization).

Note that the lone pairs are not shown in the structure:

i. Consider the carbon atoms marked in as ***a***, ***b***, ***c***, ***d***, and ***e*** in the molecule, and determine **hybridization** at each of the carbon centers. (2.5 points)

Atom	Hybridization
Carbon (<i>a</i>)	
Carbon (<i>b</i>)	
Carbon (<i>c</i>)	
Carbon (<i>d</i>)	
Carbon (<i>e</i>)	

ii. Now, consider the sulfur atom (S) marked as ***f***, nitrogen (N) atom (***g***), and oxygen(O) atoms (***h*** and ***i***), and determine their **hybridization**. (2 points)

Atom	Hybridization
Sulfur (<i>f</i>)	
Nitrogen (<i>g</i>)	
Oxygen (<i>h</i>)	
Oxygen (<i>i</i>)	

iii. **Describe** all the bonds formed by the carbon atoms ***a*** and ***b***, using sigma and pi designation as appropriate. (2 points)

Atom	Bonds formed (use sigma and pi as appropriate)
Carbon (<i>a</i>)	
Carbon (<i>b</i>)	

iv. How many total sp^3 -s sigma bonds are in the molecule? (1 point)

v. Determine **electronic** and **molecular geometry** at carbon (C) labeled ***e*** and sulfur (S) labeled ***f*** centres. (2 points)

Atom	Electronic Geometry	Molecular Geometry
Carbon (<i>e</i>)		
Sulfur (<i>f</i>)		

vi. Circle the value from the choices given below, that describes the **C-S-C bond angle at sulfur** (S) marked ***f***: (0.5 points)

>90° to <109.5°

109.5°

>109.5 to <120°

120°

>120° to <180°



Question 8 12 points

Using the **valence shell orbitals only** draw a **complete** molecular orbital (MO) diagram (with no s-p mixing) for the molecule **PS**, in the space provided below. **(6 points)**

Assume that 3s and 3p atomic orbitals will form molecular orbitals like those formed by the combination of 2s and 2p atomic orbitals that we learnt in class, and there is no s-p mixing.

Show **energy arrow**, **atomic orbitals** involved, **atom labels**, the **molecular orbitals** that are formed, **filled electrons** in the MO diagram etc. Make sure to label all the atomic and molecular orbitals.

Using the MO diagram above, answer the following questions:

- (i) Which one of the following will have the highest bond order? Circle the correct choice
 A) PS^{1-} B) PS C) PS^{1+}
- (ii) Which one of the following will have the longest bond length? Circle the correct choice
 A) PS^{1-} B) PS C) PS^{1+}
- (iii) Which of the following will be diamagnetic? Circle the correct choice.
 A) PS^{1-} B) PS C) PS^{1+}
- (iv) Draw Lewis dot structure of the molecule with the shortest bond length (based on the MO diagram) from among the three (PS^{1-} , PS, PS^{1+}), and indicate the bond order calculated from the Lewis dot structure. **(3 points)**

Lewis Dot Structure

Bond Order



CBD5B3E0-0F9F-4AEF-8864-65DB08E14808

midterm-exam-93388

#426 Page 10 of 10

SCRAP PAGE

**CHEMISTRY 110 –Midterm Exam****Thursday November 7, 2024, 6:30 – 8:30 pm**

Examiners: Pallavi Sirjoosingh and Ashok Kakkar

First name (please write as legibly as possible within the boxes)

--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--

Last name

--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--

Student ID number

--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--

This booklet has 8 questions on 10 pages (The last page can be used as a scrap page).

The staple must NOT be removed from the question booklet. Your intact question booklet must be returned at the end of the exam.

A separate booklet contains a periodic table, and datasheet.

Write your answers directly under the questions in the space provided. Do not write on the QR codes or near the top edges (as it may be cut off).

Please write **legibly and dark enough** for the scan to be readable. For students who use English as a second language, dictionaries are permitted. Calculators are allowed.

DO NOT ASK QUESTIONS OF THE INVIGILATORS about interpretations of the exam questions. If you have a concern about a question, please summarize it at the top of this page. All concerns will be evaluated after the exam. GOOD LUCK!

**Question 1 8 points** (*Assume all transitions are possible*)

a. 6 points Calculate the **energy (in kJ/mol)** required to ionize (completely remove) the electron from the **n=3** level of **He⁺ ion** (He: Helium). Show your work including equations. (3 sig. figs.)

b. 2 points

(I) Select the largest (longest) wavelength from the three choices below:

- i. Wavelength of photon to *completely remove* an electron from *1s orbital of Hydrogen atom*
- ii. Wavelength of photon to *completely remove* an electron from *1s orbital of Boron atom (B)*
- iii. Wavelength of photon to *completely remove* an electron from *1s orbital of Boron cation (B⁺)*

i or ii or iii

(II) Select the smallest (shortest) wavelength from the three choices below (*for H atom*):

- i. Wavelength of photon *emitted* for an electron transitioning from *2p to 1s orbital*
- ii. Wavelength of photon *absorbed* for an electron transitioning from *2s to 4p orbital*
- iii. Wavelength of photon *absorbed* for an electron transitioning from *1s to 3p orbital*

i or ii or iii



Question 2 8 points



A Chem110 student plots the graph shown on the left, from data obtained for the kinetic energy (eV) of the ejected electron versus frequency (s^{-1}) of the incident light, for an unknown metal. ($1\text{eV} = 1.6022 \times 10^{-19} \text{ J}$)

Based on the plot, answer the questions **a to c** below:

a. 1 point On the plot above, label the **threshold frequency**.

b. 2 points What is the value of the work potential (in J) of the unknown metal?

c. 3 points What is the *longest* wavelength (in nm) required to eject an electron from this metal's surface? Show your work including equation(s) used. (3 sig figs)

d. 2 points In the space provided below, plot the following

(Assume frequency is *above* threshold frequency):

- Kinetic energy* of an ejected electron versus the *intensity* of the incident light (left)
- Number of electrons* ejected versus the *frequency* of the incident light (right)



**Question 3** **6 points****a. 2 points**

In the ground state of Ge (Germanium; $Z = 32$) atom, **name** the following orbitals:

i. Orbital containing the electron that requires the maximum energy to be removed.

ii. Orbital containing the electron that experiences the minimum effective nuclear charge.

b. 2 points

An unknown anion (A^-) has the following *excited* electronic configuration: $[Ar] 4s^2 3d^3 4p^1$

(Ar: Argon; $Z = 18$)

Give the **condensed ground state electronic configuration** of the atom, **A** and the cation, **A^{2+}** .
Show your work. (Assume all electronic transitions are possible)

Electronic Configuration of A:

Electronic Configuration of A^{2+} :

c. 2 points

In a single atom, determine how many electrons can have the following set of quantum numbers?

(i) $n = 5; l = 3$

In a single atom, determine how many electrons can have the following set of quantum numbers?

(i) $n = 3; l = 2; m_s = +1/2$

**Question 4** **8 points**

Rank the following from 1 to 4 in **increasing** order. Provide the ranking in the boxes provided.

Note: Atomic number (Z) are provided for relevant atoms

a) Increasing (lowest to highest) order of *ionization energy*: (1= lowest; 4 = highest)

Mg

Mg: Z = 12

Rb

Rb: Z = 37

Cs

Cs: Z = 55

K

K : Z = 19

b) Increasing (lowest to highest) order of *bond angle*: (1= lowest; 4 = highest)

H₂OCCl₄XeF₄

HCN

c) Increasing (smallest to largest) order of *ionic size* (1= smallest; 4 = largest)

S²⁻

S: Z = 16

O²⁻

O: Z = 8

Be²⁺

Be: Z = 4

Na⁺

Na: Z = 11

d) Increasing (lowest to highest) order of *bond energy* (1= lowest; 4 = highest)

P-F

P-Cl

P-I

P-Br

**Question 5** **12 points**

a. 9 points For the molecules HClO_4 and TeF_6 , draw the most stable *Lewis structure* showing appropriate shape and *dashed/wedged bonds*, based on VSEPR. Determine the **molecular geometry** around the central atom. For HClO_4 , label the *most polar bond*.



b. 3 points (Note: Lone pairs are not shown in structures below)

(i) For the three *resonance* structures (1, 2 and 3) shown below, answer the following question:



(I) Circle the *most contributing* resonance structure.

Structure 1

Structure 2

Structure 3

(ii) For the three structures given (4, 5, and 6) below, answer the following questions:



(I) Circle all the structures that represent *resonance structures* of each other:

Structure 4

Structure 5

Structure 6

(II) Among the resonance structures, circle the *most contributing* resonance structure.

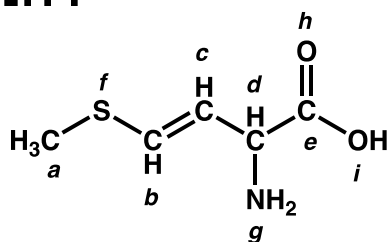
Structure 4

Structure 5

Structure 6

**Question 6** **6 points**Using the data given below, calculate the **lattice energy** for SrO(s).

	ΔH° (kJ/mol)
$\text{Sr}(s) \longrightarrow \text{Sr}(g)$	+180
$\text{Sr}(g) \longrightarrow \text{Sr}^{2+}(g) + 2e^-$	+1468
$\text{O}_2(g) \longrightarrow 2\text{O}(g)$	+500
$\text{O}(g) + 2e^- \longrightarrow \text{O}^{2-}(g)$	+600
$\text{Sr}(s) + \frac{1}{2}\text{O}_2(g) \longrightarrow \text{SrO}(s)$	-550

**Question 7 10 points**

Consider the molecule given on the left, and answer questions (i) to (vi) using chemical bonding theories that we have learnt in class (valence bond theory, hybridization).

Note that the lone pairs are not shown in the structure:

i. Consider the carbon atoms marked in as ***a***, ***b***, ***c***, ***d***, and ***e*** in the molecule, and determine **hybridization** at each of the carbon centers. (2.5 points)

Atom	Hybridization
Carbon (<i>a</i>)	
Carbon (<i>b</i>)	
Carbon (<i>c</i>)	
Carbon (<i>d</i>)	
Carbon (<i>e</i>)	

ii. Now, consider the sulfur atom (S) marked as ***f***, nitrogen (N) atom (***g***), and oxygen(O) atoms (***h*** and ***i***), and determine their **hybridization**. (2 points)

Atom	Hybridization
Sulfur (<i>f</i>)	
Nitrogen (<i>g</i>)	
Oxygen (<i>h</i>)	
Oxygen (<i>i</i>)	

iii. **Describe** all the bonds formed by the carbon atoms ***a*** and ***b***, using sigma and pi designation as appropriate. (2 points)

Atom	Bonds formed (use sigma and pi as appropriate)
Carbon (<i>a</i>)	
Carbon (<i>b</i>)	

iv. How many total sp^3 -s sigma bonds are in the molecule? (1 point)

v. Determine **electronic** and **molecular geometry** at carbon (C) labeled ***e*** and sulfur (S) labeled ***f*** centres. (2 points)

Atom	Electronic Geometry	Molecular Geometry
Carbon (<i>e</i>)		
Sulfur (<i>f</i>)		

vi. Circle the value from the choices given below, that describes the **C-S-C bond angle at sulfur** (S) marked ***f***: (0.5 points)

>90° to <109.5°

109.5°

>109.5 to <120°

120°

>120° to <180°

**Question 8 12 points**

Using the **valence shell orbitals only** draw a **complete** molecular orbital (MO) diagram (with no s-p mixing) for the molecule **PS**, in the space provided below. **(6 points)**

Assume that 3s and 3p atomic orbitals will form molecular orbitals like those formed by the combination of 2s and 2p atomic orbitals that we learnt in class, and there is no s-p mixing.

Show **energy arrow**, **atomic orbitals** involved, **atom labels**, the **molecular orbitals** that are formed, **filled electrons** in the MO diagram etc. Make sure to label all the atomic and molecular orbitals.

Using the MO diagram above, answer the following questions:

- (i) Which one of the following will have the highest bond order? Circle the correct choice
A) PS^{1-} B) PS C) PS^{1+}
- (ii) Which one of the following will have the longest bond length? Circle the correct choice
A) PS^{1-} B) PS C) PS^{1+}
- (iii) Which of the following will be diamagnetic? Circle the correct choice.
A) PS^{1-} B) PS C) PS^{1+}
- (iv) Draw Lewis dot structure of the molecule with the shortest bond length (based on the MO diagram) from among the three (PS^{1-} , PS, PS^{1+}), and indicate the bond order calculated from the Lewis dot structure. **(3 points)**

Lewis Dot Structure

Bond Order



EF1AADB0-5CD4-467D-AADE-DD5280C5B4B2

midterm-exam-93388

#427 Page 10 of 10

SCRAP PAGE

**CHEMISTRY 110 –Midterm Exam****Thursday November 7, 2024, 6:30 – 8:30 pm**

Examiners: Pallavi Sirjoosingh and Ashok Kakkar

First name (please write as legibly as possible within the boxes)

--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--

Last name

--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--

Student ID number

--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--

This booklet has 8 questions on 10 pages (The last page can be used as a scrap page).

The staple must NOT be removed from the question booklet. Your intact question booklet must be returned at the end of the exam.

A separate booklet contains a periodic table, and datasheet.

Write your answers directly under the questions in the space provided. Do not write on the QR codes or near the top edges (as it may be cut off).

Please write **legibly and dark enough** for the scan to be readable. For students who use English as a second language, dictionaries are permitted. Calculators are allowed.

DO NOT ASK QUESTIONS OF THE INVIGILATORS about interpretations of the exam questions. If you have a concern about a question, please summarize it at the top of this page. All concerns will be evaluated after the exam. GOOD LUCK!

**Question 1 8 points** (*Assume all transitions are possible*)

a. 6 points Calculate the **energy (in kJ/mol)** required to ionize (completely remove) the electron from the **$n=3$** level of **He^+ ion** (He: Helium). Show your work including equations. (3 sig. figs.)

b. 2 points

(I) Select the largest (longest) wavelength from the three choices below:

- i. Wavelength of photon to *completely remove* an electron from *1s orbital of Hydrogen atom*
- ii. Wavelength of photon to *completely remove* an electron from *1s orbital of Boron atom (B)*
- iii. Wavelength of photon to *completely remove* an electron from *1s orbital of Boron cation (B^+)*

i or ii or iii

(II) Select the smallest (shortest) wavelength from the three choices below (*for H atom*):

- i. Wavelength of photon *emitted* for an electron transitioning from *2p to 1s orbital*
- ii. Wavelength of photon *absorbed* for an electron transitioning from *2s to 4p orbital*
- iii. Wavelength of photon *absorbed* for an electron transitioning from *1s to 3p orbital*

i or ii or iii



Question 2 8 points



A Chem110 student plots the graph shown on the left, from data obtained for the kinetic energy (eV) of the ejected electron versus frequency (s^{-1}) of the incident light, for an unknown metal. ($1\text{eV} = 1.6022 \times 10^{-19} \text{ J}$)

Based on the plot, answer the questions **a to c** below:

a. 1 point On the plot above, label the **threshold frequency**.

b. 2 points What is the value of the work potential (in J) of the unknown metal?

c. 3 points What is the *longest* wavelength (in nm) required to eject an electron from this metal's surface? Show your work including equation(s) used. (3 sig figs)

d. 2 points In the space provided below, plot the following

(Assume frequency is *above* threshold frequency):

- i. *Kinetic energy* of an ejected electron versus the *intensity* of the incident light (left)
- ii. *Number of electrons* ejected versus the *frequency* of the incident light (right)



**Question 3** **6 points****a. 2 points**

In the ground state of Ge (Germanium; $Z = 32$) atom, **name** the following orbitals:

i. Orbital containing the electron that requires the maximum energy to be removed.

ii. Orbital containing the electron that experiences the minimum effective nuclear charge.

b. 2 points

An unknown anion (A^-) has the following *excited* electronic configuration: $[Ar] 4s^2 3d^3 4p^1$

(Ar: Argon; $Z = 18$)

Give the **condensed ground state electronic configuration** of the atom, **A** and the cation, A^{2+} .
Show your work. (Assume all electronic transitions are possible)

Electronic Configuration of A:

Electronic Configuration of A^{2+} :

c. 2 points

In a single atom, determine how many electrons can have the following set of quantum numbers?

(i) $n = 5; l = 3$

In a single atom, determine how many electrons can have the following set of quantum numbers?

(i) $n = 3; l = 2; m_s = +1/2$

**Question 4** **8 points**

Rank the following from 1 to 4 in **increasing** order. Provide the ranking in the boxes provided.

Note: Atomic number (Z) are provided for relevant atoms

a) Increasing (lowest to highest) order of *ionization energy*: (1= lowest; 4 = highest)

Mg

Mg: Z = 12

Rb

Rb: Z = 37

Cs

Cs: Z = 55

K

K : Z = 19

b) Increasing (lowest to highest) order of *bond angle*: (1= lowest; 4 = highest)

H₂OCCl₄XeF₄

HCN

c) Increasing (smallest to largest) order of *ionic size* (1= smallest; 4 = largest)

S²⁻

S: Z = 16

O²⁻

O: Z = 8

Be²⁺

Be: Z = 4

Na⁺

Na: Z = 11

d) Increasing (lowest to highest) order of *bond energy* (1= lowest; 4 = highest)

P-F

P-Cl

P-I

P-Br

**Question 5** **12 points**

a. 9 points For the molecules HClO_4 and TeF_6 , draw the most stable *Lewis structure* showing appropriate shape and *dashed/wedged bonds*, based on VSEPR. Determine the **molecular geometry** around the central atom. For HClO_4 , label the *most polar bond*.



b. 3 points (Note: Lone pairs are not shown in structures below)

(i) For the three *resonance* structures (1, 2 and 3) shown below, answer the following question:



(I) Circle the *most contributing* resonance structure.

Structure 1

Structure 2

Structure 3

(ii) For the three structures given (4, 5, and 6) below, answer the following questions:



(I) Circle all the structures that represent *resonance structures* of each other:

Structure 4

Structure 5

Structure 6

(II) Among the resonance structures, circle the *most contributing* resonance structure.

Structure 4

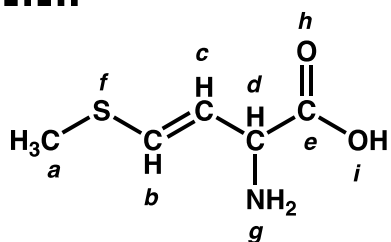
Structure 5

Structure 6

**Question 6** **6 points**

Using the data given below, calculate the **lattice energy** for SrO(s).

	ΔH° (kJ/mol)
$\text{Sr}(s) \longrightarrow \text{Sr}(g)$	+180
$\text{Sr}(g) \longrightarrow \text{Sr}^{2+}(g) + 2e^-$	+1468
$\text{O}_2(g) \longrightarrow 2\text{O}(g)$	+500
$\text{O}(g) + 2e^- \longrightarrow \text{O}^{2-}(g)$	+600
$\text{Sr}(s) + \frac{1}{2}\text{O}_2(g) \longrightarrow \text{SrO}(s)$	-550

**Question 7 10 points**

Consider the molecule given on the left, and answer questions (i) to (vi) using chemical bonding theories that we have learnt in class (valence bond theory, hybridization).

Note that the lone pairs are not shown in the structure:

i. Consider the carbon atoms marked in as ***a***, ***b***, ***c***, ***d***, and ***e*** in the molecule, and determine **hybridization** at each of the carbon centers. (2.5 points)

Atom	Hybridization
Carbon (<i>a</i>)	
Carbon (<i>b</i>)	
Carbon (<i>c</i>)	
Carbon (<i>d</i>)	
Carbon (<i>e</i>)	

ii. Now, consider the sulfur atom (S) marked as ***f***, nitrogen (N) atom (***g***), and oxygen(O) atoms (***h*** and ***i***), and determine their **hybridization**. (2 points)

Atom	Hybridization
Sulfur (<i>f</i>)	
Nitrogen (<i>g</i>)	
Oxygen (<i>h</i>)	
Oxygen (<i>i</i>)	

iii. **Describe** all the bonds formed by the carbon atoms ***a*** and ***b***, using sigma and pi designation as appropriate. (2 points)

Atom	Bonds formed (use sigma and pi as appropriate)
Carbon (<i>a</i>)	
Carbon (<i>b</i>)	

iv. How many total sp^3 -s sigma bonds are in the molecule? (1 point)

v. Determine **electronic** and **molecular geometry** at carbon (C) labeled ***e*** and sulfur (S) labeled ***f*** centres. (2 points)

Atom	Electronic Geometry	Molecular Geometry
Carbon (<i>e</i>)		
Sulfur (<i>f</i>)		

vi. Circle the value from the choices given below, that describes the **C-S-C bond angle at sulfur** (S) marked ***f***: (0.5 points)

>90° to <109.5°

109.5°

>109.5 to <120°

120°

>120° to <180°

**Question 8 12 points**

Using the **valence shell orbitals only** draw a **complete** molecular orbital (MO) diagram (with no s-p mixing) for the molecule **PS**, in the space provided below. **(6 points)**

Assume that 3s and 3p atomic orbitals will form molecular orbitals like those formed by the combination of 2s and 2p atomic orbitals that we learnt in class, and there is no s-p mixing.

Show **energy arrow**, **atomic orbitals** involved, **atom labels**, the **molecular orbitals** that are formed, **filled electrons** in the MO diagram etc. Make sure to label all the atomic and molecular orbitals.

Using the MO diagram above, answer the following questions:

- (i) Which one of the following will have the highest bond order? Circle the correct choice
A) PS^{1-} B) PS C) PS^{1+}
- (ii) Which one of the following will have the longest bond length? Circle the correct choice
A) PS^{1-} B) PS C) PS^{1+}
- (iii) Which of the following will be diamagnetic? Circle the correct choice.
A) PS^{1-} B) PS C) PS^{1+}
- (iv) Draw Lewis dot structure of the molecule with the shortest bond length (based on the MO diagram) from among the three (PS^{1-} , PS, PS^{1+}), and indicate the bond order calculated from the Lewis dot structure. **(3 points)**

Lewis Dot Structure

Bond Order



B42952DC-12BE-4C07-BE86-98C4F28EFF71

midterm-exam-93388

#428 Page 10 of 10

SCRAP PAGE

**CHEMISTRY 110 –Midterm Exam****Thursday November 7, 2024, 6:30 – 8:30 pm**

Examiners: Pallavi Sirjoosingh and Ashok Kakkar

First name (please write as legibly as possible within the boxes)

--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--

Last name

--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--

Student ID number

--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--

This booklet has 8 questions on 10 pages (The last page can be used as a scrap page).

The staple must NOT be removed from the question booklet. Your intact question booklet must be returned at the end of the exam.

A separate booklet contains a periodic table, and datasheet.

Write your answers directly under the questions in the space provided. Do not write on the QR codes or near the top edges (as it may be cut off).

Please write **legibly and dark enough** for the scan to be readable. For students who use English as a second language, dictionaries are permitted. Calculators are allowed.

DO NOT ASK QUESTIONS OF THE INVIGILATORS about interpretations of the exam questions. If you have a concern about a question, please summarize it at the top of this page. All concerns will be evaluated after the exam. GOOD LUCK!

**Question 1 8 points** (*Assume all transitions are possible*)

a. 6 points Calculate the **energy (in kJ/mol)** required to ionize (completely remove) the electron from the **$n=3$** level of **He^+ ion** (He: Helium). Show your work including equations. (3 sig. figs.)

b. 2 points

(I) Select the largest (longest) wavelength from the three choices below:

- i. Wavelength of photon to *completely remove* an electron from *1s orbital of Hydrogen atom*
- ii. Wavelength of photon to *completely remove* an electron from *1s orbital of Boron atom (B)*
- iii. Wavelength of photon to *completely remove* an electron from *1s orbital of Boron cation (B^+)*

i or ii or iii

(II) Select the smallest (shortest) wavelength from the three choices below (*for H atom*):

- i. Wavelength of photon *emitted* for an electron transitioning from *2p to 1s orbital*
- ii. Wavelength of photon *absorbed* for an electron transitioning from *2s to 4p orbital*
- iii. Wavelength of photon *absorbed* for an electron transitioning from *1s to 3p orbital*

i or ii or iii



Question 2 8 points



A Chem110 student plots the graph shown on the left, from data obtained for the kinetic energy (eV) of the ejected electron versus frequency (s^{-1}) of the incident light, for an unknown metal. ($1\text{eV} = 1.6022 \times 10^{-19} \text{ J}$)

Based on the plot, answer the questions **a to c** below:

a. 1 point On the plot above, label the **threshold frequency**.

b. 2 points What is the value of the work potential (in J) of the unknown metal?

c. 3 points What is the *longest* wavelength (in nm) required to eject an electron from this metal's surface? Show your work including equation(s) used. (3 sig figs)

d. 2 points In the space provided below, plot the following

(Assume frequency is *above* threshold frequency):

- i. *Kinetic energy* of an ejected electron versus the *intensity* of the incident light (left)
- ii. *Number of electrons* ejected versus the *frequency* of the incident light (right)



**Question 3** **6 points****a. 2 points**

In the ground state of Ge (Germanium; $Z = 32$) atom, **name** the following orbitals:

i. Orbital containing the electron that requires the maximum energy to be removed.

ii. Orbital containing the electron that experiences the minimum effective nuclear charge.

b. 2 points

An unknown anion (A^-) has the following *excited* electronic configuration: $[Ar] 4s^2 3d^3 4p^1$

(Ar: Argon; $Z = 18$)

Give the **condensed ground state electronic configuration** of the atom, **A** and the cation, **A^{2+}** .
Show your work. (Assume all electronic transitions are possible)

Electronic Configuration of A:

Electronic Configuration of A^{2+} :

c. 2 points

In a single atom, determine how many electrons can have the following set of quantum numbers?

(i) $n = 5; l = 3$

In a single atom, determine how many electrons can have the following set of quantum numbers?

(i) $n = 3; l = 2; m_s = +1/2$

**Question 4** **8 points**

Rank the following from 1 to 4 in **increasing** order. Provide the ranking in the boxes provided.

Note: Atomic number (Z) are provided for relevant atoms

a) Increasing (lowest to highest) order of *ionization energy*: (1= lowest; 4 = highest)

Mg

Mg: Z = 12

Rb

Rb: Z = 37

Cs

Cs: Z = 55

K

K : Z = 19

b) Increasing (lowest to highest) order of *bond angle*: (1= lowest; 4 = highest)

H₂OCCl₄XeF₄

HCN

c) Increasing (smallest to largest) order of *ionic size* (1= smallest; 4 = largest)

S²⁻

S: Z = 16

O²⁻

O: Z = 8

Be²⁺

Be: Z = 4

Na⁺

Na: Z = 11

d) Increasing (lowest to highest) order of *bond energy* (1= lowest; 4 = highest)

P-F

P-Cl

P-I

P-Br

**Question 5** **12 points**

a. 9 points For the molecules HClO_4 and TeF_6 , draw the most stable *Lewis structure* showing appropriate shape and *dashed/wedged bonds*, based on VSEPR. Determine the **molecular geometry** around the central atom. For HClO_4 , label the *most polar bond*.

HClO_4 (there are no O-O bonds)

Label the most polar bond

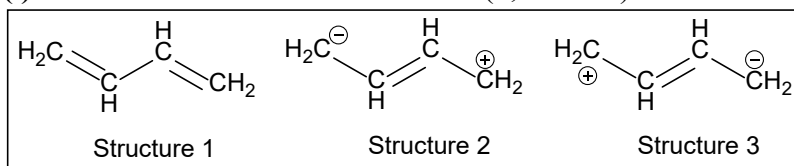
Molecular Geometry around Cl

TeF_6

Molecular Geometry around Te

b. 3 points (Note: Lone pairs are not shown in structures below)

(i) For the three *resonance* structures (1, 2 and 3) shown below, answer the following question:



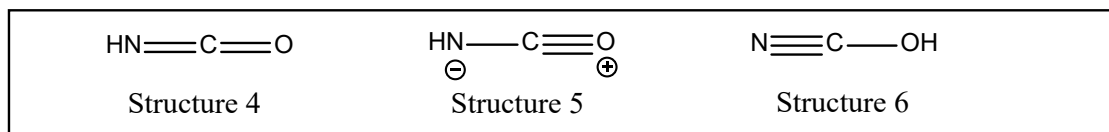
(I) Circle the *most contributing* resonance structure.

Structure 1

Structure 2

Structure 3

(ii) For the three structures given (4, 5, and 6) below, answer the following questions:



(I) Circle all the structures that represent *resonance structures* of each other:

Structure 4

Structure 5

Structure 6

(II) Among the resonance structures, circle the *most contributing* resonance structure.

Structure 4

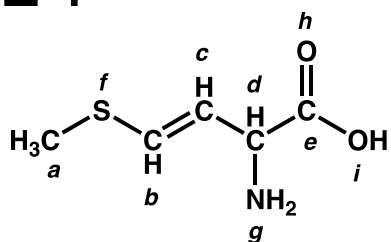
Structure 5

Structure 6

**Question 6** **6 points**

Using the data given below, calculate the **lattice energy** for SrO(s).

	ΔH° (kJ/mol)
$\text{Sr}(s) \longrightarrow \text{Sr}(g)$	+180
$\text{Sr}(g) \longrightarrow \text{Sr}^{2+}(g) + 2e^-$	+1468
$\text{O}_2(g) \longrightarrow 2\text{O}(g)$	+500
$\text{O}(g) + 2e^- \longrightarrow \text{O}^{2-}(g)$	+600
$\text{Sr}(s) + \frac{1}{2}\text{O}_2(g) \longrightarrow \text{SrO}(s)$	-550

**Question 7 10 points**

Consider the molecule given on the left, and answer questions (i) to (vi) using chemical bonding theories that we have learnt in class (valence bond theory, hybridization).

Note that the lone pairs are not shown in the structure:

i. Consider the carbon atoms marked in as ***a***, ***b***, ***c***, ***d***, and ***e*** in the molecule, and determine **hybridization** at each of the carbon centers. (2.5 points)

Atom	Hybridization
Carbon (<i>a</i>)	
Carbon (<i>b</i>)	
Carbon (<i>c</i>)	
Carbon (<i>d</i>)	
Carbon (<i>e</i>)	

ii. Now, consider the sulfur atom (S) marked as ***f***, nitrogen (N) atom (***g***), and oxygen(O) atoms (***h*** and ***i***), and determine their **hybridization**. (2 points)

Atom	Hybridization
Sulfur (<i>f</i>)	
Nitrogen (<i>g</i>)	
Oxygen (<i>h</i>)	
Oxygen (<i>i</i>)	

iii. **Describe** all the bonds formed by the carbon atoms ***a*** and ***b***, using sigma and pi designation as appropriate. (2 points)

Atom	Bonds formed (use sigma and pi as appropriate)
Carbon (<i>a</i>)	
Carbon (<i>b</i>)	

iv. How many total sp^3 -s sigma bonds are in the molecule? (1 point)

v. Determine **electronic** and **molecular geometry** at carbon (C) labeled ***e*** and sulfur (S) labeled ***f*** centres. (2 points)

Atom	Electronic Geometry	Molecular Geometry
Carbon (<i>e</i>)		
Sulfur (<i>f</i>)		

vi. Circle the value from the choices given below, that describes the **C-S-C bond angle at sulfur** (S) marked ***f***: (0.5 points)

>90° to <109.5° 109.5° >109.5 to <120° 120° >120° to <180°

**Question 8 12 points**

Using the **valence shell orbitals only** draw a **complete** molecular orbital (MO) diagram (with no s-p mixing) for the molecule **PS**, in the space provided below. **(6 points)**

Assume that 3s and 3p atomic orbitals will form molecular orbitals like those formed by the combination of 2s and 2p atomic orbitals that we learnt in class, and there is no s-p mixing.

Show **energy arrow**, **atomic orbitals** involved, **atom labels**, the **molecular orbitals** that are formed, **filled electrons** in the MO diagram etc. Make sure to label all the atomic and molecular orbitals.

Using the MO diagram above, answer the following questions:

- (i) Which one of the following will have the highest bond order? Circle the correct choice
A) PS^{1-} B) PS C) PS^{1+}
- (ii) Which one of the following will have the longest bond length? Circle the correct choice
A) PS^{1-} B) PS C) PS^{1+}
- (iii) Which of the following will be diamagnetic? Circle the correct choice.
A) PS^{1-} B) PS C) PS^{1+}
- (iv) Draw Lewis dot structure of the molecule with the shortest bond length (based on the MO diagram) from among the three (PS^{1-} , PS, PS^{1+}), and indicate the bond order calculated from the Lewis dot structure. **(3 points)**

Lewis Dot Structure

Bond Order



AD9E96AF-0B16-4F07-BE98-09401748E4EA

midterm-exam-93388

#429 Page 10 of 10

SCRAP PAGE

**CHEMISTRY 110 –Midterm Exam****Thursday November 7, 2024, 6:30 – 8:30 pm**

Examiners: Pallavi Sirjoosingh and Ashok Kakkar

First name (please write as legibly as possible within the boxes)

--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--

Last name

--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--

Student ID number

--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--

This booklet has 8 questions on 10 pages (The last page can be used as a scrap page).

The staple must NOT be removed from the question booklet. Your intact question booklet must be returned at the end of the exam.

A separate booklet contains a periodic table, and datasheet.

Write your answers directly under the questions in the space provided. Do not write on the QR codes or near the top edges (as it may be cut off).

Please write **legibly and dark enough** for the scan to be readable. For students who use English as a second language, dictionaries are permitted. Calculators are allowed.

DO NOT ASK QUESTIONS OF THE INVIGILATORS about interpretations of the exam questions. If you have a concern about a question, please summarize it at the top of this page. All concerns will be evaluated after the exam. GOOD LUCK!

**Question 1 8 points** (*Assume all transitions are possible*)

a. 6 points Calculate the **energy (in kJ/mol)** required to ionize (completely remove) the electron from the **$n=3$** level of **He^+ ion** (He: Helium). Show your work including equations. (3 sig. figs.)

b. 2 points

(I) Select the largest (longest) wavelength from the three choices below:

- i. Wavelength of photon to *completely remove* an electron from *1s orbital of Hydrogen atom*
- ii. Wavelength of photon to *completely remove* an electron from *1s orbital of Boron atom (B)*
- iii. Wavelength of photon to *completely remove* an electron from *1s orbital of Boron cation (B^+)*

i or ii or iii

(II) Select the smallest (shortest) wavelength from the three choices below (*for H atom*):

- i. Wavelength of photon *emitted* for an electron transitioning from *2p to 1s orbital*
- ii. Wavelength of photon *absorbed* for an electron transitioning from *2s to 4p orbital*
- iii. Wavelength of photon *absorbed* for an electron transitioning from *1s to 3p orbital*

i or ii or iii



Question 2 8 points



A Chem110 student plots the graph shown on the left, from data obtained for the kinetic energy (eV) of the ejected electron versus frequency (s^{-1}) of the incident light, for an unknown metal. ($1\text{eV} = 1.6022 \times 10^{-19} \text{ J}$)

Based on the plot, answer the questions **a to c** below:

a. 1 point On the plot above, label the **threshold frequency**.

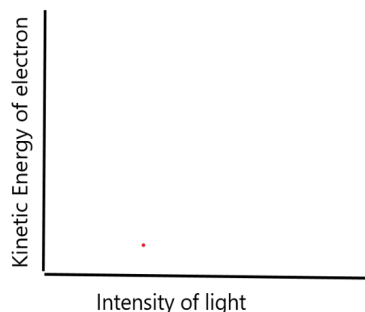
b. 2 points What is the value of the work potential (in J) of the unknown metal?

c. 3 points What is the *longest* wavelength (in nm) required to eject an electron from this metal's surface? Show your work including equation(s) used. (3 sig figs)

d. 2 points In the space provided below, plot the following

(Assume frequency is *above* threshold frequency):

- Kinetic energy* of an ejected electron versus the *intensity* of the incident light (left)
- Number of electrons* ejected versus the *frequency* of the incident light (right)



**Question 3** **6 points****a. 2 points**

In the ground state of Ge (Germanium; $Z = 32$) atom, **name** the following orbitals:

i. Orbital containing the electron that requires the maximum energy to be removed.

ii. Orbital containing the electron that experiences the minimum effective nuclear charge.

b. 2 points

An unknown anion (A^-) has the following *excited* electronic configuration: $[Ar] 4s^2 3d^3 4p^1$

(Ar: Argon; $Z = 18$)

Give the **condensed ground state electronic configuration** of the atom, **A** and the cation, **A^{2+}** .
Show your work. (Assume all electronic transitions are possible)

Electronic Configuration of A:

Electronic Configuration of A^{2+} :

c. 2 points

In a single atom, determine how many electrons can have the following set of quantum numbers?

(i) $n = 5; l = 3$

In a single atom, determine how many electrons can have the following set of quantum numbers?

(i) $n = 3; l = 2; m_s = +1/2$

**Question 4** **8 points**

Rank the following from 1 to 4 in **increasing** order. Provide the ranking in the boxes provided.

Note: Atomic number (Z) are provided for relevant atoms

a) Increasing (lowest to highest) order of *ionization energy*: (1= lowest; 4 = highest)

Mg

Mg: Z = 12

Rb

Rb: Z = 37

Cs

Cs: Z = 55

K

K : Z = 19

b) Increasing (lowest to highest) order of *bond angle*: (1= lowest; 4 = highest)

H₂OCCl₄XeF₄

HCN

c) Increasing (smallest to largest) order of *ionic size* (1= smallest; 4 = largest)

S²⁻

S: Z = 16

O²⁻

O: Z = 8

Be²⁺

Be: Z = 4

Na⁺

Na: Z = 11

d) Increasing (lowest to highest) order of *bond energy* (1= lowest; 4 = highest)

P-F

P-Cl

P-I

P-Br

**Question 5** **12 points**

a. 9 points For the molecules HClO_4 and TeF_6 , draw the most stable *Lewis structure* showing appropriate shape and *dashed/wedged bonds*, based on VSEPR. Determine the **molecular geometry** around the central atom. For HClO_4 , label the *most polar bond*.



b. 3 points (Note: Lone pairs are not shown in structures below)

(i) For the three *resonance* structures (1, 2 and 3) shown below, answer the following question:



(I) Circle the *most contributing* resonance structure.

Structure 1

Structure 2

Structure 3

(ii) For the three structures given (4, 5, and 6) below, answer the following questions:



(I) Circle all the structures that represent *resonance structures* of each other:

Structure 4

Structure 5

Structure 6

(II) Among the resonance structures, circle the *most contributing* resonance structure.

Structure 4

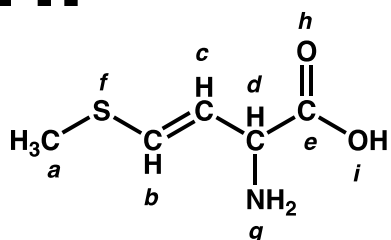
Structure 5

Structure 6

**Question 6** **6 points**

Using the data given below, calculate the **lattice energy** for SrO(s).

	ΔH° (kJ/mol)
$\text{Sr}(s) \longrightarrow \text{Sr}(g)$	+180
$\text{Sr}(g) \longrightarrow \text{Sr}^{2+}(g) + 2e^-$	+1468
$\text{O}_2(g) \longrightarrow 2\text{O}(g)$	+500
$\text{O}(g) + 2e^- \longrightarrow \text{O}^{2-}(g)$	+600
$\text{Sr}(s) + \frac{1}{2}\text{O}_2(g) \longrightarrow \text{SrO}(s)$	-550

**Question 7 10 points**

Consider the molecule given on the left, and answer questions (i) to (vi) using chemical bonding theories that we have learnt in class (valence bond theory, hybridization).

Note that the lone pairs are not shown in the structure:

i. Consider the carbon atoms marked in as ***a***, ***b***, ***c***, ***d***, and ***e*** in the molecule, and determine **hybridization** at each of the carbon centers. (2.5 points)

Atom	Hybridization
Carbon (<i>a</i>)	
Carbon (<i>b</i>)	
Carbon (<i>c</i>)	
Carbon (<i>d</i>)	
Carbon (<i>e</i>)	

ii. Now, consider the sulfur atom (S) marked as ***f***, nitrogen (N) atom (***g***), and oxygen(O) atoms (***h*** and ***i***), and determine their **hybridization**. (2 points)

Atom	Hybridization
Sulfur (<i>f</i>)	
Nitrogen (<i>g</i>)	
Oxygen (<i>h</i>)	
Oxygen (<i>i</i>)	

iii. **Describe** all the bonds formed by the carbon atoms ***a*** and ***b***, using sigma and pi designation as appropriate. (2 points)

Atom	Bonds formed (use sigma and pi as appropriate)
Carbon (<i>a</i>)	
Carbon (<i>b</i>)	

iv. How many total sp^3 -s sigma bonds are in the molecule? (1 point)

v. Determine **electronic** and **molecular geometry** at carbon (C) labeled ***e*** and sulfur (S) labeled ***f*** centres. (2 points)

Atom	Electronic Geometry	Molecular Geometry
Carbon (<i>e</i>)		
Sulfur (<i>f</i>)		

vi. Circle the value from the choices given below, that describes the **C-S-C bond angle at sulfur** (S) marked ***f***: (0.5 points)

>90° to <109.5° 109.5° >109.5 to <120° 120° >120° to <180°



Question 8 12 points

Using the **valence shell orbitals only** draw a **complete** molecular orbital (MO) diagram (with no s-p mixing) for the molecule **PS**, in the space provided below. **(6 points)**

Assume that 3s and 3p atomic orbitals will form molecular orbitals like those formed by the combination of 2s and 2p atomic orbitals that we learnt in class, and there is no s-p mixing.

Show **energy arrow**, **atomic orbitals** involved, **atom labels**, the **molecular orbitals** that are formed, **filled electrons** in the MO diagram etc. Make sure to label all the atomic and molecular orbitals.

Using the MO diagram above, answer the following questions:

- (i) Which one of the following will have the highest bond order? Circle the correct choice
 A) PS^{1-} B) PS C) PS^{1+}
- (ii) Which one of the following will have the longest bond length? Circle the correct choice
 A) PS^{1-} B) PS C) PS^{1+}
- (iii) Which of the following will be diamagnetic? Circle the correct choice.
 A) PS^{1-} B) PS C) PS^{1+}
- (iv) Draw Lewis dot structure of the molecule with the shortest bond length (based on the MO diagram) from among the three (PS^{1-} , PS, PS^{1+}), and indicate the bond order calculated from the Lewis dot structure. **(3 points)**

Lewis Dot Structure

Bond Order



883527DE-C3BA-4269-819C-97C56810490B

midterm-exam-93388

#430 Page 10 of 10

SCRAP PAGE

**CHEMISTRY 110 –Midterm Exam****Thursday November 7, 2024, 6:30 – 8:30 pm****Examiners: Pallavi Sirjoosingh and Ashok Kakkar**

First name (please write as legibly as possible within the boxes)

--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--

Last name

--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--

Student ID number

--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--

This booklet has 8 questions on 10 pages (The last page can be used as a scrap page).

The staple must NOT be removed from the question booklet. Your intact question booklet must be returned at the end of the exam.

A separate booklet contains a periodic table, and datasheet.

Write your answers directly under the questions in the space provided. Do not write on the QR codes or near the top edges (as it may be cut off).

Please write **legibly and dark enough** for the scan to be readable. For students who use English as a second language, dictionaries are permitted. Calculators are allowed.

DO NOT ASK QUESTIONS OF THE INVIGILATORS about interpretations of the exam questions. If you have a concern about a question, please summarize it at the top of this page. All concerns will be evaluated after the exam. GOOD LUCK!

**Question 1 8 points** (*Assume all transitions are possible*)

a. 6 points Calculate the **energy (in kJ/mol)** required to ionize (completely remove) the electron from the **$n=3$** level of **He^+ ion** (He: Helium). Show your work including equations. (3 sig. figs.)

b. 2 points

(I) Select the largest (longest) wavelength from the three choices below:

- i. Wavelength of photon to *completely remove* an electron from *1s orbital of Hydrogen atom*
- ii. Wavelength of photon to *completely remove* an electron from *1s orbital of Boron atom (B)*
- iii. Wavelength of photon to *completely remove* an electron from *1s orbital of Boron cation (B^+)*

i or ii or iii

(II) Select the smallest (shortest) wavelength from the three choices below (*for H atom*):

- i. Wavelength of photon *emitted* for an electron transitioning from *2p to 1s orbital*
- ii. Wavelength of photon *absorbed* for an electron transitioning from *2s to 4p orbital*
- iii. Wavelength of photon *absorbed* for an electron transitioning from *1s to 3p orbital*

i or ii or iii



Question 2 8 points



A Chem110 student plots the graph shown on the left, from data obtained for the kinetic energy (eV) of the ejected electron versus frequency (s^{-1}) of the incident light, for an unknown metal. ($1\text{eV} = 1.6022 \times 10^{-19} \text{ J}$)

Based on the plot, answer the questions **a to c** below:

a. 1 point On the plot above, label the **threshold frequency**.

b. 2 points What is the value of the work potential (in J) of the unknown metal?

c. 3 points What is the *longest* wavelength (in nm) required to eject an electron from this metal's surface? Show your work including equation(s) used. (3 sig figs)

d. 2 points In the space provided below, plot the following

(Assume frequency is *above* threshold frequency):

- i. *Kinetic energy* of an ejected electron versus the *intensity* of the incident light (left)
- ii. *Number of electrons* ejected versus the *frequency* of the incident light (right)



**Question 3** **6 points****a. 2 points**

In the ground state of Ge (Germanium; $Z = 32$) atom, **name** the following orbitals:

i. Orbital containing the electron that requires the maximum energy to be removed.

ii. Orbital containing the electron that experiences the minimum effective nuclear charge.

b. 2 points

An unknown anion (A^-) has the following *excited* electronic configuration: $[Ar] 4s^2 3d^3 4p^1$

(Ar: Argon; $Z = 18$)

Give the **condensed ground state electronic configuration** of the atom, **A** and the cation, **A^{2+}** .
Show your work. (Assume all electronic transitions are possible)

Electronic Configuration of A:

Electronic Configuration of A^{2+} :

c. 2 points

In a single atom, determine how many electrons can have the following set of quantum numbers?

(i) $n = 5; l = 3$

In a single atom, determine how many electrons can have the following set of quantum numbers?

(i) $n = 3; l = 2; m_s = +1/2$

**Question 4** **8 points**

Rank the following from 1 to 4 in **increasing** order. Provide the ranking in the boxes provided.

Note: Atomic number (Z) are provided for relevant atoms

a) Increasing (lowest to highest) order of *ionization energy*: (1= lowest; 4 = highest)

Mg

Mg: Z = 12

Rb

Rb: Z = 37

Cs

Cs: Z = 55

K

K : Z = 19

b) Increasing (lowest to highest) order of *bond angle*: (1= lowest; 4 = highest)

H₂OCCl₄XeF₄

HCN

c) Increasing (smallest to largest) order of *ionic size* (1= smallest; 4 = largest)

S²⁻

S: Z = 16

O²⁻

O: Z = 8

Be²⁺

Be: Z = 4

Na⁺

Na: Z = 11

d) Increasing (lowest to highest) order of *bond energy* (1= lowest; 4 = highest)

P-F

P-Cl

P-I

P-Br

**Question 5** **12 points**

a. 9 points For the molecules HClO_4 and TeF_6 , draw the most stable *Lewis structure* showing appropriate shape and *dashed/wedged bonds*, based on VSEPR. Determine the **molecular geometry** around the central atom. For HClO_4 , label the *most polar bond*.



b. 3 points (Note: Lone pairs are not shown in structures below)

(i) For the three *resonance* structures (1, 2 and 3) shown below, answer the following question:



(I) Circle the *most contributing* resonance structure.

Structure 1

Structure 2

Structure 3

(ii) For the three structures given (4, 5, and 6) below, answer the following questions:



(I) Circle all the structures that represent *resonance structures* of each other:

Structure 4

Structure 5

Structure 6

(II) Among the resonance structures, circle the *most contributing* resonance structure.

Structure 4

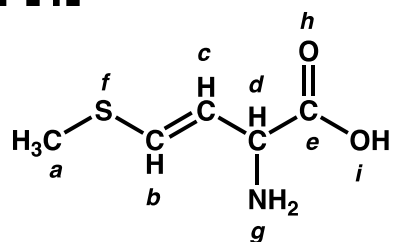
Structure 5

Structure 6

**Question 6** **6 points**

Using the data given below, calculate the **lattice energy** for SrO(s).

	ΔH° (kJ/mol)
$\text{Sr}(s) \longrightarrow \text{Sr}(g)$	+180
$\text{Sr}(g) \longrightarrow \text{Sr}^{2+}(g) + 2e^-$	+1468
$\text{O}_2(g) \longrightarrow 2\text{O}(g)$	+500
$\text{O}(g) + 2e^- \longrightarrow \text{O}^{2-}(g)$	+600
$\text{Sr}(s) + \frac{1}{2}\text{O}_2(g) \longrightarrow \text{SrO}(s)$	-550

**Question 7 10 points**

Consider the molecule given on the left, and answer questions (i) to (vi) using chemical bonding theories that we have learnt in class (valence bond theory, hybridization).

Note that the lone pairs are not shown in the structure:

i. Consider the carbon atoms marked in as ***a***, ***b***, ***c***, ***d***, and ***e*** in the molecule, and determine **hybridization** at each of the carbon centers. (2.5 points)

Atom	Hybridization
Carbon (<i>a</i>)	
Carbon (<i>b</i>)	
Carbon (<i>c</i>)	
Carbon (<i>d</i>)	
Carbon (<i>e</i>)	

ii. Now, consider the sulfur atom (S) marked as ***f***, nitrogen (N) atom (***g***), and oxygen(O) atoms (***h*** and ***i***), and determine their **hybridization**. (2 points)

Atom	Hybridization
Sulfur (<i>f</i>)	
Nitrogen (<i>g</i>)	
Oxygen (<i>h</i>)	
Oxygen (<i>i</i>)	

iii. **Describe** all the bonds formed by the carbon atoms ***a*** and ***b***, using sigma and pi designation as appropriate. (2 points)

Atom	Bonds formed (use sigma and pi as appropriate)
Carbon (<i>a</i>)	
Carbon (<i>b</i>)	

iv. How many total sp^3 -s sigma bonds are in the molecule? (1 point)

v. Determine **electronic** and **molecular geometry** at carbon (C) labeled ***e*** and sulfur (S) labeled ***f*** centres. (2 points)

Atom	Electronic Geometry	Molecular Geometry
Carbon (<i>e</i>)		
Sulfur (<i>f</i>)		

vi. Circle the value from the choices given below, that describes the **C-S-C bond angle at sulfur** (S) marked ***f***: (0.5 points)

>90° to <109.5°

109.5°

>109.5 to <120°

120°

>120° to <180°


Question 8 12 points

Using the **valence shell orbitals only** draw a **complete** molecular orbital (MO) diagram (with no s-p mixing) for the molecule **PS**, in the space provided below. **(6 points)**

Assume that 3s and 3p atomic orbitals will form molecular orbitals like those formed by the combination of 2s and 2p atomic orbitals that we learnt in class, and there is no s-p mixing.

Show **energy arrow**, **atomic orbitals** involved, **atom labels**, the **molecular orbitals** that are formed, **filled electrons** in the MO diagram etc. Make sure to **label all the atomic and molecular orbitals**.

Using the MO diagram above, answer the following questions:

- (i) Which one of the following will have the highest bond order? Circle the correct choice
 A) PS^{1-} B) PS C) PS^{1+}
- (ii) Which one of the following will have the longest bond length? Circle the correct choice
 A) PS^{1-} B) PS C) PS^{1+}
- (iii) Which of the following will be diamagnetic? Circle the correct choice.
 A) PS^{1-} B) PS C) PS^{1+}
- (iv) Draw Lewis dot structure of the molecule with the shortest bond length (based on the MO diagram) from among the three (PS^{1-} , PS, PS^{1+}), and indicate the bond order calculated from the Lewis dot structure. **(3 points)**

Lewis Dot Structure

Bond Order



34E500A3-9F54-4B2F-B812-E2D74D119CFB

midterm-exam-93388

#431 Page 10 of 10

SCRAP PAGE

**CHEMISTRY 110 –Midterm Exam****Thursday November 7, 2024, 6:30 – 8:30 pm**

Examiners: Pallavi Sirjoosingh and Ashok Kakkar

First name (please write as legibly as possible within the boxes)

--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--

Last name

--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--

Student ID number

--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--

This booklet has 8 questions on 10 pages (The last page can be used as a scrap page).

The staple must NOT be removed from the question booklet. Your intact question booklet must be returned at the end of the exam.

A separate booklet contains a periodic table, and datasheet.

Write your answers directly under the questions in the space provided. Do not write on the QR codes or near the top edges (as it may be cut off).

Please write **legibly and dark enough** for the scan to be readable. For students who use English as a second language, dictionaries are permitted. Calculators are allowed.

DO NOT ASK QUESTIONS OF THE INVIGILATORS about interpretations of the exam questions. If you have a concern about a question, please summarize it at the top of this page. All concerns will be evaluated after the exam. GOOD LUCK!

**Question 1 8 points** (*Assume all transitions are possible*)

a. 6 points Calculate the **energy (in kJ/mol)** required to ionize (completely remove) the electron from the **n=3** level of **He⁺ ion** (He: Helium). Show your work including equations. (3 sig. figs.)

b. 2 points

(I) Select the largest (longest) wavelength from the three choices below:

- i. Wavelength of photon to *completely remove* an electron from *1s orbital of Hydrogen atom*
- ii. Wavelength of photon to *completely remove* an electron from *1s orbital of Boron atom (B)*
- iii. Wavelength of photon to *completely remove* an electron from *1s orbital of Boron cation (B⁺)*

i or ii or iii

(II) Select the smallest (shortest) wavelength from the three choices below (*for H atom*):

- i. Wavelength of photon *emitted* for an electron transitioning from *2p to 1s orbital*
- ii. Wavelength of photon *absorbed* for an electron transitioning from *2s to 4p orbital*
- iii. Wavelength of photon *absorbed* for an electron transitioning from *1s to 3p orbital*

i or ii or iii



Question 2 8 points



A Chem110 student plots the graph shown on the left, from data obtained for the kinetic energy (eV) of the ejected electron versus frequency (s^{-1}) of the incident light, for an unknown metal. ($1\text{eV} = 1.6022 \times 10^{-19} \text{ J}$)

Based on the plot, answer the questions **a to c** below:

a. 1 point On the plot above, label the **threshold frequency**.

b. 2 points What is the value of the work potential (in J) of the unknown metal?

c. 3 points What is the *longest* wavelength (in nm) required to eject an electron from this metal's surface? Show your work including equation(s) used. (3 sig figs)

d. 2 points In the space provided below, plot the following

(Assume frequency is *above* threshold frequency):

- i. *Kinetic energy* of an ejected electron versus the *intensity* of the incident light (left)
- ii. *Number of electrons* ejected versus the *frequency* of the incident light (right)



**Question 3** **6 points****a. 2 points**

In the ground state of Ge (Germanium; $Z = 32$) atom, **name** the following orbitals:

i. Orbital containing the electron that requires the maximum energy to be removed.

ii. Orbital containing the electron that experiences the minimum effective nuclear charge.

b. 2 points

An unknown anion (A^-) has the following *excited* electronic configuration: $[Ar] 4s^2 3d^3 4p^1$

(Ar: Argon; $Z = 18$)

Give the **condensed ground state electronic configuration** of the atom, **A** and the cation, **A^{2+}** .
Show your work. (Assume all electronic transitions are possible)

Electronic Configuration of A:

Electronic Configuration of A^{2+} :

c. 2 points

In a single atom, determine how many electrons can have the following set of quantum numbers?

(i) $n = 5; l = 3$

In a single atom, determine how many electrons can have the following set of quantum numbers?

(i) $n = 3; l = 2; m_s = +1/2$

**Question 4** **8 points**

Rank the following from 1 to 4 in **increasing** order. Provide the ranking in the boxes provided.

Note: Atomic number (Z) are provided for relevant atoms

a) Increasing (lowest to highest) order of *ionization energy*: (1= lowest; 4 = highest)

Mg

Mg: Z = 12

Rb

Rb: Z = 37

Cs

Cs: Z = 55

K

K : Z = 19

b) Increasing (lowest to highest) order of *bond angle*: (1= lowest; 4 = highest)

H₂OCCl₄XeF₄

HCN

c) Increasing (smallest to largest) order of *ionic size* (1= smallest; 4 = largest)

S²⁻

S: Z = 16

O²⁻

O: Z = 8

Be²⁺

Be: Z = 4

Na⁺

Na: Z = 11

d) Increasing (lowest to highest) order of *bond energy* (1= lowest; 4 = highest)

P-F

P-Cl

P-I

P-Br

**Question 5** **12 points**

a. 9 points For the molecules HClO_4 and TeF_6 , draw the most stable *Lewis structure* showing appropriate shape and *dashed/wedged bonds*, based on VSEPR. Determine the **molecular geometry** around the central atom. For HClO_4 , label the *most polar bond*.



b. 3 points (Note: Lone pairs are not shown in structures below)

(i) For the three *resonance* structures (1, 2 and 3) shown below, answer the following question:



(I) Circle the *most contributing* resonance structure.

Structure 1

Structure 2

Structure 3

(ii) For the three structures given (4, 5, and 6) below, answer the following questions:



(I) Circle all the structures that represent *resonance structures* of each other:

Structure 4

Structure 5

Structure 6

(II) Among the resonance structures, circle the *most contributing* resonance structure.

Structure 4

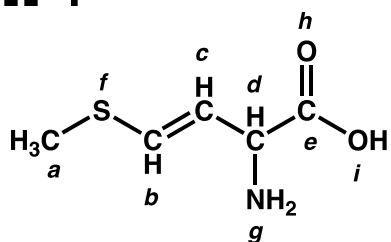
Structure 5

Structure 6

**Question 6** **6 points**

Using the data given below, calculate the **lattice energy** for SrO(s).

	ΔH° (kJ/mol)
$\text{Sr}(s) \longrightarrow \text{Sr}(g)$	+180
$\text{Sr}(g) \longrightarrow \text{Sr}^{2+}(g) + 2e^-$	+1468
$\text{O}_2(g) \longrightarrow 2\text{O}(g)$	+500
$\text{O}(g) + 2e^- \longrightarrow \text{O}^{2-}(g)$	+600
$\text{Sr}(s) + \frac{1}{2}\text{O}_2(g) \longrightarrow \text{SrO}(s)$	-550

**Question 7 10 points**

Consider the molecule given on the left, and answer questions (i) to (vi) using chemical bonding theories that we have learnt in class (valence bond theory, hybridization).

Note that the lone pairs are not shown in the structure:

i. Consider the carbon atoms marked in as ***a***, ***b***, ***c***, ***d***, and ***e*** in the molecule, and determine **hybridization** at each of the carbon centers. (2.5 points)

Atom	Hybridization
Carbon (<i>a</i>)	
Carbon (<i>b</i>)	
Carbon (<i>c</i>)	
Carbon (<i>d</i>)	
Carbon (<i>e</i>)	

ii. Now, consider the sulfur atom (S) marked as ***f***, nitrogen (N) atom (***g***), and oxygen(O) atoms (***h*** and ***i***), and determine their **hybridization**. (2 points)

Atom	Hybridization
Sulfur (<i>f</i>)	
Nitrogen (<i>g</i>)	
Oxygen (<i>h</i>)	
Oxygen (<i>i</i>)	

iii. **Describe** all the bonds formed by the carbon atoms ***a*** and ***b***, using sigma and pi designation as appropriate. (2 points)

Atom	Bonds formed (use sigma and pi as appropriate)
Carbon (<i>a</i>)	
Carbon (<i>b</i>)	

iv. How many total sp^3 -s sigma bonds are in the molecule? (1 point)

v. Determine **electronic** and **molecular geometry** at carbon (C) labeled ***e*** and sulfur (S) labeled ***f*** centres. (2 points)

Atom	Electronic Geometry	Molecular Geometry
Carbon (<i>e</i>)		
Sulfur (<i>f</i>)		

vi. Circle the value from the choices given below, that describes the **C-S-C bond angle at sulfur** (S) marked ***f***: (0.5 points)

$>90^\circ$ to $<109.5^\circ$ 109.5° >109.5 to $<120^\circ$ 120° $>120^\circ$ to $<180^\circ$


Question 8 12 points

Using the **valence shell orbitals only** draw a **complete** molecular orbital (MO) diagram (with no s-p mixing) for the molecule **PS**, in the space provided below. **(6 points)**

Assume that 3s and 3p atomic orbitals will form molecular orbitals like those formed by the combination of 2s and 2p atomic orbitals that we learnt in class, and there is no s-p mixing.

Show **energy arrow**, **atomic orbitals** involved, **atom labels**, the **molecular orbitals** that are formed, **filled electrons** in the MO diagram etc. Make sure to **label all the atomic and molecular orbitals**.

Using the MO diagram above, answer the following questions:

- (i) Which one of the following will have the highest bond order? Circle the correct choice
 A) PS^{1-} B) PS C) PS^{1+}
- (ii) Which one of the following will have the longest bond length? Circle the correct choice
 A) PS^{1-} B) PS C) PS^{1+}
- (iii) Which of the following will be diamagnetic? Circle the correct choice.
 A) PS^{1-} B) PS C) PS^{1+}
- (iv) Draw Lewis dot structure of the molecule with the shortest bond length (based on the MO diagram) from among the three (PS^{1-} , PS, PS^{1+}), and indicate the bond order calculated from the Lewis dot structure. **(3 points)**

Lewis Dot Structure

Bond Order



92923C49-8C3D-4C5E-B913-84F078432E7C

midterm-exam-93388

#432 Page 10 of 10

SCRAP PAGE

**CHEMISTRY 110 –Midterm Exam****Thursday November 7, 2024, 6:30 – 8:30 pm**

Examiners: Pallavi Sirjoosingh and Ashok Kakkar

First name (please write as legibly as possible within the boxes)

--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--

Last name

--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--

Student ID number

--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--

This booklet has 8 questions on 10 pages (The last page can be used as a scrap page).

The staple must NOT be removed from the question booklet. Your intact question booklet must be returned at the end of the exam.

A separate booklet contains a periodic table, and datasheet.

Write your answers directly under the questions in the space provided. Do not write on the QR codes or near the top edges (as it may be cut off).

Please write **legibly and dark enough** for the scan to be readable. For students who use English as a second language, dictionaries are permitted. Calculators are allowed.

DO NOT ASK QUESTIONS OF THE INVIGILATORS about interpretations of the exam questions. If you have a concern about a question, please summarize it at the top of this page. All concerns will be evaluated after the exam. GOOD LUCK!

**Question 1 8 points** (*Assume all transitions are possible*)

a. 6 points Calculate the **energy (in kJ/mol)** required to ionize (completely remove) the electron from the **n=3** level of **He⁺ ion** (He: Helium). Show your work including equations. (3 sig. figs.)

b. 2 points

(I) Select the largest (longest) wavelength from the three choices below:

- i. Wavelength of photon to *completely remove* an electron from *1s orbital of Hydrogen atom*
- ii. Wavelength of photon to *completely remove* an electron from *1s orbital of Boron atom (B)*
- iii. Wavelength of photon to *completely remove* an electron from *1s orbital of Boron cation (B⁺)*

i or ii or iii

(II) Select the smallest (shortest) wavelength from the three choices below (*for H atom*):

- i. Wavelength of photon *emitted* for an electron transitioning from *2p to 1s orbital*
- ii. Wavelength of photon *absorbed* for an electron transitioning from *2s to 4p orbital*
- iii. Wavelength of photon *absorbed* for an electron transitioning from *1s to 3p orbital*

i or ii or iii



Question 2 8 points



A Chem110 student plots the graph shown on the left, from data obtained for the kinetic energy (eV) of the ejected electron versus frequency (s^{-1}) of the incident light, for an unknown metal. ($1\text{eV} = 1.6022 \times 10^{-19} \text{ J}$)

Based on the plot, answer the questions **a to c** below:

a. 1 point On the plot above, label the **threshold frequency**.

b. 2 points What is the value of the work potential (in J) of the unknown metal?

c. 3 points What is the *longest* wavelength (in nm) required to eject an electron from this metal's surface? Show your work including equation(s) used. (3 sig figs)

d. 2 points In the space provided below, plot the following

(Assume frequency is *above* threshold frequency):

- Kinetic energy* of an ejected electron versus the *intensity* of the incident light (left)
- Number of electrons* ejected versus the *frequency* of the incident light (right)



**Question 3** **6 points****a. 2 points**

In the ground state of Ge (Germanium; $Z = 32$) atom, **name** the following orbitals:

i. Orbital containing the electron that requires the maximum energy to be removed.

ii. Orbital containing the electron that experiences the minimum effective nuclear charge.

b. 2 points

An unknown anion (A^-) has the following *excited* electronic configuration: $[Ar] 4s^2 3d^3 4p^1$

(Ar: Argon; $Z = 18$)

Give the **condensed ground state electronic configuration** of the atom, **A** and the cation, **A^{2+}** .
Show your work. (Assume all electronic transitions are possible)

Electronic Configuration of A:

Electronic Configuration of A^{2+} :

c. 2 points

In a single atom, determine how many electrons can have the following set of quantum numbers?

(i) $n = 5; l = 3$

In a single atom, determine how many electrons can have the following set of quantum numbers?

(i) $n = 3; l = 2; m_s = +1/2$

**Question 4** **8 points**

Rank the following from 1 to 4 in **increasing** order. Provide the ranking in the boxes provided.

Note: Atomic number (Z) are provided for relevant atoms

a) Increasing (lowest to highest) order of *ionization energy*: (1= lowest; 4 = highest)

Mg

Mg: Z = 12

Rb

Rb: Z = 37

Cs

Cs: Z = 55

K

K : Z = 19

b) Increasing (lowest to highest) order of *bond angle*: (1= lowest; 4 = highest)

H₂OCCl₄XeF₄

HCN

c) Increasing (smallest to largest) order of *ionic size* (1= smallest; 4 = largest)

S²⁻

S: Z = 16

O²⁻

O: Z = 8

Be²⁺

Be: Z = 4

Na⁺

Na: Z = 11

d) Increasing (lowest to highest) order of *bond energy* (1= lowest; 4 = highest)

P-F

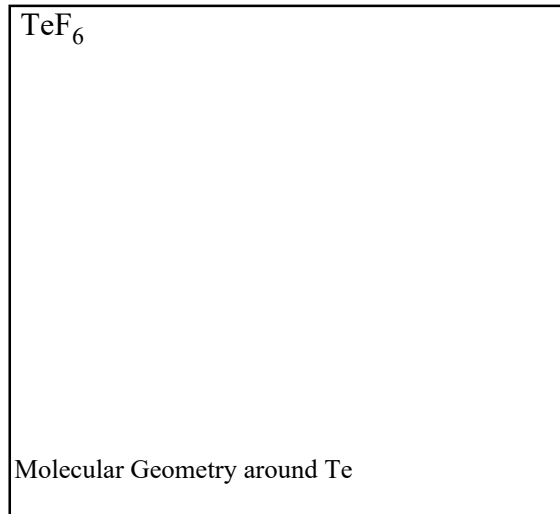
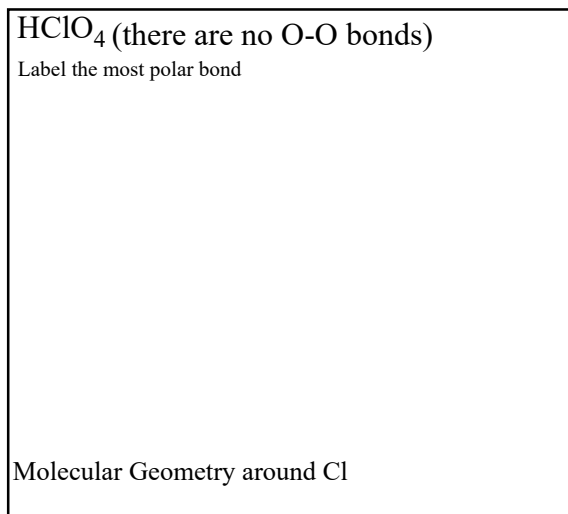
P-Cl

P-I

P-Br

**Question 5** **12 points**

a. 9 points For the molecules HClO_4 and TeF_6 , draw the most stable *Lewis structure* showing appropriate shape and *dashed/wedged bonds*, based on VSEPR. Determine the **molecular geometry** around the central atom. For HClO_4 , label the *most polar bond*.



b. 3 points (Note: Lone pairs are not shown in structures below)

(i) For the three *resonance* structures (1, 2 and 3) shown below, answer the following question:



(I) Circle the *most contributing* resonance structure.

Structure 1

Structure 2

Structure 3

(ii) For the three structures given (4, 5, and 6) below, answer the following questions:



(I) Circle all the structures that represent *resonance structures* of each other:

Structure 4

Structure 5

Structure 6

(II) Among the resonance structures, circle the *most contributing* resonance structure.

Structure 4

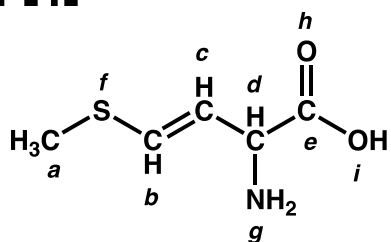
Structure 5

Structure 6

**Question 6** **6 points**

Using the data given below, calculate the **lattice energy** for SrO(s).

	ΔH° (kJ/mol)
$\text{Sr}(s) \longrightarrow \text{Sr}(g)$	+180
$\text{Sr}(g) \longrightarrow \text{Sr}^{2+}(g) + 2e^-$	+1468
$\text{O}_2(g) \longrightarrow 2\text{O}(g)$	+500
$\text{O}(g) + 2e^- \longrightarrow \text{O}^{2-}(g)$	+600
$\text{Sr}(s) + \frac{1}{2}\text{O}_2(g) \longrightarrow \text{SrO}(s)$	-550

**Question 7 10 points**

Consider the molecule given on the left, and answer questions (i) to (vi) using chemical bonding theories that we have learnt in class (valence bond theory, hybridization).

Note that the lone pairs are not shown in the structure:

i. Consider the carbon atoms marked in as ***a***, ***b***, ***c***, ***d***, and ***e*** in the molecule, and determine **hybridization** at each of the carbon centers. (2.5 points)

Atom	Hybridization
Carbon (<i>a</i>)	
Carbon (<i>b</i>)	
Carbon (<i>c</i>)	
Carbon (<i>d</i>)	
Carbon (<i>e</i>)	

ii. Now, consider the sulfur atom (S) marked as ***f***, nitrogen (N) atom (***g***), and oxygen(O) atoms (***h*** and ***i***), and determine their **hybridization**. (2 points)

Atom	Hybridization
Sulfur (<i>f</i>)	
Nitrogen (<i>g</i>)	
Oxygen (<i>h</i>)	
Oxygen (<i>i</i>)	

iii. **Describe** all the bonds formed by the carbon atoms ***a*** and ***b***, using sigma and pi designation as appropriate. (2 points)

Atom	Bonds formed (use sigma and pi as appropriate)
Carbon (<i>a</i>)	
Carbon (<i>b</i>)	

iv. How many total sp^3 -s sigma bonds are in the molecule? (1 point)

v. Determine **electronic** and **molecular geometry** at carbon (C) labeled ***e*** and sulfur (S) labeled ***f*** centres. (2 points)

Atom	Electronic Geometry	Molecular Geometry
Carbon (<i>e</i>)		
Sulfur (<i>f</i>)		

vi. Circle the value from the choices given below, that describes the **C-S-C bond angle at sulfur** (S) marked ***f***: (0.5 points)

>90° to <109.5° 109.5° >109.5 to <120° 120° >120° to <180°

**Question 8 12 points**

Using the **valence shell orbitals only** draw a **complete** molecular orbital (MO) diagram (with no s-p mixing) for the molecule **PS**, in the space provided below. **(6 points)**

Assume that 3s and 3p atomic orbitals will form molecular orbitals like those formed by the combination of 2s and 2p atomic orbitals that we learnt in class, and there is no s-p mixing.

Show **energy arrow**, **atomic orbitals** involved, **atom labels**, the **molecular orbitals** that are formed, **filled electrons** in the MO diagram etc. Make sure to label all the atomic and molecular orbitals.

Using the MO diagram above, answer the following questions:

- (i) Which one of the following will have the highest bond order? Circle the correct choice
A) PS^{1-} B) PS C) PS^{1+}
- (ii) Which one of the following will have the longest bond length? Circle the correct choice
A) PS^{1-} B) PS C) PS^{1+}
- (iii) Which of the following will be diamagnetic? Circle the correct choice.
A) PS^{1-} B) PS C) PS^{1+}
- (iv) Draw Lewis dot structure of the molecule with the shortest bond length (based on the MO diagram) from among the three (PS^{1-} , PS, PS^{1+}), and indicate the bond order calculated from the Lewis dot structure. **(3 points)**

Lewis Dot Structure

Bond Order



28CC6988-4EDF-43E5-ACD9-BC9CDFE984B4

midterm-exam-93388

#433 Page 10 of 10

SCRAP PAGE

**CHEMISTRY 110 –Midterm Exam****Thursday November 7, 2024, 6:30 – 8:30 pm**

Examiners: Pallavi Sirjoosingh and Ashok Kakkar

First name (please write as legibly as possible within the boxes)

--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--

Last name

--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--

Student ID number

--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--

This booklet has 8 questions on 10 pages (The last page can be used as a scrap page).

The staple must NOT be removed from the question booklet. Your intact question booklet must be returned at the end of the exam.

A separate booklet contains a periodic table, and datasheet.

Write your answers directly under the questions in the space provided. Do not write on the QR codes or near the top edges (as it may be cut off).

Please write **legibly and dark enough** for the scan to be readable. For students who use English as a second language, dictionaries are permitted. Calculators are allowed.

DO NOT ASK QUESTIONS OF THE INVIGILATORS about interpretations of the exam questions. If you have a concern about a question, please summarize it at the top of this page. All concerns will be evaluated after the exam. GOOD LUCK!

**Question 1 8 points** (*Assume all transitions are possible*)

a. 6 points Calculate the **energy (in kJ/mol)** required to ionize (completely remove) the electron from the **n=3** level of **He⁺ ion** (He: Helium). Show your work including equations. (3 sig. figs.)

b. 2 points

(I) Select the largest (longest) wavelength from the three choices below:

- i. Wavelength of photon to *completely remove* an electron from *1s orbital of Hydrogen atom*
- ii. Wavelength of photon to *completely remove* an electron from *1s orbital of Boron atom (B)*
- iii. Wavelength of photon to *completely remove* an electron from *1s orbital of Boron cation (B⁺)*

i or ii or iii

(II) Select the smallest (shortest) wavelength from the three choices below (*for H atom*):

- i. Wavelength of photon *emitted* for an electron transitioning from *2p to 1s orbital*
- ii. Wavelength of photon *absorbed* for an electron transitioning from *2s to 4p orbital*
- iii. Wavelength of photon *absorbed* for an electron transitioning from *1s to 3p orbital*

i or ii or iii



Question 2 8 points



A Chem110 student plots the graph shown on the left, from data obtained for the kinetic energy (eV) of the ejected electron versus frequency (s^{-1}) of the incident light, for an unknown metal. ($1\text{eV} = 1.6022 \times 10^{-19} \text{ J}$)

Based on the plot, answer the questions **a to c** below:

a. 1 point On the plot above, label the **threshold frequency**.

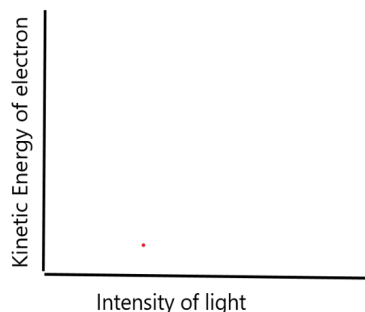
b. 2 points What is the value of the work potential (in J) of the unknown metal?

c. 3 points What is the *longest* wavelength (in nm) required to eject an electron from this metal's surface? Show your work including equation(s) used. (3 sig figs)

d. 2 points In the space provided below, plot the following

(Assume frequency is *above* threshold frequency):

- Kinetic energy* of an ejected electron versus the *intensity* of the incident light (left)
- Number of electrons* ejected versus the *frequency* of the incident light (right)



**Question 3** **6 points****a. 2 points**

In the ground state of Ge (Germanium; $Z = 32$) atom, **name** the following orbitals:

i. Orbital containing the electron that requires the maximum energy to be removed.

ii. Orbital containing the electron that experiences the minimum effective nuclear charge.

b. 2 points

An unknown anion (A^-) has the following *excited* electronic configuration: $[Ar] 4s^2 3d^3 4p^1$

(Ar: Argon; $Z = 18$)

Give the **condensed ground state electronic configuration** of the atom, **A** and the cation, **A^{2+}** .
Show your work. (Assume all electronic transitions are possible)

Electronic Configuration of A:

Electronic Configuration of A^{2+} :

c. 2 points

In a single atom, determine how many electrons can have the following set of quantum numbers?

(i) $n = 5; l = 3$

In a single atom, determine how many electrons can have the following set of quantum numbers?

(i) $n = 3; l = 2; m_s = +1/2$

**Question 4** **8 points**

Rank the following from 1 to 4 in **increasing** order. Provide the ranking in the boxes provided.

Note: Atomic number (Z) are provided for relevant atoms

a) Increasing (lowest to highest) order of *ionization energy*: (1= lowest; 4 = highest)

Mg

Mg: Z = 12

Rb

Rb: Z = 37

Cs

Cs: Z = 55

K

K : Z = 19

b) Increasing (lowest to highest) order of *bond angle*: (1= lowest; 4 = highest)

H₂OCCl₄XeF₄

HCN

c) Increasing (smallest to largest) order of *ionic size* (1= smallest; 4 = largest)

S²⁻

S: Z = 16

O²⁻

O: Z = 8

Be²⁺

Be: Z = 4

Na⁺

Na: Z = 11

d) Increasing (lowest to highest) order of *bond energy* (1= lowest; 4 = highest)

P-F

P-Cl

P-I

P-Br

**Question 5** **12 points**

a. 9 points For the molecules HClO_4 and TeF_6 , draw the most stable *Lewis structure* showing appropriate shape and *dashed/wedged bonds*, based on VSEPR. Determine the **molecular geometry** around the central atom. For HClO_4 , label the *most polar bond*.



b. 3 points (Note: Lone pairs are not shown in structures below)

(i) For the three *resonance* structures (1, 2 and 3) shown below, answer the following question:



(I) Circle the *most contributing* resonance structure.

Structure 1

Structure 2

Structure 3

(ii) For the three structures given (4, 5, and 6) below, answer the following questions:



(I) Circle all the structures that represent *resonance structures* of each other:

Structure 4

Structure 5

Structure 6

(II) Among the resonance structures, circle the *most contributing* resonance structure.

Structure 4

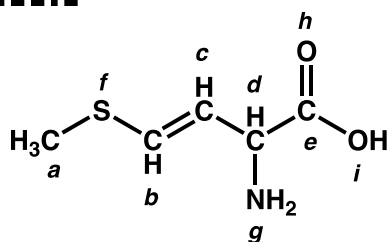
Structure 5

Structure 6

**Question 6** **6 points**

Using the data given below, calculate the **lattice energy** for SrO(s).

	ΔH° (kJ/mol)
$\text{Sr}(s) \longrightarrow \text{Sr}(g)$	+180
$\text{Sr}(g) \longrightarrow \text{Sr}^{2+}(g) + 2e^-$	+1468
$\text{O}_2(g) \longrightarrow 2\text{O}(g)$	+500
$\text{O}(g) + 2e^- \longrightarrow \text{O}^{2-}(g)$	+600
$\text{Sr}(s) + \frac{1}{2}\text{O}_2(g) \longrightarrow \text{SrO}(s)$	-550

**Question 7 10 points**

Consider the molecule given on the left, and answer questions (i) to (vi) using chemical bonding theories that we have learnt in class (valence bond theory, hybridization).

Note that the lone pairs are not shown in the structure:

i. Consider the carbon atoms marked in as ***a***, ***b***, ***c***, ***d***, and ***e*** in the molecule, and determine **hybridization** at each of the carbon centers. (2.5 points)

Atom	Hybridization
Carbon (<i>a</i>)	
Carbon (<i>b</i>)	
Carbon (<i>c</i>)	
Carbon (<i>d</i>)	
Carbon (<i>e</i>)	

ii. Now, consider the sulfur atom (S) marked as ***f***, nitrogen (N) atom (***g***), and oxygen(O) atoms (***h*** and ***i***), and determine their **hybridization**. (2 points)

Atom	Hybridization
Sulfur (<i>f</i>)	
Nitrogen (<i>g</i>)	
Oxygen (<i>h</i>)	
Oxygen (<i>i</i>)	

iii. **Describe** all the bonds formed by the carbon atoms ***a*** and ***b***, using sigma and pi designation as appropriate. (2 points)

Atom	Bonds formed (use sigma and pi as appropriate)
Carbon (<i>a</i>)	
Carbon (<i>b</i>)	

iv. How many total sp^3 -s sigma bonds are in the molecule? (1 point)

v. Determine **electronic** and **molecular geometry** at carbon (C) labeled ***e*** and sulfur (S) labeled ***f*** centres. (2 points)

Atom	Electronic Geometry	Molecular Geometry
Carbon (<i>e</i>)		
Sulfur (<i>f</i>)		

vi. Circle the value from the choices given below, that describes the **C-S-C bond angle at sulfur** (S) marked ***f***: (0.5 points)

>90° to <109.5° 109.5° >109.5 to <120° 120° >120° to <180°

**Question 8 12 points**

Using the **valence shell orbitals only** draw a **complete** molecular orbital (MO) diagram (with no s-p mixing) for the molecule **PS**, in the space provided below. **(6 points)**

Assume that 3s and 3p atomic orbitals will form molecular orbitals like those formed by the combination of 2s and 2p atomic orbitals that we learnt in class, and there is no s-p mixing.

Show **energy arrow**, **atomic orbitals** involved, **atom labels**, the **molecular orbitals** that are formed, **filled electrons** in the MO diagram etc. Make sure to label all the atomic and molecular orbitals.

Using the MO diagram above, answer the following questions:

- (i) Which one of the following will have the highest bond order? Circle the correct choice
A) PS^{1-} B) PS C) PS^{1+}
- (ii) Which one of the following will have the longest bond length? Circle the correct choice
A) PS^{1-} B) PS C) PS^{1+}
- (iii) Which of the following will be diamagnetic? Circle the correct choice.
A) PS^{1-} B) PS C) PS^{1+}
- (iv) Draw Lewis dot structure of the molecule with the shortest bond length (based on the MO diagram) from among the three (PS^{1-} , PS, PS^{1+}), and indicate the bond order calculated from the Lewis dot structure. **(3 points)**

Lewis Dot Structure

Bond Order



14D99C34-D489-4630-9C46-D7F65165EE53

midterm-exam-93388

#434 Page 10 of 10

SCRAP PAGE

**CHEMISTRY 110 –Midterm Exam****Thursday November 7, 2024, 6:30 – 8:30 pm**

Examiners: Pallavi Sirjoosingh and Ashok Kakkar

First name (please write as legibly as possible within the boxes)

--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--

Last name

--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--

Student ID number

--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--

This booklet has 8 questions on 10 pages (The last page can be used as a scrap page).

The staple must NOT be removed from the question booklet. Your intact question booklet must be returned at the end of the exam.

A separate booklet contains a periodic table, and datasheet.

Write your answers directly under the questions in the space provided. Do not write on the QR codes or near the top edges (as it may be cut off).

Please write **legibly and dark enough** for the scan to be readable. For students who use English as a second language, dictionaries are permitted. Calculators are allowed.

DO NOT ASK QUESTIONS OF THE INVIGILATORS about interpretations of the exam questions. If you have a concern about a question, please summarize it at the top of this page. All concerns will be evaluated after the exam. GOOD LUCK!

**Question 1 8 points** (*Assume all transitions are possible*)

a. 6 points Calculate the **energy (in kJ/mol)** required to ionize (completely remove) the electron from the **n=3** level of **He⁺ ion** (He: Helium). Show your work including equations. (3 sig. figs.)

b. 2 points

(I) Select the largest (longest) wavelength from the three choices below:

- i. Wavelength of photon to *completely remove* an electron from *1s orbital of Hydrogen atom*
- ii. Wavelength of photon to *completely remove* an electron from *1s orbital of Boron atom (B)*
- iii. Wavelength of photon to *completely remove* an electron from *1s orbital of Boron cation (B⁺)*

i or ii or iii

(II) Select the smallest (shortest) wavelength from the three choices below (*for H atom*):

- i. Wavelength of photon *emitted* for an electron transitioning from *2p to 1s orbital*
- ii. Wavelength of photon *absorbed* for an electron transitioning from *2s to 4p orbital*
- iii. Wavelength of photon *absorbed* for an electron transitioning from *1s to 3p orbital*

i or ii or iii



Question 2 8 points



A Chem110 student plots the graph shown on the left, from data obtained for the kinetic energy (eV) of the ejected electron versus frequency (s^{-1}) of the incident light, for an unknown metal. ($1\text{eV} = 1.6022 \times 10^{-19} \text{ J}$)

Based on the plot, answer the questions **a to c** below:

a. 1 point On the plot above, label the **threshold frequency**.

b. 2 points What is the value of the work potential (in J) of the unknown metal?

c. 3 points What is the *longest* wavelength (in nm) required to eject an electron from this metal's surface? Show your work including equation(s) used. (3 sig figs)

d. 2 points In the space provided below, plot the following

(Assume frequency is *above* threshold frequency):

- Kinetic energy* of an ejected electron versus the *intensity* of the incident light (left)
- Number of electrons* ejected versus the *frequency* of the incident light (right)



**Question 3** **6 points****a. 2 points**

In the ground state of Ge (Germanium; $Z = 32$) atom, **name** the following orbitals:

i. Orbital containing the electron that requires the maximum energy to be removed.

ii. Orbital containing the electron that experiences the minimum effective nuclear charge.

b. 2 points

An unknown anion (A^-) has the following *excited* electronic configuration: $[Ar] 4s^2 3d^3 4p^1$

(Ar: Argon; $Z = 18$)

Give the **condensed ground state electronic configuration** of the atom, **A** and the cation, **A^{2+}** .
Show your work. (Assume all electronic transitions are possible)

Electronic Configuration of A:

Electronic Configuration of A^{2+} :

c. 2 points

In a single atom, determine how many electrons can have the following set of quantum numbers?

(i) $n = 5; l = 3$

In a single atom, determine how many electrons can have the following set of quantum numbers?

(i) $n = 3; l = 2; m_s = +1/2$

**Question 4** **8 points**

Rank the following from 1 to 4 in **increasing** order. Provide the ranking in the boxes provided.

Note: Atomic number (Z) are provided for relevant atoms

a) Increasing (lowest to highest) order of *ionization energy*: (1= lowest; 4 = highest)

Mg

Mg: Z = 12

Rb

Rb: Z = 37

Cs

Cs: Z = 55

K

K : Z = 19

b) Increasing (lowest to highest) order of *bond angle*: (1= lowest; 4 = highest)

H₂OCCl₄XeF₄

HCN

c) Increasing (smallest to largest) order of *ionic size* (1= smallest; 4 = largest)

S²⁻

S: Z = 16

O²⁻

O: Z = 8

Be²⁺

Be: Z = 4

Na⁺

Na: Z = 11

d) Increasing (lowest to highest) order of *bond energy* (1= lowest; 4 = highest)

P-F

P-Cl

P-I

P-Br

**Question 5** **12 points**

a. 9 points For the molecules HClO_4 and TeF_6 , draw the most stable *Lewis structure* showing appropriate shape and *dashed/wedged bonds*, based on VSEPR. Determine the **molecular geometry** around the central atom. For HClO_4 , label the *most polar bond*.

HClO_4 (there are no O-O bonds)

Label the most polar bond

Molecular Geometry around Cl

TeF_6

Molecular Geometry around Te

b. 3 points (Note: Lone pairs are not shown in structures below)

(i) For the three *resonance* structures (1, 2 and 3) shown below, answer the following question:



(I) Circle the *most contributing* resonance structure.

Structure 1

Structure 2

Structure 3

(ii) For the three structures given (4, 5, and 6) below, answer the following questions:



(I) Circle all the structures that represent *resonance structures* of each other:

Structure 4

Structure 5

Structure 6

(II) Among the resonance structures, circle the *most contributing* resonance structure.

Structure 4

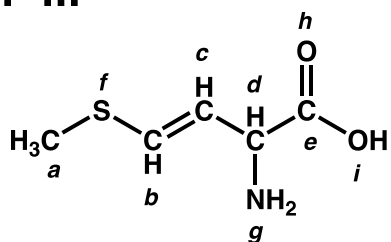
Structure 5

Structure 6

**Question 6** **6 points**

Using the data given below, calculate the **lattice energy** for SrO(s).

	ΔH° (kJ/mol)
$\text{Sr}(s) \longrightarrow \text{Sr}(g)$	+180
$\text{Sr}(g) \longrightarrow \text{Sr}^{2+}(g) + 2e^-$	+1468
$\text{O}_2(g) \longrightarrow 2\text{O}(g)$	+500
$\text{O}(g) + 2e^- \longrightarrow \text{O}^{2-}(g)$	+600
$\text{Sr}(s) + \frac{1}{2}\text{O}_2(g) \longrightarrow \text{SrO}(s)$	-550

**Question 7 10 points**

Consider the molecule given on the left, and answer questions (i) to (vi) using chemical bonding theories that we have learnt in class (valence bond theory, hybridization).

Note that the lone pairs are not shown in the structure:

i. Consider the carbon atoms marked in as ***a***, ***b***, ***c***, ***d***, and ***e*** in the molecule, and determine **hybridization** at each of the carbon centers. (2.5 points)

Atom	Hybridization
Carbon (<i>a</i>)	
Carbon (<i>b</i>)	
Carbon (<i>c</i>)	
Carbon (<i>d</i>)	
Carbon (<i>e</i>)	

ii. Now, consider the sulfur atom (S) marked as ***f***, nitrogen (N) atom (***g***), and oxygen(O) atoms (***h*** and ***i***), and determine their **hybridization**. (2 points)

Atom	Hybridization
Sulfur (<i>f</i>)	
Nitrogen (<i>g</i>)	
Oxygen (<i>h</i>)	
Oxygen (<i>i</i>)	

iii. **Describe** all the bonds formed by the carbon atoms ***a*** and ***b***, using sigma and pi designation as appropriate. (2 points)

Atom	Bonds formed (use sigma and pi as appropriate)
Carbon (<i>a</i>)	
Carbon (<i>b</i>)	

iv. How many total sp^3 -s sigma bonds are in the molecule? (1 point)

v. Determine **electronic** and **molecular geometry** at carbon (C) labeled ***e*** and sulfur (S) labeled ***f*** centres. (2 points)

Atom	Electronic Geometry	Molecular Geometry
Carbon (<i>e</i>)		
Sulfur (<i>f</i>)		

vi. Circle the value from the choices given below, that describes the **C-S-C bond angle at sulfur** (S) marked ***f***: (0.5 points)

$>90^\circ$ to $<109.5^\circ$ 109.5° >109.5 to $<120^\circ$ 120° $>120^\circ$ to $<180^\circ$


Question 8 12 points

Using the **valence shell orbitals only** draw a **complete** molecular orbital (MO) diagram (with no s-p mixing) for the molecule **PS**, in the space provided below. **(6 points)**

Assume that 3s and 3p atomic orbitals will form molecular orbitals like those formed by the combination of 2s and 2p atomic orbitals that we learnt in class, and there is no s-p mixing.

Show **energy arrow**, **atomic orbitals** involved, **atom labels**, the **molecular orbitals** that are formed, **filled electrons** in the MO diagram etc. Make sure to label all the atomic and molecular orbitals.

Using the MO diagram above, answer the following questions:

- (i) Which one of the following will have the highest bond order? Circle the correct choice
 A) PS^{1-} B) PS C) PS^{1+}
- (ii) Which one of the following will have the longest bond length? Circle the correct choice
 A) PS^{1-} B) PS C) PS^{1+}
- (iii) Which of the following will be diamagnetic? Circle the correct choice.
 A) PS^{1-} B) PS C) PS^{1+}
- (iv) Draw Lewis dot structure of the molecule with the shortest bond length (based on the MO diagram) from among the three (PS^{1-} , PS, PS^{1+}), and indicate the bond order calculated from the Lewis dot structure. **(3 points)**

Lewis Dot Structure

Bond Order



7D1981A0-F979-4F78-B58E-DF2AB5E27F22

midterm-exam-93388

#435 Page 10 of 10

SCRAP PAGE

**CHEMISTRY 110 –Midterm Exam****Thursday November 7, 2024, 6:30 – 8:30 pm****Examiners: Pallavi Sirjoosingh and Ashok Kakkar**

First name (please write as legibly as possible within the boxes)

--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--

Last name

--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--

Student ID number

--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--

This booklet has 8 questions on 10 pages (The last page can be used as a scrap page).

The staple must NOT be removed from the question booklet. Your intact question booklet must be returned at the end of the exam.

A separate booklet contains a periodic table, and datasheet.

Write your answers directly under the questions in the space provided. Do not write on the QR codes or near the top edges (as it may be cut off).

Please write **legibly and dark enough** for the scan to be readable. For students who use English as a second language, dictionaries are permitted. Calculators are allowed.

DO NOT ASK QUESTIONS OF THE INVIGILATORS about interpretations of the exam questions. If you have a concern about a question, please summarize it at the top of this page. All concerns will be evaluated after the exam. GOOD LUCK!

**Question 1 8 points** (*Assume all transitions are possible*)

a. 6 points Calculate the **energy (in kJ/mol)** required to ionize (completely remove) the electron from the **n=3** level of **He⁺ ion** (He: Helium). Show your work including equations. (3 sig. figs.)

b. 2 points

(I) Select the largest (longest) wavelength from the three choices below:

- i. Wavelength of photon to *completely remove* an electron from *1s orbital of Hydrogen atom*
- ii. Wavelength of photon to *completely remove* an electron from *1s orbital of Boron atom (B)*
- iii. Wavelength of photon to *completely remove* an electron from *1s orbital of Boron cation (B⁺)*

i or ii or iii

(II) Select the smallest (shortest) wavelength from the three choices below (*for H atom*):

- i. Wavelength of photon *emitted* for an electron transitioning from *2p to 1s orbital*
- ii. Wavelength of photon *absorbed* for an electron transitioning from *2s to 4p orbital*
- iii. Wavelength of photon *absorbed* for an electron transitioning from *1s to 3p orbital*

i or ii or iii



Question 2 8 points



A Chem110 student plots the graph shown on the left, from data obtained for the kinetic energy (eV) of the ejected electron versus frequency (s^{-1}) of the incident light, for an unknown metal. ($1\text{eV} = 1.6022 \times 10^{-19} \text{ J}$)

Based on the plot, answer the questions **a to c** below:

a. 1 point On the plot above, label the **threshold frequency**.

b. 2 points What is the value of the work potential (in J) of the unknown metal?

c. 3 points What is the *longest* wavelength (in nm) required to eject an electron from this metal's surface? Show your work including equation(s) used. (3 sig figs)

d. 2 points In the space provided below, plot the following

(Assume frequency is *above* threshold frequency):

- Kinetic energy* of an ejected electron versus the *intensity* of the incident light (left)
- Number of electrons* ejected versus the *frequency* of the incident light (right)



**Question 3** **6 points****a. 2 points**

In the ground state of Ge (Germanium; $Z = 32$) atom, **name** the following orbitals:

i. Orbital containing the electron that requires the maximum energy to be removed.

ii. Orbital containing the electron that experiences the minimum effective nuclear charge.

b. 2 points

An unknown anion (A^-) has the following *excited* electronic configuration: $[Ar] 4s^2 3d^3 4p^1$

(Ar: Argon; $Z = 18$)

Give the **condensed ground state electronic configuration** of the atom, **A** and the cation, **A^{2+}** .
Show your work. (Assume all electronic transitions are possible)

Electronic Configuration of A:

Electronic Configuration of A^{2+} :

c. 2 points

In a single atom, determine how many electrons can have the following set of quantum numbers?

(i) $n = 5; l = 3$

In a single atom, determine how many electrons can have the following set of quantum numbers?

(i) $n = 3; l = 2; m_s = +1/2$

**Question 4** **8 points**

Rank the following from 1 to 4 in **increasing** order. Provide the ranking in the boxes provided.

Note: Atomic number (Z) are provided for relevant atoms

a) Increasing (lowest to highest) order of *ionization energy*: (1= lowest; 4 = highest)

Mg

Mg: Z = 12

Rb

Rb: Z = 37

Cs

Cs: Z = 55

K

K : Z = 19

b) Increasing (lowest to highest) order of *bond angle*: (1= lowest; 4 = highest)

H₂OCCl₄XeF₄

HCN

c) Increasing (smallest to largest) order of *ionic size* (1= smallest; 4 = largest)

S²⁻

S: Z = 16

O²⁻

O: Z = 8

Be²⁺

Be: Z = 4

Na⁺

Na: Z = 11

d) Increasing (lowest to highest) order of *bond energy* (1= lowest; 4 = highest)

P-F

P-Cl

P-I

P-Br

**Question 5** **12 points**

a. 9 points For the molecules HClO_4 and TeF_6 , draw the most stable *Lewis structure* showing appropriate shape and *dashed/wedged bonds*, based on VSEPR. Determine the **molecular geometry** around the central atom. For HClO_4 , label the *most polar bond*.



b. 3 points (Note: Lone pairs are not shown in structures below)

(i) For the three *resonance* structures (1, 2 and 3) shown below, answer the following question:



(I) Circle the *most contributing* resonance structure.

Structure 1

Structure 2

Structure 3

(ii) For the three structures given (4, 5, and 6) below, answer the following questions:



(I) Circle all the structures that represent *resonance structures* of each other:

Structure 4

Structure 5

Structure 6

(II) Among the resonance structures, circle the *most contributing* resonance structure.

Structure 4

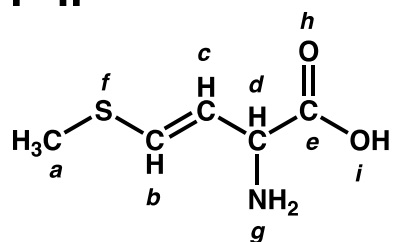
Structure 5

Structure 6

**Question 6** **6 points**

Using the data given below, calculate the **lattice energy** for SrO(s).

	ΔH° (kJ/mol)
$\text{Sr}(s) \longrightarrow \text{Sr}(g)$	+180
$\text{Sr}(g) \longrightarrow \text{Sr}^{2+}(g) + 2e^-$	+1468
$\text{O}_2(g) \longrightarrow 2\text{O}(g)$	+500
$\text{O}(g) + 2e^- \longrightarrow \text{O}^{2-}(g)$	+600
$\text{Sr}(s) + \frac{1}{2}\text{O}_2(g) \longrightarrow \text{SrO}(s)$	-550

**Question 7 10 points**

Consider the molecule given on the left, and answer questions (i) to (vi) using chemical bonding theories that we have learnt in class (valence bond theory, hybridization).

Note that the lone pairs are not shown in the structure:

i. Consider the carbon atoms marked in as ***a***, ***b***, ***c***, ***d***, and ***e*** in the molecule, and determine **hybridization** at each of the carbon centers. (2.5 points)

Atom	Hybridization
Carbon (<i>a</i>)	
Carbon (<i>b</i>)	
Carbon (<i>c</i>)	
Carbon (<i>d</i>)	
Carbon (<i>e</i>)	

ii. Now, consider the sulfur atom (S) marked as ***f***, nitrogen (N) atom (***g***), and oxygen(O) atoms (***h*** and ***i***), and determine their **hybridization**. (2 points)

Atom	Hybridization
Sulfur (<i>f</i>)	
Nitrogen (<i>g</i>)	
Oxygen (<i>h</i>)	
Oxygen (<i>i</i>)	

iii. **Describe** all the bonds formed by the carbon atoms ***a*** and ***b***, using sigma and pi designation as appropriate. (2 points)

Atom	Bonds formed (use sigma and pi as appropriate)
Carbon (<i>a</i>)	
Carbon (<i>b</i>)	

iv. How many total sp^3 -s sigma bonds are in the molecule? (1 point)

v. Determine **electronic** and **molecular geometry** at carbon (C) labeled ***e*** and sulfur (S) labeled ***f*** centres. (2 points)

Atom	Electronic Geometry	Molecular Geometry
Carbon (<i>e</i>)		
Sulfur (<i>f</i>)		

vi. Circle the value from the choices given below, that describes the **C-S-C bond angle at sulfur** (S) marked ***f***: (0.5 points)

>90° to <109.5° 109.5° >109.5 to <120° 120° >120° to <180°

**Question 8 12 points**

Using the **valence shell orbitals only** draw a **complete** molecular orbital (MO) diagram (with no s-p mixing) for the molecule **PS**, in the space provided below. **(6 points)**

Assume that 3s and 3p atomic orbitals will form molecular orbitals like those formed by the combination of 2s and 2p atomic orbitals that we learnt in class, and there is no s-p mixing.

Show **energy arrow**, **atomic orbitals** involved, **atom labels**, the **molecular orbitals** that are formed, **filled electrons** in the MO diagram etc. Make sure to label all the atomic and molecular orbitals.

Using the MO diagram above, answer the following questions:

- (i) Which one of the following will have the highest bond order? Circle the correct choice
A) PS^{1-} B) PS C) PS^{1+}
- (ii) Which one of the following will have the longest bond length? Circle the correct choice
A) PS^{1-} B) PS C) PS^{1+}
- (iii) Which of the following will be diamagnetic? Circle the correct choice.
A) PS^{1-} B) PS C) PS^{1+}
- (iv) Draw Lewis dot structure of the molecule with the shortest bond length (based on the MO diagram) from among the three (PS^{1-} , PS, PS^{1+}), and indicate the bond order calculated from the Lewis dot structure. **(3 points)**

Lewis Dot Structure

Bond Order



BEC61413-4318-47C1-A0EB-88F2BCF4669E

midterm-exam-93388

#436 Page 10 of 10

SCRAP PAGE

**CHEMISTRY 110 –Midterm Exam****Thursday November 7, 2024, 6:30 – 8:30 pm**

Examiners: Pallavi Sirjoosingh and Ashok Kakkar

First name (please write as legibly as possible within the boxes)

--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--

Last name

--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--

Student ID number

--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--

This booklet has 8 questions on 10 pages (The last page can be used as a scrap page).

The staple must NOT be removed from the question booklet. Your intact question booklet must be returned at the end of the exam.

A separate booklet contains a periodic table, and datasheet.

Write your answers directly under the questions in the space provided. Do not write on the QR codes or near the top edges (as it may be cut off).

Please write **legibly and dark enough** for the scan to be readable. For students who use English as a second language, dictionaries are permitted. Calculators are allowed.

DO NOT ASK QUESTIONS OF THE INVIGILATORS about interpretations of the exam questions. If you have a concern about a question, please summarize it at the top of this page. All concerns will be evaluated after the exam. GOOD LUCK!

**Question 1 8 points** (*Assume all transitions are possible*)

a. 6 points Calculate the **energy (in kJ/mol)** required to ionize (completely remove) the electron from the **n=3** level of **He⁺ ion** (He: Helium). Show your work including equations. (3 sig. figs.)

b. 2 points

(I) Select the largest (longest) wavelength from the three choices below:

- i. Wavelength of photon to *completely remove* an electron from *1s orbital of Hydrogen atom*
- ii. Wavelength of photon to *completely remove* an electron from *1s orbital of Boron atom (B)*
- iii. Wavelength of photon to *completely remove* an electron from *1s orbital of Boron cation (B⁺)*

i or ii or iii

(II) Select the smallest (shortest) wavelength from the three choices below (*for H atom*):

- i. Wavelength of photon *emitted* for an electron transitioning from *2p to 1s orbital*
- ii. Wavelength of photon *absorbed* for an electron transitioning from *2s to 4p orbital*
- iii. Wavelength of photon *absorbed* for an electron transitioning from *1s to 3p orbital*

i or ii or iii



Question 2 8 points



A Chem110 student plots the graph shown on the left, from data obtained for the kinetic energy (eV) of the ejected electron versus frequency (s^{-1}) of the incident light, for an unknown metal. ($1\text{eV} = 1.6022 \times 10^{-19} \text{ J}$)

Based on the plot, answer the questions **a to c** below:

a. 1 point On the plot above, label the **threshold frequency**.

b. 2 points What is the value of the work potential (in J) of the unknown metal?

c. 3 points What is the *longest* wavelength (in nm) required to eject an electron from this metal's surface? Show your work including equation(s) used. (3 sig figs)

d. 2 points In the space provided below, plot the following

(Assume frequency is *above* threshold frequency):

- i. *Kinetic energy* of an ejected electron versus the *intensity* of the incident light (left)
- ii. *Number of electrons* ejected versus the *frequency* of the incident light (right)



**Question 3** **6 points****a. 2 points**

In the ground state of Ge (Germanium; $Z = 32$) atom, **name** the following orbitals:

i. Orbital containing the electron that requires the maximum energy to be removed.

ii. Orbital containing the electron that experiences the minimum effective nuclear charge.

b. 2 points

An unknown anion (A^-) has the following *excited* electronic configuration: $[Ar] 4s^2 3d^3 4p^1$

(Ar: Argon; $Z = 18$)

Give the **condensed ground state electronic configuration** of the atom, **A** and the cation, **A^{2+}** .
Show your work. (Assume all electronic transitions are possible)

Electronic Configuration of A:

Electronic Configuration of A^{2+} :

c. 2 points

In a single atom, determine how many electrons can have the following set of quantum numbers?

(i) $n = 5; l = 3$

In a single atom, determine how many electrons can have the following set of quantum numbers?

(i) $n = 3; l = 2; m_s = +1/2$

**Question 4** **8 points**

Rank the following from 1 to 4 in **increasing** order. Provide the ranking in the boxes provided.

Note: Atomic number (Z) are provided for relevant atoms

a) Increasing (lowest to highest) order of *ionization energy*: (1= lowest; 4 = highest)

Mg

Mg: Z = 12

Rb

Rb: Z = 37

Cs

Cs: Z = 55

K

K : Z = 19

b) Increasing (lowest to highest) order of *bond angle*: (1= lowest; 4 = highest)

H₂OCCl₄XeF₄

HCN

c) Increasing (smallest to largest) order of *ionic size* (1= smallest; 4 = largest)

S²⁻

S: Z = 16

O²⁻

O: Z = 8

Be²⁺

Be: Z = 4

Na⁺

Na: Z = 11

d) Increasing (lowest to highest) order of *bond energy* (1= lowest; 4 = highest)

P-F

P-Cl

P-I

P-Br

**Question 5** **12 points**

a. 9 points For the molecules HClO_4 and TeF_6 , draw the most stable *Lewis structure* showing appropriate shape and *dashed/wedged bonds*, based on VSEPR. Determine the **molecular geometry** around the central atom. For HClO_4 , label the *most polar bond*.



b. 3 points (Note: Lone pairs are not shown in structures below)

(i) For the three *resonance* structures (1, 2 and 3) shown below, answer the following question:



(I) Circle the *most contributing* resonance structure.

Structure 1

Structure 2

Structure 3

(ii) For the three structures given (4, 5, and 6) below, answer the following questions:



(I) Circle all the structures that represent *resonance structures* of each other:

Structure 4

Structure 5

Structure 6

(II) Among the resonance structures, circle the *most contributing* resonance structure.

Structure 4

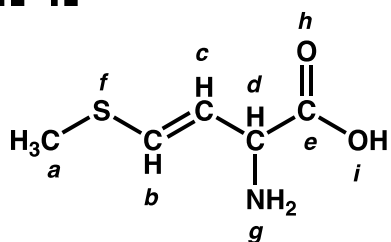
Structure 5

Structure 6

**Question 6** **6 points**

Using the data given below, calculate the **lattice energy** for SrO(s).

	ΔH° (kJ/mol)
$\text{Sr}(s) \longrightarrow \text{Sr}(g)$	+180
$\text{Sr}(g) \longrightarrow \text{Sr}^{2+}(g) + 2e^-$	+1468
$\text{O}_2(g) \longrightarrow 2\text{O}(g)$	+500
$\text{O}(g) + 2e^- \longrightarrow \text{O}^{2-}(g)$	+600
$\text{Sr}(s) + \frac{1}{2}\text{O}_2(g) \longrightarrow \text{SrO}(s)$	-550

**Question 7 10 points**

Consider the molecule given on the left, and answer questions (i) to (vi) using chemical bonding theories that we have learnt in class (valence bond theory, hybridization).

Note that the lone pairs are not shown in the structure:

i. Consider the carbon atoms marked in as ***a***, ***b***, ***c***, ***d***, and ***e*** in the molecule, and determine **hybridization** at each of the carbon centers. (2.5 points)

Atom	Hybridization
Carbon (<i>a</i>)	
Carbon (<i>b</i>)	
Carbon (<i>c</i>)	
Carbon (<i>d</i>)	
Carbon (<i>e</i>)	

ii. Now, consider the sulfur atom (S) marked as ***f***, nitrogen (N) atom (***g***), and oxygen(O) atoms (***h*** and ***i***), and determine their **hybridization**. (2 points)

Atom	Hybridization
Sulfur (<i>f</i>)	
Nitrogen (<i>g</i>)	
Oxygen (<i>h</i>)	
Oxygen (<i>i</i>)	

iii. **Describe** all the bonds formed by the carbon atoms ***a*** and ***b***, using sigma and pi designation as appropriate. (2 points)

Atom	Bonds formed (use sigma and pi as appropriate)
Carbon (<i>a</i>)	
Carbon (<i>b</i>)	

iv. How many total sp^3 -s sigma bonds are in the molecule? (1 point)

v. Determine **electronic** and **molecular geometry** at carbon (C) labeled ***e*** and sulfur (S) labeled ***f*** centres. (2 points)

Atom	Electronic Geometry	Molecular Geometry
Carbon (<i>e</i>)		
Sulfur (<i>f</i>)		

vi. Circle the value from the choices given below, that describes the **C-S-C bond angle at sulfur** (S) marked ***f***: (0.5 points)

$>90^\circ$ to $<109.5^\circ$ 109.5° >109.5 to $<120^\circ$ 120° $>120^\circ$ to $<180^\circ$


Question 8 12 points

Using the **valence shell orbitals only** draw a **complete** molecular orbital (MO) diagram (with no s-p mixing) for the molecule **PS**, in the space provided below. **(6 points)**

Assume that 3s and 3p atomic orbitals will form molecular orbitals like those formed by the combination of 2s and 2p atomic orbitals that we learnt in class, and there is no s-p mixing.

Show **energy arrow**, **atomic orbitals** involved, **atom labels**, the **molecular orbitals** that are formed, **filled electrons** in the MO diagram etc. Make sure to **label all the atomic and molecular orbitals**.

Using the MO diagram above, answer the following questions:

- (i) Which one of the following will have the highest bond order? Circle the correct choice
 A) PS^{1-} B) PS C) PS^{1+}
- (ii) Which one of the following will have the longest bond length? Circle the correct choice
 A) PS^{1-} B) PS C) PS^{1+}
- (iii) Which of the following will be diamagnetic? Circle the correct choice.
 A) PS^{1-} B) PS C) PS^{1+}
- (iv) Draw Lewis dot structure of the molecule with the shortest bond length (based on the MO diagram) from among the three (PS^{1-} , PS, PS^{1+}), and indicate the bond order calculated from the Lewis dot structure. **(3 points)**

Lewis Dot Structure

Bond Order



41FC37A8-23FE-497C-8E64-E8F6A898F11D

midterm-exam-93388

#437 Page 10 of 10

SCRAP PAGE

**CHEMISTRY 110 –Midterm Exam****Thursday November 7, 2024, 6:30 – 8:30 pm**

Examiners: Pallavi Sirjoosingh and Ashok Kakkar

First name (please write as legibly as possible within the boxes)

--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--

Last name

--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--

Student ID number

--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--

This booklet has 8 questions on 10 pages (The last page can be used as a scrap page).

The staple must NOT be removed from the question booklet. Your intact question booklet must be returned at the end of the exam.

A separate booklet contains a periodic table, and datasheet.

Write your answers directly under the questions in the space provided. Do not write on the QR codes or near the top edges (as it may be cut off).

Please write **legibly and dark enough** for the scan to be readable. For students who use English as a second language, dictionaries are permitted. Calculators are allowed.

DO NOT ASK QUESTIONS OF THE INVIGILATORS about interpretations of the exam questions. If you have a concern about a question, please summarize it at the top of this page. All concerns will be evaluated after the exam. GOOD LUCK!

**Question 1 8 points** (*Assume all transitions are possible*)

a. 6 points Calculate the **energy (in kJ/mol)** required to ionize (completely remove) the electron from the **n=3** level of **He⁺ ion** (He: Helium). Show your work including equations. (3 sig. figs.)

b. 2 points

(I) Select the largest (longest) wavelength from the three choices below:

- i. Wavelength of photon to *completely remove* an electron from *1s orbital of Hydrogen atom*
- ii. Wavelength of photon to *completely remove* an electron from *1s orbital of Boron atom (B)*
- iii. Wavelength of photon to *completely remove* an electron from *1s orbital of Boron cation (B⁺)*

i or ii or iii

(II) Select the smallest (shortest) wavelength from the three choices below (*for H atom*):

- i. Wavelength of photon *emitted* for an electron transitioning from *2p to 1s orbital*
- ii. Wavelength of photon *absorbed* for an electron transitioning from *2s to 4p orbital*
- iii. Wavelength of photon *absorbed* for an electron transitioning from *1s to 3p orbital*

i or ii or iii



Question 2 8 points



A Chem110 student plots the graph shown on the left, from data obtained for the kinetic energy (eV) of the ejected electron versus frequency (s^{-1}) of the incident light, for an unknown metal. ($1\text{eV} = 1.6022 \times 10^{-19} \text{ J}$)

Based on the plot, answer the questions **a to c** below:

a. 1 point On the plot above, label the **threshold frequency**.

b. 2 points What is the value of the work potential (in J) of the unknown metal?

c. 3 points What is the *longest* wavelength (in nm) required to eject an electron from this metal's surface? Show your work including equation(s) used. (3 sig figs)

d. 2 points In the space provided below, plot the following

(Assume frequency is *above* threshold frequency):

- Kinetic energy* of an ejected electron versus the *intensity* of the incident light (left)
- Number of electrons* ejected versus the *frequency* of the incident light (right)



**Question 3** **6 points****a. 2 points**

In the ground state of Ge (Germanium; $Z = 32$) atom, **name** the following orbitals:

i. Orbital containing the electron that requires the maximum energy to be removed.

ii. Orbital containing the electron that experiences the minimum effective nuclear charge.

b. 2 points

An unknown anion (A^-) has the following *excited* electronic configuration: $[Ar] 4s^2 3d^3 4p^1$

(Ar: Argon; $Z = 18$)

Give the **condensed ground state electronic configuration** of the atom, **A** and the cation, **A^{2+}** .
Show your work. (Assume all electronic transitions are possible)

Electronic Configuration of A:

Electronic Configuration of A^{2+} :

c. 2 points

In a single atom, determine how many electrons can have the following set of quantum numbers?

(i) $n = 5; l = 3$

In a single atom, determine how many electrons can have the following set of quantum numbers?

(i) $n = 3; l = 2; m_s = +1/2$

**Question 4** **8 points**

Rank the following from 1 to 4 in **increasing** order. Provide the ranking in the boxes provided.

Note: Atomic number (Z) are provided for relevant atoms

a) Increasing (lowest to highest) order of *ionization energy*: (1= lowest; 4 = highest)

Mg

Mg: Z = 12

Rb

Rb: Z = 37

Cs

Cs: Z = 55

K

K : Z = 19

b) Increasing (lowest to highest) order of *bond angle*: (1= lowest; 4 = highest)

H₂OCCl₄XeF₄

HCN

c) Increasing (smallest to largest) order of *ionic size* (1= smallest; 4 = largest)

S²⁻

S: Z = 16

O²⁻

O: Z = 8

Be²⁺

Be: Z = 4

Na⁺

Na: Z = 11

d) Increasing (lowest to highest) order of *bond energy* (1= lowest; 4 = highest)

P-F

P-Cl

P-I

P-Br

**Question 5** **12 points**

a. 9 points For the molecules HClO_4 and TeF_6 , draw the most stable *Lewis structure* showing appropriate shape and *dashed/wedged bonds*, based on VSEPR. Determine the **molecular geometry** around the central atom. For HClO_4 , label the *most polar bond*.



b. 3 points (Note: Lone pairs are not shown in structures below)

(i) For the three *resonance* structures (1, 2 and 3) shown below, answer the following question:



(I) Circle the *most contributing* resonance structure.

Structure 1

Structure 2

Structure 3

(ii) For the three structures given (4, 5, and 6) below, answer the following questions:



(I) Circle all the structures that represent *resonance structures* of each other:

Structure 4

Structure 5

Structure 6

(II) Among the resonance structures, circle the *most contributing* resonance structure.

Structure 4

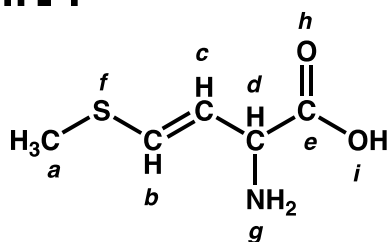
Structure 5

Structure 6

**Question 6** **6 points**

Using the data given below, calculate the **lattice energy** for SrO(s).

	ΔH° (kJ/mol)
$\text{Sr}(s) \longrightarrow \text{Sr}(g)$	+180
$\text{Sr}(g) \longrightarrow \text{Sr}^{2+}(g) + 2e^-$	+1468
$\text{O}_2(g) \longrightarrow 2\text{O}(g)$	+500
$\text{O}(g) + 2e^- \longrightarrow \text{O}^{2-}(g)$	+600
$\text{Sr}(s) + \frac{1}{2}\text{O}_2(g) \longrightarrow \text{SrO}(s)$	-550

**Question 7 10 points**

Consider the molecule given on the left, and answer questions (i) to (vi) using chemical bonding theories that we have learnt in class (valence bond theory, hybridization).

Note that the lone pairs are not shown in the structure:

i. Consider the carbon atoms marked in as ***a***, ***b***, ***c***, ***d***, and ***e*** in the molecule, and determine **hybridization** at each of the carbon centers. (2.5 points)

Atom	Hybridization
Carbon (<i>a</i>)	
Carbon (<i>b</i>)	
Carbon (<i>c</i>)	
Carbon (<i>d</i>)	
Carbon (<i>e</i>)	

ii. Now, consider the sulfur atom (S) marked as ***f***, nitrogen (N) atom (***g***), and oxygen(O) atoms (***h*** and ***i***), and determine their **hybridization**. (2 points)

Atom	Hybridization
Sulfur (<i>f</i>)	
Nitrogen (<i>g</i>)	
Oxygen (<i>h</i>)	
Oxygen (<i>i</i>)	

iii. **Describe** all the bonds formed by the carbon atoms ***a*** and ***b***, using sigma and pi designation as appropriate. (2 points)

Atom	Bonds formed (use sigma and pi as appropriate)
Carbon (<i>a</i>)	
Carbon (<i>b</i>)	

iv. How many total sp^3 -s sigma bonds are in the molecule? (1 point)

v. Determine **electronic** and **molecular geometry** at carbon (C) labeled ***e*** and sulfur (S) labeled ***f*** centres. (2 points)

Atom	Electronic Geometry	Molecular Geometry
Carbon (<i>e</i>)		
Sulfur (<i>f</i>)		

vi. Circle the value from the choices given below, that describes the **C-S-C bond angle at sulfur** (S) marked ***f***: (0.5 points)

>90° to <109.5°

109.5°

>109.5 to <120°

120°

>120° to <180°


Question 8 12 points

Using the **valence shell orbitals only** draw a **complete** molecular orbital (MO) diagram (with no s-p mixing) for the molecule **PS**, in the space provided below. **(6 points)**

Assume that 3s and 3p atomic orbitals will form molecular orbitals like those formed by the combination of 2s and 2p atomic orbitals that we learnt in class, and there is no s-p mixing.

Show **energy arrow**, **atomic orbitals** involved, **atom labels**, the **molecular orbitals** that are formed, **filled electrons** in the MO diagram etc. Make sure to label all the atomic and molecular orbitals.

Using the MO diagram above, answer the following questions:

- (i) Which one of the following will have the highest bond order? Circle the correct choice
 A) PS^{1-} B) PS C) PS^{1+}
- (ii) Which one of the following will have the longest bond length? Circle the correct choice
 A) PS^{1-} B) PS C) PS^{1+}
- (iii) Which of the following will be diamagnetic? Circle the correct choice.
 A) PS^{1-} B) PS C) PS^{1+}
- (iv) Draw Lewis dot structure of the molecule with the shortest bond length (based on the MO diagram) from among the three (PS^{1-} , PS, PS^{1+}), and indicate the bond order calculated from the Lewis dot structure. **(3 points)**

Lewis Dot Structure

Bond Order



9E86871E-DA41-4893-8BE1-B39EE414DDC9

midterm-exam-93388

#438 Page 10 of 10

SCRAP PAGE

**CHEMISTRY 110 –Midterm Exam****Thursday November 7, 2024, 6:30 – 8:30 pm****Examiners: Pallavi Sirjoosingh and Ashok Kakkar**

First name (please write as legibly as possible within the boxes)

--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--

Last name

--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--

Student ID number

--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--

This booklet has 8 questions on 10 pages (The last page can be used as a scrap page).

The staple must NOT be removed from the question booklet. Your intact question booklet must be returned at the end of the exam.

A separate booklet contains a periodic table, and datasheet.

Write your answers directly under the questions in the space provided. Do not write on the QR codes or near the top edges (as it may be cut off).

Please write **legibly and dark enough** for the scan to be readable. For students who use English as a second language, dictionaries are permitted. Calculators are allowed.

DO NOT ASK QUESTIONS OF THE INVIGILATORS about interpretations of the exam questions. If you have a concern about a question, please summarize it at the top of this page. All concerns will be evaluated after the exam. GOOD LUCK!

**Question 1 8 points** (*Assume all transitions are possible*)

a. 6 points Calculate the **energy (in kJ/mol)** required to ionize (completely remove) the electron from the **$n=3$** level of **He^+ ion** (He: Helium). Show your work including equations. (3 sig. figs.)

b. 2 points

(I) Select the largest (longest) wavelength from the three choices below:

- i. Wavelength of photon to *completely remove* an electron from *1s orbital of Hydrogen atom*
- ii. Wavelength of photon to *completely remove* an electron from *1s orbital of Boron atom (B)*
- iii. Wavelength of photon to *completely remove* an electron from *1s orbital of Boron cation (B^+)*

i or ii or iii

(II) Select the smallest (shortest) wavelength from the three choices below (*for H atom*):

- i. Wavelength of photon *emitted* for an electron transitioning from *2p to 1s orbital*
- ii. Wavelength of photon *absorbed* for an electron transitioning from *2s to 4p orbital*
- iii. Wavelength of photon *absorbed* for an electron transitioning from *1s to 3p orbital*

i or ii or iii



Question 2 8 points



A Chem110 student plots the graph shown on the left, from data obtained for the kinetic energy (eV) of the ejected electron versus frequency (s^{-1}) of the incident light, for an unknown metal. ($1\text{eV} = 1.6022 \times 10^{-19} \text{ J}$)

Based on the plot, answer the questions **a to c** below:

a. 1 point On the plot above, label the **threshold frequency**.

b. 2 points What is the value of the work potential (in J) of the unknown metal?

c. 3 points What is the *longest* wavelength (in nm) required to eject an electron from this metal's surface? Show your work including equation(s) used. (3 sig figs)

d. 2 points In the space provided below, plot the following

(Assume frequency is *above* threshold frequency):

- Kinetic energy* of an ejected electron versus the *intensity* of the incident light (left)
- Number of electrons* ejected versus the *frequency* of the incident light (right)



**Question 3** **6 points****a. 2 points**

In the ground state of Ge (Germanium; $Z = 32$) atom, **name** the following orbitals:

i. Orbital containing the electron that requires the maximum energy to be removed.

ii. Orbital containing the electron that experiences the minimum effective nuclear charge.

b. 2 points

An unknown anion (A^-) has the following *excited* electronic configuration: $[Ar] 4s^2 3d^3 4p^1$

(Ar: Argon; $Z = 18$)

Give the **condensed ground state electronic configuration** of the atom, **A** and the cation, **A^{2+}** .
Show your work. (Assume all electronic transitions are possible)

Electronic Configuration of A:

Electronic Configuration of A^{2+} :

c. 2 points

In a single atom, determine how many electrons can have the following set of quantum numbers?

(i) $n = 5; l = 3$

In a single atom, determine how many electrons can have the following set of quantum numbers?

(i) $n = 3; l = 2; m_s = +1/2$

**Question 4** **8 points**

Rank the following from 1 to 4 in **increasing** order. Provide the ranking in the boxes provided.

Note: Atomic number (Z) are provided for relevant atoms

a) Increasing (lowest to highest) order of *ionization energy*: (1= lowest; 4 = highest)

Mg

Mg: Z = 12

Rb

Rb: Z = 37

Cs

Cs: Z = 55

K

K : Z = 19

b) Increasing (lowest to highest) order of *bond angle*: (1= lowest; 4 = highest)

H₂OCCl₄XeF₄

HCN

c) Increasing (smallest to largest) order of *ionic size* (1= smallest; 4 = largest)

S²⁻

S: Z = 16

O²⁻

O: Z = 8

Be²⁺

Be: Z = 4

Na⁺

Na: Z = 11

d) Increasing (lowest to highest) order of *bond energy* (1= lowest; 4 = highest)

P-F

P-Cl

P-I

P-Br

**Question 5** **12 points**

a. 9 points For the molecules HClO_4 and TeF_6 , draw the most stable *Lewis structure* showing appropriate shape and *dashed/wedged bonds*, based on VSEPR. Determine the **molecular geometry** around the central atom. For HClO_4 , label the *most polar bond*.



b. 3 points (Note: Lone pairs are not shown in structures below)

(i) For the three *resonance* structures (1, 2 and 3) shown below, answer the following question:



(I) Circle the *most contributing* resonance structure.

Structure 1

Structure 2

Structure 3

(ii) For the three structures given (4, 5, and 6) below, answer the following questions:



(I) Circle all the structures that represent *resonance structures* of each other:

Structure 4

Structure 5

Structure 6

(II) Among the resonance structures, circle the *most contributing* resonance structure.

Structure 4

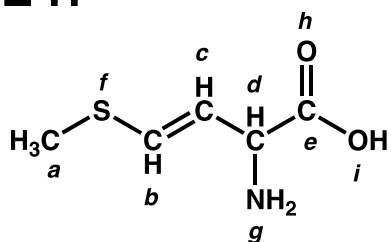
Structure 5

Structure 6

**Question 6** **6 points**

Using the data given below, calculate the **lattice energy** for SrO(s).

	ΔH° (kJ/mol)
$\text{Sr}(s) \longrightarrow \text{Sr}(g)$	+180
$\text{Sr}(g) \longrightarrow \text{Sr}^{2+}(g) + 2e^-$	+1468
$\text{O}_2(g) \longrightarrow 2\text{O}(g)$	+500
$\text{O}(g) + 2e^- \longrightarrow \text{O}^{2-}(g)$	+600
$\text{Sr}(s) + \frac{1}{2}\text{O}_2(g) \longrightarrow \text{SrO}(s)$	-550

**Question 7 10 points**

Consider the molecule given on the left, and answer questions (i) to (vi) using chemical bonding theories that we have learnt in class (valence bond theory, hybridization).

Note that the lone pairs are not shown in the structure:

i. Consider the carbon atoms marked in as ***a***, ***b***, ***c***, ***d***, and ***e*** in the molecule, and determine **hybridization** at each of the carbon centers. (2.5 points)

Atom	Hybridization
Carbon (<i>a</i>)	
Carbon (<i>b</i>)	
Carbon (<i>c</i>)	
Carbon (<i>d</i>)	
Carbon (<i>e</i>)	

ii. Now, consider the sulfur atom (S) marked as ***f***, nitrogen (N) atom (***g***), and oxygen(O) atoms (***h*** and ***i***), and determine their **hybridization**. (2 points)

Atom	Hybridization
Sulfur (<i>f</i>)	
Nitrogen (<i>g</i>)	
Oxygen (<i>h</i>)	
Oxygen (<i>i</i>)	

iii. **Describe** all the bonds formed by the carbon atoms ***a*** and ***b***, using sigma and pi designation as appropriate. (2 points)

Atom	Bonds formed (use sigma and pi as appropriate)
Carbon (<i>a</i>)	
Carbon (<i>b</i>)	

iv. How many total sp^3 -s sigma bonds are in the molecule? (1 point)

v. Determine **electronic** and **molecular geometry** at carbon (C) labeled ***e*** and sulfur (S) labeled ***f*** centres. (2 points)

Atom	Electronic Geometry	Molecular Geometry
Carbon (<i>e</i>)		
Sulfur (<i>f</i>)		

vi. Circle the value from the choices given below, that describes the **C-S-C bond angle at sulfur** (S) marked ***f***: (0.5 points)

>90° to <109.5° 109.5° >109.5 to <120° 120° >120° to <180°

**Question 8 12 points**

Using the **valence shell orbitals only** draw a **complete** molecular orbital (MO) diagram (with no s-p mixing) for the molecule **PS**, in the space provided below. **(6 points)**

Assume that 3s and 3p atomic orbitals will form molecular orbitals like those formed by the combination of 2s and 2p atomic orbitals that we learnt in class, and there is no s-p mixing.

Show **energy arrow**, **atomic orbitals** involved, **atom labels**, the **molecular orbitals** that are formed, **filled electrons** in the MO diagram etc. Make sure to label all the atomic and molecular orbitals.

Using the MO diagram above, answer the following questions:

- (i) Which one of the following will have the highest bond order? Circle the correct choice
A) PS^{1-} B) PS C) PS^{1+}
- (ii) Which one of the following will have the longest bond length? Circle the correct choice
A) PS^{1-} B) PS C) PS^{1+}
- (iii) Which of the following will be diamagnetic? Circle the correct choice.
A) PS^{1-} B) PS C) PS^{1+}
- (iv) Draw Lewis dot structure of the molecule with the shortest bond length (based on the MO diagram) from among the three (PS^{1-} , PS, PS^{1+}), and indicate the bond order calculated from the Lewis dot structure. **(3 points)**

Lewis Dot Structure

Bond Order



94E684F1-5485-4658-873E-0B3F19155895

midterm-exam-93388

#439 Page 10 of 10

SCRAP PAGE

**CHEMISTRY 110 –Midterm Exam****Thursday November 7, 2024, 6:30 – 8:30 pm**

Examiners: Pallavi Sirjoosingh and Ashok Kakkar

First name (please write as legibly as possible within the boxes)

--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--

Last name

--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--

Student ID number

--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--

This booklet has 8 questions on 10 pages (The last page can be used as a scrap page).

The staple must NOT be removed from the question booklet. Your intact question booklet must be returned at the end of the exam.

A separate booklet contains a periodic table, and datasheet.

Write your answers directly under the questions in the space provided. Do not write on the QR codes or near the top edges (as it may be cut off).

Please write **legibly and dark enough** for the scan to be readable. For students who use English as a second language, dictionaries are permitted. Calculators are allowed.

DO NOT ASK QUESTIONS OF THE INVIGILATORS about interpretations of the exam questions. If you have a concern about a question, please summarize it at the top of this page. All concerns will be evaluated after the exam. GOOD LUCK!

**Question 1 8 points** (*Assume all transitions are possible*)

a. 6 points Calculate the **energy (in kJ/mol)** required to ionize (completely remove) the electron from the **$n=3$** level of **He^+ ion** (He: Helium). Show your work including equations. (3 sig. figs.)

b. 2 points

(I) Select the largest (longest) wavelength from the three choices below:

- i. Wavelength of photon to *completely remove* an electron from *1s orbital of Hydrogen atom*
- ii. Wavelength of photon to *completely remove* an electron from *1s orbital of Boron atom (B)*
- iii. Wavelength of photon to *completely remove* an electron from *1s orbital of Boron cation (B^+)*

i or ii or iii

(II) Select the smallest (shortest) wavelength from the three choices below (*for H atom*):

- i. Wavelength of photon *emitted* for an electron transitioning from *2p to 1s orbital*
- ii. Wavelength of photon *absorbed* for an electron transitioning from *2s to 4p orbital*
- iii. Wavelength of photon *absorbed* for an electron transitioning from *1s to 3p orbital*

i or ii or iii



Question 2 8 points



A Chem110 student plots the graph shown on the left, from data obtained for the kinetic energy (eV) of the ejected electron versus frequency (s^{-1}) of the incident light, for an unknown metal. ($1\text{eV} = 1.6022 \times 10^{-19} \text{ J}$)

Based on the plot, answer the questions **a to c** below:

a. 1 point On the plot above, label the **threshold frequency**.

b. 2 points What is the value of the work potential (in J) of the unknown metal?

c. 3 points What is the *longest* wavelength (in nm) required to eject an electron from this metal's surface? Show your work including equation(s) used. (3 sig figs)

d. 2 points In the space provided below, plot the following

(Assume frequency is *above* threshold frequency):

- Kinetic energy* of an ejected electron versus the *intensity* of the incident light (left)
- Number of electrons* ejected versus the *frequency* of the incident light (right)



**Question 3** **6 points****a. 2 points**

In the ground state of Ge (Germanium; $Z = 32$) atom, **name** the following orbitals:

i. Orbital containing the electron that requires the maximum energy to be removed.

ii. Orbital containing the electron that experiences the minimum effective nuclear charge.

b. 2 points

An unknown anion (A^-) has the following *excited* electronic configuration: $[Ar] 4s^2 3d^3 4p^1$

(Ar: Argon; $Z = 18$)

Give the **condensed ground state electronic configuration** of the atom, **A** and the cation, A^{2+} .
Show your work. (Assume all electronic transitions are possible)

Electronic Configuration of A:

Electronic Configuration of A^{2+} :

c. 2 points

In a single atom, determine how many electrons can have the following set of quantum numbers?

(i) $n = 5; l = 3$

In a single atom, determine how many electrons can have the following set of quantum numbers?

(i) $n = 3; l = 2; m_s = +1/2$

**Question 4** **8 points**

Rank the following from 1 to 4 in **increasing** order. Provide the ranking in the boxes provided.

Note: Atomic number (Z) are provided for relevant atoms

a) Increasing (lowest to highest) order of *ionization energy*: (1= lowest; 4 = highest)

Mg

Mg: Z = 12

Rb

Rb: Z = 37

Cs

Cs: Z = 55

K

K : Z = 19

b) Increasing (lowest to highest) order of *bond angle*: (1= lowest; 4 = highest)

H₂OCCl₄XeF₄

HCN

c) Increasing (smallest to largest) order of *ionic size* (1= smallest; 4 = largest)

S²⁻

S: Z = 16

O²⁻

O: Z = 8

Be²⁺

Be: Z = 4

Na⁺

Na: Z = 11

d) Increasing (lowest to highest) order of *bond energy* (1= lowest; 4 = highest)

P-F

P-Cl

P-I

P-Br

**Question 5** **12 points**

a. 9 points For the molecules HClO_4 and TeF_6 , draw the most stable *Lewis structure* showing appropriate shape and *dashed/wedged bonds*, based on VSEPR. Determine the **molecular geometry** around the central atom. For HClO_4 , label the *most polar bond*.



b. 3 points (Note: Lone pairs are not shown in structures below)

(i) For the three *resonance* structures (1, 2 and 3) shown below, answer the following question:



(I) Circle the *most contributing* resonance structure.

Structure 1

Structure 2

Structure 3

(ii) For the three structures given (4, 5, and 6) below, answer the following questions:



(I) Circle all the structures that represent *resonance structures* of each other:

Structure 4

Structure 5

Structure 6

(II) Among the resonance structures, circle the *most contributing* resonance structure.

Structure 4

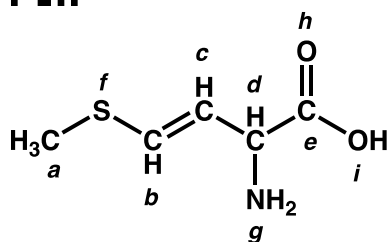
Structure 5

Structure 6

**Question 6** **6 points**

Using the data given below, calculate the **lattice energy** for SrO(s).

	ΔH° (kJ/mol)
$\text{Sr}(s) \longrightarrow \text{Sr}(g)$	+180
$\text{Sr}(g) \longrightarrow \text{Sr}^{2+}(g) + 2e^-$	+1468
$\text{O}_2(g) \longrightarrow 2\text{O}(g)$	+500
$\text{O}(g) + 2e^- \longrightarrow \text{O}^{2-}(g)$	+600
$\text{Sr}(s) + \frac{1}{2}\text{O}_2(g) \longrightarrow \text{SrO}(s)$	-550

**Question 7 10 points**

Consider the molecule given on the left, and answer questions (i) to (vi) using chemical bonding theories that we have learnt in class (valence bond theory, hybridization).

Note that the lone pairs are not shown in the structure:

i. Consider the carbon atoms marked in as ***a***, ***b***, ***c***, ***d***, and ***e*** in the molecule, and determine **hybridization** at each of the carbon centers. (2.5 points)

Atom	Hybridization
Carbon (<i>a</i>)	
Carbon (<i>b</i>)	
Carbon (<i>c</i>)	
Carbon (<i>d</i>)	
Carbon (<i>e</i>)	

ii. Now, consider the sulfur atom (S) marked as ***f***, nitrogen (N) atom (***g***), and oxygen(O) atoms (***h*** and ***i***), and determine their **hybridization**. (2 points)

Atom	Hybridization
Sulfur (<i>f</i>)	
Nitrogen (<i>g</i>)	
Oxygen (<i>h</i>)	
Oxygen (<i>i</i>)	

iii. **Describe** all the bonds formed by the carbon atoms ***a*** and ***b***, using sigma and pi designation as appropriate. (2 points)

Atom	Bonds formed (use sigma and pi as appropriate)
Carbon (<i>a</i>)	
Carbon (<i>b</i>)	

iv. How many total sp^3 -s sigma bonds are in the molecule? (1 point)

v. Determine **electronic** and **molecular geometry** at carbon (C) labeled ***e*** and sulfur (S) labeled ***f*** centres. (2 points)

Atom	Electronic Geometry	Molecular Geometry
Carbon (<i>e</i>)		
Sulfur (<i>f</i>)		

vi. Circle the value from the choices given below, that describes the **C-S-C bond angle at sulfur** (S) marked ***f***: (0.5 points)

>90° to <109.5° 109.5° >109.5 to <120° 120° >120° to <180°


Question 8 12 points

Using the **valence shell orbitals only** draw a **complete** molecular orbital (MO) diagram (with no s-p mixing) for the molecule **PS**, in the space provided below. **(6 points)**

Assume that 3s and 3p atomic orbitals will form molecular orbitals like those formed by the combination of 2s and 2p atomic orbitals that we learnt in class, and there is no s-p mixing.

Show **energy arrow**, **atomic orbitals** involved, **atom labels**, the **molecular orbitals** that are formed, **filled electrons** in the MO diagram etc. Make sure to label all the atomic and molecular orbitals.

Using the MO diagram above, answer the following questions:

- (i) Which one of the following will have the highest bond order? Circle the correct choice
 A) PS^{1-} B) PS C) PS^{1+}
- (ii) Which one of the following will have the longest bond length? Circle the correct choice
 A) PS^{1-} B) PS C) PS^{1+}
- (iii) Which of the following will be diamagnetic? Circle the correct choice.
 A) PS^{1-} B) PS C) PS^{1+}
- (iv) Draw Lewis dot structure of the molecule with the shortest bond length (based on the MO diagram) from among the three (PS^{1-} , PS, PS^{1+}), and indicate the bond order calculated from the Lewis dot structure. **(3 points)**

Lewis Dot Structure

Bond Order



7F7E89A8-0791-4D40-8B84-677DBD123CCD

midterm-exam-93388

#440 Page 10 of 10

SCRAP PAGE

**CHEMISTRY 110 –Midterm Exam****Thursday November 7, 2024, 6:30 – 8:30 pm****Examiners: Pallavi Sirjoosingh and Ashok Kakkar**

First name (please write as legibly as possible within the boxes)

--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--

Last name

--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--

Student ID number

--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--

This booklet has 8 questions on 10 pages (The last page can be used as a scrap page).

The staple must NOT be removed from the question booklet. Your intact question booklet must be returned at the end of the exam.

A separate booklet contains a periodic table, and datasheet.

Write your answers directly under the questions in the space provided. Do not write on the QR codes or near the top edges (as it may be cut off).

Please write **legibly and dark enough** for the scan to be readable. For students who use English as a second language, dictionaries are permitted. Calculators are allowed.

DO NOT ASK QUESTIONS OF THE INVIGILATORS about interpretations of the exam questions. If you have a concern about a question, please summarize it at the top of this page. All concerns will be evaluated after the exam. GOOD LUCK!

**Question 1 8 points** (*Assume all transitions are possible*)

a. 6 points Calculate the **energy (in kJ/mol)** required to ionize (completely remove) the electron from the **n=3** level of **He⁺ ion** (He: Helium). Show your work including equations. (3 sig. figs.)

b. 2 points

(I) Select the largest (longest) wavelength from the three choices below:

- i. Wavelength of photon to *completely remove* an electron from *1s orbital of Hydrogen atom*
- ii. Wavelength of photon to *completely remove* an electron from *1s orbital of Boron atom (B)*
- iii. Wavelength of photon to *completely remove* an electron from *1s orbital of Boron cation (B⁺)*

i or ii or iii

(II) Select the smallest (shortest) wavelength from the three choices below (*for H atom*):

- i. Wavelength of photon *emitted* for an electron transitioning from *2p to 1s orbital*
- ii. Wavelength of photon *absorbed* for an electron transitioning from *2s to 4p orbital*
- iii. Wavelength of photon *absorbed* for an electron transitioning from *1s to 3p orbital*

i or ii or iii



Question 2 8 points



A Chem110 student plots the graph shown on the left, from data obtained for the kinetic energy (eV) of the ejected electron versus frequency (s^{-1}) of the incident light, for an unknown metal. ($1\text{eV} = 1.6022 \times 10^{-19} \text{ J}$)

Based on the plot, answer the questions **a to c** below:

a. 1 point On the plot above, label the **threshold frequency**.

b. 2 points What is the value of the work potential (in J) of the unknown metal?

c. 3 points What is the *longest* wavelength (in nm) required to eject an electron from this metal's surface? Show your work including equation(s) used. (3 sig figs)

d. 2 points In the space provided below, plot the following

(Assume frequency is *above* threshold frequency):

- i. *Kinetic energy* of an ejected electron versus the *intensity* of the incident light (left)
- ii. *Number of electrons* ejected versus the *frequency* of the incident light (right)



**Question 3** **6 points****a. 2 points**

In the ground state of Ge (Germanium; $Z = 32$) atom, **name** the following orbitals:

i. Orbital containing the electron that requires the maximum energy to be removed.

ii. Orbital containing the electron that experiences the minimum effective nuclear charge.

b. 2 points

An unknown anion (A^-) has the following *excited* electronic configuration: $[Ar] 4s^2 3d^3 4p^1$

(Ar: Argon; $Z = 18$)

Give the **condensed ground state electronic configuration** of the atom, **A** and the cation, **A^{2+}** .
Show your work. (Assume all electronic transitions are possible)

Electronic Configuration of A:

Electronic Configuration of A^{2+} :

c. 2 points

In a single atom, determine how many electrons can have the following set of quantum numbers?

(i) $n = 5; l = 3$

In a single atom, determine how many electrons can have the following set of quantum numbers?

(i) $n = 3; l = 2; m_s = +1/2$

**Question 4** **8 points**

Rank the following from 1 to 4 in **increasing** order. Provide the ranking in the boxes provided.

Note: Atomic number (Z) are provided for relevant atoms

a) Increasing (lowest to highest) order of *ionization energy*: (1= lowest; 4 = highest)

Mg

Mg: Z = 12

Rb

Rb: Z = 37

Cs

Cs: Z = 55

K

K : Z = 19

b) Increasing (lowest to highest) order of *bond angle*: (1= lowest; 4 = highest)

H₂OCCl₄XeF₄

HCN

c) Increasing (smallest to largest) order of *ionic size* (1= smallest; 4 = largest)

S²⁻

S: Z = 16

O²⁻

O: Z = 8

Be²⁺

Be: Z = 4

Na⁺

Na: Z = 11

d) Increasing (lowest to highest) order of *bond energy* (1= lowest; 4 = highest)

P-F

P-Cl

P-I

P-Br

**Question 5** **12 points**

a. 9 points For the molecules HClO_4 and TeF_6 , draw the most stable *Lewis structure* showing appropriate shape and *dashed/wedged bonds*, based on VSEPR. Determine the **molecular geometry** around the central atom. For HClO_4 , label the *most polar bond*.



b. 3 points (Note: Lone pairs are not shown in structures below)

(i) For the three *resonance* structures (1, 2 and 3) shown below, answer the following question:



(I) Circle the *most contributing* resonance structure.

Structure 1

Structure 2

Structure 3

(ii) For the three structures given (4, 5, and 6) below, answer the following questions:



(I) Circle all the structures that represent *resonance structures* of each other:

Structure 4

Structure 5

Structure 6

(II) Among the resonance structures, circle the *most contributing* resonance structure.

Structure 4

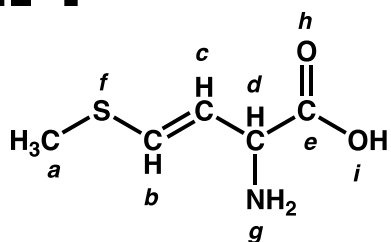
Structure 5

Structure 6

**Question 6** **6 points**

Using the data given below, calculate the **lattice energy** for SrO(s).

	ΔH° (kJ/mol)
$\text{Sr}(s) \longrightarrow \text{Sr}(g)$	+180
$\text{Sr}(g) \longrightarrow \text{Sr}^{2+}(g) + 2e^-$	+1468
$\text{O}_2(g) \longrightarrow 2\text{O}(g)$	+500
$\text{O}(g) + 2e^- \longrightarrow \text{O}^{2-}(g)$	+600
$\text{Sr}(s) + \frac{1}{2}\text{O}_2(g) \longrightarrow \text{SrO}(s)$	-550

**Question 7 10 points**

Consider the molecule given on the left, and answer questions (i) to (vi) using chemical bonding theories that we have learnt in class (valence bond theory, hybridization).

Note that the lone pairs are not shown in the structure:

i. Consider the carbon atoms marked in as ***a***, ***b***, ***c***, ***d***, and ***e*** in the molecule, and determine **hybridization** at each of the carbon centers. (2.5 points)

Atom	Hybridization
Carbon (<i>a</i>)	
Carbon (<i>b</i>)	
Carbon (<i>c</i>)	
Carbon (<i>d</i>)	
Carbon (<i>e</i>)	

ii. Now, consider the sulfur atom (S) marked as ***f***, nitrogen (N) atom (***g***), and oxygen(O) atoms (***h*** and ***i***), and determine their **hybridization**. (2 points)

Atom	Hybridization
Sulfur (<i>f</i>)	
Nitrogen (<i>g</i>)	
Oxygen (<i>h</i>)	
Oxygen (<i>i</i>)	

iii. **Describe** all the bonds formed by the carbon atoms ***a*** and ***b***, using sigma and pi designation as appropriate. (2 points)

Atom	Bonds formed (use sigma and pi as appropriate)
Carbon (<i>a</i>)	
Carbon (<i>b</i>)	

iv. How many total sp^3 -s sigma bonds are in the molecule? (1 point)

v. Determine **electronic** and **molecular geometry** at carbon (C) labeled ***e*** and sulfur (S) labeled ***f*** centres. (2 points)

Atom	Electronic Geometry	Molecular Geometry
Carbon (<i>e</i>)		
Sulfur (<i>f</i>)		

vi. Circle the value from the choices given below, that describes the **C-S-C bond angle at sulfur** (S) marked ***f***: (0.5 points)

$>90^\circ$ to $<109.5^\circ$ 109.5° >109.5 to $<120^\circ$ 120° $>120^\circ$ to $<180^\circ$


Question 8 12 points

Using the **valence shell orbitals only** draw a **complete** molecular orbital (MO) diagram (with no s-p mixing) for the molecule **PS**, in the space provided below. **(6 points)**

Assume that 3s and 3p atomic orbitals will form molecular orbitals like those formed by the combination of 2s and 2p atomic orbitals that we learnt in class, and there is no s-p mixing.

Show **energy arrow**, **atomic orbitals** involved, **atom labels**, the **molecular orbitals** that are formed, **filled electrons** in the MO diagram etc. Make sure to label all the atomic and molecular orbitals.

Using the MO diagram above, answer the following questions:

- (i) Which one of the following will have the highest bond order? Circle the correct choice
 A) PS^{1-} B) PS C) PS^{1+}
- (ii) Which one of the following will have the longest bond length? Circle the correct choice
 A) PS^{1-} B) PS C) PS^{1+}
- (iii) Which of the following will be diamagnetic? Circle the correct choice.
 A) PS^{1-} B) PS C) PS^{1+}
- (iv) Draw Lewis dot structure of the molecule with the shortest bond length (based on the MO diagram) from among the three (PS^{1-} , PS, PS^{1+}), and indicate the bond order calculated from the Lewis dot structure. **(3 points)**

Lewis Dot Structure

Bond Order



42C69375-6D1F-4C6D-9BEE-4CDF906A752F

midterm-exam-93388

#441 Page 10 of 10

SCRAP PAGE

**CHEMISTRY 110 –Midterm Exam****Thursday November 7, 2024, 6:30 – 8:30 pm**

Examiners: Pallavi Sirjoosingh and Ashok Kakkar

First name (please write as legibly as possible within the boxes)

--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--

Last name

--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--

Student ID number

--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--

This booklet has 8 questions on 10 pages (The last page can be used as a scrap page).

The staple must NOT be removed from the question booklet. Your intact question booklet must be returned at the end of the exam.

A separate booklet contains a periodic table, and datasheet.

Write your answers directly under the questions in the space provided. Do not write on the QR codes or near the top edges (as it may be cut off).

Please write **legibly and dark enough** for the scan to be readable. For students who use English as a second language, dictionaries are permitted. Calculators are allowed.

DO NOT ASK QUESTIONS OF THE INVIGILATORS about interpretations of the exam questions. If you have a concern about a question, please summarize it at the top of this page. All concerns will be evaluated after the exam. GOOD LUCK!

**Question 1 8 points** (*Assume all transitions are possible*)

a. 6 points Calculate the **energy (in kJ/mol)** required to ionize (completely remove) the electron from the **$n=3$** level of **He^+ ion** (He: Helium). Show your work including equations. (3 sig. figs.)

b. 2 points

(I) Select the largest (longest) wavelength from the three choices below:

- i. Wavelength of photon to *completely remove* an electron from *1s orbital of Hydrogen atom*
- ii. Wavelength of photon to *completely remove* an electron from *1s orbital of Boron atom (B)*
- iii. Wavelength of photon to *completely remove* an electron from *1s orbital of Boron cation (B^+)*

i or ii or iii

(II) Select the smallest (shortest) wavelength from the three choices below (*for H atom*):

- i. Wavelength of photon *emitted* for an electron transitioning from *2p to 1s orbital*
- ii. Wavelength of photon *absorbed* for an electron transitioning from *2s to 4p orbital*
- iii. Wavelength of photon *absorbed* for an electron transitioning from *1s to 3p orbital*

i or ii or iii



Question 2 8 points



A Chem110 student plots the graph shown on the left, from data obtained for the kinetic energy (eV) of the ejected electron versus frequency (s^{-1}) of the incident light, for an unknown metal. ($1\text{eV} = 1.6022 \times 10^{-19} \text{ J}$)

Based on the plot, answer the questions **a to c** below:

a. 1 point On the plot above, label the **threshold frequency**.

b. 2 points What is the value of the work potential (in J) of the unknown metal?

c. 3 points What is the *longest* wavelength (in nm) required to eject an electron from this metal's surface? Show your work including equation(s) used. (3 sig figs)

d. 2 points In the space provided below, plot the following

(Assume frequency is *above* threshold frequency):

- Kinetic energy* of an ejected electron versus the *intensity* of the incident light (left)
- Number of electrons* ejected versus the *frequency* of the incident light (right)



**Question 3** **6 points****a. 2 points**

In the ground state of Ge (Germanium; $Z = 32$) atom, **name** the following orbitals:

i. Orbital containing the electron that requires the maximum energy to be removed.

ii. Orbital containing the electron that experiences the minimum effective nuclear charge.

b. 2 points

An unknown anion (A^-) has the following *excited* electronic configuration: $[Ar] 4s^2 3d^3 4p^1$

(Ar: Argon; $Z = 18$)

Give the **condensed ground state electronic configuration** of the atom, **A** and the cation, **A^{2+}** .
Show your work. (Assume all electronic transitions are possible)

Electronic Configuration of A:

Electronic Configuration of A^{2+} :

c. 2 points

In a single atom, determine how many electrons can have the following set of quantum numbers?

(i) $n = 5; l = 3$

In a single atom, determine how many electrons can have the following set of quantum numbers?

(i) $n = 3; l = 2; m_s = +1/2$

**Question 4** **8 points**

Rank the following from 1 to 4 in **increasing** order. Provide the ranking in the boxes provided.

Note: Atomic number (Z) are provided for relevant atoms

a) Increasing (lowest to highest) order of *ionization energy*: (1= lowest; 4 = highest)

Mg

Mg: Z = 12

Rb

Rb: Z = 37

Cs

Cs: Z = 55

K

K : Z = 19

b) Increasing (lowest to highest) order of *bond angle*: (1= lowest; 4 = highest)

H₂OCCl₄XeF₄

HCN

c) Increasing (smallest to largest) order of *ionic size* (1= smallest; 4 = largest)

S²⁻

S: Z = 16

O²⁻

O: Z = 8

Be²⁺

Be: Z = 4

Na⁺

Na: Z = 11

d) Increasing (lowest to highest) order of *bond energy* (1= lowest; 4 = highest)

P-F

P-Cl

P-I

P-Br

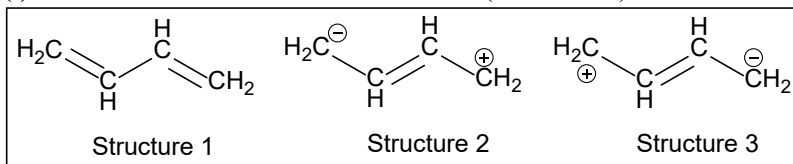
**Question 5** **12 points**

a. 9 points For the molecules HClO_4 and TeF_6 , draw the most stable *Lewis structure* showing appropriate shape and *dashed/wedged bonds*, based on VSEPR. Determine the **molecular geometry** around the central atom. For HClO_4 , label the *most polar bond*.



b. 3 points (Note: Lone pairs are not shown in structures below)

(i) For the three *resonance* structures (1, 2 and 3) shown below, answer the following question:



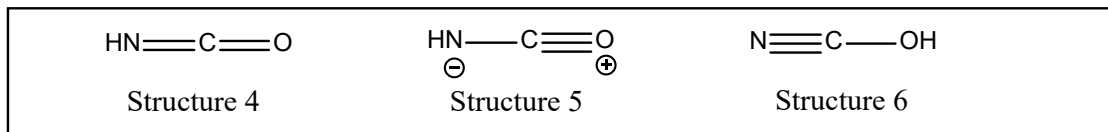
(I) Circle the *most contributing* resonance structure.

Structure 1

Structure 2

Structure 3

(ii) For the three structures given (4, 5, and 6) below, answer the following questions:



(I) Circle all the structures that represent *resonance structures* of each other:

Structure 4

Structure 5

Structure 6

(II) Among the resonance structures, circle the *most contributing* resonance structure.

Structure 4

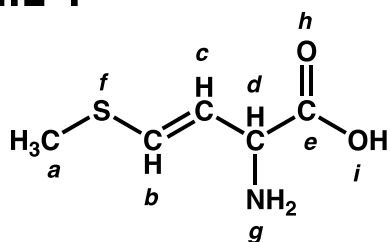
Structure 5

Structure 6

**Question 6** **6 points**

Using the data given below, calculate the **lattice energy** for SrO(s).

	ΔH° (kJ/mol)
$\text{Sr}(s) \longrightarrow \text{Sr}(g)$	+180
$\text{Sr}(g) \longrightarrow \text{Sr}^{2+}(g) + 2e^-$	+1468
$\text{O}_2(g) \longrightarrow 2\text{O}(g)$	+500
$\text{O}(g) + 2e^- \longrightarrow \text{O}^{2-}(g)$	+600
$\text{Sr}(s) + \frac{1}{2}\text{O}_2(g) \longrightarrow \text{SrO}(s)$	-550

**Question 7 10 points**

Consider the molecule given on the left, and answer questions (i) to (vi) using chemical bonding theories that we have learnt in class (valence bond theory, hybridization).

Note that the lone pairs are not shown in the structure:

i. Consider the carbon atoms marked in as ***a***, ***b***, ***c***, ***d***, and ***e*** in the molecule, and determine **hybridization** at each of the carbon centers. (2.5 points)

Atom	Hybridization
Carbon (<i>a</i>)	
Carbon (<i>b</i>)	
Carbon (<i>c</i>)	
Carbon (<i>d</i>)	
Carbon (<i>e</i>)	

ii. Now, consider the sulfur atom (S) marked as ***f***, nitrogen (N) atom (***g***), and oxygen(O) atoms (***h*** and ***i***), and determine their **hybridization**. (2 points)

Atom	Hybridization
Sulfur (<i>f</i>)	
Nitrogen (<i>g</i>)	
Oxygen (<i>h</i>)	
Oxygen (<i>i</i>)	

iii. **Describe** all the bonds formed by the carbon atoms ***a*** and ***b***, using sigma and pi designation as appropriate. (2 points)

Atom	Bonds formed (use sigma and pi as appropriate)
Carbon (<i>a</i>)	
Carbon (<i>b</i>)	

iv. How many total sp^3 -s sigma bonds are in the molecule? (1 point)

v. Determine **electronic** and **molecular geometry** at carbon (C) labeled ***e*** and sulfur (S) labeled ***f*** centres. (2 points)

Atom	Electronic Geometry	Molecular Geometry
Carbon (<i>e</i>)		
Sulfur (<i>f</i>)		

vi. Circle the value from the choices given below, that describes the **C-S-C bond angle at sulfur** (S) marked ***f***: (0.5 points)

>90° to <109.5° 109.5° >109.5 to <120° 120° >120° to <180°


Question 8 12 points

Using the **valence shell orbitals only** draw a **complete** molecular orbital (MO) diagram (with no s-p mixing) for the molecule **PS**, in the space provided below. **(6 points)**

Assume that 3s and 3p atomic orbitals will form molecular orbitals like those formed by the combination of 2s and 2p atomic orbitals that we learnt in class, and there is no s-p mixing.

Show **energy arrow**, **atomic orbitals** involved, **atom labels**, the **molecular orbitals** that are formed, **filled electrons** in the MO diagram etc. Make sure to label all the atomic and molecular orbitals.

Using the MO diagram above, answer the following questions:

- (i) Which one of the following will have the highest bond order? Circle the correct choice
 A) PS^{1-} B) PS C) PS^{1+}
- (ii) Which one of the following will have the longest bond length? Circle the correct choice
 A) PS^{1-} B) PS C) PS^{1+}
- (iii) Which of the following will be diamagnetic? Circle the correct choice.
 A) PS^{1-} B) PS C) PS^{1+}
- (iv) Draw Lewis dot structure of the molecule with the shortest bond length (based on the MO diagram) from among the three (PS^{1-} , PS, PS^{1+}), and indicate the bond order calculated from the Lewis dot structure. **(3 points)**

Lewis Dot Structure

Bond Order



A0634FAB-63EB-4CEA-924B-A60D68C47405

midterm-exam-93388

#442 Page 10 of 10

SCRAP PAGE

**CHEMISTRY 110 –Midterm Exam****Thursday November 7, 2024, 6:30 – 8:30 pm****Examiners: Pallavi Sirjoosingh and Ashok Kakkar**

First name (please write as legibly as possible within the boxes)

--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--

Last name

--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--

Student ID number

--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--

This booklet has 8 questions on 10 pages (The last page can be used as a scrap page).

The staple must NOT be removed from the question booklet. Your intact question booklet must be returned at the end of the exam.

A separate booklet contains a periodic table, and datasheet.

Write your answers directly under the questions in the space provided. Do not write on the QR codes or near the top edges (as it may be cut off).

Please write **legibly and dark enough** for the scan to be readable. For students who use English as a second language, dictionaries are permitted. Calculators are allowed.

DO NOT ASK QUESTIONS OF THE INVIGILATORS about interpretations of the exam questions. If you have a concern about a question, please summarize it at the top of this page. All concerns will be evaluated after the exam. GOOD LUCK!

**Question 1 8 points** (*Assume all transitions are possible*)

a. 6 points Calculate the **energy (in kJ/mol)** required to ionize (completely remove) the electron from the **n=3** level of **He⁺ ion** (He: Helium). Show your work including equations. (3 sig. figs.)

b. 2 points

(I) Select the largest (longest) wavelength from the three choices below:

- i. Wavelength of photon to *completely remove* an electron from *1s orbital of Hydrogen atom*
- ii. Wavelength of photon to *completely remove* an electron from *1s orbital of Boron atom (B)*
- iii. Wavelength of photon to *completely remove* an electron from *1s orbital of Boron cation (B⁺)*

i or ii or iii

(II) Select the smallest (shortest) wavelength from the three choices below (*for H atom*):

- i. Wavelength of photon *emitted* for an electron transitioning from *2p to 1s orbital*
- ii. Wavelength of photon *absorbed* for an electron transitioning from *2s to 4p orbital*
- iii. Wavelength of photon *absorbed* for an electron transitioning from *1s to 3p orbital*

i or ii or iii



Question 2 8 points



A Chem110 student plots the graph shown on the left, from data obtained for the kinetic energy (eV) of the ejected electron versus frequency (s^{-1}) of the incident light, for an unknown metal. ($1\text{eV} = 1.6022 \times 10^{-19} \text{ J}$)

Based on the plot, answer the questions **a to c** below:

a. 1 point On the plot above, label the **threshold frequency**.

b. 2 points What is the value of the work potential (in J) of the unknown metal?

c. 3 points What is the *longest* wavelength (in nm) required to eject an electron from this metal's surface? Show your work including equation(s) used. (3 sig figs)

d. 2 points In the space provided below, plot the following

(Assume frequency is *above* threshold frequency):

- i. *Kinetic energy* of an ejected electron versus the *intensity* of the incident light (left)
- ii. *Number of electrons* ejected versus the *frequency* of the incident light (right)



**Question 3** **6 points****a. 2 points**

In the ground state of Ge (Germanium; $Z = 32$) atom, **name** the following orbitals:

i. Orbital containing the electron that requires the maximum energy to be removed.

ii. Orbital containing the electron that experiences the minimum effective nuclear charge.

b. 2 points

An unknown anion (A^-) has the following *excited* electronic configuration: $[Ar] 4s^2 3d^3 4p^1$

(Ar: Argon; $Z = 18$)

Give the **condensed ground state electronic configuration** of the atom, **A** and the cation, **A^{2+}** .
Show your work. (Assume all electronic transitions are possible)

Electronic Configuration of A:

Electronic Configuration of A^{2+} :

c. 2 points

In a single atom, determine how many electrons can have the following set of quantum numbers?

(i) $n = 5; l = 3$

In a single atom, determine how many electrons can have the following set of quantum numbers?

(i) $n = 3; l = 2; m_s = +1/2$

**Question 4** **8 points**

Rank the following from 1 to 4 in **increasing** order. Provide the ranking in the boxes provided.

Note: Atomic number (Z) are provided for relevant atoms

a) Increasing (lowest to highest) order of *ionization energy*: (1= lowest; 4 = highest)

Mg

Mg: Z = 12

Rb

Rb: Z = 37

Cs

Cs: Z = 55

K

K : Z = 19

b) Increasing (lowest to highest) order of *bond angle*: (1= lowest; 4 = highest)

H₂OCCl₄XeF₄

HCN

c) Increasing (smallest to largest) order of *ionic size* (1= smallest; 4 = largest)

S²⁻

S: Z = 16

O²⁻

O: Z = 8

Be²⁺

Be: Z = 4

Na⁺

Na: Z = 11

d) Increasing (lowest to highest) order of *bond energy* (1= lowest; 4 = highest)

P-F

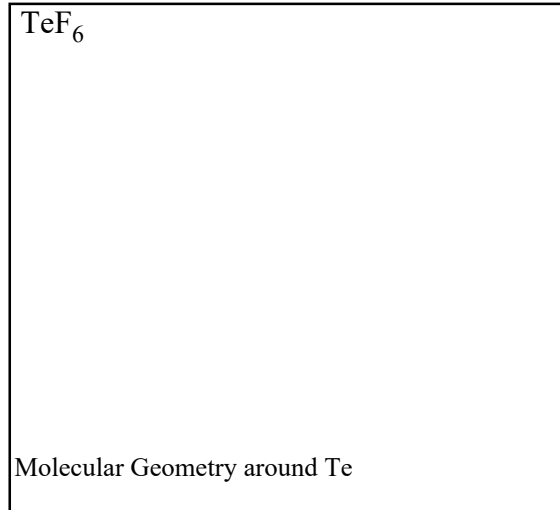
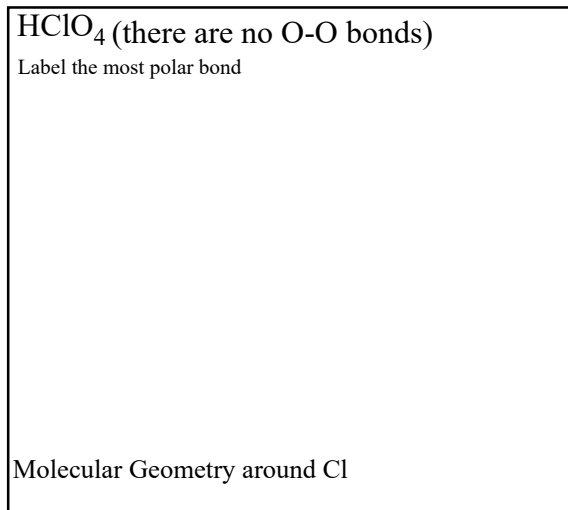
P-Cl

P-I

P-Br

**Question 5** **12 points**

a. 9 points For the molecules HClO_4 and TeF_6 , draw the most stable *Lewis structure* showing appropriate shape and *dashed/wedged bonds*, based on VSEPR. Determine the **molecular geometry** around the central atom. For HClO_4 , label the *most polar bond*.



b. 3 points (Note: Lone pairs are not shown in structures below)

(i) For the three *resonance* structures (1, 2 and 3) shown below, answer the following question:



(I) Circle the *most contributing* resonance structure.

Structure 1

Structure 2

Structure 3

(ii) For the three structures given (4, 5, and 6) below, answer the following questions:



(I) Circle all the structures that represent *resonance structures* of each other:

Structure 4

Structure 5

Structure 6

(II) Among the resonance structures, circle the *most contributing* resonance structure.

Structure 4

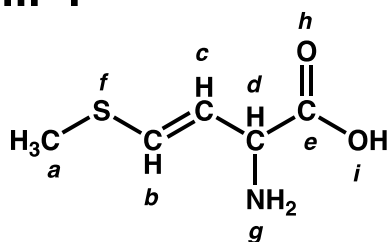
Structure 5

Structure 6

**Question 6** **6 points**

Using the data given below, calculate the **lattice energy** for SrO(s).

	ΔH° (kJ/mol)
$\text{Sr}(s) \longrightarrow \text{Sr}(g)$	+180
$\text{Sr}(g) \longrightarrow \text{Sr}^{2+}(g) + 2e^-$	+1468
$\text{O}_2(g) \longrightarrow 2\text{O}(g)$	+500
$\text{O}(g) + 2e^- \longrightarrow \text{O}^{2-}(g)$	+600
$\text{Sr}(s) + \frac{1}{2}\text{O}_2(g) \longrightarrow \text{SrO}(s)$	-550

**Question 7 10 points**

Consider the molecule given on the left, and answer questions (i) to (vi) using chemical bonding theories that we have learnt in class (valence bond theory, hybridization).

Note that the lone pairs are not shown in the structure:

i. Consider the carbon atoms marked in as ***a***, ***b***, ***c***, ***d***, and ***e*** in the molecule, and determine **hybridization** at each of the carbon centers. (2.5 points)

Atom	Hybridization
Carbon (<i>a</i>)	
Carbon (<i>b</i>)	
Carbon (<i>c</i>)	
Carbon (<i>d</i>)	
Carbon (<i>e</i>)	

ii. Now, consider the sulfur atom (S) marked as ***f***, nitrogen (N) atom (***g***), and oxygen(O) atoms (***h*** and ***i***), and determine their **hybridization**. (2 points)

Atom	Hybridization
Sulfur (<i>f</i>)	
Nitrogen (<i>g</i>)	
Oxygen (<i>h</i>)	
Oxygen (<i>i</i>)	

iii. **Describe** all the bonds formed by the carbon atoms ***a*** and ***b***, using sigma and pi designation as appropriate. (2 points)

Atom	Bonds formed (use sigma and pi as appropriate)
Carbon (<i>a</i>)	
Carbon (<i>b</i>)	

iv. How many total sp^3 -s sigma bonds are in the molecule? (1 point)

v. Determine **electronic** and **molecular geometry** at carbon (C) labeled ***e*** and sulfur (S) labeled ***f*** centres. (2 points)

Atom	Electronic Geometry	Molecular Geometry
Carbon (<i>e</i>)		
Sulfur (<i>f</i>)		

vi. Circle the value from the choices given below, that describes the **C-S-C bond angle at sulfur** (S) marked ***f***: (0.5 points)

>90° to <109.5°

109.5°

>109.5 to <120°

120°

>120° to <180°

**Question 8 12 points**

Using the **valence shell orbitals only** draw a **complete** molecular orbital (MO) diagram (with no s-p mixing) for the molecule **PS**, in the space provided below. **(6 points)**

Assume that 3s and 3p atomic orbitals will form molecular orbitals like those formed by the combination of 2s and 2p atomic orbitals that we learnt in class, and there is no s-p mixing.

Show **energy arrow**, **atomic orbitals** involved, **atom labels**, the **molecular orbitals** that are formed, **filled electrons** in the MO diagram etc. Make sure to label all the atomic and molecular orbitals.

Using the MO diagram above, answer the following questions:

- (i) Which one of the following will have the highest bond order? Circle the correct choice
A) PS^{1-} B) PS C) PS^{1+}
- (ii) Which one of the following will have the longest bond length? Circle the correct choice
A) PS^{1-} B) PS C) PS^{1+}
- (iii) Which of the following will be diamagnetic? Circle the correct choice.
A) PS^{1-} B) PS C) PS^{1+}
- (iv) Draw Lewis dot structure of the molecule with the shortest bond length (based on the MO diagram) from among the three (PS^{1-} , PS, PS^{1+}), and indicate the bond order calculated from the Lewis dot structure. **(3 points)**

Lewis Dot Structure

Bond Order



89F4C46B-65E7-450C-B69C-215624F2E431

midterm-exam-93388

#443 Page 10 of 10

SCRAP PAGE

**CHEMISTRY 110 –Midterm Exam****Thursday November 7, 2024, 6:30 – 8:30 pm**

Examiners: Pallavi Sirjoosingh and Ashok Kakkar

First name (please write as legibly as possible within the boxes)

--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--

Last name

--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--

Student ID number

--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--

This booklet has 8 questions on 10 pages (The last page can be used as a scrap page).

The staple must NOT be removed from the question booklet. Your intact question booklet must be returned at the end of the exam.

A separate booklet contains a periodic table, and datasheet.

Write your answers directly under the questions in the space provided. Do not write on the QR codes or near the top edges (as it may be cut off).

Please write **legibly and dark enough** for the scan to be readable. For students who use English as a second language, dictionaries are permitted. Calculators are allowed.

DO NOT ASK QUESTIONS OF THE INVIGILATORS about interpretations of the exam questions. If you have a concern about a question, please summarize it at the top of this page. All concerns will be evaluated after the exam. GOOD LUCK!

**Question 1 8 points** (*Assume all transitions are possible*)

a. 6 points Calculate the **energy (in kJ/mol)** required to ionize (completely remove) the electron from the **n=3** level of **He⁺ ion** (He: Helium). Show your work including equations. (3 sig. figs.)

b. 2 points

(I) Select the largest (longest) wavelength from the three choices below:

- i. Wavelength of photon to *completely remove* an electron from *1s orbital of Hydrogen atom*
- ii. Wavelength of photon to *completely remove* an electron from *1s orbital of Boron atom (B)*
- iii. Wavelength of photon to *completely remove* an electron from *1s orbital of Boron cation (B⁺)*

i or ii or iii

(II) Select the smallest (shortest) wavelength from the three choices below (*for H atom*):

- i. Wavelength of photon *emitted* for an electron transitioning from *2p to 1s orbital*
- ii. Wavelength of photon *absorbed* for an electron transitioning from *2s to 4p orbital*
- iii. Wavelength of photon *absorbed* for an electron transitioning from *1s to 3p orbital*

i or ii or iii



Question 2 8 points



A Chem110 student plots the graph shown on the left, from data obtained for the kinetic energy (eV) of the ejected electron versus frequency (s^{-1}) of the incident light, for an unknown metal. ($1\text{eV} = 1.6022 \times 10^{-19} \text{ J}$)

Based on the plot, answer the questions **a to c** below:

a. 1 point On the plot above, label the **threshold frequency**.

b. 2 points What is the value of the work potential (in J) of the unknown metal?

c. 3 points What is the *longest* wavelength (in nm) required to eject an electron from this metal's surface? Show your work including equation(s) used. (3 sig figs)

d. 2 points In the space provided below, plot the following

(Assume frequency is *above* threshold frequency):

- i. *Kinetic energy* of an ejected electron versus the *intensity* of the incident light (left)
- ii. *Number of electrons* ejected versus the *frequency* of the incident light (right)



**Question 3** **6 points****a. 2 points**

In the ground state of Ge (Germanium; $Z = 32$) atom, **name** the following orbitals:

i. Orbital containing the electron that requires the maximum energy to be removed.

ii. Orbital containing the electron that experiences the minimum effective nuclear charge.

b. 2 points

An unknown anion (A^-) has the following *excited* electronic configuration: $[Ar] 4s^2 3d^3 4p^1$

(Ar: Argon; $Z = 18$)

Give the **condensed ground state electronic configuration** of the atom, **A** and the cation, **A^{2+}** .
Show your work. (Assume all electronic transitions are possible)

Electronic Configuration of A:

Electronic Configuration of A^{2+} :

c. 2 points

In a single atom, determine how many electrons can have the following set of quantum numbers?

(i) $n = 5; l = 3$

In a single atom, determine how many electrons can have the following set of quantum numbers?

(i) $n = 3; l = 2; m_s = +1/2$

**Question 4** **8 points**

Rank the following from 1 to 4 in **increasing** order. Provide the ranking in the boxes provided.

Note: Atomic number (Z) are provided for relevant atoms

a) Increasing (lowest to highest) order of *ionization energy*: (1= lowest; 4 = highest)

Mg

Mg: Z = 12

Rb

Rb: Z = 37

Cs

Cs: Z = 55

K

K : Z = 19

b) Increasing (lowest to highest) order of *bond angle*: (1= lowest; 4 = highest)

H₂OCCl₄XeF₄

HCN

c) Increasing (smallest to largest) order of *ionic size* (1= smallest; 4 = largest)

S²⁻

S: Z = 16

O²⁻

O: Z = 8

Be²⁺

Be: Z = 4

Na⁺

Na: Z = 11

d) Increasing (lowest to highest) order of *bond energy* (1= lowest; 4 = highest)

P-F

P-Cl

P-I

P-Br

**Question 5** **12 points**

a. 9 points For the molecules HClO_4 and TeF_6 , draw the most stable *Lewis structure* showing appropriate shape and *dashed/wedged bonds*, based on VSEPR. Determine the **molecular geometry** around the central atom. For HClO_4 , label the *most polar bond*.



b. 3 points (Note: Lone pairs are not shown in structures below)

(i) For the three *resonance* structures (1, 2 and 3) shown below, answer the following question:



(I) Circle the *most contributing* resonance structure.

Structure 1

Structure 2

Structure 3

(ii) For the three structures given (4, 5, and 6) below, answer the following questions:



(I) Circle all the structures that represent *resonance structures* of each other:

Structure 4

Structure 5

Structure 6

(II) Among the resonance structures, circle the *most contributing* resonance structure.

Structure 4

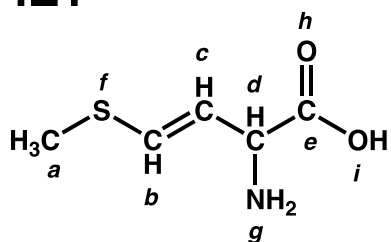
Structure 5

Structure 6

**Question 6** **6 points**

Using the data given below, calculate the **lattice energy** for SrO(s).

	ΔH° (kJ/mol)
$\text{Sr}(s) \longrightarrow \text{Sr}(g)$	+180
$\text{Sr}(g) \longrightarrow \text{Sr}^{2+}(g) + 2e^-$	+1468
$\text{O}_2(g) \longrightarrow 2\text{O}(g)$	+500
$\text{O}(g) + 2e^- \longrightarrow \text{O}^{2-}(g)$	+600
$\text{Sr}(s) + \frac{1}{2}\text{O}_2(g) \longrightarrow \text{SrO}(s)$	-550

**Question 7 10 points**

Consider the molecule given on the left, and answer questions (i) to (vi) using chemical bonding theories that we have learnt in class (valence bond theory, hybridization).

Note that the lone pairs are not shown in the structure:

i. Consider the carbon atoms marked in as ***a***, ***b***, ***c***, ***d***, and ***e*** in the molecule, and determine **hybridization** at each of the carbon centers. (2.5 points)

Atom	Hybridization
Carbon (<i>a</i>)	
Carbon (<i>b</i>)	
Carbon (<i>c</i>)	
Carbon (<i>d</i>)	
Carbon (<i>e</i>)	

ii. Now, consider the sulfur atom (S) marked as ***f***, nitrogen (N) atom (***g***), and oxygen(O) atoms (***h*** and ***i***), and determine their **hybridization**. (2 points)

Atom	Hybridization
Sulfur (<i>f</i>)	
Nitrogen (<i>g</i>)	
Oxygen (<i>h</i>)	
Oxygen (<i>i</i>)	

iii. **Describe** all the bonds formed by the carbon atoms ***a*** and ***b***, using sigma and pi designation as appropriate. (2 points)

Atom	Bonds formed (use sigma and pi as appropriate)
Carbon (<i>a</i>)	
Carbon (<i>b</i>)	

iv. How many total sp^3 -s sigma bonds are in the molecule? (1 point)

v. Determine **electronic** and **molecular geometry** at carbon (C) labeled ***e*** and sulfur (S) labeled ***f*** centres. (2 points)

Atom	Electronic Geometry	Molecular Geometry
Carbon (<i>e</i>)		
Sulfur (<i>f</i>)		

vi. Circle the value from the choices given below, that describes the **C-S-C bond angle at sulfur** (S) marked ***f***: (0.5 points)

>90° to <109.5° 109.5° >109.5 to <120° 120° >120° to <180°


Question 8 12 points

Using the **valence shell orbitals only** draw a **complete** molecular orbital (MO) diagram (with no s-p mixing) for the molecule **PS**, in the space provided below. **(6 points)**

Assume that 3s and 3p atomic orbitals will form molecular orbitals like those formed by the combination of 2s and 2p atomic orbitals that we learnt in class, and there is no s-p mixing.

Show **energy arrow**, **atomic orbitals** involved, **atom labels**, the **molecular orbitals** that are formed, **filled electrons** in the MO diagram etc. Make sure to label all the atomic and molecular orbitals.

Using the MO diagram above, answer the following questions:

- (i) Which one of the following will have the highest bond order? Circle the correct choice
 A) PS^{1-} B) PS C) PS^{1+}
- (ii) Which one of the following will have the longest bond length? Circle the correct choice
 A) PS^{1-} B) PS C) PS^{1+}
- (iii) Which of the following will be diamagnetic? Circle the correct choice.
 A) PS^{1-} B) PS C) PS^{1+}
- (iv) Draw Lewis dot structure of the molecule with the shortest bond length (based on the MO diagram) from among the three (PS^{1-} , PS, PS^{1+}), and indicate the bond order calculated from the Lewis dot structure. **(3 points)**

Lewis Dot Structure

Bond Order



AE6D1B62-C6EB-4217-858C-AF926A600C74

midterm-exam-93388

#444 Page 10 of 10

SCRAP PAGE

**CHEMISTRY 110 –Midterm Exam****Thursday November 7, 2024, 6:30 – 8:30 pm****Examiners: Pallavi Sirjoosingh and Ashok Kakkar**

First name (please write as legibly as possible within the boxes)

--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--

Last name

--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--

Student ID number

--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--

This booklet has 8 questions on 10 pages (The last page can be used as a scrap page).

The staple must NOT be removed from the question booklet. Your intact question booklet must be returned at the end of the exam.

A separate booklet contains a periodic table, and datasheet.

Write your answers directly under the questions in the space provided. Do not write on the QR codes or near the top edges (as it may be cut off).

Please write **legibly and dark enough** for the scan to be readable. For students who use English as a second language, dictionaries are permitted. Calculators are allowed.

DO NOT ASK QUESTIONS OF THE INVIGILATORS about interpretations of the exam questions. If you have a concern about a question, please summarize it at the top of this page. All concerns will be evaluated after the exam. GOOD LUCK!

**Question 1 8 points** (*Assume all transitions are possible*)

a. 6 points Calculate the **energy (in kJ/mol)** required to ionize (completely remove) the electron from the **n=3** level of **He⁺ ion** (He: Helium). Show your work including equations. (3 sig. figs.)

b. 2 points

(I) Select the largest (longest) wavelength from the three choices below:

- i. Wavelength of photon to *completely remove* an electron from *1s orbital of Hydrogen atom*
- ii. Wavelength of photon to *completely remove* an electron from *1s orbital of Boron atom (B)*
- iii. Wavelength of photon to *completely remove* an electron from *1s orbital of Boron cation (B⁺)*

i or ii or iii

(II) Select the smallest (shortest) wavelength from the three choices below (*for H atom*):

- i. Wavelength of photon *emitted* for an electron transitioning from *2p to 1s orbital*
- ii. Wavelength of photon *absorbed* for an electron transitioning from *2s to 4p orbital*
- iii. Wavelength of photon *absorbed* for an electron transitioning from *1s to 3p orbital*

i or ii or iii



Question 2 8 points



A Chem110 student plots the graph shown on the left, from data obtained for the kinetic energy (eV) of the ejected electron versus frequency (s^{-1}) of the incident light, for an unknown metal. ($1\text{eV} = 1.6022 \times 10^{-19} \text{ J}$)

Based on the plot, answer the questions **a to c** below:

a. 1 point On the plot above, label the **threshold frequency**.

b. 2 points What is the value of the work potential (in J) of the unknown metal?

c. 3 points What is the *longest* wavelength (in nm) required to eject an electron from this metal's surface? Show your work including equation(s) used. (3 sig figs)

d. 2 points In the space provided below, plot the following

(Assume frequency is *above* threshold frequency):

- Kinetic energy* of an ejected electron versus the *intensity* of the incident light (left)
- Number of electrons* ejected versus the *frequency* of the incident light (right)



**Question 3** **6 points****a. 2 points**

In the ground state of Ge (Germanium; $Z = 32$) atom, **name** the following orbitals:

i. Orbital containing the electron that requires the maximum energy to be removed.

ii. Orbital containing the electron that experiences the minimum effective nuclear charge.

b. 2 points

An unknown anion (A^-) has the following *excited* electronic configuration: $[Ar] 4s^2 3d^3 4p^1$

(Ar: Argon; $Z = 18$)

Give the **condensed ground state electronic configuration** of the atom, **A** and the cation, **A^{2+}** .
Show your work. (Assume all electronic transitions are possible)

Electronic Configuration of A:

Electronic Configuration of A^{2+} :

c. 2 points

In a single atom, determine how many electrons can have the following set of quantum numbers?

(i) $n = 5; l = 3$

In a single atom, determine how many electrons can have the following set of quantum numbers?

(i) $n = 3; l = 2; m_s = +1/2$

**Question 4** **8 points**

Rank the following from 1 to 4 in **increasing** order. Provide the ranking in the boxes provided.

Note: Atomic number (Z) are provided for relevant atoms

a) Increasing (lowest to highest) order of *ionization energy*: (1= lowest; 4 = highest)

Mg

Mg: Z = 12

Rb

Rb: Z = 37

Cs

Cs: Z = 55

K

K : Z = 19

b) Increasing (lowest to highest) order of *bond angle*: (1= lowest; 4 = highest)

H₂OCCl₄XeF₄

HCN

c) Increasing (smallest to largest) order of *ionic size* (1= smallest; 4 = largest)

S²⁻

S: Z = 16

O²⁻

O: Z = 8

Be²⁺

Be: Z = 4

Na⁺

Na: Z = 11

d) Increasing (lowest to highest) order of *bond energy* (1= lowest; 4 = highest)

P-F

P-Cl

P-I

P-Br

**Question 5** **12 points**

a. 9 points For the molecules HClO_4 and TeF_6 , draw the most stable *Lewis structure* showing appropriate shape and *dashed/wedged bonds*, based on VSEPR. Determine the **molecular geometry** around the central atom. For HClO_4 , label the *most polar bond*.



b. 3 points (Note: Lone pairs are not shown in structures below)

(i) For the three *resonance* structures (1, 2 and 3) shown below, answer the following question:



(I) Circle the *most contributing* resonance structure.

Structure 1

Structure 2

Structure 3

(ii) For the three structures given (4, 5, and 6) below, answer the following questions:



(I) Circle all the structures that represent *resonance structures* of each other:

Structure 4

Structure 5

Structure 6

(II) Among the resonance structures, circle the *most contributing* resonance structure.

Structure 4

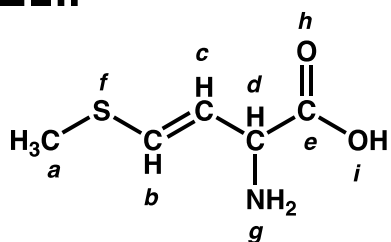
Structure 5

Structure 6

**Question 6** **6 points**

Using the data given below, calculate the **lattice energy** for SrO(s).

	ΔH° (kJ/mol)
$\text{Sr}(s) \longrightarrow \text{Sr}(g)$	+180
$\text{Sr}(g) \longrightarrow \text{Sr}^{2+}(g) + 2e^-$	+1468
$\text{O}_2(g) \longrightarrow 2\text{O}(g)$	+500
$\text{O}(g) + 2e^- \longrightarrow \text{O}^{2-}(g)$	+600
$\text{Sr}(s) + \frac{1}{2}\text{O}_2(g) \longrightarrow \text{SrO}(s)$	-550

**Question 7 10 points**

Consider the molecule given on the left, and answer questions (i) to (vi) using chemical bonding theories that we have learnt in class (valence bond theory, hybridization).

Note that the lone pairs are not shown in the structure:

i. Consider the carbon atoms marked in as ***a***, ***b***, ***c***, ***d***, and ***e*** in the molecule, and determine **hybridization** at each of the carbon centers. (2.5 points)

Atom	Hybridization
Carbon (<i>a</i>)	
Carbon (<i>b</i>)	
Carbon (<i>c</i>)	
Carbon (<i>d</i>)	
Carbon (<i>e</i>)	

ii. Now, consider the sulfur atom (S) marked as ***f***, nitrogen (N) atom (***g***), and oxygen(O) atoms (***h*** and ***i***), and determine their **hybridization**. (2 points)

Atom	Hybridization
Sulfur (<i>f</i>)	
Nitrogen (<i>g</i>)	
Oxygen (<i>h</i>)	
Oxygen (<i>i</i>)	

iii. **Describe** all the bonds formed by the carbon atoms ***a*** and ***b***, using sigma and pi designation as appropriate. (2 points)

Atom	Bonds formed (use sigma and pi as appropriate)
Carbon (<i>a</i>)	
Carbon (<i>b</i>)	

iv. How many total sp^3 -s sigma bonds are in the molecule? (1 point)

v. Determine **electronic** and **molecular geometry** at carbon (C) labeled ***e*** and sulfur (S) labeled ***f*** centres. (2 points)

Atom	Electronic Geometry	Molecular Geometry
Carbon (<i>e</i>)		
Sulfur (<i>f</i>)		

vi. Circle the value from the choices given below, that describes the **C-S-C bond angle at sulfur** (S) marked ***f***: (0.5 points)

>90° to <109.5° 109.5° >109.5 to <120° 120° >120° to <180°


Question 8 12 points

Using the **valence shell orbitals only** draw a **complete** molecular orbital (MO) diagram (with no s-p mixing) for the molecule **PS**, in the space provided below. **(6 points)**

Assume that 3s and 3p atomic orbitals will form molecular orbitals like those formed by the combination of 2s and 2p atomic orbitals that we learnt in class, and there is no s-p mixing.

Show **energy arrow**, **atomic orbitals** involved, **atom labels**, the **molecular orbitals** that are formed, **filled electrons** in the MO diagram etc. Make sure to label all the atomic and molecular orbitals.

Using the MO diagram above, answer the following questions:

- (i) Which one of the following will have the highest bond order? Circle the correct choice
 A) PS^{1-} B) PS C) PS^{1+}
- (ii) Which one of the following will have the longest bond length? Circle the correct choice
 A) PS^{1-} B) PS C) PS^{1+}
- (iii) Which of the following will be diamagnetic? Circle the correct choice.
 A) PS^{1-} B) PS C) PS^{1+}
- (iv) Draw Lewis dot structure of the molecule with the shortest bond length (based on the MO diagram) from among the three (PS^{1-} , PS, PS^{1+}), and indicate the bond order calculated from the Lewis dot structure. **(3 points)**

Lewis Dot Structure

Bond Order



4B0903AA-4450-46F9-8478-E5B1BB9EB4AB

midterm-exam-93388

#445 Page 10 of 10

SCRAP PAGE

**CHEMISTRY 110 –Midterm Exam****Thursday November 7, 2024, 6:30 – 8:30 pm****Examiners: Pallavi Sirjoosingh and Ashok Kakkar**

First name (please write as legibly as possible within the boxes)

--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--

Last name

--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--

Student ID number

--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--

This booklet has 8 questions on 10 pages (The last page can be used as a scrap page).

The staple must NOT be removed from the question booklet. Your intact question booklet must be returned at the end of the exam.

A separate booklet contains a periodic table, and datasheet.

Write your answers directly under the questions in the space provided. Do not write on the QR codes or near the top edges (as it may be cut off).

Please write **legibly and dark enough** for the scan to be readable. For students who use English as a second language, dictionaries are permitted. Calculators are allowed.

DO NOT ASK QUESTIONS OF THE INVIGILATORS about interpretations of the exam questions. If you have a concern about a question, please summarize it at the top of this page. All concerns will be evaluated after the exam. GOOD LUCK!

**Question 1 8 points** (*Assume all transitions are possible*)

a. 6 points Calculate the **energy (in kJ/mol)** required to ionize (completely remove) the electron from the **n=3** level of **He⁺ ion** (He: Helium). Show your work including equations. (3 sig. figs.)

b. 2 points

(I) Select the largest (longest) wavelength from the three choices below:

- i. Wavelength of photon to *completely remove* an electron from *1s orbital of Hydrogen atom*
- ii. Wavelength of photon to *completely remove* an electron from *1s orbital of Boron atom (B)*
- iii. Wavelength of photon to *completely remove* an electron from *1s orbital of Boron cation (B⁺)*

i or ii or iii

(II) Select the smallest (shortest) wavelength from the three choices below (*for H atom*):

- i. Wavelength of photon *emitted* for an electron transitioning from *2p to 1s orbital*
- ii. Wavelength of photon *absorbed* for an electron transitioning from *2s to 4p orbital*
- iii. Wavelength of photon *absorbed* for an electron transitioning from *1s to 3p orbital*

i or ii or iii



Question 2 8 points



A Chem110 student plots the graph shown on the left, from data obtained for the kinetic energy (eV) of the ejected electron versus frequency (s^{-1}) of the incident light, for an unknown metal. ($1\text{eV} = 1.6022 \times 10^{-19} \text{ J}$)

Based on the plot, answer the questions **a to c** below:

a. 1 point On the plot above, label the **threshold frequency**.

b. 2 points What is the value of the work potential (in J) of the unknown metal?

c. 3 points What is the *longest* wavelength (in nm) required to eject an electron from this metal's surface? Show your work including equation(s) used. (3 sig figs)

d. 2 points In the space provided below, plot the following

(Assume frequency is *above* threshold frequency):

- Kinetic energy* of an ejected electron versus the *intensity* of the incident light (left)
- Number of electrons* ejected versus the *frequency* of the incident light (right)



**Question 3** **6 points****a. 2 points**

In the ground state of Ge (Germanium; $Z = 32$) atom, **name** the following orbitals:

i. Orbital containing the electron that requires the maximum energy to be removed.

ii. Orbital containing the electron that experiences the minimum effective nuclear charge.

b. 2 points

An unknown anion (A^-) has the following *excited* electronic configuration: $[Ar] 4s^2 3d^3 4p^1$

(Ar: Argon; $Z = 18$)

Give the **condensed ground state electronic configuration** of the atom, **A** and the cation, **A^{2+}** .
Show your work. (Assume all electronic transitions are possible)

Electronic Configuration of A:

Electronic Configuration of A^{2+} :

c. 2 points

In a single atom, determine how many electrons can have the following set of quantum numbers?

(i) $n = 5; l = 3$

In a single atom, determine how many electrons can have the following set of quantum numbers?

(i) $n = 3; l = 2; m_s = +1/2$

**Question 4** **8 points**

Rank the following from 1 to 4 in **increasing** order. Provide the ranking in the boxes provided.

Note: Atomic number (Z) are provided for relevant atoms

a) Increasing (lowest to highest) order of *ionization energy*: (1= lowest; 4 = highest)

Mg

Mg: Z = 12

Rb

Rb: Z = 37

Cs

Cs: Z = 55

K

K : Z = 19

b) Increasing (lowest to highest) order of *bond angle*: (1= lowest; 4 = highest)

H₂OCCl₄XeF₄

HCN

c) Increasing (smallest to largest) order of *ionic size* (1= smallest; 4 = largest)

S²⁻

S: Z = 16

O²⁻

O: Z = 8

Be²⁺

Be: Z = 4

Na⁺

Na: Z = 11

d) Increasing (lowest to highest) order of *bond energy* (1= lowest; 4 = highest)

P-F

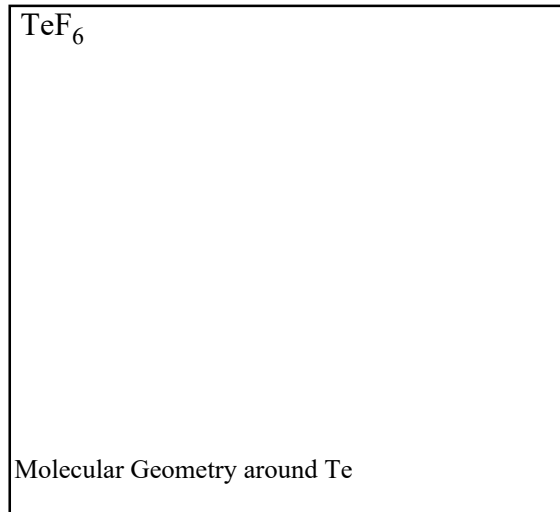
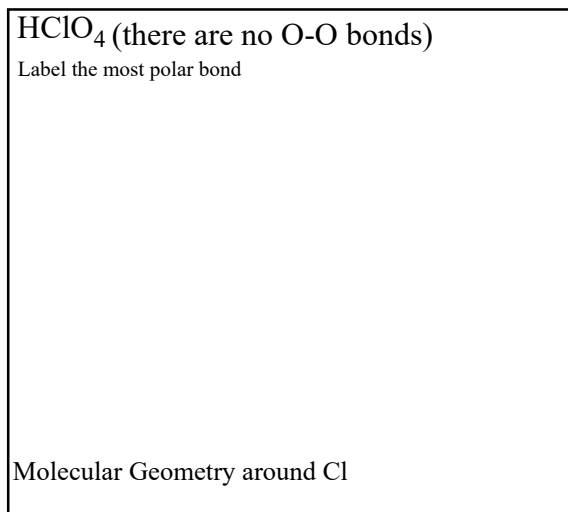
P-Cl

P-I

P-Br

**Question 5** **12 points**

a. 9 points For the molecules HClO_4 and TeF_6 , draw the most stable *Lewis structure* showing appropriate shape and *dashed/wedged bonds*, based on VSEPR. Determine the **molecular geometry** around the central atom. For HClO_4 , label the *most polar bond*.



b. 3 points (Note: Lone pairs are not shown in structures below)

(i) For the three *resonance* structures (1, 2 and 3) shown below, answer the following question:



(I) Circle the *most contributing* resonance structure.

Structure 1

Structure 2

Structure 3

(ii) For the three structures given (4, 5, and 6) below, answer the following questions:



(I) Circle all the structures that represent *resonance structures* of each other:

Structure 4

Structure 5

Structure 6

(II) Among the resonance structures, circle the *most contributing* resonance structure.

Structure 4

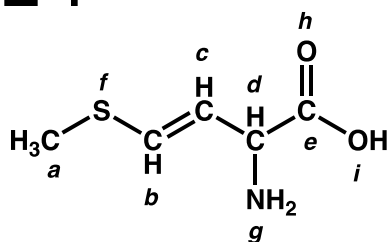
Structure 5

Structure 6

**Question 6** **6 points**

Using the data given below, calculate the **lattice energy** for SrO(s).

	ΔH° (kJ/mol)
$\text{Sr}(s) \longrightarrow \text{Sr}(g)$	+180
$\text{Sr}(g) \longrightarrow \text{Sr}^{2+}(g) + 2e^-$	+1468
$\text{O}_2(g) \longrightarrow 2\text{O}(g)$	+500
$\text{O}(g) + 2e^- \longrightarrow \text{O}^{2-}(g)$	+600
$\text{Sr}(s) + \frac{1}{2}\text{O}_2(g) \longrightarrow \text{SrO}(s)$	-550

**Question 7 10 points**

Consider the molecule given on the left, and answer questions (i) to (vi) using chemical bonding theories that we have learnt in class (valence bond theory, hybridization).

Note that the lone pairs are not shown in the structure:

i. Consider the carbon atoms marked in as ***a***, ***b***, ***c***, ***d***, and ***e*** in the molecule, and determine **hybridization** at each of the carbon centers. (2.5 points)

Atom	Hybridization
Carbon (<i>a</i>)	
Carbon (<i>b</i>)	
Carbon (<i>c</i>)	
Carbon (<i>d</i>)	
Carbon (<i>e</i>)	

ii. Now, consider the sulfur atom (S) marked as ***f***, nitrogen (N) atom (***g***), and oxygen(O) atoms (***h*** and ***i***), and determine their **hybridization**. (2 points)

Atom	Hybridization
Sulfur (<i>f</i>)	
Nitrogen (<i>g</i>)	
Oxygen (<i>h</i>)	
Oxygen (<i>i</i>)	

iii. **Describe** all the bonds formed by the carbon atoms ***a*** and ***b***, using sigma and pi designation as appropriate. (2 points)

Atom	Bonds formed (use sigma and pi as appropriate)
Carbon (<i>a</i>)	
Carbon (<i>b</i>)	

iv. How many total sp^3 -s sigma bonds are in the molecule? (1 point)

v. Determine **electronic** and **molecular geometry** at carbon (C) labeled ***e*** and sulfur (S) labeled ***f*** centres. (2 points)

Atom	Electronic Geometry	Molecular Geometry
Carbon (<i>e</i>)		
Sulfur (<i>f</i>)		

vi. Circle the value from the choices given below, that describes the **C-S-C bond angle at sulfur** (S) marked ***f***: (0.5 points)

>90° to <109.5°

109.5°

>109.5 to <120°

120°

>120° to <180°

**Question 8 12 points**

Using the **valence shell orbitals only** draw a **complete** molecular orbital (MO) diagram (with no s-p mixing) for the molecule **PS**, in the space provided below. **(6 points)**

Assume that 3s and 3p atomic orbitals will form molecular orbitals like those formed by the combination of 2s and 2p atomic orbitals that we learnt in class, and there is no s-p mixing.

Show **energy arrow**, **atomic orbitals** involved, **atom labels**, the **molecular orbitals** that are formed, **filled electrons** in the MO diagram etc. Make sure to label all the atomic and molecular orbitals.

Using the MO diagram above, answer the following questions:

- (i) Which one of the following will have the highest bond order? Circle the correct choice
A) PS^{1-} B) PS C) PS^{1+}
- (ii) Which one of the following will have the longest bond length? Circle the correct choice
A) PS^{1-} B) PS C) PS^{1+}
- (iii) Which of the following will be diamagnetic? Circle the correct choice.
A) PS^{1-} B) PS C) PS^{1+}
- (iv) Draw Lewis dot structure of the molecule with the shortest bond length (based on the MO diagram) from among the three (PS^{1-} , PS, PS^{1+}), and indicate the bond order calculated from the Lewis dot structure. **(3 points)**

Lewis Dot Structure

Bond Order



50B087DA-7748-4C02-98DB-B76E5E0F4E48

midterm-exam-93388

#446 Page 10 of 10

SCRAP PAGE

**CHEMISTRY 110 –Midterm Exam****Thursday November 7, 2024, 6:30 – 8:30 pm**

Examiners: Pallavi Sirjoosingh and Ashok Kakkar

First name (please write as legibly as possible within the boxes)

--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--

Last name

--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--

Student ID number

--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--

This booklet has 8 questions on 10 pages (The last page can be used as a scrap page).

The staple must NOT be removed from the question booklet. Your intact question booklet must be returned at the end of the exam.

A separate booklet contains a periodic table, and datasheet.

Write your answers directly under the questions in the space provided. Do not write on the QR codes or near the top edges (as it may be cut off).

Please write **legibly and dark enough** for the scan to be readable. For students who use English as a second language, dictionaries are permitted. Calculators are allowed.

DO NOT ASK QUESTIONS OF THE INVIGILATORS about interpretations of the exam questions. If you have a concern about a question, please summarize it at the top of this page. All concerns will be evaluated after the exam. GOOD LUCK!

**Question 1 8 points** (*Assume all transitions are possible*)

a. 6 points Calculate the **energy (in kJ/mol)** required to ionize (completely remove) the electron from the **$n=3$** level of **He^+ ion** (He: Helium). Show your work including equations. (3 sig. figs.)

b. 2 points

(I) Select the largest (longest) wavelength from the three choices below:

- i. Wavelength of photon to *completely remove* an electron from *1s orbital of Hydrogen atom*
- ii. Wavelength of photon to *completely remove* an electron from *1s orbital of Boron atom (B)*
- iii. Wavelength of photon to *completely remove* an electron from *1s orbital of Boron cation (B^+)*

i or ii or iii

(II) Select the smallest (shortest) wavelength from the three choices below (*for H atom*):

- i. Wavelength of photon *emitted* for an electron transitioning from *2p to 1s orbital*
- ii. Wavelength of photon *absorbed* for an electron transitioning from *2s to 4p orbital*
- iii. Wavelength of photon *absorbed* for an electron transitioning from *1s to 3p orbital*

i or ii or iii



Question 2 8 points



A Chem110 student plots the graph shown on the left, from data obtained for the kinetic energy (eV) of the ejected electron versus frequency (s^{-1}) of the incident light, for an unknown metal. ($1\text{eV} = 1.6022 \times 10^{-19} \text{ J}$)

Based on the plot, answer the questions **a to c** below:

a. 1 point On the plot above, label the **threshold frequency**.

b. 2 points What is the value of the work potential (in J) of the unknown metal?

c. 3 points What is the *longest* wavelength (in nm) required to eject an electron from this metal's surface? Show your work including equation(s) used. (3 sig figs)

d. 2 points In the space provided below, plot the following

(Assume frequency is *above* threshold frequency):

- Kinetic energy* of an ejected electron versus the *intensity* of the incident light (left)
- Number of electrons* ejected versus the *frequency* of the incident light (right)



**Question 3** **6 points****a. 2 points**

In the ground state of Ge (Germanium; $Z = 32$) atom, **name** the following orbitals:

i. Orbital containing the electron that requires the maximum energy to be removed.

ii. Orbital containing the electron that experiences the minimum effective nuclear charge.

b. 2 points

An unknown anion (A^-) has the following *excited* electronic configuration: $[Ar] 4s^2 3d^3 4p^1$

(Ar: Argon; $Z = 18$)

Give the **condensed ground state electronic configuration** of the atom, **A** and the cation, **A^{2+}** .
Show your work. (Assume all electronic transitions are possible)

Electronic Configuration of A:

Electronic Configuration of A^{2+} :

c. 2 points

In a single atom, determine how many electrons can have the following set of quantum numbers?

(i) $n = 5; l = 3$

In a single atom, determine how many electrons can have the following set of quantum numbers?

(i) $n = 3; l = 2; m_s = +1/2$

**Question 4** **8 points**

Rank the following from 1 to 4 in **increasing** order. Provide the ranking in the boxes provided.

Note: Atomic number (Z) are provided for relevant atoms

a) Increasing (lowest to highest) order of *ionization energy*: (1= lowest; 4 = highest)

Mg

Mg: Z = 12

Rb

Rb: Z = 37

Cs

Cs: Z = 55

K

K : Z = 19

b) Increasing (lowest to highest) order of *bond angle*: (1= lowest; 4 = highest)

H₂OCCl₄XeF₄

HCN

c) Increasing (smallest to largest) order of *ionic size* (1= smallest; 4 = largest)

S²⁻

S: Z = 16

O²⁻

O: Z = 8

Be²⁺

Be: Z = 4

Na⁺

Na: Z = 11

d) Increasing (lowest to highest) order of *bond energy* (1= lowest; 4 = highest)

P-F

P-Cl

P-I

P-Br

**Question 5** **12 points**

a. 9 points For the molecules HClO_4 and TeF_6 , draw the most stable *Lewis structure* showing appropriate shape and *dashed/wedged bonds*, based on VSEPR. Determine the **molecular geometry** around the central atom. For HClO_4 , label the *most polar bond*.



b. 3 points (Note: Lone pairs are not shown in structures below)

(i) For the three *resonance* structures (1, 2 and 3) shown below, answer the following question:



(I) Circle the *most contributing* resonance structure.

Structure 1

Structure 2

Structure 3

(ii) For the three structures given (4, 5, and 6) below, answer the following questions:



(I) Circle all the structures that represent *resonance structures* of each other:

Structure 4

Structure 5

Structure 6

(II) Among the resonance structures, circle the *most contributing* resonance structure.

Structure 4

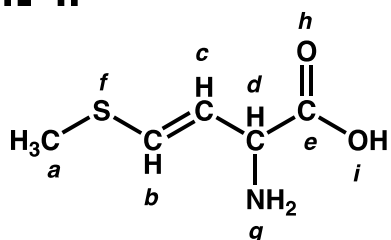
Structure 5

Structure 6

**Question 6** **6 points**

Using the data given below, calculate the **lattice energy** for SrO(s).

	ΔH° (kJ/mol)
$\text{Sr}(s) \longrightarrow \text{Sr}(g)$	+180
$\text{Sr}(g) \longrightarrow \text{Sr}^{2+}(g) + 2e^-$	+1468
$\text{O}_2(g) \longrightarrow 2\text{O}(g)$	+500
$\text{O}(g) + 2e^- \longrightarrow \text{O}^{2-}(g)$	+600
$\text{Sr}(s) + \frac{1}{2}\text{O}_2(g) \longrightarrow \text{SrO}(s)$	-550

**Question 7 10 points**

Consider the molecule given on the left, and answer questions (i) to (vi) using chemical bonding theories that we have learnt in class (valence bond theory, hybridization).

Note that the lone pairs are not shown in the structure:

i. Consider the carbon atoms marked in as ***a***, ***b***, ***c***, ***d***, and ***e*** in the molecule, and determine **hybridization** at each of the carbon centers. (2.5 points)

Atom	Hybridization
Carbon (<i>a</i>)	
Carbon (<i>b</i>)	
Carbon (<i>c</i>)	
Carbon (<i>d</i>)	
Carbon (<i>e</i>)	

ii. Now, consider the sulfur atom (S) marked as ***f***, nitrogen (N) atom (***g***), and oxygen(O) atoms (***h*** and ***i***), and determine their **hybridization**. (2 points)

Atom	Hybridization
Sulfur (<i>f</i>)	
Nitrogen (<i>g</i>)	
Oxygen (<i>h</i>)	
Oxygen (<i>i</i>)	

iii. **Describe** all the bonds formed by the carbon atoms ***a*** and ***b***, using sigma and pi designation as appropriate. (2 points)

Atom	Bonds formed (use sigma and pi as appropriate)
Carbon (<i>a</i>)	
Carbon (<i>b</i>)	

iv. How many total sp^3 -s sigma bonds are in the molecule? (1 point)

v. Determine **electronic** and **molecular geometry** at carbon (C) labeled ***e*** and sulfur (S) labeled ***f*** centres. (2 points)

Atom	Electronic Geometry	Molecular Geometry
Carbon (<i>e</i>)		
Sulfur (<i>f</i>)		

vi. Circle the value from the choices given below, that describes the **C-S-C bond angle at sulfur** (S) marked ***f***: (0.5 points)

>90° to <109.5°

109.5°

>109.5 to <120°

120°

>120° to <180°


Question 8 12 points

Using the **valence shell orbitals only** draw a **complete** molecular orbital (MO) diagram (with no s-p mixing) for the molecule **PS**, in the space provided below. **(6 points)**

Assume that 3s and 3p atomic orbitals will form molecular orbitals like those formed by the combination of 2s and 2p atomic orbitals that we learnt in class, and there is no s-p mixing.

Show **energy arrow**, **atomic orbitals** involved, **atom labels**, the **molecular orbitals** that are formed, **filled electrons** in the MO diagram etc. Make sure to **label all the atomic and molecular orbitals**.

Using the MO diagram above, answer the following questions:

- (i) Which one of the following will have the highest bond order? Circle the correct choice
 A) PS^{1-} B) PS C) PS^{1+}
- (ii) Which one of the following will have the longest bond length? Circle the correct choice
 A) PS^{1-} B) PS C) PS^{1+}
- (iii) Which of the following will be diamagnetic? Circle the correct choice.
 A) PS^{1-} B) PS C) PS^{1+}
- (iv) Draw Lewis dot structure of the molecule with the shortest bond length (based on the MO diagram) from among the three (PS^{1-} , PS, PS^{1+}), and indicate the bond order calculated from the Lewis dot structure. **(3 points)**

Lewis Dot Structure

Bond Order



24BBEA20-EE27-4294-B7B7-BC5023623AB5

midterm-exam-93388

#447 Page 10 of 10

SCRAP PAGE

**CHEMISTRY 110 –Midterm Exam****Thursday November 7, 2024, 6:30 – 8:30 pm**

Examiners: Pallavi Sirjoosingh and Ashok Kakkar

First name (please write as legibly as possible within the boxes)

--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--

Last name

--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--

Student ID number

--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--

This booklet has 8 questions on 10 pages (The last page can be used as a scrap page).

The staple must NOT be removed from the question booklet. Your intact question booklet must be returned at the end of the exam.

A separate booklet contains a periodic table, and datasheet.

Write your answers directly under the questions in the space provided. Do not write on the QR codes or near the top edges (as it may be cut off).

Please write **legibly and dark enough** for the scan to be readable. For students who use English as a second language, dictionaries are permitted. Calculators are allowed.

DO NOT ASK QUESTIONS OF THE INVIGILATORS about interpretations of the exam questions. If you have a concern about a question, please summarize it at the top of this page. All concerns will be evaluated after the exam. GOOD LUCK!

**Question 1 8 points** (*Assume all transitions are possible*)

a. 6 points Calculate the **energy (in kJ/mol)** required to ionize (completely remove) the electron from the **n=3** level of **He⁺ ion** (He: Helium). Show your work including equations. (3 sig. figs.)

b. 2 points

(I) Select the largest (longest) wavelength from the three choices below:

- i. Wavelength of photon to *completely remove* an electron from *1s orbital of Hydrogen atom*
- ii. Wavelength of photon to *completely remove* an electron from *1s orbital of Boron atom (B)*
- iii. Wavelength of photon to *completely remove* an electron from *1s orbital of Boron cation (B⁺)*

i or ii or iii

(II) Select the smallest (shortest) wavelength from the three choices below (*for H atom*):

- i. Wavelength of photon *emitted* for an electron transitioning from *2p to 1s orbital*
- ii. Wavelength of photon *absorbed* for an electron transitioning from *2s to 4p orbital*
- iii. Wavelength of photon *absorbed* for an electron transitioning from *1s to 3p orbital*

i or ii or iii



Question 2 8 points



A Chem110 student plots the graph shown on the left, from data obtained for the kinetic energy (eV) of the ejected electron versus frequency (s^{-1}) of the incident light, for an unknown metal. ($1\text{eV} = 1.6022 \times 10^{-19} \text{ J}$)

Based on the plot, answer the questions **a to c** below:

a. 1 point On the plot above, label the **threshold frequency**.

b. 2 points What is the value of the work potential (in J) of the unknown metal?

c. 3 points What is the *longest* wavelength (in nm) required to eject an electron from this metal's surface? Show your work including equation(s) used. (3 sig figs)

d. 2 points In the space provided below, plot the following

(Assume frequency is *above* threshold frequency):

- Kinetic energy* of an ejected electron versus the *intensity* of the incident light (left)
- Number of electrons* ejected versus the *frequency* of the incident light (right)



**Question 3** **6 points****a. 2 points**

In the ground state of Ge (Germanium; $Z = 32$) atom, **name** the following orbitals:

i. Orbital containing the electron that requires the maximum energy to be removed.

ii. Orbital containing the electron that experiences the minimum effective nuclear charge.

b. 2 points

An unknown anion (A^-) has the following *excited* electronic configuration: $[Ar] 4s^2 3d^3 4p^1$

(Ar: Argon; $Z = 18$)

Give the **condensed ground state electronic configuration** of the atom, **A** and the cation, **A^{2+}** .
Show your work. (Assume all electronic transitions are possible)

Electronic Configuration of A:

Electronic Configuration of A^{2+} :

c. 2 points

In a single atom, determine how many electrons can have the following set of quantum numbers?

(i) $n = 5; l = 3$

In a single atom, determine how many electrons can have the following set of quantum numbers?

(i) $n = 3; l = 2; m_s = +1/2$

**Question 4** **8 points**

Rank the following from 1 to 4 in **increasing** order. Provide the ranking in the boxes provided.

Note: Atomic number (Z) are provided for relevant atoms

a) Increasing (lowest to highest) order of *ionization energy*: (1= lowest; 4 = highest)

Mg

Mg: Z = 12

Rb

Rb: Z = 37

Cs

Cs: Z = 55

K

K : Z = 19

b) Increasing (lowest to highest) order of *bond angle*: (1= lowest; 4 = highest)

H₂OCCl₄XeF₄

HCN

c) Increasing (smallest to largest) order of *ionic size* (1= smallest; 4 = largest)

S²⁻

S: Z = 16

O²⁻

O: Z = 8

Be²⁺

Be: Z = 4

Na⁺

Na: Z = 11

d) Increasing (lowest to highest) order of *bond energy* (1= lowest; 4 = highest)

P-F

P-Cl

P-I

P-Br

**Question 5** **12 points**

a. 9 points For the molecules HClO_4 and TeF_6 , draw the most stable *Lewis structure* showing appropriate shape and *dashed/wedged bonds*, based on VSEPR. Determine the **molecular geometry** around the central atom. For HClO_4 , label the *most polar bond*.



b. 3 points (Note: Lone pairs are not shown in structures below)

(i) For the three *resonance* structures (1, 2 and 3) shown below, answer the following question:



(I) Circle the *most contributing* resonance structure.

Structure 1

Structure 2

Structure 3

(ii) For the three structures given (4, 5, and 6) below, answer the following questions:



(I) Circle all the structures that represent *resonance structures* of each other:

Structure 4

Structure 5

Structure 6

(II) Among the resonance structures, circle the *most contributing* resonance structure.

Structure 4

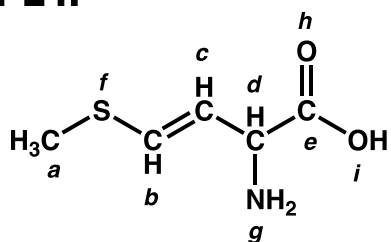
Structure 5

Structure 6

**Question 6** **6 points**

Using the data given below, calculate the **lattice energy** for SrO(s).

	ΔH° (kJ/mol)
$\text{Sr}(s) \longrightarrow \text{Sr}(g)$	+180
$\text{Sr}(g) \longrightarrow \text{Sr}^{2+}(g) + 2e^-$	+1468
$\text{O}_2(g) \longrightarrow 2\text{O}(g)$	+500
$\text{O}(g) + 2e^- \longrightarrow \text{O}^{2-}(g)$	+600
$\text{Sr}(s) + \frac{1}{2}\text{O}_2(g) \longrightarrow \text{SrO}(s)$	-550

**Question 7 10 points**

Consider the molecule given on the left, and answer questions (i) to (vi) using chemical bonding theories that we have learnt in class (valence bond theory, hybridization).

Note that the lone pairs are not shown in the structure:

i. Consider the carbon atoms marked in as ***a***, ***b***, ***c***, ***d***, and ***e*** in the molecule, and determine **hybridization** at each of the carbon centers. (2.5 points)

Atom	Hybridization
Carbon (<i>a</i>)	
Carbon (<i>b</i>)	
Carbon (<i>c</i>)	
Carbon (<i>d</i>)	
Carbon (<i>e</i>)	

ii. Now, consider the sulfur atom (S) marked as ***f***, nitrogen (N) atom (***g***), and oxygen(O) atoms (***h*** and ***i***), and determine their **hybridization**. (2 points)

Atom	Hybridization
Sulfur (<i>f</i>)	
Nitrogen (<i>g</i>)	
Oxygen (<i>h</i>)	
Oxygen (<i>i</i>)	

iii. **Describe** all the bonds formed by the carbon atoms ***a*** and ***b***, using sigma and pi designation as appropriate. (2 points)

Atom	Bonds formed (use sigma and pi as appropriate)
Carbon (<i>a</i>)	
Carbon (<i>b</i>)	

iv. How many total sp^3 -s sigma bonds are in the molecule? (1 point)

v. Determine **electronic** and **molecular geometry** at carbon (C) labeled ***e*** and sulfur (S) labeled ***f*** centres. (2 points)

Atom	Electronic Geometry	Molecular Geometry
Carbon (<i>e</i>)		
Sulfur (<i>f</i>)		

vi. Circle the value from the choices given below, that describes the **C-S-C bond angle at sulfur** (S) marked ***f***: (0.5 points)

$>90^\circ$ to $<109.5^\circ$ 109.5° >109.5 to $<120^\circ$ 120° $>120^\circ$ to $<180^\circ$

**Question 8 12 points**

Using the **valence shell orbitals only** draw a **complete** molecular orbital (MO) diagram (with no s-p mixing) for the molecule **PS**, in the space provided below. **(6 points)**

Assume that 3s and 3p atomic orbitals will form molecular orbitals like those formed by the combination of 2s and 2p atomic orbitals that we learnt in class, and there is no s-p mixing.

Show **energy arrow**, **atomic orbitals** involved, **atom labels**, the **molecular orbitals** that are formed, **filled electrons** in the MO diagram etc. Make sure to label all the atomic and molecular orbitals.

Using the MO diagram above, answer the following questions:

- (i) Which one of the following will have the highest bond order? Circle the correct choice
A) PS^{1-} B) PS C) PS^{1+}
- (ii) Which one of the following will have the longest bond length? Circle the correct choice
A) PS^{1-} B) PS C) PS^{1+}
- (iii) Which of the following will be diamagnetic? Circle the correct choice.
A) PS^{1-} B) PS C) PS^{1+}
- (iv) Draw Lewis dot structure of the molecule with the shortest bond length (based on the MO diagram) from among the three (PS^{1-} , PS, PS^{1+}), and indicate the bond order calculated from the Lewis dot structure. **(3 points)**

Lewis Dot Structure

Bond Order



3561244E-0DD0-4EF8-B2F7-6CF0915BA565

midterm-exam-93388

#448 Page 10 of 10

SCRAP PAGE

**CHEMISTRY 110 –Midterm Exam****Thursday November 7, 2024, 6:30 – 8:30 pm**

Examiners: Pallavi Sirjoosingh and Ashok Kakkar

First name (please write as legibly as possible within the boxes)

--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--

Last name

--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--

Student ID number

--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--

This booklet has 8 questions on 10 pages (The last page can be used as a scrap page).

The staple must NOT be removed from the question booklet. Your intact question booklet must be returned at the end of the exam.

A separate booklet contains a periodic table, and datasheet.

Write your answers directly under the questions in the space provided. Do not write on the QR codes or near the top edges (as it may be cut off).

Please write **legibly and dark enough** for the scan to be readable. For students who use English as a second language, dictionaries are permitted. Calculators are allowed.

DO NOT ASK QUESTIONS OF THE INVIGILATORS about interpretations of the exam questions. If you have a concern about a question, please summarize it at the top of this page. All concerns will be evaluated after the exam. GOOD LUCK!

**Question 1 8 points** (*Assume all transitions are possible*)

a. 6 points Calculate the **energy (in kJ/mol)** required to ionize (completely remove) the electron from the **$n=3$** level of **He^+ ion** (He: Helium). Show your work including equations. (3 sig. figs.)

b. 2 points

(I) Select the largest (longest) wavelength from the three choices below:

- i. Wavelength of photon to *completely remove* an electron from *1s orbital of Hydrogen atom*
- ii. Wavelength of photon to *completely remove* an electron from *1s orbital of Boron atom (B)*
- iii. Wavelength of photon to *completely remove* an electron from *1s orbital of Boron cation (B^+)*

i or ii or iii

(II) Select the smallest (shortest) wavelength from the three choices below (*for H atom*):

- i. Wavelength of photon *emitted* for an electron transitioning from *2p to 1s orbital*
- ii. Wavelength of photon *absorbed* for an electron transitioning from *2s to 4p orbital*
- iii. Wavelength of photon *absorbed* for an electron transitioning from *1s to 3p orbital*

i or ii or iii



Question 2 8 points



A Chem110 student plots the graph shown on the left, from data obtained for the kinetic energy (eV) of the ejected electron versus frequency (s^{-1}) of the incident light, for an unknown metal. ($1\text{eV} = 1.6022 \times 10^{-19} \text{ J}$)

Based on the plot, answer the questions **a to c** below:

a. 1 point On the plot above, label the **threshold frequency**.

b. 2 points What is the value of the work potential (in J) of the unknown metal?

c. 3 points What is the *longest* wavelength (in nm) required to eject an electron from this metal's surface? Show your work including equation(s) used. (3 sig figs)

d. 2 points In the space provided below, plot the following

(Assume frequency is *above* threshold frequency):

- i. *Kinetic energy* of an ejected electron versus the *intensity* of the incident light (left)
- ii. *Number of electrons* ejected versus the *frequency* of the incident light (right)



**Question 3** **6 points****a. 2 points**

In the ground state of Ge (Germanium; $Z = 32$) atom, **name** the following orbitals:

i. Orbital containing the electron that requires the maximum energy to be removed.

ii. Orbital containing the electron that experiences the minimum effective nuclear charge.

b. 2 points

An unknown anion (A^-) has the following *excited* electronic configuration: $[Ar] 4s^2 3d^3 4p^1$

(Ar: Argon; $Z = 18$)

Give the **condensed ground state electronic configuration** of the atom, **A** and the cation, **A^{2+}** .
Show your work. (Assume all electronic transitions are possible)

Electronic Configuration of A:

Electronic Configuration of A^{2+} :

c. 2 points

In a single atom, determine how many electrons can have the following set of quantum numbers?

(i) $n = 5; l = 3$

In a single atom, determine how many electrons can have the following set of quantum numbers?

(i) $n = 3; l = 2; m_s = +1/2$

**Question 4** **8 points**

Rank the following from 1 to 4 in **increasing** order. Provide the ranking in the boxes provided.

Note: Atomic number (Z) are provided for relevant atoms

a) Increasing (lowest to highest) order of *ionization energy*: (1= lowest; 4 = highest)

Mg

Mg: Z = 12

Rb

Rb: Z = 37

Cs

Cs: Z = 55

K

K : Z = 19

b) Increasing (lowest to highest) order of *bond angle*: (1= lowest; 4 = highest)

H₂OCCl₄XeF₄

HCN

c) Increasing (smallest to largest) order of *ionic size* (1= smallest; 4 = largest)

S²⁻

S: Z = 16

O²⁻

O: Z = 8

Be²⁺

Be: Z = 4

Na⁺

Na: Z = 11

d) Increasing (lowest to highest) order of *bond energy* (1= lowest; 4 = highest)

P-F

P-Cl

P-I

P-Br

**Question 5** **12 points**

a. 9 points For the molecules HClO_4 and TeF_6 , draw the most stable *Lewis structure* showing appropriate shape and *dashed/wedged bonds*, based on VSEPR. Determine the **molecular geometry** around the central atom. For HClO_4 , label the *most polar bond*.



b. 3 points (Note: Lone pairs are not shown in structures below)

(i) For the three *resonance* structures (1, 2 and 3) shown below, answer the following question:



(I) Circle the *most contributing* resonance structure.

Structure 1

Structure 2

Structure 3

(ii) For the three structures given (4, 5, and 6) below, answer the following questions:



(I) Circle all the structures that represent *resonance structures* of each other:

Structure 4

Structure 5

Structure 6

(II) Among the resonance structures, circle the *most contributing* resonance structure.

Structure 4

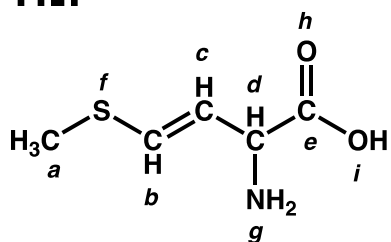
Structure 5

Structure 6

**Question 6** **6 points**

Using the data given below, calculate the **lattice energy** for SrO(s).

	ΔH° (kJ/mol)
$\text{Sr}(s) \longrightarrow \text{Sr}(g)$	+180
$\text{Sr}(g) \longrightarrow \text{Sr}^{2+}(g) + 2e^-$	+1468
$\text{O}_2(g) \longrightarrow 2\text{O}(g)$	+500
$\text{O}(g) + 2e^- \longrightarrow \text{O}^{2-}(g)$	+600
$\text{Sr}(s) + \frac{1}{2}\text{O}_2(g) \longrightarrow \text{SrO}(s)$	-550

**Question 7 10 points**

Consider the molecule given on the left, and answer questions (i) to (vi) using chemical bonding theories that we have learnt in class (valence bond theory, hybridization).

Note that the lone pairs are not shown in the structure:

i. Consider the carbon atoms marked in as ***a***, ***b***, ***c***, ***d***, and ***e*** in the molecule, and determine **hybridization** at each of the carbon centers. (2.5 points)

Atom	Hybridization
Carbon (<i>a</i>)	
Carbon (<i>b</i>)	
Carbon (<i>c</i>)	
Carbon (<i>d</i>)	
Carbon (<i>e</i>)	

ii. Now, consider the sulfur atom (S) marked as ***f***, nitrogen (N) atom (***g***), and oxygen(O) atoms (***h*** and ***i***), and determine their **hybridization**. (2 points)

Atom	Hybridization
Sulfur (<i>f</i>)	
Nitrogen (<i>g</i>)	
Oxygen (<i>h</i>)	
Oxygen (<i>i</i>)	

iii. **Describe** all the bonds formed by the carbon atoms ***a*** and ***b***, using sigma and pi designation as appropriate. (2 points)

Atom	Bonds formed (use sigma and pi as appropriate)
Carbon (<i>a</i>)	
Carbon (<i>b</i>)	

iv. How many total sp^3 -s sigma bonds are in the molecule? (1 point)

v. Determine **electronic** and **molecular geometry** at carbon (C) labeled ***e*** and sulfur (S) labeled ***f*** centres. (2 points)

Atom	Electronic Geometry	Molecular Geometry
Carbon (<i>e</i>)		
Sulfur (<i>f</i>)		

vi. Circle the value from the choices given below, that describes the **C-S-C bond angle at sulfur** (S) marked ***f***: (0.5 points)

>90° to <109.5° 109.5° >109.5 to <120° 120° >120° to <180°



Question 8 12 points

Using the **valence shell orbitals only** draw a **complete** molecular orbital (MO) diagram (with no s-p mixing) for the molecule **PS**, in the space provided below. **(6 points)**

Assume that 3s and 3p atomic orbitals will form molecular orbitals like those formed by the combination of 2s and 2p atomic orbitals that we learnt in class, and there is no s-p mixing.

Show **energy arrow**, **atomic orbitals** involved, **atom labels**, the **molecular orbitals** that are formed, **filled electrons** in the MO diagram etc. Make sure to label all the atomic and molecular orbitals.

Using the MO diagram above, answer the following questions:

- (i) Which one of the following will have the highest bond order? Circle the correct choice
 A) PS^{1-} B) PS C) PS^{1+}
- (ii) Which one of the following will have the longest bond length? Circle the correct choice
 A) PS^{1-} B) PS C) PS^{1+}
- (iii) Which of the following will be diamagnetic? Circle the correct choice.
 A) PS^{1-} B) PS C) PS^{1+}
- (iv) Draw Lewis dot structure of the molecule with the shortest bond length (based on the MO diagram) from among the three (PS^{1-} , PS, PS^{1+}), and indicate the bond order calculated from the Lewis dot structure. **(3 points)**

Lewis Dot Structure

Bond Order



D8D1209C-49C0-46F6-8C78-598BF5A9144B

midterm-exam-93388

#449 Page 10 of 10

SCRAP PAGE

**CHEMISTRY 110 –Midterm Exam****Thursday November 7, 2024, 6:30 – 8:30 pm**

Examiners: Pallavi Sirjoosingh and Ashok Kakkar

First name (please write as legibly as possible within the boxes)

--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--

Last name

--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--

Student ID number

--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--

This booklet has 8 questions on 10 pages (The last page can be used as a scrap page).

The staple must NOT be removed from the question booklet. Your intact question booklet must be returned at the end of the exam.

A separate booklet contains a periodic table, and datasheet.

Write your answers directly under the questions in the space provided. Do not write on the QR codes or near the top edges (as it may be cut off).

Please write **legibly and dark enough** for the scan to be readable. For students who use English as a second language, dictionaries are permitted. Calculators are allowed.

DO NOT ASK QUESTIONS OF THE INVIGILATORS about interpretations of the exam questions. If you have a concern about a question, please summarize it at the top of this page. All concerns will be evaluated after the exam. GOOD LUCK!

**Question 1 8 points** (*Assume all transitions are possible*)

a. 6 points Calculate the **energy (in kJ/mol)** required to ionize (completely remove) the electron from the **n=3** level of **He⁺ ion** (He: Helium). Show your work including equations. (3 sig. figs.)

b. 2 points

(I) Select the largest (longest) wavelength from the three choices below:

- i. Wavelength of photon to *completely remove* an electron from *1s orbital of Hydrogen atom*
- ii. Wavelength of photon to *completely remove* an electron from *1s orbital of Boron atom (B)*
- iii. Wavelength of photon to *completely remove* an electron from *1s orbital of Boron cation (B⁺)*

i or ii or iii

(II) Select the smallest (shortest) wavelength from the three choices below (*for H atom*):

- i. Wavelength of photon *emitted* for an electron transitioning from *2p to 1s orbital*
- ii. Wavelength of photon *absorbed* for an electron transitioning from *2s to 4p orbital*
- iii. Wavelength of photon *absorbed* for an electron transitioning from *1s to 3p orbital*

i or ii or iii



Question 2 8 points



A Chem110 student plots the graph shown on the left, from data obtained for the kinetic energy (eV) of the ejected electron versus frequency (s^{-1}) of the incident light, for an unknown metal. ($1\text{eV} = 1.6022 \times 10^{-19} \text{ J}$)

Based on the plot, answer the questions **a to c** below:

a. 1 point On the plot above, label the **threshold frequency**.

b. 2 points What is the value of the work potential (in J) of the unknown metal?

c. 3 points What is the *longest* wavelength (in nm) required to eject an electron from this metal's surface? Show your work including equation(s) used. (3 sig figs)

d. 2 points In the space provided below, plot the following

(Assume frequency is *above* threshold frequency):

- Kinetic energy* of an ejected electron versus the *intensity* of the incident light (left)
- Number of electrons* ejected versus the *frequency* of the incident light (right)



**Question 3** **6 points****a. 2 points**

In the ground state of Ge (Germanium; $Z = 32$) atom, **name** the following orbitals:

i. Orbital containing the electron that requires the maximum energy to be removed.

ii. Orbital containing the electron that experiences the minimum effective nuclear charge.

b. 2 points

An unknown anion (A^-) has the following *excited* electronic configuration: $[Ar] 4s^2 3d^3 4p^1$

(Ar: Argon; $Z = 18$)

Give the **condensed ground state electronic configuration** of the atom, **A** and the cation, **A^{2+}** .
Show your work. (Assume all electronic transitions are possible)

Electronic Configuration of A:

Electronic Configuration of A^{2+} :

c. 2 points

In a single atom, determine how many electrons can have the following set of quantum numbers?

(i) $n = 5; l = 3$

In a single atom, determine how many electrons can have the following set of quantum numbers?

(i) $n = 3; l = 2; m_s = +1/2$

**Question 4** **8 points**

Rank the following from 1 to 4 in **increasing** order. Provide the ranking in the boxes provided.

Note: Atomic number (Z) are provided for relevant atoms

a) Increasing (lowest to highest) order of *ionization energy*: (1= lowest; 4 = highest)

Mg

Mg: Z = 12

Rb

Rb: Z = 37

Cs

Cs: Z = 55

K

K : Z = 19

b) Increasing (lowest to highest) order of *bond angle*: (1= lowest; 4 = highest)

H₂OCCl₄XeF₄

HCN

c) Increasing (smallest to largest) order of *ionic size* (1= smallest; 4 = largest)

S²⁻

S: Z = 16

O²⁻

O: Z = 8

Be²⁺

Be: Z = 4

Na⁺

Na: Z = 11

d) Increasing (lowest to highest) order of *bond energy* (1= lowest; 4 = highest)

P-F

P-Cl

P-I

P-Br

**Question 5** **12 points**

a. 9 points For the molecules HClO_4 and TeF_6 , draw the most stable *Lewis structure* showing appropriate shape and *dashed/wedged bonds*, based on VSEPR. Determine the **molecular geometry** around the central atom. For HClO_4 , label the *most polar bond*.

HClO_4 (there are no O-O bonds)

Label the most polar bond

Molecular Geometry around Cl

TeF_6

Molecular Geometry around Te

b. 3 points (Note: Lone pairs are not shown in structures below)

(i) For the three *resonance* structures (1, 2 and 3) shown below, answer the following question:



(I) Circle the *most contributing* resonance structure.

Structure 1

Structure 2

Structure 3

(ii) For the three structures given (4, 5, and 6) below, answer the following questions:



(I) Circle all the structures that represent *resonance structures* of each other:

Structure 4

Structure 5

Structure 6

(II) Among the resonance structures, circle the *most contributing* resonance structure.

Structure 4

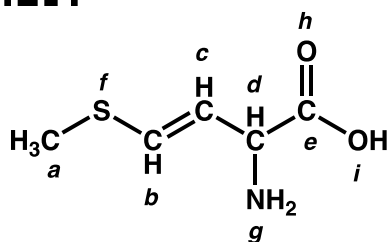
Structure 5

Structure 6

**Question 6** **6 points**

Using the data given below, calculate the **lattice energy** for SrO(s).

	ΔH° (kJ/mol)
$\text{Sr}(s) \longrightarrow \text{Sr}(g)$	+180
$\text{Sr}(g) \longrightarrow \text{Sr}^{2+}(g) + 2e^-$	+1468
$\text{O}_2(g) \longrightarrow 2\text{O}(g)$	+500
$\text{O}(g) + 2e^- \longrightarrow \text{O}^{2-}(g)$	+600
$\text{Sr}(s) + \frac{1}{2}\text{O}_2(g) \longrightarrow \text{SrO}(s)$	-550

**Question 7 10 points**

Consider the molecule given on the left, and answer questions (i) to (vi) using chemical bonding theories that we have learnt in class (valence bond theory, hybridization).

Note that the lone pairs are not shown in the structure:

i. Consider the carbon atoms marked in as ***a***, ***b***, ***c***, ***d***, and ***e*** in the molecule, and determine **hybridization** at each of the carbon centers. (2.5 points)

Atom	Hybridization
Carbon (<i>a</i>)	
Carbon (<i>b</i>)	
Carbon (<i>c</i>)	
Carbon (<i>d</i>)	
Carbon (<i>e</i>)	

ii. Now, consider the sulfur atom (S) marked as ***f***, nitrogen (N) atom (***g***), and oxygen(O) atoms (***h*** and ***i***), and determine their **hybridization**. (2 points)

Atom	Hybridization
Sulfur (<i>f</i>)	
Nitrogen (<i>g</i>)	
Oxygen (<i>h</i>)	
Oxygen (<i>i</i>)	

iii. **Describe** all the bonds formed by the carbon atoms ***a*** and ***b***, using sigma and pi designation as appropriate. (2 points)

Atom	Bonds formed (use sigma and pi as appropriate)
Carbon (<i>a</i>)	
Carbon (<i>b</i>)	

iv. How many total sp^3 -s sigma bonds are in the molecule? (1 point)

v. Determine **electronic** and **molecular geometry** at carbon (C) labeled ***e*** and sulfur (S) labeled ***f*** centres. (2 points)

Atom	Electronic Geometry	Molecular Geometry
Carbon (<i>e</i>)		
Sulfur (<i>f</i>)		

vi. Circle the value from the choices given below, that describes the **C-S-C bond angle at sulfur** (S) marked ***f***: (0.5 points)

>90° to <109.5° 109.5° >109.5 to <120° 120° >120° to <180°


Question 8 12 points

Using the **valence shell orbitals only** draw a **complete** molecular orbital (MO) diagram (with no s-p mixing) for the molecule **PS**, in the space provided below. **(6 points)**

Assume that 3s and 3p atomic orbitals will form molecular orbitals like those formed by the combination of 2s and 2p atomic orbitals that we learnt in class, and there is no s-p mixing.

Show **energy arrow**, **atomic orbitals** involved, **atom labels**, the **molecular orbitals** that are formed, **filled electrons** in the MO diagram etc. Make sure to label all the atomic and molecular orbitals.

Using the MO diagram above, answer the following questions:

- (i) Which one of the following will have the highest bond order? Circle the correct choice
 A) PS^{1-} B) PS C) PS^{1+}
- (ii) Which one of the following will have the longest bond length? Circle the correct choice
 A) PS^{1-} B) PS C) PS^{1+}
- (iii) Which of the following will be diamagnetic? Circle the correct choice.
 A) PS^{1-} B) PS C) PS^{1+}
- (iv) Draw Lewis dot structure of the molecule with the shortest bond length (based on the MO diagram) from among the three (PS^{1-} , PS, PS^{1+}), and indicate the bond order calculated from the Lewis dot structure. **(3 points)**

Lewis Dot Structure

Bond Order



E07DF9F1-43EC-4E6D-AF04-C7652CB7CF21

midterm-exam-93388

#450 Page 10 of 10

SCRAP PAGE

**CHEMISTRY 110 –Midterm Exam****Thursday November 7, 2024, 6:30 – 8:30 pm****Examiners: Pallavi Sirjoosingh and Ashok Kakkar**

First name (please write as legibly as possible within the boxes)

--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--

Last name

--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--

Student ID number

--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--

This booklet has 8 questions on 10 pages (The last page can be used as a scrap page).

The staple must NOT be removed from the question booklet. Your intact question booklet must be returned at the end of the exam.

A separate booklet contains a periodic table, and datasheet.

Write your answers directly under the questions in the space provided. Do not write on the QR codes or near the top edges (as it may be cut off).

Please write **legibly and dark enough** for the scan to be readable. For students who use English as a second language, dictionaries are permitted. Calculators are allowed.

DO NOT ASK QUESTIONS OF THE INVIGILATORS about interpretations of the exam questions. If you have a concern about a question, please summarize it at the top of this page. All concerns will be evaluated after the exam. GOOD LUCK!

**Question 1 8 points** (*Assume all transitions are possible*)

a. 6 points Calculate the **energy (in kJ/mol)** required to ionize (completely remove) the electron from the **n=3** level of **He⁺ ion** (He: Helium). Show your work including equations. (3 sig. figs.)

b. 2 points

(I) Select the largest (longest) wavelength from the three choices below:

- i. Wavelength of photon to *completely remove* an electron from *1s orbital of Hydrogen atom*
- ii. Wavelength of photon to *completely remove* an electron from *1s orbital of Boron atom (B)*
- iii. Wavelength of photon to *completely remove* an electron from *1s orbital of Boron cation (B⁺)*

i or ii or iii

(II) Select the smallest (shortest) wavelength from the three choices below (*for H atom*):

- i. Wavelength of photon *emitted* for an electron transitioning from *2p to 1s orbital*
- ii. Wavelength of photon *absorbed* for an electron transitioning from *2s to 4p orbital*
- iii. Wavelength of photon *absorbed* for an electron transitioning from *1s to 3p orbital*

i or ii or iii



Question 2 8 points



A Chem110 student plots the graph shown on the left, from data obtained for the kinetic energy (eV) of the ejected electron versus frequency (s^{-1}) of the incident light, for an unknown metal. ($1\text{eV} = 1.6022 \times 10^{-19} \text{ J}$)

Based on the plot, answer the questions **a to c** below:

a. 1 point On the plot above, label the **threshold frequency**.

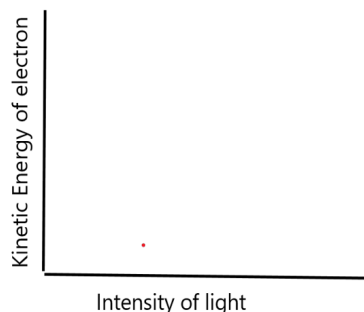
b. 2 points What is the value of the work potential (in J) of the unknown metal?

c. 3 points What is the *longest* wavelength (in nm) required to eject an electron from this metal's surface? Show your work including equation(s) used. (3 sig figs)

d. 2 points In the space provided below, plot the following

(Assume frequency is *above* threshold frequency):

- Kinetic energy* of an ejected electron versus the *intensity* of the incident light (left)
- Number of electrons* ejected versus the *frequency* of the incident light (right)



**Question 3** **6 points****a. 2 points**

In the ground state of Ge (Germanium; $Z = 32$) atom, **name** the following orbitals:

i. Orbital containing the electron that requires the maximum energy to be removed.

ii. Orbital containing the electron that experiences the minimum effective nuclear charge.

b. 2 points

An unknown anion (A^-) has the following *excited* electronic configuration: $[Ar] 4s^2 3d^3 4p^1$

(Ar: Argon; $Z = 18$)

Give the **condensed ground state electronic configuration** of the atom, **A** and the cation, **A^{2+}** .
Show your work. (Assume all electronic transitions are possible)

Electronic Configuration of A:

Electronic Configuration of A^{2+} :

c. 2 points

In a single atom, determine how many electrons can have the following set of quantum numbers?

(i) $n = 5; l = 3$

In a single atom, determine how many electrons can have the following set of quantum numbers?

(i) $n = 3; l = 2; m_s = +1/2$

**Question 4** **8 points**

Rank the following from 1 to 4 in **increasing** order. Provide the ranking in the boxes provided.

Note: Atomic number (Z) are provided for relevant atoms

a) Increasing (lowest to highest) order of *ionization energy*: (1= lowest; 4 = highest)

Mg

Mg: Z = 12

Rb

Rb: Z = 37

Cs

Cs: Z = 55

K

K : Z = 19

b) Increasing (lowest to highest) order of *bond angle*: (1= lowest; 4 = highest)

H₂OCCl₄XeF₄

HCN

c) Increasing (smallest to largest) order of *ionic size* (1= smallest; 4 = largest)

S²⁻

S: Z = 16

O²⁻

O: Z = 8

Be²⁺

Be: Z = 4

Na⁺

Na: Z = 11

d) Increasing (lowest to highest) order of *bond energy* (1= lowest; 4 = highest)

P-F

P-Cl

P-I

P-Br

**Question 5** **12 points**

a. 9 points For the molecules HClO_4 and TeF_6 , draw the most stable *Lewis structure* showing appropriate shape and *dashed/wedged bonds*, based on VSEPR. Determine the **molecular geometry** around the central atom. For HClO_4 , label the *most polar bond*.



b. 3 points (Note: Lone pairs are not shown in structures below)

(i) For the three *resonance* structures (1, 2 and 3) shown below, answer the following question:



(I) Circle the *most contributing* resonance structure.

Structure 1

Structure 2

Structure 3

(ii) For the three structures given (4, 5, and 6) below, answer the following questions:



(I) Circle all the structures that represent *resonance structures* of each other:

Structure 4

Structure 5

Structure 6

(II) Among the resonance structures, circle the *most contributing* resonance structure.

Structure 4

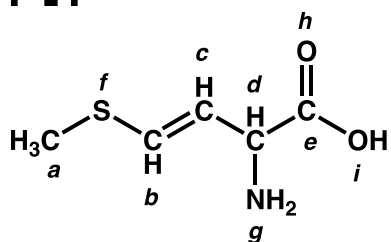
Structure 5

Structure 6

**Question 6** **6 points**

Using the data given below, calculate the **lattice energy** for SrO(s).

	ΔH° (kJ/mol)
$\text{Sr}(s) \longrightarrow \text{Sr}(g)$	+180
$\text{Sr}(g) \longrightarrow \text{Sr}^{2+}(g) + 2e^-$	+1468
$\text{O}_2(g) \longrightarrow 2\text{O}(g)$	+500
$\text{O}(g) + 2e^- \longrightarrow \text{O}^{2-}(g)$	+600
$\text{Sr}(s) + \frac{1}{2}\text{O}_2(g) \longrightarrow \text{SrO}(s)$	-550

**Question 7 10 points**

Consider the molecule given on the left, and answer questions (i) to (vi) using chemical bonding theories that we have learnt in class (valence bond theory, hybridization).

Note that the lone pairs are not shown in the structure:

i. Consider the carbon atoms marked in as ***a***, ***b***, ***c***, ***d***, and ***e*** in the molecule, and determine **hybridization** at each of the carbon centers. (2.5 points)

Atom	Hybridization
Carbon (<i>a</i>)	
Carbon (<i>b</i>)	
Carbon (<i>c</i>)	
Carbon (<i>d</i>)	
Carbon (<i>e</i>)	

ii. Now, consider the sulfur atom (S) marked as ***f***, nitrogen (N) atom (***g***), and oxygen(O) atoms (***h*** and ***i***), and determine their **hybridization**. (2 points)

Atom	Hybridization
Sulfur (<i>f</i>)	
Nitrogen (<i>g</i>)	
Oxygen (<i>h</i>)	
Oxygen (<i>i</i>)	

iii. **Describe** all the bonds formed by the carbon atoms ***a*** and ***b***, using sigma and pi designation as appropriate. (2 points)

Atom	Bonds formed (use sigma and pi as appropriate)
Carbon (<i>a</i>)	
Carbon (<i>b</i>)	

iv. How many total sp^3 -s sigma bonds are in the molecule? (1 point)

v. Determine **electronic** and **molecular geometry** at carbon (C) labeled ***e*** and sulfur (S) labeled ***f*** centres. (2 points)

Atom	Electronic Geometry	Molecular Geometry
Carbon (<i>e</i>)		
Sulfur (<i>f</i>)		

vi. Circle the value from the choices given below, that describes the **C-S-C bond angle at sulfur** (S) marked ***f***: (0.5 points)

>90° to <109.5° 109.5° >109.5 to <120° 120° >120° to <180°


Question 8 12 points

Using the **valence shell orbitals only** draw a **complete** molecular orbital (MO) diagram (with no s-p mixing) for the molecule **PS**, in the space provided below. **(6 points)**

Assume that 3s and 3p atomic orbitals will form molecular orbitals like those formed by the combination of 2s and 2p atomic orbitals that we learnt in class, and there is no s-p mixing.

Show **energy arrow**, **atomic orbitals** involved, **atom labels**, the **molecular orbitals** that are formed, **filled electrons** in the MO diagram etc. Make sure to label all the atomic and molecular orbitals.

Using the MO diagram above, answer the following questions:

- (i) Which one of the following will have the highest bond order? Circle the correct choice
 A) PS^{1-} B) PS C) PS^{1+}
- (ii) Which one of the following will have the longest bond length? Circle the correct choice
 A) PS^{1-} B) PS C) PS^{1+}
- (iii) Which of the following will be diamagnetic? Circle the correct choice.
 A) PS^{1-} B) PS C) PS^{1+}
- (iv) Draw Lewis dot structure of the molecule with the shortest bond length (based on the MO diagram) from among the three (PS^{1-} , PS, PS^{1+}), and indicate the bond order calculated from the Lewis dot structure. **(3 points)**

Lewis Dot Structure

Bond Order



6C10E946-C328-42C1-A7C3-55E9B122D425

midterm-exam-93388

#451 Page 10 of 10

SCRAP PAGE

**CHEMISTRY 110 –Midterm Exam****Thursday November 7, 2024, 6:30 – 8:30 pm****Examiners: Pallavi Sirjoosingh and Ashok Kakkar**

First name (please write as legibly as possible within the boxes)

--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--

Last name

--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--

Student ID number

--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--

This booklet has 8 questions on 10 pages (The last page can be used as a scrap page).

The staple must NOT be removed from the question booklet. Your intact question booklet must be returned at the end of the exam.

A separate booklet contains a periodic table, and datasheet.

Write your answers directly under the questions in the space provided. Do not write on the QR codes or near the top edges (as it may be cut off).

Please write **legibly and dark enough** for the scan to be readable. For students who use English as a second language, dictionaries are permitted. Calculators are allowed.

DO NOT ASK QUESTIONS OF THE INVIGILATORS about interpretations of the exam questions. If you have a concern about a question, please summarize it at the top of this page. All concerns will be evaluated after the exam. GOOD LUCK!

**Question 1 8 points** (*Assume all transitions are possible*)

a. 6 points Calculate the **energy (in kJ/mol)** required to ionize (completely remove) the electron from the **$n=3$** level of **He^+ ion** (He: Helium). Show your work including equations. (3 sig. figs.)

b. 2 points

(I) Select the largest (longest) wavelength from the three choices below:

- i. Wavelength of photon to *completely remove* an electron from *1s orbital of Hydrogen atom*
- ii. Wavelength of photon to *completely remove* an electron from *1s orbital of Boron atom (B)*
- iii. Wavelength of photon to *completely remove* an electron from *1s orbital of Boron cation (B^+)*

i or ii or iii

(II) Select the smallest (shortest) wavelength from the three choices below (*for H atom*):

- i. Wavelength of photon *emitted* for an electron transitioning from *2p to 1s orbital*
- ii. Wavelength of photon *absorbed* for an electron transitioning from *2s to 4p orbital*
- iii. Wavelength of photon *absorbed* for an electron transitioning from *1s to 3p orbital*

i or ii or iii



Question 2 8 points



A Chem110 student plots the graph shown on the left, from data obtained for the kinetic energy (eV) of the ejected electron versus frequency (s^{-1}) of the incident light, for an unknown metal. ($1\text{eV} = 1.6022 \times 10^{-19} \text{ J}$)

Based on the plot, answer the questions **a to c** below:

a. 1 point On the plot above, label the **threshold frequency**.

b. 2 points What is the value of the work potential (in J) of the unknown metal?

c. 3 points What is the *longest* wavelength (in nm) required to eject an electron from this metal's surface? Show your work including equation(s) used. (3 sig figs)

d. 2 points In the space provided below, plot the following

(Assume frequency is *above* threshold frequency):

- Kinetic energy* of an ejected electron versus the *intensity* of the incident light (left)
- Number of electrons* ejected versus the *frequency* of the incident light (right)



**Question 3** **6 points****a. 2 points**

In the ground state of Ge (Germanium; $Z = 32$) atom, **name** the following orbitals:

i. Orbital containing the electron that requires the maximum energy to be removed.

ii. Orbital containing the electron that experiences the minimum effective nuclear charge.

b. 2 points

An unknown anion (A^-) has the following *excited* electronic configuration: $[Ar] 4s^2 3d^3 4p^1$

(Ar: Argon; $Z = 18$)

Give the **condensed ground state electronic configuration** of the atom, **A** and the cation, **A^{2+}** .
Show your work. (Assume all electronic transitions are possible)

Electronic Configuration of A:

Electronic Configuration of A^{2+} :

c. 2 points

In a single atom, determine how many electrons can have the following set of quantum numbers?

(i) $n = 5; l = 3$

In a single atom, determine how many electrons can have the following set of quantum numbers?

(i) $n = 3; l = 2; m_s = +1/2$

**Question 4** **8 points**

Rank the following from 1 to 4 in **increasing** order. Provide the ranking in the boxes provided.

Note: Atomic number (Z) are provided for relevant atoms

a) Increasing (lowest to highest) order of *ionization energy*: (1= lowest; 4 = highest)

Mg

Mg: Z = 12

Rb

Rb: Z = 37

Cs

Cs: Z = 55

K

K : Z = 19

b) Increasing (lowest to highest) order of *bond angle*: (1= lowest; 4 = highest)

H₂OCCl₄XeF₄

HCN

c) Increasing (smallest to largest) order of *ionic size* (1= smallest; 4 = largest)

S²⁻

S: Z = 16

O²⁻

O: Z = 8

Be²⁺

Be: Z = 4

Na⁺

Na: Z = 11

d) Increasing (lowest to highest) order of *bond energy* (1= lowest; 4 = highest)

P-F

P-Cl

P-I

P-Br

**Question 5** **12 points**

a. 9 points For the molecules HClO_4 and TeF_6 , draw the most stable *Lewis structure* showing appropriate shape and *dashed/wedged bonds*, based on VSEPR. Determine the **molecular geometry** around the central atom. For HClO_4 , label the *most polar bond*.



b. 3 points (Note: Lone pairs are not shown in structures below)

(i) For the three *resonance* structures (1, 2 and 3) shown below, answer the following question:



(I) Circle the *most contributing* resonance structure.

Structure 1

Structure 2

Structure 3

(ii) For the three structures given (4, 5, and 6) below, answer the following questions:



(I) Circle all the structures that represent *resonance structures* of each other:

Structure 4

Structure 5

Structure 6

(II) Among the resonance structures, circle the *most contributing* resonance structure.

Structure 4

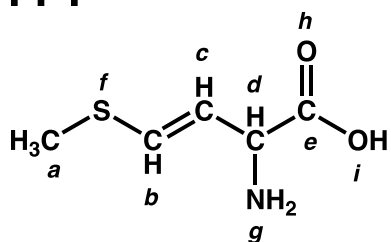
Structure 5

Structure 6

**Question 6** **6 points**

Using the data given below, calculate the **lattice energy** for SrO(s).

	ΔH° (kJ/mol)
$\text{Sr}(s) \longrightarrow \text{Sr}(g)$	+180
$\text{Sr}(g) \longrightarrow \text{Sr}^{2+}(g) + 2e^-$	+1468
$\text{O}_2(g) \longrightarrow 2\text{O}(g)$	+500
$\text{O}(g) + 2e^- \longrightarrow \text{O}^{2-}(g)$	+600
$\text{Sr}(s) + \frac{1}{2}\text{O}_2(g) \longrightarrow \text{SrO}(s)$	-550

**Question 7 10 points**

Consider the molecule given on the left, and answer questions (i) to (vi) using chemical bonding theories that we have learnt in class (valence bond theory, hybridization).

Note that the lone pairs are not shown in the structure:

i. Consider the carbon atoms marked in as ***a***, ***b***, ***c***, ***d***, and ***e*** in the molecule, and determine **hybridization** at each of the carbon centers. (2.5 points)

Atom	Hybridization
Carbon (<i>a</i>)	
Carbon (<i>b</i>)	
Carbon (<i>c</i>)	
Carbon (<i>d</i>)	
Carbon (<i>e</i>)	

ii. Now, consider the sulfur atom (S) marked as ***f***, nitrogen (N) atom (***g***), and oxygen(O) atoms (***h*** and ***i***), and determine their **hybridization**. (2 points)

Atom	Hybridization
Sulfur (<i>f</i>)	
Nitrogen (<i>g</i>)	
Oxygen (<i>h</i>)	
Oxygen (<i>i</i>)	

iii. **Describe** all the bonds formed by the carbon atoms ***a*** and ***b***, using sigma and pi designation as appropriate. (2 points)

Atom	Bonds formed (use sigma and pi as appropriate)
Carbon (<i>a</i>)	
Carbon (<i>b</i>)	

iv. How many total sp^3 -s sigma bonds are in the molecule? (1 point)

v. Determine **electronic** and **molecular geometry** at carbon (C) labeled ***e*** and sulfur (S) labeled ***f*** centres. (2 points)

Atom	Electronic Geometry	Molecular Geometry
Carbon (<i>e</i>)		
Sulfur (<i>f</i>)		

vi. Circle the value from the choices given below, that describes the **C-S-C bond angle at sulfur** (S) marked ***f***: (0.5 points)

>90° to <109.5° 109.5° >109.5 to <120° 120° >120° to <180°

**Question 8 12 points**

Using the **valence shell orbitals only** draw a **complete** molecular orbital (MO) diagram (with no s-p mixing) for the molecule **PS**, in the space provided below. **(6 points)**

Assume that 3s and 3p atomic orbitals will form molecular orbitals like those formed by the combination of 2s and 2p atomic orbitals that we learnt in class, and there is no s-p mixing.

Show **energy arrow**, **atomic orbitals** involved, **atom labels**, the **molecular orbitals** that are formed, **filled electrons** in the MO diagram etc. Make sure to **label all the atomic and molecular orbitals**.

Using the MO diagram above, answer the following questions:

- (i) Which one of the following will have the highest bond order? Circle the correct choice
A) PS^{1-} B) PS C) PS^{1+}
- (ii) Which one of the following will have the longest bond length? Circle the correct choice
A) PS^{1-} B) PS C) PS^{1+}
- (iii) Which of the following will be diamagnetic? Circle the correct choice.
A) PS^{1-} B) PS C) PS^{1+}
- (iv) Draw Lewis dot structure of the molecule with the shortest bond length (based on the MO diagram) from among the three (PS^{1-} , PS, PS^{1+}), and indicate the bond order calculated from the Lewis dot structure. **(3 points)**

Lewis Dot Structure

Bond Order



05BAF6AE-B8F3-4BA4-BB1F-B906D3263CD1

midterm-exam-93388

#452 Page 10 of 10

SCRAP PAGE

**CHEMISTRY 110 –Midterm Exam****Thursday November 7, 2024, 6:30 – 8:30 pm****Examiners: Pallavi Sirjoosingh and Ashok Kakkar**

First name (please write as legibly as possible within the boxes)

--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--

Last name

--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--

Student ID number

--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--

This booklet has 8 questions on 10 pages (The last page can be used as a scrap page).

The staple must NOT be removed from the question booklet. Your intact question booklet must be returned at the end of the exam.

A separate booklet contains a periodic table, and datasheet.

Write your answers directly under the questions in the space provided. Do not write on the QR codes or near the top edges (as it may be cut off).

Please write **legibly and dark enough** for the scan to be readable. For students who use English as a second language, dictionaries are permitted. Calculators are allowed.

DO NOT ASK QUESTIONS OF THE INVIGILATORS about interpretations of the exam questions. If you have a concern about a question, please summarize it at the top of this page. All concerns will be evaluated after the exam. GOOD LUCK!

**Question 1 8 points** (*Assume all transitions are possible*)

a. 6 points Calculate the **energy (in kJ/mol)** required to ionize (completely remove) the electron from the **n=3** level of **He⁺ ion** (He: Helium). Show your work including equations. (3 sig. figs.)

b. 2 points

(I) Select the largest (longest) wavelength from the three choices below:

- i. Wavelength of photon to *completely remove* an electron from *1s orbital of Hydrogen atom*
- ii. Wavelength of photon to *completely remove* an electron from *1s orbital of Boron atom (B)*
- iii. Wavelength of photon to *completely remove* an electron from *1s orbital of Boron cation (B⁺)*

i or ii or iii

(II) Select the smallest (shortest) wavelength from the three choices below (*for H atom*):

- i. Wavelength of photon *emitted* for an electron transitioning from *2p to 1s orbital*
- ii. Wavelength of photon *absorbed* for an electron transitioning from *2s to 4p orbital*
- iii. Wavelength of photon *absorbed* for an electron transitioning from *1s to 3p orbital*

i or ii or iii



Question 2 8 points



A Chem110 student plots the graph shown on the left, from data obtained for the kinetic energy (eV) of the ejected electron versus frequency (s^{-1}) of the incident light, for an unknown metal. ($1\text{eV} = 1.6022 \times 10^{-19} \text{ J}$)

Based on the plot, answer the questions **a to c** below:

a. 1 point On the plot above, label the **threshold frequency**.

b. 2 points What is the value of the work potential (in J) of the unknown metal?

c. 3 points What is the *longest* wavelength (in nm) required to eject an electron from this metal's surface? Show your work including equation(s) used. (3 sig figs)

d. 2 points In the space provided below, plot the following

(Assume frequency is *above* threshold frequency):

- Kinetic energy* of an ejected electron versus the *intensity* of the incident light (left)
- Number of electrons* ejected versus the *frequency* of the incident light (right)



**Question 3** **6 points****a. 2 points**

In the ground state of Ge (Germanium; $Z = 32$) atom, **name** the following orbitals:

i. Orbital containing the electron that requires the maximum energy to be removed.

ii. Orbital containing the electron that experiences the minimum effective nuclear charge.

b. 2 points

An unknown anion (A^-) has the following *excited* electronic configuration: $[Ar] 4s^2 3d^3 4p^1$

(Ar: Argon; $Z = 18$)

Give the **condensed ground state electronic configuration** of the atom, **A** and the cation, **A^{2+}** .
Show your work. (Assume all electronic transitions are possible)

Electronic Configuration of A:

Electronic Configuration of A^{2+} :

c. 2 points

In a single atom, determine how many electrons can have the following set of quantum numbers?

(i) $n = 5; l = 3$

In a single atom, determine how many electrons can have the following set of quantum numbers?

(i) $n = 3; l = 2; m_s = +1/2$

**Question 4** **8 points**

Rank the following from 1 to 4 in **increasing** order. Provide the ranking in the boxes provided.

Note: Atomic number (Z) are provided for relevant atoms

a) Increasing (lowest to highest) order of *ionization energy*: (1= lowest; 4 = highest)

Mg

Mg: Z = 12

Rb

Rb: Z = 37

Cs

Cs: Z = 55

K

K : Z = 19

b) Increasing (lowest to highest) order of *bond angle*: (1= lowest; 4 = highest)

H₂OCCl₄XeF₄

HCN

c) Increasing (smallest to largest) order of *ionic size* (1= smallest; 4 = largest)

S²⁻

S: Z = 16

O²⁻

O: Z = 8

Be²⁺

Be: Z = 4

Na⁺

Na: Z = 11

d) Increasing (lowest to highest) order of *bond energy* (1= lowest; 4 = highest)

P-F

P-Cl

P-I

P-Br

**Question 5** **12 points**

a. 9 points For the molecules HClO_4 and TeF_6 , draw the most stable *Lewis structure* showing appropriate shape and *dashed/wedged bonds*, based on VSEPR. Determine the **molecular geometry** around the central atom. For HClO_4 , label the *most polar bond*.

HClO_4 (there are no O-O bonds)

Label the most polar bond

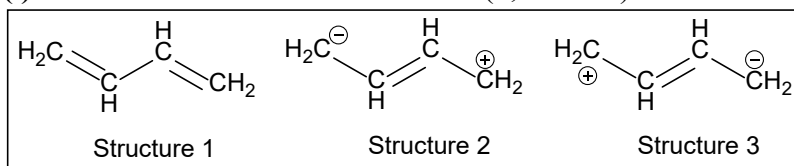
Molecular Geometry around Cl

TeF_6

Molecular Geometry around Te

b. 3 points (Note: Lone pairs are not shown in structures below)

(i) For the three *resonance* structures (1, 2 and 3) shown below, answer the following question:



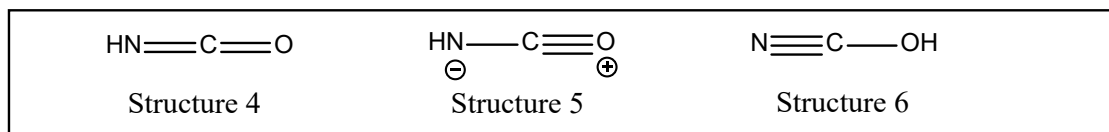
(I) Circle the *most contributing* resonance structure.

Structure 1

Structure 2

Structure 3

(ii) For the three structures given (4, 5, and 6) below, answer the following questions:



(I) Circle all the structures that represent *resonance structures* of each other:

Structure 4

Structure 5

Structure 6

(II) Among the resonance structures, circle the *most contributing* resonance structure.

Structure 4

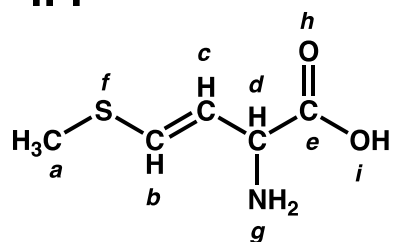
Structure 5

Structure 6

**Question 6** **6 points**

Using the data given below, calculate the **lattice energy** for SrO(s).

	ΔH° (kJ/mol)
$\text{Sr}(s) \longrightarrow \text{Sr}(g)$	+180
$\text{Sr}(g) \longrightarrow \text{Sr}^{2+}(g) + 2e^-$	+1468
$\text{O}_2(g) \longrightarrow 2\text{O}(g)$	+500
$\text{O}(g) + 2e^- \longrightarrow \text{O}^{2-}(g)$	+600
$\text{Sr}(s) + \frac{1}{2}\text{O}_2(g) \longrightarrow \text{SrO}(s)$	-550

**Question 7 10 points**

Consider the molecule given on the left, and answer questions (i) to (vi) using chemical bonding theories that we have learnt in class (valence bond theory, hybridization).

Note that the lone pairs are not shown in the structure:

i. Consider the carbon atoms marked in as ***a***, ***b***, ***c***, ***d***, and ***e*** in the molecule, and determine **hybridization** at each of the carbon centers. (2.5 points)

Atom	Hybridization
Carbon (<i>a</i>)	
Carbon (<i>b</i>)	
Carbon (<i>c</i>)	
Carbon (<i>d</i>)	
Carbon (<i>e</i>)	

ii. Now, consider the sulfur atom (S) marked as ***f***, nitrogen (N) atom (***g***), and oxygen(O) atoms (***h*** and ***i***), and determine their **hybridization**. (2 points)

Atom	Hybridization
Sulfur (<i>f</i>)	
Nitrogen (<i>g</i>)	
Oxygen (<i>h</i>)	
Oxygen (<i>i</i>)	

iii. **Describe** all the bonds formed by the carbon atoms ***a*** and ***b***, using sigma and pi designation as appropriate. (2 points)

Atom	Bonds formed (use sigma and pi as appropriate)
Carbon (<i>a</i>)	
Carbon (<i>b</i>)	

iv. How many total sp^3 -s sigma bonds are in the molecule? (1 point)

v. Determine **electronic** and **molecular geometry** at carbon (C) labeled ***e*** and sulfur (S) labeled ***f*** centres. (2 points)

Atom	Electronic Geometry	Molecular Geometry
Carbon (<i>e</i>)		
Sulfur (<i>f</i>)		

vi. Circle the value from the choices given below, that describes the **C-S-C bond angle at sulfur** (S) marked ***f***: (0.5 points)

>90° to <109.5°

109.5°

>109.5 to <120°

120°

>120° to <180°

**Question 8 12 points**

Using the **valence shell orbitals only** draw a **complete** molecular orbital (MO) diagram (with no s-p mixing) for the molecule **PS**, in the space provided below. **(6 points)**

Assume that 3s and 3p atomic orbitals will form molecular orbitals like those formed by the combination of 2s and 2p atomic orbitals that we learnt in class, and there is no s-p mixing.

Show **energy arrow**, **atomic orbitals** involved, **atom labels**, the **molecular orbitals** that are formed, **filled electrons** in the MO diagram etc. Make sure to **label all the atomic and molecular orbitals**.

Using the MO diagram above, answer the following questions:

- (i) Which one of the following will have the highest bond order? Circle the correct choice
A) PS^{1-} B) PS C) PS^{1+}
- (ii) Which one of the following will have the longest bond length? Circle the correct choice
A) PS^{1-} B) PS C) PS^{1+}
- (iii) Which of the following will be diamagnetic? Circle the correct choice.
A) PS^{1-} B) PS C) PS^{1+}
- (iv) Draw Lewis dot structure of the molecule with the shortest bond length (based on the MO diagram) from among the three (PS^{1-} , PS, PS^{1+}), and indicate the bond order calculated from the Lewis dot structure. **(3 points)**

Lewis Dot Structure

Bond Order



B80BAA7D-3BC6-4F75-9885-F234674870D2

midterm-exam-93388

#453 Page 10 of 10

SCRAP PAGE

**CHEMISTRY 110 –Midterm Exam****Thursday November 7, 2024, 6:30 – 8:30 pm**

Examiners: Pallavi Sirjoosingh and Ashok Kakkar

First name (please write as legibly as possible within the boxes)

--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--

Last name

--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--

Student ID number

--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--

This booklet has 8 questions on 10 pages (The last page can be used as a scrap page).

The staple must NOT be removed from the question booklet. Your intact question booklet must be returned at the end of the exam.

A separate booklet contains a periodic table, and datasheet.

Write your answers directly under the questions in the space provided. Do not write on the QR codes or near the top edges (as it may be cut off).

Please write **legibly and dark enough** for the scan to be readable. For students who use English as a second language, dictionaries are permitted. Calculators are allowed.

DO NOT ASK QUESTIONS OF THE INVIGILATORS about interpretations of the exam questions. If you have a concern about a question, please summarize it at the top of this page. All concerns will be evaluated after the exam. GOOD LUCK!

**Question 1 8 points** (*Assume all transitions are possible*)

a. 6 points Calculate the **energy (in kJ/mol)** required to ionize (completely remove) the electron from the **$n=3$** level of **He^+ ion** (He: Helium). Show your work including equations. (3 sig. figs.)

b. 2 points

(I) Select the largest (longest) wavelength from the three choices below:

- i. Wavelength of photon to *completely remove* an electron from *1s orbital of Hydrogen atom*
- ii. Wavelength of photon to *completely remove* an electron from *1s orbital of Boron atom (B)*
- iii. Wavelength of photon to *completely remove* an electron from *1s orbital of Boron cation (B^+)*

i or ii or iii

(II) Select the smallest (shortest) wavelength from the three choices below (*for H atom*):

- i. Wavelength of photon *emitted* for an electron transitioning from *2p to 1s orbital*
- ii. Wavelength of photon *absorbed* for an electron transitioning from *2s to 4p orbital*
- iii. Wavelength of photon *absorbed* for an electron transitioning from *1s to 3p orbital*

i or ii or iii



Question 2 8 points



A Chem110 student plots the graph shown on the left, from data obtained for the kinetic energy (eV) of the ejected electron versus frequency (s^{-1}) of the incident light, for an unknown metal. ($1\text{eV} = 1.6022 \times 10^{-19} \text{ J}$)

Based on the plot, answer the questions **a to c** below:

a. 1 point On the plot above, label the **threshold frequency**.

b. 2 points What is the value of the work potential (in J) of the unknown metal?

c. 3 points What is the *longest* wavelength (in nm) required to eject an electron from this metal's surface? Show your work including equation(s) used. (3 sig figs)

d. 2 points In the space provided below, plot the following

(Assume frequency is *above* threshold frequency):

- Kinetic energy* of an ejected electron versus the *intensity* of the incident light (left)
- Number of electrons* ejected versus the *frequency* of the incident light (right)



**Question 3** **6 points****a. 2 points**

In the ground state of Ge (Germanium; $Z = 32$) atom, **name** the following orbitals:

i. Orbital containing the electron that requires the maximum energy to be removed.

ii. Orbital containing the electron that experiences the minimum effective nuclear charge.

b. 2 points

An unknown anion (A^-) has the following *excited* electronic configuration: $[Ar] 4s^2 3d^3 4p^1$

(Ar: Argon; $Z = 18$)

Give the **condensed ground state electronic configuration** of the atom, **A** and the cation, **A^{2+}** .
Show your work. (Assume all electronic transitions are possible)

Electronic Configuration of A:

Electronic Configuration of A^{2+} :

c. 2 points

In a single atom, determine how many electrons can have the following set of quantum numbers?

(i) $n = 5; l = 3$

In a single atom, determine how many electrons can have the following set of quantum numbers?

(i) $n = 3; l = 2; m_s = +1/2$

**Question 4** **8 points**

Rank the following from 1 to 4 in **increasing** order. Provide the ranking in the boxes provided.

Note: Atomic number (Z) are provided for relevant atoms

a) Increasing (lowest to highest) order of *ionization energy*: (1= lowest; 4 = highest)

Mg

Mg: Z = 12

Rb

Rb: Z = 37

Cs

Cs: Z = 55

K

K : Z = 19

b) Increasing (lowest to highest) order of *bond angle*: (1= lowest; 4 = highest)

H₂OCCl₄XeF₄

HCN

c) Increasing (smallest to largest) order of *ionic size* (1= smallest; 4 = largest)

S²⁻

S: Z = 16

O²⁻

O: Z = 8

Be²⁺

Be: Z = 4

Na⁺

Na: Z = 11

d) Increasing (lowest to highest) order of *bond energy* (1= lowest; 4 = highest)

P-F

P-Cl

P-I

P-Br

**Question 5** **12 points**

a. 9 points For the molecules HClO_4 and TeF_6 , draw the most stable *Lewis structure* showing appropriate shape and *dashed/wedged bonds*, based on VSEPR. Determine the **molecular geometry** around the central atom. For HClO_4 , label the *most polar bond*.



b. 3 points (Note: Lone pairs are not shown in structures below)

(i) For the three *resonance* structures (1, 2 and 3) shown below, answer the following question:



(I) Circle the *most contributing* resonance structure.

Structure 1

Structure 2

Structure 3

(ii) For the three structures given (4, 5, and 6) below, answer the following questions:



(I) Circle all the structures that represent *resonance structures* of each other:

Structure 4

Structure 5

Structure 6

(II) Among the resonance structures, circle the *most contributing* resonance structure.

Structure 4

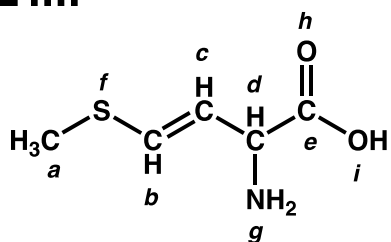
Structure 5

Structure 6

**Question 6** **6 points**

Using the data given below, calculate the **lattice energy** for SrO(s).

	ΔH° (kJ/mol)
$\text{Sr}(s) \longrightarrow \text{Sr}(g)$	+180
$\text{Sr}(g) \longrightarrow \text{Sr}^{2+}(g) + 2e^-$	+1468
$\text{O}_2(g) \longrightarrow 2\text{O}(g)$	+500
$\text{O}(g) + 2e^- \longrightarrow \text{O}^{2-}(g)$	+600
$\text{Sr}(s) + \frac{1}{2}\text{O}_2(g) \longrightarrow \text{SrO}(s)$	-550

**Question 7 10 points**

Consider the molecule given on the left, and answer questions (i) to (vi) using chemical bonding theories that we have learnt in class (valence bond theory, hybridization).

Note that the lone pairs are not shown in the structure:

i. Consider the carbon atoms marked in as ***a***, ***b***, ***c***, ***d***, and ***e*** in the molecule, and determine **hybridization** at each of the carbon centers. (2.5 points)

Atom	Hybridization
Carbon (<i>a</i>)	
Carbon (<i>b</i>)	
Carbon (<i>c</i>)	
Carbon (<i>d</i>)	
Carbon (<i>e</i>)	

ii. Now, consider the sulfur atom (S) marked as ***f***, nitrogen (N) atom (***g***), and oxygen(O) atoms (***h*** and ***i***), and determine their **hybridization**. (2 points)

Atom	Hybridization
Sulfur (<i>f</i>)	
Nitrogen (<i>g</i>)	
Oxygen (<i>h</i>)	
Oxygen (<i>i</i>)	

iii. **Describe** all the bonds formed by the carbon atoms ***a*** and ***b***, using sigma and pi designation as appropriate. (2 points)

Atom	Bonds formed (use sigma and pi as appropriate)
Carbon (<i>a</i>)	
Carbon (<i>b</i>)	

iv. How many total sp^3 -s sigma bonds are in the molecule? (1 point)

v. Determine **electronic** and **molecular geometry** at carbon (C) labeled ***e*** and sulfur (S) labeled ***f*** centres. (2 points)

Atom	Electronic Geometry	Molecular Geometry
Carbon (<i>e</i>)		
Sulfur (<i>f</i>)		

vi. Circle the value from the choices given below, that describes the **C-S-C bond angle at sulfur** (S) marked ***f***: (0.5 points)

>90° to <109.5° 109.5° >109.5 to <120° 120° >120° to <180°


Question 8 12 points

Using the **valence shell orbitals only** draw a **complete** molecular orbital (MO) diagram (with no s-p mixing) for the molecule **PS**, in the space provided below. **(6 points)**

Assume that 3s and 3p atomic orbitals will form molecular orbitals like those formed by the combination of 2s and 2p atomic orbitals that we learnt in class, and there is no s-p mixing.

Show **energy arrow**, **atomic orbitals** involved, **atom labels**, the **molecular orbitals** that are formed, **filled electrons** in the MO diagram etc. Make sure to label all the atomic and molecular orbitals.

Using the MO diagram above, answer the following questions:

- (i) Which one of the following will have the highest bond order? Circle the correct choice
 A) PS^{1-} B) PS C) PS^{1+}
- (ii) Which one of the following will have the longest bond length? Circle the correct choice
 A) PS^{1-} B) PS C) PS^{1+}
- (iii) Which of the following will be diamagnetic? Circle the correct choice.
 A) PS^{1-} B) PS C) PS^{1+}
- (iv) Draw Lewis dot structure of the molecule with the shortest bond length (based on the MO diagram) from among the three (PS^{1-} , PS, PS^{1+}), and indicate the bond order calculated from the Lewis dot structure. **(3 points)**

Lewis Dot Structure

Bond Order



38A6819C-8FB3-49F8-91F2-5AB2ECF88FD2

midterm-exam-93388

#454 Page 10 of 10

SCRAP PAGE

**CHEMISTRY 110 –Midterm Exam****Thursday November 7, 2024, 6:30 – 8:30 pm**

Examiners: Pallavi Sirjoosingh and Ashok Kakkar

First name (please write as legibly as possible within the boxes)

--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--

Last name

--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--

Student ID number

--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--

This booklet has 8 questions on 10 pages (The last page can be used as a scrap page).

The staple must NOT be removed from the question booklet. Your intact question booklet must be returned at the end of the exam.

A separate booklet contains a periodic table, and datasheet.

Write your answers directly under the questions in the space provided. Do not write on the QR codes or near the top edges (as it may be cut off).

Please write **legibly and dark enough** for the scan to be readable. For students who use English as a second language, dictionaries are permitted. Calculators are allowed.

DO NOT ASK QUESTIONS OF THE INVIGILATORS about interpretations of the exam questions. If you have a concern about a question, please summarize it at the top of this page. All concerns will be evaluated after the exam. GOOD LUCK!

**Question 1 8 points** (*Assume all transitions are possible*)

a. 6 points Calculate the **energy (in kJ/mol)** required to ionize (completely remove) the electron from the **$n=3$** level of **He^+ ion** (He: Helium). Show your work including equations. (3 sig. figs.)

b. 2 points

(I) Select the largest (longest) wavelength from the three choices below:

- i. Wavelength of photon to *completely remove* an electron from *1s orbital of Hydrogen atom*
- ii. Wavelength of photon to *completely remove* an electron from *1s orbital of Boron atom (B)*
- iii. Wavelength of photon to *completely remove* an electron from *1s orbital of Boron cation (B^+)*

i or ii or iii

(II) Select the smallest (shortest) wavelength from the three choices below (*for H atom*):

- i. Wavelength of photon *emitted* for an electron transitioning from *2p to 1s orbital*
- ii. Wavelength of photon *absorbed* for an electron transitioning from *2s to 4p orbital*
- iii. Wavelength of photon *absorbed* for an electron transitioning from *1s to 3p orbital*

i or ii or iii



Question 2 8 points



A Chem110 student plots the graph shown on the left, from data obtained for the kinetic energy (eV) of the ejected electron versus frequency (s^{-1}) of the incident light, for an unknown metal. ($1\text{eV} = 1.6022 \times 10^{-19} \text{ J}$)

Based on the plot, answer the questions **a to c** below:

a. 1 point On the plot above, label the **threshold frequency**.

b. 2 points What is the value of the work potential (in J) of the unknown metal?

c. 3 points What is the *longest* wavelength (in nm) required to eject an electron from this metal's surface? Show your work including equation(s) used. (3 sig figs)

d. 2 points In the space provided below, plot the following

(Assume frequency is *above* threshold frequency):

- i. *Kinetic energy* of an ejected electron versus the *intensity* of the incident light (left)
- ii. *Number of electrons* ejected versus the *frequency* of the incident light (right)



**Question 3** **6 points****a. 2 points**

In the ground state of Ge (Germanium; $Z = 32$) atom, **name** the following orbitals:

i. Orbital containing the electron that requires the maximum energy to be removed.

ii. Orbital containing the electron that experiences the minimum effective nuclear charge.

b. 2 points

An unknown anion (A^-) has the following *excited* electronic configuration: $[Ar] 4s^2 3d^3 4p^1$

(Ar: Argon; $Z = 18$)

Give the **condensed ground state electronic configuration** of the atom, **A** and the cation, **A^{2+}** .
Show your work. (Assume all electronic transitions are possible)

Electronic Configuration of A:

Electronic Configuration of A^{2+} :

c. 2 points

In a single atom, determine how many electrons can have the following set of quantum numbers?

(i) $n = 5; l = 3$

In a single atom, determine how many electrons can have the following set of quantum numbers?

(i) $n = 3; l = 2; m_s = +1/2$

**Question 4** **8 points**

Rank the following from 1 to 4 in **increasing** order. Provide the ranking in the boxes provided.

Note: Atomic number (Z) are provided for relevant atoms

a) Increasing (lowest to highest) order of *ionization energy*: (1= lowest; 4 = highest)

Mg

Mg: Z = 12

Rb

Rb: Z = 37

Cs

Cs: Z = 55

K

K : Z = 19

b) Increasing (lowest to highest) order of *bond angle*: (1= lowest; 4 = highest)

H₂OCCl₄XeF₄

HCN

c) Increasing (smallest to largest) order of *ionic size* (1= smallest; 4 = largest)

S²⁻

S: Z = 16

O²⁻

O: Z = 8

Be²⁺

Be: Z = 4

Na⁺

Na: Z = 11

d) Increasing (lowest to highest) order of *bond energy* (1= lowest; 4 = highest)

P-F

P-Cl

P-I

P-Br

**Question 5** **12 points**

a. 9 points For the molecules HClO_4 and TeF_6 , draw the most stable *Lewis structure* showing appropriate shape and *dashed/wedged bonds*, based on VSEPR. Determine the **molecular geometry** around the central atom. For HClO_4 , label the *most polar bond*.



b. 3 points (Note: Lone pairs are not shown in structures below)

(i) For the three *resonance* structures (1, 2 and 3) shown below, answer the following question:



(I) Circle the *most contributing* resonance structure.

Structure 1

Structure 2

Structure 3

(ii) For the three structures given (4, 5, and 6) below, answer the following questions:



(I) Circle all the structures that represent *resonance structures* of each other:

Structure 4

Structure 5

Structure 6

(II) Among the resonance structures, circle the *most contributing* resonance structure.

Structure 4

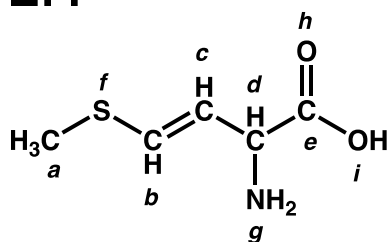
Structure 5

Structure 6

**Question 6** **6 points**

Using the data given below, calculate the **lattice energy** for SrO(s).

	ΔH° (kJ/mol)
$\text{Sr}(s) \longrightarrow \text{Sr}(g)$	+180
$\text{Sr}(g) \longrightarrow \text{Sr}^{2+}(g) + 2e^-$	+1468
$\text{O}_2(g) \longrightarrow 2\text{O}(g)$	+500
$\text{O}(g) + 2e^- \longrightarrow \text{O}^{2-}(g)$	+600
$\text{Sr}(s) + \frac{1}{2}\text{O}_2(g) \longrightarrow \text{SrO}(s)$	-550

**Question 7 10 points**

Consider the molecule given on the left, and answer questions (i) to (vi) using chemical bonding theories that we have learnt in class (valence bond theory, hybridization).

Note that the lone pairs are not shown in the structure:

i. Consider the carbon atoms marked in as ***a***, ***b***, ***c***, ***d***, and ***e*** in the molecule, and determine **hybridization** at each of the carbon centers. (2.5 points)

Atom	Hybridization
Carbon (<i>a</i>)	
Carbon (<i>b</i>)	
Carbon (<i>c</i>)	
Carbon (<i>d</i>)	
Carbon (<i>e</i>)	

ii. Now, consider the sulfur atom (S) marked as ***f***, nitrogen (N) atom (***g***), and oxygen(O) atoms (***h*** and ***i***), and determine their **hybridization**. (2 points)

Atom	Hybridization
Sulfur (<i>f</i>)	
Nitrogen (<i>g</i>)	
Oxygen (<i>h</i>)	
Oxygen (<i>i</i>)	

iii. **Describe** all the bonds formed by the carbon atoms ***a*** and ***b***, using sigma and pi designation as appropriate. (2 points)

Atom	Bonds formed (use sigma and pi as appropriate)
Carbon (<i>a</i>)	
Carbon (<i>b</i>)	

iv. How many total sp^3 -s sigma bonds are in the molecule? (1 point)

v. Determine **electronic** and **molecular geometry** at carbon (C) labeled ***e*** and sulfur (S) labeled ***f*** centres. (2 points)

Atom	Electronic Geometry	Molecular Geometry
Carbon (<i>e</i>)		
Sulfur (<i>f</i>)		

vi. Circle the value from the choices given below, that describes the **C-S-C bond angle at sulfur** (S) marked ***f***: (0.5 points)

>90° to <109.5° 109.5° >109.5 to <120° 120° >120° to <180°


Question 8 12 points

Using the **valence shell orbitals only** draw a **complete** molecular orbital (MO) diagram (with no s-p mixing) for the molecule **PS**, in the space provided below. **(6 points)**

Assume that 3s and 3p atomic orbitals will form molecular orbitals like those formed by the combination of 2s and 2p atomic orbitals that we learnt in class, and there is no s-p mixing.

Show **energy arrow**, **atomic orbitals** involved, **atom labels**, the **molecular orbitals** that are formed, **filled electrons** in the MO diagram etc. Make sure to label all the atomic and molecular orbitals.

Using the MO diagram above, answer the following questions:

- (i) Which one of the following will have the highest bond order? Circle the correct choice
 A) PS^{1-} B) PS C) PS^{1+}
- (ii) Which one of the following will have the longest bond length? Circle the correct choice
 A) PS^{1-} B) PS C) PS^{1+}
- (iii) Which of the following will be diamagnetic? Circle the correct choice.
 A) PS^{1-} B) PS C) PS^{1+}
- (iv) Draw Lewis dot structure of the molecule with the shortest bond length (based on the MO diagram) from among the three (PS^{1-} , PS, PS^{1+}), and indicate the bond order calculated from the Lewis dot structure. **(3 points)**

Lewis Dot Structure

Bond Order



74C8AADD-9151-4BA7-A97C-6A1C89FC6A17

midterm-exam-93388

#455 Page 10 of 10

SCRAP PAGE

**CHEMISTRY 110 –Midterm Exam****Thursday November 7, 2024, 6:30 – 8:30 pm****Examiners: Pallavi Sirjoosingh and Ashok Kakkar**

First name (please write as legibly as possible within the boxes)

--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--

Last name

--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--

Student ID number

--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--

This booklet has 8 questions on 10 pages (The last page can be used as a scrap page).

The staple must NOT be removed from the question booklet. Your intact question booklet must be returned at the end of the exam.

A separate booklet contains a periodic table, and datasheet.

Write your answers directly under the questions in the space provided. Do not write on the QR codes or near the top edges (as it may be cut off).

Please write **legibly and dark enough** for the scan to be readable. For students who use English as a second language, dictionaries are permitted. Calculators are allowed.

DO NOT ASK QUESTIONS OF THE INVIGILATORS about interpretations of the exam questions. If you have a concern about a question, please summarize it at the top of this page. All concerns will be evaluated after the exam. GOOD LUCK!

**Question 1 8 points** (*Assume all transitions are possible*)

a. 6 points Calculate the **energy (in kJ/mol)** required to ionize (completely remove) the electron from the **n=3** level of **He⁺ ion** (He: Helium). Show your work including equations. (3 sig. figs.)

b. 2 points

(I) Select the largest (longest) wavelength from the three choices below:

- i. Wavelength of photon to *completely remove* an electron from *1s orbital of Hydrogen atom*
- ii. Wavelength of photon to *completely remove* an electron from *1s orbital of Boron atom (B)*
- iii. Wavelength of photon to *completely remove* an electron from *1s orbital of Boron cation (B⁺)*

i or ii or iii

(II) Select the smallest (shortest) wavelength from the three choices below (*for H atom*):

- i. Wavelength of photon *emitted* for an electron transitioning from *2p to 1s orbital*
- ii. Wavelength of photon *absorbed* for an electron transitioning from *2s to 4p orbital*
- iii. Wavelength of photon *absorbed* for an electron transitioning from *1s to 3p orbital*

i or ii or iii



Question 2 8 points



A Chem110 student plots the graph shown on the left, from data obtained for the kinetic energy (eV) of the ejected electron versus frequency (s^{-1}) of the incident light, for an unknown metal. ($1\text{eV} = 1.6022 \times 10^{-19} \text{ J}$)

Based on the plot, answer the questions **a to c** below:

a. 1 point On the plot above, label the **threshold frequency**.

b. 2 points What is the value of the work potential (in J) of the unknown metal?

c. 3 points What is the *longest* wavelength (in nm) required to eject an electron from this metal's surface? Show your work including equation(s) used. (3 sig figs)

d. 2 points In the space provided below, plot the following

(Assume frequency is *above* threshold frequency):

- Kinetic energy* of an ejected electron versus the *intensity* of the incident light (left)
- Number of electrons* ejected versus the *frequency* of the incident light (right)



**Question 3** **6 points****a. 2 points**

In the ground state of Ge (Germanium; $Z = 32$) atom, **name** the following orbitals:

i. Orbital containing the electron that requires the maximum energy to be removed.

ii. Orbital containing the electron that experiences the minimum effective nuclear charge.

b. 2 points

An unknown anion (A^-) has the following *excited* electronic configuration: $[Ar] 4s^2 3d^3 4p^1$

(Ar: Argon; $Z = 18$)

Give the **condensed ground state electronic configuration** of the atom, **A** and the cation, **A^{2+}** .
Show your work. (Assume all electronic transitions are possible)

Electronic Configuration of A:

Electronic Configuration of A^{2+} :

c. 2 points

In a single atom, determine how many electrons can have the following set of quantum numbers?

(i) $n = 5; l = 3$

In a single atom, determine how many electrons can have the following set of quantum numbers?

(i) $n = 3; l = 2; m_s = +1/2$

**Question 4** **8 points**

Rank the following from 1 to 4 in **increasing** order. Provide the ranking in the boxes provided.

Note: Atomic number (Z) are provided for relevant atoms

a) Increasing (lowest to highest) order of *ionization energy*: (1= lowest; 4 = highest)

Mg

Mg: Z = 12

Rb

Rb: Z = 37

Cs

Cs: Z = 55

K

K : Z = 19

b) Increasing (lowest to highest) order of *bond angle*: (1= lowest; 4 = highest)

H₂OCCl₄XeF₄

HCN

c) Increasing (smallest to largest) order of *ionic size* (1= smallest; 4 = largest)

S²⁻

S: Z = 16

O²⁻

O: Z = 8

Be²⁺

Be: Z = 4

Na⁺

Na: Z = 11

d) Increasing (lowest to highest) order of *bond energy* (1= lowest; 4 = highest)

P-F

P-Cl

P-I

P-Br

**Question 5** **12 points**

a. 9 points For the molecules HClO_4 and TeF_6 , draw the most stable *Lewis structure* showing appropriate shape and *dashed/wedged bonds*, based on VSEPR. Determine the **molecular geometry** around the central atom. For HClO_4 , label the *most polar bond*.



b. 3 points (Note: Lone pairs are not shown in structures below)

(i) For the three *resonance* structures (1, 2 and 3) shown below, answer the following question:



(I) Circle the *most contributing* resonance structure.

Structure 1

Structure 2

Structure 3

(ii) For the three structures given (4, 5, and 6) below, answer the following questions:



(I) Circle all the structures that represent *resonance structures* of each other:

Structure 4

Structure 5

Structure 6

(II) Among the resonance structures, circle the *most contributing* resonance structure.

Structure 4

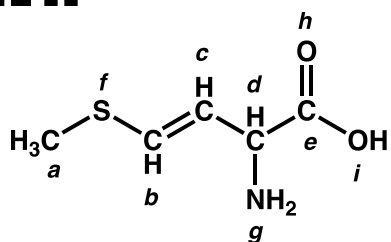
Structure 5

Structure 6

**Question 6** **6 points**

Using the data given below, calculate the **lattice energy** for SrO(s).

	ΔH° (kJ/mol)
$\text{Sr}(s) \longrightarrow \text{Sr}(g)$	+180
$\text{Sr}(g) \longrightarrow \text{Sr}^{2+}(g) + 2e^-$	+1468
$\text{O}_2(g) \longrightarrow 2\text{O}(g)$	+500
$\text{O}(g) + 2e^- \longrightarrow \text{O}^{2-}(g)$	+600
$\text{Sr}(s) + \frac{1}{2}\text{O}_2(g) \longrightarrow \text{SrO}(s)$	-550

**Question 7 10 points**

Consider the molecule given on the left, and answer questions (i) to (vi) using chemical bonding theories that we have learnt in class (valence bond theory, hybridization).

Note that the lone pairs are not shown in the structure:

i. Consider the carbon atoms marked in as ***a***, ***b***, ***c***, ***d***, and ***e*** in the molecule, and determine **hybridization** at each of the carbon centers. (2.5 points)

Atom	Hybridization
Carbon (<i>a</i>)	
Carbon (<i>b</i>)	
Carbon (<i>c</i>)	
Carbon (<i>d</i>)	
Carbon (<i>e</i>)	

ii. Now, consider the sulfur atom (S) marked as ***f***, nitrogen (N) atom (***g***), and oxygen(O) atoms (***h*** and ***i***), and determine their **hybridization**. (2 points)

Atom	Hybridization
Sulfur (<i>f</i>)	
Nitrogen (<i>g</i>)	
Oxygen (<i>h</i>)	
Oxygen (<i>i</i>)	

iii. **Describe** all the bonds formed by the carbon atoms ***a*** and ***b***, using sigma and pi designation as appropriate. (2 points)

Atom	Bonds formed (use sigma and pi as appropriate)
Carbon (<i>a</i>)	
Carbon (<i>b</i>)	

iv. How many total sp^3 -s sigma bonds are in the molecule? (1 point)

v. Determine **electronic** and **molecular geometry** at carbon (C) labeled ***e*** and sulfur (S) labeled ***f*** centres. (2 points)

Atom	Electronic Geometry	Molecular Geometry
Carbon (<i>e</i>)		
Sulfur (<i>f</i>)		

vi. Circle the value from the choices given below, that describes the **C-S-C bond angle at sulfur** (S) marked ***f***: (0.5 points)

>90° to <109.5°

109.5°

>109.5 to <120°

120°

>120° to <180°


Question 8 12 points

Using the **valence shell orbitals only** draw a **complete** molecular orbital (MO) diagram (with no s-p mixing) for the molecule **PS**, in the space provided below. **(6 points)**

Assume that 3s and 3p atomic orbitals will form molecular orbitals like those formed by the combination of 2s and 2p atomic orbitals that we learnt in class, and there is no s-p mixing.

Show **energy arrow**, **atomic orbitals** involved, **atom labels**, the **molecular orbitals** that are formed, **filled electrons** in the MO diagram etc. Make sure to label all the atomic and molecular orbitals.

Using the MO diagram above, answer the following questions:

- (i) Which one of the following will have the highest bond order? Circle the correct choice
 A) PS^{1-} B) PS C) PS^{1+}
- (ii) Which one of the following will have the longest bond length? Circle the correct choice
 A) PS^{1-} B) PS C) PS^{1+}
- (iii) Which of the following will be diamagnetic? Circle the correct choice.
 A) PS^{1-} B) PS C) PS^{1+}
- (iv) Draw Lewis dot structure of the molecule with the shortest bond length (based on the MO diagram) from among the three (PS^{1-} , PS, PS^{1+}), and indicate the bond order calculated from the Lewis dot structure. **(3 points)**

Lewis Dot Structure

Bond Order



00CF52AD-A8FF-4215-BEF2-49AB354E6FE1

midterm-exam-93388

#456 Page 10 of 10

SCRAP PAGE

**CHEMISTRY 110 –Midterm Exam****Thursday November 7, 2024, 6:30 – 8:30 pm**

Examiners: Pallavi Sirjoosingh and Ashok Kakkar

First name (please write as legibly as possible within the boxes)

--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--

Last name

--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--

Student ID number

--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--

This booklet has 8 questions on 10 pages (The last page can be used as a scrap page).

The staple must NOT be removed from the question booklet. Your intact question booklet must be returned at the end of the exam.

A separate booklet contains a periodic table, and datasheet.

Write your answers directly under the questions in the space provided. Do not write on the QR codes or near the top edges (as it may be cut off).

Please write **legibly and dark enough** for the scan to be readable. For students who use English as a second language, dictionaries are permitted. Calculators are allowed.

DO NOT ASK QUESTIONS OF THE INVIGILATORS about interpretations of the exam questions. If you have a concern about a question, please summarize it at the top of this page. All concerns will be evaluated after the exam. GOOD LUCK!

**Question 1 8 points** (*Assume all transitions are possible*)

a. 6 points Calculate the **energy (in kJ/mol)** required to ionize (completely remove) the electron from the **n=3** level of **He⁺ ion** (He: Helium). Show your work including equations. (3 sig. figs.)

b. 2 points

(I) Select the largest (longest) wavelength from the three choices below:

- i. Wavelength of photon to *completely remove* an electron from *1s orbital of Hydrogen atom*
- ii. Wavelength of photon to *completely remove* an electron from *1s orbital of Boron atom (B)*
- iii. Wavelength of photon to *completely remove* an electron from *1s orbital of Boron cation (B⁺)*

i or ii or iii

(II) Select the smallest (shortest) wavelength from the three choices below (*for H atom*):

- i. Wavelength of photon *emitted* for an electron transitioning from *2p to 1s orbital*
- ii. Wavelength of photon *absorbed* for an electron transitioning from *2s to 4p orbital*
- iii. Wavelength of photon *absorbed* for an electron transitioning from *1s to 3p orbital*

i or ii or iii



Question 2 8 points



A Chem110 student plots the graph shown on the left, from data obtained for the kinetic energy (eV) of the ejected electron versus frequency (s^{-1}) of the incident light, for an unknown metal. ($1\text{eV} = 1.6022 \times 10^{-19} \text{ J}$)

Based on the plot, answer the questions **a to c** below:

a. 1 point On the plot above, label the **threshold frequency**.

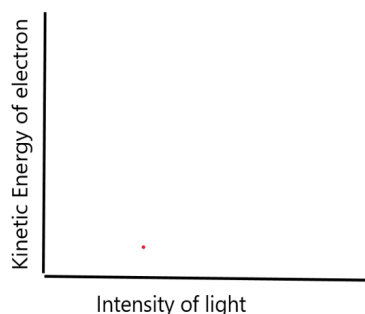
b. 2 points What is the value of the work potential (in J) of the unknown metal?

c. 3 points What is the *longest* wavelength (in nm) required to eject an electron from this metal's surface? Show your work including equation(s) used. (3 sig figs)

d. 2 points In the space provided below, plot the following

(Assume frequency is *above* threshold frequency):

- Kinetic energy* of an ejected electron versus the *intensity* of the incident light (left)
- Number of electrons* ejected versus the *frequency* of the incident light (right)



**Question 3** **6 points****a. 2 points**

In the ground state of Ge (Germanium; $Z = 32$) atom, **name** the following orbitals:

i. Orbital containing the electron that requires the maximum energy to be removed.

ii. Orbital containing the electron that experiences the minimum effective nuclear charge.

b. 2 points

An unknown anion (A^-) has the following *excited* electronic configuration: $[Ar] 4s^2 3d^3 4p^1$

(Ar: Argon; $Z = 18$)

Give the **condensed ground state electronic configuration** of the atom, **A** and the cation, **A^{2+}** .
Show your work. (Assume all electronic transitions are possible)

Electronic Configuration of A:

Electronic Configuration of A^{2+} :

c. 2 points

In a single atom, determine how many electrons can have the following set of quantum numbers?

(i) $n = 5; l = 3$

In a single atom, determine how many electrons can have the following set of quantum numbers?

(i) $n = 3; l = 2; m_s = +1/2$

**Question 4** **8 points**

Rank the following from 1 to 4 in **increasing** order. Provide the ranking in the boxes provided.

Note: Atomic number (Z) are provided for relevant atoms

a) Increasing (lowest to highest) order of *ionization energy*: (1= lowest; 4 = highest)

Mg

Mg: Z = 12

Rb

Rb: Z = 37

Cs

Cs: Z = 55

K

K : Z = 19

b) Increasing (lowest to highest) order of *bond angle*: (1= lowest; 4 = highest)

H₂OCCl₄XeF₄

HCN

c) Increasing (smallest to largest) order of *ionic size* (1= smallest; 4 = largest)

S²⁻

S: Z = 16

O²⁻

O: Z = 8

Be²⁺

Be: Z = 4

Na⁺

Na: Z = 11

d) Increasing (lowest to highest) order of *bond energy* (1= lowest; 4 = highest)

P-F

P-Cl

P-I

P-Br

**Question 5** **12 points**

a. 9 points For the molecules HClO_4 and TeF_6 , draw the most stable *Lewis structure* showing appropriate shape and *dashed/wedged bonds*, based on VSEPR. Determine the **molecular geometry** around the central atom. For HClO_4 , label the *most polar bond*.



b. 3 points (Note: Lone pairs are not shown in structures below)

(i) For the three *resonance* structures (1, 2 and 3) shown below, answer the following question:



(I) Circle the *most contributing* resonance structure.

Structure 1

Structure 2

Structure 3

(ii) For the three structures given (4, 5, and 6) below, answer the following questions:



(I) Circle all the structures that represent *resonance structures* of each other:

Structure 4

Structure 5

Structure 6

(II) Among the resonance structures, circle the *most contributing* resonance structure.

Structure 4

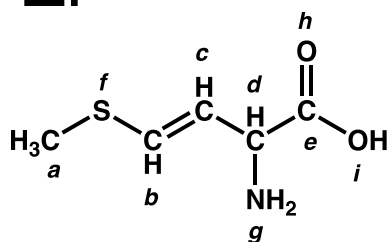
Structure 5

Structure 6

**Question 6** **6 points**

Using the data given below, calculate the **lattice energy** for SrO(s).

	ΔH° (kJ/mol)
$\text{Sr}(s) \longrightarrow \text{Sr}(g)$	+180
$\text{Sr}(g) \longrightarrow \text{Sr}^{2+}(g) + 2e^-$	+1468
$\text{O}_2(g) \longrightarrow 2\text{O}(g)$	+500
$\text{O}(g) + 2e^- \longrightarrow \text{O}^{2-}(g)$	+600
$\text{Sr}(s) + \frac{1}{2}\text{O}_2(g) \longrightarrow \text{SrO}(s)$	-550

**Question 7 10 points**

Consider the molecule given on the left, and answer questions (i) to (vi) using chemical bonding theories that we have learnt in class (valence bond theory, hybridization).

Note that the lone pairs are not shown in the structure:

i. Consider the carbon atoms marked in as ***a***, ***b***, ***c***, ***d***, and ***e*** in the molecule, and determine **hybridization** at each of the carbon centers. (2.5 points)

Atom	Hybridization
Carbon (<i>a</i>)	
Carbon (<i>b</i>)	
Carbon (<i>c</i>)	
Carbon (<i>d</i>)	
Carbon (<i>e</i>)	

ii. Now, consider the sulfur atom (S) marked as ***f***, nitrogen (N) atom (***g***), and oxygen(O) atoms (***h*** and ***i***), and determine their **hybridization**. (2 points)

Atom	Hybridization
Sulfur (<i>f</i>)	
Nitrogen (<i>g</i>)	
Oxygen (<i>h</i>)	
Oxygen (<i>i</i>)	

iii. **Describe** all the bonds formed by the carbon atoms ***a*** and ***b***, using sigma and pi designation as appropriate. (2 points)

Atom	Bonds formed (use sigma and pi as appropriate)
Carbon (<i>a</i>)	
Carbon (<i>b</i>)	

iv. How many total sp^3 -s sigma bonds are in the molecule? (1 point)

v. Determine **electronic** and **molecular geometry** at carbon (C) labeled ***e*** and sulfur (S) labeled ***f*** centres. (2 points)

Atom	Electronic Geometry	Molecular Geometry
Carbon (<i>e</i>)		
Sulfur (<i>f</i>)		

vi. Circle the value from the choices given below, that describes the **C-S-C bond angle at sulfur** (S) marked ***f***: (0.5 points)

>90° to <109.5° 109.5° >109.5 to <120° 120° >120° to <180°



Question 8 12 points

Using the **valence shell orbitals only** draw a **complete** molecular orbital (MO) diagram (with no s-p mixing) for the molecule **PS**, in the space provided below. **(6 points)**

Assume that 3s and 3p atomic orbitals will form molecular orbitals like those formed by the combination of 2s and 2p atomic orbitals that we learnt in class, and there is no s-p mixing.

Show **energy arrow**, **atomic orbitals** involved, **atom labels**, the **molecular orbitals** that are formed, **filled electrons** in the MO diagram etc. Make sure to label all the atomic and molecular orbitals.

Using the MO diagram above, answer the following questions:

- (i) Which one of the following will have the highest bond order? Circle the correct choice
 A) PS^{1-} B) PS C) PS^{1+}
- (ii) Which one of the following will have the longest bond length? Circle the correct choice
 A) PS^{1-} B) PS C) PS^{1+}
- (iii) Which of the following will be diamagnetic? Circle the correct choice.
 A) PS^{1-} B) PS C) PS^{1+}
- (iv) Draw Lewis dot structure of the molecule with the shortest bond length (based on the MO diagram) from among the three (PS^{1-} , PS, PS^{1+}), and indicate the bond order calculated from the Lewis dot structure. **(3 points)**

Lewis Dot Structure

Bond Order



9E31167E-F27E-47B3-A00F-74F09478AA64

midterm-exam-93388

#457 Page 10 of 10

SCRAP PAGE

**CHEMISTRY 110 –Midterm Exam****Thursday November 7, 2024, 6:30 – 8:30 pm****Examiners: Pallavi Sirjoosingh and Ashok Kakkar**

First name (please write as legibly as possible within the boxes)

--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--

Last name

--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--

Student ID number

--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--

This booklet has 8 questions on 10 pages (The last page can be used as a scrap page).

The staple must NOT be removed from the question booklet. Your intact question booklet must be returned at the end of the exam.

A separate booklet contains a periodic table, and datasheet.

Write your answers directly under the questions in the space provided. Do not write on the QR codes or near the top edges (as it may be cut off).

Please write **legibly and dark enough** for the scan to be readable. For students who use English as a second language, dictionaries are permitted. Calculators are allowed.

DO NOT ASK QUESTIONS OF THE INVIGILATORS about interpretations of the exam questions. If you have a concern about a question, please summarize it at the top of this page. All concerns will be evaluated after the exam. GOOD LUCK!

**Question 1 8 points** (*Assume all transitions are possible*)

a. 6 points Calculate the **energy (in kJ/mol)** required to ionize (completely remove) the electron from the **n=3** level of **He⁺ ion** (He: Helium). Show your work including equations. (3 sig. figs.)

b. 2 points

(I) Select the largest (longest) wavelength from the three choices below:

- i. Wavelength of photon to *completely remove* an electron from *1s orbital of Hydrogen atom*
- ii. Wavelength of photon to *completely remove* an electron from *1s orbital of Boron atom (B)*
- iii. Wavelength of photon to *completely remove* an electron from *1s orbital of Boron cation (B⁺)*

i or ii or iii

(II) Select the smallest (shortest) wavelength from the three choices below (*for H atom*):

- i. Wavelength of photon *emitted* for an electron transitioning from *2p to 1s orbital*
- ii. Wavelength of photon *absorbed* for an electron transitioning from *2s to 4p orbital*
- iii. Wavelength of photon *absorbed* for an electron transitioning from *1s to 3p orbital*

i or ii or iii



Question 2 8 points



A Chem110 student plots the graph shown on the left, from data obtained for the kinetic energy (eV) of the ejected electron versus frequency (s^{-1}) of the incident light, for an unknown metal. ($1\text{eV} = 1.6022 \times 10^{-19} \text{ J}$)

Based on the plot, answer the questions **a to c** below:

a. 1 point On the plot above, label the **threshold frequency**.

b. 2 points What is the value of the work potential (in J) of the unknown metal?

c. 3 points What is the *longest* wavelength (in nm) required to eject an electron from this metal's surface? Show your work including equation(s) used. (3 sig figs)

d. 2 points In the space provided below, plot the following

(Assume frequency is *above* threshold frequency):

- i. *Kinetic energy* of an ejected electron versus the *intensity* of the incident light (left)
- ii. *Number of electrons* ejected versus the *frequency* of the incident light (right)



**Question 3** **6 points****a. 2 points**

In the ground state of Ge (Germanium; $Z = 32$) atom, **name** the following orbitals:

i. Orbital containing the electron that requires the maximum energy to be removed.

ii. Orbital containing the electron that experiences the minimum effective nuclear charge.

b. 2 points

An unknown anion (A^-) has the following *excited* electronic configuration: $[Ar] 4s^2 3d^3 4p^1$

(Ar: Argon; $Z = 18$)

Give the **condensed ground state electronic configuration** of the atom, **A** and the cation, **A^{2+}** .
Show your work. (Assume all electronic transitions are possible)

Electronic Configuration of A:

Electronic Configuration of A^{2+} :

c. 2 points

In a single atom, determine how many electrons can have the following set of quantum numbers?

(i) $n = 5; l = 3$

In a single atom, determine how many electrons can have the following set of quantum numbers?

(i) $n = 3; l = 2; m_s = +1/2$

**Question 4** **8 points**

Rank the following from 1 to 4 in **increasing** order. Provide the ranking in the boxes provided.

Note: Atomic number (Z) are provided for relevant atoms

a) Increasing (lowest to highest) order of *ionization energy*: (1= lowest; 4 = highest)

Mg

Mg: Z = 12

Rb

Rb: Z = 37

Cs

Cs: Z = 55

K

K : Z = 19

b) Increasing (lowest to highest) order of *bond angle*: (1= lowest; 4 = highest)

H₂OCCl₄XeF₄

HCN

c) Increasing (smallest to largest) order of *ionic size* (1= smallest; 4 = largest)

S²⁻

S: Z = 16

O²⁻

O: Z = 8

Be²⁺

Be: Z = 4

Na⁺

Na: Z = 11

d) Increasing (lowest to highest) order of *bond energy* (1= lowest; 4 = highest)

P-F

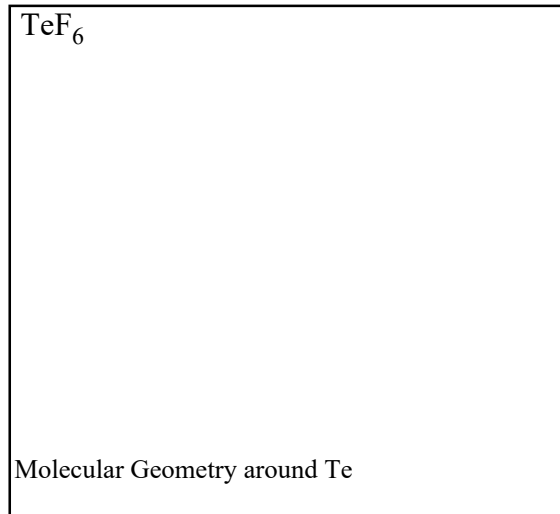
P-Cl

P-I

P-Br

**Question 5** **12 points**

a. 9 points For the molecules HClO_4 and TeF_6 , draw the most stable *Lewis structure* showing appropriate shape and *dashed/wedged bonds*, based on VSEPR. Determine the **molecular geometry** around the central atom. For HClO_4 , label the *most polar bond*.



b. 3 points (Note: Lone pairs are not shown in structures below)

(i) For the three *resonance* structures (1, 2 and 3) shown below, answer the following question:



(I) Circle the *most contributing* resonance structure.

Structure 1

Structure 2

Structure 3

(ii) For the three structures given (4, 5, and 6) below, answer the following questions:



(I) Circle all the structures that represent *resonance structures* of each other:

Structure 4

Structure 5

Structure 6

(II) Among the resonance structures, circle the *most contributing* resonance structure.

Structure 4

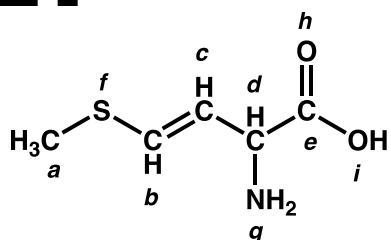
Structure 5

Structure 6

**Question 6** **6 points**

Using the data given below, calculate the **lattice energy** for SrO(s).

	ΔH° (kJ/mol)
$\text{Sr}(s) \longrightarrow \text{Sr}(g)$	+180
$\text{Sr}(g) \longrightarrow \text{Sr}^{2+}(g) + 2e^-$	+1468
$\text{O}_2(g) \longrightarrow 2\text{O}(g)$	+500
$\text{O}(g) + 2e^- \longrightarrow \text{O}^{2-}(g)$	+600
$\text{Sr}(s) + \frac{1}{2}\text{O}_2(g) \longrightarrow \text{SrO}(s)$	-550

**Question 7 10 points**

Consider the molecule given on the left, and answer questions (i) to (vi) using chemical bonding theories that we have learnt in class (valence bond theory, hybridization).

Note that the lone pairs are not shown in the structure:

i. Consider the carbon atoms marked in as ***a***, ***b***, ***c***, ***d***, and ***e*** in the molecule, and determine **hybridization** at each of the carbon centers. (2.5 points)

Atom	Hybridization
Carbon (<i>a</i>)	
Carbon (<i>b</i>)	
Carbon (<i>c</i>)	
Carbon (<i>d</i>)	
Carbon (<i>e</i>)	

ii. Now, consider the sulfur atom (S) marked as ***f***, nitrogen (N) atom (***g***), and oxygen(O) atoms (***h*** and ***i***), and determine their **hybridization**. (2 points)

Atom	Hybridization
Sulfur (<i>f</i>)	
Nitrogen (<i>g</i>)	
Oxygen (<i>h</i>)	
Oxygen (<i>i</i>)	

iii. **Describe** all the bonds formed by the carbon atoms ***a*** and ***b***, using sigma and pi designation as appropriate. (2 points)

Atom	Bonds formed (use sigma and pi as appropriate)
Carbon (<i>a</i>)	
Carbon (<i>b</i>)	

iv. How many total sp^3 -s sigma bonds are in the molecule? (1 point)

v. Determine **electronic** and **molecular geometry** at carbon (C) labeled ***e*** and sulfur (S) labeled ***f*** centres. (2 points)

Atom	Electronic Geometry	Molecular Geometry
Carbon (<i>e</i>)		
Sulfur (<i>f</i>)		

vi. Circle the value from the choices given below, that describes the **C-S-C bond angle at sulfur** (S) marked ***f***: (0.5 points)

$>90^\circ$ to $<109.5^\circ$ 109.5° >109.5 to $<120^\circ$ 120° $>120^\circ$ to $<180^\circ$


Question 8 12 points

Using the **valence shell orbitals only** draw a **complete** molecular orbital (MO) diagram (with no s-p mixing) for the molecule **PS**, in the space provided below. **(6 points)**

Assume that 3s and 3p atomic orbitals will form molecular orbitals like those formed by the combination of 2s and 2p atomic orbitals that we learnt in class, and there is no s-p mixing.

Show **energy arrow**, **atomic orbitals** involved, **atom labels**, the **molecular orbitals** that are formed, **filled electrons** in the MO diagram etc. Make sure to label all the atomic and molecular orbitals.

Using the MO diagram above, answer the following questions:

- (i) Which one of the following will have the highest bond order? Circle the correct choice
 A) PS^{1-} B) PS C) PS^{1+}
- (ii) Which one of the following will have the longest bond length? Circle the correct choice
 A) PS^{1-} B) PS C) PS^{1+}
- (iii) Which of the following will be diamagnetic? Circle the correct choice.
 A) PS^{1-} B) PS C) PS^{1+}
- (iv) Draw Lewis dot structure of the molecule with the shortest bond length (based on the MO diagram) from among the three (PS^{1-} , PS, PS^{1+}), and indicate the bond order calculated from the Lewis dot structure. **(3 points)**

Lewis Dot Structure

Bond Order



B5CD8FEE-AB05-4D52-8576-C0B4A59014C3

midterm-exam-93388

#458 Page 10 of 10

SCRAP PAGE

**CHEMISTRY 110 –Midterm Exam****Thursday November 7, 2024, 6:30 – 8:30 pm****Examiners: Pallavi Sirjoosingh and Ashok Kakkar**

First name (please write as legibly as possible within the boxes)

--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--

Last name

--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--

Student ID number

--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--

This booklet has 8 questions on 10 pages (The last page can be used as a scrap page).

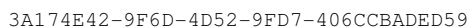
The staple must NOT be removed from the question booklet. Your intact question booklet must be returned at the end of the exam.

A separate booklet contains a periodic table, and datasheet.

Write your answers directly under the questions in the space provided. Do not write on the QR codes or near the top edges (as it may be cut off).

Please write **legibly and dark enough** for the scan to be readable. For students who use English as a second language, dictionaries are permitted. Calculators are allowed.

DO NOT ASK QUESTIONS OF THE INVIGILATORS about interpretations of the exam questions. If you have a concern about a question, please summarize it at the top of this page. All concerns will be evaluated after the exam. GOOD LUCK!



midterm-exam-93388

#459 Page 2 of 10

a. 6 points Calculate the **energy (in kJ/mol)** required to ionize (completely remove) the electron from the **n=3** level of **He⁺ ion** (He: Helium). Show your work including equations. (3 sig. figs.)

(I) Select the largest (longest) wavelength from the three choices below:

- Wavelength of photon to *completely remove* an electron from *1s orbital of Hydrogen atom*
- Wavelength of photon to *completely remove* an electron from *1s orbital of Boron atom (B)*
- Wavelength of photon to *completely remove* an electron from *1s orbital of Boron cation (B⁺)*

i
or
ii
or
iii

(II) Select the smallest (shortest) wavelength from the three choices below (*for H atom*):

- Wavelength of photon *emitted* for an electron transitioning from $2p$ to $1s$ orbital
- Wavelength of photon *absorbed* for an electron transitioning from $2s$ to $4p$ orbital
- Wavelength of photon *absorbed* for an electron transitioning from $1s$ to $3p$ orbital

i or **ii** or **iii**



Question 2 8 points



A Chem110 student plots the graph shown on the left, from data obtained for the kinetic energy (eV) of the ejected electron versus frequency (s^{-1}) of the incident light, for an unknown metal. ($1\text{eV} = 1.6022 \times 10^{-19} \text{ J}$)

Based on the plot, answer the questions **a to c** below:

a. 1 point On the plot above, label the **threshold frequency**.

b. 2 points What is the value of the work potential (in J) of the unknown metal?

c. 3 points What is the *longest* wavelength (in nm) required to eject an electron from this metal's surface? Show your work including equation(s) used. (3 sig figs)

d. 2 points In the space provided below, plot the following

(Assume frequency is *above* threshold frequency):

- i. *Kinetic energy* of an ejected electron versus the *intensity* of the incident light (left)
- ii. *Number of electrons* ejected versus the *frequency* of the incident light (right)



**Question 3** **6 points****a. 2 points**

In the ground state of Ge (Germanium; $Z = 32$) atom, **name** the following orbitals:

i. Orbital containing the electron that requires the maximum energy to be removed.

ii. Orbital containing the electron that experiences the minimum effective nuclear charge.

b. 2 points

An unknown anion (A^-) has the following *excited* electronic configuration: $[Ar] 4s^2 3d^3 4p^1$

(Ar: Argon; $Z = 18$)

Give the **condensed ground state electronic configuration** of the atom, **A** and the cation, **A^{2+}** .
Show your work. (Assume all electronic transitions are possible)

Electronic Configuration of A:

Electronic Configuration of A^{2+} :

c. 2 points

In a single atom, determine how many electrons can have the following set of quantum numbers?

(i) $n = 5; l = 3$

In a single atom, determine how many electrons can have the following set of quantum numbers?

(i) $n = 3; l = 2; m_s = +1/2$

**Question 4** **8 points**

Rank the following from 1 to 4 in **increasing** order. Provide the ranking in the boxes provided.

Note: Atomic number (Z) are provided for relevant atoms

a) Increasing (lowest to highest) order of *ionization energy*: (1= lowest; 4 = highest)

Mg

Mg: Z = 12

Rb

Rb: Z = 37

Cs

Cs: Z = 55

K

K : Z = 19

b) Increasing (lowest to highest) order of *bond angle*: (1= lowest; 4 = highest)

H₂OCCl₄XeF₄

HCN

c) Increasing (smallest to largest) order of *ionic size* (1= smallest; 4 = largest)

S²⁻

S: Z = 16

O²⁻

O: Z = 8

Be²⁺

Be: Z = 4

Na⁺

Na: Z = 11

d) Increasing (lowest to highest) order of *bond energy* (1= lowest; 4 = highest)

P-F

P-Cl

P-I

P-Br

**Question 5** **12 points**

a. 9 points For the molecules HClO_4 and TeF_6 , draw the most stable *Lewis structure* showing appropriate shape and *dashed/wedged bonds*, based on VSEPR. Determine the **molecular geometry** around the central atom. For HClO_4 , label the *most polar bond*.



b. 3 points (Note: Lone pairs are not shown in structures below)

(i) For the three *resonance* structures (1, 2 and 3) shown below, answer the following question:



(I) Circle the *most contributing* resonance structure.

Structure 1

Structure 2

Structure 3

(ii) For the three structures given (4, 5, and 6) below, answer the following questions:



(I) Circle all the structures that represent *resonance structures* of each other:

Structure 4

Structure 5

Structure 6

(II) Among the resonance structures, circle the *most contributing* resonance structure.

Structure 4

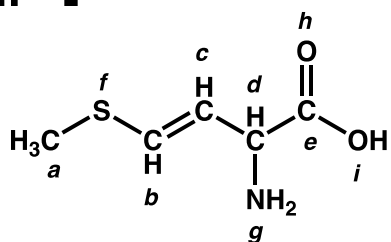
Structure 5

Structure 6

**Question 6** **6 points**

Using the data given below, calculate the **lattice energy** for SrO(s).

	ΔH° (kJ/mol)
$\text{Sr}(s) \longrightarrow \text{Sr}(g)$	+180
$\text{Sr}(g) \longrightarrow \text{Sr}^{2+}(g) + 2e^-$	+1468
$\text{O}_2(g) \longrightarrow 2\text{O}(g)$	+500
$\text{O}(g) + 2e^- \longrightarrow \text{O}^{2-}(g)$	+600
$\text{Sr}(s) + \frac{1}{2}\text{O}_2(g) \longrightarrow \text{SrO}(s)$	-550

**Question 7 10 points**

Consider the molecule given on the left, and answer questions (i) to (vi) using chemical bonding theories that we have learnt in class (valence bond theory, hybridization).

Note that the lone pairs are not shown in the structure:

i. Consider the carbon atoms marked in as ***a***, ***b***, ***c***, ***d***, and ***e*** in the molecule, and determine **hybridization** at each of the carbon centers. (2.5 points)

Atom	Hybridization
Carbon (<i>a</i>)	
Carbon (<i>b</i>)	
Carbon (<i>c</i>)	
Carbon (<i>d</i>)	
Carbon (<i>e</i>)	

ii. Now, consider the sulfur atom (S) marked as ***f***, nitrogen (N) atom (***g***), and oxygen(O) atoms (***h*** and ***i***), and determine their **hybridization**. (2 points)

Atom	Hybridization
Sulfur (<i>f</i>)	
Nitrogen (<i>g</i>)	
Oxygen (<i>h</i>)	
Oxygen (<i>i</i>)	

iii. **Describe** all the bonds formed by the carbon atoms ***a*** and ***b***, using sigma and pi designation as appropriate. (2 points)

Atom	Bonds formed (use sigma and pi as appropriate)
Carbon (<i>a</i>)	
Carbon (<i>b</i>)	

iv. How many total sp^3 -s sigma bonds are in the molecule? (1 point)

v. Determine **electronic** and **molecular geometry** at carbon (C) labeled ***e*** and sulfur (S) labeled ***f*** centres. (2 points)

Atom	Electronic Geometry	Molecular Geometry
Carbon (<i>e</i>)		
Sulfur (<i>f</i>)		

vi. Circle the value from the choices given below, that describes the **C-S-C bond angle at sulfur** (S) marked ***f***: (0.5 points)

>90° to <109.5° 109.5° >109.5 to <120° 120° >120° to <180°

**Question 8 12 points**

Using the **valence shell orbitals only** draw a **complete** molecular orbital (MO) diagram (with no s-p mixing) for the molecule **PS**, in the space provided below. **(6 points)**

Assume that 3s and 3p atomic orbitals will form molecular orbitals like those formed by the combination of 2s and 2p atomic orbitals that we learnt in class, and there is no s-p mixing.

Show **energy arrow**, **atomic orbitals** involved, **atom labels**, the **molecular orbitals** that are formed, **filled electrons** in the MO diagram etc. Make sure to label all the atomic and molecular orbitals.

Using the MO diagram above, answer the following questions:

- (i) Which one of the following will have the highest bond order? Circle the correct choice
A) PS^{1-} B) PS C) PS^{1+}
- (ii) Which one of the following will have the longest bond length? Circle the correct choice
A) PS^{1-} B) PS C) PS^{1+}
- (iii) Which of the following will be diamagnetic? Circle the correct choice.
A) PS^{1-} B) PS C) PS^{1+}
- (iv) Draw Lewis dot structure of the molecule with the shortest bond length (based on the MO diagram) from among the three (PS^{1-} , PS, PS^{1+}), and indicate the bond order calculated from the Lewis dot structure. **(3 points)**

Lewis Dot Structure

Bond Order



7D50E8DE-F169-4967-A9ED-7CE9052666B9

midterm-exam-93388

#459 Page 10 of 10

SCRAP PAGE

**CHEMISTRY 110 –Midterm Exam****Thursday November 7, 2024, 6:30 – 8:30 pm****Examiners: Pallavi Sirjoosingh and Ashok Kakkar**

First name (please write as legibly as possible within the boxes)

--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--

Last name

--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--

Student ID number

--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--

This booklet has 8 questions on 10 pages (The last page can be used as a scrap page).

The staple must NOT be removed from the question booklet. Your intact question booklet must be returned at the end of the exam.

A separate booklet contains a periodic table, and datasheet.

Write your answers directly under the questions in the space provided. Do not write on the QR codes or near the top edges (as it may be cut off).

Please write **legibly and dark enough** for the scan to be readable. For students who use English as a second language, dictionaries are permitted. Calculators are allowed.

DO NOT ASK QUESTIONS OF THE INVIGILATORS about interpretations of the exam questions. If you have a concern about a question, please summarize it at the top of this page. All concerns will be evaluated after the exam. GOOD LUCK!

**Question 1 8 points** (*Assume all transitions are possible*)

a. 6 points Calculate the **energy (in kJ/mol)** required to ionize (completely remove) the electron from the **$n=3$** level of **He^+ ion** (He: Helium). Show your work including equations. (3 sig. figs.)

b. 2 points

(I) Select the largest (longest) wavelength from the three choices below:

- i. Wavelength of photon to *completely remove* an electron from *1s orbital of Hydrogen atom*
- ii. Wavelength of photon to *completely remove* an electron from *1s orbital of Boron atom (B)*
- iii. Wavelength of photon to *completely remove* an electron from *1s orbital of Boron cation (B^+)*

i or ii or iii

(II) Select the smallest (shortest) wavelength from the three choices below (*for H atom*):

- i. Wavelength of photon *emitted* for an electron transitioning from *2p to 1s orbital*
- ii. Wavelength of photon *absorbed* for an electron transitioning from *2s to 4p orbital*
- iii. Wavelength of photon *absorbed* for an electron transitioning from *1s to 3p orbital*

i or ii or iii



Question 2 8 points



A Chem110 student plots the graph shown on the left, from data obtained for the kinetic energy (eV) of the ejected electron versus frequency (s^{-1}) of the incident light, for an unknown metal. ($1\text{eV} = 1.6022 \times 10^{-19} \text{ J}$)

Based on the plot, answer the questions **a to c** below:

a. 1 point On the plot above, label the **threshold frequency**.

b. 2 points What is the value of the work potential (in J) of the unknown metal?

c. 3 points What is the *longest* wavelength (in nm) required to eject an electron from this metal's surface? Show your work including equation(s) used. (3 sig figs)

d. 2 points In the space provided below, plot the following

(Assume frequency is *above* threshold frequency):

- i. *Kinetic energy* of an ejected electron versus the *intensity* of the incident light (left)
- ii. *Number of electrons* ejected versus the *frequency* of the incident light (right)



**Question 3** **6 points****a. 2 points**

In the ground state of Ge (Germanium; $Z = 32$) atom, **name** the following orbitals:

i. Orbital containing the electron that requires the maximum energy to be removed.

ii. Orbital containing the electron that experiences the minimum effective nuclear charge.

b. 2 points

An unknown anion (A^-) has the following *excited* electronic configuration: $[Ar] 4s^2 3d^3 4p^1$

(Ar: Argon; $Z = 18$)

Give the **condensed ground state electronic configuration** of the atom, **A** and the cation, **A^{2+}** .
Show your work. (Assume all electronic transitions are possible)

Electronic Configuration of A:

Electronic Configuration of A^{2+} :

c. 2 points

In a single atom, determine how many electrons can have the following set of quantum numbers?

(i) $n = 5; l = 3$

In a single atom, determine how many electrons can have the following set of quantum numbers?

(i) $n = 3; l = 2; m_s = +1/2$

**Question 4** **8 points**

Rank the following from 1 to 4 in **increasing** order. Provide the ranking in the boxes provided.

Note: Atomic number (Z) are provided for relevant atoms

a) Increasing (lowest to highest) order of *ionization energy*: (1= lowest; 4 = highest)

Mg

Mg: Z = 12

Rb

Rb: Z = 37

Cs

Cs: Z = 55

K

K : Z = 19

b) Increasing (lowest to highest) order of *bond angle*: (1= lowest; 4 = highest)

H₂OCCl₄XeF₄

HCN

c) Increasing (smallest to largest) order of *ionic size* (1= smallest; 4 = largest)

S²⁻

S: Z = 16

O²⁻

O: Z = 8

Be²⁺

Be: Z = 4

Na⁺

Na: Z = 11

d) Increasing (lowest to highest) order of *bond energy* (1= lowest; 4 = highest)

P-F

P-Cl

P-I

P-Br

**Question 5** **12 points**

a. 9 points For the molecules HClO_4 and TeF_6 , draw the most stable *Lewis structure* showing appropriate shape and *dashed/wedged bonds*, based on VSEPR. Determine the **molecular geometry** around the central atom. For HClO_4 , label the *most polar bond*.



b. 3 points (Note: Lone pairs are not shown in structures below)

(i) For the three *resonance* structures (1, 2 and 3) shown below, answer the following question:



(I) Circle the *most contributing* resonance structure.

Structure 1

Structure 2

Structure 3

(ii) For the three structures given (4, 5, and 6) below, answer the following questions:



(I) Circle all the structures that represent *resonance structures* of each other:

Structure 4

Structure 5

Structure 6

(II) Among the resonance structures, circle the *most contributing* resonance structure.

Structure 4

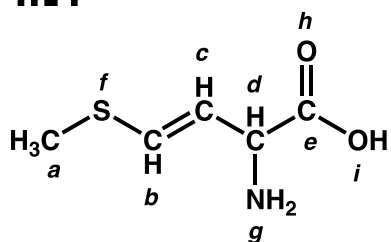
Structure 5

Structure 6

**Question 6** **6 points**

Using the data given below, calculate the **lattice energy** for SrO(s).

	ΔH° (kJ/mol)
$\text{Sr}(s) \longrightarrow \text{Sr}(g)$	+180
$\text{Sr}(g) \longrightarrow \text{Sr}^{2+}(g) + 2e^-$	+1468
$\text{O}_2(g) \longrightarrow 2\text{O}(g)$	+500
$\text{O}(g) + 2e^- \longrightarrow \text{O}^{2-}(g)$	+600
$\text{Sr}(s) + \frac{1}{2}\text{O}_2(g) \longrightarrow \text{SrO}(s)$	-550

**Question 7 10 points**

Consider the molecule given on the left, and answer questions (i) to (vi) using chemical bonding theories that we have learnt in class (valence bond theory, hybridization).

Note that the lone pairs are not shown in the structure:

i. Consider the carbon atoms marked in as ***a***, ***b***, ***c***, ***d***, and ***e*** in the molecule, and determine **hybridization** at each of the carbon centers. (2.5 points)

Atom	Hybridization
Carbon (<i>a</i>)	
Carbon (<i>b</i>)	
Carbon (<i>c</i>)	
Carbon (<i>d</i>)	
Carbon (<i>e</i>)	

ii. Now, consider the sulfur atom (S) marked as ***f***, nitrogen (N) atom (***g***), and oxygen(O) atoms (***h*** and ***i***), and determine their **hybridization**. (2 points)

Atom	Hybridization
Sulfur (<i>f</i>)	
Nitrogen (<i>g</i>)	
Oxygen (<i>h</i>)	
Oxygen (<i>i</i>)	

iii. **Describe** all the bonds formed by the carbon atoms ***a*** and ***b***, using sigma and pi designation as appropriate. (2 points)

Atom	Bonds formed (use sigma and pi as appropriate)
Carbon (<i>a</i>)	
Carbon (<i>b</i>)	

iv. How many total sp^3 -s sigma bonds are in the molecule? (1 point)

v. Determine **electronic** and **molecular geometry** at carbon (C) labeled ***e*** and sulfur (S) labeled ***f*** centres. (2 points)

Atom	Electronic Geometry	Molecular Geometry
Carbon (<i>e</i>)		
Sulfur (<i>f</i>)		

vi. Circle the value from the choices given below, that describes the **C-S-C bond angle at sulfur** (S) marked ***f***: (0.5 points)

>90° to <109.5° 109.5° >109.5 to <120° 120° >120° to <180°


Question 8 12 points

Using the **valence shell orbitals only** draw a **complete** molecular orbital (MO) diagram (with no s-p mixing) for the molecule **PS**, in the space provided below. **(6 points)**

Assume that 3s and 3p atomic orbitals will form molecular orbitals like those formed by the combination of 2s and 2p atomic orbitals that we learnt in class, and there is no s-p mixing.

Show **energy arrow**, **atomic orbitals** involved, **atom labels**, the **molecular orbitals** that are formed, **filled electrons** in the MO diagram etc. Make sure to label all the atomic and molecular orbitals.

Using the MO diagram above, answer the following questions:

- (i) Which one of the following will have the highest bond order? Circle the correct choice
 A) PS^{1-} B) PS C) PS^{1+}
- (ii) Which one of the following will have the longest bond length? Circle the correct choice
 A) PS^{1-} B) PS C) PS^{1+}
- (iii) Which of the following will be diamagnetic? Circle the correct choice.
 A) PS^{1-} B) PS C) PS^{1+}
- (iv) Draw Lewis dot structure of the molecule with the shortest bond length (based on the MO diagram) from among the three (PS^{1-} , PS, PS^{1+}), and indicate the bond order calculated from the Lewis dot structure. **(3 points)**

Lewis Dot Structure

Bond Order



84982DAD-269D-4480-9CF1-0709F75FC4F4

midterm-exam-93388

#460 Page 10 of 10

SCRAP PAGE

**CHEMISTRY 110 –Midterm Exam****Thursday November 7, 2024, 6:30 – 8:30 pm****Examiners: Pallavi Sirjoosingh and Ashok Kakkar**

First name (please write as legibly as possible within the boxes)

--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--

Last name

--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--

Student ID number

--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--

This booklet has 8 questions on 10 pages (The last page can be used as a scrap page).

The staple must NOT be removed from the question booklet. Your intact question booklet must be returned at the end of the exam.

A separate booklet contains a periodic table, and datasheet.

Write your answers directly under the questions in the space provided. Do not write on the QR codes or near the top edges (as it may be cut off).

Please write **legibly and dark enough** for the scan to be readable. For students who use English as a second language, dictionaries are permitted. Calculators are allowed.

DO NOT ASK QUESTIONS OF THE INVIGILATORS about interpretations of the exam questions. If you have a concern about a question, please summarize it at the top of this page. All concerns will be evaluated after the exam. GOOD LUCK!

**Question 1 8 points** (*Assume all transitions are possible*)

a. 6 points Calculate the **energy (in kJ/mol)** required to ionize (completely remove) the electron from the **n=3** level of **He⁺ ion** (He: Helium). Show your work including equations. (3 sig. figs.)

b. 2 points

(I) Select the largest (longest) wavelength from the three choices below:

- i. Wavelength of photon to *completely remove* an electron from *1s orbital of Hydrogen atom*
- ii. Wavelength of photon to *completely remove* an electron from *1s orbital of Boron atom (B)*
- iii. Wavelength of photon to *completely remove* an electron from *1s orbital of Boron cation (B⁺)*

i or ii or iii

(II) Select the smallest (shortest) wavelength from the three choices below (*for H atom*):

- i. Wavelength of photon *emitted* for an electron transitioning from *2p to 1s orbital*
- ii. Wavelength of photon *absorbed* for an electron transitioning from *2s to 4p orbital*
- iii. Wavelength of photon *absorbed* for an electron transitioning from *1s to 3p orbital*

i or ii or iii



Question 2 8 points



A Chem110 student plots the graph shown on the left, from data obtained for the kinetic energy (eV) of the ejected electron versus frequency (s^{-1}) of the incident light, for an unknown metal. ($1\text{eV} = 1.6022 \times 10^{-19} \text{ J}$)

Based on the plot, answer the questions **a to c** below:

a. 1 point On the plot above, label the **threshold frequency**.

b. 2 points What is the value of the work potential (in J) of the unknown metal?

c. 3 points What is the *longest* wavelength (in nm) required to eject an electron from this metal's surface? Show your work including equation(s) used. (3 sig figs)

d. 2 points In the space provided below, plot the following

(Assume frequency is *above* threshold frequency):

- i. *Kinetic energy* of an ejected electron versus the *intensity* of the incident light (left)
- ii. *Number of electrons* ejected versus the *frequency* of the incident light (right)



**Question 3** **6 points****a. 2 points**

In the ground state of Ge (Germanium; $Z = 32$) atom, **name** the following orbitals:

i. Orbital containing the electron that requires the maximum energy to be removed.

ii. Orbital containing the electron that experiences the minimum effective nuclear charge.

b. 2 points

An unknown anion (A^-) has the following *excited* electronic configuration: $[Ar] 4s^2 3d^3 4p^1$

(Ar: Argon; $Z = 18$)

Give the **condensed ground state electronic configuration** of the atom, **A** and the cation, **A^{2+}** .
Show your work. (Assume all electronic transitions are possible)

Electronic Configuration of A:

Electronic Configuration of A^{2+} :

c. 2 points

In a single atom, determine how many electrons can have the following set of quantum numbers?

(i) $n = 5; l = 3$

In a single atom, determine how many electrons can have the following set of quantum numbers?

(i) $n = 3; l = 2; m_s = +1/2$

**Question 4** **8 points**

Rank the following from 1 to 4 in **increasing** order. Provide the ranking in the boxes provided.

Note: Atomic number (Z) are provided for relevant atoms

a) Increasing (lowest to highest) order of *ionization energy*: (1= lowest; 4 = highest)

Mg

Mg: Z = 12

Rb

Rb: Z = 37

Cs

Cs: Z = 55

K

K : Z = 19

b) Increasing (lowest to highest) order of *bond angle*: (1= lowest; 4 = highest)

H₂OCCl₄XeF₄

HCN

c) Increasing (smallest to largest) order of *ionic size* (1= smallest; 4 = largest)

S²⁻

S: Z = 16

O²⁻

O: Z = 8

Be²⁺

Be: Z = 4

Na⁺

Na: Z = 11

d) Increasing (lowest to highest) order of *bond energy* (1= lowest; 4 = highest)

P-F

P-Cl

P-I

P-Br

**Question 5** **12 points**

a. 9 points For the molecules HClO_4 and TeF_6 , draw the most stable *Lewis structure* showing appropriate shape and *dashed/wedged bonds*, based on VSEPR. Determine the **molecular geometry** around the central atom. For HClO_4 , label the *most polar bond*.



b. 3 points (Note: Lone pairs are not shown in structures below)

(i) For the three *resonance* structures (1, 2 and 3) shown below, answer the following question:



(I) Circle the *most contributing* resonance structure.

Structure 1

Structure 2

Structure 3

(ii) For the three structures given (4, 5, and 6) below, answer the following questions:



(I) Circle all the structures that represent *resonance structures* of each other:

Structure 4

Structure 5

Structure 6

(II) Among the resonance structures, circle the *most contributing* resonance structure.

Structure 4

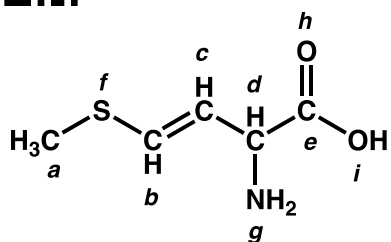
Structure 5

Structure 6

**Question 6** **6 points**

Using the data given below, calculate the **lattice energy** for SrO(s).

	ΔH° (kJ/mol)
$\text{Sr}(s) \longrightarrow \text{Sr}(g)$	+180
$\text{Sr}(g) \longrightarrow \text{Sr}^{2+}(g) + 2e^-$	+1468
$\text{O}_2(g) \longrightarrow 2\text{O}(g)$	+500
$\text{O}(g) + 2e^- \longrightarrow \text{O}^{2-}(g)$	+600
$\text{Sr}(s) + \frac{1}{2}\text{O}_2(g) \longrightarrow \text{SrO}(s)$	-550

**Question 7 10 points**

Consider the molecule given on the left, and answer questions (i) to (vi) using chemical bonding theories that we have learnt in class (valence bond theory, hybridization).

Note that the lone pairs are not shown in the structure:

i. Consider the carbon atoms marked in as ***a***, ***b***, ***c***, ***d***, and ***e*** in the molecule, and determine **hybridization** at each of the carbon centers. (2.5 points)

Atom	Hybridization
Carbon (<i>a</i>)	
Carbon (<i>b</i>)	
Carbon (<i>c</i>)	
Carbon (<i>d</i>)	
Carbon (<i>e</i>)	

ii. Now, consider the sulfur atom (S) marked as ***f***, nitrogen (N) atom (***g***), and oxygen(O) atoms (***h*** and ***i***), and determine their **hybridization**. (2 points)

Atom	Hybridization
Sulfur (<i>f</i>)	
Nitrogen (<i>g</i>)	
Oxygen (<i>h</i>)	
Oxygen (<i>i</i>)	

iii. **Describe** all the bonds formed by the carbon atoms ***a*** and ***b***, using sigma and pi designation as appropriate. (2 points)

Atom	Bonds formed (use sigma and pi as appropriate)
Carbon (<i>a</i>)	
Carbon (<i>b</i>)	

iv. How many total sp^3 -s sigma bonds are in the molecule? (1 point)

v. Determine **electronic** and **molecular geometry** at carbon (C) labeled ***e*** and sulfur (S) labeled ***f*** centres. (2 points)

Atom	Electronic Geometry	Molecular Geometry
Carbon (<i>e</i>)		
Sulfur (<i>f</i>)		

vi. Circle the value from the choices given below, that describes the **C-S-C bond angle at sulfur** (S) marked ***f***: (0.5 points)

>90° to <109.5° 109.5° >109.5 to <120° 120° >120° to <180°

**Question 8 12 points**

Using the **valence shell orbitals only** draw a **complete** molecular orbital (MO) diagram (with no s-p mixing) for the molecule **PS**, in the space provided below. **(6 points)**

Assume that 3s and 3p atomic orbitals will form molecular orbitals like those formed by the combination of 2s and 2p atomic orbitals that we learnt in class, and there is no s-p mixing.

Show **energy arrow**, **atomic orbitals** involved, **atom labels**, the **molecular orbitals** that are formed, **filled electrons** in the MO diagram etc. Make sure to label all the atomic and molecular orbitals.

Using the MO diagram above, answer the following questions:

- (i) Which one of the following will have the highest bond order? Circle the correct choice
A) PS^{1-} B) PS C) PS^{1+}
- (ii) Which one of the following will have the longest bond length? Circle the correct choice
A) PS^{1-} B) PS C) PS^{1+}
- (iii) Which of the following will be diamagnetic? Circle the correct choice.
A) PS^{1-} B) PS C) PS^{1+}
- (iv) Draw Lewis dot structure of the molecule with the shortest bond length (based on the MO diagram) from among the three (PS^{1-} , PS, PS^{1+}), and indicate the bond order calculated from the Lewis dot structure. **(3 points)**

Lewis Dot Structure

Bond Order



764556C7-2605-4427-8D09-A6BF38BB5E03

midterm-exam-93388

#461 Page 10 of 10

SCRAP PAGE

**CHEMISTRY 110 –Midterm Exam****Thursday November 7, 2024, 6:30 – 8:30 pm**

Examiners: Pallavi Sirjoosingh and Ashok Kakkar

First name (please write as legibly as possible within the boxes)

--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--

Last name

--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--

Student ID number

--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--

This booklet has 8 questions on 10 pages (The last page can be used as a scrap page).

The staple must NOT be removed from the question booklet. Your intact question booklet must be returned at the end of the exam.

A separate booklet contains a periodic table, and datasheet.

Write your answers directly under the questions in the space provided. Do not write on the QR codes or near the top edges (as it may be cut off).

Please write **legibly and dark enough** for the scan to be readable. For students who use English as a second language, dictionaries are permitted. Calculators are allowed.

DO NOT ASK QUESTIONS OF THE INVIGILATORS about interpretations of the exam questions. If you have a concern about a question, please summarize it at the top of this page. All concerns will be evaluated after the exam. GOOD LUCK!

**Question 1 8 points** (*Assume all transitions are possible*)

a. 6 points Calculate the **energy (in kJ/mol)** required to ionize (completely remove) the electron from the **$n=3$** level of **He^+ ion** (He: Helium). Show your work including equations. (3 sig. figs.)

b. 2 points

(I) Select the largest (longest) wavelength from the three choices below:

- i. Wavelength of photon to *completely remove* an electron from *1s orbital of Hydrogen atom*
- ii. Wavelength of photon to *completely remove* an electron from *1s orbital of Boron atom (B)*
- iii. Wavelength of photon to *completely remove* an electron from *1s orbital of Boron cation (B^+)*

i or ii or iii

(II) Select the smallest (shortest) wavelength from the three choices below (*for H atom*):

- i. Wavelength of photon *emitted* for an electron transitioning from *2p to 1s orbital*
- ii. Wavelength of photon *absorbed* for an electron transitioning from *2s to 4p orbital*
- iii. Wavelength of photon *absorbed* for an electron transitioning from *1s to 3p orbital*

i or ii or iii



Question 2 8 points



A Chem110 student plots the graph shown on the left, from data obtained for the kinetic energy (eV) of the ejected electron versus frequency (s^{-1}) of the incident light, for an unknown metal. ($1\text{eV} = 1.6022 \times 10^{-19} \text{ J}$)

Based on the plot, answer the questions **a to c** below:

a. 1 point On the plot above, label the **threshold frequency**.

b. 2 points What is the value of the work potential (in J) of the unknown metal?

c. 3 points What is the *longest* wavelength (in nm) required to eject an electron from this metal's surface? Show your work including equation(s) used. (3 sig figs)

d. 2 points In the space provided below, plot the following

(Assume frequency is *above* threshold frequency):

- i. *Kinetic energy* of an ejected electron versus the *intensity* of the incident light (left)
- ii. *Number of electrons* ejected versus the *frequency* of the incident light (right)



**Question 3** **6 points****a. 2 points**

In the ground state of Ge (Germanium; $Z = 32$) atom, **name** the following orbitals:

i. Orbital containing the electron that requires the maximum energy to be removed.

ii. Orbital containing the electron that experiences the minimum effective nuclear charge.

b. 2 points

An unknown anion (A^-) has the following *excited* electronic configuration: $[Ar] 4s^2 3d^3 4p^1$

(Ar: Argon; $Z = 18$)

Give the **condensed ground state electronic configuration** of the atom, **A** and the cation, **A^{2+}** .
Show your work. (Assume all electronic transitions are possible)

Electronic Configuration of A:

Electronic Configuration of A^{2+} :

c. 2 points

In a single atom, determine how many electrons can have the following set of quantum numbers?

(i) $n = 5; l = 3$

In a single atom, determine how many electrons can have the following set of quantum numbers?

(i) $n = 3; l = 2; m_s = +1/2$

**Question 4** **8 points**

Rank the following from 1 to 4 in **increasing** order. Provide the ranking in the boxes provided.

Note: Atomic number (Z) are provided for relevant atoms

a) Increasing (lowest to highest) order of *ionization energy*: (1= lowest; 4 = highest)

Mg

Mg: Z = 12

Rb

Rb: Z = 37

Cs

Cs: Z = 55

K

K : Z = 19

b) Increasing (lowest to highest) order of *bond angle*: (1= lowest; 4 = highest)

H₂OCCl₄XeF₄

HCN

c) Increasing (smallest to largest) order of *ionic size* (1= smallest; 4 = largest)

S²⁻

S: Z = 16

O²⁻

O: Z = 8

Be²⁺

Be: Z = 4

Na⁺

Na: Z = 11

d) Increasing (lowest to highest) order of *bond energy* (1= lowest; 4 = highest)

P-F

P-Cl

P-I

P-Br

**Question 5** **12 points**

a. 9 points For the molecules HClO_4 and TeF_6 , draw the most stable *Lewis structure* showing appropriate shape and *dashed/wedged bonds*, based on VSEPR. Determine the **molecular geometry** around the central atom. For HClO_4 , label the *most polar bond*.

HClO_4 (there are no O-O bonds)

Label the most polar bond

Molecular Geometry around Cl

TeF_6

Molecular Geometry around Te

b. 3 points (Note: Lone pairs are not shown in structures below)

(i) For the three *resonance* structures (1, 2 and 3) shown below, answer the following question:



(I) Circle the *most contributing* resonance structure.

Structure 1

Structure 2

Structure 3

(ii) For the three structures given (4, 5, and 6) below, answer the following questions:



(I) Circle all the structures that represent *resonance structures* of each other:

Structure 4

Structure 5

Structure 6

(II) Among the resonance structures, circle the *most contributing* resonance structure.

Structure 4

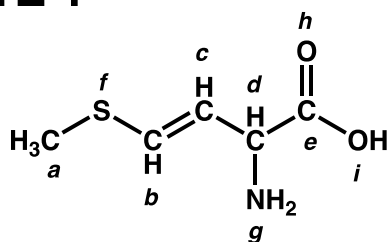
Structure 5

Structure 6

**Question 6** **6 points**

Using the data given below, calculate the **lattice energy** for SrO(s).

	ΔH° (kJ/mol)
$\text{Sr}(s) \longrightarrow \text{Sr}(g)$	+180
$\text{Sr}(g) \longrightarrow \text{Sr}^{2+}(g) + 2e^-$	+1468
$\text{O}_2(g) \longrightarrow 2\text{O}(g)$	+500
$\text{O}(g) + 2e^- \longrightarrow \text{O}^{2-}(g)$	+600
$\text{Sr}(s) + \frac{1}{2}\text{O}_2(g) \longrightarrow \text{SrO}(s)$	-550

**Question 7 10 points**

Consider the molecule given on the left, and answer questions (i) to (vi) using chemical bonding theories that we have learnt in class (valence bond theory, hybridization).

Note that the lone pairs are not shown in the structure:

i. Consider the carbon atoms marked in as ***a***, ***b***, ***c***, ***d***, and ***e*** in the molecule, and determine **hybridization** at each of the carbon centers. (2.5 points)

Atom	Hybridization
Carbon (<i>a</i>)	
Carbon (<i>b</i>)	
Carbon (<i>c</i>)	
Carbon (<i>d</i>)	
Carbon (<i>e</i>)	

ii. Now, consider the sulfur atom (S) marked as ***f***, nitrogen (N) atom (***g***), and oxygen(O) atoms (***h*** and ***i***), and determine their **hybridization**. (2 points)

Atom	Hybridization
Sulfur (<i>f</i>)	
Nitrogen (<i>g</i>)	
Oxygen (<i>h</i>)	
Oxygen (<i>i</i>)	

iii. **Describe** all the bonds formed by the carbon atoms ***a*** and ***b***, using sigma and pi designation as appropriate. (2 points)

Atom	Bonds formed (use sigma and pi as appropriate)
Carbon (<i>a</i>)	
Carbon (<i>b</i>)	

iv. How many total sp^3 -s sigma bonds are in the molecule? (1 point)

v. Determine **electronic** and **molecular geometry** at carbon (C) labeled ***e*** and sulfur (S) labeled ***f*** centres. (2 points)

Atom	Electronic Geometry	Molecular Geometry
Carbon (<i>e</i>)		
Sulfur (<i>f</i>)		

vi. Circle the value from the choices given below, that describes the **C-S-C bond angle at sulfur** (S) marked ***f***: (0.5 points)

$>90^\circ$ to $<109.5^\circ$ 109.5° >109.5 to $<120^\circ$ 120° $>120^\circ$ to $<180^\circ$


Question 8 12 points

Using the **valence shell orbitals only** draw a **complete** molecular orbital (MO) diagram (with no s-p mixing) for the molecule **PS**, in the space provided below. **(6 points)**

Assume that 3s and 3p atomic orbitals will form molecular orbitals like those formed by the combination of 2s and 2p atomic orbitals that we learnt in class, and there is no s-p mixing.

Show **energy arrow**, **atomic orbitals** involved, **atom labels**, the **molecular orbitals** that are formed, **filled electrons** in the MO diagram etc. Make sure to label all the atomic and molecular orbitals.

Using the MO diagram above, answer the following questions:

- (i) Which one of the following will have the highest bond order? Circle the correct choice
 A) PS^{1-} B) PS C) PS^{1+}
- (ii) Which one of the following will have the longest bond length? Circle the correct choice
 A) PS^{1-} B) PS C) PS^{1+}
- (iii) Which of the following will be diamagnetic? Circle the correct choice.
 A) PS^{1-} B) PS C) PS^{1+}
- (iv) Draw Lewis dot structure of the molecule with the shortest bond length (based on the MO diagram) from among the three (PS^{1-} , PS, PS^{1+}), and indicate the bond order calculated from the Lewis dot structure. **(3 points)**

Lewis Dot Structure

Bond Order



DC3A272F-4BF4-430F-9E7B-D852FFE1973D

midterm-exam-93388

#462 Page 10 of 10

SCRAP PAGE

**CHEMISTRY 110 –Midterm Exam****Thursday November 7, 2024, 6:30 – 8:30 pm**

Examiners: Pallavi Sirjoosingh and Ashok Kakkar

First name (please write as legibly as possible within the boxes)

--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--

Last name

--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--

Student ID number

--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--

This booklet has 8 questions on 10 pages (The last page can be used as a scrap page).

The staple must NOT be removed from the question booklet. Your intact question booklet must be returned at the end of the exam.

A separate booklet contains a periodic table, and datasheet.

Write your answers directly under the questions in the space provided. Do not write on the QR codes or near the top edges (as it may be cut off).

Please write **legibly and dark enough** for the scan to be readable. For students who use English as a second language, dictionaries are permitted. Calculators are allowed.

DO NOT ASK QUESTIONS OF THE INVIGILATORS about interpretations of the exam questions. If you have a concern about a question, please summarize it at the top of this page. All concerns will be evaluated after the exam. GOOD LUCK!

**Question 1 8 points** (*Assume all transitions are possible*)

a. 6 points Calculate the **energy (in kJ/mol)** required to ionize (completely remove) the electron from the **n=3** level of **He⁺ ion** (He: Helium). Show your work including equations. (3 sig. figs.)

b. 2 points

(I) Select the largest (longest) wavelength from the three choices below:

- i. Wavelength of photon to *completely remove* an electron from *1s orbital of Hydrogen atom*
- ii. Wavelength of photon to *completely remove* an electron from *1s orbital of Boron atom (B)*
- iii. Wavelength of photon to *completely remove* an electron from *1s orbital of Boron cation (B⁺)*

i or ii or iii

(II) Select the smallest (shortest) wavelength from the three choices below (*for H atom*):

- i. Wavelength of photon *emitted* for an electron transitioning from *2p to 1s orbital*
- ii. Wavelength of photon *absorbed* for an electron transitioning from *2s to 4p orbital*
- iii. Wavelength of photon *absorbed* for an electron transitioning from *1s to 3p orbital*

i or ii or iii



Question 2 8 points



A Chem110 student plots the graph shown on the left, from data obtained for the kinetic energy (eV) of the ejected electron versus frequency (s^{-1}) of the incident light, for an unknown metal. ($1\text{eV} = 1.6022 \times 10^{-19} \text{ J}$)

Based on the plot, answer the questions **a to c** below:

a. 1 point On the plot above, label the **threshold frequency**.

b. 2 points What is the value of the work potential (in J) of the unknown metal?

c. 3 points What is the *longest* wavelength (in nm) required to eject an electron from this metal's surface? Show your work including equation(s) used. (3 sig figs)

d. 2 points In the space provided below, plot the following

(Assume frequency is *above* threshold frequency):

- i. *Kinetic energy* of an ejected electron versus the *intensity* of the incident light (left)
- ii. *Number of electrons* ejected versus the *frequency* of the incident light (right)



**Question 3** **6 points****a. 2 points**

In the ground state of Ge (Germanium; $Z = 32$) atom, **name** the following orbitals:

i. Orbital containing the electron that requires the maximum energy to be removed.

ii. Orbital containing the electron that experiences the minimum effective nuclear charge.

b. 2 points

An unknown anion (A^-) has the following *excited* electronic configuration: $[Ar] 4s^2 3d^3 4p^1$

(Ar: Argon; $Z = 18$)

Give the **condensed ground state electronic configuration** of the atom, **A** and the cation, A^{2+} .
Show your work. (Assume all electronic transitions are possible)

Electronic Configuration of A:

Electronic Configuration of A^{2+} :

c. 2 points

In a single atom, determine how many electrons can have the following set of quantum numbers?

(i) $n = 5; l = 3$

In a single atom, determine how many electrons can have the following set of quantum numbers?

(i) $n = 3; l = 2; m_s = +1/2$

**Question 4** **8 points**

Rank the following from 1 to 4 in **increasing** order. Provide the ranking in the boxes provided.

Note: Atomic number (Z) are provided for relevant atoms

a) Increasing (lowest to highest) order of *ionization energy*: (1= lowest; 4 = highest)

Mg

Mg: Z = 12

Rb

Rb: Z = 37

Cs

Cs: Z = 55

K

K : Z = 19

b) Increasing (lowest to highest) order of *bond angle*: (1= lowest; 4 = highest)

H₂OCCl₄XeF₄

HCN

c) Increasing (smallest to largest) order of *ionic size* (1= smallest; 4 = largest)

S²⁻

S: Z = 16

O²⁻

O: Z = 8

Be²⁺

Be: Z = 4

Na⁺

Na: Z = 11

d) Increasing (lowest to highest) order of *bond energy* (1= lowest; 4 = highest)

P-F

P-Cl

P-I

P-Br

**Question 5** **12 points**

a. 9 points For the molecules HClO_4 and TeF_6 , draw the most stable *Lewis structure* showing appropriate shape and *dashed/wedged bonds*, based on VSEPR. Determine the **molecular geometry** around the central atom. For HClO_4 , label the *most polar bond*.



b. 3 points (Note: Lone pairs are not shown in structures below)

(i) For the three *resonance* structures (1, 2 and 3) shown below, answer the following question:



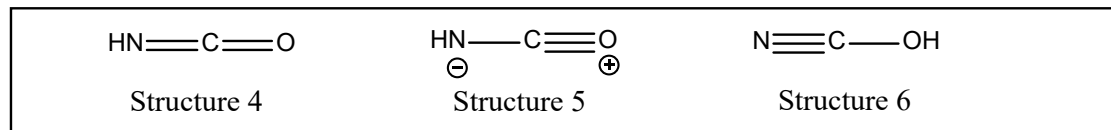
(I) Circle the *most contributing* resonance structure.

Structure 1

Structure 2

Structure 3

(ii) For the three structures given (4, 5, and 6) below, answer the following questions:



(I) Circle all the structures that represent *resonance structures* of each other:

Structure 4

Structure 5

Structure 6

(II) Among the resonance structures, circle the *most contributing* resonance structure.

Structure 4

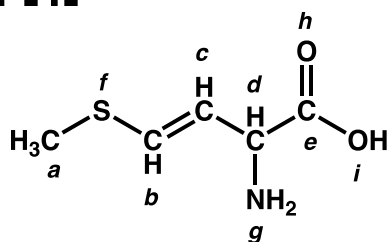
Structure 5

Structure 6

**Question 6** **6 points**

Using the data given below, calculate the **lattice energy** for SrO(s).

	ΔH° (kJ/mol)
$\text{Sr}(s) \longrightarrow \text{Sr}(g)$	+180
$\text{Sr}(g) \longrightarrow \text{Sr}^{2+}(g) + 2e^-$	+1468
$\text{O}_2(g) \longrightarrow 2\text{O}(g)$	+500
$\text{O}(g) + 2e^- \longrightarrow \text{O}^{2-}(g)$	+600
$\text{Sr}(s) + \frac{1}{2}\text{O}_2(g) \longrightarrow \text{SrO}(s)$	-550

**Question 7 10 points**

Consider the molecule given on the left, and answer questions (i) to (vi) using chemical bonding theories that we have learnt in class (valence bond theory, hybridization).

Note that the lone pairs are not shown in the structure:

i. Consider the carbon atoms marked in as ***a***, ***b***, ***c***, ***d***, and ***e*** in the molecule, and determine **hybridization** at each of the carbon centers. (2.5 points)

Atom	Hybridization
Carbon (<i>a</i>)	
Carbon (<i>b</i>)	
Carbon (<i>c</i>)	
Carbon (<i>d</i>)	
Carbon (<i>e</i>)	

ii. Now, consider the sulfur atom (S) marked as ***f***, nitrogen (N) atom (***g***), and oxygen(O) atoms (***h*** and ***i***), and determine their **hybridization**. (2 points)

Atom	Hybridization
Sulfur (<i>f</i>)	
Nitrogen (<i>g</i>)	
Oxygen (<i>h</i>)	
Oxygen (<i>i</i>)	

iii. **Describe** all the bonds formed by the carbon atoms ***a*** and ***b***, using sigma and pi designation as appropriate. (2 points)

Atom	Bonds formed (use sigma and pi as appropriate)
Carbon (<i>a</i>)	
Carbon (<i>b</i>)	

iv. How many total sp^3 -s sigma bonds are in the molecule? (1 point)

v. Determine **electronic** and **molecular geometry** at carbon (C) labeled ***e*** and sulfur (S) labeled ***f*** centres. (2 points)

Atom	Electronic Geometry	Molecular Geometry
Carbon (<i>e</i>)		
Sulfur (<i>f</i>)		

vi. Circle the value from the choices given below, that describes the **C-S-C bond angle at sulfur** (S) marked ***f***: (0.5 points)

>90° to <109.5°

109.5°

>109.5 to <120°

120°

>120° to <180°


Question 8 12 points

Using the **valence shell orbitals only** draw a **complete** molecular orbital (MO) diagram (with no s-p mixing) for the molecule **PS**, in the space provided below. **(6 points)**

Assume that 3s and 3p atomic orbitals will form molecular orbitals like those formed by the combination of 2s and 2p atomic orbitals that we learnt in class, and there is no s-p mixing.

Show **energy arrow**, **atomic orbitals** involved, **atom labels**, the **molecular orbitals** that are formed, **filled electrons** in the MO diagram etc. Make sure to **label all the atomic and molecular orbitals**.

Using the MO diagram above, answer the following questions:

- (i) Which one of the following will have the highest bond order? Circle the correct choice
 A) PS^{1-} B) PS C) PS^{1+}
- (ii) Which one of the following will have the longest bond length? Circle the correct choice
 A) PS^{1-} B) PS C) PS^{1+}
- (iii) Which of the following will be diamagnetic? Circle the correct choice.
 A) PS^{1-} B) PS C) PS^{1+}
- (iv) Draw Lewis dot structure of the molecule with the shortest bond length (based on the MO diagram) from among the three (PS^{1-} , PS, PS^{1+}), and indicate the bond order calculated from the Lewis dot structure. **(3 points)**

Lewis Dot Structure

Bond Order



CA7DDCF3-9EE6-4F64-B881-2F111D299A4B

midterm-exam-93388

#463 Page 10 of 10

SCRAP PAGE

**CHEMISTRY 110 –Midterm Exam****Thursday November 7, 2024, 6:30 – 8:30 pm**

Examiners: Pallavi Sirjoosingh and Ashok Kakkar

First name (please write as legibly as possible within the boxes)

--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--

Last name

--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--

Student ID number

--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--

This booklet has 8 questions on 10 pages (The last page can be used as a scrap page).

The staple must NOT be removed from the question booklet. Your intact question booklet must be returned at the end of the exam.

A separate booklet contains a periodic table, and datasheet.

Write your answers directly under the questions in the space provided. Do not write on the QR codes or near the top edges (as it may be cut off).

Please write **legibly and dark enough** for the scan to be readable. For students who use English as a second language, dictionaries are permitted. Calculators are allowed.

DO NOT ASK QUESTIONS OF THE INVIGILATORS about interpretations of the exam questions. If you have a concern about a question, please summarize it at the top of this page. All concerns will be evaluated after the exam. GOOD LUCK!

**Question 1 8 points** (*Assume all transitions are possible*)

a. 6 points Calculate the **energy (in kJ/mol)** required to ionize (completely remove) the electron from the **n=3** level of **He⁺ ion** (He: Helium). Show your work including equations. (3 sig. figs.)

b. 2 points

(I) Select the largest (longest) wavelength from the three choices below:

- i. Wavelength of photon to *completely remove* an electron from *1s orbital of Hydrogen atom*
- ii. Wavelength of photon to *completely remove* an electron from *1s orbital of Boron atom (B)*
- iii. Wavelength of photon to *completely remove* an electron from *1s orbital of Boron cation (B⁺)*

i or ii or iii

(II) Select the smallest (shortest) wavelength from the three choices below (*for H atom*):

- i. Wavelength of photon *emitted* for an electron transitioning from *2p to 1s orbital*
- ii. Wavelength of photon *absorbed* for an electron transitioning from *2s to 4p orbital*
- iii. Wavelength of photon *absorbed* for an electron transitioning from *1s to 3p orbital*

i or ii or iii



Question 2 8 points



A Chem110 student plots the graph shown on the left, from data obtained for the kinetic energy (eV) of the ejected electron versus frequency (s^{-1}) of the incident light, for an unknown metal. ($1\text{eV} = 1.6022 \times 10^{-19} \text{ J}$)

Based on the plot, answer the questions **a to c** below:

a. 1 point On the plot above, label the **threshold frequency**.

b. 2 points What is the value of the work potential (in J) of the unknown metal?

c. 3 points What is the *longest* wavelength (in nm) required to eject an electron from this metal's surface? Show your work including equation(s) used. (3 sig figs)

d. 2 points In the space provided below, plot the following

(Assume frequency is *above* threshold frequency):

- Kinetic energy* of an ejected electron versus the *intensity* of the incident light (left)
- Number of electrons* ejected versus the *frequency* of the incident light (right)



**Question 3** **6 points****a. 2 points**

In the ground state of Ge (Germanium; $Z = 32$) atom, **name** the following orbitals:

i. Orbital containing the electron that requires the maximum energy to be removed.

ii. Orbital containing the electron that experiences the minimum effective nuclear charge.

b. 2 points

An unknown anion (A^-) has the following *excited* electronic configuration: $[Ar] 4s^2 3d^3 4p^1$

(Ar: Argon; $Z = 18$)

Give the **condensed ground state electronic configuration** of the atom, **A** and the cation, **A^{2+}** .
Show your work. (Assume all electronic transitions are possible)

Electronic Configuration of A:

Electronic Configuration of A^{2+} :

c. 2 points

In a single atom, determine how many electrons can have the following set of quantum numbers?

(i) $n = 5; l = 3$

In a single atom, determine how many electrons can have the following set of quantum numbers?

(i) $n = 3; l = 2; m_s = +1/2$

**Question 4** **8 points**

Rank the following from 1 to 4 in **increasing** order. Provide the ranking in the boxes provided.

Note: Atomic number (Z) are provided for relevant atoms

a) Increasing (lowest to highest) order of *ionization energy*: (1= lowest; 4 = highest)

Mg

Mg: Z = 12

Rb

Rb: Z = 37

Cs

Cs: Z = 55

K

K : Z = 19

b) Increasing (lowest to highest) order of *bond angle*: (1= lowest; 4 = highest)

H₂OCCl₄XeF₄

HCN

c) Increasing (smallest to largest) order of *ionic size* (1= smallest; 4 = largest)

S²⁻

S: Z = 16

O²⁻

O: Z = 8

Be²⁺

Be: Z = 4

Na⁺

Na: Z = 11

d) Increasing (lowest to highest) order of *bond energy* (1= lowest; 4 = highest)

P-F

P-Cl

P-I

P-Br

**Question 5** **12 points**

a. 9 points For the molecules HClO_4 and TeF_6 , draw the most stable *Lewis structure* showing appropriate shape and *dashed/wedged bonds*, based on VSEPR. Determine the **molecular geometry** around the central atom. For HClO_4 , label the *most polar bond*.



b. 3 points (Note: Lone pairs are not shown in structures below)

(i) For the three *resonance* structures (1, 2 and 3) shown below, answer the following question:



(I) Circle the *most contributing* resonance structure.

Structure 1

Structure 2

Structure 3

(ii) For the three structures given (4, 5, and 6) below, answer the following questions:



(I) Circle all the structures that represent *resonance structures* of each other:

Structure 4

Structure 5

Structure 6

(II) Among the resonance structures, circle the *most contributing* resonance structure.

Structure 4

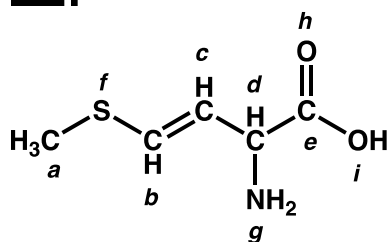
Structure 5

Structure 6

**Question 6** **6 points**

Using the data given below, calculate the **lattice energy** for SrO(s).

	ΔH° (kJ/mol)
$\text{Sr}(s) \longrightarrow \text{Sr}(g)$	+180
$\text{Sr}(g) \longrightarrow \text{Sr}^{2+}(g) + 2e^-$	+1468
$\text{O}_2(g) \longrightarrow 2\text{O}(g)$	+500
$\text{O}(g) + 2e^- \longrightarrow \text{O}^{2-}(g)$	+600
$\text{Sr}(s) + \frac{1}{2}\text{O}_2(g) \longrightarrow \text{SrO}(s)$	-550

**Question 7 10 points**

Consider the molecule given on the left, and answer questions (i) to (vi) using chemical bonding theories that we have learnt in class (valence bond theory, hybridization).

Note that the lone pairs are not shown in the structure:

i. Consider the carbon atoms marked in as ***a***, ***b***, ***c***, ***d***, and ***e*** in the molecule, and determine **hybridization** at each of the carbon centers. (2.5 points)

Atom	Hybridization
Carbon (<i>a</i>)	
Carbon (<i>b</i>)	
Carbon (<i>c</i>)	
Carbon (<i>d</i>)	
Carbon (<i>e</i>)	

ii. Now, consider the sulfur atom (S) marked as ***f***, nitrogen (N) atom (***g***), and oxygen(O) atoms (***h*** and ***i***), and determine their **hybridization**. (2 points)

Atom	Hybridization
Sulfur (<i>f</i>)	
Nitrogen (<i>g</i>)	
Oxygen (<i>h</i>)	
Oxygen (<i>i</i>)	

iii. **Describe** all the bonds formed by the carbon atoms ***a*** and ***b***, using sigma and pi designation as appropriate. (2 points)

Atom	Bonds formed (use sigma and pi as appropriate)
Carbon (<i>a</i>)	
Carbon (<i>b</i>)	

iv. How many total sp^3 -s sigma bonds are in the molecule? (1 point)

v. Determine **electronic** and **molecular geometry** at carbon (C) labeled ***e*** and sulfur (S) labeled ***f*** centres. (2 points)

Atom	Electronic Geometry	Molecular Geometry
Carbon (<i>e</i>)		
Sulfur (<i>f</i>)		

vi. Circle the value from the choices given below, that describes the **C-S-C bond angle at sulfur** (S) marked ***f***: (0.5 points)

>90° to <109.5° 109.5° >109.5 to <120° 120° >120° to <180°


Question 8 12 points

Using the **valence shell orbitals only** draw a **complete** molecular orbital (MO) diagram (with no s-p mixing) for the molecule **PS**, in the space provided below. **(6 points)**

Assume that 3s and 3p atomic orbitals will form molecular orbitals like those formed by the combination of 2s and 2p atomic orbitals that we learnt in class, and there is no s-p mixing.

Show **energy arrow**, **atomic orbitals** involved, **atom labels**, the **molecular orbitals** that are formed, **filled electrons** in the MO diagram etc. Make sure to label all the atomic and molecular orbitals.

Using the MO diagram above, answer the following questions:

- (i) Which one of the following will have the highest bond order? Circle the correct choice
 A) PS^{1-} B) PS C) PS^{1+}
- (ii) Which one of the following will have the longest bond length? Circle the correct choice
 A) PS^{1-} B) PS C) PS^{1+}
- (iii) Which of the following will be diamagnetic? Circle the correct choice.
 A) PS^{1-} B) PS C) PS^{1+}
- (iv) Draw Lewis dot structure of the molecule with the shortest bond length (based on the MO diagram) from among the three (PS^{1-} , PS, PS^{1+}), and indicate the bond order calculated from the Lewis dot structure. **(3 points)**

Lewis Dot Structure

Bond Order



0612B94B-7F6B-4DD8-A163-04396D82D7C4

midterm-exam-93388

#464 Page 10 of 10

SCRAP PAGE

**CHEMISTRY 110 –Midterm Exam****Thursday November 7, 2024, 6:30 – 8:30 pm****Examiners: Pallavi Sirjoosingh and Ashok Kakkar**

First name (please write as legibly as possible within the boxes)

--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--

Last name

--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--

Student ID number

--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--

This booklet has 8 questions on 10 pages (The last page can be used as a scrap page).

The staple must NOT be removed from the question booklet. Your intact question booklet must be returned at the end of the exam.

A separate booklet contains a periodic table, and datasheet.

Write your answers directly under the questions in the space provided. Do not write on the QR codes or near the top edges (as it may be cut off).

Please write **legibly and dark enough** for the scan to be readable. For students who use English as a second language, dictionaries are permitted. Calculators are allowed.

DO NOT ASK QUESTIONS OF THE INVIGILATORS about interpretations of the exam questions. If you have a concern about a question, please summarize it at the top of this page. All concerns will be evaluated after the exam. GOOD LUCK!

**Question 1 8 points** (*Assume all transitions are possible*)

a. 6 points Calculate the **energy (in kJ/mol)** required to ionize (completely remove) the electron from the **n=3** level of **He⁺ ion** (He: Helium). Show your work including equations. (3 sig. figs.)

b. 2 points

(I) Select the largest (longest) wavelength from the three choices below:

- i. Wavelength of photon to *completely remove* an electron from *1s orbital of Hydrogen atom*
- ii. Wavelength of photon to *completely remove* an electron from *1s orbital of Boron atom (B)*
- iii. Wavelength of photon to *completely remove* an electron from *1s orbital of Boron cation (B⁺)*

i or ii or iii

(II) Select the smallest (shortest) wavelength from the three choices below (*for H atom*):

- i. Wavelength of photon *emitted* for an electron transitioning from *2p to 1s orbital*
- ii. Wavelength of photon *absorbed* for an electron transitioning from *2s to 4p orbital*
- iii. Wavelength of photon *absorbed* for an electron transitioning from *1s to 3p orbital*

i or ii or iii



Question 2 8 points



A Chem110 student plots the graph shown on the left, from data obtained for the kinetic energy (eV) of the ejected electron versus frequency (s^{-1}) of the incident light, for an unknown metal. ($1\text{eV} = 1.6022 \times 10^{-19} \text{ J}$)

Based on the plot, answer the questions **a to c** below:

a. 1 point On the plot above, label the **threshold frequency**.

b. 2 points What is the value of the work potential (in J) of the unknown metal?

c. 3 points What is the *longest* wavelength (in nm) required to eject an electron from this metal's surface? Show your work including equation(s) used. (3 sig figs)

d. 2 points In the space provided below, plot the following

(Assume frequency is *above* threshold frequency):

- i. *Kinetic energy* of an ejected electron versus the *intensity* of the incident light (left)
- ii. *Number of electrons* ejected versus the *frequency* of the incident light (right)



**Question 3** **6 points****a. 2 points**

In the ground state of Ge (Germanium; $Z = 32$) atom, **name** the following orbitals:

i. Orbital containing the electron that requires the maximum energy to be removed.

ii. Orbital containing the electron that experiences the minimum effective nuclear charge.

b. 2 points

An unknown anion (A^-) has the following *excited* electronic configuration: $[Ar] 4s^2 3d^3 4p^1$

(Ar: Argon; $Z = 18$)

Give the **condensed ground state electronic configuration** of the atom, **A** and the cation, **A^{2+}** .
Show your work. (Assume all electronic transitions are possible)

Electronic Configuration of A:

Electronic Configuration of A^{2+} :

c. 2 points

In a single atom, determine how many electrons can have the following set of quantum numbers?

(i) $n = 5; l = 3$

In a single atom, determine how many electrons can have the following set of quantum numbers?

(i) $n = 3; l = 2; m_s = +1/2$

**Question 4** **8 points**

Rank the following from 1 to 4 in **increasing** order. Provide the ranking in the boxes provided.

Note: Atomic number (Z) are provided for relevant atoms

a) Increasing (lowest to highest) order of *ionization energy*: (1= lowest; 4 = highest)

Mg

Mg: Z = 12

Rb

Rb: Z = 37

Cs

Cs: Z = 55

K

K : Z = 19

b) Increasing (lowest to highest) order of *bond angle*: (1= lowest; 4 = highest)

H₂OCCl₄XeF₄

HCN

c) Increasing (smallest to largest) order of *ionic size* (1= smallest; 4 = largest)

S²⁻

S: Z = 16

O²⁻

O: Z = 8

Be²⁺

Be: Z = 4

Na⁺

Na: Z = 11

d) Increasing (lowest to highest) order of *bond energy* (1= lowest; 4 = highest)

P-F

P-Cl

P-I

P-Br

**Question 5** **12 points**

a. 9 points For the molecules HClO_4 and TeF_6 , draw the most stable *Lewis structure* showing appropriate shape and *dashed/wedged bonds*, based on VSEPR. Determine the **molecular geometry** around the central atom. For HClO_4 , label the *most polar bond*.



b. 3 points (Note: Lone pairs are not shown in structures below)

(i) For the three *resonance* structures (1, 2 and 3) shown below, answer the following question:



(I) Circle the *most contributing* resonance structure.

Structure 1

Structure 2

Structure 3

(ii) For the three structures given (4, 5, and 6) below, answer the following questions:



(I) Circle all the structures that represent *resonance structures* of each other:

Structure 4

Structure 5

Structure 6

(II) Among the resonance structures, circle the *most contributing* resonance structure.

Structure 4

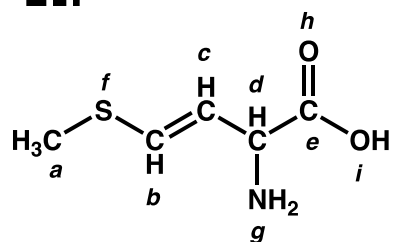
Structure 5

Structure 6

**Question 6** **6 points**

Using the data given below, calculate the **lattice energy** for SrO(s).

	ΔH° (kJ/mol)
$\text{Sr}(s) \longrightarrow \text{Sr}(g)$	+180
$\text{Sr}(g) \longrightarrow \text{Sr}^{2+}(g) + 2e^-$	+1468
$\text{O}_2(g) \longrightarrow 2\text{O}(g)$	+500
$\text{O}(g) + 2e^- \longrightarrow \text{O}^{2-}(g)$	+600
$\text{Sr}(s) + \frac{1}{2}\text{O}_2(g) \longrightarrow \text{SrO}(s)$	-550

**Question 7 10 points**

Consider the molecule given on the left, and answer questions (i) to (vi) using chemical bonding theories that we have learnt in class (valence bond theory, hybridization).

Note that the lone pairs are not shown in the structure:

i. Consider the carbon atoms marked in as ***a***, ***b***, ***c***, ***d***, and ***e*** in the molecule, and determine **hybridization** at each of the carbon centers. (2.5 points)

Atom	Hybridization
Carbon (<i>a</i>)	
Carbon (<i>b</i>)	
Carbon (<i>c</i>)	
Carbon (<i>d</i>)	
Carbon (<i>e</i>)	

ii. Now, consider the sulfur atom (S) marked as ***f***, nitrogen (N) atom (***g***), and oxygen(O) atoms (***h*** and ***i***), and determine their **hybridization**. (2 points)

Atom	Hybridization
Sulfur (<i>f</i>)	
Nitrogen (<i>g</i>)	
Oxygen (<i>h</i>)	
Oxygen (<i>i</i>)	

iii. **Describe** all the bonds formed by the carbon atoms ***a*** and ***b***, using sigma and pi designation as appropriate. (2 points)

Atom	Bonds formed (use sigma and pi as appropriate)
Carbon (<i>a</i>)	
Carbon (<i>b</i>)	

iv. How many total sp^3 -s sigma bonds are in the molecule? (1 point)

v. Determine **electronic** and **molecular geometry** at carbon (C) labeled ***e*** and sulfur (S) labeled ***f*** centres. (2 points)

Atom	Electronic Geometry	Molecular Geometry
Carbon (<i>e</i>)		
Sulfur (<i>f</i>)		

vi. Circle the value from the choices given below, that describes the **C-S-C bond angle at sulfur** (S) marked ***f***: (0.5 points)

>90° to <109.5° 109.5° >109.5 to <120° 120° >120° to <180°



Question 8 12 points

Using the **valence shell orbitals only** draw a **complete** molecular orbital (MO) diagram (with no s-p mixing) for the molecule **PS**, in the space provided below. **(6 points)**

Assume that 3s and 3p atomic orbitals will form molecular orbitals like those formed by the combination of 2s and 2p atomic orbitals that we learnt in class, and there is no s-p mixing.

Show **energy arrow**, **atomic orbitals** involved, **atom labels**, the **molecular orbitals** that are formed, **filled electrons** in the MO diagram etc. Make sure to label all the atomic and molecular orbitals.

Using the MO diagram above, answer the following questions:

- (i) Which one of the following will have the highest bond order? Circle the correct choice
 A) PS^{1-} B) PS C) PS^{1+}
- (ii) Which one of the following will have the longest bond length? Circle the correct choice
 A) PS^{1-} B) PS C) PS^{1+}
- (iii) Which of the following will be diamagnetic? Circle the correct choice.
 A) PS^{1-} B) PS C) PS^{1+}
- (iv) Draw Lewis dot structure of the molecule with the shortest bond length (based on the MO diagram) from among the three (PS^{1-} , PS, PS^{1+}), and indicate the bond order calculated from the Lewis dot structure. **(3 points)**

Lewis Dot Structure

Bond Order



ACA9D6BF-4408-4371-896A-23C0E91187FB

midterm-exam-93388

#465 Page 10 of 10

SCRAP PAGE

**CHEMISTRY 110 –Midterm Exam****Thursday November 7, 2024, 6:30 – 8:30 pm**

Examiners: Pallavi Sirjoosingh and Ashok Kakkar

First name (please write as legibly as possible within the boxes)

--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--

Last name

--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--

Student ID number

--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--

This booklet has 8 questions on 10 pages (The last page can be used as a scrap page).

The staple must NOT be removed from the question booklet. Your intact question booklet must be returned at the end of the exam.

A separate booklet contains a periodic table, and datasheet.

Write your answers directly under the questions in the space provided. Do not write on the QR codes or near the top edges (as it may be cut off).

Please write **legibly and dark enough** for the scan to be readable. For students who use English as a second language, dictionaries are permitted. Calculators are allowed.

DO NOT ASK QUESTIONS OF THE INVIGILATORS about interpretations of the exam questions. If you have a concern about a question, please summarize it at the top of this page. All concerns will be evaluated after the exam. GOOD LUCK!

**Question 1 8 points** (*Assume all transitions are possible*)

a. 6 points Calculate the **energy (in kJ/mol)** required to ionize (completely remove) the electron from the **n=3** level of **He⁺ ion** (He: Helium). Show your work including equations. (3 sig. figs.)

b. 2 points

(I) Select the largest (longest) wavelength from the three choices below:

- i. Wavelength of photon to *completely remove* an electron from *1s orbital of Hydrogen atom*
- ii. Wavelength of photon to *completely remove* an electron from *1s orbital of Boron atom (B)*
- iii. Wavelength of photon to *completely remove* an electron from *1s orbital of Boron cation (B⁺)*

i or ii or iii

(II) Select the smallest (shortest) wavelength from the three choices below (*for H atom*):

- i. Wavelength of photon *emitted* for an electron transitioning from *2p to 1s orbital*
- ii. Wavelength of photon *absorbed* for an electron transitioning from *2s to 4p orbital*
- iii. Wavelength of photon *absorbed* for an electron transitioning from *1s to 3p orbital*

i or ii or iii



Question 2 8 points



A Chem110 student plots the graph shown on the left, from data obtained for the kinetic energy (eV) of the ejected electron versus frequency (s^{-1}) of the incident light, for an unknown metal. ($1\text{eV} = 1.6022 \times 10^{-19} \text{ J}$)

Based on the plot, answer the questions **a to c** below:

a. 1 point On the plot above, label the **threshold frequency**.

b. 2 points What is the value of the work potential (in J) of the unknown metal?

c. 3 points What is the *longest* wavelength (in nm) required to eject an electron from this metal's surface? Show your work including equation(s) used. (3 sig figs)

d. 2 points In the space provided below, plot the following

(Assume frequency is *above* threshold frequency):

- Kinetic energy* of an ejected electron versus the *intensity* of the incident light (left)
- Number of electrons* ejected versus the *frequency* of the incident light (right)



**Question 3** **6 points****a. 2 points**

In the ground state of Ge (Germanium; $Z = 32$) atom, **name** the following orbitals:

i. Orbital containing the electron that requires the maximum energy to be removed.

ii. Orbital containing the electron that experiences the minimum effective nuclear charge.

b. 2 points

An unknown anion (A^-) has the following *excited* electronic configuration: $[Ar] 4s^2 3d^3 4p^1$

(Ar: Argon; $Z = 18$)

Give the **condensed ground state electronic configuration** of the atom, **A** and the cation, **A^{2+}** .
Show your work. (Assume all electronic transitions are possible)

Electronic Configuration of A:

Electronic Configuration of A^{2+} :

c. 2 points

In a single atom, determine how many electrons can have the following set of quantum numbers?

(i) $n = 5; l = 3$

In a single atom, determine how many electrons can have the following set of quantum numbers?

(i) $n = 3; l = 2; m_s = +1/2$

**Question 4** **8 points**

Rank the following from 1 to 4 in **increasing** order. Provide the ranking in the boxes provided.

Note: Atomic number (Z) are provided for relevant atoms

a) Increasing (lowest to highest) order of *ionization energy*: (1= lowest; 4 = highest)

Mg

Mg: Z = 12

Rb

Rb: Z = 37

Cs

Cs: Z = 55

K

K : Z = 19

b) Increasing (lowest to highest) order of *bond angle*: (1= lowest; 4 = highest)

H₂OCCl₄XeF₄

HCN

c) Increasing (smallest to largest) order of *ionic size* (1= smallest; 4 = largest)

S²⁻

S: Z = 16

O²⁻

O: Z = 8

Be²⁺

Be: Z = 4

Na⁺

Na: Z = 11

d) Increasing (lowest to highest) order of *bond energy* (1= lowest; 4 = highest)

P-F

P-Cl

P-I

P-Br

**Question 5** **12 points**

a. 9 points For the molecules HClO_4 and TeF_6 , draw the most stable *Lewis structure* showing appropriate shape and *dashed/wedged bonds*, based on VSEPR. Determine the **molecular geometry** around the central atom. For HClO_4 , label the *most polar bond*.



b. 3 points (Note: Lone pairs are not shown in structures below)

(i) For the three *resonance* structures (1, 2 and 3) shown below, answer the following question:



(I) Circle the *most contributing* resonance structure.

Structure 1

Structure 2

Structure 3

(ii) For the three structures given (4, 5, and 6) below, answer the following questions:



(I) Circle all the structures that represent *resonance structures* of each other:

Structure 4

Structure 5

Structure 6

(II) Among the resonance structures, circle the *most contributing* resonance structure.

Structure 4

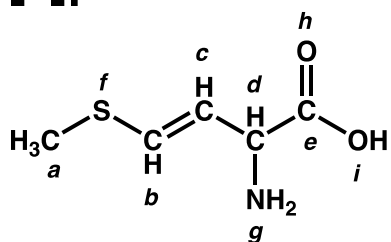
Structure 5

Structure 6

**Question 6** **6 points**

Using the data given below, calculate the **lattice energy** for SrO(s).

	ΔH° (kJ/mol)
$\text{Sr}(s) \longrightarrow \text{Sr}(g)$	+180
$\text{Sr}(g) \longrightarrow \text{Sr}^{2+}(g) + 2e^-$	+1468
$\text{O}_2(g) \longrightarrow 2\text{O}(g)$	+500
$\text{O}(g) + 2e^- \longrightarrow \text{O}^{2-}(g)$	+600
$\text{Sr}(s) + \frac{1}{2}\text{O}_2(g) \longrightarrow \text{SrO}(s)$	-550

**Question 7 10 points**

Consider the molecule given on the left, and answer questions (i) to (vi) using chemical bonding theories that we have learnt in class (valence bond theory, hybridization).

Note that the lone pairs are not shown in the structure:

i. Consider the carbon atoms marked in as ***a***, ***b***, ***c***, ***d***, and ***e*** in the molecule, and determine **hybridization** at each of the carbon centers. (2.5 points)

Atom	Hybridization
Carbon (<i>a</i>)	
Carbon (<i>b</i>)	
Carbon (<i>c</i>)	
Carbon (<i>d</i>)	
Carbon (<i>e</i>)	

ii. Now, consider the sulfur atom (S) marked as ***f***, nitrogen (N) atom (***g***), and oxygen(O) atoms (***h*** and ***i***), and determine their **hybridization**. (2 points)

Atom	Hybridization
Sulfur (<i>f</i>)	
Nitrogen (<i>g</i>)	
Oxygen (<i>h</i>)	
Oxygen (<i>i</i>)	

iii. **Describe** all the bonds formed by the carbon atoms ***a*** and ***b***, using sigma and pi designation as appropriate. (2 points)

Atom	Bonds formed (use sigma and pi as appropriate)
Carbon (<i>a</i>)	
Carbon (<i>b</i>)	

iv. How many total sp^3 -s sigma bonds are in the molecule? (1 point)

v. Determine **electronic** and **molecular geometry** at carbon (C) labeled ***e*** and sulfur (S) labeled ***f*** centres. (2 points)

Atom	Electronic Geometry	Molecular Geometry
Carbon (<i>e</i>)		
Sulfur (<i>f</i>)		

vi. Circle the value from the choices given below, that describes the **C-S-C bond angle at sulfur** (S) marked ***f***: (0.5 points)

>90° to <109.5° 109.5° >109.5 to <120° 120° >120° to <180°


Question 8 12 points

Using the **valence shell orbitals only** draw a **complete** molecular orbital (MO) diagram (with no s-p mixing) for the molecule **PS**, in the space provided below. **(6 points)**

Assume that 3s and 3p atomic orbitals will form molecular orbitals like those formed by the combination of 2s and 2p atomic orbitals that we learnt in class, and there is no s-p mixing.

Show **energy arrow**, **atomic orbitals** involved, **atom labels**, the **molecular orbitals** that are formed, **filled electrons** in the MO diagram etc. Make sure to label all the atomic and molecular orbitals.

Using the MO diagram above, answer the following questions:

- (i) Which one of the following will have the highest bond order? Circle the correct choice
 A) PS^{1-} B) PS C) PS^{1+}
- (ii) Which one of the following will have the longest bond length? Circle the correct choice
 A) PS^{1-} B) PS C) PS^{1+}
- (iii) Which of the following will be diamagnetic? Circle the correct choice.
 A) PS^{1-} B) PS C) PS^{1+}
- (iv) Draw Lewis dot structure of the molecule with the shortest bond length (based on the MO diagram) from among the three (PS^{1-} , PS, PS^{1+}), and indicate the bond order calculated from the Lewis dot structure. **(3 points)**

Lewis Dot Structure

Bond Order



344D3608-CBC2-4ED8-997B-09845D796468

midterm-exam-93388

#466 Page 10 of 10

SCRAP PAGE

**CHEMISTRY 110 –Midterm Exam****Thursday November 7, 2024, 6:30 – 8:30 pm**

Examiners: Pallavi Sirjoosingh and Ashok Kakkar

First name (please write as legibly as possible within the boxes)

--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--

Last name

--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--

Student ID number

--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--

This booklet has 8 questions on 10 pages (The last page can be used as a scrap page).

The staple must NOT be removed from the question booklet. Your intact question booklet must be returned at the end of the exam.

A separate booklet contains a periodic table, and datasheet.

Write your answers directly under the questions in the space provided. Do not write on the QR codes or near the top edges (as it may be cut off).

Please write **legibly and dark enough** for the scan to be readable. For students who use English as a second language, dictionaries are permitted. Calculators are allowed.

DO NOT ASK QUESTIONS OF THE INVIGILATORS about interpretations of the exam questions. If you have a concern about a question, please summarize it at the top of this page. All concerns will be evaluated after the exam. GOOD LUCK!

**Question 1 8 points** (*Assume all transitions are possible*)

a. 6 points Calculate the **energy (in kJ/mol)** required to ionize (completely remove) the electron from the **n=3** level of **He⁺ ion** (He: Helium). Show your work including equations. (3 sig. figs.)

b. 2 points

(I) Select the largest (longest) wavelength from the three choices below:

- i. Wavelength of photon to *completely remove* an electron from *1s orbital of Hydrogen atom*
- ii. Wavelength of photon to *completely remove* an electron from *1s orbital of Boron atom (B)*
- iii. Wavelength of photon to *completely remove* an electron from *1s orbital of Boron cation (B⁺)*

i or ii or iii

(II) Select the smallest (shortest) wavelength from the three choices below (*for H atom*):

- i. Wavelength of photon *emitted* for an electron transitioning from *2p to 1s orbital*
- ii. Wavelength of photon *absorbed* for an electron transitioning from *2s to 4p orbital*
- iii. Wavelength of photon *absorbed* for an electron transitioning from *1s to 3p orbital*

i or ii or iii



Question 2 8 points



A Chem110 student plots the graph shown on the left, from data obtained for the kinetic energy (eV) of the ejected electron versus frequency (s^{-1}) of the incident light, for an unknown metal. ($1\text{eV} = 1.6022 \times 10^{-19} \text{ J}$)

Based on the plot, answer the questions **a to c** below:

a. 1 point On the plot above, label the **threshold frequency**.

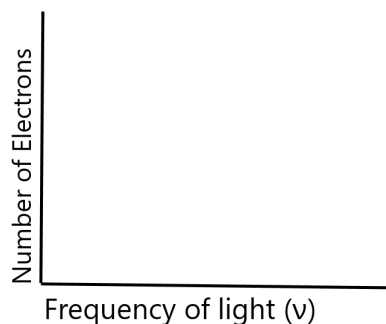
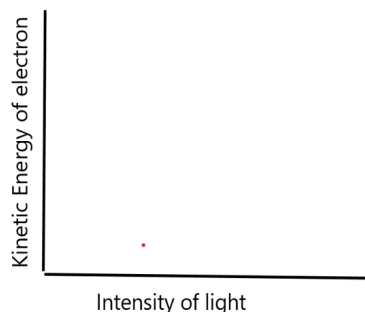
b. 2 points What is the value of the work potential (in J) of the unknown metal?

c. 3 points What is the *longest* wavelength (in nm) required to eject an electron from this metal's surface? Show your work including equation(s) used. (3 sig figs)

d. 2 points In the space provided below, plot the following

(Assume frequency is *above* threshold frequency):

- Kinetic energy* of an ejected electron versus the *intensity* of the incident light (left)
- Number of electrons* ejected versus the *frequency* of the incident light (right)



**Question 3** **6 points****a. 2 points**

In the ground state of Ge (Germanium; $Z = 32$) atom, **name** the following orbitals:

i. Orbital containing the electron that requires the maximum energy to be removed.

ii. Orbital containing the electron that experiences the minimum effective nuclear charge.

b. 2 points

An unknown anion (A^-) has the following *excited* electronic configuration: $[Ar] 4s^2 3d^3 4p^1$

(Ar: Argon; $Z = 18$)

Give the **condensed ground state electronic configuration** of the atom, **A** and the cation, **A^{2+}** .
Show your work. (Assume all electronic transitions are possible)

Electronic Configuration of A:

Electronic Configuration of A^{2+} :

c. 2 points

In a single atom, determine how many electrons can have the following set of quantum numbers?

(i) $n = 5; l = 3$

In a single atom, determine how many electrons can have the following set of quantum numbers?

(i) $n = 3; l = 2; m_s = +1/2$

**Question 4** **8 points**

Rank the following from 1 to 4 in **increasing** order. Provide the ranking in the boxes provided.

Note: Atomic number (Z) are provided for relevant atoms

a) Increasing (lowest to highest) order of *ionization energy*: (1= lowest; 4 = highest)

Mg

Mg: Z = 12

Rb

Rb: Z = 37

Cs

Cs: Z = 55

K

K : Z = 19

b) Increasing (lowest to highest) order of *bond angle*: (1= lowest; 4 = highest)

H₂OCCl₄XeF₄

HCN

c) Increasing (smallest to largest) order of *ionic size* (1= smallest; 4 = largest)

S²⁻

S: Z = 16

O²⁻

O: Z = 8

Be²⁺

Be: Z = 4

Na⁺

Na: Z = 11

d) Increasing (lowest to highest) order of *bond energy* (1= lowest; 4 = highest)

P-F

P-Cl

P-I

P-Br

**Question 5** **12 points**

a. 9 points For the molecules HClO_4 and TeF_6 , draw the most stable *Lewis structure* showing appropriate shape and *dashed/wedged bonds*, based on VSEPR. Determine the **molecular geometry** around the central atom. For HClO_4 , label the *most polar bond*.



b. 3 points (Note: Lone pairs are not shown in structures below)

(i) For the three *resonance* structures (1, 2 and 3) shown below, answer the following question:



(I) Circle the *most contributing* resonance structure.

Structure 1

Structure 2

Structure 3

(ii) For the three structures given (4, 5, and 6) below, answer the following questions:



(I) Circle all the structures that represent *resonance structures* of each other:

Structure 4

Structure 5

Structure 6

(II) Among the resonance structures, circle the *most contributing* resonance structure.

Structure 4

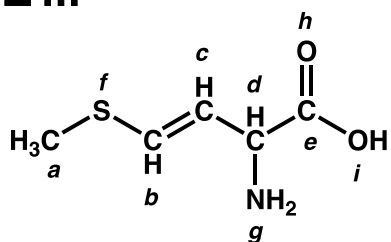
Structure 5

Structure 6

**Question 6** **6 points**

Using the data given below, calculate the **lattice energy** for SrO(s).

	ΔH° (kJ/mol)
$\text{Sr}(s) \longrightarrow \text{Sr}(g)$	+180
$\text{Sr}(g) \longrightarrow \text{Sr}^{2+}(g) + 2e^-$	+1468
$\text{O}_2(g) \longrightarrow 2\text{O}(g)$	+500
$\text{O}(g) + 2e^- \longrightarrow \text{O}^{2-}(g)$	+600
$\text{Sr}(s) + \frac{1}{2}\text{O}_2(g) \longrightarrow \text{SrO}(s)$	-550

**Question 7 10 points**

Consider the molecule given on the left, and answer questions (i) to (vi) using chemical bonding theories that we have learnt in class (valence bond theory, hybridization).

Note that the lone pairs are not shown in the structure:

i. Consider the carbon atoms marked in as ***a***, ***b***, ***c***, ***d***, and ***e*** in the molecule, and determine **hybridization** at each of the carbon centers. (2.5 points)

Atom	Hybridization
Carbon (<i>a</i>)	
Carbon (<i>b</i>)	
Carbon (<i>c</i>)	
Carbon (<i>d</i>)	
Carbon (<i>e</i>)	

ii. Now, consider the sulfur atom (S) marked as ***f***, nitrogen (N) atom (***g***), and oxygen(O) atoms (***h*** and ***i***), and determine their **hybridization**. (2 points)

Atom	Hybridization
Sulfur (<i>f</i>)	
Nitrogen (<i>g</i>)	
Oxygen (<i>h</i>)	
Oxygen (<i>i</i>)	

iii. **Describe** all the bonds formed by the carbon atoms ***a*** and ***b***, using sigma and pi designation as appropriate. (2 points)

Atom	Bonds formed (use sigma and pi as appropriate)
Carbon (<i>a</i>)	
Carbon (<i>b</i>)	

iv. How many total sp^3 -s sigma bonds are in the molecule? (1 point)

v. Determine **electronic** and **molecular geometry** at carbon (C) labeled ***e*** and sulfur (S) labeled ***f*** centres. (2 points)

Atom	Electronic Geometry	Molecular Geometry
Carbon (<i>e</i>)		
Sulfur (<i>f</i>)		

vi. Circle the value from the choices given below, that describes the **C-S-C bond angle at sulfur** (S) marked ***f***: (0.5 points)

>90° to <109.5°

109.5°

>109.5 to <120°

120°

>120° to <180°


Question 8 12 points

Using the **valence shell orbitals only** draw a **complete** molecular orbital (MO) diagram (with no s-p mixing) for the molecule **PS**, in the space provided below. **(6 points)**

Assume that 3s and 3p atomic orbitals will form molecular orbitals like those formed by the combination of 2s and 2p atomic orbitals that we learnt in class, and there is no s-p mixing.

Show **energy arrow**, **atomic orbitals** involved, **atom labels**, the **molecular orbitals** that are formed, **filled electrons** in the MO diagram etc. Make sure to label all the atomic and molecular orbitals.

Using the MO diagram above, answer the following questions:

- (i) Which one of the following will have the highest bond order? Circle the correct choice
 A) PS^{1-} B) PS C) PS^{1+}
- (ii) Which one of the following will have the longest bond length? Circle the correct choice
 A) PS^{1-} B) PS C) PS^{1+}
- (iii) Which of the following will be diamagnetic? Circle the correct choice.
 A) PS^{1-} B) PS C) PS^{1+}
- (iv) Draw Lewis dot structure of the molecule with the shortest bond length (based on the MO diagram) from among the three (PS^{1-} , PS, PS^{1+}), and indicate the bond order calculated from the Lewis dot structure. **(3 points)**

Lewis Dot Structure

Bond Order



589D0465-618C-4B6E-8DEB-7137D1D3B4A6

midterm-exam-93388

#467 Page 10 of 10

SCRAP PAGE

**CHEMISTRY 110 –Midterm Exam****Thursday November 7, 2024, 6:30 – 8:30 pm**

Examiners: Pallavi Sirjoosingh and Ashok Kakkar

First name (please write as legibly as possible within the boxes)

--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--

Last name

--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--

Student ID number

--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--

This booklet has 8 questions on 10 pages (The last page can be used as a scrap page).

The staple must NOT be removed from the question booklet. Your intact question booklet must be returned at the end of the exam.

A separate booklet contains a periodic table, and datasheet.

Write your answers directly under the questions in the space provided. Do not write on the QR codes or near the top edges (as it may be cut off).

Please write **legibly and dark enough** for the scan to be readable. For students who use English as a second language, dictionaries are permitted. Calculators are allowed.

DO NOT ASK QUESTIONS OF THE INVIGILATORS about interpretations of the exam questions. If you have a concern about a question, please summarize it at the top of this page. All concerns will be evaluated after the exam. GOOD LUCK!

**Question 1 8 points** (*Assume all transitions are possible*)

a. 6 points Calculate the **energy (in kJ/mol)** required to ionize (completely remove) the electron from the **$n=3$** level of **He^+ ion** (He: Helium). Show your work including equations. (3 sig. figs.)

b. 2 points

(I) Select the largest (longest) wavelength from the three choices below:

- i. Wavelength of photon to *completely remove* an electron from *1s orbital of Hydrogen atom*
- ii. Wavelength of photon to *completely remove* an electron from *1s orbital of Boron atom (B)*
- iii. Wavelength of photon to *completely remove* an electron from *1s orbital of Boron cation (B^+)*

i or ii or iii

(II) Select the smallest (shortest) wavelength from the three choices below (*for H atom*):

- i. Wavelength of photon *emitted* for an electron transitioning from *2p to 1s orbital*
- ii. Wavelength of photon *absorbed* for an electron transitioning from *2s to 4p orbital*
- iii. Wavelength of photon *absorbed* for an electron transitioning from *1s to 3p orbital*

i or ii or iii



Question 2 8 points



A Chem110 student plots the graph shown on the left, from data obtained for the kinetic energy (eV) of the ejected electron versus frequency (s^{-1}) of the incident light, for an unknown metal. ($1\text{eV} = 1.6022 \times 10^{-19} \text{ J}$)

Based on the plot, answer the questions **a to c** below:

a. 1 point On the plot above, label the **threshold frequency**.

b. 2 points What is the value of the work potential (in J) of the unknown metal?

c. 3 points What is the *longest* wavelength (in nm) required to eject an electron from this metal's surface? Show your work including equation(s) used. (3 sig figs)

d. 2 points In the space provided below, plot the following

(Assume frequency is *above* threshold frequency):

- Kinetic energy* of an ejected electron versus the *intensity* of the incident light (left)
- Number of electrons* ejected versus the *frequency* of the incident light (right)



**Question 3** **6 points****a. 2 points**

In the ground state of Ge (Germanium; $Z = 32$) atom, **name** the following orbitals:

i. Orbital containing the electron that requires the maximum energy to be removed.

ii. Orbital containing the electron that experiences the minimum effective nuclear charge.

b. 2 points

An unknown anion (A^-) has the following *excited* electronic configuration: $[Ar] 4s^2 3d^3 4p^1$

(Ar: Argon; $Z = 18$)

Give the **condensed ground state electronic configuration** of the atom, **A** and the cation, **A^{2+}** .
Show your work. (Assume all electronic transitions are possible)

Electronic Configuration of A:

Electronic Configuration of A^{2+} :

c. 2 points

In a single atom, determine how many electrons can have the following set of quantum numbers?

(i) $n = 5; l = 3$

In a single atom, determine how many electrons can have the following set of quantum numbers?

(i) $n = 3; l = 2; m_s = +1/2$

**Question 4** **8 points**

Rank the following from 1 to 4 in **increasing** order. Provide the ranking in the boxes provided.

Note: Atomic number (Z) are provided for relevant atoms

a) Increasing (lowest to highest) order of *ionization energy*: (1= lowest; 4 = highest)

Mg

Mg: Z = 12

Rb

Rb: Z = 37

Cs

Cs: Z = 55

K

K : Z = 19

b) Increasing (lowest to highest) order of *bond angle*: (1= lowest; 4 = highest)

H₂OCCl₄XeF₄

HCN

c) Increasing (smallest to largest) order of *ionic size* (1= smallest; 4 = largest)

S²⁻

S: Z = 16

O²⁻

O: Z = 8

Be²⁺

Be: Z = 4

Na⁺

Na: Z = 11

d) Increasing (lowest to highest) order of *bond energy* (1= lowest; 4 = highest)

P-F

P-Cl

P-I

P-Br

**Question 5** **12 points**

a. 9 points For the molecules HClO_4 and TeF_6 , draw the most stable *Lewis structure* showing appropriate shape and *dashed/wedged bonds*, based on VSEPR. Determine the **molecular geometry** around the central atom. For HClO_4 , label the *most polar bond*.



b. 3 points (Note: Lone pairs are not shown in structures below)

(i) For the three *resonance* structures (1, 2 and 3) shown below, answer the following question:



(I) Circle the *most contributing* resonance structure.

Structure 1

Structure 2

Structure 3

(ii) For the three structures given (4, 5, and 6) below, answer the following questions:



(I) Circle all the structures that represent *resonance structures* of each other:

Structure 4

Structure 5

Structure 6

(II) Among the resonance structures, circle the *most contributing* resonance structure.

Structure 4

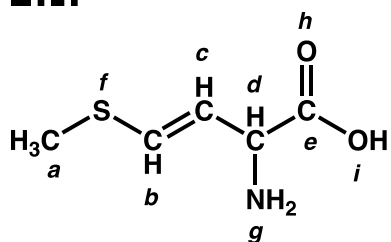
Structure 5

Structure 6

**Question 6** **6 points**

Using the data given below, calculate the **lattice energy** for SrO(s).

	ΔH° (kJ/mol)
$\text{Sr}(s) \longrightarrow \text{Sr}(g)$	+180
$\text{Sr}(g) \longrightarrow \text{Sr}^{2+}(g) + 2e^-$	+1468
$\text{O}_2(g) \longrightarrow 2\text{O}(g)$	+500
$\text{O}(g) + 2e^- \longrightarrow \text{O}^{2-}(g)$	+600
$\text{Sr}(s) + \frac{1}{2}\text{O}_2(g) \longrightarrow \text{SrO}(s)$	-550

**Question 7 10 points**

Consider the molecule given on the left, and answer questions (i) to (vi) using chemical bonding theories that we have learnt in class (valence bond theory, hybridization).

Note that the lone pairs are not shown in the structure:

i. Consider the carbon atoms marked in as ***a***, ***b***, ***c***, ***d***, and ***e*** in the molecule, and determine **hybridization** at each of the carbon centers. (2.5 points)

Atom	Hybridization
Carbon (<i>a</i>)	
Carbon (<i>b</i>)	
Carbon (<i>c</i>)	
Carbon (<i>d</i>)	
Carbon (<i>e</i>)	

ii. Now, consider the sulfur atom (S) marked as ***f***, nitrogen (N) atom (***g***), and oxygen(O) atoms (***h*** and ***i***), and determine their **hybridization**. (2 points)

Atom	Hybridization
Sulfur (<i>f</i>)	
Nitrogen (<i>g</i>)	
Oxygen (<i>h</i>)	
Oxygen (<i>i</i>)	

iii. **Describe** all the bonds formed by the carbon atoms ***a*** and ***b***, using sigma and pi designation as appropriate. (2 points)

Atom	Bonds formed (use sigma and pi as appropriate)
Carbon (<i>a</i>)	
Carbon (<i>b</i>)	

iv. How many total sp^3 -s sigma bonds are in the molecule? (1 point)

v. Determine **electronic** and **molecular geometry** at carbon (C) labeled ***e*** and sulfur (S) labeled ***f*** centres. (2 points)

Atom	Electronic Geometry	Molecular Geometry
Carbon (<i>e</i>)		
Sulfur (<i>f</i>)		

vi. Circle the value from the choices given below, that describes the **C-S-C bond angle at sulfur** (S) marked ***f***: (0.5 points)

>90° to <109.5° 109.5° >109.5 to <120° 120° >120° to <180°

**Question 8 12 points**

Using the **valence shell orbitals only** draw a **complete** molecular orbital (MO) diagram (with no s-p mixing) for the molecule **PS**, in the space provided below. **(6 points)**

Assume that 3s and 3p atomic orbitals will form molecular orbitals like those formed by the combination of 2s and 2p atomic orbitals that we learnt in class, and there is no s-p mixing.

Show **energy arrow**, **atomic orbitals** involved, **atom labels**, the **molecular orbitals** that are formed, **filled electrons** in the MO diagram etc. Make sure to label all the atomic and molecular orbitals.

Using the MO diagram above, answer the following questions:

- (i) Which one of the following will have the highest bond order? Circle the correct choice
A) PS^{1-} B) PS C) PS^{1+}
- (ii) Which one of the following will have the longest bond length? Circle the correct choice
A) PS^{1-} B) PS C) PS^{1+}
- (iii) Which of the following will be diamagnetic? Circle the correct choice.
A) PS^{1-} B) PS C) PS^{1+}
- (iv) Draw Lewis dot structure of the molecule with the shortest bond length (based on the MO diagram) from among the three (PS^{1-} , PS, PS^{1+}), and indicate the bond order calculated from the Lewis dot structure. **(3 points)**

Lewis Dot Structure

Bond Order



EA0770F4-174E-4CD9-B65C-0D7536BFE638

midterm-exam-93388

#468 Page 10 of 10

SCRAP PAGE

**CHEMISTRY 110 –Midterm Exam****Thursday November 7, 2024, 6:30 – 8:30 pm****Examiners: Pallavi Sirjoosingh and Ashok Kakkar**

First name (please write as legibly as possible within the boxes)

--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--

Last name

--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--

Student ID number

--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--

This booklet has 8 questions on 10 pages (The last page can be used as a scrap page).

The staple must NOT be removed from the question booklet. Your intact question booklet must be returned at the end of the exam.

A separate booklet contains a periodic table, and datasheet.

Write your answers directly under the questions in the space provided. Do not write on the QR codes or near the top edges (as it may be cut off).

Please write **legibly and dark enough** for the scan to be readable. For students who use English as a second language, dictionaries are permitted. Calculators are allowed.

DO NOT ASK QUESTIONS OF THE INVIGILATORS about interpretations of the exam questions. If you have a concern about a question, please summarize it at the top of this page. All concerns will be evaluated after the exam. GOOD LUCK!

**Question 1 8 points** (*Assume all transitions are possible*)

a. 6 points Calculate the **energy (in kJ/mol)** required to ionize (completely remove) the electron from the **n=3** level of **He⁺ ion** (He: Helium). Show your work including equations. (3 sig. figs.)

b. 2 points

(I) Select the largest (longest) wavelength from the three choices below:

- i. Wavelength of photon to *completely remove* an electron from *1s orbital of Hydrogen atom*
- ii. Wavelength of photon to *completely remove* an electron from *1s orbital of Boron atom (B)*
- iii. Wavelength of photon to *completely remove* an electron from *1s orbital of Boron cation (B⁺)*

i or ii or iii

(II) Select the smallest (shortest) wavelength from the three choices below (*for H atom*):

- i. Wavelength of photon *emitted* for an electron transitioning from *2p to 1s orbital*
- ii. Wavelength of photon *absorbed* for an electron transitioning from *2s to 4p orbital*
- iii. Wavelength of photon *absorbed* for an electron transitioning from *1s to 3p orbital*

i or ii or iii



Question 2 8 points



A Chem110 student plots the graph shown on the left, from data obtained for the kinetic energy (eV) of the ejected electron versus frequency (s^{-1}) of the incident light, for an unknown metal. ($1\text{eV} = 1.6022 \times 10^{-19} \text{ J}$)

Based on the plot, answer the questions **a to c** below:

a. 1 point On the plot above, label the **threshold frequency**.

b. 2 points What is the value of the work potential (in J) of the unknown metal?

c. 3 points What is the *longest* wavelength (in nm) required to eject an electron from this metal's surface? Show your work including equation(s) used. (3 sig figs)

d. 2 points In the space provided below, plot the following

(Assume frequency is *above* threshold frequency):

- i. *Kinetic energy* of an ejected electron versus the *intensity* of the incident light (left)
- ii. *Number of electrons* ejected versus the *frequency* of the incident light (right)



**Question 3** **6 points****a. 2 points**

In the ground state of Ge (Germanium; $Z = 32$) atom, **name** the following orbitals:

i. Orbital containing the electron that requires the maximum energy to be removed.

ii. Orbital containing the electron that experiences the minimum effective nuclear charge.

b. 2 points

An unknown anion (A^-) has the following *excited* electronic configuration: $[Ar] 4s^2 3d^3 4p^1$

(Ar: Argon; $Z = 18$)

Give the **condensed ground state electronic configuration** of the atom, **A** and the cation, A^{2+} .
Show your work. (Assume all electronic transitions are possible)

Electronic Configuration of A:

Electronic Configuration of A^{2+} :

c. 2 points

In a single atom, determine how many electrons can have the following set of quantum numbers?

(i) $n = 5; l = 3$

In a single atom, determine how many electrons can have the following set of quantum numbers?

(i) $n = 3; l = 2; m_s = +1/2$

**Question 4** **8 points**

Rank the following from 1 to 4 in **increasing** order. Provide the ranking in the boxes provided.

Note: Atomic number (Z) are provided for relevant atoms

a) Increasing (lowest to highest) order of *ionization energy*: (1= lowest; 4 = highest)

Mg

Mg: Z = 12

Rb

Rb: Z = 37

Cs

Cs: Z = 55

K

K : Z = 19

b) Increasing (lowest to highest) order of *bond angle*: (1= lowest; 4 = highest)

H₂OCCl₄XeF₄

HCN

c) Increasing (smallest to largest) order of *ionic size* (1= smallest; 4 = largest)

S²⁻

S: Z = 16

O²⁻

O: Z = 8

Be²⁺

Be: Z = 4

Na⁺

Na: Z = 11

d) Increasing (lowest to highest) order of *bond energy* (1= lowest; 4 = highest)

P-F

P-Cl

P-I

P-Br

**Question 5** **12 points**

a. 9 points For the molecules HClO_4 and TeF_6 , draw the most stable *Lewis structure* showing appropriate shape and *dashed/wedged bonds*, based on VSEPR. Determine the **molecular geometry** around the central atom. For HClO_4 , label the *most polar bond*.



b. 3 points (Note: Lone pairs are not shown in structures below)

(i) For the three *resonance* structures (1, 2 and 3) shown below, answer the following question:



(I) Circle the *most contributing* resonance structure.

Structure 1

Structure 2

Structure 3

(ii) For the three structures given (4, 5, and 6) below, answer the following questions:



(I) Circle all the structures that represent *resonance structures* of each other:

Structure 4

Structure 5

Structure 6

(II) Among the resonance structures, circle the *most contributing* resonance structure.

Structure 4

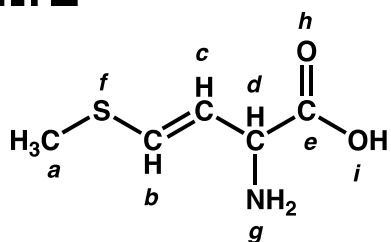
Structure 5

Structure 6

**Question 6** **6 points**

Using the data given below, calculate the **lattice energy** for SrO(s).

	ΔH° (kJ/mol)
$\text{Sr}(s) \longrightarrow \text{Sr}(g)$	+180
$\text{Sr}(g) \longrightarrow \text{Sr}^{2+}(g) + 2e^-$	+1468
$\text{O}_2(g) \longrightarrow 2\text{O}(g)$	+500
$\text{O}(g) + 2e^- \longrightarrow \text{O}^{2-}(g)$	+600
$\text{Sr}(s) + \frac{1}{2}\text{O}_2(g) \longrightarrow \text{SrO}(s)$	-550

**Question 7 10 points**

Consider the molecule given on the left, and answer questions (i) to (vi) using chemical bonding theories that we have learnt in class (valence bond theory, hybridization).

Note that the lone pairs are not shown in the structure:

i. Consider the carbon atoms marked in as ***a***, ***b***, ***c***, ***d***, and ***e*** in the molecule, and determine **hybridization** at each of the carbon centers. (2.5 points)

Atom	Hybridization
Carbon (<i>a</i>)	
Carbon (<i>b</i>)	
Carbon (<i>c</i>)	
Carbon (<i>d</i>)	
Carbon (<i>e</i>)	

ii. Now, consider the sulfur atom (S) marked as ***f***, nitrogen (N) atom (***g***), and oxygen(O) atoms (***h*** and ***i***), and determine their **hybridization**. (2 points)

Atom	Hybridization
Sulfur (<i>f</i>)	
Nitrogen (<i>g</i>)	
Oxygen (<i>h</i>)	
Oxygen (<i>i</i>)	

iii. **Describe** all the bonds formed by the carbon atoms ***a*** and ***b***, using sigma and pi designation as appropriate. (2 points)

Atom	Bonds formed (use sigma and pi as appropriate)
Carbon (<i>a</i>)	
Carbon (<i>b</i>)	

iv. How many total sp^3 -s sigma bonds are in the molecule? (1 point)

v. Determine **electronic** and **molecular geometry** at carbon (C) labeled ***e*** and sulfur (S) labeled ***f*** centres. (2 points)

Atom	Electronic Geometry	Molecular Geometry
Carbon (<i>e</i>)		
Sulfur (<i>f</i>)		

vi. Circle the value from the choices given below, that describes the **C-S-C bond angle at sulfur** (S) marked ***f***: (0.5 points)

>90° to <109.5° 109.5° >109.5 to <120° 120° >120° to <180°



Question 8 12 points

Using the **valence shell orbitals only** draw a **complete** molecular orbital (MO) diagram (with no s-p mixing) for the molecule **PS**, in the space provided below. **(6 points)**

Assume that 3s and 3p atomic orbitals will form molecular orbitals like those formed by the combination of 2s and 2p atomic orbitals that we learnt in class, and there is no s-p mixing.

Show **energy arrow**, **atomic orbitals** involved, **atom labels**, the **molecular orbitals** that are formed, **filled electrons** in the MO diagram etc. Make sure to label all the atomic and molecular orbitals.

Using the MO diagram above, answer the following questions:

- (i) Which one of the following will have the highest bond order? Circle the correct choice
 A) PS^{1-} B) PS C) PS^{1+}
- (ii) Which one of the following will have the longest bond length? Circle the correct choice
 A) PS^{1-} B) PS C) PS^{1+}
- (iii) Which of the following will be diamagnetic? Circle the correct choice.
 A) PS^{1-} B) PS C) PS^{1+}
- (iv) Draw Lewis dot structure of the molecule with the shortest bond length (based on the MO diagram) from among the three (PS^{1-} , PS, PS^{1+}), and indicate the bond order calculated from the Lewis dot structure. **(3 points)**

Lewis Dot Structure

Bond Order



74B2280D-693C-456C-833A-43C9BE5A47A2

midterm-exam-93388

#469 Page 10 of 10

SCRAP PAGE

**CHEMISTRY 110 –Midterm Exam****Thursday November 7, 2024, 6:30 – 8:30 pm**

Examiners: Pallavi Sirjoosingh and Ashok Kakkar

First name (please write as legibly as possible within the boxes)

--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--

Last name

--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--

Student ID number

--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--

This booklet has 8 questions on 10 pages (The last page can be used as a scrap page).

The staple must NOT be removed from the question booklet. Your intact question booklet must be returned at the end of the exam.

A separate booklet contains a periodic table, and datasheet.

Write your answers directly under the questions in the space provided. Do not write on the QR codes or near the top edges (as it may be cut off).

Please write **legibly and dark enough** for the scan to be readable. For students who use English as a second language, dictionaries are permitted. Calculators are allowed.

DO NOT ASK QUESTIONS OF THE INVIGILATORS about interpretations of the exam questions. If you have a concern about a question, please summarize it at the top of this page. All concerns will be evaluated after the exam. GOOD LUCK!

**Question 1 8 points** (*Assume all transitions are possible*)

a. 6 points Calculate the **energy (in kJ/mol)** required to ionize (completely remove) the electron from the **n=3** level of **He⁺ ion** (He: Helium). Show your work including equations. (3 sig. figs.)

b. 2 points

(I) Select the largest (longest) wavelength from the three choices below:

- i. Wavelength of photon to *completely remove* an electron from *1s orbital of Hydrogen atom*
- ii. Wavelength of photon to *completely remove* an electron from *1s orbital of Boron atom (B)*
- iii. Wavelength of photon to *completely remove* an electron from *1s orbital of Boron cation (B⁺)*

i or ii or iii

(II) Select the smallest (shortest) wavelength from the three choices below (*for H atom*):

- i. Wavelength of photon *emitted* for an electron transitioning from *2p to 1s orbital*
- ii. Wavelength of photon *absorbed* for an electron transitioning from *2s to 4p orbital*
- iii. Wavelength of photon *absorbed* for an electron transitioning from *1s to 3p orbital*

i or ii or iii



Question 2 8 points



A Chem110 student plots the graph shown on the left, from data obtained for the kinetic energy (eV) of the ejected electron versus frequency (s^{-1}) of the incident light, for an unknown metal. ($1\text{eV} = 1.6022 \times 10^{-19} \text{ J}$)

Based on the plot, answer the questions **a to c** below:

a. 1 point On the plot above, label the **threshold frequency**.

b. 2 points What is the value of the work potential (in J) of the unknown metal?

c. 3 points What is the *longest* wavelength (in nm) required to eject an electron from this metal's surface? Show your work including equation(s) used. (3 sig figs)

d. 2 points In the space provided below, plot the following

(Assume frequency is *above* threshold frequency):

- i. *Kinetic energy* of an ejected electron versus the *intensity* of the incident light (left)
- ii. *Number of electrons* ejected versus the *frequency* of the incident light (right)



**Question 3** **6 points****a. 2 points**

In the ground state of Ge (Germanium; $Z = 32$) atom, **name** the following orbitals:

i. Orbital containing the electron that requires the maximum energy to be removed.

ii. Orbital containing the electron that experiences the minimum effective nuclear charge.

b. 2 points

An unknown anion (A^-) has the following *excited* electronic configuration: $[Ar] 4s^2 3d^3 4p^1$

(Ar: Argon; $Z = 18$)

Give the **condensed ground state electronic configuration** of the atom, **A** and the cation, **A^{2+}** .
Show your work. (Assume all electronic transitions are possible)

Electronic Configuration of A:

Electronic Configuration of A^{2+} :

c. 2 points

In a single atom, determine how many electrons can have the following set of quantum numbers?

(i) $n = 5; l = 3$

In a single atom, determine how many electrons can have the following set of quantum numbers?

(i) $n = 3; l = 2; m_s = +1/2$

**Question 4** **8 points**

Rank the following from 1 to 4 in **increasing** order. Provide the ranking in the boxes provided.

Note: Atomic number (Z) are provided for relevant atoms

a) Increasing (lowest to highest) order of *ionization energy*: (1= lowest; 4 = highest)

Mg

Mg: Z = 12

Rb

Rb: Z = 37

Cs

Cs: Z = 55

K

K : Z = 19

b) Increasing (lowest to highest) order of *bond angle*: (1= lowest; 4 = highest)

H₂OCCl₄XeF₄

HCN

c) Increasing (smallest to largest) order of *ionic size* (1= smallest; 4 = largest)

S²⁻

S: Z = 16

O²⁻

O: Z = 8

Be²⁺

Be: Z = 4

Na⁺

Na: Z = 11

d) Increasing (lowest to highest) order of *bond energy* (1= lowest; 4 = highest)

P-F

P-Cl

P-I

P-Br

**Question 5** **12 points**

a. 9 points For the molecules HClO_4 and TeF_6 , draw the most stable *Lewis structure* showing appropriate shape and *dashed/wedged bonds*, based on VSEPR. Determine the **molecular geometry** around the central atom. For HClO_4 , label the *most polar bond*.



b. 3 points (Note: Lone pairs are not shown in structures below)

(i) For the three *resonance* structures (1, 2 and 3) shown below, answer the following question:



(I) Circle the *most contributing* resonance structure.

Structure 1

Structure 2

Structure 3

(ii) For the three structures given (4, 5, and 6) below, answer the following questions:



(I) Circle all the structures that represent *resonance structures* of each other:

Structure 4

Structure 5

Structure 6

(II) Among the resonance structures, circle the *most contributing* resonance structure.

Structure 4

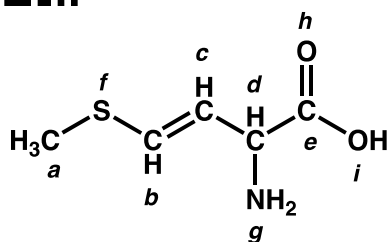
Structure 5

Structure 6

**Question 6** **6 points**

Using the data given below, calculate the **lattice energy** for SrO(s).

	ΔH° (kJ/mol)
$\text{Sr}(s) \longrightarrow \text{Sr}(g)$	+180
$\text{Sr}(g) \longrightarrow \text{Sr}^{2+}(g) + 2e^-$	+1468
$\text{O}_2(g) \longrightarrow 2\text{O}(g)$	+500
$\text{O}(g) + 2e^- \longrightarrow \text{O}^{2-}(g)$	+600
$\text{Sr}(s) + \frac{1}{2}\text{O}_2(g) \longrightarrow \text{SrO}(s)$	-550

**Question 7 10 points**

Consider the molecule given on the left, and answer questions (i) to (vi) using chemical bonding theories that we have learnt in class (valence bond theory, hybridization).

Note that the lone pairs are not shown in the structure:

i. Consider the carbon atoms marked in as ***a***, ***b***, ***c***, ***d***, and ***e*** in the molecule, and determine **hybridization** at each of the carbon centers. (2.5 points)

Atom	Hybridization
Carbon (<i>a</i>)	
Carbon (<i>b</i>)	
Carbon (<i>c</i>)	
Carbon (<i>d</i>)	
Carbon (<i>e</i>)	

ii. Now, consider the sulfur atom (S) marked as ***f***, nitrogen (N) atom (***g***), and oxygen(O) atoms (***h*** and ***i***), and determine their **hybridization**. (2 points)

Atom	Hybridization
Sulfur (<i>f</i>)	
Nitrogen (<i>g</i>)	
Oxygen (<i>h</i>)	
Oxygen (<i>i</i>)	

iii. **Describe** all the bonds formed by the carbon atoms ***a*** and ***b***, using sigma and pi designation as appropriate. (2 points)

Atom	Bonds formed (use sigma and pi as appropriate)
Carbon (<i>a</i>)	
Carbon (<i>b</i>)	

iv. How many total sp^3 -s sigma bonds are in the molecule? (1 point)

v. Determine **electronic** and **molecular geometry** at carbon (C) labeled ***e*** and sulfur (S) labeled ***f*** centres. (2 points)

Atom	Electronic Geometry	Molecular Geometry
Carbon (<i>e</i>)		
Sulfur (<i>f</i>)		

vi. Circle the value from the choices given below, that describes the **C-S-C bond angle at sulfur** (S) marked ***f***: (0.5 points)

>90° to <109.5° 109.5° >109.5 to <120° 120° >120° to <180°


Question 8 12 points

Using the **valence shell orbitals only** draw a **complete** molecular orbital (MO) diagram (with no s-p mixing) for the molecule **PS**, in the space provided below. **(6 points)**

Assume that 3s and 3p atomic orbitals will form molecular orbitals like those formed by the combination of 2s and 2p atomic orbitals that we learnt in class, and there is no s-p mixing.

Show **energy arrow**, **atomic orbitals** involved, **atom labels**, the **molecular orbitals** that are formed, **filled electrons** in the MO diagram etc. Make sure to label all the atomic and molecular orbitals.

Using the MO diagram above, answer the following questions:

- (i) Which one of the following will have the highest bond order? Circle the correct choice
 A) PS^{1-} B) PS C) PS^{1+}
- (ii) Which one of the following will have the longest bond length? Circle the correct choice
 A) PS^{1-} B) PS C) PS^{1+}
- (iii) Which of the following will be diamagnetic? Circle the correct choice.
 A) PS^{1-} B) PS C) PS^{1+}
- (iv) Draw Lewis dot structure of the molecule with the shortest bond length (based on the MO diagram) from among the three (PS^{1-} , PS, PS^{1+}), and indicate the bond order calculated from the Lewis dot structure. **(3 points)**

Lewis Dot Structure

Bond Order



A497FF31-443D-4927-BB1B-633B41930DFF

midterm-exam-93388

#470 Page 10 of 10

SCRAP PAGE

**CHEMISTRY 110 –Midterm Exam****Thursday November 7, 2024, 6:30 – 8:30 pm**

Examiners: Pallavi Sirjoosingh and Ashok Kakkar

First name (please write as legibly as possible within the boxes)

--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--

Last name

--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--

Student ID number

--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--

This booklet has 8 questions on 10 pages (The last page can be used as a scrap page).

The staple must NOT be removed from the question booklet. Your intact question booklet must be returned at the end of the exam.

A separate booklet contains a periodic table, and datasheet.

Write your answers directly under the questions in the space provided. Do not write on the QR codes or near the top edges (as it may be cut off).

Please write **legibly and dark enough** for the scan to be readable. For students who use English as a second language, dictionaries are permitted. Calculators are allowed.

DO NOT ASK QUESTIONS OF THE INVIGILATORS about interpretations of the exam questions. If you have a concern about a question, please summarize it at the top of this page. All concerns will be evaluated after the exam. GOOD LUCK!

**Question 1 8 points** (*Assume all transitions are possible*)

a. 6 points Calculate the **energy (in kJ/mol)** required to ionize (completely remove) the electron from the **n=3** level of **He⁺ ion** (He: Helium). Show your work including equations. (3 sig. figs.)

b. 2 points

(I) Select the largest (longest) wavelength from the three choices below:

- i. Wavelength of photon to *completely remove* an electron from *1s orbital of Hydrogen atom*
- ii. Wavelength of photon to *completely remove* an electron from *1s orbital of Boron atom (B)*
- iii. Wavelength of photon to *completely remove* an electron from *1s orbital of Boron cation (B⁺)*

i or ii or iii

(II) Select the smallest (shortest) wavelength from the three choices below (*for H atom*):

- i. Wavelength of photon *emitted* for an electron transitioning from *2p to 1s orbital*
- ii. Wavelength of photon *absorbed* for an electron transitioning from *2s to 4p orbital*
- iii. Wavelength of photon *absorbed* for an electron transitioning from *1s to 3p orbital*

i or ii or iii



Question 2 8 points



A Chem110 student plots the graph shown on the left, from data obtained for the kinetic energy (eV) of the ejected electron versus frequency (s^{-1}) of the incident light, for an unknown metal. ($1\text{eV} = 1.6022 \times 10^{-19} \text{ J}$)

Based on the plot, answer the questions **a to c** below:

a. 1 point On the plot above, label the **threshold frequency**.

b. 2 points What is the value of the work potential (in J) of the unknown metal?

c. 3 points What is the *longest* wavelength (in nm) required to eject an electron from this metal's surface? Show your work including equation(s) used. (3 sig figs)

d. 2 points In the space provided below, plot the following

(Assume frequency is *above* threshold frequency):

- Kinetic energy* of an ejected electron versus the *intensity* of the incident light (left)
- Number of electrons* ejected versus the *frequency* of the incident light (right)



**Question 3** **6 points****a. 2 points**

In the ground state of Ge (Germanium; $Z = 32$) atom, **name** the following orbitals:

i. Orbital containing the electron that requires the maximum energy to be removed.

ii. Orbital containing the electron that experiences the minimum effective nuclear charge.

b. 2 points

An unknown anion (A^-) has the following *excited* electronic configuration: $[Ar] 4s^2 3d^3 4p^1$

(Ar: Argon; $Z = 18$)

Give the **condensed ground state electronic configuration** of the atom, **A** and the cation, **A^{2+}** .
Show your work. (Assume all electronic transitions are possible)

Electronic Configuration of A:

Electronic Configuration of A^{2+} :

c. 2 points

In a single atom, determine how many electrons can have the following set of quantum numbers?

(i) $n = 5; l = 3$

In a single atom, determine how many electrons can have the following set of quantum numbers?

(i) $n = 3; l = 2; m_s = +1/2$

**Question 4** **8 points**

Rank the following from 1 to 4 in **increasing** order. Provide the ranking in the boxes provided.

Note: Atomic number (Z) are provided for relevant atoms

a) Increasing (lowest to highest) order of *ionization energy*: (1= lowest; 4 = highest)

Mg

Mg: Z = 12

Rb

Rb: Z = 37

Cs

Cs: Z = 55

K

K : Z = 19

b) Increasing (lowest to highest) order of *bond angle*: (1= lowest; 4 = highest)

H₂OCCl₄XeF₄

HCN

c) Increasing (smallest to largest) order of *ionic size* (1= smallest; 4 = largest)

S²⁻

S: Z = 16

O²⁻

O: Z = 8

Be²⁺

Be: Z = 4

Na⁺

Na: Z = 11

d) Increasing (lowest to highest) order of *bond energy* (1= lowest; 4 = highest)

P-F

P-Cl

P-I

P-Br

**Question 5** **12 points**

a. 9 points For the molecules HClO_4 and TeF_6 , draw the most stable *Lewis structure* showing appropriate shape and *dashed/wedged bonds*, based on VSEPR. Determine the **molecular geometry** around the central atom. For HClO_4 , label the *most polar bond*.



b. 3 points (Note: Lone pairs are not shown in structures below)

(i) For the three *resonance* structures (1, 2 and 3) shown below, answer the following question:



(I) Circle the *most contributing* resonance structure.

Structure 1

Structure 2

Structure 3

(ii) For the three structures given (4, 5, and 6) below, answer the following questions:



(I) Circle all the structures that represent *resonance structures* of each other:

Structure 4

Structure 5

Structure 6

(II) Among the resonance structures, circle the *most contributing* resonance structure.

Structure 4

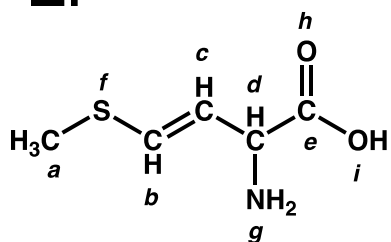
Structure 5

Structure 6

**Question 6** **6 points**

Using the data given below, calculate the **lattice energy** for SrO(s).

	ΔH° (kJ/mol)
$\text{Sr}(s) \longrightarrow \text{Sr}(g)$	+180
$\text{Sr}(g) \longrightarrow \text{Sr}^{2+}(g) + 2e^-$	+1468
$\text{O}_2(g) \longrightarrow 2\text{O}(g)$	+500
$\text{O}(g) + 2e^- \longrightarrow \text{O}^{2-}(g)$	+600
$\text{Sr}(s) + \frac{1}{2}\text{O}_2(g) \longrightarrow \text{SrO}(s)$	-550

**Question 7 10 points**

Consider the molecule given on the left, and answer questions (i) to (vi) using chemical bonding theories that we have learnt in class (valence bond theory, hybridization).

Note that the lone pairs are not shown in the structure:

i. Consider the carbon atoms marked in as ***a***, ***b***, ***c***, ***d***, and ***e*** in the molecule, and determine **hybridization** at each of the carbon centers. (2.5 points)

Atom	Hybridization
Carbon (<i>a</i>)	
Carbon (<i>b</i>)	
Carbon (<i>c</i>)	
Carbon (<i>d</i>)	
Carbon (<i>e</i>)	

ii. Now, consider the sulfur atom (S) marked as ***f***, nitrogen (N) atom (***g***), and oxygen(O) atoms (***h*** and ***i***), and determine their **hybridization**. (2 points)

Atom	Hybridization
Sulfur (<i>f</i>)	
Nitrogen (<i>g</i>)	
Oxygen (<i>h</i>)	
Oxygen (<i>i</i>)	

iii. **Describe** all the bonds formed by the carbon atoms ***a*** and ***b***, using sigma and pi designation as appropriate. (2 points)

Atom	Bonds formed (use sigma and pi as appropriate)
Carbon (<i>a</i>)	
Carbon (<i>b</i>)	

iv. How many total sp^3 -s sigma bonds are in the molecule? (1 point)

v. Determine **electronic** and **molecular geometry** at carbon (C) labeled ***e*** and sulfur (S) labeled ***f*** centres. (2 points)

Atom	Electronic Geometry	Molecular Geometry
Carbon (<i>e</i>)		
Sulfur (<i>f</i>)		

vi. Circle the value from the choices given below, that describes the **C-S-C bond angle at sulfur** (S) marked ***f***: (0.5 points)

>90° to <109.5° 109.5° >109.5 to <120° 120° >120° to <180°


Question 8 12 points

Using the **valence shell orbitals only** draw a **complete** molecular orbital (MO) diagram (with no s-p mixing) for the molecule **PS**, in the space provided below. **(6 points)**

Assume that 3s and 3p atomic orbitals will form molecular orbitals like those formed by the combination of 2s and 2p atomic orbitals that we learnt in class, and there is no s-p mixing.

Show **energy arrow**, **atomic orbitals** involved, **atom labels**, the **molecular orbitals** that are formed, **filled electrons** in the MO diagram etc. Make sure to label all the atomic and molecular orbitals.

Using the MO diagram above, answer the following questions:

- (i) Which one of the following will have the highest bond order? Circle the correct choice
 A) PS^{1-} B) PS C) PS^{1+}
- (ii) Which one of the following will have the longest bond length? Circle the correct choice
 A) PS^{1-} B) PS C) PS^{1+}
- (iii) Which of the following will be diamagnetic? Circle the correct choice.
 A) PS^{1-} B) PS C) PS^{1+}
- (iv) Draw Lewis dot structure of the molecule with the shortest bond length (based on the MO diagram) from among the three (PS^{1-} , PS, PS^{1+}), and indicate the bond order calculated from the Lewis dot structure. **(3 points)**

Lewis Dot Structure

Bond Order



08C93D0A-7DE1-4ED7-9023-9446313A37A5

midterm-exam-93388

#471 Page 10 of 10

SCRAP PAGE

**CHEMISTRY 110 –Midterm Exam****Thursday November 7, 2024, 6:30 – 8:30 pm**

Examiners: Pallavi Sirjoosingh and Ashok Kakkar

First name (please write as legibly as possible within the boxes)

--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--

Last name

--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--

Student ID number

--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--

This booklet has 8 questions on 10 pages (The last page can be used as a scrap page).

The staple must NOT be removed from the question booklet. Your intact question booklet must be returned at the end of the exam.

A separate booklet contains a periodic table, and datasheet.

Write your answers directly under the questions in the space provided. Do not write on the QR codes or near the top edges (as it may be cut off).

Please write **legibly and dark enough** for the scan to be readable. For students who use English as a second language, dictionaries are permitted. Calculators are allowed.

DO NOT ASK QUESTIONS OF THE INVIGILATORS about interpretations of the exam questions. If you have a concern about a question, please summarize it at the top of this page. All concerns will be evaluated after the exam. GOOD LUCK!

**Question 1 8 points** (*Assume all transitions are possible*)

a. 6 points Calculate the **energy (in kJ/mol)** required to ionize (completely remove) the electron from the **n=3** level of **He⁺ ion** (He: Helium). Show your work including equations. (3 sig. figs.)

b. 2 points

(I) Select the largest (longest) wavelength from the three choices below:

- i. Wavelength of photon to *completely remove* an electron from *1s orbital of Hydrogen atom*
- ii. Wavelength of photon to *completely remove* an electron from *1s orbital of Boron atom (B)*
- iii. Wavelength of photon to *completely remove* an electron from *1s orbital of Boron cation (B⁺)*

i or ii or iii

(II) Select the smallest (shortest) wavelength from the three choices below (*for H atom*):

- i. Wavelength of photon *emitted* for an electron transitioning from *2p to 1s orbital*
- ii. Wavelength of photon *absorbed* for an electron transitioning from *2s to 4p orbital*
- iii. Wavelength of photon *absorbed* for an electron transitioning from *1s to 3p orbital*

i or ii or iii



Question 2 8 points



A Chem110 student plots the graph shown on the left, from data obtained for the kinetic energy (eV) of the ejected electron versus frequency (s^{-1}) of the incident light, for an unknown metal. ($1\text{eV} = 1.6022 \times 10^{-19} \text{ J}$)

Based on the plot, answer the questions **a to c** below:

a. 1 point On the plot above, label the **threshold frequency**.

b. 2 points What is the value of the work potential (in J) of the unknown metal?

c. 3 points What is the *longest* wavelength (in nm) required to eject an electron from this metal's surface? Show your work including equation(s) used. (3 sig figs)

d. 2 points In the space provided below, plot the following

(Assume frequency is *above* threshold frequency):

- i. *Kinetic energy* of an ejected electron versus the *intensity* of the incident light (left)
- ii. *Number of electrons* ejected versus the *frequency* of the incident light (right)



**Question 3** **6 points****a. 2 points**

In the ground state of Ge (Germanium; $Z = 32$) atom, **name** the following orbitals:

i. Orbital containing the electron that requires the maximum energy to be removed.

ii. Orbital containing the electron that experiences the minimum effective nuclear charge.

b. 2 points

An unknown anion (A^-) has the following *excited* electronic configuration: $[Ar] 4s^2 3d^3 4p^1$

(Ar: Argon; $Z = 18$)

Give the **condensed ground state electronic configuration** of the atom, **A** and the cation, **A^{2+}** .
Show your work. (Assume all electronic transitions are possible)

Electronic Configuration of A:

Electronic Configuration of A^{2+} :

c. 2 points

In a single atom, determine how many electrons can have the following set of quantum numbers?

(i) $n = 5; l = 3$

In a single atom, determine how many electrons can have the following set of quantum numbers?

(i) $n = 3; l = 2; m_s = +1/2$

**Question 4** **8 points**

Rank the following from 1 to 4 in **increasing** order. Provide the ranking in the boxes provided.

Note: Atomic number (Z) are provided for relevant atoms

a) Increasing (lowest to highest) order of *ionization energy*: (1= lowest; 4 = highest)

Mg

Mg: Z = 12

Rb

Rb: Z = 37

Cs

Cs: Z = 55

K

K : Z = 19

b) Increasing (lowest to highest) order of *bond angle*: (1= lowest; 4 = highest)

H₂OCCl₄XeF₄

HCN

c) Increasing (smallest to largest) order of *ionic size* (1= smallest; 4 = largest)

S²⁻

S: Z = 16

O²⁻

O: Z = 8

Be²⁺

Be: Z = 4

Na⁺

Na: Z = 11

d) Increasing (lowest to highest) order of *bond energy* (1= lowest; 4 = highest)

P-F

P-Cl

P-I

P-Br

**Question 5** **12 points**

a. 9 points For the molecules HClO_4 and TeF_6 , draw the most stable *Lewis structure* showing appropriate shape and *dashed/wedged bonds*, based on VSEPR. Determine the **molecular geometry** around the central atom. For HClO_4 , label the *most polar bond*.



b. 3 points (Note: Lone pairs are not shown in structures below)

(i) For the three *resonance* structures (1, 2 and 3) shown below, answer the following question:



(I) Circle the *most contributing* resonance structure.

Structure 1

Structure 2

Structure 3

(ii) For the three structures given (4, 5, and 6) below, answer the following questions:



(I) Circle all the structures that represent *resonance structures* of each other:

Structure 4

Structure 5

Structure 6

(II) Among the resonance structures, circle the *most contributing* resonance structure.

Structure 4

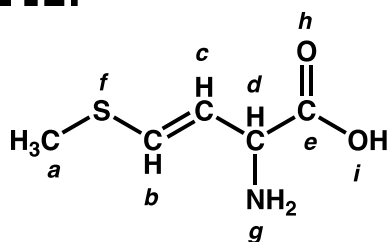
Structure 5

Structure 6

**Question 6** **6 points**

Using the data given below, calculate the **lattice energy** for SrO(s).

	ΔH° (kJ/mol)
$\text{Sr}(s) \longrightarrow \text{Sr}(g)$	+180
$\text{Sr}(g) \longrightarrow \text{Sr}^{2+}(g) + 2e^-$	+1468
$\text{O}_2(g) \longrightarrow 2\text{O}(g)$	+500
$\text{O}(g) + 2e^- \longrightarrow \text{O}^{2-}(g)$	+600
$\text{Sr}(s) + \frac{1}{2}\text{O}_2(g) \longrightarrow \text{SrO}(s)$	-550

**Question 7 10 points**

Consider the molecule given on the left, and answer questions (i) to (vi) using chemical bonding theories that we have learnt in class (valence bond theory, hybridization).

Note that the lone pairs are not shown in the structure:

i. Consider the carbon atoms marked in as ***a***, ***b***, ***c***, ***d***, and ***e*** in the molecule, and determine **hybridization** at each of the carbon centers. (2.5 points)

Atom	Hybridization
Carbon (<i>a</i>)	
Carbon (<i>b</i>)	
Carbon (<i>c</i>)	
Carbon (<i>d</i>)	
Carbon (<i>e</i>)	

ii. Now, consider the sulfur atom (S) marked as ***f***, nitrogen (N) atom (***g***), and oxygen(O) atoms (***h*** and ***i***), and determine their **hybridization**. (2 points)

Atom	Hybridization
Sulfur (<i>f</i>)	
Nitrogen (<i>g</i>)	
Oxygen (<i>h</i>)	
Oxygen (<i>i</i>)	

iii. **Describe** all the bonds formed by the carbon atoms ***a*** and ***b***, using sigma and pi designation as appropriate. (2 points)

Atom	Bonds formed (use sigma and pi as appropriate)
Carbon (<i>a</i>)	
Carbon (<i>b</i>)	

iv. How many total sp^3 -s sigma bonds are in the molecule? (1 point)

v. Determine **electronic** and **molecular geometry** at carbon (C) labeled ***e*** and sulfur (S) labeled ***f*** centres. (2 points)

Atom	Electronic Geometry	Molecular Geometry
Carbon (<i>e</i>)		
Sulfur (<i>f</i>)		

vi. Circle the value from the choices given below, that describes the **C-S-C bond angle at sulfur** (S) marked ***f***: (0.5 points)

$>90^\circ$ to $<109.5^\circ$ 109.5° >109.5 to $<120^\circ$ 120° $>120^\circ$ to $<180^\circ$


Question 8 12 points

Using the **valence shell orbitals only** draw a **complete** molecular orbital (MO) diagram (with no s-p mixing) for the molecule **PS**, in the space provided below. **(6 points)**

Assume that 3s and 3p atomic orbitals will form molecular orbitals like those formed by the combination of 2s and 2p atomic orbitals that we learnt in class, and there is no s-p mixing.

Show **energy arrow**, **atomic orbitals** involved, **atom labels**, the **molecular orbitals** that are formed, **filled electrons** in the MO diagram etc. Make sure to label all the atomic and molecular orbitals.

Using the MO diagram above, answer the following questions:

- (i) Which one of the following will have the highest bond order? Circle the correct choice
 A) PS^{1-} B) PS C) PS^{1+}
- (ii) Which one of the following will have the longest bond length? Circle the correct choice
 A) PS^{1-} B) PS C) PS^{1+}
- (iii) Which of the following will be diamagnetic? Circle the correct choice.
 A) PS^{1-} B) PS C) PS^{1+}
- (iv) Draw Lewis dot structure of the molecule with the shortest bond length (based on the MO diagram) from among the three (PS^{1-} , PS, PS^{1+}), and indicate the bond order calculated from the Lewis dot structure. **(3 points)**

Lewis Dot Structure

Bond Order



D52F030E-9559-4B8D-857C-CAFCAC235BE1

midterm-exam-93388

#472 Page 10 of 10

SCRAP PAGE

**CHEMISTRY 110 –Midterm Exam****Thursday November 7, 2024, 6:30 – 8:30 pm**

Examiners: Pallavi Sirjoosingh and Ashok Kakkar

First name (please write as legibly as possible within the boxes)

--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--

Last name

--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--

Student ID number

--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--

This booklet has 8 questions on 10 pages (The last page can be used as a scrap page).

The staple must NOT be removed from the question booklet. Your intact question booklet must be returned at the end of the exam.

A separate booklet contains a periodic table, and datasheet.

Write your answers directly under the questions in the space provided. Do not write on the QR codes or near the top edges (as it may be cut off).

Please write **legibly and dark enough** for the scan to be readable. For students who use English as a second language, dictionaries are permitted. Calculators are allowed.

DO NOT ASK QUESTIONS OF THE INVIGILATORS about interpretations of the exam questions. If you have a concern about a question, please summarize it at the top of this page. All concerns will be evaluated after the exam. GOOD LUCK!

**Question 1 8 points** (*Assume all transitions are possible*)

a. 6 points Calculate the **energy (in kJ/mol)** required to ionize (completely remove) the electron from the **$n=3$** level of **He^+ ion** (He: Helium). Show your work including equations. (3 sig. figs.)

b. 2 points

(I) Select the largest (longest) wavelength from the three choices below:

- i. Wavelength of photon to *completely remove* an electron from *1s orbital of Hydrogen atom*
- ii. Wavelength of photon to *completely remove* an electron from *1s orbital of Boron atom (B)*
- iii. Wavelength of photon to *completely remove* an electron from *1s orbital of Boron cation (B^+)*

i or ii or iii

(II) Select the smallest (shortest) wavelength from the three choices below (*for H atom*):

- i. Wavelength of photon *emitted* for an electron transitioning from *2p to 1s orbital*
- ii. Wavelength of photon *absorbed* for an electron transitioning from *2s to 4p orbital*
- iii. Wavelength of photon *absorbed* for an electron transitioning from *1s to 3p orbital*

i or ii or iii



Question 2 8 points



A Chem110 student plots the graph shown on the left, from data obtained for the kinetic energy (eV) of the ejected electron versus frequency (s^{-1}) of the incident light, for an unknown metal. ($1\text{eV} = 1.6022 \times 10^{-19} \text{ J}$)

Based on the plot, answer the questions **a to c** below:

a. 1 point On the plot above, label the **threshold frequency**.

b. 2 points What is the value of the work potential (in J) of the unknown metal?

c. 3 points What is the *longest* wavelength (in nm) required to eject an electron from this metal's surface? Show your work including equation(s) used. (3 sig figs)

d. 2 points In the space provided below, plot the following

(Assume frequency is *above* threshold frequency):

- i. *Kinetic energy* of an ejected electron versus the *intensity* of the incident light (left)
- ii. *Number of electrons* ejected versus the *frequency* of the incident light (right)



**Question 3** **6 points****a. 2 points**

In the ground state of Ge (Germanium; $Z = 32$) atom, **name** the following orbitals:

i. Orbital containing the electron that requires the maximum energy to be removed.

ii. Orbital containing the electron that experiences the minimum effective nuclear charge.

b. 2 points

An unknown anion (A^-) has the following *excited* electronic configuration: $[Ar] 4s^2 3d^3 4p^1$

(Ar: Argon; $Z = 18$)

Give the **condensed ground state electronic configuration** of the atom, **A** and the cation, **A^{2+}** .
Show your work. (Assume all electronic transitions are possible)

Electronic Configuration of A:

Electronic Configuration of A^{2+} :

c. 2 points

In a single atom, determine how many electrons can have the following set of quantum numbers?

(i) $n = 5; l = 3$

In a single atom, determine how many electrons can have the following set of quantum numbers?

(i) $n = 3; l = 2; m_s = +1/2$

**Question 4** **8 points**

Rank the following from 1 to 4 in **increasing** order. Provide the ranking in the boxes provided.

Note: Atomic number (Z) are provided for relevant atoms

a) Increasing (lowest to highest) order of *ionization energy*: (1= lowest; 4 = highest)

Mg

Mg: Z = 12

Rb

Rb: Z = 37

Cs

Cs: Z = 55

K

K : Z = 19

b) Increasing (lowest to highest) order of *bond angle*: (1= lowest; 4 = highest)

H₂OCCl₄XeF₄

HCN

c) Increasing (smallest to largest) order of *ionic size* (1= smallest; 4 = largest)

S²⁻

S: Z = 16

O²⁻

O: Z = 8

Be²⁺

Be: Z = 4

Na⁺

Na: Z = 11

d) Increasing (lowest to highest) order of *bond energy* (1= lowest; 4 = highest)

P-F

P-Cl

P-I

P-Br

**Question 5** **12 points**

a. 9 points For the molecules HClO_4 and TeF_6 , draw the most stable *Lewis structure* showing appropriate shape and *dashed/wedged bonds*, based on VSEPR. Determine the **molecular geometry** around the central atom. For HClO_4 , label the *most polar bond*.



b. 3 points (Note: Lone pairs are not shown in structures below)

(i) For the three *resonance* structures (1, 2 and 3) shown below, answer the following question:



(I) Circle the *most contributing* resonance structure.

Structure 1

Structure 2

Structure 3

(ii) For the three structures given (4, 5, and 6) below, answer the following questions:



(I) Circle all the structures that represent *resonance structures* of each other:

Structure 4

Structure 5

Structure 6

(II) Among the resonance structures, circle the *most contributing* resonance structure.

Structure 4

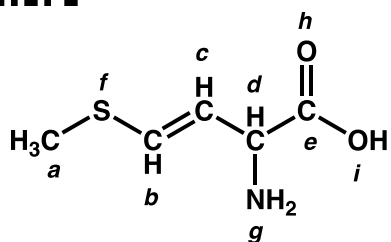
Structure 5

Structure 6

**Question 6** **6 points**

Using the data given below, calculate the **lattice energy** for SrO(s).

	ΔH° (kJ/mol)
$\text{Sr}(s) \longrightarrow \text{Sr}(g)$	+180
$\text{Sr}(g) \longrightarrow \text{Sr}^{2+}(g) + 2e^-$	+1468
$\text{O}_2(g) \longrightarrow 2\text{O}(g)$	+500
$\text{O}(g) + 2e^- \longrightarrow \text{O}^{2-}(g)$	+600
$\text{Sr}(s) + \frac{1}{2}\text{O}_2(g) \longrightarrow \text{SrO}(s)$	-550

**Question 7 10 points**

Consider the molecule given on the left, and answer questions (i) to (vi) using chemical bonding theories that we have learnt in class (valence bond theory, hybridization).

Note that the lone pairs are not shown in the structure:

i. Consider the carbon atoms marked in as ***a***, ***b***, ***c***, ***d***, and ***e*** in the molecule, and determine **hybridization** at each of the carbon centers. (2.5 points)

Atom	Hybridization
Carbon (<i>a</i>)	
Carbon (<i>b</i>)	
Carbon (<i>c</i>)	
Carbon (<i>d</i>)	
Carbon (<i>e</i>)	

ii. Now, consider the sulfur atom (S) marked as ***f***, nitrogen (N) atom (***g***), and oxygen(O) atoms (***h*** and ***i***), and determine their **hybridization**. (2 points)

Atom	Hybridization
Sulfur (<i>f</i>)	
Nitrogen (<i>g</i>)	
Oxygen (<i>h</i>)	
Oxygen (<i>i</i>)	

iii. **Describe** all the bonds formed by the carbon atoms ***a*** and ***b***, using sigma and pi designation as appropriate. (2 points)

Atom	Bonds formed (use sigma and pi as appropriate)
Carbon (<i>a</i>)	
Carbon (<i>b</i>)	

iv. How many total sp^3 -s sigma bonds are in the molecule? (1 point)

v. Determine **electronic** and **molecular geometry** at carbon (C) labeled ***e*** and sulfur (S) labeled ***f*** centres. (2 points)

Atom	Electronic Geometry	Molecular Geometry
Carbon (<i>e</i>)		
Sulfur (<i>f</i>)		

vi. Circle the value from the choices given below, that describes the **C-S-C bond angle at sulfur** (S) marked ***f***: (0.5 points)

>90° to <109.5° 109.5° >109.5 to <120° 120° >120° to <180°


Question 8 12 points

Using the **valence shell orbitals only** draw a **complete** molecular orbital (MO) diagram (with no s-p mixing) for the molecule **PS**, in the space provided below. **(6 points)**

Assume that 3s and 3p atomic orbitals will form molecular orbitals like those formed by the combination of 2s and 2p atomic orbitals that we learnt in class, and there is no s-p mixing.

Show **energy arrow**, **atomic orbitals** involved, **atom labels**, the **molecular orbitals** that are formed, **filled electrons** in the MO diagram etc. Make sure to label all the atomic and molecular orbitals.

Using the MO diagram above, answer the following questions:

- (i) Which one of the following will have the highest bond order? Circle the correct choice
 A) PS^{1-} B) PS C) PS^{1+}
- (ii) Which one of the following will have the longest bond length? Circle the correct choice
 A) PS^{1-} B) PS C) PS^{1+}
- (iii) Which of the following will be diamagnetic? Circle the correct choice.
 A) PS^{1-} B) PS C) PS^{1+}
- (iv) Draw Lewis dot structure of the molecule with the shortest bond length (based on the MO diagram) from among the three (PS^{1-} , PS, PS^{1+}), and indicate the bond order calculated from the Lewis dot structure. **(3 points)**

Lewis Dot Structure

Bond Order



F5E660C2-6ED8-4A49-9D06-DC5E52ED9B4A

midterm-exam-93388

#473 Page 10 of 10

SCRAP PAGE

**CHEMISTRY 110 –Midterm Exam****Thursday November 7, 2024, 6:30 – 8:30 pm****Examiners: Pallavi Sirjoosingh and Ashok Kakkar**

First name (please write as legibly as possible within the boxes)

--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--

Last name

--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--

Student ID number

--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--

This booklet has 8 questions on 10 pages (The last page can be used as a scrap page).

The staple must NOT be removed from the question booklet. Your intact question booklet must be returned at the end of the exam.

A separate booklet contains a periodic table, and datasheet.

Write your answers directly under the questions in the space provided. Do not write on the QR codes or near the top edges (as it may be cut off).

Please write **legibly and dark enough** for the scan to be readable. For students who use English as a second language, dictionaries are permitted. Calculators are allowed.

DO NOT ASK QUESTIONS OF THE INVIGILATORS about interpretations of the exam questions. If you have a concern about a question, please summarize it at the top of this page. All concerns will be evaluated after the exam. GOOD LUCK!

**Question 1 8 points** (*Assume all transitions are possible*)

a. 6 points Calculate the **energy (in kJ/mol)** required to ionize (completely remove) the electron from the **n=3** level of **He⁺ ion** (He: Helium). Show your work including equations. (3 sig. figs.)

b. 2 points

(I) Select the largest (longest) wavelength from the three choices below:

- i. Wavelength of photon to *completely remove* an electron from *1s orbital of Hydrogen atom*
- ii. Wavelength of photon to *completely remove* an electron from *1s orbital of Boron atom (B)*
- iii. Wavelength of photon to *completely remove* an electron from *1s orbital of Boron cation (B⁺)*

i or ii or iii

(II) Select the smallest (shortest) wavelength from the three choices below (*for H atom*):

- i. Wavelength of photon *emitted* for an electron transitioning from *2p to 1s orbital*
- ii. Wavelength of photon *absorbed* for an electron transitioning from *2s to 4p orbital*
- iii. Wavelength of photon *absorbed* for an electron transitioning from *1s to 3p orbital*

i or ii or iii



Question 2 8 points



A Chem110 student plots the graph shown on the left, from data obtained for the kinetic energy (eV) of the ejected electron versus frequency (s^{-1}) of the incident light, for an unknown metal. ($1\text{eV} = 1.6022 \times 10^{-19} \text{ J}$)

Based on the plot, answer the questions **a to c** below:

a. 1 point On the plot above, label the **threshold frequency**.

b. 2 points What is the value of the work potential (in J) of the unknown metal?

c. 3 points What is the *longest* wavelength (in nm) required to eject an electron from this metal's surface? Show your work including equation(s) used. (3 sig figs)

d. 2 points In the space provided below, plot the following

(Assume frequency is *above* threshold frequency):

- Kinetic energy* of an ejected electron versus the *intensity* of the incident light (left)
- Number of electrons* ejected versus the *frequency* of the incident light (right)



**Question 3** **6 points****a. 2 points**

In the ground state of Ge (Germanium; $Z = 32$) atom, **name** the following orbitals:

i. Orbital containing the electron that requires the maximum energy to be removed.

ii. Orbital containing the electron that experiences the minimum effective nuclear charge.

b. 2 points

An unknown anion (A^-) has the following *excited* electronic configuration: $[Ar] 4s^2 3d^3 4p^1$

(Ar: Argon; $Z = 18$)

Give the **condensed ground state electronic configuration** of the atom, **A** and the cation, **A^{2+}** .
Show your work. (Assume all electronic transitions are possible)

Electronic Configuration of A:

Electronic Configuration of A^{2+} :

c. 2 points

In a single atom, determine how many electrons can have the following set of quantum numbers?

(i) $n = 5; l = 3$

In a single atom, determine how many electrons can have the following set of quantum numbers?

(i) $n = 3; l = 2; m_s = +1/2$

**Question 4** **8 points**

Rank the following from 1 to 4 in **increasing** order. Provide the ranking in the boxes provided.

Note: Atomic number (Z) are provided for relevant atoms

a) Increasing (lowest to highest) order of *ionization energy*: (1= lowest; 4 = highest)

Mg

Mg: Z = 12

Rb

Rb: Z = 37

Cs

Cs: Z = 55

K

K : Z = 19

b) Increasing (lowest to highest) order of *bond angle*: (1= lowest; 4 = highest)

H₂OCCl₄XeF₄

HCN

c) Increasing (smallest to largest) order of *ionic size* (1= smallest; 4 = largest)

S²⁻

S: Z = 16

O²⁻

O: Z = 8

Be²⁺

Be: Z = 4

Na⁺

Na: Z = 11

d) Increasing (lowest to highest) order of *bond energy* (1= lowest; 4 = highest)

P-F

P-Cl

P-I

P-Br

**Question 5** **12 points**

a. 9 points For the molecules HClO_4 and TeF_6 , draw the most stable *Lewis structure* showing appropriate shape and *dashed/wedged bonds*, based on VSEPR. Determine the **molecular geometry** around the central atom. For HClO_4 , label the *most polar bond*.



b. 3 points (Note: Lone pairs are not shown in structures below)

(i) For the three *resonance* structures (1, 2 and 3) shown below, answer the following question:



(I) Circle the *most contributing* resonance structure.

Structure 1

Structure 2

Structure 3

(ii) For the three structures given (4, 5, and 6) below, answer the following questions:



(I) Circle all the structures that represent *resonance structures* of each other:

Structure 4

Structure 5

Structure 6

(II) Among the resonance structures, circle the *most contributing* resonance structure.

Structure 4

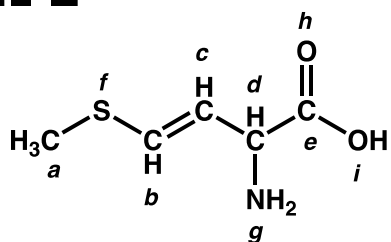
Structure 5

Structure 6

**Question 6** **6 points**

Using the data given below, calculate the **lattice energy** for SrO(s).

	ΔH° (kJ/mol)
$\text{Sr}(s) \longrightarrow \text{Sr}(g)$	+180
$\text{Sr}(g) \longrightarrow \text{Sr}^{2+}(g) + 2e^-$	+1468
$\text{O}_2(g) \longrightarrow 2\text{O}(g)$	+500
$\text{O}(g) + 2e^- \longrightarrow \text{O}^{2-}(g)$	+600
$\text{Sr}(s) + \frac{1}{2}\text{O}_2(g) \longrightarrow \text{SrO}(s)$	-550

**Question 7 10 points**

Consider the molecule given on the left, and answer questions (i) to (vi) using chemical bonding theories that we have learnt in class (valence bond theory, hybridization).

Note that the lone pairs are not shown in the structure:

i. Consider the carbon atoms marked in as ***a***, ***b***, ***c***, ***d***, and ***e*** in the molecule, and determine **hybridization** at each of the carbon centers. (2.5 points)

Atom	Hybridization
Carbon (<i>a</i>)	
Carbon (<i>b</i>)	
Carbon (<i>c</i>)	
Carbon (<i>d</i>)	
Carbon (<i>e</i>)	

ii. Now, consider the sulfur atom (S) marked as ***f***, nitrogen (N) atom (***g***), and oxygen(O) atoms (***h*** and ***i***), and determine their **hybridization**. (2 points)

Atom	Hybridization
Sulfur (<i>f</i>)	
Nitrogen (<i>g</i>)	
Oxygen (<i>h</i>)	
Oxygen (<i>i</i>)	

iii. **Describe** all the bonds formed by the carbon atoms ***a*** and ***b***, using sigma and pi designation as appropriate. (2 points)

Atom	Bonds formed (use sigma and pi as appropriate)
Carbon (<i>a</i>)	
Carbon (<i>b</i>)	

iv. How many total sp^3 -s sigma bonds are in the molecule? (1 point)

v. Determine **electronic** and **molecular geometry** at carbon (C) labeled ***e*** and sulfur (S) labeled ***f*** centres. (2 points)

Atom	Electronic Geometry	Molecular Geometry
Carbon (<i>e</i>)		
Sulfur (<i>f</i>)		

vi. Circle the value from the choices given below, that describes the **C-S-C bond angle at sulfur** (S) marked ***f***: (0.5 points)

>90° to <109.5° 109.5° >109.5 to <120° 120° >120° to <180°

**Question 8 12 points**

Using the **valence shell orbitals only** draw a **complete** molecular orbital (MO) diagram (with no s-p mixing) for the molecule **PS**, in the space provided below. **(6 points)**

Assume that 3s and 3p atomic orbitals will form molecular orbitals like those formed by the combination of 2s and 2p atomic orbitals that we learnt in class, and there is no s-p mixing.

Show **energy arrow**, **atomic orbitals** involved, **atom labels**, the **molecular orbitals** that are formed, **filled electrons** in the MO diagram etc. Make sure to label all the atomic and molecular orbitals.

Using the MO diagram above, answer the following questions:

- (i) Which one of the following will have the highest bond order? Circle the correct choice
A) PS^{1-} B) PS C) PS^{1+}
- (ii) Which one of the following will have the longest bond length? Circle the correct choice
A) PS^{1-} B) PS C) PS^{1+}
- (iii) Which of the following will be diamagnetic? Circle the correct choice.
A) PS^{1-} B) PS C) PS^{1+}
- (iv) Draw Lewis dot structure of the molecule with the shortest bond length (based on the MO diagram) from among the three (PS^{1-} , PS, PS^{1+}), and indicate the bond order calculated from the Lewis dot structure. **(3 points)**

Lewis Dot Structure

Bond Order



5D0DB1B6-D918-4FB2-99B2-A71073C0F61E

midterm-exam-93388

#474 Page 10 of 10

SCRAP PAGE

**CHEMISTRY 110 –Midterm Exam****Thursday November 7, 2024, 6:30 – 8:30 pm**

Examiners: Pallavi Sirjoosingh and Ashok Kakkar

First name (please write as legibly as possible within the boxes)

--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--

Last name

--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--

Student ID number

--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--

This booklet has 8 questions on 10 pages (The last page can be used as a scrap page).

The staple must NOT be removed from the question booklet. Your intact question booklet must be returned at the end of the exam.

A separate booklet contains a periodic table, and datasheet.

Write your answers directly under the questions in the space provided. Do not write on the QR codes or near the top edges (as it may be cut off).

Please write **legibly and dark enough** for the scan to be readable. For students who use English as a second language, dictionaries are permitted. Calculators are allowed.

DO NOT ASK QUESTIONS OF THE INVIGILATORS about interpretations of the exam questions. If you have a concern about a question, please summarize it at the top of this page. All concerns will be evaluated after the exam. GOOD LUCK!

**Question 1 8 points** (*Assume all transitions are possible*)

a. 6 points Calculate the **energy (in kJ/mol)** required to ionize (completely remove) the electron from the **$n=3$** level of **He^+ ion** (He: Helium). Show your work including equations. (3 sig. figs.)

b. 2 points

(I) Select the largest (longest) wavelength from the three choices below:

- i. Wavelength of photon to *completely remove* an electron from *1s orbital of Hydrogen atom*
- ii. Wavelength of photon to *completely remove* an electron from *1s orbital of Boron atom (B)*
- iii. Wavelength of photon to *completely remove* an electron from *1s orbital of Boron cation (B^+)*

i or ii or iii

(II) Select the smallest (shortest) wavelength from the three choices below (*for H atom*):

- i. Wavelength of photon *emitted* for an electron transitioning from *2p to 1s orbital*
- ii. Wavelength of photon *absorbed* for an electron transitioning from *2s to 4p orbital*
- iii. Wavelength of photon *absorbed* for an electron transitioning from *1s to 3p orbital*

i or ii or iii



Question 2 8 points



A Chem110 student plots the graph shown on the left, from data obtained for the kinetic energy (eV) of the ejected electron versus frequency (s^{-1}) of the incident light, for an unknown metal. ($1\text{eV} = 1.6022 \times 10^{-19} \text{ J}$)

Based on the plot, answer the questions **a to c** below:

a. 1 point On the plot above, label the **threshold frequency**.

b. 2 points What is the value of the work potential (in J) of the unknown metal?

c. 3 points What is the *longest* wavelength (in nm) required to eject an electron from this metal's surface? Show your work including equation(s) used. (3 sig figs)

d. 2 points In the space provided below, plot the following

(Assume frequency is *above* threshold frequency):

- Kinetic energy* of an ejected electron versus the *intensity* of the incident light (left)
- Number of electrons* ejected versus the *frequency* of the incident light (right)



**Question 3** **6 points****a. 2 points**

In the ground state of Ge (Germanium; $Z = 32$) atom, **name** the following orbitals:

i. Orbital containing the electron that requires the maximum energy to be removed.

ii. Orbital containing the electron that experiences the minimum effective nuclear charge.

b. 2 points

An unknown anion (A^-) has the following *excited* electronic configuration: $[Ar] 4s^2 3d^3 4p^1$

(Ar: Argon; $Z = 18$)

Give the **condensed ground state electronic configuration** of the atom, **A** and the cation, **A^{2+}** .
Show your work. (Assume all electronic transitions are possible)

Electronic Configuration of A:

Electronic Configuration of A^{2+} :

c. 2 points

In a single atom, determine how many electrons can have the following set of quantum numbers?

(i) $n = 5; l = 3$

In a single atom, determine how many electrons can have the following set of quantum numbers?

(i) $n = 3; l = 2; m_s = +1/2$

**Question 4** **8 points**

Rank the following from 1 to 4 in **increasing** order. Provide the ranking in the boxes provided.

Note: Atomic number (Z) are provided for relevant atoms

a) Increasing (lowest to highest) order of *ionization energy*: (1= lowest; 4 = highest)

Mg

Mg: Z = 12

Rb

Rb: Z = 37

Cs

Cs: Z = 55

K

K : Z = 19

b) Increasing (lowest to highest) order of *bond angle*: (1= lowest; 4 = highest)

H₂OCCl₄XeF₄

HCN

c) Increasing (smallest to largest) order of *ionic size* (1= smallest; 4 = largest)

S²⁻

S: Z = 16

O²⁻

O: Z = 8

Be²⁺

Be: Z = 4

Na⁺

Na: Z = 11

d) Increasing (lowest to highest) order of *bond energy* (1= lowest; 4 = highest)

P-F

P-Cl

P-I

P-Br

**Question 5** **12 points**

a. 9 points For the molecules HClO_4 and TeF_6 , draw the most stable *Lewis structure* showing appropriate shape and *dashed/wedged bonds*, based on VSEPR. Determine the **molecular geometry** around the central atom. For HClO_4 , label the *most polar bond*.



b. 3 points (Note: Lone pairs are not shown in structures below)

(i) For the three *resonance* structures (1, 2 and 3) shown below, answer the following question:



(I) Circle the *most contributing* resonance structure.

Structure 1

Structure 2

Structure 3

(ii) For the three structures given (4, 5, and 6) below, answer the following questions:



(I) Circle all the structures that represent *resonance structures* of each other:

Structure 4

Structure 5

Structure 6

(II) Among the resonance structures, circle the *most contributing* resonance structure.

Structure 4

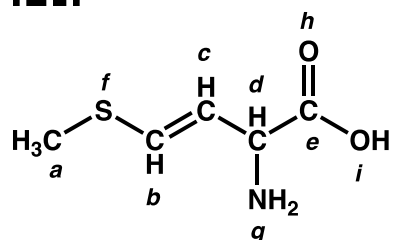
Structure 5

Structure 6

**Question 6** **6 points**

Using the data given below, calculate the **lattice energy** for SrO(s).

	ΔH° (kJ/mol)
$\text{Sr}(s) \longrightarrow \text{Sr}(g)$	+180
$\text{Sr}(g) \longrightarrow \text{Sr}^{2+}(g) + 2e^-$	+1468
$\text{O}_2(g) \longrightarrow 2\text{O}(g)$	+500
$\text{O}(g) + 2e^- \longrightarrow \text{O}^{2-}(g)$	+600
$\text{Sr}(s) + \frac{1}{2}\text{O}_2(g) \longrightarrow \text{SrO}(s)$	-550

**Question 7 10 points**

Consider the molecule given on the left, and answer questions (i) to (vi) using chemical bonding theories that we have learnt in class (valence bond theory, hybridization).

Note that the lone pairs are not shown in the structure:

i. Consider the carbon atoms marked in as ***a***, ***b***, ***c***, ***d***, and ***e*** in the molecule, and determine **hybridization** at each of the carbon centers. (2.5 points)

Atom	Hybridization
Carbon (<i>a</i>)	
Carbon (<i>b</i>)	
Carbon (<i>c</i>)	
Carbon (<i>d</i>)	
Carbon (<i>e</i>)	

ii. Now, consider the sulfur atom (S) marked as ***f***, nitrogen (N) atom (***g***), and oxygen(O) atoms (***h*** and ***i***), and determine their **hybridization**. (2 points)

Atom	Hybridization
Sulfur (<i>f</i>)	
Nitrogen (<i>g</i>)	
Oxygen (<i>h</i>)	
Oxygen (<i>i</i>)	

iii. **Describe** all the bonds formed by the carbon atoms ***a*** and ***b***, using sigma and pi designation as appropriate. (2 points)

Atom	Bonds formed (use sigma and pi as appropriate)
Carbon (<i>a</i>)	
Carbon (<i>b</i>)	

iv. How many total sp^3 -s sigma bonds are in the molecule? (1 point)

v. Determine **electronic** and **molecular geometry** at carbon (C) labeled ***e*** and sulfur (S) labeled ***f*** centres. (2 points)

Atom	Electronic Geometry	Molecular Geometry
Carbon (<i>e</i>)		
Sulfur (<i>f</i>)		

vi. Circle the value from the choices given below, that describes the **C-S-C bond angle at sulfur** (S) marked ***f***: (0.5 points)

>90° to <109.5° 109.5° >109.5 to <120° 120° >120° to <180°


Question 8 12 points

Using the **valence shell orbitals only** draw a **complete** molecular orbital (MO) diagram (with no s-p mixing) for the molecule **PS**, in the space provided below. **(6 points)**

Assume that 3s and 3p atomic orbitals will form molecular orbitals like those formed by the combination of 2s and 2p atomic orbitals that we learnt in class, and there is no s-p mixing.

Show **energy arrow**, **atomic orbitals** involved, **atom labels**, the **molecular orbitals** that are formed, **filled electrons** in the MO diagram etc. Make sure to label all the atomic and molecular orbitals.

Using the MO diagram above, answer the following questions:

- (i) Which one of the following will have the highest bond order? Circle the correct choice
 A) PS^{1-} B) PS C) PS^{1+}
- (ii) Which one of the following will have the longest bond length? Circle the correct choice
 A) PS^{1-} B) PS C) PS^{1+}
- (iii) Which of the following will be diamagnetic? Circle the correct choice.
 A) PS^{1-} B) PS C) PS^{1+}
- (iv) Draw Lewis dot structure of the molecule with the shortest bond length (based on the MO diagram) from among the three (PS^{1-} , PS, PS^{1+}), and indicate the bond order calculated from the Lewis dot structure. **(3 points)**

Lewis Dot Structure

Bond Order



775AB12E-A916-4118-A02A-CDE72A2CC726

midterm-exam-93388

#475 Page 10 of 10

SCRAP PAGE

**CHEMISTRY 110 –Midterm Exam****Thursday November 7, 2024, 6:30 – 8:30 pm**

Examiners: Pallavi Sirjoosingh and Ashok Kakkar

First name (please write as legibly as possible within the boxes)

--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--

Last name

--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--

Student ID number

--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--

This booklet has 8 questions on 10 pages (The last page can be used as a scrap page).

The staple must NOT be removed from the question booklet. Your intact question booklet must be returned at the end of the exam.

A separate booklet contains a periodic table, and datasheet.

Write your answers directly under the questions in the space provided. Do not write on the QR codes or near the top edges (as it may be cut off).

Please write **legibly and dark enough** for the scan to be readable. For students who use English as a second language, dictionaries are permitted. Calculators are allowed.

DO NOT ASK QUESTIONS OF THE INVIGILATORS about interpretations of the exam questions. If you have a concern about a question, please summarize it at the top of this page. All concerns will be evaluated after the exam. GOOD LUCK!

**Question 1 8 points** (*Assume all transitions are possible*)

a. 6 points Calculate the **energy (in kJ/mol)** required to ionize (completely remove) the electron from the **$n=3$** level of **He^+ ion** (He: Helium). Show your work including equations. (3 sig. figs.)

b. 2 points

(I) Select the largest (longest) wavelength from the three choices below:

- i. Wavelength of photon to *completely remove* an electron from *1s orbital of Hydrogen atom*
- ii. Wavelength of photon to *completely remove* an electron from *1s orbital of Boron atom (B)*
- iii. Wavelength of photon to *completely remove* an electron from *1s orbital of Boron cation (B^+)*

i or ii or iii

(II) Select the smallest (shortest) wavelength from the three choices below (*for H atom*):

- i. Wavelength of photon *emitted* for an electron transitioning from *2p to 1s orbital*
- ii. Wavelength of photon *absorbed* for an electron transitioning from *2s to 4p orbital*
- iii. Wavelength of photon *absorbed* for an electron transitioning from *1s to 3p orbital*

i or ii or iii



Question 2 8 points



A Chem110 student plots the graph shown on the left, from data obtained for the kinetic energy (eV) of the ejected electron versus frequency (s^{-1}) of the incident light, for an unknown metal. ($1\text{eV} = 1.6022 \times 10^{-19} \text{ J}$)

Based on the plot, answer the questions **a to c** below:

a. 1 point On the plot above, label the **threshold frequency**.

b. 2 points What is the value of the work potential (in J) of the unknown metal?

c. 3 points What is the *longest* wavelength (in nm) required to eject an electron from this metal's surface? Show your work including equation(s) used. (3 sig figs)

d. 2 points In the space provided below, plot the following

(Assume frequency is *above* threshold frequency):

- i. *Kinetic energy* of an ejected electron versus the *intensity* of the incident light (left)
- ii. *Number of electrons* ejected versus the *frequency* of the incident light (right)



**Question 3** **6 points****a. 2 points**

In the ground state of Ge (Germanium; $Z = 32$) atom, **name** the following orbitals:

i. Orbital containing the electron that requires the maximum energy to be removed.

ii. Orbital containing the electron that experiences the minimum effective nuclear charge.

b. 2 points

An unknown anion (A^-) has the following *excited* electronic configuration: $[Ar] 4s^2 3d^3 4p^1$

(Ar: Argon; $Z = 18$)

Give the **condensed ground state electronic configuration** of the atom, **A** and the cation, **A^{2+}** .
Show your work. (Assume all electronic transitions are possible)

Electronic Configuration of A:

Electronic Configuration of A^{2+} :

c. 2 points

In a single atom, determine how many electrons can have the following set of quantum numbers?

(i) $n = 5; l = 3$

In a single atom, determine how many electrons can have the following set of quantum numbers?

(i) $n = 3; l = 2; m_s = +1/2$

**Question 4** **8 points**

Rank the following from 1 to 4 in **increasing** order. Provide the ranking in the boxes provided.

Note: Atomic number (Z) are provided for relevant atoms

a) Increasing (lowest to highest) order of *ionization energy*: (1= lowest; 4 = highest)

Mg

Mg: Z = 12

Rb

Rb: Z = 37

Cs

Cs: Z = 55

K

K : Z = 19

b) Increasing (lowest to highest) order of *bond angle*: (1= lowest; 4 = highest)

H₂OCCl₄XeF₄

HCN

c) Increasing (smallest to largest) order of *ionic size* (1= smallest; 4 = largest)

S²⁻

S: Z = 16

O²⁻

O: Z = 8

Be²⁺

Be: Z = 4

Na⁺

Na: Z = 11

d) Increasing (lowest to highest) order of *bond energy* (1= lowest; 4 = highest)

P-F

P-Cl

P-I

P-Br

**Question 5** **12 points**

a. 9 points For the molecules HClO_4 and TeF_6 , draw the most stable *Lewis structure* showing appropriate shape and *dashed/wedged bonds*, based on VSEPR. Determine the **molecular geometry** around the central atom. For HClO_4 , label the *most polar bond*.



b. 3 points (Note: Lone pairs are not shown in structures below)

(i) For the three *resonance* structures (1, 2 and 3) shown below, answer the following question:



(I) Circle the *most contributing* resonance structure.

Structure 1

Structure 2

Structure 3

(ii) For the three structures given (4, 5, and 6) below, answer the following questions:



(I) Circle all the structures that represent *resonance structures* of each other:

Structure 4

Structure 5

Structure 6

(II) Among the resonance structures, circle the *most contributing* resonance structure.

Structure 4

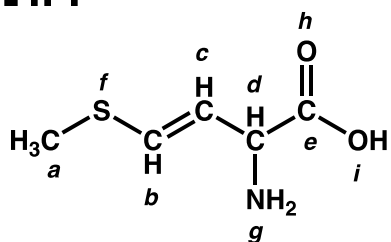
Structure 5

Structure 6

**Question 6** **6 points**

Using the data given below, calculate the **lattice energy** for SrO(s).

	ΔH° (kJ/mol)
$\text{Sr}(s) \longrightarrow \text{Sr}(g)$	+180
$\text{Sr}(g) \longrightarrow \text{Sr}^{2+}(g) + 2e^-$	+1468
$\text{O}_2(g) \longrightarrow 2\text{O}(g)$	+500
$\text{O}(g) + 2e^- \longrightarrow \text{O}^{2-}(g)$	+600
$\text{Sr}(s) + \frac{1}{2}\text{O}_2(g) \longrightarrow \text{SrO}(s)$	-550

**Question 7 10 points**

Consider the molecule given on the left, and answer questions (i) to (vi) using chemical bonding theories that we have learnt in class (valence bond theory, hybridization).

Note that the lone pairs are not shown in the structure:

i. Consider the carbon atoms marked in as ***a***, ***b***, ***c***, ***d***, and ***e*** in the molecule, and determine **hybridization** at each of the carbon centers. (2.5 points)

Atom	Hybridization
Carbon (<i>a</i>)	
Carbon (<i>b</i>)	
Carbon (<i>c</i>)	
Carbon (<i>d</i>)	
Carbon (<i>e</i>)	

ii. Now, consider the sulfur atom (S) marked as ***f***, nitrogen (N) atom (***g***), and oxygen(O) atoms (***h*** and ***i***), and determine their **hybridization**. (2 points)

Atom	Hybridization
Sulfur (<i>f</i>)	
Nitrogen (<i>g</i>)	
Oxygen (<i>h</i>)	
Oxygen (<i>i</i>)	

iii. **Describe** all the bonds formed by the carbon atoms ***a*** and ***b***, using sigma and pi designation as appropriate. (2 points)

Atom	Bonds formed (use sigma and pi as appropriate)
Carbon (<i>a</i>)	
Carbon (<i>b</i>)	

iv. How many total sp^3 -s sigma bonds are in the molecule? (1 point)

v. Determine **electronic** and **molecular geometry** at carbon (C) labeled ***e*** and sulfur (S) labeled ***f*** centres. (2 points)

Atom	Electronic Geometry	Molecular Geometry
Carbon (<i>e</i>)		
Sulfur (<i>f</i>)		

vi. Circle the value from the choices given below, that describes the **C-S-C bond angle at sulfur** (S) marked ***f***: (0.5 points)

>90° to <109.5° 109.5° >109.5 to <120° 120° >120° to <180°


Question 8 12 points

Using the **valence shell orbitals only** draw a **complete** molecular orbital (MO) diagram (with no s-p mixing) for the molecule **PS**, in the space provided below. **(6 points)**

Assume that 3s and 3p atomic orbitals will form molecular orbitals like those formed by the combination of 2s and 2p atomic orbitals that we learnt in class, and there is no s-p mixing.

Show **energy arrow**, **atomic orbitals** involved, **atom labels**, the **molecular orbitals** that are formed, **filled electrons** in the MO diagram etc. Make sure to label all the atomic and molecular orbitals.

Using the MO diagram above, answer the following questions:

- (i) Which one of the following will have the highest bond order? Circle the correct choice
 A) PS^{1-} B) PS C) PS^{1+}
- (ii) Which one of the following will have the longest bond length? Circle the correct choice
 A) PS^{1-} B) PS C) PS^{1+}
- (iii) Which of the following will be diamagnetic? Circle the correct choice.
 A) PS^{1-} B) PS C) PS^{1+}
- (iv) Draw Lewis dot structure of the molecule with the shortest bond length (based on the MO diagram) from among the three (PS^{1-} , PS, PS^{1+}), and indicate the bond order calculated from the Lewis dot structure. **(3 points)**

Lewis Dot Structure

Bond Order



C6926119-C277-43CF-B174-3477F80B33E3

midterm-exam-93388

#476 Page 10 of 10

SCRAP PAGE

**CHEMISTRY 110 –Midterm Exam****Thursday November 7, 2024, 6:30 – 8:30 pm****Examiners: Pallavi Sirjoosingh and Ashok Kakkar**

First name (please write as legibly as possible within the boxes)

--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--

Last name

--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--

Student ID number

--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--

This booklet has 8 questions on 10 pages (The last page can be used as a scrap page).

The staple must NOT be removed from the question booklet. Your intact question booklet must be returned at the end of the exam.

A separate booklet contains a periodic table, and datasheet.

Write your answers directly under the questions in the space provided. Do not write on the QR codes or near the top edges (as it may be cut off).

Please write **legibly and dark enough** for the scan to be readable. For students who use English as a second language, dictionaries are permitted. Calculators are allowed.

DO NOT ASK QUESTIONS OF THE INVIGILATORS about interpretations of the exam questions. If you have a concern about a question, please summarize it at the top of this page. All concerns will be evaluated after the exam. GOOD LUCK!

**Question 1 8 points** (*Assume all transitions are possible*)

a. 6 points Calculate the **energy (in kJ/mol)** required to ionize (completely remove) the electron from the **n=3** level of **He⁺ ion** (He: Helium). Show your work including equations. (3 sig. figs.)

b. 2 points

(I) Select the largest (longest) wavelength from the three choices below:

- i. Wavelength of photon to *completely remove* an electron from *1s orbital of Hydrogen atom*
- ii. Wavelength of photon to *completely remove* an electron from *1s orbital of Boron atom (B)*
- iii. Wavelength of photon to *completely remove* an electron from *1s orbital of Boron cation (B⁺)*

i or ii or iii

(II) Select the smallest (shortest) wavelength from the three choices below (*for H atom*):

- i. Wavelength of photon *emitted* for an electron transitioning from *2p to 1s orbital*
- ii. Wavelength of photon *absorbed* for an electron transitioning from *2s to 4p orbital*
- iii. Wavelength of photon *absorbed* for an electron transitioning from *1s to 3p orbital*

i or ii or iii



Question 2 8 points



A Chem110 student plots the graph shown on the left, from data obtained for the kinetic energy (eV) of the ejected electron versus frequency (s^{-1}) of the incident light, for an unknown metal. ($1\text{eV} = 1.6022 \times 10^{-19} \text{ J}$)

Based on the plot, answer the questions **a to c** below:

a. 1 point On the plot above, label the **threshold frequency**.

b. 2 points What is the value of the work potential (in J) of the unknown metal?

c. 3 points What is the *longest* wavelength (in nm) required to eject an electron from this metal's surface? Show your work including equation(s) used. (3 sig figs)

d. 2 points In the space provided below, plot the following

(Assume frequency is *above* threshold frequency):

- i. *Kinetic energy* of an ejected electron versus the *intensity* of the incident light (left)
- ii. *Number of electrons* ejected versus the *frequency* of the incident light (right)



**Question 3** **6 points****a. 2 points**

In the ground state of Ge (Germanium; $Z = 32$) atom, **name** the following orbitals:

i. Orbital containing the electron that requires the maximum energy to be removed.

ii. Orbital containing the electron that experiences the minimum effective nuclear charge.

b. 2 points

An unknown anion (A^-) has the following *excited* electronic configuration: $[Ar] 4s^2 3d^3 4p^1$

(Ar: Argon; $Z = 18$)

Give the **condensed ground state electronic configuration** of the atom, **A** and the cation, **A^{2+}** .
Show your work. (Assume all electronic transitions are possible)

Electronic Configuration of A:

Electronic Configuration of A^{2+} :

c. 2 points

In a single atom, determine how many electrons can have the following set of quantum numbers?

(i) $n = 5; l = 3$

In a single atom, determine how many electrons can have the following set of quantum numbers?

(i) $n = 3; l = 2; m_s = +1/2$

**Question 4** **8 points**

Rank the following from 1 to 4 in **increasing** order. Provide the ranking in the boxes provided.

Note: Atomic number (Z) are provided for relevant atoms

a) Increasing (lowest to highest) order of *ionization energy*: (1= lowest; 4 = highest)

Mg

Mg: Z = 12

Rb

Rb: Z = 37

Cs

Cs: Z = 55

K

K : Z = 19

b) Increasing (lowest to highest) order of *bond angle*: (1= lowest; 4 = highest)

H₂OCCl₄XeF₄

HCN

c) Increasing (smallest to largest) order of *ionic size* (1= smallest; 4 = largest)

S²⁻

S: Z = 16

O²⁻

O: Z = 8

Be²⁺

Be: Z = 4

Na⁺

Na: Z = 11

d) Increasing (lowest to highest) order of *bond energy* (1= lowest; 4 = highest)

P-F

P-Cl

P-I

P-Br

**Question 5** **12 points**

a. 9 points For the molecules HClO_4 and TeF_6 , draw the most stable *Lewis structure* showing appropriate shape and *dashed/wedged bonds*, based on VSEPR. Determine the **molecular geometry** around the central atom. For HClO_4 , label the *most polar bond*.



b. 3 points (Note: Lone pairs are not shown in structures below)

(i) For the three *resonance* structures (1, 2 and 3) shown below, answer the following question:



(I) Circle the *most contributing* resonance structure.

Structure 1

Structure 2

Structure 3

(ii) For the three structures given (4, 5, and 6) below, answer the following questions:



(I) Circle all the structures that represent *resonance structures* of each other:

Structure 4

Structure 5

Structure 6

(II) Among the resonance structures, circle the *most contributing* resonance structure.

Structure 4

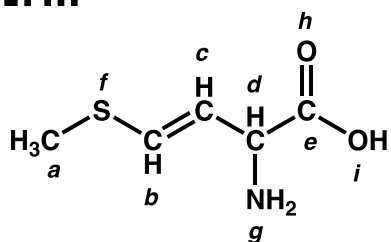
Structure 5

Structure 6

**Question 6** **6 points**

Using the data given below, calculate the **lattice energy** for SrO(s).

	ΔH° (kJ/mol)
$\text{Sr}(s) \longrightarrow \text{Sr}(g)$	+180
$\text{Sr}(g) \longrightarrow \text{Sr}^{2+}(g) + 2e^-$	+1468
$\text{O}_2(g) \longrightarrow 2\text{O}(g)$	+500
$\text{O}(g) + 2e^- \longrightarrow \text{O}^{2-}(g)$	+600
$\text{Sr}(s) + \frac{1}{2}\text{O}_2(g) \longrightarrow \text{SrO}(s)$	-550

**Question 7 10 points**

Consider the molecule given on the left, and answer questions (i) to (vi) using chemical bonding theories that we have learnt in class (valence bond theory, hybridization).

Note that the lone pairs are not shown in the structure:

i. Consider the carbon atoms marked in as ***a***, ***b***, ***c***, ***d***, and ***e*** in the molecule, and determine **hybridization** at each of the carbon centers. (2.5 points)

Atom	Hybridization
Carbon (<i>a</i>)	
Carbon (<i>b</i>)	
Carbon (<i>c</i>)	
Carbon (<i>d</i>)	
Carbon (<i>e</i>)	

ii. Now, consider the sulfur atom (S) marked as ***f***, nitrogen (N) atom (***g***), and oxygen(O) atoms (***h*** and ***i***), and determine their **hybridization**. (2 points)

Atom	Hybridization
Sulfur (<i>f</i>)	
Nitrogen (<i>g</i>)	
Oxygen (<i>h</i>)	
Oxygen (<i>i</i>)	

iii. **Describe** all the bonds formed by the carbon atoms ***a*** and ***b***, using sigma and pi designation as appropriate. (2 points)

Atom	Bonds formed (use sigma and pi as appropriate)
Carbon (<i>a</i>)	
Carbon (<i>b</i>)	

iv. How many total sp^3 -s sigma bonds are in the molecule? (1 point)

v. Determine **electronic** and **molecular geometry** at carbon (C) labeled ***e*** and sulfur (S) labeled ***f*** centres. (2 points)

Atom	Electronic Geometry	Molecular Geometry
Carbon (<i>e</i>)		
Sulfur (<i>f</i>)		

vi. Circle the value from the choices given below, that describes the **C-S-C bond angle at sulfur** (S) marked ***f***: (0.5 points)

>90° to <109.5° 109.5° >109.5 to <120° 120° >120° to <180°


Question 8 12 points

Using the **valence shell orbitals only** draw a **complete** molecular orbital (MO) diagram (with no s-p mixing) for the molecule **PS**, in the space provided below. **(6 points)**

Assume that 3s and 3p atomic orbitals will form molecular orbitals like those formed by the combination of 2s and 2p atomic orbitals that we learnt in class, and there is no s-p mixing.

Show **energy arrow**, **atomic orbitals** involved, **atom labels**, the **molecular orbitals** that are formed, **filled electrons** in the MO diagram etc. Make sure to label all the atomic and molecular orbitals.

Using the MO diagram above, answer the following questions:

- (i) Which one of the following will have the highest bond order? Circle the correct choice
 A) PS^{1-} B) PS C) PS^{1+}
- (ii) Which one of the following will have the longest bond length? Circle the correct choice
 A) PS^{1-} B) PS C) PS^{1+}
- (iii) Which of the following will be diamagnetic? Circle the correct choice.
 A) PS^{1-} B) PS C) PS^{1+}
- (iv) Draw Lewis dot structure of the molecule with the shortest bond length (based on the MO diagram) from among the three (PS^{1-} , PS, PS^{1+}), and indicate the bond order calculated from the Lewis dot structure. **(3 points)**

Lewis Dot Structure

Bond Order



45D36353-4C3B-43A9-B132-0931D2F6B8B4

midterm-exam-93388

#477 Page 10 of 10

SCRAP PAGE

**CHEMISTRY 110 –Midterm Exam****Thursday November 7, 2024, 6:30 – 8:30 pm**

Examiners: Pallavi Sirjoosingh and Ashok Kakkar

First name (please write as legibly as possible within the boxes)

--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--

Last name

--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--

Student ID number

--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--

This booklet has 8 questions on 10 pages (The last page can be used as a scrap page).

The staple must NOT be removed from the question booklet. Your intact question booklet must be returned at the end of the exam.

A separate booklet contains a periodic table, and datasheet.

Write your answers directly under the questions in the space provided. Do not write on the QR codes or near the top edges (as it may be cut off).

Please write **legibly and dark enough** for the scan to be readable. For students who use English as a second language, dictionaries are permitted. Calculators are allowed.

DO NOT ASK QUESTIONS OF THE INVIGILATORS about interpretations of the exam questions. If you have a concern about a question, please summarize it at the top of this page. All concerns will be evaluated after the exam. GOOD LUCK!

**Question 1 8 points** (*Assume all transitions are possible*)

a. 6 points Calculate the **energy (in kJ/mol)** required to ionize (completely remove) the electron from the **$n=3$** level of **He^+ ion** (He: Helium). Show your work including equations. (3 sig. figs.)

b. 2 points

(I) Select the largest (longest) wavelength from the three choices below:

- i. Wavelength of photon to *completely remove* an electron from *1s orbital of Hydrogen atom*
- ii. Wavelength of photon to *completely remove* an electron from *1s orbital of Boron atom (B)*
- iii. Wavelength of photon to *completely remove* an electron from *1s orbital of Boron cation (B^+)*

i or ii or iii

(II) Select the smallest (shortest) wavelength from the three choices below (*for H atom*):

- i. Wavelength of photon *emitted* for an electron transitioning from *2p to 1s orbital*
- ii. Wavelength of photon *absorbed* for an electron transitioning from *2s to 4p orbital*
- iii. Wavelength of photon *absorbed* for an electron transitioning from *1s to 3p orbital*

i or ii or iii



Question 2 8 points



A Chem110 student plots the graph shown on the left, from data obtained for the kinetic energy (eV) of the ejected electron versus frequency (s^{-1}) of the incident light, for an unknown metal. ($1\text{eV} = 1.6022 \times 10^{-19} \text{ J}$)

Based on the plot, answer the questions **a to c** below:

a. 1 point On the plot above, label the **threshold frequency**.

b. 2 points What is the value of the work potential (in J) of the unknown metal?

c. 3 points What is the *longest* wavelength (in nm) required to eject an electron from this metal's surface? Show your work including equation(s) used. (3 sig figs)

d. 2 points In the space provided below, plot the following

(Assume frequency is *above* threshold frequency):

- Kinetic energy* of an ejected electron versus the *intensity* of the incident light (left)
- Number of electrons* ejected versus the *frequency* of the incident light (right)



**Question 3** **6 points****a. 2 points**

In the ground state of Ge (Germanium; $Z = 32$) atom, **name** the following orbitals:

i. Orbital containing the electron that requires the maximum energy to be removed.

ii. Orbital containing the electron that experiences the minimum effective nuclear charge.

b. 2 points

An unknown anion (A^-) has the following *excited* electronic configuration: $[Ar] 4s^2 3d^3 4p^1$

(Ar: Argon; $Z = 18$)

Give the **condensed ground state electronic configuration** of the atom, **A** and the cation, **A^{2+}** .
Show your work. (Assume all electronic transitions are possible)

Electronic Configuration of A:

Electronic Configuration of A^{2+} :

c. 2 points

In a single atom, determine how many electrons can have the following set of quantum numbers?

(i) $n = 5; l = 3$

In a single atom, determine how many electrons can have the following set of quantum numbers?

(i) $n = 3; l = 2; m_s = +1/2$

**Question 4** **8 points**

Rank the following from 1 to 4 in **increasing** order. Provide the ranking in the boxes provided.

Note: Atomic number (Z) are provided for relevant atoms

a) Increasing (lowest to highest) order of *ionization energy*: (1= lowest; 4 = highest)

Mg

Mg: Z = 12

Rb

Rb: Z = 37

Cs

Cs: Z = 55

K

K : Z = 19

b) Increasing (lowest to highest) order of *bond angle*: (1= lowest; 4 = highest)

H₂OCCl₄XeF₄

HCN

c) Increasing (smallest to largest) order of *ionic size* (1= smallest; 4 = largest)

S²⁻

S: Z = 16

O²⁻

O: Z = 8

Be²⁺

Be: Z = 4

Na⁺

Na: Z = 11

d) Increasing (lowest to highest) order of *bond energy* (1= lowest; 4 = highest)

P-F

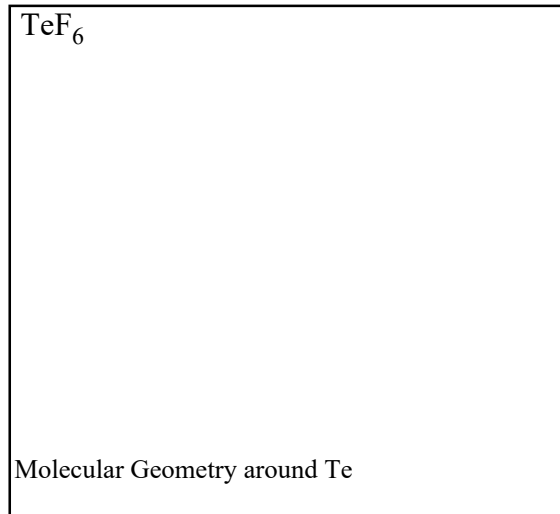
P-Cl

P-I

P-Br

**Question 5** **12 points**

a. 9 points For the molecules HClO_4 and TeF_6 , draw the most stable *Lewis structure* showing appropriate shape and *dashed/wedged bonds*, based on VSEPR. Determine the **molecular geometry** around the central atom. For HClO_4 , label the *most polar bond*.



b. 3 points (Note: Lone pairs are not shown in structures below)

(i) For the three *resonance* structures (1, 2 and 3) shown below, answer the following question:



(I) Circle the *most contributing* resonance structure.

Structure 1

Structure 2

Structure 3

(ii) For the three structures given (4, 5, and 6) below, answer the following questions:



(I) Circle all the structures that represent *resonance structures* of each other:

Structure 4

Structure 5

Structure 6

(II) Among the resonance structures, circle the *most contributing* resonance structure.

Structure 4

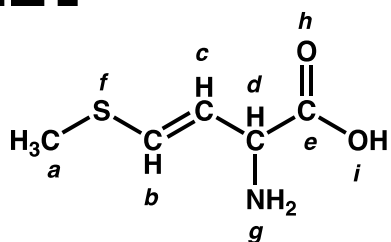
Structure 5

Structure 6

**Question 6** **6 points**

Using the data given below, calculate the **lattice energy** for SrO(s).

	ΔH° (kJ/mol)
$\text{Sr}(s) \longrightarrow \text{Sr}(g)$	+180
$\text{Sr}(g) \longrightarrow \text{Sr}^{2+}(g) + 2e^-$	+1468
$\text{O}_2(g) \longrightarrow 2\text{O}(g)$	+500
$\text{O}(g) + 2e^- \longrightarrow \text{O}^{2-}(g)$	+600
$\text{Sr}(s) + \frac{1}{2}\text{O}_2(g) \longrightarrow \text{SrO}(s)$	-550

**Question 7 10 points**

Consider the molecule given on the left, and answer questions (i) to (vi) using chemical bonding theories that we have learnt in class (valence bond theory, hybridization).

Note that the lone pairs are not shown in the structure:

i. Consider the carbon atoms marked in as ***a***, ***b***, ***c***, ***d***, and ***e*** in the molecule, and determine **hybridization** at each of the carbon centers. (2.5 points)

Atom	Hybridization
Carbon (<i>a</i>)	
Carbon (<i>b</i>)	
Carbon (<i>c</i>)	
Carbon (<i>d</i>)	
Carbon (<i>e</i>)	

ii. Now, consider the sulfur atom (S) marked as ***f***, nitrogen (N) atom (***g***), and oxygen(O) atoms (***h*** and ***i***), and determine their **hybridization**. (2 points)

Atom	Hybridization
Sulfur (<i>f</i>)	
Nitrogen (<i>g</i>)	
Oxygen (<i>h</i>)	
Oxygen (<i>i</i>)	

iii. **Describe** all the bonds formed by the carbon atoms ***a*** and ***b***, using sigma and pi designation as appropriate. (2 points)

Atom	Bonds formed (use sigma and pi as appropriate)
Carbon (<i>a</i>)	
Carbon (<i>b</i>)	

iv. How many total sp^3 -s sigma bonds are in the molecule? (1 point)

v. Determine **electronic** and **molecular geometry** at carbon (C) labeled ***e*** and sulfur (S) labeled ***f*** centres. (2 points)

Atom	Electronic Geometry	Molecular Geometry
Carbon (<i>e</i>)		
Sulfur (<i>f</i>)		

vi. Circle the value from the choices given below, that describes the **C-S-C bond angle at sulfur** (S) marked ***f***: (0.5 points)

>90° to <109.5° 109.5° >109.5 to <120° 120° >120° to <180°



Question 8 12 points

Using the **valence shell orbitals only** draw a **complete** molecular orbital (MO) diagram (with no s-p mixing) for the molecule **PS**, in the space provided below. **(6 points)**

Assume that 3s and 3p atomic orbitals will form molecular orbitals like those formed by the combination of 2s and 2p atomic orbitals that we learnt in class, and there is no s-p mixing.

Show **energy arrow**, **atomic orbitals** involved, **atom labels**, the **molecular orbitals** that are formed, **filled electrons** in the MO diagram etc. Make sure to label all the atomic and molecular orbitals.

Using the MO diagram above, answer the following questions:

- (i) Which one of the following will have the highest bond order? Circle the correct choice
 A) PS^{1-} B) PS C) PS^{1+}
- (ii) Which one of the following will have the longest bond length? Circle the correct choice
 A) PS^{1-} B) PS C) PS^{1+}
- (iii) Which of the following will be diamagnetic? Circle the correct choice.
 A) PS^{1-} B) PS C) PS^{1+}
- (iv) Draw Lewis dot structure of the molecule with the shortest bond length (based on the MO diagram) from among the three (PS^{1-} , PS, PS^{1+}), and indicate the bond order calculated from the Lewis dot structure. **(3 points)**

Lewis Dot Structure

Bond Order



FBD0F67A-6B10-485B-A2FE-D3F995EFDE1A

midterm-exam-93388

#478 Page 10 of 10

SCRAP PAGE

**CHEMISTRY 110 –Midterm Exam****Thursday November 7, 2024, 6:30 – 8:30 pm****Examiners: Pallavi Sirjoosingh and Ashok Kakkar**

First name (please write as legibly as possible within the boxes)

--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--

Last name

--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--

Student ID number

--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--

This booklet has 8 questions on 10 pages (The last page can be used as a scrap page).

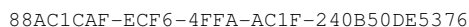
The staple must NOT be removed from the question booklet. Your intact question booklet must be returned at the end of the exam.

A separate booklet contains a periodic table, and datasheet.

Write your answers directly under the questions in the space provided. Do not write on the QR codes or near the top edges (as it may be cut off).

Please write **legibly and dark enough** for the scan to be readable. For students who use English as a second language, dictionaries are permitted. Calculators are allowed.

DO NOT ASK QUESTIONS OF THE INVIGILATORS about interpretations of the exam questions. If you have a concern about a question, please summarize it at the top of this page. All concerns will be evaluated after the exam. GOOD LUCK!



midterm-exam-93388

#479 Page 2 of 10

a. 6 points Calculate the **energy (in kJ/mol)** required to ionize (completely remove) the electron from the **n=3** level of **He⁺ ion** (He: Helium). Show your work including equations. (3 sig. figs.)

(I) Select the largest (longest) wavelength from the three choices below:

- Wavelength of photon to *completely remove* an electron from *1s orbital of Hydrogen atom*
- Wavelength of photon to *completely remove* an electron from *1s orbital of Boron atom (B)*
- Wavelength of photon to *completely remove* an electron from *1s orbital of Boron cation (B⁺)*

i
or
ii
or
iii

(II) Select the smallest (shortest) wavelength from the three choices below (*for H atom*):

- Wavelength of photon *emitted* for an electron transitioning from $2p$ to $1s$ orbital
- Wavelength of photon *absorbed* for an electron transitioning from $2s$ to $4p$ orbital
- Wavelength of photon *absorbed* for an electron transitioning from $1s$ to $3p$ orbital

i or ii or iii



Question 2 8 points



A Chem110 student plots the graph shown on the left, from data obtained for the kinetic energy (eV) of the ejected electron versus frequency (s^{-1}) of the incident light, for an unknown metal. ($1\text{eV} = 1.6022 \times 10^{-19} \text{ J}$)

Based on the plot, answer the questions **a to c** below:

a. 1 point On the plot above, label the **threshold frequency**.

b. 2 points What is the value of the work potential (in J) of the unknown metal?

c. 3 points What is the *longest* wavelength (in nm) required to eject an electron from this metal's surface? Show your work including equation(s) used. (3 sig figs)

d. 2 points In the space provided below, plot the following

(Assume frequency is *above* threshold frequency):

- i. *Kinetic energy* of an ejected electron versus the *intensity* of the incident light (left)
- ii. *Number of electrons* ejected versus the *frequency* of the incident light (right)



**Question 3** **6 points****a. 2 points**

In the ground state of Ge (Germanium; $Z = 32$) atom, **name** the following orbitals:

i. Orbital containing the electron that requires the maximum energy to be removed.

ii. Orbital containing the electron that experiences the minimum effective nuclear charge.

b. 2 points

An unknown anion (A^-) has the following *excited* electronic configuration: $[Ar] 4s^2 3d^3 4p^1$

(Ar: Argon; $Z = 18$)

Give the **condensed ground state electronic configuration** of the atom, **A** and the cation, **A^{2+}** .
Show your work. (Assume all electronic transitions are possible)

Electronic Configuration of A:

Electronic Configuration of A^{2+} :

c. 2 points

In a single atom, determine how many electrons can have the following set of quantum numbers?

(i) $n = 5; l = 3$

In a single atom, determine how many electrons can have the following set of quantum numbers?

(i) $n = 3; l = 2; m_s = +1/2$

**Question 4** **8 points**

Rank the following from 1 to 4 in **increasing** order. Provide the ranking in the boxes provided.

Note: Atomic number (Z) are provided for relevant atoms

a) Increasing (lowest to highest) order of *ionization energy*: (1= lowest; 4 = highest)

Mg

Mg: Z = 12

Rb

Rb: Z = 37

Cs

Cs: Z = 55

K

K : Z = 19

b) Increasing (lowest to highest) order of *bond angle*: (1= lowest; 4 = highest)

H₂OCCl₄XeF₄

HCN

c) Increasing (smallest to largest) order of *ionic size* (1= smallest; 4 = largest)

S²⁻

S: Z = 16

O²⁻

O: Z = 8

Be²⁺

Be: Z = 4

Na⁺

Na: Z = 11

d) Increasing (lowest to highest) order of *bond energy* (1= lowest; 4 = highest)

P-F

P-Cl

P-I

P-Br

**Question 5** **12 points**

a. 9 points For the molecules HClO_4 and TeF_6 , draw the most stable *Lewis structure* showing appropriate shape and *dashed/wedged bonds*, based on VSEPR. Determine the **molecular geometry** around the central atom. For HClO_4 , label the *most polar bond*.



b. 3 points (Note: Lone pairs are not shown in structures below)

(i) For the three *resonance* structures (1, 2 and 3) shown below, answer the following question:



(I) Circle the *most contributing* resonance structure.

Structure 1

Structure 2

Structure 3

(ii) For the three structures given (4, 5, and 6) below, answer the following questions:



(I) Circle all the structures that represent *resonance structures* of each other:

Structure 4

Structure 5

Structure 6

(II) Among the resonance structures, circle the *most contributing* resonance structure.

Structure 4

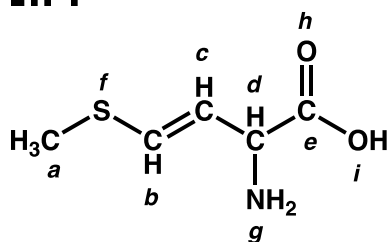
Structure 5

Structure 6

**Question 6** **6 points**

Using the data given below, calculate the **lattice energy** for SrO(s).

	ΔH° (kJ/mol)
$\text{Sr}(s) \longrightarrow \text{Sr}(g)$	+180
$\text{Sr}(g) \longrightarrow \text{Sr}^{2+}(g) + 2e^-$	+1468
$\text{O}_2(g) \longrightarrow 2\text{O}(g)$	+500
$\text{O}(g) + 2e^- \longrightarrow \text{O}^{2-}(g)$	+600
$\text{Sr}(s) + \frac{1}{2}\text{O}_2(g) \longrightarrow \text{SrO}(s)$	-550

**Question 7 10 points**

Consider the molecule given on the left, and answer questions (i) to (vi) using chemical bonding theories that we have learnt in class (valence bond theory, hybridization).

Note that the lone pairs are not shown in the structure:

i. Consider the carbon atoms marked in as ***a***, ***b***, ***c***, ***d***, and ***e*** in the molecule, and determine **hybridization** at each of the carbon centers. (2.5 points)

Atom	Hybridization
Carbon (<i>a</i>)	
Carbon (<i>b</i>)	
Carbon (<i>c</i>)	
Carbon (<i>d</i>)	
Carbon (<i>e</i>)	

ii. Now, consider the sulfur atom (S) marked as ***f***, nitrogen (N) atom (***g***), and oxygen(O) atoms (***h*** and ***i***), and determine their **hybridization**. (2 points)

Atom	Hybridization
Sulfur (<i>f</i>)	
Nitrogen (<i>g</i>)	
Oxygen (<i>h</i>)	
Oxygen (<i>i</i>)	

iii. **Describe** all the bonds formed by the carbon atoms ***a*** and ***b***, using sigma and pi designation as appropriate. (2 points)

Atom	Bonds formed (use sigma and pi as appropriate)
Carbon (<i>a</i>)	
Carbon (<i>b</i>)	

iv. How many total sp^3 -s sigma bonds are in the molecule? (1 point)

v. Determine **electronic** and **molecular geometry** at carbon (C) labeled ***e*** and sulfur (S) labeled ***f*** centres. (2 points)

Atom	Electronic Geometry	Molecular Geometry
Carbon (<i>e</i>)		
Sulfur (<i>f</i>)		

vi. Circle the value from the choices given below, that describes the **C-S-C bond angle at sulfur** (S) marked ***f***: (0.5 points)

>90° to <109.5° 109.5° >109.5 to <120° 120° >120° to <180°


Question 8 12 points

Using the **valence shell orbitals only** draw a **complete** molecular orbital (MO) diagram (with no s-p mixing) for the molecule **PS**, in the space provided below. **(6 points)**

Assume that 3s and 3p atomic orbitals will form molecular orbitals like those formed by the combination of 2s and 2p atomic orbitals that we learnt in class, and there is no s-p mixing.

Show **energy arrow**, **atomic orbitals** involved, **atom labels**, the **molecular orbitals** that are formed, **filled electrons** in the MO diagram etc. Make sure to label all the atomic and molecular orbitals.

Using the MO diagram above, answer the following questions:

- (i) Which one of the following will have the highest bond order? Circle the correct choice
 A) PS^{1-} B) PS C) PS^{1+}
- (ii) Which one of the following will have the longest bond length? Circle the correct choice
 A) PS^{1-} B) PS C) PS^{1+}
- (iii) Which of the following will be diamagnetic? Circle the correct choice.
 A) PS^{1-} B) PS C) PS^{1+}
- (iv) Draw Lewis dot structure of the molecule with the shortest bond length (based on the MO diagram) from among the three (PS^{1-} , PS, PS^{1+}), and indicate the bond order calculated from the Lewis dot structure. **(3 points)**

Lewis Dot Structure

Bond Order



CC52608A-DCAB-4147-94FA-0422AE8ECBAA

midterm-exam-93388

#479 Page 10 of 10

SCRAP PAGE

**CHEMISTRY 110 –Midterm Exam****Thursday November 7, 2024, 6:30 – 8:30 pm****Examiners: Pallavi Sirjoosingh and Ashok Kakkar**

First name (please write as legibly as possible within the boxes)

--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--

Last name

--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--

Student ID number

--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--

This booklet has 8 questions on 10 pages (The last page can be used as a scrap page).

The staple must NOT be removed from the question booklet. Your intact question booklet must be returned at the end of the exam.

A separate booklet contains a periodic table, and datasheet.

Write your answers directly under the questions in the space provided. Do not write on the QR codes or near the top edges (as it may be cut off).

Please write **legibly and dark enough** for the scan to be readable. For students who use English as a second language, dictionaries are permitted. Calculators are allowed.

DO NOT ASK QUESTIONS OF THE INVIGILATORS about interpretations of the exam questions. If you have a concern about a question, please summarize it at the top of this page. All concerns will be evaluated after the exam. GOOD LUCK!

**Question 1 8 points** (*Assume all transitions are possible*)

a. 6 points Calculate the **energy (in kJ/mol)** required to ionize (completely remove) the electron from the **n=3** level of **He⁺ ion** (He: Helium). Show your work including equations. (3 sig. figs.)

b. 2 points

(I) Select the largest (longest) wavelength from the three choices below:

- i. Wavelength of photon to *completely remove* an electron from *1s orbital of Hydrogen atom*
- ii. Wavelength of photon to *completely remove* an electron from *1s orbital of Boron atom (B)*
- iii. Wavelength of photon to *completely remove* an electron from *1s orbital of Boron cation (B⁺)*

i or ii or iii

(II) Select the smallest (shortest) wavelength from the three choices below (*for H atom*):

- i. Wavelength of photon *emitted* for an electron transitioning from *2p to 1s orbital*
- ii. Wavelength of photon *absorbed* for an electron transitioning from *2s to 4p orbital*
- iii. Wavelength of photon *absorbed* for an electron transitioning from *1s to 3p orbital*

i or ii or iii



Question 2 8 points



A Chem110 student plots the graph shown on the left, from data obtained for the kinetic energy (eV) of the ejected electron versus frequency (s^{-1}) of the incident light, for an unknown metal. ($1\text{eV} = 1.6022 \times 10^{-19} \text{ J}$)

Based on the plot, answer the questions **a to c** below:

a. 1 point On the plot above, label the **threshold frequency**.

b. 2 points What is the value of the work potential (in J) of the unknown metal?

c. 3 points What is the *longest* wavelength (in nm) required to eject an electron from this metal's surface? Show your work including equation(s) used. (3 sig figs)

d. 2 points In the space provided below, plot the following

(Assume frequency is *above* threshold frequency):

- i. *Kinetic energy* of an ejected electron versus the *intensity* of the incident light (left)
- ii. *Number of electrons* ejected versus the *frequency* of the incident light (right)



**Question 3** **6 points****a. 2 points**

In the ground state of Ge (Germanium; $Z = 32$) atom, **name** the following orbitals:

i. Orbital containing the electron that requires the maximum energy to be removed.

ii. Orbital containing the electron that experiences the minimum effective nuclear charge.

b. 2 points

An unknown anion (A^-) has the following *excited* electronic configuration: $[Ar] 4s^2 3d^3 4p^1$

(Ar: Argon; $Z = 18$)

Give the **condensed ground state electronic configuration** of the atom, **A** and the cation, **A^{2+}** .
Show your work. (Assume all electronic transitions are possible)

Electronic Configuration of A:

Electronic Configuration of A^{2+} :

c. 2 points

In a single atom, determine how many electrons can have the following set of quantum numbers?

(i) $n = 5; l = 3$

In a single atom, determine how many electrons can have the following set of quantum numbers?

(i) $n = 3; l = 2; m_s = +1/2$

**Question 4** **8 points**

Rank the following from 1 to 4 in **increasing** order. Provide the ranking in the boxes provided.

Note: Atomic number (Z) are provided for relevant atoms

a) Increasing (lowest to highest) order of *ionization energy*: (1= lowest; 4 = highest)

Mg

Mg: Z = 12

Rb

Rb: Z = 37

Cs

Cs: Z = 55

K

K : Z = 19

b) Increasing (lowest to highest) order of *bond angle*: (1= lowest; 4 = highest)

H₂OCCl₄XeF₄

HCN

c) Increasing (smallest to largest) order of *ionic size* (1= smallest; 4 = largest)

S²⁻

S: Z = 16

O²⁻

O: Z = 8

Be²⁺

Be: Z = 4

Na⁺

Na: Z = 11

d) Increasing (lowest to highest) order of *bond energy* (1= lowest; 4 = highest)

P-F

P-Cl

P-I

P-Br

**Question 5** **12 points**

a. 9 points For the molecules HClO_4 and TeF_6 , draw the most stable *Lewis structure* showing appropriate shape and *dashed/wedged bonds*, based on VSEPR. Determine the **molecular geometry** around the central atom. For HClO_4 , label the *most polar bond*.



b. 3 points (Note: Lone pairs are not shown in structures below)

(i) For the three *resonance* structures (1, 2 and 3) shown below, answer the following question:



(I) Circle the *most contributing* resonance structure.

Structure 1

Structure 2

Structure 3

(ii) For the three structures given (4, 5, and 6) below, answer the following questions:



(I) Circle all the structures that represent *resonance structures* of each other:

Structure 4

Structure 5

Structure 6

(II) Among the resonance structures, circle the *most contributing* resonance structure.

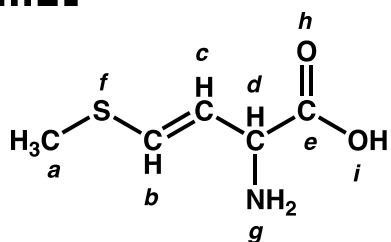
Structure 4

Structure 5

Structure 6

**Question 6** **6 points**Using the data given below, calculate the **lattice energy** for SrO(s).

	ΔH° (kJ/mol)
$\text{Sr}(s) \longrightarrow \text{Sr}(g)$	+180
$\text{Sr}(g) \longrightarrow \text{Sr}^{2+}(g) + 2e^-$	+1468
$\text{O}_2(g) \longrightarrow 2\text{O}(g)$	+500
$\text{O}(g) + 2e^- \longrightarrow \text{O}^{2-}(g)$	+600
$\text{Sr}(s) + \frac{1}{2}\text{O}_2(g) \longrightarrow \text{SrO}(s)$	-550

**Question 7 10 points**

Consider the molecule given on the left, and answer questions (i) to (vi) using chemical bonding theories that we have learnt in class (valence bond theory, hybridization).

Note that the lone pairs are not shown in the structure:

i. Consider the carbon atoms marked in as ***a***, ***b***, ***c***, ***d***, and ***e*** in the molecule, and determine **hybridization** at each of the carbon centers. (2.5 points)

Atom	Hybridization
Carbon (<i>a</i>)	
Carbon (<i>b</i>)	
Carbon (<i>c</i>)	
Carbon (<i>d</i>)	
Carbon (<i>e</i>)	

ii. Now, consider the sulfur atom (S) marked as ***f***, nitrogen (N) atom (***g***), and oxygen(O) atoms (***h*** and ***i***), and determine their **hybridization**. (2 points)

Atom	Hybridization
Sulfur (<i>f</i>)	
Nitrogen (<i>g</i>)	
Oxygen (<i>h</i>)	
Oxygen (<i>i</i>)	

iii. **Describe** all the bonds formed by the carbon atoms ***a*** and ***b***, using sigma and pi designation as appropriate. (2 points)

Atom	Bonds formed (use sigma and pi as appropriate)
Carbon (<i>a</i>)	
Carbon (<i>b</i>)	

iv. How many total sp^3 -s sigma bonds are in the molecule? (1 point)

v. Determine **electronic** and **molecular geometry** at carbon (C) labeled ***e*** and sulfur (S) labeled ***f*** centres. (2 points)

Atom	Electronic Geometry	Molecular Geometry
Carbon (<i>e</i>)		
Sulfur (<i>f</i>)		

vi. Circle the value from the choices given below, that describes the **C-S-C bond angle at sulfur** (S) marked ***f***: (0.5 points)

>90° to <109.5° 109.5° >109.5 to <120° 120° >120° to <180°


Question 8 12 points

Using the **valence shell orbitals only** draw a **complete** molecular orbital (MO) diagram (with no s-p mixing) for the molecule **PS**, in the space provided below. **(6 points)**

Assume that 3s and 3p atomic orbitals will form molecular orbitals like those formed by the combination of 2s and 2p atomic orbitals that we learnt in class, and there is no s-p mixing.

Show **energy arrow**, **atomic orbitals** involved, **atom labels**, the **molecular orbitals** that are formed, **filled electrons** in the MO diagram etc. Make sure to label all the atomic and molecular orbitals.

Using the MO diagram above, answer the following questions:

- (i) Which one of the following will have the highest bond order? Circle the correct choice
 A) PS^{1-} B) PS C) PS^{1+}
- (ii) Which one of the following will have the longest bond length? Circle the correct choice
 A) PS^{1-} B) PS C) PS^{1+}
- (iii) Which of the following will be diamagnetic? Circle the correct choice.
 A) PS^{1-} B) PS C) PS^{1+}
- (iv) Draw Lewis dot structure of the molecule with the shortest bond length (based on the MO diagram) from among the three (PS^{1-} , PS, PS^{1+}), and indicate the bond order calculated from the Lewis dot structure. **(3 points)**

Lewis Dot Structure

Bond Order



6865CD74-1C54-4A49-BF88-3FB4E2AB653C

midterm-exam-93388

#480 Page 10 of 10

SCRAP PAGE

**CHEMISTRY 110 –Midterm Exam****Thursday November 7, 2024, 6:30 – 8:30 pm**

Examiners: Pallavi Sirjoosingh and Ashok Kakkar

First name (please write as legibly as possible within the boxes)

--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--

Last name

--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--

Student ID number

--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--

This booklet has 8 questions on 10 pages (The last page can be used as a scrap page).

The staple must NOT be removed from the question booklet. Your intact question booklet must be returned at the end of the exam.

A separate booklet contains a periodic table, and datasheet.

Write your answers directly under the questions in the space provided. Do not write on the QR codes or near the top edges (as it may be cut off).

Please write **legibly and dark enough** for the scan to be readable. For students who use English as a second language, dictionaries are permitted. Calculators are allowed.

DO NOT ASK QUESTIONS OF THE INVIGILATORS about interpretations of the exam questions. If you have a concern about a question, please summarize it at the top of this page. All concerns will be evaluated after the exam. GOOD LUCK!

**Question 1 8 points** (*Assume all transitions are possible*)

a. 6 points Calculate the **energy (in kJ/mol)** required to ionize (completely remove) the electron from the **n=3** level of **He⁺ ion** (He: Helium). Show your work including equations. (3 sig. figs.)

b. 2 points

(I) Select the largest (longest) wavelength from the three choices below:

- i. Wavelength of photon to *completely remove* an electron from *1s orbital of Hydrogen atom*
- ii. Wavelength of photon to *completely remove* an electron from *1s orbital of Boron atom (B)*
- iii. Wavelength of photon to *completely remove* an electron from *1s orbital of Boron cation (B⁺)*

i or ii or iii

(II) Select the smallest (shortest) wavelength from the three choices below (*for H atom*):

- i. Wavelength of photon *emitted* for an electron transitioning from *2p to 1s orbital*
- ii. Wavelength of photon *absorbed* for an electron transitioning from *2s to 4p orbital*
- iii. Wavelength of photon *absorbed* for an electron transitioning from *1s to 3p orbital*

i or ii or iii



Question 2 8 points



A Chem110 student plots the graph shown on the left, from data obtained for the kinetic energy (eV) of the ejected electron versus frequency (s^{-1}) of the incident light, for an unknown metal. ($1\text{eV} = 1.6022 \times 10^{-19} \text{ J}$)

Based on the plot, answer the questions **a to c** below:

a. 1 point On the plot above, label the **threshold frequency**.

b. 2 points What is the value of the work potential (in J) of the unknown metal?

c. 3 points What is the *longest* wavelength (in nm) required to eject an electron from this metal's surface? Show your work including equation(s) used. (3 sig figs)

d. 2 points In the space provided below, plot the following

(Assume frequency is *above* threshold frequency):

- Kinetic energy* of an ejected electron versus the *intensity* of the incident light (left)
- Number of electrons* ejected versus the *frequency* of the incident light (right)



**Question 3** **6 points****a. 2 points**

In the ground state of Ge (Germanium; $Z = 32$) atom, **name** the following orbitals:

i. Orbital containing the electron that requires the maximum energy to be removed.

ii. Orbital containing the electron that experiences the minimum effective nuclear charge.

b. 2 points

An unknown anion (A^-) has the following *excited* electronic configuration: $[Ar] 4s^2 3d^3 4p^1$

(Ar: Argon; $Z = 18$)

Give the **condensed ground state electronic configuration** of the atom, **A** and the cation, **A^{2+}** .
Show your work. (Assume all electronic transitions are possible)

Electronic Configuration of A:

Electronic Configuration of A^{2+} :

c. 2 points

In a single atom, determine how many electrons can have the following set of quantum numbers?

(i) $n = 5; l = 3$

In a single atom, determine how many electrons can have the following set of quantum numbers?

(i) $n = 3; l = 2; m_s = +1/2$

**Question 4** **8 points**

Rank the following from 1 to 4 in **increasing** order. Provide the ranking in the boxes provided.

Note: Atomic number (Z) are provided for relevant atoms

a) Increasing (lowest to highest) order of *ionization energy*: (1= lowest; 4 = highest)

Mg

Mg: Z = 12

Rb

Rb: Z = 37

Cs

Cs: Z = 55

K

K : Z = 19

b) Increasing (lowest to highest) order of *bond angle*: (1= lowest; 4 = highest)

H₂OCCl₄XeF₄

HCN

c) Increasing (smallest to largest) order of *ionic size* (1= smallest; 4 = largest)

S²⁻

S: Z = 16

O²⁻

O: Z = 8

Be²⁺

Be: Z = 4

Na⁺

Na: Z = 11

d) Increasing (lowest to highest) order of *bond energy* (1= lowest; 4 = highest)

P-F

P-Cl

P-I

P-Br

**Question 5** **12 points**

a. 9 points For the molecules HClO_4 and TeF_6 , draw the most stable *Lewis structure* showing appropriate shape and *dashed/wedged bonds*, based on VSEPR. Determine the **molecular geometry** around the central atom. For HClO_4 , label the *most polar bond*.



b. 3 points (Note: Lone pairs are not shown in structures below)

(i) For the three *resonance* structures (1, 2 and 3) shown below, answer the following question:



(I) Circle the *most contributing* resonance structure.

Structure 1

Structure 2

Structure 3

(ii) For the three structures given (4, 5, and 6) below, answer the following questions:



(I) Circle all the structures that represent *resonance structures* of each other:

Structure 4

Structure 5

Structure 6

(II) Among the resonance structures, circle the *most contributing* resonance structure.

Structure 4

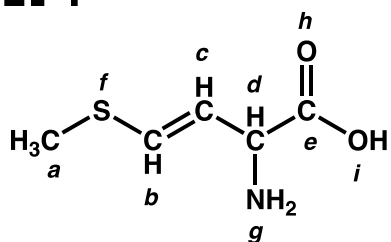
Structure 5

Structure 6

**Question 6** **6 points**

Using the data given below, calculate the **lattice energy** for SrO(s).

	ΔH° (kJ/mol)
$\text{Sr}(s) \longrightarrow \text{Sr}(g)$	+180
$\text{Sr}(g) \longrightarrow \text{Sr}^{2+}(g) + 2e^-$	+1468
$\text{O}_2(g) \longrightarrow 2\text{O}(g)$	+500
$\text{O}(g) + 2e^- \longrightarrow \text{O}^{2-}(g)$	+600
$\text{Sr}(s) + \frac{1}{2}\text{O}_2(g) \longrightarrow \text{SrO}(s)$	-550

**Question 7 10 points**

Consider the molecule given on the left, and answer questions (i) to (vi) using chemical bonding theories that we have learnt in class (valence bond theory, hybridization).

Note that the lone pairs are not shown in the structure:

i. Consider the carbon atoms marked in as ***a***, ***b***, ***c***, ***d***, and ***e*** in the molecule, and determine **hybridization** at each of the carbon centers. (2.5 points)

Atom	Hybridization
Carbon (<i>a</i>)	
Carbon (<i>b</i>)	
Carbon (<i>c</i>)	
Carbon (<i>d</i>)	
Carbon (<i>e</i>)	

ii. Now, consider the sulfur atom (S) marked as ***f***, nitrogen (N) atom (***g***), and oxygen(O) atoms (***h*** and ***i***), and determine their **hybridization**. (2 points)

Atom	Hybridization
Sulfur (<i>f</i>)	
Nitrogen (<i>g</i>)	
Oxygen (<i>h</i>)	
Oxygen (<i>i</i>)	

iii. **Describe** all the bonds formed by the carbon atoms ***a*** and ***b***, using sigma and pi designation as appropriate. (2 points)

Atom	Bonds formed (use sigma and pi as appropriate)
Carbon (<i>a</i>)	
Carbon (<i>b</i>)	

iv. How many total sp^3 -s sigma bonds are in the molecule? (1 point)

v. Determine **electronic** and **molecular geometry** at carbon (C) labeled ***e*** and sulfur (S) labeled ***f*** centres. (2 points)

Atom	Electronic Geometry	Molecular Geometry
Carbon (<i>e</i>)		
Sulfur (<i>f</i>)		

vi. Circle the value from the choices given below, that describes the **C-S-C bond angle at sulfur** (S) marked ***f***: (0.5 points)

$>90^\circ$ to $<109.5^\circ$ 109.5° >109.5 to $<120^\circ$ 120° $>120^\circ$ to $<180^\circ$


Question 8 12 points

Using the **valence shell orbitals only** draw a **complete** molecular orbital (MO) diagram (with no s-p mixing) for the molecule **PS**, in the space provided below. **(6 points)**

Assume that 3s and 3p atomic orbitals will form molecular orbitals like those formed by the combination of 2s and 2p atomic orbitals that we learnt in class, and there is no s-p mixing.

Show **energy arrow**, **atomic orbitals** involved, **atom labels**, the **molecular orbitals** that are formed, **filled electrons** in the MO diagram etc. Make sure to label all the atomic and molecular orbitals.

Using the MO diagram above, answer the following questions:

- (i) Which one of the following will have the highest bond order? Circle the correct choice
 A) PS^{1-} B) PS C) PS^{1+}
- (ii) Which one of the following will have the longest bond length? Circle the correct choice
 A) PS^{1-} B) PS C) PS^{1+}
- (iii) Which of the following will be diamagnetic? Circle the correct choice.
 A) PS^{1-} B) PS C) PS^{1+}
- (iv) Draw Lewis dot structure of the molecule with the shortest bond length (based on the MO diagram) from among the three (PS^{1-} , PS, PS^{1+}), and indicate the bond order calculated from the Lewis dot structure. **(3 points)**

Lewis Dot Structure

Bond Order



66269602-6870-4868-82E4-B3629A27F20A

midterm-exam-93388

#481 Page 10 of 10

SCRAP PAGE

**CHEMISTRY 110 –Midterm Exam****Thursday November 7, 2024, 6:30 – 8:30 pm**

Examiners: Pallavi Sirjoosingh and Ashok Kakkar

First name (please write as legibly as possible within the boxes)

--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--

Last name

--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--

Student ID number

--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--

This booklet has 8 questions on 10 pages (The last page can be used as a scrap page).

The staple must NOT be removed from the question booklet. Your intact question booklet must be returned at the end of the exam.

A separate booklet contains a periodic table, and datasheet.

Write your answers directly under the questions in the space provided. Do not write on the QR codes or near the top edges (as it may be cut off).

Please write **legibly and dark enough** for the scan to be readable. For students who use English as a second language, dictionaries are permitted. Calculators are allowed.

DO NOT ASK QUESTIONS OF THE INVIGILATORS about interpretations of the exam questions. If you have a concern about a question, please summarize it at the top of this page. All concerns will be evaluated after the exam. GOOD LUCK!

**Question 1 8 points** (*Assume all transitions are possible*)

a. 6 points Calculate the **energy (in kJ/mol)** required to ionize (completely remove) the electron from the **n=3** level of **He⁺ ion** (He: Helium). Show your work including equations. (3 sig. figs.)

b. 2 points

(I) Select the largest (longest) wavelength from the three choices below:

- i. Wavelength of photon to *completely remove* an electron from *1s orbital of Hydrogen atom*
- ii. Wavelength of photon to *completely remove* an electron from *1s orbital of Boron atom (B)*
- iii. Wavelength of photon to *completely remove* an electron from *1s orbital of Boron cation (B⁺)*

i or ii or iii

(II) Select the smallest (shortest) wavelength from the three choices below (*for H atom*):

- i. Wavelength of photon *emitted* for an electron transitioning from *2p to 1s orbital*
- ii. Wavelength of photon *absorbed* for an electron transitioning from *2s to 4p orbital*
- iii. Wavelength of photon *absorbed* for an electron transitioning from *1s to 3p orbital*

i or ii or iii



Question 2 8 points



A Chem110 student plots the graph shown on the left, from data obtained for the kinetic energy (eV) of the ejected electron versus frequency (s^{-1}) of the incident light, for an unknown metal. ($1\text{eV} = 1.6022 \times 10^{-19} \text{ J}$)

Based on the plot, answer the questions **a to c** below:

a. 1 point On the plot above, label the **threshold frequency**.

b. 2 points What is the value of the work potential (in J) of the unknown metal?

c. 3 points What is the *longest* wavelength (in nm) required to eject an electron from this metal's surface? Show your work including equation(s) used. (3 sig figs)

d. 2 points In the space provided below, plot the following

(Assume frequency is *above* threshold frequency):

- Kinetic energy* of an ejected electron versus the *intensity* of the incident light (left)
- Number of electrons* ejected versus the *frequency* of the incident light (right)



**Question 3** **6 points****a. 2 points**

In the ground state of Ge (Germanium; $Z = 32$) atom, **name** the following orbitals:

i. Orbital containing the electron that requires the maximum energy to be removed.

ii. Orbital containing the electron that experiences the minimum effective nuclear charge.

b. 2 points

An unknown anion (A^-) has the following *excited* electronic configuration: $[Ar] 4s^2 3d^3 4p^1$

(Ar: Argon; $Z = 18$)

Give the **condensed ground state electronic configuration** of the atom, **A** and the cation, **A^{2+}** .
Show your work. (Assume all electronic transitions are possible)

Electronic Configuration of A:

Electronic Configuration of A^{2+} :

c. 2 points

In a single atom, determine how many electrons can have the following set of quantum numbers?

(i) $n = 5; l = 3$

In a single atom, determine how many electrons can have the following set of quantum numbers?

(i) $n = 3; l = 2; m_s = +1/2$

**Question 4** **8 points**

Rank the following from 1 to 4 in **increasing** order. Provide the ranking in the boxes provided.

Note: Atomic number (Z) are provided for relevant atoms

a) Increasing (lowest to highest) order of *ionization energy*: (1= lowest; 4 = highest)

Mg

Mg: Z = 12

Rb

Rb: Z = 37

Cs

Cs: Z = 55

K

K : Z = 19

b) Increasing (lowest to highest) order of *bond angle*: (1= lowest; 4 = highest)

H₂OCCl₄XeF₄

HCN

c) Increasing (smallest to largest) order of *ionic size* (1= smallest; 4 = largest)

S²⁻

S: Z = 16

O²⁻

O: Z = 8

Be²⁺

Be: Z = 4

Na⁺

Na: Z = 11

d) Increasing (lowest to highest) order of *bond energy* (1= lowest; 4 = highest)

P-F

P-Cl

P-I

P-Br

**Question 5** **12 points**

a. 9 points For the molecules HClO_4 and TeF_6 , draw the most stable *Lewis structure* showing appropriate shape and *dashed/wedged bonds*, based on VSEPR. Determine the **molecular geometry** around the central atom. For HClO_4 , label the *most polar bond*.



b. 3 points (Note: Lone pairs are not shown in structures below)

(i) For the three *resonance* structures (1, 2 and 3) shown below, answer the following question:



(I) Circle the *most contributing* resonance structure.

Structure 1

Structure 2

Structure 3

(ii) For the three structures given (4, 5, and 6) below, answer the following questions:



(I) Circle all the structures that represent *resonance structures* of each other:

Structure 4

Structure 5

Structure 6

(II) Among the resonance structures, circle the *most contributing* resonance structure.

Structure 4

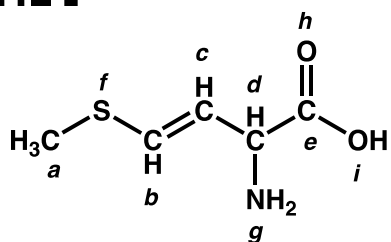
Structure 5

Structure 6

**Question 6** **6 points**

Using the data given below, calculate the **lattice energy** for SrO(s).

	ΔH° (kJ/mol)
$\text{Sr}(s) \longrightarrow \text{Sr}(g)$	+180
$\text{Sr}(g) \longrightarrow \text{Sr}^{2+}(g) + 2e^-$	+1468
$\text{O}_2(g) \longrightarrow 2\text{O}(g)$	+500
$\text{O}(g) + 2e^- \longrightarrow \text{O}^{2-}(g)$	+600
$\text{Sr}(s) + \frac{1}{2}\text{O}_2(g) \longrightarrow \text{SrO}(s)$	-550

**Question 7 10 points**

Consider the molecule given on the left, and answer questions (i) to (vi) using chemical bonding theories that we have learnt in class (valence bond theory, hybridization).

Note that the lone pairs are not shown in the structure:

i. Consider the carbon atoms marked in as ***a***, ***b***, ***c***, ***d***, and ***e*** in the molecule, and determine **hybridization** at each of the carbon centers. (2.5 points)

Atom	Hybridization
Carbon (<i>a</i>)	
Carbon (<i>b</i>)	
Carbon (<i>c</i>)	
Carbon (<i>d</i>)	
Carbon (<i>e</i>)	

ii. Now, consider the sulfur atom (S) marked as ***f***, nitrogen (N) atom (***g***), and oxygen(O) atoms (***h*** and ***i***), and determine their **hybridization**. (2 points)

Atom	Hybridization
Sulfur (<i>f</i>)	
Nitrogen (<i>g</i>)	
Oxygen (<i>h</i>)	
Oxygen (<i>i</i>)	

iii. **Describe** all the bonds formed by the carbon atoms ***a*** and ***b***, using sigma and pi designation as appropriate. (2 points)

Atom	Bonds formed (use sigma and pi as appropriate)
Carbon (<i>a</i>)	
Carbon (<i>b</i>)	

iv. How many total sp^3 -s sigma bonds are in the molecule? (1 point)

v. Determine **electronic** and **molecular geometry** at carbon (C) labeled ***e*** and sulfur (S) labeled ***f*** centres. (2 points)

Atom	Electronic Geometry	Molecular Geometry
Carbon (<i>e</i>)		
Sulfur (<i>f</i>)		

vi. Circle the value from the choices given below, that describes the **C-S-C bond angle at sulfur** (S) marked ***f***: (0.5 points)

>90° to <109.5° 109.5° >109.5 to <120° 120° >120° to <180°


Question 8 12 points

Using the **valence shell orbitals only** draw a **complete** molecular orbital (MO) diagram (with no s-p mixing) for the molecule **PS**, in the space provided below. **(6 points)**

Assume that 3s and 3p atomic orbitals will form molecular orbitals like those formed by the combination of 2s and 2p atomic orbitals that we learnt in class, and there is no s-p mixing.

Show **energy arrow**, **atomic orbitals** involved, **atom labels**, the **molecular orbitals** that are formed, **filled electrons** in the MO diagram etc. Make sure to label all the atomic and molecular orbitals.

Using the MO diagram above, answer the following questions:

- (i) Which one of the following will have the highest bond order? Circle the correct choice
 A) PS^{1-} B) PS C) PS^{1+}
- (ii) Which one of the following will have the longest bond length? Circle the correct choice
 A) PS^{1-} B) PS C) PS^{1+}
- (iii) Which of the following will be diamagnetic? Circle the correct choice.
 A) PS^{1-} B) PS C) PS^{1+}
- (iv) Draw Lewis dot structure of the molecule with the shortest bond length (based on the MO diagram) from among the three (PS^{1-} , PS, PS^{1+}), and indicate the bond order calculated from the Lewis dot structure. **(3 points)**

Lewis Dot Structure

Bond Order



FB6838B9-DDF2-445A-A3F2-96978FA60001

midterm-exam-93388

#482 Page 10 of 10

SCRAP PAGE

**CHEMISTRY 110 –Midterm Exam****Thursday November 7, 2024, 6:30 – 8:30 pm****Examiners: Pallavi Sirjoosingh and Ashok Kakkar**

First name (please write as legibly as possible within the boxes)

--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--

Last name

--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--

Student ID number

--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--

This booklet has 8 questions on 10 pages (The last page can be used as a scrap page).

The staple must NOT be removed from the question booklet. Your intact question booklet must be returned at the end of the exam.

A separate booklet contains a periodic table, and datasheet.

Write your answers directly under the questions in the space provided. Do not write on the QR codes or near the top edges (as it may be cut off).

Please write **legibly and dark enough** for the scan to be readable. For students who use English as a second language, dictionaries are permitted. Calculators are allowed.

DO NOT ASK QUESTIONS OF THE INVIGILATORS about interpretations of the exam questions. If you have a concern about a question, please summarize it at the top of this page. All concerns will be evaluated after the exam. GOOD LUCK!

**Question 1 8 points** (*Assume all transitions are possible*)

a. 6 points Calculate the **energy (in kJ/mol)** required to ionize (completely remove) the electron from the **n=3** level of **He⁺ ion** (He: Helium). Show your work including equations. (3 sig. figs.)

b. 2 points

(I) Select the largest (longest) wavelength from the three choices below:

- i. Wavelength of photon to *completely remove* an electron from *1s orbital of Hydrogen atom*
- ii. Wavelength of photon to *completely remove* an electron from *1s orbital of Boron atom (B)*
- iii. Wavelength of photon to *completely remove* an electron from *1s orbital of Boron cation (B⁺)*

i or ii or iii

(II) Select the smallest (shortest) wavelength from the three choices below (*for H atom*):

- i. Wavelength of photon *emitted* for an electron transitioning from *2p to 1s orbital*
- ii. Wavelength of photon *absorbed* for an electron transitioning from *2s to 4p orbital*
- iii. Wavelength of photon *absorbed* for an electron transitioning from *1s to 3p orbital*

i or ii or iii



Question 2 8 points



A Chem110 student plots the graph shown on the left, from data obtained for the kinetic energy (eV) of the ejected electron versus frequency (s^{-1}) of the incident light, for an unknown metal. ($1\text{eV} = 1.6022 \times 10^{-19} \text{ J}$)

Based on the plot, answer the questions **a to c** below:

a. 1 point On the plot above, label the **threshold frequency**.

b. 2 points What is the value of the work potential (in J) of the unknown metal?

c. 3 points What is the *longest* wavelength (in nm) required to eject an electron from this metal's surface? Show your work including equation(s) used. (3 sig figs)

d. 2 points In the space provided below, plot the following

(Assume frequency is *above* threshold frequency):

- i. *Kinetic energy* of an ejected electron versus the *intensity* of the incident light (left)
- ii. *Number of electrons* ejected versus the *frequency* of the incident light (right)



**Question 3** **6 points****a. 2 points**

In the ground state of Ge (Germanium; $Z = 32$) atom, **name** the following orbitals:

i. Orbital containing the electron that requires the maximum energy to be removed.

ii. Orbital containing the electron that experiences the minimum effective nuclear charge.

b. 2 points

An unknown anion (A^-) has the following *excited* electronic configuration: $[Ar] 4s^2 3d^3 4p^1$

(Ar: Argon; $Z = 18$)

Give the **condensed ground state electronic configuration** of the atom, **A** and the cation, **A^{2+}** .
Show your work. (Assume all electronic transitions are possible)

Electronic Configuration of A:

Electronic Configuration of A^{2+} :

c. 2 points

In a single atom, determine how many electrons can have the following set of quantum numbers?

(i) $n = 5; l = 3$

In a single atom, determine how many electrons can have the following set of quantum numbers?

(i) $n = 3; l = 2; m_s = +1/2$

**Question 4** **8 points**

Rank the following from 1 to 4 in **increasing** order. Provide the ranking in the boxes provided.

Note: Atomic number (Z) are provided for relevant atoms

a) Increasing (lowest to highest) order of *ionization energy*: (1= lowest; 4 = highest)

Mg

Mg: Z = 12

Rb

Rb: Z = 37

Cs

Cs: Z = 55

K

K : Z = 19

b) Increasing (lowest to highest) order of *bond angle*: (1= lowest; 4 = highest)

H₂OCCl₄XeF₄

HCN

c) Increasing (smallest to largest) order of *ionic size* (1= smallest; 4 = largest)

S²⁻

S: Z = 16

O²⁻

O: Z = 8

Be²⁺

Be: Z = 4

Na⁺

Na: Z = 11

d) Increasing (lowest to highest) order of *bond energy* (1= lowest; 4 = highest)

P-F

P-Cl

P-I

P-Br

**Question 5** **12 points**

a. 9 points For the molecules HClO_4 and TeF_6 , draw the most stable *Lewis structure* showing appropriate shape and *dashed/wedged bonds*, based on VSEPR. Determine the **molecular geometry** around the central atom. For HClO_4 , label the *most polar bond*.



b. 3 points (Note: Lone pairs are not shown in structures below)

(i) For the three *resonance* structures (1, 2 and 3) shown below, answer the following question:



(I) Circle the *most contributing* resonance structure.

Structure 1

Structure 2

Structure 3

(ii) For the three structures given (4, 5, and 6) below, answer the following questions:



(I) Circle all the structures that represent *resonance structures* of each other:

Structure 4

Structure 5

Structure 6

(II) Among the resonance structures, circle the *most contributing* resonance structure.

Structure 4

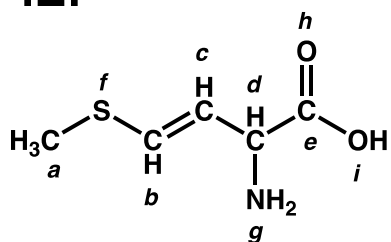
Structure 5

Structure 6

**Question 6** **6 points**

Using the data given below, calculate the **lattice energy** for SrO(s).

	ΔH° (kJ/mol)
$\text{Sr}(s) \longrightarrow \text{Sr}(g)$	+180
$\text{Sr}(g) \longrightarrow \text{Sr}^{2+}(g) + 2e^-$	+1468
$\text{O}_2(g) \longrightarrow 2\text{O}(g)$	+500
$\text{O}(g) + 2e^- \longrightarrow \text{O}^{2-}(g)$	+600
$\text{Sr}(s) + \frac{1}{2}\text{O}_2(g) \longrightarrow \text{SrO}(s)$	-550

**Question 7 10 points**

Consider the molecule given on the left, and answer questions (i) to (vi) using chemical bonding theories that we have learnt in class (valence bond theory, hybridization).

Note that the lone pairs are not shown in the structure:

i. Consider the carbon atoms marked in as ***a***, ***b***, ***c***, ***d***, and ***e*** in the molecule, and determine **hybridization** at each of the carbon centers. (2.5 points)

Atom	Hybridization
Carbon (<i>a</i>)	
Carbon (<i>b</i>)	
Carbon (<i>c</i>)	
Carbon (<i>d</i>)	
Carbon (<i>e</i>)	

ii. Now, consider the sulfur atom (S) marked as ***f***, nitrogen (N) atom (***g***), and oxygen(O) atoms (***h*** and ***i***), and determine their **hybridization**. (2 points)

Atom	Hybridization
Sulfur (<i>f</i>)	
Nitrogen (<i>g</i>)	
Oxygen (<i>h</i>)	
Oxygen (<i>i</i>)	

iii. **Describe** all the bonds formed by the carbon atoms ***a*** and ***b***, using sigma and pi designation as appropriate. (2 points)

Atom	Bonds formed (use sigma and pi as appropriate)
Carbon (<i>a</i>)	
Carbon (<i>b</i>)	

iv. How many total sp^3 -s sigma bonds are in the molecule? (1 point)

v. Determine **electronic** and **molecular geometry** at carbon (C) labeled ***e*** and sulfur (S) labeled ***f*** centres. (2 points)

Atom	Electronic Geometry	Molecular Geometry
Carbon (<i>e</i>)		
Sulfur (<i>f</i>)		

vi. Circle the value from the choices given below, that describes the **C-S-C bond angle at sulfur** (S) marked ***f***: (0.5 points)

$>90^\circ$ to $<109.5^\circ$ 109.5° >109.5 to $<120^\circ$ 120° $>120^\circ$ to $<180^\circ$


Question 8 12 points

Using the **valence shell orbitals only** draw a **complete** molecular orbital (MO) diagram (with no s-p mixing) for the molecule **PS**, in the space provided below. **(6 points)**

Assume that 3s and 3p atomic orbitals will form molecular orbitals like those formed by the combination of 2s and 2p atomic orbitals that we learnt in class, and there is no s-p mixing.

Show **energy arrow**, **atomic orbitals** involved, **atom labels**, the **molecular orbitals** that are formed, **filled electrons** in the MO diagram etc. Make sure to label all the atomic and molecular orbitals.

Using the MO diagram above, answer the following questions:

- (i) Which one of the following will have the highest bond order? Circle the correct choice
 A) PS^{1-} B) PS C) PS^{1+}
- (ii) Which one of the following will have the longest bond length? Circle the correct choice
 A) PS^{1-} B) PS C) PS^{1+}
- (iii) Which of the following will be diamagnetic? Circle the correct choice.
 A) PS^{1-} B) PS C) PS^{1+}
- (iv) Draw Lewis dot structure of the molecule with the shortest bond length (based on the MO diagram) from among the three (PS^{1-} , PS, PS^{1+}), and indicate the bond order calculated from the Lewis dot structure. **(3 points)**

Lewis Dot Structure

Bond Order



54C10593-F0E9-457E-9ED4-7085C2E42A52

midterm-exam-93388

#483 Page 10 of 10

SCRAP PAGE

**CHEMISTRY 110 –Midterm Exam****Thursday November 7, 2024, 6:30 – 8:30 pm****Examiners: Pallavi Sirjoosingh and Ashok Kakkar**

First name (please write as legibly as possible within the boxes)

--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--

Last name

--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--

Student ID number

--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--

This booklet has 8 questions on 10 pages (The last page can be used as a scrap page).

The staple must NOT be removed from the question booklet. Your intact question booklet must be returned at the end of the exam.

A separate booklet contains a periodic table, and datasheet.

Write your answers directly under the questions in the space provided. Do not write on the QR codes or near the top edges (as it may be cut off).

Please write **legibly and dark enough** for the scan to be readable. For students who use English as a second language, dictionaries are permitted. Calculators are allowed.

DO NOT ASK QUESTIONS OF THE INVIGILATORS about interpretations of the exam questions. If you have a concern about a question, please summarize it at the top of this page. All concerns will be evaluated after the exam. GOOD LUCK!

**Question 1 8 points** (*Assume all transitions are possible*)

a. 6 points Calculate the **energy (in kJ/mol)** required to ionize (completely remove) the electron from the **n=3** level of **He⁺ ion** (He: Helium). Show your work including equations. (3 sig. figs.)

b. 2 points

(I) Select the largest (longest) wavelength from the three choices below:

- i. Wavelength of photon to *completely remove* an electron from *1s orbital of Hydrogen atom*
- ii. Wavelength of photon to *completely remove* an electron from *1s orbital of Boron atom (B)*
- iii. Wavelength of photon to *completely remove* an electron from *1s orbital of Boron cation (B⁺)*

i or ii or iii

(II) Select the smallest (shortest) wavelength from the three choices below (*for H atom*):

- i. Wavelength of photon *emitted* for an electron transitioning from *2p to 1s orbital*
- ii. Wavelength of photon *absorbed* for an electron transitioning from *2s to 4p orbital*
- iii. Wavelength of photon *absorbed* for an electron transitioning from *1s to 3p orbital*

i or ii or iii



Question 2 8 points



A Chem110 student plots the graph shown on the left, from data obtained for the kinetic energy (eV) of the ejected electron versus frequency (s^{-1}) of the incident light, for an unknown metal. ($1\text{eV} = 1.6022 \times 10^{-19} \text{ J}$)

Based on the plot, answer the questions **a to c** below:

a. 1 point On the plot above, label the **threshold frequency**.

b. 2 points What is the value of the work potential (in J) of the unknown metal?

c. 3 points What is the *longest* wavelength (in nm) required to eject an electron from this metal's surface? Show your work including equation(s) used. (3 sig figs)

d. 2 points In the space provided below, plot the following

(Assume frequency is *above* threshold frequency):

- Kinetic energy* of an ejected electron versus the *intensity* of the incident light (left)
- Number of electrons* ejected versus the *frequency* of the incident light (right)



**Question 3** **6 points****a. 2 points**

In the ground state of Ge (Germanium; $Z = 32$) atom, **name** the following orbitals:

i. Orbital containing the electron that requires the maximum energy to be removed.

ii. Orbital containing the electron that experiences the minimum effective nuclear charge.

b. 2 points

An unknown anion (A^-) has the following *excited* electronic configuration: $[Ar] 4s^2 3d^3 4p^1$

(Ar: Argon; $Z = 18$)

Give the **condensed ground state electronic configuration** of the atom, **A** and the cation, **A^{2+}** .
Show your work. (Assume all electronic transitions are possible)

Electronic Configuration of A:

Electronic Configuration of A^{2+} :

c. 2 points

In a single atom, determine how many electrons can have the following set of quantum numbers?

(i) $n = 5; l = 3$

In a single atom, determine how many electrons can have the following set of quantum numbers?

(i) $n = 3; l = 2; m_s = +1/2$

**Question 4** **8 points**

Rank the following from 1 to 4 in **increasing** order. Provide the ranking in the boxes provided.

Note: Atomic number (Z) are provided for relevant atoms

a) Increasing (lowest to highest) order of *ionization energy*: (1= lowest; 4 = highest)

Mg

Mg: Z = 12

Rb

Rb: Z = 37

Cs

Cs: Z = 55

K

K : Z = 19

b) Increasing (lowest to highest) order of *bond angle*: (1= lowest; 4 = highest)

H₂OCCl₄XeF₄

HCN

c) Increasing (smallest to largest) order of *ionic size* (1= smallest; 4 = largest)

S²⁻

S: Z = 16

O²⁻

O: Z = 8

Be²⁺

Be: Z = 4

Na⁺

Na: Z = 11

d) Increasing (lowest to highest) order of *bond energy* (1= lowest; 4 = highest)

P-F

P-Cl

P-I

P-Br

**Question 5** **12 points**

a. 9 points For the molecules HClO_4 and TeF_6 , draw the most stable *Lewis structure* showing appropriate shape and *dashed/wedged bonds*, based on VSEPR. Determine the **molecular geometry** around the central atom. For HClO_4 , label the *most polar bond*.



b. 3 points (Note: Lone pairs are not shown in structures below)

(i) For the three *resonance* structures (1, 2 and 3) shown below, answer the following question:



(I) Circle the *most contributing* resonance structure.

Structure 1

Structure 2

Structure 3

(ii) For the three structures given (4, 5, and 6) below, answer the following questions:



(I) Circle all the structures that represent *resonance structures* of each other:

Structure 4

Structure 5

Structure 6

(II) Among the resonance structures, circle the *most contributing* resonance structure.

Structure 4

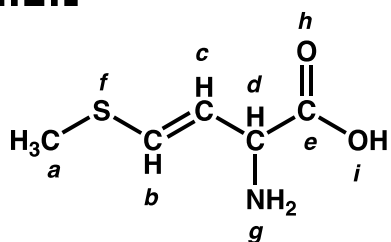
Structure 5

Structure 6

**Question 6** **6 points**

Using the data given below, calculate the **lattice energy** for SrO(s).

	ΔH° (kJ/mol)
$\text{Sr}(s) \longrightarrow \text{Sr}(g)$	+180
$\text{Sr}(g) \longrightarrow \text{Sr}^{2+}(g) + 2e^-$	+1468
$\text{O}_2(g) \longrightarrow 2\text{O}(g)$	+500
$\text{O}(g) + 2e^- \longrightarrow \text{O}^{2-}(g)$	+600
$\text{Sr}(s) + \frac{1}{2}\text{O}_2(g) \longrightarrow \text{SrO}(s)$	-550

**Question 7 10 points**

Consider the molecule given on the left, and answer questions (i) to (vi) using chemical bonding theories that we have learnt in class (valence bond theory, hybridization).

Note that the lone pairs are not shown in the structure:

i. Consider the carbon atoms marked in as ***a***, ***b***, ***c***, ***d***, and ***e*** in the molecule, and determine **hybridization** at each of the carbon centers. (2.5 points)

Atom	Hybridization
Carbon (<i>a</i>)	
Carbon (<i>b</i>)	
Carbon (<i>c</i>)	
Carbon (<i>d</i>)	
Carbon (<i>e</i>)	

ii. Now, consider the sulfur atom (S) marked as ***f***, nitrogen (N) atom (***g***), and oxygen(O) atoms (***h*** and ***i***), and determine their **hybridization**. (2 points)

Atom	Hybridization
Sulfur (<i>f</i>)	
Nitrogen (<i>g</i>)	
Oxygen (<i>h</i>)	
Oxygen (<i>i</i>)	

iii. **Describe** all the bonds formed by the carbon atoms ***a*** and ***b***, using sigma and pi designation as appropriate. (2 points)

Atom	Bonds formed (use sigma and pi as appropriate)
Carbon (<i>a</i>)	
Carbon (<i>b</i>)	

iv. How many total sp^3 -s sigma bonds are in the molecule? (1 point)

v. Determine **electronic** and **molecular geometry** at carbon (C) labeled ***e*** and sulfur (S) labeled ***f*** centres. (2 points)

Atom	Electronic Geometry	Molecular Geometry
Carbon (<i>e</i>)		
Sulfur (<i>f</i>)		

vi. Circle the value from the choices given below, that describes the **C-S-C bond angle at sulfur** (S) marked ***f***: (0.5 points)

$>90^\circ$ to $<109.5^\circ$ 109.5° >109.5 to $<120^\circ$ 120° $>120^\circ$ to $<180^\circ$


Question 8 12 points

Using the **valence shell orbitals only** draw a **complete** molecular orbital (MO) diagram (with no s-p mixing) for the molecule **PS**, in the space provided below. **(6 points)**

Assume that 3s and 3p atomic orbitals will form molecular orbitals like those formed by the combination of 2s and 2p atomic orbitals that we learnt in class, and there is no s-p mixing.

Show **energy arrow**, **atomic orbitals** involved, **atom labels**, the **molecular orbitals** that are formed, **filled electrons** in the MO diagram etc. Make sure to label all the atomic and molecular orbitals.

Using the MO diagram above, answer the following questions:

- (i) Which one of the following will have the highest bond order? Circle the correct choice
 A) PS^{1-} B) PS C) PS^{1+}
- (ii) Which one of the following will have the longest bond length? Circle the correct choice
 A) PS^{1-} B) PS C) PS^{1+}
- (iii) Which of the following will be diamagnetic? Circle the correct choice.
 A) PS^{1-} B) PS C) PS^{1+}
- (iv) Draw Lewis dot structure of the molecule with the shortest bond length (based on the MO diagram) from among the three (PS^{1-} , PS, PS^{1+}), and indicate the bond order calculated from the Lewis dot structure. **(3 points)**

Lewis Dot Structure

Bond Order



A6EB7C0E-B691-4D74-A75B-08912A881B58

midterm-exam-93388

#484 Page 10 of 10

SCRAP PAGE

**CHEMISTRY 110 –Midterm Exam****Thursday November 7, 2024, 6:30 – 8:30 pm**

Examiners: Pallavi Sirjoosingh and Ashok Kakkar

First name (please write as legibly as possible within the boxes)

--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--

Last name

--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--

Student ID number

--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--

This booklet has 8 questions on 10 pages (The last page can be used as a scrap page).

The staple must NOT be removed from the question booklet. Your intact question booklet must be returned at the end of the exam.

A separate booklet contains a periodic table, and datasheet.

Write your answers directly under the questions in the space provided. Do not write on the QR codes or near the top edges (as it may be cut off).

Please write **legibly and dark enough** for the scan to be readable. For students who use English as a second language, dictionaries are permitted. Calculators are allowed.

DO NOT ASK QUESTIONS OF THE INVIGILATORS about interpretations of the exam questions. If you have a concern about a question, please summarize it at the top of this page. All concerns will be evaluated after the exam. GOOD LUCK!

**Question 1 8 points** (*Assume all transitions are possible*)

a. 6 points Calculate the **energy (in kJ/mol)** required to ionize (completely remove) the electron from the **$n=3$** level of **He^+ ion** (He: Helium). Show your work including equations. (3 sig. figs.)

b. 2 points

(I) Select the largest (longest) wavelength from the three choices below:

- i. Wavelength of photon to *completely remove* an electron from *1s orbital of Hydrogen atom*
- ii. Wavelength of photon to *completely remove* an electron from *1s orbital of Boron atom (B)*
- iii. Wavelength of photon to *completely remove* an electron from *1s orbital of Boron cation (B^+)*

i or ii or iii

(II) Select the smallest (shortest) wavelength from the three choices below (*for H atom*):

- i. Wavelength of photon *emitted* for an electron transitioning from *2p to 1s orbital*
- ii. Wavelength of photon *absorbed* for an electron transitioning from *2s to 4p orbital*
- iii. Wavelength of photon *absorbed* for an electron transitioning from *1s to 3p orbital*

i or ii or iii



Question 2 8 points



A Chem110 student plots the graph shown on the left, from data obtained for the kinetic energy (eV) of the ejected electron versus frequency (s^{-1}) of the incident light, for an unknown metal. ($1\text{eV} = 1.6022 \times 10^{-19} \text{ J}$)

Based on the plot, answer the questions **a to c** below:

a. 1 point On the plot above, label the **threshold frequency**.

b. 2 points What is the value of the work potential (in J) of the unknown metal?

c. 3 points What is the *longest* wavelength (in nm) required to eject an electron from this metal's surface? Show your work including equation(s) used. (3 sig figs)

d. 2 points In the space provided below, plot the following

(Assume frequency is *above* threshold frequency):

- Kinetic energy* of an ejected electron versus the *intensity* of the incident light (left)
- Number of electrons* ejected versus the *frequency* of the incident light (right)



**Question 3** **6 points****a. 2 points**

In the ground state of Ge (Germanium; $Z = 32$) atom, **name** the following orbitals:

i. Orbital containing the electron that requires the maximum energy to be removed.

ii. Orbital containing the electron that experiences the minimum effective nuclear charge.

b. 2 points

An unknown anion (A^-) has the following *excited* electronic configuration: $[Ar] 4s^2 3d^3 4p^1$

(Ar: Argon; $Z = 18$)

Give the **condensed ground state electronic configuration** of the atom, **A** and the cation, **A^{2+}** .
Show your work. (Assume all electronic transitions are possible)

Electronic Configuration of A:

Electronic Configuration of A^{2+} :

c. 2 points

In a single atom, determine how many electrons can have the following set of quantum numbers?

(i) $n = 5; l = 3$

In a single atom, determine how many electrons can have the following set of quantum numbers?

(i) $n = 3; l = 2; m_s = +1/2$

**Question 4** **8 points**

Rank the following from 1 to 4 in **increasing** order. Provide the ranking in the boxes provided.

Note: Atomic number (Z) are provided for relevant atoms

a) Increasing (lowest to highest) order of *ionization energy*: (1= lowest; 4 = highest)

Mg

Mg: Z = 12

Rb

Rb: Z = 37

Cs

Cs: Z = 55

K

K : Z = 19

b) Increasing (lowest to highest) order of *bond angle*: (1= lowest; 4 = highest)

H₂OCCl₄XeF₄

HCN

c) Increasing (smallest to largest) order of *ionic size* (1= smallest; 4 = largest)

S²⁻

S: Z = 16

O²⁻

O: Z = 8

Be²⁺

Be: Z = 4

Na⁺

Na: Z = 11

d) Increasing (lowest to highest) order of *bond energy* (1= lowest; 4 = highest)

P-F

P-Cl

P-I

P-Br

**Question 5** **12 points**

a. 9 points For the molecules HClO_4 and TeF_6 , draw the most stable *Lewis structure* showing appropriate shape and *dashed/wedged bonds*, based on VSEPR. Determine the **molecular geometry** around the central atom. For HClO_4 , label the *most polar bond*.



b. 3 points (Note: Lone pairs are not shown in structures below)

(i) For the three *resonance* structures (1, 2 and 3) shown below, answer the following question:



(I) Circle the *most contributing* resonance structure.

Structure 1

Structure 2

Structure 3

(ii) For the three structures given (4, 5, and 6) below, answer the following questions:



(I) Circle all the structures that represent *resonance structures* of each other:

Structure 4

Structure 5

Structure 6

(II) Among the resonance structures, circle the *most contributing* resonance structure.

Structure 4

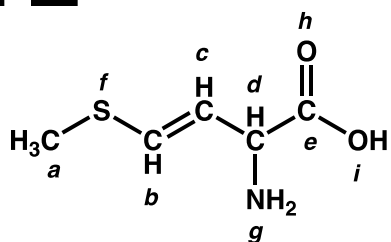
Structure 5

Structure 6

**Question 6** **6 points**

Using the data given below, calculate the **lattice energy** for SrO(s).

	ΔH° (kJ/mol)
$\text{Sr}(s) \longrightarrow \text{Sr}(g)$	+180
$\text{Sr}(g) \longrightarrow \text{Sr}^{2+}(g) + 2e^-$	+1468
$\text{O}_2(g) \longrightarrow 2\text{O}(g)$	+500
$\text{O}(g) + 2e^- \longrightarrow \text{O}^{2-}(g)$	+600
$\text{Sr}(s) + \frac{1}{2}\text{O}_2(g) \longrightarrow \text{SrO}(s)$	-550

**Question 7 10 points**

Consider the molecule given on the left, and answer questions (i) to (vi) using chemical bonding theories that we have learnt in class (valence bond theory, hybridization).

Note that the lone pairs are not shown in the structure:

i. Consider the carbon atoms marked in as ***a***, ***b***, ***c***, ***d***, and ***e*** in the molecule, and determine **hybridization** at each of the carbon centers. (2.5 points)

Atom	Hybridization
Carbon (<i>a</i>)	
Carbon (<i>b</i>)	
Carbon (<i>c</i>)	
Carbon (<i>d</i>)	
Carbon (<i>e</i>)	

ii. Now, consider the sulfur atom (S) marked as ***f***, nitrogen (N) atom (***g***), and oxygen(O) atoms (***h*** and ***i***), and determine their **hybridization**. (2 points)

Atom	Hybridization
Sulfur (<i>f</i>)	
Nitrogen (<i>g</i>)	
Oxygen (<i>h</i>)	
Oxygen (<i>i</i>)	

iii. **Describe** all the bonds formed by the carbon atoms ***a*** and ***b***, using sigma and pi designation as appropriate. (2 points)

Atom	Bonds formed (use sigma and pi as appropriate)
Carbon (<i>a</i>)	
Carbon (<i>b</i>)	

iv. How many total sp^3 -s sigma bonds are in the molecule? (1 point)

v. Determine **electronic** and **molecular geometry** at carbon (C) labeled ***e*** and sulfur (S) labeled ***f*** centres. (2 points)

Atom	Electronic Geometry	Molecular Geometry
Carbon (<i>e</i>)		
Sulfur (<i>f</i>)		

vi. Circle the value from the choices given below, that describes the **C-S-C bond angle at sulfur** (S) marked ***f***: (0.5 points)

$>90^\circ$ to $<109.5^\circ$ 109.5° >109.5 to $<120^\circ$ 120° $>120^\circ$ to $<180^\circ$


Question 8 12 points

Using the **valence shell orbitals only** draw a **complete** molecular orbital (MO) diagram (with no s-p mixing) for the molecule **PS**, in the space provided below. **(6 points)**

Assume that 3s and 3p atomic orbitals will form molecular orbitals like those formed by the combination of 2s and 2p atomic orbitals that we learnt in class, and there is no s-p mixing.

Show **energy arrow**, **atomic orbitals** involved, **atom labels**, the **molecular orbitals** that are formed, **filled electrons** in the MO diagram etc. Make sure to label all the atomic and molecular orbitals.

Using the MO diagram above, answer the following questions:

- (i) Which one of the following will have the highest bond order? Circle the correct choice
 A) PS^{1-} B) PS C) PS^{1+}
- (ii) Which one of the following will have the longest bond length? Circle the correct choice
 A) PS^{1-} B) PS C) PS^{1+}
- (iii) Which of the following will be diamagnetic? Circle the correct choice.
 A) PS^{1-} B) PS C) PS^{1+}
- (iv) Draw Lewis dot structure of the molecule with the shortest bond length (based on the MO diagram) from among the three (PS^{1-} , PS, PS^{1+}), and indicate the bond order calculated from the Lewis dot structure. **(3 points)**

Lewis Dot Structure

Bond Order



A0AA800F-FC06-453F-BEE2-B539A1393AA8

midterm-exam-93388

#485 Page 10 of 10

SCRAP PAGE

**CHEMISTRY 110 –Midterm Exam****Thursday November 7, 2024, 6:30 – 8:30 pm**

Examiners: Pallavi Sirjoosingh and Ashok Kakkar

First name (please write as legibly as possible within the boxes)

--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--

Last name

--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--

Student ID number

--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--

This booklet has 8 questions on 10 pages (The last page can be used as a scrap page).

The staple must NOT be removed from the question booklet. Your intact question booklet must be returned at the end of the exam.

A separate booklet contains a periodic table, and datasheet.

Write your answers directly under the questions in the space provided. Do not write on the QR codes or near the top edges (as it may be cut off).

Please write **legibly and dark enough** for the scan to be readable. For students who use English as a second language, dictionaries are permitted. Calculators are allowed.

DO NOT ASK QUESTIONS OF THE INVIGILATORS about interpretations of the exam questions. If you have a concern about a question, please summarize it at the top of this page. All concerns will be evaluated after the exam. GOOD LUCK!

**Question 1 8 points** (*Assume all transitions are possible*)

a. 6 points Calculate the **energy (in kJ/mol)** required to ionize (completely remove) the electron from the **$n=3$** level of **He^+ ion** (He: Helium). Show your work including equations. (3 sig. figs.)

b. 2 points

(I) Select the largest (longest) wavelength from the three choices below:

- i. Wavelength of photon to *completely remove* an electron from *1s orbital of Hydrogen atom*
- ii. Wavelength of photon to *completely remove* an electron from *1s orbital of Boron atom (B)*
- iii. Wavelength of photon to *completely remove* an electron from *1s orbital of Boron cation (B^+)*

i or ii or iii

(II) Select the smallest (shortest) wavelength from the three choices below (*for H atom*):

- i. Wavelength of photon *emitted* for an electron transitioning from *2p to 1s orbital*
- ii. Wavelength of photon *absorbed* for an electron transitioning from *2s to 4p orbital*
- iii. Wavelength of photon *absorbed* for an electron transitioning from *1s to 3p orbital*

i or ii or iii



Question 2 8 points



A Chem110 student plots the graph shown on the left, from data obtained for the kinetic energy (eV) of the ejected electron versus frequency (s^{-1}) of the incident light, for an unknown metal. ($1\text{eV} = 1.6022 \times 10^{-19} \text{ J}$)

Based on the plot, answer the questions **a to c** below:

a. 1 point On the plot above, label the **threshold frequency**.

b. 2 points What is the value of the work potential (in J) of the unknown metal?

c. 3 points What is the *longest* wavelength (in nm) required to eject an electron from this metal's surface? Show your work including equation(s) used. (3 sig figs)

d. 2 points In the space provided below, plot the following

(Assume frequency is *above* threshold frequency):

- i. *Kinetic energy* of an ejected electron versus the *intensity* of the incident light (left)
- ii. *Number of electrons* ejected versus the *frequency* of the incident light (right)



**Question 3** **6 points****a. 2 points**

In the ground state of Ge (Germanium; $Z = 32$) atom, **name** the following orbitals:

i. Orbital containing the electron that requires the maximum energy to be removed.

ii. Orbital containing the electron that experiences the minimum effective nuclear charge.

b. 2 points

An unknown anion (A^-) has the following *excited* electronic configuration: $[Ar] 4s^2 3d^3 4p^1$

(Ar: Argon; $Z = 18$)

Give the **condensed ground state electronic configuration** of the atom, **A** and the cation, **A^{2+}** .
Show your work. (Assume all electronic transitions are possible)

Electronic Configuration of A:

Electronic Configuration of A^{2+} :

c. 2 points

In a single atom, determine how many electrons can have the following set of quantum numbers?

(i) $n = 5; l = 3$

In a single atom, determine how many electrons can have the following set of quantum numbers?

(i) $n = 3; l = 2; m_s = +1/2$

**Question 4** **8 points**

Rank the following from 1 to 4 in **increasing** order. Provide the ranking in the boxes provided.

Note: Atomic number (Z) are provided for relevant atoms

a) Increasing (lowest to highest) order of *ionization energy*: (1= lowest; 4 = highest)

Mg

Mg: Z = 12

Rb

Rb: Z = 37

Cs

Cs: Z = 55

K

K : Z = 19

b) Increasing (lowest to highest) order of *bond angle*: (1= lowest; 4 = highest)

H₂OCCl₄XeF₄

HCN

c) Increasing (smallest to largest) order of *ionic size* (1= smallest; 4 = largest)

S²⁻

S: Z = 16

O²⁻

O: Z = 8

Be²⁺

Be: Z = 4

Na⁺

Na: Z = 11

d) Increasing (lowest to highest) order of *bond energy* (1= lowest; 4 = highest)

P-F

P-Cl

P-I

P-Br

**Question 5** **12 points**

a. 9 points For the molecules HClO_4 and TeF_6 , draw the most stable *Lewis structure* showing appropriate shape and *dashed/wedged bonds*, based on VSEPR. Determine the **molecular geometry** around the central atom. For HClO_4 , label the *most polar bond*.



b. 3 points (Note: Lone pairs are not shown in structures below)

(i) For the three *resonance* structures (1, 2 and 3) shown below, answer the following question:



(I) Circle the *most contributing* resonance structure.

Structure 1

Structure 2

Structure 3

(ii) For the three structures given (4, 5, and 6) below, answer the following questions:



(I) Circle all the structures that represent *resonance structures* of each other:

Structure 4

Structure 5

Structure 6

(II) Among the resonance structures, circle the *most contributing* resonance structure.

Structure 4

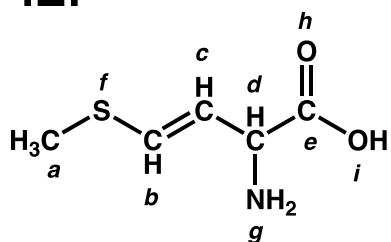
Structure 5

Structure 6

**Question 6** **6 points**

Using the data given below, calculate the **lattice energy** for SrO(s).

	ΔH° (kJ/mol)
$\text{Sr}(s) \longrightarrow \text{Sr}(g)$	+180
$\text{Sr}(g) \longrightarrow \text{Sr}^{2+}(g) + 2e^-$	+1468
$\text{O}_2(g) \longrightarrow 2\text{O}(g)$	+500
$\text{O}(g) + 2e^- \longrightarrow \text{O}^{2-}(g)$	+600
$\text{Sr}(s) + \frac{1}{2}\text{O}_2(g) \longrightarrow \text{SrO}(s)$	-550

**Question 7 10 points**

Consider the molecule given on the left, and answer questions (i) to (vi) using chemical bonding theories that we have learnt in class (valence bond theory, hybridization).

Note that the lone pairs are not shown in the structure:

i. Consider the carbon atoms marked in as ***a***, ***b***, ***c***, ***d***, and ***e*** in the molecule, and determine **hybridization** at each of the carbon centers. (2.5 points)

Atom	Hybridization
Carbon (<i>a</i>)	
Carbon (<i>b</i>)	
Carbon (<i>c</i>)	
Carbon (<i>d</i>)	
Carbon (<i>e</i>)	

ii. Now, consider the sulfur atom (S) marked as ***f***, nitrogen (N) atom (***g***), and oxygen(O) atoms (***h*** and ***i***), and determine their **hybridization**. (2 points)

Atom	Hybridization
Sulfur (<i>f</i>)	
Nitrogen (<i>g</i>)	
Oxygen (<i>h</i>)	
Oxygen (<i>i</i>)	

iii. **Describe** all the bonds formed by the carbon atoms ***a*** and ***b***, using sigma and pi designation as appropriate. (2 points)

Atom	Bonds formed (use sigma and pi as appropriate)
Carbon (<i>a</i>)	
Carbon (<i>b</i>)	

iv. How many total sp^3 -s sigma bonds are in the molecule? (1 point)

v. Determine **electronic** and **molecular geometry** at carbon (C) labeled ***e*** and sulfur (S) labeled ***f*** centres. (2 points)

Atom	Electronic Geometry	Molecular Geometry
Carbon (<i>e</i>)		
Sulfur (<i>f</i>)		

vi. Circle the value from the choices given below, that describes the **C-S-C bond angle at sulfur** (S) marked ***f***: (0.5 points)

$>90^\circ$ to $<109.5^\circ$ 109.5° >109.5 to $<120^\circ$ 120° $>120^\circ$ to $<180^\circ$


Question 8 12 points

Using the **valence shell orbitals only** draw a **complete** molecular orbital (MO) diagram (with no s-p mixing) for the molecule **PS**, in the space provided below. **(6 points)**

Assume that 3s and 3p atomic orbitals will form molecular orbitals like those formed by the combination of 2s and 2p atomic orbitals that we learnt in class, and there is no s-p mixing.

Show **energy arrow**, **atomic orbitals** involved, **atom labels**, the **molecular orbitals** that are formed, **filled electrons** in the MO diagram etc. Make sure to label all the atomic and molecular orbitals.

Using the MO diagram above, answer the following questions:

- (i) Which one of the following will have the highest bond order? Circle the correct choice
 A) PS^{1-} B) PS C) PS^{1+}
- (ii) Which one of the following will have the longest bond length? Circle the correct choice
 A) PS^{1-} B) PS C) PS^{1+}
- (iii) Which of the following will be diamagnetic? Circle the correct choice.
 A) PS^{1-} B) PS C) PS^{1+}
- (iv) Draw Lewis dot structure of the molecule with the shortest bond length (based on the MO diagram) from among the three (PS^{1-} , PS, PS^{1+}), and indicate the bond order calculated from the Lewis dot structure. **(3 points)**

Lewis Dot Structure

Bond Order



23881C6E-EFEE-4132-96C0-8C175FA4A781

midterm-exam-93388

#486 Page 10 of 10

SCRAP PAGE

**CHEMISTRY 110 –Midterm Exam****Thursday November 7, 2024, 6:30 – 8:30 pm****Examiners: Pallavi Sirjoosingh and Ashok Kakkar**

First name (please write as legibly as possible within the boxes)

--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--

Last name

--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--

Student ID number

--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--

This booklet has 8 questions on 10 pages (The last page can be used as a scrap page).

The staple must NOT be removed from the question booklet. Your intact question booklet must be returned at the end of the exam.

A separate booklet contains a periodic table, and datasheet.

Write your answers directly under the questions in the space provided. Do not write on the QR codes or near the top edges (as it may be cut off).

Please write **legibly and dark enough** for the scan to be readable. For students who use English as a second language, dictionaries are permitted. Calculators are allowed.

DO NOT ASK QUESTIONS OF THE INVIGILATORS about interpretations of the exam questions. If you have a concern about a question, please summarize it at the top of this page. All concerns will be evaluated after the exam. GOOD LUCK!

**Question 1 8 points** (*Assume all transitions are possible*)

a. 6 points Calculate the **energy (in kJ/mol)** required to ionize (completely remove) the electron from the **$n=3$** level of **He^+ ion** (He: Helium). Show your work including equations. (3 sig. figs.)

b. 2 points

(I) Select the largest (longest) wavelength from the three choices below:

- i. Wavelength of photon to *completely remove* an electron from *1s orbital of Hydrogen atom*
- ii. Wavelength of photon to *completely remove* an electron from *1s orbital of Boron atom (B)*
- iii. Wavelength of photon to *completely remove* an electron from *1s orbital of Boron cation (B^+)*

i or ii or iii

(II) Select the smallest (shortest) wavelength from the three choices below (*for H atom*):

- i. Wavelength of photon *emitted* for an electron transitioning from *2p to 1s orbital*
- ii. Wavelength of photon *absorbed* for an electron transitioning from *2s to 4p orbital*
- iii. Wavelength of photon *absorbed* for an electron transitioning from *1s to 3p orbital*

i or ii or iii



Question 2 8 points



A Chem110 student plots the graph shown on the left, from data obtained for the kinetic energy (eV) of the ejected electron versus frequency (s^{-1}) of the incident light, for an unknown metal. ($1\text{eV} = 1.6022 \times 10^{-19} \text{ J}$)

Based on the plot, answer the questions **a to c** below:

a. 1 point On the plot above, label the **threshold frequency**.

b. 2 points What is the value of the work potential (in J) of the unknown metal?

c. 3 points What is the *longest* wavelength (in nm) required to eject an electron from this metal's surface? Show your work including equation(s) used. (3 sig figs)

d. 2 points In the space provided below, plot the following

(Assume frequency is *above* threshold frequency):

- Kinetic energy* of an ejected electron versus the *intensity* of the incident light (left)
- Number of electrons* ejected versus the *frequency* of the incident light (right)



**Question 3** **6 points****a. 2 points**

In the ground state of Ge (Germanium; $Z = 32$) atom, **name** the following orbitals:

i. Orbital containing the electron that requires the maximum energy to be removed.

ii. Orbital containing the electron that experiences the minimum effective nuclear charge.

b. 2 points

An unknown anion (A^-) has the following *excited* electronic configuration: $[Ar] 4s^2 3d^3 4p^1$

(Ar: Argon; $Z = 18$)

Give the **condensed ground state electronic configuration** of the atom, **A** and the cation, **A^{2+}** .
Show your work. (Assume all electronic transitions are possible)

Electronic Configuration of A:

Electronic Configuration of A^{2+} :

c. 2 points

In a single atom, determine how many electrons can have the following set of quantum numbers?

(i) $n = 5; l = 3$

In a single atom, determine how many electrons can have the following set of quantum numbers?

(i) $n = 3; l = 2; m_s = +1/2$

**Question 4** **8 points**

Rank the following from 1 to 4 in **increasing** order. Provide the ranking in the boxes provided.

Note: Atomic number (Z) are provided for relevant atoms

a) Increasing (lowest to highest) order of *ionization energy*: (1= lowest; 4 = highest)

Mg

Mg: Z = 12

Rb

Rb: Z = 37

Cs

Cs: Z = 55

K

K : Z = 19

b) Increasing (lowest to highest) order of *bond angle*: (1= lowest; 4 = highest)

H₂OCCl₄XeF₄

HCN

c) Increasing (smallest to largest) order of *ionic size* (1= smallest; 4 = largest)

S²⁻

S: Z = 16

O²⁻

O: Z = 8

Be²⁺

Be: Z = 4

Na⁺

Na: Z = 11

d) Increasing (lowest to highest) order of *bond energy* (1= lowest; 4 = highest)

P-F

P-Cl

P-I

P-Br

**Question 5** **12 points**

a. 9 points For the molecules HClO_4 and TeF_6 , draw the most stable *Lewis structure* showing appropriate shape and *dashed/wedged bonds*, based on VSEPR. Determine the **molecular geometry** around the central atom. For HClO_4 , label the *most polar bond*.



b. 3 points (Note: Lone pairs are not shown in structures below)

(i) For the three *resonance* structures (1, 2 and 3) shown below, answer the following question:



(I) Circle the *most contributing* resonance structure.

Structure 1

Structure 2

Structure 3

(ii) For the three structures given (4, 5, and 6) below, answer the following questions:



(I) Circle all the structures that represent *resonance structures* of each other:

Structure 4

Structure 5

Structure 6

(II) Among the resonance structures, circle the *most contributing* resonance structure.

Structure 4

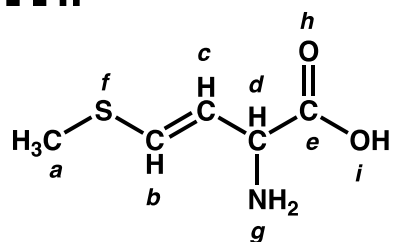
Structure 5

Structure 6

**Question 6** **6 points**

Using the data given below, calculate the **lattice energy** for SrO(s).

	ΔH° (kJ/mol)
$\text{Sr}(s) \longrightarrow \text{Sr}(g)$	+180
$\text{Sr}(g) \longrightarrow \text{Sr}^{2+}(g) + 2e^-$	+1468
$\text{O}_2(g) \longrightarrow 2\text{O}(g)$	+500
$\text{O}(g) + 2e^- \longrightarrow \text{O}^{2-}(g)$	+600
$\text{Sr}(s) + \frac{1}{2}\text{O}_2(g) \longrightarrow \text{SrO}(s)$	-550

**Question 7 10 points**

Consider the molecule given on the left, and answer questions (i) to (vi) using chemical bonding theories that we have learnt in class (valence bond theory, hybridization).

Note that the lone pairs are not shown in the structure:

i. Consider the carbon atoms marked in as ***a***, ***b***, ***c***, ***d***, and ***e*** in the molecule, and determine **hybridization** at each of the carbon centers. (2.5 points)

Atom	Hybridization
Carbon (<i>a</i>)	
Carbon (<i>b</i>)	
Carbon (<i>c</i>)	
Carbon (<i>d</i>)	
Carbon (<i>e</i>)	

ii. Now, consider the sulfur atom (S) marked as ***f***, nitrogen (N) atom (***g***), and oxygen(O) atoms (***h*** and ***i***), and determine their **hybridization**. (2 points)

Atom	Hybridization
Sulfur (<i>f</i>)	
Nitrogen (<i>g</i>)	
Oxygen (<i>h</i>)	
Oxygen (<i>i</i>)	

iii. **Describe** all the bonds formed by the carbon atoms ***a*** and ***b***, using sigma and pi designation as appropriate. (2 points)

Atom	Bonds formed (use sigma and pi as appropriate)
Carbon (<i>a</i>)	
Carbon (<i>b</i>)	

iv. How many total sp^3 -s sigma bonds are in the molecule? (1 point)

v. Determine **electronic** and **molecular geometry** at carbon (C) labeled ***e*** and sulfur (S) labeled ***f*** centres. (2 points)

Atom	Electronic Geometry	Molecular Geometry
Carbon (<i>e</i>)		
Sulfur (<i>f</i>)		

vi. Circle the value from the choices given below, that describes the **C-S-C bond angle at sulfur** (S) marked ***f***: (0.5 points)

>90° to <109.5° 109.5° >109.5 to <120° 120° >120° to <180°


Question 8 12 points

Using the **valence shell orbitals only** draw a **complete** molecular orbital (MO) diagram (with no s-p mixing) for the molecule **PS**, in the space provided below. **(6 points)**

Assume that 3s and 3p atomic orbitals will form molecular orbitals like those formed by the combination of 2s and 2p atomic orbitals that we learnt in class, and there is no s-p mixing.

Show **energy arrow**, **atomic orbitals** involved, **atom labels**, the **molecular orbitals** that are formed, **filled electrons** in the MO diagram etc. Make sure to label all the atomic and molecular orbitals.

Using the MO diagram above, answer the following questions:

- (i) Which one of the following will have the highest bond order? Circle the correct choice
 A) PS^{1-} B) PS C) PS^{1+}
- (ii) Which one of the following will have the longest bond length? Circle the correct choice
 A) PS^{1-} B) PS C) PS^{1+}
- (iii) Which of the following will be diamagnetic? Circle the correct choice.
 A) PS^{1-} B) PS C) PS^{1+}
- (iv) Draw Lewis dot structure of the molecule with the shortest bond length (based on the MO diagram) from among the three (PS^{1-} , PS, PS^{1+}), and indicate the bond order calculated from the Lewis dot structure. **(3 points)**

Lewis Dot Structure

Bond Order



D0802BCA-DFC8-4EDF-BEAB-6723C3DD6DE5

midterm-exam-93388

#487 Page 10 of 10

SCRAP PAGE

**CHEMISTRY 110 –Midterm Exam****Thursday November 7, 2024, 6:30 – 8:30 pm****Examiners: Pallavi Sirjoosingh and Ashok Kakkar**

First name (please write as legibly as possible within the boxes)

--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--

Last name

--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--

Student ID number

--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--

This booklet has 8 questions on 10 pages (The last page can be used as a scrap page).

The staple must NOT be removed from the question booklet. Your intact question booklet must be returned at the end of the exam.

A separate booklet contains a periodic table, and datasheet.

Write your answers directly under the questions in the space provided. Do not write on the QR codes or near the top edges (as it may be cut off).

Please write **legibly and dark enough** for the scan to be readable. For students who use English as a second language, dictionaries are permitted. Calculators are allowed.

DO NOT ASK QUESTIONS OF THE INVIGILATORS about interpretations of the exam questions. If you have a concern about a question, please summarize it at the top of this page. All concerns will be evaluated after the exam. GOOD LUCK!

**Question 1 8 points** (*Assume all transitions are possible*)

a. 6 points Calculate the **energy (in kJ/mol)** required to ionize (completely remove) the electron from the **n=3** level of **He⁺ ion** (He: Helium). Show your work including equations. (3 sig. figs.)

b. 2 points

(I) Select the largest (longest) wavelength from the three choices below:

- i. Wavelength of photon to *completely remove* an electron from *1s orbital of Hydrogen atom*
- ii. Wavelength of photon to *completely remove* an electron from *1s orbital of Boron atom (B)*
- iii. Wavelength of photon to *completely remove* an electron from *1s orbital of Boron cation (B⁺)*

i or ii or iii

(II) Select the smallest (shortest) wavelength from the three choices below (*for H atom*):

- i. Wavelength of photon *emitted* for an electron transitioning from *2p to 1s orbital*
- ii. Wavelength of photon *absorbed* for an electron transitioning from *2s to 4p orbital*
- iii. Wavelength of photon *absorbed* for an electron transitioning from *1s to 3p orbital*

i or ii or iii



Question 2 8 points



A Chem110 student plots the graph shown on the left, from data obtained for the kinetic energy (eV) of the ejected electron versus frequency (s^{-1}) of the incident light, for an unknown metal. ($1\text{eV} = 1.6022 \times 10^{-19} \text{ J}$)

Based on the plot, answer the questions **a to c** below:

a. 1 point On the plot above, label the **threshold frequency**.

b. 2 points What is the value of the work potential (in J) of the unknown metal?

c. 3 points What is the *longest* wavelength (in nm) required to eject an electron from this metal's surface? Show your work including equation(s) used. (3 sig figs)

d. 2 points In the space provided below, plot the following

(Assume frequency is *above* threshold frequency):

- i. *Kinetic energy* of an ejected electron versus the *intensity* of the incident light (left)
- ii. *Number of electrons* ejected versus the *frequency* of the incident light (right)



**Question 3** **6 points****a. 2 points**

In the ground state of Ge (Germanium; $Z = 32$) atom, **name** the following orbitals:

i. Orbital containing the electron that requires the maximum energy to be removed.

ii. Orbital containing the electron that experiences the minimum effective nuclear charge.

b. 2 points

An unknown anion (A^-) has the following *excited* electronic configuration: $[Ar] 4s^2 3d^3 4p^1$

(Ar: Argon; $Z = 18$)

Give the **condensed ground state electronic configuration** of the atom, **A** and the cation, A^{2+} .
Show your work. (Assume all electronic transitions are possible)

Electronic Configuration of A:

Electronic Configuration of A^{2+} :

c. 2 points

In a single atom, determine how many electrons can have the following set of quantum numbers?

(i) $n = 5; l = 3$

In a single atom, determine how many electrons can have the following set of quantum numbers?

(i) $n = 3; l = 2; m_s = +1/2$

**Question 4** **8 points**

Rank the following from 1 to 4 in **increasing** order. Provide the ranking in the boxes provided.

Note: Atomic number (Z) are provided for relevant atoms

a) Increasing (lowest to highest) order of *ionization energy*: (1= lowest; 4 = highest)

Mg

Mg: Z = 12

Rb

Rb: Z = 37

Cs

Cs: Z = 55

K

K : Z = 19

b) Increasing (lowest to highest) order of *bond angle*: (1= lowest; 4 = highest)

H₂OCCl₄XeF₄

HCN

c) Increasing (smallest to largest) order of *ionic size* (1= smallest; 4 = largest)

S²⁻

S: Z = 16

O²⁻

O: Z = 8

Be²⁺

Be: Z = 4

Na⁺

Na: Z = 11

d) Increasing (lowest to highest) order of *bond energy* (1= lowest; 4 = highest)

P-F

P-Cl

P-I

P-Br

**Question 5** **12 points**

a. 9 points For the molecules HClO_4 and TeF_6 , draw the most stable *Lewis structure* showing appropriate shape and *dashed/wedged bonds*, based on VSEPR. Determine the **molecular geometry** around the central atom. For HClO_4 , label the *most polar bond*.



b. 3 points (Note: Lone pairs are not shown in structures below)

(i) For the three *resonance* structures (1, 2 and 3) shown below, answer the following question:



(I) Circle the *most contributing* resonance structure.

Structure 1

Structure 2

Structure 3

(ii) For the three structures given (4, 5, and 6) below, answer the following questions:



(I) Circle all the structures that represent *resonance structures* of each other:

Structure 4

Structure 5

Structure 6

(II) Among the resonance structures, circle the *most contributing* resonance structure.

Structure 4

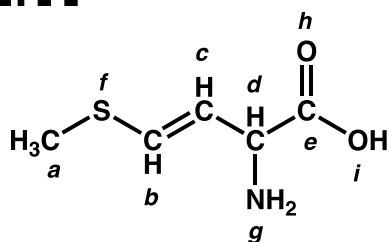
Structure 5

Structure 6

**Question 6** **6 points**

Using the data given below, calculate the **lattice energy** for SrO(s).

	ΔH° (kJ/mol)
$\text{Sr}(s) \longrightarrow \text{Sr}(g)$	+180
$\text{Sr}(g) \longrightarrow \text{Sr}^{2+}(g) + 2e^-$	+1468
$\text{O}_2(g) \longrightarrow 2\text{O}(g)$	+500
$\text{O}(g) + 2e^- \longrightarrow \text{O}^{2-}(g)$	+600
$\text{Sr}(s) + \frac{1}{2}\text{O}_2(g) \longrightarrow \text{SrO}(s)$	-550

**Question 7 10 points**

Consider the molecule given on the left, and answer questions (i) to (vi) using chemical bonding theories that we have learnt in class (valence bond theory, hybridization).

Note that the lone pairs are not shown in the structure:

i. Consider the carbon atoms marked in as ***a***, ***b***, ***c***, ***d***, and ***e*** in the molecule, and determine **hybridization** at each of the carbon centers. (2.5 points)

Atom	Hybridization
Carbon (<i>a</i>)	
Carbon (<i>b</i>)	
Carbon (<i>c</i>)	
Carbon (<i>d</i>)	
Carbon (<i>e</i>)	

ii. Now, consider the sulfur atom (S) marked as ***f***, nitrogen (N) atom (***g***), and oxygen(O) atoms (***h*** and ***i***), and determine their **hybridization**. (2 points)

Atom	Hybridization
Sulfur (<i>f</i>)	
Nitrogen (<i>g</i>)	
Oxygen (<i>h</i>)	
Oxygen (<i>i</i>)	

iii. **Describe** all the bonds formed by the carbon atoms ***a*** and ***b***, using sigma and pi designation as appropriate. (2 points)

Atom	Bonds formed (use sigma and pi as appropriate)
Carbon (<i>a</i>)	
Carbon (<i>b</i>)	

iv. How many total sp^3 -s sigma bonds are in the molecule? (1 point)

v. Determine **electronic** and **molecular geometry** at carbon (C) labeled ***e*** and sulfur (S) labeled ***f*** centres. (2 points)

Atom	Electronic Geometry	Molecular Geometry
Carbon (<i>e</i>)		
Sulfur (<i>f</i>)		

vi. Circle the value from the choices given below, that describes the **C-S-C bond angle at sulfur** (S) marked ***f***: (0.5 points)

>90° to <109.5°

109.5°

>109.5 to <120°

120°

>120° to <180°


Question 8 12 points

Using the **valence shell orbitals only** draw a **complete** molecular orbital (MO) diagram (with no s-p mixing) for the molecule **PS**, in the space provided below. **(6 points)**

Assume that 3s and 3p atomic orbitals will form molecular orbitals like those formed by the combination of 2s and 2p atomic orbitals that we learnt in class, and there is no s-p mixing.

Show **energy arrow**, **atomic orbitals** involved, **atom labels**, the **molecular orbitals** that are formed, **filled electrons** in the MO diagram etc. Make sure to label all the atomic and molecular orbitals.

Using the MO diagram above, answer the following questions:

- (i) Which one of the following will have the highest bond order? Circle the correct choice
 A) PS^{1-} B) PS C) PS^{1+}
- (ii) Which one of the following will have the longest bond length? Circle the correct choice
 A) PS^{1-} B) PS C) PS^{1+}
- (iii) Which of the following will be diamagnetic? Circle the correct choice.
 A) PS^{1-} B) PS C) PS^{1+}
- (iv) Draw Lewis dot structure of the molecule with the shortest bond length (based on the MO diagram) from among the three (PS^{1-} , PS, PS^{1+}), and indicate the bond order calculated from the Lewis dot structure. **(3 points)**

Lewis Dot Structure

Bond Order



93258789-A647-4E30-89B0-43F0C8B194FD

midterm-exam-93388

#488 Page 10 of 10

SCRAP PAGE

**CHEMISTRY 110 –Midterm Exam****Thursday November 7, 2024, 6:30 – 8:30 pm**

Examiners: Pallavi Sirjoosingh and Ashok Kakkar

First name (please write as legibly as possible within the boxes)

--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--

Last name

--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--

Student ID number

--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--

This booklet has 8 questions on 10 pages (The last page can be used as a scrap page).

The staple must NOT be removed from the question booklet. Your intact question booklet must be returned at the end of the exam.

A separate booklet contains a periodic table, and datasheet.

Write your answers directly under the questions in the space provided. Do not write on the QR codes or near the top edges (as it may be cut off).

Please write **legibly and dark enough** for the scan to be readable. For students who use English as a second language, dictionaries are permitted. Calculators are allowed.

DO NOT ASK QUESTIONS OF THE INVIGILATORS about interpretations of the exam questions. If you have a concern about a question, please summarize it at the top of this page. All concerns will be evaluated after the exam. GOOD LUCK!

**Question 1 8 points** (*Assume all transitions are possible*)

a. 6 points Calculate the **energy (in kJ/mol)** required to ionize (completely remove) the electron from the **n=3** level of **He⁺ ion** (He: Helium). Show your work including equations. (3 sig. figs.)

b. 2 points

(I) Select the largest (longest) wavelength from the three choices below:

- i. Wavelength of photon to *completely remove* an electron from *1s orbital of Hydrogen atom*
- ii. Wavelength of photon to *completely remove* an electron from *1s orbital of Boron atom (B)*
- iii. Wavelength of photon to *completely remove* an electron from *1s orbital of Boron cation (B⁺)*

i or ii or iii

(II) Select the smallest (shortest) wavelength from the three choices below (*for H atom*):

- i. Wavelength of photon *emitted* for an electron transitioning from *2p to 1s orbital*
- ii. Wavelength of photon *absorbed* for an electron transitioning from *2s to 4p orbital*
- iii. Wavelength of photon *absorbed* for an electron transitioning from *1s to 3p orbital*

i or ii or iii



Question 2 8 points



A Chem110 student plots the graph shown on the left, from data obtained for the kinetic energy (eV) of the ejected electron versus frequency (s^{-1}) of the incident light, for an unknown metal. ($1\text{eV} = 1.6022 \times 10^{-19} \text{ J}$)

Based on the plot, answer the questions **a to c** below:

a. 1 point On the plot above, label the **threshold frequency**.

b. 2 points What is the value of the work potential (in J) of the unknown metal?

c. 3 points What is the *longest* wavelength (in nm) required to eject an electron from this metal's surface? Show your work including equation(s) used. (3 sig figs)

d. 2 points In the space provided below, plot the following

(Assume frequency is *above* threshold frequency):

- i. *Kinetic energy* of an ejected electron versus the *intensity* of the incident light (left)
- ii. *Number of electrons* ejected versus the *frequency* of the incident light (right)



**Question 3** **6 points****a. 2 points**

In the ground state of Ge (Germanium; $Z = 32$) atom, **name** the following orbitals:

i. Orbital containing the electron that requires the maximum energy to be removed.

ii. Orbital containing the electron that experiences the minimum effective nuclear charge.

b. 2 points

An unknown anion (A^-) has the following *excited* electronic configuration: $[Ar] 4s^2 3d^3 4p^1$

(Ar: Argon; $Z = 18$)

Give the **condensed ground state electronic configuration** of the atom, **A** and the cation, **A^{2+}** .
Show your work. (Assume all electronic transitions are possible)

Electronic Configuration of A:

Electronic Configuration of A^{2+} :

c. 2 points

In a single atom, determine how many electrons can have the following set of quantum numbers?

(i) $n = 5; l = 3$

In a single atom, determine how many electrons can have the following set of quantum numbers?

(i) $n = 3; l = 2; m_s = +1/2$

**Question 4** **8 points**

Rank the following from 1 to 4 in **increasing** order. Provide the ranking in the boxes provided.

Note: Atomic number (Z) are provided for relevant atoms

a) Increasing (lowest to highest) order of *ionization energy*: (1= lowest; 4 = highest)

Mg

Mg: Z = 12

Rb

Rb: Z = 37

Cs

Cs: Z = 55

K

K : Z = 19

b) Increasing (lowest to highest) order of *bond angle*: (1= lowest; 4 = highest)

H₂OCCl₄XeF₄

HCN

c) Increasing (smallest to largest) order of *ionic size* (1= smallest; 4 = largest)

S²⁻

S: Z = 16

O²⁻

O: Z = 8

Be²⁺

Be: Z = 4

Na⁺

Na: Z = 11

d) Increasing (lowest to highest) order of *bond energy* (1= lowest; 4 = highest)

P-F

P-Cl

P-I

P-Br

**Question 5** **12 points**

a. 9 points For the molecules HClO_4 and TeF_6 , draw the most stable *Lewis structure* showing appropriate shape and *dashed/wedged bonds*, based on VSEPR. Determine the **molecular geometry** around the central atom. For HClO_4 , label the *most polar bond*.



b. 3 points (Note: Lone pairs are not shown in structures below)

(i) For the three *resonance* structures (1, 2 and 3) shown below, answer the following question:



(I) Circle the *most contributing* resonance structure.

Structure 1

Structure 2

Structure 3

(ii) For the three structures given (4, 5, and 6) below, answer the following questions:



(I) Circle all the structures that represent *resonance structures* of each other:

Structure 4

Structure 5

Structure 6

(II) Among the resonance structures, circle the *most contributing* resonance structure.

Structure 4

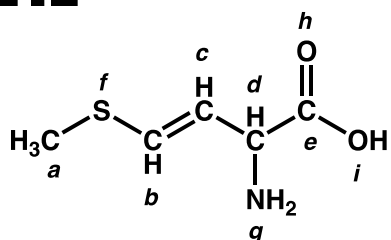
Structure 5

Structure 6

**Question 6** **6 points**

Using the data given below, calculate the **lattice energy** for SrO(s).

	ΔH° (kJ/mol)
$\text{Sr}(s) \longrightarrow \text{Sr}(g)$	+180
$\text{Sr}(g) \longrightarrow \text{Sr}^{2+}(g) + 2e^-$	+1468
$\text{O}_2(g) \longrightarrow 2\text{O}(g)$	+500
$\text{O}(g) + 2e^- \longrightarrow \text{O}^{2-}(g)$	+600
$\text{Sr}(s) + \frac{1}{2}\text{O}_2(g) \longrightarrow \text{SrO}(s)$	-550

**Question 7 10 points**

Consider the molecule given on the left, and answer questions (i) to (vi) using chemical bonding theories that we have learnt in class (valence bond theory, hybridization).

Note that the lone pairs are not shown in the structure:

i. Consider the carbon atoms marked in as ***a***, ***b***, ***c***, ***d***, and ***e*** in the molecule, and determine **hybridization** at each of the carbon centers. (2.5 points)

Atom	Hybridization
Carbon (<i>a</i>)	
Carbon (<i>b</i>)	
Carbon (<i>c</i>)	
Carbon (<i>d</i>)	
Carbon (<i>e</i>)	

ii. Now, consider the sulfur atom (S) marked as ***f***, nitrogen (N) atom (***g***), and oxygen(O) atoms (***h*** and ***i***), and determine their **hybridization**. (2 points)

Atom	Hybridization
Sulfur (<i>f</i>)	
Nitrogen (<i>g</i>)	
Oxygen (<i>h</i>)	
Oxygen (<i>i</i>)	

iii. **Describe** all the bonds formed by the carbon atoms ***a*** and ***b***, using sigma and pi designation as appropriate. (2 points)

Atom	Bonds formed (use sigma and pi as appropriate)
Carbon (<i>a</i>)	
Carbon (<i>b</i>)	

iv. How many total sp^3 -s sigma bonds are in the molecule? (1 point)

v. Determine **electronic** and **molecular geometry** at carbon (C) labeled ***e*** and sulfur (S) labeled ***f*** centres. (2 points)

Atom	Electronic Geometry	Molecular Geometry
Carbon (<i>e</i>)		
Sulfur (<i>f</i>)		

vi. Circle the value from the choices given below, that describes the **C-S-C bond angle at sulfur** (S) marked ***f***: (0.5 points)

$>90^\circ$ to $<109.5^\circ$ 109.5° >109.5 to $<120^\circ$ 120° $>120^\circ$ to $<180^\circ$


Question 8 12 points

Using the **valence shell orbitals only** draw a **complete** molecular orbital (MO) diagram (with no s-p mixing) for the molecule **PS**, in the space provided below. **(6 points)**

Assume that 3s and 3p atomic orbitals will form molecular orbitals like those formed by the combination of 2s and 2p atomic orbitals that we learnt in class, and there is no s-p mixing.

Show **energy arrow**, **atomic orbitals** involved, **atom labels**, the **molecular orbitals** that are formed, **filled electrons** in the MO diagram etc. Make sure to label all the atomic and molecular orbitals.

Using the MO diagram above, answer the following questions:

- (i) Which one of the following will have the highest bond order? Circle the correct choice
 A) PS^{1-} B) PS C) PS^{1+}
- (ii) Which one of the following will have the longest bond length? Circle the correct choice
 A) PS^{1-} B) PS C) PS^{1+}
- (iii) Which of the following will be diamagnetic? Circle the correct choice.
 A) PS^{1-} B) PS C) PS^{1+}
- (iv) Draw Lewis dot structure of the molecule with the shortest bond length (based on the MO diagram) from among the three (PS^{1-} , PS, PS^{1+}), and indicate the bond order calculated from the Lewis dot structure. **(3 points)**

Lewis Dot Structure

Bond Order



D4BA19C6-DAFB-4B1C-B3E6-E918F8A13F49

midterm-exam-93388

#489 Page 10 of 10

SCRAP PAGE

**CHEMISTRY 110 –Midterm Exam****Thursday November 7, 2024, 6:30 – 8:30 pm**

Examiners: Pallavi Sirjoosingh and Ashok Kakkar

First name (please write as legibly as possible within the boxes)

--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--

Last name

--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--

Student ID number

--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--

This booklet has 8 questions on 10 pages (The last page can be used as a scrap page).

The staple must NOT be removed from the question booklet. Your intact question booklet must be returned at the end of the exam.

A separate booklet contains a periodic table, and datasheet.

Write your answers directly under the questions in the space provided. Do not write on the QR codes or near the top edges (as it may be cut off).

Please write **legibly and dark enough** for the scan to be readable. For students who use English as a second language, dictionaries are permitted. Calculators are allowed.

DO NOT ASK QUESTIONS OF THE INVIGILATORS about interpretations of the exam questions. If you have a concern about a question, please summarize it at the top of this page. All concerns will be evaluated after the exam. GOOD LUCK!

**Question 1 8 points** (*Assume all transitions are possible*)

a. 6 points Calculate the **energy (in kJ/mol)** required to ionize (completely remove) the electron from the **$n=3$** level of **He^+ ion** (He: Helium). Show your work including equations. (3 sig. figs.)

b. 2 points

(I) Select the largest (longest) wavelength from the three choices below:

- i. Wavelength of photon to *completely remove* an electron from *1s orbital of Hydrogen atom*
- ii. Wavelength of photon to *completely remove* an electron from *1s orbital of Boron atom (B)*
- iii. Wavelength of photon to *completely remove* an electron from *1s orbital of Boron cation (B^+)*

i or ii or iii

(II) Select the smallest (shortest) wavelength from the three choices below (*for H atom*):

- i. Wavelength of photon *emitted* for an electron transitioning from *2p to 1s orbital*
- ii. Wavelength of photon *absorbed* for an electron transitioning from *2s to 4p orbital*
- iii. Wavelength of photon *absorbed* for an electron transitioning from *1s to 3p orbital*

i or ii or iii



Question 2 8 points



A Chem110 student plots the graph shown on the left, from data obtained for the kinetic energy (eV) of the ejected electron versus frequency (s^{-1}) of the incident light, for an unknown metal. ($1\text{eV} = 1.6022 \times 10^{-19} \text{ J}$)

Based on the plot, answer the questions **a to c** below:

a. 1 point On the plot above, label the **threshold frequency**.

b. 2 points What is the value of the work potential (in J) of the unknown metal?

c. 3 points What is the *longest* wavelength (in nm) required to eject an electron from this metal's surface? Show your work including equation(s) used. (3 sig figs)

d. 2 points In the space provided below, plot the following

(Assume frequency is *above* threshold frequency):

- i. *Kinetic energy* of an ejected electron versus the *intensity* of the incident light (left)
- ii. *Number of electrons* ejected versus the *frequency* of the incident light (right)



**Question 3** **6 points****a. 2 points**

In the ground state of Ge (Germanium; $Z = 32$) atom, **name** the following orbitals:

i. Orbital containing the electron that requires the maximum energy to be removed.

ii. Orbital containing the electron that experiences the minimum effective nuclear charge.

b. 2 points

An unknown anion (A^-) has the following *excited* electronic configuration: $[Ar] 4s^2 3d^3 4p^1$

(Ar: Argon; $Z = 18$)

Give the **condensed ground state electronic configuration** of the atom, **A** and the cation, **A^{2+}** .
Show your work. (Assume all electronic transitions are possible)

Electronic Configuration of A:

Electronic Configuration of A^{2+} :

c. 2 points

In a single atom, determine how many electrons can have the following set of quantum numbers?

(i) $n = 5; l = 3$

In a single atom, determine how many electrons can have the following set of quantum numbers?

(i) $n = 3; l = 2; m_s = +1/2$

**Question 4** **8 points**

Rank the following from 1 to 4 in **increasing** order. Provide the ranking in the boxes provided.

Note: Atomic number (Z) are provided for relevant atoms

a) Increasing (lowest to highest) order of *ionization energy*: (1= lowest; 4 = highest)

Mg

Mg: Z = 12

Rb

Rb: Z = 37

Cs

Cs: Z = 55

K

K : Z = 19

b) Increasing (lowest to highest) order of *bond angle*: (1= lowest; 4 = highest)

H₂OCCl₄XeF₄

HCN

c) Increasing (smallest to largest) order of *ionic size* (1= smallest; 4 = largest)

S²⁻

S: Z = 16

O²⁻

O: Z = 8

Be²⁺

Be: Z = 4

Na⁺

Na: Z = 11

d) Increasing (lowest to highest) order of *bond energy* (1= lowest; 4 = highest)

P-F

P-Cl

P-I

P-Br

**Question 5** **12 points**

a. 9 points For the molecules HClO_4 and TeF_6 , draw the most stable *Lewis structure* showing appropriate shape and *dashed/wedged bonds*, based on VSEPR. Determine the **molecular geometry** around the central atom. For HClO_4 , label the *most polar bond*.



b. 3 points (Note: Lone pairs are not shown in structures below)

(i) For the three *resonance* structures (1, 2 and 3) shown below, answer the following question:



(I) Circle the *most contributing* resonance structure.

Structure 1

Structure 2

Structure 3

(ii) For the three structures given (4, 5, and 6) below, answer the following questions:



(I) Circle all the structures that represent *resonance structures* of each other:

Structure 4

Structure 5

Structure 6

(II) Among the resonance structures, circle the *most contributing* resonance structure.

Structure 4

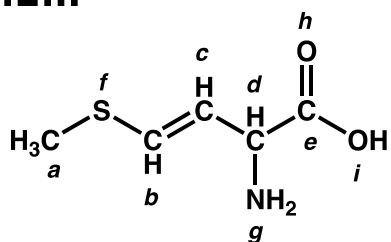
Structure 5

Structure 6

**Question 6** **6 points**

Using the data given below, calculate the **lattice energy** for SrO(s).

	ΔH° (kJ/mol)
$\text{Sr}(s) \longrightarrow \text{Sr}(g)$	+180
$\text{Sr}(g) \longrightarrow \text{Sr}^{2+}(g) + 2e^-$	+1468
$\text{O}_2(g) \longrightarrow 2\text{O}(g)$	+500
$\text{O}(g) + 2e^- \longrightarrow \text{O}^{2-}(g)$	+600
$\text{Sr}(s) + \frac{1}{2}\text{O}_2(g) \longrightarrow \text{SrO}(s)$	-550

**Question 7 10 points**

Consider the molecule given on the left, and answer questions (i) to (vi) using chemical bonding theories that we have learnt in class (valence bond theory, hybridization).

Note that the lone pairs are not shown in the structure:

i. Consider the carbon atoms marked in as ***a***, ***b***, ***c***, ***d***, and ***e*** in the molecule, and determine **hybridization** at each of the carbon centers. (2.5 points)

Atom	Hybridization
Carbon (<i>a</i>)	
Carbon (<i>b</i>)	
Carbon (<i>c</i>)	
Carbon (<i>d</i>)	
Carbon (<i>e</i>)	

ii. Now, consider the sulfur atom (S) marked as ***f***, nitrogen (N) atom (***g***), and oxygen(O) atoms (***h*** and ***i***), and determine their **hybridization**. (2 points)

Atom	Hybridization
Sulfur (<i>f</i>)	
Nitrogen (<i>g</i>)	
Oxygen (<i>h</i>)	
Oxygen (<i>i</i>)	

iii. **Describe** all the bonds formed by the carbon atoms ***a*** and ***b***, using sigma and pi designation as appropriate. (2 points)

Atom	Bonds formed (use sigma and pi as appropriate)
Carbon (<i>a</i>)	
Carbon (<i>b</i>)	

iv. How many total sp^3 -s sigma bonds are in the molecule? (1 point)

v. Determine **electronic** and **molecular geometry** at carbon (C) labeled ***e*** and sulfur (S) labeled ***f*** centres. (2 points)

Atom	Electronic Geometry	Molecular Geometry
Carbon (<i>e</i>)		
Sulfur (<i>f</i>)		

vi. Circle the value from the choices given below, that describes the **C-S-C bond angle at sulfur** (S) marked ***f***: (0.5 points)

>90° to <109.5° 109.5° >109.5 to <120° 120° >120° to <180°

**Question 8 12 points**

Using the **valence shell orbitals only** draw a **complete** molecular orbital (MO) diagram (with no s-p mixing) for the molecule **PS**, in the space provided below. **(6 points)**

Assume that 3s and 3p atomic orbitals will form molecular orbitals like those formed by the combination of 2s and 2p atomic orbitals that we learnt in class, and there is no s-p mixing.

Show **energy arrow**, **atomic orbitals** involved, **atom labels**, the **molecular orbitals** that are formed, **filled electrons** in the MO diagram etc. Make sure to label all the atomic and molecular orbitals.

Using the MO diagram above, answer the following questions:

- (i) Which one of the following will have the highest bond order? Circle the correct choice
A) PS^{1-} B) PS C) PS^{1+}
- (ii) Which one of the following will have the longest bond length? Circle the correct choice
A) PS^{1-} B) PS C) PS^{1+}
- (iii) Which of the following will be diamagnetic? Circle the correct choice.
A) PS^{1-} B) PS C) PS^{1+}
- (iv) Draw Lewis dot structure of the molecule with the shortest bond length (based on the MO diagram) from among the three (PS^{1-} , PS, PS^{1+}), and indicate the bond order calculated from the Lewis dot structure. **(3 points)**

Lewis Dot Structure

Bond Order



A4D4997E-DAD2-47F3-9A50-A7A2957361D1

midterm-exam-93388

#490 Page 10 of 10

SCRAP PAGE

**CHEMISTRY 110 –Midterm Exam****Thursday November 7, 2024, 6:30 – 8:30 pm****Examiners: Pallavi Sirjoosingh and Ashok Kakkar**

First name (please write as legibly as possible within the boxes)

--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--

Last name

--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--

Student ID number

--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--

This booklet has 8 questions on 10 pages (The last page can be used as a scrap page).

The staple must NOT be removed from the question booklet. Your intact question booklet must be returned at the end of the exam.

A separate booklet contains a periodic table, and datasheet.

Write your answers directly under the questions in the space provided. Do not write on the QR codes or near the top edges (as it may be cut off).

Please write **legibly and dark enough** for the scan to be readable. For students who use English as a second language, dictionaries are permitted. Calculators are allowed.

DO NOT ASK QUESTIONS OF THE INVIGILATORS about interpretations of the exam questions. If you have a concern about a question, please summarize it at the top of this page. All concerns will be evaluated after the exam. GOOD LUCK!

**Question 1 8 points** (*Assume all transitions are possible*)

a. 6 points Calculate the **energy (in kJ/mol)** required to ionize (completely remove) the electron from the **n=3** level of **He⁺ ion** (He: Helium). Show your work including equations. (3 sig. figs.)

b. 2 points

(I) Select the largest (longest) wavelength from the three choices below:

- i. Wavelength of photon to *completely remove* an electron from *1s orbital of Hydrogen atom*
- ii. Wavelength of photon to *completely remove* an electron from *1s orbital of Boron atom (B)*
- iii. Wavelength of photon to *completely remove* an electron from *1s orbital of Boron cation (B⁺)*

i or ii or iii

(II) Select the smallest (shortest) wavelength from the three choices below (*for H atom*):

- i. Wavelength of photon *emitted* for an electron transitioning from *2p to 1s orbital*
- ii. Wavelength of photon *absorbed* for an electron transitioning from *2s to 4p orbital*
- iii. Wavelength of photon *absorbed* for an electron transitioning from *1s to 3p orbital*

i or ii or iii



Question 2 8 points



A Chem110 student plots the graph shown on the left, from data obtained for the kinetic energy (eV) of the ejected electron versus frequency (s^{-1}) of the incident light, for an unknown metal. ($1\text{eV} = 1.6022 \times 10^{-19} \text{ J}$)

Based on the plot, answer the questions **a to c** below:

a. 1 point On the plot above, label the **threshold frequency**.

b. 2 points What is the value of the work potential (in J) of the unknown metal?

c. 3 points What is the *longest* wavelength (in nm) required to eject an electron from this metal's surface? Show your work including equation(s) used. (3 sig figs)

d. 2 points In the space provided below, plot the following

(Assume frequency is *above* threshold frequency):

- Kinetic energy* of an ejected electron versus the *intensity* of the incident light (left)
- Number of electrons* ejected versus the *frequency* of the incident light (right)



**Question 3** **6 points****a. 2 points**

In the ground state of Ge (Germanium; $Z = 32$) atom, **name** the following orbitals:

i. Orbital containing the electron that requires the maximum energy to be removed.

ii. Orbital containing the electron that experiences the minimum effective nuclear charge.

b. 2 points

An unknown anion (A^-) has the following *excited* electronic configuration: $[Ar] 4s^2 3d^3 4p^1$

(Ar: Argon; $Z = 18$)

Give the **condensed ground state electronic configuration** of the atom, **A** and the cation, **A^{2+}** .
Show your work. (Assume all electronic transitions are possible)

Electronic Configuration of A:

Electronic Configuration of A^{2+} :

c. 2 points

In a single atom, determine how many electrons can have the following set of quantum numbers?

(i) $n = 5; l = 3$

In a single atom, determine how many electrons can have the following set of quantum numbers?

(i) $n = 3; l = 2; m_s = +1/2$

**Question 4** **8 points**

Rank the following from 1 to 4 in **increasing** order. Provide the ranking in the boxes provided.

Note: Atomic number (Z) are provided for relevant atoms

a) Increasing (lowest to highest) order of *ionization energy*: (1= lowest; 4 = highest)

Mg

Mg: Z = 12

Rb

Rb: Z = 37

Cs

Cs: Z = 55

K

K : Z = 19

b) Increasing (lowest to highest) order of *bond angle*: (1= lowest; 4 = highest)

H₂OCCl₄XeF₄

HCN

c) Increasing (smallest to largest) order of *ionic size* (1= smallest; 4 = largest)

S²⁻

S: Z = 16

O²⁻

O: Z = 8

Be²⁺

Be: Z = 4

Na⁺

Na: Z = 11

d) Increasing (lowest to highest) order of *bond energy* (1= lowest; 4 = highest)

P-F

P-Cl

P-I

P-Br

**Question 5** **12 points**

a. 9 points For the molecules HClO_4 and TeF_6 , draw the most stable *Lewis structure* showing appropriate shape and *dashed/wedged bonds*, based on VSEPR. Determine the **molecular geometry** around the central atom. For HClO_4 , label the *most polar bond*.

HClO_4 (there are no O-O bonds)

Label the most polar bond

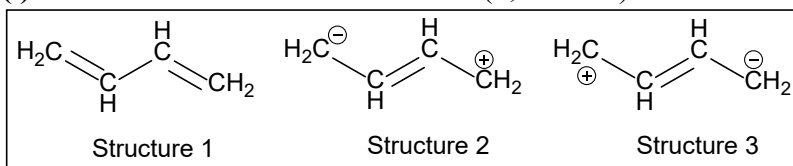
Molecular Geometry around Cl

TeF_6

Molecular Geometry around Te

b. 3 points (Note: Lone pairs are not shown in structures below)

(i) For the three *resonance* structures (1, 2 and 3) shown below, answer the following question:



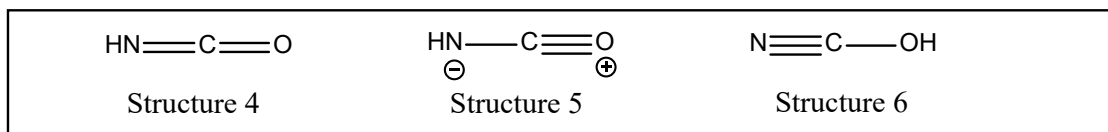
(I) Circle the *most contributing* resonance structure.

Structure 1

Structure 2

Structure 3

(ii) For the three structures given (4, 5, and 6) below, answer the following questions:



(I) Circle all the structures that represent *resonance structures* of each other:

Structure 4

Structure 5

Structure 6

(II) Among the resonance structures, circle the *most contributing* resonance structure.

Structure 4

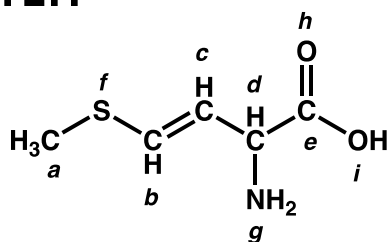
Structure 5

Structure 6

**Question 6** **6 points**

Using the data given below, calculate the **lattice energy** for SrO(s).

	ΔH° (kJ/mol)
$\text{Sr}(s) \longrightarrow \text{Sr}(g)$	+180
$\text{Sr}(g) \longrightarrow \text{Sr}^{2+}(g) + 2e^-$	+1468
$\text{O}_2(g) \longrightarrow 2\text{O}(g)$	+500
$\text{O}(g) + 2e^- \longrightarrow \text{O}^{2-}(g)$	+600
$\text{Sr}(s) + \frac{1}{2}\text{O}_2(g) \longrightarrow \text{SrO}(s)$	-550

**Question 7 10 points**

Consider the molecule given on the left, and answer questions (i) to (vi) using chemical bonding theories that we have learnt in class (valence bond theory, hybridization).

Note that the lone pairs are not shown in the structure:

i. Consider the carbon atoms marked in as ***a***, ***b***, ***c***, ***d***, and ***e*** in the molecule, and determine **hybridization** at each of the carbon centers. (2.5 points)

Atom	Hybridization
Carbon (<i>a</i>)	
Carbon (<i>b</i>)	
Carbon (<i>c</i>)	
Carbon (<i>d</i>)	
Carbon (<i>e</i>)	

ii. Now, consider the sulfur atom (S) marked as ***f***, nitrogen (N) atom (***g***), and oxygen(O) atoms (***h*** and ***i***), and determine their **hybridization**. (2 points)

Atom	Hybridization
Sulfur (<i>f</i>)	
Nitrogen (<i>g</i>)	
Oxygen (<i>h</i>)	
Oxygen (<i>i</i>)	

iii. **Describe** all the bonds formed by the carbon atoms ***a*** and ***b***, using sigma and pi designation as appropriate. (2 points)

Atom	Bonds formed (use sigma and pi as appropriate)
Carbon (<i>a</i>)	
Carbon (<i>b</i>)	

iv. How many total sp^3 -s sigma bonds are in the molecule? (1 point)

v. Determine **electronic** and **molecular geometry** at carbon (C) labeled ***e*** and sulfur (S) labeled ***f*** centres. (2 points)

Atom	Electronic Geometry	Molecular Geometry
Carbon (<i>e</i>)		
Sulfur (<i>f</i>)		

vi. Circle the value from the choices given below, that describes the **C-S-C bond angle at sulfur** (S) marked ***f***: (0.5 points)

>90° to <109.5° 109.5° >109.5 to <120° 120° >120° to <180°

**Question 8 12 points**

Using the **valence shell orbitals only** draw a **complete** molecular orbital (MO) diagram (with no s-p mixing) for the molecule **PS**, in the space provided below. **(6 points)**

Assume that 3s and 3p atomic orbitals will form molecular orbitals like those formed by the combination of 2s and 2p atomic orbitals that we learnt in class, and there is no s-p mixing.

Show **energy arrow**, **atomic orbitals** involved, **atom labels**, the **molecular orbitals** that are formed, **filled electrons** in the MO diagram etc. Make sure to label all the atomic and molecular orbitals.

Using the MO diagram above, answer the following questions:

- (i) Which one of the following will have the highest bond order? Circle the correct choice
A) PS^{1-} B) PS C) PS^{1+}
- (ii) Which one of the following will have the longest bond length? Circle the correct choice
A) PS^{1-} B) PS C) PS^{1+}
- (iii) Which of the following will be diamagnetic? Circle the correct choice.
A) PS^{1-} B) PS C) PS^{1+}
- (iv) Draw Lewis dot structure of the molecule with the shortest bond length (based on the MO diagram) from among the three (PS^{1-} , PS, PS^{1+}), and indicate the bond order calculated from the Lewis dot structure. **(3 points)**

Lewis Dot Structure

Bond Order



C13F661D-9DDE-4541-9C8A-4325D3D28068

midterm-exam-93388

#491 Page 10 of 10

SCRAP PAGE

**CHEMISTRY 110 –Midterm Exam****Thursday November 7, 2024, 6:30 – 8:30 pm****Examiners: Pallavi Sirjoosingh and Ashok Kakkar**

First name (please write as legibly as possible within the boxes)

--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--

Last name

--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--

Student ID number

--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--

This booklet has 8 questions on 10 pages (The last page can be used as a scrap page).

The staple must NOT be removed from the question booklet. Your intact question booklet must be returned at the end of the exam.

A separate booklet contains a periodic table, and datasheet.

Write your answers directly under the questions in the space provided. Do not write on the QR codes or near the top edges (as it may be cut off).

Please write **legibly and dark enough** for the scan to be readable. For students who use English as a second language, dictionaries are permitted. Calculators are allowed.

DO NOT ASK QUESTIONS OF THE INVIGILATORS about interpretations of the exam questions. If you have a concern about a question, please summarize it at the top of this page. All concerns will be evaluated after the exam. GOOD LUCK!

**Question 1 8 points** (*Assume all transitions are possible*)

a. 6 points Calculate the **energy (in kJ/mol)** required to ionize (completely remove) the electron from the **$n=3$** level of **He^+ ion** (He: Helium). Show your work including equations. (3 sig. figs.)

b. 2 points

(I) Select the largest (longest) wavelength from the three choices below:

- i. Wavelength of photon to *completely remove* an electron from *1s orbital of Hydrogen atom*
- ii. Wavelength of photon to *completely remove* an electron from *1s orbital of Boron atom (B)*
- iii. Wavelength of photon to *completely remove* an electron from *1s orbital of Boron cation (B^+)*

i or ii or iii

(II) Select the smallest (shortest) wavelength from the three choices below (*for H atom*):

- i. Wavelength of photon *emitted* for an electron transitioning from *2p to 1s orbital*
- ii. Wavelength of photon *absorbed* for an electron transitioning from *2s to 4p orbital*
- iii. Wavelength of photon *absorbed* for an electron transitioning from *1s to 3p orbital*

i or ii or iii



Question 2 8 points



A Chem110 student plots the graph shown on the left, from data obtained for the kinetic energy (eV) of the ejected electron versus frequency (s^{-1}) of the incident light, for an unknown metal. ($1\text{eV} = 1.6022 \times 10^{-19} \text{ J}$)

Based on the plot, answer the questions **a to c** below:

a. 1 point On the plot above, label the **threshold frequency**.

b. 2 points What is the value of the work potential (in J) of the unknown metal?

c. 3 points What is the *longest* wavelength (in nm) required to eject an electron from this metal's surface? Show your work including equation(s) used. (3 sig figs)

d. 2 points In the space provided below, plot the following

(Assume frequency is *above* threshold frequency):

- Kinetic energy* of an ejected electron versus the *intensity* of the incident light (left)
- Number of electrons* ejected versus the *frequency* of the incident light (right)



**Question 3** **6 points****a. 2 points**

In the ground state of Ge (Germanium; $Z = 32$) atom, **name** the following orbitals:

i. Orbital containing the electron that requires the maximum energy to be removed.

ii. Orbital containing the electron that experiences the minimum effective nuclear charge.

b. 2 points

An unknown anion (A^-) has the following *excited* electronic configuration: $[Ar] 4s^2 3d^3 4p^1$

(Ar: Argon; $Z = 18$)

Give the **condensed ground state electronic configuration** of the atom, **A** and the cation, **A^{2+}** .
Show your work. (Assume all electronic transitions are possible)

Electronic Configuration of A:

Electronic Configuration of A^{2+} :

c. 2 points

In a single atom, determine how many electrons can have the following set of quantum numbers?

(i) $n = 5; l = 3$

In a single atom, determine how many electrons can have the following set of quantum numbers?

(i) $n = 3; l = 2; m_s = +1/2$

**Question 4** **8 points**

Rank the following from 1 to 4 in **increasing** order. Provide the ranking in the boxes provided.

Note: Atomic number (Z) are provided for relevant atoms

a) Increasing (lowest to highest) order of *ionization energy*: (1= lowest; 4 = highest)

Mg

Mg: Z = 12

Rb

Rb: Z = 37

Cs

Cs: Z = 55

K

K : Z = 19

b) Increasing (lowest to highest) order of *bond angle*: (1= lowest; 4 = highest)

H₂OCCl₄XeF₄

HCN

c) Increasing (smallest to largest) order of *ionic size* (1= smallest; 4 = largest)

S²⁻

S: Z = 16

O²⁻

O: Z = 8

Be²⁺

Be: Z = 4

Na⁺

Na: Z = 11

d) Increasing (lowest to highest) order of *bond energy* (1= lowest; 4 = highest)

P-F

P-Cl

P-I

P-Br

**Question 5** **12 points**

a. 9 points For the molecules HClO_4 and TeF_6 , draw the most stable *Lewis structure* showing appropriate shape and *dashed/wedged bonds*, based on VSEPR. Determine the **molecular geometry** around the central atom. For HClO_4 , label the *most polar bond*.



b. 3 points (Note: Lone pairs are not shown in structures below)

(i) For the three *resonance* structures (1, 2 and 3) shown below, answer the following question:



(I) Circle the *most contributing* resonance structure.

Structure 1

Structure 2

Structure 3

(ii) For the three structures given (4, 5, and 6) below, answer the following questions:



(I) Circle all the structures that represent *resonance structures* of each other:

Structure 4

Structure 5

Structure 6

(II) Among the resonance structures, circle the *most contributing* resonance structure.

Structure 4

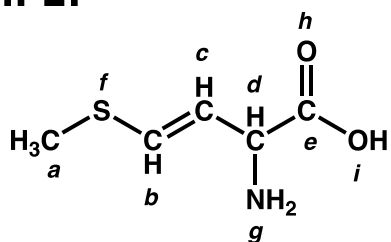
Structure 5

Structure 6

**Question 6** **6 points**

Using the data given below, calculate the **lattice energy** for SrO(s).

	ΔH° (kJ/mol)
$\text{Sr}(s) \longrightarrow \text{Sr}(g)$	+180
$\text{Sr}(g) \longrightarrow \text{Sr}^{2+}(g) + 2e^-$	+1468
$\text{O}_2(g) \longrightarrow 2\text{O}(g)$	+500
$\text{O}(g) + 2e^- \longrightarrow \text{O}^{2-}(g)$	+600
$\text{Sr}(s) + \frac{1}{2}\text{O}_2(g) \longrightarrow \text{SrO}(s)$	-550

**Question 7 10 points**

Consider the molecule given on the left, and answer questions (i) to (vi) using chemical bonding theories that we have learnt in class (valence bond theory, hybridization).

Note that the lone pairs are not shown in the structure:

i. Consider the carbon atoms marked in as ***a***, ***b***, ***c***, ***d***, and ***e*** in the molecule, and determine **hybridization** at each of the carbon centers. (2.5 points)

Atom	Hybridization
Carbon (<i>a</i>)	
Carbon (<i>b</i>)	
Carbon (<i>c</i>)	
Carbon (<i>d</i>)	
Carbon (<i>e</i>)	

ii. Now, consider the sulfur atom (S) marked as ***f***, nitrogen (N) atom (***g***), and oxygen(O) atoms (***h*** and ***i***), and determine their **hybridization**. (2 points)

Atom	Hybridization
Sulfur (<i>f</i>)	
Nitrogen (<i>g</i>)	
Oxygen (<i>h</i>)	
Oxygen (<i>i</i>)	

iii. **Describe** all the bonds formed by the carbon atoms ***a*** and ***b***, using sigma and pi designation as appropriate. (2 points)

Atom	Bonds formed (use sigma and pi as appropriate)
Carbon (<i>a</i>)	
Carbon (<i>b</i>)	

iv. How many total sp^3 -s sigma bonds are in the molecule? (1 point)

v. Determine **electronic** and **molecular geometry** at carbon (C) labeled ***e*** and sulfur (S) labeled ***f*** centres. (2 points)

Atom	Electronic Geometry	Molecular Geometry
Carbon (<i>e</i>)		
Sulfur (<i>f</i>)		

vi. Circle the value from the choices given below, that describes the **C-S-C bond angle at sulfur** (S) marked ***f***: (0.5 points)

>90° to <109.5° 109.5° >109.5 to <120° 120° >120° to <180°


Question 8 12 points

Using the **valence shell orbitals only** draw a **complete** molecular orbital (MO) diagram (with no s-p mixing) for the molecule **PS**, in the space provided below. **(6 points)**

Assume that 3s and 3p atomic orbitals will form molecular orbitals like those formed by the combination of 2s and 2p atomic orbitals that we learnt in class, and there is no s-p mixing.

Show **energy arrow**, **atomic orbitals** involved, **atom labels**, the **molecular orbitals** that are formed, **filled electrons** in the MO diagram etc. Make sure to label all the atomic and molecular orbitals.

Using the MO diagram above, answer the following questions:

- (i) Which one of the following will have the highest bond order? Circle the correct choice
 A) PS^{1-} B) PS C) PS^{1+}
- (ii) Which one of the following will have the longest bond length? Circle the correct choice
 A) PS^{1-} B) PS C) PS^{1+}
- (iii) Which of the following will be diamagnetic? Circle the correct choice.
 A) PS^{1-} B) PS C) PS^{1+}
- (iv) Draw Lewis dot structure of the molecule with the shortest bond length (based on the MO diagram) from among the three (PS^{1-} , PS, PS^{1+}), and indicate the bond order calculated from the Lewis dot structure. **(3 points)**

Lewis Dot Structure

Bond Order



B06E6A6E-5C84-43F7-ABB7-9B1642DE0F9F

midterm-exam-93388

#492 Page 10 of 10

SCRAP PAGE

**CHEMISTRY 110 –Midterm Exam****Thursday November 7, 2024, 6:30 – 8:30 pm****Examiners: Pallavi Sirjoosingh and Ashok Kakkar**

First name (please write as legibly as possible within the boxes)

--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--

Last name

--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--

Student ID number

--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--

This booklet has 8 questions on 10 pages (The last page can be used as a scrap page).

The staple must NOT be removed from the question booklet. Your intact question booklet must be returned at the end of the exam.

A separate booklet contains a periodic table, and datasheet.

Write your answers directly under the questions in the space provided. Do not write on the QR codes or near the top edges (as it may be cut off).

Please write **legibly and dark enough** for the scan to be readable. For students who use English as a second language, dictionaries are permitted. Calculators are allowed.

DO NOT ASK QUESTIONS OF THE INVIGILATORS about interpretations of the exam questions. If you have a concern about a question, please summarize it at the top of this page. All concerns will be evaluated after the exam. GOOD LUCK!

**Question 1 8 points** (*Assume all transitions are possible*)

a. 6 points Calculate the **energy (in kJ/mol)** required to ionize (completely remove) the electron from the **n=3** level of **He⁺ ion** (He: Helium). Show your work including equations. (3 sig. figs.)

b. 2 points

(I) Select the largest (longest) wavelength from the three choices below:

- i. Wavelength of photon to *completely remove* an electron from *1s orbital of Hydrogen atom*
- ii. Wavelength of photon to *completely remove* an electron from *1s orbital of Boron atom (B)*
- iii. Wavelength of photon to *completely remove* an electron from *1s orbital of Boron cation (B⁺)*

i or ii or iii

(II) Select the smallest (shortest) wavelength from the three choices below (*for H atom*):

- i. Wavelength of photon *emitted* for an electron transitioning from *2p to 1s orbital*
- ii. Wavelength of photon *absorbed* for an electron transitioning from *2s to 4p orbital*
- iii. Wavelength of photon *absorbed* for an electron transitioning from *1s to 3p orbital*

i or ii or iii



Question 2 8 points



A Chem110 student plots the graph shown on the left, from data obtained for the kinetic energy (eV) of the ejected electron versus frequency (s^{-1}) of the incident light, for an unknown metal. ($1\text{eV} = 1.6022 \times 10^{-19} \text{ J}$)

Based on the plot, answer the questions **a to c** below:

a. 1 point On the plot above, label the **threshold frequency**.

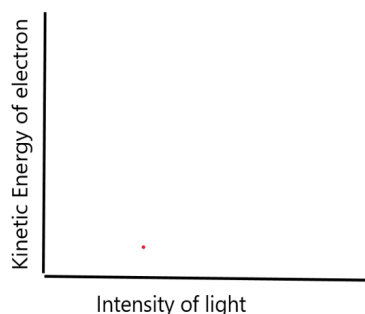
b. 2 points What is the value of the work potential (in J) of the unknown metal?

c. 3 points What is the *longest* wavelength (in nm) required to eject an electron from this metal's surface? Show your work including equation(s) used. (3 sig figs)

d. 2 points In the space provided below, plot the following

(Assume frequency is *above* threshold frequency):

- Kinetic energy* of an ejected electron versus the *intensity* of the incident light (left)
- Number of electrons* ejected versus the *frequency* of the incident light (right)



**Question 3** **6 points****a. 2 points**

In the ground state of Ge (Germanium; $Z = 32$) atom, **name** the following orbitals:

i. Orbital containing the electron that requires the maximum energy to be removed.

ii. Orbital containing the electron that experiences the minimum effective nuclear charge.

b. 2 points

An unknown anion (A^-) has the following *excited* electronic configuration: $[Ar] 4s^2 3d^3 4p^1$

(Ar: Argon; $Z = 18$)

Give the **condensed ground state electronic configuration** of the atom, **A** and the cation, **A^{2+}** .
Show your work. (Assume all electronic transitions are possible)

Electronic Configuration of A:

Electronic Configuration of A^{2+} :

c. 2 points

In a single atom, determine how many electrons can have the following set of quantum numbers?

(i) $n = 5; l = 3$

In a single atom, determine how many electrons can have the following set of quantum numbers?

(i) $n = 3; l = 2; m_s = +1/2$

**Question 4** **8 points**

Rank the following from 1 to 4 in **increasing** order. Provide the ranking in the boxes provided.

Note: Atomic number (Z) are provided for relevant atoms

a) Increasing (lowest to highest) order of *ionization energy*: (1= lowest; 4 = highest)

Mg

Mg: Z = 12

Rb

Rb: Z = 37

Cs

Cs: Z = 55

K

K : Z = 19

b) Increasing (lowest to highest) order of *bond angle*: (1= lowest; 4 = highest)

H₂OCCl₄XeF₄

HCN

c) Increasing (smallest to largest) order of *ionic size* (1= smallest; 4 = largest)

S²⁻

S: Z = 16

O²⁻

O: Z = 8

Be²⁺

Be: Z = 4

Na⁺

Na: Z = 11

d) Increasing (lowest to highest) order of *bond energy* (1= lowest; 4 = highest)

P-F

P-Cl

P-I

P-Br

**Question 5** **12 points**

a. 9 points For the molecules HClO_4 and TeF_6 , draw the most stable *Lewis structure* showing appropriate shape and *dashed/wedged bonds*, based on VSEPR. Determine the **molecular geometry** around the central atom. For HClO_4 , label the *most polar bond*.



b. 3 points (Note: Lone pairs are not shown in structures below)

(i) For the three *resonance* structures (1, 2 and 3) shown below, answer the following question:



(I) Circle the *most contributing* resonance structure.

Structure 1

Structure 2

Structure 3

(ii) For the three structures given (4, 5, and 6) below, answer the following questions:



(I) Circle all the structures that represent *resonance structures* of each other:

Structure 4

Structure 5

Structure 6

(II) Among the resonance structures, circle the *most contributing* resonance structure.

Structure 4

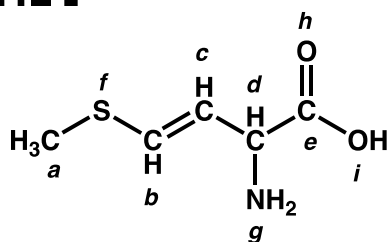
Structure 5

Structure 6

**Question 6** **6 points**

Using the data given below, calculate the **lattice energy** for SrO(s).

	ΔH° (kJ/mol)
$\text{Sr}(s) \longrightarrow \text{Sr}(g)$	+180
$\text{Sr}(g) \longrightarrow \text{Sr}^{2+}(g) + 2e^-$	+1468
$\text{O}_2(g) \longrightarrow 2\text{O}(g)$	+500
$\text{O}(g) + 2e^- \longrightarrow \text{O}^{2-}(g)$	+600
$\text{Sr}(s) + \frac{1}{2}\text{O}_2(g) \longrightarrow \text{SrO}(s)$	-550

**Question 7 10 points**

Consider the molecule given on the left, and answer questions (i) to (vi) using chemical bonding theories that we have learnt in class (valence bond theory, hybridization).

Note that the lone pairs are not shown in the structure:

i. Consider the carbon atoms marked in as ***a***, ***b***, ***c***, ***d***, and ***e*** in the molecule, and determine **hybridization** at each of the carbon centers. (2.5 points)

Atom	Hybridization
Carbon (<i>a</i>)	
Carbon (<i>b</i>)	
Carbon (<i>c</i>)	
Carbon (<i>d</i>)	
Carbon (<i>e</i>)	

ii. Now, consider the sulfur atom (S) marked as ***f***, nitrogen (N) atom (***g***), and oxygen(O) atoms (***h*** and ***i***), and determine their **hybridization**. (2 points)

Atom	Hybridization
Sulfur (<i>f</i>)	
Nitrogen (<i>g</i>)	
Oxygen (<i>h</i>)	
Oxygen (<i>i</i>)	

iii. **Describe** all the bonds formed by the carbon atoms ***a*** and ***b***, using sigma and pi designation as appropriate. (2 points)

Atom	Bonds formed (use sigma and pi as appropriate)
Carbon (<i>a</i>)	
Carbon (<i>b</i>)	

iv. How many total sp^3 -s sigma bonds are in the molecule? (1 point)

v. Determine **electronic** and **molecular geometry** at carbon (C) labeled ***e*** and sulfur (S) labeled ***f*** centres. (2 points)

Atom	Electronic Geometry	Molecular Geometry
Carbon (<i>e</i>)		
Sulfur (<i>f</i>)		

vi. Circle the value from the choices given below, that describes the **C-S-C bond angle at sulfur** (S) marked ***f***: (0.5 points)

>90° to <109.5°

109.5°

>109.5 to <120°

120°

>120° to <180°

**Question 8 12 points**

Using the **valence shell orbitals only** draw a **complete** molecular orbital (MO) diagram (with no s-p mixing) for the molecule **PS**, in the space provided below. **(6 points)**

Assume that 3s and 3p atomic orbitals will form molecular orbitals like those formed by the combination of 2s and 2p atomic orbitals that we learnt in class, and there is no s-p mixing.

Show **energy arrow**, **atomic orbitals** involved, **atom labels**, the **molecular orbitals** that are formed, **filled electrons** in the MO diagram etc. Make sure to label all the atomic and molecular orbitals.

Using the MO diagram above, answer the following questions:

- (i) Which one of the following will have the highest bond order? Circle the correct choice
A) PS^{1-} B) PS C) PS^{1+}
- (ii) Which one of the following will have the longest bond length? Circle the correct choice
A) PS^{1-} B) PS C) PS^{1+}
- (iii) Which of the following will be diamagnetic? Circle the correct choice.
A) PS^{1-} B) PS C) PS^{1+}
- (iv) Draw Lewis dot structure of the molecule with the shortest bond length (based on the MO diagram) from among the three (PS^{1-} , PS, PS^{1+}), and indicate the bond order calculated from the Lewis dot structure. **(3 points)**

Lewis Dot Structure

Bond Order



B5739A60-79B0-489E-94BA-C9EE4F4C93F6

midterm-exam-93388

#493 Page 10 of 10

SCRAP PAGE

**CHEMISTRY 110 –Midterm Exam****Thursday November 7, 2024, 6:30 – 8:30 pm**

Examiners: Pallavi Sirjoosingh and Ashok Kakkar

First name (please write as legibly as possible within the boxes)

--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--

Last name

--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--

Student ID number

--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--

This booklet has 8 questions on 10 pages (The last page can be used as a scrap page).

The staple must NOT be removed from the question booklet. Your intact question booklet must be returned at the end of the exam.

A separate booklet contains a periodic table, and datasheet.

Write your answers directly under the questions in the space provided. Do not write on the QR codes or near the top edges (as it may be cut off).

Please write **legibly and dark enough** for the scan to be readable. For students who use English as a second language, dictionaries are permitted. Calculators are allowed.

DO NOT ASK QUESTIONS OF THE INVIGILATORS about interpretations of the exam questions. If you have a concern about a question, please summarize it at the top of this page. All concerns will be evaluated after the exam. GOOD LUCK!

**Question 1 8 points** (*Assume all transitions are possible*)

a. 6 points Calculate the **energy (in kJ/mol)** required to ionize (completely remove) the electron from the **$n=3$** level of **He^+ ion** (He: Helium). Show your work including equations. (3 sig. figs.)

b. 2 points

(I) Select the largest (longest) wavelength from the three choices below:

- i. Wavelength of photon to *completely remove* an electron from *1s orbital of Hydrogen atom*
- ii. Wavelength of photon to *completely remove* an electron from *1s orbital of Boron atom (B)*
- iii. Wavelength of photon to *completely remove* an electron from *1s orbital of Boron cation (B^+)*

i or ii or iii

(II) Select the smallest (shortest) wavelength from the three choices below (*for H atom*):

- i. Wavelength of photon *emitted* for an electron transitioning from *2p to 1s orbital*
- ii. Wavelength of photon *absorbed* for an electron transitioning from *2s to 4p orbital*
- iii. Wavelength of photon *absorbed* for an electron transitioning from *1s to 3p orbital*

i or ii or iii



Question 2 8 points



A Chem110 student plots the graph shown on the left, from data obtained for the kinetic energy (eV) of the ejected electron versus frequency (s^{-1}) of the incident light, for an unknown metal. ($1\text{eV} = 1.6022 \times 10^{-19} \text{ J}$)

Based on the plot, answer the questions **a to c** below:

a. 1 point On the plot above, label the **threshold frequency**.

b. 2 points What is the value of the work potential (in J) of the unknown metal?

c. 3 points What is the *longest* wavelength (in nm) required to eject an electron from this metal's surface? Show your work including equation(s) used. (3 sig figs)

d. 2 points In the space provided below, plot the following

(Assume frequency is *above* threshold frequency):

- i. *Kinetic energy* of an ejected electron versus the *intensity* of the incident light (left)
- ii. *Number of electrons* ejected versus the *frequency* of the incident light (right)



**Question 3** **6 points****a. 2 points**

In the ground state of Ge (Germanium; $Z = 32$) atom, **name** the following orbitals:

i. Orbital containing the electron that requires the maximum energy to be removed.

ii. Orbital containing the electron that experiences the minimum effective nuclear charge.

b. 2 points

An unknown anion (A^-) has the following *excited* electronic configuration: $[Ar] 4s^2 3d^3 4p^1$

(Ar: Argon; $Z = 18$)

Give the **condensed ground state electronic configuration** of the atom, **A** and the cation, **A^{2+}** .
Show your work. (Assume all electronic transitions are possible)

Electronic Configuration of A:

Electronic Configuration of A^{2+} :

c. 2 points

In a single atom, determine how many electrons can have the following set of quantum numbers?

(i) $n = 5; l = 3$

In a single atom, determine how many electrons can have the following set of quantum numbers?

(i) $n = 3; l = 2; m_s = +1/2$

**Question 4** **8 points**

Rank the following from 1 to 4 in **increasing** order. Provide the ranking in the boxes provided.

Note: Atomic number (Z) are provided for relevant atoms

a) Increasing (lowest to highest) order of *ionization energy*: (1= lowest; 4 = highest)

Mg

Mg: Z = 12

Rb

Rb: Z = 37

Cs

Cs: Z = 55

K

K : Z = 19

b) Increasing (lowest to highest) order of *bond angle*: (1= lowest; 4 = highest)

H₂OCCl₄XeF₄

HCN

c) Increasing (smallest to largest) order of *ionic size* (1= smallest; 4 = largest)

S²⁻

S: Z = 16

O²⁻

O: Z = 8

Be²⁺

Be: Z = 4

Na⁺

Na: Z = 11

d) Increasing (lowest to highest) order of *bond energy* (1= lowest; 4 = highest)

P-F

P-Cl

P-I

P-Br

**Question 5** **12 points**

a. 9 points For the molecules HClO_4 and TeF_6 , draw the most stable *Lewis structure* showing appropriate shape and *dashed/wedged bonds*, based on VSEPR. Determine the **molecular geometry** around the central atom. For HClO_4 , label the *most polar bond*.

HClO_4 (there are no O-O bonds)

Label the most polar bond

Molecular Geometry around Cl

TeF_6

Molecular Geometry around Te

b. 3 points (Note: Lone pairs are not shown in structures below)

(i) For the three *resonance* structures (1, 2 and 3) shown below, answer the following question:



(I) Circle the *most contributing* resonance structure.

Structure 1

Structure 2

Structure 3

(ii) For the three structures given (4, 5, and 6) below, answer the following questions:



(I) Circle all the structures that represent *resonance structures* of each other:

Structure 4

Structure 5

Structure 6

(II) Among the resonance structures, circle the *most contributing* resonance structure.

Structure 4

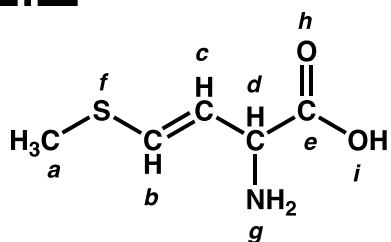
Structure 5

Structure 6

**Question 6** **6 points**

Using the data given below, calculate the **lattice energy** for SrO(s).

	ΔH° (kJ/mol)
$\text{Sr}(s) \longrightarrow \text{Sr}(g)$	+180
$\text{Sr}(g) \longrightarrow \text{Sr}^{2+}(g) + 2e^-$	+1468
$\text{O}_2(g) \longrightarrow 2\text{O}(g)$	+500
$\text{O}(g) + 2e^- \longrightarrow \text{O}^{2-}(g)$	+600
$\text{Sr}(s) + \frac{1}{2}\text{O}_2(g) \longrightarrow \text{SrO}(s)$	-550

**Question 7 10 points**

Consider the molecule given on the left, and answer questions (i) to (vi) using chemical bonding theories that we have learnt in class (valence bond theory, hybridization).

Note that the lone pairs are not shown in the structure:

i. Consider the carbon atoms marked in as ***a***, ***b***, ***c***, ***d***, and ***e*** in the molecule, and determine **hybridization** at each of the carbon centers. (2.5 points)

Atom	Hybridization
Carbon (<i>a</i>)	
Carbon (<i>b</i>)	
Carbon (<i>c</i>)	
Carbon (<i>d</i>)	
Carbon (<i>e</i>)	

ii. Now, consider the sulfur atom (S) marked as ***f***, nitrogen (N) atom (***g***), and oxygen(O) atoms (***h*** and ***i***), and determine their **hybridization**. (2 points)

Atom	Hybridization
Sulfur (<i>f</i>)	
Nitrogen (<i>g</i>)	
Oxygen (<i>h</i>)	
Oxygen (<i>i</i>)	

iii. **Describe** all the bonds formed by the carbon atoms ***a*** and ***b***, using sigma and pi designation as appropriate. (2 points)

Atom	Bonds formed (use sigma and pi as appropriate)
Carbon (<i>a</i>)	
Carbon (<i>b</i>)	

iv. How many total sp^3 -s sigma bonds are in the molecule? (1 point)

v. Determine **electronic** and **molecular geometry** at carbon (C) labeled ***e*** and sulfur (S) labeled ***f*** centres. (2 points)

Atom	Electronic Geometry	Molecular Geometry
Carbon (<i>e</i>)		
Sulfur (<i>f</i>)		

vi. Circle the value from the choices given below, that describes the **C-S-C bond angle at sulfur** (S) marked ***f***: (0.5 points)

>90° to <109.5° 109.5° >109.5 to <120° 120° >120° to <180°

**Question 8 12 points**

Using the **valence shell orbitals only** draw a **complete** molecular orbital (MO) diagram (with no s-p mixing) for the molecule **PS**, in the space provided below. **(6 points)**

Assume that 3s and 3p atomic orbitals will form molecular orbitals like those formed by the combination of 2s and 2p atomic orbitals that we learnt in class, and there is no s-p mixing.

Show **energy arrow**, **atomic orbitals** involved, **atom labels**, the **molecular orbitals** that are formed, **filled electrons** in the MO diagram etc. Make sure to label all the atomic and molecular orbitals.

Using the MO diagram above, answer the following questions:

- (i) Which one of the following will have the highest bond order? Circle the correct choice
A) PS^{1-} B) PS C) PS^{1+}
- (ii) Which one of the following will have the longest bond length? Circle the correct choice
A) PS^{1-} B) PS C) PS^{1+}
- (iii) Which of the following will be diamagnetic? Circle the correct choice.
A) PS^{1-} B) PS C) PS^{1+}
- (iv) Draw Lewis dot structure of the molecule with the shortest bond length (based on the MO diagram) from among the three (PS^{1-} , PS, PS^{1+}), and indicate the bond order calculated from the Lewis dot structure. **(3 points)**

Lewis Dot Structure

Bond Order



A23AC959-4193-423B-B1E7-6A57F19F16B5

midterm-exam-93388

#494 Page 10 of 10

SCRAP PAGE

**CHEMISTRY 110 –Midterm Exam****Thursday November 7, 2024, 6:30 – 8:30 pm****Examiners: Pallavi Sirjoosingh and Ashok Kakkar**

First name (please write as legibly as possible within the boxes)

--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--

Last name

--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--

Student ID number

--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--

This booklet has 8 questions on 10 pages (The last page can be used as a scrap page).

The staple must NOT be removed from the question booklet. Your intact question booklet must be returned at the end of the exam.

A separate booklet contains a periodic table, and datasheet.

Write your answers directly under the questions in the space provided. Do not write on the QR codes or near the top edges (as it may be cut off).

Please write **legibly and dark enough** for the scan to be readable. For students who use English as a second language, dictionaries are permitted. Calculators are allowed.

DO NOT ASK QUESTIONS OF THE INVIGILATORS about interpretations of the exam questions. If you have a concern about a question, please summarize it at the top of this page. All concerns will be evaluated after the exam. GOOD LUCK!

**Question 1 8 points** (*Assume all transitions are possible*)

a. 6 points Calculate the **energy (in kJ/mol)** required to ionize (completely remove) the electron from the **n=3** level of **He⁺ ion** (He: Helium). Show your work including equations. (3 sig. figs.)

b. 2 points

(I) Select the largest (longest) wavelength from the three choices below:

- i. Wavelength of photon to *completely remove* an electron from *1s orbital of Hydrogen atom*
- ii. Wavelength of photon to *completely remove* an electron from *1s orbital of Boron atom (B)*
- iii. Wavelength of photon to *completely remove* an electron from *1s orbital of Boron cation (B⁺)*

i or ii or iii

(II) Select the smallest (shortest) wavelength from the three choices below (*for H atom*):

- i. Wavelength of photon *emitted* for an electron transitioning from *2p to 1s orbital*
- ii. Wavelength of photon *absorbed* for an electron transitioning from *2s to 4p orbital*
- iii. Wavelength of photon *absorbed* for an electron transitioning from *1s to 3p orbital*

i or ii or iii



Question 2 8 points



A Chem110 student plots the graph shown on the left, from data obtained for the kinetic energy (eV) of the ejected electron versus frequency (s^{-1}) of the incident light, for an unknown metal. ($1\text{eV} = 1.6022 \times 10^{-19} \text{ J}$)

Based on the plot, answer the questions **a to c** below:

a. 1 point On the plot above, label the **threshold frequency**.

b. 2 points What is the value of the work potential (in J) of the unknown metal?

c. 3 points What is the *longest* wavelength (in nm) required to eject an electron from this metal's surface? Show your work including equation(s) used. (3 sig figs)

d. 2 points In the space provided below, plot the following

(Assume frequency is *above* threshold frequency):

- i. *Kinetic energy* of an ejected electron versus the *intensity* of the incident light (left)
- ii. *Number of electrons* ejected versus the *frequency* of the incident light (right)



**Question 3** **6 points****a. 2 points**

In the ground state of Ge (Germanium; $Z = 32$) atom, **name** the following orbitals:

i. Orbital containing the electron that requires the maximum energy to be removed.

ii. Orbital containing the electron that experiences the minimum effective nuclear charge.

b. 2 points

An unknown anion (A^-) has the following *excited* electronic configuration: $[Ar] 4s^2 3d^3 4p^1$

(Ar: Argon; $Z = 18$)

Give the **condensed ground state electronic configuration** of the atom, **A** and the cation, **A^{2+}** .
Show your work. (Assume all electronic transitions are possible)

Electronic Configuration of A:

Electronic Configuration of A^{2+} :

c. 2 points

In a single atom, determine how many electrons can have the following set of quantum numbers?

(i) $n = 5; l = 3$

In a single atom, determine how many electrons can have the following set of quantum numbers?

(i) $n = 3; l = 2; m_s = +1/2$

**Question 4** **8 points**

Rank the following from 1 to 4 in **increasing** order. Provide the ranking in the boxes provided.

Note: Atomic number (Z) are provided for relevant atoms

a) Increasing (lowest to highest) order of *ionization energy*: (1= lowest; 4 = highest)

Mg

Mg: Z = 12

Rb

Rb: Z = 37

Cs

Cs: Z = 55

K

K : Z = 19

b) Increasing (lowest to highest) order of *bond angle*: (1= lowest; 4 = highest)

H₂OCCl₄XeF₄

HCN

c) Increasing (smallest to largest) order of *ionic size* (1= smallest; 4 = largest)

S²⁻

S: Z = 16

O²⁻

O: Z = 8

Be²⁺

Be: Z = 4

Na⁺

Na: Z = 11

d) Increasing (lowest to highest) order of *bond energy* (1= lowest; 4 = highest)

P-F

P-Cl

P-I

P-Br

**Question 5** **12 points**

a. 9 points For the molecules HClO_4 and TeF_6 , draw the most stable *Lewis structure* showing appropriate shape and *dashed/wedged bonds*, based on VSEPR. Determine the **molecular geometry** around the central atom. For HClO_4 , label the *most polar bond*.



b. 3 points (Note: Lone pairs are not shown in structures below)

(i) For the three *resonance* structures (1, 2 and 3) shown below, answer the following question:



(I) Circle the *most contributing* resonance structure.

Structure 1

Structure 2

Structure 3

(ii) For the three structures given (4, 5, and 6) below, answer the following questions:



(I) Circle all the structures that represent *resonance structures* of each other:

Structure 4

Structure 5

Structure 6

(II) Among the resonance structures, circle the *most contributing* resonance structure.

Structure 4

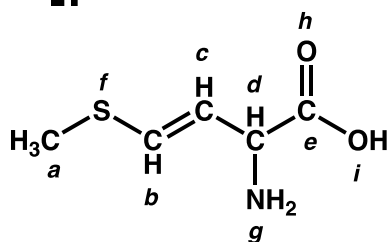
Structure 5

Structure 6

**Question 6** **6 points**

Using the data given below, calculate the **lattice energy** for SrO(s).

	ΔH° (kJ/mol)
$\text{Sr}(s) \longrightarrow \text{Sr}(g)$	+180
$\text{Sr}(g) \longrightarrow \text{Sr}^{2+}(g) + 2e^-$	+1468
$\text{O}_2(g) \longrightarrow 2\text{O}(g)$	+500
$\text{O}(g) + 2e^- \longrightarrow \text{O}^{2-}(g)$	+600
$\text{Sr}(s) + \frac{1}{2}\text{O}_2(g) \longrightarrow \text{SrO}(s)$	-550

**Question 7 10 points**

Consider the molecule given on the left, and answer questions (i) to (vi) using chemical bonding theories that we have learnt in class (valence bond theory, hybridization).

Note that the lone pairs are not shown in the structure:

i. Consider the carbon atoms marked in as ***a***, ***b***, ***c***, ***d***, and ***e*** in the molecule, and determine **hybridization** at each of the carbon centers. (2.5 points)

Atom	Hybridization
Carbon (<i>a</i>)	
Carbon (<i>b</i>)	
Carbon (<i>c</i>)	
Carbon (<i>d</i>)	
Carbon (<i>e</i>)	

ii. Now, consider the sulfur atom (S) marked as ***f***, nitrogen (N) atom (***g***), and oxygen(O) atoms (***h*** and ***i***), and determine their **hybridization**. (2 points)

Atom	Hybridization
Sulfur (<i>f</i>)	
Nitrogen (<i>g</i>)	
Oxygen (<i>h</i>)	
Oxygen (<i>i</i>)	

iii. **Describe** all the bonds formed by the carbon atoms ***a*** and ***b***, using sigma and pi designation as appropriate. (2 points)

Atom	Bonds formed (use sigma and pi as appropriate)
Carbon (<i>a</i>)	
Carbon (<i>b</i>)	

iv. How many total sp^3 -s sigma bonds are in the molecule? (1 point)

v. Determine **electronic** and **molecular geometry** at carbon (C) labeled ***e*** and sulfur (S) labeled ***f*** centres. (2 points)

Atom	Electronic Geometry	Molecular Geometry
Carbon (<i>e</i>)		
Sulfur (<i>f</i>)		

vi. Circle the value from the choices given below, that describes the **C-S-C bond angle at sulfur** (S) marked ***f***: (0.5 points)

>90° to <109.5° 109.5° >109.5 to <120° 120° >120° to <180°

**Question 8 12 points**

Using the **valence shell orbitals only** draw a **complete** molecular orbital (MO) diagram (with no s-p mixing) for the molecule **PS**, in the space provided below. **(6 points)**

Assume that 3s and 3p atomic orbitals will form molecular orbitals like those formed by the combination of 2s and 2p atomic orbitals that we learnt in class, and there is no s-p mixing.

Show **energy arrow**, **atomic orbitals** involved, **atom labels**, the **molecular orbitals** that are formed, **filled electrons** in the MO diagram etc. Make sure to label all the atomic and molecular orbitals.

Using the MO diagram above, answer the following questions:

- (i) Which one of the following will have the highest bond order? Circle the correct choice
A) PS^{1-} B) PS C) PS^{1+}
- (ii) Which one of the following will have the longest bond length? Circle the correct choice
A) PS^{1-} B) PS C) PS^{1+}
- (iii) Which of the following will be diamagnetic? Circle the correct choice.
A) PS^{1-} B) PS C) PS^{1+}
- (iv) Draw Lewis dot structure of the molecule with the shortest bond length (based on the MO diagram) from among the three (PS^{1-} , PS, PS^{1+}), and indicate the bond order calculated from the Lewis dot structure. **(3 points)**

Lewis Dot Structure

Bond Order



C0832619-00A0-4168-9AE9-4841034997B5

midterm-exam-93388

#495 Page 10 of 10

SCRAP PAGE

**CHEMISTRY 110 –Midterm Exam****Thursday November 7, 2024, 6:30 – 8:30 pm****Examiners: Pallavi Sirjoosingh and Ashok Kakkar**

First name (please write as legibly as possible within the boxes)

--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--

Last name

--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--

Student ID number

--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--

This booklet has 8 questions on 10 pages (The last page can be used as a scrap page).

The staple must NOT be removed from the question booklet. Your intact question booklet must be returned at the end of the exam.

A separate booklet contains a periodic table, and datasheet.

Write your answers directly under the questions in the space provided. Do not write on the QR codes or near the top edges (as it may be cut off).

Please write **legibly and dark enough** for the scan to be readable. For students who use English as a second language, dictionaries are permitted. Calculators are allowed.

DO NOT ASK QUESTIONS OF THE INVIGILATORS about interpretations of the exam questions. If you have a concern about a question, please summarize it at the top of this page. All concerns will be evaluated after the exam. GOOD LUCK!

**Question 1 8 points** (*Assume all transitions are possible*)

a. 6 points Calculate the **energy (in kJ/mol)** required to ionize (completely remove) the electron from the **$n=3$** level of **He^+ ion** (He: Helium). Show your work including equations. (3 sig. figs.)

b. 2 points

(I) Select the largest (longest) wavelength from the three choices below:

- i. Wavelength of photon to *completely remove* an electron from *1s orbital of Hydrogen atom*
- ii. Wavelength of photon to *completely remove* an electron from *1s orbital of Boron atom (B)*
- iii. Wavelength of photon to *completely remove* an electron from *1s orbital of Boron cation (B^+)*

i or ii or iii

(II) Select the smallest (shortest) wavelength from the three choices below (*for H atom*):

- i. Wavelength of photon *emitted* for an electron transitioning from *2p to 1s orbital*
- ii. Wavelength of photon *absorbed* for an electron transitioning from *2s to 4p orbital*
- iii. Wavelength of photon *absorbed* for an electron transitioning from *1s to 3p orbital*

i or ii or iii



Question 2 8 points



A Chem110 student plots the graph shown on the left, from data obtained for the kinetic energy (eV) of the ejected electron versus frequency (s^{-1}) of the incident light, for an unknown metal. ($1\text{eV} = 1.6022 \times 10^{-19} \text{ J}$)

Based on the plot, answer the questions **a to c** below:

a. 1 point On the plot above, label the **threshold frequency**.

b. 2 points What is the value of the work potential (in J) of the unknown metal?

c. 3 points What is the *longest* wavelength (in nm) required to eject an electron from this metal's surface? Show your work including equation(s) used. (3 sig figs)

d. 2 points In the space provided below, plot the following

(Assume frequency is *above* threshold frequency):

- i. *Kinetic energy* of an ejected electron versus the *intensity* of the incident light (left)
- ii. *Number of electrons* ejected versus the *frequency* of the incident light (right)



**Question 3** **6 points****a. 2 points**

In the ground state of Ge (Germanium; $Z = 32$) atom, **name** the following orbitals:

i. Orbital containing the electron that requires the maximum energy to be removed.

ii. Orbital containing the electron that experiences the minimum effective nuclear charge.

b. 2 points

An unknown anion (A^-) has the following *excited* electronic configuration: $[Ar] 4s^2 3d^3 4p^1$

(Ar: Argon; $Z = 18$)

Give the **condensed ground state electronic configuration** of the atom, **A** and the cation, **A^{2+}** .
Show your work. (Assume all electronic transitions are possible)

Electronic Configuration of A:

Electronic Configuration of A^{2+} :

c. 2 points

In a single atom, determine how many electrons can have the following set of quantum numbers?

(i) $n = 5; l = 3$

In a single atom, determine how many electrons can have the following set of quantum numbers?

(i) $n = 3; l = 2; m_s = +1/2$

**Question 4** **8 points**

Rank the following from 1 to 4 in **increasing** order. Provide the ranking in the boxes provided.

Note: Atomic number (Z) are provided for relevant atoms

a) Increasing (lowest to highest) order of *ionization energy*: (1= lowest; 4 = highest)

Mg

Mg: Z = 12

Rb

Rb: Z = 37

Cs

Cs: Z = 55

K

K : Z = 19

b) Increasing (lowest to highest) order of *bond angle*: (1= lowest; 4 = highest)

H₂OCCl₄XeF₄

HCN

c) Increasing (smallest to largest) order of *ionic size* (1= smallest; 4 = largest)

S²⁻

S: Z = 16

O²⁻

O: Z = 8

Be²⁺

Be: Z = 4

Na⁺

Na: Z = 11

d) Increasing (lowest to highest) order of *bond energy* (1= lowest; 4 = highest)

P-F

P-Cl

P-I

P-Br

**Question 5** **12 points**

a. 9 points For the molecules HClO_4 and TeF_6 , draw the most stable *Lewis structure* showing appropriate shape and *dashed/wedged bonds*, based on VSEPR. Determine the **molecular geometry** around the central atom. For HClO_4 , label the *most polar bond*.

HClO_4 (there are no O-O bonds)

Label the most polar bond

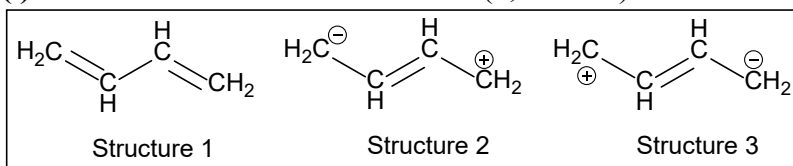
Molecular Geometry around Cl

TeF_6

Molecular Geometry around Te

b. 3 points (Note: Lone pairs are not shown in structures below)

(i) For the three *resonance* structures (1, 2 and 3) shown below, answer the following question:



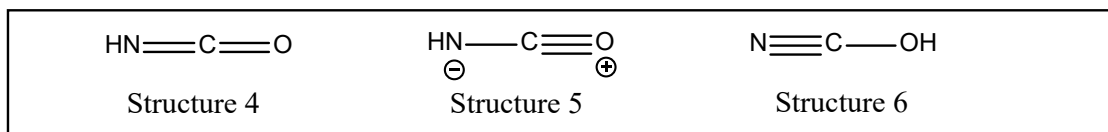
(I) Circle the *most contributing* resonance structure.

Structure 1

Structure 2

Structure 3

(ii) For the three structures given (4, 5, and 6) below, answer the following questions:



(I) Circle all the structures that represent *resonance structures* of each other:

Structure 4

Structure 5

Structure 6

(II) Among the resonance structures, circle the *most contributing* resonance structure.

Structure 4

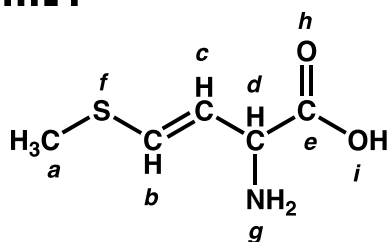
Structure 5

Structure 6

**Question 6** **6 points**

Using the data given below, calculate the **lattice energy** for SrO(s).

	ΔH° (kJ/mol)
$\text{Sr}(s) \longrightarrow \text{Sr}(g)$	+180
$\text{Sr}(g) \longrightarrow \text{Sr}^{2+}(g) + 2e^-$	+1468
$\text{O}_2(g) \longrightarrow 2\text{O}(g)$	+500
$\text{O}(g) + 2e^- \longrightarrow \text{O}^{2-}(g)$	+600
$\text{Sr}(s) + \frac{1}{2}\text{O}_2(g) \longrightarrow \text{SrO}(s)$	-550

**Question 7 10 points**

Consider the molecule given on the left, and answer questions (i) to (vi) using chemical bonding theories that we have learnt in class (valence bond theory, hybridization).

Note that the lone pairs are not shown in the structure:

i. Consider the carbon atoms marked in as ***a***, ***b***, ***c***, ***d***, and ***e*** in the molecule, and determine **hybridization** at each of the carbon centers. (2.5 points)

Atom	Hybridization
Carbon (<i>a</i>)	
Carbon (<i>b</i>)	
Carbon (<i>c</i>)	
Carbon (<i>d</i>)	
Carbon (<i>e</i>)	

ii. Now, consider the sulfur atom (S) marked as ***f***, nitrogen (N) atom (***g***), and oxygen(O) atoms (***h*** and ***i***), and determine their **hybridization**. (2 points)

Atom	Hybridization
Sulfur (<i>f</i>)	
Nitrogen (<i>g</i>)	
Oxygen (<i>h</i>)	
Oxygen (<i>i</i>)	

iii. **Describe** all the bonds formed by the carbon atoms ***a*** and ***b***, using sigma and pi designation as appropriate. (2 points)

Atom	Bonds formed (use sigma and pi as appropriate)
Carbon (<i>a</i>)	
Carbon (<i>b</i>)	

iv. How many total sp^3 -s sigma bonds are in the molecule? (1 point)

v. Determine **electronic** and **molecular geometry** at carbon (C) labeled ***e*** and sulfur (S) labeled ***f*** centres. (2 points)

Atom	Electronic Geometry	Molecular Geometry
Carbon (<i>e</i>)		
Sulfur (<i>f</i>)		

vi. Circle the value from the choices given below, that describes the **C-S-C bond angle at sulfur** (S) marked ***f***: (0.5 points)

$>90^\circ$ to $<109.5^\circ$ 109.5° >109.5 to $<120^\circ$ 120° $>120^\circ$ to $<180^\circ$


Question 8 12 points

Using the **valence shell orbitals only** draw a **complete** molecular orbital (MO) diagram (with no s-p mixing) for the molecule **PS**, in the space provided below. **(6 points)**

Assume that 3s and 3p atomic orbitals will form molecular orbitals like those formed by the combination of 2s and 2p atomic orbitals that we learnt in class, and there is no s-p mixing.

Show **energy arrow**, **atomic orbitals** involved, **atom labels**, the **molecular orbitals** that are formed, **filled electrons** in the MO diagram etc. Make sure to label all the atomic and molecular orbitals.

Using the MO diagram above, answer the following questions:

- (i) Which one of the following will have the highest bond order? Circle the correct choice
 A) PS^{1-} B) PS C) PS^{1+}
- (ii) Which one of the following will have the longest bond length? Circle the correct choice
 A) PS^{1-} B) PS C) PS^{1+}
- (iii) Which of the following will be diamagnetic? Circle the correct choice.
 A) PS^{1-} B) PS C) PS^{1+}
- (iv) Draw Lewis dot structure of the molecule with the shortest bond length (based on the MO diagram) from among the three (PS^{1-} , PS, PS^{1+}), and indicate the bond order calculated from the Lewis dot structure. **(3 points)**

Lewis Dot Structure

Bond Order



23B6DF79-6EF1-4AE2-A9FC-0D1D4DC00359

midterm-exam-93388

#496 Page 10 of 10

SCRAP PAGE

**CHEMISTRY 110 –Midterm Exam****Thursday November 7, 2024, 6:30 – 8:30 pm****Examiners: Pallavi Sirjoosingh and Ashok Kakkar**

First name (please write as legibly as possible within the boxes)

--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--

Last name

--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--

Student ID number

--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--

This booklet has 8 questions on 10 pages (The last page can be used as a scrap page).

The staple must NOT be removed from the question booklet. Your intact question booklet must be returned at the end of the exam.

A separate booklet contains a periodic table, and datasheet.

Write your answers directly under the questions in the space provided. Do not write on the QR codes or near the top edges (as it may be cut off).

Please write **legibly and dark enough** for the scan to be readable. For students who use English as a second language, dictionaries are permitted. Calculators are allowed.

DO NOT ASK QUESTIONS OF THE INVIGILATORS about interpretations of the exam questions. If you have a concern about a question, please summarize it at the top of this page. All concerns will be evaluated after the exam. GOOD LUCK!

**Question 1 8 points** (*Assume all transitions are possible*)

a. 6 points Calculate the **energy (in kJ/mol)** required to ionize (completely remove) the electron from the **n=3** level of **He⁺ ion** (He: Helium). Show your work including equations. (3 sig. figs.)

b. 2 points

(I) Select the largest (longest) wavelength from the three choices below:

- i. Wavelength of photon to *completely remove* an electron from *1s orbital of Hydrogen atom*
- ii. Wavelength of photon to *completely remove* an electron from *1s orbital of Boron atom (B)*
- iii. Wavelength of photon to *completely remove* an electron from *1s orbital of Boron cation (B⁺)*

i or ii or iii

(II) Select the smallest (shortest) wavelength from the three choices below (*for H atom*):

- i. Wavelength of photon *emitted* for an electron transitioning from *2p to 1s orbital*
- ii. Wavelength of photon *absorbed* for an electron transitioning from *2s to 4p orbital*
- iii. Wavelength of photon *absorbed* for an electron transitioning from *1s to 3p orbital*

i or ii or iii



Question 2 8 points



A Chem110 student plots the graph shown on the left, from data obtained for the kinetic energy (eV) of the ejected electron versus frequency (s^{-1}) of the incident light, for an unknown metal. ($1\text{eV} = 1.6022 \times 10^{-19} \text{ J}$)

Based on the plot, answer the questions **a to c** below:

a. 1 point On the plot above, label the **threshold frequency**.

b. 2 points What is the value of the work potential (in J) of the unknown metal?

c. 3 points What is the *longest* wavelength (in nm) required to eject an electron from this metal's surface? Show your work including equation(s) used. (3 sig figs)

d. 2 points In the space provided below, plot the following

(Assume frequency is *above* threshold frequency):

- Kinetic energy* of an ejected electron versus the *intensity* of the incident light (left)
- Number of electrons* ejected versus the *frequency* of the incident light (right)



**Question 3** **6 points****a. 2 points**

In the ground state of Ge (Germanium; $Z = 32$) atom, **name** the following orbitals:

i. Orbital containing the electron that requires the maximum energy to be removed.

ii. Orbital containing the electron that experiences the minimum effective nuclear charge.

b. 2 points

An unknown anion (A^-) has the following *excited* electronic configuration: $[Ar] 4s^2 3d^3 4p^1$

(Ar: Argon; $Z = 18$)

Give the **condensed ground state electronic configuration** of the atom, **A** and the cation, **A^{2+}** .
Show your work. (Assume all electronic transitions are possible)

Electronic Configuration of A:

Electronic Configuration of A^{2+} :

c. 2 points

In a single atom, determine how many electrons can have the following set of quantum numbers?

(i) $n = 5; l = 3$

In a single atom, determine how many electrons can have the following set of quantum numbers?

(i) $n = 3; l = 2; m_s = +1/2$

**Question 4** **8 points**

Rank the following from 1 to 4 in **increasing** order. Provide the ranking in the boxes provided.

Note: Atomic number (Z) are provided for relevant atoms

a) Increasing (lowest to highest) order of *ionization energy*: (1= lowest; 4 = highest)

Mg

Mg: Z = 12

Rb

Rb: Z = 37

Cs

Cs: Z = 55

K

K : Z = 19

b) Increasing (lowest to highest) order of *bond angle*: (1= lowest; 4 = highest)

H₂OCCl₄XeF₄

HCN

c) Increasing (smallest to largest) order of *ionic size* (1= smallest; 4 = largest)

S²⁻

S: Z = 16

O²⁻

O: Z = 8

Be²⁺

Be: Z = 4

Na⁺

Na: Z = 11

d) Increasing (lowest to highest) order of *bond energy* (1= lowest; 4 = highest)

P-F

P-Cl

P-I

P-Br

**Question 5** **12 points**

a. 9 points For the molecules HClO_4 and TeF_6 , draw the most stable *Lewis structure* showing appropriate shape and *dashed/wedged bonds*, based on VSEPR. Determine the **molecular geometry** around the central atom. For HClO_4 , label the *most polar bond*.



b. 3 points (Note: Lone pairs are not shown in structures below)

(i) For the three *resonance* structures (1, 2 and 3) shown below, answer the following question:



(I) Circle the *most contributing* resonance structure.

Structure 1

Structure 2

Structure 3

(ii) For the three structures given (4, 5, and 6) below, answer the following questions:



(I) Circle all the structures that represent *resonance structures* of each other:

Structure 4

Structure 5

Structure 6

(II) Among the resonance structures, circle the *most contributing* resonance structure.

Structure 4

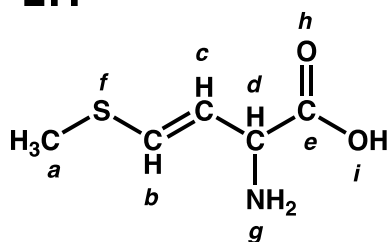
Structure 5

Structure 6

**Question 6** **6 points**

Using the data given below, calculate the **lattice energy** for SrO(s).

	ΔH° (kJ/mol)
$\text{Sr}(s) \longrightarrow \text{Sr}(g)$	+180
$\text{Sr}(g) \longrightarrow \text{Sr}^{2+}(g) + 2e^-$	+1468
$\text{O}_2(g) \longrightarrow 2\text{O}(g)$	+500
$\text{O}(g) + 2e^- \longrightarrow \text{O}^{2-}(g)$	+600
$\text{Sr}(s) + \frac{1}{2}\text{O}_2(g) \longrightarrow \text{SrO}(s)$	-550

**Question 7 10 points**

Consider the molecule given on the left, and answer questions (i) to (vi) using chemical bonding theories that we have learnt in class (valence bond theory, hybridization).

Note that the lone pairs are not shown in the structure:

i. Consider the carbon atoms marked in as ***a***, ***b***, ***c***, ***d***, and ***e*** in the molecule, and determine **hybridization** at each of the carbon centers. (2.5 points)

Atom	Hybridization
Carbon (<i>a</i>)	
Carbon (<i>b</i>)	
Carbon (<i>c</i>)	
Carbon (<i>d</i>)	
Carbon (<i>e</i>)	

ii. Now, consider the sulfur atom (S) marked as ***f***, nitrogen (N) atom (***g***), and oxygen(O) atoms (***h*** and ***i***), and determine their **hybridization**. (2 points)

Atom	Hybridization
Sulfur (<i>f</i>)	
Nitrogen (<i>g</i>)	
Oxygen (<i>h</i>)	
Oxygen (<i>i</i>)	

iii. **Describe** all the bonds formed by the carbon atoms ***a*** and ***b***, using sigma and pi designation as appropriate. (2 points)

Atom	Bonds formed (use sigma and pi as appropriate)
Carbon (<i>a</i>)	
Carbon (<i>b</i>)	

iv. How many total sp^3 -s sigma bonds are in the molecule? (1 point)

v. Determine **electronic** and **molecular geometry** at carbon (C) labeled ***e*** and sulfur (S) labeled ***f*** centres. (2 points)

Atom	Electronic Geometry	Molecular Geometry
Carbon (<i>e</i>)		
Sulfur (<i>f</i>)		

vi. Circle the value from the choices given below, that describes the **C-S-C bond angle at sulfur** (S) marked ***f***: (0.5 points)

>90° to <109.5° 109.5° >109.5 to <120° 120° >120° to <180°


Question 8 12 points

Using the **valence shell orbitals only** draw a **complete** molecular orbital (MO) diagram (with no s-p mixing) for the molecule **PS**, in the space provided below. **(6 points)**

Assume that 3s and 3p atomic orbitals will form molecular orbitals like those formed by the combination of 2s and 2p atomic orbitals that we learnt in class, and there is no s-p mixing.

Show **energy arrow**, **atomic orbitals** involved, **atom labels**, the **molecular orbitals** that are formed, **filled electrons** in the MO diagram etc. Make sure to label all the atomic and molecular orbitals.

Using the MO diagram above, answer the following questions:

- (i) Which one of the following will have the highest bond order? Circle the correct choice
 A) PS^{1-} B) PS C) PS^{1+}
- (ii) Which one of the following will have the longest bond length? Circle the correct choice
 A) PS^{1-} B) PS C) PS^{1+}
- (iii) Which of the following will be diamagnetic? Circle the correct choice.
 A) PS^{1-} B) PS C) PS^{1+}
- (iv) Draw Lewis dot structure of the molecule with the shortest bond length (based on the MO diagram) from among the three (PS^{1-} , PS, PS^{1+}), and indicate the bond order calculated from the Lewis dot structure. **(3 points)**

Lewis Dot Structure

Bond Order



0E614BB3-D42D-4901-8AD4-70BA4D21387C

midterm-exam-93388

#497 Page 10 of 10

SCRAP PAGE

**CHEMISTRY 110 –Midterm Exam****Thursday November 7, 2024, 6:30 – 8:30 pm**

Examiners: Pallavi Sirjoosingh and Ashok Kakkar

First name (please write as legibly as possible within the boxes)

--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--

Last name

--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--

Student ID number

--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--

This booklet has 8 questions on 10 pages (The last page can be used as a scrap page).

The staple must NOT be removed from the question booklet. Your intact question booklet must be returned at the end of the exam.

A separate booklet contains a periodic table, and datasheet.

Write your answers directly under the questions in the space provided. Do not write on the QR codes or near the top edges (as it may be cut off).

Please write **legibly and dark enough** for the scan to be readable. For students who use English as a second language, dictionaries are permitted. Calculators are allowed.

DO NOT ASK QUESTIONS OF THE INVIGILATORS about interpretations of the exam questions. If you have a concern about a question, please summarize it at the top of this page. All concerns will be evaluated after the exam. GOOD LUCK!

**Question 1 8 points** (*Assume all transitions are possible*)

a. 6 points Calculate the **energy (in kJ/mol)** required to ionize (completely remove) the electron from the **$n=3$** level of **He^+ ion** (He: Helium). Show your work including equations. (3 sig. figs.)

b. 2 points

(I) Select the largest (longest) wavelength from the three choices below:

- i. Wavelength of photon to *completely remove* an electron from *1s orbital of Hydrogen atom*
- ii. Wavelength of photon to *completely remove* an electron from *1s orbital of Boron atom (B)*
- iii. Wavelength of photon to *completely remove* an electron from *1s orbital of Boron cation (B^+)*

i or ii or iii

(II) Select the smallest (shortest) wavelength from the three choices below (*for H atom*):

- i. Wavelength of photon *emitted* for an electron transitioning from *2p to 1s orbital*
- ii. Wavelength of photon *absorbed* for an electron transitioning from *2s to 4p orbital*
- iii. Wavelength of photon *absorbed* for an electron transitioning from *1s to 3p orbital*

i or ii or iii



Question 2 8 points



A Chem110 student plots the graph shown on the left, from data obtained for the kinetic energy (eV) of the ejected electron versus frequency (s^{-1}) of the incident light, for an unknown metal. ($1\text{eV} = 1.6022 \times 10^{-19} \text{ J}$)

Based on the plot, answer the questions **a to c** below:

a. 1 point On the plot above, label the **threshold frequency**.

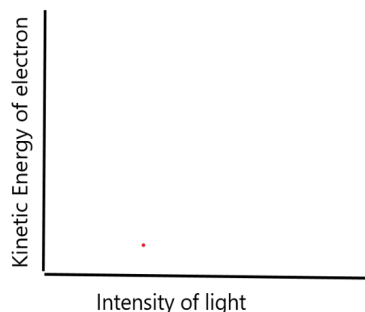
b. 2 points What is the value of the work potential (in J) of the unknown metal?

c. 3 points What is the *longest* wavelength (in nm) required to eject an electron from this metal's surface? Show your work including equation(s) used. (3 sig figs)

d. 2 points In the space provided below, plot the following

(Assume frequency is *above* threshold frequency):

- Kinetic energy* of an ejected electron versus the *intensity* of the incident light (left)
- Number of electrons* ejected versus the *frequency* of the incident light (right)



**Question 3** **6 points****a. 2 points**

In the ground state of Ge (Germanium; $Z = 32$) atom, **name** the following orbitals:

i. Orbital containing the electron that requires the maximum energy to be removed.

ii. Orbital containing the electron that experiences the minimum effective nuclear charge.

b. 2 points

An unknown anion (A^-) has the following *excited* electronic configuration: $[Ar] 4s^2 3d^3 4p^1$

(Ar: Argon; $Z = 18$)

Give the **condensed ground state electronic configuration** of the atom, **A** and the cation, **A^{2+}** .
Show your work. (Assume all electronic transitions are possible)

Electronic Configuration of A:

Electronic Configuration of A^{2+} :

c. 2 points

In a single atom, determine how many electrons can have the following set of quantum numbers?

(i) $n = 5; l = 3$

In a single atom, determine how many electrons can have the following set of quantum numbers?

(i) $n = 3; l = 2; m_s = +1/2$

**Question 4** **8 points**

Rank the following from 1 to 4 in **increasing** order. Provide the ranking in the boxes provided.

Note: Atomic number (Z) are provided for relevant atoms

a) Increasing (lowest to highest) order of *ionization energy*: (1= lowest; 4 = highest)

Mg

Mg: Z = 12

Rb

Rb: Z = 37

Cs

Cs: Z = 55

K

K : Z = 19

b) Increasing (lowest to highest) order of *bond angle*: (1= lowest; 4 = highest)

H₂OCCl₄XeF₄

HCN

c) Increasing (smallest to largest) order of *ionic size* (1= smallest; 4 = largest)

S²⁻

S: Z = 16

O²⁻

O: Z = 8

Be²⁺

Be: Z = 4

Na⁺

Na: Z = 11

d) Increasing (lowest to highest) order of *bond energy* (1= lowest; 4 = highest)

P-F

P-Cl

P-I

P-Br

**Question 5** **12 points**

a. 9 points For the molecules HClO_4 and TeF_6 , draw the most stable *Lewis structure* showing appropriate shape and *dashed/wedged bonds*, based on VSEPR. Determine the **molecular geometry** around the central atom. For HClO_4 , label the *most polar bond*.

HClO_4 (there are no O-O bonds)

Label the most polar bond

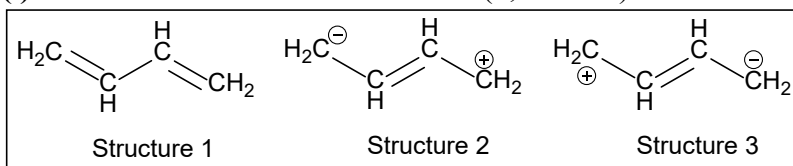
Molecular Geometry around Cl

TeF_6

Molecular Geometry around Te

b. 3 points (Note: Lone pairs are not shown in structures below)

(i) For the three *resonance* structures (1, 2 and 3) shown below, answer the following question:



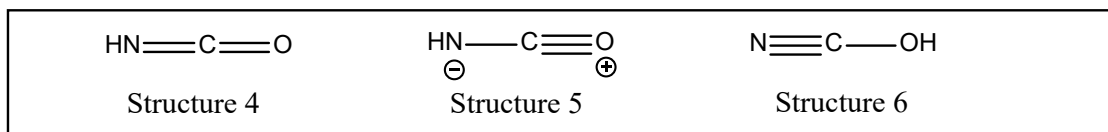
(I) Circle the *most contributing* resonance structure.

Structure 1

Structure 2

Structure 3

(ii) For the three structures given (4, 5, and 6) below, answer the following questions:



(I) Circle all the structures that represent *resonance structures* of each other:

Structure 4

Structure 5

Structure 6

(II) Among the resonance structures, circle the *most contributing* resonance structure.

Structure 4

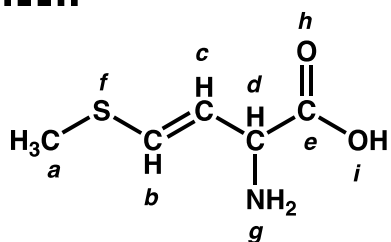
Structure 5

Structure 6

**Question 6** **6 points**

Using the data given below, calculate the **lattice energy** for SrO(s).

	ΔH° (kJ/mol)
$\text{Sr}(s) \longrightarrow \text{Sr}(g)$	+180
$\text{Sr}(g) \longrightarrow \text{Sr}^{2+}(g) + 2e^-$	+1468
$\text{O}_2(g) \longrightarrow 2\text{O}(g)$	+500
$\text{O}(g) + 2e^- \longrightarrow \text{O}^{2-}(g)$	+600
$\text{Sr}(s) + \frac{1}{2}\text{O}_2(g) \longrightarrow \text{SrO}(s)$	-550

**Question 7 10 points**

Consider the molecule given on the left, and answer questions (i) to (vi) using chemical bonding theories that we have learnt in class (valence bond theory, hybridization).

Note that the lone pairs are not shown in the structure:

i. Consider the carbon atoms marked in as ***a***, ***b***, ***c***, ***d***, and ***e*** in the molecule, and determine **hybridization** at each of the carbon centers. (2.5 points)

Atom	Hybridization
Carbon (<i>a</i>)	
Carbon (<i>b</i>)	
Carbon (<i>c</i>)	
Carbon (<i>d</i>)	
Carbon (<i>e</i>)	

ii. Now, consider the sulfur atom (S) marked as ***f***, nitrogen (N) atom (***g***), and oxygen(O) atoms (***h*** and ***i***), and determine their **hybridization**. (2 points)

Atom	Hybridization
Sulfur (<i>f</i>)	
Nitrogen (<i>g</i>)	
Oxygen (<i>h</i>)	
Oxygen (<i>i</i>)	

iii. **Describe** all the bonds formed by the carbon atoms ***a*** and ***b***, using sigma and pi designation as appropriate. (2 points)

Atom	Bonds formed (use sigma and pi as appropriate)
Carbon (<i>a</i>)	
Carbon (<i>b</i>)	

iv. How many total sp^3 -s sigma bonds are in the molecule? (1 point)

v. Determine **electronic** and **molecular geometry** at carbon (C) labeled ***e*** and sulfur (S) labeled ***f*** centres. (2 points)

Atom	Electronic Geometry	Molecular Geometry
Carbon (<i>e</i>)		
Sulfur (<i>f</i>)		

vi. Circle the value from the choices given below, that describes the **C-S-C bond angle at sulfur** (S) marked ***f***: (0.5 points)

>90° to <109.5°

109.5°

>109.5 to <120°

120°

>120° to <180°

**Question 8 12 points**

Using the **valence shell orbitals only** draw a **complete** molecular orbital (MO) diagram (with no s-p mixing) for the molecule **PS**, in the space provided below. **(6 points)**

Assume that 3s and 3p atomic orbitals will form molecular orbitals like those formed by the combination of 2s and 2p atomic orbitals that we learnt in class, and there is no s-p mixing.

Show **energy arrow**, **atomic orbitals** involved, **atom labels**, the **molecular orbitals** that are formed, **filled electrons** in the MO diagram etc. Make sure to label all the atomic and molecular orbitals.

Using the MO diagram above, answer the following questions:

- (i) Which one of the following will have the highest bond order? Circle the correct choice
A) PS^{1-} B) PS C) PS^{1+}
- (ii) Which one of the following will have the longest bond length? Circle the correct choice
A) PS^{1-} B) PS C) PS^{1+}
- (iii) Which of the following will be diamagnetic? Circle the correct choice.
A) PS^{1-} B) PS C) PS^{1+}
- (iv) Draw Lewis dot structure of the molecule with the shortest bond length (based on the MO diagram) from among the three (PS^{1-} , PS, PS^{1+}), and indicate the bond order calculated from the Lewis dot structure. **(3 points)**

Lewis Dot Structure

Bond Order



56A0F140-C377-4CAB-A2CE-AD1852C6A6CB

midterm-exam-93388

#498 Page 10 of 10

SCRAP PAGE

**CHEMISTRY 110 –Midterm Exam****Thursday November 7, 2024, 6:30 – 8:30 pm****Examiners: Pallavi Sirjoosingh and Ashok Kakkar**

First name (please write as legibly as possible within the boxes)

--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--

Last name

--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--

Student ID number

--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--

This booklet has 8 questions on 10 pages (The last page can be used as a scrap page).

The staple must NOT be removed from the question booklet. Your intact question booklet must be returned at the end of the exam.

A separate booklet contains a periodic table, and datasheet.

Write your answers directly under the questions in the space provided. Do not write on the QR codes or near the top edges (as it may be cut off).

Please write **legibly and dark enough** for the scan to be readable. For students who use English as a second language, dictionaries are permitted. Calculators are allowed.

DO NOT ASK QUESTIONS OF THE INVIGILATORS about interpretations of the exam questions. If you have a concern about a question, please summarize it at the top of this page. All concerns will be evaluated after the exam. GOOD LUCK!

**Question 1 8 points** (*Assume all transitions are possible*)

a. 6 points Calculate the **energy (in kJ/mol)** required to ionize (completely remove) the electron from the **$n=3$** level of **He^+ ion** (He: Helium). Show your work including equations. (3 sig. figs.)

b. 2 points

(I) Select the largest (longest) wavelength from the three choices below:

- i. Wavelength of photon to *completely remove* an electron from *1s orbital of Hydrogen atom*
- ii. Wavelength of photon to *completely remove* an electron from *1s orbital of Boron atom (B)*
- iii. Wavelength of photon to *completely remove* an electron from *1s orbital of Boron cation (B^+)*

i or ii or iii

(II) Select the smallest (shortest) wavelength from the three choices below (*for H atom*):

- i. Wavelength of photon *emitted* for an electron transitioning from *2p to 1s orbital*
- ii. Wavelength of photon *absorbed* for an electron transitioning from *2s to 4p orbital*
- iii. Wavelength of photon *absorbed* for an electron transitioning from *1s to 3p orbital*

i or ii or iii



Question 2 8 points



A Chem110 student plots the graph shown on the left, from data obtained for the kinetic energy (eV) of the ejected electron versus frequency (s^{-1}) of the incident light, for an unknown metal. ($1\text{eV} = 1.6022 \times 10^{-19} \text{ J}$)

Based on the plot, answer the questions **a to c** below:

a. 1 point On the plot above, label the **threshold frequency**.

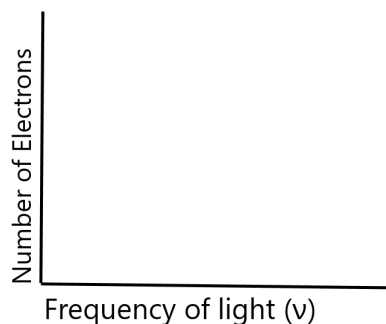
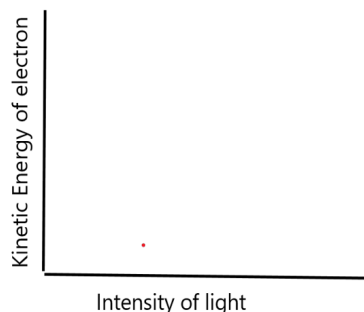
b. 2 points What is the value of the work potential (in J) of the unknown metal?

c. 3 points What is the *longest* wavelength (in nm) required to eject an electron from this metal's surface? Show your work including equation(s) used. (3 sig figs)

d. 2 points In the space provided below, plot the following

(Assume frequency is *above* threshold frequency):

- i. *Kinetic energy* of an ejected electron versus the *intensity* of the incident light (left)
- ii. *Number of electrons* ejected versus the *frequency* of the incident light (right)



**Question 3** **6 points****a. 2 points**

In the ground state of Ge (Germanium; $Z = 32$) atom, **name** the following orbitals:

i. Orbital containing the electron that requires the maximum energy to be removed.

ii. Orbital containing the electron that experiences the minimum effective nuclear charge.

b. 2 points

An unknown anion (A^-) has the following *excited* electronic configuration: $[Ar] 4s^2 3d^3 4p^1$

(Ar: Argon; $Z = 18$)

Give the **condensed ground state electronic configuration** of the atom, **A** and the cation, **A^{2+}** .
Show your work. (Assume all electronic transitions are possible)

Electronic Configuration of A:

Electronic Configuration of A^{2+} :

c. 2 points

In a single atom, determine how many electrons can have the following set of quantum numbers?

(i) $n = 5; l = 3$

In a single atom, determine how many electrons can have the following set of quantum numbers?

(i) $n = 3; l = 2; m_s = +1/2$

**Question 4** **8 points**

Rank the following from 1 to 4 in **increasing** order. Provide the ranking in the boxes provided.

Note: Atomic number (Z) are provided for relevant atoms

a) Increasing (lowest to highest) order of *ionization energy*: (1= lowest; 4 = highest)

Mg

Mg: Z = 12

Rb

Rb: Z = 37

Cs

Cs: Z = 55

K

K : Z = 19

b) Increasing (lowest to highest) order of *bond angle*: (1= lowest; 4 = highest)

H₂OCCl₄XeF₄

HCN

c) Increasing (smallest to largest) order of *ionic size* (1= smallest; 4 = largest)

S²⁻

S: Z = 16

O²⁻

O: Z = 8

Be²⁺

Be: Z = 4

Na⁺

Na: Z = 11

d) Increasing (lowest to highest) order of *bond energy* (1= lowest; 4 = highest)

P-F

P-Cl

P-I

P-Br

**Question 5** **12 points**

a. 9 points For the molecules HClO_4 and TeF_6 , draw the most stable *Lewis structure* showing appropriate shape and *dashed/wedged bonds*, based on VSEPR. Determine the **molecular geometry** around the central atom. For HClO_4 , label the *most polar bond*.



b. 3 points (Note: Lone pairs are not shown in structures below)

(i) For the three *resonance* structures (1, 2 and 3) shown below, answer the following question:



(I) Circle the *most contributing* resonance structure.

Structure 1

Structure 2

Structure 3

(ii) For the three structures given (4, 5, and 6) below, answer the following questions:



(I) Circle all the structures that represent *resonance structures* of each other:

Structure 4

Structure 5

Structure 6

(II) Among the resonance structures, circle the *most contributing* resonance structure.

Structure 4

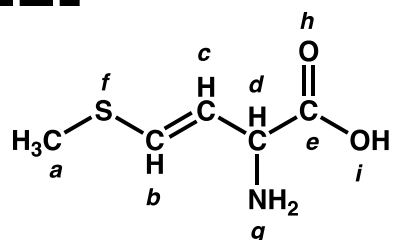
Structure 5

Structure 6

**Question 6** **6 points**

Using the data given below, calculate the **lattice energy** for SrO(s).

	ΔH° (kJ/mol)
$\text{Sr}(s) \longrightarrow \text{Sr}(g)$	+180
$\text{Sr}(g) \longrightarrow \text{Sr}^{2+}(g) + 2e^-$	+1468
$\text{O}_2(g) \longrightarrow 2\text{O}(g)$	+500
$\text{O}(g) + 2e^- \longrightarrow \text{O}^{2-}(g)$	+600
$\text{Sr}(s) + \frac{1}{2}\text{O}_2(g) \longrightarrow \text{SrO}(s)$	-550

**Question 7 10 points**

Consider the molecule given on the left, and answer questions (i) to (vi) using chemical bonding theories that we have learnt in class (valence bond theory, hybridization).

Note that the lone pairs are not shown in the structure:

i. Consider the carbon atoms marked in as ***a***, ***b***, ***c***, ***d***, and ***e*** in the molecule, and determine **hybridization** at each of the carbon centers. (2.5 points)

Atom	Hybridization
Carbon (<i>a</i>)	
Carbon (<i>b</i>)	
Carbon (<i>c</i>)	
Carbon (<i>d</i>)	
Carbon (<i>e</i>)	

ii. Now, consider the sulfur atom (S) marked as ***f***, nitrogen (N) atom (***g***), and oxygen(O) atoms (***h*** and ***i***), and determine their **hybridization**. (2 points)

Atom	Hybridization
Sulfur (<i>f</i>)	
Nitrogen (<i>g</i>)	
Oxygen (<i>h</i>)	
Oxygen (<i>i</i>)	

iii. **Describe** all the bonds formed by the carbon atoms ***a*** and ***b***, using sigma and pi designation as appropriate. (2 points)

Atom	Bonds formed (use sigma and pi as appropriate)
Carbon (<i>a</i>)	
Carbon (<i>b</i>)	

iv. How many total sp^3 -s sigma bonds are in the molecule? (1 point)

v. Determine **electronic** and **molecular geometry** at carbon (C) labeled ***e*** and sulfur (S) labeled ***f*** centres. (2 points)

Atom	Electronic Geometry	Molecular Geometry
Carbon (<i>e</i>)		
Sulfur (<i>f</i>)		

vi. Circle the value from the choices given below, that describes the **C-S-C bond angle at sulfur** (S) marked ***f***: (0.5 points)

>90° to <109.5° 109.5° >109.5 to <120° 120° >120° to <180°


Question 8 12 points

Using the **valence shell orbitals only** draw a **complete** molecular orbital (MO) diagram (with no s-p mixing) for the molecule **PS**, in the space provided below. **(6 points)**

Assume that 3s and 3p atomic orbitals will form molecular orbitals like those formed by the combination of 2s and 2p atomic orbitals that we learnt in class, and there is no s-p mixing.

Show **energy arrow**, **atomic orbitals** involved, **atom labels**, the **molecular orbitals** that are formed, **filled electrons** in the MO diagram etc. Make sure to label all the atomic and molecular orbitals.

Using the MO diagram above, answer the following questions:

- (i) Which one of the following will have the highest bond order? Circle the correct choice
 A) PS^{1-} B) PS C) PS^{1+}
- (ii) Which one of the following will have the longest bond length? Circle the correct choice
 A) PS^{1-} B) PS C) PS^{1+}
- (iii) Which of the following will be diamagnetic? Circle the correct choice.
 A) PS^{1-} B) PS C) PS^{1+}
- (iv) Draw Lewis dot structure of the molecule with the shortest bond length (based on the MO diagram) from among the three (PS^{1-} , PS, PS^{1+}), and indicate the bond order calculated from the Lewis dot structure. **(3 points)**

Lewis Dot Structure

Bond Order



16568F79-CF14-4571-A840-D7E9AD989217

midterm-exam-93388

#499 Page 10 of 10

SCRAP PAGE

**CHEMISTRY 110 –Midterm Exam****Thursday November 7, 2024, 6:30 – 8:30 pm**

Examiners: Pallavi Sirjoosingh and Ashok Kakkar

First name (please write as legibly as possible within the boxes)

--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--

Last name

--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--

Student ID number

--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--

This booklet has 8 questions on 10 pages (The last page can be used as a scrap page).

The staple must NOT be removed from the question booklet. Your intact question booklet must be returned at the end of the exam.

A separate booklet contains a periodic table, and datasheet.

Write your answers directly under the questions in the space provided. Do not write on the QR codes or near the top edges (as it may be cut off).

Please write **legibly and dark enough** for the scan to be readable. For students who use English as a second language, dictionaries are permitted. Calculators are allowed.

DO NOT ASK QUESTIONS OF THE INVIGILATORS about interpretations of the exam questions. If you have a concern about a question, please summarize it at the top of this page. All concerns will be evaluated after the exam. GOOD LUCK!

**Question 1 8 points** (*Assume all transitions are possible*)

a. 6 points Calculate the **energy (in kJ/mol)** required to ionize (completely remove) the electron from the **$n=3$** level of **He^+ ion** (He: Helium). Show your work including equations. (3 sig. figs.)

b. 2 points

(I) Select the largest (longest) wavelength from the three choices below:

- i. Wavelength of photon to *completely remove* an electron from *1s orbital of Hydrogen atom*
- ii. Wavelength of photon to *completely remove* an electron from *1s orbital of Boron atom (B)*
- iii. Wavelength of photon to *completely remove* an electron from *1s orbital of Boron cation (B^+)*

i or ii or iii

(II) Select the smallest (shortest) wavelength from the three choices below (*for H atom*):

- i. Wavelength of photon *emitted* for an electron transitioning from *2p to 1s orbital*
- ii. Wavelength of photon *absorbed* for an electron transitioning from *2s to 4p orbital*
- iii. Wavelength of photon *absorbed* for an electron transitioning from *1s to 3p orbital*

i or ii or iii



Question 2 8 points



A Chem110 student plots the graph shown on the left, from data obtained for the kinetic energy (eV) of the ejected electron versus frequency (s^{-1}) of the incident light, for an unknown metal. ($1\text{eV} = 1.6022 \times 10^{-19} \text{ J}$)

Based on the plot, answer the questions **a to c** below:

a. 1 point On the plot above, label the **threshold frequency**.

b. 2 points What is the value of the work potential (in J) of the unknown metal?

c. 3 points What is the *longest* wavelength (in nm) required to eject an electron from this metal's surface? Show your work including equation(s) used. (3 sig figs)

d. 2 points In the space provided below, plot the following

(Assume frequency is *above* threshold frequency):

- Kinetic energy* of an ejected electron versus the *intensity* of the incident light (left)
- Number of electrons* ejected versus the *frequency* of the incident light (right)



**Question 3** **6 points****a. 2 points**

In the ground state of Ge (Germanium; $Z = 32$) atom, **name** the following orbitals:

i. Orbital containing the electron that requires the maximum energy to be removed.

ii. Orbital containing the electron that experiences the minimum effective nuclear charge.

b. 2 points

An unknown anion (A^-) has the following *excited* electronic configuration: $[Ar] 4s^2 3d^3 4p^1$

(Ar: Argon; $Z = 18$)

Give the **condensed ground state electronic configuration** of the atom, **A** and the cation, **A^{2+}** .
Show your work. (Assume all electronic transitions are possible)

Electronic Configuration of A:

Electronic Configuration of A^{2+} :

c. 2 points

In a single atom, determine how many electrons can have the following set of quantum numbers?

(i) $n = 5; l = 3$

In a single atom, determine how many electrons can have the following set of quantum numbers?

(i) $n = 3; l = 2; m_s = +1/2$

**Question 4** **8 points**

Rank the following from 1 to 4 in **increasing** order. Provide the ranking in the boxes provided.

Note: Atomic number (Z) are provided for relevant atoms

a) Increasing (lowest to highest) order of *ionization energy*: (1= lowest; 4 = highest)

Mg

Mg: Z = 12

Rb

Rb: Z = 37

Cs

Cs: Z = 55

K

K : Z = 19

b) Increasing (lowest to highest) order of *bond angle*: (1= lowest; 4 = highest)

H₂OCCl₄XeF₄

HCN

c) Increasing (smallest to largest) order of *ionic size* (1= smallest; 4 = largest)

S²⁻

S: Z = 16

O²⁻

O: Z = 8

Be²⁺

Be: Z = 4

Na⁺

Na: Z = 11

d) Increasing (lowest to highest) order of *bond energy* (1= lowest; 4 = highest)

P-F

P-Cl

P-I

P-Br

**Question 5** **12 points**

a. 9 points For the molecules HClO_4 and TeF_6 , draw the most stable *Lewis structure* showing appropriate shape and *dashed/wedged bonds*, based on VSEPR. Determine the **molecular geometry** around the central atom. For HClO_4 , label the *most polar bond*.



b. 3 points (Note: Lone pairs are not shown in structures below)

(i) For the three *resonance* structures (1, 2 and 3) shown below, answer the following question:



(I) Circle the *most contributing* resonance structure.

Structure 1

Structure 2

Structure 3

(ii) For the three structures given (4, 5, and 6) below, answer the following questions:



(I) Circle all the structures that represent *resonance structures* of each other:

Structure 4

Structure 5

Structure 6

(II) Among the resonance structures, circle the *most contributing* resonance structure.

Structure 4

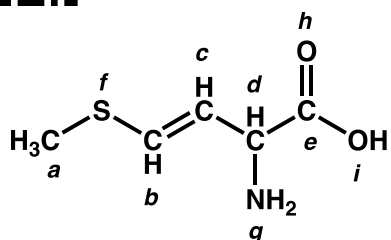
Structure 5

Structure 6

**Question 6** **6 points**

Using the data given below, calculate the **lattice energy** for SrO(s).

	ΔH° (kJ/mol)
$\text{Sr}(s) \longrightarrow \text{Sr}(g)$	+180
$\text{Sr}(g) \longrightarrow \text{Sr}^{2+}(g) + 2e^-$	+1468
$\text{O}_2(g) \longrightarrow 2\text{O}(g)$	+500
$\text{O}(g) + 2e^- \longrightarrow \text{O}^{2-}(g)$	+600
$\text{Sr}(s) + \frac{1}{2}\text{O}_2(g) \longrightarrow \text{SrO}(s)$	-550

**Question 7 10 points**

Consider the molecule given on the left, and answer questions (i) to (vi) using chemical bonding theories that we have learnt in class (valence bond theory, hybridization).

Note that the lone pairs are not shown in the structure:

i. Consider the carbon atoms marked in as ***a***, ***b***, ***c***, ***d***, and ***e*** in the molecule, and determine **hybridization** at each of the carbon centers. (2.5 points)

Atom	Hybridization
Carbon (<i>a</i>)	
Carbon (<i>b</i>)	
Carbon (<i>c</i>)	
Carbon (<i>d</i>)	
Carbon (<i>e</i>)	

ii. Now, consider the sulfur atom (S) marked as ***f***, nitrogen (N) atom (***g***), and oxygen(O) atoms (***h*** and ***i***), and determine their **hybridization**. (2 points)

Atom	Hybridization
Sulfur (<i>f</i>)	
Nitrogen (<i>g</i>)	
Oxygen (<i>h</i>)	
Oxygen (<i>i</i>)	

iii. **Describe** all the bonds formed by the carbon atoms ***a*** and ***b***, using sigma and pi designation as appropriate. (2 points)

Atom	Bonds formed (use sigma and pi as appropriate)
Carbon (<i>a</i>)	
Carbon (<i>b</i>)	

iv. How many total sp^3 -s sigma bonds are in the molecule? (1 point)

v. Determine **electronic** and **molecular geometry** at carbon (C) labeled ***e*** and sulfur (S) labeled ***f*** centres. (2 points)

Atom	Electronic Geometry	Molecular Geometry
Carbon (<i>e</i>)		
Sulfur (<i>f</i>)		

vi. Circle the value from the choices given below, that describes the **C-S-C bond angle at sulfur** (S) marked ***f***: (0.5 points)

>90° to <109.5°

109.5°

>109.5 to <120°

120°

>120° to <180°

**Question 8 12 points**

Using the **valence shell orbitals only** draw a **complete** molecular orbital (MO) diagram (with no s-p mixing) for the molecule **PS**, in the space provided below. **(6 points)**

Assume that 3s and 3p atomic orbitals will form molecular orbitals like those formed by the combination of 2s and 2p atomic orbitals that we learnt in class, and there is no s-p mixing.

Show **energy arrow**, **atomic orbitals** involved, **atom labels**, the **molecular orbitals** that are formed, **filled electrons** in the MO diagram etc. Make sure to label all the atomic and molecular orbitals.

Using the MO diagram above, answer the following questions:

- (i) Which one of the following will have the highest bond order? Circle the correct choice
A) PS^{1-} B) PS C) PS^{1+}
- (ii) Which one of the following will have the longest bond length? Circle the correct choice
A) PS^{1-} B) PS C) PS^{1+}
- (iii) Which of the following will be diamagnetic? Circle the correct choice.
A) PS^{1-} B) PS C) PS^{1+}
- (iv) Draw Lewis dot structure of the molecule with the shortest bond length (based on the MO diagram) from among the three (PS^{1-} , PS, PS^{1+}), and indicate the bond order calculated from the Lewis dot structure. **(3 points)**

Lewis Dot Structure

Bond Order



9B18B9F5-5CDE-475F-96FC-C662709CCAC4

midterm-exam-93388

#500 Page 10 of 10

SCRAP PAGE

**CHEMISTRY 110 –Midterm Exam****Thursday November 7, 2024, 6:30 – 8:30 pm****Examiners: Pallavi Sirjoosingh and Ashok Kakkar**

First name (please write as legibly as possible within the boxes)

--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--

Last name

--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--

Student ID number

--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--

This booklet has 8 questions on 10 pages (The last page can be used as a scrap page).

The staple must NOT be removed from the question booklet. Your intact question booklet must be returned at the end of the exam.

A separate booklet contains a periodic table, and datasheet.

Write your answers directly under the questions in the space provided. Do not write on the QR codes or near the top edges (as it may be cut off).

Please write **legibly and dark enough** for the scan to be readable. For students who use English as a second language, dictionaries are permitted. Calculators are allowed.

DO NOT ASK QUESTIONS OF THE INVIGILATORS about interpretations of the exam questions. If you have a concern about a question, please summarize it at the top of this page. All concerns will be evaluated after the exam. GOOD LUCK!

**Question 1 8 points** (*Assume all transitions are possible*)

a. 6 points Calculate the **energy (in kJ/mol)** required to ionize (completely remove) the electron from the **$n=3$** level of **He^+ ion** (He: Helium). Show your work including equations. (3 sig. figs.)

b. 2 points

(I) Select the largest (longest) wavelength from the three choices below:

- i. Wavelength of photon to *completely remove* an electron from *1s orbital of Hydrogen atom*
- ii. Wavelength of photon to *completely remove* an electron from *1s orbital of Boron atom (B)*
- iii. Wavelength of photon to *completely remove* an electron from *1s orbital of Boron cation (B^+)*

i or ii or iii

(II) Select the smallest (shortest) wavelength from the three choices below (*for H atom*):

- i. Wavelength of photon *emitted* for an electron transitioning from *2p to 1s orbital*
- ii. Wavelength of photon *absorbed* for an electron transitioning from *2s to 4p orbital*
- iii. Wavelength of photon *absorbed* for an electron transitioning from *1s to 3p orbital*

i or ii or iii



Question 2 8 points



A Chem110 student plots the graph shown on the left, from data obtained for the kinetic energy (eV) of the ejected electron versus frequency (s^{-1}) of the incident light, for an unknown metal. ($1\text{eV} = 1.6022 \times 10^{-19} \text{ J}$)

Based on the plot, answer the questions **a to c** below:

a. 1 point On the plot above, label the **threshold frequency**.

b. 2 points What is the value of the work potential (in J) of the unknown metal?

c. 3 points What is the *longest* wavelength (in nm) required to eject an electron from this metal's surface? Show your work including equation(s) used. (3 sig figs)

d. 2 points In the space provided below, plot the following

(Assume frequency is *above* threshold frequency):

- Kinetic energy* of an ejected electron versus the *intensity* of the incident light (left)
- Number of electrons* ejected versus the *frequency* of the incident light (right)



**Question 3** **6 points****a. 2 points**

In the ground state of Ge (Germanium; $Z = 32$) atom, **name** the following orbitals:

i. Orbital containing the electron that requires the maximum energy to be removed.

ii. Orbital containing the electron that experiences the minimum effective nuclear charge.

b. 2 points

An unknown anion (A^-) has the following *excited* electronic configuration: $[Ar] 4s^2 3d^3 4p^1$

(Ar: Argon; $Z = 18$)

Give the **condensed ground state electronic configuration** of the atom, **A** and the cation, **A^{2+}** .
Show your work. (Assume all electronic transitions are possible)

Electronic Configuration of A:

Electronic Configuration of A^{2+} :

c. 2 points

In a single atom, determine how many electrons can have the following set of quantum numbers?

(i) $n = 5; l = 3$

In a single atom, determine how many electrons can have the following set of quantum numbers?

(i) $n = 3; l = 2; m_s = +1/2$

**Question 4** **8 points**

Rank the following from 1 to 4 in **increasing** order. Provide the ranking in the boxes provided.

Note: Atomic number (Z) are provided for relevant atoms

a) Increasing (lowest to highest) order of *ionization energy*: (1= lowest; 4 = highest)

Mg

Mg: Z = 12

Rb

Rb: Z = 37

Cs

Cs: Z = 55

K

K : Z = 19

b) Increasing (lowest to highest) order of *bond angle*: (1= lowest; 4 = highest)

H₂OCCl₄XeF₄

HCN

c) Increasing (smallest to largest) order of *ionic size* (1= smallest; 4 = largest)

S²⁻

S: Z = 16

O²⁻

O: Z = 8

Be²⁺

Be: Z = 4

Na⁺

Na: Z = 11

d) Increasing (lowest to highest) order of *bond energy* (1= lowest; 4 = highest)

P-F

P-Cl

P-I

P-Br

**Question 5** **12 points**

a. 9 points For the molecules HClO_4 and TeF_6 , draw the most stable *Lewis structure* showing appropriate shape and *dashed/wedged bonds*, based on VSEPR. Determine the **molecular geometry** around the central atom. For HClO_4 , label the *most polar bond*.

HClO_4 (there are no O-O bonds)

Label the most polar bond

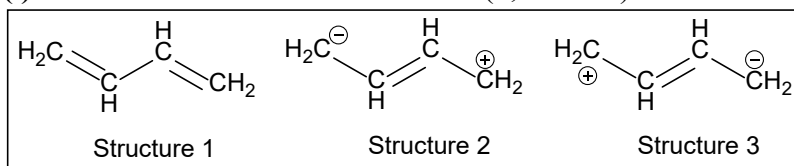
Molecular Geometry around Cl

TeF_6

Molecular Geometry around Te

b. 3 points (Note: Lone pairs are not shown in structures below)

(i) For the three *resonance* structures (1, 2 and 3) shown below, answer the following question:



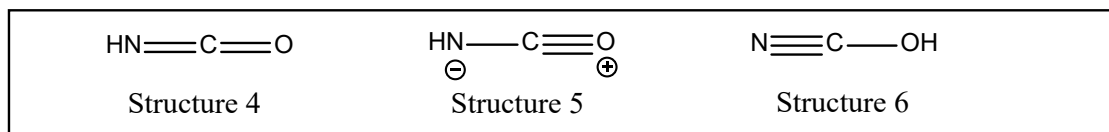
(I) Circle the *most contributing* resonance structure.

Structure 1

Structure 2

Structure 3

(ii) For the three structures given (4, 5, and 6) below, answer the following questions:



(I) Circle all the structures that represent *resonance structures* of each other:

Structure 4

Structure 5

Structure 6

(II) Among the resonance structures, circle the *most contributing* resonance structure.

Structure 4

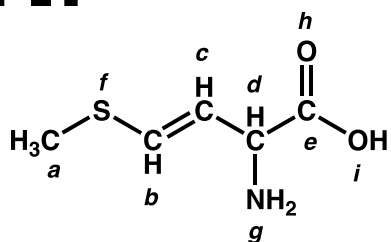
Structure 5

Structure 6

**Question 6** **6 points**

Using the data given below, calculate the **lattice energy** for SrO(s).

	ΔH° (kJ/mol)
$\text{Sr}(s) \longrightarrow \text{Sr}(g)$	+180
$\text{Sr}(g) \longrightarrow \text{Sr}^{2+}(g) + 2e^-$	+1468
$\text{O}_2(g) \longrightarrow 2\text{O}(g)$	+500
$\text{O}(g) + 2e^- \longrightarrow \text{O}^{2-}(g)$	+600
$\text{Sr}(s) + \frac{1}{2}\text{O}_2(g) \longrightarrow \text{SrO}(s)$	-550

**Question 7 10 points**

Consider the molecule given on the left, and answer questions (i) to (vi) using chemical bonding theories that we have learnt in class (valence bond theory, hybridization).

Note that the lone pairs are not shown in the structure:

i. Consider the carbon atoms marked in as ***a***, ***b***, ***c***, ***d***, and ***e*** in the molecule, and determine **hybridization** at each of the carbon centers. (2.5 points)

Atom	Hybridization
Carbon (<i>a</i>)	
Carbon (<i>b</i>)	
Carbon (<i>c</i>)	
Carbon (<i>d</i>)	
Carbon (<i>e</i>)	

ii. Now, consider the sulfur atom (S) marked as ***f***, nitrogen (N) atom (***g***), and oxygen(O) atoms (***h*** and ***i***), and determine their **hybridization**. (2 points)

Atom	Hybridization
Sulfur (<i>f</i>)	
Nitrogen (<i>g</i>)	
Oxygen (<i>h</i>)	
Oxygen (<i>i</i>)	

iii. **Describe** all the bonds formed by the carbon atoms ***a*** and ***b***, using sigma and pi designation as appropriate. (2 points)

Atom	Bonds formed (use sigma and pi as appropriate)
Carbon (<i>a</i>)	
Carbon (<i>b</i>)	

iv. How many total sp^3 -s sigma bonds are in the molecule? (1 point)

v. Determine **electronic** and **molecular geometry** at carbon (C) labeled ***e*** and sulfur (S) labeled ***f*** centres. (2 points)

Atom	Electronic Geometry	Molecular Geometry
Carbon (<i>e</i>)		
Sulfur (<i>f</i>)		

vi. Circle the value from the choices given below, that describes the **C-S-C bond angle at sulfur** (S) marked ***f***: (0.5 points)

>90° to <109.5° 109.5° >109.5 to <120° 120° >120° to <180°


Question 8 12 points

Using the **valence shell orbitals only** draw a **complete** molecular orbital (MO) diagram (with no s-p mixing) for the molecule **PS**, in the space provided below. **(6 points)**

Assume that 3s and 3p atomic orbitals will form molecular orbitals like those formed by the combination of 2s and 2p atomic orbitals that we learnt in class, and there is no s-p mixing.

Show **energy arrow**, **atomic orbitals** involved, **atom labels**, the **molecular orbitals** that are formed, **filled electrons** in the MO diagram etc. Make sure to label all the atomic and molecular orbitals.

Using the MO diagram above, answer the following questions:

- (i) Which one of the following will have the highest bond order? Circle the correct choice
 A) PS^{1-} B) PS C) PS^{1+}
- (ii) Which one of the following will have the longest bond length? Circle the correct choice
 A) PS^{1-} B) PS C) PS^{1+}
- (iii) Which of the following will be diamagnetic? Circle the correct choice.
 A) PS^{1-} B) PS C) PS^{1+}
- (iv) Draw Lewis dot structure of the molecule with the shortest bond length (based on the MO diagram) from among the three (PS^{1-} , PS, PS^{1+}), and indicate the bond order calculated from the Lewis dot structure. **(3 points)**

Lewis Dot Structure

Bond Order



A9704C64-76AC-48D2-9E6E-4F3794905A23

midterm-exam-93388

#501 Page 10 of 10

SCRAP PAGE

**CHEMISTRY 110 –Midterm Exam****Thursday November 7, 2024, 6:30 – 8:30 pm**

Examiners: Pallavi Sirjoosingh and Ashok Kakkar

First name (please write as legibly as possible within the boxes)

--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--

Last name

--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--

Student ID number

--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--

This booklet has 8 questions on 10 pages (The last page can be used as a scrap page).

The staple must NOT be removed from the question booklet. Your intact question booklet must be returned at the end of the exam.

A separate booklet contains a periodic table, and datasheet.

Write your answers directly under the questions in the space provided. Do not write on the QR codes or near the top edges (as it may be cut off).

Please write **legibly and dark enough** for the scan to be readable. For students who use English as a second language, dictionaries are permitted. Calculators are allowed.

DO NOT ASK QUESTIONS OF THE INVIGILATORS about interpretations of the exam questions. If you have a concern about a question, please summarize it at the top of this page. All concerns will be evaluated after the exam. GOOD LUCK!

**Question 1 8 points** (*Assume all transitions are possible*)

a. 6 points Calculate the **energy (in kJ/mol)** required to ionize (completely remove) the electron from the **n=3** level of **He⁺ ion** (He: Helium). Show your work including equations. (3 sig. figs.)

b. 2 points

(I) Select the largest (longest) wavelength from the three choices below:

- i. Wavelength of photon to *completely remove* an electron from *1s orbital of Hydrogen atom*
- ii. Wavelength of photon to *completely remove* an electron from *1s orbital of Boron atom (B)*
- iii. Wavelength of photon to *completely remove* an electron from *1s orbital of Boron cation (B⁺)*

i or ii or iii

(II) Select the smallest (shortest) wavelength from the three choices below (*for H atom*):

- i. Wavelength of photon *emitted* for an electron transitioning from *2p to 1s orbital*
- ii. Wavelength of photon *absorbed* for an electron transitioning from *2s to 4p orbital*
- iii. Wavelength of photon *absorbed* for an electron transitioning from *1s to 3p orbital*

i or ii or iii



Question 2 8 points



A Chem110 student plots the graph shown on the left, from data obtained for the kinetic energy (eV) of the ejected electron versus frequency (s^{-1}) of the incident light, for an unknown metal. ($1\text{eV} = 1.6022 \times 10^{-19} \text{ J}$)

Based on the plot, answer the questions **a to c** below:

a. 1 point On the plot above, label the **threshold frequency**.

b. 2 points What is the value of the work potential (in J) of the unknown metal?

c. 3 points What is the *longest* wavelength (in nm) required to eject an electron from this metal's surface? Show your work including equation(s) used. (3 sig figs)

d. 2 points In the space provided below, plot the following

(Assume frequency is *above* threshold frequency):

- Kinetic energy* of an ejected electron versus the *intensity* of the incident light (left)
- Number of electrons* ejected versus the *frequency* of the incident light (right)



**Question 3** **6 points****a. 2 points**

In the ground state of Ge (Germanium; $Z = 32$) atom, **name** the following orbitals:

i. Orbital containing the electron that requires the maximum energy to be removed.

ii. Orbital containing the electron that experiences the minimum effective nuclear charge.

b. 2 points

An unknown anion (A^-) has the following *excited* electronic configuration: $[Ar] 4s^2 3d^3 4p^1$

(Ar: Argon; $Z = 18$)

Give the **condensed ground state electronic configuration** of the atom, **A** and the cation, **A^{2+}** .
Show your work. (Assume all electronic transitions are possible)

Electronic Configuration of A:

Electronic Configuration of A^{2+} :

c. 2 points

In a single atom, determine how many electrons can have the following set of quantum numbers?

(i) $n = 5; l = 3$

In a single atom, determine how many electrons can have the following set of quantum numbers?

(i) $n = 3; l = 2; m_s = +1/2$

**Question 4** **8 points**

Rank the following from 1 to 4 in **increasing** order. Provide the ranking in the boxes provided.

Note: Atomic number (Z) are provided for relevant atoms

a) Increasing (lowest to highest) order of *ionization energy*: (1= lowest; 4 = highest)

Mg

Mg: Z = 12

Rb

Rb: Z = 37

Cs

Cs: Z = 55

K

K : Z = 19

b) Increasing (lowest to highest) order of *bond angle*: (1= lowest; 4 = highest)

H₂OCCl₄XeF₄

HCN

c) Increasing (smallest to largest) order of *ionic size* (1= smallest; 4 = largest)

S²⁻

S: Z = 16

O²⁻

O: Z = 8

Be²⁺

Be: Z = 4

Na⁺

Na: Z = 11

d) Increasing (lowest to highest) order of *bond energy* (1= lowest; 4 = highest)

P-F

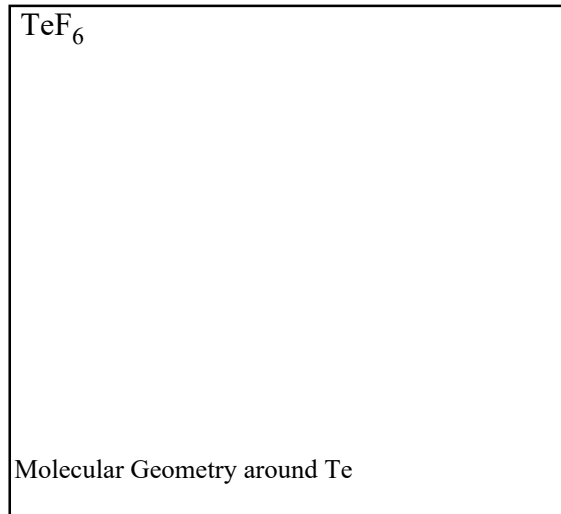
P-Cl

P-I

P-Br

**Question 5** **12 points**

a. 9 points For the molecules HClO_4 and TeF_6 , draw the most stable *Lewis structure* showing appropriate shape and *dashed/wedged bonds*, based on VSEPR. Determine the **molecular geometry** around the central atom. For HClO_4 , label the *most polar bond*.



b. 3 points (Note: Lone pairs are not shown in structures below)

(i) For the three *resonance* structures (1, 2 and 3) shown below, answer the following question:



(I) Circle the *most contributing* resonance structure.

Structure 1

Structure 2

Structure 3

(ii) For the three structures given (4, 5, and 6) below, answer the following questions:



(I) Circle all the structures that represent *resonance structures* of each other:

Structure 4

Structure 5

Structure 6

(II) Among the resonance structures, circle the *most contributing* resonance structure.

Structure 4

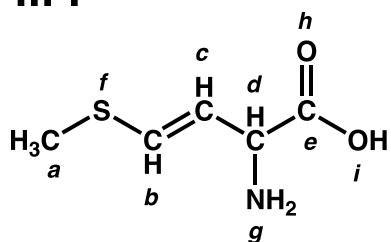
Structure 5

Structure 6

**Question 6** **6 points**

Using the data given below, calculate the **lattice energy** for SrO(s).

	ΔH° (kJ/mol)
$\text{Sr}(s) \longrightarrow \text{Sr}(g)$	+180
$\text{Sr}(g) \longrightarrow \text{Sr}^{2+}(g) + 2e^-$	+1468
$\text{O}_2(g) \longrightarrow 2\text{O}(g)$	+500
$\text{O}(g) + 2e^- \longrightarrow \text{O}^{2-}(g)$	+600
$\text{Sr}(s) + \frac{1}{2}\text{O}_2(g) \longrightarrow \text{SrO}(s)$	-550

**Question 7 10 points**

Consider the molecule given on the left, and answer questions (i) to (vi) using chemical bonding theories that we have learnt in class (valence bond theory, hybridization).

Note that the lone pairs are not shown in the structure:

i. Consider the carbon atoms marked in as ***a***, ***b***, ***c***, ***d***, and ***e*** in the molecule, and determine **hybridization** at each of the carbon centers. (2.5 points)

Atom	Hybridization
Carbon (<i>a</i>)	
Carbon (<i>b</i>)	
Carbon (<i>c</i>)	
Carbon (<i>d</i>)	
Carbon (<i>e</i>)	

ii. Now, consider the sulfur atom (S) marked as ***f***, nitrogen (N) atom (***g***), and oxygen(O) atoms (***h*** and ***i***), and determine their **hybridization**. (2 points)

Atom	Hybridization
Sulfur (<i>f</i>)	
Nitrogen (<i>g</i>)	
Oxygen (<i>h</i>)	
Oxygen (<i>i</i>)	

iii. **Describe** all the bonds formed by the carbon atoms ***a*** and ***b***, using sigma and pi designation as appropriate. (2 points)

Atom	Bonds formed (use sigma and pi as appropriate)
Carbon (<i>a</i>)	
Carbon (<i>b</i>)	

iv. How many total sp^3 -s sigma bonds are in the molecule? (1 point)

v. Determine **electronic** and **molecular geometry** at carbon (C) labeled ***e*** and sulfur (S) labeled ***f*** centres. (2 points)

Atom	Electronic Geometry	Molecular Geometry
Carbon (<i>e</i>)		
Sulfur (<i>f</i>)		

vi. Circle the value from the choices given below, that describes the **C-S-C bond angle at sulfur** (S) marked ***f***: (0.5 points)

>90° to <109.5°

109.5°

>109.5 to <120°

120°

>120° to <180°


Question 8 12 points

Using the **valence shell orbitals only** draw a **complete** molecular orbital (MO) diagram (with no s-p mixing) for the molecule **PS**, in the space provided below. **(6 points)**

Assume that 3s and 3p atomic orbitals will form molecular orbitals like those formed by the combination of 2s and 2p atomic orbitals that we learnt in class, and there is no s-p mixing.

Show **energy arrow**, **atomic orbitals** involved, **atom labels**, the **molecular orbitals** that are formed, **filled electrons** in the MO diagram etc. Make sure to label all the atomic and molecular orbitals.

Using the MO diagram above, answer the following questions:

- (i) Which one of the following will have the highest bond order? Circle the correct choice
 A) PS^{1-} B) PS C) PS^{1+}
- (ii) Which one of the following will have the longest bond length? Circle the correct choice
 A) PS^{1-} B) PS C) PS^{1+}
- (iii) Which of the following will be diamagnetic? Circle the correct choice.
 A) PS^{1-} B) PS C) PS^{1+}
- (iv) Draw Lewis dot structure of the molecule with the shortest bond length (based on the MO diagram) from among the three (PS^{1-} , PS, PS^{1+}), and indicate the bond order calculated from the Lewis dot structure. **(3 points)**

Lewis Dot Structure

Bond Order



04F5C3CF-7F3C-4590-BA1A-16ACE8E9811B

midterm-exam-93388

#502 Page 10 of 10

SCRAP PAGE

**CHEMISTRY 110 –Midterm Exam****Thursday November 7, 2024, 6:30 – 8:30 pm**

Examiners: Pallavi Sirjoosingh and Ashok Kakkar

First name (please write as legibly as possible within the boxes)

--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--

Last name

--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--

Student ID number

--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--

This booklet has 8 questions on 10 pages (The last page can be used as a scrap page).

The staple must NOT be removed from the question booklet. Your intact question booklet must be returned at the end of the exam.

A separate booklet contains a periodic table, and datasheet.

Write your answers directly under the questions in the space provided. Do not write on the QR codes or near the top edges (as it may be cut off).

Please write **legibly and dark enough** for the scan to be readable. For students who use English as a second language, dictionaries are permitted. Calculators are allowed.

DO NOT ASK QUESTIONS OF THE INVIGILATORS about interpretations of the exam questions. If you have a concern about a question, please summarize it at the top of this page. All concerns will be evaluated after the exam. GOOD LUCK!

**Question 1 8 points** (*Assume all transitions are possible*)

a. 6 points Calculate the **energy (in kJ/mol)** required to ionize (completely remove) the electron from the **n=3** level of **He⁺ ion** (He: Helium). Show your work including equations. (3 sig. figs.)

b. 2 points

(I) Select the largest (longest) wavelength from the three choices below:

- i. Wavelength of photon to *completely remove* an electron from *1s orbital of Hydrogen atom*
- ii. Wavelength of photon to *completely remove* an electron from *1s orbital of Boron atom (B)*
- iii. Wavelength of photon to *completely remove* an electron from *1s orbital of Boron cation (B⁺)*

i or ii or iii

(II) Select the smallest (shortest) wavelength from the three choices below (*for H atom*):

- i. Wavelength of photon *emitted* for an electron transitioning from *2p to 1s orbital*
- ii. Wavelength of photon *absorbed* for an electron transitioning from *2s to 4p orbital*
- iii. Wavelength of photon *absorbed* for an electron transitioning from *1s to 3p orbital*

i or ii or iii



Question 2 8 points



A Chem110 student plots the graph shown on the left, from data obtained for the kinetic energy (eV) of the ejected electron versus frequency (s^{-1}) of the incident light, for an unknown metal. ($1\text{eV} = 1.6022 \times 10^{-19} \text{ J}$)

Based on the plot, answer the questions **a to c** below:

a. 1 point On the plot above, label the **threshold frequency**.

b. 2 points What is the value of the work potential (in J) of the unknown metal?

c. 3 points What is the *longest* wavelength (in nm) required to eject an electron from this metal's surface? Show your work including equation(s) used. (3 sig figs)

d. 2 points In the space provided below, plot the following

(Assume frequency is *above* threshold frequency):

- Kinetic energy* of an ejected electron versus the *intensity* of the incident light (left)
- Number of electrons* ejected versus the *frequency* of the incident light (right)



**Question 3** **6 points****a. 2 points**

In the ground state of Ge (Germanium; $Z = 32$) atom, **name** the following orbitals:

i. Orbital containing the electron that requires the maximum energy to be removed.

ii. Orbital containing the electron that experiences the minimum effective nuclear charge.

b. 2 points

An unknown anion (A^-) has the following *excited* electronic configuration: $[Ar] 4s^2 3d^3 4p^1$

(Ar: Argon; $Z = 18$)

Give the **condensed ground state electronic configuration** of the atom, **A** and the cation, **A^{2+}** .
Show your work. (Assume all electronic transitions are possible)

Electronic Configuration of A:

Electronic Configuration of A^{2+} :

c. 2 points

In a single atom, determine how many electrons can have the following set of quantum numbers?

(i) $n = 5; l = 3$

In a single atom, determine how many electrons can have the following set of quantum numbers?

(i) $n = 3; l = 2; m_s = +1/2$

**Question 4** **8 points**

Rank the following from 1 to 4 in **increasing** order. Provide the ranking in the boxes provided.

Note: Atomic number (Z) are provided for relevant atoms

a) Increasing (lowest to highest) order of *ionization energy*: (1= lowest; 4 = highest)

Mg

Mg: Z = 12

Rb

Rb: Z = 37

Cs

Cs: Z = 55

K

K : Z = 19

b) Increasing (lowest to highest) order of *bond angle*: (1= lowest; 4 = highest)

H₂OCCl₄XeF₄

HCN

c) Increasing (smallest to largest) order of *ionic size* (1= smallest; 4 = largest)

S²⁻

S: Z = 16

O²⁻

O: Z = 8

Be²⁺

Be: Z = 4

Na⁺

Na: Z = 11

d) Increasing (lowest to highest) order of *bond energy* (1= lowest; 4 = highest)

P-F

P-Cl

P-I

P-Br

**Question 5** **12 points**

a. 9 points For the molecules HClO_4 and TeF_6 , draw the most stable *Lewis structure* showing appropriate shape and *dashed/wedged bonds*, based on VSEPR. Determine the **molecular geometry** around the central atom. For HClO_4 , label the *most polar bond*.

HClO_4 (there are no O-O bonds)

Label the most polar bond

Molecular Geometry around Cl

TeF_6

Molecular Geometry around Te

b. 3 points (Note: Lone pairs are not shown in structures below)

(i) For the three *resonance* structures (1, 2 and 3) shown below, answer the following question:



(I) Circle the *most contributing* resonance structure.

Structure 1

Structure 2

Structure 3

(ii) For the three structures given (4, 5, and 6) below, answer the following questions:



(I) Circle all the structures that represent *resonance structures* of each other:

Structure 4

Structure 5

Structure 6

(II) Among the resonance structures, circle the *most contributing* resonance structure.

Structure 4

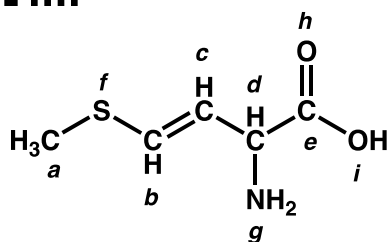
Structure 5

Structure 6

**Question 6** **6 points**

Using the data given below, calculate the **lattice energy** for SrO(s).

	ΔH° (kJ/mol)
$\text{Sr}(s) \longrightarrow \text{Sr}(g)$	+180
$\text{Sr}(g) \longrightarrow \text{Sr}^{2+}(g) + 2e^-$	+1468
$\text{O}_2(g) \longrightarrow 2\text{O}(g)$	+500
$\text{O}(g) + 2e^- \longrightarrow \text{O}^{2-}(g)$	+600
$\text{Sr}(s) + \frac{1}{2}\text{O}_2(g) \longrightarrow \text{SrO}(s)$	-550

**Question 7 10 points**

Consider the molecule given on the left, and answer questions (i) to (vi) using chemical bonding theories that we have learnt in class (valence bond theory, hybridization).

Note that the lone pairs are not shown in the structure:

i. Consider the carbon atoms marked in as ***a***, ***b***, ***c***, ***d***, and ***e*** in the molecule, and determine **hybridization** at each of the carbon centers. (2.5 points)

Atom	Hybridization
Carbon (<i>a</i>)	
Carbon (<i>b</i>)	
Carbon (<i>c</i>)	
Carbon (<i>d</i>)	
Carbon (<i>e</i>)	

ii. Now, consider the sulfur atom (S) marked as ***f***, nitrogen (N) atom (***g***), and oxygen(O) atoms (***h*** and ***i***), and determine their **hybridization**. (2 points)

Atom	Hybridization
Sulfur (<i>f</i>)	
Nitrogen (<i>g</i>)	
Oxygen (<i>h</i>)	
Oxygen (<i>i</i>)	

iii. **Describe** all the bonds formed by the carbon atoms ***a*** and ***b***, using sigma and pi designation as appropriate. (2 points)

Atom	Bonds formed (use sigma and pi as appropriate)
Carbon (<i>a</i>)	
Carbon (<i>b</i>)	

iv. How many total sp^3 -s sigma bonds are in the molecule? (1 point)

v. Determine **electronic** and **molecular geometry** at carbon (C) labeled ***e*** and sulfur (S) labeled ***f*** centres. (2 points)

Atom	Electronic Geometry	Molecular Geometry
Carbon (<i>e</i>)		
Sulfur (<i>f</i>)		

vi. Circle the value from the choices given below, that describes the **C-S-C bond angle at sulfur** (S) marked ***f***: (0.5 points)

>90° to <109.5° 109.5° >109.5 to <120° 120° >120° to <180°


Question 8 12 points

Using the **valence shell orbitals only** draw a **complete** molecular orbital (MO) diagram (with no s-p mixing) for the molecule **PS**, in the space provided below. **(6 points)**

Assume that 3s and 3p atomic orbitals will form molecular orbitals like those formed by the combination of 2s and 2p atomic orbitals that we learnt in class, and there is no s-p mixing.

Show **energy arrow**, **atomic orbitals** involved, **atom labels**, the **molecular orbitals** that are formed, **filled electrons** in the MO diagram etc. Make sure to label all the atomic and molecular orbitals.

Using the MO diagram above, answer the following questions:

- (i) Which one of the following will have the highest bond order? Circle the correct choice
 A) PS^{1-} B) PS C) PS^{1+}
- (ii) Which one of the following will have the longest bond length? Circle the correct choice
 A) PS^{1-} B) PS C) PS^{1+}
- (iii) Which of the following will be diamagnetic? Circle the correct choice.
 A) PS^{1-} B) PS C) PS^{1+}
- (iv) Draw Lewis dot structure of the molecule with the shortest bond length (based on the MO diagram) from among the three (PS^{1-} , PS, PS^{1+}), and indicate the bond order calculated from the Lewis dot structure. **(3 points)**

Lewis Dot Structure

Bond Order



8C2D90D2-A152-4B0C-B6A9-DA60D9591301

midterm-exam-93388

#503 Page 10 of 10

SCRAP PAGE

**CHEMISTRY 110 –Midterm Exam****Thursday November 7, 2024, 6:30 – 8:30 pm**

Examiners: Pallavi Sirjoosingh and Ashok Kakkar

First name (please write as legibly as possible within the boxes)

--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--

Last name

--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--

Student ID number

--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--

This booklet has 8 questions on 10 pages (The last page can be used as a scrap page).

The staple must NOT be removed from the question booklet. Your intact question booklet must be returned at the end of the exam.

A separate booklet contains a periodic table, and datasheet.

Write your answers directly under the questions in the space provided. Do not write on the QR codes or near the top edges (as it may be cut off).

Please write **legibly and dark enough** for the scan to be readable. For students who use English as a second language, dictionaries are permitted. Calculators are allowed.

DO NOT ASK QUESTIONS OF THE INVIGILATORS about interpretations of the exam questions. If you have a concern about a question, please summarize it at the top of this page. All concerns will be evaluated after the exam. GOOD LUCK!

**Question 1 8 points** (*Assume all transitions are possible*)

a. 6 points Calculate the **energy (in kJ/mol)** required to ionize (completely remove) the electron from the **n=3** level of **He⁺ ion** (He: Helium). Show your work including equations. (3 sig. figs.)

b. 2 points

(I) Select the largest (longest) wavelength from the three choices below:

- i. Wavelength of photon to *completely remove* an electron from *1s orbital of Hydrogen atom*
- ii. Wavelength of photon to *completely remove* an electron from *1s orbital of Boron atom (B)*
- iii. Wavelength of photon to *completely remove* an electron from *1s orbital of Boron cation (B⁺)*

i or ii or iii

(II) Select the smallest (shortest) wavelength from the three choices below (*for H atom*):

- i. Wavelength of photon *emitted* for an electron transitioning from *2p to 1s orbital*
- ii. Wavelength of photon *absorbed* for an electron transitioning from *2s to 4p orbital*
- iii. Wavelength of photon *absorbed* for an electron transitioning from *1s to 3p orbital*

i or ii or iii



Question 2 8 points



A Chem110 student plots the graph shown on the left, from data obtained for the kinetic energy (eV) of the ejected electron versus frequency (s^{-1}) of the incident light, for an unknown metal. ($1\text{eV} = 1.6022 \times 10^{-19} \text{ J}$)

Based on the plot, answer the questions **a to c** below:

a. 1 point On the plot above, label the **threshold frequency**.

b. 2 points What is the value of the work potential (in J) of the unknown metal?

c. 3 points What is the *longest* wavelength (in nm) required to eject an electron from this metal's surface? Show your work including equation(s) used. (3 sig figs)

d. 2 points In the space provided below, plot the following

(Assume frequency is *above* threshold frequency):

- Kinetic energy* of an ejected electron versus the *intensity* of the incident light (left)
- Number of electrons* ejected versus the *frequency* of the incident light (right)



**Question 3** **6 points****a. 2 points**

In the ground state of Ge (Germanium; $Z = 32$) atom, **name** the following orbitals:

i. Orbital containing the electron that requires the maximum energy to be removed.

ii. Orbital containing the electron that experiences the minimum effective nuclear charge.

b. 2 points

An unknown anion (A^-) has the following *excited* electronic configuration: $[Ar] 4s^2 3d^3 4p^1$

(Ar: Argon; $Z = 18$)

Give the **condensed ground state electronic configuration** of the atom, **A** and the cation, **A^{2+}** .
Show your work. (Assume all electronic transitions are possible)

Electronic Configuration of A:

Electronic Configuration of A^{2+} :

c. 2 points

In a single atom, determine how many electrons can have the following set of quantum numbers?

(i) $n = 5; l = 3$

In a single atom, determine how many electrons can have the following set of quantum numbers?

(i) $n = 3; l = 2; m_s = +1/2$

**Question 4** **8 points**

Rank the following from 1 to 4 in **increasing** order. Provide the ranking in the boxes provided.

Note: Atomic number (Z) are provided for relevant atoms

a) Increasing (lowest to highest) order of *ionization energy*: (1= lowest; 4 = highest)

Mg

Mg: Z = 12

Rb

Rb: Z = 37

Cs

Cs: Z = 55

K

K : Z = 19

b) Increasing (lowest to highest) order of *bond angle*: (1= lowest; 4 = highest)

H₂OCCl₄XeF₄

HCN

c) Increasing (smallest to largest) order of *ionic size* (1= smallest; 4 = largest)

S²⁻

S: Z = 16

O²⁻

O: Z = 8

Be²⁺

Be: Z = 4

Na⁺

Na: Z = 11

d) Increasing (lowest to highest) order of *bond energy* (1= lowest; 4 = highest)

P-F

P-Cl

P-I

P-Br

**Question 5** **12 points**

a. 9 points For the molecules HClO_4 and TeF_6 , draw the most stable *Lewis structure* showing appropriate shape and *dashed/wedged bonds*, based on VSEPR. Determine the **molecular geometry** around the central atom. For HClO_4 , label the *most polar bond*.

HClO_4 (there are no O-O bonds)

Label the most polar bond

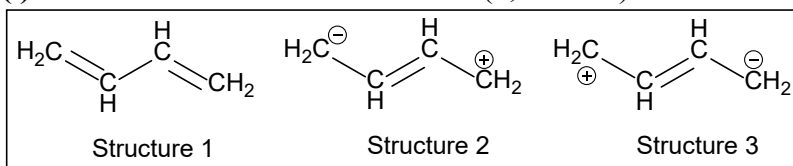
Molecular Geometry around Cl

TeF_6

Molecular Geometry around Te

b. 3 points (Note: Lone pairs are not shown in structures below)

(i) For the three *resonance* structures (1, 2 and 3) shown below, answer the following question:



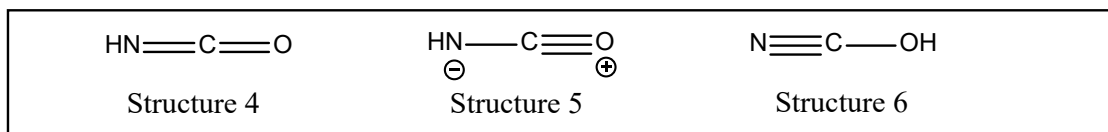
(I) Circle the *most contributing* resonance structure.

Structure 1

Structure 2

Structure 3

(ii) For the three structures given (4, 5, and 6) below, answer the following questions:



(I) Circle all the structures that represent *resonance structures* of each other:

Structure 4

Structure 5

Structure 6

(II) Among the resonance structures, circle the *most contributing* resonance structure.

Structure 4

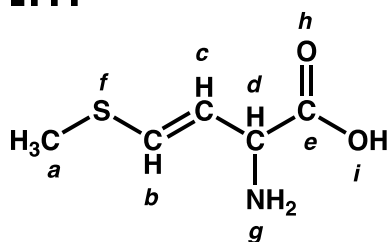
Structure 5

Structure 6

**Question 6** **6 points**

Using the data given below, calculate the **lattice energy** for SrO(s).

	ΔH° (kJ/mol)
$\text{Sr}(s) \longrightarrow \text{Sr}(g)$	+180
$\text{Sr}(g) \longrightarrow \text{Sr}^{2+}(g) + 2e^-$	+1468
$\text{O}_2(g) \longrightarrow 2\text{O}(g)$	+500
$\text{O}(g) + 2e^- \longrightarrow \text{O}^{2-}(g)$	+600
$\text{Sr}(s) + \frac{1}{2}\text{O}_2(g) \longrightarrow \text{SrO}(s)$	-550

**Question 7 10 points**

Consider the molecule given on the left, and answer questions (i) to (vi) using chemical bonding theories that we have learnt in class (valence bond theory, hybridization).

Note that the lone pairs are not shown in the structure:

i. Consider the carbon atoms marked in as ***a***, ***b***, ***c***, ***d***, and ***e*** in the molecule, and determine **hybridization** at each of the carbon centers. (2.5 points)

Atom	Hybridization
Carbon (<i>a</i>)	
Carbon (<i>b</i>)	
Carbon (<i>c</i>)	
Carbon (<i>d</i>)	
Carbon (<i>e</i>)	

ii. Now, consider the sulfur atom (S) marked as ***f***, nitrogen (N) atom (***g***), and oxygen(O) atoms (***h*** and ***i***), and determine their **hybridization**. (2 points)

Atom	Hybridization
Sulfur (<i>f</i>)	
Nitrogen (<i>g</i>)	
Oxygen (<i>h</i>)	
Oxygen (<i>i</i>)	

iii. **Describe** all the bonds formed by the carbon atoms ***a*** and ***b***, using sigma and pi designation as appropriate. (2 points)

Atom	Bonds formed (use sigma and pi as appropriate)
Carbon (<i>a</i>)	
Carbon (<i>b</i>)	

iv. How many total sp^3 -s sigma bonds are in the molecule? (1 point)

v. Determine **electronic** and **molecular geometry** at carbon (C) labeled ***e*** and sulfur (S) labeled ***f*** centres. (2 points)

Atom	Electronic Geometry	Molecular Geometry
Carbon (<i>e</i>)		
Sulfur (<i>f</i>)		

vi. Circle the value from the choices given below, that describes the **C-S-C bond angle at sulfur** (S) marked ***f***: (0.5 points)

>90° to <109.5° 109.5° >109.5 to <120° 120° >120° to <180°


Question 8 12 points

Using the **valence shell orbitals only** draw a **complete** molecular orbital (MO) diagram (with no s-p mixing) for the molecule **PS**, in the space provided below. **(6 points)**

Assume that 3s and 3p atomic orbitals will form molecular orbitals like those formed by the combination of 2s and 2p atomic orbitals that we learnt in class, and there is no s-p mixing.

Show **energy arrow**, **atomic orbitals** involved, **atom labels**, the **molecular orbitals** that are formed, **filled electrons** in the MO diagram etc. Make sure to label all the atomic and molecular orbitals.

Using the MO diagram above, answer the following questions:

- (i) Which one of the following will have the highest bond order? Circle the correct choice
 A) PS^{1-} B) PS C) PS^{1+}
- (ii) Which one of the following will have the longest bond length? Circle the correct choice
 A) PS^{1-} B) PS C) PS^{1+}
- (iii) Which of the following will be diamagnetic? Circle the correct choice.
 A) PS^{1-} B) PS C) PS^{1+}
- (iv) Draw Lewis dot structure of the molecule with the shortest bond length (based on the MO diagram) from among the three (PS^{1-} , PS, PS^{1+}), and indicate the bond order calculated from the Lewis dot structure. **(3 points)**

Lewis Dot Structure

Bond Order



DD47D1AA-A479-4DC5-AA89-5630F1FD5606

midterm-exam-93388

#504 Page 10 of 10

SCRAP PAGE

**CHEMISTRY 110 –Midterm Exam****Thursday November 7, 2024, 6:30 – 8:30 pm**

Examiners: Pallavi Sirjoosingh and Ashok Kakkar

First name (please write as legibly as possible within the boxes)

--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--

Last name

--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--

Student ID number

--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--

This booklet has 8 questions on 10 pages (The last page can be used as a scrap page).

The staple must NOT be removed from the question booklet. Your intact question booklet must be returned at the end of the exam.

A separate booklet contains a periodic table, and datasheet.

Write your answers directly under the questions in the space provided. Do not write on the QR codes or near the top edges (as it may be cut off).

Please write **legibly and dark enough** for the scan to be readable. For students who use English as a second language, dictionaries are permitted. Calculators are allowed.

DO NOT ASK QUESTIONS OF THE INVIGILATORS about interpretations of the exam questions. If you have a concern about a question, please summarize it at the top of this page. All concerns will be evaluated after the exam. GOOD LUCK!

**Question 1 8 points** (*Assume all transitions are possible*)

a. 6 points Calculate the **energy (in kJ/mol)** required to ionize (completely remove) the electron from the **n=3** level of **He⁺ ion** (He: Helium). Show your work including equations. (3 sig. figs.)

b. 2 points

(I) Select the largest (longest) wavelength from the three choices below:

- i. Wavelength of photon to *completely remove* an electron from *1s orbital of Hydrogen atom*
- ii. Wavelength of photon to *completely remove* an electron from *1s orbital of Boron atom (B)*
- iii. Wavelength of photon to *completely remove* an electron from *1s orbital of Boron cation (B⁺)*

i or ii or iii

(II) Select the smallest (shortest) wavelength from the three choices below (*for H atom*):

- i. Wavelength of photon *emitted* for an electron transitioning from *2p to 1s orbital*
- ii. Wavelength of photon *absorbed* for an electron transitioning from *2s to 4p orbital*
- iii. Wavelength of photon *absorbed* for an electron transitioning from *1s to 3p orbital*

i or ii or iii



Question 2 8 points



A Chem110 student plots the graph shown on the left, from data obtained for the kinetic energy (eV) of the ejected electron versus frequency (s^{-1}) of the incident light, for an unknown metal. ($1\text{eV} = 1.6022 \times 10^{-19} \text{ J}$)

Based on the plot, answer the questions **a to c** below:

a. 1 point On the plot above, label the **threshold frequency**.

b. 2 points What is the value of the work potential (in J) of the unknown metal?

c. 3 points What is the *longest* wavelength (in nm) required to eject an electron from this metal's surface? Show your work including equation(s) used. (3 sig figs)

d. 2 points In the space provided below, plot the following

(Assume frequency is *above* threshold frequency):

- i. *Kinetic energy* of an ejected electron versus the *intensity* of the incident light (left)
- ii. *Number of electrons* ejected versus the *frequency* of the incident light (right)



**Question 3** **6 points****a. 2 points**

In the ground state of Ge (Germanium; $Z = 32$) atom, **name** the following orbitals:

i. Orbital containing the electron that requires the maximum energy to be removed.

ii. Orbital containing the electron that experiences the minimum effective nuclear charge.

b. 2 points

An unknown anion (A^-) has the following *excited* electronic configuration: $[Ar] 4s^2 3d^3 4p^1$

(Ar: Argon; $Z = 18$)

Give the **condensed ground state electronic configuration** of the atom, **A** and the cation, **A^{2+}** .
Show your work. (Assume all electronic transitions are possible)

Electronic Configuration of A:

Electronic Configuration of A^{2+} :

c. 2 points

In a single atom, determine how many electrons can have the following set of quantum numbers?

(i) $n = 5; l = 3$

In a single atom, determine how many electrons can have the following set of quantum numbers?

(i) $n = 3; l = 2; m_s = +1/2$

**Question 4** **8 points**

Rank the following from 1 to 4 in **increasing** order. Provide the ranking in the boxes provided.

Note: Atomic number (Z) are provided for relevant atoms

a) Increasing (lowest to highest) order of *ionization energy*: (1= lowest; 4 = highest)

Mg

Mg: Z = 12

Rb

Rb: Z = 37

Cs

Cs: Z = 55

K

K : Z = 19

b) Increasing (lowest to highest) order of *bond angle*: (1= lowest; 4 = highest)

H₂OCCl₄XeF₄

HCN

c) Increasing (smallest to largest) order of *ionic size* (1= smallest; 4 = largest)

S²⁻

S: Z = 16

O²⁻

O: Z = 8

Be²⁺

Be: Z = 4

Na⁺

Na: Z = 11

d) Increasing (lowest to highest) order of *bond energy* (1= lowest; 4 = highest)

P-F

P-Cl

P-I

P-Br

**Question 5** **12 points**

a. 9 points For the molecules HClO_4 and TeF_6 , draw the most stable *Lewis structure* showing appropriate shape and *dashed/wedged bonds*, based on VSEPR. Determine the **molecular geometry** around the central atom. For HClO_4 , label the *most polar bond*.



b. 3 points (Note: Lone pairs are not shown in structures below)

(i) For the three *resonance* structures (1, 2 and 3) shown below, answer the following question:



(I) Circle the *most contributing* resonance structure.

Structure 1

Structure 2

Structure 3

(ii) For the three structures given (4, 5, and 6) below, answer the following questions:



(I) Circle all the structures that represent *resonance structures* of each other:

Structure 4

Structure 5

Structure 6

(II) Among the resonance structures, circle the *most contributing* resonance structure.

Structure 4

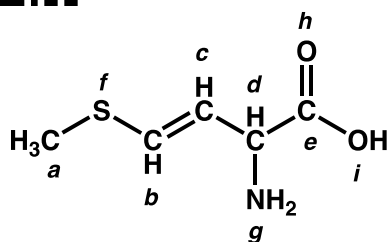
Structure 5

Structure 6

**Question 6** **6 points**

Using the data given below, calculate the **lattice energy** for SrO(s).

	ΔH° (kJ/mol)
$\text{Sr}(s) \longrightarrow \text{Sr}(g)$	+180
$\text{Sr}(g) \longrightarrow \text{Sr}^{2+}(g) + 2e^-$	+1468
$\text{O}_2(g) \longrightarrow 2\text{O}(g)$	+500
$\text{O}(g) + 2e^- \longrightarrow \text{O}^{2-}(g)$	+600
$\text{Sr}(s) + \frac{1}{2}\text{O}_2(g) \longrightarrow \text{SrO}(s)$	-550

**Question 7 10 points**

Consider the molecule given on the left, and answer questions (i) to (vi) using chemical bonding theories that we have learnt in class (valence bond theory, hybridization).

Note that the lone pairs are not shown in the structure:

i. Consider the carbon atoms marked in as ***a***, ***b***, ***c***, ***d***, and ***e*** in the molecule, and determine **hybridization** at each of the carbon centers. (2.5 points)

Atom	Hybridization
Carbon (<i>a</i>)	
Carbon (<i>b</i>)	
Carbon (<i>c</i>)	
Carbon (<i>d</i>)	
Carbon (<i>e</i>)	

ii. Now, consider the sulfur atom (S) marked as ***f***, nitrogen (N) atom (***g***), and oxygen(O) atoms (***h*** and ***i***), and determine their **hybridization**. (2 points)

Atom	Hybridization
Sulfur (<i>f</i>)	
Nitrogen (<i>g</i>)	
Oxygen (<i>h</i>)	
Oxygen (<i>i</i>)	

iii. **Describe** all the bonds formed by the carbon atoms ***a*** and ***b***, using sigma and pi designation as appropriate. (2 points)

Atom	Bonds formed (use sigma and pi as appropriate)
Carbon (<i>a</i>)	
Carbon (<i>b</i>)	

iv. How many total sp^3 -s sigma bonds are in the molecule? (1 point)

v. Determine **electronic** and **molecular geometry** at carbon (C) labeled ***e*** and sulfur (S) labeled ***f*** centres. (2 points)

Atom	Electronic Geometry	Molecular Geometry
Carbon (<i>e</i>)		
Sulfur (<i>f</i>)		

vi. Circle the value from the choices given below, that describes the **C-S-C bond angle at sulfur** (S) marked ***f***: (0.5 points)

>90° to <109.5°

109.5°

>109.5 to <120°

120°

>120° to <180°

**Question 8 12 points**

Using the **valence shell orbitals only** draw a **complete** molecular orbital (MO) diagram (with no s-p mixing) for the molecule **PS**, in the space provided below. **(6 points)**

Assume that 3s and 3p atomic orbitals will form molecular orbitals like those formed by the combination of 2s and 2p atomic orbitals that we learnt in class, and there is no s-p mixing.

Show **energy arrow**, **atomic orbitals** involved, **atom labels**, the **molecular orbitals** that are formed, **filled electrons** in the MO diagram etc. Make sure to label all the atomic and molecular orbitals.

Using the MO diagram above, answer the following questions:

- (i) Which one of the following will have the highest bond order? Circle the correct choice
A) PS^{1-} B) PS C) PS^{1+}
- (ii) Which one of the following will have the longest bond length? Circle the correct choice
A) PS^{1-} B) PS C) PS^{1+}
- (iii) Which of the following will be diamagnetic? Circle the correct choice.
A) PS^{1-} B) PS C) PS^{1+}
- (iv) Draw Lewis dot structure of the molecule with the shortest bond length (based on the MO diagram) from among the three (PS^{1-} , PS, PS^{1+}), and indicate the bond order calculated from the Lewis dot structure. **(3 points)**

Lewis Dot Structure

Bond Order



055FC9E1-D0AE-43A4-B696-4E0E87301377

midterm-exam-93388

#505 Page 10 of 10

SCRAP PAGE

**CHEMISTRY 110 –Midterm Exam****Thursday November 7, 2024, 6:30 – 8:30 pm**

Examiners: Pallavi Sirjoosingh and Ashok Kakkar

First name (please write as legibly as possible within the boxes)

--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--

Last name

--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--

Student ID number

--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--

This booklet has 8 questions on 10 pages (The last page can be used as a scrap page).

The staple must NOT be removed from the question booklet. Your intact question booklet must be returned at the end of the exam.

A separate booklet contains a periodic table, and datasheet.

Write your answers directly under the questions in the space provided. Do not write on the QR codes or near the top edges (as it may be cut off).

Please write **legibly and dark enough** for the scan to be readable. For students who use English as a second language, dictionaries are permitted. Calculators are allowed.

DO NOT ASK QUESTIONS OF THE INVIGILATORS about interpretations of the exam questions. If you have a concern about a question, please summarize it at the top of this page. All concerns will be evaluated after the exam. GOOD LUCK!

**Question 1 8 points** (*Assume all transitions are possible*)

a. 6 points Calculate the **energy (in kJ/mol)** required to ionize (completely remove) the electron from the **n=3** level of **He⁺ ion** (He: Helium). Show your work including equations. (3 sig. figs.)

b. 2 points

(I) Select the largest (longest) wavelength from the three choices below:

- i. Wavelength of photon to *completely remove* an electron from *1s orbital of Hydrogen atom*
- ii. Wavelength of photon to *completely remove* an electron from *1s orbital of Boron atom (B)*
- iii. Wavelength of photon to *completely remove* an electron from *1s orbital of Boron cation (B⁺)*

i or ii or iii

(II) Select the smallest (shortest) wavelength from the three choices below (*for H atom*):

- i. Wavelength of photon *emitted* for an electron transitioning from *2p to 1s orbital*
- ii. Wavelength of photon *absorbed* for an electron transitioning from *2s to 4p orbital*
- iii. Wavelength of photon *absorbed* for an electron transitioning from *1s to 3p orbital*

i or ii or iii



Question 2 8 points



A Chem110 student plots the graph shown on the left, from data obtained for the kinetic energy (eV) of the ejected electron versus frequency (s^{-1}) of the incident light, for an unknown metal. ($1\text{eV} = 1.6022 \times 10^{-19} \text{ J}$)

Based on the plot, answer the questions **a to c** below:

a. 1 point On the plot above, label the **threshold frequency**.

b. 2 points What is the value of the work potential (in J) of the unknown metal?

c. 3 points What is the *longest* wavelength (in nm) required to eject an electron from this metal's surface? Show your work including equation(s) used. (3 sig figs)

d. 2 points In the space provided below, plot the following

(Assume frequency is *above* threshold frequency):

- i. *Kinetic energy* of an ejected electron versus the *intensity* of the incident light (left)
- ii. *Number of electrons* ejected versus the *frequency* of the incident light (right)



**Question 3** **6 points****a. 2 points**

In the ground state of Ge (Germanium; $Z = 32$) atom, **name** the following orbitals:

i. Orbital containing the electron that requires the maximum energy to be removed.

ii. Orbital containing the electron that experiences the minimum effective nuclear charge.

b. 2 points

An unknown anion (A^-) has the following *excited* electronic configuration: $[Ar] 4s^2 3d^3 4p^1$

(Ar: Argon; $Z = 18$)

Give the **condensed ground state electronic configuration** of the atom, **A** and the cation, **A^{2+}** .
Show your work. (Assume all electronic transitions are possible)

Electronic Configuration of A:

Electronic Configuration of A^{2+} :

c. 2 points

In a single atom, determine how many electrons can have the following set of quantum numbers?

(i) $n = 5; l = 3$

In a single atom, determine how many electrons can have the following set of quantum numbers?

(i) $n = 3; l = 2; m_s = +1/2$

**Question 4** **8 points**

Rank the following from 1 to 4 in **increasing** order. Provide the ranking in the boxes provided.

Note: Atomic number (Z) are provided for relevant atoms

a) Increasing (lowest to highest) order of *ionization energy*: (1= lowest; 4 = highest)

Mg

Mg: Z = 12

Rb

Rb: Z = 37

Cs

Cs: Z = 55

K

K : Z = 19

b) Increasing (lowest to highest) order of *bond angle*: (1= lowest; 4 = highest)

H₂OCCl₄XeF₄

HCN

c) Increasing (smallest to largest) order of *ionic size* (1= smallest; 4 = largest)

S²⁻

S: Z = 16

O²⁻

O: Z = 8

Be²⁺

Be: Z = 4

Na⁺

Na: Z = 11

d) Increasing (lowest to highest) order of *bond energy* (1= lowest; 4 = highest)

P-F

P-Cl

P-I

P-Br

**Question 5** **12 points**

a. 9 points For the molecules HClO_4 and TeF_6 , draw the most stable *Lewis structure* showing appropriate shape and *dashed/wedged bonds*, based on VSEPR. Determine the **molecular geometry** around the central atom. For HClO_4 , label the *most polar bond*.



b. 3 points (Note: Lone pairs are not shown in structures below)

(i) For the three *resonance* structures (1, 2 and 3) shown below, answer the following question:



(I) Circle the *most contributing* resonance structure.

Structure 1

Structure 2

Structure 3

(ii) For the three structures given (4, 5, and 6) below, answer the following questions:



(I) Circle all the structures that represent *resonance structures* of each other:

Structure 4

Structure 5

Structure 6

(II) Among the resonance structures, circle the *most contributing* resonance structure.

Structure 4

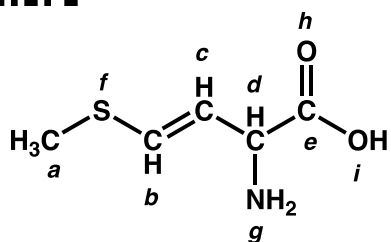
Structure 5

Structure 6

**Question 6** **6 points**

Using the data given below, calculate the **lattice energy** for SrO(s).

	ΔH° (kJ/mol)
$\text{Sr}(s) \longrightarrow \text{Sr}(g)$	+180
$\text{Sr}(g) \longrightarrow \text{Sr}^{2+}(g) + 2e^-$	+1468
$\text{O}_2(g) \longrightarrow 2\text{O}(g)$	+500
$\text{O}(g) + 2e^- \longrightarrow \text{O}^{2-}(g)$	+600
$\text{Sr}(s) + \frac{1}{2}\text{O}_2(g) \longrightarrow \text{SrO}(s)$	-550

**Question 7 10 points**

Consider the molecule given on the left, and answer questions (i) to (vi) using chemical bonding theories that we have learnt in class (valence bond theory, hybridization).

Note that the lone pairs are not shown in the structure:

i. Consider the carbon atoms marked in as ***a***, ***b***, ***c***, ***d***, and ***e*** in the molecule, and determine **hybridization** at each of the carbon centers. (2.5 points)

Atom	Hybridization
Carbon (<i>a</i>)	
Carbon (<i>b</i>)	
Carbon (<i>c</i>)	
Carbon (<i>d</i>)	
Carbon (<i>e</i>)	

ii. Now, consider the sulfur atom (S) marked as ***f***, nitrogen (N) atom (***g***), and oxygen(O) atoms (***h*** and ***i***), and determine their **hybridization**. (2 points)

Atom	Hybridization
Sulfur (<i>f</i>)	
Nitrogen (<i>g</i>)	
Oxygen (<i>h</i>)	
Oxygen (<i>i</i>)	

iii. **Describe** all the bonds formed by the carbon atoms ***a*** and ***b***, using sigma and pi designation as appropriate. (2 points)

Atom	Bonds formed (use sigma and pi as appropriate)
Carbon (<i>a</i>)	
Carbon (<i>b</i>)	

iv. How many total sp^3 -s sigma bonds are in the molecule? (1 point)

v. Determine **electronic** and **molecular geometry** at carbon (C) labeled ***e*** and sulfur (S) labeled ***f*** centres. (2 points)

Atom	Electronic Geometry	Molecular Geometry
Carbon (<i>e</i>)		
Sulfur (<i>f</i>)		

vi. Circle the value from the choices given below, that describes the **C-S-C bond angle at sulfur** (S) marked ***f***: (0.5 points)

>90° to <109.5° 109.5° >109.5 to <120° 120° >120° to <180°

**Question 8 12 points**

Using the **valence shell orbitals only** draw a **complete** molecular orbital (MO) diagram (with no s-p mixing) for the molecule **PS**, in the space provided below. **(6 points)**

Assume that 3s and 3p atomic orbitals will form molecular orbitals like those formed by the combination of 2s and 2p atomic orbitals that we learnt in class, and there is no s-p mixing.

Show **energy arrow**, **atomic orbitals** involved, **atom labels**, the **molecular orbitals** that are formed, **filled electrons** in the MO diagram etc. Make sure to label all the atomic and molecular orbitals.

Using the MO diagram above, answer the following questions:

- (i) Which one of the following will have the highest bond order? Circle the correct choice
A) PS^{1-} B) PS C) PS^{1+}
- (ii) Which one of the following will have the longest bond length? Circle the correct choice
A) PS^{1-} B) PS C) PS^{1+}
- (iii) Which of the following will be diamagnetic? Circle the correct choice.
A) PS^{1-} B) PS C) PS^{1+}
- (iv) Draw Lewis dot structure of the molecule with the shortest bond length (based on the MO diagram) from among the three (PS^{1-} , PS, PS^{1+}), and indicate the bond order calculated from the Lewis dot structure. **(3 points)**

Lewis Dot Structure

Bond Order



ADA838CB-D2F1-4401-9F1C-A7637FA1786B

midterm-exam-93388

#506 Page 10 of 10

SCRAP PAGE

**CHEMISTRY 110 –Midterm Exam****Thursday November 7, 2024, 6:30 – 8:30 pm**

Examiners: Pallavi Sirjoosingh and Ashok Kakkar

First name (please write as legibly as possible within the boxes)

--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--

Last name

--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--

Student ID number

--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--

This booklet has 8 questions on 10 pages (The last page can be used as a scrap page).

The staple must NOT be removed from the question booklet. Your intact question booklet must be returned at the end of the exam.

A separate booklet contains a periodic table, and datasheet.

Write your answers directly under the questions in the space provided. Do not write on the QR codes or near the top edges (as it may be cut off).

Please write **legibly and dark enough** for the scan to be readable. For students who use English as a second language, dictionaries are permitted. Calculators are allowed.

DO NOT ASK QUESTIONS OF THE INVIGILATORS about interpretations of the exam questions. If you have a concern about a question, please summarize it at the top of this page. All concerns will be evaluated after the exam. GOOD LUCK!

**Question 1 8 points** (*Assume all transitions are possible*)

a. 6 points Calculate the **energy (in kJ/mol)** required to ionize (completely remove) the electron from the **$n=3$** level of **He^+ ion** (He: Helium). Show your work including equations. (3 sig. figs.)

b. 2 points

(I) Select the largest (longest) wavelength from the three choices below:

- i. Wavelength of photon to *completely remove* an electron from *1s orbital of Hydrogen atom*
- ii. Wavelength of photon to *completely remove* an electron from *1s orbital of Boron atom (B)*
- iii. Wavelength of photon to *completely remove* an electron from *1s orbital of Boron cation (B^+)*

i or ii or iii

(II) Select the smallest (shortest) wavelength from the three choices below (*for H atom*):

- i. Wavelength of photon *emitted* for an electron transitioning from *2p to 1s orbital*
- ii. Wavelength of photon *absorbed* for an electron transitioning from *2s to 4p orbital*
- iii. Wavelength of photon *absorbed* for an electron transitioning from *1s to 3p orbital*

i or ii or iii



Question 2 8 points



A Chem110 student plots the graph shown on the left, from data obtained for the kinetic energy (eV) of the ejected electron versus frequency (s^{-1}) of the incident light, for an unknown metal. ($1\text{eV} = 1.6022 \times 10^{-19} \text{ J}$)

Based on the plot, answer the questions **a to c** below:

a. 1 point On the plot above, label the **threshold frequency**.

b. 2 points What is the value of the work potential (in J) of the unknown metal?

c. 3 points What is the *longest* wavelength (in nm) required to eject an electron from this metal's surface? Show your work including equation(s) used. (3 sig figs)

d. 2 points In the space provided below, plot the following

(Assume frequency is *above* threshold frequency):

- Kinetic energy* of an ejected electron versus the *intensity* of the incident light (left)
- Number of electrons* ejected versus the *frequency* of the incident light (right)



**Question 3** **6 points****a. 2 points**

In the ground state of Ge (Germanium; $Z = 32$) atom, **name** the following orbitals:

i. Orbital containing the electron that requires the maximum energy to be removed.

ii. Orbital containing the electron that experiences the minimum effective nuclear charge.

b. 2 points

An unknown anion (A^-) has the following *excited* electronic configuration: $[Ar] 4s^2 3d^3 4p^1$

(Ar: Argon; $Z = 18$)

Give the **condensed ground state electronic configuration** of the atom, **A** and the cation, **A^{2+}** .
Show your work. (Assume all electronic transitions are possible)

Electronic Configuration of A:

Electronic Configuration of A^{2+} :

c. 2 points

In a single atom, determine how many electrons can have the following set of quantum numbers?

(i) $n = 5; l = 3$

In a single atom, determine how many electrons can have the following set of quantum numbers?

(i) $n = 3; l = 2; m_s = +1/2$

**Question 4** **8 points**

Rank the following from 1 to 4 in **increasing** order. Provide the ranking in the boxes provided.

Note: Atomic number (Z) are provided for relevant atoms

a) Increasing (lowest to highest) order of *ionization energy*: (1= lowest; 4 = highest)

Mg

Mg: Z = 12

Rb

Rb: Z = 37

Cs

Cs: Z = 55

K

K : Z = 19

b) Increasing (lowest to highest) order of *bond angle*: (1= lowest; 4 = highest)

H₂OCCl₄XeF₄

HCN

c) Increasing (smallest to largest) order of *ionic size* (1= smallest; 4 = largest)

S²⁻

S: Z = 16

O²⁻

O: Z = 8

Be²⁺

Be: Z = 4

Na⁺

Na: Z = 11

d) Increasing (lowest to highest) order of *bond energy* (1= lowest; 4 = highest)

P-F

P-Cl

P-I

P-Br

**Question 5** **12 points**

a. 9 points For the molecules HClO_4 and TeF_6 , draw the most stable *Lewis structure* showing appropriate shape and *dashed/wedged bonds*, based on VSEPR. Determine the **molecular geometry** around the central atom. For HClO_4 , label the *most polar bond*.



b. 3 points (Note: Lone pairs are not shown in structures below)

(i) For the three *resonance* structures (1, 2 and 3) shown below, answer the following question:



(I) Circle the *most contributing* resonance structure.

Structure 1

Structure 2

Structure 3

(ii) For the three structures given (4, 5, and 6) below, answer the following questions:



(I) Circle all the structures that represent *resonance structures* of each other:

Structure 4

Structure 5

Structure 6

(II) Among the resonance structures, circle the *most contributing* resonance structure.

Structure 4

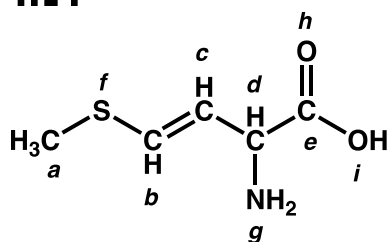
Structure 5

Structure 6

**Question 6** **6 points**

Using the data given below, calculate the **lattice energy** for SrO(s).

	ΔH° (kJ/mol)
$\text{Sr}(s) \longrightarrow \text{Sr}(g)$	+180
$\text{Sr}(g) \longrightarrow \text{Sr}^{2+}(g) + 2e^-$	+1468
$\text{O}_2(g) \longrightarrow 2\text{O}(g)$	+500
$\text{O}(g) + 2e^- \longrightarrow \text{O}^{2-}(g)$	+600
$\text{Sr}(s) + \frac{1}{2}\text{O}_2(g) \longrightarrow \text{SrO}(s)$	-550

**Question 7 10 points**

Consider the molecule given on the left, and answer questions (i) to (vi) using chemical bonding theories that we have learnt in class (valence bond theory, hybridization).

Note that the lone pairs are not shown in the structure:

i. Consider the carbon atoms marked in as ***a***, ***b***, ***c***, ***d***, and ***e*** in the molecule, and determine **hybridization** at each of the carbon centers. (2.5 points)

Atom	Hybridization
Carbon (<i>a</i>)	
Carbon (<i>b</i>)	
Carbon (<i>c</i>)	
Carbon (<i>d</i>)	
Carbon (<i>e</i>)	

ii. Now, consider the sulfur atom (S) marked as ***f***, nitrogen (N) atom (***g***), and oxygen(O) atoms (***h*** and ***i***), and determine their **hybridization**. (2 points)

Atom	Hybridization
Sulfur (<i>f</i>)	
Nitrogen (<i>g</i>)	
Oxygen (<i>h</i>)	
Oxygen (<i>i</i>)	

iii. **Describe** all the bonds formed by the carbon atoms ***a*** and ***b***, using sigma and pi designation as appropriate. (2 points)

Atom	Bonds formed (use sigma and pi as appropriate)
Carbon (<i>a</i>)	
Carbon (<i>b</i>)	

iv. How many total sp^3 -s sigma bonds are in the molecule? (1 point)

v. Determine **electronic** and **molecular geometry** at carbon (C) labeled ***e*** and sulfur (S) labeled ***f*** centres. (2 points)

Atom	Electronic Geometry	Molecular Geometry
Carbon (<i>e</i>)		
Sulfur (<i>f</i>)		

vi. Circle the value from the choices given below, that describes the **C-S-C bond angle at sulfur** (S) marked ***f***: (0.5 points)

>90° to <109.5°

109.5°

>109.5 to <120°

120°

>120° to <180°


Question 8 12 points

Using the **valence shell orbitals only** draw a **complete** molecular orbital (MO) diagram (with no s-p mixing) for the molecule **PS**, in the space provided below. **(6 points)**

Assume that 3s and 3p atomic orbitals will form molecular orbitals like those formed by the combination of 2s and 2p atomic orbitals that we learnt in class, and there is no s-p mixing.

Show **energy arrow**, **atomic orbitals** involved, **atom labels**, the **molecular orbitals** that are formed, **filled electrons** in the MO diagram etc. Make sure to label all the atomic and molecular orbitals.

Using the MO diagram above, answer the following questions:

- (i) Which one of the following will have the highest bond order? Circle the correct choice
 A) PS^{1-} B) PS C) PS^{1+}
- (ii) Which one of the following will have the longest bond length? Circle the correct choice
 A) PS^{1-} B) PS C) PS^{1+}
- (iii) Which of the following will be diamagnetic? Circle the correct choice.
 A) PS^{1-} B) PS C) PS^{1+}
- (iv) Draw Lewis dot structure of the molecule with the shortest bond length (based on the MO diagram) from among the three (PS^{1-} , PS, PS^{1+}), and indicate the bond order calculated from the Lewis dot structure. **(3 points)**

Lewis Dot Structure

Bond Order



4E7FFB9E-1928-4CA6-A354-0E8C6ACF5319

midterm-exam-93388

#507 Page 10 of 10

SCRAP PAGE

**CHEMISTRY 110 –Midterm Exam****Thursday November 7, 2024, 6:30 – 8:30 pm**

Examiners: Pallavi Sirjoosingh and Ashok Kakkar

First name (please write as legibly as possible within the boxes)

--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--

Last name

--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--

Student ID number

--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--

This booklet has 8 questions on 10 pages (The last page can be used as a scrap page).

The staple must NOT be removed from the question booklet. Your intact question booklet must be returned at the end of the exam.

A separate booklet contains a periodic table, and datasheet.

Write your answers directly under the questions in the space provided. Do not write on the QR codes or near the top edges (as it may be cut off).

Please write **legibly and dark enough** for the scan to be readable. For students who use English as a second language, dictionaries are permitted. Calculators are allowed.

DO NOT ASK QUESTIONS OF THE INVIGILATORS about interpretations of the exam questions. If you have a concern about a question, please summarize it at the top of this page. All concerns will be evaluated after the exam. GOOD LUCK!

**Question 1 8 points** (*Assume all transitions are possible*)

a. 6 points Calculate the **energy (in kJ/mol)** required to ionize (completely remove) the electron from the **n=3** level of **He⁺ ion** (He: Helium). Show your work including equations. (3 sig. figs.)

b. 2 points

(I) Select the largest (longest) wavelength from the three choices below:

- i. Wavelength of photon to *completely remove* an electron from *1s orbital of Hydrogen atom*
- ii. Wavelength of photon to *completely remove* an electron from *1s orbital of Boron atom (B)*
- iii. Wavelength of photon to *completely remove* an electron from *1s orbital of Boron cation (B⁺)*

i or ii or iii

(II) Select the smallest (shortest) wavelength from the three choices below (*for H atom*):

- i. Wavelength of photon *emitted* for an electron transitioning from *2p to 1s orbital*
- ii. Wavelength of photon *absorbed* for an electron transitioning from *2s to 4p orbital*
- iii. Wavelength of photon *absorbed* for an electron transitioning from *1s to 3p orbital*

i or ii or iii



Question 2 8 points



A Chem110 student plots the graph shown on the left, from data obtained for the kinetic energy (eV) of the ejected electron versus frequency (s^{-1}) of the incident light, for an unknown metal. ($1\text{eV} = 1.6022 \times 10^{-19} \text{ J}$)

Based on the plot, answer the questions **a to c** below:

a. 1 point On the plot above, label the **threshold frequency**.

b. 2 points What is the value of the work potential (in J) of the unknown metal?

c. 3 points What is the *longest* wavelength (in nm) required to eject an electron from this metal's surface? Show your work including equation(s) used. (3 sig figs)

d. 2 points In the space provided below, plot the following

(Assume frequency is *above* threshold frequency):

- i. *Kinetic energy* of an ejected electron versus the *intensity* of the incident light (left)
- ii. *Number of electrons* ejected versus the *frequency* of the incident light (right)



**Question 3** **6 points****a. 2 points**

In the ground state of Ge (Germanium; $Z = 32$) atom, **name** the following orbitals:

i. Orbital containing the electron that requires the maximum energy to be removed.

ii. Orbital containing the electron that experiences the minimum effective nuclear charge.

b. 2 points

An unknown anion (A^-) has the following *excited* electronic configuration: $[Ar] 4s^2 3d^3 4p^1$

(Ar: Argon; $Z = 18$)

Give the **condensed ground state electronic configuration** of the atom, **A** and the cation, A^{2+} .
Show your work. (Assume all electronic transitions are possible)

Electronic Configuration of A:

Electronic Configuration of A^{2+} :

c. 2 points

In a single atom, determine how many electrons can have the following set of quantum numbers?

(i) $n = 5; l = 3$

In a single atom, determine how many electrons can have the following set of quantum numbers?

(i) $n = 3; l = 2; m_s = +1/2$

**Question 4** **8 points**

Rank the following from 1 to 4 in **increasing** order. Provide the ranking in the boxes provided.

Note: Atomic number (Z) are provided for relevant atoms

a) Increasing (lowest to highest) order of *ionization energy*: (1= lowest; 4 = highest)

Mg

Mg: Z = 12

Rb

Rb: Z = 37

Cs

Cs: Z = 55

K

K : Z = 19

b) Increasing (lowest to highest) order of *bond angle*: (1= lowest; 4 = highest)

H₂OCCl₄XeF₄

HCN

c) Increasing (smallest to largest) order of *ionic size* (1= smallest; 4 = largest)

S²⁻

S: Z = 16

O²⁻

O: Z = 8

Be²⁺

Be: Z = 4

Na⁺

Na: Z = 11

d) Increasing (lowest to highest) order of *bond energy* (1= lowest; 4 = highest)

P-F

P-Cl

P-I

P-Br

**Question 5** **12 points**

a. 9 points For the molecules HClO_4 and TeF_6 , draw the most stable *Lewis structure* showing appropriate shape and *dashed/wedged bonds*, based on VSEPR. Determine the **molecular geometry** around the central atom. For HClO_4 , label the *most polar bond*.



b. 3 points (Note: Lone pairs are not shown in structures below)

(i) For the three *resonance* structures (1, 2 and 3) shown below, answer the following question:



(I) Circle the *most contributing* resonance structure.

Structure 1

Structure 2

Structure 3

(ii) For the three structures given (4, 5, and 6) below, answer the following questions:



(I) Circle all the structures that represent *resonance structures* of each other:

Structure 4

Structure 5

Structure 6

(II) Among the resonance structures, circle the *most contributing* resonance structure.

Structure 4

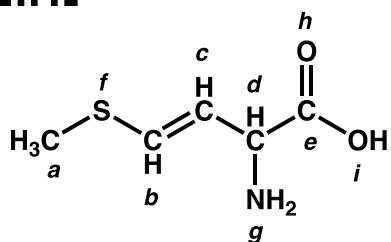
Structure 5

Structure 6

**Question 6** **6 points**

Using the data given below, calculate the **lattice energy** for SrO(s).

	ΔH° (kJ/mol)
$\text{Sr}(s) \longrightarrow \text{Sr}(g)$	+180
$\text{Sr}(g) \longrightarrow \text{Sr}^{2+}(g) + 2e^-$	+1468
$\text{O}_2(g) \longrightarrow 2\text{O}(g)$	+500
$\text{O}(g) + 2e^- \longrightarrow \text{O}^{2-}(g)$	+600
$\text{Sr}(s) + \frac{1}{2}\text{O}_2(g) \longrightarrow \text{SrO}(s)$	-550

**Question 7 10 points**

Consider the molecule given on the left, and answer questions (i) to (vi) using chemical bonding theories that we have learnt in class (valence bond theory, hybridization).

Note that the lone pairs are not shown in the structure:

i. Consider the carbon atoms marked in as ***a***, ***b***, ***c***, ***d***, and ***e*** in the molecule, and determine **hybridization** at each of the carbon centers. (2.5 points)

Atom	Hybridization
Carbon (<i>a</i>)	
Carbon (<i>b</i>)	
Carbon (<i>c</i>)	
Carbon (<i>d</i>)	
Carbon (<i>e</i>)	

ii. Now, consider the sulfur atom (S) marked as ***f***, nitrogen (N) atom (***g***), and oxygen(O) atoms (***h*** and ***i***), and determine their **hybridization**. (2 points)

Atom	Hybridization
Sulfur (<i>f</i>)	
Nitrogen (<i>g</i>)	
Oxygen (<i>h</i>)	
Oxygen (<i>i</i>)	

iii. **Describe** all the bonds formed by the carbon atoms ***a*** and ***b***, using sigma and pi designation as appropriate. (2 points)

Atom	Bonds formed (use sigma and pi as appropriate)
Carbon (<i>a</i>)	
Carbon (<i>b</i>)	

iv. How many total sp^3 -s sigma bonds are in the molecule? (1 point)

v. Determine **electronic** and **molecular geometry** at carbon (C) labeled ***e*** and sulfur (S) labeled ***f*** centres. (2 points)

Atom	Electronic Geometry	Molecular Geometry
Carbon (<i>e</i>)		
Sulfur (<i>f</i>)		

vi. Circle the value from the choices given below, that describes the **C-S-C bond angle at sulfur** (S) marked ***f***: (0.5 points)

>90° to <109.5°

109.5°

>109.5 to <120°

120°

>120° to <180°


Question 8 12 points

Using the **valence shell orbitals only** draw a **complete** molecular orbital (MO) diagram (with no s-p mixing) for the molecule **PS**, in the space provided below. **(6 points)**

Assume that 3s and 3p atomic orbitals will form molecular orbitals like those formed by the combination of 2s and 2p atomic orbitals that we learnt in class, and there is no s-p mixing.

Show **energy arrow**, **atomic orbitals** involved, **atom labels**, the **molecular orbitals** that are formed, **filled electrons** in the MO diagram etc. Make sure to **label all the atomic and molecular orbitals**.

Using the MO diagram above, answer the following questions:

- (i) Which one of the following will have the highest bond order? Circle the correct choice
 A) PS^{1-} B) PS C) PS^{1+}
- (ii) Which one of the following will have the longest bond length? Circle the correct choice
 A) PS^{1-} B) PS C) PS^{1+}
- (iii) Which of the following will be diamagnetic? Circle the correct choice.
 A) PS^{1-} B) PS C) PS^{1+}
- (iv) Draw Lewis dot structure of the molecule with the shortest bond length (based on the MO diagram) from among the three (PS^{1-} , PS, PS^{1+}), and indicate the bond order calculated from the Lewis dot structure. **(3 points)**

Lewis Dot Structure

Bond Order



55234E2B-0FDB-4B86-8082-B24989B14E30

midterm-exam-93388

#508 Page 10 of 10

SCRAP PAGE

**CHEMISTRY 110 –Midterm Exam****Thursday November 7, 2024, 6:30 – 8:30 pm**

Examiners: Pallavi Sirjoosingh and Ashok Kakkar

First name (please write as legibly as possible within the boxes)

--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--

Last name

--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--

Student ID number

--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--

This booklet has 8 questions on 10 pages (The last page can be used as a scrap page).

The staple must NOT be removed from the question booklet. Your intact question booklet must be returned at the end of the exam.

A separate booklet contains a periodic table, and datasheet.

Write your answers directly under the questions in the space provided. Do not write on the QR codes or near the top edges (as it may be cut off).

Please write **legibly and dark enough** for the scan to be readable. For students who use English as a second language, dictionaries are permitted. Calculators are allowed.

DO NOT ASK QUESTIONS OF THE INVIGILATORS about interpretations of the exam questions. If you have a concern about a question, please summarize it at the top of this page. All concerns will be evaluated after the exam. GOOD LUCK!

**Question 1 8 points** (*Assume all transitions are possible*)

a. 6 points Calculate the **energy (in kJ/mol)** required to ionize (completely remove) the electron from the **n=3** level of **He⁺ ion** (He: Helium). Show your work including equations. (3 sig. figs.)

b. 2 points

(I) Select the largest (longest) wavelength from the three choices below:

- i. Wavelength of photon to *completely remove* an electron from *1s orbital of Hydrogen atom*
- ii. Wavelength of photon to *completely remove* an electron from *1s orbital of Boron atom (B)*
- iii. Wavelength of photon to *completely remove* an electron from *1s orbital of Boron cation (B⁺)*

i or ii or iii

(II) Select the smallest (shortest) wavelength from the three choices below (*for H atom*):

- i. Wavelength of photon *emitted* for an electron transitioning from *2p to 1s orbital*
- ii. Wavelength of photon *absorbed* for an electron transitioning from *2s to 4p orbital*
- iii. Wavelength of photon *absorbed* for an electron transitioning from *1s to 3p orbital*

i or ii or iii



Question 2 8 points



A Chem110 student plots the graph shown on the left, from data obtained for the kinetic energy (eV) of the ejected electron versus frequency (s^{-1}) of the incident light, for an unknown metal. ($1\text{eV} = 1.6022 \times 10^{-19} \text{ J}$)

Based on the plot, answer the questions **a to c** below:

a. 1 point On the plot above, label the **threshold frequency**.

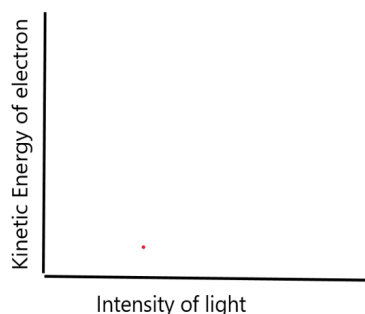
b. 2 points What is the value of the work potential (in J) of the unknown metal?

c. 3 points What is the *longest* wavelength (in nm) required to eject an electron from this metal's surface? Show your work including equation(s) used. (3 sig figs)

d. 2 points In the space provided below, plot the following

(Assume frequency is *above* threshold frequency):

- Kinetic energy* of an ejected electron versus the *intensity* of the incident light (left)
- Number of electrons* ejected versus the *frequency* of the incident light (right)



**Question 3** **6 points****a. 2 points**

In the ground state of Ge (Germanium; $Z = 32$) atom, **name** the following orbitals:

i. Orbital containing the electron that requires the maximum energy to be removed.

ii. Orbital containing the electron that experiences the minimum effective nuclear charge.

b. 2 points

An unknown anion (A^-) has the following *excited* electronic configuration: $[Ar] 4s^2 3d^3 4p^1$

(Ar: Argon; $Z = 18$)

Give the **condensed ground state electronic configuration** of the atom, **A** and the cation, **A^{2+}** .
Show your work. (Assume all electronic transitions are possible)

Electronic Configuration of A:

Electronic Configuration of A^{2+} :

c. 2 points

In a single atom, determine how many electrons can have the following set of quantum numbers?

(i) $n = 5; l = 3$

In a single atom, determine how many electrons can have the following set of quantum numbers?

(i) $n = 3; l = 2; m_s = +1/2$

**Question 4** **8 points**

Rank the following from 1 to 4 in **increasing** order. Provide the ranking in the boxes provided.

Note: Atomic number (Z) are provided for relevant atoms

a) Increasing (lowest to highest) order of *ionization energy*: (1= lowest; 4 = highest)

Mg

Mg: Z = 12

Rb

Rb: Z = 37

Cs

Cs: Z = 55

K

K : Z = 19

b) Increasing (lowest to highest) order of *bond angle*: (1= lowest; 4 = highest)

H₂OCCl₄XeF₄

HCN

c) Increasing (smallest to largest) order of *ionic size* (1= smallest; 4 = largest)

S²⁻

S: Z = 16

O²⁻

O: Z = 8

Be²⁺

Be: Z = 4

Na⁺

Na: Z = 11

d) Increasing (lowest to highest) order of *bond energy* (1= lowest; 4 = highest)

P-F

P-Cl

P-I

P-Br

**Question 5** **12 points**

a. 9 points For the molecules HClO_4 and TeF_6 , draw the most stable *Lewis structure* showing appropriate shape and *dashed/wedged bonds*, based on VSEPR. Determine the **molecular geometry** around the central atom. For HClO_4 , label the *most polar bond*.



b. 3 points (Note: Lone pairs are not shown in structures below)

(i) For the three *resonance* structures (1, 2 and 3) shown below, answer the following question:



(I) Circle the *most contributing* resonance structure.

Structure 1

Structure 2

Structure 3

(ii) For the three structures given (4, 5, and 6) below, answer the following questions:



(I) Circle all the structures that represent *resonance structures* of each other:

Structure 4

Structure 5

Structure 6

(II) Among the resonance structures, circle the *most contributing* resonance structure.

Structure 4

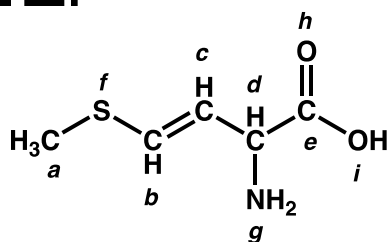
Structure 5

Structure 6

**Question 6** **6 points**

Using the data given below, calculate the **lattice energy** for SrO(s).

	ΔH° (kJ/mol)
$\text{Sr}(s) \longrightarrow \text{Sr}(g)$	+180
$\text{Sr}(g) \longrightarrow \text{Sr}^{2+}(g) + 2e^-$	+1468
$\text{O}_2(g) \longrightarrow 2\text{O}(g)$	+500
$\text{O}(g) + 2e^- \longrightarrow \text{O}^{2-}(g)$	+600
$\text{Sr}(s) + \frac{1}{2}\text{O}_2(g) \longrightarrow \text{SrO}(s)$	-550

**Question 7 10 points**

Consider the molecule given on the left, and answer questions (i) to (vi) using chemical bonding theories that we have learnt in class (valence bond theory, hybridization).

Note that the lone pairs are not shown in the structure:

i. Consider the carbon atoms marked in as ***a***, ***b***, ***c***, ***d***, and ***e*** in the molecule, and determine **hybridization** at each of the carbon centers. (2.5 points)

Atom	Hybridization
Carbon (<i>a</i>)	
Carbon (<i>b</i>)	
Carbon (<i>c</i>)	
Carbon (<i>d</i>)	
Carbon (<i>e</i>)	

ii. Now, consider the sulfur atom (S) marked as ***f***, nitrogen (N) atom (***g***), and oxygen(O) atoms (***h*** and ***i***), and determine their **hybridization**. (2 points)

Atom	Hybridization
Sulfur (<i>f</i>)	
Nitrogen (<i>g</i>)	
Oxygen (<i>h</i>)	
Oxygen (<i>i</i>)	

iii. **Describe** all the bonds formed by the carbon atoms ***a*** and ***b***, using sigma and pi designation as appropriate. (2 points)

Atom	Bonds formed (use sigma and pi as appropriate)
Carbon (<i>a</i>)	
Carbon (<i>b</i>)	

iv. How many total sp^3 -s sigma bonds are in the molecule? (1 point)

v. Determine **electronic** and **molecular geometry** at carbon (C) labeled ***e*** and sulfur (S) labeled ***f*** centres. (2 points)

Atom	Electronic Geometry	Molecular Geometry
Carbon (<i>e</i>)		
Sulfur (<i>f</i>)		

vi. Circle the value from the choices given below, that describes the **C-S-C bond angle at sulfur** (S) marked ***f***: (0.5 points)

>90° to <109.5° 109.5° >109.5 to <120° 120° >120° to <180°

**Question 8 12 points**

Using the **valence shell orbitals only** draw a **complete** molecular orbital (MO) diagram (with no s-p mixing) for the molecule **PS**, in the space provided below. **(6 points)**

Assume that 3s and 3p atomic orbitals will form molecular orbitals like those formed by the combination of 2s and 2p atomic orbitals that we learnt in class, and there is no s-p mixing.

Show **energy arrow**, **atomic orbitals** involved, **atom labels**, the **molecular orbitals** that are formed, **filled electrons** in the MO diagram etc. Make sure to label all the atomic and molecular orbitals.

Using the MO diagram above, answer the following questions:

- (i) Which one of the following will have the highest bond order? Circle the correct choice
A) PS^{1-} B) PS C) PS^{1+}
- (ii) Which one of the following will have the longest bond length? Circle the correct choice
A) PS^{1-} B) PS C) PS^{1+}
- (iii) Which of the following will be diamagnetic? Circle the correct choice.
A) PS^{1-} B) PS C) PS^{1+}
- (iv) Draw Lewis dot structure of the molecule with the shortest bond length (based on the MO diagram) from among the three (PS^{1-} , PS, PS^{1+}), and indicate the bond order calculated from the Lewis dot structure. **(3 points)**

Lewis Dot Structure

Bond Order



4CE71079-5B7E-4A42-A25D-C382493617D8

midterm-exam-93388

#509 Page 10 of 10

SCRAP PAGE

**CHEMISTRY 110 –Midterm Exam****Thursday November 7, 2024, 6:30 – 8:30 pm****Examiners: Pallavi Sirjoosingh and Ashok Kakkar**

First name (please write as legibly as possible within the boxes)

--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--

Last name

--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--

Student ID number

--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--

This booklet has 8 questions on 10 pages (The last page can be used as a scrap page).

The staple must NOT be removed from the question booklet. Your intact question booklet must be returned at the end of the exam.

A separate booklet contains a periodic table, and datasheet.

Write your answers directly under the questions in the space provided. Do not write on the QR codes or near the top edges (as it may be cut off).

Please write **legibly and dark enough** for the scan to be readable. For students who use English as a second language, dictionaries are permitted. Calculators are allowed.

DO NOT ASK QUESTIONS OF THE INVIGILATORS about interpretations of the exam questions. If you have a concern about a question, please summarize it at the top of this page. All concerns will be evaluated after the exam. GOOD LUCK!

**Question 1 8 points** (*Assume all transitions are possible*)

a. 6 points Calculate the **energy (in kJ/mol)** required to ionize (completely remove) the electron from the **n=3** level of **He⁺ ion** (He: Helium). Show your work including equations. (3 sig. figs.)

b. 2 points

(I) Select the largest (longest) wavelength from the three choices below:

- i. Wavelength of photon to *completely remove* an electron from *1s orbital of Hydrogen atom*
- ii. Wavelength of photon to *completely remove* an electron from *1s orbital of Boron atom (B)*
- iii. Wavelength of photon to *completely remove* an electron from *1s orbital of Boron cation (B⁺)*

i or ii or iii

(II) Select the smallest (shortest) wavelength from the three choices below (*for H atom*):

- i. Wavelength of photon *emitted* for an electron transitioning from *2p to 1s orbital*
- ii. Wavelength of photon *absorbed* for an electron transitioning from *2s to 4p orbital*
- iii. Wavelength of photon *absorbed* for an electron transitioning from *1s to 3p orbital*

i or ii or iii



Question 2 8 points



A Chem110 student plots the graph shown on the left, from data obtained for the kinetic energy (eV) of the ejected electron versus frequency (s^{-1}) of the incident light, for an unknown metal. ($1\text{eV} = 1.6022 \times 10^{-19} \text{ J}$)

Based on the plot, answer the questions **a to c** below:

a. 1 point On the plot above, label the **threshold frequency**.

b. 2 points What is the value of the work potential (in J) of the unknown metal?

c. 3 points What is the *longest* wavelength (in nm) required to eject an electron from this metal's surface? Show your work including equation(s) used. (3 sig figs)

d. 2 points In the space provided below, plot the following

(Assume frequency is *above* threshold frequency):

- i. *Kinetic energy* of an ejected electron versus the *intensity* of the incident light (left)
- ii. *Number of electrons* ejected versus the *frequency* of the incident light (right)



**Question 3** **6 points****a. 2 points**

In the ground state of Ge (Germanium; $Z = 32$) atom, **name** the following orbitals:

i. Orbital containing the electron that requires the maximum energy to be removed.

ii. Orbital containing the electron that experiences the minimum effective nuclear charge.

b. 2 points

An unknown anion (A^-) has the following *excited* electronic configuration: $[Ar] 4s^2 3d^3 4p^1$

(Ar: Argon; $Z = 18$)

Give the **condensed ground state electronic configuration** of the atom, **A** and the cation, **A^{2+}** .
Show your work. (Assume all electronic transitions are possible)

Electronic Configuration of A:

Electronic Configuration of A^{2+} :

c. 2 points

In a single atom, determine how many electrons can have the following set of quantum numbers?

(i) $n = 5; l = 3$

In a single atom, determine how many electrons can have the following set of quantum numbers?

(i) $n = 3; l = 2; m_s = +1/2$

**Question 4** **8 points**

Rank the following from 1 to 4 in **increasing** order. Provide the ranking in the boxes provided.

Note: Atomic number (Z) are provided for relevant atoms

a) Increasing (lowest to highest) order of *ionization energy*: (1= lowest; 4 = highest)

Mg

Mg: Z = 12

Rb

Rb: Z = 37

Cs

Cs: Z = 55

K

K : Z = 19

b) Increasing (lowest to highest) order of *bond angle*: (1= lowest; 4 = highest)

H₂OCCl₄XeF₄

HCN

c) Increasing (smallest to largest) order of *ionic size* (1= smallest; 4 = largest)

S²⁻

S: Z = 16

O²⁻

O: Z = 8

Be²⁺

Be: Z = 4

Na⁺

Na: Z = 11

d) Increasing (lowest to highest) order of *bond energy* (1= lowest; 4 = highest)

P-F

P-Cl

P-I

P-Br

**Question 5** **12 points**

a. 9 points For the molecules HClO_4 and TeF_6 , draw the most stable *Lewis structure* showing appropriate shape and *dashed/wedged bonds*, based on VSEPR. Determine the **molecular geometry** around the central atom. For HClO_4 , label the *most polar bond*.



b. 3 points (Note: Lone pairs are not shown in structures below)

(i) For the three *resonance* structures (1, 2 and 3) shown below, answer the following question:



(I) Circle the *most contributing* resonance structure.

Structure 1

Structure 2

Structure 3

(ii) For the three structures given (4, 5, and 6) below, answer the following questions:



(I) Circle all the structures that represent *resonance structures* of each other:

Structure 4

Structure 5

Structure 6

(II) Among the resonance structures, circle the *most contributing* resonance structure.

Structure 4

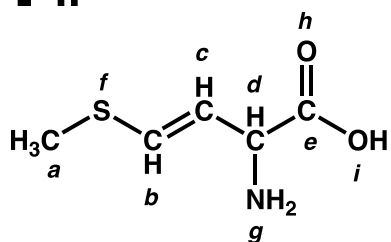
Structure 5

Structure 6

**Question 6** **6 points**

Using the data given below, calculate the **lattice energy** for SrO(s).

	ΔH° (kJ/mol)
$\text{Sr}(s) \longrightarrow \text{Sr}(g)$	+180
$\text{Sr}(g) \longrightarrow \text{Sr}^{2+}(g) + 2e^-$	+1468
$\text{O}_2(g) \longrightarrow 2\text{O}(g)$	+500
$\text{O}(g) + 2e^- \longrightarrow \text{O}^{2-}(g)$	+600
$\text{Sr}(s) + \frac{1}{2}\text{O}_2(g) \longrightarrow \text{SrO}(s)$	-550

**Question 7 10 points**

Consider the molecule given on the left, and answer questions (i) to (vi) using chemical bonding theories that we have learnt in class (valence bond theory, hybridization).

Note that the lone pairs are not shown in the structure:

i. Consider the carbon atoms marked in as ***a***, ***b***, ***c***, ***d***, and ***e*** in the molecule, and determine **hybridization** at each of the carbon centers. (2.5 points)

Atom	Hybridization
Carbon (<i>a</i>)	
Carbon (<i>b</i>)	
Carbon (<i>c</i>)	
Carbon (<i>d</i>)	
Carbon (<i>e</i>)	

ii. Now, consider the sulfur atom (S) marked as ***f***, nitrogen (N) atom (***g***), and oxygen(O) atoms (***h*** and ***i***), and determine their **hybridization**. (2 points)

Atom	Hybridization
Sulfur (<i>f</i>)	
Nitrogen (<i>g</i>)	
Oxygen (<i>h</i>)	
Oxygen (<i>i</i>)	

iii. **Describe** all the bonds formed by the carbon atoms ***a*** and ***b***, using sigma and pi designation as appropriate. (2 points)

Atom	Bonds formed (use sigma and pi as appropriate)
Carbon (<i>a</i>)	
Carbon (<i>b</i>)	

iv. How many total sp^3 -s sigma bonds are in the molecule? (1 point)

v. Determine **electronic** and **molecular geometry** at carbon (C) labeled ***e*** and sulfur (S) labeled ***f*** centres. (2 points)

Atom	Electronic Geometry	Molecular Geometry
Carbon (<i>e</i>)		
Sulfur (<i>f</i>)		

vi. Circle the value from the choices given below, that describes the **C-S-C bond angle at sulfur** (S) marked ***f***: (0.5 points)

>90° to <109.5° 109.5° >109.5 to <120° 120° >120° to <180°


Question 8 12 points

Using the **valence shell orbitals only** draw a **complete** molecular orbital (MO) diagram (with no s-p mixing) for the molecule **PS**, in the space provided below. **(6 points)**

Assume that 3s and 3p atomic orbitals will form molecular orbitals like those formed by the combination of 2s and 2p atomic orbitals that we learnt in class, and there is no s-p mixing.

Show **energy arrow**, **atomic orbitals** involved, **atom labels**, the **molecular orbitals** that are formed, **filled electrons** in the MO diagram etc. Make sure to label all the atomic and molecular orbitals.

Using the MO diagram above, answer the following questions:

- (i) Which one of the following will have the highest bond order? Circle the correct choice
 A) PS^{1-} B) PS C) PS^{1+}
- (ii) Which one of the following will have the longest bond length? Circle the correct choice
 A) PS^{1-} B) PS C) PS^{1+}
- (iii) Which of the following will be diamagnetic? Circle the correct choice.
 A) PS^{1-} B) PS C) PS^{1+}
- (iv) Draw Lewis dot structure of the molecule with the shortest bond length (based on the MO diagram) from among the three (PS^{1-} , PS, PS^{1+}), and indicate the bond order calculated from the Lewis dot structure. **(3 points)**

Lewis Dot Structure

Bond Order



CE3D8167-BBE7-4C9A-AD7D-8F29614FEC9F

midterm-exam-93388

#510 Page 10 of 10

SCRAP PAGE

**CHEMISTRY 110 –Midterm Exam****Thursday November 7, 2024, 6:30 – 8:30 pm**

Examiners: Pallavi Sirjoosingh and Ashok Kakkar

First name (please write as legibly as possible within the boxes)

--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--

Last name

--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--

Student ID number

--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--

This booklet has 8 questions on 10 pages (The last page can be used as a scrap page).

The staple must NOT be removed from the question booklet. Your intact question booklet must be returned at the end of the exam.

A separate booklet contains a periodic table, and datasheet.

Write your answers directly under the questions in the space provided. Do not write on the QR codes or near the top edges (as it may be cut off).

Please write **legibly and dark enough** for the scan to be readable. For students who use English as a second language, dictionaries are permitted. Calculators are allowed.

DO NOT ASK QUESTIONS OF THE INVIGILATORS about interpretations of the exam questions. If you have a concern about a question, please summarize it at the top of this page. All concerns will be evaluated after the exam. GOOD LUCK!

**Question 1 8 points** (*Assume all transitions are possible*)

a. 6 points Calculate the **energy (in kJ/mol)** required to ionize (completely remove) the electron from the **n=3** level of **He⁺ ion** (He: Helium). Show your work including equations. (3 sig. figs.)

b. 2 points

(I) Select the largest (longest) wavelength from the three choices below:

- i. Wavelength of photon to *completely remove* an electron from *1s orbital of Hydrogen atom*
- ii. Wavelength of photon to *completely remove* an electron from *1s orbital of Boron atom (B)*
- iii. Wavelength of photon to *completely remove* an electron from *1s orbital of Boron cation (B⁺)*

i or ii or iii

(II) Select the smallest (shortest) wavelength from the three choices below (*for H atom*):

- i. Wavelength of photon *emitted* for an electron transitioning from *2p to 1s orbital*
- ii. Wavelength of photon *absorbed* for an electron transitioning from *2s to 4p orbital*
- iii. Wavelength of photon *absorbed* for an electron transitioning from *1s to 3p orbital*

i or ii or iii



Question 2 8 points



A Chem110 student plots the graph shown on the left, from data obtained for the kinetic energy (eV) of the ejected electron versus frequency (s^{-1}) of the incident light, for an unknown metal. ($1\text{eV} = 1.6022 \times 10^{-19} \text{ J}$)

Based on the plot, answer the questions **a to c** below:

a. 1 point On the plot above, label the **threshold frequency**.

b. 2 points What is the value of the work potential (in J) of the unknown metal?

c. 3 points What is the *longest* wavelength (in nm) required to eject an electron from this metal's surface? Show your work including equation(s) used. (3 sig figs)

d. 2 points In the space provided below, plot the following

(Assume frequency is *above* threshold frequency):

- i. *Kinetic energy* of an ejected electron versus the *intensity* of the incident light (left)
- ii. *Number of electrons* ejected versus the *frequency* of the incident light (right)



**Question 3** **6 points****a. 2 points**

In the ground state of Ge (Germanium; $Z = 32$) atom, **name** the following orbitals:

i. Orbital containing the electron that requires the maximum energy to be removed.

ii. Orbital containing the electron that experiences the minimum effective nuclear charge.

b. 2 points

An unknown anion (A^-) has the following *excited* electronic configuration: $[Ar] 4s^2 3d^3 4p^1$

(Ar: Argon; $Z = 18$)

Give the **condensed ground state electronic configuration** of the atom, **A** and the cation, **A^{2+}** .
Show your work. (Assume all electronic transitions are possible)

Electronic Configuration of A:

Electronic Configuration of A^{2+} :

c. 2 points

In a single atom, determine how many electrons can have the following set of quantum numbers?

(i) $n = 5; l = 3$

In a single atom, determine how many electrons can have the following set of quantum numbers?

(i) $n = 3; l = 2; m_s = +1/2$

**Question 4** **8 points**

Rank the following from 1 to 4 in **increasing** order. Provide the ranking in the boxes provided.

Note: Atomic number (Z) are provided for relevant atoms

a) Increasing (lowest to highest) order of *ionization energy*: (1= lowest; 4 = highest)

Mg

Mg: Z = 12

Rb

Rb: Z = 37

Cs

Cs: Z = 55

K

K : Z = 19

b) Increasing (lowest to highest) order of *bond angle*: (1= lowest; 4 = highest)

H₂OCCl₄XeF₄

HCN

c) Increasing (smallest to largest) order of *ionic size* (1= smallest; 4 = largest)

S²⁻

S: Z = 16

O²⁻

O: Z = 8

Be²⁺

Be: Z = 4

Na⁺

Na: Z = 11

d) Increasing (lowest to highest) order of *bond energy* (1= lowest; 4 = highest)

P-F

P-Cl

P-I

P-Br

**Question 5** **12 points**

a. 9 points For the molecules HClO_4 and TeF_6 , draw the most stable *Lewis structure* showing appropriate shape and *dashed/wedged bonds*, based on VSEPR. Determine the **molecular geometry** around the central atom. For HClO_4 , label the *most polar bond*.



b. 3 points (Note: Lone pairs are not shown in structures below)

(i) For the three *resonance* structures (1, 2 and 3) shown below, answer the following question:



(I) Circle the *most contributing* resonance structure.

Structure 1

Structure 2

Structure 3

(ii) For the three structures given (4, 5, and 6) below, answer the following questions:



(I) Circle all the structures that represent *resonance structures* of each other:

Structure 4

Structure 5

Structure 6

(II) Among the resonance structures, circle the *most contributing* resonance structure.

Structure 4

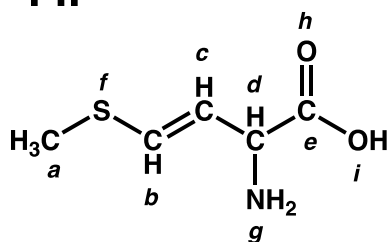
Structure 5

Structure 6

**Question 6** **6 points**

Using the data given below, calculate the **lattice energy** for SrO(s).

	ΔH° (kJ/mol)
$\text{Sr}(s) \longrightarrow \text{Sr}(g)$	+180
$\text{Sr}(g) \longrightarrow \text{Sr}^{2+}(g) + 2e^-$	+1468
$\text{O}_2(g) \longrightarrow 2\text{O}(g)$	+500
$\text{O}(g) + 2e^- \longrightarrow \text{O}^{2-}(g)$	+600
$\text{Sr}(s) + \frac{1}{2}\text{O}_2(g) \longrightarrow \text{SrO}(s)$	-550

**Question 7 10 points**

Consider the molecule given on the left, and answer questions (i) to (vi) using chemical bonding theories that we have learnt in class (valence bond theory, hybridization).

Note that the lone pairs are not shown in the structure:

i. Consider the carbon atoms marked in as ***a***, ***b***, ***c***, ***d***, and ***e*** in the molecule, and determine **hybridization** at each of the carbon centers. (2.5 points)

Atom	Hybridization
Carbon (<i>a</i>)	
Carbon (<i>b</i>)	
Carbon (<i>c</i>)	
Carbon (<i>d</i>)	
Carbon (<i>e</i>)	

ii. Now, consider the sulfur atom (S) marked as ***f***, nitrogen (N) atom (***g***), and oxygen(O) atoms (***h*** and ***i***), and determine their **hybridization**. (2 points)

Atom	Hybridization
Sulfur (<i>f</i>)	
Nitrogen (<i>g</i>)	
Oxygen (<i>h</i>)	
Oxygen (<i>i</i>)	

iii. **Describe** all the bonds formed by the carbon atoms ***a*** and ***b***, using sigma and pi designation as appropriate. (2 points)

Atom	Bonds formed (use sigma and pi as appropriate)
Carbon (<i>a</i>)	
Carbon (<i>b</i>)	

iv. How many total sp^3 -s sigma bonds are in the molecule? (1 point)

v. Determine **electronic** and **molecular geometry** at carbon (C) labeled ***e*** and sulfur (S) labeled ***f*** centres. (2 points)

Atom	Electronic Geometry	Molecular Geometry
Carbon (<i>e</i>)		
Sulfur (<i>f</i>)		

vi. Circle the value from the choices given below, that describes the **C-S-C bond angle at sulfur** (S) marked ***f***: (0.5 points)

>90° to <109.5° 109.5° >109.5 to <120° 120° >120° to <180°

**Question 8 12 points**

Using the **valence shell orbitals only** draw a **complete** molecular orbital (MO) diagram (with no s-p mixing) for the molecule **PS**, in the space provided below. **(6 points)**

Assume that 3s and 3p atomic orbitals will form molecular orbitals like those formed by the combination of 2s and 2p atomic orbitals that we learnt in class, and there is no s-p mixing.

Show **energy arrow**, **atomic orbitals** involved, **atom labels**, the **molecular orbitals** that are formed, **filled electrons** in the MO diagram etc. Make sure to label all the atomic and molecular orbitals.

Using the MO diagram above, answer the following questions:

- (i) Which one of the following will have the highest bond order? Circle the correct choice
A) PS^{1-} B) PS C) PS^{1+}
- (ii) Which one of the following will have the longest bond length? Circle the correct choice
A) PS^{1-} B) PS C) PS^{1+}
- (iii) Which of the following will be diamagnetic? Circle the correct choice.
A) PS^{1-} B) PS C) PS^{1+}
- (iv) Draw Lewis dot structure of the molecule with the shortest bond length (based on the MO diagram) from among the three (PS^{1-} , PS, PS^{1+}), and indicate the bond order calculated from the Lewis dot structure. **(3 points)**

Lewis Dot Structure

Bond Order



028EA5F1-255A-4D29-BFA5-60BE0CEDDE12

midterm-exam-93388

#511 Page 10 of 10

SCRAP PAGE

**CHEMISTRY 110 –Midterm Exam****Thursday November 7, 2024, 6:30 – 8:30 pm****Examiners: Pallavi Sirjoosingh and Ashok Kakkar**

First name (please write as legibly as possible within the boxes)

--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--

Last name

--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--

Student ID number

--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--

This booklet has 8 questions on 10 pages (The last page can be used as a scrap page).

The staple must NOT be removed from the question booklet. Your intact question booklet must be returned at the end of the exam.

A separate booklet contains a periodic table, and datasheet.

Write your answers directly under the questions in the space provided. Do not write on the QR codes or near the top edges (as it may be cut off).

Please write **legibly and dark enough** for the scan to be readable. For students who use English as a second language, dictionaries are permitted. Calculators are allowed.

DO NOT ASK QUESTIONS OF THE INVIGILATORS about interpretations of the exam questions. If you have a concern about a question, please summarize it at the top of this page. All concerns will be evaluated after the exam. GOOD LUCK!

**Question 1 8 points** (*Assume all transitions are possible*)

a. 6 points Calculate the **energy (in kJ/mol)** required to ionize (completely remove) the electron from the **n=3** level of **He⁺ ion** (He: Helium). Show your work including equations. (3 sig. figs.)

b. 2 points

(I) Select the largest (longest) wavelength from the three choices below:

- i. Wavelength of photon to *completely remove* an electron from *1s orbital of Hydrogen atom*
- ii. Wavelength of photon to *completely remove* an electron from *1s orbital of Boron atom (B)*
- iii. Wavelength of photon to *completely remove* an electron from *1s orbital of Boron cation (B⁺)*

i or ii or iii

(II) Select the smallest (shortest) wavelength from the three choices below (*for H atom*):

- i. Wavelength of photon *emitted* for an electron transitioning from *2p to 1s orbital*
- ii. Wavelength of photon *absorbed* for an electron transitioning from *2s to 4p orbital*
- iii. Wavelength of photon *absorbed* for an electron transitioning from *1s to 3p orbital*

i or ii or iii



Question 2 8 points



A Chem110 student plots the graph shown on the left, from data obtained for the kinetic energy (eV) of the ejected electron versus frequency (s^{-1}) of the incident light, for an unknown metal. ($1\text{eV} = 1.6022 \times 10^{-19} \text{ J}$)

Based on the plot, answer the questions **a to c** below:

a. 1 point On the plot above, label the **threshold frequency**.

b. 2 points What is the value of the work potential (in J) of the unknown metal?

c. 3 points What is the *longest* wavelength (in nm) required to eject an electron from this metal's surface? Show your work including equation(s) used. (3 sig figs)

d. 2 points In the space provided below, plot the following

(Assume frequency is *above* threshold frequency):

- Kinetic energy* of an ejected electron versus the *intensity* of the incident light (left)
- Number of electrons* ejected versus the *frequency* of the incident light (right)



**Question 3** **6 points****a. 2 points**

In the ground state of Ge (Germanium; $Z = 32$) atom, **name** the following orbitals:

i. Orbital containing the electron that requires the maximum energy to be removed.

ii. Orbital containing the electron that experiences the minimum effective nuclear charge.

b. 2 points

An unknown anion (A^-) has the following *excited* electronic configuration: $[Ar] 4s^2 3d^3 4p^1$

(Ar: Argon; $Z = 18$)

Give the **condensed ground state electronic configuration** of the atom, **A** and the cation, **A^{2+}** .
Show your work. (Assume all electronic transitions are possible)

Electronic Configuration of A:

Electronic Configuration of A^{2+} :

c. 2 points

In a single atom, determine how many electrons can have the following set of quantum numbers?

(i) $n = 5; l = 3$

In a single atom, determine how many electrons can have the following set of quantum numbers?

(i) $n = 3; l = 2; m_s = +1/2$

**Question 4** **8 points**

Rank the following from 1 to 4 in **increasing** order. Provide the ranking in the boxes provided.

Note: Atomic number (Z) are provided for relevant atoms

a) Increasing (lowest to highest) order of *ionization energy*: (1= lowest; 4 = highest)

Mg

Mg: Z = 12

Rb

Rb: Z = 37

Cs

Cs: Z = 55

K

K : Z = 19

b) Increasing (lowest to highest) order of *bond angle*: (1= lowest; 4 = highest)

H₂OCCl₄XeF₄

HCN

c) Increasing (smallest to largest) order of *ionic size* (1= smallest; 4 = largest)

S²⁻

S: Z = 16

O²⁻

O: Z = 8

Be²⁺

Be: Z = 4

Na⁺

Na: Z = 11

d) Increasing (lowest to highest) order of *bond energy* (1= lowest; 4 = highest)

P-F

P-Cl

P-I

P-Br

**Question 5** **12 points**

a. 9 points For the molecules HClO_4 and TeF_6 , draw the most stable *Lewis structure* showing appropriate shape and *dashed/wedged bonds*, based on VSEPR. Determine the **molecular geometry** around the central atom. For HClO_4 , label the *most polar bond*.



b. 3 points (Note: Lone pairs are not shown in structures below)

(i) For the three *resonance* structures (1, 2 and 3) shown below, answer the following question:



(I) Circle the *most contributing* resonance structure.

Structure 1

Structure 2

Structure 3

(ii) For the three structures given (4, 5, and 6) below, answer the following questions:



(I) Circle all the structures that represent *resonance structures* of each other:

Structure 4

Structure 5

Structure 6

(II) Among the resonance structures, circle the *most contributing* resonance structure.

Structure 4

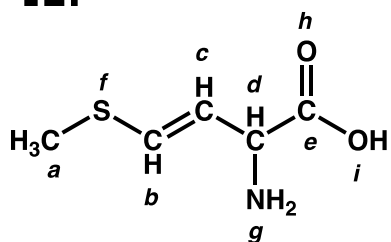
Structure 5

Structure 6

**Question 6** **6 points**

Using the data given below, calculate the **lattice energy** for SrO(s).

	ΔH° (kJ/mol)
$\text{Sr}(s) \longrightarrow \text{Sr}(g)$	+180
$\text{Sr}(g) \longrightarrow \text{Sr}^{2+}(g) + 2e^-$	+1468
$\text{O}_2(g) \longrightarrow 2\text{O}(g)$	+500
$\text{O}(g) + 2e^- \longrightarrow \text{O}^{2-}(g)$	+600
$\text{Sr}(s) + \frac{1}{2}\text{O}_2(g) \longrightarrow \text{SrO}(s)$	-550

**Question 7 10 points**

Consider the molecule given on the left, and answer questions (i) to (vi) using chemical bonding theories that we have learnt in class (valence bond theory, hybridization).

Note that the lone pairs are not shown in the structure:

i. Consider the carbon atoms marked in as ***a***, ***b***, ***c***, ***d***, and ***e*** in the molecule, and determine **hybridization** at each of the carbon centers. (2.5 points)

Atom	Hybridization
Carbon (<i>a</i>)	
Carbon (<i>b</i>)	
Carbon (<i>c</i>)	
Carbon (<i>d</i>)	
Carbon (<i>e</i>)	

ii. Now, consider the sulfur atom (S) marked as ***f***, nitrogen (N) atom (***g***), and oxygen(O) atoms (***h*** and ***i***), and determine their **hybridization**. (2 points)

Atom	Hybridization
Sulfur (<i>f</i>)	
Nitrogen (<i>g</i>)	
Oxygen (<i>h</i>)	
Oxygen (<i>i</i>)	

iii. **Describe** all the bonds formed by the carbon atoms ***a*** and ***b***, using sigma and pi designation as appropriate. (2 points)

Atom	Bonds formed (use sigma and pi as appropriate)
Carbon (<i>a</i>)	
Carbon (<i>b</i>)	

iv. How many total sp^3 -s sigma bonds are in the molecule? (1 point)

v. Determine **electronic** and **molecular geometry** at carbon (C) labeled ***e*** and sulfur (S) labeled ***f*** centres. (2 points)

Atom	Electronic Geometry	Molecular Geometry
Carbon (<i>e</i>)		
Sulfur (<i>f</i>)		

vi. Circle the value from the choices given below, that describes the **C-S-C bond angle at sulfur** (S) marked ***f***: (0.5 points)

$>90^\circ$ to $<109.5^\circ$ 109.5° >109.5 to $<120^\circ$ 120° $>120^\circ$ to $<180^\circ$


Question 8 12 points

Using the **valence shell orbitals only** draw a **complete** molecular orbital (MO) diagram (with no s-p mixing) for the molecule **PS**, in the space provided below. **(6 points)**

Assume that 3s and 3p atomic orbitals will form molecular orbitals like those formed by the combination of 2s and 2p atomic orbitals that we learnt in class, and there is no s-p mixing.

Show **energy arrow**, **atomic orbitals** involved, **atom labels**, the **molecular orbitals** that are formed, **filled electrons** in the MO diagram etc. Make sure to label all the atomic and molecular orbitals.

Using the MO diagram above, answer the following questions:

- (i) Which one of the following will have the highest bond order? Circle the correct choice
 A) PS^{1-} B) PS C) PS^{1+}
- (ii) Which one of the following will have the longest bond length? Circle the correct choice
 A) PS^{1-} B) PS C) PS^{1+}
- (iii) Which of the following will be diamagnetic? Circle the correct choice.
 A) PS^{1-} B) PS C) PS^{1+}
- (iv) Draw Lewis dot structure of the molecule with the shortest bond length (based on the MO diagram) from among the three (PS^{1-} , PS, PS^{1+}), and indicate the bond order calculated from the Lewis dot structure. **(3 points)**

Lewis Dot Structure

Bond Order



09C8148E-4074-4E9A-BC2A-FBD168A56273

midterm-exam-93388

#512 Page 10 of 10

SCRAP PAGE

**CHEMISTRY 110 –Midterm Exam****Thursday November 7, 2024, 6:30 – 8:30 pm**

Examiners: Pallavi Sirjoosingh and Ashok Kakkar

First name (please write as legibly as possible within the boxes)

--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--

Last name

--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--

Student ID number

--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--

This booklet has 8 questions on 10 pages (The last page can be used as a scrap page).

The staple must NOT be removed from the question booklet. Your intact question booklet must be returned at the end of the exam.

A separate booklet contains a periodic table, and datasheet.

Write your answers directly under the questions in the space provided. Do not write on the QR codes or near the top edges (as it may be cut off).

Please write **legibly and dark enough** for the scan to be readable. For students who use English as a second language, dictionaries are permitted. Calculators are allowed.

DO NOT ASK QUESTIONS OF THE INVIGILATORS about interpretations of the exam questions. If you have a concern about a question, please summarize it at the top of this page. All concerns will be evaluated after the exam. GOOD LUCK!

**Question 1 8 points** (*Assume all transitions are possible*)

a. 6 points Calculate the **energy (in kJ/mol)** required to ionize (completely remove) the electron from the **n=3** level of **He⁺ ion** (He: Helium). Show your work including equations. (3 sig. figs.)

b. 2 points

(I) Select the largest (longest) wavelength from the three choices below:

- i. Wavelength of photon to *completely remove* an electron from *1s orbital of Hydrogen atom*
- ii. Wavelength of photon to *completely remove* an electron from *1s orbital of Boron atom (B)*
- iii. Wavelength of photon to *completely remove* an electron from *1s orbital of Boron cation (B⁺)*

i or ii or iii

(II) Select the smallest (shortest) wavelength from the three choices below (*for H atom*):

- i. Wavelength of photon *emitted* for an electron transitioning from *2p to 1s orbital*
- ii. Wavelength of photon *absorbed* for an electron transitioning from *2s to 4p orbital*
- iii. Wavelength of photon *absorbed* for an electron transitioning from *1s to 3p orbital*

i or ii or iii



Question 2 8 points



A Chem110 student plots the graph shown on the left, from data obtained for the kinetic energy (eV) of the ejected electron versus frequency (s^{-1}) of the incident light, for an unknown metal. ($1\text{eV} = 1.6022 \times 10^{-19} \text{ J}$)

Based on the plot, answer the questions **a to c** below:

a. 1 point On the plot above, label the **threshold frequency**.

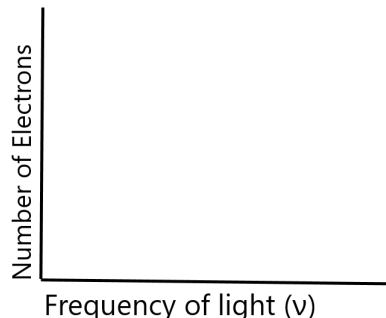
b. 2 points What is the value of the work potential (in J) of the unknown metal?

c. 3 points What is the *longest* wavelength (in nm) required to eject an electron from this metal's surface? Show your work including equation(s) used. (3 sig figs)

d. 2 points In the space provided below, plot the following

(Assume frequency is *above* threshold frequency):

- Kinetic energy* of an ejected electron versus the *intensity* of the incident light (left)
- Number of electrons* ejected versus the *frequency* of the incident light (right)



**Question 3** **6 points****a. 2 points**

In the ground state of Ge (Germanium; $Z = 32$) atom, **name** the following orbitals:

i. Orbital containing the electron that requires the maximum energy to be removed.

ii. Orbital containing the electron that experiences the minimum effective nuclear charge.

b. 2 points

An unknown anion (A^-) has the following *excited* electronic configuration: $[Ar] 4s^2 3d^3 4p^1$

(Ar: Argon; $Z = 18$)

Give the **condensed ground state electronic configuration** of the atom, **A** and the cation, **A^{2+}** .
Show your work. (Assume all electronic transitions are possible)

Electronic Configuration of A:

Electronic Configuration of A^{2+} :

c. 2 points

In a single atom, determine how many electrons can have the following set of quantum numbers?

(i) $n = 5; l = 3$

In a single atom, determine how many electrons can have the following set of quantum numbers?

(i) $n = 3; l = 2; m_s = +1/2$

**Question 4** **8 points**

Rank the following from 1 to 4 in **increasing** order. Provide the ranking in the boxes provided.

Note: Atomic number (Z) are provided for relevant atoms

a) Increasing (lowest to highest) order of *ionization energy*: (1= lowest; 4 = highest)

Mg

Mg: Z = 12

Rb

Rb: Z = 37

Cs

Cs: Z = 55

K

K : Z = 19

b) Increasing (lowest to highest) order of *bond angle*: (1= lowest; 4 = highest)

H₂OCCl₄XeF₄

HCN

c) Increasing (smallest to largest) order of *ionic size* (1= smallest; 4 = largest)

S²⁻

S: Z = 16

O²⁻

O: Z = 8

Be²⁺

Be: Z = 4

Na⁺

Na: Z = 11

d) Increasing (lowest to highest) order of *bond energy* (1= lowest; 4 = highest)

P-F

P-Cl

P-I

P-Br

**Question 5** **12 points**

a. 9 points For the molecules HClO_4 and TeF_6 , draw the most stable *Lewis structure* showing appropriate shape and *dashed/wedged bonds*, based on VSEPR. Determine the **molecular geometry** around the central atom. For HClO_4 , label the *most polar bond*.



b. 3 points (Note: Lone pairs are not shown in structures below)

(i) For the three *resonance* structures (1, 2 and 3) shown below, answer the following question:



(I) Circle the *most contributing* resonance structure.

Structure 1

Structure 2

Structure 3

(ii) For the three structures given (4, 5, and 6) below, answer the following questions:



(I) Circle all the structures that represent *resonance structures* of each other:

Structure 4

Structure 5

Structure 6

(II) Among the resonance structures, circle the *most contributing* resonance structure.

Structure 4

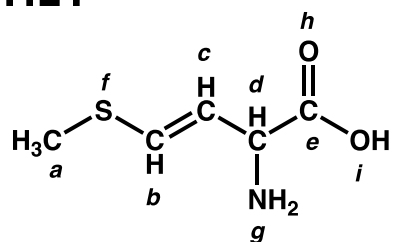
Structure 5

Structure 6

**Question 6** **6 points**

Using the data given below, calculate the **lattice energy** for SrO(s).

	ΔH° (kJ/mol)
$\text{Sr}(s) \longrightarrow \text{Sr}(g)$	+180
$\text{Sr}(g) \longrightarrow \text{Sr}^{2+}(g) + 2e^-$	+1468
$\text{O}_2(g) \longrightarrow 2\text{O}(g)$	+500
$\text{O}(g) + 2e^- \longrightarrow \text{O}^{2-}(g)$	+600
$\text{Sr}(s) + \frac{1}{2}\text{O}_2(g) \longrightarrow \text{SrO}(s)$	-550

**Question 7 10 points**

Consider the molecule given on the left, and answer questions (i) to (vi) using chemical bonding theories that we have learnt in class (valence bond theory, hybridization).

Note that the lone pairs are not shown in the structure:

i. Consider the carbon atoms marked in as ***a***, ***b***, ***c***, ***d***, and ***e*** in the molecule, and determine **hybridization** at each of the carbon centers. (2.5 points)

Atom	Hybridization
Carbon (<i>a</i>)	
Carbon (<i>b</i>)	
Carbon (<i>c</i>)	
Carbon (<i>d</i>)	
Carbon (<i>e</i>)	

ii. Now, consider the sulfur atom (S) marked as ***f***, nitrogen (N) atom (***g***), and oxygen(O) atoms (***h*** and ***i***), and determine their **hybridization**. (2 points)

Atom	Hybridization
Sulfur (<i>f</i>)	
Nitrogen (<i>g</i>)	
Oxygen (<i>h</i>)	
Oxygen (<i>i</i>)	

iii. **Describe** all the bonds formed by the carbon atoms ***a*** and ***b***, using sigma and pi designation as appropriate. (2 points)

Atom	Bonds formed (use sigma and pi as appropriate)
Carbon (<i>a</i>)	
Carbon (<i>b</i>)	

iv. How many total sp^3 -s sigma bonds are in the molecule? (1 point)

v. Determine **electronic** and **molecular geometry** at carbon (C) labeled ***e*** and sulfur (S) labeled ***f*** centres. (2 points)

Atom	Electronic Geometry	Molecular Geometry
Carbon (<i>e</i>)		
Sulfur (<i>f</i>)		

vi. Circle the value from the choices given below, that describes the **C-S-C bond angle at sulfur** (S) marked ***f***: (0.5 points)

>90° to <109.5° 109.5° >109.5 to <120° 120° >120° to <180°


Question 8 12 points

Using the **valence shell orbitals only** draw a **complete** molecular orbital (MO) diagram (with no s-p mixing) for the molecule **PS**, in the space provided below. **(6 points)**

Assume that 3s and 3p atomic orbitals will form molecular orbitals like those formed by the combination of 2s and 2p atomic orbitals that we learnt in class, and there is no s-p mixing.

Show **energy arrow**, **atomic orbitals** involved, **atom labels**, the **molecular orbitals** that are formed, **filled electrons** in the MO diagram etc. Make sure to label all the atomic and molecular orbitals.

Using the MO diagram above, answer the following questions:

- (i) Which one of the following will have the highest bond order? Circle the correct choice
 A) PS^{1-} B) PS C) PS^{1+}
- (ii) Which one of the following will have the longest bond length? Circle the correct choice
 A) PS^{1-} B) PS C) PS^{1+}
- (iii) Which of the following will be diamagnetic? Circle the correct choice.
 A) PS^{1-} B) PS C) PS^{1+}
- (iv) Draw Lewis dot structure of the molecule with the shortest bond length (based on the MO diagram) from among the three (PS^{1-} , PS, PS^{1+}), and indicate the bond order calculated from the Lewis dot structure. **(3 points)**

Lewis Dot Structure

Bond Order



DF46C0B6-7D5D-40FD-BFDE-5D087436BB89

midterm-exam-93388

#513 Page 10 of 10

SCRAP PAGE

**CHEMISTRY 110 –Midterm Exam****Thursday November 7, 2024, 6:30 – 8:30 pm****Examiners: Pallavi Sirjoosingh and Ashok Kakkar**

First name (please write as legibly as possible within the boxes)

--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--

Last name

--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--

Student ID number

--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--

This booklet has 8 questions on 10 pages (The last page can be used as a scrap page).

The staple must NOT be removed from the question booklet. Your intact question booklet must be returned at the end of the exam.

A separate booklet contains a periodic table, and datasheet.

Write your answers directly under the questions in the space provided. Do not write on the QR codes or near the top edges (as it may be cut off).

Please write **legibly and dark enough** for the scan to be readable. For students who use English as a second language, dictionaries are permitted. Calculators are allowed.

DO NOT ASK QUESTIONS OF THE INVIGILATORS about interpretations of the exam questions. If you have a concern about a question, please summarize it at the top of this page. All concerns will be evaluated after the exam. GOOD LUCK!

**Question 1 8 points** (*Assume all transitions are possible*)

a. 6 points Calculate the **energy (in kJ/mol)** required to ionize (completely remove) the electron from the **n=3** level of **He⁺ ion** (He: Helium). Show your work including equations. (3 sig. figs.)

b. 2 points

(I) Select the largest (longest) wavelength from the three choices below:

- i. Wavelength of photon to *completely remove* an electron from *1s orbital of Hydrogen atom*
- ii. Wavelength of photon to *completely remove* an electron from *1s orbital of Boron atom (B)*
- iii. Wavelength of photon to *completely remove* an electron from *1s orbital of Boron cation (B⁺)*

i or ii or iii

(II) Select the smallest (shortest) wavelength from the three choices below (*for H atom*):

- i. Wavelength of photon *emitted* for an electron transitioning from *2p to 1s orbital*
- ii. Wavelength of photon *absorbed* for an electron transitioning from *2s to 4p orbital*
- iii. Wavelength of photon *absorbed* for an electron transitioning from *1s to 3p orbital*

i or ii or iii



Question 2 8 points



A Chem110 student plots the graph shown on the left, from data obtained for the kinetic energy (eV) of the ejected electron versus frequency (s^{-1}) of the incident light, for an unknown metal. ($1\text{eV} = 1.6022 \times 10^{-19} \text{ J}$)

Based on the plot, answer the questions **a to c** below:

a. 1 point On the plot above, label the **threshold frequency**.

b. 2 points What is the value of the work potential (in J) of the unknown metal?

c. 3 points What is the *longest* wavelength (in nm) required to eject an electron from this metal's surface? Show your work including equation(s) used. (3 sig figs)

d. 2 points In the space provided below, plot the following

(Assume frequency is *above* threshold frequency):

- i. *Kinetic energy* of an ejected electron versus the *intensity* of the incident light (left)
- ii. *Number of electrons* ejected versus the *frequency* of the incident light (right)



**Question 3** **6 points****a. 2 points**

In the ground state of Ge (Germanium; $Z = 32$) atom, **name** the following orbitals:

i. Orbital containing the electron that requires the maximum energy to be removed.

ii. Orbital containing the electron that experiences the minimum effective nuclear charge.

b. 2 points

An unknown anion (A^-) has the following *excited* electronic configuration: $[Ar] 4s^2 3d^3 4p^1$

(Ar: Argon; $Z = 18$)

Give the **condensed ground state electronic configuration** of the atom, **A** and the cation, **A^{2+}** .
Show your work. (Assume all electronic transitions are possible)

Electronic Configuration of A:

Electronic Configuration of A^{2+} :

c. 2 points

In a single atom, determine how many electrons can have the following set of quantum numbers?

(i) $n = 5; l = 3$

In a single atom, determine how many electrons can have the following set of quantum numbers?

(i) $n = 3; l = 2; m_s = +1/2$

**Question 4** **8 points**

Rank the following from 1 to 4 in **increasing** order. Provide the ranking in the boxes provided.

Note: Atomic number (Z) are provided for relevant atoms

a) Increasing (lowest to highest) order of *ionization energy*: (1= lowest; 4 = highest)

Mg

Mg: Z = 12

Rb

Rb: Z = 37

Cs

Cs: Z = 55

K

K : Z = 19

b) Increasing (lowest to highest) order of *bond angle*: (1= lowest; 4 = highest)

H₂OCCl₄XeF₄

HCN

c) Increasing (smallest to largest) order of *ionic size* (1= smallest; 4 = largest)

S²⁻

S: Z = 16

O²⁻

O: Z = 8

Be²⁺

Be: Z = 4

Na⁺

Na: Z = 11

d) Increasing (lowest to highest) order of *bond energy* (1= lowest; 4 = highest)

P-F

P-Cl

P-I

P-Br

**Question 5** **12 points**

a. 9 points For the molecules HClO_4 and TeF_6 , draw the most stable *Lewis structure* showing appropriate shape and *dashed/wedged bonds*, based on VSEPR. Determine the **molecular geometry** around the central atom. For HClO_4 , label the *most polar bond*.



b. 3 points (Note: Lone pairs are not shown in structures below)

(i) For the three *resonance* structures (1, 2 and 3) shown below, answer the following question:



(I) Circle the *most contributing* resonance structure.

Structure 1

Structure 2

Structure 3

(ii) For the three structures given (4, 5, and 6) below, answer the following questions:



(I) Circle all the structures that represent *resonance structures* of each other:

Structure 4

Structure 5

Structure 6

(II) Among the resonance structures, circle the *most contributing* resonance structure.

Structure 4

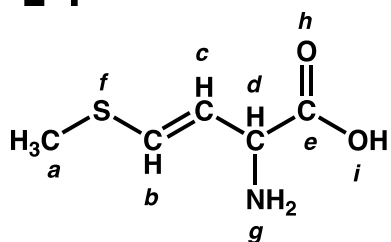
Structure 5

Structure 6

**Question 6** **6 points**

Using the data given below, calculate the **lattice energy** for SrO(s).

	ΔH° (kJ/mol)
$\text{Sr}(s) \longrightarrow \text{Sr}(g)$	+180
$\text{Sr}(g) \longrightarrow \text{Sr}^{2+}(g) + 2e^-$	+1468
$\text{O}_2(g) \longrightarrow 2\text{O}(g)$	+500
$\text{O}(g) + 2e^- \longrightarrow \text{O}^{2-}(g)$	+600
$\text{Sr}(s) + \frac{1}{2}\text{O}_2(g) \longrightarrow \text{SrO}(s)$	-550

**Question 7 10 points**

Consider the molecule given on the left, and answer questions (i) to (vi) using chemical bonding theories that we have learnt in class (valence bond theory, hybridization).

Note that the lone pairs are not shown in the structure:

i. Consider the carbon atoms marked in as ***a***, ***b***, ***c***, ***d***, and ***e*** in the molecule, and determine **hybridization** at each of the carbon centers. (2.5 points)

Atom	Hybridization
Carbon (<i>a</i>)	
Carbon (<i>b</i>)	
Carbon (<i>c</i>)	
Carbon (<i>d</i>)	
Carbon (<i>e</i>)	

ii. Now, consider the sulfur atom (S) marked as ***f***, nitrogen (N) atom (***g***), and oxygen(O) atoms (***h*** and ***i***), and determine their **hybridization**. (2 points)

Atom	Hybridization
Sulfur (<i>f</i>)	
Nitrogen (<i>g</i>)	
Oxygen (<i>h</i>)	
Oxygen (<i>i</i>)	

iii. **Describe** all the bonds formed by the carbon atoms ***a*** and ***b***, using sigma and pi designation as appropriate. (2 points)

Atom	Bonds formed (use sigma and pi as appropriate)
Carbon (<i>a</i>)	
Carbon (<i>b</i>)	

iv. How many total sp^3 -s sigma bonds are in the molecule? (1 point)

v. Determine **electronic** and **molecular geometry** at carbon (C) labeled ***e*** and sulfur (S) labeled ***f*** centres. (2 points)

Atom	Electronic Geometry	Molecular Geometry
Carbon (<i>e</i>)		
Sulfur (<i>f</i>)		

vi. Circle the value from the choices given below, that describes the **C-S-C bond angle at sulfur** (S) marked ***f***: (0.5 points)

>90° to <109.5° 109.5° >109.5 to <120° 120° >120° to <180°

**Question 8 12 points**

Using the **valence shell orbitals only** draw a **complete** molecular orbital (MO) diagram (with no s-p mixing) for the molecule **PS**, in the space provided below. **(6 points)**

Assume that 3s and 3p atomic orbitals will form molecular orbitals like those formed by the combination of 2s and 2p atomic orbitals that we learnt in class, and there is no s-p mixing.

Show **energy arrow**, **atomic orbitals** involved, **atom labels**, the **molecular orbitals** that are formed, **filled electrons** in the MO diagram etc. Make sure to label all the atomic and molecular orbitals.

Using the MO diagram above, answer the following questions:

- (i) Which one of the following will have the highest bond order? Circle the correct choice
A) PS^{1-} B) PS C) PS^{1+}
- (ii) Which one of the following will have the longest bond length? Circle the correct choice
A) PS^{1-} B) PS C) PS^{1+}
- (iii) Which of the following will be diamagnetic? Circle the correct choice.
A) PS^{1-} B) PS C) PS^{1+}
- (iv) Draw Lewis dot structure of the molecule with the shortest bond length (based on the MO diagram) from among the three (PS^{1-} , PS, PS^{1+}), and indicate the bond order calculated from the Lewis dot structure. **(3 points)**

Lewis Dot Structure

Bond Order



0A5755BB-24EF-432E-94C7-E73C2F659ACD

midterm-exam-93388

#514 Page 10 of 10

SCRAP PAGE

**CHEMISTRY 110 –Midterm Exam****Thursday November 7, 2024, 6:30 – 8:30 pm****Examiners: Pallavi Sirjoosingh and Ashok Kakkar**

First name (please write as legibly as possible within the boxes)

--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--

Last name

--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--

Student ID number

--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--

This booklet has 8 questions on 10 pages (The last page can be used as a scrap page).

The staple must NOT be removed from the question booklet. Your intact question booklet must be returned at the end of the exam.

A separate booklet contains a periodic table, and datasheet.

Write your answers directly under the questions in the space provided. Do not write on the QR codes or near the top edges (as it may be cut off).

Please write **legibly and dark enough** for the scan to be readable. For students who use English as a second language, dictionaries are permitted. Calculators are allowed.

DO NOT ASK QUESTIONS OF THE INVIGILATORS about interpretations of the exam questions. If you have a concern about a question, please summarize it at the top of this page. All concerns will be evaluated after the exam. GOOD LUCK!

**Question 1 8 points** (*Assume all transitions are possible*)

a. 6 points Calculate the **energy (in kJ/mol)** required to ionize (completely remove) the electron from the **$n=3$** level of **He^+ ion** (He: Helium). Show your work including equations. (3 sig. figs.)

b. 2 points

(I) Select the largest (longest) wavelength from the three choices below:

- i. Wavelength of photon to *completely remove* an electron from *1s orbital of Hydrogen atom*
- ii. Wavelength of photon to *completely remove* an electron from *1s orbital of Boron atom (B)*
- iii. Wavelength of photon to *completely remove* an electron from *1s orbital of Boron cation (B^+)*

i or ii or iii

(II) Select the smallest (shortest) wavelength from the three choices below (*for H atom*):

- i. Wavelength of photon *emitted* for an electron transitioning from *2p to 1s orbital*
- ii. Wavelength of photon *absorbed* for an electron transitioning from *2s to 4p orbital*
- iii. Wavelength of photon *absorbed* for an electron transitioning from *1s to 3p orbital*

i or ii or iii



Question 2 8 points



A Chem110 student plots the graph shown on the left, from data obtained for the kinetic energy (eV) of the ejected electron versus frequency (s^{-1}) of the incident light, for an unknown metal. ($1\text{eV} = 1.6022 \times 10^{-19} \text{ J}$)

Based on the plot, answer the questions **a to c** below:

a. 1 point On the plot above, label the **threshold frequency**.

b. 2 points What is the value of the work potential (in J) of the unknown metal?

c. 3 points What is the *longest* wavelength (in nm) required to eject an electron from this metal's surface? Show your work including equation(s) used. (3 sig figs)

d. 2 points In the space provided below, plot the following

(Assume frequency is *above* threshold frequency):

- i. *Kinetic energy* of an ejected electron versus the *intensity* of the incident light (left)
- ii. *Number of electrons* ejected versus the *frequency* of the incident light (right)



**Question 3** **6 points****a. 2 points**

In the ground state of Ge (Germanium; $Z = 32$) atom, **name** the following orbitals:

i. Orbital containing the electron that requires the maximum energy to be removed.

ii. Orbital containing the electron that experiences the minimum effective nuclear charge.

b. 2 points

An unknown anion (A^-) has the following *excited* electronic configuration: $[Ar] 4s^2 3d^3 4p^1$

(Ar: Argon; $Z = 18$)

Give the **condensed ground state electronic configuration** of the atom, **A** and the cation, **A^{2+}** .
Show your work. (Assume all electronic transitions are possible)

Electronic Configuration of A:

Electronic Configuration of A^{2+} :

c. 2 points

In a single atom, determine how many electrons can have the following set of quantum numbers?

(i) $n = 5; l = 3$

In a single atom, determine how many electrons can have the following set of quantum numbers?

(i) $n = 3; l = 2; m_s = +1/2$

**Question 4** **8 points**

Rank the following from 1 to 4 in **increasing** order. Provide the ranking in the boxes provided.

Note: Atomic number (Z) are provided for relevant atoms

a) Increasing (lowest to highest) order of *ionization energy*: (1= lowest; 4 = highest)

Mg

Mg: Z = 12

Rb

Rb: Z = 37

Cs

Cs: Z = 55

K

K : Z = 19

b) Increasing (lowest to highest) order of *bond angle*: (1= lowest; 4 = highest)

H₂OCCl₄XeF₄

HCN

c) Increasing (smallest to largest) order of *ionic size* (1= smallest; 4 = largest)

S²⁻

S: Z = 16

O²⁻

O: Z = 8

Be²⁺

Be: Z = 4

Na⁺

Na: Z = 11

d) Increasing (lowest to highest) order of *bond energy* (1= lowest; 4 = highest)

P-F

P-Cl

P-I

P-Br

**Question 5** **12 points**

a. 9 points For the molecules HClO_4 and TeF_6 , draw the most stable *Lewis structure* showing appropriate shape and *dashed/wedged bonds*, based on VSEPR. Determine the **molecular geometry** around the central atom. For HClO_4 , label the *most polar bond*.



b. 3 points (Note: Lone pairs are not shown in structures below)

(i) For the three *resonance* structures (1, 2 and 3) shown below, answer the following question:



(I) Circle the *most contributing* resonance structure.

Structure 1

Structure 2

Structure 3

(ii) For the three structures given (4, 5, and 6) below, answer the following questions:



(I) Circle all the structures that represent *resonance structures* of each other:

Structure 4

Structure 5

Structure 6

(II) Among the resonance structures, circle the *most contributing* resonance structure.

Structure 4

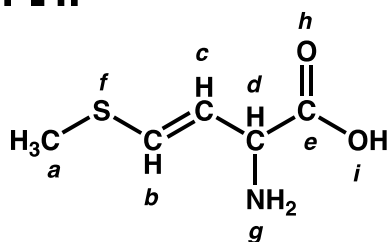
Structure 5

Structure 6

**Question 6** **6 points**

Using the data given below, calculate the **lattice energy** for SrO(s).

	ΔH° (kJ/mol)
$\text{Sr}(s) \longrightarrow \text{Sr}(g)$	+180
$\text{Sr}(g) \longrightarrow \text{Sr}^{2+}(g) + 2e^-$	+1468
$\text{O}_2(g) \longrightarrow 2\text{O}(g)$	+500
$\text{O}(g) + 2e^- \longrightarrow \text{O}^{2-}(g)$	+600
$\text{Sr}(s) + \frac{1}{2}\text{O}_2(g) \longrightarrow \text{SrO}(s)$	-550

**Question 7 10 points**

Consider the molecule given on the left, and answer questions (i) to (vi) using chemical bonding theories that we have learnt in class (valence bond theory, hybridization).

Note that the lone pairs are not shown in the structure:

i. Consider the carbon atoms marked in as ***a***, ***b***, ***c***, ***d***, and ***e*** in the molecule, and determine **hybridization** at each of the carbon centers. (2.5 points)

Atom	Hybridization
Carbon (<i>a</i>)	
Carbon (<i>b</i>)	
Carbon (<i>c</i>)	
Carbon (<i>d</i>)	
Carbon (<i>e</i>)	

ii. Now, consider the sulfur atom (S) marked as ***f***, nitrogen (N) atom (***g***), and oxygen(O) atoms (***h*** and ***i***), and determine their **hybridization**. (2 points)

Atom	Hybridization
Sulfur (<i>f</i>)	
Nitrogen (<i>g</i>)	
Oxygen (<i>h</i>)	
Oxygen (<i>i</i>)	

iii. **Describe** all the bonds formed by the carbon atoms ***a*** and ***b***, using sigma and pi designation as appropriate. (2 points)

Atom	Bonds formed (use sigma and pi as appropriate)
Carbon (<i>a</i>)	
Carbon (<i>b</i>)	

iv. How many total sp^3 -s sigma bonds are in the molecule? (1 point)

v. Determine **electronic** and **molecular geometry** at carbon (C) labeled ***e*** and sulfur (S) labeled ***f*** centres. (2 points)

Atom	Electronic Geometry	Molecular Geometry
Carbon (<i>e</i>)		
Sulfur (<i>f</i>)		

vi. Circle the value from the choices given below, that describes the **C-S-C bond angle at sulfur** (S) marked ***f***: (0.5 points)

>90° to <109.5° 109.5° >109.5 to <120° 120° >120° to <180°


Question 8 12 points

Using the **valence shell orbitals only** draw a **complete** molecular orbital (MO) diagram (with no s-p mixing) for the molecule **PS**, in the space provided below. **(6 points)**

Assume that 3s and 3p atomic orbitals will form molecular orbitals like those formed by the combination of 2s and 2p atomic orbitals that we learnt in class, and there is no s-p mixing.

Show **energy arrow**, **atomic orbitals** involved, **atom labels**, the **molecular orbitals** that are formed, **filled electrons** in the MO diagram etc. Make sure to label all the atomic and molecular orbitals.

Using the MO diagram above, answer the following questions:

- (i) Which one of the following will have the highest bond order? Circle the correct choice
 A) PS^{1-} B) PS C) PS^{1+}
- (ii) Which one of the following will have the longest bond length? Circle the correct choice
 A) PS^{1-} B) PS C) PS^{1+}
- (iii) Which of the following will be diamagnetic? Circle the correct choice.
 A) PS^{1-} B) PS C) PS^{1+}
- (iv) Draw Lewis dot structure of the molecule with the shortest bond length (based on the MO diagram) from among the three (PS^{1-} , PS, PS^{1+}), and indicate the bond order calculated from the Lewis dot structure. **(3 points)**

Lewis Dot Structure

Bond Order



0F3CE5B3-1D0F-4788-927C-5B02C2121A7F

midterm-exam-93388

#515 Page 10 of 10

SCRAP PAGE

**CHEMISTRY 110 –Midterm Exam****Thursday November 7, 2024, 6:30 – 8:30 pm**

Examiners: Pallavi Sirjoosingh and Ashok Kakkar

First name (please write as legibly as possible within the boxes)

--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--

Last name

--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--

Student ID number

--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--

This booklet has 8 questions on 10 pages (The last page can be used as a scrap page).

The staple must NOT be removed from the question booklet. Your intact question booklet must be returned at the end of the exam.

A separate booklet contains a periodic table, and datasheet.

Write your answers directly under the questions in the space provided. Do not write on the QR codes or near the top edges (as it may be cut off).

Please write **legibly and dark enough** for the scan to be readable. For students who use English as a second language, dictionaries are permitted. Calculators are allowed.

DO NOT ASK QUESTIONS OF THE INVIGILATORS about interpretations of the exam questions. If you have a concern about a question, please summarize it at the top of this page. All concerns will be evaluated after the exam. GOOD LUCK!

**Question 1 8 points** (*Assume all transitions are possible*)

a. 6 points Calculate the **energy (in kJ/mol)** required to ionize (completely remove) the electron from the **n=3** level of **He⁺ ion** (He: Helium). Show your work including equations. (3 sig. figs.)

b. 2 points

(I) Select the largest (longest) wavelength from the three choices below:

- i. Wavelength of photon to *completely remove* an electron from *1s orbital of Hydrogen atom*
- ii. Wavelength of photon to *completely remove* an electron from *1s orbital of Boron atom (B)*
- iii. Wavelength of photon to *completely remove* an electron from *1s orbital of Boron cation (B⁺)*

i or ii or iii

(II) Select the smallest (shortest) wavelength from the three choices below (*for H atom*):

- i. Wavelength of photon *emitted* for an electron transitioning from *2p to 1s orbital*
- ii. Wavelength of photon *absorbed* for an electron transitioning from *2s to 4p orbital*
- iii. Wavelength of photon *absorbed* for an electron transitioning from *1s to 3p orbital*

i or ii or iii



Question 2 8 points



A Chem110 student plots the graph shown on the left, from data obtained for the kinetic energy (eV) of the ejected electron versus frequency (s^{-1}) of the incident light, for an unknown metal. ($1\text{eV} = 1.6022 \times 10^{-19} \text{ J}$)

Based on the plot, answer the questions **a to c** below:

a. 1 point On the plot above, label the **threshold frequency**.

b. 2 points What is the value of the work potential (in J) of the unknown metal?

c. 3 points What is the *longest* wavelength (in nm) required to eject an electron from this metal's surface? Show your work including equation(s) used. (3 sig figs)

d. 2 points In the space provided below, plot the following

(Assume frequency is *above* threshold frequency):

- Kinetic energy* of an ejected electron versus the *intensity* of the incident light (left)
- Number of electrons* ejected versus the *frequency* of the incident light (right)



**Question 3** **6 points****a. 2 points**

In the ground state of Ge (Germanium; $Z = 32$) atom, **name** the following orbitals:

i. Orbital containing the electron that requires the maximum energy to be removed.

ii. Orbital containing the electron that experiences the minimum effective nuclear charge.

b. 2 points

An unknown anion (A^-) has the following *excited* electronic configuration: $[Ar] 4s^2 3d^3 4p^1$

(Ar: Argon; $Z = 18$)

Give the **condensed ground state electronic configuration** of the atom, **A** and the cation, **A^{2+}** .
Show your work. (Assume all electronic transitions are possible)

Electronic Configuration of A:

Electronic Configuration of A^{2+} :

c. 2 points

In a single atom, determine how many electrons can have the following set of quantum numbers?

(i) $n = 5; l = 3$

In a single atom, determine how many electrons can have the following set of quantum numbers?

(i) $n = 3; l = 2; m_s = +1/2$

**Question 4** **8 points**

Rank the following from 1 to 4 in **increasing** order. Provide the ranking in the boxes provided.

Note: Atomic number (Z) are provided for relevant atoms

a) Increasing (lowest to highest) order of *ionization energy*: (1= lowest; 4 = highest)

Mg

Mg: Z = 12

Rb

Rb: Z = 37

Cs

Cs: Z = 55

K

K : Z = 19

b) Increasing (lowest to highest) order of *bond angle*: (1= lowest; 4 = highest)

H₂OCCl₄XeF₄

HCN

c) Increasing (smallest to largest) order of *ionic size* (1= smallest; 4 = largest)

S²⁻

S: Z = 16

O²⁻

O: Z = 8

Be²⁺

Be: Z = 4

Na⁺

Na: Z = 11

d) Increasing (lowest to highest) order of *bond energy* (1= lowest; 4 = highest)

P-F

P-Cl

P-I

P-Br

**Question 5** **12 points**

a. 9 points For the molecules HClO_4 and TeF_6 , draw the most stable *Lewis structure* showing appropriate shape and *dashed/wedged bonds*, based on VSEPR. Determine the **molecular geometry** around the central atom. For HClO_4 , label the *most polar bond*.

HClO_4 (there are no O-O bonds)

Label the most polar bond

Molecular Geometry around Cl

TeF_6

Molecular Geometry around Te

b. 3 points (Note: Lone pairs are not shown in structures below)

(i) For the three *resonance* structures (1, 2 and 3) shown below, answer the following question:



(I) Circle the *most contributing* resonance structure.

Structure 1

Structure 2

Structure 3

(ii) For the three structures given (4, 5, and 6) below, answer the following questions:



(I) Circle all the structures that represent *resonance structures* of each other:

Structure 4

Structure 5

Structure 6

(II) Among the resonance structures, circle the *most contributing* resonance structure.

Structure 4

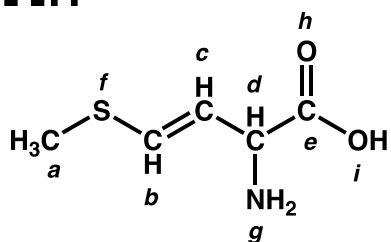
Structure 5

Structure 6

**Question 6** **6 points**

Using the data given below, calculate the **lattice energy** for SrO(s).

	ΔH° (kJ/mol)
$\text{Sr}(s) \longrightarrow \text{Sr}(g)$	+180
$\text{Sr}(g) \longrightarrow \text{Sr}^{2+}(g) + 2e^-$	+1468
$\text{O}_2(g) \longrightarrow 2\text{O}(g)$	+500
$\text{O}(g) + 2e^- \longrightarrow \text{O}^{2-}(g)$	+600
$\text{Sr}(s) + \frac{1}{2}\text{O}_2(g) \longrightarrow \text{SrO}(s)$	-550

**Question 7 10 points**

Consider the molecule given on the left, and answer questions (i) to (vi) using chemical bonding theories that we have learnt in class (valence bond theory, hybridization).

Note that the lone pairs are not shown in the structure:

i. Consider the carbon atoms marked in as ***a***, ***b***, ***c***, ***d***, and ***e*** in the molecule, and determine **hybridization** at each of the carbon centers. (2.5 points)

Atom	Hybridization
Carbon (<i>a</i>)	
Carbon (<i>b</i>)	
Carbon (<i>c</i>)	
Carbon (<i>d</i>)	
Carbon (<i>e</i>)	

ii. Now, consider the sulfur atom (S) marked as ***f***, nitrogen (N) atom (***g***), and oxygen(O) atoms (***h*** and ***i***), and determine their **hybridization**. (2 points)

Atom	Hybridization
Sulfur (<i>f</i>)	
Nitrogen (<i>g</i>)	
Oxygen (<i>h</i>)	
Oxygen (<i>i</i>)	

iii. **Describe** all the bonds formed by the carbon atoms ***a*** and ***b***, using sigma and pi designation as appropriate. (2 points)

Atom	Bonds formed (use sigma and pi as appropriate)
Carbon (<i>a</i>)	
Carbon (<i>b</i>)	

iv. How many total sp^3 -s sigma bonds are in the molecule? (1 point)

v. Determine **electronic** and **molecular geometry** at carbon (C) labeled ***e*** and sulfur (S) labeled ***f*** centres. (2 points)

Atom	Electronic Geometry	Molecular Geometry
Carbon (<i>e</i>)		
Sulfur (<i>f</i>)		

vi. Circle the value from the choices given below, that describes the **C-S-C bond angle at sulfur** (S) marked ***f***: (0.5 points)

>90° to <109.5° 109.5° >109.5 to <120° 120° >120° to <180°


Question 8 12 points

Using the **valence shell orbitals only** draw a **complete** molecular orbital (MO) diagram (with no s-p mixing) for the molecule **PS**, in the space provided below. **(6 points)**

Assume that 3s and 3p atomic orbitals will form molecular orbitals like those formed by the combination of 2s and 2p atomic orbitals that we learnt in class, and there is no s-p mixing.

Show **energy arrow**, **atomic orbitals** involved, **atom labels**, the **molecular orbitals** that are formed, **filled electrons** in the MO diagram etc. Make sure to label all the atomic and molecular orbitals.

Using the MO diagram above, answer the following questions:

- (i) Which one of the following will have the highest bond order? Circle the correct choice
 A) PS^{1-} B) PS C) PS^{1+}
- (ii) Which one of the following will have the longest bond length? Circle the correct choice
 A) PS^{1-} B) PS C) PS^{1+}
- (iii) Which of the following will be diamagnetic? Circle the correct choice.
 A) PS^{1-} B) PS C) PS^{1+}
- (iv) Draw Lewis dot structure of the molecule with the shortest bond length (based on the MO diagram) from among the three (PS^{1-} , PS, PS^{1+}), and indicate the bond order calculated from the Lewis dot structure. **(3 points)**

Lewis Dot Structure

Bond Order



1747934D-A1A1-498C-8C31-6CCDC0903D1B

midterm-exam-93388

#516 Page 10 of 10

SCRAP PAGE

**CHEMISTRY 110 –Midterm Exam****Thursday November 7, 2024, 6:30 – 8:30 pm**

Examiners: Pallavi Sirjoosingh and Ashok Kakkar

First name (please write as legibly as possible within the boxes)

--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--

Last name

--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--

Student ID number

--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--

This booklet has 8 questions on 10 pages (The last page can be used as a scrap page).

The staple must NOT be removed from the question booklet. Your intact question booklet must be returned at the end of the exam.

A separate booklet contains a periodic table, and datasheet.

Write your answers directly under the questions in the space provided. Do not write on the QR codes or near the top edges (as it may be cut off).

Please write **legibly and dark enough** for the scan to be readable. For students who use English as a second language, dictionaries are permitted. Calculators are allowed.

DO NOT ASK QUESTIONS OF THE INVIGILATORS about interpretations of the exam questions. If you have a concern about a question, please summarize it at the top of this page. All concerns will be evaluated after the exam. GOOD LUCK!

**Question 1 8 points** (*Assume all transitions are possible*)

a. 6 points Calculate the **energy (in kJ/mol)** required to ionize (completely remove) the electron from the **$n=3$** level of **He^+ ion** (He: Helium). Show your work including equations. (3 sig. figs.)

b. 2 points

(I) Select the largest (longest) wavelength from the three choices below:

- i. Wavelength of photon to *completely remove* an electron from *1s orbital of Hydrogen atom*
- ii. Wavelength of photon to *completely remove* an electron from *1s orbital of Boron atom (B)*
- iii. Wavelength of photon to *completely remove* an electron from *1s orbital of Boron cation (B^+)*

i or ii or iii

(II) Select the smallest (shortest) wavelength from the three choices below (*for H atom*):

- i. Wavelength of photon *emitted* for an electron transitioning from *2p to 1s orbital*
- ii. Wavelength of photon *absorbed* for an electron transitioning from *2s to 4p orbital*
- iii. Wavelength of photon *absorbed* for an electron transitioning from *1s to 3p orbital*

i or ii or iii



Question 2 8 points



A Chem110 student plots the graph shown on the left, from data obtained for the kinetic energy (eV) of the ejected electron versus frequency (s^{-1}) of the incident light, for an unknown metal. ($1\text{eV} = 1.6022 \times 10^{-19} \text{ J}$)

Based on the plot, answer the questions **a to c** below:

a. 1 point On the plot above, label the **threshold frequency**.

b. 2 points What is the value of the work potential (in J) of the unknown metal?

c. 3 points What is the *longest* wavelength (in nm) required to eject an electron from this metal's surface? Show your work including equation(s) used. (3 sig figs)

d. 2 points In the space provided below, plot the following

(Assume frequency is *above* threshold frequency):

- Kinetic energy* of an ejected electron versus the *intensity* of the incident light (left)
- Number of electrons* ejected versus the *frequency* of the incident light (right)



**Question 3** **6 points****a. 2 points**

In the ground state of Ge (Germanium; $Z = 32$) atom, **name** the following orbitals:

i. Orbital containing the electron that requires the maximum energy to be removed.

ii. Orbital containing the electron that experiences the minimum effective nuclear charge.

b. 2 points

An unknown anion (A^-) has the following *excited* electronic configuration: $[Ar] 4s^2 3d^3 4p^1$

(Ar: Argon; $Z = 18$)

Give the **condensed ground state electronic configuration** of the atom, **A** and the cation, **A^{2+}** .
Show your work. (Assume all electronic transitions are possible)

Electronic Configuration of A:

Electronic Configuration of A^{2+} :

c. 2 points

In a single atom, determine how many electrons can have the following set of quantum numbers?

(i) $n = 5; l = 3$

In a single atom, determine how many electrons can have the following set of quantum numbers?

(i) $n = 3; l = 2; m_s = +1/2$

**Question 4** **8 points**

Rank the following from 1 to 4 in **increasing** order. Provide the ranking in the boxes provided.

Note: Atomic number (Z) are provided for relevant atoms

a) Increasing (lowest to highest) order of *ionization energy*: (1= lowest; 4 = highest)

Mg

Mg: Z = 12

Rb

Rb: Z = 37

Cs

Cs: Z = 55

K

K : Z = 19

b) Increasing (lowest to highest) order of *bond angle*: (1= lowest; 4 = highest)

H₂OCCl₄XeF₄

HCN

c) Increasing (smallest to largest) order of *ionic size* (1= smallest; 4 = largest)

S²⁻

S: Z = 16

O²⁻

O: Z = 8

Be²⁺

Be: Z = 4

Na⁺

Na: Z = 11

d) Increasing (lowest to highest) order of *bond energy* (1= lowest; 4 = highest)

P-F

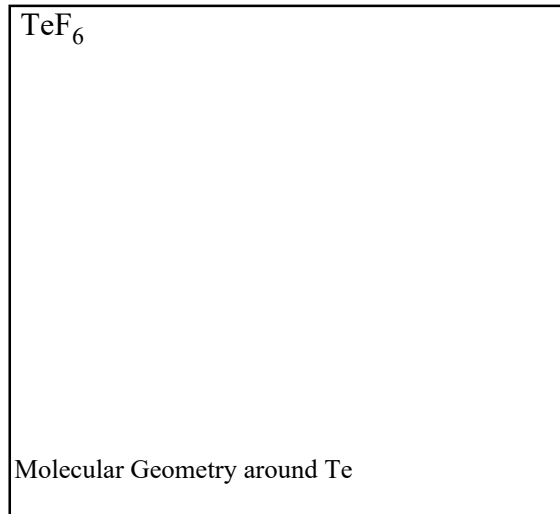
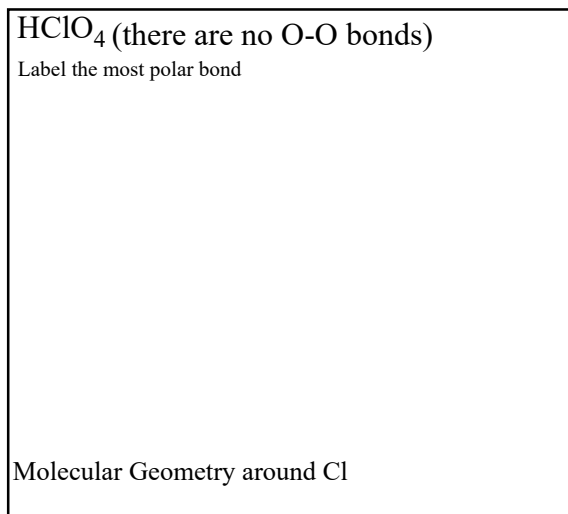
P-Cl

P-I

P-Br

**Question 5** **12 points**

a. 9 points For the molecules HClO_4 and TeF_6 , draw the most stable *Lewis structure* showing appropriate shape and *dashed/wedged bonds*, based on VSEPR. Determine the **molecular geometry** around the central atom. For HClO_4 , label the *most polar bond*.



b. 3 points (Note: Lone pairs are not shown in structures below)

(i) For the three *resonance* structures (1, 2 and 3) shown below, answer the following question:



(I) Circle the *most contributing* resonance structure.

Structure 1

Structure 2

Structure 3

(ii) For the three structures given (4, 5, and 6) below, answer the following questions:



(I) Circle all the structures that represent *resonance structures* of each other:

Structure 4

Structure 5

Structure 6

(II) Among the resonance structures, circle the *most contributing* resonance structure.

Structure 4

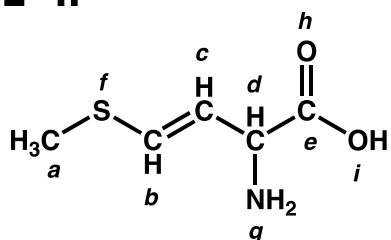
Structure 5

Structure 6

**Question 6** **6 points**

Using the data given below, calculate the **lattice energy** for SrO(s).

	ΔH° (kJ/mol)
$\text{Sr}(s) \longrightarrow \text{Sr}(g)$	+180
$\text{Sr}(g) \longrightarrow \text{Sr}^{2+}(g) + 2e^-$	+1468
$\text{O}_2(g) \longrightarrow 2\text{O}(g)$	+500
$\text{O}(g) + 2e^- \longrightarrow \text{O}^{2-}(g)$	+600
$\text{Sr}(s) + \frac{1}{2}\text{O}_2(g) \longrightarrow \text{SrO}(s)$	-550

**Question 7 10 points**

Consider the molecule given on the left, and answer questions (i) to (vi) using chemical bonding theories that we have learnt in class (valence bond theory, hybridization).

Note that the lone pairs are not shown in the structure:

i. Consider the carbon atoms marked in as ***a***, ***b***, ***c***, ***d***, and ***e*** in the molecule, and determine **hybridization** at each of the carbon centers. (2.5 points)

Atom	Hybridization
Carbon (<i>a</i>)	
Carbon (<i>b</i>)	
Carbon (<i>c</i>)	
Carbon (<i>d</i>)	
Carbon (<i>e</i>)	

ii. Now, consider the sulfur atom (S) marked as ***f***, nitrogen (N) atom (***g***), and oxygen(O) atoms (***h*** and ***i***), and determine their **hybridization**. (2 points)

Atom	Hybridization
Sulfur (<i>f</i>)	
Nitrogen (<i>g</i>)	
Oxygen (<i>h</i>)	
Oxygen (<i>i</i>)	

iii. **Describe** all the bonds formed by the carbon atoms ***a*** and ***b***, using sigma and pi designation as appropriate. (2 points)

Atom	Bonds formed (use sigma and pi as appropriate)
Carbon (<i>a</i>)	
Carbon (<i>b</i>)	

iv. How many total sp^3 -s sigma bonds are in the molecule? (1 point)

v. Determine **electronic** and **molecular geometry** at carbon (C) labeled ***e*** and sulfur (S) labeled ***f*** centres. (2 points)

Atom	Electronic Geometry	Molecular Geometry
Carbon (<i>e</i>)		
Sulfur (<i>f</i>)		

vi. Circle the value from the choices given below, that describes the **C-S-C bond angle at sulfur** (S) marked ***f***: (0.5 points)

>90° to <109.5° 109.5° >109.5 to <120° 120° >120° to <180°


Question 8 12 points

Using the **valence shell orbitals only** draw a **complete** molecular orbital (MO) diagram (with no s-p mixing) for the molecule **PS**, in the space provided below. **(6 points)**

Assume that 3s and 3p atomic orbitals will form molecular orbitals like those formed by the combination of 2s and 2p atomic orbitals that we learnt in class, and there is no s-p mixing.

Show **energy arrow**, **atomic orbitals** involved, **atom labels**, the **molecular orbitals** that are formed, **filled electrons** in the MO diagram etc. Make sure to label all the atomic and molecular orbitals.

Using the MO diagram above, answer the following questions:

- (i) Which one of the following will have the highest bond order? Circle the correct choice
 A) PS^{1-} B) PS C) PS^{1+}
- (ii) Which one of the following will have the longest bond length? Circle the correct choice
 A) PS^{1-} B) PS C) PS^{1+}
- (iii) Which of the following will be diamagnetic? Circle the correct choice.
 A) PS^{1-} B) PS C) PS^{1+}
- (iv) Draw Lewis dot structure of the molecule with the shortest bond length (based on the MO diagram) from among the three (PS^{1-} , PS, PS^{1+}), and indicate the bond order calculated from the Lewis dot structure. **(3 points)**

Lewis Dot Structure

Bond Order



756E79DE-5619-4211-A189-1C578967A821

midterm-exam-93388

#517 Page 10 of 10

SCRAP PAGE

**CHEMISTRY 110 –Midterm Exam****Thursday November 7, 2024, 6:30 – 8:30 pm**

Examiners: Pallavi Sirjoosingh and Ashok Kakkar

First name (please write as legibly as possible within the boxes)

--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--

Last name

--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--

Student ID number

--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--

This booklet has 8 questions on 10 pages (The last page can be used as a scrap page).

The staple must NOT be removed from the question booklet. Your intact question booklet must be returned at the end of the exam.

A separate booklet contains a periodic table, and datasheet.

Write your answers directly under the questions in the space provided. Do not write on the QR codes or near the top edges (as it may be cut off).

Please write **legibly and dark enough** for the scan to be readable. For students who use English as a second language, dictionaries are permitted. Calculators are allowed.

DO NOT ASK QUESTIONS OF THE INVIGILATORS about interpretations of the exam questions. If you have a concern about a question, please summarize it at the top of this page. All concerns will be evaluated after the exam. GOOD LUCK!

**Question 1 8 points** (*Assume all transitions are possible*)

a. 6 points Calculate the **energy (in kJ/mol)** required to ionize (completely remove) the electron from the **n=3** level of **He⁺ ion** (He: Helium). Show your work including equations. (3 sig. figs.)

b. 2 points

(I) Select the largest (longest) wavelength from the three choices below:

- i. Wavelength of photon to *completely remove* an electron from *1s orbital of Hydrogen atom*
- ii. Wavelength of photon to *completely remove* an electron from *1s orbital of Boron atom (B)*
- iii. Wavelength of photon to *completely remove* an electron from *1s orbital of Boron cation (B⁺)*

i or ii or iii

(II) Select the smallest (shortest) wavelength from the three choices below (*for H atom*):

- i. Wavelength of photon *emitted* for an electron transitioning from *2p to 1s orbital*
- ii. Wavelength of photon *absorbed* for an electron transitioning from *2s to 4p orbital*
- iii. Wavelength of photon *absorbed* for an electron transitioning from *1s to 3p orbital*

i or ii or iii



Question 2 8 points



A Chem110 student plots the graph shown on the left, from data obtained for the kinetic energy (eV) of the ejected electron versus frequency (s^{-1}) of the incident light, for an unknown metal. ($1\text{eV} = 1.6022 \times 10^{-19} \text{ J}$)

Based on the plot, answer the questions **a to c** below:

a. 1 point On the plot above, label the **threshold frequency**.

b. 2 points What is the value of the work potential (in J) of the unknown metal?

c. 3 points What is the *longest* wavelength (in nm) required to eject an electron from this metal's surface? Show your work including equation(s) used. (3 sig figs)

d. 2 points In the space provided below, plot the following

(Assume frequency is *above* threshold frequency):

- Kinetic energy* of an ejected electron versus the *intensity* of the incident light (left)
- Number of electrons* ejected versus the *frequency* of the incident light (right)



**Question 3** **6 points****a. 2 points**

In the ground state of Ge (Germanium; $Z = 32$) atom, **name** the following orbitals:

i. Orbital containing the electron that requires the maximum energy to be removed.

ii. Orbital containing the electron that experiences the minimum effective nuclear charge.

b. 2 points

An unknown anion (A^-) has the following *excited* electronic configuration: $[Ar] 4s^2 3d^3 4p^1$

(Ar: Argon; $Z = 18$)

Give the **condensed ground state electronic configuration** of the atom, **A** and the cation, **A^{2+}** .
Show your work. (Assume all electronic transitions are possible)

Electronic Configuration of A:

Electronic Configuration of A^{2+} :

c. 2 points

In a single atom, determine how many electrons can have the following set of quantum numbers?

(i) $n = 5; l = 3$

In a single atom, determine how many electrons can have the following set of quantum numbers?

(i) $n = 3; l = 2; m_s = +1/2$

**Question 4** **8 points**

Rank the following from 1 to 4 in **increasing** order. Provide the ranking in the boxes provided.

Note: Atomic number (Z) are provided for relevant atoms

a) Increasing (lowest to highest) order of *ionization energy*: (1= lowest; 4 = highest)

Mg

Mg: Z = 12

Rb

Rb: Z = 37

Cs

Cs: Z = 55

K

K : Z = 19

b) Increasing (lowest to highest) order of *bond angle*: (1= lowest; 4 = highest)

H₂OCCl₄XeF₄

HCN

c) Increasing (smallest to largest) order of *ionic size* (1= smallest; 4 = largest)

S²⁻

S: Z = 16

O²⁻

O: Z = 8

Be²⁺

Be: Z = 4

Na⁺

Na: Z = 11

d) Increasing (lowest to highest) order of *bond energy* (1= lowest; 4 = highest)

P-F

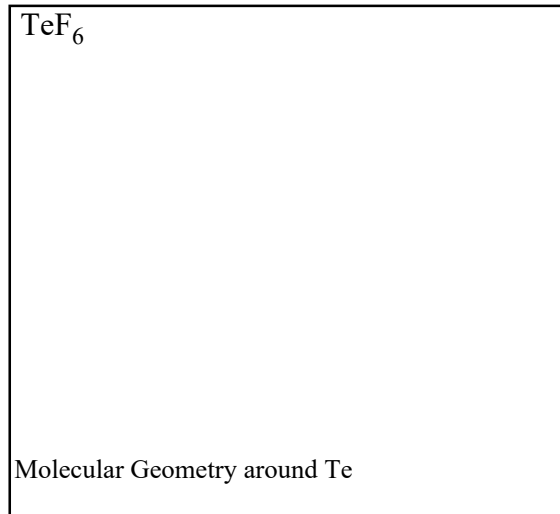
P-Cl

P-I

P-Br

**Question 5** **12 points**

a. 9 points For the molecules HClO_4 and TeF_6 , draw the most stable *Lewis structure* showing appropriate shape and *dashed/wedged bonds*, based on VSEPR. Determine the **molecular geometry** around the central atom. For HClO_4 , label the *most polar bond*.



b. 3 points (Note: Lone pairs are not shown in structures below)

(i) For the three *resonance* structures (1, 2 and 3) shown below, answer the following question:



(I) Circle the *most contributing* resonance structure.

Structure 1

Structure 2

Structure 3

(ii) For the three structures given (4, 5, and 6) below, answer the following questions:



(I) Circle all the structures that represent *resonance structures* of each other:

Structure 4

Structure 5

Structure 6

(II) Among the resonance structures, circle the *most contributing* resonance structure.

Structure 4

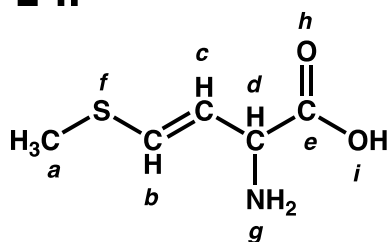
Structure 5

Structure 6

**Question 6** **6 points**

Using the data given below, calculate the **lattice energy** for SrO(s).

	ΔH° (kJ/mol)
$\text{Sr}(s) \longrightarrow \text{Sr}(g)$	+180
$\text{Sr}(g) \longrightarrow \text{Sr}^{2+}(g) + 2e^-$	+1468
$\text{O}_2(g) \longrightarrow 2\text{O}(g)$	+500
$\text{O}(g) + 2e^- \longrightarrow \text{O}^{2-}(g)$	+600
$\text{Sr}(s) + \frac{1}{2}\text{O}_2(g) \longrightarrow \text{SrO}(s)$	-550

**Question 7 10 points**

Consider the molecule given on the left, and answer questions (i) to (vi) using chemical bonding theories that we have learnt in class (valence bond theory, hybridization).

Note that the lone pairs are not shown in the structure:

i. Consider the carbon atoms marked in as ***a***, ***b***, ***c***, ***d***, and ***e*** in the molecule, and determine **hybridization** at each of the carbon centers. (2.5 points)

Atom	Hybridization
Carbon (<i>a</i>)	
Carbon (<i>b</i>)	
Carbon (<i>c</i>)	
Carbon (<i>d</i>)	
Carbon (<i>e</i>)	

ii. Now, consider the sulfur atom (S) marked as ***f***, nitrogen (N) atom (***g***), and oxygen(O) atoms (***h*** and ***i***), and determine their **hybridization**. (2 points)

Atom	Hybridization
Sulfur (<i>f</i>)	
Nitrogen (<i>g</i>)	
Oxygen (<i>h</i>)	
Oxygen (<i>i</i>)	

iii. **Describe** all the bonds formed by the carbon atoms ***a*** and ***b***, using sigma and pi designation as appropriate. (2 points)

Atom	Bonds formed (use sigma and pi as appropriate)
Carbon (<i>a</i>)	
Carbon (<i>b</i>)	

iv. How many total sp^3 -s sigma bonds are in the molecule? (1 point)

v. Determine **electronic** and **molecular geometry** at carbon (C) labeled ***e*** and sulfur (S) labeled ***f*** centres. (2 points)

Atom	Electronic Geometry	Molecular Geometry
Carbon (<i>e</i>)		
Sulfur (<i>f</i>)		

vi. Circle the value from the choices given below, that describes the **C-S-C bond angle at sulfur** (S) marked ***f***: (0.5 points)

>90° to <109.5° 109.5° >109.5 to <120° 120° >120° to <180°


Question 8 12 points

Using the **valence shell orbitals only** draw a **complete** molecular orbital (MO) diagram (with no s-p mixing) for the molecule **PS**, in the space provided below. **(6 points)**

Assume that 3s and 3p atomic orbitals will form molecular orbitals like those formed by the combination of 2s and 2p atomic orbitals that we learnt in class, and there is no s-p mixing.

Show **energy arrow**, **atomic orbitals** involved, **atom labels**, the **molecular orbitals** that are formed, **filled electrons** in the MO diagram etc. Make sure to label all the atomic and molecular orbitals.

Using the MO diagram above, answer the following questions:

- (i) Which one of the following will have the highest bond order? Circle the correct choice
 A) PS^{1-} B) PS C) PS^{1+}
- (ii) Which one of the following will have the longest bond length? Circle the correct choice
 A) PS^{1-} B) PS C) PS^{1+}
- (iii) Which of the following will be diamagnetic? Circle the correct choice.
 A) PS^{1-} B) PS C) PS^{1+}
- (iv) Draw Lewis dot structure of the molecule with the shortest bond length (based on the MO diagram) from among the three (PS^{1-} , PS, PS^{1+}), and indicate the bond order calculated from the Lewis dot structure. **(3 points)**

Lewis Dot Structure

Bond Order



602A7A6A-0956-4F4E-96A9-6AE9244714C6

midterm-exam-93388

#518 Page 10 of 10

SCRAP PAGE

**CHEMISTRY 110 –Midterm Exam****Thursday November 7, 2024, 6:30 – 8:30 pm****Examiners: Pallavi Sirjoosingh and Ashok Kakkar**

First name (please write as legibly as possible within the boxes)

--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--

Last name

--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--

Student ID number

--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--

This booklet has 8 questions on 10 pages (The last page can be used as a scrap page).

The staple must NOT be removed from the question booklet. Your intact question booklet must be returned at the end of the exam.

A separate booklet contains a periodic table, and datasheet.

Write your answers directly under the questions in the space provided. Do not write on the QR codes or near the top edges (as it may be cut off).

Please write **legibly and dark enough** for the scan to be readable. For students who use English as a second language, dictionaries are permitted. Calculators are allowed.

DO NOT ASK QUESTIONS OF THE INVIGILATORS about interpretations of the exam questions. If you have a concern about a question, please summarize it at the top of this page. All concerns will be evaluated after the exam. GOOD LUCK!

**Question 1 8 points** (*Assume all transitions are possible*)

a. 6 points Calculate the **energy (in kJ/mol)** required to ionize (completely remove) the electron from the **n=3** level of **He⁺ ion** (He: Helium). Show your work including equations. (3 sig. figs.)

b. 2 points

(I) Select the largest (longest) wavelength from the three choices below:

- i. Wavelength of photon to *completely remove* an electron from *1s orbital of Hydrogen atom*
- ii. Wavelength of photon to *completely remove* an electron from *1s orbital of Boron atom (B)*
- iii. Wavelength of photon to *completely remove* an electron from *1s orbital of Boron cation (B⁺)*

i or ii or iii

(II) Select the smallest (shortest) wavelength from the three choices below (*for H atom*):

- i. Wavelength of photon *emitted* for an electron transitioning from *2p to 1s orbital*
- ii. Wavelength of photon *absorbed* for an electron transitioning from *2s to 4p orbital*
- iii. Wavelength of photon *absorbed* for an electron transitioning from *1s to 3p orbital*

i or ii or iii



Question 2 8 points



A Chem110 student plots the graph shown on the left, from data obtained for the kinetic energy (eV) of the ejected electron versus frequency (s^{-1}) of the incident light, for an unknown metal. ($1\text{eV} = 1.6022 \times 10^{-19} \text{ J}$)

Based on the plot, answer the questions **a to c** below:

a. 1 point On the plot above, label the **threshold frequency**.

b. 2 points What is the value of the work potential (in J) of the unknown metal?

c. 3 points What is the *longest* wavelength (in nm) required to eject an electron from this metal's surface? Show your work including equation(s) used. (3 sig figs)

d. 2 points In the space provided below, plot the following

(Assume frequency is *above* threshold frequency):

- i. *Kinetic energy* of an ejected electron versus the *intensity* of the incident light (left)
- ii. *Number of electrons* ejected versus the *frequency* of the incident light (right)



**Question 3** **6 points****a. 2 points**

In the ground state of Ge (Germanium; $Z = 32$) atom, **name** the following orbitals:

i. Orbital containing the electron that requires the maximum energy to be removed.

ii. Orbital containing the electron that experiences the minimum effective nuclear charge.

b. 2 points

An unknown anion (A^-) has the following *excited* electronic configuration: $[Ar] 4s^2 3d^3 4p^1$

(Ar: Argon; $Z = 18$)

Give the **condensed ground state electronic configuration** of the atom, **A** and the cation, **A^{2+}** .
Show your work. (Assume all electronic transitions are possible)

Electronic Configuration of A:

Electronic Configuration of A^{2+} :

c. 2 points

In a single atom, determine how many electrons can have the following set of quantum numbers?

(i) $n = 5; l = 3$

In a single atom, determine how many electrons can have the following set of quantum numbers?

(i) $n = 3; l = 2; m_s = +1/2$

**Question 4** **8 points**

Rank the following from 1 to 4 in **increasing** order. Provide the ranking in the boxes provided.

Note: Atomic number (Z) are provided for relevant atoms

a) Increasing (lowest to highest) order of *ionization energy*: (1= lowest; 4 = highest)

Mg

Mg: Z = 12

Rb

Rb: Z = 37

Cs

Cs: Z = 55

K

K : Z = 19

b) Increasing (lowest to highest) order of *bond angle*: (1= lowest; 4 = highest)

H₂OCCl₄XeF₄

HCN

c) Increasing (smallest to largest) order of *ionic size* (1= smallest; 4 = largest)

S²⁻

S: Z = 16

O²⁻

O: Z = 8

Be²⁺

Be: Z = 4

Na⁺

Na: Z = 11

d) Increasing (lowest to highest) order of *bond energy* (1= lowest; 4 = highest)

P-F

P-Cl

P-I

P-Br

**Question 5** **12 points**

a. 9 points For the molecules HClO_4 and TeF_6 , draw the most stable *Lewis structure* showing appropriate shape and *dashed/wedged bonds*, based on VSEPR. Determine the **molecular geometry** around the central atom. For HClO_4 , label the *most polar bond*.

HClO_4 (there are no O-O bonds)

Label the most polar bond

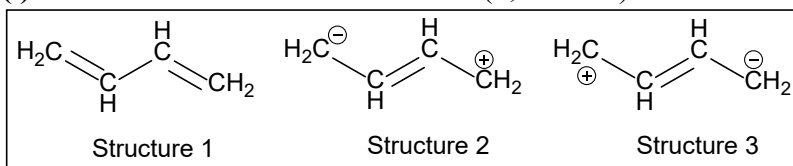
Molecular Geometry around Cl

TeF_6

Molecular Geometry around Te

b. 3 points (Note: Lone pairs are not shown in structures below)

(i) For the three *resonance* structures (1, 2 and 3) shown below, answer the following question:



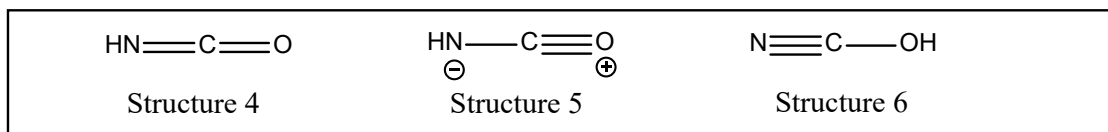
(I) Circle the *most contributing* resonance structure.

Structure 1

Structure 2

Structure 3

(ii) For the three structures given (4, 5, and 6) below, answer the following questions:



(I) Circle all the structures that represent *resonance structures* of each other:

Structure 4

Structure 5

Structure 6

(II) Among the resonance structures, circle the *most contributing* resonance structure.

Structure 4

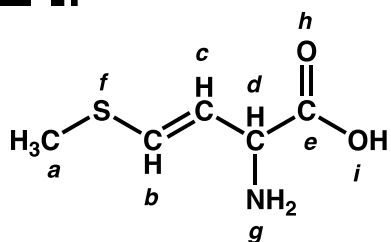
Structure 5

Structure 6

**Question 6** **6 points**

Using the data given below, calculate the **lattice energy** for SrO(s).

	ΔH° (kJ/mol)
$\text{Sr}(s) \longrightarrow \text{Sr}(g)$	+180
$\text{Sr}(g) \longrightarrow \text{Sr}^{2+}(g) + 2e^-$	+1468
$\text{O}_2(g) \longrightarrow 2\text{O}(g)$	+500
$\text{O}(g) + 2e^- \longrightarrow \text{O}^{2-}(g)$	+600
$\text{Sr}(s) + \frac{1}{2}\text{O}_2(g) \longrightarrow \text{SrO}(s)$	-550

**Question 7 10 points**

Consider the molecule given on the left, and answer questions (i) to (vi) using chemical bonding theories that we have learnt in class (valence bond theory, hybridization).

Note that the lone pairs are not shown in the structure:

i. Consider the carbon atoms marked in as ***a***, ***b***, ***c***, ***d***, and ***e*** in the molecule, and determine **hybridization** at each of the carbon centers. (2.5 points)

Atom	Hybridization
Carbon (<i>a</i>)	
Carbon (<i>b</i>)	
Carbon (<i>c</i>)	
Carbon (<i>d</i>)	
Carbon (<i>e</i>)	

ii. Now, consider the sulfur atom (S) marked as ***f***, nitrogen (N) atom (***g***), and oxygen(O) atoms (***h*** and ***i***), and determine their **hybridization**. (2 points)

Atom	Hybridization
Sulfur (<i>f</i>)	
Nitrogen (<i>g</i>)	
Oxygen (<i>h</i>)	
Oxygen (<i>i</i>)	

iii. **Describe** all the bonds formed by the carbon atoms ***a*** and ***b***, using sigma and pi designation as appropriate. (2 points)

Atom	Bonds formed (use sigma and pi as appropriate)
Carbon (<i>a</i>)	
Carbon (<i>b</i>)	

iv. How many total sp^3 -s sigma bonds are in the molecule? (1 point)

v. Determine **electronic** and **molecular geometry** at carbon (C) labeled ***e*** and sulfur (S) labeled ***f*** centres. (2 points)

Atom	Electronic Geometry	Molecular Geometry
Carbon (<i>e</i>)		
Sulfur (<i>f</i>)		

vi. Circle the value from the choices given below, that describes the **C-S-C bond angle at sulfur** (S) marked ***f***: (0.5 points)

>90° to <109.5° 109.5° >109.5 to <120° 120° >120° to <180°


Question 8 12 points

Using the **valence shell orbitals only** draw a **complete** molecular orbital (MO) diagram (with no s-p mixing) for the molecule **PS**, in the space provided below. **(6 points)**

Assume that 3s and 3p atomic orbitals will form molecular orbitals like those formed by the combination of 2s and 2p atomic orbitals that we learnt in class, and there is no s-p mixing.

Show **energy arrow**, **atomic orbitals** involved, **atom labels**, the **molecular orbitals** that are formed, **filled electrons** in the MO diagram etc. Make sure to label all the atomic and molecular orbitals.

Using the MO diagram above, answer the following questions:

- (i) Which one of the following will have the highest bond order? Circle the correct choice
 A) PS^{1-} B) PS C) PS^{1+}
- (ii) Which one of the following will have the longest bond length? Circle the correct choice
 A) PS^{1-} B) PS C) PS^{1+}
- (iii) Which of the following will be diamagnetic? Circle the correct choice.
 A) PS^{1-} B) PS C) PS^{1+}
- (iv) Draw Lewis dot structure of the molecule with the shortest bond length (based on the MO diagram) from among the three (PS^{1-} , PS, PS^{1+}), and indicate the bond order calculated from the Lewis dot structure. **(3 points)**

Lewis Dot Structure

Bond Order



2013DE7C-4A9B-47D1-A373-93BE269B25C4

midterm-exam-93388

#519 Page 10 of 10

SCRAP PAGE

**CHEMISTRY 110 –Midterm Exam****Thursday November 7, 2024, 6:30 – 8:30 pm****Examiners: Pallavi Sirjoosingh and Ashok Kakkar**

First name (please write as legibly as possible within the boxes)

--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--

Last name

--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--

Student ID number

--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--

This booklet has 8 questions on 10 pages (The last page can be used as a scrap page).

The staple must NOT be removed from the question booklet. Your intact question booklet must be returned at the end of the exam.

A separate booklet contains a periodic table, and datasheet.

Write your answers directly under the questions in the space provided. Do not write on the QR codes or near the top edges (as it may be cut off).

Please write **legibly and dark enough** for the scan to be readable. For students who use English as a second language, dictionaries are permitted. Calculators are allowed.

DO NOT ASK QUESTIONS OF THE INVIGILATORS about interpretations of the exam questions. If you have a concern about a question, please summarize it at the top of this page. All concerns will be evaluated after the exam. GOOD LUCK!

**Question 1 8 points** (*Assume all transitions are possible*)

a. 6 points Calculate the **energy (in kJ/mol)** required to ionize (completely remove) the electron from the **n=3** level of **He⁺ ion** (He: Helium). Show your work including equations. (3 sig. figs.)

b. 2 points

(I) Select the largest (longest) wavelength from the three choices below:

- i. Wavelength of photon to *completely remove* an electron from *1s orbital of Hydrogen atom*
- ii. Wavelength of photon to *completely remove* an electron from *1s orbital of Boron atom (B)*
- iii. Wavelength of photon to *completely remove* an electron from *1s orbital of Boron cation (B⁺)*

i or ii or iii

(II) Select the smallest (shortest) wavelength from the three choices below (*for H atom*):

- i. Wavelength of photon *emitted* for an electron transitioning from *2p to 1s orbital*
- ii. Wavelength of photon *absorbed* for an electron transitioning from *2s to 4p orbital*
- iii. Wavelength of photon *absorbed* for an electron transitioning from *1s to 3p orbital*

i or ii or iii



Question 2 8 points



A Chem110 student plots the graph shown on the left, from data obtained for the kinetic energy (eV) of the ejected electron versus frequency (s^{-1}) of the incident light, for an unknown metal. ($1\text{eV} = 1.6022 \times 10^{-19} \text{ J}$)

Based on the plot, answer the questions **a to c** below:

a. 1 point On the plot above, label the **threshold frequency**.

b. 2 points What is the value of the work potential (in J) of the unknown metal?

c. 3 points What is the *longest* wavelength (in nm) required to eject an electron from this metal's surface? Show your work including equation(s) used. (3 sig figs)

d. 2 points In the space provided below, plot the following

(Assume frequency is *above* threshold frequency):

- i. *Kinetic energy* of an ejected electron versus the *intensity* of the incident light (left)
- ii. *Number of electrons* ejected versus the *frequency* of the incident light (right)



**Question 3** **6 points****a. 2 points**

In the ground state of Ge (Germanium; $Z = 32$) atom, **name** the following orbitals:

i. Orbital containing the electron that requires the maximum energy to be removed.

ii. Orbital containing the electron that experiences the minimum effective nuclear charge.

b. 2 points

An unknown anion (A^-) has the following *excited* electronic configuration: $[Ar] 4s^2 3d^3 4p^1$

(Ar: Argon; $Z = 18$)

Give the **condensed ground state electronic configuration** of the atom, **A** and the cation, **A^{2+}** .
Show your work. (Assume all electronic transitions are possible)

Electronic Configuration of A:

Electronic Configuration of A^{2+} :

c. 2 points

In a single atom, determine how many electrons can have the following set of quantum numbers?

(i) $n = 5; l = 3$

In a single atom, determine how many electrons can have the following set of quantum numbers?

(i) $n = 3; l = 2; m_s = +1/2$

**Question 4** **8 points**

Rank the following from 1 to 4 in **increasing** order. Provide the ranking in the boxes provided.

Note: Atomic number (Z) are provided for relevant atoms

a) Increasing (lowest to highest) order of *ionization energy*: (1= lowest; 4 = highest)

Mg

Mg: Z = 12

Rb

Rb: Z = 37

Cs

Cs: Z = 55

K

K : Z = 19

b) Increasing (lowest to highest) order of *bond angle*: (1= lowest; 4 = highest)

H₂OCCl₄XeF₄

HCN

c) Increasing (smallest to largest) order of *ionic size* (1= smallest; 4 = largest)

S²⁻

S: Z = 16

O²⁻

O: Z = 8

Be²⁺

Be: Z = 4

Na⁺

Na: Z = 11

d) Increasing (lowest to highest) order of *bond energy* (1= lowest; 4 = highest)

P-F

P-Cl

P-I

P-Br

**Question 5** **12 points**

a. 9 points For the molecules HClO_4 and TeF_6 , draw the most stable *Lewis structure* showing appropriate shape and *dashed/wedged bonds*, based on VSEPR. Determine the **molecular geometry** around the central atom. For HClO_4 , label the *most polar bond*.



b. 3 points (Note: Lone pairs are not shown in structures below)

(i) For the three *resonance* structures (1, 2 and 3) shown below, answer the following question:



(I) Circle the *most contributing* resonance structure.

Structure 1

Structure 2

Structure 3

(ii) For the three structures given (4, 5, and 6) below, answer the following questions:



(I) Circle all the structures that represent *resonance structures* of each other:

Structure 4

Structure 5

Structure 6

(II) Among the resonance structures, circle the *most contributing* resonance structure.

Structure 4

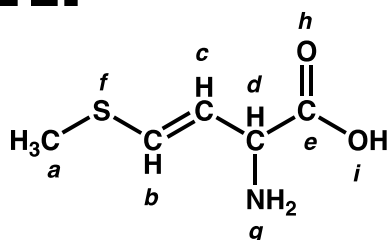
Structure 5

Structure 6

**Question 6** **6 points**

Using the data given below, calculate the **lattice energy** for SrO(s).

	ΔH° (kJ/mol)
$\text{Sr}(s) \longrightarrow \text{Sr}(g)$	+180
$\text{Sr}(g) \longrightarrow \text{Sr}^{2+}(g) + 2e^-$	+1468
$\text{O}_2(g) \longrightarrow 2\text{O}(g)$	+500
$\text{O}(g) + 2e^- \longrightarrow \text{O}^{2-}(g)$	+600
$\text{Sr}(s) + \frac{1}{2}\text{O}_2(g) \longrightarrow \text{SrO}(s)$	-550

**Question 7 10 points**

Consider the molecule given on the left, and answer questions (i) to (vi) using chemical bonding theories that we have learnt in class (valence bond theory, hybridization).

Note that the lone pairs are not shown in the structure:

i. Consider the carbon atoms marked in as ***a***, ***b***, ***c***, ***d***, and ***e*** in the molecule, and determine **hybridization** at each of the carbon centers. (2.5 points)

Atom	Hybridization
Carbon (<i>a</i>)	
Carbon (<i>b</i>)	
Carbon (<i>c</i>)	
Carbon (<i>d</i>)	
Carbon (<i>e</i>)	

ii. Now, consider the sulfur atom (S) marked as ***f***, nitrogen (N) atom (***g***), and oxygen(O) atoms (***h*** and ***i***), and determine their **hybridization**. (2 points)

Atom	Hybridization
Sulfur (<i>f</i>)	
Nitrogen (<i>g</i>)	
Oxygen (<i>h</i>)	
Oxygen (<i>i</i>)	

iii. **Describe** all the bonds formed by the carbon atoms ***a*** and ***b***, using sigma and pi designation as appropriate. (2 points)

Atom	Bonds formed (use sigma and pi as appropriate)
Carbon (<i>a</i>)	
Carbon (<i>b</i>)	

iv. How many total sp^3 -s sigma bonds are in the molecule? (1 point)

v. Determine **electronic** and **molecular geometry** at carbon (C) labeled ***e*** and sulfur (S) labeled ***f*** centres. (2 points)

Atom	Electronic Geometry	Molecular Geometry
Carbon (<i>e</i>)		
Sulfur (<i>f</i>)		

vi. Circle the value from the choices given below, that describes the **C-S-C bond angle at sulfur** (S) marked ***f***: (0.5 points)

>90° to <109.5° 109.5° >109.5 to <120° 120° >120° to <180°

**Question 8 12 points**

Using the **valence shell orbitals only** draw a **complete** molecular orbital (MO) diagram (with no s-p mixing) for the molecule **PS**, in the space provided below. **(6 points)**

Assume that 3s and 3p atomic orbitals will form molecular orbitals like those formed by the combination of 2s and 2p atomic orbitals that we learnt in class, and there is no s-p mixing.

Show **energy arrow**, **atomic orbitals** involved, **atom labels**, the **molecular orbitals** that are formed, **filled electrons** in the MO diagram etc. Make sure to label all the atomic and molecular orbitals.

Using the MO diagram above, answer the following questions:

- (i) Which one of the following will have the highest bond order? Circle the correct choice
A) PS^{1-} B) PS C) PS^{1+}
- (ii) Which one of the following will have the longest bond length? Circle the correct choice
A) PS^{1-} B) PS C) PS^{1+}
- (iii) Which of the following will be diamagnetic? Circle the correct choice.
A) PS^{1-} B) PS C) PS^{1+}
- (iv) Draw Lewis dot structure of the molecule with the shortest bond length (based on the MO diagram) from among the three (PS^{1-} , PS, PS^{1+}), and indicate the bond order calculated from the Lewis dot structure. **(3 points)**

Lewis Dot Structure

Bond Order



11621758-9107-42E0-A1B1-74C8E9F4DFA2

midterm-exam-93388

#520 Page 10 of 10

SCRAP PAGE

**CHEMISTRY 110 –Midterm Exam****Thursday November 7, 2024, 6:30 – 8:30 pm****Examiners: Pallavi Sirjoosingh and Ashok Kakkar**

First name (please write as legibly as possible within the boxes)

--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--

Last name

--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--

Student ID number

--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--

This booklet has 8 questions on 10 pages (The last page can be used as a scrap page).

The staple must NOT be removed from the question booklet. Your intact question booklet must be returned at the end of the exam.

A separate booklet contains a periodic table, and datasheet.

Write your answers directly under the questions in the space provided. Do not write on the QR codes or near the top edges (as it may be cut off).

Please write **legibly and dark enough** for the scan to be readable. For students who use English as a second language, dictionaries are permitted. Calculators are allowed.

DO NOT ASK QUESTIONS OF THE INVIGILATORS about interpretations of the exam questions. If you have a concern about a question, please summarize it at the top of this page. All concerns will be evaluated after the exam. GOOD LUCK!

**Question 1 8 points** (*Assume all transitions are possible*)

a. 6 points Calculate the **energy (in kJ/mol)** required to ionize (completely remove) the electron from the **n=3** level of **He⁺ ion** (He: Helium). Show your work including equations. (3 sig. figs.)

b. 2 points

(I) Select the largest (longest) wavelength from the three choices below:

- i. Wavelength of photon to *completely remove* an electron from *1s orbital of Hydrogen atom*
- ii. Wavelength of photon to *completely remove* an electron from *1s orbital of Boron atom (B)*
- iii. Wavelength of photon to *completely remove* an electron from *1s orbital of Boron cation (B⁺)*

i or ii or iii

(II) Select the smallest (shortest) wavelength from the three choices below (*for H atom*):

- i. Wavelength of photon *emitted* for an electron transitioning from *2p to 1s orbital*
- ii. Wavelength of photon *absorbed* for an electron transitioning from *2s to 4p orbital*
- iii. Wavelength of photon *absorbed* for an electron transitioning from *1s to 3p orbital*

i or ii or iii



Question 2 8 points



A Chem110 student plots the graph shown on the left, from data obtained for the kinetic energy (eV) of the ejected electron versus frequency (s^{-1}) of the incident light, for an unknown metal. ($1\text{eV} = 1.6022 \times 10^{-19} \text{ J}$)

Based on the plot, answer the questions **a to c** below:

a. 1 point On the plot above, label the **threshold frequency**.

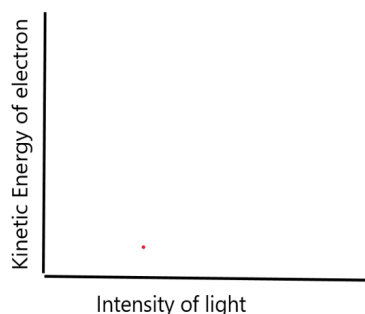
b. 2 points What is the value of the work potential (in J) of the unknown metal?

c. 3 points What is the *longest* wavelength (in nm) required to eject an electron from this metal's surface? Show your work including equation(s) used. (3 sig figs)

d. 2 points In the space provided below, plot the following

(Assume frequency is *above* threshold frequency):

- Kinetic energy* of an ejected electron versus the *intensity* of the incident light (left)
- Number of electrons* ejected versus the *frequency* of the incident light (right)



**Question 3** **6 points****a. 2 points**

In the ground state of Ge (Germanium; $Z = 32$) atom, **name** the following orbitals:

i. Orbital containing the electron that requires the maximum energy to be removed.

ii. Orbital containing the electron that experiences the minimum effective nuclear charge.

b. 2 points

An unknown anion (A^-) has the following *excited* electronic configuration: $[Ar] 4s^2 3d^3 4p^1$

(Ar: Argon; $Z = 18$)

Give the **condensed ground state electronic configuration** of the atom, **A** and the cation, **A^{2+}** .
Show your work. (Assume all electronic transitions are possible)

Electronic Configuration of A:

Electronic Configuration of A^{2+} :

c. 2 points

In a single atom, determine how many electrons can have the following set of quantum numbers?

(i) $n = 5; l = 3$

In a single atom, determine how many electrons can have the following set of quantum numbers?

(i) $n = 3; l = 2; m_s = +1/2$

**Question 4** **8 points**

Rank the following from 1 to 4 in **increasing** order. Provide the ranking in the boxes provided.

Note: Atomic number (Z) are provided for relevant atoms

a) Increasing (lowest to highest) order of *ionization energy*: (1= lowest; 4 = highest)

Mg

Mg: Z = 12

Rb

Rb: Z = 37

Cs

Cs: Z = 55

K

K : Z = 19

b) Increasing (lowest to highest) order of *bond angle*: (1= lowest; 4 = highest)

H₂OCCl₄XeF₄

HCN

c) Increasing (smallest to largest) order of *ionic size* (1= smallest; 4 = largest)

S²⁻

S: Z = 16

O²⁻

O: Z = 8

Be²⁺

Be: Z = 4

Na⁺

Na: Z = 11

d) Increasing (lowest to highest) order of *bond energy* (1= lowest; 4 = highest)

P-F

P-Cl

P-I

P-Br

**Question 5** **12 points**

a. 9 points For the molecules HClO_4 and TeF_6 , draw the most stable *Lewis structure* showing appropriate shape and *dashed/wedged bonds*, based on VSEPR. Determine the **molecular geometry** around the central atom. For HClO_4 , label the *most polar bond*.



b. 3 points (Note: Lone pairs are not shown in structures below)

(i) For the three *resonance* structures (1, 2 and 3) shown below, answer the following question:



(I) Circle the *most contributing* resonance structure.

Structure 1

Structure 2

Structure 3

(ii) For the three structures given (4, 5, and 6) below, answer the following questions:



(I) Circle all the structures that represent *resonance structures* of each other:

Structure 4

Structure 5

Structure 6

(II) Among the resonance structures, circle the *most contributing* resonance structure.

Structure 4

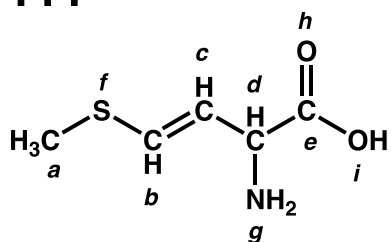
Structure 5

Structure 6

**Question 6** **6 points**

Using the data given below, calculate the **lattice energy** for SrO(s).

	ΔH° (kJ/mol)
$\text{Sr}(s) \longrightarrow \text{Sr}(g)$	+180
$\text{Sr}(g) \longrightarrow \text{Sr}^{2+}(g) + 2e^-$	+1468
$\text{O}_2(g) \longrightarrow 2\text{O}(g)$	+500
$\text{O}(g) + 2e^- \longrightarrow \text{O}^{2-}(g)$	+600
$\text{Sr}(s) + \frac{1}{2}\text{O}_2(g) \longrightarrow \text{SrO}(s)$	-550

**Question 7 10 points**

Consider the molecule given on the left, and answer questions (i) to (vi) using chemical bonding theories that we have learnt in class (valence bond theory, hybridization).

Note that the lone pairs are not shown in the structure:

i. Consider the carbon atoms marked in as ***a***, ***b***, ***c***, ***d***, and ***e*** in the molecule, and determine **hybridization** at each of the carbon centers. (2.5 points)

Atom	Hybridization
Carbon (<i>a</i>)	
Carbon (<i>b</i>)	
Carbon (<i>c</i>)	
Carbon (<i>d</i>)	
Carbon (<i>e</i>)	

ii. Now, consider the sulfur atom (S) marked as ***f***, nitrogen (N) atom (***g***), and oxygen(O) atoms (***h*** and ***i***), and determine their **hybridization**. (2 points)

Atom	Hybridization
Sulfur (<i>f</i>)	
Nitrogen (<i>g</i>)	
Oxygen (<i>h</i>)	
Oxygen (<i>i</i>)	

iii. **Describe** all the bonds formed by the carbon atoms ***a*** and ***b***, using sigma and pi designation as appropriate. (2 points)

Atom	Bonds formed (use sigma and pi as appropriate)
Carbon (<i>a</i>)	
Carbon (<i>b</i>)	

iv. How many total sp^3 -s sigma bonds are in the molecule? (1 point)

v. Determine **electronic** and **molecular geometry** at carbon (C) labeled ***e*** and sulfur (S) labeled ***f*** centres. (2 points)

Atom	Electronic Geometry	Molecular Geometry
Carbon (<i>e</i>)		
Sulfur (<i>f</i>)		

vi. Circle the value from the choices given below, that describes the **C-S-C bond angle at sulfur** (S) marked ***f***: (0.5 points)

>90° to <109.5° 109.5° >109.5 to <120° 120° >120° to <180°



Question 8 12 points

Using the **valence shell orbitals only** draw a **complete** molecular orbital (MO) diagram (with no s-p mixing) for the molecule **PS**, in the space provided below. **(6 points)**

Assume that 3s and 3p atomic orbitals will form molecular orbitals like those formed by the combination of 2s and 2p atomic orbitals that we learnt in class, and there is no s-p mixing.

Show **energy arrow**, **atomic orbitals** involved, **atom labels**, the **molecular orbitals** that are formed, **filled electrons** in the MO diagram etc. Make sure to label all the atomic and molecular orbitals.

Using the MO diagram above, answer the following questions:

- (i) Which one of the following will have the highest bond order? Circle the correct choice
 A) PS^{1-} B) PS C) PS^{1+}
- (ii) Which one of the following will have the longest bond length? Circle the correct choice
 A) PS^{1-} B) PS C) PS^{1+}
- (iii) Which of the following will be diamagnetic? Circle the correct choice.
 A) PS^{1-} B) PS C) PS^{1+}
- (iv) Draw Lewis dot structure of the molecule with the shortest bond length (based on the MO diagram) from among the three (PS^{1-} , PS, PS^{1+}), and indicate the bond order calculated from the Lewis dot structure. **(3 points)**

Lewis Dot Structure

Bond Order



D25D2F3F-2550-4F64-A686-7B589B0751B0

midterm-exam-93388

#521 Page 10 of 10

SCRAP PAGE

**CHEMISTRY 110 –Midterm Exam****Thursday November 7, 2024, 6:30 – 8:30 pm****Examiners: Pallavi Sirjoosingh and Ashok Kakkar**

First name (please write as legibly as possible within the boxes)

--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--

Last name

--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--

Student ID number

--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--

This booklet has 8 questions on 10 pages (The last page can be used as a scrap page).

The staple must NOT be removed from the question booklet. Your intact question booklet must be returned at the end of the exam.

A separate booklet contains a periodic table, and datasheet.

Write your answers directly under the questions in the space provided. Do not write on the QR codes or near the top edges (as it may be cut off).

Please write **legibly and dark enough** for the scan to be readable. For students who use English as a second language, dictionaries are permitted. Calculators are allowed.

DO NOT ASK QUESTIONS OF THE INVIGILATORS about interpretations of the exam questions. If you have a concern about a question, please summarize it at the top of this page. All concerns will be evaluated after the exam. GOOD LUCK!

**Question 1 8 points** (*Assume all transitions are possible*)

a. 6 points Calculate the **energy (in kJ/mol)** required to ionize (completely remove) the electron from the **$n=3$** level of **He^+ ion** (He: Helium). Show your work including equations. (3 sig. figs.)

b. 2 points

(I) Select the largest (longest) wavelength from the three choices below:

- i. Wavelength of photon to *completely remove* an electron from *1s orbital of Hydrogen atom*
- ii. Wavelength of photon to *completely remove* an electron from *1s orbital of Boron atom (B)*
- iii. Wavelength of photon to *completely remove* an electron from *1s orbital of Boron cation (B^+)*

i or ii or iii

(II) Select the smallest (shortest) wavelength from the three choices below (*for H atom*):

- i. Wavelength of photon *emitted* for an electron transitioning from *2p to 1s orbital*
- ii. Wavelength of photon *absorbed* for an electron transitioning from *2s to 4p orbital*
- iii. Wavelength of photon *absorbed* for an electron transitioning from *1s to 3p orbital*

i or ii or iii



Question 2 8 points



A Chem110 student plots the graph shown on the left, from data obtained for the kinetic energy (eV) of the ejected electron versus frequency (s^{-1}) of the incident light, for an unknown metal. ($1\text{eV} = 1.6022 \times 10^{-19} \text{ J}$)

Based on the plot, answer the questions **a to c** below:

a. 1 point On the plot above, label the **threshold frequency**.

b. 2 points What is the value of the work potential (in J) of the unknown metal?

c. 3 points What is the *longest* wavelength (in nm) required to eject an electron from this metal's surface? Show your work including equation(s) used. (3 sig figs)

d. 2 points In the space provided below, plot the following

(Assume frequency is *above* threshold frequency):

- Kinetic energy* of an ejected electron versus the *intensity* of the incident light (left)
- Number of electrons* ejected versus the *frequency* of the incident light (right)



**Question 3** **6 points****a. 2 points**

In the ground state of Ge (Germanium; $Z = 32$) atom, **name** the following orbitals:

i. Orbital containing the electron that requires the maximum energy to be removed.

ii. Orbital containing the electron that experiences the minimum effective nuclear charge.

b. 2 points

An unknown anion (A^-) has the following *excited* electronic configuration: $[Ar] 4s^2 3d^3 4p^1$

(Ar: Argon; $Z = 18$)

Give the **condensed ground state electronic configuration** of the atom, **A** and the cation, **A^{2+}** .
Show your work. (Assume all electronic transitions are possible)

Electronic Configuration of A:

Electronic Configuration of A^{2+} :

c. 2 points

In a single atom, determine how many electrons can have the following set of quantum numbers?

(i) $n = 5; l = 3$

In a single atom, determine how many electrons can have the following set of quantum numbers?

(i) $n = 3; l = 2; m_s = +1/2$

**Question 4** **8 points**

Rank the following from 1 to 4 in **increasing** order. Provide the ranking in the boxes provided.

Note: Atomic number (Z) are provided for relevant atoms

a) Increasing (lowest to highest) order of *ionization energy*: (1= lowest; 4 = highest)

Mg

Mg: Z = 12

Rb

Rb: Z = 37

Cs

Cs: Z = 55

K

K : Z = 19

b) Increasing (lowest to highest) order of *bond angle*: (1= lowest; 4 = highest)

H₂OCCl₄XeF₄

HCN

c) Increasing (smallest to largest) order of *ionic size* (1= smallest; 4 = largest)

S²⁻

S: Z = 16

O²⁻

O: Z = 8

Be²⁺

Be: Z = 4

Na⁺

Na: Z = 11

d) Increasing (lowest to highest) order of *bond energy* (1= lowest; 4 = highest)

P-F

P-Cl

P-I

P-Br

**Question 5** **12 points**

a. 9 points For the molecules HClO_4 and TeF_6 , draw the most stable *Lewis structure* showing appropriate shape and *dashed/wedged bonds*, based on VSEPR. Determine the **molecular geometry** around the central atom. For HClO_4 , label the *most polar bond*.

HClO_4 (there are no O-O bonds)

Label the most polar bond

Molecular Geometry around Cl

TeF_6

Molecular Geometry around Te

b. 3 points (Note: Lone pairs are not shown in structures below)

(i) For the three *resonance* structures (1, 2 and 3) shown below, answer the following question:



(I) Circle the *most contributing* resonance structure.

Structure 1

Structure 2

Structure 3

(ii) For the three structures given (4, 5, and 6) below, answer the following questions:



(I) Circle all the structures that represent *resonance structures* of each other:

Structure 4

Structure 5

Structure 6

(II) Among the resonance structures, circle the *most contributing* resonance structure.

Structure 4

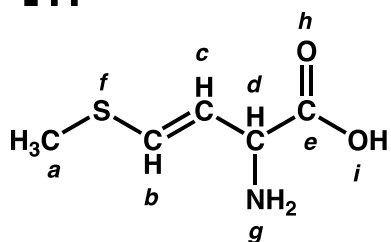
Structure 5

Structure 6

**Question 6** **6 points**

Using the data given below, calculate the **lattice energy** for SrO(s).

	ΔH° (kJ/mol)
$\text{Sr}(s) \longrightarrow \text{Sr}(g)$	+180
$\text{Sr}(g) \longrightarrow \text{Sr}^{2+}(g) + 2e^-$	+1468
$\text{O}_2(g) \longrightarrow 2\text{O}(g)$	+500
$\text{O}(g) + 2e^- \longrightarrow \text{O}^{2-}(g)$	+600
$\text{Sr}(s) + \frac{1}{2}\text{O}_2(g) \longrightarrow \text{SrO}(s)$	-550

**Question 7 10 points**

Consider the molecule given on the left, and answer questions (i) to (vi) using chemical bonding theories that we have learnt in class (valence bond theory, hybridization).

Note that the lone pairs are not shown in the structure:

i. Consider the carbon atoms marked in as ***a***, ***b***, ***c***, ***d***, and ***e*** in the molecule, and determine **hybridization** at each of the carbon centers. (2.5 points)

Atom	Hybridization
Carbon (<i>a</i>)	
Carbon (<i>b</i>)	
Carbon (<i>c</i>)	
Carbon (<i>d</i>)	
Carbon (<i>e</i>)	

ii. Now, consider the sulfur atom (S) marked as ***f***, nitrogen (N) atom (***g***), and oxygen(O) atoms (***h*** and ***i***), and determine their **hybridization**. (2 points)

Atom	Hybridization
Sulfur (<i>f</i>)	
Nitrogen (<i>g</i>)	
Oxygen (<i>h</i>)	
Oxygen (<i>i</i>)	

iii. **Describe** all the bonds formed by the carbon atoms ***a*** and ***b***, using sigma and pi designation as appropriate. (2 points)

Atom	Bonds formed (use sigma and pi as appropriate)
Carbon (<i>a</i>)	
Carbon (<i>b</i>)	

iv. How many total sp^3 -s sigma bonds are in the molecule? (1 point)

v. Determine **electronic** and **molecular geometry** at carbon (C) labeled ***e*** and sulfur (S) labeled ***f*** centres. (2 points)

Atom	Electronic Geometry	Molecular Geometry
Carbon (<i>e</i>)		
Sulfur (<i>f</i>)		

vi. Circle the value from the choices given below, that describes the **C-S-C bond angle at sulfur** (S) marked ***f***: (0.5 points)

>90° to <109.5° 109.5° >109.5 to <120° 120° >120° to <180°

**Question 8 12 points**

Using the **valence shell orbitals only** draw a **complete** molecular orbital (MO) diagram (with no s-p mixing) for the molecule **PS**, in the space provided below. **(6 points)**

Assume that 3s and 3p atomic orbitals will form molecular orbitals like those formed by the combination of 2s and 2p atomic orbitals that we learnt in class, and there is no s-p mixing.

Show **energy arrow**, **atomic orbitals** involved, **atom labels**, the **molecular orbitals** that are formed, **filled electrons** in the MO diagram etc. Make sure to label all the atomic and molecular orbitals.

Using the MO diagram above, answer the following questions:

- (i) Which one of the following will have the highest bond order? Circle the correct choice
A) PS^{1-} B) PS C) PS^{1+}
- (ii) Which one of the following will have the longest bond length? Circle the correct choice
A) PS^{1-} B) PS C) PS^{1+}
- (iii) Which of the following will be diamagnetic? Circle the correct choice.
A) PS^{1-} B) PS C) PS^{1+}
- (iv) Draw Lewis dot structure of the molecule with the shortest bond length (based on the MO diagram) from among the three (PS^{1-} , PS, PS^{1+}), and indicate the bond order calculated from the Lewis dot structure. **(3 points)**

Lewis Dot Structure

Bond Order



BABF075E-B693-458F-A5A3-E848EABC94C0

midterm-exam-93388

#522 Page 10 of 10

SCRAP PAGE

**CHEMISTRY 110 –Midterm Exam****Thursday November 7, 2024, 6:30 – 8:30 pm****Examiners: Pallavi Sirjoosingh and Ashok Kakkar**

First name (please write as legibly as possible within the boxes)

--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--

Last name

--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--

Student ID number

--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--

This booklet has 8 questions on 10 pages (The last page can be used as a scrap page).

The staple must NOT be removed from the question booklet. Your intact question booklet must be returned at the end of the exam.

A separate booklet contains a periodic table, and datasheet.

Write your answers directly under the questions in the space provided. Do not write on the QR codes or near the top edges (as it may be cut off).

Please write **legibly and dark enough** for the scan to be readable. For students who use English as a second language, dictionaries are permitted. Calculators are allowed.

DO NOT ASK QUESTIONS OF THE INVIGILATORS about interpretations of the exam questions. If you have a concern about a question, please summarize it at the top of this page. All concerns will be evaluated after the exam. GOOD LUCK!

**Question 1 8 points** (*Assume all transitions are possible*)

a. 6 points Calculate the **energy (in kJ/mol)** required to ionize (completely remove) the electron from the **n=3** level of **He⁺ ion** (He: Helium). Show your work including equations. (3 sig. figs.)

b. 2 points

(I) Select the largest (longest) wavelength from the three choices below:

- i. Wavelength of photon to *completely remove* an electron from *1s orbital of Hydrogen atom*
- ii. Wavelength of photon to *completely remove* an electron from *1s orbital of Boron atom (B)*
- iii. Wavelength of photon to *completely remove* an electron from *1s orbital of Boron cation (B⁺)*

i or ii or iii

(II) Select the smallest (shortest) wavelength from the three choices below (*for H atom*):

- i. Wavelength of photon *emitted* for an electron transitioning from *2p to 1s orbital*
- ii. Wavelength of photon *absorbed* for an electron transitioning from *2s to 4p orbital*
- iii. Wavelength of photon *absorbed* for an electron transitioning from *1s to 3p orbital*

i or ii or iii



Question 2 8 points



A Chem110 student plots the graph shown on the left, from data obtained for the kinetic energy (eV) of the ejected electron versus frequency (s^{-1}) of the incident light, for an unknown metal. ($1\text{eV} = 1.6022 \times 10^{-19} \text{ J}$)

Based on the plot, answer the questions **a to c** below:

a. 1 point On the plot above, label the **threshold frequency**.

b. 2 points What is the value of the work potential (in J) of the unknown metal?

c. 3 points What is the *longest* wavelength (in nm) required to eject an electron from this metal's surface? Show your work including equation(s) used. (3 sig figs)

d. 2 points In the space provided below, plot the following

(Assume frequency is *above* threshold frequency):

- i. *Kinetic energy* of an ejected electron versus the *intensity* of the incident light (left)
- ii. *Number of electrons* ejected versus the *frequency* of the incident light (right)



**Question 3** **6 points****a. 2 points**

In the ground state of Ge (Germanium; $Z = 32$) atom, **name** the following orbitals:

i. Orbital containing the electron that requires the maximum energy to be removed.

ii. Orbital containing the electron that experiences the minimum effective nuclear charge.

b. 2 points

An unknown anion (A^-) has the following *excited* electronic configuration: $[Ar] 4s^2 3d^3 4p^1$

(Ar: Argon; $Z = 18$)

Give the **condensed ground state electronic configuration** of the atom, **A** and the cation, **A^{2+}** .
Show your work. (Assume all electronic transitions are possible)

Electronic Configuration of A:

Electronic Configuration of A^{2+} :

c. 2 points

In a single atom, determine how many electrons can have the following set of quantum numbers?

(i) $n = 5; l = 3$

In a single atom, determine how many electrons can have the following set of quantum numbers?

(i) $n = 3; l = 2; m_s = +1/2$

**Question 4** **8 points**

Rank the following from 1 to 4 in **increasing** order. Provide the ranking in the boxes provided.

Note: Atomic number (Z) are provided for relevant atoms

a) Increasing (lowest to highest) order of *ionization energy*: (1= lowest; 4 = highest)

Mg

Mg: Z = 12

Rb

Rb: Z = 37

Cs

Cs: Z = 55

K

K : Z = 19

b) Increasing (lowest to highest) order of *bond angle*: (1= lowest; 4 = highest)

H₂OCCl₄XeF₄

HCN

c) Increasing (smallest to largest) order of *ionic size* (1= smallest; 4 = largest)

S²⁻

S: Z = 16

O²⁻

O: Z = 8

Be²⁺

Be: Z = 4

Na⁺

Na: Z = 11

d) Increasing (lowest to highest) order of *bond energy* (1= lowest; 4 = highest)

P-F

P-Cl

P-I

P-Br

**Question 5** **12 points**

a. 9 points For the molecules HClO_4 and TeF_6 , draw the most stable *Lewis structure* showing appropriate shape and *dashed/wedged bonds*, based on VSEPR. Determine the **molecular geometry** around the central atom. For HClO_4 , label the *most polar bond*.



b. 3 points (Note: Lone pairs are not shown in structures below)

(i) For the three *resonance* structures (1, 2 and 3) shown below, answer the following question:



(I) Circle the *most contributing* resonance structure.

Structure 1

Structure 2

Structure 3

(ii) For the three structures given (4, 5, and 6) below, answer the following questions:



(I) Circle all the structures that represent *resonance structures* of each other:

Structure 4

Structure 5

Structure 6

(II) Among the resonance structures, circle the *most contributing* resonance structure.

Structure 4

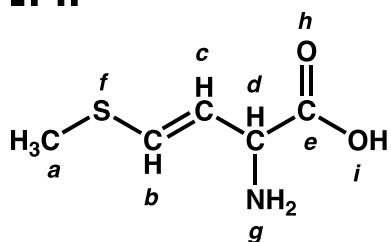
Structure 5

Structure 6

**Question 6** **6 points**

Using the data given below, calculate the **lattice energy** for SrO(s).

	ΔH° (kJ/mol)
$\text{Sr}(s) \longrightarrow \text{Sr}(g)$	+180
$\text{Sr}(g) \longrightarrow \text{Sr}^{2+}(g) + 2e^-$	+1468
$\text{O}_2(g) \longrightarrow 2\text{O}(g)$	+500
$\text{O}(g) + 2e^- \longrightarrow \text{O}^{2-}(g)$	+600
$\text{Sr}(s) + \frac{1}{2}\text{O}_2(g) \longrightarrow \text{SrO}(s)$	-550

**Question 7 10 points**

Consider the molecule given on the left, and answer questions (i) to (vi) using chemical bonding theories that we have learnt in class (valence bond theory, hybridization).

Note that the lone pairs are not shown in the structure:

i. Consider the carbon atoms marked in as ***a***, ***b***, ***c***, ***d***, and ***e*** in the molecule, and determine **hybridization** at each of the carbon centers. (2.5 points)

Atom	Hybridization
Carbon (<i>a</i>)	
Carbon (<i>b</i>)	
Carbon (<i>c</i>)	
Carbon (<i>d</i>)	
Carbon (<i>e</i>)	

ii. Now, consider the sulfur atom (S) marked as ***f***, nitrogen (N) atom (***g***), and oxygen(O) atoms (***h*** and ***i***), and determine their **hybridization**. (2 points)

Atom	Hybridization
Sulfur (<i>f</i>)	
Nitrogen (<i>g</i>)	
Oxygen (<i>h</i>)	
Oxygen (<i>i</i>)	

iii. **Describe** all the bonds formed by the carbon atoms ***a*** and ***b***, using sigma and pi designation as appropriate. (2 points)

Atom	Bonds formed (use sigma and pi as appropriate)
Carbon (<i>a</i>)	
Carbon (<i>b</i>)	

iv. How many total sp^3 -s sigma bonds are in the molecule? (1 point)

v. Determine **electronic** and **molecular geometry** at carbon (C) labeled ***e*** and sulfur (S) labeled ***f*** centres. (2 points)

Atom	Electronic Geometry	Molecular Geometry
Carbon (<i>e</i>)		
Sulfur (<i>f</i>)		

vi. Circle the value from the choices given below, that describes the **C-S-C bond angle at sulfur** (S) marked ***f***: (0.5 points)

>90° to <109.5° 109.5° >109.5 to <120° 120° >120° to <180°

**Question 8 12 points**

Using the **valence shell orbitals only** draw a **complete** molecular orbital (MO) diagram (with no s-p mixing) for the molecule **PS**, in the space provided below. **(6 points)**

Assume that 3s and 3p atomic orbitals will form molecular orbitals like those formed by the combination of 2s and 2p atomic orbitals that we learnt in class, and there is no s-p mixing.

Show **energy arrow**, **atomic orbitals** involved, **atom labels**, the **molecular orbitals** that are formed, **filled electrons** in the MO diagram etc. Make sure to label all the atomic and molecular orbitals.

Using the MO diagram above, answer the following questions:

- (i) Which one of the following will have the highest bond order? Circle the correct choice
A) PS^{1-} B) PS C) PS^{1+}
- (ii) Which one of the following will have the longest bond length? Circle the correct choice
A) PS^{1-} B) PS C) PS^{1+}
- (iii) Which of the following will be diamagnetic? Circle the correct choice.
A) PS^{1-} B) PS C) PS^{1+}
- (iv) Draw Lewis dot structure of the molecule with the shortest bond length (based on the MO diagram) from among the three (PS^{1-} , PS, PS^{1+}), and indicate the bond order calculated from the Lewis dot structure. **(3 points)**

Lewis Dot Structure

Bond Order



4CD48DB1-815E-4EE9-A227-FE418728C2C0

midterm-exam-93388

#523 Page 10 of 10

SCRAP PAGE

**CHEMISTRY 110 –Midterm Exam****Thursday November 7, 2024, 6:30 – 8:30 pm****Examiners: Pallavi Sirjoosingh and Ashok Kakkar**

First name (please write as legibly as possible within the boxes)

--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--

Last name

--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--

Student ID number

--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--

This booklet has 8 questions on 10 pages (The last page can be used as a scrap page).

The staple must NOT be removed from the question booklet. Your intact question booklet must be returned at the end of the exam.

A separate booklet contains a periodic table, and datasheet.

Write your answers directly under the questions in the space provided. Do not write on the QR codes or near the top edges (as it may be cut off).

Please write **legibly and dark enough** for the scan to be readable. For students who use English as a second language, dictionaries are permitted. Calculators are allowed.

DO NOT ASK QUESTIONS OF THE INVIGILATORS about interpretations of the exam questions. If you have a concern about a question, please summarize it at the top of this page. All concerns will be evaluated after the exam. GOOD LUCK!

**Question 1 8 points** (*Assume all transitions are possible*)

a. 6 points Calculate the **energy (in kJ/mol)** required to ionize (completely remove) the electron from the **n=3** level of **He⁺ ion** (He: Helium). Show your work including equations. (3 sig. figs.)

b. 2 points

(I) Select the largest (longest) wavelength from the three choices below:

- i. Wavelength of photon to *completely remove* an electron from *1s orbital of Hydrogen atom*
- ii. Wavelength of photon to *completely remove* an electron from *1s orbital of Boron atom (B)*
- iii. Wavelength of photon to *completely remove* an electron from *1s orbital of Boron cation (B⁺)*

i or ii or iii

(II) Select the smallest (shortest) wavelength from the three choices below (*for H atom*):

- i. Wavelength of photon *emitted* for an electron transitioning from *2p to 1s orbital*
- ii. Wavelength of photon *absorbed* for an electron transitioning from *2s to 4p orbital*
- iii. Wavelength of photon *absorbed* for an electron transitioning from *1s to 3p orbital*

i or ii or iii



Question 2 8 points



A Chem110 student plots the graph shown on the left, from data obtained for the kinetic energy (eV) of the ejected electron versus frequency (s^{-1}) of the incident light, for an unknown metal. ($1\text{eV} = 1.6022 \times 10^{-19} \text{ J}$)

Based on the plot, answer the questions **a to c** below:

a. 1 point On the plot above, label the **threshold frequency**.

b. 2 points What is the value of the work potential (in J) of the unknown metal?

c. 3 points What is the *longest* wavelength (in nm) required to eject an electron from this metal's surface? Show your work including equation(s) used. (3 sig figs)

d. 2 points In the space provided below, plot the following

(Assume frequency is *above* threshold frequency):

- i. *Kinetic energy* of an ejected electron versus the *intensity* of the incident light (left)
- ii. *Number of electrons* ejected versus the *frequency* of the incident light (right)



**Question 3** **6 points****a. 2 points**

In the ground state of Ge (Germanium; $Z = 32$) atom, **name** the following orbitals:

i. Orbital containing the electron that requires the maximum energy to be removed.

ii. Orbital containing the electron that experiences the minimum effective nuclear charge.

b. 2 points

An unknown anion (A^-) has the following *excited* electronic configuration: $[Ar] 4s^2 3d^3 4p^1$

(Ar: Argon; $Z = 18$)

Give the **condensed ground state electronic configuration** of the atom, **A** and the cation, **A^{2+}** .
Show your work. (Assume all electronic transitions are possible)

Electronic Configuration of A:

Electronic Configuration of A^{2+} :

c. 2 points

In a single atom, determine how many electrons can have the following set of quantum numbers?

(i) $n = 5; l = 3$

In a single atom, determine how many electrons can have the following set of quantum numbers?

(i) $n = 3; l = 2; m_s = +1/2$

**Question 4** **8 points**

Rank the following from 1 to 4 in **increasing** order. Provide the ranking in the boxes provided.

Note: Atomic number (Z) are provided for relevant atoms

a) Increasing (lowest to highest) order of *ionization energy*: (1= lowest; 4 = highest)

Mg

Mg: Z = 12

Rb

Rb: Z = 37

Cs

Cs: Z = 55

K

K : Z = 19

b) Increasing (lowest to highest) order of *bond angle*: (1= lowest; 4 = highest)

H₂OCCl₄XeF₄

HCN

c) Increasing (smallest to largest) order of *ionic size* (1= smallest; 4 = largest)

S²⁻

S: Z = 16

O²⁻

O: Z = 8

Be²⁺

Be: Z = 4

Na⁺

Na: Z = 11

d) Increasing (lowest to highest) order of *bond energy* (1= lowest; 4 = highest)

P-F

P-Cl

P-I

P-Br

**Question 5** **12 points**

a. 9 points For the molecules HClO_4 and TeF_6 , draw the most stable *Lewis structure* showing appropriate shape and *dashed/wedged bonds*, based on VSEPR. Determine the **molecular geometry** around the central atom. For HClO_4 , label the *most polar bond*.



b. 3 points (Note: Lone pairs are not shown in structures below)

(i) For the three *resonance* structures (1, 2 and 3) shown below, answer the following question:



(I) Circle the *most contributing* resonance structure.

Structure 1

Structure 2

Structure 3

(ii) For the three structures given (4, 5, and 6) below, answer the following questions:



(I) Circle all the structures that represent *resonance structures* of each other:

Structure 4

Structure 5

Structure 6

(II) Among the resonance structures, circle the *most contributing* resonance structure.

Structure 4

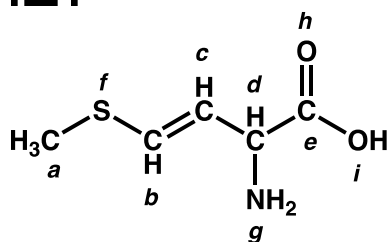
Structure 5

Structure 6

**Question 6** **6 points**

Using the data given below, calculate the **lattice energy** for SrO(s).

	ΔH° (kJ/mol)
$\text{Sr}(s) \longrightarrow \text{Sr}(g)$	+180
$\text{Sr}(g) \longrightarrow \text{Sr}^{2+}(g) + 2e^-$	+1468
$\text{O}_2(g) \longrightarrow 2\text{O}(g)$	+500
$\text{O}(g) + 2e^- \longrightarrow \text{O}^{2-}(g)$	+600
$\text{Sr}(s) + \frac{1}{2}\text{O}_2(g) \longrightarrow \text{SrO}(s)$	-550

**Question 7 10 points**

Consider the molecule given on the left, and answer questions (i) to (vi) using chemical bonding theories that we have learnt in class (valence bond theory, hybridization).

Note that the lone pairs are not shown in the structure:

i. Consider the carbon atoms marked in as ***a***, ***b***, ***c***, ***d***, and ***e*** in the molecule, and determine **hybridization** at each of the carbon centers. (2.5 points)

Atom	Hybridization
Carbon (<i>a</i>)	
Carbon (<i>b</i>)	
Carbon (<i>c</i>)	
Carbon (<i>d</i>)	
Carbon (<i>e</i>)	

ii. Now, consider the sulfur atom (S) marked as ***f***, nitrogen (N) atom (***g***), and oxygen(O) atoms (***h*** and ***i***), and determine their **hybridization**. (2 points)

Atom	Hybridization
Sulfur (<i>f</i>)	
Nitrogen (<i>g</i>)	
Oxygen (<i>h</i>)	
Oxygen (<i>i</i>)	

iii. **Describe** all the bonds formed by the carbon atoms ***a*** and ***b***, using sigma and pi designation as appropriate. (2 points)

Atom	Bonds formed (use sigma and pi as appropriate)
Carbon (<i>a</i>)	
Carbon (<i>b</i>)	

iv. How many total sp^3 -s sigma bonds are in the molecule? (1 point)

v. Determine **electronic** and **molecular geometry** at carbon (C) labeled ***e*** and sulfur (S) labeled ***f*** centres. (2 points)

Atom	Electronic Geometry	Molecular Geometry
Carbon (<i>e</i>)		
Sulfur (<i>f</i>)		

vi. Circle the value from the choices given below, that describes the **C-S-C bond angle at sulfur** (S) marked ***f***: (0.5 points)

>90° to <109.5° 109.5° >109.5 to <120° 120° >120° to <180°


Question 8 12 points

Using the **valence shell orbitals only** draw a **complete** molecular orbital (MO) diagram (with no s-p mixing) for the molecule **PS**, in the space provided below. **(6 points)**

Assume that 3s and 3p atomic orbitals will form molecular orbitals like those formed by the combination of 2s and 2p atomic orbitals that we learnt in class, and there is no s-p mixing.

Show **energy arrow**, **atomic orbitals** involved, **atom labels**, the **molecular orbitals** that are formed, **filled electrons** in the MO diagram etc. Make sure to label all the atomic and molecular orbitals.

Using the MO diagram above, answer the following questions:

- (i) Which one of the following will have the highest bond order? Circle the correct choice
 A) PS^{1-} B) PS C) PS^{1+}
- (ii) Which one of the following will have the longest bond length? Circle the correct choice
 A) PS^{1-} B) PS C) PS^{1+}
- (iii) Which of the following will be diamagnetic? Circle the correct choice.
 A) PS^{1-} B) PS C) PS^{1+}
- (iv) Draw Lewis dot structure of the molecule with the shortest bond length (based on the MO diagram) from among the three (PS^{1-} , PS, PS^{1+}), and indicate the bond order calculated from the Lewis dot structure. **(3 points)**

Lewis Dot Structure

Bond Order



C94D3FB5-9D79-4B9A-A72C-FBFAAC8AF1D0

midterm-exam-93388

#524 Page 10 of 10

SCRAP PAGE

**CHEMISTRY 110 –Midterm Exam****Thursday November 7, 2024, 6:30 – 8:30 pm****Examiners: Pallavi Sirjoosingh and Ashok Kakkar**

First name (please write as legibly as possible within the boxes)

--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--

Last name

--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--

Student ID number

--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--

This booklet has 8 questions on 10 pages (The last page can be used as a scrap page).

The staple must NOT be removed from the question booklet. Your intact question booklet must be returned at the end of the exam.

A separate booklet contains a periodic table, and datasheet.

Write your answers directly under the questions in the space provided. Do not write on the QR codes or near the top edges (as it may be cut off).

Please write **legibly and dark enough** for the scan to be readable. For students who use English as a second language, dictionaries are permitted. Calculators are allowed.

DO NOT ASK QUESTIONS OF THE INVIGILATORS about interpretations of the exam questions. If you have a concern about a question, please summarize it at the top of this page. All concerns will be evaluated after the exam. GOOD LUCK!

**Question 1 8 points** (*Assume all transitions are possible*)

a. 6 points Calculate the **energy (in kJ/mol)** required to ionize (completely remove) the electron from the **$n=3$** level of **He^+ ion** (He: Helium). Show your work including equations. (3 sig. figs.)

b. 2 points

(I) Select the largest (longest) wavelength from the three choices below:

- i. Wavelength of photon to *completely remove* an electron from *1s orbital of Hydrogen atom*
- ii. Wavelength of photon to *completely remove* an electron from *1s orbital of Boron atom (B)*
- iii. Wavelength of photon to *completely remove* an electron from *1s orbital of Boron cation (B^+)*

i or ii or iii

(II) Select the smallest (shortest) wavelength from the three choices below (*for H atom*):

- i. Wavelength of photon *emitted* for an electron transitioning from *2p to 1s orbital*
- ii. Wavelength of photon *absorbed* for an electron transitioning from *2s to 4p orbital*
- iii. Wavelength of photon *absorbed* for an electron transitioning from *1s to 3p orbital*

i or ii or iii



Question 2 8 points



A Chem110 student plots the graph shown on the left, from data obtained for the kinetic energy (eV) of the ejected electron versus frequency (s^{-1}) of the incident light, for an unknown metal. ($1\text{eV} = 1.6022 \times 10^{-19} \text{ J}$)

Based on the plot, answer the questions **a to c** below:

a. 1 point On the plot above, label the **threshold frequency**.

b. 2 points What is the value of the work potential (in J) of the unknown metal?

c. 3 points What is the *longest* wavelength (in nm) required to eject an electron from this metal's surface? Show your work including equation(s) used. (3 sig figs)

d. 2 points In the space provided below, plot the following

(Assume frequency is *above* threshold frequency):

- i. *Kinetic energy* of an ejected electron versus the *intensity* of the incident light (left)
- ii. *Number of electrons* ejected versus the *frequency* of the incident light (right)



**Question 3** **6 points****a. 2 points**

In the ground state of Ge (Germanium; $Z = 32$) atom, **name** the following orbitals:

i. Orbital containing the electron that requires the maximum energy to be removed.

ii. Orbital containing the electron that experiences the minimum effective nuclear charge.

b. 2 points

An unknown anion (A^-) has the following *excited* electronic configuration: $[Ar] 4s^2 3d^3 4p^1$

(Ar: Argon; $Z = 18$)

Give the **condensed ground state electronic configuration** of the atom, **A** and the cation, **A^{2+}** .
Show your work. (Assume all electronic transitions are possible)

Electronic Configuration of A:

Electronic Configuration of A^{2+} :

c. 2 points

In a single atom, determine how many electrons can have the following set of quantum numbers?

(i) $n = 5; l = 3$

In a single atom, determine how many electrons can have the following set of quantum numbers?

(i) $n = 3; l = 2; m_s = +1/2$

**Question 4** **8 points**

Rank the following from 1 to 4 in **increasing** order. Provide the ranking in the boxes provided.

Note: Atomic number (Z) are provided for relevant atoms

a) Increasing (lowest to highest) order of *ionization energy*: (1= lowest; 4 = highest)

Mg

Mg: Z = 12

Rb

Rb: Z = 37

Cs

Cs: Z = 55

K

K : Z = 19

b) Increasing (lowest to highest) order of *bond angle*: (1= lowest; 4 = highest)

H₂OCCl₄XeF₄

HCN

c) Increasing (smallest to largest) order of *ionic size* (1= smallest; 4 = largest)

S²⁻

S: Z = 16

O²⁻

O: Z = 8

Be²⁺

Be: Z = 4

Na⁺

Na: Z = 11

d) Increasing (lowest to highest) order of *bond energy* (1= lowest; 4 = highest)

P-F

P-Cl

P-I

P-Br

**Question 5** **12 points**

a. 9 points For the molecules HClO_4 and TeF_6 , draw the most stable *Lewis structure* showing appropriate shape and *dashed/wedged bonds*, based on VSEPR. Determine the **molecular geometry** around the central atom. For HClO_4 , label the *most polar bond*.



b. 3 points (Note: Lone pairs are not shown in structures below)

(i) For the three *resonance* structures (1, 2 and 3) shown below, answer the following question:



(I) Circle the *most contributing* resonance structure.

Structure 1

Structure 2

Structure 3

(ii) For the three structures given (4, 5, and 6) below, answer the following questions:



(I) Circle all the structures that represent *resonance structures* of each other:

Structure 4

Structure 5

Structure 6

(II) Among the resonance structures, circle the *most contributing* resonance structure.

Structure 4

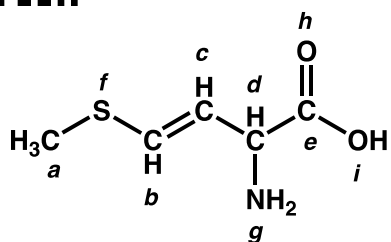
Structure 5

Structure 6

**Question 6** **6 points**

Using the data given below, calculate the **lattice energy** for SrO(s).

	ΔH° (kJ/mol)
$\text{Sr}(s) \longrightarrow \text{Sr}(g)$	+180
$\text{Sr}(g) \longrightarrow \text{Sr}^{2+}(g) + 2e^-$	+1468
$\text{O}_2(g) \longrightarrow 2\text{O}(g)$	+500
$\text{O}(g) + 2e^- \longrightarrow \text{O}^{2-}(g)$	+600
$\text{Sr}(s) + \frac{1}{2}\text{O}_2(g) \longrightarrow \text{SrO}(s)$	-550

**Question 7 10 points**

Consider the molecule given on the left, and answer questions (i) to (vi) using chemical bonding theories that we have learnt in class (valence bond theory, hybridization).

Note that the lone pairs are not shown in the structure:

i. Consider the carbon atoms marked in as ***a***, ***b***, ***c***, ***d***, and ***e*** in the molecule, and determine **hybridization** at each of the carbon centers. (2.5 points)

Atom	Hybridization
Carbon (<i>a</i>)	
Carbon (<i>b</i>)	
Carbon (<i>c</i>)	
Carbon (<i>d</i>)	
Carbon (<i>e</i>)	

ii. Now, consider the sulfur atom (S) marked as ***f***, nitrogen (N) atom (***g***), and oxygen(O) atoms (***h*** and ***i***), and determine their **hybridization**. (2 points)

Atom	Hybridization
Sulfur (<i>f</i>)	
Nitrogen (<i>g</i>)	
Oxygen (<i>h</i>)	
Oxygen (<i>i</i>)	

iii. **Describe** all the bonds formed by the carbon atoms ***a*** and ***b***, using sigma and pi designation as appropriate. (2 points)

Atom	Bonds formed (use sigma and pi as appropriate)
Carbon (<i>a</i>)	
Carbon (<i>b</i>)	

iv. How many total sp^3 -s sigma bonds are in the molecule? (1 point)

v. Determine **electronic** and **molecular geometry** at carbon (C) labeled ***e*** and sulfur (S) labeled ***f*** centres. (2 points)

Atom	Electronic Geometry	Molecular Geometry
Carbon (<i>e</i>)		
Sulfur (<i>f</i>)		

vi. Circle the value from the choices given below, that describes the **C-S-C bond angle at sulfur** (S) marked ***f***: (0.5 points)

>90° to <109.5° 109.5° >109.5 to <120° 120° >120° to <180°


Question 8 12 points

Using the **valence shell orbitals only** draw a **complete** molecular orbital (MO) diagram (with no s-p mixing) for the molecule **PS**, in the space provided below. **(6 points)**

Assume that 3s and 3p atomic orbitals will form molecular orbitals like those formed by the combination of 2s and 2p atomic orbitals that we learnt in class, and there is no s-p mixing.

Show **energy arrow**, **atomic orbitals** involved, **atom labels**, the **molecular orbitals** that are formed, **filled electrons** in the MO diagram etc. Make sure to label all the atomic and molecular orbitals.

Using the MO diagram above, answer the following questions:

- (i) Which one of the following will have the highest bond order? Circle the correct choice
 A) PS^{1-} B) PS C) PS^{1+}
- (ii) Which one of the following will have the longest bond length? Circle the correct choice
 A) PS^{1-} B) PS C) PS^{1+}
- (iii) Which of the following will be diamagnetic? Circle the correct choice.
 A) PS^{1-} B) PS C) PS^{1+}
- (iv) Draw Lewis dot structure of the molecule with the shortest bond length (based on the MO diagram) from among the three (PS^{1-} , PS, PS^{1+}), and indicate the bond order calculated from the Lewis dot structure. **(3 points)**

Lewis Dot Structure

Bond Order



F5F41A11-566C-4F43-BD28-2B396377CC4D

midterm-exam-93388

#525 Page 10 of 10

SCRAP PAGE

**CHEMISTRY 110 –Midterm Exam****Thursday November 7, 2024, 6:30 – 8:30 pm**

Examiners: Pallavi Sirjoosingh and Ashok Kakkar

First name (please write as legibly as possible within the boxes)

--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--

Last name

--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--

Student ID number

--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--

This booklet has 8 questions on 10 pages (The last page can be used as a scrap page).

The staple must NOT be removed from the question booklet. Your intact question booklet must be returned at the end of the exam.

A separate booklet contains a periodic table, and datasheet.

Write your answers directly under the questions in the space provided. Do not write on the QR codes or near the top edges (as it may be cut off).

Please write **legibly and dark enough** for the scan to be readable. For students who use English as a second language, dictionaries are permitted. Calculators are allowed.

DO NOT ASK QUESTIONS OF THE INVIGILATORS about interpretations of the exam questions. If you have a concern about a question, please summarize it at the top of this page. All concerns will be evaluated after the exam. GOOD LUCK!

**Question 1 8 points** (*Assume all transitions are possible*)

a. 6 points Calculate the **energy (in kJ/mol)** required to ionize (completely remove) the electron from the **n=3** level of **He⁺ ion** (He: Helium). Show your work including equations. (3 sig. figs.)

b. 2 points

(I) Select the largest (longest) wavelength from the three choices below:

- i. Wavelength of photon to *completely remove* an electron from *1s orbital of Hydrogen atom*
- ii. Wavelength of photon to *completely remove* an electron from *1s orbital of Boron atom (B)*
- iii. Wavelength of photon to *completely remove* an electron from *1s orbital of Boron cation (B⁺)*

i or ii or iii

(II) Select the smallest (shortest) wavelength from the three choices below (*for H atom*):

- i. Wavelength of photon *emitted* for an electron transitioning from *2p to 1s orbital*
- ii. Wavelength of photon *absorbed* for an electron transitioning from *2s to 4p orbital*
- iii. Wavelength of photon *absorbed* for an electron transitioning from *1s to 3p orbital*

i or ii or iii



Question 2 8 points



A Chem110 student plots the graph shown on the left, from data obtained for the kinetic energy (eV) of the ejected electron versus frequency (s^{-1}) of the incident light, for an unknown metal. ($1\text{eV} = 1.6022 \times 10^{-19} \text{ J}$)

Based on the plot, answer the questions **a to c** below:

a. 1 point On the plot above, label the **threshold frequency**.

b. 2 points What is the value of the work potential (in J) of the unknown metal?

c. 3 points What is the *longest* wavelength (in nm) required to eject an electron from this metal's surface? Show your work including equation(s) used. (3 sig figs)

d. 2 points In the space provided below, plot the following

(Assume frequency is *above* threshold frequency):

- Kinetic energy* of an ejected electron versus the *intensity* of the incident light (left)
- Number of electrons* ejected versus the *frequency* of the incident light (right)



**Question 3** **6 points****a. 2 points**

In the ground state of Ge (Germanium; $Z = 32$) atom, **name** the following orbitals:

i. Orbital containing the electron that requires the maximum energy to be removed.

ii. Orbital containing the electron that experiences the minimum effective nuclear charge.

b. 2 points

An unknown anion (A^-) has the following *excited* electronic configuration: $[Ar] 4s^2 3d^3 4p^1$

(Ar: Argon; $Z = 18$)

Give the **condensed ground state electronic configuration** of the atom, **A** and the cation, **A^{2+}** .
Show your work. (Assume all electronic transitions are possible)

Electronic Configuration of A:

Electronic Configuration of A^{2+} :

c. 2 points

In a single atom, determine how many electrons can have the following set of quantum numbers?

(i) $n = 5; l = 3$

In a single atom, determine how many electrons can have the following set of quantum numbers?

(i) $n = 3; l = 2; m_s = +1/2$

**Question 4** **8 points**

Rank the following from 1 to 4 in **increasing** order. Provide the ranking in the boxes provided.

Note: Atomic number (Z) are provided for relevant atoms

a) Increasing (lowest to highest) order of *ionization energy*: (1= lowest; 4 = highest)

Mg

Mg: Z = 12

Rb

Rb: Z = 37

Cs

Cs: Z = 55

K

K : Z = 19

b) Increasing (lowest to highest) order of *bond angle*: (1= lowest; 4 = highest)

H₂OCCl₄XeF₄

HCN

c) Increasing (smallest to largest) order of *ionic size* (1= smallest; 4 = largest)

S²⁻

S: Z = 16

O²⁻

O: Z = 8

Be²⁺

Be: Z = 4

Na⁺

Na: Z = 11

d) Increasing (lowest to highest) order of *bond energy* (1= lowest; 4 = highest)

P-F

P-Cl

P-I

P-Br

**Question 5** **12 points**

a. 9 points For the molecules HClO_4 and TeF_6 , draw the most stable *Lewis structure* showing appropriate shape and *dashed/wedged bonds*, based on VSEPR. Determine the **molecular geometry** around the central atom. For HClO_4 , label the *most polar bond*.



b. 3 points (Note: Lone pairs are not shown in structures below)

(i) For the three *resonance* structures (1, 2 and 3) shown below, answer the following question:



(I) Circle the *most contributing* resonance structure.

Structure 1

Structure 2

Structure 3

(ii) For the three structures given (4, 5, and 6) below, answer the following questions:



(I) Circle all the structures that represent *resonance structures* of each other:

Structure 4

Structure 5

Structure 6

(II) Among the resonance structures, circle the *most contributing* resonance structure.

Structure 4

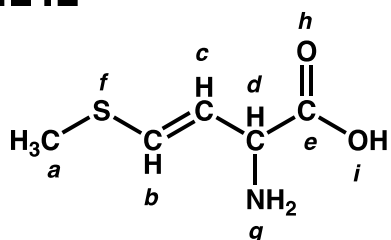
Structure 5

Structure 6

**Question 6** **6 points**

Using the data given below, calculate the **lattice energy** for SrO(s).

	ΔH° (kJ/mol)
$\text{Sr}(s) \longrightarrow \text{Sr}(g)$	+180
$\text{Sr}(g) \longrightarrow \text{Sr}^{2+}(g) + 2e^-$	+1468
$\text{O}_2(g) \longrightarrow 2\text{O}(g)$	+500
$\text{O}(g) + 2e^- \longrightarrow \text{O}^{2-}(g)$	+600
$\text{Sr}(s) + \frac{1}{2}\text{O}_2(g) \longrightarrow \text{SrO}(s)$	-550

**Question 7 10 points**

Consider the molecule given on the left, and answer questions (i) to (vi) using chemical bonding theories that we have learnt in class (valence bond theory, hybridization).

Note that the lone pairs are not shown in the structure:

i. Consider the carbon atoms marked in as ***a***, ***b***, ***c***, ***d***, and ***e*** in the molecule, and determine **hybridization** at each of the carbon centers. (2.5 points)

Atom	Hybridization
Carbon (<i>a</i>)	
Carbon (<i>b</i>)	
Carbon (<i>c</i>)	
Carbon (<i>d</i>)	
Carbon (<i>e</i>)	

ii. Now, consider the sulfur atom (S) marked as ***f***, nitrogen (N) atom (***g***), and oxygen(O) atoms (***h*** and ***i***), and determine their **hybridization**. (2 points)

Atom	Hybridization
Sulfur (<i>f</i>)	
Nitrogen (<i>g</i>)	
Oxygen (<i>h</i>)	
Oxygen (<i>i</i>)	

iii. **Describe** all the bonds formed by the carbon atoms ***a*** and ***b***, using sigma and pi designation as appropriate. (2 points)

Atom	Bonds formed (use sigma and pi as appropriate)
Carbon (<i>a</i>)	
Carbon (<i>b</i>)	

iv. How many total sp^3 -s sigma bonds are in the molecule? (1 point)

v. Determine **electronic** and **molecular geometry** at carbon (C) labeled ***e*** and sulfur (S) labeled ***f*** centres. (2 points)

Atom	Electronic Geometry	Molecular Geometry
Carbon (<i>e</i>)		
Sulfur (<i>f</i>)		

vi. Circle the value from the choices given below, that describes the **C-S-C bond angle at sulfur** (S) marked ***f***: (0.5 points)

>90° to <109.5°

109.5°

>109.5 to <120°

120°

>120° to <180°

**Question 8 12 points**

Using the **valence shell orbitals only** draw a **complete** molecular orbital (MO) diagram (with no s-p mixing) for the molecule **PS**, in the space provided below. **(6 points)**

Assume that 3s and 3p atomic orbitals will form molecular orbitals like those formed by the combination of 2s and 2p atomic orbitals that we learnt in class, and there is no s-p mixing.

Show **energy arrow**, **atomic orbitals** involved, **atom labels**, the **molecular orbitals** that are formed, **filled electrons** in the MO diagram etc. Make sure to label all the atomic and molecular orbitals.

Using the MO diagram above, answer the following questions:

- (i) Which one of the following will have the highest bond order? Circle the correct choice
A) PS^{1-} B) PS C) PS^{1+}
- (ii) Which one of the following will have the longest bond length? Circle the correct choice
A) PS^{1-} B) PS C) PS^{1+}
- (iii) Which of the following will be diamagnetic? Circle the correct choice.
A) PS^{1-} B) PS C) PS^{1+}
- (iv) Draw Lewis dot structure of the molecule with the shortest bond length (based on the MO diagram) from among the three (PS^{1-} , PS, PS^{1+}), and indicate the bond order calculated from the Lewis dot structure. **(3 points)**

Lewis Dot Structure

Bond Order



47A993C1-2AA7-45B1-80B0-9142DE0F17EE

midterm-exam-93388

#526 Page 10 of 10

SCRAP PAGE

**CHEMISTRY 110 –Midterm Exam****Thursday November 7, 2024, 6:30 – 8:30 pm**

Examiners: Pallavi Sirjoosingh and Ashok Kakkar

First name (please write as legibly as possible within the boxes)

--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--

Last name

--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--

Student ID number

--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--

This booklet has 8 questions on 10 pages (The last page can be used as a scrap page).

The staple must NOT be removed from the question booklet. Your intact question booklet must be returned at the end of the exam.

A separate booklet contains a periodic table, and datasheet.

Write your answers directly under the questions in the space provided. Do not write on the QR codes or near the top edges (as it may be cut off).

Please write **legibly and dark enough** for the scan to be readable. For students who use English as a second language, dictionaries are permitted. Calculators are allowed.

DO NOT ASK QUESTIONS OF THE INVIGILATORS about interpretations of the exam questions. If you have a concern about a question, please summarize it at the top of this page. All concerns will be evaluated after the exam. GOOD LUCK!

**Question 1 8 points** (*Assume all transitions are possible*)

a. 6 points Calculate the **energy (in kJ/mol)** required to ionize (completely remove) the electron from the **n=3** level of **He⁺ ion** (He: Helium). Show your work including equations. (3 sig. figs.)

b. 2 points

(I) Select the largest (longest) wavelength from the three choices below:

- i. Wavelength of photon to *completely remove* an electron from *1s orbital of Hydrogen atom*
- ii. Wavelength of photon to *completely remove* an electron from *1s orbital of Boron atom (B)*
- iii. Wavelength of photon to *completely remove* an electron from *1s orbital of Boron cation (B⁺)*

i or ii or iii

(II) Select the smallest (shortest) wavelength from the three choices below (*for H atom*):

- i. Wavelength of photon *emitted* for an electron transitioning from *2p to 1s orbital*
- ii. Wavelength of photon *absorbed* for an electron transitioning from *2s to 4p orbital*
- iii. Wavelength of photon *absorbed* for an electron transitioning from *1s to 3p orbital*

i or ii or iii



Question 2 8 points



A Chem110 student plots the graph shown on the left, from data obtained for the kinetic energy (eV) of the ejected electron versus frequency (s^{-1}) of the incident light, for an unknown metal. ($1\text{eV} = 1.6022 \times 10^{-19} \text{ J}$)

Based on the plot, answer the questions **a to c** below:

a. 1 point On the plot above, label the **threshold frequency**.

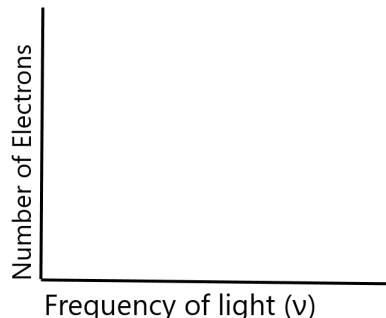
b. 2 points What is the value of the work potential (in J) of the unknown metal?

c. 3 points What is the *longest* wavelength (in nm) required to eject an electron from this metal's surface? Show your work including equation(s) used. (3 sig figs)

d. 2 points In the space provided below, plot the following

(Assume frequency is *above* threshold frequency):

- i. *Kinetic energy* of an ejected electron versus the *intensity* of the incident light (left)
- ii. *Number of electrons* ejected versus the *frequency* of the incident light (right)



**Question 3** **6 points****a. 2 points**

In the ground state of Ge (Germanium; $Z = 32$) atom, **name** the following orbitals:

i. Orbital containing the electron that requires the maximum energy to be removed.

ii. Orbital containing the electron that experiences the minimum effective nuclear charge.

b. 2 points

An unknown anion (A^-) has the following *excited* electronic configuration: $[Ar] 4s^2 3d^3 4p^1$

(Ar: Argon; $Z = 18$)

Give the **condensed ground state electronic configuration** of the atom, **A** and the cation, **A^{2+}** .
Show your work. (Assume all electronic transitions are possible)

Electronic Configuration of A:

Electronic Configuration of A^{2+} :

c. 2 points

In a single atom, determine how many electrons can have the following set of quantum numbers?

(i) $n = 5; l = 3$

In a single atom, determine how many electrons can have the following set of quantum numbers?

(i) $n = 3; l = 2; m_s = +1/2$

**Question 4** **8 points**

Rank the following from 1 to 4 in **increasing** order. Provide the ranking in the boxes provided.

Note: Atomic number (Z) are provided for relevant atoms

a) Increasing (lowest to highest) order of *ionization energy*: (1= lowest; 4 = highest)

Mg

Mg: Z = 12

Rb

Rb: Z = 37

Cs

Cs: Z = 55

K

K : Z = 19

b) Increasing (lowest to highest) order of *bond angle*: (1= lowest; 4 = highest)

H₂OCCl₄XeF₄

HCN

c) Increasing (smallest to largest) order of *ionic size* (1= smallest; 4 = largest)

S²⁻

S: Z = 16

O²⁻

O: Z = 8

Be²⁺

Be: Z = 4

Na⁺

Na: Z = 11

d) Increasing (lowest to highest) order of *bond energy* (1= lowest; 4 = highest)

P-F

P-Cl

P-I

P-Br

**Question 5** **12 points**

a. 9 points For the molecules HClO_4 and TeF_6 , draw the most stable *Lewis structure* showing appropriate shape and *dashed/wedged bonds*, based on VSEPR. Determine the **molecular geometry** around the central atom. For HClO_4 , label the *most polar bond*.



b. 3 points (Note: Lone pairs are not shown in structures below)

(i) For the three *resonance* structures (1, 2 and 3) shown below, answer the following question:



(I) Circle the *most contributing* resonance structure.

Structure 1

Structure 2

Structure 3

(ii) For the three structures given (4, 5, and 6) below, answer the following questions:



(I) Circle all the structures that represent *resonance structures* of each other:

Structure 4

Structure 5

Structure 6

(II) Among the resonance structures, circle the *most contributing* resonance structure.

Structure 4

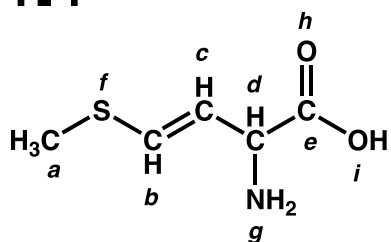
Structure 5

Structure 6

**Question 6** **6 points**

Using the data given below, calculate the **lattice energy** for SrO(s).

	ΔH° (kJ/mol)
$\text{Sr}(s) \longrightarrow \text{Sr}(g)$	+180
$\text{Sr}(g) \longrightarrow \text{Sr}^{2+}(g) + 2e^-$	+1468
$\text{O}_2(g) \longrightarrow 2\text{O}(g)$	+500
$\text{O}(g) + 2e^- \longrightarrow \text{O}^{2-}(g)$	+600
$\text{Sr}(s) + \frac{1}{2}\text{O}_2(g) \longrightarrow \text{SrO}(s)$	-550

**Question 7 10 points**

Consider the molecule given on the left, and answer questions (i) to (vi) using chemical bonding theories that we have learnt in class (valence bond theory, hybridization).

Note that the lone pairs are not shown in the structure:

i. Consider the carbon atoms marked in as ***a***, ***b***, ***c***, ***d***, and ***e*** in the molecule, and determine **hybridization** at each of the carbon centers. (2.5 points)

Atom	Hybridization
Carbon (<i>a</i>)	
Carbon (<i>b</i>)	
Carbon (<i>c</i>)	
Carbon (<i>d</i>)	
Carbon (<i>e</i>)	

ii. Now, consider the sulfur atom (S) marked as ***f***, nitrogen (N) atom (***g***), and oxygen(O) atoms (***h*** and ***i***), and determine their **hybridization**. (2 points)

Atom	Hybridization
Sulfur (<i>f</i>)	
Nitrogen (<i>g</i>)	
Oxygen (<i>h</i>)	
Oxygen (<i>i</i>)	

iii. **Describe** all the bonds formed by the carbon atoms ***a*** and ***b***, using sigma and pi designation as appropriate. (2 points)

Atom	Bonds formed (use sigma and pi as appropriate)
Carbon (<i>a</i>)	
Carbon (<i>b</i>)	

iv. How many total sp^3 -s sigma bonds are in the molecule? (1 point)

v. Determine **electronic** and **molecular geometry** at carbon (C) labeled ***e*** and sulfur (S) labeled ***f*** centres. (2 points)

Atom	Electronic Geometry	Molecular Geometry
Carbon (<i>e</i>)		
Sulfur (<i>f</i>)		

vi. Circle the value from the choices given below, that describes the **C-S-C bond angle at sulfur** (S) marked ***f***: (0.5 points)

>90° to <109.5° 109.5° >109.5 to <120° 120° >120° to <180°


Question 8 12 points

Using the **valence shell orbitals only** draw a **complete** molecular orbital (MO) diagram (with no s-p mixing) for the molecule **PS**, in the space provided below. **(6 points)**

Assume that 3s and 3p atomic orbitals will form molecular orbitals like those formed by the combination of 2s and 2p atomic orbitals that we learnt in class, and there is no s-p mixing.

Show **energy arrow**, **atomic orbitals** involved, **atom labels**, the **molecular orbitals** that are formed, **filled electrons** in the MO diagram etc. Make sure to label all the atomic and molecular orbitals.

Using the MO diagram above, answer the following questions:

- (i) Which one of the following will have the highest bond order? Circle the correct choice
 A) PS^{1-} B) PS C) PS^{1+}
- (ii) Which one of the following will have the longest bond length? Circle the correct choice
 A) PS^{1-} B) PS C) PS^{1+}
- (iii) Which of the following will be diamagnetic? Circle the correct choice.
 A) PS^{1-} B) PS C) PS^{1+}
- (iv) Draw Lewis dot structure of the molecule with the shortest bond length (based on the MO diagram) from among the three (PS^{1-} , PS, PS^{1+}), and indicate the bond order calculated from the Lewis dot structure. **(3 points)**

Lewis Dot Structure

Bond Order



24CC7093-CC18-4AC6-B41B-01F709F75622

midterm-exam-93388

#527 Page 10 of 10

SCRAP PAGE

**CHEMISTRY 110 –Midterm Exam****Thursday November 7, 2024, 6:30 – 8:30 pm**

Examiners: Pallavi Sirjoosingh and Ashok Kakkar

First name (please write as legibly as possible within the boxes)

--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--

Last name

--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--

Student ID number

--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--

This booklet has 8 questions on 10 pages (The last page can be used as a scrap page).

The staple must NOT be removed from the question booklet. Your intact question booklet must be returned at the end of the exam.

A separate booklet contains a periodic table, and datasheet.

Write your answers directly under the questions in the space provided. Do not write on the QR codes or near the top edges (as it may be cut off).

Please write **legibly and dark enough** for the scan to be readable. For students who use English as a second language, dictionaries are permitted. Calculators are allowed.

DO NOT ASK QUESTIONS OF THE INVIGILATORS about interpretations of the exam questions. If you have a concern about a question, please summarize it at the top of this page. All concerns will be evaluated after the exam. GOOD LUCK!

**Question 1 8 points** (*Assume all transitions are possible*)

a. 6 points Calculate the **energy (in kJ/mol)** required to ionize (completely remove) the electron from the **n=3** level of **He⁺ ion** (He: Helium). Show your work including equations. (3 sig. figs.)

b. 2 points

(I) Select the largest (longest) wavelength from the three choices below:

- i. Wavelength of photon to *completely remove* an electron from *1s orbital of Hydrogen atom*
- ii. Wavelength of photon to *completely remove* an electron from *1s orbital of Boron atom (B)*
- iii. Wavelength of photon to *completely remove* an electron from *1s orbital of Boron cation (B⁺)*

i or ii or iii

(II) Select the smallest (shortest) wavelength from the three choices below (*for H atom*):

- i. Wavelength of photon *emitted* for an electron transitioning from *2p to 1s orbital*
- ii. Wavelength of photon *absorbed* for an electron transitioning from *2s to 4p orbital*
- iii. Wavelength of photon *absorbed* for an electron transitioning from *1s to 3p orbital*

i or ii or iii



Question 2 8 points



A Chem110 student plots the graph shown on the left, from data obtained for the kinetic energy (eV) of the ejected electron versus frequency (s^{-1}) of the incident light, for an unknown metal. ($1\text{eV} = 1.6022 \times 10^{-19} \text{ J}$)

Based on the plot, answer the questions **a to c** below:

a. 1 point On the plot above, label the **threshold frequency**.

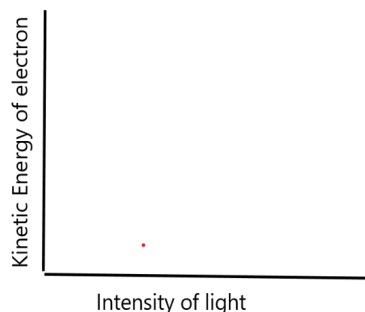
b. 2 points What is the value of the work potential (in J) of the unknown metal?

c. 3 points What is the *longest* wavelength (in nm) required to eject an electron from this metal's surface? Show your work including equation(s) used. (3 sig figs)

d. 2 points In the space provided below, plot the following

(Assume frequency is *above* threshold frequency):

- i. *Kinetic energy* of an ejected electron versus the *intensity* of the incident light (left)
- ii. *Number of electrons* ejected versus the *frequency* of the incident light (right)



**Question 3** **6 points****a. 2 points**

In the ground state of Ge (Germanium; $Z = 32$) atom, **name** the following orbitals:

i. Orbital containing the electron that requires the maximum energy to be removed.

ii. Orbital containing the electron that experiences the minimum effective nuclear charge.

b. 2 points

An unknown anion (A^-) has the following *excited* electronic configuration: $[Ar] 4s^2 3d^3 4p^1$

(Ar: Argon; $Z = 18$)

Give the **condensed ground state electronic configuration** of the atom, **A** and the cation, **A^{2+}** .
Show your work. (Assume all electronic transitions are possible)

Electronic Configuration of A:

Electronic Configuration of A^{2+} :

c. 2 points

In a single atom, determine how many electrons can have the following set of quantum numbers?

(i) $n = 5; l = 3$

In a single atom, determine how many electrons can have the following set of quantum numbers?

(i) $n = 3; l = 2; m_s = +1/2$

**Question 4** **8 points**

Rank the following from 1 to 4 in **increasing** order. Provide the ranking in the boxes provided.

Note: Atomic number (Z) are provided for relevant atoms

a) Increasing (lowest to highest) order of *ionization energy*: (1= lowest; 4 = highest)

Mg

Mg: Z = 12

Rb

Rb: Z = 37

Cs

Cs: Z = 55

K

K : Z = 19

b) Increasing (lowest to highest) order of *bond angle*: (1= lowest; 4 = highest)

H₂OCCl₄XeF₄

HCN

c) Increasing (smallest to largest) order of *ionic size* (1= smallest; 4 = largest)

S²⁻

S: Z = 16

O²⁻

O: Z = 8

Be²⁺

Be: Z = 4

Na⁺

Na: Z = 11

d) Increasing (lowest to highest) order of *bond energy* (1= lowest; 4 = highest)

P-F

P-Cl

P-I

P-Br

**Question 5** **12 points**

a. 9 points For the molecules HClO_4 and TeF_6 , draw the most stable *Lewis structure* showing appropriate shape and *dashed/wedged bonds*, based on VSEPR. Determine the **molecular geometry** around the central atom. For HClO_4 , label the *most polar bond*.



b. 3 points (Note: Lone pairs are not shown in structures below)

(i) For the three *resonance* structures (1, 2 and 3) shown below, answer the following question:



(I) Circle the *most contributing* resonance structure.

Structure 1

Structure 2

Structure 3

(ii) For the three structures given (4, 5, and 6) below, answer the following questions:



(I) Circle all the structures that represent *resonance structures* of each other:

Structure 4

Structure 5

Structure 6

(II) Among the resonance structures, circle the *most contributing* resonance structure.

Structure 4

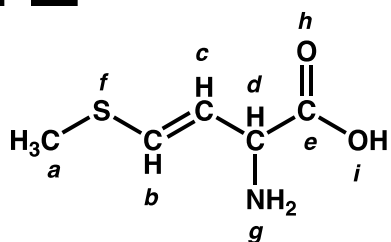
Structure 5

Structure 6

**Question 6** **6 points**

Using the data given below, calculate the **lattice energy** for SrO(s).

	ΔH° (kJ/mol)
$\text{Sr}(s) \longrightarrow \text{Sr}(g)$	+180
$\text{Sr}(g) \longrightarrow \text{Sr}^{2+}(g) + 2e^-$	+1468
$\text{O}_2(g) \longrightarrow 2\text{O}(g)$	+500
$\text{O}(g) + 2e^- \longrightarrow \text{O}^{2-}(g)$	+600
$\text{Sr}(s) + \frac{1}{2}\text{O}_2(g) \longrightarrow \text{SrO}(s)$	-550

**Question 7 10 points**

Consider the molecule given on the left, and answer questions (i) to (vi) using chemical bonding theories that we have learnt in class (valence bond theory, hybridization).

Note that the lone pairs are not shown in the structure:

i. Consider the carbon atoms marked in as ***a***, ***b***, ***c***, ***d***, and ***e*** in the molecule, and determine **hybridization** at each of the carbon centers. (2.5 points)

Atom	Hybridization
Carbon (<i>a</i>)	
Carbon (<i>b</i>)	
Carbon (<i>c</i>)	
Carbon (<i>d</i>)	
Carbon (<i>e</i>)	

ii. Now, consider the sulfur atom (S) marked as ***f***, nitrogen (N) atom (***g***), and oxygen(O) atoms (***h*** and ***i***), and determine their **hybridization**. (2 points)

Atom	Hybridization
Sulfur (<i>f</i>)	
Nitrogen (<i>g</i>)	
Oxygen (<i>h</i>)	
Oxygen (<i>i</i>)	

iii. **Describe** all the bonds formed by the carbon atoms ***a*** and ***b***, using sigma and pi designation as appropriate. (2 points)

Atom	Bonds formed (use sigma and pi as appropriate)
Carbon (<i>a</i>)	
Carbon (<i>b</i>)	

iv. How many total sp^3 -s sigma bonds are in the molecule? (1 point)

v. Determine **electronic** and **molecular geometry** at carbon (C) labeled ***e*** and sulfur (S) labeled ***f*** centres. (2 points)

Atom	Electronic Geometry	Molecular Geometry
Carbon (<i>e</i>)		
Sulfur (<i>f</i>)		

vi. Circle the value from the choices given below, that describes the **C-S-C bond angle at sulfur** (S) marked ***f***: (0.5 points)

>90° to <109.5°

109.5°

>109.5 to <120°

120°

>120° to <180°


Question 8 12 points

Using the **valence shell orbitals only** draw a **complete** molecular orbital (MO) diagram (with no s-p mixing) for the molecule **PS**, in the space provided below. **(6 points)**

Assume that 3s and 3p atomic orbitals will form molecular orbitals like those formed by the combination of 2s and 2p atomic orbitals that we learnt in class, and there is no s-p mixing.

Show **energy arrow**, **atomic orbitals** involved, **atom labels**, the **molecular orbitals** that are formed, **filled electrons** in the MO diagram etc. Make sure to label all the atomic and molecular orbitals.

Using the MO diagram above, answer the following questions:

- (i) Which one of the following will have the highest bond order? Circle the correct choice
 A) PS^{1-} B) PS C) PS^{1+}
- (ii) Which one of the following will have the longest bond length? Circle the correct choice
 A) PS^{1-} B) PS C) PS^{1+}
- (iii) Which of the following will be diamagnetic? Circle the correct choice.
 A) PS^{1-} B) PS C) PS^{1+}
- (iv) Draw Lewis dot structure of the molecule with the shortest bond length (based on the MO diagram) from among the three (PS^{1-} , PS, PS^{1+}), and indicate the bond order calculated from the Lewis dot structure. **(3 points)**

Lewis Dot Structure

Bond Order



E72E49F4-DF8D-4E5A-B710-947FA61C60A1

midterm-exam-93388

#528 Page 10 of 10

SCRAP PAGE

**CHEMISTRY 110 –Midterm Exam****Thursday November 7, 2024, 6:30 – 8:30 pm****Examiners: Pallavi Sirjoosingh and Ashok Kakkar**

First name (please write as legibly as possible within the boxes)

--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--

Last name

--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--

Student ID number

--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--

This booklet has 8 questions on 10 pages (The last page can be used as a scrap page).

The staple must NOT be removed from the question booklet. Your intact question booklet must be returned at the end of the exam.

A separate booklet contains a periodic table, and datasheet.

Write your answers directly under the questions in the space provided. Do not write on the QR codes or near the top edges (as it may be cut off).

Please write **legibly and dark enough** for the scan to be readable. For students who use English as a second language, dictionaries are permitted. Calculators are allowed.

DO NOT ASK QUESTIONS OF THE INVIGILATORS about interpretations of the exam questions. If you have a concern about a question, please summarize it at the top of this page. All concerns will be evaluated after the exam. GOOD LUCK!

**Question 1 8 points** (*Assume all transitions are possible*)

a. 6 points Calculate the **energy (in kJ/mol)** required to ionize (completely remove) the electron from the **n=3** level of **He⁺ ion** (He: Helium). Show your work including equations. (3 sig. figs.)

b. 2 points

(I) Select the largest (longest) wavelength from the three choices below:

- i. Wavelength of photon to *completely remove* an electron from *1s orbital of Hydrogen atom*
- ii. Wavelength of photon to *completely remove* an electron from *1s orbital of Boron atom (B)*
- iii. Wavelength of photon to *completely remove* an electron from *1s orbital of Boron cation (B⁺)*

i or ii or iii

(II) Select the smallest (shortest) wavelength from the three choices below (*for H atom*):

- i. Wavelength of photon *emitted* for an electron transitioning from *2p to 1s orbital*
- ii. Wavelength of photon *absorbed* for an electron transitioning from *2s to 4p orbital*
- iii. Wavelength of photon *absorbed* for an electron transitioning from *1s to 3p orbital*

i or ii or iii



Question 2 8 points



A Chem110 student plots the graph shown on the left, from data obtained for the kinetic energy (eV) of the ejected electron versus frequency (s^{-1}) of the incident light, for an unknown metal. ($1\text{eV} = 1.6022 \times 10^{-19} \text{ J}$)

Based on the plot, answer the questions **a to c** below:

a. 1 point On the plot above, label the **threshold frequency**.

b. 2 points What is the value of the work potential (in J) of the unknown metal?

c. 3 points What is the *longest* wavelength (in nm) required to eject an electron from this metal's surface? Show your work including equation(s) used. (3 sig figs)

d. 2 points In the space provided below, plot the following

(Assume frequency is *above* threshold frequency):

- Kinetic energy* of an ejected electron versus the *intensity* of the incident light (left)
- Number of electrons* ejected versus the *frequency* of the incident light (right)



**Question 3** **6 points****a. 2 points**

In the ground state of Ge (Germanium; $Z = 32$) atom, **name** the following orbitals:

i. Orbital containing the electron that requires the maximum energy to be removed.

ii. Orbital containing the electron that experiences the minimum effective nuclear charge.

b. 2 points

An unknown anion (A^-) has the following *excited* electronic configuration: $[Ar] 4s^2 3d^3 4p^1$

(Ar: Argon; $Z = 18$)

Give the **condensed ground state electronic configuration** of the atom, **A** and the cation, **A^{2+}** .
Show your work. (Assume all electronic transitions are possible)

Electronic Configuration of A:

Electronic Configuration of A^{2+} :

c. 2 points

In a single atom, determine how many electrons can have the following set of quantum numbers?

(i) $n = 5; l = 3$

In a single atom, determine how many electrons can have the following set of quantum numbers?

(i) $n = 3; l = 2; m_s = +1/2$

**Question 4** **8 points**

Rank the following from 1 to 4 in **increasing** order. Provide the ranking in the boxes provided.

Note: Atomic number (Z) are provided for relevant atoms

a) Increasing (lowest to highest) order of *ionization energy*: (1= lowest; 4 = highest)

Mg

Mg: Z = 12

Rb

Rb: Z = 37

Cs

Cs: Z = 55

K

K : Z = 19

b) Increasing (lowest to highest) order of *bond angle*: (1= lowest; 4 = highest)

H₂OCCl₄XeF₄

HCN

c) Increasing (smallest to largest) order of *ionic size* (1= smallest; 4 = largest)

S²⁻

S: Z = 16

O²⁻

O: Z = 8

Be²⁺

Be: Z = 4

Na⁺

Na: Z = 11

d) Increasing (lowest to highest) order of *bond energy* (1= lowest; 4 = highest)

P-F

P-Cl

P-I

P-Br

**Question 5** **12 points**

a. 9 points For the molecules HClO_4 and TeF_6 , draw the most stable *Lewis structure* showing appropriate shape and *dashed/wedged bonds*, based on VSEPR. Determine the **molecular geometry** around the central atom. For HClO_4 , label the *most polar bond*.



b. 3 points (Note: Lone pairs are not shown in structures below)

(i) For the three *resonance* structures (1, 2 and 3) shown below, answer the following question:



(I) Circle the *most contributing* resonance structure.

Structure 1

Structure 2

Structure 3

(ii) For the three structures given (4, 5, and 6) below, answer the following questions:



(I) Circle all the structures that represent *resonance structures* of each other:

Structure 4

Structure 5

Structure 6

(II) Among the resonance structures, circle the *most contributing* resonance structure.

Structure 4

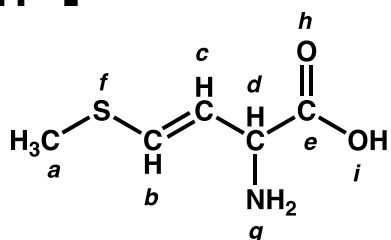
Structure 5

Structure 6

**Question 6** **6 points**

Using the data given below, calculate the **lattice energy** for SrO(s).

	ΔH° (kJ/mol)
$\text{Sr}(s) \longrightarrow \text{Sr}(g)$	+180
$\text{Sr}(g) \longrightarrow \text{Sr}^{2+}(g) + 2e^-$	+1468
$\text{O}_2(g) \longrightarrow 2\text{O}(g)$	+500
$\text{O}(g) + 2e^- \longrightarrow \text{O}^{2-}(g)$	+600
$\text{Sr}(s) + \frac{1}{2}\text{O}_2(g) \longrightarrow \text{SrO}(s)$	-550

**Question 7 10 points**

Consider the molecule given on the left, and answer questions (i) to (vi) using chemical bonding theories that we have learnt in class (valence bond theory, hybridization).

Note that the lone pairs are not shown in the structure:

i. Consider the carbon atoms marked in as ***a***, ***b***, ***c***, ***d***, and ***e*** in the molecule, and determine **hybridization** at each of the carbon centers. (2.5 points)

Atom	Hybridization
Carbon (<i>a</i>)	
Carbon (<i>b</i>)	
Carbon (<i>c</i>)	
Carbon (<i>d</i>)	
Carbon (<i>e</i>)	

ii. Now, consider the sulfur atom (S) marked as ***f***, nitrogen (N) atom (***g***), and oxygen(O) atoms (***h*** and ***i***), and determine their **hybridization**. (2 points)

Atom	Hybridization
Sulfur (<i>f</i>)	
Nitrogen (<i>g</i>)	
Oxygen (<i>h</i>)	
Oxygen (<i>i</i>)	

iii. **Describe** all the bonds formed by the carbon atoms ***a*** and ***b***, using sigma and pi designation as appropriate. (2 points)

Atom	Bonds formed (use sigma and pi as appropriate)
Carbon (<i>a</i>)	
Carbon (<i>b</i>)	

iv. How many total sp^3 -s sigma bonds are in the molecule? (1 point)

v. Determine **electronic** and **molecular geometry** at carbon (C) labeled ***e*** and sulfur (S) labeled ***f*** centres. (2 points)

Atom	Electronic Geometry	Molecular Geometry
Carbon (<i>e</i>)		
Sulfur (<i>f</i>)		

vi. Circle the value from the choices given below, that describes the **C-S-C bond angle at sulfur** (S) marked ***f***: (0.5 points)

$>90^\circ$ to $<109.5^\circ$ 109.5° >109.5 to $<120^\circ$ 120° $>120^\circ$ to $<180^\circ$

**Question 8 12 points**

Using the **valence shell orbitals only** draw a **complete** molecular orbital (MO) diagram (with no s-p mixing) for the molecule **PS**, in the space provided below. **(6 points)**

Assume that 3s and 3p atomic orbitals will form molecular orbitals like those formed by the combination of 2s and 2p atomic orbitals that we learnt in class, and there is no s-p mixing.

Show **energy arrow**, **atomic orbitals** involved, **atom labels**, the **molecular orbitals** that are formed, **filled electrons** in the MO diagram etc. Make sure to label all the atomic and molecular orbitals.

Using the MO diagram above, answer the following questions:

- (i) Which one of the following will have the highest bond order? Circle the correct choice
A) PS^{1-} B) PS C) PS^{1+}
- (ii) Which one of the following will have the longest bond length? Circle the correct choice
A) PS^{1-} B) PS C) PS^{1+}
- (iii) Which of the following will be diamagnetic? Circle the correct choice.
A) PS^{1-} B) PS C) PS^{1+}
- (iv) Draw Lewis dot structure of the molecule with the shortest bond length (based on the MO diagram) from among the three (PS^{1-} , PS, PS^{1+}), and indicate the bond order calculated from the Lewis dot structure. **(3 points)**

Lewis Dot Structure

Bond Order



844A6BA6-E98D-4CC1-8C06-B240474CB364

midterm-exam-93388

#529 Page 10 of 10

SCRAP PAGE

**CHEMISTRY 110 –Midterm Exam****Thursday November 7, 2024, 6:30 – 8:30 pm**

Examiners: Pallavi Sirjoosingh and Ashok Kakkar

First name (please write as legibly as possible within the boxes)

--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--

Last name

--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--

Student ID number

--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--

This booklet has 8 questions on 10 pages (The last page can be used as a scrap page).

The staple must NOT be removed from the question booklet. Your intact question booklet must be returned at the end of the exam.

A separate booklet contains a periodic table, and datasheet.

Write your answers directly under the questions in the space provided. Do not write on the QR codes or near the top edges (as it may be cut off).

Please write **legibly and dark enough** for the scan to be readable. For students who use English as a second language, dictionaries are permitted. Calculators are allowed.

DO NOT ASK QUESTIONS OF THE INVIGILATORS about interpretations of the exam questions. If you have a concern about a question, please summarize it at the top of this page. All concerns will be evaluated after the exam. GOOD LUCK!

**Question 1 8 points** (*Assume all transitions are possible*)

a. 6 points Calculate the **energy (in kJ/mol)** required to ionize (completely remove) the electron from the **n=3** level of **He⁺ ion** (He: Helium). Show your work including equations. (3 sig. figs.)

b. 2 points

(I) Select the largest (longest) wavelength from the three choices below:

- i. Wavelength of photon to *completely remove* an electron from *1s orbital of Hydrogen atom*
- ii. Wavelength of photon to *completely remove* an electron from *1s orbital of Boron atom (B)*
- iii. Wavelength of photon to *completely remove* an electron from *1s orbital of Boron cation (B⁺)*

i or ii or iii

(II) Select the smallest (shortest) wavelength from the three choices below (*for H atom*):

- i. Wavelength of photon *emitted* for an electron transitioning from *2p to 1s orbital*
- ii. Wavelength of photon *absorbed* for an electron transitioning from *2s to 4p orbital*
- iii. Wavelength of photon *absorbed* for an electron transitioning from *1s to 3p orbital*

i or ii or iii



Question 2 8 points



A Chem110 student plots the graph shown on the left, from data obtained for the kinetic energy (eV) of the ejected electron versus frequency (s^{-1}) of the incident light, for an unknown metal. ($1\text{eV} = 1.6022 \times 10^{-19} \text{ J}$)

Based on the plot, answer the questions **a to c** below:

a. 1 point On the plot above, label the **threshold frequency**.

b. 2 points What is the value of the work potential (in J) of the unknown metal?

c. 3 points What is the *longest* wavelength (in nm) required to eject an electron from this metal's surface? Show your work including equation(s) used. (3 sig figs)

d. 2 points In the space provided below, plot the following

(Assume frequency is *above* threshold frequency):

- Kinetic energy* of an ejected electron versus the *intensity* of the incident light (left)
- Number of electrons* ejected versus the *frequency* of the incident light (right)



**Question 3** **6 points****a. 2 points**

In the ground state of Ge (Germanium; $Z = 32$) atom, **name** the following orbitals:

i. Orbital containing the electron that requires the maximum energy to be removed.

ii. Orbital containing the electron that experiences the minimum effective nuclear charge.

b. 2 points

An unknown anion (A^-) has the following *excited* electronic configuration: $[Ar] 4s^2 3d^3 4p^1$

(Ar: Argon; $Z = 18$)

Give the **condensed ground state electronic configuration** of the atom, **A** and the cation, **A^{2+}** .
Show your work. (Assume all electronic transitions are possible)

Electronic Configuration of A:

Electronic Configuration of A^{2+} :

c. 2 points

In a single atom, determine how many electrons can have the following set of quantum numbers?

(i) $n = 5; l = 3$

In a single atom, determine how many electrons can have the following set of quantum numbers?

(i) $n = 3; l = 2; m_s = +1/2$

**Question 4** **8 points**

Rank the following from 1 to 4 in **increasing** order. Provide the ranking in the boxes provided.

Note: Atomic number (Z) are provided for relevant atoms

a) Increasing (lowest to highest) order of *ionization energy*: (1= lowest; 4 = highest)

Mg

Mg: Z = 12

Rb

Rb: Z = 37

Cs

Cs: Z = 55

K

K : Z = 19

b) Increasing (lowest to highest) order of *bond angle*: (1= lowest; 4 = highest)

H₂OCCl₄XeF₄

HCN

c) Increasing (smallest to largest) order of *ionic size* (1= smallest; 4 = largest)

S²⁻

S: Z = 16

O²⁻

O: Z = 8

Be²⁺

Be: Z = 4

Na⁺

Na: Z = 11

d) Increasing (lowest to highest) order of *bond energy* (1= lowest; 4 = highest)

P-F

P-Cl

P-I

P-Br

**Question 5** **12 points**

a. 9 points For the molecules HClO_4 and TeF_6 , draw the most stable *Lewis structure* showing appropriate shape and *dashed/wedged bonds*, based on VSEPR. Determine the **molecular geometry** around the central atom. For HClO_4 , label the *most polar bond*.



b. 3 points (Note: Lone pairs are not shown in structures below)

(i) For the three *resonance* structures (1, 2 and 3) shown below, answer the following question:



(I) Circle the *most contributing* resonance structure.

Structure 1

Structure 2

Structure 3

(ii) For the three structures given (4, 5, and 6) below, answer the following questions:



(I) Circle all the structures that represent *resonance structures* of each other:

Structure 4

Structure 5

Structure 6

(II) Among the resonance structures, circle the *most contributing* resonance structure.

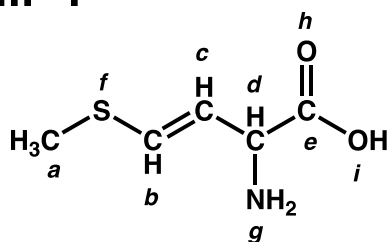
Structure 4

Structure 5

Structure 6

**Question 6** **6 points**Using the data given below, calculate the **lattice energy** for SrO(s).

	ΔH° (kJ/mol)
$\text{Sr}(s) \longrightarrow \text{Sr}(g)$	+180
$\text{Sr}(g) \longrightarrow \text{Sr}^{2+}(g) + 2e^-$	+1468
$\text{O}_2(g) \longrightarrow 2\text{O}(g)$	+500
$\text{O}(g) + 2e^- \longrightarrow \text{O}^{2-}(g)$	+600
$\text{Sr}(s) + \frac{1}{2}\text{O}_2(g) \longrightarrow \text{SrO}(s)$	-550

**Question 7 10 points**

Consider the molecule given on the left, and answer questions (i) to (vi) using chemical bonding theories that we have learnt in class (valence bond theory, hybridization).

Note that the lone pairs are not shown in the structure:

i. Consider the carbon atoms marked in as ***a***, ***b***, ***c***, ***d***, and ***e*** in the molecule, and determine **hybridization** at each of the carbon centers. (2.5 points)

Atom	Hybridization
Carbon (<i>a</i>)	
Carbon (<i>b</i>)	
Carbon (<i>c</i>)	
Carbon (<i>d</i>)	
Carbon (<i>e</i>)	

ii. Now, consider the sulfur atom (S) marked as ***f***, nitrogen (N) atom (***g***), and oxygen(O) atoms (***h*** and ***i***), and determine their **hybridization**. (2 points)

Atom	Hybridization
Sulfur (<i>f</i>)	
Nitrogen (<i>g</i>)	
Oxygen (<i>h</i>)	
Oxygen (<i>i</i>)	

iii. **Describe** all the bonds formed by the carbon atoms ***a*** and ***b***, using sigma and pi designation as appropriate. (2 points)

Atom	Bonds formed (use sigma and pi as appropriate)
Carbon (<i>a</i>)	
Carbon (<i>b</i>)	

iv. How many total sp^3 -s sigma bonds are in the molecule? (1 point)

v. Determine **electronic** and **molecular geometry** at carbon (C) labeled ***e*** and sulfur (S) labeled ***f*** centres. (2 points)

Atom	Electronic Geometry	Molecular Geometry
Carbon (<i>e</i>)		
Sulfur (<i>f</i>)		

vi. Circle the value from the choices given below, that describes the **C-S-C bond angle at sulfur** (S) marked ***f***: (0.5 points)

>90° to <109.5° 109.5° >109.5 to <120° 120° >120° to <180°


Question 8 12 points

Using the **valence shell orbitals only** draw a **complete** molecular orbital (MO) diagram (with no s-p mixing) for the molecule **PS**, in the space provided below. **(6 points)**

Assume that 3s and 3p atomic orbitals will form molecular orbitals like those formed by the combination of 2s and 2p atomic orbitals that we learnt in class, and there is no s-p mixing.

Show **energy arrow**, **atomic orbitals** involved, **atom labels**, the **molecular orbitals** that are formed, **filled electrons** in the MO diagram etc. Make sure to label all the atomic and molecular orbitals.

Using the MO diagram above, answer the following questions:

- (i) Which one of the following will have the highest bond order? Circle the correct choice
 A) PS^{1-} B) PS C) PS^{1+}
- (ii) Which one of the following will have the longest bond length? Circle the correct choice
 A) PS^{1-} B) PS C) PS^{1+}
- (iii) Which of the following will be diamagnetic? Circle the correct choice.
 A) PS^{1-} B) PS C) PS^{1+}
- (iv) Draw Lewis dot structure of the molecule with the shortest bond length (based on the MO diagram) from among the three (PS^{1-} , PS, PS^{1+}), and indicate the bond order calculated from the Lewis dot structure. **(3 points)**

Lewis Dot Structure

Bond Order



6D25BFC4-A6A1-4A1A-ADAE-C0B69F9B36A3

midterm-exam-93388

#530 Page 10 of 10

SCRAP PAGE

**CHEMISTRY 110 –Midterm Exam****Thursday November 7, 2024, 6:30 – 8:30 pm**

Examiners: Pallavi Sirjoosingh and Ashok Kakkar

First name (please write as legibly as possible within the boxes)

--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--

Last name

--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--

Student ID number

--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--

This booklet has 8 questions on 10 pages (The last page can be used as a scrap page).

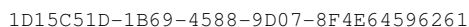
The staple must NOT be removed from the question booklet. Your intact question booklet must be returned at the end of the exam.

A separate booklet contains a periodic table, and datasheet.

Write your answers directly under the questions in the space provided. Do not write on the QR codes or near the top edges (as it may be cut off).

Please write **legibly and dark enough** for the scan to be readable. For students who use English as a second language, dictionaries are permitted. Calculators are allowed.

DO NOT ASK QUESTIONS OF THE INVIGILATORS about interpretations of the exam questions. If you have a concern about a question, please summarize it at the top of this page. All concerns will be evaluated after the exam. GOOD LUCK!



midterm-exam-93388

#531 Page 2 of 10

a. 6 points Calculate the **energy (in kJ/mol)** required to ionize (completely remove) the electron from the **n=3** level of **He⁺ ion** (He: Helium). Show your work including equations. (3 sig. figs.)

(I) Select the largest (longest) wavelength from the three choices below:

- Wavelength of photon to *completely remove* an electron from *1s orbital of Hydrogen atom*
- Wavelength of photon to *completely remove* an electron from *1s orbital of Boron atom (B)*
- Wavelength of photon to *completely remove* an electron from *1s orbital of Boron cation (B⁺)*

i **or** **ii** **or** **iii**

(II) Select the **smallest (shortest) wavelength from the three choices below (*for H atom*):**

- Wavelength of photon *emitted* for an electron transitioning from $2p$ to $1s$ orbital
- Wavelength of photon *absorbed* for an electron transitioning from $2s$ to $4p$ orbital
- Wavelength of photon *absorbed* for an electron transitioning from $1s$ to $3p$ orbital

[illegible]



Question 2 8 points



A Chem110 student plots the graph shown on the left, from data obtained for the kinetic energy (eV) of the ejected electron versus frequency (s^{-1}) of the incident light, for an unknown metal. ($1\text{eV} = 1.6022 \times 10^{-19} \text{ J}$)

Based on the plot, answer the questions **a to c** below:

a. 1 point On the plot above, label the **threshold frequency**.

b. 2 points What is the value of the work potential (in J) of the unknown metal?

c. 3 points What is the *longest* wavelength (in nm) required to eject an electron from this metal's surface? Show your work including equation(s) used. (3 sig figs)

d. 2 points In the space provided below, plot the following

(Assume frequency is *above* threshold frequency):

- Kinetic energy* of an ejected electron versus the *intensity* of the incident light (left)
- Number of electrons* ejected versus the *frequency* of the incident light (right)



**Question 3** **6 points****a. 2 points**

In the ground state of Ge (Germanium; $Z = 32$) atom, **name** the following orbitals:

i. Orbital containing the electron that requires the maximum energy to be removed.

ii. Orbital containing the electron that experiences the minimum effective nuclear charge.

b. 2 points

An unknown anion (A^-) has the following *excited* electronic configuration: $[Ar] 4s^2 3d^3 4p^1$

(Ar: Argon; $Z = 18$)

Give the **condensed ground state electronic configuration** of the atom, **A** and the cation, **A^{2+}** .
Show your work. (Assume all electronic transitions are possible)

Electronic Configuration of A:

Electronic Configuration of A^{2+} :

c. 2 points

In a single atom, determine how many electrons can have the following set of quantum numbers?

(i) $n = 5; l = 3$

In a single atom, determine how many electrons can have the following set of quantum numbers?

(i) $n = 3; l = 2; m_s = +1/2$

**Question 4** **8 points**

Rank the following from 1 to 4 in **increasing** order. Provide the ranking in the boxes provided.

Note: Atomic number (Z) are provided for relevant atoms

a) Increasing (lowest to highest) order of *ionization energy*: (1= lowest; 4 = highest)

Mg

Mg: Z = 12

Rb

Rb: Z = 37

Cs

Cs: Z = 55

K

K : Z = 19

b) Increasing (lowest to highest) order of *bond angle*: (1= lowest; 4 = highest)

H₂OCCl₄XeF₄

HCN

c) Increasing (smallest to largest) order of *ionic size* (1= smallest; 4 = largest)

S²⁻

S: Z = 16

O²⁻

O: Z = 8

Be²⁺

Be: Z = 4

Na⁺

Na: Z = 11

d) Increasing (lowest to highest) order of *bond energy* (1= lowest; 4 = highest)

P-F

P-Cl

P-I

P-Br

**Question 5** **12 points**

a. 9 points For the molecules HClO_4 and TeF_6 , draw the most stable *Lewis structure* showing appropriate shape and *dashed/wedged bonds*, based on VSEPR. Determine the **molecular geometry** around the central atom. For HClO_4 , label the *most polar bond*.

HClO_4 (there are no O-O bonds)

Label the most polar bond

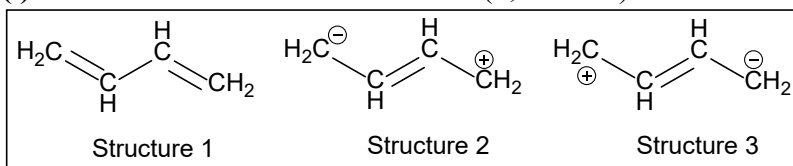
Molecular Geometry around Cl

TeF_6

Molecular Geometry around Te

b. 3 points (Note: Lone pairs are not shown in structures below)

(i) For the three *resonance* structures (1, 2 and 3) shown below, answer the following question:



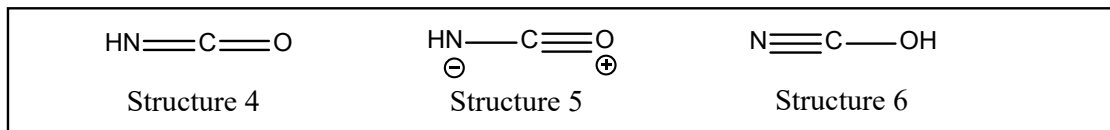
(I) Circle the *most contributing* resonance structure.

Structure 1

Structure 2

Structure 3

(ii) For the three structures given (4, 5, and 6) below, answer the following questions:



(I) Circle all the structures that represent *resonance structures* of each other:

Structure 4

Structure 5

Structure 6

(II) Among the resonance structures, circle the *most contributing* resonance structure.

Structure 4

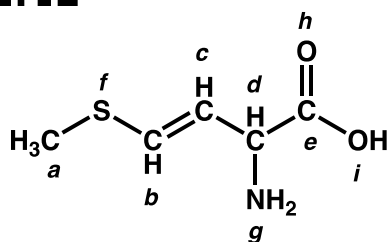
Structure 5

Structure 6

**Question 6** **6 points**

Using the data given below, calculate the **lattice energy** for SrO(s).

	ΔH° (kJ/mol)
$\text{Sr}(s) \longrightarrow \text{Sr}(g)$	+180
$\text{Sr}(g) \longrightarrow \text{Sr}^{2+}(g) + 2e^-$	+1468
$\text{O}_2(g) \longrightarrow 2\text{O}(g)$	+500
$\text{O}(g) + 2e^- \longrightarrow \text{O}^{2-}(g)$	+600
$\text{Sr}(s) + \frac{1}{2}\text{O}_2(g) \longrightarrow \text{SrO}(s)$	-550

**Question 7 10 points**

Consider the molecule given on the left, and answer questions (i) to (vi) using chemical bonding theories that we have learnt in class (valence bond theory, hybridization).

Note that the lone pairs are not shown in the structure:

i. Consider the carbon atoms marked in as ***a***, ***b***, ***c***, ***d***, and ***e*** in the molecule, and determine **hybridization** at each of the carbon centers. (2.5 points)

Atom	Hybridization
Carbon (<i>a</i>)	
Carbon (<i>b</i>)	
Carbon (<i>c</i>)	
Carbon (<i>d</i>)	
Carbon (<i>e</i>)	

ii. Now, consider the sulfur atom (S) marked as ***f***, nitrogen (N) atom (***g***), and oxygen(O) atoms (***h*** and ***i***), and determine their **hybridization**. (2 points)

Atom	Hybridization
Sulfur (<i>f</i>)	
Nitrogen (<i>g</i>)	
Oxygen (<i>h</i>)	
Oxygen (<i>i</i>)	

iii. **Describe** all the bonds formed by the carbon atoms ***a*** and ***b***, using sigma and pi designation as appropriate. (2 points)

Atom	Bonds formed (use sigma and pi as appropriate)
Carbon (<i>a</i>)	
Carbon (<i>b</i>)	

iv. How many total sp^3 -s sigma bonds are in the molecule? (1 point)

v. Determine **electronic** and **molecular geometry** at carbon (C) labeled ***e*** and sulfur (S) labeled ***f*** centres. (2 points)

Atom	Electronic Geometry	Molecular Geometry
Carbon (<i>e</i>)		
Sulfur (<i>f</i>)		

vi. Circle the value from the choices given below, that describes the **C-S-C bond angle at sulfur** (S) marked ***f***: (0.5 points)

>90° to <109.5° 109.5° >109.5 to <120° 120° >120° to <180°

**Question 8 12 points**

Using the **valence shell orbitals only** draw a **complete** molecular orbital (MO) diagram (with no s-p mixing) for the molecule **PS**, in the space provided below. **(6 points)**

Assume that 3s and 3p atomic orbitals will form molecular orbitals like those formed by the combination of 2s and 2p atomic orbitals that we learnt in class, and there is no s-p mixing.

Show **energy arrow**, **atomic orbitals** involved, **atom labels**, the **molecular orbitals** that are formed, **filled electrons** in the MO diagram etc. Make sure to label all the atomic and molecular orbitals.

Using the MO diagram above, answer the following questions:

- (i) Which one of the following will have the highest bond order? Circle the correct choice
A) PS^{1-} B) PS C) PS^{1+}
- (ii) Which one of the following will have the longest bond length? Circle the correct choice
A) PS^{1-} B) PS C) PS^{1+}
- (iii) Which of the following will be diamagnetic? Circle the correct choice.
A) PS^{1-} B) PS C) PS^{1+}
- (iv) Draw Lewis dot structure of the molecule with the shortest bond length (based on the MO diagram) from among the three (PS^{1-} , PS, PS^{1+}), and indicate the bond order calculated from the Lewis dot structure. **(3 points)**

Lewis Dot Structure

Bond Order



2B148424-5464-4728-86B6-7BDBBCBF7B8B

midterm-exam-93388

#531 Page 10 of 10

SCRAP PAGE

**CHEMISTRY 110 –Midterm Exam****Thursday November 7, 2024, 6:30 – 8:30 pm**

Examiners: Pallavi Sirjoosingh and Ashok Kakkar

First name (please write as legibly as possible within the boxes)

--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--

Last name

--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--

Student ID number

--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--

This booklet has 8 questions on 10 pages (The last page can be used as a scrap page).

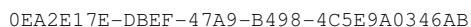
The staple must NOT be removed from the question booklet. Your intact question booklet must be returned at the end of the exam.

A separate booklet contains a periodic table, and datasheet.

Write your answers directly under the questions in the space provided. Do not write on the QR codes or near the top edges (as it may be cut off).

Please write **legibly and dark enough** for the scan to be readable. For students who use English as a second language, dictionaries are permitted. Calculators are allowed.

DO NOT ASK QUESTIONS OF THE INVIGILATORS about interpretations of the exam questions. If you have a concern about a question, please summarize it at the top of this page. All concerns will be evaluated after the exam. GOOD LUCK!



#532 Page 2 of 10

a. 6 points Calculate the **energy (in kJ/mol)** required to ionize (completely remove) the electron from the **n=3** level of **He⁺ ion** (He: Helium). Show your work including equations. (3 sig. figs.)

- Wavelength of photon to *completely remove* an electron from *1s orbital of Hydrogen atom*
- Wavelength of photon to *completely remove* an electron from *1s orbital of Boron atom (B)*
- Wavelength of photon to *completely remove* an electron from *1s orbital of Boron cation (B⁺)*

- Wavelength of photon *emitted* for an electron transitioning from $2p$ to $1s$ orbital
- Wavelength of photon *absorbed* for an electron transitioning from $2s$ to $4p$ orbital
- Wavelength of photon *absorbed* for an electron transitioning from $1s$ to $3p$ orbital

i or ii or iii



Question 2 8 points



A Chem110 student plots the graph shown on the left, from data obtained for the kinetic energy (eV) of the ejected electron versus frequency (s^{-1}) of the incident light, for an unknown metal. ($1\text{eV} = 1.6022 \times 10^{-19} \text{ J}$)

Based on the plot, answer the questions **a to c** below:

a. 1 point On the plot above, label the **threshold frequency**.

b. 2 points What is the value of the work potential (in J) of the unknown metal?

c. 3 points What is the *longest* wavelength (in nm) required to eject an electron from this metal's surface? Show your work including equation(s) used. (3 sig figs)

d. 2 points In the space provided below, plot the following

(Assume frequency is *above* threshold frequency):

- Kinetic energy* of an ejected electron versus the *intensity* of the incident light (left)
- Number of electrons* ejected versus the *frequency* of the incident light (right)



**Question 3** **6 points****a. 2 points**

In the ground state of Ge (Germanium; $Z = 32$) atom, **name** the following orbitals:

i. Orbital containing the electron that requires the maximum energy to be removed.

ii. Orbital containing the electron that experiences the minimum effective nuclear charge.

b. 2 points

An unknown anion (A^-) has the following *excited* electronic configuration: $[Ar] 4s^2 3d^3 4p^1$

(Ar: Argon; $Z = 18$)

Give the **condensed ground state electronic configuration** of the atom, **A** and the cation, **A^{2+}** .
Show your work. (Assume all electronic transitions are possible)

Electronic Configuration of A:

Electronic Configuration of A^{2+} :

c. 2 points

In a single atom, determine how many electrons can have the following set of quantum numbers?

(i) $n = 5; l = 3$

In a single atom, determine how many electrons can have the following set of quantum numbers?

(i) $n = 3; l = 2; m_s = +1/2$

**Question 4** **8 points**

Rank the following from 1 to 4 in **increasing** order. Provide the ranking in the boxes provided.

Note: Atomic number (Z) are provided for relevant atoms

a) Increasing (lowest to highest) order of *ionization energy*: (1= lowest; 4 = highest)

Mg

Mg: Z = 12

Rb

Rb: Z = 37

Cs

Cs: Z = 55

K

K : Z = 19

b) Increasing (lowest to highest) order of *bond angle*: (1= lowest; 4 = highest)

H₂OCCl₄XeF₄

HCN

c) Increasing (smallest to largest) order of *ionic size* (1= smallest; 4 = largest)

S²⁻

S: Z = 16

O²⁻

O: Z = 8

Be²⁺

Be: Z = 4

Na⁺

Na: Z = 11

d) Increasing (lowest to highest) order of *bond energy* (1= lowest; 4 = highest)

P-F

P-Cl

P-I

P-Br

**Question 5** **12 points**

a. 9 points For the molecules HClO_4 and TeF_6 , draw the most stable *Lewis structure* showing appropriate shape and *dashed/wedged bonds*, based on VSEPR. Determine the **molecular geometry** around the central atom. For HClO_4 , label the *most polar bond*.



b. 3 points (Note: Lone pairs are not shown in structures below)

(i) For the three *resonance* structures (1, 2 and 3) shown below, answer the following question:



(I) Circle the *most contributing* resonance structure.

Structure 1

Structure 2

Structure 3

(ii) For the three structures given (4, 5, and 6) below, answer the following questions:



(I) Circle all the structures that represent *resonance structures* of each other:

Structure 4

Structure 5

Structure 6

(II) Among the resonance structures, circle the *most contributing* resonance structure.

Structure 4

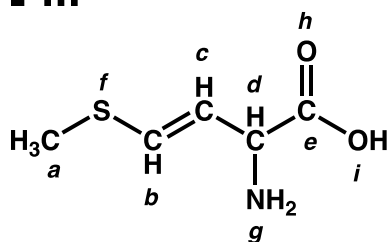
Structure 5

Structure 6

**Question 6** **6 points**

Using the data given below, calculate the **lattice energy** for SrO(s).

	ΔH° (kJ/mol)
$\text{Sr}(s) \longrightarrow \text{Sr}(g)$	+180
$\text{Sr}(g) \longrightarrow \text{Sr}^{2+}(g) + 2e^-$	+1468
$\text{O}_2(g) \longrightarrow 2\text{O}(g)$	+500
$\text{O}(g) + 2e^- \longrightarrow \text{O}^{2-}(g)$	+600
$\text{Sr}(s) + \frac{1}{2}\text{O}_2(g) \longrightarrow \text{SrO}(s)$	-550

**Question 7 10 points**

Consider the molecule given on the left, and answer questions (i) to (vi) using chemical bonding theories that we have learnt in class (valence bond theory, hybridization).

Note that the lone pairs are not shown in the structure:

i. Consider the carbon atoms marked in as ***a***, ***b***, ***c***, ***d***, and ***e*** in the molecule, and determine **hybridization** at each of the carbon centers. (2.5 points)

Atom	Hybridization
Carbon (<i>a</i>)	
Carbon (<i>b</i>)	
Carbon (<i>c</i>)	
Carbon (<i>d</i>)	
Carbon (<i>e</i>)	

ii. Now, consider the sulfur atom (S) marked as ***f***, nitrogen (N) atom (***g***), and oxygen(O) atoms (***h*** and ***i***), and determine their **hybridization**. (2 points)

Atom	Hybridization
Sulfur (<i>f</i>)	
Nitrogen (<i>g</i>)	
Oxygen (<i>h</i>)	
Oxygen (<i>i</i>)	

iii. **Describe** all the bonds formed by the carbon atoms ***a*** and ***b***, using sigma and pi designation as appropriate. (2 points)

Atom	Bonds formed (use sigma and pi as appropriate)
Carbon (<i>a</i>)	
Carbon (<i>b</i>)	

iv. How many total sp^3 -s sigma bonds are in the molecule? (1 point)

v. Determine **electronic** and **molecular geometry** at carbon (C) labeled ***e*** and sulfur (S) labeled ***f*** centres. (2 points)

Atom	Electronic Geometry	Molecular Geometry
Carbon (<i>e</i>)		
Sulfur (<i>f</i>)		

vi. Circle the value from the choices given below, that describes the **C-S-C bond angle at sulfur** (S) marked ***f***: (0.5 points)

>90° to <109.5° 109.5° >109.5 to <120° 120° >120° to <180°


Question 8 12 points

Using the **valence shell orbitals only** draw a **complete** molecular orbital (MO) diagram (with no s-p mixing) for the molecule **PS**, in the space provided below. **(6 points)**

Assume that 3s and 3p atomic orbitals will form molecular orbitals like those formed by the combination of 2s and 2p atomic orbitals that we learnt in class, and there is no s-p mixing.

Show **energy arrow**, **atomic orbitals** involved, **atom labels**, the **molecular orbitals** that are formed, **filled electrons** in the MO diagram etc. Make sure to **label all the atomic and molecular orbitals**.

Using the MO diagram above, answer the following questions:

- (i) Which one of the following will have the highest bond order? Circle the correct choice
 A) PS^{1-} B) PS C) PS^{1+}
- (ii) Which one of the following will have the longest bond length? Circle the correct choice
 A) PS^{1-} B) PS C) PS^{1+}
- (iii) Which of the following will be diamagnetic? Circle the correct choice.
 A) PS^{1-} B) PS C) PS^{1+}
- (iv) Draw Lewis dot structure of the molecule with the shortest bond length (based on the MO diagram) from among the three (PS^{1-} , PS, PS^{1+}), and indicate the bond order calculated from the Lewis dot structure. **(3 points)**

Lewis Dot Structure

Bond Order



CABE9E2D-3A23-4CF6-9B53-4044156B1811

midterm-exam-93388

#532 Page 10 of 10

SCRAP PAGE

**CHEMISTRY 110 –Midterm Exam****Thursday November 7, 2024, 6:30 – 8:30 pm****Examiners: Pallavi Sirjoosingh and Ashok Kakkar**

First name (please write as legibly as possible within the boxes)

--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--

Last name

--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--

Student ID number

--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--

This booklet has 8 questions on 10 pages (The last page can be used as a scrap page).

The staple must NOT be removed from the question booklet. Your intact question booklet must be returned at the end of the exam.

A separate booklet contains a periodic table, and datasheet.

Write your answers directly under the questions in the space provided. Do not write on the QR codes or near the top edges (as it may be cut off).

Please write **legibly and dark enough** for the scan to be readable. For students who use English as a second language, dictionaries are permitted. Calculators are allowed.

DO NOT ASK QUESTIONS OF THE INVIGILATORS about interpretations of the exam questions. If you have a concern about a question, please summarize it at the top of this page. All concerns will be evaluated after the exam. GOOD LUCK!

**Question 1 8 points** (*Assume all transitions are possible*)

a. 6 points Calculate the **energy (in kJ/mol)** required to ionize (completely remove) the electron from the **$n=3$** level of **He^+ ion** (He: Helium). Show your work including equations. (3 sig. figs.)

b. 2 points

(I) Select the largest (longest) wavelength from the three choices below:

- i. Wavelength of photon to *completely remove* an electron from *1s orbital of Hydrogen atom*
- ii. Wavelength of photon to *completely remove* an electron from *1s orbital of Boron atom (B)*
- iii. Wavelength of photon to *completely remove* an electron from *1s orbital of Boron cation (B^+)*

i or ii or iii

(II) Select the smallest (shortest) wavelength from the three choices below (*for H atom*):

- i. Wavelength of photon *emitted* for an electron transitioning from *2p to 1s orbital*
- ii. Wavelength of photon *absorbed* for an electron transitioning from *2s to 4p orbital*
- iii. Wavelength of photon *absorbed* for an electron transitioning from *1s to 3p orbital*

i or ii or iii



Question 2 8 points



A Chem110 student plots the graph shown on the left, from data obtained for the kinetic energy (eV) of the ejected electron versus frequency (s^{-1}) of the incident light, for an unknown metal. ($1\text{eV} = 1.6022 \times 10^{-19} \text{ J}$)

Based on the plot, answer the questions **a to c** below:

a. 1 point On the plot above, label the **threshold frequency**.

b. 2 points What is the value of the work potential (in J) of the unknown metal?

c. 3 points What is the *longest* wavelength (in nm) required to eject an electron from this metal's surface? Show your work including equation(s) used. (3 sig figs)

d. 2 points In the space provided below, plot the following

(Assume frequency is *above* threshold frequency):

- Kinetic energy* of an ejected electron versus the *intensity* of the incident light (left)
- Number of electrons* ejected versus the *frequency* of the incident light (right)



**Question 3** **6 points****a. 2 points**

In the ground state of Ge (Germanium; $Z = 32$) atom, **name** the following orbitals:

i. Orbital containing the electron that requires the maximum energy to be removed.

ii. Orbital containing the electron that experiences the minimum effective nuclear charge.

b. 2 points

An unknown anion (A^-) has the following *excited* electronic configuration: $[Ar] 4s^2 3d^3 4p^1$

(Ar: Argon; $Z = 18$)

Give the **condensed ground state electronic configuration** of the atom, **A** and the cation, **A^{2+}** .
Show your work. (Assume all electronic transitions are possible)

Electronic Configuration of A:

Electronic Configuration of A^{2+} :

c. 2 points

In a single atom, determine how many electrons can have the following set of quantum numbers?

(i) $n = 5; l = 3$

In a single atom, determine how many electrons can have the following set of quantum numbers?

(i) $n = 3; l = 2; m_s = +1/2$

**Question 4** **8 points**

Rank the following from 1 to 4 in **increasing** order. Provide the ranking in the boxes provided.

Note: Atomic number (Z) are provided for relevant atoms

a) Increasing (lowest to highest) order of *ionization energy*: (1= lowest; 4 = highest)

Mg

Mg: Z = 12

Rb

Rb: Z = 37

Cs

Cs: Z = 55

K

K : Z = 19

b) Increasing (lowest to highest) order of *bond angle*: (1= lowest; 4 = highest)

H₂OCCl₄XeF₄

HCN

c) Increasing (smallest to largest) order of *ionic size* (1= smallest; 4 = largest)

S²⁻

S: Z = 16

O²⁻

O: Z = 8

Be²⁺

Be: Z = 4

Na⁺

Na: Z = 11

d) Increasing (lowest to highest) order of *bond energy* (1= lowest; 4 = highest)

P-F

P-Cl

P-I

P-Br

**Question 5** **12 points**

a. 9 points For the molecules HClO_4 and TeF_6 , draw the most stable *Lewis structure* showing appropriate shape and *dashed/wedged bonds*, based on VSEPR. Determine the **molecular geometry** around the central atom. For HClO_4 , label the *most polar bond*.



b. 3 points (Note: Lone pairs are not shown in structures below)

(i) For the three *resonance* structures (1, 2 and 3) shown below, answer the following question:



(I) Circle the *most contributing* resonance structure.

Structure 1

Structure 2

Structure 3

(ii) For the three structures given (4, 5, and 6) below, answer the following questions:



(I) Circle all the structures that represent *resonance structures* of each other:

Structure 4

Structure 5

Structure 6

(II) Among the resonance structures, circle the *most contributing* resonance structure.

Structure 4

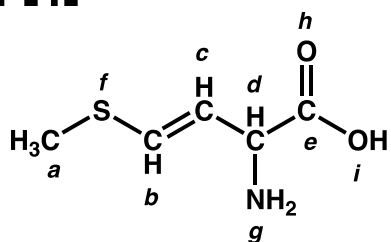
Structure 5

Structure 6

**Question 6** **6 points**

Using the data given below, calculate the **lattice energy** for SrO(s).

	ΔH° (kJ/mol)
$\text{Sr}(s) \longrightarrow \text{Sr}(g)$	+180
$\text{Sr}(g) \longrightarrow \text{Sr}^{2+}(g) + 2e^-$	+1468
$\text{O}_2(g) \longrightarrow 2\text{O}(g)$	+500
$\text{O}(g) + 2e^- \longrightarrow \text{O}^{2-}(g)$	+600
$\text{Sr}(s) + \frac{1}{2}\text{O}_2(g) \longrightarrow \text{SrO}(s)$	-550

**Question 7 10 points**

Consider the molecule given on the left, and answer questions (i) to (vi) using chemical bonding theories that we have learnt in class (valence bond theory, hybridization).

Note that the lone pairs are not shown in the structure:

i. Consider the carbon atoms marked in as ***a***, ***b***, ***c***, ***d***, and ***e*** in the molecule, and determine **hybridization** at each of the carbon centers. (2.5 points)

Atom	Hybridization
Carbon (<i>a</i>)	
Carbon (<i>b</i>)	
Carbon (<i>c</i>)	
Carbon (<i>d</i>)	
Carbon (<i>e</i>)	

ii. Now, consider the sulfur atom (S) marked as ***f***, nitrogen (N) atom (***g***), and oxygen(O) atoms (***h*** and ***i***), and determine their **hybridization**. (2 points)

Atom	Hybridization
Sulfur (<i>f</i>)	
Nitrogen (<i>g</i>)	
Oxygen (<i>h</i>)	
Oxygen (<i>i</i>)	

iii. **Describe** all the bonds formed by the carbon atoms ***a*** and ***b***, using sigma and pi designation as appropriate. (2 points)

Atom	Bonds formed (use sigma and pi as appropriate)
Carbon (<i>a</i>)	
Carbon (<i>b</i>)	

iv. How many total sp^3 -s sigma bonds are in the molecule? (1 point)

v. Determine **electronic** and **molecular geometry** at carbon (C) labeled ***e*** and sulfur (S) labeled ***f*** centres. (2 points)

Atom	Electronic Geometry	Molecular Geometry
Carbon (<i>e</i>)		
Sulfur (<i>f</i>)		

vi. Circle the value from the choices given below, that describes the **C-S-C bond angle at sulfur** (S) marked ***f***: (0.5 points)

>90° to <109.5° 109.5° >109.5 to <120° 120° >120° to <180°


Question 8 12 points

Using the **valence shell orbitals only** draw a **complete** molecular orbital (MO) diagram (with no s-p mixing) for the molecule **PS**, in the space provided below. **(6 points)**

Assume that 3s and 3p atomic orbitals will form molecular orbitals like those formed by the combination of 2s and 2p atomic orbitals that we learnt in class, and there is no s-p mixing.

Show **energy arrow**, **atomic orbitals** involved, **atom labels**, the **molecular orbitals** that are formed, **filled electrons** in the MO diagram etc. Make sure to label all the atomic and molecular orbitals.

Using the MO diagram above, answer the following questions:

- (i) Which one of the following will have the highest bond order? Circle the correct choice
 A) PS^{1-} B) PS C) PS^{1+}
- (ii) Which one of the following will have the longest bond length? Circle the correct choice
 A) PS^{1-} B) PS C) PS^{1+}
- (iii) Which of the following will be diamagnetic? Circle the correct choice.
 A) PS^{1-} B) PS C) PS^{1+}
- (iv) Draw Lewis dot structure of the molecule with the shortest bond length (based on the MO diagram) from among the three (PS^{1-} , PS, PS^{1+}), and indicate the bond order calculated from the Lewis dot structure. **(3 points)**

Lewis Dot Structure

Bond Order



F5FEC6BF-8EBF-4DB6-897A-BDB435E43974

midterm-exam-93388

#533 Page 10 of 10

SCRAP PAGE

**CHEMISTRY 110 –Midterm Exam****Thursday November 7, 2024, 6:30 – 8:30 pm****Examiners: Pallavi Sirjoosingh and Ashok Kakkar**

First name (please write as legibly as possible within the boxes)

--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--

Last name

--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--

Student ID number

--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--

This booklet has 8 questions on 10 pages (The last page can be used as a scrap page).

The staple must NOT be removed from the question booklet. Your intact question booklet must be returned at the end of the exam.

A separate booklet contains a periodic table, and datasheet.

Write your answers directly under the questions in the space provided. Do not write on the QR codes or near the top edges (as it may be cut off).

Please write **legibly and dark enough** for the scan to be readable. For students who use English as a second language, dictionaries are permitted. Calculators are allowed.

DO NOT ASK QUESTIONS OF THE INVIGILATORS about interpretations of the exam questions. If you have a concern about a question, please summarize it at the top of this page. All concerns will be evaluated after the exam. GOOD LUCK!

**Question 1 8 points** (*Assume all transitions are possible*)

a. 6 points Calculate the **energy (in kJ/mol)** required to ionize (completely remove) the electron from the **n=3** level of **He⁺ ion** (He: Helium). Show your work including equations. (3 sig. figs.)

b. 2 points

(I) Select the largest (longest) wavelength from the three choices below:

- i. Wavelength of photon to *completely remove* an electron from *1s orbital of Hydrogen atom*
- ii. Wavelength of photon to *completely remove* an electron from *1s orbital of Boron atom (B)*
- iii. Wavelength of photon to *completely remove* an electron from *1s orbital of Boron cation (B⁺)*

i or ii or iii

(II) Select the smallest (shortest) wavelength from the three choices below (*for H atom*):

- i. Wavelength of photon *emitted* for an electron transitioning from *2p to 1s orbital*
- ii. Wavelength of photon *absorbed* for an electron transitioning from *2s to 4p orbital*
- iii. Wavelength of photon *absorbed* for an electron transitioning from *1s to 3p orbital*

i or ii or iii



Question 2 8 points



A Chem110 student plots the graph shown on the left, from data obtained for the kinetic energy (eV) of the ejected electron versus frequency (s^{-1}) of the incident light, for an unknown metal. ($1\text{eV} = 1.6022 \times 10^{-19} \text{ J}$)

Based on the plot, answer the questions **a to c** below:

a. 1 point On the plot above, label the **threshold frequency**.

b. 2 points What is the value of the work potential (in J) of the unknown metal?

c. 3 points What is the *longest* wavelength (in nm) required to eject an electron from this metal's surface? Show your work including equation(s) used. (3 sig figs)

d. 2 points In the space provided below, plot the following

(Assume frequency is *above* threshold frequency):

- Kinetic energy* of an ejected electron versus the *intensity* of the incident light (left)
- Number of electrons* ejected versus the *frequency* of the incident light (right)



**Question 3** **6 points****a. 2 points**

In the ground state of Ge (Germanium; $Z = 32$) atom, **name** the following orbitals:

i. Orbital containing the electron that requires the maximum energy to be removed.

ii. Orbital containing the electron that experiences the minimum effective nuclear charge.

b. 2 points

An unknown anion (A^-) has the following *excited* electronic configuration: $[Ar] 4s^2 3d^3 4p^1$

(Ar: Argon; $Z = 18$)

Give the **condensed ground state electronic configuration** of the atom, **A** and the cation, **A^{2+}** .
Show your work. (Assume all electronic transitions are possible)

Electronic Configuration of A:

Electronic Configuration of A^{2+} :

c. 2 points

In a single atom, determine how many electrons can have the following set of quantum numbers?

(i) $n = 5; l = 3$

In a single atom, determine how many electrons can have the following set of quantum numbers?

(i) $n = 3; l = 2; m_s = +1/2$

**Question 4** **8 points**

Rank the following from 1 to 4 in **increasing** order. Provide the ranking in the boxes provided.

Note: Atomic number (Z) are provided for relevant atoms

a) Increasing (lowest to highest) order of *ionization energy*: (1= lowest; 4 = highest)

Mg

Mg: Z = 12

Rb

Rb: Z = 37

Cs

Cs: Z = 55

K

K : Z = 19

b) Increasing (lowest to highest) order of *bond angle*: (1= lowest; 4 = highest)

H₂OCCl₄XeF₄

HCN

c) Increasing (smallest to largest) order of *ionic size* (1= smallest; 4 = largest)

S²⁻

S: Z = 16

O²⁻

O: Z = 8

Be²⁺

Be: Z = 4

Na⁺

Na: Z = 11

d) Increasing (lowest to highest) order of *bond energy* (1= lowest; 4 = highest)

P-F

P-Cl

P-I

P-Br

**Question 5** **12 points**

a. 9 points For the molecules HClO_4 and TeF_6 , draw the most stable *Lewis structure* showing appropriate shape and *dashed/wedged bonds*, based on VSEPR. Determine the **molecular geometry** around the central atom. For HClO_4 , label the *most polar bond*.



b. 3 points (Note: Lone pairs are not shown in structures below)

(i) For the three *resonance* structures (1, 2 and 3) shown below, answer the following question:



(I) Circle the *most contributing* resonance structure.

Structure 1

Structure 2

Structure 3

(ii) For the three structures given (4, 5, and 6) below, answer the following questions:



(I) Circle all the structures that represent *resonance structures* of each other:

Structure 4

Structure 5

Structure 6

(II) Among the resonance structures, circle the *most contributing* resonance structure.

Structure 4

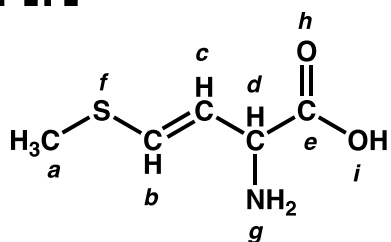
Structure 5

Structure 6

**Question 6** **6 points**

Using the data given below, calculate the **lattice energy** for SrO(s).

	ΔH° (kJ/mol)
$\text{Sr}(s) \longrightarrow \text{Sr}(g)$	+180
$\text{Sr}(g) \longrightarrow \text{Sr}^{2+}(g) + 2e^-$	+1468
$\text{O}_2(g) \longrightarrow 2\text{O}(g)$	+500
$\text{O}(g) + 2e^- \longrightarrow \text{O}^{2-}(g)$	+600
$\text{Sr}(s) + \frac{1}{2}\text{O}_2(g) \longrightarrow \text{SrO}(s)$	-550

**Question 7 10 points**

Consider the molecule given on the left, and answer questions (i) to (vi) using chemical bonding theories that we have learnt in class (valence bond theory, hybridization).

Note that the lone pairs are not shown in the structure:

i. Consider the carbon atoms marked in as ***a***, ***b***, ***c***, ***d***, and ***e*** in the molecule, and determine **hybridization** at each of the carbon centers. (2.5 points)

Atom	Hybridization
Carbon (<i>a</i>)	
Carbon (<i>b</i>)	
Carbon (<i>c</i>)	
Carbon (<i>d</i>)	
Carbon (<i>e</i>)	

ii. Now, consider the sulfur atom (S) marked as ***f***, nitrogen (N) atom (***g***), and oxygen(O) atoms (***h*** and ***i***), and determine their **hybridization**. (2 points)

Atom	Hybridization
Sulfur (<i>f</i>)	
Nitrogen (<i>g</i>)	
Oxygen (<i>h</i>)	
Oxygen (<i>i</i>)	

iii. **Describe** all the bonds formed by the carbon atoms ***a*** and ***b***, using sigma and pi designation as appropriate. (2 points)

Atom	Bonds formed (use sigma and pi as appropriate)
Carbon (<i>a</i>)	
Carbon (<i>b</i>)	

iv. How many total sp^3 -s sigma bonds are in the molecule? (1 point)

v. Determine **electronic** and **molecular geometry** at carbon (C) labeled ***e*** and sulfur (S) labeled ***f*** centres. (2 points)

Atom	Electronic Geometry	Molecular Geometry
Carbon (<i>e</i>)		
Sulfur (<i>f</i>)		

vi. Circle the value from the choices given below, that describes the **C-S-C bond angle at sulfur** (S) marked ***f***: (0.5 points)

>90° to <109.5° 109.5° >109.5 to <120° 120° >120° to <180°

**Question 8 12 points**

Using the **valence shell orbitals only** draw a **complete** molecular orbital (MO) diagram (with no s-p mixing) for the molecule **PS**, in the space provided below. **(6 points)**

Assume that 3s and 3p atomic orbitals will form molecular orbitals like those formed by the combination of 2s and 2p atomic orbitals that we learnt in class, and there is no s-p mixing.

Show **energy arrow**, **atomic orbitals** involved, **atom labels**, the **molecular orbitals** that are formed, **filled electrons** in the MO diagram etc. Make sure to label all the atomic and molecular orbitals.

Using the MO diagram above, answer the following questions:

- (i) Which one of the following will have the highest bond order? Circle the correct choice
A) PS^{1-} B) PS C) PS^{1+}
- (ii) Which one of the following will have the longest bond length? Circle the correct choice
A) PS^{1-} B) PS C) PS^{1+}
- (iii) Which of the following will be diamagnetic? Circle the correct choice.
A) PS^{1-} B) PS C) PS^{1+}
- (iv) Draw Lewis dot structure of the molecule with the shortest bond length (based on the MO diagram) from among the three (PS^{1-} , PS, PS^{1+}), and indicate the bond order calculated from the Lewis dot structure. **(3 points)**

Lewis Dot Structure

Bond Order



8C090531-DFED-4B09-89EF-65E9D0345FB5

midterm-exam-93388

#534 Page 10 of 10

SCRAP PAGE

**CHEMISTRY 110 –Midterm Exam****Thursday November 7, 2024, 6:30 – 8:30 pm****Examiners: Pallavi Sirjoosingh and Ashok Kakkar**

First name (please write as legibly as possible within the boxes)

--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--

Last name

--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--

Student ID number

--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--

This booklet has 8 questions on 10 pages (The last page can be used as a scrap page).

The staple must NOT be removed from the question booklet. Your intact question booklet must be returned at the end of the exam.

A separate booklet contains a periodic table, and datasheet.

Write your answers directly under the questions in the space provided. Do not write on the QR codes or near the top edges (as it may be cut off).

Please write **legibly and dark enough** for the scan to be readable. For students who use English as a second language, dictionaries are permitted. Calculators are allowed.

DO NOT ASK QUESTIONS OF THE INVIGILATORS about interpretations of the exam questions. If you have a concern about a question, please summarize it at the top of this page. All concerns will be evaluated after the exam. GOOD LUCK!

**Question 1 8 points** (*Assume all transitions are possible*)

a. 6 points Calculate the **energy (in kJ/mol)** required to ionize (completely remove) the electron from the **n=3** level of **He⁺ ion** (He: Helium). Show your work including equations. (3 sig. figs.)

b. 2 points

(I) Select the largest (longest) wavelength from the three choices below:

- i. Wavelength of photon to *completely remove* an electron from *1s orbital of Hydrogen atom*
- ii. Wavelength of photon to *completely remove* an electron from *1s orbital of Boron atom (B)*
- iii. Wavelength of photon to *completely remove* an electron from *1s orbital of Boron cation (B⁺)*

i or ii or iii

(II) Select the smallest (shortest) wavelength from the three choices below (*for H atom*):

- i. Wavelength of photon *emitted* for an electron transitioning from *2p to 1s orbital*
- ii. Wavelength of photon *absorbed* for an electron transitioning from *2s to 4p orbital*
- iii. Wavelength of photon *absorbed* for an electron transitioning from *1s to 3p orbital*

i or ii or iii



Question 2 8 points



A Chem110 student plots the graph shown on the left, from data obtained for the kinetic energy (eV) of the ejected electron versus frequency (s^{-1}) of the incident light, for an unknown metal. ($1\text{eV} = 1.6022 \times 10^{-19} \text{ J}$)

Based on the plot, answer the questions **a to c** below:

a. 1 point On the plot above, label the **threshold frequency**.

b. 2 points What is the value of the work potential (in J) of the unknown metal?

c. 3 points What is the *longest* wavelength (in nm) required to eject an electron from this metal's surface? Show your work including equation(s) used. (3 sig figs)

d. 2 points In the space provided below, plot the following

(Assume frequency is *above* threshold frequency):

- i. *Kinetic energy* of an ejected electron versus the *intensity* of the incident light (left)
- ii. *Number of electrons* ejected versus the *frequency* of the incident light (right)



**Question 3** **6 points****a. 2 points**

In the ground state of Ge (Germanium; $Z = 32$) atom, **name** the following orbitals:

i. Orbital containing the electron that requires the maximum energy to be removed.

ii. Orbital containing the electron that experiences the minimum effective nuclear charge.

b. 2 points

An unknown anion (A^-) has the following *excited* electronic configuration: $[Ar] 4s^2 3d^3 4p^1$

(Ar: Argon; $Z = 18$)

Give the **condensed ground state electronic configuration** of the atom, **A** and the cation, **A^{2+}** .
Show your work. (Assume all electronic transitions are possible)

Electronic Configuration of A:

Electronic Configuration of A^{2+} :

c. 2 points

In a single atom, determine how many electrons can have the following set of quantum numbers?

(i) $n = 5; l = 3$

In a single atom, determine how many electrons can have the following set of quantum numbers?

(i) $n = 3; l = 2; m_s = +1/2$

**Question 4** **8 points**

Rank the following from 1 to 4 in **increasing** order. Provide the ranking in the boxes provided.

Note: Atomic number (Z) are provided for relevant atoms

a) Increasing (lowest to highest) order of *ionization energy*: (1= lowest; 4 = highest)

Mg

Mg: Z = 12

Rb

Rb: Z = 37

Cs

Cs: Z = 55

K

K : Z = 19

b) Increasing (lowest to highest) order of *bond angle*: (1= lowest; 4 = highest)

H₂OCCl₄XeF₄

HCN

c) Increasing (smallest to largest) order of *ionic size* (1= smallest; 4 = largest)

S²⁻

S: Z = 16

O²⁻

O: Z = 8

Be²⁺

Be: Z = 4

Na⁺

Na: Z = 11

d) Increasing (lowest to highest) order of *bond energy* (1= lowest; 4 = highest)

P-F

P-Cl

P-I

P-Br

**Question 5** **12 points**

a. 9 points For the molecules HClO_4 and TeF_6 , draw the most stable *Lewis structure* showing appropriate shape and *dashed/wedged bonds*, based on VSEPR. Determine the **molecular geometry** around the central atom. For HClO_4 , label the *most polar bond*.



b. 3 points (Note: Lone pairs are not shown in structures below)

(i) For the three *resonance* structures (1, 2 and 3) shown below, answer the following question:



(I) Circle the *most contributing* resonance structure.

Structure 1

Structure 2

Structure 3

(ii) For the three structures given (4, 5, and 6) below, answer the following questions:



(I) Circle all the structures that represent *resonance structures* of each other:

Structure 4

Structure 5

Structure 6

(II) Among the resonance structures, circle the *most contributing* resonance structure.

Structure 4

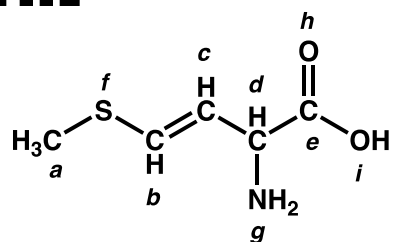
Structure 5

Structure 6

**Question 6** **6 points**

Using the data given below, calculate the **lattice energy** for SrO(s).

	ΔH° (kJ/mol)
$\text{Sr}(s) \longrightarrow \text{Sr}(g)$	+180
$\text{Sr}(g) \longrightarrow \text{Sr}^{2+}(g) + 2e^-$	+1468
$\text{O}_2(g) \longrightarrow 2\text{O}(g)$	+500
$\text{O}(g) + 2e^- \longrightarrow \text{O}^{2-}(g)$	+600
$\text{Sr}(s) + \frac{1}{2}\text{O}_2(g) \longrightarrow \text{SrO}(s)$	-550

**Question 7 10 points**

Consider the molecule given on the left, and answer questions (i) to (vi) using chemical bonding theories that we have learnt in class (valence bond theory, hybridization).

Note that the lone pairs are not shown in the structure:

i. Consider the carbon atoms marked in as ***a***, ***b***, ***c***, ***d***, and ***e*** in the molecule, and determine **hybridization** at each of the carbon centers. (2.5 points)

Atom	Hybridization
Carbon (<i>a</i>)	
Carbon (<i>b</i>)	
Carbon (<i>c</i>)	
Carbon (<i>d</i>)	
Carbon (<i>e</i>)	

ii. Now, consider the sulfur atom (S) marked as ***f***, nitrogen (N) atom (***g***), and oxygen(O) atoms (***h*** and ***i***), and determine their **hybridization**. (2 points)

Atom	Hybridization
Sulfur (<i>f</i>)	
Nitrogen (<i>g</i>)	
Oxygen (<i>h</i>)	
Oxygen (<i>i</i>)	

iii. **Describe** all the bonds formed by the carbon atoms ***a*** and ***b***, using sigma and pi designation as appropriate. (2 points)

Atom	Bonds formed (use sigma and pi as appropriate)
Carbon (<i>a</i>)	
Carbon (<i>b</i>)	

iv. How many total sp^3 -s sigma bonds are in the molecule? (1 point)

v. Determine **electronic** and **molecular geometry** at carbon (C) labeled ***e*** and sulfur (S) labeled ***f*** centres. (2 points)

Atom	Electronic Geometry	Molecular Geometry
Carbon (<i>e</i>)		
Sulfur (<i>f</i>)		

vi. Circle the value from the choices given below, that describes the **C-S-C bond angle at sulfur** (S) marked ***f***: (0.5 points)

>90° to <109.5° 109.5° >109.5 to <120° 120° >120° to <180°


Question 8 12 points

Using the **valence shell orbitals only** draw a **complete** molecular orbital (MO) diagram (with no s-p mixing) for the molecule **PS**, in the space provided below. **(6 points)**

Assume that 3s and 3p atomic orbitals will form molecular orbitals like those formed by the combination of 2s and 2p atomic orbitals that we learnt in class, and there is no s-p mixing.

Show **energy arrow**, **atomic orbitals** involved, **atom labels**, the **molecular orbitals** that are formed, **filled electrons** in the MO diagram etc. Make sure to label all the atomic and molecular orbitals.

Using the MO diagram above, answer the following questions:

- (i) Which one of the following will have the highest bond order? Circle the correct choice
 A) PS^{1-} B) PS C) PS^{1+}
- (ii) Which one of the following will have the longest bond length? Circle the correct choice
 A) PS^{1-} B) PS C) PS^{1+}
- (iii) Which of the following will be diamagnetic? Circle the correct choice.
 A) PS^{1-} B) PS C) PS^{1+}
- (iv) Draw Lewis dot structure of the molecule with the shortest bond length (based on the MO diagram) from among the three (PS^{1-} , PS, PS^{1+}), and indicate the bond order calculated from the Lewis dot structure. **(3 points)**

Lewis Dot Structure

Bond Order



D7E85CE8-989C-4EC7-B1D0-BB368B40B063

midterm-exam-93388

#535 Page 10 of 10

SCRAP PAGE

**CHEMISTRY 110 –Midterm Exam****Thursday November 7, 2024, 6:30 – 8:30 pm**

Examiners: Pallavi Sirjoosingh and Ashok Kakkar

First name (please write as legibly as possible within the boxes)

--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--

Last name

--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--

Student ID number

--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--

This booklet has 8 questions on 10 pages (The last page can be used as a scrap page).

The staple must NOT be removed from the question booklet. Your intact question booklet must be returned at the end of the exam.

A separate booklet contains a periodic table, and datasheet.

Write your answers directly under the questions in the space provided. Do not write on the QR codes or near the top edges (as it may be cut off).

Please write **legibly and dark enough** for the scan to be readable. For students who use English as a second language, dictionaries are permitted. Calculators are allowed.

DO NOT ASK QUESTIONS OF THE INVIGILATORS about interpretations of the exam questions. If you have a concern about a question, please summarize it at the top of this page. All concerns will be evaluated after the exam. GOOD LUCK!

**Question 1 8 points** (*Assume all transitions are possible*)

a. 6 points Calculate the **energy (in kJ/mol)** required to ionize (completely remove) the electron from the **n=3** level of **He⁺ ion** (He: Helium). Show your work including equations. (3 sig. figs.)

b. 2 points

(I) Select the largest (longest) wavelength from the three choices below:

- i. Wavelength of photon to *completely remove* an electron from *1s orbital of Hydrogen atom*
- ii. Wavelength of photon to *completely remove* an electron from *1s orbital of Boron atom (B)*
- iii. Wavelength of photon to *completely remove* an electron from *1s orbital of Boron cation (B⁺)*

i or ii or iii

(II) Select the smallest (shortest) wavelength from the three choices below (*for H atom*):

- i. Wavelength of photon *emitted* for an electron transitioning from *2p to 1s orbital*
- ii. Wavelength of photon *absorbed* for an electron transitioning from *2s to 4p orbital*
- iii. Wavelength of photon *absorbed* for an electron transitioning from *1s to 3p orbital*

i or ii or iii



Question 2 8 points



A Chem110 student plots the graph shown on the left, from data obtained for the kinetic energy (eV) of the ejected electron versus frequency (s^{-1}) of the incident light, for an unknown metal. ($1\text{eV} = 1.6022 \times 10^{-19} \text{ J}$)

Based on the plot, answer the questions **a to c** below:

a. 1 point On the plot above, label the **threshold frequency**.

b. 2 points What is the value of the work potential (in J) of the unknown metal?

c. 3 points What is the *longest* wavelength (in nm) required to eject an electron from this metal's surface? Show your work including equation(s) used. (3 sig figs)

d. 2 points In the space provided below, plot the following

(Assume frequency is *above* threshold frequency):

- i. *Kinetic energy* of an ejected electron versus the *intensity* of the incident light (left)
- ii. *Number of electrons* ejected versus the *frequency* of the incident light (right)



**Question 3** **6 points****a. 2 points**

In the ground state of Ge (Germanium; $Z = 32$) atom, **name** the following orbitals:

i. Orbital containing the electron that requires the maximum energy to be removed.

ii. Orbital containing the electron that experiences the minimum effective nuclear charge.

b. 2 points

An unknown anion (A^-) has the following *excited* electronic configuration: $[Ar] 4s^2 3d^3 4p^1$

(Ar: Argon; $Z = 18$)

Give the **condensed ground state electronic configuration** of the atom, **A** and the cation, **A^{2+}** .
Show your work. (Assume all electronic transitions are possible)

Electronic Configuration of A:

Electronic Configuration of A^{2+} :

c. 2 points

In a single atom, determine how many electrons can have the following set of quantum numbers?

(i) $n = 5; l = 3$

In a single atom, determine how many electrons can have the following set of quantum numbers?

(i) $n = 3; l = 2; m_s = +1/2$

**Question 4** **8 points**

Rank the following from 1 to 4 in **increasing** order. Provide the ranking in the boxes provided.

Note: Atomic number (Z) are provided for relevant atoms

a) Increasing (lowest to highest) order of *ionization energy*: (1= lowest; 4 = highest)

Mg

Mg: Z = 12

Rb

Rb: Z = 37

Cs

Cs: Z = 55

K

K : Z = 19

b) Increasing (lowest to highest) order of *bond angle*: (1= lowest; 4 = highest)

H₂OCCl₄XeF₄

HCN

c) Increasing (smallest to largest) order of *ionic size* (1= smallest; 4 = largest)

S²⁻

S: Z = 16

O²⁻

O: Z = 8

Be²⁺

Be: Z = 4

Na⁺

Na: Z = 11

d) Increasing (lowest to highest) order of *bond energy* (1= lowest; 4 = highest)

P-F

P-Cl

P-I

P-Br

**Question 5** **12 points**

a. 9 points For the molecules HClO_4 and TeF_6 , draw the most stable *Lewis structure* showing appropriate shape and *dashed/wedged bonds*, based on VSEPR. Determine the **molecular geometry** around the central atom. For HClO_4 , label the *most polar bond*.



b. 3 points (Note: Lone pairs are not shown in structures below)

(i) For the three *resonance* structures (1, 2 and 3) shown below, answer the following question:



(I) Circle the *most contributing* resonance structure.

Structure 1

Structure 2

Structure 3

(ii) For the three structures given (4, 5, and 6) below, answer the following questions:



(I) Circle all the structures that represent *resonance structures* of each other:

Structure 4

Structure 5

Structure 6

(II) Among the resonance structures, circle the *most contributing* resonance structure.

Structure 4

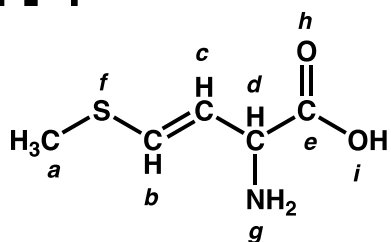
Structure 5

Structure 6

**Question 6** **6 points**

Using the data given below, calculate the **lattice energy** for SrO(s).

	ΔH° (kJ/mol)
$\text{Sr}(s) \longrightarrow \text{Sr}(g)$	+180
$\text{Sr}(g) \longrightarrow \text{Sr}^{2+}(g) + 2e^-$	+1468
$\text{O}_2(g) \longrightarrow 2\text{O}(g)$	+500
$\text{O}(g) + 2e^- \longrightarrow \text{O}^{2-}(g)$	+600
$\text{Sr}(s) + \frac{1}{2}\text{O}_2(g) \longrightarrow \text{SrO}(s)$	-550

**Question 7 10 points**

Consider the molecule given on the left, and answer questions (i) to (vi) using chemical bonding theories that we have learnt in class (valence bond theory, hybridization).

Note that the lone pairs are not shown in the structure:

i. Consider the carbon atoms marked in as ***a***, ***b***, ***c***, ***d***, and ***e*** in the molecule, and determine **hybridization** at each of the carbon centers. (2.5 points)

Atom	Hybridization
Carbon (<i>a</i>)	
Carbon (<i>b</i>)	
Carbon (<i>c</i>)	
Carbon (<i>d</i>)	
Carbon (<i>e</i>)	

ii. Now, consider the sulfur atom (S) marked as ***f***, nitrogen (N) atom (***g***), and oxygen(O) atoms (***h*** and ***i***), and determine their **hybridization**. (2 points)

Atom	Hybridization
Sulfur (<i>f</i>)	
Nitrogen (<i>g</i>)	
Oxygen (<i>h</i>)	
Oxygen (<i>i</i>)	

iii. **Describe** all the bonds formed by the carbon atoms ***a*** and ***b***, using sigma and pi designation as appropriate. (2 points)

Atom	Bonds formed (use sigma and pi as appropriate)
Carbon (<i>a</i>)	
Carbon (<i>b</i>)	

iv. How many total sp^3 -s sigma bonds are in the molecule? (1 point)

v. Determine **electronic** and **molecular geometry** at carbon (C) labeled ***e*** and sulfur (S) labeled ***f*** centres. (2 points)

Atom	Electronic Geometry	Molecular Geometry
Carbon (<i>e</i>)		
Sulfur (<i>f</i>)		

vi. Circle the value from the choices given below, that describes the **C-S-C bond angle at sulfur** (S) marked ***f***: (0.5 points)

>90° to <109.5°

109.5°

>109.5 to <120°

120°

>120° to <180°



Question 8 12 points

Using the **valence shell orbitals only** draw a **complete** molecular orbital (MO) diagram (with no s-p mixing) for the molecule **PS**, in the space provided below. **(6 points)**

Assume that 3s and 3p atomic orbitals will form molecular orbitals like those formed by the combination of 2s and 2p atomic orbitals that we learnt in class, and there is no s-p mixing.

Show **energy arrow**, **atomic orbitals** involved, **atom labels**, the **molecular orbitals** that are formed, **filled electrons** in the MO diagram etc. Make sure to label all the atomic and molecular orbitals.

Using the MO diagram above, answer the following questions:

- (i) Which one of the following will have the highest bond order? Circle the correct choice
 A) PS^{1-} B) PS C) PS^{1+}
- (ii) Which one of the following will have the longest bond length? Circle the correct choice
 A) PS^{1-} B) PS C) PS^{1+}
- (iii) Which of the following will be diamagnetic? Circle the correct choice.
 A) PS^{1-} B) PS C) PS^{1+}
- (iv) Draw Lewis dot structure of the molecule with the shortest bond length (based on the MO diagram) from among the three (PS^{1-} , PS, PS^{1+}), and indicate the bond order calculated from the Lewis dot structure. **(3 points)**

Lewis Dot Structure

Bond Order



45F220A5-6B2E-4AEF-AA9B-CD51534BD5E2

midterm-exam-93388

#536 Page 10 of 10

SCRAP PAGE

**CHEMISTRY 110 –Midterm Exam****Thursday November 7, 2024, 6:30 – 8:30 pm****Examiners: Pallavi Sirjoosingh and Ashok Kakkar**

First name (please write as legibly as possible within the boxes)

--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--

Last name

--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--

Student ID number

--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--

This booklet has 8 questions on 10 pages (The last page can be used as a scrap page).

The staple must NOT be removed from the question booklet. Your intact question booklet must be returned at the end of the exam.

A separate booklet contains a periodic table, and datasheet.

Write your answers directly under the questions in the space provided. Do not write on the QR codes or near the top edges (as it may be cut off).

Please write **legibly and dark enough** for the scan to be readable. For students who use English as a second language, dictionaries are permitted. Calculators are allowed.

DO NOT ASK QUESTIONS OF THE INVIGILATORS about interpretations of the exam questions. If you have a concern about a question, please summarize it at the top of this page. All concerns will be evaluated after the exam. GOOD LUCK!

**Question 1 8 points** (*Assume all transitions are possible*)

a. 6 points Calculate the **energy (in kJ/mol)** required to ionize (completely remove) the electron from the **$n=3$** level of **He^+ ion** (He: Helium). Show your work including equations. (3 sig. figs.)

b. 2 points

(I) Select the largest (longest) wavelength from the three choices below:

- i. Wavelength of photon to *completely remove* an electron from *1s orbital of Hydrogen atom*
- ii. Wavelength of photon to *completely remove* an electron from *1s orbital of Boron atom (B)*
- iii. Wavelength of photon to *completely remove* an electron from *1s orbital of Boron cation (B^+)*

i or ii or iii

(II) Select the smallest (shortest) wavelength from the three choices below (*for H atom*):

- i. Wavelength of photon *emitted* for an electron transitioning from *2p to 1s orbital*
- ii. Wavelength of photon *absorbed* for an electron transitioning from *2s to 4p orbital*
- iii. Wavelength of photon *absorbed* for an electron transitioning from *1s to 3p orbital*

i or ii or iii



Question 2 8 points



A Chem110 student plots the graph shown on the left, from data obtained for the kinetic energy (eV) of the ejected electron versus frequency (s^{-1}) of the incident light, for an unknown metal. ($1\text{eV} = 1.6022 \times 10^{-19} \text{ J}$)

Based on the plot, answer the questions **a to c** below:

a. 1 point On the plot above, label the **threshold frequency**.

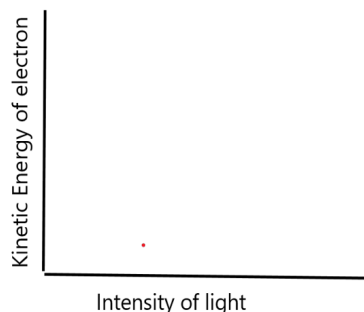
b. 2 points What is the value of the work potential (in J) of the unknown metal?

c. 3 points What is the *longest* wavelength (in nm) required to eject an electron from this metal's surface? Show your work including equation(s) used. (3 sig figs)

d. 2 points In the space provided below, plot the following

(Assume frequency is *above* threshold frequency):

- i. *Kinetic energy* of an ejected electron versus the *intensity* of the incident light (left)
- ii. *Number of electrons* ejected versus the *frequency* of the incident light (right)



**Question 3** **6 points****a. 2 points**

In the ground state of Ge (Germanium; $Z = 32$) atom, **name** the following orbitals:

i. Orbital containing the electron that requires the maximum energy to be removed.

ii. Orbital containing the electron that experiences the minimum effective nuclear charge.

b. 2 points

An unknown anion (A^-) has the following *excited* electronic configuration: $[Ar] 4s^2 3d^3 4p^1$

(Ar: Argon; $Z = 18$)

Give the **condensed ground state electronic configuration** of the atom, **A** and the cation, **A^{2+}** .
Show your work. (Assume all electronic transitions are possible)

Electronic Configuration of A:

Electronic Configuration of A^{2+} :

c. 2 points

In a single atom, determine how many electrons can have the following set of quantum numbers?

(i) $n = 5; l = 3$

In a single atom, determine how many electrons can have the following set of quantum numbers?

(i) $n = 3; l = 2; m_s = +1/2$

**Question 4** **8 points**

Rank the following from 1 to 4 in **increasing** order. Provide the ranking in the boxes provided.

Note: Atomic number (Z) are provided for relevant atoms

a) Increasing (lowest to highest) order of *ionization energy*: (1= lowest; 4 = highest)

Mg

Mg: Z = 12

Rb

Rb: Z = 37

Cs

Cs: Z = 55

K

K : Z = 19

b) Increasing (lowest to highest) order of *bond angle*: (1= lowest; 4 = highest)

H₂OCCl₄XeF₄

HCN

c) Increasing (smallest to largest) order of *ionic size* (1= smallest; 4 = largest)

S²⁻

S: Z = 16

O²⁻

O: Z = 8

Be²⁺

Be: Z = 4

Na⁺

Na: Z = 11

d) Increasing (lowest to highest) order of *bond energy* (1= lowest; 4 = highest)

P-F

P-Cl

P-I

P-Br

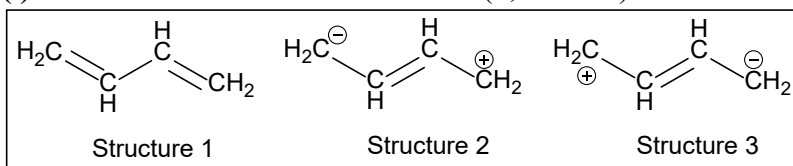
**Question 5** **12 points**

a. 9 points For the molecules HClO_4 and TeF_6 , draw the most stable *Lewis structure* showing appropriate shape and *dashed/wedged bonds*, based on VSEPR. Determine the **molecular geometry** around the central atom. For HClO_4 , label the *most polar bond*.



b. 3 points (Note: Lone pairs are not shown in structures below)

(i) For the three *resonance* structures (1, 2 and 3) shown below, answer the following question:



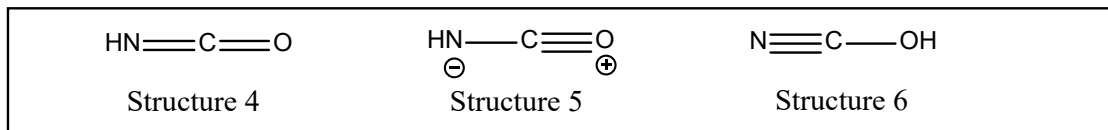
(I) Circle the *most contributing* resonance structure.

Structure 1

Structure 2

Structure 3

(ii) For the three structures given (4, 5, and 6) below, answer the following questions:



(I) Circle all the structures that represent *resonance structures* of each other:

Structure 4

Structure 5

Structure 6

(II) Among the resonance structures, circle the *most contributing* resonance structure.

Structure 4

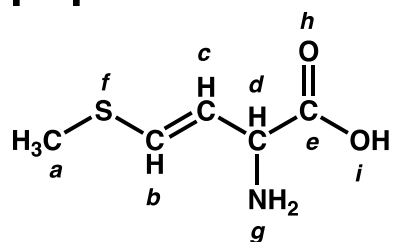
Structure 5

Structure 6

**Question 6** **6 points**

Using the data given below, calculate the **lattice energy** for SrO(s).

	ΔH° (kJ/mol)
$\text{Sr}(s) \longrightarrow \text{Sr}(g)$	+180
$\text{Sr}(g) \longrightarrow \text{Sr}^{2+}(g) + 2e^-$	+1468
$\text{O}_2(g) \longrightarrow 2\text{O}(g)$	+500
$\text{O}(g) + 2e^- \longrightarrow \text{O}^{2-}(g)$	+600
$\text{Sr}(s) + \frac{1}{2}\text{O}_2(g) \longrightarrow \text{SrO}(s)$	-550

**Question 7 10 points**

Consider the molecule given on the left, and answer questions (i) to (vi) using chemical bonding theories that we have learnt in class (valence bond theory, hybridization).

Note that the lone pairs are not shown in the structure:

i. Consider the carbon atoms marked in as ***a***, ***b***, ***c***, ***d***, and ***e*** in the molecule, and determine **hybridization** at each of the carbon centers. (2.5 points)

Atom	Hybridization
Carbon (<i>a</i>)	
Carbon (<i>b</i>)	
Carbon (<i>c</i>)	
Carbon (<i>d</i>)	
Carbon (<i>e</i>)	

ii. Now, consider the sulfur atom (S) marked as ***f***, nitrogen (N) atom (***g***), and oxygen(O) atoms (***h*** and ***i***), and determine their **hybridization**. (2 points)

Atom	Hybridization
Sulfur (<i>f</i>)	
Nitrogen (<i>g</i>)	
Oxygen (<i>h</i>)	
Oxygen (<i>i</i>)	

iii. **Describe** all the bonds formed by the carbon atoms ***a*** and ***b***, using sigma and pi designation as appropriate. (2 points)

Atom	Bonds formed (use sigma and pi as appropriate)
Carbon (<i>a</i>)	
Carbon (<i>b</i>)	

iv. How many total sp^3 -s sigma bonds are in the molecule? (1 point)

v. Determine **electronic** and **molecular geometry** at carbon (C) labeled ***e*** and sulfur (S) labeled ***f*** centres. (2 points)

Atom	Electronic Geometry	Molecular Geometry
Carbon (<i>e</i>)		
Sulfur (<i>f</i>)		

vi. Circle the value from the choices given below, that describes the **C-S-C bond angle at sulfur** (S) marked ***f***: (0.5 points)

>90° to <109.5°

109.5°

>109.5 to <120°

120°

>120° to <180°

**Question 8 12 points**

Using the **valence shell orbitals only** draw a **complete** molecular orbital (MO) diagram (with no s-p mixing) for the molecule **PS**, in the space provided below. **(6 points)**

Assume that 3s and 3p atomic orbitals will form molecular orbitals like those formed by the combination of 2s and 2p atomic orbitals that we learnt in class, and there is no s-p mixing.

Show **energy arrow**, **atomic orbitals** involved, **atom labels**, the **molecular orbitals** that are formed, **filled electrons** in the MO diagram etc. Make sure to **label all the atomic and molecular orbitals**.

Using the MO diagram above, answer the following questions:

- (i) Which one of the following will have the highest bond order? Circle the correct choice
A) PS^{1-} B) PS C) PS^{1+}
- (ii) Which one of the following will have the longest bond length? Circle the correct choice
A) PS^{1-} B) PS C) PS^{1+}
- (iii) Which of the following will be diamagnetic? Circle the correct choice.
A) PS^{1-} B) PS C) PS^{1+}
- (iv) Draw Lewis dot structure of the molecule with the shortest bond length (based on the MO diagram) from among the three (PS^{1-} , PS, PS^{1+}), and indicate the bond order calculated from the Lewis dot structure. **(3 points)**

Lewis Dot Structure

Bond Order



DD04FCA2-55DB-4375-8417-70374BDC8091

midterm-exam-93388

#537 Page 10 of 10

SCRAP PAGE

**CHEMISTRY 110 –Midterm Exam****Thursday November 7, 2024, 6:30 – 8:30 pm****Examiners: Pallavi Sirjoosingh and Ashok Kakkar**

First name (please write as legibly as possible within the boxes)

--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--

Last name

--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--

Student ID number

--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--

This booklet has 8 questions on 10 pages (The last page can be used as a scrap page).

The staple must NOT be removed from the question booklet. Your intact question booklet must be returned at the end of the exam.

A separate booklet contains a periodic table, and datasheet.

Write your answers directly under the questions in the space provided. Do not write on the QR codes or near the top edges (as it may be cut off).

Please write **legibly and dark enough** for the scan to be readable. For students who use English as a second language, dictionaries are permitted. Calculators are allowed.

DO NOT ASK QUESTIONS OF THE INVIGILATORS about interpretations of the exam questions. If you have a concern about a question, please summarize it at the top of this page. All concerns will be evaluated after the exam. GOOD LUCK!

**Question 1 8 points** (*Assume all transitions are possible*)

a. 6 points Calculate the **energy (in kJ/mol)** required to ionize (completely remove) the electron from the **$n=3$** level of **He^+ ion** (He: Helium). Show your work including equations. (3 sig. figs.)

b. 2 points

(I) Select the largest (longest) wavelength from the three choices below:

- i. Wavelength of photon to *completely remove* an electron from *1s orbital of Hydrogen atom*
- ii. Wavelength of photon to *completely remove* an electron from *1s orbital of Boron atom (B)*
- iii. Wavelength of photon to *completely remove* an electron from *1s orbital of Boron cation (B^+)*

i or ii or iii

(II) Select the smallest (shortest) wavelength from the three choices below (*for H atom*):

- i. Wavelength of photon *emitted* for an electron transitioning from *2p to 1s orbital*
- ii. Wavelength of photon *absorbed* for an electron transitioning from *2s to 4p orbital*
- iii. Wavelength of photon *absorbed* for an electron transitioning from *1s to 3p orbital*

i or ii or iii



Question 2 8 points



A Chem110 student plots the graph shown on the left, from data obtained for the kinetic energy (eV) of the ejected electron versus frequency (s^{-1}) of the incident light, for an unknown metal. ($1\text{eV} = 1.6022 \times 10^{-19} \text{ J}$)

Based on the plot, answer the questions **a to c** below:

a. 1 point On the plot above, label the **threshold frequency**.

b. 2 points What is the value of the work potential (in J) of the unknown metal?

c. 3 points What is the *longest* wavelength (in nm) required to eject an electron from this metal's surface? Show your work including equation(s) used. (3 sig figs)

d. 2 points In the space provided below, plot the following

(Assume frequency is *above* threshold frequency):

- i. *Kinetic energy* of an ejected electron versus the *intensity* of the incident light (left)
- ii. *Number of electrons* ejected versus the *frequency* of the incident light (right)



**Question 3** **6 points****a. 2 points**

In the ground state of Ge (Germanium; $Z = 32$) atom, **name** the following orbitals:

i. Orbital containing the electron that requires the maximum energy to be removed.

ii. Orbital containing the electron that experiences the minimum effective nuclear charge.

b. 2 points

An unknown anion (A^-) has the following *excited* electronic configuration: $[Ar] 4s^2 3d^3 4p^1$

(Ar: Argon; $Z = 18$)

Give the **condensed ground state electronic configuration** of the atom, **A** and the cation, **A^{2+}** .
Show your work. (Assume all electronic transitions are possible)

Electronic Configuration of A:

Electronic Configuration of A^{2+} :

c. 2 points

In a single atom, determine how many electrons can have the following set of quantum numbers?

(i) $n = 5; l = 3$

In a single atom, determine how many electrons can have the following set of quantum numbers?

(i) $n = 3; l = 2; m_s = +1/2$

**Question 4** **8 points**

Rank the following from 1 to 4 in **increasing** order. Provide the ranking in the boxes provided.

Note: Atomic number (Z) are provided for relevant atoms

a) Increasing (lowest to highest) order of *ionization energy*: (1= lowest; 4 = highest)

Mg

Mg: Z = 12

Rb

Rb: Z = 37

Cs

Cs: Z = 55

K

K : Z = 19

b) Increasing (lowest to highest) order of *bond angle*: (1= lowest; 4 = highest)

H₂OCCl₄XeF₄

HCN

c) Increasing (smallest to largest) order of *ionic size* (1= smallest; 4 = largest)

S²⁻

S: Z = 16

O²⁻

O: Z = 8

Be²⁺

Be: Z = 4

Na⁺

Na: Z = 11

d) Increasing (lowest to highest) order of *bond energy* (1= lowest; 4 = highest)

P-F

P-Cl

P-I

P-Br

**Question 5** **12 points**

a. 9 points For the molecules HClO_4 and TeF_6 , draw the most stable *Lewis structure* showing appropriate shape and *dashed/wedged bonds*, based on VSEPR. Determine the **molecular geometry** around the central atom. For HClO_4 , label the *most polar bond*.



b. 3 points (Note: Lone pairs are not shown in structures below)

(i) For the three *resonance* structures (1, 2 and 3) shown below, answer the following question:



(I) Circle the *most contributing* resonance structure.

Structure 1

Structure 2

Structure 3

(ii) For the three structures given (4, 5, and 6) below, answer the following questions:



(I) Circle all the structures that represent *resonance structures* of each other:

Structure 4

Structure 5

Structure 6

(II) Among the resonance structures, circle the *most contributing* resonance structure.

Structure 4

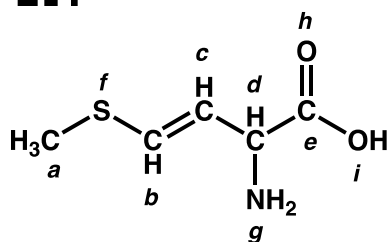
Structure 5

Structure 6

**Question 6** **6 points**

Using the data given below, calculate the **lattice energy** for SrO(s).

	ΔH° (kJ/mol)
$\text{Sr}(s) \longrightarrow \text{Sr}(g)$	+180
$\text{Sr}(g) \longrightarrow \text{Sr}^{2+}(g) + 2e^-$	+1468
$\text{O}_2(g) \longrightarrow 2\text{O}(g)$	+500
$\text{O}(g) + 2e^- \longrightarrow \text{O}^{2-}(g)$	+600
$\text{Sr}(s) + \frac{1}{2}\text{O}_2(g) \longrightarrow \text{SrO}(s)$	-550

**Question 7 10 points**

Consider the molecule given on the left, and answer questions (i) to (vi) using chemical bonding theories that we have learnt in class (valence bond theory, hybridization).

Note that the lone pairs are not shown in the structure:

i. Consider the carbon atoms marked in as ***a***, ***b***, ***c***, ***d***, and ***e*** in the molecule, and determine **hybridization** at each of the carbon centers. (2.5 points)

Atom	Hybridization
Carbon (<i>a</i>)	
Carbon (<i>b</i>)	
Carbon (<i>c</i>)	
Carbon (<i>d</i>)	
Carbon (<i>e</i>)	

ii. Now, consider the sulfur atom (S) marked as ***f***, nitrogen (N) atom (***g***), and oxygen(O) atoms (***h*** and ***i***), and determine their **hybridization**. (2 points)

Atom	Hybridization
Sulfur (<i>f</i>)	
Nitrogen (<i>g</i>)	
Oxygen (<i>h</i>)	
Oxygen (<i>i</i>)	

iii. **Describe** all the bonds formed by the carbon atoms ***a*** and ***b***, using sigma and pi designation as appropriate. (2 points)

Atom	Bonds formed (use sigma and pi as appropriate)
Carbon (<i>a</i>)	
Carbon (<i>b</i>)	

iv. How many total sp^3 -s sigma bonds are in the molecule? (1 point)

v. Determine **electronic** and **molecular geometry** at carbon (C) labeled ***e*** and sulfur (S) labeled ***f*** centres. (2 points)

Atom	Electronic Geometry	Molecular Geometry
Carbon (<i>e</i>)		
Sulfur (<i>f</i>)		

vi. Circle the value from the choices given below, that describes the **C-S-C bond angle at sulfur** (S) marked ***f***: (0.5 points)

>90° to <109.5° 109.5° >109.5 to <120° 120° >120° to <180°


Question 8 12 points

Using the **valence shell orbitals only** draw a **complete** molecular orbital (MO) diagram (with no s-p mixing) for the molecule **PS**, in the space provided below. **(6 points)**

Assume that 3s and 3p atomic orbitals will form molecular orbitals like those formed by the combination of 2s and 2p atomic orbitals that we learnt in class, and there is no s-p mixing.

Show **energy arrow**, **atomic orbitals** involved, **atom labels**, the **molecular orbitals** that are formed, **filled electrons** in the MO diagram etc. Make sure to label all the atomic and molecular orbitals.

Using the MO diagram above, answer the following questions:

- (i) Which one of the following will have the highest bond order? Circle the correct choice
 A) PS^{1-} B) PS C) PS^{1+}
- (ii) Which one of the following will have the longest bond length? Circle the correct choice
 A) PS^{1-} B) PS C) PS^{1+}
- (iii) Which of the following will be diamagnetic? Circle the correct choice.
 A) PS^{1-} B) PS C) PS^{1+}
- (iv) Draw Lewis dot structure of the molecule with the shortest bond length (based on the MO diagram) from among the three (PS^{1-} , PS, PS^{1+}), and indicate the bond order calculated from the Lewis dot structure. **(3 points)**

Lewis Dot Structure

Bond Order



6FF1E5DC-A147-4BEF-80C2-F570CFEAA6C3

midterm-exam-93388

#538 Page 10 of 10

SCRAP PAGE

**CHEMISTRY 110 –Midterm Exam****Thursday November 7, 2024, 6:30 – 8:30 pm**

Examiners: Pallavi Sirjoosingh and Ashok Kakkar

First name (please write as legibly as possible within the boxes)

--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--

Last name

--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--

Student ID number

--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--

This booklet has 8 questions on 10 pages (The last page can be used as a scrap page).

The staple must NOT be removed from the question booklet. Your intact question booklet must be returned at the end of the exam.

A separate booklet contains a periodic table, and datasheet.

Write your answers directly under the questions in the space provided. Do not write on the QR codes or near the top edges (as it may be cut off).

Please write **legibly and dark enough** for the scan to be readable. For students who use English as a second language, dictionaries are permitted. Calculators are allowed.

DO NOT ASK QUESTIONS OF THE INVIGILATORS about interpretations of the exam questions. If you have a concern about a question, please summarize it at the top of this page. All concerns will be evaluated after the exam. GOOD LUCK!

**Question 1 8 points** (*Assume all transitions are possible*)

a. 6 points Calculate the **energy (in kJ/mol)** required to ionize (completely remove) the electron from the **$n=3$** level of **He^+ ion** (He: Helium). Show your work including equations. (3 sig. figs.)

b. 2 points

(I) Select the largest (longest) wavelength from the three choices below:

- i. Wavelength of photon to *completely remove* an electron from *1s orbital of Hydrogen atom*
- ii. Wavelength of photon to *completely remove* an electron from *1s orbital of Boron atom (B)*
- iii. Wavelength of photon to *completely remove* an electron from *1s orbital of Boron cation (B^+)*

i or ii or iii

(II) Select the smallest (shortest) wavelength from the three choices below (*for H atom*):

- i. Wavelength of photon *emitted* for an electron transitioning from *2p to 1s orbital*
- ii. Wavelength of photon *absorbed* for an electron transitioning from *2s to 4p orbital*
- iii. Wavelength of photon *absorbed* for an electron transitioning from *1s to 3p orbital*

i or ii or iii



Question 2 8 points



A Chem110 student plots the graph shown on the left, from data obtained for the kinetic energy (eV) of the ejected electron versus frequency (s^{-1}) of the incident light, for an unknown metal. ($1\text{eV} = 1.6022 \times 10^{-19} \text{ J}$)

Based on the plot, answer the questions **a to c** below:

a. 1 point On the plot above, label the **threshold frequency**.

b. 2 points What is the value of the work potential (in J) of the unknown metal?

c. 3 points What is the *longest* wavelength (in nm) required to eject an electron from this metal's surface? Show your work including equation(s) used. (3 sig figs)

d. 2 points In the space provided below, plot the following

(Assume frequency is *above* threshold frequency):

- Kinetic energy* of an ejected electron versus the *intensity* of the incident light (left)
- Number of electrons* ejected versus the *frequency* of the incident light (right)



**Question 3** **6 points****a. 2 points**

In the ground state of Ge (Germanium; $Z = 32$) atom, **name** the following orbitals:

i. Orbital containing the electron that requires the maximum energy to be removed.

ii. Orbital containing the electron that experiences the minimum effective nuclear charge.

b. 2 points

An unknown anion (A^-) has the following *excited* electronic configuration: $[Ar] 4s^2 3d^3 4p^1$

(Ar: Argon; $Z = 18$)

Give the **condensed ground state electronic configuration** of the atom, **A** and the cation, **A^{2+}** .
Show your work. (Assume all electronic transitions are possible)

Electronic Configuration of A:

Electronic Configuration of A^{2+} :

c. 2 points

In a single atom, determine how many electrons can have the following set of quantum numbers?

(i) $n = 5; l = 3$

In a single atom, determine how many electrons can have the following set of quantum numbers?

(i) $n = 3; l = 2; m_s = +1/2$

**Question 4** **8 points**

Rank the following from 1 to 4 in **increasing** order. Provide the ranking in the boxes provided.

Note: Atomic number (Z) are provided for relevant atoms

a) Increasing (lowest to highest) order of *ionization energy*: (1= lowest; 4 = highest)

Mg

Mg: Z = 12

Rb

Rb: Z = 37

Cs

Cs: Z = 55

K

K : Z = 19

b) Increasing (lowest to highest) order of *bond angle*: (1= lowest; 4 = highest)

H₂OCCl₄XeF₄

HCN

c) Increasing (smallest to largest) order of *ionic size* (1= smallest; 4 = largest)

S²⁻

S: Z = 16

O²⁻

O: Z = 8

Be²⁺

Be: Z = 4

Na⁺

Na: Z = 11

d) Increasing (lowest to highest) order of *bond energy* (1= lowest; 4 = highest)

P-F

P-Cl

P-I

P-Br

**Question 5** **12 points**

a. 9 points For the molecules HClO_4 and TeF_6 , draw the most stable *Lewis structure* showing appropriate shape and *dashed/wedged bonds*, based on VSEPR. Determine the **molecular geometry** around the central atom. For HClO_4 , label the *most polar bond*.



b. 3 points (Note: Lone pairs are not shown in structures below)

(i) For the three *resonance* structures (1, 2 and 3) shown below, answer the following question:



(I) Circle the *most contributing* resonance structure.

Structure 1

Structure 2

Structure 3

(ii) For the three structures given (4, 5, and 6) below, answer the following questions:



(I) Circle all the structures that represent *resonance structures* of each other:

Structure 4

Structure 5

Structure 6

(II) Among the resonance structures, circle the *most contributing* resonance structure.

Structure 4

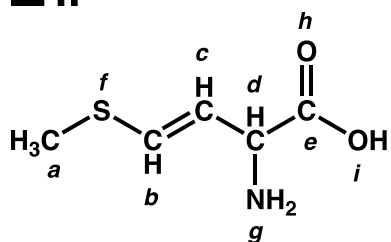
Structure 5

Structure 6

**Question 6** **6 points**

Using the data given below, calculate the **lattice energy** for SrO(s).

	ΔH° (kJ/mol)
$\text{Sr}(s) \longrightarrow \text{Sr}(g)$	+180
$\text{Sr}(g) \longrightarrow \text{Sr}^{2+}(g) + 2e^-$	+1468
$\text{O}_2(g) \longrightarrow 2\text{O}(g)$	+500
$\text{O}(g) + 2e^- \longrightarrow \text{O}^{2-}(g)$	+600
$\text{Sr}(s) + \frac{1}{2}\text{O}_2(g) \longrightarrow \text{SrO}(s)$	-550

**Question 7 10 points**

Consider the molecule given on the left, and answer questions (i) to (vi) using chemical bonding theories that we have learnt in class (valence bond theory, hybridization).

Note that the lone pairs are not shown in the structure:

i. Consider the carbon atoms marked in as ***a***, ***b***, ***c***, ***d***, and ***e*** in the molecule, and determine **hybridization** at each of the carbon centers. (2.5 points)

Atom	Hybridization
Carbon (<i>a</i>)	
Carbon (<i>b</i>)	
Carbon (<i>c</i>)	
Carbon (<i>d</i>)	
Carbon (<i>e</i>)	

ii. Now, consider the sulfur atom (S) marked as ***f***, nitrogen (N) atom (***g***), and oxygen(O) atoms (***h*** and ***i***), and determine their **hybridization**. (2 points)

Atom	Hybridization
Sulfur (<i>f</i>)	
Nitrogen (<i>g</i>)	
Oxygen (<i>h</i>)	
Oxygen (<i>i</i>)	

iii. **Describe** all the bonds formed by the carbon atoms ***a*** and ***b***, using sigma and pi designation as appropriate. (2 points)

Atom	Bonds formed (use sigma and pi as appropriate)
Carbon (<i>a</i>)	
Carbon (<i>b</i>)	

iv. How many total sp^3 -s sigma bonds are in the molecule? (1 point)

v. Determine **electronic** and **molecular geometry** at carbon (C) labeled ***e*** and sulfur (S) labeled ***f*** centres. (2 points)

Atom	Electronic Geometry	Molecular Geometry
Carbon (<i>e</i>)		
Sulfur (<i>f</i>)		

vi. Circle the value from the choices given below, that describes the **C-S-C bond angle at sulfur** (S) marked ***f***: (0.5 points)

>90° to <109.5° 109.5° >109.5 to <120° 120° >120° to <180°

**Question 8 12 points**

Using the **valence shell orbitals only** draw a **complete** molecular orbital (MO) diagram (with no s-p mixing) for the molecule **PS**, in the space provided below. **(6 points)**

Assume that 3s and 3p atomic orbitals will form molecular orbitals like those formed by the combination of 2s and 2p atomic orbitals that we learnt in class, and there is no s-p mixing.

Show **energy arrow**, **atomic orbitals** involved, **atom labels**, the **molecular orbitals** that are formed, **filled electrons** in the MO diagram etc. Make sure to label all the atomic and molecular orbitals.

Using the MO diagram above, answer the following questions:

- (i) Which one of the following will have the highest bond order? Circle the correct choice
A) PS^{1-} B) PS C) PS^{1+}
- (ii) Which one of the following will have the longest bond length? Circle the correct choice
A) PS^{1-} B) PS C) PS^{1+}
- (iii) Which of the following will be diamagnetic? Circle the correct choice.
A) PS^{1-} B) PS C) PS^{1+}
- (iv) Draw Lewis dot structure of the molecule with the shortest bond length (based on the MO diagram) from among the three (PS^{1-} , PS, PS^{1+}), and indicate the bond order calculated from the Lewis dot structure. **(3 points)**

Lewis Dot Structure

Bond Order



D721D6FE-E5EE-4460-8372-951DA3F132DB

midterm-exam-93388

#539 Page 10 of 10

SCRAP PAGE

**CHEMISTRY 110 –Midterm Exam****Thursday November 7, 2024, 6:30 – 8:30 pm**

Examiners: Pallavi Sirjoosingh and Ashok Kakkar

First name (please write as legibly as possible within the boxes)

--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--

Last name

--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--

Student ID number

--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--

This booklet has 8 questions on 10 pages (The last page can be used as a scrap page).

The staple must NOT be removed from the question booklet. Your intact question booklet must be returned at the end of the exam.

A separate booklet contains a periodic table, and datasheet.

Write your answers directly under the questions in the space provided. Do not write on the QR codes or near the top edges (as it may be cut off).

Please write **legibly and dark enough** for the scan to be readable. For students who use English as a second language, dictionaries are permitted. Calculators are allowed.

DO NOT ASK QUESTIONS OF THE INVIGILATORS about interpretations of the exam questions. If you have a concern about a question, please summarize it at the top of this page. All concerns will be evaluated after the exam. GOOD LUCK!

**Question 1 8 points** (*Assume all transitions are possible*)

a. 6 points Calculate the **energy (in kJ/mol)** required to ionize (completely remove) the electron from the **$n=3$** level of **He^+ ion** (He: Helium). Show your work including equations. (3 sig. figs.)

b. 2 points

(I) Select the largest (longest) wavelength from the three choices below:

- i. Wavelength of photon to *completely remove* an electron from *1s orbital of Hydrogen atom*
- ii. Wavelength of photon to *completely remove* an electron from *1s orbital of Boron atom (B)*
- iii. Wavelength of photon to *completely remove* an electron from *1s orbital of Boron cation (B^+)*

i or ii or iii

(II) Select the smallest (shortest) wavelength from the three choices below (*for H atom*):

- i. Wavelength of photon *emitted* for an electron transitioning from *2p to 1s orbital*
- ii. Wavelength of photon *absorbed* for an electron transitioning from *2s to 4p orbital*
- iii. Wavelength of photon *absorbed* for an electron transitioning from *1s to 3p orbital*

i or ii or iii



Question 2 8 points



A Chem110 student plots the graph shown on the left, from data obtained for the kinetic energy (eV) of the ejected electron versus frequency (s^{-1}) of the incident light, for an unknown metal. ($1\text{eV} = 1.6022 \times 10^{-19} \text{ J}$)

Based on the plot, answer the questions **a to c** below:

a. 1 point On the plot above, label the **threshold frequency**.

b. 2 points What is the value of the work potential (in J) of the unknown metal?

c. 3 points What is the *longest* wavelength (in nm) required to eject an electron from this metal's surface? Show your work including equation(s) used. (3 sig figs)

d. 2 points In the space provided below, plot the following

(Assume frequency is *above* threshold frequency):

- i. *Kinetic energy* of an ejected electron versus the *intensity* of the incident light (left)
- ii. *Number of electrons* ejected versus the *frequency* of the incident light (right)



**Question 3** **6 points****a. 2 points**

In the ground state of Ge (Germanium; $Z = 32$) atom, **name** the following orbitals:

i. Orbital containing the electron that requires the maximum energy to be removed.

ii. Orbital containing the electron that experiences the minimum effective nuclear charge.

b. 2 points

An unknown anion (A^-) has the following *excited* electronic configuration: $[Ar] 4s^2 3d^3 4p^1$

(Ar: Argon; $Z = 18$)

Give the **condensed ground state electronic configuration** of the atom, **A** and the cation, **A^{2+}** .
Show your work. (Assume all electronic transitions are possible)

Electronic Configuration of A:

Electronic Configuration of A^{2+} :

c. 2 points

In a single atom, determine how many electrons can have the following set of quantum numbers?

(i) $n = 5; l = 3$

In a single atom, determine how many electrons can have the following set of quantum numbers?

(i) $n = 3; l = 2; m_s = +1/2$

**Question 4** **8 points**

Rank the following from 1 to 4 in **increasing** order. Provide the ranking in the boxes provided.

Note: Atomic number (Z) are provided for relevant atoms

a) Increasing (lowest to highest) order of *ionization energy*: (1= lowest; 4 = highest)

Mg

Mg: Z = 12

Rb

Rb: Z = 37

Cs

Cs: Z = 55

K

K : Z = 19

b) Increasing (lowest to highest) order of *bond angle*: (1= lowest; 4 = highest)

H₂OCCl₄XeF₄

HCN

c) Increasing (smallest to largest) order of *ionic size* (1= smallest; 4 = largest)

S²⁻

S: Z = 16

O²⁻

O: Z = 8

Be²⁺

Be: Z = 4

Na⁺

Na: Z = 11

d) Increasing (lowest to highest) order of *bond energy* (1= lowest; 4 = highest)

P-F

P-Cl

P-I

P-Br

**Question 5** **12 points**

a. 9 points For the molecules HClO_4 and TeF_6 , draw the most stable *Lewis structure* showing appropriate shape and *dashed/wedged bonds*, based on VSEPR. Determine the **molecular geometry** around the central atom. For HClO_4 , label the *most polar bond*.



b. 3 points (Note: Lone pairs are not shown in structures below)

(i) For the three *resonance* structures (1, 2 and 3) shown below, answer the following question:



(I) Circle the *most contributing* resonance structure.

Structure 1

Structure 2

Structure 3

(ii) For the three structures given (4, 5, and 6) below, answer the following questions:



(I) Circle all the structures that represent *resonance structures* of each other:

Structure 4

Structure 5

Structure 6

(II) Among the resonance structures, circle the *most contributing* resonance structure.

Structure 4

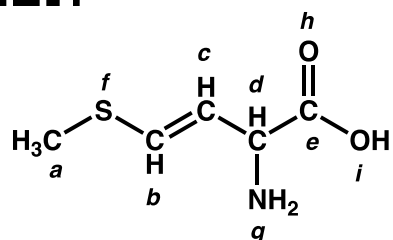
Structure 5

Structure 6

**Question 6** **6 points**

Using the data given below, calculate the **lattice energy** for SrO(s).

	ΔH° (kJ/mol)
$\text{Sr}(s) \longrightarrow \text{Sr}(g)$	+180
$\text{Sr}(g) \longrightarrow \text{Sr}^{2+}(g) + 2e^-$	+1468
$\text{O}_2(g) \longrightarrow 2\text{O}(g)$	+500
$\text{O}(g) + 2e^- \longrightarrow \text{O}^{2-}(g)$	+600
$\text{Sr}(s) + \frac{1}{2}\text{O}_2(g) \longrightarrow \text{SrO}(s)$	-550

**Question 7 10 points**

Consider the molecule given on the left, and answer questions (i) to (vi) using chemical bonding theories that we have learnt in class (valence bond theory, hybridization).

Note that the lone pairs are not shown in the structure:

i. Consider the carbon atoms marked in as ***a***, ***b***, ***c***, ***d***, and ***e*** in the molecule, and determine **hybridization** at each of the carbon centers. (2.5 points)

Atom	Hybridization
Carbon (<i>a</i>)	
Carbon (<i>b</i>)	
Carbon (<i>c</i>)	
Carbon (<i>d</i>)	
Carbon (<i>e</i>)	

ii. Now, consider the sulfur atom (S) marked as ***f***, nitrogen (N) atom (***g***), and oxygen(O) atoms (***h*** and ***i***), and determine their **hybridization**. (2 points)

Atom	Hybridization
Sulfur (<i>f</i>)	
Nitrogen (<i>g</i>)	
Oxygen (<i>h</i>)	
Oxygen (<i>i</i>)	

iii. **Describe** all the bonds formed by the carbon atoms ***a*** and ***b***, using sigma and pi designation as appropriate. (2 points)

Atom	Bonds formed (use sigma and pi as appropriate)
Carbon (<i>a</i>)	
Carbon (<i>b</i>)	

iv. How many total sp^3 -s sigma bonds are in the molecule? (1 point)

v. Determine **electronic** and **molecular geometry** at carbon (C) labeled ***e*** and sulfur (S) labeled ***f*** centres. (2 points)

Atom	Electronic Geometry	Molecular Geometry
Carbon (<i>e</i>)		
Sulfur (<i>f</i>)		

vi. Circle the value from the choices given below, that describes the **C-S-C bond angle at sulfur** (S) marked ***f***: (0.5 points)

>90° to <109.5° 109.5° >109.5 to <120° 120° >120° to <180°

**Question 8 12 points**

Using the **valence shell orbitals only** draw a **complete** molecular orbital (MO) diagram (with no s-p mixing) for the molecule **PS**, in the space provided below. **(6 points)**

Assume that 3s and 3p atomic orbitals will form molecular orbitals like those formed by the combination of 2s and 2p atomic orbitals that we learnt in class, and there is no s-p mixing.

Show **energy arrow**, **atomic orbitals** involved, **atom labels**, the **molecular orbitals** that are formed, **filled electrons** in the MO diagram etc. Make sure to label all the atomic and molecular orbitals.

Using the MO diagram above, answer the following questions:

- (i) Which one of the following will have the highest bond order? Circle the correct choice
A) PS^{1-} B) PS C) PS^{1+}
- (ii) Which one of the following will have the longest bond length? Circle the correct choice
A) PS^{1-} B) PS C) PS^{1+}
- (iii) Which of the following will be diamagnetic? Circle the correct choice.
A) PS^{1-} B) PS C) PS^{1+}
- (iv) Draw Lewis dot structure of the molecule with the shortest bond length (based on the MO diagram) from among the three (PS^{1-} , PS, PS^{1+}), and indicate the bond order calculated from the Lewis dot structure. **(3 points)**

Lewis Dot Structure

Bond Order



009A2C34-1641-479D-9C4D-5E4B8AE6FFD6

midterm-exam-93388

#540 Page 10 of 10

SCRAP PAGE

**CHEMISTRY 110 –Midterm Exam****Thursday November 7, 2024, 6:30 – 8:30 pm****Examiners: Pallavi Sirjoosingh and Ashok Kakkar**

First name (please write as legibly as possible within the boxes)

--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--

Last name

--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--

Student ID number

--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--

This booklet has 8 questions on 10 pages (The last page can be used as a scrap page).

The staple must NOT be removed from the question booklet. Your intact question booklet must be returned at the end of the exam.

A separate booklet contains a periodic table, and datasheet.

Write your answers directly under the questions in the space provided. Do not write on the QR codes or near the top edges (as it may be cut off).

Please write **legibly and dark enough** for the scan to be readable. For students who use English as a second language, dictionaries are permitted. Calculators are allowed.

DO NOT ASK QUESTIONS OF THE INVIGILATORS about interpretations of the exam questions. If you have a concern about a question, please summarize it at the top of this page. All concerns will be evaluated after the exam. GOOD LUCK!

**Question 1 8 points** (*Assume all transitions are possible*)

a. 6 points Calculate the **energy (in kJ/mol)** required to ionize (completely remove) the electron from the **n=3** level of **He⁺ ion** (He: Helium). Show your work including equations. (3 sig. figs.)

b. 2 points

(I) Select the largest (longest) wavelength from the three choices below:

- i. Wavelength of photon to *completely remove* an electron from *1s orbital of Hydrogen atom*
- ii. Wavelength of photon to *completely remove* an electron from *1s orbital of Boron atom (B)*
- iii. Wavelength of photon to *completely remove* an electron from *1s orbital of Boron cation (B⁺)*

i or ii or iii

(II) Select the smallest (shortest) wavelength from the three choices below (*for H atom*):

- i. Wavelength of photon *emitted* for an electron transitioning from *2p to 1s orbital*
- ii. Wavelength of photon *absorbed* for an electron transitioning from *2s to 4p orbital*
- iii. Wavelength of photon *absorbed* for an electron transitioning from *1s to 3p orbital*

i or ii or iii



Question 2 8 points



A Chem110 student plots the graph shown on the left, from data obtained for the kinetic energy (eV) of the ejected electron versus frequency (s^{-1}) of the incident light, for an unknown metal. ($1\text{eV} = 1.6022 \times 10^{-19} \text{ J}$)

Based on the plot, answer the questions **a to c** below:

a. 1 point On the plot above, label the **threshold frequency**.

b. 2 points What is the value of the work potential (in J) of the unknown metal?

c. 3 points What is the *longest* wavelength (in nm) required to eject an electron from this metal's surface? Show your work including equation(s) used. (3 sig figs)

d. 2 points In the space provided below, plot the following

(Assume frequency is *above* threshold frequency):

- i. *Kinetic energy* of an ejected electron versus the *intensity* of the incident light (left)
- ii. *Number of electrons* ejected versus the *frequency* of the incident light (right)



**Question 3** **6 points****a. 2 points**

In the ground state of Ge (Germanium; $Z = 32$) atom, **name** the following orbitals:

i. Orbital containing the electron that requires the maximum energy to be removed.

ii. Orbital containing the electron that experiences the minimum effective nuclear charge.

b. 2 points

An unknown anion (A^-) has the following *excited* electronic configuration: $[Ar] 4s^2 3d^3 4p^1$

(Ar: Argon; $Z = 18$)

Give the **condensed ground state electronic configuration** of the atom, **A** and the cation, **A^{2+}** .
Show your work. (Assume all electronic transitions are possible)

Electronic Configuration of A:

Electronic Configuration of A^{2+} :

c. 2 points

In a single atom, determine how many electrons can have the following set of quantum numbers?

(i) $n = 5; l = 3$

In a single atom, determine how many electrons can have the following set of quantum numbers?

(i) $n = 3; l = 2; m_s = +1/2$

**Question 4** **8 points**

Rank the following from 1 to 4 in **increasing** order. Provide the ranking in the boxes provided.

Note: Atomic number (Z) are provided for relevant atoms

a) Increasing (lowest to highest) order of *ionization energy*: (1= lowest; 4 = highest)

Mg

Mg: Z = 12

Rb

Rb: Z = 37

Cs

Cs: Z = 55

K

K : Z = 19

b) Increasing (lowest to highest) order of *bond angle*: (1= lowest; 4 = highest)

H₂OCCl₄XeF₄

HCN

c) Increasing (smallest to largest) order of *ionic size* (1= smallest; 4 = largest)

S²⁻

S: Z = 16

O²⁻

O: Z = 8

Be²⁺

Be: Z = 4

Na⁺

Na: Z = 11

d) Increasing (lowest to highest) order of *bond energy* (1= lowest; 4 = highest)

P-F

P-Cl

P-I

P-Br

**Question 5** **12 points**

a. 9 points For the molecules HClO_4 and TeF_6 , draw the most stable *Lewis structure* showing appropriate shape and *dashed/wedged bonds*, based on VSEPR. Determine the **molecular geometry** around the central atom. For HClO_4 , label the *most polar bond*.



b. 3 points (Note: Lone pairs are not shown in structures below)

(i) For the three *resonance* structures (1, 2 and 3) shown below, answer the following question:



(I) Circle the *most contributing* resonance structure.

Structure 1

Structure 2

Structure 3

(ii) For the three structures given (4, 5, and 6) below, answer the following questions:



(I) Circle all the structures that represent *resonance structures* of each other:

Structure 4

Structure 5

Structure 6

(II) Among the resonance structures, circle the *most contributing* resonance structure.

Structure 4

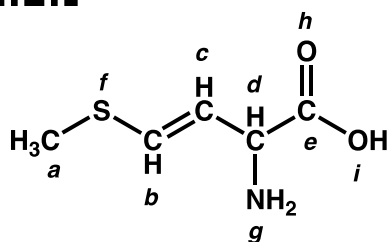
Structure 5

Structure 6

**Question 6** **6 points**

Using the data given below, calculate the **lattice energy** for SrO(s).

	ΔH° (kJ/mol)
$\text{Sr}(s) \longrightarrow \text{Sr}(g)$	+180
$\text{Sr}(g) \longrightarrow \text{Sr}^{2+}(g) + 2e^-$	+1468
$\text{O}_2(g) \longrightarrow 2\text{O}(g)$	+500
$\text{O}(g) + 2e^- \longrightarrow \text{O}^{2-}(g)$	+600
$\text{Sr}(s) + \frac{1}{2}\text{O}_2(g) \longrightarrow \text{SrO}(s)$	-550

**Question 7 10 points**

Consider the molecule given on the left, and answer questions (i) to (vi) using chemical bonding theories that we have learnt in class (valence bond theory, hybridization).

Note that the lone pairs are not shown in the structure:

i. Consider the carbon atoms marked in as ***a***, ***b***, ***c***, ***d***, and ***e*** in the molecule, and determine **hybridization** at each of the carbon centers. (2.5 points)

Atom	Hybridization
Carbon (<i>a</i>)	
Carbon (<i>b</i>)	
Carbon (<i>c</i>)	
Carbon (<i>d</i>)	
Carbon (<i>e</i>)	

ii. Now, consider the sulfur atom (S) marked as ***f***, nitrogen (N) atom (***g***), and oxygen(O) atoms (***h*** and ***i***), and determine their **hybridization**. (2 points)

Atom	Hybridization
Sulfur (<i>f</i>)	
Nitrogen (<i>g</i>)	
Oxygen (<i>h</i>)	
Oxygen (<i>i</i>)	

iii. **Describe** all the bonds formed by the carbon atoms ***a*** and ***b***, using sigma and pi designation as appropriate. (2 points)

Atom	Bonds formed (use sigma and pi as appropriate)
Carbon (<i>a</i>)	
Carbon (<i>b</i>)	

iv. How many total sp^3 -s sigma bonds are in the molecule? (1 point)

v. Determine **electronic** and **molecular geometry** at carbon (C) labeled ***e*** and sulfur (S) labeled ***f*** centres. (2 points)

Atom	Electronic Geometry	Molecular Geometry
Carbon (<i>e</i>)		
Sulfur (<i>f</i>)		

vi. Circle the value from the choices given below, that describes the **C-S-C bond angle at sulfur** (S) marked ***f***: (0.5 points)

>90° to <109.5° 109.5° >109.5 to <120° 120° >120° to <180°


Question 8 12 points

Using the **valence shell orbitals only** draw a **complete** molecular orbital (MO) diagram (with no s-p mixing) for the molecule **PS**, in the space provided below. **(6 points)**

Assume that 3s and 3p atomic orbitals will form molecular orbitals like those formed by the combination of 2s and 2p atomic orbitals that we learnt in class, and there is no s-p mixing.

Show **energy arrow**, **atomic orbitals** involved, **atom labels**, the **molecular orbitals** that are formed, **filled electrons** in the MO diagram etc. Make sure to label all the atomic and molecular orbitals.

Using the MO diagram above, answer the following questions:

- (i) Which one of the following will have the highest bond order? Circle the correct choice
 A) PS^{1-} B) PS C) PS^{1+}
- (ii) Which one of the following will have the longest bond length? Circle the correct choice
 A) PS^{1-} B) PS C) PS^{1+}
- (iii) Which of the following will be diamagnetic? Circle the correct choice.
 A) PS^{1-} B) PS C) PS^{1+}
- (iv) Draw Lewis dot structure of the molecule with the shortest bond length (based on the MO diagram) from among the three (PS^{1-} , PS, PS^{1+}), and indicate the bond order calculated from the Lewis dot structure. **(3 points)**

Lewis Dot Structure

Bond Order



6112FE01-4E0F-465F-8D66-DF5FEEAF6DC4

midterm-exam-93388

#541 Page 10 of 10

SCRAP PAGE

**CHEMISTRY 110 –Midterm Exam****Thursday November 7, 2024, 6:30 – 8:30 pm**

Examiners: Pallavi Sirjoosingh and Ashok Kakkar

First name (please write as legibly as possible within the boxes)

--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--

Last name

--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--

Student ID number

--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--

This booklet has 8 questions on 10 pages (The last page can be used as a scrap page).

The staple must NOT be removed from the question booklet. Your intact question booklet must be returned at the end of the exam.

A separate booklet contains a periodic table, and datasheet.

Write your answers directly under the questions in the space provided. Do not write on the QR codes or near the top edges (as it may be cut off).

Please write **legibly and dark enough** for the scan to be readable. For students who use English as a second language, dictionaries are permitted. Calculators are allowed.

DO NOT ASK QUESTIONS OF THE INVIGILATORS about interpretations of the exam questions. If you have a concern about a question, please summarize it at the top of this page. All concerns will be evaluated after the exam. GOOD LUCK!

**Question 1 8 points** (*Assume all transitions are possible*)

a. 6 points Calculate the **energy (in kJ/mol)** required to ionize (completely remove) the electron from the **$n=3$** level of **He^+ ion** (He: Helium). Show your work including equations. (3 sig. figs.)

b. 2 points

(I) Select the largest (longest) wavelength from the three choices below:

- i. Wavelength of photon to *completely remove* an electron from *1s orbital of Hydrogen atom*
- ii. Wavelength of photon to *completely remove* an electron from *1s orbital of Boron atom (B)*
- iii. Wavelength of photon to *completely remove* an electron from *1s orbital of Boron cation (B^+)*

i or ii or iii

(II) Select the smallest (shortest) wavelength from the three choices below (*for H atom*):

- i. Wavelength of photon *emitted* for an electron transitioning from *2p to 1s orbital*
- ii. Wavelength of photon *absorbed* for an electron transitioning from *2s to 4p orbital*
- iii. Wavelength of photon *absorbed* for an electron transitioning from *1s to 3p orbital*

i or ii or iii



Question 2 8 points



A Chem110 student plots the graph shown on the left, from data obtained for the kinetic energy (eV) of the ejected electron versus frequency (s^{-1}) of the incident light, for an unknown metal. ($1\text{eV} = 1.6022 \times 10^{-19} \text{ J}$)

Based on the plot, answer the questions **a to c** below:

a. 1 point On the plot above, label the **threshold frequency**.

b. 2 points What is the value of the work potential (in J) of the unknown metal?

c. 3 points What is the *longest* wavelength (in nm) required to eject an electron from this metal's surface? Show your work including equation(s) used. (3 sig figs)

d. 2 points In the space provided below, plot the following

(Assume frequency is *above* threshold frequency):

- Kinetic energy* of an ejected electron versus the *intensity* of the incident light (left)
- Number of electrons* ejected versus the *frequency* of the incident light (right)



**Question 3** **6 points****a. 2 points**

In the ground state of Ge (Germanium; $Z = 32$) atom, **name** the following orbitals:

i. Orbital containing the electron that requires the maximum energy to be removed.

ii. Orbital containing the electron that experiences the minimum effective nuclear charge.

b. 2 points

An unknown anion (A^-) has the following *excited* electronic configuration: $[Ar] 4s^2 3d^3 4p^1$

(Ar: Argon; $Z = 18$)

Give the **condensed ground state electronic configuration** of the atom, **A** and the cation, **A^{2+}** .
Show your work. (Assume all electronic transitions are possible)

Electronic Configuration of A:

Electronic Configuration of A^{2+} :

c. 2 points

In a single atom, determine how many electrons can have the following set of quantum numbers?

(i) $n = 5; l = 3$

In a single atom, determine how many electrons can have the following set of quantum numbers?

(i) $n = 3; l = 2; m_s = +1/2$

**Question 4** **8 points**

Rank the following from 1 to 4 in **increasing** order. Provide the ranking in the boxes provided.

Note: Atomic number (Z) are provided for relevant atoms

a) Increasing (lowest to highest) order of *ionization energy*: (1= lowest; 4 = highest)

Mg

Mg: Z = 12

Rb

Rb: Z = 37

Cs

Cs: Z = 55

K

K : Z = 19

b) Increasing (lowest to highest) order of *bond angle*: (1= lowest; 4 = highest)

H₂OCCl₄XeF₄

HCN

c) Increasing (smallest to largest) order of *ionic size* (1= smallest; 4 = largest)

S²⁻

S: Z = 16

O²⁻

O: Z = 8

Be²⁺

Be: Z = 4

Na⁺

Na: Z = 11

d) Increasing (lowest to highest) order of *bond energy* (1= lowest; 4 = highest)

P-F

P-Cl

P-I

P-Br

**Question 5** **12 points**

a. 9 points For the molecules HClO_4 and TeF_6 , draw the most stable *Lewis structure* showing appropriate shape and *dashed/wedged bonds*, based on VSEPR. Determine the **molecular geometry** around the central atom. For HClO_4 , label the *most polar bond*.



b. 3 points (Note: Lone pairs are not shown in structures below)

(i) For the three *resonance* structures (1, 2 and 3) shown below, answer the following question:



(I) Circle the *most contributing* resonance structure.

Structure 1

Structure 2

Structure 3

(ii) For the three structures given (4, 5, and 6) below, answer the following questions:



(I) Circle all the structures that represent *resonance structures* of each other:

Structure 4

Structure 5

Structure 6

(II) Among the resonance structures, circle the *most contributing* resonance structure.

Structure 4

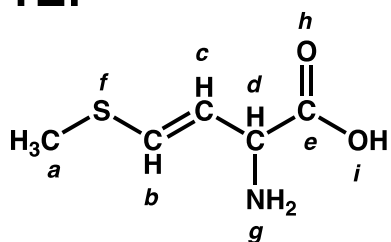
Structure 5

Structure 6

**Question 6** **6 points**

Using the data given below, calculate the **lattice energy** for SrO(s).

	ΔH° (kJ/mol)
$\text{Sr}(s) \longrightarrow \text{Sr}(g)$	+180
$\text{Sr}(g) \longrightarrow \text{Sr}^{2+}(g) + 2e^-$	+1468
$\text{O}_2(g) \longrightarrow 2\text{O}(g)$	+500
$\text{O}(g) + 2e^- \longrightarrow \text{O}^{2-}(g)$	+600
$\text{Sr}(s) + \frac{1}{2}\text{O}_2(g) \longrightarrow \text{SrO}(s)$	-550

**Question 7 10 points**

Consider the molecule given on the left, and answer questions (i) to (vi) using chemical bonding theories that we have learnt in class (valence bond theory, hybridization).

Note that the lone pairs are not shown in the structure:

i. Consider the carbon atoms marked in as ***a***, ***b***, ***c***, ***d***, and ***e*** in the molecule, and determine **hybridization** at each of the carbon centers. (2.5 points)

Atom	Hybridization
Carbon (<i>a</i>)	
Carbon (<i>b</i>)	
Carbon (<i>c</i>)	
Carbon (<i>d</i>)	
Carbon (<i>e</i>)	

ii. Now, consider the sulfur atom (S) marked as ***f***, nitrogen (N) atom (***g***), and oxygen(O) atoms (***h*** and ***i***), and determine their **hybridization**. (2 points)

Atom	Hybridization
Sulfur (<i>f</i>)	
Nitrogen (<i>g</i>)	
Oxygen (<i>h</i>)	
Oxygen (<i>i</i>)	

iii. **Describe** all the bonds formed by the carbon atoms ***a*** and ***b***, using sigma and pi designation as appropriate. (2 points)

Atom	Bonds formed (use sigma and pi as appropriate)
Carbon (<i>a</i>)	
Carbon (<i>b</i>)	

iv. How many total sp^3 -s sigma bonds are in the molecule? (1 point)

v. Determine **electronic** and **molecular geometry** at carbon (C) labeled ***e*** and sulfur (S) labeled ***f*** centres. (2 points)

Atom	Electronic Geometry	Molecular Geometry
Carbon (<i>e</i>)		
Sulfur (<i>f</i>)		

vi. Circle the value from the choices given below, that describes the **C-S-C bond angle at sulfur** (S) marked ***f***: (0.5 points)

>90° to <109.5° 109.5° >109.5 to <120° 120° >120° to <180°


Question 8 12 points

Using the **valence shell orbitals only** draw a **complete** molecular orbital (MO) diagram (with no s-p mixing) for the molecule **PS**, in the space provided below. **(6 points)**

Assume that 3s and 3p atomic orbitals will form molecular orbitals like those formed by the combination of 2s and 2p atomic orbitals that we learnt in class, and there is no s-p mixing.

Show **energy arrow**, **atomic orbitals** involved, **atom labels**, the **molecular orbitals** that are formed, **filled electrons** in the MO diagram etc. Make sure to label all the atomic and molecular orbitals.

Using the MO diagram above, answer the following questions:

- (i) Which one of the following will have the highest bond order? Circle the correct choice
 A) PS^{1-} B) PS C) PS^{1+}
- (ii) Which one of the following will have the longest bond length? Circle the correct choice
 A) PS^{1-} B) PS C) PS^{1+}
- (iii) Which of the following will be diamagnetic? Circle the correct choice.
 A) PS^{1-} B) PS C) PS^{1+}
- (iv) Draw Lewis dot structure of the molecule with the shortest bond length (based on the MO diagram) from among the three (PS^{1-} , PS, PS^{1+}), and indicate the bond order calculated from the Lewis dot structure. **(3 points)**

Lewis Dot Structure

Bond Order



16DD12CC-EB66-4514-BA9F-81ED9F5F674C

midterm-exam-93388

#542 Page 10 of 10

SCRAP PAGE

**CHEMISTRY 110 –Midterm Exam****Thursday November 7, 2024, 6:30 – 8:30 pm**

Examiners: Pallavi Sirjoosingh and Ashok Kakkar

First name (please write as legibly as possible within the boxes)

--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--

Last name

--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--

Student ID number

--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--

This booklet has 8 questions on 10 pages (The last page can be used as a scrap page).

The staple must NOT be removed from the question booklet. Your intact question booklet must be returned at the end of the exam.

A separate booklet contains a periodic table, and datasheet.

Write your answers directly under the questions in the space provided. Do not write on the QR codes or near the top edges (as it may be cut off).

Please write **legibly and dark enough** for the scan to be readable. For students who use English as a second language, dictionaries are permitted. Calculators are allowed.

DO NOT ASK QUESTIONS OF THE INVIGILATORS about interpretations of the exam questions. If you have a concern about a question, please summarize it at the top of this page. All concerns will be evaluated after the exam. GOOD LUCK!

**Question 1 8 points** (*Assume all transitions are possible*)

a. 6 points Calculate the **energy (in kJ/mol)** required to ionize (completely remove) the electron from the **n=3** level of **He⁺ ion** (He: Helium). Show your work including equations. (3 sig. figs.)

b. 2 points

(I) Select the largest (longest) wavelength from the three choices below:

- i. Wavelength of photon to *completely remove* an electron from *1s orbital of Hydrogen atom*
- ii. Wavelength of photon to *completely remove* an electron from *1s orbital of Boron atom (B)*
- iii. Wavelength of photon to *completely remove* an electron from *1s orbital of Boron cation (B⁺)*

i or ii or iii

(II) Select the smallest (shortest) wavelength from the three choices below (*for H atom*):

- i. Wavelength of photon *emitted* for an electron transitioning from *2p to 1s orbital*
- ii. Wavelength of photon *absorbed* for an electron transitioning from *2s to 4p orbital*
- iii. Wavelength of photon *absorbed* for an electron transitioning from *1s to 3p orbital*

i or ii or iii



Question 2 8 points



A Chem110 student plots the graph shown on the left, from data obtained for the kinetic energy (eV) of the ejected electron versus frequency (s^{-1}) of the incident light, for an unknown metal. ($1\text{eV} = 1.6022 \times 10^{-19} \text{ J}$)

Based on the plot, answer the questions **a to c** below:

a. 1 point On the plot above, label the **threshold frequency**.

b. 2 points What is the value of the work potential (in J) of the unknown metal?

c. 3 points What is the *longest* wavelength (in nm) required to eject an electron from this metal's surface? Show your work including equation(s) used. (3 sig figs)

d. 2 points In the space provided below, plot the following

(Assume frequency is *above* threshold frequency):

- Kinetic energy* of an ejected electron versus the *intensity* of the incident light (left)
- Number of electrons* ejected versus the *frequency* of the incident light (right)



**Question 3** **6 points****a. 2 points**

In the ground state of Ge (Germanium; $Z = 32$) atom, **name** the following orbitals:

i. Orbital containing the electron that requires the maximum energy to be removed.

ii. Orbital containing the electron that experiences the minimum effective nuclear charge.

b. 2 points

An unknown anion (A^-) has the following *excited* electronic configuration: $[Ar] 4s^2 3d^3 4p^1$

(Ar: Argon; $Z = 18$)

Give the **condensed ground state electronic configuration** of the atom, **A** and the cation, **A^{2+}** .
Show your work. (Assume all electronic transitions are possible)

Electronic Configuration of A:

Electronic Configuration of A^{2+} :

c. 2 points

In a single atom, determine how many electrons can have the following set of quantum numbers?

(i) $n = 5; l = 3$

In a single atom, determine how many electrons can have the following set of quantum numbers?

(i) $n = 3; l = 2; m_s = +1/2$

**Question 4** **8 points**

Rank the following from 1 to 4 in **increasing** order. Provide the ranking in the boxes provided.

Note: Atomic number (Z) are provided for relevant atoms

a) Increasing (lowest to highest) order of *ionization energy*: (1= lowest; 4 = highest)

Mg

Mg: Z = 12

Rb

Rb: Z = 37

Cs

Cs: Z = 55

K

K : Z = 19

b) Increasing (lowest to highest) order of *bond angle*: (1= lowest; 4 = highest)

H₂OCCl₄XeF₄

HCN

c) Increasing (smallest to largest) order of *ionic size* (1= smallest; 4 = largest)

S²⁻

S: Z = 16

O²⁻

O: Z = 8

Be²⁺

Be: Z = 4

Na⁺

Na: Z = 11

d) Increasing (lowest to highest) order of *bond energy* (1= lowest; 4 = highest)

P-F

P-Cl

P-I

P-Br

**Question 5** **12 points**

a. 9 points For the molecules HClO_4 and TeF_6 , draw the most stable *Lewis structure* showing appropriate shape and *dashed/wedged bonds*, based on VSEPR. Determine the **molecular geometry** around the central atom. For HClO_4 , label the *most polar bond*.



b. 3 points (Note: Lone pairs are not shown in structures below)

(i) For the three *resonance* structures (1, 2 and 3) shown below, answer the following question:



(I) Circle the *most contributing* resonance structure.

Structure 1

Structure 2

Structure 3

(ii) For the three structures given (4, 5, and 6) below, answer the following questions:



(I) Circle all the structures that represent *resonance structures* of each other:

Structure 4

Structure 5

Structure 6

(II) Among the resonance structures, circle the *most contributing* resonance structure.

Structure 4

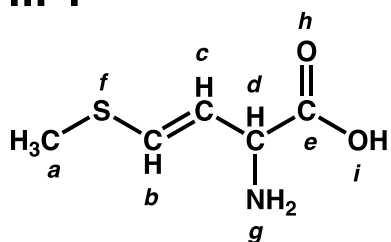
Structure 5

Structure 6

**Question 6** **6 points**

Using the data given below, calculate the **lattice energy** for SrO(s).

	ΔH° (kJ/mol)
$\text{Sr}(s) \longrightarrow \text{Sr}(g)$	+180
$\text{Sr}(g) \longrightarrow \text{Sr}^{2+}(g) + 2e^-$	+1468
$\text{O}_2(g) \longrightarrow 2\text{O}(g)$	+500
$\text{O}(g) + 2e^- \longrightarrow \text{O}^{2-}(g)$	+600
$\text{Sr}(s) + \frac{1}{2}\text{O}_2(g) \longrightarrow \text{SrO}(s)$	-550

**Question 7 10 points**

Consider the molecule given on the left, and answer questions (i) to (vi) using chemical bonding theories that we have learnt in class (valence bond theory, hybridization).

Note that the lone pairs are not shown in the structure:

i. Consider the carbon atoms marked in as ***a***, ***b***, ***c***, ***d***, and ***e*** in the molecule, and determine **hybridization** at each of the carbon centers. (2.5 points)

Atom	Hybridization
Carbon (<i>a</i>)	
Carbon (<i>b</i>)	
Carbon (<i>c</i>)	
Carbon (<i>d</i>)	
Carbon (<i>e</i>)	

ii. Now, consider the sulfur atom (S) marked as ***f***, nitrogen (N) atom (***g***), and oxygen(O) atoms (***h*** and ***i***), and determine their **hybridization**. (2 points)

Atom	Hybridization
Sulfur (<i>f</i>)	
Nitrogen (<i>g</i>)	
Oxygen (<i>h</i>)	
Oxygen (<i>i</i>)	

iii. **Describe** all the bonds formed by the carbon atoms ***a*** and ***b***, using sigma and pi designation as appropriate. (2 points)

Atom	Bonds formed (use sigma and pi as appropriate)
Carbon (<i>a</i>)	
Carbon (<i>b</i>)	

iv. How many total sp^3 -s sigma bonds are in the molecule? (1 point)

v. Determine **electronic** and **molecular geometry** at carbon (C) labeled ***e*** and sulfur (S) labeled ***f*** centres. (2 points)

Atom	Electronic Geometry	Molecular Geometry
Carbon (<i>e</i>)		
Sulfur (<i>f</i>)		

vi. Circle the value from the choices given below, that describes the **C-S-C bond angle at sulfur** (S) marked ***f***: (0.5 points)

>90° to <109.5° 109.5° >109.5 to <120° 120° >120° to <180°

**Question 8 12 points**

Using the **valence shell orbitals only** draw a **complete** molecular orbital (MO) diagram (with no s-p mixing) for the molecule **PS**, in the space provided below. **(6 points)**

Assume that 3s and 3p atomic orbitals will form molecular orbitals like those formed by the combination of 2s and 2p atomic orbitals that we learnt in class, and there is no s-p mixing.

Show **energy arrow**, **atomic orbitals** involved, **atom labels**, the **molecular orbitals** that are formed, **filled electrons** in the MO diagram etc. Make sure to label all the atomic and molecular orbitals.

Using the MO diagram above, answer the following questions:

- (i) Which one of the following will have the highest bond order? Circle the correct choice
A) PS^{1-} B) PS C) PS^{1+}
- (ii) Which one of the following will have the longest bond length? Circle the correct choice
A) PS^{1-} B) PS C) PS^{1+}
- (iii) Which of the following will be diamagnetic? Circle the correct choice.
A) PS^{1-} B) PS C) PS^{1+}
- (iv) Draw Lewis dot structure of the molecule with the shortest bond length (based on the MO diagram) from among the three (PS^{1-} , PS, PS^{1+}), and indicate the bond order calculated from the Lewis dot structure. **(3 points)**

Lewis Dot Structure

Bond Order



F84607F4-BFD6-4435-A0CA-42F08518B2AC

midterm-exam-93388

#543 Page 10 of 10

SCRAP PAGE

**CHEMISTRY 110 –Midterm Exam****Thursday November 7, 2024, 6:30 – 8:30 pm**

Examiners: Pallavi Sirjoosingh and Ashok Kakkar

First name (please write as legibly as possible within the boxes)

--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--

Last name

--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--

Student ID number

--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--

This booklet has 8 questions on 10 pages (The last page can be used as a scrap page).

The staple must NOT be removed from the question booklet. Your intact question booklet must be returned at the end of the exam.

A separate booklet contains a periodic table, and datasheet.

Write your answers directly under the questions in the space provided. Do not write on the QR codes or near the top edges (as it may be cut off).

Please write **legibly and dark enough** for the scan to be readable. For students who use English as a second language, dictionaries are permitted. Calculators are allowed.

DO NOT ASK QUESTIONS OF THE INVIGILATORS about interpretations of the exam questions. If you have a concern about a question, please summarize it at the top of this page. All concerns will be evaluated after the exam. GOOD LUCK!

**Question 1 8 points** (*Assume all transitions are possible*)

a. 6 points Calculate the **energy (in kJ/mol)** required to ionize (completely remove) the electron from the **$n=3$** level of **He^+ ion** (He: Helium). Show your work including equations. (3 sig. figs.)

b. 2 points

(I) Select the largest (longest) wavelength from the three choices below:

- i. Wavelength of photon to *completely remove* an electron from *1s orbital of Hydrogen atom*
- ii. Wavelength of photon to *completely remove* an electron from *1s orbital of Boron atom (B)*
- iii. Wavelength of photon to *completely remove* an electron from *1s orbital of Boron cation (B^+)*

i or ii or iii

(II) Select the smallest (shortest) wavelength from the three choices below (*for H atom*):

- i. Wavelength of photon *emitted* for an electron transitioning from *2p to 1s orbital*
- ii. Wavelength of photon *absorbed* for an electron transitioning from *2s to 4p orbital*
- iii. Wavelength of photon *absorbed* for an electron transitioning from *1s to 3p orbital*

i or ii or iii



Question 2 8 points



A Chem110 student plots the graph shown on the left, from data obtained for the kinetic energy (eV) of the ejected electron versus frequency (s^{-1}) of the incident light, for an unknown metal. ($1\text{eV} = 1.6022 \times 10^{-19} \text{ J}$)

Based on the plot, answer the questions **a to c** below:

a. 1 point On the plot above, label the **threshold frequency**.

b. 2 points What is the value of the work potential (in J) of the unknown metal?

c. 3 points What is the *longest* wavelength (in nm) required to eject an electron from this metal's surface? Show your work including equation(s) used. (3 sig figs)

d. 2 points In the space provided below, plot the following

(Assume frequency is *above* threshold frequency):

- i. *Kinetic energy* of an ejected electron versus the *intensity* of the incident light (left)
- ii. *Number of electrons* ejected versus the *frequency* of the incident light (right)



**Question 3** **6 points****a. 2 points**

In the ground state of Ge (Germanium; $Z = 32$) atom, **name** the following orbitals:

i. Orbital containing the electron that requires the maximum energy to be removed.

ii. Orbital containing the electron that experiences the minimum effective nuclear charge.

b. 2 points

An unknown anion (A^-) has the following *excited* electronic configuration: $[Ar] 4s^2 3d^3 4p^1$

(Ar: Argon; $Z = 18$)

Give the **condensed ground state electronic configuration** of the atom, **A** and the cation, **A^{2+}** .
Show your work. (Assume all electronic transitions are possible)

Electronic Configuration of A:

Electronic Configuration of A^{2+} :

c. 2 points

In a single atom, determine how many electrons can have the following set of quantum numbers?

(i) $n = 5; l = 3$

In a single atom, determine how many electrons can have the following set of quantum numbers?

(i) $n = 3; l = 2; m_s = +1/2$

**Question 4** **8 points**

Rank the following from 1 to 4 in **increasing** order. Provide the ranking in the boxes provided.

Note: Atomic number (Z) are provided for relevant atoms

a) Increasing (lowest to highest) order of *ionization energy*: (1= lowest; 4 = highest)

Mg

Mg: Z = 12

Rb

Rb: Z = 37

Cs

Cs: Z = 55

K

K : Z = 19

b) Increasing (lowest to highest) order of *bond angle*: (1= lowest; 4 = highest)

H₂OCCl₄XeF₄

HCN

c) Increasing (smallest to largest) order of *ionic size* (1= smallest; 4 = largest)

S²⁻

S: Z = 16

O²⁻

O: Z = 8

Be²⁺

Be: Z = 4

Na⁺

Na: Z = 11

d) Increasing (lowest to highest) order of *bond energy* (1= lowest; 4 = highest)

P-F

P-Cl

P-I

P-Br

**Question 5** **12 points**

a. 9 points For the molecules HClO_4 and TeF_6 , draw the most stable *Lewis structure* showing appropriate shape and *dashed/wedged bonds*, based on VSEPR. Determine the **molecular geometry** around the central atom. For HClO_4 , label the *most polar bond*.



b. 3 points (Note: Lone pairs are not shown in structures below)

(i) For the three *resonance* structures (1, 2 and 3) shown below, answer the following question:



(I) Circle the *most contributing* resonance structure.

Structure 1

Structure 2

Structure 3

(ii) For the three structures given (4, 5, and 6) below, answer the following questions:



(I) Circle all the structures that represent *resonance structures* of each other:

Structure 4

Structure 5

Structure 6

(II) Among the resonance structures, circle the *most contributing* resonance structure.

Structure 4

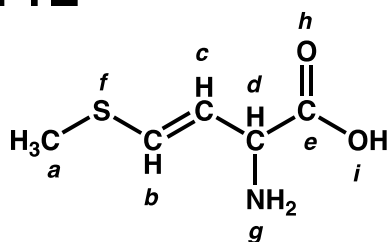
Structure 5

Structure 6

**Question 6** **6 points**

Using the data given below, calculate the **lattice energy** for SrO(s).

	ΔH° (kJ/mol)
$\text{Sr}(s) \longrightarrow \text{Sr}(g)$	+180
$\text{Sr}(g) \longrightarrow \text{Sr}^{2+}(g) + 2e^-$	+1468
$\text{O}_2(g) \longrightarrow 2\text{O}(g)$	+500
$\text{O}(g) + 2e^- \longrightarrow \text{O}^{2-}(g)$	+600
$\text{Sr}(s) + \frac{1}{2}\text{O}_2(g) \longrightarrow \text{SrO}(s)$	-550

**Question 7 10 points**

Consider the molecule given on the left, and answer questions (i) to (vi) using chemical bonding theories that we have learnt in class (valence bond theory, hybridization).

Note that the lone pairs are not shown in the structure:

i. Consider the carbon atoms marked in as ***a***, ***b***, ***c***, ***d***, and ***e*** in the molecule, and determine **hybridization** at each of the carbon centers. (2.5 points)

Atom	Hybridization
Carbon (<i>a</i>)	
Carbon (<i>b</i>)	
Carbon (<i>c</i>)	
Carbon (<i>d</i>)	
Carbon (<i>e</i>)	

ii. Now, consider the sulfur atom (S) marked as ***f***, nitrogen (N) atom (***g***), and oxygen(O) atoms (***h*** and ***i***), and determine their **hybridization**. (2 points)

Atom	Hybridization
Sulfur (<i>f</i>)	
Nitrogen (<i>g</i>)	
Oxygen (<i>h</i>)	
Oxygen (<i>i</i>)	

iii. **Describe** all the bonds formed by the carbon atoms ***a*** and ***b***, using sigma and pi designation as appropriate. (2 points)

Atom	Bonds formed (use sigma and pi as appropriate)
Carbon (<i>a</i>)	
Carbon (<i>b</i>)	

iv. How many total sp^3 -s sigma bonds are in the molecule? (1 point)

v. Determine **electronic** and **molecular geometry** at carbon (C) labeled ***e*** and sulfur (S) labeled ***f*** centres. (2 points)

Atom	Electronic Geometry	Molecular Geometry
Carbon (<i>e</i>)		
Sulfur (<i>f</i>)		

vi. Circle the value from the choices given below, that describes the **C-S-C bond angle at sulfur** (S) marked ***f***: (0.5 points)

>90° to <109.5° 109.5° >109.5 to <120° 120° >120° to <180°


Question 8 12 points

Using the **valence shell orbitals only** draw a **complete** molecular orbital (MO) diagram (with no s-p mixing) for the molecule **PS**, in the space provided below. **(6 points)**

Assume that 3s and 3p atomic orbitals will form molecular orbitals like those formed by the combination of 2s and 2p atomic orbitals that we learnt in class, and there is no s-p mixing.

Show **energy arrow**, **atomic orbitals** involved, **atom labels**, the **molecular orbitals** that are formed, **filled electrons** in the MO diagram etc. Make sure to label all the atomic and molecular orbitals.

Using the MO diagram above, answer the following questions:

- (i) Which one of the following will have the highest bond order? Circle the correct choice
 A) PS^{1-} B) PS C) PS^{1+}
- (ii) Which one of the following will have the longest bond length? Circle the correct choice
 A) PS^{1-} B) PS C) PS^{1+}
- (iii) Which of the following will be diamagnetic? Circle the correct choice.
 A) PS^{1-} B) PS C) PS^{1+}
- (iv) Draw Lewis dot structure of the molecule with the shortest bond length (based on the MO diagram) from among the three (PS^{1-} , PS, PS^{1+}), and indicate the bond order calculated from the Lewis dot structure. **(3 points)**

Lewis Dot Structure

Bond Order



0C533C0C-9D99-4FF9-9F14-12EE0CA1135D

midterm-exam-93388

#544 Page 10 of 10

SCRAP PAGE

**CHEMISTRY 110 –Midterm Exam****Thursday November 7, 2024, 6:30 – 8:30 pm**

Examiners: Pallavi Sirjoosingh and Ashok Kakkar

First name (please write as legibly as possible within the boxes)

--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--

Last name

--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--

Student ID number

--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--

This booklet has 8 questions on 10 pages (The last page can be used as a scrap page).

The staple must NOT be removed from the question booklet. Your intact question booklet must be returned at the end of the exam.

A separate booklet contains a periodic table, and datasheet.

Write your answers directly under the questions in the space provided. Do not write on the QR codes or near the top edges (as it may be cut off).

Please write **legibly and dark enough** for the scan to be readable. For students who use English as a second language, dictionaries are permitted. Calculators are allowed.

DO NOT ASK QUESTIONS OF THE INVIGILATORS about interpretations of the exam questions. If you have a concern about a question, please summarize it at the top of this page. All concerns will be evaluated after the exam. GOOD LUCK!

**Question 1 8 points** (*Assume all transitions are possible*)

a. 6 points Calculate the **energy (in kJ/mol)** required to ionize (completely remove) the electron from the **$n=3$** level of **He^+ ion** (He: Helium). Show your work including equations. (3 sig. figs.)

b. 2 points

(I) Select the largest (longest) wavelength from the three choices below:

- i. Wavelength of photon to *completely remove* an electron from *1s orbital of Hydrogen atom*
- ii. Wavelength of photon to *completely remove* an electron from *1s orbital of Boron atom (B)*
- iii. Wavelength of photon to *completely remove* an electron from *1s orbital of Boron cation (B^+)*

i or ii or iii

(II) Select the smallest (shortest) wavelength from the three choices below (*for H atom*):

- i. Wavelength of photon *emitted* for an electron transitioning from *2p to 1s orbital*
- ii. Wavelength of photon *absorbed* for an electron transitioning from *2s to 4p orbital*
- iii. Wavelength of photon *absorbed* for an electron transitioning from *1s to 3p orbital*

i or ii or iii



Question 2 8 points



A Chem110 student plots the graph shown on the left, from data obtained for the kinetic energy (eV) of the ejected electron versus frequency (s^{-1}) of the incident light, for an unknown metal. ($1\text{eV} = 1.6022 \times 10^{-19} \text{ J}$)

Based on the plot, answer the questions **a to c** below:

a. 1 point On the plot above, label the **threshold frequency**.

b. 2 points What is the value of the work potential (in J) of the unknown metal?

c. 3 points What is the *longest* wavelength (in nm) required to eject an electron from this metal's surface? Show your work including equation(s) used. (3 sig figs)

d. 2 points In the space provided below, plot the following

(Assume frequency is *above* threshold frequency):

- i. *Kinetic energy* of an ejected electron versus the *intensity* of the incident light (left)
- ii. *Number of electrons* ejected versus the *frequency* of the incident light (right)



**Question 3** **6 points****a. 2 points**

In the ground state of Ge (Germanium; $Z = 32$) atom, **name** the following orbitals:

i. Orbital containing the electron that requires the maximum energy to be removed.

ii. Orbital containing the electron that experiences the minimum effective nuclear charge.

b. 2 points

An unknown anion (A^-) has the following *excited* electronic configuration: $[Ar] 4s^2 3d^3 4p^1$

(Ar: Argon; $Z = 18$)

Give the **condensed ground state electronic configuration** of the atom, **A** and the cation, **A^{2+}** .
Show your work. (Assume all electronic transitions are possible)

Electronic Configuration of A:

Electronic Configuration of A^{2+} :

c. 2 points

In a single atom, determine how many electrons can have the following set of quantum numbers?

(i) $n = 5; l = 3$

In a single atom, determine how many electrons can have the following set of quantum numbers?

(i) $n = 3; l = 2; m_s = +1/2$

**Question 4** **8 points**

Rank the following from 1 to 4 in **increasing** order. Provide the ranking in the boxes provided.

Note: Atomic number (Z) are provided for relevant atoms

a) Increasing (lowest to highest) order of *ionization energy*: (1= lowest; 4 = highest)

Mg

Mg: Z = 12

Rb

Rb: Z = 37

Cs

Cs: Z = 55

K

K : Z = 19

b) Increasing (lowest to highest) order of *bond angle*: (1= lowest; 4 = highest)

H₂OCCl₄XeF₄

HCN

c) Increasing (smallest to largest) order of *ionic size* (1= smallest; 4 = largest)

S²⁻

S: Z = 16

O²⁻

O: Z = 8

Be²⁺

Be: Z = 4

Na⁺

Na: Z = 11

d) Increasing (lowest to highest) order of *bond energy* (1= lowest; 4 = highest)

P-F

P-Cl

P-I

P-Br

**Question 5** **12 points**

a. 9 points For the molecules HClO_4 and TeF_6 , draw the most stable *Lewis structure* showing appropriate shape and *dashed/wedged bonds*, based on VSEPR. Determine the **molecular geometry** around the central atom. For HClO_4 , label the *most polar bond*.



b. 3 points (Note: Lone pairs are not shown in structures below)

(i) For the three *resonance* structures (1, 2 and 3) shown below, answer the following question:



(I) Circle the *most contributing* resonance structure.

Structure 1

Structure 2

Structure 3

(ii) For the three structures given (4, 5, and 6) below, answer the following questions:



(I) Circle all the structures that represent *resonance structures* of each other:

Structure 4

Structure 5

Structure 6

(II) Among the resonance structures, circle the *most contributing* resonance structure.

Structure 4

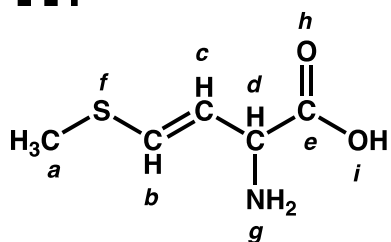
Structure 5

Structure 6

**Question 6** **6 points**

Using the data given below, calculate the **lattice energy** for SrO(s).

	ΔH° (kJ/mol)
$\text{Sr}(s) \longrightarrow \text{Sr}(g)$	+180
$\text{Sr}(g) \longrightarrow \text{Sr}^{2+}(g) + 2e^-$	+1468
$\text{O}_2(g) \longrightarrow 2\text{O}(g)$	+500
$\text{O}(g) + 2e^- \longrightarrow \text{O}^{2-}(g)$	+600
$\text{Sr}(s) + \frac{1}{2}\text{O}_2(g) \longrightarrow \text{SrO}(s)$	-550

**Question 7 10 points**

Consider the molecule given on the left, and answer questions (i) to (vi) using chemical bonding theories that we have learnt in class (valence bond theory, hybridization).

Note that the lone pairs are not shown in the structure:

i. Consider the carbon atoms marked in as ***a***, ***b***, ***c***, ***d***, and ***e*** in the molecule, and determine **hybridization** at each of the carbon centers. (2.5 points)

Atom	Hybridization
Carbon (<i>a</i>)	
Carbon (<i>b</i>)	
Carbon (<i>c</i>)	
Carbon (<i>d</i>)	
Carbon (<i>e</i>)	

ii. Now, consider the sulfur atom (S) marked as ***f***, nitrogen (N) atom (***g***), and oxygen(O) atoms (***h*** and ***i***), and determine their **hybridization**. (2 points)

Atom	Hybridization
Sulfur (<i>f</i>)	
Nitrogen (<i>g</i>)	
Oxygen (<i>h</i>)	
Oxygen (<i>i</i>)	

iii. **Describe** all the bonds formed by the carbon atoms ***a*** and ***b***, using sigma and pi designation as appropriate. (2 points)

Atom	Bonds formed (use sigma and pi as appropriate)
Carbon (<i>a</i>)	
Carbon (<i>b</i>)	

iv. How many total sp^3 -s sigma bonds are in the molecule? (1 point)

v. Determine **electronic** and **molecular geometry** at carbon (C) labeled ***e*** and sulfur (S) labeled ***f*** centres. (2 points)

Atom	Electronic Geometry	Molecular Geometry
Carbon (<i>e</i>)		
Sulfur (<i>f</i>)		

vi. Circle the value from the choices given below, that describes the **C-S-C bond angle at sulfur** (S) marked ***f***: (0.5 points)

>90° to <109.5° 109.5° >109.5 to <120° 120° >120° to <180°


Question 8 12 points

Using the **valence shell orbitals only** draw a **complete** molecular orbital (MO) diagram (with no s-p mixing) for the molecule **PS**, in the space provided below. **(6 points)**

Assume that 3s and 3p atomic orbitals will form molecular orbitals like those formed by the combination of 2s and 2p atomic orbitals that we learnt in class, and there is no s-p mixing.

Show **energy arrow**, **atomic orbitals** involved, **atom labels**, the **molecular orbitals** that are formed, **filled electrons** in the MO diagram etc. Make sure to label all the atomic and molecular orbitals.

Using the MO diagram above, answer the following questions:

- (i) Which one of the following will have the highest bond order? Circle the correct choice
 A) PS^{1-} B) PS C) PS^{1+}
- (ii) Which one of the following will have the longest bond length? Circle the correct choice
 A) PS^{1-} B) PS C) PS^{1+}
- (iii) Which of the following will be diamagnetic? Circle the correct choice.
 A) PS^{1-} B) PS C) PS^{1+}
- (iv) Draw Lewis dot structure of the molecule with the shortest bond length (based on the MO diagram) from among the three (PS^{1-} , PS, PS^{1+}), and indicate the bond order calculated from the Lewis dot structure. **(3 points)**

Lewis Dot Structure

Bond Order



23019162-11AA-4614-B935-D016D3257021

midterm-exam-93388

#545 Page 10 of 10

SCRAP PAGE

**CHEMISTRY 110 –Midterm Exam****Thursday November 7, 2024, 6:30 – 8:30 pm**

Examiners: Pallavi Sirjoosingh and Ashok Kakkar

First name (please write as legibly as possible within the boxes)

--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--

Last name

--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--

Student ID number

--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--

This booklet has 8 questions on 10 pages (The last page can be used as a scrap page).

The staple must NOT be removed from the question booklet. Your intact question booklet must be returned at the end of the exam.

A separate booklet contains a periodic table, and datasheet.

Write your answers directly under the questions in the space provided. Do not write on the QR codes or near the top edges (as it may be cut off).

Please write **legibly and dark enough** for the scan to be readable. For students who use English as a second language, dictionaries are permitted. Calculators are allowed.

DO NOT ASK QUESTIONS OF THE INVIGILATORS about interpretations of the exam questions. If you have a concern about a question, please summarize it at the top of this page. All concerns will be evaluated after the exam. GOOD LUCK!

**Question 1 8 points** (*Assume all transitions are possible*)

a. 6 points Calculate the **energy (in kJ/mol)** required to ionize (completely remove) the electron from the **n=3** level of **He⁺ ion** (He: Helium). Show your work including equations. (3 sig. figs.)

b. 2 points

(I) Select the largest (longest) wavelength from the three choices below:

- i. Wavelength of photon to *completely remove* an electron from *1s orbital of Hydrogen atom*
- ii. Wavelength of photon to *completely remove* an electron from *1s orbital of Boron atom (B)*
- iii. Wavelength of photon to *completely remove* an electron from *1s orbital of Boron cation (B⁺)*

i or ii or iii

(II) Select the smallest (shortest) wavelength from the three choices below (*for H atom*):

- i. Wavelength of photon *emitted* for an electron transitioning from *2p to 1s orbital*
- ii. Wavelength of photon *absorbed* for an electron transitioning from *2s to 4p orbital*
- iii. Wavelength of photon *absorbed* for an electron transitioning from *1s to 3p orbital*

i or ii or iii



Question 2 8 points



A Chem110 student plots the graph shown on the left, from data obtained for the kinetic energy (eV) of the ejected electron versus frequency (s^{-1}) of the incident light, for an unknown metal. ($1\text{eV} = 1.6022 \times 10^{-19} \text{ J}$)

Based on the plot, answer the questions **a to c** below:

a. 1 point On the plot above, label the **threshold frequency**.

b. 2 points What is the value of the work potential (in J) of the unknown metal?

c. 3 points What is the *longest* wavelength (in nm) required to eject an electron from this metal's surface? Show your work including equation(s) used. (3 sig figs)

d. 2 points In the space provided below, plot the following

(Assume frequency is *above* threshold frequency):

- Kinetic energy* of an ejected electron versus the *intensity* of the incident light (left)
- Number of electrons* ejected versus the *frequency* of the incident light (right)



**Question 3** **6 points****a. 2 points**

In the ground state of Ge (Germanium; $Z = 32$) atom, **name** the following orbitals:

i. Orbital containing the electron that requires the maximum energy to be removed.

ii. Orbital containing the electron that experiences the minimum effective nuclear charge.

b. 2 points

An unknown anion (A^-) has the following *excited* electronic configuration: $[Ar] 4s^2 3d^3 4p^1$

(Ar: Argon; $Z = 18$)

Give the **condensed ground state electronic configuration** of the atom, **A** and the cation, **A^{2+}** .
Show your work. (Assume all electronic transitions are possible)

Electronic Configuration of A:

Electronic Configuration of A^{2+} :

c. 2 points

In a single atom, determine how many electrons can have the following set of quantum numbers?

(i) $n = 5; l = 3$

In a single atom, determine how many electrons can have the following set of quantum numbers?

(i) $n = 3; l = 2; m_s = +1/2$

**Question 4** **8 points**

Rank the following from 1 to 4 in **increasing** order. Provide the ranking in the boxes provided.

Note: Atomic number (Z) are provided for relevant atoms

a) Increasing (lowest to highest) order of *ionization energy*: (1= lowest; 4 = highest)

Mg

Mg: Z = 12

Rb

Rb: Z = 37

Cs

Cs: Z = 55

K

K : Z = 19

b) Increasing (lowest to highest) order of *bond angle*: (1= lowest; 4 = highest)

H₂OCCl₄XeF₄

HCN

c) Increasing (smallest to largest) order of *ionic size* (1= smallest; 4 = largest)

S²⁻

S: Z = 16

O²⁻

O: Z = 8

Be²⁺

Be: Z = 4

Na⁺

Na: Z = 11

d) Increasing (lowest to highest) order of *bond energy* (1= lowest; 4 = highest)

P-F

P-Cl

P-I

P-Br

**Question 5** **12 points**

a. 9 points For the molecules HClO_4 and TeF_6 , draw the most stable *Lewis structure* showing appropriate shape and *dashed/wedged bonds*, based on VSEPR. Determine the **molecular geometry** around the central atom. For HClO_4 , label the *most polar bond*.



b. 3 points (Note: Lone pairs are not shown in structures below)

(i) For the three *resonance* structures (1, 2 and 3) shown below, answer the following question:



(I) Circle the *most contributing* resonance structure.

Structure 1

Structure 2

Structure 3

(ii) For the three structures given (4, 5, and 6) below, answer the following questions:



(I) Circle all the structures that represent *resonance structures* of each other:

Structure 4

Structure 5

Structure 6

(II) Among the resonance structures, circle the *most contributing* resonance structure.

Structure 4

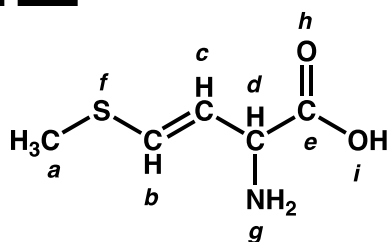
Structure 5

Structure 6

**Question 6** **6 points**

Using the data given below, calculate the **lattice energy** for SrO(s).

	ΔH° (kJ/mol)
$\text{Sr}(s) \longrightarrow \text{Sr}(g)$	+180
$\text{Sr}(g) \longrightarrow \text{Sr}^{2+}(g) + 2e^-$	+1468
$\text{O}_2(g) \longrightarrow 2\text{O}(g)$	+500
$\text{O}(g) + 2e^- \longrightarrow \text{O}^{2-}(g)$	+600
$\text{Sr}(s) + \frac{1}{2}\text{O}_2(g) \longrightarrow \text{SrO}(s)$	-550

**Question 7 10 points**

Consider the molecule given on the left, and answer questions (i) to (vi) using chemical bonding theories that we have learnt in class (valence bond theory, hybridization).

Note that the lone pairs are not shown in the structure:

i. Consider the carbon atoms marked in as ***a***, ***b***, ***c***, ***d***, and ***e*** in the molecule, and determine **hybridization** at each of the carbon centers. (2.5 points)

Atom	Hybridization
Carbon (<i>a</i>)	
Carbon (<i>b</i>)	
Carbon (<i>c</i>)	
Carbon (<i>d</i>)	
Carbon (<i>e</i>)	

ii. Now, consider the sulfur atom (S) marked as ***f***, nitrogen (N) atom (***g***), and oxygen(O) atoms (***h*** and ***i***), and determine their **hybridization**. (2 points)

Atom	Hybridization
Sulfur (<i>f</i>)	
Nitrogen (<i>g</i>)	
Oxygen (<i>h</i>)	
Oxygen (<i>i</i>)	

iii. **Describe** all the bonds formed by the carbon atoms ***a*** and ***b***, using sigma and pi designation as appropriate. (2 points)

Atom	Bonds formed (use sigma and pi as appropriate)
Carbon (<i>a</i>)	
Carbon (<i>b</i>)	

iv. How many total sp^3 -s sigma bonds are in the molecule? (1 point)

v. Determine **electronic** and **molecular geometry** at carbon (C) labeled ***e*** and sulfur (S) labeled ***f*** centres. (2 points)

Atom	Electronic Geometry	Molecular Geometry
Carbon (<i>e</i>)		
Sulfur (<i>f</i>)		

vi. Circle the value from the choices given below, that describes the **C-S-C bond angle at sulfur** (S) marked ***f***: (0.5 points)

$>90^\circ$ to $<109.5^\circ$ 109.5° >109.5 to $<120^\circ$ 120° $>120^\circ$ to $<180^\circ$


Question 8 12 points

Using the **valence shell orbitals only** draw a **complete** molecular orbital (MO) diagram (with no s-p mixing) for the molecule **PS**, in the space provided below. **(6 points)**

Assume that 3s and 3p atomic orbitals will form molecular orbitals like those formed by the combination of 2s and 2p atomic orbitals that we learnt in class, and there is no s-p mixing.

Show **energy arrow**, **atomic orbitals** involved, **atom labels**, the **molecular orbitals** that are formed, **filled electrons** in the MO diagram etc. Make sure to **label all the atomic and molecular orbitals**.

Using the MO diagram above, answer the following questions:

- (i) Which one of the following will have the highest bond order? Circle the correct choice
 A) PS^{1-} B) PS C) PS^{1+}
- (ii) Which one of the following will have the longest bond length? Circle the correct choice
 A) PS^{1-} B) PS C) PS^{1+}
- (iii) Which of the following will be diamagnetic? Circle the correct choice.
 A) PS^{1-} B) PS C) PS^{1+}
- (iv) Draw Lewis dot structure of the molecule with the shortest bond length (based on the MO diagram) from among the three (PS^{1-} , PS, PS^{1+}), and indicate the bond order calculated from the Lewis dot structure. **(3 points)**

Lewis Dot Structure

Bond Order



4D96FF77-430B-483D-B670-E6548EF937B0

midterm-exam-93388

#546 Page 10 of 10

SCRAP PAGE

**CHEMISTRY 110 –Midterm Exam****Thursday November 7, 2024, 6:30 – 8:30 pm**

Examiners: Pallavi Sirjoosingh and Ashok Kakkar

First name (please write as legibly as possible within the boxes)

--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--

Last name

--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--

Student ID number

--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--

This booklet has 8 questions on 10 pages (The last page can be used as a scrap page).

The staple must NOT be removed from the question booklet. Your intact question booklet must be returned at the end of the exam.

A separate booklet contains a periodic table, and datasheet.

Write your answers directly under the questions in the space provided. Do not write on the QR codes or near the top edges (as it may be cut off).

Please write **legibly and dark enough** for the scan to be readable. For students who use English as a second language, dictionaries are permitted. Calculators are allowed.

DO NOT ASK QUESTIONS OF THE INVIGILATORS about interpretations of the exam questions. If you have a concern about a question, please summarize it at the top of this page. All concerns will be evaluated after the exam. GOOD LUCK!

**Question 1 8 points** (*Assume all transitions are possible*)

a. 6 points Calculate the **energy (in kJ/mol)** required to ionize (completely remove) the electron from the **n=3** level of **He⁺ ion** (He: Helium). Show your work including equations. (3 sig. figs.)

b. 2 points

(I) Select the largest (longest) wavelength from the three choices below:

- i. Wavelength of photon to *completely remove* an electron from *1s orbital of Hydrogen atom*
- ii. Wavelength of photon to *completely remove* an electron from *1s orbital of Boron atom (B)*
- iii. Wavelength of photon to *completely remove* an electron from *1s orbital of Boron cation (B⁺)*

i or ii or iii

(II) Select the smallest (shortest) wavelength from the three choices below (*for H atom*):

- i. Wavelength of photon *emitted* for an electron transitioning from *2p to 1s orbital*
- ii. Wavelength of photon *absorbed* for an electron transitioning from *2s to 4p orbital*
- iii. Wavelength of photon *absorbed* for an electron transitioning from *1s to 3p orbital*

i or ii or iii



Question 2 8 points



A Chem110 student plots the graph shown on the left, from data obtained for the kinetic energy (eV) of the ejected electron versus frequency (s^{-1}) of the incident light, for an unknown metal. ($1\text{eV} = 1.6022 \times 10^{-19} \text{ J}$)

Based on the plot, answer the questions **a to c** below:

a. 1 point On the plot above, label the **threshold frequency**.

b. 2 points What is the value of the work potential (in J) of the unknown metal?

c. 3 points What is the *longest* wavelength (in nm) required to eject an electron from this metal's surface? Show your work including equation(s) used. (3 sig figs)

d. 2 points In the space provided below, plot the following

(Assume frequency is *above* threshold frequency):

- Kinetic energy* of an ejected electron versus the *intensity* of the incident light (left)
- Number of electrons* ejected versus the *frequency* of the incident light (right)



**Question 3** **6 points****a. 2 points**

In the ground state of Ge (Germanium; $Z = 32$) atom, **name** the following orbitals:

i. Orbital containing the electron that requires the maximum energy to be removed.

ii. Orbital containing the electron that experiences the minimum effective nuclear charge.

b. 2 points

An unknown anion (A^-) has the following *excited* electronic configuration: $[Ar] 4s^2 3d^3 4p^1$

(Ar: Argon; $Z = 18$)

Give the **condensed ground state electronic configuration** of the atom, **A** and the cation, **A^{2+}** .
Show your work. (Assume all electronic transitions are possible)

Electronic Configuration of A:

Electronic Configuration of A^{2+} :

c. 2 points

In a single atom, determine how many electrons can have the following set of quantum numbers?

(i) $n = 5; l = 3$

In a single atom, determine how many electrons can have the following set of quantum numbers?

(i) $n = 3; l = 2; m_s = +1/2$

**Question 4** **8 points**

Rank the following from 1 to 4 in **increasing** order. Provide the ranking in the boxes provided.

Note: Atomic number (Z) are provided for relevant atoms

a) Increasing (lowest to highest) order of *ionization energy*: (1= lowest; 4 = highest)

Mg

Mg: Z = 12

Rb

Rb: Z = 37

Cs

Cs: Z = 55

K

K : Z = 19

b) Increasing (lowest to highest) order of *bond angle*: (1= lowest; 4 = highest)

H₂OCCl₄XeF₄

HCN

c) Increasing (smallest to largest) order of *ionic size* (1= smallest; 4 = largest)

S²⁻

S: Z = 16

O²⁻

O: Z = 8

Be²⁺

Be: Z = 4

Na⁺

Na: Z = 11

d) Increasing (lowest to highest) order of *bond energy* (1= lowest; 4 = highest)

P-F

P-Cl

P-I

P-Br

**Question 5** **12 points**

a. 9 points For the molecules HClO_4 and TeF_6 , draw the most stable *Lewis structure* showing appropriate shape and *dashed/wedged bonds*, based on VSEPR. Determine the **molecular geometry** around the central atom. For HClO_4 , label the *most polar bond*.



b. 3 points (Note: Lone pairs are not shown in structures below)

(i) For the three *resonance* structures (1, 2 and 3) shown below, answer the following question:



(I) Circle the *most contributing* resonance structure.

Structure 1

Structure 2

Structure 3

(ii) For the three structures given (4, 5, and 6) below, answer the following questions:



(I) Circle all the structures that represent *resonance structures* of each other:

Structure 4

Structure 5

Structure 6

(II) Among the resonance structures, circle the *most contributing* resonance structure.

Structure 4

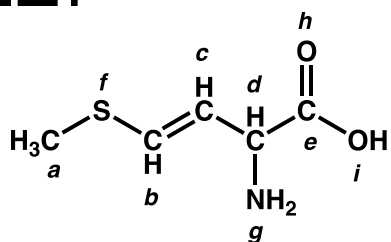
Structure 5

Structure 6

**Question 6** **6 points**

Using the data given below, calculate the **lattice energy** for SrO(s).

	ΔH° (kJ/mol)
$\text{Sr}(s) \longrightarrow \text{Sr}(g)$	+180
$\text{Sr}(g) \longrightarrow \text{Sr}^{2+}(g) + 2e^-$	+1468
$\text{O}_2(g) \longrightarrow 2\text{O}(g)$	+500
$\text{O}(g) + 2e^- \longrightarrow \text{O}^{2-}(g)$	+600
$\text{Sr}(s) + \frac{1}{2}\text{O}_2(g) \longrightarrow \text{SrO}(s)$	-550

**Question 7 10 points**

Consider the molecule given on the left, and answer questions (i) to (vi) using chemical bonding theories that we have learnt in class (valence bond theory, hybridization).

Note that the lone pairs are not shown in the structure:

i. Consider the carbon atoms marked in as ***a***, ***b***, ***c***, ***d***, and ***e*** in the molecule, and determine **hybridization** at each of the carbon centers. (2.5 points)

Atom	Hybridization
Carbon (<i>a</i>)	
Carbon (<i>b</i>)	
Carbon (<i>c</i>)	
Carbon (<i>d</i>)	
Carbon (<i>e</i>)	

ii. Now, consider the sulfur atom (S) marked as ***f***, nitrogen (N) atom (***g***), and oxygen(O) atoms (***h*** and ***i***), and determine their **hybridization**. (2 points)

Atom	Hybridization
Sulfur (<i>f</i>)	
Nitrogen (<i>g</i>)	
Oxygen (<i>h</i>)	
Oxygen (<i>i</i>)	

iii. **Describe** all the bonds formed by the carbon atoms ***a*** and ***b***, using sigma and pi designation as appropriate. (2 points)

Atom	Bonds formed (use sigma and pi as appropriate)
Carbon (<i>a</i>)	
Carbon (<i>b</i>)	

iv. How many total sp^3 -s sigma bonds are in the molecule? (1 point)

v. Determine **electronic** and **molecular geometry** at carbon (C) labeled ***e*** and sulfur (S) labeled ***f*** centres. (2 points)

Atom	Electronic Geometry	Molecular Geometry
Carbon (<i>e</i>)		
Sulfur (<i>f</i>)		

vi. Circle the value from the choices given below, that describes the **C-S-C bond angle at sulfur** (S) marked ***f***: (0.5 points)

>90° to <109.5° 109.5° >109.5 to <120° 120° >120° to <180°


Question 8 12 points

Using the **valence shell orbitals only** draw a **complete** molecular orbital (MO) diagram (with no s-p mixing) for the molecule **PS**, in the space provided below. **(6 points)**

Assume that 3s and 3p atomic orbitals will form molecular orbitals like those formed by the combination of 2s and 2p atomic orbitals that we learnt in class, and there is no s-p mixing.

Show **energy arrow**, **atomic orbitals** involved, **atom labels**, the **molecular orbitals** that are formed, **filled electrons** in the MO diagram etc. Make sure to label all the atomic and molecular orbitals.

Using the MO diagram above, answer the following questions:

- (i) Which one of the following will have the highest bond order? Circle the correct choice
 A) PS^{1-} B) PS C) PS^{1+}
- (ii) Which one of the following will have the longest bond length? Circle the correct choice
 A) PS^{1-} B) PS C) PS^{1+}
- (iii) Which of the following will be diamagnetic? Circle the correct choice.
 A) PS^{1-} B) PS C) PS^{1+}
- (iv) Draw Lewis dot structure of the molecule with the shortest bond length (based on the MO diagram) from among the three (PS^{1-} , PS, PS^{1+}), and indicate the bond order calculated from the Lewis dot structure. **(3 points)**

Lewis Dot Structure

Bond Order



BA664E47-A251-4721-8385-C629FA96A552

midterm-exam-93388

#547 Page 10 of 10

SCRAP PAGE

**CHEMISTRY 110 –Midterm Exam****Thursday November 7, 2024, 6:30 – 8:30 pm**

Examiners: Pallavi Sirjoosingh and Ashok Kakkar

First name (please write as legibly as possible within the boxes)

--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--

Last name

--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--

Student ID number

--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--

This booklet has 8 questions on 10 pages (The last page can be used as a scrap page).

The staple must NOT be removed from the question booklet. Your intact question booklet must be returned at the end of the exam.

A separate booklet contains a periodic table, and datasheet.

Write your answers directly under the questions in the space provided. Do not write on the QR codes or near the top edges (as it may be cut off).

Please write **legibly and dark enough** for the scan to be readable. For students who use English as a second language, dictionaries are permitted. Calculators are allowed.

DO NOT ASK QUESTIONS OF THE INVIGILATORS about interpretations of the exam questions. If you have a concern about a question, please summarize it at the top of this page. All concerns will be evaluated after the exam. GOOD LUCK!

**Question 1 8 points** (*Assume all transitions are possible*)

a. 6 points Calculate the **energy (in kJ/mol)** required to ionize (completely remove) the electron from the **n=3** level of **He⁺ ion** (He: Helium). Show your work including equations. (3 sig. figs.)

b. 2 points

(I) Select the largest (longest) wavelength from the three choices below:

- i. Wavelength of photon to *completely remove* an electron from *1s orbital of Hydrogen atom*
- ii. Wavelength of photon to *completely remove* an electron from *1s orbital of Boron atom (B)*
- iii. Wavelength of photon to *completely remove* an electron from *1s orbital of Boron cation (B⁺)*

i or ii or iii

(II) Select the smallest (shortest) wavelength from the three choices below (*for H atom*):

- i. Wavelength of photon *emitted* for an electron transitioning from *2p to 1s orbital*
- ii. Wavelength of photon *absorbed* for an electron transitioning from *2s to 4p orbital*
- iii. Wavelength of photon *absorbed* for an electron transitioning from *1s to 3p orbital*

i or ii or iii



Question 2 8 points



A Chem110 student plots the graph shown on the left, from data obtained for the kinetic energy (eV) of the ejected electron versus frequency (s^{-1}) of the incident light, for an unknown metal. ($1\text{eV} = 1.6022 \times 10^{-19} \text{ J}$)

Based on the plot, answer the questions **a to c** below:

a. 1 point On the plot above, label the **threshold frequency**.

b. 2 points What is the value of the work potential (in J) of the unknown metal?

c. 3 points What is the *longest* wavelength (in nm) required to eject an electron from this metal's surface? Show your work including equation(s) used. (3 sig figs)

d. 2 points In the space provided below, plot the following

(Assume frequency is *above* threshold frequency):

- Kinetic energy* of an ejected electron versus the *intensity* of the incident light (left)
- Number of electrons* ejected versus the *frequency* of the incident light (right)



**Question 3** **6 points****a. 2 points**

In the ground state of Ge (Germanium; $Z = 32$) atom, **name** the following orbitals:

i. Orbital containing the electron that requires the maximum energy to be removed.

ii. Orbital containing the electron that experiences the minimum effective nuclear charge.

b. 2 points

An unknown anion (A^-) has the following *excited* electronic configuration: $[Ar] 4s^2 3d^3 4p^1$

(Ar: Argon; $Z = 18$)

Give the **condensed ground state electronic configuration** of the atom, **A** and the cation, **A^{2+}** .
Show your work. (Assume all electronic transitions are possible)

Electronic Configuration of A:

Electronic Configuration of A^{2+} :

c. 2 points

In a single atom, determine how many electrons can have the following set of quantum numbers?

(i) $n = 5; l = 3$

In a single atom, determine how many electrons can have the following set of quantum numbers?

(i) $n = 3; l = 2; m_s = +1/2$

**Question 4** **8 points**

Rank the following from 1 to 4 in **increasing** order. Provide the ranking in the boxes provided.

Note: Atomic number (Z) are provided for relevant atoms

a) Increasing (lowest to highest) order of *ionization energy*: (1= lowest; 4 = highest)

Mg

Mg: Z = 12

Rb

Rb: Z = 37

Cs

Cs: Z = 55

K

K : Z = 19

b) Increasing (lowest to highest) order of *bond angle*: (1= lowest; 4 = highest)

H₂OCCl₄XeF₄

HCN

c) Increasing (smallest to largest) order of *ionic size* (1= smallest; 4 = largest)

S²⁻

S: Z = 16

O²⁻

O: Z = 8

Be²⁺

Be: Z = 4

Na⁺

Na: Z = 11

d) Increasing (lowest to highest) order of *bond energy* (1= lowest; 4 = highest)

P-F

P-Cl

P-I

P-Br

**Question 5** **12 points**

a. 9 points For the molecules HClO_4 and TeF_6 , draw the most stable *Lewis structure* showing appropriate shape and *dashed/wedged bonds*, based on VSEPR. Determine the **molecular geometry** around the central atom. For HClO_4 , label the *most polar bond*.



b. 3 points (Note: Lone pairs are not shown in structures below)

(i) For the three *resonance* structures (1, 2 and 3) shown below, answer the following question:



(I) Circle the *most contributing* resonance structure.

Structure 1

Structure 2

Structure 3

(ii) For the three structures given (4, 5, and 6) below, answer the following questions:



(I) Circle all the structures that represent *resonance structures* of each other:

Structure 4

Structure 5

Structure 6

(II) Among the resonance structures, circle the *most contributing* resonance structure.

Structure 4

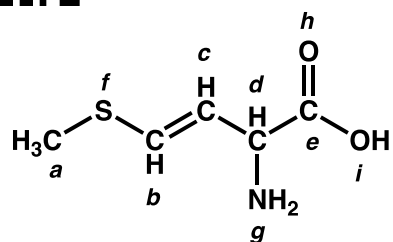
Structure 5

Structure 6

**Question 6** **6 points**

Using the data given below, calculate the **lattice energy** for SrO(s).

	ΔH° (kJ/mol)
$\text{Sr}(s) \longrightarrow \text{Sr}(g)$	+180
$\text{Sr}(g) \longrightarrow \text{Sr}^{2+}(g) + 2e^-$	+1468
$\text{O}_2(g) \longrightarrow 2\text{O}(g)$	+500
$\text{O}(g) + 2e^- \longrightarrow \text{O}^{2-}(g)$	+600
$\text{Sr}(s) + \frac{1}{2}\text{O}_2(g) \longrightarrow \text{SrO}(s)$	-550

**Question 7 10 points**

Consider the molecule given on the left, and answer questions (i) to (vi) using chemical bonding theories that we have learnt in class (valence bond theory, hybridization).

Note that the lone pairs are not shown in the structure:

i. Consider the carbon atoms marked in as ***a***, ***b***, ***c***, ***d***, and ***e*** in the molecule, and determine **hybridization** at each of the carbon centers. (2.5 points)

Atom	Hybridization
Carbon (<i>a</i>)	
Carbon (<i>b</i>)	
Carbon (<i>c</i>)	
Carbon (<i>d</i>)	
Carbon (<i>e</i>)	

ii. Now, consider the sulfur atom (S) marked as ***f***, nitrogen (N) atom (***g***), and oxygen(O) atoms (***h*** and ***i***), and determine their **hybridization**. (2 points)

Atom	Hybridization
Sulfur (<i>f</i>)	
Nitrogen (<i>g</i>)	
Oxygen (<i>h</i>)	
Oxygen (<i>i</i>)	

iii. **Describe** all the bonds formed by the carbon atoms ***a*** and ***b***, using sigma and pi designation as appropriate. (2 points)

Atom	Bonds formed (use sigma and pi as appropriate)
Carbon (<i>a</i>)	
Carbon (<i>b</i>)	

iv. How many total sp^3 -s sigma bonds are in the molecule? (1 point)

v. Determine **electronic** and **molecular geometry** at carbon (C) labeled ***e*** and sulfur (S) labeled ***f*** centres. (2 points)

Atom	Electronic Geometry	Molecular Geometry
Carbon (<i>e</i>)		
Sulfur (<i>f</i>)		

vi. Circle the value from the choices given below, that describes the **C-S-C bond angle at sulfur** (S) marked ***f***: (0.5 points)

>90° to <109.5° 109.5° >109.5 to <120° 120° >120° to <180°

**Question 8 12 points**

Using the **valence shell orbitals only** draw a **complete** molecular orbital (MO) diagram (with no s-p mixing) for the molecule **PS**, in the space provided below. **(6 points)**

Assume that 3s and 3p atomic orbitals will form molecular orbitals like those formed by the combination of 2s and 2p atomic orbitals that we learnt in class, and there is no s-p mixing.

Show **energy arrow**, **atomic orbitals** involved, **atom labels**, the **molecular orbitals** that are formed, **filled electrons** in the MO diagram etc. Make sure to label all the atomic and molecular orbitals.

Using the MO diagram above, answer the following questions:

- (i) Which one of the following will have the highest bond order? Circle the correct choice
A) PS^{1-} B) PS C) PS^{1+}
- (ii) Which one of the following will have the longest bond length? Circle the correct choice
A) PS^{1-} B) PS C) PS^{1+}
- (iii) Which of the following will be diamagnetic? Circle the correct choice.
A) PS^{1-} B) PS C) PS^{1+}
- (iv) Draw Lewis dot structure of the molecule with the shortest bond length (based on the MO diagram) from among the three (PS^{1-} , PS, PS^{1+}), and indicate the bond order calculated from the Lewis dot structure. **(3 points)**

Lewis Dot Structure

Bond Order



C934EDE8-C44F-4329-9A7F-9F0BB1476705

midterm-exam-93388

#548 Page 10 of 10

SCRAP PAGE

**CHEMISTRY 110 –Midterm Exam****Thursday November 7, 2024, 6:30 – 8:30 pm**

Examiners: Pallavi Sirjoosingh and Ashok Kakkar

First name (please write as legibly as possible within the boxes)

--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--

Last name

--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--

Student ID number

--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--

This booklet has 8 questions on 10 pages (The last page can be used as a scrap page).

The staple must NOT be removed from the question booklet. Your intact question booklet must be returned at the end of the exam.

A separate booklet contains a periodic table, and datasheet.

Write your answers directly under the questions in the space provided. Do not write on the QR codes or near the top edges (as it may be cut off).

Please write **legibly and dark enough** for the scan to be readable. For students who use English as a second language, dictionaries are permitted. Calculators are allowed.

DO NOT ASK QUESTIONS OF THE INVIGILATORS about interpretations of the exam questions. If you have a concern about a question, please summarize it at the top of this page. All concerns will be evaluated after the exam. GOOD LUCK!

**Question 1 8 points** (*Assume all transitions are possible*)

a. 6 points Calculate the **energy (in kJ/mol)** required to ionize (completely remove) the electron from the **n=3** level of **He⁺ ion** (He: Helium). Show your work including equations. (3 sig. figs.)

b. 2 points

(I) Select the largest (longest) wavelength from the three choices below:

- i. Wavelength of photon to *completely remove* an electron from *1s orbital of Hydrogen atom*
- ii. Wavelength of photon to *completely remove* an electron from *1s orbital of Boron atom (B)*
- iii. Wavelength of photon to *completely remove* an electron from *1s orbital of Boron cation (B⁺)*

i or ii or iii

(II) Select the smallest (shortest) wavelength from the three choices below (*for H atom*):

- i. Wavelength of photon *emitted* for an electron transitioning from *2p to 1s orbital*
- ii. Wavelength of photon *absorbed* for an electron transitioning from *2s to 4p orbital*
- iii. Wavelength of photon *absorbed* for an electron transitioning from *1s to 3p orbital*

i or ii or iii



Question 2 8 points



A Chem110 student plots the graph shown on the left, from data obtained for the kinetic energy (eV) of the ejected electron versus frequency (s^{-1}) of the incident light, for an unknown metal. ($1\text{eV} = 1.6022 \times 10^{-19} \text{ J}$)

Based on the plot, answer the questions **a to c** below:

a. 1 point On the plot above, label the **threshold frequency**.

b. 2 points What is the value of the work potential (in J) of the unknown metal?

c. 3 points What is the *longest* wavelength (in nm) required to eject an electron from this metal's surface? Show your work including equation(s) used. (3 sig figs)

d. 2 points In the space provided below, plot the following

(Assume frequency is *above* threshold frequency):

- i. *Kinetic energy* of an ejected electron versus the *intensity* of the incident light (left)
- ii. *Number of electrons* ejected versus the *frequency* of the incident light (right)



**Question 3** **6 points****a. 2 points**

In the ground state of Ge (Germanium; $Z = 32$) atom, **name** the following orbitals:

i. Orbital containing the electron that requires the maximum energy to be removed.

ii. Orbital containing the electron that experiences the minimum effective nuclear charge.

b. 2 points

An unknown anion (A^-) has the following *excited* electronic configuration: $[Ar] 4s^2 3d^3 4p^1$

(Ar: Argon; $Z = 18$)

Give the **condensed ground state electronic configuration** of the atom, **A** and the cation, **A^{2+}** .
Show your work. (Assume all electronic transitions are possible)

Electronic Configuration of A:

Electronic Configuration of A^{2+} :

c. 2 points

In a single atom, determine how many electrons can have the following set of quantum numbers?

(i) $n = 5; l = 3$

In a single atom, determine how many electrons can have the following set of quantum numbers?

(i) $n = 3; l = 2; m_s = +1/2$

**Question 4** **8 points**

Rank the following from 1 to 4 in **increasing** order. Provide the ranking in the boxes provided.

Note: Atomic number (Z) are provided for relevant atoms

a) Increasing (lowest to highest) order of *ionization energy*: (1= lowest; 4 = highest)

Mg

Mg: Z = 12

Rb

Rb: Z = 37

Cs

Cs: Z = 55

K

K : Z = 19

b) Increasing (lowest to highest) order of *bond angle*: (1= lowest; 4 = highest)

H₂OCCl₄XeF₄

HCN

c) Increasing (smallest to largest) order of *ionic size* (1= smallest; 4 = largest)

S²⁻

S: Z = 16

O²⁻

O: Z = 8

Be²⁺

Be: Z = 4

Na⁺

Na: Z = 11

d) Increasing (lowest to highest) order of *bond energy* (1= lowest; 4 = highest)

P-F

P-Cl

P-I

P-Br

**Question 5** **12 points**

a. 9 points For the molecules HClO_4 and TeF_6 , draw the most stable *Lewis structure* showing appropriate shape and *dashed/wedged bonds*, based on VSEPR. Determine the **molecular geometry** around the central atom. For HClO_4 , label the *most polar bond*.



b. 3 points (Note: Lone pairs are not shown in structures below)

(i) For the three *resonance* structures (1, 2 and 3) shown below, answer the following question:



(I) Circle the *most contributing* resonance structure.

Structure 1

Structure 2

Structure 3

(ii) For the three structures given (4, 5, and 6) below, answer the following questions:



(I) Circle all the structures that represent *resonance structures* of each other:

Structure 4

Structure 5

Structure 6

(II) Among the resonance structures, circle the *most contributing* resonance structure.

Structure 4

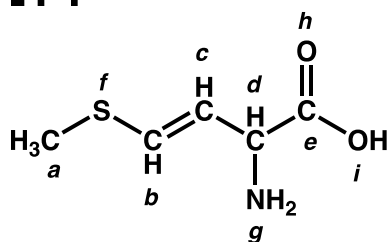
Structure 5

Structure 6

**Question 6** **6 points**

Using the data given below, calculate the **lattice energy** for SrO(s).

	ΔH° (kJ/mol)
$\text{Sr}(s) \longrightarrow \text{Sr}(g)$	+180
$\text{Sr}(g) \longrightarrow \text{Sr}^{2+}(g) + 2e^-$	+1468
$\text{O}_2(g) \longrightarrow 2\text{O}(g)$	+500
$\text{O}(g) + 2e^- \longrightarrow \text{O}^{2-}(g)$	+600
$\text{Sr}(s) + \frac{1}{2}\text{O}_2(g) \longrightarrow \text{SrO}(s)$	-550

**Question 7 10 points**

Consider the molecule given on the left, and answer questions (i) to (vi) using chemical bonding theories that we have learnt in class (valence bond theory, hybridization).

Note that the lone pairs are not shown in the structure:

i. Consider the carbon atoms marked in as ***a***, ***b***, ***c***, ***d***, and ***e*** in the molecule, and determine **hybridization** at each of the carbon centers. (2.5 points)

Atom	Hybridization
Carbon (<i>a</i>)	
Carbon (<i>b</i>)	
Carbon (<i>c</i>)	
Carbon (<i>d</i>)	
Carbon (<i>e</i>)	

ii. Now, consider the sulfur atom (S) marked as ***f***, nitrogen (N) atom (***g***), and oxygen(O) atoms (***h*** and ***i***), and determine their **hybridization**. (2 points)

Atom	Hybridization
Sulfur (<i>f</i>)	
Nitrogen (<i>g</i>)	
Oxygen (<i>h</i>)	
Oxygen (<i>i</i>)	

iii. **Describe** all the bonds formed by the carbon atoms ***a*** and ***b***, using sigma and pi designation as appropriate. (2 points)

Atom	Bonds formed (use sigma and pi as appropriate)
Carbon (<i>a</i>)	
Carbon (<i>b</i>)	

iv. How many total sp^3 -s sigma bonds are in the molecule? (1 point)

v. Determine **electronic** and **molecular geometry** at carbon (C) labeled ***e*** and sulfur (S) labeled ***f*** centres. (2 points)

Atom	Electronic Geometry	Molecular Geometry
Carbon (<i>e</i>)		
Sulfur (<i>f</i>)		

vi. Circle the value from the choices given below, that describes the **C-S-C bond angle at sulfur** (S) marked ***f***: (0.5 points)

>90° to <109.5° 109.5° >109.5 to <120° 120° >120° to <180°


Question 8 12 points

Using the **valence shell orbitals only** draw a **complete** molecular orbital (MO) diagram (with no s-p mixing) for the molecule **PS**, in the space provided below. **(6 points)**

Assume that 3s and 3p atomic orbitals will form molecular orbitals like those formed by the combination of 2s and 2p atomic orbitals that we learnt in class, and there is no s-p mixing.

Show **energy arrow**, **atomic orbitals** involved, **atom labels**, the **molecular orbitals** that are formed, **filled electrons** in the MO diagram etc. Make sure to **label all the atomic and molecular orbitals**.

Using the MO diagram above, answer the following questions:

- (i) Which one of the following will have the highest bond order? Circle the correct choice
 A) PS^{1-} B) PS C) PS^{1+}
- (ii) Which one of the following will have the longest bond length? Circle the correct choice
 A) PS^{1-} B) PS C) PS^{1+}
- (iii) Which of the following will be diamagnetic? Circle the correct choice.
 A) PS^{1-} B) PS C) PS^{1+}
- (iv) Draw Lewis dot structure of the molecule with the shortest bond length (based on the MO diagram) from among the three (PS^{1-} , PS, PS^{1+}), and indicate the bond order calculated from the Lewis dot structure. **(3 points)**

Lewis Dot Structure

Bond Order



21E4400F-9AFB-4D9B-996C-7C130FE6AF6C

midterm-exam-93388

#549 Page 10 of 10

SCRAP PAGE

**CHEMISTRY 110 –Midterm Exam****Thursday November 7, 2024, 6:30 – 8:30 pm****Examiners: Pallavi Sirjoosingh and Ashok Kakkar**

First name (please write as legibly as possible within the boxes)

--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--

Last name

--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--

Student ID number

--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--

This booklet has 8 questions on 10 pages (The last page can be used as a scrap page).

The staple must NOT be removed from the question booklet. Your intact question booklet must be returned at the end of the exam.

A separate booklet contains a periodic table, and datasheet.

Write your answers directly under the questions in the space provided. Do not write on the QR codes or near the top edges (as it may be cut off).

Please write **legibly and dark enough** for the scan to be readable. For students who use English as a second language, dictionaries are permitted. Calculators are allowed.

DO NOT ASK QUESTIONS OF THE INVIGILATORS about interpretations of the exam questions. If you have a concern about a question, please summarize it at the top of this page. All concerns will be evaluated after the exam. GOOD LUCK!

**Question 1 8 points** (*Assume all transitions are possible*)

a. 6 points Calculate the **energy (in kJ/mol)** required to ionize (completely remove) the electron from the **$n=3$** level of **He^+ ion** (He: Helium). Show your work including equations. (3 sig. figs.)

b. 2 points

(I) Select the largest (longest) wavelength from the three choices below:

- i. Wavelength of photon to *completely remove* an electron from *1s orbital of Hydrogen atom*
- ii. Wavelength of photon to *completely remove* an electron from *1s orbital of Boron atom (B)*
- iii. Wavelength of photon to *completely remove* an electron from *1s orbital of Boron cation (B^+)*

i or ii or iii

(II) Select the smallest (shortest) wavelength from the three choices below (*for H atom*):

- i. Wavelength of photon *emitted* for an electron transitioning from *2p to 1s orbital*
- ii. Wavelength of photon *absorbed* for an electron transitioning from *2s to 4p orbital*
- iii. Wavelength of photon *absorbed* for an electron transitioning from *1s to 3p orbital*

i or ii or iii



Question 2 8 points



A Chem110 student plots the graph shown on the left, from data obtained for the kinetic energy (eV) of the ejected electron versus frequency (s^{-1}) of the incident light, for an unknown metal. ($1\text{eV} = 1.6022 \times 10^{-19} \text{ J}$)

Based on the plot, answer the questions **a to c** below:

a. 1 point On the plot above, label the **threshold frequency**.

b. 2 points What is the value of the work potential (in J) of the unknown metal?

c. 3 points What is the *longest* wavelength (in nm) required to eject an electron from this metal's surface? Show your work including equation(s) used. (3 sig figs)

d. 2 points In the space provided below, plot the following

(Assume frequency is *above* threshold frequency):

- i. *Kinetic energy* of an ejected electron versus the *intensity* of the incident light (left)
- ii. *Number of electrons* ejected versus the *frequency* of the incident light (right)



**Question 3** **6 points****a. 2 points**

In the ground state of Ge (Germanium; $Z = 32$) atom, **name** the following orbitals:

i. Orbital containing the electron that requires the maximum energy to be removed.

ii. Orbital containing the electron that experiences the minimum effective nuclear charge.

b. 2 points

An unknown anion (A^-) has the following *excited* electronic configuration: $[Ar] 4s^2 3d^3 4p^1$

(Ar: Argon; $Z = 18$)

Give the **condensed ground state electronic configuration** of the atom, **A** and the cation, **A^{2+}** .
Show your work. (Assume all electronic transitions are possible)

Electronic Configuration of A:

Electronic Configuration of A^{2+} :

c. 2 points

In a single atom, determine how many electrons can have the following set of quantum numbers?

(i) $n = 5; l = 3$

In a single atom, determine how many electrons can have the following set of quantum numbers?

(i) $n = 3; l = 2; m_s = +1/2$

**Question 4** **8 points**

Rank the following from 1 to 4 in **increasing** order. Provide the ranking in the boxes provided.

Note: Atomic number (Z) are provided for relevant atoms

a) Increasing (lowest to highest) order of *ionization energy*: (1= lowest; 4 = highest)

Mg

Mg: Z = 12

Rb

Rb: Z = 37

Cs

Cs: Z = 55

K

K : Z = 19

b) Increasing (lowest to highest) order of *bond angle*: (1= lowest; 4 = highest)

H₂OCCl₄XeF₄

HCN

c) Increasing (smallest to largest) order of *ionic size* (1= smallest; 4 = largest)

S²⁻

S: Z = 16

O²⁻

O: Z = 8

Be²⁺

Be: Z = 4

Na⁺

Na: Z = 11

d) Increasing (lowest to highest) order of *bond energy* (1= lowest; 4 = highest)

P-F

P-Cl

P-I

P-Br

**Question 5** **12 points**

a. 9 points For the molecules HClO_4 and TeF_6 , draw the most stable *Lewis structure* showing appropriate shape and *dashed/wedged bonds*, based on VSEPR. Determine the **molecular geometry** around the central atom. For HClO_4 , label the *most polar bond*.



b. 3 points (Note: Lone pairs are not shown in structures below)

(i) For the three *resonance* structures (1, 2 and 3) shown below, answer the following question:



(I) Circle the *most contributing* resonance structure.

Structure 1

Structure 2

Structure 3

(ii) For the three structures given (4, 5, and 6) below, answer the following questions:



(I) Circle all the structures that represent *resonance structures* of each other:

Structure 4

Structure 5

Structure 6

(II) Among the resonance structures, circle the *most contributing* resonance structure.

Structure 4

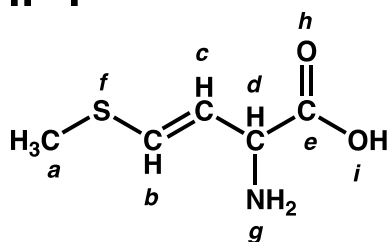
Structure 5

Structure 6

**Question 6** **6 points**

Using the data given below, calculate the **lattice energy** for SrO(s).

	ΔH° (kJ/mol)
$\text{Sr}(s) \longrightarrow \text{Sr}(g)$	+180
$\text{Sr}(g) \longrightarrow \text{Sr}^{2+}(g) + 2e^-$	+1468
$\text{O}_2(g) \longrightarrow 2\text{O}(g)$	+500
$\text{O}(g) + 2e^- \longrightarrow \text{O}^{2-}(g)$	+600
$\text{Sr}(s) + \frac{1}{2}\text{O}_2(g) \longrightarrow \text{SrO}(s)$	-550

**Question 7 10 points**

Consider the molecule given on the left, and answer questions (i) to (vi) using chemical bonding theories that we have learnt in class (valence bond theory, hybridization).

Note that the lone pairs are not shown in the structure:

i. Consider the carbon atoms marked in as ***a***, ***b***, ***c***, ***d***, and ***e*** in the molecule, and determine **hybridization** at each of the carbon centers. (2.5 points)

Atom	Hybridization
Carbon (<i>a</i>)	
Carbon (<i>b</i>)	
Carbon (<i>c</i>)	
Carbon (<i>d</i>)	
Carbon (<i>e</i>)	

ii. Now, consider the sulfur atom (S) marked as ***f***, nitrogen (N) atom (***g***), and oxygen(O) atoms (***h*** and ***i***), and determine their **hybridization**. (2 points)

Atom	Hybridization
Sulfur (<i>f</i>)	
Nitrogen (<i>g</i>)	
Oxygen (<i>h</i>)	
Oxygen (<i>i</i>)	

iii. **Describe** all the bonds formed by the carbon atoms ***a*** and ***b***, using sigma and pi designation as appropriate. (2 points)

Atom	Bonds formed (use sigma and pi as appropriate)
Carbon (<i>a</i>)	
Carbon (<i>b</i>)	

iv. How many total sp^3 -s sigma bonds are in the molecule? (1 point)

v. Determine **electronic** and **molecular geometry** at carbon (C) labeled ***e*** and sulfur (S) labeled ***f*** centres. (2 points)

Atom	Electronic Geometry	Molecular Geometry
Carbon (<i>e</i>)		
Sulfur (<i>f</i>)		

vi. Circle the value from the choices given below, that describes the **C-S-C bond angle at sulfur** (S) marked ***f***: (0.5 points)

>90° to <109.5° 109.5° >109.5 to <120° 120° >120° to <180°

**Question 8 12 points**

Using the **valence shell orbitals only** draw a **complete** molecular orbital (MO) diagram (with no s-p mixing) for the molecule **PS**, in the space provided below. **(6 points)**

Assume that 3s and 3p atomic orbitals will form molecular orbitals like those formed by the combination of 2s and 2p atomic orbitals that we learnt in class, and there is no s-p mixing.

Show **energy arrow**, **atomic orbitals** involved, **atom labels**, the **molecular orbitals** that are formed, **filled electrons** in the MO diagram etc. Make sure to label all the atomic and molecular orbitals.

Using the MO diagram above, answer the following questions:

- (i) Which one of the following will have the highest bond order? Circle the correct choice
A) PS^{1-} B) PS C) PS^{1+}
- (ii) Which one of the following will have the longest bond length? Circle the correct choice
A) PS^{1-} B) PS C) PS^{1+}
- (iii) Which of the following will be diamagnetic? Circle the correct choice.
A) PS^{1-} B) PS C) PS^{1+}
- (iv) Draw Lewis dot structure of the molecule with the shortest bond length (based on the MO diagram) from among the three (PS^{1-} , PS, PS^{1+}), and indicate the bond order calculated from the Lewis dot structure. **(3 points)**

Lewis Dot Structure

Bond Order



B729D09E-C956-4A14-8A90-4B8CEC189C7B

midterm-exam-93388

#550 Page 10 of 10

SCRAP PAGE

**CHEMISTRY 110 –Midterm Exam****Thursday November 7, 2024, 6:30 – 8:30 pm**

Examiners: Pallavi Sirjoosingh and Ashok Kakkar

First name (please write as legibly as possible within the boxes)

--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--

Last name

--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--

Student ID number

--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--

This booklet has 8 questions on 10 pages (The last page can be used as a scrap page).

The staple must NOT be removed from the question booklet. Your intact question booklet must be returned at the end of the exam.

A separate booklet contains a periodic table, and datasheet.

Write your answers directly under the questions in the space provided. Do not write on the QR codes or near the top edges (as it may be cut off).

Please write **legibly and dark enough** for the scan to be readable. For students who use English as a second language, dictionaries are permitted. Calculators are allowed.

DO NOT ASK QUESTIONS OF THE INVIGILATORS about interpretations of the exam questions. If you have a concern about a question, please summarize it at the top of this page. All concerns will be evaluated after the exam. GOOD LUCK!

**Question 1 8 points** (*Assume all transitions are possible*)

a. 6 points Calculate the **energy (in kJ/mol)** required to ionize (completely remove) the electron from the **$n=3$** level of **He^+ ion** (He: Helium). Show your work including equations. (3 sig. figs.)

b. 2 points

(I) Select the largest (longest) wavelength from the three choices below:

- i. Wavelength of photon to *completely remove* an electron from *1s orbital of Hydrogen atom*
- ii. Wavelength of photon to *completely remove* an electron from *1s orbital of Boron atom (B)*
- iii. Wavelength of photon to *completely remove* an electron from *1s orbital of Boron cation (B^+)*

i or ii or iii

(II) Select the smallest (shortest) wavelength from the three choices below (*for H atom*):

- i. Wavelength of photon *emitted* for an electron transitioning from *2p to 1s orbital*
- ii. Wavelength of photon *absorbed* for an electron transitioning from *2s to 4p orbital*
- iii. Wavelength of photon *absorbed* for an electron transitioning from *1s to 3p orbital*

i or ii or iii



Question 2 8 points



A Chem110 student plots the graph shown on the left, from data obtained for the kinetic energy (eV) of the ejected electron versus frequency (s^{-1}) of the incident light, for an unknown metal. ($1\text{eV} = 1.6022 \times 10^{-19} \text{ J}$)

Based on the plot, answer the questions **a to c** below:

a. 1 point On the plot above, label the **threshold frequency**.

b. 2 points What is the value of the work potential (in J) of the unknown metal?

c. 3 points What is the *longest* wavelength (in nm) required to eject an electron from this metal's surface? Show your work including equation(s) used. (3 sig figs)

d. 2 points In the space provided below, plot the following

(Assume frequency is *above* threshold frequency):

- Kinetic energy* of an ejected electron versus the *intensity* of the incident light (left)
- Number of electrons* ejected versus the *frequency* of the incident light (right)



**Question 3** **6 points****a. 2 points**

In the ground state of Ge (Germanium; $Z = 32$) atom, **name** the following orbitals:

i. Orbital containing the electron that requires the maximum energy to be removed.

ii. Orbital containing the electron that experiences the minimum effective nuclear charge.

b. 2 points

An unknown anion (A^-) has the following *excited* electronic configuration: $[Ar] 4s^2 3d^3 4p^1$

(Ar: Argon; $Z = 18$)

Give the **condensed ground state electronic configuration** of the atom, **A** and the cation, A^{2+} .
Show your work. (Assume all electronic transitions are possible)

Electronic Configuration of A:

Electronic Configuration of A^{2+} :

c. 2 points

In a single atom, determine how many electrons can have the following set of quantum numbers?

(i) $n = 5; l = 3$

In a single atom, determine how many electrons can have the following set of quantum numbers?

(i) $n = 3; l = 2; m_s = +1/2$

**Question 4** **8 points**

Rank the following from 1 to 4 in **increasing** order. Provide the ranking in the boxes provided.

Note: Atomic number (Z) are provided for relevant atoms

a) Increasing (lowest to highest) order of *ionization energy*: (1= lowest; 4 = highest)

Mg

Mg: Z = 12

Rb

Rb: Z = 37

Cs

Cs: Z = 55

K

K : Z = 19

b) Increasing (lowest to highest) order of *bond angle*: (1= lowest; 4 = highest)

H₂OCCl₄XeF₄

HCN

c) Increasing (smallest to largest) order of *ionic size* (1= smallest; 4 = largest)

S²⁻

S: Z = 16

O²⁻

O: Z = 8

Be²⁺

Be: Z = 4

Na⁺

Na: Z = 11

d) Increasing (lowest to highest) order of *bond energy* (1= lowest; 4 = highest)

P-F

P-Cl

P-I

P-Br

**Question 5** **12 points**

a. 9 points For the molecules HClO_4 and TeF_6 , draw the most stable *Lewis structure* showing appropriate shape and *dashed/wedged bonds*, based on VSEPR. Determine the **molecular geometry** around the central atom. For HClO_4 , label the *most polar bond*.



b. 3 points (Note: Lone pairs are not shown in structures below)

(i) For the three *resonance* structures (1, 2 and 3) shown below, answer the following question:



(I) Circle the *most contributing* resonance structure.

Structure 1

Structure 2

Structure 3

(ii) For the three structures given (4, 5, and 6) below, answer the following questions:



(I) Circle all the structures that represent *resonance structures* of each other:

Structure 4

Structure 5

Structure 6

(II) Among the resonance structures, circle the *most contributing* resonance structure.

Structure 4

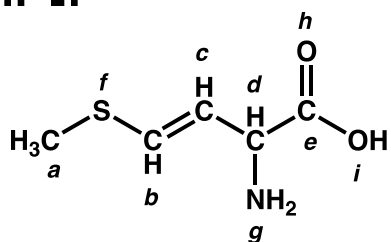
Structure 5

Structure 6

**Question 6** **6 points**

Using the data given below, calculate the **lattice energy** for SrO(s).

	ΔH° (kJ/mol)
$\text{Sr}(s) \longrightarrow \text{Sr}(g)$	+180
$\text{Sr}(g) \longrightarrow \text{Sr}^{2+}(g) + 2e^-$	+1468
$\text{O}_2(g) \longrightarrow 2\text{O}(g)$	+500
$\text{O}(g) + 2e^- \longrightarrow \text{O}^{2-}(g)$	+600
$\text{Sr}(s) + \frac{1}{2}\text{O}_2(g) \longrightarrow \text{SrO}(s)$	-550

**Question 7 10 points**

Consider the molecule given on the left, and answer questions (i) to (vi) using chemical bonding theories that we have learnt in class (valence bond theory, hybridization).

Note that the lone pairs are not shown in the structure:

i. Consider the carbon atoms marked in as ***a***, ***b***, ***c***, ***d***, and ***e*** in the molecule, and determine **hybridization** at each of the carbon centers. (2.5 points)

Atom	Hybridization
Carbon (<i>a</i>)	
Carbon (<i>b</i>)	
Carbon (<i>c</i>)	
Carbon (<i>d</i>)	
Carbon (<i>e</i>)	

ii. Now, consider the sulfur atom (S) marked as ***f***, nitrogen (N) atom (***g***), and oxygen(O) atoms (***h*** and ***i***), and determine their **hybridization**. (2 points)

Atom	Hybridization
Sulfur (<i>f</i>)	
Nitrogen (<i>g</i>)	
Oxygen (<i>h</i>)	
Oxygen (<i>i</i>)	

iii. **Describe** all the bonds formed by the carbon atoms ***a*** and ***b***, using sigma and pi designation as appropriate. (2 points)

Atom	Bonds formed (use sigma and pi as appropriate)
Carbon (<i>a</i>)	
Carbon (<i>b</i>)	

iv. How many total sp^3 -s sigma bonds are in the molecule? (1 point)

v. Determine **electronic** and **molecular geometry** at carbon (C) labeled ***e*** and sulfur (S) labeled ***f*** centres. (2 points)

Atom	Electronic Geometry	Molecular Geometry
Carbon (<i>e</i>)		
Sulfur (<i>f</i>)		

vi. Circle the value from the choices given below, that describes the **C-S-C bond angle at sulfur** (S) marked ***f***: (0.5 points)

>90° to <109.5°

109.5°

>109.5 to <120°

120°

>120° to <180°


Question 8 12 points

Using the **valence shell orbitals only** draw a **complete** molecular orbital (MO) diagram (with no s-p mixing) for the molecule **PS**, in the space provided below. **(6 points)**

Assume that 3s and 3p atomic orbitals will form molecular orbitals like those formed by the combination of 2s and 2p atomic orbitals that we learnt in class, and there is no s-p mixing.

Show **energy arrow**, **atomic orbitals** involved, **atom labels**, the **molecular orbitals** that are formed, **filled electrons** in the MO diagram etc. Make sure to label all the atomic and molecular orbitals.

Using the MO diagram above, answer the following questions:

- (i) Which one of the following will have the highest bond order? Circle the correct choice
 A) PS^{1-} B) PS C) PS^{1+}
- (ii) Which one of the following will have the longest bond length? Circle the correct choice
 A) PS^{1-} B) PS C) PS^{1+}
- (iii) Which of the following will be diamagnetic? Circle the correct choice.
 A) PS^{1-} B) PS C) PS^{1+}
- (iv) Draw Lewis dot structure of the molecule with the shortest bond length (based on the MO diagram) from among the three (PS^{1-} , PS, PS^{1+}), and indicate the bond order calculated from the Lewis dot structure. **(3 points)**

Lewis Dot Structure

Bond Order



83263A0D-6F5B-4674-9AF0-B64341846D26

midterm-exam-93388

#551 Page 10 of 10

SCRAP PAGE

**CHEMISTRY 110 –Midterm Exam****Thursday November 7, 2024, 6:30 – 8:30 pm****Examiners: Pallavi Sirjoosingh and Ashok Kakkar**

First name (please write as legibly as possible within the boxes)

--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--

Last name

--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--

Student ID number

--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--

This booklet has 8 questions on 10 pages (The last page can be used as a scrap page).

The staple must NOT be removed from the question booklet. Your intact question booklet must be returned at the end of the exam.

A separate booklet contains a periodic table, and datasheet.

Write your answers directly under the questions in the space provided. Do not write on the QR codes or near the top edges (as it may be cut off).

Please write **legibly and dark enough** for the scan to be readable. For students who use English as a second language, dictionaries are permitted. Calculators are allowed.

DO NOT ASK QUESTIONS OF THE INVIGILATORS about interpretations of the exam questions. If you have a concern about a question, please summarize it at the top of this page. All concerns will be evaluated after the exam. GOOD LUCK!

**Question 1 8 points** (*Assume all transitions are possible*)

a. 6 points Calculate the **energy (in kJ/mol)** required to ionize (completely remove) the electron from the **n=3** level of **He⁺ ion** (He: Helium). Show your work including equations. (3 sig. figs.)

b. 2 points

(I) Select the largest (longest) wavelength from the three choices below:

- i. Wavelength of photon to *completely remove* an electron from *1s orbital of Hydrogen atom*
- ii. Wavelength of photon to *completely remove* an electron from *1s orbital of Boron atom (B)*
- iii. Wavelength of photon to *completely remove* an electron from *1s orbital of Boron cation (B⁺)*

i or ii or iii

(II) Select the smallest (shortest) wavelength from the three choices below (*for H atom*):

- i. Wavelength of photon *emitted* for an electron transitioning from *2p to 1s orbital*
- ii. Wavelength of photon *absorbed* for an electron transitioning from *2s to 4p orbital*
- iii. Wavelength of photon *absorbed* for an electron transitioning from *1s to 3p orbital*

i or ii or iii



Question 2 8 points



A Chem110 student plots the graph shown on the left, from data obtained for the kinetic energy (eV) of the ejected electron versus frequency (s^{-1}) of the incident light, for an unknown metal. ($1\text{eV} = 1.6022 \times 10^{-19} \text{ J}$)

Based on the plot, answer the questions **a to c** below:

a. 1 point On the plot above, label the **threshold frequency**.

b. 2 points What is the value of the work potential (in J) of the unknown metal?

c. 3 points What is the *longest* wavelength (in nm) required to eject an electron from this metal's surface? Show your work including equation(s) used. (3 sig figs)

d. 2 points In the space provided below, plot the following

(Assume frequency is *above* threshold frequency):

- i. *Kinetic energy* of an ejected electron versus the *intensity* of the incident light (left)
- ii. *Number of electrons* ejected versus the *frequency* of the incident light (right)



**Question 3** **6 points****a. 2 points**

In the ground state of Ge (Germanium; $Z = 32$) atom, **name** the following orbitals:

i. Orbital containing the electron that requires the maximum energy to be removed.

ii. Orbital containing the electron that experiences the minimum effective nuclear charge.

b. 2 points

An unknown anion (A^-) has the following *excited* electronic configuration: $[Ar] 4s^2 3d^3 4p^1$

(Ar: Argon; $Z = 18$)

Give the **condensed ground state electronic configuration** of the atom, **A** and the cation, **A^{2+}** .
Show your work. (Assume all electronic transitions are possible)

Electronic Configuration of A:

Electronic Configuration of A^{2+} :

c. 2 points

In a single atom, determine how many electrons can have the following set of quantum numbers?

(i) $n = 5; l = 3$

In a single atom, determine how many electrons can have the following set of quantum numbers?

(i) $n = 3; l = 2; m_s = +1/2$

**Question 4** **8 points**

Rank the following from 1 to 4 in **increasing** order. Provide the ranking in the boxes provided.

Note: Atomic number (Z) are provided for relevant atoms

a) Increasing (lowest to highest) order of *ionization energy*: (1= lowest; 4 = highest)

Mg

Mg: Z = 12

Rb

Rb: Z = 37

Cs

Cs: Z = 55

K

K : Z = 19

b) Increasing (lowest to highest) order of *bond angle*: (1= lowest; 4 = highest)

H₂OCCl₄XeF₄

HCN

c) Increasing (smallest to largest) order of *ionic size* (1= smallest; 4 = largest)

S²⁻

S: Z = 16

O²⁻

O: Z = 8

Be²⁺

Be: Z = 4

Na⁺

Na: Z = 11

d) Increasing (lowest to highest) order of *bond energy* (1= lowest; 4 = highest)

P-F

P-Cl

P-I

P-Br

**Question 5** **12 points**

a. 9 points For the molecules HClO_4 and TeF_6 , draw the most stable *Lewis structure* showing appropriate shape and *dashed/wedged bonds*, based on VSEPR. Determine the **molecular geometry** around the central atom. For HClO_4 , label the *most polar bond*.



b. 3 points (Note: Lone pairs are not shown in structures below)

(i) For the three *resonance* structures (1, 2 and 3) shown below, answer the following question:



(I) Circle the *most contributing* resonance structure.

Structure 1

Structure 2

Structure 3

(ii) For the three structures given (4, 5, and 6) below, answer the following questions:



(I) Circle all the structures that represent *resonance structures* of each other:

Structure 4

Structure 5

Structure 6

(II) Among the resonance structures, circle the *most contributing* resonance structure.

Structure 4

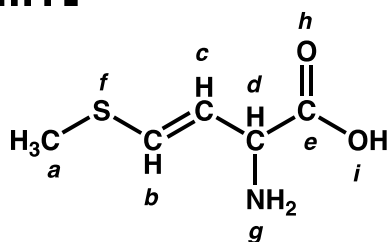
Structure 5

Structure 6

**Question 6** **6 points**

Using the data given below, calculate the **lattice energy** for SrO(s).

	ΔH° (kJ/mol)
$\text{Sr}(s) \longrightarrow \text{Sr}(g)$	+180
$\text{Sr}(g) \longrightarrow \text{Sr}^{2+}(g) + 2e^-$	+1468
$\text{O}_2(g) \longrightarrow 2\text{O}(g)$	+500
$\text{O}(g) + 2e^- \longrightarrow \text{O}^{2-}(g)$	+600
$\text{Sr}(s) + \frac{1}{2}\text{O}_2(g) \longrightarrow \text{SrO}(s)$	-550

**Question 7 10 points**

Consider the molecule given on the left, and answer questions (i) to (vi) using chemical bonding theories that we have learnt in class (valence bond theory, hybridization).

Note that the lone pairs are not shown in the structure:

i. Consider the carbon atoms marked in as ***a***, ***b***, ***c***, ***d***, and ***e*** in the molecule, and determine **hybridization** at each of the carbon centers. (2.5 points)

Atom	Hybridization
Carbon (<i>a</i>)	
Carbon (<i>b</i>)	
Carbon (<i>c</i>)	
Carbon (<i>d</i>)	
Carbon (<i>e</i>)	

ii. Now, consider the sulfur atom (S) marked as ***f***, nitrogen (N) atom (***g***), and oxygen(O) atoms (***h*** and ***i***), and determine their **hybridization**. (2 points)

Atom	Hybridization
Sulfur (<i>f</i>)	
Nitrogen (<i>g</i>)	
Oxygen (<i>h</i>)	
Oxygen (<i>i</i>)	

iii. **Describe** all the bonds formed by the carbon atoms ***a*** and ***b***, using sigma and pi designation as appropriate. (2 points)

Atom	Bonds formed (use sigma and pi as appropriate)
Carbon (<i>a</i>)	
Carbon (<i>b</i>)	

iv. How many total sp^3 -s sigma bonds are in the molecule? (1 point)

v. Determine **electronic** and **molecular geometry** at carbon (C) labeled ***e*** and sulfur (S) labeled ***f*** centres. (2 points)

Atom	Electronic Geometry	Molecular Geometry
Carbon (<i>e</i>)		
Sulfur (<i>f</i>)		

vi. Circle the value from the choices given below, that describes the **C-S-C bond angle at sulfur** (S) marked ***f***: (0.5 points)

>90° to <109.5°

109.5°

>109.5 to <120°

120°

>120° to <180°


Question 8 12 points

Using the **valence shell orbitals only** draw a **complete** molecular orbital (MO) diagram (with no s-p mixing) for the molecule **PS**, in the space provided below. **(6 points)**

Assume that 3s and 3p atomic orbitals will form molecular orbitals like those formed by the combination of 2s and 2p atomic orbitals that we learnt in class, and there is no s-p mixing.

Show **energy arrow**, **atomic orbitals** involved, **atom labels**, the **molecular orbitals** that are formed, **filled electrons** in the MO diagram etc. Make sure to label all the atomic and molecular orbitals.

Using the MO diagram above, answer the following questions:

- (i) Which one of the following will have the highest bond order? Circle the correct choice
 A) PS^{1-} B) PS C) PS^{1+}
- (ii) Which one of the following will have the longest bond length? Circle the correct choice
 A) PS^{1-} B) PS C) PS^{1+}
- (iii) Which of the following will be diamagnetic? Circle the correct choice.
 A) PS^{1-} B) PS C) PS^{1+}
- (iv) Draw Lewis dot structure of the molecule with the shortest bond length (based on the MO diagram) from among the three (PS^{1-} , PS, PS^{1+}), and indicate the bond order calculated from the Lewis dot structure. **(3 points)**

Lewis Dot Structure

Bond Order



C5B3E450-EC7C-42C3-B20B-69CA6484E8C6

midterm-exam-93388

#552 Page 10 of 10

SCRAP PAGE

**CHEMISTRY 110 –Midterm Exam****Thursday November 7, 2024, 6:30 – 8:30 pm**

Examiners: Pallavi Sirjoosingh and Ashok Kakkar

First name (please write as legibly as possible within the boxes)

--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--

Last name

--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--

Student ID number

--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--

This booklet has 8 questions on 10 pages (The last page can be used as a scrap page).

The staple must NOT be removed from the question booklet. Your intact question booklet must be returned at the end of the exam.

A separate booklet contains a periodic table, and datasheet.

Write your answers directly under the questions in the space provided. Do not write on the QR codes or near the top edges (as it may be cut off).

Please write **legibly and dark enough** for the scan to be readable. For students who use English as a second language, dictionaries are permitted. Calculators are allowed.

DO NOT ASK QUESTIONS OF THE INVIGILATORS about interpretations of the exam questions. If you have a concern about a question, please summarize it at the top of this page. All concerns will be evaluated after the exam. GOOD LUCK!

**Question 1 8 points** (*Assume all transitions are possible*)

a. 6 points Calculate the **energy (in kJ/mol)** required to ionize (completely remove) the electron from the **n=3** level of **He⁺ ion** (He: Helium). Show your work including equations. (3 sig. figs.)

b. 2 points

(I) Select the largest (longest) wavelength from the three choices below:

- i. Wavelength of photon to *completely remove* an electron from *1s orbital of Hydrogen atom*
- ii. Wavelength of photon to *completely remove* an electron from *1s orbital of Boron atom (B)*
- iii. Wavelength of photon to *completely remove* an electron from *1s orbital of Boron cation (B⁺)*

i or ii or iii

(II) Select the smallest (shortest) wavelength from the three choices below (*for H atom*):

- i. Wavelength of photon *emitted* for an electron transitioning from *2p to 1s orbital*
- ii. Wavelength of photon *absorbed* for an electron transitioning from *2s to 4p orbital*
- iii. Wavelength of photon *absorbed* for an electron transitioning from *1s to 3p orbital*

i or ii or iii



Question 2 8 points



A Chem110 student plots the graph shown on the left, from data obtained for the kinetic energy (eV) of the ejected electron versus frequency (s^{-1}) of the incident light, for an unknown metal. ($1\text{eV} = 1.6022 \times 10^{-19} \text{ J}$)

Based on the plot, answer the questions **a to c** below:

a. 1 point On the plot above, label the **threshold frequency**.

b. 2 points What is the value of the work potential (in J) of the unknown metal?

c. 3 points What is the *longest* wavelength (in nm) required to eject an electron from this metal's surface? Show your work including equation(s) used. (3 sig figs)

d. 2 points In the space provided below, plot the following

(Assume frequency is *above* threshold frequency):

- i. *Kinetic energy* of an ejected electron versus the *intensity* of the incident light (left)
- ii. *Number of electrons* ejected versus the *frequency* of the incident light (right)



**Question 3** **6 points****a. 2 points**

In the ground state of Ge (Germanium; $Z = 32$) atom, **name** the following orbitals:

i. Orbital containing the electron that requires the maximum energy to be removed.

ii. Orbital containing the electron that experiences the minimum effective nuclear charge.

b. 2 points

An unknown anion (A^-) has the following *excited* electronic configuration: $[Ar] 4s^2 3d^3 4p^1$

(Ar: Argon; $Z = 18$)

Give the **condensed ground state electronic configuration** of the atom, **A** and the cation, **A^{2+}** .
Show your work. (Assume all electronic transitions are possible)

Electronic Configuration of A:

Electronic Configuration of A^{2+} :

c. 2 points

In a single atom, determine how many electrons can have the following set of quantum numbers?

(i) $n = 5; l = 3$

In a single atom, determine how many electrons can have the following set of quantum numbers?

(i) $n = 3; l = 2; m_s = +1/2$

**Question 4** **8 points**

Rank the following from 1 to 4 in **increasing** order. Provide the ranking in the boxes provided.

Note: Atomic number (Z) are provided for relevant atoms

a) Increasing (lowest to highest) order of *ionization energy*: (1= lowest; 4 = highest)

Mg

Mg: Z = 12

Rb

Rb: Z = 37

Cs

Cs: Z = 55

K

K : Z = 19

b) Increasing (lowest to highest) order of *bond angle*: (1= lowest; 4 = highest)

H₂OCCl₄XeF₄

HCN

c) Increasing (smallest to largest) order of *ionic size* (1= smallest; 4 = largest)

S²⁻

S: Z = 16

O²⁻

O: Z = 8

Be²⁺

Be: Z = 4

Na⁺

Na: Z = 11

d) Increasing (lowest to highest) order of *bond energy* (1= lowest; 4 = highest)

P-F

P-Cl

P-I

P-Br

**Question 5** **12 points**

a. 9 points For the molecules HClO_4 and TeF_6 , draw the most stable *Lewis structure* showing appropriate shape and *dashed/wedged bonds*, based on VSEPR. Determine the **molecular geometry** around the central atom. For HClO_4 , label the *most polar bond*.

HClO_4 (there are no O-O bonds)

Label the most polar bond

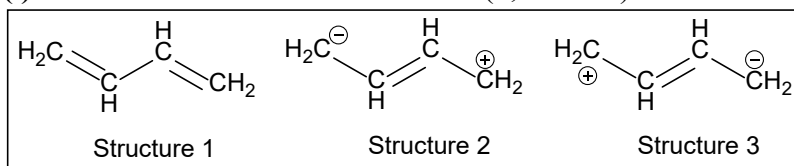
Molecular Geometry around Cl

TeF_6

Molecular Geometry around Te

b. 3 points (Note: Lone pairs are not shown in structures below)

(i) For the three *resonance* structures (1, 2 and 3) shown below, answer the following question:



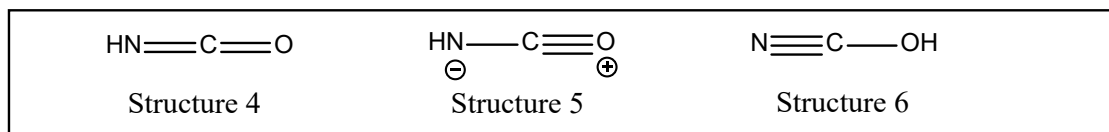
(I) Circle the *most contributing* resonance structure.

Structure 1

Structure 2

Structure 3

(ii) For the three structures given (4, 5, and 6) below, answer the following questions:



(I) Circle all the structures that represent *resonance structures* of each other:

Structure 4

Structure 5

Structure 6

(II) Among the resonance structures, circle the *most contributing* resonance structure.

Structure 4

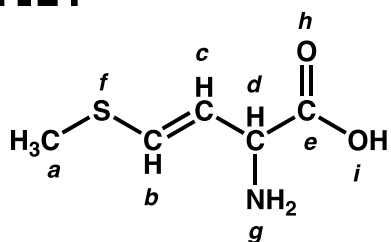
Structure 5

Structure 6

**Question 6** **6 points**

Using the data given below, calculate the **lattice energy** for SrO(s).

	ΔH° (kJ/mol)
$\text{Sr}(s) \longrightarrow \text{Sr}(g)$	+180
$\text{Sr}(g) \longrightarrow \text{Sr}^{2+}(g) + 2e^-$	+1468
$\text{O}_2(g) \longrightarrow 2\text{O}(g)$	+500
$\text{O}(g) + 2e^- \longrightarrow \text{O}^{2-}(g)$	+600
$\text{Sr}(s) + \frac{1}{2}\text{O}_2(g) \longrightarrow \text{SrO}(s)$	-550

**Question 7 10 points**

Consider the molecule given on the left, and answer questions (i) to (vi) using chemical bonding theories that we have learnt in class (valence bond theory, hybridization).

Note that the lone pairs are not shown in the structure:

i. Consider the carbon atoms marked in as ***a***, ***b***, ***c***, ***d***, and ***e*** in the molecule, and determine **hybridization** at each of the carbon centers. (2.5 points)

Atom	Hybridization
Carbon (<i>a</i>)	
Carbon (<i>b</i>)	
Carbon (<i>c</i>)	
Carbon (<i>d</i>)	
Carbon (<i>e</i>)	

ii. Now, consider the sulfur atom (S) marked as ***f***, nitrogen (N) atom (***g***), and oxygen(O) atoms (***h*** and ***i***), and determine their **hybridization**. (2 points)

Atom	Hybridization
Sulfur (<i>f</i>)	
Nitrogen (<i>g</i>)	
Oxygen (<i>h</i>)	
Oxygen (<i>i</i>)	

iii. **Describe** all the bonds formed by the carbon atoms ***a*** and ***b***, using sigma and pi designation as appropriate. (2 points)

Atom	Bonds formed (use sigma and pi as appropriate)
Carbon (<i>a</i>)	
Carbon (<i>b</i>)	

iv. How many total sp^3 -s sigma bonds are in the molecule? (1 point)

v. Determine **electronic** and **molecular geometry** at carbon (C) labeled ***e*** and sulfur (S) labeled ***f*** centres. (2 points)

Atom	Electronic Geometry	Molecular Geometry
Carbon (<i>e</i>)		
Sulfur (<i>f</i>)		

vi. Circle the value from the choices given below, that describes the **C-S-C bond angle at sulfur** (S) marked ***f***: (0.5 points)

>90° to <109.5° 109.5° >109.5 to <120° 120° >120° to <180°


Question 8 12 points

Using the **valence shell orbitals only** draw a **complete** molecular orbital (MO) diagram (with no s-p mixing) for the molecule **PS**, in the space provided below. **(6 points)**

Assume that 3s and 3p atomic orbitals will form molecular orbitals like those formed by the combination of 2s and 2p atomic orbitals that we learnt in class, and there is no s-p mixing.

Show **energy arrow**, **atomic orbitals** involved, **atom labels**, the **molecular orbitals** that are formed, **filled electrons** in the MO diagram etc. Make sure to label all the atomic and molecular orbitals.

Using the MO diagram above, answer the following questions:

- (i) Which one of the following will have the highest bond order? Circle the correct choice
 A) PS^{1-} B) PS C) PS^{1+}
- (ii) Which one of the following will have the longest bond length? Circle the correct choice
 A) PS^{1-} B) PS C) PS^{1+}
- (iii) Which of the following will be diamagnetic? Circle the correct choice.
 A) PS^{1-} B) PS C) PS^{1+}
- (iv) Draw Lewis dot structure of the molecule with the shortest bond length (based on the MO diagram) from among the three (PS^{1-} , PS, PS^{1+}), and indicate the bond order calculated from the Lewis dot structure. **(3 points)**

Lewis Dot Structure

Bond Order



F798F12C-D914-42E9-BCCF-FE12805096D9

midterm-exam-93388

#553 Page 10 of 10

SCRAP PAGE

**CHEMISTRY 110 –Midterm Exam****Thursday November 7, 2024, 6:30 – 8:30 pm****Examiners: Pallavi Sirjoosingh and Ashok Kakkar**

First name (please write as legibly as possible within the boxes)

--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--

Last name

--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--

Student ID number

--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--

This booklet has 8 questions on 10 pages (The last page can be used as a scrap page).

The staple must NOT be removed from the question booklet. Your intact question booklet must be returned at the end of the exam.

A separate booklet contains a periodic table, and datasheet.

Write your answers directly under the questions in the space provided. Do not write on the QR codes or near the top edges (as it may be cut off).

Please write **legibly and dark enough** for the scan to be readable. For students who use English as a second language, dictionaries are permitted. Calculators are allowed.

DO NOT ASK QUESTIONS OF THE INVIGILATORS about interpretations of the exam questions. If you have a concern about a question, please summarize it at the top of this page. All concerns will be evaluated after the exam. GOOD LUCK!

**Question 1 8 points** (*Assume all transitions are possible*)

a. 6 points Calculate the **energy (in kJ/mol)** required to ionize (completely remove) the electron from the **$n=3$** level of **He^+ ion** (He: Helium). Show your work including equations. (3 sig. figs.)

b. 2 points

(I) Select the largest (longest) wavelength from the three choices below:

- i. Wavelength of photon to *completely remove* an electron from *1s orbital of Hydrogen atom*
- ii. Wavelength of photon to *completely remove* an electron from *1s orbital of Boron atom (B)*
- iii. Wavelength of photon to *completely remove* an electron from *1s orbital of Boron cation (B^+)*

i or ii or iii

(II) Select the smallest (shortest) wavelength from the three choices below (*for H atom*):

- i. Wavelength of photon *emitted* for an electron transitioning from *2p to 1s orbital*
- ii. Wavelength of photon *absorbed* for an electron transitioning from *2s to 4p orbital*
- iii. Wavelength of photon *absorbed* for an electron transitioning from *1s to 3p orbital*

i or ii or iii



Question 2 8 points



A Chem110 student plots the graph shown on the left, from data obtained for the kinetic energy (eV) of the ejected electron versus frequency (s^{-1}) of the incident light, for an unknown metal. ($1\text{eV} = 1.6022 \times 10^{-19} \text{ J}$)

Based on the plot, answer the questions **a to c** below:

a. 1 point On the plot above, label the **threshold frequency**.

b. 2 points What is the value of the work potential (in J) of the unknown metal?

c. 3 points What is the *longest* wavelength (in nm) required to eject an electron from this metal's surface? Show your work including equation(s) used. (3 sig figs)

d. 2 points In the space provided below, plot the following

(Assume frequency is *above* threshold frequency):

- Kinetic energy* of an ejected electron versus the *intensity* of the incident light (left)
- Number of electrons* ejected versus the *frequency* of the incident light (right)



**Question 3** **6 points****a. 2 points**

In the ground state of Ge (Germanium; $Z = 32$) atom, **name** the following orbitals:

i. Orbital containing the electron that requires the maximum energy to be removed.

ii. Orbital containing the electron that experiences the minimum effective nuclear charge.

b. 2 points

An unknown anion (A^-) has the following *excited* electronic configuration: $[Ar] 4s^2 3d^3 4p^1$

(Ar: Argon; $Z = 18$)

Give the **condensed ground state electronic configuration** of the atom, **A** and the cation, **A^{2+}** .
Show your work. (Assume all electronic transitions are possible)

Electronic Configuration of A:

Electronic Configuration of A^{2+} :

c. 2 points

In a single atom, determine how many electrons can have the following set of quantum numbers?

(i) $n = 5; l = 3$

In a single atom, determine how many electrons can have the following set of quantum numbers?

(i) $n = 3; l = 2; m_s = +1/2$

**Question 4** **8 points**

Rank the following from 1 to 4 in **increasing** order. Provide the ranking in the boxes provided.

Note: Atomic number (Z) are provided for relevant atoms

a) Increasing (lowest to highest) order of *ionization energy*: (1= lowest; 4 = highest)

Mg

Mg: Z = 12

Rb

Rb: Z = 37

Cs

Cs: Z = 55

K

K : Z = 19

b) Increasing (lowest to highest) order of *bond angle*: (1= lowest; 4 = highest)

H₂OCCl₄XeF₄

HCN

c) Increasing (smallest to largest) order of *ionic size* (1= smallest; 4 = largest)

S²⁻

S: Z = 16

O²⁻

O: Z = 8

Be²⁺

Be: Z = 4

Na⁺

Na: Z = 11

d) Increasing (lowest to highest) order of *bond energy* (1= lowest; 4 = highest)

P-F

P-Cl

P-I

P-Br

**Question 5** **12 points**

a. 9 points For the molecules HClO_4 and TeF_6 , draw the most stable *Lewis structure* showing appropriate shape and *dashed/wedged bonds*, based on VSEPR. Determine the **molecular geometry** around the central atom. For HClO_4 , label the *most polar bond*.

HClO_4 (there are no O-O bonds)

Label the most polar bond

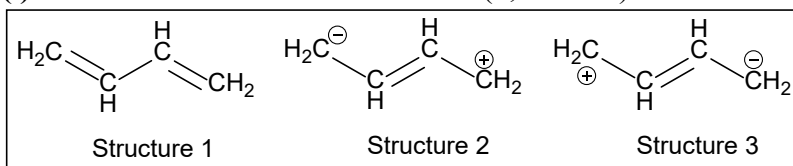
Molecular Geometry around Cl

TeF_6

Molecular Geometry around Te

b. 3 points (Note: Lone pairs are not shown in structures below)

(i) For the three *resonance* structures (1, 2 and 3) shown below, answer the following question:



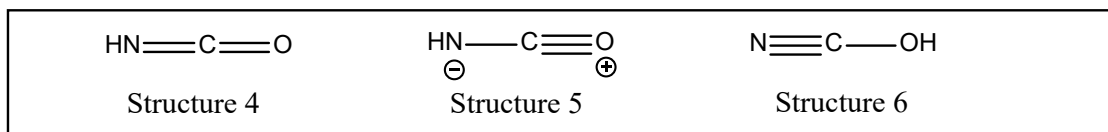
(I) Circle the *most contributing* resonance structure.

Structure 1

Structure 2

Structure 3

(ii) For the three structures given (4, 5, and 6) below, answer the following questions:



(I) Circle all the structures that represent *resonance structures* of each other:

Structure 4

Structure 5

Structure 6

(II) Among the resonance structures, circle the *most contributing* resonance structure.

Structure 4

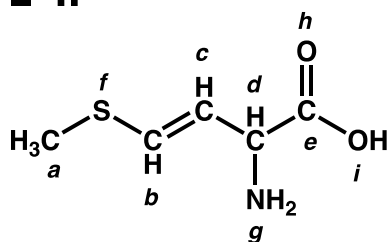
Structure 5

Structure 6

**Question 6** **6 points**

Using the data given below, calculate the **lattice energy** for SrO(s).

	ΔH° (kJ/mol)
$\text{Sr}(s) \longrightarrow \text{Sr}(g)$	+180
$\text{Sr}(g) \longrightarrow \text{Sr}^{2+}(g) + 2e^-$	+1468
$\text{O}_2(g) \longrightarrow 2\text{O}(g)$	+500
$\text{O}(g) + 2e^- \longrightarrow \text{O}^{2-}(g)$	+600
$\text{Sr}(s) + \frac{1}{2}\text{O}_2(g) \longrightarrow \text{SrO}(s)$	-550

**Question 7 10 points**

Consider the molecule given on the left, and answer questions (i) to (vi) using chemical bonding theories that we have learnt in class (valence bond theory, hybridization).

Note that the lone pairs are not shown in the structure:

i. Consider the carbon atoms marked in as ***a***, ***b***, ***c***, ***d***, and ***e*** in the molecule, and determine **hybridization** at each of the carbon centers. (2.5 points)

Atom	Hybridization
Carbon (<i>a</i>)	
Carbon (<i>b</i>)	
Carbon (<i>c</i>)	
Carbon (<i>d</i>)	
Carbon (<i>e</i>)	

ii. Now, consider the sulfur atom (S) marked as ***f***, nitrogen (N) atom (***g***), and oxygen(O) atoms (***h*** and ***i***), and determine their **hybridization**. (2 points)

Atom	Hybridization
Sulfur (<i>f</i>)	
Nitrogen (<i>g</i>)	
Oxygen (<i>h</i>)	
Oxygen (<i>i</i>)	

iii. **Describe** all the bonds formed by the carbon atoms ***a*** and ***b***, using sigma and pi designation as appropriate. (2 points)

Atom	Bonds formed (use sigma and pi as appropriate)
Carbon (<i>a</i>)	
Carbon (<i>b</i>)	

iv. How many total sp^3 -s sigma bonds are in the molecule? (1 point)

v. Determine **electronic** and **molecular geometry** at carbon (C) labeled ***e*** and sulfur (S) labeled ***f*** centres. (2 points)

Atom	Electronic Geometry	Molecular Geometry
Carbon (<i>e</i>)		
Sulfur (<i>f</i>)		

vi. Circle the value from the choices given below, that describes the **C-S-C bond angle at sulfur** (S) marked ***f***: (0.5 points)

>90° to <109.5°

109.5°

>109.5 to <120°

120°

>120° to <180°

**Question 8 12 points**

Using the **valence shell orbitals only** draw a **complete** molecular orbital (MO) diagram (with no s-p mixing) for the molecule **PS**, in the space provided below. **(6 points)**

Assume that 3s and 3p atomic orbitals will form molecular orbitals like those formed by the combination of 2s and 2p atomic orbitals that we learnt in class, and there is no s-p mixing.

Show **energy arrow**, **atomic orbitals** involved, **atom labels**, the **molecular orbitals** that are formed, **filled electrons** in the MO diagram etc. Make sure to label all the atomic and molecular orbitals.

Using the MO diagram above, answer the following questions:

- (i) Which one of the following will have the highest bond order? Circle the correct choice
A) PS^{1-} B) PS C) PS^{1+}
- (ii) Which one of the following will have the longest bond length? Circle the correct choice
A) PS^{1-} B) PS C) PS^{1+}
- (iii) Which of the following will be diamagnetic? Circle the correct choice.
A) PS^{1-} B) PS C) PS^{1+}
- (iv) Draw Lewis dot structure of the molecule with the shortest bond length (based on the MO diagram) from among the three (PS^{1-} , PS, PS^{1+}), and indicate the bond order calculated from the Lewis dot structure. **(3 points)**

Lewis Dot Structure

Bond Order



631BC936-0F27-4CDD-9D74-6AE2E52248A1

midterm-exam-93388

#554 Page 10 of 10

SCRAP PAGE

**CHEMISTRY 110 –Midterm Exam****Thursday November 7, 2024, 6:30 – 8:30 pm**

Examiners: Pallavi Sirjoosingh and Ashok Kakkar

First name (please write as legibly as possible within the boxes)

--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--

Last name

--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--

Student ID number

--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--

This booklet has 8 questions on 10 pages (The last page can be used as a scrap page).

The staple must NOT be removed from the question booklet. Your intact question booklet must be returned at the end of the exam.

A separate booklet contains a periodic table, and datasheet.

Write your answers directly under the questions in the space provided. Do not write on the QR codes or near the top edges (as it may be cut off).

Please write **legibly and dark enough** for the scan to be readable. For students who use English as a second language, dictionaries are permitted. Calculators are allowed.

DO NOT ASK QUESTIONS OF THE INVIGILATORS about interpretations of the exam questions. If you have a concern about a question, please summarize it at the top of this page. All concerns will be evaluated after the exam. GOOD LUCK!

**Question 1 8 points** (*Assume all transitions are possible*)

a. 6 points Calculate the **energy (in kJ/mol)** required to ionize (completely remove) the electron from the **$n=3$** level of **He^+ ion** (He: Helium). Show your work including equations. (3 sig. figs.)

b. 2 points

(I) **Select** the **largest (longest)** wavelength from the three choices below:

- i. Wavelength of photon to *completely remove* an electron from *1s orbital of Hydrogen atom*
- ii. Wavelength of photon to *completely remove* an electron from *1s orbital of Boron atom (B)*
- iii. Wavelength of photon to *completely remove* an electron from *1s orbital of Boron cation (B^+)*

i **or** **ii** **or** **iii**

(II) **Select** the **smallest (shortest)** wavelength from the three choices below (*for H atom*):

- i. Wavelength of photon *emitted* for an electron transitioning from *2p to 1s orbital*
- ii. Wavelength of photon *absorbed* for an electron transitioning from *2s to 4p orbital*
- iii. Wavelength of photon *absorbed* for an electron transitioning from *1s to 3p orbital*

i **or** **ii** **or** **iii**



Question 2 8 points



A Chem110 student plots the graph shown on the left, from data obtained for the kinetic energy (eV) of the ejected electron versus frequency (s^{-1}) of the incident light, for an unknown metal. ($1\text{eV} = 1.6022 \times 10^{-19} \text{ J}$)

Based on the plot, answer the questions **a to c** below:

a. 1 point On the plot above, label the **threshold frequency**.

b. 2 points What is the value of the work potential (in J) of the unknown metal?

c. 3 points What is the *longest* wavelength (in nm) required to eject an electron from this metal's surface? Show your work including equation(s) used. (3 sig figs)

d. 2 points In the space provided below, plot the following

(Assume frequency is *above* threshold frequency):

- i. *Kinetic energy* of an ejected electron versus the *intensity* of the incident light (left)
- ii. *Number of electrons* ejected versus the *frequency* of the incident light (right)



**Question 3** **6 points****a. 2 points**

In the ground state of Ge (Germanium; $Z = 32$) atom, **name** the following orbitals:

i. Orbital containing the electron that requires the maximum energy to be removed.

ii. Orbital containing the electron that experiences the minimum effective nuclear charge.

b. 2 points

An unknown anion (A^-) has the following *excited* electronic configuration: $[Ar] 4s^2 3d^3 4p^1$

(Ar: Argon; $Z = 18$)

Give the **condensed ground state electronic configuration** of the atom, **A** and the cation, **A^{2+}** .
Show your work. (Assume all electronic transitions are possible)

Electronic Configuration of A:

Electronic Configuration of A^{2+} :

c. 2 points

In a single atom, determine how many electrons can have the following set of quantum numbers?

(i) $n = 5; l = 3$

In a single atom, determine how many electrons can have the following set of quantum numbers?

(i) $n = 3; l = 2; m_s = +1/2$

**Question 4** **8 points**

Rank the following from 1 to 4 in **increasing** order. Provide the ranking in the boxes provided.

Note: Atomic number (Z) are provided for relevant atoms

a) Increasing (lowest to highest) order of *ionization energy*: (1= lowest; 4 = highest)

Mg

Mg: Z = 12

Rb

Rb: Z = 37

Cs

Cs: Z = 55

K

K : Z = 19

b) Increasing (lowest to highest) order of *bond angle*: (1= lowest; 4 = highest)

H₂OCCl₄XeF₄

HCN

c) Increasing (smallest to largest) order of *ionic size* (1= smallest; 4 = largest)

S²⁻

S: Z = 16

O²⁻

O: Z = 8

Be²⁺

Be: Z = 4

Na⁺

Na: Z = 11

d) Increasing (lowest to highest) order of *bond energy* (1= lowest; 4 = highest)

P-F

P-Cl

P-I

P-Br

**Question 5** **12 points**

a. 9 points For the molecules HClO_4 and TeF_6 , draw the most stable *Lewis structure* showing appropriate shape and *dashed/wedged bonds*, based on VSEPR. Determine the **molecular geometry** around the central atom. For HClO_4 , label the *most polar bond*.



b. 3 points (Note: Lone pairs are not shown in structures below)

(i) For the three *resonance* structures (1, 2 and 3) shown below, answer the following question:



(I) Circle the *most contributing* resonance structure.

Structure 1

Structure 2

Structure 3

(ii) For the three structures given (4, 5, and 6) below, answer the following questions:



(I) Circle all the structures that represent *resonance structures* of each other:

Structure 4

Structure 5

Structure 6

(II) Among the resonance structures, circle the *most contributing* resonance structure.

Structure 4

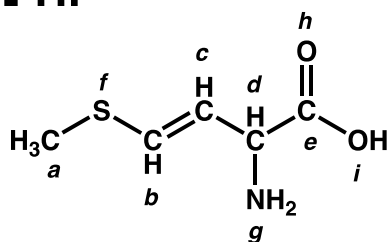
Structure 5

Structure 6

**Question 6** **6 points**

Using the data given below, calculate the **lattice energy** for SrO(s).

	ΔH° (kJ/mol)
$\text{Sr}(s) \longrightarrow \text{Sr}(g)$	+180
$\text{Sr}(g) \longrightarrow \text{Sr}^{2+}(g) + 2e^-$	+1468
$\text{O}_2(g) \longrightarrow 2\text{O}(g)$	+500
$\text{O}(g) + 2e^- \longrightarrow \text{O}^{2-}(g)$	+600
$\text{Sr}(s) + \frac{1}{2}\text{O}_2(g) \longrightarrow \text{SrO}(s)$	-550

**Question 7 10 points**

Consider the molecule given on the left, and answer questions (i) to (vi) using chemical bonding theories that we have learnt in class (valence bond theory, hybridization).

Note that the lone pairs are not shown in the structure:

i. Consider the carbon atoms marked in as ***a***, ***b***, ***c***, ***d***, and ***e*** in the molecule, and determine **hybridization** at each of the carbon centers. (2.5 points)

Atom	Hybridization
Carbon (<i>a</i>)	
Carbon (<i>b</i>)	
Carbon (<i>c</i>)	
Carbon (<i>d</i>)	
Carbon (<i>e</i>)	

ii. Now, consider the sulfur atom (S) marked as ***f***, nitrogen (N) atom (***g***), and oxygen(O) atoms (***h*** and ***i***), and determine their **hybridization**. (2 points)

Atom	Hybridization
Sulfur (<i>f</i>)	
Nitrogen (<i>g</i>)	
Oxygen (<i>h</i>)	
Oxygen (<i>i</i>)	

iii. **Describe** all the bonds formed by the carbon atoms ***a*** and ***b***, using sigma and pi designation as appropriate. (2 points)

Atom	Bonds formed (use sigma and pi as appropriate)
Carbon (<i>a</i>)	
Carbon (<i>b</i>)	

iv. How many total sp^3 -s sigma bonds are in the molecule? (1 point)

v. Determine **electronic** and **molecular geometry** at carbon (C) labeled ***e*** and sulfur (S) labeled ***f*** centres. (2 points)

Atom	Electronic Geometry	Molecular Geometry
Carbon (<i>e</i>)		
Sulfur (<i>f</i>)		

vi. Circle the value from the choices given below, that describes the **C-S-C bond angle at sulfur** (S) marked ***f***: (0.5 points)

>90° to <109.5° 109.5° >109.5 to <120° 120° >120° to <180°


Question 8 12 points

Using the **valence shell orbitals only** draw a **complete** molecular orbital (MO) diagram (with no s-p mixing) for the molecule **PS**, in the space provided below. **(6 points)**

Assume that 3s and 3p atomic orbitals will form molecular orbitals like those formed by the combination of 2s and 2p atomic orbitals that we learnt in class, and there is no s-p mixing.

Show **energy arrow**, **atomic orbitals** involved, **atom labels**, the **molecular orbitals** that are formed, **filled electrons** in the MO diagram etc. Make sure to **label all the atomic and molecular orbitals**.

Using the MO diagram above, answer the following questions:

- (i) Which one of the following will have the highest bond order? Circle the correct choice
 A) PS^{1-} B) PS C) PS^{1+}
- (ii) Which one of the following will have the longest bond length? Circle the correct choice
 A) PS^{1-} B) PS C) PS^{1+}
- (iii) Which of the following will be diamagnetic? Circle the correct choice.
 A) PS^{1-} B) PS C) PS^{1+}
- (iv) Draw Lewis dot structure of the molecule with the shortest bond length (based on the MO diagram) from among the three (PS^{1-} , PS, PS^{1+}), and indicate the bond order calculated from the Lewis dot structure. **(3 points)**

Lewis Dot Structure

Bond Order



55E50A7B-0D72-46D4-89A1-EE39CA447861

midterm-exam-93388

#555 Page 10 of 10

SCRAP PAGE

**CHEMISTRY 110 –Midterm Exam****Thursday November 7, 2024, 6:30 – 8:30 pm****Examiners: Pallavi Sirjoosingh and Ashok Kakkar**

First name (please write as legibly as possible within the boxes)

--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--

Last name

--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--

Student ID number

--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--

This booklet has 8 questions on 10 pages (The last page can be used as a scrap page).

The staple must NOT be removed from the question booklet. Your intact question booklet must be returned at the end of the exam.

A separate booklet contains a periodic table, and datasheet.

Write your answers directly under the questions in the space provided. Do not write on the QR codes or near the top edges (as it may be cut off).

Please write **legibly and dark enough** for the scan to be readable. For students who use English as a second language, dictionaries are permitted. Calculators are allowed.

DO NOT ASK QUESTIONS OF THE INVIGILATORS about interpretations of the exam questions. If you have a concern about a question, please summarize it at the top of this page. All concerns will be evaluated after the exam. GOOD LUCK!

**Question 1 8 points** (*Assume all transitions are possible*)

a. 6 points Calculate the **energy (in kJ/mol)** required to ionize (completely remove) the electron from the **n=3** level of **He⁺ ion** (He: Helium). Show your work including equations. (3 sig. figs.)

b. 2 points

(I) Select the largest (longest) wavelength from the three choices below:

- i. Wavelength of photon to *completely remove* an electron from *1s orbital of Hydrogen atom*
- ii. Wavelength of photon to *completely remove* an electron from *1s orbital of Boron atom (B)*
- iii. Wavelength of photon to *completely remove* an electron from *1s orbital of Boron cation (B⁺)*

i or ii or iii

(II) Select the smallest (shortest) wavelength from the three choices below (*for H atom*):

- i. Wavelength of photon *emitted* for an electron transitioning from *2p to 1s orbital*
- ii. Wavelength of photon *absorbed* for an electron transitioning from *2s to 4p orbital*
- iii. Wavelength of photon *absorbed* for an electron transitioning from *1s to 3p orbital*

i or ii or iii



Question 2 8 points



A Chem110 student plots the graph shown on the left, from data obtained for the kinetic energy (eV) of the ejected electron versus frequency (s^{-1}) of the incident light, for an unknown metal. ($1\text{eV} = 1.6022 \times 10^{-19} \text{ J}$)

Based on the plot, answer the questions **a to c** below:

a. 1 point On the plot above, label the **threshold frequency**.

b. 2 points What is the value of the work potential (in J) of the unknown metal?

c. 3 points What is the *longest* wavelength (in nm) required to eject an electron from this metal's surface? Show your work including equation(s) used. (3 sig figs)

d. 2 points In the space provided below, plot the following

(Assume frequency is *above* threshold frequency):

- Kinetic energy* of an ejected electron versus the *intensity* of the incident light (left)
- Number of electrons* ejected versus the *frequency* of the incident light (right)



**Question 3** **6 points****a. 2 points**

In the ground state of Ge (Germanium; $Z = 32$) atom, **name** the following orbitals:

i. Orbital containing the electron that requires the maximum energy to be removed.

ii. Orbital containing the electron that experiences the minimum effective nuclear charge.

b. 2 points

An unknown anion (A^-) has the following *excited* electronic configuration: $[Ar] 4s^2 3d^3 4p^1$

(Ar: Argon; $Z = 18$)

Give the **condensed ground state electronic configuration** of the atom, **A** and the cation, **A^{2+}** .
Show your work. (Assume all electronic transitions are possible)

Electronic Configuration of A:

Electronic Configuration of A^{2+} :

c. 2 points

In a single atom, determine how many electrons can have the following set of quantum numbers?

(i) $n = 5; l = 3$

In a single atom, determine how many electrons can have the following set of quantum numbers?

(i) $n = 3; l = 2; m_s = +1/2$

**Question 4** **8 points**

Rank the following from 1 to 4 in **increasing** order. Provide the ranking in the boxes provided.

Note: Atomic number (Z) are provided for relevant atoms

a) Increasing (lowest to highest) order of *ionization energy*: (1= lowest; 4 = highest)

Mg

Mg: Z = 12

Rb

Rb: Z = 37

Cs

Cs: Z = 55

K

K : Z = 19

b) Increasing (lowest to highest) order of *bond angle*: (1= lowest; 4 = highest)

H₂OCCl₄XeF₄

HCN

c) Increasing (smallest to largest) order of *ionic size* (1= smallest; 4 = largest)

S²⁻

S: Z = 16

O²⁻

O: Z = 8

Be²⁺

Be: Z = 4

Na⁺

Na: Z = 11

d) Increasing (lowest to highest) order of *bond energy* (1= lowest; 4 = highest)

P-F

P-Cl

P-I

P-Br

**Question 5** **12 points**

a. 9 points For the molecules HClO_4 and TeF_6 , draw the most stable *Lewis structure* showing appropriate shape and *dashed/wedged bonds*, based on VSEPR. Determine the **molecular geometry** around the central atom. For HClO_4 , label the *most polar bond*.



b. 3 points (Note: Lone pairs are not shown in structures below)

(i) For the three *resonance* structures (1, 2 and 3) shown below, answer the following question:



(I) Circle the *most contributing* resonance structure.

Structure 1

Structure 2

Structure 3

(ii) For the three structures given (4, 5, and 6) below, answer the following questions:



(I) Circle all the structures that represent *resonance structures* of each other:

Structure 4

Structure 5

Structure 6

(II) Among the resonance structures, circle the *most contributing* resonance structure.

Structure 4

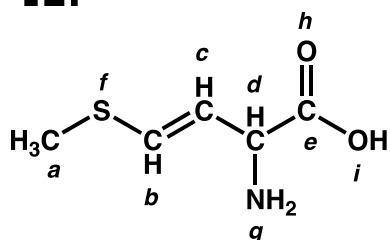
Structure 5

Structure 6

**Question 6** **6 points**

Using the data given below, calculate the **lattice energy** for SrO(s).

	ΔH° (kJ/mol)
$\text{Sr}(s) \longrightarrow \text{Sr}(g)$	+180
$\text{Sr}(g) \longrightarrow \text{Sr}^{2+}(g) + 2e^-$	+1468
$\text{O}_2(g) \longrightarrow 2\text{O}(g)$	+500
$\text{O}(g) + 2e^- \longrightarrow \text{O}^{2-}(g)$	+600
$\text{Sr}(s) + \frac{1}{2}\text{O}_2(g) \longrightarrow \text{SrO}(s)$	-550

**Question 7 10 points**

Consider the molecule given on the left, and answer questions (i) to (vi) using chemical bonding theories that we have learnt in class (valence bond theory, hybridization).

Note that the lone pairs are not shown in the structure:

i. Consider the carbon atoms marked in as ***a***, ***b***, ***c***, ***d***, and ***e*** in the molecule, and determine **hybridization** at each of the carbon centers. (2.5 points)

Atom	Hybridization
Carbon (<i>a</i>)	
Carbon (<i>b</i>)	
Carbon (<i>c</i>)	
Carbon (<i>d</i>)	
Carbon (<i>e</i>)	

ii. Now, consider the sulfur atom (S) marked as ***f***, nitrogen (N) atom (***g***), and oxygen(O) atoms (***h*** and ***i***), and determine their **hybridization**. (2 points)

Atom	Hybridization
Sulfur (<i>f</i>)	
Nitrogen (<i>g</i>)	
Oxygen (<i>h</i>)	
Oxygen (<i>i</i>)	

iii. **Describe** all the bonds formed by the carbon atoms ***a*** and ***b***, using sigma and pi designation as appropriate. (2 points)

Atom	Bonds formed (use sigma and pi as appropriate)
Carbon (<i>a</i>)	
Carbon (<i>b</i>)	

iv. How many total sp^3 -s sigma bonds are in the molecule? (1 point)

v. Determine **electronic** and **molecular geometry** at carbon (C) labeled ***e*** and sulfur (S) labeled ***f*** centres. (2 points)

Atom	Electronic Geometry	Molecular Geometry
Carbon (<i>e</i>)		
Sulfur (<i>f</i>)		

vi. Circle the value from the choices given below, that describes the **C-S-C bond angle at sulfur** (S) marked ***f***: (0.5 points)

$>90^\circ$ to $<109.5^\circ$ 109.5° >109.5 to $<120^\circ$ 120° $>120^\circ$ to $<180^\circ$


Question 8 12 points

Using the **valence shell orbitals only** draw a **complete** molecular orbital (MO) diagram (with no s-p mixing) for the molecule **PS**, in the space provided below. **(6 points)**

Assume that 3s and 3p atomic orbitals will form molecular orbitals like those formed by the combination of 2s and 2p atomic orbitals that we learnt in class, and there is no s-p mixing.

Show **energy arrow**, **atomic orbitals** involved, **atom labels**, the **molecular orbitals** that are formed, **filled electrons** in the MO diagram etc. Make sure to label all the atomic and molecular orbitals.

Using the MO diagram above, answer the following questions:

- (i) Which one of the following will have the highest bond order? Circle the correct choice
 A) PS^{1-} B) PS C) PS^{1+}
- (ii) Which one of the following will have the longest bond length? Circle the correct choice
 A) PS^{1-} B) PS C) PS^{1+}
- (iii) Which of the following will be diamagnetic? Circle the correct choice.
 A) PS^{1-} B) PS C) PS^{1+}
- (iv) Draw Lewis dot structure of the molecule with the shortest bond length (based on the MO diagram) from among the three (PS^{1-} , PS, PS^{1+}), and indicate the bond order calculated from the Lewis dot structure. **(3 points)**

Lewis Dot Structure

Bond Order



AC92C19F-31E0-4835-B310-DF3AC67F7F5A

midterm-exam-93388

#556 Page 10 of 10

SCRAP PAGE

**CHEMISTRY 110 –Midterm Exam****Thursday November 7, 2024, 6:30 – 8:30 pm****Examiners: Pallavi Sirjoosingh and Ashok Kakkar**

First name (please write as legibly as possible within the boxes)

--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--

Last name

--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--

Student ID number

--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--

This booklet has 8 questions on 10 pages (The last page can be used as a scrap page).

The staple must NOT be removed from the question booklet. Your intact question booklet must be returned at the end of the exam.

A separate booklet contains a periodic table, and datasheet.

Write your answers directly under the questions in the space provided. Do not write on the QR codes or near the top edges (as it may be cut off).

Please write **legibly and dark enough** for the scan to be readable. For students who use English as a second language, dictionaries are permitted. Calculators are allowed.

DO NOT ASK QUESTIONS OF THE INVIGILATORS about interpretations of the exam questions. If you have a concern about a question, please summarize it at the top of this page. All concerns will be evaluated after the exam. GOOD LUCK!

**Question 1 8 points** (*Assume all transitions are possible*)

a. 6 points Calculate the **energy (in kJ/mol)** required to ionize (completely remove) the electron from the **n=3** level of **He⁺ ion** (He: Helium). Show your work including equations. (3 sig. figs.)

b. 2 points

(I) Select the largest (longest) wavelength from the three choices below:

- i. Wavelength of photon to *completely remove* an electron from *1s orbital of Hydrogen atom*
- ii. Wavelength of photon to *completely remove* an electron from *1s orbital of Boron atom (B)*
- iii. Wavelength of photon to *completely remove* an electron from *1s orbital of Boron cation (B⁺)*

i or ii or iii

(II) Select the smallest (shortest) wavelength from the three choices below (*for H atom*):

- i. Wavelength of photon *emitted* for an electron transitioning from *2p to 1s orbital*
- ii. Wavelength of photon *absorbed* for an electron transitioning from *2s to 4p orbital*
- iii. Wavelength of photon *absorbed* for an electron transitioning from *1s to 3p orbital*

i or ii or iii



Question 2 8 points



A Chem110 student plots the graph shown on the left, from data obtained for the kinetic energy (eV) of the ejected electron versus frequency (s^{-1}) of the incident light, for an unknown metal. ($1\text{eV} = 1.6022 \times 10^{-19} \text{ J}$)

Based on the plot, answer the questions **a to c** below:

a. 1 point On the plot above, label the **threshold frequency**.

b. 2 points What is the value of the work potential (in J) of the unknown metal?

c. 3 points What is the *longest* wavelength (in nm) required to eject an electron from this metal's surface? Show your work including equation(s) used. (3 sig figs)

d. 2 points In the space provided below, plot the following

(Assume frequency is *above* threshold frequency):

- Kinetic energy* of an ejected electron versus the *intensity* of the incident light (left)
- Number of electrons* ejected versus the *frequency* of the incident light (right)



**Question 3** **6 points****a. 2 points**

In the ground state of Ge (Germanium; $Z = 32$) atom, **name** the following orbitals:

i. Orbital containing the electron that requires the maximum energy to be removed.

ii. Orbital containing the electron that experiences the minimum effective nuclear charge.

b. 2 points

An unknown anion (A^-) has the following *excited* electronic configuration: $[Ar] 4s^2 3d^3 4p^1$

(Ar: Argon; $Z = 18$)

Give the **condensed ground state electronic configuration** of the atom, **A** and the cation, **A^{2+}** .
Show your work. (Assume all electronic transitions are possible)

Electronic Configuration of A:

Electronic Configuration of A^{2+} :

c. 2 points

In a single atom, determine how many electrons can have the following set of quantum numbers?

(i) $n = 5; l = 3$

In a single atom, determine how many electrons can have the following set of quantum numbers?

(i) $n = 3; l = 2; m_s = +1/2$

**Question 4** **8 points**

Rank the following from 1 to 4 in **increasing** order. Provide the ranking in the boxes provided.

Note: Atomic number (Z) are provided for relevant atoms

a) Increasing (lowest to highest) order of *ionization energy*: (1= lowest; 4 = highest)

Mg

Mg: Z = 12

Rb

Rb: Z = 37

Cs

Cs: Z = 55

K

K : Z = 19

b) Increasing (lowest to highest) order of *bond angle*: (1= lowest; 4 = highest)

H₂OCCl₄XeF₄

HCN

c) Increasing (smallest to largest) order of *ionic size* (1= smallest; 4 = largest)

S²⁻

S: Z = 16

O²⁻

O: Z = 8

Be²⁺

Be: Z = 4

Na⁺

Na: Z = 11

d) Increasing (lowest to highest) order of *bond energy* (1= lowest; 4 = highest)

P-F

P-Cl

P-I

P-Br

**Question 5** **12 points**

a. 9 points For the molecules HClO_4 and TeF_6 , draw the most stable *Lewis structure* showing appropriate shape and *dashed/wedged bonds*, based on VSEPR. Determine the **molecular geometry** around the central atom. For HClO_4 , label the *most polar bond*.



b. 3 points (Note: Lone pairs are not shown in structures below)

(i) For the three *resonance* structures (1, 2 and 3) shown below, answer the following question:



(I) Circle the *most contributing* resonance structure.

Structure 1

Structure 2

Structure 3

(ii) For the three structures given (4, 5, and 6) below, answer the following questions:



(I) Circle all the structures that represent *resonance structures* of each other:

Structure 4

Structure 5

Structure 6

(II) Among the resonance structures, circle the *most contributing* resonance structure.

Structure 4

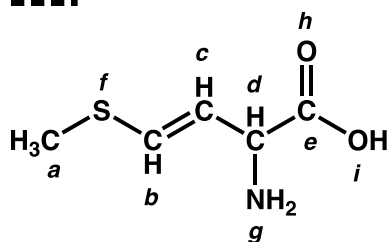
Structure 5

Structure 6

**Question 6** **6 points**

Using the data given below, calculate the **lattice energy** for SrO(s).

	ΔH° (kJ/mol)
$\text{Sr}(s) \longrightarrow \text{Sr}(g)$	+180
$\text{Sr}(g) \longrightarrow \text{Sr}^{2+}(g) + 2e^-$	+1468
$\text{O}_2(g) \longrightarrow 2\text{O}(g)$	+500
$\text{O}(g) + 2e^- \longrightarrow \text{O}^{2-}(g)$	+600
$\text{Sr}(s) + \frac{1}{2}\text{O}_2(g) \longrightarrow \text{SrO}(s)$	-550

**Question 7 10 points**

Consider the molecule given on the left, and answer questions (i) to (vi) using chemical bonding theories that we have learnt in class (valence bond theory, hybridization).

Note that the lone pairs are not shown in the structure:

i. Consider the carbon atoms marked in as ***a***, ***b***, ***c***, ***d***, and ***e*** in the molecule, and determine **hybridization** at each of the carbon centers. (2.5 points)

Atom	Hybridization
Carbon (<i>a</i>)	
Carbon (<i>b</i>)	
Carbon (<i>c</i>)	
Carbon (<i>d</i>)	
Carbon (<i>e</i>)	

ii. Now, consider the sulfur atom (S) marked as ***f***, nitrogen (N) atom (***g***), and oxygen(O) atoms (***h*** and ***i***), and determine their **hybridization**. (2 points)

Atom	Hybridization
Sulfur (<i>f</i>)	
Nitrogen (<i>g</i>)	
Oxygen (<i>h</i>)	
Oxygen (<i>i</i>)	

iii. **Describe** all the bonds formed by the carbon atoms ***a*** and ***b***, using sigma and pi designation as appropriate. (2 points)

Atom	Bonds formed (use sigma and pi as appropriate)
Carbon (<i>a</i>)	
Carbon (<i>b</i>)	

iv. How many total sp^3 -s sigma bonds are in the molecule? (1 point)

v. Determine **electronic** and **molecular geometry** at carbon (C) labeled ***e*** and sulfur (S) labeled ***f*** centres. (2 points)

Atom	Electronic Geometry	Molecular Geometry
Carbon (<i>e</i>)		
Sulfur (<i>f</i>)		

vi. Circle the value from the choices given below, that describes the **C-S-C bond angle at sulfur** (S) marked ***f***: (0.5 points)

>90° to <109.5° 109.5° >109.5 to <120° 120° >120° to <180°



Question 8 12 points

Using the **valence shell orbitals only** draw a **complete** molecular orbital (MO) diagram (with no s-p mixing) for the molecule **PS**, in the space provided below. **(6 points)**

Assume that 3s and 3p atomic orbitals will form molecular orbitals like those formed by the combination of 2s and 2p atomic orbitals that we learnt in class, and there is no s-p mixing.

Show **energy arrow**, **atomic orbitals** involved, **atom labels**, the **molecular orbitals** that are formed, **filled electrons** in the MO diagram etc. Make sure to label all the atomic and molecular orbitals.

Using the MO diagram above, answer the following questions:

- (i) Which one of the following will have the highest bond order? Circle the correct choice
 A) PS^{1-} B) PS C) PS^{1+}
- (ii) Which one of the following will have the longest bond length? Circle the correct choice
 A) PS^{1-} B) PS C) PS^{1+}
- (iii) Which of the following will be diamagnetic? Circle the correct choice.
 A) PS^{1-} B) PS C) PS^{1+}
- (iv) Draw Lewis dot structure of the molecule with the shortest bond length (based on the MO diagram) from among the three (PS^{1-} , PS, PS^{1+}), and indicate the bond order calculated from the Lewis dot structure. **(3 points)**

Lewis Dot Structure

Bond Order



E1BD9C6B-448F-4675-BEF7-DDDFA5F80417

midterm-exam-93388

#557 Page 10 of 10

SCRAP PAGE

**CHEMISTRY 110 –Midterm Exam****Thursday November 7, 2024, 6:30 – 8:30 pm**

Examiners: Pallavi Sirjoosingh and Ashok Kakkar

First name (please write as legibly as possible within the boxes)

--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--

Last name

--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--

Student ID number

--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--

This booklet has 8 questions on 10 pages (The last page can be used as a scrap page).

The staple must NOT be removed from the question booklet. Your intact question booklet must be returned at the end of the exam.

A separate booklet contains a periodic table, and datasheet.

Write your answers directly under the questions in the space provided. Do not write on the QR codes or near the top edges (as it may be cut off).

Please write **legibly and dark enough** for the scan to be readable. For students who use English as a second language, dictionaries are permitted. Calculators are allowed.

DO NOT ASK QUESTIONS OF THE INVIGILATORS about interpretations of the exam questions. If you have a concern about a question, please summarize it at the top of this page. All concerns will be evaluated after the exam. GOOD LUCK!

**Question 1 8 points** (*Assume all transitions are possible*)

a. 6 points Calculate the **energy (in kJ/mol)** required to ionize (completely remove) the electron from the **$n=3$** level of **He^+ ion** (He: Helium). Show your work including equations. (3 sig. figs.)

b. 2 points

(I) Select the largest (longest) wavelength from the three choices below:

- i. Wavelength of photon to *completely remove* an electron from *1s orbital of Hydrogen atom*
- ii. Wavelength of photon to *completely remove* an electron from *1s orbital of Boron atom (B)*
- iii. Wavelength of photon to *completely remove* an electron from *1s orbital of Boron cation (B^+)*

i or ii or iii

(II) Select the smallest (shortest) wavelength from the three choices below (*for H atom*):

- i. Wavelength of photon *emitted* for an electron transitioning from *2p to 1s orbital*
- ii. Wavelength of photon *absorbed* for an electron transitioning from *2s to 4p orbital*
- iii. Wavelength of photon *absorbed* for an electron transitioning from *1s to 3p orbital*

i or ii or iii



Question 2 8 points



A Chem110 student plots the graph shown on the left, from data obtained for the kinetic energy (eV) of the ejected electron versus frequency (s^{-1}) of the incident light, for an unknown metal. ($1\text{eV} = 1.6022 \times 10^{-19} \text{ J}$)

Based on the plot, answer the questions **a to c** below:

a. 1 point On the plot above, label the **threshold frequency**.

b. 2 points What is the value of the work potential (in J) of the unknown metal?

c. 3 points What is the *longest* wavelength (in nm) required to eject an electron from this metal's surface? Show your work including equation(s) used. (3 sig figs)

d. 2 points In the space provided below, plot the following

(Assume frequency is *above* threshold frequency):

- i. *Kinetic energy* of an ejected electron versus the *intensity* of the incident light (left)
- ii. *Number of electrons* ejected versus the *frequency* of the incident light (right)



**Question 3** **6 points****a. 2 points**

In the ground state of Ge (Germanium; $Z = 32$) atom, **name** the following orbitals:

i. Orbital containing the electron that requires the maximum energy to be removed.

ii. Orbital containing the electron that experiences the minimum effective nuclear charge.

b. 2 points

An unknown anion (A^-) has the following *excited* electronic configuration: $[Ar] 4s^2 3d^3 4p^1$

(Ar: Argon; $Z = 18$)

Give the **condensed ground state electronic configuration** of the atom, **A** and the cation, **A^{2+}** .
Show your work. (Assume all electronic transitions are possible)

Electronic Configuration of A:

Electronic Configuration of A^{2+} :

c. 2 points

In a single atom, determine how many electrons can have the following set of quantum numbers?

(i) $n = 5; l = 3$

In a single atom, determine how many electrons can have the following set of quantum numbers?

(i) $n = 3; l = 2; m_s = +1/2$

**Question 4** **8 points**

Rank the following from 1 to 4 in **increasing** order. Provide the ranking in the boxes provided.

Note: Atomic number (Z) are provided for relevant atoms

a) Increasing (lowest to highest) order of *ionization energy*: (1= lowest; 4 = highest)

Mg

Mg: Z = 12

Rb

Rb: Z = 37

Cs

Cs: Z = 55

K

K : Z = 19

b) Increasing (lowest to highest) order of *bond angle*: (1= lowest; 4 = highest)

H₂OCCl₄XeF₄

HCN

c) Increasing (smallest to largest) order of *ionic size* (1= smallest; 4 = largest)

S²⁻

S: Z = 16

O²⁻

O: Z = 8

Be²⁺

Be: Z = 4

Na⁺

Na: Z = 11

d) Increasing (lowest to highest) order of *bond energy* (1= lowest; 4 = highest)

P-F

P-Cl

P-I

P-Br

**Question 5** **12 points**

a. 9 points For the molecules HClO_4 and TeF_6 , draw the most stable *Lewis structure* showing appropriate shape and *dashed/wedged bonds*, based on VSEPR. Determine the **molecular geometry** around the central atom. For HClO_4 , label the *most polar bond*.



b. 3 points (Note: Lone pairs are not shown in structures below)

(i) For the three *resonance* structures (1, 2 and 3) shown below, answer the following question:



(I) Circle the *most contributing* resonance structure.

Structure 1

Structure 2

Structure 3

(ii) For the three structures given (4, 5, and 6) below, answer the following questions:



(I) Circle all the structures that represent *resonance structures* of each other:

Structure 4

Structure 5

Structure 6

(II) Among the resonance structures, circle the *most contributing* resonance structure.

Structure 4

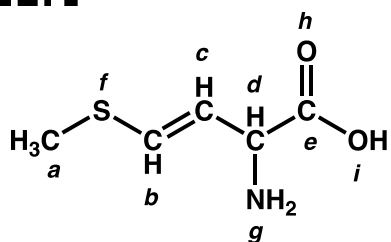
Structure 5

Structure 6

**Question 6** **6 points**

Using the data given below, calculate the **lattice energy** for SrO(s).

	ΔH° (kJ/mol)
$\text{Sr}(s) \longrightarrow \text{Sr}(g)$	+180
$\text{Sr}(g) \longrightarrow \text{Sr}^{2+}(g) + 2e^-$	+1468
$\text{O}_2(g) \longrightarrow 2\text{O}(g)$	+500
$\text{O}(g) + 2e^- \longrightarrow \text{O}^{2-}(g)$	+600
$\text{Sr}(s) + \frac{1}{2}\text{O}_2(g) \longrightarrow \text{SrO}(s)$	-550

**Question 7 10 points**

Consider the molecule given on the left, and answer questions (i) to (vi) using chemical bonding theories that we have learnt in class (valence bond theory, hybridization).

Note that the lone pairs are not shown in the structure:

i. Consider the carbon atoms marked in as ***a***, ***b***, ***c***, ***d***, and ***e*** in the molecule, and determine **hybridization** at each of the carbon centers. (2.5 points)

Atom	Hybridization
Carbon (<i>a</i>)	
Carbon (<i>b</i>)	
Carbon (<i>c</i>)	
Carbon (<i>d</i>)	
Carbon (<i>e</i>)	

ii. Now, consider the sulfur atom (S) marked as ***f***, nitrogen (N) atom (***g***), and oxygen(O) atoms (***h*** and ***i***), and determine their **hybridization**. (2 points)

Atom	Hybridization
Sulfur (<i>f</i>)	
Nitrogen (<i>g</i>)	
Oxygen (<i>h</i>)	
Oxygen (<i>i</i>)	

iii. **Describe** all the bonds formed by the carbon atoms ***a*** and ***b***, using sigma and pi designation as appropriate. (2 points)

Atom	Bonds formed (use sigma and pi as appropriate)
Carbon (<i>a</i>)	
Carbon (<i>b</i>)	

iv. How many total sp^3 -s sigma bonds are in the molecule? (1 point)

v. Determine **electronic** and **molecular geometry** at carbon (C) labeled ***e*** and sulfur (S) labeled ***f*** centres. (2 points)

Atom	Electronic Geometry	Molecular Geometry
Carbon (<i>e</i>)		
Sulfur (<i>f</i>)		

vi. Circle the value from the choices given below, that describes the **C-S-C bond angle at sulfur** (S) marked ***f***: (0.5 points)

>90° to <109.5°

109.5°

>109.5 to <120°

120°

>120° to <180°

**Question 8 12 points**

Using the **valence shell orbitals only** draw a **complete** molecular orbital (MO) diagram (with no s-p mixing) for the molecule **PS**, in the space provided below. **(6 points)**

Assume that 3s and 3p atomic orbitals will form molecular orbitals like those formed by the combination of 2s and 2p atomic orbitals that we learnt in class, and there is no s-p mixing.

Show **energy arrow**, **atomic orbitals** involved, **atom labels**, the **molecular orbitals** that are formed, **filled electrons** in the MO diagram etc. Make sure to label all the atomic and molecular orbitals.

Using the MO diagram above, answer the following questions:

- (i) Which one of the following will have the highest bond order? Circle the correct choice
A) PS^{1-} B) PS C) PS^{1+}
- (ii) Which one of the following will have the longest bond length? Circle the correct choice
A) PS^{1-} B) PS C) PS^{1+}
- (iii) Which of the following will be diamagnetic? Circle the correct choice.
A) PS^{1-} B) PS C) PS^{1+}
- (iv) Draw Lewis dot structure of the molecule with the shortest bond length (based on the MO diagram) from among the three (PS^{1-} , PS, PS^{1+}), and indicate the bond order calculated from the Lewis dot structure. **(3 points)**

Lewis Dot Structure

Bond Order



9BD1613E-3F9E-458C-ADC7-8B53B82E051F

midterm-exam-93388

#558 Page 10 of 10

SCRAP PAGE

**CHEMISTRY 110 –Midterm Exam****Thursday November 7, 2024, 6:30 – 8:30 pm**

Examiners: Pallavi Sirjoosingh and Ashok Kakkar

First name (please write as legibly as possible within the boxes)

--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--

Last name

--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--

Student ID number

--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--

This booklet has 8 questions on 10 pages (The last page can be used as a scrap page).

The staple must NOT be removed from the question booklet. Your intact question booklet must be returned at the end of the exam.

A separate booklet contains a periodic table, and datasheet.

Write your answers directly under the questions in the space provided. Do not write on the QR codes or near the top edges (as it may be cut off).

Please write **legibly and dark enough** for the scan to be readable. For students who use English as a second language, dictionaries are permitted. Calculators are allowed.

DO NOT ASK QUESTIONS OF THE INVIGILATORS about interpretations of the exam questions. If you have a concern about a question, please summarize it at the top of this page. All concerns will be evaluated after the exam. GOOD LUCK!

**Question 1 8 points** (*Assume all transitions are possible*)

a. 6 points Calculate the **energy (in kJ/mol)** required to ionize (completely remove) the electron from the **$n=3$** level of **He^+ ion** (He: Helium). Show your work including equations. (3 sig. figs.)

b. 2 points

(I) Select the largest (longest) wavelength from the three choices below:

- i. Wavelength of photon to *completely remove* an electron from *1s orbital of Hydrogen atom*
- ii. Wavelength of photon to *completely remove* an electron from *1s orbital of Boron atom (B)*
- iii. Wavelength of photon to *completely remove* an electron from *1s orbital of Boron cation (B^+)*

i or ii or iii

(II) Select the smallest (shortest) wavelength from the three choices below (*for H atom*):

- i. Wavelength of photon *emitted* for an electron transitioning from *2p to 1s orbital*
- ii. Wavelength of photon *absorbed* for an electron transitioning from *2s to 4p orbital*
- iii. Wavelength of photon *absorbed* for an electron transitioning from *1s to 3p orbital*

i or ii or iii



Question 2 8 points



A Chem110 student plots the graph shown on the left, from data obtained for the kinetic energy (eV) of the ejected electron versus frequency (s^{-1}) of the incident light, for an unknown metal. ($1\text{eV} = 1.6022 \times 10^{-19} \text{ J}$)

Based on the plot, answer the questions **a to c** below:

a. 1 point On the plot above, label the **threshold frequency**.

b. 2 points What is the value of the work potential (in J) of the unknown metal?

c. 3 points What is the *longest* wavelength (in nm) required to eject an electron from this metal's surface? Show your work including equation(s) used. (3 sig figs)

d. 2 points In the space provided below, plot the following

(Assume frequency is *above* threshold frequency):

- i. *Kinetic energy* of an ejected electron versus the *intensity* of the incident light (left)
- ii. *Number of electrons* ejected versus the *frequency* of the incident light (right)



**Question 3** **6 points****a. 2 points**

In the ground state of Ge (Germanium; $Z = 32$) atom, **name** the following orbitals:

i. Orbital containing the electron that requires the maximum energy to be removed.

ii. Orbital containing the electron that experiences the minimum effective nuclear charge.

b. 2 points

An unknown anion (A^-) has the following *excited* electronic configuration: $[Ar] 4s^2 3d^3 4p^1$

(Ar: Argon; $Z = 18$)

Give the **condensed ground state electronic configuration** of the atom, **A** and the cation, **A^{2+}** .
Show your work. (Assume all electronic transitions are possible)

Electronic Configuration of A:

Electronic Configuration of A^{2+} :

c. 2 points

In a single atom, determine how many electrons can have the following set of quantum numbers?

(i) $n = 5; l = 3$

In a single atom, determine how many electrons can have the following set of quantum numbers?

(i) $n = 3; l = 2; m_s = +1/2$

**Question 4** **8 points**

Rank the following from 1 to 4 in **increasing** order. Provide the ranking in the boxes provided.

Note: Atomic number (Z) are provided for relevant atoms

a) Increasing (lowest to highest) order of *ionization energy*: (1= lowest; 4 = highest)

Mg

Mg: Z = 12

Rb

Rb: Z = 37

Cs

Cs: Z = 55

K

K : Z = 19

b) Increasing (lowest to highest) order of *bond angle*: (1= lowest; 4 = highest)

H₂OCCl₄XeF₄

HCN

c) Increasing (smallest to largest) order of *ionic size* (1= smallest; 4 = largest)

S²⁻

S: Z = 16

O²⁻

O: Z = 8

Be²⁺

Be: Z = 4

Na⁺

Na: Z = 11

d) Increasing (lowest to highest) order of *bond energy* (1= lowest; 4 = highest)

P-F

P-Cl

P-I

P-Br

**Question 5** **12 points**

a. 9 points For the molecules HClO_4 and TeF_6 , draw the most stable *Lewis structure* showing appropriate shape and *dashed/wedged bonds*, based on VSEPR. Determine the **molecular geometry** around the central atom. For HClO_4 , label the *most polar bond*.



b. 3 points (Note: Lone pairs are not shown in structures below)

(i) For the three *resonance* structures (1, 2 and 3) shown below, answer the following question:



(I) Circle the *most contributing* resonance structure.

Structure 1

Structure 2

Structure 3

(ii) For the three structures given (4, 5, and 6) below, answer the following questions:



(I) Circle all the structures that represent *resonance structures* of each other:

Structure 4

Structure 5

Structure 6

(II) Among the resonance structures, circle the *most contributing* resonance structure.

Structure 4

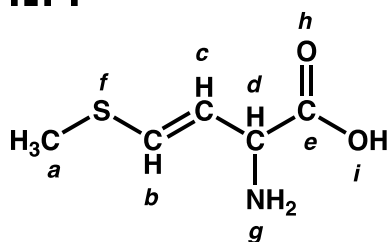
Structure 5

Structure 6

**Question 6** **6 points**

Using the data given below, calculate the **lattice energy** for SrO(s).

	ΔH° (kJ/mol)
$\text{Sr}(s) \longrightarrow \text{Sr}(g)$	+180
$\text{Sr}(g) \longrightarrow \text{Sr}^{2+}(g) + 2e^-$	+1468
$\text{O}_2(g) \longrightarrow 2\text{O}(g)$	+500
$\text{O}(g) + 2e^- \longrightarrow \text{O}^{2-}(g)$	+600
$\text{Sr}(s) + \frac{1}{2}\text{O}_2(g) \longrightarrow \text{SrO}(s)$	-550

**Question 7 10 points**

Consider the molecule given on the left, and answer questions (i) to (vi) using chemical bonding theories that we have learnt in class (valence bond theory, hybridization).

Note that the lone pairs are not shown in the structure:

i. Consider the carbon atoms marked in as ***a***, ***b***, ***c***, ***d***, and ***e*** in the molecule, and determine **hybridization** at each of the carbon centers. (2.5 points)

Atom	Hybridization
Carbon (<i>a</i>)	
Carbon (<i>b</i>)	
Carbon (<i>c</i>)	
Carbon (<i>d</i>)	
Carbon (<i>e</i>)	

ii. Now, consider the sulfur atom (S) marked as ***f***, nitrogen (N) atom (***g***), and oxygen(O) atoms (***h*** and ***i***), and determine their **hybridization**. (2 points)

Atom	Hybridization
Sulfur (<i>f</i>)	
Nitrogen (<i>g</i>)	
Oxygen (<i>h</i>)	
Oxygen (<i>i</i>)	

iii. **Describe** all the bonds formed by the carbon atoms ***a*** and ***b***, using sigma and pi designation as appropriate. (2 points)

Atom	Bonds formed (use sigma and pi as appropriate)
Carbon (<i>a</i>)	
Carbon (<i>b</i>)	

iv. How many total sp^3 -s sigma bonds are in the molecule? (1 point)

v. Determine **electronic** and **molecular geometry** at carbon (C) labeled ***e*** and sulfur (S) labeled ***f*** centres. (2 points)

Atom	Electronic Geometry	Molecular Geometry
Carbon (<i>e</i>)		
Sulfur (<i>f</i>)		

vi. Circle the value from the choices given below, that describes the **C-S-C bond angle at sulfur** (S) marked ***f***: (0.5 points)

$>90^\circ$ to $<109.5^\circ$ 109.5° >109.5 to $<120^\circ$ 120° $>120^\circ$ to $<180^\circ$

**Question 8 12 points**

Using the **valence shell orbitals only** draw a **complete** molecular orbital (MO) diagram (with no s-p mixing) for the molecule **PS**, in the space provided below. **(6 points)**

Assume that 3s and 3p atomic orbitals will form molecular orbitals like those formed by the combination of 2s and 2p atomic orbitals that we learnt in class, and there is no s-p mixing.

Show **energy arrow**, **atomic orbitals** involved, **atom labels**, the **molecular orbitals** that are formed, **filled electrons** in the MO diagram etc. Make sure to label all the atomic and molecular orbitals.

Using the MO diagram above, answer the following questions:

- (i) Which one of the following will have the highest bond order? Circle the correct choice
A) PS^{1-} B) PS C) PS^{1+}
- (ii) Which one of the following will have the longest bond length? Circle the correct choice
A) PS^{1-} B) PS C) PS^{1+}
- (iii) Which of the following will be diamagnetic? Circle the correct choice.
A) PS^{1-} B) PS C) PS^{1+}
- (iv) Draw Lewis dot structure of the molecule with the shortest bond length (based on the MO diagram) from among the three (PS^{1-} , PS, PS^{1+}), and indicate the bond order calculated from the Lewis dot structure. **(3 points)**

Lewis Dot Structure

Bond Order



2C5FD43B-11EF-4FCC-914A-D7E80D6C66DC

midterm-exam-93388

#559 Page 10 of 10

SCRAP PAGE

**CHEMISTRY 110 –Midterm Exam****Thursday November 7, 2024, 6:30 – 8:30 pm**

Examiners: Pallavi Sirjoosingh and Ashok Kakkar

First name (please write as legibly as possible within the boxes)

--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--

Last name

--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--

Student ID number

--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--

This booklet has 8 questions on 10 pages (The last page can be used as a scrap page).

The staple must NOT be removed from the question booklet. Your intact question booklet must be returned at the end of the exam.

A separate booklet contains a periodic table, and datasheet.

Write your answers directly under the questions in the space provided. Do not write on the QR codes or near the top edges (as it may be cut off).

Please write **legibly and dark enough** for the scan to be readable. For students who use English as a second language, dictionaries are permitted. Calculators are allowed.

DO NOT ASK QUESTIONS OF THE INVIGILATORS about interpretations of the exam questions. If you have a concern about a question, please summarize it at the top of this page. All concerns will be evaluated after the exam. GOOD LUCK!

**Question 1 8 points** (*Assume all transitions are possible*)

a. 6 points Calculate the **energy (in kJ/mol)** required to ionize (completely remove) the electron from the **$n=3$** level of **He^+ ion** (He: Helium). Show your work including equations. (3 sig. figs.)

b. 2 points

(I) Select the largest (longest) wavelength from the three choices below:

- i. Wavelength of photon to *completely remove* an electron from *1s orbital of Hydrogen atom*
- ii. Wavelength of photon to *completely remove* an electron from *1s orbital of Boron atom (B)*
- iii. Wavelength of photon to *completely remove* an electron from *1s orbital of Boron cation (B^+)*

i or ii or iii

(II) Select the smallest (shortest) wavelength from the three choices below (*for H atom*):

- i. Wavelength of photon *emitted* for an electron transitioning from *2p to 1s orbital*
- ii. Wavelength of photon *absorbed* for an electron transitioning from *2s to 4p orbital*
- iii. Wavelength of photon *absorbed* for an electron transitioning from *1s to 3p orbital*

i or ii or iii



Question 2 8 points



A Chem110 student plots the graph shown on the left, from data obtained for the kinetic energy (eV) of the ejected electron versus frequency (s^{-1}) of the incident light, for an unknown metal. ($1\text{eV} = 1.6022 \times 10^{-19} \text{ J}$)

Based on the plot, answer the questions **a to c** below:

a. 1 point On the plot above, label the **threshold frequency**.

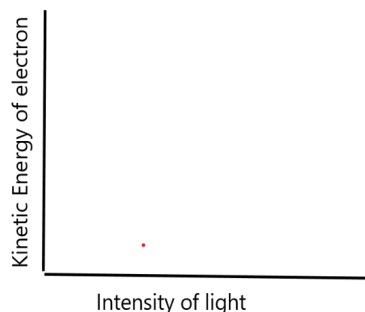
b. 2 points What is the value of the work potential (in J) of the unknown metal?

c. 3 points What is the *longest* wavelength (in nm) required to eject an electron from this metal's surface? Show your work including equation(s) used. (3 sig figs)

d. 2 points In the space provided below, plot the following

(Assume frequency is *above* threshold frequency):

- i. *Kinetic energy* of an ejected electron versus the *intensity* of the incident light (left)
- ii. *Number of electrons* ejected versus the *frequency* of the incident light (right)



**Question 3** **6 points****a. 2 points**

In the ground state of Ge (Germanium; $Z = 32$) atom, **name** the following orbitals:

i. Orbital containing the electron that requires the maximum energy to be removed.

ii. Orbital containing the electron that experiences the minimum effective nuclear charge.

b. 2 points

An unknown anion (A^-) has the following *excited* electronic configuration: $[Ar] 4s^2 3d^3 4p^1$

(Ar: Argon; $Z = 18$)

Give the **condensed ground state electronic configuration** of the atom, **A** and the cation, **A^{2+}** .
Show your work. (Assume all electronic transitions are possible)

Electronic Configuration of A:

Electronic Configuration of A^{2+} :

c. 2 points

In a single atom, determine how many electrons can have the following set of quantum numbers?

(i) $n = 5; l = 3$

In a single atom, determine how many electrons can have the following set of quantum numbers?

(i) $n = 3; l = 2; m_s = +1/2$

**Question 4** **8 points**

Rank the following from 1 to 4 in **increasing** order. Provide the ranking in the boxes provided.

Note: Atomic number (Z) are provided for relevant atoms

a) Increasing (lowest to highest) order of *ionization energy*: (1= lowest; 4 = highest)

Mg

Mg: Z = 12

Rb

Rb: Z = 37

Cs

Cs: Z = 55

K

K : Z = 19

b) Increasing (lowest to highest) order of *bond angle*: (1= lowest; 4 = highest)

H₂OCCl₄XeF₄

HCN

c) Increasing (smallest to largest) order of *ionic size* (1= smallest; 4 = largest)

S²⁻

S: Z = 16

O²⁻

O: Z = 8

Be²⁺

Be: Z = 4

Na⁺

Na: Z = 11

d) Increasing (lowest to highest) order of *bond energy* (1= lowest; 4 = highest)

P-F

P-Cl

P-I

P-Br

**Question 5** **12 points**

a. 9 points For the molecules HClO_4 and TeF_6 , draw the most stable *Lewis structure* showing appropriate shape and *dashed/wedged bonds*, based on VSEPR. Determine the **molecular geometry** around the central atom. For HClO_4 , label the *most polar bond*.

HClO_4 (there are no O-O bonds)

Label the most polar bond

Molecular Geometry around Cl

TeF_6

Molecular Geometry around Te

b. 3 points (Note: Lone pairs are not shown in structures below)

(i) For the three *resonance* structures (1, 2 and 3) shown below, answer the following question:



(I) Circle the *most contributing* resonance structure.

Structure 1

Structure 2

Structure 3

(ii) For the three structures given (4, 5, and 6) below, answer the following questions:



(I) Circle all the structures that represent *resonance structures* of each other:

Structure 4

Structure 5

Structure 6

(II) Among the resonance structures, circle the *most contributing* resonance structure.

Structure 4

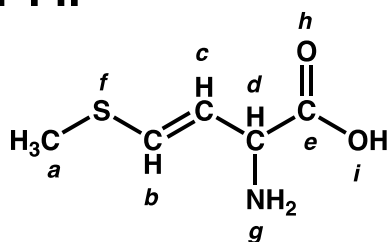
Structure 5

Structure 6

**Question 6** **6 points**

Using the data given below, calculate the **lattice energy** for SrO(s).

	ΔH° (kJ/mol)
$\text{Sr}(s) \longrightarrow \text{Sr}(g)$	+180
$\text{Sr}(g) \longrightarrow \text{Sr}^{2+}(g) + 2e^-$	+1468
$\text{O}_2(g) \longrightarrow 2\text{O}(g)$	+500
$\text{O}(g) + 2e^- \longrightarrow \text{O}^{2-}(g)$	+600
$\text{Sr}(s) + \frac{1}{2}\text{O}_2(g) \longrightarrow \text{SrO}(s)$	-550

**Question 7 10 points**

Consider the molecule given on the left, and answer questions (i) to (vi) using chemical bonding theories that we have learnt in class (valence bond theory, hybridization).

Note that the lone pairs are not shown in the structure:

i. Consider the carbon atoms marked in as ***a***, ***b***, ***c***, ***d***, and ***e*** in the molecule, and determine **hybridization** at each of the carbon centers. (2.5 points)

Atom	Hybridization
Carbon (<i>a</i>)	
Carbon (<i>b</i>)	
Carbon (<i>c</i>)	
Carbon (<i>d</i>)	
Carbon (<i>e</i>)	

ii. Now, consider the sulfur atom (S) marked as ***f***, nitrogen (N) atom (***g***), and oxygen(O) atoms (***h*** and ***i***), and determine their **hybridization**. (2 points)

Atom	Hybridization
Sulfur (<i>f</i>)	
Nitrogen (<i>g</i>)	
Oxygen (<i>h</i>)	
Oxygen (<i>i</i>)	

iii. **Describe** all the bonds formed by the carbon atoms ***a*** and ***b***, using sigma and pi designation as appropriate. (2 points)

Atom	Bonds formed (use sigma and pi as appropriate)
Carbon (<i>a</i>)	
Carbon (<i>b</i>)	

iv. How many total sp^3 -s sigma bonds are in the molecule? (1 point)

v. Determine **electronic** and **molecular geometry** at carbon (C) labeled ***e*** and sulfur (S) labeled ***f*** centres. (2 points)

Atom	Electronic Geometry	Molecular Geometry
Carbon (<i>e</i>)		
Sulfur (<i>f</i>)		

vi. Circle the value from the choices given below, that describes the **C-S-C bond angle at sulfur** (S) marked ***f***: (0.5 points)

>90° to <109.5° 109.5° >109.5 to <120° 120° >120° to <180°


Question 8 12 points

Using the **valence shell orbitals only** draw a **complete** molecular orbital (MO) diagram (with no s-p mixing) for the molecule **PS**, in the space provided below. **(6 points)**

Assume that 3s and 3p atomic orbitals will form molecular orbitals like those formed by the combination of 2s and 2p atomic orbitals that we learnt in class, and there is no s-p mixing.

Show **energy arrow**, **atomic orbitals** involved, **atom labels**, the **molecular orbitals** that are formed, **filled electrons** in the MO diagram etc. Make sure to label all the atomic and molecular orbitals.

Using the MO diagram above, answer the following questions:

- (i) Which one of the following will have the highest bond order? Circle the correct choice
 A) PS^{1-} B) PS C) PS^{1+}
- (ii) Which one of the following will have the longest bond length? Circle the correct choice
 A) PS^{1-} B) PS C) PS^{1+}
- (iii) Which of the following will be diamagnetic? Circle the correct choice.
 A) PS^{1-} B) PS C) PS^{1+}
- (iv) Draw Lewis dot structure of the molecule with the shortest bond length (based on the MO diagram) from among the three (PS^{1-} , PS, PS^{1+}), and indicate the bond order calculated from the Lewis dot structure. **(3 points)**

Lewis Dot Structure

Bond Order



10D4394D-5E93-4B87-B49F-384C563FA931

midterm-exam-93388

#560 Page 10 of 10

SCRAP PAGE

**CHEMISTRY 110 –Midterm Exam****Thursday November 7, 2024, 6:30 – 8:30 pm**

Examiners: Pallavi Sirjoosingh and Ashok Kakkar

First name (please write as legibly as possible within the boxes)

--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--

Last name

--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--

Student ID number

--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--

This booklet has 8 questions on 10 pages (The last page can be used as a scrap page).

The staple must NOT be removed from the question booklet. Your intact question booklet must be returned at the end of the exam.

A separate booklet contains a periodic table, and datasheet.

Write your answers directly under the questions in the space provided. Do not write on the QR codes or near the top edges (as it may be cut off).

Please write **legibly and dark enough** for the scan to be readable. For students who use English as a second language, dictionaries are permitted. Calculators are allowed.

DO NOT ASK QUESTIONS OF THE INVIGILATORS about interpretations of the exam questions. If you have a concern about a question, please summarize it at the top of this page. All concerns will be evaluated after the exam. GOOD LUCK!

**Question 1 8 points** (*Assume all transitions are possible*)

a. 6 points Calculate the **energy (in kJ/mol)** required to ionize (completely remove) the electron from the **$n=3$** level of **He^+ ion** (He: Helium). Show your work including equations. (3 sig. figs.)

b. 2 points

(I) Select the largest (longest) wavelength from the three choices below:

- i. Wavelength of photon to *completely remove* an electron from *1s orbital of Hydrogen atom*
- ii. Wavelength of photon to *completely remove* an electron from *1s orbital of Boron atom (B)*
- iii. Wavelength of photon to *completely remove* an electron from *1s orbital of Boron cation (B^+)*

i or ii or iii

(II) Select the smallest (shortest) wavelength from the three choices below (*for H atom*):

- i. Wavelength of photon *emitted* for an electron transitioning from *2p to 1s orbital*
- ii. Wavelength of photon *absorbed* for an electron transitioning from *2s to 4p orbital*
- iii. Wavelength of photon *absorbed* for an electron transitioning from *1s to 3p orbital*

i or ii or iii



Question 2 8 points



A Chem110 student plots the graph shown on the left, from data obtained for the kinetic energy (eV) of the ejected electron versus frequency (s^{-1}) of the incident light, for an unknown metal. ($1\text{eV} = 1.6022 \times 10^{-19} \text{ J}$)

Based on the plot, answer the questions **a to c** below:

a. 1 point On the plot above, label the **threshold frequency**.

b. 2 points What is the value of the work potential (in J) of the unknown metal?

c. 3 points What is the *longest* wavelength (in nm) required to eject an electron from this metal's surface? Show your work including equation(s) used. (3 sig figs)

d. 2 points In the space provided below, plot the following

(Assume frequency is *above* threshold frequency):

- Kinetic energy* of an ejected electron versus the *intensity* of the incident light (left)
- Number of electrons* ejected versus the *frequency* of the incident light (right)



**Question 3** **6 points****a. 2 points**

In the ground state of Ge (Germanium; $Z = 32$) atom, **name** the following orbitals:

i. Orbital containing the electron that requires the maximum energy to be removed.

ii. Orbital containing the electron that experiences the minimum effective nuclear charge.

b. 2 points

An unknown anion (A^-) has the following *excited* electronic configuration: $[Ar] 4s^2 3d^3 4p^1$

(Ar: Argon; $Z = 18$)

Give the **condensed ground state electronic configuration** of the atom, **A** and the cation, A^{2+} .
Show your work. (Assume all electronic transitions are possible)

Electronic Configuration of A:

Electronic Configuration of A^{2+} :

c. 2 points

In a single atom, determine how many electrons can have the following set of quantum numbers?

(i) $n = 5; l = 3$

In a single atom, determine how many electrons can have the following set of quantum numbers?

(i) $n = 3; l = 2; m_s = +1/2$

**Question 4** **8 points**

Rank the following from 1 to 4 in **increasing** order. Provide the ranking in the boxes provided.

Note: Atomic number (Z) are provided for relevant atoms

a) Increasing (lowest to highest) order of *ionization energy*: (1= lowest; 4 = highest)

Mg

Mg: Z = 12

Rb

Rb: Z = 37

Cs

Cs: Z = 55

K

K : Z = 19

b) Increasing (lowest to highest) order of *bond angle*: (1= lowest; 4 = highest)

H₂OCCl₄XeF₄

HCN

c) Increasing (smallest to largest) order of *ionic size* (1= smallest; 4 = largest)

S²⁻

S: Z = 16

O²⁻

O: Z = 8

Be²⁺

Be: Z = 4

Na⁺

Na: Z = 11

d) Increasing (lowest to highest) order of *bond energy* (1= lowest; 4 = highest)

P-F

P-Cl

P-I

P-Br

**Question 5** **12 points**

a. 9 points For the molecules HClO_4 and TeF_6 , draw the most stable *Lewis structure* showing appropriate shape and *dashed/wedged bonds*, based on VSEPR. Determine the **molecular geometry** around the central atom. For HClO_4 , label the *most polar bond*.



b. 3 points (Note: Lone pairs are not shown in structures below)

(i) For the three *resonance* structures (1, 2 and 3) shown below, answer the following question:



(I) Circle the *most contributing* resonance structure.

Structure 1

Structure 2

Structure 3

(ii) For the three structures given (4, 5, and 6) below, answer the following questions:



(I) Circle all the structures that represent *resonance structures* of each other:

Structure 4

Structure 5

Structure 6

(II) Among the resonance structures, circle the *most contributing* resonance structure.

Structure 4

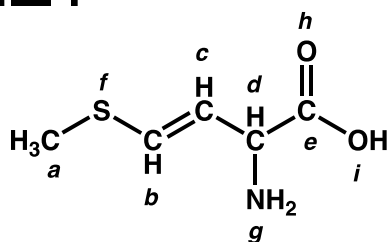
Structure 5

Structure 6

**Question 6** **6 points**

Using the data given below, calculate the **lattice energy** for SrO(s).

	ΔH° (kJ/mol)
$\text{Sr}(s) \longrightarrow \text{Sr}(g)$	+180
$\text{Sr}(g) \longrightarrow \text{Sr}^{2+}(g) + 2e^-$	+1468
$\text{O}_2(g) \longrightarrow 2\text{O}(g)$	+500
$\text{O}(g) + 2e^- \longrightarrow \text{O}^{2-}(g)$	+600
$\text{Sr}(s) + \frac{1}{2}\text{O}_2(g) \longrightarrow \text{SrO}(s)$	-550

**Question 7 10 points**

Consider the molecule given on the left, and answer questions (i) to (vi) using chemical bonding theories that we have learnt in class (valence bond theory, hybridization).

Note that the lone pairs are not shown in the structure:

i. Consider the carbon atoms marked in as ***a***, ***b***, ***c***, ***d***, and ***e*** in the molecule, and determine **hybridization** at each of the carbon centers. (2.5 points)

Atom	Hybridization
Carbon (<i>a</i>)	
Carbon (<i>b</i>)	
Carbon (<i>c</i>)	
Carbon (<i>d</i>)	
Carbon (<i>e</i>)	

ii. Now, consider the sulfur atom (S) marked as ***f***, nitrogen (N) atom (***g***), and oxygen(O) atoms (***h*** and ***i***), and determine their **hybridization**. (2 points)

Atom	Hybridization
Sulfur (<i>f</i>)	
Nitrogen (<i>g</i>)	
Oxygen (<i>h</i>)	
Oxygen (<i>i</i>)	

iii. **Describe** all the bonds formed by the carbon atoms ***a*** and ***b***, using sigma and pi designation as appropriate. (2 points)

Atom	Bonds formed (use sigma and pi as appropriate)
Carbon (<i>a</i>)	
Carbon (<i>b</i>)	

iv. How many total sp^3 -s sigma bonds are in the molecule? (1 point)

v. Determine **electronic** and **molecular geometry** at carbon (C) labeled ***e*** and sulfur (S) labeled ***f*** centres. (2 points)

Atom	Electronic Geometry	Molecular Geometry
Carbon (<i>e</i>)		
Sulfur (<i>f</i>)		

vi. Circle the value from the choices given below, that describes the **C-S-C bond angle at sulfur** (S) marked ***f***: (0.5 points)

>90° to <109.5°

109.5°

>109.5 to <120°

120°

>120° to <180°


Question 8 12 points

Using the **valence shell orbitals only** draw a **complete** molecular orbital (MO) diagram (with no s-p mixing) for the molecule **PS**, in the space provided below. **(6 points)**

Assume that 3s and 3p atomic orbitals will form molecular orbitals like those formed by the combination of 2s and 2p atomic orbitals that we learnt in class, and there is no s-p mixing.

Show **energy arrow**, **atomic orbitals** involved, **atom labels**, the **molecular orbitals** that are formed, **filled electrons** in the MO diagram etc. Make sure to **label all the atomic and molecular orbitals**.

Using the MO diagram above, answer the following questions:

- (i) Which one of the following will have the highest bond order? Circle the correct choice
 A) PS^{1-} B) PS C) PS^{1+}
- (ii) Which one of the following will have the longest bond length? Circle the correct choice
 A) PS^{1-} B) PS C) PS^{1+}
- (iii) Which of the following will be diamagnetic? Circle the correct choice.
 A) PS^{1-} B) PS C) PS^{1+}
- (iv) Draw Lewis dot structure of the molecule with the shortest bond length (based on the MO diagram) from among the three (PS^{1-} , PS, PS^{1+}), and indicate the bond order calculated from the Lewis dot structure. **(3 points)**

Lewis Dot Structure

Bond Order



2EEF4E15-C365-4BD6-989E-54F8E30FE2C4

midterm-exam-93388

#561 Page 10 of 10

SCRAP PAGE

**CHEMISTRY 110 –Midterm Exam****Thursday November 7, 2024, 6:30 – 8:30 pm**

Examiners: Pallavi Sirjoosingh and Ashok Kakkar

First name (please write as legibly as possible within the boxes)

--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--

Last name

--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--

Student ID number

--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--

This booklet has 8 questions on 10 pages (The last page can be used as a scrap page).

The staple must NOT be removed from the question booklet. Your intact question booklet must be returned at the end of the exam.

A separate booklet contains a periodic table, and datasheet.

Write your answers directly under the questions in the space provided. Do not write on the QR codes or near the top edges (as it may be cut off).

Please write **legibly and dark enough** for the scan to be readable. For students who use English as a second language, dictionaries are permitted. Calculators are allowed.

DO NOT ASK QUESTIONS OF THE INVIGILATORS about interpretations of the exam questions. If you have a concern about a question, please summarize it at the top of this page. All concerns will be evaluated after the exam. GOOD LUCK!

**Question 1 8 points** (*Assume all transitions are possible*)

a. 6 points Calculate the **energy (in kJ/mol)** required to ionize (completely remove) the electron from the **n=3** level of **He⁺ ion** (He: Helium). Show your work including equations. (3 sig. figs.)

b. 2 points

(I) Select the largest (longest) wavelength from the three choices below:

- i. Wavelength of photon to *completely remove* an electron from *1s orbital of Hydrogen atom*
- ii. Wavelength of photon to *completely remove* an electron from *1s orbital of Boron atom (B)*
- iii. Wavelength of photon to *completely remove* an electron from *1s orbital of Boron cation (B⁺)*

i or ii or iii

(II) Select the smallest (shortest) wavelength from the three choices below (*for H atom*):

- i. Wavelength of photon *emitted* for an electron transitioning from *2p to 1s orbital*
- ii. Wavelength of photon *absorbed* for an electron transitioning from *2s to 4p orbital*
- iii. Wavelength of photon *absorbed* for an electron transitioning from *1s to 3p orbital*

i or ii or iii



Question 2 8 points



A Chem110 student plots the graph shown on the left, from data obtained for the kinetic energy (eV) of the ejected electron versus frequency (s^{-1}) of the incident light, for an unknown metal. ($1\text{eV} = 1.6022 \times 10^{-19} \text{ J}$)

Based on the plot, answer the questions **a to c** below:

a. 1 point On the plot above, label the **threshold frequency**.

b. 2 points What is the value of the work potential (in J) of the unknown metal?

c. 3 points What is the *longest* wavelength (in nm) required to eject an electron from this metal's surface? Show your work including equation(s) used. (3 sig figs)

d. 2 points In the space provided below, plot the following

(Assume frequency is *above* threshold frequency):

- Kinetic energy* of an ejected electron versus the *intensity* of the incident light (left)
- Number of electrons* ejected versus the *frequency* of the incident light (right)



**Question 3** **6 points****a. 2 points**

In the ground state of Ge (Germanium; $Z = 32$) atom, **name** the following orbitals:

i. Orbital containing the electron that requires the maximum energy to be removed.

ii. Orbital containing the electron that experiences the minimum effective nuclear charge.

b. 2 points

An unknown anion (A^-) has the following *excited* electronic configuration: $[Ar] 4s^2 3d^3 4p^1$

(Ar: Argon; $Z = 18$)

Give the **condensed ground state electronic configuration** of the atom, **A** and the cation, **A^{2+}** .
Show your work. (Assume all electronic transitions are possible)

Electronic Configuration of A:

Electronic Configuration of A^{2+} :

c. 2 points

In a single atom, determine how many electrons can have the following set of quantum numbers?

(i) $n = 5; l = 3$

In a single atom, determine how many electrons can have the following set of quantum numbers?

(i) $n = 3; l = 2; m_s = +1/2$

**Question 4** **8 points**

Rank the following from 1 to 4 in **increasing** order. Provide the ranking in the boxes provided.

Note: Atomic number (Z) are provided for relevant atoms

a) Increasing (lowest to highest) order of *ionization energy*: (1= lowest; 4 = highest)

Mg

Mg: Z = 12

Rb

Rb: Z = 37

Cs

Cs: Z = 55

K

K : Z = 19

b) Increasing (lowest to highest) order of *bond angle*: (1= lowest; 4 = highest)

H₂OCCl₄XeF₄

HCN

c) Increasing (smallest to largest) order of *ionic size* (1= smallest; 4 = largest)

S²⁻

S: Z = 16

O²⁻

O: Z = 8

Be²⁺

Be: Z = 4

Na⁺

Na: Z = 11

d) Increasing (lowest to highest) order of *bond energy* (1= lowest; 4 = highest)

P-F

P-Cl

P-I

P-Br

**Question 5** **12 points**

a. 9 points For the molecules HClO_4 and TeF_6 , draw the most stable *Lewis structure* showing appropriate shape and *dashed/wedged bonds*, based on VSEPR. Determine the **molecular geometry** around the central atom. For HClO_4 , label the *most polar bond*.



b. 3 points (Note: Lone pairs are not shown in structures below)

(i) For the three *resonance* structures (1, 2 and 3) shown below, answer the following question:



(I) Circle the *most contributing* resonance structure.

Structure 1

Structure 2

Structure 3

(ii) For the three structures given (4, 5, and 6) below, answer the following questions:



(I) Circle all the structures that represent *resonance structures* of each other:

Structure 4

Structure 5

Structure 6

(II) Among the resonance structures, circle the *most contributing* resonance structure.

Structure 4

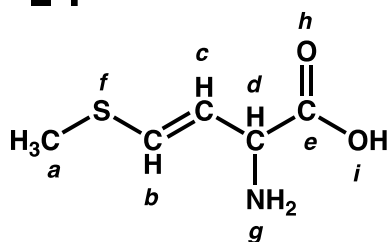
Structure 5

Structure 6

**Question 6** **6 points**

Using the data given below, calculate the **lattice energy** for SrO(s).

	ΔH° (kJ/mol)
$\text{Sr}(s) \longrightarrow \text{Sr}(g)$	+180
$\text{Sr}(g) \longrightarrow \text{Sr}^{2+}(g) + 2e^-$	+1468
$\text{O}_2(g) \longrightarrow 2\text{O}(g)$	+500
$\text{O}(g) + 2e^- \longrightarrow \text{O}^{2-}(g)$	+600
$\text{Sr}(s) + \frac{1}{2}\text{O}_2(g) \longrightarrow \text{SrO}(s)$	-550

**Question 7 10 points**

Consider the molecule given on the left, and answer questions (i) to (vi) using chemical bonding theories that we have learnt in class (valence bond theory, hybridization).

Note that the lone pairs are not shown in the structure:

i. Consider the carbon atoms marked in as ***a***, ***b***, ***c***, ***d***, and ***e*** in the molecule, and determine **hybridization** at each of the carbon centers. (2.5 points)

Atom	Hybridization
Carbon (<i>a</i>)	
Carbon (<i>b</i>)	
Carbon (<i>c</i>)	
Carbon (<i>d</i>)	
Carbon (<i>e</i>)	

ii. Now, consider the sulfur atom (S) marked as ***f***, nitrogen (N) atom (***g***), and oxygen(O) atoms (***h*** and ***i***), and determine their **hybridization**. (2 points)

Atom	Hybridization
Sulfur (<i>f</i>)	
Nitrogen (<i>g</i>)	
Oxygen (<i>h</i>)	
Oxygen (<i>i</i>)	

iii. **Describe** all the bonds formed by the carbon atoms ***a*** and ***b***, using sigma and pi designation as appropriate. (2 points)

Atom	Bonds formed (use sigma and pi as appropriate)
Carbon (<i>a</i>)	
Carbon (<i>b</i>)	

iv. How many total sp^3 -s sigma bonds are in the molecule? (1 point)

v. Determine **electronic** and **molecular geometry** at carbon (C) labeled ***e*** and sulfur (S) labeled ***f*** centres. (2 points)

Atom	Electronic Geometry	Molecular Geometry
Carbon (<i>e</i>)		
Sulfur (<i>f</i>)		

vi. Circle the value from the choices given below, that describes the **C-S-C bond angle at sulfur** (S) marked ***f***: (0.5 points)

>90° to <109.5° 109.5° >109.5 to <120° 120° >120° to <180°


Question 8 12 points

Using the **valence shell orbitals only** draw a **complete** molecular orbital (MO) diagram (with no s-p mixing) for the molecule **PS**, in the space provided below. **(6 points)**

Assume that 3s and 3p atomic orbitals will form molecular orbitals like those formed by the combination of 2s and 2p atomic orbitals that we learnt in class, and there is no s-p mixing.

Show **energy arrow**, **atomic orbitals** involved, **atom labels**, the **molecular orbitals** that are formed, **filled electrons** in the MO diagram etc. Make sure to label all the atomic and molecular orbitals.

Using the MO diagram above, answer the following questions:

- (i) Which one of the following will have the highest bond order? Circle the correct choice
 A) PS^{1-} B) PS C) PS^{1+}
- (ii) Which one of the following will have the longest bond length? Circle the correct choice
 A) PS^{1-} B) PS C) PS^{1+}
- (iii) Which of the following will be diamagnetic? Circle the correct choice.
 A) PS^{1-} B) PS C) PS^{1+}
- (iv) Draw Lewis dot structure of the molecule with the shortest bond length (based on the MO diagram) from among the three (PS^{1-} , PS, PS^{1+}), and indicate the bond order calculated from the Lewis dot structure. **(3 points)**

Lewis Dot Structure

Bond Order



25728C02-B3FC-4A12-AD68-DFF83AFD2CD8

midterm-exam-93388

#562 Page 10 of 10

SCRAP PAGE

**CHEMISTRY 110 –Midterm Exam****Thursday November 7, 2024, 6:30 – 8:30 pm****Examiners: Pallavi Sirjoosingh and Ashok Kakkar**

First name (please write as legibly as possible within the boxes)

--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--

Last name

--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--

Student ID number

--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--

This booklet has 8 questions on 10 pages (The last page can be used as a scrap page).

The staple must NOT be removed from the question booklet. Your intact question booklet must be returned at the end of the exam.

A separate booklet contains a periodic table, and datasheet.

Write your answers directly under the questions in the space provided. Do not write on the QR codes or near the top edges (as it may be cut off).

Please write **legibly and dark enough** for the scan to be readable. For students who use English as a second language, dictionaries are permitted. Calculators are allowed.

DO NOT ASK QUESTIONS OF THE INVIGILATORS about interpretations of the exam questions. If you have a concern about a question, please summarize it at the top of this page. All concerns will be evaluated after the exam. GOOD LUCK!

**Question 1 8 points** (*Assume all transitions are possible*)

a. 6 points Calculate the **energy (in kJ/mol)** required to ionize (completely remove) the electron from the **n=3** level of **He⁺ ion** (He: Helium). Show your work including equations. (3 sig. figs.)

b. 2 points

(I) Select the largest (longest) wavelength from the three choices below:

- i. Wavelength of photon to *completely remove* an electron from *1s orbital of Hydrogen atom*
- ii. Wavelength of photon to *completely remove* an electron from *1s orbital of Boron atom (B)*
- iii. Wavelength of photon to *completely remove* an electron from *1s orbital of Boron cation (B⁺)*

i or ii or iii

(II) Select the smallest (shortest) wavelength from the three choices below (*for H atom*):

- i. Wavelength of photon *emitted* for an electron transitioning from *2p to 1s orbital*
- ii. Wavelength of photon *absorbed* for an electron transitioning from *2s to 4p orbital*
- iii. Wavelength of photon *absorbed* for an electron transitioning from *1s to 3p orbital*

i or ii or iii



Question 2 8 points



A Chem110 student plots the graph shown on the left, from data obtained for the kinetic energy (eV) of the ejected electron versus frequency (s^{-1}) of the incident light, for an unknown metal. ($1\text{eV} = 1.6022 \times 10^{-19} \text{ J}$)

Based on the plot, answer the questions **a to c** below:

a. 1 point On the plot above, label the **threshold frequency**.

b. 2 points What is the value of the work potential (in J) of the unknown metal?

c. 3 points What is the *longest* wavelength (in nm) required to eject an electron from this metal's surface? Show your work including equation(s) used. (3 sig figs)

d. 2 points In the space provided below, plot the following

(Assume frequency is *above* threshold frequency):

- i. *Kinetic energy* of an ejected electron versus the *intensity* of the incident light (left)
- ii. *Number of electrons* ejected versus the *frequency* of the incident light (right)



**Question 3** **6 points****a. 2 points**

In the ground state of Ge (Germanium; $Z = 32$) atom, **name** the following orbitals:

i. Orbital containing the electron that requires the maximum energy to be removed.

ii. Orbital containing the electron that experiences the minimum effective nuclear charge.

b. 2 points

An unknown anion (A^-) has the following *excited* electronic configuration: $[Ar] 4s^2 3d^3 4p^1$

(Ar: Argon; $Z = 18$)

Give the **condensed ground state electronic configuration** of the atom, **A** and the cation, A^{2+} .
Show your work. (Assume all electronic transitions are possible)

Electronic Configuration of A:

Electronic Configuration of A^{2+} :

c. 2 points

In a single atom, determine how many electrons can have the following set of quantum numbers?

(i) $n = 5; l = 3$

In a single atom, determine how many electrons can have the following set of quantum numbers?

(i) $n = 3; l = 2; m_s = +1/2$

**Question 4** **8 points**

Rank the following from 1 to 4 in **increasing** order. Provide the ranking in the boxes provided.

Note: Atomic number (Z) are provided for relevant atoms

a) Increasing (lowest to highest) order of *ionization energy*: (1= lowest; 4 = highest)

Mg

Mg: Z = 12

Rb

Rb: Z = 37

Cs

Cs: Z = 55

K

K : Z = 19

b) Increasing (lowest to highest) order of *bond angle*: (1= lowest; 4 = highest)

H₂OCCl₄XeF₄

HCN

c) Increasing (smallest to largest) order of *ionic size* (1= smallest; 4 = largest)

S²⁻

S: Z = 16

O²⁻

O: Z = 8

Be²⁺

Be: Z = 4

Na⁺

Na: Z = 11

d) Increasing (lowest to highest) order of *bond energy* (1= lowest; 4 = highest)

P-F

P-Cl

P-I

P-Br

**Question 5** **12 points**

a. 9 points For the molecules HClO_4 and TeF_6 , draw the most stable *Lewis structure* showing appropriate shape and *dashed/wedged bonds*, based on VSEPR. Determine the **molecular geometry** around the central atom. For HClO_4 , label the *most polar bond*.



b. 3 points (Note: Lone pairs are not shown in structures below)

(i) For the three *resonance* structures (1, 2 and 3) shown below, answer the following question:



(I) Circle the *most contributing* resonance structure.

Structure 1

Structure 2

Structure 3

(ii) For the three structures given (4, 5, and 6) below, answer the following questions:



(I) Circle all the structures that represent *resonance structures* of each other:

Structure 4

Structure 5

Structure 6

(II) Among the resonance structures, circle the *most contributing* resonance structure.

Structure 4

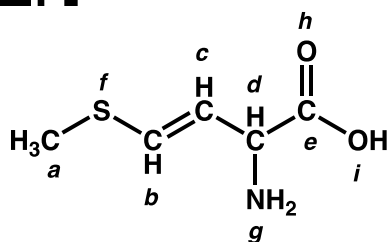
Structure 5

Structure 6

**Question 6** **6 points**

Using the data given below, calculate the **lattice energy** for SrO(s).

	ΔH° (kJ/mol)
$\text{Sr}(s) \longrightarrow \text{Sr}(g)$	+180
$\text{Sr}(g) \longrightarrow \text{Sr}^{2+}(g) + 2e^-$	+1468
$\text{O}_2(g) \longrightarrow 2\text{O}(g)$	+500
$\text{O}(g) + 2e^- \longrightarrow \text{O}^{2-}(g)$	+600
$\text{Sr}(s) + \frac{1}{2}\text{O}_2(g) \longrightarrow \text{SrO}(s)$	-550

**Question 7 10 points**

Consider the molecule given on the left, and answer questions (i) to (vi) using chemical bonding theories that we have learnt in class (valence bond theory, hybridization).

Note that the lone pairs are not shown in the structure:

i. Consider the carbon atoms marked in as ***a***, ***b***, ***c***, ***d***, and ***e*** in the molecule, and determine **hybridization** at each of the carbon centers. (2.5 points)

Atom	Hybridization
Carbon (<i>a</i>)	
Carbon (<i>b</i>)	
Carbon (<i>c</i>)	
Carbon (<i>d</i>)	
Carbon (<i>e</i>)	

ii. Now, consider the sulfur atom (S) marked as ***f***, nitrogen (N) atom (***g***), and oxygen(O) atoms (***h*** and ***i***), and determine their **hybridization**. (2 points)

Atom	Hybridization
Sulfur (<i>f</i>)	
Nitrogen (<i>g</i>)	
Oxygen (<i>h</i>)	
Oxygen (<i>i</i>)	

iii. **Describe** all the bonds formed by the carbon atoms ***a*** and ***b***, using sigma and pi designation as appropriate. (2 points)

Atom	Bonds formed (use sigma and pi as appropriate)
Carbon (<i>a</i>)	
Carbon (<i>b</i>)	

iv. How many total sp^3 -s sigma bonds are in the molecule? (1 point)

v. Determine **electronic** and **molecular geometry** at carbon (C) labeled ***e*** and sulfur (S) labeled ***f*** centres. (2 points)

Atom	Electronic Geometry	Molecular Geometry
Carbon (<i>e</i>)		
Sulfur (<i>f</i>)		

vi. Circle the value from the choices given below, that describes the **C-S-C bond angle at sulfur** (S) marked ***f***: (0.5 points)

>90° to <109.5° 109.5° >109.5 to <120° 120° >120° to <180°


Question 8 12 points

Using the **valence shell orbitals only** draw a **complete** molecular orbital (MO) diagram (with no s-p mixing) for the molecule **PS**, in the space provided below. **(6 points)**

Assume that 3s and 3p atomic orbitals will form molecular orbitals like those formed by the combination of 2s and 2p atomic orbitals that we learnt in class, and there is no s-p mixing.

Show **energy arrow**, **atomic orbitals** involved, **atom labels**, the **molecular orbitals** that are formed, **filled electrons** in the MO diagram etc. Make sure to label all the atomic and molecular orbitals.

Using the MO diagram above, answer the following questions:

- (i) Which one of the following will have the highest bond order? Circle the correct choice
 A) PS^{1-} B) PS C) PS^{1+}
- (ii) Which one of the following will have the longest bond length? Circle the correct choice
 A) PS^{1-} B) PS C) PS^{1+}
- (iii) Which of the following will be diamagnetic? Circle the correct choice.
 A) PS^{1-} B) PS C) PS^{1+}
- (iv) Draw Lewis dot structure of the molecule with the shortest bond length (based on the MO diagram) from among the three (PS^{1-} , PS, PS^{1+}), and indicate the bond order calculated from the Lewis dot structure. **(3 points)**

Lewis Dot Structure

Bond Order



39249607-79E4-457B-ADC6-4F17F899BCD2

midterm-exam-93388

#563 Page 10 of 10

SCRAP PAGE

**CHEMISTRY 110 –Midterm Exam****Thursday November 7, 2024, 6:30 – 8:30 pm**

Examiners: Pallavi Sirjoosingh and Ashok Kakkar

First name (please write as legibly as possible within the boxes)

--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--

Last name

--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--

Student ID number

--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--

This booklet has 8 questions on 10 pages (The last page can be used as a scrap page).

The staple must NOT be removed from the question booklet. Your intact question booklet must be returned at the end of the exam.

A separate booklet contains a periodic table, and datasheet.

Write your answers directly under the questions in the space provided. Do not write on the QR codes or near the top edges (as it may be cut off).

Please write **legibly and dark enough** for the scan to be readable. For students who use English as a second language, dictionaries are permitted. Calculators are allowed.

DO NOT ASK QUESTIONS OF THE INVIGILATORS about interpretations of the exam questions. If you have a concern about a question, please summarize it at the top of this page. All concerns will be evaluated after the exam. GOOD LUCK!

**Question 1 8 points** (*Assume all transitions are possible*)

a. 6 points Calculate the **energy (in kJ/mol)** required to ionize (completely remove) the electron from the **n=3** level of **He⁺ ion** (He: Helium). Show your work including equations. (3 sig. figs.)

b. 2 points

(I) Select the largest (longest) wavelength from the three choices below:

- i. Wavelength of photon to *completely remove* an electron from *1s orbital of Hydrogen atom*
- ii. Wavelength of photon to *completely remove* an electron from *1s orbital of Boron atom (B)*
- iii. Wavelength of photon to *completely remove* an electron from *1s orbital of Boron cation (B⁺)*

i or ii or iii

(II) Select the smallest (shortest) wavelength from the three choices below (*for H atom*):

- i. Wavelength of photon *emitted* for an electron transitioning from *2p to 1s orbital*
- ii. Wavelength of photon *absorbed* for an electron transitioning from *2s to 4p orbital*
- iii. Wavelength of photon *absorbed* for an electron transitioning from *1s to 3p orbital*

i or ii or iii



Question 2 8 points



A Chem110 student plots the graph shown on the left, from data obtained for the kinetic energy (eV) of the ejected electron versus frequency (s^{-1}) of the incident light, for an unknown metal. ($1\text{eV} = 1.6022 \times 10^{-19} \text{ J}$)

Based on the plot, answer the questions **a to c** below:

a. 1 point On the plot above, label the **threshold frequency**.

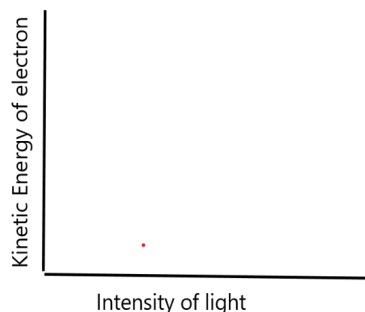
b. 2 points What is the value of the work potential (in J) of the unknown metal?

c. 3 points What is the *longest* wavelength (in nm) required to eject an electron from this metal's surface? Show your work including equation(s) used. (3 sig figs)

d. 2 points In the space provided below, plot the following

(Assume frequency is *above* threshold frequency):

- i. *Kinetic energy* of an ejected electron versus the *intensity* of the incident light (left)
- ii. *Number of electrons* ejected versus the *frequency* of the incident light (right)



**Question 3** **6 points****a. 2 points**

In the ground state of Ge (Germanium; $Z = 32$) atom, **name** the following orbitals:

i. Orbital containing the electron that requires the maximum energy to be removed.

ii. Orbital containing the electron that experiences the minimum effective nuclear charge.

b. 2 points

An unknown anion (A^-) has the following *excited* electronic configuration: $[Ar] 4s^2 3d^3 4p^1$

(Ar: Argon; $Z = 18$)

Give the **condensed ground state electronic configuration** of the atom, **A** and the cation, **A^{2+}** .
Show your work. (Assume all electronic transitions are possible)

Electronic Configuration of A:

Electronic Configuration of A^{2+} :

c. 2 points

In a single atom, determine how many electrons can have the following set of quantum numbers?

(i) $n = 5; l = 3$

In a single atom, determine how many electrons can have the following set of quantum numbers?

(i) $n = 3; l = 2; m_s = +1/2$

**Question 4** **8 points**

Rank the following from 1 to 4 in **increasing** order. Provide the ranking in the boxes provided.

Note: Atomic number (Z) are provided for relevant atoms

a) Increasing (lowest to highest) order of *ionization energy*: (1= lowest; 4 = highest)

Mg

Mg: Z = 12

Rb

Rb: Z = 37

Cs

Cs: Z = 55

K

K : Z = 19

b) Increasing (lowest to highest) order of *bond angle*: (1= lowest; 4 = highest)

H₂OCCl₄XeF₄

HCN

c) Increasing (smallest to largest) order of *ionic size* (1= smallest; 4 = largest)

S²⁻

S: Z = 16

O²⁻

O: Z = 8

Be²⁺

Be: Z = 4

Na⁺

Na: Z = 11

d) Increasing (lowest to highest) order of *bond energy* (1= lowest; 4 = highest)

P-F

P-Cl

P-I

P-Br

**Question 5** **12 points**

a. 9 points For the molecules HClO_4 and TeF_6 , draw the most stable *Lewis structure* showing appropriate shape and *dashed/wedged bonds*, based on VSEPR. Determine the **molecular geometry** around the central atom. For HClO_4 , label the *most polar bond*.



b. 3 points (Note: Lone pairs are not shown in structures below)

(i) For the three *resonance* structures (1, 2 and 3) shown below, answer the following question:



(I) Circle the *most contributing* resonance structure.

Structure 1

Structure 2

Structure 3

(ii) For the three structures given (4, 5, and 6) below, answer the following questions:



(I) Circle all the structures that represent *resonance structures* of each other:

Structure 4

Structure 5

Structure 6

(II) Among the resonance structures, circle the *most contributing* resonance structure.

Structure 4

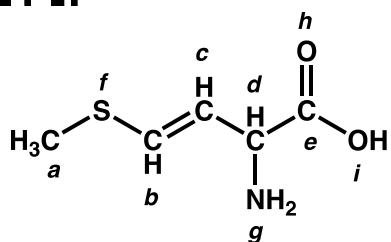
Structure 5

Structure 6

**Question 6** **6 points**

Using the data given below, calculate the **lattice energy** for SrO(s).

	ΔH° (kJ/mol)
$\text{Sr}(s) \longrightarrow \text{Sr}(g)$	+180
$\text{Sr}(g) \longrightarrow \text{Sr}^{2+}(g) + 2e^-$	+1468
$\text{O}_2(g) \longrightarrow 2\text{O}(g)$	+500
$\text{O}(g) + 2e^- \longrightarrow \text{O}^{2-}(g)$	+600
$\text{Sr}(s) + \frac{1}{2}\text{O}_2(g) \longrightarrow \text{SrO}(s)$	-550

**Question 7 10 points**

Consider the molecule given on the left, and answer questions (i) to (vi) using chemical bonding theories that we have learnt in class (valence bond theory, hybridization).

Note that the lone pairs are not shown in the structure:

i. Consider the carbon atoms marked in as ***a***, ***b***, ***c***, ***d***, and ***e*** in the molecule, and determine **hybridization** at each of the carbon centers. (2.5 points)

Atom	Hybridization
Carbon (<i>a</i>)	
Carbon (<i>b</i>)	
Carbon (<i>c</i>)	
Carbon (<i>d</i>)	
Carbon (<i>e</i>)	

ii. Now, consider the sulfur atom (S) marked as ***f***, nitrogen (N) atom (***g***), and oxygen(O) atoms (***h*** and ***i***), and determine their **hybridization**. (2 points)

Atom	Hybridization
Sulfur (<i>f</i>)	
Nitrogen (<i>g</i>)	
Oxygen (<i>h</i>)	
Oxygen (<i>i</i>)	

iii. **Describe** all the bonds formed by the carbon atoms ***a*** and ***b***, using sigma and pi designation as appropriate. (2 points)

Atom	Bonds formed (use sigma and pi as appropriate)
Carbon (<i>a</i>)	
Carbon (<i>b</i>)	

iv. How many total sp^3 -s sigma bonds are in the molecule? (1 point)

v. Determine **electronic** and **molecular geometry** at carbon (C) labeled ***e*** and sulfur (S) labeled ***f*** centres. (2 points)

Atom	Electronic Geometry	Molecular Geometry
Carbon (<i>e</i>)		
Sulfur (<i>f</i>)		

vi. Circle the value from the choices given below, that describes the **C-S-C bond angle at sulfur** (S) marked ***f***: (0.5 points)

>90° to <109.5° 109.5° >109.5 to <120° 120° >120° to <180°

**Question 8 12 points**

Using the **valence shell orbitals only** draw a **complete** molecular orbital (MO) diagram (with no s-p mixing) for the molecule **PS**, in the space provided below. **(6 points)**

Assume that 3s and 3p atomic orbitals will form molecular orbitals like those formed by the combination of 2s and 2p atomic orbitals that we learnt in class, and there is no s-p mixing.

Show **energy arrow**, **atomic orbitals** involved, **atom labels**, the **molecular orbitals** that are formed, **filled electrons** in the MO diagram etc. Make sure to label all the atomic and molecular orbitals.

Using the MO diagram above, answer the following questions:

- (i) Which one of the following will have the highest bond order? Circle the correct choice
A) PS^{1-} B) PS C) PS^{1+}
- (ii) Which one of the following will have the longest bond length? Circle the correct choice
A) PS^{1-} B) PS C) PS^{1+}
- (iii) Which of the following will be diamagnetic? Circle the correct choice.
A) PS^{1-} B) PS C) PS^{1+}
- (iv) Draw Lewis dot structure of the molecule with the shortest bond length (based on the MO diagram) from among the three (PS^{1-} , PS, PS^{1+}), and indicate the bond order calculated from the Lewis dot structure. **(3 points)**

Lewis Dot Structure

Bond Order



F4C5AA74-4513-44A5-9E8C-9970BF74944F

midterm-exam-93388

#564 Page 10 of 10

SCRAP PAGE

**CHEMISTRY 110 –Midterm Exam****Thursday November 7, 2024, 6:30 – 8:30 pm**

Examiners: Pallavi Sirjoosingh and Ashok Kakkar

First name (please write as legibly as possible within the boxes)

--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--

Last name

--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--

Student ID number

--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--

This booklet has 8 questions on 10 pages (The last page can be used as a scrap page).

The staple must NOT be removed from the question booklet. Your intact question booklet must be returned at the end of the exam.

A separate booklet contains a periodic table, and datasheet.

Write your answers directly under the questions in the space provided. Do not write on the QR codes or near the top edges (as it may be cut off).

Please write **legibly and dark enough** for the scan to be readable. For students who use English as a second language, dictionaries are permitted. Calculators are allowed.

DO NOT ASK QUESTIONS OF THE INVIGILATORS about interpretations of the exam questions. If you have a concern about a question, please summarize it at the top of this page. All concerns will be evaluated after the exam. GOOD LUCK!

**Question 1 8 points** (*Assume all transitions are possible*)

a. 6 points Calculate the **energy (in kJ/mol)** required to ionize (completely remove) the electron from the **n=3** level of **He⁺ ion** (He: Helium). Show your work including equations. (3 sig. figs.)

b. 2 points

(I) Select the largest (longest) wavelength from the three choices below:

- i. Wavelength of photon to *completely remove* an electron from *1s orbital of Hydrogen atom*
- ii. Wavelength of photon to *completely remove* an electron from *1s orbital of Boron atom (B)*
- iii. Wavelength of photon to *completely remove* an electron from *1s orbital of Boron cation (B⁺)*

i or ii or iii

(II) Select the smallest (shortest) wavelength from the three choices below (*for H atom*):

- i. Wavelength of photon *emitted* for an electron transitioning from *2p to 1s orbital*
- ii. Wavelength of photon *absorbed* for an electron transitioning from *2s to 4p orbital*
- iii. Wavelength of photon *absorbed* for an electron transitioning from *1s to 3p orbital*

i or ii or iii



Question 2 8 points



A Chem110 student plots the graph shown on the left, from data obtained for the kinetic energy (eV) of the ejected electron versus frequency (s^{-1}) of the incident light, for an unknown metal. ($1\text{eV} = 1.6022 \times 10^{-19} \text{ J}$)

Based on the plot, answer the questions **a to c** below:

a. 1 point On the plot above, label the **threshold frequency**.

b. 2 points What is the value of the work potential (in J) of the unknown metal?

c. 3 points What is the *longest* wavelength (in nm) required to eject an electron from this metal's surface? Show your work including equation(s) used. (3 sig figs)

d. 2 points In the space provided below, plot the following

(Assume frequency is *above* threshold frequency):

- i. *Kinetic energy* of an ejected electron versus the *intensity* of the incident light (left)
- ii. *Number of electrons* ejected versus the *frequency* of the incident light (right)



**Question 3** **6 points****a. 2 points**

In the ground state of Ge (Germanium; $Z = 32$) atom, **name** the following orbitals:

i. Orbital containing the electron that requires the maximum energy to be removed.

ii. Orbital containing the electron that experiences the minimum effective nuclear charge.

b. 2 points

An unknown anion (A^-) has the following *excited* electronic configuration: $[Ar] 4s^2 3d^3 4p^1$

(Ar: Argon; $Z = 18$)

Give the **condensed ground state electronic configuration** of the atom, **A** and the cation, **A^{2+}** .
Show your work. (Assume all electronic transitions are possible)

Electronic Configuration of A:

Electronic Configuration of A^{2+} :

c. 2 points

In a single atom, determine how many electrons can have the following set of quantum numbers?

(i) $n = 5; l = 3$

In a single atom, determine how many electrons can have the following set of quantum numbers?

(i) $n = 3; l = 2; m_s = +1/2$

**Question 4** **8 points**

Rank the following from 1 to 4 in **increasing** order. Provide the ranking in the boxes provided.

Note: Atomic number (Z) are provided for relevant atoms

a) Increasing (lowest to highest) order of *ionization energy*: (1= lowest; 4 = highest)

Mg

Mg: Z = 12

Rb

Rb: Z = 37

Cs

Cs: Z = 55

K

K : Z = 19

b) Increasing (lowest to highest) order of *bond angle*: (1= lowest; 4 = highest)

H₂OCCl₄XeF₄

HCN

c) Increasing (smallest to largest) order of *ionic size* (1= smallest; 4 = largest)

S²⁻

S: Z = 16

O²⁻

O: Z = 8

Be²⁺

Be: Z = 4

Na⁺

Na: Z = 11

d) Increasing (lowest to highest) order of *bond energy* (1= lowest; 4 = highest)

P-F

P-Cl

P-I

P-Br

**Question 5** **12 points**

a. 9 points For the molecules HClO_4 and TeF_6 , draw the most stable *Lewis structure* showing appropriate shape and *dashed/wedged bonds*, based on VSEPR. Determine the **molecular geometry** around the central atom. For HClO_4 , label the *most polar bond*.



b. 3 points (Note: Lone pairs are not shown in structures below)

(i) For the three *resonance* structures (1, 2 and 3) shown below, answer the following question:



(I) Circle the *most contributing* resonance structure.

Structure 1

Structure 2

Structure 3

(ii) For the three structures given (4, 5, and 6) below, answer the following questions:



(I) Circle all the structures that represent *resonance structures* of each other:

Structure 4

Structure 5

Structure 6

(II) Among the resonance structures, circle the *most contributing* resonance structure.

Structure 4

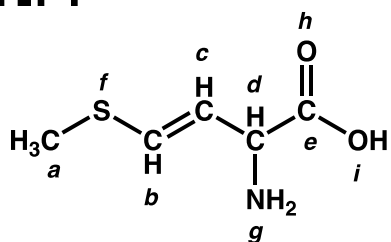
Structure 5

Structure 6

**Question 6** **6 points**

Using the data given below, calculate the **lattice energy** for SrO(s).

	ΔH° (kJ/mol)
$\text{Sr}(s) \longrightarrow \text{Sr}(g)$	+180
$\text{Sr}(g) \longrightarrow \text{Sr}^{2+}(g) + 2e^-$	+1468
$\text{O}_2(g) \longrightarrow 2\text{O}(g)$	+500
$\text{O}(g) + 2e^- \longrightarrow \text{O}^{2-}(g)$	+600
$\text{Sr}(s) + \frac{1}{2}\text{O}_2(g) \longrightarrow \text{SrO}(s)$	-550

**Question 7 10 points**

Consider the molecule given on the left, and answer questions (i) to (vi) using chemical bonding theories that we have learnt in class (valence bond theory, hybridization).

Note that the lone pairs are not shown in the structure:

i. Consider the carbon atoms marked in as ***a***, ***b***, ***c***, ***d***, and ***e*** in the molecule, and determine **hybridization** at each of the carbon centers. (2.5 points)

Atom	Hybridization
Carbon (<i>a</i>)	
Carbon (<i>b</i>)	
Carbon (<i>c</i>)	
Carbon (<i>d</i>)	
Carbon (<i>e</i>)	

ii. Now, consider the sulfur atom (S) marked as ***f***, nitrogen (N) atom (***g***), and oxygen(O) atoms (***h*** and ***i***), and determine their **hybridization**. (2 points)

Atom	Hybridization
Sulfur (<i>f</i>)	
Nitrogen (<i>g</i>)	
Oxygen (<i>h</i>)	
Oxygen (<i>i</i>)	

iii. **Describe** all the bonds formed by the carbon atoms ***a*** and ***b***, using sigma and pi designation as appropriate. (2 points)

Atom	Bonds formed (use sigma and pi as appropriate)
Carbon (<i>a</i>)	
Carbon (<i>b</i>)	

iv. How many total sp^3 -s sigma bonds are in the molecule? (1 point)

v. Determine **electronic** and **molecular geometry** at carbon (C) labeled ***e*** and sulfur (S) labeled ***f*** centres. (2 points)

Atom	Electronic Geometry	Molecular Geometry
Carbon (<i>e</i>)		
Sulfur (<i>f</i>)		

vi. Circle the value from the choices given below, that describes the **C-S-C bond angle at sulfur** (S) marked ***f***: (0.5 points)

>90° to <109.5° 109.5° >109.5 to <120° 120° >120° to <180°



Question 8 12 points

Using the **valence shell orbitals only** draw a **complete** molecular orbital (MO) diagram (with no s-p mixing) for the molecule **PS**, in the space provided below. **(6 points)**

Assume that 3s and 3p atomic orbitals will form molecular orbitals like those formed by the combination of 2s and 2p atomic orbitals that we learnt in class, and there is no s-p mixing.

Show **energy arrow**, **atomic orbitals** involved, **atom labels**, the **molecular orbitals** that are formed, **filled electrons** in the MO diagram etc. Make sure to label all the atomic and molecular orbitals.

Using the MO diagram above, answer the following questions:

- (i) Which one of the following will have the highest bond order? Circle the correct choice
 A) PS^{1-} B) PS C) PS^{1+}
- (ii) Which one of the following will have the longest bond length? Circle the correct choice
 A) PS^{1-} B) PS C) PS^{1+}
- (iii) Which of the following will be diamagnetic? Circle the correct choice.
 A) PS^{1-} B) PS C) PS^{1+}
- (iv) Draw Lewis dot structure of the molecule with the shortest bond length (based on the MO diagram) from among the three (PS^{1-} , PS, PS^{1+}), and indicate the bond order calculated from the Lewis dot structure. **(3 points)**

Lewis Dot Structure

Bond Order



3B89BFEB-A3D9-48EE-8BB4-2746900712EB

midterm-exam-93388

#565 Page 10 of 10

SCRAP PAGE

**CHEMISTRY 110 –Midterm Exam****Thursday November 7, 2024, 6:30 – 8:30 pm****Examiners: Pallavi Sirjoosingh and Ashok Kakkar**

First name (please write as legibly as possible within the boxes)

--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--

Last name

--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--

Student ID number

--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--

This booklet has 8 questions on 10 pages (The last page can be used as a scrap page).

The staple must NOT be removed from the question booklet. Your intact question booklet must be returned at the end of the exam.

A separate booklet contains a periodic table, and datasheet.

Write your answers directly under the questions in the space provided. Do not write on the QR codes or near the top edges (as it may be cut off).

Please write **legibly and dark enough** for the scan to be readable. For students who use English as a second language, dictionaries are permitted. Calculators are allowed.

DO NOT ASK QUESTIONS OF THE INVIGILATORS about interpretations of the exam questions. If you have a concern about a question, please summarize it at the top of this page. All concerns will be evaluated after the exam. GOOD LUCK!

**Question 1 8 points** (*Assume all transitions are possible*)

a. 6 points Calculate the **energy (in kJ/mol)** required to ionize (completely remove) the electron from the **n=3** level of **He⁺ ion** (He: Helium). Show your work including equations. (3 sig. figs.)

b. 2 points

(I) Select the largest (longest) wavelength from the three choices below:

- i. Wavelength of photon to *completely remove* an electron from *1s orbital of Hydrogen atom*
- ii. Wavelength of photon to *completely remove* an electron from *1s orbital of Boron atom (B)*
- iii. Wavelength of photon to *completely remove* an electron from *1s orbital of Boron cation (B⁺)*

i or ii or iii

(II) Select the smallest (shortest) wavelength from the three choices below (*for H atom*):

- i. Wavelength of photon *emitted* for an electron transitioning from *2p to 1s orbital*
- ii. Wavelength of photon *absorbed* for an electron transitioning from *2s to 4p orbital*
- iii. Wavelength of photon *absorbed* for an electron transitioning from *1s to 3p orbital*

i or ii or iii



Question 2 8 points



A Chem110 student plots the graph shown on the left, from data obtained for the kinetic energy (eV) of the ejected electron versus frequency (s^{-1}) of the incident light, for an unknown metal. ($1\text{eV} = 1.6022 \times 10^{-19} \text{ J}$)

Based on the plot, answer the questions **a to c** below:

a. 1 point On the plot above, label the **threshold frequency**.

b. 2 points What is the value of the work potential (in J) of the unknown metal?

c. 3 points What is the *longest* wavelength (in nm) required to eject an electron from this metal's surface? Show your work including equation(s) used. (3 sig figs)

d. 2 points In the space provided below, plot the following

(Assume frequency is *above* threshold frequency):

- Kinetic energy* of an ejected electron versus the *intensity* of the incident light (left)
- Number of electrons* ejected versus the *frequency* of the incident light (right)



**Question 3** **6 points****a. 2 points**

In the ground state of Ge (Germanium; $Z = 32$) atom, **name** the following orbitals:

i. Orbital containing the electron that requires the maximum energy to be removed.

ii. Orbital containing the electron that experiences the minimum effective nuclear charge.

b. 2 points

An unknown anion (A^-) has the following *excited* electronic configuration: $[Ar] 4s^2 3d^3 4p^1$

(Ar: Argon; $Z = 18$)

Give the **condensed ground state electronic configuration** of the atom, **A** and the cation, **A^{2+}** .
Show your work. (Assume all electronic transitions are possible)

Electronic Configuration of A:

Electronic Configuration of A^{2+} :

c. 2 points

In a single atom, determine how many electrons can have the following set of quantum numbers?

(i) $n = 5; l = 3$

In a single atom, determine how many electrons can have the following set of quantum numbers?

(i) $n = 3; l = 2; m_s = +1/2$

**Question 4** **8 points**

Rank the following from 1 to 4 in **increasing** order. Provide the ranking in the boxes provided.

Note: Atomic number (Z) are provided for relevant atoms

a) Increasing (lowest to highest) order of *ionization energy*: (1= lowest; 4 = highest)

Mg

Mg: Z = 12

Rb

Rb: Z = 37

Cs

Cs: Z = 55

K

K : Z = 19

b) Increasing (lowest to highest) order of *bond angle*: (1= lowest; 4 = highest)

H₂OCCl₄XeF₄

HCN

c) Increasing (smallest to largest) order of *ionic size* (1= smallest; 4 = largest)

S²⁻

S: Z = 16

O²⁻

O: Z = 8

Be²⁺

Be: Z = 4

Na⁺

Na: Z = 11

d) Increasing (lowest to highest) order of *bond energy* (1= lowest; 4 = highest)

P-F

P-Cl

P-I

P-Br

**Question 5** **12 points**

a. 9 points For the molecules HClO_4 and TeF_6 , draw the most stable *Lewis structure* showing appropriate shape and *dashed/wedged bonds*, based on VSEPR. Determine the **molecular geometry** around the central atom. For HClO_4 , label the *most polar bond*.



b. 3 points (Note: Lone pairs are not shown in structures below)

(i) For the three *resonance* structures (1, 2 and 3) shown below, answer the following question:



(I) Circle the *most contributing* resonance structure.

Structure 1

Structure 2

Structure 3

(ii) For the three structures given (4, 5, and 6) below, answer the following questions:



(I) Circle all the structures that represent *resonance structures* of each other:

Structure 4

Structure 5

Structure 6

(II) Among the resonance structures, circle the *most contributing* resonance structure.

Structure 4

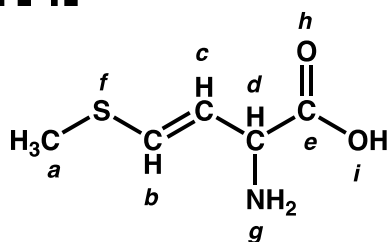
Structure 5

Structure 6

**Question 6** **6 points**

Using the data given below, calculate the **lattice energy** for SrO(s).

	ΔH° (kJ/mol)
$\text{Sr}(s) \longrightarrow \text{Sr}(g)$	+180
$\text{Sr}(g) \longrightarrow \text{Sr}^{2+}(g) + 2e^-$	+1468
$\text{O}_2(g) \longrightarrow 2\text{O}(g)$	+500
$\text{O}(g) + 2e^- \longrightarrow \text{O}^{2-}(g)$	+600
$\text{Sr}(s) + \frac{1}{2}\text{O}_2(g) \longrightarrow \text{SrO}(s)$	-550

**Question 7 10 points**

Consider the molecule given on the left, and answer questions (i) to (vi) using chemical bonding theories that we have learnt in class (valence bond theory, hybridization).

Note that the lone pairs are not shown in the structure:

i. Consider the carbon atoms marked in as ***a***, ***b***, ***c***, ***d***, and ***e*** in the molecule, and determine **hybridization** at each of the carbon centers. (2.5 points)

Atom	Hybridization
Carbon (<i>a</i>)	
Carbon (<i>b</i>)	
Carbon (<i>c</i>)	
Carbon (<i>d</i>)	
Carbon (<i>e</i>)	

ii. Now, consider the sulfur atom (S) marked as ***f***, nitrogen (N) atom (***g***), and oxygen(O) atoms (***h*** and ***i***), and determine their **hybridization**. (2 points)

Atom	Hybridization
Sulfur (<i>f</i>)	
Nitrogen (<i>g</i>)	
Oxygen (<i>h</i>)	
Oxygen (<i>i</i>)	

iii. **Describe** all the bonds formed by the carbon atoms ***a*** and ***b***, using sigma and pi designation as appropriate. (2 points)

Atom	Bonds formed (use sigma and pi as appropriate)
Carbon (<i>a</i>)	
Carbon (<i>b</i>)	

iv. How many total sp^3 -s sigma bonds are in the molecule? (1 point)

v. Determine **electronic** and **molecular geometry** at carbon (C) labeled ***e*** and sulfur (S) labeled ***f*** centres. (2 points)

Atom	Electronic Geometry	Molecular Geometry
Carbon (<i>e</i>)		
Sulfur (<i>f</i>)		

vi. Circle the value from the choices given below, that describes the **C-S-C bond angle at sulfur** (S) marked ***f***: (0.5 points)

>90° to <109.5° 109.5° >109.5 to <120° 120° >120° to <180°

**Question 8 12 points**

Using the **valence shell orbitals only** draw a **complete** molecular orbital (MO) diagram (with no s-p mixing) for the molecule **PS**, in the space provided below. **(6 points)**

Assume that 3s and 3p atomic orbitals will form molecular orbitals like those formed by the combination of 2s and 2p atomic orbitals that we learnt in class, and there is no s-p mixing.

Show **energy arrow**, **atomic orbitals** involved, **atom labels**, the **molecular orbitals** that are formed, **filled electrons** in the MO diagram etc. Make sure to label all the atomic and molecular orbitals.

Using the MO diagram above, answer the following questions:

- (i) Which one of the following will have the highest bond order? Circle the correct choice
A) PS^{1-} B) PS C) PS^{1+}
- (ii) Which one of the following will have the longest bond length? Circle the correct choice
A) PS^{1-} B) PS C) PS^{1+}
- (iii) Which of the following will be diamagnetic? Circle the correct choice.
A) PS^{1-} B) PS C) PS^{1+}
- (iv) Draw Lewis dot structure of the molecule with the shortest bond length (based on the MO diagram) from among the three (PS^{1-} , PS, PS^{1+}), and indicate the bond order calculated from the Lewis dot structure. **(3 points)**

Lewis Dot Structure

Bond Order



31F05B38-C93A-4045-B4BC-47122AE5E826

midterm-exam-93388

#566 Page 10 of 10

SCRAP PAGE

**CHEMISTRY 110 –Midterm Exam****Thursday November 7, 2024, 6:30 – 8:30 pm**

Examiners: Pallavi Sirjoosingh and Ashok Kakkar

First name (please write as legibly as possible within the boxes)

--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--

Last name

--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--

Student ID number

--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--

This booklet has 8 questions on 10 pages (The last page can be used as a scrap page).

The staple must NOT be removed from the question booklet. Your intact question booklet must be returned at the end of the exam.

A separate booklet contains a periodic table, and datasheet.

Write your answers directly under the questions in the space provided. Do not write on the QR codes or near the top edges (as it may be cut off).

Please write **legibly and dark enough** for the scan to be readable. For students who use English as a second language, dictionaries are permitted. Calculators are allowed.

DO NOT ASK QUESTIONS OF THE INVIGILATORS about interpretations of the exam questions. If you have a concern about a question, please summarize it at the top of this page. All concerns will be evaluated after the exam. GOOD LUCK!

**Question 1 8 points** (*Assume all transitions are possible*)

a. 6 points Calculate the **energy (in kJ/mol)** required to ionize (completely remove) the electron from the **n=3** level of **He⁺ ion** (He: Helium). Show your work including equations. (3 sig. figs.)

b. 2 points

(I) Select the largest (longest) wavelength from the three choices below:

- i. Wavelength of photon to *completely remove* an electron from *1s orbital of Hydrogen atom*
- ii. Wavelength of photon to *completely remove* an electron from *1s orbital of Boron atom (B)*
- iii. Wavelength of photon to *completely remove* an electron from *1s orbital of Boron cation (B⁺)*

i or ii or iii

(II) Select the smallest (shortest) wavelength from the three choices below (*for H atom*):

- i. Wavelength of photon *emitted* for an electron transitioning from *2p to 1s orbital*
- ii. Wavelength of photon *absorbed* for an electron transitioning from *2s to 4p orbital*
- iii. Wavelength of photon *absorbed* for an electron transitioning from *1s to 3p orbital*

i or ii or iii



Question 2 8 points



A Chem110 student plots the graph shown on the left, from data obtained for the kinetic energy (eV) of the ejected electron versus frequency (s^{-1}) of the incident light, for an unknown metal. ($1\text{eV} = 1.6022 \times 10^{-19} \text{ J}$)

Based on the plot, answer the questions **a to c** below:

a. 1 point On the plot above, label the **threshold frequency**.

b. 2 points What is the value of the work potential (in J) of the unknown metal?

c. 3 points What is the *longest* wavelength (in nm) required to eject an electron from this metal's surface? Show your work including equation(s) used. (3 sig figs)

d. 2 points In the space provided below, plot the following

(Assume frequency is *above* threshold frequency):

- i. *Kinetic energy* of an ejected electron versus the *intensity* of the incident light (left)
- ii. *Number of electrons* ejected versus the *frequency* of the incident light (right)



**Question 3** **6 points****a. 2 points**

In the ground state of Ge (Germanium; $Z = 32$) atom, **name** the following orbitals:

i. Orbital containing the electron that requires the maximum energy to be removed.

ii. Orbital containing the electron that experiences the minimum effective nuclear charge.

b. 2 points

An unknown anion (A^-) has the following *excited* electronic configuration: $[Ar] 4s^2 3d^3 4p^1$

(Ar: Argon; $Z = 18$)

Give the **condensed ground state electronic configuration** of the atom, **A** and the cation, **A^{2+}** .
Show your work. (Assume all electronic transitions are possible)

Electronic Configuration of A:

Electronic Configuration of A^{2+} :

c. 2 points

In a single atom, determine how many electrons can have the following set of quantum numbers?

(i) $n = 5; l = 3$

In a single atom, determine how many electrons can have the following set of quantum numbers?

(i) $n = 3; l = 2; m_s = +1/2$

**Question 4** **8 points**

Rank the following from 1 to 4 in **increasing** order. Provide the ranking in the boxes provided.

Note: Atomic number (Z) are provided for relevant atoms

a) Increasing (lowest to highest) order of *ionization energy*: (1= lowest; 4 = highest)

Mg

Mg: Z = 12

Rb

Rb: Z = 37

Cs

Cs: Z = 55

K

K : Z = 19

b) Increasing (lowest to highest) order of *bond angle*: (1= lowest; 4 = highest)

H₂OCCl₄XeF₄

HCN

c) Increasing (smallest to largest) order of *ionic size* (1= smallest; 4 = largest)

S²⁻

S: Z = 16

O²⁻

O: Z = 8

Be²⁺

Be: Z = 4

Na⁺

Na: Z = 11

d) Increasing (lowest to highest) order of *bond energy* (1= lowest; 4 = highest)

P-F

P-Cl

P-I

P-Br

**Question 5** **12 points**

a. 9 points For the molecules HClO_4 and TeF_6 , draw the most stable *Lewis structure* showing appropriate shape and *dashed/wedged bonds*, based on VSEPR. Determine the **molecular geometry** around the central atom. For HClO_4 , label the *most polar bond*.

HClO_4 (there are no O-O bonds)

Label the most polar bond

Molecular Geometry around Cl

TeF_6

Molecular Geometry around Te

b. 3 points (Note: Lone pairs are not shown in structures below)

(i) For the three *resonance* structures (1, 2 and 3) shown below, answer the following question:



(I) Circle the *most contributing* resonance structure.

Structure 1

Structure 2

Structure 3

(ii) For the three structures given (4, 5, and 6) below, answer the following questions:



(I) Circle all the structures that represent *resonance structures* of each other:

Structure 4

Structure 5

Structure 6

(II) Among the resonance structures, circle the *most contributing* resonance structure.

Structure 4

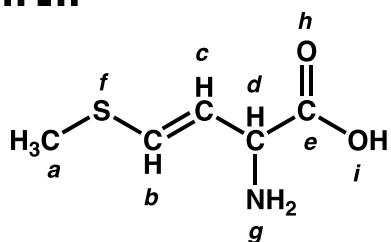
Structure 5

Structure 6

**Question 6** **6 points**

Using the data given below, calculate the **lattice energy** for SrO(s).

	ΔH° (kJ/mol)
$\text{Sr}(s) \longrightarrow \text{Sr}(g)$	+180
$\text{Sr}(g) \longrightarrow \text{Sr}^{2+}(g) + 2e^-$	+1468
$\text{O}_2(g) \longrightarrow 2\text{O}(g)$	+500
$\text{O}(g) + 2e^- \longrightarrow \text{O}^{2-}(g)$	+600
$\text{Sr}(s) + \frac{1}{2}\text{O}_2(g) \longrightarrow \text{SrO}(s)$	-550

**Question 7 10 points**

Consider the molecule given on the left, and answer questions (i) to (vi) using chemical bonding theories that we have learnt in class (valence bond theory, hybridization).

Note that the lone pairs are not shown in the structure:

i. Consider the carbon atoms marked in as ***a***, ***b***, ***c***, ***d***, and ***e*** in the molecule, and determine **hybridization** at each of the carbon centers. (2.5 points)

Atom	Hybridization
Carbon (<i>a</i>)	
Carbon (<i>b</i>)	
Carbon (<i>c</i>)	
Carbon (<i>d</i>)	
Carbon (<i>e</i>)	

ii. Now, consider the sulfur atom (S) marked as ***f***, nitrogen (N) atom (***g***), and oxygen(O) atoms (***h*** and ***i***), and determine their **hybridization**. (2 points)

Atom	Hybridization
Sulfur (<i>f</i>)	
Nitrogen (<i>g</i>)	
Oxygen (<i>h</i>)	
Oxygen (<i>i</i>)	

iii. **Describe** all the bonds formed by the carbon atoms ***a*** and ***b***, using sigma and pi designation as appropriate. (2 points)

Atom	Bonds formed (use sigma and pi as appropriate)
Carbon (<i>a</i>)	
Carbon (<i>b</i>)	

iv. How many total sp^3 -s sigma bonds are in the molecule? (1 point)

v. Determine **electronic** and **molecular geometry** at carbon (C) labeled ***e*** and sulfur (S) labeled ***f*** centres. (2 points)

Atom	Electronic Geometry	Molecular Geometry
Carbon (<i>e</i>)		
Sulfur (<i>f</i>)		

vi. Circle the value from the choices given below, that describes the **C-S-C bond angle at sulfur** (S) marked ***f***: (0.5 points)

>90° to <109.5°

109.5°

>109.5 to <120°

120°

>120° to <180°


Question 8 12 points

Using the **valence shell orbitals only** draw a **complete** molecular orbital (MO) diagram (with no s-p mixing) for the molecule **PS**, in the space provided below. **(6 points)**

Assume that 3s and 3p atomic orbitals will form molecular orbitals like those formed by the combination of 2s and 2p atomic orbitals that we learnt in class, and there is no s-p mixing.

Show **energy arrow**, **atomic orbitals** involved, **atom labels**, the **molecular orbitals** that are formed, **filled electrons** in the MO diagram etc. Make sure to label all the atomic and molecular orbitals.

Using the MO diagram above, answer the following questions:

- (i) Which one of the following will have the highest bond order? Circle the correct choice
 A) PS^{1-} B) PS C) PS^{1+}
- (ii) Which one of the following will have the longest bond length? Circle the correct choice
 A) PS^{1-} B) PS C) PS^{1+}
- (iii) Which of the following will be diamagnetic? Circle the correct choice.
 A) PS^{1-} B) PS C) PS^{1+}
- (iv) Draw Lewis dot structure of the molecule with the shortest bond length (based on the MO diagram) from among the three (PS^{1-} , PS, PS^{1+}), and indicate the bond order calculated from the Lewis dot structure. **(3 points)**

Lewis Dot Structure

Bond Order



2F129B8E-58D2-49E7-93AB-DD7C1F07B321

midterm-exam-93388

#567 Page 10 of 10

SCRAP PAGE

**CHEMISTRY 110 –Midterm Exam****Thursday November 7, 2024, 6:30 – 8:30 pm**

Examiners: Pallavi Sirjoosingh and Ashok Kakkar

First name (please write as legibly as possible within the boxes)

--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--

Last name

--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--

Student ID number

--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--

This booklet has 8 questions on 10 pages (The last page can be used as a scrap page).

The staple must NOT be removed from the question booklet. Your intact question booklet must be returned at the end of the exam.

A separate booklet contains a periodic table, and datasheet.

Write your answers directly under the questions in the space provided. Do not write on the QR codes or near the top edges (as it may be cut off).

Please write **legibly and dark enough** for the scan to be readable. For students who use English as a second language, dictionaries are permitted. Calculators are allowed.

DO NOT ASK QUESTIONS OF THE INVIGILATORS about interpretations of the exam questions. If you have a concern about a question, please summarize it at the top of this page. All concerns will be evaluated after the exam. GOOD LUCK!

**Question 1 8 points** (*Assume all transitions are possible*)

a. 6 points Calculate the **energy (in kJ/mol)** required to ionize (completely remove) the electron from the **$n=3$** level of **He^+ ion** (He: Helium). Show your work including equations. (3 sig. figs.)

b. 2 points

(I) Select the largest (longest) wavelength from the three choices below:

- i. Wavelength of photon to *completely remove* an electron from *1s orbital of Hydrogen atom*
- ii. Wavelength of photon to *completely remove* an electron from *1s orbital of Boron atom (B)*
- iii. Wavelength of photon to *completely remove* an electron from *1s orbital of Boron cation (B^+)*

i or ii or iii

(II) Select the smallest (shortest) wavelength from the three choices below (*for H atom*):

- i. Wavelength of photon *emitted* for an electron transitioning from *2p to 1s orbital*
- ii. Wavelength of photon *absorbed* for an electron transitioning from *2s to 4p orbital*
- iii. Wavelength of photon *absorbed* for an electron transitioning from *1s to 3p orbital*

i or ii or iii



Question 2 8 points



A Chem110 student plots the graph shown on the left, from data obtained for the kinetic energy (eV) of the ejected electron versus frequency (s^{-1}) of the incident light, for an unknown metal. ($1\text{eV} = 1.6022 \times 10^{-19} \text{ J}$)

Based on the plot, answer the questions **a to c** below:

a. 1 point On the plot above, label the **threshold frequency**.

b. 2 points What is the value of the work potential (in J) of the unknown metal?

c. 3 points What is the *longest* wavelength (in nm) required to eject an electron from this metal's surface? Show your work including equation(s) used. (3 sig figs)

d. 2 points In the space provided below, plot the following

(Assume frequency is *above* threshold frequency):

- Kinetic energy* of an ejected electron versus the *intensity* of the incident light (left)
- Number of electrons* ejected versus the *frequency* of the incident light (right)



**Question 3** **6 points****a. 2 points**

In the ground state of Ge (Germanium; $Z = 32$) atom, **name** the following orbitals:

i. Orbital containing the electron that requires the maximum energy to be removed.

ii. Orbital containing the electron that experiences the minimum effective nuclear charge.

b. 2 points

An unknown anion (A^-) has the following *excited* electronic configuration: $[Ar] 4s^2 3d^3 4p^1$

(Ar: Argon; $Z = 18$)

Give the **condensed ground state electronic configuration** of the atom, **A** and the cation, **A^{2+}** .
Show your work. (Assume all electronic transitions are possible)

Electronic Configuration of A:

Electronic Configuration of A^{2+} :

c. 2 points

In a single atom, determine how many electrons can have the following set of quantum numbers?

(i) $n = 5; l = 3$

In a single atom, determine how many electrons can have the following set of quantum numbers?

(i) $n = 3; l = 2; m_s = +1/2$

**Question 4** **8 points**

Rank the following from 1 to 4 in **increasing** order. Provide the ranking in the boxes provided.

Note: Atomic number (Z) are provided for relevant atoms

a) Increasing (lowest to highest) order of *ionization energy*: (1= lowest; 4 = highest)

Mg

Mg: Z = 12

Rb

Rb: Z = 37

Cs

Cs: Z = 55

K

K : Z = 19

b) Increasing (lowest to highest) order of *bond angle*: (1= lowest; 4 = highest)

H₂OCCl₄XeF₄

HCN

c) Increasing (smallest to largest) order of *ionic size* (1= smallest; 4 = largest)

S²⁻

S: Z = 16

O²⁻

O: Z = 8

Be²⁺

Be: Z = 4

Na⁺

Na: Z = 11

d) Increasing (lowest to highest) order of *bond energy* (1= lowest; 4 = highest)

P-F

P-Cl

P-I

P-Br

**Question 5** **12 points**

a. 9 points For the molecules HClO_4 and TeF_6 , draw the most stable *Lewis structure* showing appropriate shape and *dashed/wedged bonds*, based on VSEPR. Determine the **molecular geometry** around the central atom. For HClO_4 , label the *most polar bond*.



b. 3 points (Note: Lone pairs are not shown in structures below)

(i) For the three *resonance* structures (1, 2 and 3) shown below, answer the following question:



(I) Circle the *most contributing* resonance structure.

Structure 1

Structure 2

Structure 3

(ii) For the three structures given (4, 5, and 6) below, answer the following questions:



(I) Circle all the structures that represent *resonance structures* of each other:

Structure 4

Structure 5

Structure 6

(II) Among the resonance structures, circle the *most contributing* resonance structure.

Structure 4

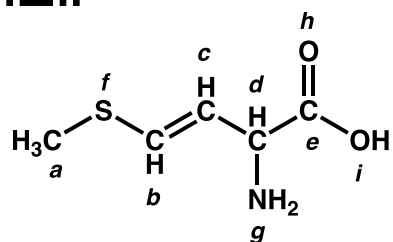
Structure 5

Structure 6

**Question 6** **6 points**

Using the data given below, calculate the **lattice energy** for SrO(s).

	ΔH° (kJ/mol)
$\text{Sr}(s) \longrightarrow \text{Sr}(g)$	+180
$\text{Sr}(g) \longrightarrow \text{Sr}^{2+}(g) + 2e^-$	+1468
$\text{O}_2(g) \longrightarrow 2\text{O}(g)$	+500
$\text{O}(g) + 2e^- \longrightarrow \text{O}^{2-}(g)$	+600
$\text{Sr}(s) + \frac{1}{2}\text{O}_2(g) \longrightarrow \text{SrO}(s)$	-550

**Question 7 10 points**

Consider the molecule given on the left, and answer questions (i) to (vi) using chemical bonding theories that we have learnt in class (valence bond theory, hybridization).

Note that the lone pairs are not shown in the structure:

i. Consider the carbon atoms marked in as ***a***, ***b***, ***c***, ***d***, and ***e*** in the molecule, and determine **hybridization** at each of the carbon centers. (2.5 points)

Atom	Hybridization
Carbon (<i>a</i>)	
Carbon (<i>b</i>)	
Carbon (<i>c</i>)	
Carbon (<i>d</i>)	
Carbon (<i>e</i>)	

ii. Now, consider the sulfur atom (S) marked as ***f***, nitrogen (N) atom (***g***), and oxygen(O) atoms (***h*** and ***i***), and determine their **hybridization**. (2 points)

Atom	Hybridization
Sulfur (<i>f</i>)	
Nitrogen (<i>g</i>)	
Oxygen (<i>h</i>)	
Oxygen (<i>i</i>)	

iii. **Describe** all the bonds formed by the carbon atoms ***a*** and ***b***, using sigma and pi designation as appropriate. (2 points)

Atom	Bonds formed (use sigma and pi as appropriate)
Carbon (<i>a</i>)	
Carbon (<i>b</i>)	

iv. How many total sp^3 -s sigma bonds are in the molecule? (1 point)

v. Determine **electronic** and **molecular geometry** at carbon (C) labeled ***e*** and sulfur (S) labeled ***f*** centres. (2 points)

Atom	Electronic Geometry	Molecular Geometry
Carbon (<i>e</i>)		
Sulfur (<i>f</i>)		

vi. Circle the value from the choices given below, that describes the **C-S-C bond angle at sulfur** (S) marked ***f***: (0.5 points)

>90° to <109.5° 109.5° >109.5 to <120° 120° >120° to <180°


Question 8 12 points

Using the **valence shell orbitals only** draw a **complete** molecular orbital (MO) diagram (with no s-p mixing) for the molecule **PS**, in the space provided below. **(6 points)**

Assume that 3s and 3p atomic orbitals will form molecular orbitals like those formed by the combination of 2s and 2p atomic orbitals that we learnt in class, and there is no s-p mixing.

Show **energy arrow**, **atomic orbitals** involved, **atom labels**, the **molecular orbitals** that are formed, **filled electrons** in the MO diagram etc. Make sure to label all the atomic and molecular orbitals.

Using the MO diagram above, answer the following questions:

- (i) Which one of the following will have the highest bond order? Circle the correct choice
 A) PS^{1-} B) PS C) PS^{1+}
- (ii) Which one of the following will have the longest bond length? Circle the correct choice
 A) PS^{1-} B) PS C) PS^{1+}
- (iii) Which of the following will be diamagnetic? Circle the correct choice.
 A) PS^{1-} B) PS C) PS^{1+}
- (iv) Draw Lewis dot structure of the molecule with the shortest bond length (based on the MO diagram) from among the three (PS^{1-} , PS, PS^{1+}), and indicate the bond order calculated from the Lewis dot structure. **(3 points)**

Lewis Dot Structure

Bond Order



8CFEBCC6-30D9-495E-A403-B6A7F0C26A4C

midterm-exam-93388

#568 Page 10 of 10

SCRAP PAGE

**CHEMISTRY 110 –Midterm Exam****Thursday November 7, 2024, 6:30 – 8:30 pm****Examiners: Pallavi Sirjoosingh and Ashok Kakkar**

First name (please write as legibly as possible within the boxes)

--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--

Last name

--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--

Student ID number

--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--

This booklet has 8 questions on 10 pages (The last page can be used as a scrap page).

The staple must NOT be removed from the question booklet. Your intact question booklet must be returned at the end of the exam.

A separate booklet contains a periodic table, and datasheet.

Write your answers directly under the questions in the space provided. Do not write on the QR codes or near the top edges (as it may be cut off).

Please write **legibly and dark enough** for the scan to be readable. For students who use English as a second language, dictionaries are permitted. Calculators are allowed.

DO NOT ASK QUESTIONS OF THE INVIGILATORS about interpretations of the exam questions. If you have a concern about a question, please summarize it at the top of this page. All concerns will be evaluated after the exam. GOOD LUCK!

**Question 1 8 points** (*Assume all transitions are possible*)

a. 6 points Calculate the **energy (in kJ/mol)** required to ionize (completely remove) the electron from the **n=3** level of **He⁺ ion** (He: Helium). Show your work including equations. (3 sig. figs.)

b. 2 points

(I) Select the largest (longest) wavelength from the three choices below:

- i. Wavelength of photon to *completely remove* an electron from *1s orbital of Hydrogen atom*
- ii. Wavelength of photon to *completely remove* an electron from *1s orbital of Boron atom (B)*
- iii. Wavelength of photon to *completely remove* an electron from *1s orbital of Boron cation (B⁺)*

i or ii or iii

(II) Select the smallest (shortest) wavelength from the three choices below (*for H atom*):

- i. Wavelength of photon *emitted* for an electron transitioning from *2p to 1s orbital*
- ii. Wavelength of photon *absorbed* for an electron transitioning from *2s to 4p orbital*
- iii. Wavelength of photon *absorbed* for an electron transitioning from *1s to 3p orbital*

i or ii or iii



Question 2 8 points



A Chem110 student plots the graph shown on the left, from data obtained for the kinetic energy (eV) of the ejected electron versus frequency (s^{-1}) of the incident light, for an unknown metal. ($1\text{eV} = 1.6022 \times 10^{-19} \text{ J}$)

Based on the plot, answer the questions **a to c** below:

a. 1 point On the plot above, label the **threshold frequency**.

b. 2 points What is the value of the work potential (in J) of the unknown metal?

c. 3 points What is the *longest* wavelength (in nm) required to eject an electron from this metal's surface? Show your work including equation(s) used. (3 sig figs)

d. 2 points In the space provided below, plot the following

(Assume frequency is *above* threshold frequency):

- Kinetic energy* of an ejected electron versus the *intensity* of the incident light (left)
- Number of electrons* ejected versus the *frequency* of the incident light (right)



**Question 3** **6 points****a. 2 points**

In the ground state of Ge (Germanium; $Z = 32$) atom, **name** the following orbitals:

i. Orbital containing the electron that requires the maximum energy to be removed.

ii. Orbital containing the electron that experiences the minimum effective nuclear charge.

b. 2 points

An unknown anion (A^-) has the following *excited* electronic configuration: $[Ar] 4s^2 3d^3 4p^1$

(Ar: Argon; $Z = 18$)

Give the **condensed ground state electronic configuration** of the atom, **A** and the cation, **A^{2+}** .
Show your work. (Assume all electronic transitions are possible)

Electronic Configuration of A:

Electronic Configuration of A^{2+} :

c. 2 points

In a single atom, determine how many electrons can have the following set of quantum numbers?

(i) $n = 5; l = 3$

In a single atom, determine how many electrons can have the following set of quantum numbers?

(i) $n = 3; l = 2; m_s = +1/2$

**Question 4** **8 points**

Rank the following from 1 to 4 in **increasing** order. Provide the ranking in the boxes provided.

Note: Atomic number (Z) are provided for relevant atoms

a) Increasing (lowest to highest) order of *ionization energy*: (1= lowest; 4 = highest)

Mg

Mg: Z = 12

Rb

Rb: Z = 37

Cs

Cs: Z = 55

K

K : Z = 19

b) Increasing (lowest to highest) order of *bond angle*: (1= lowest; 4 = highest)

H₂OCCl₄XeF₄

HCN

c) Increasing (smallest to largest) order of *ionic size* (1= smallest; 4 = largest)

S²⁻

S: Z = 16

O²⁻

O: Z = 8

Be²⁺

Be: Z = 4

Na⁺

Na: Z = 11

d) Increasing (lowest to highest) order of *bond energy* (1= lowest; 4 = highest)

P-F

P-Cl

P-I

P-Br

**Question 5** **12 points**

a. 9 points For the molecules HClO_4 and TeF_6 , draw the most stable *Lewis structure* showing appropriate shape and *dashed/wedged bonds*, based on VSEPR. Determine the **molecular geometry** around the central atom. For HClO_4 , label the *most polar bond*.



b. 3 points (Note: Lone pairs are not shown in structures below)

(i) For the three *resonance* structures (1, 2 and 3) shown below, answer the following question:



(I) Circle the *most contributing* resonance structure.

Structure 1

Structure 2

Structure 3

(ii) For the three structures given (4, 5, and 6) below, answer the following questions:



(I) Circle all the structures that represent *resonance structures* of each other:

Structure 4

Structure 5

Structure 6

(II) Among the resonance structures, circle the *most contributing* resonance structure.

Structure 4

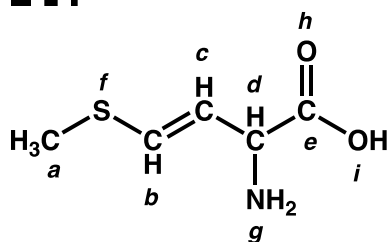
Structure 5

Structure 6

**Question 6** **6 points**

Using the data given below, calculate the **lattice energy** for SrO(s).

	ΔH° (kJ/mol)
$\text{Sr}(s) \longrightarrow \text{Sr}(g)$	+180
$\text{Sr}(g) \longrightarrow \text{Sr}^{2+}(g) + 2e^-$	+1468
$\text{O}_2(g) \longrightarrow 2\text{O}(g)$	+500
$\text{O}(g) + 2e^- \longrightarrow \text{O}^{2-}(g)$	+600
$\text{Sr}(s) + \frac{1}{2}\text{O}_2(g) \longrightarrow \text{SrO}(s)$	-550

**Question 7 10 points**

Consider the molecule given on the left, and answer questions (i) to (vi) using chemical bonding theories that we have learnt in class (valence bond theory, hybridization).

Note that the lone pairs are not shown in the structure:

i. Consider the carbon atoms marked in as ***a***, ***b***, ***c***, ***d***, and ***e*** in the molecule, and determine **hybridization** at each of the carbon centers. (2.5 points)

Atom	Hybridization
Carbon (<i>a</i>)	
Carbon (<i>b</i>)	
Carbon (<i>c</i>)	
Carbon (<i>d</i>)	
Carbon (<i>e</i>)	

ii. Now, consider the sulfur atom (S) marked as ***f***, nitrogen (N) atom (***g***), and oxygen(O) atoms (***h*** and ***i***), and determine their **hybridization**. (2 points)

Atom	Hybridization
Sulfur (<i>f</i>)	
Nitrogen (<i>g</i>)	
Oxygen (<i>h</i>)	
Oxygen (<i>i</i>)	

iii. **Describe** all the bonds formed by the carbon atoms ***a*** and ***b***, using sigma and pi designation as appropriate. (2 points)

Atom	Bonds formed (use sigma and pi as appropriate)
Carbon (<i>a</i>)	
Carbon (<i>b</i>)	

iv. How many total sp^3 -s sigma bonds are in the molecule? (1 point)

v. Determine **electronic** and **molecular geometry** at carbon (C) labeled ***e*** and sulfur (S) labeled ***f*** centres. (2 points)

Atom	Electronic Geometry	Molecular Geometry
Carbon (<i>e</i>)		
Sulfur (<i>f</i>)		

vi. Circle the value from the choices given below, that describes the **C-S-C bond angle at sulfur** (S) marked ***f***: (0.5 points)

>90° to <109.5° 109.5° >109.5 to <120° 120° >120° to <180°

**Question 8 12 points**

Using the **valence shell orbitals only** draw a **complete** molecular orbital (MO) diagram (with no s-p mixing) for the molecule **PS**, in the space provided below. **(6 points)**

Assume that 3s and 3p atomic orbitals will form molecular orbitals like those formed by the combination of 2s and 2p atomic orbitals that we learnt in class, and there is no s-p mixing.

Show **energy arrow**, **atomic orbitals** involved, **atom labels**, the **molecular orbitals** that are formed, **filled electrons** in the MO diagram etc. Make sure to label all the atomic and molecular orbitals.

Using the MO diagram above, answer the following questions:

- (i) Which one of the following will have the highest bond order? Circle the correct choice
A) PS^{1-} B) PS C) PS^{1+}
- (ii) Which one of the following will have the longest bond length? Circle the correct choice
A) PS^{1-} B) PS C) PS^{1+}
- (iii) Which of the following will be diamagnetic? Circle the correct choice.
A) PS^{1-} B) PS C) PS^{1+}
- (iv) Draw Lewis dot structure of the molecule with the shortest bond length (based on the MO diagram) from among the three (PS^{1-} , PS, PS^{1+}), and indicate the bond order calculated from the Lewis dot structure. **(3 points)**

Lewis Dot Structure

Bond Order



CD213E62-8763-40A0-A7D4-E2A4E8D22198

midterm-exam-93388

#569 Page 10 of 10

SCRAP PAGE

**CHEMISTRY 110 –Midterm Exam****Thursday November 7, 2024, 6:30 – 8:30 pm****Examiners: Pallavi Sirjoosingh and Ashok Kakkar**

First name (please write as legibly as possible within the boxes)

--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--

Last name

--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--

Student ID number

--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--

This booklet has 8 questions on 10 pages (The last page can be used as a scrap page).

The staple must NOT be removed from the question booklet. Your intact question booklet must be returned at the end of the exam.

A separate booklet contains a periodic table, and datasheet.

Write your answers directly under the questions in the space provided. Do not write on the QR codes or near the top edges (as it may be cut off).

Please write **legibly and dark enough** for the scan to be readable. For students who use English as a second language, dictionaries are permitted. Calculators are allowed.

DO NOT ASK QUESTIONS OF THE INVIGILATORS about interpretations of the exam questions. If you have a concern about a question, please summarize it at the top of this page. All concerns will be evaluated after the exam. GOOD LUCK!

**Question 1 8 points** (*Assume all transitions are possible*)

a. 6 points Calculate the **energy (in kJ/mol)** required to ionize (completely remove) the electron from the **$n=3$** level of **He^+ ion** (He: Helium). Show your work including equations. (3 sig. figs.)

b. 2 points

(I) Select the largest (longest) wavelength from the three choices below:

- i. Wavelength of photon to *completely remove* an electron from *1s orbital of Hydrogen atom*
- ii. Wavelength of photon to *completely remove* an electron from *1s orbital of Boron atom (B)*
- iii. Wavelength of photon to *completely remove* an electron from *1s orbital of Boron cation (B^+)*

i or ii or iii

(II) Select the smallest (shortest) wavelength from the three choices below (*for H atom*):

- i. Wavelength of photon *emitted* for an electron transitioning from *2p to 1s orbital*
- ii. Wavelength of photon *absorbed* for an electron transitioning from *2s to 4p orbital*
- iii. Wavelength of photon *absorbed* for an electron transitioning from *1s to 3p orbital*

i or ii or iii



Question 2 8 points



A Chem110 student plots the graph shown on the left, from data obtained for the kinetic energy (eV) of the ejected electron versus frequency (s^{-1}) of the incident light, for an unknown metal. ($1\text{eV} = 1.6022 \times 10^{-19} \text{ J}$)

Based on the plot, answer the questions **a to c** below:

a. 1 point On the plot above, label the **threshold frequency**.

b. 2 points What is the value of the work potential (in J) of the unknown metal?

c. 3 points What is the *longest* wavelength (in nm) required to eject an electron from this metal's surface? Show your work including equation(s) used. (3 sig figs)

d. 2 points In the space provided below, plot the following

(Assume frequency is *above* threshold frequency):

- i. *Kinetic energy* of an ejected electron versus the *intensity* of the incident light (left)
- ii. *Number of electrons* ejected versus the *frequency* of the incident light (right)



**Question 3** **6 points****a. 2 points**

In the ground state of Ge (Germanium; $Z = 32$) atom, **name** the following orbitals:

i. Orbital containing the electron that requires the maximum energy to be removed.

ii. Orbital containing the electron that experiences the minimum effective nuclear charge.

b. 2 points

An unknown anion (A^-) has the following *excited* electronic configuration: $[Ar] 4s^2 3d^3 4p^1$

(Ar: Argon; $Z = 18$)

Give the **condensed ground state electronic configuration** of the atom, **A** and the cation, **A^{2+}** .
Show your work. (Assume all electronic transitions are possible)

Electronic Configuration of A:

Electronic Configuration of A^{2+} :

c. 2 points

In a single atom, determine how many electrons can have the following set of quantum numbers?

(i) $n = 5; l = 3$

In a single atom, determine how many electrons can have the following set of quantum numbers?

(i) $n = 3; l = 2; m_s = +1/2$

**Question 4** **8 points**

Rank the following from 1 to 4 in **increasing** order. Provide the ranking in the boxes provided.

Note: Atomic number (Z) are provided for relevant atoms

a) Increasing (lowest to highest) order of *ionization energy*: (1= lowest; 4 = highest)

Mg

Mg: Z = 12

Rb

Rb: Z = 37

Cs

Cs: Z = 55

K

K : Z = 19

b) Increasing (lowest to highest) order of *bond angle*: (1= lowest; 4 = highest)

H₂OCCl₄XeF₄

HCN

c) Increasing (smallest to largest) order of *ionic size* (1= smallest; 4 = largest)

S²⁻

S: Z = 16

O²⁻

O: Z = 8

Be²⁺

Be: Z = 4

Na⁺

Na: Z = 11

d) Increasing (lowest to highest) order of *bond energy* (1= lowest; 4 = highest)

P-F

P-Cl

P-I

P-Br

**Question 5** **12 points**

a. 9 points For the molecules HClO_4 and TeF_6 , draw the most stable *Lewis structure* showing appropriate shape and *dashed/wedged bonds*, based on VSEPR. Determine the **molecular geometry** around the central atom. For HClO_4 , label the *most polar bond*.



b. 3 points (Note: Lone pairs are not shown in structures below)

(i) For the three *resonance* structures (1, 2 and 3) shown below, answer the following question:



(I) Circle the *most contributing* resonance structure.

Structure 1

Structure 2

Structure 3

(ii) For the three structures given (4, 5, and 6) below, answer the following questions:



(I) Circle all the structures that represent *resonance structures* of each other:

Structure 4

Structure 5

Structure 6

(II) Among the resonance structures, circle the *most contributing* resonance structure.

Structure 4

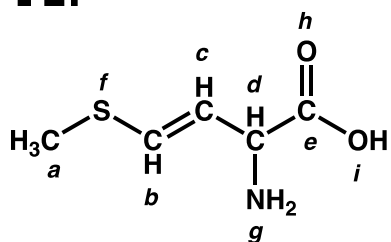
Structure 5

Structure 6

**Question 6** **6 points**

Using the data given below, calculate the **lattice energy** for SrO(s).

	ΔH° (kJ/mol)
$\text{Sr}(s) \longrightarrow \text{Sr}(g)$	+180
$\text{Sr}(g) \longrightarrow \text{Sr}^{2+}(g) + 2e^-$	+1468
$\text{O}_2(g) \longrightarrow 2\text{O}(g)$	+500
$\text{O}(g) + 2e^- \longrightarrow \text{O}^{2-}(g)$	+600
$\text{Sr}(s) + \frac{1}{2}\text{O}_2(g) \longrightarrow \text{SrO}(s)$	-550

**Question 7 10 points**

Consider the molecule given on the left, and answer questions (i) to (vi) using chemical bonding theories that we have learnt in class (valence bond theory, hybridization).

Note that the lone pairs are not shown in the structure:

i. Consider the carbon atoms marked in as ***a***, ***b***, ***c***, ***d***, and ***e*** in the molecule, and determine **hybridization** at each of the carbon centers. (2.5 points)

Atom	Hybridization
Carbon (<i>a</i>)	
Carbon (<i>b</i>)	
Carbon (<i>c</i>)	
Carbon (<i>d</i>)	
Carbon (<i>e</i>)	

ii. Now, consider the sulfur atom (S) marked as ***f***, nitrogen (N) atom (***g***), and oxygen(O) atoms (***h*** and ***i***), and determine their **hybridization**. (2 points)

Atom	Hybridization
Sulfur (<i>f</i>)	
Nitrogen (<i>g</i>)	
Oxygen (<i>h</i>)	
Oxygen (<i>i</i>)	

iii. **Describe** all the bonds formed by the carbon atoms ***a*** and ***b***, using sigma and pi designation as appropriate. (2 points)

Atom	Bonds formed (use sigma and pi as appropriate)
Carbon (<i>a</i>)	
Carbon (<i>b</i>)	

iv. How many total sp^3 -s sigma bonds are in the molecule? (1 point)

v. Determine **electronic** and **molecular geometry** at carbon (C) labeled ***e*** and sulfur (S) labeled ***f*** centres. (2 points)

Atom	Electronic Geometry	Molecular Geometry
Carbon (<i>e</i>)		
Sulfur (<i>f</i>)		

vi. Circle the value from the choices given below, that describes the **C-S-C bond angle at sulfur** (S) marked ***f***: (0.5 points)

>90° to <109.5° 109.5° >109.5 to <120° 120° >120° to <180°



Question 8 12 points

Using the **valence shell orbitals only** draw a **complete** molecular orbital (MO) diagram (with no s-p mixing) for the molecule **PS**, in the space provided below. **(6 points)**

Assume that 3s and 3p atomic orbitals will form molecular orbitals like those formed by the combination of 2s and 2p atomic orbitals that we learnt in class, and there is no s-p mixing.

Show **energy arrow**, **atomic orbitals** involved, **atom labels**, the **molecular orbitals** that are formed, **filled electrons** in the MO diagram etc. Make sure to label all the atomic and molecular orbitals.

Using the MO diagram above, answer the following questions:

- (i) Which one of the following will have the highest bond order? Circle the correct choice
 A) PS^{1-} B) PS C) PS^{1+}
- (ii) Which one of the following will have the longest bond length? Circle the correct choice
 A) PS^{1-} B) PS C) PS^{1+}
- (iii) Which of the following will be diamagnetic? Circle the correct choice.
 A) PS^{1-} B) PS C) PS^{1+}
- (iv) Draw Lewis dot structure of the molecule with the shortest bond length (based on the MO diagram) from among the three (PS^{1-} , PS, PS^{1+}), and indicate the bond order calculated from the Lewis dot structure. **(3 points)**

Lewis Dot Structure

Bond Order



D850BE0A-241B-4F5D-A358-769AF38FBED3

midterm-exam-93388

#570 Page 10 of 10

SCRAP PAGE

**CHEMISTRY 110 –Midterm Exam****Thursday November 7, 2024, 6:30 – 8:30 pm****Examiners: Pallavi Sirjoosingh and Ashok Kakkar**

First name (please write as legibly as possible within the boxes)

--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--

Last name

--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--

Student ID number

--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--

This booklet has 8 questions on 10 pages (The last page can be used as a scrap page).

The staple must NOT be removed from the question booklet. Your intact question booklet must be returned at the end of the exam.

A separate booklet contains a periodic table, and datasheet.

Write your answers directly under the questions in the space provided. Do not write on the QR codes or near the top edges (as it may be cut off).

Please write **legibly and dark enough** for the scan to be readable. For students who use English as a second language, dictionaries are permitted. Calculators are allowed.

DO NOT ASK QUESTIONS OF THE INVIGILATORS about interpretations of the exam questions. If you have a concern about a question, please summarize it at the top of this page. All concerns will be evaluated after the exam. GOOD LUCK!

**Question 1 8 points** (*Assume all transitions are possible*)

a. 6 points Calculate the **energy (in kJ/mol)** required to ionize (completely remove) the electron from the **$n=3$** level of **He^+ ion** (He: Helium). Show your work including equations. (3 sig. figs.)

b. 2 points

(I) Select the largest (longest) wavelength from the three choices below:

- i. Wavelength of photon to *completely remove* an electron from *1s orbital of Hydrogen atom*
- ii. Wavelength of photon to *completely remove* an electron from *1s orbital of Boron atom (B)*
- iii. Wavelength of photon to *completely remove* an electron from *1s orbital of Boron cation (B^+)*

i or ii or iii

(II) Select the smallest (shortest) wavelength from the three choices below (*for H atom*):

- i. Wavelength of photon *emitted* for an electron transitioning from *2p to 1s orbital*
- ii. Wavelength of photon *absorbed* for an electron transitioning from *2s to 4p orbital*
- iii. Wavelength of photon *absorbed* for an electron transitioning from *1s to 3p orbital*

i or ii or iii



Question 2 8 points



A Chem110 student plots the graph shown on the left, from data obtained for the kinetic energy (eV) of the ejected electron versus frequency (s^{-1}) of the incident light, for an unknown metal. ($1\text{eV} = 1.6022 \times 10^{-19} \text{ J}$)

Based on the plot, answer the questions **a to c** below:

a. 1 point On the plot above, label the **threshold frequency**.

b. 2 points What is the value of the work potential (in J) of the unknown metal?

c. 3 points What is the *longest* wavelength (in nm) required to eject an electron from this metal's surface? Show your work including equation(s) used. (3 sig figs)

d. 2 points In the space provided below, plot the following

(Assume frequency is *above* threshold frequency):

- Kinetic energy* of an ejected electron versus the *intensity* of the incident light (left)
- Number of electrons* ejected versus the *frequency* of the incident light (right)



**Question 3** **6 points****a. 2 points**

In the ground state of Ge (Germanium; $Z = 32$) atom, **name** the following orbitals:

i. Orbital containing the electron that requires the maximum energy to be removed.

ii. Orbital containing the electron that experiences the minimum effective nuclear charge.

b. 2 points

An unknown anion (A^-) has the following *excited* electronic configuration: $[Ar] 4s^2 3d^3 4p^1$

(Ar: Argon; $Z = 18$)

Give the **condensed ground state electronic configuration** of the atom, **A** and the cation, **A^{2+}** .
Show your work. (Assume all electronic transitions are possible)

Electronic Configuration of A:

Electronic Configuration of A^{2+} :

c. 2 points

In a single atom, determine how many electrons can have the following set of quantum numbers?

(i) $n = 5; l = 3$

In a single atom, determine how many electrons can have the following set of quantum numbers?

(i) $n = 3; l = 2; m_s = +1/2$

**Question 4** **8 points**

Rank the following from 1 to 4 in **increasing** order. Provide the ranking in the boxes provided.

Note: Atomic number (Z) are provided for relevant atoms

a) Increasing (lowest to highest) order of *ionization energy*: (1= lowest; 4 = highest)

Mg

Mg: Z = 12

Rb

Rb: Z = 37

Cs

Cs: Z = 55

K

K : Z = 19

b) Increasing (lowest to highest) order of *bond angle*: (1= lowest; 4 = highest)

H₂OCCl₄XeF₄

HCN

c) Increasing (smallest to largest) order of *ionic size* (1= smallest; 4 = largest)

S²⁻

S: Z = 16

O²⁻

O: Z = 8

Be²⁺

Be: Z = 4

Na⁺

Na: Z = 11

d) Increasing (lowest to highest) order of *bond energy* (1= lowest; 4 = highest)

P-F

P-Cl

P-I

P-Br

**Question 5** **12 points**

a. 9 points For the molecules HClO_4 and TeF_6 , draw the most stable *Lewis structure* showing appropriate shape and *dashed/wedged bonds*, based on VSEPR. Determine the **molecular geometry** around the central atom. For HClO_4 , label the *most polar bond*.



b. 3 points (Note: Lone pairs are not shown in structures below)

(i) For the three *resonance* structures (1, 2 and 3) shown below, answer the following question:



(I) Circle the *most contributing* resonance structure.

Structure 1

Structure 2

Structure 3

(ii) For the three structures given (4, 5, and 6) below, answer the following questions:



(I) Circle all the structures that represent *resonance structures* of each other:

Structure 4

Structure 5

Structure 6

(II) Among the resonance structures, circle the *most contributing* resonance structure.

Structure 4

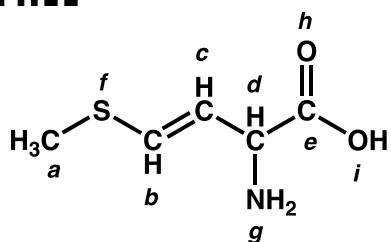
Structure 5

Structure 6

**Question 6** **6 points**

Using the data given below, calculate the **lattice energy** for SrO(s).

	ΔH° (kJ/mol)
$\text{Sr}(s) \longrightarrow \text{Sr}(g)$	+180
$\text{Sr}(g) \longrightarrow \text{Sr}^{2+}(g) + 2e^-$	+1468
$\text{O}_2(g) \longrightarrow 2\text{O}(g)$	+500
$\text{O}(g) + 2e^- \longrightarrow \text{O}^{2-}(g)$	+600
$\text{Sr}(s) + \frac{1}{2}\text{O}_2(g) \longrightarrow \text{SrO}(s)$	-550

**Question 7 10 points**

Consider the molecule given on the left, and answer questions (i) to (vi) using chemical bonding theories that we have learnt in class (valence bond theory, hybridization).

Note that the lone pairs are not shown in the structure:

i. Consider the carbon atoms marked in as ***a***, ***b***, ***c***, ***d***, and ***e*** in the molecule, and determine **hybridization** at each of the carbon centers. (2.5 points)

Atom	Hybridization
Carbon (<i>a</i>)	
Carbon (<i>b</i>)	
Carbon (<i>c</i>)	
Carbon (<i>d</i>)	
Carbon (<i>e</i>)	

ii. Now, consider the sulfur atom (S) marked as ***f***, nitrogen (N) atom (***g***), and oxygen(O) atoms (***h*** and ***i***), and determine their **hybridization**. (2 points)

Atom	Hybridization
Sulfur (<i>f</i>)	
Nitrogen (<i>g</i>)	
Oxygen (<i>h</i>)	
Oxygen (<i>i</i>)	

iii. **Describe** all the bonds formed by the carbon atoms ***a*** and ***b***, using sigma and pi designation as appropriate. (2 points)

Atom	Bonds formed (use sigma and pi as appropriate)
Carbon (<i>a</i>)	
Carbon (<i>b</i>)	

iv. How many total sp^3 -s sigma bonds are in the molecule? (1 point)

v. Determine **electronic** and **molecular geometry** at carbon (C) labeled ***e*** and sulfur (S) labeled ***f*** centres. (2 points)

Atom	Electronic Geometry	Molecular Geometry
Carbon (<i>e</i>)		
Sulfur (<i>f</i>)		

vi. Circle the value from the choices given below, that describes the **C-S-C bond angle at sulfur** (S) marked ***f***: (0.5 points)

>90° to <109.5°

109.5°

>109.5 to <120°

120°

>120° to <180°


Question 8 12 points

Using the **valence shell orbitals only** draw a **complete** molecular orbital (MO) diagram (with no s-p mixing) for the molecule **PS**, in the space provided below. **(6 points)**

Assume that 3s and 3p atomic orbitals will form molecular orbitals like those formed by the combination of 2s and 2p atomic orbitals that we learnt in class, and there is no s-p mixing.

Show **energy arrow**, **atomic orbitals** involved, **atom labels**, the **molecular orbitals** that are formed, **filled electrons** in the MO diagram etc. Make sure to **label all the atomic and molecular orbitals**.

Using the MO diagram above, answer the following questions:

- (i) Which one of the following will have the highest bond order? Circle the correct choice
 A) PS^{1-} B) PS C) PS^{1+}
- (ii) Which one of the following will have the longest bond length? Circle the correct choice
 A) PS^{1-} B) PS C) PS^{1+}
- (iii) Which of the following will be diamagnetic? Circle the correct choice.
 A) PS^{1-} B) PS C) PS^{1+}
- (iv) Draw Lewis dot structure of the molecule with the shortest bond length (based on the MO diagram) from among the three (PS^{1-} , PS, PS^{1+}), and indicate the bond order calculated from the Lewis dot structure. **(3 points)**

Lewis Dot Structure

Bond Order



D178C1F9-E0BA-4A1A-BCE3-DABF853E77CD

midterm-exam-93388

#571 Page 10 of 10

SCRAP PAGE

**CHEMISTRY 110 –Midterm Exam****Thursday November 7, 2024, 6:30 – 8:30 pm**

Examiners: Pallavi Sirjoosingh and Ashok Kakkar

First name (please write as legibly as possible within the boxes)

--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--

Last name

--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--

Student ID number

--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--

This booklet has 8 questions on 10 pages (The last page can be used as a scrap page).

The staple must NOT be removed from the question booklet. Your intact question booklet must be returned at the end of the exam.

A separate booklet contains a periodic table, and datasheet.

Write your answers directly under the questions in the space provided. Do not write on the QR codes or near the top edges (as it may be cut off).

Please write **legibly and dark enough** for the scan to be readable. For students who use English as a second language, dictionaries are permitted. Calculators are allowed.

DO NOT ASK QUESTIONS OF THE INVIGILATORS about interpretations of the exam questions. If you have a concern about a question, please summarize it at the top of this page. All concerns will be evaluated after the exam. GOOD LUCK!

**Question 1 8 points** (*Assume all transitions are possible*)

a. 6 points Calculate the **energy (in kJ/mol)** required to ionize (completely remove) the electron from the **n=3** level of **He⁺ ion** (He: Helium). Show your work including equations. (3 sig. figs.)

b. 2 points

(I) Select the largest (longest) wavelength from the three choices below:

- i. Wavelength of photon to *completely remove* an electron from *1s orbital of Hydrogen atom*
- ii. Wavelength of photon to *completely remove* an electron from *1s orbital of Boron atom (B)*
- iii. Wavelength of photon to *completely remove* an electron from *1s orbital of Boron cation (B⁺)*

i or ii or iii

(II) Select the smallest (shortest) wavelength from the three choices below (*for H atom*):

- i. Wavelength of photon *emitted* for an electron transitioning from *2p to 1s orbital*
- ii. Wavelength of photon *absorbed* for an electron transitioning from *2s to 4p orbital*
- iii. Wavelength of photon *absorbed* for an electron transitioning from *1s to 3p orbital*

i or ii or iii



Question 2 8 points



A Chem110 student plots the graph shown on the left, from data obtained for the kinetic energy (eV) of the ejected electron versus frequency (s^{-1}) of the incident light, for an unknown metal. ($1\text{eV} = 1.6022 \times 10^{-19} \text{ J}$)

Based on the plot, answer the questions **a to c** below:

a. 1 point On the plot above, label the **threshold frequency**.

b. 2 points What is the value of the work potential (in J) of the unknown metal?

c. 3 points What is the *longest* wavelength (in nm) required to eject an electron from this metal's surface? Show your work including equation(s) used. (3 sig figs)

d. 2 points In the space provided below, plot the following

(Assume frequency is *above* threshold frequency):

- i. *Kinetic energy* of an ejected electron versus the *intensity* of the incident light (left)
- ii. *Number of electrons* ejected versus the *frequency* of the incident light (right)



**Question 3** **6 points****a. 2 points**

In the ground state of Ge (Germanium; $Z = 32$) atom, **name** the following orbitals:

i. Orbital containing the electron that requires the maximum energy to be removed.

ii. Orbital containing the electron that experiences the minimum effective nuclear charge.

b. 2 points

An unknown anion (A^-) has the following *excited* electronic configuration: $[Ar] 4s^2 3d^3 4p^1$

(Ar: Argon; $Z = 18$)

Give the **condensed ground state electronic configuration** of the atom, **A** and the cation, **A^{2+}** .
Show your work. (Assume all electronic transitions are possible)

Electronic Configuration of A:

Electronic Configuration of A^{2+} :

c. 2 points

In a single atom, determine how many electrons can have the following set of quantum numbers?

(i) $n = 5; l = 3$

In a single atom, determine how many electrons can have the following set of quantum numbers?

(i) $n = 3; l = 2; m_s = +1/2$

**Question 4** **8 points**

Rank the following from 1 to 4 in **increasing** order. Provide the ranking in the boxes provided.

Note: Atomic number (Z) are provided for relevant atoms

a) Increasing (lowest to highest) order of *ionization energy*: (1= lowest; 4 = highest)

Mg

Mg: Z = 12

Rb

Rb: Z = 37

Cs

Cs: Z = 55

K

K : Z = 19

b) Increasing (lowest to highest) order of *bond angle*: (1= lowest; 4 = highest)

H₂OCCl₄XeF₄

HCN

c) Increasing (smallest to largest) order of *ionic size* (1= smallest; 4 = largest)

S²⁻

S: Z = 16

O²⁻

O: Z = 8

Be²⁺

Be: Z = 4

Na⁺

Na: Z = 11

d) Increasing (lowest to highest) order of *bond energy* (1= lowest; 4 = highest)

P-F

P-Cl

P-I

P-Br

**Question 5** **12 points**

a. 9 points For the molecules HClO_4 and TeF_6 , draw the most stable *Lewis structure* showing appropriate shape and *dashed/wedged bonds*, based on VSEPR. Determine the **molecular geometry** around the central atom. For HClO_4 , label the *most polar bond*.



b. 3 points (Note: Lone pairs are not shown in structures below)

(i) For the three *resonance* structures (1, 2 and 3) shown below, answer the following question:



(I) Circle the *most contributing* resonance structure.

Structure 1

Structure 2

Structure 3

(ii) For the three structures given (4, 5, and 6) below, answer the following questions:



(I) Circle all the structures that represent *resonance structures* of each other:

Structure 4

Structure 5

Structure 6

(II) Among the resonance structures, circle the *most contributing* resonance structure.

Structure 4

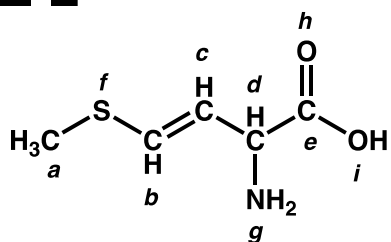
Structure 5

Structure 6

**Question 6** **6 points**

Using the data given below, calculate the **lattice energy** for SrO(s).

	ΔH° (kJ/mol)
$\text{Sr}(s) \longrightarrow \text{Sr}(g)$	+180
$\text{Sr}(g) \longrightarrow \text{Sr}^{2+}(g) + 2e^-$	+1468
$\text{O}_2(g) \longrightarrow 2\text{O}(g)$	+500
$\text{O}(g) + 2e^- \longrightarrow \text{O}^{2-}(g)$	+600
$\text{Sr}(s) + \frac{1}{2}\text{O}_2(g) \longrightarrow \text{SrO}(s)$	-550

**Question 7 10 points**

Consider the molecule given on the left, and answer questions (i) to (vi) using chemical bonding theories that we have learnt in class (valence bond theory, hybridization).

Note that the lone pairs are not shown in the structure:

i. Consider the carbon atoms marked in as ***a***, ***b***, ***c***, ***d***, and ***e*** in the molecule, and determine **hybridization** at each of the carbon centers. (2.5 points)

Atom	Hybridization
Carbon (<i>a</i>)	
Carbon (<i>b</i>)	
Carbon (<i>c</i>)	
Carbon (<i>d</i>)	
Carbon (<i>e</i>)	

ii. Now, consider the sulfur atom (S) marked as ***f***, nitrogen (N) atom (***g***), and oxygen(O) atoms (***h*** and ***i***), and determine their **hybridization**. (2 points)

Atom	Hybridization
Sulfur (<i>f</i>)	
Nitrogen (<i>g</i>)	
Oxygen (<i>h</i>)	
Oxygen (<i>i</i>)	

iii. **Describe** all the bonds formed by the carbon atoms ***a*** and ***b***, using sigma and pi designation as appropriate. (2 points)

Atom	Bonds formed (use sigma and pi as appropriate)
Carbon (<i>a</i>)	
Carbon (<i>b</i>)	

iv. How many total sp^3 -s sigma bonds are in the molecule? (1 point)

v. Determine **electronic** and **molecular geometry** at carbon (C) labeled ***e*** and sulfur (S) labeled ***f*** centres. (2 points)

Atom	Electronic Geometry	Molecular Geometry
Carbon (<i>e</i>)		
Sulfur (<i>f</i>)		

vi. Circle the value from the choices given below, that describes the **C-S-C bond angle at sulfur** (S) marked ***f***: (0.5 points)

>90° to <109.5°

109.5°

>109.5 to <120°

120°

>120° to <180°

**Question 8 12 points**

Using the **valence shell orbitals only** draw a **complete** molecular orbital (MO) diagram (with no s-p mixing) for the molecule **PS**, in the space provided below. **(6 points)**

Assume that 3s and 3p atomic orbitals will form molecular orbitals like those formed by the combination of 2s and 2p atomic orbitals that we learnt in class, and there is no s-p mixing.

Show **energy arrow**, **atomic orbitals** involved, **atom labels**, the **molecular orbitals** that are formed, **filled electrons** in the MO diagram etc. Make sure to label all the atomic and molecular orbitals.

Using the MO diagram above, answer the following questions:

- (i) Which one of the following will have the highest bond order? Circle the correct choice
A) PS^{1-} B) PS C) PS^{1+}
- (ii) Which one of the following will have the longest bond length? Circle the correct choice
A) PS^{1-} B) PS C) PS^{1+}
- (iii) Which of the following will be diamagnetic? Circle the correct choice.
A) PS^{1-} B) PS C) PS^{1+}
- (iv) Draw Lewis dot structure of the molecule with the shortest bond length (based on the MO diagram) from among the three (PS^{1-} , PS, PS^{1+}), and indicate the bond order calculated from the Lewis dot structure. **(3 points)**

Lewis Dot Structure

Bond Order



416E8D92-3AC0-42F6-A260-5413B217A63E

midterm-exam-93388

#572 Page 10 of 10

SCRAP PAGE

**CHEMISTRY 110 –Midterm Exam****Thursday November 7, 2024, 6:30 – 8:30 pm****Examiners: Pallavi Sirjoosingh and Ashok Kakkar**

First name (please write as legibly as possible within the boxes)

--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--

Last name

--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--

Student ID number

--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--

This booklet has 8 questions on 10 pages (The last page can be used as a scrap page).

The staple must NOT be removed from the question booklet. Your intact question booklet must be returned at the end of the exam.

A separate booklet contains a periodic table, and datasheet.

Write your answers directly under the questions in the space provided. Do not write on the QR codes or near the top edges (as it may be cut off).

Please write **legibly and dark enough** for the scan to be readable. For students who use English as a second language, dictionaries are permitted. Calculators are allowed.

DO NOT ASK QUESTIONS OF THE INVIGILATORS about interpretations of the exam questions. If you have a concern about a question, please summarize it at the top of this page. All concerns will be evaluated after the exam. GOOD LUCK!

**Question 1 8 points** (*Assume all transitions are possible*)

a. 6 points Calculate the **energy (in kJ/mol)** required to ionize (completely remove) the electron from the **n=3** level of **He⁺ ion** (He: Helium). Show your work including equations. (3 sig. figs.)

b. 2 points

(I) Select the largest (longest) wavelength from the three choices below:

- i. Wavelength of photon to *completely remove* an electron from *1s orbital of Hydrogen atom*
- ii. Wavelength of photon to *completely remove* an electron from *1s orbital of Boron atom (B)*
- iii. Wavelength of photon to *completely remove* an electron from *1s orbital of Boron cation (B⁺)*

i or ii or iii

(II) Select the smallest (shortest) wavelength from the three choices below (*for H atom*):

- i. Wavelength of photon *emitted* for an electron transitioning from *2p to 1s orbital*
- ii. Wavelength of photon *absorbed* for an electron transitioning from *2s to 4p orbital*
- iii. Wavelength of photon *absorbed* for an electron transitioning from *1s to 3p orbital*

i or ii or iii



Question 2 8 points



A Chem110 student plots the graph shown on the left, from data obtained for the kinetic energy (eV) of the ejected electron versus frequency (s^{-1}) of the incident light, for an unknown metal. ($1\text{eV} = 1.6022 \times 10^{-19} \text{ J}$)

Based on the plot, answer the questions **a to c** below:

a. 1 point On the plot above, label the **threshold frequency**.

b. 2 points What is the value of the work potential (in J) of the unknown metal?

c. 3 points What is the *longest* wavelength (in nm) required to eject an electron from this metal's surface? Show your work including equation(s) used. (3 sig figs)

d. 2 points In the space provided below, plot the following

(Assume frequency is *above* threshold frequency):

- i. *Kinetic energy* of an ejected electron versus the *intensity* of the incident light (left)
- ii. *Number of electrons* ejected versus the *frequency* of the incident light (right)



**Question 3** **6 points****a. 2 points**

In the ground state of Ge (Germanium; $Z = 32$) atom, **name** the following orbitals:

i. Orbital containing the electron that requires the maximum energy to be removed.

ii. Orbital containing the electron that experiences the minimum effective nuclear charge.

b. 2 points

An unknown anion (A^-) has the following *excited* electronic configuration: $[Ar] 4s^2 3d^3 4p^1$

(Ar: Argon; $Z = 18$)

Give the **condensed ground state electronic configuration** of the atom, **A** and the cation, **A^{2+}** .
Show your work. (Assume all electronic transitions are possible)

Electronic Configuration of A:

Electronic Configuration of A^{2+} :

c. 2 points

In a single atom, determine how many electrons can have the following set of quantum numbers?

(i) $n = 5; l = 3$

In a single atom, determine how many electrons can have the following set of quantum numbers?

(i) $n = 3; l = 2; m_s = +1/2$

**Question 4** **8 points**

Rank the following from 1 to 4 in **increasing** order. Provide the ranking in the boxes provided.

Note: Atomic number (Z) are provided for relevant atoms

a) Increasing (lowest to highest) order of *ionization energy*: (1= lowest; 4 = highest)

Mg

Mg: Z = 12

Rb

Rb: Z = 37

Cs

Cs: Z = 55

K

K : Z = 19

b) Increasing (lowest to highest) order of *bond angle*: (1= lowest; 4 = highest)

H₂OCCl₄XeF₄

HCN

c) Increasing (smallest to largest) order of *ionic size* (1= smallest; 4 = largest)

S²⁻

S: Z = 16

O²⁻

O: Z = 8

Be²⁺

Be: Z = 4

Na⁺

Na: Z = 11

d) Increasing (lowest to highest) order of *bond energy* (1= lowest; 4 = highest)

P-F

P-Cl

P-I

P-Br

**Question 5** **12 points**

a. 9 points For the molecules HClO_4 and TeF_6 , draw the most stable *Lewis structure* showing appropriate shape and *dashed/wedged bonds*, based on VSEPR. Determine the **molecular geometry** around the central atom. For HClO_4 , label the *most polar bond*.



b. 3 points (Note: Lone pairs are not shown in structures below)

(i) For the three *resonance* structures (1, 2 and 3) shown below, answer the following question:



(I) Circle the *most contributing* resonance structure.

Structure 1

Structure 2

Structure 3

(ii) For the three structures given (4, 5, and 6) below, answer the following questions:



(I) Circle all the structures that represent *resonance structures* of each other:

Structure 4

Structure 5

Structure 6

(II) Among the resonance structures, circle the *most contributing* resonance structure.

Structure 4

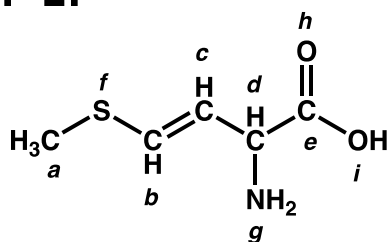
Structure 5

Structure 6

**Question 6** **6 points**

Using the data given below, calculate the **lattice energy** for SrO(s).

	ΔH° (kJ/mol)
$\text{Sr}(s) \longrightarrow \text{Sr}(g)$	+180
$\text{Sr}(g) \longrightarrow \text{Sr}^{2+}(g) + 2e^-$	+1468
$\text{O}_2(g) \longrightarrow 2\text{O}(g)$	+500
$\text{O}(g) + 2e^- \longrightarrow \text{O}^{2-}(g)$	+600
$\text{Sr}(s) + \frac{1}{2}\text{O}_2(g) \longrightarrow \text{SrO}(s)$	-550

**Question 7 10 points**

Consider the molecule given on the left, and answer questions (i) to (vi) using chemical bonding theories that we have learnt in class (valence bond theory, hybridization).

Note that the lone pairs are not shown in the structure:

i. Consider the carbon atoms marked in as ***a***, ***b***, ***c***, ***d***, and ***e*** in the molecule, and determine **hybridization** at each of the carbon centers. (2.5 points)

Atom	Hybridization
Carbon (<i>a</i>)	
Carbon (<i>b</i>)	
Carbon (<i>c</i>)	
Carbon (<i>d</i>)	
Carbon (<i>e</i>)	

ii. Now, consider the sulfur atom (S) marked as ***f***, nitrogen (N) atom (***g***), and oxygen(O) atoms (***h*** and ***i***), and determine their **hybridization**. (2 points)

Atom	Hybridization
Sulfur (<i>f</i>)	
Nitrogen (<i>g</i>)	
Oxygen (<i>h</i>)	
Oxygen (<i>i</i>)	

iii. **Describe** all the bonds formed by the carbon atoms ***a*** and ***b***, using sigma and pi designation as appropriate. (2 points)

Atom	Bonds formed (use sigma and pi as appropriate)
Carbon (<i>a</i>)	
Carbon (<i>b</i>)	

iv. How many total sp^3 -s sigma bonds are in the molecule? (1 point)

v. Determine **electronic** and **molecular geometry** at carbon (C) labeled ***e*** and sulfur (S) labeled ***f*** centres. (2 points)

Atom	Electronic Geometry	Molecular Geometry
Carbon (<i>e</i>)		
Sulfur (<i>f</i>)		

vi. Circle the value from the choices given below, that describes the **C-S-C bond angle at sulfur** (S) marked ***f***: (0.5 points)

>90° to <109.5°

109.5°

>109.5 to <120°

120°

>120° to <180°


Question 8 12 points

Using the **valence shell orbitals only** draw a **complete** molecular orbital (MO) diagram (with no s-p mixing) for the molecule **PS**, in the space provided below. **(6 points)**

Assume that 3s and 3p atomic orbitals will form molecular orbitals like those formed by the combination of 2s and 2p atomic orbitals that we learnt in class, and there is no s-p mixing.

Show **energy arrow**, **atomic orbitals** involved, **atom labels**, the **molecular orbitals** that are formed, **filled electrons** in the MO diagram etc. Make sure to label all the atomic and molecular orbitals.

Using the MO diagram above, answer the following questions:

- (i) Which one of the following will have the highest bond order? Circle the correct choice
 A) PS^{1-} B) PS C) PS^{1+}
- (ii) Which one of the following will have the longest bond length? Circle the correct choice
 A) PS^{1-} B) PS C) PS^{1+}
- (iii) Which of the following will be diamagnetic? Circle the correct choice.
 A) PS^{1-} B) PS C) PS^{1+}
- (iv) Draw Lewis dot structure of the molecule with the shortest bond length (based on the MO diagram) from among the three (PS^{1-} , PS, PS^{1+}), and indicate the bond order calculated from the Lewis dot structure. **(3 points)**

Lewis Dot Structure

Bond Order



E8C81C65-342D-456E-ABF2-7EA3606DF5F2

midterm-exam-93388

#573 Page 10 of 10

SCRAP PAGE

**CHEMISTRY 110 –Midterm Exam****Thursday November 7, 2024, 6:30 – 8:30 pm****Examiners: Pallavi Sirjoosingh and Ashok Kakkar**

First name (please write as legibly as possible within the boxes)

--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--

Last name

--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--

Student ID number

--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--

This booklet has 8 questions on 10 pages (The last page can be used as a scrap page).

The staple must NOT be removed from the question booklet. Your intact question booklet must be returned at the end of the exam.

A separate booklet contains a periodic table, and datasheet.

Write your answers directly under the questions in the space provided. Do not write on the QR codes or near the top edges (as it may be cut off).

Please write **legibly and dark enough** for the scan to be readable. For students who use English as a second language, dictionaries are permitted. Calculators are allowed.

DO NOT ASK QUESTIONS OF THE INVIGILATORS about interpretations of the exam questions. If you have a concern about a question, please summarize it at the top of this page. All concerns will be evaluated after the exam. GOOD LUCK!

**Question 1 8 points** (*Assume all transitions are possible*)

a. 6 points Calculate the **energy (in kJ/mol)** required to ionize (completely remove) the electron from the **$n=3$** level of **He^+ ion** (He: Helium). Show your work including equations. (3 sig. figs.)

b. 2 points

(I) Select the largest (longest) wavelength from the three choices below:

- i. Wavelength of photon to *completely remove* an electron from *1s orbital of Hydrogen atom*
- ii. Wavelength of photon to *completely remove* an electron from *1s orbital of Boron atom (B)*
- iii. Wavelength of photon to *completely remove* an electron from *1s orbital of Boron cation (B^+)*

i or ii or iii

(II) Select the smallest (shortest) wavelength from the three choices below (*for H atom*):

- i. Wavelength of photon *emitted* for an electron transitioning from *2p to 1s orbital*
- ii. Wavelength of photon *absorbed* for an electron transitioning from *2s to 4p orbital*
- iii. Wavelength of photon *absorbed* for an electron transitioning from *1s to 3p orbital*

i or ii or iii



Question 2 8 points



A Chem110 student plots the graph shown on the left, from data obtained for the kinetic energy (eV) of the ejected electron versus frequency (s^{-1}) of the incident light, for an unknown metal. ($1\text{eV} = 1.6022 \times 10^{-19} \text{ J}$)

Based on the plot, answer the questions **a to c** below:

a. 1 point On the plot above, label the **threshold frequency**.

b. 2 points What is the value of the work potential (in J) of the unknown metal?

c. 3 points What is the *longest* wavelength (in nm) required to eject an electron from this metal's surface? Show your work including equation(s) used. (3 sig figs)

d. 2 points In the space provided below, plot the following

(Assume frequency is *above* threshold frequency):

- i. *Kinetic energy* of an ejected electron versus the *intensity* of the incident light (left)
- ii. *Number of electrons* ejected versus the *frequency* of the incident light (right)



**Question 3** **6 points****a. 2 points**

In the ground state of Ge (Germanium; $Z = 32$) atom, **name** the following orbitals:

i. Orbital containing the electron that requires the maximum energy to be removed.

ii. Orbital containing the electron that experiences the minimum effective nuclear charge.

b. 2 points

An unknown anion (A^-) has the following *excited* electronic configuration: $[Ar] 4s^2 3d^3 4p^1$

(Ar: Argon; $Z = 18$)

Give the **condensed ground state electronic configuration** of the atom, **A** and the cation, **A^{2+}** .
Show your work. (Assume all electronic transitions are possible)

Electronic Configuration of A:

Electronic Configuration of A^{2+} :

c. 2 points

In a single atom, determine how many electrons can have the following set of quantum numbers?

(i) $n = 5; l = 3$

In a single atom, determine how many electrons can have the following set of quantum numbers?

(i) $n = 3; l = 2; m_s = +1/2$

**Question 4** **8 points**

Rank the following from 1 to 4 in **increasing** order. Provide the ranking in the boxes provided.

Note: Atomic number (Z) are provided for relevant atoms

a) Increasing (lowest to highest) order of *ionization energy*: (1= lowest; 4 = highest)

Mg

Mg: Z = 12

Rb

Rb: Z = 37

Cs

Cs: Z = 55

K

K : Z = 19

b) Increasing (lowest to highest) order of *bond angle*: (1= lowest; 4 = highest)

H₂OCCl₄XeF₄

HCN

c) Increasing (smallest to largest) order of *ionic size* (1= smallest; 4 = largest)

S²⁻

S: Z = 16

O²⁻

O: Z = 8

Be²⁺

Be: Z = 4

Na⁺

Na: Z = 11

d) Increasing (lowest to highest) order of *bond energy* (1= lowest; 4 = highest)

P-F

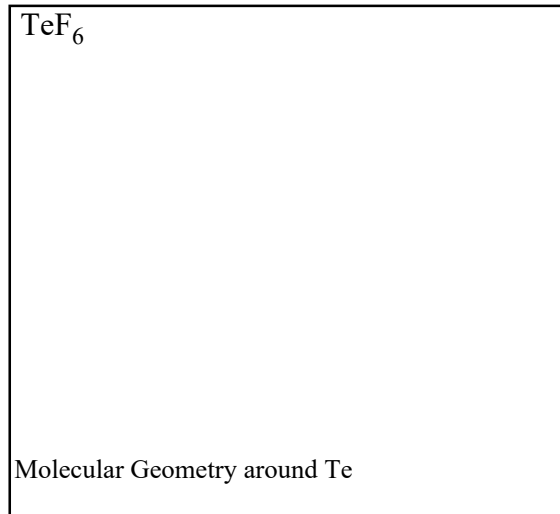
P-Cl

P-I

P-Br

**Question 5** **12 points**

a. 9 points For the molecules HClO_4 and TeF_6 , draw the most stable *Lewis structure* showing appropriate shape and *dashed/wedged bonds*, based on VSEPR. Determine the **molecular geometry** around the central atom. For HClO_4 , label the *most polar bond*.



b. 3 points (Note: Lone pairs are not shown in structures below)

(i) For the three *resonance* structures (1, 2 and 3) shown below, answer the following question:



(I) Circle the *most contributing* resonance structure.

Structure 1

Structure 2

Structure 3

(ii) For the three structures given (4, 5, and 6) below, answer the following questions:



(I) Circle all the structures that represent *resonance structures* of each other:

Structure 4

Structure 5

Structure 6

(II) Among the resonance structures, circle the *most contributing* resonance structure.

Structure 4

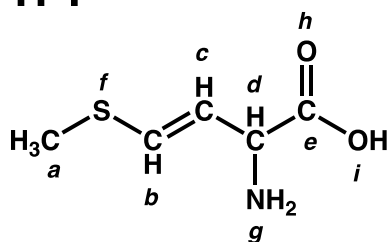
Structure 5

Structure 6

**Question 6** **6 points**

Using the data given below, calculate the **lattice energy** for SrO(s).

	ΔH° (kJ/mol)
$\text{Sr}(s) \longrightarrow \text{Sr}(g)$	+180
$\text{Sr}(g) \longrightarrow \text{Sr}^{2+}(g) + 2e^-$	+1468
$\text{O}_2(g) \longrightarrow 2\text{O}(g)$	+500
$\text{O}(g) + 2e^- \longrightarrow \text{O}^{2-}(g)$	+600
$\text{Sr}(s) + \frac{1}{2}\text{O}_2(g) \longrightarrow \text{SrO}(s)$	-550

**Question 7 10 points**

Consider the molecule given on the left, and answer questions (i) to (vi) using chemical bonding theories that we have learnt in class (valence bond theory, hybridization).

Note that the lone pairs are not shown in the structure:

i. Consider the carbon atoms marked in as ***a***, ***b***, ***c***, ***d***, and ***e*** in the molecule, and determine **hybridization** at each of the carbon centers. (2.5 points)

Atom	Hybridization
Carbon (<i>a</i>)	
Carbon (<i>b</i>)	
Carbon (<i>c</i>)	
Carbon (<i>d</i>)	
Carbon (<i>e</i>)	

ii. Now, consider the sulfur atom (S) marked as ***f***, nitrogen (N) atom (***g***), and oxygen(O) atoms (***h*** and ***i***), and determine their **hybridization**. (2 points)

Atom	Hybridization
Sulfur (<i>f</i>)	
Nitrogen (<i>g</i>)	
Oxygen (<i>h</i>)	
Oxygen (<i>i</i>)	

iii. **Describe** all the bonds formed by the carbon atoms ***a*** and ***b***, using sigma and pi designation as appropriate. (2 points)

Atom	Bonds formed (use sigma and pi as appropriate)
Carbon (<i>a</i>)	
Carbon (<i>b</i>)	

iv. How many total sp^3 -s sigma bonds are in the molecule? (1 point)

v. Determine **electronic** and **molecular geometry** at carbon (C) labeled ***e*** and sulfur (S) labeled ***f*** centres. (2 points)

Atom	Electronic Geometry	Molecular Geometry
Carbon (<i>e</i>)		
Sulfur (<i>f</i>)		

vi. Circle the value from the choices given below, that describes the **C-S-C bond angle at sulfur** (S) marked ***f***: (0.5 points)

>90° to <109.5° 109.5° >109.5 to <120° 120° >120° to <180°


Question 8 12 points

Using the **valence shell orbitals only** draw a **complete** molecular orbital (MO) diagram (with no s-p mixing) for the molecule **PS**, in the space provided below. **(6 points)**

Assume that 3s and 3p atomic orbitals will form molecular orbitals like those formed by the combination of 2s and 2p atomic orbitals that we learnt in class, and there is no s-p mixing.

Show **energy arrow**, **atomic orbitals** involved, **atom labels**, the **molecular orbitals** that are formed, **filled electrons** in the MO diagram etc. Make sure to label all the atomic and molecular orbitals.

Using the MO diagram above, answer the following questions:

- (i) Which one of the following will have the highest bond order? Circle the correct choice
 A) PS^{1-} B) PS C) PS^{1+}
- (ii) Which one of the following will have the longest bond length? Circle the correct choice
 A) PS^{1-} B) PS C) PS^{1+}
- (iii) Which of the following will be diamagnetic? Circle the correct choice.
 A) PS^{1-} B) PS C) PS^{1+}
- (iv) Draw Lewis dot structure of the molecule with the shortest bond length (based on the MO diagram) from among the three (PS^{1-} , PS, PS^{1+}), and indicate the bond order calculated from the Lewis dot structure. **(3 points)**

Lewis Dot Structure

Bond Order



A5DF1800-4FF3-4DFD-AC65-F7B3643B8C60

midterm-exam-93388

#574 Page 10 of 10

SCRAP PAGE

**CHEMISTRY 110 –Midterm Exam****Thursday November 7, 2024, 6:30 – 8:30 pm**

Examiners: Pallavi Sirjoosingh and Ashok Kakkar

First name (please write as legibly as possible within the boxes)

--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--

Last name

--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--

Student ID number

--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--

This booklet has 8 questions on 10 pages (The last page can be used as a scrap page).

The staple must NOT be removed from the question booklet. Your intact question booklet must be returned at the end of the exam.

A separate booklet contains a periodic table, and datasheet.

Write your answers directly under the questions in the space provided. Do not write on the QR codes or near the top edges (as it may be cut off).

Please write **legibly and dark enough** for the scan to be readable. For students who use English as a second language, dictionaries are permitted. Calculators are allowed.

DO NOT ASK QUESTIONS OF THE INVIGILATORS about interpretations of the exam questions. If you have a concern about a question, please summarize it at the top of this page. All concerns will be evaluated after the exam. GOOD LUCK!

**Question 1 8 points** (*Assume all transitions are possible*)

a. 6 points Calculate the **energy (in kJ/mol)** required to ionize (completely remove) the electron from the **$n=3$** level of **He^+ ion** (He: Helium). Show your work including equations. (3 sig. figs.)

b. 2 points

(I) Select the largest (longest) wavelength from the three choices below:

- i. Wavelength of photon to *completely remove* an electron from *1s orbital of Hydrogen atom*
- ii. Wavelength of photon to *completely remove* an electron from *1s orbital of Boron atom (B)*
- iii. Wavelength of photon to *completely remove* an electron from *1s orbital of Boron cation (B^+)*

i or ii or iii

(II) Select the smallest (shortest) wavelength from the three choices below (*for H atom*):

- i. Wavelength of photon *emitted* for an electron transitioning from *2p to 1s orbital*
- ii. Wavelength of photon *absorbed* for an electron transitioning from *2s to 4p orbital*
- iii. Wavelength of photon *absorbed* for an electron transitioning from *1s to 3p orbital*

i or ii or iii



Question 2 8 points



A Chem110 student plots the graph shown on the left, from data obtained for the kinetic energy (eV) of the ejected electron versus frequency (s^{-1}) of the incident light, for an unknown metal. ($1\text{eV} = 1.6022 \times 10^{-19} \text{ J}$)

Based on the plot, answer the questions **a to c** below:

a. 1 point On the plot above, label the **threshold frequency**.

b. 2 points What is the value of the work potential (in J) of the unknown metal?

c. 3 points What is the *longest* wavelength (in nm) required to eject an electron from this metal's surface? Show your work including equation(s) used. (3 sig figs)

d. 2 points In the space provided below, plot the following

(Assume frequency is *above* threshold frequency):

- Kinetic energy* of an ejected electron versus the *intensity* of the incident light (left)
- Number of electrons* ejected versus the *frequency* of the incident light (right)



**Question 3** **6 points****a. 2 points**

In the ground state of Ge (Germanium; $Z = 32$) atom, **name** the following orbitals:

i. Orbital containing the electron that requires the maximum energy to be removed.

ii. Orbital containing the electron that experiences the minimum effective nuclear charge.

b. 2 points

An unknown anion (A^-) has the following *excited* electronic configuration: $[Ar] 4s^2 3d^3 4p^1$

(Ar: Argon; $Z = 18$)

Give the **condensed ground state electronic configuration** of the atom, **A** and the cation, **A^{2+}** .
Show your work. (Assume all electronic transitions are possible)

Electronic Configuration of A:

Electronic Configuration of A^{2+} :

c. 2 points

In a single atom, determine how many electrons can have the following set of quantum numbers?

(i) $n = 5; l = 3$

In a single atom, determine how many electrons can have the following set of quantum numbers?

(i) $n = 3; l = 2; m_s = +1/2$

**Question 4** **8 points**

Rank the following from 1 to 4 in **increasing** order. Provide the ranking in the boxes provided.

Note: Atomic number (Z) are provided for relevant atoms

a) Increasing (lowest to highest) order of *ionization energy*: (1= lowest; 4 = highest)

Mg

Mg: Z = 12

Rb

Rb: Z = 37

Cs

Cs: Z = 55

K

K : Z = 19

b) Increasing (lowest to highest) order of *bond angle*: (1= lowest; 4 = highest)

H₂OCCl₄XeF₄

HCN

c) Increasing (smallest to largest) order of *ionic size* (1= smallest; 4 = largest)

S²⁻

S: Z = 16

O²⁻

O: Z = 8

Be²⁺

Be: Z = 4

Na⁺

Na: Z = 11

d) Increasing (lowest to highest) order of *bond energy* (1= lowest; 4 = highest)

P-F

P-Cl

P-I

P-Br

**Question 5** **12 points**

a. 9 points For the molecules HClO_4 and TeF_6 , draw the most stable *Lewis structure* showing appropriate shape and *dashed/wedged bonds*, based on VSEPR. Determine the **molecular geometry** around the central atom. For HClO_4 , label the *most polar bond*.



b. 3 points (Note: Lone pairs are not shown in structures below)

(i) For the three *resonance* structures (1, 2 and 3) shown below, answer the following question:



(I) Circle the *most contributing* resonance structure.

Structure 1

Structure 2

Structure 3

(ii) For the three structures given (4, 5, and 6) below, answer the following questions:



(I) Circle all the structures that represent *resonance structures* of each other:

Structure 4

Structure 5

Structure 6

(II) Among the resonance structures, circle the *most contributing* resonance structure.

Structure 4

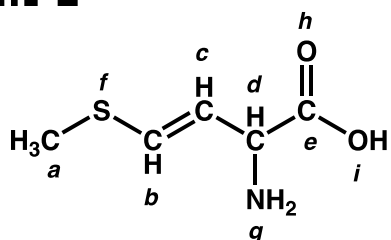
Structure 5

Structure 6

**Question 6** **6 points**

Using the data given below, calculate the **lattice energy** for SrO(s).

	ΔH° (kJ/mol)
$\text{Sr}(s) \longrightarrow \text{Sr}(g)$	+180
$\text{Sr}(g) \longrightarrow \text{Sr}^{2+}(g) + 2e^-$	+1468
$\text{O}_2(g) \longrightarrow 2\text{O}(g)$	+500
$\text{O}(g) + 2e^- \longrightarrow \text{O}^{2-}(g)$	+600
$\text{Sr}(s) + \frac{1}{2}\text{O}_2(g) \longrightarrow \text{SrO}(s)$	-550

**Question 7 10 points**

Consider the molecule given on the left, and answer questions (i) to (vi) using chemical bonding theories that we have learnt in class (valence bond theory, hybridization).

Note that the lone pairs are not shown in the structure:

i. Consider the carbon atoms marked in as ***a***, ***b***, ***c***, ***d***, and ***e*** in the molecule, and determine **hybridization** at each of the carbon centers. (2.5 points)

Atom	Hybridization
Carbon (<i>a</i>)	
Carbon (<i>b</i>)	
Carbon (<i>c</i>)	
Carbon (<i>d</i>)	
Carbon (<i>e</i>)	

ii. Now, consider the sulfur atom (S) marked as ***f***, nitrogen (N) atom (***g***), and oxygen(O) atoms (***h*** and ***i***), and determine their **hybridization**. (2 points)

Atom	Hybridization
Sulfur (<i>f</i>)	
Nitrogen (<i>g</i>)	
Oxygen (<i>h</i>)	
Oxygen (<i>i</i>)	

iii. **Describe** all the bonds formed by the carbon atoms ***a*** and ***b***, using sigma and pi designation as appropriate. (2 points)

Atom	Bonds formed (use sigma and pi as appropriate)
Carbon (<i>a</i>)	
Carbon (<i>b</i>)	

iv. How many total sp^3 -s sigma bonds are in the molecule? (1 point)

v. Determine **electronic** and **molecular geometry** at carbon (C) labeled ***e*** and sulfur (S) labeled ***f*** centres. (2 points)

Atom	Electronic Geometry	Molecular Geometry
Carbon (<i>e</i>)		
Sulfur (<i>f</i>)		

vi. Circle the value from the choices given below, that describes the **C-S-C bond angle at sulfur** (S) marked ***f***: (0.5 points)

$>90^\circ$ to $<109.5^\circ$ 109.5° >109.5 to $<120^\circ$ 120° $>120^\circ$ to $<180^\circ$



Question 8 12 points

Using the **valence shell orbitals only** draw a **complete** molecular orbital (MO) diagram (with no s-p mixing) for the molecule **PS**, in the space provided below. **(6 points)**

Assume that 3s and 3p atomic orbitals will form molecular orbitals like those formed by the combination of 2s and 2p atomic orbitals that we learnt in class, and there is no s-p mixing.

Show **energy arrow**, **atomic orbitals** involved, **atom labels**, the **molecular orbitals** that are formed, **filled electrons** in the MO diagram etc. Make sure to label all the atomic and molecular orbitals.

Using the MO diagram above, answer the following questions:

- (i) Which one of the following will have the highest bond order? Circle the correct choice
 A) PS^{1-} B) PS C) PS^{1+}
- (ii) Which one of the following will have the longest bond length? Circle the correct choice
 A) PS^{1-} B) PS C) PS^{1+}
- (iii) Which of the following will be diamagnetic? Circle the correct choice.
 A) PS^{1-} B) PS C) PS^{1+}
- (iv) Draw Lewis dot structure of the molecule with the shortest bond length (based on the MO diagram) from among the three (PS^{1-} , PS, PS^{1+}), and indicate the bond order calculated from the Lewis dot structure. **(3 points)**

Lewis Dot Structure

Bond Order



62568C12-FBF0-4BE8-B26F-E024F445E2E1

midterm-exam-93388

#575 Page 10 of 10

SCRAP PAGE

**CHEMISTRY 110 –Midterm Exam****Thursday November 7, 2024, 6:30 – 8:30 pm**

Examiners: Pallavi Sirjoosingh and Ashok Kakkar

First name (please write as legibly as possible within the boxes)

--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--

Last name

--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--

Student ID number

--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--

This booklet has 8 questions on 10 pages (The last page can be used as a scrap page).

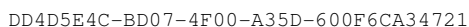
The staple must NOT be removed from the question booklet. Your intact question booklet must be returned at the end of the exam.

A separate booklet contains a periodic table, and datasheet.

Write your answers directly under the questions in the space provided. Do not write on the QR codes or near the top edges (as it may be cut off).

Please write **legibly and dark enough** for the scan to be readable. For students who use English as a second language, dictionaries are permitted. Calculators are allowed.

DO NOT ASK QUESTIONS OF THE INVIGILATORS about interpretations of the exam questions. If you have a concern about a question, please summarize it at the top of this page. All concerns will be evaluated after the exam. GOOD LUCK!



midterm-exam-93388

#576 Page 2 of 10

a. 6 points Calculate the **energy (in kJ/mol)** required to ionize (completely remove) the electron from the **n=3** level of **He⁺ ion** (He: Helium). Show your work including equations. (3 sig. figs.)

(I) Select the largest (longest) wavelength from the three choices below:

- Wavelength of photon to *completely remove* an electron from *1s orbital of Hydrogen atom*
- Wavelength of photon to *completely remove* an electron from *1s orbital of Boron atom (B)*
- Wavelength of photon to *completely remove* an electron from *1s orbital of Boron cation (B⁺)*

i
or
ii
or
iii

(II) Select the smallest (shortest) wavelength from the three choices below (*for H atom*):

- Wavelength of photon *emitted* for an electron transitioning from $2p$ to $1s$ orbital
- Wavelength of photon *absorbed* for an electron transitioning from $2s$ to $4p$ orbital
- Wavelength of photon *absorbed* for an electron transitioning from $1s$ to $3p$ orbital

i **or** **ii** **or** **iii**



Question 2 8 points



A Chem110 student plots the graph shown on the left, from data obtained for the kinetic energy (eV) of the ejected electron versus frequency (s^{-1}) of the incident light, for an unknown metal. ($1\text{eV} = 1.6022 \times 10^{-19} \text{ J}$)

Based on the plot, answer the questions **a to c** below:

a. 1 point On the plot above, label the **threshold frequency**.

b. 2 points What is the value of the work potential (in J) of the unknown metal?

c. 3 points What is the *longest* wavelength (in nm) required to eject an electron from this metal's surface? Show your work including equation(s) used. (3 sig figs)

d. 2 points In the space provided below, plot the following

(Assume frequency is *above* threshold frequency):

- Kinetic energy* of an ejected electron versus the *intensity* of the incident light (left)
- Number of electrons* ejected versus the *frequency* of the incident light (right)



**Question 3** **6 points****a. 2 points**

In the ground state of Ge (Germanium; $Z = 32$) atom, **name** the following orbitals:

i. Orbital containing the electron that requires the maximum energy to be removed.

ii. Orbital containing the electron that experiences the minimum effective nuclear charge.

b. 2 points

An unknown anion (A^-) has the following *excited* electronic configuration: $[Ar] 4s^2 3d^3 4p^1$

(Ar: Argon; $Z = 18$)

Give the **condensed ground state electronic configuration** of the atom, **A** and the cation, **A^{2+}** .
Show your work. (Assume all electronic transitions are possible)

Electronic Configuration of A:

Electronic Configuration of A^{2+} :

c. 2 points

In a single atom, determine how many electrons can have the following set of quantum numbers?

(i) $n = 5; l = 3$

In a single atom, determine how many electrons can have the following set of quantum numbers?

(i) $n = 3; l = 2; m_s = +1/2$

**Question 4** **8 points**

Rank the following from 1 to 4 in **increasing** order. Provide the ranking in the boxes provided.

Note: Atomic number (Z) are provided for relevant atoms

a) Increasing (lowest to highest) order of *ionization energy*: (1= lowest; 4 = highest)

Mg

Mg: Z = 12

Rb

Rb: Z = 37

Cs

Cs: Z = 55

K

K : Z = 19

b) Increasing (lowest to highest) order of *bond angle*: (1= lowest; 4 = highest)

H₂OCCl₄XeF₄

HCN

c) Increasing (smallest to largest) order of *ionic size* (1= smallest; 4 = largest)

S²⁻

S: Z = 16

O²⁻

O: Z = 8

Be²⁺

Be: Z = 4

Na⁺

Na: Z = 11

d) Increasing (lowest to highest) order of *bond energy* (1= lowest; 4 = highest)

P-F

P-Cl

P-I

P-Br

**Question 5** **12 points**

a. 9 points For the molecules HClO_4 and TeF_6 , draw the most stable *Lewis structure* showing appropriate shape and *dashed/wedged bonds*, based on VSEPR. Determine the **molecular geometry** around the central atom. For HClO_4 , label the *most polar bond*.



b. 3 points (Note: Lone pairs are not shown in structures below)

(i) For the three *resonance* structures (1, 2 and 3) shown below, answer the following question:



(I) Circle the *most contributing* resonance structure.

Structure 1

Structure 2

Structure 3

(ii) For the three structures given (4, 5, and 6) below, answer the following questions:



(I) Circle all the structures that represent *resonance structures* of each other:

Structure 4

Structure 5

Structure 6

(II) Among the resonance structures, circle the *most contributing* resonance structure.

Structure 4

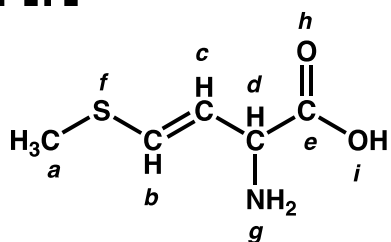
Structure 5

Structure 6

**Question 6** **6 points**

Using the data given below, calculate the **lattice energy** for SrO(s).

	ΔH° (kJ/mol)
$\text{Sr}(s) \longrightarrow \text{Sr}(g)$	+180
$\text{Sr}(g) \longrightarrow \text{Sr}^{2+}(g) + 2e^-$	+1468
$\text{O}_2(g) \longrightarrow 2\text{O}(g)$	+500
$\text{O}(g) + 2e^- \longrightarrow \text{O}^{2-}(g)$	+600
$\text{Sr}(s) + \frac{1}{2}\text{O}_2(g) \longrightarrow \text{SrO}(s)$	-550

**Question 7 10 points**

Consider the molecule given on the left, and answer questions (i) to (vi) using chemical bonding theories that we have learnt in class (valence bond theory, hybridization).

Note that the lone pairs are not shown in the structure:

i. Consider the carbon atoms marked in as ***a***, ***b***, ***c***, ***d***, and ***e*** in the molecule, and determine **hybridization** at each of the carbon centers. (2.5 points)

Atom	Hybridization
Carbon (<i>a</i>)	
Carbon (<i>b</i>)	
Carbon (<i>c</i>)	
Carbon (<i>d</i>)	
Carbon (<i>e</i>)	

ii. Now, consider the sulfur atom (S) marked as ***f***, nitrogen (N) atom (***g***), and oxygen(O) atoms (***h*** and ***i***), and determine their **hybridization**. (2 points)

Atom	Hybridization
Sulfur (<i>f</i>)	
Nitrogen (<i>g</i>)	
Oxygen (<i>h</i>)	
Oxygen (<i>i</i>)	

iii. **Describe** all the bonds formed by the carbon atoms ***a*** and ***b***, using sigma and pi designation as appropriate. (2 points)

Atom	Bonds formed (use sigma and pi as appropriate)
Carbon (<i>a</i>)	
Carbon (<i>b</i>)	

iv. How many total sp^3 -s sigma bonds are in the molecule? (1 point)

v. Determine **electronic** and **molecular geometry** at carbon (C) labeled ***e*** and sulfur (S) labeled ***f*** centres. (2 points)

Atom	Electronic Geometry	Molecular Geometry
Carbon (<i>e</i>)		
Sulfur (<i>f</i>)		

vi. Circle the value from the choices given below, that describes the **C-S-C bond angle at sulfur** (S) marked ***f***: (0.5 points)

>90° to <109.5°

109.5°

>109.5 to <120°

120°

>120° to <180°



Question 8 12 points

Using the **valence shell orbitals only** draw a **complete** molecular orbital (MO) diagram (with no s-p mixing) for the molecule **PS**, in the space provided below. **(6 points)**

Assume that 3s and 3p atomic orbitals will form molecular orbitals like those formed by the combination of 2s and 2p atomic orbitals that we learnt in class, and there is no s-p mixing.

Show **energy arrow**, **atomic orbitals** involved, **atom labels**, the **molecular orbitals** that are formed, **filled electrons** in the MO diagram etc. Make sure to label all the atomic and molecular orbitals.

Using the MO diagram above, answer the following questions:

- (i) Which one of the following will have the highest bond order? Circle the correct choice
 A) PS^{1-} B) PS C) PS^{1+}
- (ii) Which one of the following will have the longest bond length? Circle the correct choice
 A) PS^{1-} B) PS C) PS^{1+}
- (iii) Which of the following will be diamagnetic? Circle the correct choice.
 A) PS^{1-} B) PS C) PS^{1+}
- (iv) Draw Lewis dot structure of the molecule with the shortest bond length (based on the MO diagram) from among the three (PS^{1-} , PS, PS^{1+}), and indicate the bond order calculated from the Lewis dot structure. **(3 points)**

Lewis Dot Structure

Bond Order



F3AB97DB-0A4A-445C-8BED-F7DC348E0927

midterm-exam-93388

#576 Page 10 of 10

SCRAP PAGE

**CHEMISTRY 110 –Midterm Exam****Thursday November 7, 2024, 6:30 – 8:30 pm**

Examiners: Pallavi Sirjoosingh and Ashok Kakkar

First name (please write as legibly as possible within the boxes)

--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--

Last name

--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--

Student ID number

--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--

This booklet has 8 questions on 10 pages (The last page can be used as a scrap page).

The staple must NOT be removed from the question booklet. Your intact question booklet must be returned at the end of the exam.

A separate booklet contains a periodic table, and datasheet.

Write your answers directly under the questions in the space provided. Do not write on the QR codes or near the top edges (as it may be cut off).

Please write **legibly and dark enough** for the scan to be readable. For students who use English as a second language, dictionaries are permitted. Calculators are allowed.

DO NOT ASK QUESTIONS OF THE INVIGILATORS about interpretations of the exam questions. If you have a concern about a question, please summarize it at the top of this page. All concerns will be evaluated after the exam. GOOD LUCK!

**Question 1 8 points** (*Assume all transitions are possible*)

a. 6 points Calculate the **energy (in kJ/mol)** required to ionize (completely remove) the electron from the **n=3** level of **He⁺ ion** (He: Helium). Show your work including equations. (3 sig. figs.)

b. 2 points

(I) Select the largest (longest) wavelength from the three choices below:

- i. Wavelength of photon to *completely remove* an electron from *1s orbital of Hydrogen atom*
- ii. Wavelength of photon to *completely remove* an electron from *1s orbital of Boron atom (B)*
- iii. Wavelength of photon to *completely remove* an electron from *1s orbital of Boron cation (B⁺)*

i or ii or iii

(II) Select the smallest (shortest) wavelength from the three choices below (*for H atom*):

- i. Wavelength of photon *emitted* for an electron transitioning from *2p to 1s orbital*
- ii. Wavelength of photon *absorbed* for an electron transitioning from *2s to 4p orbital*
- iii. Wavelength of photon *absorbed* for an electron transitioning from *1s to 3p orbital*

i or ii or iii



Question 2 8 points



A Chem110 student plots the graph shown on the left, from data obtained for the kinetic energy (eV) of the ejected electron versus frequency (s^{-1}) of the incident light, for an unknown metal. ($1\text{eV} = 1.6022 \times 10^{-19} \text{ J}$)

Based on the plot, answer the questions **a to c** below:

a. 1 point On the plot above, label the **threshold frequency**.

b. 2 points What is the value of the work potential (in J) of the unknown metal?

c. 3 points What is the *longest* wavelength (in nm) required to eject an electron from this metal's surface? Show your work including equation(s) used. (3 sig figs)

d. 2 points In the space provided below, plot the following

(Assume frequency is *above* threshold frequency):

- i. *Kinetic energy* of an ejected electron versus the *intensity* of the incident light (left)
- ii. *Number of electrons* ejected versus the *frequency* of the incident light (right)



**Question 3** **6 points****a. 2 points**

In the ground state of Ge (Germanium; $Z = 32$) atom, **name** the following orbitals:

i. Orbital containing the electron that requires the maximum energy to be removed.

ii. Orbital containing the electron that experiences the minimum effective nuclear charge.

b. 2 points

An unknown anion (A^-) has the following *excited* electronic configuration: $[Ar] 4s^2 3d^3 4p^1$

(Ar: Argon; $Z = 18$)

Give the **condensed ground state electronic configuration** of the atom, **A** and the cation, **A^{2+}** .
Show your work. (Assume all electronic transitions are possible)

Electronic Configuration of A:

Electronic Configuration of A^{2+} :

c. 2 points

In a single atom, determine how many electrons can have the following set of quantum numbers?

(i) $n = 5; l = 3$

In a single atom, determine how many electrons can have the following set of quantum numbers?

(i) $n = 3; l = 2; m_s = +1/2$

**Question 4** **8 points**

Rank the following from 1 to 4 in **increasing** order. Provide the ranking in the boxes provided.

Note: Atomic number (Z) are provided for relevant atoms

a) Increasing (lowest to highest) order of *ionization energy*: (1= lowest; 4 = highest)

Mg

Mg: Z = 12

Rb

Rb: Z = 37

Cs

Cs: Z = 55

K

K : Z = 19

b) Increasing (lowest to highest) order of *bond angle*: (1= lowest; 4 = highest)

H₂OCCl₄XeF₄

HCN

c) Increasing (smallest to largest) order of *ionic size* (1= smallest; 4 = largest)

S²⁻

S: Z = 16

O²⁻

O: Z = 8

Be²⁺

Be: Z = 4

Na⁺

Na: Z = 11

d) Increasing (lowest to highest) order of *bond energy* (1= lowest; 4 = highest)

P-F

P-Cl

P-I

P-Br

**Question 5** **12 points**

a. 9 points For the molecules HClO_4 and TeF_6 , draw the most stable *Lewis structure* showing appropriate shape and *dashed/wedged bonds*, based on VSEPR. Determine the **molecular geometry** around the central atom. For HClO_4 , label the *most polar bond*.



b. 3 points (Note: Lone pairs are not shown in structures below)

(i) For the three *resonance* structures (1, 2 and 3) shown below, answer the following question:



(I) Circle the *most contributing* resonance structure.

Structure 1

Structure 2

Structure 3

(ii) For the three structures given (4, 5, and 6) below, answer the following questions:



(I) Circle all the structures that represent *resonance structures* of each other:

Structure 4

Structure 5

Structure 6

(II) Among the resonance structures, circle the *most contributing* resonance structure.

Structure 4

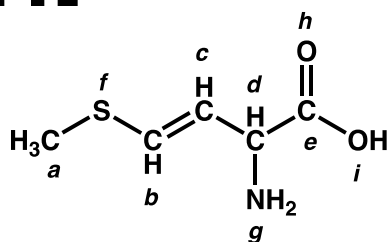
Structure 5

Structure 6

**Question 6** **6 points**

Using the data given below, calculate the **lattice energy** for SrO(s).

	ΔH° (kJ/mol)
$\text{Sr}(s) \longrightarrow \text{Sr}(g)$	+180
$\text{Sr}(g) \longrightarrow \text{Sr}^{2+}(g) + 2e^-$	+1468
$\text{O}_2(g) \longrightarrow 2\text{O}(g)$	+500
$\text{O}(g) + 2e^- \longrightarrow \text{O}^{2-}(g)$	+600
$\text{Sr}(s) + \frac{1}{2}\text{O}_2(g) \longrightarrow \text{SrO}(s)$	-550

**Question 7 10 points**

Consider the molecule given on the left, and answer questions (i) to (vi) using chemical bonding theories that we have learnt in class (valence bond theory, hybridization).

Note that the lone pairs are not shown in the structure:

i. Consider the carbon atoms marked in as ***a***, ***b***, ***c***, ***d***, and ***e*** in the molecule, and determine **hybridization** at each of the carbon centers. (2.5 points)

Atom	Hybridization
Carbon (<i>a</i>)	
Carbon (<i>b</i>)	
Carbon (<i>c</i>)	
Carbon (<i>d</i>)	
Carbon (<i>e</i>)	

ii. Now, consider the sulfur atom (S) marked as ***f***, nitrogen (N) atom (***g***), and oxygen(O) atoms (***h*** and ***i***), and determine their **hybridization**. (2 points)

Atom	Hybridization
Sulfur (<i>f</i>)	
Nitrogen (<i>g</i>)	
Oxygen (<i>h</i>)	
Oxygen (<i>i</i>)	

iii. **Describe** all the bonds formed by the carbon atoms ***a*** and ***b***, using sigma and pi designation as appropriate. (2 points)

Atom	Bonds formed (use sigma and pi as appropriate)
Carbon (<i>a</i>)	
Carbon (<i>b</i>)	

iv. How many total sp^3 -s sigma bonds are in the molecule? (1 point)

v. Determine **electronic** and **molecular geometry** at carbon (C) labeled ***e*** and sulfur (S) labeled ***f*** centres. (2 points)

Atom	Electronic Geometry	Molecular Geometry
Carbon (<i>e</i>)		
Sulfur (<i>f</i>)		

vi. Circle the value from the choices given below, that describes the **C-S-C bond angle at sulfur** (S) marked ***f***: (0.5 points)

>90° to <109.5° 109.5° >109.5 to <120° 120° >120° to <180°


Question 8 12 points

Using the **valence shell orbitals only** draw a **complete** molecular orbital (MO) diagram (with no s-p mixing) for the molecule **PS**, in the space provided below. **(6 points)**

Assume that 3s and 3p atomic orbitals will form molecular orbitals like those formed by the combination of 2s and 2p atomic orbitals that we learnt in class, and there is no s-p mixing.

Show **energy arrow**, **atomic orbitals** involved, **atom labels**, the **molecular orbitals** that are formed, **filled electrons** in the MO diagram etc. Make sure to label all the atomic and molecular orbitals.

Using the MO diagram above, answer the following questions:

- (i) Which one of the following will have the highest bond order? Circle the correct choice
 A) PS^{1-} B) PS C) PS^{1+}
- (ii) Which one of the following will have the longest bond length? Circle the correct choice
 A) PS^{1-} B) PS C) PS^{1+}
- (iii) Which of the following will be diamagnetic? Circle the correct choice.
 A) PS^{1-} B) PS C) PS^{1+}
- (iv) Draw Lewis dot structure of the molecule with the shortest bond length (based on the MO diagram) from among the three (PS^{1-} , PS, PS^{1+}), and indicate the bond order calculated from the Lewis dot structure. **(3 points)**

Lewis Dot Structure

Bond Order



B5ABFA39-64B1-44D1-913F-658CEB42DE8B

midterm-exam-93388

#577 Page 10 of 10

SCRAP PAGE

**CHEMISTRY 110 –Midterm Exam****Thursday November 7, 2024, 6:30 – 8:30 pm**

Examiners: Pallavi Sirjoosingh and Ashok Kakkar

First name (please write as legibly as possible within the boxes)

--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--

Last name

--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--

Student ID number

--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--

This booklet has 8 questions on 10 pages (The last page can be used as a scrap page).

The staple must NOT be removed from the question booklet. Your intact question booklet must be returned at the end of the exam.

A separate booklet contains a periodic table, and datasheet.

Write your answers directly under the questions in the space provided. Do not write on the QR codes or near the top edges (as it may be cut off).

Please write **legibly and dark enough** for the scan to be readable. For students who use English as a second language, dictionaries are permitted. Calculators are allowed.

DO NOT ASK QUESTIONS OF THE INVIGILATORS about interpretations of the exam questions. If you have a concern about a question, please summarize it at the top of this page. All concerns will be evaluated after the exam. GOOD LUCK!

**Question 1 8 points** (*Assume all transitions are possible*)

a. 6 points Calculate the **energy (in kJ/mol)** required to ionize (completely remove) the electron from the **$n=3$** level of **He^+ ion** (He: Helium). Show your work including equations. (3 sig. figs.)

b. 2 points

(I) Select the largest (longest) wavelength from the three choices below:

- i. Wavelength of photon to *completely remove* an electron from *1s orbital of Hydrogen atom*
- ii. Wavelength of photon to *completely remove* an electron from *1s orbital of Boron atom (B)*
- iii. Wavelength of photon to *completely remove* an electron from *1s orbital of Boron cation (B^+)*

i or ii or iii

(II) Select the smallest (shortest) wavelength from the three choices below (*for H atom*):

- i. Wavelength of photon *emitted* for an electron transitioning from *2p to 1s orbital*
- ii. Wavelength of photon *absorbed* for an electron transitioning from *2s to 4p orbital*
- iii. Wavelength of photon *absorbed* for an electron transitioning from *1s to 3p orbital*

i or ii or iii



Question 2 8 points



A Chem110 student plots the graph shown on the left, from data obtained for the kinetic energy (eV) of the ejected electron versus frequency (s^{-1}) of the incident light, for an unknown metal. ($1\text{eV} = 1.6022 \times 10^{-19} \text{ J}$)

Based on the plot, answer the questions **a to c** below:

a. 1 point On the plot above, label the **threshold frequency**.

b. 2 points What is the value of the work potential (in J) of the unknown metal?

c. 3 points What is the *longest* wavelength (in nm) required to eject an electron from this metal's surface? Show your work including equation(s) used. (3 sig figs)

d. 2 points In the space provided below, plot the following

(Assume frequency is *above* threshold frequency):

- Kinetic energy* of an ejected electron versus the *intensity* of the incident light (left)
- Number of electrons* ejected versus the *frequency* of the incident light (right)



**Question 3** **6 points****a. 2 points**

In the ground state of Ge (Germanium; $Z = 32$) atom, **name** the following orbitals:

i. Orbital containing the electron that requires the maximum energy to be removed.

ii. Orbital containing the electron that experiences the minimum effective nuclear charge.

b. 2 points

An unknown anion (A^-) has the following *excited* electronic configuration: $[Ar] 4s^2 3d^3 4p^1$

(Ar: Argon; $Z = 18$)

Give the **condensed ground state electronic configuration** of the atom, **A** and the cation, **A^{2+}** .
Show your work. (Assume all electronic transitions are possible)

Electronic Configuration of A:

Electronic Configuration of A^{2+} :

c. 2 points

In a single atom, determine how many electrons can have the following set of quantum numbers?

(i) $n = 5; l = 3$

In a single atom, determine how many electrons can have the following set of quantum numbers?

(i) $n = 3; l = 2; m_s = +1/2$

**Question 4** **8 points**

Rank the following from 1 to 4 in **increasing** order. Provide the ranking in the boxes provided.

Note: Atomic number (Z) are provided for relevant atoms

a) Increasing (lowest to highest) order of *ionization energy*: (1= lowest; 4 = highest)

Mg

Mg: Z = 12

Rb

Rb: Z = 37

Cs

Cs: Z = 55

K

K : Z = 19

b) Increasing (lowest to highest) order of *bond angle*: (1= lowest; 4 = highest)

H₂OCCl₄XeF₄

HCN

c) Increasing (smallest to largest) order of *ionic size* (1= smallest; 4 = largest)

S²⁻

S: Z = 16

O²⁻

O: Z = 8

Be²⁺

Be: Z = 4

Na⁺

Na: Z = 11

d) Increasing (lowest to highest) order of *bond energy* (1= lowest; 4 = highest)

P-F

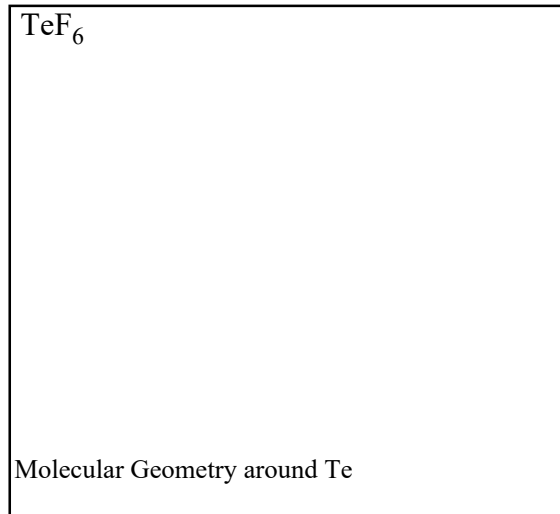
P-Cl

P-I

P-Br

**Question 5** **12 points**

a. 9 points For the molecules HClO_4 and TeF_6 , draw the most stable *Lewis structure* showing appropriate shape and *dashed/wedged bonds*, based on VSEPR. Determine the **molecular geometry** around the central atom. For HClO_4 , label the *most polar bond*.



b. 3 points (Note: Lone pairs are not shown in structures below)

(i) For the three *resonance* structures (1, 2 and 3) shown below, answer the following question:



(I) Circle the *most contributing* resonance structure.

Structure 1

Structure 2

Structure 3

(ii) For the three structures given (4, 5, and 6) below, answer the following questions:



(I) Circle all the structures that represent *resonance structures* of each other:

Structure 4

Structure 5

Structure 6

(II) Among the resonance structures, circle the *most contributing* resonance structure.

Structure 4

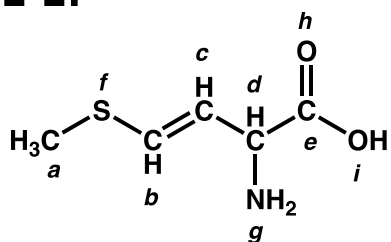
Structure 5

Structure 6

**Question 6** **6 points**

Using the data given below, calculate the **lattice energy** for SrO(s).

	ΔH° (kJ/mol)
$\text{Sr}(s) \longrightarrow \text{Sr}(g)$	+180
$\text{Sr}(g) \longrightarrow \text{Sr}^{2+}(g) + 2e^-$	+1468
$\text{O}_2(g) \longrightarrow 2\text{O}(g)$	+500
$\text{O}(g) + 2e^- \longrightarrow \text{O}^{2-}(g)$	+600
$\text{Sr}(s) + \frac{1}{2}\text{O}_2(g) \longrightarrow \text{SrO}(s)$	-550

**Question 7 10 points**

Consider the molecule given on the left, and answer questions (i) to (vi) using chemical bonding theories that we have learnt in class (valence bond theory, hybridization).

Note that the lone pairs are not shown in the structure:

i. Consider the carbon atoms marked in as ***a***, ***b***, ***c***, ***d***, and ***e*** in the molecule, and determine **hybridization** at each of the carbon centers. (2.5 points)

Atom	Hybridization
Carbon (<i>a</i>)	
Carbon (<i>b</i>)	
Carbon (<i>c</i>)	
Carbon (<i>d</i>)	
Carbon (<i>e</i>)	

ii. Now, consider the sulfur atom (S) marked as ***f***, nitrogen (N) atom (***g***), and oxygen(O) atoms (***h*** and ***i***), and determine their **hybridization**. (2 points)

Atom	Hybridization
Sulfur (<i>f</i>)	
Nitrogen (<i>g</i>)	
Oxygen (<i>h</i>)	
Oxygen (<i>i</i>)	

iii. **Describe** all the bonds formed by the carbon atoms ***a*** and ***b***, using sigma and pi designation as appropriate. (2 points)

Atom	Bonds formed (use sigma and pi as appropriate)
Carbon (<i>a</i>)	
Carbon (<i>b</i>)	

iv. How many total sp^3 -s sigma bonds are in the molecule? (1 point)

v. Determine **electronic** and **molecular geometry** at carbon (C) labeled ***e*** and sulfur (S) labeled ***f*** centres. (2 points)

Atom	Electronic Geometry	Molecular Geometry
Carbon (<i>e</i>)		
Sulfur (<i>f</i>)		

vi. Circle the value from the choices given below, that describes the **C-S-C bond angle at sulfur** (S) marked ***f***: (0.5 points)

>90° to <109.5° 109.5° >109.5 to <120° 120° >120° to <180°


Question 8 12 points

Using the **valence shell orbitals only** draw a **complete** molecular orbital (MO) diagram (with no s-p mixing) for the molecule **PS**, in the space provided below. **(6 points)**

Assume that 3s and 3p atomic orbitals will form molecular orbitals like those formed by the combination of 2s and 2p atomic orbitals that we learnt in class, and there is no s-p mixing.

Show **energy arrow**, **atomic orbitals** involved, **atom labels**, the **molecular orbitals** that are formed, **filled electrons** in the MO diagram etc. Make sure to label all the atomic and molecular orbitals.

Using the MO diagram above, answer the following questions:

- (i) Which one of the following will have the highest bond order? Circle the correct choice
 A) PS^{1-} B) PS C) PS^{1+}
- (ii) Which one of the following will have the longest bond length? Circle the correct choice
 A) PS^{1-} B) PS C) PS^{1+}
- (iii) Which of the following will be diamagnetic? Circle the correct choice.
 A) PS^{1-} B) PS C) PS^{1+}
- (iv) Draw Lewis dot structure of the molecule with the shortest bond length (based on the MO diagram) from among the three (PS^{1-} , PS, PS^{1+}), and indicate the bond order calculated from the Lewis dot structure. **(3 points)**

Lewis Dot Structure

Bond Order



8475128C-443C-4F5B-9886-6286A82EFC77

midterm-exam-93388

#578 Page 10 of 10

SCRAP PAGE

**CHEMISTRY 110 –Midterm Exam****Thursday November 7, 2024, 6:30 – 8:30 pm**

Examiners: Pallavi Sirjoosingh and Ashok Kakkar

First name (please write as legibly as possible within the boxes)

--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--

Last name

--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--

Student ID number

--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--

This booklet has 8 questions on 10 pages (The last page can be used as a scrap page).

The staple must NOT be removed from the question booklet. Your intact question booklet must be returned at the end of the exam.

A separate booklet contains a periodic table, and datasheet.

Write your answers directly under the questions in the space provided. Do not write on the QR codes or near the top edges (as it may be cut off).

Please write **legibly and dark enough** for the scan to be readable. For students who use English as a second language, dictionaries are permitted. Calculators are allowed.

DO NOT ASK QUESTIONS OF THE INVIGILATORS about interpretations of the exam questions. If you have a concern about a question, please summarize it at the top of this page. All concerns will be evaluated after the exam. GOOD LUCK!

**Question 1 8 points** (*Assume all transitions are possible*)

a. 6 points Calculate the **energy (in kJ/mol)** required to ionize (completely remove) the electron from the **$n=3$** level of **He^+ ion** (He: Helium). Show your work including equations. (3 sig. figs.)

b. 2 points

(I) Select the largest (longest) wavelength from the three choices below:

- i. Wavelength of photon to *completely remove* an electron from *1s orbital of Hydrogen atom*
- ii. Wavelength of photon to *completely remove* an electron from *1s orbital of Boron atom (B)*
- iii. Wavelength of photon to *completely remove* an electron from *1s orbital of Boron cation (B^+)*

i or ii or iii

(II) Select the smallest (shortest) wavelength from the three choices below (*for H atom*):

- i. Wavelength of photon *emitted* for an electron transitioning from *2p to 1s orbital*
- ii. Wavelength of photon *absorbed* for an electron transitioning from *2s to 4p orbital*
- iii. Wavelength of photon *absorbed* for an electron transitioning from *1s to 3p orbital*

i or ii or iii



Question 2 8 points



A Chem110 student plots the graph shown on the left, from data obtained for the kinetic energy (eV) of the ejected electron versus frequency (s^{-1}) of the incident light, for an unknown metal. ($1\text{eV} = 1.6022 \times 10^{-19} \text{ J}$)

Based on the plot, answer the questions **a to c** below:

a. 1 point On the plot above, label the **threshold frequency**.

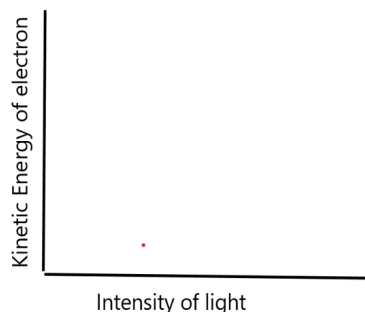
b. 2 points What is the value of the work potential (in J) of the unknown metal?

c. 3 points What is the *longest* wavelength (in nm) required to eject an electron from this metal's surface? Show your work including equation(s) used. (3 sig figs)

d. 2 points In the space provided below, plot the following

(Assume frequency is *above* threshold frequency):

- i. *Kinetic energy* of an ejected electron versus the *intensity* of the incident light (left)
- ii. *Number of electrons* ejected versus the *frequency* of the incident light (right)



**Question 3** **6 points****a. 2 points**

In the ground state of Ge (Germanium; $Z = 32$) atom, **name** the following orbitals:

i. Orbital containing the electron that requires the maximum energy to be removed.

ii. Orbital containing the electron that experiences the minimum effective nuclear charge.

b. 2 points

An unknown anion (A^-) has the following *excited* electronic configuration: $[Ar] 4s^2 3d^3 4p^1$

(Ar: Argon; $Z = 18$)

Give the **condensed ground state electronic configuration** of the atom, **A** and the cation, **A^{2+}** .
Show your work. (Assume all electronic transitions are possible)

Electronic Configuration of A:

Electronic Configuration of A^{2+} :

c. 2 points

In a single atom, determine how many electrons can have the following set of quantum numbers?

(i) $n = 5; l = 3$

In a single atom, determine how many electrons can have the following set of quantum numbers?

(i) $n = 3; l = 2; m_s = +1/2$

**Question 4** **8 points**

Rank the following from 1 to 4 in **increasing** order. Provide the ranking in the boxes provided.

Note: Atomic number (Z) are provided for relevant atoms

a) Increasing (lowest to highest) order of *ionization energy*: (1= lowest; 4 = highest)

Mg

Mg: Z = 12

Rb

Rb: Z = 37

Cs

Cs: Z = 55

K

K : Z = 19

b) Increasing (lowest to highest) order of *bond angle*: (1= lowest; 4 = highest)

H₂OCCl₄XeF₄

HCN

c) Increasing (smallest to largest) order of *ionic size* (1= smallest; 4 = largest)

S²⁻

S: Z = 16

O²⁻

O: Z = 8

Be²⁺

Be: Z = 4

Na⁺

Na: Z = 11

d) Increasing (lowest to highest) order of *bond energy* (1= lowest; 4 = highest)

P-F

P-Cl

P-I

P-Br

**Question 5** **12 points**

a. 9 points For the molecules HClO_4 and TeF_6 , draw the most stable *Lewis structure* showing appropriate shape and *dashed/wedged bonds*, based on VSEPR. Determine the **molecular geometry** around the central atom. For HClO_4 , label the *most polar bond*.



b. 3 points (Note: Lone pairs are not shown in structures below)

(i) For the three *resonance* structures (1, 2 and 3) shown below, answer the following question:



(I) Circle the *most contributing* resonance structure.

Structure 1

Structure 2

Structure 3

(ii) For the three structures given (4, 5, and 6) below, answer the following questions:



(I) Circle all the structures that represent *resonance structures* of each other:

Structure 4

Structure 5

Structure 6

(II) Among the resonance structures, circle the *most contributing* resonance structure.

Structure 4

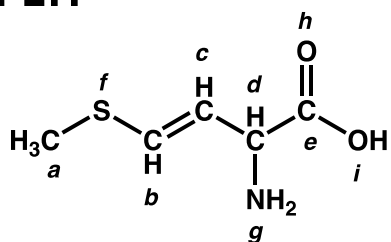
Structure 5

Structure 6

**Question 6** **6 points**

Using the data given below, calculate the **lattice energy** for SrO(s).

	ΔH° (kJ/mol)
$\text{Sr}(s) \longrightarrow \text{Sr}(g)$	+180
$\text{Sr}(g) \longrightarrow \text{Sr}^{2+}(g) + 2e^-$	+1468
$\text{O}_2(g) \longrightarrow 2\text{O}(g)$	+500
$\text{O}(g) + 2e^- \longrightarrow \text{O}^{2-}(g)$	+600
$\text{Sr}(s) + \frac{1}{2}\text{O}_2(g) \longrightarrow \text{SrO}(s)$	-550

**Question 7 10 points**

Consider the molecule given on the left, and answer questions (i) to (vi) using chemical bonding theories that we have learnt in class (valence bond theory, hybridization).

Note that the lone pairs are not shown in the structure:

i. Consider the carbon atoms marked in as ***a***, ***b***, ***c***, ***d***, and ***e*** in the molecule, and determine **hybridization** at each of the carbon centers. (2.5 points)

Atom	Hybridization
Carbon (<i>a</i>)	
Carbon (<i>b</i>)	
Carbon (<i>c</i>)	
Carbon (<i>d</i>)	
Carbon (<i>e</i>)	

ii. Now, consider the sulfur atom (S) marked as ***f***, nitrogen (N) atom (***g***), and oxygen(O) atoms (***h*** and ***i***), and determine their **hybridization**. (2 points)

Atom	Hybridization
Sulfur (<i>f</i>)	
Nitrogen (<i>g</i>)	
Oxygen (<i>h</i>)	
Oxygen (<i>i</i>)	

iii. **Describe** all the bonds formed by the carbon atoms ***a*** and ***b***, using sigma and pi designation as appropriate. (2 points)

Atom	Bonds formed (use sigma and pi as appropriate)
Carbon (<i>a</i>)	
Carbon (<i>b</i>)	

iv. How many total sp^3 -s sigma bonds are in the molecule? (1 point)

v. Determine **electronic** and **molecular geometry** at carbon (C) labeled ***e*** and sulfur (S) labeled ***f*** centres. (2 points)

Atom	Electronic Geometry	Molecular Geometry
Carbon (<i>e</i>)		
Sulfur (<i>f</i>)		

vi. Circle the value from the choices given below, that describes the **C-S-C bond angle at sulfur** (S) marked ***f***: (0.5 points)

>90° to <109.5°

109.5°

>109.5 to <120°

120°

>120° to <180°


Question 8 12 points

Using the **valence shell orbitals only** draw a **complete** molecular orbital (MO) diagram (with no s-p mixing) for the molecule **PS**, in the space provided below. **(6 points)**

Assume that 3s and 3p atomic orbitals will form molecular orbitals like those formed by the combination of 2s and 2p atomic orbitals that we learnt in class, and there is no s-p mixing.

Show **energy arrow**, **atomic orbitals** involved, **atom labels**, the **molecular orbitals** that are formed, **filled electrons** in the MO diagram etc. Make sure to **label all the atomic and molecular orbitals**.

Using the MO diagram above, answer the following questions:

- (i) Which one of the following will have the highest bond order? Circle the correct choice
 A) PS^{1-} B) PS C) PS^{1+}
- (ii) Which one of the following will have the longest bond length? Circle the correct choice
 A) PS^{1-} B) PS C) PS^{1+}
- (iii) Which of the following will be diamagnetic? Circle the correct choice.
 A) PS^{1-} B) PS C) PS^{1+}
- (iv) Draw Lewis dot structure of the molecule with the shortest bond length (based on the MO diagram) from among the three (PS^{1-} , PS, PS^{1+}), and indicate the bond order calculated from the Lewis dot structure. **(3 points)**

Lewis Dot Structure

Bond Order



83B7F18D-1D6E-4147-8766-E2A1D0C69582

midterm-exam-93388

#579 Page 10 of 10

SCRAP PAGE

**CHEMISTRY 110 –Midterm Exam****Thursday November 7, 2024, 6:30 – 8:30 pm**

Examiners: Pallavi Sirjoosingh and Ashok Kakkar

First name (please write as legibly as possible within the boxes)

--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--

Last name

--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--

Student ID number

--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--

This booklet has 8 questions on 10 pages (The last page can be used as a scrap page).

The staple must NOT be removed from the question booklet. Your intact question booklet must be returned at the end of the exam.

A separate booklet contains a periodic table, and datasheet.

Write your answers directly under the questions in the space provided. Do not write on the QR codes or near the top edges (as it may be cut off).

Please write **legibly and dark enough** for the scan to be readable. For students who use English as a second language, dictionaries are permitted. Calculators are allowed.

DO NOT ASK QUESTIONS OF THE INVIGILATORS about interpretations of the exam questions. If you have a concern about a question, please summarize it at the top of this page. All concerns will be evaluated after the exam. GOOD LUCK!

**Question 1 8 points** (*Assume all transitions are possible*)

a. 6 points Calculate the **energy (in kJ/mol)** required to ionize (completely remove) the electron from the **n=3** level of **He⁺ ion** (He: Helium). Show your work including equations. (3 sig. figs.)

b. 2 points

(I) Select the largest (longest) wavelength from the three choices below:

- i. Wavelength of photon to *completely remove* an electron from *1s orbital of Hydrogen atom*
- ii. Wavelength of photon to *completely remove* an electron from *1s orbital of Boron atom (B)*
- iii. Wavelength of photon to *completely remove* an electron from *1s orbital of Boron cation (B⁺)*

i or ii or iii

(II) Select the smallest (shortest) wavelength from the three choices below (*for H atom*):

- i. Wavelength of photon *emitted* for an electron transitioning from *2p to 1s orbital*
- ii. Wavelength of photon *absorbed* for an electron transitioning from *2s to 4p orbital*
- iii. Wavelength of photon *absorbed* for an electron transitioning from *1s to 3p orbital*

i or ii or iii



Question 2 8 points



A Chem110 student plots the graph shown on the left, from data obtained for the kinetic energy (eV) of the ejected electron versus frequency (s^{-1}) of the incident light, for an unknown metal. ($1\text{eV} = 1.6022 \times 10^{-19} \text{ J}$)

Based on the plot, answer the questions **a to c** below:

a. 1 point On the plot above, label the **threshold frequency**.

b. 2 points What is the value of the work potential (in J) of the unknown metal?

c. 3 points What is the *longest* wavelength (in nm) required to eject an electron from this metal's surface? Show your work including equation(s) used. (3 sig figs)

d. 2 points In the space provided below, plot the following

(Assume frequency is *above* threshold frequency):

- i. *Kinetic energy* of an ejected electron versus the *intensity* of the incident light (left)
- ii. *Number of electrons* ejected versus the *frequency* of the incident light (right)



**Question 3** **6 points****a. 2 points**

In the ground state of Ge (Germanium; $Z = 32$) atom, **name** the following orbitals:

i. Orbital containing the electron that requires the maximum energy to be removed.

ii. Orbital containing the electron that experiences the minimum effective nuclear charge.

b. 2 points

An unknown anion (A^-) has the following *excited* electronic configuration: $[Ar] 4s^2 3d^3 4p^1$

(Ar: Argon; $Z = 18$)

Give the **condensed ground state electronic configuration** of the atom, **A** and the cation, **A^{2+}** .
Show your work. (Assume all electronic transitions are possible)

Electronic Configuration of A:

Electronic Configuration of A^{2+} :

c. 2 points

In a single atom, determine how many electrons can have the following set of quantum numbers?

(i) $n = 5; l = 3$

In a single atom, determine how many electrons can have the following set of quantum numbers?

(i) $n = 3; l = 2; m_s = +1/2$

**Question 4** **8 points**

Rank the following from 1 to 4 in **increasing** order. Provide the ranking in the boxes provided.

Note: Atomic number (Z) are provided for relevant atoms

a) Increasing (lowest to highest) order of *ionization energy*: (1= lowest; 4 = highest)

Mg

Mg: Z = 12

Rb

Rb: Z = 37

Cs

Cs: Z = 55

K

K : Z = 19

b) Increasing (lowest to highest) order of *bond angle*: (1= lowest; 4 = highest)

H₂OCCl₄XeF₄

HCN

c) Increasing (smallest to largest) order of *ionic size* (1= smallest; 4 = largest)

S²⁻

S: Z = 16

O²⁻

O: Z = 8

Be²⁺

Be: Z = 4

Na⁺

Na: Z = 11

d) Increasing (lowest to highest) order of *bond energy* (1= lowest; 4 = highest)

P-F

P-Cl

P-I

P-Br

**Question 5** **12 points**

a. 9 points For the molecules HClO_4 and TeF_6 , draw the most stable *Lewis structure* showing appropriate shape and *dashed/wedged bonds*, based on VSEPR. Determine the **molecular geometry** around the central atom. For HClO_4 , label the *most polar bond*.



b. 3 points (Note: Lone pairs are not shown in structures below)

(i) For the three *resonance* structures (1, 2 and 3) shown below, answer the following question:



(I) Circle the *most contributing* resonance structure.

Structure 1

Structure 2

Structure 3

(ii) For the three structures given (4, 5, and 6) below, answer the following questions:



(I) Circle all the structures that represent *resonance structures* of each other:

Structure 4

Structure 5

Structure 6

(II) Among the resonance structures, circle the *most contributing* resonance structure.

Structure 4

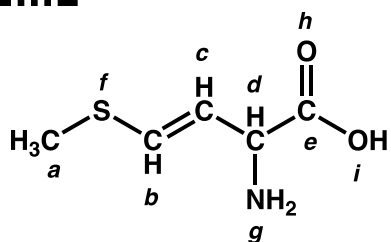
Structure 5

Structure 6

**Question 6** **6 points**

Using the data given below, calculate the **lattice energy** for SrO(s).

	ΔH° (kJ/mol)
$\text{Sr}(s) \longrightarrow \text{Sr}(g)$	+180
$\text{Sr}(g) \longrightarrow \text{Sr}^{2+}(g) + 2e^-$	+1468
$\text{O}_2(g) \longrightarrow 2\text{O}(g)$	+500
$\text{O}(g) + 2e^- \longrightarrow \text{O}^{2-}(g)$	+600
$\text{Sr}(s) + \frac{1}{2}\text{O}_2(g) \longrightarrow \text{SrO}(s)$	-550

**Question 7 10 points**

Consider the molecule given on the left, and answer questions (i) to (vi) using chemical bonding theories that we have learnt in class (valence bond theory, hybridization).

Note that the lone pairs are not shown in the structure:

i. Consider the carbon atoms marked in as ***a***, ***b***, ***c***, ***d***, and ***e*** in the molecule, and determine **hybridization** at each of the carbon centers. (2.5 points)

Atom	Hybridization
Carbon (<i>a</i>)	
Carbon (<i>b</i>)	
Carbon (<i>c</i>)	
Carbon (<i>d</i>)	
Carbon (<i>e</i>)	

ii. Now, consider the sulfur atom (S) marked as ***f***, nitrogen (N) atom (***g***), and oxygen(O) atoms (***h*** and ***i***), and determine their **hybridization**. (2 points)

Atom	Hybridization
Sulfur (<i>f</i>)	
Nitrogen (<i>g</i>)	
Oxygen (<i>h</i>)	
Oxygen (<i>i</i>)	

iii. **Describe** all the bonds formed by the carbon atoms ***a*** and ***b***, using sigma and pi designation as appropriate. (2 points)

Atom	Bonds formed (use sigma and pi as appropriate)
Carbon (<i>a</i>)	
Carbon (<i>b</i>)	

iv. How many total sp^3 -s sigma bonds are in the molecule? (1 point)

v. Determine **electronic** and **molecular geometry** at carbon (C) labeled ***e*** and sulfur (S) labeled ***f*** centres. (2 points)

Atom	Electronic Geometry	Molecular Geometry
Carbon (<i>e</i>)		
Sulfur (<i>f</i>)		

vi. Circle the value from the choices given below, that describes the **C-S-C bond angle at sulfur** (S) marked ***f***: (0.5 points)

$>90^\circ$ to $<109.5^\circ$ 109.5° >109.5 to $<120^\circ$ 120° $>120^\circ$ to $<180^\circ$


Question 8 12 points

Using the **valence shell orbitals only** draw a **complete** molecular orbital (MO) diagram (with no s-p mixing) for the molecule **PS**, in the space provided below. **(6 points)**

Assume that 3s and 3p atomic orbitals will form molecular orbitals like those formed by the combination of 2s and 2p atomic orbitals that we learnt in class, and there is no s-p mixing.

Show **energy arrow**, **atomic orbitals** involved, **atom labels**, the **molecular orbitals** that are formed, **filled electrons** in the MO diagram etc. Make sure to label all the atomic and molecular orbitals.

Using the MO diagram above, answer the following questions:

- (i) Which one of the following will have the highest bond order? Circle the correct choice
 A) PS^{1-} B) PS C) PS^{1+}
- (ii) Which one of the following will have the longest bond length? Circle the correct choice
 A) PS^{1-} B) PS C) PS^{1+}
- (iii) Which of the following will be diamagnetic? Circle the correct choice.
 A) PS^{1-} B) PS C) PS^{1+}
- (iv) Draw Lewis dot structure of the molecule with the shortest bond length (based on the MO diagram) from among the three (PS^{1-} , PS, PS^{1+}), and indicate the bond order calculated from the Lewis dot structure. **(3 points)**

Lewis Dot Structure

Bond Order



3C88C606-FF5F-4C50-802E-15D39E6BD757

midterm-exam-93388

#580 Page 10 of 10

SCRAP PAGE

**CHEMISTRY 110 –Midterm Exam****Thursday November 7, 2024, 6:30 – 8:30 pm****Examiners: Pallavi Sirjoosingh and Ashok Kakkar**

First name (please write as legibly as possible within the boxes)

--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--

Last name

--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--

Student ID number

--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--

This booklet has 8 questions on 10 pages (The last page can be used as a scrap page).

The staple must NOT be removed from the question booklet. Your intact question booklet must be returned at the end of the exam.

A separate booklet contains a periodic table, and datasheet.

Write your answers directly under the questions in the space provided. Do not write on the QR codes or near the top edges (as it may be cut off).

Please write **legibly and dark enough** for the scan to be readable. For students who use English as a second language, dictionaries are permitted. Calculators are allowed.

DO NOT ASK QUESTIONS OF THE INVIGILATORS about interpretations of the exam questions. If you have a concern about a question, please summarize it at the top of this page. All concerns will be evaluated after the exam. GOOD LUCK!

**Question 1 8 points** (*Assume all transitions are possible*)

a. 6 points Calculate the **energy (in kJ/mol)** required to ionize (completely remove) the electron from the **n=3** level of **He⁺ ion** (He: Helium). Show your work including equations. (3 sig. figs.)

b. 2 points

(I) Select the largest (longest) wavelength from the three choices below:

- i. Wavelength of photon to *completely remove* an electron from *1s orbital of Hydrogen atom*
- ii. Wavelength of photon to *completely remove* an electron from *1s orbital of Boron atom (B)*
- iii. Wavelength of photon to *completely remove* an electron from *1s orbital of Boron cation (B⁺)*

i or ii or iii

(II) Select the smallest (shortest) wavelength from the three choices below (*for H atom*):

- i. Wavelength of photon *emitted* for an electron transitioning from *2p to 1s orbital*
- ii. Wavelength of photon *absorbed* for an electron transitioning from *2s to 4p orbital*
- iii. Wavelength of photon *absorbed* for an electron transitioning from *1s to 3p orbital*

i or ii or iii



Question 2 8 points



A Chem110 student plots the graph shown on the left, from data obtained for the kinetic energy (eV) of the ejected electron versus frequency (s^{-1}) of the incident light, for an unknown metal. ($1\text{eV} = 1.6022 \times 10^{-19} \text{ J}$)

Based on the plot, answer the questions **a to c** below:

a. 1 point On the plot above, label the **threshold frequency**.

b. 2 points What is the value of the work potential (in J) of the unknown metal?

c. 3 points What is the *longest* wavelength (in nm) required to eject an electron from this metal's surface? Show your work including equation(s) used. (3 sig figs)

d. 2 points In the space provided below, plot the following

(Assume frequency is *above* threshold frequency):

- Kinetic energy* of an ejected electron versus the *intensity* of the incident light (left)
- Number of electrons* ejected versus the *frequency* of the incident light (right)



**Question 3** **6 points****a. 2 points**

In the ground state of Ge (Germanium; $Z = 32$) atom, **name** the following orbitals:

i. Orbital containing the electron that requires the maximum energy to be removed.

ii. Orbital containing the electron that experiences the minimum effective nuclear charge.

b. 2 points

An unknown anion (A^-) has the following *excited* electronic configuration: $[Ar] 4s^2 3d^3 4p^1$

(Ar: Argon; $Z = 18$)

Give the **condensed ground state electronic configuration** of the atom, **A** and the cation, **A^{2+}** .
Show your work. (Assume all electronic transitions are possible)

Electronic Configuration of A:

Electronic Configuration of A^{2+} :

c. 2 points

In a single atom, determine how many electrons can have the following set of quantum numbers?

(i) $n = 5; l = 3$

In a single atom, determine how many electrons can have the following set of quantum numbers?

(i) $n = 3; l = 2; m_s = +1/2$

**Question 4** **8 points**

Rank the following from 1 to 4 in **increasing** order. Provide the ranking in the boxes provided.

Note: Atomic number (Z) are provided for relevant atoms

a) Increasing (lowest to highest) order of *ionization energy*: (1= lowest; 4 = highest)

Mg

Mg: Z = 12

Rb

Rb: Z = 37

Cs

Cs: Z = 55

K

K : Z = 19

b) Increasing (lowest to highest) order of *bond angle*: (1= lowest; 4 = highest)

H₂OCCl₄XeF₄

HCN

c) Increasing (smallest to largest) order of *ionic size* (1= smallest; 4 = largest)

S²⁻

S: Z = 16

O²⁻

O: Z = 8

Be²⁺

Be: Z = 4

Na⁺

Na: Z = 11

d) Increasing (lowest to highest) order of *bond energy* (1= lowest; 4 = highest)

P-F

P-Cl

P-I

P-Br

**Question 5** **12 points**

a. 9 points For the molecules HClO_4 and TeF_6 , draw the most stable *Lewis structure* showing appropriate shape and *dashed/wedged bonds*, based on VSEPR. Determine the **molecular geometry** around the central atom. For HClO_4 , label the *most polar bond*.



b. 3 points (Note: Lone pairs are not shown in structures below)

(i) For the three *resonance* structures (1, 2 and 3) shown below, answer the following question:



(I) Circle the *most contributing* resonance structure.

Structure 1

Structure 2

Structure 3

(ii) For the three structures given (4, 5, and 6) below, answer the following questions:



(I) Circle all the structures that represent *resonance structures* of each other:

Structure 4

Structure 5

Structure 6

(II) Among the resonance structures, circle the *most contributing* resonance structure.

Structure 4

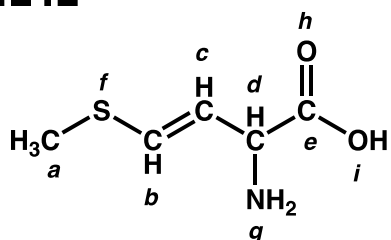
Structure 5

Structure 6

**Question 6** **6 points**

Using the data given below, calculate the **lattice energy** for SrO(s).

	ΔH° (kJ/mol)
$\text{Sr}(s) \longrightarrow \text{Sr}(g)$	+180
$\text{Sr}(g) \longrightarrow \text{Sr}^{2+}(g) + 2e^-$	+1468
$\text{O}_2(g) \longrightarrow 2\text{O}(g)$	+500
$\text{O}(g) + 2e^- \longrightarrow \text{O}^{2-}(g)$	+600
$\text{Sr}(s) + \frac{1}{2}\text{O}_2(g) \longrightarrow \text{SrO}(s)$	-550

**Question 7 10 points**

Consider the molecule given on the left, and answer questions (i) to (vi) using chemical bonding theories that we have learnt in class (valence bond theory, hybridization).

Note that the lone pairs are not shown in the structure:

i. Consider the carbon atoms marked in as ***a***, ***b***, ***c***, ***d***, and ***e*** in the molecule, and determine **hybridization** at each of the carbon centers. (2.5 points)

Atom	Hybridization
Carbon (<i>a</i>)	
Carbon (<i>b</i>)	
Carbon (<i>c</i>)	
Carbon (<i>d</i>)	
Carbon (<i>e</i>)	

ii. Now, consider the sulfur atom (S) marked as ***f***, nitrogen (N) atom (***g***), and oxygen(O) atoms (***h*** and ***i***), and determine their **hybridization**. (2 points)

Atom	Hybridization
Sulfur (<i>f</i>)	
Nitrogen (<i>g</i>)	
Oxygen (<i>h</i>)	
Oxygen (<i>i</i>)	

iii. **Describe** all the bonds formed by the carbon atoms ***a*** and ***b***, using sigma and pi designation as appropriate. (2 points)

Atom	Bonds formed (use sigma and pi as appropriate)
Carbon (<i>a</i>)	
Carbon (<i>b</i>)	

iv. How many total sp^3 -s sigma bonds are in the molecule? (1 point)

v. Determine **electronic** and **molecular geometry** at carbon (C) labeled ***e*** and sulfur (S) labeled ***f*** centres. (2 points)

Atom	Electronic Geometry	Molecular Geometry
Carbon (<i>e</i>)		
Sulfur (<i>f</i>)		

vi. Circle the value from the choices given below, that describes the **C-S-C bond angle at sulfur** (S) marked ***f***: (0.5 points)

>90° to <109.5°

109.5°

>109.5 to <120°

120°

>120° to <180°

**Question 8 12 points**

Using the **valence shell orbitals only** draw a **complete** molecular orbital (MO) diagram (with no s-p mixing) for the molecule **PS**, in the space provided below. **(6 points)**

Assume that 3s and 3p atomic orbitals will form molecular orbitals like those formed by the combination of 2s and 2p atomic orbitals that we learnt in class, and there is no s-p mixing.

Show **energy arrow**, **atomic orbitals** involved, **atom labels**, the **molecular orbitals** that are formed, **filled electrons** in the MO diagram etc. Make sure to label all the atomic and molecular orbitals.

Using the MO diagram above, answer the following questions:

- (i) Which one of the following will have the highest bond order? Circle the correct choice
A) PS^{1-} B) PS C) PS^{1+}
- (ii) Which one of the following will have the longest bond length? Circle the correct choice
A) PS^{1-} B) PS C) PS^{1+}
- (iii) Which of the following will be diamagnetic? Circle the correct choice.
A) PS^{1-} B) PS C) PS^{1+}
- (iv) Draw Lewis dot structure of the molecule with the shortest bond length (based on the MO diagram) from among the three (PS^{1-} , PS, PS^{1+}), and indicate the bond order calculated from the Lewis dot structure. **(3 points)**

Lewis Dot Structure

Bond Order



8882A623-0DD3-48D8-8E53-3C468F096BA6

midterm-exam-93388

#581 Page 10 of 10

SCRAP PAGE

**CHEMISTRY 110 –Midterm Exam****Thursday November 7, 2024, 6:30 – 8:30 pm**

Examiners: Pallavi Sirjoosingh and Ashok Kakkar

First name (please write as legibly as possible within the boxes)

--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--

Last name

--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--

Student ID number

--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--

This booklet has 8 questions on 10 pages (The last page can be used as a scrap page).

The staple must NOT be removed from the question booklet. Your intact question booklet must be returned at the end of the exam.

A separate booklet contains a periodic table, and datasheet.

Write your answers directly under the questions in the space provided. Do not write on the QR codes or near the top edges (as it may be cut off).

Please write **legibly and dark enough** for the scan to be readable. For students who use English as a second language, dictionaries are permitted. Calculators are allowed.

DO NOT ASK QUESTIONS OF THE INVIGILATORS about interpretations of the exam questions. If you have a concern about a question, please summarize it at the top of this page. All concerns will be evaluated after the exam. GOOD LUCK!

**Question 1 8 points** (*Assume all transitions are possible*)

a. 6 points Calculate the **energy (in kJ/mol)** required to ionize (completely remove) the electron from the **n=3** level of **He⁺ ion** (He: Helium). Show your work including equations. (3 sig. figs.)

b. 2 points

(I) Select the largest (longest) wavelength from the three choices below:

- i. Wavelength of photon to *completely remove* an electron from *1s orbital of Hydrogen atom*
- ii. Wavelength of photon to *completely remove* an electron from *1s orbital of Boron atom (B)*
- iii. Wavelength of photon to *completely remove* an electron from *1s orbital of Boron cation (B⁺)*

i or ii or iii

(II) Select the smallest (shortest) wavelength from the three choices below (*for H atom*):

- i. Wavelength of photon *emitted* for an electron transitioning from *2p to 1s orbital*
- ii. Wavelength of photon *absorbed* for an electron transitioning from *2s to 4p orbital*
- iii. Wavelength of photon *absorbed* for an electron transitioning from *1s to 3p orbital*

i or ii or iii



Question 2 8 points



A Chem110 student plots the graph shown on the left, from data obtained for the kinetic energy (eV) of the ejected electron versus frequency (s^{-1}) of the incident light, for an unknown metal. ($1\text{eV} = 1.6022 \times 10^{-19} \text{ J}$)

Based on the plot, answer the questions **a to c** below:

a. 1 point On the plot above, label the **threshold frequency**.

b. 2 points What is the value of the work potential (in J) of the unknown metal?

c. 3 points What is the *longest* wavelength (in nm) required to eject an electron from this metal's surface? Show your work including equation(s) used. (3 sig figs)

d. 2 points In the space provided below, plot the following

(Assume frequency is *above* threshold frequency):

- Kinetic energy* of an ejected electron versus the *intensity* of the incident light (left)
- Number of electrons* ejected versus the *frequency* of the incident light (right)



**Question 3** **6 points****a. 2 points**

In the ground state of Ge (Germanium; $Z = 32$) atom, **name** the following orbitals:

i. Orbital containing the electron that requires the maximum energy to be removed.

ii. Orbital containing the electron that experiences the minimum effective nuclear charge.

b. 2 points

An unknown anion (A^-) has the following *excited* electronic configuration: $[Ar] 4s^2 3d^3 4p^1$

(Ar: Argon; $Z = 18$)

Give the **condensed ground state electronic configuration** of the atom, **A** and the cation, **A^{2+}** .
Show your work. (Assume all electronic transitions are possible)

Electronic Configuration of A:

Electronic Configuration of A^{2+} :

c. 2 points

In a single atom, determine how many electrons can have the following set of quantum numbers?

(i) $n = 5; l = 3$

In a single atom, determine how many electrons can have the following set of quantum numbers?

(i) $n = 3; l = 2; m_s = +1/2$

**Question 4** **8 points**

Rank the following from 1 to 4 in **increasing** order. Provide the ranking in the boxes provided.

Note: Atomic number (Z) are provided for relevant atoms

a) Increasing (lowest to highest) order of *ionization energy*: (1= lowest; 4 = highest)

Mg

Mg: Z = 12

Rb

Rb: Z = 37

Cs

Cs: Z = 55

K

K : Z = 19

b) Increasing (lowest to highest) order of *bond angle*: (1= lowest; 4 = highest)

H₂OCCl₄XeF₄

HCN

c) Increasing (smallest to largest) order of *ionic size* (1= smallest; 4 = largest)

S²⁻

S: Z = 16

O²⁻

O: Z = 8

Be²⁺

Be: Z = 4

Na⁺

Na: Z = 11

d) Increasing (lowest to highest) order of *bond energy* (1= lowest; 4 = highest)

P-F

P-Cl

P-I

P-Br

**Question 5** **12 points**

a. 9 points For the molecules HClO_4 and TeF_6 , draw the most stable *Lewis structure* showing appropriate shape and *dashed/wedged bonds*, based on VSEPR. Determine the **molecular geometry** around the central atom. For HClO_4 , label the *most polar bond*.

HClO_4 (there are no O-O bonds)

Label the most polar bond

Molecular Geometry around Cl

TeF_6

Molecular Geometry around Te

b. 3 points (Note: Lone pairs are not shown in structures below)

(i) For the three *resonance* structures (1, 2 and 3) shown below, answer the following question:



(I) Circle the *most contributing* resonance structure.

Structure 1

Structure 2

Structure 3

(ii) For the three structures given (4, 5, and 6) below, answer the following questions:



(I) Circle all the structures that represent *resonance structures* of each other:

Structure 4

Structure 5

Structure 6

(II) Among the resonance structures, circle the *most contributing* resonance structure.

Structure 4

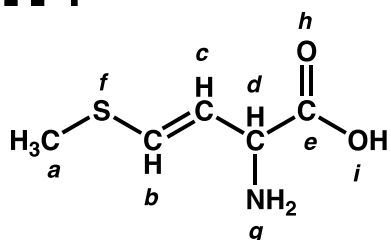
Structure 5

Structure 6

**Question 6** **6 points**

Using the data given below, calculate the **lattice energy** for SrO(s).

	ΔH° (kJ/mol)
$\text{Sr}(s) \longrightarrow \text{Sr}(g)$	+180
$\text{Sr}(g) \longrightarrow \text{Sr}^{2+}(g) + 2e^-$	+1468
$\text{O}_2(g) \longrightarrow 2\text{O}(g)$	+500
$\text{O}(g) + 2e^- \longrightarrow \text{O}^{2-}(g)$	+600
$\text{Sr}(s) + \frac{1}{2}\text{O}_2(g) \longrightarrow \text{SrO}(s)$	-550

**Question 7 10 points**

Consider the molecule given on the left, and answer questions (i) to (vi) using chemical bonding theories that we have learnt in class (valence bond theory, hybridization).

Note that the lone pairs are not shown in the structure:

i. Consider the carbon atoms marked in as ***a***, ***b***, ***c***, ***d***, and ***e*** in the molecule, and determine **hybridization** at each of the carbon centers. (2.5 points)

Atom	Hybridization
Carbon (<i>a</i>)	
Carbon (<i>b</i>)	
Carbon (<i>c</i>)	
Carbon (<i>d</i>)	
Carbon (<i>e</i>)	

ii. Now, consider the sulfur atom (S) marked as ***f***, nitrogen (N) atom (***g***), and oxygen(O) atoms (***h*** and ***i***), and determine their **hybridization**. (2 points)

Atom	Hybridization
Sulfur (<i>f</i>)	
Nitrogen (<i>g</i>)	
Oxygen (<i>h</i>)	
Oxygen (<i>i</i>)	

iii. **Describe** all the bonds formed by the carbon atoms ***a*** and ***b***, using sigma and pi designation as appropriate. (2 points)

Atom	Bonds formed (use sigma and pi as appropriate)
Carbon (<i>a</i>)	
Carbon (<i>b</i>)	

iv. How many total sp^3 -s sigma bonds are in the molecule? (1 point)

v. Determine **electronic** and **molecular geometry** at carbon (C) labeled ***e*** and sulfur (S) labeled ***f*** centres. (2 points)

Atom	Electronic Geometry	Molecular Geometry
Carbon (<i>e</i>)		
Sulfur (<i>f</i>)		

vi. Circle the value from the choices given below, that describes the **C-S-C bond angle at sulfur** (S) marked ***f***: (0.5 points)

$>90^\circ$ to $<109.5^\circ$ 109.5° >109.5 to $<120^\circ$ 120° $>120^\circ$ to $<180^\circ$


Question 8 12 points

Using the **valence shell orbitals only** draw a **complete** molecular orbital (MO) diagram (with no s-p mixing) for the molecule **PS**, in the space provided below. **(6 points)**

Assume that 3s and 3p atomic orbitals will form molecular orbitals like those formed by the combination of 2s and 2p atomic orbitals that we learnt in class, and there is no s-p mixing.

Show **energy arrow**, **atomic orbitals** involved, **atom labels**, the **molecular orbitals** that are formed, **filled electrons** in the MO diagram etc. Make sure to label all the atomic and molecular orbitals.

Using the MO diagram above, answer the following questions:

- (i) Which one of the following will have the highest bond order? Circle the correct choice
 A) PS^{1-} B) PS C) PS^{1+}
- (ii) Which one of the following will have the longest bond length? Circle the correct choice
 A) PS^{1-} B) PS C) PS^{1+}
- (iii) Which of the following will be diamagnetic? Circle the correct choice.
 A) PS^{1-} B) PS C) PS^{1+}
- (iv) Draw Lewis dot structure of the molecule with the shortest bond length (based on the MO diagram) from among the three (PS^{1-} , PS, PS^{1+}), and indicate the bond order calculated from the Lewis dot structure. **(3 points)**

Lewis Dot Structure

Bond Order



550DE8C7-1924-4BC6-BA7E-3F239E3C7956

midterm-exam-93388

#582 Page 10 of 10

SCRAP PAGE

**CHEMISTRY 110 –Midterm Exam****Thursday November 7, 2024, 6:30 – 8:30 pm****Examiners: Pallavi Sirjoosingh and Ashok Kakkar**

First name (please write as legibly as possible within the boxes)

--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--

Last name

--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--

Student ID number

--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--

This booklet has 8 questions on 10 pages (The last page can be used as a scrap page).

The staple must NOT be removed from the question booklet. Your intact question booklet must be returned at the end of the exam.

A separate booklet contains a periodic table, and datasheet.

Write your answers directly under the questions in the space provided. Do not write on the QR codes or near the top edges (as it may be cut off).

Please write **legibly and dark enough** for the scan to be readable. For students who use English as a second language, dictionaries are permitted. Calculators are allowed.

DO NOT ASK QUESTIONS OF THE INVIGILATORS about interpretations of the exam questions. If you have a concern about a question, please summarize it at the top of this page. All concerns will be evaluated after the exam. GOOD LUCK!

**Question 1 8 points** (*Assume all transitions are possible*)

a. 6 points Calculate the **energy (in kJ/mol)** required to ionize (completely remove) the electron from the **n=3** level of **He⁺ ion** (He: Helium). Show your work including equations. (3 sig. figs.)

b. 2 points

(I) Select the largest (longest) wavelength from the three choices below:

- i. Wavelength of photon to *completely remove* an electron from *1s orbital of Hydrogen atom*
- ii. Wavelength of photon to *completely remove* an electron from *1s orbital of Boron atom (B)*
- iii. Wavelength of photon to *completely remove* an electron from *1s orbital of Boron cation (B⁺)*

i or ii or iii

(II) Select the smallest (shortest) wavelength from the three choices below (*for H atom*):

- i. Wavelength of photon *emitted* for an electron transitioning from *2p to 1s orbital*
- ii. Wavelength of photon *absorbed* for an electron transitioning from *2s to 4p orbital*
- iii. Wavelength of photon *absorbed* for an electron transitioning from *1s to 3p orbital*

i or ii or iii



Question 2 8 points



A Chem110 student plots the graph shown on the left, from data obtained for the kinetic energy (eV) of the ejected electron versus frequency (s^{-1}) of the incident light, for an unknown metal. ($1\text{eV} = 1.6022 \times 10^{-19} \text{ J}$)

Based on the plot, answer the questions **a to c** below:

a. 1 point On the plot above, label the **threshold frequency**.

b. 2 points What is the value of the work potential (in J) of the unknown metal?

c. 3 points What is the *longest* wavelength (in nm) required to eject an electron from this metal's surface? Show your work including equation(s) used. (3 sig figs)

d. 2 points In the space provided below, plot the following

(Assume frequency is *above* threshold frequency):

- i. *Kinetic energy* of an ejected electron versus the *intensity* of the incident light (left)
- ii. *Number of electrons* ejected versus the *frequency* of the incident light (right)



**Question 3** **6 points****a. 2 points**

In the ground state of Ge (Germanium; $Z = 32$) atom, **name** the following orbitals:

i. Orbital containing the electron that requires the maximum energy to be removed.

ii. Orbital containing the electron that experiences the minimum effective nuclear charge.

b. 2 points

An unknown anion (A^-) has the following *excited* electronic configuration: $[Ar] 4s^2 3d^3 4p^1$

(Ar: Argon; $Z = 18$)

Give the **condensed ground state electronic configuration** of the atom, **A** and the cation, **A^{2+}** .
Show your work. (Assume all electronic transitions are possible)

Electronic Configuration of A:

Electronic Configuration of A^{2+} :

c. 2 points

In a single atom, determine how many electrons can have the following set of quantum numbers?

(i) $n = 5; l = 3$

In a single atom, determine how many electrons can have the following set of quantum numbers?

(i) $n = 3; l = 2; m_s = +1/2$

**Question 4** **8 points**

Rank the following from 1 to 4 in **increasing** order. Provide the ranking in the boxes provided.

Note: Atomic number (Z) are provided for relevant atoms

a) Increasing (lowest to highest) order of *ionization energy*: (1= lowest; 4 = highest)

Mg

Mg: Z = 12

Rb

Rb: Z = 37

Cs

Cs: Z = 55

K

K : Z = 19

b) Increasing (lowest to highest) order of *bond angle*: (1= lowest; 4 = highest)

H₂OCCl₄XeF₄

HCN

c) Increasing (smallest to largest) order of *ionic size* (1= smallest; 4 = largest)

S²⁻

S: Z = 16

O²⁻

O: Z = 8

Be²⁺

Be: Z = 4

Na⁺

Na: Z = 11

d) Increasing (lowest to highest) order of *bond energy* (1= lowest; 4 = highest)

P-F

P-Cl

P-I

P-Br

**Question 5** **12 points**

a. 9 points For the molecules HClO_4 and TeF_6 , draw the most stable *Lewis structure* showing appropriate shape and *dashed/wedged bonds*, based on VSEPR. Determine the **molecular geometry** around the central atom. For HClO_4 , label the *most polar bond*.



b. 3 points (Note: Lone pairs are not shown in structures below)

(i) For the three *resonance* structures (1, 2 and 3) shown below, answer the following question:



(I) Circle the *most contributing* resonance structure.

Structure 1

Structure 2

Structure 3

(ii) For the three structures given (4, 5, and 6) below, answer the following questions:



(I) Circle all the structures that represent *resonance structures* of each other:

Structure 4

Structure 5

Structure 6

(II) Among the resonance structures, circle the *most contributing* resonance structure.

Structure 4

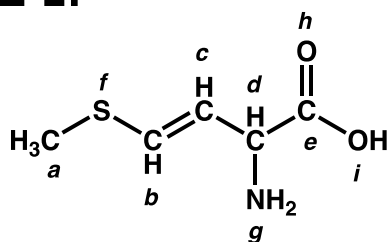
Structure 5

Structure 6

**Question 6** **6 points**

Using the data given below, calculate the **lattice energy** for SrO(s).

	ΔH° (kJ/mol)
$\text{Sr}(s) \longrightarrow \text{Sr}(g)$	+180
$\text{Sr}(g) \longrightarrow \text{Sr}^{2+}(g) + 2e^-$	+1468
$\text{O}_2(g) \longrightarrow 2\text{O}(g)$	+500
$\text{O}(g) + 2e^- \longrightarrow \text{O}^{2-}(g)$	+600
$\text{Sr}(s) + \frac{1}{2}\text{O}_2(g) \longrightarrow \text{SrO}(s)$	-550

**Question 7 10 points**

Consider the molecule given on the left, and answer questions (i) to (vi) using chemical bonding theories that we have learnt in class (valence bond theory, hybridization).

Note that the lone pairs are not shown in the structure:

i. Consider the carbon atoms marked in as ***a***, ***b***, ***c***, ***d***, and ***e*** in the molecule, and determine **hybridization** at each of the carbon centers. (2.5 points)

Atom	Hybridization
Carbon (<i>a</i>)	
Carbon (<i>b</i>)	
Carbon (<i>c</i>)	
Carbon (<i>d</i>)	
Carbon (<i>e</i>)	

ii. Now, consider the sulfur atom (S) marked as ***f***, nitrogen (N) atom (***g***), and oxygen(O) atoms (***h*** and ***i***), and determine their **hybridization**. (2 points)

Atom	Hybridization
Sulfur (<i>f</i>)	
Nitrogen (<i>g</i>)	
Oxygen (<i>h</i>)	
Oxygen (<i>i</i>)	

iii. **Describe** all the bonds formed by the carbon atoms ***a*** and ***b***, using sigma and pi designation as appropriate. (2 points)

Atom	Bonds formed (use sigma and pi as appropriate)
Carbon (<i>a</i>)	
Carbon (<i>b</i>)	

iv. How many total sp^3 -s sigma bonds are in the molecule? (1 point)

v. Determine **electronic** and **molecular geometry** at carbon (C) labeled ***e*** and sulfur (S) labeled ***f*** centres. (2 points)

Atom	Electronic Geometry	Molecular Geometry
Carbon (<i>e</i>)		
Sulfur (<i>f</i>)		

vi. Circle the value from the choices given below, that describes the **C-S-C bond angle at sulfur** (S) marked ***f***: (0.5 points)

>90° to <109.5° 109.5° >109.5 to <120° 120° >120° to <180°


Question 8 12 points

Using the **valence shell orbitals only** draw a **complete** molecular orbital (MO) diagram (with no s-p mixing) for the molecule **PS**, in the space provided below. **(6 points)**

Assume that 3s and 3p atomic orbitals will form molecular orbitals like those formed by the combination of 2s and 2p atomic orbitals that we learnt in class, and there is no s-p mixing.

Show **energy arrow**, **atomic orbitals** involved, **atom labels**, the **molecular orbitals** that are formed, **filled electrons** in the MO diagram etc. Make sure to **label all the atomic and molecular orbitals**.

Using the MO diagram above, answer the following questions:

- (i) Which one of the following will have the highest bond order? Circle the correct choice
 A) PS^{1-} B) PS C) PS^{1+}
- (ii) Which one of the following will have the longest bond length? Circle the correct choice
 A) PS^{1-} B) PS C) PS^{1+}
- (iii) Which of the following will be diamagnetic? Circle the correct choice.
 A) PS^{1-} B) PS C) PS^{1+}
- (iv) Draw Lewis dot structure of the molecule with the shortest bond length (based on the MO diagram) from among the three (PS^{1-} , PS, PS^{1+}), and indicate the bond order calculated from the Lewis dot structure. **(3 points)**

Lewis Dot Structure

Bond Order



1231CA84-0B78-4A8F-96EA-7D2735B4A1BF

midterm-exam-93388

#583 Page 10 of 10

SCRAP PAGE

**CHEMISTRY 110 –Midterm Exam****Thursday November 7, 2024, 6:30 – 8:30 pm**

Examiners: Pallavi Sirjoosingh and Ashok Kakkar

First name (please write as legibly as possible within the boxes)

--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--

Last name

--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--

Student ID number

--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--

This booklet has 8 questions on 10 pages (The last page can be used as a scrap page).

The staple must NOT be removed from the question booklet. Your intact question booklet must be returned at the end of the exam.

A separate booklet contains a periodic table, and datasheet.

Write your answers directly under the questions in the space provided. Do not write on the QR codes or near the top edges (as it may be cut off).

Please write **legibly and dark enough** for the scan to be readable. For students who use English as a second language, dictionaries are permitted. Calculators are allowed.

DO NOT ASK QUESTIONS OF THE INVIGILATORS about interpretations of the exam questions. If you have a concern about a question, please summarize it at the top of this page. All concerns will be evaluated after the exam. GOOD LUCK!

**Question 1 8 points** (*Assume all transitions are possible*)

a. 6 points Calculate the **energy (in kJ/mol)** required to ionize (completely remove) the electron from the **$n=3$** level of **He^+ ion** (He: Helium). Show your work including equations. (3 sig. figs.)

b. 2 points

(I) Select the largest (longest) wavelength from the three choices below:

- i. Wavelength of photon to *completely remove* an electron from *1s orbital of Hydrogen atom*
- ii. Wavelength of photon to *completely remove* an electron from *1s orbital of Boron atom (B)*
- iii. Wavelength of photon to *completely remove* an electron from *1s orbital of Boron cation (B^+)*

i or ii or iii

(II) Select the smallest (shortest) wavelength from the three choices below (*for H atom*):

- i. Wavelength of photon *emitted* for an electron transitioning from *2p to 1s orbital*
- ii. Wavelength of photon *absorbed* for an electron transitioning from *2s to 4p orbital*
- iii. Wavelength of photon *absorbed* for an electron transitioning from *1s to 3p orbital*

i or ii or iii



Question 2 8 points



A Chem110 student plots the graph shown on the left, from data obtained for the kinetic energy (eV) of the ejected electron versus frequency (s^{-1}) of the incident light, for an unknown metal. ($1\text{eV} = 1.6022 \times 10^{-19} \text{ J}$)

Based on the plot, answer the questions **a to c** below:

a. 1 point On the plot above, label the **threshold frequency**.

b. 2 points What is the value of the work potential (in J) of the unknown metal?

c. 3 points What is the *longest* wavelength (in nm) required to eject an electron from this metal's surface? Show your work including equation(s) used. (3 sig figs)

d. 2 points In the space provided below, plot the following

(Assume frequency is *above* threshold frequency):

- i. *Kinetic energy* of an ejected electron versus the *intensity* of the incident light (left)
- ii. *Number of electrons* ejected versus the *frequency* of the incident light (right)



**Question 3** **6 points****a. 2 points**

In the ground state of Ge (Germanium; $Z = 32$) atom, **name** the following orbitals:

i. Orbital containing the electron that requires the maximum energy to be removed.

ii. Orbital containing the electron that experiences the minimum effective nuclear charge.

b. 2 points

An unknown anion (A^-) has the following *excited* electronic configuration: $[Ar] 4s^2 3d^3 4p^1$

(Ar: Argon; $Z = 18$)

Give the **condensed ground state electronic configuration** of the atom, **A** and the cation, **A^{2+}** .
Show your work. (Assume all electronic transitions are possible)

Electronic Configuration of A:

Electronic Configuration of A^{2+} :

c. 2 points

In a single atom, determine how many electrons can have the following set of quantum numbers?

(i) $n = 5; l = 3$

In a single atom, determine how many electrons can have the following set of quantum numbers?

(i) $n = 3; l = 2; m_s = +1/2$

**Question 4** **8 points**

Rank the following from 1 to 4 in **increasing** order. Provide the ranking in the boxes provided.

Note: Atomic number (Z) are provided for relevant atoms

a) Increasing (lowest to highest) order of *ionization energy*: (1= lowest; 4 = highest)

Mg

Mg: Z = 12

Rb

Rb: Z = 37

Cs

Cs: Z = 55

K

K : Z = 19

b) Increasing (lowest to highest) order of *bond angle*: (1= lowest; 4 = highest)

H₂OCCl₄XeF₄

HCN

c) Increasing (smallest to largest) order of *ionic size* (1= smallest; 4 = largest)

S²⁻

S: Z = 16

O²⁻

O: Z = 8

Be²⁺

Be: Z = 4

Na⁺

Na: Z = 11

d) Increasing (lowest to highest) order of *bond energy* (1= lowest; 4 = highest)

P-F

P-Cl

P-I

P-Br

**Question 5** **12 points**

a. 9 points For the molecules HClO_4 and TeF_6 , draw the most stable *Lewis structure* showing appropriate shape and *dashed/wedged bonds*, based on VSEPR. Determine the **molecular geometry** around the central atom. For HClO_4 , label the *most polar bond*.



b. 3 points (Note: Lone pairs are not shown in structures below)

(i) For the three *resonance* structures (1, 2 and 3) shown below, answer the following question:



(I) Circle the *most contributing* resonance structure.

Structure 1

Structure 2

Structure 3

(ii) For the three structures given (4, 5, and 6) below, answer the following questions:



(I) Circle all the structures that represent *resonance structures* of each other:

Structure 4

Structure 5

Structure 6

(II) Among the resonance structures, circle the *most contributing* resonance structure.

Structure 4

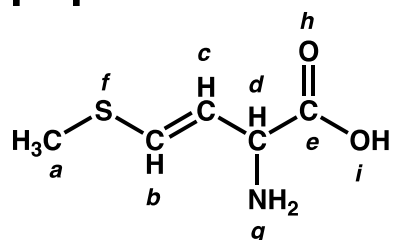
Structure 5

Structure 6

**Question 6** **6 points**

Using the data given below, calculate the **lattice energy** for SrO(s).

	ΔH° (kJ/mol)
$\text{Sr}(s) \longrightarrow \text{Sr}(g)$	+180
$\text{Sr}(g) \longrightarrow \text{Sr}^{2+}(g) + 2e^-$	+1468
$\text{O}_2(g) \longrightarrow 2\text{O}(g)$	+500
$\text{O}(g) + 2e^- \longrightarrow \text{O}^{2-}(g)$	+600
$\text{Sr}(s) + \frac{1}{2}\text{O}_2(g) \longrightarrow \text{SrO}(s)$	-550

**Question 7 10 points**

Consider the molecule given on the left, and answer questions (i) to (vi) using chemical bonding theories that we have learnt in class (valence bond theory, hybridization).

Note that the lone pairs are not shown in the structure:

i. Consider the carbon atoms marked in as ***a***, ***b***, ***c***, ***d***, and ***e*** in the molecule, and determine **hybridization** at each of the carbon centers. (2.5 points)

Atom	Hybridization
Carbon (<i>a</i>)	
Carbon (<i>b</i>)	
Carbon (<i>c</i>)	
Carbon (<i>d</i>)	
Carbon (<i>e</i>)	

ii. Now, consider the sulfur atom (S) marked as ***f***, nitrogen (N) atom (***g***), and oxygen(O) atoms (***h*** and ***i***), and determine their **hybridization**. (2 points)

Atom	Hybridization
Sulfur (<i>f</i>)	
Nitrogen (<i>g</i>)	
Oxygen (<i>h</i>)	
Oxygen (<i>i</i>)	

iii. **Describe** all the bonds formed by the carbon atoms ***a*** and ***b***, using sigma and pi designation as appropriate. (2 points)

Atom	Bonds formed (use sigma and pi as appropriate)
Carbon (<i>a</i>)	
Carbon (<i>b</i>)	

iv. How many total sp^3 -s sigma bonds are in the molecule? (1 point)

v. Determine **electronic** and **molecular geometry** at carbon (C) labeled ***e*** and sulfur (S) labeled ***f*** centres. (2 points)

Atom	Electronic Geometry	Molecular Geometry
Carbon (<i>e</i>)		
Sulfur (<i>f</i>)		

vi. Circle the value from the choices given below, that describes the **C-S-C bond angle at sulfur** (S) marked ***f***: (0.5 points)

>90° to <109.5° 109.5° >109.5 to <120° 120° >120° to <180°

**Question 8 12 points**

Using the **valence shell orbitals only** draw a **complete** molecular orbital (MO) diagram (with no s-p mixing) for the molecule **PS**, in the space provided below. **(6 points)**

Assume that 3s and 3p atomic orbitals will form molecular orbitals like those formed by the combination of 2s and 2p atomic orbitals that we learnt in class, and there is no s-p mixing.

Show **energy arrow**, **atomic orbitals** involved, **atom labels**, the **molecular orbitals** that are formed, **filled electrons** in the MO diagram etc. Make sure to label all the atomic and molecular orbitals.

Using the MO diagram above, answer the following questions:

- (i) Which one of the following will have the highest bond order? Circle the correct choice
A) PS^{1-} B) PS C) PS^{1+}
- (ii) Which one of the following will have the longest bond length? Circle the correct choice
A) PS^{1-} B) PS C) PS^{1+}
- (iii) Which of the following will be diamagnetic? Circle the correct choice.
A) PS^{1-} B) PS C) PS^{1+}
- (iv) Draw Lewis dot structure of the molecule with the shortest bond length (based on the MO diagram) from among the three (PS^{1-} , PS, PS^{1+}), and indicate the bond order calculated from the Lewis dot structure. **(3 points)**

Lewis Dot Structure

Bond Order



334CA723-706F-4D7F-81AA-CBEAED873FC8

midterm-exam-93388

#584 Page 10 of 10

SCRAP PAGE

**CHEMISTRY 110 –Midterm Exam****Thursday November 7, 2024, 6:30 – 8:30 pm****Examiners: Pallavi Sirjoosingh and Ashok Kakkar**

First name (please write as legibly as possible within the boxes)

--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--

Last name

--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--

Student ID number

--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--

This booklet has 8 questions on 10 pages (The last page can be used as a scrap page).

The staple must NOT be removed from the question booklet. Your intact question booklet must be returned at the end of the exam.

A separate booklet contains a periodic table, and datasheet.

Write your answers directly under the questions in the space provided. Do not write on the QR codes or near the top edges (as it may be cut off).

Please write **legibly and dark enough** for the scan to be readable. For students who use English as a second language, dictionaries are permitted. Calculators are allowed.

DO NOT ASK QUESTIONS OF THE INVIGILATORS about interpretations of the exam questions. If you have a concern about a question, please summarize it at the top of this page. All concerns will be evaluated after the exam. GOOD LUCK!

**Question 1 8 points** (*Assume all transitions are possible*)

a. 6 points Calculate the **energy (in kJ/mol)** required to ionize (completely remove) the electron from the **$n=3$** level of **He^+ ion** (He: Helium). Show your work including equations. (3 sig. figs.)

b. 2 points

(I) Select the largest (longest) wavelength from the three choices below:

- i. Wavelength of photon to *completely remove* an electron from *1s orbital of Hydrogen atom*
- ii. Wavelength of photon to *completely remove* an electron from *1s orbital of Boron atom (B)*
- iii. Wavelength of photon to *completely remove* an electron from *1s orbital of Boron cation (B^+)*

i or ii or iii

(II) Select the smallest (shortest) wavelength from the three choices below (*for H atom*):

- i. Wavelength of photon *emitted* for an electron transitioning from *2p to 1s orbital*
- ii. Wavelength of photon *absorbed* for an electron transitioning from *2s to 4p orbital*
- iii. Wavelength of photon *absorbed* for an electron transitioning from *1s to 3p orbital*

i or ii or iii



Question 2 8 points



A Chem110 student plots the graph shown on the left, from data obtained for the kinetic energy (eV) of the ejected electron versus frequency (s^{-1}) of the incident light, for an unknown metal. ($1\text{eV} = 1.6022 \times 10^{-19} \text{ J}$)

Based on the plot, answer the questions **a to c** below:

a. 1 point On the plot above, label the **threshold frequency**.

b. 2 points What is the value of the work potential (in J) of the unknown metal?

c. 3 points What is the *longest* wavelength (in nm) required to eject an electron from this metal's surface? Show your work including equation(s) used. (3 sig figs)

d. 2 points In the space provided below, plot the following

(Assume frequency is *above* threshold frequency):

- i. *Kinetic energy* of an ejected electron versus the *intensity* of the incident light (left)
- ii. *Number of electrons* ejected versus the *frequency* of the incident light (right)



**Question 3** **6 points****a. 2 points**

In the ground state of Ge (Germanium; $Z = 32$) atom, **name** the following orbitals:

i. Orbital containing the electron that requires the maximum energy to be removed.

ii. Orbital containing the electron that experiences the minimum effective nuclear charge.

b. 2 points

An unknown anion (A^-) has the following *excited* electronic configuration: $[Ar] 4s^2 3d^3 4p^1$

(Ar: Argon; $Z = 18$)

Give the **condensed ground state electronic configuration** of the atom, **A** and the cation, **A^{2+}** .
Show your work. (Assume all electronic transitions are possible)

Electronic Configuration of A:

Electronic Configuration of A^{2+} :

c. 2 points

In a single atom, determine how many electrons can have the following set of quantum numbers?

(i) $n = 5; l = 3$

In a single atom, determine how many electrons can have the following set of quantum numbers?

(i) $n = 3; l = 2; m_s = +1/2$

**Question 4** **8 points**

Rank the following from 1 to 4 in **increasing** order. Provide the ranking in the boxes provided.

Note: Atomic number (Z) are provided for relevant atoms

a) Increasing (lowest to highest) order of *ionization energy*: (1= lowest; 4 = highest)

Mg

Mg: Z = 12

Rb

Rb: Z = 37

Cs

Cs: Z = 55

K

K : Z = 19

b) Increasing (lowest to highest) order of *bond angle*: (1= lowest; 4 = highest)

H₂OCCl₄XeF₄

HCN

c) Increasing (smallest to largest) order of *ionic size* (1= smallest; 4 = largest)

S²⁻

S: Z = 16

O²⁻

O: Z = 8

Be²⁺

Be: Z = 4

Na⁺

Na: Z = 11

d) Increasing (lowest to highest) order of *bond energy* (1= lowest; 4 = highest)

P-F

P-Cl

P-I

P-Br

**Question 5** **12 points**

a. 9 points For the molecules HClO_4 and TeF_6 , draw the most stable *Lewis structure* showing appropriate shape and *dashed/wedged bonds*, based on VSEPR. Determine the **molecular geometry** around the central atom. For HClO_4 , label the *most polar bond*.

HClO_4 (there are no O-O bonds)

Label the most polar bond

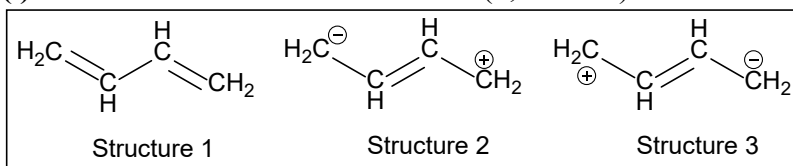
Molecular Geometry around Cl

TeF_6

Molecular Geometry around Te

b. 3 points (Note: Lone pairs are not shown in structures below)

(i) For the three *resonance* structures (1, 2 and 3) shown below, answer the following question:



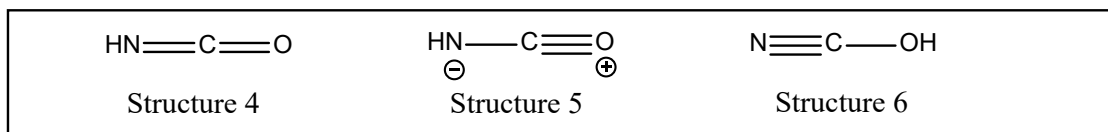
(I) Circle the *most contributing* resonance structure.

Structure 1

Structure 2

Structure 3

(ii) For the three structures given (4, 5, and 6) below, answer the following questions:



(I) Circle all the structures that represent *resonance structures* of each other:

Structure 4

Structure 5

Structure 6

(II) Among the resonance structures, circle the *most contributing* resonance structure.

Structure 4

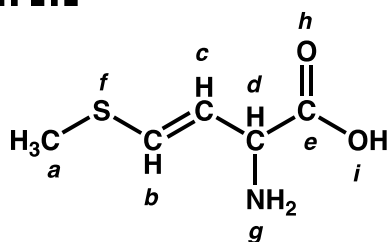
Structure 5

Structure 6

**Question 6** **6 points**

Using the data given below, calculate the **lattice energy** for SrO(s).

	ΔH° (kJ/mol)
$\text{Sr}(s) \longrightarrow \text{Sr}(g)$	+180
$\text{Sr}(g) \longrightarrow \text{Sr}^{2+}(g) + 2e^-$	+1468
$\text{O}_2(g) \longrightarrow 2\text{O}(g)$	+500
$\text{O}(g) + 2e^- \longrightarrow \text{O}^{2-}(g)$	+600
$\text{Sr}(s) + \frac{1}{2}\text{O}_2(g) \longrightarrow \text{SrO}(s)$	-550

**Question 7 10 points**

Consider the molecule given on the left, and answer questions (i) to (vi) using chemical bonding theories that we have learnt in class (valence bond theory, hybridization).

Note that the lone pairs are not shown in the structure:

i. Consider the carbon atoms marked in as ***a***, ***b***, ***c***, ***d***, and ***e*** in the molecule, and determine **hybridization** at each of the carbon centers. (2.5 points)

Atom	Hybridization
Carbon (<i>a</i>)	
Carbon (<i>b</i>)	
Carbon (<i>c</i>)	
Carbon (<i>d</i>)	
Carbon (<i>e</i>)	

ii. Now, consider the sulfur atom (S) marked as ***f***, nitrogen (N) atom (***g***), and oxygen(O) atoms (***h*** and ***i***), and determine their **hybridization**. (2 points)

Atom	Hybridization
Sulfur (<i>f</i>)	
Nitrogen (<i>g</i>)	
Oxygen (<i>h</i>)	
Oxygen (<i>i</i>)	

iii. **Describe** all the bonds formed by the carbon atoms ***a*** and ***b***, using sigma and pi designation as appropriate. (2 points)

Atom	Bonds formed (use sigma and pi as appropriate)
Carbon (<i>a</i>)	
Carbon (<i>b</i>)	

iv. How many total sp^3 -s sigma bonds are in the molecule? (1 point)

v. Determine **electronic** and **molecular geometry** at carbon (C) labeled ***e*** and sulfur (S) labeled ***f*** centres. (2 points)

Atom	Electronic Geometry	Molecular Geometry
Carbon (<i>e</i>)		
Sulfur (<i>f</i>)		

vi. Circle the value from the choices given below, that describes the **C-S-C bond angle at sulfur** (S) marked ***f***: (0.5 points)

>90° to <109.5°

109.5°

>109.5 to <120°

120°

>120° to <180°


Question 8 12 points

Using the **valence shell orbitals only** draw a **complete** molecular orbital (MO) diagram (with no s-p mixing) for the molecule **PS**, in the space provided below. **(6 points)**

Assume that 3s and 3p atomic orbitals will form molecular orbitals like those formed by the combination of 2s and 2p atomic orbitals that we learnt in class, and there is no s-p mixing.

Show **energy arrow**, **atomic orbitals** involved, **atom labels**, the **molecular orbitals** that are formed, **filled electrons** in the MO diagram etc. Make sure to **label all the atomic and molecular orbitals**.

Using the MO diagram above, answer the following questions:

- (i) Which one of the following will have the highest bond order? Circle the correct choice
 A) PS^{1-} B) PS C) PS^{1+}
- (ii) Which one of the following will have the longest bond length? Circle the correct choice
 A) PS^{1-} B) PS C) PS^{1+}
- (iii) Which of the following will be diamagnetic? Circle the correct choice.
 A) PS^{1-} B) PS C) PS^{1+}
- (iv) Draw Lewis dot structure of the molecule with the shortest bond length (based on the MO diagram) from among the three (PS^{1-} , PS, PS^{1+}), and indicate the bond order calculated from the Lewis dot structure. **(3 points)**

Lewis Dot Structure

Bond Order



AC0EF116-C003-4520-8937-5B9682A6EC0D

midterm-exam-93388

#585 Page 10 of 10

SCRAP PAGE

**CHEMISTRY 110 –Midterm Exam****Thursday November 7, 2024, 6:30 – 8:30 pm**

Examiners: Pallavi Sirjoosingh and Ashok Kakkar

First name (please write as legibly as possible within the boxes)

--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--

Last name

--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--

Student ID number

--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--

This booklet has 8 questions on 10 pages (The last page can be used as a scrap page).

The staple must NOT be removed from the question booklet. Your intact question booklet must be returned at the end of the exam.

A separate booklet contains a periodic table, and datasheet.

Write your answers directly under the questions in the space provided. Do not write on the QR codes or near the top edges (as it may be cut off).

Please write **legibly and dark enough** for the scan to be readable. For students who use English as a second language, dictionaries are permitted. Calculators are allowed.

DO NOT ASK QUESTIONS OF THE INVIGILATORS about interpretations of the exam questions. If you have a concern about a question, please summarize it at the top of this page. All concerns will be evaluated after the exam. GOOD LUCK!

**Question 1 8 points** (*Assume all transitions are possible*)

a. 6 points Calculate the **energy (in kJ/mol)** required to ionize (completely remove) the electron from the **n=3** level of **He⁺ ion** (He: Helium). Show your work including equations. (3 sig. figs.)

b. 2 points

(I) Select the largest (longest) wavelength from the three choices below:

- i. Wavelength of photon to *completely remove* an electron from *1s orbital of Hydrogen atom*
- ii. Wavelength of photon to *completely remove* an electron from *1s orbital of Boron atom (B)*
- iii. Wavelength of photon to *completely remove* an electron from *1s orbital of Boron cation (B⁺)*

i or ii or iii

(II) Select the smallest (shortest) wavelength from the three choices below (*for H atom*):

- i. Wavelength of photon *emitted* for an electron transitioning from *2p to 1s orbital*
- ii. Wavelength of photon *absorbed* for an electron transitioning from *2s to 4p orbital*
- iii. Wavelength of photon *absorbed* for an electron transitioning from *1s to 3p orbital*

i or ii or iii



Question 2 8 points



A Chem110 student plots the graph shown on the left, from data obtained for the kinetic energy (eV) of the ejected electron versus frequency (s^{-1}) of the incident light, for an unknown metal. ($1\text{eV} = 1.6022 \times 10^{-19} \text{ J}$)

Based on the plot, answer the questions **a to c** below:

a. 1 point On the plot above, label the **threshold frequency**.

b. 2 points What is the value of the work potential (in J) of the unknown metal?

c. 3 points What is the *longest* wavelength (in nm) required to eject an electron from this metal's surface? Show your work including equation(s) used. (3 sig figs)

d. 2 points In the space provided below, plot the following

(Assume frequency is *above* threshold frequency):

- Kinetic energy* of an ejected electron versus the *intensity* of the incident light (left)
- Number of electrons* ejected versus the *frequency* of the incident light (right)



**Question 3** **6 points****a. 2 points**

In the ground state of Ge (Germanium; $Z = 32$) atom, **name** the following orbitals:

i. Orbital containing the electron that requires the maximum energy to be removed.

ii. Orbital containing the electron that experiences the minimum effective nuclear charge.

b. 2 points

An unknown anion (A^-) has the following *excited* electronic configuration: $[Ar] 4s^2 3d^3 4p^1$

(Ar: Argon; $Z = 18$)

Give the **condensed ground state electronic configuration** of the atom, **A** and the cation, **A^{2+}** .
Show your work. (Assume all electronic transitions are possible)

Electronic Configuration of A:

Electronic Configuration of A^{2+} :

c. 2 points

In a single atom, determine how many electrons can have the following set of quantum numbers?

(i) $n = 5; l = 3$

In a single atom, determine how many electrons can have the following set of quantum numbers?

(i) $n = 3; l = 2; m_s = +1/2$

**Question 4** **8 points**

Rank the following from 1 to 4 in **increasing** order. Provide the ranking in the boxes provided.

Note: Atomic number (Z) are provided for relevant atoms

a) Increasing (lowest to highest) order of *ionization energy*: (1= lowest; 4 = highest)

Mg

Mg: Z = 12

Rb

Rb: Z = 37

Cs

Cs: Z = 55

K

K : Z = 19

b) Increasing (lowest to highest) order of *bond angle*: (1= lowest; 4 = highest)

H₂OCCl₄XeF₄

HCN

c) Increasing (smallest to largest) order of *ionic size* (1= smallest; 4 = largest)

S²⁻

S: Z = 16

O²⁻

O: Z = 8

Be²⁺

Be: Z = 4

Na⁺

Na: Z = 11

d) Increasing (lowest to highest) order of *bond energy* (1= lowest; 4 = highest)

P-F

P-Cl

P-I

P-Br

**Question 5** **12 points**

a. 9 points For the molecules HClO_4 and TeF_6 , draw the most stable *Lewis structure* showing appropriate shape and *dashed/wedged bonds*, based on VSEPR. Determine the **molecular geometry** around the central atom. For HClO_4 , label the *most polar bond*.

HClO_4 (there are no O-O bonds)

Label the most polar bond

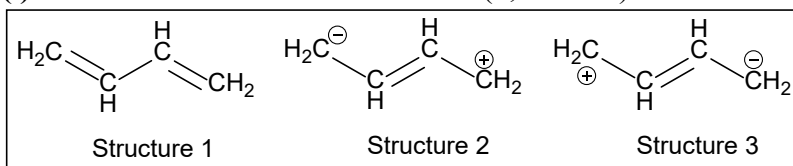
Molecular Geometry around Cl

TeF_6

Molecular Geometry around Te

b. 3 points (Note: Lone pairs are not shown in structures below)

(i) For the three *resonance* structures (1, 2 and 3) shown below, answer the following question:



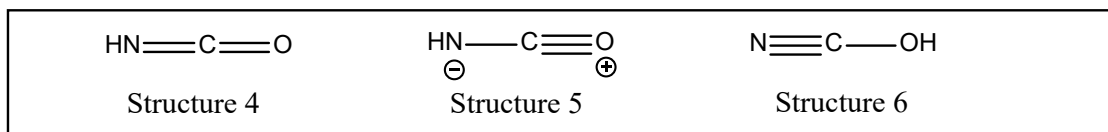
(I) Circle the *most contributing* resonance structure.

Structure 1

Structure 2

Structure 3

(ii) For the three structures given (4, 5, and 6) below, answer the following questions:



(I) Circle all the structures that represent *resonance structures* of each other:

Structure 4

Structure 5

Structure 6

(II) Among the resonance structures, circle the *most contributing* resonance structure.

Structure 4

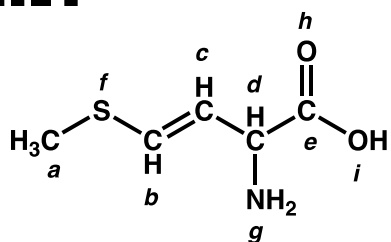
Structure 5

Structure 6

**Question 6** **6 points**

Using the data given below, calculate the **lattice energy** for SrO(s).

	ΔH° (kJ/mol)
$\text{Sr}(s) \longrightarrow \text{Sr}(g)$	+180
$\text{Sr}(g) \longrightarrow \text{Sr}^{2+}(g) + 2e^-$	+1468
$\text{O}_2(g) \longrightarrow 2\text{O}(g)$	+500
$\text{O}(g) + 2e^- \longrightarrow \text{O}^{2-}(g)$	+600
$\text{Sr}(s) + \frac{1}{2}\text{O}_2(g) \longrightarrow \text{SrO}(s)$	-550

**Question 7 10 points**

Consider the molecule given on the left, and answer questions (i) to (vi) using chemical bonding theories that we have learnt in class (valence bond theory, hybridization).

Note that the lone pairs are not shown in the structure:

i. Consider the carbon atoms marked in as ***a***, ***b***, ***c***, ***d***, and ***e*** in the molecule, and determine **hybridization** at each of the carbon centers. (2.5 points)

Atom	Hybridization
Carbon (<i>a</i>)	
Carbon (<i>b</i>)	
Carbon (<i>c</i>)	
Carbon (<i>d</i>)	
Carbon (<i>e</i>)	

ii. Now, consider the sulfur atom (S) marked as ***f***, nitrogen (N) atom (***g***), and oxygen(O) atoms (***h*** and ***i***), and determine their **hybridization**. (2 points)

Atom	Hybridization
Sulfur (<i>f</i>)	
Nitrogen (<i>g</i>)	
Oxygen (<i>h</i>)	
Oxygen (<i>i</i>)	

iii. **Describe** all the bonds formed by the carbon atoms ***a*** and ***b***, using sigma and pi designation as appropriate. (2 points)

Atom	Bonds formed (use sigma and pi as appropriate)
Carbon (<i>a</i>)	
Carbon (<i>b</i>)	

iv. How many total sp^3 -s sigma bonds are in the molecule? (1 point)

v. Determine **electronic** and **molecular geometry** at carbon (C) labeled ***e*** and sulfur (S) labeled ***f*** centres. (2 points)

Atom	Electronic Geometry	Molecular Geometry
Carbon (<i>e</i>)		
Sulfur (<i>f</i>)		

vi. Circle the value from the choices given below, that describes the **C-S-C bond angle at sulfur** (S) marked ***f***: (0.5 points)

>90° to <109.5°

109.5°

>109.5 to <120°

120°

>120° to <180°


Question 8 12 points

Using the **valence shell orbitals only** draw a **complete** molecular orbital (MO) diagram (with no s-p mixing) for the molecule **PS**, in the space provided below. **(6 points)**

Assume that 3s and 3p atomic orbitals will form molecular orbitals like those formed by the combination of 2s and 2p atomic orbitals that we learnt in class, and there is no s-p mixing.

Show **energy arrow**, **atomic orbitals** involved, **atom labels**, the **molecular orbitals** that are formed, **filled electrons** in the MO diagram etc. Make sure to label all the atomic and molecular orbitals.

Using the MO diagram above, answer the following questions:

- (i) Which one of the following will have the highest bond order? Circle the correct choice
 A) PS^{1-} B) PS C) PS^{1+}
- (ii) Which one of the following will have the longest bond length? Circle the correct choice
 A) PS^{1-} B) PS C) PS^{1+}
- (iii) Which of the following will be diamagnetic? Circle the correct choice.
 A) PS^{1-} B) PS C) PS^{1+}
- (iv) Draw Lewis dot structure of the molecule with the shortest bond length (based on the MO diagram) from among the three (PS^{1-} , PS, PS^{1+}), and indicate the bond order calculated from the Lewis dot structure. **(3 points)**

Lewis Dot Structure

Bond Order



8A4AD63B-B8C6-4CB0-9C02-C38ED38D3E4D

midterm-exam-93388

#586 Page 10 of 10

SCRAP PAGE

**CHEMISTRY 110 –Midterm Exam****Thursday November 7, 2024, 6:30 – 8:30 pm****Examiners: Pallavi Sirjoosingh and Ashok Kakkar**

First name (please write as legibly as possible within the boxes)

--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--

Last name

--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--

Student ID number

--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--

This booklet has 8 questions on 10 pages (The last page can be used as a scrap page).

The staple must NOT be removed from the question booklet. Your intact question booklet must be returned at the end of the exam.

A separate booklet contains a periodic table, and datasheet.

Write your answers directly under the questions in the space provided. Do not write on the QR codes or near the top edges (as it may be cut off).

Please write **legibly and dark enough** for the scan to be readable. For students who use English as a second language, dictionaries are permitted. Calculators are allowed.

DO NOT ASK QUESTIONS OF THE INVIGILATORS about interpretations of the exam questions. If you have a concern about a question, please summarize it at the top of this page. All concerns will be evaluated after the exam. GOOD LUCK!

**Question 1 8 points** (*Assume all transitions are possible*)

a. 6 points Calculate the **energy (in kJ/mol)** required to ionize (completely remove) the electron from the **n=3** level of **He⁺ ion** (He: Helium). Show your work including equations. (3 sig. figs.)

b. 2 points

(I) Select the largest (longest) wavelength from the three choices below:

- i. Wavelength of photon to *completely remove* an electron from *1s orbital of Hydrogen atom*
- ii. Wavelength of photon to *completely remove* an electron from *1s orbital of Boron atom (B)*
- iii. Wavelength of photon to *completely remove* an electron from *1s orbital of Boron cation (B⁺)*

i or ii or iii

(II) Select the smallest (shortest) wavelength from the three choices below (*for H atom*):

- i. Wavelength of photon *emitted* for an electron transitioning from *2p to 1s orbital*
- ii. Wavelength of photon *absorbed* for an electron transitioning from *2s to 4p orbital*
- iii. Wavelength of photon *absorbed* for an electron transitioning from *1s to 3p orbital*

i or ii or iii



Question 2 8 points



A Chem110 student plots the graph shown on the left, from data obtained for the kinetic energy (eV) of the ejected electron versus frequency (s^{-1}) of the incident light, for an unknown metal. ($1\text{eV} = 1.6022 \times 10^{-19} \text{ J}$)

Based on the plot, answer the questions **a to c** below:

a. 1 point On the plot above, label the **threshold frequency**.

b. 2 points What is the value of the work potential (in J) of the unknown metal?

c. 3 points What is the *longest* wavelength (in nm) required to eject an electron from this metal's surface? Show your work including equation(s) used. (3 sig figs)

d. 2 points In the space provided below, plot the following

(Assume frequency is *above* threshold frequency):

- Kinetic energy* of an ejected electron versus the *intensity* of the incident light (left)
- Number of electrons* ejected versus the *frequency* of the incident light (right)



**Question 3** **6 points****a. 2 points**

In the ground state of Ge (Germanium; $Z = 32$) atom, **name** the following orbitals:

i. Orbital containing the electron that requires the maximum energy to be removed.

ii. Orbital containing the electron that experiences the minimum effective nuclear charge.

b. 2 points

An unknown anion (A^-) has the following *excited* electronic configuration: $[Ar] 4s^2 3d^3 4p^1$

(Ar: Argon; $Z = 18$)

Give the **condensed ground state electronic configuration** of the atom, **A** and the cation, **A^{2+}** .
Show your work. (Assume all electronic transitions are possible)

Electronic Configuration of A:

Electronic Configuration of A^{2+} :

c. 2 points

In a single atom, determine how many electrons can have the following set of quantum numbers?

(i) $n = 5; l = 3$

In a single atom, determine how many electrons can have the following set of quantum numbers?

(i) $n = 3; l = 2; m_s = +1/2$

**Question 4** **8 points**

Rank the following from 1 to 4 in **increasing** order. Provide the ranking in the boxes provided.

Note: Atomic number (Z) are provided for relevant atoms

a) Increasing (lowest to highest) order of *ionization energy*: (1= lowest; 4 = highest)

Mg

Mg: Z = 12

Rb

Rb: Z = 37

Cs

Cs: Z = 55

K

K : Z = 19

b) Increasing (lowest to highest) order of *bond angle*: (1= lowest; 4 = highest)

H₂OCCl₄XeF₄

HCN

c) Increasing (smallest to largest) order of *ionic size* (1= smallest; 4 = largest)

S²⁻

S: Z = 16

O²⁻

O: Z = 8

Be²⁺

Be: Z = 4

Na⁺

Na: Z = 11

d) Increasing (lowest to highest) order of *bond energy* (1= lowest; 4 = highest)

P-F

P-Cl

P-I

P-Br

**Question 5** **12 points**

a. 9 points For the molecules HClO_4 and TeF_6 , draw the most stable *Lewis structure* showing appropriate shape and *dashed/wedged bonds*, based on VSEPR. Determine the **molecular geometry** around the central atom. For HClO_4 , label the *most polar bond*.



b. 3 points (Note: Lone pairs are not shown in structures below)

(i) For the three *resonance* structures (1, 2 and 3) shown below, answer the following question:



(I) Circle the *most contributing* resonance structure.

Structure 1

Structure 2

Structure 3

(ii) For the three structures given (4, 5, and 6) below, answer the following questions:



(I) Circle all the structures that represent *resonance structures* of each other:

Structure 4

Structure 5

Structure 6

(II) Among the resonance structures, circle the *most contributing* resonance structure.

Structure 4

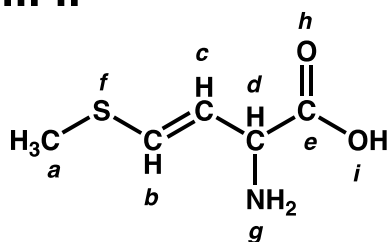
Structure 5

Structure 6

**Question 6** **6 points**

Using the data given below, calculate the **lattice energy** for SrO(s).

	ΔH° (kJ/mol)
$\text{Sr}(s) \longrightarrow \text{Sr}(g)$	+180
$\text{Sr}(g) \longrightarrow \text{Sr}^{2+}(g) + 2e^-$	+1468
$\text{O}_2(g) \longrightarrow 2\text{O}(g)$	+500
$\text{O}(g) + 2e^- \longrightarrow \text{O}^{2-}(g)$	+600
$\text{Sr}(s) + \frac{1}{2}\text{O}_2(g) \longrightarrow \text{SrO}(s)$	-550

**Question 7 10 points**

Consider the molecule given on the left, and answer questions (i) to (vi) using chemical bonding theories that we have learnt in class (valence bond theory, hybridization).

Note that the lone pairs are not shown in the structure:

i. Consider the carbon atoms marked in as ***a***, ***b***, ***c***, ***d***, and ***e*** in the molecule, and determine **hybridization** at each of the carbon centers. (2.5 points)

Atom	Hybridization
Carbon (<i>a</i>)	
Carbon (<i>b</i>)	
Carbon (<i>c</i>)	
Carbon (<i>d</i>)	
Carbon (<i>e</i>)	

ii. Now, consider the sulfur atom (S) marked as ***f***, nitrogen (N) atom (***g***), and oxygen(O) atoms (***h*** and ***i***), and determine their **hybridization**. (2 points)

Atom	Hybridization
Sulfur (<i>f</i>)	
Nitrogen (<i>g</i>)	
Oxygen (<i>h</i>)	
Oxygen (<i>i</i>)	

iii. **Describe** all the bonds formed by the carbon atoms ***a*** and ***b***, using sigma and pi designation as appropriate. (2 points)

Atom	Bonds formed (use sigma and pi as appropriate)
Carbon (<i>a</i>)	
Carbon (<i>b</i>)	

iv. How many total sp^3 -s sigma bonds are in the molecule? (1 point)

v. Determine **electronic** and **molecular geometry** at carbon (C) labeled ***e*** and sulfur (S) labeled ***f*** centres. (2 points)

Atom	Electronic Geometry	Molecular Geometry
Carbon (<i>e</i>)		
Sulfur (<i>f</i>)		

vi. Circle the value from the choices given below, that describes the **C-S-C bond angle at sulfur** (S) marked ***f***: (0.5 points)

>90° to <109.5° 109.5° >109.5 to <120° 120° >120° to <180°


Question 8 12 points

Using the **valence shell orbitals only** draw a **complete** molecular orbital (MO) diagram (with no s-p mixing) for the molecule **PS**, in the space provided below. **(6 points)**

Assume that 3s and 3p atomic orbitals will form molecular orbitals like those formed by the combination of 2s and 2p atomic orbitals that we learnt in class, and there is no s-p mixing.

Show **energy arrow**, **atomic orbitals** involved, **atom labels**, the **molecular orbitals** that are formed, **filled electrons** in the MO diagram etc. Make sure to label all the atomic and molecular orbitals.

Using the MO diagram above, answer the following questions:

- (i) Which one of the following will have the highest bond order? Circle the correct choice
 A) PS^{1-} B) PS C) PS^{1+}
- (ii) Which one of the following will have the longest bond length? Circle the correct choice
 A) PS^{1-} B) PS C) PS^{1+}
- (iii) Which of the following will be diamagnetic? Circle the correct choice.
 A) PS^{1-} B) PS C) PS^{1+}
- (iv) Draw Lewis dot structure of the molecule with the shortest bond length (based on the MO diagram) from among the three (PS^{1-} , PS, PS^{1+}), and indicate the bond order calculated from the Lewis dot structure. **(3 points)**

Lewis Dot Structure

Bond Order



3FE12BFE-9551-4892-94C7-CE858BC459CF

midterm-exam-93388

#587 Page 10 of 10

SCRAP PAGE

**CHEMISTRY 110 –Midterm Exam****Thursday November 7, 2024, 6:30 – 8:30 pm****Examiners: Pallavi Sirjoosingh and Ashok Kakkar**

First name (please write as legibly as possible within the boxes)

--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--

Last name

--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--

Student ID number

--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--

This booklet has 8 questions on 10 pages (The last page can be used as a scrap page).

The staple must NOT be removed from the question booklet. Your intact question booklet must be returned at the end of the exam.

A separate booklet contains a periodic table, and datasheet.

Write your answers directly under the questions in the space provided. Do not write on the QR codes or near the top edges (as it may be cut off).

Please write **legibly and dark enough** for the scan to be readable. For students who use English as a second language, dictionaries are permitted. Calculators are allowed.

DO NOT ASK QUESTIONS OF THE INVIGILATORS about interpretations of the exam questions. If you have a concern about a question, please summarize it at the top of this page. All concerns will be evaluated after the exam. GOOD LUCK!

**Question 1 8 points** (*Assume all transitions are possible*)

a. 6 points Calculate the **energy (in kJ/mol)** required to ionize (completely remove) the electron from the **n=3** level of **He⁺ ion** (He: Helium). Show your work including equations. (3 sig. figs.)

b. 2 points

(I) Select the largest (longest) wavelength from the three choices below:

- i. Wavelength of photon to *completely remove* an electron from *1s orbital of Hydrogen atom*
- ii. Wavelength of photon to *completely remove* an electron from *1s orbital of Boron atom (B)*
- iii. Wavelength of photon to *completely remove* an electron from *1s orbital of Boron cation (B⁺)*

i or ii or iii

(II) Select the smallest (shortest) wavelength from the three choices below (*for H atom*):

- i. Wavelength of photon *emitted* for an electron transitioning from *2p to 1s orbital*
- ii. Wavelength of photon *absorbed* for an electron transitioning from *2s to 4p orbital*
- iii. Wavelength of photon *absorbed* for an electron transitioning from *1s to 3p orbital*

i or ii or iii



Question 2 8 points



A Chem110 student plots the graph shown on the left, from data obtained for the kinetic energy (eV) of the ejected electron versus frequency (s^{-1}) of the incident light, for an unknown metal. ($1\text{eV} = 1.6022 \times 10^{-19} \text{ J}$)

Based on the plot, answer the questions **a to c** below:

a. 1 point On the plot above, label the **threshold frequency**.

b. 2 points What is the value of the work potential (in J) of the unknown metal?

c. 3 points What is the *longest* wavelength (in nm) required to eject an electron from this metal's surface? Show your work including equation(s) used. (3 sig figs)

d. 2 points In the space provided below, plot the following

(Assume frequency is *above* threshold frequency):

- Kinetic energy* of an ejected electron versus the *intensity* of the incident light (left)
- Number of electrons* ejected versus the *frequency* of the incident light (right)



**Question 3** **6 points****a. 2 points**

In the ground state of Ge (Germanium; $Z = 32$) atom, **name** the following orbitals:

i. Orbital containing the electron that requires the maximum energy to be removed.

ii. Orbital containing the electron that experiences the minimum effective nuclear charge.

b. 2 points

An unknown anion (A^-) has the following *excited* electronic configuration: $[Ar] 4s^2 3d^3 4p^1$

(Ar: Argon; $Z = 18$)

Give the **condensed ground state electronic configuration** of the atom, **A** and the cation, **A^{2+}** .
Show your work. (Assume all electronic transitions are possible)

Electronic Configuration of A:

Electronic Configuration of A^{2+} :

c. 2 points

In a single atom, determine how many electrons can have the following set of quantum numbers?

(i) $n = 5; l = 3$

In a single atom, determine how many electrons can have the following set of quantum numbers?

(i) $n = 3; l = 2; m_s = +1/2$

**Question 4** **8 points**

Rank the following from 1 to 4 in **increasing** order. Provide the ranking in the boxes provided.

Note: Atomic number (Z) are provided for relevant atoms

a) Increasing (lowest to highest) order of *ionization energy*: (1= lowest; 4 = highest)

Mg

Mg: Z = 12

Rb

Rb: Z = 37

Cs

Cs: Z = 55

K

K : Z = 19

b) Increasing (lowest to highest) order of *bond angle*: (1= lowest; 4 = highest)

H₂OCCl₄XeF₄

HCN

c) Increasing (smallest to largest) order of *ionic size* (1= smallest; 4 = largest)

S²⁻

S: Z = 16

O²⁻

O: Z = 8

Be²⁺

Be: Z = 4

Na⁺

Na: Z = 11

d) Increasing (lowest to highest) order of *bond energy* (1= lowest; 4 = highest)

P-F

P-Cl

P-I

P-Br

**Question 5** **12 points**

a. 9 points For the molecules HClO_4 and TeF_6 , draw the most stable *Lewis structure* showing appropriate shape and *dashed/wedged bonds*, based on VSEPR. Determine the **molecular geometry** around the central atom. For HClO_4 , label the *most polar bond*.



b. 3 points (Note: Lone pairs are not shown in structures below)

(i) For the three *resonance* structures (1, 2 and 3) shown below, answer the following question:



(I) Circle the *most contributing* resonance structure.

Structure 1

Structure 2

Structure 3

(ii) For the three structures given (4, 5, and 6) below, answer the following questions:



(I) Circle all the structures that represent *resonance structures* of each other:

Structure 4

Structure 5

Structure 6

(II) Among the resonance structures, circle the *most contributing* resonance structure.

Structure 4

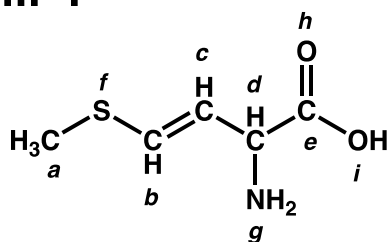
Structure 5

Structure 6

**Question 6** **6 points**

Using the data given below, calculate the **lattice energy** for SrO(s).

	ΔH° (kJ/mol)
$\text{Sr}(s) \longrightarrow \text{Sr}(g)$	+180
$\text{Sr}(g) \longrightarrow \text{Sr}^{2+}(g) + 2e^-$	+1468
$\text{O}_2(g) \longrightarrow 2\text{O}(g)$	+500
$\text{O}(g) + 2e^- \longrightarrow \text{O}^{2-}(g)$	+600
$\text{Sr}(s) + \frac{1}{2}\text{O}_2(g) \longrightarrow \text{SrO}(s)$	-550

**Question 7 10 points**

Consider the molecule given on the left, and answer questions (i) to (vi) using chemical bonding theories that we have learnt in class (valence bond theory, hybridization).

Note that the lone pairs are not shown in the structure:

i. Consider the carbon atoms marked in as ***a***, ***b***, ***c***, ***d***, and ***e*** in the molecule, and determine **hybridization** at each of the carbon centers. (2.5 points)

Atom	Hybridization
Carbon (<i>a</i>)	
Carbon (<i>b</i>)	
Carbon (<i>c</i>)	
Carbon (<i>d</i>)	
Carbon (<i>e</i>)	

ii. Now, consider the sulfur atom (S) marked as ***f***, nitrogen (N) atom (***g***), and oxygen(O) atoms (***h*** and ***i***), and determine their **hybridization**. (2 points)

Atom	Hybridization
Sulfur (<i>f</i>)	
Nitrogen (<i>g</i>)	
Oxygen (<i>h</i>)	
Oxygen (<i>i</i>)	

iii. **Describe** all the bonds formed by the carbon atoms ***a*** and ***b***, using sigma and pi designation as appropriate. (2 points)

Atom	Bonds formed (use sigma and pi as appropriate)
Carbon (<i>a</i>)	
Carbon (<i>b</i>)	

iv. How many total sp^3 -s sigma bonds are in the molecule? (1 point)

v. Determine **electronic** and **molecular geometry** at carbon (C) labeled ***e*** and sulfur (S) labeled ***f*** centres. (2 points)

Atom	Electronic Geometry	Molecular Geometry
Carbon (<i>e</i>)		
Sulfur (<i>f</i>)		

vi. Circle the value from the choices given below, that describes the **C-S-C bond angle at sulfur** (S) marked ***f***: (0.5 points)

>90° to <109.5°

109.5°

>109.5 to <120°

120°

>120° to <180°


Question 8 12 points

Using the **valence shell orbitals only** draw a **complete** molecular orbital (MO) diagram (with no s-p mixing) for the molecule **PS**, in the space provided below. **(6 points)**

Assume that 3s and 3p atomic orbitals will form molecular orbitals like those formed by the combination of 2s and 2p atomic orbitals that we learnt in class, and there is no s-p mixing.

Show **energy arrow**, **atomic orbitals** involved, **atom labels**, the **molecular orbitals** that are formed, **filled electrons** in the MO diagram etc. Make sure to label all the atomic and molecular orbitals.

Using the MO diagram above, answer the following questions:

- (i) Which one of the following will have the highest bond order? Circle the correct choice
 A) PS^{1-} B) PS C) PS^{1+}
- (ii) Which one of the following will have the longest bond length? Circle the correct choice
 A) PS^{1-} B) PS C) PS^{1+}
- (iii) Which of the following will be diamagnetic? Circle the correct choice.
 A) PS^{1-} B) PS C) PS^{1+}
- (iv) Draw Lewis dot structure of the molecule with the shortest bond length (based on the MO diagram) from among the three (PS^{1-} , PS, PS^{1+}), and indicate the bond order calculated from the Lewis dot structure. **(3 points)**

Lewis Dot Structure

Bond Order



528E7D45-FE54-4F3D-AB69-D5589D38B768

midterm-exam-93388

#588 Page 10 of 10

SCRAP PAGE

**CHEMISTRY 110 –Midterm Exam****Thursday November 7, 2024, 6:30 – 8:30 pm****Examiners: Pallavi Sirjoosingh and Ashok Kakkar**

First name (please write as legibly as possible within the boxes)

--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--

Last name

--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--

Student ID number

--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--

This booklet has 8 questions on 10 pages (The last page can be used as a scrap page).

The staple must NOT be removed from the question booklet. Your intact question booklet must be returned at the end of the exam.

A separate booklet contains a periodic table, and datasheet.

Write your answers directly under the questions in the space provided. Do not write on the QR codes or near the top edges (as it may be cut off).

Please write **legibly and dark enough** for the scan to be readable. For students who use English as a second language, dictionaries are permitted. Calculators are allowed.

DO NOT ASK QUESTIONS OF THE INVIGILATORS about interpretations of the exam questions. If you have a concern about a question, please summarize it at the top of this page. All concerns will be evaluated after the exam. GOOD LUCK!

**Question 1 8 points** (*Assume all transitions are possible*)

a. 6 points Calculate the **energy (in kJ/mol)** required to ionize (completely remove) the electron from the **n=3** level of **He⁺ ion** (He: Helium). Show your work including equations. (3 sig. figs.)

b. 2 points

(I) Select the largest (longest) wavelength from the three choices below:

- i. Wavelength of photon to *completely remove* an electron from *1s orbital of Hydrogen atom*
- ii. Wavelength of photon to *completely remove* an electron from *1s orbital of Boron atom (B)*
- iii. Wavelength of photon to *completely remove* an electron from *1s orbital of Boron cation (B⁺)*

i or ii or iii

(II) Select the smallest (shortest) wavelength from the three choices below (*for H atom*):

- i. Wavelength of photon *emitted* for an electron transitioning from *2p to 1s orbital*
- ii. Wavelength of photon *absorbed* for an electron transitioning from *2s to 4p orbital*
- iii. Wavelength of photon *absorbed* for an electron transitioning from *1s to 3p orbital*

i or ii or iii



Question 2 8 points



A Chem110 student plots the graph shown on the left, from data obtained for the kinetic energy (eV) of the ejected electron versus frequency (s^{-1}) of the incident light, for an unknown metal. ($1\text{eV} = 1.6022 \times 10^{-19} \text{ J}$)

Based on the plot, answer the questions **a to c** below:

a. 1 point On the plot above, label the **threshold frequency**.

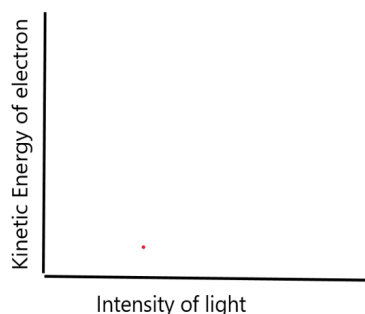
b. 2 points What is the value of the work potential (in J) of the unknown metal?

c. 3 points What is the *longest* wavelength (in nm) required to eject an electron from this metal's surface? Show your work including equation(s) used. (3 sig figs)

d. 2 points In the space provided below, plot the following

(Assume frequency is *above* threshold frequency):

- Kinetic energy* of an ejected electron versus the *intensity* of the incident light (left)
- Number of electrons* ejected versus the *frequency* of the incident light (right)



**Question 3** **6 points****a. 2 points**

In the ground state of Ge (Germanium; $Z = 32$) atom, **name** the following orbitals:

i. Orbital containing the electron that requires the maximum energy to be removed.

ii. Orbital containing the electron that experiences the minimum effective nuclear charge.

b. 2 points

An unknown anion (A^-) has the following *excited* electronic configuration: $[Ar] 4s^2 3d^3 4p^1$

(Ar: Argon; $Z = 18$)

Give the **condensed ground state electronic configuration** of the atom, **A** and the cation, **A^{2+}** .
Show your work. (Assume all electronic transitions are possible)

Electronic Configuration of A:

Electronic Configuration of A^{2+} :

c. 2 points

In a single atom, determine how many electrons can have the following set of quantum numbers?

(i) $n = 5; l = 3$

In a single atom, determine how many electrons can have the following set of quantum numbers?

(i) $n = 3; l = 2; m_s = +1/2$

**Question 4** **8 points**

Rank the following from 1 to 4 in **increasing** order. Provide the ranking in the boxes provided.

Note: Atomic number (Z) are provided for relevant atoms

a) Increasing (lowest to highest) order of *ionization energy*: (1= lowest; 4 = highest)

Mg

Mg: Z = 12

Rb

Rb: Z = 37

Cs

Cs: Z = 55

K

K : Z = 19

b) Increasing (lowest to highest) order of *bond angle*: (1= lowest; 4 = highest)

H₂OCCl₄XeF₄

HCN

c) Increasing (smallest to largest) order of *ionic size* (1= smallest; 4 = largest)

S²⁻

S: Z = 16

O²⁻

O: Z = 8

Be²⁺

Be: Z = 4

Na⁺

Na: Z = 11

d) Increasing (lowest to highest) order of *bond energy* (1= lowest; 4 = highest)

P-F

P-Cl

P-I

P-Br

**Question 5** **12 points**

a. 9 points For the molecules HClO_4 and TeF_6 , draw the most stable *Lewis structure* showing appropriate shape and *dashed/wedged bonds*, based on VSEPR. Determine the **molecular geometry** around the central atom. For HClO_4 , label the *most polar bond*.



b. 3 points (Note: Lone pairs are not shown in structures below)

(i) For the three *resonance* structures (1, 2 and 3) shown below, answer the following question:



(I) Circle the *most contributing* resonance structure.

Structure 1

Structure 2

Structure 3

(ii) For the three structures given (4, 5, and 6) below, answer the following questions:



(I) Circle all the structures that represent *resonance structures* of each other:

Structure 4

Structure 5

Structure 6

(II) Among the resonance structures, circle the *most contributing* resonance structure.

Structure 4

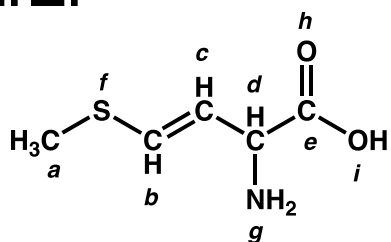
Structure 5

Structure 6

**Question 6** **6 points**

Using the data given below, calculate the **lattice energy** for SrO(s).

	ΔH° (kJ/mol)
$\text{Sr}(s) \longrightarrow \text{Sr}(g)$	+180
$\text{Sr}(g) \longrightarrow \text{Sr}^{2+}(g) + 2e^-$	+1468
$\text{O}_2(g) \longrightarrow 2\text{O}(g)$	+500
$\text{O}(g) + 2e^- \longrightarrow \text{O}^{2-}(g)$	+600
$\text{Sr}(s) + \frac{1}{2}\text{O}_2(g) \longrightarrow \text{SrO}(s)$	-550

**Question 7 10 points**

Consider the molecule given on the left, and answer questions (i) to (vi) using chemical bonding theories that we have learnt in class (valence bond theory, hybridization).

Note that the lone pairs are not shown in the structure:

i. Consider the carbon atoms marked in as ***a***, ***b***, ***c***, ***d***, and ***e*** in the molecule, and determine **hybridization** at each of the carbon centers. (2.5 points)

Atom	Hybridization
Carbon (<i>a</i>)	
Carbon (<i>b</i>)	
Carbon (<i>c</i>)	
Carbon (<i>d</i>)	
Carbon (<i>e</i>)	

ii. Now, consider the sulfur atom (S) marked as ***f***, nitrogen (N) atom (***g***), and oxygen(O) atoms (***h*** and ***i***), and determine their **hybridization**. (2 points)

Atom	Hybridization
Sulfur (<i>f</i>)	
Nitrogen (<i>g</i>)	
Oxygen (<i>h</i>)	
Oxygen (<i>i</i>)	

iii. **Describe** all the bonds formed by the carbon atoms ***a*** and ***b***, using sigma and pi designation as appropriate. (2 points)

Atom	Bonds formed (use sigma and pi as appropriate)
Carbon (<i>a</i>)	
Carbon (<i>b</i>)	

iv. How many total sp^3 -s sigma bonds are in the molecule? (1 point)

v. Determine **electronic** and **molecular geometry** at carbon (C) labeled ***e*** and sulfur (S) labeled ***f*** centres. (2 points)

Atom	Electronic Geometry	Molecular Geometry
Carbon (<i>e</i>)		
Sulfur (<i>f</i>)		

vi. Circle the value from the choices given below, that describes the **C-S-C bond angle at sulfur** (S) marked ***f***: (0.5 points)

>90° to <109.5° 109.5° >109.5 to <120° 120° >120° to <180°


Question 8 12 points

Using the **valence shell orbitals only** draw a **complete** molecular orbital (MO) diagram (with no s-p mixing) for the molecule **PS**, in the space provided below. **(6 points)**

Assume that 3s and 3p atomic orbitals will form molecular orbitals like those formed by the combination of 2s and 2p atomic orbitals that we learnt in class, and there is no s-p mixing.

Show **energy arrow**, **atomic orbitals** involved, **atom labels**, the **molecular orbitals** that are formed, **filled electrons** in the MO diagram etc. Make sure to label all the atomic and molecular orbitals.

Using the MO diagram above, answer the following questions:

- (i) Which one of the following will have the highest bond order? Circle the correct choice
 A) PS^{1-} B) PS C) PS^{1+}
- (ii) Which one of the following will have the longest bond length? Circle the correct choice
 A) PS^{1-} B) PS C) PS^{1+}
- (iii) Which of the following will be diamagnetic? Circle the correct choice.
 A) PS^{1-} B) PS C) PS^{1+}
- (iv) Draw Lewis dot structure of the molecule with the shortest bond length (based on the MO diagram) from among the three (PS^{1-} , PS, PS^{1+}), and indicate the bond order calculated from the Lewis dot structure. **(3 points)**

Lewis Dot Structure

Bond Order



95DCD0D7-3150-4324-8B0C-209E50EE33DF

midterm-exam-93388

#589 Page 10 of 10

SCRAP PAGE

**CHEMISTRY 110 –Midterm Exam****Thursday November 7, 2024, 6:30 – 8:30 pm**

Examiners: Pallavi Sirjoosingh and Ashok Kakkar

First name (please write as legibly as possible within the boxes)

--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--

Last name

--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--

Student ID number

--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--

This booklet has 8 questions on 10 pages (The last page can be used as a scrap page).

The staple must NOT be removed from the question booklet. Your intact question booklet must be returned at the end of the exam.

A separate booklet contains a periodic table, and datasheet.

Write your answers directly under the questions in the space provided. Do not write on the QR codes or near the top edges (as it may be cut off).

Please write **legibly and dark enough** for the scan to be readable. For students who use English as a second language, dictionaries are permitted. Calculators are allowed.

DO NOT ASK QUESTIONS OF THE INVIGILATORS about interpretations of the exam questions. If you have a concern about a question, please summarize it at the top of this page. All concerns will be evaluated after the exam. GOOD LUCK!

**Question 1 8 points** (*Assume all transitions are possible*)

a. 6 points Calculate the **energy (in kJ/mol)** required to ionize (completely remove) the electron from the **n=3** level of **He⁺ ion** (He: Helium). Show your work including equations. (3 sig. figs.)

b. 2 points

(I) Select the largest (longest) wavelength from the three choices below:

- i. Wavelength of photon to *completely remove* an electron from *1s orbital of Hydrogen atom*
- ii. Wavelength of photon to *completely remove* an electron from *1s orbital of Boron atom (B)*
- iii. Wavelength of photon to *completely remove* an electron from *1s orbital of Boron cation (B⁺)*

i or ii or iii

(II) Select the smallest (shortest) wavelength from the three choices below (*for H atom*):

- i. Wavelength of photon *emitted* for an electron transitioning from *2p to 1s orbital*
- ii. Wavelength of photon *absorbed* for an electron transitioning from *2s to 4p orbital*
- iii. Wavelength of photon *absorbed* for an electron transitioning from *1s to 3p orbital*

i or ii or iii



Question 2 8 points



A Chem110 student plots the graph shown on the left, from data obtained for the kinetic energy (eV) of the ejected electron versus frequency (s^{-1}) of the incident light, for an unknown metal. ($1\text{eV} = 1.6022 \times 10^{-19} \text{ J}$)

Based on the plot, answer the questions **a to c** below:

a. 1 point On the plot above, label the **threshold frequency**.

b. 2 points What is the value of the work potential (in J) of the unknown metal?

c. 3 points What is the *longest* wavelength (in nm) required to eject an electron from this metal's surface? Show your work including equation(s) used. (3 sig figs)

d. 2 points In the space provided below, plot the following

(Assume frequency is *above* threshold frequency):

- i. *Kinetic energy* of an ejected electron versus the *intensity* of the incident light (left)
- ii. *Number of electrons* ejected versus the *frequency* of the incident light (right)



**Question 3** **6 points****a. 2 points**

In the ground state of Ge (Germanium; $Z = 32$) atom, **name** the following orbitals:

i. Orbital containing the electron that requires the maximum energy to be removed.

ii. Orbital containing the electron that experiences the minimum effective nuclear charge.

b. 2 points

An unknown anion (A^-) has the following *excited* electronic configuration: $[Ar] 4s^2 3d^3 4p^1$

(Ar: Argon; $Z = 18$)

Give the **condensed ground state electronic configuration** of the atom, **A** and the cation, **A^{2+}** .
Show your work. (Assume all electronic transitions are possible)

Electronic Configuration of A:

Electronic Configuration of A^{2+} :

c. 2 points

In a single atom, determine how many electrons can have the following set of quantum numbers?

(i) $n = 5; l = 3$

In a single atom, determine how many electrons can have the following set of quantum numbers?

(i) $n = 3; l = 2; m_s = +1/2$

**Question 4** **8 points**

Rank the following from 1 to 4 in **increasing** order. Provide the ranking in the boxes provided.

Note: Atomic number (Z) are provided for relevant atoms

a) Increasing (lowest to highest) order of *ionization energy*: (1= lowest; 4 = highest)

Mg

Mg: Z = 12

Rb

Rb: Z = 37

Cs

Cs: Z = 55

K

K : Z = 19

b) Increasing (lowest to highest) order of *bond angle*: (1= lowest; 4 = highest)

H₂OCCl₄XeF₄

HCN

c) Increasing (smallest to largest) order of *ionic size* (1= smallest; 4 = largest)

S²⁻

S: Z = 16

O²⁻

O: Z = 8

Be²⁺

Be: Z = 4

Na⁺

Na: Z = 11

d) Increasing (lowest to highest) order of *bond energy* (1= lowest; 4 = highest)

P-F

P-Cl

P-I

P-Br

**Question 5** **12 points**

a. 9 points For the molecules HClO_4 and TeF_6 , draw the most stable *Lewis structure* showing appropriate shape and *dashed/wedged bonds*, based on VSEPR. Determine the **molecular geometry** around the central atom. For HClO_4 , label the *most polar bond*.



b. 3 points (Note: Lone pairs are not shown in structures below)

(i) For the three *resonance* structures (1, 2 and 3) shown below, answer the following question:



(I) Circle the *most contributing* resonance structure.

Structure 1

Structure 2

Structure 3

(ii) For the three structures given (4, 5, and 6) below, answer the following questions:



(I) Circle all the structures that represent *resonance structures* of each other:

Structure 4

Structure 5

Structure 6

(II) Among the resonance structures, circle the *most contributing* resonance structure.

Structure 4

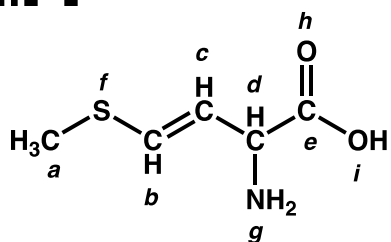
Structure 5

Structure 6

**Question 6** **6 points**

Using the data given below, calculate the **lattice energy** for SrO(s).

	ΔH° (kJ/mol)
$\text{Sr}(s) \longrightarrow \text{Sr}(g)$	+180
$\text{Sr}(g) \longrightarrow \text{Sr}^{2+}(g) + 2e^-$	+1468
$\text{O}_2(g) \longrightarrow 2\text{O}(g)$	+500
$\text{O}(g) + 2e^- \longrightarrow \text{O}^{2-}(g)$	+600
$\text{Sr}(s) + \frac{1}{2}\text{O}_2(g) \longrightarrow \text{SrO}(s)$	-550

**Question 7 10 points**

Consider the molecule given on the left, and answer questions (i) to (vi) using chemical bonding theories that we have learnt in class (valence bond theory, hybridization).

Note that the lone pairs are not shown in the structure:

i. Consider the carbon atoms marked in as ***a***, ***b***, ***c***, ***d***, and ***e*** in the molecule, and determine **hybridization** at each of the carbon centers. (2.5 points)

Atom	Hybridization
Carbon (<i>a</i>)	
Carbon (<i>b</i>)	
Carbon (<i>c</i>)	
Carbon (<i>d</i>)	
Carbon (<i>e</i>)	

ii. Now, consider the sulfur atom (S) marked as ***f***, nitrogen (N) atom (***g***), and oxygen(O) atoms (***h*** and ***i***), and determine their **hybridization**. (2 points)

Atom	Hybridization
Sulfur (<i>f</i>)	
Nitrogen (<i>g</i>)	
Oxygen (<i>h</i>)	
Oxygen (<i>i</i>)	

iii. **Describe** all the bonds formed by the carbon atoms ***a*** and ***b***, using sigma and pi designation as appropriate. (2 points)

Atom	Bonds formed (use sigma and pi as appropriate)
Carbon (<i>a</i>)	
Carbon (<i>b</i>)	

iv. How many total sp^3 -s sigma bonds are in the molecule? (1 point)

v. Determine **electronic** and **molecular geometry** at carbon (C) labeled ***e*** and sulfur (S) labeled ***f*** centres. (2 points)

Atom	Electronic Geometry	Molecular Geometry
Carbon (<i>e</i>)		
Sulfur (<i>f</i>)		

vi. Circle the value from the choices given below, that describes the **C-S-C bond angle at sulfur** (S) marked ***f***: (0.5 points)

>90° to <109.5° 109.5° >109.5 to <120° 120° >120° to <180°


Question 8 12 points

Using the **valence shell orbitals only** draw a **complete** molecular orbital (MO) diagram (with no s-p mixing) for the molecule **PS**, in the space provided below. **(6 points)**

Assume that 3s and 3p atomic orbitals will form molecular orbitals like those formed by the combination of 2s and 2p atomic orbitals that we learnt in class, and there is no s-p mixing.

Show **energy arrow**, **atomic orbitals** involved, **atom labels**, the **molecular orbitals** that are formed, **filled electrons** in the MO diagram etc. Make sure to label all the atomic and molecular orbitals.

Using the MO diagram above, answer the following questions:

- (i) Which one of the following will have the highest bond order? Circle the correct choice
 A) PS^{1-} B) PS C) PS^{1+}
- (ii) Which one of the following will have the longest bond length? Circle the correct choice
 A) PS^{1-} B) PS C) PS^{1+}
- (iii) Which of the following will be diamagnetic? Circle the correct choice.
 A) PS^{1-} B) PS C) PS^{1+}
- (iv) Draw Lewis dot structure of the molecule with the shortest bond length (based on the MO diagram) from among the three (PS^{1-} , PS, PS^{1+}), and indicate the bond order calculated from the Lewis dot structure. **(3 points)**

Lewis Dot Structure

Bond Order



0304AE83-D35A-4A53-8CA4-8CF55C8793E9

midterm-exam-93388

#590 Page 10 of 10

SCRAP PAGE

**CHEMISTRY 110 –Midterm Exam****Thursday November 7, 2024, 6:30 – 8:30 pm****Examiners: Pallavi Sirjoosingh and Ashok Kakkar**

First name (please write as legibly as possible within the boxes)

--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--

Last name

--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--

Student ID number

--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--

This booklet has 8 questions on 10 pages (The last page can be used as a scrap page).

The staple must NOT be removed from the question booklet. Your intact question booklet must be returned at the end of the exam.

A separate booklet contains a periodic table, and datasheet.

Write your answers directly under the questions in the space provided. Do not write on the QR codes or near the top edges (as it may be cut off).

Please write **legibly and dark enough** for the scan to be readable. For students who use English as a second language, dictionaries are permitted. Calculators are allowed.

DO NOT ASK QUESTIONS OF THE INVIGILATORS about interpretations of the exam questions. If you have a concern about a question, please summarize it at the top of this page. All concerns will be evaluated after the exam. GOOD LUCK!

**Question 1 8 points** (*Assume all transitions are possible*)

a. 6 points Calculate the **energy (in kJ/mol)** required to ionize (completely remove) the electron from the **n=3** level of **He⁺ ion** (He: Helium). Show your work including equations. (3 sig. figs.)

b. 2 points

(I) Select the largest (longest) wavelength from the three choices below:

- i. Wavelength of photon to *completely remove* an electron from *1s orbital of Hydrogen atom*
- ii. Wavelength of photon to *completely remove* an electron from *1s orbital of Boron atom (B)*
- iii. Wavelength of photon to *completely remove* an electron from *1s orbital of Boron cation (B⁺)*

i or ii or iii

(II) Select the smallest (shortest) wavelength from the three choices below (*for H atom*):

- i. Wavelength of photon *emitted* for an electron transitioning from *2p to 1s orbital*
- ii. Wavelength of photon *absorbed* for an electron transitioning from *2s to 4p orbital*
- iii. Wavelength of photon *absorbed* for an electron transitioning from *1s to 3p orbital*

i or ii or iii



Question 2 8 points



A Chem110 student plots the graph shown on the left, from data obtained for the kinetic energy (eV) of the ejected electron versus frequency (s^{-1}) of the incident light, for an unknown metal. ($1\text{eV} = 1.6022 \times 10^{-19} \text{ J}$)

Based on the plot, answer the questions **a to c** below:

a. 1 point On the plot above, label the **threshold frequency**.

b. 2 points What is the value of the work potential (in J) of the unknown metal?

c. 3 points What is the *longest* wavelength (in nm) required to eject an electron from this metal's surface? Show your work including equation(s) used. (3 sig figs)

d. 2 points In the space provided below, plot the following

(Assume frequency is *above* threshold frequency):

- i. *Kinetic energy* of an ejected electron versus the *intensity* of the incident light (left)
- ii. *Number of electrons* ejected versus the *frequency* of the incident light (right)



**Question 3** **6 points****a. 2 points**

In the ground state of Ge (Germanium; $Z = 32$) atom, **name** the following orbitals:

i. Orbital containing the electron that requires the maximum energy to be removed.

ii. Orbital containing the electron that experiences the minimum effective nuclear charge.

b. 2 points

An unknown anion (A^-) has the following *excited* electronic configuration: $[Ar] 4s^2 3d^3 4p^1$

(Ar: Argon; $Z = 18$)

Give the **condensed ground state electronic configuration** of the atom, **A** and the cation, **A^{2+}** .
Show your work. (Assume all electronic transitions are possible)

Electronic Configuration of A:

Electronic Configuration of A^{2+} :

c. 2 points

In a single atom, determine how many electrons can have the following set of quantum numbers?

(i) $n = 5; l = 3$

In a single atom, determine how many electrons can have the following set of quantum numbers?

(i) $n = 3; l = 2; m_s = +1/2$

**Question 4** **8 points**

Rank the following from 1 to 4 in **increasing** order. Provide the ranking in the boxes provided.

Note: Atomic number (Z) are provided for relevant atoms

a) Increasing (lowest to highest) order of *ionization energy*: (1= lowest; 4 = highest)

Mg

Mg: Z = 12

Rb

Rb: Z = 37

Cs

Cs: Z = 55

K

K : Z = 19

b) Increasing (lowest to highest) order of *bond angle*: (1= lowest; 4 = highest)

H₂OCCl₄XeF₄

HCN

c) Increasing (smallest to largest) order of *ionic size* (1= smallest; 4 = largest)

S²⁻

S: Z = 16

O²⁻

O: Z = 8

Be²⁺

Be: Z = 4

Na⁺

Na: Z = 11

d) Increasing (lowest to highest) order of *bond energy* (1= lowest; 4 = highest)

P-F

P-Cl

P-I

P-Br

**Question 5** **12 points**

a. 9 points For the molecules HClO_4 and TeF_6 , draw the most stable *Lewis structure* showing appropriate shape and *dashed/wedged bonds*, based on VSEPR. Determine the **molecular geometry** around the central atom. For HClO_4 , label the *most polar bond*.



b. 3 points (Note: Lone pairs are not shown in structures below)

(i) For the three *resonance* structures (1, 2 and 3) shown below, answer the following question:



(I) Circle the *most contributing* resonance structure.

Structure 1

Structure 2

Structure 3

(ii) For the three structures given (4, 5, and 6) below, answer the following questions:



(I) Circle all the structures that represent *resonance structures* of each other:

Structure 4

Structure 5

Structure 6

(II) Among the resonance structures, circle the *most contributing* resonance structure.

Structure 4

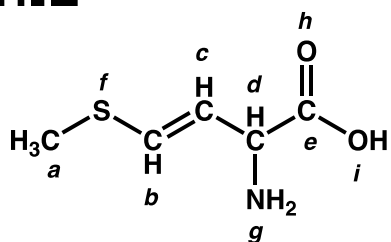
Structure 5

Structure 6

**Question 6** **6 points**

Using the data given below, calculate the **lattice energy** for SrO(s).

	ΔH° (kJ/mol)
$\text{Sr}(s) \longrightarrow \text{Sr}(g)$	+180
$\text{Sr}(g) \longrightarrow \text{Sr}^{2+}(g) + 2e^-$	+1468
$\text{O}_2(g) \longrightarrow 2\text{O}(g)$	+500
$\text{O}(g) + 2e^- \longrightarrow \text{O}^{2-}(g)$	+600
$\text{Sr}(s) + \frac{1}{2}\text{O}_2(g) \longrightarrow \text{SrO}(s)$	-550

**Question 7 10 points**

Consider the molecule given on the left, and answer questions (i) to (vi) using chemical bonding theories that we have learnt in class (valence bond theory, hybridization).

Note that the lone pairs are not shown in the structure:

i. Consider the carbon atoms marked in as ***a***, ***b***, ***c***, ***d***, and ***e*** in the molecule, and determine **hybridization** at each of the carbon centers. (2.5 points)

Atom	Hybridization
Carbon (<i>a</i>)	
Carbon (<i>b</i>)	
Carbon (<i>c</i>)	
Carbon (<i>d</i>)	
Carbon (<i>e</i>)	

ii. Now, consider the sulfur atom (S) marked as ***f***, nitrogen (N) atom (***g***), and oxygen(O) atoms (***h*** and ***i***), and determine their **hybridization**. (2 points)

Atom	Hybridization
Sulfur (<i>f</i>)	
Nitrogen (<i>g</i>)	
Oxygen (<i>h</i>)	
Oxygen (<i>i</i>)	

iii. **Describe** all the bonds formed by the carbon atoms ***a*** and ***b***, using sigma and pi designation as appropriate. (2 points)

Atom	Bonds formed (use sigma and pi as appropriate)
Carbon (<i>a</i>)	
Carbon (<i>b</i>)	

iv. How many total sp^3 -s sigma bonds are in the molecule? (1 point)

v. Determine **electronic** and **molecular geometry** at carbon (C) labeled ***e*** and sulfur (S) labeled ***f*** centres. (2 points)

Atom	Electronic Geometry	Molecular Geometry
Carbon (<i>e</i>)		
Sulfur (<i>f</i>)		

vi. Circle the value from the choices given below, that describes the **C-S-C bond angle at sulfur** (S) marked ***f***: (0.5 points)

>90° to <109.5° 109.5° >109.5 to <120° 120° >120° to <180°


Question 8 12 points

Using the **valence shell orbitals only** draw a **complete** molecular orbital (MO) diagram (with no s-p mixing) for the molecule **PS**, in the space provided below. **(6 points)**

Assume that 3s and 3p atomic orbitals will form molecular orbitals like those formed by the combination of 2s and 2p atomic orbitals that we learnt in class, and there is no s-p mixing.

Show **energy arrow**, **atomic orbitals** involved, **atom labels**, the **molecular orbitals** that are formed, **filled electrons** in the MO diagram etc. Make sure to label all the atomic and molecular orbitals.

Using the MO diagram above, answer the following questions:

- (i) Which one of the following will have the highest bond order? Circle the correct choice
 A) PS^{1-} B) PS C) PS^{1+}
- (ii) Which one of the following will have the longest bond length? Circle the correct choice
 A) PS^{1-} B) PS C) PS^{1+}
- (iii) Which of the following will be diamagnetic? Circle the correct choice.
 A) PS^{1-} B) PS C) PS^{1+}
- (iv) Draw Lewis dot structure of the molecule with the shortest bond length (based on the MO diagram) from among the three (PS^{1-} , PS, PS^{1+}), and indicate the bond order calculated from the Lewis dot structure. **(3 points)**

Lewis Dot Structure

Bond Order



38C7482A-8F43-462A-982B-27E88DF2F855

midterm-exam-93388

#591 Page 10 of 10

SCRAP PAGE

**CHEMISTRY 110 –Midterm Exam****Thursday November 7, 2024, 6:30 – 8:30 pm**

Examiners: Pallavi Sirjoosingh and Ashok Kakkar

First name (please write as legibly as possible within the boxes)

--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--

Last name

--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--

Student ID number

--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--

This booklet has 8 questions on 10 pages (The last page can be used as a scrap page).

The staple must NOT be removed from the question booklet. Your intact question booklet must be returned at the end of the exam.

A separate booklet contains a periodic table, and datasheet.

Write your answers directly under the questions in the space provided. Do not write on the QR codes or near the top edges (as it may be cut off).

Please write **legibly and dark enough** for the scan to be readable. For students who use English as a second language, dictionaries are permitted. Calculators are allowed.

DO NOT ASK QUESTIONS OF THE INVIGILATORS about interpretations of the exam questions. If you have a concern about a question, please summarize it at the top of this page. All concerns will be evaluated after the exam. GOOD LUCK!

**Question 1 8 points** (*Assume all transitions are possible*)

a. 6 points Calculate the **energy (in kJ/mol)** required to ionize (completely remove) the electron from the **$n=3$** level of **He^+ ion** (He: Helium). Show your work including equations. (3 sig. figs.)

b. 2 points

(I) Select the largest (longest) wavelength from the three choices below:

- i. Wavelength of photon to *completely remove* an electron from *1s orbital of Hydrogen atom*
- ii. Wavelength of photon to *completely remove* an electron from *1s orbital of Boron atom (B)*
- iii. Wavelength of photon to *completely remove* an electron from *1s orbital of Boron cation (B^+)*

i or ii or iii

(II) Select the smallest (shortest) wavelength from the three choices below (*for H atom*):

- i. Wavelength of photon *emitted* for an electron transitioning from *2p to 1s orbital*
- ii. Wavelength of photon *absorbed* for an electron transitioning from *2s to 4p orbital*
- iii. Wavelength of photon *absorbed* for an electron transitioning from *1s to 3p orbital*

i or ii or iii



Question 2 8 points



A Chem110 student plots the graph shown on the left, from data obtained for the kinetic energy (eV) of the ejected electron versus frequency (s^{-1}) of the incident light, for an unknown metal. ($1\text{eV} = 1.6022 \times 10^{-19} \text{ J}$)

Based on the plot, answer the questions **a to c** below:

a. 1 point On the plot above, label the **threshold frequency**.

b. 2 points What is the value of the work potential (in J) of the unknown metal?

c. 3 points What is the *longest* wavelength (in nm) required to eject an electron from this metal's surface? Show your work including equation(s) used. (3 sig figs)

d. 2 points In the space provided below, plot the following

(Assume frequency is *above* threshold frequency):

- i. *Kinetic energy* of an ejected electron versus the *intensity* of the incident light (left)
- ii. *Number of electrons* ejected versus the *frequency* of the incident light (right)



**Question 3** **6 points****a. 2 points**

In the ground state of Ge (Germanium; $Z = 32$) atom, **name** the following orbitals:

i. Orbital containing the electron that requires the maximum energy to be removed.

ii. Orbital containing the electron that experiences the minimum effective nuclear charge.

b. 2 points

An unknown anion (A^-) has the following *excited* electronic configuration: $[Ar] 4s^2 3d^3 4p^1$

(Ar: Argon; $Z = 18$)

Give the **condensed ground state electronic configuration** of the atom, **A** and the cation, A^{2+} .
Show your work. (Assume all electronic transitions are possible)

Electronic Configuration of A:

Electronic Configuration of A^{2+} :

c. 2 points

In a single atom, determine how many electrons can have the following set of quantum numbers?

(i) $n = 5; l = 3$

In a single atom, determine how many electrons can have the following set of quantum numbers?

(i) $n = 3; l = 2; m_s = +1/2$

**Question 4** **8 points**

Rank the following from 1 to 4 in **increasing** order. Provide the ranking in the boxes provided.

Note: Atomic number (Z) are provided for relevant atoms

a) Increasing (lowest to highest) order of *ionization energy*: (1= lowest; 4 = highest)

Mg

Mg: Z = 12

Rb

Rb: Z = 37

Cs

Cs: Z = 55

K

K : Z = 19

b) Increasing (lowest to highest) order of *bond angle*: (1= lowest; 4 = highest)

H₂OCCl₄XeF₄

HCN

c) Increasing (smallest to largest) order of *ionic size* (1= smallest; 4 = largest)

S²⁻

S: Z = 16

O²⁻

O: Z = 8

Be²⁺

Be: Z = 4

Na⁺

Na: Z = 11

d) Increasing (lowest to highest) order of *bond energy* (1= lowest; 4 = highest)

P-F

P-Cl

P-I

P-Br

**Question 5** **12 points**

a. 9 points For the molecules HClO_4 and TeF_6 , draw the most stable *Lewis structure* showing appropriate shape and *dashed/wedged bonds*, based on VSEPR. Determine the **molecular geometry** around the central atom. For HClO_4 , label the *most polar bond*.

HClO_4 (there are no O-O bonds)

Label the most polar bond

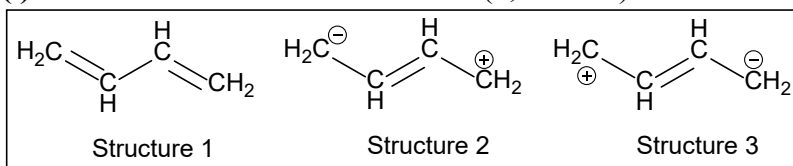
Molecular Geometry around Cl

TeF_6

Molecular Geometry around Te

b. 3 points (Note: Lone pairs are not shown in structures below)

(i) For the three *resonance* structures (1, 2 and 3) shown below, answer the following question:



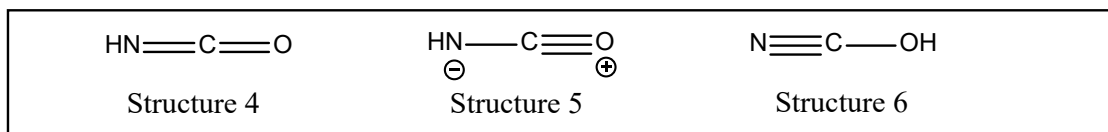
(I) Circle the *most contributing* resonance structure.

Structure 1

Structure 2

Structure 3

(ii) For the three structures given (4, 5, and 6) below, answer the following questions:



(I) Circle all the structures that represent *resonance structures* of each other:

Structure 4

Structure 5

Structure 6

(II) Among the resonance structures, circle the *most contributing* resonance structure.

Structure 4

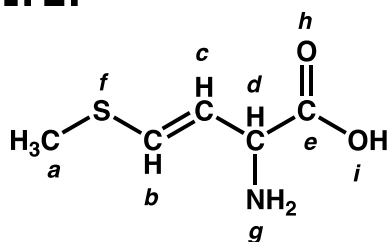
Structure 5

Structure 6

**Question 6** **6 points**

Using the data given below, calculate the **lattice energy** for SrO(s).

	ΔH° (kJ/mol)
$\text{Sr}(s) \longrightarrow \text{Sr}(g)$	+180
$\text{Sr}(g) \longrightarrow \text{Sr}^{2+}(g) + 2e^-$	+1468
$\text{O}_2(g) \longrightarrow 2\text{O}(g)$	+500
$\text{O}(g) + 2e^- \longrightarrow \text{O}^{2-}(g)$	+600
$\text{Sr}(s) + \frac{1}{2}\text{O}_2(g) \longrightarrow \text{SrO}(s)$	-550

**Question 7 10 points**

Consider the molecule given on the left, and answer questions (i) to (vi) using chemical bonding theories that we have learnt in class (valence bond theory, hybridization).

Note that the lone pairs are not shown in the structure:

i. Consider the carbon atoms marked in as ***a***, ***b***, ***c***, ***d***, and ***e*** in the molecule, and determine **hybridization** at each of the carbon centers. (2.5 points)

Atom	Hybridization
Carbon (<i>a</i>)	
Carbon (<i>b</i>)	
Carbon (<i>c</i>)	
Carbon (<i>d</i>)	
Carbon (<i>e</i>)	

ii. Now, consider the sulfur atom (S) marked as ***f***, nitrogen (N) atom (***g***), and oxygen(O) atoms (***h*** and ***i***), and determine their **hybridization**. (2 points)

Atom	Hybridization
Sulfur (<i>f</i>)	
Nitrogen (<i>g</i>)	
Oxygen (<i>h</i>)	
Oxygen (<i>i</i>)	

iii. **Describe** all the bonds formed by the carbon atoms ***a*** and ***b***, using sigma and pi designation as appropriate. (2 points)

Atom	Bonds formed (use sigma and pi as appropriate)
Carbon (<i>a</i>)	
Carbon (<i>b</i>)	

iv. How many total sp^3 -s sigma bonds are in the molecule? (1 point)

v. Determine **electronic** and **molecular geometry** at carbon (C) labeled ***e*** and sulfur (S) labeled ***f*** centres. (2 points)

Atom	Electronic Geometry	Molecular Geometry
Carbon (<i>e</i>)		
Sulfur (<i>f</i>)		

vi. Circle the value from the choices given below, that describes the **C-S-C bond angle at sulfur** (S) marked ***f***: (0.5 points)

>90° to <109.5°

109.5°

>109.5 to <120°

120°

>120° to <180°


Question 8 12 points

Using the **valence shell orbitals only** draw a **complete** molecular orbital (MO) diagram (with no s-p mixing) for the molecule **PS**, in the space provided below. **(6 points)**

Assume that 3s and 3p atomic orbitals will form molecular orbitals like those formed by the combination of 2s and 2p atomic orbitals that we learnt in class, and there is no s-p mixing.

Show **energy arrow**, **atomic orbitals** involved, **atom labels**, the **molecular orbitals** that are formed, **filled electrons** in the MO diagram etc. Make sure to **label all the atomic and molecular orbitals**.

Using the MO diagram above, answer the following questions:

- (i) Which one of the following will have the highest bond order? Circle the correct choice
 A) PS^{1-} B) PS C) PS^{1+}
- (ii) Which one of the following will have the longest bond length? Circle the correct choice
 A) PS^{1-} B) PS C) PS^{1+}
- (iii) Which of the following will be diamagnetic? Circle the correct choice.
 A) PS^{1-} B) PS C) PS^{1+}
- (iv) Draw Lewis dot structure of the molecule with the shortest bond length (based on the MO diagram) from among the three (PS^{1-} , PS, PS^{1+}), and indicate the bond order calculated from the Lewis dot structure. **(3 points)**

Lewis Dot Structure

Bond Order



F5B91FEE-67EA-400C-AF34-45E7438434E9

midterm-exam-93388

#592 Page 10 of 10

SCRAP PAGE

**CHEMISTRY 110 –Midterm Exam****Thursday November 7, 2024, 6:30 – 8:30 pm**

Examiners: Pallavi Sirjoosingh and Ashok Kakkar

First name (please write as legibly as possible within the boxes)

--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--

Last name

--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--

Student ID number

--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--

This booklet has 8 questions on 10 pages (The last page can be used as a scrap page).

The staple must NOT be removed from the question booklet. Your intact question booklet must be returned at the end of the exam.

A separate booklet contains a periodic table, and datasheet.

Write your answers directly under the questions in the space provided. Do not write on the QR codes or near the top edges (as it may be cut off).

Please write **legibly and dark enough** for the scan to be readable. For students who use English as a second language, dictionaries are permitted. Calculators are allowed.

DO NOT ASK QUESTIONS OF THE INVIGILATORS about interpretations of the exam questions. If you have a concern about a question, please summarize it at the top of this page. All concerns will be evaluated after the exam. GOOD LUCK!

**Question 1 8 points** (*Assume all transitions are possible*)

a. 6 points Calculate the **energy (in kJ/mol)** required to ionize (completely remove) the electron from the **n=3** level of **He⁺ ion** (He: Helium). Show your work including equations. (3 sig. figs.)

b. 2 points

(I) Select the largest (longest) wavelength from the three choices below:

- i. Wavelength of photon to *completely remove* an electron from *1s orbital of Hydrogen atom*
- ii. Wavelength of photon to *completely remove* an electron from *1s orbital of Boron atom (B)*
- iii. Wavelength of photon to *completely remove* an electron from *1s orbital of Boron cation (B⁺)*

i or ii or iii

(II) Select the smallest (shortest) wavelength from the three choices below (*for H atom*):

- i. Wavelength of photon *emitted* for an electron transitioning from *2p to 1s orbital*
- ii. Wavelength of photon *absorbed* for an electron transitioning from *2s to 4p orbital*
- iii. Wavelength of photon *absorbed* for an electron transitioning from *1s to 3p orbital*

i or ii or iii



Question 2 8 points



A Chem110 student plots the graph shown on the left, from data obtained for the kinetic energy (eV) of the ejected electron versus frequency (s^{-1}) of the incident light, for an unknown metal. ($1\text{eV} = 1.6022 \times 10^{-19} \text{ J}$)

Based on the plot, answer the questions **a to c** below:

a. 1 point On the plot above, label the **threshold frequency**.

b. 2 points What is the value of the work potential (in J) of the unknown metal?

c. 3 points What is the *longest* wavelength (in nm) required to eject an electron from this metal's surface? Show your work including equation(s) used. (3 sig figs)

d. 2 points In the space provided below, plot the following

(Assume frequency is *above* threshold frequency):

- i. *Kinetic energy* of an ejected electron versus the *intensity* of the incident light (left)
- ii. *Number of electrons* ejected versus the *frequency* of the incident light (right)



**Question 3** **6 points****a. 2 points**

In the ground state of Ge (Germanium; $Z = 32$) atom, **name** the following orbitals:

i. Orbital containing the electron that requires the maximum energy to be removed.

ii. Orbital containing the electron that experiences the minimum effective nuclear charge.

b. 2 points

An unknown anion (A^-) has the following *excited* electronic configuration: $[Ar] 4s^2 3d^3 4p^1$

(Ar: Argon; $Z = 18$)

Give the **condensed ground state electronic configuration** of the atom, **A** and the cation, **A^{2+}** .
Show your work. (Assume all electronic transitions are possible)

Electronic Configuration of A:

Electronic Configuration of A^{2+} :

c. 2 points

In a single atom, determine how many electrons can have the following set of quantum numbers?

(i) $n = 5; l = 3$

In a single atom, determine how many electrons can have the following set of quantum numbers?

(i) $n = 3; l = 2; m_s = +1/2$

**Question 4** **8 points**

Rank the following from 1 to 4 in **increasing** order. Provide the ranking in the boxes provided.

Note: Atomic number (Z) are provided for relevant atoms

a) Increasing (lowest to highest) order of *ionization energy*: (1= lowest; 4 = highest)

Mg

Mg: Z = 12

Rb

Rb: Z = 37

Cs

Cs: Z = 55

K

K : Z = 19

b) Increasing (lowest to highest) order of *bond angle*: (1= lowest; 4 = highest)

H₂OCCl₄XeF₄

HCN

c) Increasing (smallest to largest) order of *ionic size* (1= smallest; 4 = largest)

S²⁻

S: Z = 16

O²⁻

O: Z = 8

Be²⁺

Be: Z = 4

Na⁺

Na: Z = 11

d) Increasing (lowest to highest) order of *bond energy* (1= lowest; 4 = highest)

P-F

P-Cl

P-I

P-Br

**Question 5** **12 points**

a. 9 points For the molecules HClO_4 and TeF_6 , draw the most stable *Lewis structure* showing appropriate shape and *dashed/wedged bonds*, based on VSEPR. Determine the **molecular geometry** around the central atom. For HClO_4 , label the *most polar bond*.

HClO_4 (there are no O-O bonds)

Label the most polar bond

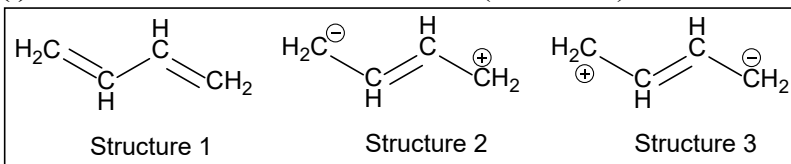
Molecular Geometry around Cl

TeF_6

Molecular Geometry around Te

b. 3 points (Note: Lone pairs are not shown in structures below)

(i) For the three *resonance* structures (1, 2 and 3) shown below, answer the following question:



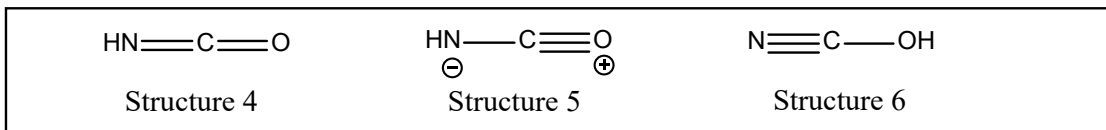
(I) Circle the *most contributing* resonance structure.

Structure 1

Structure 2

Structure 3

(ii) For the three structures given (4, 5, and 6) below, answer the following questions:



(I) Circle all the structures that represent *resonance structures* of each other:

Structure 4

Structure 5

Structure 6

(II) Among the resonance structures, circle the *most contributing* resonance structure.

Structure 4

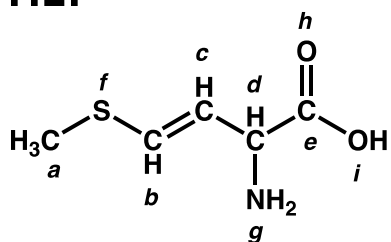
Structure 5

Structure 6

**Question 6** **6 points**

Using the data given below, calculate the **lattice energy** for SrO(s).

	ΔH° (kJ/mol)
$\text{Sr}(s) \longrightarrow \text{Sr}(g)$	+180
$\text{Sr}(g) \longrightarrow \text{Sr}^{2+}(g) + 2e^-$	+1468
$\text{O}_2(g) \longrightarrow 2\text{O}(g)$	+500
$\text{O}(g) + 2e^- \longrightarrow \text{O}^{2-}(g)$	+600
$\text{Sr}(s) + \frac{1}{2}\text{O}_2(g) \longrightarrow \text{SrO}(s)$	-550

**Question 7 10 points**

Consider the molecule given on the left, and answer questions (i) to (vi) using chemical bonding theories that we have learnt in class (valence bond theory, hybridization).

Note that the lone pairs are not shown in the structure:

i. Consider the carbon atoms marked in as ***a***, ***b***, ***c***, ***d***, and ***e*** in the molecule, and determine **hybridization** at each of the carbon centers. (2.5 points)

Atom	Hybridization
Carbon (<i>a</i>)	
Carbon (<i>b</i>)	
Carbon (<i>c</i>)	
Carbon (<i>d</i>)	
Carbon (<i>e</i>)	

ii. Now, consider the sulfur atom (S) marked as ***f***, nitrogen (N) atom (***g***), and oxygen(O) atoms (***h*** and ***i***), and determine their **hybridization**. (2 points)

Atom	Hybridization
Sulfur (<i>f</i>)	
Nitrogen (<i>g</i>)	
Oxygen (<i>h</i>)	
Oxygen (<i>i</i>)	

iii. **Describe** all the bonds formed by the carbon atoms ***a*** and ***b***, using sigma and pi designation as appropriate. (2 points)

Atom	Bonds formed (use sigma and pi as appropriate)
Carbon (<i>a</i>)	
Carbon (<i>b</i>)	

iv. How many total sp^3 -s sigma bonds are in the molecule? (1 point)

v. Determine **electronic** and **molecular geometry** at carbon (C) labeled ***e*** and sulfur (S) labeled ***f*** centres. (2 points)

Atom	Electronic Geometry	Molecular Geometry
Carbon (<i>e</i>)		
Sulfur (<i>f</i>)		

vi. Circle the value from the choices given below, that describes the **C-S-C bond angle at sulfur** (S) marked ***f***: (0.5 points)

>90° to <109.5°

109.5°

>109.5 to <120°

120°

>120° to <180°


Question 8 12 points

Using the **valence shell orbitals only** draw a **complete** molecular orbital (MO) diagram (with no s-p mixing) for the molecule **PS**, in the space provided below. **(6 points)**

Assume that 3s and 3p atomic orbitals will form molecular orbitals like those formed by the combination of 2s and 2p atomic orbitals that we learnt in class, and there is no s-p mixing.

Show **energy arrow**, **atomic orbitals** involved, **atom labels**, the **molecular orbitals** that are formed, **filled electrons** in the MO diagram etc. Make sure to label all the atomic and molecular orbitals.

Using the MO diagram above, answer the following questions:

- (i) Which one of the following will have the highest bond order? Circle the correct choice
 A) PS^{1-} B) PS C) PS^{1+}
- (ii) Which one of the following will have the longest bond length? Circle the correct choice
 A) PS^{1-} B) PS C) PS^{1+}
- (iii) Which of the following will be diamagnetic? Circle the correct choice.
 A) PS^{1-} B) PS C) PS^{1+}
- (iv) Draw Lewis dot structure of the molecule with the shortest bond length (based on the MO diagram) from among the three (PS^{1-} , PS, PS^{1+}), and indicate the bond order calculated from the Lewis dot structure. **(3 points)**

Lewis Dot Structure

Bond Order



81354594-37F1-4411-9507-63A8CABC5372

midterm-exam-93388

#593 Page 10 of 10

SCRAP PAGE

**CHEMISTRY 110 –Midterm Exam****Thursday November 7, 2024, 6:30 – 8:30 pm****Examiners: Pallavi Sirjoosingh and Ashok Kakkar**

First name (please write as legibly as possible within the boxes)

--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--

Last name

--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--

Student ID number

--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--

This booklet has 8 questions on 10 pages (The last page can be used as a scrap page).

The staple must NOT be removed from the question booklet. Your intact question booklet must be returned at the end of the exam.

A separate booklet contains a periodic table, and datasheet.

Write your answers directly under the questions in the space provided. Do not write on the QR codes or near the top edges (as it may be cut off).

Please write **legibly and dark enough** for the scan to be readable. For students who use English as a second language, dictionaries are permitted. Calculators are allowed.

DO NOT ASK QUESTIONS OF THE INVIGILATORS about interpretations of the exam questions. If you have a concern about a question, please summarize it at the top of this page. All concerns will be evaluated after the exam. GOOD LUCK!

**Question 1 8 points** (*Assume all transitions are possible*)

a. 6 points Calculate the **energy (in kJ/mol)** required to ionize (completely remove) the electron from the **n=3** level of **He⁺ ion** (He: Helium). Show your work including equations. (3 sig. figs.)

b. 2 points

(I) Select the largest (longest) wavelength from the three choices below:

- i. Wavelength of photon to *completely remove* an electron from *1s orbital of Hydrogen atom*
- ii. Wavelength of photon to *completely remove* an electron from *1s orbital of Boron atom (B)*
- iii. Wavelength of photon to *completely remove* an electron from *1s orbital of Boron cation (B⁺)*

i or ii or iii

(II) Select the smallest (shortest) wavelength from the three choices below (*for H atom*):

- i. Wavelength of photon *emitted* for an electron transitioning from *2p to 1s orbital*
- ii. Wavelength of photon *absorbed* for an electron transitioning from *2s to 4p orbital*
- iii. Wavelength of photon *absorbed* for an electron transitioning from *1s to 3p orbital*

i or ii or iii



Question 2 8 points



A Chem110 student plots the graph shown on the left, from data obtained for the kinetic energy (eV) of the ejected electron versus frequency (s^{-1}) of the incident light, for an unknown metal. ($1\text{eV} = 1.6022 \times 10^{-19} \text{ J}$)

Based on the plot, answer the questions **a to c** below:

a. 1 point On the plot above, label the **threshold frequency**.

b. 2 points What is the value of the work potential (in J) of the unknown metal?

c. 3 points What is the *longest* wavelength (in nm) required to eject an electron from this metal's surface? Show your work including equation(s) used. (3 sig figs)

d. 2 points In the space provided below, plot the following

(Assume frequency is *above* threshold frequency):

- i. *Kinetic energy* of an ejected electron versus the *intensity* of the incident light (left)
- ii. *Number of electrons* ejected versus the *frequency* of the incident light (right)



**Question 3** **6 points****a. 2 points**

In the ground state of Ge (Germanium; $Z = 32$) atom, **name** the following orbitals:

i. Orbital containing the electron that requires the maximum energy to be removed.

ii. Orbital containing the electron that experiences the minimum effective nuclear charge.

b. 2 points

An unknown anion (A^-) has the following *excited* electronic configuration: $[Ar] 4s^2 3d^3 4p^1$

(Ar: Argon; $Z = 18$)

Give the **condensed ground state electronic configuration** of the atom, **A** and the cation, **A^{2+}** .
Show your work. (Assume all electronic transitions are possible)

Electronic Configuration of A:

Electronic Configuration of A^{2+} :

c. 2 points

In a single atom, determine how many electrons can have the following set of quantum numbers?

(i) $n = 5; l = 3$

In a single atom, determine how many electrons can have the following set of quantum numbers?

(i) $n = 3; l = 2; m_s = +1/2$

**Question 4** **8 points**

Rank the following from 1 to 4 in **increasing** order. Provide the ranking in the boxes provided.

Note: Atomic number (Z) are provided for relevant atoms

a) Increasing (lowest to highest) order of *ionization energy*: (1= lowest; 4 = highest)

Mg

Mg: Z = 12

Rb

Rb: Z = 37

Cs

Cs: Z = 55

K

K : Z = 19

b) Increasing (lowest to highest) order of *bond angle*: (1= lowest; 4 = highest)

H₂OCCl₄XeF₄

HCN

c) Increasing (smallest to largest) order of *ionic size* (1= smallest; 4 = largest)

S²⁻

S: Z = 16

O²⁻

O: Z = 8

Be²⁺

Be: Z = 4

Na⁺

Na: Z = 11

d) Increasing (lowest to highest) order of *bond energy* (1= lowest; 4 = highest)

P-F

P-Cl

P-I

P-Br

**Question 5** **12 points**

a. 9 points For the molecules HClO_4 and TeF_6 , draw the most stable *Lewis structure* showing appropriate shape and *dashed/wedged bonds*, based on VSEPR. Determine the **molecular geometry** around the central atom. For HClO_4 , label the *most polar bond*.



b. 3 points (Note: Lone pairs are not shown in structures below)

(i) For the three *resonance* structures (1, 2 and 3) shown below, answer the following question:



(I) Circle the *most contributing* resonance structure.

Structure 1

Structure 2

Structure 3

(ii) For the three structures given (4, 5, and 6) below, answer the following questions:



(I) Circle all the structures that represent *resonance structures* of each other:

Structure 4

Structure 5

Structure 6

(II) Among the resonance structures, circle the *most contributing* resonance structure.

Structure 4

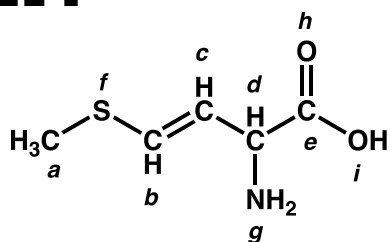
Structure 5

Structure 6

**Question 6** **6 points**

Using the data given below, calculate the **lattice energy** for SrO(s).

	ΔH° (kJ/mol)
$\text{Sr}(s) \longrightarrow \text{Sr}(g)$	+180
$\text{Sr}(g) \longrightarrow \text{Sr}^{2+}(g) + 2e^-$	+1468
$\text{O}_2(g) \longrightarrow 2\text{O}(g)$	+500
$\text{O}(g) + 2e^- \longrightarrow \text{O}^{2-}(g)$	+600
$\text{Sr}(s) + \frac{1}{2}\text{O}_2(g) \longrightarrow \text{SrO}(s)$	-550

**Question 7 10 points**

Consider the molecule given on the left, and answer questions (i) to (vi) using chemical bonding theories that we have learnt in class (valence bond theory, hybridization).

Note that the lone pairs are not shown in the structure:

i. Consider the carbon atoms marked in as ***a***, ***b***, ***c***, ***d***, and ***e*** in the molecule, and determine **hybridization** at each of the carbon centers. (2.5 points)

Atom	Hybridization
Carbon (<i>a</i>)	
Carbon (<i>b</i>)	
Carbon (<i>c</i>)	
Carbon (<i>d</i>)	
Carbon (<i>e</i>)	

ii. Now, consider the sulfur atom (S) marked as ***f***, nitrogen (N) atom (***g***), and oxygen(O) atoms (***h*** and ***i***), and determine their **hybridization**. (2 points)

Atom	Hybridization
Sulfur (<i>f</i>)	
Nitrogen (<i>g</i>)	
Oxygen (<i>h</i>)	
Oxygen (<i>i</i>)	

iii. **Describe** all the bonds formed by the carbon atoms ***a*** and ***b***, using sigma and pi designation as appropriate. (2 points)

Atom	Bonds formed (use sigma and pi as appropriate)
Carbon (<i>a</i>)	
Carbon (<i>b</i>)	

iv. How many total sp^3 -s sigma bonds are in the molecule? (1 point)

v. Determine **electronic** and **molecular geometry** at carbon (C) labeled ***e*** and sulfur (S) labeled ***f*** centres. (2 points)

Atom	Electronic Geometry	Molecular Geometry
Carbon (<i>e</i>)		
Sulfur (<i>f</i>)		

vi. Circle the value from the choices given below, that describes the **C-S-C bond angle at sulfur** (S) marked ***f***: (0.5 points)

>90° to <109.5° 109.5° >109.5 to <120° 120° >120° to <180°


Question 8 12 points

Using the **valence shell orbitals only** draw a **complete** molecular orbital (MO) diagram (with no s-p mixing) for the molecule **PS**, in the space provided below. **(6 points)**

Assume that 3s and 3p atomic orbitals will form molecular orbitals like those formed by the combination of 2s and 2p atomic orbitals that we learnt in class, and there is no s-p mixing.

Show **energy arrow**, **atomic orbitals** involved, **atom labels**, the **molecular orbitals** that are formed, **filled electrons** in the MO diagram etc. Make sure to **label all the atomic and molecular orbitals**.

Using the MO diagram above, answer the following questions:

- (i) Which one of the following will have the highest bond order? Circle the correct choice
 A) PS^{1-} B) PS C) PS^{1+}
- (ii) Which one of the following will have the longest bond length? Circle the correct choice
 A) PS^{1-} B) PS C) PS^{1+}
- (iii) Which of the following will be diamagnetic? Circle the correct choice.
 A) PS^{1-} B) PS C) PS^{1+}
- (iv) Draw Lewis dot structure of the molecule with the shortest bond length (based on the MO diagram) from among the three (PS^{1-} , PS, PS^{1+}), and indicate the bond order calculated from the Lewis dot structure. **(3 points)**

Lewis Dot Structure

Bond Order



598EBA9F-B95F-476B-884D-58E4E0344CF7

midterm-exam-93388

#594 Page 10 of 10

SCRAP PAGE

**CHEMISTRY 110 –Midterm Exam****Thursday November 7, 2024, 6:30 – 8:30 pm****Examiners: Pallavi Sirjoosingh and Ashok Kakkar**

First name (please write as legibly as possible within the boxes)

--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--

Last name

--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--

Student ID number

--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--

This booklet has 8 questions on 10 pages (The last page can be used as a scrap page).

The staple must NOT be removed from the question booklet. Your intact question booklet must be returned at the end of the exam.

A separate booklet contains a periodic table, and datasheet.

Write your answers directly under the questions in the space provided. Do not write on the QR codes or near the top edges (as it may be cut off).

Please write **legibly and dark enough** for the scan to be readable. For students who use English as a second language, dictionaries are permitted. Calculators are allowed.

DO NOT ASK QUESTIONS OF THE INVIGILATORS about interpretations of the exam questions. If you have a concern about a question, please summarize it at the top of this page. All concerns will be evaluated after the exam. GOOD LUCK!

**Question 1 8 points** (*Assume all transitions are possible*)

a. 6 points Calculate the **energy (in kJ/mol)** required to ionize (completely remove) the electron from the **$n=3$** level of **He^+ ion** (He: Helium). Show your work including equations. (3 sig. figs.)

b. 2 points

(I) Select the largest (longest) wavelength from the three choices below:

- i. Wavelength of photon to *completely remove* an electron from *1s orbital of Hydrogen atom*
- ii. Wavelength of photon to *completely remove* an electron from *1s orbital of Boron atom (B)*
- iii. Wavelength of photon to *completely remove* an electron from *1s orbital of Boron cation (B^+)*

i or ii or iii

(II) Select the smallest (shortest) wavelength from the three choices below (*for H atom*):

- i. Wavelength of photon *emitted* for an electron transitioning from *2p to 1s orbital*
- ii. Wavelength of photon *absorbed* for an electron transitioning from *2s to 4p orbital*
- iii. Wavelength of photon *absorbed* for an electron transitioning from *1s to 3p orbital*

i or ii or iii



Question 2 8 points



A Chem110 student plots the graph shown on the left, from data obtained for the kinetic energy (eV) of the ejected electron versus frequency (s^{-1}) of the incident light, for an unknown metal. ($1\text{eV} = 1.6022 \times 10^{-19} \text{ J}$)

Based on the plot, answer the questions **a to c** below:

a. 1 point On the plot above, label the **threshold frequency**.

b. 2 points What is the value of the work potential (in J) of the unknown metal?

c. 3 points What is the *longest* wavelength (in nm) required to eject an electron from this metal's surface? Show your work including equation(s) used. (3 sig figs)

d. 2 points In the space provided below, plot the following

(Assume frequency is *above* threshold frequency):

- Kinetic energy* of an ejected electron versus the *intensity* of the incident light (left)
- Number of electrons* ejected versus the *frequency* of the incident light (right)



**Question 3** **6 points****a. 2 points**

In the ground state of Ge (Germanium; $Z = 32$) atom, **name** the following orbitals:

i. Orbital containing the electron that requires the maximum energy to be removed.

ii. Orbital containing the electron that experiences the minimum effective nuclear charge.

b. 2 points

An unknown anion (A^-) has the following *excited* electronic configuration: $[Ar] 4s^2 3d^3 4p^1$

(Ar: Argon; $Z = 18$)

Give the **condensed ground state electronic configuration** of the atom, **A** and the cation, **A^{2+}** .
Show your work. (Assume all electronic transitions are possible)

Electronic Configuration of A:

Electronic Configuration of A^{2+} :

c. 2 points

In a single atom, determine how many electrons can have the following set of quantum numbers?

(i) $n = 5; l = 3$

In a single atom, determine how many electrons can have the following set of quantum numbers?

(i) $n = 3; l = 2; m_s = +1/2$

**Question 4** **8 points**

Rank the following from 1 to 4 in **increasing** order. Provide the ranking in the boxes provided.

Note: Atomic number (Z) are provided for relevant atoms

a) Increasing (lowest to highest) order of *ionization energy*: (1= lowest; 4 = highest)

Mg

Mg: Z = 12

Rb

Rb: Z = 37

Cs

Cs: Z = 55

K

K : Z = 19

b) Increasing (lowest to highest) order of *bond angle*: (1= lowest; 4 = highest)

H₂OCCl₄XeF₄

HCN

c) Increasing (smallest to largest) order of *ionic size* (1= smallest; 4 = largest)

S²⁻

S: Z = 16

O²⁻

O: Z = 8

Be²⁺

Be: Z = 4

Na⁺

Na: Z = 11

d) Increasing (lowest to highest) order of *bond energy* (1= lowest; 4 = highest)

P-F

P-Cl

P-I

P-Br

**Question 5** **12 points**

a. 9 points For the molecules HClO_4 and TeF_6 , draw the most stable *Lewis structure* showing appropriate shape and *dashed/wedged bonds*, based on VSEPR. Determine the **molecular geometry** around the central atom. For HClO_4 , label the *most polar bond*.



b. 3 points (Note: Lone pairs are not shown in structures below)

(i) For the three *resonance* structures (1, 2 and 3) shown below, answer the following question:



(I) Circle the *most contributing* resonance structure.

Structure 1

Structure 2

Structure 3

(ii) For the three structures given (4, 5, and 6) below, answer the following questions:



(I) Circle all the structures that represent *resonance structures* of each other:

Structure 4

Structure 5

Structure 6

(II) Among the resonance structures, circle the *most contributing* resonance structure.

Structure 4

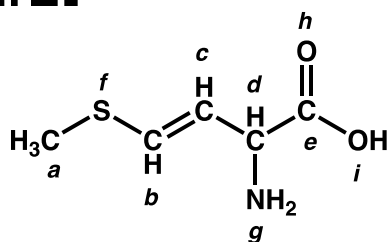
Structure 5

Structure 6

**Question 6** **6 points**

Using the data given below, calculate the **lattice energy** for SrO(s).

	ΔH° (kJ/mol)
$\text{Sr}(s) \longrightarrow \text{Sr}(g)$	+180
$\text{Sr}(g) \longrightarrow \text{Sr}^{2+}(g) + 2e^-$	+1468
$\text{O}_2(g) \longrightarrow 2\text{O}(g)$	+500
$\text{O}(g) + 2e^- \longrightarrow \text{O}^{2-}(g)$	+600
$\text{Sr}(s) + \frac{1}{2}\text{O}_2(g) \longrightarrow \text{SrO}(s)$	-550

**Question 7 10 points**

Consider the molecule given on the left, and answer questions (i) to (vi) using chemical bonding theories that we have learnt in class (valence bond theory, hybridization).

Note that the lone pairs are not shown in the structure:

i. Consider the carbon atoms marked in as ***a***, ***b***, ***c***, ***d***, and ***e*** in the molecule, and determine **hybridization** at each of the carbon centers. (2.5 points)

Atom	Hybridization
Carbon (<i>a</i>)	
Carbon (<i>b</i>)	
Carbon (<i>c</i>)	
Carbon (<i>d</i>)	
Carbon (<i>e</i>)	

ii. Now, consider the sulfur atom (S) marked as ***f***, nitrogen (N) atom (***g***), and oxygen(O) atoms (***h*** and ***i***), and determine their **hybridization**. (2 points)

Atom	Hybridization
Sulfur (<i>f</i>)	
Nitrogen (<i>g</i>)	
Oxygen (<i>h</i>)	
Oxygen (<i>i</i>)	

iii. **Describe** all the bonds formed by the carbon atoms ***a*** and ***b***, using sigma and pi designation as appropriate. (2 points)

Atom	Bonds formed (use sigma and pi as appropriate)
Carbon (<i>a</i>)	
Carbon (<i>b</i>)	

iv. How many total sp^3 -s sigma bonds are in the molecule? (1 point)

v. Determine **electronic** and **molecular geometry** at carbon (C) labeled ***e*** and sulfur (S) labeled ***f*** centres. (2 points)

Atom	Electronic Geometry	Molecular Geometry
Carbon (<i>e</i>)		
Sulfur (<i>f</i>)		

vi. Circle the value from the choices given below, that describes the **C-S-C bond angle at sulfur** (S) marked ***f***: (0.5 points)

>90° to <109.5° 109.5° >109.5 to <120° 120° >120° to <180°


Question 8 12 points

Using the **valence shell orbitals only** draw a **complete** molecular orbital (MO) diagram (with no s-p mixing) for the molecule **PS**, in the space provided below. **(6 points)**

Assume that 3s and 3p atomic orbitals will form molecular orbitals like those formed by the combination of 2s and 2p atomic orbitals that we learnt in class, and there is no s-p mixing.

Show **energy arrow**, **atomic orbitals** involved, **atom labels**, the **molecular orbitals** that are formed, **filled electrons** in the MO diagram etc. Make sure to label all the atomic and molecular orbitals.

Using the MO diagram above, answer the following questions:

- (i) Which one of the following will have the highest bond order? Circle the correct choice
 A) PS^{1-} B) PS C) PS^{1+}
- (ii) Which one of the following will have the longest bond length? Circle the correct choice
 A) PS^{1-} B) PS C) PS^{1+}
- (iii) Which of the following will be diamagnetic? Circle the correct choice.
 A) PS^{1-} B) PS C) PS^{1+}
- (iv) Draw Lewis dot structure of the molecule with the shortest bond length (based on the MO diagram) from among the three (PS^{1-} , PS, PS^{1+}), and indicate the bond order calculated from the Lewis dot structure. **(3 points)**

Lewis Dot Structure

Bond Order



AB8E0675-D216-4E76-AA73-6C163EAA7F31

midterm-exam-93388

#595 Page 10 of 10

SCRAP PAGE

**CHEMISTRY 110 –Midterm Exam****Thursday November 7, 2024, 6:30 – 8:30 pm****Examiners: Pallavi Sirjoosingh and Ashok Kakkar**

First name (please write as legibly as possible within the boxes)

--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--

Last name

--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--

Student ID number

--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--

This booklet has 8 questions on 10 pages (The last page can be used as a scrap page).

The staple must NOT be removed from the question booklet. Your intact question booklet must be returned at the end of the exam.

A separate booklet contains a periodic table, and datasheet.

Write your answers directly under the questions in the space provided. Do not write on the QR codes or near the top edges (as it may be cut off).

Please write **legibly and dark enough** for the scan to be readable. For students who use English as a second language, dictionaries are permitted. Calculators are allowed.

DO NOT ASK QUESTIONS OF THE INVIGILATORS about interpretations of the exam questions. If you have a concern about a question, please summarize it at the top of this page. All concerns will be evaluated after the exam. GOOD LUCK!

**Question 1 8 points** (*Assume all transitions are possible*)

a. 6 points Calculate the **energy (in kJ/mol)** required to ionize (completely remove) the electron from the **$n=3$** level of **He^+ ion** (He: Helium). Show your work including equations. (3 sig. figs.)

b. 2 points

(I) Select the largest (longest) wavelength from the three choices below:

- i. Wavelength of photon to *completely remove* an electron from *1s orbital of Hydrogen atom*
- ii. Wavelength of photon to *completely remove* an electron from *1s orbital of Boron atom (B)*
- iii. Wavelength of photon to *completely remove* an electron from *1s orbital of Boron cation (B^+)*

i or ii or iii

(II) Select the smallest (shortest) wavelength from the three choices below (*for H atom*):

- i. Wavelength of photon *emitted* for an electron transitioning from *2p to 1s orbital*
- ii. Wavelength of photon *absorbed* for an electron transitioning from *2s to 4p orbital*
- iii. Wavelength of photon *absorbed* for an electron transitioning from *1s to 3p orbital*

i or ii or iii



Question 2 8 points



A Chem110 student plots the graph shown on the left, from data obtained for the kinetic energy (eV) of the ejected electron versus frequency (s^{-1}) of the incident light, for an unknown metal. ($1\text{eV} = 1.6022 \times 10^{-19} \text{ J}$)

Based on the plot, answer the questions **a to c** below:

a. 1 point On the plot above, label the **threshold frequency**.

b. 2 points What is the value of the work potential (in J) of the unknown metal?

c. 3 points What is the *longest* wavelength (in nm) required to eject an electron from this metal's surface? Show your work including equation(s) used. (3 sig figs)

d. 2 points In the space provided below, plot the following

(Assume frequency is *above* threshold frequency):

- i. *Kinetic energy* of an ejected electron versus the *intensity* of the incident light (left)
- ii. *Number of electrons* ejected versus the *frequency* of the incident light (right)



**Question 3** **6 points****a. 2 points**

In the ground state of Ge (Germanium; $Z = 32$) atom, **name** the following orbitals:

i. Orbital containing the electron that requires the maximum energy to be removed.

ii. Orbital containing the electron that experiences the minimum effective nuclear charge.

b. 2 points

An unknown anion (A^-) has the following *excited* electronic configuration: $[Ar] 4s^2 3d^3 4p^1$

(Ar: Argon; $Z = 18$)

Give the **condensed ground state electronic configuration** of the atom, **A** and the cation, **A^{2+}** .
Show your work. (Assume all electronic transitions are possible)

Electronic Configuration of A:

Electronic Configuration of A^{2+} :

c. 2 points

In a single atom, determine how many electrons can have the following set of quantum numbers?

(i) $n = 5; l = 3$

In a single atom, determine how many electrons can have the following set of quantum numbers?

(i) $n = 3; l = 2; m_s = +1/2$

**Question 4** **8 points**

Rank the following from 1 to 4 in **increasing** order. Provide the ranking in the boxes provided.

Note: Atomic number (Z) are provided for relevant atoms

a) Increasing (lowest to highest) order of *ionization energy*: (1= lowest; 4 = highest)

Mg

Mg: Z = 12

Rb

Rb: Z = 37

Cs

Cs: Z = 55

K

K : Z = 19

b) Increasing (lowest to highest) order of *bond angle*: (1= lowest; 4 = highest)

H₂OCCl₄XeF₄

HCN

c) Increasing (smallest to largest) order of *ionic size* (1= smallest; 4 = largest)

S²⁻

S: Z = 16

O²⁻

O: Z = 8

Be²⁺

Be: Z = 4

Na⁺

Na: Z = 11

d) Increasing (lowest to highest) order of *bond energy* (1= lowest; 4 = highest)

P-F

P-Cl

P-I

P-Br

**Question 5** **12 points**

a. 9 points For the molecules HClO_4 and TeF_6 , draw the most stable *Lewis structure* showing appropriate shape and *dashed/wedged bonds*, based on VSEPR. Determine the **molecular geometry** around the central atom. For HClO_4 , label the *most polar bond*.



b. 3 points (Note: Lone pairs are not shown in structures below)

(i) For the three *resonance* structures (1, 2 and 3) shown below, answer the following question:



(I) Circle the *most contributing* resonance structure.

Structure 1

Structure 2

Structure 3

(ii) For the three structures given (4, 5, and 6) below, answer the following questions:



(I) Circle all the structures that represent *resonance structures* of each other:

Structure 4

Structure 5

Structure 6

(II) Among the resonance structures, circle the *most contributing* resonance structure.

Structure 4

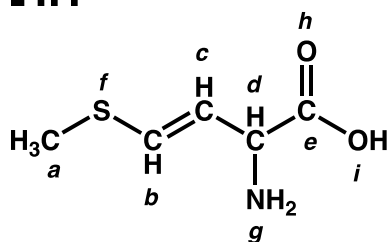
Structure 5

Structure 6

**Question 6** **6 points**

Using the data given below, calculate the **lattice energy** for SrO(s).

	ΔH° (kJ/mol)
$\text{Sr}(s) \longrightarrow \text{Sr}(g)$	+180
$\text{Sr}(g) \longrightarrow \text{Sr}^{2+}(g) + 2e^-$	+1468
$\text{O}_2(g) \longrightarrow 2\text{O}(g)$	+500
$\text{O}(g) + 2e^- \longrightarrow \text{O}^{2-}(g)$	+600
$\text{Sr}(s) + \frac{1}{2}\text{O}_2(g) \longrightarrow \text{SrO}(s)$	-550

**Question 7 10 points**

Consider the molecule given on the left, and answer questions (i) to (vi) using chemical bonding theories that we have learnt in class (valence bond theory, hybridization).

Note that the lone pairs are not shown in the structure:

i. Consider the carbon atoms marked in as ***a***, ***b***, ***c***, ***d***, and ***e*** in the molecule, and determine **hybridization** at each of the carbon centers. (2.5 points)

Atom	Hybridization
Carbon (<i>a</i>)	
Carbon (<i>b</i>)	
Carbon (<i>c</i>)	
Carbon (<i>d</i>)	
Carbon (<i>e</i>)	

ii. Now, consider the sulfur atom (S) marked as ***f***, nitrogen (N) atom (***g***), and oxygen(O) atoms (***h*** and ***i***), and determine their **hybridization**. (2 points)

Atom	Hybridization
Sulfur (<i>f</i>)	
Nitrogen (<i>g</i>)	
Oxygen (<i>h</i>)	
Oxygen (<i>i</i>)	

iii. **Describe** all the bonds formed by the carbon atoms ***a*** and ***b***, using sigma and pi designation as appropriate. (2 points)

Atom	Bonds formed (use sigma and pi as appropriate)
Carbon (<i>a</i>)	
Carbon (<i>b</i>)	

iv. How many total sp^3 -s sigma bonds are in the molecule? (1 point)

v. Determine **electronic** and **molecular geometry** at carbon (C) labeled ***e*** and sulfur (S) labeled ***f*** centres. (2 points)

Atom	Electronic Geometry	Molecular Geometry
Carbon (<i>e</i>)		
Sulfur (<i>f</i>)		

vi. Circle the value from the choices given below, that describes the **C-S-C bond angle at sulfur** (S) marked ***f***: (0.5 points)

>90° to <109.5°

109.5°

>109.5 to <120°

120°

>120° to <180°

**Question 8 12 points**

Using the **valence shell orbitals only** draw a **complete** molecular orbital (MO) diagram (with no s-p mixing) for the molecule **PS**, in the space provided below. **(6 points)**

Assume that 3s and 3p atomic orbitals will form molecular orbitals like those formed by the combination of 2s and 2p atomic orbitals that we learnt in class, and there is no s-p mixing.

Show **energy arrow**, **atomic orbitals** involved, **atom labels**, the **molecular orbitals** that are formed, **filled electrons** in the MO diagram etc. Make sure to label all the atomic and molecular orbitals.

Using the MO diagram above, answer the following questions:

- (i) Which one of the following will have the highest bond order? Circle the correct choice
A) PS^{1-} B) PS C) PS^{1+}
- (ii) Which one of the following will have the longest bond length? Circle the correct choice
A) PS^{1-} B) PS C) PS^{1+}
- (iii) Which of the following will be diamagnetic? Circle the correct choice.
A) PS^{1-} B) PS C) PS^{1+}
- (iv) Draw Lewis dot structure of the molecule with the shortest bond length (based on the MO diagram) from among the three (PS^{1-} , PS, PS^{1+}), and indicate the bond order calculated from the Lewis dot structure. **(3 points)**

Lewis Dot Structure

Bond Order



F4A328A0-0AEF-453E-B2FE-C66B753170C6

midterm-exam-93388

#596 Page 10 of 10

SCRAP PAGE

**CHEMISTRY 110 –Midterm Exam****Thursday November 7, 2024, 6:30 – 8:30 pm**

Examiners: Pallavi Sirjoosingh and Ashok Kakkar

First name (please write as legibly as possible within the boxes)

--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--

Last name

--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--

Student ID number

--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--

This booklet has 8 questions on 10 pages (The last page can be used as a scrap page).

The staple must NOT be removed from the question booklet. Your intact question booklet must be returned at the end of the exam.

A separate booklet contains a periodic table, and datasheet.

Write your answers directly under the questions in the space provided. Do not write on the QR codes or near the top edges (as it may be cut off).

Please write **legibly and dark enough** for the scan to be readable. For students who use English as a second language, dictionaries are permitted. Calculators are allowed.

DO NOT ASK QUESTIONS OF THE INVIGILATORS about interpretations of the exam questions. If you have a concern about a question, please summarize it at the top of this page. All concerns will be evaluated after the exam. GOOD LUCK!

**Question 1 8 points** (*Assume all transitions are possible*)

a. 6 points Calculate the **energy (in kJ/mol)** required to ionize (completely remove) the electron from the **n=3** level of **He⁺ ion** (He: Helium). Show your work including equations. (3 sig. figs.)

b. 2 points

(I) Select the largest (longest) wavelength from the three choices below:

- i. Wavelength of photon to *completely remove* an electron from *1s orbital of Hydrogen atom*
- ii. Wavelength of photon to *completely remove* an electron from *1s orbital of Boron atom (B)*
- iii. Wavelength of photon to *completely remove* an electron from *1s orbital of Boron cation (B⁺)*

i or ii or iii

(II) Select the smallest (shortest) wavelength from the three choices below (*for H atom*):

- i. Wavelength of photon *emitted* for an electron transitioning from *2p to 1s orbital*
- ii. Wavelength of photon *absorbed* for an electron transitioning from *2s to 4p orbital*
- iii. Wavelength of photon *absorbed* for an electron transitioning from *1s to 3p orbital*

i or ii or iii



Question 2 8 points



A Chem110 student plots the graph shown on the left, from data obtained for the kinetic energy (eV) of the ejected electron versus frequency (s^{-1}) of the incident light, for an unknown metal. ($1\text{eV} = 1.6022 \times 10^{-19} \text{ J}$)

Based on the plot, answer the questions **a to c** below:

a. 1 point On the plot above, label the **threshold frequency**.

b. 2 points What is the value of the work potential (in J) of the unknown metal?

c. 3 points What is the *longest* wavelength (in nm) required to eject an electron from this metal's surface? Show your work including equation(s) used. (3 sig figs)

d. 2 points In the space provided below, plot the following

(Assume frequency is *above* threshold frequency):

- i. *Kinetic energy* of an ejected electron versus the *intensity* of the incident light (left)
- ii. *Number of electrons* ejected versus the *frequency* of the incident light (right)



**Question 3** **6 points****a. 2 points**

In the ground state of Ge (Germanium; $Z = 32$) atom, **name** the following orbitals:

i. Orbital containing the electron that requires the maximum energy to be removed.

ii. Orbital containing the electron that experiences the minimum effective nuclear charge.

b. 2 points

An unknown anion (A^-) has the following *excited* electronic configuration: $[Ar] 4s^2 3d^3 4p^1$

(Ar: Argon; $Z = 18$)

Give the **condensed ground state electronic configuration** of the atom, **A** and the cation, **A^{2+}** .
Show your work. (Assume all electronic transitions are possible)

Electronic Configuration of A:

Electronic Configuration of A^{2+} :

c. 2 points

In a single atom, determine how many electrons can have the following set of quantum numbers?

(i) $n = 5; l = 3$

In a single atom, determine how many electrons can have the following set of quantum numbers?

(i) $n = 3; l = 2; m_s = +1/2$

**Question 4** **8 points**

Rank the following from 1 to 4 in **increasing** order. Provide the ranking in the boxes provided.

Note: Atomic number (Z) are provided for relevant atoms

a) Increasing (lowest to highest) order of *ionization energy*: (1= lowest; 4 = highest)

Mg

Mg: Z = 12

Rb

Rb: Z = 37

Cs

Cs: Z = 55

K

K : Z = 19

b) Increasing (lowest to highest) order of *bond angle*: (1= lowest; 4 = highest)

H₂OCCl₄XeF₄

HCN

c) Increasing (smallest to largest) order of *ionic size* (1= smallest; 4 = largest)

S²⁻

S: Z = 16

O²⁻

O: Z = 8

Be²⁺

Be: Z = 4

Na⁺

Na: Z = 11

d) Increasing (lowest to highest) order of *bond energy* (1= lowest; 4 = highest)

P-F

P-Cl

P-I

P-Br

**Question 5** **12 points**

a. 9 points For the molecules HClO_4 and TeF_6 , draw the most stable *Lewis structure* showing appropriate shape and *dashed/wedged bonds*, based on VSEPR. Determine the **molecular geometry** around the central atom. For HClO_4 , label the *most polar bond*.



b. 3 points (Note: Lone pairs are not shown in structures below)

(i) For the three *resonance* structures (1, 2 and 3) shown below, answer the following question:



(I) Circle the *most contributing* resonance structure.

Structure 1

Structure 2

Structure 3

(ii) For the three structures given (4, 5, and 6) below, answer the following questions:



(I) Circle all the structures that represent *resonance structures* of each other:

Structure 4

Structure 5

Structure 6

(II) Among the resonance structures, circle the *most contributing* resonance structure.

Structure 4

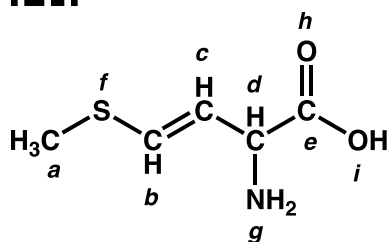
Structure 5

Structure 6

**Question 6** **6 points**

Using the data given below, calculate the **lattice energy** for SrO(s).

	ΔH° (kJ/mol)
$\text{Sr}(s) \longrightarrow \text{Sr}(g)$	+180
$\text{Sr}(g) \longrightarrow \text{Sr}^{2+}(g) + 2e^-$	+1468
$\text{O}_2(g) \longrightarrow 2\text{O}(g)$	+500
$\text{O}(g) + 2e^- \longrightarrow \text{O}^{2-}(g)$	+600
$\text{Sr}(s) + \frac{1}{2}\text{O}_2(g) \longrightarrow \text{SrO}(s)$	-550

**Question 7 10 points**

Consider the molecule given on the left, and answer questions (i) to (vi) using chemical bonding theories that we have learnt in class (valence bond theory, hybridization).

Note that the lone pairs are not shown in the structure:

i. Consider the carbon atoms marked in as ***a***, ***b***, ***c***, ***d***, and ***e*** in the molecule, and determine **hybridization** at each of the carbon centers. (2.5 points)

Atom	Hybridization
Carbon (<i>a</i>)	
Carbon (<i>b</i>)	
Carbon (<i>c</i>)	
Carbon (<i>d</i>)	
Carbon (<i>e</i>)	

ii. Now, consider the sulfur atom (S) marked as ***f***, nitrogen (N) atom (***g***), and oxygen(O) atoms (***h*** and ***i***), and determine their **hybridization**. (2 points)

Atom	Hybridization
Sulfur (<i>f</i>)	
Nitrogen (<i>g</i>)	
Oxygen (<i>h</i>)	
Oxygen (<i>i</i>)	

iii. **Describe** all the bonds formed by the carbon atoms ***a*** and ***b***, using sigma and pi designation as appropriate. (2 points)

Atom	Bonds formed (use sigma and pi as appropriate)
Carbon (<i>a</i>)	
Carbon (<i>b</i>)	

iv. How many total sp^3 -s sigma bonds are in the molecule? (1 point)

v. Determine **electronic** and **molecular geometry** at carbon (C) labeled ***e*** and sulfur (S) labeled ***f*** centres. (2 points)

Atom	Electronic Geometry	Molecular Geometry
Carbon (<i>e</i>)		
Sulfur (<i>f</i>)		

vi. Circle the value from the choices given below, that describes the **C-S-C bond angle at sulfur** (S) marked ***f***: (0.5 points)

>90° to <109.5°

109.5°

>109.5 to <120°

120°

>120° to <180°


Question 8 12 points

Using the **valence shell orbitals only** draw a **complete** molecular orbital (MO) diagram (with no s-p mixing) for the molecule **PS**, in the space provided below. **(6 points)**

Assume that 3s and 3p atomic orbitals will form molecular orbitals like those formed by the combination of 2s and 2p atomic orbitals that we learnt in class, and there is no s-p mixing.

Show **energy arrow**, **atomic orbitals** involved, **atom labels**, the **molecular orbitals** that are formed, **filled electrons** in the MO diagram etc. Make sure to label all the atomic and molecular orbitals.

Using the MO diagram above, answer the following questions:

- (i) Which one of the following will have the highest bond order? Circle the correct choice
 A) PS^{1-} B) PS C) PS^{1+}
- (ii) Which one of the following will have the longest bond length? Circle the correct choice
 A) PS^{1-} B) PS C) PS^{1+}
- (iii) Which of the following will be diamagnetic? Circle the correct choice.
 A) PS^{1-} B) PS C) PS^{1+}
- (iv) Draw Lewis dot structure of the molecule with the shortest bond length (based on the MO diagram) from among the three (PS^{1-} , PS, PS^{1+}), and indicate the bond order calculated from the Lewis dot structure. **(3 points)**

Lewis Dot Structure

Bond Order



E5DA71DE-ACDA-49F9-811A-C1E8162D1ED0

midterm-exam-93388

#597 Page 10 of 10

SCRAP PAGE

**CHEMISTRY 110 –Midterm Exam****Thursday November 7, 2024, 6:30 – 8:30 pm**

Examiners: Pallavi Sirjoosingh and Ashok Kakkar

First name (please write as legibly as possible within the boxes)

--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--

Last name

--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--

Student ID number

--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--

This booklet has 8 questions on 10 pages (The last page can be used as a scrap page).

The staple must NOT be removed from the question booklet. Your intact question booklet must be returned at the end of the exam.

A separate booklet contains a periodic table, and datasheet.

Write your answers directly under the questions in the space provided. Do not write on the QR codes or near the top edges (as it may be cut off).

Please write **legibly and dark enough** for the scan to be readable. For students who use English as a second language, dictionaries are permitted. Calculators are allowed.

DO NOT ASK QUESTIONS OF THE INVIGILATORS about interpretations of the exam questions. If you have a concern about a question, please summarize it at the top of this page. All concerns will be evaluated after the exam. GOOD LUCK!

**Question 1 8 points** (*Assume all transitions are possible*)

a. 6 points Calculate the **energy (in kJ/mol)** required to ionize (completely remove) the electron from the **$n=3$** level of **He^+ ion** (He: Helium). Show your work including equations. (3 sig. figs.)

b. 2 points

(I) Select the largest (longest) wavelength from the three choices below:

- i. Wavelength of photon to *completely remove* an electron from *1s orbital of Hydrogen atom*
- ii. Wavelength of photon to *completely remove* an electron from *1s orbital of Boron atom (B)*
- iii. Wavelength of photon to *completely remove* an electron from *1s orbital of Boron cation (B^+)*

i or ii or iii

(II) Select the smallest (shortest) wavelength from the three choices below (*for H atom*):

- i. Wavelength of photon *emitted* for an electron transitioning from *2p to 1s orbital*
- ii. Wavelength of photon *absorbed* for an electron transitioning from *2s to 4p orbital*
- iii. Wavelength of photon *absorbed* for an electron transitioning from *1s to 3p orbital*

i or ii or iii



Question 2 8 points



A Chem110 student plots the graph shown on the left, from data obtained for the kinetic energy (eV) of the ejected electron versus frequency (s^{-1}) of the incident light, for an unknown metal. ($1\text{eV} = 1.6022 \times 10^{-19} \text{ J}$)

Based on the plot, answer the questions **a to c** below:

a. 1 point On the plot above, label the **threshold frequency**.

b. 2 points What is the value of the work potential (in J) of the unknown metal?

c. 3 points What is the *longest* wavelength (in nm) required to eject an electron from this metal's surface? Show your work including equation(s) used. (3 sig figs)

d. 2 points In the space provided below, plot the following

(Assume frequency is *above* threshold frequency):

- i. *Kinetic energy* of an ejected electron versus the *intensity* of the incident light (left)
- ii. *Number of electrons* ejected versus the *frequency* of the incident light (right)



**Question 3** **6 points****a. 2 points**

In the ground state of Ge (Germanium; $Z = 32$) atom, **name** the following orbitals:

i. Orbital containing the electron that requires the maximum energy to be removed.

ii. Orbital containing the electron that experiences the minimum effective nuclear charge.

b. 2 points

An unknown anion (A^-) has the following *excited* electronic configuration: $[Ar] 4s^2 3d^3 4p^1$

(Ar: Argon; $Z = 18$)

Give the **condensed ground state electronic configuration** of the atom, **A** and the cation, **A^{2+}** .
Show your work. (Assume all electronic transitions are possible)

Electronic Configuration of A:

Electronic Configuration of A^{2+} :

c. 2 points

In a single atom, determine how many electrons can have the following set of quantum numbers?

(i) $n = 5; l = 3$

In a single atom, determine how many electrons can have the following set of quantum numbers?

(i) $n = 3; l = 2; m_s = +1/2$

**Question 4** **8 points**

Rank the following from 1 to 4 in **increasing** order. Provide the ranking in the boxes provided.

Note: Atomic number (Z) are provided for relevant atoms

a) Increasing (lowest to highest) order of *ionization energy*: (1= lowest; 4 = highest)

Mg

Mg: Z = 12

Rb

Rb: Z = 37

Cs

Cs: Z = 55

K

K : Z = 19

b) Increasing (lowest to highest) order of *bond angle*: (1= lowest; 4 = highest)

H₂OCCl₄XeF₄

HCN

c) Increasing (smallest to largest) order of *ionic size* (1= smallest; 4 = largest)

S²⁻

S: Z = 16

O²⁻

O: Z = 8

Be²⁺

Be: Z = 4

Na⁺

Na: Z = 11

d) Increasing (lowest to highest) order of *bond energy* (1= lowest; 4 = highest)

P-F

P-Cl

P-I

P-Br

**Question 5** **12 points**

a. 9 points For the molecules HClO_4 and TeF_6 , draw the most stable *Lewis structure* showing appropriate shape and *dashed/wedged bonds*, based on VSEPR. Determine the **molecular geometry** around the central atom. For HClO_4 , label the *most polar bond*.



b. 3 points (Note: Lone pairs are not shown in structures below)

(i) For the three *resonance* structures (1, 2 and 3) shown below, answer the following question:



(I) Circle the *most contributing* resonance structure.

Structure 1

Structure 2

Structure 3

(ii) For the three structures given (4, 5, and 6) below, answer the following questions:



(I) Circle all the structures that represent *resonance structures* of each other:

Structure 4

Structure 5

Structure 6

(II) Among the resonance structures, circle the *most contributing* resonance structure.

Structure 4

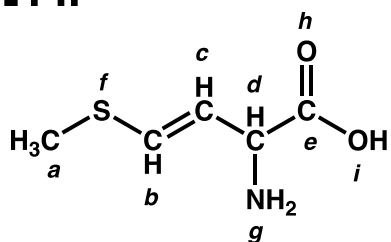
Structure 5

Structure 6

**Question 6** **6 points**

Using the data given below, calculate the **lattice energy** for SrO(s).

	ΔH° (kJ/mol)
$\text{Sr}(s) \longrightarrow \text{Sr}(g)$	+180
$\text{Sr}(g) \longrightarrow \text{Sr}^{2+}(g) + 2e^-$	+1468
$\text{O}_2(g) \longrightarrow 2\text{O}(g)$	+500
$\text{O}(g) + 2e^- \longrightarrow \text{O}^{2-}(g)$	+600
$\text{Sr}(s) + \frac{1}{2}\text{O}_2(g) \longrightarrow \text{SrO}(s)$	-550

**Question 7 10 points**

Consider the molecule given on the left, and answer questions (i) to (vi) using chemical bonding theories that we have learnt in class (valence bond theory, hybridization).

Note that the lone pairs are not shown in the structure:

i. Consider the carbon atoms marked in as ***a***, ***b***, ***c***, ***d***, and ***e*** in the molecule, and determine **hybridization** at each of the carbon centers. (2.5 points)

Atom	Hybridization
Carbon (<i>a</i>)	
Carbon (<i>b</i>)	
Carbon (<i>c</i>)	
Carbon (<i>d</i>)	
Carbon (<i>e</i>)	

ii. Now, consider the sulfur atom (S) marked as ***f***, nitrogen (N) atom (***g***), and oxygen(O) atoms (***h*** and ***i***), and determine their **hybridization**. (2 points)

Atom	Hybridization
Sulfur (<i>f</i>)	
Nitrogen (<i>g</i>)	
Oxygen (<i>h</i>)	
Oxygen (<i>i</i>)	

iii. **Describe** all the bonds formed by the carbon atoms ***a*** and ***b***, using sigma and pi designation as appropriate. (2 points)

Atom	Bonds formed (use sigma and pi as appropriate)
Carbon (<i>a</i>)	
Carbon (<i>b</i>)	

iv. How many total sp^3 -s sigma bonds are in the molecule? (1 point)

v. Determine **electronic** and **molecular geometry** at carbon (C) labeled ***e*** and sulfur (S) labeled ***f*** centres. (2 points)

Atom	Electronic Geometry	Molecular Geometry
Carbon (<i>e</i>)		
Sulfur (<i>f</i>)		

vi. Circle the value from the choices given below, that describes the **C-S-C bond angle at sulfur** (S) marked ***f***: (0.5 points)

$>90^\circ$ to $<109.5^\circ$ 109.5° >109.5 to $<120^\circ$ 120° $>120^\circ$ to $<180^\circ$


Question 8 12 points

Using the **valence shell orbitals only** draw a **complete** molecular orbital (MO) diagram (with no s-p mixing) for the molecule **PS**, in the space provided below. **(6 points)**

Assume that 3s and 3p atomic orbitals will form molecular orbitals like those formed by the combination of 2s and 2p atomic orbitals that we learnt in class, and there is no s-p mixing.

Show **energy arrow**, **atomic orbitals** involved, **atom labels**, the **molecular orbitals** that are formed, **filled electrons** in the MO diagram etc. Make sure to label all the atomic and molecular orbitals.

Using the MO diagram above, answer the following questions:

- (i) Which one of the following will have the highest bond order? Circle the correct choice
 A) PS^{1-} B) PS C) PS^{1+}
- (ii) Which one of the following will have the longest bond length? Circle the correct choice
 A) PS^{1-} B) PS C) PS^{1+}
- (iii) Which of the following will be diamagnetic? Circle the correct choice.
 A) PS^{1-} B) PS C) PS^{1+}
- (iv) Draw Lewis dot structure of the molecule with the shortest bond length (based on the MO diagram) from among the three (PS^{1-} , PS, PS^{1+}), and indicate the bond order calculated from the Lewis dot structure. **(3 points)**

Lewis Dot Structure

Bond Order



2C13E9CA-F083-420D-9C18-81F3C2A7870D

midterm-exam-93388

#598 Page 10 of 10

SCRAP PAGE

**CHEMISTRY 110 –Midterm Exam****Thursday November 7, 2024, 6:30 – 8:30 pm**

Examiners: Pallavi Sirjoosingh and Ashok Kakkar

First name (please write as legibly as possible within the boxes)

--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--

Last name

--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--

Student ID number

--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--

This booklet has 8 questions on 10 pages (The last page can be used as a scrap page).

The staple must NOT be removed from the question booklet. Your intact question booklet must be returned at the end of the exam.

A separate booklet contains a periodic table, and datasheet.

Write your answers directly under the questions in the space provided. Do not write on the QR codes or near the top edges (as it may be cut off).

Please write **legibly and dark enough** for the scan to be readable. For students who use English as a second language, dictionaries are permitted. Calculators are allowed.

DO NOT ASK QUESTIONS OF THE INVIGILATORS about interpretations of the exam questions. If you have a concern about a question, please summarize it at the top of this page. All concerns will be evaluated after the exam. GOOD LUCK!

**Question 1 8 points** (*Assume all transitions are possible*)

a. 6 points Calculate the **energy (in kJ/mol)** required to ionize (completely remove) the electron from the **$n=3$** level of **He^+ ion** (He: Helium). Show your work including equations. (3 sig. figs.)

b. 2 points

(I) Select the largest (longest) wavelength from the three choices below:

- i. Wavelength of photon to *completely remove* an electron from *1s orbital of Hydrogen atom*
- ii. Wavelength of photon to *completely remove* an electron from *1s orbital of Boron atom (B)*
- iii. Wavelength of photon to *completely remove* an electron from *1s orbital of Boron cation (B^+)*

i or ii or iii

(II) Select the smallest (shortest) wavelength from the three choices below (*for H atom*):

- i. Wavelength of photon *emitted* for an electron transitioning from *2p to 1s orbital*
- ii. Wavelength of photon *absorbed* for an electron transitioning from *2s to 4p orbital*
- iii. Wavelength of photon *absorbed* for an electron transitioning from *1s to 3p orbital*

i or ii or iii



Question 2 8 points



A Chem110 student plots the graph shown on the left, from data obtained for the kinetic energy (eV) of the ejected electron versus frequency (s^{-1}) of the incident light, for an unknown metal. ($1\text{eV} = 1.6022 \times 10^{-19} \text{ J}$)

Based on the plot, answer the questions **a to c** below:

a. 1 point On the plot above, label the **threshold frequency**.

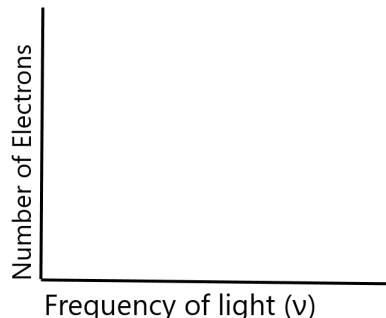
b. 2 points What is the value of the work potential (in J) of the unknown metal?

c. 3 points What is the *longest* wavelength (in nm) required to eject an electron from this metal's surface? Show your work including equation(s) used. (3 sig figs)

d. 2 points In the space provided below, plot the following

(Assume frequency is *above* threshold frequency):

- Kinetic energy* of an ejected electron versus the *intensity* of the incident light (left)
- Number of electrons* ejected versus the *frequency* of the incident light (right)



**Question 3** **6 points****a. 2 points**

In the ground state of Ge (Germanium; $Z = 32$) atom, **name** the following orbitals:

i. Orbital containing the electron that requires the maximum energy to be removed.

ii. Orbital containing the electron that experiences the minimum effective nuclear charge.

b. 2 points

An unknown anion (A^-) has the following *excited* electronic configuration: $[Ar] 4s^2 3d^3 4p^1$

(Ar: Argon; $Z = 18$)

Give the **condensed ground state electronic configuration** of the atom, **A** and the cation, **A^{2+}** .
Show your work. (Assume all electronic transitions are possible)

Electronic Configuration of A:

Electronic Configuration of A^{2+} :

c. 2 points

In a single atom, determine how many electrons can have the following set of quantum numbers?

(i) $n = 5; l = 3$

In a single atom, determine how many electrons can have the following set of quantum numbers?

(i) $n = 3; l = 2; m_s = +1/2$

**Question 4** **8 points**

Rank the following from 1 to 4 in **increasing** order. Provide the ranking in the boxes provided.

Note: Atomic number (Z) are provided for relevant atoms

a) Increasing (lowest to highest) order of *ionization energy*: (1= lowest; 4 = highest)

Mg

Mg: Z = 12

Rb

Rb: Z = 37

Cs

Cs: Z = 55

K

K : Z = 19

b) Increasing (lowest to highest) order of *bond angle*: (1= lowest; 4 = highest)

H₂OCCl₄XeF₄

HCN

c) Increasing (smallest to largest) order of *ionic size* (1= smallest; 4 = largest)

S²⁻

S: Z = 16

O²⁻

O: Z = 8

Be²⁺

Be: Z = 4

Na⁺

Na: Z = 11

d) Increasing (lowest to highest) order of *bond energy* (1= lowest; 4 = highest)

P-F

P-Cl

P-I

P-Br

**Question 5** **12 points**

a. 9 points For the molecules HClO_4 and TeF_6 , draw the most stable *Lewis structure* showing appropriate shape and *dashed/wedged bonds*, based on VSEPR. Determine the **molecular geometry** around the central atom. For HClO_4 , label the *most polar bond*.

HClO_4 (there are no O-O bonds)

Label the most polar bond

Molecular Geometry around Cl

TeF_6

Molecular Geometry around Te

b. 3 points (Note: Lone pairs are not shown in structures below)

(i) For the three *resonance* structures (1, 2 and 3) shown below, answer the following question:



(I) Circle the *most contributing* resonance structure.

Structure 1

Structure 2

Structure 3

(ii) For the three structures given (4, 5, and 6) below, answer the following questions:



(I) Circle all the structures that represent *resonance structures* of each other:

Structure 4

Structure 5

Structure 6

(II) Among the resonance structures, circle the *most contributing* resonance structure.

Structure 4

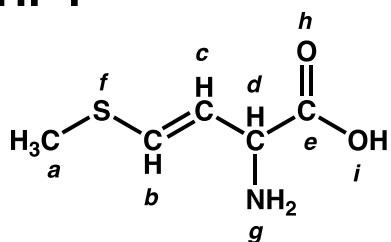
Structure 5

Structure 6

**Question 6** **6 points**

Using the data given below, calculate the **lattice energy** for SrO(s).

	ΔH° (kJ/mol)
$\text{Sr}(s) \longrightarrow \text{Sr}(g)$	+180
$\text{Sr}(g) \longrightarrow \text{Sr}^{2+}(g) + 2e^-$	+1468
$\text{O}_2(g) \longrightarrow 2\text{O}(g)$	+500
$\text{O}(g) + 2e^- \longrightarrow \text{O}^{2-}(g)$	+600
$\text{Sr}(s) + \frac{1}{2}\text{O}_2(g) \longrightarrow \text{SrO}(s)$	-550

**Question 7 10 points**

Consider the molecule given on the left, and answer questions (i) to (vi) using chemical bonding theories that we have learnt in class (valence bond theory, hybridization).

Note that the lone pairs are not shown in the structure:

i. Consider the carbon atoms marked in as ***a***, ***b***, ***c***, ***d***, and ***e*** in the molecule, and determine **hybridization** at each of the carbon centers. (2.5 points)

Atom	Hybridization
Carbon (<i>a</i>)	
Carbon (<i>b</i>)	
Carbon (<i>c</i>)	
Carbon (<i>d</i>)	
Carbon (<i>e</i>)	

ii. Now, consider the sulfur atom (S) marked as ***f***, nitrogen (N) atom (***g***), and oxygen(O) atoms (***h*** and ***i***), and determine their **hybridization**. (2 points)

Atom	Hybridization
Sulfur (<i>f</i>)	
Nitrogen (<i>g</i>)	
Oxygen (<i>h</i>)	
Oxygen (<i>i</i>)	

iii. **Describe** all the bonds formed by the carbon atoms ***a*** and ***b***, using sigma and pi designation as appropriate. (2 points)

Atom	Bonds formed (use sigma and pi as appropriate)
Carbon (<i>a</i>)	
Carbon (<i>b</i>)	

iv. How many total sp^3 -s sigma bonds are in the molecule? (1 point)

v. Determine **electronic** and **molecular geometry** at carbon (C) labeled ***e*** and sulfur (S) labeled ***f*** centres. (2 points)

Atom	Electronic Geometry	Molecular Geometry
Carbon (<i>e</i>)		
Sulfur (<i>f</i>)		

vi. Circle the value from the choices given below, that describes the **C-S-C bond angle at sulfur** (S) marked ***f***: (0.5 points)

>90° to <109.5° 109.5° >109.5 to <120° 120° >120° to <180°


Question 8 12 points

Using the **valence shell orbitals only** draw a **complete** molecular orbital (MO) diagram (with no s-p mixing) for the molecule **PS**, in the space provided below. **(6 points)**

Assume that 3s and 3p atomic orbitals will form molecular orbitals like those formed by the combination of 2s and 2p atomic orbitals that we learnt in class, and there is no s-p mixing.

Show **energy arrow**, **atomic orbitals** involved, **atom labels**, the **molecular orbitals** that are formed, **filled electrons** in the MO diagram etc. Make sure to label all the atomic and molecular orbitals.

Using the MO diagram above, answer the following questions:

- (i) Which one of the following will have the highest bond order? Circle the correct choice
 A) PS^{1-} B) PS C) PS^{1+}
- (ii) Which one of the following will have the longest bond length? Circle the correct choice
 A) PS^{1-} B) PS C) PS^{1+}
- (iii) Which of the following will be diamagnetic? Circle the correct choice.
 A) PS^{1-} B) PS C) PS^{1+}
- (iv) Draw Lewis dot structure of the molecule with the shortest bond length (based on the MO diagram) from among the three (PS^{1-} , PS, PS^{1+}), and indicate the bond order calculated from the Lewis dot structure. **(3 points)**

Lewis Dot Structure

Bond Order



B79F5DEA-D22A-483A-95CB-35F4CD0AA561

midterm-exam-93388

#599 Page 10 of 10

SCRAP PAGE

**CHEMISTRY 110 –Midterm Exam****Thursday November 7, 2024, 6:30 – 8:30 pm****Examiners: Pallavi Sirjoosingh and Ashok Kakkar**

First name (please write as legibly as possible within the boxes)

--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--

Last name

--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--

Student ID number

--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--

This booklet has 8 questions on 10 pages (The last page can be used as a scrap page).

The staple must NOT be removed from the question booklet. Your intact question booklet must be returned at the end of the exam.

A separate booklet contains a periodic table, and datasheet.

Write your answers directly under the questions in the space provided. Do not write on the QR codes or near the top edges (as it may be cut off).

Please write **legibly and dark enough** for the scan to be readable. For students who use English as a second language, dictionaries are permitted. Calculators are allowed.

DO NOT ASK QUESTIONS OF THE INVIGILATORS about interpretations of the exam questions. If you have a concern about a question, please summarize it at the top of this page. All concerns will be evaluated after the exam. GOOD LUCK!

**Question 1 8 points** (*Assume all transitions are possible*)

a. 6 points Calculate the **energy (in kJ/mol)** required to ionize (completely remove) the electron from the **n=3** level of **He⁺ ion** (He: Helium). Show your work including equations. (3 sig. figs.)

b. 2 points

(I) Select the largest (longest) wavelength from the three choices below:

- i. Wavelength of photon to *completely remove* an electron from *1s orbital of Hydrogen atom*
- ii. Wavelength of photon to *completely remove* an electron from *1s orbital of Boron atom (B)*
- iii. Wavelength of photon to *completely remove* an electron from *1s orbital of Boron cation (B⁺)*

i or ii or iii

(II) Select the smallest (shortest) wavelength from the three choices below (*for H atom*):

- i. Wavelength of photon *emitted* for an electron transitioning from *2p to 1s orbital*
- ii. Wavelength of photon *absorbed* for an electron transitioning from *2s to 4p orbital*
- iii. Wavelength of photon *absorbed* for an electron transitioning from *1s to 3p orbital*

i or ii or iii



Question 2 8 points



A Chem110 student plots the graph shown on the left, from data obtained for the kinetic energy (eV) of the ejected electron versus frequency (s^{-1}) of the incident light, for an unknown metal. ($1\text{eV} = 1.6022 \times 10^{-19} \text{ J}$)

Based on the plot, answer the questions **a to c** below:

a. 1 point On the plot above, label the **threshold frequency**.

b. 2 points What is the value of the work potential (in J) of the unknown metal?

c. 3 points What is the *longest* wavelength (in nm) required to eject an electron from this metal's surface? Show your work including equation(s) used. (3 sig figs)

d. 2 points In the space provided below, plot the following

(Assume frequency is *above* threshold frequency):

- Kinetic energy* of an ejected electron versus the *intensity* of the incident light (left)
- Number of electrons* ejected versus the *frequency* of the incident light (right)



**Question 3** **6 points****a. 2 points**

In the ground state of Ge (Germanium; $Z = 32$) atom, **name** the following orbitals:

i. Orbital containing the electron that requires the maximum energy to be removed.

ii. Orbital containing the electron that experiences the minimum effective nuclear charge.

b. 2 points

An unknown anion (A^-) has the following *excited* electronic configuration: $[Ar] 4s^2 3d^3 4p^1$

(Ar: Argon; $Z = 18$)

Give the **condensed ground state electronic configuration** of the atom, **A** and the cation, **A^{2+}** .
Show your work. (Assume all electronic transitions are possible)

Electronic Configuration of A:

Electronic Configuration of A^{2+} :

c. 2 points

In a single atom, determine how many electrons can have the following set of quantum numbers?

(i) $n = 5; l = 3$

In a single atom, determine how many electrons can have the following set of quantum numbers?

(i) $n = 3; l = 2; m_s = +1/2$

**Question 4** **8 points**

Rank the following from 1 to 4 in **increasing** order. Provide the ranking in the boxes provided.

Note: Atomic number (Z) are provided for relevant atoms

a) Increasing (lowest to highest) order of *ionization energy*: (1= lowest; 4 = highest)

Mg

Mg: Z = 12

Rb

Rb: Z = 37

Cs

Cs: Z = 55

K

K : Z = 19

b) Increasing (lowest to highest) order of *bond angle*: (1= lowest; 4 = highest)

H₂OCCl₄XeF₄

HCN

c) Increasing (smallest to largest) order of *ionic size* (1= smallest; 4 = largest)

S²⁻

S: Z = 16

O²⁻

O: Z = 8

Be²⁺

Be: Z = 4

Na⁺

Na: Z = 11

d) Increasing (lowest to highest) order of *bond energy* (1= lowest; 4 = highest)

P-F

P-Cl

P-I

P-Br

**Question 5** **12 points**

a. 9 points For the molecules HClO_4 and TeF_6 , draw the most stable *Lewis structure* showing appropriate shape and *dashed/wedged bonds*, based on VSEPR. Determine the **molecular geometry** around the central atom. For HClO_4 , label the *most polar bond*.

HClO_4 (there are no O-O bonds)

Label the most polar bond

Molecular Geometry around Cl

TeF_6

Molecular Geometry around Te

b. 3 points (Note: Lone pairs are not shown in structures below)

(i) For the three *resonance* structures (1, 2 and 3) shown below, answer the following question:



(I) Circle the *most contributing* resonance structure.

Structure 1

Structure 2

Structure 3

(ii) For the three structures given (4, 5, and 6) below, answer the following questions:



(I) Circle all the structures that represent *resonance structures* of each other:

Structure 4

Structure 5

Structure 6

(II) Among the resonance structures, circle the *most contributing* resonance structure.

Structure 4

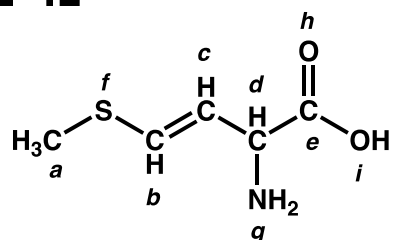
Structure 5

Structure 6

**Question 6** **6 points**

Using the data given below, calculate the **lattice energy** for SrO(s).

	ΔH° (kJ/mol)
$\text{Sr}(s) \longrightarrow \text{Sr}(g)$	+180
$\text{Sr}(g) \longrightarrow \text{Sr}^{2+}(g) + 2e^-$	+1468
$\text{O}_2(g) \longrightarrow 2\text{O}(g)$	+500
$\text{O}(g) + 2e^- \longrightarrow \text{O}^{2-}(g)$	+600
$\text{Sr}(s) + \frac{1}{2}\text{O}_2(g) \longrightarrow \text{SrO}(s)$	-550

**Question 7 10 points**

Consider the molecule given on the left, and answer questions (i) to (vi) using chemical bonding theories that we have learnt in class (valence bond theory, hybridization).

Note that the lone pairs are not shown in the structure:

i. Consider the carbon atoms marked in as ***a***, ***b***, ***c***, ***d***, and ***e*** in the molecule, and determine **hybridization** at each of the carbon centers. (2.5 points)

Atom	Hybridization
Carbon (<i>a</i>)	
Carbon (<i>b</i>)	
Carbon (<i>c</i>)	
Carbon (<i>d</i>)	
Carbon (<i>e</i>)	

ii. Now, consider the sulfur atom (S) marked as ***f***, nitrogen (N) atom (***g***), and oxygen(O) atoms (***h*** and ***i***), and determine their **hybridization**. (2 points)

Atom	Hybridization
Sulfur (<i>f</i>)	
Nitrogen (<i>g</i>)	
Oxygen (<i>h</i>)	
Oxygen (<i>i</i>)	

iii. **Describe** all the bonds formed by the carbon atoms ***a*** and ***b***, using sigma and pi designation as appropriate. (2 points)

Atom	Bonds formed (use sigma and pi as appropriate)
Carbon (<i>a</i>)	
Carbon (<i>b</i>)	

iv. How many total sp^3 -s sigma bonds are in the molecule? (1 point)

v. Determine **electronic** and **molecular geometry** at carbon (C) labeled ***e*** and sulfur (S) labeled ***f*** centres. (2 points)

Atom	Electronic Geometry	Molecular Geometry
Carbon (<i>e</i>)		
Sulfur (<i>f</i>)		

vi. Circle the value from the choices given below, that describes the **C-S-C bond angle at sulfur** (S) marked ***f***: (0.5 points)

>90° to <109.5° 109.5° >109.5 to <120° 120° >120° to <180°


Question 8 12 points

Using the **valence shell orbitals only** draw a **complete** molecular orbital (MO) diagram (with no s-p mixing) for the molecule **PS**, in the space provided below. **(6 points)**

Assume that 3s and 3p atomic orbitals will form molecular orbitals like those formed by the combination of 2s and 2p atomic orbitals that we learnt in class, and there is no s-p mixing.

Show **energy arrow**, **atomic orbitals** involved, **atom labels**, the **molecular orbitals** that are formed, **filled electrons** in the MO diagram etc. Make sure to label all the atomic and molecular orbitals.

Using the MO diagram above, answer the following questions:

- (i) Which one of the following will have the highest bond order? Circle the correct choice
 A) PS^{1-} B) PS C) PS^{1+}
- (ii) Which one of the following will have the longest bond length? Circle the correct choice
 A) PS^{1-} B) PS C) PS^{1+}
- (iii) Which of the following will be diamagnetic? Circle the correct choice.
 A) PS^{1-} B) PS C) PS^{1+}
- (iv) Draw Lewis dot structure of the molecule with the shortest bond length (based on the MO diagram) from among the three (PS^{1-} , PS, PS^{1+}), and indicate the bond order calculated from the Lewis dot structure. **(3 points)**

Lewis Dot Structure

Bond Order



9CF9D384-9D84-4F2C-8354-4D484483A66B

midterm-exam-93388

#600 Page 10 of 10

SCRAP PAGE

**CHEMISTRY 110 –Midterm Exam****Thursday November 7, 2024, 6:30 – 8:30 pm**

Examiners: Pallavi Sirjoosingh and Ashok Kakkar

First name (please write as legibly as possible within the boxes)

--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--

Last name

--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--

Student ID number

--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--

This booklet has 8 questions on 10 pages (The last page can be used as a scrap page).

The staple must NOT be removed from the question booklet. Your intact question booklet must be returned at the end of the exam.

A separate booklet contains a periodic table, and datasheet.

Write your answers directly under the questions in the space provided. Do not write on the QR codes or near the top edges (as it may be cut off).

Please write **legibly and dark enough** for the scan to be readable. For students who use English as a second language, dictionaries are permitted. Calculators are allowed.

DO NOT ASK QUESTIONS OF THE INVIGILATORS about interpretations of the exam questions. If you have a concern about a question, please summarize it at the top of this page. All concerns will be evaluated after the exam. GOOD LUCK!

**Question 1 8 points** (*Assume all transitions are possible*)

a. 6 points Calculate the **energy (in kJ/mol)** required to ionize (completely remove) the electron from the **$n=3$** level of **He^+ ion** (He: Helium). Show your work including equations. (3 sig. figs.)

b. 2 points

(I) Select the largest (longest) wavelength from the three choices below:

- i. Wavelength of photon to *completely remove* an electron from *1s orbital of Hydrogen atom*
- ii. Wavelength of photon to *completely remove* an electron from *1s orbital of Boron atom (B)*
- iii. Wavelength of photon to *completely remove* an electron from *1s orbital of Boron cation (B^+)*

i or ii or iii

(II) Select the smallest (shortest) wavelength from the three choices below (*for H atom*):

- i. Wavelength of photon *emitted* for an electron transitioning from *2p to 1s orbital*
- ii. Wavelength of photon *absorbed* for an electron transitioning from *2s to 4p orbital*
- iii. Wavelength of photon *absorbed* for an electron transitioning from *1s to 3p orbital*

i or ii or iii



Question 2 8 points



A Chem110 student plots the graph shown on the left, from data obtained for the kinetic energy (eV) of the ejected electron versus frequency (s^{-1}) of the incident light, for an unknown metal. ($1\text{eV} = 1.6022 \times 10^{-19} \text{ J}$)

Based on the plot, answer the questions **a to c** below:

a. 1 point On the plot above, label the **threshold frequency**.

b. 2 points What is the value of the work potential (in J) of the unknown metal?

c. 3 points What is the *longest* wavelength (in nm) required to eject an electron from this metal's surface? Show your work including equation(s) used. (3 sig figs)

d. 2 points In the space provided below, plot the following

(Assume frequency is *above* threshold frequency):

- i. *Kinetic energy* of an ejected electron versus the *intensity* of the incident light (left)
- ii. *Number of electrons* ejected versus the *frequency* of the incident light (right)



**Question 3** **6 points****a. 2 points**

In the ground state of Ge (Germanium; $Z = 32$) atom, **name** the following orbitals:

i. Orbital containing the electron that requires the maximum energy to be removed.

ii. Orbital containing the electron that experiences the minimum effective nuclear charge.

b. 2 points

An unknown anion (A^-) has the following *excited* electronic configuration: $[Ar] 4s^2 3d^3 4p^1$

(Ar: Argon; $Z = 18$)

Give the **condensed ground state electronic configuration** of the atom, **A** and the cation, **A^{2+}** .
Show your work. (Assume all electronic transitions are possible)

Electronic Configuration of A:

Electronic Configuration of A^{2+} :

c. 2 points

In a single atom, determine how many electrons can have the following set of quantum numbers?

(i) $n = 5; l = 3$

In a single atom, determine how many electrons can have the following set of quantum numbers?

(i) $n = 3; l = 2; m_s = +1/2$

**Question 4** **8 points**

Rank the following from 1 to 4 in **increasing** order. Provide the ranking in the boxes provided.

Note: Atomic number (Z) are provided for relevant atoms

a) Increasing (lowest to highest) order of *ionization energy*: (1= lowest; 4 = highest)

Mg

Mg: Z = 12

Rb

Rb: Z = 37

Cs

Cs: Z = 55

K

K : Z = 19

b) Increasing (lowest to highest) order of *bond angle*: (1= lowest; 4 = highest)

H₂OCCl₄XeF₄

HCN

c) Increasing (smallest to largest) order of *ionic size* (1= smallest; 4 = largest)

S²⁻

S: Z = 16

O²⁻

O: Z = 8

Be²⁺

Be: Z = 4

Na⁺

Na: Z = 11

d) Increasing (lowest to highest) order of *bond energy* (1= lowest; 4 = highest)

P-F

P-Cl

P-I

P-Br

**Question 5** **12 points**

a. 9 points For the molecules HClO_4 and TeF_6 , draw the most stable *Lewis structure* showing appropriate shape and *dashed/wedged bonds*, based on VSEPR. Determine the **molecular geometry** around the central atom. For HClO_4 , label the *most polar bond*.



b. 3 points (Note: Lone pairs are not shown in structures below)

(i) For the three *resonance* structures (1, 2 and 3) shown below, answer the following question:



(I) Circle the *most contributing* resonance structure.

Structure 1

Structure 2

Structure 3

(ii) For the three structures given (4, 5, and 6) below, answer the following questions:



(I) Circle all the structures that represent *resonance structures* of each other:

Structure 4

Structure 5

Structure 6

(II) Among the resonance structures, circle the *most contributing* resonance structure.

Structure 4

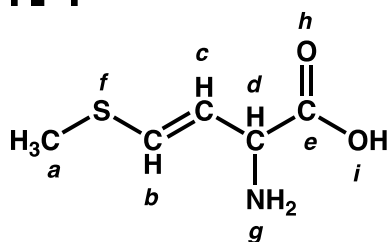
Structure 5

Structure 6

**Question 6** **6 points**

Using the data given below, calculate the **lattice energy** for SrO(s).

	ΔH° (kJ/mol)
$\text{Sr}(s) \longrightarrow \text{Sr}(g)$	+180
$\text{Sr}(g) \longrightarrow \text{Sr}^{2+}(g) + 2e^-$	+1468
$\text{O}_2(g) \longrightarrow 2\text{O}(g)$	+500
$\text{O}(g) + 2e^- \longrightarrow \text{O}^{2-}(g)$	+600
$\text{Sr}(s) + \frac{1}{2}\text{O}_2(g) \longrightarrow \text{SrO}(s)$	-550

**Question 7 10 points**

Consider the molecule given on the left, and answer questions (i) to (vi) using chemical bonding theories that we have learnt in class (valence bond theory, hybridization).

Note that the lone pairs are not shown in the structure:

i. Consider the carbon atoms marked in as ***a***, ***b***, ***c***, ***d***, and ***e*** in the molecule, and determine **hybridization** at each of the carbon centers. (2.5 points)

Atom	Hybridization
Carbon (<i>a</i>)	
Carbon (<i>b</i>)	
Carbon (<i>c</i>)	
Carbon (<i>d</i>)	
Carbon (<i>e</i>)	

ii. Now, consider the sulfur atom (S) marked as ***f***, nitrogen (N) atom (***g***), and oxygen(O) atoms (***h*** and ***i***), and determine their **hybridization**. (2 points)

Atom	Hybridization
Sulfur (<i>f</i>)	
Nitrogen (<i>g</i>)	
Oxygen (<i>h</i>)	
Oxygen (<i>i</i>)	

iii. **Describe** all the bonds formed by the carbon atoms ***a*** and ***b***, using sigma and pi designation as appropriate. (2 points)

Atom	Bonds formed (use sigma and pi as appropriate)
Carbon (<i>a</i>)	
Carbon (<i>b</i>)	

iv. How many total sp^3 -s sigma bonds are in the molecule? (1 point)

v. Determine **electronic** and **molecular geometry** at carbon (C) labeled ***e*** and sulfur (S) labeled ***f*** centres. (2 points)

Atom	Electronic Geometry	Molecular Geometry
Carbon (<i>e</i>)		
Sulfur (<i>f</i>)		

vi. Circle the value from the choices given below, that describes the **C-S-C bond angle at sulfur** (S) marked ***f***: (0.5 points)

>90° to <109.5° 109.5° >109.5 to <120° 120° >120° to <180°


Question 8 12 points

Using the **valence shell orbitals only** draw a **complete** molecular orbital (MO) diagram (with no s-p mixing) for the molecule **PS**, in the space provided below. **(6 points)**

Assume that 3s and 3p atomic orbitals will form molecular orbitals like those formed by the combination of 2s and 2p atomic orbitals that we learnt in class, and there is no s-p mixing.

Show **energy arrow**, **atomic orbitals** involved, **atom labels**, the **molecular orbitals** that are formed, **filled electrons** in the MO diagram etc. Make sure to label all the atomic and molecular orbitals.

Using the MO diagram above, answer the following questions:

- (i) Which one of the following will have the highest bond order? Circle the correct choice
 A) PS^{1-} B) PS C) PS^{1+}
- (ii) Which one of the following will have the longest bond length? Circle the correct choice
 A) PS^{1-} B) PS C) PS^{1+}
- (iii) Which of the following will be diamagnetic? Circle the correct choice.
 A) PS^{1-} B) PS C) PS^{1+}
- (iv) Draw Lewis dot structure of the molecule with the shortest bond length (based on the MO diagram) from among the three (PS^{1-} , PS, PS^{1+}), and indicate the bond order calculated from the Lewis dot structure. **(3 points)**

Lewis Dot Structure

Bond Order



932FFF01-3A1F-4F6D-AE10-B91013B83917

midterm-exam-93388

#601 Page 10 of 10

SCRAP PAGE

**CHEMISTRY 110 –Midterm Exam****Thursday November 7, 2024, 6:30 – 8:30 pm**

Examiners: Pallavi Sirjoosingh and Ashok Kakkar

First name (please write as legibly as possible within the boxes)

--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--

Last name

--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--

Student ID number

--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--

This booklet has 8 questions on 10 pages (The last page can be used as a scrap page).

The staple must NOT be removed from the question booklet. Your intact question booklet must be returned at the end of the exam.

A separate booklet contains a periodic table, and datasheet.

Write your answers directly under the questions in the space provided. Do not write on the QR codes or near the top edges (as it may be cut off).

Please write **legibly and dark enough** for the scan to be readable. For students who use English as a second language, dictionaries are permitted. Calculators are allowed.

DO NOT ASK QUESTIONS OF THE INVIGILATORS about interpretations of the exam questions. If you have a concern about a question, please summarize it at the top of this page. All concerns will be evaluated after the exam. GOOD LUCK!

**Question 1 8 points** (*Assume all transitions are possible*)

a. 6 points Calculate the **energy (in kJ/mol)** required to ionize (completely remove) the electron from the **$n=3$** level of **He^+ ion** (He: Helium). Show your work including equations. (3 sig. figs.)

b. 2 points

(I) Select the largest (longest) wavelength from the three choices below:

- i. Wavelength of photon to *completely remove* an electron from *1s orbital of Hydrogen atom*
- ii. Wavelength of photon to *completely remove* an electron from *1s orbital of Boron atom (B)*
- iii. Wavelength of photon to *completely remove* an electron from *1s orbital of Boron cation (B^+)*

i or ii or iii

(II) Select the smallest (shortest) wavelength from the three choices below (*for H atom*):

- i. Wavelength of photon *emitted* for an electron transitioning from *2p to 1s orbital*
- ii. Wavelength of photon *absorbed* for an electron transitioning from *2s to 4p orbital*
- iii. Wavelength of photon *absorbed* for an electron transitioning from *1s to 3p orbital*

i or ii or iii



Question 2 8 points



A Chem110 student plots the graph shown on the left, from data obtained for the kinetic energy (eV) of the ejected electron versus frequency (s^{-1}) of the incident light, for an unknown metal. ($1\text{eV} = 1.6022 \times 10^{-19} \text{ J}$)

Based on the plot, answer the questions **a to c** below:

a. 1 point On the plot above, label the **threshold frequency**.

b. 2 points What is the value of the work potential (in J) of the unknown metal?

c. 3 points What is the *longest* wavelength (in nm) required to eject an electron from this metal's surface? Show your work including equation(s) used. (3 sig figs)

d. 2 points In the space provided below, plot the following

(Assume frequency is *above* threshold frequency):

- Kinetic energy* of an ejected electron versus the *intensity* of the incident light (left)
- Number of electrons* ejected versus the *frequency* of the incident light (right)



**Question 3** **6 points****a. 2 points**

In the ground state of Ge (Germanium; $Z = 32$) atom, **name** the following orbitals:

i. Orbital containing the electron that requires the maximum energy to be removed.

ii. Orbital containing the electron that experiences the minimum effective nuclear charge.

b. 2 points

An unknown anion (A^-) has the following *excited* electronic configuration: $[Ar] 4s^2 3d^3 4p^1$

(Ar: Argon; $Z = 18$)

Give the **condensed ground state electronic configuration** of the atom, **A** and the cation, **A^{2+}** .
Show your work. (Assume all electronic transitions are possible)

Electronic Configuration of A:

Electronic Configuration of A^{2+} :

c. 2 points

In a single atom, determine how many electrons can have the following set of quantum numbers?

(i) $n = 5; l = 3$

In a single atom, determine how many electrons can have the following set of quantum numbers?

(i) $n = 3; l = 2; m_s = +1/2$

**Question 4** **8 points**

Rank the following from 1 to 4 in **increasing** order. Provide the ranking in the boxes provided.

Note: Atomic number (Z) are provided for relevant atoms

a) Increasing (lowest to highest) order of *ionization energy*: (1= lowest; 4 = highest)

Mg

Mg: Z = 12

Rb

Rb: Z = 37

Cs

Cs: Z = 55

K

K : Z = 19

b) Increasing (lowest to highest) order of *bond angle*: (1= lowest; 4 = highest)

H₂OCCl₄XeF₄

HCN

c) Increasing (smallest to largest) order of *ionic size* (1= smallest; 4 = largest)

S²⁻

S: Z = 16

O²⁻

O: Z = 8

Be²⁺

Be: Z = 4

Na⁺

Na: Z = 11

d) Increasing (lowest to highest) order of *bond energy* (1= lowest; 4 = highest)

P-F

P-Cl

P-I

P-Br

**Question 5** **12 points**

a. 9 points For the molecules HClO_4 and TeF_6 , draw the most stable *Lewis structure* showing appropriate shape and *dashed/wedged bonds*, based on VSEPR. Determine the **molecular geometry** around the central atom. For HClO_4 , label the *most polar bond*.

HClO_4 (there are no O-O bonds)

Label the most polar bond

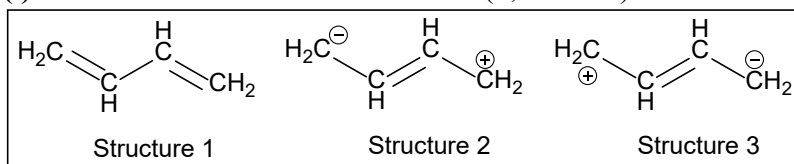
Molecular Geometry around Cl

TeF_6

Molecular Geometry around Te

b. 3 points (Note: Lone pairs are not shown in structures below)

(i) For the three *resonance* structures (1, 2 and 3) shown below, answer the following question:



(I) Circle the *most contributing* resonance structure.

Structure 1

Structure 2

Structure 3

(ii) For the three structures given (4, 5, and 6) below, answer the following questions:



(I) Circle all the structures that represent *resonance structures* of each other:

Structure 4

Structure 5

Structure 6

(II) Among the resonance structures, circle the *most contributing* resonance structure.

Structure 4

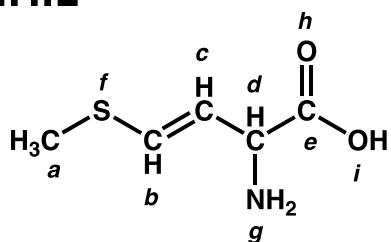
Structure 5

Structure 6

**Question 6** **6 points**

Using the data given below, calculate the **lattice energy** for SrO(s).

	ΔH° (kJ/mol)
$\text{Sr}(s) \longrightarrow \text{Sr}(g)$	+180
$\text{Sr}(g) \longrightarrow \text{Sr}^{2+}(g) + 2e^-$	+1468
$\text{O}_2(g) \longrightarrow 2\text{O}(g)$	+500
$\text{O}(g) + 2e^- \longrightarrow \text{O}^{2-}(g)$	+600
$\text{Sr}(s) + \frac{1}{2}\text{O}_2(g) \longrightarrow \text{SrO}(s)$	-550

**Question 7 10 points**

Consider the molecule given on the left, and answer questions (i) to (vi) using chemical bonding theories that we have learnt in class (valence bond theory, hybridization).

Note that the lone pairs are not shown in the structure:

i. Consider the carbon atoms marked in as ***a***, ***b***, ***c***, ***d***, and ***e*** in the molecule, and determine **hybridization** at each of the carbon centers. (2.5 points)

Atom	Hybridization
Carbon (<i>a</i>)	
Carbon (<i>b</i>)	
Carbon (<i>c</i>)	
Carbon (<i>d</i>)	
Carbon (<i>e</i>)	

ii. Now, consider the sulfur atom (S) marked as ***f***, nitrogen (N) atom (***g***), and oxygen(O) atoms (***h*** and ***i***), and determine their **hybridization**. (2 points)

Atom	Hybridization
Sulfur (<i>f</i>)	
Nitrogen (<i>g</i>)	
Oxygen (<i>h</i>)	
Oxygen (<i>i</i>)	

iii. **Describe** all the bonds formed by the carbon atoms ***a*** and ***b***, using sigma and pi designation as appropriate. (2 points)

Atom	Bonds formed (use sigma and pi as appropriate)
Carbon (<i>a</i>)	
Carbon (<i>b</i>)	

iv. How many total sp^3 -s sigma bonds are in the molecule? (1 point)

v. Determine **electronic** and **molecular geometry** at carbon (C) labeled ***e*** and sulfur (S) labeled ***f*** centres. (2 points)

Atom	Electronic Geometry	Molecular Geometry
Carbon (<i>e</i>)		
Sulfur (<i>f</i>)		

vi. Circle the value from the choices given below, that describes the **C-S-C bond angle at sulfur** (S) marked ***f***: (0.5 points)

>90° to <109.5° 109.5° >109.5 to <120° 120° >120° to <180°


Question 8 12 points

Using the **valence shell orbitals only** draw a **complete** molecular orbital (MO) diagram (with no s-p mixing) for the molecule **PS**, in the space provided below. **(6 points)**

Assume that 3s and 3p atomic orbitals will form molecular orbitals like those formed by the combination of 2s and 2p atomic orbitals that we learnt in class, and there is no s-p mixing.

Show **energy arrow**, **atomic orbitals** involved, **atom labels**, the **molecular orbitals** that are formed, **filled electrons** in the MO diagram etc. Make sure to label all the atomic and molecular orbitals.

Using the MO diagram above, answer the following questions:

- (i) Which one of the following will have the highest bond order? Circle the correct choice
 A) PS^{1-} B) PS C) PS^{1+}
- (ii) Which one of the following will have the longest bond length? Circle the correct choice
 A) PS^{1-} B) PS C) PS^{1+}
- (iii) Which of the following will be diamagnetic? Circle the correct choice.
 A) PS^{1-} B) PS C) PS^{1+}
- (iv) Draw Lewis dot structure of the molecule with the shortest bond length (based on the MO diagram) from among the three (PS^{1-} , PS, PS^{1+}), and indicate the bond order calculated from the Lewis dot structure. **(3 points)**

Lewis Dot Structure

Bond Order



9A39ECE9-46B1-47E3-B641-404A9C9A4CDF

midterm-exam-93388

#602 Page 10 of 10

SCRAP PAGE

**CHEMISTRY 110 –Midterm Exam****Thursday November 7, 2024, 6:30 – 8:30 pm****Examiners: Pallavi Sirjoosingh and Ashok Kakkar**

First name (please write as legibly as possible within the boxes)

--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--

Last name

--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--

Student ID number

--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--

This booklet has 8 questions on 10 pages (The last page can be used as a scrap page).

The staple must NOT be removed from the question booklet. Your intact question booklet must be returned at the end of the exam.

A separate booklet contains a periodic table, and datasheet.

Write your answers directly under the questions in the space provided. Do not write on the QR codes or near the top edges (as it may be cut off).

Please write **legibly and dark enough** for the scan to be readable. For students who use English as a second language, dictionaries are permitted. Calculators are allowed.

DO NOT ASK QUESTIONS OF THE INVIGILATORS about interpretations of the exam questions. If you have a concern about a question, please summarize it at the top of this page. All concerns will be evaluated after the exam. GOOD LUCK!

**Question 1 8 points** (*Assume all transitions are possible*)

a. 6 points Calculate the **energy (in kJ/mol)** required to ionize (completely remove) the electron from the **n=3** level of **He⁺ ion** (He: Helium). Show your work including equations. (3 sig. figs.)

b. 2 points

(I) Select the largest (longest) wavelength from the three choices below:

- i. Wavelength of photon to *completely remove* an electron from *1s orbital of Hydrogen atom*
- ii. Wavelength of photon to *completely remove* an electron from *1s orbital of Boron atom (B)*
- iii. Wavelength of photon to *completely remove* an electron from *1s orbital of Boron cation (B⁺)*

i or ii or iii

(II) Select the smallest (shortest) wavelength from the three choices below (*for H atom*):

- i. Wavelength of photon *emitted* for an electron transitioning from *2p to 1s orbital*
- ii. Wavelength of photon *absorbed* for an electron transitioning from *2s to 4p orbital*
- iii. Wavelength of photon *absorbed* for an electron transitioning from *1s to 3p orbital*

i or ii or iii



Question 2 8 points



A Chem110 student plots the graph shown on the left, from data obtained for the kinetic energy (eV) of the ejected electron versus frequency (s^{-1}) of the incident light, for an unknown metal. ($1\text{eV} = 1.6022 \times 10^{-19} \text{ J}$)

Based on the plot, answer the questions **a to c** below:

a. 1 point On the plot above, label the **threshold frequency**.

b. 2 points What is the value of the work potential (in J) of the unknown metal?

c. 3 points What is the *longest* wavelength (in nm) required to eject an electron from this metal's surface? Show your work including equation(s) used. (3 sig figs)

d. 2 points In the space provided below, plot the following

(Assume frequency is *above* threshold frequency):

- Kinetic energy* of an ejected electron versus the *intensity* of the incident light (left)
- Number of electrons* ejected versus the *frequency* of the incident light (right)



**Question 3** **6 points****a. 2 points**

In the ground state of Ge (Germanium; $Z = 32$) atom, **name** the following orbitals:

i. Orbital containing the electron that requires the maximum energy to be removed.

ii. Orbital containing the electron that experiences the minimum effective nuclear charge.

b. 2 points

An unknown anion (A^-) has the following *excited* electronic configuration: $[Ar] 4s^2 3d^3 4p^1$

(Ar: Argon; $Z = 18$)

Give the **condensed ground state electronic configuration** of the atom, **A** and the cation, **A^{2+}** .
Show your work. (Assume all electronic transitions are possible)

Electronic Configuration of A:

Electronic Configuration of A^{2+} :

c. 2 points

In a single atom, determine how many electrons can have the following set of quantum numbers?

(i) $n = 5; l = 3$

In a single atom, determine how many electrons can have the following set of quantum numbers?

(i) $n = 3; l = 2; m_s = +1/2$

**Question 4** **8 points**

Rank the following from 1 to 4 in **increasing** order. Provide the ranking in the boxes provided.

Note: Atomic number (Z) are provided for relevant atoms

a) Increasing (lowest to highest) order of *ionization energy*: (1= lowest; 4 = highest)

Mg

Mg: Z = 12

Rb

Rb: Z = 37

Cs

Cs: Z = 55

K

K : Z = 19

b) Increasing (lowest to highest) order of *bond angle*: (1= lowest; 4 = highest)

H₂OCCl₄XeF₄

HCN

c) Increasing (smallest to largest) order of *ionic size* (1= smallest; 4 = largest)

S²⁻

S: Z = 16

O²⁻

O: Z = 8

Be²⁺

Be: Z = 4

Na⁺

Na: Z = 11

d) Increasing (lowest to highest) order of *bond energy* (1= lowest; 4 = highest)

P-F

P-Cl

P-I

P-Br

**Question 5** **12 points**

a. 9 points For the molecules HClO_4 and TeF_6 , draw the most stable *Lewis structure* showing appropriate shape and *dashed/wedged bonds*, based on VSEPR. Determine the **molecular geometry** around the central atom. For HClO_4 , label the *most polar bond*.



b. 3 points (Note: Lone pairs are not shown in structures below)

(i) For the three *resonance* structures (1, 2 and 3) shown below, answer the following question:



(I) Circle the *most contributing* resonance structure.

Structure 1

Structure 2

Structure 3

(ii) For the three structures given (4, 5, and 6) below, answer the following questions:



(I) Circle all the structures that represent *resonance structures* of each other:

Structure 4

Structure 5

Structure 6

(II) Among the resonance structures, circle the *most contributing* resonance structure.

Structure 4

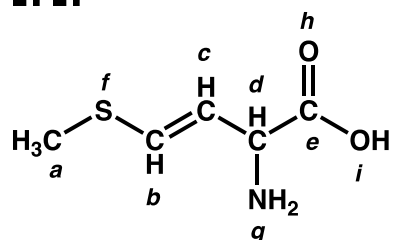
Structure 5

Structure 6

**Question 6** **6 points**

Using the data given below, calculate the **lattice energy** for SrO(s).

	ΔH° (kJ/mol)
$\text{Sr}(s) \longrightarrow \text{Sr}(g)$	+180
$\text{Sr}(g) \longrightarrow \text{Sr}^{2+}(g) + 2e^-$	+1468
$\text{O}_2(g) \longrightarrow 2\text{O}(g)$	+500
$\text{O}(g) + 2e^- \longrightarrow \text{O}^{2-}(g)$	+600
$\text{Sr}(s) + \frac{1}{2}\text{O}_2(g) \longrightarrow \text{SrO}(s)$	-550

**Question 7 10 points**

Consider the molecule given on the left, and answer questions (i) to (vi) using chemical bonding theories that we have learnt in class (valence bond theory, hybridization).

Note that the lone pairs are not shown in the structure:

i. Consider the carbon atoms marked in as ***a***, ***b***, ***c***, ***d***, and ***e*** in the molecule, and determine **hybridization** at each of the carbon centers. (2.5 points)

Atom	Hybridization
Carbon (<i>a</i>)	
Carbon (<i>b</i>)	
Carbon (<i>c</i>)	
Carbon (<i>d</i>)	
Carbon (<i>e</i>)	

ii. Now, consider the sulfur atom (S) marked as ***f***, nitrogen (N) atom (***g***), and oxygen(O) atoms (***h*** and ***i***), and determine their **hybridization**. (2 points)

Atom	Hybridization
Sulfur (<i>f</i>)	
Nitrogen (<i>g</i>)	
Oxygen (<i>h</i>)	
Oxygen (<i>i</i>)	

iii. **Describe** all the bonds formed by the carbon atoms ***a*** and ***b***, using sigma and pi designation as appropriate. (2 points)

Atom	Bonds formed (use sigma and pi as appropriate)
Carbon (<i>a</i>)	
Carbon (<i>b</i>)	

iv. How many total sp^3 -s sigma bonds are in the molecule? (1 point)

v. Determine **electronic** and **molecular geometry** at carbon (C) labeled ***e*** and sulfur (S) labeled ***f*** centres. (2 points)

Atom	Electronic Geometry	Molecular Geometry
Carbon (<i>e</i>)		
Sulfur (<i>f</i>)		

vi. Circle the value from the choices given below, that describes the **C-S-C bond angle at sulfur** (S) marked ***f***: (0.5 points)

$>90^\circ$ to $<109.5^\circ$ 109.5° >109.5 to $<120^\circ$ 120° $>120^\circ$ to $<180^\circ$

**Question 8 12 points**

Using the **valence shell orbitals only** draw a **complete** molecular orbital (MO) diagram (with no s-p mixing) for the molecule **PS**, in the space provided below. **(6 points)**

Assume that 3s and 3p atomic orbitals will form molecular orbitals like those formed by the combination of 2s and 2p atomic orbitals that we learnt in class, and there is no s-p mixing.

Show **energy arrow**, **atomic orbitals** involved, **atom labels**, the **molecular orbitals** that are formed, **filled electrons** in the MO diagram etc. Make sure to label all the atomic and molecular orbitals.

Using the MO diagram above, answer the following questions:

- (i) Which one of the following will have the highest bond order? Circle the correct choice
A) PS^{1-} B) PS C) PS^{1+}
- (ii) Which one of the following will have the longest bond length? Circle the correct choice
A) PS^{1-} B) PS C) PS^{1+}
- (iii) Which of the following will be diamagnetic? Circle the correct choice.
A) PS^{1-} B) PS C) PS^{1+}
- (iv) Draw Lewis dot structure of the molecule with the shortest bond length (based on the MO diagram) from among the three (PS^{1-} , PS, PS^{1+}), and indicate the bond order calculated from the Lewis dot structure. **(3 points)**

Lewis Dot Structure

Bond Order



C9C7D770-B6EB-4A91-A6C7-5453176A4E5A

midterm-exam-93388

#603 Page 10 of 10

SCRAP PAGE

**CHEMISTRY 110 –Midterm Exam****Thursday November 7, 2024, 6:30 – 8:30 pm**

Examiners: Pallavi Sirjoosingh and Ashok Kakkar

First name (please write as legibly as possible within the boxes)

--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--

Last name

--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--

Student ID number

--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--

This booklet has 8 questions on 10 pages (The last page can be used as a scrap page).

The staple must NOT be removed from the question booklet. Your intact question booklet must be returned at the end of the exam.

A separate booklet contains a periodic table, and datasheet.

Write your answers directly under the questions in the space provided. Do not write on the QR codes or near the top edges (as it may be cut off).

Please write **legibly and dark enough** for the scan to be readable. For students who use English as a second language, dictionaries are permitted. Calculators are allowed.

DO NOT ASK QUESTIONS OF THE INVIGILATORS about interpretations of the exam questions. If you have a concern about a question, please summarize it at the top of this page. All concerns will be evaluated after the exam. GOOD LUCK!

**Question 1 8 points** (*Assume all transitions are possible*)

a. 6 points Calculate the **energy (in kJ/mol)** required to ionize (completely remove) the electron from the **$n=3$** level of **He^+ ion** (He: Helium). Show your work including equations. (3 sig. figs.)

b. 2 points

(I) Select the largest (longest) wavelength from the three choices below:

- i. Wavelength of photon to *completely remove* an electron from *1s orbital of Hydrogen atom*
- ii. Wavelength of photon to *completely remove* an electron from *1s orbital of Boron atom (B)*
- iii. Wavelength of photon to *completely remove* an electron from *1s orbital of Boron cation (B^+)*

i or ii or iii

(II) Select the smallest (shortest) wavelength from the three choices below (*for H atom*):

- i. Wavelength of photon *emitted* for an electron transitioning from *2p to 1s orbital*
- ii. Wavelength of photon *absorbed* for an electron transitioning from *2s to 4p orbital*
- iii. Wavelength of photon *absorbed* for an electron transitioning from *1s to 3p orbital*

i or ii or iii



Question 2 8 points



A Chem110 student plots the graph shown on the left, from data obtained for the kinetic energy (eV) of the ejected electron versus frequency (s^{-1}) of the incident light, for an unknown metal. ($1\text{eV} = 1.6022 \times 10^{-19} \text{ J}$)

Based on the plot, answer the questions **a to c** below:

a. 1 point On the plot above, label the **threshold frequency**.

b. 2 points What is the value of the work potential (in J) of the unknown metal?

c. 3 points What is the *longest* wavelength (in nm) required to eject an electron from this metal's surface? Show your work including equation(s) used. (3 sig figs)

d. 2 points In the space provided below, plot the following

(Assume frequency is *above* threshold frequency):

- Kinetic energy* of an ejected electron versus the *intensity* of the incident light (left)
- Number of electrons* ejected versus the *frequency* of the incident light (right)



**Question 3** **6 points****a. 2 points**

In the ground state of Ge (Germanium; $Z = 32$) atom, **name** the following orbitals:

i. Orbital containing the electron that requires the maximum energy to be removed.

ii. Orbital containing the electron that experiences the minimum effective nuclear charge.

b. 2 points

An unknown anion (A^-) has the following *excited* electronic configuration: $[Ar] 4s^2 3d^3 4p^1$

(Ar: Argon; $Z = 18$)

Give the **condensed ground state electronic configuration** of the atom, **A** and the cation, **A^{2+}** .
Show your work. (Assume all electronic transitions are possible)

Electronic Configuration of A:

Electronic Configuration of A^{2+} :

c. 2 points

In a single atom, determine how many electrons can have the following set of quantum numbers?

(i) $n = 5; l = 3$

In a single atom, determine how many electrons can have the following set of quantum numbers?

(i) $n = 3; l = 2; m_s = +1/2$

**Question 4** **8 points**

Rank the following from 1 to 4 in **increasing** order. Provide the ranking in the boxes provided.

Note: Atomic number (Z) are provided for relevant atoms

a) Increasing (lowest to highest) order of *ionization energy*: (1= lowest; 4 = highest)

Mg

Mg: Z = 12

Rb

Rb: Z = 37

Cs

Cs: Z = 55

K

K : Z = 19

b) Increasing (lowest to highest) order of *bond angle*: (1= lowest; 4 = highest)

H₂OCCl₄XeF₄

HCN

c) Increasing (smallest to largest) order of *ionic size* (1= smallest; 4 = largest)

S²⁻

S: Z = 16

O²⁻

O: Z = 8

Be²⁺

Be: Z = 4

Na⁺

Na: Z = 11

d) Increasing (lowest to highest) order of *bond energy* (1= lowest; 4 = highest)

P-F

P-Cl

P-I

P-Br

**Question 5** **12 points**

a. 9 points For the molecules HClO_4 and TeF_6 , draw the most stable *Lewis structure* showing appropriate shape and *dashed/wedged bonds*, based on VSEPR. Determine the **molecular geometry** around the central atom. For HClO_4 , label the *most polar bond*.



b. 3 points (Note: Lone pairs are not shown in structures below)

(i) For the three *resonance* structures (1, 2 and 3) shown below, answer the following question:



(I) Circle the *most contributing* resonance structure.

Structure 1

Structure 2

Structure 3

(ii) For the three structures given (4, 5, and 6) below, answer the following questions:



(I) Circle all the structures that represent *resonance structures* of each other:

Structure 4

Structure 5

Structure 6

(II) Among the resonance structures, circle the *most contributing* resonance structure.

Structure 4

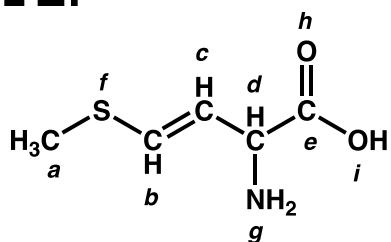
Structure 5

Structure 6

**Question 6** **6 points**

Using the data given below, calculate the **lattice energy** for SrO(s).

	ΔH° (kJ/mol)
$\text{Sr}(s) \longrightarrow \text{Sr}(g)$	+180
$\text{Sr}(g) \longrightarrow \text{Sr}^{2+}(g) + 2e^-$	+1468
$\text{O}_2(g) \longrightarrow 2\text{O}(g)$	+500
$\text{O}(g) + 2e^- \longrightarrow \text{O}^{2-}(g)$	+600
$\text{Sr}(s) + \frac{1}{2}\text{O}_2(g) \longrightarrow \text{SrO}(s)$	-550

**Question 7 10 points**

Consider the molecule given on the left, and answer questions (i) to (vi) using chemical bonding theories that we have learnt in class (valence bond theory, hybridization).

Note that the lone pairs are not shown in the structure:

i. Consider the carbon atoms marked in as ***a***, ***b***, ***c***, ***d***, and ***e*** in the molecule, and determine **hybridization** at each of the carbon centers. (2.5 points)

Atom	Hybridization
Carbon (<i>a</i>)	
Carbon (<i>b</i>)	
Carbon (<i>c</i>)	
Carbon (<i>d</i>)	
Carbon (<i>e</i>)	

ii. Now, consider the sulfur atom (S) marked as ***f***, nitrogen (N) atom (***g***), and oxygen(O) atoms (***h*** and ***i***), and determine their **hybridization**. (2 points)

Atom	Hybridization
Sulfur (<i>f</i>)	
Nitrogen (<i>g</i>)	
Oxygen (<i>h</i>)	
Oxygen (<i>i</i>)	

iii. **Describe** all the bonds formed by the carbon atoms ***a*** and ***b***, using sigma and pi designation as appropriate. (2 points)

Atom	Bonds formed (use sigma and pi as appropriate)
Carbon (<i>a</i>)	
Carbon (<i>b</i>)	

iv. How many total sp^3 -s sigma bonds are in the molecule? (1 point)

v. Determine **electronic** and **molecular geometry** at carbon (C) labeled ***e*** and sulfur (S) labeled ***f*** centres. (2 points)

Atom	Electronic Geometry	Molecular Geometry
Carbon (<i>e</i>)		
Sulfur (<i>f</i>)		

vi. Circle the value from the choices given below, that describes the **C-S-C bond angle at sulfur** (S) marked ***f***: (0.5 points)

>90° to <109.5° 109.5° >109.5 to <120° 120° >120° to <180°


Question 8 12 points

Using the **valence shell orbitals only** draw a **complete** molecular orbital (MO) diagram (with no s-p mixing) for the molecule **PS**, in the space provided below. **(6 points)**

Assume that 3s and 3p atomic orbitals will form molecular orbitals like those formed by the combination of 2s and 2p atomic orbitals that we learnt in class, and there is no s-p mixing.

Show **energy arrow**, **atomic orbitals** involved, **atom labels**, the **molecular orbitals** that are formed, **filled electrons** in the MO diagram etc. Make sure to label all the atomic and molecular orbitals.

Using the MO diagram above, answer the following questions:

- (i) Which one of the following will have the highest bond order? Circle the correct choice
 A) PS^{1-} B) PS C) PS^{1+}
- (ii) Which one of the following will have the longest bond length? Circle the correct choice
 A) PS^{1-} B) PS C) PS^{1+}
- (iii) Which of the following will be diamagnetic? Circle the correct choice.
 A) PS^{1-} B) PS C) PS^{1+}
- (iv) Draw Lewis dot structure of the molecule with the shortest bond length (based on the MO diagram) from among the three (PS^{1-} , PS, PS^{1+}), and indicate the bond order calculated from the Lewis dot structure. **(3 points)**

Lewis Dot Structure

Bond Order



B586C6D2-3425-4A8D-BF29-54DC22A37A57

midterm-exam-93388

#604 Page 10 of 10

SCRAP PAGE

**CHEMISTRY 110 –Midterm Exam****Thursday November 7, 2024, 6:30 – 8:30 pm**

Examiners: Pallavi Sirjoosingh and Ashok Kakkar

First name (please write as legibly as possible within the boxes)

--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--

Last name

--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--

Student ID number

--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--

This booklet has 8 questions on 10 pages (The last page can be used as a scrap page).

The staple must NOT be removed from the question booklet. Your intact question booklet must be returned at the end of the exam.

A separate booklet contains a periodic table, and datasheet.

Write your answers directly under the questions in the space provided. Do not write on the QR codes or near the top edges (as it may be cut off).

Please write **legibly and dark enough** for the scan to be readable. For students who use English as a second language, dictionaries are permitted. Calculators are allowed.

DO NOT ASK QUESTIONS OF THE INVIGILATORS about interpretations of the exam questions. If you have a concern about a question, please summarize it at the top of this page. All concerns will be evaluated after the exam. GOOD LUCK!

**Question 1 8 points** (*Assume all transitions are possible*)

a. 6 points Calculate the **energy (in kJ/mol)** required to ionize (completely remove) the electron from the **n=3** level of **He⁺ ion** (He: Helium). Show your work including equations. (3 sig. figs.)

b. 2 points

(I) Select the largest (longest) wavelength from the three choices below:

- i. Wavelength of photon to *completely remove* an electron from *1s orbital of Hydrogen atom*
- ii. Wavelength of photon to *completely remove* an electron from *1s orbital of Boron atom (B)*
- iii. Wavelength of photon to *completely remove* an electron from *1s orbital of Boron cation (B⁺)*

i or ii or iii

(II) Select the smallest (shortest) wavelength from the three choices below (*for H atom*):

- i. Wavelength of photon *emitted* for an electron transitioning from *2p to 1s orbital*
- ii. Wavelength of photon *absorbed* for an electron transitioning from *2s to 4p orbital*
- iii. Wavelength of photon *absorbed* for an electron transitioning from *1s to 3p orbital*

i or ii or iii



Question 2 8 points



A Chem110 student plots the graph shown on the left, from data obtained for the kinetic energy (eV) of the ejected electron versus frequency (s^{-1}) of the incident light, for an unknown metal. ($1\text{eV} = 1.6022 \times 10^{-19} \text{ J}$)

Based on the plot, answer the questions **a to c** below:

a. 1 point On the plot above, label the **threshold frequency**.

b. 2 points What is the value of the work potential (in J) of the unknown metal?

c. 3 points What is the *longest* wavelength (in nm) required to eject an electron from this metal's surface? Show your work including equation(s) used. (3 sig figs)

d. 2 points In the space provided below, plot the following

(Assume frequency is *above* threshold frequency):

- i. *Kinetic energy* of an ejected electron versus the *intensity* of the incident light (left)
- ii. *Number of electrons* ejected versus the *frequency* of the incident light (right)



**Question 3** **6 points****a. 2 points**

In the ground state of Ge (Germanium; $Z = 32$) atom, **name** the following orbitals:

i. Orbital containing the electron that requires the maximum energy to be removed.

ii. Orbital containing the electron that experiences the minimum effective nuclear charge.

b. 2 points

An unknown anion (A^-) has the following *excited* electronic configuration: $[Ar] 4s^2 3d^3 4p^1$

(Ar: Argon; $Z = 18$)

Give the **condensed ground state electronic configuration** of the atom, **A** and the cation, **A^{2+}** .
Show your work. (Assume all electronic transitions are possible)

Electronic Configuration of A:

Electronic Configuration of A^{2+} :

c. 2 points

In a single atom, determine how many electrons can have the following set of quantum numbers?

(i) $n = 5; l = 3$

In a single atom, determine how many electrons can have the following set of quantum numbers?

(i) $n = 3; l = 2; m_s = +1/2$

**Question 4** **8 points**

Rank the following from 1 to 4 in **increasing** order. Provide the ranking in the boxes provided.

Note: Atomic number (Z) are provided for relevant atoms

a) Increasing (lowest to highest) order of *ionization energy*: (1= lowest; 4 = highest)

Mg

Mg: Z = 12

Rb

Rb: Z = 37

Cs

Cs: Z = 55

K

K : Z = 19

b) Increasing (lowest to highest) order of *bond angle*: (1= lowest; 4 = highest)

H₂OCCl₄XeF₄

HCN

c) Increasing (smallest to largest) order of *ionic size* (1= smallest; 4 = largest)

S²⁻

S: Z = 16

O²⁻

O: Z = 8

Be²⁺

Be: Z = 4

Na⁺

Na: Z = 11

d) Increasing (lowest to highest) order of *bond energy* (1= lowest; 4 = highest)

P-F

P-Cl

P-I

P-Br

**Question 5** **12 points**

a. 9 points For the molecules HClO_4 and TeF_6 , draw the most stable *Lewis structure* showing appropriate shape and *dashed/wedged bonds*, based on VSEPR. Determine the **molecular geometry** around the central atom. For HClO_4 , label the *most polar bond*.



b. 3 points (Note: Lone pairs are not shown in structures below)

(i) For the three *resonance* structures (1, 2 and 3) shown below, answer the following question:



(I) Circle the *most contributing* resonance structure.

Structure 1

Structure 2

Structure 3

(ii) For the three structures given (4, 5, and 6) below, answer the following questions:



(I) Circle all the structures that represent *resonance structures* of each other:

Structure 4

Structure 5

Structure 6

(II) Among the resonance structures, circle the *most contributing* resonance structure.

Structure 4

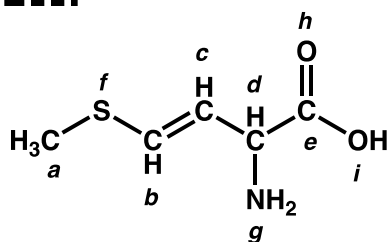
Structure 5

Structure 6

**Question 6** **6 points**

Using the data given below, calculate the **lattice energy** for SrO(s).

	ΔH° (kJ/mol)
$\text{Sr}(s) \longrightarrow \text{Sr}(g)$	+180
$\text{Sr}(g) \longrightarrow \text{Sr}^{2+}(g) + 2e^-$	+1468
$\text{O}_2(g) \longrightarrow 2\text{O}(g)$	+500
$\text{O}(g) + 2e^- \longrightarrow \text{O}^{2-}(g)$	+600
$\text{Sr}(s) + \frac{1}{2}\text{O}_2(g) \longrightarrow \text{SrO}(s)$	-550

**Question 7 10 points**

Consider the molecule given on the left, and answer questions (i) to (vi) using chemical bonding theories that we have learnt in class (valence bond theory, hybridization).

Note that the lone pairs are not shown in the structure:

i. Consider the carbon atoms marked in as ***a***, ***b***, ***c***, ***d***, and ***e*** in the molecule, and determine **hybridization** at each of the carbon centers. (2.5 points)

Atom	Hybridization
Carbon (<i>a</i>)	
Carbon (<i>b</i>)	
Carbon (<i>c</i>)	
Carbon (<i>d</i>)	
Carbon (<i>e</i>)	

ii. Now, consider the sulfur atom (S) marked as ***f***, nitrogen (N) atom (***g***), and oxygen(O) atoms (***h*** and ***i***), and determine their **hybridization**. (2 points)

Atom	Hybridization
Sulfur (<i>f</i>)	
Nitrogen (<i>g</i>)	
Oxygen (<i>h</i>)	
Oxygen (<i>i</i>)	

iii. **Describe** all the bonds formed by the carbon atoms ***a*** and ***b***, using sigma and pi designation as appropriate. (2 points)

Atom	Bonds formed (use sigma and pi as appropriate)
Carbon (<i>a</i>)	
Carbon (<i>b</i>)	

iv. How many total sp^3 -s sigma bonds are in the molecule? (1 point)

v. Determine **electronic** and **molecular geometry** at carbon (C) labeled ***e*** and sulfur (S) labeled ***f*** centres. (2 points)

Atom	Electronic Geometry	Molecular Geometry
Carbon (<i>e</i>)		
Sulfur (<i>f</i>)		

vi. Circle the value from the choices given below, that describes the **C-S-C bond angle at sulfur** (S) marked ***f***: (0.5 points)

$>90^\circ$ to $<109.5^\circ$ 109.5° >109.5 to $<120^\circ$ 120° $>120^\circ$ to $<180^\circ$


Question 8 12 points

Using the **valence shell orbitals only** draw a **complete** molecular orbital (MO) diagram (with no s-p mixing) for the molecule **PS**, in the space provided below. **(6 points)**

Assume that 3s and 3p atomic orbitals will form molecular orbitals like those formed by the combination of 2s and 2p atomic orbitals that we learnt in class, and there is no s-p mixing.

Show **energy arrow**, **atomic orbitals** involved, **atom labels**, the **molecular orbitals** that are formed, **filled electrons** in the MO diagram etc. Make sure to label all the atomic and molecular orbitals.

Using the MO diagram above, answer the following questions:

- (i) Which one of the following will have the highest bond order? Circle the correct choice
 A) PS^{1-} B) PS C) PS^{1+}
- (ii) Which one of the following will have the longest bond length? Circle the correct choice
 A) PS^{1-} B) PS C) PS^{1+}
- (iii) Which of the following will be diamagnetic? Circle the correct choice.
 A) PS^{1-} B) PS C) PS^{1+}
- (iv) Draw Lewis dot structure of the molecule with the shortest bond length (based on the MO diagram) from among the three (PS^{1-} , PS, PS^{1+}), and indicate the bond order calculated from the Lewis dot structure. **(3 points)**

Lewis Dot Structure

Bond Order



819C41DE-9A1A-4D72-B10B-C1701BCEECFF

midterm-exam-93388

#605 Page 10 of 10

SCRAP PAGE

**CHEMISTRY 110 –Midterm Exam****Thursday November 7, 2024, 6:30 – 8:30 pm**

Examiners: Pallavi Sirjoosingh and Ashok Kakkar

First name (please write as legibly as possible within the boxes)

--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--

Last name

--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--

Student ID number

--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--

This booklet has 8 questions on 10 pages (The last page can be used as a scrap page).

The staple must NOT be removed from the question booklet. Your intact question booklet must be returned at the end of the exam.

A separate booklet contains a periodic table, and datasheet.

Write your answers directly under the questions in the space provided. Do not write on the QR codes or near the top edges (as it may be cut off).

Please write **legibly and dark enough** for the scan to be readable. For students who use English as a second language, dictionaries are permitted. Calculators are allowed.

DO NOT ASK QUESTIONS OF THE INVIGILATORS about interpretations of the exam questions. If you have a concern about a question, please summarize it at the top of this page. All concerns will be evaluated after the exam. GOOD LUCK!

**Question 1 8 points** (*Assume all transitions are possible*)

a. 6 points Calculate the **energy (in kJ/mol)** required to ionize (completely remove) the electron from the **$n=3$** level of **He^+ ion** (He: Helium). Show your work including equations. (3 sig. figs.)

b. 2 points

(I) **Select** the **largest (longest)** wavelength from the three choices below:

- i. Wavelength of photon to *completely remove* an electron from *1s orbital of Hydrogen atom*
- ii. Wavelength of photon to *completely remove* an electron from *1s orbital of Boron atom (B)*
- iii. Wavelength of photon to *completely remove* an electron from *1s orbital of Boron cation (B^+)*

i **or** **ii** **or** **iii**

(II) **Select** the **smallest (shortest)** wavelength from the three choices below (*for H atom*):

- i. Wavelength of photon *emitted* for an electron transitioning from *2p to 1s orbital*
- ii. Wavelength of photon *absorbed* for an electron transitioning from *2s to 4p orbital*
- iii. Wavelength of photon *absorbed* for an electron transitioning from *1s to 3p orbital*

i **or** **ii** **or** **iii**



Question 2 8 points



A Chem110 student plots the graph shown on the left, from data obtained for the kinetic energy (eV) of the ejected electron versus frequency (s^{-1}) of the incident light, for an unknown metal. ($1\text{eV} = 1.6022 \times 10^{-19} \text{ J}$)

Based on the plot, answer the questions **a to c** below:

a. 1 point On the plot above, label the **threshold frequency**.

b. 2 points What is the value of the work potential (in J) of the unknown metal?

c. 3 points What is the *longest* wavelength (in nm) required to eject an electron from this metal's surface? Show your work including equation(s) used. (3 sig figs)

d. 2 points In the space provided below, plot the following

(Assume frequency is *above* threshold frequency):

- i. *Kinetic energy* of an ejected electron versus the *intensity* of the incident light (left)
- ii. *Number of electrons* ejected versus the *frequency* of the incident light (right)



**Question 3** **6 points****a. 2 points**

In the ground state of Ge (Germanium; $Z = 32$) atom, **name** the following orbitals:

i. Orbital containing the electron that requires the maximum energy to be removed.

ii. Orbital containing the electron that experiences the minimum effective nuclear charge.

b. 2 points

An unknown anion (A^-) has the following *excited* electronic configuration: $[Ar] 4s^2 3d^3 4p^1$

(Ar: Argon; $Z = 18$)

Give the **condensed ground state electronic configuration** of the atom, **A** and the cation, A^{2+} .
Show your work. (Assume all electronic transitions are possible)

Electronic Configuration of A:

Electronic Configuration of A^{2+} :

c. 2 points

In a single atom, determine how many electrons can have the following set of quantum numbers?

(i) $n = 5; l = 3$

In a single atom, determine how many electrons can have the following set of quantum numbers?

(i) $n = 3; l = 2; m_s = +1/2$

**Question 4** **8 points**

Rank the following from 1 to 4 in **increasing** order. Provide the ranking in the boxes provided.

Note: Atomic number (Z) are provided for relevant atoms

a) Increasing (lowest to highest) order of *ionization energy*: (1= lowest; 4 = highest)

Mg

Mg: Z = 12

Rb

Rb: Z = 37

Cs

Cs: Z = 55

K

K : Z = 19

b) Increasing (lowest to highest) order of *bond angle*: (1= lowest; 4 = highest)

H₂OCCl₄XeF₄

HCN

c) Increasing (smallest to largest) order of *ionic size* (1= smallest; 4 = largest)

S²⁻

S: Z = 16

O²⁻

O: Z = 8

Be²⁺

Be: Z = 4

Na⁺

Na: Z = 11

d) Increasing (lowest to highest) order of *bond energy* (1= lowest; 4 = highest)

P-F

P-Cl

P-I

P-Br

**Question 5** **12 points**

a. 9 points For the molecules HClO_4 and TeF_6 , draw the most stable *Lewis structure* showing appropriate shape and *dashed/wedged bonds*, based on VSEPR. Determine the **molecular geometry** around the central atom. For HClO_4 , label the *most polar bond*.



b. 3 points (Note: Lone pairs are not shown in structures below)

(i) For the three *resonance* structures (1, 2 and 3) shown below, answer the following question:



(I) Circle the *most contributing* resonance structure.

Structure 1

Structure 2

Structure 3

(ii) For the three structures given (4, 5, and 6) below, answer the following questions:



(I) Circle all the structures that represent *resonance structures* of each other:

Structure 4

Structure 5

Structure 6

(II) Among the resonance structures, circle the *most contributing* resonance structure.

Structure 4

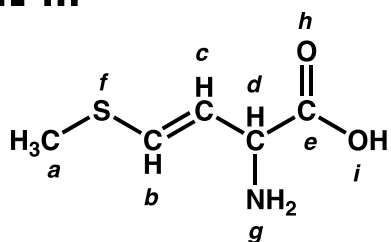
Structure 5

Structure 6

**Question 6** **6 points**

Using the data given below, calculate the **lattice energy** for SrO(s).

	ΔH° (kJ/mol)
$\text{Sr}(s) \longrightarrow \text{Sr}(g)$	+180
$\text{Sr}(g) \longrightarrow \text{Sr}^{2+}(g) + 2e^-$	+1468
$\text{O}_2(g) \longrightarrow 2\text{O}(g)$	+500
$\text{O}(g) + 2e^- \longrightarrow \text{O}^{2-}(g)$	+600
$\text{Sr}(s) + \frac{1}{2}\text{O}_2(g) \longrightarrow \text{SrO}(s)$	-550

**Question 7 10 points**

Consider the molecule given on the left, and answer questions (i) to (vi) using chemical bonding theories that we have learnt in class (valence bond theory, hybridization).

Note that the lone pairs are not shown in the structure:

i. Consider the carbon atoms marked in as ***a***, ***b***, ***c***, ***d***, and ***e*** in the molecule, and determine **hybridization** at each of the carbon centers. (2.5 points)

Atom	Hybridization
Carbon (<i>a</i>)	
Carbon (<i>b</i>)	
Carbon (<i>c</i>)	
Carbon (<i>d</i>)	
Carbon (<i>e</i>)	

ii. Now, consider the sulfur atom (S) marked as ***f***, nitrogen (N) atom (***g***), and oxygen(O) atoms (***h*** and ***i***), and determine their **hybridization**. (2 points)

Atom	Hybridization
Sulfur (<i>f</i>)	
Nitrogen (<i>g</i>)	
Oxygen (<i>h</i>)	
Oxygen (<i>i</i>)	

iii. **Describe** all the bonds formed by the carbon atoms ***a*** and ***b***, using sigma and pi designation as appropriate. (2 points)

Atom	Bonds formed (use sigma and pi as appropriate)
Carbon (<i>a</i>)	
Carbon (<i>b</i>)	

iv. How many total sp^3 -s sigma bonds are in the molecule? (1 point)

v. Determine **electronic** and **molecular geometry** at carbon (C) labeled ***e*** and sulfur (S) labeled ***f*** centres. (2 points)

Atom	Electronic Geometry	Molecular Geometry
Carbon (<i>e</i>)		
Sulfur (<i>f</i>)		

vi. Circle the value from the choices given below, that describes the **C-S-C bond angle at sulfur** (S) marked ***f***: (0.5 points)

>90° to <109.5°

109.5°

>109.5 to <120°

120°

>120° to <180°


Question 8 12 points

Using the **valence shell orbitals only** draw a **complete** molecular orbital (MO) diagram (with no s-p mixing) for the molecule **PS**, in the space provided below. **(6 points)**

Assume that 3s and 3p atomic orbitals will form molecular orbitals like those formed by the combination of 2s and 2p atomic orbitals that we learnt in class, and there is no s-p mixing.

Show **energy arrow**, **atomic orbitals** involved, **atom labels**, the **molecular orbitals** that are formed, **filled electrons** in the MO diagram etc. Make sure to label all the atomic and molecular orbitals.

Using the MO diagram above, answer the following questions:

- (i) Which one of the following will have the highest bond order? Circle the correct choice
 A) PS^{1-} B) PS C) PS^{1+}
- (ii) Which one of the following will have the longest bond length? Circle the correct choice
 A) PS^{1-} B) PS C) PS^{1+}
- (iii) Which of the following will be diamagnetic? Circle the correct choice.
 A) PS^{1-} B) PS C) PS^{1+}
- (iv) Draw Lewis dot structure of the molecule with the shortest bond length (based on the MO diagram) from among the three (PS^{1-} , PS, PS^{1+}), and indicate the bond order calculated from the Lewis dot structure. **(3 points)**

Lewis Dot Structure

Bond Order



C15E6277-0999-495A-83CA-5D850EC36D03

midterm-exam-93388

#606 Page 10 of 10

SCRAP PAGE

**CHEMISTRY 110 –Midterm Exam****Thursday November 7, 2024, 6:30 – 8:30 pm**

Examiners: Pallavi Sirjoosingh and Ashok Kakkar

First name (please write as legibly as possible within the boxes)

--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--

Last name

--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--

Student ID number

--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--

This booklet has 8 questions on 10 pages (The last page can be used as a scrap page).

The staple must NOT be removed from the question booklet. Your intact question booklet must be returned at the end of the exam.

A separate booklet contains a periodic table, and datasheet.

Write your answers directly under the questions in the space provided. Do not write on the QR codes or near the top edges (as it may be cut off).

Please write **legibly and dark enough** for the scan to be readable. For students who use English as a second language, dictionaries are permitted. Calculators are allowed.

DO NOT ASK QUESTIONS OF THE INVIGILATORS about interpretations of the exam questions. If you have a concern about a question, please summarize it at the top of this page. All concerns will be evaluated after the exam. GOOD LUCK!

**Question 1 8 points** (*Assume all transitions are possible*)

a. 6 points Calculate the **energy (in kJ/mol)** required to ionize (completely remove) the electron from the **$n=3$** level of **He^+ ion** (He: Helium). Show your work including equations. (3 sig. figs.)

b. 2 points

(I) Select the largest (longest) wavelength from the three choices below:

- i. Wavelength of photon to *completely remove* an electron from *1s orbital of Hydrogen atom*
- ii. Wavelength of photon to *completely remove* an electron from *1s orbital of Boron atom (B)*
- iii. Wavelength of photon to *completely remove* an electron from *1s orbital of Boron cation (B^+)*

i or ii or iii

(II) Select the smallest (shortest) wavelength from the three choices below (*for H atom*):

- i. Wavelength of photon *emitted* for an electron transitioning from *2p to 1s orbital*
- ii. Wavelength of photon *absorbed* for an electron transitioning from *2s to 4p orbital*
- iii. Wavelength of photon *absorbed* for an electron transitioning from *1s to 3p orbital*

i or ii or iii



Question 2 8 points



A Chem110 student plots the graph shown on the left, from data obtained for the kinetic energy (eV) of the ejected electron versus frequency (s^{-1}) of the incident light, for an unknown metal. ($1\text{eV} = 1.6022 \times 10^{-19} \text{ J}$)

Based on the plot, answer the questions **a to c** below:

a. 1 point On the plot above, label the **threshold frequency**.

b. 2 points What is the value of the work potential (in J) of the unknown metal?

c. 3 points What is the *longest* wavelength (in nm) required to eject an electron from this metal's surface? Show your work including equation(s) used. (3 sig figs)

d. 2 points In the space provided below, plot the following

(Assume frequency is *above* threshold frequency):

- i. *Kinetic energy* of an ejected electron versus the *intensity* of the incident light (left)
- ii. *Number of electrons* ejected versus the *frequency* of the incident light (right)



**Question 3** **6 points****a. 2 points**

In the ground state of Ge (Germanium; $Z = 32$) atom, **name** the following orbitals:

i. Orbital containing the electron that requires the maximum energy to be removed.

ii. Orbital containing the electron that experiences the minimum effective nuclear charge.

b. 2 points

An unknown anion (A^-) has the following *excited* electronic configuration: $[Ar] 4s^2 3d^3 4p^1$

(Ar: Argon; $Z = 18$)

Give the **condensed ground state electronic configuration** of the atom, **A** and the cation, A^{2+} .
Show your work. (Assume all electronic transitions are possible)

Electronic Configuration of A:

Electronic Configuration of A^{2+} :

c. 2 points

In a single atom, determine how many electrons can have the following set of quantum numbers?

(i) $n = 5; l = 3$

In a single atom, determine how many electrons can have the following set of quantum numbers?

(i) $n = 3; l = 2; m_s = +1/2$

**Question 4** **8 points**

Rank the following from 1 to 4 in **increasing** order. Provide the ranking in the boxes provided.

Note: Atomic number (Z) are provided for relevant atoms

a) Increasing (lowest to highest) order of *ionization energy*: (1= lowest; 4 = highest)

Mg

Mg: Z = 12

Rb

Rb: Z = 37

Cs

Cs: Z = 55

K

K : Z = 19

b) Increasing (lowest to highest) order of *bond angle*: (1= lowest; 4 = highest)

H₂OCCl₄XeF₄

HCN

c) Increasing (smallest to largest) order of *ionic size* (1= smallest; 4 = largest)

S²⁻

S: Z = 16

O²⁻

O: Z = 8

Be²⁺

Be: Z = 4

Na⁺

Na: Z = 11

d) Increasing (lowest to highest) order of *bond energy* (1= lowest; 4 = highest)

P-F

P-Cl

P-I

P-Br

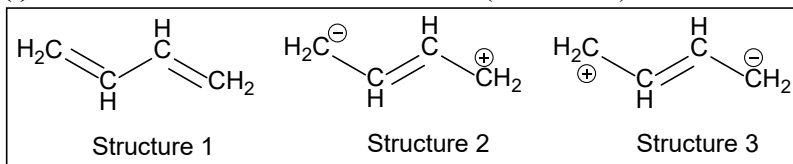
**Question 5** **12 points**

a. 9 points For the molecules HClO_4 and TeF_6 , draw the most stable *Lewis structure* showing appropriate shape and *dashed/wedged bonds*, based on VSEPR. Determine the **molecular geometry** around the central atom. For HClO_4 , label the *most polar bond*.



b. 3 points (Note: Lone pairs are not shown in structures below)

(i) For the three *resonance* structures (1, 2 and 3) shown below, answer the following question:



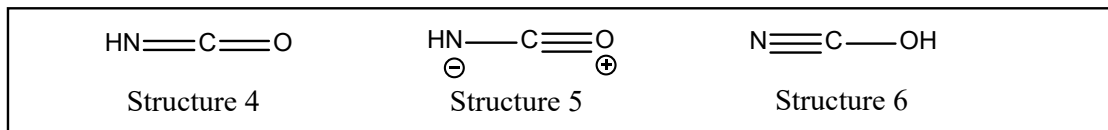
(I) Circle the *most contributing* resonance structure.

Structure 1

Structure 2

Structure 3

(ii) For the three structures given (4, 5, and 6) below, answer the following questions:



(I) Circle all the structures that represent *resonance structures* of each other:

Structure 4

Structure 5

Structure 6

(II) Among the resonance structures, circle the *most contributing* resonance structure.

Structure 4

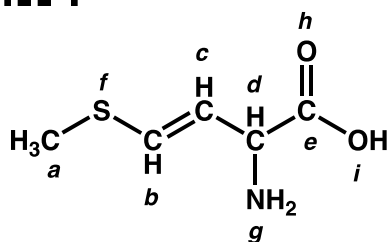
Structure 5

Structure 6

**Question 6** **6 points**

Using the data given below, calculate the **lattice energy** for SrO(s).

	ΔH° (kJ/mol)
$\text{Sr}(s) \longrightarrow \text{Sr}(g)$	+180
$\text{Sr}(g) \longrightarrow \text{Sr}^{2+}(g) + 2e^-$	+1468
$\text{O}_2(g) \longrightarrow 2\text{O}(g)$	+500
$\text{O}(g) + 2e^- \longrightarrow \text{O}^{2-}(g)$	+600
$\text{Sr}(s) + \frac{1}{2}\text{O}_2(g) \longrightarrow \text{SrO}(s)$	-550

**Question 7 10 points**

Consider the molecule given on the left, and answer questions (i) to (vi) using chemical bonding theories that we have learnt in class (valence bond theory, hybridization).

Note that the lone pairs are not shown in the structure:

i. Consider the carbon atoms marked in as ***a***, ***b***, ***c***, ***d***, and ***e*** in the molecule, and determine **hybridization** at each of the carbon centers. (2.5 points)

Atom	Hybridization
Carbon (<i>a</i>)	
Carbon (<i>b</i>)	
Carbon (<i>c</i>)	
Carbon (<i>d</i>)	
Carbon (<i>e</i>)	

ii. Now, consider the sulfur atom (S) marked as ***f***, nitrogen (N) atom (***g***), and oxygen(O) atoms (***h*** and ***i***), and determine their **hybridization**. (2 points)

Atom	Hybridization
Sulfur (<i>f</i>)	
Nitrogen (<i>g</i>)	
Oxygen (<i>h</i>)	
Oxygen (<i>i</i>)	

iii. **Describe** all the bonds formed by the carbon atoms ***a*** and ***b***, using sigma and pi designation as appropriate. (2 points)

Atom	Bonds formed (use sigma and pi as appropriate)
Carbon (<i>a</i>)	
Carbon (<i>b</i>)	

iv. How many total sp^3 -s sigma bonds are in the molecule? (1 point)

v. Determine **electronic** and **molecular geometry** at carbon (C) labeled ***e*** and sulfur (S) labeled ***f*** centres. (2 points)

Atom	Electronic Geometry	Molecular Geometry
Carbon (<i>e</i>)		
Sulfur (<i>f</i>)		

vi. Circle the value from the choices given below, that describes the **C-S-C bond angle at sulfur** (S) marked ***f***: (0.5 points)

>90° to <109.5° 109.5° >109.5 to <120° 120° >120° to <180°


Question 8 12 points

Using the **valence shell orbitals only** draw a **complete** molecular orbital (MO) diagram (with no s-p mixing) for the molecule **PS**, in the space provided below. **(6 points)**

Assume that 3s and 3p atomic orbitals will form molecular orbitals like those formed by the combination of 2s and 2p atomic orbitals that we learnt in class, and there is no s-p mixing.

Show **energy arrow**, **atomic orbitals** involved, **atom labels**, the **molecular orbitals** that are formed, **filled electrons** in the MO diagram etc. Make sure to label all the atomic and molecular orbitals.

Using the MO diagram above, answer the following questions:

- (i) Which one of the following will have the highest bond order? Circle the correct choice
 A) PS^{1-} B) PS C) PS^{1+}
- (ii) Which one of the following will have the longest bond length? Circle the correct choice
 A) PS^{1-} B) PS C) PS^{1+}
- (iii) Which of the following will be diamagnetic? Circle the correct choice.
 A) PS^{1-} B) PS C) PS^{1+}
- (iv) Draw Lewis dot structure of the molecule with the shortest bond length (based on the MO diagram) from among the three (PS^{1-} , PS, PS^{1+}), and indicate the bond order calculated from the Lewis dot structure. **(3 points)**

Lewis Dot Structure

Bond Order



8B0896D8-1BB6-4AAB-AD07-84EA9B2EC62D

midterm-exam-93388

#607 Page 10 of 10

SCRAP PAGE

**CHEMISTRY 110 –Midterm Exam****Thursday November 7, 2024, 6:30 – 8:30 pm**

Examiners: Pallavi Sirjoosingh and Ashok Kakkar

First name (please write as legibly as possible within the boxes)

--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--

Last name

--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--

Student ID number

--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--

This booklet has 8 questions on 10 pages (The last page can be used as a scrap page).

The staple must NOT be removed from the question booklet. Your intact question booklet must be returned at the end of the exam.

A separate booklet contains a periodic table, and datasheet.

Write your answers directly under the questions in the space provided. Do not write on the QR codes or near the top edges (as it may be cut off).

Please write **legibly and dark enough** for the scan to be readable. For students who use English as a second language, dictionaries are permitted. Calculators are allowed.

DO NOT ASK QUESTIONS OF THE INVIGILATORS about interpretations of the exam questions. If you have a concern about a question, please summarize it at the top of this page. All concerns will be evaluated after the exam. GOOD LUCK!

**Question 1 8 points** (*Assume all transitions are possible*)

a. 6 points Calculate the **energy (in kJ/mol)** required to ionize (completely remove) the electron from the **$n=3$** level of **He^+ ion** (He: Helium). Show your work including equations. (3 sig. figs.)

b. 2 points

(I) Select the largest (longest) wavelength from the three choices below:

- i. Wavelength of photon to *completely remove* an electron from *1s orbital of Hydrogen atom*
- ii. Wavelength of photon to *completely remove* an electron from *1s orbital of Boron atom (B)*
- iii. Wavelength of photon to *completely remove* an electron from *1s orbital of Boron cation (B^+)*

i or ii or iii

(II) Select the smallest (shortest) wavelength from the three choices below (*for H atom*):

- i. Wavelength of photon *emitted* for an electron transitioning from *2p to 1s orbital*
- ii. Wavelength of photon *absorbed* for an electron transitioning from *2s to 4p orbital*
- iii. Wavelength of photon *absorbed* for an electron transitioning from *1s to 3p orbital*

i or ii or iii



Question 2 8 points



A Chem110 student plots the graph shown on the left, from data obtained for the kinetic energy (eV) of the ejected electron versus frequency (s^{-1}) of the incident light, for an unknown metal. ($1\text{eV} = 1.6022 \times 10^{-19} \text{ J}$)

Based on the plot, answer the questions **a to c** below:

a. 1 point On the plot above, label the **threshold frequency**.

b. 2 points What is the value of the work potential (in J) of the unknown metal?

c. 3 points What is the *longest* wavelength (in nm) required to eject an electron from this metal's surface? Show your work including equation(s) used. (3 sig figs)

d. 2 points In the space provided below, plot the following

(Assume frequency is *above* threshold frequency):

- i. *Kinetic energy* of an ejected electron versus the *intensity* of the incident light (left)
- ii. *Number of electrons* ejected versus the *frequency* of the incident light (right)



**Question 3** **6 points****a. 2 points**

In the ground state of Ge (Germanium; $Z = 32$) atom, **name** the following orbitals:

i. Orbital containing the electron that requires the maximum energy to be removed.

ii. Orbital containing the electron that experiences the minimum effective nuclear charge.

b. 2 points

An unknown anion (A^-) has the following *excited* electronic configuration: $[Ar] 4s^2 3d^3 4p^1$

(Ar: Argon; $Z = 18$)

Give the **condensed ground state electronic configuration** of the atom, **A** and the cation, **A^{2+}** .
Show your work. (Assume all electronic transitions are possible)

Electronic Configuration of A:

Electronic Configuration of A^{2+} :

c. 2 points

In a single atom, determine how many electrons can have the following set of quantum numbers?

(i) $n = 5; l = 3$

In a single atom, determine how many electrons can have the following set of quantum numbers?

(i) $n = 3; l = 2; m_s = +1/2$

**Question 4** **8 points**

Rank the following from 1 to 4 in **increasing** order. Provide the ranking in the boxes provided.

Note: Atomic number (Z) are provided for relevant atoms

a) Increasing (lowest to highest) order of *ionization energy*: (1= lowest; 4 = highest)

Mg

Mg: Z = 12

Rb

Rb: Z = 37

Cs

Cs: Z = 55

K

K : Z = 19

b) Increasing (lowest to highest) order of *bond angle*: (1= lowest; 4 = highest)

H₂OCCl₄XeF₄

HCN

c) Increasing (smallest to largest) order of *ionic size* (1= smallest; 4 = largest)

S²⁻

S: Z = 16

O²⁻

O: Z = 8

Be²⁺

Be: Z = 4

Na⁺

Na: Z = 11

d) Increasing (lowest to highest) order of *bond energy* (1= lowest; 4 = highest)

P-F

P-Cl

P-I

P-Br

**Question 5** **12 points**

a. 9 points For the molecules HClO_4 and TeF_6 , draw the most stable *Lewis structure* showing appropriate shape and *dashed/wedged bonds*, based on VSEPR. Determine the **molecular geometry** around the central atom. For HClO_4 , label the *most polar bond*.



b. 3 points (Note: Lone pairs are not shown in structures below)

(i) For the three *resonance* structures (1, 2 and 3) shown below, answer the following question:



(I) Circle the *most contributing* resonance structure.

Structure 1

Structure 2

Structure 3

(ii) For the three structures given (4, 5, and 6) below, answer the following questions:



(I) Circle all the structures that represent *resonance structures* of each other:

Structure 4

Structure 5

Structure 6

(II) Among the resonance structures, circle the *most contributing* resonance structure.

Structure 4

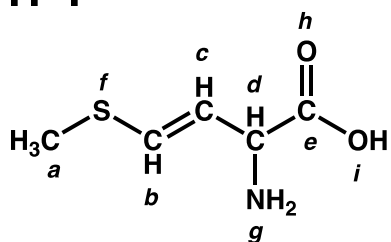
Structure 5

Structure 6

**Question 6** **6 points**

Using the data given below, calculate the **lattice energy** for SrO(s).

	ΔH° (kJ/mol)
$\text{Sr}(s) \longrightarrow \text{Sr}(g)$	+180
$\text{Sr}(g) \longrightarrow \text{Sr}^{2+}(g) + 2e^-$	+1468
$\text{O}_2(g) \longrightarrow 2\text{O}(g)$	+500
$\text{O}(g) + 2e^- \longrightarrow \text{O}^{2-}(g)$	+600
$\text{Sr}(s) + \frac{1}{2}\text{O}_2(g) \longrightarrow \text{SrO}(s)$	-550

**Question 7 10 points**

Consider the molecule given on the left, and answer questions (i) to (vi) using chemical bonding theories that we have learnt in class (valence bond theory, hybridization).

Note that the lone pairs are not shown in the structure:

i. Consider the carbon atoms marked in as ***a***, ***b***, ***c***, ***d***, and ***e*** in the molecule, and determine **hybridization** at each of the carbon centers. (2.5 points)

Atom	Hybridization
Carbon (<i>a</i>)	
Carbon (<i>b</i>)	
Carbon (<i>c</i>)	
Carbon (<i>d</i>)	
Carbon (<i>e</i>)	

ii. Now, consider the sulfur atom (S) marked as ***f***, nitrogen (N) atom (***g***), and oxygen(O) atoms (***h*** and ***i***), and determine their **hybridization**. (2 points)

Atom	Hybridization
Sulfur (<i>f</i>)	
Nitrogen (<i>g</i>)	
Oxygen (<i>h</i>)	
Oxygen (<i>i</i>)	

iii. **Describe** all the bonds formed by the carbon atoms ***a*** and ***b***, using sigma and pi designation as appropriate. (2 points)

Atom	Bonds formed (use sigma and pi as appropriate)
Carbon (<i>a</i>)	
Carbon (<i>b</i>)	

iv. How many total sp^3 -s sigma bonds are in the molecule? (1 point)

v. Determine **electronic** and **molecular geometry** at carbon (C) labeled ***e*** and sulfur (S) labeled ***f*** centres. (2 points)

Atom	Electronic Geometry	Molecular Geometry
Carbon (<i>e</i>)		
Sulfur (<i>f</i>)		

vi. Circle the value from the choices given below, that describes the **C-S-C bond angle at sulfur** (S) marked ***f***: (0.5 points)

>90° to <109.5° 109.5° >109.5 to <120° 120° >120° to <180°


Question 8 12 points

Using the **valence shell orbitals only** draw a **complete** molecular orbital (MO) diagram (with no s-p mixing) for the molecule **PS**, in the space provided below. **(6 points)**

Assume that 3s and 3p atomic orbitals will form molecular orbitals like those formed by the combination of 2s and 2p atomic orbitals that we learnt in class, and there is no s-p mixing.

Show **energy arrow**, **atomic orbitals** involved, **atom labels**, the **molecular orbitals** that are formed, **filled electrons** in the MO diagram etc. Make sure to label all the atomic and molecular orbitals.

Using the MO diagram above, answer the following questions:

- (i) Which one of the following will have the highest bond order? Circle the correct choice
 A) PS^{1-} B) PS C) PS^{1+}
- (ii) Which one of the following will have the longest bond length? Circle the correct choice
 A) PS^{1-} B) PS C) PS^{1+}
- (iii) Which of the following will be diamagnetic? Circle the correct choice.
 A) PS^{1-} B) PS C) PS^{1+}
- (iv) Draw Lewis dot structure of the molecule with the shortest bond length (based on the MO diagram) from among the three (PS^{1-} , PS, PS^{1+}), and indicate the bond order calculated from the Lewis dot structure. **(3 points)**

Lewis Dot Structure

Bond Order



CC26AE42-3DB0-4DF5-9626-4BAF73EFBA4E

midterm-exam-93388

#608 Page 10 of 10

SCRAP PAGE

**CHEMISTRY 110 –Midterm Exam****Thursday November 7, 2024, 6:30 – 8:30 pm****Examiners: Pallavi Sirjoosingh and Ashok Kakkar**

First name (please write as legibly as possible within the boxes)

--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--

Last name

--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--

Student ID number

--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--

This booklet has 8 questions on 10 pages (The last page can be used as a scrap page).

The staple must NOT be removed from the question booklet. Your intact question booklet must be returned at the end of the exam.

A separate booklet contains a periodic table, and datasheet.

Write your answers directly under the questions in the space provided. Do not write on the QR codes or near the top edges (as it may be cut off).

Please write **legibly and dark enough** for the scan to be readable. For students who use English as a second language, dictionaries are permitted. Calculators are allowed.

DO NOT ASK QUESTIONS OF THE INVIGILATORS about interpretations of the exam questions. If you have a concern about a question, please summarize it at the top of this page. All concerns will be evaluated after the exam. GOOD LUCK!

**Question 1 8 points** (*Assume all transitions are possible*)

a. 6 points Calculate the **energy (in kJ/mol)** required to ionize (completely remove) the electron from the **n=3** level of **He⁺ ion** (He: Helium). Show your work including equations. (3 sig. figs.)

b. 2 points

(I) Select the largest (longest) wavelength from the three choices below:

- i. Wavelength of photon to *completely remove* an electron from *1s orbital of Hydrogen atom*
- ii. Wavelength of photon to *completely remove* an electron from *1s orbital of Boron atom (B)*
- iii. Wavelength of photon to *completely remove* an electron from *1s orbital of Boron cation (B⁺)*

i or ii or iii

(II) Select the smallest (shortest) wavelength from the three choices below (*for H atom*):

- i. Wavelength of photon *emitted* for an electron transitioning from *2p to 1s orbital*
- ii. Wavelength of photon *absorbed* for an electron transitioning from *2s to 4p orbital*
- iii. Wavelength of photon *absorbed* for an electron transitioning from *1s to 3p orbital*

i or ii or iii



Question 2 8 points



A Chem110 student plots the graph shown on the left, from data obtained for the kinetic energy (eV) of the ejected electron versus frequency (s^{-1}) of the incident light, for an unknown metal. ($1\text{eV} = 1.6022 \times 10^{-19} \text{ J}$)

Based on the plot, answer the questions **a to c** below:

a. 1 point On the plot above, label the **threshold frequency**.

b. 2 points What is the value of the work potential (in J) of the unknown metal?

c. 3 points What is the *longest* wavelength (in nm) required to eject an electron from this metal's surface? Show your work including equation(s) used. (3 sig figs)

d. 2 points In the space provided below, plot the following

(Assume frequency is *above* threshold frequency):

- Kinetic energy* of an ejected electron versus the *intensity* of the incident light (left)
- Number of electrons* ejected versus the *frequency* of the incident light (right)



**Question 3** **6 points****a. 2 points**

In the ground state of Ge (Germanium; $Z = 32$) atom, **name** the following orbitals:

i. Orbital containing the electron that requires the maximum energy to be removed.

ii. Orbital containing the electron that experiences the minimum effective nuclear charge.

b. 2 points

An unknown anion (A^-) has the following *excited* electronic configuration: $[Ar] 4s^2 3d^3 4p^1$

(Ar: Argon; $Z = 18$)

Give the **condensed ground state electronic configuration** of the atom, **A** and the cation, **A^{2+}** .
Show your work. (Assume all electronic transitions are possible)

Electronic Configuration of A:

Electronic Configuration of A^{2+} :

c. 2 points

In a single atom, determine how many electrons can have the following set of quantum numbers?

(i) $n = 5; l = 3$

In a single atom, determine how many electrons can have the following set of quantum numbers?

(i) $n = 3; l = 2; m_s = +1/2$

**Question 4** **8 points**

Rank the following from 1 to 4 in **increasing** order. Provide the ranking in the boxes provided.

Note: Atomic number (Z) are provided for relevant atoms

a) Increasing (lowest to highest) order of *ionization energy*: (1= lowest; 4 = highest)

Mg

Mg: Z = 12

Rb

Rb: Z = 37

Cs

Cs: Z = 55

K

K : Z = 19

b) Increasing (lowest to highest) order of *bond angle*: (1= lowest; 4 = highest)

H₂OCCl₄XeF₄

HCN

c) Increasing (smallest to largest) order of *ionic size* (1= smallest; 4 = largest)

S²⁻

S: Z = 16

O²⁻

O: Z = 8

Be²⁺

Be: Z = 4

Na⁺

Na: Z = 11

d) Increasing (lowest to highest) order of *bond energy* (1= lowest; 4 = highest)

P-F

P-Cl

P-I

P-Br

**Question 5** **12 points**

a. 9 points For the molecules HClO_4 and TeF_6 , draw the most stable *Lewis structure* showing appropriate shape and *dashed/wedged bonds*, based on VSEPR. Determine the **molecular geometry** around the central atom. For HClO_4 , label the *most polar bond*.

HClO_4 (there are no O-O bonds)

Label the most polar bond

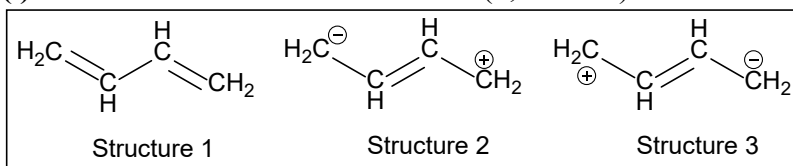
Molecular Geometry around Cl

TeF_6

Molecular Geometry around Te

b. 3 points (Note: Lone pairs are not shown in structures below)

(i) For the three *resonance* structures (1, 2 and 3) shown below, answer the following question:



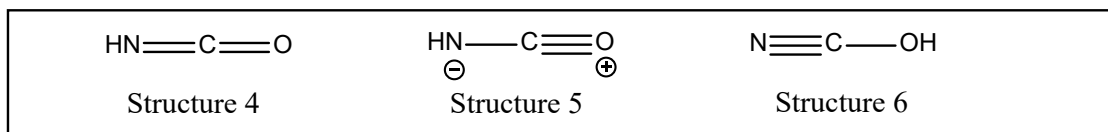
(I) Circle the *most contributing* resonance structure.

Structure 1

Structure 2

Structure 3

(ii) For the three structures given (4, 5, and 6) below, answer the following questions:



(I) Circle all the structures that represent *resonance structures* of each other:

Structure 4

Structure 5

Structure 6

(II) Among the resonance structures, circle the *most contributing* resonance structure.

Structure 4

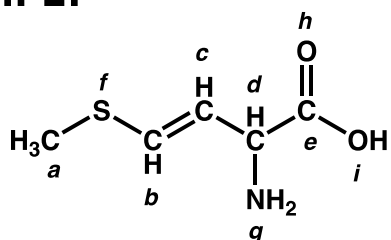
Structure 5

Structure 6

**Question 6** **6 points**

Using the data given below, calculate the **lattice energy** for SrO(s).

	ΔH° (kJ/mol)
$\text{Sr}(s) \longrightarrow \text{Sr}(g)$	+180
$\text{Sr}(g) \longrightarrow \text{Sr}^{2+}(g) + 2e^-$	+1468
$\text{O}_2(g) \longrightarrow 2\text{O}(g)$	+500
$\text{O}(g) + 2e^- \longrightarrow \text{O}^{2-}(g)$	+600
$\text{Sr}(s) + \frac{1}{2}\text{O}_2(g) \longrightarrow \text{SrO}(s)$	-550

**Question 7 10 points**

Consider the molecule given on the left, and answer questions (i) to (vi) using chemical bonding theories that we have learnt in class (valence bond theory, hybridization).

Note that the lone pairs are not shown in the structure:

i. Consider the carbon atoms marked in as ***a***, ***b***, ***c***, ***d***, and ***e*** in the molecule, and determine **hybridization** at each of the carbon centers. (2.5 points)

Atom	Hybridization
Carbon (<i>a</i>)	
Carbon (<i>b</i>)	
Carbon (<i>c</i>)	
Carbon (<i>d</i>)	
Carbon (<i>e</i>)	

ii. Now, consider the sulfur atom (S) marked as ***f***, nitrogen (N) atom (***g***), and oxygen(O) atoms (***h*** and ***i***), and determine their **hybridization**. (2 points)

Atom	Hybridization
Sulfur (<i>f</i>)	
Nitrogen (<i>g</i>)	
Oxygen (<i>h</i>)	
Oxygen (<i>i</i>)	

iii. **Describe** all the bonds formed by the carbon atoms ***a*** and ***b***, using sigma and pi designation as appropriate. (2 points)

Atom	Bonds formed (use sigma and pi as appropriate)
Carbon (<i>a</i>)	
Carbon (<i>b</i>)	

iv. How many total sp^3 -s sigma bonds are in the molecule? (1 point)

v. Determine **electronic** and **molecular geometry** at carbon (C) labeled ***e*** and sulfur (S) labeled ***f*** centres. (2 points)

Atom	Electronic Geometry	Molecular Geometry
Carbon (<i>e</i>)		
Sulfur (<i>f</i>)		

vi. Circle the value from the choices given below, that describes the **C-S-C bond angle at sulfur** (S) marked ***f***: (0.5 points)

>90° to <109.5° 109.5° >109.5 to <120° 120° >120° to <180°


Question 8 12 points

Using the **valence shell orbitals only** draw a **complete** molecular orbital (MO) diagram (with no s-p mixing) for the molecule **PS**, in the space provided below. **(6 points)**

Assume that 3s and 3p atomic orbitals will form molecular orbitals like those formed by the combination of 2s and 2p atomic orbitals that we learnt in class, and there is no s-p mixing.

Show **energy arrow**, **atomic orbitals** involved, **atom labels**, the **molecular orbitals** that are formed, **filled electrons** in the MO diagram etc. Make sure to label all the atomic and molecular orbitals.

Using the MO diagram above, answer the following questions:

- (i) Which one of the following will have the highest bond order? Circle the correct choice
 A) PS^{1-} B) PS C) PS^{1+}
- (ii) Which one of the following will have the longest bond length? Circle the correct choice
 A) PS^{1-} B) PS C) PS^{1+}
- (iii) Which of the following will be diamagnetic? Circle the correct choice.
 A) PS^{1-} B) PS C) PS^{1+}
- (iv) Draw Lewis dot structure of the molecule with the shortest bond length (based on the MO diagram) from among the three (PS^{1-} , PS, PS^{1+}), and indicate the bond order calculated from the Lewis dot structure. **(3 points)**

Lewis Dot Structure

Bond Order



A2626B38-7CBB-462A-BC46-D2D992D5C1E4

midterm-exam-93388

#609 Page 10 of 10

SCRAP PAGE

**CHEMISTRY 110 –Midterm Exam****Thursday November 7, 2024, 6:30 – 8:30 pm**

Examiners: Pallavi Sirjoosingh and Ashok Kakkar

First name (please write as legibly as possible within the boxes)

--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--

Last name

--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--

Student ID number

--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--

This booklet has 8 questions on 10 pages (The last page can be used as a scrap page).

The staple must NOT be removed from the question booklet. Your intact question booklet must be returned at the end of the exam.

A separate booklet contains a periodic table, and datasheet.

Write your answers directly under the questions in the space provided. Do not write on the QR codes or near the top edges (as it may be cut off).

Please write **legibly and dark enough** for the scan to be readable. For students who use English as a second language, dictionaries are permitted. Calculators are allowed.

DO NOT ASK QUESTIONS OF THE INVIGILATORS about interpretations of the exam questions. If you have a concern about a question, please summarize it at the top of this page. All concerns will be evaluated after the exam. GOOD LUCK!

**Question 1 8 points** (*Assume all transitions are possible*)

a. 6 points Calculate the **energy (in kJ/mol)** required to ionize (completely remove) the electron from the **$n=3$** level of **He^+ ion** (He: Helium). Show your work including equations. (3 sig. figs.)

b. 2 points

(I) Select the largest (longest) wavelength from the three choices below:

- i. Wavelength of photon to *completely remove* an electron from *1s orbital of Hydrogen atom*
- ii. Wavelength of photon to *completely remove* an electron from *1s orbital of Boron atom (B)*
- iii. Wavelength of photon to *completely remove* an electron from *1s orbital of Boron cation (B^+)*

i or ii or iii

(II) Select the smallest (shortest) wavelength from the three choices below (*for H atom*):

- i. Wavelength of photon *emitted* for an electron transitioning from *2p to 1s orbital*
- ii. Wavelength of photon *absorbed* for an electron transitioning from *2s to 4p orbital*
- iii. Wavelength of photon *absorbed* for an electron transitioning from *1s to 3p orbital*

i or ii or iii



Question 2 8 points



A Chem110 student plots the graph shown on the left, from data obtained for the kinetic energy (eV) of the ejected electron versus frequency (s^{-1}) of the incident light, for an unknown metal. ($1\text{eV} = 1.6022 \times 10^{-19} \text{ J}$)

Based on the plot, answer the questions **a to c** below:

a. 1 point On the plot above, label the **threshold frequency**.

b. 2 points What is the value of the work potential (in J) of the unknown metal?

c. 3 points What is the *longest* wavelength (in nm) required to eject an electron from this metal's surface? Show your work including equation(s) used. (3 sig figs)

d. 2 points In the space provided below, plot the following

(Assume frequency is *above* threshold frequency):

- i. *Kinetic energy* of an ejected electron versus the *intensity* of the incident light (left)
- ii. *Number of electrons* ejected versus the *frequency* of the incident light (right)



**Question 3** **6 points****a. 2 points**

In the ground state of Ge (Germanium; $Z = 32$) atom, **name** the following orbitals:

i. Orbital containing the electron that requires the maximum energy to be removed.

ii. Orbital containing the electron that experiences the minimum effective nuclear charge.

b. 2 points

An unknown anion (A^-) has the following *excited* electronic configuration: $[Ar] 4s^2 3d^3 4p^1$

(Ar: Argon; $Z = 18$)

Give the **condensed ground state electronic configuration** of the atom, **A** and the cation, **A^{2+}** .
Show your work. (Assume all electronic transitions are possible)

Electronic Configuration of A:

Electronic Configuration of A^{2+} :

c. 2 points

In a single atom, determine how many electrons can have the following set of quantum numbers?

(i) $n = 5; l = 3$

In a single atom, determine how many electrons can have the following set of quantum numbers?

(i) $n = 3; l = 2; m_s = +1/2$

**Question 4** **8 points**

Rank the following from 1 to 4 in **increasing** order. Provide the ranking in the boxes provided.

Note: Atomic number (Z) are provided for relevant atoms

a) Increasing (lowest to highest) order of *ionization energy*: (1= lowest; 4 = highest)

Mg

Mg: Z = 12

Rb

Rb: Z = 37

Cs

Cs: Z = 55

K

K : Z = 19

b) Increasing (lowest to highest) order of *bond angle*: (1= lowest; 4 = highest)

H₂OCCl₄XeF₄

HCN

c) Increasing (smallest to largest) order of *ionic size* (1= smallest; 4 = largest)

S²⁻

S: Z = 16

O²⁻

O: Z = 8

Be²⁺

Be: Z = 4

Na⁺

Na: Z = 11

d) Increasing (lowest to highest) order of *bond energy* (1= lowest; 4 = highest)

P-F

P-Cl

P-I

P-Br

**Question 5** **12 points**

a. 9 points For the molecules HClO_4 and TeF_6 , draw the most stable *Lewis structure* showing appropriate shape and *dashed/wedged bonds*, based on VSEPR. Determine the **molecular geometry** around the central atom. For HClO_4 , label the *most polar bond*.



b. 3 points (Note: Lone pairs are not shown in structures below)

(i) For the three *resonance* structures (1, 2 and 3) shown below, answer the following question:



(I) Circle the *most contributing* resonance structure.

Structure 1

Structure 2

Structure 3

(ii) For the three structures given (4, 5, and 6) below, answer the following questions:



(I) Circle all the structures that represent *resonance structures* of each other:

Structure 4

Structure 5

Structure 6

(II) Among the resonance structures, circle the *most contributing* resonance structure.

Structure 4

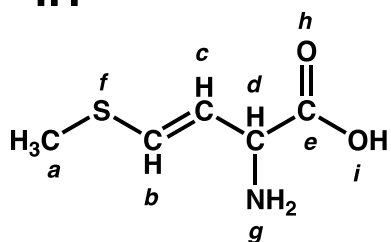
Structure 5

Structure 6

**Question 6** **6 points**

Using the data given below, calculate the **lattice energy** for SrO(s).

	ΔH° (kJ/mol)
$\text{Sr}(s) \longrightarrow \text{Sr}(g)$	+180
$\text{Sr}(g) \longrightarrow \text{Sr}^{2+}(g) + 2e^-$	+1468
$\text{O}_2(g) \longrightarrow 2\text{O}(g)$	+500
$\text{O}(g) + 2e^- \longrightarrow \text{O}^{2-}(g)$	+600
$\text{Sr}(s) + \frac{1}{2}\text{O}_2(g) \longrightarrow \text{SrO}(s)$	-550

**Question 7 10 points**

Consider the molecule given on the left, and answer questions (i) to (vi) using chemical bonding theories that we have learnt in class (valence bond theory, hybridization).

Note that the lone pairs are not shown in the structure:

i. Consider the carbon atoms marked in as ***a***, ***b***, ***c***, ***d***, and ***e*** in the molecule, and determine **hybridization** at each of the carbon centers. (2.5 points)

Atom	Hybridization
Carbon (<i>a</i>)	
Carbon (<i>b</i>)	
Carbon (<i>c</i>)	
Carbon (<i>d</i>)	
Carbon (<i>e</i>)	

ii. Now, consider the sulfur atom (S) marked as ***f***, nitrogen (N) atom (***g***), and oxygen(O) atoms (***h*** and ***i***), and determine their **hybridization**. (2 points)

Atom	Hybridization
Sulfur (<i>f</i>)	
Nitrogen (<i>g</i>)	
Oxygen (<i>h</i>)	
Oxygen (<i>i</i>)	

iii. **Describe** all the bonds formed by the carbon atoms ***a*** and ***b***, using sigma and pi designation as appropriate. (2 points)

Atom	Bonds formed (use sigma and pi as appropriate)
Carbon (<i>a</i>)	
Carbon (<i>b</i>)	

iv. How many total sp^3 -s sigma bonds are in the molecule? (1 point)

v. Determine **electronic** and **molecular geometry** at carbon (C) labeled ***e*** and sulfur (S) labeled ***f*** centres. (2 points)

Atom	Electronic Geometry	Molecular Geometry
Carbon (<i>e</i>)		
Sulfur (<i>f</i>)		

vi. Circle the value from the choices given below, that describes the **C-S-C bond angle at sulfur** (S) marked ***f***: (0.5 points)

>90° to <109.5° 109.5° >109.5 to <120° 120° >120° to <180°


Question 8 12 points

Using the **valence shell orbitals only** draw a **complete** molecular orbital (MO) diagram (with no s-p mixing) for the molecule **PS**, in the space provided below. **(6 points)**

Assume that 3s and 3p atomic orbitals will form molecular orbitals like those formed by the combination of 2s and 2p atomic orbitals that we learnt in class, and there is no s-p mixing.

Show **energy arrow**, **atomic orbitals** involved, **atom labels**, the **molecular orbitals** that are formed, **filled electrons** in the MO diagram etc. Make sure to label all the atomic and molecular orbitals.

Using the MO diagram above, answer the following questions:

- (i) Which one of the following will have the highest bond order? Circle the correct choice
 A) PS^{1-} B) PS C) PS^{1+}
- (ii) Which one of the following will have the longest bond length? Circle the correct choice
 A) PS^{1-} B) PS C) PS^{1+}
- (iii) Which of the following will be diamagnetic? Circle the correct choice.
 A) PS^{1-} B) PS C) PS^{1+}
- (iv) Draw Lewis dot structure of the molecule with the shortest bond length (based on the MO diagram) from among the three (PS^{1-} , PS, PS^{1+}), and indicate the bond order calculated from the Lewis dot structure. **(3 points)**

Lewis Dot Structure

Bond Order



2F40A403-FFF1-49B9-B5FE-5493C0CCD03E

midterm-exam-93388

#610 Page 10 of 10

SCRAP PAGE

**CHEMISTRY 110 –Midterm Exam****Thursday November 7, 2024, 6:30 – 8:30 pm****Examiners: Pallavi Sirjoosingh and Ashok Kakkar**

First name (please write as legibly as possible within the boxes)

--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--

Last name

--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--

Student ID number

--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--

This booklet has 8 questions on 10 pages (The last page can be used as a scrap page).

The staple must NOT be removed from the question booklet. Your intact question booklet must be returned at the end of the exam.

A separate booklet contains a periodic table, and datasheet.

Write your answers directly under the questions in the space provided. Do not write on the QR codes or near the top edges (as it may be cut off).

Please write **legibly and dark enough** for the scan to be readable. For students who use English as a second language, dictionaries are permitted. Calculators are allowed.

DO NOT ASK QUESTIONS OF THE INVIGILATORS about interpretations of the exam questions. If you have a concern about a question, please summarize it at the top of this page. All concerns will be evaluated after the exam. GOOD LUCK!

**Question 1 8 points** (*Assume all transitions are possible*)

a. 6 points Calculate the **energy (in kJ/mol)** required to ionize (completely remove) the electron from the **$n=3$** level of **He^+ ion** (He: Helium). Show your work including equations. (3 sig. figs.)

b. 2 points

(I) Select the largest (longest) wavelength from the three choices below:

- i. Wavelength of photon to *completely remove* an electron from *1s orbital of Hydrogen atom*
- ii. Wavelength of photon to *completely remove* an electron from *1s orbital of Boron atom (B)*
- iii. Wavelength of photon to *completely remove* an electron from *1s orbital of Boron cation (B^+)*

i or ii or iii

(II) Select the smallest (shortest) wavelength from the three choices below (*for H atom*):

- i. Wavelength of photon *emitted* for an electron transitioning from *2p to 1s orbital*
- ii. Wavelength of photon *absorbed* for an electron transitioning from *2s to 4p orbital*
- iii. Wavelength of photon *absorbed* for an electron transitioning from *1s to 3p orbital*

i or ii or iii



Question 2 8 points



A Chem110 student plots the graph shown on the left, from data obtained for the kinetic energy (eV) of the ejected electron versus frequency (s^{-1}) of the incident light, for an unknown metal. ($1\text{eV} = 1.6022 \times 10^{-19} \text{ J}$)

Based on the plot, answer the questions **a to c** below:

a. 1 point On the plot above, label the **threshold frequency**.

b. 2 points What is the value of the work potential (in J) of the unknown metal?

c. 3 points What is the *longest* wavelength (in nm) required to eject an electron from this metal's surface? Show your work including equation(s) used. (3 sig figs)

d. 2 points In the space provided below, plot the following

(Assume frequency is *above* threshold frequency):

- Kinetic energy* of an ejected electron versus the *intensity* of the incident light (left)
- Number of electrons* ejected versus the *frequency* of the incident light (right)



**Question 3** **6 points****a. 2 points**

In the ground state of Ge (Germanium; $Z = 32$) atom, **name** the following orbitals:

i. Orbital containing the electron that requires the maximum energy to be removed.

ii. Orbital containing the electron that experiences the minimum effective nuclear charge.

b. 2 points

An unknown anion (A^-) has the following *excited* electronic configuration: $[Ar] 4s^2 3d^3 4p^1$

(Ar: Argon; $Z = 18$)

Give the **condensed ground state electronic configuration** of the atom, **A** and the cation, **A^{2+}** .
Show your work. (Assume all electronic transitions are possible)

Electronic Configuration of A:

Electronic Configuration of A^{2+} :

c. 2 points

In a single atom, determine how many electrons can have the following set of quantum numbers?

(i) $n = 5; l = 3$

In a single atom, determine how many electrons can have the following set of quantum numbers?

(i) $n = 3; l = 2; m_s = +1/2$

**Question 4** **8 points**

Rank the following from 1 to 4 in **increasing** order. Provide the ranking in the boxes provided.

Note: Atomic number (Z) are provided for relevant atoms

a) Increasing (lowest to highest) order of *ionization energy*: (1= lowest; 4 = highest)

Mg

Mg: Z = 12

Rb

Rb: Z = 37

Cs

Cs: Z = 55

K

K : Z = 19

b) Increasing (lowest to highest) order of *bond angle*: (1= lowest; 4 = highest)

H₂OCCl₄XeF₄

HCN

c) Increasing (smallest to largest) order of *ionic size* (1= smallest; 4 = largest)

S²⁻

S: Z = 16

O²⁻

O: Z = 8

Be²⁺

Be: Z = 4

Na⁺

Na: Z = 11

d) Increasing (lowest to highest) order of *bond energy* (1= lowest; 4 = highest)

P-F

P-Cl

P-I

P-Br

**Question 5** **12 points**

a. 9 points For the molecules HClO_4 and TeF_6 , draw the most stable *Lewis structure* showing appropriate shape and *dashed/wedged bonds*, based on VSEPR. Determine the **molecular geometry** around the central atom. For HClO_4 , label the *most polar bond*.



b. 3 points (Note: Lone pairs are not shown in structures below)

(i) For the three *resonance* structures (1, 2 and 3) shown below, answer the following question:



(I) Circle the *most contributing* resonance structure.

Structure 1

Structure 2

Structure 3

(ii) For the three structures given (4, 5, and 6) below, answer the following questions:



(I) Circle all the structures that represent *resonance structures* of each other:

Structure 4

Structure 5

Structure 6

(II) Among the resonance structures, circle the *most contributing* resonance structure.

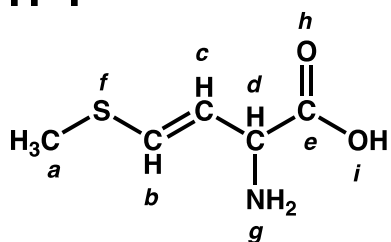
Structure 4

Structure 5

Structure 6

**Question 6** **6 points**Using the data given below, calculate the **lattice energy** for SrO(s).

	ΔH° (kJ/mol)
$\text{Sr}(s) \longrightarrow \text{Sr}(g)$	+180
$\text{Sr}(g) \longrightarrow \text{Sr}^{2+}(g) + 2e^-$	+1468
$\text{O}_2(g) \longrightarrow 2\text{O}(g)$	+500
$\text{O}(g) + 2e^- \longrightarrow \text{O}^{2-}(g)$	+600
$\text{Sr}(s) + \frac{1}{2}\text{O}_2(g) \longrightarrow \text{SrO}(s)$	-550

**Question 7 10 points**

Consider the molecule given on the left, and answer questions (i) to (vi) using chemical bonding theories that we have learnt in class (valence bond theory, hybridization).

Note that the lone pairs are not shown in the structure:

i. Consider the carbon atoms marked in as ***a***, ***b***, ***c***, ***d***, and ***e*** in the molecule, and determine **hybridization** at each of the carbon centers. (2.5 points)

Atom	Hybridization
Carbon (<i>a</i>)	
Carbon (<i>b</i>)	
Carbon (<i>c</i>)	
Carbon (<i>d</i>)	
Carbon (<i>e</i>)	

ii. Now, consider the sulfur atom (S) marked as ***f***, nitrogen (N) atom (***g***), and oxygen(O) atoms (***h*** and ***i***), and determine their **hybridization**. (2 points)

Atom	Hybridization
Sulfur (<i>f</i>)	
Nitrogen (<i>g</i>)	
Oxygen (<i>h</i>)	
Oxygen (<i>i</i>)	

iii. **Describe** all the bonds formed by the carbon atoms ***a*** and ***b***, using sigma and pi designation as appropriate. (2 points)

Atom	Bonds formed (use sigma and pi as appropriate)
Carbon (<i>a</i>)	
Carbon (<i>b</i>)	

iv. How many total sp^3 -s sigma bonds are in the molecule? (1 point)

v. Determine **electronic** and **molecular geometry** at carbon (C) labeled ***e*** and sulfur (S) labeled ***f*** centres. (2 points)

Atom	Electronic Geometry	Molecular Geometry
Carbon (<i>e</i>)		
Sulfur (<i>f</i>)		

vi. Circle the value from the choices given below, that describes the **C-S-C bond angle at sulfur** (S) marked ***f***: (0.5 points)

>90° to <109.5° 109.5° >109.5 to <120° 120° >120° to <180°

**Question 8 12 points**

Using the **valence shell orbitals only** draw a **complete** molecular orbital (MO) diagram (with no s-p mixing) for the molecule **PS**, in the space provided below. **(6 points)**

Assume that 3s and 3p atomic orbitals will form molecular orbitals like those formed by the combination of 2s and 2p atomic orbitals that we learnt in class, and there is no s-p mixing.

Show **energy arrow**, **atomic orbitals** involved, **atom labels**, the **molecular orbitals** that are formed, **filled electrons** in the MO diagram etc. Make sure to label all the atomic and molecular orbitals.

Using the MO diagram above, answer the following questions:

- (i) Which one of the following will have the highest bond order? Circle the correct choice
A) PS^{1-} B) PS C) PS^{1+}
- (ii) Which one of the following will have the longest bond length? Circle the correct choice
A) PS^{1-} B) PS C) PS^{1+}
- (iii) Which of the following will be diamagnetic? Circle the correct choice.
A) PS^{1-} B) PS C) PS^{1+}
- (iv) Draw Lewis dot structure of the molecule with the shortest bond length (based on the MO diagram) from among the three (PS^{1-} , PS, PS^{1+}), and indicate the bond order calculated from the Lewis dot structure. **(3 points)**

Lewis Dot Structure

Bond Order



DB24107E-B4C2-4414-BF64-0B55BB2AEE17

midterm-exam-93388

#611 Page 10 of 10

SCRAP PAGE

**CHEMISTRY 110 –Midterm Exam****Thursday November 7, 2024, 6:30 – 8:30 pm**

Examiners: Pallavi Sirjoosingh and Ashok Kakkar

First name (please write as legibly as possible within the boxes)

--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--

Last name

--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--

Student ID number

--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--

This booklet has 8 questions on 10 pages (The last page can be used as a scrap page).

The staple must NOT be removed from the question booklet. Your intact question booklet must be returned at the end of the exam.

A separate booklet contains a periodic table, and datasheet.

Write your answers directly under the questions in the space provided. Do not write on the QR codes or near the top edges (as it may be cut off).

Please write **legibly and dark enough** for the scan to be readable. For students who use English as a second language, dictionaries are permitted. Calculators are allowed.

DO NOT ASK QUESTIONS OF THE INVIGILATORS about interpretations of the exam questions. If you have a concern about a question, please summarize it at the top of this page. All concerns will be evaluated after the exam. GOOD LUCK!

**Question 1 8 points** (*Assume all transitions are possible*)

a. 6 points Calculate the **energy (in kJ/mol)** required to ionize (completely remove) the electron from the **n=3** level of **He⁺ ion** (He: Helium). Show your work including equations. (3 sig. figs.)

b. 2 points

(I) Select the largest (longest) wavelength from the three choices below:

- i. Wavelength of photon to *completely remove* an electron from *1s orbital of Hydrogen atom*
- ii. Wavelength of photon to *completely remove* an electron from *1s orbital of Boron atom (B)*
- iii. Wavelength of photon to *completely remove* an electron from *1s orbital of Boron cation (B⁺)*

i or ii or iii

(II) Select the smallest (shortest) wavelength from the three choices below (*for H atom*):

- i. Wavelength of photon *emitted* for an electron transitioning from *2p to 1s orbital*
- ii. Wavelength of photon *absorbed* for an electron transitioning from *2s to 4p orbital*
- iii. Wavelength of photon *absorbed* for an electron transitioning from *1s to 3p orbital*

i or ii or iii



Question 2 8 points



A Chem110 student plots the graph shown on the left, from data obtained for the kinetic energy (eV) of the ejected electron versus frequency (s^{-1}) of the incident light, for an unknown metal. ($1\text{eV} = 1.6022 \times 10^{-19} \text{ J}$)

Based on the plot, answer the questions **a to c** below:

a. 1 point On the plot above, label the **threshold frequency**.

b. 2 points What is the value of the work potential (in J) of the unknown metal?

c. 3 points What is the *longest* wavelength (in nm) required to eject an electron from this metal's surface? Show your work including equation(s) used. (3 sig figs)

d. 2 points In the space provided below, plot the following

(Assume frequency is *above* threshold frequency):

- i. *Kinetic energy* of an ejected electron versus the *intensity* of the incident light (left)
- ii. *Number of electrons* ejected versus the *frequency* of the incident light (right)



**Question 3** **6 points****a. 2 points**

In the ground state of Ge (Germanium; $Z = 32$) atom, **name** the following orbitals:

i. Orbital containing the electron that requires the maximum energy to be removed.

ii. Orbital containing the electron that experiences the minimum effective nuclear charge.

b. 2 points

An unknown anion (A^-) has the following *excited* electronic configuration: $[Ar] 4s^2 3d^3 4p^1$

(Ar: Argon; $Z = 18$)

Give the **condensed ground state electronic configuration** of the atom, **A** and the cation, **A^{2+}** .
Show your work. (Assume all electronic transitions are possible)

Electronic Configuration of A:

Electronic Configuration of A^{2+} :

c. 2 points

In a single atom, determine how many electrons can have the following set of quantum numbers?

(i) $n = 5; l = 3$

In a single atom, determine how many electrons can have the following set of quantum numbers?

(i) $n = 3; l = 2; m_s = +1/2$

**Question 4** **8 points**

Rank the following from 1 to 4 in **increasing** order. Provide the ranking in the boxes provided.

Note: Atomic number (Z) are provided for relevant atoms

a) Increasing (lowest to highest) order of *ionization energy*: (1= lowest; 4 = highest)

Mg

Mg: Z = 12

Rb

Rb: Z = 37

Cs

Cs: Z = 55

K

K : Z = 19

b) Increasing (lowest to highest) order of *bond angle*: (1= lowest; 4 = highest)

H₂OCCl₄XeF₄

HCN

c) Increasing (smallest to largest) order of *ionic size* (1= smallest; 4 = largest)

S²⁻

S: Z = 16

O²⁻

O: Z = 8

Be²⁺

Be: Z = 4

Na⁺

Na: Z = 11

d) Increasing (lowest to highest) order of *bond energy* (1= lowest; 4 = highest)

P-F

P-Cl

P-I

P-Br

**Question 5** **12 points**

a. 9 points For the molecules HClO_4 and TeF_6 , draw the most stable *Lewis structure* showing appropriate shape and *dashed/wedged bonds*, based on VSEPR. Determine the **molecular geometry** around the central atom. For HClO_4 , label the *most polar bond*.



b. 3 points (Note: Lone pairs are not shown in structures below)

(i) For the three *resonance* structures (1, 2 and 3) shown below, answer the following question:



(I) Circle the *most contributing* resonance structure.

Structure 1

Structure 2

Structure 3

(ii) For the three structures given (4, 5, and 6) below, answer the following questions:



(I) Circle all the structures that represent *resonance structures* of each other:

Structure 4

Structure 5

Structure 6

(II) Among the resonance structures, circle the *most contributing* resonance structure.

Structure 4

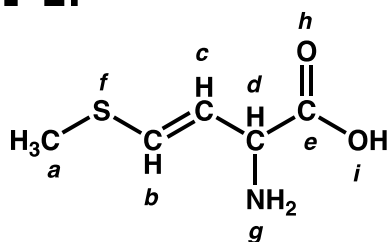
Structure 5

Structure 6

**Question 6** **6 points**

Using the data given below, calculate the **lattice energy** for SrO(s).

	ΔH° (kJ/mol)
$\text{Sr}(s) \longrightarrow \text{Sr}(g)$	+180
$\text{Sr}(g) \longrightarrow \text{Sr}^{2+}(g) + 2e^-$	+1468
$\text{O}_2(g) \longrightarrow 2\text{O}(g)$	+500
$\text{O}(g) + 2e^- \longrightarrow \text{O}^{2-}(g)$	+600
$\text{Sr}(s) + \frac{1}{2}\text{O}_2(g) \longrightarrow \text{SrO}(s)$	-550

**Question 7 10 points**

Consider the molecule given on the left, and answer questions (i) to (vi) using chemical bonding theories that we have learnt in class (valence bond theory, hybridization).

Note that the lone pairs are not shown in the structure:

i. Consider the carbon atoms marked in as ***a***, ***b***, ***c***, ***d***, and ***e*** in the molecule, and determine **hybridization** at each of the carbon centers. (2.5 points)

Atom	Hybridization
Carbon (<i>a</i>)	
Carbon (<i>b</i>)	
Carbon (<i>c</i>)	
Carbon (<i>d</i>)	
Carbon (<i>e</i>)	

ii. Now, consider the sulfur atom (S) marked as ***f***, nitrogen (N) atom (***g***), and oxygen(O) atoms (***h*** and ***i***), and determine their **hybridization**. (2 points)

Atom	Hybridization
Sulfur (<i>f</i>)	
Nitrogen (<i>g</i>)	
Oxygen (<i>h</i>)	
Oxygen (<i>i</i>)	

iii. **Describe** all the bonds formed by the carbon atoms ***a*** and ***b***, using sigma and pi designation as appropriate. (2 points)

Atom	Bonds formed (use sigma and pi as appropriate)
Carbon (<i>a</i>)	
Carbon (<i>b</i>)	

iv. How many total sp^3 -s sigma bonds are in the molecule? (1 point)

v. Determine **electronic** and **molecular geometry** at carbon (C) labeled ***e*** and sulfur (S) labeled ***f*** centres. (2 points)

Atom	Electronic Geometry	Molecular Geometry
Carbon (<i>e</i>)		
Sulfur (<i>f</i>)		

vi. Circle the value from the choices given below, that describes the **C-S-C bond angle at sulfur** (S) marked ***f***: (0.5 points)

$>90^\circ$ to $<109.5^\circ$ 109.5° >109.5 to $<120^\circ$ 120° $>120^\circ$ to $<180^\circ$


Question 8 12 points

Using the **valence shell orbitals only** draw a **complete** molecular orbital (MO) diagram (with no s-p mixing) for the molecule **PS**, in the space provided below. **(6 points)**

Assume that 3s and 3p atomic orbitals will form molecular orbitals like those formed by the combination of 2s and 2p atomic orbitals that we learnt in class, and there is no s-p mixing.

Show **energy arrow**, **atomic orbitals** involved, **atom labels**, the **molecular orbitals** that are formed, **filled electrons** in the MO diagram etc. Make sure to label all the atomic and molecular orbitals.

Using the MO diagram above, answer the following questions:

- (i) Which one of the following will have the highest bond order? Circle the correct choice
 A) PS^{1-} B) PS C) PS^{1+}
- (ii) Which one of the following will have the longest bond length? Circle the correct choice
 A) PS^{1-} B) PS C) PS^{1+}
- (iii) Which of the following will be diamagnetic? Circle the correct choice.
 A) PS^{1-} B) PS C) PS^{1+}
- (iv) Draw Lewis dot structure of the molecule with the shortest bond length (based on the MO diagram) from among the three (PS^{1-} , PS, PS^{1+}), and indicate the bond order calculated from the Lewis dot structure. **(3 points)**

Lewis Dot Structure

Bond Order



A3811DA7-0C71-4B5C-A853-F69E557E4102

midterm-exam-93388

#612 Page 10 of 10

SCRAP PAGE

**CHEMISTRY 110 –Midterm Exam****Thursday November 7, 2024, 6:30 – 8:30 pm****Examiners: Pallavi Sirjoosingh and Ashok Kakkar**

First name (please write as legibly as possible within the boxes)

--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--

Last name

--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--

Student ID number

--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--

This booklet has 8 questions on 10 pages (The last page can be used as a scrap page).

The staple must NOT be removed from the question booklet. Your intact question booklet must be returned at the end of the exam.

A separate booklet contains a periodic table, and datasheet.

Write your answers directly under the questions in the space provided. Do not write on the QR codes or near the top edges (as it may be cut off).

Please write **legibly and dark enough** for the scan to be readable. For students who use English as a second language, dictionaries are permitted. Calculators are allowed.

DO NOT ASK QUESTIONS OF THE INVIGILATORS about interpretations of the exam questions. If you have a concern about a question, please summarize it at the top of this page. All concerns will be evaluated after the exam. GOOD LUCK!

**Question 1 8 points** (*Assume all transitions are possible*)

a. 6 points Calculate the **energy (in kJ/mol)** required to ionize (completely remove) the electron from the **n=3** level of **He⁺ ion** (He: Helium). Show your work including equations. (3 sig. figs.)

b. 2 points

(I) Select the largest (longest) wavelength from the three choices below:

- i. Wavelength of photon to *completely remove* an electron from *1s orbital of Hydrogen atom*
- ii. Wavelength of photon to *completely remove* an electron from *1s orbital of Boron atom (B)*
- iii. Wavelength of photon to *completely remove* an electron from *1s orbital of Boron cation (B⁺)*

i or ii or iii

(II) Select the smallest (shortest) wavelength from the three choices below (*for H atom*):

- i. Wavelength of photon *emitted* for an electron transitioning from *2p to 1s orbital*
- ii. Wavelength of photon *absorbed* for an electron transitioning from *2s to 4p orbital*
- iii. Wavelength of photon *absorbed* for an electron transitioning from *1s to 3p orbital*

i or ii or iii



Question 2 8 points



A Chem110 student plots the graph shown on the left, from data obtained for the kinetic energy (eV) of the ejected electron versus frequency (s^{-1}) of the incident light, for an unknown metal. ($1\text{eV} = 1.6022 \times 10^{-19} \text{ J}$)

Based on the plot, answer the questions **a to c** below:

a. 1 point On the plot above, label the **threshold frequency**.

b. 2 points What is the value of the work potential (in J) of the unknown metal?

c. 3 points What is the *longest* wavelength (in nm) required to eject an electron from this metal's surface? Show your work including equation(s) used. (3 sig figs)

d. 2 points In the space provided below, plot the following

(Assume frequency is *above* threshold frequency):

- i. *Kinetic energy* of an ejected electron versus the *intensity* of the incident light (left)
- ii. *Number of electrons* ejected versus the *frequency* of the incident light (right)



**Question 3** **6 points****a. 2 points**

In the ground state of Ge (Germanium; $Z = 32$) atom, **name** the following orbitals:

i. Orbital containing the electron that requires the maximum energy to be removed.

ii. Orbital containing the electron that experiences the minimum effective nuclear charge.

b. 2 points

An unknown anion (A^-) has the following *excited* electronic configuration: $[Ar] 4s^2 3d^3 4p^1$

(Ar: Argon; $Z = 18$)

Give the **condensed ground state electronic configuration** of the atom, **A** and the cation, **A^{2+}** .
Show your work. (Assume all electronic transitions are possible)

Electronic Configuration of A:

Electronic Configuration of A^{2+} :

c. 2 points

In a single atom, determine how many electrons can have the following set of quantum numbers?

(i) $n = 5; l = 3$

In a single atom, determine how many electrons can have the following set of quantum numbers?

(i) $n = 3; l = 2; m_s = +1/2$

**Question 4** **8 points**

Rank the following from 1 to 4 in **increasing** order. Provide the ranking in the boxes provided.

Note: Atomic number (Z) are provided for relevant atoms

a) Increasing (lowest to highest) order of *ionization energy*: (1= lowest; 4 = highest)

Mg

Mg: Z = 12

Rb

Rb: Z = 37

Cs

Cs: Z = 55

K

K : Z = 19

b) Increasing (lowest to highest) order of *bond angle*: (1= lowest; 4 = highest)

H₂OCCl₄XeF₄

HCN

c) Increasing (smallest to largest) order of *ionic size* (1= smallest; 4 = largest)

S²⁻

S: Z = 16

O²⁻

O: Z = 8

Be²⁺

Be: Z = 4

Na⁺

Na: Z = 11

d) Increasing (lowest to highest) order of *bond energy* (1= lowest; 4 = highest)

P-F

P-Cl

P-I

P-Br

**Question 5** **12 points**

a. 9 points For the molecules HClO_4 and TeF_6 , draw the most stable *Lewis structure* showing appropriate shape and *dashed/wedged bonds*, based on VSEPR. Determine the **molecular geometry** around the central atom. For HClO_4 , label the *most polar bond*.



b. 3 points (Note: Lone pairs are not shown in structures below)

(i) For the three *resonance* structures (1, 2 and 3) shown below, answer the following question:



(I) Circle the *most contributing* resonance structure.

Structure 1

Structure 2

Structure 3

(ii) For the three structures given (4, 5, and 6) below, answer the following questions:



(I) Circle all the structures that represent *resonance structures* of each other:

Structure 4

Structure 5

Structure 6

(II) Among the resonance structures, circle the *most contributing* resonance structure.

Structure 4

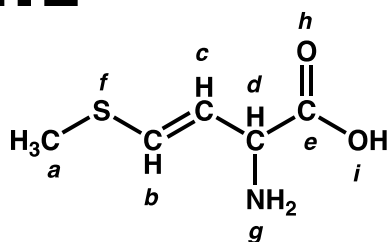
Structure 5

Structure 6

**Question 6** **6 points**

Using the data given below, calculate the **lattice energy** for SrO(s).

	ΔH° (kJ/mol)
$\text{Sr}(s) \longrightarrow \text{Sr}(g)$	+180
$\text{Sr}(g) \longrightarrow \text{Sr}^{2+}(g) + 2e^-$	+1468
$\text{O}_2(g) \longrightarrow 2\text{O}(g)$	+500
$\text{O}(g) + 2e^- \longrightarrow \text{O}^{2-}(g)$	+600
$\text{Sr}(s) + \frac{1}{2}\text{O}_2(g) \longrightarrow \text{SrO}(s)$	-550

**Question 7 10 points**

Consider the molecule given on the left, and answer questions (i) to (vi) using chemical bonding theories that we have learnt in class (valence bond theory, hybridization).

Note that the lone pairs are not shown in the structure:

i. Consider the carbon atoms marked in as ***a***, ***b***, ***c***, ***d***, and ***e*** in the molecule, and determine **hybridization** at each of the carbon centers. (2.5 points)

Atom	Hybridization
Carbon (<i>a</i>)	
Carbon (<i>b</i>)	
Carbon (<i>c</i>)	
Carbon (<i>d</i>)	
Carbon (<i>e</i>)	

ii. Now, consider the sulfur atom (S) marked as ***f***, nitrogen (N) atom (***g***), and oxygen(O) atoms (***h*** and ***i***), and determine their **hybridization**. (2 points)

Atom	Hybridization
Sulfur (<i>f</i>)	
Nitrogen (<i>g</i>)	
Oxygen (<i>h</i>)	
Oxygen (<i>i</i>)	

iii. **Describe** all the bonds formed by the carbon atoms ***a*** and ***b***, using sigma and pi designation as appropriate. (2 points)

Atom	Bonds formed (use sigma and pi as appropriate)
Carbon (<i>a</i>)	
Carbon (<i>b</i>)	

iv. How many total sp^3 -s sigma bonds are in the molecule? (1 point)

v. Determine **electronic** and **molecular geometry** at carbon (C) labeled ***e*** and sulfur (S) labeled ***f*** centres. (2 points)

Atom	Electronic Geometry	Molecular Geometry
Carbon (<i>e</i>)		
Sulfur (<i>f</i>)		

vi. Circle the value from the choices given below, that describes the **C-S-C bond angle at sulfur** (S) marked ***f***: (0.5 points)

>90° to <109.5° 109.5° >109.5 to <120° 120° >120° to <180°


Question 8 12 points

Using the **valence shell orbitals only** draw a **complete** molecular orbital (MO) diagram (with no s-p mixing) for the molecule **PS**, in the space provided below. **(6 points)**

Assume that 3s and 3p atomic orbitals will form molecular orbitals like those formed by the combination of 2s and 2p atomic orbitals that we learnt in class, and there is no s-p mixing.

Show **energy arrow**, **atomic orbitals** involved, **atom labels**, the **molecular orbitals** that are formed, **filled electrons** in the MO diagram etc. Make sure to label all the atomic and molecular orbitals.

Using the MO diagram above, answer the following questions:

- (i) Which one of the following will have the highest bond order? Circle the correct choice
 A) PS^{1-} B) PS C) PS^{1+}
- (ii) Which one of the following will have the longest bond length? Circle the correct choice
 A) PS^{1-} B) PS C) PS^{1+}
- (iii) Which of the following will be diamagnetic? Circle the correct choice.
 A) PS^{1-} B) PS C) PS^{1+}
- (iv) Draw Lewis dot structure of the molecule with the shortest bond length (based on the MO diagram) from among the three (PS^{1-} , PS, PS^{1+}), and indicate the bond order calculated from the Lewis dot structure. **(3 points)**

Lewis Dot Structure

Bond Order



AC83750C-1F86-4DA3-B599-6634DF500C80

midterm-exam-93388

#613 Page 10 of 10

SCRAP PAGE

**CHEMISTRY 110 –Midterm Exam****Thursday November 7, 2024, 6:30 – 8:30 pm****Examiners: Pallavi Sirjoosingh and Ashok Kakkar**

First name (please write as legibly as possible within the boxes)

--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--

Last name

--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--

Student ID number

--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--

This booklet has 8 questions on 10 pages (The last page can be used as a scrap page).

The staple must NOT be removed from the question booklet. Your intact question booklet must be returned at the end of the exam.

A separate booklet contains a periodic table, and datasheet.

Write your answers directly under the questions in the space provided. Do not write on the QR codes or near the top edges (as it may be cut off).

Please write **legibly and dark enough** for the scan to be readable. For students who use English as a second language, dictionaries are permitted. Calculators are allowed.

DO NOT ASK QUESTIONS OF THE INVIGILATORS about interpretations of the exam questions. If you have a concern about a question, please summarize it at the top of this page. All concerns will be evaluated after the exam. GOOD LUCK!

**Question 1 8 points** (*Assume all transitions are possible*)

a. 6 points Calculate the **energy (in kJ/mol)** required to ionize (completely remove) the electron from the **$n=3$** level of **He^+ ion** (He: Helium). Show your work including equations. (3 sig. figs.)

b. 2 points

(I) Select the largest (longest) wavelength from the three choices below:

- i. Wavelength of photon to *completely remove* an electron from *1s orbital of Hydrogen atom*
- ii. Wavelength of photon to *completely remove* an electron from *1s orbital of Boron atom (B)*
- iii. Wavelength of photon to *completely remove* an electron from *1s orbital of Boron cation (B^+)*

i or ii or iii

(II) Select the smallest (shortest) wavelength from the three choices below (*for H atom*):

- i. Wavelength of photon *emitted* for an electron transitioning from *2p to 1s orbital*
- ii. Wavelength of photon *absorbed* for an electron transitioning from *2s to 4p orbital*
- iii. Wavelength of photon *absorbed* for an electron transitioning from *1s to 3p orbital*

i or ii or iii



Question 2 8 points



A Chem110 student plots the graph shown on the left, from data obtained for the kinetic energy (eV) of the ejected electron versus frequency (s^{-1}) of the incident light, for an unknown metal. ($1\text{eV} = 1.6022 \times 10^{-19} \text{ J}$)

Based on the plot, answer the questions **a to c** below:

a. 1 point On the plot above, label the **threshold frequency**.

b. 2 points What is the value of the work potential (in J) of the unknown metal?

c. 3 points What is the *longest* wavelength (in nm) required to eject an electron from this metal's surface? Show your work including equation(s) used. (3 sig figs)

d. 2 points In the space provided below, plot the following

(Assume frequency is *above* threshold frequency):

- i. *Kinetic energy* of an ejected electron versus the *intensity* of the incident light (left)
- ii. *Number of electrons* ejected versus the *frequency* of the incident light (right)



**Question 3** **6 points****a. 2 points**

In the ground state of Ge (Germanium; $Z = 32$) atom, **name** the following orbitals:

i. Orbital containing the electron that requires the maximum energy to be removed.

ii. Orbital containing the electron that experiences the minimum effective nuclear charge.

b. 2 points

An unknown anion (A^-) has the following *excited* electronic configuration: $[Ar] 4s^2 3d^3 4p^1$

(Ar: Argon; $Z = 18$)

Give the **condensed ground state electronic configuration** of the atom, **A** and the cation, **A^{2+}** .
Show your work. (Assume all electronic transitions are possible)

Electronic Configuration of A:

Electronic Configuration of A^{2+} :

c. 2 points

In a single atom, determine how many electrons can have the following set of quantum numbers?

(i) $n = 5; l = 3$

In a single atom, determine how many electrons can have the following set of quantum numbers?

(i) $n = 3; l = 2; m_s = +1/2$

**Question 4** **8 points**

Rank the following from 1 to 4 in **increasing** order. Provide the ranking in the boxes provided.

Note: Atomic number (Z) are provided for relevant atoms

a) Increasing (lowest to highest) order of *ionization energy*: (1= lowest; 4 = highest)

Mg

Mg: Z = 12

Rb

Rb: Z = 37

Cs

Cs: Z = 55

K

K : Z = 19

b) Increasing (lowest to highest) order of *bond angle*: (1= lowest; 4 = highest)

H₂OCCl₄XeF₄

HCN

c) Increasing (smallest to largest) order of *ionic size* (1= smallest; 4 = largest)

S²⁻

S: Z = 16

O²⁻

O: Z = 8

Be²⁺

Be: Z = 4

Na⁺

Na: Z = 11

d) Increasing (lowest to highest) order of *bond energy* (1= lowest; 4 = highest)

P-F

P-Cl

P-I

P-Br

**Question 5** **12 points**

a. 9 points For the molecules HClO_4 and TeF_6 , draw the most stable *Lewis structure* showing appropriate shape and *dashed/wedged bonds*, based on VSEPR. Determine the **molecular geometry** around the central atom. For HClO_4 , label the *most polar bond*.



b. 3 points (Note: Lone pairs are not shown in structures below)

(i) For the three *resonance* structures (1, 2 and 3) shown below, answer the following question:



(I) Circle the *most contributing* resonance structure.

Structure 1

Structure 2

Structure 3

(ii) For the three structures given (4, 5, and 6) below, answer the following questions:



(I) Circle all the structures that represent *resonance structures* of each other:

Structure 4

Structure 5

Structure 6

(II) Among the resonance structures, circle the *most contributing* resonance structure.

Structure 4

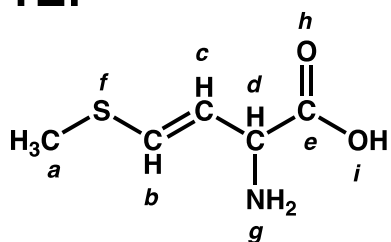
Structure 5

Structure 6

**Question 6** **6 points**

Using the data given below, calculate the **lattice energy** for SrO(s).

	ΔH° (kJ/mol)
$\text{Sr}(s) \longrightarrow \text{Sr}(g)$	+180
$\text{Sr}(g) \longrightarrow \text{Sr}^{2+}(g) + 2e^-$	+1468
$\text{O}_2(g) \longrightarrow 2\text{O}(g)$	+500
$\text{O}(g) + 2e^- \longrightarrow \text{O}^{2-}(g)$	+600
$\text{Sr}(s) + \frac{1}{2}\text{O}_2(g) \longrightarrow \text{SrO}(s)$	-550

**Question 7 10 points**

Consider the molecule given on the left, and answer questions (i) to (vi) using chemical bonding theories that we have learnt in class (valence bond theory, hybridization).

Note that the lone pairs are not shown in the structure:

i. Consider the carbon atoms marked in as ***a***, ***b***, ***c***, ***d***, and ***e*** in the molecule, and determine **hybridization** at each of the carbon centers. (2.5 points)

Atom	Hybridization
Carbon (<i>a</i>)	
Carbon (<i>b</i>)	
Carbon (<i>c</i>)	
Carbon (<i>d</i>)	
Carbon (<i>e</i>)	

ii. Now, consider the sulfur atom (S) marked as ***f***, nitrogen (N) atom (***g***), and oxygen(O) atoms (***h*** and ***i***), and determine their **hybridization**. (2 points)

Atom	Hybridization
Sulfur (<i>f</i>)	
Nitrogen (<i>g</i>)	
Oxygen (<i>h</i>)	
Oxygen (<i>i</i>)	

iii. **Describe** all the bonds formed by the carbon atoms ***a*** and ***b***, using sigma and pi designation as appropriate. (2 points)

Atom	Bonds formed (use sigma and pi as appropriate)
Carbon (<i>a</i>)	
Carbon (<i>b</i>)	

iv. How many total sp^3 -s sigma bonds are in the molecule? (1 point)

v. Determine **electronic** and **molecular geometry** at carbon (C) labeled ***e*** and sulfur (S) labeled ***f*** centres. (2 points)

Atom	Electronic Geometry	Molecular Geometry
Carbon (<i>e</i>)		
Sulfur (<i>f</i>)		

vi. Circle the value from the choices given below, that describes the **C-S-C bond angle at sulfur** (S) marked ***f***: (0.5 points)

>90° to <109.5° 109.5° >109.5 to <120° 120° >120° to <180°


Question 8 12 points

Using the **valence shell orbitals only** draw a **complete** molecular orbital (MO) diagram (with no s-p mixing) for the molecule **PS**, in the space provided below. **(6 points)**

Assume that 3s and 3p atomic orbitals will form molecular orbitals like those formed by the combination of 2s and 2p atomic orbitals that we learnt in class, and there is no s-p mixing.

Show **energy arrow**, **atomic orbitals** involved, **atom labels**, the **molecular orbitals** that are formed, **filled electrons** in the MO diagram etc. Make sure to label all the atomic and molecular orbitals.

Using the MO diagram above, answer the following questions:

- (i) Which one of the following will have the highest bond order? Circle the correct choice
 A) PS^{1-} B) PS C) PS^{1+}
- (ii) Which one of the following will have the longest bond length? Circle the correct choice
 A) PS^{1-} B) PS C) PS^{1+}
- (iii) Which of the following will be diamagnetic? Circle the correct choice.
 A) PS^{1-} B) PS C) PS^{1+}
- (iv) Draw Lewis dot structure of the molecule with the shortest bond length (based on the MO diagram) from among the three (PS^{1-} , PS, PS^{1+}), and indicate the bond order calculated from the Lewis dot structure. **(3 points)**

Lewis Dot Structure

Bond Order



1F3E8D34-C781-4916-A061-AFB9D0CE3E25

midterm-exam-93388

#614 Page 10 of 10

SCRAP PAGE

**CHEMISTRY 110 –Midterm Exam****Thursday November 7, 2024, 6:30 – 8:30 pm****Examiners: Pallavi Sirjoosingh and Ashok Kakkar**

First name (please write as legibly as possible within the boxes)

--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--

Last name

--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--

Student ID number

--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--

This booklet has 8 questions on 10 pages (The last page can be used as a scrap page).

The staple must NOT be removed from the question booklet. Your intact question booklet must be returned at the end of the exam.

A separate booklet contains a periodic table, and datasheet.

Write your answers directly under the questions in the space provided. Do not write on the QR codes or near the top edges (as it may be cut off).

Please write **legibly and dark enough** for the scan to be readable. For students who use English as a second language, dictionaries are permitted. Calculators are allowed.

DO NOT ASK QUESTIONS OF THE INVIGILATORS about interpretations of the exam questions. If you have a concern about a question, please summarize it at the top of this page. All concerns will be evaluated after the exam. GOOD LUCK!

**Question 1 8 points** (*Assume all transitions are possible*)

a. 6 points Calculate the **energy (in kJ/mol)** required to ionize (completely remove) the electron from the **n=3** level of **He⁺ ion** (He: Helium). Show your work including equations. (3 sig. figs.)

b. 2 points

(I) Select the largest (longest) wavelength from the three choices below:

- i. Wavelength of photon to *completely remove* an electron from *1s orbital of Hydrogen atom*
- ii. Wavelength of photon to *completely remove* an electron from *1s orbital of Boron atom (B)*
- iii. Wavelength of photon to *completely remove* an electron from *1s orbital of Boron cation (B⁺)*

i or ii or iii

(II) Select the smallest (shortest) wavelength from the three choices below (*for H atom*):

- i. Wavelength of photon *emitted* for an electron transitioning from *2p to 1s orbital*
- ii. Wavelength of photon *absorbed* for an electron transitioning from *2s to 4p orbital*
- iii. Wavelength of photon *absorbed* for an electron transitioning from *1s to 3p orbital*

i or ii or iii



Question 2 8 points



A Chem110 student plots the graph shown on the left, from data obtained for the kinetic energy (eV) of the ejected electron versus frequency (s^{-1}) of the incident light, for an unknown metal. ($1\text{eV} = 1.6022 \times 10^{-19} \text{ J}$)

Based on the plot, answer the questions **a to c** below:

a. 1 point On the plot above, label the **threshold frequency**.

b. 2 points What is the value of the work potential (in J) of the unknown metal?

c. 3 points What is the *longest* wavelength (in nm) required to eject an electron from this metal's surface? Show your work including equation(s) used. (3 sig figs)

d. 2 points In the space provided below, plot the following

(Assume frequency is *above* threshold frequency):

- Kinetic energy* of an ejected electron versus the *intensity* of the incident light (left)
- Number of electrons* ejected versus the *frequency* of the incident light (right)



**Question 3** **6 points****a. 2 points**

In the ground state of Ge (Germanium; $Z = 32$) atom, **name** the following orbitals:

i. Orbital containing the electron that requires the maximum energy to be removed.

ii. Orbital containing the electron that experiences the minimum effective nuclear charge.

b. 2 points

An unknown anion (A^-) has the following *excited* electronic configuration: $[Ar] 4s^2 3d^3 4p^1$

(Ar: Argon; $Z = 18$)

Give the **condensed ground state electronic configuration** of the atom, **A** and the cation, **A^{2+}** .
Show your work. (Assume all electronic transitions are possible)

Electronic Configuration of A:

Electronic Configuration of A^{2+} :

c. 2 points

In a single atom, determine how many electrons can have the following set of quantum numbers?

(i) $n = 5; l = 3$

In a single atom, determine how many electrons can have the following set of quantum numbers?

(i) $n = 3; l = 2; m_s = +1/2$

**Question 4** **8 points**

Rank the following from 1 to 4 in **increasing** order. Provide the ranking in the boxes provided.

Note: Atomic number (Z) are provided for relevant atoms

a) Increasing (lowest to highest) order of *ionization energy*: (1= lowest; 4 = highest)

Mg

Mg: Z = 12

Rb

Rb: Z = 37

Cs

Cs: Z = 55

K

K : Z = 19

b) Increasing (lowest to highest) order of *bond angle*: (1= lowest; 4 = highest)

H₂OCCl₄XeF₄

HCN

c) Increasing (smallest to largest) order of *ionic size* (1= smallest; 4 = largest)

S²⁻

S: Z = 16

O²⁻

O: Z = 8

Be²⁺

Be: Z = 4

Na⁺

Na: Z = 11

d) Increasing (lowest to highest) order of *bond energy* (1= lowest; 4 = highest)

P-F

P-Cl

P-I

P-Br

**Question 5** **12 points**

a. 9 points For the molecules HClO_4 and TeF_6 , draw the most stable *Lewis structure* showing appropriate shape and *dashed/wedged bonds*, based on VSEPR. Determine the **molecular geometry** around the central atom. For HClO_4 , label the *most polar bond*.



b. 3 points (Note: Lone pairs are not shown in structures below)

(i) For the three *resonance* structures (1, 2 and 3) shown below, answer the following question:



(I) Circle the *most contributing* resonance structure.

Structure 1

Structure 2

Structure 3

(ii) For the three structures given (4, 5, and 6) below, answer the following questions:



(I) Circle all the structures that represent *resonance structures* of each other:

Structure 4

Structure 5

Structure 6

(II) Among the resonance structures, circle the *most contributing* resonance structure.

Structure 4

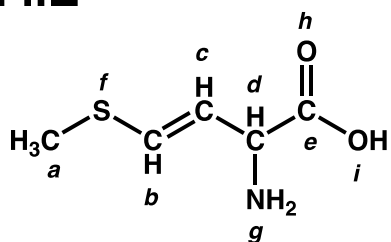
Structure 5

Structure 6

**Question 6** **6 points**

Using the data given below, calculate the **lattice energy** for SrO(s).

	ΔH° (kJ/mol)
$\text{Sr}(s) \longrightarrow \text{Sr}(g)$	+180
$\text{Sr}(g) \longrightarrow \text{Sr}^{2+}(g) + 2e^-$	+1468
$\text{O}_2(g) \longrightarrow 2\text{O}(g)$	+500
$\text{O}(g) + 2e^- \longrightarrow \text{O}^{2-}(g)$	+600
$\text{Sr}(s) + \frac{1}{2}\text{O}_2(g) \longrightarrow \text{SrO}(s)$	-550

**Question 7 10 points**

Consider the molecule given on the left, and answer questions (i) to (vi) using chemical bonding theories that we have learnt in class (valence bond theory, hybridization).

Note that the lone pairs are not shown in the structure:

i. Consider the carbon atoms marked in as ***a***, ***b***, ***c***, ***d***, and ***e*** in the molecule, and determine **hybridization** at each of the carbon centers. (2.5 points)

Atom	Hybridization
Carbon (<i>a</i>)	
Carbon (<i>b</i>)	
Carbon (<i>c</i>)	
Carbon (<i>d</i>)	
Carbon (<i>e</i>)	

ii. Now, consider the sulfur atom (S) marked as ***f***, nitrogen (N) atom (***g***), and oxygen(O) atoms (***h*** and ***i***), and determine their **hybridization**. (2 points)

Atom	Hybridization
Sulfur (<i>f</i>)	
Nitrogen (<i>g</i>)	
Oxygen (<i>h</i>)	
Oxygen (<i>i</i>)	

iii. **Describe** all the bonds formed by the carbon atoms ***a*** and ***b***, using sigma and pi designation as appropriate. (2 points)

Atom	Bonds formed (use sigma and pi as appropriate)
Carbon (<i>a</i>)	
Carbon (<i>b</i>)	

iv. How many total sp^3 -s sigma bonds are in the molecule? (1 point)

v. Determine **electronic** and **molecular geometry** at carbon (C) labeled ***e*** and sulfur (S) labeled ***f*** centres. (2 points)

Atom	Electronic Geometry	Molecular Geometry
Carbon (<i>e</i>)		
Sulfur (<i>f</i>)		

vi. Circle the value from the choices given below, that describes the **C-S-C bond angle at sulfur** (S) marked ***f***: (0.5 points)

$>90^\circ$ to $<109.5^\circ$ 109.5° >109.5 to $<120^\circ$ 120° $>120^\circ$ to $<180^\circ$

**Question 8 12 points**

Using the **valence shell orbitals only** draw a **complete** molecular orbital (MO) diagram (with no s-p mixing) for the molecule **PS**, in the space provided below. **(6 points)**

Assume that 3s and 3p atomic orbitals will form molecular orbitals like those formed by the combination of 2s and 2p atomic orbitals that we learnt in class, and there is no s-p mixing.

Show **energy arrow**, **atomic orbitals** involved, **atom labels**, the **molecular orbitals** that are formed, **filled electrons** in the MO diagram etc. Make sure to label all the atomic and molecular orbitals.

Using the MO diagram above, answer the following questions:

- (i) Which one of the following will have the highest bond order? Circle the correct choice
A) PS^{1-} B) PS C) PS^{1+}
- (ii) Which one of the following will have the longest bond length? Circle the correct choice
A) PS^{1-} B) PS C) PS^{1+}
- (iii) Which of the following will be diamagnetic? Circle the correct choice.
A) PS^{1-} B) PS C) PS^{1+}
- (iv) Draw Lewis dot structure of the molecule with the shortest bond length (based on the MO diagram) from among the three (PS^{1-} , PS, PS^{1+}), and indicate the bond order calculated from the Lewis dot structure. **(3 points)**

Lewis Dot Structure

Bond Order



21421E3B-7C38-4766-BCD8-A1F8A29508BE

midterm-exam-93388

#615 Page 10 of 10

SCRAP PAGE

**CHEMISTRY 110 –Midterm Exam****Thursday November 7, 2024, 6:30 – 8:30 pm**

Examiners: Pallavi Sirjoosingh and Ashok Kakkar

First name (please write as legibly as possible within the boxes)

--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--

Last name

--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--

Student ID number

--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--

This booklet has 8 questions on 10 pages (The last page can be used as a scrap page).

The staple must NOT be removed from the question booklet. Your intact question booklet must be returned at the end of the exam.

A separate booklet contains a periodic table, and datasheet.

Write your answers directly under the questions in the space provided. Do not write on the QR codes or near the top edges (as it may be cut off).

Please write **legibly and dark enough** for the scan to be readable. For students who use English as a second language, dictionaries are permitted. Calculators are allowed.

DO NOT ASK QUESTIONS OF THE INVIGILATORS about interpretations of the exam questions. If you have a concern about a question, please summarize it at the top of this page. All concerns will be evaluated after the exam. GOOD LUCK!

**Question 1 8 points** (*Assume all transitions are possible*)

a. 6 points Calculate the **energy (in kJ/mol)** required to ionize (completely remove) the electron from the **n=3** level of **He⁺ ion** (He: Helium). Show your work including equations. (3 sig. figs.)

b. 2 points

(I) Select the largest (longest) wavelength from the three choices below:

- i. Wavelength of photon to *completely remove* an electron from *1s orbital of Hydrogen atom*
- ii. Wavelength of photon to *completely remove* an electron from *1s orbital of Boron atom (B)*
- iii. Wavelength of photon to *completely remove* an electron from *1s orbital of Boron cation (B⁺)*

i or ii or iii

(II) Select the smallest (shortest) wavelength from the three choices below (*for H atom*):

- i. Wavelength of photon *emitted* for an electron transitioning from *2p to 1s orbital*
- ii. Wavelength of photon *absorbed* for an electron transitioning from *2s to 4p orbital*
- iii. Wavelength of photon *absorbed* for an electron transitioning from *1s to 3p orbital*

i or ii or iii



Question 2 8 points



A Chem110 student plots the graph shown on the left, from data obtained for the kinetic energy (eV) of the ejected electron versus frequency (s^{-1}) of the incident light, for an unknown metal. ($1\text{eV} = 1.6022 \times 10^{-19} \text{ J}$)

Based on the plot, answer the questions **a to c** below:

a. 1 point On the plot above, label the **threshold frequency**.

b. 2 points What is the value of the work potential (in J) of the unknown metal?

c. 3 points What is the *longest* wavelength (in nm) required to eject an electron from this metal's surface? Show your work including equation(s) used. (3 sig figs)

d. 2 points In the space provided below, plot the following

(Assume frequency is *above* threshold frequency):

- i. *Kinetic energy* of an ejected electron versus the *intensity* of the incident light (left)
- ii. *Number of electrons* ejected versus the *frequency* of the incident light (right)



**Question 3** **6 points****a. 2 points**

In the ground state of Ge (Germanium; $Z = 32$) atom, **name** the following orbitals:

i. Orbital containing the electron that requires the maximum energy to be removed.

ii. Orbital containing the electron that experiences the minimum effective nuclear charge.

b. 2 points

An unknown anion (A^-) has the following *excited* electronic configuration: $[Ar] 4s^2 3d^3 4p^1$

(Ar: Argon; $Z = 18$)

Give the **condensed ground state electronic configuration** of the atom, **A** and the cation, **A^{2+}** .
Show your work. (Assume all electronic transitions are possible)

Electronic Configuration of A:

Electronic Configuration of A^{2+} :

c. 2 points

In a single atom, determine how many electrons can have the following set of quantum numbers?

(i) $n = 5; l = 3$

In a single atom, determine how many electrons can have the following set of quantum numbers?

(i) $n = 3; l = 2; m_s = +1/2$

**Question 4** **8 points**

Rank the following from 1 to 4 in **increasing** order. Provide the ranking in the boxes provided.

Note: Atomic number (Z) are provided for relevant atoms

a) Increasing (lowest to highest) order of *ionization energy*: (1= lowest; 4 = highest)

Mg

Mg: Z = 12

Rb

Rb: Z = 37

Cs

Cs: Z = 55

K

K : Z = 19

b) Increasing (lowest to highest) order of *bond angle*: (1= lowest; 4 = highest)

H₂OCCl₄XeF₄

HCN

c) Increasing (smallest to largest) order of *ionic size* (1= smallest; 4 = largest)

S²⁻

S: Z = 16

O²⁻

O: Z = 8

Be²⁺

Be: Z = 4

Na⁺

Na: Z = 11

d) Increasing (lowest to highest) order of *bond energy* (1= lowest; 4 = highest)

P-F

P-Cl

P-I

P-Br

**Question 5** **12 points**

a. 9 points For the molecules HClO_4 and TeF_6 , draw the most stable *Lewis structure* showing appropriate shape and *dashed/wedged bonds*, based on VSEPR. Determine the **molecular geometry** around the central atom. For HClO_4 , label the *most polar bond*.



b. 3 points (Note: Lone pairs are not shown in structures below)

(i) For the three *resonance* structures (1, 2 and 3) shown below, answer the following question:



(I) Circle the *most contributing* resonance structure.

Structure 1

Structure 2

Structure 3

(ii) For the three structures given (4, 5, and 6) below, answer the following questions:



(I) Circle all the structures that represent *resonance structures* of each other:

Structure 4

Structure 5

Structure 6

(II) Among the resonance structures, circle the *most contributing* resonance structure.

Structure 4

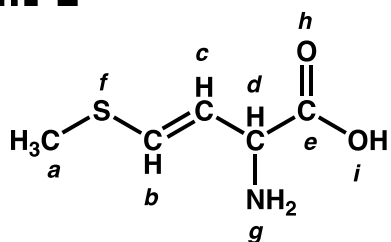
Structure 5

Structure 6

**Question 6** **6 points**

Using the data given below, calculate the **lattice energy** for SrO(s).

	ΔH° (kJ/mol)
$\text{Sr}(s) \longrightarrow \text{Sr}(g)$	+180
$\text{Sr}(g) \longrightarrow \text{Sr}^{2+}(g) + 2e^-$	+1468
$\text{O}_2(g) \longrightarrow 2\text{O}(g)$	+500
$\text{O}(g) + 2e^- \longrightarrow \text{O}^{2-}(g)$	+600
$\text{Sr}(s) + \frac{1}{2}\text{O}_2(g) \longrightarrow \text{SrO}(s)$	-550

**Question 7 10 points**

Consider the molecule given on the left, and answer questions (i) to (vi) using chemical bonding theories that we have learnt in class (valence bond theory, hybridization).

Note that the lone pairs are not shown in the structure:

i. Consider the carbon atoms marked in as ***a***, ***b***, ***c***, ***d***, and ***e*** in the molecule, and determine **hybridization** at each of the carbon centers. (2.5 points)

Atom	Hybridization
Carbon (<i>a</i>)	
Carbon (<i>b</i>)	
Carbon (<i>c</i>)	
Carbon (<i>d</i>)	
Carbon (<i>e</i>)	

ii. Now, consider the sulfur atom (S) marked as ***f***, nitrogen (N) atom (***g***), and oxygen(O) atoms (***h*** and ***i***), and determine their **hybridization**. (2 points)

Atom	Hybridization
Sulfur (<i>f</i>)	
Nitrogen (<i>g</i>)	
Oxygen (<i>h</i>)	
Oxygen (<i>i</i>)	

iii. **Describe** all the bonds formed by the carbon atoms ***a*** and ***b***, using sigma and pi designation as appropriate. (2 points)

Atom	Bonds formed (use sigma and pi as appropriate)
Carbon (<i>a</i>)	
Carbon (<i>b</i>)	

iv. How many total sp^3 -s sigma bonds are in the molecule? (1 point)

v. Determine **electronic** and **molecular geometry** at carbon (C) labeled ***e*** and sulfur (S) labeled ***f*** centres. (2 points)

Atom	Electronic Geometry	Molecular Geometry
Carbon (<i>e</i>)		
Sulfur (<i>f</i>)		

vi. Circle the value from the choices given below, that describes the **C-S-C bond angle at sulfur** (S) marked ***f***: (0.5 points)

>90° to <109.5° 109.5° >109.5 to <120° 120° >120° to <180°


Question 8 12 points

Using the **valence shell orbitals only** draw a **complete** molecular orbital (MO) diagram (with no s-p mixing) for the molecule **PS**, in the space provided below. **(6 points)**

Assume that 3s and 3p atomic orbitals will form molecular orbitals like those formed by the combination of 2s and 2p atomic orbitals that we learnt in class, and there is no s-p mixing.

Show **energy arrow**, **atomic orbitals** involved, **atom labels**, the **molecular orbitals** that are formed, **filled electrons** in the MO diagram etc. Make sure to label all the atomic and molecular orbitals.

Using the MO diagram above, answer the following questions:

- (i) Which one of the following will have the highest bond order? Circle the correct choice
 A) PS^{1-} B) PS C) PS^{1+}
- (ii) Which one of the following will have the longest bond length? Circle the correct choice
 A) PS^{1-} B) PS C) PS^{1+}
- (iii) Which of the following will be diamagnetic? Circle the correct choice.
 A) PS^{1-} B) PS C) PS^{1+}
- (iv) Draw Lewis dot structure of the molecule with the shortest bond length (based on the MO diagram) from among the three (PS^{1-} , PS, PS^{1+}), and indicate the bond order calculated from the Lewis dot structure. **(3 points)**

Lewis Dot Structure

Bond Order



EA6440FD-4BFE-4229-95AE-71E78C44EE2F

midterm-exam-93388

#616 Page 10 of 10

SCRAP PAGE

**CHEMISTRY 110 –Midterm Exam****Thursday November 7, 2024, 6:30 – 8:30 pm****Examiners: Pallavi Sirjoosingh and Ashok Kakkar**

First name (please write as legibly as possible within the boxes)

--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--

Last name

--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--

Student ID number

--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--

This booklet has 8 questions on 10 pages (The last page can be used as a scrap page).

The staple must NOT be removed from the question booklet. Your intact question booklet must be returned at the end of the exam.

A separate booklet contains a periodic table, and datasheet.

Write your answers directly under the questions in the space provided. Do not write on the QR codes or near the top edges (as it may be cut off).

Please write **legibly and dark enough** for the scan to be readable. For students who use English as a second language, dictionaries are permitted. Calculators are allowed.

DO NOT ASK QUESTIONS OF THE INVIGILATORS about interpretations of the exam questions. If you have a concern about a question, please summarize it at the top of this page. All concerns will be evaluated after the exam. GOOD LUCK!

**Question 1 8 points** (*Assume all transitions are possible*)

a. 6 points Calculate the **energy (in kJ/mol)** required to ionize (completely remove) the electron from the **n=3** level of **He⁺ ion** (He: Helium). Show your work including equations. (3 sig. figs.)

b. 2 points

(I) Select the largest (longest) wavelength from the three choices below:

- i. Wavelength of photon to *completely remove* an electron from *1s orbital of Hydrogen atom*
- ii. Wavelength of photon to *completely remove* an electron from *1s orbital of Boron atom (B)*
- iii. Wavelength of photon to *completely remove* an electron from *1s orbital of Boron cation (B⁺)*

i or ii or iii

(II) Select the smallest (shortest) wavelength from the three choices below (*for H atom*):

- i. Wavelength of photon *emitted* for an electron transitioning from *2p to 1s orbital*
- ii. Wavelength of photon *absorbed* for an electron transitioning from *2s to 4p orbital*
- iii. Wavelength of photon *absorbed* for an electron transitioning from *1s to 3p orbital*

i or ii or iii



Question 2 8 points



A Chem110 student plots the graph shown on the left, from data obtained for the kinetic energy (eV) of the ejected electron versus frequency (s^{-1}) of the incident light, for an unknown metal. ($1\text{eV} = 1.6022 \times 10^{-19} \text{ J}$)

Based on the plot, answer the questions **a to c** below:

a. 1 point On the plot above, label the **threshold frequency**.

b. 2 points What is the value of the work potential (in J) of the unknown metal?

c. 3 points What is the *longest* wavelength (in nm) required to eject an electron from this metal's surface? Show your work including equation(s) used. (3 sig figs)

d. 2 points In the space provided below, plot the following

(Assume frequency is *above* threshold frequency):

- i. *Kinetic energy* of an ejected electron versus the *intensity* of the incident light (left)
- ii. *Number of electrons* ejected versus the *frequency* of the incident light (right)



**Question 3** **6 points****a. 2 points**

In the ground state of Ge (Germanium; $Z = 32$) atom, **name** the following orbitals:

i. Orbital containing the electron that requires the maximum energy to be removed.

ii. Orbital containing the electron that experiences the minimum effective nuclear charge.

b. 2 points

An unknown anion (A^-) has the following *excited* electronic configuration: $[Ar] 4s^2 3d^3 4p^1$

(Ar: Argon; $Z = 18$)

Give the **condensed ground state electronic configuration** of the atom, **A** and the cation, **A^{2+}** .
Show your work. (Assume all electronic transitions are possible)

Electronic Configuration of A:

Electronic Configuration of A^{2+} :

c. 2 points

In a single atom, determine how many electrons can have the following set of quantum numbers?

(i) $n = 5; l = 3$

In a single atom, determine how many electrons can have the following set of quantum numbers?

(i) $n = 3; l = 2; m_s = +1/2$

**Question 4** **8 points**

Rank the following from 1 to 4 in **increasing** order. Provide the ranking in the boxes provided.

Note: Atomic number (Z) are provided for relevant atoms

a) Increasing (lowest to highest) order of *ionization energy*: (1= lowest; 4 = highest)

Mg

Mg: Z = 12

Rb

Rb: Z = 37

Cs

Cs: Z = 55

K

K : Z = 19

b) Increasing (lowest to highest) order of *bond angle*: (1= lowest; 4 = highest)

H₂OCCl₄XeF₄

HCN

c) Increasing (smallest to largest) order of *ionic size* (1= smallest; 4 = largest)

S²⁻

S: Z = 16

O²⁻

O: Z = 8

Be²⁺

Be: Z = 4

Na⁺

Na: Z = 11

d) Increasing (lowest to highest) order of *bond energy* (1= lowest; 4 = highest)

P-F

P-Cl

P-I

P-Br

**Question 5** **12 points**

a. 9 points For the molecules HClO_4 and TeF_6 , draw the most stable *Lewis structure* showing appropriate shape and *dashed/wedged bonds*, based on VSEPR. Determine the **molecular geometry** around the central atom. For HClO_4 , label the *most polar bond*.

HClO_4 (there are no O-O bonds)

Label the most polar bond

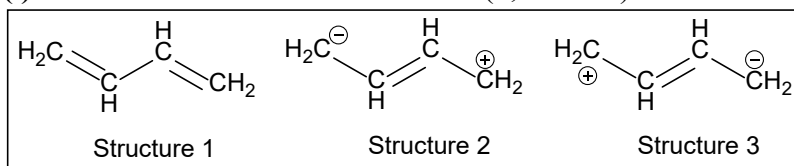
Molecular Geometry around Cl

TeF_6

Molecular Geometry around Te

b. 3 points (Note: Lone pairs are not shown in structures below)

(i) For the three *resonance* structures (1, 2 and 3) shown below, answer the following question:



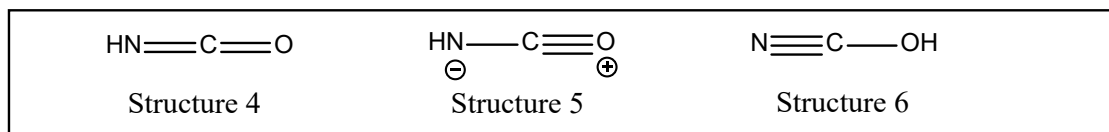
(I) Circle the *most contributing* resonance structure.

Structure 1

Structure 2

Structure 3

(ii) For the three structures given (4, 5, and 6) below, answer the following questions:



(I) Circle all the structures that represent *resonance structures* of each other:

Structure 4

Structure 5

Structure 6

(II) Among the resonance structures, circle the *most contributing* resonance structure.

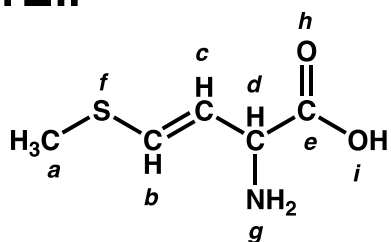
Structure 4

Structure 5

Structure 6

**Question 6** **6 points**Using the data given below, calculate the **lattice energy** for SrO(s).

	ΔH° (kJ/mol)
$\text{Sr}(s) \longrightarrow \text{Sr}(g)$	+180
$\text{Sr}(g) \longrightarrow \text{Sr}^{2+}(g) + 2e^-$	+1468
$\text{O}_2(g) \longrightarrow 2\text{O}(g)$	+500
$\text{O}(g) + 2e^- \longrightarrow \text{O}^{2-}(g)$	+600
$\text{Sr}(s) + \frac{1}{2}\text{O}_2(g) \longrightarrow \text{SrO}(s)$	-550

**Question 7 10 points**

Consider the molecule given on the left, and answer questions (i) to (vi) using chemical bonding theories that we have learnt in class (valence bond theory, hybridization).

Note that the lone pairs are not shown in the structure:

i. Consider the carbon atoms marked in as ***a***, ***b***, ***c***, ***d***, and ***e*** in the molecule, and determine **hybridization** at each of the carbon centers. (2.5 points)

Atom	Hybridization
Carbon (<i>a</i>)	
Carbon (<i>b</i>)	
Carbon (<i>c</i>)	
Carbon (<i>d</i>)	
Carbon (<i>e</i>)	

ii. Now, consider the sulfur atom (S) marked as ***f***, nitrogen (N) atom (***g***), and oxygen(O) atoms (***h*** and ***i***), and determine their **hybridization**. (2 points)

Atom	Hybridization
Sulfur (<i>f</i>)	
Nitrogen (<i>g</i>)	
Oxygen (<i>h</i>)	
Oxygen (<i>i</i>)	

iii. **Describe** all the bonds formed by the carbon atoms ***a*** and ***b***, using sigma and pi designation as appropriate. (2 points)

Atom	Bonds formed (use sigma and pi as appropriate)
Carbon (<i>a</i>)	
Carbon (<i>b</i>)	

iv. How many total sp^3 -s sigma bonds are in the molecule? (1 point)

v. Determine **electronic** and **molecular geometry** at carbon (C) labeled ***e*** and sulfur (S) labeled ***f*** centres. (2 points)

Atom	Electronic Geometry	Molecular Geometry
Carbon (<i>e</i>)		
Sulfur (<i>f</i>)		

vi. Circle the value from the choices given below, that describes the **C-S-C bond angle at sulfur** (S) marked ***f***: (0.5 points)

>90° to <109.5° 109.5° >109.5 to <120° 120° >120° to <180°


Question 8 12 points

Using the **valence shell orbitals only** draw a **complete** molecular orbital (MO) diagram (with no s-p mixing) for the molecule **PS**, in the space provided below. **(6 points)**

Assume that 3s and 3p atomic orbitals will form molecular orbitals like those formed by the combination of 2s and 2p atomic orbitals that we learnt in class, and there is no s-p mixing.

Show **energy arrow**, **atomic orbitals** involved, **atom labels**, the **molecular orbitals** that are formed, **filled electrons** in the MO diagram etc. Make sure to label all the atomic and molecular orbitals.

Using the MO diagram above, answer the following questions:

- (i) Which one of the following will have the highest bond order? Circle the correct choice
 A) PS^{1-} B) PS C) PS^{1+}
- (ii) Which one of the following will have the longest bond length? Circle the correct choice
 A) PS^{1-} B) PS C) PS^{1+}
- (iii) Which of the following will be diamagnetic? Circle the correct choice.
 A) PS^{1-} B) PS C) PS^{1+}
- (iv) Draw Lewis dot structure of the molecule with the shortest bond length (based on the MO diagram) from among the three (PS^{1-} , PS, PS^{1+}), and indicate the bond order calculated from the Lewis dot structure. **(3 points)**

Lewis Dot Structure

Bond Order



B92A5474-1004-485C-860B-4424676613EF

midterm-exam-93388

#617 Page 10 of 10

SCRAP PAGE

**CHEMISTRY 110 –Midterm Exam****Thursday November 7, 2024, 6:30 – 8:30 pm**

Examiners: Pallavi Sirjoosingh and Ashok Kakkar

First name (please write as legibly as possible within the boxes)

--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--

Last name

--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--

Student ID number

--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--

This booklet has 8 questions on 10 pages (The last page can be used as a scrap page).

The staple must NOT be removed from the question booklet. Your intact question booklet must be returned at the end of the exam.

A separate booklet contains a periodic table, and datasheet.

Write your answers directly under the questions in the space provided. Do not write on the QR codes or near the top edges (as it may be cut off).

Please write **legibly and dark enough** for the scan to be readable. For students who use English as a second language, dictionaries are permitted. Calculators are allowed.

DO NOT ASK QUESTIONS OF THE INVIGILATORS about interpretations of the exam questions. If you have a concern about a question, please summarize it at the top of this page. All concerns will be evaluated after the exam. GOOD LUCK!

**Question 1 8 points** (*Assume all transitions are possible*)

a. 6 points Calculate the **energy (in kJ/mol)** required to ionize (completely remove) the electron from the **$n=3$** level of **He^+ ion** (He: Helium). Show your work including equations. (3 sig. figs.)

b. 2 points

(I) **Select** the **largest (longest)** wavelength from the three choices below:

- i. Wavelength of photon to *completely remove* an electron from *1s orbital of Hydrogen atom*
- ii. Wavelength of photon to *completely remove* an electron from *1s orbital of Boron atom (B)*
- iii. Wavelength of photon to *completely remove* an electron from *1s orbital of Boron cation (B^+)*

i **or** **ii** **or** **iii**

(II) **Select** the **smallest (shortest)** wavelength from the three choices below (*for H atom*):

- i. Wavelength of photon *emitted* for an electron transitioning from *2p to 1s orbital*
- ii. Wavelength of photon *absorbed* for an electron transitioning from *2s to 4p orbital*
- iii. Wavelength of photon *absorbed* for an electron transitioning from *1s to 3p orbital*

i **or** **ii** **or** **iii**



Question 2 8 points



A Chem110 student plots the graph shown on the left, from data obtained for the kinetic energy (eV) of the ejected electron versus frequency (s^{-1}) of the incident light, for an unknown metal. ($1\text{eV} = 1.6022 \times 10^{-19} \text{ J}$)

Based on the plot, answer the questions **a to c** below:

a. 1 point On the plot above, label the **threshold frequency**.

b. 2 points What is the value of the work potential (in J) of the unknown metal?

c. 3 points What is the *longest* wavelength (in nm) required to eject an electron from this metal's surface? Show your work including equation(s) used. (3 sig figs)

d. 2 points In the space provided below, plot the following

(Assume frequency is *above* threshold frequency):

- Kinetic energy* of an ejected electron versus the *intensity* of the incident light (left)
- Number of electrons* ejected versus the *frequency* of the incident light (right)



**Question 3** **6 points****a. 2 points**

In the ground state of Ge (Germanium; $Z = 32$) atom, **name** the following orbitals:

i. Orbital containing the electron that requires the maximum energy to be removed.

ii. Orbital containing the electron that experiences the minimum effective nuclear charge.

b. 2 points

An unknown anion (A^-) has the following *excited* electronic configuration: $[Ar] 4s^2 3d^3 4p^1$

(Ar: Argon; $Z = 18$)

Give the **condensed ground state electronic configuration** of the atom, **A** and the cation, **A^{2+}** .
Show your work. (Assume all electronic transitions are possible)

Electronic Configuration of A:

Electronic Configuration of A^{2+} :

c. 2 points

In a single atom, determine how many electrons can have the following set of quantum numbers?

(i) $n = 5; l = 3$

In a single atom, determine how many electrons can have the following set of quantum numbers?

(i) $n = 3; l = 2; m_s = +1/2$

**Question 4** **8 points**

Rank the following from 1 to 4 in **increasing** order. Provide the ranking in the boxes provided.

Note: Atomic number (Z) are provided for relevant atoms

a) Increasing (lowest to highest) order of *ionization energy*: (1= lowest; 4 = highest)

Mg

Mg: Z = 12

Rb

Rb: Z = 37

Cs

Cs: Z = 55

K

K : Z = 19

b) Increasing (lowest to highest) order of *bond angle*: (1= lowest; 4 = highest)

H₂OCCl₄XeF₄

HCN

c) Increasing (smallest to largest) order of *ionic size* (1= smallest; 4 = largest)

S²⁻

S: Z = 16

O²⁻

O: Z = 8

Be²⁺

Be: Z = 4

Na⁺

Na: Z = 11

d) Increasing (lowest to highest) order of *bond energy* (1= lowest; 4 = highest)

P-F

P-Cl

P-I

P-Br

**Question 5** **12 points**

a. 9 points For the molecules HClO_4 and TeF_6 , draw the most stable *Lewis structure* showing appropriate shape and *dashed/wedged bonds*, based on VSEPR. Determine the **molecular geometry** around the central atom. For HClO_4 , label the *most polar bond*.



b. 3 points (Note: Lone pairs are not shown in structures below)

(i) For the three *resonance* structures (1, 2 and 3) shown below, answer the following question:



(I) Circle the *most contributing* resonance structure.

Structure 1

Structure 2

Structure 3

(ii) For the three structures given (4, 5, and 6) below, answer the following questions:



(I) Circle all the structures that represent *resonance structures* of each other:

Structure 4

Structure 5

Structure 6

(II) Among the resonance structures, circle the *most contributing* resonance structure.

Structure 4

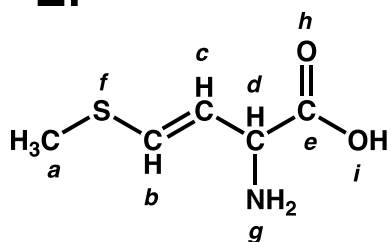
Structure 5

Structure 6

**Question 6** **6 points**

Using the data given below, calculate the **lattice energy** for SrO(s).

	ΔH° (kJ/mol)
$\text{Sr}(s) \longrightarrow \text{Sr}(g)$	+180
$\text{Sr}(g) \longrightarrow \text{Sr}^{2+}(g) + 2e^-$	+1468
$\text{O}_2(g) \longrightarrow 2\text{O}(g)$	+500
$\text{O}(g) + 2e^- \longrightarrow \text{O}^{2-}(g)$	+600
$\text{Sr}(s) + \frac{1}{2}\text{O}_2(g) \longrightarrow \text{SrO}(s)$	-550

**Question 7 10 points**

Consider the molecule given on the left, and answer questions (i) to (vi) using chemical bonding theories that we have learnt in class (valence bond theory, hybridization).

Note that the lone pairs are not shown in the structure:

i. Consider the carbon atoms marked in as ***a***, ***b***, ***c***, ***d***, and ***e*** in the molecule, and determine **hybridization** at each of the carbon centers. (2.5 points)

Atom	Hybridization
Carbon (<i>a</i>)	
Carbon (<i>b</i>)	
Carbon (<i>c</i>)	
Carbon (<i>d</i>)	
Carbon (<i>e</i>)	

ii. Now, consider the sulfur atom (S) marked as ***f***, nitrogen (N) atom (***g***), and oxygen(O) atoms (***h*** and ***i***), and determine their **hybridization**. (2 points)

Atom	Hybridization
Sulfur (<i>f</i>)	
Nitrogen (<i>g</i>)	
Oxygen (<i>h</i>)	
Oxygen (<i>i</i>)	

iii. **Describe** all the bonds formed by the carbon atoms ***a*** and ***b***, using sigma and pi designation as appropriate. (2 points)

Atom	Bonds formed (use sigma and pi as appropriate)
Carbon (<i>a</i>)	
Carbon (<i>b</i>)	

iv. How many total sp^3 -s sigma bonds are in the molecule? (1 point)

v. Determine **electronic** and **molecular geometry** at carbon (C) labeled ***e*** and sulfur (S) labeled ***f*** centres. (2 points)

Atom	Electronic Geometry	Molecular Geometry
Carbon (<i>e</i>)		
Sulfur (<i>f</i>)		

vi. Circle the value from the choices given below, that describes the **C-S-C bond angle at sulfur** (S) marked ***f***: (0.5 points)

$>90^\circ$ to $<109.5^\circ$ 109.5° >109.5 to $<120^\circ$ 120° $>120^\circ$ to $<180^\circ$


Question 8 12 points

Using the **valence shell orbitals only** draw a **complete** molecular orbital (MO) diagram (with no s-p mixing) for the molecule **PS**, in the space provided below. **(6 points)**

Assume that 3s and 3p atomic orbitals will form molecular orbitals like those formed by the combination of 2s and 2p atomic orbitals that we learnt in class, and there is no s-p mixing.

Show **energy arrow**, **atomic orbitals** involved, **atom labels**, the **molecular orbitals** that are formed, **filled electrons** in the MO diagram etc. Make sure to label all the atomic and molecular orbitals.

Using the MO diagram above, answer the following questions:

- (i) Which one of the following will have the highest bond order? Circle the correct choice
 A) PS^{1-} B) PS C) PS^{1+}
- (ii) Which one of the following will have the longest bond length? Circle the correct choice
 A) PS^{1-} B) PS C) PS^{1+}
- (iii) Which of the following will be diamagnetic? Circle the correct choice.
 A) PS^{1-} B) PS C) PS^{1+}
- (iv) Draw Lewis dot structure of the molecule with the shortest bond length (based on the MO diagram) from among the three (PS^{1-} , PS, PS^{1+}), and indicate the bond order calculated from the Lewis dot structure. **(3 points)**

Lewis Dot Structure

Bond Order



0C67BE3E-51E2-41ED-8882-13B2DBB1321B

midterm-exam-93388

#618 Page 10 of 10

SCRAP PAGE

**CHEMISTRY 110 –Midterm Exam****Thursday November 7, 2024, 6:30 – 8:30 pm****Examiners: Pallavi Sirjoosingh and Ashok Kakkar**

First name (please write as legibly as possible within the boxes)

--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--

Last name

--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--

Student ID number

--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--

This booklet has 8 questions on 10 pages (The last page can be used as a scrap page).

The staple must NOT be removed from the question booklet. Your intact question booklet must be returned at the end of the exam.

A separate booklet contains a periodic table, and datasheet.

Write your answers directly under the questions in the space provided. Do not write on the QR codes or near the top edges (as it may be cut off).

Please write **legibly and dark enough** for the scan to be readable. For students who use English as a second language, dictionaries are permitted. Calculators are allowed.

DO NOT ASK QUESTIONS OF THE INVIGILATORS about interpretations of the exam questions. If you have a concern about a question, please summarize it at the top of this page. All concerns will be evaluated after the exam. GOOD LUCK!

**Question 1 8 points** (*Assume all transitions are possible*)

a. 6 points Calculate the **energy (in kJ/mol)** required to ionize (completely remove) the electron from the **n=3** level of **He⁺ ion** (He: Helium). Show your work including equations. (3 sig. figs.)

b. 2 points

(I) Select the largest (longest) wavelength from the three choices below:

- i. Wavelength of photon to *completely remove* an electron from *1s orbital of Hydrogen atom*
- ii. Wavelength of photon to *completely remove* an electron from *1s orbital of Boron atom (B)*
- iii. Wavelength of photon to *completely remove* an electron from *1s orbital of Boron cation (B⁺)*

i or ii or iii

(II) Select the smallest (shortest) wavelength from the three choices below (*for H atom*):

- i. Wavelength of photon *emitted* for an electron transitioning from *2p to 1s orbital*
- ii. Wavelength of photon *absorbed* for an electron transitioning from *2s to 4p orbital*
- iii. Wavelength of photon *absorbed* for an electron transitioning from *1s to 3p orbital*

i or ii or iii



Question 2 8 points



A Chem110 student plots the graph shown on the left, from data obtained for the kinetic energy (eV) of the ejected electron versus frequency (s^{-1}) of the incident light, for an unknown metal. ($1\text{eV} = 1.6022 \times 10^{-19} \text{ J}$)

Based on the plot, answer the questions **a to c** below:

a. 1 point On the plot above, label the **threshold frequency**.

b. 2 points What is the value of the work potential (in J) of the unknown metal?

c. 3 points What is the *longest* wavelength (in nm) required to eject an electron from this metal's surface? Show your work including equation(s) used. (3 sig figs)

d. 2 points In the space provided below, plot the following

(Assume frequency is *above* threshold frequency):

- Kinetic energy* of an ejected electron versus the *intensity* of the incident light (left)
- Number of electrons* ejected versus the *frequency* of the incident light (right)



**Question 3** **6 points****a. 2 points**

In the ground state of Ge (Germanium; $Z = 32$) atom, **name** the following orbitals:

i. Orbital containing the electron that requires the maximum energy to be removed.

ii. Orbital containing the electron that experiences the minimum effective nuclear charge.

b. 2 points

An unknown anion (A^-) has the following *excited* electronic configuration: $[Ar] 4s^2 3d^3 4p^1$

(Ar: Argon; $Z = 18$)

Give the **condensed ground state electronic configuration** of the atom, **A** and the cation, **A^{2+}** .
Show your work. (Assume all electronic transitions are possible)

Electronic Configuration of A:

Electronic Configuration of A^{2+} :

c. 2 points

In a single atom, determine how many electrons can have the following set of quantum numbers?

(i) $n = 5; l = 3$

In a single atom, determine how many electrons can have the following set of quantum numbers?

(i) $n = 3; l = 2; m_s = +1/2$

**Question 4** **8 points**

Rank the following from 1 to 4 in **increasing** order. Provide the ranking in the boxes provided.

Note: Atomic number (Z) are provided for relevant atoms

a) Increasing (lowest to highest) order of *ionization energy*: (1= lowest; 4 = highest)

Mg

Mg: Z = 12

Rb

Rb: Z = 37

Cs

Cs: Z = 55

K

K : Z = 19

b) Increasing (lowest to highest) order of *bond angle*: (1= lowest; 4 = highest)

H₂OCCl₄XeF₄

HCN

c) Increasing (smallest to largest) order of *ionic size* (1= smallest; 4 = largest)

S²⁻

S: Z = 16

O²⁻

O: Z = 8

Be²⁺

Be: Z = 4

Na⁺

Na: Z = 11

d) Increasing (lowest to highest) order of *bond energy* (1= lowest; 4 = highest)

P-F

P-Cl

P-I

P-Br

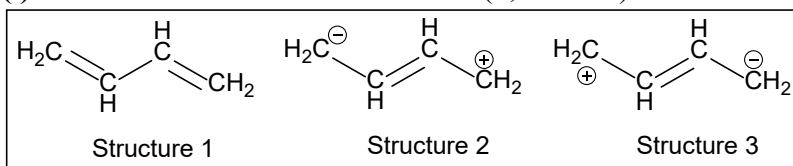
**Question 5** **12 points**

a. 9 points For the molecules HClO_4 and TeF_6 , draw the most stable *Lewis structure* showing appropriate shape and *dashed/wedged bonds*, based on VSEPR. Determine the **molecular geometry** around the central atom. For HClO_4 , label the *most polar bond*.



b. 3 points (Note: Lone pairs are not shown in structures below)

(i) For the three *resonance* structures (1, 2 and 3) shown below, answer the following question:



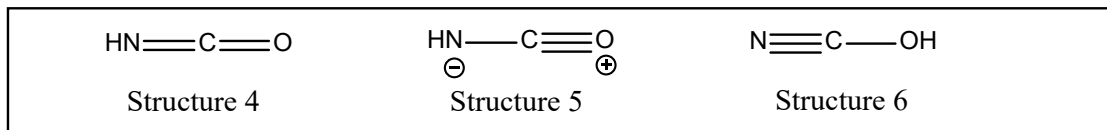
(I) Circle the *most contributing* resonance structure.

Structure 1

Structure 2

Structure 3

(ii) For the three structures given (4, 5, and 6) below, answer the following questions:



(I) Circle all the structures that represent *resonance structures* of each other:

Structure 4

Structure 5

Structure 6

(II) Among the resonance structures, circle the *most contributing* resonance structure.

Structure 4

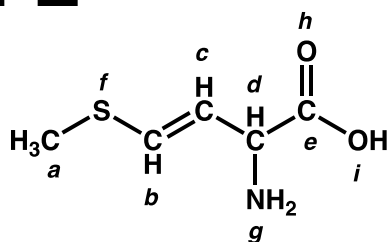
Structure 5

Structure 6

**Question 6** **6 points**

Using the data given below, calculate the **lattice energy** for SrO(s).

	ΔH° (kJ/mol)
$\text{Sr}(s) \longrightarrow \text{Sr}(g)$	+180
$\text{Sr}(g) \longrightarrow \text{Sr}^{2+}(g) + 2e^-$	+1468
$\text{O}_2(g) \longrightarrow 2\text{O}(g)$	+500
$\text{O}(g) + 2e^- \longrightarrow \text{O}^{2-}(g)$	+600
$\text{Sr}(s) + \frac{1}{2}\text{O}_2(g) \longrightarrow \text{SrO}(s)$	-550

**Question 7 10 points**

Consider the molecule given on the left, and answer questions (i) to (vi) using chemical bonding theories that we have learnt in class (valence bond theory, hybridization).

Note that the lone pairs are not shown in the structure:

i. Consider the carbon atoms marked in as ***a***, ***b***, ***c***, ***d***, and ***e*** in the molecule, and determine **hybridization** at each of the carbon centers. (2.5 points)

Atom	Hybridization
Carbon (<i>a</i>)	
Carbon (<i>b</i>)	
Carbon (<i>c</i>)	
Carbon (<i>d</i>)	
Carbon (<i>e</i>)	

ii. Now, consider the sulfur atom (S) marked as ***f***, nitrogen (N) atom (***g***), and oxygen(O) atoms (***h*** and ***i***), and determine their **hybridization**. (2 points)

Atom	Hybridization
Sulfur (<i>f</i>)	
Nitrogen (<i>g</i>)	
Oxygen (<i>h</i>)	
Oxygen (<i>i</i>)	

iii. **Describe** all the bonds formed by the carbon atoms ***a*** and ***b***, using sigma and pi designation as appropriate. (2 points)

Atom	Bonds formed (use sigma and pi as appropriate)
Carbon (<i>a</i>)	
Carbon (<i>b</i>)	

iv. How many total sp^3 -s sigma bonds are in the molecule? (1 point)

v. Determine **electronic** and **molecular geometry** at carbon (C) labeled ***e*** and sulfur (S) labeled ***f*** centres. (2 points)

Atom	Electronic Geometry	Molecular Geometry
Carbon (<i>e</i>)		
Sulfur (<i>f</i>)		

vi. Circle the value from the choices given below, that describes the **C-S-C bond angle at sulfur** (S) marked ***f***: (0.5 points)

>90° to <109.5° 109.5° >109.5 to <120° 120° >120° to <180°


Question 8 12 points

Using the **valence shell orbitals only** draw a **complete** molecular orbital (MO) diagram (with no s-p mixing) for the molecule **PS**, in the space provided below. **(6 points)**

Assume that 3s and 3p atomic orbitals will form molecular orbitals like those formed by the combination of 2s and 2p atomic orbitals that we learnt in class, and there is no s-p mixing.

Show **energy arrow**, **atomic orbitals** involved, **atom labels**, the **molecular orbitals** that are formed, **filled electrons** in the MO diagram etc. Make sure to label all the atomic and molecular orbitals.

Using the MO diagram above, answer the following questions:

- (i) Which one of the following will have the highest bond order? Circle the correct choice
 A) PS^{1-} B) PS C) PS^{1+}
- (ii) Which one of the following will have the longest bond length? Circle the correct choice
 A) PS^{1-} B) PS C) PS^{1+}
- (iii) Which of the following will be diamagnetic? Circle the correct choice.
 A) PS^{1-} B) PS C) PS^{1+}
- (iv) Draw Lewis dot structure of the molecule with the shortest bond length (based on the MO diagram) from among the three (PS^{1-} , PS, PS^{1+}), and indicate the bond order calculated from the Lewis dot structure. **(3 points)**

Lewis Dot Structure

Bond Order



64C0E4E4-7C6C-4403-B74F-F4A33C3BA370

midterm-exam-93388

#619 Page 10 of 10

SCRAP PAGE

**CHEMISTRY 110 –Midterm Exam****Thursday November 7, 2024, 6:30 – 8:30 pm****Examiners: Pallavi Sirjoosingh and Ashok Kakkar**

First name (please write as legibly as possible within the boxes)

--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--

Last name

--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--

Student ID number

--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--

This booklet has 8 questions on 10 pages (The last page can be used as a scrap page).

The staple must NOT be removed from the question booklet. Your intact question booklet must be returned at the end of the exam.

A separate booklet contains a periodic table, and datasheet.

Write your answers directly under the questions in the space provided. Do not write on the QR codes or near the top edges (as it may be cut off).

Please write **legibly and dark enough** for the scan to be readable. For students who use English as a second language, dictionaries are permitted. Calculators are allowed.

DO NOT ASK QUESTIONS OF THE INVIGILATORS about interpretations of the exam questions. If you have a concern about a question, please summarize it at the top of this page. All concerns will be evaluated after the exam. GOOD LUCK!

**Question 1 8 points** (*Assume all transitions are possible*)

a. 6 points Calculate the **energy (in kJ/mol)** required to ionize (completely remove) the electron from the **n=3** level of **He⁺ ion** (He: Helium). Show your work including equations. (3 sig. figs.)

b. 2 points

(I) Select the largest (longest) wavelength from the three choices below:

- i. Wavelength of photon to *completely remove* an electron from *1s orbital of Hydrogen atom*
- ii. Wavelength of photon to *completely remove* an electron from *1s orbital of Boron atom (B)*
- iii. Wavelength of photon to *completely remove* an electron from *1s orbital of Boron cation (B⁺)*

i or ii or iii

(II) Select the smallest (shortest) wavelength from the three choices below (*for H atom*):

- i. Wavelength of photon *emitted* for an electron transitioning from *2p to 1s orbital*
- ii. Wavelength of photon *absorbed* for an electron transitioning from *2s to 4p orbital*
- iii. Wavelength of photon *absorbed* for an electron transitioning from *1s to 3p orbital*

i or ii or iii



Question 2 8 points



A Chem110 student plots the graph shown on the left, from data obtained for the kinetic energy (eV) of the ejected electron versus frequency (s^{-1}) of the incident light, for an unknown metal. ($1\text{eV} = 1.6022 \times 10^{-19} \text{ J}$)

Based on the plot, answer the questions **a to c** below:

a. 1 point On the plot above, label the **threshold frequency**.

b. 2 points What is the value of the work potential (in J) of the unknown metal?

c. 3 points What is the *longest* wavelength (in nm) required to eject an electron from this metal's surface? Show your work including equation(s) used. (3 sig figs)

d. 2 points In the space provided below, plot the following

(Assume frequency is *above* threshold frequency):

- i. *Kinetic energy* of an ejected electron versus the *intensity* of the incident light (left)
- ii. *Number of electrons* ejected versus the *frequency* of the incident light (right)



**Question 3** **6 points****a. 2 points**

In the ground state of Ge (Germanium; $Z = 32$) atom, **name** the following orbitals:

i. Orbital containing the electron that requires the maximum energy to be removed.

ii. Orbital containing the electron that experiences the minimum effective nuclear charge.

b. 2 points

An unknown anion (A^-) has the following *excited* electronic configuration: $[Ar] 4s^2 3d^3 4p^1$

(Ar: Argon; $Z = 18$)

Give the **condensed ground state electronic configuration** of the atom, **A** and the cation, **A^{2+}** .
Show your work. (Assume all electronic transitions are possible)

Electronic Configuration of A:

Electronic Configuration of A^{2+} :

c. 2 points

In a single atom, determine how many electrons can have the following set of quantum numbers?

(i) $n = 5; l = 3$

In a single atom, determine how many electrons can have the following set of quantum numbers?

(i) $n = 3; l = 2; m_s = +1/2$

**Question 4** **8 points**

Rank the following from 1 to 4 in **increasing** order. Provide the ranking in the boxes provided.

Note: Atomic number (Z) are provided for relevant atoms

a) Increasing (lowest to highest) order of *ionization energy*: (1= lowest; 4 = highest)

Mg

Mg: Z = 12

Rb

Rb: Z = 37

Cs

Cs: Z = 55

K

K : Z = 19

b) Increasing (lowest to highest) order of *bond angle*: (1= lowest; 4 = highest)

H₂OCCl₄XeF₄

HCN

c) Increasing (smallest to largest) order of *ionic size* (1= smallest; 4 = largest)

S²⁻

S: Z = 16

O²⁻

O: Z = 8

Be²⁺

Be: Z = 4

Na⁺

Na: Z = 11

d) Increasing (lowest to highest) order of *bond energy* (1= lowest; 4 = highest)

P-F

P-Cl

P-I

P-Br

**Question 5** **12 points**

a. 9 points For the molecules HClO_4 and TeF_6 , draw the most stable *Lewis structure* showing appropriate shape and *dashed/wedged bonds*, based on VSEPR. Determine the **molecular geometry** around the central atom. For HClO_4 , label the *most polar bond*.



b. 3 points (Note: Lone pairs are not shown in structures below)

(i) For the three *resonance* structures (1, 2 and 3) shown below, answer the following question:



(I) Circle the *most contributing* resonance structure.

Structure 1

Structure 2

Structure 3

(ii) For the three structures given (4, 5, and 6) below, answer the following questions:



(I) Circle all the structures that represent *resonance structures* of each other:

Structure 4

Structure 5

Structure 6

(II) Among the resonance structures, circle the *most contributing* resonance structure.

Structure 4

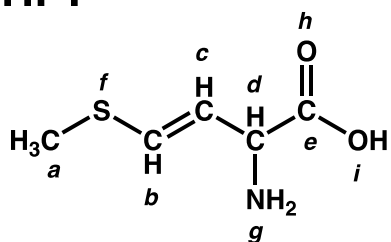
Structure 5

Structure 6

**Question 6** **6 points**

Using the data given below, calculate the **lattice energy** for SrO(s).

	ΔH° (kJ/mol)
$\text{Sr}(s) \longrightarrow \text{Sr}(g)$	+180
$\text{Sr}(g) \longrightarrow \text{Sr}^{2+}(g) + 2e^-$	+1468
$\text{O}_2(g) \longrightarrow 2\text{O}(g)$	+500
$\text{O}(g) + 2e^- \longrightarrow \text{O}^{2-}(g)$	+600
$\text{Sr}(s) + \frac{1}{2}\text{O}_2(g) \longrightarrow \text{SrO}(s)$	-550

**Question 7 10 points**

Consider the molecule given on the left, and answer questions (i) to (vi) using chemical bonding theories that we have learnt in class (valence bond theory, hybridization).

Note that the lone pairs are not shown in the structure:

i. Consider the carbon atoms marked in as ***a***, ***b***, ***c***, ***d***, and ***e*** in the molecule, and determine **hybridization** at each of the carbon centers. (2.5 points)

Atom	Hybridization
Carbon (<i>a</i>)	
Carbon (<i>b</i>)	
Carbon (<i>c</i>)	
Carbon (<i>d</i>)	
Carbon (<i>e</i>)	

ii. Now, consider the sulfur atom (S) marked as ***f***, nitrogen (N) atom (***g***), and oxygen(O) atoms (***h*** and ***i***), and determine their **hybridization**. (2 points)

Atom	Hybridization
Sulfur (<i>f</i>)	
Nitrogen (<i>g</i>)	
Oxygen (<i>h</i>)	
Oxygen (<i>i</i>)	

iii. **Describe** all the bonds formed by the carbon atoms ***a*** and ***b***, using sigma and pi designation as appropriate. (2 points)

Atom	Bonds formed (use sigma and pi as appropriate)
Carbon (<i>a</i>)	
Carbon (<i>b</i>)	

iv. How many total sp^3 -s sigma bonds are in the molecule? (1 point)

v. Determine **electronic** and **molecular geometry** at carbon (C) labeled ***e*** and sulfur (S) labeled ***f*** centres. (2 points)

Atom	Electronic Geometry	Molecular Geometry
Carbon (<i>e</i>)		
Sulfur (<i>f</i>)		

vi. Circle the value from the choices given below, that describes the **C-S-C bond angle at sulfur** (S) marked ***f***: (0.5 points)

>90° to <109.5° 109.5° >109.5 to <120° 120° >120° to <180°

**Question 8 12 points**

Using the **valence shell orbitals only** draw a **complete** molecular orbital (MO) diagram (with no s-p mixing) for the molecule **PS**, in the space provided below. **(6 points)**

Assume that 3s and 3p atomic orbitals will form molecular orbitals like those formed by the combination of 2s and 2p atomic orbitals that we learnt in class, and there is no s-p mixing.

Show **energy arrow**, **atomic orbitals** involved, **atom labels**, the **molecular orbitals** that are formed, **filled electrons** in the MO diagram etc. Make sure to label all the atomic and molecular orbitals.

Using the MO diagram above, answer the following questions:

- (i) Which one of the following will have the highest bond order? Circle the correct choice
A) PS^{1-} B) PS C) PS^{1+}
- (ii) Which one of the following will have the longest bond length? Circle the correct choice
A) PS^{1-} B) PS C) PS^{1+}
- (iii) Which of the following will be diamagnetic? Circle the correct choice.
A) PS^{1-} B) PS C) PS^{1+}
- (iv) Draw Lewis dot structure of the molecule with the shortest bond length (based on the MO diagram) from among the three (PS^{1-} , PS, PS^{1+}), and indicate the bond order calculated from the Lewis dot structure. **(3 points)**

Lewis Dot Structure

Bond Order



FC0867A2-0BE2-4F6B-B15C-E619A32EBA75

midterm-exam-93388

#620 Page 10 of 10

SCRAP PAGE

**CHEMISTRY 110 –Midterm Exam****Thursday November 7, 2024, 6:30 – 8:30 pm****Examiners: Pallavi Sirjoosingh and Ashok Kakkar**

First name (please write as legibly as possible within the boxes)

--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--

Last name

--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--

Student ID number

--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--

This booklet has 8 questions on 10 pages (The last page can be used as a scrap page).

The staple must NOT be removed from the question booklet. Your intact question booklet must be returned at the end of the exam.

A separate booklet contains a periodic table, and datasheet.

Write your answers directly under the questions in the space provided. Do not write on the QR codes or near the top edges (as it may be cut off).

Please write **legibly and dark enough** for the scan to be readable. For students who use English as a second language, dictionaries are permitted. Calculators are allowed.

DO NOT ASK QUESTIONS OF THE INVIGILATORS about interpretations of the exam questions. If you have a concern about a question, please summarize it at the top of this page. All concerns will be evaluated after the exam. GOOD LUCK!

**Question 1 8 points** (*Assume all transitions are possible*)

a. 6 points Calculate the **energy (in kJ/mol)** required to ionize (completely remove) the electron from the **$n=3$** level of **He^+ ion** (He: Helium). Show your work including equations. (3 sig. figs.)

b. 2 points

(I) Select the largest (longest) wavelength from the three choices below:

- i. Wavelength of photon to *completely remove* an electron from *1s orbital of Hydrogen atom*
- ii. Wavelength of photon to *completely remove* an electron from *1s orbital of Boron atom (B)*
- iii. Wavelength of photon to *completely remove* an electron from *1s orbital of Boron cation (B^+)*

i or ii or iii

(II) Select the smallest (shortest) wavelength from the three choices below (*for H atom*):

- i. Wavelength of photon *emitted* for an electron transitioning from *2p to 1s orbital*
- ii. Wavelength of photon *absorbed* for an electron transitioning from *2s to 4p orbital*
- iii. Wavelength of photon *absorbed* for an electron transitioning from *1s to 3p orbital*

i or ii or iii



Question 2 8 points



A Chem110 student plots the graph shown on the left, from data obtained for the kinetic energy (eV) of the ejected electron versus frequency (s^{-1}) of the incident light, for an unknown metal. ($1\text{eV} = 1.6022 \times 10^{-19} \text{ J}$)

Based on the plot, answer the questions **a to c** below:

a. 1 point On the plot above, label the **threshold frequency**.

b. 2 points What is the value of the work potential (in J) of the unknown metal?

c. 3 points What is the *longest* wavelength (in nm) required to eject an electron from this metal's surface? Show your work including equation(s) used. (3 sig figs)

d. 2 points In the space provided below, plot the following

(Assume frequency is *above* threshold frequency):

- Kinetic energy* of an ejected electron versus the *intensity* of the incident light (left)
- Number of electrons* ejected versus the *frequency* of the incident light (right)



**Question 3** **6 points****a. 2 points**

In the ground state of Ge (Germanium; $Z = 32$) atom, **name** the following orbitals:

i. Orbital containing the electron that requires the maximum energy to be removed.

ii. Orbital containing the electron that experiences the minimum effective nuclear charge.

b. 2 points

An unknown anion (A^-) has the following *excited* electronic configuration: $[Ar] 4s^2 3d^3 4p^1$

(Ar: Argon; $Z = 18$)

Give the **condensed ground state electronic configuration** of the atom, **A** and the cation, **A^{2+}** .
Show your work. (Assume all electronic transitions are possible)

Electronic Configuration of A:

Electronic Configuration of A^{2+} :

c. 2 points

In a single atom, determine how many electrons can have the following set of quantum numbers?

(i) $n = 5; l = 3$

In a single atom, determine how many electrons can have the following set of quantum numbers?

(i) $n = 3; l = 2; m_s = +1/2$

**Question 4** **8 points**

Rank the following from 1 to 4 in **increasing** order. Provide the ranking in the boxes provided.

Note: Atomic number (Z) are provided for relevant atoms

a) Increasing (lowest to highest) order of *ionization energy*: (1= lowest; 4 = highest)

Mg

Mg: Z = 12

Rb

Rb: Z = 37

Cs

Cs: Z = 55

K

K : Z = 19

b) Increasing (lowest to highest) order of *bond angle*: (1= lowest; 4 = highest)

H₂OCCl₄XeF₄

HCN

c) Increasing (smallest to largest) order of *ionic size* (1= smallest; 4 = largest)

S²⁻

S: Z = 16

O²⁻

O: Z = 8

Be²⁺

Be: Z = 4

Na⁺

Na: Z = 11

d) Increasing (lowest to highest) order of *bond energy* (1= lowest; 4 = highest)

P-F

P-Cl

P-I

P-Br

**Question 5** **12 points**

a. 9 points For the molecules HClO_4 and TeF_6 , draw the most stable *Lewis structure* showing appropriate shape and *dashed/wedged bonds*, based on VSEPR. Determine the **molecular geometry** around the central atom. For HClO_4 , label the *most polar bond*.



b. 3 points (Note: Lone pairs are not shown in structures below)

(i) For the three *resonance* structures (1, 2 and 3) shown below, answer the following question:



(I) Circle the *most contributing* resonance structure.

Structure 1

Structure 2

Structure 3

(ii) For the three structures given (4, 5, and 6) below, answer the following questions:



(I) Circle all the structures that represent *resonance structures* of each other:

Structure 4

Structure 5

Structure 6

(II) Among the resonance structures, circle the *most contributing* resonance structure.

Structure 4

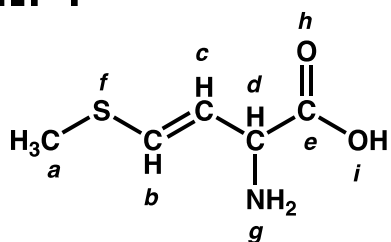
Structure 5

Structure 6

**Question 6** **6 points**

Using the data given below, calculate the **lattice energy** for SrO(s).

	ΔH° (kJ/mol)
$\text{Sr}(s) \longrightarrow \text{Sr}(g)$	+180
$\text{Sr}(g) \longrightarrow \text{Sr}^{2+}(g) + 2e^-$	+1468
$\text{O}_2(g) \longrightarrow 2\text{O}(g)$	+500
$\text{O}(g) + 2e^- \longrightarrow \text{O}^{2-}(g)$	+600
$\text{Sr}(s) + \frac{1}{2}\text{O}_2(g) \longrightarrow \text{SrO}(s)$	-550

**Question 7 10 points**

Consider the molecule given on the left, and answer questions (i) to (vi) using chemical bonding theories that we have learnt in class (valence bond theory, hybridization).

Note that the lone pairs are not shown in the structure:

i. Consider the carbon atoms marked in as ***a***, ***b***, ***c***, ***d***, and ***e*** in the molecule, and determine **hybridization** at each of the carbon centers. (2.5 points)

Atom	Hybridization
Carbon (<i>a</i>)	
Carbon (<i>b</i>)	
Carbon (<i>c</i>)	
Carbon (<i>d</i>)	
Carbon (<i>e</i>)	

ii. Now, consider the sulfur atom (S) marked as ***f***, nitrogen (N) atom (***g***), and oxygen(O) atoms (***h*** and ***i***), and determine their **hybridization**. (2 points)

Atom	Hybridization
Sulfur (<i>f</i>)	
Nitrogen (<i>g</i>)	
Oxygen (<i>h</i>)	
Oxygen (<i>i</i>)	

iii. **Describe** all the bonds formed by the carbon atoms ***a*** and ***b***, using sigma and pi designation as appropriate. (2 points)

Atom	Bonds formed (use sigma and pi as appropriate)
Carbon (<i>a</i>)	
Carbon (<i>b</i>)	

iv. How many total sp^3 -s sigma bonds are in the molecule? (1 point)

v. Determine **electronic** and **molecular geometry** at carbon (C) labeled ***e*** and sulfur (S) labeled ***f*** centres. (2 points)

Atom	Electronic Geometry	Molecular Geometry
Carbon (<i>e</i>)		
Sulfur (<i>f</i>)		

vi. Circle the value from the choices given below, that describes the **C-S-C bond angle at sulfur** (S) marked ***f***: (0.5 points)

>90° to <109.5° 109.5° >109.5 to <120° 120° >120° to <180°

**Question 8 12 points**

Using the **valence shell orbitals only** draw a **complete** molecular orbital (MO) diagram (with no s-p mixing) for the molecule **PS**, in the space provided below. **(6 points)**

Assume that 3s and 3p atomic orbitals will form molecular orbitals like those formed by the combination of 2s and 2p atomic orbitals that we learnt in class, and there is no s-p mixing.

Show **energy arrow**, **atomic orbitals** involved, **atom labels**, the **molecular orbitals** that are formed, **filled electrons** in the MO diagram etc. Make sure to label all the atomic and molecular orbitals.

Using the MO diagram above, answer the following questions:

- (i) Which one of the following will have the highest bond order? Circle the correct choice
A) PS^{1-} B) PS C) PS^{1+}
- (ii) Which one of the following will have the longest bond length? Circle the correct choice
A) PS^{1-} B) PS C) PS^{1+}
- (iii) Which of the following will be diamagnetic? Circle the correct choice.
A) PS^{1-} B) PS C) PS^{1+}
- (iv) Draw Lewis dot structure of the molecule with the shortest bond length (based on the MO diagram) from among the three (PS^{1-} , PS, PS^{1+}), and indicate the bond order calculated from the Lewis dot structure. **(3 points)**

Lewis Dot Structure

Bond Order



599A9F44-FFAB-42BA-B4A8-1BD19024A484

midterm-exam-93388

#621 Page 10 of 10

SCRAP PAGE

**CHEMISTRY 110 –Midterm Exam****Thursday November 7, 2024, 6:30 – 8:30 pm****Examiners: Pallavi Sirjoosingh and Ashok Kakkar**

First name (please write as legibly as possible within the boxes)

--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--

Last name

--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--

Student ID number

--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--

This booklet has 8 questions on 10 pages (The last page can be used as a scrap page).

The staple must NOT be removed from the question booklet. Your intact question booklet must be returned at the end of the exam.

A separate booklet contains a periodic table, and datasheet.

Write your answers directly under the questions in the space provided. Do not write on the QR codes or near the top edges (as it may be cut off).

Please write **legibly and dark enough** for the scan to be readable. For students who use English as a second language, dictionaries are permitted. Calculators are allowed.

DO NOT ASK QUESTIONS OF THE INVIGILATORS about interpretations of the exam questions. If you have a concern about a question, please summarize it at the top of this page. All concerns will be evaluated after the exam. GOOD LUCK!

**Question 1 8 points** (*Assume all transitions are possible*)

a. 6 points Calculate the **energy (in kJ/mol)** required to ionize (completely remove) the electron from the **$n=3$** level of **He^+ ion** (He: Helium). Show your work including equations. (3 sig. figs.)

b. 2 points

(I) Select the largest (longest) wavelength from the three choices below:

- i. Wavelength of photon to *completely remove* an electron from *1s orbital of Hydrogen atom*
- ii. Wavelength of photon to *completely remove* an electron from *1s orbital of Boron atom (B)*
- iii. Wavelength of photon to *completely remove* an electron from *1s orbital of Boron cation (B^+)*

i or ii or iii

(II) Select the smallest (shortest) wavelength from the three choices below (*for H atom*):

- i. Wavelength of photon *emitted* for an electron transitioning from *2p to 1s orbital*
- ii. Wavelength of photon *absorbed* for an electron transitioning from *2s to 4p orbital*
- iii. Wavelength of photon *absorbed* for an electron transitioning from *1s to 3p orbital*

i or ii or iii



Question 2 8 points



A Chem110 student plots the graph shown on the left, from data obtained for the kinetic energy (eV) of the ejected electron versus frequency (s^{-1}) of the incident light, for an unknown metal. ($1\text{eV} = 1.6022 \times 10^{-19} \text{ J}$)

Based on the plot, answer the questions **a to c** below:

a. 1 point On the plot above, label the **threshold frequency**.

b. 2 points What is the value of the work potential (in J) of the unknown metal?

c. 3 points What is the *longest* wavelength (in nm) required to eject an electron from this metal's surface? Show your work including equation(s) used. (3 sig figs)

d. 2 points In the space provided below, plot the following

(Assume frequency is *above* threshold frequency):

- Kinetic energy* of an ejected electron versus the *intensity* of the incident light (left)
- Number of electrons* ejected versus the *frequency* of the incident light (right)



**Question 3** **6 points****a. 2 points**

In the ground state of Ge (Germanium; $Z = 32$) atom, **name** the following orbitals:

i. Orbital containing the electron that requires the maximum energy to be removed.

ii. Orbital containing the electron that experiences the minimum effective nuclear charge.

b. 2 points

An unknown anion (A^-) has the following *excited* electronic configuration: $[Ar] 4s^2 3d^3 4p^1$

(Ar: Argon; $Z = 18$)

Give the **condensed ground state electronic configuration** of the atom, **A** and the cation, **A^{2+}** .
Show your work. (Assume all electronic transitions are possible)

Electronic Configuration of A:

Electronic Configuration of A^{2+} :

c. 2 points

In a single atom, determine how many electrons can have the following set of quantum numbers?

(i) $n = 5; l = 3$

In a single atom, determine how many electrons can have the following set of quantum numbers?

(i) $n = 3; l = 2; m_s = +1/2$

**Question 4** **8 points**

Rank the following from 1 to 4 in **increasing** order. Provide the ranking in the boxes provided.

Note: Atomic number (Z) are provided for relevant atoms

a) Increasing (lowest to highest) order of *ionization energy*: (1= lowest; 4 = highest)

Mg

Mg: Z = 12

Rb

Rb: Z = 37

Cs

Cs: Z = 55

K

K : Z = 19

b) Increasing (lowest to highest) order of *bond angle*: (1= lowest; 4 = highest)

H₂OCCl₄XeF₄

HCN

c) Increasing (smallest to largest) order of *ionic size* (1= smallest; 4 = largest)

S²⁻

S: Z = 16

O²⁻

O: Z = 8

Be²⁺

Be: Z = 4

Na⁺

Na: Z = 11

d) Increasing (lowest to highest) order of *bond energy* (1= lowest; 4 = highest)

P-F

P-Cl

P-I

P-Br

**Question 5** **12 points**

a. 9 points For the molecules HClO_4 and TeF_6 , draw the most stable *Lewis structure* showing appropriate shape and *dashed/wedged bonds*, based on VSEPR. Determine the **molecular geometry** around the central atom. For HClO_4 , label the *most polar bond*.



b. 3 points (Note: Lone pairs are not shown in structures below)

(i) For the three *resonance* structures (1, 2 and 3) shown below, answer the following question:



(I) Circle the *most contributing* resonance structure.

Structure 1

Structure 2

Structure 3

(ii) For the three structures given (4, 5, and 6) below, answer the following questions:



(I) Circle all the structures that represent *resonance structures* of each other:

Structure 4

Structure 5

Structure 6

(II) Among the resonance structures, circle the *most contributing* resonance structure.

Structure 4

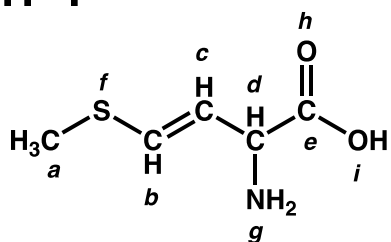
Structure 5

Structure 6

**Question 6** **6 points**

Using the data given below, calculate the **lattice energy** for SrO(s).

	ΔH° (kJ/mol)
$\text{Sr}(s) \longrightarrow \text{Sr}(g)$	+180
$\text{Sr}(g) \longrightarrow \text{Sr}^{2+}(g) + 2e^-$	+1468
$\text{O}_2(g) \longrightarrow 2\text{O}(g)$	+500
$\text{O}(g) + 2e^- \longrightarrow \text{O}^{2-}(g)$	+600
$\text{Sr}(s) + \frac{1}{2}\text{O}_2(g) \longrightarrow \text{SrO}(s)$	-550

**Question 7 10 points**

Consider the molecule given on the left, and answer questions (i) to (vi) using chemical bonding theories that we have learnt in class (valence bond theory, hybridization).

Note that the lone pairs are not shown in the structure:

i. Consider the carbon atoms marked in as ***a***, ***b***, ***c***, ***d***, and ***e*** in the molecule, and determine **hybridization** at each of the carbon centers. (2.5 points)

Atom	Hybridization
Carbon (<i>a</i>)	
Carbon (<i>b</i>)	
Carbon (<i>c</i>)	
Carbon (<i>d</i>)	
Carbon (<i>e</i>)	

ii. Now, consider the sulfur atom (S) marked as ***f***, nitrogen (N) atom (***g***), and oxygen(O) atoms (***h*** and ***i***), and determine their **hybridization**. (2 points)

Atom	Hybridization
Sulfur (<i>f</i>)	
Nitrogen (<i>g</i>)	
Oxygen (<i>h</i>)	
Oxygen (<i>i</i>)	

iii. **Describe** all the bonds formed by the carbon atoms ***a*** and ***b***, using sigma and pi designation as appropriate. (2 points)

Atom	Bonds formed (use sigma and pi as appropriate)
Carbon (<i>a</i>)	
Carbon (<i>b</i>)	

iv. How many total sp^3 -s sigma bonds are in the molecule? (1 point)

v. Determine **electronic** and **molecular geometry** at carbon (C) labeled ***e*** and sulfur (S) labeled ***f*** centres. (2 points)

Atom	Electronic Geometry	Molecular Geometry
Carbon (<i>e</i>)		
Sulfur (<i>f</i>)		

vi. Circle the value from the choices given below, that describes the **C-S-C bond angle at sulfur** (S) marked ***f***: (0.5 points)

>90° to <109.5° 109.5° >109.5 to <120° 120° >120° to <180°

**Question 8 12 points**

Using the **valence shell orbitals only** draw a **complete** molecular orbital (MO) diagram (with no s-p mixing) for the molecule **PS**, in the space provided below. **(6 points)**

Assume that 3s and 3p atomic orbitals will form molecular orbitals like those formed by the combination of 2s and 2p atomic orbitals that we learnt in class, and there is no s-p mixing.

Show **energy arrow**, **atomic orbitals** involved, **atom labels**, the **molecular orbitals** that are formed, **filled electrons** in the MO diagram etc. Make sure to label all the atomic and molecular orbitals.

Using the MO diagram above, answer the following questions:

- (i) Which one of the following will have the highest bond order? Circle the correct choice
A) PS^{1-} B) PS C) PS^{1+}
- (ii) Which one of the following will have the longest bond length? Circle the correct choice
A) PS^{1-} B) PS C) PS^{1+}
- (iii) Which of the following will be diamagnetic? Circle the correct choice.
A) PS^{1-} B) PS C) PS^{1+}
- (iv) Draw Lewis dot structure of the molecule with the shortest bond length (based on the MO diagram) from among the three (PS^{1-} , PS, PS^{1+}), and indicate the bond order calculated from the Lewis dot structure. **(3 points)**

Lewis Dot Structure

Bond Order



569BED81-C3B7-4EBE-A65E-B3D63B4E64CC

midterm-exam-93388

#622 Page 10 of 10

SCRAP PAGE

**CHEMISTRY 110 –Midterm Exam****Thursday November 7, 2024, 6:30 – 8:30 pm****Examiners: Pallavi Sirjoosingh and Ashok Kakkar**

First name (please write as legibly as possible within the boxes)

--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--

Last name

--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--

Student ID number

--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--

This booklet has 8 questions on 10 pages (The last page can be used as a scrap page).

The staple must NOT be removed from the question booklet. Your intact question booklet must be returned at the end of the exam.

A separate booklet contains a periodic table, and datasheet.

Write your answers directly under the questions in the space provided. Do not write on the QR codes or near the top edges (as it may be cut off).

Please write **legibly and dark enough** for the scan to be readable. For students who use English as a second language, dictionaries are permitted. Calculators are allowed.

DO NOT ASK QUESTIONS OF THE INVIGILATORS about interpretations of the exam questions. If you have a concern about a question, please summarize it at the top of this page. All concerns will be evaluated after the exam. GOOD LUCK!

**Question 1 8 points** (*Assume all transitions are possible*)

a. 6 points Calculate the **energy (in kJ/mol)** required to ionize (completely remove) the electron from the **n=3** level of **He⁺ ion** (He: Helium). Show your work including equations. (3 sig. figs.)

b. 2 points

(I) Select the largest (longest) wavelength from the three choices below:

- i. Wavelength of photon to *completely remove* an electron from *1s orbital of Hydrogen atom*
- ii. Wavelength of photon to *completely remove* an electron from *1s orbital of Boron atom (B)*
- iii. Wavelength of photon to *completely remove* an electron from *1s orbital of Boron cation (B⁺)*

i or ii or iii

(II) Select the smallest (shortest) wavelength from the three choices below (*for H atom*):

- i. Wavelength of photon *emitted* for an electron transitioning from *2p to 1s orbital*
- ii. Wavelength of photon *absorbed* for an electron transitioning from *2s to 4p orbital*
- iii. Wavelength of photon *absorbed* for an electron transitioning from *1s to 3p orbital*

i or ii or iii



Question 2 8 points



A Chem110 student plots the graph shown on the left, from data obtained for the kinetic energy (eV) of the ejected electron versus frequency (s^{-1}) of the incident light, for an unknown metal. ($1\text{eV} = 1.6022 \times 10^{-19} \text{ J}$)

Based on the plot, answer the questions **a to c** below:

a. 1 point On the plot above, label the **threshold frequency**.

b. 2 points What is the value of the work potential (in J) of the unknown metal?

c. 3 points What is the *longest* wavelength (in nm) required to eject an electron from this metal's surface? Show your work including equation(s) used. (3 sig figs)

d. 2 points In the space provided below, plot the following

(Assume frequency is *above* threshold frequency):

- i. *Kinetic energy* of an ejected electron versus the *intensity* of the incident light (left)
- ii. *Number of electrons* ejected versus the *frequency* of the incident light (right)



**Question 3** **6 points****a. 2 points**

In the ground state of Ge (Germanium; $Z = 32$) atom, **name** the following orbitals:

i. Orbital containing the electron that requires the maximum energy to be removed.

ii. Orbital containing the electron that experiences the minimum effective nuclear charge.

b. 2 points

An unknown anion (A^-) has the following *excited* electronic configuration: $[Ar] 4s^2 3d^3 4p^1$

(Ar: Argon; $Z = 18$)

Give the **condensed ground state electronic configuration** of the atom, **A** and the cation, A^{2+} .
Show your work. (Assume all electronic transitions are possible)

Electronic Configuration of A:

Electronic Configuration of A^{2+} :

c. 2 points

In a single atom, determine how many electrons can have the following set of quantum numbers?

(i) $n = 5; l = 3$

In a single atom, determine how many electrons can have the following set of quantum numbers?

(i) $n = 3; l = 2; m_s = +1/2$

**Question 4** **8 points**

Rank the following from 1 to 4 in **increasing** order. Provide the ranking in the boxes provided.

Note: Atomic number (Z) are provided for relevant atoms

a) Increasing (lowest to highest) order of *ionization energy*: (1= lowest; 4 = highest)

Mg

Mg: Z = 12

Rb

Rb: Z = 37

Cs

Cs: Z = 55

K

K : Z = 19

b) Increasing (lowest to highest) order of *bond angle*: (1= lowest; 4 = highest)

H₂OCCl₄XeF₄

HCN

c) Increasing (smallest to largest) order of *ionic size* (1= smallest; 4 = largest)

S²⁻

S: Z = 16

O²⁻

O: Z = 8

Be²⁺

Be: Z = 4

Na⁺

Na: Z = 11

d) Increasing (lowest to highest) order of *bond energy* (1= lowest; 4 = highest)

P-F

P-Cl

P-I

P-Br

**Question 5** **12 points**

a. 9 points For the molecules HClO_4 and TeF_6 , draw the most stable *Lewis structure* showing appropriate shape and *dashed/wedged bonds*, based on VSEPR. Determine the **molecular geometry** around the central atom. For HClO_4 , label the *most polar bond*.



b. 3 points (Note: Lone pairs are not shown in structures below)

(i) For the three *resonance* structures (1, 2 and 3) shown below, answer the following question:



(I) Circle the *most contributing* resonance structure.

Structure 1

Structure 2

Structure 3

(ii) For the three structures given (4, 5, and 6) below, answer the following questions:



(I) Circle all the structures that represent *resonance structures* of each other:

Structure 4

Structure 5

Structure 6

(II) Among the resonance structures, circle the *most contributing* resonance structure.

Structure 4

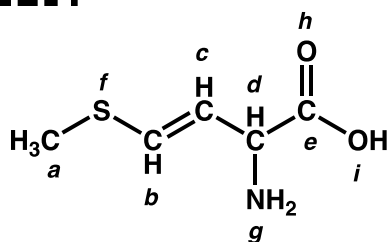
Structure 5

Structure 6

**Question 6** **6 points**

Using the data given below, calculate the **lattice energy** for SrO(s).

	ΔH° (kJ/mol)
$\text{Sr}(s) \longrightarrow \text{Sr}(g)$	+180
$\text{Sr}(g) \longrightarrow \text{Sr}^{2+}(g) + 2e^-$	+1468
$\text{O}_2(g) \longrightarrow 2\text{O}(g)$	+500
$\text{O}(g) + 2e^- \longrightarrow \text{O}^{2-}(g)$	+600
$\text{Sr}(s) + \frac{1}{2}\text{O}_2(g) \longrightarrow \text{SrO}(s)$	-550

**Question 7 10 points**

Consider the molecule given on the left, and answer questions (i) to (vi) using chemical bonding theories that we have learnt in class (valence bond theory, hybridization).

Note that the lone pairs are not shown in the structure:

i. Consider the carbon atoms marked in as ***a***, ***b***, ***c***, ***d***, and ***e*** in the molecule, and determine **hybridization** at each of the carbon centers. (2.5 points)

Atom	Hybridization
Carbon (<i>a</i>)	
Carbon (<i>b</i>)	
Carbon (<i>c</i>)	
Carbon (<i>d</i>)	
Carbon (<i>e</i>)	

ii. Now, consider the sulfur atom (S) marked as ***f***, nitrogen (N) atom (***g***), and oxygen(O) atoms (***h*** and ***i***), and determine their **hybridization**. (2 points)

Atom	Hybridization
Sulfur (<i>f</i>)	
Nitrogen (<i>g</i>)	
Oxygen (<i>h</i>)	
Oxygen (<i>i</i>)	

iii. **Describe** all the bonds formed by the carbon atoms ***a*** and ***b***, using sigma and pi designation as appropriate. (2 points)

Atom	Bonds formed (use sigma and pi as appropriate)
Carbon (<i>a</i>)	
Carbon (<i>b</i>)	

iv. How many total sp^3 -s sigma bonds are in the molecule? (1 point)

v. Determine **electronic** and **molecular geometry** at carbon (C) labeled ***e*** and sulfur (S) labeled ***f*** centres. (2 points)

Atom	Electronic Geometry	Molecular Geometry
Carbon (<i>e</i>)		
Sulfur (<i>f</i>)		

vi. Circle the value from the choices given below, that describes the **C-S-C bond angle at sulfur** (S) marked ***f***: (0.5 points)

>90° to <109.5° 109.5° >109.5 to <120° 120° >120° to <180°

**Question 8 12 points**

Using the **valence shell orbitals only** draw a **complete** molecular orbital (MO) diagram (with no s-p mixing) for the molecule **PS**, in the space provided below. **(6 points)**

Assume that 3s and 3p atomic orbitals will form molecular orbitals like those formed by the combination of 2s and 2p atomic orbitals that we learnt in class, and there is no s-p mixing.

Show **energy arrow**, **atomic orbitals** involved, **atom labels**, the **molecular orbitals** that are formed, **filled electrons** in the MO diagram etc. Make sure to label all the atomic and molecular orbitals.

Using the MO diagram above, answer the following questions:

- (i) Which one of the following will have the highest bond order? Circle the correct choice
A) PS^{1-} B) PS C) PS^{1+}
- (ii) Which one of the following will have the longest bond length? Circle the correct choice
A) PS^{1-} B) PS C) PS^{1+}
- (iii) Which of the following will be diamagnetic? Circle the correct choice.
A) PS^{1-} B) PS C) PS^{1+}
- (iv) Draw Lewis dot structure of the molecule with the shortest bond length (based on the MO diagram) from among the three (PS^{1-} , PS, PS^{1+}), and indicate the bond order calculated from the Lewis dot structure. **(3 points)**

Lewis Dot Structure

Bond Order



CA1409AD-C7AF-495D-91A5-073DDD906E18

midterm-exam-93388

#623 Page 10 of 10

SCRAP PAGE

**CHEMISTRY 110 –Midterm Exam****Thursday November 7, 2024, 6:30 – 8:30 pm****Examiners: Pallavi Sirjoosingh and Ashok Kakkar**

First name (please write as legibly as possible within the boxes)

--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--

Last name

--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--

Student ID number

--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--

This booklet has 8 questions on 10 pages (The last page can be used as a scrap page).

The staple must NOT be removed from the question booklet. Your intact question booklet must be returned at the end of the exam.

A separate booklet contains a periodic table, and datasheet.

Write your answers directly under the questions in the space provided. Do not write on the QR codes or near the top edges (as it may be cut off).

Please write **legibly and dark enough** for the scan to be readable. For students who use English as a second language, dictionaries are permitted. Calculators are allowed.

DO NOT ASK QUESTIONS OF THE INVIGILATORS about interpretations of the exam questions. If you have a concern about a question, please summarize it at the top of this page. All concerns will be evaluated after the exam. GOOD LUCK!

**Question 1 8 points** (*Assume all transitions are possible*)

a. 6 points Calculate the **energy (in kJ/mol)** required to ionize (completely remove) the electron from the **n=3** level of **He⁺ ion** (He: Helium). Show your work including equations. (3 sig. figs.)

b. 2 points

(I) Select the largest (longest) wavelength from the three choices below:

- i. Wavelength of photon to *completely remove* an electron from *1s orbital of Hydrogen atom*
- ii. Wavelength of photon to *completely remove* an electron from *1s orbital of Boron atom (B)*
- iii. Wavelength of photon to *completely remove* an electron from *1s orbital of Boron cation (B⁺)*

i or ii or iii

(II) Select the smallest (shortest) wavelength from the three choices below (*for H atom*):

- i. Wavelength of photon *emitted* for an electron transitioning from *2p to 1s orbital*
- ii. Wavelength of photon *absorbed* for an electron transitioning from *2s to 4p orbital*
- iii. Wavelength of photon *absorbed* for an electron transitioning from *1s to 3p orbital*

i or ii or iii



Question 2 8 points



A Chem110 student plots the graph shown on the left, from data obtained for the kinetic energy (eV) of the ejected electron versus frequency (s^{-1}) of the incident light, for an unknown metal. ($1\text{eV} = 1.6022 \times 10^{-19} \text{ J}$)

Based on the plot, answer the questions **a to c** below:

a. 1 point On the plot above, label the **threshold frequency**.

b. 2 points What is the value of the work potential (in J) of the unknown metal?

c. 3 points What is the *longest* wavelength (in nm) required to eject an electron from this metal's surface? Show your work including equation(s) used. (3 sig figs)

d. 2 points In the space provided below, plot the following

(Assume frequency is *above* threshold frequency):

- Kinetic energy* of an ejected electron versus the *intensity* of the incident light (left)
- Number of electrons* ejected versus the *frequency* of the incident light (right)



**Question 3** **6 points****a. 2 points**

In the ground state of Ge (Germanium; $Z = 32$) atom, **name** the following orbitals:

i. Orbital containing the electron that requires the maximum energy to be removed.

ii. Orbital containing the electron that experiences the minimum effective nuclear charge.

b. 2 points

An unknown anion (A^-) has the following *excited* electronic configuration: $[Ar] 4s^2 3d^3 4p^1$

(Ar: Argon; $Z = 18$)

Give the **condensed ground state electronic configuration** of the atom, **A** and the cation, **A^{2+}** .
Show your work. (Assume all electronic transitions are possible)

Electronic Configuration of A:

Electronic Configuration of A^{2+} :

c. 2 points

In a single atom, determine how many electrons can have the following set of quantum numbers?

(i) $n = 5; l = 3$

In a single atom, determine how many electrons can have the following set of quantum numbers?

(i) $n = 3; l = 2; m_s = +1/2$

**Question 4** **8 points**

Rank the following from 1 to 4 in **increasing** order. Provide the ranking in the boxes provided.

Note: Atomic number (Z) are provided for relevant atoms

a) Increasing (lowest to highest) order of *ionization energy*: (1= lowest; 4 = highest)

Mg

Mg: Z = 12

Rb

Rb: Z = 37

Cs

Cs: Z = 55

K

K : Z = 19

b) Increasing (lowest to highest) order of *bond angle*: (1= lowest; 4 = highest)

H₂OCCl₄XeF₄

HCN

c) Increasing (smallest to largest) order of *ionic size* (1= smallest; 4 = largest)

S²⁻

S: Z = 16

O²⁻

O: Z = 8

Be²⁺

Be: Z = 4

Na⁺

Na: Z = 11

d) Increasing (lowest to highest) order of *bond energy* (1= lowest; 4 = highest)

P-F

P-Cl

P-I

P-Br

**Question 5** **12 points**

a. 9 points For the molecules HClO_4 and TeF_6 , draw the most stable *Lewis structure* showing appropriate shape and *dashed/wedged bonds*, based on VSEPR. Determine the **molecular geometry** around the central atom. For HClO_4 , label the *most polar bond*.

HClO_4 (there are no O-O bonds)

Label the most polar bond

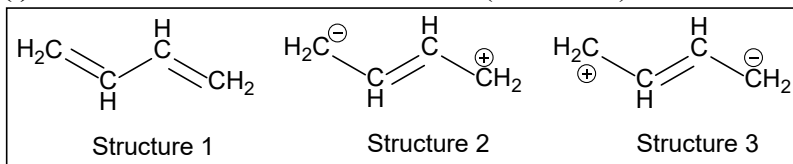
Molecular Geometry around Cl

TeF_6

Molecular Geometry around Te

b. 3 points (Note: Lone pairs are not shown in structures below)

(i) For the three *resonance* structures (1, 2 and 3) shown below, answer the following question:



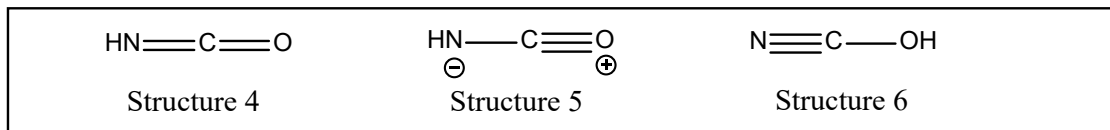
(I) Circle the *most contributing* resonance structure.

Structure 1

Structure 2

Structure 3

(ii) For the three structures given (4, 5, and 6) below, answer the following questions:



(I) Circle all the structures that represent *resonance structures* of each other:

Structure 4

Structure 5

Structure 6

(II) Among the resonance structures, circle the *most contributing* resonance structure.

Structure 4

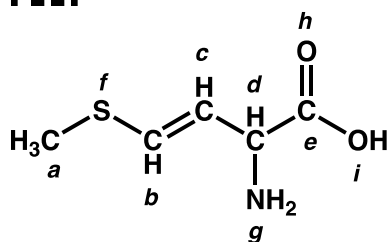
Structure 5

Structure 6

**Question 6** **6 points**

Using the data given below, calculate the **lattice energy** for SrO(s).

	ΔH° (kJ/mol)
$\text{Sr}(s) \longrightarrow \text{Sr}(g)$	+180
$\text{Sr}(g) \longrightarrow \text{Sr}^{2+}(g) + 2e^-$	+1468
$\text{O}_2(g) \longrightarrow 2\text{O}(g)$	+500
$\text{O}(g) + 2e^- \longrightarrow \text{O}^{2-}(g)$	+600
$\text{Sr}(s) + \frac{1}{2}\text{O}_2(g) \longrightarrow \text{SrO}(s)$	-550

**Question 7 10 points**

Consider the molecule given on the left, and answer questions (i) to (vi) using chemical bonding theories that we have learnt in class (valence bond theory, hybridization).

Note that the lone pairs are not shown in the structure:

i. Consider the carbon atoms marked in as ***a***, ***b***, ***c***, ***d***, and ***e*** in the molecule, and determine **hybridization** at each of the carbon centers. (2.5 points)

Atom	Hybridization
Carbon (<i>a</i>)	
Carbon (<i>b</i>)	
Carbon (<i>c</i>)	
Carbon (<i>d</i>)	
Carbon (<i>e</i>)	

ii. Now, consider the sulfur atom (S) marked as ***f***, nitrogen (N) atom (***g***), and oxygen(O) atoms (***h*** and ***i***), and determine their **hybridization**. (2 points)

Atom	Hybridization
Sulfur (<i>f</i>)	
Nitrogen (<i>g</i>)	
Oxygen (<i>h</i>)	
Oxygen (<i>i</i>)	

iii. **Describe** all the bonds formed by the carbon atoms ***a*** and ***b***, using sigma and pi designation as appropriate. (2 points)

Atom	Bonds formed (use sigma and pi as appropriate)
Carbon (<i>a</i>)	
Carbon (<i>b</i>)	

iv. How many total sp^3 -s sigma bonds are in the molecule? (1 point)

v. Determine **electronic** and **molecular geometry** at carbon (C) labeled ***e*** and sulfur (S) labeled ***f*** centres. (2 points)

Atom	Electronic Geometry	Molecular Geometry
Carbon (<i>e</i>)		
Sulfur (<i>f</i>)		

vi. Circle the value from the choices given below, that describes the **C-S-C bond angle at sulfur** (S) marked ***f***: (0.5 points)

>90° to <109.5°

109.5°

>109.5 to <120°

120°

>120° to <180°


Question 8 12 points

Using the **valence shell orbitals only** draw a **complete** molecular orbital (MO) diagram (with no s-p mixing) for the molecule **PS**, in the space provided below. **(6 points)**

Assume that 3s and 3p atomic orbitals will form molecular orbitals like those formed by the combination of 2s and 2p atomic orbitals that we learnt in class, and there is no s-p mixing.

Show **energy arrow**, **atomic orbitals** involved, **atom labels**, the **molecular orbitals** that are formed, **filled electrons** in the MO diagram etc. Make sure to label all the atomic and molecular orbitals.

Using the MO diagram above, answer the following questions:

- (i) Which one of the following will have the highest bond order? Circle the correct choice
 A) PS^{1-} B) PS C) PS^{1+}
- (ii) Which one of the following will have the longest bond length? Circle the correct choice
 A) PS^{1-} B) PS C) PS^{1+}
- (iii) Which of the following will be diamagnetic? Circle the correct choice.
 A) PS^{1-} B) PS C) PS^{1+}
- (iv) Draw Lewis dot structure of the molecule with the shortest bond length (based on the MO diagram) from among the three (PS^{1-} , PS, PS^{1+}), and indicate the bond order calculated from the Lewis dot structure. **(3 points)**

Lewis Dot Structure

Bond Order



3CB8A522-B55C-47DB-80C1-FF7C4E854FCA

midterm-exam-93388

#624 Page 10 of 10

SCRAP PAGE

**CHEMISTRY 110 –Midterm Exam****Thursday November 7, 2024, 6:30 – 8:30 pm****Examiners: Pallavi Sirjoosingh and Ashok Kakkar**

First name (please write as legibly as possible within the boxes)

--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--

Last name

--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--

Student ID number

--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--

This booklet has 8 questions on 10 pages (The last page can be used as a scrap page).

The staple must NOT be removed from the question booklet. Your intact question booklet must be returned at the end of the exam.

A separate booklet contains a periodic table, and datasheet.

Write your answers directly under the questions in the space provided. Do not write on the QR codes or near the top edges (as it may be cut off).

Please write **legibly and dark enough** for the scan to be readable. For students who use English as a second language, dictionaries are permitted. Calculators are allowed.

DO NOT ASK QUESTIONS OF THE INVIGILATORS about interpretations of the exam questions. If you have a concern about a question, please summarize it at the top of this page. All concerns will be evaluated after the exam. GOOD LUCK!

**Question 1 8 points** (*Assume all transitions are possible*)

a. 6 points Calculate the **energy (in kJ/mol)** required to ionize (completely remove) the electron from the **$n=3$** level of **He^+ ion** (He: Helium). Show your work including equations. (3 sig. figs.)

b. 2 points

(I) Select the largest (longest) wavelength from the three choices below:

- i. Wavelength of photon to *completely remove* an electron from *1s orbital of Hydrogen atom*
- ii. Wavelength of photon to *completely remove* an electron from *1s orbital of Boron atom (B)*
- iii. Wavelength of photon to *completely remove* an electron from *1s orbital of Boron cation (B^+)*

i or ii or iii

(II) Select the smallest (shortest) wavelength from the three choices below (*for H atom*):

- i. Wavelength of photon *emitted* for an electron transitioning from *2p to 1s orbital*
- ii. Wavelength of photon *absorbed* for an electron transitioning from *2s to 4p orbital*
- iii. Wavelength of photon *absorbed* for an electron transitioning from *1s to 3p orbital*

i or ii or iii



Question 2 8 points



A Chem110 student plots the graph shown on the left, from data obtained for the kinetic energy (eV) of the ejected electron versus frequency (s^{-1}) of the incident light, for an unknown metal. ($1\text{eV} = 1.6022 \times 10^{-19} \text{ J}$)

Based on the plot, answer the questions **a to c** below:

a. 1 point On the plot above, label the **threshold frequency**.

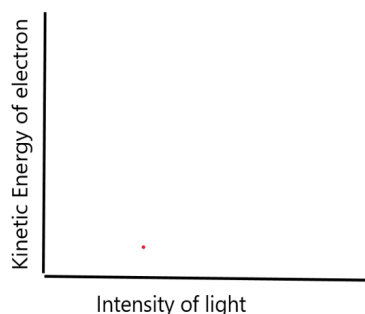
b. 2 points What is the value of the work potential (in J) of the unknown metal?

c. 3 points What is the *longest* wavelength (in nm) required to eject an electron from this metal's surface? Show your work including equation(s) used. (3 sig figs)

d. 2 points In the space provided below, plot the following

(Assume frequency is *above* threshold frequency):

- i. *Kinetic energy* of an ejected electron versus the *intensity* of the incident light (left)
- ii. *Number of electrons* ejected versus the *frequency* of the incident light (right)



**Question 3** **6 points****a. 2 points**

In the ground state of Ge (Germanium; $Z = 32$) atom, **name** the following orbitals:

i. Orbital containing the electron that requires the maximum energy to be removed.

ii. Orbital containing the electron that experiences the minimum effective nuclear charge.

b. 2 points

An unknown anion (A^-) has the following *excited* electronic configuration: $[Ar] 4s^2 3d^3 4p^1$

(Ar: Argon; $Z = 18$)

Give the **condensed ground state electronic configuration** of the atom, **A** and the cation, **A^{2+}** .
Show your work. (Assume all electronic transitions are possible)

Electronic Configuration of A:

Electronic Configuration of A^{2+} :

c. 2 points

In a single atom, determine how many electrons can have the following set of quantum numbers?

(i) $n = 5; l = 3$

In a single atom, determine how many electrons can have the following set of quantum numbers?

(i) $n = 3; l = 2; m_s = +1/2$

**Question 4** **8 points**

Rank the following from 1 to 4 in **increasing** order. Provide the ranking in the boxes provided.

Note: Atomic number (Z) are provided for relevant atoms

a) Increasing (lowest to highest) order of *ionization energy*: (1= lowest; 4 = highest)

Mg

Mg: Z = 12

Rb

Rb: Z = 37

Cs

Cs: Z = 55

K

K : Z = 19

b) Increasing (lowest to highest) order of *bond angle*: (1= lowest; 4 = highest)

H₂OCCl₄XeF₄

HCN

c) Increasing (smallest to largest) order of *ionic size* (1= smallest; 4 = largest)

S²⁻

S: Z = 16

O²⁻

O: Z = 8

Be²⁺

Be: Z = 4

Na⁺

Na: Z = 11

d) Increasing (lowest to highest) order of *bond energy* (1= lowest; 4 = highest)

P-F

P-Cl

P-I

P-Br

**Question 5** **12 points**

a. 9 points For the molecules HClO_4 and TeF_6 , draw the most stable *Lewis structure* showing appropriate shape and *dashed/wedged bonds*, based on VSEPR. Determine the **molecular geometry** around the central atom. For HClO_4 , label the *most polar bond*.



b. 3 points (Note: Lone pairs are not shown in structures below)

(i) For the three *resonance* structures (1, 2 and 3) shown below, answer the following question:



(I) Circle the *most contributing* resonance structure.

Structure 1

Structure 2

Structure 3

(ii) For the three structures given (4, 5, and 6) below, answer the following questions:



(I) Circle all the structures that represent *resonance structures* of each other:

Structure 4

Structure 5

Structure 6

(II) Among the resonance structures, circle the *most contributing* resonance structure.

Structure 4

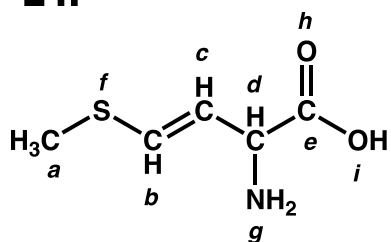
Structure 5

Structure 6

**Question 6** **6 points**

Using the data given below, calculate the **lattice energy** for SrO(s).

	ΔH° (kJ/mol)
$\text{Sr}(s) \longrightarrow \text{Sr}(g)$	+180
$\text{Sr}(g) \longrightarrow \text{Sr}^{2+}(g) + 2e^-$	+1468
$\text{O}_2(g) \longrightarrow 2\text{O}(g)$	+500
$\text{O}(g) + 2e^- \longrightarrow \text{O}^{2-}(g)$	+600
$\text{Sr}(s) + \frac{1}{2}\text{O}_2(g) \longrightarrow \text{SrO}(s)$	-550

**Question 7 10 points**

Consider the molecule given on the left, and answer questions (i) to (vi) using chemical bonding theories that we have learnt in class (valence bond theory, hybridization).

Note that the lone pairs are not shown in the structure:

i. Consider the carbon atoms marked in as ***a***, ***b***, ***c***, ***d***, and ***e*** in the molecule, and determine **hybridization** at each of the carbon centers. (2.5 points)

Atom	Hybridization
Carbon (<i>a</i>)	
Carbon (<i>b</i>)	
Carbon (<i>c</i>)	
Carbon (<i>d</i>)	
Carbon (<i>e</i>)	

ii. Now, consider the sulfur atom (S) marked as ***f***, nitrogen (N) atom (***g***), and oxygen(O) atoms (***h*** and ***i***), and determine their **hybridization**. (2 points)

Atom	Hybridization
Sulfur (<i>f</i>)	
Nitrogen (<i>g</i>)	
Oxygen (<i>h</i>)	
Oxygen (<i>i</i>)	

iii. **Describe** all the bonds formed by the carbon atoms ***a*** and ***b***, using sigma and pi designation as appropriate. (2 points)

Atom	Bonds formed (use sigma and pi as appropriate)
Carbon (<i>a</i>)	
Carbon (<i>b</i>)	

iv. How many total sp^3 -s sigma bonds are in the molecule? (1 point)

v. Determine **electronic** and **molecular geometry** at carbon (C) labeled ***e*** and sulfur (S) labeled ***f*** centres. (2 points)

Atom	Electronic Geometry	Molecular Geometry
Carbon (<i>e</i>)		
Sulfur (<i>f</i>)		

vi. Circle the value from the choices given below, that describes the **C-S-C bond angle at sulfur** (S) marked ***f***: (0.5 points)

$>90^\circ$ to $<109.5^\circ$ 109.5° >109.5 to $<120^\circ$ 120° $>120^\circ$ to $<180^\circ$



Question 8 12 points

Using the **valence shell orbitals only** draw a **complete** molecular orbital (MO) diagram (with no s-p mixing) for the molecule **PS**, in the space provided below. **(6 points)**

Assume that 3s and 3p atomic orbitals will form molecular orbitals like those formed by the combination of 2s and 2p atomic orbitals that we learnt in class, and there is no s-p mixing.

Show **energy arrow**, **atomic orbitals** involved, **atom labels**, the **molecular orbitals** that are formed, **filled electrons** in the MO diagram etc. Make sure to label all the atomic and molecular orbitals.

Using the MO diagram above, answer the following questions:

- (i) Which one of the following will have the highest bond order? Circle the correct choice
 A) PS^{1-} B) PS C) PS^{1+}
- (ii) Which one of the following will have the longest bond length? Circle the correct choice
 A) PS^{1-} B) PS C) PS^{1+}
- (iii) Which of the following will be diamagnetic? Circle the correct choice.
 A) PS^{1-} B) PS C) PS^{1+}
- (iv) Draw Lewis dot structure of the molecule with the shortest bond length (based on the MO diagram) from among the three (PS^{1-} , PS, PS^{1+}), and indicate the bond order calculated from the Lewis dot structure. **(3 points)**

Lewis Dot Structure

Bond Order



EC0A0A17-3471-49F6-90FB-2D4A4BD8F9ED

midterm-exam-93388

#625 Page 10 of 10

SCRAP PAGE

**CHEMISTRY 110 –Midterm Exam****Thursday November 7, 2024, 6:30 – 8:30 pm**

Examiners: Pallavi Sirjoosingh and Ashok Kakkar

First name (please write as legibly as possible within the boxes)

--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--

Last name

--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--

Student ID number

--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--

This booklet has 8 questions on 10 pages (The last page can be used as a scrap page).

The staple must NOT be removed from the question booklet. Your intact question booklet must be returned at the end of the exam.

A separate booklet contains a periodic table, and datasheet.

Write your answers directly under the questions in the space provided. Do not write on the QR codes or near the top edges (as it may be cut off).

Please write **legibly and dark enough** for the scan to be readable. For students who use English as a second language, dictionaries are permitted. Calculators are allowed.

DO NOT ASK QUESTIONS OF THE INVIGILATORS about interpretations of the exam questions. If you have a concern about a question, please summarize it at the top of this page. All concerns will be evaluated after the exam. GOOD LUCK!

**Question 1 8 points** (*Assume all transitions are possible*)

a. 6 points Calculate the **energy (in kJ/mol)** required to ionize (completely remove) the electron from the **n=3** level of **He⁺ ion** (He: Helium). Show your work including equations. (3 sig. figs.)

b. 2 points

(I) Select the largest (longest) wavelength from the three choices below:

- i. Wavelength of photon to *completely remove* an electron from *1s orbital of Hydrogen atom*
- ii. Wavelength of photon to *completely remove* an electron from *1s orbital of Boron atom (B)*
- iii. Wavelength of photon to *completely remove* an electron from *1s orbital of Boron cation (B⁺)*

i or ii or iii

(II) Select the smallest (shortest) wavelength from the three choices below (*for H atom*):

- i. Wavelength of photon *emitted* for an electron transitioning from *2p to 1s orbital*
- ii. Wavelength of photon *absorbed* for an electron transitioning from *2s to 4p orbital*
- iii. Wavelength of photon *absorbed* for an electron transitioning from *1s to 3p orbital*

i or ii or iii



Question 2 8 points



A Chem110 student plots the graph shown on the left, from data obtained for the kinetic energy (eV) of the ejected electron versus frequency (s^{-1}) of the incident light, for an unknown metal. ($1\text{eV} = 1.6022 \times 10^{-19} \text{ J}$)

Based on the plot, answer the questions **a to c** below:

a. 1 point On the plot above, label the **threshold frequency**.

b. 2 points What is the value of the work potential (in J) of the unknown metal?

c. 3 points What is the *longest* wavelength (in nm) required to eject an electron from this metal's surface? Show your work including equation(s) used. (3 sig figs)

d. 2 points In the space provided below, plot the following

(Assume frequency is *above* threshold frequency):

- Kinetic energy* of an ejected electron versus the *intensity* of the incident light (left)
- Number of electrons* ejected versus the *frequency* of the incident light (right)



**Question 3** **6 points****a. 2 points**

In the ground state of Ge (Germanium; $Z = 32$) atom, **name** the following orbitals:

i. Orbital containing the electron that requires the maximum energy to be removed.

ii. Orbital containing the electron that experiences the minimum effective nuclear charge.

b. 2 points

An unknown anion (A^-) has the following *excited* electronic configuration: $[Ar] 4s^2 3d^3 4p^1$

(Ar: Argon; $Z = 18$)

Give the **condensed ground state electronic configuration** of the atom, **A** and the cation, **A^{2+}** .
Show your work. (Assume all electronic transitions are possible)

Electronic Configuration of A:

Electronic Configuration of A^{2+} :

c. 2 points

In a single atom, determine how many electrons can have the following set of quantum numbers?

(i) $n = 5; l = 3$

In a single atom, determine how many electrons can have the following set of quantum numbers?

(i) $n = 3; l = 2; m_s = +1/2$

**Question 4** **8 points**

Rank the following from 1 to 4 in **increasing** order. Provide the ranking in the boxes provided.

Note: Atomic number (Z) are provided for relevant atoms

a) Increasing (lowest to highest) order of *ionization energy*: (1= lowest; 4 = highest)

Mg

Mg: Z = 12

Rb

Rb: Z = 37

Cs

Cs: Z = 55

K

K : Z = 19

b) Increasing (lowest to highest) order of *bond angle*: (1= lowest; 4 = highest)

H₂OCCl₄XeF₄

HCN

c) Increasing (smallest to largest) order of *ionic size* (1= smallest; 4 = largest)

S²⁻

S: Z = 16

O²⁻

O: Z = 8

Be²⁺

Be: Z = 4

Na⁺

Na: Z = 11

d) Increasing (lowest to highest) order of *bond energy* (1= lowest; 4 = highest)

P-F

P-Cl

P-I

P-Br

**Question 5** **12 points**

a. 9 points For the molecules HClO_4 and TeF_6 , draw the most stable *Lewis structure* showing appropriate shape and *dashed/wedged bonds*, based on VSEPR. Determine the **molecular geometry** around the central atom. For HClO_4 , label the *most polar bond*.



b. 3 points (Note: Lone pairs are not shown in structures below)

(i) For the three *resonance* structures (1, 2 and 3) shown below, answer the following question:



(I) Circle the *most contributing* resonance structure.

Structure 1

Structure 2

Structure 3

(ii) For the three structures given (4, 5, and 6) below, answer the following questions:



(I) Circle all the structures that represent *resonance structures* of each other:

Structure 4

Structure 5

Structure 6

(II) Among the resonance structures, circle the *most contributing* resonance structure.

Structure 4

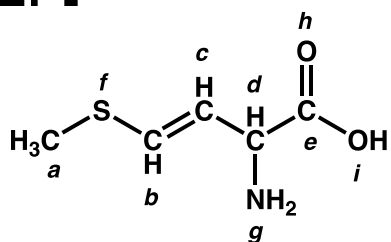
Structure 5

Structure 6

**Question 6** **6 points**

Using the data given below, calculate the **lattice energy** for SrO(s).

	ΔH° (kJ/mol)
$\text{Sr}(s) \longrightarrow \text{Sr}(g)$	+180
$\text{Sr}(g) \longrightarrow \text{Sr}^{2+}(g) + 2e^-$	+1468
$\text{O}_2(g) \longrightarrow 2\text{O}(g)$	+500
$\text{O}(g) + 2e^- \longrightarrow \text{O}^{2-}(g)$	+600
$\text{Sr}(s) + \frac{1}{2}\text{O}_2(g) \longrightarrow \text{SrO}(s)$	-550

**Question 7 10 points**

Consider the molecule given on the left, and answer questions (i) to (vi) using chemical bonding theories that we have learnt in class (valence bond theory, hybridization).

Note that the lone pairs are not shown in the structure:

i. Consider the carbon atoms marked in as ***a***, ***b***, ***c***, ***d***, and ***e*** in the molecule, and determine **hybridization** at each of the carbon centers. (2.5 points)

Atom	Hybridization
Carbon (<i>a</i>)	
Carbon (<i>b</i>)	
Carbon (<i>c</i>)	
Carbon (<i>d</i>)	
Carbon (<i>e</i>)	

ii. Now, consider the sulfur atom (S) marked as ***f***, nitrogen (N) atom (***g***), and oxygen(O) atoms (***h*** and ***i***), and determine their **hybridization**. (2 points)

Atom	Hybridization
Sulfur (<i>f</i>)	
Nitrogen (<i>g</i>)	
Oxygen (<i>h</i>)	
Oxygen (<i>i</i>)	

iii. **Describe** all the bonds formed by the carbon atoms ***a*** and ***b***, using sigma and pi designation as appropriate. (2 points)

Atom	Bonds formed (use sigma and pi as appropriate)
Carbon (<i>a</i>)	
Carbon (<i>b</i>)	

iv. How many total sp^3 -s sigma bonds are in the molecule? (1 point)

v. Determine **electronic** and **molecular geometry** at carbon (C) labeled ***e*** and sulfur (S) labeled ***f*** centres. (2 points)

Atom	Electronic Geometry	Molecular Geometry
Carbon (<i>e</i>)		
Sulfur (<i>f</i>)		

vi. Circle the value from the choices given below, that describes the **C-S-C bond angle at sulfur** (S) marked ***f***: (0.5 points)

>90° to <109.5°

109.5°

>109.5 to <120°

120°

>120° to <180°



Question 8 12 points

Using the **valence shell orbitals only** draw a **complete** molecular orbital (MO) diagram (with no s-p mixing) for the molecule **PS**, in the space provided below. **(6 points)**

Assume that 3s and 3p atomic orbitals will form molecular orbitals like those formed by the combination of 2s and 2p atomic orbitals that we learnt in class, and there is no s-p mixing.

Show **energy arrow**, **atomic orbitals** involved, **atom labels**, the **molecular orbitals** that are formed, **filled electrons** in the MO diagram etc. Make sure to label all the atomic and molecular orbitals.

Using the MO diagram above, answer the following questions:

- (i) Which one of the following will have the highest bond order? Circle the correct choice
 A) PS^{1-} B) PS C) PS^{1+}
- (ii) Which one of the following will have the longest bond length? Circle the correct choice
 A) PS^{1-} B) PS C) PS^{1+}
- (iii) Which of the following will be diamagnetic? Circle the correct choice.
 A) PS^{1-} B) PS C) PS^{1+}
- (iv) Draw Lewis dot structure of the molecule with the shortest bond length (based on the MO diagram) from among the three (PS^{1-} , PS, PS^{1+}), and indicate the bond order calculated from the Lewis dot structure. **(3 points)**

Lewis Dot Structure

Bond Order



229AE767-7258-4E68-A76B-FB122FA07D49

midterm-exam-93388

#626 Page 10 of 10

SCRAP PAGE

**CHEMISTRY 110 –Midterm Exam****Thursday November 7, 2024, 6:30 – 8:30 pm****Examiners: Pallavi Sirjoosingh and Ashok Kakkar**

First name (please write as legibly as possible within the boxes)

--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--

Last name

--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--

Student ID number

--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--

This booklet has 8 questions on 10 pages (The last page can be used as a scrap page).

The staple must NOT be removed from the question booklet. Your intact question booklet must be returned at the end of the exam.

A separate booklet contains a periodic table, and datasheet.

Write your answers directly under the questions in the space provided. Do not write on the QR codes or near the top edges (as it may be cut off).

Please write **legibly and dark enough** for the scan to be readable. For students who use English as a second language, dictionaries are permitted. Calculators are allowed.

DO NOT ASK QUESTIONS OF THE INVIGILATORS about interpretations of the exam questions. If you have a concern about a question, please summarize it at the top of this page. All concerns will be evaluated after the exam. GOOD LUCK!

**Question 1 8 points** (*Assume all transitions are possible*)

a. 6 points Calculate the **energy (in kJ/mol)** required to ionize (completely remove) the electron from the **$n=3$** level of **He^+ ion** (He: Helium). Show your work including equations. (3 sig. figs.)

b. 2 points

(I) Select the largest (longest) wavelength from the three choices below:

- i. Wavelength of photon to *completely remove* an electron from *1s orbital of Hydrogen atom*
- ii. Wavelength of photon to *completely remove* an electron from *1s orbital of Boron atom (B)*
- iii. Wavelength of photon to *completely remove* an electron from *1s orbital of Boron cation (B^+)*

i or ii or iii

(II) Select the smallest (shortest) wavelength from the three choices below (*for H atom*):

- i. Wavelength of photon *emitted* for an electron transitioning from *2p to 1s orbital*
- ii. Wavelength of photon *absorbed* for an electron transitioning from *2s to 4p orbital*
- iii. Wavelength of photon *absorbed* for an electron transitioning from *1s to 3p orbital*

i or ii or iii



Question 2 8 points



A Chem110 student plots the graph shown on the left, from data obtained for the kinetic energy (eV) of the ejected electron versus frequency (s^{-1}) of the incident light, for an unknown metal. ($1\text{eV} = 1.6022 \times 10^{-19} \text{ J}$)

Based on the plot, answer the questions **a to c** below:

a. 1 point On the plot above, label the **threshold frequency**.

b. 2 points What is the value of the work potential (in J) of the unknown metal?

c. 3 points What is the *longest* wavelength (in nm) required to eject an electron from this metal's surface? Show your work including equation(s) used. (3 sig figs)

d. 2 points In the space provided below, plot the following

(Assume frequency is *above* threshold frequency):

- Kinetic energy* of an ejected electron versus the *intensity* of the incident light (left)
- Number of electrons* ejected versus the *frequency* of the incident light (right)



**Question 3** **6 points****a. 2 points**

In the ground state of Ge (Germanium; $Z = 32$) atom, **name** the following orbitals:

i. Orbital containing the electron that requires the maximum energy to be removed.

ii. Orbital containing the electron that experiences the minimum effective nuclear charge.

b. 2 points

An unknown anion (A^-) has the following *excited* electronic configuration: $[Ar] 4s^2 3d^3 4p^1$

(Ar: Argon; $Z = 18$)

Give the **condensed ground state electronic configuration** of the atom, **A** and the cation, **A^{2+}** .
Show your work. (Assume all electronic transitions are possible)

Electronic Configuration of A:

Electronic Configuration of A^{2+} :

c. 2 points

In a single atom, determine how many electrons can have the following set of quantum numbers?

(i) $n = 5; l = 3$

In a single atom, determine how many electrons can have the following set of quantum numbers?

(i) $n = 3; l = 2; m_s = +1/2$

**Question 4** **8 points**

Rank the following from 1 to 4 in **increasing** order. Provide the ranking in the boxes provided.

Note: Atomic number (Z) are provided for relevant atoms

a) Increasing (lowest to highest) order of *ionization energy*: (1= lowest; 4 = highest)

Mg

Mg: Z = 12

Rb

Rb: Z = 37

Cs

Cs: Z = 55

K

K : Z = 19

b) Increasing (lowest to highest) order of *bond angle*: (1= lowest; 4 = highest)

H₂OCCl₄XeF₄

HCN

c) Increasing (smallest to largest) order of *ionic size* (1= smallest; 4 = largest)

S²⁻

S: Z = 16

O²⁻

O: Z = 8

Be²⁺

Be: Z = 4

Na⁺

Na: Z = 11

d) Increasing (lowest to highest) order of *bond energy* (1= lowest; 4 = highest)

P-F

P-Cl

P-I

P-Br

**Question 5** **12 points**

a. 9 points For the molecules HClO_4 and TeF_6 , draw the most stable *Lewis structure* showing appropriate shape and *dashed/wedged bonds*, based on VSEPR. Determine the **molecular geometry** around the central atom. For HClO_4 , label the *most polar bond*.



b. 3 points (Note: Lone pairs are not shown in structures below)

(i) For the three *resonance* structures (1, 2 and 3) shown below, answer the following question:



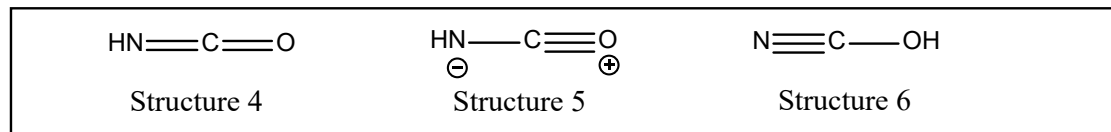
(I) Circle the *most contributing* resonance structure.

Structure 1

Structure 2

Structure 3

(ii) For the three structures given (4, 5, and 6) below, answer the following questions:



(I) Circle all the structures that represent *resonance structures* of each other:

Structure 4

Structure 5

Structure 6

(II) Among the resonance structures, circle the *most contributing* resonance structure.

Structure 4

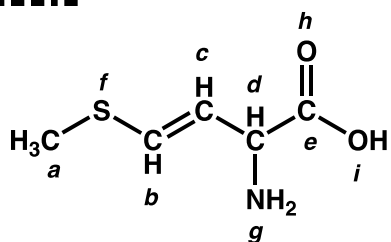
Structure 5

Structure 6

**Question 6** **6 points**

Using the data given below, calculate the **lattice energy** for SrO(s).

	ΔH° (kJ/mol)
$\text{Sr}(s) \longrightarrow \text{Sr}(g)$	+180
$\text{Sr}(g) \longrightarrow \text{Sr}^{2+}(g) + 2e^-$	+1468
$\text{O}_2(g) \longrightarrow 2\text{O}(g)$	+500
$\text{O}(g) + 2e^- \longrightarrow \text{O}^{2-}(g)$	+600
$\text{Sr}(s) + \frac{1}{2}\text{O}_2(g) \longrightarrow \text{SrO}(s)$	-550

**Question 7 10 points**

Consider the molecule given on the left, and answer questions (i) to (vi) using chemical bonding theories that we have learnt in class (valence bond theory, hybridization).

Note that the lone pairs are not shown in the structure:

i. Consider the carbon atoms marked in as ***a***, ***b***, ***c***, ***d***, and ***e*** in the molecule, and determine **hybridization** at each of the carbon centers. (2.5 points)

Atom	Hybridization
Carbon (<i>a</i>)	
Carbon (<i>b</i>)	
Carbon (<i>c</i>)	
Carbon (<i>d</i>)	
Carbon (<i>e</i>)	

ii. Now, consider the sulfur atom (S) marked as ***f***, nitrogen (N) atom (***g***), and oxygen(O) atoms (***h*** and ***i***), and determine their **hybridization**. (2 points)

Atom	Hybridization
Sulfur (<i>f</i>)	
Nitrogen (<i>g</i>)	
Oxygen (<i>h</i>)	
Oxygen (<i>i</i>)	

iii. **Describe** all the bonds formed by the carbon atoms ***a*** and ***b***, using sigma and pi designation as appropriate. (2 points)

Atom	Bonds formed (use sigma and pi as appropriate)
Carbon (<i>a</i>)	
Carbon (<i>b</i>)	

iv. How many total sp^3 -s sigma bonds are in the molecule? (1 point)

v. Determine **electronic** and **molecular geometry** at carbon (C) labeled ***e*** and sulfur (S) labeled ***f*** centres. (2 points)

Atom	Electronic Geometry	Molecular Geometry
Carbon (<i>e</i>)		
Sulfur (<i>f</i>)		

vi. Circle the value from the choices given below, that describes the **C-S-C bond angle at sulfur** (S) marked ***f***: (0.5 points)

$>90^\circ$ to $<109.5^\circ$ 109.5° >109.5 to $<120^\circ$ 120° $>120^\circ$ to $<180^\circ$


Question 8 12 points

Using the **valence shell orbitals only** draw a **complete** molecular orbital (MO) diagram (with no s-p mixing) for the molecule **PS**, in the space provided below. **(6 points)**

Assume that 3s and 3p atomic orbitals will form molecular orbitals like those formed by the combination of 2s and 2p atomic orbitals that we learnt in class, and there is no s-p mixing.

Show **energy arrow**, **atomic orbitals** involved, **atom labels**, the **molecular orbitals** that are formed, **filled electrons** in the MO diagram etc. Make sure to label all the atomic and molecular orbitals.

Using the MO diagram above, answer the following questions:

- (i) Which one of the following will have the highest bond order? Circle the correct choice
 A) PS^{1-} B) PS C) PS^{1+}
- (ii) Which one of the following will have the longest bond length? Circle the correct choice
 A) PS^{1-} B) PS C) PS^{1+}
- (iii) Which of the following will be diamagnetic? Circle the correct choice.
 A) PS^{1-} B) PS C) PS^{1+}
- (iv) Draw Lewis dot structure of the molecule with the shortest bond length (based on the MO diagram) from among the three (PS^{1-} , PS, PS^{1+}), and indicate the bond order calculated from the Lewis dot structure. **(3 points)**

Lewis Dot Structure

Bond Order



316DBEF5-EAB6-4E72-9743-D3E3C30D5EA3

midterm-exam-93388

#627 Page 10 of 10

SCRAP PAGE

**CHEMISTRY 110 –Midterm Exam****Thursday November 7, 2024, 6:30 – 8:30 pm****Examiners: Pallavi Sirjoosingh and Ashok Kakkar**

First name (please write as legibly as possible within the boxes)

--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--

Last name

--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--

Student ID number

--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--

This booklet has 8 questions on 10 pages (The last page can be used as a scrap page).

The staple must NOT be removed from the question booklet. Your intact question booklet must be returned at the end of the exam.

A separate booklet contains a periodic table, and datasheet.

Write your answers directly under the questions in the space provided. Do not write on the QR codes or near the top edges (as it may be cut off).

Please write **legibly and dark enough** for the scan to be readable. For students who use English as a second language, dictionaries are permitted. Calculators are allowed.

DO NOT ASK QUESTIONS OF THE INVIGILATORS about interpretations of the exam questions. If you have a concern about a question, please summarize it at the top of this page. All concerns will be evaluated after the exam. GOOD LUCK!

**Question 1 8 points** (*Assume all transitions are possible*)

a. 6 points Calculate the **energy (in kJ/mol)** required to ionize (completely remove) the electron from the **n=3** level of **He⁺ ion** (He: Helium). Show your work including equations. (3 sig. figs.)

b. 2 points

(I) Select the largest (longest) wavelength from the three choices below:

- i. Wavelength of photon to *completely remove* an electron from *1s orbital of Hydrogen atom*
- ii. Wavelength of photon to *completely remove* an electron from *1s orbital of Boron atom (B)*
- iii. Wavelength of photon to *completely remove* an electron from *1s orbital of Boron cation (B⁺)*

i or ii or iii

(II) Select the smallest (shortest) wavelength from the three choices below (*for H atom*):

- i. Wavelength of photon *emitted* for an electron transitioning from *2p to 1s orbital*
- ii. Wavelength of photon *absorbed* for an electron transitioning from *2s to 4p orbital*
- iii. Wavelength of photon *absorbed* for an electron transitioning from *1s to 3p orbital*

i or ii or iii



Question 2 8 points



A Chem110 student plots the graph shown on the left, from data obtained for the kinetic energy (eV) of the ejected electron versus frequency (s^{-1}) of the incident light, for an unknown metal. ($1\text{eV} = 1.6022 \times 10^{-19} \text{ J}$)

Based on the plot, answer the questions **a to c** below:

a. 1 point On the plot above, label the **threshold frequency**.

b. 2 points What is the value of the work potential (in J) of the unknown metal?

c. 3 points What is the *longest* wavelength (in nm) required to eject an electron from this metal's surface? Show your work including equation(s) used. (3 sig figs)

d. 2 points In the space provided below, plot the following

(Assume frequency is *above* threshold frequency):

- Kinetic energy* of an ejected electron versus the *intensity* of the incident light (left)
- Number of electrons* ejected versus the *frequency* of the incident light (right)



**Question 3** **6 points****a. 2 points**

In the ground state of Ge (Germanium; $Z = 32$) atom, **name** the following orbitals:

i. Orbital containing the electron that requires the maximum energy to be removed.

ii. Orbital containing the electron that experiences the minimum effective nuclear charge.

b. 2 points

An unknown anion (A^-) has the following *excited* electronic configuration: $[Ar] 4s^2 3d^3 4p^1$

(Ar: Argon; $Z = 18$)

Give the **condensed ground state electronic configuration** of the atom, **A** and the cation, **A^{2+}** .
Show your work. (Assume all electronic transitions are possible)

Electronic Configuration of A:

Electronic Configuration of A^{2+} :

c. 2 points

In a single atom, determine how many electrons can have the following set of quantum numbers?

(i) $n = 5; l = 3$

In a single atom, determine how many electrons can have the following set of quantum numbers?

(i) $n = 3; l = 2; m_s = +1/2$

**Question 4** **8 points**

Rank the following from 1 to 4 in **increasing** order. Provide the ranking in the boxes provided.

Note: Atomic number (Z) are provided for relevant atoms

a) Increasing (lowest to highest) order of *ionization energy*: (1= lowest; 4 = highest)

Mg

Mg: Z = 12

Rb

Rb: Z = 37

Cs

Cs: Z = 55

K

K : Z = 19

b) Increasing (lowest to highest) order of *bond angle*: (1= lowest; 4 = highest)

H₂OCCl₄XeF₄

HCN

c) Increasing (smallest to largest) order of *ionic size* (1= smallest; 4 = largest)

S²⁻

S: Z = 16

O²⁻

O: Z = 8

Be²⁺

Be: Z = 4

Na⁺

Na: Z = 11

d) Increasing (lowest to highest) order of *bond energy* (1= lowest; 4 = highest)

P-F

P-Cl

P-I

P-Br

**Question 5** **12 points**

a. 9 points For the molecules HClO_4 and TeF_6 , draw the most stable *Lewis structure* showing appropriate shape and *dashed/wedged bonds*, based on VSEPR. Determine the **molecular geometry** around the central atom. For HClO_4 , label the *most polar bond*.



b. 3 points (Note: Lone pairs are not shown in structures below)

(i) For the three *resonance* structures (1, 2 and 3) shown below, answer the following question:



(I) Circle the *most contributing* resonance structure.

Structure 1

Structure 2

Structure 3

(ii) For the three structures given (4, 5, and 6) below, answer the following questions:



(I) Circle all the structures that represent *resonance structures* of each other:

Structure 4

Structure 5

Structure 6

(II) Among the resonance structures, circle the *most contributing* resonance structure.

Structure 4

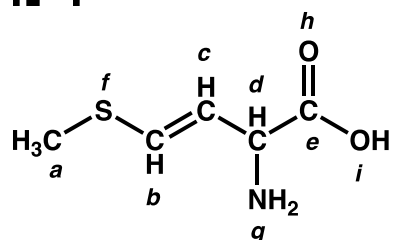
Structure 5

Structure 6

**Question 6** **6 points**

Using the data given below, calculate the **lattice energy** for SrO(s).

	ΔH° (kJ/mol)
$\text{Sr}(s) \longrightarrow \text{Sr}(g)$	+180
$\text{Sr}(g) \longrightarrow \text{Sr}^{2+}(g) + 2e^-$	+1468
$\text{O}_2(g) \longrightarrow 2\text{O}(g)$	+500
$\text{O}(g) + 2e^- \longrightarrow \text{O}^{2-}(g)$	+600
$\text{Sr}(s) + \frac{1}{2}\text{O}_2(g) \longrightarrow \text{SrO}(s)$	-550

**Question 7 10 points**

Consider the molecule given on the left, and answer questions (i) to (vi) using chemical bonding theories that we have learnt in class (valence bond theory, hybridization).

Note that the lone pairs are not shown in the structure:

i. Consider the carbon atoms marked in as ***a***, ***b***, ***c***, ***d***, and ***e*** in the molecule, and determine **hybridization** at each of the carbon centers. (2.5 points)

Atom	Hybridization
Carbon (<i>a</i>)	
Carbon (<i>b</i>)	
Carbon (<i>c</i>)	
Carbon (<i>d</i>)	
Carbon (<i>e</i>)	

ii. Now, consider the sulfur atom (S) marked as ***f***, nitrogen (N) atom (***g***), and oxygen(O) atoms (***h*** and ***i***), and determine their **hybridization**. (2 points)

Atom	Hybridization
Sulfur (<i>f</i>)	
Nitrogen (<i>g</i>)	
Oxygen (<i>h</i>)	
Oxygen (<i>i</i>)	

iii. **Describe** all the bonds formed by the carbon atoms ***a*** and ***b***, using sigma and pi designation as appropriate. (2 points)

Atom	Bonds formed (use sigma and pi as appropriate)
Carbon (<i>a</i>)	
Carbon (<i>b</i>)	

iv. How many total sp^3 -s sigma bonds are in the molecule? (1 point)

v. Determine **electronic** and **molecular geometry** at carbon (C) labeled ***e*** and sulfur (S) labeled ***f*** centres. (2 points)

Atom	Electronic Geometry	Molecular Geometry
Carbon (<i>e</i>)		
Sulfur (<i>f</i>)		

vi. Circle the value from the choices given below, that describes the **C-S-C bond angle at sulfur** (S) marked ***f***: (0.5 points)

>90° to <109.5°

109.5°

>109.5 to <120°

120°

>120° to <180°

**Question 8 12 points**

Using the **valence shell orbitals only** draw a **complete** molecular orbital (MO) diagram (with no s-p mixing) for the molecule **PS**, in the space provided below. **(6 points)**

Assume that 3s and 3p atomic orbitals will form molecular orbitals like those formed by the combination of 2s and 2p atomic orbitals that we learnt in class, and there is no s-p mixing.

Show **energy arrow**, **atomic orbitals** involved, **atom labels**, the **molecular orbitals** that are formed, **filled electrons** in the MO diagram etc. Make sure to label all the atomic and molecular orbitals.

Using the MO diagram above, answer the following questions:

- (i) Which one of the following will have the highest bond order? Circle the correct choice
A) PS^{1-} B) PS C) PS^{1+}
- (ii) Which one of the following will have the longest bond length? Circle the correct choice
A) PS^{1-} B) PS C) PS^{1+}
- (iii) Which of the following will be diamagnetic? Circle the correct choice.
A) PS^{1-} B) PS C) PS^{1+}
- (iv) Draw Lewis dot structure of the molecule with the shortest bond length (based on the MO diagram) from among the three (PS^{1-} , PS, PS^{1+}), and indicate the bond order calculated from the Lewis dot structure. **(3 points)**

Lewis Dot Structure

Bond Order



A7D4CAAE-6152-4130-B107-4DAF2A981805

midterm-exam-93388

#628 Page 10 of 10

SCRAP PAGE

**CHEMISTRY 110 –Midterm Exam****Thursday November 7, 2024, 6:30 – 8:30 pm**

Examiners: Pallavi Sirjoosingh and Ashok Kakkar

First name (please write as legibly as possible within the boxes)

--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--

Last name

--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--

Student ID number

--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--

This booklet has 8 questions on 10 pages (The last page can be used as a scrap page).

The staple must NOT be removed from the question booklet. Your intact question booklet must be returned at the end of the exam.

A separate booklet contains a periodic table, and datasheet.

Write your answers directly under the questions in the space provided. Do not write on the QR codes or near the top edges (as it may be cut off).

Please write **legibly and dark enough** for the scan to be readable. For students who use English as a second language, dictionaries are permitted. Calculators are allowed.

DO NOT ASK QUESTIONS OF THE INVIGILATORS about interpretations of the exam questions. If you have a concern about a question, please summarize it at the top of this page. All concerns will be evaluated after the exam. GOOD LUCK!

**Question 1 8 points** (*Assume all transitions are possible*)

a. 6 points Calculate the **energy (in kJ/mol)** required to ionize (completely remove) the electron from the **n=3** level of **He⁺ ion** (He: Helium). Show your work including equations. (3 sig. figs.)

b. 2 points

(I) Select the largest (longest) wavelength from the three choices below:

- i. Wavelength of photon to *completely remove* an electron from *1s orbital of Hydrogen atom*
- ii. Wavelength of photon to *completely remove* an electron from *1s orbital of Boron atom (B)*
- iii. Wavelength of photon to *completely remove* an electron from *1s orbital of Boron cation (B⁺)*

i or ii or iii

(II) Select the smallest (shortest) wavelength from the three choices below (*for H atom*):

- i. Wavelength of photon *emitted* for an electron transitioning from *2p to 1s orbital*
- ii. Wavelength of photon *absorbed* for an electron transitioning from *2s to 4p orbital*
- iii. Wavelength of photon *absorbed* for an electron transitioning from *1s to 3p orbital*

i or ii or iii



Question 2 8 points



A Chem110 student plots the graph shown on the left, from data obtained for the kinetic energy (eV) of the ejected electron versus frequency (s^{-1}) of the incident light, for an unknown metal. ($1\text{eV} = 1.6022 \times 10^{-19} \text{ J}$)

Based on the plot, answer the questions **a to c** below:

a. 1 point On the plot above, label the **threshold frequency**.

b. 2 points What is the value of the work potential (in J) of the unknown metal?

c. 3 points What is the *longest* wavelength (in nm) required to eject an electron from this metal's surface? Show your work including equation(s) used. (3 sig figs)

d. 2 points In the space provided below, plot the following

(Assume frequency is *above* threshold frequency):

- Kinetic energy* of an ejected electron versus the *intensity* of the incident light (left)
- Number of electrons* ejected versus the *frequency* of the incident light (right)



**Question 3** **6 points****a. 2 points**

In the ground state of Ge (Germanium; $Z = 32$) atom, **name** the following orbitals:

i. Orbital containing the electron that requires the maximum energy to be removed.

ii. Orbital containing the electron that experiences the minimum effective nuclear charge.

b. 2 points

An unknown anion (A^-) has the following *excited* electronic configuration: $[Ar] 4s^2 3d^3 4p^1$

(Ar: Argon; $Z = 18$)

Give the **condensed ground state electronic configuration** of the atom, **A** and the cation, **A^{2+}** .
Show your work. (Assume all electronic transitions are possible)

Electronic Configuration of A:

Electronic Configuration of A^{2+} :

c. 2 points

In a single atom, determine how many electrons can have the following set of quantum numbers?

(i) $n = 5; l = 3$

In a single atom, determine how many electrons can have the following set of quantum numbers?

(i) $n = 3; l = 2; m_s = +1/2$

**Question 4** **8 points**

Rank the following from 1 to 4 in **increasing** order. Provide the ranking in the boxes provided.

Note: Atomic number (Z) are provided for relevant atoms

a) Increasing (lowest to highest) order of *ionization energy*: (1= lowest; 4 = highest)

Mg

Mg: Z = 12

Rb

Rb: Z = 37

Cs

Cs: Z = 55

K

K : Z = 19

b) Increasing (lowest to highest) order of *bond angle*: (1= lowest; 4 = highest)

H₂OCCl₄XeF₄

HCN

c) Increasing (smallest to largest) order of *ionic size* (1= smallest; 4 = largest)

S²⁻

S: Z = 16

O²⁻

O: Z = 8

Be²⁺

Be: Z = 4

Na⁺

Na: Z = 11

d) Increasing (lowest to highest) order of *bond energy* (1= lowest; 4 = highest)

P-F

P-Cl

P-I

P-Br

**Question 5** **12 points**

a. 9 points For the molecules HClO_4 and TeF_6 , draw the most stable *Lewis structure* showing appropriate shape and *dashed/wedged bonds*, based on VSEPR. Determine the **molecular geometry** around the central atom. For HClO_4 , label the *most polar bond*.



b. 3 points (Note: Lone pairs are not shown in structures below)

(i) For the three *resonance* structures (1, 2 and 3) shown below, answer the following question:



(I) Circle the *most contributing* resonance structure.

Structure 1

Structure 2

Structure 3

(ii) For the three structures given (4, 5, and 6) below, answer the following questions:



(I) Circle all the structures that represent *resonance structures* of each other:

Structure 4

Structure 5

Structure 6

(II) Among the resonance structures, circle the *most contributing* resonance structure.

Structure 4

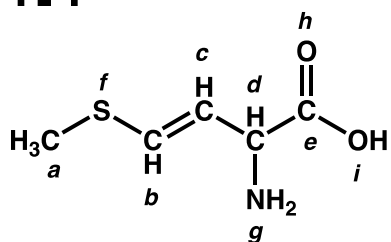
Structure 5

Structure 6

**Question 6** **6 points**

Using the data given below, calculate the **lattice energy** for SrO(s).

	ΔH° (kJ/mol)
$\text{Sr}(s) \longrightarrow \text{Sr}(g)$	+180
$\text{Sr}(g) \longrightarrow \text{Sr}^{2+}(g) + 2e^-$	+1468
$\text{O}_2(g) \longrightarrow 2\text{O}(g)$	+500
$\text{O}(g) + 2e^- \longrightarrow \text{O}^{2-}(g)$	+600
$\text{Sr}(s) + \frac{1}{2}\text{O}_2(g) \longrightarrow \text{SrO}(s)$	-550

**Question 7 10 points**

Consider the molecule given on the left, and answer questions (i) to (vi) using chemical bonding theories that we have learnt in class (valence bond theory, hybridization).

Note that the lone pairs are not shown in the structure:

i. Consider the carbon atoms marked in as ***a***, ***b***, ***c***, ***d***, and ***e*** in the molecule, and determine **hybridization** at each of the carbon centers. (2.5 points)

Atom	Hybridization
Carbon (<i>a</i>)	
Carbon (<i>b</i>)	
Carbon (<i>c</i>)	
Carbon (<i>d</i>)	
Carbon (<i>e</i>)	

ii. Now, consider the sulfur atom (S) marked as ***f***, nitrogen (N) atom (***g***), and oxygen(O) atoms (***h*** and ***i***), and determine their **hybridization**. (2 points)

Atom	Hybridization
Sulfur (<i>f</i>)	
Nitrogen (<i>g</i>)	
Oxygen (<i>h</i>)	
Oxygen (<i>i</i>)	

iii. **Describe** all the bonds formed by the carbon atoms ***a*** and ***b***, using sigma and pi designation as appropriate. (2 points)

Atom	Bonds formed (use sigma and pi as appropriate)
Carbon (<i>a</i>)	
Carbon (<i>b</i>)	

iv. How many total sp^3 -s sigma bonds are in the molecule? (1 point)

v. Determine **electronic** and **molecular geometry** at carbon (C) labeled ***e*** and sulfur (S) labeled ***f*** centres. (2 points)

Atom	Electronic Geometry	Molecular Geometry
Carbon (<i>e</i>)		
Sulfur (<i>f</i>)		

vi. Circle the value from the choices given below, that describes the **C-S-C bond angle at sulfur** (S) marked ***f***: (0.5 points)

>90° to <109.5°

109.5°

>109.5 to <120°

120°

>120° to <180°


Question 8 12 points

Using the **valence shell orbitals only** draw a **complete** molecular orbital (MO) diagram (with no s-p mixing) for the molecule **PS**, in the space provided below. **(6 points)**

Assume that 3s and 3p atomic orbitals will form molecular orbitals like those formed by the combination of 2s and 2p atomic orbitals that we learnt in class, and there is no s-p mixing.

Show **energy arrow**, **atomic orbitals** involved, **atom labels**, the **molecular orbitals** that are formed, **filled electrons** in the MO diagram etc. Make sure to label all the atomic and molecular orbitals.

Using the MO diagram above, answer the following questions:

- (i) Which one of the following will have the highest bond order? Circle the correct choice
 A) PS^{1-} B) PS C) PS^{1+}
- (ii) Which one of the following will have the longest bond length? Circle the correct choice
 A) PS^{1-} B) PS C) PS^{1+}
- (iii) Which of the following will be diamagnetic? Circle the correct choice.
 A) PS^{1-} B) PS C) PS^{1+}
- (iv) Draw Lewis dot structure of the molecule with the shortest bond length (based on the MO diagram) from among the three (PS^{1-} , PS, PS^{1+}), and indicate the bond order calculated from the Lewis dot structure. **(3 points)**

Lewis Dot Structure

Bond Order



BA0E2584-CE6C-4544-AC23-562A695ECEBD

midterm-exam-93388

#629 Page 10 of 10

SCRAP PAGE

**CHEMISTRY 110 –Midterm Exam****Thursday November 7, 2024, 6:30 – 8:30 pm****Examiners: Pallavi Sirjoosingh and Ashok Kakkar**

First name (please write as legibly as possible within the boxes)

--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--

Last name

--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--

Student ID number

--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--

This booklet has 8 questions on 10 pages (The last page can be used as a scrap page).

The staple must NOT be removed from the question booklet. Your intact question booklet must be returned at the end of the exam.

A separate booklet contains a periodic table, and datasheet.

Write your answers directly under the questions in the space provided. Do not write on the QR codes or near the top edges (as it may be cut off).

Please write **legibly and dark enough** for the scan to be readable. For students who use English as a second language, dictionaries are permitted. Calculators are allowed.

DO NOT ASK QUESTIONS OF THE INVIGILATORS about interpretations of the exam questions. If you have a concern about a question, please summarize it at the top of this page. All concerns will be evaluated after the exam. GOOD LUCK!

**Question 1 8 points** (*Assume all transitions are possible*)

a. 6 points Calculate the **energy (in kJ/mol)** required to ionize (completely remove) the electron from the **n=3** level of **He⁺ ion** (He: Helium). Show your work including equations. (3 sig. figs.)

b. 2 points

(I) Select the largest (longest) wavelength from the three choices below:

- i. Wavelength of photon to *completely remove* an electron from *1s orbital of Hydrogen atom*
- ii. Wavelength of photon to *completely remove* an electron from *1s orbital of Boron atom (B)*
- iii. Wavelength of photon to *completely remove* an electron from *1s orbital of Boron cation (B⁺)*

i or ii or iii

(II) Select the smallest (shortest) wavelength from the three choices below (*for H atom*):

- i. Wavelength of photon *emitted* for an electron transitioning from *2p to 1s orbital*
- ii. Wavelength of photon *absorbed* for an electron transitioning from *2s to 4p orbital*
- iii. Wavelength of photon *absorbed* for an electron transitioning from *1s to 3p orbital*

i or ii or iii



Question 2 8 points



A Chem110 student plots the graph shown on the left, from data obtained for the kinetic energy (eV) of the ejected electron versus frequency (s^{-1}) of the incident light, for an unknown metal. ($1\text{eV} = 1.6022 \times 10^{-19} \text{ J}$)

Based on the plot, answer the questions **a to c** below:

a. 1 point On the plot above, label the **threshold frequency**.

b. 2 points What is the value of the work potential (in J) of the unknown metal?

c. 3 points What is the *longest* wavelength (in nm) required to eject an electron from this metal's surface? Show your work including equation(s) used. (3 sig figs)

d. 2 points In the space provided below, plot the following

(Assume frequency is *above* threshold frequency):

- i. *Kinetic energy* of an ejected electron versus the *intensity* of the incident light (left)
- ii. *Number of electrons* ejected versus the *frequency* of the incident light (right)



**Question 3** **6 points****a. 2 points**

In the ground state of Ge (Germanium; $Z = 32$) atom, **name** the following orbitals:

i. Orbital containing the electron that requires the maximum energy to be removed.

ii. Orbital containing the electron that experiences the minimum effective nuclear charge.

b. 2 points

An unknown anion (A^-) has the following *excited* electronic configuration: $[Ar] 4s^2 3d^3 4p^1$

(Ar: Argon; $Z = 18$)

Give the **condensed ground state electronic configuration** of the atom, **A** and the cation, **A^{2+}** .
Show your work. (Assume all electronic transitions are possible)

Electronic Configuration of A:

Electronic Configuration of A^{2+} :

c. 2 points

In a single atom, determine how many electrons can have the following set of quantum numbers?

(i) $n = 5; l = 3$

In a single atom, determine how many electrons can have the following set of quantum numbers?

(i) $n = 3; l = 2; m_s = +1/2$

**Question 4** **8 points**

Rank the following from 1 to 4 in **increasing** order. Provide the ranking in the boxes provided.

Note: Atomic number (Z) are provided for relevant atoms

a) Increasing (lowest to highest) order of *ionization energy*: (1= lowest; 4 = highest)

Mg

Mg: Z = 12

Rb

Rb: Z = 37

Cs

Cs: Z = 55

K

K : Z = 19

b) Increasing (lowest to highest) order of *bond angle*: (1= lowest; 4 = highest)

H₂OCCl₄XeF₄

HCN

c) Increasing (smallest to largest) order of *ionic size* (1= smallest; 4 = largest)

S²⁻

S: Z = 16

O²⁻

O: Z = 8

Be²⁺

Be: Z = 4

Na⁺

Na: Z = 11

d) Increasing (lowest to highest) order of *bond energy* (1= lowest; 4 = highest)

P-F

P-Cl

P-I

P-Br

**Question 5** **12 points**

a. 9 points For the molecules HClO_4 and TeF_6 , draw the most stable *Lewis structure* showing appropriate shape and *dashed/wedged bonds*, based on VSEPR. Determine the **molecular geometry** around the central atom. For HClO_4 , label the *most polar bond*.



b. 3 points (Note: Lone pairs are not shown in structures below)

(i) For the three *resonance* structures (1, 2 and 3) shown below, answer the following question:



(I) Circle the *most contributing* resonance structure.

Structure 1

Structure 2

Structure 3

(ii) For the three structures given (4, 5, and 6) below, answer the following questions:



(I) Circle all the structures that represent *resonance structures* of each other:

Structure 4

Structure 5

Structure 6

(II) Among the resonance structures, circle the *most contributing* resonance structure.

Structure 4

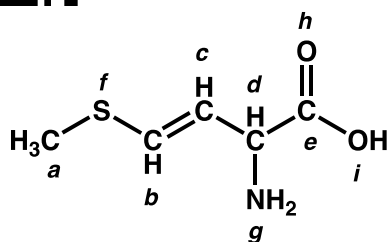
Structure 5

Structure 6

**Question 6** **6 points**

Using the data given below, calculate the **lattice energy** for SrO(s).

	ΔH° (kJ/mol)
$\text{Sr}(s) \longrightarrow \text{Sr}(g)$	+180
$\text{Sr}(g) \longrightarrow \text{Sr}^{2+}(g) + 2e^-$	+1468
$\text{O}_2(g) \longrightarrow 2\text{O}(g)$	+500
$\text{O}(g) + 2e^- \longrightarrow \text{O}^{2-}(g)$	+600
$\text{Sr}(s) + \frac{1}{2}\text{O}_2(g) \longrightarrow \text{SrO}(s)$	-550

**Question 7 10 points**

Consider the molecule given on the left, and answer questions (i) to (vi) using chemical bonding theories that we have learnt in class (valence bond theory, hybridization).

Note that the lone pairs are not shown in the structure:

i. Consider the carbon atoms marked in as ***a***, ***b***, ***c***, ***d***, and ***e*** in the molecule, and determine **hybridization** at each of the carbon centers. (2.5 points)

Atom	Hybridization
Carbon (<i>a</i>)	
Carbon (<i>b</i>)	
Carbon (<i>c</i>)	
Carbon (<i>d</i>)	
Carbon (<i>e</i>)	

ii. Now, consider the sulfur atom (S) marked as ***f***, nitrogen (N) atom (***g***), and oxygen(O) atoms (***h*** and ***i***), and determine their **hybridization**. (2 points)

Atom	Hybridization
Sulfur (<i>f</i>)	
Nitrogen (<i>g</i>)	
Oxygen (<i>h</i>)	
Oxygen (<i>i</i>)	

iii. **Describe** all the bonds formed by the carbon atoms ***a*** and ***b***, using sigma and pi designation as appropriate. (2 points)

Atom	Bonds formed (use sigma and pi as appropriate)
Carbon (<i>a</i>)	
Carbon (<i>b</i>)	

iv. How many total sp^3 -s sigma bonds are in the molecule? (1 point)

v. Determine **electronic** and **molecular geometry** at carbon (C) labeled ***e*** and sulfur (S) labeled ***f*** centres. (2 points)

Atom	Electronic Geometry	Molecular Geometry
Carbon (<i>e</i>)		
Sulfur (<i>f</i>)		

vi. Circle the value from the choices given below, that describes the **C-S-C bond angle at sulfur** (S) marked ***f***: (0.5 points)

$>90^\circ$ to $<109.5^\circ$ 109.5° >109.5 to $<120^\circ$ 120° $>120^\circ$ to $<180^\circ$


Question 8 12 points

Using the **valence shell orbitals only** draw a **complete** molecular orbital (MO) diagram (with no s-p mixing) for the molecule **PS**, in the space provided below. **(6 points)**

Assume that 3s and 3p atomic orbitals will form molecular orbitals like those formed by the combination of 2s and 2p atomic orbitals that we learnt in class, and there is no s-p mixing.

Show **energy arrow**, **atomic orbitals** involved, **atom labels**, the **molecular orbitals** that are formed, **filled electrons** in the MO diagram etc. Make sure to label all the atomic and molecular orbitals.

Using the MO diagram above, answer the following questions:

- (i) Which one of the following will have the highest bond order? Circle the correct choice
 A) PS^{1-} B) PS C) PS^{1+}
- (ii) Which one of the following will have the longest bond length? Circle the correct choice
 A) PS^{1-} B) PS C) PS^{1+}
- (iii) Which of the following will be diamagnetic? Circle the correct choice.
 A) PS^{1-} B) PS C) PS^{1+}
- (iv) Draw Lewis dot structure of the molecule with the shortest bond length (based on the MO diagram) from among the three (PS^{1-} , PS, PS^{1+}), and indicate the bond order calculated from the Lewis dot structure. **(3 points)**

Lewis Dot Structure

Bond Order



865AF66D-AEDF-4A30-8CB2-C60ED16653FE

midterm-exam-93388

#630 Page 10 of 10

SCRAP PAGE

**CHEMISTRY 110 –Midterm Exam****Thursday November 7, 2024, 6:30 – 8:30 pm****Examiners: Pallavi Sirjoosingh and Ashok Kakkar**

First name (please write as legibly as possible within the boxes)

--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--

Last name

--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--

Student ID number

--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--

This booklet has 8 questions on 10 pages (The last page can be used as a scrap page).

The staple must NOT be removed from the question booklet. Your intact question booklet must be returned at the end of the exam.

A separate booklet contains a periodic table, and datasheet.

Write your answers directly under the questions in the space provided. Do not write on the QR codes or near the top edges (as it may be cut off).

Please write **legibly and dark enough** for the scan to be readable. For students who use English as a second language, dictionaries are permitted. Calculators are allowed.

DO NOT ASK QUESTIONS OF THE INVIGILATORS about interpretations of the exam questions. If you have a concern about a question, please summarize it at the top of this page. All concerns will be evaluated after the exam. GOOD LUCK!

**Question 1 8 points** (*Assume all transitions are possible*)

a. 6 points Calculate the **energy (in kJ/mol)** required to ionize (completely remove) the electron from the **n=3** level of **He⁺ ion** (He: Helium). Show your work including equations. (3 sig. figs.)

b. 2 points

(I) Select the largest (longest) wavelength from the three choices below:

- i. Wavelength of photon to *completely remove* an electron from *1s orbital of Hydrogen atom*
- ii. Wavelength of photon to *completely remove* an electron from *1s orbital of Boron atom (B)*
- iii. Wavelength of photon to *completely remove* an electron from *1s orbital of Boron cation (B⁺)*

i or ii or iii

(II) Select the smallest (shortest) wavelength from the three choices below (*for H atom*):

- i. Wavelength of photon *emitted* for an electron transitioning from *2p to 1s orbital*
- ii. Wavelength of photon *absorbed* for an electron transitioning from *2s to 4p orbital*
- iii. Wavelength of photon *absorbed* for an electron transitioning from *1s to 3p orbital*

i or ii or iii



Question 2 8 points



A Chem110 student plots the graph shown on the left, from data obtained for the kinetic energy (eV) of the ejected electron versus frequency (s^{-1}) of the incident light, for an unknown metal. ($1\text{eV} = 1.6022 \times 10^{-19} \text{ J}$)

Based on the plot, answer the questions **a to c** below:

a. 1 point On the plot above, label the **threshold frequency**.

b. 2 points What is the value of the work potential (in J) of the unknown metal?

c. 3 points What is the *longest* wavelength (in nm) required to eject an electron from this metal's surface? Show your work including equation(s) used. (3 sig figs)

d. 2 points In the space provided below, plot the following

(Assume frequency is *above* threshold frequency):

- Kinetic energy* of an ejected electron versus the *intensity* of the incident light (left)
- Number of electrons* ejected versus the *frequency* of the incident light (right)



**Question 3** **6 points****a. 2 points**

In the ground state of Ge (Germanium; $Z = 32$) atom, **name** the following orbitals:

i. Orbital containing the electron that requires the maximum energy to be removed.

ii. Orbital containing the electron that experiences the minimum effective nuclear charge.

b. 2 points

An unknown anion (A^-) has the following *excited* electronic configuration: $[Ar] 4s^2 3d^3 4p^1$

(Ar: Argon; $Z = 18$)

Give the **condensed ground state electronic configuration** of the atom, **A** and the cation, **A^{2+}** .
Show your work. (Assume all electronic transitions are possible)

Electronic Configuration of A:

Electronic Configuration of A^{2+} :

c. 2 points

In a single atom, determine how many electrons can have the following set of quantum numbers?

(i) $n = 5; l = 3$

In a single atom, determine how many electrons can have the following set of quantum numbers?

(i) $n = 3; l = 2; m_s = +1/2$

**Question 4** **8 points**

Rank the following from 1 to 4 in **increasing** order. Provide the ranking in the boxes provided.

Note: Atomic number (Z) are provided for relevant atoms

a) Increasing (lowest to highest) order of *ionization energy*: (1= lowest; 4 = highest)

Mg

Mg: Z = 12

Rb

Rb: Z = 37

Cs

Cs: Z = 55

K

K : Z = 19

b) Increasing (lowest to highest) order of *bond angle*: (1= lowest; 4 = highest)

H₂OCCl₄XeF₄

HCN

c) Increasing (smallest to largest) order of *ionic size* (1= smallest; 4 = largest)

S²⁻

S: Z = 16

O²⁻

O: Z = 8

Be²⁺

Be: Z = 4

Na⁺

Na: Z = 11

d) Increasing (lowest to highest) order of *bond energy* (1= lowest; 4 = highest)

P-F

P-Cl

P-I

P-Br

**Question 5** **12 points**

a. 9 points For the molecules HClO_4 and TeF_6 , draw the most stable *Lewis structure* showing appropriate shape and *dashed/wedged bonds*, based on VSEPR. Determine the **molecular geometry** around the central atom. For HClO_4 , label the *most polar bond*.



b. 3 points (Note: Lone pairs are not shown in structures below)

(i) For the three *resonance* structures (1, 2 and 3) shown below, answer the following question:



(I) Circle the *most contributing* resonance structure.

Structure 1

Structure 2

Structure 3

(ii) For the three structures given (4, 5, and 6) below, answer the following questions:



(I) Circle all the structures that represent *resonance structures* of each other:

Structure 4

Structure 5

Structure 6

(II) Among the resonance structures, circle the *most contributing* resonance structure.

Structure 4

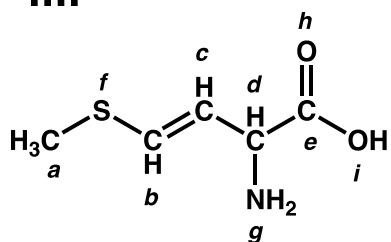
Structure 5

Structure 6

**Question 6** **6 points**

Using the data given below, calculate the **lattice energy** for SrO(s).

	ΔH° (kJ/mol)
$\text{Sr}(s) \longrightarrow \text{Sr}(g)$	+180
$\text{Sr}(g) \longrightarrow \text{Sr}^{2+}(g) + 2e^-$	+1468
$\text{O}_2(g) \longrightarrow 2\text{O}(g)$	+500
$\text{O}(g) + 2e^- \longrightarrow \text{O}^{2-}(g)$	+600
$\text{Sr}(s) + \frac{1}{2}\text{O}_2(g) \longrightarrow \text{SrO}(s)$	-550

**Question 7 10 points**

Consider the molecule given on the left, and answer questions (i) to (vi) using chemical bonding theories that we have learnt in class (valence bond theory, hybridization).

Note that the lone pairs are not shown in the structure:

i. Consider the carbon atoms marked in as ***a***, ***b***, ***c***, ***d***, and ***e*** in the molecule, and determine **hybridization** at each of the carbon centers. (2.5 points)

Atom	Hybridization
Carbon (<i>a</i>)	
Carbon (<i>b</i>)	
Carbon (<i>c</i>)	
Carbon (<i>d</i>)	
Carbon (<i>e</i>)	

ii. Now, consider the sulfur atom (S) marked as ***f***, nitrogen (N) atom (***g***), and oxygen(O) atoms (***h*** and ***i***), and determine their **hybridization**. (2 points)

Atom	Hybridization
Sulfur (<i>f</i>)	
Nitrogen (<i>g</i>)	
Oxygen (<i>h</i>)	
Oxygen (<i>i</i>)	

iii. **Describe** all the bonds formed by the carbon atoms ***a*** and ***b***, using sigma and pi designation as appropriate. (2 points)

Atom	Bonds formed (use sigma and pi as appropriate)
Carbon (<i>a</i>)	
Carbon (<i>b</i>)	

iv. How many total sp^3 -s sigma bonds are in the molecule? (1 point)

v. Determine **electronic** and **molecular geometry** at carbon (C) labeled ***e*** and sulfur (S) labeled ***f*** centres. (2 points)

Atom	Electronic Geometry	Molecular Geometry
Carbon (<i>e</i>)		
Sulfur (<i>f</i>)		

vi. Circle the value from the choices given below, that describes the **C-S-C bond angle at sulfur** (S) marked ***f***: (0.5 points)

>90° to <109.5° 109.5° >109.5 to <120° 120° >120° to <180°


Question 8 12 points

Using the **valence shell orbitals only** draw a **complete** molecular orbital (MO) diagram (with no s-p mixing) for the molecule **PS**, in the space provided below. **(6 points)**

Assume that 3s and 3p atomic orbitals will form molecular orbitals like those formed by the combination of 2s and 2p atomic orbitals that we learnt in class, and there is no s-p mixing.

Show **energy arrow**, **atomic orbitals** involved, **atom labels**, the **molecular orbitals** that are formed, **filled electrons** in the MO diagram etc. Make sure to label all the atomic and molecular orbitals.

Using the MO diagram above, answer the following questions:

- (i) Which one of the following will have the highest bond order? Circle the correct choice
 A) PS^{1-} B) PS C) PS^{1+}
- (ii) Which one of the following will have the longest bond length? Circle the correct choice
 A) PS^{1-} B) PS C) PS^{1+}
- (iii) Which of the following will be diamagnetic? Circle the correct choice.
 A) PS^{1-} B) PS C) PS^{1+}
- (iv) Draw Lewis dot structure of the molecule with the shortest bond length (based on the MO diagram) from among the three (PS^{1-} , PS, PS^{1+}), and indicate the bond order calculated from the Lewis dot structure. **(3 points)**

Lewis Dot Structure

Bond Order



FE302E3D-AAFE-4E70-952A-CCE3FBAC63AD

midterm-exam-93388

#631 Page 10 of 10

SCRAP PAGE

**CHEMISTRY 110 –Midterm Exam****Thursday November 7, 2024, 6:30 – 8:30 pm****Examiners: Pallavi Sirjoosingh and Ashok Kakkar**

First name (please write as legibly as possible within the boxes)

--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--

Last name

--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--

Student ID number

--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--

This booklet has 8 questions on 10 pages (The last page can be used as a scrap page).

The staple must NOT be removed from the question booklet. Your intact question booklet must be returned at the end of the exam.

A separate booklet contains a periodic table, and datasheet.

Write your answers directly under the questions in the space provided. Do not write on the QR codes or near the top edges (as it may be cut off).

Please write **legibly and dark enough** for the scan to be readable. For students who use English as a second language, dictionaries are permitted. Calculators are allowed.

DO NOT ASK QUESTIONS OF THE INVIGILATORS about interpretations of the exam questions. If you have a concern about a question, please summarize it at the top of this page. All concerns will be evaluated after the exam. GOOD LUCK!

**Question 1 8 points** (*Assume all transitions are possible*)

a. 6 points Calculate the **energy (in kJ/mol)** required to ionize (completely remove) the electron from the **n=3** level of **He⁺ ion** (He: Helium). Show your work including equations. (3 sig. figs.)

b. 2 points

(I) Select the largest (longest) wavelength from the three choices below:

- i. Wavelength of photon to *completely remove* an electron from *1s orbital of Hydrogen atom*
- ii. Wavelength of photon to *completely remove* an electron from *1s orbital of Boron atom (B)*
- iii. Wavelength of photon to *completely remove* an electron from *1s orbital of Boron cation (B⁺)*

i or ii or iii

(II) Select the smallest (shortest) wavelength from the three choices below (*for H atom*):

- i. Wavelength of photon *emitted* for an electron transitioning from *2p to 1s orbital*
- ii. Wavelength of photon *absorbed* for an electron transitioning from *2s to 4p orbital*
- iii. Wavelength of photon *absorbed* for an electron transitioning from *1s to 3p orbital*

i or ii or iii



Question 2 8 points



A Chem110 student plots the graph shown on the left, from data obtained for the kinetic energy (eV) of the ejected electron versus frequency (s^{-1}) of the incident light, for an unknown metal. ($1\text{eV} = 1.6022 \times 10^{-19} \text{ J}$)

Based on the plot, answer the questions **a to c** below:

a. 1 point On the plot above, label the **threshold frequency**.

b. 2 points What is the value of the work potential (in J) of the unknown metal?

c. 3 points What is the *longest* wavelength (in nm) required to eject an electron from this metal's surface? Show your work including equation(s) used. (3 sig figs)

d. 2 points In the space provided below, plot the following

(Assume frequency is *above* threshold frequency):

- i. *Kinetic energy* of an ejected electron versus the *intensity* of the incident light (left)
- ii. *Number of electrons* ejected versus the *frequency* of the incident light (right)



**Question 3** **6 points****a. 2 points**

In the ground state of Ge (Germanium; $Z = 32$) atom, **name** the following orbitals:

i. Orbital containing the electron that requires the maximum energy to be removed.

ii. Orbital containing the electron that experiences the minimum effective nuclear charge.

b. 2 points

An unknown anion (A^-) has the following *excited* electronic configuration: $[Ar] 4s^2 3d^3 4p^1$

(Ar: Argon; $Z = 18$)

Give the **condensed ground state electronic configuration** of the atom, **A** and the cation, **A^{2+}** .
Show your work. (Assume all electronic transitions are possible)

Electronic Configuration of A:

Electronic Configuration of A^{2+} :

c. 2 points

In a single atom, determine how many electrons can have the following set of quantum numbers?

(i) $n = 5; l = 3$

In a single atom, determine how many electrons can have the following set of quantum numbers?

(i) $n = 3; l = 2; m_s = +1/2$

**Question 4** **8 points**

Rank the following from 1 to 4 in **increasing** order. Provide the ranking in the boxes provided.

Note: Atomic number (Z) are provided for relevant atoms

a) Increasing (lowest to highest) order of *ionization energy*: (1= lowest; 4 = highest)

Mg

Mg: Z = 12

Rb

Rb: Z = 37

Cs

Cs: Z = 55

K

K : Z = 19

b) Increasing (lowest to highest) order of *bond angle*: (1= lowest; 4 = highest)

H₂OCCl₄XeF₄

HCN

c) Increasing (smallest to largest) order of *ionic size* (1= smallest; 4 = largest)

S²⁻

S: Z = 16

O²⁻

O: Z = 8

Be²⁺

Be: Z = 4

Na⁺

Na: Z = 11

d) Increasing (lowest to highest) order of *bond energy* (1= lowest; 4 = highest)

P-F

P-Cl

P-I

P-Br

**Question 5** **12 points**

a. 9 points For the molecules HClO_4 and TeF_6 , draw the most stable *Lewis structure* showing appropriate shape and *dashed/wedged bonds*, based on VSEPR. Determine the **molecular geometry** around the central atom. For HClO_4 , label the *most polar bond*.



b. 3 points (Note: Lone pairs are not shown in structures below)

(i) For the three *resonance* structures (1, 2 and 3) shown below, answer the following question:



(I) Circle the *most contributing* resonance structure.

Structure 1

Structure 2

Structure 3

(ii) For the three structures given (4, 5, and 6) below, answer the following questions:



(I) Circle all the structures that represent *resonance structures* of each other:

Structure 4

Structure 5

Structure 6

(II) Among the resonance structures, circle the *most contributing* resonance structure.

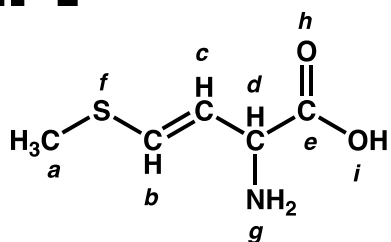
Structure 4

Structure 5

Structure 6

**Question 6** **6 points**Using the data given below, calculate the **lattice energy** for SrO(s).

	ΔH° (kJ/mol)
$\text{Sr}(s) \longrightarrow \text{Sr}(g)$	+180
$\text{Sr}(g) \longrightarrow \text{Sr}^{2+}(g) + 2e^-$	+1468
$\text{O}_2(g) \longrightarrow 2\text{O}(g)$	+500
$\text{O}(g) + 2e^- \longrightarrow \text{O}^{2-}(g)$	+600
$\text{Sr}(s) + \frac{1}{2}\text{O}_2(g) \longrightarrow \text{SrO}(s)$	-550

**Question 7 10 points**

Consider the molecule given on the left, and answer questions (i) to (vi) using chemical bonding theories that we have learnt in class (valence bond theory, hybridization).

Note that the lone pairs are not shown in the structure:

i. Consider the carbon atoms marked in as ***a***, ***b***, ***c***, ***d***, and ***e*** in the molecule, and determine **hybridization** at each of the carbon centers. (2.5 points)

Atom	Hybridization
Carbon (<i>a</i>)	
Carbon (<i>b</i>)	
Carbon (<i>c</i>)	
Carbon (<i>d</i>)	
Carbon (<i>e</i>)	

ii. Now, consider the sulfur atom (S) marked as ***f***, nitrogen (N) atom (***g***), and oxygen(O) atoms (***h*** and ***i***), and determine their **hybridization**. (2 points)

Atom	Hybridization
Sulfur (<i>f</i>)	
Nitrogen (<i>g</i>)	
Oxygen (<i>h</i>)	
Oxygen (<i>i</i>)	

iii. **Describe** all the bonds formed by the carbon atoms ***a*** and ***b***, using sigma and pi designation as appropriate. (2 points)

Atom	Bonds formed (use sigma and pi as appropriate)
Carbon (<i>a</i>)	
Carbon (<i>b</i>)	

iv. How many total sp^3 -s sigma bonds are in the molecule? (1 point)

v. Determine **electronic** and **molecular geometry** at carbon (C) labeled ***e*** and sulfur (S) labeled ***f*** centres. (2 points)

Atom	Electronic Geometry	Molecular Geometry
Carbon (<i>e</i>)		
Sulfur (<i>f</i>)		

vi. Circle the value from the choices given below, that describes the **C-S-C bond angle at sulfur** (S) marked ***f***: (0.5 points)

>90° to <109.5° 109.5° >109.5 to <120° 120° >120° to <180°

**Question 8 12 points**

Using the **valence shell orbitals only** draw a **complete** molecular orbital (MO) diagram (with no s-p mixing) for the molecule **PS**, in the space provided below. **(6 points)**

Assume that 3s and 3p atomic orbitals will form molecular orbitals like those formed by the combination of 2s and 2p atomic orbitals that we learnt in class, and there is no s-p mixing.

Show **energy arrow**, **atomic orbitals** involved, **atom labels**, the **molecular orbitals** that are formed, **filled electrons** in the MO diagram etc. Make sure to label all the atomic and molecular orbitals.

Using the MO diagram above, answer the following questions:

- (i) Which one of the following will have the highest bond order? Circle the correct choice
A) PS^{1-} B) PS C) PS^{1+}
- (ii) Which one of the following will have the longest bond length? Circle the correct choice
A) PS^{1-} B) PS C) PS^{1+}
- (iii) Which of the following will be diamagnetic? Circle the correct choice.
A) PS^{1-} B) PS C) PS^{1+}
- (iv) Draw Lewis dot structure of the molecule with the shortest bond length (based on the MO diagram) from among the three (PS^{1-} , PS, PS^{1+}), and indicate the bond order calculated from the Lewis dot structure. **(3 points)**

Lewis Dot Structure

Bond Order



1D02D2F9-1BBB-4C4E-9F87-967487377152

midterm-exam-93388

#632 Page 10 of 10

SCRAP PAGE

**CHEMISTRY 110 –Midterm Exam****Thursday November 7, 2024, 6:30 – 8:30 pm****Examiners: Pallavi Sirjoosingh and Ashok Kakkar**

First name (please write as legibly as possible within the boxes)

--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--

Last name

--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--

Student ID number

--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--

This booklet has 8 questions on 10 pages (The last page can be used as a scrap page).

The staple must NOT be removed from the question booklet. Your intact question booklet must be returned at the end of the exam.

A separate booklet contains a periodic table, and datasheet.

Write your answers directly under the questions in the space provided. Do not write on the QR codes or near the top edges (as it may be cut off).

Please write **legibly and dark enough** for the scan to be readable. For students who use English as a second language, dictionaries are permitted. Calculators are allowed.

DO NOT ASK QUESTIONS OF THE INVIGILATORS about interpretations of the exam questions. If you have a concern about a question, please summarize it at the top of this page. All concerns will be evaluated after the exam. GOOD LUCK!

**Question 1 8 points** (*Assume all transitions are possible*)

a. 6 points Calculate the **energy (in kJ/mol)** required to ionize (completely remove) the electron from the **$n=3$** level of **He^+ ion** (He: Helium). Show your work including equations. (3 sig. figs.)

b. 2 points

(I) Select the largest (longest) wavelength from the three choices below:

- i. Wavelength of photon to *completely remove* an electron from *1s orbital of Hydrogen atom*
- ii. Wavelength of photon to *completely remove* an electron from *1s orbital of Boron atom (B)*
- iii. Wavelength of photon to *completely remove* an electron from *1s orbital of Boron cation (B^+)*

i or ii or iii

(II) Select the smallest (shortest) wavelength from the three choices below (*for H atom*):

- i. Wavelength of photon *emitted* for an electron transitioning from *2p to 1s orbital*
- ii. Wavelength of photon *absorbed* for an electron transitioning from *2s to 4p orbital*
- iii. Wavelength of photon *absorbed* for an electron transitioning from *1s to 3p orbital*

i or ii or iii



Question 2 8 points



A Chem110 student plots the graph shown on the left, from data obtained for the kinetic energy (eV) of the ejected electron versus frequency (s^{-1}) of the incident light, for an unknown metal. ($1\text{eV} = 1.6022 \times 10^{-19} \text{ J}$)

Based on the plot, answer the questions **a to c** below:

a. 1 point On the plot above, label the **threshold frequency**.

b. 2 points What is the value of the work potential (in J) of the unknown metal?

c. 3 points What is the *longest* wavelength (in nm) required to eject an electron from this metal's surface? Show your work including equation(s) used. (3 sig figs)

d. 2 points In the space provided below, plot the following

(Assume frequency is *above* threshold frequency):

- i. *Kinetic energy* of an ejected electron versus the *intensity* of the incident light (left)
- ii. *Number of electrons* ejected versus the *frequency* of the incident light (right)



**Question 3** **6 points****a. 2 points**

In the ground state of Ge (Germanium; $Z = 32$) atom, **name** the following orbitals:

i. Orbital containing the electron that requires the maximum energy to be removed.

ii. Orbital containing the electron that experiences the minimum effective nuclear charge.

b. 2 points

An unknown anion (A^-) has the following *excited* electronic configuration: $[Ar] 4s^2 3d^3 4p^1$

(Ar: Argon; $Z = 18$)

Give the **condensed ground state electronic configuration** of the atom, **A** and the cation, **A^{2+}** .
Show your work. (Assume all electronic transitions are possible)

Electronic Configuration of A:

Electronic Configuration of A^{2+} :

c. 2 points

In a single atom, determine how many electrons can have the following set of quantum numbers?

(i) $n = 5; l = 3$

In a single atom, determine how many electrons can have the following set of quantum numbers?

(i) $n = 3; l = 2; m_s = +1/2$

**Question 4** **8 points**

Rank the following from 1 to 4 in **increasing** order. Provide the ranking in the boxes provided.

Note: Atomic number (Z) are provided for relevant atoms

a) Increasing (lowest to highest) order of *ionization energy*: (1= lowest; 4 = highest)

Mg

Mg: Z = 12

Rb

Rb: Z = 37

Cs

Cs: Z = 55

K

K : Z = 19

b) Increasing (lowest to highest) order of *bond angle*: (1= lowest; 4 = highest)

H₂OCCl₄XeF₄

HCN

c) Increasing (smallest to largest) order of *ionic size* (1= smallest; 4 = largest)

S²⁻

S: Z = 16

O²⁻

O: Z = 8

Be²⁺

Be: Z = 4

Na⁺

Na: Z = 11

d) Increasing (lowest to highest) order of *bond energy* (1= lowest; 4 = highest)

P-F

P-Cl

P-I

P-Br

**Question 5** **12 points**

a. 9 points For the molecules HClO_4 and TeF_6 , draw the most stable *Lewis structure* showing appropriate shape and *dashed/wedged bonds*, based on VSEPR. Determine the **molecular geometry** around the central atom. For HClO_4 , label the *most polar bond*.



b. 3 points (Note: Lone pairs are not shown in structures below)

(i) For the three *resonance* structures (1, 2 and 3) shown below, answer the following question:



(I) Circle the *most contributing* resonance structure.

Structure 1

Structure 2

Structure 3

(ii) For the three structures given (4, 5, and 6) below, answer the following questions:



(I) Circle all the structures that represent *resonance structures* of each other:

Structure 4

Structure 5

Structure 6

(II) Among the resonance structures, circle the *most contributing* resonance structure.

Structure 4

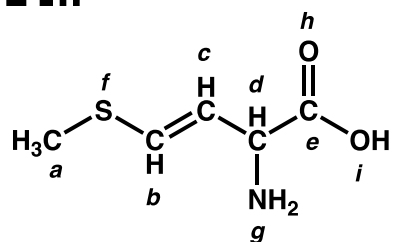
Structure 5

Structure 6

**Question 6** **6 points**

Using the data given below, calculate the **lattice energy** for SrO(s).

	ΔH° (kJ/mol)
$\text{Sr}(s) \longrightarrow \text{Sr}(g)$	+180
$\text{Sr}(g) \longrightarrow \text{Sr}^{2+}(g) + 2e^-$	+1468
$\text{O}_2(g) \longrightarrow 2\text{O}(g)$	+500
$\text{O}(g) + 2e^- \longrightarrow \text{O}^{2-}(g)$	+600
$\text{Sr}(s) + \frac{1}{2}\text{O}_2(g) \longrightarrow \text{SrO}(s)$	-550

**Question 7 10 points**

Consider the molecule given on the left, and answer questions (i) to (vi) using chemical bonding theories that we have learnt in class (valence bond theory, hybridization).

Note that the lone pairs are not shown in the structure:

i. Consider the carbon atoms marked in as ***a***, ***b***, ***c***, ***d***, and ***e*** in the molecule, and determine **hybridization** at each of the carbon centers. (2.5 points)

Atom	Hybridization
Carbon (<i>a</i>)	
Carbon (<i>b</i>)	
Carbon (<i>c</i>)	
Carbon (<i>d</i>)	
Carbon (<i>e</i>)	

ii. Now, consider the sulfur atom (S) marked as ***f***, nitrogen (N) atom (***g***), and oxygen(O) atoms (***h*** and ***i***), and determine their **hybridization**. (2 points)

Atom	Hybridization
Sulfur (<i>f</i>)	
Nitrogen (<i>g</i>)	
Oxygen (<i>h</i>)	
Oxygen (<i>i</i>)	

iii. **Describe** all the bonds formed by the carbon atoms ***a*** and ***b***, using sigma and pi designation as appropriate. (2 points)

Atom	Bonds formed (use sigma and pi as appropriate)
Carbon (<i>a</i>)	
Carbon (<i>b</i>)	

iv. How many total sp^3 -s sigma bonds are in the molecule? (1 point)

v. Determine **electronic** and **molecular geometry** at carbon (C) labeled ***e*** and sulfur (S) labeled ***f*** centres. (2 points)

Atom	Electronic Geometry	Molecular Geometry
Carbon (<i>e</i>)		
Sulfur (<i>f</i>)		

vi. Circle the value from the choices given below, that describes the **C-S-C bond angle at sulfur** (S) marked ***f***: (0.5 points)

>90° to <109.5° 109.5° >109.5 to <120° 120° >120° to <180°



Question 8 12 points

Using the **valence shell orbitals only** draw a **complete** molecular orbital (MO) diagram (with no s-p mixing) for the molecule **PS**, in the space provided below. **(6 points)**

Assume that 3s and 3p atomic orbitals will form molecular orbitals like those formed by the combination of 2s and 2p atomic orbitals that we learnt in class, and there is no s-p mixing.

Show **energy arrow**, **atomic orbitals** involved, **atom labels**, the **molecular orbitals** that are formed, **filled electrons** in the MO diagram etc. Make sure to label all the atomic and molecular orbitals.

Using the MO diagram above, answer the following questions:

- (i) Which one of the following will have the highest bond order? Circle the correct choice
 A) PS^{1-} B) PS C) PS^{1+}
- (ii) Which one of the following will have the longest bond length? Circle the correct choice
 A) PS^{1-} B) PS C) PS^{1+}
- (iii) Which of the following will be diamagnetic? Circle the correct choice.
 A) PS^{1-} B) PS C) PS^{1+}
- (iv) Draw Lewis dot structure of the molecule with the shortest bond length (based on the MO diagram) from among the three (PS^{1-} , PS, PS^{1+}), and indicate the bond order calculated from the Lewis dot structure. **(3 points)**

Lewis Dot Structure

Bond Order



4036BEA5-58DF-4B40-91DD-ED5DFE444864

midterm-exam-93388

#633 Page 10 of 10

SCRAP PAGE

**CHEMISTRY 110 –Midterm Exam****Thursday November 7, 2024, 6:30 – 8:30 pm****Examiners: Pallavi Sirjoosingh and Ashok Kakkar**

First name (please write as legibly as possible within the boxes)

--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--

Last name

--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--

Student ID number

--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--

This booklet has 8 questions on 10 pages (The last page can be used as a scrap page).

The staple must NOT be removed from the question booklet. Your intact question booklet must be returned at the end of the exam.

A separate booklet contains a periodic table, and datasheet.

Write your answers directly under the questions in the space provided. Do not write on the QR codes or near the top edges (as it may be cut off).

Please write **legibly and dark enough** for the scan to be readable. For students who use English as a second language, dictionaries are permitted. Calculators are allowed.

DO NOT ASK QUESTIONS OF THE INVIGILATORS about interpretations of the exam questions. If you have a concern about a question, please summarize it at the top of this page. All concerns will be evaluated after the exam. GOOD LUCK!

**Question 1 8 points** (*Assume all transitions are possible*)

a. 6 points Calculate the **energy (in kJ/mol)** required to ionize (completely remove) the electron from the **$n=3$** level of **He^+ ion** (He: Helium). Show your work including equations. (3 sig. figs.)

b. 2 points

(I) Select the largest (longest) wavelength from the three choices below:

- i. Wavelength of photon to *completely remove* an electron from *1s orbital of Hydrogen atom*
- ii. Wavelength of photon to *completely remove* an electron from *1s orbital of Boron atom (B)*
- iii. Wavelength of photon to *completely remove* an electron from *1s orbital of Boron cation (B^+)*

i or ii or iii

(II) Select the smallest (shortest) wavelength from the three choices below (*for H atom*):

- i. Wavelength of photon *emitted* for an electron transitioning from *2p to 1s orbital*
- ii. Wavelength of photon *absorbed* for an electron transitioning from *2s to 4p orbital*
- iii. Wavelength of photon *absorbed* for an electron transitioning from *1s to 3p orbital*

i or ii or iii



Question 2 8 points



A Chem110 student plots the graph shown on the left, from data obtained for the kinetic energy (eV) of the ejected electron versus frequency (s^{-1}) of the incident light, for an unknown metal. ($1\text{eV} = 1.6022 \times 10^{-19} \text{ J}$)

Based on the plot, answer the questions **a to c** below:

a. 1 point On the plot above, label the **threshold frequency**.

b. 2 points What is the value of the work potential (in J) of the unknown metal?

c. 3 points What is the *longest* wavelength (in nm) required to eject an electron from this metal's surface? Show your work including equation(s) used. (3 sig figs)

d. 2 points In the space provided below, plot the following

(Assume frequency is *above* threshold frequency):

- Kinetic energy* of an ejected electron versus the *intensity* of the incident light (left)
- Number of electrons* ejected versus the *frequency* of the incident light (right)



**Question 3** **6 points****a. 2 points**

In the ground state of Ge (Germanium; $Z = 32$) atom, **name** the following orbitals:

i. Orbital containing the electron that requires the maximum energy to be removed.

ii. Orbital containing the electron that experiences the minimum effective nuclear charge.

b. 2 points

An unknown anion (A^-) has the following *excited* electronic configuration: $[Ar] 4s^2 3d^3 4p^1$

(Ar: Argon; $Z = 18$)

Give the **condensed ground state electronic configuration** of the atom, **A** and the cation, **A^{2+}** .
Show your work. (Assume all electronic transitions are possible)

Electronic Configuration of A:

Electronic Configuration of A^{2+} :

c. 2 points

In a single atom, determine how many electrons can have the following set of quantum numbers?

(i) $n = 5; l = 3$

In a single atom, determine how many electrons can have the following set of quantum numbers?

(i) $n = 3; l = 2; m_s = +1/2$

**Question 4** **8 points**

Rank the following from 1 to 4 in **increasing** order. Provide the ranking in the boxes provided.

Note: Atomic number (Z) are provided for relevant atoms

a) Increasing (lowest to highest) order of *ionization energy*: (1= lowest; 4 = highest)

Mg

Mg: Z = 12

Rb

Rb: Z = 37

Cs

Cs: Z = 55

K

K : Z = 19

b) Increasing (lowest to highest) order of *bond angle*: (1= lowest; 4 = highest)

H₂OCCl₄XeF₄

HCN

c) Increasing (smallest to largest) order of *ionic size* (1= smallest; 4 = largest)

S²⁻

S: Z = 16

O²⁻

O: Z = 8

Be²⁺

Be: Z = 4

Na⁺

Na: Z = 11

d) Increasing (lowest to highest) order of *bond energy* (1= lowest; 4 = highest)

P-F

P-Cl

P-I

P-Br

**Question 5** **12 points**

a. 9 points For the molecules HClO_4 and TeF_6 , draw the most stable *Lewis structure* showing appropriate shape and *dashed/wedged bonds*, based on VSEPR. Determine the **molecular geometry** around the central atom. For HClO_4 , label the *most polar bond*.



b. 3 points (Note: Lone pairs are not shown in structures below)

(i) For the three *resonance* structures (1, 2 and 3) shown below, answer the following question:



(I) Circle the *most contributing* resonance structure.

Structure 1

Structure 2

Structure 3

(ii) For the three structures given (4, 5, and 6) below, answer the following questions:



(I) Circle all the structures that represent *resonance structures* of each other:

Structure 4

Structure 5

Structure 6

(II) Among the resonance structures, circle the *most contributing* resonance structure.

Structure 4

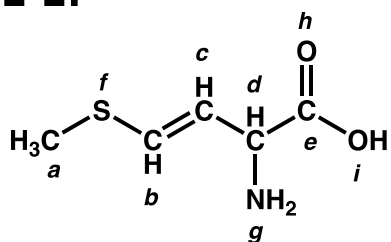
Structure 5

Structure 6

**Question 6** **6 points**

Using the data given below, calculate the **lattice energy** for SrO(s).

	ΔH° (kJ/mol)
$\text{Sr}(s) \longrightarrow \text{Sr}(g)$	+180
$\text{Sr}(g) \longrightarrow \text{Sr}^{2+}(g) + 2e^-$	+1468
$\text{O}_2(g) \longrightarrow 2\text{O}(g)$	+500
$\text{O}(g) + 2e^- \longrightarrow \text{O}^{2-}(g)$	+600
$\text{Sr}(s) + \frac{1}{2}\text{O}_2(g) \longrightarrow \text{SrO}(s)$	-550

**Question 7 10 points**

Consider the molecule given on the left, and answer questions (i) to (vi) using chemical bonding theories that we have learnt in class (valence bond theory, hybridization).

Note that the lone pairs are not shown in the structure:

i. Consider the carbon atoms marked in as ***a***, ***b***, ***c***, ***d***, and ***e*** in the molecule, and determine **hybridization** at each of the carbon centers. (2.5 points)

Atom	Hybridization
Carbon (<i>a</i>)	
Carbon (<i>b</i>)	
Carbon (<i>c</i>)	
Carbon (<i>d</i>)	
Carbon (<i>e</i>)	

ii. Now, consider the sulfur atom (S) marked as ***f***, nitrogen (N) atom (***g***), and oxygen(O) atoms (***h*** and ***i***), and determine their **hybridization**. (2 points)

Atom	Hybridization
Sulfur (<i>f</i>)	
Nitrogen (<i>g</i>)	
Oxygen (<i>h</i>)	
Oxygen (<i>i</i>)	

iii. **Describe** all the bonds formed by the carbon atoms ***a*** and ***b***, using sigma and pi designation as appropriate. (2 points)

Atom	Bonds formed (use sigma and pi as appropriate)
Carbon (<i>a</i>)	
Carbon (<i>b</i>)	

iv. How many total sp^3 -s sigma bonds are in the molecule? (1 point)

v. Determine **electronic** and **molecular geometry** at carbon (C) labeled ***e*** and sulfur (S) labeled ***f*** centres. (2 points)

Atom	Electronic Geometry	Molecular Geometry
Carbon (<i>e</i>)		
Sulfur (<i>f</i>)		

vi. Circle the value from the choices given below, that describes the **C-S-C bond angle at sulfur** (S) marked ***f***: (0.5 points)

$>90^\circ$ to $<109.5^\circ$ 109.5° >109.5 to $<120^\circ$ 120° $>120^\circ$ to $<180^\circ$


Question 8 12 points

Using the **valence shell orbitals only** draw a **complete** molecular orbital (MO) diagram (with no s-p mixing) for the molecule **PS**, in the space provided below. **(6 points)**

Assume that 3s and 3p atomic orbitals will form molecular orbitals like those formed by the combination of 2s and 2p atomic orbitals that we learnt in class, and there is no s-p mixing.

Show **energy arrow**, **atomic orbitals** involved, **atom labels**, the **molecular orbitals** that are formed, **filled electrons** in the MO diagram etc. Make sure to label all the atomic and molecular orbitals.

Using the MO diagram above, answer the following questions:

- (i) Which one of the following will have the highest bond order? Circle the correct choice
 A) PS^{1-} B) PS C) PS^{1+}
- (ii) Which one of the following will have the longest bond length? Circle the correct choice
 A) PS^{1-} B) PS C) PS^{1+}
- (iii) Which of the following will be diamagnetic? Circle the correct choice.
 A) PS^{1-} B) PS C) PS^{1+}
- (iv) Draw Lewis dot structure of the molecule with the shortest bond length (based on the MO diagram) from among the three (PS^{1-} , PS, PS^{1+}), and indicate the bond order calculated from the Lewis dot structure. **(3 points)**

Lewis Dot Structure

Bond Order



F907DAFC-752E-487D-95B2-FF5B25C07D04

midterm-exam-93388

#634 Page 10 of 10

SCRAP PAGE

**CHEMISTRY 110 –Midterm Exam****Thursday November 7, 2024, 6:30 – 8:30 pm****Examiners: Pallavi Sirjoosingh and Ashok Kakkar**

First name (please write as legibly as possible within the boxes)

--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--

Last name

--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--

Student ID number

--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--

This booklet has 8 questions on 10 pages (The last page can be used as a scrap page).

The staple must NOT be removed from the question booklet. Your intact question booklet must be returned at the end of the exam.

A separate booklet contains a periodic table, and datasheet.

Write your answers directly under the questions in the space provided. Do not write on the QR codes or near the top edges (as it may be cut off).

Please write **legibly and dark enough** for the scan to be readable. For students who use English as a second language, dictionaries are permitted. Calculators are allowed.

DO NOT ASK QUESTIONS OF THE INVIGILATORS about interpretations of the exam questions. If you have a concern about a question, please summarize it at the top of this page. All concerns will be evaluated after the exam. GOOD LUCK!

**Question 1 8 points** (*Assume all transitions are possible*)

a. 6 points Calculate the **energy (in kJ/mol)** required to ionize (completely remove) the electron from the **$n=3$** level of **He^+ ion** (He: Helium). Show your work including equations. (3 sig. figs.)

b. 2 points

(I) Select the largest (longest) wavelength from the three choices below:

- i. Wavelength of photon to *completely remove* an electron from *1s orbital of Hydrogen atom*
- ii. Wavelength of photon to *completely remove* an electron from *1s orbital of Boron atom (B)*
- iii. Wavelength of photon to *completely remove* an electron from *1s orbital of Boron cation (B^+)*

i or ii or iii

(II) Select the smallest (shortest) wavelength from the three choices below (*for H atom*):

- i. Wavelength of photon *emitted* for an electron transitioning from *2p to 1s orbital*
- ii. Wavelength of photon *absorbed* for an electron transitioning from *2s to 4p orbital*
- iii. Wavelength of photon *absorbed* for an electron transitioning from *1s to 3p orbital*

i or ii or iii



Question 2 8 points



A Chem110 student plots the graph shown on the left, from data obtained for the kinetic energy (eV) of the ejected electron versus frequency (s^{-1}) of the incident light, for an unknown metal. ($1\text{eV} = 1.6022 \times 10^{-19} \text{ J}$)

Based on the plot, answer the questions **a to c** below:

a. 1 point On the plot above, label the **threshold frequency**.

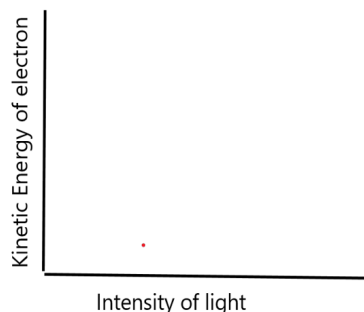
b. 2 points What is the value of the work potential (in J) of the unknown metal?

c. 3 points What is the *longest* wavelength (in nm) required to eject an electron from this metal's surface? Show your work including equation(s) used. (3 sig figs)

d. 2 points In the space provided below, plot the following

(Assume frequency is *above* threshold frequency):

- Kinetic energy* of an ejected electron versus the *intensity* of the incident light (left)
- Number of electrons* ejected versus the *frequency* of the incident light (right)



**Question 3** **6 points****a. 2 points**

In the ground state of Ge (Germanium; $Z = 32$) atom, **name** the following orbitals:

i. Orbital containing the electron that requires the maximum energy to be removed.

ii. Orbital containing the electron that experiences the minimum effective nuclear charge.

b. 2 points

An unknown anion (A^-) has the following *excited* electronic configuration: $[Ar] 4s^2 3d^3 4p^1$

(Ar: Argon; $Z = 18$)

Give the **condensed ground state electronic configuration** of the atom, **A** and the cation, **A^{2+}** .
Show your work. (Assume all electronic transitions are possible)

Electronic Configuration of A:

Electronic Configuration of A^{2+} :

c. 2 points

In a single atom, determine how many electrons can have the following set of quantum numbers?

(i) $n = 5; l = 3$

In a single atom, determine how many electrons can have the following set of quantum numbers?

(i) $n = 3; l = 2; m_s = +1/2$

**Question 4** **8 points**

Rank the following from 1 to 4 in **increasing** order. Provide the ranking in the boxes provided.

Note: Atomic number (Z) are provided for relevant atoms

a) Increasing (lowest to highest) order of *ionization energy*: (1= lowest; 4 = highest)

Mg

Mg: Z = 12

Rb

Rb: Z = 37

Cs

Cs: Z = 55

K

K : Z = 19

b) Increasing (lowest to highest) order of *bond angle*: (1= lowest; 4 = highest)

H₂OCCl₄XeF₄

HCN

c) Increasing (smallest to largest) order of *ionic size* (1= smallest; 4 = largest)

S²⁻

S: Z = 16

O²⁻

O: Z = 8

Be²⁺

Be: Z = 4

Na⁺

Na: Z = 11

d) Increasing (lowest to highest) order of *bond energy* (1= lowest; 4 = highest)

P-F

P-Cl

P-I

P-Br

**Question 5** **12 points**

a. 9 points For the molecules HClO_4 and TeF_6 , draw the most stable *Lewis structure* showing appropriate shape and *dashed/wedged bonds*, based on VSEPR. Determine the **molecular geometry** around the central atom. For HClO_4 , label the *most polar bond*.



b. 3 points (Note: Lone pairs are not shown in structures below)

(i) For the three *resonance* structures (1, 2 and 3) shown below, answer the following question:



(I) Circle the *most contributing* resonance structure.

Structure 1

Structure 2

Structure 3

(ii) For the three structures given (4, 5, and 6) below, answer the following questions:



(I) Circle all the structures that represent *resonance structures* of each other:

Structure 4

Structure 5

Structure 6

(II) Among the resonance structures, circle the *most contributing* resonance structure.

Structure 4

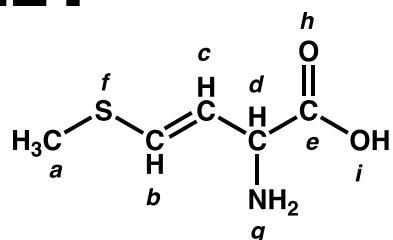
Structure 5

Structure 6

**Question 6** **6 points**

Using the data given below, calculate the **lattice energy** for SrO(s).

	ΔH° (kJ/mol)
$\text{Sr}(s) \longrightarrow \text{Sr}(g)$	+180
$\text{Sr}(g) \longrightarrow \text{Sr}^{2+}(g) + 2e^-$	+1468
$\text{O}_2(g) \longrightarrow 2\text{O}(g)$	+500
$\text{O}(g) + 2e^- \longrightarrow \text{O}^{2-}(g)$	+600
$\text{Sr}(s) + \frac{1}{2}\text{O}_2(g) \longrightarrow \text{SrO}(s)$	-550

**Question 7 10 points**

Consider the molecule given on the left, and answer questions (i) to (vi) using chemical bonding theories that we have learnt in class (valence bond theory, hybridization).

Note that the lone pairs are not shown in the structure:

i. Consider the carbon atoms marked in as ***a***, ***b***, ***c***, ***d***, and ***e*** in the molecule, and determine **hybridization** at each of the carbon centers. (2.5 points)

Atom	Hybridization
Carbon (<i>a</i>)	
Carbon (<i>b</i>)	
Carbon (<i>c</i>)	
Carbon (<i>d</i>)	
Carbon (<i>e</i>)	

ii. Now, consider the sulfur atom (S) marked as ***f***, nitrogen (N) atom (***g***), and oxygen(O) atoms (***h*** and ***i***), and determine their **hybridization**. (2 points)

Atom	Hybridization
Sulfur (<i>f</i>)	
Nitrogen (<i>g</i>)	
Oxygen (<i>h</i>)	
Oxygen (<i>i</i>)	

iii. **Describe** all the bonds formed by the carbon atoms ***a*** and ***b***, using sigma and pi designation as appropriate. (2 points)

Atom	Bonds formed (use sigma and pi as appropriate)
Carbon (<i>a</i>)	
Carbon (<i>b</i>)	

iv. How many total sp^3 -s sigma bonds are in the molecule? (1 point)

v. Determine **electronic** and **molecular geometry** at carbon (C) labeled ***e*** and sulfur (S) labeled ***f*** centres. (2 points)

Atom	Electronic Geometry	Molecular Geometry
Carbon (<i>e</i>)		
Sulfur (<i>f</i>)		

vi. Circle the value from the choices given below, that describes the **C-S-C bond angle at sulfur** (S) marked ***f***: (0.5 points)

>90° to <109.5° 109.5° >109.5 to <120° 120° >120° to <180°

**Question 8 12 points**

Using the **valence shell orbitals only** draw a **complete** molecular orbital (MO) diagram (with no s-p mixing) for the molecule **PS**, in the space provided below. **(6 points)**

Assume that 3s and 3p atomic orbitals will form molecular orbitals like those formed by the combination of 2s and 2p atomic orbitals that we learnt in class, and there is no s-p mixing.

Show **energy arrow**, **atomic orbitals** involved, **atom labels**, the **molecular orbitals** that are formed, **filled electrons** in the MO diagram etc. Make sure to label all the atomic and molecular orbitals.

Using the MO diagram above, answer the following questions:

- (i) Which one of the following will have the highest bond order? Circle the correct choice
A) PS^{1-} B) PS C) PS^{1+}
- (ii) Which one of the following will have the longest bond length? Circle the correct choice
A) PS^{1-} B) PS C) PS^{1+}
- (iii) Which of the following will be diamagnetic? Circle the correct choice.
A) PS^{1-} B) PS C) PS^{1+}
- (iv) Draw Lewis dot structure of the molecule with the shortest bond length (based on the MO diagram) from among the three (PS^{1-} , PS, PS^{1+}), and indicate the bond order calculated from the Lewis dot structure. **(3 points)**

Lewis Dot Structure

Bond Order



073FFCD4-5702-4ACD-8D12-E552B3F4323B

midterm-exam-93388

#635 Page 10 of 10

SCRAP PAGE

**CHEMISTRY 110 –Midterm Exam****Thursday November 7, 2024, 6:30 – 8:30 pm****Examiners: Pallavi Sirjoosingh and Ashok Kakkar**

First name (please write as legibly as possible within the boxes)

--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--

Last name

--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--

Student ID number

--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--

This booklet has 8 questions on 10 pages (The last page can be used as a scrap page).

The staple must NOT be removed from the question booklet. Your intact question booklet must be returned at the end of the exam.

A separate booklet contains a periodic table, and datasheet.

Write your answers directly under the questions in the space provided. Do not write on the QR codes or near the top edges (as it may be cut off).

Please write **legibly and dark enough** for the scan to be readable. For students who use English as a second language, dictionaries are permitted. Calculators are allowed.

DO NOT ASK QUESTIONS OF THE INVIGILATORS about interpretations of the exam questions. If you have a concern about a question, please summarize it at the top of this page. All concerns will be evaluated after the exam. GOOD LUCK!

**Question 1 8 points** (*Assume all transitions are possible*)

a. 6 points Calculate the **energy (in kJ/mol)** required to ionize (completely remove) the electron from the **n=3** level of **He⁺ ion** (He: Helium). Show your work including equations. (3 sig. figs.)

b. 2 points

(I) Select the largest (longest) wavelength from the three choices below:

- i. Wavelength of photon to *completely remove* an electron from *1s orbital of Hydrogen atom*
- ii. Wavelength of photon to *completely remove* an electron from *1s orbital of Boron atom (B)*
- iii. Wavelength of photon to *completely remove* an electron from *1s orbital of Boron cation (B⁺)*

i or ii or iii

(II) Select the smallest (shortest) wavelength from the three choices below (*for H atom*):

- i. Wavelength of photon *emitted* for an electron transitioning from *2p to 1s orbital*
- ii. Wavelength of photon *absorbed* for an electron transitioning from *2s to 4p orbital*
- iii. Wavelength of photon *absorbed* for an electron transitioning from *1s to 3p orbital*

i or ii or iii



Question 2 8 points



A Chem110 student plots the graph shown on the left, from data obtained for the kinetic energy (eV) of the ejected electron versus frequency (s^{-1}) of the incident light, for an unknown metal. ($1\text{eV} = 1.6022 \times 10^{-19} \text{ J}$)

Based on the plot, answer the questions **a to c** below:

a. 1 point On the plot above, label the **threshold frequency**.

b. 2 points What is the value of the work potential (in J) of the unknown metal?

c. 3 points What is the *longest* wavelength (in nm) required to eject an electron from this metal's surface? Show your work including equation(s) used. (3 sig figs)

d. 2 points In the space provided below, plot the following

(Assume frequency is *above* threshold frequency):

- i. *Kinetic energy* of an ejected electron versus the *intensity* of the incident light (left)
- ii. *Number of electrons* ejected versus the *frequency* of the incident light (right)



**Question 3** **6 points****a. 2 points**

In the ground state of Ge (Germanium; $Z = 32$) atom, **name** the following orbitals:

i. Orbital containing the electron that requires the maximum energy to be removed.

ii. Orbital containing the electron that experiences the minimum effective nuclear charge.

b. 2 points

An unknown anion (A^-) has the following *excited* electronic configuration: $[Ar] 4s^2 3d^3 4p^1$

(Ar: Argon; $Z = 18$)

Give the **condensed ground state electronic configuration** of the atom, **A** and the cation, **A^{2+}** .
Show your work. (Assume all electronic transitions are possible)

Electronic Configuration of A:

Electronic Configuration of A^{2+} :

c. 2 points

In a single atom, determine how many electrons can have the following set of quantum numbers?

(i) $n = 5; l = 3$

In a single atom, determine how many electrons can have the following set of quantum numbers?

(i) $n = 3; l = 2; m_s = +1/2$

**Question 4** **8 points**

Rank the following from 1 to 4 in **increasing** order. Provide the ranking in the boxes provided.

Note: Atomic number (Z) are provided for relevant atoms

a) Increasing (lowest to highest) order of *ionization energy*: (1= lowest; 4 = highest)

Mg

Mg: Z = 12

Rb

Rb: Z = 37

Cs

Cs: Z = 55

K

K : Z = 19

b) Increasing (lowest to highest) order of *bond angle*: (1= lowest; 4 = highest)

H₂OCCl₄XeF₄

HCN

c) Increasing (smallest to largest) order of *ionic size* (1= smallest; 4 = largest)

S²⁻

S: Z = 16

O²⁻

O: Z = 8

Be²⁺

Be: Z = 4

Na⁺

Na: Z = 11

d) Increasing (lowest to highest) order of *bond energy* (1= lowest; 4 = highest)

P-F

P-Cl

P-I

P-Br

**Question 5** **12 points**

a. 9 points For the molecules HClO_4 and TeF_6 , draw the most stable *Lewis structure* showing appropriate shape and *dashed/wedged bonds*, based on VSEPR. Determine the **molecular geometry** around the central atom. For HClO_4 , label the *most polar bond*.



b. 3 points (Note: Lone pairs are not shown in structures below)

(i) For the three *resonance* structures (1, 2 and 3) shown below, answer the following question:



(I) Circle the *most contributing* resonance structure.

Structure 1

Structure 2

Structure 3

(ii) For the three structures given (4, 5, and 6) below, answer the following questions:



(I) Circle all the structures that represent *resonance structures* of each other:

Structure 4

Structure 5

Structure 6

(II) Among the resonance structures, circle the *most contributing* resonance structure.

Structure 4

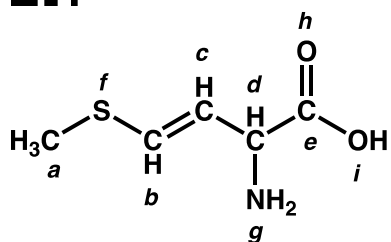
Structure 5

Structure 6

**Question 6** **6 points**

Using the data given below, calculate the **lattice energy** for SrO(s).

	ΔH° (kJ/mol)
$\text{Sr}(s) \longrightarrow \text{Sr}(g)$	+180
$\text{Sr}(g) \longrightarrow \text{Sr}^{2+}(g) + 2e^-$	+1468
$\text{O}_2(g) \longrightarrow 2\text{O}(g)$	+500
$\text{O}(g) + 2e^- \longrightarrow \text{O}^{2-}(g)$	+600
$\text{Sr}(s) + \frac{1}{2}\text{O}_2(g) \longrightarrow \text{SrO}(s)$	-550

**Question 7 10 points**

Consider the molecule given on the left, and answer questions (i) to (vi) using chemical bonding theories that we have learnt in class (valence bond theory, hybridization).

Note that the lone pairs are not shown in the structure:

i. Consider the carbon atoms marked in as ***a***, ***b***, ***c***, ***d***, and ***e*** in the molecule, and determine **hybridization** at each of the carbon centers. (2.5 points)

Atom	Hybridization
Carbon (<i>a</i>)	
Carbon (<i>b</i>)	
Carbon (<i>c</i>)	
Carbon (<i>d</i>)	
Carbon (<i>e</i>)	

ii. Now, consider the sulfur atom (S) marked as ***f***, nitrogen (N) atom (***g***), and oxygen(O) atoms (***h*** and ***i***), and determine their **hybridization**. (2 points)

Atom	Hybridization
Sulfur (<i>f</i>)	
Nitrogen (<i>g</i>)	
Oxygen (<i>h</i>)	
Oxygen (<i>i</i>)	

iii. **Describe** all the bonds formed by the carbon atoms ***a*** and ***b***, using sigma and pi designation as appropriate. (2 points)

Atom	Bonds formed (use sigma and pi as appropriate)
Carbon (<i>a</i>)	
Carbon (<i>b</i>)	

iv. How many total sp^3 -s sigma bonds are in the molecule? (1 point)

v. Determine **electronic** and **molecular geometry** at carbon (C) labeled ***e*** and sulfur (S) labeled ***f*** centres. (2 points)

Atom	Electronic Geometry	Molecular Geometry
Carbon (<i>e</i>)		
Sulfur (<i>f</i>)		

vi. Circle the value from the choices given below, that describes the **C-S-C bond angle at sulfur** (S) marked ***f***: (0.5 points)

>90° to <109.5° 109.5° >109.5 to <120° 120° >120° to <180°


Question 8 12 points

Using the **valence shell orbitals only** draw a **complete** molecular orbital (MO) diagram (with no s-p mixing) for the molecule **PS**, in the space provided below. **(6 points)**

Assume that 3s and 3p atomic orbitals will form molecular orbitals like those formed by the combination of 2s and 2p atomic orbitals that we learnt in class, and there is no s-p mixing.

Show **energy arrow**, **atomic orbitals** involved, **atom labels**, the **molecular orbitals** that are formed, **filled electrons** in the MO diagram etc. Make sure to label all the atomic and molecular orbitals.

Using the MO diagram above, answer the following questions:

- (i) Which one of the following will have the highest bond order? Circle the correct choice
 A) PS^{1-} B) PS C) PS^{1+}
- (ii) Which one of the following will have the longest bond length? Circle the correct choice
 A) PS^{1-} B) PS C) PS^{1+}
- (iii) Which of the following will be diamagnetic? Circle the correct choice.
 A) PS^{1-} B) PS C) PS^{1+}
- (iv) Draw Lewis dot structure of the molecule with the shortest bond length (based on the MO diagram) from among the three (PS^{1-} , PS, PS^{1+}), and indicate the bond order calculated from the Lewis dot structure. **(3 points)**

Lewis Dot Structure

Bond Order



A59D32EE-6BF5-4A27-A8C3-10D14BAD1E3D

midterm-exam-93388

#636 Page 10 of 10

SCRAP PAGE

**CHEMISTRY 110 –Midterm Exam****Thursday November 7, 2024, 6:30 – 8:30 pm**

Examiners: Pallavi Sirjoosingh and Ashok Kakkar

First name (please write as legibly as possible within the boxes)

--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--

Last name

--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--

Student ID number

--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--

This booklet has 8 questions on 10 pages (The last page can be used as a scrap page).

The staple must NOT be removed from the question booklet. Your intact question booklet must be returned at the end of the exam.

A separate booklet contains a periodic table, and datasheet.

Write your answers directly under the questions in the space provided. Do not write on the QR codes or near the top edges (as it may be cut off).

Please write **legibly and dark enough** for the scan to be readable. For students who use English as a second language, dictionaries are permitted. Calculators are allowed.

DO NOT ASK QUESTIONS OF THE INVIGILATORS about interpretations of the exam questions. If you have a concern about a question, please summarize it at the top of this page. All concerns will be evaluated after the exam. GOOD LUCK!

**Question 1 8 points** (*Assume all transitions are possible*)

a. 6 points Calculate the **energy (in kJ/mol)** required to ionize (completely remove) the electron from the **n=3** level of **He⁺ ion** (He: Helium). Show your work including equations. (3 sig. figs.)

b. 2 points

(I) Select the largest (longest) wavelength from the three choices below:

- i. Wavelength of photon to *completely remove* an electron from *1s orbital of Hydrogen atom*
- ii. Wavelength of photon to *completely remove* an electron from *1s orbital of Boron atom (B)*
- iii. Wavelength of photon to *completely remove* an electron from *1s orbital of Boron cation (B⁺)*

i or ii or iii

(II) Select the smallest (shortest) wavelength from the three choices below (*for H atom*):

- i. Wavelength of photon *emitted* for an electron transitioning from *2p to 1s orbital*
- ii. Wavelength of photon *absorbed* for an electron transitioning from *2s to 4p orbital*
- iii. Wavelength of photon *absorbed* for an electron transitioning from *1s to 3p orbital*

i or ii or iii



Question 2 8 points



A Chem110 student plots the graph shown on the left, from data obtained for the kinetic energy (eV) of the ejected electron versus frequency (s^{-1}) of the incident light, for an unknown metal. ($1\text{eV} = 1.6022 \times 10^{-19} \text{ J}$)

Based on the plot, answer the questions **a to c** below:

a. 1 point On the plot above, label the **threshold frequency**.

b. 2 points What is the value of the work potential (in J) of the unknown metal?

c. 3 points What is the *longest* wavelength (in nm) required to eject an electron from this metal's surface? Show your work including equation(s) used. (3 sig figs)

d. 2 points In the space provided below, plot the following

(Assume frequency is *above* threshold frequency):

- i. *Kinetic energy* of an ejected electron versus the *intensity* of the incident light (left)
- ii. *Number of electrons* ejected versus the *frequency* of the incident light (right)



**Question 3** **6 points****a. 2 points**

In the ground state of Ge (Germanium; $Z = 32$) atom, **name** the following orbitals:

i. Orbital containing the electron that requires the maximum energy to be removed.

ii. Orbital containing the electron that experiences the minimum effective nuclear charge.

b. 2 points

An unknown anion (A^-) has the following *excited* electronic configuration: $[Ar] 4s^2 3d^3 4p^1$

(Ar: Argon; $Z = 18$)

Give the **condensed ground state electronic configuration** of the atom, **A** and the cation, **A^{2+}** .
Show your work. (Assume all electronic transitions are possible)

Electronic Configuration of A:

Electronic Configuration of A^{2+} :

c. 2 points

In a single atom, determine how many electrons can have the following set of quantum numbers?

(i) $n = 5; l = 3$

In a single atom, determine how many electrons can have the following set of quantum numbers?

(i) $n = 3; l = 2; m_s = +1/2$

**Question 4** **8 points**

Rank the following from 1 to 4 in **increasing** order. Provide the ranking in the boxes provided.

Note: Atomic number (Z) are provided for relevant atoms

a) Increasing (lowest to highest) order of *ionization energy*: (1= lowest; 4 = highest)

Mg

Mg: Z = 12

Rb

Rb: Z = 37

Cs

Cs: Z = 55

K

K : Z = 19

b) Increasing (lowest to highest) order of *bond angle*: (1= lowest; 4 = highest)

H₂OCCl₄XeF₄

HCN

c) Increasing (smallest to largest) order of *ionic size* (1= smallest; 4 = largest)

S²⁻

S: Z = 16

O²⁻

O: Z = 8

Be²⁺

Be: Z = 4

Na⁺

Na: Z = 11

d) Increasing (lowest to highest) order of *bond energy* (1= lowest; 4 = highest)

P-F

P-Cl

P-I

P-Br

**Question 5** **12 points**

a. 9 points For the molecules HClO_4 and TeF_6 , draw the most stable *Lewis structure* showing appropriate shape and *dashed/wedged bonds*, based on VSEPR. Determine the **molecular geometry** around the central atom. For HClO_4 , label the *most polar bond*.



b. 3 points (Note: Lone pairs are not shown in structures below)

(i) For the three *resonance* structures (1, 2 and 3) shown below, answer the following question:



(I) Circle the *most contributing* resonance structure.

Structure 1

Structure 2

Structure 3

(ii) For the three structures given (4, 5, and 6) below, answer the following questions:



(I) Circle all the structures that represent *resonance structures* of each other:

Structure 4

Structure 5

Structure 6

(II) Among the resonance structures, circle the *most contributing* resonance structure.

Structure 4

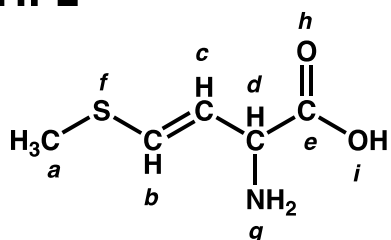
Structure 5

Structure 6

**Question 6** **6 points**

Using the data given below, calculate the **lattice energy** for SrO(s).

	ΔH° (kJ/mol)
$\text{Sr}(s) \longrightarrow \text{Sr}(g)$	+180
$\text{Sr}(g) \longrightarrow \text{Sr}^{2+}(g) + 2e^-$	+1468
$\text{O}_2(g) \longrightarrow 2\text{O}(g)$	+500
$\text{O}(g) + 2e^- \longrightarrow \text{O}^{2-}(g)$	+600
$\text{Sr}(s) + \frac{1}{2}\text{O}_2(g) \longrightarrow \text{SrO}(s)$	-550

**Question 7 10 points**

Consider the molecule given on the left, and answer questions (i) to (vi) using chemical bonding theories that we have learnt in class (valence bond theory, hybridization).

Note that the lone pairs are not shown in the structure:

i. Consider the carbon atoms marked in as ***a***, ***b***, ***c***, ***d***, and ***e*** in the molecule, and determine **hybridization** at each of the carbon centers. (2.5 points)

Atom	Hybridization
Carbon (<i>a</i>)	
Carbon (<i>b</i>)	
Carbon (<i>c</i>)	
Carbon (<i>d</i>)	
Carbon (<i>e</i>)	

ii. Now, consider the sulfur atom (S) marked as ***f***, nitrogen (N) atom (***g***), and oxygen(O) atoms (***h*** and ***i***), and determine their **hybridization**. (2 points)

Atom	Hybridization
Sulfur (<i>f</i>)	
Nitrogen (<i>g</i>)	
Oxygen (<i>h</i>)	
Oxygen (<i>i</i>)	

iii. **Describe** all the bonds formed by the carbon atoms ***a*** and ***b***, using sigma and pi designation as appropriate. (2 points)

Atom	Bonds formed (use sigma and pi as appropriate)
Carbon (<i>a</i>)	
Carbon (<i>b</i>)	

iv. How many total sp^3 -s sigma bonds are in the molecule? (1 point)

v. Determine **electronic** and **molecular geometry** at carbon (C) labeled ***e*** and sulfur (S) labeled ***f*** centres. (2 points)

Atom	Electronic Geometry	Molecular Geometry
Carbon (<i>e</i>)		
Sulfur (<i>f</i>)		

vi. Circle the value from the choices given below, that describes the **C-S-C bond angle at sulfur** (S) marked ***f***: (0.5 points)

>90° to <109.5° 109.5° >109.5 to <120° 120° >120° to <180°

**Question 8 12 points**

Using the **valence shell orbitals only** draw a **complete** molecular orbital (MO) diagram (with no s-p mixing) for the molecule **PS**, in the space provided below. **(6 points)**

Assume that 3s and 3p atomic orbitals will form molecular orbitals like those formed by the combination of 2s and 2p atomic orbitals that we learnt in class, and there is no s-p mixing.

Show **energy arrow**, **atomic orbitals** involved, **atom labels**, the **molecular orbitals** that are formed, **filled electrons** in the MO diagram etc. Make sure to label all the atomic and molecular orbitals.

Using the MO diagram above, answer the following questions:

- (i) Which one of the following will have the highest bond order? Circle the correct choice
A) PS^{1-} B) PS C) PS^{1+}
- (ii) Which one of the following will have the longest bond length? Circle the correct choice
A) PS^{1-} B) PS C) PS^{1+}
- (iii) Which of the following will be diamagnetic? Circle the correct choice.
A) PS^{1-} B) PS C) PS^{1+}
- (iv) Draw Lewis dot structure of the molecule with the shortest bond length (based on the MO diagram) from among the three (PS^{1-} , PS, PS^{1+}), and indicate the bond order calculated from the Lewis dot structure. **(3 points)**

Lewis Dot Structure

Bond Order



00337B25-039F-4853-BC16-0823E5AF7C91

midterm-exam-93388

#637 Page 10 of 10

SCRAP PAGE

**CHEMISTRY 110 –Midterm Exam****Thursday November 7, 2024, 6:30 – 8:30 pm**

Examiners: Pallavi Sirjoosingh and Ashok Kakkar

First name (please write as legibly as possible within the boxes)

--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--

Last name

--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--

Student ID number

--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--

This booklet has 8 questions on 10 pages (The last page can be used as a scrap page).

The staple must NOT be removed from the question booklet. Your intact question booklet must be returned at the end of the exam.

A separate booklet contains a periodic table, and datasheet.

Write your answers directly under the questions in the space provided. Do not write on the QR codes or near the top edges (as it may be cut off).

Please write **legibly and dark enough** for the scan to be readable. For students who use English as a second language, dictionaries are permitted. Calculators are allowed.

DO NOT ASK QUESTIONS OF THE INVIGILATORS about interpretations of the exam questions. If you have a concern about a question, please summarize it at the top of this page. All concerns will be evaluated after the exam. GOOD LUCK!

**Question 1 8 points** (*Assume all transitions are possible*)

a. 6 points Calculate the **energy (in kJ/mol)** required to ionize (completely remove) the electron from the **$n=3$** level of **He^+ ion** (He: Helium). Show your work including equations. (3 sig. figs.)

b. 2 points

(I) Select the largest (longest) wavelength from the three choices below:

- i. Wavelength of photon to *completely remove* an electron from *1s orbital of Hydrogen atom*
- ii. Wavelength of photon to *completely remove* an electron from *1s orbital of Boron atom (B)*
- iii. Wavelength of photon to *completely remove* an electron from *1s orbital of Boron cation (B^+)*

i or ii or iii

(II) Select the smallest (shortest) wavelength from the three choices below (*for H atom*):

- i. Wavelength of photon *emitted* for an electron transitioning from *2p to 1s orbital*
- ii. Wavelength of photon *absorbed* for an electron transitioning from *2s to 4p orbital*
- iii. Wavelength of photon *absorbed* for an electron transitioning from *1s to 3p orbital*

i or ii or iii



Question 2 8 points



A Chem110 student plots the graph shown on the left, from data obtained for the kinetic energy (eV) of the ejected electron versus frequency (s^{-1}) of the incident light, for an unknown metal. ($1\text{eV} = 1.6022 \times 10^{-19} \text{ J}$)

Based on the plot, answer the questions **a to c** below:

a. 1 point On the plot above, label the **threshold frequency**.

b. 2 points What is the value of the work potential (in J) of the unknown metal?

c. 3 points What is the *longest* wavelength (in nm) required to eject an electron from this metal's surface? Show your work including equation(s) used. (3 sig figs)

d. 2 points In the space provided below, plot the following

(Assume frequency is *above* threshold frequency):

- i. *Kinetic energy* of an ejected electron versus the *intensity* of the incident light (left)
- ii. *Number of electrons* ejected versus the *frequency* of the incident light (right)



**Question 3** **6 points****a. 2 points**

In the ground state of Ge (Germanium; $Z = 32$) atom, **name** the following orbitals:

i. Orbital containing the electron that requires the maximum energy to be removed.

ii. Orbital containing the electron that experiences the minimum effective nuclear charge.

b. 2 points

An unknown anion (A^-) has the following *excited* electronic configuration: $[Ar] 4s^2 3d^3 4p^1$

(Ar: Argon; $Z = 18$)

Give the **condensed ground state electronic configuration** of the atom, **A** and the cation, **A^{2+}** .
Show your work. (Assume all electronic transitions are possible)

Electronic Configuration of A:

Electronic Configuration of A^{2+} :

c. 2 points

In a single atom, determine how many electrons can have the following set of quantum numbers?

(i) $n = 5; l = 3$

In a single atom, determine how many electrons can have the following set of quantum numbers?

(i) $n = 3; l = 2; m_s = +1/2$

**Question 4** **8 points**

Rank the following from 1 to 4 in **increasing** order. Provide the ranking in the boxes provided.

Note: Atomic number (Z) are provided for relevant atoms

a) Increasing (lowest to highest) order of *ionization energy*: (1= lowest; 4 = highest)

Mg

Mg: Z = 12

Rb

Rb: Z = 37

Cs

Cs: Z = 55

K

K : Z = 19

b) Increasing (lowest to highest) order of *bond angle*: (1= lowest; 4 = highest)

H₂OCCl₄XeF₄

HCN

c) Increasing (smallest to largest) order of *ionic size* (1= smallest; 4 = largest)

S²⁻

S: Z = 16

O²⁻

O: Z = 8

Be²⁺

Be: Z = 4

Na⁺

Na: Z = 11

d) Increasing (lowest to highest) order of *bond energy* (1= lowest; 4 = highest)

P-F

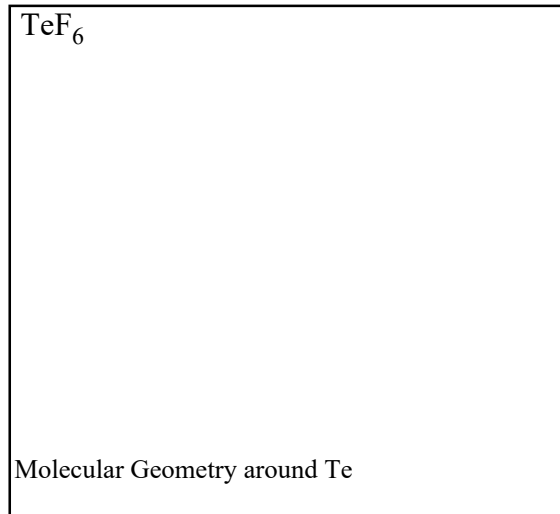
P-Cl

P-I

P-Br

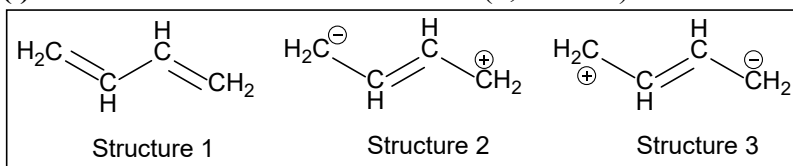
**Question 5** **12 points**

a. 9 points For the molecules HClO_4 and TeF_6 , draw the most stable *Lewis structure* showing appropriate shape and *dashed/wedged bonds*, based on VSEPR. Determine the **molecular geometry** around the central atom. For HClO_4 , label the *most polar bond*.



b. 3 points (Note: Lone pairs are not shown in structures below)

(i) For the three *resonance* structures (1, 2 and 3) shown below, answer the following question:



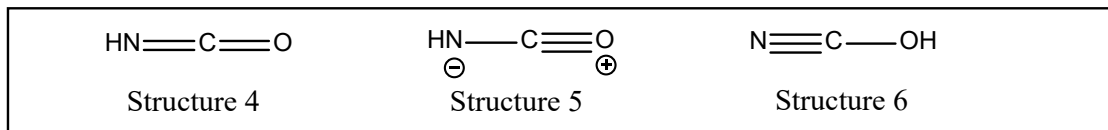
(I) Circle the *most contributing* resonance structure.

Structure 1

Structure 2

Structure 3

(ii) For the three structures given (4, 5, and 6) below, answer the following questions:



(I) Circle all the structures that represent *resonance structures* of each other:

Structure 4

Structure 5

Structure 6

(II) Among the resonance structures, circle the *most contributing* resonance structure.

Structure 4

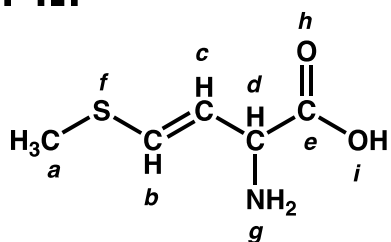
Structure 5

Structure 6

**Question 6** **6 points**

Using the data given below, calculate the **lattice energy** for SrO(s).

	ΔH° (kJ/mol)
$\text{Sr}(s) \longrightarrow \text{Sr}(g)$	+180
$\text{Sr}(g) \longrightarrow \text{Sr}^{2+}(g) + 2e^-$	+1468
$\text{O}_2(g) \longrightarrow 2\text{O}(g)$	+500
$\text{O}(g) + 2e^- \longrightarrow \text{O}^{2-}(g)$	+600
$\text{Sr}(s) + \frac{1}{2}\text{O}_2(g) \longrightarrow \text{SrO}(s)$	-550

**Question 7 10 points**

Consider the molecule given on the left, and answer questions (i) to (vi) using chemical bonding theories that we have learnt in class (valence bond theory, hybridization).

Note that the lone pairs are not shown in the structure:

i. Consider the carbon atoms marked in as ***a***, ***b***, ***c***, ***d***, and ***e*** in the molecule, and determine **hybridization** at each of the carbon centers. (2.5 points)

Atom	Hybridization
Carbon (<i>a</i>)	
Carbon (<i>b</i>)	
Carbon (<i>c</i>)	
Carbon (<i>d</i>)	
Carbon (<i>e</i>)	

ii. Now, consider the sulfur atom (S) marked as ***f***, nitrogen (N) atom (***g***), and oxygen(O) atoms (***h*** and ***i***), and determine their **hybridization**. (2 points)

Atom	Hybridization
Sulfur (<i>f</i>)	
Nitrogen (<i>g</i>)	
Oxygen (<i>h</i>)	
Oxygen (<i>i</i>)	

iii. **Describe** all the bonds formed by the carbon atoms ***a*** and ***b***, using sigma and pi designation as appropriate. (2 points)

Atom	Bonds formed (use sigma and pi as appropriate)
Carbon (<i>a</i>)	
Carbon (<i>b</i>)	

iv. How many total sp^3 -s sigma bonds are in the molecule? (1 point)

v. Determine **electronic** and **molecular geometry** at carbon (C) labeled ***e*** and sulfur (S) labeled ***f*** centres. (2 points)

Atom	Electronic Geometry	Molecular Geometry
Carbon (<i>e</i>)		
Sulfur (<i>f</i>)		

vi. Circle the value from the choices given below, that describes the **C-S-C bond angle at sulfur** (S) marked ***f***: (0.5 points)

>90° to <109.5° 109.5° >109.5 to <120° 120° >120° to <180°


Question 8 12 points

Using the **valence shell orbitals only** draw a **complete** molecular orbital (MO) diagram (with no s-p mixing) for the molecule **PS**, in the space provided below. **(6 points)**

Assume that 3s and 3p atomic orbitals will form molecular orbitals like those formed by the combination of 2s and 2p atomic orbitals that we learnt in class, and there is no s-p mixing.

Show **energy arrow**, **atomic orbitals** involved, **atom labels**, the **molecular orbitals** that are formed, **filled electrons** in the MO diagram etc. Make sure to label all the atomic and molecular orbitals.

Using the MO diagram above, answer the following questions:

- (i) Which one of the following will have the highest bond order? Circle the correct choice
 A) PS^{1-} B) PS C) PS^{1+}
- (ii) Which one of the following will have the longest bond length? Circle the correct choice
 A) PS^{1-} B) PS C) PS^{1+}
- (iii) Which of the following will be diamagnetic? Circle the correct choice.
 A) PS^{1-} B) PS C) PS^{1+}
- (iv) Draw Lewis dot structure of the molecule with the shortest bond length (based on the MO diagram) from among the three (PS^{1-} , PS, PS^{1+}), and indicate the bond order calculated from the Lewis dot structure. **(3 points)**

Lewis Dot Structure

Bond Order



4DC9DDE9-7178-4694-BC28-E6210F3E2BF8

midterm-exam-93388

#638 Page 10 of 10

SCRAP PAGE

**CHEMISTRY 110 –Midterm Exam****Thursday November 7, 2024, 6:30 – 8:30 pm**

Examiners: Pallavi Sirjoosingh and Ashok Kakkar

First name (please write as legibly as possible within the boxes)

--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--

Last name

--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--

Student ID number

--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--

This booklet has 8 questions on 10 pages (The last page can be used as a scrap page).

The staple must NOT be removed from the question booklet. Your intact question booklet must be returned at the end of the exam.

A separate booklet contains a periodic table, and datasheet.

Write your answers directly under the questions in the space provided. Do not write on the QR codes or near the top edges (as it may be cut off).

Please write **legibly and dark enough** for the scan to be readable. For students who use English as a second language, dictionaries are permitted. Calculators are allowed.

DO NOT ASK QUESTIONS OF THE INVIGILATORS about interpretations of the exam questions. If you have a concern about a question, please summarize it at the top of this page. All concerns will be evaluated after the exam. GOOD LUCK!

**Question 1 8 points** (*Assume all transitions are possible*)

a. 6 points Calculate the **energy (in kJ/mol)** required to ionize (completely remove) the electron from the **$n=3$** level of **He^+ ion** (He: Helium). Show your work including equations. (3 sig. figs.)

b. 2 points

(I) Select the largest (longest) wavelength from the three choices below:

- i. Wavelength of photon to *completely remove* an electron from *1s orbital of Hydrogen atom*
- ii. Wavelength of photon to *completely remove* an electron from *1s orbital of Boron atom (B)*
- iii. Wavelength of photon to *completely remove* an electron from *1s orbital of Boron cation (B^+)*

i or ii or iii

(II) Select the smallest (shortest) wavelength from the three choices below (*for H atom*):

- i. Wavelength of photon *emitted* for an electron transitioning from *2p to 1s orbital*
- ii. Wavelength of photon *absorbed* for an electron transitioning from *2s to 4p orbital*
- iii. Wavelength of photon *absorbed* for an electron transitioning from *1s to 3p orbital*

i or ii or iii



Question 2 8 points



A Chem110 student plots the graph shown on the left, from data obtained for the kinetic energy (eV) of the ejected electron versus frequency (s^{-1}) of the incident light, for an unknown metal. ($1\text{eV} = 1.6022 \times 10^{-19} \text{ J}$)

Based on the plot, answer the questions **a to c** below:

a. 1 point On the plot above, label the **threshold frequency**.

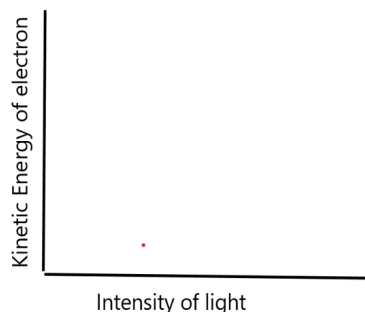
b. 2 points What is the value of the work potential (in J) of the unknown metal?

c. 3 points What is the *longest* wavelength (in nm) required to eject an electron from this metal's surface? Show your work including equation(s) used. (3 sig figs)

d. 2 points In the space provided below, plot the following

(Assume frequency is *above* threshold frequency):

- Kinetic energy* of an ejected electron versus the *intensity* of the incident light (left)
- Number of electrons* ejected versus the *frequency* of the incident light (right)



**Question 3** **6 points****a. 2 points**

In the ground state of Ge (Germanium; $Z = 32$) atom, **name** the following orbitals:

i. Orbital containing the electron that requires the maximum energy to be removed.

ii. Orbital containing the electron that experiences the minimum effective nuclear charge.

b. 2 points

An unknown anion (A^-) has the following *excited* electronic configuration: $[Ar] 4s^2 3d^3 4p^1$

(Ar: Argon; $Z = 18$)

Give the **condensed ground state electronic configuration** of the atom, **A** and the cation, **A^{2+}** .
Show your work. (Assume all electronic transitions are possible)

Electronic Configuration of A:

Electronic Configuration of A^{2+} :

c. 2 points

In a single atom, determine how many electrons can have the following set of quantum numbers?

(i) $n = 5; l = 3$

In a single atom, determine how many electrons can have the following set of quantum numbers?

(i) $n = 3; l = 2; m_s = +1/2$

**Question 4** **8 points**

Rank the following from 1 to 4 in **increasing** order. Provide the ranking in the boxes provided.

Note: Atomic number (Z) are provided for relevant atoms

a) Increasing (lowest to highest) order of *ionization energy*: (1= lowest; 4 = highest)

Mg

Mg: Z = 12

Rb

Rb: Z = 37

Cs

Cs: Z = 55

K

K : Z = 19

b) Increasing (lowest to highest) order of *bond angle*: (1= lowest; 4 = highest)

H₂OCCl₄XeF₄

HCN

c) Increasing (smallest to largest) order of *ionic size* (1= smallest; 4 = largest)

S²⁻

S: Z = 16

O²⁻

O: Z = 8

Be²⁺

Be: Z = 4

Na⁺

Na: Z = 11

d) Increasing (lowest to highest) order of *bond energy* (1= lowest; 4 = highest)

P-F

P-Cl

P-I

P-Br

**Question 5** **12 points**

a. 9 points For the molecules HClO_4 and TeF_6 , draw the most stable *Lewis structure* showing appropriate shape and *dashed/wedged bonds*, based on VSEPR. Determine the **molecular geometry** around the central atom. For HClO_4 , label the *most polar bond*.



b. 3 points (Note: Lone pairs are not shown in structures below)

(i) For the three *resonance* structures (1, 2 and 3) shown below, answer the following question:



(I) Circle the *most contributing* resonance structure.

Structure 1

Structure 2

Structure 3

(ii) For the three structures given (4, 5, and 6) below, answer the following questions:



(I) Circle all the structures that represent *resonance structures* of each other:

Structure 4

Structure 5

Structure 6

(II) Among the resonance structures, circle the *most contributing* resonance structure.

Structure 4

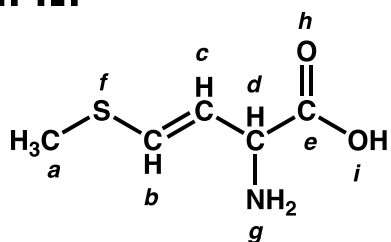
Structure 5

Structure 6

**Question 6** **6 points**

Using the data given below, calculate the **lattice energy** for SrO(s).

	ΔH° (kJ/mol)
$\text{Sr}(s) \longrightarrow \text{Sr}(g)$	+180
$\text{Sr}(g) \longrightarrow \text{Sr}^{2+}(g) + 2e^-$	+1468
$\text{O}_2(g) \longrightarrow 2\text{O}(g)$	+500
$\text{O}(g) + 2e^- \longrightarrow \text{O}^{2-}(g)$	+600
$\text{Sr}(s) + \frac{1}{2}\text{O}_2(g) \longrightarrow \text{SrO}(s)$	-550

**Question 7 10 points**

Consider the molecule given on the left, and answer questions (i) to (vi) using chemical bonding theories that we have learnt in class (valence bond theory, hybridization).

Note that the lone pairs are not shown in the structure:

i. Consider the carbon atoms marked in as ***a***, ***b***, ***c***, ***d***, and ***e*** in the molecule, and determine **hybridization** at each of the carbon centers. (2.5 points)

Atom	Hybridization
Carbon (<i>a</i>)	
Carbon (<i>b</i>)	
Carbon (<i>c</i>)	
Carbon (<i>d</i>)	
Carbon (<i>e</i>)	

ii. Now, consider the sulfur atom (S) marked as ***f***, nitrogen (N) atom (***g***), and oxygen(O) atoms (***h*** and ***i***), and determine their **hybridization**. (2 points)

Atom	Hybridization
Sulfur (<i>f</i>)	
Nitrogen (<i>g</i>)	
Oxygen (<i>h</i>)	
Oxygen (<i>i</i>)	

iii. **Describe** all the bonds formed by the carbon atoms ***a*** and ***b***, using sigma and pi designation as appropriate. (2 points)

Atom	Bonds formed (use sigma and pi as appropriate)
Carbon (<i>a</i>)	
Carbon (<i>b</i>)	

iv. How many total sp^3 -s sigma bonds are in the molecule? (1 point)

v. Determine **electronic** and **molecular geometry** at carbon (C) labeled ***e*** and sulfur (S) labeled ***f*** centres. (2 points)

Atom	Electronic Geometry	Molecular Geometry
Carbon (<i>e</i>)		
Sulfur (<i>f</i>)		

vi. Circle the value from the choices given below, that describes the **C-S-C bond angle at sulfur** (S) marked ***f***: (0.5 points)

>90° to <109.5° 109.5° >109.5 to <120° 120° >120° to <180°


Question 8 12 points

Using the **valence shell orbitals only** draw a **complete** molecular orbital (MO) diagram (with no s-p mixing) for the molecule **PS**, in the space provided below. **(6 points)**

Assume that 3s and 3p atomic orbitals will form molecular orbitals like those formed by the combination of 2s and 2p atomic orbitals that we learnt in class, and there is no s-p mixing.

Show **energy arrow**, **atomic orbitals** involved, **atom labels**, the **molecular orbitals** that are formed, **filled electrons** in the MO diagram etc. Make sure to label all the atomic and molecular orbitals.

Using the MO diagram above, answer the following questions:

- (i) Which one of the following will have the highest bond order? Circle the correct choice
 A) PS^{1-} B) PS C) PS^{1+}
- (ii) Which one of the following will have the longest bond length? Circle the correct choice
 A) PS^{1-} B) PS C) PS^{1+}
- (iii) Which of the following will be diamagnetic? Circle the correct choice.
 A) PS^{1-} B) PS C) PS^{1+}
- (iv) Draw Lewis dot structure of the molecule with the shortest bond length (based on the MO diagram) from among the three (PS^{1-} , PS, PS^{1+}), and indicate the bond order calculated from the Lewis dot structure. **(3 points)**

Lewis Dot Structure

Bond Order



9FEB73F2-ADC2-4587-B3EA-A32EBFA3A20B

midterm-exam-93388

#639 Page 10 of 10

SCRAP PAGE

**CHEMISTRY 110 –Midterm Exam****Thursday November 7, 2024, 6:30 – 8:30 pm****Examiners: Pallavi Sirjoosingh and Ashok Kakkar**

First name (please write as legibly as possible within the boxes)

--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--

Last name

--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--

Student ID number

--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--

This booklet has 8 questions on 10 pages (The last page can be used as a scrap page).

The staple must NOT be removed from the question booklet. Your intact question booklet must be returned at the end of the exam.

A separate booklet contains a periodic table, and datasheet.

Write your answers directly under the questions in the space provided. Do not write on the QR codes or near the top edges (as it may be cut off).

Please write **legibly and dark enough** for the scan to be readable. For students who use English as a second language, dictionaries are permitted. Calculators are allowed.

DO NOT ASK QUESTIONS OF THE INVIGILATORS about interpretations of the exam questions. If you have a concern about a question, please summarize it at the top of this page. All concerns will be evaluated after the exam. GOOD LUCK!

**Question 1 8 points** (*Assume all transitions are possible*)

a. 6 points Calculate the **energy (in kJ/mol)** required to ionize (completely remove) the electron from the **n=3** level of **He⁺ ion** (He: Helium). Show your work including equations. (3 sig. figs.)

b. 2 points

(I) Select the largest (longest) wavelength from the three choices below:

- i. Wavelength of photon to *completely remove* an electron from *1s orbital of Hydrogen atom*
- ii. Wavelength of photon to *completely remove* an electron from *1s orbital of Boron atom (B)*
- iii. Wavelength of photon to *completely remove* an electron from *1s orbital of Boron cation (B⁺)*

i or ii or iii

(II) Select the smallest (shortest) wavelength from the three choices below (*for H atom*):

- i. Wavelength of photon *emitted* for an electron transitioning from *2p to 1s orbital*
- ii. Wavelength of photon *absorbed* for an electron transitioning from *2s to 4p orbital*
- iii. Wavelength of photon *absorbed* for an electron transitioning from *1s to 3p orbital*

i or ii or iii



Question 2 8 points



A Chem110 student plots the graph shown on the left, from data obtained for the kinetic energy (eV) of the ejected electron versus frequency (s^{-1}) of the incident light, for an unknown metal. ($1\text{eV} = 1.6022 \times 10^{-19} \text{ J}$)

Based on the plot, answer the questions **a to c** below:

a. 1 point On the plot above, label the **threshold frequency**.

b. 2 points What is the value of the work potential (in J) of the unknown metal?

c. 3 points What is the *longest* wavelength (in nm) required to eject an electron from this metal's surface? Show your work including equation(s) used. (3 sig figs)

d. 2 points In the space provided below, plot the following

(Assume frequency is *above* threshold frequency):

- i. *Kinetic energy* of an ejected electron versus the *intensity* of the incident light (left)
- ii. *Number of electrons* ejected versus the *frequency* of the incident light (right)



**Question 3** **6 points****a. 2 points**

In the ground state of Ge (Germanium; $Z = 32$) atom, **name** the following orbitals:

i. Orbital containing the electron that requires the maximum energy to be removed.

ii. Orbital containing the electron that experiences the minimum effective nuclear charge.

b. 2 points

An unknown anion (A^-) has the following *excited* electronic configuration: $[Ar] 4s^2 3d^3 4p^1$

(Ar: Argon; $Z = 18$)

Give the **condensed ground state electronic configuration** of the atom, **A** and the cation, **A^{2+}** .
Show your work. (Assume all electronic transitions are possible)

Electronic Configuration of A:

Electronic Configuration of A^{2+} :

c. 2 points

In a single atom, determine how many electrons can have the following set of quantum numbers?

(i) $n = 5; l = 3$

In a single atom, determine how many electrons can have the following set of quantum numbers?

(i) $n = 3; l = 2; m_s = +1/2$

**Question 4** **8 points**

Rank the following from 1 to 4 in **increasing** order. Provide the ranking in the boxes provided.

Note: Atomic number (Z) are provided for relevant atoms

a) Increasing (lowest to highest) order of *ionization energy*: (1= lowest; 4 = highest)

Mg

Mg: Z = 12

Rb

Rb: Z = 37

Cs

Cs: Z = 55

K

K : Z = 19

b) Increasing (lowest to highest) order of *bond angle*: (1= lowest; 4 = highest)

H₂OCCl₄XeF₄

HCN

c) Increasing (smallest to largest) order of *ionic size* (1= smallest; 4 = largest)

S²⁻

S: Z = 16

O²⁻

O: Z = 8

Be²⁺

Be: Z = 4

Na⁺

Na: Z = 11

d) Increasing (lowest to highest) order of *bond energy* (1= lowest; 4 = highest)

P-F

P-Cl

P-I

P-Br

**Question 5** **12 points**

a. 9 points For the molecules HClO_4 and TeF_6 , draw the most stable *Lewis structure* showing appropriate shape and *dashed/wedged bonds*, based on VSEPR. Determine the **molecular geometry** around the central atom. For HClO_4 , label the *most polar bond*.



b. 3 points (Note: Lone pairs are not shown in structures below)

(i) For the three *resonance* structures (1, 2 and 3) shown below, answer the following question:



(I) Circle the *most contributing* resonance structure.

Structure 1

Structure 2

Structure 3

(ii) For the three structures given (4, 5, and 6) below, answer the following questions:



(I) Circle all the structures that represent *resonance structures* of each other:

Structure 4

Structure 5

Structure 6

(II) Among the resonance structures, circle the *most contributing* resonance structure.

Structure 4

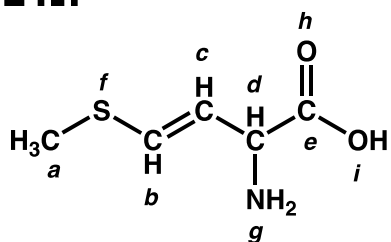
Structure 5

Structure 6

**Question 6** **6 points**

Using the data given below, calculate the **lattice energy** for SrO(s).

	ΔH° (kJ/mol)
$\text{Sr}(s) \longrightarrow \text{Sr}(g)$	+180
$\text{Sr}(g) \longrightarrow \text{Sr}^{2+}(g) + 2e^-$	+1468
$\text{O}_2(g) \longrightarrow 2\text{O}(g)$	+500
$\text{O}(g) + 2e^- \longrightarrow \text{O}^{2-}(g)$	+600
$\text{Sr}(s) + \frac{1}{2}\text{O}_2(g) \longrightarrow \text{SrO}(s)$	-550

**Question 7 10 points**

Consider the molecule given on the left, and answer questions (i) to (vi) using chemical bonding theories that we have learnt in class (valence bond theory, hybridization).

Note that the lone pairs are not shown in the structure:

i. Consider the carbon atoms marked in as ***a***, ***b***, ***c***, ***d***, and ***e*** in the molecule, and determine **hybridization** at each of the carbon centers. (2.5 points)

Atom	Hybridization
Carbon (<i>a</i>)	
Carbon (<i>b</i>)	
Carbon (<i>c</i>)	
Carbon (<i>d</i>)	
Carbon (<i>e</i>)	

ii. Now, consider the sulfur atom (S) marked as ***f***, nitrogen (N) atom (***g***), and oxygen(O) atoms (***h*** and ***i***), and determine their **hybridization**. (2 points)

Atom	Hybridization
Sulfur (<i>f</i>)	
Nitrogen (<i>g</i>)	
Oxygen (<i>h</i>)	
Oxygen (<i>i</i>)	

iii. **Describe** all the bonds formed by the carbon atoms ***a*** and ***b***, using sigma and pi designation as appropriate. (2 points)

Atom	Bonds formed (use sigma and pi as appropriate)
Carbon (<i>a</i>)	
Carbon (<i>b</i>)	

iv. How many total sp^3 -s sigma bonds are in the molecule? (1 point)

v. Determine **electronic** and **molecular geometry** at carbon (C) labeled ***e*** and sulfur (S) labeled ***f*** centres. (2 points)

Atom	Electronic Geometry	Molecular Geometry
Carbon (<i>e</i>)		
Sulfur (<i>f</i>)		

vi. Circle the value from the choices given below, that describes the **C-S-C bond angle at sulfur** (S) marked ***f***: (0.5 points)

>90° to <109.5° 109.5° >109.5 to <120° 120° >120° to <180°


Question 8 12 points

Using the **valence shell orbitals only** draw a **complete** molecular orbital (MO) diagram (with no s-p mixing) for the molecule **PS**, in the space provided below. **(6 points)**

Assume that 3s and 3p atomic orbitals will form molecular orbitals like those formed by the combination of 2s and 2p atomic orbitals that we learnt in class, and there is no s-p mixing.

Show **energy arrow**, **atomic orbitals** involved, **atom labels**, the **molecular orbitals** that are formed, **filled electrons** in the MO diagram etc. Make sure to label all the atomic and molecular orbitals.

Using the MO diagram above, answer the following questions:

- (i) Which one of the following will have the highest bond order? Circle the correct choice
 A) PS^{1-} B) PS C) PS^{1+}
- (ii) Which one of the following will have the longest bond length? Circle the correct choice
 A) PS^{1-} B) PS C) PS^{1+}
- (iii) Which of the following will be diamagnetic? Circle the correct choice.
 A) PS^{1-} B) PS C) PS^{1+}
- (iv) Draw Lewis dot structure of the molecule with the shortest bond length (based on the MO diagram) from among the three (PS^{1-} , PS, PS^{1+}), and indicate the bond order calculated from the Lewis dot structure. **(3 points)**

Lewis Dot Structure

Bond Order



506FF6A2-83FC-4119-ADF3-75FFB813D9A1

midterm-exam-93388

#640 Page 10 of 10

SCRAP PAGE

**CHEMISTRY 110 –Midterm Exam****Thursday November 7, 2024, 6:30 – 8:30 pm**

Examiners: Pallavi Sirjoosingh and Ashok Kakkar

First name (please write as legibly as possible within the boxes)

--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--

Last name

--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--

Student ID number

--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--

This booklet has 8 questions on 10 pages (The last page can be used as a scrap page).

The staple must NOT be removed from the question booklet. Your intact question booklet must be returned at the end of the exam.

A separate booklet contains a periodic table, and datasheet.

Write your answers directly under the questions in the space provided. Do not write on the QR codes or near the top edges (as it may be cut off).

Please write **legibly and dark enough** for the scan to be readable. For students who use English as a second language, dictionaries are permitted. Calculators are allowed.

DO NOT ASK QUESTIONS OF THE INVIGILATORS about interpretations of the exam questions. If you have a concern about a question, please summarize it at the top of this page. All concerns will be evaluated after the exam. GOOD LUCK!

**Question 1 8 points** (*Assume all transitions are possible*)

a. 6 points Calculate the **energy (in kJ/mol)** required to ionize (completely remove) the electron from the **$n=3$** level of **He^+ ion** (He: Helium). Show your work including equations. (3 sig. figs.)

b. 2 points

(I) Select the largest (longest) wavelength from the three choices below:

- i. Wavelength of photon to *completely remove* an electron from *1s orbital of Hydrogen atom*
- ii. Wavelength of photon to *completely remove* an electron from *1s orbital of Boron atom (B)*
- iii. Wavelength of photon to *completely remove* an electron from *1s orbital of Boron cation (B^+)*

i or ii or iii

(II) Select the smallest (shortest) wavelength from the three choices below (*for H atom*):

- i. Wavelength of photon *emitted* for an electron transitioning from *2p to 1s orbital*
- ii. Wavelength of photon *absorbed* for an electron transitioning from *2s to 4p orbital*
- iii. Wavelength of photon *absorbed* for an electron transitioning from *1s to 3p orbital*

i or ii or iii



Question 2 8 points



A Chem110 student plots the graph shown on the left, from data obtained for the kinetic energy (eV) of the ejected electron versus frequency (s^{-1}) of the incident light, for an unknown metal. ($1\text{eV} = 1.6022 \times 10^{-19} \text{ J}$)

Based on the plot, answer the questions **a to c** below:

a. 1 point On the plot above, label the **threshold frequency**.

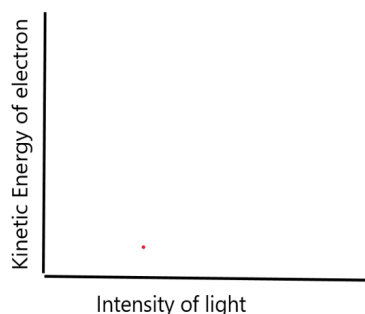
b. 2 points What is the value of the work potential (in J) of the unknown metal?

c. 3 points What is the *longest* wavelength (in nm) required to eject an electron from this metal's surface? Show your work including equation(s) used. (3 sig figs)

d. 2 points In the space provided below, plot the following

(Assume frequency is *above* threshold frequency):

- i. *Kinetic energy* of an ejected electron versus the *intensity* of the incident light (left)
- ii. *Number of electrons* ejected versus the *frequency* of the incident light (right)



**Question 3** **6 points****a. 2 points**

In the ground state of Ge (Germanium; $Z = 32$) atom, **name** the following orbitals:

i. Orbital containing the electron that requires the maximum energy to be removed.

ii. Orbital containing the electron that experiences the minimum effective nuclear charge.

b. 2 points

An unknown anion (A^-) has the following *excited* electronic configuration: $[Ar] 4s^2 3d^3 4p^1$

(Ar: Argon; $Z = 18$)

Give the **condensed ground state electronic configuration** of the atom, **A** and the cation, **A^{2+}** .
Show your work. (Assume all electronic transitions are possible)

Electronic Configuration of A:

Electronic Configuration of A^{2+} :

c. 2 points

In a single atom, determine how many electrons can have the following set of quantum numbers?

(i) $n = 5; l = 3$

In a single atom, determine how many electrons can have the following set of quantum numbers?

(i) $n = 3; l = 2; m_s = +1/2$

**Question 4** **8 points**

Rank the following from 1 to 4 in **increasing** order. Provide the ranking in the boxes provided.

Note: Atomic number (Z) are provided for relevant atoms

a) Increasing (lowest to highest) order of *ionization energy*: (1= lowest; 4 = highest)

Mg

Mg: Z = 12

Rb

Rb: Z = 37

Cs

Cs: Z = 55

K

K : Z = 19

b) Increasing (lowest to highest) order of *bond angle*: (1= lowest; 4 = highest)

H₂OCCl₄XeF₄

HCN

c) Increasing (smallest to largest) order of *ionic size* (1= smallest; 4 = largest)

S²⁻

S: Z = 16

O²⁻

O: Z = 8

Be²⁺

Be: Z = 4

Na⁺

Na: Z = 11

d) Increasing (lowest to highest) order of *bond energy* (1= lowest; 4 = highest)

P-F

P-Cl

P-I

P-Br

**Question 5** **12 points**

a. 9 points For the molecules HClO_4 and TeF_6 , draw the most stable *Lewis structure* showing appropriate shape and *dashed/wedged bonds*, based on VSEPR. Determine the **molecular geometry** around the central atom. For HClO_4 , label the *most polar bond*.



b. 3 points (Note: Lone pairs are not shown in structures below)

(i) For the three *resonance* structures (1, 2 and 3) shown below, answer the following question:



(I) Circle the *most contributing* resonance structure.

Structure 1

Structure 2

Structure 3

(ii) For the three structures given (4, 5, and 6) below, answer the following questions:



(I) Circle all the structures that represent *resonance structures* of each other:

Structure 4

Structure 5

Structure 6

(II) Among the resonance structures, circle the *most contributing* resonance structure.

Structure 4

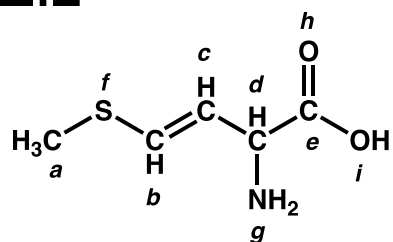
Structure 5

Structure 6

**Question 6** **6 points**

Using the data given below, calculate the **lattice energy** for SrO(s).

	ΔH° (kJ/mol)
$\text{Sr}(s) \longrightarrow \text{Sr}(g)$	+180
$\text{Sr}(g) \longrightarrow \text{Sr}^{2+}(g) + 2e^-$	+1468
$\text{O}_2(g) \longrightarrow 2\text{O}(g)$	+500
$\text{O}(g) + 2e^- \longrightarrow \text{O}^{2-}(g)$	+600
$\text{Sr}(s) + \frac{1}{2}\text{O}_2(g) \longrightarrow \text{SrO}(s)$	-550

**Question 7 10 points**

Consider the molecule given on the left, and answer questions (i) to (vi) using chemical bonding theories that we have learnt in class (valence bond theory, hybridization).

Note that the lone pairs are not shown in the structure:

i. Consider the carbon atoms marked in as ***a***, ***b***, ***c***, ***d***, and ***e*** in the molecule, and determine **hybridization** at each of the carbon centers. (2.5 points)

Atom	Hybridization
Carbon (<i>a</i>)	
Carbon (<i>b</i>)	
Carbon (<i>c</i>)	
Carbon (<i>d</i>)	
Carbon (<i>e</i>)	

ii. Now, consider the sulfur atom (S) marked as ***f***, nitrogen (N) atom (***g***), and oxygen(O) atoms (***h*** and ***i***), and determine their **hybridization**. (2 points)

Atom	Hybridization
Sulfur (<i>f</i>)	
Nitrogen (<i>g</i>)	
Oxygen (<i>h</i>)	
Oxygen (<i>i</i>)	

iii. **Describe** all the bonds formed by the carbon atoms ***a*** and ***b***, using sigma and pi designation as appropriate. (2 points)

Atom	Bonds formed (use sigma and pi as appropriate)
Carbon (<i>a</i>)	
Carbon (<i>b</i>)	

iv. How many total sp^3 -s sigma bonds are in the molecule? (1 point)

v. Determine **electronic** and **molecular geometry** at carbon (C) labeled ***e*** and sulfur (S) labeled ***f*** centres. (2 points)

Atom	Electronic Geometry	Molecular Geometry
Carbon (<i>e</i>)		
Sulfur (<i>f</i>)		

vi. Circle the value from the choices given below, that describes the **C-S-C bond angle at sulfur** (S) marked ***f***: (0.5 points)

>90° to <109.5° 109.5° >109.5 to <120° 120° >120° to <180°


Question 8 12 points

Using the **valence shell orbitals only** draw a **complete** molecular orbital (MO) diagram (with no s-p mixing) for the molecule **PS**, in the space provided below. **(6 points)**

Assume that 3s and 3p atomic orbitals will form molecular orbitals like those formed by the combination of 2s and 2p atomic orbitals that we learnt in class, and there is no s-p mixing.

Show **energy arrow**, **atomic orbitals** involved, **atom labels**, the **molecular orbitals** that are formed, **filled electrons** in the MO diagram etc. Make sure to label all the atomic and molecular orbitals.

Using the MO diagram above, answer the following questions:

- (i) Which one of the following will have the highest bond order? Circle the correct choice
 A) PS^{1-} B) PS C) PS^{1+}
- (ii) Which one of the following will have the longest bond length? Circle the correct choice
 A) PS^{1-} B) PS C) PS^{1+}
- (iii) Which of the following will be diamagnetic? Circle the correct choice.
 A) PS^{1-} B) PS C) PS^{1+}
- (iv) Draw Lewis dot structure of the molecule with the shortest bond length (based on the MO diagram) from among the three (PS^{1-} , PS, PS^{1+}), and indicate the bond order calculated from the Lewis dot structure. **(3 points)**

Lewis Dot Structure

Bond Order



B727B3FD-776E-44DF-AAD8-96990704837A

midterm-exam-93388

#641 Page 10 of 10

SCRAP PAGE

**CHEMISTRY 110 –Midterm Exam****Thursday November 7, 2024, 6:30 – 8:30 pm****Examiners: Pallavi Sirjoosingh and Ashok Kakkar**

First name (please write as legibly as possible within the boxes)

--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--

Last name

--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--

Student ID number

--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--

This booklet has 8 questions on 10 pages (The last page can be used as a scrap page).

The staple must NOT be removed from the question booklet. Your intact question booklet must be returned at the end of the exam.

A separate booklet contains a periodic table, and datasheet.

Write your answers directly under the questions in the space provided. Do not write on the QR codes or near the top edges (as it may be cut off).

Please write **legibly and dark enough** for the scan to be readable. For students who use English as a second language, dictionaries are permitted. Calculators are allowed.

DO NOT ASK QUESTIONS OF THE INVIGILATORS about interpretations of the exam questions. If you have a concern about a question, please summarize it at the top of this page. All concerns will be evaluated after the exam. GOOD LUCK!

**Question 1 8 points** (*Assume all transitions are possible*)

a. 6 points Calculate the **energy (in kJ/mol)** required to ionize (completely remove) the electron from the **n=3** level of **He⁺ ion** (He: Helium). Show your work including equations. (3 sig. figs.)

b. 2 points

(I) Select the largest (longest) wavelength from the three choices below:

- i. Wavelength of photon to *completely remove* an electron from *1s orbital of Hydrogen atom*
- ii. Wavelength of photon to *completely remove* an electron from *1s orbital of Boron atom (B)*
- iii. Wavelength of photon to *completely remove* an electron from *1s orbital of Boron cation (B⁺)*

i or ii or iii

(II) Select the smallest (shortest) wavelength from the three choices below (*for H atom*):

- i. Wavelength of photon *emitted* for an electron transitioning from *2p to 1s orbital*
- ii. Wavelength of photon *absorbed* for an electron transitioning from *2s to 4p orbital*
- iii. Wavelength of photon *absorbed* for an electron transitioning from *1s to 3p orbital*

i or ii or iii



Question 2 8 points



A Chem110 student plots the graph shown on the left, from data obtained for the kinetic energy (eV) of the ejected electron versus frequency (s^{-1}) of the incident light, for an unknown metal. ($1\text{eV} = 1.6022 \times 10^{-19} \text{ J}$)

Based on the plot, answer the questions **a to c** below:

a. 1 point On the plot above, label the **threshold frequency**.

b. 2 points What is the value of the work potential (in J) of the unknown metal?

c. 3 points What is the *longest* wavelength (in nm) required to eject an electron from this metal's surface? Show your work including equation(s) used. (3 sig figs)

d. 2 points In the space provided below, plot the following

(Assume frequency is *above* threshold frequency):

- i. *Kinetic energy* of an ejected electron versus the *intensity* of the incident light (left)
- ii. *Number of electrons* ejected versus the *frequency* of the incident light (right)



**Question 3** **6 points****a. 2 points**

In the ground state of Ge (Germanium; $Z = 32$) atom, **name** the following orbitals:

i. Orbital containing the electron that requires the maximum energy to be removed.

ii. Orbital containing the electron that experiences the minimum effective nuclear charge.

b. 2 points

An unknown anion (A^-) has the following *excited* electronic configuration: $[Ar] 4s^2 3d^3 4p^1$

(Ar: Argon; $Z = 18$)

Give the **condensed ground state electronic configuration** of the atom, **A** and the cation, **A^{2+}** .
Show your work. (Assume all electronic transitions are possible)

Electronic Configuration of A:

Electronic Configuration of A^{2+} :

c. 2 points

In a single atom, determine how many electrons can have the following set of quantum numbers?

(i) $n = 5; l = 3$

In a single atom, determine how many electrons can have the following set of quantum numbers?

(i) $n = 3; l = 2; m_s = +1/2$

**Question 4** **8 points**

Rank the following from 1 to 4 in **increasing** order. Provide the ranking in the boxes provided.

Note: Atomic number (Z) are provided for relevant atoms

a) Increasing (lowest to highest) order of *ionization energy*: (1= lowest; 4 = highest)

Mg

Mg: Z = 12

Rb

Rb: Z = 37

Cs

Cs: Z = 55

K

K : Z = 19

b) Increasing (lowest to highest) order of *bond angle*: (1= lowest; 4 = highest)

H₂OCCl₄XeF₄

HCN

c) Increasing (smallest to largest) order of *ionic size* (1= smallest; 4 = largest)

S²⁻

S: Z = 16

O²⁻

O: Z = 8

Be²⁺

Be: Z = 4

Na⁺

Na: Z = 11

d) Increasing (lowest to highest) order of *bond energy* (1= lowest; 4 = highest)

P-F

P-Cl

P-I

P-Br

**Question 5** **12 points**

a. 9 points For the molecules HClO_4 and TeF_6 , draw the most stable *Lewis structure* showing appropriate shape and *dashed/wedged bonds*, based on VSEPR. Determine the **molecular geometry** around the central atom. For HClO_4 , label the *most polar bond*.



b. 3 points (Note: Lone pairs are not shown in structures below)

(i) For the three *resonance* structures (1, 2 and 3) shown below, answer the following question:



(I) Circle the *most contributing* resonance structure.

Structure 1

Structure 2

Structure 3

(ii) For the three structures given (4, 5, and 6) below, answer the following questions:



(I) Circle all the structures that represent *resonance structures* of each other:

Structure 4

Structure 5

Structure 6

(II) Among the resonance structures, circle the *most contributing* resonance structure.

Structure 4

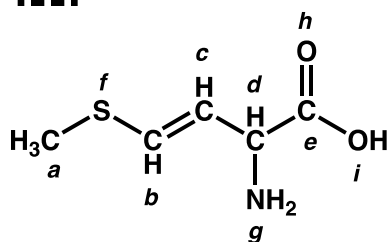
Structure 5

Structure 6

**Question 6** **6 points**

Using the data given below, calculate the **lattice energy** for SrO(s).

	ΔH° (kJ/mol)
$\text{Sr}(s) \longrightarrow \text{Sr}(g)$	+180
$\text{Sr}(g) \longrightarrow \text{Sr}^{2+}(g) + 2e^-$	+1468
$\text{O}_2(g) \longrightarrow 2\text{O}(g)$	+500
$\text{O}(g) + 2e^- \longrightarrow \text{O}^{2-}(g)$	+600
$\text{Sr}(s) + \frac{1}{2}\text{O}_2(g) \longrightarrow \text{SrO}(s)$	-550

**Question 7 10 points**

Consider the molecule given on the left, and answer questions (i) to (vi) using chemical bonding theories that we have learnt in class (valence bond theory, hybridization).

Note that the lone pairs are not shown in the structure:

i. Consider the carbon atoms marked in as ***a***, ***b***, ***c***, ***d***, and ***e*** in the molecule, and determine **hybridization** at each of the carbon centers. (2.5 points)

Atom	Hybridization
Carbon (<i>a</i>)	
Carbon (<i>b</i>)	
Carbon (<i>c</i>)	
Carbon (<i>d</i>)	
Carbon (<i>e</i>)	

ii. Now, consider the sulfur atom (S) marked as ***f***, nitrogen (N) atom (***g***), and oxygen(O) atoms (***h*** and ***i***), and determine their **hybridization**. (2 points)

Atom	Hybridization
Sulfur (<i>f</i>)	
Nitrogen (<i>g</i>)	
Oxygen (<i>h</i>)	
Oxygen (<i>i</i>)	

iii. **Describe** all the bonds formed by the carbon atoms ***a*** and ***b***, using sigma and pi designation as appropriate. (2 points)

Atom	Bonds formed (use sigma and pi as appropriate)
Carbon (<i>a</i>)	
Carbon (<i>b</i>)	

iv. How many total sp^3 -s sigma bonds are in the molecule? (1 point)

v. Determine **electronic** and **molecular geometry** at carbon (C) labeled ***e*** and sulfur (S) labeled ***f*** centres. (2 points)

Atom	Electronic Geometry	Molecular Geometry
Carbon (<i>e</i>)		
Sulfur (<i>f</i>)		

vi. Circle the value from the choices given below, that describes the **C-S-C bond angle at sulfur** (S) marked ***f***: (0.5 points)

>90° to <109.5° 109.5° >109.5 to <120° 120° >120° to <180°


Question 8 12 points

Using the **valence shell orbitals only** draw a **complete** molecular orbital (MO) diagram (with no s-p mixing) for the molecule **PS**, in the space provided below. **(6 points)**

Assume that 3s and 3p atomic orbitals will form molecular orbitals like those formed by the combination of 2s and 2p atomic orbitals that we learnt in class, and there is no s-p mixing.

Show **energy arrow**, **atomic orbitals** involved, **atom labels**, the **molecular orbitals** that are formed, **filled electrons** in the MO diagram etc. Make sure to **label all the atomic and molecular orbitals**.

Using the MO diagram above, answer the following questions:

- (i) Which one of the following will have the highest bond order? Circle the correct choice
 A) PS^{1-} B) PS C) PS^{1+}
- (ii) Which one of the following will have the longest bond length? Circle the correct choice
 A) PS^{1-} B) PS C) PS^{1+}
- (iii) Which of the following will be diamagnetic? Circle the correct choice.
 A) PS^{1-} B) PS C) PS^{1+}
- (iv) Draw Lewis dot structure of the molecule with the shortest bond length (based on the MO diagram) from among the three (PS^{1-} , PS, PS^{1+}), and indicate the bond order calculated from the Lewis dot structure. **(3 points)**

Lewis Dot Structure

Bond Order



8A34E51F-04A1-4037-B4E8-E02D0C1A4DDE

midterm-exam-93388

#642 Page 10 of 10

SCRAP PAGE

**CHEMISTRY 110 –Midterm Exam****Thursday November 7, 2024, 6:30 – 8:30 pm**

Examiners: Pallavi Sirjoosingh and Ashok Kakkar

First name (please write as legibly as possible within the boxes)

--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--

Last name

--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--

Student ID number

--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--

This booklet has 8 questions on 10 pages (The last page can be used as a scrap page).

The staple must NOT be removed from the question booklet. Your intact question booklet must be returned at the end of the exam.

A separate booklet contains a periodic table, and datasheet.

Write your answers directly under the questions in the space provided. Do not write on the QR codes or near the top edges (as it may be cut off).

Please write **legibly and dark enough** for the scan to be readable. For students who use English as a second language, dictionaries are permitted. Calculators are allowed.

DO NOT ASK QUESTIONS OF THE INVIGILATORS about interpretations of the exam questions. If you have a concern about a question, please summarize it at the top of this page. All concerns will be evaluated after the exam. GOOD LUCK!

**Question 1 8 points** (*Assume all transitions are possible*)

a. 6 points Calculate the **energy (in kJ/mol)** required to ionize (completely remove) the electron from the **n=3** level of **He⁺ ion** (He: Helium). Show your work including equations. (3 sig. figs.)

b. 2 points

(I) Select the largest (longest) wavelength from the three choices below:

- i. Wavelength of photon to *completely remove* an electron from *1s orbital of Hydrogen atom*
- ii. Wavelength of photon to *completely remove* an electron from *1s orbital of Boron atom (B)*
- iii. Wavelength of photon to *completely remove* an electron from *1s orbital of Boron cation (B⁺)*

i or ii or iii

(II) Select the smallest (shortest) wavelength from the three choices below (*for H atom*):

- i. Wavelength of photon *emitted* for an electron transitioning from *2p to 1s orbital*
- ii. Wavelength of photon *absorbed* for an electron transitioning from *2s to 4p orbital*
- iii. Wavelength of photon *absorbed* for an electron transitioning from *1s to 3p orbital*

i or ii or iii



Question 2 8 points



A Chem110 student plots the graph shown on the left, from data obtained for the kinetic energy (eV) of the ejected electron versus frequency (s^{-1}) of the incident light, for an unknown metal. ($1\text{eV} = 1.6022 \times 10^{-19} \text{ J}$)

Based on the plot, answer the questions **a to c** below:

a. 1 point On the plot above, label the **threshold frequency**.

b. 2 points What is the value of the work potential (in J) of the unknown metal?

c. 3 points What is the *longest* wavelength (in nm) required to eject an electron from this metal's surface? Show your work including equation(s) used. (3 sig figs)

d. 2 points In the space provided below, plot the following

(Assume frequency is *above* threshold frequency):

- Kinetic energy* of an ejected electron versus the *intensity* of the incident light (left)
- Number of electrons* ejected versus the *frequency* of the incident light (right)



**Question 3** **6 points****a. 2 points**

In the ground state of Ge (Germanium; $Z = 32$) atom, **name** the following orbitals:

i. Orbital containing the electron that requires the maximum energy to be removed.

ii. Orbital containing the electron that experiences the minimum effective nuclear charge.

b. 2 points

An unknown anion (A^-) has the following *excited* electronic configuration: $[Ar] 4s^2 3d^3 4p^1$

(Ar: Argon; $Z = 18$)

Give the **condensed ground state electronic configuration** of the atom, **A** and the cation, **A^{2+}** .
Show your work. (Assume all electronic transitions are possible)

Electronic Configuration of A:

Electronic Configuration of A^{2+} :

c. 2 points

In a single atom, determine how many electrons can have the following set of quantum numbers?

(i) $n = 5; l = 3$

In a single atom, determine how many electrons can have the following set of quantum numbers?

(i) $n = 3; l = 2; m_s = +1/2$

**Question 4** **8 points**

Rank the following from 1 to 4 in **increasing** order. Provide the ranking in the boxes provided.

Note: Atomic number (Z) are provided for relevant atoms

a) Increasing (lowest to highest) order of *ionization energy*: (1= lowest; 4 = highest)

Mg

Mg: Z = 12

Rb

Rb: Z = 37

Cs

Cs: Z = 55

K

K : Z = 19

b) Increasing (lowest to highest) order of *bond angle*: (1= lowest; 4 = highest)

H₂OCCl₄XeF₄

HCN

c) Increasing (smallest to largest) order of *ionic size* (1= smallest; 4 = largest)

S²⁻

S: Z = 16

O²⁻

O: Z = 8

Be²⁺

Be: Z = 4

Na⁺

Na: Z = 11

d) Increasing (lowest to highest) order of *bond energy* (1= lowest; 4 = highest)

P-F

P-Cl

P-I

P-Br

**Question 5** **12 points**

a. 9 points For the molecules HClO_4 and TeF_6 , draw the most stable *Lewis structure* showing appropriate shape and *dashed/wedged bonds*, based on VSEPR. Determine the **molecular geometry** around the central atom. For HClO_4 , label the *most polar bond*.



b. 3 points (Note: Lone pairs are not shown in structures below)

(i) For the three *resonance* structures (1, 2 and 3) shown below, answer the following question:



(I) Circle the *most contributing* resonance structure.

Structure 1

Structure 2

Structure 3

(ii) For the three structures given (4, 5, and 6) below, answer the following questions:



(I) Circle all the structures that represent *resonance structures* of each other:

Structure 4

Structure 5

Structure 6

(II) Among the resonance structures, circle the *most contributing* resonance structure.

Structure 4

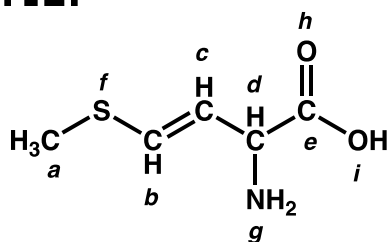
Structure 5

Structure 6

**Question 6** **6 points**

Using the data given below, calculate the **lattice energy** for SrO(s).

	ΔH° (kJ/mol)
$\text{Sr}(s) \longrightarrow \text{Sr}(g)$	+180
$\text{Sr}(g) \longrightarrow \text{Sr}^{2+}(g) + 2e^-$	+1468
$\text{O}_2(g) \longrightarrow 2\text{O}(g)$	+500
$\text{O}(g) + 2e^- \longrightarrow \text{O}^{2-}(g)$	+600
$\text{Sr}(s) + \frac{1}{2}\text{O}_2(g) \longrightarrow \text{SrO}(s)$	-550

**Question 7 10 points**

Consider the molecule given on the left, and answer questions (i) to (vi) using chemical bonding theories that we have learnt in class (valence bond theory, hybridization).

Note that the lone pairs are not shown in the structure:

i. Consider the carbon atoms marked in as ***a***, ***b***, ***c***, ***d***, and ***e*** in the molecule, and determine **hybridization** at each of the carbon centers. (2.5 points)

Atom	Hybridization
Carbon (<i>a</i>)	
Carbon (<i>b</i>)	
Carbon (<i>c</i>)	
Carbon (<i>d</i>)	
Carbon (<i>e</i>)	

ii. Now, consider the sulfur atom (S) marked as ***f***, nitrogen (N) atom (***g***), and oxygen(O) atoms (***h*** and ***i***), and determine their **hybridization**. (2 points)

Atom	Hybridization
Sulfur (<i>f</i>)	
Nitrogen (<i>g</i>)	
Oxygen (<i>h</i>)	
Oxygen (<i>i</i>)	

iii. **Describe** all the bonds formed by the carbon atoms ***a*** and ***b***, using sigma and pi designation as appropriate. (2 points)

Atom	Bonds formed (use sigma and pi as appropriate)
Carbon (<i>a</i>)	
Carbon (<i>b</i>)	

iv. How many total sp^3 -s sigma bonds are in the molecule? (1 point)

v. Determine **electronic** and **molecular geometry** at carbon (C) labeled ***e*** and sulfur (S) labeled ***f*** centres. (2 points)

Atom	Electronic Geometry	Molecular Geometry
Carbon (<i>e</i>)		
Sulfur (<i>f</i>)		

vi. Circle the value from the choices given below, that describes the **C-S-C bond angle at sulfur** (S) marked ***f***: (0.5 points)

$>90^\circ$ to $<109.5^\circ$ 109.5° >109.5 to $<120^\circ$ 120° $>120^\circ$ to $<180^\circ$

**Question 8 12 points**

Using the **valence shell orbitals only** draw a **complete** molecular orbital (MO) diagram (with no s-p mixing) for the molecule **PS**, in the space provided below. **(6 points)**

Assume that 3s and 3p atomic orbitals will form molecular orbitals like those formed by the combination of 2s and 2p atomic orbitals that we learnt in class, and there is no s-p mixing.

Show **energy arrow**, **atomic orbitals** involved, **atom labels**, the **molecular orbitals** that are formed, **filled electrons** in the MO diagram etc. Make sure to label all the atomic and molecular orbitals.

Using the MO diagram above, answer the following questions:

- (i) Which one of the following will have the highest bond order? Circle the correct choice
A) PS^{1-} B) PS C) PS^{1+}
- (ii) Which one of the following will have the longest bond length? Circle the correct choice
A) PS^{1-} B) PS C) PS^{1+}
- (iii) Which of the following will be diamagnetic? Circle the correct choice.
A) PS^{1-} B) PS C) PS^{1+}
- (iv) Draw Lewis dot structure of the molecule with the shortest bond length (based on the MO diagram) from among the three (PS^{1-} , PS, PS^{1+}), and indicate the bond order calculated from the Lewis dot structure. **(3 points)**

Lewis Dot Structure

Bond Order



9B66F3AF-31FC-4FD7-9F8D-376BB1399D19

midterm-exam-93388

#643 Page 10 of 10

SCRAP PAGE

**CHEMISTRY 110 –Midterm Exam****Thursday November 7, 2024, 6:30 – 8:30 pm**

Examiners: Pallavi Sirjoosingh and Ashok Kakkar

First name (please write as legibly as possible within the boxes)

--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--

Last name

--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--

Student ID number

--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--

This booklet has 8 questions on 10 pages (The last page can be used as a scrap page).

The staple must NOT be removed from the question booklet. Your intact question booklet must be returned at the end of the exam.

A separate booklet contains a periodic table, and datasheet.

Write your answers directly under the questions in the space provided. Do not write on the QR codes or near the top edges (as it may be cut off).

Please write **legibly and dark enough** for the scan to be readable. For students who use English as a second language, dictionaries are permitted. Calculators are allowed.

DO NOT ASK QUESTIONS OF THE INVIGILATORS about interpretations of the exam questions. If you have a concern about a question, please summarize it at the top of this page. All concerns will be evaluated after the exam. GOOD LUCK!

**Question 1 8 points** (*Assume all transitions are possible*)

a. 6 points Calculate the **energy (in kJ/mol)** required to ionize (completely remove) the electron from the **$n=3$** level of **He^+ ion** (He: Helium). Show your work including equations. (3 sig. figs.)

b. 2 points

(I) Select the largest (longest) wavelength from the three choices below:

- i. Wavelength of photon to *completely remove* an electron from *1s orbital of Hydrogen atom*
- ii. Wavelength of photon to *completely remove* an electron from *1s orbital of Boron atom (B)*
- iii. Wavelength of photon to *completely remove* an electron from *1s orbital of Boron cation (B^+)*

i or ii or iii

(II) Select the smallest (shortest) wavelength from the three choices below (*for H atom*):

- i. Wavelength of photon *emitted* for an electron transitioning from *2p to 1s orbital*
- ii. Wavelength of photon *absorbed* for an electron transitioning from *2s to 4p orbital*
- iii. Wavelength of photon *absorbed* for an electron transitioning from *1s to 3p orbital*

i or ii or iii



Question 2 8 points



A Chem110 student plots the graph shown on the left, from data obtained for the kinetic energy (eV) of the ejected electron versus frequency (s^{-1}) of the incident light, for an unknown metal. ($1\text{eV} = 1.6022 \times 10^{-19} \text{ J}$)

Based on the plot, answer the questions **a to c** below:

a. 1 point On the plot above, label the **threshold frequency**.

b. 2 points What is the value of the work potential (in J) of the unknown metal?

c. 3 points What is the *longest* wavelength (in nm) required to eject an electron from this metal's surface? Show your work including equation(s) used. (3 sig figs)

d. 2 points In the space provided below, plot the following

(Assume frequency is *above* threshold frequency):

- i. *Kinetic energy* of an ejected electron versus the *intensity* of the incident light (left)
- ii. *Number of electrons* ejected versus the *frequency* of the incident light (right)



**Question 3** **6 points****a. 2 points**

In the ground state of Ge (Germanium; $Z = 32$) atom, **name** the following orbitals:

i. Orbital containing the electron that requires the maximum energy to be removed.

ii. Orbital containing the electron that experiences the minimum effective nuclear charge.

b. 2 points

An unknown anion (A^-) has the following *excited* electronic configuration: $[Ar] 4s^2 3d^3 4p^1$

(Ar: Argon; $Z = 18$)

Give the **condensed ground state electronic configuration** of the atom, **A** and the cation, **A^{2+}** .
Show your work. (Assume all electronic transitions are possible)

Electronic Configuration of A:

Electronic Configuration of A^{2+} :

c. 2 points

In a single atom, determine how many electrons can have the following set of quantum numbers?

(i) $n = 5; l = 3$

In a single atom, determine how many electrons can have the following set of quantum numbers?

(i) $n = 3; l = 2; m_s = +1/2$

**Question 4** **8 points**

Rank the following from 1 to 4 in **increasing** order. Provide the ranking in the boxes provided.

Note: Atomic number (Z) are provided for relevant atoms

a) Increasing (lowest to highest) order of *ionization energy*: (1= lowest; 4 = highest)

Mg

Mg: Z = 12

Rb

Rb: Z = 37

Cs

Cs: Z = 55

K

K : Z = 19

b) Increasing (lowest to highest) order of *bond angle*: (1= lowest; 4 = highest)

H₂OCCl₄XeF₄

HCN

c) Increasing (smallest to largest) order of *ionic size* (1= smallest; 4 = largest)

S²⁻

S: Z = 16

O²⁻

O: Z = 8

Be²⁺

Be: Z = 4

Na⁺

Na: Z = 11

d) Increasing (lowest to highest) order of *bond energy* (1= lowest; 4 = highest)

P-F

P-Cl

P-I

P-Br

**Question 5** **12 points**

a. 9 points For the molecules HClO_4 and TeF_6 , draw the most stable *Lewis structure* showing appropriate shape and *dashed/wedged bonds*, based on VSEPR. Determine the **molecular geometry** around the central atom. For HClO_4 , label the *most polar bond*.



b. 3 points (Note: Lone pairs are not shown in structures below)

(i) For the three *resonance* structures (1, 2 and 3) shown below, answer the following question:



(I) Circle the *most contributing* resonance structure.

Structure 1

Structure 2

Structure 3

(ii) For the three structures given (4, 5, and 6) below, answer the following questions:



(I) Circle all the structures that represent *resonance structures* of each other:

Structure 4

Structure 5

Structure 6

(II) Among the resonance structures, circle the *most contributing* resonance structure.

Structure 4

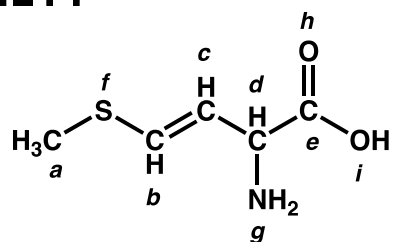
Structure 5

Structure 6

**Question 6** **6 points**

Using the data given below, calculate the **lattice energy** for SrO(s).

	ΔH° (kJ/mol)
$\text{Sr}(s) \longrightarrow \text{Sr}(g)$	+180
$\text{Sr}(g) \longrightarrow \text{Sr}^{2+}(g) + 2e^-$	+1468
$\text{O}_2(g) \longrightarrow 2\text{O}(g)$	+500
$\text{O}(g) + 2e^- \longrightarrow \text{O}^{2-}(g)$	+600
$\text{Sr}(s) + \frac{1}{2}\text{O}_2(g) \longrightarrow \text{SrO}(s)$	-550

**Question 7 10 points**

Consider the molecule given on the left, and answer questions (i) to (vi) using chemical bonding theories that we have learnt in class (valence bond theory, hybridization).

Note that the lone pairs are not shown in the structure:

i. Consider the carbon atoms marked in as ***a***, ***b***, ***c***, ***d***, and ***e*** in the molecule, and determine **hybridization** at each of the carbon centers. (2.5 points)

Atom	Hybridization
Carbon (<i>a</i>)	
Carbon (<i>b</i>)	
Carbon (<i>c</i>)	
Carbon (<i>d</i>)	
Carbon (<i>e</i>)	

ii. Now, consider the sulfur atom (S) marked as ***f***, nitrogen (N) atom (***g***), and oxygen(O) atoms (***h*** and ***i***), and determine their **hybridization**. (2 points)

Atom	Hybridization
Sulfur (<i>f</i>)	
Nitrogen (<i>g</i>)	
Oxygen (<i>h</i>)	
Oxygen (<i>i</i>)	

iii. **Describe** all the bonds formed by the carbon atoms ***a*** and ***b***, using sigma and pi designation as appropriate. (2 points)

Atom	Bonds formed (use sigma and pi as appropriate)
Carbon (<i>a</i>)	
Carbon (<i>b</i>)	

iv. How many total sp^3 -s sigma bonds are in the molecule? (1 point)

v. Determine **electronic** and **molecular geometry** at carbon (C) labeled ***e*** and sulfur (S) labeled ***f*** centres. (2 points)

Atom	Electronic Geometry	Molecular Geometry
Carbon (<i>e</i>)		
Sulfur (<i>f</i>)		

vi. Circle the value from the choices given below, that describes the **C-S-C bond angle at sulfur** (S) marked ***f***: (0.5 points)

$>90^\circ$ to $<109.5^\circ$ 109.5° >109.5 to $<120^\circ$ 120° $>120^\circ$ to $<180^\circ$


Question 8 12 points

Using the **valence shell orbitals only** draw a **complete** molecular orbital (MO) diagram (with no s-p mixing) for the molecule **PS**, in the space provided below. **(6 points)**

Assume that 3s and 3p atomic orbitals will form molecular orbitals like those formed by the combination of 2s and 2p atomic orbitals that we learnt in class, and there is no s-p mixing.

Show **energy arrow**, **atomic orbitals** involved, **atom labels**, the **molecular orbitals** that are formed, **filled electrons** in the MO diagram etc. Make sure to **label all the atomic and molecular orbitals**.

Using the MO diagram above, answer the following questions:

- (i) Which one of the following will have the highest bond order? Circle the correct choice
 A) PS^{1-} B) PS C) PS^{1+}
- (ii) Which one of the following will have the longest bond length? Circle the correct choice
 A) PS^{1-} B) PS C) PS^{1+}
- (iii) Which of the following will be diamagnetic? Circle the correct choice.
 A) PS^{1-} B) PS C) PS^{1+}
- (iv) Draw Lewis dot structure of the molecule with the shortest bond length (based on the MO diagram) from among the three (PS^{1-} , PS, PS^{1+}), and indicate the bond order calculated from the Lewis dot structure. **(3 points)**

Lewis Dot Structure

Bond Order



8F3F0C0D-6070-4000-BDB5-32D652235371

midterm-exam-93388

#644 Page 10 of 10

SCRAP PAGE

**CHEMISTRY 110 –Midterm Exam****Thursday November 7, 2024, 6:30 – 8:30 pm**

Examiners: Pallavi Sirjoosingh and Ashok Kakkar

First name (please write as legibly as possible within the boxes)

--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--

Last name

--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--

Student ID number

--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--

This booklet has 8 questions on 10 pages (The last page can be used as a scrap page).

The staple must NOT be removed from the question booklet. Your intact question booklet must be returned at the end of the exam.

A separate booklet contains a periodic table, and datasheet.

Write your answers directly under the questions in the space provided. Do not write on the QR codes or near the top edges (as it may be cut off).

Please write **legibly and dark enough** for the scan to be readable. For students who use English as a second language, dictionaries are permitted. Calculators are allowed.

DO NOT ASK QUESTIONS OF THE INVIGILATORS about interpretations of the exam questions. If you have a concern about a question, please summarize it at the top of this page. All concerns will be evaluated after the exam. GOOD LUCK!

**Question 1 8 points** (*Assume all transitions are possible*)

a. 6 points Calculate the **energy (in kJ/mol)** required to ionize (completely remove) the electron from the **$n=3$** level of **He^+ ion** (He: Helium). Show your work including equations. (3 sig. figs.)

b. 2 points

(I) Select the largest (longest) wavelength from the three choices below:

- i. Wavelength of photon to *completely remove* an electron from *1s orbital of Hydrogen atom*
- ii. Wavelength of photon to *completely remove* an electron from *1s orbital of Boron atom (B)*
- iii. Wavelength of photon to *completely remove* an electron from *1s orbital of Boron cation (B^+)*

i or ii or iii

(II) Select the smallest (shortest) wavelength from the three choices below (*for H atom*):

- i. Wavelength of photon *emitted* for an electron transitioning from *2p to 1s orbital*
- ii. Wavelength of photon *absorbed* for an electron transitioning from *2s to 4p orbital*
- iii. Wavelength of photon *absorbed* for an electron transitioning from *1s to 3p orbital*

i or ii or iii



Question 2 8 points



A Chem110 student plots the graph shown on the left, from data obtained for the kinetic energy (eV) of the ejected electron versus frequency (s^{-1}) of the incident light, for an unknown metal. ($1\text{eV} = 1.6022 \times 10^{-19} \text{ J}$)

Based on the plot, answer the questions **a to c** below:

a. 1 point On the plot above, label the **threshold frequency**.

b. 2 points What is the value of the work potential (in J) of the unknown metal?

c. 3 points What is the *longest* wavelength (in nm) required to eject an electron from this metal's surface? Show your work including equation(s) used. (3 sig figs)

d. 2 points In the space provided below, plot the following

(Assume frequency is *above* threshold frequency):

- Kinetic energy* of an ejected electron versus the *intensity* of the incident light (left)
- Number of electrons* ejected versus the *frequency* of the incident light (right)



**Question 3** **6 points****a. 2 points**

In the ground state of Ge (Germanium; $Z = 32$) atom, **name** the following orbitals:

i. Orbital containing the electron that requires the maximum energy to be removed.

ii. Orbital containing the electron that experiences the minimum effective nuclear charge.

b. 2 points

An unknown anion (A^-) has the following *excited* electronic configuration: $[Ar] 4s^2 3d^3 4p^1$

(Ar: Argon; $Z = 18$)

Give the **condensed ground state electronic configuration** of the atom, **A** and the cation, **A^{2+}** .
Show your work. (Assume all electronic transitions are possible)

Electronic Configuration of A:

Electronic Configuration of A^{2+} :

c. 2 points

In a single atom, determine how many electrons can have the following set of quantum numbers?

(i) $n = 5; l = 3$

In a single atom, determine how many electrons can have the following set of quantum numbers?

(i) $n = 3; l = 2; m_s = +1/2$

**Question 4** **8 points**

Rank the following from 1 to 4 in **increasing** order. Provide the ranking in the boxes provided.

Note: Atomic number (Z) are provided for relevant atoms

a) Increasing (lowest to highest) order of *ionization energy*: (1= lowest; 4 = highest)

Mg

Mg: Z = 12

Rb

Rb: Z = 37

Cs

Cs: Z = 55

K

K : Z = 19

b) Increasing (lowest to highest) order of *bond angle*: (1= lowest; 4 = highest)

H₂OCCl₄XeF₄

HCN

c) Increasing (smallest to largest) order of *ionic size* (1= smallest; 4 = largest)

S²⁻

S: Z = 16

O²⁻

O: Z = 8

Be²⁺

Be: Z = 4

Na⁺

Na: Z = 11

d) Increasing (lowest to highest) order of *bond energy* (1= lowest; 4 = highest)

P-F

P-Cl

P-I

P-Br

**Question 5** **12 points**

a. 9 points For the molecules HClO_4 and TeF_6 , draw the most stable *Lewis structure* showing appropriate shape and *dashed/wedged bonds*, based on VSEPR. Determine the **molecular geometry** around the central atom. For HClO_4 , label the *most polar bond*.



b. 3 points (Note: Lone pairs are not shown in structures below)

(i) For the three *resonance* structures (1, 2 and 3) shown below, answer the following question:



(I) Circle the *most contributing* resonance structure.

Structure 1

Structure 2

Structure 3

(ii) For the three structures given (4, 5, and 6) below, answer the following questions:



(I) Circle all the structures that represent *resonance structures* of each other:

Structure 4

Structure 5

Structure 6

(II) Among the resonance structures, circle the *most contributing* resonance structure.

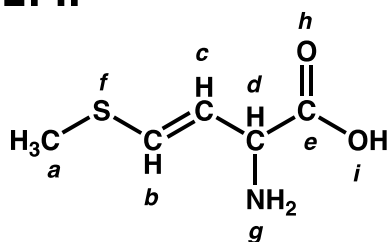
Structure 4

Structure 5

Structure 6

**Question 6** **6 points**Using the data given below, calculate the **lattice energy** for SrO(s).

	ΔH° (kJ/mol)
$\text{Sr}(s) \longrightarrow \text{Sr}(g)$	+180
$\text{Sr}(g) \longrightarrow \text{Sr}^{2+}(g) + 2e^-$	+1468
$\text{O}_2(g) \longrightarrow 2\text{O}(g)$	+500
$\text{O}(g) + 2e^- \longrightarrow \text{O}^{2-}(g)$	+600
$\text{Sr}(s) + \frac{1}{2}\text{O}_2(g) \longrightarrow \text{SrO}(s)$	-550

**Question 7 10 points**

Consider the molecule given on the left, and answer questions (i) to (vi) using chemical bonding theories that we have learnt in class (valence bond theory, hybridization).

Note that the lone pairs are not shown in the structure:

i. Consider the carbon atoms marked in as ***a***, ***b***, ***c***, ***d***, and ***e*** in the molecule, and determine **hybridization** at each of the carbon centers. (2.5 points)

Atom	Hybridization
Carbon (<i>a</i>)	
Carbon (<i>b</i>)	
Carbon (<i>c</i>)	
Carbon (<i>d</i>)	
Carbon (<i>e</i>)	

ii. Now, consider the sulfur atom (S) marked as ***f***, nitrogen (N) atom (***g***), and oxygen(O) atoms (***h*** and ***i***), and determine their **hybridization**. (2 points)

Atom	Hybridization
Sulfur (<i>f</i>)	
Nitrogen (<i>g</i>)	
Oxygen (<i>h</i>)	
Oxygen (<i>i</i>)	

iii. **Describe** all the bonds formed by the carbon atoms ***a*** and ***b***, using sigma and pi designation as appropriate. (2 points)

Atom	Bonds formed (use sigma and pi as appropriate)
Carbon (<i>a</i>)	
Carbon (<i>b</i>)	

iv. How many total sp^3 -s sigma bonds are in the molecule? (1 point)

v. Determine **electronic** and **molecular geometry** at carbon (C) labeled ***e*** and sulfur (S) labeled ***f*** centres. (2 points)

Atom	Electronic Geometry	Molecular Geometry
Carbon (<i>e</i>)		
Sulfur (<i>f</i>)		

vi. Circle the value from the choices given below, that describes the **C-S-C bond angle at sulfur** (S) marked ***f***: (0.5 points)

>90° to <109.5° 109.5° >109.5 to <120° 120° >120° to <180°


Question 8 12 points

Using the **valence shell orbitals only** draw a **complete** molecular orbital (MO) diagram (with no s-p mixing) for the molecule **PS**, in the space provided below. **(6 points)**

Assume that 3s and 3p atomic orbitals will form molecular orbitals like those formed by the combination of 2s and 2p atomic orbitals that we learnt in class, and there is no s-p mixing.

Show **energy arrow**, **atomic orbitals** involved, **atom labels**, the **molecular orbitals** that are formed, **filled electrons** in the MO diagram etc. Make sure to label all the atomic and molecular orbitals.

Using the MO diagram above, answer the following questions:

- (i) Which one of the following will have the highest bond order? Circle the correct choice
 A) PS^{1-} B) PS C) PS^{1+}
- (ii) Which one of the following will have the longest bond length? Circle the correct choice
 A) PS^{1-} B) PS C) PS^{1+}
- (iii) Which of the following will be diamagnetic? Circle the correct choice.
 A) PS^{1-} B) PS C) PS^{1+}
- (iv) Draw Lewis dot structure of the molecule with the shortest bond length (based on the MO diagram) from among the three (PS^{1-} , PS, PS^{1+}), and indicate the bond order calculated from the Lewis dot structure. **(3 points)**

Lewis Dot Structure

Bond Order



3A45ED7B-9A24-4DB0-8C5A-72BB2F334550

midterm-exam-93388

#645 Page 10 of 10

SCRAP PAGE

**CHEMISTRY 110 –Midterm Exam****Thursday November 7, 2024, 6:30 – 8:30 pm**

Examiners: Pallavi Sirjoosingh and Ashok Kakkar

First name (please write as legibly as possible within the boxes)

--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--

Last name

--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--

Student ID number

--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--

This booklet has 8 questions on 10 pages (The last page can be used as a scrap page).

The staple must NOT be removed from the question booklet. Your intact question booklet must be returned at the end of the exam.

A separate booklet contains a periodic table, and datasheet.

Write your answers directly under the questions in the space provided. Do not write on the QR codes or near the top edges (as it may be cut off).

Please write **legibly and dark enough** for the scan to be readable. For students who use English as a second language, dictionaries are permitted. Calculators are allowed.

DO NOT ASK QUESTIONS OF THE INVIGILATORS about interpretations of the exam questions. If you have a concern about a question, please summarize it at the top of this page. All concerns will be evaluated after the exam. GOOD LUCK!

**Question 1 8 points** (*Assume all transitions are possible*)

a. 6 points Calculate the **energy (in kJ/mol)** required to ionize (completely remove) the electron from the **$n=3$** level of **He^+ ion** (He: Helium). Show your work including equations. (3 sig. figs.)

b. 2 points

(I) Select the largest (longest) wavelength from the three choices below:

- i. Wavelength of photon to *completely remove* an electron from *1s orbital of Hydrogen atom*
- ii. Wavelength of photon to *completely remove* an electron from *1s orbital of Boron atom (B)*
- iii. Wavelength of photon to *completely remove* an electron from *1s orbital of Boron cation (B^+)*

i or ii or iii

(II) Select the smallest (shortest) wavelength from the three choices below (*for H atom*):

- i. Wavelength of photon *emitted* for an electron transitioning from *2p to 1s orbital*
- ii. Wavelength of photon *absorbed* for an electron transitioning from *2s to 4p orbital*
- iii. Wavelength of photon *absorbed* for an electron transitioning from *1s to 3p orbital*

i or ii or iii



Question 2 8 points



A Chem110 student plots the graph shown on the left, from data obtained for the kinetic energy (eV) of the ejected electron versus frequency (s^{-1}) of the incident light, for an unknown metal. ($1\text{eV} = 1.6022 \times 10^{-19} \text{ J}$)

Based on the plot, answer the questions **a to c** below:

a. 1 point On the plot above, label the **threshold frequency**.

b. 2 points What is the value of the work potential (in J) of the unknown metal?

c. 3 points What is the *longest* wavelength (in nm) required to eject an electron from this metal's surface? Show your work including equation(s) used. (3 sig figs)

d. 2 points In the space provided below, plot the following

(Assume frequency is *above* threshold frequency):

- i. *Kinetic energy* of an ejected electron versus the *intensity* of the incident light (left)
- ii. *Number of electrons* ejected versus the *frequency* of the incident light (right)



**Question 3** **6 points****a. 2 points**

In the ground state of Ge (Germanium; $Z = 32$) atom, **name** the following orbitals:

i. Orbital containing the electron that requires the maximum energy to be removed.

ii. Orbital containing the electron that experiences the minimum effective nuclear charge.

b. 2 points

An unknown anion (A^-) has the following *excited* electronic configuration: $[Ar] 4s^2 3d^3 4p^1$

(Ar: Argon; $Z = 18$)

Give the **condensed ground state electronic configuration** of the atom, **A** and the cation, **A^{2+}** .
Show your work. (Assume all electronic transitions are possible)

Electronic Configuration of A:

Electronic Configuration of A^{2+} :

c. 2 points

In a single atom, determine how many electrons can have the following set of quantum numbers?

(i) $n = 5; l = 3$

In a single atom, determine how many electrons can have the following set of quantum numbers?

(i) $n = 3; l = 2; m_s = +1/2$

**Question 4** **8 points**

Rank the following from 1 to 4 in **increasing** order. Provide the ranking in the boxes provided.

Note: Atomic number (Z) are provided for relevant atoms

a) Increasing (lowest to highest) order of *ionization energy*: (1= lowest; 4 = highest)

Mg

Mg: Z = 12

Rb

Rb: Z = 37

Cs

Cs: Z = 55

K

K : Z = 19

b) Increasing (lowest to highest) order of *bond angle*: (1= lowest; 4 = highest)

H₂OCCl₄XeF₄

HCN

c) Increasing (smallest to largest) order of *ionic size* (1= smallest; 4 = largest)

S²⁻

S: Z = 16

O²⁻

O: Z = 8

Be²⁺

Be: Z = 4

Na⁺

Na: Z = 11

d) Increasing (lowest to highest) order of *bond energy* (1= lowest; 4 = highest)

P-F

P-Cl

P-I

P-Br

**Question 5** **12 points**

a. 9 points For the molecules HClO_4 and TeF_6 , draw the most stable *Lewis structure* showing appropriate shape and *dashed/wedged bonds*, based on VSEPR. Determine the **molecular geometry** around the central atom. For HClO_4 , label the *most polar bond*.



b. 3 points (Note: Lone pairs are not shown in structures below)

(i) For the three *resonance* structures (1, 2 and 3) shown below, answer the following question:



(I) Circle the *most contributing* resonance structure.

Structure 1

Structure 2

Structure 3

(ii) For the three structures given (4, 5, and 6) below, answer the following questions:



(I) Circle all the structures that represent *resonance structures* of each other:

Structure 4

Structure 5

Structure 6

(II) Among the resonance structures, circle the *most contributing* resonance structure.

Structure 4

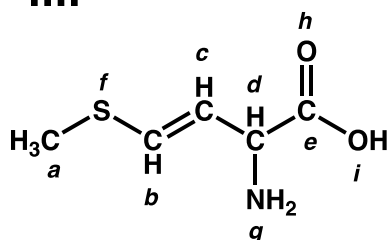
Structure 5

Structure 6

**Question 6** **6 points**

Using the data given below, calculate the **lattice energy** for SrO(s).

	ΔH° (kJ/mol)
$\text{Sr}(s) \longrightarrow \text{Sr}(g)$	+180
$\text{Sr}(g) \longrightarrow \text{Sr}^{2+}(g) + 2e^-$	+1468
$\text{O}_2(g) \longrightarrow 2\text{O}(g)$	+500
$\text{O}(g) + 2e^- \longrightarrow \text{O}^{2-}(g)$	+600
$\text{Sr}(s) + \frac{1}{2}\text{O}_2(g) \longrightarrow \text{SrO}(s)$	-550

**Question 7 10 points**

Consider the molecule given on the left, and answer questions (i) to (vi) using chemical bonding theories that we have learnt in class (valence bond theory, hybridization).

Note that the lone pairs are not shown in the structure:

i. Consider the carbon atoms marked in as ***a***, ***b***, ***c***, ***d***, and ***e*** in the molecule, and determine **hybridization** at each of the carbon centers. (2.5 points)

Atom	Hybridization
Carbon (<i>a</i>)	
Carbon (<i>b</i>)	
Carbon (<i>c</i>)	
Carbon (<i>d</i>)	
Carbon (<i>e</i>)	

ii. Now, consider the sulfur atom (S) marked as ***f***, nitrogen (N) atom (***g***), and oxygen(O) atoms (***h*** and ***i***), and determine their **hybridization**. (2 points)

Atom	Hybridization
Sulfur (<i>f</i>)	
Nitrogen (<i>g</i>)	
Oxygen (<i>h</i>)	
Oxygen (<i>i</i>)	

iii. **Describe** all the bonds formed by the carbon atoms ***a*** and ***b***, using sigma and pi designation as appropriate. (2 points)

Atom	Bonds formed (use sigma and pi as appropriate)
Carbon (<i>a</i>)	
Carbon (<i>b</i>)	

iv. How many total sp^3 -s sigma bonds are in the molecule? (1 point)

v. Determine **electronic** and **molecular geometry** at carbon (C) labeled ***e*** and sulfur (S) labeled ***f*** centres. (2 points)

Atom	Electronic Geometry	Molecular Geometry
Carbon (<i>e</i>)		
Sulfur (<i>f</i>)		

vi. Circle the value from the choices given below, that describes the **C-S-C bond angle at sulfur** (S) marked ***f***: (0.5 points)

>90° to <109.5° 109.5° >109.5 to <120° 120° >120° to <180°

**Question 8 12 points**

Using the **valence shell orbitals only** draw a **complete** molecular orbital (MO) diagram (with no s-p mixing) for the molecule **PS**, in the space provided below. **(6 points)**

Assume that 3s and 3p atomic orbitals will form molecular orbitals like those formed by the combination of 2s and 2p atomic orbitals that we learnt in class, and there is no s-p mixing.

Show **energy arrow**, **atomic orbitals** involved, **atom labels**, the **molecular orbitals** that are formed, **filled electrons** in the MO diagram etc. Make sure to label all the atomic and molecular orbitals.

Using the MO diagram above, answer the following questions:

- (i) Which one of the following will have the highest bond order? Circle the correct choice
A) PS^{1-} B) PS C) PS^{1+}
- (ii) Which one of the following will have the longest bond length? Circle the correct choice
A) PS^{1-} B) PS C) PS^{1+}
- (iii) Which of the following will be diamagnetic? Circle the correct choice.
A) PS^{1-} B) PS C) PS^{1+}
- (iv) Draw Lewis dot structure of the molecule with the shortest bond length (based on the MO diagram) from among the three (PS^{1-} , PS, PS^{1+}), and indicate the bond order calculated from the Lewis dot structure. **(3 points)**

Lewis Dot Structure

Bond Order



1AE98A14-E340-4026-BDDC-CAF6390BD734

midterm-exam-93388

#646 Page 10 of 10

SCRAP PAGE

**CHEMISTRY 110 –Midterm Exam****Thursday November 7, 2024, 6:30 – 8:30 pm****Examiners: Pallavi Sirjoosingh and Ashok Kakkar**

First name (please write as legibly as possible within the boxes)

--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--

Last name

--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--

Student ID number

--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--

This booklet has 8 questions on 10 pages (The last page can be used as a scrap page).

The staple must NOT be removed from the question booklet. Your intact question booklet must be returned at the end of the exam.

A separate booklet contains a periodic table, and datasheet.

Write your answers directly under the questions in the space provided. Do not write on the QR codes or near the top edges (as it may be cut off).

Please write **legibly and dark enough** for the scan to be readable. For students who use English as a second language, dictionaries are permitted. Calculators are allowed.

DO NOT ASK QUESTIONS OF THE INVIGILATORS about interpretations of the exam questions. If you have a concern about a question, please summarize it at the top of this page. All concerns will be evaluated after the exam. GOOD LUCK!

**Question 1 8 points** (*Assume all transitions are possible*)

a. 6 points Calculate the **energy (in kJ/mol)** required to ionize (completely remove) the electron from the **n=3** level of **He⁺ ion** (He: Helium). Show your work including equations. (3 sig. figs.)

b. 2 points

(I) Select the largest (longest) wavelength from the three choices below:

- i. Wavelength of photon to *completely remove* an electron from *1s orbital of Hydrogen atom*
- ii. Wavelength of photon to *completely remove* an electron from *1s orbital of Boron atom (B)*
- iii. Wavelength of photon to *completely remove* an electron from *1s orbital of Boron cation (B⁺)*

i or ii or iii

(II) Select the smallest (shortest) wavelength from the three choices below (*for H atom*):

- i. Wavelength of photon *emitted* for an electron transitioning from *2p to 1s orbital*
- ii. Wavelength of photon *absorbed* for an electron transitioning from *2s to 4p orbital*
- iii. Wavelength of photon *absorbed* for an electron transitioning from *1s to 3p orbital*

i or ii or iii



Question 2 8 points



A Chem110 student plots the graph shown on the left, from data obtained for the kinetic energy (eV) of the ejected electron versus frequency (s^{-1}) of the incident light, for an unknown metal. ($1\text{eV} = 1.6022 \times 10^{-19} \text{ J}$)

Based on the plot, answer the questions **a to c** below:

a. 1 point On the plot above, label the **threshold frequency**.

b. 2 points What is the value of the work potential (in J) of the unknown metal?

c. 3 points What is the *longest* wavelength (in nm) required to eject an electron from this metal's surface? Show your work including equation(s) used. (3 sig figs)

d. 2 points In the space provided below, plot the following

(Assume frequency is *above* threshold frequency):

- Kinetic energy* of an ejected electron versus the *intensity* of the incident light (left)
- Number of electrons* ejected versus the *frequency* of the incident light (right)



**Question 3** **6 points****a. 2 points**

In the ground state of Ge (Germanium; $Z = 32$) atom, **name** the following orbitals:

i. Orbital containing the electron that requires the maximum energy to be removed.

ii. Orbital containing the electron that experiences the minimum effective nuclear charge.

b. 2 points

An unknown anion (A^-) has the following *excited* electronic configuration: $[Ar] 4s^2 3d^3 4p^1$

(Ar: Argon; $Z = 18$)

Give the **condensed ground state electronic configuration** of the atom, **A** and the cation, **A^{2+}** .
Show your work. (Assume all electronic transitions are possible)

Electronic Configuration of A:

Electronic Configuration of A^{2+} :

c. 2 points

In a single atom, determine how many electrons can have the following set of quantum numbers?

(i) $n = 5; l = 3$

In a single atom, determine how many electrons can have the following set of quantum numbers?

(i) $n = 3; l = 2; m_s = +1/2$

**Question 4** **8 points**

Rank the following from 1 to 4 in **increasing** order. Provide the ranking in the boxes provided.

Note: Atomic number (Z) are provided for relevant atoms

a) Increasing (lowest to highest) order of *ionization energy*: (1= lowest; 4 = highest)

Mg

Mg: Z = 12

Rb

Rb: Z = 37

Cs

Cs: Z = 55

K

K : Z = 19

b) Increasing (lowest to highest) order of *bond angle*: (1= lowest; 4 = highest)

H₂OCCl₄XeF₄

HCN

c) Increasing (smallest to largest) order of *ionic size* (1= smallest; 4 = largest)

S²⁻

S: Z = 16

O²⁻

O: Z = 8

Be²⁺

Be: Z = 4

Na⁺

Na: Z = 11

d) Increasing (lowest to highest) order of *bond energy* (1= lowest; 4 = highest)

P-F

P-Cl

P-I

P-Br

**Question 5** **12 points**

a. 9 points For the molecules HClO_4 and TeF_6 , draw the most stable *Lewis structure* showing appropriate shape and *dashed/wedged bonds*, based on VSEPR. Determine the **molecular geometry** around the central atom. For HClO_4 , label the *most polar bond*.



b. 3 points (Note: Lone pairs are not shown in structures below)

(i) For the three *resonance* structures (1, 2 and 3) shown below, answer the following question:



(I) Circle the *most contributing* resonance structure.

Structure 1

Structure 2

Structure 3

(ii) For the three structures given (4, 5, and 6) below, answer the following questions:



(I) Circle all the structures that represent *resonance structures* of each other:

Structure 4

Structure 5

Structure 6

(II) Among the resonance structures, circle the *most contributing* resonance structure.

Structure 4

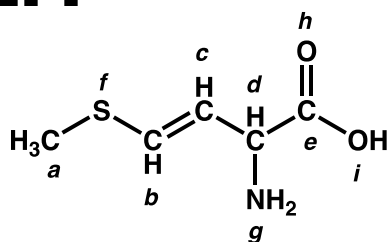
Structure 5

Structure 6

**Question 6** **6 points**

Using the data given below, calculate the **lattice energy** for SrO(s).

	ΔH° (kJ/mol)
$\text{Sr}(s) \longrightarrow \text{Sr}(g)$	+180
$\text{Sr}(g) \longrightarrow \text{Sr}^{2+}(g) + 2e^-$	+1468
$\text{O}_2(g) \longrightarrow 2\text{O}(g)$	+500
$\text{O}(g) + 2e^- \longrightarrow \text{O}^{2-}(g)$	+600
$\text{Sr}(s) + \frac{1}{2}\text{O}_2(g) \longrightarrow \text{SrO}(s)$	-550

**Question 7 10 points**

Consider the molecule given on the left, and answer questions (i) to (vi) using chemical bonding theories that we have learnt in class (valence bond theory, hybridization).

Note that the lone pairs are not shown in the structure:

i. Consider the carbon atoms marked in as ***a***, ***b***, ***c***, ***d***, and ***e*** in the molecule, and determine **hybridization** at each of the carbon centers. (2.5 points)

Atom	Hybridization
Carbon (<i>a</i>)	
Carbon (<i>b</i>)	
Carbon (<i>c</i>)	
Carbon (<i>d</i>)	
Carbon (<i>e</i>)	

ii. Now, consider the sulfur atom (S) marked as ***f***, nitrogen (N) atom (***g***), and oxygen(O) atoms (***h*** and ***i***), and determine their **hybridization**. (2 points)

Atom	Hybridization
Sulfur (<i>f</i>)	
Nitrogen (<i>g</i>)	
Oxygen (<i>h</i>)	
Oxygen (<i>i</i>)	

iii. **Describe** all the bonds formed by the carbon atoms ***a*** and ***b***, using sigma and pi designation as appropriate. (2 points)

Atom	Bonds formed (use sigma and pi as appropriate)
Carbon (<i>a</i>)	
Carbon (<i>b</i>)	

iv. How many total sp^3 -s sigma bonds are in the molecule? (1 point)

v. Determine **electronic** and **molecular geometry** at carbon (C) labeled ***e*** and sulfur (S) labeled ***f*** centres. (2 points)

Atom	Electronic Geometry	Molecular Geometry
Carbon (<i>e</i>)		
Sulfur (<i>f</i>)		

vi. Circle the value from the choices given below, that describes the **C-S-C bond angle at sulfur** (S) marked ***f***: (0.5 points)

$>90^\circ$ to $<109.5^\circ$ 109.5° >109.5 to $<120^\circ$ 120° $>120^\circ$ to $<180^\circ$


Question 8 12 points

Using the **valence shell orbitals only** draw a **complete** molecular orbital (MO) diagram (with no s-p mixing) for the molecule **PS**, in the space provided below. **(6 points)**

Assume that 3s and 3p atomic orbitals will form molecular orbitals like those formed by the combination of 2s and 2p atomic orbitals that we learnt in class, and there is no s-p mixing.

Show **energy arrow**, **atomic orbitals** involved, **atom labels**, the **molecular orbitals** that are formed, **filled electrons** in the MO diagram etc. Make sure to label all the atomic and molecular orbitals.

Using the MO diagram above, answer the following questions:

- (i) Which one of the following will have the highest bond order? Circle the correct choice
 A) PS^{1-} B) PS C) PS^{1+}
- (ii) Which one of the following will have the longest bond length? Circle the correct choice
 A) PS^{1-} B) PS C) PS^{1+}
- (iii) Which of the following will be diamagnetic? Circle the correct choice.
 A) PS^{1-} B) PS C) PS^{1+}
- (iv) Draw Lewis dot structure of the molecule with the shortest bond length (based on the MO diagram) from among the three (PS^{1-} , PS, PS^{1+}), and indicate the bond order calculated from the Lewis dot structure. **(3 points)**

Lewis Dot Structure

Bond Order



1314F0F2-3800-4AF2-871E-1A7F883AB69C

midterm-exam-93388

#647 Page 10 of 10

SCRAP PAGE

**CHEMISTRY 110 –Midterm Exam****Thursday November 7, 2024, 6:30 – 8:30 pm****Examiners: Pallavi Sirjoosingh and Ashok Kakkar**

First name (please write as legibly as possible within the boxes)

--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--

Last name

--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--

Student ID number

--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--

This booklet has 8 questions on 10 pages (The last page can be used as a scrap page).

The staple must NOT be removed from the question booklet. Your intact question booklet must be returned at the end of the exam.

A separate booklet contains a periodic table, and datasheet.

Write your answers directly under the questions in the space provided. Do not write on the QR codes or near the top edges (as it may be cut off).

Please write **legibly and dark enough** for the scan to be readable. For students who use English as a second language, dictionaries are permitted. Calculators are allowed.

DO NOT ASK QUESTIONS OF THE INVIGILATORS about interpretations of the exam questions. If you have a concern about a question, please summarize it at the top of this page. All concerns will be evaluated after the exam. GOOD LUCK!

**Question 1 8 points** (*Assume all transitions are possible*)

a. 6 points Calculate the **energy (in kJ/mol)** required to ionize (completely remove) the electron from the **n=3** level of **He⁺ ion** (He: Helium). Show your work including equations. (3 sig. figs.)

b. 2 points

(I) Select the largest (longest) wavelength from the three choices below:

- i. Wavelength of photon to *completely remove* an electron from *1s orbital of Hydrogen atom*
- ii. Wavelength of photon to *completely remove* an electron from *1s orbital of Boron atom (B)*
- iii. Wavelength of photon to *completely remove* an electron from *1s orbital of Boron cation (B⁺)*

i or ii or iii

(II) Select the smallest (shortest) wavelength from the three choices below (*for H atom*):

- i. Wavelength of photon *emitted* for an electron transitioning from *2p to 1s orbital*
- ii. Wavelength of photon *absorbed* for an electron transitioning from *2s to 4p orbital*
- iii. Wavelength of photon *absorbed* for an electron transitioning from *1s to 3p orbital*

i or ii or iii



Question 2 8 points



A Chem110 student plots the graph shown on the left, from data obtained for the kinetic energy (eV) of the ejected electron versus frequency (s^{-1}) of the incident light, for an unknown metal. ($1\text{eV} = 1.6022 \times 10^{-19} \text{ J}$)

Based on the plot, answer the questions **a to c** below:

a. 1 point On the plot above, label the **threshold frequency**.

b. 2 points What is the value of the work potential (in J) of the unknown metal?

c. 3 points What is the *longest* wavelength (in nm) required to eject an electron from this metal's surface? Show your work including equation(s) used. (3 sig figs)

d. 2 points In the space provided below, plot the following

(Assume frequency is *above* threshold frequency):

- i. *Kinetic energy* of an ejected electron versus the *intensity* of the incident light (left)
- ii. *Number of electrons* ejected versus the *frequency* of the incident light (right)



**Question 3** **6 points****a. 2 points**

In the ground state of Ge (Germanium; $Z = 32$) atom, **name** the following orbitals:

i. Orbital containing the electron that requires the maximum energy to be removed.

ii. Orbital containing the electron that experiences the minimum effective nuclear charge.

b. 2 points

An unknown anion (A^-) has the following *excited* electronic configuration: $[Ar] 4s^2 3d^3 4p^1$

(Ar: Argon; $Z = 18$)

Give the **condensed ground state electronic configuration** of the atom, **A** and the cation, **A^{2+}** .
Show your work. (Assume all electronic transitions are possible)

Electronic Configuration of A:

Electronic Configuration of A^{2+} :

c. 2 points

In a single atom, determine how many electrons can have the following set of quantum numbers?

(i) $n = 5; l = 3$

In a single atom, determine how many electrons can have the following set of quantum numbers?

(i) $n = 3; l = 2; m_s = +1/2$

**Question 4** **8 points**

Rank the following from 1 to 4 in **increasing** order. Provide the ranking in the boxes provided.

Note: Atomic number (Z) are provided for relevant atoms

a) Increasing (lowest to highest) order of *ionization energy*: (1= lowest; 4 = highest)

Mg

Mg: Z = 12

Rb

Rb: Z = 37

Cs

Cs: Z = 55

K

K : Z = 19

b) Increasing (lowest to highest) order of *bond angle*: (1= lowest; 4 = highest)

H₂OCCl₄XeF₄

HCN

c) Increasing (smallest to largest) order of *ionic size* (1= smallest; 4 = largest)

S²⁻

S: Z = 16

O²⁻

O: Z = 8

Be²⁺

Be: Z = 4

Na⁺

Na: Z = 11

d) Increasing (lowest to highest) order of *bond energy* (1= lowest; 4 = highest)

P-F

P-Cl

P-I

P-Br

**Question 5** **12 points**

a. 9 points For the molecules HClO_4 and TeF_6 , draw the most stable *Lewis structure* showing appropriate shape and *dashed/wedged bonds*, based on VSEPR. Determine the **molecular geometry** around the central atom. For HClO_4 , label the *most polar bond*.



b. 3 points (Note: Lone pairs are not shown in structures below)

(i) For the three *resonance* structures (1, 2 and 3) shown below, answer the following question:



(I) Circle the *most contributing* resonance structure.

Structure 1

Structure 2

Structure 3

(ii) For the three structures given (4, 5, and 6) below, answer the following questions:



(I) Circle all the structures that represent *resonance structures* of each other:

Structure 4

Structure 5

Structure 6

(II) Among the resonance structures, circle the *most contributing* resonance structure.

Structure 4

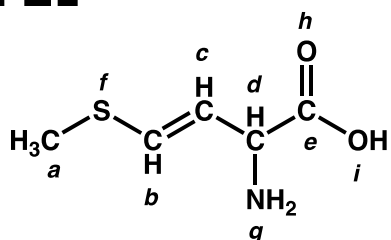
Structure 5

Structure 6

**Question 6** **6 points**

Using the data given below, calculate the **lattice energy** for SrO(s).

	ΔH° (kJ/mol)
$\text{Sr}(s) \longrightarrow \text{Sr}(g)$	+180
$\text{Sr}(g) \longrightarrow \text{Sr}^{2+}(g) + 2e^-$	+1468
$\text{O}_2(g) \longrightarrow 2\text{O}(g)$	+500
$\text{O}(g) + 2e^- \longrightarrow \text{O}^{2-}(g)$	+600
$\text{Sr}(s) + \frac{1}{2}\text{O}_2(g) \longrightarrow \text{SrO}(s)$	-550

**Question 7 10 points**

Consider the molecule given on the left, and answer questions (i) to (vi) using chemical bonding theories that we have learnt in class (valence bond theory, hybridization).

Note that the lone pairs are not shown in the structure:

i. Consider the carbon atoms marked in as ***a***, ***b***, ***c***, ***d***, and ***e*** in the molecule, and determine **hybridization** at each of the carbon centers. (2.5 points)

Atom	Hybridization
Carbon (<i>a</i>)	
Carbon (<i>b</i>)	
Carbon (<i>c</i>)	
Carbon (<i>d</i>)	
Carbon (<i>e</i>)	

ii. Now, consider the sulfur atom (S) marked as ***f***, nitrogen (N) atom (***g***), and oxygen(O) atoms (***h*** and ***i***), and determine their **hybridization**. (2 points)

Atom	Hybridization
Sulfur (<i>f</i>)	
Nitrogen (<i>g</i>)	
Oxygen (<i>h</i>)	
Oxygen (<i>i</i>)	

iii. **Describe** all the bonds formed by the carbon atoms ***a*** and ***b***, using sigma and pi designation as appropriate. (2 points)

Atom	Bonds formed (use sigma and pi as appropriate)
Carbon (<i>a</i>)	
Carbon (<i>b</i>)	

iv. How many total sp^3 -s sigma bonds are in the molecule? (1 point)

v. Determine **electronic** and **molecular geometry** at carbon (C) labeled ***e*** and sulfur (S) labeled ***f*** centres. (2 points)

Atom	Electronic Geometry	Molecular Geometry
Carbon (<i>e</i>)		
Sulfur (<i>f</i>)		

vi. Circle the value from the choices given below, that describes the **C-S-C bond angle at sulfur** (S) marked ***f***: (0.5 points)

>90° to <109.5° 109.5° >109.5 to <120° 120° >120° to <180°


Question 8 12 points

Using the **valence shell orbitals only** draw a **complete** molecular orbital (MO) diagram (with no s-p mixing) for the molecule **PS**, in the space provided below. **(6 points)**

Assume that 3s and 3p atomic orbitals will form molecular orbitals like those formed by the combination of 2s and 2p atomic orbitals that we learnt in class, and there is no s-p mixing.

Show **energy arrow**, **atomic orbitals** involved, **atom labels**, the **molecular orbitals** that are formed, **filled electrons** in the MO diagram etc. Make sure to label all the atomic and molecular orbitals.

Using the MO diagram above, answer the following questions:

- (i) Which one of the following will have the highest bond order? Circle the correct choice
 A) PS^{1-} B) PS C) PS^{1+}
- (ii) Which one of the following will have the longest bond length? Circle the correct choice
 A) PS^{1-} B) PS C) PS^{1+}
- (iii) Which of the following will be diamagnetic? Circle the correct choice.
 A) PS^{1-} B) PS C) PS^{1+}
- (iv) Draw Lewis dot structure of the molecule with the shortest bond length (based on the MO diagram) from among the three (PS^{1-} , PS, PS^{1+}), and indicate the bond order calculated from the Lewis dot structure. **(3 points)**

Lewis Dot Structure

Bond Order



EB6CF9A7-ABE8-4DF0-A04E-5AE2C003A7A2

midterm-exam-93388

#648 Page 10 of 10

SCRAP PAGE

**CHEMISTRY 110 –Midterm Exam****Thursday November 7, 2024, 6:30 – 8:30 pm**

Examiners: Pallavi Sirjoosingh and Ashok Kakkar

First name (please write as legibly as possible within the boxes)

--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--

Last name

--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--

Student ID number

--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--

This booklet has 8 questions on 10 pages (The last page can be used as a scrap page).

The staple must NOT be removed from the question booklet. Your intact question booklet must be returned at the end of the exam.

A separate booklet contains a periodic table, and datasheet.

Write your answers directly under the questions in the space provided. Do not write on the QR codes or near the top edges (as it may be cut off).

Please write **legibly and dark enough** for the scan to be readable. For students who use English as a second language, dictionaries are permitted. Calculators are allowed.

DO NOT ASK QUESTIONS OF THE INVIGILATORS about interpretations of the exam questions. If you have a concern about a question, please summarize it at the top of this page. All concerns will be evaluated after the exam. GOOD LUCK!

**Question 1 8 points** (*Assume all transitions are possible*)

a. 6 points Calculate the **energy (in kJ/mol)** required to ionize (completely remove) the electron from the **$n=3$** level of **He^+ ion** (He: Helium). Show your work including equations. (3 sig. figs.)

b. 2 points

(I) Select the largest (longest) wavelength from the three choices below:

- i. Wavelength of photon to *completely remove* an electron from *1s orbital of Hydrogen atom*
- ii. Wavelength of photon to *completely remove* an electron from *1s orbital of Boron atom (B)*
- iii. Wavelength of photon to *completely remove* an electron from *1s orbital of Boron cation (B^+)*

i or ii or iii

(II) Select the smallest (shortest) wavelength from the three choices below (*for H atom*):

- i. Wavelength of photon *emitted* for an electron transitioning from *2p to 1s orbital*
- ii. Wavelength of photon *absorbed* for an electron transitioning from *2s to 4p orbital*
- iii. Wavelength of photon *absorbed* for an electron transitioning from *1s to 3p orbital*

i or ii or iii



Question 2 8 points



A Chem110 student plots the graph shown on the left, from data obtained for the kinetic energy (eV) of the ejected electron versus frequency (s^{-1}) of the incident light, for an unknown metal. ($1\text{eV} = 1.6022 \times 10^{-19} \text{ J}$)

Based on the plot, answer the questions **a to c** below:

a. 1 point On the plot above, label the **threshold frequency**.

b. 2 points What is the value of the work potential (in J) of the unknown metal?

c. 3 points What is the *longest* wavelength (in nm) required to eject an electron from this metal's surface? Show your work including equation(s) used. (3 sig figs)

d. 2 points In the space provided below, plot the following

(Assume frequency is *above* threshold frequency):

- Kinetic energy* of an ejected electron versus the *intensity* of the incident light (left)
- Number of electrons* ejected versus the *frequency* of the incident light (right)



**Question 3** **6 points****a. 2 points**

In the ground state of Ge (Germanium; $Z = 32$) atom, **name** the following orbitals:

i. Orbital containing the electron that requires the maximum energy to be removed.

ii. Orbital containing the electron that experiences the minimum effective nuclear charge.

b. 2 points

An unknown anion (A^-) has the following *excited* electronic configuration: $[Ar] 4s^2 3d^3 4p^1$

(Ar: Argon; $Z = 18$)

Give the **condensed ground state electronic configuration** of the atom, **A** and the cation, **A^{2+}** .
Show your work. (Assume all electronic transitions are possible)

Electronic Configuration of A:

Electronic Configuration of A^{2+} :

c. 2 points

In a single atom, determine how many electrons can have the following set of quantum numbers?

(i) $n = 5; l = 3$

In a single atom, determine how many electrons can have the following set of quantum numbers?

(i) $n = 3; l = 2; m_s = +1/2$

**Question 4** **8 points**

Rank the following from 1 to 4 in **increasing** order. Provide the ranking in the boxes provided.

Note: Atomic number (Z) are provided for relevant atoms

a) Increasing (lowest to highest) order of *ionization energy*: (1= lowest; 4 = highest)

Mg

Mg: Z = 12

Rb

Rb: Z = 37

Cs

Cs: Z = 55

K

K : Z = 19

b) Increasing (lowest to highest) order of *bond angle*: (1= lowest; 4 = highest)

H₂OCCl₄XeF₄

HCN

c) Increasing (smallest to largest) order of *ionic size* (1= smallest; 4 = largest)

S²⁻

S: Z = 16

O²⁻

O: Z = 8

Be²⁺

Be: Z = 4

Na⁺

Na: Z = 11

d) Increasing (lowest to highest) order of *bond energy* (1= lowest; 4 = highest)

P-F

P-Cl

P-I

P-Br

**Question 5** **12 points**

a. 9 points For the molecules HClO_4 and TeF_6 , draw the most stable *Lewis structure* showing appropriate shape and *dashed/wedged bonds*, based on VSEPR. Determine the **molecular geometry** around the central atom. For HClO_4 , label the *most polar bond*.



b. 3 points (Note: Lone pairs are not shown in structures below)

(i) For the three *resonance* structures (1, 2 and 3) shown below, answer the following question:



(I) Circle the *most contributing* resonance structure.

Structure 1

Structure 2

Structure 3

(ii) For the three structures given (4, 5, and 6) below, answer the following questions:



(I) Circle all the structures that represent *resonance structures* of each other:

Structure 4

Structure 5

Structure 6

(II) Among the resonance structures, circle the *most contributing* resonance structure.

Structure 4

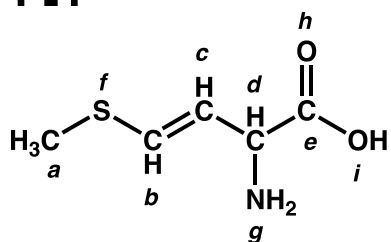
Structure 5

Structure 6

**Question 6** **6 points**

Using the data given below, calculate the **lattice energy** for SrO(s).

	ΔH° (kJ/mol)
$\text{Sr}(s) \longrightarrow \text{Sr}(g)$	+180
$\text{Sr}(g) \longrightarrow \text{Sr}^{2+}(g) + 2e^-$	+1468
$\text{O}_2(g) \longrightarrow 2\text{O}(g)$	+500
$\text{O}(g) + 2e^- \longrightarrow \text{O}^{2-}(g)$	+600
$\text{Sr}(s) + \frac{1}{2}\text{O}_2(g) \longrightarrow \text{SrO}(s)$	-550

**Question 7 10 points**

Consider the molecule given on the left, and answer questions (i) to (vi) using chemical bonding theories that we have learnt in class (valence bond theory, hybridization).

Note that the lone pairs are not shown in the structure:

i. Consider the carbon atoms marked in as ***a***, ***b***, ***c***, ***d***, and ***e*** in the molecule, and determine **hybridization** at each of the carbon centers. (2.5 points)

Atom	Hybridization
Carbon (<i>a</i>)	
Carbon (<i>b</i>)	
Carbon (<i>c</i>)	
Carbon (<i>d</i>)	
Carbon (<i>e</i>)	

ii. Now, consider the sulfur atom (S) marked as ***f***, nitrogen (N) atom (***g***), and oxygen(O) atoms (***h*** and ***i***), and determine their **hybridization**. (2 points)

Atom	Hybridization
Sulfur (<i>f</i>)	
Nitrogen (<i>g</i>)	
Oxygen (<i>h</i>)	
Oxygen (<i>i</i>)	

iii. **Describe** all the bonds formed by the carbon atoms ***a*** and ***b***, using sigma and pi designation as appropriate. (2 points)

Atom	Bonds formed (use sigma and pi as appropriate)
Carbon (<i>a</i>)	
Carbon (<i>b</i>)	

iv. How many total sp^3 -s sigma bonds are in the molecule? (1 point)

v. Determine **electronic** and **molecular geometry** at carbon (C) labeled ***e*** and sulfur (S) labeled ***f*** centres. (2 points)

Atom	Electronic Geometry	Molecular Geometry
Carbon (<i>e</i>)		
Sulfur (<i>f</i>)		

vi. Circle the value from the choices given below, that describes the **C-S-C bond angle at sulfur** (S) marked ***f***: (0.5 points)

>90° to <109.5° 109.5° >109.5 to <120° 120° >120° to <180°


Question 8 12 points

Using the **valence shell orbitals only** draw a **complete** molecular orbital (MO) diagram (with no s-p mixing) for the molecule **PS**, in the space provided below. **(6 points)**

Assume that 3s and 3p atomic orbitals will form molecular orbitals like those formed by the combination of 2s and 2p atomic orbitals that we learnt in class, and there is no s-p mixing.

Show **energy arrow**, **atomic orbitals** involved, **atom labels**, the **molecular orbitals** that are formed, **filled electrons** in the MO diagram etc. Make sure to label all the atomic and molecular orbitals.

Using the MO diagram above, answer the following questions:

- (i) Which one of the following will have the highest bond order? Circle the correct choice
 A) PS^{1-} B) PS C) PS^{1+}
- (ii) Which one of the following will have the longest bond length? Circle the correct choice
 A) PS^{1-} B) PS C) PS^{1+}
- (iii) Which of the following will be diamagnetic? Circle the correct choice.
 A) PS^{1-} B) PS C) PS^{1+}
- (iv) Draw Lewis dot structure of the molecule with the shortest bond length (based on the MO diagram) from among the three (PS^{1-} , PS, PS^{1+}), and indicate the bond order calculated from the Lewis dot structure. **(3 points)**

Lewis Dot Structure

Bond Order



F6D0508C-EADD-4829-B3A6-E7309E9178BD

midterm-exam-93388

#649 Page 10 of 10

SCRAP PAGE

**CHEMISTRY 110 –Midterm Exam****Thursday November 7, 2024, 6:30 – 8:30 pm****Examiners: Pallavi Sirjoosingh and Ashok Kakkar**

First name (please write as legibly as possible within the boxes)

--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--

Last name

--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--

Student ID number

--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--

This booklet has 8 questions on 10 pages (The last page can be used as a scrap page).

The staple must NOT be removed from the question booklet. Your intact question booklet must be returned at the end of the exam.

A separate booklet contains a periodic table, and datasheet.

Write your answers directly under the questions in the space provided. Do not write on the QR codes or near the top edges (as it may be cut off).

Please write **legibly and dark enough** for the scan to be readable. For students who use English as a second language, dictionaries are permitted. Calculators are allowed.

DO NOT ASK QUESTIONS OF THE INVIGILATORS about interpretations of the exam questions. If you have a concern about a question, please summarize it at the top of this page. All concerns will be evaluated after the exam. GOOD LUCK!

**Question 1 8 points** (*Assume all transitions are possible*)

a. 6 points Calculate the **energy (in kJ/mol)** required to ionize (completely remove) the electron from the **n=3** level of **He⁺ ion** (He: Helium). Show your work including equations. (3 sig. figs.)

b. 2 points

(I) Select the largest (longest) wavelength from the three choices below:

- i. Wavelength of photon to *completely remove* an electron from *1s orbital of Hydrogen atom*
- ii. Wavelength of photon to *completely remove* an electron from *1s orbital of Boron atom (B)*
- iii. Wavelength of photon to *completely remove* an electron from *1s orbital of Boron cation (B⁺)*

i or ii or iii

(II) Select the smallest (shortest) wavelength from the three choices below (*for H atom*):

- i. Wavelength of photon *emitted* for an electron transitioning from *2p to 1s orbital*
- ii. Wavelength of photon *absorbed* for an electron transitioning from *2s to 4p orbital*
- iii. Wavelength of photon *absorbed* for an electron transitioning from *1s to 3p orbital*

i or ii or iii



Question 2 8 points



A Chem110 student plots the graph shown on the left, from data obtained for the kinetic energy (eV) of the ejected electron versus frequency (s^{-1}) of the incident light, for an unknown metal. ($1\text{eV} = 1.6022 \times 10^{-19} \text{ J}$)

Based on the plot, answer the questions **a to c** below:

a. 1 point On the plot above, label the **threshold frequency**.

b. 2 points What is the value of the work potential (in J) of the unknown metal?

c. 3 points What is the *longest* wavelength (in nm) required to eject an electron from this metal's surface? Show your work including equation(s) used. (3 sig figs)

d. 2 points In the space provided below, plot the following

(Assume frequency is *above* threshold frequency):

- i. *Kinetic energy* of an ejected electron versus the *intensity* of the incident light (left)
- ii. *Number of electrons* ejected versus the *frequency* of the incident light (right)



**Question 3** **6 points****a. 2 points**

In the ground state of Ge (Germanium; $Z = 32$) atom, **name** the following orbitals:

i. Orbital containing the electron that requires the maximum energy to be removed.

ii. Orbital containing the electron that experiences the minimum effective nuclear charge.

b. 2 points

An unknown anion (A^-) has the following *excited* electronic configuration: $[Ar] 4s^2 3d^3 4p^1$

(Ar: Argon; $Z = 18$)

Give the **condensed ground state electronic configuration** of the atom, **A** and the cation, **A^{2+}** .
Show your work. (Assume all electronic transitions are possible)

Electronic Configuration of A:

Electronic Configuration of A^{2+} :

c. 2 points

In a single atom, determine how many electrons can have the following set of quantum numbers?

(i) $n = 5; l = 3$

In a single atom, determine how many electrons can have the following set of quantum numbers?

(i) $n = 3; l = 2; m_s = +1/2$

**Question 4** **8 points**

Rank the following from 1 to 4 in **increasing** order. Provide the ranking in the boxes provided.

Note: Atomic number (Z) are provided for relevant atoms

a) Increasing (lowest to highest) order of *ionization energy*: (1= lowest; 4 = highest)

Mg

Mg: Z = 12

Rb

Rb: Z = 37

Cs

Cs: Z = 55

K

K : Z = 19

b) Increasing (lowest to highest) order of *bond angle*: (1= lowest; 4 = highest)

H₂OCCl₄XeF₄

HCN

c) Increasing (smallest to largest) order of *ionic size* (1= smallest; 4 = largest)

S²⁻

S: Z = 16

O²⁻

O: Z = 8

Be²⁺

Be: Z = 4

Na⁺

Na: Z = 11

d) Increasing (lowest to highest) order of *bond energy* (1= lowest; 4 = highest)

P-F

P-Cl

P-I

P-Br

**Question 5** **12 points**

a. 9 points For the molecules HClO_4 and TeF_6 , draw the most stable *Lewis structure* showing appropriate shape and *dashed/wedged bonds*, based on VSEPR. Determine the **molecular geometry** around the central atom. For HClO_4 , label the *most polar bond*.



b. 3 points (Note: Lone pairs are not shown in structures below)

(i) For the three *resonance* structures (1, 2 and 3) shown below, answer the following question:



(I) Circle the *most contributing* resonance structure.

Structure 1

Structure 2

Structure 3

(ii) For the three structures given (4, 5, and 6) below, answer the following questions:



(I) Circle all the structures that represent *resonance structures* of each other:

Structure 4

Structure 5

Structure 6

(II) Among the resonance structures, circle the *most contributing* resonance structure.

Structure 4

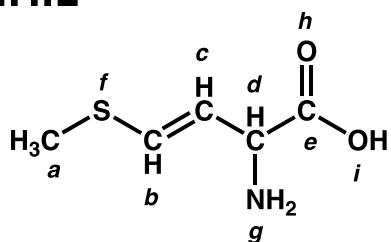
Structure 5

Structure 6

**Question 6** **6 points**

Using the data given below, calculate the **lattice energy** for SrO(s).

	ΔH° (kJ/mol)
$\text{Sr}(s) \longrightarrow \text{Sr}(g)$	+180
$\text{Sr}(g) \longrightarrow \text{Sr}^{2+}(g) + 2e^-$	+1468
$\text{O}_2(g) \longrightarrow 2\text{O}(g)$	+500
$\text{O}(g) + 2e^- \longrightarrow \text{O}^{2-}(g)$	+600
$\text{Sr}(s) + \frac{1}{2}\text{O}_2(g) \longrightarrow \text{SrO}(s)$	-550

**Question 7 10 points**

Consider the molecule given on the left, and answer questions (i) to (vi) using chemical bonding theories that we have learnt in class (valence bond theory, hybridization).

Note that the lone pairs are not shown in the structure:

i. Consider the carbon atoms marked in as ***a***, ***b***, ***c***, ***d***, and ***e*** in the molecule, and determine **hybridization** at each of the carbon centers. (2.5 points)

Atom	Hybridization
Carbon (<i>a</i>)	
Carbon (<i>b</i>)	
Carbon (<i>c</i>)	
Carbon (<i>d</i>)	
Carbon (<i>e</i>)	

ii. Now, consider the sulfur atom (S) marked as ***f***, nitrogen (N) atom (***g***), and oxygen(O) atoms (***h*** and ***i***), and determine their **hybridization**. (2 points)

Atom	Hybridization
Sulfur (<i>f</i>)	
Nitrogen (<i>g</i>)	
Oxygen (<i>h</i>)	
Oxygen (<i>i</i>)	

iii. **Describe** all the bonds formed by the carbon atoms ***a*** and ***b***, using sigma and pi designation as appropriate. (2 points)

Atom	Bonds formed (use sigma and pi as appropriate)
Carbon (<i>a</i>)	
Carbon (<i>b</i>)	

iv. How many total sp^3 -s sigma bonds are in the molecule? (1 point)

v. Determine **electronic** and **molecular geometry** at carbon (C) labeled ***e*** and sulfur (S) labeled ***f*** centres. (2 points)

Atom	Electronic Geometry	Molecular Geometry
Carbon (<i>e</i>)		
Sulfur (<i>f</i>)		

vi. Circle the value from the choices given below, that describes the **C-S-C bond angle at sulfur** (S) marked ***f***: (0.5 points)

>90° to <109.5° 109.5° >109.5 to <120° 120° >120° to <180°

**Question 8 12 points**

Using the **valence shell orbitals only** draw a **complete** molecular orbital (MO) diagram (with no s-p mixing) for the molecule **PS**, in the space provided below. **(6 points)**

Assume that 3s and 3p atomic orbitals will form molecular orbitals like those formed by the combination of 2s and 2p atomic orbitals that we learnt in class, and there is no s-p mixing.

Show **energy arrow**, **atomic orbitals** involved, **atom labels**, the **molecular orbitals** that are formed, **filled electrons** in the MO diagram etc. Make sure to label all the atomic and molecular orbitals.

Using the MO diagram above, answer the following questions:

- (i) Which one of the following will have the highest bond order? Circle the correct choice
A) PS^{1-} B) PS C) PS^{1+}
- (ii) Which one of the following will have the longest bond length? Circle the correct choice
A) PS^{1-} B) PS C) PS^{1+}
- (iii) Which of the following will be diamagnetic? Circle the correct choice.
A) PS^{1-} B) PS C) PS^{1+}
- (iv) Draw Lewis dot structure of the molecule with the shortest bond length (based on the MO diagram) from among the three (PS^{1-} , PS, PS^{1+}), and indicate the bond order calculated from the Lewis dot structure. **(3 points)**

Lewis Dot Structure

Bond Order



F3903B0E-0D4D-4A0E-9454-FDFD2ABF3F3C

midterm-exam-93388

#650 Page 10 of 10

SCRAP PAGE

**CHEMISTRY 110 –Midterm Exam****Thursday November 7, 2024, 6:30 – 8:30 pm**

Examiners: Pallavi Sirjoosingh and Ashok Kakkar

First name (please write as legibly as possible within the boxes)

--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--

Last name

--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--

Student ID number

--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--

This booklet has 8 questions on 10 pages (The last page can be used as a scrap page).

The staple must NOT be removed from the question booklet. Your intact question booklet must be returned at the end of the exam.

A separate booklet contains a periodic table, and datasheet.

Write your answers directly under the questions in the space provided. Do not write on the QR codes or near the top edges (as it may be cut off).

Please write **legibly and dark enough** for the scan to be readable. For students who use English as a second language, dictionaries are permitted. Calculators are allowed.

DO NOT ASK QUESTIONS OF THE INVIGILATORS about interpretations of the exam questions. If you have a concern about a question, please summarize it at the top of this page. All concerns will be evaluated after the exam. GOOD LUCK!

**Question 1 8 points** (*Assume all transitions are possible*)

a. 6 points Calculate the **energy (in kJ/mol)** required to ionize (completely remove) the electron from the **$n=3$** level of **He^+ ion** (He: Helium). Show your work including equations. (3 sig. figs.)

b. 2 points

(I) Select the largest (longest) wavelength from the three choices below:

- i. Wavelength of photon to *completely remove* an electron from *1s orbital of Hydrogen atom*
- ii. Wavelength of photon to *completely remove* an electron from *1s orbital of Boron atom (B)*
- iii. Wavelength of photon to *completely remove* an electron from *1s orbital of Boron cation (B^+)*

i or ii or iii

(II) Select the smallest (shortest) wavelength from the three choices below (*for H atom*):

- i. Wavelength of photon *emitted* for an electron transitioning from *2p to 1s orbital*
- ii. Wavelength of photon *absorbed* for an electron transitioning from *2s to 4p orbital*
- iii. Wavelength of photon *absorbed* for an electron transitioning from *1s to 3p orbital*

i or ii or iii



Question 2 8 points



A Chem110 student plots the graph shown on the left, from data obtained for the kinetic energy (eV) of the ejected electron versus frequency (s^{-1}) of the incident light, for an unknown metal. ($1\text{eV} = 1.6022 \times 10^{-19} \text{ J}$)

Based on the plot, answer the questions **a to c** below:

a. 1 point On the plot above, label the **threshold frequency**.

b. 2 points What is the value of the work potential (in J) of the unknown metal?

c. 3 points What is the *longest* wavelength (in nm) required to eject an electron from this metal's surface? Show your work including equation(s) used. (3 sig figs)

d. 2 points In the space provided below, plot the following

(Assume frequency is *above* threshold frequency):

- Kinetic energy* of an ejected electron versus the *intensity* of the incident light (left)
- Number of electrons* ejected versus the *frequency* of the incident light (right)



**Question 3** **6 points****a. 2 points**

In the ground state of Ge (Germanium; $Z = 32$) atom, **name** the following orbitals:

i. Orbital containing the electron that requires the maximum energy to be removed.

ii. Orbital containing the electron that experiences the minimum effective nuclear charge.

b. 2 points

An unknown anion (A^-) has the following *excited* electronic configuration: $[Ar] 4s^2 3d^3 4p^1$

(Ar: Argon; $Z = 18$)

Give the **condensed ground state electronic configuration** of the atom, **A** and the cation, **A^{2+}** .
Show your work. (Assume all electronic transitions are possible)

Electronic Configuration of A:

Electronic Configuration of A^{2+} :

c. 2 points

In a single atom, determine how many electrons can have the following set of quantum numbers?

(i) $n = 5; l = 3$

In a single atom, determine how many electrons can have the following set of quantum numbers?

(i) $n = 3; l = 2; m_s = +1/2$

**Question 4** **8 points**

Rank the following from 1 to 4 in **increasing** order. Provide the ranking in the boxes provided.

Note: Atomic number (Z) are provided for relevant atoms

a) Increasing (lowest to highest) order of *ionization energy*: (1= lowest; 4 = highest)

Mg

Mg: Z = 12

Rb

Rb: Z = 37

Cs

Cs: Z = 55

K

K : Z = 19

b) Increasing (lowest to highest) order of *bond angle*: (1= lowest; 4 = highest)

H₂OCCl₄XeF₄

HCN

c) Increasing (smallest to largest) order of *ionic size* (1= smallest; 4 = largest)

S²⁻

S: Z = 16

O²⁻

O: Z = 8

Be²⁺

Be: Z = 4

Na⁺

Na: Z = 11

d) Increasing (lowest to highest) order of *bond energy* (1= lowest; 4 = highest)

P-F

P-Cl

P-I

P-Br

**Question 5** **12 points**

a. 9 points For the molecules HClO_4 and TeF_6 , draw the most stable *Lewis structure* showing appropriate shape and *dashed/wedged bonds*, based on VSEPR. Determine the **molecular geometry** around the central atom. For HClO_4 , label the *most polar bond*.



b. 3 points (Note: Lone pairs are not shown in structures below)

(i) For the three *resonance* structures (1, 2 and 3) shown below, answer the following question:



(I) Circle the *most contributing* resonance structure.

Structure 1

Structure 2

Structure 3

(ii) For the three structures given (4, 5, and 6) below, answer the following questions:



(I) Circle all the structures that represent *resonance structures* of each other:

Structure 4

Structure 5

Structure 6

(II) Among the resonance structures, circle the *most contributing* resonance structure.

Structure 4

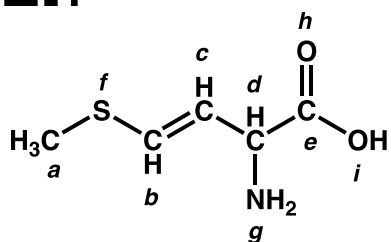
Structure 5

Structure 6

**Question 6** **6 points**

Using the data given below, calculate the **lattice energy** for SrO(s).

	ΔH° (kJ/mol)
$\text{Sr}(s) \longrightarrow \text{Sr}(g)$	+180
$\text{Sr}(g) \longrightarrow \text{Sr}^{2+}(g) + 2e^-$	+1468
$\text{O}_2(g) \longrightarrow 2\text{O}(g)$	+500
$\text{O}(g) + 2e^- \longrightarrow \text{O}^{2-}(g)$	+600
$\text{Sr}(s) + \frac{1}{2}\text{O}_2(g) \longrightarrow \text{SrO}(s)$	-550

**Question 7 10 points**

Consider the molecule given on the left, and answer questions (i) to (vi) using chemical bonding theories that we have learnt in class (valence bond theory, hybridization).

Note that the lone pairs are not shown in the structure:

i. Consider the carbon atoms marked in as ***a***, ***b***, ***c***, ***d***, and ***e*** in the molecule, and determine **hybridization** at each of the carbon centers. (2.5 points)

Atom	Hybridization
Carbon (<i>a</i>)	
Carbon (<i>b</i>)	
Carbon (<i>c</i>)	
Carbon (<i>d</i>)	
Carbon (<i>e</i>)	

ii. Now, consider the sulfur atom (S) marked as ***f***, nitrogen (N) atom (***g***), and oxygen(O) atoms (***h*** and ***i***), and determine their **hybridization**. (2 points)

Atom	Hybridization
Sulfur (<i>f</i>)	
Nitrogen (<i>g</i>)	
Oxygen (<i>h</i>)	
Oxygen (<i>i</i>)	

iii. **Describe** all the bonds formed by the carbon atoms ***a*** and ***b***, using sigma and pi designation as appropriate. (2 points)

Atom	Bonds formed (use sigma and pi as appropriate)
Carbon (<i>a</i>)	
Carbon (<i>b</i>)	

iv. How many total sp^3 -s sigma bonds are in the molecule? (1 point)

v. Determine **electronic** and **molecular geometry** at carbon (C) labeled ***e*** and sulfur (S) labeled ***f*** centres. (2 points)

Atom	Electronic Geometry	Molecular Geometry
Carbon (<i>e</i>)		
Sulfur (<i>f</i>)		

vi. Circle the value from the choices given below, that describes the **C-S-C bond angle at sulfur** (S) marked ***f***: (0.5 points)

>90° to <109.5° 109.5° >109.5 to <120° 120° >120° to <180°

**Question 8 12 points**

Using the **valence shell orbitals only** draw a **complete** molecular orbital (MO) diagram (with no s-p mixing) for the molecule **PS**, in the space provided below. **(6 points)**

Assume that 3s and 3p atomic orbitals will form molecular orbitals like those formed by the combination of 2s and 2p atomic orbitals that we learnt in class, and there is no s-p mixing.

Show **energy arrow**, **atomic orbitals** involved, **atom labels**, the **molecular orbitals** that are formed, **filled electrons** in the MO diagram etc. Make sure to label all the atomic and molecular orbitals.

Using the MO diagram above, answer the following questions:

- (i) Which one of the following will have the highest bond order? Circle the correct choice
A) PS^{1-} B) PS C) PS^{1+}
- (ii) Which one of the following will have the longest bond length? Circle the correct choice
A) PS^{1-} B) PS C) PS^{1+}
- (iii) Which of the following will be diamagnetic? Circle the correct choice.
A) PS^{1-} B) PS C) PS^{1+}
- (iv) Draw Lewis dot structure of the molecule with the shortest bond length (based on the MO diagram) from among the three (PS^{1-} , PS, PS^{1+}), and indicate the bond order calculated from the Lewis dot structure. **(3 points)**

Lewis Dot Structure

Bond Order



B1778756-0F53-45AF-9CFF-D3FE7251D2A4

midterm-exam-93388

#651 Page 10 of 10

SCRAP PAGE

**CHEMISTRY 110 –Midterm Exam****Thursday November 7, 2024, 6:30 – 8:30 pm**

Examiners: Pallavi Sirjoosingh and Ashok Kakkar

First name (please write as legibly as possible within the boxes)

--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--

Last name

--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--

Student ID number

--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--

This booklet has 8 questions on 10 pages (The last page can be used as a scrap page).

The staple must NOT be removed from the question booklet. Your intact question booklet must be returned at the end of the exam.

A separate booklet contains a periodic table, and datasheet.

Write your answers directly under the questions in the space provided. Do not write on the QR codes or near the top edges (as it may be cut off).

Please write **legibly and dark enough** for the scan to be readable. For students who use English as a second language, dictionaries are permitted. Calculators are allowed.

DO NOT ASK QUESTIONS OF THE INVIGILATORS about interpretations of the exam questions. If you have a concern about a question, please summarize it at the top of this page. All concerns will be evaluated after the exam. GOOD LUCK!

**Question 1 8 points** (*Assume all transitions are possible*)

a. 6 points Calculate the **energy (in kJ/mol)** required to ionize (completely remove) the electron from the **n=3** level of **He⁺ ion** (He: Helium). Show your work including equations. (3 sig. figs.)

b. 2 points

(I) Select the largest (longest) wavelength from the three choices below:

- i. Wavelength of photon to *completely remove* an electron from *1s orbital of Hydrogen atom*
- ii. Wavelength of photon to *completely remove* an electron from *1s orbital of Boron atom (B)*
- iii. Wavelength of photon to *completely remove* an electron from *1s orbital of Boron cation (B⁺)*

i or ii or iii

(II) Select the smallest (shortest) wavelength from the three choices below (*for H atom*):

- i. Wavelength of photon *emitted* for an electron transitioning from *2p to 1s orbital*
- ii. Wavelength of photon *absorbed* for an electron transitioning from *2s to 4p orbital*
- iii. Wavelength of photon *absorbed* for an electron transitioning from *1s to 3p orbital*

i or ii or iii



Question 2 8 points



A Chem110 student plots the graph shown on the left, from data obtained for the kinetic energy (eV) of the ejected electron versus frequency (s^{-1}) of the incident light, for an unknown metal. ($1\text{eV} = 1.6022 \times 10^{-19} \text{ J}$)

Based on the plot, answer the questions **a to c** below:

a. 1 point On the plot above, label the **threshold frequency**.

b. 2 points What is the value of the work potential (in J) of the unknown metal?

c. 3 points What is the *longest* wavelength (in nm) required to eject an electron from this metal's surface? Show your work including equation(s) used. (3 sig figs)

d. 2 points In the space provided below, plot the following

(Assume frequency is *above* threshold frequency):

- i. *Kinetic energy* of an ejected electron versus the *intensity* of the incident light (left)
- ii. *Number of electrons* ejected versus the *frequency* of the incident light (right)



**Question 3** **6 points****a. 2 points**

In the ground state of Ge (Germanium; $Z = 32$) atom, **name** the following orbitals:

i. Orbital containing the electron that requires the maximum energy to be removed.

ii. Orbital containing the electron that experiences the minimum effective nuclear charge.

b. 2 points

An unknown anion (A^-) has the following *excited* electronic configuration: $[Ar] 4s^2 3d^3 4p^1$

(Ar: Argon; $Z = 18$)

Give the **condensed ground state electronic configuration** of the atom, **A** and the cation, **A^{2+}** .
Show your work. (Assume all electronic transitions are possible)

Electronic Configuration of A:

Electronic Configuration of A^{2+} :

c. 2 points

In a single atom, determine how many electrons can have the following set of quantum numbers?

(i) $n = 5; l = 3$

In a single atom, determine how many electrons can have the following set of quantum numbers?

(i) $n = 3; l = 2; m_s = +1/2$

**Question 4** **8 points**

Rank the following from 1 to 4 in **increasing** order. Provide the ranking in the boxes provided.

Note: Atomic number (Z) are provided for relevant atoms

a) Increasing (lowest to highest) order of *ionization energy*: (1= lowest; 4 = highest)

Mg

Mg: Z = 12

Rb

Rb: Z = 37

Cs

Cs: Z = 55

K

K : Z = 19

b) Increasing (lowest to highest) order of *bond angle*: (1= lowest; 4 = highest)

H₂OCCl₄XeF₄

HCN

c) Increasing (smallest to largest) order of *ionic size* (1= smallest; 4 = largest)

S²⁻

S: Z = 16

O²⁻

O: Z = 8

Be²⁺

Be: Z = 4

Na⁺

Na: Z = 11

d) Increasing (lowest to highest) order of *bond energy* (1= lowest; 4 = highest)

P-F

P-Cl

P-I

P-Br

**Question 5** **12 points**

a. 9 points For the molecules HClO_4 and TeF_6 , draw the most stable *Lewis structure* showing appropriate shape and *dashed/wedged bonds*, based on VSEPR. Determine the **molecular geometry** around the central atom. For HClO_4 , label the *most polar bond*.



b. 3 points (Note: Lone pairs are not shown in structures below)

(i) For the three *resonance* structures (1, 2 and 3) shown below, answer the following question:



(I) Circle the *most contributing* resonance structure.

Structure 1

Structure 2

Structure 3

(ii) For the three structures given (4, 5, and 6) below, answer the following questions:



(I) Circle all the structures that represent *resonance structures* of each other:

Structure 4

Structure 5

Structure 6

(II) Among the resonance structures, circle the *most contributing* resonance structure.

Structure 4

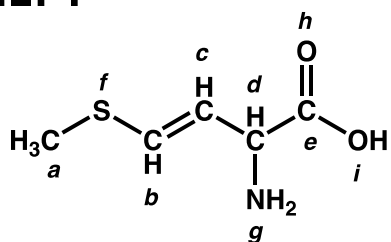
Structure 5

Structure 6

**Question 6** **6 points**

Using the data given below, calculate the **lattice energy** for SrO(s).

	ΔH° (kJ/mol)
$\text{Sr}(s) \longrightarrow \text{Sr}(g)$	+180
$\text{Sr}(g) \longrightarrow \text{Sr}^{2+}(g) + 2e^-$	+1468
$\text{O}_2(g) \longrightarrow 2\text{O}(g)$	+500
$\text{O}(g) + 2e^- \longrightarrow \text{O}^{2-}(g)$	+600
$\text{Sr}(s) + \frac{1}{2}\text{O}_2(g) \longrightarrow \text{SrO}(s)$	-550

**Question 7 10 points**

Consider the molecule given on the left, and answer questions (i) to (vi) using chemical bonding theories that we have learnt in class (valence bond theory, hybridization).

Note that the lone pairs are not shown in the structure:

i. Consider the carbon atoms marked in as ***a***, ***b***, ***c***, ***d***, and ***e*** in the molecule, and determine **hybridization** at each of the carbon centers. (2.5 points)

Atom	Hybridization
Carbon (<i>a</i>)	
Carbon (<i>b</i>)	
Carbon (<i>c</i>)	
Carbon (<i>d</i>)	
Carbon (<i>e</i>)	

ii. Now, consider the sulfur atom (S) marked as ***f***, nitrogen (N) atom (***g***), and oxygen(O) atoms (***h*** and ***i***), and determine their **hybridization**. (2 points)

Atom	Hybridization
Sulfur (<i>f</i>)	
Nitrogen (<i>g</i>)	
Oxygen (<i>h</i>)	
Oxygen (<i>i</i>)	

iii. **Describe** all the bonds formed by the carbon atoms ***a*** and ***b***, using sigma and pi designation as appropriate. (2 points)

Atom	Bonds formed (use sigma and pi as appropriate)
Carbon (<i>a</i>)	
Carbon (<i>b</i>)	

iv. How many total sp^3 -s sigma bonds are in the molecule? (1 point)

v. Determine **electronic** and **molecular geometry** at carbon (C) labeled ***e*** and sulfur (S) labeled ***f*** centres. (2 points)

Atom	Electronic Geometry	Molecular Geometry
Carbon (<i>e</i>)		
Sulfur (<i>f</i>)		

vi. Circle the value from the choices given below, that describes the **C-S-C bond angle at sulfur** (S) marked ***f***: (0.5 points)

>90° to <109.5°

109.5°

>109.5 to <120°

120°

>120° to <180°


Question 8 12 points

Using the **valence shell orbitals only** draw a **complete** molecular orbital (MO) diagram (with no s-p mixing) for the molecule **PS**, in the space provided below. **(6 points)**

Assume that 3s and 3p atomic orbitals will form molecular orbitals like those formed by the combination of 2s and 2p atomic orbitals that we learnt in class, and there is no s-p mixing.

Show **energy arrow**, **atomic orbitals** involved, **atom labels**, the **molecular orbitals** that are formed, **filled electrons** in the MO diagram etc. Make sure to label all the atomic and molecular orbitals.

Using the MO diagram above, answer the following questions:

- (i) Which one of the following will have the highest bond order? Circle the correct choice
 A) PS^{1-} B) PS C) PS^{1+}
- (ii) Which one of the following will have the longest bond length? Circle the correct choice
 A) PS^{1-} B) PS C) PS^{1+}
- (iii) Which of the following will be diamagnetic? Circle the correct choice.
 A) PS^{1-} B) PS C) PS^{1+}
- (iv) Draw Lewis dot structure of the molecule with the shortest bond length (based on the MO diagram) from among the three (PS^{1-} , PS, PS^{1+}), and indicate the bond order calculated from the Lewis dot structure. **(3 points)**

Lewis Dot Structure

Bond Order



CDA133F4-500B-44D7-99AE-14769D51CA5B

midterm-exam-93388

#652 Page 10 of 10

SCRAP PAGE

**CHEMISTRY 110 –Midterm Exam****Thursday November 7, 2024, 6:30 – 8:30 pm**

Examiners: Pallavi Sirjoosingh and Ashok Kakkar

First name (please write as legibly as possible within the boxes)

--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--

Last name

--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--

Student ID number

--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--

This booklet has 8 questions on 10 pages (The last page can be used as a scrap page).

The staple must NOT be removed from the question booklet. Your intact question booklet must be returned at the end of the exam.

A separate booklet contains a periodic table, and datasheet.

Write your answers directly under the questions in the space provided. Do not write on the QR codes or near the top edges (as it may be cut off).

Please write **legibly and dark enough** for the scan to be readable. For students who use English as a second language, dictionaries are permitted. Calculators are allowed.

DO NOT ASK QUESTIONS OF THE INVIGILATORS about interpretations of the exam questions. If you have a concern about a question, please summarize it at the top of this page. All concerns will be evaluated after the exam. GOOD LUCK!

**Question 1 8 points** (*Assume all transitions are possible*)

a. 6 points Calculate the **energy (in kJ/mol)** required to ionize (completely remove) the electron from the **n=3** level of **He⁺ ion** (He: Helium). Show your work including equations. (3 sig. figs.)

b. 2 points

(I) Select the largest (longest) wavelength from the three choices below:

- i. Wavelength of photon to *completely remove* an electron from *1s orbital of Hydrogen atom*
- ii. Wavelength of photon to *completely remove* an electron from *1s orbital of Boron atom (B)*
- iii. Wavelength of photon to *completely remove* an electron from *1s orbital of Boron cation (B⁺)*

i or ii or iii

(II) Select the smallest (shortest) wavelength from the three choices below (*for H atom*):

- i. Wavelength of photon *emitted* for an electron transitioning from *2p to 1s orbital*
- ii. Wavelength of photon *absorbed* for an electron transitioning from *2s to 4p orbital*
- iii. Wavelength of photon *absorbed* for an electron transitioning from *1s to 3p orbital*

i or ii or iii



Question 2 8 points



A Chem110 student plots the graph shown on the left, from data obtained for the kinetic energy (eV) of the ejected electron versus frequency (s^{-1}) of the incident light, for an unknown metal. ($1\text{eV} = 1.6022 \times 10^{-19} \text{ J}$)

Based on the plot, answer the questions **a to c** below:

a. 1 point On the plot above, label the **threshold frequency**.

b. 2 points What is the value of the work potential (in J) of the unknown metal?

c. 3 points What is the *longest* wavelength (in nm) required to eject an electron from this metal's surface? Show your work including equation(s) used. (3 sig figs)

d. 2 points In the space provided below, plot the following

(Assume frequency is *above* threshold frequency):

- Kinetic energy* of an ejected electron versus the *intensity* of the incident light (left)
- Number of electrons* ejected versus the *frequency* of the incident light (right)



**Question 3** **6 points****a. 2 points**

In the ground state of Ge (Germanium; $Z = 32$) atom, **name** the following orbitals:

i. Orbital containing the electron that requires the maximum energy to be removed.

ii. Orbital containing the electron that experiences the minimum effective nuclear charge.

b. 2 points

An unknown anion (A^-) has the following *excited* electronic configuration: $[Ar] 4s^2 3d^3 4p^1$

(Ar: Argon; $Z = 18$)

Give the **condensed ground state electronic configuration** of the atom, **A** and the cation, **A^{2+}** .
Show your work. (Assume all electronic transitions are possible)

Electronic Configuration of A:

Electronic Configuration of A^{2+} :

c. 2 points

In a single atom, determine how many electrons can have the following set of quantum numbers?

(i) $n = 5; l = 3$

In a single atom, determine how many electrons can have the following set of quantum numbers?

(i) $n = 3; l = 2; m_s = +1/2$

**Question 4** **8 points**

Rank the following from 1 to 4 in **increasing** order. Provide the ranking in the boxes provided.

Note: Atomic number (Z) are provided for relevant atoms

a) Increasing (lowest to highest) order of *ionization energy*: (1= lowest; 4 = highest)

Mg

Mg: Z = 12

Rb

Rb: Z = 37

Cs

Cs: Z = 55

K

K : Z = 19

b) Increasing (lowest to highest) order of *bond angle*: (1= lowest; 4 = highest)

H₂OCCl₄XeF₄

HCN

c) Increasing (smallest to largest) order of *ionic size* (1= smallest; 4 = largest)

S²⁻

S: Z = 16

O²⁻

O: Z = 8

Be²⁺

Be: Z = 4

Na⁺

Na: Z = 11

d) Increasing (lowest to highest) order of *bond energy* (1= lowest; 4 = highest)

P-F

P-Cl

P-I

P-Br

**Question 5** **12 points**

a. 9 points For the molecules HClO_4 and TeF_6 , draw the most stable *Lewis structure* showing appropriate shape and *dashed/wedged bonds*, based on VSEPR. Determine the **molecular geometry** around the central atom. For HClO_4 , label the *most polar bond*.



b. 3 points (Note: Lone pairs are not shown in structures below)

(i) For the three *resonance* structures (1, 2 and 3) shown below, answer the following question:



(I) Circle the *most contributing* resonance structure.

Structure 1

Structure 2

Structure 3

(ii) For the three structures given (4, 5, and 6) below, answer the following questions:



(I) Circle all the structures that represent *resonance structures* of each other:

Structure 4

Structure 5

Structure 6

(II) Among the resonance structures, circle the *most contributing* resonance structure.

Structure 4

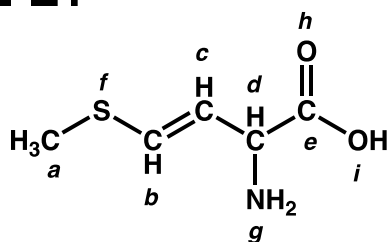
Structure 5

Structure 6

**Question 6** **6 points**

Using the data given below, calculate the **lattice energy** for SrO(s).

	ΔH° (kJ/mol)
$\text{Sr}(s) \longrightarrow \text{Sr}(g)$	+180
$\text{Sr}(g) \longrightarrow \text{Sr}^{2+}(g) + 2e^-$	+1468
$\text{O}_2(g) \longrightarrow 2\text{O}(g)$	+500
$\text{O}(g) + 2e^- \longrightarrow \text{O}^{2-}(g)$	+600
$\text{Sr}(s) + \frac{1}{2}\text{O}_2(g) \longrightarrow \text{SrO}(s)$	-550

**Question 7 10 points**

Consider the molecule given on the left, and answer questions (i) to (vi) using chemical bonding theories that we have learnt in class (valence bond theory, hybridization).

Note that the lone pairs are not shown in the structure:

i. Consider the carbon atoms marked in as ***a***, ***b***, ***c***, ***d***, and ***e*** in the molecule, and determine **hybridization** at each of the carbon centers. (2.5 points)

Atom	Hybridization
Carbon (<i>a</i>)	
Carbon (<i>b</i>)	
Carbon (<i>c</i>)	
Carbon (<i>d</i>)	
Carbon (<i>e</i>)	

ii. Now, consider the sulfur atom (S) marked as ***f***, nitrogen (N) atom (***g***), and oxygen(O) atoms (***h*** and ***i***), and determine their **hybridization**. (2 points)

Atom	Hybridization
Sulfur (<i>f</i>)	
Nitrogen (<i>g</i>)	
Oxygen (<i>h</i>)	
Oxygen (<i>i</i>)	

iii. **Describe** all the bonds formed by the carbon atoms ***a*** and ***b***, using sigma and pi designation as appropriate. (2 points)

Atom	Bonds formed (use sigma and pi as appropriate)
Carbon (<i>a</i>)	
Carbon (<i>b</i>)	

iv. How many total sp^3 -s sigma bonds are in the molecule? (1 point)

v. Determine **electronic** and **molecular geometry** at carbon (C) labeled ***e*** and sulfur (S) labeled ***f*** centres. (2 points)

Atom	Electronic Geometry	Molecular Geometry
Carbon (<i>e</i>)		
Sulfur (<i>f</i>)		

vi. Circle the value from the choices given below, that describes the **C-S-C bond angle at sulfur** (S) marked ***f***: (0.5 points)

>90° to <109.5° 109.5° >109.5 to <120° 120° >120° to <180°

**Question 8 12 points**

Using the **valence shell orbitals only** draw a **complete** molecular orbital (MO) diagram (with no s-p mixing) for the molecule **PS**, in the space provided below. **(6 points)**

Assume that 3s and 3p atomic orbitals will form molecular orbitals like those formed by the combination of 2s and 2p atomic orbitals that we learnt in class, and there is no s-p mixing.

Show **energy arrow**, **atomic orbitals** involved, **atom labels**, the **molecular orbitals** that are formed, **filled electrons** in the MO diagram etc. Make sure to label all the atomic and molecular orbitals.

Using the MO diagram above, answer the following questions:

- (i) Which one of the following will have the highest bond order? Circle the correct choice
A) PS^{1-} B) PS C) PS^{1+}
- (ii) Which one of the following will have the longest bond length? Circle the correct choice
A) PS^{1-} B) PS C) PS^{1+}
- (iii) Which of the following will be diamagnetic? Circle the correct choice.
A) PS^{1-} B) PS C) PS^{1+}
- (iv) Draw Lewis dot structure of the molecule with the shortest bond length (based on the MO diagram) from among the three (PS^{1-} , PS, PS^{1+}), and indicate the bond order calculated from the Lewis dot structure. **(3 points)**

Lewis Dot Structure

Bond Order



32E1CBF8-960A-46FE-8222-903810CB676F

midterm-exam-93388

#653 Page 10 of 10

SCRAP PAGE

**CHEMISTRY 110 –Midterm Exam****Thursday November 7, 2024, 6:30 – 8:30 pm**

Examiners: Pallavi Sirjoosingh and Ashok Kakkar

First name (please write as legibly as possible within the boxes)

--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--

Last name

--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--

Student ID number

--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--

This booklet has 8 questions on 10 pages (The last page can be used as a scrap page).

The staple must NOT be removed from the question booklet. Your intact question booklet must be returned at the end of the exam.

A separate booklet contains a periodic table, and datasheet.

Write your answers directly under the questions in the space provided. Do not write on the QR codes or near the top edges (as it may be cut off).

Please write **legibly and dark enough** for the scan to be readable. For students who use English as a second language, dictionaries are permitted. Calculators are allowed.

DO NOT ASK QUESTIONS OF THE INVIGILATORS about interpretations of the exam questions. If you have a concern about a question, please summarize it at the top of this page. All concerns will be evaluated after the exam. GOOD LUCK!

**Question 1 8 points** (*Assume all transitions are possible*)

a. 6 points Calculate the **energy (in kJ/mol)** required to ionize (completely remove) the electron from the **$n=3$** level of **He^+ ion** (He: Helium). Show your work including equations. (3 sig. figs.)

b. 2 points

(I) Select the largest (longest) wavelength from the three choices below:

- i. Wavelength of photon to *completely remove* an electron from *1s orbital of Hydrogen atom*
- ii. Wavelength of photon to *completely remove* an electron from *1s orbital of Boron atom (B)*
- iii. Wavelength of photon to *completely remove* an electron from *1s orbital of Boron cation (B^+)*

i or ii or iii

(II) Select the smallest (shortest) wavelength from the three choices below (*for H atom*):

- i. Wavelength of photon *emitted* for an electron transitioning from *2p to 1s orbital*
- ii. Wavelength of photon *absorbed* for an electron transitioning from *2s to 4p orbital*
- iii. Wavelength of photon *absorbed* for an electron transitioning from *1s to 3p orbital*

i or ii or iii



Question 2 8 points



A Chem110 student plots the graph shown on the left, from data obtained for the kinetic energy (eV) of the ejected electron versus frequency (s^{-1}) of the incident light, for an unknown metal. ($1\text{eV} = 1.6022 \times 10^{-19} \text{ J}$)

Based on the plot, answer the questions **a to c** below:

a. 1 point On the plot above, label the **threshold frequency**.

b. 2 points What is the value of the work potential (in J) of the unknown metal?

c. 3 points What is the *longest* wavelength (in nm) required to eject an electron from this metal's surface? Show your work including equation(s) used. (3 sig figs)

d. 2 points In the space provided below, plot the following

(Assume frequency is *above* threshold frequency):

- i. *Kinetic energy* of an ejected electron versus the *intensity* of the incident light (left)
- ii. *Number of electrons* ejected versus the *frequency* of the incident light (right)



**Question 3** **6 points****a. 2 points**

In the ground state of Ge (Germanium; $Z = 32$) atom, **name** the following orbitals:

i. Orbital containing the electron that requires the maximum energy to be removed.

ii. Orbital containing the electron that experiences the minimum effective nuclear charge.

b. 2 points

An unknown anion (A^-) has the following *excited* electronic configuration: $[Ar] 4s^2 3d^3 4p^1$

(Ar: Argon; $Z = 18$)

Give the **condensed ground state electronic configuration** of the atom, **A** and the cation, **A^{2+}** .
Show your work. (Assume all electronic transitions are possible)

Electronic Configuration of A:

Electronic Configuration of A^{2+} :

c. 2 points

In a single atom, determine how many electrons can have the following set of quantum numbers?

(i) $n = 5; l = 3$

In a single atom, determine how many electrons can have the following set of quantum numbers?

(i) $n = 3; l = 2; m_s = +1/2$

**Question 4** **8 points**

Rank the following from 1 to 4 in **increasing** order. Provide the ranking in the boxes provided.

Note: Atomic number (Z) are provided for relevant atoms

a) Increasing (lowest to highest) order of *ionization energy*: (1= lowest; 4 = highest)

Mg

Mg: Z = 12

Rb

Rb: Z = 37

Cs

Cs: Z = 55

K

K : Z = 19

b) Increasing (lowest to highest) order of *bond angle*: (1= lowest; 4 = highest)

H₂OCCl₄XeF₄

HCN

c) Increasing (smallest to largest) order of *ionic size* (1= smallest; 4 = largest)

S²⁻

S: Z = 16

O²⁻

O: Z = 8

Be²⁺

Be: Z = 4

Na⁺

Na: Z = 11

d) Increasing (lowest to highest) order of *bond energy* (1= lowest; 4 = highest)

P-F

P-Cl

P-I

P-Br

**Question 5** **12 points**

a. 9 points For the molecules HClO_4 and TeF_6 , draw the most stable *Lewis structure* showing appropriate shape and *dashed/wedged bonds*, based on VSEPR. Determine the **molecular geometry** around the central atom. For HClO_4 , label the *most polar bond*.



b. 3 points (Note: Lone pairs are not shown in structures below)

(i) For the three *resonance* structures (1, 2 and 3) shown below, answer the following question:



(I) Circle the *most contributing* resonance structure.

Structure 1

Structure 2

Structure 3

(ii) For the three structures given (4, 5, and 6) below, answer the following questions:



(I) Circle all the structures that represent *resonance structures* of each other:

Structure 4

Structure 5

Structure 6

(II) Among the resonance structures, circle the *most contributing* resonance structure.

Structure 4

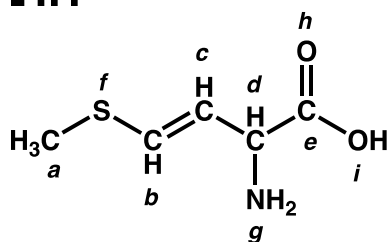
Structure 5

Structure 6

**Question 6** **6 points**

Using the data given below, calculate the **lattice energy** for SrO(s).

	ΔH° (kJ/mol)
$\text{Sr}(s) \longrightarrow \text{Sr}(g)$	+180
$\text{Sr}(g) \longrightarrow \text{Sr}^{2+}(g) + 2e^-$	+1468
$\text{O}_2(g) \longrightarrow 2\text{O}(g)$	+500
$\text{O}(g) + 2e^- \longrightarrow \text{O}^{2-}(g)$	+600
$\text{Sr}(s) + \frac{1}{2}\text{O}_2(g) \longrightarrow \text{SrO}(s)$	-550

**Question 7 10 points**

Consider the molecule given on the left, and answer questions (i) to (vi) using chemical bonding theories that we have learnt in class (valence bond theory, hybridization).

Note that the lone pairs are not shown in the structure:

i. Consider the carbon atoms marked in as ***a***, ***b***, ***c***, ***d***, and ***e*** in the molecule, and determine **hybridization** at each of the carbon centers. (2.5 points)

Atom	Hybridization
Carbon (<i>a</i>)	
Carbon (<i>b</i>)	
Carbon (<i>c</i>)	
Carbon (<i>d</i>)	
Carbon (<i>e</i>)	

ii. Now, consider the sulfur atom (S) marked as ***f***, nitrogen (N) atom (***g***), and oxygen(O) atoms (***h*** and ***i***), and determine their **hybridization**. (2 points)

Atom	Hybridization
Sulfur (<i>f</i>)	
Nitrogen (<i>g</i>)	
Oxygen (<i>h</i>)	
Oxygen (<i>i</i>)	

iii. **Describe** all the bonds formed by the carbon atoms ***a*** and ***b***, using sigma and pi designation as appropriate. (2 points)

Atom	Bonds formed (use sigma and pi as appropriate)
Carbon (<i>a</i>)	
Carbon (<i>b</i>)	

iv. How many total sp^3 -s sigma bonds are in the molecule? (1 point)

v. Determine **electronic** and **molecular geometry** at carbon (C) labeled ***e*** and sulfur (S) labeled ***f*** centres. (2 points)

Atom	Electronic Geometry	Molecular Geometry
Carbon (<i>e</i>)		
Sulfur (<i>f</i>)		

vi. Circle the value from the choices given below, that describes the **C-S-C bond angle at sulfur** (S) marked ***f***: (0.5 points)

>90° to <109.5° 109.5° >109.5 to <120° 120° >120° to <180°



Question 8 12 points

Using the **valence shell orbitals only** draw a **complete** molecular orbital (MO) diagram (with no s-p mixing) for the molecule **PS**, in the space provided below. **(6 points)**

Assume that 3s and 3p atomic orbitals will form molecular orbitals like those formed by the combination of 2s and 2p atomic orbitals that we learnt in class, and there is no s-p mixing.

Show **energy arrow**, **atomic orbitals** involved, **atom labels**, the **molecular orbitals** that are formed, **filled electrons** in the MO diagram etc. Make sure to label all the atomic and molecular orbitals.

Using the MO diagram above, answer the following questions:

- (i) Which one of the following will have the highest bond order? Circle the correct choice
 A) PS^{1-} B) PS C) PS^{1+}
- (ii) Which one of the following will have the longest bond length? Circle the correct choice
 A) PS^{1-} B) PS C) PS^{1+}
- (iii) Which of the following will be diamagnetic? Circle the correct choice.
 A) PS^{1-} B) PS C) PS^{1+}
- (iv) Draw Lewis dot structure of the molecule with the shortest bond length (based on the MO diagram) from among the three (PS^{1-} , PS, PS^{1+}), and indicate the bond order calculated from the Lewis dot structure. **(3 points)**

Lewis Dot Structure

Bond Order



03DFC792-9D12-47C5-9CD4-3988E7E600E8

midterm-exam-93388

#654 Page 10 of 10

SCRAP PAGE

**CHEMISTRY 110 –Midterm Exam****Thursday November 7, 2024, 6:30 – 8:30 pm****Examiners: Pallavi Sirjoosingh and Ashok Kakkar**

First name (please write as legibly as possible within the boxes)

--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--

Last name

--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--

Student ID number

--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--

This booklet has 8 questions on 10 pages (The last page can be used as a scrap page).

The staple must NOT be removed from the question booklet. Your intact question booklet must be returned at the end of the exam.

A separate booklet contains a periodic table, and datasheet.

Write your answers directly under the questions in the space provided. Do not write on the QR codes or near the top edges (as it may be cut off).

Please write **legibly and dark enough** for the scan to be readable. For students who use English as a second language, dictionaries are permitted. Calculators are allowed.

DO NOT ASK QUESTIONS OF THE INVIGILATORS about interpretations of the exam questions. If you have a concern about a question, please summarize it at the top of this page. All concerns will be evaluated after the exam. GOOD LUCK!

**Question 1 8 points** (*Assume all transitions are possible*)

a. 6 points Calculate the **energy (in kJ/mol)** required to ionize (completely remove) the electron from the **n=3** level of **He⁺ ion** (He: Helium). Show your work including equations. (3 sig. figs.)

b. 2 points

(I) Select the largest (longest) wavelength from the three choices below:

- i. Wavelength of photon to *completely remove* an electron from *1s orbital of Hydrogen atom*
- ii. Wavelength of photon to *completely remove* an electron from *1s orbital of Boron atom (B)*
- iii. Wavelength of photon to *completely remove* an electron from *1s orbital of Boron cation (B⁺)*

i or ii or iii

(II) Select the smallest (shortest) wavelength from the three choices below (*for H atom*):

- i. Wavelength of photon *emitted* for an electron transitioning from *2p to 1s orbital*
- ii. Wavelength of photon *absorbed* for an electron transitioning from *2s to 4p orbital*
- iii. Wavelength of photon *absorbed* for an electron transitioning from *1s to 3p orbital*

i or ii or iii



Question 2 8 points



A Chem110 student plots the graph shown on the left, from data obtained for the kinetic energy (eV) of the ejected electron versus frequency (s^{-1}) of the incident light, for an unknown metal. ($1\text{eV} = 1.6022 \times 10^{-19} \text{ J}$)

Based on the plot, answer the questions **a to c** below:

a. 1 point On the plot above, label the **threshold frequency**.

b. 2 points What is the value of the work potential (in J) of the unknown metal?

c. 3 points What is the *longest* wavelength (in nm) required to eject an electron from this metal's surface? Show your work including equation(s) used. (3 sig figs)

d. 2 points In the space provided below, plot the following

(Assume frequency is *above* threshold frequency):

- Kinetic energy* of an ejected electron versus the *intensity* of the incident light (left)
- Number of electrons* ejected versus the *frequency* of the incident light (right)



**Question 3** **6 points****a. 2 points**

In the ground state of Ge (Germanium; $Z = 32$) atom, **name** the following orbitals:

i. Orbital containing the electron that requires the maximum energy to be removed.

ii. Orbital containing the electron that experiences the minimum effective nuclear charge.

b. 2 points

An unknown anion (A^-) has the following *excited* electronic configuration: $[Ar] 4s^2 3d^3 4p^1$

(Ar: Argon; $Z = 18$)

Give the **condensed ground state electronic configuration** of the atom, **A** and the cation, **A^{2+}** .
Show your work. (Assume all electronic transitions are possible)

Electronic Configuration of A:

Electronic Configuration of A^{2+} :

c. 2 points

In a single atom, determine how many electrons can have the following set of quantum numbers?

(i) $n = 5; l = 3$

In a single atom, determine how many electrons can have the following set of quantum numbers?

(i) $n = 3; l = 2; m_s = +1/2$

**Question 4** **8 points**

Rank the following from 1 to 4 in **increasing** order. Provide the ranking in the boxes provided.

Note: Atomic number (Z) are provided for relevant atoms

a) Increasing (lowest to highest) order of *ionization energy*: (1= lowest; 4 = highest)

Mg

Mg: Z = 12

Rb

Rb: Z = 37

Cs

Cs: Z = 55

K

K : Z = 19

b) Increasing (lowest to highest) order of *bond angle*: (1= lowest; 4 = highest)

H₂OCCl₄XeF₄

HCN

c) Increasing (smallest to largest) order of *ionic size* (1= smallest; 4 = largest)

S²⁻

S: Z = 16

O²⁻

O: Z = 8

Be²⁺

Be: Z = 4

Na⁺

Na: Z = 11

d) Increasing (lowest to highest) order of *bond energy* (1= lowest; 4 = highest)

P-F

P-Cl

P-I

P-Br

**Question 5** **12 points**

a. 9 points For the molecules HClO_4 and TeF_6 , draw the most stable *Lewis structure* showing appropriate shape and *dashed/wedged bonds*, based on VSEPR. Determine the **molecular geometry** around the central atom. For HClO_4 , label the *most polar bond*.



b. 3 points (Note: Lone pairs are not shown in structures below)

(i) For the three *resonance* structures (1, 2 and 3) shown below, answer the following question:



(I) Circle the *most contributing* resonance structure.

Structure 1

Structure 2

Structure 3

(ii) For the three structures given (4, 5, and 6) below, answer the following questions:



(I) Circle all the structures that represent *resonance structures* of each other:

Structure 4

Structure 5

Structure 6

(II) Among the resonance structures, circle the *most contributing* resonance structure.

Structure 4

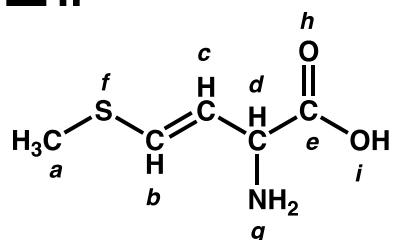
Structure 5

Structure 6

**Question 6** **6 points**

Using the data given below, calculate the **lattice energy** for SrO(s).

	ΔH° (kJ/mol)
$\text{Sr}(s) \longrightarrow \text{Sr}(g)$	+180
$\text{Sr}(g) \longrightarrow \text{Sr}^{2+}(g) + 2e^-$	+1468
$\text{O}_2(g) \longrightarrow 2\text{O}(g)$	+500
$\text{O}(g) + 2e^- \longrightarrow \text{O}^{2-}(g)$	+600
$\text{Sr}(s) + \frac{1}{2}\text{O}_2(g) \longrightarrow \text{SrO}(s)$	-550

**Question 7 10 points**

Consider the molecule given on the left, and answer questions (i) to (vi) using chemical bonding theories that we have learnt in class (valence bond theory, hybridization).

Note that the lone pairs are not shown in the structure:

i. Consider the carbon atoms marked in as ***a***, ***b***, ***c***, ***d***, and ***e*** in the molecule, and determine **hybridization** at each of the carbon centers. (2.5 points)

Atom	Hybridization
Carbon (<i>a</i>)	
Carbon (<i>b</i>)	
Carbon (<i>c</i>)	
Carbon (<i>d</i>)	
Carbon (<i>e</i>)	

ii. Now, consider the sulfur atom (S) marked as ***f***, nitrogen (N) atom (***g***), and oxygen(O) atoms (***h*** and ***i***), and determine their **hybridization**. (2 points)

Atom	Hybridization
Sulfur (<i>f</i>)	
Nitrogen (<i>g</i>)	
Oxygen (<i>h</i>)	
Oxygen (<i>i</i>)	

iii. **Describe** all the bonds formed by the carbon atoms ***a*** and ***b***, using sigma and pi designation as appropriate. (2 points)

Atom	Bonds formed (use sigma and pi as appropriate)
Carbon (<i>a</i>)	
Carbon (<i>b</i>)	

iv. How many total sp^3 -s sigma bonds are in the molecule? (1 point)

v. Determine **electronic** and **molecular geometry** at carbon (C) labeled ***e*** and sulfur (S) labeled ***f*** centres. (2 points)

Atom	Electronic Geometry	Molecular Geometry
Carbon (<i>e</i>)		
Sulfur (<i>f</i>)		

vi. Circle the value from the choices given below, that describes the **C-S-C bond angle at sulfur** (S) marked ***f***: (0.5 points)

>90° to <109.5°

109.5°

>109.5 to <120°

120°

>120° to <180°


Question 8 12 points

Using the **valence shell orbitals only** draw a **complete** molecular orbital (MO) diagram (with no s-p mixing) for the molecule **PS**, in the space provided below. **(6 points)**

Assume that 3s and 3p atomic orbitals will form molecular orbitals like those formed by the combination of 2s and 2p atomic orbitals that we learnt in class, and there is no s-p mixing.

Show **energy arrow**, **atomic orbitals** involved, **atom labels**, the **molecular orbitals** that are formed, **filled electrons** in the MO diagram etc. Make sure to label all the atomic and molecular orbitals.

Using the MO diagram above, answer the following questions:

- (i) Which one of the following will have the highest bond order? Circle the correct choice
 A) PS^{1-} B) PS C) PS^{1+}
- (ii) Which one of the following will have the longest bond length? Circle the correct choice
 A) PS^{1-} B) PS C) PS^{1+}
- (iii) Which of the following will be diamagnetic? Circle the correct choice.
 A) PS^{1-} B) PS C) PS^{1+}
- (iv) Draw Lewis dot structure of the molecule with the shortest bond length (based on the MO diagram) from among the three (PS^{1-} , PS, PS^{1+}), and indicate the bond order calculated from the Lewis dot structure. **(3 points)**

Lewis Dot Structure

Bond Order



581A332F-2A1E-48A0-9CE6-B89D60A9BF1E

midterm-exam-93388

#655 Page 10 of 10

SCRAP PAGE

**CHEMISTRY 110 –Midterm Exam****Thursday November 7, 2024, 6:30 – 8:30 pm****Examiners: Pallavi Sirjoosingh and Ashok Kakkar**

First name (please write as legibly as possible within the boxes)

--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--

Last name

--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--

Student ID number

--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--

This booklet has 8 questions on 10 pages (The last page can be used as a scrap page).

The staple must NOT be removed from the question booklet. Your intact question booklet must be returned at the end of the exam.

A separate booklet contains a periodic table, and datasheet.

Write your answers directly under the questions in the space provided. Do not write on the QR codes or near the top edges (as it may be cut off).

Please write **legibly and dark enough** for the scan to be readable. For students who use English as a second language, dictionaries are permitted. Calculators are allowed.

DO NOT ASK QUESTIONS OF THE INVIGILATORS about interpretations of the exam questions. If you have a concern about a question, please summarize it at the top of this page. All concerns will be evaluated after the exam. GOOD LUCK!

**Question 1 8 points** (*Assume all transitions are possible*)

a. 6 points Calculate the **energy (in kJ/mol)** required to ionize (completely remove) the electron from the **$n=3$** level of **He^+ ion** (He: Helium). Show your work including equations. (3 sig. figs.)

b. 2 points

(I) Select the largest (longest) wavelength from the three choices below:

- i. Wavelength of photon to *completely remove* an electron from *1s orbital of Hydrogen atom*
- ii. Wavelength of photon to *completely remove* an electron from *1s orbital of Boron atom (B)*
- iii. Wavelength of photon to *completely remove* an electron from *1s orbital of Boron cation (B^+)*

i or ii or iii

(II) Select the smallest (shortest) wavelength from the three choices below (*for H atom*):

- i. Wavelength of photon *emitted* for an electron transitioning from *2p to 1s orbital*
- ii. Wavelength of photon *absorbed* for an electron transitioning from *2s to 4p orbital*
- iii. Wavelength of photon *absorbed* for an electron transitioning from *1s to 3p orbital*

i or ii or iii



Question 2 8 points



A Chem110 student plots the graph shown on the left, from data obtained for the kinetic energy (eV) of the ejected electron versus frequency (s^{-1}) of the incident light, for an unknown metal. ($1\text{eV} = 1.6022 \times 10^{-19} \text{ J}$)

Based on the plot, answer the questions **a to c** below:

a. 1 point On the plot above, label the **threshold frequency**.

b. 2 points What is the value of the work potential (in J) of the unknown metal?

c. 3 points What is the *longest* wavelength (in nm) required to eject an electron from this metal's surface? Show your work including equation(s) used. (3 sig figs)

d. 2 points In the space provided below, plot the following

(Assume frequency is *above* threshold frequency):

- i. *Kinetic energy* of an ejected electron versus the *intensity* of the incident light (left)
- ii. *Number of electrons* ejected versus the *frequency* of the incident light (right)



**Question 3** **6 points****a. 2 points**

In the ground state of Ge (Germanium; $Z = 32$) atom, **name** the following orbitals:

i. Orbital containing the electron that requires the maximum energy to be removed.

ii. Orbital containing the electron that experiences the minimum effective nuclear charge.

b. 2 points

An unknown anion (A^-) has the following *excited* electronic configuration: $[Ar] 4s^2 3d^3 4p^1$

(Ar: Argon; $Z = 18$)

Give the **condensed ground state electronic configuration** of the atom, **A** and the cation, **A^{2+}** .
Show your work. (Assume all electronic transitions are possible)

Electronic Configuration of A:

Electronic Configuration of A^{2+} :

c. 2 points

In a single atom, determine how many electrons can have the following set of quantum numbers?

(i) $n = 5; l = 3$

In a single atom, determine how many electrons can have the following set of quantum numbers?

(i) $n = 3; l = 2; m_s = +1/2$

**Question 4** **8 points**

Rank the following from 1 to 4 in **increasing** order. Provide the ranking in the boxes provided.

Note: Atomic number (Z) are provided for relevant atoms

a) Increasing (lowest to highest) order of *ionization energy*: (1= lowest; 4 = highest)

Mg

Mg: Z = 12

Rb

Rb: Z = 37

Cs

Cs: Z = 55

K

K : Z = 19

b) Increasing (lowest to highest) order of *bond angle*: (1= lowest; 4 = highest)

H₂OCCl₄XeF₄

HCN

c) Increasing (smallest to largest) order of *ionic size* (1= smallest; 4 = largest)

S²⁻

S: Z = 16

O²⁻

O: Z = 8

Be²⁺

Be: Z = 4

Na⁺

Na: Z = 11

d) Increasing (lowest to highest) order of *bond energy* (1= lowest; 4 = highest)

P-F

P-Cl

P-I

P-Br

**Question 5** **12 points**

a. 9 points For the molecules HClO_4 and TeF_6 , draw the most stable *Lewis structure* showing appropriate shape and *dashed/wedged bonds*, based on VSEPR. Determine the **molecular geometry** around the central atom. For HClO_4 , label the *most polar bond*.



b. 3 points (Note: Lone pairs are not shown in structures below)

(i) For the three *resonance* structures (1, 2 and 3) shown below, answer the following question:



(I) Circle the *most contributing* resonance structure.

Structure 1

Structure 2

Structure 3

(ii) For the three structures given (4, 5, and 6) below, answer the following questions:



(I) Circle all the structures that represent *resonance structures* of each other:

Structure 4

Structure 5

Structure 6

(II) Among the resonance structures, circle the *most contributing* resonance structure.

Structure 4

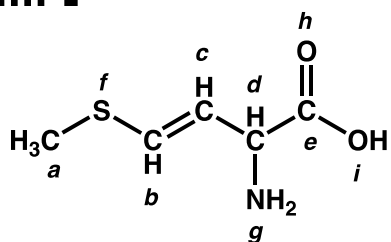
Structure 5

Structure 6

**Question 6** **6 points**

Using the data given below, calculate the **lattice energy** for SrO(s).

	ΔH° (kJ/mol)
$\text{Sr}(s) \longrightarrow \text{Sr}(g)$	+180
$\text{Sr}(g) \longrightarrow \text{Sr}^{2+}(g) + 2e^-$	+1468
$\text{O}_2(g) \longrightarrow 2\text{O}(g)$	+500
$\text{O}(g) + 2e^- \longrightarrow \text{O}^{2-}(g)$	+600
$\text{Sr}(s) + \frac{1}{2}\text{O}_2(g) \longrightarrow \text{SrO}(s)$	-550

**Question 7 10 points**

Consider the molecule given on the left, and answer questions (i) to (vi) using chemical bonding theories that we have learnt in class (valence bond theory, hybridization).

Note that the lone pairs are not shown in the structure:

i. Consider the carbon atoms marked in as ***a***, ***b***, ***c***, ***d***, and ***e*** in the molecule, and determine **hybridization** at each of the carbon centers. (2.5 points)

Atom	Hybridization
Carbon (<i>a</i>)	
Carbon (<i>b</i>)	
Carbon (<i>c</i>)	
Carbon (<i>d</i>)	
Carbon (<i>e</i>)	

ii. Now, consider the sulfur atom (S) marked as ***f***, nitrogen (N) atom (***g***), and oxygen(O) atoms (***h*** and ***i***), and determine their **hybridization**. (2 points)

Atom	Hybridization
Sulfur (<i>f</i>)	
Nitrogen (<i>g</i>)	
Oxygen (<i>h</i>)	
Oxygen (<i>i</i>)	

iii. **Describe** all the bonds formed by the carbon atoms ***a*** and ***b***, using sigma and pi designation as appropriate. (2 points)

Atom	Bonds formed (use sigma and pi as appropriate)
Carbon (<i>a</i>)	
Carbon (<i>b</i>)	

iv. How many total sp^3 -s sigma bonds are in the molecule? (1 point)

v. Determine **electronic** and **molecular geometry** at carbon (C) labeled ***e*** and sulfur (S) labeled ***f*** centres. (2 points)

Atom	Electronic Geometry	Molecular Geometry
Carbon (<i>e</i>)		
Sulfur (<i>f</i>)		

vi. Circle the value from the choices given below, that describes the **C-S-C bond angle at sulfur** (S) marked ***f***: (0.5 points)

>90° to <109.5° 109.5° >109.5 to <120° 120° >120° to <180°


Question 8 12 points

Using the **valence shell orbitals only** draw a **complete** molecular orbital (MO) diagram (with no s-p mixing) for the molecule **PS**, in the space provided below. **(6 points)**

Assume that 3s and 3p atomic orbitals will form molecular orbitals like those formed by the combination of 2s and 2p atomic orbitals that we learnt in class, and there is no s-p mixing.

Show **energy arrow**, **atomic orbitals** involved, **atom labels**, the **molecular orbitals** that are formed, **filled electrons** in the MO diagram etc. Make sure to label all the atomic and molecular orbitals.

Using the MO diagram above, answer the following questions:

- (i) Which one of the following will have the highest bond order? Circle the correct choice
 A) PS^{1-} B) PS C) PS^{1+}
- (ii) Which one of the following will have the longest bond length? Circle the correct choice
 A) PS^{1-} B) PS C) PS^{1+}
- (iii) Which of the following will be diamagnetic? Circle the correct choice.
 A) PS^{1-} B) PS C) PS^{1+}
- (iv) Draw Lewis dot structure of the molecule with the shortest bond length (based on the MO diagram) from among the three (PS^{1-} , PS, PS^{1+}), and indicate the bond order calculated from the Lewis dot structure. **(3 points)**

Lewis Dot Structure

Bond Order



ADFBF80E-6701-4915-9A2B-620A2E8779E2

midterm-exam-93388

#656 Page 10 of 10

SCRAP PAGE

**CHEMISTRY 110 –Midterm Exam****Thursday November 7, 2024, 6:30 – 8:30 pm****Examiners: Pallavi Sirjoosingh and Ashok Kakkar**

First name (please write as legibly as possible within the boxes)

--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--

Last name

--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--

Student ID number

--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--

This booklet has 8 questions on 10 pages (The last page can be used as a scrap page).

The staple must NOT be removed from the question booklet. Your intact question booklet must be returned at the end of the exam.

A separate booklet contains a periodic table, and datasheet.

Write your answers directly under the questions in the space provided. Do not write on the QR codes or near the top edges (as it may be cut off).

Please write **legibly and dark enough** for the scan to be readable. For students who use English as a second language, dictionaries are permitted. Calculators are allowed.

DO NOT ASK QUESTIONS OF THE INVIGILATORS about interpretations of the exam questions. If you have a concern about a question, please summarize it at the top of this page. All concerns will be evaluated after the exam. GOOD LUCK!

**Question 1 8 points** (*Assume all transitions are possible*)

a. 6 points Calculate the **energy (in kJ/mol)** required to ionize (completely remove) the electron from the **n=3** level of **He⁺ ion** (He: Helium). Show your work including equations. (3 sig. figs.)

b. 2 points

(I) Select the largest (longest) wavelength from the three choices below:

- i. Wavelength of photon to *completely remove* an electron from *1s orbital of Hydrogen atom*
- ii. Wavelength of photon to *completely remove* an electron from *1s orbital of Boron atom (B)*
- iii. Wavelength of photon to *completely remove* an electron from *1s orbital of Boron cation (B⁺)*

i or ii or iii

(II) Select the smallest (shortest) wavelength from the three choices below (*for H atom*):

- i. Wavelength of photon *emitted* for an electron transitioning from *2p to 1s orbital*
- ii. Wavelength of photon *absorbed* for an electron transitioning from *2s to 4p orbital*
- iii. Wavelength of photon *absorbed* for an electron transitioning from *1s to 3p orbital*

i or ii or iii



Question 2 8 points



A Chem110 student plots the graph shown on the left, from data obtained for the kinetic energy (eV) of the ejected electron versus frequency (s^{-1}) of the incident light, for an unknown metal. ($1\text{eV} = 1.6022 \times 10^{-19} \text{ J}$)

Based on the plot, answer the questions **a to c** below:

a. 1 point On the plot above, label the **threshold frequency**.

b. 2 points What is the value of the work potential (in J) of the unknown metal?

c. 3 points What is the *longest* wavelength (in nm) required to eject an electron from this metal's surface? Show your work including equation(s) used. (3 sig figs)

d. 2 points In the space provided below, plot the following

(Assume frequency is *above* threshold frequency):

- Kinetic energy* of an ejected electron versus the *intensity* of the incident light (left)
- Number of electrons* ejected versus the *frequency* of the incident light (right)



**Question 3** **6 points****a. 2 points**

In the ground state of Ge (Germanium; $Z = 32$) atom, **name** the following orbitals:

i. Orbital containing the electron that requires the maximum energy to be removed.

ii. Orbital containing the electron that experiences the minimum effective nuclear charge.

b. 2 points

An unknown anion (A^-) has the following *excited* electronic configuration: $[Ar] 4s^2 3d^3 4p^1$

(Ar: Argon; $Z = 18$)

Give the **condensed ground state electronic configuration** of the atom, **A** and the cation, **A^{2+}** .
Show your work. (Assume all electronic transitions are possible)

Electronic Configuration of A:

Electronic Configuration of A^{2+} :

c. 2 points

In a single atom, determine how many electrons can have the following set of quantum numbers?

(i) $n = 5; l = 3$

In a single atom, determine how many electrons can have the following set of quantum numbers?

(i) $n = 3; l = 2; m_s = +1/2$

**Question 4** **8 points**

Rank the following from 1 to 4 in **increasing** order. Provide the ranking in the boxes provided.

Note: Atomic number (Z) are provided for relevant atoms

a) Increasing (lowest to highest) order of *ionization energy*: (1= lowest; 4 = highest)

Mg

Mg: Z = 12

Rb

Rb: Z = 37

Cs

Cs: Z = 55

K

K : Z = 19

b) Increasing (lowest to highest) order of *bond angle*: (1= lowest; 4 = highest)

H₂OCCl₄XeF₄

HCN

c) Increasing (smallest to largest) order of *ionic size* (1= smallest; 4 = largest)

S²⁻

S: Z = 16

O²⁻

O: Z = 8

Be²⁺

Be: Z = 4

Na⁺

Na: Z = 11

d) Increasing (lowest to highest) order of *bond energy* (1= lowest; 4 = highest)

P-F

P-Cl

P-I

P-Br

**Question 5** **12 points**

a. 9 points For the molecules HClO_4 and TeF_6 , draw the most stable *Lewis structure* showing appropriate shape and *dashed/wedged bonds*, based on VSEPR. Determine the **molecular geometry** around the central atom. For HClO_4 , label the *most polar bond*.



b. 3 points (Note: Lone pairs are not shown in structures below)

(i) For the three *resonance* structures (1, 2 and 3) shown below, answer the following question:



(I) Circle the *most contributing* resonance structure.

Structure 1

Structure 2

Structure 3

(ii) For the three structures given (4, 5, and 6) below, answer the following questions:



(I) Circle all the structures that represent *resonance structures* of each other:

Structure 4

Structure 5

Structure 6

(II) Among the resonance structures, circle the *most contributing* resonance structure.

Structure 4

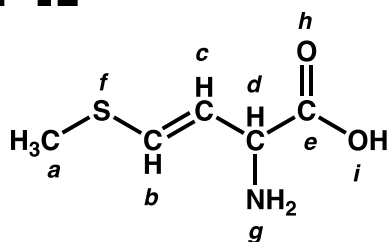
Structure 5

Structure 6

**Question 6** **6 points**

Using the data given below, calculate the **lattice energy** for SrO(s).

	ΔH° (kJ/mol)
$\text{Sr}(s) \longrightarrow \text{Sr}(g)$	+180
$\text{Sr}(g) \longrightarrow \text{Sr}^{2+}(g) + 2e^-$	+1468
$\text{O}_2(g) \longrightarrow 2\text{O}(g)$	+500
$\text{O}(g) + 2e^- \longrightarrow \text{O}^{2-}(g)$	+600
$\text{Sr}(s) + \frac{1}{2}\text{O}_2(g) \longrightarrow \text{SrO}(s)$	-550

**Question 7 10 points**

Consider the molecule given on the left, and answer questions (i) to (vi) using chemical bonding theories that we have learnt in class (valence bond theory, hybridization).

Note that the lone pairs are not shown in the structure:

i. Consider the carbon atoms marked in as ***a***, ***b***, ***c***, ***d***, and ***e*** in the molecule, and determine **hybridization** at each of the carbon centers. (2.5 points)

Atom	Hybridization
Carbon (<i>a</i>)	
Carbon (<i>b</i>)	
Carbon (<i>c</i>)	
Carbon (<i>d</i>)	
Carbon (<i>e</i>)	

ii. Now, consider the sulfur atom (S) marked as ***f***, nitrogen (N) atom (***g***), and oxygen(O) atoms (***h*** and ***i***), and determine their **hybridization**. (2 points)

Atom	Hybridization
Sulfur (<i>f</i>)	
Nitrogen (<i>g</i>)	
Oxygen (<i>h</i>)	
Oxygen (<i>i</i>)	

iii. **Describe** all the bonds formed by the carbon atoms ***a*** and ***b***, using sigma and pi designation as appropriate. (2 points)

Atom	Bonds formed (use sigma and pi as appropriate)
Carbon (<i>a</i>)	
Carbon (<i>b</i>)	

iv. How many total sp^3 -s sigma bonds are in the molecule? (1 point)

v. Determine **electronic** and **molecular geometry** at carbon (C) labeled ***e*** and sulfur (S) labeled ***f*** centres. (2 points)

Atom	Electronic Geometry	Molecular Geometry
Carbon (<i>e</i>)		
Sulfur (<i>f</i>)		

vi. Circle the value from the choices given below, that describes the **C-S-C bond angle at sulfur** (S) marked ***f***: (0.5 points)

>90° to <109.5° 109.5° >109.5 to <120° 120° >120° to <180°

**Question 8 12 points**

Using the **valence shell orbitals only** draw a **complete** molecular orbital (MO) diagram (with no s-p mixing) for the molecule **PS**, in the space provided below. **(6 points)**

Assume that 3s and 3p atomic orbitals will form molecular orbitals like those formed by the combination of 2s and 2p atomic orbitals that we learnt in class, and there is no s-p mixing.

Show **energy arrow**, **atomic orbitals** involved, **atom labels**, the **molecular orbitals** that are formed, **filled electrons** in the MO diagram etc. Make sure to label all the atomic and molecular orbitals.

Using the MO diagram above, answer the following questions:

- (i) Which one of the following will have the highest bond order? Circle the correct choice
A) PS^{1-} B) PS C) PS^{1+}
- (ii) Which one of the following will have the longest bond length? Circle the correct choice
A) PS^{1-} B) PS C) PS^{1+}
- (iii) Which of the following will be diamagnetic? Circle the correct choice.
A) PS^{1-} B) PS C) PS^{1+}
- (iv) Draw Lewis dot structure of the molecule with the shortest bond length (based on the MO diagram) from among the three (PS^{1-} , PS, PS^{1+}), and indicate the bond order calculated from the Lewis dot structure. **(3 points)**

Lewis Dot Structure

Bond Order



A727EE0E-772D-4DBE-97E2-946894E3613F

midterm-exam-93388

#657 Page 10 of 10

SCRAP PAGE

**CHEMISTRY 110 –Midterm Exam****Thursday November 7, 2024, 6:30 – 8:30 pm**

Examiners: Pallavi Sirjoosingh and Ashok Kakkar

First name (please write as legibly as possible within the boxes)

--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--

Last name

--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--

Student ID number

--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--

This booklet has 8 questions on 10 pages (The last page can be used as a scrap page).

The staple must NOT be removed from the question booklet. Your intact question booklet must be returned at the end of the exam.

A separate booklet contains a periodic table, and datasheet.

Write your answers directly under the questions in the space provided. Do not write on the QR codes or near the top edges (as it may be cut off).

Please write **legibly and dark enough** for the scan to be readable. For students who use English as a second language, dictionaries are permitted. Calculators are allowed.

DO NOT ASK QUESTIONS OF THE INVIGILATORS about interpretations of the exam questions. If you have a concern about a question, please summarize it at the top of this page. All concerns will be evaluated after the exam. GOOD LUCK!

**Question 1 8 points** (*Assume all transitions are possible*)

a. 6 points Calculate the **energy (in kJ/mol)** required to ionize (completely remove) the electron from the **n=3** level of **He⁺ ion** (He: Helium). Show your work including equations. (3 sig. figs.)

b. 2 points

(I) Select the largest (longest) wavelength from the three choices below:

- i. Wavelength of photon to *completely remove* an electron from *1s orbital of Hydrogen atom*
- ii. Wavelength of photon to *completely remove* an electron from *1s orbital of Boron atom (B)*
- iii. Wavelength of photon to *completely remove* an electron from *1s orbital of Boron cation (B⁺)*

i or ii or iii

(II) Select the smallest (shortest) wavelength from the three choices below (*for H atom*):

- i. Wavelength of photon *emitted* for an electron transitioning from *2p to 1s orbital*
- ii. Wavelength of photon *absorbed* for an electron transitioning from *2s to 4p orbital*
- iii. Wavelength of photon *absorbed* for an electron transitioning from *1s to 3p orbital*

i or ii or iii



Question 2 8 points



A Chem110 student plots the graph shown on the left, from data obtained for the kinetic energy (eV) of the ejected electron versus frequency (s^{-1}) of the incident light, for an unknown metal. ($1\text{eV} = 1.6022 \times 10^{-19} \text{ J}$)

Based on the plot, answer the questions **a to c** below:

a. 1 point On the plot above, label the **threshold frequency**.

b. 2 points What is the value of the work potential (in J) of the unknown metal?

c. 3 points What is the *longest* wavelength (in nm) required to eject an electron from this metal's surface? Show your work including equation(s) used. (3 sig figs)

d. 2 points In the space provided below, plot the following

(Assume frequency is *above* threshold frequency):

- Kinetic energy* of an ejected electron versus the *intensity* of the incident light (left)
- Number of electrons* ejected versus the *frequency* of the incident light (right)



**Question 3** **6 points****a. 2 points**

In the ground state of Ge (Germanium; $Z = 32$) atom, **name** the following orbitals:

i. Orbital containing the electron that requires the maximum energy to be removed.

ii. Orbital containing the electron that experiences the minimum effective nuclear charge.

b. 2 points

An unknown anion (A^-) has the following *excited* electronic configuration: $[Ar] 4s^2 3d^3 4p^1$

(Ar: Argon; $Z = 18$)

Give the **condensed ground state electronic configuration** of the atom, **A** and the cation, **A^{2+}** .
Show your work. (Assume all electronic transitions are possible)

Electronic Configuration of A:

Electronic Configuration of A^{2+} :

c. 2 points

In a single atom, determine how many electrons can have the following set of quantum numbers?

(i) $n = 5; l = 3$

In a single atom, determine how many electrons can have the following set of quantum numbers?

(i) $n = 3; l = 2; m_s = +1/2$

**Question 4** **8 points**

Rank the following from 1 to 4 in **increasing** order. Provide the ranking in the boxes provided.

Note: Atomic number (Z) are provided for relevant atoms

a) Increasing (lowest to highest) order of *ionization energy*: (1= lowest; 4 = highest)

Mg

Mg: Z = 12

Rb

Rb: Z = 37

Cs

Cs: Z = 55

K

K : Z = 19

b) Increasing (lowest to highest) order of *bond angle*: (1= lowest; 4 = highest)

H₂OCCl₄XeF₄

HCN

c) Increasing (smallest to largest) order of *ionic size* (1= smallest; 4 = largest)

S²⁻

S: Z = 16

O²⁻

O: Z = 8

Be²⁺

Be: Z = 4

Na⁺

Na: Z = 11

d) Increasing (lowest to highest) order of *bond energy* (1= lowest; 4 = highest)

P-F

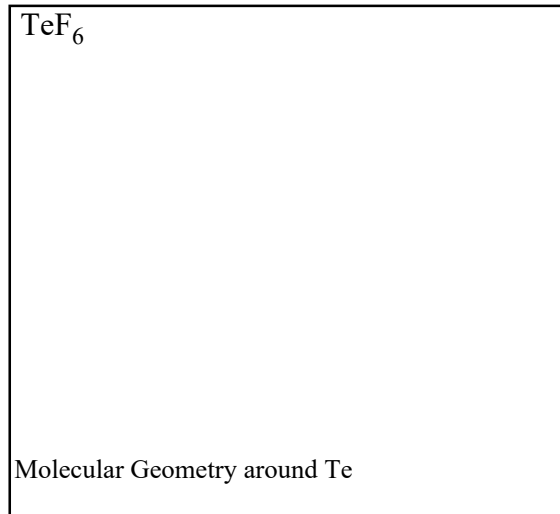
P-Cl

P-I

P-Br

**Question 5** **12 points**

a. 9 points For the molecules HClO_4 and TeF_6 , draw the most stable *Lewis structure* showing appropriate shape and *dashed/wedged bonds*, based on VSEPR. Determine the **molecular geometry** around the central atom. For HClO_4 , label the *most polar bond*.



b. 3 points (Note: Lone pairs are not shown in structures below)

(i) For the three *resonance* structures (1, 2 and 3) shown below, answer the following question:



(I) Circle the *most contributing* resonance structure.

Structure 1

Structure 2

Structure 3

(ii) For the three structures given (4, 5, and 6) below, answer the following questions:



(I) Circle all the structures that represent *resonance structures* of each other:

Structure 4

Structure 5

Structure 6

(II) Among the resonance structures, circle the *most contributing* resonance structure.

Structure 4

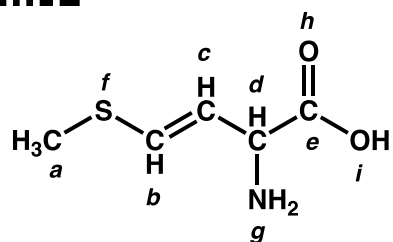
Structure 5

Structure 6

**Question 6** **6 points**

Using the data given below, calculate the **lattice energy** for SrO(s).

	ΔH° (kJ/mol)
$\text{Sr}(s) \longrightarrow \text{Sr}(g)$	+180
$\text{Sr}(g) \longrightarrow \text{Sr}^{2+}(g) + 2e^-$	+1468
$\text{O}_2(g) \longrightarrow 2\text{O}(g)$	+500
$\text{O}(g) + 2e^- \longrightarrow \text{O}^{2-}(g)$	+600
$\text{Sr}(s) + \frac{1}{2}\text{O}_2(g) \longrightarrow \text{SrO}(s)$	-550

**Question 7 10 points**

Consider the molecule given on the left, and answer questions (i) to (vi) using chemical bonding theories that we have learnt in class (valence bond theory, hybridization).

Note that the lone pairs are not shown in the structure:

i. Consider the carbon atoms marked in as ***a***, ***b***, ***c***, ***d***, and ***e*** in the molecule, and determine **hybridization** at each of the carbon centers. (2.5 points)

Atom	Hybridization
Carbon (<i>a</i>)	
Carbon (<i>b</i>)	
Carbon (<i>c</i>)	
Carbon (<i>d</i>)	
Carbon (<i>e</i>)	

ii. Now, consider the sulfur atom (S) marked as ***f***, nitrogen (N) atom (***g***), and oxygen(O) atoms (***h*** and ***i***), and determine their **hybridization**. (2 points)

Atom	Hybridization
Sulfur (<i>f</i>)	
Nitrogen (<i>g</i>)	
Oxygen (<i>h</i>)	
Oxygen (<i>i</i>)	

iii. **Describe** all the bonds formed by the carbon atoms ***a*** and ***b***, using sigma and pi designation as appropriate. (2 points)

Atom	Bonds formed (use sigma and pi as appropriate)
Carbon (<i>a</i>)	
Carbon (<i>b</i>)	

iv. How many total sp^3 -s sigma bonds are in the molecule? (1 point)

v. Determine **electronic** and **molecular geometry** at carbon (C) labeled ***e*** and sulfur (S) labeled ***f*** centres. (2 points)

Atom	Electronic Geometry	Molecular Geometry
Carbon (<i>e</i>)		
Sulfur (<i>f</i>)		

vi. Circle the value from the choices given below, that describes the **C-S-C bond angle at sulfur** (S) marked ***f***: (0.5 points)

>90° to <109.5°

109.5°

>109.5 to <120°

120°

>120° to <180°



Question 8 12 points

Using the **valence shell orbitals only** draw a **complete** molecular orbital (MO) diagram (with no s-p mixing) for the molecule **PS**, in the space provided below. **(6 points)**

Assume that 3s and 3p atomic orbitals will form molecular orbitals like those formed by the combination of 2s and 2p atomic orbitals that we learnt in class, and there is no s-p mixing.

Show **energy arrow**, **atomic orbitals** involved, **atom labels**, the **molecular orbitals** that are formed, **filled electrons** in the MO diagram etc. Make sure to label all the atomic and molecular orbitals.

Using the MO diagram above, answer the following questions:

- (i) Which one of the following will have the highest bond order? Circle the correct choice
 A) PS^{1-} B) PS C) PS^{1+}
- (ii) Which one of the following will have the longest bond length? Circle the correct choice
 A) PS^{1-} B) PS C) PS^{1+}
- (iii) Which of the following will be diamagnetic? Circle the correct choice.
 A) PS^{1-} B) PS C) PS^{1+}
- (iv) Draw Lewis dot structure of the molecule with the shortest bond length (based on the MO diagram) from among the three (PS^{1-} , PS, PS^{1+}), and indicate the bond order calculated from the Lewis dot structure. **(3 points)**

Lewis Dot Structure

Bond Order



1F8DE8C4-3D1D-4004-8A6C-5976CC0C8E52

midterm-exam-93388

#658 Page 10 of 10

SCRAP PAGE

**CHEMISTRY 110 –Midterm Exam****Thursday November 7, 2024, 6:30 – 8:30 pm****Examiners: Pallavi Sirjoosingh and Ashok Kakkar**

First name (please write as legibly as possible within the boxes)

--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--

Last name

--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--

Student ID number

--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--

This booklet has 8 questions on 10 pages (The last page can be used as a scrap page).

The staple must NOT be removed from the question booklet. Your intact question booklet must be returned at the end of the exam.

A separate booklet contains a periodic table, and datasheet.

Write your answers directly under the questions in the space provided. Do not write on the QR codes or near the top edges (as it may be cut off).

Please write **legibly and dark enough** for the scan to be readable. For students who use English as a second language, dictionaries are permitted. Calculators are allowed.

DO NOT ASK QUESTIONS OF THE INVIGILATORS about interpretations of the exam questions. If you have a concern about a question, please summarize it at the top of this page. All concerns will be evaluated after the exam. GOOD LUCK!

**Question 1 8 points** (*Assume all transitions are possible*)

a. 6 points Calculate the **energy (in kJ/mol)** required to ionize (completely remove) the electron from the **$n=3$** level of **He^+ ion** (He: Helium). Show your work including equations. (3 sig. figs.)

b. 2 points

(I) Select the largest (longest) wavelength from the three choices below:

- i. Wavelength of photon to *completely remove* an electron from *1s orbital of Hydrogen atom*
- ii. Wavelength of photon to *completely remove* an electron from *1s orbital of Boron atom (B)*
- iii. Wavelength of photon to *completely remove* an electron from *1s orbital of Boron cation (B^+)*

i or ii or iii

(II) Select the smallest (shortest) wavelength from the three choices below (*for H atom*):

- i. Wavelength of photon *emitted* for an electron transitioning from *2p to 1s orbital*
- ii. Wavelength of photon *absorbed* for an electron transitioning from *2s to 4p orbital*
- iii. Wavelength of photon *absorbed* for an electron transitioning from *1s to 3p orbital*

i or ii or iii



Question 2 8 points



A Chem110 student plots the graph shown on the left, from data obtained for the kinetic energy (eV) of the ejected electron versus frequency (s^{-1}) of the incident light, for an unknown metal. ($1\text{eV} = 1.6022 \times 10^{-19} \text{ J}$)

Based on the plot, answer the questions **a to c** below:

a. 1 point On the plot above, label the **threshold frequency**.

b. 2 points What is the value of the work potential (in J) of the unknown metal?

c. 3 points What is the *longest* wavelength (in nm) required to eject an electron from this metal's surface? Show your work including equation(s) used. (3 sig figs)

d. 2 points In the space provided below, plot the following

(Assume frequency is *above* threshold frequency):

- Kinetic energy* of an ejected electron versus the *intensity* of the incident light (left)
- Number of electrons* ejected versus the *frequency* of the incident light (right)



**Question 3** **6 points****a. 2 points**

In the ground state of Ge (Germanium; $Z = 32$) atom, **name** the following orbitals:

i. Orbital containing the electron that requires the maximum energy to be removed.

ii. Orbital containing the electron that experiences the minimum effective nuclear charge.

b. 2 points

An unknown anion (A^-) has the following *excited* electronic configuration: $[Ar] 4s^2 3d^3 4p^1$

(Ar: Argon; $Z = 18$)

Give the **condensed ground state electronic configuration** of the atom, **A** and the cation, **A^{2+}** .
Show your work. (Assume all electronic transitions are possible)

Electronic Configuration of A:

Electronic Configuration of A^{2+} :

c. 2 points

In a single atom, determine how many electrons can have the following set of quantum numbers?

(i) $n = 5; l = 3$

In a single atom, determine how many electrons can have the following set of quantum numbers?

(i) $n = 3; l = 2; m_s = +1/2$

**Question 4** **8 points**

Rank the following from 1 to 4 in **increasing** order. Provide the ranking in the boxes provided.

Note: Atomic number (Z) are provided for relevant atoms

a) Increasing (lowest to highest) order of *ionization energy*: (1= lowest; 4 = highest)

Mg

Mg: Z = 12

Rb

Rb: Z = 37

Cs

Cs: Z = 55

K

K : Z = 19

b) Increasing (lowest to highest) order of *bond angle*: (1= lowest; 4 = highest)

H₂OCCl₄XeF₄

HCN

c) Increasing (smallest to largest) order of *ionic size* (1= smallest; 4 = largest)

S²⁻

S: Z = 16

O²⁻

O: Z = 8

Be²⁺

Be: Z = 4

Na⁺

Na: Z = 11

d) Increasing (lowest to highest) order of *bond energy* (1= lowest; 4 = highest)

P-F

P-Cl

P-I

P-Br

**Question 5** **12 points**

a. 9 points For the molecules HClO_4 and TeF_6 , draw the most stable *Lewis structure* showing appropriate shape and *dashed/wedged bonds*, based on VSEPR. Determine the **molecular geometry** around the central atom. For HClO_4 , label the *most polar bond*.



b. 3 points (Note: Lone pairs are not shown in structures below)

(i) For the three *resonance* structures (1, 2 and 3) shown below, answer the following question:



(I) Circle the *most contributing* resonance structure.

Structure 1

Structure 2

Structure 3

(ii) For the three structures given (4, 5, and 6) below, answer the following questions:



(I) Circle all the structures that represent *resonance structures* of each other:

Structure 4

Structure 5

Structure 6

(II) Among the resonance structures, circle the *most contributing* resonance structure.

Structure 4

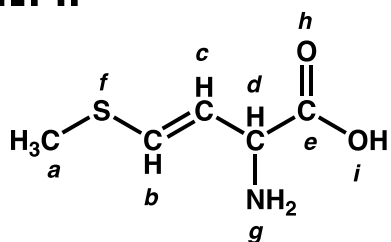
Structure 5

Structure 6

**Question 6** **6 points**

Using the data given below, calculate the **lattice energy** for SrO(s).

	ΔH° (kJ/mol)
$\text{Sr}(s) \longrightarrow \text{Sr}(g)$	+180
$\text{Sr}(g) \longrightarrow \text{Sr}^{2+}(g) + 2e^-$	+1468
$\text{O}_2(g) \longrightarrow 2\text{O}(g)$	+500
$\text{O}(g) + 2e^- \longrightarrow \text{O}^{2-}(g)$	+600
$\text{Sr}(s) + \frac{1}{2}\text{O}_2(g) \longrightarrow \text{SrO}(s)$	-550

**Question 7 10 points**

Consider the molecule given on the left, and answer questions (i) to (vi) using chemical bonding theories that we have learnt in class (valence bond theory, hybridization).

Note that the lone pairs are not shown in the structure:

i. Consider the carbon atoms marked in as ***a***, ***b***, ***c***, ***d***, and ***e*** in the molecule, and determine **hybridization** at each of the carbon centers. (2.5 points)

Atom	Hybridization
Carbon (<i>a</i>)	
Carbon (<i>b</i>)	
Carbon (<i>c</i>)	
Carbon (<i>d</i>)	
Carbon (<i>e</i>)	

ii. Now, consider the sulfur atom (S) marked as ***f***, nitrogen (N) atom (***g***), and oxygen(O) atoms (***h*** and ***i***), and determine their **hybridization**. (2 points)

Atom	Hybridization
Sulfur (<i>f</i>)	
Nitrogen (<i>g</i>)	
Oxygen (<i>h</i>)	
Oxygen (<i>i</i>)	

iii. **Describe** all the bonds formed by the carbon atoms ***a*** and ***b***, using sigma and pi designation as appropriate. (2 points)

Atom	Bonds formed (use sigma and pi as appropriate)
Carbon (<i>a</i>)	
Carbon (<i>b</i>)	

iv. How many total sp^3 -s sigma bonds are in the molecule? (1 point)

v. Determine **electronic** and **molecular geometry** at carbon (C) labeled ***e*** and sulfur (S) labeled ***f*** centres. (2 points)

Atom	Electronic Geometry	Molecular Geometry
Carbon (<i>e</i>)		
Sulfur (<i>f</i>)		

vi. Circle the value from the choices given below, that describes the **C-S-C bond angle at sulfur** (S) marked ***f***: (0.5 points)

>90° to <109.5° 109.5° >109.5 to <120° 120° >120° to <180°

**Question 8 12 points**

Using the **valence shell orbitals only** draw a **complete** molecular orbital (MO) diagram (with no s-p mixing) for the molecule **PS**, in the space provided below. **(6 points)**

Assume that 3s and 3p atomic orbitals will form molecular orbitals like those formed by the combination of 2s and 2p atomic orbitals that we learnt in class, and there is no s-p mixing.

Show **energy arrow**, **atomic orbitals** involved, **atom labels**, the **molecular orbitals** that are formed, **filled electrons** in the MO diagram etc. Make sure to label all the atomic and molecular orbitals.

Using the MO diagram above, answer the following questions:

- (i) Which one of the following will have the highest bond order? Circle the correct choice
A) PS^{1-} B) PS C) PS^{1+}
- (ii) Which one of the following will have the longest bond length? Circle the correct choice
A) PS^{1-} B) PS C) PS^{1+}
- (iii) Which of the following will be diamagnetic? Circle the correct choice.
A) PS^{1-} B) PS C) PS^{1+}
- (iv) Draw Lewis dot structure of the molecule with the shortest bond length (based on the MO diagram) from among the three (PS^{1-} , PS, PS^{1+}), and indicate the bond order calculated from the Lewis dot structure. **(3 points)**

Lewis Dot Structure

Bond Order



1AA6B771-4AFF-4251-B1C4-5E4BA8A6B4D0

midterm-exam-93388

#659 Page 10 of 10

SCRAP PAGE

**CHEMISTRY 110 –Midterm Exam****Thursday November 7, 2024, 6:30 – 8:30 pm**

Examiners: Pallavi Sirjoosingh and Ashok Kakkar

First name (please write as legibly as possible within the boxes)

--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--

Last name

--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--

Student ID number

--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--

This booklet has 8 questions on 10 pages (The last page can be used as a scrap page).

The staple must NOT be removed from the question booklet. Your intact question booklet must be returned at the end of the exam.

A separate booklet contains a periodic table, and datasheet.

Write your answers directly under the questions in the space provided. Do not write on the QR codes or near the top edges (as it may be cut off).

Please write **legibly and dark enough** for the scan to be readable. For students who use English as a second language, dictionaries are permitted. Calculators are allowed.

DO NOT ASK QUESTIONS OF THE INVIGILATORS about interpretations of the exam questions. If you have a concern about a question, please summarize it at the top of this page. All concerns will be evaluated after the exam. GOOD LUCK!

**Question 1 8 points** (*Assume all transitions are possible*)

a. 6 points Calculate the **energy (in kJ/mol)** required to ionize (completely remove) the electron from the **n=3** level of **He⁺ ion** (He: Helium). Show your work including equations. (3 sig. figs.)

b. 2 points

(I) Select the largest (longest) wavelength from the three choices below:

- i. Wavelength of photon to *completely remove* an electron from *1s orbital of Hydrogen atom*
- ii. Wavelength of photon to *completely remove* an electron from *1s orbital of Boron atom (B)*
- iii. Wavelength of photon to *completely remove* an electron from *1s orbital of Boron cation (B⁺)*

i or ii or iii

(II) Select the smallest (shortest) wavelength from the three choices below (*for H atom*):

- i. Wavelength of photon *emitted* for an electron transitioning from *2p to 1s orbital*
- ii. Wavelength of photon *absorbed* for an electron transitioning from *2s to 4p orbital*
- iii. Wavelength of photon *absorbed* for an electron transitioning from *1s to 3p orbital*

i or ii or iii



Question 2 8 points



A Chem110 student plots the graph shown on the left, from data obtained for the kinetic energy (eV) of the ejected electron versus frequency (s^{-1}) of the incident light, for an unknown metal. ($1\text{eV} = 1.6022 \times 10^{-19} \text{ J}$)

Based on the plot, answer the questions **a to c** below:

a. 1 point On the plot above, label the **threshold frequency**.

b. 2 points What is the value of the work potential (in J) of the unknown metal?

c. 3 points What is the *longest* wavelength (in nm) required to eject an electron from this metal's surface? Show your work including equation(s) used. (3 sig figs)

d. 2 points In the space provided below, plot the following

(Assume frequency is *above* threshold frequency):

- i. *Kinetic energy* of an ejected electron versus the *intensity* of the incident light (left)
- ii. *Number of electrons* ejected versus the *frequency* of the incident light (right)



**Question 3** **6 points****a. 2 points**

In the ground state of Ge (Germanium; $Z = 32$) atom, **name** the following orbitals:

i. Orbital containing the electron that requires the maximum energy to be removed.

ii. Orbital containing the electron that experiences the minimum effective nuclear charge.

b. 2 points

An unknown anion (A^-) has the following *excited* electronic configuration: $[Ar] 4s^2 3d^3 4p^1$

(Ar: Argon; $Z = 18$)

Give the **condensed ground state electronic configuration** of the atom, **A** and the cation, **A^{2+}** .
Show your work. (Assume all electronic transitions are possible)

Electronic Configuration of A:

Electronic Configuration of A^{2+} :

c. 2 points

In a single atom, determine how many electrons can have the following set of quantum numbers?

(i) $n = 5; l = 3$

In a single atom, determine how many electrons can have the following set of quantum numbers?

(i) $n = 3; l = 2; m_s = +1/2$

**Question 4** **8 points**

Rank the following from 1 to 4 in **increasing** order. Provide the ranking in the boxes provided.

Note: Atomic number (Z) are provided for relevant atoms

a) Increasing (lowest to highest) order of *ionization energy*: (1= lowest; 4 = highest)

Mg

Mg: Z = 12

Rb

Rb: Z = 37

Cs

Cs: Z = 55

K

K : Z = 19

b) Increasing (lowest to highest) order of *bond angle*: (1= lowest; 4 = highest)

H₂OCCl₄XeF₄

HCN

c) Increasing (smallest to largest) order of *ionic size* (1= smallest; 4 = largest)

S²⁻

S: Z = 16

O²⁻

O: Z = 8

Be²⁺

Be: Z = 4

Na⁺

Na: Z = 11

d) Increasing (lowest to highest) order of *bond energy* (1= lowest; 4 = highest)

P-F

P-Cl

P-I

P-Br

**Question 5** **12 points**

a. 9 points For the molecules HClO_4 and TeF_6 , draw the most stable *Lewis structure* showing appropriate shape and *dashed/wedged bonds*, based on VSEPR. Determine the **molecular geometry** around the central atom. For HClO_4 , label the *most polar bond*.



b. 3 points (Note: Lone pairs are not shown in structures below)

(i) For the three *resonance* structures (1, 2 and 3) shown below, answer the following question:



(I) Circle the *most contributing* resonance structure.

Structure 1

Structure 2

Structure 3

(ii) For the three structures given (4, 5, and 6) below, answer the following questions:



(I) Circle all the structures that represent *resonance structures* of each other:

Structure 4

Structure 5

Structure 6

(II) Among the resonance structures, circle the *most contributing* resonance structure.

Structure 4

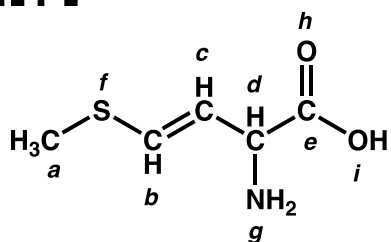
Structure 5

Structure 6

**Question 6** **6 points**

Using the data given below, calculate the **lattice energy** for SrO(s).

	ΔH° (kJ/mol)
$\text{Sr}(s) \longrightarrow \text{Sr}(g)$	+180
$\text{Sr}(g) \longrightarrow \text{Sr}^{2+}(g) + 2e^-$	+1468
$\text{O}_2(g) \longrightarrow 2\text{O}(g)$	+500
$\text{O}(g) + 2e^- \longrightarrow \text{O}^{2-}(g)$	+600
$\text{Sr}(s) + \frac{1}{2}\text{O}_2(g) \longrightarrow \text{SrO}(s)$	-550

**Question 7 10 points**

Consider the molecule given on the left, and answer questions (i) to (vi) using chemical bonding theories that we have learnt in class (valence bond theory, hybridization).

Note that the lone pairs are not shown in the structure:

i. Consider the carbon atoms marked in as ***a***, ***b***, ***c***, ***d***, and ***e*** in the molecule, and determine **hybridization** at each of the carbon centers. (2.5 points)

Atom	Hybridization
Carbon (<i>a</i>)	
Carbon (<i>b</i>)	
Carbon (<i>c</i>)	
Carbon (<i>d</i>)	
Carbon (<i>e</i>)	

ii. Now, consider the sulfur atom (S) marked as ***f***, nitrogen (N) atom (***g***), and oxygen(O) atoms (***h*** and ***i***), and determine their **hybridization**. (2 points)

Atom	Hybridization
Sulfur (<i>f</i>)	
Nitrogen (<i>g</i>)	
Oxygen (<i>h</i>)	
Oxygen (<i>i</i>)	

iii. **Describe** all the bonds formed by the carbon atoms ***a*** and ***b***, using sigma and pi designation as appropriate. (2 points)

Atom	Bonds formed (use sigma and pi as appropriate)
Carbon (<i>a</i>)	
Carbon (<i>b</i>)	

iv. How many total sp^3 -s sigma bonds are in the molecule? (1 point)

v. Determine **electronic** and **molecular geometry** at carbon (C) labeled ***e*** and sulfur (S) labeled ***f*** centres. (2 points)

Atom	Electronic Geometry	Molecular Geometry
Carbon (<i>e</i>)		
Sulfur (<i>f</i>)		

vi. Circle the value from the choices given below, that describes the **C-S-C bond angle at sulfur** (S) marked ***f***: (0.5 points)

>90° to <109.5° 109.5° >109.5 to <120° 120° >120° to <180°

**Question 8 12 points**

Using the **valence shell orbitals only** draw a **complete** molecular orbital (MO) diagram (with no s-p mixing) for the molecule **PS**, in the space provided below. **(6 points)**

Assume that 3s and 3p atomic orbitals will form molecular orbitals like those formed by the combination of 2s and 2p atomic orbitals that we learnt in class, and there is no s-p mixing.

Show **energy arrow**, **atomic orbitals** involved, **atom labels**, the **molecular orbitals** that are formed, **filled electrons** in the MO diagram etc. Make sure to label all the atomic and molecular orbitals.

Using the MO diagram above, answer the following questions:

- (i) Which one of the following will have the highest bond order? Circle the correct choice
A) PS^{1-} B) PS C) PS^{1+}
- (ii) Which one of the following will have the longest bond length? Circle the correct choice
A) PS^{1-} B) PS C) PS^{1+}
- (iii) Which of the following will be diamagnetic? Circle the correct choice.
A) PS^{1-} B) PS C) PS^{1+}
- (iv) Draw Lewis dot structure of the molecule with the shortest bond length (based on the MO diagram) from among the three (PS^{1-} , PS, PS^{1+}), and indicate the bond order calculated from the Lewis dot structure. **(3 points)**

Lewis Dot Structure

Bond Order



7B597ADE-53A0-4CC9-897A-1A2FB5808F3F

midterm-exam-93388

#660 Page 10 of 10

SCRAP PAGE

**CHEMISTRY 110 –Midterm Exam****Thursday November 7, 2024, 6:30 – 8:30 pm****Examiners: Pallavi Sirjoosingh and Ashok Kakkar**

First name (please write as legibly as possible within the boxes)

--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--

Last name

--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--

Student ID number

--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--

This booklet has 8 questions on 10 pages (The last page can be used as a scrap page).

The staple must NOT be removed from the question booklet. Your intact question booklet must be returned at the end of the exam.

A separate booklet contains a periodic table, and datasheet.

Write your answers directly under the questions in the space provided. Do not write on the QR codes or near the top edges (as it may be cut off).

Please write **legibly and dark enough** for the scan to be readable. For students who use English as a second language, dictionaries are permitted. Calculators are allowed.

DO NOT ASK QUESTIONS OF THE INVIGILATORS about interpretations of the exam questions. If you have a concern about a question, please summarize it at the top of this page. All concerns will be evaluated after the exam. GOOD LUCK!

**Question 1 8 points** (*Assume all transitions are possible*)

a. 6 points Calculate the **energy (in kJ/mol)** required to ionize (completely remove) the electron from the **$n=3$** level of **He^+ ion** (He: Helium). Show your work including equations. (3 sig. figs.)

b. 2 points

(I) Select the largest (longest) wavelength from the three choices below:

- i. Wavelength of photon to *completely remove* an electron from *1s orbital of Hydrogen atom*
- ii. Wavelength of photon to *completely remove* an electron from *1s orbital of Boron atom (B)*
- iii. Wavelength of photon to *completely remove* an electron from *1s orbital of Boron cation (B^+)*

i or ii or iii

(II) Select the smallest (shortest) wavelength from the three choices below (*for H atom*):

- i. Wavelength of photon *emitted* for an electron transitioning from *2p to 1s orbital*
- ii. Wavelength of photon *absorbed* for an electron transitioning from *2s to 4p orbital*
- iii. Wavelength of photon *absorbed* for an electron transitioning from *1s to 3p orbital*

i or ii or iii



Question 2 8 points



A Chem110 student plots the graph shown on the left, from data obtained for the kinetic energy (eV) of the ejected electron versus frequency (s^{-1}) of the incident light, for an unknown metal. ($1\text{eV} = 1.6022 \times 10^{-19} \text{ J}$)

Based on the plot, answer the questions **a to c** below:

a. 1 point On the plot above, label the **threshold frequency**.

b. 2 points What is the value of the work potential (in J) of the unknown metal?

c. 3 points What is the *longest* wavelength (in nm) required to eject an electron from this metal's surface? Show your work including equation(s) used. (3 sig figs)

d. 2 points In the space provided below, plot the following

(Assume frequency is *above* threshold frequency):

- i. *Kinetic energy* of an ejected electron versus the *intensity* of the incident light (left)
- ii. *Number of electrons* ejected versus the *frequency* of the incident light (right)



**Question 3** **6 points****a. 2 points**

In the ground state of Ge (Germanium; $Z = 32$) atom, **name** the following orbitals:

i. Orbital containing the electron that requires the maximum energy to be removed.

ii. Orbital containing the electron that experiences the minimum effective nuclear charge.

b. 2 points

An unknown anion (A^-) has the following *excited* electronic configuration: $[Ar] 4s^2 3d^3 4p^1$

(Ar: Argon; $Z = 18$)

Give the **condensed ground state electronic configuration** of the atom, **A** and the cation, **A^{2+}** .
Show your work. (Assume all electronic transitions are possible)

Electronic Configuration of A:

Electronic Configuration of A^{2+} :

c. 2 points

In a single atom, determine how many electrons can have the following set of quantum numbers?

(i) $n = 5; l = 3$

In a single atom, determine how many electrons can have the following set of quantum numbers?

(i) $n = 3; l = 2; m_s = +1/2$

**Question 4** **8 points**

Rank the following from 1 to 4 in **increasing** order. Provide the ranking in the boxes provided.

Note: Atomic number (Z) are provided for relevant atoms

a) Increasing (lowest to highest) order of *ionization energy*: (1= lowest; 4 = highest)

Mg

Mg: Z = 12

Rb

Rb: Z = 37

Cs

Cs: Z = 55

K

K : Z = 19

b) Increasing (lowest to highest) order of *bond angle*: (1= lowest; 4 = highest)

H₂OCCl₄XeF₄

HCN

c) Increasing (smallest to largest) order of *ionic size* (1= smallest; 4 = largest)

S²⁻

S: Z = 16

O²⁻

O: Z = 8

Be²⁺

Be: Z = 4

Na⁺

Na: Z = 11

d) Increasing (lowest to highest) order of *bond energy* (1= lowest; 4 = highest)

P-F

P-Cl

P-I

P-Br

**Question 5** **12 points**

a. 9 points For the molecules HClO_4 and TeF_6 , draw the most stable *Lewis structure* showing appropriate shape and *dashed/wedged bonds*, based on VSEPR. Determine the **molecular geometry** around the central atom. For HClO_4 , label the *most polar bond*.



b. 3 points (Note: Lone pairs are not shown in structures below)

(i) For the three *resonance* structures (1, 2 and 3) shown below, answer the following question:



(I) Circle the *most contributing* resonance structure.

Structure 1

Structure 2

Structure 3

(ii) For the three structures given (4, 5, and 6) below, answer the following questions:



(I) Circle all the structures that represent *resonance structures* of each other:

Structure 4

Structure 5

Structure 6

(II) Among the resonance structures, circle the *most contributing* resonance structure.

Structure 4

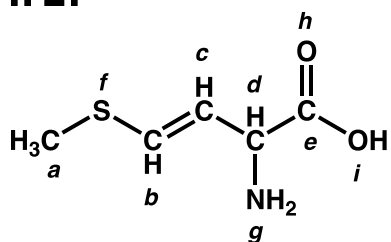
Structure 5

Structure 6

**Question 6** **6 points**

Using the data given below, calculate the **lattice energy** for SrO(s).

	ΔH° (kJ/mol)
$\text{Sr}(s) \longrightarrow \text{Sr}(g)$	+180
$\text{Sr}(g) \longrightarrow \text{Sr}^{2+}(g) + 2e^-$	+1468
$\text{O}_2(g) \longrightarrow 2\text{O}(g)$	+500
$\text{O}(g) + 2e^- \longrightarrow \text{O}^{2-}(g)$	+600
$\text{Sr}(s) + \frac{1}{2}\text{O}_2(g) \longrightarrow \text{SrO}(s)$	-550

**Question 7 10 points**

Consider the molecule given on the left, and answer questions (i) to (vi) using chemical bonding theories that we have learnt in class (valence bond theory, hybridization).

Note that the lone pairs are not shown in the structure:

i. Consider the carbon atoms marked in as ***a***, ***b***, ***c***, ***d***, and ***e*** in the molecule, and determine **hybridization** at each of the carbon centers. (2.5 points)

Atom	Hybridization
Carbon (<i>a</i>)	
Carbon (<i>b</i>)	
Carbon (<i>c</i>)	
Carbon (<i>d</i>)	
Carbon (<i>e</i>)	

ii. Now, consider the sulfur atom (S) marked as ***f***, nitrogen (N) atom (***g***), and oxygen(O) atoms (***h*** and ***i***), and determine their **hybridization**. (2 points)

Atom	Hybridization
Sulfur (<i>f</i>)	
Nitrogen (<i>g</i>)	
Oxygen (<i>h</i>)	
Oxygen (<i>i</i>)	

iii. **Describe** all the bonds formed by the carbon atoms ***a*** and ***b***, using sigma and pi designation as appropriate. (2 points)

Atom	Bonds formed (use sigma and pi as appropriate)
Carbon (<i>a</i>)	
Carbon (<i>b</i>)	

iv. How many total sp^3 -s sigma bonds are in the molecule? (1 point)

v. Determine **electronic** and **molecular geometry** at carbon (C) labeled ***e*** and sulfur (S) labeled ***f*** centres. (2 points)

Atom	Electronic Geometry	Molecular Geometry
Carbon (<i>e</i>)		
Sulfur (<i>f</i>)		

vi. Circle the value from the choices given below, that describes the **C-S-C bond angle at sulfur** (S) marked ***f***: (0.5 points)

>90° to <109.5° 109.5° >109.5 to <120° 120° >120° to <180°


Question 8 12 points

Using the **valence shell orbitals only** draw a **complete** molecular orbital (MO) diagram (with no s-p mixing) for the molecule **PS**, in the space provided below. **(6 points)**

Assume that 3s and 3p atomic orbitals will form molecular orbitals like those formed by the combination of 2s and 2p atomic orbitals that we learnt in class, and there is no s-p mixing.

Show **energy arrow**, **atomic orbitals** involved, **atom labels**, the **molecular orbitals** that are formed, **filled electrons** in the MO diagram etc. Make sure to label all the atomic and molecular orbitals.

Using the MO diagram above, answer the following questions:

- (i) Which one of the following will have the highest bond order? Circle the correct choice
 A) PS^{1-} B) PS C) PS^{1+}
- (ii) Which one of the following will have the longest bond length? Circle the correct choice
 A) PS^{1-} B) PS C) PS^{1+}
- (iii) Which of the following will be diamagnetic? Circle the correct choice.
 A) PS^{1-} B) PS C) PS^{1+}
- (iv) Draw Lewis dot structure of the molecule with the shortest bond length (based on the MO diagram) from among the three (PS^{1-} , PS, PS^{1+}), and indicate the bond order calculated from the Lewis dot structure. **(3 points)**

Lewis Dot Structure

Bond Order



EA0AE190-0093-4307-BF57-054B57879517

midterm-exam-93388

#661 Page 10 of 10

SCRAP PAGE

**CHEMISTRY 110 –Midterm Exam****Thursday November 7, 2024, 6:30 – 8:30 pm**

Examiners: Pallavi Sirjoosingh and Ashok Kakkar

First name (please write as legibly as possible within the boxes)

--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--

Last name

--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--

Student ID number

--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--

This booklet has 8 questions on 10 pages (The last page can be used as a scrap page).

The staple must NOT be removed from the question booklet. Your intact question booklet must be returned at the end of the exam.

A separate booklet contains a periodic table, and datasheet.

Write your answers directly under the questions in the space provided. Do not write on the QR codes or near the top edges (as it may be cut off).

Please write **legibly and dark enough** for the scan to be readable. For students who use English as a second language, dictionaries are permitted. Calculators are allowed.

DO NOT ASK QUESTIONS OF THE INVIGILATORS about interpretations of the exam questions. If you have a concern about a question, please summarize it at the top of this page. All concerns will be evaluated after the exam. GOOD LUCK!

**Question 1 8 points** (*Assume all transitions are possible*)

a. 6 points Calculate the **energy (in kJ/mol)** required to ionize (completely remove) the electron from the **n=3** level of **He⁺ ion** (He: Helium). Show your work including equations. (3 sig. figs.)

b. 2 points

(I) Select the largest (longest) wavelength from the three choices below:

- i. Wavelength of photon to *completely remove* an electron from *1s orbital of Hydrogen atom*
- ii. Wavelength of photon to *completely remove* an electron from *1s orbital of Boron atom (B)*
- iii. Wavelength of photon to *completely remove* an electron from *1s orbital of Boron cation (B⁺)*

i or ii or iii

(II) Select the smallest (shortest) wavelength from the three choices below (*for H atom*):

- i. Wavelength of photon *emitted* for an electron transitioning from *2p to 1s orbital*
- ii. Wavelength of photon *absorbed* for an electron transitioning from *2s to 4p orbital*
- iii. Wavelength of photon *absorbed* for an electron transitioning from *1s to 3p orbital*

i or ii or iii



Question 2 8 points



A Chem110 student plots the graph shown on the left, from data obtained for the kinetic energy (eV) of the ejected electron versus frequency (s^{-1}) of the incident light, for an unknown metal. ($1\text{eV} = 1.6022 \times 10^{-19} \text{ J}$)

Based on the plot, answer the questions **a to c** below:

a. 1 point On the plot above, label the **threshold frequency**.

b. 2 points What is the value of the work potential (in J) of the unknown metal?

c. 3 points What is the *longest* wavelength (in nm) required to eject an electron from this metal's surface? Show your work including equation(s) used. (3 sig figs)

d. 2 points In the space provided below, plot the following

(Assume frequency is *above* threshold frequency):

- i. *Kinetic energy* of an ejected electron versus the *intensity* of the incident light (left)
- ii. *Number of electrons* ejected versus the *frequency* of the incident light (right)



**Question 3** **6 points****a. 2 points**

In the ground state of Ge (Germanium; $Z = 32$) atom, **name** the following orbitals:

i. Orbital containing the electron that requires the maximum energy to be removed.

ii. Orbital containing the electron that experiences the minimum effective nuclear charge.

b. 2 points

An unknown anion (A^-) has the following *excited* electronic configuration: $[Ar] 4s^2 3d^3 4p^1$

(Ar: Argon; $Z = 18$)

Give the **condensed ground state electronic configuration** of the atom, **A** and the cation, **A^{2+}** .
Show your work. (Assume all electronic transitions are possible)

Electronic Configuration of A:

Electronic Configuration of A^{2+} :

c. 2 points

In a single atom, determine how many electrons can have the following set of quantum numbers?

(i) $n = 5; l = 3$

In a single atom, determine how many electrons can have the following set of quantum numbers?

(i) $n = 3; l = 2; m_s = +1/2$

**Question 4** **8 points**

Rank the following from 1 to 4 in **increasing** order. Provide the ranking in the boxes provided.

Note: Atomic number (Z) are provided for relevant atoms

a) Increasing (lowest to highest) order of *ionization energy*: (1= lowest; 4 = highest)

Mg

Mg: Z = 12

Rb

Rb: Z = 37

Cs

Cs: Z = 55

K

K : Z = 19

b) Increasing (lowest to highest) order of *bond angle*: (1= lowest; 4 = highest)

H₂OCCl₄XeF₄

HCN

c) Increasing (smallest to largest) order of *ionic size* (1= smallest; 4 = largest)

S²⁻

S: Z = 16

O²⁻

O: Z = 8

Be²⁺

Be: Z = 4

Na⁺

Na: Z = 11

d) Increasing (lowest to highest) order of *bond energy* (1= lowest; 4 = highest)

P-F

P-Cl

P-I

P-Br

**Question 5** **12 points**

a. 9 points For the molecules HClO_4 and TeF_6 , draw the most stable *Lewis structure* showing appropriate shape and *dashed/wedged bonds*, based on VSEPR. Determine the **molecular geometry** around the central atom. For HClO_4 , label the *most polar bond*.



b. 3 points (Note: Lone pairs are not shown in structures below)

(i) For the three *resonance* structures (1, 2 and 3) shown below, answer the following question:



(I) Circle the *most contributing* resonance structure.

Structure 1

Structure 2

Structure 3

(ii) For the three structures given (4, 5, and 6) below, answer the following questions:



(I) Circle all the structures that represent *resonance structures* of each other:

Structure 4

Structure 5

Structure 6

(II) Among the resonance structures, circle the *most contributing* resonance structure.

Structure 4

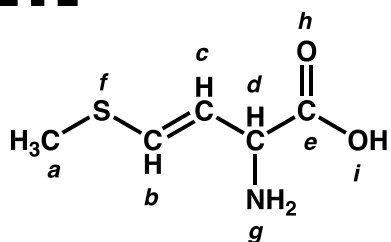
Structure 5

Structure 6

**Question 6** **6 points**

Using the data given below, calculate the **lattice energy** for SrO(s).

	ΔH° (kJ/mol)
$\text{Sr}(s) \longrightarrow \text{Sr}(g)$	+180
$\text{Sr}(g) \longrightarrow \text{Sr}^{2+}(g) + 2e^-$	+1468
$\text{O}_2(g) \longrightarrow 2\text{O}(g)$	+500
$\text{O}(g) + 2e^- \longrightarrow \text{O}^{2-}(g)$	+600
$\text{Sr}(s) + \frac{1}{2}\text{O}_2(g) \longrightarrow \text{SrO}(s)$	-550

**Question 7 10 points**

Consider the molecule given on the left, and answer questions (i) to (vi) using chemical bonding theories that we have learnt in class (valence bond theory, hybridization).

Note that the lone pairs are not shown in the structure:

i. Consider the carbon atoms marked in as ***a***, ***b***, ***c***, ***d***, and ***e*** in the molecule, and determine **hybridization** at each of the carbon centers. (2.5 points)

Atom	Hybridization
Carbon (<i>a</i>)	
Carbon (<i>b</i>)	
Carbon (<i>c</i>)	
Carbon (<i>d</i>)	
Carbon (<i>e</i>)	

ii. Now, consider the sulfur atom (S) marked as ***f***, nitrogen (N) atom (***g***), and oxygen(O) atoms (***h*** and ***i***), and determine their **hybridization**. (2 points)

Atom	Hybridization
Sulfur (<i>f</i>)	
Nitrogen (<i>g</i>)	
Oxygen (<i>h</i>)	
Oxygen (<i>i</i>)	

iii. **Describe** all the bonds formed by the carbon atoms ***a*** and ***b***, using sigma and pi designation as appropriate. (2 points)

Atom	Bonds formed (use sigma and pi as appropriate)
Carbon (<i>a</i>)	
Carbon (<i>b</i>)	

iv. How many total sp^3 -s sigma bonds are in the molecule? (1 point)

v. Determine **electronic** and **molecular geometry** at carbon (C) labeled ***e*** and sulfur (S) labeled ***f*** centres. (2 points)

Atom	Electronic Geometry	Molecular Geometry
Carbon (<i>e</i>)		
Sulfur (<i>f</i>)		

vi. Circle the value from the choices given below, that describes the **C-S-C bond angle at sulfur** (S) marked ***f***: (0.5 points)

>90° to <109.5°

109.5°

>109.5 to <120°

120°

>120° to <180°


Question 8 12 points

Using the **valence shell orbitals only** draw a **complete** molecular orbital (MO) diagram (with no s-p mixing) for the molecule **PS**, in the space provided below. **(6 points)**

Assume that 3s and 3p atomic orbitals will form molecular orbitals like those formed by the combination of 2s and 2p atomic orbitals that we learnt in class, and there is no s-p mixing.

Show **energy arrow**, **atomic orbitals** involved, **atom labels**, the **molecular orbitals** that are formed, **filled electrons** in the MO diagram etc. Make sure to label all the atomic and molecular orbitals.

Using the MO diagram above, answer the following questions:

- (i) Which one of the following will have the highest bond order? Circle the correct choice
 A) PS^{1-} B) PS C) PS^{1+}
- (ii) Which one of the following will have the longest bond length? Circle the correct choice
 A) PS^{1-} B) PS C) PS^{1+}
- (iii) Which of the following will be diamagnetic? Circle the correct choice.
 A) PS^{1-} B) PS C) PS^{1+}
- (iv) Draw Lewis dot structure of the molecule with the shortest bond length (based on the MO diagram) from among the three (PS^{1-} , PS, PS^{1+}), and indicate the bond order calculated from the Lewis dot structure. **(3 points)**

Lewis Dot Structure

Bond Order



7EA987CF-E034-4E7D-BBFB-CDB1BDB66439

midterm-exam-93388

#662 Page 10 of 10

SCRAP PAGE

**CHEMISTRY 110 –Midterm Exam****Thursday November 7, 2024, 6:30 – 8:30 pm**

Examiners: Pallavi Sirjoosingh and Ashok Kakkar

First name (please write as legibly as possible within the boxes)

--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--

Last name

--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--

Student ID number

--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--

This booklet has 8 questions on 10 pages (The last page can be used as a scrap page).

The staple must NOT be removed from the question booklet. Your intact question booklet must be returned at the end of the exam.

A separate booklet contains a periodic table, and datasheet.

Write your answers directly under the questions in the space provided. Do not write on the QR codes or near the top edges (as it may be cut off).

Please write **legibly and dark enough** for the scan to be readable. For students who use English as a second language, dictionaries are permitted. Calculators are allowed.

DO NOT ASK QUESTIONS OF THE INVIGILATORS about interpretations of the exam questions. If you have a concern about a question, please summarize it at the top of this page. All concerns will be evaluated after the exam. GOOD LUCK!

**Question 1 8 points** (*Assume all transitions are possible*)

a. 6 points Calculate the **energy (in kJ/mol)** required to ionize (completely remove) the electron from the **n=3** level of **He⁺ ion** (He: Helium). Show your work including equations. (3 sig. figs.)

b. 2 points

(I) Select the largest (longest) wavelength from the three choices below:

- i. Wavelength of photon to *completely remove* an electron from *1s orbital of Hydrogen atom*
- ii. Wavelength of photon to *completely remove* an electron from *1s orbital of Boron atom (B)*
- iii. Wavelength of photon to *completely remove* an electron from *1s orbital of Boron cation (B⁺)*

i or ii or iii

(II) Select the smallest (shortest) wavelength from the three choices below (*for H atom*):

- i. Wavelength of photon *emitted* for an electron transitioning from *2p to 1s orbital*
- ii. Wavelength of photon *absorbed* for an electron transitioning from *2s to 4p orbital*
- iii. Wavelength of photon *absorbed* for an electron transitioning from *1s to 3p orbital*

i or ii or iii



Question 2 8 points



A Chem110 student plots the graph shown on the left, from data obtained for the kinetic energy (eV) of the ejected electron versus frequency (s^{-1}) of the incident light, for an unknown metal. ($1\text{eV} = 1.6022 \times 10^{-19} \text{ J}$)

Based on the plot, answer the questions **a to c** below:

a. 1 point On the plot above, label the **threshold frequency**.

b. 2 points What is the value of the work potential (in J) of the unknown metal?

c. 3 points What is the *longest* wavelength (in nm) required to eject an electron from this metal's surface? Show your work including equation(s) used. (3 sig figs)

d. 2 points In the space provided below, plot the following

(Assume frequency is *above* threshold frequency):

- Kinetic energy* of an ejected electron versus the *intensity* of the incident light (left)
- Number of electrons* ejected versus the *frequency* of the incident light (right)



**Question 3** **6 points****a. 2 points**

In the ground state of Ge (Germanium; $Z = 32$) atom, **name** the following orbitals:

i. Orbital containing the electron that requires the maximum energy to be removed.

ii. Orbital containing the electron that experiences the minimum effective nuclear charge.

b. 2 points

An unknown anion (A^-) has the following *excited* electronic configuration: $[Ar] 4s^2 3d^3 4p^1$

(Ar: Argon; $Z = 18$)

Give the **condensed ground state electronic configuration** of the atom, **A** and the cation, **A^{2+}** .
Show your work. (Assume all electronic transitions are possible)

Electronic Configuration of A:

Electronic Configuration of A^{2+} :

c. 2 points

In a single atom, determine how many electrons can have the following set of quantum numbers?

(i) $n = 5; l = 3$

In a single atom, determine how many electrons can have the following set of quantum numbers?

(i) $n = 3; l = 2; m_s = +1/2$

**Question 4** **8 points**

Rank the following from 1 to 4 in **increasing** order. Provide the ranking in the boxes provided.

Note: Atomic number (Z) are provided for relevant atoms

a) Increasing (lowest to highest) order of *ionization energy*: (1= lowest; 4 = highest)

Mg

Mg: Z = 12

Rb

Rb: Z = 37

Cs

Cs: Z = 55

K

K : Z = 19

b) Increasing (lowest to highest) order of *bond angle*: (1= lowest; 4 = highest)

H₂OCCl₄XeF₄

HCN

c) Increasing (smallest to largest) order of *ionic size* (1= smallest; 4 = largest)

S²⁻

S: Z = 16

O²⁻

O: Z = 8

Be²⁺

Be: Z = 4

Na⁺

Na: Z = 11

d) Increasing (lowest to highest) order of *bond energy* (1= lowest; 4 = highest)

P-F

P-Cl

P-I

P-Br

**Question 5** **12 points**

a. 9 points For the molecules HClO_4 and TeF_6 , draw the most stable *Lewis structure* showing appropriate shape and *dashed/wedged bonds*, based on VSEPR. Determine the **molecular geometry** around the central atom. For HClO_4 , label the *most polar bond*.



b. 3 points (Note: Lone pairs are not shown in structures below)

(i) For the three *resonance* structures (1, 2 and 3) shown below, answer the following question:



(I) Circle the *most contributing* resonance structure.

Structure 1

Structure 2

Structure 3

(ii) For the three structures given (4, 5, and 6) below, answer the following questions:



(I) Circle all the structures that represent *resonance structures* of each other:

Structure 4

Structure 5

Structure 6

(II) Among the resonance structures, circle the *most contributing* resonance structure.

Structure 4

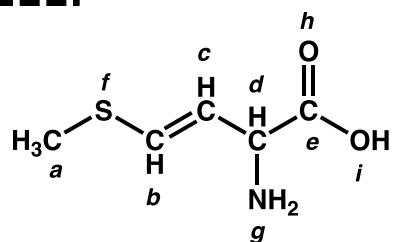
Structure 5

Structure 6

**Question 6** **6 points**

Using the data given below, calculate the **lattice energy** for SrO(s).

	ΔH° (kJ/mol)
$\text{Sr}(s) \longrightarrow \text{Sr}(g)$	+180
$\text{Sr}(g) \longrightarrow \text{Sr}^{2+}(g) + 2e^-$	+1468
$\text{O}_2(g) \longrightarrow 2\text{O}(g)$	+500
$\text{O}(g) + 2e^- \longrightarrow \text{O}^{2-}(g)$	+600
$\text{Sr}(s) + \frac{1}{2}\text{O}_2(g) \longrightarrow \text{SrO}(s)$	-550

**Question 7 10 points**

Consider the molecule given on the left, and answer questions (i) to (vi) using chemical bonding theories that we have learnt in class (valence bond theory, hybridization).

Note that the lone pairs are not shown in the structure:

i. Consider the carbon atoms marked in as ***a***, ***b***, ***c***, ***d***, and ***e*** in the molecule, and determine **hybridization** at each of the carbon centers. (2.5 points)

Atom	Hybridization
Carbon (<i>a</i>)	
Carbon (<i>b</i>)	
Carbon (<i>c</i>)	
Carbon (<i>d</i>)	
Carbon (<i>e</i>)	

ii. Now, consider the sulfur atom (S) marked as ***f***, nitrogen (N) atom (***g***), and oxygen(O) atoms (***h*** and ***i***), and determine their **hybridization**. (2 points)

Atom	Hybridization
Sulfur (<i>f</i>)	
Nitrogen (<i>g</i>)	
Oxygen (<i>h</i>)	
Oxygen (<i>i</i>)	

iii. **Describe** all the bonds formed by the carbon atoms ***a*** and ***b***, using sigma and pi designation as appropriate. (2 points)

Atom	Bonds formed (use sigma and pi as appropriate)
Carbon (<i>a</i>)	
Carbon (<i>b</i>)	

iv. How many total sp^3 -s sigma bonds are in the molecule? (1 point)

v. Determine **electronic** and **molecular geometry** at carbon (C) labeled ***e*** and sulfur (S) labeled ***f*** centres. (2 points)

Atom	Electronic Geometry	Molecular Geometry
Carbon (<i>e</i>)		
Sulfur (<i>f</i>)		

vi. Circle the value from the choices given below, that describes the **C-S-C bond angle at sulfur** (S) marked ***f***: (0.5 points)

>90° to <109.5° 109.5° >109.5 to <120° 120° >120° to <180°


Question 8 12 points

Using the **valence shell orbitals only** draw a **complete** molecular orbital (MO) diagram (with no s-p mixing) for the molecule **PS**, in the space provided below. **(6 points)**

Assume that 3s and 3p atomic orbitals will form molecular orbitals like those formed by the combination of 2s and 2p atomic orbitals that we learnt in class, and there is no s-p mixing.

Show **energy arrow**, **atomic orbitals** involved, **atom labels**, the **molecular orbitals** that are formed, **filled electrons** in the MO diagram etc. Make sure to label all the atomic and molecular orbitals.

Using the MO diagram above, answer the following questions:

- (i) Which one of the following will have the highest bond order? Circle the correct choice
 A) PS^{1-} B) PS C) PS^{1+}
- (ii) Which one of the following will have the longest bond length? Circle the correct choice
 A) PS^{1-} B) PS C) PS^{1+}
- (iii) Which of the following will be diamagnetic? Circle the correct choice.
 A) PS^{1-} B) PS C) PS^{1+}
- (iv) Draw Lewis dot structure of the molecule with the shortest bond length (based on the MO diagram) from among the three (PS^{1-} , PS, PS^{1+}), and indicate the bond order calculated from the Lewis dot structure. **(3 points)**

Lewis Dot Structure

Bond Order



99F0923D-DEF7-4B54-ADD3-70DB89D27B55

midterm-exam-93388

#663 Page 10 of 10

SCRAP PAGE

**CHEMISTRY 110 –Midterm Exam****Thursday November 7, 2024, 6:30 – 8:30 pm****Examiners: Pallavi Sirjoosingh and Ashok Kakkar**

First name (please write as legibly as possible within the boxes)

--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--

Last name

--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--

Student ID number

--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--

This booklet has 8 questions on 10 pages (The last page can be used as a scrap page).

The staple must NOT be removed from the question booklet. Your intact question booklet must be returned at the end of the exam.

A separate booklet contains a periodic table, and datasheet.

Write your answers directly under the questions in the space provided. Do not write on the QR codes or near the top edges (as it may be cut off).

Please write **legibly and dark enough** for the scan to be readable. For students who use English as a second language, dictionaries are permitted. Calculators are allowed.

DO NOT ASK QUESTIONS OF THE INVIGILATORS about interpretations of the exam questions. If you have a concern about a question, please summarize it at the top of this page. All concerns will be evaluated after the exam. GOOD LUCK!

**Question 1 8 points** (*Assume all transitions are possible*)

a. 6 points Calculate the **energy (in kJ/mol)** required to ionize (completely remove) the electron from the **$n=3$** level of **He^+ ion** (He: Helium). Show your work including equations. (3 sig. figs.)

b. 2 points

(I) Select the largest (longest) wavelength from the three choices below:

- i. Wavelength of photon to *completely remove* an electron from *1s orbital of Hydrogen atom*
- ii. Wavelength of photon to *completely remove* an electron from *1s orbital of Boron atom (B)*
- iii. Wavelength of photon to *completely remove* an electron from *1s orbital of Boron cation (B^+)*

i or ii or iii

(II) Select the smallest (shortest) wavelength from the three choices below (*for H atom*):

- i. Wavelength of photon *emitted* for an electron transitioning from *2p to 1s orbital*
- ii. Wavelength of photon *absorbed* for an electron transitioning from *2s to 4p orbital*
- iii. Wavelength of photon *absorbed* for an electron transitioning from *1s to 3p orbital*

i or ii or iii



Question 2 8 points



A Chem110 student plots the graph shown on the left, from data obtained for the kinetic energy (eV) of the ejected electron versus frequency (s^{-1}) of the incident light, for an unknown metal. ($1\text{eV} = 1.6022 \times 10^{-19} \text{ J}$)

Based on the plot, answer the questions **a to c** below:

a. 1 point On the plot above, label the **threshold frequency**.

b. 2 points What is the value of the work potential (in J) of the unknown metal?

c. 3 points What is the *longest* wavelength (in nm) required to eject an electron from this metal's surface? Show your work including equation(s) used. (3 sig figs)

d. 2 points In the space provided below, plot the following

(Assume frequency is *above* threshold frequency):

- i. *Kinetic energy* of an ejected electron versus the *intensity* of the incident light (left)
- ii. *Number of electrons* ejected versus the *frequency* of the incident light (right)



**Question 3** **6 points****a. 2 points**

In the ground state of Ge (Germanium; $Z = 32$) atom, **name** the following orbitals:

i. Orbital containing the electron that requires the maximum energy to be removed.

ii. Orbital containing the electron that experiences the minimum effective nuclear charge.

b. 2 points

An unknown anion (A^-) has the following *excited* electronic configuration: $[Ar] 4s^2 3d^3 4p^1$

(Ar: Argon; $Z = 18$)

Give the **condensed ground state electronic configuration** of the atom, **A** and the cation, **A^{2+}** .
Show your work. (Assume all electronic transitions are possible)

Electronic Configuration of A:

Electronic Configuration of A^{2+} :

c. 2 points

In a single atom, determine how many electrons can have the following set of quantum numbers?

(i) $n = 5; l = 3$

In a single atom, determine how many electrons can have the following set of quantum numbers?

(i) $n = 3; l = 2; m_s = +1/2$

**Question 4** **8 points**

Rank the following from 1 to 4 in **increasing** order. Provide the ranking in the boxes provided.

Note: Atomic number (Z) are provided for relevant atoms

a) Increasing (lowest to highest) order of *ionization energy*: (1= lowest; 4 = highest)

Mg

Mg: Z = 12

Rb

Rb: Z = 37

Cs

Cs: Z = 55

K

K : Z = 19

b) Increasing (lowest to highest) order of *bond angle*: (1= lowest; 4 = highest)

H₂OCCl₄XeF₄

HCN

c) Increasing (smallest to largest) order of *ionic size* (1= smallest; 4 = largest)

S²⁻

S: Z = 16

O²⁻

O: Z = 8

Be²⁺

Be: Z = 4

Na⁺

Na: Z = 11

d) Increasing (lowest to highest) order of *bond energy* (1= lowest; 4 = highest)

P-F

P-Cl

P-I

P-Br

**Question 5** **12 points**

a. 9 points For the molecules HClO_4 and TeF_6 , draw the most stable *Lewis structure* showing appropriate shape and *dashed/wedged bonds*, based on VSEPR. Determine the **molecular geometry** around the central atom. For HClO_4 , label the *most polar bond*.



b. 3 points (Note: Lone pairs are not shown in structures below)

(i) For the three *resonance* structures (1, 2 and 3) shown below, answer the following question:



(I) Circle the *most contributing* resonance structure.

Structure 1

Structure 2

Structure 3

(ii) For the three structures given (4, 5, and 6) below, answer the following questions:



(I) Circle all the structures that represent *resonance structures* of each other:

Structure 4

Structure 5

Structure 6

(II) Among the resonance structures, circle the *most contributing* resonance structure.

Structure 4

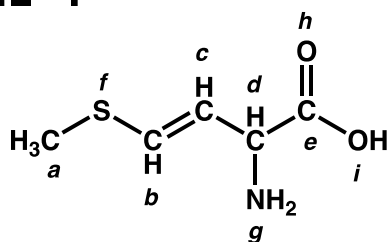
Structure 5

Structure 6

**Question 6** **6 points**

Using the data given below, calculate the **lattice energy** for SrO(s).

	ΔH° (kJ/mol)
$\text{Sr}(s) \longrightarrow \text{Sr}(g)$	+180
$\text{Sr}(g) \longrightarrow \text{Sr}^{2+}(g) + 2e^-$	+1468
$\text{O}_2(g) \longrightarrow 2\text{O}(g)$	+500
$\text{O}(g) + 2e^- \longrightarrow \text{O}^{2-}(g)$	+600
$\text{Sr}(s) + \frac{1}{2}\text{O}_2(g) \longrightarrow \text{SrO}(s)$	-550

**Question 7 10 points**

Consider the molecule given on the left, and answer questions (i) to (vi) using chemical bonding theories that we have learnt in class (valence bond theory, hybridization).

Note that the lone pairs are not shown in the structure:

i. Consider the carbon atoms marked in as ***a***, ***b***, ***c***, ***d***, and ***e*** in the molecule, and determine **hybridization** at each of the carbon centers. (2.5 points)

Atom	Hybridization
Carbon (<i>a</i>)	
Carbon (<i>b</i>)	
Carbon (<i>c</i>)	
Carbon (<i>d</i>)	
Carbon (<i>e</i>)	

ii. Now, consider the sulfur atom (S) marked as ***f***, nitrogen (N) atom (***g***), and oxygen(O) atoms (***h*** and ***i***), and determine their **hybridization**. (2 points)

Atom	Hybridization
Sulfur (<i>f</i>)	
Nitrogen (<i>g</i>)	
Oxygen (<i>h</i>)	
Oxygen (<i>i</i>)	

iii. **Describe** all the bonds formed by the carbon atoms ***a*** and ***b***, using sigma and pi designation as appropriate. (2 points)

Atom	Bonds formed (use sigma and pi as appropriate)
Carbon (<i>a</i>)	
Carbon (<i>b</i>)	

iv. How many total sp^3 -s sigma bonds are in the molecule? (1 point)

v. Determine **electronic** and **molecular geometry** at carbon (C) labeled ***e*** and sulfur (S) labeled ***f*** centres. (2 points)

Atom	Electronic Geometry	Molecular Geometry
Carbon (<i>e</i>)		
Sulfur (<i>f</i>)		

vi. Circle the value from the choices given below, that describes the **C-S-C bond angle at sulfur** (S) marked ***f***: (0.5 points)

>90° to <109.5° 109.5° >109.5 to <120° 120° >120° to <180°


Question 8 12 points

Using the **valence shell orbitals only** draw a **complete** molecular orbital (MO) diagram (with no s-p mixing) for the molecule **PS**, in the space provided below. **(6 points)**

Assume that 3s and 3p atomic orbitals will form molecular orbitals like those formed by the combination of 2s and 2p atomic orbitals that we learnt in class, and there is no s-p mixing.

Show **energy arrow**, **atomic orbitals** involved, **atom labels**, the **molecular orbitals** that are formed, **filled electrons** in the MO diagram etc. Make sure to **label all the atomic and molecular orbitals**.

Using the MO diagram above, answer the following questions:

- (i) Which one of the following will have the highest bond order? Circle the correct choice
 A) PS^{1-} B) PS C) PS^{1+}
- (ii) Which one of the following will have the longest bond length? Circle the correct choice
 A) PS^{1-} B) PS C) PS^{1+}
- (iii) Which of the following will be diamagnetic? Circle the correct choice.
 A) PS^{1-} B) PS C) PS^{1+}
- (iv) Draw Lewis dot structure of the molecule with the shortest bond length (based on the MO diagram) from among the three (PS^{1-} , PS, PS^{1+}), and indicate the bond order calculated from the Lewis dot structure. **(3 points)**

Lewis Dot Structure

Bond Order



A4265534-FFEE-4C71-B043-EA96FE0ACFA3

midterm-exam-93388

#664 Page 10 of 10

SCRAP PAGE

**CHEMISTRY 110 –Midterm Exam****Thursday November 7, 2024, 6:30 – 8:30 pm**

Examiners: Pallavi Sirjoosingh and Ashok Kakkar

First name (please write as legibly as possible within the boxes)

--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--

Last name

--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--

Student ID number

--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--

This booklet has 8 questions on 10 pages (The last page can be used as a scrap page).

The staple must NOT be removed from the question booklet. Your intact question booklet must be returned at the end of the exam.

A separate booklet contains a periodic table, and datasheet.

Write your answers directly under the questions in the space provided. Do not write on the QR codes or near the top edges (as it may be cut off).

Please write **legibly and dark enough** for the scan to be readable. For students who use English as a second language, dictionaries are permitted. Calculators are allowed.

DO NOT ASK QUESTIONS OF THE INVIGILATORS about interpretations of the exam questions. If you have a concern about a question, please summarize it at the top of this page. All concerns will be evaluated after the exam. GOOD LUCK!

**Question 1 8 points** (*Assume all transitions are possible*)

a. 6 points Calculate the **energy (in kJ/mol)** required to ionize (completely remove) the electron from the **$n=3$** level of **He^+ ion** (He: Helium). Show your work including equations. (3 sig. figs.)

b. 2 points

(I) Select the largest (longest) wavelength from the three choices below:

- i. Wavelength of photon to *completely remove* an electron from *1s orbital of Hydrogen atom*
- ii. Wavelength of photon to *completely remove* an electron from *1s orbital of Boron atom (B)*
- iii. Wavelength of photon to *completely remove* an electron from *1s orbital of Boron cation (B^+)*

i or ii or iii

(II) Select the smallest (shortest) wavelength from the three choices below (*for H atom*):

- i. Wavelength of photon *emitted* for an electron transitioning from *2p to 1s orbital*
- ii. Wavelength of photon *absorbed* for an electron transitioning from *2s to 4p orbital*
- iii. Wavelength of photon *absorbed* for an electron transitioning from *1s to 3p orbital*

i or ii or iii



Question 2 8 points



A Chem110 student plots the graph shown on the left, from data obtained for the kinetic energy (eV) of the ejected electron versus frequency (s^{-1}) of the incident light, for an unknown metal. ($1\text{eV} = 1.6022 \times 10^{-19} \text{ J}$)

Based on the plot, answer the questions **a to c** below:

a. 1 point On the plot above, label the **threshold frequency**.

b. 2 points What is the value of the work potential (in J) of the unknown metal?

c. 3 points What is the *longest* wavelength (in nm) required to eject an electron from this metal's surface? Show your work including equation(s) used. (3 sig figs)

d. 2 points In the space provided below, plot the following

(Assume frequency is *above* threshold frequency):

- Kinetic energy* of an ejected electron versus the *intensity* of the incident light (left)
- Number of electrons* ejected versus the *frequency* of the incident light (right)



**Question 3** **6 points****a. 2 points**

In the ground state of Ge (Germanium; $Z = 32$) atom, **name** the following orbitals:

i. Orbital containing the electron that requires the maximum energy to be removed.

ii. Orbital containing the electron that experiences the minimum effective nuclear charge.

b. 2 points

An unknown anion (A^-) has the following *excited* electronic configuration: $[Ar] 4s^2 3d^3 4p^1$

(Ar: Argon; $Z = 18$)

Give the **condensed ground state electronic configuration** of the atom, **A** and the cation, **A^{2+}** .
Show your work. (Assume all electronic transitions are possible)

Electronic Configuration of A:

Electronic Configuration of A^{2+} :

c. 2 points

In a single atom, determine how many electrons can have the following set of quantum numbers?

(i) $n = 5; l = 3$

In a single atom, determine how many electrons can have the following set of quantum numbers?

(i) $n = 3; l = 2; m_s = +1/2$

**Question 4** **8 points**

Rank the following from 1 to 4 in **increasing** order. Provide the ranking in the boxes provided.

Note: Atomic number (Z) are provided for relevant atoms

a) Increasing (lowest to highest) order of *ionization energy*: (1= lowest; 4 = highest)

Mg

Mg: Z = 12

Rb

Rb: Z = 37

Cs

Cs: Z = 55

K

K : Z = 19

b) Increasing (lowest to highest) order of *bond angle*: (1= lowest; 4 = highest)

H₂OCCl₄XeF₄

HCN

c) Increasing (smallest to largest) order of *ionic size* (1= smallest; 4 = largest)

S²⁻

S: Z = 16

O²⁻

O: Z = 8

Be²⁺

Be: Z = 4

Na⁺

Na: Z = 11

d) Increasing (lowest to highest) order of *bond energy* (1= lowest; 4 = highest)

P-F

P-Cl

P-I

P-Br

**Question 5** **12 points**

a. 9 points For the molecules HClO_4 and TeF_6 , draw the most stable *Lewis structure* showing appropriate shape and *dashed/wedged bonds*, based on VSEPR. Determine the **molecular geometry** around the central atom. For HClO_4 , label the *most polar bond*.



b. 3 points (Note: Lone pairs are not shown in structures below)

(i) For the three *resonance* structures (1, 2 and 3) shown below, answer the following question:



(I) Circle the *most contributing* resonance structure.

Structure 1

Structure 2

Structure 3

(ii) For the three structures given (4, 5, and 6) below, answer the following questions:



(I) Circle all the structures that represent *resonance structures* of each other:

Structure 4

Structure 5

Structure 6

(II) Among the resonance structures, circle the *most contributing* resonance structure.

Structure 4

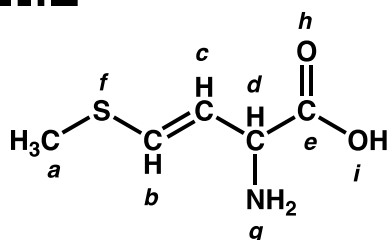
Structure 5

Structure 6

**Question 6** **6 points**

Using the data given below, calculate the **lattice energy** for SrO(s).

	ΔH° (kJ/mol)
$\text{Sr}(s) \longrightarrow \text{Sr}(g)$	+180
$\text{Sr}(g) \longrightarrow \text{Sr}^{2+}(g) + 2e^-$	+1468
$\text{O}_2(g) \longrightarrow 2\text{O}(g)$	+500
$\text{O}(g) + 2e^- \longrightarrow \text{O}^{2-}(g)$	+600
$\text{Sr}(s) + \frac{1}{2}\text{O}_2(g) \longrightarrow \text{SrO}(s)$	-550

**Question 7 10 points**

Consider the molecule given on the left, and answer questions (i) to (vi) using chemical bonding theories that we have learnt in class (valence bond theory, hybridization).

Note that the lone pairs are not shown in the structure:

i. Consider the carbon atoms marked in as ***a***, ***b***, ***c***, ***d***, and ***e*** in the molecule, and determine **hybridization** at each of the carbon centers. (2.5 points)

Atom	Hybridization
Carbon (<i>a</i>)	
Carbon (<i>b</i>)	
Carbon (<i>c</i>)	
Carbon (<i>d</i>)	
Carbon (<i>e</i>)	

ii. Now, consider the sulfur atom (S) marked as ***f***, nitrogen (N) atom (***g***), and oxygen(O) atoms (***h*** and ***i***), and determine their **hybridization**. (2 points)

Atom	Hybridization
Sulfur (<i>f</i>)	
Nitrogen (<i>g</i>)	
Oxygen (<i>h</i>)	
Oxygen (<i>i</i>)	

iii. **Describe** all the bonds formed by the carbon atoms ***a*** and ***b***, using sigma and pi designation as appropriate. (2 points)

Atom	Bonds formed (use sigma and pi as appropriate)
Carbon (<i>a</i>)	
Carbon (<i>b</i>)	

iv. How many total sp^3 -s sigma bonds are in the molecule? (1 point)

v. Determine **electronic** and **molecular geometry** at carbon (C) labeled ***e*** and sulfur (S) labeled ***f*** centres. (2 points)

Atom	Electronic Geometry	Molecular Geometry
Carbon (<i>e</i>)		
Sulfur (<i>f</i>)		

vi. Circle the value from the choices given below, that describes the **C-S-C bond angle at sulfur** (S) marked ***f***: (0.5 points)

>90° to <109.5°

109.5°

>109.5 to <120°

120°

>120° to <180°

**Question 8 12 points**

Using the **valence shell orbitals only** draw a **complete** molecular orbital (MO) diagram (with no s-p mixing) for the molecule **PS**, in the space provided below. **(6 points)**

Assume that 3s and 3p atomic orbitals will form molecular orbitals like those formed by the combination of 2s and 2p atomic orbitals that we learnt in class, and there is no s-p mixing.

Show **energy arrow**, **atomic orbitals** involved, **atom labels**, the **molecular orbitals** that are formed, **filled electrons** in the MO diagram etc. Make sure to label all the atomic and molecular orbitals.

Using the MO diagram above, answer the following questions:

- (i) Which one of the following will have the highest bond order? Circle the correct choice
A) PS^{1-} B) PS C) PS^{1+}
- (ii) Which one of the following will have the longest bond length? Circle the correct choice
A) PS^{1-} B) PS C) PS^{1+}
- (iii) Which of the following will be diamagnetic? Circle the correct choice.
A) PS^{1-} B) PS C) PS^{1+}
- (iv) Draw Lewis dot structure of the molecule with the shortest bond length (based on the MO diagram) from among the three (PS^{1-} , PS, PS^{1+}), and indicate the bond order calculated from the Lewis dot structure. **(3 points)**

Lewis Dot Structure

Bond Order



1C2216E6-B124-4B54-9B53-37DA8EAA832F

midterm-exam-93388

#665 Page 10 of 10

SCRAP PAGE

**CHEMISTRY 110 –Midterm Exam****Thursday November 7, 2024, 6:30 – 8:30 pm**

Examiners: Pallavi Sirjoosingh and Ashok Kakkar

First name (please write as legibly as possible within the boxes)

--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--

Last name

--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--

Student ID number

--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--

This booklet has 8 questions on 10 pages (The last page can be used as a scrap page).

The staple must NOT be removed from the question booklet. Your intact question booklet must be returned at the end of the exam.

A separate booklet contains a periodic table, and datasheet.

Write your answers directly under the questions in the space provided. Do not write on the QR codes or near the top edges (as it may be cut off).

Please write **legibly and dark enough** for the scan to be readable. For students who use English as a second language, dictionaries are permitted. Calculators are allowed.

DO NOT ASK QUESTIONS OF THE INVIGILATORS about interpretations of the exam questions. If you have a concern about a question, please summarize it at the top of this page. All concerns will be evaluated after the exam. GOOD LUCK!

**Question 1 8 points** (*Assume all transitions are possible*)

a. 6 points Calculate the **energy (in kJ/mol)** required to ionize (completely remove) the electron from the **n=3** level of **He⁺ ion** (He: Helium). Show your work including equations. (3 sig. figs.)

b. 2 points

(I) Select the largest (longest) wavelength from the three choices below:

- i. Wavelength of photon to *completely remove* an electron from *1s orbital of Hydrogen atom*
- ii. Wavelength of photon to *completely remove* an electron from *1s orbital of Boron atom (B)*
- iii. Wavelength of photon to *completely remove* an electron from *1s orbital of Boron cation (B⁺)*

i or ii or iii

(II) Select the smallest (shortest) wavelength from the three choices below (*for H atom*):

- i. Wavelength of photon *emitted* for an electron transitioning from *2p to 1s orbital*
- ii. Wavelength of photon *absorbed* for an electron transitioning from *2s to 4p orbital*
- iii. Wavelength of photon *absorbed* for an electron transitioning from *1s to 3p orbital*

i or ii or iii



Question 2 8 points



A Chem110 student plots the graph shown on the left, from data obtained for the kinetic energy (eV) of the ejected electron versus frequency (s^{-1}) of the incident light, for an unknown metal. ($1\text{eV} = 1.6022 \times 10^{-19} \text{ J}$)

Based on the plot, answer the questions **a to c** below:

a. 1 point On the plot above, label the **threshold frequency**.

b. 2 points What is the value of the work potential (in J) of the unknown metal?

c. 3 points What is the *longest* wavelength (in nm) required to eject an electron from this metal's surface? Show your work including equation(s) used. (3 sig figs)

d. 2 points In the space provided below, plot the following

(Assume frequency is *above* threshold frequency):

- i. *Kinetic energy* of an ejected electron versus the *intensity* of the incident light (left)
- ii. *Number of electrons* ejected versus the *frequency* of the incident light (right)



**Question 3** **6 points****a. 2 points**

In the ground state of Ge (Germanium; $Z = 32$) atom, **name** the following orbitals:

i. Orbital containing the electron that requires the maximum energy to be removed.

ii. Orbital containing the electron that experiences the minimum effective nuclear charge.

b. 2 points

An unknown anion (A^-) has the following *excited* electronic configuration: $[Ar] 4s^2 3d^3 4p^1$

(Ar: Argon; $Z = 18$)

Give the **condensed ground state electronic configuration** of the atom, **A** and the cation, **A^{2+}** .
Show your work. (Assume all electronic transitions are possible)

Electronic Configuration of A:

Electronic Configuration of A^{2+} :

c. 2 points

In a single atom, determine how many electrons can have the following set of quantum numbers?

(i) $n = 5; l = 3$

In a single atom, determine how many electrons can have the following set of quantum numbers?

(i) $n = 3; l = 2; m_s = +1/2$

**Question 4** **8 points**

Rank the following from 1 to 4 in **increasing** order. Provide the ranking in the boxes provided.

Note: Atomic number (Z) are provided for relevant atoms

a) Increasing (lowest to highest) order of *ionization energy*: (1= lowest; 4 = highest)

Mg

Mg: Z = 12

Rb

Rb: Z = 37

Cs

Cs: Z = 55

K

K : Z = 19

b) Increasing (lowest to highest) order of *bond angle*: (1= lowest; 4 = highest)

H₂OCCl₄XeF₄

HCN

c) Increasing (smallest to largest) order of *ionic size* (1= smallest; 4 = largest)

S²⁻

S: Z = 16

O²⁻

O: Z = 8

Be²⁺

Be: Z = 4

Na⁺

Na: Z = 11

d) Increasing (lowest to highest) order of *bond energy* (1= lowest; 4 = highest)

P-F

P-Cl

P-I

P-Br

**Question 5** **12 points**

a. 9 points For the molecules HClO_4 and TeF_6 , draw the most stable *Lewis structure* showing appropriate shape and *dashed/wedged bonds*, based on VSEPR. Determine the **molecular geometry** around the central atom. For HClO_4 , label the *most polar bond*.



b. 3 points (Note: Lone pairs are not shown in structures below)

(i) For the three *resonance* structures (1, 2 and 3) shown below, answer the following question:



(I) Circle the *most contributing* resonance structure.

Structure 1

Structure 2

Structure 3

(ii) For the three structures given (4, 5, and 6) below, answer the following questions:



(I) Circle all the structures that represent *resonance structures* of each other:

Structure 4

Structure 5

Structure 6

(II) Among the resonance structures, circle the *most contributing* resonance structure.

Structure 4

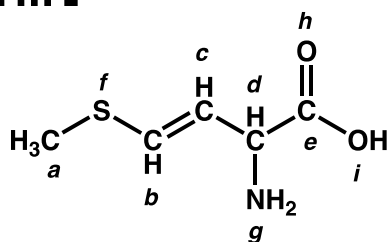
Structure 5

Structure 6

**Question 6** **6 points**

Using the data given below, calculate the **lattice energy** for SrO(s).

	ΔH° (kJ/mol)
$\text{Sr}(s) \longrightarrow \text{Sr}(g)$	+180
$\text{Sr}(g) \longrightarrow \text{Sr}^{2+}(g) + 2e^-$	+1468
$\text{O}_2(g) \longrightarrow 2\text{O}(g)$	+500
$\text{O}(g) + 2e^- \longrightarrow \text{O}^{2-}(g)$	+600
$\text{Sr}(s) + \frac{1}{2}\text{O}_2(g) \longrightarrow \text{SrO}(s)$	-550

**Question 7 10 points**

Consider the molecule given on the left, and answer questions (i) to (vi) using chemical bonding theories that we have learnt in class (valence bond theory, hybridization).

Note that the lone pairs are not shown in the structure:

i. Consider the carbon atoms marked in as ***a***, ***b***, ***c***, ***d***, and ***e*** in the molecule, and determine **hybridization** at each of the carbon centers. (2.5 points)

Atom	Hybridization
Carbon (<i>a</i>)	
Carbon (<i>b</i>)	
Carbon (<i>c</i>)	
Carbon (<i>d</i>)	
Carbon (<i>e</i>)	

ii. Now, consider the sulfur atom (S) marked as ***f***, nitrogen (N) atom (***g***), and oxygen(O) atoms (***h*** and ***i***), and determine their **hybridization**. (2 points)

Atom	Hybridization
Sulfur (<i>f</i>)	
Nitrogen (<i>g</i>)	
Oxygen (<i>h</i>)	
Oxygen (<i>i</i>)	

iii. **Describe** all the bonds formed by the carbon atoms ***a*** and ***b***, using sigma and pi designation as appropriate. (2 points)

Atom	Bonds formed (use sigma and pi as appropriate)
Carbon (<i>a</i>)	
Carbon (<i>b</i>)	

iv. How many total sp^3 -s sigma bonds are in the molecule? (1 point)

v. Determine **electronic** and **molecular geometry** at carbon (C) labeled ***e*** and sulfur (S) labeled ***f*** centres. (2 points)

Atom	Electronic Geometry	Molecular Geometry
Carbon (<i>e</i>)		
Sulfur (<i>f</i>)		

vi. Circle the value from the choices given below, that describes the **C-S-C bond angle at sulfur** (S) marked ***f***: (0.5 points)

>90° to <109.5°

109.5°

>109.5 to <120°

120°

>120° to <180°


Question 8 12 points

Using the **valence shell orbitals only** draw a **complete** molecular orbital (MO) diagram (with no s-p mixing) for the molecule **PS**, in the space provided below. **(6 points)**

Assume that 3s and 3p atomic orbitals will form molecular orbitals like those formed by the combination of 2s and 2p atomic orbitals that we learnt in class, and there is no s-p mixing.

Show **energy arrow**, **atomic orbitals** involved, **atom labels**, the **molecular orbitals** that are formed, **filled electrons** in the MO diagram etc. Make sure to label all the atomic and molecular orbitals.

Using the MO diagram above, answer the following questions:

- (i) Which one of the following will have the highest bond order? Circle the correct choice
 A) PS^{1-} B) PS C) PS^{1+}
- (ii) Which one of the following will have the longest bond length? Circle the correct choice
 A) PS^{1-} B) PS C) PS^{1+}
- (iii) Which of the following will be diamagnetic? Circle the correct choice.
 A) PS^{1-} B) PS C) PS^{1+}
- (iv) Draw Lewis dot structure of the molecule with the shortest bond length (based on the MO diagram) from among the three (PS^{1-} , PS, PS^{1+}), and indicate the bond order calculated from the Lewis dot structure. **(3 points)**

Lewis Dot Structure

Bond Order



927F9518-1A24-4D0E-87C8-D58775217DBC

midterm-exam-93388

#666 Page 10 of 10

SCRAP PAGE

**CHEMISTRY 110 –Midterm Exam****Thursday November 7, 2024, 6:30 – 8:30 pm****Examiners: Pallavi Sirjoosingh and Ashok Kakkar**

First name (please write as legibly as possible within the boxes)

--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--

Last name

--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--

Student ID number

--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--

This booklet has 8 questions on 10 pages (The last page can be used as a scrap page).

The staple must NOT be removed from the question booklet. Your intact question booklet must be returned at the end of the exam.

A separate booklet contains a periodic table, and datasheet.

Write your answers directly under the questions in the space provided. Do not write on the QR codes or near the top edges (as it may be cut off).

Please write **legibly and dark enough** for the scan to be readable. For students who use English as a second language, dictionaries are permitted. Calculators are allowed.

DO NOT ASK QUESTIONS OF THE INVIGILATORS about interpretations of the exam questions. If you have a concern about a question, please summarize it at the top of this page. All concerns will be evaluated after the exam. GOOD LUCK!

**Question 1 8 points** (*Assume all transitions are possible*)

a. 6 points Calculate the **energy (in kJ/mol)** required to ionize (completely remove) the electron from the **$n=3$** level of **He^+ ion** (He: Helium). Show your work including equations. (3 sig. figs.)

b. 2 points

(I) Select the largest (longest) wavelength from the three choices below:

- i. Wavelength of photon to *completely remove* an electron from *1s orbital of Hydrogen atom*
- ii. Wavelength of photon to *completely remove* an electron from *1s orbital of Boron atom (B)*
- iii. Wavelength of photon to *completely remove* an electron from *1s orbital of Boron cation (B^+)*

i or ii or iii

(II) Select the smallest (shortest) wavelength from the three choices below (*for H atom*):

- i. Wavelength of photon *emitted* for an electron transitioning from *2p to 1s orbital*
- ii. Wavelength of photon *absorbed* for an electron transitioning from *2s to 4p orbital*
- iii. Wavelength of photon *absorbed* for an electron transitioning from *1s to 3p orbital*

i or ii or iii



Question 2 8 points



A Chem110 student plots the graph shown on the left, from data obtained for the kinetic energy (eV) of the ejected electron versus frequency (s^{-1}) of the incident light, for an unknown metal. ($1\text{eV} = 1.6022 \times 10^{-19} \text{ J}$)

Based on the plot, answer the questions **a to c** below:

a. 1 point On the plot above, label the **threshold frequency**.

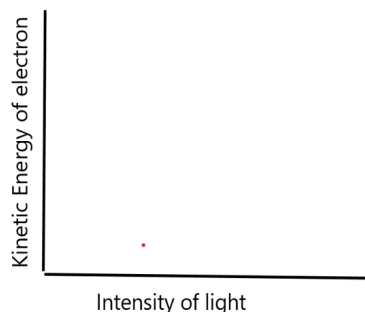
b. 2 points What is the value of the work potential (in J) of the unknown metal?

c. 3 points What is the *longest* wavelength (in nm) required to eject an electron from this metal's surface? Show your work including equation(s) used. (3 sig figs)

d. 2 points In the space provided below, plot the following

(Assume frequency is *above* threshold frequency):

- Kinetic energy* of an ejected electron versus the *intensity* of the incident light (left)
- Number of electrons* ejected versus the *frequency* of the incident light (right)



**Question 3** **6 points****a. 2 points**

In the ground state of Ge (Germanium; $Z = 32$) atom, **name** the following orbitals:

i. Orbital containing the electron that requires the maximum energy to be removed.

ii. Orbital containing the electron that experiences the minimum effective nuclear charge.

b. 2 points

An unknown anion (A^-) has the following *excited* electronic configuration: $[Ar] 4s^2 3d^3 4p^1$

(Ar: Argon; $Z = 18$)

Give the **condensed ground state electronic configuration** of the atom, **A** and the cation, **A^{2+}** .
Show your work. (Assume all electronic transitions are possible)

Electronic Configuration of A:

Electronic Configuration of A^{2+} :

c. 2 points

In a single atom, determine how many electrons can have the following set of quantum numbers?

(i) $n = 5; l = 3$

In a single atom, determine how many electrons can have the following set of quantum numbers?

(i) $n = 3; l = 2; m_s = +1/2$

**Question 4** **8 points**

Rank the following from 1 to 4 in **increasing** order. Provide the ranking in the boxes provided.

Note: Atomic number (Z) are provided for relevant atoms

a) Increasing (lowest to highest) order of *ionization energy*: (1= lowest; 4 = highest)

Mg

Mg: Z = 12

Rb

Rb: Z = 37

Cs

Cs: Z = 55

K

K : Z = 19

b) Increasing (lowest to highest) order of *bond angle*: (1= lowest; 4 = highest)

H₂OCCl₄XeF₄

HCN

c) Increasing (smallest to largest) order of *ionic size* (1= smallest; 4 = largest)

S²⁻

S: Z = 16

O²⁻

O: Z = 8

Be²⁺

Be: Z = 4

Na⁺

Na: Z = 11

d) Increasing (lowest to highest) order of *bond energy* (1= lowest; 4 = highest)

P-F

P-Cl

P-I

P-Br

**Question 5** **12 points**

a. 9 points For the molecules HClO_4 and TeF_6 , draw the most stable *Lewis structure* showing appropriate shape and *dashed/wedged bonds*, based on VSEPR. Determine the **molecular geometry** around the central atom. For HClO_4 , label the *most polar bond*.

HClO_4 (there are no O-O bonds)

Label the most polar bond

Molecular Geometry around Cl

TeF_6

Molecular Geometry around Te

b. 3 points (Note: Lone pairs are not shown in structures below)

(i) For the three *resonance* structures (1, 2 and 3) shown below, answer the following question:



(I) Circle the *most contributing* resonance structure.

Structure 1

Structure 2

Structure 3

(ii) For the three structures given (4, 5, and 6) below, answer the following questions:



(I) Circle all the structures that represent *resonance structures* of each other:

Structure 4

Structure 5

Structure 6

(II) Among the resonance structures, circle the *most contributing* resonance structure.

Structure 4

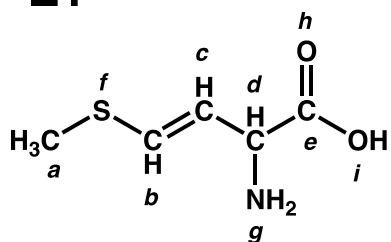
Structure 5

Structure 6

**Question 6** **6 points**

Using the data given below, calculate the **lattice energy** for SrO(s).

	ΔH° (kJ/mol)
$\text{Sr}(s) \longrightarrow \text{Sr}(g)$	+180
$\text{Sr}(g) \longrightarrow \text{Sr}^{2+}(g) + 2e^-$	+1468
$\text{O}_2(g) \longrightarrow 2\text{O}(g)$	+500
$\text{O}(g) + 2e^- \longrightarrow \text{O}^{2-}(g)$	+600
$\text{Sr}(s) + \frac{1}{2}\text{O}_2(g) \longrightarrow \text{SrO}(s)$	-550

**Question 7 10 points**

Consider the molecule given on the left, and answer questions (i) to (vi) using chemical bonding theories that we have learnt in class (valence bond theory, hybridization).

Note that the lone pairs are not shown in the structure:

i. Consider the carbon atoms marked in as ***a***, ***b***, ***c***, ***d***, and ***e*** in the molecule, and determine **hybridization** at each of the carbon centers. (2.5 points)

Atom	Hybridization
Carbon (<i>a</i>)	
Carbon (<i>b</i>)	
Carbon (<i>c</i>)	
Carbon (<i>d</i>)	
Carbon (<i>e</i>)	

ii. Now, consider the sulfur atom (S) marked as ***f***, nitrogen (N) atom (***g***), and oxygen(O) atoms (***h*** and ***i***), and determine their **hybridization**. (2 points)

Atom	Hybridization
Sulfur (<i>f</i>)	
Nitrogen (<i>g</i>)	
Oxygen (<i>h</i>)	
Oxygen (<i>i</i>)	

iii. **Describe** all the bonds formed by the carbon atoms ***a*** and ***b***, using sigma and pi designation as appropriate. (2 points)

Atom	Bonds formed (use sigma and pi as appropriate)
Carbon (<i>a</i>)	
Carbon (<i>b</i>)	

iv. How many total sp^3 -s sigma bonds are in the molecule? (1 point)

v. Determine **electronic** and **molecular geometry** at carbon (C) labeled ***e*** and sulfur (S) labeled ***f*** centres. (2 points)

Atom	Electronic Geometry	Molecular Geometry
Carbon (<i>e</i>)		
Sulfur (<i>f</i>)		

vi. Circle the value from the choices given below, that describes the **C-S-C bond angle at sulfur** (S) marked ***f***: (0.5 points)

>90° to <109.5°

109.5°

>109.5 to <120°

120°

>120° to <180°

**Question 8 12 points**

Using the **valence shell orbitals only** draw a **complete** molecular orbital (MO) diagram (with no s-p mixing) for the molecule **PS**, in the space provided below. **(6 points)**

Assume that 3s and 3p atomic orbitals will form molecular orbitals like those formed by the combination of 2s and 2p atomic orbitals that we learnt in class, and there is no s-p mixing.

Show **energy arrow**, **atomic orbitals** involved, **atom labels**, the **molecular orbitals** that are formed, **filled electrons** in the MO diagram etc. Make sure to label all the atomic and molecular orbitals.

Using the MO diagram above, answer the following questions:

- (i) Which one of the following will have the highest bond order? Circle the correct choice
A) PS^{1-} B) PS C) PS^{1+}
- (ii) Which one of the following will have the longest bond length? Circle the correct choice
A) PS^{1-} B) PS C) PS^{1+}
- (iii) Which of the following will be diamagnetic? Circle the correct choice.
A) PS^{1-} B) PS C) PS^{1+}
- (iv) Draw Lewis dot structure of the molecule with the shortest bond length (based on the MO diagram) from among the three (PS^{1-} , PS, PS^{1+}), and indicate the bond order calculated from the Lewis dot structure. **(3 points)**

Lewis Dot Structure

Bond Order



C78D9288-174D-43AC-AFAB-155FFBA3C998

midterm-exam-93388

#667 Page 10 of 10

SCRAP PAGE

**CHEMISTRY 110 –Midterm Exam****Thursday November 7, 2024, 6:30 – 8:30 pm**

Examiners: Pallavi Sirjoosingh and Ashok Kakkar

First name (please write as legibly as possible within the boxes)

--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--

Last name

--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--

Student ID number

--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--

This booklet has 8 questions on 10 pages (The last page can be used as a scrap page).

The staple must NOT be removed from the question booklet. Your intact question booklet must be returned at the end of the exam.

A separate booklet contains a periodic table, and datasheet.

Write your answers directly under the questions in the space provided. Do not write on the QR codes or near the top edges (as it may be cut off).

Please write **legibly and dark enough** for the scan to be readable. For students who use English as a second language, dictionaries are permitted. Calculators are allowed.

DO NOT ASK QUESTIONS OF THE INVIGILATORS about interpretations of the exam questions. If you have a concern about a question, please summarize it at the top of this page. All concerns will be evaluated after the exam. GOOD LUCK!

**Question 1 8 points** (*Assume all transitions are possible*)

a. 6 points Calculate the **energy (in kJ/mol)** required to ionize (completely remove) the electron from the **$n=3$** level of **He^+ ion** (He: Helium). Show your work including equations. (3 sig. figs.)

b. 2 points

(I) Select the largest (longest) wavelength from the three choices below:

- i. Wavelength of photon to *completely remove* an electron from *1s orbital of Hydrogen atom*
- ii. Wavelength of photon to *completely remove* an electron from *1s orbital of Boron atom (B)*
- iii. Wavelength of photon to *completely remove* an electron from *1s orbital of Boron cation (B^+)*

i or ii or iii

(II) Select the smallest (shortest) wavelength from the three choices below (*for H atom*):

- i. Wavelength of photon *emitted* for an electron transitioning from *2p to 1s orbital*
- ii. Wavelength of photon *absorbed* for an electron transitioning from *2s to 4p orbital*
- iii. Wavelength of photon *absorbed* for an electron transitioning from *1s to 3p orbital*

i or ii or iii



Question 2 8 points



A Chem110 student plots the graph shown on the left, from data obtained for the kinetic energy (eV) of the ejected electron versus frequency (s^{-1}) of the incident light, for an unknown metal. ($1\text{eV} = 1.6022 \times 10^{-19} \text{ J}$)

Based on the plot, answer the questions **a to c** below:

a. 1 point On the plot above, label the **threshold frequency**.

b. 2 points What is the value of the work potential (in J) of the unknown metal?

c. 3 points What is the *longest* wavelength (in nm) required to eject an electron from this metal's surface? Show your work including equation(s) used. (3 sig figs)

d. 2 points In the space provided below, plot the following

(Assume frequency is *above* threshold frequency):

- i. *Kinetic energy* of an ejected electron versus the *intensity* of the incident light (left)
- ii. *Number of electrons* ejected versus the *frequency* of the incident light (right)



**Question 3** **6 points****a. 2 points**

In the ground state of Ge (Germanium; $Z = 32$) atom, **name** the following orbitals:

i. Orbital containing the electron that requires the maximum energy to be removed.

ii. Orbital containing the electron that experiences the minimum effective nuclear charge.

b. 2 points

An unknown anion (A^-) has the following *excited* electronic configuration: $[Ar] 4s^2 3d^3 4p^1$

(Ar: Argon; $Z = 18$)

Give the **condensed ground state electronic configuration** of the atom, **A** and the cation, **A^{2+}** .
Show your work. (Assume all electronic transitions are possible)

Electronic Configuration of A:

Electronic Configuration of A^{2+} :

c. 2 points

In a single atom, determine how many electrons can have the following set of quantum numbers?

(i) $n = 5; l = 3$

In a single atom, determine how many electrons can have the following set of quantum numbers?

(i) $n = 3; l = 2; m_s = +1/2$

**Question 4** **8 points**

Rank the following from 1 to 4 in **increasing** order. Provide the ranking in the boxes provided.

Note: Atomic number (Z) are provided for relevant atoms

a) Increasing (lowest to highest) order of *ionization energy*: (1= lowest; 4 = highest)

Mg

Mg: Z = 12

Rb

Rb: Z = 37

Cs

Cs: Z = 55

K

K : Z = 19

b) Increasing (lowest to highest) order of *bond angle*: (1= lowest; 4 = highest)

H₂OCCl₄XeF₄

HCN

c) Increasing (smallest to largest) order of *ionic size* (1= smallest; 4 = largest)

S²⁻

S: Z = 16

O²⁻

O: Z = 8

Be²⁺

Be: Z = 4

Na⁺

Na: Z = 11

d) Increasing (lowest to highest) order of *bond energy* (1= lowest; 4 = highest)

P-F

P-Cl

P-I

P-Br

**Question 5** **12 points**

a. 9 points For the molecules HClO_4 and TeF_6 , draw the most stable *Lewis structure* showing appropriate shape and *dashed/wedged bonds*, based on VSEPR. Determine the **molecular geometry** around the central atom. For HClO_4 , label the *most polar bond*.



b. 3 points (Note: Lone pairs are not shown in structures below)

(i) For the three *resonance* structures (1, 2 and 3) shown below, answer the following question:



(I) Circle the *most contributing* resonance structure.

Structure 1

Structure 2

Structure 3

(ii) For the three structures given (4, 5, and 6) below, answer the following questions:



(I) Circle all the structures that represent *resonance structures* of each other:

Structure 4

Structure 5

Structure 6

(II) Among the resonance structures, circle the *most contributing* resonance structure.

Structure 4

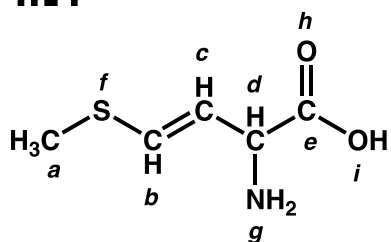
Structure 5

Structure 6

**Question 6** **6 points**

Using the data given below, calculate the **lattice energy** for SrO(s).

	ΔH° (kJ/mol)
$\text{Sr}(s) \longrightarrow \text{Sr}(g)$	+180
$\text{Sr}(g) \longrightarrow \text{Sr}^{2+}(g) + 2e^-$	+1468
$\text{O}_2(g) \longrightarrow 2\text{O}(g)$	+500
$\text{O}(g) + 2e^- \longrightarrow \text{O}^{2-}(g)$	+600
$\text{Sr}(s) + \frac{1}{2}\text{O}_2(g) \longrightarrow \text{SrO}(s)$	-550

**Question 7 10 points**

Consider the molecule given on the left, and answer questions (i) to (vi) using chemical bonding theories that we have learnt in class (valence bond theory, hybridization).

Note that the lone pairs are not shown in the structure:

i. Consider the carbon atoms marked in as ***a***, ***b***, ***c***, ***d***, and ***e*** in the molecule, and determine **hybridization** at each of the carbon centers. (2.5 points)

Atom	Hybridization
Carbon (<i>a</i>)	
Carbon (<i>b</i>)	
Carbon (<i>c</i>)	
Carbon (<i>d</i>)	
Carbon (<i>e</i>)	

ii. Now, consider the sulfur atom (S) marked as ***f***, nitrogen (N) atom (***g***), and oxygen(O) atoms (***h*** and ***i***), and determine their **hybridization**. (2 points)

Atom	Hybridization
Sulfur (<i>f</i>)	
Nitrogen (<i>g</i>)	
Oxygen (<i>h</i>)	
Oxygen (<i>i</i>)	

iii. **Describe** all the bonds formed by the carbon atoms ***a*** and ***b***, using sigma and pi designation as appropriate. (2 points)

Atom	Bonds formed (use sigma and pi as appropriate)
Carbon (<i>a</i>)	
Carbon (<i>b</i>)	

iv. How many total sp^3 -s sigma bonds are in the molecule? (1 point)

v. Determine **electronic** and **molecular geometry** at carbon (C) labeled ***e*** and sulfur (S) labeled ***f*** centres. (2 points)

Atom	Electronic Geometry	Molecular Geometry
Carbon (<i>e</i>)		
Sulfur (<i>f</i>)		

vi. Circle the value from the choices given below, that describes the **C-S-C bond angle at sulfur** (S) marked ***f***: (0.5 points)

$>90^\circ$ to $<109.5^\circ$ 109.5° >109.5 to $<120^\circ$ 120° $>120^\circ$ to $<180^\circ$


Question 8 12 points

Using the **valence shell orbitals only** draw a **complete** molecular orbital (MO) diagram (with no s-p mixing) for the molecule **PS**, in the space provided below. **(6 points)**

Assume that 3s and 3p atomic orbitals will form molecular orbitals like those formed by the combination of 2s and 2p atomic orbitals that we learnt in class, and there is no s-p mixing.

Show **energy arrow**, **atomic orbitals** involved, **atom labels**, the **molecular orbitals** that are formed, **filled electrons** in the MO diagram etc. Make sure to label all the atomic and molecular orbitals.

Using the MO diagram above, answer the following questions:

- (i) Which one of the following will have the highest bond order? Circle the correct choice
 A) PS^{1-} B) PS C) PS^{1+}
- (ii) Which one of the following will have the longest bond length? Circle the correct choice
 A) PS^{1-} B) PS C) PS^{1+}
- (iii) Which of the following will be diamagnetic? Circle the correct choice.
 A) PS^{1-} B) PS C) PS^{1+}
- (iv) Draw Lewis dot structure of the molecule with the shortest bond length (based on the MO diagram) from among the three (PS^{1-} , PS, PS^{1+}), and indicate the bond order calculated from the Lewis dot structure. **(3 points)**

Lewis Dot Structure

Bond Order



11118205-F0E6-45DB-8922-E0AFF8863F53

midterm-exam-93388

#668 Page 10 of 10

SCRAP PAGE

**CHEMISTRY 110 –Midterm Exam****Thursday November 7, 2024, 6:30 – 8:30 pm**

Examiners: Pallavi Sirjoosingh and Ashok Kakkar

First name (please write as legibly as possible within the boxes)

--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--

Last name

--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--

Student ID number

--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--

This booklet has 8 questions on 10 pages (The last page can be used as a scrap page).

The staple must NOT be removed from the question booklet. Your intact question booklet must be returned at the end of the exam.

A separate booklet contains a periodic table, and datasheet.

Write your answers directly under the questions in the space provided. Do not write on the QR codes or near the top edges (as it may be cut off).

Please write **legibly and dark enough** for the scan to be readable. For students who use English as a second language, dictionaries are permitted. Calculators are allowed.

DO NOT ASK QUESTIONS OF THE INVIGILATORS about interpretations of the exam questions. If you have a concern about a question, please summarize it at the top of this page. All concerns will be evaluated after the exam. GOOD LUCK!

**Question 1 8 points** (*Assume all transitions are possible*)

a. 6 points Calculate the **energy (in kJ/mol)** required to ionize (completely remove) the electron from the **n=3** level of **He⁺ ion** (He: Helium). Show your work including equations. (3 sig. figs.)

b. 2 points

(I) Select the largest (longest) wavelength from the three choices below:

- i. Wavelength of photon to *completely remove* an electron from *1s orbital of Hydrogen atom*
- ii. Wavelength of photon to *completely remove* an electron from *1s orbital of Boron atom (B)*
- iii. Wavelength of photon to *completely remove* an electron from *1s orbital of Boron cation (B⁺)*

i or ii or iii

(II) Select the smallest (shortest) wavelength from the three choices below (*for H atom*):

- i. Wavelength of photon *emitted* for an electron transitioning from *2p to 1s orbital*
- ii. Wavelength of photon *absorbed* for an electron transitioning from *2s to 4p orbital*
- iii. Wavelength of photon *absorbed* for an electron transitioning from *1s to 3p orbital*

i or ii or iii



Question 2 8 points



A Chem110 student plots the graph shown on the left, from data obtained for the kinetic energy (eV) of the ejected electron versus frequency (s^{-1}) of the incident light, for an unknown metal. ($1\text{eV} = 1.6022 \times 10^{-19} \text{ J}$)

Based on the plot, answer the questions **a to c** below:

a. 1 point On the plot above, label the **threshold frequency**.

b. 2 points What is the value of the work potential (in J) of the unknown metal?

c. 3 points What is the *longest* wavelength (in nm) required to eject an electron from this metal's surface? Show your work including equation(s) used. (3 sig figs)

d. 2 points In the space provided below, plot the following

(Assume frequency is *above* threshold frequency):

- i. *Kinetic energy* of an ejected electron versus the *intensity* of the incident light (left)
- ii. *Number of electrons* ejected versus the *frequency* of the incident light (right)



**Question 3** **6 points****a. 2 points**

In the ground state of Ge (Germanium; $Z = 32$) atom, **name** the following orbitals:

i. Orbital containing the electron that requires the maximum energy to be removed.

ii. Orbital containing the electron that experiences the minimum effective nuclear charge.

b. 2 points

An unknown anion (A^-) has the following *excited* electronic configuration: $[Ar] 4s^2 3d^3 4p^1$

(Ar: Argon; $Z = 18$)

Give the **condensed ground state electronic configuration** of the atom, **A** and the cation, **A^{2+}** .
Show your work. (Assume all electronic transitions are possible)

Electronic Configuration of A:

Electronic Configuration of A^{2+} :

c. 2 points

In a single atom, determine how many electrons can have the following set of quantum numbers?

(i) $n = 5; l = 3$

In a single atom, determine how many electrons can have the following set of quantum numbers?

(i) $n = 3; l = 2; m_s = +1/2$

**Question 4** **8 points**

Rank the following from 1 to 4 in **increasing** order. Provide the ranking in the boxes provided.

Note: Atomic number (Z) are provided for relevant atoms

a) Increasing (lowest to highest) order of *ionization energy*: (1= lowest; 4 = highest)

Mg

Mg: Z = 12

Rb

Rb: Z = 37

Cs

Cs: Z = 55

K

K : Z = 19

b) Increasing (lowest to highest) order of *bond angle*: (1= lowest; 4 = highest)

H₂OCCl₄XeF₄

HCN

c) Increasing (smallest to largest) order of *ionic size* (1= smallest; 4 = largest)

S²⁻

S: Z = 16

O²⁻

O: Z = 8

Be²⁺

Be: Z = 4

Na⁺

Na: Z = 11

d) Increasing (lowest to highest) order of *bond energy* (1= lowest; 4 = highest)

P-F

P-Cl

P-I

P-Br

**Question 5** **12 points**

a. 9 points For the molecules HClO_4 and TeF_6 , draw the most stable *Lewis structure* showing appropriate shape and *dashed/wedged bonds*, based on VSEPR. Determine the **molecular geometry** around the central atom. For HClO_4 , label the *most polar bond*.



b. 3 points (Note: Lone pairs are not shown in structures below)

(i) For the three *resonance* structures (1, 2 and 3) shown below, answer the following question:



(I) Circle the *most contributing* resonance structure.

Structure 1

Structure 2

Structure 3

(ii) For the three structures given (4, 5, and 6) below, answer the following questions:



(I) Circle all the structures that represent *resonance structures* of each other:

Structure 4

Structure 5

Structure 6

(II) Among the resonance structures, circle the *most contributing* resonance structure.

Structure 4

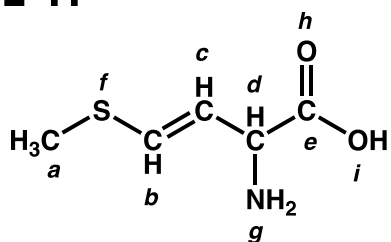
Structure 5

Structure 6

**Question 6** **6 points**

Using the data given below, calculate the **lattice energy** for SrO(s).

	ΔH° (kJ/mol)
$\text{Sr}(s) \longrightarrow \text{Sr}(g)$	+180
$\text{Sr}(g) \longrightarrow \text{Sr}^{2+}(g) + 2e^-$	+1468
$\text{O}_2(g) \longrightarrow 2\text{O}(g)$	+500
$\text{O}(g) + 2e^- \longrightarrow \text{O}^{2-}(g)$	+600
$\text{Sr}(s) + \frac{1}{2}\text{O}_2(g) \longrightarrow \text{SrO}(s)$	-550

**Question 7 10 points**

Consider the molecule given on the left, and answer questions (i) to (vi) using chemical bonding theories that we have learnt in class (valence bond theory, hybridization).

Note that the lone pairs are not shown in the structure:

i. Consider the carbon atoms marked in as ***a***, ***b***, ***c***, ***d***, and ***e*** in the molecule, and determine **hybridization** at each of the carbon centers. (2.5 points)

Atom	Hybridization
Carbon (<i>a</i>)	
Carbon (<i>b</i>)	
Carbon (<i>c</i>)	
Carbon (<i>d</i>)	
Carbon (<i>e</i>)	

ii. Now, consider the sulfur atom (S) marked as ***f***, nitrogen (N) atom (***g***), and oxygen(O) atoms (***h*** and ***i***), and determine their **hybridization**. (2 points)

Atom	Hybridization
Sulfur (<i>f</i>)	
Nitrogen (<i>g</i>)	
Oxygen (<i>h</i>)	
Oxygen (<i>i</i>)	

iii. **Describe** all the bonds formed by the carbon atoms ***a*** and ***b***, using sigma and pi designation as appropriate. (2 points)

Atom	Bonds formed (use sigma and pi as appropriate)
Carbon (<i>a</i>)	
Carbon (<i>b</i>)	

iv. How many total sp^3 -s sigma bonds are in the molecule? (1 point)

v. Determine **electronic** and **molecular geometry** at carbon (C) labeled ***e*** and sulfur (S) labeled ***f*** centres. (2 points)

Atom	Electronic Geometry	Molecular Geometry
Carbon (<i>e</i>)		
Sulfur (<i>f</i>)		

vi. Circle the value from the choices given below, that describes the **C-S-C bond angle at sulfur** (S) marked ***f***: (0.5 points)

>90° to <109.5° 109.5° >109.5 to <120° 120° >120° to <180°

**Question 8 12 points**

Using the **valence shell orbitals only** draw a **complete** molecular orbital (MO) diagram (with no s-p mixing) for the molecule **PS**, in the space provided below. **(6 points)**

Assume that 3s and 3p atomic orbitals will form molecular orbitals like those formed by the combination of 2s and 2p atomic orbitals that we learnt in class, and there is no s-p mixing.

Show **energy arrow**, **atomic orbitals** involved, **atom labels**, the **molecular orbitals** that are formed, **filled electrons** in the MO diagram etc. Make sure to label all the atomic and molecular orbitals.

Using the MO diagram above, answer the following questions:

- (i) Which one of the following will have the highest bond order? Circle the correct choice
A) PS^{1-} B) PS C) PS^{1+}
- (ii) Which one of the following will have the longest bond length? Circle the correct choice
A) PS^{1-} B) PS C) PS^{1+}
- (iii) Which of the following will be diamagnetic? Circle the correct choice.
A) PS^{1-} B) PS C) PS^{1+}
- (iv) Draw Lewis dot structure of the molecule with the shortest bond length (based on the MO diagram) from among the three (PS^{1-} , PS, PS^{1+}), and indicate the bond order calculated from the Lewis dot structure. **(3 points)**

Lewis Dot Structure

Bond Order



B6C5F98A-AD0C-44F3-A25B-C484DDD67530

midterm-exam-93388

#669 Page 10 of 10

SCRAP PAGE

**CHEMISTRY 110 –Midterm Exam****Thursday November 7, 2024, 6:30 – 8:30 pm**

Examiners: Pallavi Sirjoosingh and Ashok Kakkar

First name (please write as legibly as possible within the boxes)

--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--

Last name

--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--

Student ID number

--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--

This booklet has 8 questions on 10 pages (The last page can be used as a scrap page).

The staple must NOT be removed from the question booklet. Your intact question booklet must be returned at the end of the exam.

A separate booklet contains a periodic table, and datasheet.

Write your answers directly under the questions in the space provided. Do not write on the QR codes or near the top edges (as it may be cut off).

Please write **legibly and dark enough** for the scan to be readable. For students who use English as a second language, dictionaries are permitted. Calculators are allowed.

DO NOT ASK QUESTIONS OF THE INVIGILATORS about interpretations of the exam questions. If you have a concern about a question, please summarize it at the top of this page. All concerns will be evaluated after the exam. GOOD LUCK!

**Question 1 8 points** (*Assume all transitions are possible*)

a. 6 points Calculate the **energy (in kJ/mol)** required to ionize (completely remove) the electron from the **$n=3$** level of **He^+ ion** (He: Helium). Show your work including equations. (3 sig. figs.)

b. 2 points

(I) Select the largest (longest) wavelength from the three choices below:

- i. Wavelength of photon to *completely remove* an electron from *1s orbital of Hydrogen atom*
- ii. Wavelength of photon to *completely remove* an electron from *1s orbital of Boron atom (B)*
- iii. Wavelength of photon to *completely remove* an electron from *1s orbital of Boron cation (B^+)*

i or ii or iii

(II) Select the smallest (shortest) wavelength from the three choices below (*for H atom*):

- i. Wavelength of photon *emitted* for an electron transitioning from *2p to 1s orbital*
- ii. Wavelength of photon *absorbed* for an electron transitioning from *2s to 4p orbital*
- iii. Wavelength of photon *absorbed* for an electron transitioning from *1s to 3p orbital*

i or ii or iii



Question 2 8 points



A Chem110 student plots the graph shown on the left, from data obtained for the kinetic energy (eV) of the ejected electron versus frequency (s^{-1}) of the incident light, for an unknown metal. ($1\text{eV} = 1.6022 \times 10^{-19} \text{ J}$)

Based on the plot, answer the questions **a to c** below:

a. 1 point On the plot above, label the **threshold frequency**.

b. 2 points What is the value of the work potential (in J) of the unknown metal?

c. 3 points What is the *longest* wavelength (in nm) required to eject an electron from this metal's surface? Show your work including equation(s) used. (3 sig figs)

d. 2 points In the space provided below, plot the following

(Assume frequency is *above* threshold frequency):

- i. *Kinetic energy* of an ejected electron versus the *intensity* of the incident light (left)
- ii. *Number of electrons* ejected versus the *frequency* of the incident light (right)



**Question 3** **6 points****a. 2 points**

In the ground state of Ge (Germanium; $Z = 32$) atom, **name** the following orbitals:

i. Orbital containing the electron that requires the maximum energy to be removed.

ii. Orbital containing the electron that experiences the minimum effective nuclear charge.

b. 2 points

An unknown anion (A^-) has the following *excited* electronic configuration: $[Ar] 4s^2 3d^3 4p^1$

(Ar: Argon; $Z = 18$)

Give the **condensed ground state electronic configuration** of the atom, **A** and the cation, **A^{2+}** .
Show your work. (Assume all electronic transitions are possible)

Electronic Configuration of A:

Electronic Configuration of A^{2+} :

c. 2 points

In a single atom, determine how many electrons can have the following set of quantum numbers?

(i) $n = 5; l = 3$

In a single atom, determine how many electrons can have the following set of quantum numbers?

(i) $n = 3; l = 2; m_s = +1/2$

**Question 4** **8 points**

Rank the following from 1 to 4 in **increasing** order. Provide the ranking in the boxes provided.

Note: Atomic number (Z) are provided for relevant atoms

a) Increasing (lowest to highest) order of *ionization energy*: (1= lowest; 4 = highest)

Mg

Mg: Z = 12

Rb

Rb: Z = 37

Cs

Cs: Z = 55

K

K : Z = 19

b) Increasing (lowest to highest) order of *bond angle*: (1= lowest; 4 = highest)

H₂OCCl₄XeF₄

HCN

c) Increasing (smallest to largest) order of *ionic size* (1= smallest; 4 = largest)

S²⁻

S: Z = 16

O²⁻

O: Z = 8

Be²⁺

Be: Z = 4

Na⁺

Na: Z = 11

d) Increasing (lowest to highest) order of *bond energy* (1= lowest; 4 = highest)

P-F

P-Cl

P-I

P-Br

**Question 5** **12 points**

a. 9 points For the molecules HClO_4 and TeF_6 , draw the most stable *Lewis structure* showing appropriate shape and *dashed/wedged bonds*, based on VSEPR. Determine the **molecular geometry** around the central atom. For HClO_4 , label the *most polar bond*.

HClO_4 (there are no O-O bonds)

Label the most polar bond

Molecular Geometry around Cl

TeF_6

Molecular Geometry around Te

b. 3 points (Note: Lone pairs are not shown in structures below)

(i) For the three *resonance* structures (1, 2 and 3) shown below, answer the following question:



(I) Circle the *most contributing* resonance structure.

Structure 1

Structure 2

Structure 3

(ii) For the three structures given (4, 5, and 6) below, answer the following questions:



(I) Circle all the structures that represent *resonance structures* of each other:

Structure 4

Structure 5

Structure 6

(II) Among the resonance structures, circle the *most contributing* resonance structure.

Structure 4

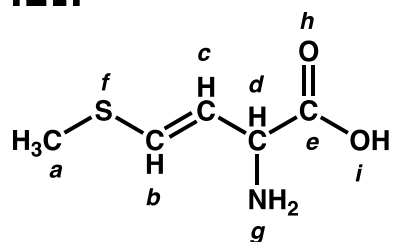
Structure 5

Structure 6

**Question 6** **6 points**

Using the data given below, calculate the **lattice energy** for SrO(s).

	ΔH° (kJ/mol)
$\text{Sr}(s) \longrightarrow \text{Sr}(g)$	+180
$\text{Sr}(g) \longrightarrow \text{Sr}^{2+}(g) + 2e^-$	+1468
$\text{O}_2(g) \longrightarrow 2\text{O}(g)$	+500
$\text{O}(g) + 2e^- \longrightarrow \text{O}^{2-}(g)$	+600
$\text{Sr}(s) + \frac{1}{2}\text{O}_2(g) \longrightarrow \text{SrO}(s)$	-550

**Question 7 10 points**

Consider the molecule given on the left, and answer questions (i) to (vi) using chemical bonding theories that we have learnt in class (valence bond theory, hybridization).

Note that the lone pairs are not shown in the structure:

i. Consider the carbon atoms marked in as ***a***, ***b***, ***c***, ***d***, and ***e*** in the molecule, and determine **hybridization** at each of the carbon centers. (2.5 points)

Atom	Hybridization
Carbon (<i>a</i>)	
Carbon (<i>b</i>)	
Carbon (<i>c</i>)	
Carbon (<i>d</i>)	
Carbon (<i>e</i>)	

ii. Now, consider the sulfur atom (S) marked as ***f***, nitrogen (N) atom (***g***), and oxygen(O) atoms (***h*** and ***i***), and determine their **hybridization**. (2 points)

Atom	Hybridization
Sulfur (<i>f</i>)	
Nitrogen (<i>g</i>)	
Oxygen (<i>h</i>)	
Oxygen (<i>i</i>)	

iii. **Describe** all the bonds formed by the carbon atoms ***a*** and ***b***, using sigma and pi designation as appropriate. (2 points)

Atom	Bonds formed (use sigma and pi as appropriate)
Carbon (<i>a</i>)	
Carbon (<i>b</i>)	

iv. How many total sp^3 -s sigma bonds are in the molecule? (1 point)

v. Determine **electronic** and **molecular geometry** at carbon (C) labeled ***e*** and sulfur (S) labeled ***f*** centres. (2 points)

Atom	Electronic Geometry	Molecular Geometry
Carbon (<i>e</i>)		
Sulfur (<i>f</i>)		

vi. Circle the value from the choices given below, that describes the **C-S-C bond angle at sulfur** (S) marked ***f***: (0.5 points)

>90° to <109.5° 109.5° >109.5 to <120° 120° >120° to <180°



Question 8 12 points

Using the **valence shell orbitals only** draw a **complete** molecular orbital (MO) diagram (with no s-p mixing) for the molecule **PS**, in the space provided below. **(6 points)**

Assume that 3s and 3p atomic orbitals will form molecular orbitals like those formed by the combination of 2s and 2p atomic orbitals that we learnt in class, and there is no s-p mixing.

Show **energy arrow**, **atomic orbitals** involved, **atom labels**, the **molecular orbitals** that are formed, **filled electrons** in the MO diagram etc. Make sure to label all the atomic and molecular orbitals.

Using the MO diagram above, answer the following questions:

- (i) Which one of the following will have the highest bond order? Circle the correct choice
 A) PS^{1-} B) PS C) PS^{1+}
- (ii) Which one of the following will have the longest bond length? Circle the correct choice
 A) PS^{1-} B) PS C) PS^{1+}
- (iii) Which of the following will be diamagnetic? Circle the correct choice.
 A) PS^{1-} B) PS C) PS^{1+}
- (iv) Draw Lewis dot structure of the molecule with the shortest bond length (based on the MO diagram) from among the three (PS^{1-} , PS, PS^{1+}), and indicate the bond order calculated from the Lewis dot structure. **(3 points)**

Lewis Dot Structure

Bond Order



B833FBA9-24BD-45D0-9910-FA12F2C14AAF

midterm-exam-93388

#670 Page 10 of 10

SCRAP PAGE

**CHEMISTRY 110 –Midterm Exam****Thursday November 7, 2024, 6:30 – 8:30 pm****Examiners: Pallavi Sirjoosingh and Ashok Kakkar**

First name (please write as legibly as possible within the boxes)

--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--

Last name

--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--

Student ID number

--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--

This booklet has 8 questions on 10 pages (The last page can be used as a scrap page).

The staple must NOT be removed from the question booklet. Your intact question booklet must be returned at the end of the exam.

A separate booklet contains a periodic table, and datasheet.

Write your answers directly under the questions in the space provided. Do not write on the QR codes or near the top edges (as it may be cut off).

Please write **legibly and dark enough** for the scan to be readable. For students who use English as a second language, dictionaries are permitted. Calculators are allowed.

DO NOT ASK QUESTIONS OF THE INVIGILATORS about interpretations of the exam questions. If you have a concern about a question, please summarize it at the top of this page. All concerns will be evaluated after the exam. GOOD LUCK!

**Question 1 8 points** (*Assume all transitions are possible*)

a. 6 points Calculate the **energy (in kJ/mol)** required to ionize (completely remove) the electron from the **$n=3$** level of **He^+ ion** (He: Helium). Show your work including equations. (3 sig. figs.)

b. 2 points

(I) Select the largest (longest) wavelength from the three choices below:

- i. Wavelength of photon to *completely remove* an electron from *1s orbital of Hydrogen atom*
- ii. Wavelength of photon to *completely remove* an electron from *1s orbital of Boron atom (B)*
- iii. Wavelength of photon to *completely remove* an electron from *1s orbital of Boron cation (B^+)*

i or ii or iii

(II) Select the smallest (shortest) wavelength from the three choices below (*for H atom*):

- i. Wavelength of photon *emitted* for an electron transitioning from *2p to 1s orbital*
- ii. Wavelength of photon *absorbed* for an electron transitioning from *2s to 4p orbital*
- iii. Wavelength of photon *absorbed* for an electron transitioning from *1s to 3p orbital*

i or ii or iii



Question 2 8 points



A Chem110 student plots the graph shown on the left, from data obtained for the kinetic energy (eV) of the ejected electron versus frequency (s^{-1}) of the incident light, for an unknown metal. ($1\text{eV} = 1.6022 \times 10^{-19} \text{ J}$)

Based on the plot, answer the questions **a to c** below:

a. 1 point On the plot above, label the **threshold frequency**.

b. 2 points What is the value of the work potential (in J) of the unknown metal?

c. 3 points What is the *longest* wavelength (in nm) required to eject an electron from this metal's surface? Show your work including equation(s) used. (3 sig figs)

d. 2 points In the space provided below, plot the following

(Assume frequency is *above* threshold frequency):

- i. *Kinetic energy* of an ejected electron versus the *intensity* of the incident light (left)
- ii. *Number of electrons* ejected versus the *frequency* of the incident light (right)



**Question 3** **6 points****a. 2 points**

In the ground state of Ge (Germanium; $Z = 32$) atom, **name** the following orbitals:

i. Orbital containing the electron that requires the maximum energy to be removed.

ii. Orbital containing the electron that experiences the minimum effective nuclear charge.

b. 2 points

An unknown anion (A^-) has the following *excited* electronic configuration: $[Ar] 4s^2 3d^3 4p^1$

(Ar: Argon; $Z = 18$)

Give the **condensed ground state electronic configuration** of the atom, **A** and the cation, **A^{2+}** .
Show your work. (Assume all electronic transitions are possible)

Electronic Configuration of A:

Electronic Configuration of A^{2+} :

c. 2 points

In a single atom, determine how many electrons can have the following set of quantum numbers?

(i) $n = 5; l = 3$

In a single atom, determine how many electrons can have the following set of quantum numbers?

(i) $n = 3; l = 2; m_s = +1/2$

**Question 4** **8 points**

Rank the following from 1 to 4 in **increasing** order. Provide the ranking in the boxes provided.

Note: Atomic number (Z) are provided for relevant atoms

a) Increasing (lowest to highest) order of *ionization energy*: (1= lowest; 4 = highest)

Mg

Mg: Z = 12

Rb

Rb: Z = 37

Cs

Cs: Z = 55

K

K : Z = 19

b) Increasing (lowest to highest) order of *bond angle*: (1= lowest; 4 = highest)

H₂OCCl₄XeF₄

HCN

c) Increasing (smallest to largest) order of *ionic size* (1= smallest; 4 = largest)

S²⁻

S: Z = 16

O²⁻

O: Z = 8

Be²⁺

Be: Z = 4

Na⁺

Na: Z = 11

d) Increasing (lowest to highest) order of *bond energy* (1= lowest; 4 = highest)

P-F

P-Cl

P-I

P-Br

**Question 5** **12 points**

a. 9 points For the molecules HClO_4 and TeF_6 , draw the most stable *Lewis structure* showing appropriate shape and *dashed/wedged bonds*, based on VSEPR. Determine the **molecular geometry** around the central atom. For HClO_4 , label the *most polar bond*.



b. 3 points (Note: Lone pairs are not shown in structures below)

(i) For the three *resonance* structures (1, 2 and 3) shown below, answer the following question:



(I) Circle the *most contributing* resonance structure.

Structure 1

Structure 2

Structure 3

(ii) For the three structures given (4, 5, and 6) below, answer the following questions:



(I) Circle all the structures that represent *resonance structures* of each other:

Structure 4

Structure 5

Structure 6

(II) Among the resonance structures, circle the *most contributing* resonance structure.

Structure 4

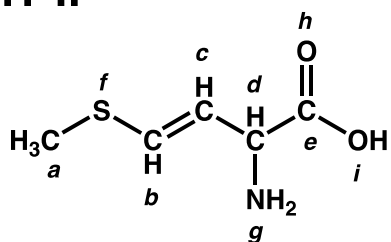
Structure 5

Structure 6

**Question 6** **6 points**

Using the data given below, calculate the **lattice energy** for SrO(s).

	ΔH° (kJ/mol)
$\text{Sr}(s) \longrightarrow \text{Sr}(g)$	+180
$\text{Sr}(g) \longrightarrow \text{Sr}^{2+}(g) + 2e^-$	+1468
$\text{O}_2(g) \longrightarrow 2\text{O}(g)$	+500
$\text{O}(g) + 2e^- \longrightarrow \text{O}^{2-}(g)$	+600
$\text{Sr}(s) + \frac{1}{2}\text{O}_2(g) \longrightarrow \text{SrO}(s)$	-550

**Question 7 10 points**

Consider the molecule given on the left, and answer questions (i) to (vi) using chemical bonding theories that we have learnt in class (valence bond theory, hybridization).

Note that the lone pairs are not shown in the structure:

i. Consider the carbon atoms marked in as ***a***, ***b***, ***c***, ***d***, and ***e*** in the molecule, and determine **hybridization** at each of the carbon centers. (2.5 points)

Atom	Hybridization
Carbon (<i>a</i>)	
Carbon (<i>b</i>)	
Carbon (<i>c</i>)	
Carbon (<i>d</i>)	
Carbon (<i>e</i>)	

ii. Now, consider the sulfur atom (S) marked as ***f***, nitrogen (N) atom (***g***), and oxygen(O) atoms (***h*** and ***i***), and determine their **hybridization**. (2 points)

Atom	Hybridization
Sulfur (<i>f</i>)	
Nitrogen (<i>g</i>)	
Oxygen (<i>h</i>)	
Oxygen (<i>i</i>)	

iii. **Describe** all the bonds formed by the carbon atoms ***a*** and ***b***, using sigma and pi designation as appropriate. (2 points)

Atom	Bonds formed (use sigma and pi as appropriate)
Carbon (<i>a</i>)	
Carbon (<i>b</i>)	

iv. How many total sp^3 -s sigma bonds are in the molecule? (1 point)

v. Determine **electronic** and **molecular geometry** at carbon (C) labeled ***e*** and sulfur (S) labeled ***f*** centres. (2 points)

Atom	Electronic Geometry	Molecular Geometry
Carbon (<i>e</i>)		
Sulfur (<i>f</i>)		

vi. Circle the value from the choices given below, that describes the **C-S-C bond angle at sulfur** (S) marked ***f***: (0.5 points)

>90° to <109.5° 109.5° >109.5 to <120° 120° >120° to <180°



Question 8 12 points

Using the **valence shell orbitals only** draw a **complete** molecular orbital (MO) diagram (with no s-p mixing) for the molecule **PS**, in the space provided below. **(6 points)**

Assume that 3s and 3p atomic orbitals will form molecular orbitals like those formed by the combination of 2s and 2p atomic orbitals that we learnt in class, and there is no s-p mixing.

Show **energy arrow**, **atomic orbitals** involved, **atom labels**, the **molecular orbitals** that are formed, **filled electrons** in the MO diagram etc. Make sure to label all the atomic and molecular orbitals.

Using the MO diagram above, answer the following questions:

- (i) Which one of the following will have the highest bond order? Circle the correct choice
 A) PS^{1-} B) PS C) PS^{1+}
- (ii) Which one of the following will have the longest bond length? Circle the correct choice
 A) PS^{1-} B) PS C) PS^{1+}
- (iii) Which of the following will be diamagnetic? Circle the correct choice.
 A) PS^{1-} B) PS C) PS^{1+}
- (iv) Draw Lewis dot structure of the molecule with the shortest bond length (based on the MO diagram) from among the three (PS^{1-} , PS, PS^{1+}), and indicate the bond order calculated from the Lewis dot structure. **(3 points)**

Lewis Dot Structure

Bond Order



D75CCE12-9084-4539-9D5E-F99A89A41EB6

midterm-exam-93388

#671 Page 10 of 10

SCRAP PAGE

**CHEMISTRY 110 –Midterm Exam****Thursday November 7, 2024, 6:30 – 8:30 pm**

Examiners: Pallavi Sirjoosingh and Ashok Kakkar

First name (please write as legibly as possible within the boxes)

--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--

Last name

--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--

Student ID number

--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--

This booklet has 8 questions on 10 pages (The last page can be used as a scrap page).

The staple must NOT be removed from the question booklet. Your intact question booklet must be returned at the end of the exam.

A separate booklet contains a periodic table, and datasheet.

Write your answers directly under the questions in the space provided. Do not write on the QR codes or near the top edges (as it may be cut off).

Please write **legibly and dark enough** for the scan to be readable. For students who use English as a second language, dictionaries are permitted. Calculators are allowed.

DO NOT ASK QUESTIONS OF THE INVIGILATORS about interpretations of the exam questions. If you have a concern about a question, please summarize it at the top of this page. All concerns will be evaluated after the exam. GOOD LUCK!

**Question 1 8 points** (*Assume all transitions are possible*)

a. 6 points Calculate the **energy (in kJ/mol)** required to ionize (completely remove) the electron from the **$n=3$** level of **He^+ ion** (He: Helium). Show your work including equations. (3 sig. figs.)

b. 2 points

(I) Select the largest (longest) wavelength from the three choices below:

- i. Wavelength of photon to *completely remove* an electron from *1s orbital of Hydrogen atom*
- ii. Wavelength of photon to *completely remove* an electron from *1s orbital of Boron atom (B)*
- iii. Wavelength of photon to *completely remove* an electron from *1s orbital of Boron cation (B^+)*

i or ii or iii

(II) Select the smallest (shortest) wavelength from the three choices below (*for H atom*):

- i. Wavelength of photon *emitted* for an electron transitioning from *2p to 1s orbital*
- ii. Wavelength of photon *absorbed* for an electron transitioning from *2s to 4p orbital*
- iii. Wavelength of photon *absorbed* for an electron transitioning from *1s to 3p orbital*

i or ii or iii



Question 2 8 points



A Chem110 student plots the graph shown on the left, from data obtained for the kinetic energy (eV) of the ejected electron versus frequency (s^{-1}) of the incident light, for an unknown metal. ($1\text{eV} = 1.6022 \times 10^{-19} \text{ J}$)

Based on the plot, answer the questions **a to c** below:

a. 1 point On the plot above, label the **threshold frequency**.

b. 2 points What is the value of the work potential (in J) of the unknown metal?

c. 3 points What is the *longest* wavelength (in nm) required to eject an electron from this metal's surface? Show your work including equation(s) used. (3 sig figs)

d. 2 points In the space provided below, plot the following

(Assume frequency is *above* threshold frequency):

- Kinetic energy* of an ejected electron versus the *intensity* of the incident light (left)
- Number of electrons* ejected versus the *frequency* of the incident light (right)



**Question 3** **6 points****a. 2 points**

In the ground state of Ge (Germanium; $Z = 32$) atom, **name** the following orbitals:

i. Orbital containing the electron that requires the maximum energy to be removed.

ii. Orbital containing the electron that experiences the minimum effective nuclear charge.

b. 2 points

An unknown anion (A^-) has the following *excited* electronic configuration: $[Ar] 4s^2 3d^3 4p^1$

(Ar: Argon; $Z = 18$)

Give the **condensed ground state electronic configuration** of the atom, **A** and the cation, **A^{2+}** .
Show your work. (Assume all electronic transitions are possible)

Electronic Configuration of A:

Electronic Configuration of A^{2+} :

c. 2 points

In a single atom, determine how many electrons can have the following set of quantum numbers?

(i) $n = 5; l = 3$

In a single atom, determine how many electrons can have the following set of quantum numbers?

(i) $n = 3; l = 2; m_s = +1/2$

**Question 4** **8 points**

Rank the following from 1 to 4 in **increasing** order. Provide the ranking in the boxes provided.

Note: Atomic number (Z) are provided for relevant atoms

a) Increasing (lowest to highest) order of *ionization energy*: (1= lowest; 4 = highest)

Mg

Mg: Z = 12

Rb

Rb: Z = 37

Cs

Cs: Z = 55

K

K : Z = 19

b) Increasing (lowest to highest) order of *bond angle*: (1= lowest; 4 = highest)

H₂OCCl₄XeF₄

HCN

c) Increasing (smallest to largest) order of *ionic size* (1= smallest; 4 = largest)

S²⁻

S: Z = 16

O²⁻

O: Z = 8

Be²⁺

Be: Z = 4

Na⁺

Na: Z = 11

d) Increasing (lowest to highest) order of *bond energy* (1= lowest; 4 = highest)

P-F

P-Cl

P-I

P-Br

**Question 5** **12 points**

a. 9 points For the molecules HClO_4 and TeF_6 , draw the most stable *Lewis structure* showing appropriate shape and *dashed/wedged bonds*, based on VSEPR. Determine the **molecular geometry** around the central atom. For HClO_4 , label the *most polar bond*.



b. 3 points (Note: Lone pairs are not shown in structures below)

(i) For the three *resonance* structures (1, 2 and 3) shown below, answer the following question:



(I) Circle the *most contributing* resonance structure.

Structure 1

Structure 2

Structure 3

(ii) For the three structures given (4, 5, and 6) below, answer the following questions:



(I) Circle all the structures that represent *resonance structures* of each other:

Structure 4

Structure 5

Structure 6

(II) Among the resonance structures, circle the *most contributing* resonance structure.

Structure 4

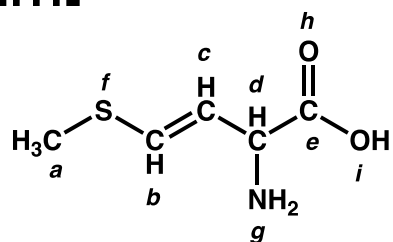
Structure 5

Structure 6

**Question 6** **6 points**

Using the data given below, calculate the **lattice energy** for SrO(s).

	ΔH° (kJ/mol)
$\text{Sr}(s) \longrightarrow \text{Sr}(g)$	+180
$\text{Sr}(g) \longrightarrow \text{Sr}^{2+}(g) + 2e^-$	+1468
$\text{O}_2(g) \longrightarrow 2\text{O}(g)$	+500
$\text{O}(g) + 2e^- \longrightarrow \text{O}^{2-}(g)$	+600
$\text{Sr}(s) + \frac{1}{2}\text{O}_2(g) \longrightarrow \text{SrO}(s)$	-550

**Question 7 10 points**

Consider the molecule given on the left, and answer questions (i) to (vi) using chemical bonding theories that we have learnt in class (valence bond theory, hybridization).

Note that the lone pairs are not shown in the structure:

i. Consider the carbon atoms marked in as ***a***, ***b***, ***c***, ***d***, and ***e*** in the molecule, and determine **hybridization** at each of the carbon centers. (2.5 points)

Atom	Hybridization
Carbon (<i>a</i>)	
Carbon (<i>b</i>)	
Carbon (<i>c</i>)	
Carbon (<i>d</i>)	
Carbon (<i>e</i>)	

ii. Now, consider the sulfur atom (S) marked as ***f***, nitrogen (N) atom (***g***), and oxygen(O) atoms (***h*** and ***i***), and determine their **hybridization**. (2 points)

Atom	Hybridization
Sulfur (<i>f</i>)	
Nitrogen (<i>g</i>)	
Oxygen (<i>h</i>)	
Oxygen (<i>i</i>)	

iii. **Describe** all the bonds formed by the carbon atoms ***a*** and ***b***, using sigma and pi designation as appropriate. (2 points)

Atom	Bonds formed (use sigma and pi as appropriate)
Carbon (<i>a</i>)	
Carbon (<i>b</i>)	

iv. How many total sp^3 -s sigma bonds are in the molecule? (1 point)

v. Determine **electronic** and **molecular geometry** at carbon (C) labeled ***e*** and sulfur (S) labeled ***f*** centres. (2 points)

Atom	Electronic Geometry	Molecular Geometry
Carbon (<i>e</i>)		
Sulfur (<i>f</i>)		

vi. Circle the value from the choices given below, that describes the **C-S-C bond angle at sulfur** (S) marked ***f***: (0.5 points)

>90° to <109.5°

109.5°

>109.5 to <120°

120°

>120° to <180°


Question 8 12 points

Using the **valence shell orbitals only** draw a **complete** molecular orbital (MO) diagram (with no s-p mixing) for the molecule **PS**, in the space provided below. **(6 points)**

Assume that 3s and 3p atomic orbitals will form molecular orbitals like those formed by the combination of 2s and 2p atomic orbitals that we learnt in class, and there is no s-p mixing.

Show **energy arrow**, **atomic orbitals** involved, **atom labels**, the **molecular orbitals** that are formed, **filled electrons** in the MO diagram etc. Make sure to label all the atomic and molecular orbitals.

Using the MO diagram above, answer the following questions:

- (i) Which one of the following will have the highest bond order? Circle the correct choice
 A) PS^{1-} B) PS C) PS^{1+}
- (ii) Which one of the following will have the longest bond length? Circle the correct choice
 A) PS^{1-} B) PS C) PS^{1+}
- (iii) Which of the following will be diamagnetic? Circle the correct choice.
 A) PS^{1-} B) PS C) PS^{1+}
- (iv) Draw Lewis dot structure of the molecule with the shortest bond length (based on the MO diagram) from among the three (PS^{1-} , PS, PS^{1+}), and indicate the bond order calculated from the Lewis dot structure. **(3 points)**

Lewis Dot Structure

Bond Order



E15D72D8-3693-4CB9-AA98-5DC6454A78B7

midterm-exam-93388

#672 Page 10 of 10

SCRAP PAGE

**CHEMISTRY 110 –Midterm Exam****Thursday November 7, 2024, 6:30 – 8:30 pm****Examiners: Pallavi Sirjoosingh and Ashok Kakkar**

First name (please write as legibly as possible within the boxes)

--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--

Last name

--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--

Student ID number

--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--

This booklet has 8 questions on 10 pages (The last page can be used as a scrap page).

The staple must NOT be removed from the question booklet. Your intact question booklet must be returned at the end of the exam.

A separate booklet contains a periodic table, and datasheet.

Write your answers directly under the questions in the space provided. Do not write on the QR codes or near the top edges (as it may be cut off).

Please write **legibly and dark enough** for the scan to be readable. For students who use English as a second language, dictionaries are permitted. Calculators are allowed.

DO NOT ASK QUESTIONS OF THE INVIGILATORS about interpretations of the exam questions. If you have a concern about a question, please summarize it at the top of this page. All concerns will be evaluated after the exam. GOOD LUCK!

**Question 1 8 points** (*Assume all transitions are possible*)

a. 6 points Calculate the **energy (in kJ/mol)** required to ionize (completely remove) the electron from the **$n=3$** level of **He^+ ion** (He: Helium). Show your work including equations. (3 sig. figs.)

b. 2 points

(I) Select the largest (longest) wavelength from the three choices below:

- i. Wavelength of photon to *completely remove* an electron from *1s orbital of Hydrogen atom*
- ii. Wavelength of photon to *completely remove* an electron from *1s orbital of Boron atom (B)*
- iii. Wavelength of photon to *completely remove* an electron from *1s orbital of Boron cation (B^+)*

i or ii or iii

(II) Select the smallest (shortest) wavelength from the three choices below (*for H atom*):

- i. Wavelength of photon *emitted* for an electron transitioning from *2p to 1s orbital*
- ii. Wavelength of photon *absorbed* for an electron transitioning from *2s to 4p orbital*
- iii. Wavelength of photon *absorbed* for an electron transitioning from *1s to 3p orbital*

i or ii or iii



Question 2 8 points



A Chem110 student plots the graph shown on the left, from data obtained for the kinetic energy (eV) of the ejected electron versus frequency (s^{-1}) of the incident light, for an unknown metal. ($1\text{eV} = 1.6022 \times 10^{-19} \text{ J}$)

Based on the plot, answer the questions **a to c** below:

a. 1 point On the plot above, label the **threshold frequency**.

b. 2 points What is the value of the work potential (in J) of the unknown metal?

c. 3 points What is the *longest* wavelength (in nm) required to eject an electron from this metal's surface? Show your work including equation(s) used. (3 sig figs)

d. 2 points In the space provided below, plot the following

(Assume frequency is *above* threshold frequency):

- Kinetic energy* of an ejected electron versus the *intensity* of the incident light (left)
- Number of electrons* ejected versus the *frequency* of the incident light (right)



**Question 3** **6 points****a. 2 points**

In the ground state of Ge (Germanium; $Z = 32$) atom, **name** the following orbitals:

i. Orbital containing the electron that requires the maximum energy to be removed.

ii. Orbital containing the electron that experiences the minimum effective nuclear charge.

b. 2 points

An unknown anion (A^-) has the following *excited* electronic configuration: $[Ar] 4s^2 3d^3 4p^1$

(Ar: Argon; $Z = 18$)

Give the **condensed ground state electronic configuration** of the atom, **A** and the cation, **A^{2+}** .
Show your work. (Assume all electronic transitions are possible)

Electronic Configuration of A:

Electronic Configuration of A^{2+} :

c. 2 points

In a single atom, determine how many electrons can have the following set of quantum numbers?

(i) $n = 5; l = 3$

In a single atom, determine how many electrons can have the following set of quantum numbers?

(i) $n = 3; l = 2; m_s = +1/2$

**Question 4** **8 points**

Rank the following from 1 to 4 in **increasing** order. Provide the ranking in the boxes provided.

Note: Atomic number (Z) are provided for relevant atoms

a) Increasing (lowest to highest) order of *ionization energy*: (1= lowest; 4 = highest)

Mg

Mg: Z = 12

Rb

Rb: Z = 37

Cs

Cs: Z = 55

K

K : Z = 19

b) Increasing (lowest to highest) order of *bond angle*: (1= lowest; 4 = highest)

H₂OCCl₄XeF₄

HCN

c) Increasing (smallest to largest) order of *ionic size* (1= smallest; 4 = largest)

S²⁻

S: Z = 16

O²⁻

O: Z = 8

Be²⁺

Be: Z = 4

Na⁺

Na: Z = 11

d) Increasing (lowest to highest) order of *bond energy* (1= lowest; 4 = highest)

P-F

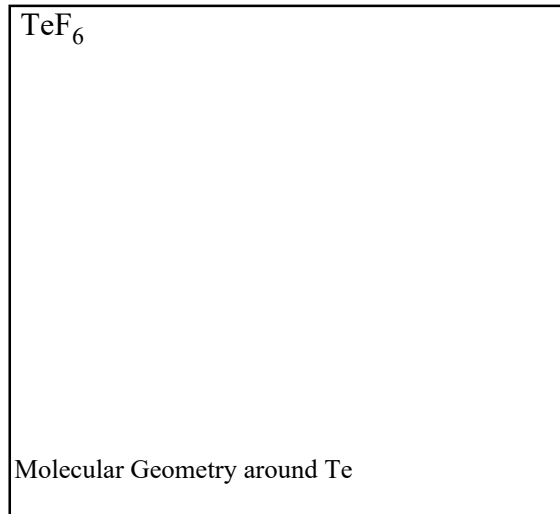
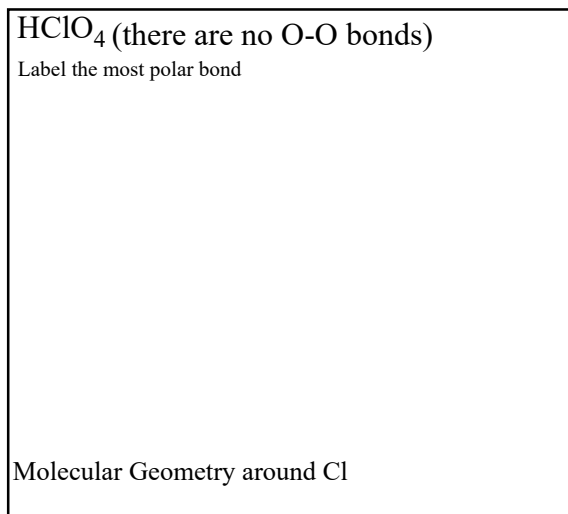
P-Cl

P-I

P-Br

**Question 5** **12 points**

a. 9 points For the molecules HClO_4 and TeF_6 , draw the most stable *Lewis structure* showing appropriate shape and *dashed/wedged bonds*, based on VSEPR. Determine the **molecular geometry** around the central atom. For HClO_4 , label the *most polar bond*.



b. 3 points (Note: Lone pairs are not shown in structures below)

(i) For the three *resonance* structures (1, 2 and 3) shown below, answer the following question:



(I) Circle the *most contributing* resonance structure.

Structure 1

Structure 2

Structure 3

(ii) For the three structures given (4, 5, and 6) below, answer the following questions:



(I) Circle all the structures that represent *resonance structures* of each other:

Structure 4

Structure 5

Structure 6

(II) Among the resonance structures, circle the *most contributing* resonance structure.

Structure 4

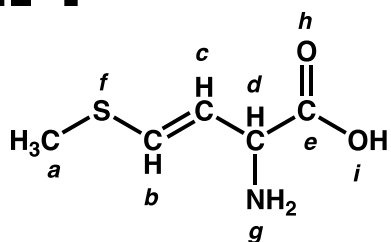
Structure 5

Structure 6

**Question 6** **6 points**

Using the data given below, calculate the **lattice energy** for SrO(s).

	ΔH° (kJ/mol)
$\text{Sr}(s) \longrightarrow \text{Sr}(g)$	+180
$\text{Sr}(g) \longrightarrow \text{Sr}^{2+}(g) + 2e^-$	+1468
$\text{O}_2(g) \longrightarrow 2\text{O}(g)$	+500
$\text{O}(g) + 2e^- \longrightarrow \text{O}^{2-}(g)$	+600
$\text{Sr}(s) + \frac{1}{2}\text{O}_2(g) \longrightarrow \text{SrO}(s)$	-550

**Question 7 10 points**

Consider the molecule given on the left, and answer questions (i) to (vi) using chemical bonding theories that we have learnt in class (valence bond theory, hybridization).

Note that the lone pairs are not shown in the structure:

i. Consider the carbon atoms marked in as ***a***, ***b***, ***c***, ***d***, and ***e*** in the molecule, and determine **hybridization** at each of the carbon centers. (2.5 points)

Atom	Hybridization
Carbon (<i>a</i>)	
Carbon (<i>b</i>)	
Carbon (<i>c</i>)	
Carbon (<i>d</i>)	
Carbon (<i>e</i>)	

ii. Now, consider the sulfur atom (S) marked as ***f***, nitrogen (N) atom (***g***), and oxygen(O) atoms (***h*** and ***i***), and determine their **hybridization**. (2 points)

Atom	Hybridization
Sulfur (<i>f</i>)	
Nitrogen (<i>g</i>)	
Oxygen (<i>h</i>)	
Oxygen (<i>i</i>)	

iii. **Describe** all the bonds formed by the carbon atoms ***a*** and ***b***, using sigma and pi designation as appropriate. (2 points)

Atom	Bonds formed (use sigma and pi as appropriate)
Carbon (<i>a</i>)	
Carbon (<i>b</i>)	

iv. How many total sp^3 -s sigma bonds are in the molecule? (1 point)

v. Determine **electronic** and **molecular geometry** at carbon (C) labeled ***e*** and sulfur (S) labeled ***f*** centres. (2 points)

Atom	Electronic Geometry	Molecular Geometry
Carbon (<i>e</i>)		
Sulfur (<i>f</i>)		

vi. Circle the value from the choices given below, that describes the **C-S-C bond angle at sulfur** (S) marked ***f***: (0.5 points)

>90° to <109.5° 109.5° >109.5 to <120° 120° >120° to <180°

**Question 8 12 points**

Using the **valence shell orbitals only** draw a **complete** molecular orbital (MO) diagram (with no s-p mixing) for the molecule **PS**, in the space provided below. **(6 points)**

Assume that 3s and 3p atomic orbitals will form molecular orbitals like those formed by the combination of 2s and 2p atomic orbitals that we learnt in class, and there is no s-p mixing.

Show **energy arrow**, **atomic orbitals** involved, **atom labels**, the **molecular orbitals** that are formed, **filled electrons** in the MO diagram etc. Make sure to **label all the atomic and molecular orbitals**.

Using the MO diagram above, answer the following questions:

- (i) Which one of the following will have the highest bond order? Circle the correct choice
A) PS^{1-} B) PS C) PS^{1+}
- (ii) Which one of the following will have the longest bond length? Circle the correct choice
A) PS^{1-} B) PS C) PS^{1+}
- (iii) Which of the following will be diamagnetic? Circle the correct choice.
A) PS^{1-} B) PS C) PS^{1+}
- (iv) Draw Lewis dot structure of the molecule with the shortest bond length (based on the MO diagram) from among the three (PS^{1-} , PS, PS^{1+}), and indicate the bond order calculated from the Lewis dot structure. **(3 points)**

Lewis Dot Structure

Bond Order



75DF5163-B3DA-43AD-AB32-F1A8C16E8C90

midterm-exam-93388

#673 Page 10 of 10

SCRAP PAGE

**CHEMISTRY 110 –Midterm Exam****Thursday November 7, 2024, 6:30 – 8:30 pm****Examiners: Pallavi Sirjoosingh and Ashok Kakkar**

First name (please write as legibly as possible within the boxes)

--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--

Last name

--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--

Student ID number

--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--

This booklet has 8 questions on 10 pages (The last page can be used as a scrap page).

The staple must NOT be removed from the question booklet. Your intact question booklet must be returned at the end of the exam.

A separate booklet contains a periodic table, and datasheet.

Write your answers directly under the questions in the space provided. Do not write on the QR codes or near the top edges (as it may be cut off).

Please write **legibly and dark enough** for the scan to be readable. For students who use English as a second language, dictionaries are permitted. Calculators are allowed.

DO NOT ASK QUESTIONS OF THE INVIGILATORS about interpretations of the exam questions. If you have a concern about a question, please summarize it at the top of this page. All concerns will be evaluated after the exam. GOOD LUCK!

**Question 1 8 points** (*Assume all transitions are possible*)

a. 6 points Calculate the **energy (in kJ/mol)** required to ionize (completely remove) the electron from the **n=3** level of **He⁺ ion** (He: Helium). Show your work including equations. (3 sig. figs.)

b. 2 points

(I) Select the largest (longest) wavelength from the three choices below:

- i. Wavelength of photon to *completely remove* an electron from *1s orbital of Hydrogen atom*
- ii. Wavelength of photon to *completely remove* an electron from *1s orbital of Boron atom (B)*
- iii. Wavelength of photon to *completely remove* an electron from *1s orbital of Boron cation (B⁺)*

i or ii or iii

(II) Select the smallest (shortest) wavelength from the three choices below (*for H atom*):

- i. Wavelength of photon *emitted* for an electron transitioning from *2p to 1s orbital*
- ii. Wavelength of photon *absorbed* for an electron transitioning from *2s to 4p orbital*
- iii. Wavelength of photon *absorbed* for an electron transitioning from *1s to 3p orbital*

i or ii or iii



Question 2 8 points



A Chem110 student plots the graph shown on the left, from data obtained for the kinetic energy (eV) of the ejected electron versus frequency (s^{-1}) of the incident light, for an unknown metal. ($1\text{eV} = 1.6022 \times 10^{-19} \text{ J}$)

Based on the plot, answer the questions **a to c** below:

a. 1 point On the plot above, label the **threshold frequency**.

b. 2 points What is the value of the work potential (in J) of the unknown metal?

c. 3 points What is the *longest* wavelength (in nm) required to eject an electron from this metal's surface? Show your work including equation(s) used. (3 sig figs)

d. 2 points In the space provided below, plot the following

(Assume frequency is *above* threshold frequency):

- i. *Kinetic energy* of an ejected electron versus the *intensity* of the incident light (left)
- ii. *Number of electrons* ejected versus the *frequency* of the incident light (right)



**Question 3** **6 points****a. 2 points**

In the ground state of Ge (Germanium; $Z = 32$) atom, **name** the following orbitals:

i. Orbital containing the electron that requires the maximum energy to be removed.

ii. Orbital containing the electron that experiences the minimum effective nuclear charge.

b. 2 points

An unknown anion (A^-) has the following *excited* electronic configuration: $[Ar] 4s^2 3d^3 4p^1$

(Ar: Argon; $Z = 18$)

Give the **condensed ground state electronic configuration** of the atom, **A** and the cation, **A^{2+}** .
Show your work. (Assume all electronic transitions are possible)

Electronic Configuration of A:

Electronic Configuration of A^{2+} :

c. 2 points

In a single atom, determine how many electrons can have the following set of quantum numbers?

(i) $n = 5; l = 3$

In a single atom, determine how many electrons can have the following set of quantum numbers?

(i) $n = 3; l = 2; m_s = +1/2$

**Question 4** **8 points**

Rank the following from 1 to 4 in **increasing** order. Provide the ranking in the boxes provided.

Note: Atomic number (Z) are provided for relevant atoms

a) Increasing (lowest to highest) order of *ionization energy*: (1= lowest; 4 = highest)

Mg

Mg: Z = 12

Rb

Rb: Z = 37

Cs

Cs: Z = 55

K

K : Z = 19

b) Increasing (lowest to highest) order of *bond angle*: (1= lowest; 4 = highest)

H₂OCCl₄XeF₄

HCN

c) Increasing (smallest to largest) order of *ionic size* (1= smallest; 4 = largest)

S²⁻

S: Z = 16

O²⁻

O: Z = 8

Be²⁺

Be: Z = 4

Na⁺

Na: Z = 11

d) Increasing (lowest to highest) order of *bond energy* (1= lowest; 4 = highest)

P-F

P-Cl

P-I

P-Br

**Question 5** **12 points**

a. 9 points For the molecules HClO_4 and TeF_6 , draw the most stable *Lewis structure* showing appropriate shape and *dashed/wedged bonds*, based on VSEPR. Determine the **molecular geometry** around the central atom. For HClO_4 , label the *most polar bond*.

HClO_4 (there are no O-O bonds)

Label the most polar bond

Molecular Geometry around Cl

TeF_6

Molecular Geometry around Te

b. 3 points (Note: Lone pairs are not shown in structures below)

(i) For the three *resonance* structures (1, 2 and 3) shown below, answer the following question:



(I) Circle the *most contributing* resonance structure.

Structure 1

Structure 2

Structure 3

(ii) For the three structures given (4, 5, and 6) below, answer the following questions:



(I) Circle all the structures that represent *resonance structures* of each other:

Structure 4

Structure 5

Structure 6

(II) Among the resonance structures, circle the *most contributing* resonance structure.

Structure 4

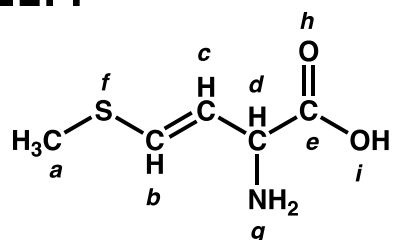
Structure 5

Structure 6

**Question 6** **6 points**

Using the data given below, calculate the **lattice energy** for SrO(s).

	ΔH° (kJ/mol)
$\text{Sr}(s) \longrightarrow \text{Sr}(g)$	+180
$\text{Sr}(g) \longrightarrow \text{Sr}^{2+}(g) + 2e^-$	+1468
$\text{O}_2(g) \longrightarrow 2\text{O}(g)$	+500
$\text{O}(g) + 2e^- \longrightarrow \text{O}^{2-}(g)$	+600
$\text{Sr}(s) + \frac{1}{2}\text{O}_2(g) \longrightarrow \text{SrO}(s)$	-550

**Question 7 10 points**

Consider the molecule given on the left, and answer questions (i) to (vi) using chemical bonding theories that we have learnt in class (valence bond theory, hybridization).

Note that the lone pairs are not shown in the structure:

i. Consider the carbon atoms marked in as ***a***, ***b***, ***c***, ***d***, and ***e*** in the molecule, and determine **hybridization** at each of the carbon centers. (2.5 points)

Atom	Hybridization
Carbon (<i>a</i>)	
Carbon (<i>b</i>)	
Carbon (<i>c</i>)	
Carbon (<i>d</i>)	
Carbon (<i>e</i>)	

ii. Now, consider the sulfur atom (S) marked as ***f***, nitrogen (N) atom (***g***), and oxygen(O) atoms (***h*** and ***i***), and determine their **hybridization**. (2 points)

Atom	Hybridization
Sulfur (<i>f</i>)	
Nitrogen (<i>g</i>)	
Oxygen (<i>h</i>)	
Oxygen (<i>i</i>)	

iii. **Describe** all the bonds formed by the carbon atoms ***a*** and ***b***, using sigma and pi designation as appropriate. (2 points)

Atom	Bonds formed (use sigma and pi as appropriate)
Carbon (<i>a</i>)	
Carbon (<i>b</i>)	

iv. How many total sp^3 -s sigma bonds are in the molecule? (1 point)

v. Determine **electronic** and **molecular geometry** at carbon (C) labeled ***e*** and sulfur (S) labeled ***f*** centres. (2 points)

Atom	Electronic Geometry	Molecular Geometry
Carbon (<i>e</i>)		
Sulfur (<i>f</i>)		

vi. Circle the value from the choices given below, that describes the **C-S-C bond angle at sulfur** (S) marked ***f***: (0.5 points)

>90° to <109.5° 109.5° >109.5 to <120° 120° >120° to <180°

**Question 8 12 points**

Using the **valence shell orbitals only** draw a **complete** molecular orbital (MO) diagram (with no s-p mixing) for the molecule **PS**, in the space provided below. **(6 points)**

Assume that 3s and 3p atomic orbitals will form molecular orbitals like those formed by the combination of 2s and 2p atomic orbitals that we learnt in class, and there is no s-p mixing.

Show **energy arrow**, **atomic orbitals** involved, **atom labels**, the **molecular orbitals** that are formed, **filled electrons** in the MO diagram etc. Make sure to label all the atomic and molecular orbitals.

Using the MO diagram above, answer the following questions:

- (i) Which one of the following will have the highest bond order? Circle the correct choice
A) PS^{1-} B) PS C) PS^{1+}
- (ii) Which one of the following will have the longest bond length? Circle the correct choice
A) PS^{1-} B) PS C) PS^{1+}
- (iii) Which of the following will be diamagnetic? Circle the correct choice.
A) PS^{1-} B) PS C) PS^{1+}
- (iv) Draw Lewis dot structure of the molecule with the shortest bond length (based on the MO diagram) from among the three (PS^{1-} , PS, PS^{1+}), and indicate the bond order calculated from the Lewis dot structure. **(3 points)**

Lewis Dot Structure

Bond Order



60432377-219F-4C38-8EE4-DCE8A36DD83E

midterm-exam-93388

#674 Page 10 of 10

SCRAP PAGE

**CHEMISTRY 110 –Midterm Exam****Thursday November 7, 2024, 6:30 – 8:30 pm****Examiners: Pallavi Sirjoosingh and Ashok Kakkar**

First name (please write as legibly as possible within the boxes)

--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--

Last name

--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--

Student ID number

--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--

This booklet has 8 questions on 10 pages (The last page can be used as a scrap page).

The staple must NOT be removed from the question booklet. Your intact question booklet must be returned at the end of the exam.

A separate booklet contains a periodic table, and datasheet.

Write your answers directly under the questions in the space provided. Do not write on the QR codes or near the top edges (as it may be cut off).

Please write **legibly and dark enough** for the scan to be readable. For students who use English as a second language, dictionaries are permitted. Calculators are allowed.

DO NOT ASK QUESTIONS OF THE INVIGILATORS about interpretations of the exam questions. If you have a concern about a question, please summarize it at the top of this page. All concerns will be evaluated after the exam. GOOD LUCK!

**Question 1 8 points** (*Assume all transitions are possible*)

a. 6 points Calculate the **energy (in kJ/mol)** required to ionize (completely remove) the electron from the **n=3** level of **He⁺ ion** (He: Helium). Show your work including equations. (3 sig. figs.)

b. 2 points

(I) Select the largest (longest) wavelength from the three choices below:

- i. Wavelength of photon to *completely remove* an electron from *1s orbital of Hydrogen atom*
- ii. Wavelength of photon to *completely remove* an electron from *1s orbital of Boron atom (B)*
- iii. Wavelength of photon to *completely remove* an electron from *1s orbital of Boron cation (B⁺)*

i or ii or iii

(II) Select the smallest (shortest) wavelength from the three choices below (*for H atom*):

- i. Wavelength of photon *emitted* for an electron transitioning from *2p to 1s orbital*
- ii. Wavelength of photon *absorbed* for an electron transitioning from *2s to 4p orbital*
- iii. Wavelength of photon *absorbed* for an electron transitioning from *1s to 3p orbital*

i or ii or iii



Question 2 8 points



A Chem110 student plots the graph shown on the left, from data obtained for the kinetic energy (eV) of the ejected electron versus frequency (s^{-1}) of the incident light, for an unknown metal. ($1\text{eV} = 1.6022 \times 10^{-19} \text{ J}$)

Based on the plot, answer the questions **a to c** below:

a. 1 point On the plot above, label the **threshold frequency**.

b. 2 points What is the value of the work potential (in J) of the unknown metal?

c. 3 points What is the *longest* wavelength (in nm) required to eject an electron from this metal's surface? Show your work including equation(s) used. (3 sig figs)

d. 2 points In the space provided below, plot the following

(Assume frequency is *above* threshold frequency):

- i. *Kinetic energy* of an ejected electron versus the *intensity* of the incident light (left)
- ii. *Number of electrons* ejected versus the *frequency* of the incident light (right)



**Question 3** **6 points****a. 2 points**

In the ground state of Ge (Germanium; $Z = 32$) atom, **name** the following orbitals:

i. Orbital containing the electron that requires the maximum energy to be removed.

ii. Orbital containing the electron that experiences the minimum effective nuclear charge.

b. 2 points

An unknown anion (A^-) has the following *excited* electronic configuration: $[Ar] 4s^2 3d^3 4p^1$

(Ar: Argon; $Z = 18$)

Give the **condensed ground state electronic configuration** of the atom, **A** and the cation, **A^{2+}** .
Show your work. (Assume all electronic transitions are possible)

Electronic Configuration of A:

Electronic Configuration of A^{2+} :

c. 2 points

In a single atom, determine how many electrons can have the following set of quantum numbers?

(i) $n = 5; l = 3$

In a single atom, determine how many electrons can have the following set of quantum numbers?

(i) $n = 3; l = 2; m_s = +1/2$

**Question 4** **8 points**

Rank the following from 1 to 4 in **increasing** order. Provide the ranking in the boxes provided.

Note: Atomic number (Z) are provided for relevant atoms

a) Increasing (lowest to highest) order of *ionization energy*: (1= lowest; 4 = highest)

Mg

Mg: Z = 12

Rb

Rb: Z = 37

Cs

Cs: Z = 55

K

K : Z = 19

b) Increasing (lowest to highest) order of *bond angle*: (1= lowest; 4 = highest)

H₂OCCl₄XeF₄

HCN

c) Increasing (smallest to largest) order of *ionic size* (1= smallest; 4 = largest)

S²⁻

S: Z = 16

O²⁻

O: Z = 8

Be²⁺

Be: Z = 4

Na⁺

Na: Z = 11

d) Increasing (lowest to highest) order of *bond energy* (1= lowest; 4 = highest)

P-F

P-Cl

P-I

P-Br

**Question 5** **12 points**

a. 9 points For the molecules HClO_4 and TeF_6 , draw the most stable *Lewis structure* showing appropriate shape and *dashed/wedged bonds*, based on VSEPR. Determine the **molecular geometry** around the central atom. For HClO_4 , label the *most polar bond*.



b. 3 points (Note: Lone pairs are not shown in structures below)

(i) For the three *resonance* structures (1, 2 and 3) shown below, answer the following question:



(I) Circle the *most contributing* resonance structure.

Structure 1

Structure 2

Structure 3

(ii) For the three structures given (4, 5, and 6) below, answer the following questions:



(I) Circle all the structures that represent *resonance structures* of each other:

Structure 4

Structure 5

Structure 6

(II) Among the resonance structures, circle the *most contributing* resonance structure.

Structure 4

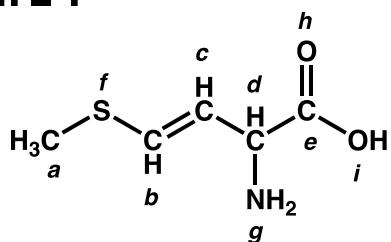
Structure 5

Structure 6

**Question 6** **6 points**

Using the data given below, calculate the **lattice energy** for SrO(s).

	ΔH° (kJ/mol)
$\text{Sr}(s) \longrightarrow \text{Sr}(g)$	+180
$\text{Sr}(g) \longrightarrow \text{Sr}^{2+}(g) + 2e^-$	+1468
$\text{O}_2(g) \longrightarrow 2\text{O}(g)$	+500
$\text{O}(g) + 2e^- \longrightarrow \text{O}^{2-}(g)$	+600
$\text{Sr}(s) + \frac{1}{2}\text{O}_2(g) \longrightarrow \text{SrO}(s)$	-550

**Question 7 10 points**

Consider the molecule given on the left, and answer questions (i) to (vi) using chemical bonding theories that we have learnt in class (valence bond theory, hybridization).

Note that the lone pairs are not shown in the structure:

i. Consider the carbon atoms marked in as ***a***, ***b***, ***c***, ***d***, and ***e*** in the molecule, and determine **hybridization** at each of the carbon centers. (2.5 points)

Atom	Hybridization
Carbon (<i>a</i>)	
Carbon (<i>b</i>)	
Carbon (<i>c</i>)	
Carbon (<i>d</i>)	
Carbon (<i>e</i>)	

ii. Now, consider the sulfur atom (S) marked as ***f***, nitrogen (N) atom (***g***), and oxygen(O) atoms (***h*** and ***i***), and determine their **hybridization**. (2 points)

Atom	Hybridization
Sulfur (<i>f</i>)	
Nitrogen (<i>g</i>)	
Oxygen (<i>h</i>)	
Oxygen (<i>i</i>)	

iii. **Describe** all the bonds formed by the carbon atoms ***a*** and ***b***, using sigma and pi designation as appropriate. (2 points)

Atom	Bonds formed (use sigma and pi as appropriate)
Carbon (<i>a</i>)	
Carbon (<i>b</i>)	

iv. How many total sp^3 -s sigma bonds are in the molecule? (1 point)

v. Determine **electronic** and **molecular geometry** at carbon (C) labeled ***e*** and sulfur (S) labeled ***f*** centres. (2 points)

Atom	Electronic Geometry	Molecular Geometry
Carbon (<i>e</i>)		
Sulfur (<i>f</i>)		

vi. Circle the value from the choices given below, that describes the **C-S-C bond angle at sulfur** (S) marked ***f***: (0.5 points)

>90° to <109.5°

109.5°

>109.5 to <120°

120°

>120° to <180°

**Question 8 12 points**

Using the **valence shell orbitals only** draw a **complete** molecular orbital (MO) diagram (with no s-p mixing) for the molecule **PS**, in the space provided below. **(6 points)**

Assume that 3s and 3p atomic orbitals will form molecular orbitals like those formed by the combination of 2s and 2p atomic orbitals that we learnt in class, and there is no s-p mixing.

Show **energy arrow**, **atomic orbitals** involved, **atom labels**, the **molecular orbitals** that are formed, **filled electrons** in the MO diagram etc. Make sure to label all the atomic and molecular orbitals.

Using the MO diagram above, answer the following questions:

- (i) Which one of the following will have the highest bond order? Circle the correct choice
A) PS^{1-} B) PS C) PS^{1+}
- (ii) Which one of the following will have the longest bond length? Circle the correct choice
A) PS^{1-} B) PS C) PS^{1+}
- (iii) Which of the following will be diamagnetic? Circle the correct choice.
A) PS^{1-} B) PS C) PS^{1+}
- (iv) Draw Lewis dot structure of the molecule with the shortest bond length (based on the MO diagram) from among the three (PS^{1-} , PS, PS^{1+}), and indicate the bond order calculated from the Lewis dot structure. **(3 points)**

Lewis Dot Structure

Bond Order



327EA387-8B39-4D5A-BB04-F1A363BD717B

midterm-exam-93388

#675 Page 10 of 10

SCRAP PAGE

**CHEMISTRY 110 –Midterm Exam****Thursday November 7, 2024, 6:30 – 8:30 pm****Examiners: Pallavi Sirjoosingh and Ashok Kakkar**

First name (please write as legibly as possible within the boxes)

--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--

Last name

--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--

Student ID number

--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--

This booklet has 8 questions on 10 pages (The last page can be used as a scrap page).

The staple must NOT be removed from the question booklet. Your intact question booklet must be returned at the end of the exam.

A separate booklet contains a periodic table, and datasheet.

Write your answers directly under the questions in the space provided. Do not write on the QR codes or near the top edges (as it may be cut off).

Please write **legibly and dark enough** for the scan to be readable. For students who use English as a second language, dictionaries are permitted. Calculators are allowed.

DO NOT ASK QUESTIONS OF THE INVIGILATORS about interpretations of the exam questions. If you have a concern about a question, please summarize it at the top of this page. All concerns will be evaluated after the exam. GOOD LUCK!

**Question 1 8 points** (*Assume all transitions are possible*)

a. 6 points Calculate the **energy (in kJ/mol)** required to ionize (completely remove) the electron from the **$n=3$** level of **He^+ ion** (He: Helium). Show your work including equations. (3 sig. figs.)

b. 2 points

(I) Select the largest (longest) wavelength from the three choices below:

- i. Wavelength of photon to *completely remove* an electron from *1s orbital of Hydrogen atom*
- ii. Wavelength of photon to *completely remove* an electron from *1s orbital of Boron atom (B)*
- iii. Wavelength of photon to *completely remove* an electron from *1s orbital of Boron cation (B^+)*

i or ii or iii

(II) Select the smallest (shortest) wavelength from the three choices below (*for H atom*):

- i. Wavelength of photon *emitted* for an electron transitioning from *2p to 1s orbital*
- ii. Wavelength of photon *absorbed* for an electron transitioning from *2s to 4p orbital*
- iii. Wavelength of photon *absorbed* for an electron transitioning from *1s to 3p orbital*

i or ii or iii



Question 2 8 points



A Chem110 student plots the graph shown on the left, from data obtained for the kinetic energy (eV) of the ejected electron versus frequency (s^{-1}) of the incident light, for an unknown metal. ($1\text{eV} = 1.6022 \times 10^{-19} \text{ J}$)

Based on the plot, answer the questions **a to c** below:

a. 1 point On the plot above, label the **threshold frequency**.

b. 2 points What is the value of the work potential (in J) of the unknown metal?

c. 3 points What is the *longest* wavelength (in nm) required to eject an electron from this metal's surface? Show your work including equation(s) used. (3 sig figs)

d. 2 points In the space provided below, plot the following

(Assume frequency is *above* threshold frequency):

- i. *Kinetic energy* of an ejected electron versus the *intensity* of the incident light (left)
- ii. *Number of electrons* ejected versus the *frequency* of the incident light (right)



**Question 3** **6 points****a. 2 points**

In the ground state of Ge (Germanium; $Z = 32$) atom, **name** the following orbitals:

i. Orbital containing the electron that requires the maximum energy to be removed.

ii. Orbital containing the electron that experiences the minimum effective nuclear charge.

b. 2 points

An unknown anion (A^-) has the following *excited* electronic configuration: $[Ar] 4s^2 3d^3 4p^1$

(Ar: Argon; $Z = 18$)

Give the **condensed ground state electronic configuration** of the atom, **A** and the cation, **A^{2+}** .
Show your work. (Assume all electronic transitions are possible)

Electronic Configuration of A:

Electronic Configuration of A^{2+} :

c. 2 points

In a single atom, determine how many electrons can have the following set of quantum numbers?

(i) $n = 5; l = 3$

In a single atom, determine how many electrons can have the following set of quantum numbers?

(i) $n = 3; l = 2; m_s = +1/2$

**Question 4** **8 points**

Rank the following from 1 to 4 in **increasing** order. Provide the ranking in the boxes provided.

Note: Atomic number (Z) are provided for relevant atoms

a) Increasing (lowest to highest) order of *ionization energy*: (1= lowest; 4 = highest)

Mg

Mg: Z = 12

Rb

Rb: Z = 37

Cs

Cs: Z = 55

K

K : Z = 19

b) Increasing (lowest to highest) order of *bond angle*: (1= lowest; 4 = highest)

H₂OCCl₄XeF₄

HCN

c) Increasing (smallest to largest) order of *ionic size* (1= smallest; 4 = largest)

S²⁻

S: Z = 16

O²⁻

O: Z = 8

Be²⁺

Be: Z = 4

Na⁺

Na: Z = 11

d) Increasing (lowest to highest) order of *bond energy* (1= lowest; 4 = highest)

P-F

P-Cl

P-I

P-Br

**Question 5** **12 points**

a. 9 points For the molecules HClO_4 and TeF_6 , draw the most stable *Lewis structure* showing appropriate shape and *dashed/wedged bonds*, based on VSEPR. Determine the **molecular geometry** around the central atom. For HClO_4 , label the *most polar bond*.

HClO_4 (there are no O-O bonds)

Label the most polar bond

Molecular Geometry around Cl

TeF_6

Molecular Geometry around Te

b. 3 points (Note: Lone pairs are not shown in structures below)

(i) For the three *resonance* structures (1, 2 and 3) shown below, answer the following question:



(I) Circle the *most contributing* resonance structure.

Structure 1

Structure 2

Structure 3

(ii) For the three structures given (4, 5, and 6) below, answer the following questions:



(I) Circle all the structures that represent *resonance structures* of each other:

Structure 4

Structure 5

Structure 6

(II) Among the resonance structures, circle the *most contributing* resonance structure.

Structure 4

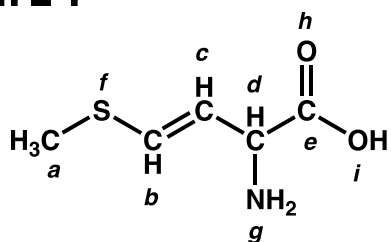
Structure 5

Structure 6

**Question 6** **6 points**

Using the data given below, calculate the **lattice energy** for SrO(s).

	ΔH° (kJ/mol)
$\text{Sr}(s) \longrightarrow \text{Sr}(g)$	+180
$\text{Sr}(g) \longrightarrow \text{Sr}^{2+}(g) + 2e^-$	+1468
$\text{O}_2(g) \longrightarrow 2\text{O}(g)$	+500
$\text{O}(g) + 2e^- \longrightarrow \text{O}^{2-}(g)$	+600
$\text{Sr}(s) + \frac{1}{2}\text{O}_2(g) \longrightarrow \text{SrO}(s)$	-550

**Question 7 10 points**

Consider the molecule given on the left, and answer questions (i) to (vi) using chemical bonding theories that we have learnt in class (valence bond theory, hybridization).

Note that the lone pairs are not shown in the structure:

i. Consider the carbon atoms marked in as ***a***, ***b***, ***c***, ***d***, and ***e*** in the molecule, and determine **hybridization** at each of the carbon centers. (2.5 points)

Atom	Hybridization
Carbon (<i>a</i>)	
Carbon (<i>b</i>)	
Carbon (<i>c</i>)	
Carbon (<i>d</i>)	
Carbon (<i>e</i>)	

ii. Now, consider the sulfur atom (S) marked as ***f***, nitrogen (N) atom (***g***), and oxygen(O) atoms (***h*** and ***i***), and determine their **hybridization**. (2 points)

Atom	Hybridization
Sulfur (<i>f</i>)	
Nitrogen (<i>g</i>)	
Oxygen (<i>h</i>)	
Oxygen (<i>i</i>)	

iii. **Describe** all the bonds formed by the carbon atoms ***a*** and ***b***, using sigma and pi designation as appropriate. (2 points)

Atom	Bonds formed (use sigma and pi as appropriate)
Carbon (<i>a</i>)	
Carbon (<i>b</i>)	

iv. How many total sp^3 -s sigma bonds are in the molecule? (1 point)

v. Determine **electronic** and **molecular geometry** at carbon (C) labeled ***e*** and sulfur (S) labeled ***f*** centres. (2 points)

Atom	Electronic Geometry	Molecular Geometry
Carbon (<i>e</i>)		
Sulfur (<i>f</i>)		

vi. Circle the value from the choices given below, that describes the **C-S-C bond angle at sulfur** (S) marked ***f***: (0.5 points)

$>90^\circ$ to $<109.5^\circ$ 109.5° >109.5 to $<120^\circ$ 120° $>120^\circ$ to $<180^\circ$

**Question 8 12 points**

Using the **valence shell orbitals only** draw a **complete** molecular orbital (MO) diagram (with no s-p mixing) for the molecule **PS**, in the space provided below. **(6 points)**

Assume that 3s and 3p atomic orbitals will form molecular orbitals like those formed by the combination of 2s and 2p atomic orbitals that we learnt in class, and there is no s-p mixing.

Show **energy arrow**, **atomic orbitals** involved, **atom labels**, the **molecular orbitals** that are formed, **filled electrons** in the MO diagram etc. Make sure to label all the atomic and molecular orbitals.

Using the MO diagram above, answer the following questions:

- (i) Which one of the following will have the highest bond order? Circle the correct choice
A) PS^{1-} B) PS C) PS^{1+}
- (ii) Which one of the following will have the longest bond length? Circle the correct choice
A) PS^{1-} B) PS C) PS^{1+}
- (iii) Which of the following will be diamagnetic? Circle the correct choice.
A) PS^{1-} B) PS C) PS^{1+}
- (iv) Draw Lewis dot structure of the molecule with the shortest bond length (based on the MO diagram) from among the three (PS^{1-} , PS, PS^{1+}), and indicate the bond order calculated from the Lewis dot structure. **(3 points)**

Lewis Dot Structure

Bond Order



82678387-BB9D-4CBA-B7BD-E7A1828495DA

midterm-exam-93388

#676 Page 10 of 10

SCRAP PAGE

**CHEMISTRY 110 –Midterm Exam****Thursday November 7, 2024, 6:30 – 8:30 pm**

Examiners: Pallavi Sirjoosingh and Ashok Kakkar

First name (please write as legibly as possible within the boxes)

--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--

Last name

--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--

Student ID number

--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--

This booklet has 8 questions on 10 pages (The last page can be used as a scrap page).

The staple must NOT be removed from the question booklet. Your intact question booklet must be returned at the end of the exam.

A separate booklet contains a periodic table, and datasheet.

Write your answers directly under the questions in the space provided. Do not write on the QR codes or near the top edges (as it may be cut off).

Please write **legibly and dark enough** for the scan to be readable. For students who use English as a second language, dictionaries are permitted. Calculators are allowed.

DO NOT ASK QUESTIONS OF THE INVIGILATORS about interpretations of the exam questions. If you have a concern about a question, please summarize it at the top of this page. All concerns will be evaluated after the exam. GOOD LUCK!

**Question 1 8 points** (*Assume all transitions are possible*)

a. 6 points Calculate the **energy (in kJ/mol)** required to ionize (completely remove) the electron from the **$n=3$** level of **He^+ ion** (He: Helium). Show your work including equations. (3 sig. figs.)

b. 2 points

(I) Select the largest (longest) wavelength from the three choices below:

- i. Wavelength of photon to *completely remove* an electron from *1s orbital of Hydrogen atom*
- ii. Wavelength of photon to *completely remove* an electron from *1s orbital of Boron atom (B)*
- iii. Wavelength of photon to *completely remove* an electron from *1s orbital of Boron cation (B^+)*

i or ii or iii

(II) Select the smallest (shortest) wavelength from the three choices below (*for H atom*):

- i. Wavelength of photon *emitted* for an electron transitioning from *2p to 1s orbital*
- ii. Wavelength of photon *absorbed* for an electron transitioning from *2s to 4p orbital*
- iii. Wavelength of photon *absorbed* for an electron transitioning from *1s to 3p orbital*

i or ii or iii



Question 2 8 points



A Chem110 student plots the graph shown on the left, from data obtained for the kinetic energy (eV) of the ejected electron versus frequency (s^{-1}) of the incident light, for an unknown metal. ($1\text{eV} = 1.6022 \times 10^{-19} \text{ J}$)

Based on the plot, answer the questions **a to c** below:

a. 1 point On the plot above, label the **threshold frequency**.

b. 2 points What is the value of the work potential (in J) of the unknown metal?

c. 3 points What is the *longest* wavelength (in nm) required to eject an electron from this metal's surface? Show your work including equation(s) used. (3 sig figs)

d. 2 points In the space provided below, plot the following

(Assume frequency is *above* threshold frequency):

- Kinetic energy* of an ejected electron versus the *intensity* of the incident light (left)
- Number of electrons* ejected versus the *frequency* of the incident light (right)



**Question 3** **6 points****a. 2 points**

In the ground state of Ge (Germanium; $Z = 32$) atom, **name** the following orbitals:

i. Orbital containing the electron that requires the maximum energy to be removed.

ii. Orbital containing the electron that experiences the minimum effective nuclear charge.

b. 2 points

An unknown anion (A^-) has the following *excited* electronic configuration: $[Ar] 4s^2 3d^3 4p^1$

(Ar: Argon; $Z = 18$)

Give the **condensed ground state electronic configuration** of the atom, **A** and the cation, **A^{2+}** .
Show your work. (Assume all electronic transitions are possible)

Electronic Configuration of A:

Electronic Configuration of A^{2+} :

c. 2 points

In a single atom, determine how many electrons can have the following set of quantum numbers?

(i) $n = 5; l = 3$

In a single atom, determine how many electrons can have the following set of quantum numbers?

(i) $n = 3; l = 2; m_s = +1/2$

**Question 4** **8 points**

Rank the following from 1 to 4 in **increasing** order. Provide the ranking in the boxes provided.

Note: Atomic number (Z) are provided for relevant atoms

a) Increasing (lowest to highest) order of *ionization energy*: (1= lowest; 4 = highest)

Mg

Mg: Z = 12

Rb

Rb: Z = 37

Cs

Cs: Z = 55

K

K : Z = 19

b) Increasing (lowest to highest) order of *bond angle*: (1= lowest; 4 = highest)

H₂OCCl₄XeF₄

HCN

c) Increasing (smallest to largest) order of *ionic size* (1= smallest; 4 = largest)

S²⁻

S: Z = 16

O²⁻

O: Z = 8

Be²⁺

Be: Z = 4

Na⁺

Na: Z = 11

d) Increasing (lowest to highest) order of *bond energy* (1= lowest; 4 = highest)

P-F

P-Cl

P-I

P-Br

**Question 5** **12 points**

a. 9 points For the molecules HClO_4 and TeF_6 , draw the most stable *Lewis structure* showing appropriate shape and *dashed/wedged bonds*, based on VSEPR. Determine the **molecular geometry** around the central atom. For HClO_4 , label the *most polar bond*.



b. 3 points (Note: Lone pairs are not shown in structures below)

(i) For the three *resonance* structures (1, 2 and 3) shown below, answer the following question:



(I) Circle the *most contributing* resonance structure.

Structure 1

Structure 2

Structure 3

(ii) For the three structures given (4, 5, and 6) below, answer the following questions:



(I) Circle all the structures that represent *resonance structures* of each other:

Structure 4

Structure 5

Structure 6

(II) Among the resonance structures, circle the *most contributing* resonance structure.

Structure 4

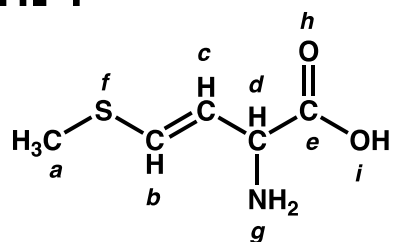
Structure 5

Structure 6

**Question 6** **6 points**

Using the data given below, calculate the **lattice energy** for SrO(s).

	ΔH° (kJ/mol)
$\text{Sr}(s) \longrightarrow \text{Sr}(g)$	+180
$\text{Sr}(g) \longrightarrow \text{Sr}^{2+}(g) + 2e^-$	+1468
$\text{O}_2(g) \longrightarrow 2\text{O}(g)$	+500
$\text{O}(g) + 2e^- \longrightarrow \text{O}^{2-}(g)$	+600
$\text{Sr}(s) + \frac{1}{2}\text{O}_2(g) \longrightarrow \text{SrO}(s)$	-550

**Question 7 10 points**

Consider the molecule given on the left, and answer questions (i) to (vi) using chemical bonding theories that we have learnt in class (valence bond theory, hybridization).

Note that the lone pairs are not shown in the structure:

i. Consider the carbon atoms marked in as ***a***, ***b***, ***c***, ***d***, and ***e*** in the molecule, and determine **hybridization** at each of the carbon centers. (2.5 points)

Atom	Hybridization
Carbon (<i>a</i>)	
Carbon (<i>b</i>)	
Carbon (<i>c</i>)	
Carbon (<i>d</i>)	
Carbon (<i>e</i>)	

ii. Now, consider the sulfur atom (S) marked as ***f***, nitrogen (N) atom (***g***), and oxygen(O) atoms (***h*** and ***i***), and determine their **hybridization**. (2 points)

Atom	Hybridization
Sulfur (<i>f</i>)	
Nitrogen (<i>g</i>)	
Oxygen (<i>h</i>)	
Oxygen (<i>i</i>)	

iii. **Describe** all the bonds formed by the carbon atoms ***a*** and ***b***, using sigma and pi designation as appropriate. (2 points)

Atom	Bonds formed (use sigma and pi as appropriate)
Carbon (<i>a</i>)	
Carbon (<i>b</i>)	

iv. How many total sp^3 -s sigma bonds are in the molecule? (1 point)

v. Determine **electronic** and **molecular geometry** at carbon (C) labeled ***e*** and sulfur (S) labeled ***f*** centres. (2 points)

Atom	Electronic Geometry	Molecular Geometry
Carbon (<i>e</i>)		
Sulfur (<i>f</i>)		

vi. Circle the value from the choices given below, that describes the **C-S-C bond angle at sulfur** (S) marked ***f***: (0.5 points)

>90° to <109.5° 109.5° >109.5 to <120° 120° >120° to <180°

**Question 8 12 points**

Using the **valence shell orbitals only** draw a **complete** molecular orbital (MO) diagram (with no s-p mixing) for the molecule **PS**, in the space provided below. **(6 points)**

Assume that 3s and 3p atomic orbitals will form molecular orbitals like those formed by the combination of 2s and 2p atomic orbitals that we learnt in class, and there is no s-p mixing.

Show **energy arrow**, **atomic orbitals** involved, **atom labels**, the **molecular orbitals** that are formed, **filled electrons** in the MO diagram etc. Make sure to label all the atomic and molecular orbitals.

Using the MO diagram above, answer the following questions:

- (i) Which one of the following will have the highest bond order? Circle the correct choice
A) PS^{1-} B) PS C) PS^{1+}
- (ii) Which one of the following will have the longest bond length? Circle the correct choice
A) PS^{1-} B) PS C) PS^{1+}
- (iii) Which of the following will be diamagnetic? Circle the correct choice.
A) PS^{1-} B) PS C) PS^{1+}
- (iv) Draw Lewis dot structure of the molecule with the shortest bond length (based on the MO diagram) from among the three (PS^{1-} , PS, PS^{1+}), and indicate the bond order calculated from the Lewis dot structure. **(3 points)**

Lewis Dot Structure

Bond Order



49F4E0DA-AB19-4A8D-B162-2E6862C5CE6F

midterm-exam-93388

#677 Page 10 of 10

SCRAP PAGE

**CHEMISTRY 110 –Midterm Exam****Thursday November 7, 2024, 6:30 – 8:30 pm**

Examiners: Pallavi Sirjoosingh and Ashok Kakkar

First name (please write as legibly as possible within the boxes)

--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--

Last name

--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--

Student ID number

--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--

This booklet has 8 questions on 10 pages (The last page can be used as a scrap page).

The staple must NOT be removed from the question booklet. Your intact question booklet must be returned at the end of the exam.

A separate booklet contains a periodic table, and datasheet.

Write your answers directly under the questions in the space provided. Do not write on the QR codes or near the top edges (as it may be cut off).

Please write **legibly and dark enough** for the scan to be readable. For students who use English as a second language, dictionaries are permitted. Calculators are allowed.

DO NOT ASK QUESTIONS OF THE INVIGILATORS about interpretations of the exam questions. If you have a concern about a question, please summarize it at the top of this page. All concerns will be evaluated after the exam. GOOD LUCK!

**Question 1 8 points** (*Assume all transitions are possible*)

a. 6 points Calculate the **energy (in kJ/mol)** required to ionize (completely remove) the electron from the **n=3** level of **He⁺ ion** (He: Helium). Show your work including equations. (3 sig. figs.)

b. 2 points

(I) Select the largest (longest) wavelength from the three choices below:

- i. Wavelength of photon to *completely remove* an electron from *1s orbital of Hydrogen atom*
- ii. Wavelength of photon to *completely remove* an electron from *1s orbital of Boron atom (B)*
- iii. Wavelength of photon to *completely remove* an electron from *1s orbital of Boron cation (B⁺)*

i or ii or iii

(II) Select the smallest (shortest) wavelength from the three choices below (*for H atom*):

- i. Wavelength of photon *emitted* for an electron transitioning from *2p to 1s orbital*
- ii. Wavelength of photon *absorbed* for an electron transitioning from *2s to 4p orbital*
- iii. Wavelength of photon *absorbed* for an electron transitioning from *1s to 3p orbital*

i or ii or iii



Question 2 8 points



A Chem110 student plots the graph shown on the left, from data obtained for the kinetic energy (eV) of the ejected electron versus frequency (s^{-1}) of the incident light, for an unknown metal. ($1\text{eV} = 1.6022 \times 10^{-19} \text{ J}$)

Based on the plot, answer the questions **a to c** below:

a. 1 point On the plot above, label the **threshold frequency**.

b. 2 points What is the value of the work potential (in J) of the unknown metal?

c. 3 points What is the *longest* wavelength (in nm) required to eject an electron from this metal's surface? Show your work including equation(s) used. (3 sig figs)

d. 2 points In the space provided below, plot the following

(Assume frequency is *above* threshold frequency):

- i. *Kinetic energy* of an ejected electron versus the *intensity* of the incident light (left)
- ii. *Number of electrons* ejected versus the *frequency* of the incident light (right)



**Question 3** **6 points****a. 2 points**

In the ground state of Ge (Germanium; $Z = 32$) atom, **name** the following orbitals:

i. Orbital containing the electron that requires the maximum energy to be removed.

ii. Orbital containing the electron that experiences the minimum effective nuclear charge.

b. 2 points

An unknown anion (A^-) has the following *excited* electronic configuration: $[Ar] 4s^2 3d^3 4p^1$

(Ar: Argon; $Z = 18$)

Give the **condensed ground state electronic configuration** of the atom, **A** and the cation, **A^{2+}** .
Show your work. (Assume all electronic transitions are possible)

Electronic Configuration of A:

Electronic Configuration of A^{2+} :

c. 2 points

In a single atom, determine how many electrons can have the following set of quantum numbers?

(i) $n = 5; l = 3$

In a single atom, determine how many electrons can have the following set of quantum numbers?

(i) $n = 3; l = 2; m_s = +1/2$

**Question 4** **8 points**

Rank the following from 1 to 4 in **increasing** order. Provide the ranking in the boxes provided.

Note: Atomic number (Z) are provided for relevant atoms

a) Increasing (lowest to highest) order of *ionization energy*: (1= lowest; 4 = highest)

Mg

Mg: Z = 12

Rb

Rb: Z = 37

Cs

Cs: Z = 55

K

K : Z = 19

b) Increasing (lowest to highest) order of *bond angle*: (1= lowest; 4 = highest)

H₂OCCl₄XeF₄

HCN

c) Increasing (smallest to largest) order of *ionic size* (1= smallest; 4 = largest)

S²⁻

S: Z = 16

O²⁻

O: Z = 8

Be²⁺

Be: Z = 4

Na⁺

Na: Z = 11

d) Increasing (lowest to highest) order of *bond energy* (1= lowest; 4 = highest)

P-F

P-Cl

P-I

P-Br

**Question 5** **12 points**

a. 9 points For the molecules HClO_4 and TeF_6 , draw the most stable *Lewis structure* showing appropriate shape and *dashed/wedged bonds*, based on VSEPR. Determine the **molecular geometry** around the central atom. For HClO_4 , label the *most polar bond*.



b. 3 points (Note: Lone pairs are not shown in structures below)

(i) For the three *resonance* structures (1, 2 and 3) shown below, answer the following question:



(I) Circle the *most contributing* resonance structure.

Structure 1

Structure 2

Structure 3

(ii) For the three structures given (4, 5, and 6) below, answer the following questions:



(I) Circle all the structures that represent *resonance structures* of each other:

Structure 4

Structure 5

Structure 6

(II) Among the resonance structures, circle the *most contributing* resonance structure.

Structure 4

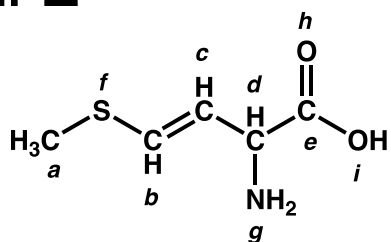
Structure 5

Structure 6

**Question 6** **6 points**

Using the data given below, calculate the **lattice energy** for SrO(s).

	ΔH° (kJ/mol)
$\text{Sr}(s) \longrightarrow \text{Sr}(g)$	+180
$\text{Sr}(g) \longrightarrow \text{Sr}^{2+}(g) + 2e^-$	+1468
$\text{O}_2(g) \longrightarrow 2\text{O}(g)$	+500
$\text{O}(g) + 2e^- \longrightarrow \text{O}^{2-}(g)$	+600
$\text{Sr}(s) + \frac{1}{2}\text{O}_2(g) \longrightarrow \text{SrO}(s)$	-550

**Question 7 10 points**

Consider the molecule given on the left, and answer questions (i) to (vi) using chemical bonding theories that we have learnt in class (valence bond theory, hybridization).

Note that the lone pairs are not shown in the structure:

i. Consider the carbon atoms marked in as ***a***, ***b***, ***c***, ***d***, and ***e*** in the molecule, and determine **hybridization** at each of the carbon centers. (2.5 points)

Atom	Hybridization
Carbon (<i>a</i>)	
Carbon (<i>b</i>)	
Carbon (<i>c</i>)	
Carbon (<i>d</i>)	
Carbon (<i>e</i>)	

ii. Now, consider the sulfur atom (S) marked as ***f***, nitrogen (N) atom (***g***), and oxygen(O) atoms (***h*** and ***i***), and determine their **hybridization**. (2 points)

Atom	Hybridization
Sulfur (<i>f</i>)	
Nitrogen (<i>g</i>)	
Oxygen (<i>h</i>)	
Oxygen (<i>i</i>)	

iii. **Describe** all the bonds formed by the carbon atoms ***a*** and ***b***, using sigma and pi designation as appropriate. (2 points)

Atom	Bonds formed (use sigma and pi as appropriate)
Carbon (<i>a</i>)	
Carbon (<i>b</i>)	

iv. How many total sp^3 -s sigma bonds are in the molecule? (1 point)

v. Determine **electronic** and **molecular geometry** at carbon (C) labeled ***e*** and sulfur (S) labeled ***f*** centres. (2 points)

Atom	Electronic Geometry	Molecular Geometry
Carbon (<i>e</i>)		
Sulfur (<i>f</i>)		

vi. Circle the value from the choices given below, that describes the **C-S-C bond angle at sulfur** (S) marked ***f***: (0.5 points)

>90° to <109.5° 109.5° >109.5 to <120° 120° >120° to <180°


Question 8 12 points

Using the **valence shell orbitals only** draw a **complete** molecular orbital (MO) diagram (with no s-p mixing) for the molecule **PS**, in the space provided below. **(6 points)**

Assume that 3s and 3p atomic orbitals will form molecular orbitals like those formed by the combination of 2s and 2p atomic orbitals that we learnt in class, and there is no s-p mixing.

Show **energy arrow**, **atomic orbitals** involved, **atom labels**, the **molecular orbitals** that are formed, **filled electrons** in the MO diagram etc. Make sure to label all the atomic and molecular orbitals.

Using the MO diagram above, answer the following questions:

- (i) Which one of the following will have the highest bond order? Circle the correct choice
 A) PS^{1-} B) PS C) PS^{1+}
- (ii) Which one of the following will have the longest bond length? Circle the correct choice
 A) PS^{1-} B) PS C) PS^{1+}
- (iii) Which of the following will be diamagnetic? Circle the correct choice.
 A) PS^{1-} B) PS C) PS^{1+}
- (iv) Draw Lewis dot structure of the molecule with the shortest bond length (based on the MO diagram) from among the three (PS^{1-} , PS, PS^{1+}), and indicate the bond order calculated from the Lewis dot structure. **(3 points)**

Lewis Dot Structure

Bond Order



E99BA217-BE7C-4FD6-AA2D-535F34DF0DCB

midterm-exam-93388

#678 Page 10 of 10

SCRAP PAGE

**CHEMISTRY 110 –Midterm Exam****Thursday November 7, 2024, 6:30 – 8:30 pm**

Examiners: Pallavi Sirjoosingh and Ashok Kakkar

First name (please write as legibly as possible within the boxes)

--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--

Last name

--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--

Student ID number

--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--

This booklet has 8 questions on 10 pages (The last page can be used as a scrap page).

The staple must NOT be removed from the question booklet. Your intact question booklet must be returned at the end of the exam.

A separate booklet contains a periodic table, and datasheet.

Write your answers directly under the questions in the space provided. Do not write on the QR codes or near the top edges (as it may be cut off).

Please write **legibly and dark enough** for the scan to be readable. For students who use English as a second language, dictionaries are permitted. Calculators are allowed.

DO NOT ASK QUESTIONS OF THE INVIGILATORS about interpretations of the exam questions. If you have a concern about a question, please summarize it at the top of this page. All concerns will be evaluated after the exam. GOOD LUCK!

**Question 1 8 points** (*Assume all transitions are possible*)

a. 6 points Calculate the **energy (in kJ/mol)** required to ionize (completely remove) the electron from the **$n=3$** level of **He^+ ion** (He: Helium). Show your work including equations. (3 sig. figs.)

b. 2 points

(I) Select the largest (longest) wavelength from the three choices below:

- i. Wavelength of photon to *completely remove* an electron from *1s orbital of Hydrogen atom*
- ii. Wavelength of photon to *completely remove* an electron from *1s orbital of Boron atom (B)*
- iii. Wavelength of photon to *completely remove* an electron from *1s orbital of Boron cation (B^+)*

i or ii or iii

(II) Select the smallest (shortest) wavelength from the three choices below (*for H atom*):

- i. Wavelength of photon *emitted* for an electron transitioning from *2p to 1s orbital*
- ii. Wavelength of photon *absorbed* for an electron transitioning from *2s to 4p orbital*
- iii. Wavelength of photon *absorbed* for an electron transitioning from *1s to 3p orbital*

i or ii or iii



Question 2 8 points



A Chem110 student plots the graph shown on the left, from data obtained for the kinetic energy (eV) of the ejected electron versus frequency (s^{-1}) of the incident light, for an unknown metal. ($1\text{eV} = 1.6022 \times 10^{-19} \text{ J}$)

Based on the plot, answer the questions **a to c** below:

a. 1 point On the plot above, label the **threshold frequency**.

b. 2 points What is the value of the work potential (in J) of the unknown metal?

c. 3 points What is the *longest* wavelength (in nm) required to eject an electron from this metal's surface? Show your work including equation(s) used. (3 sig figs)

d. 2 points In the space provided below, plot the following

(Assume frequency is *above* threshold frequency):

- i. *Kinetic energy* of an ejected electron versus the *intensity* of the incident light (left)
- ii. *Number of electrons* ejected versus the *frequency* of the incident light (right)



**Question 3** **6 points****a. 2 points**

In the ground state of Ge (Germanium; $Z = 32$) atom, **name** the following orbitals:

i. Orbital containing the electron that requires the maximum energy to be removed.

ii. Orbital containing the electron that experiences the minimum effective nuclear charge.

b. 2 points

An unknown anion (A^-) has the following *excited* electronic configuration: $[Ar] 4s^2 3d^3 4p^1$

(Ar: Argon; $Z = 18$)

Give the **condensed ground state electronic configuration** of the atom, **A** and the cation, A^{2+} .
Show your work. (Assume all electronic transitions are possible)

Electronic Configuration of A:

Electronic Configuration of A^{2+} :

c. 2 points

In a single atom, determine how many electrons can have the following set of quantum numbers?

(i) $n = 5; l = 3$

In a single atom, determine how many electrons can have the following set of quantum numbers?

(i) $n = 3; l = 2; m_s = +1/2$

**Question 4** **8 points**

Rank the following from 1 to 4 in **increasing** order. Provide the ranking in the boxes provided.

Note: Atomic number (Z) are provided for relevant atoms

a) Increasing (lowest to highest) order of *ionization energy*: (1= lowest; 4 = highest)

Mg

Mg: Z = 12

Rb

Rb: Z = 37

Cs

Cs: Z = 55

K

K : Z = 19

b) Increasing (lowest to highest) order of *bond angle*: (1= lowest; 4 = highest)

H₂OCCl₄XeF₄

HCN

c) Increasing (smallest to largest) order of *ionic size* (1= smallest; 4 = largest)

S²⁻

S: Z = 16

O²⁻

O: Z = 8

Be²⁺

Be: Z = 4

Na⁺

Na: Z = 11

d) Increasing (lowest to highest) order of *bond energy* (1= lowest; 4 = highest)

P-F

P-Cl

P-I

P-Br

**Question 5** **12 points**

a. 9 points For the molecules HClO_4 and TeF_6 , draw the most stable *Lewis structure* showing appropriate shape and *dashed/wedged bonds*, based on VSEPR. Determine the **molecular geometry** around the central atom. For HClO_4 , label the *most polar bond*.



b. 3 points (Note: Lone pairs are not shown in structures below)

(i) For the three *resonance* structures (1, 2 and 3) shown below, answer the following question:



(I) Circle the *most contributing* resonance structure.

Structure 1

Structure 2

Structure 3

(ii) For the three structures given (4, 5, and 6) below, answer the following questions:



(I) Circle all the structures that represent *resonance structures* of each other:

Structure 4

Structure 5

Structure 6

(II) Among the resonance structures, circle the *most contributing* resonance structure.

Structure 4

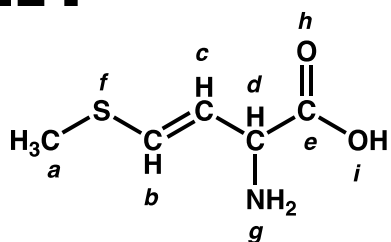
Structure 5

Structure 6

**Question 6** **6 points**

Using the data given below, calculate the **lattice energy** for SrO(s).

	ΔH° (kJ/mol)
$\text{Sr}(s) \longrightarrow \text{Sr}(g)$	+180
$\text{Sr}(g) \longrightarrow \text{Sr}^{2+}(g) + 2e^-$	+1468
$\text{O}_2(g) \longrightarrow 2\text{O}(g)$	+500
$\text{O}(g) + 2e^- \longrightarrow \text{O}^{2-}(g)$	+600
$\text{Sr}(s) + \frac{1}{2}\text{O}_2(g) \longrightarrow \text{SrO}(s)$	-550

**Question 7 10 points**

Consider the molecule given on the left, and answer questions (i) to (vi) using chemical bonding theories that we have learnt in class (valence bond theory, hybridization).

Note that the lone pairs are not shown in the structure:

i. Consider the carbon atoms marked in as ***a***, ***b***, ***c***, ***d***, and ***e*** in the molecule, and determine **hybridization** at each of the carbon centers. (2.5 points)

Atom	Hybridization
Carbon (<i>a</i>)	
Carbon (<i>b</i>)	
Carbon (<i>c</i>)	
Carbon (<i>d</i>)	
Carbon (<i>e</i>)	

ii. Now, consider the sulfur atom (S) marked as ***f***, nitrogen (N) atom (***g***), and oxygen(O) atoms (***h*** and ***i***), and determine their **hybridization**. (2 points)

Atom	Hybridization
Sulfur (<i>f</i>)	
Nitrogen (<i>g</i>)	
Oxygen (<i>h</i>)	
Oxygen (<i>i</i>)	

iii. **Describe** all the bonds formed by the carbon atoms ***a*** and ***b***, using sigma and pi designation as appropriate. (2 points)

Atom	Bonds formed (use sigma and pi as appropriate)
Carbon (<i>a</i>)	
Carbon (<i>b</i>)	

iv. How many total sp^3 -s sigma bonds are in the molecule? (1 point)

v. Determine **electronic** and **molecular geometry** at carbon (C) labeled ***e*** and sulfur (S) labeled ***f*** centres. (2 points)

Atom	Electronic Geometry	Molecular Geometry
Carbon (<i>e</i>)		
Sulfur (<i>f</i>)		

vi. Circle the value from the choices given below, that describes the **C-S-C bond angle at sulfur** (S) marked ***f***: (0.5 points)

$>90^\circ$ to $<109.5^\circ$ 109.5° >109.5 to $<120^\circ$ 120° $>120^\circ$ to $<180^\circ$


Question 8 12 points

Using the **valence shell orbitals only** draw a **complete** molecular orbital (MO) diagram (with no s-p mixing) for the molecule **PS**, in the space provided below. **(6 points)**

Assume that 3s and 3p atomic orbitals will form molecular orbitals like those formed by the combination of 2s and 2p atomic orbitals that we learnt in class, and there is no s-p mixing.

Show **energy arrow**, **atomic orbitals** involved, **atom labels**, the **molecular orbitals** that are formed, **filled electrons** in the MO diagram etc. Make sure to label all the atomic and molecular orbitals.

Using the MO diagram above, answer the following questions:

- (i) Which one of the following will have the highest bond order? Circle the correct choice
 A) PS^{1-} B) PS C) PS^{1+}
- (ii) Which one of the following will have the longest bond length? Circle the correct choice
 A) PS^{1-} B) PS C) PS^{1+}
- (iii) Which of the following will be diamagnetic? Circle the correct choice.
 A) PS^{1-} B) PS C) PS^{1+}
- (iv) Draw Lewis dot structure of the molecule with the shortest bond length (based on the MO diagram) from among the three (PS^{1-} , PS, PS^{1+}), and indicate the bond order calculated from the Lewis dot structure. **(3 points)**

Lewis Dot Structure

Bond Order



705AA323-4CAD-442B-8FB4-19E2B8F52A55

midterm-exam-93388

#679 Page 10 of 10

SCRAP PAGE

**CHEMISTRY 110 –Midterm Exam****Thursday November 7, 2024, 6:30 – 8:30 pm**

Examiners: Pallavi Sirjoosingh and Ashok Kakkar

First name (please write as legibly as possible within the boxes)

--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--

Last name

--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--

Student ID number

--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--

This booklet has 8 questions on 10 pages (The last page can be used as a scrap page).

The staple must NOT be removed from the question booklet. Your intact question booklet must be returned at the end of the exam.

A separate booklet contains a periodic table, and datasheet.

Write your answers directly under the questions in the space provided. Do not write on the QR codes or near the top edges (as it may be cut off).

Please write **legibly and dark enough** for the scan to be readable. For students who use English as a second language, dictionaries are permitted. Calculators are allowed.

DO NOT ASK QUESTIONS OF THE INVIGILATORS about interpretations of the exam questions. If you have a concern about a question, please summarize it at the top of this page. All concerns will be evaluated after the exam. GOOD LUCK!

**Question 1 8 points** (*Assume all transitions are possible*)

a. 6 points Calculate the **energy (in kJ/mol)** required to ionize (completely remove) the electron from the **n=3** level of **He⁺ ion** (He: Helium). Show your work including equations. (3 sig. figs.)

b. 2 points

(I) Select the largest (longest) wavelength from the three choices below:

- i. Wavelength of photon to *completely remove* an electron from *1s orbital of Hydrogen atom*
- ii. Wavelength of photon to *completely remove* an electron from *1s orbital of Boron atom (B)*
- iii. Wavelength of photon to *completely remove* an electron from *1s orbital of Boron cation (B⁺)*

i or ii or iii

(II) Select the smallest (shortest) wavelength from the three choices below (*for H atom*):

- i. Wavelength of photon *emitted* for an electron transitioning from *2p to 1s orbital*
- ii. Wavelength of photon *absorbed* for an electron transitioning from *2s to 4p orbital*
- iii. Wavelength of photon *absorbed* for an electron transitioning from *1s to 3p orbital*

i or ii or iii



Question 2 8 points



A Chem110 student plots the graph shown on the left, from data obtained for the kinetic energy (eV) of the ejected electron versus frequency (s^{-1}) of the incident light, for an unknown metal. ($1\text{eV} = 1.6022 \times 10^{-19} \text{ J}$)

Based on the plot, answer the questions **a to c** below:

a. 1 point On the plot above, label the **threshold frequency**.

b. 2 points What is the value of the work potential (in J) of the unknown metal?

c. 3 points What is the *longest* wavelength (in nm) required to eject an electron from this metal's surface? Show your work including equation(s) used. (3 sig figs)

d. 2 points In the space provided below, plot the following

(Assume frequency is *above* threshold frequency):

- i. *Kinetic energy* of an ejected electron versus the *intensity* of the incident light (left)
- ii. *Number of electrons* ejected versus the *frequency* of the incident light (right)



**Question 3** **6 points****a. 2 points**

In the ground state of Ge (Germanium; $Z = 32$) atom, **name** the following orbitals:

i. Orbital containing the electron that requires the maximum energy to be removed.

ii. Orbital containing the electron that experiences the minimum effective nuclear charge.

b. 2 points

An unknown anion (A^-) has the following *excited* electronic configuration: $[Ar] 4s^2 3d^3 4p^1$

(Ar: Argon; $Z = 18$)

Give the **condensed ground state electronic configuration** of the atom, **A** and the cation, **A^{2+}** .
Show your work. (Assume all electronic transitions are possible)

Electronic Configuration of A:

Electronic Configuration of A^{2+} :

c. 2 points

In a single atom, determine how many electrons can have the following set of quantum numbers?

(i) $n = 5; l = 3$

In a single atom, determine how many electrons can have the following set of quantum numbers?

(i) $n = 3; l = 2; m_s = +1/2$

**Question 4** **8 points**

Rank the following from 1 to 4 in **increasing** order. Provide the ranking in the boxes provided.

Note: Atomic number (Z) are provided for relevant atoms

a) Increasing (lowest to highest) order of *ionization energy*: (1= lowest; 4 = highest)

Mg

Mg: Z = 12

Rb

Rb: Z = 37

Cs

Cs: Z = 55

K

K : Z = 19

b) Increasing (lowest to highest) order of *bond angle*: (1= lowest; 4 = highest)

H₂OCCl₄XeF₄

HCN

c) Increasing (smallest to largest) order of *ionic size* (1= smallest; 4 = largest)

S²⁻

S: Z = 16

O²⁻

O: Z = 8

Be²⁺

Be: Z = 4

Na⁺

Na: Z = 11

d) Increasing (lowest to highest) order of *bond energy* (1= lowest; 4 = highest)

P-F

P-Cl

P-I

P-Br

**Question 5** **12 points**

a. 9 points For the molecules HClO_4 and TeF_6 , draw the most stable *Lewis structure* showing appropriate shape and *dashed/wedged bonds*, based on VSEPR. Determine the **molecular geometry** around the central atom. For HClO_4 , label the *most polar bond*.



b. 3 points (Note: Lone pairs are not shown in structures below)

(i) For the three *resonance* structures (1, 2 and 3) shown below, answer the following question:



(I) Circle the *most contributing* resonance structure.

Structure 1

Structure 2

Structure 3

(ii) For the three structures given (4, 5, and 6) below, answer the following questions:



(I) Circle all the structures that represent *resonance structures* of each other:

Structure 4

Structure 5

Structure 6

(II) Among the resonance structures, circle the *most contributing* resonance structure.

Structure 4

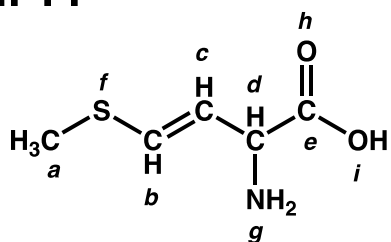
Structure 5

Structure 6

**Question 6** **6 points**

Using the data given below, calculate the **lattice energy** for SrO(s).

	ΔH° (kJ/mol)
$\text{Sr}(s) \longrightarrow \text{Sr}(g)$	+180
$\text{Sr}(g) \longrightarrow \text{Sr}^{2+}(g) + 2e^-$	+1468
$\text{O}_2(g) \longrightarrow 2\text{O}(g)$	+500
$\text{O}(g) + 2e^- \longrightarrow \text{O}^{2-}(g)$	+600
$\text{Sr}(s) + \frac{1}{2}\text{O}_2(g) \longrightarrow \text{SrO}(s)$	-550

**Question 7 10 points**

Consider the molecule given on the left, and answer questions (i) to (vi) using chemical bonding theories that we have learnt in class (valence bond theory, hybridization).

Note that the lone pairs are not shown in the structure:

i. Consider the carbon atoms marked in as ***a***, ***b***, ***c***, ***d***, and ***e*** in the molecule, and determine **hybridization** at each of the carbon centers. (2.5 points)

Atom	Hybridization
Carbon (<i>a</i>)	
Carbon (<i>b</i>)	
Carbon (<i>c</i>)	
Carbon (<i>d</i>)	
Carbon (<i>e</i>)	

ii. Now, consider the sulfur atom (S) marked as ***f***, nitrogen (N) atom (***g***), and oxygen(O) atoms (***h*** and ***i***), and determine their **hybridization**. (2 points)

Atom	Hybridization
Sulfur (<i>f</i>)	
Nitrogen (<i>g</i>)	
Oxygen (<i>h</i>)	
Oxygen (<i>i</i>)	

iii. **Describe** all the bonds formed by the carbon atoms ***a*** and ***b***, using sigma and pi designation as appropriate. (2 points)

Atom	Bonds formed (use sigma and pi as appropriate)
Carbon (<i>a</i>)	
Carbon (<i>b</i>)	

iv. How many total sp^3 -s sigma bonds are in the molecule? (1 point)

v. Determine **electronic** and **molecular geometry** at carbon (C) labeled ***e*** and sulfur (S) labeled ***f*** centres. (2 points)

Atom	Electronic Geometry	Molecular Geometry
Carbon (<i>e</i>)		
Sulfur (<i>f</i>)		

vi. Circle the value from the choices given below, that describes the **C-S-C bond angle at sulfur** (S) marked ***f***: (0.5 points)

>90° to <109.5°

109.5°

>109.5 to <120°

120°

>120° to <180°

**Question 8 12 points**

Using the **valence shell orbitals only** draw a **complete** molecular orbital (MO) diagram (with no s-p mixing) for the molecule **PS**, in the space provided below. **(6 points)**

Assume that 3s and 3p atomic orbitals will form molecular orbitals like those formed by the combination of 2s and 2p atomic orbitals that we learnt in class, and there is no s-p mixing.

Show **energy arrow**, **atomic orbitals** involved, **atom labels**, the **molecular orbitals** that are formed, **filled electrons** in the MO diagram etc. Make sure to label all the atomic and molecular orbitals.

Using the MO diagram above, answer the following questions:

- (i) Which one of the following will have the highest bond order? Circle the correct choice
A) PS^{1-} B) PS C) PS^{1+}
- (ii) Which one of the following will have the longest bond length? Circle the correct choice
A) PS^{1-} B) PS C) PS^{1+}
- (iii) Which of the following will be diamagnetic? Circle the correct choice.
A) PS^{1-} B) PS C) PS^{1+}
- (iv) Draw Lewis dot structure of the molecule with the shortest bond length (based on the MO diagram) from among the three (PS^{1-} , PS, PS^{1+}), and indicate the bond order calculated from the Lewis dot structure. **(3 points)**

Lewis Dot Structure

Bond Order



65CFBA58-B030-4870-8EE7-5D04A73A27FB

midterm-exam-93388

#680 Page 10 of 10

SCRAP PAGE

**CHEMISTRY 110 –Midterm Exam****Thursday November 7, 2024, 6:30 – 8:30 pm****Examiners: Pallavi Sirjoosingh and Ashok Kakkar**

First name (please write as legibly as possible within the boxes)

--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--

Last name

--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--

Student ID number

--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--

This booklet has 8 questions on 10 pages (The last page can be used as a scrap page).

The staple must NOT be removed from the question booklet. Your intact question booklet must be returned at the end of the exam.

A separate booklet contains a periodic table, and datasheet.

Write your answers directly under the questions in the space provided. Do not write on the QR codes or near the top edges (as it may be cut off).

Please write **legibly and dark enough** for the scan to be readable. For students who use English as a second language, dictionaries are permitted. Calculators are allowed.

DO NOT ASK QUESTIONS OF THE INVIGILATORS about interpretations of the exam questions. If you have a concern about a question, please summarize it at the top of this page. All concerns will be evaluated after the exam. GOOD LUCK!

**Question 1 8 points** (*Assume all transitions are possible*)

a. 6 points Calculate the **energy (in kJ/mol)** required to ionize (completely remove) the electron from the **n=3** level of **He⁺ ion** (He: Helium). Show your work including equations. (3 sig. figs.)

b. 2 points

(I) Select the largest (longest) wavelength from the three choices below:

- i. Wavelength of photon to *completely remove* an electron from *1s orbital of Hydrogen atom*
- ii. Wavelength of photon to *completely remove* an electron from *1s orbital of Boron atom (B)*
- iii. Wavelength of photon to *completely remove* an electron from *1s orbital of Boron cation (B⁺)*

i or ii or iii

(II) Select the smallest (shortest) wavelength from the three choices below (*for H atom*):

- i. Wavelength of photon *emitted* for an electron transitioning from *2p to 1s orbital*
- ii. Wavelength of photon *absorbed* for an electron transitioning from *2s to 4p orbital*
- iii. Wavelength of photon *absorbed* for an electron transitioning from *1s to 3p orbital*

i or ii or iii



Question 2 8 points



A Chem110 student plots the graph shown on the left, from data obtained for the kinetic energy (eV) of the ejected electron versus frequency (s^{-1}) of the incident light, for an unknown metal. ($1\text{eV} = 1.6022 \times 10^{-19} \text{ J}$)

Based on the plot, answer the questions **a to c** below:

a. 1 point On the plot above, label the **threshold frequency**.

b. 2 points What is the value of the work potential (in J) of the unknown metal?

c. 3 points What is the *longest* wavelength (in nm) required to eject an electron from this metal's surface? Show your work including equation(s) used. (3 sig figs)

d. 2 points In the space provided below, plot the following

(Assume frequency is *above* threshold frequency):

- Kinetic energy* of an ejected electron versus the *intensity* of the incident light (left)
- Number of electrons* ejected versus the *frequency* of the incident light (right)



**Question 3** **6 points****a. 2 points**

In the ground state of Ge (Germanium; $Z = 32$) atom, **name** the following orbitals:

i. Orbital containing the electron that requires the maximum energy to be removed.

ii. Orbital containing the electron that experiences the minimum effective nuclear charge.

b. 2 points

An unknown anion (A^-) has the following *excited* electronic configuration: $[Ar] 4s^2 3d^3 4p^1$

(Ar: Argon; $Z = 18$)

Give the **condensed ground state electronic configuration** of the atom, **A** and the cation, **A^{2+}** .
Show your work. (Assume all electronic transitions are possible)

Electronic Configuration of A:

Electronic Configuration of A^{2+} :

c. 2 points

In a single atom, determine how many electrons can have the following set of quantum numbers?

(i) $n = 5; l = 3$

In a single atom, determine how many electrons can have the following set of quantum numbers?

(i) $n = 3; l = 2; m_s = +1/2$

**Question 4** **8 points**

Rank the following from 1 to 4 in **increasing** order. Provide the ranking in the boxes provided.

Note: Atomic number (Z) are provided for relevant atoms

a) Increasing (lowest to highest) order of *ionization energy*: (1= lowest; 4 = highest)

Mg

Mg: Z = 12

Rb

Rb: Z = 37

Cs

Cs: Z = 55

K

K : Z = 19

b) Increasing (lowest to highest) order of *bond angle*: (1= lowest; 4 = highest)

H₂OCCl₄XeF₄

HCN

c) Increasing (smallest to largest) order of *ionic size* (1= smallest; 4 = largest)

S²⁻

S: Z = 16

O²⁻

O: Z = 8

Be²⁺

Be: Z = 4

Na⁺

Na: Z = 11

d) Increasing (lowest to highest) order of *bond energy* (1= lowest; 4 = highest)

P-F

P-Cl

P-I

P-Br

**Question 5** **12 points**

a. 9 points For the molecules HClO_4 and TeF_6 , draw the most stable *Lewis structure* showing appropriate shape and *dashed/wedged bonds*, based on VSEPR. Determine the **molecular geometry** around the central atom. For HClO_4 , label the *most polar bond*.



b. 3 points (Note: Lone pairs are not shown in structures below)

(i) For the three *resonance* structures (1, 2 and 3) shown below, answer the following question:



(I) Circle the *most contributing* resonance structure.

Structure 1

Structure 2

Structure 3

(ii) For the three structures given (4, 5, and 6) below, answer the following questions:



(I) Circle all the structures that represent *resonance structures* of each other:

Structure 4

Structure 5

Structure 6

(II) Among the resonance structures, circle the *most contributing* resonance structure.

Structure 4

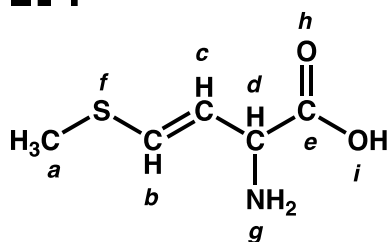
Structure 5

Structure 6

**Question 6** **6 points**

Using the data given below, calculate the **lattice energy** for SrO(s).

	ΔH° (kJ/mol)
$\text{Sr}(s) \longrightarrow \text{Sr}(g)$	+180
$\text{Sr}(g) \longrightarrow \text{Sr}^{2+}(g) + 2e^-$	+1468
$\text{O}_2(g) \longrightarrow 2\text{O}(g)$	+500
$\text{O}(g) + 2e^- \longrightarrow \text{O}^{2-}(g)$	+600
$\text{Sr}(s) + \frac{1}{2}\text{O}_2(g) \longrightarrow \text{SrO}(s)$	-550

**Question 7 10 points**

Consider the molecule given on the left, and answer questions (i) to (vi) using chemical bonding theories that we have learnt in class (valence bond theory, hybridization).

Note that the lone pairs are not shown in the structure:

i. Consider the carbon atoms marked in as ***a***, ***b***, ***c***, ***d***, and ***e*** in the molecule, and determine **hybridization** at each of the carbon centers. (2.5 points)

Atom	Hybridization
Carbon (<i>a</i>)	
Carbon (<i>b</i>)	
Carbon (<i>c</i>)	
Carbon (<i>d</i>)	
Carbon (<i>e</i>)	

ii. Now, consider the sulfur atom (S) marked as ***f***, nitrogen (N) atom (***g***), and oxygen(O) atoms (***h*** and ***i***), and determine their **hybridization**. (2 points)

Atom	Hybridization
Sulfur (<i>f</i>)	
Nitrogen (<i>g</i>)	
Oxygen (<i>h</i>)	
Oxygen (<i>i</i>)	

iii. **Describe** all the bonds formed by the carbon atoms ***a*** and ***b***, using sigma and pi designation as appropriate. (2 points)

Atom	Bonds formed (use sigma and pi as appropriate)
Carbon (<i>a</i>)	
Carbon (<i>b</i>)	

iv. How many total sp^3 -s sigma bonds are in the molecule? (1 point)

v. Determine **electronic** and **molecular geometry** at carbon (C) labeled ***e*** and sulfur (S) labeled ***f*** centres. (2 points)

Atom	Electronic Geometry	Molecular Geometry
Carbon (<i>e</i>)		
Sulfur (<i>f</i>)		

vi. Circle the value from the choices given below, that describes the **C-S-C bond angle at sulfur** (S) marked ***f***: (0.5 points)

>90° to <109.5°

109.5°

>109.5 to <120°

120°

>120° to <180°

**Question 8 12 points**

Using the **valence shell orbitals only** draw a **complete** molecular orbital (MO) diagram (with no s-p mixing) for the molecule **PS**, in the space provided below. **(6 points)**

Assume that 3s and 3p atomic orbitals will form molecular orbitals like those formed by the combination of 2s and 2p atomic orbitals that we learnt in class, and there is no s-p mixing.

Show **energy arrow**, **atomic orbitals** involved, **atom labels**, the **molecular orbitals** that are formed, **filled electrons** in the MO diagram etc. Make sure to label all the atomic and molecular orbitals.

Using the MO diagram above, answer the following questions:

- (i) Which one of the following will have the highest bond order? Circle the correct choice
A) PS^{1-} B) PS C) PS^{1+}
- (ii) Which one of the following will have the longest bond length? Circle the correct choice
A) PS^{1-} B) PS C) PS^{1+}
- (iii) Which of the following will be diamagnetic? Circle the correct choice.
A) PS^{1-} B) PS C) PS^{1+}
- (iv) Draw Lewis dot structure of the molecule with the shortest bond length (based on the MO diagram) from among the three (PS^{1-} , PS, PS^{1+}), and indicate the bond order calculated from the Lewis dot structure. **(3 points)**

Lewis Dot Structure

Bond Order



48A27C98-D43D-4F0E-9A62-2C9ABB34659B

midterm-exam-93388

#681 Page 10 of 10

SCRAP PAGE

**CHEMISTRY 110 –Midterm Exam****Thursday November 7, 2024, 6:30 – 8:30 pm****Examiners: Pallavi Sirjoosingh and Ashok Kakkar**

First name (please write as legibly as possible within the boxes)

--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--

Last name

--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--

Student ID number

--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--

This booklet has 8 questions on 10 pages (The last page can be used as a scrap page).

The staple must NOT be removed from the question booklet. Your intact question booklet must be returned at the end of the exam.

A separate booklet contains a periodic table, and datasheet.

Write your answers directly under the questions in the space provided. Do not write on the QR codes or near the top edges (as it may be cut off).

Please write **legibly and dark enough** for the scan to be readable. For students who use English as a second language, dictionaries are permitted. Calculators are allowed.

DO NOT ASK QUESTIONS OF THE INVIGILATORS about interpretations of the exam questions. If you have a concern about a question, please summarize it at the top of this page. All concerns will be evaluated after the exam. GOOD LUCK!

**Question 1 8 points** (*Assume all transitions are possible*)

a. 6 points Calculate the **energy (in kJ/mol)** required to ionize (completely remove) the electron from the **n=3** level of **He⁺ ion** (He: Helium). Show your work including equations. (3 sig. figs.)

b. 2 points

(I) Select the largest (longest) wavelength from the three choices below:

- i. Wavelength of photon to *completely remove* an electron from *1s orbital of Hydrogen atom*
- ii. Wavelength of photon to *completely remove* an electron from *1s orbital of Boron atom (B)*
- iii. Wavelength of photon to *completely remove* an electron from *1s orbital of Boron cation (B⁺)*

i or ii or iii

(II) Select the smallest (shortest) wavelength from the three choices below (*for H atom*):

- i. Wavelength of photon *emitted* for an electron transitioning from *2p to 1s orbital*
- ii. Wavelength of photon *absorbed* for an electron transitioning from *2s to 4p orbital*
- iii. Wavelength of photon *absorbed* for an electron transitioning from *1s to 3p orbital*

i or ii or iii



Question 2 8 points



A Chem110 student plots the graph shown on the left, from data obtained for the kinetic energy (eV) of the ejected electron versus frequency (s^{-1}) of the incident light, for an unknown metal. ($1\text{eV} = 1.6022 \times 10^{-19} \text{ J}$)

Based on the plot, answer the questions **a to c** below:

a. 1 point On the plot above, label the **threshold frequency**.

b. 2 points What is the value of the work potential (in J) of the unknown metal?

c. 3 points What is the *longest* wavelength (in nm) required to eject an electron from this metal's surface? Show your work including equation(s) used. (3 sig figs)

d. 2 points In the space provided below, plot the following

(Assume frequency is *above* threshold frequency):

- Kinetic energy* of an ejected electron versus the *intensity* of the incident light (left)
- Number of electrons* ejected versus the *frequency* of the incident light (right)



**Question 3** **6 points****a. 2 points**

In the ground state of Ge (Germanium; $Z = 32$) atom, **name** the following orbitals:

i. Orbital containing the electron that requires the maximum energy to be removed.

ii. Orbital containing the electron that experiences the minimum effective nuclear charge.

b. 2 points

An unknown anion (A^-) has the following *excited* electronic configuration: $[Ar] 4s^2 3d^3 4p^1$

(Ar: Argon; $Z = 18$)

Give the **condensed ground state electronic configuration** of the atom, **A** and the cation, **A^{2+}** .
Show your work. (Assume all electronic transitions are possible)

Electronic Configuration of A:

Electronic Configuration of A^{2+} :

c. 2 points

In a single atom, determine how many electrons can have the following set of quantum numbers?

(i) $n = 5; l = 3$

In a single atom, determine how many electrons can have the following set of quantum numbers?

(i) $n = 3; l = 2; m_s = +1/2$

**Question 4** **8 points**

Rank the following from 1 to 4 in **increasing** order. Provide the ranking in the boxes provided.

Note: Atomic number (Z) are provided for relevant atoms

a) Increasing (lowest to highest) order of *ionization energy*: (1= lowest; 4 = highest)

Mg

Mg: Z = 12

Rb

Rb: Z = 37

Cs

Cs: Z = 55

K

K : Z = 19

b) Increasing (lowest to highest) order of *bond angle*: (1= lowest; 4 = highest)

H₂OCCl₄XeF₄

HCN

c) Increasing (smallest to largest) order of *ionic size* (1= smallest; 4 = largest)

S²⁻

S: Z = 16

O²⁻

O: Z = 8

Be²⁺

Be: Z = 4

Na⁺

Na: Z = 11

d) Increasing (lowest to highest) order of *bond energy* (1= lowest; 4 = highest)

P-F

P-Cl

P-I

P-Br

**Question 5** **12 points**

a. 9 points For the molecules HClO_4 and TeF_6 , draw the most stable *Lewis structure* showing appropriate shape and *dashed/wedged bonds*, based on VSEPR. Determine the **molecular geometry** around the central atom. For HClO_4 , label the *most polar bond*.

HClO_4 (there are no O-O bonds)

Label the most polar bond

Molecular Geometry around Cl

TeF_6

Molecular Geometry around Te

b. 3 points (Note: Lone pairs are not shown in structures below)

(i) For the three *resonance* structures (1, 2 and 3) shown below, answer the following question:



(I) Circle the *most contributing* resonance structure.

Structure 1

Structure 2

Structure 3

(ii) For the three structures given (4, 5, and 6) below, answer the following questions:



(I) Circle all the structures that represent *resonance structures* of each other:

Structure 4

Structure 5

Structure 6

(II) Among the resonance structures, circle the *most contributing* resonance structure.

Structure 4

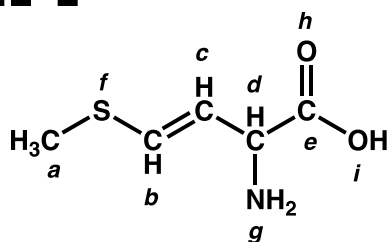
Structure 5

Structure 6

**Question 6** **6 points**

Using the data given below, calculate the **lattice energy** for SrO(s).

	ΔH° (kJ/mol)
$\text{Sr}(s) \longrightarrow \text{Sr}(g)$	+180
$\text{Sr}(g) \longrightarrow \text{Sr}^{2+}(g) + 2e^-$	+1468
$\text{O}_2(g) \longrightarrow 2\text{O}(g)$	+500
$\text{O}(g) + 2e^- \longrightarrow \text{O}^{2-}(g)$	+600
$\text{Sr}(s) + \frac{1}{2}\text{O}_2(g) \longrightarrow \text{SrO}(s)$	-550

**Question 7 10 points**

Consider the molecule given on the left, and answer questions (i) to (vi) using chemical bonding theories that we have learnt in class (valence bond theory, hybridization).

Note that the lone pairs are not shown in the structure:

i. Consider the carbon atoms marked in as ***a***, ***b***, ***c***, ***d***, and ***e*** in the molecule, and determine **hybridization** at each of the carbon centers. (2.5 points)

Atom	Hybridization
Carbon (<i>a</i>)	
Carbon (<i>b</i>)	
Carbon (<i>c</i>)	
Carbon (<i>d</i>)	
Carbon (<i>e</i>)	

ii. Now, consider the sulfur atom (S) marked as ***f***, nitrogen (N) atom (***g***), and oxygen(O) atoms (***h*** and ***i***), and determine their **hybridization**. (2 points)

Atom	Hybridization
Sulfur (<i>f</i>)	
Nitrogen (<i>g</i>)	
Oxygen (<i>h</i>)	
Oxygen (<i>i</i>)	

iii. **Describe** all the bonds formed by the carbon atoms ***a*** and ***b***, using sigma and pi designation as appropriate. (2 points)

Atom	Bonds formed (use sigma and pi as appropriate)
Carbon (<i>a</i>)	
Carbon (<i>b</i>)	

iv. How many total sp^3 -s sigma bonds are in the molecule? (1 point)

v. Determine **electronic** and **molecular geometry** at carbon (C) labeled ***e*** and sulfur (S) labeled ***f*** centres. (2 points)

Atom	Electronic Geometry	Molecular Geometry
Carbon (<i>e</i>)		
Sulfur (<i>f</i>)		

vi. Circle the value from the choices given below, that describes the **C-S-C bond angle at sulfur** (S) marked ***f***: (0.5 points)

>90° to <109.5° 109.5° >109.5 to <120° 120° >120° to <180°

**Question 8 12 points**

Using the **valence shell orbitals only** draw a **complete** molecular orbital (MO) diagram (with no s-p mixing) for the molecule **PS**, in the space provided below. **(6 points)**

Assume that 3s and 3p atomic orbitals will form molecular orbitals like those formed by the combination of 2s and 2p atomic orbitals that we learnt in class, and there is no s-p mixing.

Show **energy arrow**, **atomic orbitals** involved, **atom labels**, the **molecular orbitals** that are formed, **filled electrons** in the MO diagram etc. Make sure to **label all the atomic and molecular orbitals**.

Using the MO diagram above, answer the following questions:

- (i) Which one of the following will have the highest bond order? Circle the correct choice
A) PS^{1-} B) PS C) PS^{1+}
- (ii) Which one of the following will have the longest bond length? Circle the correct choice
A) PS^{1-} B) PS C) PS^{1+}
- (iii) Which of the following will be diamagnetic? Circle the correct choice.
A) PS^{1-} B) PS C) PS^{1+}
- (iv) Draw Lewis dot structure of the molecule with the shortest bond length (based on the MO diagram) from among the three (PS^{1-} , PS, PS^{1+}), and indicate the bond order calculated from the Lewis dot structure. **(3 points)**

Lewis Dot Structure

Bond Order



74318E36-BD8F-44AD-8308-2EB788282909

midterm-exam-93388

#682 Page 10 of 10

SCRAP PAGE

**CHEMISTRY 110 –Midterm Exam****Thursday November 7, 2024, 6:30 – 8:30 pm****Examiners: Pallavi Sirjoosingh and Ashok Kakkar**

First name (please write as legibly as possible within the boxes)

--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--

Last name

--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--

Student ID number

--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--

This booklet has 8 questions on 10 pages (The last page can be used as a scrap page).

The staple must NOT be removed from the question booklet. Your intact question booklet must be returned at the end of the exam.

A separate booklet contains a periodic table, and datasheet.

Write your answers directly under the questions in the space provided. Do not write on the QR codes or near the top edges (as it may be cut off).

Please write **legibly and dark enough** for the scan to be readable. For students who use English as a second language, dictionaries are permitted. Calculators are allowed.

DO NOT ASK QUESTIONS OF THE INVIGILATORS about interpretations of the exam questions. If you have a concern about a question, please summarize it at the top of this page. All concerns will be evaluated after the exam. GOOD LUCK!

**Question 1 8 points** (*Assume all transitions are possible*)

a. 6 points Calculate the **energy (in kJ/mol)** required to ionize (completely remove) the electron from the **n=3** level of **He⁺ ion** (He: Helium). Show your work including equations. (3 sig. figs.)

b. 2 points

(I) Select the largest (longest) wavelength from the three choices below:

- i. Wavelength of photon to *completely remove* an electron from *1s orbital of Hydrogen atom*
- ii. Wavelength of photon to *completely remove* an electron from *1s orbital of Boron atom (B)*
- iii. Wavelength of photon to *completely remove* an electron from *1s orbital of Boron cation (B⁺)*

i or ii or iii

(II) Select the smallest (shortest) wavelength from the three choices below (*for H atom*):

- i. Wavelength of photon *emitted* for an electron transitioning from *2p to 1s orbital*
- ii. Wavelength of photon *absorbed* for an electron transitioning from *2s to 4p orbital*
- iii. Wavelength of photon *absorbed* for an electron transitioning from *1s to 3p orbital*

i or ii or iii



Question 2 8 points



A Chem110 student plots the graph shown on the left, from data obtained for the kinetic energy (eV) of the ejected electron versus frequency (s^{-1}) of the incident light, for an unknown metal. ($1\text{eV} = 1.6022 \times 10^{-19} \text{ J}$)

Based on the plot, answer the questions **a to c** below:

a. 1 point On the plot above, label the **threshold frequency**.

b. 2 points What is the value of the work potential (in J) of the unknown metal?

c. 3 points What is the *longest* wavelength (in nm) required to eject an electron from this metal's surface? Show your work including equation(s) used. (3 sig figs)

d. 2 points In the space provided below, plot the following

(Assume frequency is *above* threshold frequency):

- Kinetic energy* of an ejected electron versus the *intensity* of the incident light (left)
- Number of electrons* ejected versus the *frequency* of the incident light (right)



**Question 3** **6 points****a. 2 points**

In the ground state of Ge (Germanium; $Z = 32$) atom, **name** the following orbitals:

i. Orbital containing the electron that requires the maximum energy to be removed.

ii. Orbital containing the electron that experiences the minimum effective nuclear charge.

b. 2 points

An unknown anion (A^-) has the following *excited* electronic configuration: $[Ar] 4s^2 3d^3 4p^1$

(Ar: Argon; $Z = 18$)

Give the **condensed ground state electronic configuration** of the atom, **A** and the cation, **A^{2+}** .
Show your work. (Assume all electronic transitions are possible)

Electronic Configuration of A:

Electronic Configuration of A^{2+} :

c. 2 points

In a single atom, determine how many electrons can have the following set of quantum numbers?

(i) $n = 5; l = 3$

In a single atom, determine how many electrons can have the following set of quantum numbers?

(i) $n = 3; l = 2; m_s = +1/2$

**Question 4** **8 points**

Rank the following from 1 to 4 in **increasing** order. Provide the ranking in the boxes provided.

Note: Atomic number (Z) are provided for relevant atoms

a) Increasing (lowest to highest) order of *ionization energy*: (1= lowest; 4 = highest)

Mg

Mg: Z = 12

Rb

Rb: Z = 37

Cs

Cs: Z = 55

K

K : Z = 19

b) Increasing (lowest to highest) order of *bond angle*: (1= lowest; 4 = highest)

H₂OCCl₄XeF₄

HCN

c) Increasing (smallest to largest) order of *ionic size* (1= smallest; 4 = largest)

S²⁻

S: Z = 16

O²⁻

O: Z = 8

Be²⁺

Be: Z = 4

Na⁺

Na: Z = 11

d) Increasing (lowest to highest) order of *bond energy* (1= lowest; 4 = highest)

P-F

P-Cl

P-I

P-Br

**Question 5** **12 points**

a. 9 points For the molecules HClO_4 and TeF_6 , draw the most stable *Lewis structure* showing appropriate shape and *dashed/wedged bonds*, based on VSEPR. Determine the **molecular geometry** around the central atom. For HClO_4 , label the *most polar bond*.



b. 3 points (Note: Lone pairs are not shown in structures below)

(i) For the three *resonance* structures (1, 2 and 3) shown below, answer the following question:



(I) Circle the *most contributing* resonance structure.

Structure 1

Structure 2

Structure 3

(ii) For the three structures given (4, 5, and 6) below, answer the following questions:



(I) Circle all the structures that represent *resonance structures* of each other:

Structure 4

Structure 5

Structure 6

(II) Among the resonance structures, circle the *most contributing* resonance structure.

Structure 4

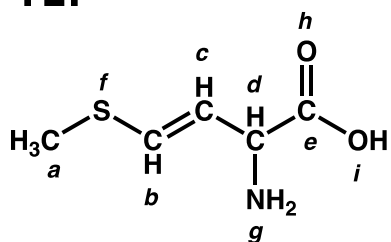
Structure 5

Structure 6

**Question 6** **6 points**

Using the data given below, calculate the **lattice energy** for SrO(s).

	ΔH° (kJ/mol)
$\text{Sr}(s) \longrightarrow \text{Sr}(g)$	+180
$\text{Sr}(g) \longrightarrow \text{Sr}^{2+}(g) + 2e^-$	+1468
$\text{O}_2(g) \longrightarrow 2\text{O}(g)$	+500
$\text{O}(g) + 2e^- \longrightarrow \text{O}^{2-}(g)$	+600
$\text{Sr}(s) + \frac{1}{2}\text{O}_2(g) \longrightarrow \text{SrO}(s)$	-550

**Question 7 10 points**

Consider the molecule given on the left, and answer questions (i) to (vi) using chemical bonding theories that we have learnt in class (valence bond theory, hybridization).

Note that the lone pairs are not shown in the structure:

i. Consider the carbon atoms marked in as ***a***, ***b***, ***c***, ***d***, and ***e*** in the molecule, and determine **hybridization** at each of the carbon centers. (2.5 points)

Atom	Hybridization
Carbon (<i>a</i>)	
Carbon (<i>b</i>)	
Carbon (<i>c</i>)	
Carbon (<i>d</i>)	
Carbon (<i>e</i>)	

ii. Now, consider the sulfur atom (S) marked as ***f***, nitrogen (N) atom (***g***), and oxygen(O) atoms (***h*** and ***i***), and determine their **hybridization**. (2 points)

Atom	Hybridization
Sulfur (<i>f</i>)	
Nitrogen (<i>g</i>)	
Oxygen (<i>h</i>)	
Oxygen (<i>i</i>)	

iii. **Describe** all the bonds formed by the carbon atoms ***a*** and ***b***, using sigma and pi designation as appropriate. (2 points)

Atom	Bonds formed (use sigma and pi as appropriate)
Carbon (<i>a</i>)	
Carbon (<i>b</i>)	

iv. How many total sp^3 -s sigma bonds are in the molecule? (1 point)

v. Determine **electronic** and **molecular geometry** at carbon (C) labeled ***e*** and sulfur (S) labeled ***f*** centres. (2 points)

Atom	Electronic Geometry	Molecular Geometry
Carbon (<i>e</i>)		
Sulfur (<i>f</i>)		

vi. Circle the value from the choices given below, that describes the **C-S-C bond angle at sulfur** (S) marked ***f***: (0.5 points)

>90° to <109.5° 109.5° >109.5 to <120° 120° >120° to <180°


Question 8 12 points

Using the **valence shell orbitals only** draw a **complete** molecular orbital (MO) diagram (with no s-p mixing) for the molecule **PS**, in the space provided below. **(6 points)**

Assume that 3s and 3p atomic orbitals will form molecular orbitals like those formed by the combination of 2s and 2p atomic orbitals that we learnt in class, and there is no s-p mixing.

Show **energy arrow**, **atomic orbitals** involved, **atom labels**, the **molecular orbitals** that are formed, **filled electrons** in the MO diagram etc. Make sure to **label all the atomic and molecular orbitals**.

Using the MO diagram above, answer the following questions:

- (i) Which one of the following will have the highest bond order? Circle the correct choice
 A) PS^{1-} B) PS C) PS^{1+}
- (ii) Which one of the following will have the longest bond length? Circle the correct choice
 A) PS^{1-} B) PS C) PS^{1+}
- (iii) Which of the following will be diamagnetic? Circle the correct choice.
 A) PS^{1-} B) PS C) PS^{1+}
- (iv) Draw Lewis dot structure of the molecule with the shortest bond length (based on the MO diagram) from among the three (PS^{1-} , PS, PS^{1+}), and indicate the bond order calculated from the Lewis dot structure. **(3 points)**

Lewis Dot Structure

Bond Order



A7F09AEE-57F1-425F-9686-34959248CCB6

midterm-exam-93388

#683 Page 10 of 10

SCRAP PAGE

**CHEMISTRY 110 –Midterm Exam****Thursday November 7, 2024, 6:30 – 8:30 pm**

Examiners: Pallavi Sirjoosingh and Ashok Kakkar

First name (please write as legibly as possible within the boxes)

--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--

Last name

--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--

Student ID number

--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--

This booklet has 8 questions on 10 pages (The last page can be used as a scrap page).

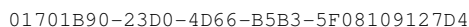
The staple must NOT be removed from the question booklet. Your intact question booklet must be returned at the end of the exam.

A separate booklet contains a periodic table, and datasheet.

Write your answers directly under the questions in the space provided. Do not write on the QR codes or near the top edges (as it may be cut off).

Please write **legibly and dark enough** for the scan to be readable. For students who use English as a second language, dictionaries are permitted. Calculators are allowed.

DO NOT ASK QUESTIONS OF THE INVIGILATORS about interpretations of the exam questions. If you have a concern about a question, please summarize it at the top of this page. All concerns will be evaluated after the exam. GOOD LUCK!



#684 Page 2 of 10

a. 6 points Calculate the **energy (in kJ/mol)** required to ionize (completely remove) the electron from the **n=3** level of **He⁺ ion** (He: Helium). Show your work including equations. (3 sig. figs.)

- Wavelength of photon to *completely remove* an electron from *1s orbital of Hydrogen atom*
- Wavelength of photon to *completely remove* an electron from *1s orbital of Boron atom (B)*
- Wavelength of photon to *completely remove* an electron from *1s orbital of Boron cation (B⁺)*

- Wavelength of photon *emitted* for an electron transitioning from $2p$ to $1s$ orbital
- Wavelength of photon *absorbed* for an electron transitioning from $2s$ to $4p$ orbital
- Wavelength of photon *absorbed* for an electron transitioning from $1s$ to $3p$ orbital

i
or
ii
or
iii



Question 2 8 points



A Chem110 student plots the graph shown on the left, from data obtained for the kinetic energy (eV) of the ejected electron versus frequency (s^{-1}) of the incident light, for an unknown metal. ($1\text{eV} = 1.6022 \times 10^{-19} \text{ J}$)

Based on the plot, answer the questions **a to c** below:

a. 1 point On the plot above, label the **threshold frequency**.

b. 2 points What is the value of the work potential (in J) of the unknown metal?

c. 3 points What is the *longest* wavelength (in nm) required to eject an electron from this metal's surface? Show your work including equation(s) used. (3 sig figs)

d. 2 points In the space provided below, plot the following

(Assume frequency is *above* threshold frequency):

- Kinetic energy* of an ejected electron versus the *intensity* of the incident light (left)
- Number of electrons* ejected versus the *frequency* of the incident light (right)



**Question 3** **6 points****a. 2 points**

In the ground state of Ge (Germanium; $Z = 32$) atom, **name** the following orbitals:

i. Orbital containing the electron that requires the maximum energy to be removed.

ii. Orbital containing the electron that experiences the minimum effective nuclear charge.

b. 2 points

An unknown anion (A^-) has the following *excited* electronic configuration: $[Ar] 4s^2 3d^3 4p^1$

(Ar: Argon; $Z = 18$)

Give the **condensed ground state electronic configuration** of the atom, **A** and the cation, **A^{2+}** .
Show your work. (Assume all electronic transitions are possible)

Electronic Configuration of A:

Electronic Configuration of A^{2+} :

c. 2 points

In a single atom, determine how many electrons can have the following set of quantum numbers?

(i) $n = 5; l = 3$

In a single atom, determine how many electrons can have the following set of quantum numbers?

(i) $n = 3; l = 2; m_s = +1/2$

**Question 4** **8 points**

Rank the following from 1 to 4 in **increasing** order. Provide the ranking in the boxes provided.

Note: Atomic number (Z) are provided for relevant atoms

a) Increasing (lowest to highest) order of *ionization energy*: (1= lowest; 4 = highest)

Mg

Mg: Z = 12

Rb

Rb: Z = 37

Cs

Cs: Z = 55

K

K : Z = 19

b) Increasing (lowest to highest) order of *bond angle*: (1= lowest; 4 = highest)

H₂OCCl₄XeF₄

HCN

c) Increasing (smallest to largest) order of *ionic size* (1= smallest; 4 = largest)

S²⁻

S: Z = 16

O²⁻

O: Z = 8

Be²⁺

Be: Z = 4

Na⁺

Na: Z = 11

d) Increasing (lowest to highest) order of *bond energy* (1= lowest; 4 = highest)

P-F

P-Cl

P-I

P-Br

**Question 5** **12 points**

a. 9 points For the molecules HClO_4 and TeF_6 , draw the most stable *Lewis structure* showing appropriate shape and *dashed/wedged bonds*, based on VSEPR. Determine the **molecular geometry** around the central atom. For HClO_4 , label the *most polar bond*.

HClO_4 (there are no O-O bonds)

Label the most polar bond

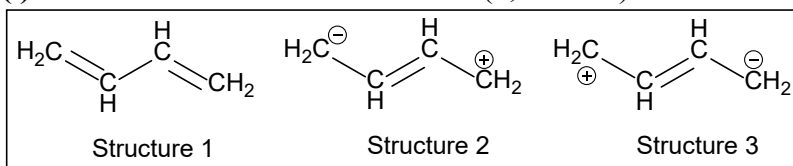
Molecular Geometry around Cl

TeF_6

Molecular Geometry around Te

b. 3 points (Note: Lone pairs are not shown in structures below)

(i) For the three *resonance* structures (1, 2 and 3) shown below, answer the following question:



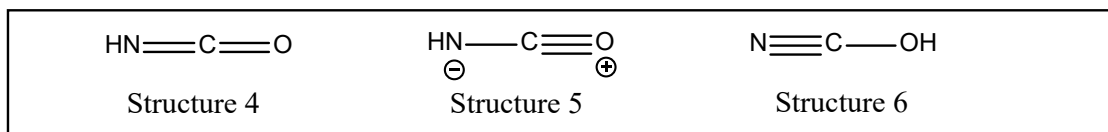
(I) Circle the *most contributing* resonance structure.

Structure 1

Structure 2

Structure 3

(ii) For the three structures given (4, 5, and 6) below, answer the following questions:



(I) Circle all the structures that represent *resonance structures* of each other:

Structure 4

Structure 5

Structure 6

(II) Among the resonance structures, circle the *most contributing* resonance structure.

Structure 4

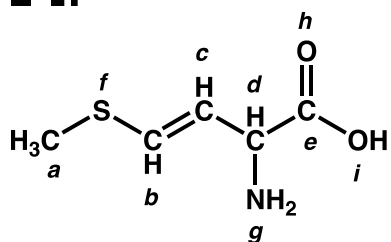
Structure 5

Structure 6

**Question 6** **6 points**

Using the data given below, calculate the **lattice energy** for SrO(s).

	ΔH° (kJ/mol)
$\text{Sr}(s) \longrightarrow \text{Sr}(g)$	+180
$\text{Sr}(g) \longrightarrow \text{Sr}^{2+}(g) + 2e^-$	+1468
$\text{O}_2(g) \longrightarrow 2\text{O}(g)$	+500
$\text{O}(g) + 2e^- \longrightarrow \text{O}^{2-}(g)$	+600
$\text{Sr}(s) + \frac{1}{2}\text{O}_2(g) \longrightarrow \text{SrO}(s)$	-550

**Question 7 10 points**

Consider the molecule given on the left, and answer questions (i) to (vi) using chemical bonding theories that we have learnt in class (valence bond theory, hybridization).

Note that the lone pairs are not shown in the structure:

i. Consider the carbon atoms marked in as ***a***, ***b***, ***c***, ***d***, and ***e*** in the molecule, and determine **hybridization** at each of the carbon centers. (2.5 points)

Atom	Hybridization
Carbon (<i>a</i>)	
Carbon (<i>b</i>)	
Carbon (<i>c</i>)	
Carbon (<i>d</i>)	
Carbon (<i>e</i>)	

ii. Now, consider the sulfur atom (S) marked as ***f***, nitrogen (N) atom (***g***), and oxygen(O) atoms (***h*** and ***i***), and determine their **hybridization**. (2 points)

Atom	Hybridization
Sulfur (<i>f</i>)	
Nitrogen (<i>g</i>)	
Oxygen (<i>h</i>)	
Oxygen (<i>i</i>)	

iii. **Describe** all the bonds formed by the carbon atoms ***a*** and ***b***, using sigma and pi designation as appropriate. (2 points)

Atom	Bonds formed (use sigma and pi as appropriate)
Carbon (<i>a</i>)	
Carbon (<i>b</i>)	

iv. How many total sp^3 -s sigma bonds are in the molecule? (1 point)

v. Determine **electronic** and **molecular geometry** at carbon (C) labeled ***e*** and sulfur (S) labeled ***f*** centres. (2 points)

Atom	Electronic Geometry	Molecular Geometry
Carbon (<i>e</i>)		
Sulfur (<i>f</i>)		

vi. Circle the value from the choices given below, that describes the **C-S-C bond angle at sulfur** (S) marked ***f***: (0.5 points)

>90° to <109.5°

109.5°

>109.5 to <120°

120°

>120° to <180°

**Question 8 12 points**

Using the **valence shell orbitals only** draw a **complete** molecular orbital (MO) diagram (with no s-p mixing) for the molecule **PS**, in the space provided below. **(6 points)**

Assume that 3s and 3p atomic orbitals will form molecular orbitals like those formed by the combination of 2s and 2p atomic orbitals that we learnt in class, and there is no s-p mixing.

Show **energy arrow**, **atomic orbitals** involved, **atom labels**, the **molecular orbitals** that are formed, **filled electrons** in the MO diagram etc. Make sure to label all the atomic and molecular orbitals.

Using the MO diagram above, answer the following questions:

- (i) Which one of the following will have the highest bond order? Circle the correct choice
A) PS^{1-} B) PS C) PS^{1+}
- (ii) Which one of the following will have the longest bond length? Circle the correct choice
A) PS^{1-} B) PS C) PS^{1+}
- (iii) Which of the following will be diamagnetic? Circle the correct choice.
A) PS^{1-} B) PS C) PS^{1+}
- (iv) Draw Lewis dot structure of the molecule with the shortest bond length (based on the MO diagram) from among the three (PS^{1-} , PS, PS^{1+}), and indicate the bond order calculated from the Lewis dot structure. **(3 points)**

Lewis Dot Structure

Bond Order



62AD0B4A-80B6-4F0C-8B82-86A828F990FC

midterm-exam-93388

#684 Page 10 of 10

SCRAP PAGE

**CHEMISTRY 110 –Midterm Exam****Thursday November 7, 2024, 6:30 – 8:30 pm****Examiners: Pallavi Sirjoosingh and Ashok Kakkar**

First name (please write as legibly as possible within the boxes)

--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--

Last name

--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--

Student ID number

--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--

This booklet has 8 questions on 10 pages (The last page can be used as a scrap page).

The staple must NOT be removed from the question booklet. Your intact question booklet must be returned at the end of the exam.

A separate booklet contains a periodic table, and datasheet.

Write your answers directly under the questions in the space provided. Do not write on the QR codes or near the top edges (as it may be cut off).

Please write **legibly and dark enough** for the scan to be readable. For students who use English as a second language, dictionaries are permitted. Calculators are allowed.

DO NOT ASK QUESTIONS OF THE INVIGILATORS about interpretations of the exam questions. If you have a concern about a question, please summarize it at the top of this page. All concerns will be evaluated after the exam. GOOD LUCK!

**Question 1 8 points** (*Assume all transitions are possible*)

a. 6 points Calculate the **energy (in kJ/mol)** required to ionize (completely remove) the electron from the **n=3** level of **He⁺ ion** (He: Helium). Show your work including equations. (3 sig. figs.)

b. 2 points

(I) Select the largest (longest) wavelength from the three choices below:

- i. Wavelength of photon to *completely remove* an electron from *1s orbital of Hydrogen atom*
- ii. Wavelength of photon to *completely remove* an electron from *1s orbital of Boron atom (B)*
- iii. Wavelength of photon to *completely remove* an electron from *1s orbital of Boron cation (B⁺)*

i or ii or iii

(II) Select the smallest (shortest) wavelength from the three choices below (*for H atom*):

- i. Wavelength of photon *emitted* for an electron transitioning from *2p to 1s orbital*
- ii. Wavelength of photon *absorbed* for an electron transitioning from *2s to 4p orbital*
- iii. Wavelength of photon *absorbed* for an electron transitioning from *1s to 3p orbital*

i or ii or iii



Question 2 8 points



A Chem110 student plots the graph shown on the left, from data obtained for the kinetic energy (eV) of the ejected electron versus frequency (s^{-1}) of the incident light, for an unknown metal. ($1\text{eV} = 1.6022 \times 10^{-19} \text{ J}$)

Based on the plot, answer the questions **a to c** below:

a. 1 point On the plot above, label the **threshold frequency**.

b. 2 points What is the value of the work potential (in J) of the unknown metal?

c. 3 points What is the *longest* wavelength (in nm) required to eject an electron from this metal's surface? Show your work including equation(s) used. (3 sig figs)

d. 2 points In the space provided below, plot the following

(Assume frequency is *above* threshold frequency):

- Kinetic energy* of an ejected electron versus the *intensity* of the incident light (left)
- Number of electrons* ejected versus the *frequency* of the incident light (right)



**Question 3** **6 points****a. 2 points**

In the ground state of Ge (Germanium; $Z = 32$) atom, **name** the following orbitals:

i. Orbital containing the electron that requires the maximum energy to be removed.

ii. Orbital containing the electron that experiences the minimum effective nuclear charge.

b. 2 points

An unknown anion (A^-) has the following *excited* electronic configuration: $[Ar] 4s^2 3d^3 4p^1$

(Ar: Argon; $Z = 18$)

Give the **condensed ground state electronic configuration** of the atom, **A** and the cation, **A^{2+}** .
Show your work. (Assume all electronic transitions are possible)

Electronic Configuration of A:

Electronic Configuration of A^{2+} :

c. 2 points

In a single atom, determine how many electrons can have the following set of quantum numbers?

(i) $n = 5; l = 3$

In a single atom, determine how many electrons can have the following set of quantum numbers?

(i) $n = 3; l = 2; m_s = +1/2$

**Question 4** **8 points**

Rank the following from 1 to 4 in **increasing** order. Provide the ranking in the boxes provided.

Note: Atomic number (Z) are provided for relevant atoms

a) Increasing (lowest to highest) order of *ionization energy*: (1= lowest; 4 = highest)

Mg

Mg: Z = 12

Rb

Rb: Z = 37

Cs

Cs: Z = 55

K

K : Z = 19

b) Increasing (lowest to highest) order of *bond angle*: (1= lowest; 4 = highest)

H₂OCCl₄XeF₄

HCN

c) Increasing (smallest to largest) order of *ionic size* (1= smallest; 4 = largest)

S²⁻

S: Z = 16

O²⁻

O: Z = 8

Be²⁺

Be: Z = 4

Na⁺

Na: Z = 11

d) Increasing (lowest to highest) order of *bond energy* (1= lowest; 4 = highest)

P-F

P-Cl

P-I

P-Br

**Question 5** **12 points**

a. 9 points For the molecules HClO_4 and TeF_6 , draw the most stable *Lewis structure* showing appropriate shape and *dashed/wedged bonds*, based on VSEPR. Determine the **molecular geometry** around the central atom. For HClO_4 , label the *most polar bond*.



b. 3 points (Note: Lone pairs are not shown in structures below)

(i) For the three *resonance* structures (1, 2 and 3) shown below, answer the following question:



(I) Circle the *most contributing* resonance structure.

Structure 1

Structure 2

Structure 3

(ii) For the three structures given (4, 5, and 6) below, answer the following questions:



(I) Circle all the structures that represent *resonance structures* of each other:

Structure 4

Structure 5

Structure 6

(II) Among the resonance structures, circle the *most contributing* resonance structure.

Structure 4

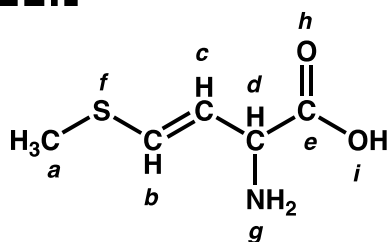
Structure 5

Structure 6

**Question 6** **6 points**

Using the data given below, calculate the **lattice energy** for SrO(s).

	ΔH° (kJ/mol)
$\text{Sr}(s) \longrightarrow \text{Sr}(g)$	+180
$\text{Sr}(g) \longrightarrow \text{Sr}^{2+}(g) + 2e^-$	+1468
$\text{O}_2(g) \longrightarrow 2\text{O}(g)$	+500
$\text{O}(g) + 2e^- \longrightarrow \text{O}^{2-}(g)$	+600
$\text{Sr}(s) + \frac{1}{2}\text{O}_2(g) \longrightarrow \text{SrO}(s)$	-550

**Question 7 10 points**

Consider the molecule given on the left, and answer questions (i) to (vi) using chemical bonding theories that we have learnt in class (valence bond theory, hybridization).

Note that the lone pairs are not shown in the structure:

i. Consider the carbon atoms marked in as ***a***, ***b***, ***c***, ***d***, and ***e*** in the molecule, and determine **hybridization** at each of the carbon centers. (2.5 points)

Atom	Hybridization
Carbon (<i>a</i>)	
Carbon (<i>b</i>)	
Carbon (<i>c</i>)	
Carbon (<i>d</i>)	
Carbon (<i>e</i>)	

ii. Now, consider the sulfur atom (S) marked as ***f***, nitrogen (N) atom (***g***), and oxygen(O) atoms (***h*** and ***i***), and determine their **hybridization**. (2 points)

Atom	Hybridization
Sulfur (<i>f</i>)	
Nitrogen (<i>g</i>)	
Oxygen (<i>h</i>)	
Oxygen (<i>i</i>)	

iii. **Describe** all the bonds formed by the carbon atoms ***a*** and ***b***, using sigma and pi designation as appropriate. (2 points)

Atom	Bonds formed (use sigma and pi as appropriate)
Carbon (<i>a</i>)	
Carbon (<i>b</i>)	

iv. How many total sp^3 -s sigma bonds are in the molecule? (1 point)

v. Determine **electronic** and **molecular geometry** at carbon (C) labeled ***e*** and sulfur (S) labeled ***f*** centres. (2 points)

Atom	Electronic Geometry	Molecular Geometry
Carbon (<i>e</i>)		
Sulfur (<i>f</i>)		

vi. Circle the value from the choices given below, that describes the **C-S-C bond angle at sulfur** (S) marked ***f***: (0.5 points)

>90° to <109.5° 109.5° >109.5 to <120° 120° >120° to <180°


Question 8 12 points

Using the **valence shell orbitals only** draw a **complete** molecular orbital (MO) diagram (with no s-p mixing) for the molecule **PS**, in the space provided below. **(6 points)**

Assume that 3s and 3p atomic orbitals will form molecular orbitals like those formed by the combination of 2s and 2p atomic orbitals that we learnt in class, and there is no s-p mixing.

Show **energy arrow**, **atomic orbitals** involved, **atom labels**, the **molecular orbitals** that are formed, **filled electrons** in the MO diagram etc. Make sure to label all the atomic and molecular orbitals.

Using the MO diagram above, answer the following questions:

- (i) Which one of the following will have the highest bond order? Circle the correct choice
 A) PS^{1-} B) PS C) PS^{1+}
- (ii) Which one of the following will have the longest bond length? Circle the correct choice
 A) PS^{1-} B) PS C) PS^{1+}
- (iii) Which of the following will be diamagnetic? Circle the correct choice.
 A) PS^{1-} B) PS C) PS^{1+}
- (iv) Draw Lewis dot structure of the molecule with the shortest bond length (based on the MO diagram) from among the three (PS^{1-} , PS, PS^{1+}), and indicate the bond order calculated from the Lewis dot structure. **(3 points)**

Lewis Dot Structure

Bond Order



5E3DD1FC-FDC3-43A7-88C8-0F628A3F8646

midterm-exam-93388

#685 Page 10 of 10

SCRAP PAGE

**CHEMISTRY 110 –Midterm Exam****Thursday November 7, 2024, 6:30 – 8:30 pm****Examiners: Pallavi Sirjoosingh and Ashok Kakkar**

First name (please write as legibly as possible within the boxes)

--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--

Last name

--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--

Student ID number

--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--

This booklet has 8 questions on 10 pages (The last page can be used as a scrap page).

The staple must NOT be removed from the question booklet. Your intact question booklet must be returned at the end of the exam.

A separate booklet contains a periodic table, and datasheet.

Write your answers directly under the questions in the space provided. Do not write on the QR codes or near the top edges (as it may be cut off).

Please write **legibly and dark enough** for the scan to be readable. For students who use English as a second language, dictionaries are permitted. Calculators are allowed.

DO NOT ASK QUESTIONS OF THE INVIGILATORS about interpretations of the exam questions. If you have a concern about a question, please summarize it at the top of this page. All concerns will be evaluated after the exam. GOOD LUCK!

**Question 1 8 points** (*Assume all transitions are possible*)

a. 6 points Calculate the **energy (in kJ/mol)** required to ionize (completely remove) the electron from the **n=3** level of **He⁺ ion** (He: Helium). Show your work including equations. (3 sig. figs.)

b. 2 points

(I) Select the largest (longest) wavelength from the three choices below:

- i. Wavelength of photon to *completely remove* an electron from *1s orbital of Hydrogen atom*
- ii. Wavelength of photon to *completely remove* an electron from *1s orbital of Boron atom (B)*
- iii. Wavelength of photon to *completely remove* an electron from *1s orbital of Boron cation (B⁺)*

i or ii or iii

(II) Select the smallest (shortest) wavelength from the three choices below (*for H atom*):

- i. Wavelength of photon *emitted* for an electron transitioning from *2p to 1s orbital*
- ii. Wavelength of photon *absorbed* for an electron transitioning from *2s to 4p orbital*
- iii. Wavelength of photon *absorbed* for an electron transitioning from *1s to 3p orbital*

i or ii or iii



Question 2 8 points



A Chem110 student plots the graph shown on the left, from data obtained for the kinetic energy (eV) of the ejected electron versus frequency (s^{-1}) of the incident light, for an unknown metal. ($1\text{eV} = 1.6022 \times 10^{-19} \text{ J}$)

Based on the plot, answer the questions **a to c** below:

a. 1 point On the plot above, label the **threshold frequency**.

b. 2 points What is the value of the work potential (in J) of the unknown metal?

c. 3 points What is the *longest* wavelength (in nm) required to eject an electron from this metal's surface? Show your work including equation(s) used. (3 sig figs)

d. 2 points In the space provided below, plot the following

(Assume frequency is *above* threshold frequency):

- Kinetic energy* of an ejected electron versus the *intensity* of the incident light (left)
- Number of electrons* ejected versus the *frequency* of the incident light (right)



**Question 3** **6 points****a. 2 points**

In the ground state of Ge (Germanium; $Z = 32$) atom, **name** the following orbitals:

i. Orbital containing the electron that requires the maximum energy to be removed.

ii. Orbital containing the electron that experiences the minimum effective nuclear charge.

b. 2 points

An unknown anion (A^-) has the following *excited* electronic configuration: $[Ar] 4s^2 3d^3 4p^1$

(Ar: Argon; $Z = 18$)

Give the **condensed ground state electronic configuration** of the atom, **A** and the cation, A^{2+} .
Show your work. (Assume all electronic transitions are possible)

Electronic Configuration of A:

Electronic Configuration of A^{2+} :

c. 2 points

In a single atom, determine how many electrons can have the following set of quantum numbers?

(i) $n = 5; l = 3$

In a single atom, determine how many electrons can have the following set of quantum numbers?

(i) $n = 3; l = 2; m_s = +1/2$

**Question 4** **8 points**

Rank the following from 1 to 4 in **increasing** order. Provide the ranking in the boxes provided.

Note: Atomic number (Z) are provided for relevant atoms

a) Increasing (lowest to highest) order of *ionization energy*: (1= lowest; 4 = highest)

Mg

Mg: Z = 12

Rb

Rb: Z = 37

Cs

Cs: Z = 55

K

K : Z = 19

b) Increasing (lowest to highest) order of *bond angle*: (1= lowest; 4 = highest)

H₂OCCl₄XeF₄

HCN

c) Increasing (smallest to largest) order of *ionic size* (1= smallest; 4 = largest)

S²⁻

S: Z = 16

O²⁻

O: Z = 8

Be²⁺

Be: Z = 4

Na⁺

Na: Z = 11

d) Increasing (lowest to highest) order of *bond energy* (1= lowest; 4 = highest)

P-F

P-Cl

P-I

P-Br

**Question 5** **12 points**

a. 9 points For the molecules HClO_4 and TeF_6 , draw the most stable *Lewis structure* showing appropriate shape and *dashed/wedged bonds*, based on VSEPR. Determine the **molecular geometry** around the central atom. For HClO_4 , label the *most polar bond*.



b. 3 points (Note: Lone pairs are not shown in structures below)

(i) For the three *resonance* structures (1, 2 and 3) shown below, answer the following question:



(I) Circle the *most contributing* resonance structure.

Structure 1

Structure 2

Structure 3

(ii) For the three structures given (4, 5, and 6) below, answer the following questions:



(I) Circle all the structures that represent *resonance structures* of each other:

Structure 4

Structure 5

Structure 6

(II) Among the resonance structures, circle the *most contributing* resonance structure.

Structure 4

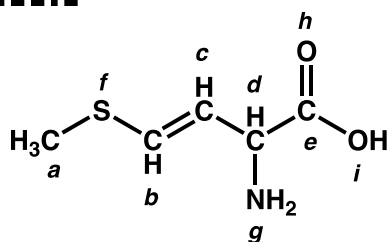
Structure 5

Structure 6

**Question 6** **6 points**

Using the data given below, calculate the **lattice energy** for SrO(s).

	ΔH° (kJ/mol)
$\text{Sr}(s) \longrightarrow \text{Sr}(g)$	+180
$\text{Sr}(g) \longrightarrow \text{Sr}^{2+}(g) + 2e^-$	+1468
$\text{O}_2(g) \longrightarrow 2\text{O}(g)$	+500
$\text{O}(g) + 2e^- \longrightarrow \text{O}^{2-}(g)$	+600
$\text{Sr}(s) + \frac{1}{2}\text{O}_2(g) \longrightarrow \text{SrO}(s)$	-550

**Question 7 10 points**

Consider the molecule given on the left, and answer questions (i) to (vi) using chemical bonding theories that we have learnt in class (valence bond theory, hybridization).

Note that the lone pairs are not shown in the structure:

i. Consider the carbon atoms marked in as ***a***, ***b***, ***c***, ***d***, and ***e*** in the molecule, and determine **hybridization** at each of the carbon centers. (2.5 points)

Atom	Hybridization
Carbon (<i>a</i>)	
Carbon (<i>b</i>)	
Carbon (<i>c</i>)	
Carbon (<i>d</i>)	
Carbon (<i>e</i>)	

ii. Now, consider the sulfur atom (S) marked as ***f***, nitrogen (N) atom (***g***), and oxygen(O) atoms (***h*** and ***i***), and determine their **hybridization**. (2 points)

Atom	Hybridization
Sulfur (<i>f</i>)	
Nitrogen (<i>g</i>)	
Oxygen (<i>h</i>)	
Oxygen (<i>i</i>)	

iii. **Describe** all the bonds formed by the carbon atoms ***a*** and ***b***, using sigma and pi designation as appropriate. (2 points)

Atom	Bonds formed (use sigma and pi as appropriate)
Carbon (<i>a</i>)	
Carbon (<i>b</i>)	

iv. How many total sp^3 -s sigma bonds are in the molecule? (1 point)

v. Determine **electronic** and **molecular geometry** at carbon (C) labeled ***e*** and sulfur (S) labeled ***f*** centres. (2 points)

Atom	Electronic Geometry	Molecular Geometry
Carbon (<i>e</i>)		
Sulfur (<i>f</i>)		

vi. Circle the value from the choices given below, that describes the **C-S-C bond angle at sulfur** (S) marked ***f***: (0.5 points)

>90° to <109.5° 109.5° >109.5 to <120° 120° >120° to <180°


Question 8 12 points

Using the **valence shell orbitals only** draw a **complete** molecular orbital (MO) diagram (with no s-p mixing) for the molecule **PS**, in the space provided below. **(6 points)**

Assume that 3s and 3p atomic orbitals will form molecular orbitals like those formed by the combination of 2s and 2p atomic orbitals that we learnt in class, and there is no s-p mixing.

Show **energy arrow**, **atomic orbitals** involved, **atom labels**, the **molecular orbitals** that are formed, **filled electrons** in the MO diagram etc. Make sure to label all the atomic and molecular orbitals.

Using the MO diagram above, answer the following questions:

- (i) Which one of the following will have the highest bond order? Circle the correct choice
 A) PS^{1-} B) PS C) PS^{1+}
- (ii) Which one of the following will have the longest bond length? Circle the correct choice
 A) PS^{1-} B) PS C) PS^{1+}
- (iii) Which of the following will be diamagnetic? Circle the correct choice.
 A) PS^{1-} B) PS C) PS^{1+}
- (iv) Draw Lewis dot structure of the molecule with the shortest bond length (based on the MO diagram) from among the three (PS^{1-} , PS, PS^{1+}), and indicate the bond order calculated from the Lewis dot structure. **(3 points)**

Lewis Dot Structure

Bond Order



1FF1F0F7-65BF-484F-8582-EDCB4AE01A1B

midterm-exam-93388

#686 Page 10 of 10

SCRAP PAGE

**CHEMISTRY 110 –Midterm Exam****Thursday November 7, 2024, 6:30 – 8:30 pm**

Examiners: Pallavi Sirjoosingh and Ashok Kakkar

First name (please write as legibly as possible within the boxes)

--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--

Last name

--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--

Student ID number

--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--

This booklet has 8 questions on 10 pages (The last page can be used as a scrap page).

The staple must NOT be removed from the question booklet. Your intact question booklet must be returned at the end of the exam.

A separate booklet contains a periodic table, and datasheet.

Write your answers directly under the questions in the space provided. Do not write on the QR codes or near the top edges (as it may be cut off).

Please write **legibly and dark enough** for the scan to be readable. For students who use English as a second language, dictionaries are permitted. Calculators are allowed.

DO NOT ASK QUESTIONS OF THE INVIGILATORS about interpretations of the exam questions. If you have a concern about a question, please summarize it at the top of this page. All concerns will be evaluated after the exam. GOOD LUCK!

**Question 1 8 points** (*Assume all transitions are possible*)

a. 6 points Calculate the **energy (in kJ/mol)** required to ionize (completely remove) the electron from the **$n=3$** level of **He^+ ion** (He: Helium). Show your work including equations. (3 sig. figs.)

b. 2 points

(I) Select the largest (longest) wavelength from the three choices below:

- i. Wavelength of photon to *completely remove* an electron from *1s orbital of Hydrogen atom*
- ii. Wavelength of photon to *completely remove* an electron from *1s orbital of Boron atom (B)*
- iii. Wavelength of photon to *completely remove* an electron from *1s orbital of Boron cation (B^+)*

i or ii or iii

(II) Select the smallest (shortest) wavelength from the three choices below (*for H atom*):

- i. Wavelength of photon *emitted* for an electron transitioning from *2p to 1s orbital*
- ii. Wavelength of photon *absorbed* for an electron transitioning from *2s to 4p orbital*
- iii. Wavelength of photon *absorbed* for an electron transitioning from *1s to 3p orbital*

i or ii or iii



Question 2 8 points



A Chem110 student plots the graph shown on the left, from data obtained for the kinetic energy (eV) of the ejected electron versus frequency (s^{-1}) of the incident light, for an unknown metal. ($1\text{eV} = 1.6022 \times 10^{-19} \text{ J}$)

Based on the plot, answer the questions **a to c** below:

a. 1 point On the plot above, label the **threshold frequency**.

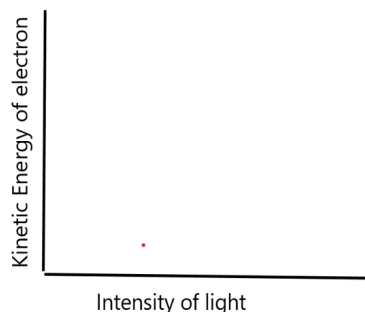
b. 2 points What is the value of the work potential (in J) of the unknown metal?

c. 3 points What is the *longest* wavelength (in nm) required to eject an electron from this metal's surface? Show your work including equation(s) used. (3 sig figs)

d. 2 points In the space provided below, plot the following

(Assume frequency is *above* threshold frequency):

- Kinetic energy* of an ejected electron versus the *intensity* of the incident light (left)
- Number of electrons* ejected versus the *frequency* of the incident light (right)



**Question 3** **6 points****a. 2 points**

In the ground state of Ge (Germanium; $Z = 32$) atom, **name** the following orbitals:

i. Orbital containing the electron that requires the maximum energy to be removed.

ii. Orbital containing the electron that experiences the minimum effective nuclear charge.

b. 2 points

An unknown anion (A^-) has the following *excited* electronic configuration: $[Ar] 4s^2 3d^3 4p^1$

(Ar: Argon; $Z = 18$)

Give the **condensed ground state electronic configuration** of the atom, **A** and the cation, **A^{2+}** .
Show your work. (Assume all electronic transitions are possible)

Electronic Configuration of A:

Electronic Configuration of A^{2+} :

c. 2 points

In a single atom, determine how many electrons can have the following set of quantum numbers?

(i) $n = 5; l = 3$

In a single atom, determine how many electrons can have the following set of quantum numbers?

(i) $n = 3; l = 2; m_s = +1/2$

**Question 4** **8 points**

Rank the following from 1 to 4 in **increasing** order. Provide the ranking in the boxes provided.

Note: Atomic number (Z) are provided for relevant atoms

a) Increasing (lowest to highest) order of *ionization energy*: (1= lowest; 4 = highest)

Mg

Mg: Z = 12

Rb

Rb: Z = 37

Cs

Cs: Z = 55

K

K : Z = 19

b) Increasing (lowest to highest) order of *bond angle*: (1= lowest; 4 = highest)

H₂OCCl₄XeF₄

HCN

c) Increasing (smallest to largest) order of *ionic size* (1= smallest; 4 = largest)

S²⁻

S: Z = 16

O²⁻

O: Z = 8

Be²⁺

Be: Z = 4

Na⁺

Na: Z = 11

d) Increasing (lowest to highest) order of *bond energy* (1= lowest; 4 = highest)

P-F

P-Cl

P-I

P-Br

**Question 5** **12 points**

a. 9 points For the molecules HClO_4 and TeF_6 , draw the most stable *Lewis structure* showing appropriate shape and *dashed/wedged bonds*, based on VSEPR. Determine the **molecular geometry** around the central atom. For HClO_4 , label the *most polar bond*.



b. 3 points (Note: Lone pairs are not shown in structures below)

(i) For the three *resonance* structures (1, 2 and 3) shown below, answer the following question:



(I) Circle the *most contributing* resonance structure.

Structure 1

Structure 2

Structure 3

(ii) For the three structures given (4, 5, and 6) below, answer the following questions:



(I) Circle all the structures that represent *resonance structures* of each other:

Structure 4

Structure 5

Structure 6

(II) Among the resonance structures, circle the *most contributing* resonance structure.

Structure 4

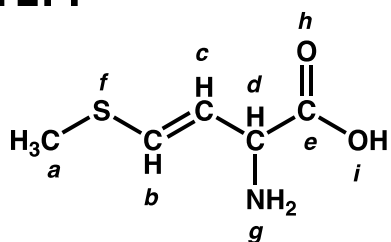
Structure 5

Structure 6

**Question 6** **6 points**

Using the data given below, calculate the **lattice energy** for SrO(s).

	ΔH° (kJ/mol)
$\text{Sr}(s) \longrightarrow \text{Sr}(g)$	+180
$\text{Sr}(g) \longrightarrow \text{Sr}^{2+}(g) + 2e^-$	+1468
$\text{O}_2(g) \longrightarrow 2\text{O}(g)$	+500
$\text{O}(g) + 2e^- \longrightarrow \text{O}^{2-}(g)$	+600
$\text{Sr}(s) + \frac{1}{2}\text{O}_2(g) \longrightarrow \text{SrO}(s)$	-550

**Question 7 10 points**

Consider the molecule given on the left, and answer questions (i) to (vi) using chemical bonding theories that we have learnt in class (valence bond theory, hybridization).

Note that the lone pairs are not shown in the structure:

i. Consider the carbon atoms marked in as ***a***, ***b***, ***c***, ***d***, and ***e*** in the molecule, and determine **hybridization** at each of the carbon centers. (2.5 points)

Atom	Hybridization
Carbon (<i>a</i>)	
Carbon (<i>b</i>)	
Carbon (<i>c</i>)	
Carbon (<i>d</i>)	
Carbon (<i>e</i>)	

ii. Now, consider the sulfur atom (S) marked as ***f***, nitrogen (N) atom (***g***), and oxygen(O) atoms (***h*** and ***i***), and determine their **hybridization**. (2 points)

Atom	Hybridization
Sulfur (<i>f</i>)	
Nitrogen (<i>g</i>)	
Oxygen (<i>h</i>)	
Oxygen (<i>i</i>)	

iii. **Describe** all the bonds formed by the carbon atoms ***a*** and ***b***, using sigma and pi designation as appropriate. (2 points)

Atom	Bonds formed (use sigma and pi as appropriate)
Carbon (<i>a</i>)	
Carbon (<i>b</i>)	

iv. How many total sp^3 -s sigma bonds are in the molecule? (1 point)

v. Determine **electronic** and **molecular geometry** at carbon (C) labeled ***e*** and sulfur (S) labeled ***f*** centres. (2 points)

Atom	Electronic Geometry	Molecular Geometry
Carbon (<i>e</i>)		
Sulfur (<i>f</i>)		

vi. Circle the value from the choices given below, that describes the **C-S-C bond angle at sulfur** (S) marked ***f***: (0.5 points)

>90° to <109.5° 109.5° >109.5 to <120° 120° >120° to <180°

**Question 8 12 points**

Using the **valence shell orbitals only** draw a **complete** molecular orbital (MO) diagram (with no s-p mixing) for the molecule **PS**, in the space provided below. **(6 points)**

Assume that 3s and 3p atomic orbitals will form molecular orbitals like those formed by the combination of 2s and 2p atomic orbitals that we learnt in class, and there is no s-p mixing.

Show **energy arrow**, **atomic orbitals** involved, **atom labels**, the **molecular orbitals** that are formed, **filled electrons** in the MO diagram etc. Make sure to label all the atomic and molecular orbitals.

Using the MO diagram above, answer the following questions:

- (i) Which one of the following will have the highest bond order? Circle the correct choice
A) PS^{1-} B) PS C) PS^{1+}
- (ii) Which one of the following will have the longest bond length? Circle the correct choice
A) PS^{1-} B) PS C) PS^{1+}
- (iii) Which of the following will be diamagnetic? Circle the correct choice.
A) PS^{1-} B) PS C) PS^{1+}
- (iv) Draw Lewis dot structure of the molecule with the shortest bond length (based on the MO diagram) from among the three (PS^{1-} , PS, PS^{1+}), and indicate the bond order calculated from the Lewis dot structure. **(3 points)**

Lewis Dot Structure

Bond Order



A6EEBECA-4671-44AA-8824-5431F56CC9F9

midterm-exam-93388

#687 Page 10 of 10

SCRAP PAGE

**CHEMISTRY 110 –Midterm Exam****Thursday November 7, 2024, 6:30 – 8:30 pm**

Examiners: Pallavi Sirjoosingh and Ashok Kakkar

First name (please write as legibly as possible within the boxes)

--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--

Last name

--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--

Student ID number

--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--

This booklet has 8 questions on 10 pages (The last page can be used as a scrap page).

The staple must NOT be removed from the question booklet. Your intact question booklet must be returned at the end of the exam.

A separate booklet contains a periodic table, and datasheet.

Write your answers directly under the questions in the space provided. Do not write on the QR codes or near the top edges (as it may be cut off).

Please write **legibly and dark enough** for the scan to be readable. For students who use English as a second language, dictionaries are permitted. Calculators are allowed.

DO NOT ASK QUESTIONS OF THE INVIGILATORS about interpretations of the exam questions. If you have a concern about a question, please summarize it at the top of this page. All concerns will be evaluated after the exam. GOOD LUCK!

**Question 1 8 points** (*Assume all transitions are possible*)

a. 6 points Calculate the **energy (in kJ/mol)** required to ionize (completely remove) the electron from the **n=3** level of **He⁺ ion** (He: Helium). Show your work including equations. (3 sig. figs.)

b. 2 points

(I) Select the largest (longest) wavelength from the three choices below:

- i. Wavelength of photon to *completely remove* an electron from *1s orbital of Hydrogen atom*
- ii. Wavelength of photon to *completely remove* an electron from *1s orbital of Boron atom (B)*
- iii. Wavelength of photon to *completely remove* an electron from *1s orbital of Boron cation (B⁺)*

i or ii or iii

(II) Select the smallest (shortest) wavelength from the three choices below (*for H atom*):

- i. Wavelength of photon *emitted* for an electron transitioning from *2p to 1s orbital*
- ii. Wavelength of photon *absorbed* for an electron transitioning from *2s to 4p orbital*
- iii. Wavelength of photon *absorbed* for an electron transitioning from *1s to 3p orbital*

i or ii or iii



Question 2 8 points



A Chem110 student plots the graph shown on the left, from data obtained for the kinetic energy (eV) of the ejected electron versus frequency (s^{-1}) of the incident light, for an unknown metal. ($1\text{eV} = 1.6022 \times 10^{-19} \text{ J}$)

Based on the plot, answer the questions **a to c** below:

a. 1 point On the plot above, label the **threshold frequency**.

b. 2 points What is the value of the work potential (in J) of the unknown metal?

c. 3 points What is the *longest* wavelength (in nm) required to eject an electron from this metal's surface? Show your work including equation(s) used. (3 sig figs)

d. 2 points In the space provided below, plot the following

(Assume frequency is *above* threshold frequency):

- i. *Kinetic energy* of an ejected electron versus the *intensity* of the incident light (left)
- ii. *Number of electrons* ejected versus the *frequency* of the incident light (right)



**Question 3** **6 points****a. 2 points**

In the ground state of Ge (Germanium; $Z = 32$) atom, **name** the following orbitals:

i. Orbital containing the electron that requires the maximum energy to be removed.

ii. Orbital containing the electron that experiences the minimum effective nuclear charge.

b. 2 points

An unknown anion (A^-) has the following *excited* electronic configuration: $[Ar] 4s^2 3d^3 4p^1$

(Ar: Argon; $Z = 18$)

Give the **condensed ground state electronic configuration** of the atom, **A** and the cation, **A^{2+}** .
Show your work. (Assume all electronic transitions are possible)

Electronic Configuration of A:

Electronic Configuration of A^{2+} :

c. 2 points

In a single atom, determine how many electrons can have the following set of quantum numbers?

(i) $n = 5; l = 3$

In a single atom, determine how many electrons can have the following set of quantum numbers?

(i) $n = 3; l = 2; m_s = +1/2$

**Question 4** **8 points**

Rank the following from 1 to 4 in **increasing** order. Provide the ranking in the boxes provided.

Note: Atomic number (Z) are provided for relevant atoms

a) Increasing (lowest to highest) order of *ionization energy*: (1= lowest; 4 = highest)

Mg

Mg: Z = 12

Rb

Rb: Z = 37

Cs

Cs: Z = 55

K

K : Z = 19

b) Increasing (lowest to highest) order of *bond angle*: (1= lowest; 4 = highest)

H₂OCCl₄XeF₄

HCN

c) Increasing (smallest to largest) order of *ionic size* (1= smallest; 4 = largest)

S²⁻

S: Z = 16

O²⁻

O: Z = 8

Be²⁺

Be: Z = 4

Na⁺

Na: Z = 11

d) Increasing (lowest to highest) order of *bond energy* (1= lowest; 4 = highest)

P-F

P-Cl

P-I

P-Br

**Question 5** **12 points**

a. 9 points For the molecules HClO_4 and TeF_6 , draw the most stable *Lewis structure* showing appropriate shape and *dashed/wedged bonds*, based on VSEPR. Determine the **molecular geometry** around the central atom. For HClO_4 , label the *most polar bond*.



b. 3 points (Note: Lone pairs are not shown in structures below)

(i) For the three *resonance* structures (1, 2 and 3) shown below, answer the following question:



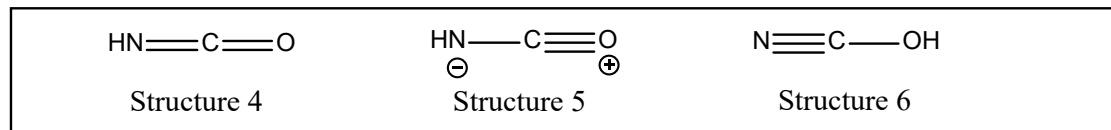
(I) Circle the *most contributing* resonance structure.

Structure 1

Structure 2

Structure 3

(ii) For the three structures given (4, 5, and 6) below, answer the following questions:



(I) Circle all the structures that represent *resonance structures* of each other:

Structure 4

Structure 5

Structure 6

(II) Among the resonance structures, circle the *most contributing* resonance structure.

Structure 4

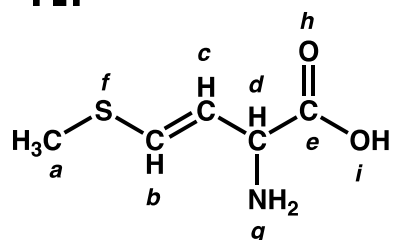
Structure 5

Structure 6

**Question 6** **6 points**

Using the data given below, calculate the **lattice energy** for SrO(s).

	ΔH° (kJ/mol)
$\text{Sr}(s) \longrightarrow \text{Sr}(g)$	+180
$\text{Sr}(g) \longrightarrow \text{Sr}^{2+}(g) + 2e^-$	+1468
$\text{O}_2(g) \longrightarrow 2\text{O}(g)$	+500
$\text{O}(g) + 2e^- \longrightarrow \text{O}^{2-}(g)$	+600
$\text{Sr}(s) + \frac{1}{2}\text{O}_2(g) \longrightarrow \text{SrO}(s)$	-550

**Question 7 10 points**

Consider the molecule given on the left, and answer questions (i) to (vi) using chemical bonding theories that we have learnt in class (valence bond theory, hybridization).

Note that the lone pairs are not shown in the structure:

i. Consider the carbon atoms marked in as ***a***, ***b***, ***c***, ***d***, and ***e*** in the molecule, and determine **hybridization** at each of the carbon centers. (2.5 points)

Atom	Hybridization
Carbon (<i>a</i>)	
Carbon (<i>b</i>)	
Carbon (<i>c</i>)	
Carbon (<i>d</i>)	
Carbon (<i>e</i>)	

ii. Now, consider the sulfur atom (S) marked as ***f***, nitrogen (N) atom (***g***), and oxygen(O) atoms (***h*** and ***i***), and determine their **hybridization**. (2 points)

Atom	Hybridization
Sulfur (<i>f</i>)	
Nitrogen (<i>g</i>)	
Oxygen (<i>h</i>)	
Oxygen (<i>i</i>)	

iii. **Describe** all the bonds formed by the carbon atoms ***a*** and ***b***, using sigma and pi designation as appropriate. (2 points)

Atom	Bonds formed (use sigma and pi as appropriate)
Carbon (<i>a</i>)	
Carbon (<i>b</i>)	

iv. How many total sp^3 -s sigma bonds are in the molecule? (1 point)

v. Determine **electronic** and **molecular geometry** at carbon (C) labeled ***e*** and sulfur (S) labeled ***f*** centres. (2 points)

Atom	Electronic Geometry	Molecular Geometry
Carbon (<i>e</i>)		
Sulfur (<i>f</i>)		

vi. Circle the value from the choices given below, that describes the **C-S-C bond angle at sulfur** (S) marked ***f***: (0.5 points)

>90° to <109.5° 109.5° >109.5 to <120° 120° >120° to <180°



Question 8 12 points

Using the **valence shell orbitals only** draw a **complete** molecular orbital (MO) diagram (with no s-p mixing) for the molecule **PS**, in the space provided below. **(6 points)**

Assume that 3s and 3p atomic orbitals will form molecular orbitals like those formed by the combination of 2s and 2p atomic orbitals that we learnt in class, and there is no s-p mixing.

Show **energy arrow**, **atomic orbitals** involved, **atom labels**, the **molecular orbitals** that are formed, **filled electrons** in the MO diagram etc. Make sure to label all the atomic and molecular orbitals.

Using the MO diagram above, answer the following questions:

- (i) Which one of the following will have the highest bond order? Circle the correct choice
 A) PS^{1-} B) PS C) PS^{1+}
- (ii) Which one of the following will have the longest bond length? Circle the correct choice
 A) PS^{1-} B) PS C) PS^{1+}
- (iii) Which of the following will be diamagnetic? Circle the correct choice.
 A) PS^{1-} B) PS C) PS^{1+}
- (iv) Draw Lewis dot structure of the molecule with the shortest bond length (based on the MO diagram) from among the three (PS^{1-} , PS, PS^{1+}), and indicate the bond order calculated from the Lewis dot structure. **(3 points)**

Lewis Dot Structure

Bond Order



DC1DAB26-1564-4EC3-BB68-2F6536558633

midterm-exam-93388

#688 Page 10 of 10

SCRAP PAGE

**CHEMISTRY 110 –Midterm Exam****Thursday November 7, 2024, 6:30 – 8:30 pm**

Examiners: Pallavi Sirjoosingh and Ashok Kakkar

First name (please write as legibly as possible within the boxes)

--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--

Last name

--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--

Student ID number

--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--

This booklet has 8 questions on 10 pages (The last page can be used as a scrap page).

The staple must NOT be removed from the question booklet. Your intact question booklet must be returned at the end of the exam.

A separate booklet contains a periodic table, and datasheet.

Write your answers directly under the questions in the space provided. Do not write on the QR codes or near the top edges (as it may be cut off).

Please write **legibly and dark enough** for the scan to be readable. For students who use English as a second language, dictionaries are permitted. Calculators are allowed.

DO NOT ASK QUESTIONS OF THE INVIGILATORS about interpretations of the exam questions. If you have a concern about a question, please summarize it at the top of this page. All concerns will be evaluated after the exam. GOOD LUCK!

**Question 1 8 points** (*Assume all transitions are possible*)

a. 6 points Calculate the **energy (in kJ/mol)** required to ionize (completely remove) the electron from the **$n=3$** level of **He^+ ion** (He: Helium). Show your work including equations. (3 sig. figs.)

b. 2 points

(I) Select the largest (longest) wavelength from the three choices below:

- i. Wavelength of photon to *completely remove* an electron from *1s orbital of Hydrogen atom*
- ii. Wavelength of photon to *completely remove* an electron from *1s orbital of Boron atom (B)*
- iii. Wavelength of photon to *completely remove* an electron from *1s orbital of Boron cation (B^+)*

i or ii or iii

(II) Select the smallest (shortest) wavelength from the three choices below (*for H atom*):

- i. Wavelength of photon *emitted* for an electron transitioning from *2p to 1s orbital*
- ii. Wavelength of photon *absorbed* for an electron transitioning from *2s to 4p orbital*
- iii. Wavelength of photon *absorbed* for an electron transitioning from *1s to 3p orbital*

i or ii or iii



Question 2 8 points



A Chem110 student plots the graph shown on the left, from data obtained for the kinetic energy (eV) of the ejected electron versus frequency (s^{-1}) of the incident light, for an unknown metal. ($1\text{eV} = 1.6022 \times 10^{-19} \text{ J}$)

Based on the plot, answer the questions **a to c** below:

a. 1 point On the plot above, label the **threshold frequency**.

b. 2 points What is the value of the work potential (in J) of the unknown metal?

c. 3 points What is the *longest* wavelength (in nm) required to eject an electron from this metal's surface? Show your work including equation(s) used. (3 sig figs)

d. 2 points In the space provided below, plot the following

(Assume frequency is *above* threshold frequency):

- Kinetic energy* of an ejected electron versus the *intensity* of the incident light (left)
- Number of electrons* ejected versus the *frequency* of the incident light (right)



**Question 3** **6 points****a. 2 points**

In the ground state of Ge (Germanium; $Z = 32$) atom, **name** the following orbitals:

i. Orbital containing the electron that requires the maximum energy to be removed.

ii. Orbital containing the electron that experiences the minimum effective nuclear charge.

b. 2 points

An unknown anion (A^-) has the following *excited* electronic configuration: $[Ar] 4s^2 3d^3 4p^1$

(Ar: Argon; $Z = 18$)

Give the **condensed ground state electronic configuration** of the atom, **A** and the cation, **A^{2+}** .
Show your work. (Assume all electronic transitions are possible)

Electronic Configuration of A:

Electronic Configuration of A^{2+} :

c. 2 points

In a single atom, determine how many electrons can have the following set of quantum numbers?

(i) $n = 5; l = 3$

In a single atom, determine how many electrons can have the following set of quantum numbers?

(i) $n = 3; l = 2; m_s = +1/2$

**Question 4** **8 points**

Rank the following from 1 to 4 in **increasing** order. Provide the ranking in the boxes provided.

Note: Atomic number (Z) are provided for relevant atoms

a) Increasing (lowest to highest) order of *ionization energy*: (1= lowest; 4 = highest)

Mg

Mg: Z = 12

Rb

Rb: Z = 37

Cs

Cs: Z = 55

K

K : Z = 19

b) Increasing (lowest to highest) order of *bond angle*: (1= lowest; 4 = highest)

H₂OCCl₄XeF₄

HCN

c) Increasing (smallest to largest) order of *ionic size* (1= smallest; 4 = largest)

S²⁻

S: Z = 16

O²⁻

O: Z = 8

Be²⁺

Be: Z = 4

Na⁺

Na: Z = 11

d) Increasing (lowest to highest) order of *bond energy* (1= lowest; 4 = highest)

P-F

P-Cl

P-I

P-Br

**Question 5** **12 points**

a. 9 points For the molecules HClO_4 and TeF_6 , draw the most stable *Lewis structure* showing appropriate shape and *dashed/wedged bonds*, based on VSEPR. Determine the **molecular geometry** around the central atom. For HClO_4 , label the *most polar bond*.



b. 3 points (Note: Lone pairs are not shown in structures below)

(i) For the three *resonance* structures (1, 2 and 3) shown below, answer the following question:



(I) Circle the *most contributing* resonance structure.

Structure 1

Structure 2

Structure 3

(ii) For the three structures given (4, 5, and 6) below, answer the following questions:



(I) Circle all the structures that represent *resonance structures* of each other:

Structure 4

Structure 5

Structure 6

(II) Among the resonance structures, circle the *most contributing* resonance structure.

Structure 4

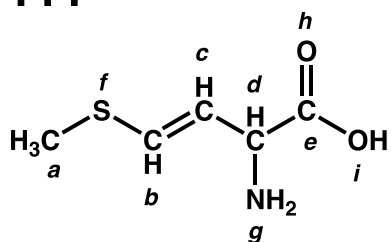
Structure 5

Structure 6

**Question 6** **6 points**

Using the data given below, calculate the **lattice energy** for SrO(s).

	ΔH° (kJ/mol)
$\text{Sr}(s) \longrightarrow \text{Sr}(g)$	+180
$\text{Sr}(g) \longrightarrow \text{Sr}^{2+}(g) + 2e^-$	+1468
$\text{O}_2(g) \longrightarrow 2\text{O}(g)$	+500
$\text{O}(g) + 2e^- \longrightarrow \text{O}^{2-}(g)$	+600
$\text{Sr}(s) + \frac{1}{2}\text{O}_2(g) \longrightarrow \text{SrO}(s)$	-550

**Question 7 10 points**

Consider the molecule given on the left, and answer questions (i) to (vi) using chemical bonding theories that we have learnt in class (valence bond theory, hybridization).

Note that the lone pairs are not shown in the structure:

i. Consider the carbon atoms marked in as ***a***, ***b***, ***c***, ***d***, and ***e*** in the molecule, and determine **hybridization** at each of the carbon centers. (2.5 points)

Atom	Hybridization
Carbon (<i>a</i>)	
Carbon (<i>b</i>)	
Carbon (<i>c</i>)	
Carbon (<i>d</i>)	
Carbon (<i>e</i>)	

ii. Now, consider the sulfur atom (S) marked as ***f***, nitrogen (N) atom (***g***), and oxygen(O) atoms (***h*** and ***i***), and determine their **hybridization**. (2 points)

Atom	Hybridization
Sulfur (<i>f</i>)	
Nitrogen (<i>g</i>)	
Oxygen (<i>h</i>)	
Oxygen (<i>i</i>)	

iii. **Describe** all the bonds formed by the carbon atoms ***a*** and ***b***, using sigma and pi designation as appropriate. (2 points)

Atom	Bonds formed (use sigma and pi as appropriate)
Carbon (<i>a</i>)	
Carbon (<i>b</i>)	

iv. How many total sp^3 -s sigma bonds are in the molecule? (1 point)

v. Determine **electronic** and **molecular geometry** at carbon (C) labeled ***e*** and sulfur (S) labeled ***f*** centres. (2 points)

Atom	Electronic Geometry	Molecular Geometry
Carbon (<i>e</i>)		
Sulfur (<i>f</i>)		

vi. Circle the value from the choices given below, that describes the **C-S-C bond angle at sulfur** (S) marked ***f***: (0.5 points)

>90° to <109.5° 109.5° >109.5 to <120° 120° >120° to <180°


Question 8 12 points

Using the **valence shell orbitals only** draw a **complete** molecular orbital (MO) diagram (with no s-p mixing) for the molecule **PS**, in the space provided below. **(6 points)**

Assume that 3s and 3p atomic orbitals will form molecular orbitals like those formed by the combination of 2s and 2p atomic orbitals that we learnt in class, and there is no s-p mixing.

Show **energy arrow**, **atomic orbitals** involved, **atom labels**, the **molecular orbitals** that are formed, **filled electrons** in the MO diagram etc. Make sure to label all the atomic and molecular orbitals.

Using the MO diagram above, answer the following questions:

- (i) Which one of the following will have the highest bond order? Circle the correct choice
 A) PS^{1-} B) PS C) PS^{1+}
- (ii) Which one of the following will have the longest bond length? Circle the correct choice
 A) PS^{1-} B) PS C) PS^{1+}
- (iii) Which of the following will be diamagnetic? Circle the correct choice.
 A) PS^{1-} B) PS C) PS^{1+}
- (iv) Draw Lewis dot structure of the molecule with the shortest bond length (based on the MO diagram) from among the three (PS^{1-} , PS, PS^{1+}), and indicate the bond order calculated from the Lewis dot structure. **(3 points)**

Lewis Dot Structure

Bond Order



ABD58CBF-9FB6-4A27-931A-6E94EC485327

midterm-exam-93388

#689 Page 10 of 10

SCRAP PAGE

**CHEMISTRY 110 –Midterm Exam****Thursday November 7, 2024, 6:30 – 8:30 pm****Examiners: Pallavi Sirjoosingh and Ashok Kakkar**

First name (please write as legibly as possible within the boxes)

--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--

Last name

--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--

Student ID number

--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--

This booklet has 8 questions on 10 pages (The last page can be used as a scrap page).

The staple must NOT be removed from the question booklet. Your intact question booklet must be returned at the end of the exam.

A separate booklet contains a periodic table, and datasheet.

Write your answers directly under the questions in the space provided. Do not write on the QR codes or near the top edges (as it may be cut off).

Please write **legibly and dark enough** for the scan to be readable. For students who use English as a second language, dictionaries are permitted. Calculators are allowed.

DO NOT ASK QUESTIONS OF THE INVIGILATORS about interpretations of the exam questions. If you have a concern about a question, please summarize it at the top of this page. All concerns will be evaluated after the exam. GOOD LUCK!

**Question 1 8 points** (*Assume all transitions are possible*)

a. 6 points Calculate the **energy (in kJ/mol)** required to ionize (completely remove) the electron from the **$n=3$** level of **He^+ ion** (He: Helium). Show your work including equations. (3 sig. figs.)

b. 2 points

(I) Select the largest (longest) wavelength from the three choices below:

- i. Wavelength of photon to *completely remove* an electron from *1s orbital of Hydrogen atom*
- ii. Wavelength of photon to *completely remove* an electron from *1s orbital of Boron atom (B)*
- iii. Wavelength of photon to *completely remove* an electron from *1s orbital of Boron cation (B^+)*

i or ii or iii

(II) Select the smallest (shortest) wavelength from the three choices below (*for H atom*):

- i. Wavelength of photon *emitted* for an electron transitioning from *2p to 1s orbital*
- ii. Wavelength of photon *absorbed* for an electron transitioning from *2s to 4p orbital*
- iii. Wavelength of photon *absorbed* for an electron transitioning from *1s to 3p orbital*

i or ii or iii



Question 2 8 points



A Chem110 student plots the graph shown on the left, from data obtained for the kinetic energy (eV) of the ejected electron versus frequency (s^{-1}) of the incident light, for an unknown metal. ($1\text{eV} = 1.6022 \times 10^{-19} \text{ J}$)

Based on the plot, answer the questions **a to c** below:

a. 1 point On the plot above, label the **threshold frequency**.

b. 2 points What is the value of the work potential (in J) of the unknown metal?

c. 3 points What is the *longest* wavelength (in nm) required to eject an electron from this metal's surface? Show your work including equation(s) used. (3 sig figs)

d. 2 points In the space provided below, plot the following

(Assume frequency is *above* threshold frequency):

- Kinetic energy* of an ejected electron versus the *intensity* of the incident light (left)
- Number of electrons* ejected versus the *frequency* of the incident light (right)



**Question 3** **6 points****a. 2 points**

In the ground state of Ge (Germanium; $Z = 32$) atom, **name** the following orbitals:

i. Orbital containing the electron that requires the maximum energy to be removed.

ii. Orbital containing the electron that experiences the minimum effective nuclear charge.

b. 2 points

An unknown anion (A^-) has the following *excited* electronic configuration: $[Ar] 4s^2 3d^3 4p^1$

(Ar: Argon; $Z = 18$)

Give the **condensed ground state electronic configuration** of the atom, **A** and the cation, **A^{2+}** .
Show your work. (Assume all electronic transitions are possible)

Electronic Configuration of A:

Electronic Configuration of A^{2+} :

c. 2 points

In a single atom, determine how many electrons can have the following set of quantum numbers?

(i) $n = 5; l = 3$

In a single atom, determine how many electrons can have the following set of quantum numbers?

(i) $n = 3; l = 2; m_s = +1/2$

**Question 4** **8 points**

Rank the following from 1 to 4 in **increasing** order. Provide the ranking in the boxes provided.

Note: Atomic number (Z) are provided for relevant atoms

a) Increasing (lowest to highest) order of *ionization energy*: (1= lowest; 4 = highest)

Mg

Mg: Z = 12

Rb

Rb: Z = 37

Cs

Cs: Z = 55

K

K : Z = 19

b) Increasing (lowest to highest) order of *bond angle*: (1= lowest; 4 = highest)

H₂OCCl₄XeF₄

HCN

c) Increasing (smallest to largest) order of *ionic size* (1= smallest; 4 = largest)

S²⁻

S: Z = 16

O²⁻

O: Z = 8

Be²⁺

Be: Z = 4

Na⁺

Na: Z = 11

d) Increasing (lowest to highest) order of *bond energy* (1= lowest; 4 = highest)

P-F

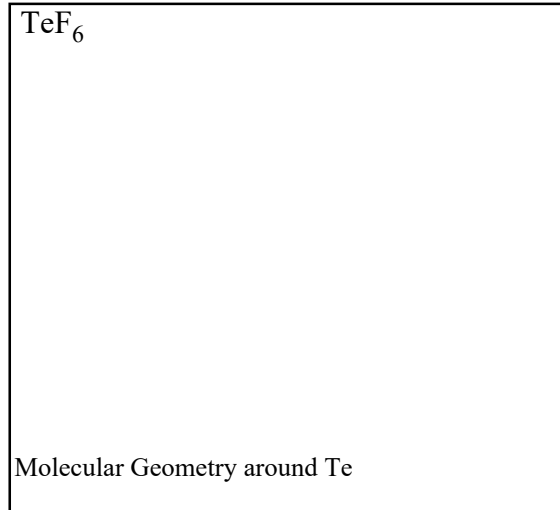
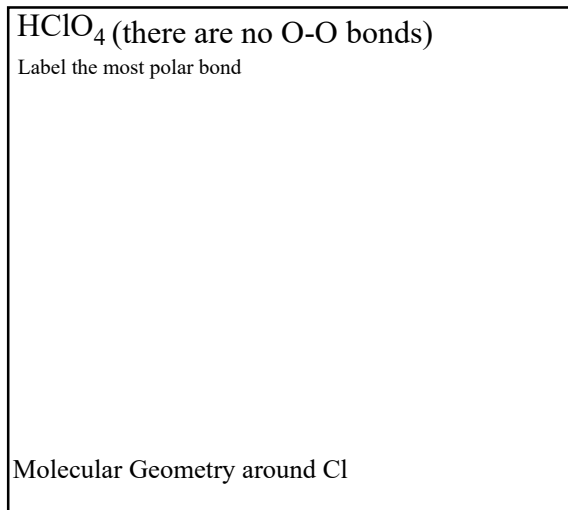
P-Cl

P-I

P-Br

**Question 5** **12 points**

a. 9 points For the molecules HClO_4 and TeF_6 , draw the most stable *Lewis structure* showing appropriate shape and *dashed/wedged bonds*, based on VSEPR. Determine the **molecular geometry** around the central atom. For HClO_4 , label the *most polar bond*.



b. 3 points (Note: Lone pairs are not shown in structures below)

(i) For the three *resonance* structures (1, 2 and 3) shown below, answer the following question:



(I) Circle the *most contributing* resonance structure.

Structure 1

Structure 2

Structure 3

(ii) For the three structures given (4, 5, and 6) below, answer the following questions:



(I) Circle all the structures that represent *resonance structures* of each other:

Structure 4

Structure 5

Structure 6

(II) Among the resonance structures, circle the *most contributing* resonance structure.

Structure 4

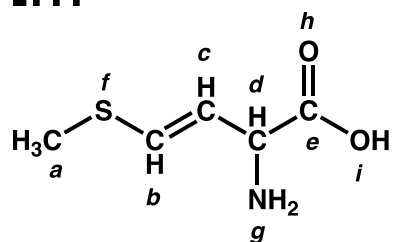
Structure 5

Structure 6

**Question 6** **6 points**

Using the data given below, calculate the **lattice energy** for SrO(s).

	ΔH° (kJ/mol)
$\text{Sr}(s) \longrightarrow \text{Sr}(g)$	+180
$\text{Sr}(g) \longrightarrow \text{Sr}^{2+}(g) + 2e^-$	+1468
$\text{O}_2(g) \longrightarrow 2\text{O}(g)$	+500
$\text{O}(g) + 2e^- \longrightarrow \text{O}^{2-}(g)$	+600
$\text{Sr}(s) + \frac{1}{2}\text{O}_2(g) \longrightarrow \text{SrO}(s)$	-550

**Question 7 10 points**

Consider the molecule given on the left, and answer questions (i) to (vi) using chemical bonding theories that we have learnt in class (valence bond theory, hybridization).

Note that the lone pairs are not shown in the structure:

i. Consider the carbon atoms marked in as ***a***, ***b***, ***c***, ***d***, and ***e*** in the molecule, and determine **hybridization** at each of the carbon centers. (2.5 points)

Atom	Hybridization
Carbon (<i>a</i>)	
Carbon (<i>b</i>)	
Carbon (<i>c</i>)	
Carbon (<i>d</i>)	
Carbon (<i>e</i>)	

ii. Now, consider the sulfur atom (S) marked as ***f***, nitrogen (N) atom (***g***), and oxygen(O) atoms (***h*** and ***i***), and determine their **hybridization**. (2 points)

Atom	Hybridization
Sulfur (<i>f</i>)	
Nitrogen (<i>g</i>)	
Oxygen (<i>h</i>)	
Oxygen (<i>i</i>)	

iii. **Describe** all the bonds formed by the carbon atoms ***a*** and ***b***, using sigma and pi designation as appropriate. (2 points)

Atom	Bonds formed (use sigma and pi as appropriate)
Carbon (<i>a</i>)	
Carbon (<i>b</i>)	

iv. How many total sp^3 -s sigma bonds are in the molecule? (1 point)

v. Determine **electronic** and **molecular geometry** at carbon (C) labeled ***e*** and sulfur (S) labeled ***f*** centres. (2 points)

Atom	Electronic Geometry	Molecular Geometry
Carbon (<i>e</i>)		
Sulfur (<i>f</i>)		

vi. Circle the value from the choices given below, that describes the **C-S-C bond angle at sulfur** (S) marked ***f***: (0.5 points)

>90° to <109.5° 109.5° >109.5 to <120° 120° >120° to <180°


Question 8 12 points

Using the **valence shell orbitals only** draw a **complete** molecular orbital (MO) diagram (with no s-p mixing) for the molecule **PS**, in the space provided below. **(6 points)**

Assume that 3s and 3p atomic orbitals will form molecular orbitals like those formed by the combination of 2s and 2p atomic orbitals that we learnt in class, and there is no s-p mixing.

Show **energy arrow**, **atomic orbitals** involved, **atom labels**, the **molecular orbitals** that are formed, **filled electrons** in the MO diagram etc. Make sure to label all the atomic and molecular orbitals.

Using the MO diagram above, answer the following questions:

- (i) Which one of the following will have the highest bond order? Circle the correct choice
 A) PS^{1-} B) PS C) PS^{1+}
- (ii) Which one of the following will have the longest bond length? Circle the correct choice
 A) PS^{1-} B) PS C) PS^{1+}
- (iii) Which of the following will be diamagnetic? Circle the correct choice.
 A) PS^{1-} B) PS C) PS^{1+}
- (iv) Draw Lewis dot structure of the molecule with the shortest bond length (based on the MO diagram) from among the three (PS^{1-} , PS, PS^{1+}), and indicate the bond order calculated from the Lewis dot structure. **(3 points)**

Lewis Dot Structure

Bond Order



94AA9364-1FB3-4999-BF1D-B7B18D9E2503

midterm-exam-93388

#690 Page 10 of 10

SCRAP PAGE

**CHEMISTRY 110 –Midterm Exam****Thursday November 7, 2024, 6:30 – 8:30 pm**

Examiners: Pallavi Sirjoosingh and Ashok Kakkar

First name (please write as legibly as possible within the boxes)

--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--

Last name

--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--

Student ID number

--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--

This booklet has 8 questions on 10 pages (The last page can be used as a scrap page).

The staple must NOT be removed from the question booklet. Your intact question booklet must be returned at the end of the exam.

A separate booklet contains a periodic table, and datasheet.

Write your answers directly under the questions in the space provided. Do not write on the QR codes or near the top edges (as it may be cut off).

Please write **legibly and dark enough** for the scan to be readable. For students who use English as a second language, dictionaries are permitted. Calculators are allowed.

DO NOT ASK QUESTIONS OF THE INVIGILATORS about interpretations of the exam questions. If you have a concern about a question, please summarize it at the top of this page. All concerns will be evaluated after the exam. GOOD LUCK!

**Question 1 8 points** (*Assume all transitions are possible*)

a. 6 points Calculate the **energy (in kJ/mol)** required to ionize (completely remove) the electron from the **n=3** level of **He⁺ ion** (He: Helium). Show your work including equations. (3 sig. figs.)

b. 2 points

(I) Select the largest (longest) wavelength from the three choices below:

- i. Wavelength of photon to *completely remove* an electron from *1s orbital of Hydrogen atom*
- ii. Wavelength of photon to *completely remove* an electron from *1s orbital of Boron atom (B)*
- iii. Wavelength of photon to *completely remove* an electron from *1s orbital of Boron cation (B⁺)*

i or ii or iii

(II) Select the smallest (shortest) wavelength from the three choices below (*for H atom*):

- i. Wavelength of photon *emitted* for an electron transitioning from *2p to 1s orbital*
- ii. Wavelength of photon *absorbed* for an electron transitioning from *2s to 4p orbital*
- iii. Wavelength of photon *absorbed* for an electron transitioning from *1s to 3p orbital*

i or ii or iii



Question 2 8 points



A Chem110 student plots the graph shown on the left, from data obtained for the kinetic energy (eV) of the ejected electron versus frequency (s^{-1}) of the incident light, for an unknown metal. ($1\text{eV} = 1.6022 \times 10^{-19} \text{ J}$)

Based on the plot, answer the questions **a to c** below:

a. 1 point On the plot above, label the **threshold frequency**.

b. 2 points What is the value of the work potential (in J) of the unknown metal?

c. 3 points What is the *longest* wavelength (in nm) required to eject an electron from this metal's surface? Show your work including equation(s) used. (3 sig figs)

d. 2 points In the space provided below, plot the following

(Assume frequency is *above* threshold frequency):

- Kinetic energy* of an ejected electron versus the *intensity* of the incident light (left)
- Number of electrons* ejected versus the *frequency* of the incident light (right)



**Question 3** **6 points****a. 2 points**

In the ground state of Ge (Germanium; $Z = 32$) atom, **name** the following orbitals:

i. Orbital containing the electron that requires the maximum energy to be removed.

ii. Orbital containing the electron that experiences the minimum effective nuclear charge.

b. 2 points

An unknown anion (A^-) has the following *excited* electronic configuration: $[Ar] 4s^2 3d^3 4p^1$

(Ar: Argon; $Z = 18$)

Give the **condensed ground state electronic configuration** of the atom, **A** and the cation, **A^{2+}** .
Show your work. (Assume all electronic transitions are possible)

Electronic Configuration of A:

Electronic Configuration of A^{2+} :

c. 2 points

In a single atom, determine how many electrons can have the following set of quantum numbers?

(i) $n = 5; l = 3$

In a single atom, determine how many electrons can have the following set of quantum numbers?

(i) $n = 3; l = 2; m_s = +1/2$

**Question 4** **8 points**

Rank the following from 1 to 4 in **increasing** order. Provide the ranking in the boxes provided.

Note: Atomic number (Z) are provided for relevant atoms

a) Increasing (lowest to highest) order of *ionization energy*: (1= lowest; 4 = highest)

Mg

Mg: Z = 12

Rb

Rb: Z = 37

Cs

Cs: Z = 55

K

K : Z = 19

b) Increasing (lowest to highest) order of *bond angle*: (1= lowest; 4 = highest)

H₂OCCl₄XeF₄

HCN

c) Increasing (smallest to largest) order of *ionic size* (1= smallest; 4 = largest)

S²⁻

S: Z = 16

O²⁻

O: Z = 8

Be²⁺

Be: Z = 4

Na⁺

Na: Z = 11

d) Increasing (lowest to highest) order of *bond energy* (1= lowest; 4 = highest)

P-F

P-Cl

P-I

P-Br

**Question 5** **12 points**

a. 9 points For the molecules HClO_4 and TeF_6 , draw the most stable *Lewis structure* showing appropriate shape and *dashed/wedged bonds*, based on VSEPR. Determine the **molecular geometry** around the central atom. For HClO_4 , label the *most polar bond*.



b. 3 points (Note: Lone pairs are not shown in structures below)

(i) For the three *resonance* structures (1, 2 and 3) shown below, answer the following question:



(I) Circle the *most contributing* resonance structure.

Structure 1

Structure 2

Structure 3

(ii) For the three structures given (4, 5, and 6) below, answer the following questions:



(I) Circle all the structures that represent *resonance structures* of each other:

Structure 4

Structure 5

Structure 6

(II) Among the resonance structures, circle the *most contributing* resonance structure.

Structure 4

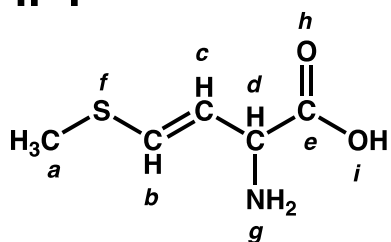
Structure 5

Structure 6

**Question 6** **6 points**

Using the data given below, calculate the **lattice energy** for SrO(s).

	ΔH° (kJ/mol)
$\text{Sr}(s) \longrightarrow \text{Sr}(g)$	+180
$\text{Sr}(g) \longrightarrow \text{Sr}^{2+}(g) + 2e^-$	+1468
$\text{O}_2(g) \longrightarrow 2\text{O}(g)$	+500
$\text{O}(g) + 2e^- \longrightarrow \text{O}^{2-}(g)$	+600
$\text{Sr}(s) + \frac{1}{2}\text{O}_2(g) \longrightarrow \text{SrO}(s)$	-550

**Question 7 10 points**

Consider the molecule given on the left, and answer questions (i) to (vi) using chemical bonding theories that we have learnt in class (valence bond theory, hybridization).

Note that the lone pairs are not shown in the structure:

i. Consider the carbon atoms marked in as ***a***, ***b***, ***c***, ***d***, and ***e*** in the molecule, and determine **hybridization** at each of the carbon centers. (2.5 points)

Atom	Hybridization
Carbon (<i>a</i>)	
Carbon (<i>b</i>)	
Carbon (<i>c</i>)	
Carbon (<i>d</i>)	
Carbon (<i>e</i>)	

ii. Now, consider the sulfur atom (S) marked as ***f***, nitrogen (N) atom (***g***), and oxygen(O) atoms (***h*** and ***i***), and determine their **hybridization**. (2 points)

Atom	Hybridization
Sulfur (<i>f</i>)	
Nitrogen (<i>g</i>)	
Oxygen (<i>h</i>)	
Oxygen (<i>i</i>)	

iii. **Describe** all the bonds formed by the carbon atoms ***a*** and ***b***, using sigma and pi designation as appropriate. (2 points)

Atom	Bonds formed (use sigma and pi as appropriate)
Carbon (<i>a</i>)	
Carbon (<i>b</i>)	

iv. How many total sp^3 -s sigma bonds are in the molecule? (1 point)

v. Determine **electronic** and **molecular geometry** at carbon (C) labeled ***e*** and sulfur (S) labeled ***f*** centres. (2 points)

Atom	Electronic Geometry	Molecular Geometry
Carbon (<i>e</i>)		
Sulfur (<i>f</i>)		

vi. Circle the value from the choices given below, that describes the **C-S-C bond angle at sulfur** (S) marked ***f***: (0.5 points)

>90° to <109.5° 109.5° >109.5 to <120° 120° >120° to <180°


Question 8 12 points

Using the **valence shell orbitals only** draw a **complete** molecular orbital (MO) diagram (with no s-p mixing) for the molecule **PS**, in the space provided below. **(6 points)**

Assume that 3s and 3p atomic orbitals will form molecular orbitals like those formed by the combination of 2s and 2p atomic orbitals that we learnt in class, and there is no s-p mixing.

Show **energy arrow**, **atomic orbitals** involved, **atom labels**, the **molecular orbitals** that are formed, **filled electrons** in the MO diagram etc. Make sure to label all the atomic and molecular orbitals.

Using the MO diagram above, answer the following questions:

- (i) Which one of the following will have the highest bond order? Circle the correct choice
 A) PS^{1-} B) PS C) PS^{1+}
- (ii) Which one of the following will have the longest bond length? Circle the correct choice
 A) PS^{1-} B) PS C) PS^{1+}
- (iii) Which of the following will be diamagnetic? Circle the correct choice.
 A) PS^{1-} B) PS C) PS^{1+}
- (iv) Draw Lewis dot structure of the molecule with the shortest bond length (based on the MO diagram) from among the three (PS^{1-} , PS, PS^{1+}), and indicate the bond order calculated from the Lewis dot structure. **(3 points)**

Lewis Dot Structure

Bond Order



2A16E016-AF9D-4AC8-8CAD-24FCE4701092

midterm-exam-93388

#691 Page 10 of 10

SCRAP PAGE

**CHEMISTRY 110 –Midterm Exam****Thursday November 7, 2024, 6:30 – 8:30 pm**

Examiners: Pallavi Sirjoosingh and Ashok Kakkar

First name (please write as legibly as possible within the boxes)

--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--

Last name

--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--

Student ID number

--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--

This booklet has 8 questions on 10 pages (The last page can be used as a scrap page).

The staple must NOT be removed from the question booklet. Your intact question booklet must be returned at the end of the exam.

A separate booklet contains a periodic table, and datasheet.

Write your answers directly under the questions in the space provided. Do not write on the QR codes or near the top edges (as it may be cut off).

Please write **legibly and dark enough** for the scan to be readable. For students who use English as a second language, dictionaries are permitted. Calculators are allowed.

DO NOT ASK QUESTIONS OF THE INVIGILATORS about interpretations of the exam questions. If you have a concern about a question, please summarize it at the top of this page. All concerns will be evaluated after the exam. GOOD LUCK!

**Question 1 8 points** (*Assume all transitions are possible*)

a. 6 points Calculate the **energy (in kJ/mol)** required to ionize (completely remove) the electron from the **$n=3$** level of **He^+ ion** (He: Helium). Show your work including equations. (3 sig. figs.)

b. 2 points

(I) Select the largest (longest) wavelength from the three choices below:

- i. Wavelength of photon to *completely remove* an electron from *1s orbital of Hydrogen atom*
- ii. Wavelength of photon to *completely remove* an electron from *1s orbital of Boron atom (B)*
- iii. Wavelength of photon to *completely remove* an electron from *1s orbital of Boron cation (B^+)*

i or ii or iii

(II) Select the smallest (shortest) wavelength from the three choices below (*for H atom*):

- i. Wavelength of photon *emitted* for an electron transitioning from *2p to 1s orbital*
- ii. Wavelength of photon *absorbed* for an electron transitioning from *2s to 4p orbital*
- iii. Wavelength of photon *absorbed* for an electron transitioning from *1s to 3p orbital*

i or ii or iii



Question 2 8 points



A Chem110 student plots the graph shown on the left, from data obtained for the kinetic energy (eV) of the ejected electron versus frequency (s^{-1}) of the incident light, for an unknown metal. ($1\text{eV} = 1.6022 \times 10^{-19} \text{ J}$)

Based on the plot, answer the questions **a to c** below:

a. 1 point On the plot above, label the **threshold frequency**.

b. 2 points What is the value of the work potential (in J) of the unknown metal?

c. 3 points What is the *longest* wavelength (in nm) required to eject an electron from this metal's surface? Show your work including equation(s) used. (3 sig figs)

d. 2 points In the space provided below, plot the following

(Assume frequency is *above* threshold frequency):

- Kinetic energy* of an ejected electron versus the *intensity* of the incident light (left)
- Number of electrons* ejected versus the *frequency* of the incident light (right)



**Question 3** **6 points****a. 2 points**

In the ground state of Ge (Germanium; $Z = 32$) atom, **name** the following orbitals:

i. Orbital containing the electron that requires the maximum energy to be removed.

ii. Orbital containing the electron that experiences the minimum effective nuclear charge.

b. 2 points

An unknown anion (A^-) has the following *excited* electronic configuration: $[Ar] 4s^2 3d^3 4p^1$

(Ar: Argon; $Z = 18$)

Give the **condensed ground state electronic configuration** of the atom, **A** and the cation, **A^{2+}** .
Show your work. (Assume all electronic transitions are possible)

Electronic Configuration of A:

Electronic Configuration of A^{2+} :

c. 2 points

In a single atom, determine how many electrons can have the following set of quantum numbers?

(i) $n = 5; l = 3$

In a single atom, determine how many electrons can have the following set of quantum numbers?

(i) $n = 3; l = 2; m_s = +1/2$

**Question 4** **8 points**

Rank the following from 1 to 4 in **increasing** order. Provide the ranking in the boxes provided.

Note: Atomic number (Z) are provided for relevant atoms

a) Increasing (lowest to highest) order of *ionization energy*: (1= lowest; 4 = highest)

Mg

Mg: Z = 12

Rb

Rb: Z = 37

Cs

Cs: Z = 55

K

K : Z = 19

b) Increasing (lowest to highest) order of *bond angle*: (1= lowest; 4 = highest)

H₂OCCl₄XeF₄

HCN

c) Increasing (smallest to largest) order of *ionic size* (1= smallest; 4 = largest)

S²⁻

S: Z = 16

O²⁻

O: Z = 8

Be²⁺

Be: Z = 4

Na⁺

Na: Z = 11

d) Increasing (lowest to highest) order of *bond energy* (1= lowest; 4 = highest)

P-F

P-Cl

P-I

P-Br

**Question 5** **12 points**

a. 9 points For the molecules HClO_4 and TeF_6 , draw the most stable *Lewis structure* showing appropriate shape and *dashed/wedged bonds*, based on VSEPR. Determine the **molecular geometry** around the central atom. For HClO_4 , label the *most polar bond*.



b. 3 points (Note: Lone pairs are not shown in structures below)

(i) For the three *resonance* structures (1, 2 and 3) shown below, answer the following question:



(I) Circle the *most contributing* resonance structure.

Structure 1

Structure 2

Structure 3

(ii) For the three structures given (4, 5, and 6) below, answer the following questions:



(I) Circle all the structures that represent *resonance structures* of each other:

Structure 4

Structure 5

Structure 6

(II) Among the resonance structures, circle the *most contributing* resonance structure.

Structure 4

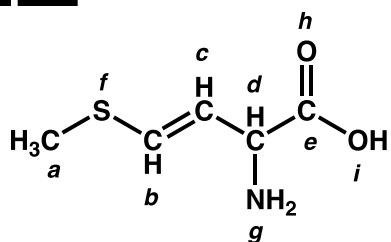
Structure 5

Structure 6

**Question 6** **6 points**

Using the data given below, calculate the **lattice energy** for SrO(s).

	ΔH° (kJ/mol)
$\text{Sr}(s) \longrightarrow \text{Sr}(g)$	+180
$\text{Sr}(g) \longrightarrow \text{Sr}^{2+}(g) + 2e^-$	+1468
$\text{O}_2(g) \longrightarrow 2\text{O}(g)$	+500
$\text{O}(g) + 2e^- \longrightarrow \text{O}^{2-}(g)$	+600
$\text{Sr}(s) + \frac{1}{2}\text{O}_2(g) \longrightarrow \text{SrO}(s)$	-550

**Question 7 10 points**

Consider the molecule given on the left, and answer questions (i) to (vi) using chemical bonding theories that we have learnt in class (valence bond theory, hybridization).

Note that the lone pairs are not shown in the structure:

i. Consider the carbon atoms marked in as ***a***, ***b***, ***c***, ***d***, and ***e*** in the molecule, and determine **hybridization** at each of the carbon centers. (2.5 points)

Atom	Hybridization
Carbon (<i>a</i>)	
Carbon (<i>b</i>)	
Carbon (<i>c</i>)	
Carbon (<i>d</i>)	
Carbon (<i>e</i>)	

ii. Now, consider the sulfur atom (S) marked as ***f***, nitrogen (N) atom (***g***), and oxygen(O) atoms (***h*** and ***i***), and determine their **hybridization**. (2 points)

Atom	Hybridization
Sulfur (<i>f</i>)	
Nitrogen (<i>g</i>)	
Oxygen (<i>h</i>)	
Oxygen (<i>i</i>)	

iii. **Describe** all the bonds formed by the carbon atoms ***a*** and ***b***, using sigma and pi designation as appropriate. (2 points)

Atom	Bonds formed (use sigma and pi as appropriate)
Carbon (<i>a</i>)	
Carbon (<i>b</i>)	

iv. How many total sp^3 -s sigma bonds are in the molecule? (1 point)

v. Determine **electronic** and **molecular geometry** at carbon (C) labeled ***e*** and sulfur (S) labeled ***f*** centres. (2 points)

Atom	Electronic Geometry	Molecular Geometry
Carbon (<i>e</i>)		
Sulfur (<i>f</i>)		

vi. Circle the value from the choices given below, that describes the **C-S-C bond angle at sulfur** (S) marked ***f***: (0.5 points)

>90° to <109.5° 109.5° >109.5 to <120° 120° >120° to <180°


Question 8 12 points

Using the **valence shell orbitals only** draw a **complete** molecular orbital (MO) diagram (with no s-p mixing) for the molecule **PS**, in the space provided below. **(6 points)**

Assume that 3s and 3p atomic orbitals will form molecular orbitals like those formed by the combination of 2s and 2p atomic orbitals that we learnt in class, and there is no s-p mixing.

Show **energy arrow**, **atomic orbitals** involved, **atom labels**, the **molecular orbitals** that are formed, **filled electrons** in the MO diagram etc. Make sure to label all the atomic and molecular orbitals.

Using the MO diagram above, answer the following questions:

- (i) Which one of the following will have the highest bond order? Circle the correct choice
 A) PS^{1-} B) PS C) PS^{1+}
- (ii) Which one of the following will have the longest bond length? Circle the correct choice
 A) PS^{1-} B) PS C) PS^{1+}
- (iii) Which of the following will be diamagnetic? Circle the correct choice.
 A) PS^{1-} B) PS C) PS^{1+}
- (iv) Draw Lewis dot structure of the molecule with the shortest bond length (based on the MO diagram) from among the three (PS^{1-} , PS, PS^{1+}), and indicate the bond order calculated from the Lewis dot structure. **(3 points)**

Lewis Dot Structure

Bond Order



B0F2DD0D-BA90-4E90-8C34-6C11390CD243

midterm-exam-93388

#692 Page 10 of 10

SCRAP PAGE

**CHEMISTRY 110 –Midterm Exam****Thursday November 7, 2024, 6:30 – 8:30 pm****Examiners: Pallavi Sirjoosingh and Ashok Kakkar**

First name (please write as legibly as possible within the boxes)

--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--

Last name

--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--

Student ID number

--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--

This booklet has 8 questions on 10 pages (The last page can be used as a scrap page).

The staple must NOT be removed from the question booklet. Your intact question booklet must be returned at the end of the exam.

A separate booklet contains a periodic table, and datasheet.

Write your answers directly under the questions in the space provided. Do not write on the QR codes or near the top edges (as it may be cut off).

Please write **legibly and dark enough** for the scan to be readable. For students who use English as a second language, dictionaries are permitted. Calculators are allowed.

DO NOT ASK QUESTIONS OF THE INVIGILATORS about interpretations of the exam questions. If you have a concern about a question, please summarize it at the top of this page. All concerns will be evaluated after the exam. GOOD LUCK!

**Question 1 8 points** (*Assume all transitions are possible*)

a. 6 points Calculate the **energy (in kJ/mol)** required to ionize (completely remove) the electron from the **$n=3$** level of **He^+ ion** (He: Helium). Show your work including equations. (3 sig. figs.)

b. 2 points

(I) Select the largest (longest) wavelength from the three choices below:

- i. Wavelength of photon to *completely remove* an electron from *1s orbital of Hydrogen atom*
- ii. Wavelength of photon to *completely remove* an electron from *1s orbital of Boron atom (B)*
- iii. Wavelength of photon to *completely remove* an electron from *1s orbital of Boron cation (B^+)*

i or ii or iii

(II) Select the smallest (shortest) wavelength from the three choices below (*for H atom*):

- i. Wavelength of photon *emitted* for an electron transitioning from *2p to 1s orbital*
- ii. Wavelength of photon *absorbed* for an electron transitioning from *2s to 4p orbital*
- iii. Wavelength of photon *absorbed* for an electron transitioning from *1s to 3p orbital*

i or ii or iii



Question 2 8 points



A Chem110 student plots the graph shown on the left, from data obtained for the kinetic energy (eV) of the ejected electron versus frequency (s^{-1}) of the incident light, for an unknown metal. ($1\text{eV} = 1.6022 \times 10^{-19} \text{ J}$)

Based on the plot, answer the questions **a to c** below:

a. 1 point On the plot above, label the **threshold frequency**.

b. 2 points What is the value of the work potential (in J) of the unknown metal?

c. 3 points What is the *longest* wavelength (in nm) required to eject an electron from this metal's surface? Show your work including equation(s) used. (3 sig figs)

d. 2 points In the space provided below, plot the following

(Assume frequency is *above* threshold frequency):

- i. *Kinetic energy* of an ejected electron versus the *intensity* of the incident light (left)
- ii. *Number of electrons* ejected versus the *frequency* of the incident light (right)



**Question 3** **6 points****a. 2 points**

In the ground state of Ge (Germanium; $Z = 32$) atom, **name** the following orbitals:

i. Orbital containing the electron that requires the maximum energy to be removed.

ii. Orbital containing the electron that experiences the minimum effective nuclear charge.

b. 2 points

An unknown anion (A^-) has the following *excited* electronic configuration: $[Ar] 4s^2 3d^3 4p^1$

(Ar: Argon; $Z = 18$)

Give the **condensed ground state electronic configuration** of the atom, **A** and the cation, A^{2+} .
Show your work. (Assume all electronic transitions are possible)

Electronic Configuration of A:

Electronic Configuration of A^{2+} :

c. 2 points

In a single atom, determine how many electrons can have the following set of quantum numbers?

(i) $n = 5; l = 3$

In a single atom, determine how many electrons can have the following set of quantum numbers?

(i) $n = 3; l = 2; m_s = +1/2$

**Question 4** **8 points**

Rank the following from 1 to 4 in **increasing** order. Provide the ranking in the boxes provided.

Note: Atomic number (Z) are provided for relevant atoms

a) Increasing (lowest to highest) order of *ionization energy*: (1= lowest; 4 = highest)

Mg

Mg: Z = 12

Rb

Rb: Z = 37

Cs

Cs: Z = 55

K

K : Z = 19

b) Increasing (lowest to highest) order of *bond angle*: (1= lowest; 4 = highest)

H₂OCCl₄XeF₄

HCN

c) Increasing (smallest to largest) order of *ionic size* (1= smallest; 4 = largest)

S²⁻

S: Z = 16

O²⁻

O: Z = 8

Be²⁺

Be: Z = 4

Na⁺

Na: Z = 11

d) Increasing (lowest to highest) order of *bond energy* (1= lowest; 4 = highest)

P-F

P-Cl

P-I

P-Br

**Question 5** **12 points**

a. 9 points For the molecules HClO_4 and TeF_6 , draw the most stable *Lewis structure* showing appropriate shape and *dashed/wedged bonds*, based on VSEPR. Determine the **molecular geometry** around the central atom. For HClO_4 , label the *most polar bond*.



b. 3 points (Note: Lone pairs are not shown in structures below)

(i) For the three *resonance* structures (1, 2 and 3) shown below, answer the following question:



(I) Circle the *most contributing* resonance structure.

Structure 1

Structure 2

Structure 3

(ii) For the three structures given (4, 5, and 6) below, answer the following questions:



(I) Circle all the structures that represent *resonance structures* of each other:

Structure 4

Structure 5

Structure 6

(II) Among the resonance structures, circle the *most contributing* resonance structure.

Structure 4

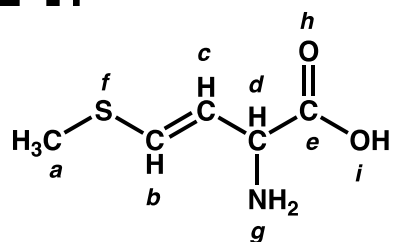
Structure 5

Structure 6

**Question 6** **6 points**

Using the data given below, calculate the **lattice energy** for SrO(s).

	ΔH° (kJ/mol)
$\text{Sr}(s) \longrightarrow \text{Sr}(g)$	+180
$\text{Sr}(g) \longrightarrow \text{Sr}^{2+}(g) + 2e^-$	+1468
$\text{O}_2(g) \longrightarrow 2\text{O}(g)$	+500
$\text{O}(g) + 2e^- \longrightarrow \text{O}^{2-}(g)$	+600
$\text{Sr}(s) + \frac{1}{2}\text{O}_2(g) \longrightarrow \text{SrO}(s)$	-550

**Question 7 10 points**

Consider the molecule given on the left, and answer questions (i) to (vi) using chemical bonding theories that we have learnt in class (valence bond theory, hybridization).

Note that the lone pairs are not shown in the structure:

i. Consider the carbon atoms marked in as ***a***, ***b***, ***c***, ***d***, and ***e*** in the molecule, and determine **hybridization** at each of the carbon centers. (2.5 points)

Atom	Hybridization
Carbon (<i>a</i>)	
Carbon (<i>b</i>)	
Carbon (<i>c</i>)	
Carbon (<i>d</i>)	
Carbon (<i>e</i>)	

ii. Now, consider the sulfur atom (S) marked as ***f***, nitrogen (N) atom (***g***), and oxygen(O) atoms (***h*** and ***i***), and determine their **hybridization**. (2 points)

Atom	Hybridization
Sulfur (<i>f</i>)	
Nitrogen (<i>g</i>)	
Oxygen (<i>h</i>)	
Oxygen (<i>i</i>)	

iii. **Describe** all the bonds formed by the carbon atoms ***a*** and ***b***, using sigma and pi designation as appropriate. (2 points)

Atom	Bonds formed (use sigma and pi as appropriate)
Carbon (<i>a</i>)	
Carbon (<i>b</i>)	

iv. How many total sp^3 -s sigma bonds are in the molecule? (1 point)

v. Determine **electronic** and **molecular geometry** at carbon (C) labeled ***e*** and sulfur (S) labeled ***f*** centres. (2 points)

Atom	Electronic Geometry	Molecular Geometry
Carbon (<i>e</i>)		
Sulfur (<i>f</i>)		

vi. Circle the value from the choices given below, that describes the **C-S-C bond angle at sulfur** (S) marked ***f***: (0.5 points)

>90° to <109.5° 109.5° >109.5 to <120° 120° >120° to <180°


Question 8 12 points

Using the **valence shell orbitals only** draw a **complete** molecular orbital (MO) diagram (with no s-p mixing) for the molecule **PS**, in the space provided below. **(6 points)**

Assume that 3s and 3p atomic orbitals will form molecular orbitals like those formed by the combination of 2s and 2p atomic orbitals that we learnt in class, and there is no s-p mixing.

Show **energy arrow**, **atomic orbitals** involved, **atom labels**, the **molecular orbitals** that are formed, **filled electrons** in the MO diagram etc. Make sure to label all the atomic and molecular orbitals.

Using the MO diagram above, answer the following questions:

- (i) Which one of the following will have the highest bond order? Circle the correct choice
 A) PS^{1-} B) PS C) PS^{1+}
- (ii) Which one of the following will have the longest bond length? Circle the correct choice
 A) PS^{1-} B) PS C) PS^{1+}
- (iii) Which of the following will be diamagnetic? Circle the correct choice.
 A) PS^{1-} B) PS C) PS^{1+}
- (iv) Draw Lewis dot structure of the molecule with the shortest bond length (based on the MO diagram) from among the three (PS^{1-} , PS, PS^{1+}), and indicate the bond order calculated from the Lewis dot structure. **(3 points)**

Lewis Dot Structure

Bond Order



CCD71FDB-1E99-4A52-BE7A-BCB73AD25CC6

midterm-exam-93388

#693 Page 10 of 10

SCRAP PAGE

**CHEMISTRY 110 –Midterm Exam****Thursday November 7, 2024, 6:30 – 8:30 pm**

Examiners: Pallavi Sirjoosingh and Ashok Kakkar

First name (please write as legibly as possible within the boxes)

--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--

Last name

--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--

Student ID number

--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--

This booklet has 8 questions on 10 pages (The last page can be used as a scrap page).

The staple must NOT be removed from the question booklet. Your intact question booklet must be returned at the end of the exam.

A separate booklet contains a periodic table, and datasheet.

Write your answers directly under the questions in the space provided. Do not write on the QR codes or near the top edges (as it may be cut off).

Please write **legibly and dark enough** for the scan to be readable. For students who use English as a second language, dictionaries are permitted. Calculators are allowed.

DO NOT ASK QUESTIONS OF THE INVIGILATORS about interpretations of the exam questions. If you have a concern about a question, please summarize it at the top of this page. All concerns will be evaluated after the exam. GOOD LUCK!

**Question 1 8 points** (*Assume all transitions are possible*)

a. 6 points Calculate the **energy (in kJ/mol)** required to ionize (completely remove) the electron from the **n=3** level of **He⁺ ion** (He: Helium). Show your work including equations. (3 sig. figs.)

b. 2 points

(I) Select the largest (longest) wavelength from the three choices below:

- i. Wavelength of photon to *completely remove* an electron from *1s orbital of Hydrogen atom*
- ii. Wavelength of photon to *completely remove* an electron from *1s orbital of Boron atom (B)*
- iii. Wavelength of photon to *completely remove* an electron from *1s orbital of Boron cation (B⁺)*

i or ii or iii

(II) Select the smallest (shortest) wavelength from the three choices below (*for H atom*):

- i. Wavelength of photon *emitted* for an electron transitioning from *2p to 1s orbital*
- ii. Wavelength of photon *absorbed* for an electron transitioning from *2s to 4p orbital*
- iii. Wavelength of photon *absorbed* for an electron transitioning from *1s to 3p orbital*

i or ii or iii



Question 2 8 points



A Chem110 student plots the graph shown on the left, from data obtained for the kinetic energy (eV) of the ejected electron versus frequency (s^{-1}) of the incident light, for an unknown metal. ($1\text{eV} = 1.6022 \times 10^{-19} \text{ J}$)

Based on the plot, answer the questions **a to c** below:

a. 1 point On the plot above, label the **threshold frequency**.

b. 2 points What is the value of the work potential (in J) of the unknown metal?

c. 3 points What is the *longest* wavelength (in nm) required to eject an electron from this metal's surface? Show your work including equation(s) used. (3 sig figs)

d. 2 points In the space provided below, plot the following

(Assume frequency is *above* threshold frequency):

- Kinetic energy* of an ejected electron versus the *intensity* of the incident light (left)
- Number of electrons* ejected versus the *frequency* of the incident light (right)



**Question 3** **6 points****a. 2 points**

In the ground state of Ge (Germanium; $Z = 32$) atom, **name** the following orbitals:

i. Orbital containing the electron that requires the maximum energy to be removed.

ii. Orbital containing the electron that experiences the minimum effective nuclear charge.

b. 2 points

An unknown anion (A^-) has the following *excited* electronic configuration: $[Ar] 4s^2 3d^3 4p^1$

(Ar: Argon; $Z = 18$)

Give the **condensed ground state electronic configuration** of the atom, **A** and the cation, **A^{2+}** .
Show your work. (Assume all electronic transitions are possible)

Electronic Configuration of A:

Electronic Configuration of A^{2+} :

c. 2 points

In a single atom, determine how many electrons can have the following set of quantum numbers?

(i) $n = 5; l = 3$

In a single atom, determine how many electrons can have the following set of quantum numbers?

(i) $n = 3; l = 2; m_s = +1/2$

**Question 4** **8 points**

Rank the following from 1 to 4 in **increasing** order. Provide the ranking in the boxes provided.

Note: Atomic number (Z) are provided for relevant atoms

a) Increasing (lowest to highest) order of *ionization energy*: (1= lowest; 4 = highest)

Mg

Mg: Z = 12

Rb

Rb: Z = 37

Cs

Cs: Z = 55

K

K : Z = 19

b) Increasing (lowest to highest) order of *bond angle*: (1= lowest; 4 = highest)

H₂OCCl₄XeF₄

HCN

c) Increasing (smallest to largest) order of *ionic size* (1= smallest; 4 = largest)

S²⁻

S: Z = 16

O²⁻

O: Z = 8

Be²⁺

Be: Z = 4

Na⁺

Na: Z = 11

d) Increasing (lowest to highest) order of *bond energy* (1= lowest; 4 = highest)

P-F

P-Cl

P-I

P-Br

**Question 5** **12 points**

a. 9 points For the molecules HClO_4 and TeF_6 , draw the most stable *Lewis structure* showing appropriate shape and *dashed/wedged bonds*, based on VSEPR. Determine the **molecular geometry** around the central atom. For HClO_4 , label the *most polar bond*.

HClO_4 (there are no O-O bonds)

Label the most polar bond

Molecular Geometry around Cl

TeF_6

Molecular Geometry around Te

b. 3 points (Note: Lone pairs are not shown in structures below)

(i) For the three *resonance* structures (1, 2 and 3) shown below, answer the following question:



(I) Circle the *most contributing* resonance structure.

Structure 1

Structure 2

Structure 3

(ii) For the three structures given (4, 5, and 6) below, answer the following questions:



(I) Circle all the structures that represent *resonance structures* of each other:

Structure 4

Structure 5

Structure 6

(II) Among the resonance structures, circle the *most contributing* resonance structure.

Structure 4

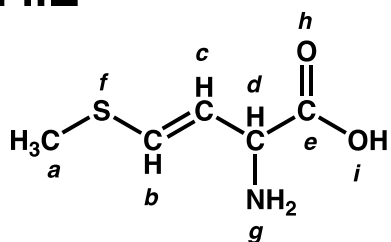
Structure 5

Structure 6

**Question 6** **6 points**

Using the data given below, calculate the **lattice energy** for SrO(s).

	ΔH° (kJ/mol)
$\text{Sr}(s) \longrightarrow \text{Sr}(g)$	+180
$\text{Sr}(g) \longrightarrow \text{Sr}^{2+}(g) + 2e^-$	+1468
$\text{O}_2(g) \longrightarrow 2\text{O}(g)$	+500
$\text{O}(g) + 2e^- \longrightarrow \text{O}^{2-}(g)$	+600
$\text{Sr}(s) + \frac{1}{2}\text{O}_2(g) \longrightarrow \text{SrO}(s)$	-550

**Question 7 10 points**

Consider the molecule given on the left, and answer questions (i) to (vi) using chemical bonding theories that we have learnt in class (valence bond theory, hybridization).

Note that the lone pairs are not shown in the structure:

i. Consider the carbon atoms marked in as ***a***, ***b***, ***c***, ***d***, and ***e*** in the molecule, and determine **hybridization** at each of the carbon centers. (2.5 points)

Atom	Hybridization
Carbon (<i>a</i>)	
Carbon (<i>b</i>)	
Carbon (<i>c</i>)	
Carbon (<i>d</i>)	
Carbon (<i>e</i>)	

ii. Now, consider the sulfur atom (S) marked as ***f***, nitrogen (N) atom (***g***), and oxygen(O) atoms (***h*** and ***i***), and determine their **hybridization**. (2 points)

Atom	Hybridization
Sulfur (<i>f</i>)	
Nitrogen (<i>g</i>)	
Oxygen (<i>h</i>)	
Oxygen (<i>i</i>)	

iii. **Describe** all the bonds formed by the carbon atoms ***a*** and ***b***, using sigma and pi designation as appropriate. (2 points)

Atom	Bonds formed (use sigma and pi as appropriate)
Carbon (<i>a</i>)	
Carbon (<i>b</i>)	

iv. How many total sp^3 -s sigma bonds are in the molecule? (1 point)

v. Determine **electronic** and **molecular geometry** at carbon (C) labeled ***e*** and sulfur (S) labeled ***f*** centres. (2 points)

Atom	Electronic Geometry	Molecular Geometry
Carbon (<i>e</i>)		
Sulfur (<i>f</i>)		

vi. Circle the value from the choices given below, that describes the **C-S-C bond angle at sulfur** (S) marked ***f***: (0.5 points)

>90° to <109.5°

109.5°

>109.5 to <120°

120°

>120° to <180°


Question 8 12 points

Using the **valence shell orbitals only** draw a **complete** molecular orbital (MO) diagram (with no s-p mixing) for the molecule **PS**, in the space provided below. **(6 points)**

Assume that 3s and 3p atomic orbitals will form molecular orbitals like those formed by the combination of 2s and 2p atomic orbitals that we learnt in class, and there is no s-p mixing.

Show **energy arrow**, **atomic orbitals** involved, **atom labels**, the **molecular orbitals** that are formed, **filled electrons** in the MO diagram etc. Make sure to label all the atomic and molecular orbitals.

Using the MO diagram above, answer the following questions:

- (i) Which one of the following will have the highest bond order? Circle the correct choice
 A) PS^{1-} B) PS C) PS^{1+}
- (ii) Which one of the following will have the longest bond length? Circle the correct choice
 A) PS^{1-} B) PS C) PS^{1+}
- (iii) Which of the following will be diamagnetic? Circle the correct choice.
 A) PS^{1-} B) PS C) PS^{1+}
- (iv) Draw Lewis dot structure of the molecule with the shortest bond length (based on the MO diagram) from among the three (PS^{1-} , PS, PS^{1+}), and indicate the bond order calculated from the Lewis dot structure. **(3 points)**

Lewis Dot Structure

Bond Order



E33B9A2E-2AB2-4945-B852-BA9E45A0C130

midterm-exam-93388

#694 Page 10 of 10

SCRAP PAGE

**CHEMISTRY 110 –Midterm Exam****Thursday November 7, 2024, 6:30 – 8:30 pm****Examiners: Pallavi Sirjoosingh and Ashok Kakkar**

First name (please write as legibly as possible within the boxes)

--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--

Last name

--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--

Student ID number

--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--

This booklet has 8 questions on 10 pages (The last page can be used as a scrap page).

The staple must NOT be removed from the question booklet. Your intact question booklet must be returned at the end of the exam.

A separate booklet contains a periodic table, and datasheet.

Write your answers directly under the questions in the space provided. Do not write on the QR codes or near the top edges (as it may be cut off).

Please write **legibly and dark enough** for the scan to be readable. For students who use English as a second language, dictionaries are permitted. Calculators are allowed.

DO NOT ASK QUESTIONS OF THE INVIGILATORS about interpretations of the exam questions. If you have a concern about a question, please summarize it at the top of this page. All concerns will be evaluated after the exam. GOOD LUCK!

**Question 1 8 points** (*Assume all transitions are possible*)

a. 6 points Calculate the **energy (in kJ/mol)** required to ionize (completely remove) the electron from the **n=3** level of **He⁺ ion** (He: Helium). Show your work including equations. (3 sig. figs.)

b. 2 points

(I) Select the largest (longest) wavelength from the three choices below:

- i. Wavelength of photon to *completely remove* an electron from *1s orbital of Hydrogen atom*
- ii. Wavelength of photon to *completely remove* an electron from *1s orbital of Boron atom (B)*
- iii. Wavelength of photon to *completely remove* an electron from *1s orbital of Boron cation (B⁺)*

i or ii or iii

(II) Select the smallest (shortest) wavelength from the three choices below (*for H atom*):

- i. Wavelength of photon *emitted* for an electron transitioning from *2p to 1s orbital*
- ii. Wavelength of photon *absorbed* for an electron transitioning from *2s to 4p orbital*
- iii. Wavelength of photon *absorbed* for an electron transitioning from *1s to 3p orbital*

i or ii or iii



Question 2 8 points



A Chem110 student plots the graph shown on the left, from data obtained for the kinetic energy (eV) of the ejected electron versus frequency (s^{-1}) of the incident light, for an unknown metal. ($1\text{eV} = 1.6022 \times 10^{-19} \text{ J}$)

Based on the plot, answer the questions **a to c** below:

a. 1 point On the plot above, label the **threshold frequency**.

b. 2 points What is the value of the work potential (in J) of the unknown metal?

c. 3 points What is the *longest* wavelength (in nm) required to eject an electron from this metal's surface? Show your work including equation(s) used. (3 sig figs)

d. 2 points In the space provided below, plot the following

(Assume frequency is *above* threshold frequency):

- i. *Kinetic energy* of an ejected electron versus the *intensity* of the incident light (left)
- ii. *Number of electrons* ejected versus the *frequency* of the incident light (right)



**Question 3** **6 points****a. 2 points**

In the ground state of Ge (Germanium; $Z = 32$) atom, **name** the following orbitals:

i. Orbital containing the electron that requires the maximum energy to be removed.

ii. Orbital containing the electron that experiences the minimum effective nuclear charge.

b. 2 points

An unknown anion (A^-) has the following *excited* electronic configuration: $[Ar] 4s^2 3d^3 4p^1$

(Ar: Argon; $Z = 18$)

Give the **condensed ground state electronic configuration** of the atom, **A** and the cation, **A^{2+}** .
Show your work. (Assume all electronic transitions are possible)

Electronic Configuration of A:

Electronic Configuration of A^{2+} :

c. 2 points

In a single atom, determine how many electrons can have the following set of quantum numbers?

(i) $n = 5; l = 3$

In a single atom, determine how many electrons can have the following set of quantum numbers?

(i) $n = 3; l = 2; m_s = +1/2$

**Question 4** **8 points**

Rank the following from 1 to 4 in **increasing** order. Provide the ranking in the boxes provided.

Note: Atomic number (Z) are provided for relevant atoms

a) Increasing (lowest to highest) order of *ionization energy*: (1= lowest; 4 = highest)

Mg

Mg: Z = 12

Rb

Rb: Z = 37

Cs

Cs: Z = 55

K

K : Z = 19

b) Increasing (lowest to highest) order of *bond angle*: (1= lowest; 4 = highest)

H₂OCCl₄XeF₄

HCN

c) Increasing (smallest to largest) order of *ionic size* (1= smallest; 4 = largest)

S²⁻

S: Z = 16

O²⁻

O: Z = 8

Be²⁺

Be: Z = 4

Na⁺

Na: Z = 11

d) Increasing (lowest to highest) order of *bond energy* (1= lowest; 4 = highest)

P-F

P-Cl

P-I

P-Br

**Question 5** **12 points**

a. 9 points For the molecules HClO_4 and TeF_6 , draw the most stable *Lewis structure* showing appropriate shape and *dashed/wedged bonds*, based on VSEPR. Determine the **molecular geometry** around the central atom. For HClO_4 , label the *most polar bond*.



b. 3 points (Note: Lone pairs are not shown in structures below)

(i) For the three *resonance* structures (1, 2 and 3) shown below, answer the following question:



(I) Circle the *most contributing* resonance structure.

Structure 1

Structure 2

Structure 3

(ii) For the three structures given (4, 5, and 6) below, answer the following questions:



(I) Circle all the structures that represent *resonance structures* of each other:

Structure 4

Structure 5

Structure 6

(II) Among the resonance structures, circle the *most contributing* resonance structure.

Structure 4

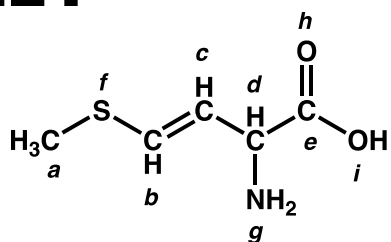
Structure 5

Structure 6

**Question 6** **6 points**

Using the data given below, calculate the **lattice energy** for SrO(s).

	ΔH° (kJ/mol)
$\text{Sr}(s) \longrightarrow \text{Sr}(g)$	+180
$\text{Sr}(g) \longrightarrow \text{Sr}^{2+}(g) + 2e^-$	+1468
$\text{O}_2(g) \longrightarrow 2\text{O}(g)$	+500
$\text{O}(g) + 2e^- \longrightarrow \text{O}^{2-}(g)$	+600
$\text{Sr}(s) + \frac{1}{2}\text{O}_2(g) \longrightarrow \text{SrO}(s)$	-550

**Question 7 10 points**

Consider the molecule given on the left, and answer questions (i) to (vi) using chemical bonding theories that we have learnt in class (valence bond theory, hybridization).

Note that the lone pairs are not shown in the structure:

i. Consider the carbon atoms marked in as ***a***, ***b***, ***c***, ***d***, and ***e*** in the molecule, and determine **hybridization** at each of the carbon centers. (2.5 points)

Atom	Hybridization
Carbon (<i>a</i>)	
Carbon (<i>b</i>)	
Carbon (<i>c</i>)	
Carbon (<i>d</i>)	
Carbon (<i>e</i>)	

ii. Now, consider the sulfur atom (S) marked as ***f***, nitrogen (N) atom (***g***), and oxygen(O) atoms (***h*** and ***i***), and determine their **hybridization**. (2 points)

Atom	Hybridization
Sulfur (<i>f</i>)	
Nitrogen (<i>g</i>)	
Oxygen (<i>h</i>)	
Oxygen (<i>i</i>)	

iii. **Describe** all the bonds formed by the carbon atoms ***a*** and ***b***, using sigma and pi designation as appropriate. (2 points)

Atom	Bonds formed (use sigma and pi as appropriate)
Carbon (<i>a</i>)	
Carbon (<i>b</i>)	

iv. How many total sp^3 -s sigma bonds are in the molecule? (1 point)

v. Determine **electronic** and **molecular geometry** at carbon (C) labeled ***e*** and sulfur (S) labeled ***f*** centres. (2 points)

Atom	Electronic Geometry	Molecular Geometry
Carbon (<i>e</i>)		
Sulfur (<i>f</i>)		

vi. Circle the value from the choices given below, that describes the **C-S-C bond angle at sulfur** (S) marked ***f***: (0.5 points)

>90° to <109.5° 109.5° >109.5 to <120° 120° >120° to <180°


Question 8 12 points

Using the **valence shell orbitals only** draw a **complete** molecular orbital (MO) diagram (with no s-p mixing) for the molecule **PS**, in the space provided below. **(6 points)**

Assume that 3s and 3p atomic orbitals will form molecular orbitals like those formed by the combination of 2s and 2p atomic orbitals that we learnt in class, and there is no s-p mixing.

Show **energy arrow**, **atomic orbitals** involved, **atom labels**, the **molecular orbitals** that are formed, **filled electrons** in the MO diagram etc. Make sure to label all the atomic and molecular orbitals.

Using the MO diagram above, answer the following questions:

- (i) Which one of the following will have the highest bond order? Circle the correct choice
 A) PS^{1-} B) PS C) PS^{1+}
- (ii) Which one of the following will have the longest bond length? Circle the correct choice
 A) PS^{1-} B) PS C) PS^{1+}
- (iii) Which of the following will be diamagnetic? Circle the correct choice.
 A) PS^{1-} B) PS C) PS^{1+}
- (iv) Draw Lewis dot structure of the molecule with the shortest bond length (based on the MO diagram) from among the three (PS^{1-} , PS, PS^{1+}), and indicate the bond order calculated from the Lewis dot structure. **(3 points)**

Lewis Dot Structure

Bond Order



2D40FEEE-C71C-4E63-ACB2-05148704AC5B

midterm-exam-93388

#695 Page 10 of 10

SCRAP PAGE

**CHEMISTRY 110 –Midterm Exam****Thursday November 7, 2024, 6:30 – 8:30 pm****Examiners: Pallavi Sirjoosingh and Ashok Kakkar**

First name (please write as legibly as possible within the boxes)

--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--

Last name

--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--

Student ID number

--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--

This booklet has 8 questions on 10 pages (The last page can be used as a scrap page).

The staple must NOT be removed from the question booklet. Your intact question booklet must be returned at the end of the exam.

A separate booklet contains a periodic table, and datasheet.

Write your answers directly under the questions in the space provided. Do not write on the QR codes or near the top edges (as it may be cut off).

Please write **legibly and dark enough** for the scan to be readable. For students who use English as a second language, dictionaries are permitted. Calculators are allowed.

DO NOT ASK QUESTIONS OF THE INVIGILATORS about interpretations of the exam questions. If you have a concern about a question, please summarize it at the top of this page. All concerns will be evaluated after the exam. GOOD LUCK!

**Question 1 8 points** (*Assume all transitions are possible*)

a. 6 points Calculate the **energy (in kJ/mol)** required to ionize (completely remove) the electron from the **n=3** level of **He⁺ ion** (He: Helium). Show your work including equations. (3 sig. figs.)

b. 2 points

(I) Select the largest (longest) wavelength from the three choices below:

- i. Wavelength of photon to *completely remove* an electron from *1s orbital of Hydrogen atom*
- ii. Wavelength of photon to *completely remove* an electron from *1s orbital of Boron atom (B)*
- iii. Wavelength of photon to *completely remove* an electron from *1s orbital of Boron cation (B⁺)*

i or ii or iii

(II) Select the smallest (shortest) wavelength from the three choices below (*for H atom*):

- i. Wavelength of photon *emitted* for an electron transitioning from *2p to 1s orbital*
- ii. Wavelength of photon *absorbed* for an electron transitioning from *2s to 4p orbital*
- iii. Wavelength of photon *absorbed* for an electron transitioning from *1s to 3p orbital*

i or ii or iii



Question 2 8 points



A Chem110 student plots the graph shown on the left, from data obtained for the kinetic energy (eV) of the ejected electron versus frequency (s^{-1}) of the incident light, for an unknown metal. ($1\text{eV} = 1.6022 \times 10^{-19} \text{ J}$)

Based on the plot, answer the questions **a to c** below:

a. 1 point On the plot above, label the **threshold frequency**.

b. 2 points What is the value of the work potential (in J) of the unknown metal?

c. 3 points What is the *longest* wavelength (in nm) required to eject an electron from this metal's surface? Show your work including equation(s) used. (3 sig figs)

d. 2 points In the space provided below, plot the following

(Assume frequency is *above* threshold frequency):

- Kinetic energy* of an ejected electron versus the *intensity* of the incident light (left)
- Number of electrons* ejected versus the *frequency* of the incident light (right)



**Question 3** **6 points****a. 2 points**

In the ground state of Ge (Germanium; $Z = 32$) atom, **name** the following orbitals:

i. Orbital containing the electron that requires the maximum energy to be removed.

ii. Orbital containing the electron that experiences the minimum effective nuclear charge.

b. 2 points

An unknown anion (A^-) has the following *excited* electronic configuration: $[Ar] 4s^2 3d^3 4p^1$

(Ar: Argon; $Z = 18$)

Give the **condensed ground state electronic configuration** of the atom, **A** and the cation, **A^{2+}** .
Show your work. (Assume all electronic transitions are possible)

Electronic Configuration of A:

Electronic Configuration of A^{2+} :

c. 2 points

In a single atom, determine how many electrons can have the following set of quantum numbers?

(i) $n = 5; l = 3$

In a single atom, determine how many electrons can have the following set of quantum numbers?

(i) $n = 3; l = 2; m_s = +1/2$

**Question 4** **8 points**

Rank the following from 1 to 4 in **increasing** order. Provide the ranking in the boxes provided.

Note: Atomic number (Z) are provided for relevant atoms

a) Increasing (lowest to highest) order of *ionization energy*: (1= lowest; 4 = highest)

Mg

Mg: Z = 12

Rb

Rb: Z = 37

Cs

Cs: Z = 55

K

K : Z = 19

b) Increasing (lowest to highest) order of *bond angle*: (1= lowest; 4 = highest)

H₂OCCl₄XeF₄

HCN

c) Increasing (smallest to largest) order of *ionic size* (1= smallest; 4 = largest)

S²⁻

S: Z = 16

O²⁻

O: Z = 8

Be²⁺

Be: Z = 4

Na⁺

Na: Z = 11

d) Increasing (lowest to highest) order of *bond energy* (1= lowest; 4 = highest)

P-F

P-Cl

P-I

P-Br

**Question 5** **12 points**

a. 9 points For the molecules HClO_4 and TeF_6 , draw the most stable *Lewis structure* showing appropriate shape and *dashed/wedged bonds*, based on VSEPR. Determine the **molecular geometry** around the central atom. For HClO_4 , label the *most polar bond*.



b. 3 points (Note: Lone pairs are not shown in structures below)

(i) For the three *resonance* structures (1, 2 and 3) shown below, answer the following question:



(I) Circle the *most contributing* resonance structure.

Structure 1

Structure 2

Structure 3

(ii) For the three structures given (4, 5, and 6) below, answer the following questions:



(I) Circle all the structures that represent *resonance structures* of each other:

Structure 4

Structure 5

Structure 6

(II) Among the resonance structures, circle the *most contributing* resonance structure.

Structure 4

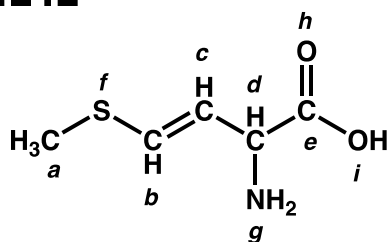
Structure 5

Structure 6

**Question 6** **6 points**

Using the data given below, calculate the **lattice energy** for SrO(s).

	ΔH° (kJ/mol)
$\text{Sr}(s) \longrightarrow \text{Sr}(g)$	+180
$\text{Sr}(g) \longrightarrow \text{Sr}^{2+}(g) + 2e^-$	+1468
$\text{O}_2(g) \longrightarrow 2\text{O}(g)$	+500
$\text{O}(g) + 2e^- \longrightarrow \text{O}^{2-}(g)$	+600
$\text{Sr}(s) + \frac{1}{2}\text{O}_2(g) \longrightarrow \text{SrO}(s)$	-550

**Question 7 10 points**

Consider the molecule given on the left, and answer questions (i) to (vi) using chemical bonding theories that we have learnt in class (valence bond theory, hybridization).

Note that the lone pairs are not shown in the structure:

i. Consider the carbon atoms marked in as ***a***, ***b***, ***c***, ***d***, and ***e*** in the molecule, and determine **hybridization** at each of the carbon centers. (2.5 points)

Atom	Hybridization
Carbon (<i>a</i>)	
Carbon (<i>b</i>)	
Carbon (<i>c</i>)	
Carbon (<i>d</i>)	
Carbon (<i>e</i>)	

ii. Now, consider the sulfur atom (S) marked as ***f***, nitrogen (N) atom (***g***), and oxygen(O) atoms (***h*** and ***i***), and determine their **hybridization**. (2 points)

Atom	Hybridization
Sulfur (<i>f</i>)	
Nitrogen (<i>g</i>)	
Oxygen (<i>h</i>)	
Oxygen (<i>i</i>)	

iii. **Describe** all the bonds formed by the carbon atoms ***a*** and ***b***, using sigma and pi designation as appropriate. (2 points)

Atom	Bonds formed (use sigma and pi as appropriate)
Carbon (<i>a</i>)	
Carbon (<i>b</i>)	

iv. How many total sp^3 -s sigma bonds are in the molecule? (1 point)

v. Determine **electronic** and **molecular geometry** at carbon (C) labeled ***e*** and sulfur (S) labeled ***f*** centres. (2 points)

Atom	Electronic Geometry	Molecular Geometry
Carbon (<i>e</i>)		
Sulfur (<i>f</i>)		

vi. Circle the value from the choices given below, that describes the **C-S-C bond angle at sulfur** (S) marked ***f***: (0.5 points)

>90° to <109.5° 109.5° >109.5 to <120° 120° >120° to <180°


Question 8 12 points

Using the **valence shell orbitals only** draw a **complete** molecular orbital (MO) diagram (with no s-p mixing) for the molecule **PS**, in the space provided below. **(6 points)**

Assume that 3s and 3p atomic orbitals will form molecular orbitals like those formed by the combination of 2s and 2p atomic orbitals that we learnt in class, and there is no s-p mixing.

Show **energy arrow**, **atomic orbitals** involved, **atom labels**, the **molecular orbitals** that are formed, **filled electrons** in the MO diagram etc. Make sure to label all the atomic and molecular orbitals.

Using the MO diagram above, answer the following questions:

- (i) Which one of the following will have the highest bond order? Circle the correct choice
 A) PS^{1-} B) PS C) PS^{1+}
- (ii) Which one of the following will have the longest bond length? Circle the correct choice
 A) PS^{1-} B) PS C) PS^{1+}
- (iii) Which of the following will be diamagnetic? Circle the correct choice.
 A) PS^{1-} B) PS C) PS^{1+}
- (iv) Draw Lewis dot structure of the molecule with the shortest bond length (based on the MO diagram) from among the three (PS^{1-} , PS, PS^{1+}), and indicate the bond order calculated from the Lewis dot structure. **(3 points)**

Lewis Dot Structure

Bond Order



C038C70B-05E2-486C-A67D-6020F39950BB

midterm-exam-93388

#696 Page 10 of 10

SCRAP PAGE

**CHEMISTRY 110 –Midterm Exam****Thursday November 7, 2024, 6:30 – 8:30 pm****Examiners: Pallavi Sirjoosingh and Ashok Kakkar**

First name (please write as legibly as possible within the boxes)

--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--

Last name

--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--

Student ID number

--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--

This booklet has 8 questions on 10 pages (The last page can be used as a scrap page).

The staple must NOT be removed from the question booklet. Your intact question booklet must be returned at the end of the exam.

A separate booklet contains a periodic table, and datasheet.

Write your answers directly under the questions in the space provided. Do not write on the QR codes or near the top edges (as it may be cut off).

Please write **legibly and dark enough** for the scan to be readable. For students who use English as a second language, dictionaries are permitted. Calculators are allowed.

DO NOT ASK QUESTIONS OF THE INVIGILATORS about interpretations of the exam questions. If you have a concern about a question, please summarize it at the top of this page. All concerns will be evaluated after the exam. GOOD LUCK!

**Question 1 8 points** (*Assume all transitions are possible*)

a. 6 points Calculate the **energy (in kJ/mol)** required to ionize (completely remove) the electron from the **n=3** level of **He⁺ ion** (He: Helium). Show your work including equations. (3 sig. figs.)

b. 2 points

(I) Select the largest (longest) wavelength from the three choices below:

- i. Wavelength of photon to *completely remove* an electron from *1s orbital of Hydrogen atom*
- ii. Wavelength of photon to *completely remove* an electron from *1s orbital of Boron atom (B)*
- iii. Wavelength of photon to *completely remove* an electron from *1s orbital of Boron cation (B⁺)*

i or ii or iii

(II) Select the smallest (shortest) wavelength from the three choices below (*for H atom*):

- i. Wavelength of photon *emitted* for an electron transitioning from *2p to 1s orbital*
- ii. Wavelength of photon *absorbed* for an electron transitioning from *2s to 4p orbital*
- iii. Wavelength of photon *absorbed* for an electron transitioning from *1s to 3p orbital*

i or ii or iii



Question 2 8 points



A Chem110 student plots the graph shown on the left, from data obtained for the kinetic energy (eV) of the ejected electron versus frequency (s^{-1}) of the incident light, for an unknown metal. ($1\text{eV} = 1.6022 \times 10^{-19} \text{ J}$)

Based on the plot, answer the questions **a to c** below:

a. 1 point On the plot above, label the **threshold frequency**.

b. 2 points What is the value of the work potential (in J) of the unknown metal?

c. 3 points What is the *longest* wavelength (in nm) required to eject an electron from this metal's surface? Show your work including equation(s) used. (3 sig figs)

d. 2 points In the space provided below, plot the following

(Assume frequency is *above* threshold frequency):

- i. *Kinetic energy* of an ejected electron versus the *intensity* of the incident light (left)
- ii. *Number of electrons* ejected versus the *frequency* of the incident light (right)



**Question 3** **6 points****a. 2 points**

In the ground state of Ge (Germanium; $Z = 32$) atom, **name** the following orbitals:

i. Orbital containing the electron that requires the maximum energy to be removed.

ii. Orbital containing the electron that experiences the minimum effective nuclear charge.

b. 2 points

An unknown anion (A^-) has the following *excited* electronic configuration: $[Ar] 4s^2 3d^3 4p^1$

(Ar: Argon; $Z = 18$)

Give the **condensed ground state electronic configuration** of the atom, **A** and the cation, **A^{2+}** .
Show your work. (Assume all electronic transitions are possible)

Electronic Configuration of A:

Electronic Configuration of A^{2+} :

c. 2 points

In a single atom, determine how many electrons can have the following set of quantum numbers?

(i) $n = 5; l = 3$

In a single atom, determine how many electrons can have the following set of quantum numbers?

(i) $n = 3; l = 2; m_s = +1/2$

**Question 4** **8 points**

Rank the following from 1 to 4 in **increasing** order. Provide the ranking in the boxes provided.

Note: Atomic number (Z) are provided for relevant atoms

a) Increasing (lowest to highest) order of *ionization energy*: (1= lowest; 4 = highest)

Mg

Mg: Z = 12

Rb

Rb: Z = 37

Cs

Cs: Z = 55

K

K : Z = 19

b) Increasing (lowest to highest) order of *bond angle*: (1= lowest; 4 = highest)

H₂OCCl₄XeF₄

HCN

c) Increasing (smallest to largest) order of *ionic size* (1= smallest; 4 = largest)

S²⁻

S: Z = 16

O²⁻

O: Z = 8

Be²⁺

Be: Z = 4

Na⁺

Na: Z = 11

d) Increasing (lowest to highest) order of *bond energy* (1= lowest; 4 = highest)

P-F

P-Cl

P-I

P-Br

**Question 5** **12 points**

a. 9 points For the molecules HClO_4 and TeF_6 , draw the most stable *Lewis structure* showing appropriate shape and *dashed/wedged bonds*, based on VSEPR. Determine the **molecular geometry** around the central atom. For HClO_4 , label the *most polar bond*.



b. 3 points (Note: Lone pairs are not shown in structures below)

(i) For the three *resonance* structures (1, 2 and 3) shown below, answer the following question:



(I) Circle the *most contributing* resonance structure.

Structure 1

Structure 2

Structure 3

(ii) For the three structures given (4, 5, and 6) below, answer the following questions:



(I) Circle all the structures that represent *resonance structures* of each other:

Structure 4

Structure 5

Structure 6

(II) Among the resonance structures, circle the *most contributing* resonance structure.

Structure 4

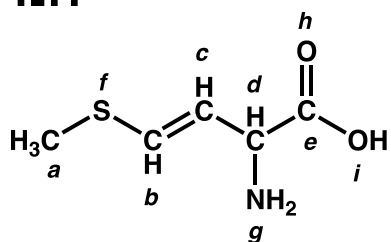
Structure 5

Structure 6

**Question 6** **6 points**

Using the data given below, calculate the **lattice energy** for SrO(s).

	ΔH° (kJ/mol)
$\text{Sr}(s) \longrightarrow \text{Sr}(g)$	+180
$\text{Sr}(g) \longrightarrow \text{Sr}^{2+}(g) + 2e^-$	+1468
$\text{O}_2(g) \longrightarrow 2\text{O}(g)$	+500
$\text{O}(g) + 2e^- \longrightarrow \text{O}^{2-}(g)$	+600
$\text{Sr}(s) + \frac{1}{2}\text{O}_2(g) \longrightarrow \text{SrO}(s)$	-550

**Question 7 10 points**

Consider the molecule given on the left, and answer questions (i) to (vi) using chemical bonding theories that we have learnt in class (valence bond theory, hybridization).

Note that the lone pairs are not shown in the structure:

i. Consider the carbon atoms marked in as ***a***, ***b***, ***c***, ***d***, and ***e*** in the molecule, and determine **hybridization** at each of the carbon centers. (2.5 points)

Atom	Hybridization
Carbon (<i>a</i>)	
Carbon (<i>b</i>)	
Carbon (<i>c</i>)	
Carbon (<i>d</i>)	
Carbon (<i>e</i>)	

ii. Now, consider the sulfur atom (S) marked as ***f***, nitrogen (N) atom (***g***), and oxygen(O) atoms (***h*** and ***i***), and determine their **hybridization**. (2 points)

Atom	Hybridization
Sulfur (<i>f</i>)	
Nitrogen (<i>g</i>)	
Oxygen (<i>h</i>)	
Oxygen (<i>i</i>)	

iii. **Describe** all the bonds formed by the carbon atoms ***a*** and ***b***, using sigma and pi designation as appropriate. (2 points)

Atom	Bonds formed (use sigma and pi as appropriate)
Carbon (<i>a</i>)	
Carbon (<i>b</i>)	

iv. How many total sp^3 -s sigma bonds are in the molecule? (1 point)

v. Determine **electronic** and **molecular geometry** at carbon (C) labeled ***e*** and sulfur (S) labeled ***f*** centres. (2 points)

Atom	Electronic Geometry	Molecular Geometry
Carbon (<i>e</i>)		
Sulfur (<i>f</i>)		

vi. Circle the value from the choices given below, that describes the **C-S-C bond angle at sulfur** (S) marked ***f***: (0.5 points)

$>90^\circ$ to $<109.5^\circ$ 109.5° >109.5 to $<120^\circ$ 120° $>120^\circ$ to $<180^\circ$


Question 8 12 points

Using the **valence shell orbitals only** draw a **complete** molecular orbital (MO) diagram (with no s-p mixing) for the molecule **PS**, in the space provided below. **(6 points)**

Assume that 3s and 3p atomic orbitals will form molecular orbitals like those formed by the combination of 2s and 2p atomic orbitals that we learnt in class, and there is no s-p mixing.

Show **energy arrow**, **atomic orbitals** involved, **atom labels**, the **molecular orbitals** that are formed, **filled electrons** in the MO diagram etc. Make sure to label all the atomic and molecular orbitals.

Using the MO diagram above, answer the following questions:

- (i) Which one of the following will have the highest bond order? Circle the correct choice
 A) PS^{1-} B) PS C) PS^{1+}
- (ii) Which one of the following will have the longest bond length? Circle the correct choice
 A) PS^{1-} B) PS C) PS^{1+}
- (iii) Which of the following will be diamagnetic? Circle the correct choice.
 A) PS^{1-} B) PS C) PS^{1+}
- (iv) Draw Lewis dot structure of the molecule with the shortest bond length (based on the MO diagram) from among the three (PS^{1-} , PS, PS^{1+}), and indicate the bond order calculated from the Lewis dot structure. **(3 points)**

Lewis Dot Structure

Bond Order



DFDFF8BA-D97E-42AF-8EB8-EC916F68F480

midterm-exam-93388

#697 Page 10 of 10

SCRAP PAGE

**CHEMISTRY 110 –Midterm Exam****Thursday November 7, 2024, 6:30 – 8:30 pm****Examiners: Pallavi Sirjoosingh and Ashok Kakkar**

First name (please write as legibly as possible within the boxes)

--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--

Last name

--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--

Student ID number

--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--

This booklet has 8 questions on 10 pages (The last page can be used as a scrap page).

The staple must NOT be removed from the question booklet. Your intact question booklet must be returned at the end of the exam.

A separate booklet contains a periodic table, and datasheet.

Write your answers directly under the questions in the space provided. Do not write on the QR codes or near the top edges (as it may be cut off).

Please write **legibly and dark enough** for the scan to be readable. For students who use English as a second language, dictionaries are permitted. Calculators are allowed.

DO NOT ASK QUESTIONS OF THE INVIGILATORS about interpretations of the exam questions. If you have a concern about a question, please summarize it at the top of this page. All concerns will be evaluated after the exam. GOOD LUCK!

**Question 1 8 points** (*Assume all transitions are possible*)

a. 6 points Calculate the **energy (in kJ/mol)** required to ionize (completely remove) the electron from the **n=3** level of **He⁺ ion** (He: Helium). Show your work including equations. (3 sig. figs.)

b. 2 points

(I) Select the largest (longest) wavelength from the three choices below:

- i. Wavelength of photon to *completely remove* an electron from *1s orbital of Hydrogen atom*
- ii. Wavelength of photon to *completely remove* an electron from *1s orbital of Boron atom (B)*
- iii. Wavelength of photon to *completely remove* an electron from *1s orbital of Boron cation (B⁺)*

i or ii or iii

(II) Select the smallest (shortest) wavelength from the three choices below (*for H atom*):

- i. Wavelength of photon *emitted* for an electron transitioning from *2p to 1s orbital*
- ii. Wavelength of photon *absorbed* for an electron transitioning from *2s to 4p orbital*
- iii. Wavelength of photon *absorbed* for an electron transitioning from *1s to 3p orbital*

i or ii or iii



Question 2 8 points



A Chem110 student plots the graph shown on the left, from data obtained for the kinetic energy (eV) of the ejected electron versus frequency (s^{-1}) of the incident light, for an unknown metal. ($1\text{eV} = 1.6022 \times 10^{-19} \text{ J}$)

Based on the plot, answer the questions **a to c** below:

a. 1 point On the plot above, label the **threshold frequency**.

b. 2 points What is the value of the work potential (in J) of the unknown metal?

c. 3 points What is the *longest* wavelength (in nm) required to eject an electron from this metal's surface? Show your work including equation(s) used. (3 sig figs)

d. 2 points In the space provided below, plot the following

(Assume frequency is *above* threshold frequency):

- Kinetic energy* of an ejected electron versus the *intensity* of the incident light (left)
- Number of electrons* ejected versus the *frequency* of the incident light (right)



**Question 3** **6 points****a. 2 points**

In the ground state of Ge (Germanium; $Z = 32$) atom, **name** the following orbitals:

i. Orbital containing the electron that requires the maximum energy to be removed.

ii. Orbital containing the electron that experiences the minimum effective nuclear charge.

b. 2 points

An unknown anion (A^-) has the following *excited* electronic configuration: $[Ar] 4s^2 3d^3 4p^1$

(Ar: Argon; $Z = 18$)

Give the **condensed ground state electronic configuration** of the atom, **A** and the cation, A^{2+} .
Show your work. (Assume all electronic transitions are possible)

Electronic Configuration of A:

Electronic Configuration of A^{2+} :

c. 2 points

In a single atom, determine how many electrons can have the following set of quantum numbers?

(i) $n = 5; l = 3$

In a single atom, determine how many electrons can have the following set of quantum numbers?

(i) $n = 3; l = 2; m_s = +1/2$

**Question 4** **8 points**

Rank the following from 1 to 4 in **increasing** order. Provide the ranking in the boxes provided.

Note: Atomic number (Z) are provided for relevant atoms

a) Increasing (lowest to highest) order of *ionization energy*: (1= lowest; 4 = highest)

Mg

Mg: Z = 12

Rb

Rb: Z = 37

Cs

Cs: Z = 55

K

K : Z = 19

b) Increasing (lowest to highest) order of *bond angle*: (1= lowest; 4 = highest)

H₂OCCl₄XeF₄

HCN

c) Increasing (smallest to largest) order of *ionic size* (1= smallest; 4 = largest)

S²⁻

S: Z = 16

O²⁻

O: Z = 8

Be²⁺

Be: Z = 4

Na⁺

Na: Z = 11

d) Increasing (lowest to highest) order of *bond energy* (1= lowest; 4 = highest)

P-F

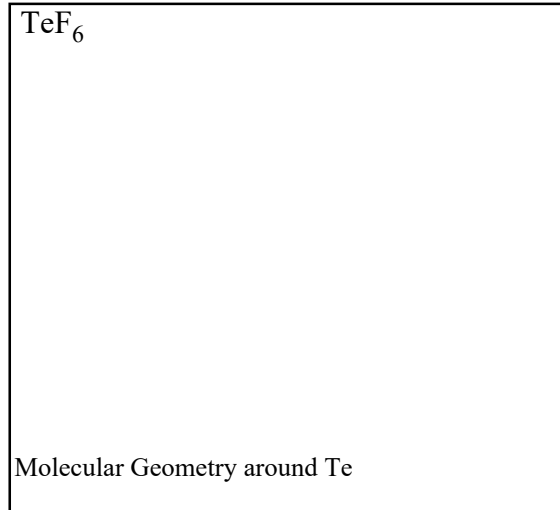
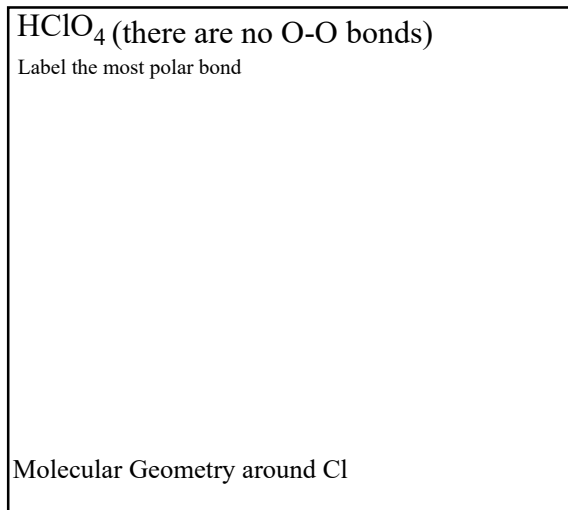
P-Cl

P-I

P-Br

**Question 5** **12 points**

a. 9 points For the molecules HClO_4 and TeF_6 , draw the most stable *Lewis structure* showing appropriate shape and *dashed/wedged bonds*, based on VSEPR. Determine the **molecular geometry** around the central atom. For HClO_4 , label the *most polar bond*.



b. 3 points (Note: Lone pairs are not shown in structures below)

(i) For the three *resonance* structures (1, 2 and 3) shown below, answer the following question:



(I) Circle the *most contributing* resonance structure.

Structure 1

Structure 2

Structure 3

(ii) For the three structures given (4, 5, and 6) below, answer the following questions:



(I) Circle all the structures that represent *resonance structures* of each other:

Structure 4

Structure 5

Structure 6

(II) Among the resonance structures, circle the *most contributing* resonance structure.

Structure 4

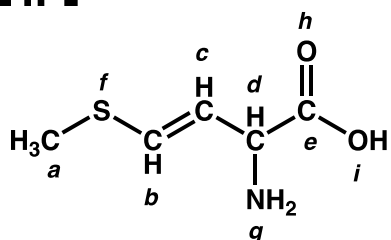
Structure 5

Structure 6

**Question 6** **6 points**

Using the data given below, calculate the **lattice energy** for SrO(s).

	ΔH° (kJ/mol)
$\text{Sr}(s) \longrightarrow \text{Sr}(g)$	+180
$\text{Sr}(g) \longrightarrow \text{Sr}^{2+}(g) + 2e^-$	+1468
$\text{O}_2(g) \longrightarrow 2\text{O}(g)$	+500
$\text{O}(g) + 2e^- \longrightarrow \text{O}^{2-}(g)$	+600
$\text{Sr}(s) + \frac{1}{2}\text{O}_2(g) \longrightarrow \text{SrO}(s)$	-550

**Question 7 10 points**

Consider the molecule given on the left, and answer questions (i) to (vi) using chemical bonding theories that we have learnt in class (valence bond theory, hybridization).

Note that the lone pairs are not shown in the structure:

i. Consider the carbon atoms marked in as ***a***, ***b***, ***c***, ***d***, and ***e*** in the molecule, and determine **hybridization** at each of the carbon centers. (2.5 points)

Atom	Hybridization
Carbon (<i>a</i>)	
Carbon (<i>b</i>)	
Carbon (<i>c</i>)	
Carbon (<i>d</i>)	
Carbon (<i>e</i>)	

ii. Now, consider the sulfur atom (S) marked as ***f***, nitrogen (N) atom (***g***), and oxygen(O) atoms (***h*** and ***i***), and determine their **hybridization**. (2 points)

Atom	Hybridization
Sulfur (<i>f</i>)	
Nitrogen (<i>g</i>)	
Oxygen (<i>h</i>)	
Oxygen (<i>i</i>)	

iii. **Describe** all the bonds formed by the carbon atoms ***a*** and ***b***, using sigma and pi designation as appropriate. (2 points)

Atom	Bonds formed (use sigma and pi as appropriate)
Carbon (<i>a</i>)	
Carbon (<i>b</i>)	

iv. How many total sp^3 -s sigma bonds are in the molecule? (1 point)

v. Determine **electronic** and **molecular geometry** at carbon (C) labeled ***e*** and sulfur (S) labeled ***f*** centres. (2 points)

Atom	Electronic Geometry	Molecular Geometry
Carbon (<i>e</i>)		
Sulfur (<i>f</i>)		

vi. Circle the value from the choices given below, that describes the **C-S-C bond angle at sulfur** (S) marked ***f***: (0.5 points)

>90° to <109.5°

109.5°

>109.5 to <120°

120°

>120° to <180°

**Question 8 12 points**

Using the **valence shell orbitals only** draw a **complete** molecular orbital (MO) diagram (with no s-p mixing) for the molecule **PS**, in the space provided below. **(6 points)**

Assume that 3s and 3p atomic orbitals will form molecular orbitals like those formed by the combination of 2s and 2p atomic orbitals that we learnt in class, and there is no s-p mixing.

Show **energy arrow**, **atomic orbitals** involved, **atom labels**, the **molecular orbitals** that are formed, **filled electrons** in the MO diagram etc. Make sure to label all the atomic and molecular orbitals.

Using the MO diagram above, answer the following questions:

- (i) Which one of the following will have the highest bond order? Circle the correct choice
A) PS^{1-} B) PS C) PS^{1+}
- (ii) Which one of the following will have the longest bond length? Circle the correct choice
A) PS^{1-} B) PS C) PS^{1+}
- (iii) Which of the following will be diamagnetic? Circle the correct choice.
A) PS^{1-} B) PS C) PS^{1+}
- (iv) Draw Lewis dot structure of the molecule with the shortest bond length (based on the MO diagram) from among the three (PS^{1-} , PS, PS^{1+}), and indicate the bond order calculated from the Lewis dot structure. **(3 points)**

Lewis Dot Structure

Bond Order



4EA2823E-376B-4819-961B-A55064DD5B6A

midterm-exam-93388

#698 Page 10 of 10

SCRAP PAGE

**CHEMISTRY 110 –Midterm Exam****Thursday November 7, 2024, 6:30 – 8:30 pm****Examiners: Pallavi Sirjoosingh and Ashok Kakkar**

First name (please write as legibly as possible within the boxes)

--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--

Last name

--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--

Student ID number

--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--

This booklet has 8 questions on 10 pages (The last page can be used as a scrap page).

The staple must NOT be removed from the question booklet. Your intact question booklet must be returned at the end of the exam.

A separate booklet contains a periodic table, and datasheet.

Write your answers directly under the questions in the space provided. Do not write on the QR codes or near the top edges (as it may be cut off).

Please write **legibly and dark enough** for the scan to be readable. For students who use English as a second language, dictionaries are permitted. Calculators are allowed.

DO NOT ASK QUESTIONS OF THE INVIGILATORS about interpretations of the exam questions. If you have a concern about a question, please summarize it at the top of this page. All concerns will be evaluated after the exam. GOOD LUCK!

**Question 1 8 points** (*Assume all transitions are possible*)

a. 6 points Calculate the **energy (in kJ/mol)** required to ionize (completely remove) the electron from the **$n=3$** level of **He^+ ion** (He: Helium). Show your work including equations. (3 sig. figs.)

b. 2 points

(I) Select the largest (longest) wavelength from the three choices below:

- i. Wavelength of photon to *completely remove* an electron from *1s orbital of Hydrogen atom*
- ii. Wavelength of photon to *completely remove* an electron from *1s orbital of Boron atom (B)*
- iii. Wavelength of photon to *completely remove* an electron from *1s orbital of Boron cation (B^+)*

i or ii or iii

(II) Select the smallest (shortest) wavelength from the three choices below (*for H atom*):

- i. Wavelength of photon *emitted* for an electron transitioning from *2p to 1s orbital*
- ii. Wavelength of photon *absorbed* for an electron transitioning from *2s to 4p orbital*
- iii. Wavelength of photon *absorbed* for an electron transitioning from *1s to 3p orbital*

i or ii or iii



Question 2 8 points



A Chem110 student plots the graph shown on the left, from data obtained for the kinetic energy (eV) of the ejected electron versus frequency (s^{-1}) of the incident light, for an unknown metal. ($1\text{eV} = 1.6022 \times 10^{-19} \text{ J}$)

Based on the plot, answer the questions **a to c** below:

a. 1 point On the plot above, label the **threshold frequency**.

b. 2 points What is the value of the work potential (in J) of the unknown metal?

c. 3 points What is the *longest* wavelength (in nm) required to eject an electron from this metal's surface? Show your work including equation(s) used. (3 sig figs)

d. 2 points In the space provided below, plot the following

(Assume frequency is *above* threshold frequency):

- i. *Kinetic energy* of an ejected electron versus the *intensity* of the incident light (left)
- ii. *Number of electrons* ejected versus the *frequency* of the incident light (right)



**Question 3** **6 points****a. 2 points**

In the ground state of Ge (Germanium; $Z = 32$) atom, **name** the following orbitals:

i. Orbital containing the electron that requires the maximum energy to be removed.

ii. Orbital containing the electron that experiences the minimum effective nuclear charge.

b. 2 points

An unknown anion (A^-) has the following *excited* electronic configuration: $[Ar] 4s^2 3d^3 4p^1$

(Ar: Argon; $Z = 18$)

Give the **condensed ground state electronic configuration** of the atom, **A** and the cation, **A^{2+}** .
Show your work. (Assume all electronic transitions are possible)

Electronic Configuration of A:

Electronic Configuration of A^{2+} :

c. 2 points

In a single atom, determine how many electrons can have the following set of quantum numbers?

(i) $n = 5; l = 3$

In a single atom, determine how many electrons can have the following set of quantum numbers?

(i) $n = 3; l = 2; m_s = +1/2$

**Question 4** **8 points**

Rank the following from 1 to 4 in **increasing** order. Provide the ranking in the boxes provided.

Note: Atomic number (Z) are provided for relevant atoms

a) Increasing (lowest to highest) order of *ionization energy*: (1= lowest; 4 = highest)

Mg

Mg: Z = 12

Rb

Rb: Z = 37

Cs

Cs: Z = 55

K

K : Z = 19

b) Increasing (lowest to highest) order of *bond angle*: (1= lowest; 4 = highest)

H₂OCCl₄XeF₄

HCN

c) Increasing (smallest to largest) order of *ionic size* (1= smallest; 4 = largest)

S²⁻

S: Z = 16

O²⁻

O: Z = 8

Be²⁺

Be: Z = 4

Na⁺

Na: Z = 11

d) Increasing (lowest to highest) order of *bond energy* (1= lowest; 4 = highest)

P-F

P-Cl

P-I

P-Br

**Question 5** **12 points**

a. 9 points For the molecules HClO_4 and TeF_6 , draw the most stable *Lewis structure* showing appropriate shape and *dashed/wedged bonds*, based on VSEPR. Determine the **molecular geometry** around the central atom. For HClO_4 , label the *most polar bond*.



b. 3 points (Note: Lone pairs are not shown in structures below)

(i) For the three *resonance* structures (1, 2 and 3) shown below, answer the following question:



(I) Circle the *most contributing* resonance structure.

Structure 1

Structure 2

Structure 3

(ii) For the three structures given (4, 5, and 6) below, answer the following questions:



(I) Circle all the structures that represent *resonance structures* of each other:

Structure 4

Structure 5

Structure 6

(II) Among the resonance structures, circle the *most contributing* resonance structure.

Structure 4

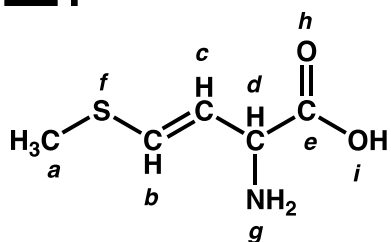
Structure 5

Structure 6

**Question 6** **6 points**

Using the data given below, calculate the **lattice energy** for SrO(s).

	ΔH° (kJ/mol)
$\text{Sr}(s) \longrightarrow \text{Sr}(g)$	+180
$\text{Sr}(g) \longrightarrow \text{Sr}^{2+}(g) + 2e^-$	+1468
$\text{O}_2(g) \longrightarrow 2\text{O}(g)$	+500
$\text{O}(g) + 2e^- \longrightarrow \text{O}^{2-}(g)$	+600
$\text{Sr}(s) + \frac{1}{2}\text{O}_2(g) \longrightarrow \text{SrO}(s)$	-550

**Question 7 10 points**

Consider the molecule given on the left, and answer questions (i) to (vi) using chemical bonding theories that we have learnt in class (valence bond theory, hybridization).

Note that the lone pairs are not shown in the structure:

i. Consider the carbon atoms marked in as ***a***, ***b***, ***c***, ***d***, and ***e*** in the molecule, and determine **hybridization** at each of the carbon centers. (2.5 points)

Atom	Hybridization
Carbon (<i>a</i>)	
Carbon (<i>b</i>)	
Carbon (<i>c</i>)	
Carbon (<i>d</i>)	
Carbon (<i>e</i>)	

ii. Now, consider the sulfur atom (S) marked as ***f***, nitrogen (N) atom (***g***), and oxygen(O) atoms (***h*** and ***i***), and determine their **hybridization**. (2 points)

Atom	Hybridization
Sulfur (<i>f</i>)	
Nitrogen (<i>g</i>)	
Oxygen (<i>h</i>)	
Oxygen (<i>i</i>)	

iii. **Describe** all the bonds formed by the carbon atoms ***a*** and ***b***, using sigma and pi designation as appropriate. (2 points)

Atom	Bonds formed (use sigma and pi as appropriate)
Carbon (<i>a</i>)	
Carbon (<i>b</i>)	

iv. How many total sp^3 -s sigma bonds are in the molecule? (1 point)

v. Determine **electronic** and **molecular geometry** at carbon (C) labeled ***e*** and sulfur (S) labeled ***f*** centres. (2 points)

Atom	Electronic Geometry	Molecular Geometry
Carbon (<i>e</i>)		
Sulfur (<i>f</i>)		

vi. Circle the value from the choices given below, that describes the **C-S-C bond angle at sulfur** (S) marked ***f***: (0.5 points)

>90° to <109.5° 109.5° >109.5 to <120° 120° >120° to <180°

**Question 8 12 points**

Using the **valence shell orbitals only** draw a **complete** molecular orbital (MO) diagram (with no s-p mixing) for the molecule **PS**, in the space provided below. **(6 points)**

Assume that 3s and 3p atomic orbitals will form molecular orbitals like those formed by the combination of 2s and 2p atomic orbitals that we learnt in class, and there is no s-p mixing.

Show **energy arrow**, **atomic orbitals** involved, **atom labels**, the **molecular orbitals** that are formed, **filled electrons** in the MO diagram etc. Make sure to label all the atomic and molecular orbitals.

Using the MO diagram above, answer the following questions:

- (i) Which one of the following will have the highest bond order? Circle the correct choice
A) PS^{1-} B) PS C) PS^{1+}
- (ii) Which one of the following will have the longest bond length? Circle the correct choice
A) PS^{1-} B) PS C) PS^{1+}
- (iii) Which of the following will be diamagnetic? Circle the correct choice.
A) PS^{1-} B) PS C) PS^{1+}
- (iv) Draw Lewis dot structure of the molecule with the shortest bond length (based on the MO diagram) from among the three (PS^{1-} , PS, PS^{1+}), and indicate the bond order calculated from the Lewis dot structure. **(3 points)**

Lewis Dot Structure

Bond Order



FB1B4881-9B74-400E-B4C4-C156ADDF254C

midterm-exam-93388

#699 Page 10 of 10

SCRAP PAGE

**CHEMISTRY 110 –Midterm Exam****Thursday November 7, 2024, 6:30 – 8:30 pm****Examiners: Pallavi Sirjoosingh and Ashok Kakkar**

First name (please write as legibly as possible within the boxes)

--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--

Last name

--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--

Student ID number

--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--

This booklet has 8 questions on 10 pages (The last page can be used as a scrap page).

The staple must NOT be removed from the question booklet. Your intact question booklet must be returned at the end of the exam.

A separate booklet contains a periodic table, and datasheet.

Write your answers directly under the questions in the space provided. Do not write on the QR codes or near the top edges (as it may be cut off).

Please write **legibly and dark enough** for the scan to be readable. For students who use English as a second language, dictionaries are permitted. Calculators are allowed.

DO NOT ASK QUESTIONS OF THE INVIGILATORS about interpretations of the exam questions. If you have a concern about a question, please summarize it at the top of this page. All concerns will be evaluated after the exam. GOOD LUCK!

**Question 1 8 points** (*Assume all transitions are possible*)

a. 6 points Calculate the **energy (in kJ/mol)** required to ionize (completely remove) the electron from the **n=3** level of **He⁺ ion** (He: Helium). Show your work including equations. (3 sig. figs.)

b. 2 points

(I) Select the largest (longest) wavelength from the three choices below:

- i. Wavelength of photon to *completely remove* an electron from *1s orbital of Hydrogen atom*
- ii. Wavelength of photon to *completely remove* an electron from *1s orbital of Boron atom (B)*
- iii. Wavelength of photon to *completely remove* an electron from *1s orbital of Boron cation (B⁺)*

i or ii or iii

(II) Select the smallest (shortest) wavelength from the three choices below (*for H atom*):

- i. Wavelength of photon *emitted* for an electron transitioning from *2p to 1s orbital*
- ii. Wavelength of photon *absorbed* for an electron transitioning from *2s to 4p orbital*
- iii. Wavelength of photon *absorbed* for an electron transitioning from *1s to 3p orbital*

i or ii or iii



Question 2 8 points



A Chem110 student plots the graph shown on the left, from data obtained for the kinetic energy (eV) of the ejected electron versus frequency (s^{-1}) of the incident light, for an unknown metal. ($1\text{eV} = 1.6022 \times 10^{-19} \text{ J}$)

Based on the plot, answer the questions **a to c** below:

a. 1 point On the plot above, label the **threshold frequency**.

b. 2 points What is the value of the work potential (in J) of the unknown metal?

c. 3 points What is the *longest* wavelength (in nm) required to eject an electron from this metal's surface? Show your work including equation(s) used. (3 sig figs)

d. 2 points In the space provided below, plot the following

(Assume frequency is *above* threshold frequency):

- i. *Kinetic energy* of an ejected electron versus the *intensity* of the incident light (left)
- ii. *Number of electrons* ejected versus the *frequency* of the incident light (right)



**Question 3** **6 points****a. 2 points**

In the ground state of Ge (Germanium; $Z = 32$) atom, **name** the following orbitals:

i. Orbital containing the electron that requires the maximum energy to be removed.

ii. Orbital containing the electron that experiences the minimum effective nuclear charge.

b. 2 points

An unknown anion (A^-) has the following *excited* electronic configuration: $[Ar] 4s^2 3d^3 4p^1$

(Ar: Argon; $Z = 18$)

Give the **condensed ground state electronic configuration** of the atom, **A** and the cation, **A^{2+}** .
Show your work. (Assume all electronic transitions are possible)

Electronic Configuration of A:

Electronic Configuration of A^{2+} :

c. 2 points

In a single atom, determine how many electrons can have the following set of quantum numbers?

(i) $n = 5; l = 3$

In a single atom, determine how many electrons can have the following set of quantum numbers?

(i) $n = 3; l = 2; m_s = +1/2$

**Question 4** **8 points**

Rank the following from 1 to 4 in **increasing** order. Provide the ranking in the boxes provided.

Note: Atomic number (Z) are provided for relevant atoms

a) Increasing (lowest to highest) order of *ionization energy*: (1= lowest; 4 = highest)

Mg

Mg: Z = 12

Rb

Rb: Z = 37

Cs

Cs: Z = 55

K

K : Z = 19

b) Increasing (lowest to highest) order of *bond angle*: (1= lowest; 4 = highest)

H₂OCCl₄XeF₄

HCN

c) Increasing (smallest to largest) order of *ionic size* (1= smallest; 4 = largest)

S²⁻

S: Z = 16

O²⁻

O: Z = 8

Be²⁺

Be: Z = 4

Na⁺

Na: Z = 11

d) Increasing (lowest to highest) order of *bond energy* (1= lowest; 4 = highest)

P-F

P-Cl

P-I

P-Br

**Question 5** **12 points**

a. 9 points For the molecules HClO_4 and TeF_6 , draw the most stable *Lewis structure* showing appropriate shape and *dashed/wedged bonds*, based on VSEPR. Determine the **molecular geometry** around the central atom. For HClO_4 , label the *most polar bond*.



b. 3 points (Note: Lone pairs are not shown in structures below)

(i) For the three *resonance* structures (1, 2 and 3) shown below, answer the following question:



(I) Circle the *most contributing* resonance structure.

Structure 1

Structure 2

Structure 3

(ii) For the three structures given (4, 5, and 6) below, answer the following questions:



(I) Circle all the structures that represent *resonance structures* of each other:

Structure 4

Structure 5

Structure 6

(II) Among the resonance structures, circle the *most contributing* resonance structure.

Structure 4

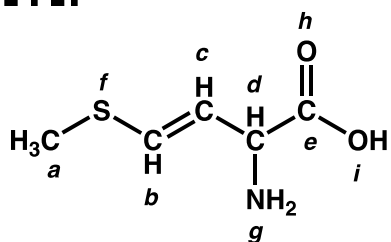
Structure 5

Structure 6

**Question 6** **6 points**

Using the data given below, calculate the **lattice energy** for SrO(s).

	ΔH° (kJ/mol)
$\text{Sr}(s) \longrightarrow \text{Sr}(g)$	+180
$\text{Sr}(g) \longrightarrow \text{Sr}^{2+}(g) + 2e^-$	+1468
$\text{O}_2(g) \longrightarrow 2\text{O}(g)$	+500
$\text{O}(g) + 2e^- \longrightarrow \text{O}^{2-}(g)$	+600
$\text{Sr}(s) + \frac{1}{2}\text{O}_2(g) \longrightarrow \text{SrO}(s)$	-550

**Question 7 10 points**

Consider the molecule given on the left, and answer questions (i) to (vi) using chemical bonding theories that we have learnt in class (valence bond theory, hybridization).

Note that the lone pairs are not shown in the structure:

i. Consider the carbon atoms marked in as ***a***, ***b***, ***c***, ***d***, and ***e*** in the molecule, and determine **hybridization** at each of the carbon centers. (2.5 points)

Atom	Hybridization
Carbon (<i>a</i>)	
Carbon (<i>b</i>)	
Carbon (<i>c</i>)	
Carbon (<i>d</i>)	
Carbon (<i>e</i>)	

ii. Now, consider the sulfur atom (S) marked as ***f***, nitrogen (N) atom (***g***), and oxygen(O) atoms (***h*** and ***i***), and determine their **hybridization**. (2 points)

Atom	Hybridization
Sulfur (<i>f</i>)	
Nitrogen (<i>g</i>)	
Oxygen (<i>h</i>)	
Oxygen (<i>i</i>)	

iii. **Describe** all the bonds formed by the carbon atoms ***a*** and ***b***, using sigma and pi designation as appropriate. (2 points)

Atom	Bonds formed (use sigma and pi as appropriate)
Carbon (<i>a</i>)	
Carbon (<i>b</i>)	

iv. How many total sp^3 -s sigma bonds are in the molecule? (1 point)

v. Determine **electronic** and **molecular geometry** at carbon (C) labeled ***e*** and sulfur (S) labeled ***f*** centres. (2 points)

Atom	Electronic Geometry	Molecular Geometry
Carbon (<i>e</i>)		
Sulfur (<i>f</i>)		

vi. Circle the value from the choices given below, that describes the **C-S-C bond angle at sulfur** (S) marked ***f***: (0.5 points)

$>90^\circ$ to $<109.5^\circ$ 109.5° >109.5 to $<120^\circ$ 120° $>120^\circ$ to $<180^\circ$

**Question 8 12 points**

Using the **valence shell orbitals only** draw a **complete** molecular orbital (MO) diagram (with no s-p mixing) for the molecule **PS**, in the space provided below. **(6 points)**

Assume that 3s and 3p atomic orbitals will form molecular orbitals like those formed by the combination of 2s and 2p atomic orbitals that we learnt in class, and there is no s-p mixing.

Show **energy arrow**, **atomic orbitals** involved, **atom labels**, the **molecular orbitals** that are formed, **filled electrons** in the MO diagram etc. Make sure to label all the atomic and molecular orbitals.

Using the MO diagram above, answer the following questions:

- (i) Which one of the following will have the highest bond order? Circle the correct choice
A) PS^{1-} B) PS C) PS^{1+}
- (ii) Which one of the following will have the longest bond length? Circle the correct choice
A) PS^{1-} B) PS C) PS^{1+}
- (iii) Which of the following will be diamagnetic? Circle the correct choice.
A) PS^{1-} B) PS C) PS^{1+}
- (iv) Draw Lewis dot structure of the molecule with the shortest bond length (based on the MO diagram) from among the three (PS^{1-} , PS, PS^{1+}), and indicate the bond order calculated from the Lewis dot structure. **(3 points)**

Lewis Dot Structure

Bond Order



AFFB04A5-B4A5-43B3-A2F4-730E08D63086

midterm-exam-93388

#700 Page 10 of 10

SCRAP PAGE

**CHEMISTRY 110 –Midterm Exam****Thursday November 7, 2024, 6:30 – 8:30 pm**

Examiners: Pallavi Sirjoosingh and Ashok Kakkar

First name (please write as legibly as possible within the boxes)

--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--

Last name

--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--

Student ID number

--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--

This booklet has 8 questions on 10 pages (The last page can be used as a scrap page).

The staple must NOT be removed from the question booklet. Your intact question booklet must be returned at the end of the exam.

A separate booklet contains a periodic table, and datasheet.

Write your answers directly under the questions in the space provided. Do not write on the QR codes or near the top edges (as it may be cut off).

Please write **legibly and dark enough** for the scan to be readable. For students who use English as a second language, dictionaries are permitted. Calculators are allowed.

DO NOT ASK QUESTIONS OF THE INVIGILATORS about interpretations of the exam questions. If you have a concern about a question, please summarize it at the top of this page. All concerns will be evaluated after the exam. GOOD LUCK!

**Question 1 8 points** (*Assume all transitions are possible*)

a. 6 points Calculate the **energy (in kJ/mol)** required to ionize (completely remove) the electron from the **n=3** level of **He⁺ ion** (He: Helium). Show your work including equations. (3 sig. figs.)

b. 2 points

(I) Select the largest (longest) wavelength from the three choices below:

- i. Wavelength of photon to *completely remove* an electron from *1s orbital of Hydrogen atom*
- ii. Wavelength of photon to *completely remove* an electron from *1s orbital of Boron atom (B)*
- iii. Wavelength of photon to *completely remove* an electron from *1s orbital of Boron cation (B⁺)*

i or ii or iii

(II) Select the smallest (shortest) wavelength from the three choices below (*for H atom*):

- i. Wavelength of photon *emitted* for an electron transitioning from *2p to 1s orbital*
- ii. Wavelength of photon *absorbed* for an electron transitioning from *2s to 4p orbital*
- iii. Wavelength of photon *absorbed* for an electron transitioning from *1s to 3p orbital*

i or ii or iii



Question 2 8 points



A Chem110 student plots the graph shown on the left, from data obtained for the kinetic energy (eV) of the ejected electron versus frequency (s^{-1}) of the incident light, for an unknown metal. ($1\text{eV} = 1.6022 \times 10^{-19} \text{ J}$)

Based on the plot, answer the questions **a to c** below:

a. 1 point On the plot above, label the **threshold frequency**.

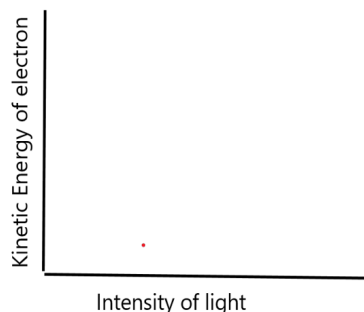
b. 2 points What is the value of the work potential (in J) of the unknown metal?

c. 3 points What is the *longest* wavelength (in nm) required to eject an electron from this metal's surface? Show your work including equation(s) used. (3 sig figs)

d. 2 points In the space provided below, plot the following

(Assume frequency is *above* threshold frequency):

- i. *Kinetic energy* of an ejected electron versus the *intensity* of the incident light (left)
- ii. *Number of electrons* ejected versus the *frequency* of the incident light (right)



**Question 3** **6 points****a. 2 points**

In the ground state of Ge (Germanium; $Z = 32$) atom, **name** the following orbitals:

i. Orbital containing the electron that requires the maximum energy to be removed.

ii. Orbital containing the electron that experiences the minimum effective nuclear charge.

b. 2 points

An unknown anion (A^-) has the following *excited* electronic configuration: $[Ar] 4s^2 3d^3 4p^1$

(Ar: Argon; $Z = 18$)

Give the **condensed ground state electronic configuration** of the atom, **A** and the cation, **A^{2+}** .
Show your work. (Assume all electronic transitions are possible)

Electronic Configuration of A:

Electronic Configuration of A^{2+} :

c. 2 points

In a single atom, determine how many electrons can have the following set of quantum numbers?

(i) $n = 5; l = 3$

In a single atom, determine how many electrons can have the following set of quantum numbers?

(i) $n = 3; l = 2; m_s = +1/2$

**Question 4** **8 points**

Rank the following from 1 to 4 in **increasing** order. Provide the ranking in the boxes provided.

Note: Atomic number (Z) are provided for relevant atoms

a) Increasing (lowest to highest) order of *ionization energy*: (1= lowest; 4 = highest)

Mg

Mg: Z = 12

Rb

Rb: Z = 37

Cs

Cs: Z = 55

K

K : Z = 19

b) Increasing (lowest to highest) order of *bond angle*: (1= lowest; 4 = highest)

H₂OCCl₄XeF₄

HCN

c) Increasing (smallest to largest) order of *ionic size* (1= smallest; 4 = largest)

S²⁻

S: Z = 16

O²⁻

O: Z = 8

Be²⁺

Be: Z = 4

Na⁺

Na: Z = 11

d) Increasing (lowest to highest) order of *bond energy* (1= lowest; 4 = highest)

P-F

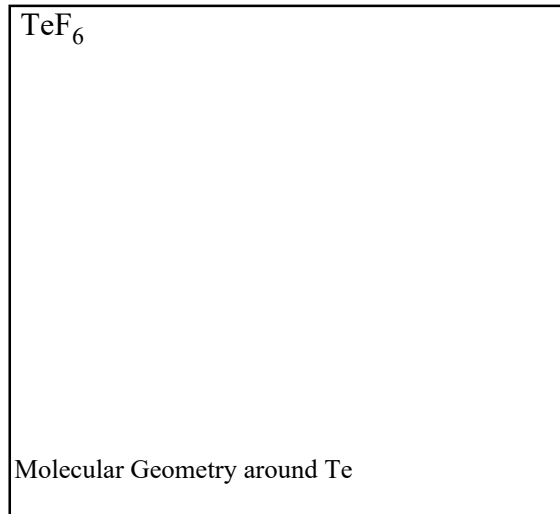
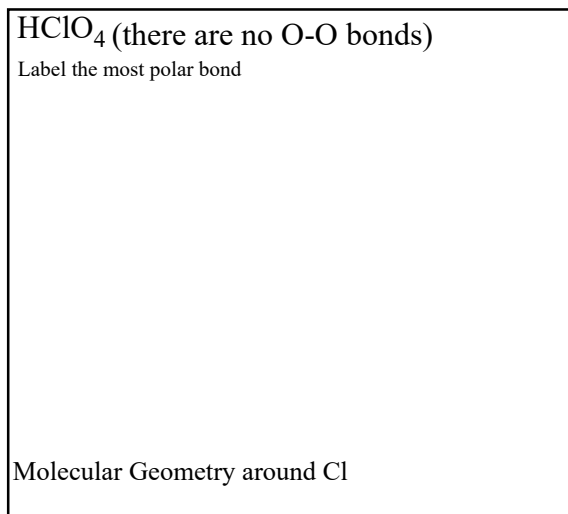
P-Cl

P-I

P-Br

**Question 5** **12 points**

a. 9 points For the molecules HClO_4 and TeF_6 , draw the most stable *Lewis structure* showing appropriate shape and *dashed/wedged bonds*, based on VSEPR. Determine the **molecular geometry** around the central atom. For HClO_4 , label the *most polar bond*.



b. 3 points (Note: Lone pairs are not shown in structures below)

(i) For the three *resonance* structures (1, 2 and 3) shown below, answer the following question:



(I) Circle the *most contributing* resonance structure.

Structure 1

Structure 2

Structure 3

(ii) For the three structures given (4, 5, and 6) below, answer the following questions:



(I) Circle all the structures that represent *resonance structures* of each other:

Structure 4

Structure 5

Structure 6

(II) Among the resonance structures, circle the *most contributing* resonance structure.

Structure 4

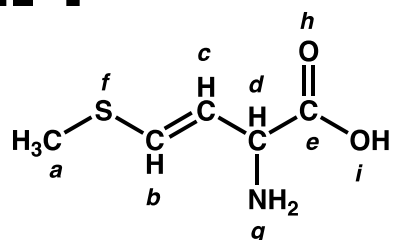
Structure 5

Structure 6

**Question 6** **6 points**

Using the data given below, calculate the **lattice energy** for SrO(s).

	ΔH° (kJ/mol)
$\text{Sr}(s) \longrightarrow \text{Sr}(g)$	+180
$\text{Sr}(g) \longrightarrow \text{Sr}^{2+}(g) + 2e^-$	+1468
$\text{O}_2(g) \longrightarrow 2\text{O}(g)$	+500
$\text{O}(g) + 2e^- \longrightarrow \text{O}^{2-}(g)$	+600
$\text{Sr}(s) + \frac{1}{2}\text{O}_2(g) \longrightarrow \text{SrO}(s)$	-550

**Question 7 10 points**

Consider the molecule given on the left, and answer questions (i) to (vi) using chemical bonding theories that we have learnt in class (valence bond theory, hybridization).

Note that the lone pairs are not shown in the structure:

i. Consider the carbon atoms marked in as ***a***, ***b***, ***c***, ***d***, and ***e*** in the molecule, and determine **hybridization** at each of the carbon centers. (2.5 points)

Atom	Hybridization
Carbon (<i>a</i>)	
Carbon (<i>b</i>)	
Carbon (<i>c</i>)	
Carbon (<i>d</i>)	
Carbon (<i>e</i>)	

ii. Now, consider the sulfur atom (S) marked as ***f***, nitrogen (N) atom (***g***), and oxygen(O) atoms (***h*** and ***i***), and determine their **hybridization**. (2 points)

Atom	Hybridization
Sulfur (<i>f</i>)	
Nitrogen (<i>g</i>)	
Oxygen (<i>h</i>)	
Oxygen (<i>i</i>)	

iii. **Describe** all the bonds formed by the carbon atoms ***a*** and ***b***, using sigma and pi designation as appropriate. (2 points)

Atom	Bonds formed (use sigma and pi as appropriate)
Carbon (<i>a</i>)	
Carbon (<i>b</i>)	

iv. How many total sp^3 -s sigma bonds are in the molecule? (1 point)

v. Determine **electronic** and **molecular geometry** at carbon (C) labeled ***e*** and sulfur (S) labeled ***f*** centres. (2 points)

Atom	Electronic Geometry	Molecular Geometry
Carbon (<i>e</i>)		
Sulfur (<i>f</i>)		

vi. Circle the value from the choices given below, that describes the **C-S-C bond angle at sulfur** (S) marked ***f***: (0.5 points)

>90° to <109.5° 109.5° >109.5 to <120° 120° >120° to <180°


Question 8 12 points

Using the **valence shell orbitals only** draw a **complete** molecular orbital (MO) diagram (with no s-p mixing) for the molecule **PS**, in the space provided below. **(6 points)**

Assume that 3s and 3p atomic orbitals will form molecular orbitals like those formed by the combination of 2s and 2p atomic orbitals that we learnt in class, and there is no s-p mixing.

Show **energy arrow**, **atomic orbitals** involved, **atom labels**, the **molecular orbitals** that are formed, **filled electrons** in the MO diagram etc. Make sure to label all the atomic and molecular orbitals.

Using the MO diagram above, answer the following questions:

- (i) Which one of the following will have the highest bond order? Circle the correct choice
 A) PS^{1-} B) PS C) PS^{1+}
- (ii) Which one of the following will have the longest bond length? Circle the correct choice
 A) PS^{1-} B) PS C) PS^{1+}
- (iii) Which of the following will be diamagnetic? Circle the correct choice.
 A) PS^{1-} B) PS C) PS^{1+}
- (iv) Draw Lewis dot structure of the molecule with the shortest bond length (based on the MO diagram) from among the three (PS^{1-} , PS, PS^{1+}), and indicate the bond order calculated from the Lewis dot structure. **(3 points)**

Lewis Dot Structure

Bond Order



517572AE-D648-4BFA-9165-4914C793B591

midterm-exam-93388

#701 Page 10 of 10

SCRAP PAGE

**CHEMISTRY 110 –Midterm Exam****Thursday November 7, 2024, 6:30 – 8:30 pm**

Examiners: Pallavi Sirjoosingh and Ashok Kakkar

First name (please write as legibly as possible within the boxes)

--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--

Last name

--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--

Student ID number

--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--

This booklet has 8 questions on 10 pages (The last page can be used as a scrap page).

The staple must NOT be removed from the question booklet. Your intact question booklet must be returned at the end of the exam.

A separate booklet contains a periodic table, and datasheet.

Write your answers directly under the questions in the space provided. Do not write on the QR codes or near the top edges (as it may be cut off).

Please write **legibly and dark enough** for the scan to be readable. For students who use English as a second language, dictionaries are permitted. Calculators are allowed.

DO NOT ASK QUESTIONS OF THE INVIGILATORS about interpretations of the exam questions. If you have a concern about a question, please summarize it at the top of this page. All concerns will be evaluated after the exam. GOOD LUCK!

**Question 1 8 points** (*Assume all transitions are possible*)

a. 6 points Calculate the **energy (in kJ/mol)** required to ionize (completely remove) the electron from the **n=3** level of **He⁺ ion** (He: Helium). Show your work including equations. (3 sig. figs.)

b. 2 points

(I) Select the largest (longest) wavelength from the three choices below:

- i. Wavelength of photon to *completely remove* an electron from *1s orbital of Hydrogen atom*
- ii. Wavelength of photon to *completely remove* an electron from *1s orbital of Boron atom (B)*
- iii. Wavelength of photon to *completely remove* an electron from *1s orbital of Boron cation (B⁺)*

i or ii or iii

(II) Select the smallest (shortest) wavelength from the three choices below (*for H atom*):

- i. Wavelength of photon *emitted* for an electron transitioning from *2p to 1s orbital*
- ii. Wavelength of photon *absorbed* for an electron transitioning from *2s to 4p orbital*
- iii. Wavelength of photon *absorbed* for an electron transitioning from *1s to 3p orbital*

i or ii or iii



Question 2 8 points



A Chem110 student plots the graph shown on the left, from data obtained for the kinetic energy (eV) of the ejected electron versus frequency (s^{-1}) of the incident light, for an unknown metal. ($1\text{eV} = 1.6022 \times 10^{-19} \text{ J}$)

Based on the plot, answer the questions **a to c** below:

a. 1 point On the plot above, label the **threshold frequency**.

b. 2 points What is the value of the work potential (in J) of the unknown metal?

c. 3 points What is the *longest* wavelength (in nm) required to eject an electron from this metal's surface? Show your work including equation(s) used. (3 sig figs)

d. 2 points In the space provided below, plot the following

(Assume frequency is *above* threshold frequency):

- i. *Kinetic energy* of an ejected electron versus the *intensity* of the incident light (left)
- ii. *Number of electrons* ejected versus the *frequency* of the incident light (right)



**Question 3** **6 points****a. 2 points**

In the ground state of Ge (Germanium; $Z = 32$) atom, **name** the following orbitals:

i. Orbital containing the electron that requires the maximum energy to be removed.

ii. Orbital containing the electron that experiences the minimum effective nuclear charge.

b. 2 points

An unknown anion (A^-) has the following *excited* electronic configuration: $[Ar] 4s^2 3d^3 4p^1$

(Ar: Argon; $Z = 18$)

Give the **condensed ground state electronic configuration** of the atom, **A** and the cation, **A^{2+}** .
Show your work. (Assume all electronic transitions are possible)

Electronic Configuration of A:

Electronic Configuration of A^{2+} :

c. 2 points

In a single atom, determine how many electrons can have the following set of quantum numbers?

(i) $n = 5; l = 3$

In a single atom, determine how many electrons can have the following set of quantum numbers?

(i) $n = 3; l = 2; m_s = +1/2$

**Question 4** **8 points**

Rank the following from 1 to 4 in **increasing** order. Provide the ranking in the boxes provided.

Note: Atomic number (Z) are provided for relevant atoms

a) Increasing (lowest to highest) order of *ionization energy*: (1= lowest; 4 = highest)

Mg

Mg: Z = 12

Rb

Rb: Z = 37

Cs

Cs: Z = 55

K

K : Z = 19

b) Increasing (lowest to highest) order of *bond angle*: (1= lowest; 4 = highest)

H₂OCCl₄XeF₄

HCN

c) Increasing (smallest to largest) order of *ionic size* (1= smallest; 4 = largest)

S²⁻

S: Z = 16

O²⁻

O: Z = 8

Be²⁺

Be: Z = 4

Na⁺

Na: Z = 11

d) Increasing (lowest to highest) order of *bond energy* (1= lowest; 4 = highest)

P-F

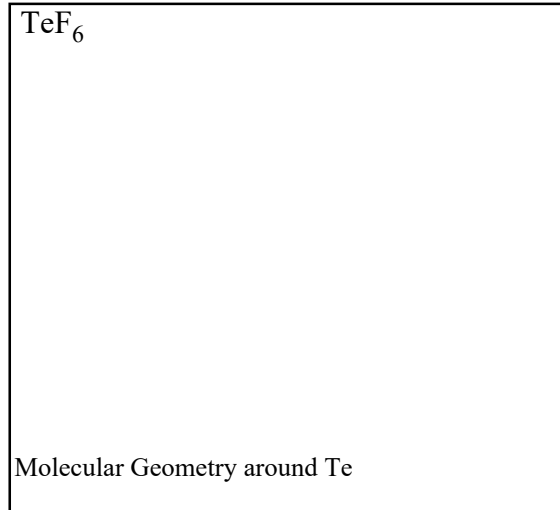
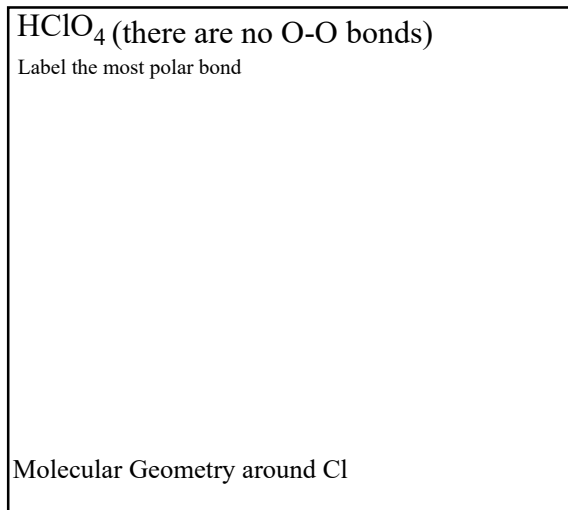
P-Cl

P-I

P-Br

**Question 5** **12 points**

a. 9 points For the molecules HClO_4 and TeF_6 , draw the most stable *Lewis structure* showing appropriate shape and *dashed/wedged bonds*, based on VSEPR. Determine the **molecular geometry** around the central atom. For HClO_4 , label the *most polar bond*.



b. 3 points (Note: Lone pairs are not shown in structures below)

(i) For the three *resonance* structures (1, 2 and 3) shown below, answer the following question:



(I) Circle the *most contributing* resonance structure.

Structure 1

Structure 2

Structure 3

(ii) For the three structures given (4, 5, and 6) below, answer the following questions:



(I) Circle all the structures that represent *resonance structures* of each other:

Structure 4

Structure 5

Structure 6

(II) Among the resonance structures, circle the *most contributing* resonance structure.

Structure 4

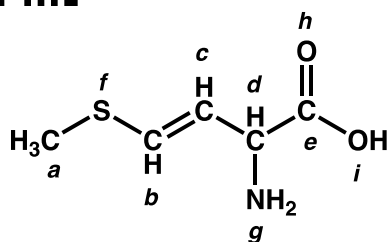
Structure 5

Structure 6

**Question 6** **6 points**

Using the data given below, calculate the **lattice energy** for SrO(s).

	ΔH° (kJ/mol)
$\text{Sr}(s) \longrightarrow \text{Sr}(g)$	+180
$\text{Sr}(g) \longrightarrow \text{Sr}^{2+}(g) + 2e^-$	+1468
$\text{O}_2(g) \longrightarrow 2\text{O}(g)$	+500
$\text{O}(g) + 2e^- \longrightarrow \text{O}^{2-}(g)$	+600
$\text{Sr}(s) + \frac{1}{2}\text{O}_2(g) \longrightarrow \text{SrO}(s)$	-550

**Question 7 10 points**

Consider the molecule given on the left, and answer questions (i) to (vi) using chemical bonding theories that we have learnt in class (valence bond theory, hybridization).

Note that the lone pairs are not shown in the structure:

i. Consider the carbon atoms marked in as ***a***, ***b***, ***c***, ***d***, and ***e*** in the molecule, and determine **hybridization** at each of the carbon centers. (2.5 points)

Atom	Hybridization
Carbon (<i>a</i>)	
Carbon (<i>b</i>)	
Carbon (<i>c</i>)	
Carbon (<i>d</i>)	
Carbon (<i>e</i>)	

ii. Now, consider the sulfur atom (S) marked as ***f***, nitrogen (N) atom (***g***), and oxygen(O) atoms (***h*** and ***i***), and determine their **hybridization**. (2 points)

Atom	Hybridization
Sulfur (<i>f</i>)	
Nitrogen (<i>g</i>)	
Oxygen (<i>h</i>)	
Oxygen (<i>i</i>)	

iii. **Describe** all the bonds formed by the carbon atoms ***a*** and ***b***, using sigma and pi designation as appropriate. (2 points)

Atom	Bonds formed (use sigma and pi as appropriate)
Carbon (<i>a</i>)	
Carbon (<i>b</i>)	

iv. How many total sp^3 -s sigma bonds are in the molecule? (1 point)

v. Determine **electronic** and **molecular geometry** at carbon (C) labeled ***e*** and sulfur (S) labeled ***f*** centres. (2 points)

Atom	Electronic Geometry	Molecular Geometry
Carbon (<i>e</i>)		
Sulfur (<i>f</i>)		

vi. Circle the value from the choices given below, that describes the **C-S-C bond angle at sulfur** (S) marked ***f***: (0.5 points)

>90° to <109.5° 109.5° >109.5 to <120° 120° >120° to <180°


Question 8 12 points

Using the **valence shell orbitals only** draw a **complete** molecular orbital (MO) diagram (with no s-p mixing) for the molecule **PS**, in the space provided below. **(6 points)**

Assume that 3s and 3p atomic orbitals will form molecular orbitals like those formed by the combination of 2s and 2p atomic orbitals that we learnt in class, and there is no s-p mixing.

Show **energy arrow**, **atomic orbitals** involved, **atom labels**, the **molecular orbitals** that are formed, **filled electrons** in the MO diagram etc. Make sure to label all the atomic and molecular orbitals.

Using the MO diagram above, answer the following questions:

- (i) Which one of the following will have the highest bond order? Circle the correct choice
 A) PS^{1-} B) PS C) PS^{1+}
- (ii) Which one of the following will have the longest bond length? Circle the correct choice
 A) PS^{1-} B) PS C) PS^{1+}
- (iii) Which of the following will be diamagnetic? Circle the correct choice.
 A) PS^{1-} B) PS C) PS^{1+}
- (iv) Draw Lewis dot structure of the molecule with the shortest bond length (based on the MO diagram) from among the three (PS^{1-} , PS, PS^{1+}), and indicate the bond order calculated from the Lewis dot structure. **(3 points)**

Lewis Dot Structure

Bond Order



1F55B27D-BEEE-497D-8753-EFA62DA833CE

midterm-exam-93388

#702 Page 10 of 10

SCRAP PAGE

**CHEMISTRY 110 –Midterm Exam****Thursday November 7, 2024, 6:30 – 8:30 pm**

Examiners: Pallavi Sirjoosingh and Ashok Kakkar

First name (please write as legibly as possible within the boxes)

--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--

Last name

--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--

Student ID number

--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--

This booklet has 8 questions on 10 pages (The last page can be used as a scrap page).

The staple must NOT be removed from the question booklet. Your intact question booklet must be returned at the end of the exam.

A separate booklet contains a periodic table, and datasheet.

Write your answers directly under the questions in the space provided. Do not write on the QR codes or near the top edges (as it may be cut off).

Please write **legibly and dark enough** for the scan to be readable. For students who use English as a second language, dictionaries are permitted. Calculators are allowed.

DO NOT ASK QUESTIONS OF THE INVIGILATORS about interpretations of the exam questions. If you have a concern about a question, please summarize it at the top of this page. All concerns will be evaluated after the exam. GOOD LUCK!

**Question 1 8 points** (*Assume all transitions are possible*)

a. 6 points Calculate the **energy (in kJ/mol)** required to ionize (completely remove) the electron from the **n=3** level of **He⁺ ion** (He: Helium). Show your work including equations. (3 sig. figs.)

b. 2 points

(I) Select the largest (longest) wavelength from the three choices below:

- i. Wavelength of photon to *completely remove* an electron from *1s orbital of Hydrogen atom*
- ii. Wavelength of photon to *completely remove* an electron from *1s orbital of Boron atom (B)*
- iii. Wavelength of photon to *completely remove* an electron from *1s orbital of Boron cation (B⁺)*

i or ii or iii

(II) Select the smallest (shortest) wavelength from the three choices below (*for H atom*):

- i. Wavelength of photon *emitted* for an electron transitioning from *2p to 1s orbital*
- ii. Wavelength of photon *absorbed* for an electron transitioning from *2s to 4p orbital*
- iii. Wavelength of photon *absorbed* for an electron transitioning from *1s to 3p orbital*

i or ii or iii



Question 2 8 points



A Chem110 student plots the graph shown on the left, from data obtained for the kinetic energy (eV) of the ejected electron versus frequency (s^{-1}) of the incident light, for an unknown metal. ($1\text{eV} = 1.6022 \times 10^{-19} \text{ J}$)

Based on the plot, answer the questions **a to c** below:

a. 1 point On the plot above, label the **threshold frequency**.

b. 2 points What is the value of the work potential (in J) of the unknown metal?

c. 3 points What is the *longest* wavelength (in nm) required to eject an electron from this metal's surface? Show your work including equation(s) used. (3 sig figs)

d. 2 points In the space provided below, plot the following

(Assume frequency is *above* threshold frequency):

- i. *Kinetic energy* of an ejected electron versus the *intensity* of the incident light (left)
- ii. *Number of electrons* ejected versus the *frequency* of the incident light (right)



**Question 3** **6 points****a. 2 points**

In the ground state of Ge (Germanium; $Z = 32$) atom, **name** the following orbitals:

i. Orbital containing the electron that requires the maximum energy to be removed.

ii. Orbital containing the electron that experiences the minimum effective nuclear charge.

b. 2 points

An unknown anion (A^-) has the following *excited* electronic configuration: $[Ar] 4s^2 3d^3 4p^1$

(Ar: Argon; $Z = 18$)

Give the **condensed ground state electronic configuration** of the atom, **A** and the cation, **A^{2+}** .
Show your work. (Assume all electronic transitions are possible)

Electronic Configuration of A:

Electronic Configuration of A^{2+} :

c. 2 points

In a single atom, determine how many electrons can have the following set of quantum numbers?

(i) $n = 5; l = 3$

In a single atom, determine how many electrons can have the following set of quantum numbers?

(i) $n = 3; l = 2; m_s = +1/2$

**Question 4** **8 points**

Rank the following from 1 to 4 in **increasing** order. Provide the ranking in the boxes provided.

Note: Atomic number (Z) are provided for relevant atoms

a) Increasing (lowest to highest) order of *ionization energy*: (1= lowest; 4 = highest)

Mg

Mg: Z = 12

Rb

Rb: Z = 37

Cs

Cs: Z = 55

K

K : Z = 19

b) Increasing (lowest to highest) order of *bond angle*: (1= lowest; 4 = highest)

H₂OCCl₄XeF₄

HCN

c) Increasing (smallest to largest) order of *ionic size* (1= smallest; 4 = largest)

S²⁻

S: Z = 16

O²⁻

O: Z = 8

Be²⁺

Be: Z = 4

Na⁺

Na: Z = 11

d) Increasing (lowest to highest) order of *bond energy* (1= lowest; 4 = highest)

P-F

P-Cl

P-I

P-Br

**Question 5** **12 points**

a. 9 points For the molecules HClO_4 and TeF_6 , draw the most stable *Lewis structure* showing appropriate shape and *dashed/wedged bonds*, based on VSEPR. Determine the **molecular geometry** around the central atom. For HClO_4 , label the *most polar bond*.



b. 3 points (Note: Lone pairs are not shown in structures below)

(i) For the three *resonance* structures (1, 2 and 3) shown below, answer the following question:



(I) Circle the *most contributing* resonance structure.

Structure 1

Structure 2

Structure 3

(ii) For the three structures given (4, 5, and 6) below, answer the following questions:



(I) Circle all the structures that represent *resonance structures* of each other:

Structure 4

Structure 5

Structure 6

(II) Among the resonance structures, circle the *most contributing* resonance structure.

Structure 4

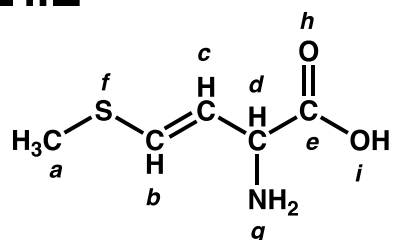
Structure 5

Structure 6

**Question 6** **6 points**

Using the data given below, calculate the **lattice energy** for SrO(s).

	ΔH° (kJ/mol)
$\text{Sr}(s) \longrightarrow \text{Sr}(g)$	+180
$\text{Sr}(g) \longrightarrow \text{Sr}^{2+}(g) + 2e^-$	+1468
$\text{O}_2(g) \longrightarrow 2\text{O}(g)$	+500
$\text{O}(g) + 2e^- \longrightarrow \text{O}^{2-}(g)$	+600
$\text{Sr}(s) + \frac{1}{2}\text{O}_2(g) \longrightarrow \text{SrO}(s)$	-550

**Question 7 10 points**

Consider the molecule given on the left, and answer questions (i) to (vi) using chemical bonding theories that we have learnt in class (valence bond theory, hybridization).

Note that the lone pairs are not shown in the structure:

i. Consider the carbon atoms marked in as ***a***, ***b***, ***c***, ***d***, and ***e*** in the molecule, and determine **hybridization** at each of the carbon centers. (2.5 points)

Atom	Hybridization
Carbon (<i>a</i>)	
Carbon (<i>b</i>)	
Carbon (<i>c</i>)	
Carbon (<i>d</i>)	
Carbon (<i>e</i>)	

ii. Now, consider the sulfur atom (S) marked as ***f***, nitrogen (N) atom (***g***), and oxygen(O) atoms (***h*** and ***i***), and determine their **hybridization**. (2 points)

Atom	Hybridization
Sulfur (<i>f</i>)	
Nitrogen (<i>g</i>)	
Oxygen (<i>h</i>)	
Oxygen (<i>i</i>)	

iii. **Describe** all the bonds formed by the carbon atoms ***a*** and ***b***, using sigma and pi designation as appropriate. (2 points)

Atom	Bonds formed (use sigma and pi as appropriate)
Carbon (<i>a</i>)	
Carbon (<i>b</i>)	

iv. How many total sp^3 -s sigma bonds are in the molecule? (1 point)

v. Determine **electronic** and **molecular geometry** at carbon (C) labeled ***e*** and sulfur (S) labeled ***f*** centres. (2 points)

Atom	Electronic Geometry	Molecular Geometry
Carbon (<i>e</i>)		
Sulfur (<i>f</i>)		

vi. Circle the value from the choices given below, that describes the **C-S-C bond angle at sulfur** (S) marked ***f***: (0.5 points)

>90° to <109.5°

109.5°

>109.5 to <120°

120°

>120° to <180°


Question 8 12 points

Using the **valence shell orbitals only** draw a **complete** molecular orbital (MO) diagram (with no s-p mixing) for the molecule **PS**, in the space provided below. **(6 points)**

Assume that 3s and 3p atomic orbitals will form molecular orbitals like those formed by the combination of 2s and 2p atomic orbitals that we learnt in class, and there is no s-p mixing.

Show **energy arrow**, **atomic orbitals** involved, **atom labels**, the **molecular orbitals** that are formed, **filled electrons** in the MO diagram etc. Make sure to label all the atomic and molecular orbitals.

Using the MO diagram above, answer the following questions:

- (i) Which one of the following will have the highest bond order? Circle the correct choice
 A) PS^{1-} B) PS C) PS^{1+}
- (ii) Which one of the following will have the longest bond length? Circle the correct choice
 A) PS^{1-} B) PS C) PS^{1+}
- (iii) Which of the following will be diamagnetic? Circle the correct choice.
 A) PS^{1-} B) PS C) PS^{1+}
- (iv) Draw Lewis dot structure of the molecule with the shortest bond length (based on the MO diagram) from among the three (PS^{1-} , PS, PS^{1+}), and indicate the bond order calculated from the Lewis dot structure. **(3 points)**

Lewis Dot Structure

Bond Order



72007D00-48BB-4BBE-B24F-E49BDC52225E

midterm-exam-93388

#703 Page 10 of 10

SCRAP PAGE

**CHEMISTRY 110 –Midterm Exam****Thursday November 7, 2024, 6:30 – 8:30 pm****Examiners: Pallavi Sirjoosingh and Ashok Kakkar**

First name (please write as legibly as possible within the boxes)

--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--

Last name

--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--

Student ID number

--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--

This booklet has 8 questions on 10 pages (The last page can be used as a scrap page).

The staple must NOT be removed from the question booklet. Your intact question booklet must be returned at the end of the exam.

A separate booklet contains a periodic table, and datasheet.

Write your answers directly under the questions in the space provided. Do not write on the QR codes or near the top edges (as it may be cut off).

Please write **legibly and dark enough** for the scan to be readable. For students who use English as a second language, dictionaries are permitted. Calculators are allowed.

DO NOT ASK QUESTIONS OF THE INVIGILATORS about interpretations of the exam questions. If you have a concern about a question, please summarize it at the top of this page. All concerns will be evaluated after the exam. GOOD LUCK!

**Question 1 8 points** (*Assume all transitions are possible*)

a. 6 points Calculate the **energy (in kJ/mol)** required to ionize (completely remove) the electron from the **n=3** level of **He⁺ ion** (He: Helium). Show your work including equations. (3 sig. figs.)

b. 2 points

(I) Select the largest (longest) wavelength from the three choices below:

- i. Wavelength of photon to *completely remove* an electron from *1s orbital of Hydrogen atom*
- ii. Wavelength of photon to *completely remove* an electron from *1s orbital of Boron atom (B)*
- iii. Wavelength of photon to *completely remove* an electron from *1s orbital of Boron cation (B⁺)*

i or ii or iii

(II) Select the smallest (shortest) wavelength from the three choices below (*for H atom*):

- i. Wavelength of photon *emitted* for an electron transitioning from *2p to 1s orbital*
- ii. Wavelength of photon *absorbed* for an electron transitioning from *2s to 4p orbital*
- iii. Wavelength of photon *absorbed* for an electron transitioning from *1s to 3p orbital*

i or ii or iii



Question 2 8 points



A Chem110 student plots the graph shown on the left, from data obtained for the kinetic energy (eV) of the ejected electron versus frequency (s^{-1}) of the incident light, for an unknown metal. ($1\text{eV} = 1.6022 \times 10^{-19} \text{ J}$)

Based on the plot, answer the questions **a to c** below:

a. 1 point On the plot above, label the **threshold frequency**.

b. 2 points What is the value of the work potential (in J) of the unknown metal?

c. 3 points What is the *longest* wavelength (in nm) required to eject an electron from this metal's surface? Show your work including equation(s) used. (3 sig figs)

d. 2 points In the space provided below, plot the following

(Assume frequency is *above* threshold frequency):

- Kinetic energy* of an ejected electron versus the *intensity* of the incident light (left)
- Number of electrons* ejected versus the *frequency* of the incident light (right)



**Question 3** **6 points****a. 2 points**

In the ground state of Ge (Germanium; $Z = 32$) atom, **name** the following orbitals:

i. Orbital containing the electron that requires the maximum energy to be removed.

ii. Orbital containing the electron that experiences the minimum effective nuclear charge.

b. 2 points

An unknown anion (A^-) has the following *excited* electronic configuration: $[Ar] 4s^2 3d^3 4p^1$

(Ar: Argon; $Z = 18$)

Give the **condensed ground state electronic configuration** of the atom, **A** and the cation, **A^{2+}** .
Show your work. (Assume all electronic transitions are possible)

Electronic Configuration of A:

Electronic Configuration of A^{2+} :

c. 2 points

In a single atom, determine how many electrons can have the following set of quantum numbers?

(i) $n = 5; l = 3$

In a single atom, determine how many electrons can have the following set of quantum numbers?

(i) $n = 3; l = 2; m_s = +1/2$

**Question 4** **8 points**

Rank the following from 1 to 4 in **increasing** order. Provide the ranking in the boxes provided.

Note: Atomic number (Z) are provided for relevant atoms

a) Increasing (lowest to highest) order of *ionization energy*: (1= lowest; 4 = highest)

Mg

Mg: Z = 12

Rb

Rb: Z = 37

Cs

Cs: Z = 55

K

K : Z = 19

b) Increasing (lowest to highest) order of *bond angle*: (1= lowest; 4 = highest)

H₂OCCl₄XeF₄

HCN

c) Increasing (smallest to largest) order of *ionic size* (1= smallest; 4 = largest)

S²⁻

S: Z = 16

O²⁻

O: Z = 8

Be²⁺

Be: Z = 4

Na⁺

Na: Z = 11

d) Increasing (lowest to highest) order of *bond energy* (1= lowest; 4 = highest)

P-F

P-Cl

P-I

P-Br

**Question 5** **12 points**

a. 9 points For the molecules HClO_4 and TeF_6 , draw the most stable *Lewis structure* showing appropriate shape and *dashed/wedged bonds*, based on VSEPR. Determine the **molecular geometry** around the central atom. For HClO_4 , label the *most polar bond*.



b. 3 points (Note: Lone pairs are not shown in structures below)

(i) For the three *resonance* structures (1, 2 and 3) shown below, answer the following question:



(I) Circle the *most contributing* resonance structure.

Structure 1

Structure 2

Structure 3

(ii) For the three structures given (4, 5, and 6) below, answer the following questions:



(I) Circle all the structures that represent *resonance structures* of each other:

Structure 4

Structure 5

Structure 6

(II) Among the resonance structures, circle the *most contributing* resonance structure.

Structure 4

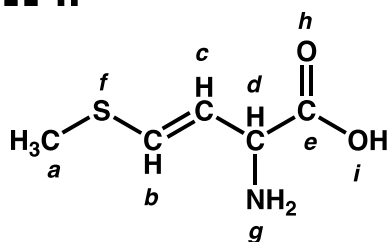
Structure 5

Structure 6

**Question 6** **6 points**

Using the data given below, calculate the **lattice energy** for SrO(s).

	ΔH° (kJ/mol)
$\text{Sr}(s) \longrightarrow \text{Sr}(g)$	+180
$\text{Sr}(g) \longrightarrow \text{Sr}^{2+}(g) + 2e^-$	+1468
$\text{O}_2(g) \longrightarrow 2\text{O}(g)$	+500
$\text{O}(g) + 2e^- \longrightarrow \text{O}^{2-}(g)$	+600
$\text{Sr}(s) + \frac{1}{2}\text{O}_2(g) \longrightarrow \text{SrO}(s)$	-550

**Question 7 10 points**

Consider the molecule given on the left, and answer questions (i) to (vi) using chemical bonding theories that we have learnt in class (valence bond theory, hybridization).

Note that the lone pairs are not shown in the structure:

i. Consider the carbon atoms marked in as ***a***, ***b***, ***c***, ***d***, and ***e*** in the molecule, and determine **hybridization** at each of the carbon centers. (2.5 points)

Atom	Hybridization
Carbon (<i>a</i>)	
Carbon (<i>b</i>)	
Carbon (<i>c</i>)	
Carbon (<i>d</i>)	
Carbon (<i>e</i>)	

ii. Now, consider the sulfur atom (S) marked as ***f***, nitrogen (N) atom (***g***), and oxygen(O) atoms (***h*** and ***i***), and determine their **hybridization**. (2 points)

Atom	Hybridization
Sulfur (<i>f</i>)	
Nitrogen (<i>g</i>)	
Oxygen (<i>h</i>)	
Oxygen (<i>i</i>)	

iii. **Describe** all the bonds formed by the carbon atoms ***a*** and ***b***, using sigma and pi designation as appropriate. (2 points)

Atom	Bonds formed (use sigma and pi as appropriate)
Carbon (<i>a</i>)	
Carbon (<i>b</i>)	

iv. How many total sp^3 -s sigma bonds are in the molecule? (1 point)

v. Determine **electronic** and **molecular geometry** at carbon (C) labeled ***e*** and sulfur (S) labeled ***f*** centres. (2 points)

Atom	Electronic Geometry	Molecular Geometry
Carbon (<i>e</i>)		
Sulfur (<i>f</i>)		

vi. Circle the value from the choices given below, that describes the **C-S-C bond angle at sulfur** (S) marked ***f***: (0.5 points)

>90° to <109.5°

109.5°

>109.5 to <120°

120°

>120° to <180°


Question 8 12 points

Using the **valence shell orbitals only** draw a **complete** molecular orbital (MO) diagram (with no s-p mixing) for the molecule **PS**, in the space provided below. **(6 points)**

Assume that 3s and 3p atomic orbitals will form molecular orbitals like those formed by the combination of 2s and 2p atomic orbitals that we learnt in class, and there is no s-p mixing.

Show **energy arrow**, **atomic orbitals** involved, **atom labels**, the **molecular orbitals** that are formed, **filled electrons** in the MO diagram etc. Make sure to label all the atomic and molecular orbitals.

Using the MO diagram above, answer the following questions:

- (i) Which one of the following will have the highest bond order? Circle the correct choice
 A) PS^{1-} B) PS C) PS^{1+}
- (ii) Which one of the following will have the longest bond length? Circle the correct choice
 A) PS^{1-} B) PS C) PS^{1+}
- (iii) Which of the following will be diamagnetic? Circle the correct choice.
 A) PS^{1-} B) PS C) PS^{1+}
- (iv) Draw Lewis dot structure of the molecule with the shortest bond length (based on the MO diagram) from among the three (PS^{1-} , PS, PS^{1+}), and indicate the bond order calculated from the Lewis dot structure. **(3 points)**

Lewis Dot Structure

Bond Order



64E5E7AA-9F11-4AAE-A688-C3B740248A57

midterm-exam-93388

#704 Page 10 of 10

SCRAP PAGE

**CHEMISTRY 110 –Midterm Exam****Thursday November 7, 2024, 6:30 – 8:30 pm**

Examiners: Pallavi Sirjoosingh and Ashok Kakkar

First name (please write as legibly as possible within the boxes)

--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--

Last name

--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--

Student ID number

--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--

This booklet has 8 questions on 10 pages (The last page can be used as a scrap page).

The staple must NOT be removed from the question booklet. Your intact question booklet must be returned at the end of the exam.

A separate booklet contains a periodic table, and datasheet.

Write your answers directly under the questions in the space provided. Do not write on the QR codes or near the top edges (as it may be cut off).

Please write **legibly and dark enough** for the scan to be readable. For students who use English as a second language, dictionaries are permitted. Calculators are allowed.

DO NOT ASK QUESTIONS OF THE INVIGILATORS about interpretations of the exam questions. If you have a concern about a question, please summarize it at the top of this page. All concerns will be evaluated after the exam. GOOD LUCK!

**Question 1 8 points** (*Assume all transitions are possible*)

a. 6 points Calculate the **energy (in kJ/mol)** required to ionize (completely remove) the electron from the **n=3** level of **He⁺ ion** (He: Helium). Show your work including equations. (3 sig. figs.)

b. 2 points

(I) Select the largest (longest) wavelength from the three choices below:

- i. Wavelength of photon to *completely remove* an electron from *1s orbital of Hydrogen atom*
- ii. Wavelength of photon to *completely remove* an electron from *1s orbital of Boron atom (B)*
- iii. Wavelength of photon to *completely remove* an electron from *1s orbital of Boron cation (B⁺)*

i or ii or iii

(II) Select the smallest (shortest) wavelength from the three choices below (*for H atom*):

- i. Wavelength of photon *emitted* for an electron transitioning from *2p to 1s orbital*
- ii. Wavelength of photon *absorbed* for an electron transitioning from *2s to 4p orbital*
- iii. Wavelength of photon *absorbed* for an electron transitioning from *1s to 3p orbital*

i or ii or iii



Question 2 8 points



A Chem110 student plots the graph shown on the left, from data obtained for the kinetic energy (eV) of the ejected electron versus frequency (s^{-1}) of the incident light, for an unknown metal. ($1\text{eV} = 1.6022 \times 10^{-19} \text{ J}$)

Based on the plot, answer the questions **a to c** below:

a. 1 point On the plot above, label the **threshold frequency**.

b. 2 points What is the value of the work potential (in J) of the unknown metal?

c. 3 points What is the *longest* wavelength (in nm) required to eject an electron from this metal's surface? Show your work including equation(s) used. (3 sig figs)

d. 2 points In the space provided below, plot the following

(Assume frequency is *above* threshold frequency):

- i. *Kinetic energy* of an ejected electron versus the *intensity* of the incident light (left)
- ii. *Number of electrons* ejected versus the *frequency* of the incident light (right)



**Question 3** **6 points****a. 2 points**

In the ground state of Ge (Germanium; $Z = 32$) atom, **name** the following orbitals:

i. Orbital containing the electron that requires the maximum energy to be removed.

ii. Orbital containing the electron that experiences the minimum effective nuclear charge.

b. 2 points

An unknown anion (A^-) has the following *excited* electronic configuration: $[Ar] 4s^2 3d^3 4p^1$

(Ar: Argon; $Z = 18$)

Give the **condensed ground state electronic configuration** of the atom, **A** and the cation, **A^{2+}** .
Show your work. (Assume all electronic transitions are possible)

Electronic Configuration of A:

Electronic Configuration of A^{2+} :

c. 2 points

In a single atom, determine how many electrons can have the following set of quantum numbers?

(i) $n = 5; l = 3$

In a single atom, determine how many electrons can have the following set of quantum numbers?

(i) $n = 3; l = 2; m_s = +1/2$

**Question 4** **8 points**

Rank the following from 1 to 4 in **increasing** order. Provide the ranking in the boxes provided.

Note: Atomic number (Z) are provided for relevant atoms

a) Increasing (lowest to highest) order of *ionization energy*: (1= lowest; 4 = highest)

Mg

Mg: Z = 12

Rb

Rb: Z = 37

Cs

Cs: Z = 55

K

K : Z = 19

b) Increasing (lowest to highest) order of *bond angle*: (1= lowest; 4 = highest)

H₂OCCl₄XeF₄

HCN

c) Increasing (smallest to largest) order of *ionic size* (1= smallest; 4 = largest)

S²⁻

S: Z = 16

O²⁻

O: Z = 8

Be²⁺

Be: Z = 4

Na⁺

Na: Z = 11

d) Increasing (lowest to highest) order of *bond energy* (1= lowest; 4 = highest)

P-F

P-Cl

P-I

P-Br

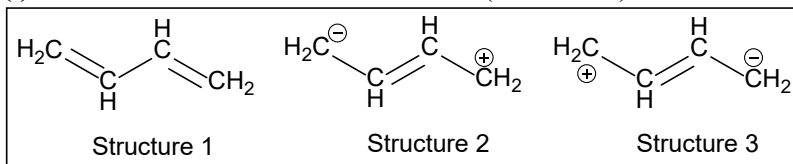
**Question 5** **12 points**

a. 9 points For the molecules HClO_4 and TeF_6 , draw the most stable *Lewis structure* showing appropriate shape and *dashed/wedged bonds*, based on VSEPR. Determine the **molecular geometry** around the central atom. For HClO_4 , label the *most polar bond*.



b. 3 points (Note: Lone pairs are not shown in structures below)

(i) For the three *resonance* structures (1, 2 and 3) shown below, answer the following question:



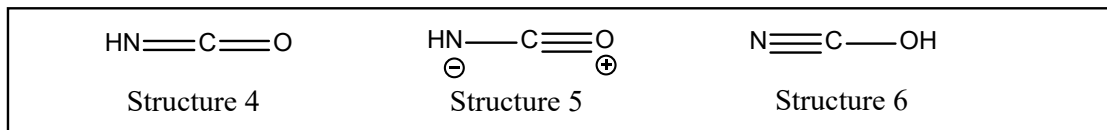
(I) Circle the *most contributing* resonance structure.

Structure 1

Structure 2

Structure 3

(ii) For the three structures given (4, 5, and 6) below, answer the following questions:



(I) Circle all the structures that represent *resonance structures* of each other:

Structure 4

Structure 5

Structure 6

(II) Among the resonance structures, circle the *most contributing* resonance structure.

Structure 4

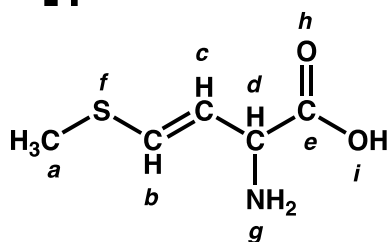
Structure 5

Structure 6

**Question 6** **6 points**

Using the data given below, calculate the **lattice energy** for SrO(s).

	ΔH° (kJ/mol)
$\text{Sr}(s) \longrightarrow \text{Sr}(g)$	+180
$\text{Sr}(g) \longrightarrow \text{Sr}^{2+}(g) + 2e^-$	+1468
$\text{O}_2(g) \longrightarrow 2\text{O}(g)$	+500
$\text{O}(g) + 2e^- \longrightarrow \text{O}^{2-}(g)$	+600
$\text{Sr}(s) + \frac{1}{2}\text{O}_2(g) \longrightarrow \text{SrO}(s)$	-550

**Question 7 10 points**

Consider the molecule given on the left, and answer questions (i) to (vi) using chemical bonding theories that we have learnt in class (valence bond theory, hybridization).

Note that the lone pairs are not shown in the structure:

i. Consider the carbon atoms marked in as ***a***, ***b***, ***c***, ***d***, and ***e*** in the molecule, and determine **hybridization** at each of the carbon centers. (2.5 points)

Atom	Hybridization
Carbon (<i>a</i>)	
Carbon (<i>b</i>)	
Carbon (<i>c</i>)	
Carbon (<i>d</i>)	
Carbon (<i>e</i>)	

ii. Now, consider the sulfur atom (S) marked as ***f***, nitrogen (N) atom (***g***), and oxygen(O) atoms (***h*** and ***i***), and determine their **hybridization**. (2 points)

Atom	Hybridization
Sulfur (<i>f</i>)	
Nitrogen (<i>g</i>)	
Oxygen (<i>h</i>)	
Oxygen (<i>i</i>)	

iii. **Describe** all the bonds formed by the carbon atoms ***a*** and ***b***, using sigma and pi designation as appropriate. (2 points)

Atom	Bonds formed (use sigma and pi as appropriate)
Carbon (<i>a</i>)	
Carbon (<i>b</i>)	

iv. How many total sp^3 -s sigma bonds are in the molecule? (1 point)

v. Determine **electronic** and **molecular geometry** at carbon (C) labeled ***e*** and sulfur (S) labeled ***f*** centres. (2 points)

Atom	Electronic Geometry	Molecular Geometry
Carbon (<i>e</i>)		
Sulfur (<i>f</i>)		

vi. Circle the value from the choices given below, that describes the **C-S-C bond angle at sulfur** (S) marked ***f***: (0.5 points)

>90° to <109.5°

109.5°

>109.5 to <120°

120°

>120° to <180°

**Question 8 12 points**

Using the **valence shell orbitals only** draw a **complete** molecular orbital (MO) diagram (with no s-p mixing) for the molecule **PS**, in the space provided below. **(6 points)**

Assume that 3s and 3p atomic orbitals will form molecular orbitals like those formed by the combination of 2s and 2p atomic orbitals that we learnt in class, and there is no s-p mixing.

Show **energy arrow**, **atomic orbitals** involved, **atom labels**, the **molecular orbitals** that are formed, **filled electrons** in the MO diagram etc. Make sure to label all the atomic and molecular orbitals.

Using the MO diagram above, answer the following questions:

- (i) Which one of the following will have the highest bond order? Circle the correct choice
A) PS^{1-} B) PS C) PS^{1+}
- (ii) Which one of the following will have the longest bond length? Circle the correct choice
A) PS^{1-} B) PS C) PS^{1+}
- (iii) Which of the following will be diamagnetic? Circle the correct choice.
A) PS^{1-} B) PS C) PS^{1+}
- (iv) Draw Lewis dot structure of the molecule with the shortest bond length (based on the MO diagram) from among the three (PS^{1-} , PS, PS^{1+}), and indicate the bond order calculated from the Lewis dot structure. **(3 points)**

Lewis Dot Structure

Bond Order



E7C78AA2-35ED-4FF9-A171-29D55A8B7C21

midterm-exam-93388

#705 Page 10 of 10

SCRAP PAGE

**CHEMISTRY 110 –Midterm Exam****Thursday November 7, 2024, 6:30 – 8:30 pm**

Examiners: Pallavi Sirjoosingh and Ashok Kakkar

First name (please write as legibly as possible within the boxes)

--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--

Last name

--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--

Student ID number

--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--

This booklet has 8 questions on 10 pages (The last page can be used as a scrap page).

The staple must NOT be removed from the question booklet. Your intact question booklet must be returned at the end of the exam.

A separate booklet contains a periodic table, and datasheet.

Write your answers directly under the questions in the space provided. Do not write on the QR codes or near the top edges (as it may be cut off).

Please write **legibly and dark enough** for the scan to be readable. For students who use English as a second language, dictionaries are permitted. Calculators are allowed.

DO NOT ASK QUESTIONS OF THE INVIGILATORS about interpretations of the exam questions. If you have a concern about a question, please summarize it at the top of this page. All concerns will be evaluated after the exam. GOOD LUCK!

**Question 1 8 points** (*Assume all transitions are possible*)

a. 6 points Calculate the **energy (in kJ/mol)** required to ionize (completely remove) the electron from the **$n=3$** level of **He^+ ion** (He: Helium). Show your work including equations. (3 sig. figs.)

b. 2 points

(I) Select the largest (longest) wavelength from the three choices below:

- i. Wavelength of photon to *completely remove* an electron from *1s orbital of Hydrogen atom*
- ii. Wavelength of photon to *completely remove* an electron from *1s orbital of Boron atom (B)*
- iii. Wavelength of photon to *completely remove* an electron from *1s orbital of Boron cation (B^+)*

i or ii or iii

(II) Select the smallest (shortest) wavelength from the three choices below (*for H atom*):

- i. Wavelength of photon *emitted* for an electron transitioning from *2p to 1s orbital*
- ii. Wavelength of photon *absorbed* for an electron transitioning from *2s to 4p orbital*
- iii. Wavelength of photon *absorbed* for an electron transitioning from *1s to 3p orbital*

i or ii or iii



Question 2 8 points



A Chem110 student plots the graph shown on the left, from data obtained for the kinetic energy (eV) of the ejected electron versus frequency (s^{-1}) of the incident light, for an unknown metal. ($1\text{eV} = 1.6022 \times 10^{-19} \text{ J}$)

Based on the plot, answer the questions **a to c** below:

a. 1 point On the plot above, label the **threshold frequency**.

b. 2 points What is the value of the work potential (in J) of the unknown metal?

c. 3 points What is the *longest* wavelength (in nm) required to eject an electron from this metal's surface? Show your work including equation(s) used. (3 sig figs)

d. 2 points In the space provided below, plot the following

(Assume frequency is *above* threshold frequency):

- Kinetic energy* of an ejected electron versus the *intensity* of the incident light (left)
- Number of electrons* ejected versus the *frequency* of the incident light (right)



**Question 3** **6 points****a. 2 points**

In the ground state of Ge (Germanium; $Z = 32$) atom, **name** the following orbitals:

i. Orbital containing the electron that requires the maximum energy to be removed.

ii. Orbital containing the electron that experiences the minimum effective nuclear charge.

b. 2 points

An unknown anion (A^-) has the following *excited* electronic configuration: $[Ar] 4s^2 3d^3 4p^1$

(Ar: Argon; $Z = 18$)

Give the **condensed ground state electronic configuration** of the atom, **A** and the cation, **A^{2+}** .
Show your work. (Assume all electronic transitions are possible)

Electronic Configuration of A:

Electronic Configuration of A^{2+} :

c. 2 points

In a single atom, determine how many electrons can have the following set of quantum numbers?

(i) $n = 5; l = 3$

In a single atom, determine how many electrons can have the following set of quantum numbers?

(i) $n = 3; l = 2; m_s = +1/2$

**Question 4** **8 points**

Rank the following from 1 to 4 in **increasing** order. Provide the ranking in the boxes provided.

Note: Atomic number (Z) are provided for relevant atoms

a) Increasing (lowest to highest) order of *ionization energy*: (1= lowest; 4 = highest)

Mg

Mg: Z = 12

Rb

Rb: Z = 37

Cs

Cs: Z = 55

K

K : Z = 19

b) Increasing (lowest to highest) order of *bond angle*: (1= lowest; 4 = highest)

H₂OCCl₄XeF₄

HCN

c) Increasing (smallest to largest) order of *ionic size* (1= smallest; 4 = largest)

S²⁻

S: Z = 16

O²⁻

O: Z = 8

Be²⁺

Be: Z = 4

Na⁺

Na: Z = 11

d) Increasing (lowest to highest) order of *bond energy* (1= lowest; 4 = highest)

P-F

P-Cl

P-I

P-Br

**Question 5** **12 points**

a. 9 points For the molecules HClO_4 and TeF_6 , draw the most stable *Lewis structure* showing appropriate shape and *dashed/wedged bonds*, based on VSEPR. Determine the **molecular geometry** around the central atom. For HClO_4 , label the *most polar bond*.

HClO_4 (there are no O-O bonds)

Label the most polar bond

Molecular Geometry around Cl

TeF_6

Molecular Geometry around Te

b. 3 points (Note: Lone pairs are not shown in structures below)

(i) For the three *resonance* structures (1, 2 and 3) shown below, answer the following question:



(I) Circle the *most contributing* resonance structure.

Structure 1

Structure 2

Structure 3

(ii) For the three structures given (4, 5, and 6) below, answer the following questions:



(I) Circle all the structures that represent *resonance structures* of each other:

Structure 4

Structure 5

Structure 6

(II) Among the resonance structures, circle the *most contributing* resonance structure.

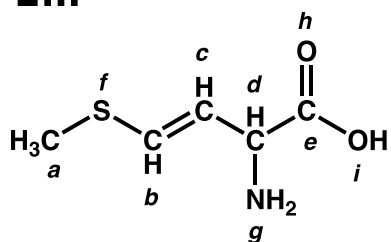
Structure 4

Structure 5

Structure 6

**Question 6** **6 points**Using the data given below, calculate the **lattice energy** for SrO(s).

	ΔH° (kJ/mol)
$\text{Sr}(s) \longrightarrow \text{Sr}(g)$	+180
$\text{Sr}(g) \longrightarrow \text{Sr}^{2+}(g) + 2e^-$	+1468
$\text{O}_2(g) \longrightarrow 2\text{O}(g)$	+500
$\text{O}(g) + 2e^- \longrightarrow \text{O}^{2-}(g)$	+600
$\text{Sr}(s) + \frac{1}{2}\text{O}_2(g) \longrightarrow \text{SrO}(s)$	-550

**Question 7 10 points**

Consider the molecule given on the left, and answer questions (i) to (vi) using chemical bonding theories that we have learnt in class (valence bond theory, hybridization).

Note that the lone pairs are not shown in the structure:

i. Consider the carbon atoms marked in as ***a***, ***b***, ***c***, ***d***, and ***e*** in the molecule, and determine **hybridization** at each of the carbon centers. (2.5 points)

Atom	Hybridization
Carbon (<i>a</i>)	
Carbon (<i>b</i>)	
Carbon (<i>c</i>)	
Carbon (<i>d</i>)	
Carbon (<i>e</i>)	

ii. Now, consider the sulfur atom (S) marked as ***f***, nitrogen (N) atom (***g***), and oxygen(O) atoms (***h*** and ***i***), and determine their **hybridization**. (2 points)

Atom	Hybridization
Sulfur (<i>f</i>)	
Nitrogen (<i>g</i>)	
Oxygen (<i>h</i>)	
Oxygen (<i>i</i>)	

iii. **Describe** all the bonds formed by the carbon atoms ***a*** and ***b***, using sigma and pi designation as appropriate. (2 points)

Atom	Bonds formed (use sigma and pi as appropriate)
Carbon (<i>a</i>)	
Carbon (<i>b</i>)	

iv. How many total sp^3 -s sigma bonds are in the molecule? (1 point)

v. Determine **electronic** and **molecular geometry** at carbon (C) labeled ***e*** and sulfur (S) labeled ***f*** centres. (2 points)

Atom	Electronic Geometry	Molecular Geometry
Carbon (<i>e</i>)		
Sulfur (<i>f</i>)		

vi. Circle the value from the choices given below, that describes the **C-S-C bond angle at sulfur** (S) marked ***f***: (0.5 points)

>90° to <109.5°

109.5°

>109.5 to <120°

120°

>120° to <180°


Question 8 12 points

Using the **valence shell orbitals only** draw a **complete** molecular orbital (MO) diagram (with no s-p mixing) for the molecule **PS**, in the space provided below. **(6 points)**

Assume that 3s and 3p atomic orbitals will form molecular orbitals like those formed by the combination of 2s and 2p atomic orbitals that we learnt in class, and there is no s-p mixing.

Show **energy arrow**, **atomic orbitals** involved, **atom labels**, the **molecular orbitals** that are formed, **filled electrons** in the MO diagram etc. Make sure to label all the atomic and molecular orbitals.

Using the MO diagram above, answer the following questions:

- (i) Which one of the following will have the highest bond order? Circle the correct choice
 A) PS^{1-} B) PS C) PS^{1+}
- (ii) Which one of the following will have the longest bond length? Circle the correct choice
 A) PS^{1-} B) PS C) PS^{1+}
- (iii) Which of the following will be diamagnetic? Circle the correct choice.
 A) PS^{1-} B) PS C) PS^{1+}
- (iv) Draw Lewis dot structure of the molecule with the shortest bond length (based on the MO diagram) from among the three (PS^{1-} , PS, PS^{1+}), and indicate the bond order calculated from the Lewis dot structure. **(3 points)**

Lewis Dot Structure

Bond Order



967F0BC9-EFD5-4C79-A21A-49B83EEB9E2D

midterm-exam-93388

#706 Page 10 of 10

SCRAP PAGE

**CHEMISTRY 110 –Midterm Exam****Thursday November 7, 2024, 6:30 – 8:30 pm****Examiners: Pallavi Sirjoosingh and Ashok Kakkar**

First name (please write as legibly as possible within the boxes)

--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--

Last name

--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--

Student ID number

--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--

This booklet has 8 questions on 10 pages (The last page can be used as a scrap page).

The staple must NOT be removed from the question booklet. Your intact question booklet must be returned at the end of the exam.

A separate booklet contains a periodic table, and datasheet.

Write your answers directly under the questions in the space provided. Do not write on the QR codes or near the top edges (as it may be cut off).

Please write **legibly and dark enough** for the scan to be readable. For students who use English as a second language, dictionaries are permitted. Calculators are allowed.

DO NOT ASK QUESTIONS OF THE INVIGILATORS about interpretations of the exam questions. If you have a concern about a question, please summarize it at the top of this page. All concerns will be evaluated after the exam. GOOD LUCK!

**Question 1 8 points** (*Assume all transitions are possible*)

a. 6 points Calculate the **energy (in kJ/mol)** required to ionize (completely remove) the electron from the **n=3** level of **He⁺ ion** (He: Helium). Show your work including equations. (3 sig. figs.)

b. 2 points

(I) Select the largest (longest) wavelength from the three choices below:

- i. Wavelength of photon to *completely remove* an electron from *1s orbital of Hydrogen atom*
- ii. Wavelength of photon to *completely remove* an electron from *1s orbital of Boron atom (B)*
- iii. Wavelength of photon to *completely remove* an electron from *1s orbital of Boron cation (B⁺)*

i or ii or iii

(II) Select the smallest (shortest) wavelength from the three choices below (*for H atom*):

- i. Wavelength of photon *emitted* for an electron transitioning from *2p to 1s orbital*
- ii. Wavelength of photon *absorbed* for an electron transitioning from *2s to 4p orbital*
- iii. Wavelength of photon *absorbed* for an electron transitioning from *1s to 3p orbital*

i or ii or iii



Question 2 8 points



A Chem110 student plots the graph shown on the left, from data obtained for the kinetic energy (eV) of the ejected electron versus frequency (s^{-1}) of the incident light, for an unknown metal. ($1\text{eV} = 1.6022 \times 10^{-19} \text{ J}$)

Based on the plot, answer the questions **a to c** below:

a. 1 point On the plot above, label the **threshold frequency**.

b. 2 points What is the value of the work potential (in J) of the unknown metal?

c. 3 points What is the *longest* wavelength (in nm) required to eject an electron from this metal's surface? Show your work including equation(s) used. (3 sig figs)

d. 2 points In the space provided below, plot the following

(Assume frequency is *above* threshold frequency):

- i. *Kinetic energy* of an ejected electron versus the *intensity* of the incident light (left)
- ii. *Number of electrons* ejected versus the *frequency* of the incident light (right)



**Question 3** **6 points****a. 2 points**

In the ground state of Ge (Germanium; $Z = 32$) atom, **name** the following orbitals:

i. Orbital containing the electron that requires the maximum energy to be removed.

ii. Orbital containing the electron that experiences the minimum effective nuclear charge.

b. 2 points

An unknown anion (A^-) has the following *excited* electronic configuration: $[Ar] 4s^2 3d^3 4p^1$

(Ar: Argon; $Z = 18$)

Give the **condensed ground state electronic configuration** of the atom, **A** and the cation, **A^{2+}** .
Show your work. (Assume all electronic transitions are possible)

Electronic Configuration of A:

Electronic Configuration of A^{2+} :

c. 2 points

In a single atom, determine how many electrons can have the following set of quantum numbers?

(i) $n = 5; l = 3$

In a single atom, determine how many electrons can have the following set of quantum numbers?

(i) $n = 3; l = 2; m_s = +1/2$

**Question 4** **8 points**

Rank the following from 1 to 4 in **increasing** order. Provide the ranking in the boxes provided.

Note: Atomic number (Z) are provided for relevant atoms

a) Increasing (lowest to highest) order of *ionization energy*: (1= lowest; 4 = highest)

Mg

Mg: Z = 12

Rb

Rb: Z = 37

Cs

Cs: Z = 55

K

K : Z = 19

b) Increasing (lowest to highest) order of *bond angle*: (1= lowest; 4 = highest)

H₂OCCl₄XeF₄

HCN

c) Increasing (smallest to largest) order of *ionic size* (1= smallest; 4 = largest)

S²⁻

S: Z = 16

O²⁻

O: Z = 8

Be²⁺

Be: Z = 4

Na⁺

Na: Z = 11

d) Increasing (lowest to highest) order of *bond energy* (1= lowest; 4 = highest)

P-F

P-Cl

P-I

P-Br

**Question 5** **12 points**

a. 9 points For the molecules HClO_4 and TeF_6 , draw the most stable *Lewis structure* showing appropriate shape and *dashed/wedged bonds*, based on VSEPR. Determine the **molecular geometry** around the central atom. For HClO_4 , label the *most polar bond*.



b. 3 points (Note: Lone pairs are not shown in structures below)

(i) For the three *resonance* structures (1, 2 and 3) shown below, answer the following question:



(I) Circle the *most contributing* resonance structure.

Structure 1

Structure 2

Structure 3

(ii) For the three structures given (4, 5, and 6) below, answer the following questions:



(I) Circle all the structures that represent *resonance structures* of each other:

Structure 4

Structure 5

Structure 6

(II) Among the resonance structures, circle the *most contributing* resonance structure.

Structure 4

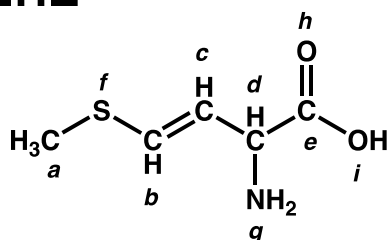
Structure 5

Structure 6

**Question 6** **6 points**

Using the data given below, calculate the **lattice energy** for SrO(s).

	ΔH° (kJ/mol)
$\text{Sr}(s) \longrightarrow \text{Sr}(g)$	+180
$\text{Sr}(g) \longrightarrow \text{Sr}^{2+}(g) + 2e^-$	+1468
$\text{O}_2(g) \longrightarrow 2\text{O}(g)$	+500
$\text{O}(g) + 2e^- \longrightarrow \text{O}^{2-}(g)$	+600
$\text{Sr}(s) + \frac{1}{2}\text{O}_2(g) \longrightarrow \text{SrO}(s)$	-550

**Question 7 10 points**

Consider the molecule given on the left, and answer questions (i) to (vi) using chemical bonding theories that we have learnt in class (valence bond theory, hybridization).

Note that the lone pairs are not shown in the structure:

i. Consider the carbon atoms marked in as ***a***, ***b***, ***c***, ***d***, and ***e*** in the molecule, and determine **hybridization** at each of the carbon centers. (2.5 points)

Atom	Hybridization
Carbon (<i>a</i>)	
Carbon (<i>b</i>)	
Carbon (<i>c</i>)	
Carbon (<i>d</i>)	
Carbon (<i>e</i>)	

ii. Now, consider the sulfur atom (S) marked as ***f***, nitrogen (N) atom (***g***), and oxygen(O) atoms (***h*** and ***i***), and determine their **hybridization**. (2 points)

Atom	Hybridization
Sulfur (<i>f</i>)	
Nitrogen (<i>g</i>)	
Oxygen (<i>h</i>)	
Oxygen (<i>i</i>)	

iii. **Describe** all the bonds formed by the carbon atoms ***a*** and ***b***, using sigma and pi designation as appropriate. (2 points)

Atom	Bonds formed (use sigma and pi as appropriate)
Carbon (<i>a</i>)	
Carbon (<i>b</i>)	

iv. How many total sp^3 -s sigma bonds are in the molecule? (1 point)

v. Determine **electronic** and **molecular geometry** at carbon (C) labeled ***e*** and sulfur (S) labeled ***f*** centres. (2 points)

Atom	Electronic Geometry	Molecular Geometry
Carbon (<i>e</i>)		
Sulfur (<i>f</i>)		

vi. Circle the value from the choices given below, that describes the **C-S-C bond angle at sulfur** (S) marked ***f***: (0.5 points)

>90° to <109.5° 109.5° >109.5 to <120° 120° >120° to <180°

**Question 8 12 points**

Using the **valence shell orbitals only** draw a **complete** molecular orbital (MO) diagram (with no s-p mixing) for the molecule **PS**, in the space provided below. **(6 points)**

Assume that 3s and 3p atomic orbitals will form molecular orbitals like those formed by the combination of 2s and 2p atomic orbitals that we learnt in class, and there is no s-p mixing.

Show **energy arrow**, **atomic orbitals** involved, **atom labels**, the **molecular orbitals** that are formed, **filled electrons** in the MO diagram etc. Make sure to label all the atomic and molecular orbitals.

Using the MO diagram above, answer the following questions:

- (i) Which one of the following will have the highest bond order? Circle the correct choice
A) PS^{1-} B) PS C) PS^{1+}
- (ii) Which one of the following will have the longest bond length? Circle the correct choice
A) PS^{1-} B) PS C) PS^{1+}
- (iii) Which of the following will be diamagnetic? Circle the correct choice.
A) PS^{1-} B) PS C) PS^{1+}
- (iv) Draw Lewis dot structure of the molecule with the shortest bond length (based on the MO diagram) from among the three (PS^{1-} , PS, PS^{1+}), and indicate the bond order calculated from the Lewis dot structure. **(3 points)**

Lewis Dot Structure

Bond Order



3E339DC6-9F3E-40CF-9EDB-95068CA6211E

midterm-exam-93388

#707 Page 10 of 10

SCRAP PAGE

**CHEMISTRY 110 –Midterm Exam****Thursday November 7, 2024, 6:30 – 8:30 pm**

Examiners: Pallavi Sirjoosingh and Ashok Kakkar

First name (please write as legibly as possible within the boxes)

--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--

Last name

--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--

Student ID number

--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--

This booklet has 8 questions on 10 pages (The last page can be used as a scrap page).

The staple must NOT be removed from the question booklet. Your intact question booklet must be returned at the end of the exam.

A separate booklet contains a periodic table, and datasheet.

Write your answers directly under the questions in the space provided. Do not write on the QR codes or near the top edges (as it may be cut off).

Please write **legibly and dark enough** for the scan to be readable. For students who use English as a second language, dictionaries are permitted. Calculators are allowed.

DO NOT ASK QUESTIONS OF THE INVIGILATORS about interpretations of the exam questions. If you have a concern about a question, please summarize it at the top of this page. All concerns will be evaluated after the exam. GOOD LUCK!

**Question 1 8 points** (*Assume all transitions are possible*)

a. 6 points Calculate the **energy (in kJ/mol)** required to ionize (completely remove) the electron from the **n=3** level of **He⁺ ion** (He: Helium). Show your work including equations. (3 sig. figs.)

b. 2 points

(I) Select the largest (longest) wavelength from the three choices below:

- i. Wavelength of photon to *completely remove* an electron from *1s orbital of Hydrogen atom*
- ii. Wavelength of photon to *completely remove* an electron from *1s orbital of Boron atom (B)*
- iii. Wavelength of photon to *completely remove* an electron from *1s orbital of Boron cation (B⁺)*

i or ii or iii

(II) Select the smallest (shortest) wavelength from the three choices below (*for H atom*):

- i. Wavelength of photon *emitted* for an electron transitioning from *2p to 1s orbital*
- ii. Wavelength of photon *absorbed* for an electron transitioning from *2s to 4p orbital*
- iii. Wavelength of photon *absorbed* for an electron transitioning from *1s to 3p orbital*

i or ii or iii



Question 2 8 points



A Chem110 student plots the graph shown on the left, from data obtained for the kinetic energy (eV) of the ejected electron versus frequency (s^{-1}) of the incident light, for an unknown metal. ($1\text{eV} = 1.6022 \times 10^{-19} \text{ J}$)

Based on the plot, answer the questions **a to c** below:

a. 1 point On the plot above, label the **threshold frequency**.

b. 2 points What is the value of the work potential (in J) of the unknown metal?

c. 3 points What is the *longest* wavelength (in nm) required to eject an electron from this metal's surface? Show your work including equation(s) used. (3 sig figs)

d. 2 points In the space provided below, plot the following

(Assume frequency is *above* threshold frequency):

- Kinetic energy* of an ejected electron versus the *intensity* of the incident light (left)
- Number of electrons* ejected versus the *frequency* of the incident light (right)



**Question 3** **6 points****a. 2 points**

In the ground state of Ge (Germanium; $Z = 32$) atom, **name** the following orbitals:

i. Orbital containing the electron that requires the maximum energy to be removed.

ii. Orbital containing the electron that experiences the minimum effective nuclear charge.

b. 2 points

An unknown anion (A^-) has the following *excited* electronic configuration: $[Ar] 4s^2 3d^3 4p^1$

(Ar: Argon; $Z = 18$)

Give the **condensed ground state electronic configuration** of the atom, **A** and the cation, **A^{2+}** .
Show your work. (Assume all electronic transitions are possible)

Electronic Configuration of A:

Electronic Configuration of A^{2+} :

c. 2 points

In a single atom, determine how many electrons can have the following set of quantum numbers?

(i) $n = 5; l = 3$

In a single atom, determine how many electrons can have the following set of quantum numbers?

(i) $n = 3; l = 2; m_s = +1/2$

**Question 4** **8 points**

Rank the following from 1 to 4 in **increasing** order. Provide the ranking in the boxes provided.

Note: Atomic number (Z) are provided for relevant atoms

a) Increasing (lowest to highest) order of *ionization energy*: (1= lowest; 4 = highest)

Mg

Mg: Z = 12

Rb

Rb: Z = 37

Cs

Cs: Z = 55

K

K : Z = 19

b) Increasing (lowest to highest) order of *bond angle*: (1= lowest; 4 = highest)

H₂OCCl₄XeF₄

HCN

c) Increasing (smallest to largest) order of *ionic size* (1= smallest; 4 = largest)

S²⁻

S: Z = 16

O²⁻

O: Z = 8

Be²⁺

Be: Z = 4

Na⁺

Na: Z = 11

d) Increasing (lowest to highest) order of *bond energy* (1= lowest; 4 = highest)

P-F

P-Cl

P-I

P-Br

**Question 5** **12 points**

a. 9 points For the molecules HClO_4 and TeF_6 , draw the most stable *Lewis structure* showing appropriate shape and *dashed/wedged bonds*, based on VSEPR. Determine the **molecular geometry** around the central atom. For HClO_4 , label the *most polar bond*.



b. 3 points (Note: Lone pairs are not shown in structures below)

(i) For the three *resonance* structures (1, 2 and 3) shown below, answer the following question:



(I) Circle the *most contributing* resonance structure.

Structure 1

Structure 2

Structure 3

(ii) For the three structures given (4, 5, and 6) below, answer the following questions:



(I) Circle all the structures that represent *resonance structures* of each other:

Structure 4

Structure 5

Structure 6

(II) Among the resonance structures, circle the *most contributing* resonance structure.

Structure 4

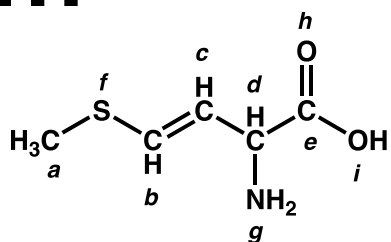
Structure 5

Structure 6

**Question 6** **6 points**

Using the data given below, calculate the **lattice energy** for SrO(s).

	ΔH° (kJ/mol)
$\text{Sr}(s) \longrightarrow \text{Sr}(g)$	+180
$\text{Sr}(g) \longrightarrow \text{Sr}^{2+}(g) + 2e^-$	+1468
$\text{O}_2(g) \longrightarrow 2\text{O}(g)$	+500
$\text{O}(g) + 2e^- \longrightarrow \text{O}^{2-}(g)$	+600
$\text{Sr}(s) + \frac{1}{2}\text{O}_2(g) \longrightarrow \text{SrO}(s)$	-550

**Question 7 10 points**

Consider the molecule given on the left, and answer questions (i) to (vi) using chemical bonding theories that we have learnt in class (valence bond theory, hybridization).

Note that the lone pairs are not shown in the structure:

i. Consider the carbon atoms marked in as ***a***, ***b***, ***c***, ***d***, and ***e*** in the molecule, and determine **hybridization** at each of the carbon centers. (2.5 points)

Atom	Hybridization
Carbon (<i>a</i>)	
Carbon (<i>b</i>)	
Carbon (<i>c</i>)	
Carbon (<i>d</i>)	
Carbon (<i>e</i>)	

ii. Now, consider the sulfur atom (S) marked as ***f***, nitrogen (N) atom (***g***), and oxygen(O) atoms (***h*** and ***i***), and determine their **hybridization**. (2 points)

Atom	Hybridization
Sulfur (<i>f</i>)	
Nitrogen (<i>g</i>)	
Oxygen (<i>h</i>)	
Oxygen (<i>i</i>)	

iii. **Describe** all the bonds formed by the carbon atoms ***a*** and ***b***, using sigma and pi designation as appropriate. (2 points)

Atom	Bonds formed (use sigma and pi as appropriate)
Carbon (<i>a</i>)	
Carbon (<i>b</i>)	

iv. How many total sp^3 -s sigma bonds are in the molecule? (1 point)

v. Determine **electronic** and **molecular geometry** at carbon (C) labeled ***e*** and sulfur (S) labeled ***f*** centres. (2 points)

Atom	Electronic Geometry	Molecular Geometry
Carbon (<i>e</i>)		
Sulfur (<i>f</i>)		

vi. Circle the value from the choices given below, that describes the **C-S-C bond angle at sulfur** (S) marked ***f***: (0.5 points)

>90° to <109.5° 109.5° >109.5 to <120° 120° >120° to <180°

**Question 8 12 points**

Using the **valence shell orbitals only** draw a **complete** molecular orbital (MO) diagram (with no s-p mixing) for the molecule **PS**, in the space provided below. **(6 points)**

Assume that 3s and 3p atomic orbitals will form molecular orbitals like those formed by the combination of 2s and 2p atomic orbitals that we learnt in class, and there is no s-p mixing.

Show **energy arrow**, **atomic orbitals** involved, **atom labels**, the **molecular orbitals** that are formed, **filled electrons** in the MO diagram etc. Make sure to label all the atomic and molecular orbitals.

Using the MO diagram above, answer the following questions:

- (i) Which one of the following will have the highest bond order? Circle the correct choice
A) PS^{1-} B) PS C) PS^{1+}
- (ii) Which one of the following will have the longest bond length? Circle the correct choice
A) PS^{1-} B) PS C) PS^{1+}
- (iii) Which of the following will be diamagnetic? Circle the correct choice.
A) PS^{1-} B) PS C) PS^{1+}
- (iv) Draw Lewis dot structure of the molecule with the shortest bond length (based on the MO diagram) from among the three (PS^{1-} , PS, PS^{1+}), and indicate the bond order calculated from the Lewis dot structure. **(3 points)**

Lewis Dot Structure

Bond Order



CDB56D32-6C81-4C60-8493-DD67DD470EA1

midterm-exam-93388

#708 Page 10 of 10

SCRAP PAGE

**CHEMISTRY 110 –Midterm Exam****Thursday November 7, 2024, 6:30 – 8:30 pm**

Examiners: Pallavi Sirjoosingh and Ashok Kakkar

First name (please write as legibly as possible within the boxes)

--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--

Last name

--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--

Student ID number

--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--

This booklet has 8 questions on 10 pages (The last page can be used as a scrap page).

The staple must NOT be removed from the question booklet. Your intact question booklet must be returned at the end of the exam.

A separate booklet contains a periodic table, and datasheet.

Write your answers directly under the questions in the space provided. Do not write on the QR codes or near the top edges (as it may be cut off).

Please write **legibly and dark enough** for the scan to be readable. For students who use English as a second language, dictionaries are permitted. Calculators are allowed.

DO NOT ASK QUESTIONS OF THE INVIGILATORS about interpretations of the exam questions. If you have a concern about a question, please summarize it at the top of this page. All concerns will be evaluated after the exam. GOOD LUCK!

**Question 1 8 points** (*Assume all transitions are possible*)

a. 6 points Calculate the **energy (in kJ/mol)** required to ionize (completely remove) the electron from the **n=3** level of **He⁺ ion** (He: Helium). Show your work including equations. (3 sig. figs.)

b. 2 points

(I) Select the largest (longest) wavelength from the three choices below:

- i. Wavelength of photon to *completely remove* an electron from *1s orbital of Hydrogen atom*
- ii. Wavelength of photon to *completely remove* an electron from *1s orbital of Boron atom (B)*
- iii. Wavelength of photon to *completely remove* an electron from *1s orbital of Boron cation (B⁺)*

i or ii or iii

(II) Select the smallest (shortest) wavelength from the three choices below (*for H atom*):

- i. Wavelength of photon *emitted* for an electron transitioning from *2p to 1s orbital*
- ii. Wavelength of photon *absorbed* for an electron transitioning from *2s to 4p orbital*
- iii. Wavelength of photon *absorbed* for an electron transitioning from *1s to 3p orbital*

i or ii or iii



Question 2 8 points



A Chem110 student plots the graph shown on the left, from data obtained for the kinetic energy (eV) of the ejected electron versus frequency (s^{-1}) of the incident light, for an unknown metal. ($1\text{eV} = 1.6022 \times 10^{-19} \text{ J}$)

Based on the plot, answer the questions **a to c** below:

a. 1 point On the plot above, label the **threshold frequency**.

b. 2 points What is the value of the work potential (in J) of the unknown metal?

c. 3 points What is the *longest* wavelength (in nm) required to eject an electron from this metal's surface? Show your work including equation(s) used. (3 sig figs)

d. 2 points In the space provided below, plot the following

(Assume frequency is *above* threshold frequency):

- i. *Kinetic energy* of an ejected electron versus the *intensity* of the incident light (left)
- ii. *Number of electrons* ejected versus the *frequency* of the incident light (right)



**Question 3** **6 points****a. 2 points**

In the ground state of Ge (Germanium; $Z = 32$) atom, **name** the following orbitals:

i. Orbital containing the electron that requires the maximum energy to be removed.

ii. Orbital containing the electron that experiences the minimum effective nuclear charge.

b. 2 points

An unknown anion (A^-) has the following *excited* electronic configuration: $[Ar] 4s^2 3d^3 4p^1$

(Ar: Argon; $Z = 18$)

Give the **condensed ground state electronic configuration** of the atom, **A** and the cation, **A^{2+}** .
Show your work. (Assume all electronic transitions are possible)

Electronic Configuration of A:

Electronic Configuration of A^{2+} :

c. 2 points

In a single atom, determine how many electrons can have the following set of quantum numbers?

(i) $n = 5; l = 3$

In a single atom, determine how many electrons can have the following set of quantum numbers?

(i) $n = 3; l = 2; m_s = +1/2$

**Question 4** **8 points**

Rank the following from 1 to 4 in **increasing** order. Provide the ranking in the boxes provided.

Note: Atomic number (Z) are provided for relevant atoms

a) Increasing (lowest to highest) order of *ionization energy*: (1= lowest; 4 = highest)

Mg

Mg: Z = 12

Rb

Rb: Z = 37

Cs

Cs: Z = 55

K

K : Z = 19

b) Increasing (lowest to highest) order of *bond angle*: (1= lowest; 4 = highest)

H₂OCCl₄XeF₄

HCN

c) Increasing (smallest to largest) order of *ionic size* (1= smallest; 4 = largest)

S²⁻

S: Z = 16

O²⁻

O: Z = 8

Be²⁺

Be: Z = 4

Na⁺

Na: Z = 11

d) Increasing (lowest to highest) order of *bond energy* (1= lowest; 4 = highest)

P-F

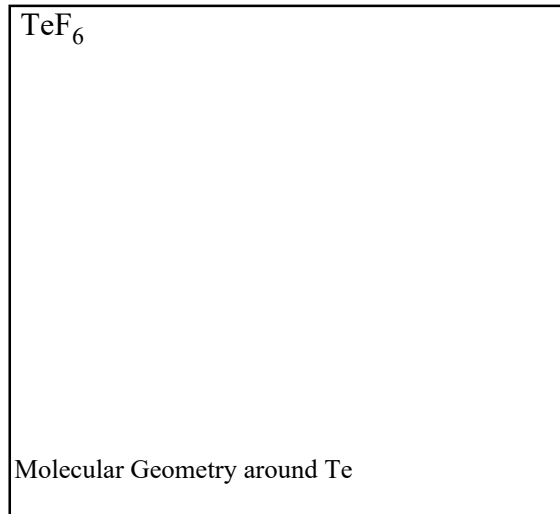
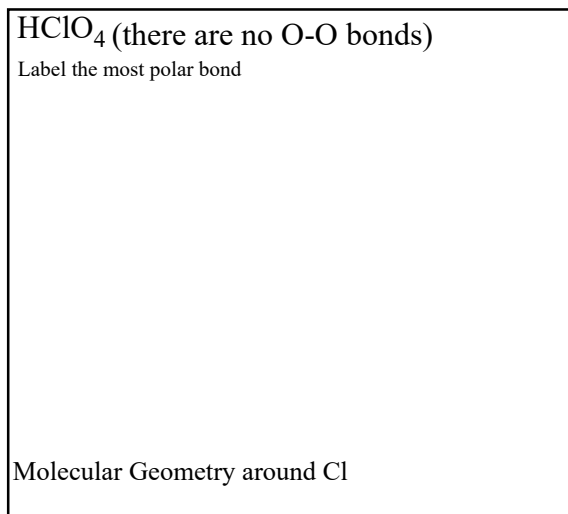
P-Cl

P-I

P-Br

**Question 5** **12 points**

a. 9 points For the molecules HClO_4 and TeF_6 , draw the most stable *Lewis structure* showing appropriate shape and *dashed/wedged bonds*, based on VSEPR. Determine the **molecular geometry** around the central atom. For HClO_4 , label the *most polar bond*.



b. 3 points (Note: Lone pairs are not shown in structures below)

(i) For the three *resonance* structures (1, 2 and 3) shown below, answer the following question:



(I) Circle the *most contributing* resonance structure.

Structure 1

Structure 2

Structure 3

(ii) For the three structures given (4, 5, and 6) below, answer the following questions:



(I) Circle all the structures that represent *resonance structures* of each other:

Structure 4

Structure 5

Structure 6

(II) Among the resonance structures, circle the *most contributing* resonance structure.

Structure 4

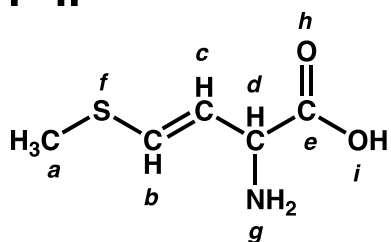
Structure 5

Structure 6

**Question 6** **6 points**

Using the data given below, calculate the **lattice energy** for SrO(s).

	ΔH° (kJ/mol)
$\text{Sr}(s) \longrightarrow \text{Sr}(g)$	+180
$\text{Sr}(g) \longrightarrow \text{Sr}^{2+}(g) + 2e^-$	+1468
$\text{O}_2(g) \longrightarrow 2\text{O}(g)$	+500
$\text{O}(g) + 2e^- \longrightarrow \text{O}^{2-}(g)$	+600
$\text{Sr}(s) + \frac{1}{2}\text{O}_2(g) \longrightarrow \text{SrO}(s)$	-550

**Question 7 10 points**

Consider the molecule given on the left, and answer questions (i) to (vi) using chemical bonding theories that we have learnt in class (valence bond theory, hybridization).

Note that the lone pairs are not shown in the structure:

i. Consider the carbon atoms marked in as ***a***, ***b***, ***c***, ***d***, and ***e*** in the molecule, and determine **hybridization** at each of the carbon centers. (2.5 points)

Atom	Hybridization
Carbon (<i>a</i>)	
Carbon (<i>b</i>)	
Carbon (<i>c</i>)	
Carbon (<i>d</i>)	
Carbon (<i>e</i>)	

ii. Now, consider the sulfur atom (S) marked as ***f***, nitrogen (N) atom (***g***), and oxygen(O) atoms (***h*** and ***i***), and determine their **hybridization**. (2 points)

Atom	Hybridization
Sulfur (<i>f</i>)	
Nitrogen (<i>g</i>)	
Oxygen (<i>h</i>)	
Oxygen (<i>i</i>)	

iii. **Describe** all the bonds formed by the carbon atoms ***a*** and ***b***, using sigma and pi designation as appropriate. (2 points)

Atom	Bonds formed (use sigma and pi as appropriate)
Carbon (<i>a</i>)	
Carbon (<i>b</i>)	

iv. How many total sp^3 -s sigma bonds are in the molecule? (1 point)

v. Determine **electronic** and **molecular geometry** at carbon (C) labeled ***e*** and sulfur (S) labeled ***f*** centres. (2 points)

Atom	Electronic Geometry	Molecular Geometry
Carbon (<i>e</i>)		
Sulfur (<i>f</i>)		

vi. Circle the value from the choices given below, that describes the **C-S-C bond angle at sulfur** (S) marked ***f***: (0.5 points)

$>90^\circ$ to $<109.5^\circ$ 109.5° >109.5 to $<120^\circ$ 120° $>120^\circ$ to $<180^\circ$


Question 8 12 points

Using the **valence shell orbitals only** draw a **complete** molecular orbital (MO) diagram (with no s-p mixing) for the molecule **PS**, in the space provided below. **(6 points)**

Assume that 3s and 3p atomic orbitals will form molecular orbitals like those formed by the combination of 2s and 2p atomic orbitals that we learnt in class, and there is no s-p mixing.

Show **energy arrow**, **atomic orbitals** involved, **atom labels**, the **molecular orbitals** that are formed, **filled electrons** in the MO diagram etc. Make sure to label all the atomic and molecular orbitals.

Using the MO diagram above, answer the following questions:

- (i) Which one of the following will have the highest bond order? Circle the correct choice
 A) PS^{1-} B) PS C) PS^{1+}
- (ii) Which one of the following will have the longest bond length? Circle the correct choice
 A) PS^{1-} B) PS C) PS^{1+}
- (iii) Which of the following will be diamagnetic? Circle the correct choice.
 A) PS^{1-} B) PS C) PS^{1+}
- (iv) Draw Lewis dot structure of the molecule with the shortest bond length (based on the MO diagram) from among the three (PS^{1-} , PS, PS^{1+}), and indicate the bond order calculated from the Lewis dot structure. **(3 points)**

Lewis Dot Structure

Bond Order



E00036C0-EE1E-4962-A3BF-691C982DB569

midterm-exam-93388

#709 Page 10 of 10

SCRAP PAGE

**CHEMISTRY 110 –Midterm Exam****Thursday November 7, 2024, 6:30 – 8:30 pm**

Examiners: Pallavi Sirjoosingh and Ashok Kakkar

First name (please write as legibly as possible within the boxes)

--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--

Last name

--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--

Student ID number

--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--

This booklet has 8 questions on 10 pages (The last page can be used as a scrap page).

The staple must NOT be removed from the question booklet. Your intact question booklet must be returned at the end of the exam.

A separate booklet contains a periodic table, and datasheet.

Write your answers directly under the questions in the space provided. Do not write on the QR codes or near the top edges (as it may be cut off).

Please write **legibly and dark enough** for the scan to be readable. For students who use English as a second language, dictionaries are permitted. Calculators are allowed.

DO NOT ASK QUESTIONS OF THE INVIGILATORS about interpretations of the exam questions. If you have a concern about a question, please summarize it at the top of this page. All concerns will be evaluated after the exam. GOOD LUCK!

**Question 1 8 points** (*Assume all transitions are possible*)

a. 6 points Calculate the **energy (in kJ/mol)** required to ionize (completely remove) the electron from the **n=3** level of **He⁺ ion** (He: Helium). Show your work including equations. (3 sig. figs.)

b. 2 points

(I) Select the largest (longest) wavelength from the three choices below:

- i. Wavelength of photon to *completely remove* an electron from *1s orbital of Hydrogen atom*
- ii. Wavelength of photon to *completely remove* an electron from *1s orbital of Boron atom (B)*
- iii. Wavelength of photon to *completely remove* an electron from *1s orbital of Boron cation (B⁺)*

i or ii or iii

(II) Select the smallest (shortest) wavelength from the three choices below (*for H atom*):

- i. Wavelength of photon *emitted* for an electron transitioning from *2p to 1s orbital*
- ii. Wavelength of photon *absorbed* for an electron transitioning from *2s to 4p orbital*
- iii. Wavelength of photon *absorbed* for an electron transitioning from *1s to 3p orbital*

i or ii or iii



Question 2 8 points



A Chem110 student plots the graph shown on the left, from data obtained for the kinetic energy (eV) of the ejected electron versus frequency (s^{-1}) of the incident light, for an unknown metal. ($1\text{eV} = 1.6022 \times 10^{-19} \text{ J}$)

Based on the plot, answer the questions **a to c** below:

a. 1 point On the plot above, label the **threshold frequency**.

b. 2 points What is the value of the work potential (in J) of the unknown metal?

c. 3 points What is the *longest* wavelength (in nm) required to eject an electron from this metal's surface? Show your work including equation(s) used. (3 sig figs)

d. 2 points In the space provided below, plot the following

(Assume frequency is *above* threshold frequency):

- i. *Kinetic energy* of an ejected electron versus the *intensity* of the incident light (left)
- ii. *Number of electrons* ejected versus the *frequency* of the incident light (right)



**Question 3** **6 points****a. 2 points**

In the ground state of Ge (Germanium; $Z = 32$) atom, **name** the following orbitals:

i. Orbital containing the electron that requires the maximum energy to be removed.

ii. Orbital containing the electron that experiences the minimum effective nuclear charge.

b. 2 points

An unknown anion (A^-) has the following *excited* electronic configuration: $[Ar] 4s^2 3d^3 4p^1$

(Ar: Argon; $Z = 18$)

Give the **condensed ground state electronic configuration** of the atom, **A** and the cation, **A^{2+}** .
Show your work. (Assume all electronic transitions are possible)

Electronic Configuration of A:

Electronic Configuration of A^{2+} :

c. 2 points

In a single atom, determine how many electrons can have the following set of quantum numbers?

(i) $n = 5; l = 3$

In a single atom, determine how many electrons can have the following set of quantum numbers?

(i) $n = 3; l = 2; m_s = +1/2$

**Question 4** **8 points**

Rank the following from 1 to 4 in **increasing** order. Provide the ranking in the boxes provided.

Note: Atomic number (Z) are provided for relevant atoms

a) Increasing (lowest to highest) order of *ionization energy*: (1= lowest; 4 = highest)

Mg

Mg: Z = 12

Rb

Rb: Z = 37

Cs

Cs: Z = 55

K

K : Z = 19

b) Increasing (lowest to highest) order of *bond angle*: (1= lowest; 4 = highest)

H₂OCCl₄XeF₄

HCN

c) Increasing (smallest to largest) order of *ionic size* (1= smallest; 4 = largest)

S²⁻

S: Z = 16

O²⁻

O: Z = 8

Be²⁺

Be: Z = 4

Na⁺

Na: Z = 11

d) Increasing (lowest to highest) order of *bond energy* (1= lowest; 4 = highest)

P-F

P-Cl

P-I

P-Br

**Question 5** **12 points**

a. 9 points For the molecules HClO_4 and TeF_6 , draw the most stable *Lewis structure* showing appropriate shape and *dashed/wedged bonds*, based on VSEPR. Determine the **molecular geometry** around the central atom. For HClO_4 , label the *most polar bond*.



b. 3 points (Note: Lone pairs are not shown in structures below)

(i) For the three *resonance* structures (1, 2 and 3) shown below, answer the following question:



(I) Circle the *most contributing* resonance structure.

Structure 1

Structure 2

Structure 3

(ii) For the three structures given (4, 5, and 6) below, answer the following questions:



(I) Circle all the structures that represent *resonance structures* of each other:

Structure 4

Structure 5

Structure 6

(II) Among the resonance structures, circle the *most contributing* resonance structure.

Structure 4

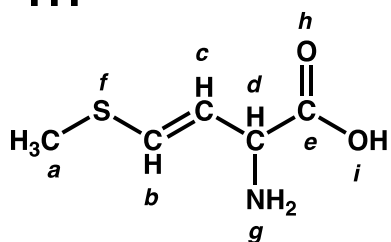
Structure 5

Structure 6

**Question 6** **6 points**

Using the data given below, calculate the **lattice energy** for SrO(s).

	ΔH° (kJ/mol)
$\text{Sr}(s) \longrightarrow \text{Sr}(g)$	+180
$\text{Sr}(g) \longrightarrow \text{Sr}^{2+}(g) + 2e^-$	+1468
$\text{O}_2(g) \longrightarrow 2\text{O}(g)$	+500
$\text{O}(g) + 2e^- \longrightarrow \text{O}^{2-}(g)$	+600
$\text{Sr}(s) + \frac{1}{2}\text{O}_2(g) \longrightarrow \text{SrO}(s)$	-550

**Question 7 10 points**

Consider the molecule given on the left, and answer questions (i) to (vi) using chemical bonding theories that we have learnt in class (valence bond theory, hybridization).

Note that the lone pairs are not shown in the structure:

i. Consider the carbon atoms marked in as ***a***, ***b***, ***c***, ***d***, and ***e*** in the molecule, and determine **hybridization** at each of the carbon centers. (2.5 points)

Atom	Hybridization
Carbon (<i>a</i>)	
Carbon (<i>b</i>)	
Carbon (<i>c</i>)	
Carbon (<i>d</i>)	
Carbon (<i>e</i>)	

ii. Now, consider the sulfur atom (S) marked as ***f***, nitrogen (N) atom (***g***), and oxygen(O) atoms (***h*** and ***i***), and determine their **hybridization**. (2 points)

Atom	Hybridization
Sulfur (<i>f</i>)	
Nitrogen (<i>g</i>)	
Oxygen (<i>h</i>)	
Oxygen (<i>i</i>)	

iii. **Describe** all the bonds formed by the carbon atoms ***a*** and ***b***, using sigma and pi designation as appropriate. (2 points)

Atom	Bonds formed (use sigma and pi as appropriate)
Carbon (<i>a</i>)	
Carbon (<i>b</i>)	

iv. How many total sp^3 -s sigma bonds are in the molecule? (1 point)

v. Determine **electronic** and **molecular geometry** at carbon (C) labeled ***e*** and sulfur (S) labeled ***f*** centres. (2 points)

Atom	Electronic Geometry	Molecular Geometry
Carbon (<i>e</i>)		
Sulfur (<i>f</i>)		

vi. Circle the value from the choices given below, that describes the **C-S-C bond angle at sulfur** (S) marked ***f***: (0.5 points)

>90° to <109.5° 109.5° >109.5 to <120° 120° >120° to <180°


Question 8 12 points

Using the **valence shell orbitals only** draw a **complete** molecular orbital (MO) diagram (with no s-p mixing) for the molecule **PS**, in the space provided below. **(6 points)**

Assume that 3s and 3p atomic orbitals will form molecular orbitals like those formed by the combination of 2s and 2p atomic orbitals that we learnt in class, and there is no s-p mixing.

Show **energy arrow**, **atomic orbitals** involved, **atom labels**, the **molecular orbitals** that are formed, **filled electrons** in the MO diagram etc. Make sure to label all the atomic and molecular orbitals.

Using the MO diagram above, answer the following questions:

- (i) Which one of the following will have the highest bond order? Circle the correct choice
 A) PS^{1-} B) PS C) PS^{1+}
- (ii) Which one of the following will have the longest bond length? Circle the correct choice
 A) PS^{1-} B) PS C) PS^{1+}
- (iii) Which of the following will be diamagnetic? Circle the correct choice.
 A) PS^{1-} B) PS C) PS^{1+}
- (iv) Draw Lewis dot structure of the molecule with the shortest bond length (based on the MO diagram) from among the three (PS^{1-} , PS, PS^{1+}), and indicate the bond order calculated from the Lewis dot structure. **(3 points)**

Lewis Dot Structure

Bond Order



C580C025-9048-44B4-8536-1C1062E95CCC

midterm-exam-93388

#710 Page 10 of 10

SCRAP PAGE

**CHEMISTRY 110 –Midterm Exam****Thursday November 7, 2024, 6:30 – 8:30 pm****Examiners: Pallavi Sirjoosingh and Ashok Kakkar**

First name (please write as legibly as possible within the boxes)

--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--

Last name

--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--

Student ID number

--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--

This booklet has 8 questions on 10 pages (The last page can be used as a scrap page).

The staple must NOT be removed from the question booklet. Your intact question booklet must be returned at the end of the exam.

A separate booklet contains a periodic table, and datasheet.

Write your answers directly under the questions in the space provided. Do not write on the QR codes or near the top edges (as it may be cut off).

Please write **legibly and dark enough** for the scan to be readable. For students who use English as a second language, dictionaries are permitted. Calculators are allowed.

DO NOT ASK QUESTIONS OF THE INVIGILATORS about interpretations of the exam questions. If you have a concern about a question, please summarize it at the top of this page. All concerns will be evaluated after the exam. GOOD LUCK!

**Question 1 8 points** (*Assume all transitions are possible*)

a. 6 points Calculate the **energy (in kJ/mol)** required to ionize (completely remove) the electron from the **n=3** level of **He⁺ ion** (He: Helium). Show your work including equations. (3 sig. figs.)

b. 2 points

(I) Select the largest (longest) wavelength from the three choices below:

- i. Wavelength of photon to *completely remove* an electron from *1s orbital of Hydrogen atom*
- ii. Wavelength of photon to *completely remove* an electron from *1s orbital of Boron atom (B)*
- iii. Wavelength of photon to *completely remove* an electron from *1s orbital of Boron cation (B⁺)*

i or ii or iii

(II) Select the smallest (shortest) wavelength from the three choices below (*for H atom*):

- i. Wavelength of photon *emitted* for an electron transitioning from *2p to 1s orbital*
- ii. Wavelength of photon *absorbed* for an electron transitioning from *2s to 4p orbital*
- iii. Wavelength of photon *absorbed* for an electron transitioning from *1s to 3p orbital*

i or ii or iii



Question 2 8 points



A Chem110 student plots the graph shown on the left, from data obtained for the kinetic energy (eV) of the ejected electron versus frequency (s^{-1}) of the incident light, for an unknown metal. ($1\text{eV} = 1.6022 \times 10^{-19} \text{ J}$)

Based on the plot, answer the questions **a to c** below:

a. 1 point On the plot above, label the **threshold frequency**.

b. 2 points What is the value of the work potential (in J) of the unknown metal?

c. 3 points What is the *longest* wavelength (in nm) required to eject an electron from this metal's surface? Show your work including equation(s) used. (3 sig figs)

d. 2 points In the space provided below, plot the following

(Assume frequency is *above* threshold frequency):

- i. *Kinetic energy* of an ejected electron versus the *intensity* of the incident light (left)
- ii. *Number of electrons* ejected versus the *frequency* of the incident light (right)



**Question 3** **6 points****a. 2 points**

In the ground state of Ge (Germanium; $Z = 32$) atom, **name** the following orbitals:

i. Orbital containing the electron that requires the maximum energy to be removed.

ii. Orbital containing the electron that experiences the minimum effective nuclear charge.

b. 2 points

An unknown anion (A^-) has the following *excited* electronic configuration: $[Ar] 4s^2 3d^3 4p^1$

(Ar: Argon; $Z = 18$)

Give the **condensed ground state electronic configuration** of the atom, **A** and the cation, **A^{2+}** .
Show your work. (Assume all electronic transitions are possible)

Electronic Configuration of A:

Electronic Configuration of A^{2+} :

c. 2 points

In a single atom, determine how many electrons can have the following set of quantum numbers?

(i) $n = 5; l = 3$

In a single atom, determine how many electrons can have the following set of quantum numbers?

(i) $n = 3; l = 2; m_s = +1/2$

**Question 4** **8 points**

Rank the following from 1 to 4 in **increasing** order. Provide the ranking in the boxes provided.

Note: Atomic number (Z) are provided for relevant atoms

a) Increasing (lowest to highest) order of *ionization energy*: (1= lowest; 4 = highest)

Mg

Mg: Z = 12

Rb

Rb: Z = 37

Cs

Cs: Z = 55

K

K : Z = 19

b) Increasing (lowest to highest) order of *bond angle*: (1= lowest; 4 = highest)

H₂OCCl₄XeF₄

HCN

c) Increasing (smallest to largest) order of *ionic size* (1= smallest; 4 = largest)

S²⁻

S: Z = 16

O²⁻

O: Z = 8

Be²⁺

Be: Z = 4

Na⁺

Na: Z = 11

d) Increasing (lowest to highest) order of *bond energy* (1= lowest; 4 = highest)

P-F

P-Cl

P-I

P-Br

**Question 5** **12 points**

a. 9 points For the molecules HClO_4 and TeF_6 , draw the most stable *Lewis structure* showing appropriate shape and *dashed/wedged bonds*, based on VSEPR. Determine the **molecular geometry** around the central atom. For HClO_4 , label the *most polar bond*.

HClO_4 (there are no O-O bonds)

Label the most polar bond

Molecular Geometry around Cl

TeF_6

Molecular Geometry around Te

b. 3 points (Note: Lone pairs are not shown in structures below)

(i) For the three *resonance* structures (1, 2 and 3) shown below, answer the following question:



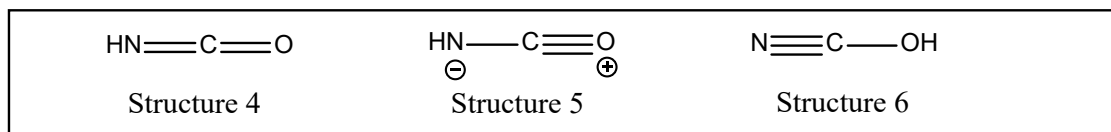
(I) Circle the *most contributing* resonance structure.

Structure 1

Structure 2

Structure 3

(ii) For the three structures given (4, 5, and 6) below, answer the following questions:



(I) Circle all the structures that represent *resonance structures* of each other:

Structure 4

Structure 5

Structure 6

(II) Among the resonance structures, circle the *most contributing* resonance structure.

Structure 4

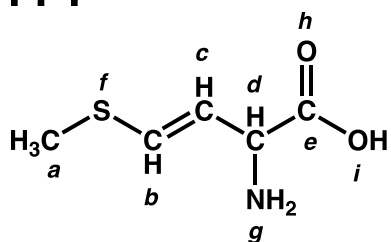
Structure 5

Structure 6

**Question 6** **6 points**

Using the data given below, calculate the **lattice energy** for SrO(s).

	ΔH° (kJ/mol)
$\text{Sr}(s) \longrightarrow \text{Sr}(g)$	+180
$\text{Sr}(g) \longrightarrow \text{Sr}^{2+}(g) + 2e^-$	+1468
$\text{O}_2(g) \longrightarrow 2\text{O}(g)$	+500
$\text{O}(g) + 2e^- \longrightarrow \text{O}^{2-}(g)$	+600
$\text{Sr}(s) + \frac{1}{2}\text{O}_2(g) \longrightarrow \text{SrO}(s)$	-550

**Question 7 10 points**

Consider the molecule given on the left, and answer questions (i) to (vi) using chemical bonding theories that we have learnt in class (valence bond theory, hybridization).

Note that the lone pairs are not shown in the structure:

i. Consider the carbon atoms marked in as ***a***, ***b***, ***c***, ***d***, and ***e*** in the molecule, and determine **hybridization** at each of the carbon centers. (2.5 points)

Atom	Hybridization
Carbon (<i>a</i>)	
Carbon (<i>b</i>)	
Carbon (<i>c</i>)	
Carbon (<i>d</i>)	
Carbon (<i>e</i>)	

ii. Now, consider the sulfur atom (S) marked as ***f***, nitrogen (N) atom (***g***), and oxygen(O) atoms (***h*** and ***i***), and determine their **hybridization**. (2 points)

Atom	Hybridization
Sulfur (<i>f</i>)	
Nitrogen (<i>g</i>)	
Oxygen (<i>h</i>)	
Oxygen (<i>i</i>)	

iii. **Describe** all the bonds formed by the carbon atoms ***a*** and ***b***, using sigma and pi designation as appropriate. (2 points)

Atom	Bonds formed (use sigma and pi as appropriate)
Carbon (<i>a</i>)	
Carbon (<i>b</i>)	

iv. How many total sp^3 -s sigma bonds are in the molecule? (1 point)

v. Determine **electronic** and **molecular geometry** at carbon (C) labeled ***e*** and sulfur (S) labeled ***f*** centres. (2 points)

Atom	Electronic Geometry	Molecular Geometry
Carbon (<i>e</i>)		
Sulfur (<i>f</i>)		

vi. Circle the value from the choices given below, that describes the **C-S-C bond angle at sulfur** (S) marked ***f***: (0.5 points)

>90° to <109.5° 109.5° >109.5 to <120° 120° >120° to <180°

**Question 8 12 points**

Using the **valence shell orbitals only** draw a **complete** molecular orbital (MO) diagram (with no s-p mixing) for the molecule **PS**, in the space provided below. **(6 points)**

Assume that 3s and 3p atomic orbitals will form molecular orbitals like those formed by the combination of 2s and 2p atomic orbitals that we learnt in class, and there is no s-p mixing.

Show **energy arrow**, **atomic orbitals** involved, **atom labels**, the **molecular orbitals** that are formed, **filled electrons** in the MO diagram etc. Make sure to label all the atomic and molecular orbitals.

Using the MO diagram above, answer the following questions:

- (i) Which one of the following will have the highest bond order? Circle the correct choice
A) PS^{1-} B) PS C) PS^{1+}
- (ii) Which one of the following will have the longest bond length? Circle the correct choice
A) PS^{1-} B) PS C) PS^{1+}
- (iii) Which of the following will be diamagnetic? Circle the correct choice.
A) PS^{1-} B) PS C) PS^{1+}
- (iv) Draw Lewis dot structure of the molecule with the shortest bond length (based on the MO diagram) from among the three (PS^{1-} , PS, PS^{1+}), and indicate the bond order calculated from the Lewis dot structure. **(3 points)**

Lewis Dot Structure

Bond Order



DEE3B96B-D774-4427-B939-7B41C14DDA80

midterm-exam-93388

#711 Page 10 of 10

SCRAP PAGE

**CHEMISTRY 110 –Midterm Exam****Thursday November 7, 2024, 6:30 – 8:30 pm**

Examiners: Pallavi Sirjoosingh and Ashok Kakkar

First name (please write as legibly as possible within the boxes)

--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--

Last name

--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--

Student ID number

--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--

This booklet has 8 questions on 10 pages (The last page can be used as a scrap page).

The staple must NOT be removed from the question booklet. Your intact question booklet must be returned at the end of the exam.

A separate booklet contains a periodic table, and datasheet.

Write your answers directly under the questions in the space provided. Do not write on the QR codes or near the top edges (as it may be cut off).

Please write **legibly and dark enough** for the scan to be readable. For students who use English as a second language, dictionaries are permitted. Calculators are allowed.

DO NOT ASK QUESTIONS OF THE INVIGILATORS about interpretations of the exam questions. If you have a concern about a question, please summarize it at the top of this page. All concerns will be evaluated after the exam. GOOD LUCK!

**Question 1 8 points** (*Assume all transitions are possible*)

a. 6 points Calculate the **energy (in kJ/mol)** required to ionize (completely remove) the electron from the **$n=3$** level of **He^+ ion** (He: Helium). Show your work including equations. (3 sig. figs.)

b. 2 points

(I) Select the largest (longest) wavelength from the three choices below:

- i. Wavelength of photon to *completely remove* an electron from *1s orbital of Hydrogen atom*
- ii. Wavelength of photon to *completely remove* an electron from *1s orbital of Boron atom (B)*
- iii. Wavelength of photon to *completely remove* an electron from *1s orbital of Boron cation (B^+)*

i or ii or iii

(II) Select the smallest (shortest) wavelength from the three choices below (*for H atom*):

- i. Wavelength of photon *emitted* for an electron transitioning from *2p to 1s orbital*
- ii. Wavelength of photon *absorbed* for an electron transitioning from *2s to 4p orbital*
- iii. Wavelength of photon *absorbed* for an electron transitioning from *1s to 3p orbital*

i or ii or iii



Question 2 8 points



A Chem110 student plots the graph shown on the left, from data obtained for the kinetic energy (eV) of the ejected electron versus frequency (s^{-1}) of the incident light, for an unknown metal. ($1\text{eV} = 1.6022 \times 10^{-19} \text{ J}$)

Based on the plot, answer the questions **a to c** below:

a. 1 point On the plot above, label the **threshold frequency**.

b. 2 points What is the value of the work potential (in J) of the unknown metal?

c. 3 points What is the *longest* wavelength (in nm) required to eject an electron from this metal's surface? Show your work including equation(s) used. (3 sig figs)

d. 2 points In the space provided below, plot the following

(Assume frequency is *above* threshold frequency):

- Kinetic energy* of an ejected electron versus the *intensity* of the incident light (left)
- Number of electrons* ejected versus the *frequency* of the incident light (right)



**Question 3** **6 points****a. 2 points**

In the ground state of Ge (Germanium; $Z = 32$) atom, **name** the following orbitals:

i. Orbital containing the electron that requires the maximum energy to be removed.

ii. Orbital containing the electron that experiences the minimum effective nuclear charge.

b. 2 points

An unknown anion (A^-) has the following *excited* electronic configuration: $[Ar] 4s^2 3d^3 4p^1$

(Ar: Argon; $Z = 18$)

Give the **condensed ground state electronic configuration** of the atom, **A** and the cation, **A^{2+}** .
Show your work. (Assume all electronic transitions are possible)

Electronic Configuration of A:

Electronic Configuration of A^{2+} :

c. 2 points

In a single atom, determine how many electrons can have the following set of quantum numbers?

(i) $n = 5; l = 3$

In a single atom, determine how many electrons can have the following set of quantum numbers?

(i) $n = 3; l = 2; m_s = +1/2$

**Question 4** **8 points**

Rank the following from 1 to 4 in **increasing** order. Provide the ranking in the boxes provided.

Note: Atomic number (Z) are provided for relevant atoms

a) Increasing (lowest to highest) order of *ionization energy*: (1= lowest; 4 = highest)

Mg

Mg: Z = 12

Rb

Rb: Z = 37

Cs

Cs: Z = 55

K

K : Z = 19

b) Increasing (lowest to highest) order of *bond angle*: (1= lowest; 4 = highest)

H₂OCCl₄XeF₄

HCN

c) Increasing (smallest to largest) order of *ionic size* (1= smallest; 4 = largest)

S²⁻

S: Z = 16

O²⁻

O: Z = 8

Be²⁺

Be: Z = 4

Na⁺

Na: Z = 11

d) Increasing (lowest to highest) order of *bond energy* (1= lowest; 4 = highest)

P-F

P-Cl

P-I

P-Br

**Question 5** **12 points**

a. 9 points For the molecules HClO_4 and TeF_6 , draw the most stable *Lewis structure* showing appropriate shape and *dashed/wedged bonds*, based on VSEPR. Determine the **molecular geometry** around the central atom. For HClO_4 , label the *most polar bond*.

HClO_4 (there are no O-O bonds)

Label the most polar bond

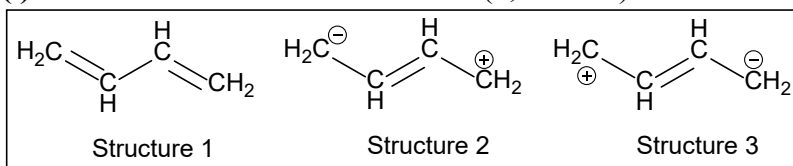
Molecular Geometry around Cl

TeF_6

Molecular Geometry around Te

b. 3 points (Note: Lone pairs are not shown in structures below)

(i) For the three *resonance* structures (1, 2 and 3) shown below, answer the following question:



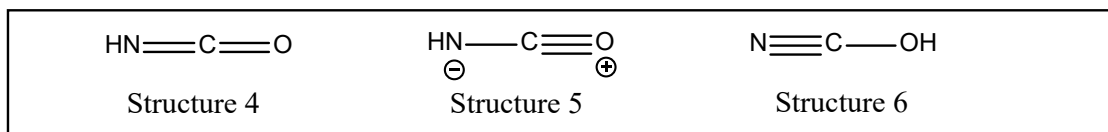
(I) Circle the *most contributing* resonance structure.

Structure 1

Structure 2

Structure 3

(ii) For the three structures given (4, 5, and 6) below, answer the following questions:



(I) Circle all the structures that represent *resonance structures* of each other:

Structure 4

Structure 5

Structure 6

(II) Among the resonance structures, circle the *most contributing* resonance structure.

Structure 4

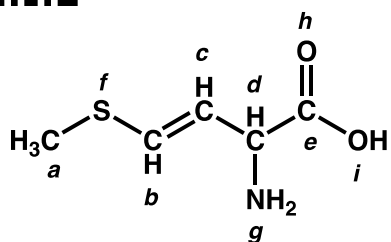
Structure 5

Structure 6

**Question 6** **6 points**

Using the data given below, calculate the **lattice energy** for SrO(s).

	ΔH° (kJ/mol)
$\text{Sr}(s) \longrightarrow \text{Sr}(g)$	+180
$\text{Sr}(g) \longrightarrow \text{Sr}^{2+}(g) + 2e^-$	+1468
$\text{O}_2(g) \longrightarrow 2\text{O}(g)$	+500
$\text{O}(g) + 2e^- \longrightarrow \text{O}^{2-}(g)$	+600
$\text{Sr}(s) + \frac{1}{2}\text{O}_2(g) \longrightarrow \text{SrO}(s)$	-550

**Question 7 10 points**

Consider the molecule given on the left, and answer questions (i) to (vi) using chemical bonding theories that we have learnt in class (valence bond theory, hybridization).

Note that the lone pairs are not shown in the structure:

i. Consider the carbon atoms marked in as ***a***, ***b***, ***c***, ***d***, and ***e*** in the molecule, and determine **hybridization** at each of the carbon centers. (2.5 points)

Atom	Hybridization
Carbon (<i>a</i>)	
Carbon (<i>b</i>)	
Carbon (<i>c</i>)	
Carbon (<i>d</i>)	
Carbon (<i>e</i>)	

ii. Now, consider the sulfur atom (S) marked as ***f***, nitrogen (N) atom (***g***), and oxygen(O) atoms (***h*** and ***i***), and determine their **hybridization**. (2 points)

Atom	Hybridization
Sulfur (<i>f</i>)	
Nitrogen (<i>g</i>)	
Oxygen (<i>h</i>)	
Oxygen (<i>i</i>)	

iii. **Describe** all the bonds formed by the carbon atoms ***a*** and ***b***, using sigma and pi designation as appropriate. (2 points)

Atom	Bonds formed (use sigma and pi as appropriate)
Carbon (<i>a</i>)	
Carbon (<i>b</i>)	

iv. How many total sp^3 -s sigma bonds are in the molecule? (1 point)

v. Determine **electronic** and **molecular geometry** at carbon (C) labeled ***e*** and sulfur (S) labeled ***f*** centres. (2 points)

Atom	Electronic Geometry	Molecular Geometry
Carbon (<i>e</i>)		
Sulfur (<i>f</i>)		

vi. Circle the value from the choices given below, that describes the **C-S-C bond angle at sulfur** (S) marked ***f***: (0.5 points)

>90° to <109.5° 109.5° >109.5 to <120° 120° >120° to <180°


Question 8 12 points

Using the **valence shell orbitals only** draw a **complete** molecular orbital (MO) diagram (with no s-p mixing) for the molecule **PS**, in the space provided below. **(6 points)**

Assume that 3s and 3p atomic orbitals will form molecular orbitals like those formed by the combination of 2s and 2p atomic orbitals that we learnt in class, and there is no s-p mixing.

Show **energy arrow**, **atomic orbitals** involved, **atom labels**, the **molecular orbitals** that are formed, **filled electrons** in the MO diagram etc. Make sure to **label all the atomic and molecular orbitals**.

Using the MO diagram above, answer the following questions:

- (i) Which one of the following will have the highest bond order? Circle the correct choice
 A) PS^{1-} B) PS C) PS^{1+}
- (ii) Which one of the following will have the longest bond length? Circle the correct choice
 A) PS^{1-} B) PS C) PS^{1+}
- (iii) Which of the following will be diamagnetic? Circle the correct choice.
 A) PS^{1-} B) PS C) PS^{1+}
- (iv) Draw Lewis dot structure of the molecule with the shortest bond length (based on the MO diagram) from among the three (PS^{1-} , PS, PS^{1+}), and indicate the bond order calculated from the Lewis dot structure. **(3 points)**

Lewis Dot Structure

Bond Order



C4DA8F2A-674D-4140-A924-8BAD41187166

midterm-exam-93388

#712 Page 10 of 10

SCRAP PAGE

**CHEMISTRY 110 –Midterm Exam****Thursday November 7, 2024, 6:30 – 8:30 pm**

Examiners: Pallavi Sirjoosingh and Ashok Kakkar

First name (please write as legibly as possible within the boxes)

--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--

Last name

--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--

Student ID number

--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--

This booklet has 8 questions on 10 pages (The last page can be used as a scrap page).

The staple must NOT be removed from the question booklet. Your intact question booklet must be returned at the end of the exam.

A separate booklet contains a periodic table, and datasheet.

Write your answers directly under the questions in the space provided. Do not write on the QR codes or near the top edges (as it may be cut off).

Please write **legibly and dark enough** for the scan to be readable. For students who use English as a second language, dictionaries are permitted. Calculators are allowed.

DO NOT ASK QUESTIONS OF THE INVIGILATORS about interpretations of the exam questions. If you have a concern about a question, please summarize it at the top of this page. All concerns will be evaluated after the exam. GOOD LUCK!

**Question 1 8 points** (*Assume all transitions are possible*)

a. 6 points Calculate the **energy (in kJ/mol)** required to ionize (completely remove) the electron from the **n=3** level of **He⁺ ion** (He: Helium). Show your work including equations. (3 sig. figs.)

b. 2 points

(I) Select the largest (longest) wavelength from the three choices below:

- i. Wavelength of photon to *completely remove* an electron from *1s orbital of Hydrogen atom*
- ii. Wavelength of photon to *completely remove* an electron from *1s orbital of Boron atom (B)*
- iii. Wavelength of photon to *completely remove* an electron from *1s orbital of Boron cation (B⁺)*

i or ii or iii

(II) Select the smallest (shortest) wavelength from the three choices below (*for H atom*):

- i. Wavelength of photon *emitted* for an electron transitioning from *2p to 1s orbital*
- ii. Wavelength of photon *absorbed* for an electron transitioning from *2s to 4p orbital*
- iii. Wavelength of photon *absorbed* for an electron transitioning from *1s to 3p orbital*

i or ii or iii



Question 2 8 points



A Chem110 student plots the graph shown on the left, from data obtained for the kinetic energy (eV) of the ejected electron versus frequency (s^{-1}) of the incident light, for an unknown metal. ($1\text{eV} = 1.6022 \times 10^{-19} \text{ J}$)

Based on the plot, answer the questions **a to c** below:

a. 1 point On the plot above, label the **threshold frequency**.

b. 2 points What is the value of the work potential (in J) of the unknown metal?

c. 3 points What is the *longest* wavelength (in nm) required to eject an electron from this metal's surface? Show your work including equation(s) used. (3 sig figs)

d. 2 points In the space provided below, plot the following

(Assume frequency is *above* threshold frequency):

- i. *Kinetic energy* of an ejected electron versus the *intensity* of the incident light (left)
- ii. *Number of electrons* ejected versus the *frequency* of the incident light (right)



**Question 3** **6 points****a. 2 points**

In the ground state of Ge (Germanium; $Z = 32$) atom, **name** the following orbitals:

i. Orbital containing the electron that requires the maximum energy to be removed.

ii. Orbital containing the electron that experiences the minimum effective nuclear charge.

b. 2 points

An unknown anion (A^-) has the following *excited* electronic configuration: $[Ar] 4s^2 3d^3 4p^1$

(Ar: Argon; $Z = 18$)

Give the **condensed ground state electronic configuration** of the atom, **A** and the cation, **A^{2+}** .
Show your work. (Assume all electronic transitions are possible)

Electronic Configuration of A:

Electronic Configuration of A^{2+} :

c. 2 points

In a single atom, determine how many electrons can have the following set of quantum numbers?

(i) $n = 5; l = 3$

In a single atom, determine how many electrons can have the following set of quantum numbers?

(i) $n = 3; l = 2; m_s = +1/2$

**Question 4** **8 points**

Rank the following from 1 to 4 in **increasing** order. Provide the ranking in the boxes provided.

Note: Atomic number (Z) are provided for relevant atoms

a) Increasing (lowest to highest) order of *ionization energy*: (1= lowest; 4 = highest)

Mg

Mg: Z = 12

Rb

Rb: Z = 37

Cs

Cs: Z = 55

K

K : Z = 19

b) Increasing (lowest to highest) order of *bond angle*: (1= lowest; 4 = highest)

H₂OCCl₄XeF₄

HCN

c) Increasing (smallest to largest) order of *ionic size* (1= smallest; 4 = largest)

S²⁻

S: Z = 16

O²⁻

O: Z = 8

Be²⁺

Be: Z = 4

Na⁺

Na: Z = 11

d) Increasing (lowest to highest) order of *bond energy* (1= lowest; 4 = highest)

P-F

P-Cl

P-I

P-Br

**Question 5** **12 points**

a. 9 points For the molecules HClO_4 and TeF_6 , draw the most stable *Lewis structure* showing appropriate shape and *dashed/wedged bonds*, based on VSEPR. Determine the **molecular geometry** around the central atom. For HClO_4 , label the *most polar bond*.



b. 3 points (Note: Lone pairs are not shown in structures below)

(i) For the three *resonance* structures (1, 2 and 3) shown below, answer the following question:



(I) Circle the *most contributing* resonance structure.

Structure 1

Structure 2

Structure 3

(ii) For the three structures given (4, 5, and 6) below, answer the following questions:



(I) Circle all the structures that represent *resonance structures* of each other:

Structure 4

Structure 5

Structure 6

(II) Among the resonance structures, circle the *most contributing* resonance structure.

Structure 4

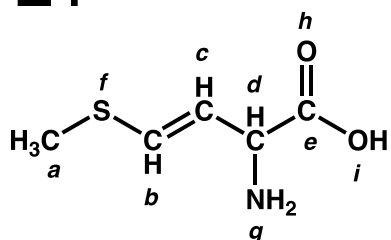
Structure 5

Structure 6

**Question 6** **6 points**

Using the data given below, calculate the **lattice energy** for SrO(s).

	ΔH° (kJ/mol)
$\text{Sr}(s) \longrightarrow \text{Sr}(g)$	+180
$\text{Sr}(g) \longrightarrow \text{Sr}^{2+}(g) + 2e^-$	+1468
$\text{O}_2(g) \longrightarrow 2\text{O}(g)$	+500
$\text{O}(g) + 2e^- \longrightarrow \text{O}^{2-}(g)$	+600
$\text{Sr}(s) + \frac{1}{2}\text{O}_2(g) \longrightarrow \text{SrO}(s)$	-550

**Question 7 10 points**

Consider the molecule given on the left, and answer questions (i) to (vi) using chemical bonding theories that we have learnt in class (valence bond theory, hybridization).

Note that the lone pairs are not shown in the structure:

i. Consider the carbon atoms marked in as ***a***, ***b***, ***c***, ***d***, and ***e*** in the molecule, and determine **hybridization** at each of the carbon centers. (2.5 points)

Atom	Hybridization
Carbon (<i>a</i>)	
Carbon (<i>b</i>)	
Carbon (<i>c</i>)	
Carbon (<i>d</i>)	
Carbon (<i>e</i>)	

ii. Now, consider the sulfur atom (S) marked as ***f***, nitrogen (N) atom (***g***), and oxygen(O) atoms (***h*** and ***i***), and determine their **hybridization**. (2 points)

Atom	Hybridization
Sulfur (<i>f</i>)	
Nitrogen (<i>g</i>)	
Oxygen (<i>h</i>)	
Oxygen (<i>i</i>)	

iii. **Describe** all the bonds formed by the carbon atoms ***a*** and ***b***, using sigma and pi designation as appropriate. (2 points)

Atom	Bonds formed (use sigma and pi as appropriate)
Carbon (<i>a</i>)	
Carbon (<i>b</i>)	

iv. How many total sp^3 -s sigma bonds are in the molecule? (1 point)

v. Determine **electronic** and **molecular geometry** at carbon (C) labeled ***e*** and sulfur (S) labeled ***f*** centres. (2 points)

Atom	Electronic Geometry	Molecular Geometry
Carbon (<i>e</i>)		
Sulfur (<i>f</i>)		

vi. Circle the value from the choices given below, that describes the **C-S-C bond angle at sulfur** (S) marked ***f***: (0.5 points)

>90° to <109.5°

109.5°

>109.5 to <120°

120°

>120° to <180°

**Question 8 12 points**

Using the **valence shell orbitals only** draw a **complete** molecular orbital (MO) diagram (with no s-p mixing) for the molecule **PS**, in the space provided below. **(6 points)**

Assume that 3s and 3p atomic orbitals will form molecular orbitals like those formed by the combination of 2s and 2p atomic orbitals that we learnt in class, and there is no s-p mixing.

Show **energy arrow**, **atomic orbitals** involved, **atom labels**, the **molecular orbitals** that are formed, **filled electrons** in the MO diagram etc. Make sure to label all the atomic and molecular orbitals.

Using the MO diagram above, answer the following questions:

- (i) Which one of the following will have the highest bond order? Circle the correct choice
A) PS^{1-} B) PS C) PS^{1+}
- (ii) Which one of the following will have the longest bond length? Circle the correct choice
A) PS^{1-} B) PS C) PS^{1+}
- (iii) Which of the following will be diamagnetic? Circle the correct choice.
A) PS^{1-} B) PS C) PS^{1+}
- (iv) Draw Lewis dot structure of the molecule with the shortest bond length (based on the MO diagram) from among the three (PS^{1-} , PS, PS^{1+}), and indicate the bond order calculated from the Lewis dot structure. **(3 points)**

Lewis Dot Structure

Bond Order



62F1FC04-587E-4427-AE36-22B22F15C13B

midterm-exam-93388

#713 Page 10 of 10

SCRAP PAGE

**CHEMISTRY 110 –Midterm Exam****Thursday November 7, 2024, 6:30 – 8:30 pm****Examiners: Pallavi Sirjoosingh and Ashok Kakkar**

First name (please write as legibly as possible within the boxes)

--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--

Last name

--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--

Student ID number

--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--

This booklet has 8 questions on 10 pages (The last page can be used as a scrap page).

The staple must NOT be removed from the question booklet. Your intact question booklet must be returned at the end of the exam.

A separate booklet contains a periodic table, and datasheet.

Write your answers directly under the questions in the space provided. Do not write on the QR codes or near the top edges (as it may be cut off).

Please write **legibly and dark enough** for the scan to be readable. For students who use English as a second language, dictionaries are permitted. Calculators are allowed.

DO NOT ASK QUESTIONS OF THE INVIGILATORS about interpretations of the exam questions. If you have a concern about a question, please summarize it at the top of this page. All concerns will be evaluated after the exam. GOOD LUCK!

**Question 1 8 points** (*Assume all transitions are possible*)

a. 6 points Calculate the **energy (in kJ/mol)** required to ionize (completely remove) the electron from the **$n=3$** level of **He^+ ion** (He: Helium). Show your work including equations. (3 sig. figs.)

b. 2 points

(I) Select the largest (longest) wavelength from the three choices below:

- i. Wavelength of photon to *completely remove* an electron from *1s orbital of Hydrogen atom*
- ii. Wavelength of photon to *completely remove* an electron from *1s orbital of Boron atom (B)*
- iii. Wavelength of photon to *completely remove* an electron from *1s orbital of Boron cation (B^+)*

i or ii or iii

(II) Select the smallest (shortest) wavelength from the three choices below (*for H atom*):

- i. Wavelength of photon *emitted* for an electron transitioning from *2p to 1s orbital*
- ii. Wavelength of photon *absorbed* for an electron transitioning from *2s to 4p orbital*
- iii. Wavelength of photon *absorbed* for an electron transitioning from *1s to 3p orbital*

i or ii or iii



Question 2 8 points



A Chem110 student plots the graph shown on the left, from data obtained for the kinetic energy (eV) of the ejected electron versus frequency (s^{-1}) of the incident light, for an unknown metal. ($1\text{eV} = 1.6022 \times 10^{-19} \text{ J}$)

Based on the plot, answer the questions **a to c** below:

a. 1 point On the plot above, label the **threshold frequency**.

b. 2 points What is the value of the work potential (in J) of the unknown metal?

c. 3 points What is the *longest* wavelength (in nm) required to eject an electron from this metal's surface? Show your work including equation(s) used. (3 sig figs)

d. 2 points In the space provided below, plot the following

(Assume frequency is *above* threshold frequency):

- i. *Kinetic energy* of an ejected electron versus the *intensity* of the incident light (left)
- ii. *Number of electrons* ejected versus the *frequency* of the incident light (right)



**Question 3** **6 points****a. 2 points**

In the ground state of Ge (Germanium; $Z = 32$) atom, **name** the following orbitals:

i. Orbital containing the electron that requires the maximum energy to be removed.

ii. Orbital containing the electron that experiences the minimum effective nuclear charge.

b. 2 points

An unknown anion (A^-) has the following *excited* electronic configuration: $[Ar] 4s^2 3d^3 4p^1$

(Ar: Argon; $Z = 18$)

Give the **condensed ground state electronic configuration** of the atom, **A** and the cation, **A^{2+}** .
Show your work. (Assume all electronic transitions are possible)

Electronic Configuration of A:

Electronic Configuration of A^{2+} :

c. 2 points

In a single atom, determine how many electrons can have the following set of quantum numbers?

(i) $n = 5; l = 3$

In a single atom, determine how many electrons can have the following set of quantum numbers?

(i) $n = 3; l = 2; m_s = +1/2$

**Question 4** **8 points**

Rank the following from 1 to 4 in **increasing** order. Provide the ranking in the boxes provided.

Note: Atomic number (Z) are provided for relevant atoms

a) Increasing (lowest to highest) order of *ionization energy*: (1= lowest; 4 = highest)

Mg

Mg: Z = 12

Rb

Rb: Z = 37

Cs

Cs: Z = 55

K

K : Z = 19

b) Increasing (lowest to highest) order of *bond angle*: (1= lowest; 4 = highest)

H₂OCCl₄XeF₄

HCN

c) Increasing (smallest to largest) order of *ionic size* (1= smallest; 4 = largest)

S²⁻

S: Z = 16

O²⁻

O: Z = 8

Be²⁺

Be: Z = 4

Na⁺

Na: Z = 11

d) Increasing (lowest to highest) order of *bond energy* (1= lowest; 4 = highest)

P-F

P-Cl

P-I

P-Br

**Question 5** **12 points**

a. 9 points For the molecules HClO_4 and TeF_6 , draw the most stable *Lewis structure* showing appropriate shape and *dashed/wedged bonds*, based on VSEPR. Determine the **molecular geometry** around the central atom. For HClO_4 , label the *most polar bond*.



b. 3 points (Note: Lone pairs are not shown in structures below)

(i) For the three *resonance* structures (1, 2 and 3) shown below, answer the following question:



(I) Circle the *most contributing* resonance structure.

Structure 1

Structure 2

Structure 3

(ii) For the three structures given (4, 5, and 6) below, answer the following questions:



(I) Circle all the structures that represent *resonance structures* of each other:

Structure 4

Structure 5

Structure 6

(II) Among the resonance structures, circle the *most contributing* resonance structure.

Structure 4

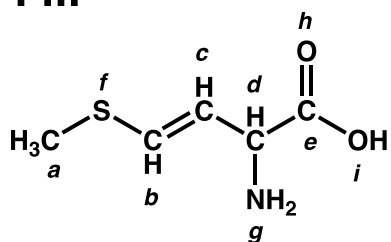
Structure 5

Structure 6

**Question 6** **6 points**

Using the data given below, calculate the **lattice energy** for SrO(s).

	ΔH° (kJ/mol)
$\text{Sr}(s) \longrightarrow \text{Sr}(g)$	+180
$\text{Sr}(g) \longrightarrow \text{Sr}^{2+}(g) + 2e^-$	+1468
$\text{O}_2(g) \longrightarrow 2\text{O}(g)$	+500
$\text{O}(g) + 2e^- \longrightarrow \text{O}^{2-}(g)$	+600
$\text{Sr}(s) + \frac{1}{2}\text{O}_2(g) \longrightarrow \text{SrO}(s)$	-550

**Question 7 10 points**

Consider the molecule given on the left, and answer questions (i) to (vi) using chemical bonding theories that we have learnt in class (valence bond theory, hybridization).

Note that the lone pairs are not shown in the structure:

i. Consider the carbon atoms marked in as ***a***, ***b***, ***c***, ***d***, and ***e*** in the molecule, and determine **hybridization** at each of the carbon centers. (2.5 points)

Atom	Hybridization
Carbon (<i>a</i>)	
Carbon (<i>b</i>)	
Carbon (<i>c</i>)	
Carbon (<i>d</i>)	
Carbon (<i>e</i>)	

ii. Now, consider the sulfur atom (S) marked as ***f***, nitrogen (N) atom (***g***), and oxygen(O) atoms (***h*** and ***i***), and determine their **hybridization**. (2 points)

Atom	Hybridization
Sulfur (<i>f</i>)	
Nitrogen (<i>g</i>)	
Oxygen (<i>h</i>)	
Oxygen (<i>i</i>)	

iii. **Describe** all the bonds formed by the carbon atoms ***a*** and ***b***, using sigma and pi designation as appropriate. (2 points)

Atom	Bonds formed (use sigma and pi as appropriate)
Carbon (<i>a</i>)	
Carbon (<i>b</i>)	

iv. How many total sp^3 -s sigma bonds are in the molecule? (1 point)

v. Determine **electronic** and **molecular geometry** at carbon (C) labeled ***e*** and sulfur (S) labeled ***f*** centres. (2 points)

Atom	Electronic Geometry	Molecular Geometry
Carbon (<i>e</i>)		
Sulfur (<i>f</i>)		

vi. Circle the value from the choices given below, that describes the **C-S-C bond angle at sulfur** (S) marked ***f***: (0.5 points)

>90° to <109.5°

109.5°

>109.5 to <120°

120°

>120° to <180°


Question 8 12 points

Using the **valence shell orbitals only** draw a **complete** molecular orbital (MO) diagram (with no s-p mixing) for the molecule **PS**, in the space provided below. **(6 points)**

Assume that 3s and 3p atomic orbitals will form molecular orbitals like those formed by the combination of 2s and 2p atomic orbitals that we learnt in class, and there is no s-p mixing.

Show **energy arrow**, **atomic orbitals** involved, **atom labels**, the **molecular orbitals** that are formed, **filled electrons** in the MO diagram etc. Make sure to **label all the atomic and molecular orbitals**.

Using the MO diagram above, answer the following questions:

- (i) Which one of the following will have the highest bond order? Circle the correct choice
 A) PS^{1-} B) PS C) PS^{1+}
- (ii) Which one of the following will have the longest bond length? Circle the correct choice
 A) PS^{1-} B) PS C) PS^{1+}
- (iii) Which of the following will be diamagnetic? Circle the correct choice.
 A) PS^{1-} B) PS C) PS^{1+}
- (iv) Draw Lewis dot structure of the molecule with the shortest bond length (based on the MO diagram) from among the three (PS^{1-} , PS, PS^{1+}), and indicate the bond order calculated from the Lewis dot structure. **(3 points)**

Lewis Dot Structure

Bond Order



4636FBA5-5DA6-4267-A6FD-DD0B629D975D

midterm-exam-93388

#714 Page 10 of 10

SCRAP PAGE

**CHEMISTRY 110 –Midterm Exam****Thursday November 7, 2024, 6:30 – 8:30 pm****Examiners: Pallavi Sirjoosingh and Ashok Kakkar**

First name (please write as legibly as possible within the boxes)

--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--

Last name

--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--

Student ID number

--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--

This booklet has 8 questions on 10 pages (The last page can be used as a scrap page).

The staple must NOT be removed from the question booklet. Your intact question booklet must be returned at the end of the exam.

A separate booklet contains a periodic table, and datasheet.

Write your answers directly under the questions in the space provided. Do not write on the QR codes or near the top edges (as it may be cut off).

Please write **legibly and dark enough** for the scan to be readable. For students who use English as a second language, dictionaries are permitted. Calculators are allowed.

DO NOT ASK QUESTIONS OF THE INVIGILATORS about interpretations of the exam questions. If you have a concern about a question, please summarize it at the top of this page. All concerns will be evaluated after the exam. GOOD LUCK!

**Question 1 8 points** (*Assume all transitions are possible*)

a. 6 points Calculate the **energy (in kJ/mol)** required to ionize (completely remove) the electron from the **n=3** level of **He⁺ ion** (He: Helium). Show your work including equations. (3 sig. figs.)

b. 2 points

(I) Select the largest (longest) wavelength from the three choices below:

- i. Wavelength of photon to *completely remove* an electron from *1s orbital of Hydrogen atom*
- ii. Wavelength of photon to *completely remove* an electron from *1s orbital of Boron atom (B)*
- iii. Wavelength of photon to *completely remove* an electron from *1s orbital of Boron cation (B⁺)*

i or ii or iii

(II) Select the smallest (shortest) wavelength from the three choices below (*for H atom*):

- i. Wavelength of photon *emitted* for an electron transitioning from *2p to 1s orbital*
- ii. Wavelength of photon *absorbed* for an electron transitioning from *2s to 4p orbital*
- iii. Wavelength of photon *absorbed* for an electron transitioning from *1s to 3p orbital*

i or ii or iii



Question 2 8 points



A Chem110 student plots the graph shown on the left, from data obtained for the kinetic energy (eV) of the ejected electron versus frequency (s^{-1}) of the incident light, for an unknown metal. ($1\text{eV} = 1.6022 \times 10^{-19} \text{ J}$)

Based on the plot, answer the questions **a to c** below:

a. 1 point On the plot above, label the **threshold frequency**.

b. 2 points What is the value of the work potential (in J) of the unknown metal?

c. 3 points What is the *longest* wavelength (in nm) required to eject an electron from this metal's surface? Show your work including equation(s) used. (3 sig figs)

d. 2 points In the space provided below, plot the following

(Assume frequency is *above* threshold frequency):

- Kinetic energy* of an ejected electron versus the *intensity* of the incident light (left)
- Number of electrons* ejected versus the *frequency* of the incident light (right)



**Question 3** **6 points****a. 2 points**

In the ground state of Ge (Germanium; $Z = 32$) atom, **name** the following orbitals:

i. Orbital containing the electron that requires the maximum energy to be removed.

ii. Orbital containing the electron that experiences the minimum effective nuclear charge.

b. 2 points

An unknown anion (A^-) has the following *excited* electronic configuration: $[Ar] 4s^2 3d^3 4p^1$

(Ar: Argon; $Z = 18$)

Give the **condensed ground state electronic configuration** of the atom, **A** and the cation, **A^{2+}** .
Show your work. (Assume all electronic transitions are possible)

Electronic Configuration of A:

Electronic Configuration of A^{2+} :

c. 2 points

In a single atom, determine how many electrons can have the following set of quantum numbers?

(i) $n = 5; l = 3$

In a single atom, determine how many electrons can have the following set of quantum numbers?

(i) $n = 3; l = 2; m_s = +1/2$

**Question 4** **8 points**

Rank the following from 1 to 4 in **increasing** order. Provide the ranking in the boxes provided.

Note: Atomic number (Z) are provided for relevant atoms

a) Increasing (lowest to highest) order of *ionization energy*: (1= lowest; 4 = highest)

Mg

Mg: Z = 12

Rb

Rb: Z = 37

Cs

Cs: Z = 55

K

K : Z = 19

b) Increasing (lowest to highest) order of *bond angle*: (1= lowest; 4 = highest)

H₂OCCl₄XeF₄

HCN

c) Increasing (smallest to largest) order of *ionic size* (1= smallest; 4 = largest)

S²⁻

S: Z = 16

O²⁻

O: Z = 8

Be²⁺

Be: Z = 4

Na⁺

Na: Z = 11

d) Increasing (lowest to highest) order of *bond energy* (1= lowest; 4 = highest)

P-F

P-Cl

P-I

P-Br

**Question 5** **12 points**

a. 9 points For the molecules HClO_4 and TeF_6 , draw the most stable *Lewis structure* showing appropriate shape and *dashed/wedged bonds*, based on VSEPR. Determine the **molecular geometry** around the central atom. For HClO_4 , label the *most polar bond*.



b. 3 points (Note: Lone pairs are not shown in structures below)

(i) For the three *resonance* structures (1, 2 and 3) shown below, answer the following question:



(I) Circle the *most contributing* resonance structure.

Structure 1

Structure 2

Structure 3

(ii) For the three structures given (4, 5, and 6) below, answer the following questions:



(I) Circle all the structures that represent *resonance structures* of each other:

Structure 4

Structure 5

Structure 6

(II) Among the resonance structures, circle the *most contributing* resonance structure.

Structure 4

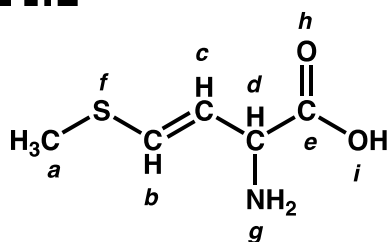
Structure 5

Structure 6

**Question 6** **6 points**

Using the data given below, calculate the **lattice energy** for SrO(s).

	ΔH° (kJ/mol)
$\text{Sr}(s) \longrightarrow \text{Sr}(g)$	+180
$\text{Sr}(g) \longrightarrow \text{Sr}^{2+}(g) + 2e^-$	+1468
$\text{O}_2(g) \longrightarrow 2\text{O}(g)$	+500
$\text{O}(g) + 2e^- \longrightarrow \text{O}^{2-}(g)$	+600
$\text{Sr}(s) + \frac{1}{2}\text{O}_2(g) \longrightarrow \text{SrO}(s)$	-550

**Question 7 10 points**

Consider the molecule given on the left, and answer questions (i) to (vi) using chemical bonding theories that we have learnt in class (valence bond theory, hybridization).

Note that the lone pairs are not shown in the structure:

i. Consider the carbon atoms marked in as ***a***, ***b***, ***c***, ***d***, and ***e*** in the molecule, and determine **hybridization** at each of the carbon centers. (2.5 points)

Atom	Hybridization
Carbon (<i>a</i>)	
Carbon (<i>b</i>)	
Carbon (<i>c</i>)	
Carbon (<i>d</i>)	
Carbon (<i>e</i>)	

ii. Now, consider the sulfur atom (S) marked as ***f***, nitrogen (N) atom (***g***), and oxygen(O) atoms (***h*** and ***i***), and determine their **hybridization**. (2 points)

Atom	Hybridization
Sulfur (<i>f</i>)	
Nitrogen (<i>g</i>)	
Oxygen (<i>h</i>)	
Oxygen (<i>i</i>)	

iii. **Describe** all the bonds formed by the carbon atoms ***a*** and ***b***, using sigma and pi designation as appropriate. (2 points)

Atom	Bonds formed (use sigma and pi as appropriate)
Carbon (<i>a</i>)	
Carbon (<i>b</i>)	

iv. How many total sp^3 -s sigma bonds are in the molecule? (1 point)

v. Determine **electronic** and **molecular geometry** at carbon (C) labeled ***e*** and sulfur (S) labeled ***f*** centres. (2 points)

Atom	Electronic Geometry	Molecular Geometry
Carbon (<i>e</i>)		
Sulfur (<i>f</i>)		

vi. Circle the value from the choices given below, that describes the **C-S-C bond angle at sulfur** (S) marked ***f***: (0.5 points)

>90° to <109.5° 109.5° >109.5 to <120° 120° >120° to <180°

**Question 8 12 points**

Using the **valence shell orbitals only** draw a **complete** molecular orbital (MO) diagram (with no s-p mixing) for the molecule **PS**, in the space provided below. **(6 points)**

Assume that 3s and 3p atomic orbitals will form molecular orbitals like those formed by the combination of 2s and 2p atomic orbitals that we learnt in class, and there is no s-p mixing.

Show **energy arrow**, **atomic orbitals** involved, **atom labels**, the **molecular orbitals** that are formed, **filled electrons** in the MO diagram etc. Make sure to label all the atomic and molecular orbitals.

Using the MO diagram above, answer the following questions:

- (i) Which one of the following will have the highest bond order? Circle the correct choice
A) PS^{1-} B) PS C) PS^{1+}
- (ii) Which one of the following will have the longest bond length? Circle the correct choice
A) PS^{1-} B) PS C) PS^{1+}
- (iii) Which of the following will be diamagnetic? Circle the correct choice.
A) PS^{1-} B) PS C) PS^{1+}
- (iv) Draw Lewis dot structure of the molecule with the shortest bond length (based on the MO diagram) from among the three (PS^{1-} , PS, PS^{1+}), and indicate the bond order calculated from the Lewis dot structure. **(3 points)**

Lewis Dot Structure

Bond Order



7BA0C1AF-F0DA-4853-8322-EDC4AA5B6C4D

midterm-exam-93388

#715 Page 10 of 10

SCRAP PAGE

**CHEMISTRY 110 –Midterm Exam****Thursday November 7, 2024, 6:30 – 8:30 pm****Examiners: Pallavi Sirjoosingh and Ashok Kakkar**

First name (please write as legibly as possible within the boxes)

--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--

Last name

--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--

Student ID number

--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--

This booklet has 8 questions on 10 pages (The last page can be used as a scrap page).

The staple must NOT be removed from the question booklet. Your intact question booklet must be returned at the end of the exam.

A separate booklet contains a periodic table, and datasheet.

Write your answers directly under the questions in the space provided. Do not write on the QR codes or near the top edges (as it may be cut off).

Please write **legibly and dark enough** for the scan to be readable. For students who use English as a second language, dictionaries are permitted. Calculators are allowed.

DO NOT ASK QUESTIONS OF THE INVIGILATORS about interpretations of the exam questions. If you have a concern about a question, please summarize it at the top of this page. All concerns will be evaluated after the exam. GOOD LUCK!

**Question 1 8 points** (*Assume all transitions are possible*)

a. 6 points Calculate the **energy (in kJ/mol)** required to ionize (completely remove) the electron from the **n=3** level of **He⁺ ion** (He: Helium). Show your work including equations. (3 sig. figs.)

b. 2 points

(I) Select the largest (longest) wavelength from the three choices below:

- i. Wavelength of photon to *completely remove* an electron from *1s orbital of Hydrogen atom*
- ii. Wavelength of photon to *completely remove* an electron from *1s orbital of Boron atom (B)*
- iii. Wavelength of photon to *completely remove* an electron from *1s orbital of Boron cation (B⁺)*

i or ii or iii

(II) Select the smallest (shortest) wavelength from the three choices below (*for H atom*):

- i. Wavelength of photon *emitted* for an electron transitioning from *2p to 1s orbital*
- ii. Wavelength of photon *absorbed* for an electron transitioning from *2s to 4p orbital*
- iii. Wavelength of photon *absorbed* for an electron transitioning from *1s to 3p orbital*

i or ii or iii



Question 2 8 points



A Chem110 student plots the graph shown on the left, from data obtained for the kinetic energy (eV) of the ejected electron versus frequency (s^{-1}) of the incident light, for an unknown metal. ($1\text{eV} = 1.6022 \times 10^{-19} \text{ J}$)

Based on the plot, answer the questions **a to c** below:

a. 1 point On the plot above, label the **threshold frequency**.

b. 2 points What is the value of the work potential (in J) of the unknown metal?

c. 3 points What is the *longest* wavelength (in nm) required to eject an electron from this metal's surface? Show your work including equation(s) used. (3 sig figs)

d. 2 points In the space provided below, plot the following

(Assume frequency is *above* threshold frequency):

- i. *Kinetic energy* of an ejected electron versus the *intensity* of the incident light (left)
- ii. *Number of electrons* ejected versus the *frequency* of the incident light (right)



**Question 3** **6 points****a. 2 points**

In the ground state of Ge (Germanium; $Z = 32$) atom, **name** the following orbitals:

i. Orbital containing the electron that requires the maximum energy to be removed.

ii. Orbital containing the electron that experiences the minimum effective nuclear charge.

b. 2 points

An unknown anion (A^-) has the following *excited* electronic configuration: $[Ar] 4s^2 3d^3 4p^1$

(Ar: Argon; $Z = 18$)

Give the **condensed ground state electronic configuration** of the atom, **A** and the cation, **A^{2+}** .
Show your work. (Assume all electronic transitions are possible)

Electronic Configuration of A:

Electronic Configuration of A^{2+} :

c. 2 points

In a single atom, determine how many electrons can have the following set of quantum numbers?

(i) $n = 5; l = 3$

In a single atom, determine how many electrons can have the following set of quantum numbers?

(i) $n = 3; l = 2; m_s = +1/2$

**Question 4** **8 points**

Rank the following from 1 to 4 in **increasing** order. Provide the ranking in the boxes provided.

Note: Atomic number (Z) are provided for relevant atoms

a) Increasing (lowest to highest) order of *ionization energy*: (1= lowest; 4 = highest)

Mg

Mg: Z = 12

Rb

Rb: Z = 37

Cs

Cs: Z = 55

K

K : Z = 19

b) Increasing (lowest to highest) order of *bond angle*: (1= lowest; 4 = highest)

H₂OCCl₄XeF₄

HCN

c) Increasing (smallest to largest) order of *ionic size* (1= smallest; 4 = largest)

S²⁻

S: Z = 16

O²⁻

O: Z = 8

Be²⁺

Be: Z = 4

Na⁺

Na: Z = 11

d) Increasing (lowest to highest) order of *bond energy* (1= lowest; 4 = highest)

P-F

P-Cl

P-I

P-Br

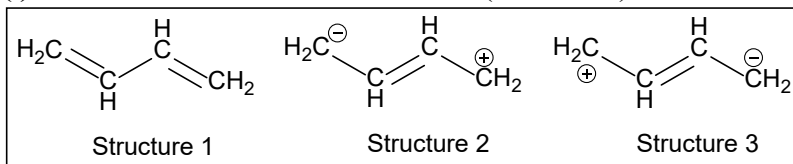
**Question 5** **12 points**

a. 9 points For the molecules HClO_4 and TeF_6 , draw the most stable *Lewis structure* showing appropriate shape and *dashed/wedged bonds*, based on VSEPR. Determine the **molecular geometry** around the central atom. For HClO_4 , label the *most polar bond*.



b. 3 points (Note: Lone pairs are not shown in structures below)

(i) For the three *resonance* structures (1, 2 and 3) shown below, answer the following question:



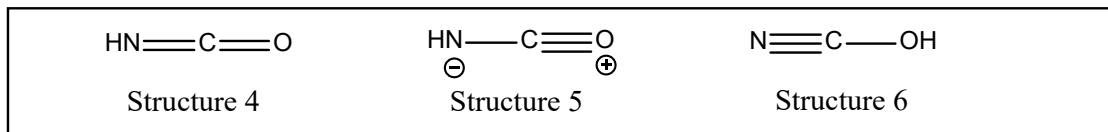
(I) Circle the *most contributing* resonance structure.

Structure 1

Structure 2

Structure 3

(ii) For the three structures given (4, 5, and 6) below, answer the following questions:



(I) Circle all the structures that represent *resonance structures* of each other:

Structure 4

Structure 5

Structure 6

(II) Among the resonance structures, circle the *most contributing* resonance structure.

Structure 4

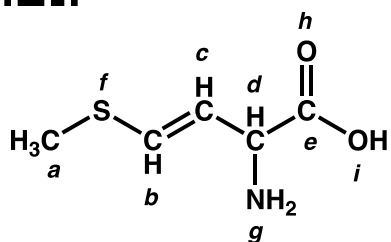
Structure 5

Structure 6

**Question 6** **6 points**

Using the data given below, calculate the **lattice energy** for SrO(s).

	ΔH° (kJ/mol)
$\text{Sr}(s) \longrightarrow \text{Sr}(g)$	+180
$\text{Sr}(g) \longrightarrow \text{Sr}^{2+}(g) + 2e^-$	+1468
$\text{O}_2(g) \longrightarrow 2\text{O}(g)$	+500
$\text{O}(g) + 2e^- \longrightarrow \text{O}^{2-}(g)$	+600
$\text{Sr}(s) + \frac{1}{2}\text{O}_2(g) \longrightarrow \text{SrO}(s)$	-550

**Question 7 10 points**

Consider the molecule given on the left, and answer questions (i) to (vi) using chemical bonding theories that we have learnt in class (valence bond theory, hybridization).

Note that the lone pairs are not shown in the structure:

i. Consider the carbon atoms marked in as ***a***, ***b***, ***c***, ***d***, and ***e*** in the molecule, and determine **hybridization** at each of the carbon centers. (2.5 points)

Atom	Hybridization
Carbon (<i>a</i>)	
Carbon (<i>b</i>)	
Carbon (<i>c</i>)	
Carbon (<i>d</i>)	
Carbon (<i>e</i>)	

ii. Now, consider the sulfur atom (S) marked as ***f***, nitrogen (N) atom (***g***), and oxygen(O) atoms (***h*** and ***i***), and determine their **hybridization**. (2 points)

Atom	Hybridization
Sulfur (<i>f</i>)	
Nitrogen (<i>g</i>)	
Oxygen (<i>h</i>)	
Oxygen (<i>i</i>)	

iii. **Describe** all the bonds formed by the carbon atoms ***a*** and ***b***, using sigma and pi designation as appropriate. (2 points)

Atom	Bonds formed (use sigma and pi as appropriate)
Carbon (<i>a</i>)	
Carbon (<i>b</i>)	

iv. How many total sp^3 -s sigma bonds are in the molecule? (1 point)

v. Determine **electronic** and **molecular geometry** at carbon (C) labeled ***e*** and sulfur (S) labeled ***f*** centres. (2 points)

Atom	Electronic Geometry	Molecular Geometry
Carbon (<i>e</i>)		
Sulfur (<i>f</i>)		

vi. Circle the value from the choices given below, that describes the **C-S-C bond angle at sulfur** (S) marked ***f***: (0.5 points)

>90° to <109.5° 109.5° >109.5 to <120° 120° >120° to <180°

**Question 8 12 points**

Using the **valence shell orbitals only** draw a **complete** molecular orbital (MO) diagram (with no s-p mixing) for the molecule **PS**, in the space provided below. **(6 points)**

Assume that 3s and 3p atomic orbitals will form molecular orbitals like those formed by the combination of 2s and 2p atomic orbitals that we learnt in class, and there is no s-p mixing.

Show **energy arrow**, **atomic orbitals** involved, **atom labels**, the **molecular orbitals** that are formed, **filled electrons** in the MO diagram etc. Make sure to label all the atomic and molecular orbitals.

Using the MO diagram above, answer the following questions:

- (i) Which one of the following will have the highest bond order? Circle the correct choice
A) PS^{1-} B) PS C) PS^{1+}
- (ii) Which one of the following will have the longest bond length? Circle the correct choice
A) PS^{1-} B) PS C) PS^{1+}
- (iii) Which of the following will be diamagnetic? Circle the correct choice.
A) PS^{1-} B) PS C) PS^{1+}
- (iv) Draw Lewis dot structure of the molecule with the shortest bond length (based on the MO diagram) from among the three (PS^{1-} , PS, PS^{1+}), and indicate the bond order calculated from the Lewis dot structure. **(3 points)**

Lewis Dot Structure

Bond Order



AB440B15-28EA-4F52-9CF0-7513C6CA6B1B

midterm-exam-93388

#716 Page 10 of 10

SCRAP PAGE

**CHEMISTRY 110 –Midterm Exam****Thursday November 7, 2024, 6:30 – 8:30 pm**

Examiners: Pallavi Sirjoosingh and Ashok Kakkar

First name (please write as legibly as possible within the boxes)

--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--

Last name

--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--

Student ID number

--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--

This booklet has 8 questions on 10 pages (The last page can be used as a scrap page).

The staple must NOT be removed from the question booklet. Your intact question booklet must be returned at the end of the exam.

A separate booklet contains a periodic table, and datasheet.

Write your answers directly under the questions in the space provided. Do not write on the QR codes or near the top edges (as it may be cut off).

Please write **legibly and dark enough** for the scan to be readable. For students who use English as a second language, dictionaries are permitted. Calculators are allowed.

DO NOT ASK QUESTIONS OF THE INVIGILATORS about interpretations of the exam questions. If you have a concern about a question, please summarize it at the top of this page. All concerns will be evaluated after the exam. GOOD LUCK!

**Question 1 8 points** (*Assume all transitions are possible*)

a. 6 points Calculate the **energy (in kJ/mol)** required to ionize (completely remove) the electron from the **n=3** level of **He⁺ ion** (He: Helium). Show your work including equations. (3 sig. figs.)

b. 2 points

(I) Select the largest (longest) wavelength from the three choices below:

- i. Wavelength of photon to *completely remove* an electron from *1s orbital of Hydrogen atom*
- ii. Wavelength of photon to *completely remove* an electron from *1s orbital of Boron atom (B)*
- iii. Wavelength of photon to *completely remove* an electron from *1s orbital of Boron cation (B⁺)*

i or ii or iii

(II) Select the smallest (shortest) wavelength from the three choices below (*for H atom*):

- i. Wavelength of photon *emitted* for an electron transitioning from *2p to 1s orbital*
- ii. Wavelength of photon *absorbed* for an electron transitioning from *2s to 4p orbital*
- iii. Wavelength of photon *absorbed* for an electron transitioning from *1s to 3p orbital*

i or ii or iii



Question 2 8 points



A Chem110 student plots the graph shown on the left, from data obtained for the kinetic energy (eV) of the ejected electron versus frequency (s^{-1}) of the incident light, for an unknown metal. ($1\text{eV} = 1.6022 \times 10^{-19} \text{ J}$)

Based on the plot, answer the questions **a to c** below:

a. 1 point On the plot above, label the **threshold frequency**.

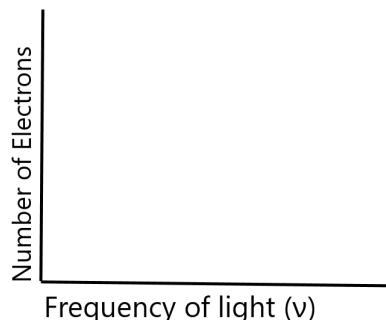
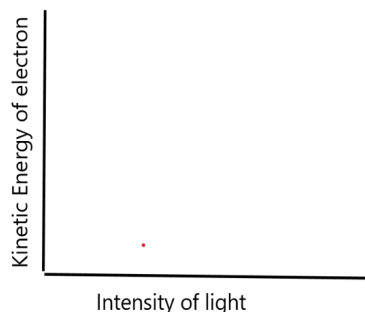
b. 2 points What is the value of the work potential (in J) of the unknown metal?

c. 3 points What is the *longest* wavelength (in nm) required to eject an electron from this metal's surface? Show your work including equation(s) used. (3 sig figs)

d. 2 points In the space provided below, plot the following

(Assume frequency is *above* threshold frequency):

- Kinetic energy* of an ejected electron versus the *intensity* of the incident light (left)
- Number of electrons* ejected versus the *frequency* of the incident light (right)



**Question 3** **6 points****a. 2 points**

In the ground state of Ge (Germanium; $Z = 32$) atom, **name** the following orbitals:

i. Orbital containing the electron that requires the maximum energy to be removed.

ii. Orbital containing the electron that experiences the minimum effective nuclear charge.

b. 2 points

An unknown anion (A^-) has the following *excited* electronic configuration: $[Ar] 4s^2 3d^3 4p^1$

(Ar: Argon; $Z = 18$)

Give the **condensed ground state electronic configuration** of the atom, **A** and the cation, **A^{2+}** .
Show your work. (Assume all electronic transitions are possible)

Electronic Configuration of A:

Electronic Configuration of A^{2+} :

c. 2 points

In a single atom, determine how many electrons can have the following set of quantum numbers?

(i) $n = 5; l = 3$

In a single atom, determine how many electrons can have the following set of quantum numbers?

(i) $n = 3; l = 2; m_s = +1/2$

**Question 4** **8 points**

Rank the following from 1 to 4 in **increasing** order. Provide the ranking in the boxes provided.

Note: Atomic number (Z) are provided for relevant atoms

a) Increasing (lowest to highest) order of *ionization energy*: (1= lowest; 4 = highest)

Mg

Mg: Z = 12

Rb

Rb: Z = 37

Cs

Cs: Z = 55

K

K : Z = 19

b) Increasing (lowest to highest) order of *bond angle*: (1= lowest; 4 = highest)

H₂OCCl₄XeF₄

HCN

c) Increasing (smallest to largest) order of *ionic size* (1= smallest; 4 = largest)

S²⁻

S: Z = 16

O²⁻

O: Z = 8

Be²⁺

Be: Z = 4

Na⁺

Na: Z = 11

d) Increasing (lowest to highest) order of *bond energy* (1= lowest; 4 = highest)

P-F

P-Cl

P-I

P-Br

**Question 5** **12 points**

a. 9 points For the molecules HClO_4 and TeF_6 , draw the most stable *Lewis structure* showing appropriate shape and *dashed/wedged bonds*, based on VSEPR. Determine the **molecular geometry** around the central atom. For HClO_4 , label the *most polar bond*.



b. 3 points (Note: Lone pairs are not shown in structures below)

(i) For the three *resonance* structures (1, 2 and 3) shown below, answer the following question:



(I) Circle the *most contributing* resonance structure.

Structure 1

Structure 2

Structure 3

(ii) For the three structures given (4, 5, and 6) below, answer the following questions:



(I) Circle all the structures that represent *resonance structures* of each other:

Structure 4

Structure 5

Structure 6

(II) Among the resonance structures, circle the *most contributing* resonance structure.

Structure 4

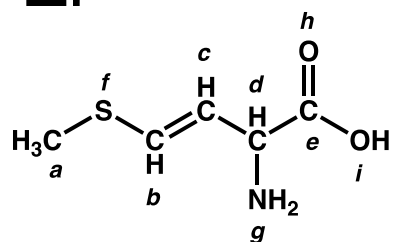
Structure 5

Structure 6

**Question 6** **6 points**

Using the data given below, calculate the **lattice energy** for SrO(s).

	ΔH° (kJ/mol)
$\text{Sr}(s) \longrightarrow \text{Sr}(g)$	+180
$\text{Sr}(g) \longrightarrow \text{Sr}^{2+}(g) + 2e^-$	+1468
$\text{O}_2(g) \longrightarrow 2\text{O}(g)$	+500
$\text{O}(g) + 2e^- \longrightarrow \text{O}^{2-}(g)$	+600
$\text{Sr}(s) + \frac{1}{2}\text{O}_2(g) \longrightarrow \text{SrO}(s)$	-550

**Question 7 10 points**

Consider the molecule given on the left, and answer questions (i) to (vi) using chemical bonding theories that we have learnt in class (valence bond theory, hybridization).

Note that the lone pairs are not shown in the structure:

i. Consider the carbon atoms marked in as ***a***, ***b***, ***c***, ***d***, and ***e*** in the molecule, and determine **hybridization** at each of the carbon centers. (2.5 points)

Atom	Hybridization
Carbon (<i>a</i>)	
Carbon (<i>b</i>)	
Carbon (<i>c</i>)	
Carbon (<i>d</i>)	
Carbon (<i>e</i>)	

ii. Now, consider the sulfur atom (S) marked as ***f***, nitrogen (N) atom (***g***), and oxygen(O) atoms (***h*** and ***i***), and determine their **hybridization**. (2 points)

Atom	Hybridization
Sulfur (<i>f</i>)	
Nitrogen (<i>g</i>)	
Oxygen (<i>h</i>)	
Oxygen (<i>i</i>)	

iii. **Describe** all the bonds formed by the carbon atoms ***a*** and ***b***, using sigma and pi designation as appropriate. (2 points)

Atom	Bonds formed (use sigma and pi as appropriate)
Carbon (<i>a</i>)	
Carbon (<i>b</i>)	

iv. How many total sp^3 -s sigma bonds are in the molecule? (1 point)

v. Determine **electronic** and **molecular geometry** at carbon (C) labeled ***e*** and sulfur (S) labeled ***f*** centres. (2 points)

Atom	Electronic Geometry	Molecular Geometry
Carbon (<i>e</i>)		
Sulfur (<i>f</i>)		

vi. Circle the value from the choices given below, that describes the **C-S-C bond angle at sulfur** (S) marked ***f***: (0.5 points)

$>90^\circ$ to $<109.5^\circ$ 109.5° >109.5 to $<120^\circ$ 120° $>120^\circ$ to $<180^\circ$


Question 8 12 points

Using the **valence shell orbitals only** draw a **complete** molecular orbital (MO) diagram (with no s-p mixing) for the molecule **PS**, in the space provided below. **(6 points)**

Assume that 3s and 3p atomic orbitals will form molecular orbitals like those formed by the combination of 2s and 2p atomic orbitals that we learnt in class, and there is no s-p mixing.

Show **energy arrow**, **atomic orbitals** involved, **atom labels**, the **molecular orbitals** that are formed, **filled electrons** in the MO diagram etc. Make sure to **label all the atomic and molecular orbitals**.

Using the MO diagram above, answer the following questions:

- (i) Which one of the following will have the highest bond order? Circle the correct choice
 A) PS^{1-} B) PS C) PS^{1+}
- (ii) Which one of the following will have the longest bond length? Circle the correct choice
 A) PS^{1-} B) PS C) PS^{1+}
- (iii) Which of the following will be diamagnetic? Circle the correct choice.
 A) PS^{1-} B) PS C) PS^{1+}
- (iv) Draw Lewis dot structure of the molecule with the shortest bond length (based on the MO diagram) from among the three (PS^{1-} , PS, PS^{1+}), and indicate the bond order calculated from the Lewis dot structure. **(3 points)**

Lewis Dot Structure

Bond Order



7EA4603F-C91C-4FA1-9376-260381DC4F11

midterm-exam-93388

#717 Page 10 of 10

SCRAP PAGE

**CHEMISTRY 110 –Midterm Exam****Thursday November 7, 2024, 6:30 – 8:30 pm****Examiners: Pallavi Sirjoosingh and Ashok Kakkar**

First name (please write as legibly as possible within the boxes)

--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--

Last name

--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--

Student ID number

--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--

This booklet has 8 questions on 10 pages (The last page can be used as a scrap page).

The staple must NOT be removed from the question booklet. Your intact question booklet must be returned at the end of the exam.

A separate booklet contains a periodic table, and datasheet.

Write your answers directly under the questions in the space provided. Do not write on the QR codes or near the top edges (as it may be cut off).

Please write **legibly and dark enough** for the scan to be readable. For students who use English as a second language, dictionaries are permitted. Calculators are allowed.

DO NOT ASK QUESTIONS OF THE INVIGILATORS about interpretations of the exam questions. If you have a concern about a question, please summarize it at the top of this page. All concerns will be evaluated after the exam. GOOD LUCK!

**Question 1 8 points** (*Assume all transitions are possible*)

a. 6 points Calculate the **energy (in kJ/mol)** required to ionize (completely remove) the electron from the **n=3** level of **He⁺ ion** (He: Helium). Show your work including equations. (3 sig. figs.)

b. 2 points

(I) Select the largest (longest) wavelength from the three choices below:

- i. Wavelength of photon to *completely remove* an electron from *1s orbital of Hydrogen atom*
- ii. Wavelength of photon to *completely remove* an electron from *1s orbital of Boron atom (B)*
- iii. Wavelength of photon to *completely remove* an electron from *1s orbital of Boron cation (B⁺)*

i or ii or iii

(II) Select the smallest (shortest) wavelength from the three choices below (*for H atom*):

- i. Wavelength of photon *emitted* for an electron transitioning from *2p to 1s orbital*
- ii. Wavelength of photon *absorbed* for an electron transitioning from *2s to 4p orbital*
- iii. Wavelength of photon *absorbed* for an electron transitioning from *1s to 3p orbital*

i or ii or iii



Question 2 8 points



A Chem110 student plots the graph shown on the left, from data obtained for the kinetic energy (eV) of the ejected electron versus frequency (s^{-1}) of the incident light, for an unknown metal. ($1\text{eV} = 1.6022 \times 10^{-19} \text{ J}$)

Based on the plot, answer the questions **a to c** below:

a. 1 point On the plot above, label the **threshold frequency**.

b. 2 points What is the value of the work potential (in J) of the unknown metal?

c. 3 points What is the *longest* wavelength (in nm) required to eject an electron from this metal's surface? Show your work including equation(s) used. (3 sig figs)

d. 2 points In the space provided below, plot the following

(Assume frequency is *above* threshold frequency):

- Kinetic energy* of an ejected electron versus the *intensity* of the incident light (left)
- Number of electrons* ejected versus the *frequency* of the incident light (right)



**Question 3** **6 points****a. 2 points**

In the ground state of Ge (Germanium; $Z = 32$) atom, **name** the following orbitals:

i. Orbital containing the electron that requires the maximum energy to be removed.

ii. Orbital containing the electron that experiences the minimum effective nuclear charge.

b. 2 points

An unknown anion (A^-) has the following *excited* electronic configuration: $[Ar] 4s^2 3d^3 4p^1$

(Ar: Argon; $Z = 18$)

Give the **condensed ground state electronic configuration** of the atom, **A** and the cation, **A^{2+}** .
Show your work. (Assume all electronic transitions are possible)

Electronic Configuration of A:

Electronic Configuration of A^{2+} :

c. 2 points

In a single atom, determine how many electrons can have the following set of quantum numbers?

(i) $n = 5; l = 3$

In a single atom, determine how many electrons can have the following set of quantum numbers?

(i) $n = 3; l = 2; m_s = +1/2$

**Question 4** **8 points**

Rank the following from 1 to 4 in **increasing** order. Provide the ranking in the boxes provided.

Note: Atomic number (Z) are provided for relevant atoms

a) Increasing (lowest to highest) order of *ionization energy*: (1= lowest; 4 = highest)

Mg

Mg: Z = 12

Rb

Rb: Z = 37

Cs

Cs: Z = 55

K

K : Z = 19

b) Increasing (lowest to highest) order of *bond angle*: (1= lowest; 4 = highest)

H₂OCCl₄XeF₄

HCN

c) Increasing (smallest to largest) order of *ionic size* (1= smallest; 4 = largest)

S²⁻

S: Z = 16

O²⁻

O: Z = 8

Be²⁺

Be: Z = 4

Na⁺

Na: Z = 11

d) Increasing (lowest to highest) order of *bond energy* (1= lowest; 4 = highest)

P-F

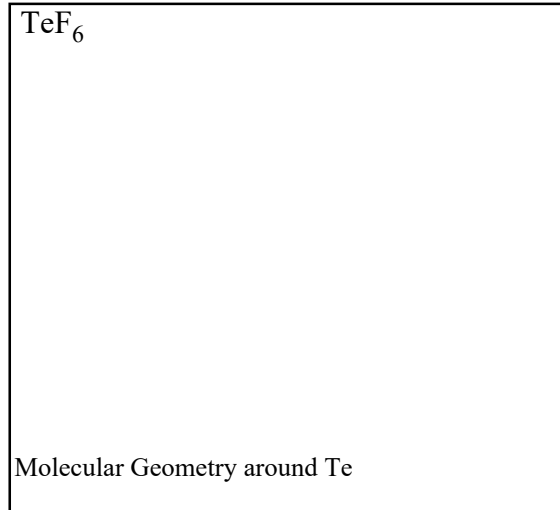
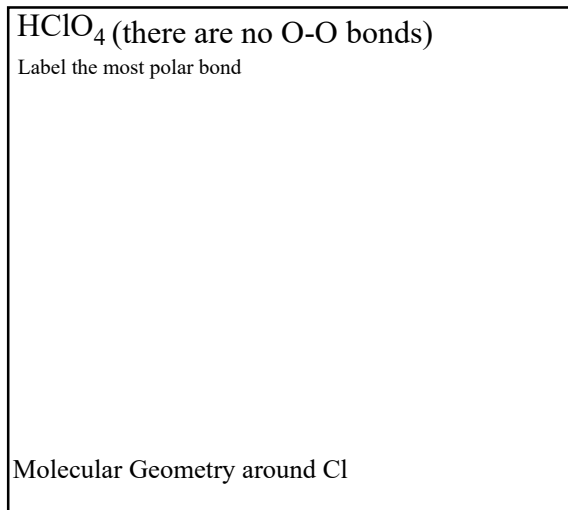
P-Cl

P-I

P-Br

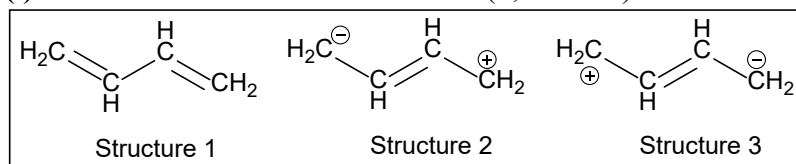
**Question 5** **12 points**

a. 9 points For the molecules HClO_4 and TeF_6 , draw the most stable *Lewis structure* showing appropriate shape and *dashed/wedged bonds*, based on VSEPR. Determine the **molecular geometry** around the central atom. For HClO_4 , label the *most polar bond*.



b. 3 points (Note: Lone pairs are not shown in structures below)

(i) For the three *resonance* structures (1, 2 and 3) shown below, answer the following question:



(I) Circle the *most contributing* resonance structure.

Structure 1

Structure 2

Structure 3

(ii) For the three structures given (4, 5, and 6) below, answer the following questions:



(I) Circle all the structures that represent *resonance structures* of each other:

Structure 4

Structure 5

Structure 6

(II) Among the resonance structures, circle the *most contributing* resonance structure.

Structure 4

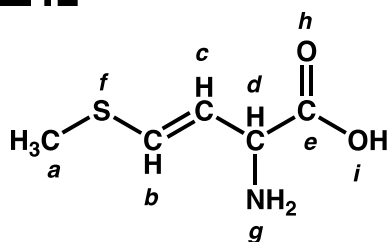
Structure 5

Structure 6

**Question 6** **6 points**

Using the data given below, calculate the **lattice energy** for SrO(s).

	ΔH° (kJ/mol)
$\text{Sr}(s) \longrightarrow \text{Sr}(g)$	+180
$\text{Sr}(g) \longrightarrow \text{Sr}^{2+}(g) + 2e^-$	+1468
$\text{O}_2(g) \longrightarrow 2\text{O}(g)$	+500
$\text{O}(g) + 2e^- \longrightarrow \text{O}^{2-}(g)$	+600
$\text{Sr}(s) + \frac{1}{2}\text{O}_2(g) \longrightarrow \text{SrO}(s)$	-550

**Question 7 10 points**

Consider the molecule given on the left, and answer questions (i) to (vi) using chemical bonding theories that we have learnt in class (valence bond theory, hybridization).

Note that the lone pairs are not shown in the structure:

i. Consider the carbon atoms marked in as ***a***, ***b***, ***c***, ***d***, and ***e*** in the molecule, and determine **hybridization** at each of the carbon centers. (2.5 points)

Atom	Hybridization
Carbon (<i>a</i>)	
Carbon (<i>b</i>)	
Carbon (<i>c</i>)	
Carbon (<i>d</i>)	
Carbon (<i>e</i>)	

ii. Now, consider the sulfur atom (S) marked as ***f***, nitrogen (N) atom (***g***), and oxygen(O) atoms (***h*** and ***i***), and determine their **hybridization**. (2 points)

Atom	Hybridization
Sulfur (<i>f</i>)	
Nitrogen (<i>g</i>)	
Oxygen (<i>h</i>)	
Oxygen (<i>i</i>)	

iii. **Describe** all the bonds formed by the carbon atoms ***a*** and ***b***, using sigma and pi designation as appropriate. (2 points)

Atom	Bonds formed (use sigma and pi as appropriate)
Carbon (<i>a</i>)	
Carbon (<i>b</i>)	

iv. How many total sp^3 -s sigma bonds are in the molecule? (1 point)

v. Determine **electronic** and **molecular geometry** at carbon (C) labeled ***e*** and sulfur (S) labeled ***f*** centres. (2 points)

Atom	Electronic Geometry	Molecular Geometry
Carbon (<i>e</i>)		
Sulfur (<i>f</i>)		

vi. Circle the value from the choices given below, that describes the **C-S-C bond angle at sulfur** (S) marked ***f***: (0.5 points)

>90° to <109.5° 109.5° >109.5 to <120° 120° >120° to <180°


Question 8 12 points

Using the **valence shell orbitals only** draw a **complete** molecular orbital (MO) diagram (with no s-p mixing) for the molecule **PS**, in the space provided below. **(6 points)**

Assume that 3s and 3p atomic orbitals will form molecular orbitals like those formed by the combination of 2s and 2p atomic orbitals that we learnt in class, and there is no s-p mixing.

Show **energy arrow**, **atomic orbitals** involved, **atom labels**, the **molecular orbitals** that are formed, **filled electrons** in the MO diagram etc. Make sure to label all the atomic and molecular orbitals.

Using the MO diagram above, answer the following questions:

- (i) Which one of the following will have the highest bond order? Circle the correct choice
 A) PS^{1-} B) PS C) PS^{1+}
- (ii) Which one of the following will have the longest bond length? Circle the correct choice
 A) PS^{1-} B) PS C) PS^{1+}
- (iii) Which of the following will be diamagnetic? Circle the correct choice.
 A) PS^{1-} B) PS C) PS^{1+}
- (iv) Draw Lewis dot structure of the molecule with the shortest bond length (based on the MO diagram) from among the three (PS^{1-} , PS, PS^{1+}), and indicate the bond order calculated from the Lewis dot structure. **(3 points)**

Lewis Dot Structure

Bond Order



3B637403-4289-4E40-87B1-2C0F4F46EC84

midterm-exam-93388

#718 Page 10 of 10

SCRAP PAGE

**CHEMISTRY 110 –Midterm Exam****Thursday November 7, 2024, 6:30 – 8:30 pm**

Examiners: Pallavi Sirjoosingh and Ashok Kakkar

First name (please write as legibly as possible within the boxes)

--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--

Last name

--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--

Student ID number

--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--

This booklet has 8 questions on 10 pages (The last page can be used as a scrap page).

The staple must NOT be removed from the question booklet. Your intact question booklet must be returned at the end of the exam.

A separate booklet contains a periodic table, and datasheet.

Write your answers directly under the questions in the space provided. Do not write on the QR codes or near the top edges (as it may be cut off).

Please write **legibly and dark enough** for the scan to be readable. For students who use English as a second language, dictionaries are permitted. Calculators are allowed.

DO NOT ASK QUESTIONS OF THE INVIGILATORS about interpretations of the exam questions. If you have a concern about a question, please summarize it at the top of this page. All concerns will be evaluated after the exam. GOOD LUCK!

**Question 1 8 points** (*Assume all transitions are possible*)

a. 6 points Calculate the **energy (in kJ/mol)** required to ionize (completely remove) the electron from the **n=3** level of **He⁺ ion** (He: Helium). Show your work including equations. (3 sig. figs.)

b. 2 points

(I) Select the largest (longest) wavelength from the three choices below:

- i. Wavelength of photon to *completely remove* an electron from *1s orbital of Hydrogen atom*
- ii. Wavelength of photon to *completely remove* an electron from *1s orbital of Boron atom (B)*
- iii. Wavelength of photon to *completely remove* an electron from *1s orbital of Boron cation (B⁺)*

i or ii or iii

(II) Select the smallest (shortest) wavelength from the three choices below (*for H atom*):

- i. Wavelength of photon *emitted* for an electron transitioning from *2p to 1s orbital*
- ii. Wavelength of photon *absorbed* for an electron transitioning from *2s to 4p orbital*
- iii. Wavelength of photon *absorbed* for an electron transitioning from *1s to 3p orbital*

i or ii or iii



Question 2 8 points



A Chem110 student plots the graph shown on the left, from data obtained for the kinetic energy (eV) of the ejected electron versus frequency (s^{-1}) of the incident light, for an unknown metal. ($1\text{eV} = 1.6022 \times 10^{-19} \text{ J}$)

Based on the plot, answer the questions **a to c** below:

a. 1 point On the plot above, label the **threshold frequency**.

b. 2 points What is the value of the work potential (in J) of the unknown metal?

c. 3 points What is the *longest* wavelength (in nm) required to eject an electron from this metal's surface? Show your work including equation(s) used. (3 sig figs)

d. 2 points In the space provided below, plot the following

(Assume frequency is *above* threshold frequency):

- i. *Kinetic energy* of an ejected electron versus the *intensity* of the incident light (left)
- ii. *Number of electrons* ejected versus the *frequency* of the incident light (right)



**Question 3** **6 points****a. 2 points**

In the ground state of Ge (Germanium; $Z = 32$) atom, **name** the following orbitals:

i. Orbital containing the electron that requires the maximum energy to be removed.

ii. Orbital containing the electron that experiences the minimum effective nuclear charge.

b. 2 points

An unknown anion (A^-) has the following *excited* electronic configuration: $[Ar] 4s^2 3d^3 4p^1$

(Ar: Argon; $Z = 18$)

Give the **condensed ground state electronic configuration** of the atom, **A** and the cation, **A^{2+}** .
Show your work. (Assume all electronic transitions are possible)

Electronic Configuration of A:

Electronic Configuration of A^{2+} :

c. 2 points

In a single atom, determine how many electrons can have the following set of quantum numbers?

(i) $n = 5; l = 3$

In a single atom, determine how many electrons can have the following set of quantum numbers?

(i) $n = 3; l = 2; m_s = +1/2$

**Question 4** **8 points**

Rank the following from 1 to 4 in **increasing** order. Provide the ranking in the boxes provided.

Note: Atomic number (Z) are provided for relevant atoms

a) Increasing (lowest to highest) order of *ionization energy*: (1= lowest; 4 = highest)

Mg

Mg: Z = 12

Rb

Rb: Z = 37

Cs

Cs: Z = 55

K

K : Z = 19

b) Increasing (lowest to highest) order of *bond angle*: (1= lowest; 4 = highest)

H₂OCCl₄XeF₄

HCN

c) Increasing (smallest to largest) order of *ionic size* (1= smallest; 4 = largest)

S²⁻

S: Z = 16

O²⁻

O: Z = 8

Be²⁺

Be: Z = 4

Na⁺

Na: Z = 11

d) Increasing (lowest to highest) order of *bond energy* (1= lowest; 4 = highest)

P-F

P-Cl

P-I

P-Br

**Question 5** **12 points**

a. 9 points For the molecules HClO_4 and TeF_6 , draw the most stable *Lewis structure* showing appropriate shape and *dashed/wedged bonds*, based on VSEPR. Determine the **molecular geometry** around the central atom. For HClO_4 , label the *most polar bond*.



b. 3 points (Note: Lone pairs are not shown in structures below)

(i) For the three *resonance* structures (1, 2 and 3) shown below, answer the following question:



(I) Circle the *most contributing* resonance structure.

Structure 1

Structure 2

Structure 3

(ii) For the three structures given (4, 5, and 6) below, answer the following questions:



(I) Circle all the structures that represent *resonance structures* of each other:

Structure 4

Structure 5

Structure 6

(II) Among the resonance structures, circle the *most contributing* resonance structure.

Structure 4

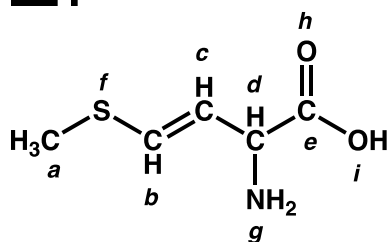
Structure 5

Structure 6

**Question 6** **6 points**

Using the data given below, calculate the **lattice energy** for SrO(s).

	ΔH° (kJ/mol)
$\text{Sr}(s) \longrightarrow \text{Sr}(g)$	+180
$\text{Sr}(g) \longrightarrow \text{Sr}^{2+}(g) + 2e^-$	+1468
$\text{O}_2(g) \longrightarrow 2\text{O}(g)$	+500
$\text{O}(g) + 2e^- \longrightarrow \text{O}^{2-}(g)$	+600
$\text{Sr}(s) + \frac{1}{2}\text{O}_2(g) \longrightarrow \text{SrO}(s)$	-550

**Question 7 10 points**

Consider the molecule given on the left, and answer questions (i) to (vi) using chemical bonding theories that we have learnt in class (valence bond theory, hybridization).

Note that the lone pairs are not shown in the structure:

i. Consider the carbon atoms marked in as ***a***, ***b***, ***c***, ***d***, and ***e*** in the molecule, and determine **hybridization** at each of the carbon centers. (2.5 points)

Atom	Hybridization
Carbon (<i>a</i>)	
Carbon (<i>b</i>)	
Carbon (<i>c</i>)	
Carbon (<i>d</i>)	
Carbon (<i>e</i>)	

ii. Now, consider the sulfur atom (S) marked as ***f***, nitrogen (N) atom (***g***), and oxygen(O) atoms (***h*** and ***i***), and determine their **hybridization**. (2 points)

Atom	Hybridization
Sulfur (<i>f</i>)	
Nitrogen (<i>g</i>)	
Oxygen (<i>h</i>)	
Oxygen (<i>i</i>)	

iii. **Describe** all the bonds formed by the carbon atoms ***a*** and ***b***, using sigma and pi designation as appropriate. (2 points)

Atom	Bonds formed (use sigma and pi as appropriate)
Carbon (<i>a</i>)	
Carbon (<i>b</i>)	

iv. How many total sp^3 -s sigma bonds are in the molecule? (1 point)

v. Determine **electronic** and **molecular geometry** at carbon (C) labeled ***e*** and sulfur (S) labeled ***f*** centres. (2 points)

Atom	Electronic Geometry	Molecular Geometry
Carbon (<i>e</i>)		
Sulfur (<i>f</i>)		

vi. Circle the value from the choices given below, that describes the **C-S-C bond angle at sulfur** (S) marked ***f***: (0.5 points)

$>90^\circ$ to $<109.5^\circ$ 109.5° >109.5 to $<120^\circ$ 120° $>120^\circ$ to $<180^\circ$


Question 8 12 points

Using the **valence shell orbitals only** draw a **complete** molecular orbital (MO) diagram (with no s-p mixing) for the molecule **PS**, in the space provided below. **(6 points)**

Assume that 3s and 3p atomic orbitals will form molecular orbitals like those formed by the combination of 2s and 2p atomic orbitals that we learnt in class, and there is no s-p mixing.

Show **energy arrow**, **atomic orbitals** involved, **atom labels**, the **molecular orbitals** that are formed, **filled electrons** in the MO diagram etc. Make sure to label all the atomic and molecular orbitals.

Using the MO diagram above, answer the following questions:

- (i) Which one of the following will have the highest bond order? Circle the correct choice
 A) PS^{1-} B) PS C) PS^{1+}
- (ii) Which one of the following will have the longest bond length? Circle the correct choice
 A) PS^{1-} B) PS C) PS^{1+}
- (iii) Which of the following will be diamagnetic? Circle the correct choice.
 A) PS^{1-} B) PS C) PS^{1+}
- (iv) Draw Lewis dot structure of the molecule with the shortest bond length (based on the MO diagram) from among the three (PS^{1-} , PS, PS^{1+}), and indicate the bond order calculated from the Lewis dot structure. **(3 points)**

Lewis Dot Structure

Bond Order



81138AE1-902B-41E8-BC21-7EE11A78AE66

midterm-exam-93388

#719 Page 10 of 10

SCRAP PAGE

**CHEMISTRY 110 –Midterm Exam****Thursday November 7, 2024, 6:30 – 8:30 pm****Examiners: Pallavi Sirjoosingh and Ashok Kakkar**

First name (please write as legibly as possible within the boxes)

--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--

Last name

--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--

Student ID number

--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--

This booklet has 8 questions on 10 pages (The last page can be used as a scrap page).

The staple must NOT be removed from the question booklet. Your intact question booklet must be returned at the end of the exam.

A separate booklet contains a periodic table, and datasheet.

Write your answers directly under the questions in the space provided. Do not write on the QR codes or near the top edges (as it may be cut off).

Please write **legibly and dark enough** for the scan to be readable. For students who use English as a second language, dictionaries are permitted. Calculators are allowed.

DO NOT ASK QUESTIONS OF THE INVIGILATORS about interpretations of the exam questions. If you have a concern about a question, please summarize it at the top of this page. All concerns will be evaluated after the exam. GOOD LUCK!

**Question 1 8 points** (*Assume all transitions are possible*)

a. 6 points Calculate the **energy (in kJ/mol)** required to ionize (completely remove) the electron from the **n=3** level of **He⁺ ion** (He: Helium). Show your work including equations. (3 sig. figs.)

b. 2 points

(I) Select the largest (longest) wavelength from the three choices below:

- i. Wavelength of photon to *completely remove* an electron from *1s orbital of Hydrogen atom*
- ii. Wavelength of photon to *completely remove* an electron from *1s orbital of Boron atom (B)*
- iii. Wavelength of photon to *completely remove* an electron from *1s orbital of Boron cation (B⁺)*

i or ii or iii

(II) Select the smallest (shortest) wavelength from the three choices below (*for H atom*):

- i. Wavelength of photon *emitted* for an electron transitioning from *2p to 1s orbital*
- ii. Wavelength of photon *absorbed* for an electron transitioning from *2s to 4p orbital*
- iii. Wavelength of photon *absorbed* for an electron transitioning from *1s to 3p orbital*

i or ii or iii



Question 2 8 points



A Chem110 student plots the graph shown on the left, from data obtained for the kinetic energy (eV) of the ejected electron versus frequency (s^{-1}) of the incident light, for an unknown metal. ($1\text{eV} = 1.6022 \times 10^{-19} \text{ J}$)

Based on the plot, answer the questions **a to c** below:

a. 1 point On the plot above, label the **threshold frequency**.

b. 2 points What is the value of the work potential (in J) of the unknown metal?

c. 3 points What is the *longest* wavelength (in nm) required to eject an electron from this metal's surface? Show your work including equation(s) used. (3 sig figs)

d. 2 points In the space provided below, plot the following

(Assume frequency is *above* threshold frequency):

- i. *Kinetic energy* of an ejected electron versus the *intensity* of the incident light (left)
- ii. *Number of electrons* ejected versus the *frequency* of the incident light (right)



**Question 3** **6 points****a. 2 points**

In the ground state of Ge (Germanium; $Z = 32$) atom, **name** the following orbitals:

i. Orbital containing the electron that requires the maximum energy to be removed.

ii. Orbital containing the electron that experiences the minimum effective nuclear charge.

b. 2 points

An unknown anion (A^-) has the following *excited* electronic configuration: $[Ar] 4s^2 3d^3 4p^1$

(Ar: Argon; $Z = 18$)

Give the **condensed ground state electronic configuration** of the atom, **A** and the cation, **A^{2+}** .
Show your work. (Assume all electronic transitions are possible)

Electronic Configuration of A:

Electronic Configuration of A^{2+} :

c. 2 points

In a single atom, determine how many electrons can have the following set of quantum numbers?

(i) $n = 5; l = 3$

In a single atom, determine how many electrons can have the following set of quantum numbers?

(i) $n = 3; l = 2; m_s = +1/2$

**Question 4** **8 points**

Rank the following from 1 to 4 in **increasing** order. Provide the ranking in the boxes provided.

Note: Atomic number (Z) are provided for relevant atoms

a) Increasing (lowest to highest) order of *ionization energy*: (1= lowest; 4 = highest)

Mg

Mg: Z = 12

Rb

Rb: Z = 37

Cs

Cs: Z = 55

K

K : Z = 19

b) Increasing (lowest to highest) order of *bond angle*: (1= lowest; 4 = highest)

H₂OCCl₄XeF₄

HCN

c) Increasing (smallest to largest) order of *ionic size* (1= smallest; 4 = largest)

S²⁻

S: Z = 16

O²⁻

O: Z = 8

Be²⁺

Be: Z = 4

Na⁺

Na: Z = 11

d) Increasing (lowest to highest) order of *bond energy* (1= lowest; 4 = highest)

P-F

P-Cl

P-I

P-Br

**Question 5** **12 points**

a. 9 points For the molecules HClO_4 and TeF_6 , draw the most stable *Lewis structure* showing appropriate shape and *dashed/wedged bonds*, based on VSEPR. Determine the **molecular geometry** around the central atom. For HClO_4 , label the *most polar bond*.



b. 3 points (Note: Lone pairs are not shown in structures below)

(i) For the three *resonance* structures (1, 2 and 3) shown below, answer the following question:



(I) Circle the *most contributing* resonance structure.

Structure 1

Structure 2

Structure 3

(ii) For the three structures given (4, 5, and 6) below, answer the following questions:



(I) Circle all the structures that represent *resonance structures* of each other:

Structure 4

Structure 5

Structure 6

(II) Among the resonance structures, circle the *most contributing* resonance structure.

Structure 4

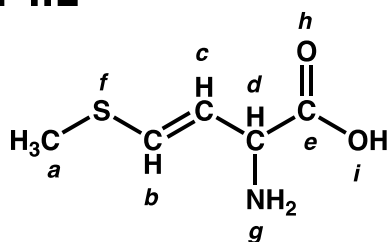
Structure 5

Structure 6

**Question 6** **6 points**

Using the data given below, calculate the **lattice energy** for SrO(s).

	ΔH° (kJ/mol)
$\text{Sr}(s) \longrightarrow \text{Sr}(g)$	+180
$\text{Sr}(g) \longrightarrow \text{Sr}^{2+}(g) + 2e^-$	+1468
$\text{O}_2(g) \longrightarrow 2\text{O}(g)$	+500
$\text{O}(g) + 2e^- \longrightarrow \text{O}^{2-}(g)$	+600
$\text{Sr}(s) + \frac{1}{2}\text{O}_2(g) \longrightarrow \text{SrO}(s)$	-550

**Question 7 10 points**

Consider the molecule given on the left, and answer questions (i) to (vi) using chemical bonding theories that we have learnt in class (valence bond theory, hybridization).

Note that the lone pairs are not shown in the structure:

i. Consider the carbon atoms marked in as ***a***, ***b***, ***c***, ***d***, and ***e*** in the molecule, and determine **hybridization** at each of the carbon centers. (2.5 points)

Atom	Hybridization
Carbon (<i>a</i>)	
Carbon (<i>b</i>)	
Carbon (<i>c</i>)	
Carbon (<i>d</i>)	
Carbon (<i>e</i>)	

ii. Now, consider the sulfur atom (S) marked as ***f***, nitrogen (N) atom (***g***), and oxygen(O) atoms (***h*** and ***i***), and determine their **hybridization**. (2 points)

Atom	Hybridization
Sulfur (<i>f</i>)	
Nitrogen (<i>g</i>)	
Oxygen (<i>h</i>)	
Oxygen (<i>i</i>)	

iii. **Describe** all the bonds formed by the carbon atoms ***a*** and ***b***, using sigma and pi designation as appropriate. (2 points)

Atom	Bonds formed (use sigma and pi as appropriate)
Carbon (<i>a</i>)	
Carbon (<i>b</i>)	

iv. How many total sp^3 -s sigma bonds are in the molecule? (1 point)

v. Determine **electronic** and **molecular geometry** at carbon (C) labeled ***e*** and sulfur (S) labeled ***f*** centres. (2 points)

Atom	Electronic Geometry	Molecular Geometry
Carbon (<i>e</i>)		
Sulfur (<i>f</i>)		

vi. Circle the value from the choices given below, that describes the **C-S-C bond angle at sulfur** (S) marked ***f***: (0.5 points)

>90° to <109.5°

109.5°

>109.5 to <120°

120°

>120° to <180°


Question 8 12 points

Using the **valence shell orbitals only** draw a **complete** molecular orbital (MO) diagram (with no s-p mixing) for the molecule **PS**, in the space provided below. **(6 points)**

Assume that 3s and 3p atomic orbitals will form molecular orbitals like those formed by the combination of 2s and 2p atomic orbitals that we learnt in class, and there is no s-p mixing.

Show **energy arrow**, **atomic orbitals** involved, **atom labels**, the **molecular orbitals** that are formed, **filled electrons** in the MO diagram etc. Make sure to **label all the atomic and molecular orbitals**.

Using the MO diagram above, answer the following questions:

- (i) Which one of the following will have the highest bond order? Circle the correct choice
 A) PS^{1-} B) PS C) PS^{1+}
- (ii) Which one of the following will have the longest bond length? Circle the correct choice
 A) PS^{1-} B) PS C) PS^{1+}
- (iii) Which of the following will be diamagnetic? Circle the correct choice.
 A) PS^{1-} B) PS C) PS^{1+}
- (iv) Draw Lewis dot structure of the molecule with the shortest bond length (based on the MO diagram) from among the three (PS^{1-} , PS, PS^{1+}), and indicate the bond order calculated from the Lewis dot structure. **(3 points)**

Lewis Dot Structure

Bond Order



79E2F8A6-5CC8-441F-8F65-E4382B5CF648

midterm-exam-93388

#720 Page 10 of 10

SCRAP PAGE

**CHEMISTRY 110 –Midterm Exam****Thursday November 7, 2024, 6:30 – 8:30 pm****Examiners: Pallavi Sirjoosingh and Ashok Kakkar**

First name (please write as legibly as possible within the boxes)

--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--

Last name

--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--

Student ID number

--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--

This booklet has 8 questions on 10 pages (The last page can be used as a scrap page).

The staple must NOT be removed from the question booklet. Your intact question booklet must be returned at the end of the exam.

A separate booklet contains a periodic table, and datasheet.

Write your answers directly under the questions in the space provided. Do not write on the QR codes or near the top edges (as it may be cut off).

Please write **legibly and dark enough** for the scan to be readable. For students who use English as a second language, dictionaries are permitted. Calculators are allowed.

DO NOT ASK QUESTIONS OF THE INVIGILATORS about interpretations of the exam questions. If you have a concern about a question, please summarize it at the top of this page. All concerns will be evaluated after the exam. GOOD LUCK!

**Question 1 8 points** (*Assume all transitions are possible*)

a. 6 points Calculate the **energy (in kJ/mol)** required to ionize (completely remove) the electron from the **n=3** level of **He⁺ ion** (He: Helium). Show your work including equations. (3 sig. figs.)

b. 2 points

(I) Select the largest (longest) wavelength from the three choices below:

- i. Wavelength of photon to *completely remove* an electron from *1s orbital of Hydrogen atom*
- ii. Wavelength of photon to *completely remove* an electron from *1s orbital of Boron atom (B)*
- iii. Wavelength of photon to *completely remove* an electron from *1s orbital of Boron cation (B⁺)*

i or ii or iii

(II) Select the smallest (shortest) wavelength from the three choices below (*for H atom*):

- i. Wavelength of photon *emitted* for an electron transitioning from *2p to 1s orbital*
- ii. Wavelength of photon *absorbed* for an electron transitioning from *2s to 4p orbital*
- iii. Wavelength of photon *absorbed* for an electron transitioning from *1s to 3p orbital*

i or ii or iii



Question 2 8 points



A Chem110 student plots the graph shown on the left, from data obtained for the kinetic energy (eV) of the ejected electron versus frequency (s^{-1}) of the incident light, for an unknown metal. ($1\text{eV} = 1.6022 \times 10^{-19} \text{ J}$)

Based on the plot, answer the questions **a to c** below:

a. 1 point On the plot above, label the **threshold frequency**.

b. 2 points What is the value of the work potential (in J) of the unknown metal?

c. 3 points What is the *longest* wavelength (in nm) required to eject an electron from this metal's surface? Show your work including equation(s) used. (3 sig figs)

d. 2 points In the space provided below, plot the following

(Assume frequency is *above* threshold frequency):

- i. *Kinetic energy* of an ejected electron versus the *intensity* of the incident light (left)
- ii. *Number of electrons* ejected versus the *frequency* of the incident light (right)



**Question 3** **6 points****a. 2 points**

In the ground state of Ge (Germanium; $Z = 32$) atom, **name** the following orbitals:

i. Orbital containing the electron that requires the maximum energy to be removed.

ii. Orbital containing the electron that experiences the minimum effective nuclear charge.

b. 2 points

An unknown anion (A^-) has the following *excited* electronic configuration: $[Ar] 4s^2 3d^3 4p^1$

(Ar: Argon; $Z = 18$)

Give the **condensed ground state electronic configuration** of the atom, **A** and the cation, **A^{2+}** .
Show your work. (Assume all electronic transitions are possible)

Electronic Configuration of A:

Electronic Configuration of A^{2+} :

c. 2 points

In a single atom, determine how many electrons can have the following set of quantum numbers?

(i) $n = 5; l = 3$

In a single atom, determine how many electrons can have the following set of quantum numbers?

(i) $n = 3; l = 2; m_s = +1/2$

**Question 4** **8 points**

Rank the following from 1 to 4 in **increasing** order. Provide the ranking in the boxes provided.

Note: Atomic number (Z) are provided for relevant atoms

a) Increasing (lowest to highest) order of *ionization energy*: (1= lowest; 4 = highest)

Mg

Mg: Z = 12

Rb

Rb: Z = 37

Cs

Cs: Z = 55

K

K : Z = 19

b) Increasing (lowest to highest) order of *bond angle*: (1= lowest; 4 = highest)

H₂OCCl₄XeF₄

HCN

c) Increasing (smallest to largest) order of *ionic size* (1= smallest; 4 = largest)

S²⁻

S: Z = 16

O²⁻

O: Z = 8

Be²⁺

Be: Z = 4

Na⁺

Na: Z = 11

d) Increasing (lowest to highest) order of *bond energy* (1= lowest; 4 = highest)

P-F

P-Cl

P-I

P-Br

**Question 5** **12 points**

a. 9 points For the molecules HClO_4 and TeF_6 , draw the most stable *Lewis structure* showing appropriate shape and *dashed/wedged bonds*, based on VSEPR. Determine the **molecular geometry** around the central atom. For HClO_4 , label the *most polar bond*.



b. 3 points (Note: Lone pairs are not shown in structures below)

(i) For the three *resonance* structures (1, 2 and 3) shown below, answer the following question:



(I) Circle the *most contributing* resonance structure.

Structure 1

Structure 2

Structure 3

(ii) For the three structures given (4, 5, and 6) below, answer the following questions:



(I) Circle all the structures that represent *resonance structures* of each other:

Structure 4

Structure 5

Structure 6

(II) Among the resonance structures, circle the *most contributing* resonance structure.

Structure 4

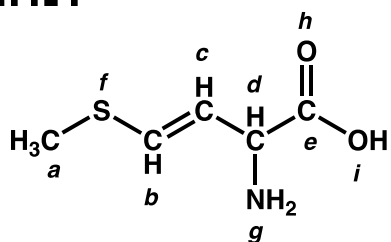
Structure 5

Structure 6

**Question 6** **6 points**

Using the data given below, calculate the **lattice energy** for SrO(s).

	ΔH° (kJ/mol)
$\text{Sr}(s) \longrightarrow \text{Sr}(g)$	+180
$\text{Sr}(g) \longrightarrow \text{Sr}^{2+}(g) + 2e^-$	+1468
$\text{O}_2(g) \longrightarrow 2\text{O}(g)$	+500
$\text{O}(g) + 2e^- \longrightarrow \text{O}^{2-}(g)$	+600
$\text{Sr}(s) + \frac{1}{2}\text{O}_2(g) \longrightarrow \text{SrO}(s)$	-550

**Question 7 10 points**

Consider the molecule given on the left, and answer questions (i) to (vi) using chemical bonding theories that we have learnt in class (valence bond theory, hybridization).

Note that the lone pairs are not shown in the structure:

i. Consider the carbon atoms marked in as ***a***, ***b***, ***c***, ***d***, and ***e*** in the molecule, and determine **hybridization** at each of the carbon centers. (2.5 points)

Atom	Hybridization
Carbon (<i>a</i>)	
Carbon (<i>b</i>)	
Carbon (<i>c</i>)	
Carbon (<i>d</i>)	
Carbon (<i>e</i>)	

ii. Now, consider the sulfur atom (S) marked as ***f***, nitrogen (N) atom (***g***), and oxygen(O) atoms (***h*** and ***i***), and determine their **hybridization**. (2 points)

Atom	Hybridization
Sulfur (<i>f</i>)	
Nitrogen (<i>g</i>)	
Oxygen (<i>h</i>)	
Oxygen (<i>i</i>)	

iii. **Describe** all the bonds formed by the carbon atoms ***a*** and ***b***, using sigma and pi designation as appropriate. (2 points)

Atom	Bonds formed (use sigma and pi as appropriate)
Carbon (<i>a</i>)	
Carbon (<i>b</i>)	

iv. How many total sp^3 -s sigma bonds are in the molecule? (1 point)

v. Determine **electronic** and **molecular geometry** at carbon (C) labeled ***e*** and sulfur (S) labeled ***f*** centres. (2 points)

Atom	Electronic Geometry	Molecular Geometry
Carbon (<i>e</i>)		
Sulfur (<i>f</i>)		

vi. Circle the value from the choices given below, that describes the **C-S-C bond angle at sulfur** (S) marked ***f***: (0.5 points)

>90° to <109.5° 109.5° >109.5 to <120° 120° >120° to <180°



Question 8 12 points

Using the **valence shell orbitals only** draw a **complete** molecular orbital (MO) diagram (with no s-p mixing) for the molecule **PS**, in the space provided below. **(6 points)**

Assume that 3s and 3p atomic orbitals will form molecular orbitals like those formed by the combination of 2s and 2p atomic orbitals that we learnt in class, and there is no s-p mixing.

Show **energy arrow**, **atomic orbitals** involved, **atom labels**, the **molecular orbitals** that are formed, **filled electrons** in the MO diagram etc. Make sure to label all the atomic and molecular orbitals.

Using the MO diagram above, answer the following questions:

- (i) Which one of the following will have the highest bond order? Circle the correct choice
 A) PS^{1-} B) PS C) PS^{1+}
- (ii) Which one of the following will have the longest bond length? Circle the correct choice
 A) PS^{1-} B) PS C) PS^{1+}
- (iii) Which of the following will be diamagnetic? Circle the correct choice.
 A) PS^{1-} B) PS C) PS^{1+}
- (iv) Draw Lewis dot structure of the molecule with the shortest bond length (based on the MO diagram) from among the three (PS^{1-} , PS, PS^{1+}), and indicate the bond order calculated from the Lewis dot structure. **(3 points)**

Lewis Dot Structure

Bond Order



987093BF-F399-46A4-9F9A-F3084CD85F28

midterm-exam-93388

#721 Page 10 of 10

SCRAP PAGE

**CHEMISTRY 110 –Midterm Exam****Thursday November 7, 2024, 6:30 – 8:30 pm****Examiners: Pallavi Sirjoosingh and Ashok Kakkar**

First name (please write as legibly as possible within the boxes)

--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--

Last name

--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--

Student ID number

--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--

This booklet has 8 questions on 10 pages (The last page can be used as a scrap page).

The staple must NOT be removed from the question booklet. Your intact question booklet must be returned at the end of the exam.

A separate booklet contains a periodic table, and datasheet.

Write your answers directly under the questions in the space provided. Do not write on the QR codes or near the top edges (as it may be cut off).

Please write **legibly and dark enough** for the scan to be readable. For students who use English as a second language, dictionaries are permitted. Calculators are allowed.

DO NOT ASK QUESTIONS OF THE INVIGILATORS about interpretations of the exam questions. If you have a concern about a question, please summarize it at the top of this page. All concerns will be evaluated after the exam. GOOD LUCK!

**Question 1 8 points** (*Assume all transitions are possible*)

a. 6 points Calculate the **energy (in kJ/mol)** required to ionize (completely remove) the electron from the **n=3** level of **He⁺ ion** (He: Helium). Show your work including equations. (3 sig. figs.)

b. 2 points

(I) Select the largest (longest) wavelength from the three choices below:

- i. Wavelength of photon to *completely remove* an electron from *1s orbital of Hydrogen atom*
- ii. Wavelength of photon to *completely remove* an electron from *1s orbital of Boron atom (B)*
- iii. Wavelength of photon to *completely remove* an electron from *1s orbital of Boron cation (B⁺)*

i

or

ii

or

iii

(II) Select the smallest (shortest) wavelength from the three choices below (*for H atom*):

- i. Wavelength of photon *emitted* for an electron transitioning from *2p to 1s orbital*
- ii. Wavelength of photon *absorbed* for an electron transitioning from *2s to 4p orbital*
- iii. Wavelength of photon *absorbed* for an electron transitioning from *1s to 3p orbital*

i

or

ii

or

iii



Question 2 8 points



A Chem110 student plots the graph shown on the left, from data obtained for the kinetic energy (eV) of the ejected electron versus frequency (s^{-1}) of the incident light, for an unknown metal. ($1\text{eV} = 1.6022 \times 10^{-19} \text{ J}$)

Based on the plot, answer the questions **a to c** below:

a. 1 point On the plot above, label the **threshold frequency**.

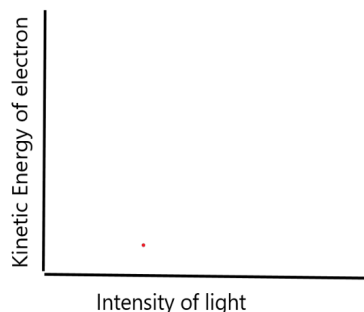
b. 2 points What is the value of the work potential (in J) of the unknown metal?

c. 3 points What is the *longest* wavelength (in nm) required to eject an electron from this metal's surface? Show your work including equation(s) used. (3 sig figs)

d. 2 points In the space provided below, plot the following

(Assume frequency is *above* threshold frequency):

- Kinetic energy* of an ejected electron versus the *intensity* of the incident light (left)
- Number of electrons* ejected versus the *frequency* of the incident light (right)



**Question 3** **6 points****a. 2 points**

In the ground state of Ge (Germanium; $Z = 32$) atom, **name** the following orbitals:

i. Orbital containing the electron that requires the maximum energy to be removed.

ii. Orbital containing the electron that experiences the minimum effective nuclear charge.

b. 2 points

An unknown anion (A^-) has the following *excited* electronic configuration: $[Ar] 4s^2 3d^3 4p^1$

(Ar: Argon; $Z = 18$)

Give the **condensed ground state electronic configuration** of the atom, **A** and the cation, **A^{2+}** .
Show your work. (Assume all electronic transitions are possible)

Electronic Configuration of A:

Electronic Configuration of A^{2+} :

c. 2 points

In a single atom, determine how many electrons can have the following set of quantum numbers?

(i) $n = 5; l = 3$

In a single atom, determine how many electrons can have the following set of quantum numbers?

(i) $n = 3; l = 2; m_s = +1/2$

**Question 4** **8 points**

Rank the following from 1 to 4 in **increasing** order. Provide the ranking in the boxes provided.

Note: Atomic number (Z) are provided for relevant atoms

a) Increasing (lowest to highest) order of *ionization energy*: (1= lowest; 4 = highest)

Mg

Mg: Z = 12

Rb

Rb: Z = 37

Cs

Cs: Z = 55

K

K : Z = 19

b) Increasing (lowest to highest) order of *bond angle*: (1= lowest; 4 = highest)

H₂OCCl₄XeF₄

HCN

c) Increasing (smallest to largest) order of *ionic size* (1= smallest; 4 = largest)

S²⁻

S: Z = 16

O²⁻

O: Z = 8

Be²⁺

Be: Z = 4

Na⁺

Na: Z = 11

d) Increasing (lowest to highest) order of *bond energy* (1= lowest; 4 = highest)

P-F

P-Cl

P-I

P-Br

**Question 5** **12 points**

a. 9 points For the molecules HClO_4 and TeF_6 , draw the most stable *Lewis structure* showing appropriate shape and *dashed/wedged bonds*, based on VSEPR. Determine the **molecular geometry** around the central atom. For HClO_4 , label the *most polar bond*.

HClO_4 (there are no O-O bonds)

Label the most polar bond

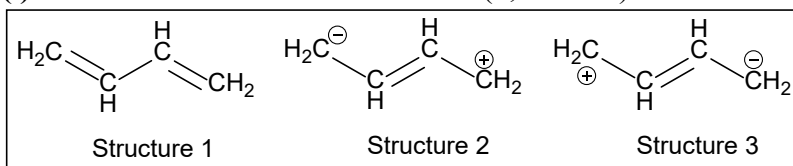
Molecular Geometry around Cl

TeF_6

Molecular Geometry around Te

b. 3 points (Note: Lone pairs are not shown in structures below)

(i) For the three *resonance* structures (1, 2 and 3) shown below, answer the following question:



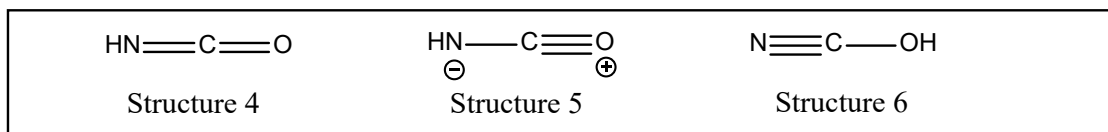
(I) Circle the *most contributing* resonance structure.

Structure 1

Structure 2

Structure 3

(ii) For the three structures given (4, 5, and 6) below, answer the following questions:



(I) Circle all the structures that represent *resonance structures* of each other:

Structure 4

Structure 5

Structure 6

(II) Among the resonance structures, circle the *most contributing* resonance structure.

Structure 4

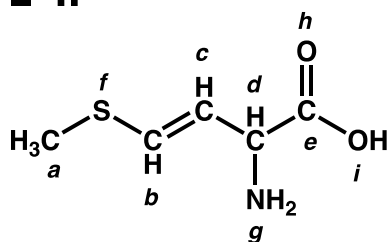
Structure 5

Structure 6

**Question 6** **6 points**

Using the data given below, calculate the **lattice energy** for SrO(s).

	ΔH° (kJ/mol)
$\text{Sr}(s) \longrightarrow \text{Sr}(g)$	+180
$\text{Sr}(g) \longrightarrow \text{Sr}^{2+}(g) + 2e^-$	+1468
$\text{O}_2(g) \longrightarrow 2\text{O}(g)$	+500
$\text{O}(g) + 2e^- \longrightarrow \text{O}^{2-}(g)$	+600
$\text{Sr}(s) + \frac{1}{2}\text{O}_2(g) \longrightarrow \text{SrO}(s)$	-550

**Question 7 10 points**

Consider the molecule given on the left, and answer questions (i) to (vi) using chemical bonding theories that we have learnt in class (valence bond theory, hybridization).

Note that the lone pairs are not shown in the structure:

i. Consider the carbon atoms marked in as ***a***, ***b***, ***c***, ***d***, and ***e*** in the molecule, and determine **hybridization** at each of the carbon centers. (2.5 points)

Atom	Hybridization
Carbon (<i>a</i>)	
Carbon (<i>b</i>)	
Carbon (<i>c</i>)	
Carbon (<i>d</i>)	
Carbon (<i>e</i>)	

ii. Now, consider the sulfur atom (S) marked as ***f***, nitrogen (N) atom (***g***), and oxygen(O) atoms (***h*** and ***i***), and determine their **hybridization**. (2 points)

Atom	Hybridization
Sulfur (<i>f</i>)	
Nitrogen (<i>g</i>)	
Oxygen (<i>h</i>)	
Oxygen (<i>i</i>)	

iii. **Describe** all the bonds formed by the carbon atoms ***a*** and ***b***, using sigma and pi designation as appropriate. (2 points)

Atom	Bonds formed (use sigma and pi as appropriate)
Carbon (<i>a</i>)	
Carbon (<i>b</i>)	

iv. How many total sp^3 -s sigma bonds are in the molecule? (1 point)

v. Determine **electronic** and **molecular geometry** at carbon (C) labeled ***e*** and sulfur (S) labeled ***f*** centres. (2 points)

Atom	Electronic Geometry	Molecular Geometry
Carbon (<i>e</i>)		
Sulfur (<i>f</i>)		

vi. Circle the value from the choices given below, that describes the **C-S-C bond angle at sulfur** (S) marked ***f***: (0.5 points)

>90° to <109.5° 109.5° >109.5 to <120° 120° >120° to <180°


Question 8 12 points

Using the **valence shell orbitals only** draw a **complete** molecular orbital (MO) diagram (with no s-p mixing) for the molecule **PS**, in the space provided below. **(6 points)**

Assume that 3s and 3p atomic orbitals will form molecular orbitals like those formed by the combination of 2s and 2p atomic orbitals that we learnt in class, and there is no s-p mixing.

Show **energy arrow**, **atomic orbitals** involved, **atom labels**, the **molecular orbitals** that are formed, **filled electrons** in the MO diagram etc. Make sure to **label all the atomic and molecular orbitals**.

Using the MO diagram above, answer the following questions:

- (i) Which one of the following will have the highest bond order? Circle the correct choice
 A) PS^{1-} B) PS C) PS^{1+}
- (ii) Which one of the following will have the longest bond length? Circle the correct choice
 A) PS^{1-} B) PS C) PS^{1+}
- (iii) Which of the following will be diamagnetic? Circle the correct choice.
 A) PS^{1-} B) PS C) PS^{1+}
- (iv) Draw Lewis dot structure of the molecule with the shortest bond length (based on the MO diagram) from among the three (PS^{1-} , PS, PS^{1+}), and indicate the bond order calculated from the Lewis dot structure. **(3 points)**

Lewis Dot Structure

Bond Order



50C466A3-E4E9-4AC6-B5F9-0BF94676DC47

midterm-exam-93388

#722 Page 10 of 10

SCRAP PAGE

**CHEMISTRY 110 –Midterm Exam****Thursday November 7, 2024, 6:30 – 8:30 pm****Examiners: Pallavi Sirjoosingh and Ashok Kakkar**

First name (please write as legibly as possible within the boxes)

--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--

Last name

--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--

Student ID number

--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--

This booklet has 8 questions on 10 pages (The last page can be used as a scrap page).

The staple must NOT be removed from the question booklet. Your intact question booklet must be returned at the end of the exam.

A separate booklet contains a periodic table, and datasheet.

Write your answers directly under the questions in the space provided. Do not write on the QR codes or near the top edges (as it may be cut off).

Please write **legibly and dark enough** for the scan to be readable. For students who use English as a second language, dictionaries are permitted. Calculators are allowed.

DO NOT ASK QUESTIONS OF THE INVIGILATORS about interpretations of the exam questions. If you have a concern about a question, please summarize it at the top of this page. All concerns will be evaluated after the exam. GOOD LUCK!

**Question 1 8 points** (*Assume all transitions are possible*)

a. 6 points Calculate the **energy (in kJ/mol)** required to ionize (completely remove) the electron from the **n=3** level of **He⁺ ion** (He: Helium). Show your work including equations. (3 sig. figs.)

b. 2 points

(I) Select the largest (longest) wavelength from the three choices below:

- i. Wavelength of photon to *completely remove* an electron from *1s orbital of Hydrogen atom*
- ii. Wavelength of photon to *completely remove* an electron from *1s orbital of Boron atom (B)*
- iii. Wavelength of photon to *completely remove* an electron from *1s orbital of Boron cation (B⁺)*

i or ii or iii

(II) Select the smallest (shortest) wavelength from the three choices below (*for H atom*):

- i. Wavelength of photon *emitted* for an electron transitioning from *2p to 1s orbital*
- ii. Wavelength of photon *absorbed* for an electron transitioning from *2s to 4p orbital*
- iii. Wavelength of photon *absorbed* for an electron transitioning from *1s to 3p orbital*

i or ii or iii



Question 2 8 points



A Chem110 student plots the graph shown on the left, from data obtained for the kinetic energy (eV) of the ejected electron versus frequency (s^{-1}) of the incident light, for an unknown metal. ($1\text{eV} = 1.6022 \times 10^{-19} \text{ J}$)

Based on the plot, answer the questions **a to c** below:

a. 1 point On the plot above, label the **threshold frequency**.

b. 2 points What is the value of the work potential (in J) of the unknown metal?

c. 3 points What is the *longest* wavelength (in nm) required to eject an electron from this metal's surface? Show your work including equation(s) used. (3 sig figs)

d. 2 points In the space provided below, plot the following

(Assume frequency is *above* threshold frequency):

- Kinetic energy* of an ejected electron versus the *intensity* of the incident light (left)
- Number of electrons* ejected versus the *frequency* of the incident light (right)



**Question 3** **6 points****a. 2 points**

In the ground state of Ge (Germanium; $Z = 32$) atom, **name** the following orbitals:

i. Orbital containing the electron that requires the maximum energy to be removed.

ii. Orbital containing the electron that experiences the minimum effective nuclear charge.

b. 2 points

An unknown anion (A^-) has the following *excited* electronic configuration: $[Ar] 4s^2 3d^3 4p^1$

(Ar: Argon; $Z = 18$)

Give the **condensed ground state electronic configuration** of the atom, **A** and the cation, A^{2+} .
Show your work. (Assume all electronic transitions are possible)

Electronic Configuration of A:

Electronic Configuration of A^{2+} :

c. 2 points

In a single atom, determine how many electrons can have the following set of quantum numbers?

(i) $n = 5; l = 3$

In a single atom, determine how many electrons can have the following set of quantum numbers?

(i) $n = 3; l = 2; m_s = +1/2$

**Question 4** **8 points**

Rank the following from 1 to 4 in **increasing** order. Provide the ranking in the boxes provided.

Note: Atomic number (Z) are provided for relevant atoms

a) Increasing (lowest to highest) order of *ionization energy*: (1= lowest; 4 = highest)

Mg

Mg: Z = 12

Rb

Rb: Z = 37

Cs

Cs: Z = 55

K

K : Z = 19

b) Increasing (lowest to highest) order of *bond angle*: (1= lowest; 4 = highest)

H₂OCCl₄XeF₄

HCN

c) Increasing (smallest to largest) order of *ionic size* (1= smallest; 4 = largest)

S²⁻

S: Z = 16

O²⁻

O: Z = 8

Be²⁺

Be: Z = 4

Na⁺

Na: Z = 11

d) Increasing (lowest to highest) order of *bond energy* (1= lowest; 4 = highest)

P-F

P-Cl

P-I

P-Br

**Question 5** **12 points**

a. 9 points For the molecules HClO_4 and TeF_6 , draw the most stable *Lewis structure* showing appropriate shape and *dashed/wedged bonds*, based on VSEPR. Determine the **molecular geometry** around the central atom. For HClO_4 , label the *most polar bond*.

HClO_4 (there are no O-O bonds)

Label the most polar bond

Molecular Geometry around Cl

TeF_6

Molecular Geometry around Te

b. 3 points (Note: Lone pairs are not shown in structures below)

(i) For the three *resonance* structures (1, 2 and 3) shown below, answer the following question:



(I) Circle the *most contributing* resonance structure.

Structure 1

Structure 2

Structure 3

(ii) For the three structures given (4, 5, and 6) below, answer the following questions:



(I) Circle all the structures that represent *resonance structures* of each other:

Structure 4

Structure 5

Structure 6

(II) Among the resonance structures, circle the *most contributing* resonance structure.

Structure 4

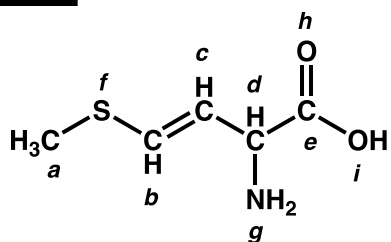
Structure 5

Structure 6

**Question 6** **6 points**

Using the data given below, calculate the **lattice energy** for SrO(s).

	ΔH° (kJ/mol)
$\text{Sr}(s) \longrightarrow \text{Sr}(g)$	+180
$\text{Sr}(g) \longrightarrow \text{Sr}^{2+}(g) + 2e^-$	+1468
$\text{O}_2(g) \longrightarrow 2\text{O}(g)$	+500
$\text{O}(g) + 2e^- \longrightarrow \text{O}^{2-}(g)$	+600
$\text{Sr}(s) + \frac{1}{2}\text{O}_2(g) \longrightarrow \text{SrO}(s)$	-550

**Question 7 10 points**

Consider the molecule given on the left, and answer questions (i) to (vi) using chemical bonding theories that we have learnt in class (valence bond theory, hybridization).

Note that the lone pairs are not shown in the structure:

i. Consider the carbon atoms marked in as ***a***, ***b***, ***c***, ***d***, and ***e*** in the molecule, and determine **hybridization** at each of the carbon centers. (2.5 points)

Atom	Hybridization
Carbon (<i>a</i>)	
Carbon (<i>b</i>)	
Carbon (<i>c</i>)	
Carbon (<i>d</i>)	
Carbon (<i>e</i>)	

ii. Now, consider the sulfur atom (S) marked as ***f***, nitrogen (N) atom (***g***), and oxygen(O) atoms (***h*** and ***i***), and determine their **hybridization**. (2 points)

Atom	Hybridization
Sulfur (<i>f</i>)	
Nitrogen (<i>g</i>)	
Oxygen (<i>h</i>)	
Oxygen (<i>i</i>)	

iii. **Describe** all the bonds formed by the carbon atoms ***a*** and ***b***, using sigma and pi designation as appropriate. (2 points)

Atom	Bonds formed (use sigma and pi as appropriate)
Carbon (<i>a</i>)	
Carbon (<i>b</i>)	

iv. How many total sp^3 -s sigma bonds are in the molecule? (1 point)

v. Determine **electronic** and **molecular geometry** at carbon (C) labeled ***e*** and sulfur (S) labeled ***f*** centres. (2 points)

Atom	Electronic Geometry	Molecular Geometry
Carbon (<i>e</i>)		
Sulfur (<i>f</i>)		

vi. Circle the value from the choices given below, that describes the **C-S-C bond angle at sulfur** (S) marked ***f***: (0.5 points)

>90° to <109.5°

109.5°

>109.5 to <120°

120°

>120° to <180°


Question 8 12 points

Using the **valence shell orbitals only** draw a **complete** molecular orbital (MO) diagram (with no s-p mixing) for the molecule **PS**, in the space provided below. **(6 points)**

Assume that 3s and 3p atomic orbitals will form molecular orbitals like those formed by the combination of 2s and 2p atomic orbitals that we learnt in class, and there is no s-p mixing.

Show **energy arrow**, **atomic orbitals** involved, **atom labels**, the **molecular orbitals** that are formed, **filled electrons** in the MO diagram etc. Make sure to label all the atomic and molecular orbitals.

Using the MO diagram above, answer the following questions:

- (i) Which one of the following will have the highest bond order? Circle the correct choice
 A) PS^{1-} B) PS C) PS^{1+}
- (ii) Which one of the following will have the longest bond length? Circle the correct choice
 A) PS^{1-} B) PS C) PS^{1+}
- (iii) Which of the following will be diamagnetic? Circle the correct choice.
 A) PS^{1-} B) PS C) PS^{1+}
- (iv) Draw Lewis dot structure of the molecule with the shortest bond length (based on the MO diagram) from among the three (PS^{1-} , PS, PS^{1+}), and indicate the bond order calculated from the Lewis dot structure. **(3 points)**

Lewis Dot Structure

Bond Order



B9698C2A-3ECA-44F0-B8B9-A85F38DD7DC1

midterm-exam-93388

#723 Page 10 of 10

SCRAP PAGE

**CHEMISTRY 110 –Midterm Exam****Thursday November 7, 2024, 6:30 – 8:30 pm**

Examiners: Pallavi Sirjoosingh and Ashok Kakkar

First name (please write as legibly as possible within the boxes)

--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--

Last name

--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--

Student ID number

--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--

This booklet has 8 questions on 10 pages (The last page can be used as a scrap page).

The staple must NOT be removed from the question booklet. Your intact question booklet must be returned at the end of the exam.

A separate booklet contains a periodic table, and datasheet.

Write your answers directly under the questions in the space provided. Do not write on the QR codes or near the top edges (as it may be cut off).

Please write **legibly and dark enough** for the scan to be readable. For students who use English as a second language, dictionaries are permitted. Calculators are allowed.

DO NOT ASK QUESTIONS OF THE INVIGILATORS about interpretations of the exam questions. If you have a concern about a question, please summarize it at the top of this page. All concerns will be evaluated after the exam. GOOD LUCK!

**Question 1 8 points** (*Assume all transitions are possible*)

a. 6 points Calculate the **energy (in kJ/mol)** required to ionize (completely remove) the electron from the **n=3** level of **He⁺ ion** (He: Helium). Show your work including equations. (3 sig. figs.)

b. 2 points

(I) Select the largest (longest) wavelength from the three choices below:

- i. Wavelength of photon to *completely remove* an electron from *1s orbital of Hydrogen atom*
- ii. Wavelength of photon to *completely remove* an electron from *1s orbital of Boron atom (B)*
- iii. Wavelength of photon to *completely remove* an electron from *1s orbital of Boron cation (B⁺)*

i or ii or iii

(II) Select the smallest (shortest) wavelength from the three choices below (*for H atom*):

- i. Wavelength of photon *emitted* for an electron transitioning from *2p to 1s orbital*
- ii. Wavelength of photon *absorbed* for an electron transitioning from *2s to 4p orbital*
- iii. Wavelength of photon *absorbed* for an electron transitioning from *1s to 3p orbital*

i or ii or iii



Question 2 8 points



A Chem110 student plots the graph shown on the left, from data obtained for the kinetic energy (eV) of the ejected electron versus frequency (s^{-1}) of the incident light, for an unknown metal. ($1\text{eV} = 1.6022 \times 10^{-19} \text{ J}$)

Based on the plot, answer the questions **a to c** below:

a. 1 point On the plot above, label the **threshold frequency**.

b. 2 points What is the value of the work potential (in J) of the unknown metal?

c. 3 points What is the *longest* wavelength (in nm) required to eject an electron from this metal's surface? Show your work including equation(s) used. (3 sig figs)

d. 2 points In the space provided below, plot the following

(Assume frequency is *above* threshold frequency):

- Kinetic energy* of an ejected electron versus the *intensity* of the incident light (left)
- Number of electrons* ejected versus the *frequency* of the incident light (right)



**Question 3** **6 points****a. 2 points**

In the ground state of Ge (Germanium; $Z = 32$) atom, **name** the following orbitals:

i. Orbital containing the electron that requires the maximum energy to be removed.

ii. Orbital containing the electron that experiences the minimum effective nuclear charge.

b. 2 points

An unknown anion (A^-) has the following *excited* electronic configuration: $[Ar] 4s^2 3d^3 4p^1$

(Ar: Argon; $Z = 18$)

Give the **condensed ground state electronic configuration** of the atom, **A** and the cation, **A^{2+}** .
Show your work. (Assume all electronic transitions are possible)

Electronic Configuration of A:

Electronic Configuration of A^{2+} :

c. 2 points

In a single atom, determine how many electrons can have the following set of quantum numbers?

(i) $n = 5; l = 3$

In a single atom, determine how many electrons can have the following set of quantum numbers?

(i) $n = 3; l = 2; m_s = +1/2$

**Question 4** **8 points**

Rank the following from 1 to 4 in **increasing** order. Provide the ranking in the boxes provided.

Note: Atomic number (Z) are provided for relevant atoms

a) Increasing (lowest to highest) order of *ionization energy*: (1= lowest; 4 = highest)

Mg

Mg: Z = 12

Rb

Rb: Z = 37

Cs

Cs: Z = 55

K

K : Z = 19

b) Increasing (lowest to highest) order of *bond angle*: (1= lowest; 4 = highest)

H₂OCCl₄XeF₄

HCN

c) Increasing (smallest to largest) order of *ionic size* (1= smallest; 4 = largest)

S²⁻

S: Z = 16

O²⁻

O: Z = 8

Be²⁺

Be: Z = 4

Na⁺

Na: Z = 11

d) Increasing (lowest to highest) order of *bond energy* (1= lowest; 4 = highest)

P-F

P-Cl

P-I

P-Br

**Question 5** **12 points**

a. 9 points For the molecules HClO_4 and TeF_6 , draw the most stable *Lewis structure* showing appropriate shape and *dashed/wedged bonds*, based on VSEPR. Determine the **molecular geometry** around the central atom. For HClO_4 , label the *most polar bond*.



b. 3 points (Note: Lone pairs are not shown in structures below)

(i) For the three *resonance* structures (1, 2 and 3) shown below, answer the following question:



(I) Circle the *most contributing* resonance structure.

Structure 1

Structure 2

Structure 3

(ii) For the three structures given (4, 5, and 6) below, answer the following questions:



(I) Circle all the structures that represent *resonance structures* of each other:

Structure 4

Structure 5

Structure 6

(II) Among the resonance structures, circle the *most contributing* resonance structure.

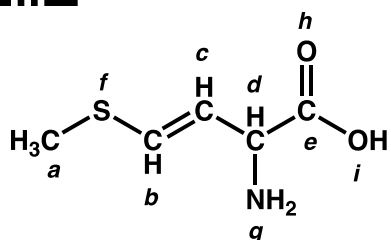
Structure 4

Structure 5

Structure 6

**Question 6** **6 points**Using the data given below, calculate the **lattice energy** for SrO(s).

	ΔH° (kJ/mol)
$\text{Sr}(s) \longrightarrow \text{Sr}(g)$	+180
$\text{Sr}(g) \longrightarrow \text{Sr}^{2+}(g) + 2e^-$	+1468
$\text{O}_2(g) \longrightarrow 2\text{O}(g)$	+500
$\text{O}(g) + 2e^- \longrightarrow \text{O}^{2-}(g)$	+600
$\text{Sr}(s) + \frac{1}{2}\text{O}_2(g) \longrightarrow \text{SrO}(s)$	-550

**Question 7 10 points**

Consider the molecule given on the left, and answer questions (i) to (vi) using chemical bonding theories that we have learnt in class (valence bond theory, hybridization).

Note that the lone pairs are not shown in the structure:

i. Consider the carbon atoms marked in as ***a***, ***b***, ***c***, ***d***, and ***e*** in the molecule, and determine **hybridization** at each of the carbon centers. (2.5 points)

Atom	Hybridization
Carbon (<i>a</i>)	
Carbon (<i>b</i>)	
Carbon (<i>c</i>)	
Carbon (<i>d</i>)	
Carbon (<i>e</i>)	

ii. Now, consider the sulfur atom (S) marked as ***f***, nitrogen (N) atom (***g***), and oxygen(O) atoms (***h*** and ***i***), and determine their **hybridization**. (2 points)

Atom	Hybridization
Sulfur (<i>f</i>)	
Nitrogen (<i>g</i>)	
Oxygen (<i>h</i>)	
Oxygen (<i>i</i>)	

iii. **Describe** all the bonds formed by the carbon atoms ***a*** and ***b***, using sigma and pi designation as appropriate. (2 points)

Atom	Bonds formed (use sigma and pi as appropriate)
Carbon (<i>a</i>)	
Carbon (<i>b</i>)	

iv. How many total sp^3 -s sigma bonds are in the molecule? (1 point)

v. Determine **electronic** and **molecular geometry** at carbon (C) labeled ***e*** and sulfur (S) labeled ***f*** centres. (2 points)

Atom	Electronic Geometry	Molecular Geometry
Carbon (<i>e</i>)		
Sulfur (<i>f</i>)		

vi. Circle the value from the choices given below, that describes the **C-S-C bond angle at sulfur** (S) marked ***f***: (0.5 points)

>90° to <109.5°

109.5°

>109.5 to <120°

120°

>120° to <180°


Question 8 12 points

Using the **valence shell orbitals only** draw a **complete** molecular orbital (MO) diagram (with no s-p mixing) for the molecule **PS**, in the space provided below. **(6 points)**

Assume that 3s and 3p atomic orbitals will form molecular orbitals like those formed by the combination of 2s and 2p atomic orbitals that we learnt in class, and there is no s-p mixing.

Show **energy arrow**, **atomic orbitals** involved, **atom labels**, the **molecular orbitals** that are formed, **filled electrons** in the MO diagram etc. Make sure to label all the atomic and molecular orbitals.

Using the MO diagram above, answer the following questions:

- (i) Which one of the following will have the highest bond order? Circle the correct choice
 A) PS^{1-} B) PS C) PS^{1+}
- (ii) Which one of the following will have the longest bond length? Circle the correct choice
 A) PS^{1-} B) PS C) PS^{1+}
- (iii) Which of the following will be diamagnetic? Circle the correct choice.
 A) PS^{1-} B) PS C) PS^{1+}
- (iv) Draw Lewis dot structure of the molecule with the shortest bond length (based on the MO diagram) from among the three (PS^{1-} , PS, PS^{1+}), and indicate the bond order calculated from the Lewis dot structure. **(3 points)**

Lewis Dot Structure

Bond Order



4B6959AD-F522-4B3B-BBFE-89A5086BE665

midterm-exam-93388

#724 Page 10 of 10

SCRAP PAGE

**CHEMISTRY 110 –Midterm Exam****Thursday November 7, 2024, 6:30 – 8:30 pm****Examiners: Pallavi Sirjoosingh and Ashok Kakkar**

First name (please write as legibly as possible within the boxes)

--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--

Last name

--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--

Student ID number

--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--

This booklet has 8 questions on 10 pages (The last page can be used as a scrap page).

The staple must NOT be removed from the question booklet. Your intact question booklet must be returned at the end of the exam.

A separate booklet contains a periodic table, and datasheet.

Write your answers directly under the questions in the space provided. Do not write on the QR codes or near the top edges (as it may be cut off).

Please write **legibly and dark enough** for the scan to be readable. For students who use English as a second language, dictionaries are permitted. Calculators are allowed.

DO NOT ASK QUESTIONS OF THE INVIGILATORS about interpretations of the exam questions. If you have a concern about a question, please summarize it at the top of this page. All concerns will be evaluated after the exam. GOOD LUCK!

**Question 1 8 points** (*Assume all transitions are possible*)

a. 6 points Calculate the **energy (in kJ/mol)** required to ionize (completely remove) the electron from the **$n=3$** level of **He^+ ion** (He: Helium). Show your work including equations. (3 sig. figs.)

b. 2 points

(I) Select the largest (longest) wavelength from the three choices below:

- i. Wavelength of photon to *completely remove* an electron from *1s orbital of Hydrogen atom*
- ii. Wavelength of photon to *completely remove* an electron from *1s orbital of Boron atom (B)*
- iii. Wavelength of photon to *completely remove* an electron from *1s orbital of Boron cation (B^+)*

i or ii or iii

(II) Select the smallest (shortest) wavelength from the three choices below (*for H atom*):

- i. Wavelength of photon *emitted* for an electron transitioning from *2p to 1s orbital*
- ii. Wavelength of photon *absorbed* for an electron transitioning from *2s to 4p orbital*
- iii. Wavelength of photon *absorbed* for an electron transitioning from *1s to 3p orbital*

i or ii or iii



Question 2 8 points



A Chem110 student plots the graph shown on the left, from data obtained for the kinetic energy (eV) of the ejected electron versus frequency (s^{-1}) of the incident light, for an unknown metal. ($1\text{eV} = 1.6022 \times 10^{-19} \text{ J}$)

Based on the plot, answer the questions **a to c** below:

a. 1 point On the plot above, label the **threshold frequency**.

b. 2 points What is the value of the work potential (in J) of the unknown metal?

c. 3 points What is the *longest* wavelength (in nm) required to eject an electron from this metal's surface? Show your work including equation(s) used. (3 sig figs)

d. 2 points In the space provided below, plot the following

(Assume frequency is *above* threshold frequency):

- Kinetic energy* of an ejected electron versus the *intensity* of the incident light (left)
- Number of electrons* ejected versus the *frequency* of the incident light (right)



**Question 3** **6 points****a. 2 points**

In the ground state of Ge (Germanium; $Z = 32$) atom, **name** the following orbitals:

i. Orbital containing the electron that requires the maximum energy to be removed.

ii. Orbital containing the electron that experiences the minimum effective nuclear charge.

b. 2 points

An unknown anion (A^-) has the following *excited* electronic configuration: $[Ar] 4s^2 3d^3 4p^1$

(Ar: Argon; $Z = 18$)

Give the **condensed ground state electronic configuration** of the atom, **A** and the cation, **A^{2+}** .
Show your work. (Assume all electronic transitions are possible)

Electronic Configuration of A:

Electronic Configuration of A^{2+} :

c. 2 points

In a single atom, determine how many electrons can have the following set of quantum numbers?

(i) $n = 5; l = 3$

In a single atom, determine how many electrons can have the following set of quantum numbers?

(i) $n = 3; l = 2; m_s = +1/2$

**Question 4** **8 points**

Rank the following from 1 to 4 in **increasing** order. Provide the ranking in the boxes provided.

Note: Atomic number (Z) are provided for relevant atoms

a) Increasing (lowest to highest) order of *ionization energy*: (1= lowest; 4 = highest)

Mg

Mg: Z = 12

Rb

Rb: Z = 37

Cs

Cs: Z = 55

K

K : Z = 19

b) Increasing (lowest to highest) order of *bond angle*: (1= lowest; 4 = highest)

H₂OCCl₄XeF₄

HCN

c) Increasing (smallest to largest) order of *ionic size* (1= smallest; 4 = largest)

S²⁻

S: Z = 16

O²⁻

O: Z = 8

Be²⁺

Be: Z = 4

Na⁺

Na: Z = 11

d) Increasing (lowest to highest) order of *bond energy* (1= lowest; 4 = highest)

P-F

P-Cl

P-I

P-Br

**Question 5** **12 points**

a. 9 points For the molecules HClO_4 and TeF_6 , draw the most stable *Lewis structure* showing appropriate shape and *dashed/wedged bonds*, based on VSEPR. Determine the **molecular geometry** around the central atom. For HClO_4 , label the *most polar bond*.



b. 3 points (Note: Lone pairs are not shown in structures below)

(i) For the three *resonance* structures (1, 2 and 3) shown below, answer the following question:



(I) Circle the *most contributing* resonance structure.

Structure 1

Structure 2

Structure 3

(ii) For the three structures given (4, 5, and 6) below, answer the following questions:



(I) Circle all the structures that represent *resonance structures* of each other:

Structure 4

Structure 5

Structure 6

(II) Among the resonance structures, circle the *most contributing* resonance structure.

Structure 4

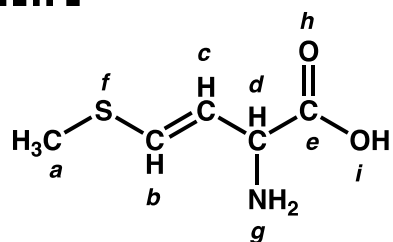
Structure 5

Structure 6

**Question 6** **6 points**

Using the data given below, calculate the **lattice energy** for SrO(s).

	ΔH° (kJ/mol)
$\text{Sr}(s) \longrightarrow \text{Sr}(g)$	+180
$\text{Sr}(g) \longrightarrow \text{Sr}^{2+}(g) + 2e^-$	+1468
$\text{O}_2(g) \longrightarrow 2\text{O}(g)$	+500
$\text{O}(g) + 2e^- \longrightarrow \text{O}^{2-}(g)$	+600
$\text{Sr}(s) + \frac{1}{2}\text{O}_2(g) \longrightarrow \text{SrO}(s)$	-550

**Question 7 10 points**

Consider the molecule given on the left, and answer questions (i) to (vi) using chemical bonding theories that we have learnt in class (valence bond theory, hybridization).

Note that the lone pairs are not shown in the structure:

i. Consider the carbon atoms marked in as ***a***, ***b***, ***c***, ***d***, and ***e*** in the molecule, and determine **hybridization** at each of the carbon centers. (2.5 points)

Atom	Hybridization
Carbon (<i>a</i>)	
Carbon (<i>b</i>)	
Carbon (<i>c</i>)	
Carbon (<i>d</i>)	
Carbon (<i>e</i>)	

ii. Now, consider the sulfur atom (S) marked as ***f***, nitrogen (N) atom (***g***), and oxygen(O) atoms (***h*** and ***i***), and determine their **hybridization**. (2 points)

Atom	Hybridization
Sulfur (<i>f</i>)	
Nitrogen (<i>g</i>)	
Oxygen (<i>h</i>)	
Oxygen (<i>i</i>)	

iii. **Describe** all the bonds formed by the carbon atoms ***a*** and ***b***, using sigma and pi designation as appropriate. (2 points)

Atom	Bonds formed (use sigma and pi as appropriate)
Carbon (<i>a</i>)	
Carbon (<i>b</i>)	

iv. How many total sp^3 -s sigma bonds are in the molecule? (1 point)

v. Determine **electronic** and **molecular geometry** at carbon (C) labeled ***e*** and sulfur (S) labeled ***f*** centres. (2 points)

Atom	Electronic Geometry	Molecular Geometry
Carbon (<i>e</i>)		
Sulfur (<i>f</i>)		

vi. Circle the value from the choices given below, that describes the **C-S-C bond angle at sulfur** (S) marked ***f***: (0.5 points)

>90° to <109.5° 109.5° >109.5 to <120° 120° >120° to <180°


Question 8 12 points

Using the **valence shell orbitals only** draw a **complete** molecular orbital (MO) diagram (with no s-p mixing) for the molecule **PS**, in the space provided below. **(6 points)**

Assume that 3s and 3p atomic orbitals will form molecular orbitals like those formed by the combination of 2s and 2p atomic orbitals that we learnt in class, and there is no s-p mixing.

Show **energy arrow**, **atomic orbitals** involved, **atom labels**, the **molecular orbitals** that are formed, **filled electrons** in the MO diagram etc. Make sure to label all the atomic and molecular orbitals.

Using the MO diagram above, answer the following questions:

- (i) Which one of the following will have the highest bond order? Circle the correct choice
 A) PS^{1-} B) PS C) PS^{1+}
- (ii) Which one of the following will have the longest bond length? Circle the correct choice
 A) PS^{1-} B) PS C) PS^{1+}
- (iii) Which of the following will be diamagnetic? Circle the correct choice.
 A) PS^{1-} B) PS C) PS^{1+}
- (iv) Draw Lewis dot structure of the molecule with the shortest bond length (based on the MO diagram) from among the three (PS^{1-} , PS, PS^{1+}), and indicate the bond order calculated from the Lewis dot structure. **(3 points)**

Lewis Dot Structure

Bond Order



0E1A963A-DE3B-4600-B1C0-E6BFD43E095C

midterm-exam-93388

#725 Page 10 of 10

SCRAP PAGE

**CHEMISTRY 110 –Midterm Exam****Thursday November 7, 2024, 6:30 – 8:30 pm****Examiners: Pallavi Sirjoosingh and Ashok Kakkar**

First name (please write as legibly as possible within the boxes)

--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--

Last name

--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--

Student ID number

--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--

This booklet has 8 questions on 10 pages (The last page can be used as a scrap page).

The staple must NOT be removed from the question booklet. Your intact question booklet must be returned at the end of the exam.

A separate booklet contains a periodic table, and datasheet.

Write your answers directly under the questions in the space provided. Do not write on the QR codes or near the top edges (as it may be cut off).

Please write **legibly and dark enough** for the scan to be readable. For students who use English as a second language, dictionaries are permitted. Calculators are allowed.

DO NOT ASK QUESTIONS OF THE INVIGILATORS about interpretations of the exam questions. If you have a concern about a question, please summarize it at the top of this page. All concerns will be evaluated after the exam. GOOD LUCK!

**Question 1 8 points** (*Assume all transitions are possible*)

a. 6 points Calculate the **energy (in kJ/mol)** required to ionize (completely remove) the electron from the **n=3** level of **He⁺ ion** (He: Helium). Show your work including equations. (3 sig. figs.)

b. 2 points

(I) Select the largest (longest) wavelength from the three choices below:

- i. Wavelength of photon to *completely remove* an electron from *1s orbital of Hydrogen atom*
- ii. Wavelength of photon to *completely remove* an electron from *1s orbital of Boron atom (B)*
- iii. Wavelength of photon to *completely remove* an electron from *1s orbital of Boron cation (B⁺)*

i or ii or iii

(II) Select the smallest (shortest) wavelength from the three choices below (*for H atom*):

- i. Wavelength of photon *emitted* for an electron transitioning from *2p to 1s orbital*
- ii. Wavelength of photon *absorbed* for an electron transitioning from *2s to 4p orbital*
- iii. Wavelength of photon *absorbed* for an electron transitioning from *1s to 3p orbital*

i or ii or iii



Question 2 8 points



A Chem110 student plots the graph shown on the left, from data obtained for the kinetic energy (eV) of the ejected electron versus frequency (s^{-1}) of the incident light, for an unknown metal. ($1\text{eV} = 1.6022 \times 10^{-19} \text{ J}$)

Based on the plot, answer the questions **a to c** below:

a. 1 point On the plot above, label the **threshold frequency**.

b. 2 points What is the value of the work potential (in J) of the unknown metal?

c. 3 points What is the *longest* wavelength (in nm) required to eject an electron from this metal's surface? Show your work including equation(s) used. (3 sig figs)

d. 2 points In the space provided below, plot the following

(Assume frequency is *above* threshold frequency):

- i. *Kinetic energy* of an ejected electron versus the *intensity* of the incident light (left)
- ii. *Number of electrons* ejected versus the *frequency* of the incident light (right)



**Question 3** **6 points****a. 2 points**

In the ground state of Ge (Germanium; $Z = 32$) atom, **name** the following orbitals:

i. Orbital containing the electron that requires the maximum energy to be removed.

ii. Orbital containing the electron that experiences the minimum effective nuclear charge.

b. 2 points

An unknown anion (A^-) has the following *excited* electronic configuration: $[Ar] 4s^2 3d^3 4p^1$

(Ar: Argon; $Z = 18$)

Give the **condensed ground state electronic configuration** of the atom, **A** and the cation, **A^{2+}** .
Show your work. (Assume all electronic transitions are possible)

Electronic Configuration of A:

Electronic Configuration of A^{2+} :

c. 2 points

In a single atom, determine how many electrons can have the following set of quantum numbers?

(i) $n = 5; l = 3$

In a single atom, determine how many electrons can have the following set of quantum numbers?

(i) $n = 3; l = 2; m_s = +1/2$

**Question 4** **8 points**

Rank the following from 1 to 4 in **increasing** order. Provide the ranking in the boxes provided.

Note: Atomic number (Z) are provided for relevant atoms

a) Increasing (lowest to highest) order of *ionization energy*: (1= lowest; 4 = highest)

Mg

Mg: Z = 12

Rb

Rb: Z = 37

Cs

Cs: Z = 55

K

K : Z = 19

b) Increasing (lowest to highest) order of *bond angle*: (1= lowest; 4 = highest)

H₂OCCl₄XeF₄

HCN

c) Increasing (smallest to largest) order of *ionic size* (1= smallest; 4 = largest)

S²⁻

S: Z = 16

O²⁻

O: Z = 8

Be²⁺

Be: Z = 4

Na⁺

Na: Z = 11

d) Increasing (lowest to highest) order of *bond energy* (1= lowest; 4 = highest)

P-F

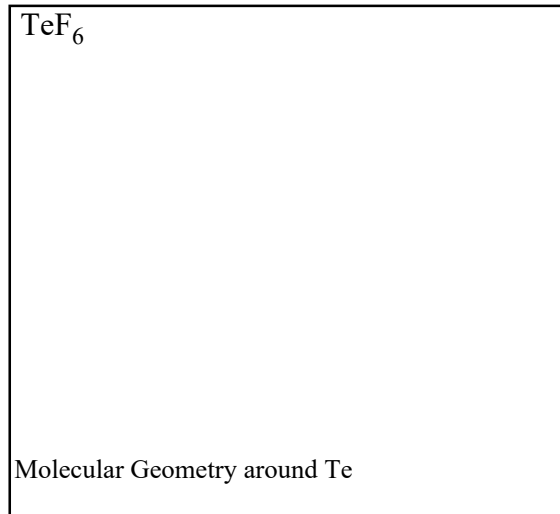
P-Cl

P-I

P-Br

**Question 5** **12 points**

a. 9 points For the molecules HClO_4 and TeF_6 , draw the most stable *Lewis structure* showing appropriate shape and *dashed/wedged bonds*, based on VSEPR. Determine the **molecular geometry** around the central atom. For HClO_4 , label the *most polar bond*.



b. 3 points (Note: Lone pairs are not shown in structures below)

(i) For the three *resonance* structures (1, 2 and 3) shown below, answer the following question:



(I) Circle the *most contributing* resonance structure.

Structure 1

Structure 2

Structure 3

(ii) For the three structures given (4, 5, and 6) below, answer the following questions:



(I) Circle all the structures that represent *resonance structures* of each other:

Structure 4

Structure 5

Structure 6

(II) Among the resonance structures, circle the *most contributing* resonance structure.

Structure 4

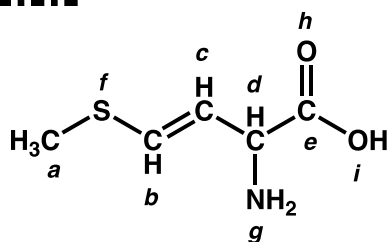
Structure 5

Structure 6

**Question 6** **6 points**

Using the data given below, calculate the **lattice energy** for SrO(s).

	ΔH° (kJ/mol)
$\text{Sr}(s) \longrightarrow \text{Sr}(g)$	+180
$\text{Sr}(g) \longrightarrow \text{Sr}^{2+}(g) + 2e^-$	+1468
$\text{O}_2(g) \longrightarrow 2\text{O}(g)$	+500
$\text{O}(g) + 2e^- \longrightarrow \text{O}^{2-}(g)$	+600
$\text{Sr}(s) + \frac{1}{2}\text{O}_2(g) \longrightarrow \text{SrO}(s)$	-550

**Question 7 10 points**

Consider the molecule given on the left, and answer questions (i) to (vi) using chemical bonding theories that we have learnt in class (valence bond theory, hybridization).

Note that the lone pairs are not shown in the structure:

i. Consider the carbon atoms marked in as ***a***, ***b***, ***c***, ***d***, and ***e*** in the molecule, and determine **hybridization** at each of the carbon centers. (2.5 points)

Atom	Hybridization
Carbon (<i>a</i>)	
Carbon (<i>b</i>)	
Carbon (<i>c</i>)	
Carbon (<i>d</i>)	
Carbon (<i>e</i>)	

ii. Now, consider the sulfur atom (S) marked as ***f***, nitrogen (N) atom (***g***), and oxygen(O) atoms (***h*** and ***i***), and determine their **hybridization**. (2 points)

Atom	Hybridization
Sulfur (<i>f</i>)	
Nitrogen (<i>g</i>)	
Oxygen (<i>h</i>)	
Oxygen (<i>i</i>)	

iii. **Describe** all the bonds formed by the carbon atoms ***a*** and ***b***, using sigma and pi designation as appropriate. (2 points)

Atom	Bonds formed (use sigma and pi as appropriate)
Carbon (<i>a</i>)	
Carbon (<i>b</i>)	

iv. How many total sp^3 -s sigma bonds are in the molecule? (1 point)

v. Determine **electronic** and **molecular geometry** at carbon (C) labeled ***e*** and sulfur (S) labeled ***f*** centres. (2 points)

Atom	Electronic Geometry	Molecular Geometry
Carbon (<i>e</i>)		
Sulfur (<i>f</i>)		

vi. Circle the value from the choices given below, that describes the **C-S-C bond angle at sulfur** (S) marked ***f***: (0.5 points)

$>90^\circ$ to $<109.5^\circ$ 109.5° >109.5 to $<120^\circ$ 120° $>120^\circ$ to $<180^\circ$


Question 8 12 points

Using the **valence shell orbitals only** draw a **complete** molecular orbital (MO) diagram (with no s-p mixing) for the molecule **PS**, in the space provided below. **(6 points)**

Assume that 3s and 3p atomic orbitals will form molecular orbitals like those formed by the combination of 2s and 2p atomic orbitals that we learnt in class, and there is no s-p mixing.

Show **energy arrow**, **atomic orbitals** involved, **atom labels**, the **molecular orbitals** that are formed, **filled electrons** in the MO diagram etc. Make sure to label all the atomic and molecular orbitals.

Using the MO diagram above, answer the following questions:

- (i) Which one of the following will have the highest bond order? Circle the correct choice
 A) PS^{1-} B) PS C) PS^{1+}
- (ii) Which one of the following will have the longest bond length? Circle the correct choice
 A) PS^{1-} B) PS C) PS^{1+}
- (iii) Which of the following will be diamagnetic? Circle the correct choice.
 A) PS^{1-} B) PS C) PS^{1+}
- (iv) Draw Lewis dot structure of the molecule with the shortest bond length (based on the MO diagram) from among the three (PS^{1-} , PS, PS^{1+}), and indicate the bond order calculated from the Lewis dot structure. **(3 points)**

Lewis Dot Structure

Bond Order



EE749497-67F2-41B3-B42D-F3BC8744EFC4

midterm-exam-93388

#726 Page 10 of 10

SCRAP PAGE

**CHEMISTRY 110 –Midterm Exam****Thursday November 7, 2024, 6:30 – 8:30 pm**

Examiners: Pallavi Sirjoosingh and Ashok Kakkar

First name (please write as legibly as possible within the boxes)

--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--

Last name

--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--

Student ID number

--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--

This booklet has 8 questions on 10 pages (The last page can be used as a scrap page).

The staple must NOT be removed from the question booklet. Your intact question booklet must be returned at the end of the exam.

A separate booklet contains a periodic table, and datasheet.

Write your answers directly under the questions in the space provided. Do not write on the QR codes or near the top edges (as it may be cut off).

Please write **legibly and dark enough** for the scan to be readable. For students who use English as a second language, dictionaries are permitted. Calculators are allowed.

DO NOT ASK QUESTIONS OF THE INVIGILATORS about interpretations of the exam questions. If you have a concern about a question, please summarize it at the top of this page. All concerns will be evaluated after the exam. GOOD LUCK!

**Question 1 8 points** (*Assume all transitions are possible*)

a. 6 points Calculate the **energy (in kJ/mol)** required to ionize (completely remove) the electron from the **$n=3$** level of **He^+ ion** (He: Helium). Show your work including equations. (3 sig. figs.)

b. 2 points

(I) Select the largest (longest) wavelength from the three choices below:

- i. Wavelength of photon to *completely remove* an electron from *1s orbital of Hydrogen atom*
- ii. Wavelength of photon to *completely remove* an electron from *1s orbital of Boron atom (B)*
- iii. Wavelength of photon to *completely remove* an electron from *1s orbital of Boron cation (B^+)*

i or ii or iii

(II) Select the smallest (shortest) wavelength from the three choices below (*for H atom*):

- i. Wavelength of photon *emitted* for an electron transitioning from *2p to 1s orbital*
- ii. Wavelength of photon *absorbed* for an electron transitioning from *2s to 4p orbital*
- iii. Wavelength of photon *absorbed* for an electron transitioning from *1s to 3p orbital*

i or ii or iii



Question 2 8 points



A Chem110 student plots the graph shown on the left, from data obtained for the kinetic energy (eV) of the ejected electron versus frequency (s^{-1}) of the incident light, for an unknown metal. ($1\text{eV} = 1.6022 \times 10^{-19} \text{ J}$)

Based on the plot, answer the questions **a to c** below:

a. 1 point On the plot above, label the **threshold frequency**.

b. 2 points What is the value of the work potential (in J) of the unknown metal?

c. 3 points What is the *longest* wavelength (in nm) required to eject an electron from this metal's surface? Show your work including equation(s) used. (3 sig figs)

d. 2 points In the space provided below, plot the following

(Assume frequency is *above* threshold frequency):

- i. *Kinetic energy* of an ejected electron versus the *intensity* of the incident light (left)
- ii. *Number of electrons* ejected versus the *frequency* of the incident light (right)



**Question 3** **6 points****a. 2 points**

In the ground state of Ge (Germanium; $Z = 32$) atom, **name** the following orbitals:

i. Orbital containing the electron that requires the maximum energy to be removed.

ii. Orbital containing the electron that experiences the minimum effective nuclear charge.

b. 2 points

An unknown anion (A^-) has the following *excited* electronic configuration: $[Ar] 4s^2 3d^3 4p^1$

(Ar: Argon; $Z = 18$)

Give the **condensed ground state electronic configuration** of the atom, **A** and the cation, **A^{2+}** .
Show your work. (Assume all electronic transitions are possible)

Electronic Configuration of A:

Electronic Configuration of A^{2+} :

c. 2 points

In a single atom, determine how many electrons can have the following set of quantum numbers?

(i) $n = 5; l = 3$

In a single atom, determine how many electrons can have the following set of quantum numbers?

(i) $n = 3; l = 2; m_s = +1/2$

**Question 4** **8 points**

Rank the following from 1 to 4 in **increasing** order. Provide the ranking in the boxes provided.

Note: Atomic number (Z) are provided for relevant atoms

a) Increasing (lowest to highest) order of *ionization energy*: (1= lowest; 4 = highest)

Mg

Mg: Z = 12

Rb

Rb: Z = 37

Cs

Cs: Z = 55

K

K : Z = 19

b) Increasing (lowest to highest) order of *bond angle*: (1= lowest; 4 = highest)

H₂OCCl₄XeF₄

HCN

c) Increasing (smallest to largest) order of *ionic size* (1= smallest; 4 = largest)

S²⁻

S: Z = 16

O²⁻

O: Z = 8

Be²⁺

Be: Z = 4

Na⁺

Na: Z = 11

d) Increasing (lowest to highest) order of *bond energy* (1= lowest; 4 = highest)

P-F

P-Cl

P-I

P-Br

**Question 5** **12 points**

a. 9 points For the molecules HClO_4 and TeF_6 , draw the most stable *Lewis structure* showing appropriate shape and *dashed/wedged bonds*, based on VSEPR. Determine the **molecular geometry** around the central atom. For HClO_4 , label the *most polar bond*.



b. 3 points (Note: Lone pairs are not shown in structures below)

(i) For the three *resonance* structures (1, 2 and 3) shown below, answer the following question:



(I) Circle the *most contributing* resonance structure.

Structure 1

Structure 2

Structure 3

(ii) For the three structures given (4, 5, and 6) below, answer the following questions:



(I) Circle all the structures that represent *resonance structures* of each other:

Structure 4

Structure 5

Structure 6

(II) Among the resonance structures, circle the *most contributing* resonance structure.

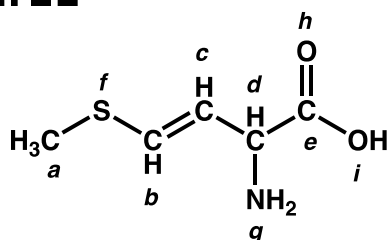
Structure 4

Structure 5

Structure 6

**Question 6** **6 points**Using the data given below, calculate the **lattice energy** for SrO(s).

	ΔH° (kJ/mol)
$\text{Sr}(s) \longrightarrow \text{Sr}(g)$	+180
$\text{Sr}(g) \longrightarrow \text{Sr}^{2+}(g) + 2e^-$	+1468
$\text{O}_2(g) \longrightarrow 2\text{O}(g)$	+500
$\text{O}(g) + 2e^- \longrightarrow \text{O}^{2-}(g)$	+600
$\text{Sr}(s) + \frac{1}{2}\text{O}_2(g) \longrightarrow \text{SrO}(s)$	-550

**Question 7 10 points**

Consider the molecule given on the left, and answer questions (i) to (vi) using chemical bonding theories that we have learnt in class (valence bond theory, hybridization).

Note that the lone pairs are not shown in the structure:

i. Consider the carbon atoms marked in as ***a***, ***b***, ***c***, ***d***, and ***e*** in the molecule, and determine **hybridization** at each of the carbon centers. (2.5 points)

Atom	Hybridization
Carbon (<i>a</i>)	
Carbon (<i>b</i>)	
Carbon (<i>c</i>)	
Carbon (<i>d</i>)	
Carbon (<i>e</i>)	

ii. Now, consider the sulfur atom (S) marked as ***f***, nitrogen (N) atom (***g***), and oxygen(O) atoms (***h*** and ***i***), and determine their **hybridization**. (2 points)

Atom	Hybridization
Sulfur (<i>f</i>)	
Nitrogen (<i>g</i>)	
Oxygen (<i>h</i>)	
Oxygen (<i>i</i>)	

iii. **Describe** all the bonds formed by the carbon atoms ***a*** and ***b***, using sigma and pi designation as appropriate. (2 points)

Atom	Bonds formed (use sigma and pi as appropriate)
Carbon (<i>a</i>)	
Carbon (<i>b</i>)	

iv. How many total sp^3 -s sigma bonds are in the molecule? (1 point)

v. Determine **electronic** and **molecular geometry** at carbon (C) labeled ***e*** and sulfur (S) labeled ***f*** centres. (2 points)

Atom	Electronic Geometry	Molecular Geometry
Carbon (<i>e</i>)		
Sulfur (<i>f</i>)		

vi. Circle the value from the choices given below, that describes the **C-S-C bond angle at sulfur** (S) marked ***f***: (0.5 points)

>90° to <109.5° 109.5° >109.5 to <120° 120° >120° to <180°


Question 8 12 points

Using the **valence shell orbitals only** draw a **complete** molecular orbital (MO) diagram (with no s-p mixing) for the molecule **PS**, in the space provided below. **(6 points)**

Assume that 3s and 3p atomic orbitals will form molecular orbitals like those formed by the combination of 2s and 2p atomic orbitals that we learnt in class, and there is no s-p mixing.

Show **energy arrow**, **atomic orbitals** involved, **atom labels**, the **molecular orbitals** that are formed, **filled electrons** in the MO diagram etc. Make sure to label all the atomic and molecular orbitals.

Using the MO diagram above, answer the following questions:

- (i) Which one of the following will have the highest bond order? Circle the correct choice
 A) PS^{1-} B) PS C) PS^{1+}
- (ii) Which one of the following will have the longest bond length? Circle the correct choice
 A) PS^{1-} B) PS C) PS^{1+}
- (iii) Which of the following will be diamagnetic? Circle the correct choice.
 A) PS^{1-} B) PS C) PS^{1+}
- (iv) Draw Lewis dot structure of the molecule with the shortest bond length (based on the MO diagram) from among the three (PS^{1-} , PS, PS^{1+}), and indicate the bond order calculated from the Lewis dot structure. **(3 points)**

Lewis Dot Structure

Bond Order



8E7630A7-CDCB-480B-9159-DFBDDD2DC143

midterm-exam-93388

#727 Page 10 of 10

SCRAP PAGE

**CHEMISTRY 110 –Midterm Exam****Thursday November 7, 2024, 6:30 – 8:30 pm****Examiners: Pallavi Sirjoosingh and Ashok Kakkar**

First name (please write as legibly as possible within the boxes)

--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--

Last name

--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--

Student ID number

--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--

This booklet has 8 questions on 10 pages (The last page can be used as a scrap page).

The staple must NOT be removed from the question booklet. Your intact question booklet must be returned at the end of the exam.

A separate booklet contains a periodic table, and datasheet.

Write your answers directly under the questions in the space provided. Do not write on the QR codes or near the top edges (as it may be cut off).

Please write **legibly and dark enough** for the scan to be readable. For students who use English as a second language, dictionaries are permitted. Calculators are allowed.

DO NOT ASK QUESTIONS OF THE INVIGILATORS about interpretations of the exam questions. If you have a concern about a question, please summarize it at the top of this page. All concerns will be evaluated after the exam. GOOD LUCK!

**Question 1 8 points** (*Assume all transitions are possible*)

a. 6 points Calculate the **energy (in kJ/mol)** required to ionize (completely remove) the electron from the **n=3** level of **He⁺ ion** (He: Helium). Show your work including equations. (3 sig. figs.)

b. 2 points

(I) Select the largest (longest) wavelength from the three choices below:

- i. Wavelength of photon to *completely remove* an electron from *1s orbital of Hydrogen atom*
- ii. Wavelength of photon to *completely remove* an electron from *1s orbital of Boron atom (B)*
- iii. Wavelength of photon to *completely remove* an electron from *1s orbital of Boron cation (B⁺)*

i or ii or iii

(II) Select the smallest (shortest) wavelength from the three choices below (*for H atom*):

- i. Wavelength of photon *emitted* for an electron transitioning from *2p to 1s orbital*
- ii. Wavelength of photon *absorbed* for an electron transitioning from *2s to 4p orbital*
- iii. Wavelength of photon *absorbed* for an electron transitioning from *1s to 3p orbital*

i or ii or iii



Question 2 8 points



A Chem110 student plots the graph shown on the left, from data obtained for the kinetic energy (eV) of the ejected electron versus frequency (s^{-1}) of the incident light, for an unknown metal. ($1\text{eV} = 1.6022 \times 10^{-19} \text{ J}$)

Based on the plot, answer the questions **a to c** below:

a. 1 point On the plot above, label the **threshold frequency**.

b. 2 points What is the value of the work potential (in J) of the unknown metal?

c. 3 points What is the *longest* wavelength (in nm) required to eject an electron from this metal's surface? Show your work including equation(s) used. (3 sig figs)

d. 2 points In the space provided below, plot the following

(Assume frequency is *above* threshold frequency):

- i. *Kinetic energy* of an ejected electron versus the *intensity* of the incident light (left)
- ii. *Number of electrons* ejected versus the *frequency* of the incident light (right)



**Question 3** **6 points****a. 2 points**

In the ground state of Ge (Germanium; $Z = 32$) atom, **name** the following orbitals:

i. Orbital containing the electron that requires the maximum energy to be removed.

ii. Orbital containing the electron that experiences the minimum effective nuclear charge.

b. 2 points

An unknown anion (A^-) has the following *excited* electronic configuration: $[Ar] 4s^2 3d^3 4p^1$

(Ar: Argon; $Z = 18$)

Give the **condensed ground state electronic configuration** of the atom, **A** and the cation, **A^{2+}** .
Show your work. (Assume all electronic transitions are possible)

Electronic Configuration of A:

Electronic Configuration of A^{2+} :

c. 2 points

In a single atom, determine how many electrons can have the following set of quantum numbers?

(i) $n = 5; l = 3$

In a single atom, determine how many electrons can have the following set of quantum numbers?

(i) $n = 3; l = 2; m_s = +1/2$

**Question 4** **8 points**

Rank the following from 1 to 4 in **increasing** order. Provide the ranking in the boxes provided.

Note: Atomic number (Z) are provided for relevant atoms

a) Increasing (lowest to highest) order of *ionization energy*: (1= lowest; 4 = highest)

Mg

Mg: Z = 12

Rb

Rb: Z = 37

Cs

Cs: Z = 55

K

K : Z = 19

b) Increasing (lowest to highest) order of *bond angle*: (1= lowest; 4 = highest)

H₂OCCl₄XeF₄

HCN

c) Increasing (smallest to largest) order of *ionic size* (1= smallest; 4 = largest)

S²⁻

S: Z = 16

O²⁻

O: Z = 8

Be²⁺

Be: Z = 4

Na⁺

Na: Z = 11

d) Increasing (lowest to highest) order of *bond energy* (1= lowest; 4 = highest)

P-F

P-Cl

P-I

P-Br

**Question 5** **12 points**

a. 9 points For the molecules HClO_4 and TeF_6 , draw the most stable *Lewis structure* showing appropriate shape and *dashed/wedged bonds*, based on VSEPR. Determine the **molecular geometry** around the central atom. For HClO_4 , label the *most polar bond*.



b. 3 points (Note: Lone pairs are not shown in structures below)

(i) For the three *resonance* structures (1, 2 and 3) shown below, answer the following question:



(I) Circle the *most contributing* resonance structure.

Structure 1

Structure 2

Structure 3

(ii) For the three structures given (4, 5, and 6) below, answer the following questions:



(I) Circle all the structures that represent *resonance structures* of each other:

Structure 4

Structure 5

Structure 6

(II) Among the resonance structures, circle the *most contributing* resonance structure.

Structure 4

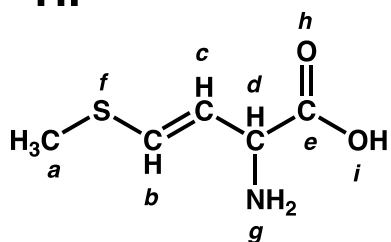
Structure 5

Structure 6

**Question 6** **6 points**

Using the data given below, calculate the **lattice energy** for SrO(s).

	ΔH° (kJ/mol)
$\text{Sr}(s) \longrightarrow \text{Sr}(g)$	+180
$\text{Sr}(g) \longrightarrow \text{Sr}^{2+}(g) + 2e^-$	+1468
$\text{O}_2(g) \longrightarrow 2\text{O}(g)$	+500
$\text{O}(g) + 2e^- \longrightarrow \text{O}^{2-}(g)$	+600
$\text{Sr}(s) + \frac{1}{2}\text{O}_2(g) \longrightarrow \text{SrO}(s)$	-550

**Question 7 10 points**

Consider the molecule given on the left, and answer questions (i) to (vi) using chemical bonding theories that we have learnt in class (valence bond theory, hybridization).

Note that the lone pairs are not shown in the structure:

i. Consider the carbon atoms marked in as ***a***, ***b***, ***c***, ***d***, and ***e*** in the molecule, and determine **hybridization** at each of the carbon centers. (2.5 points)

Atom	Hybridization
Carbon (<i>a</i>)	
Carbon (<i>b</i>)	
Carbon (<i>c</i>)	
Carbon (<i>d</i>)	
Carbon (<i>e</i>)	

ii. Now, consider the sulfur atom (S) marked as ***f***, nitrogen (N) atom (***g***), and oxygen(O) atoms (***h*** and ***i***), and determine their **hybridization**. (2 points)

Atom	Hybridization
Sulfur (<i>f</i>)	
Nitrogen (<i>g</i>)	
Oxygen (<i>h</i>)	
Oxygen (<i>i</i>)	

iii. **Describe** all the bonds formed by the carbon atoms ***a*** and ***b***, using sigma and pi designation as appropriate. (2 points)

Atom	Bonds formed (use sigma and pi as appropriate)
Carbon (<i>a</i>)	
Carbon (<i>b</i>)	

iv. How many total sp^3 -s sigma bonds are in the molecule? (1 point)

v. Determine **electronic** and **molecular geometry** at carbon (C) labeled ***e*** and sulfur (S) labeled ***f*** centres. (2 points)

Atom	Electronic Geometry	Molecular Geometry
Carbon (<i>e</i>)		
Sulfur (<i>f</i>)		

vi. Circle the value from the choices given below, that describes the **C-S-C bond angle at sulfur** (S) marked ***f***: (0.5 points)

>90° to <109.5°

109.5°

>109.5 to <120°

120°

>120° to <180°



Question 8 12 points

Using the **valence shell orbitals only** draw a **complete** molecular orbital (MO) diagram (with no s-p mixing) for the molecule **PS**, in the space provided below. **(6 points)**

Assume that 3s and 3p atomic orbitals will form molecular orbitals like those formed by the combination of 2s and 2p atomic orbitals that we learnt in class, and there is no s-p mixing.

Show **energy arrow**, **atomic orbitals** involved, **atom labels**, the **molecular orbitals** that are formed, **filled electrons** in the MO diagram etc. Make sure to label all the atomic and molecular orbitals.

Using the MO diagram above, answer the following questions:

- (i) Which one of the following will have the highest bond order? Circle the correct choice
 A) PS^{1-} B) PS C) PS^{1+}
- (ii) Which one of the following will have the longest bond length? Circle the correct choice
 A) PS^{1-} B) PS C) PS^{1+}
- (iii) Which of the following will be diamagnetic? Circle the correct choice.
 A) PS^{1-} B) PS C) PS^{1+}
- (iv) Draw Lewis dot structure of the molecule with the shortest bond length (based on the MO diagram) from among the three (PS^{1-} , PS, PS^{1+}), and indicate the bond order calculated from the Lewis dot structure. **(3 points)**

Lewis Dot Structure

Bond Order



AF93DCA4-10A3-4810-8AE7-FB60432EA94D

midterm-exam-93388

#728 Page 10 of 10

SCRAP PAGE

**CHEMISTRY 110 –Midterm Exam****Thursday November 7, 2024, 6:30 – 8:30 pm****Examiners: Pallavi Sirjoosingh and Ashok Kakkar**

First name (please write as legibly as possible within the boxes)

--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--

Last name

--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--

Student ID number

--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--

This booklet has 8 questions on 10 pages (The last page can be used as a scrap page).

The staple must NOT be removed from the question booklet. Your intact question booklet must be returned at the end of the exam.

A separate booklet contains a periodic table, and datasheet.

Write your answers directly under the questions in the space provided. Do not write on the QR codes or near the top edges (as it may be cut off).

Please write **legibly and dark enough** for the scan to be readable. For students who use English as a second language, dictionaries are permitted. Calculators are allowed.

DO NOT ASK QUESTIONS OF THE INVIGILATORS about interpretations of the exam questions. If you have a concern about a question, please summarize it at the top of this page. All concerns will be evaluated after the exam. GOOD LUCK!

**Question 1 8 points** (*Assume all transitions are possible*)

a. 6 points Calculate the **energy (in kJ/mol)** required to ionize (completely remove) the electron from the **$n=3$** level of **He^+ ion** (He: Helium). Show your work including equations. (3 sig. figs.)

b. 2 points

(I) Select the largest (longest) wavelength from the three choices below:

- i. Wavelength of photon to *completely remove* an electron from *1s orbital of Hydrogen atom*
- ii. Wavelength of photon to *completely remove* an electron from *1s orbital of Boron atom (B)*
- iii. Wavelength of photon to *completely remove* an electron from *1s orbital of Boron cation (B^+)*

i or ii or iii

(II) Select the smallest (shortest) wavelength from the three choices below (*for H atom*):

- i. Wavelength of photon *emitted* for an electron transitioning from *2p to 1s orbital*
- ii. Wavelength of photon *absorbed* for an electron transitioning from *2s to 4p orbital*
- iii. Wavelength of photon *absorbed* for an electron transitioning from *1s to 3p orbital*

i or ii or iii



Question 2 8 points



A Chem110 student plots the graph shown on the left, from data obtained for the kinetic energy (eV) of the ejected electron versus frequency (s^{-1}) of the incident light, for an unknown metal. ($1\text{eV} = 1.6022 \times 10^{-19} \text{ J}$)

Based on the plot, answer the questions **a to c** below:

a. 1 point On the plot above, label the **threshold frequency**.

b. 2 points What is the value of the work potential (in J) of the unknown metal?

c. 3 points What is the *longest* wavelength (in nm) required to eject an electron from this metal's surface? Show your work including equation(s) used. (3 sig figs)

d. 2 points In the space provided below, plot the following

(Assume frequency is *above* threshold frequency):

- i. *Kinetic energy* of an ejected electron versus the *intensity* of the incident light (left)
- ii. *Number of electrons* ejected versus the *frequency* of the incident light (right)



**Question 3** **6 points****a. 2 points**

In the ground state of Ge (Germanium; $Z = 32$) atom, **name** the following orbitals:

i. Orbital containing the electron that requires the maximum energy to be removed.

ii. Orbital containing the electron that experiences the minimum effective nuclear charge.

b. 2 points

An unknown anion (A^-) has the following *excited* electronic configuration: $[Ar] 4s^2 3d^3 4p^1$

(Ar: Argon; $Z = 18$)

Give the **condensed ground state electronic configuration** of the atom, **A** and the cation, **A^{2+}** .
Show your work. (Assume all electronic transitions are possible)

Electronic Configuration of A:

Electronic Configuration of A^{2+} :

c. 2 points

In a single atom, determine how many electrons can have the following set of quantum numbers?

(i) $n = 5; l = 3$

In a single atom, determine how many electrons can have the following set of quantum numbers?

(i) $n = 3; l = 2; m_s = +1/2$

**Question 4** **8 points**

Rank the following from 1 to 4 in **increasing** order. Provide the ranking in the boxes provided.

Note: Atomic number (Z) are provided for relevant atoms

a) Increasing (lowest to highest) order of *ionization energy*: (1= lowest; 4 = highest)

Mg

Mg: Z = 12

Rb

Rb: Z = 37

Cs

Cs: Z = 55

K

K : Z = 19

b) Increasing (lowest to highest) order of *bond angle*: (1= lowest; 4 = highest)

H₂OCCl₄XeF₄

HCN

c) Increasing (smallest to largest) order of *ionic size* (1= smallest; 4 = largest)

S²⁻

S: Z = 16

O²⁻

O: Z = 8

Be²⁺

Be: Z = 4

Na⁺

Na: Z = 11

d) Increasing (lowest to highest) order of *bond energy* (1= lowest; 4 = highest)

P-F

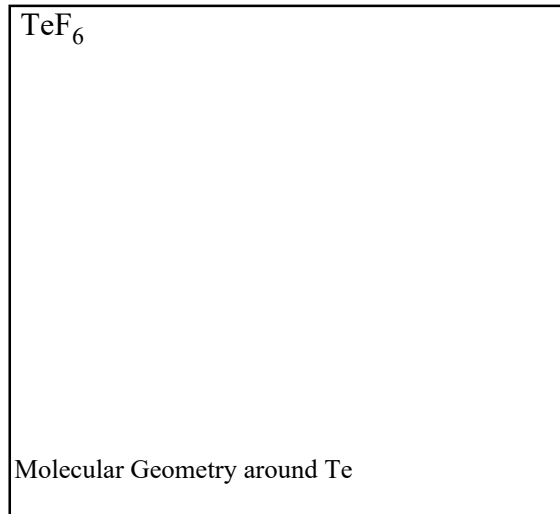
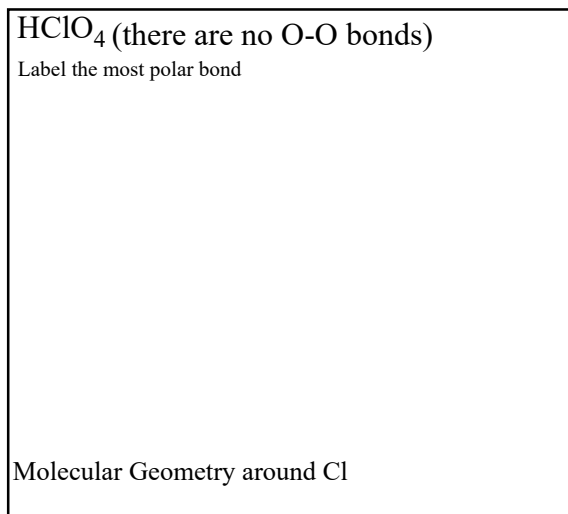
P-Cl

P-I

P-Br

**Question 5** **12 points**

a. 9 points For the molecules HClO_4 and TeF_6 , draw the most stable *Lewis structure* showing appropriate shape and *dashed/wedged bonds*, based on VSEPR. Determine the **molecular geometry** around the central atom. For HClO_4 , label the *most polar bond*.



b. 3 points (Note: Lone pairs are not shown in structures below)

(i) For the three *resonance* structures (1, 2 and 3) shown below, answer the following question:



(I) Circle the *most contributing* resonance structure.

Structure 1

Structure 2

Structure 3

(ii) For the three structures given (4, 5, and 6) below, answer the following questions:



(I) Circle all the structures that represent *resonance structures* of each other:

Structure 4

Structure 5

Structure 6

(II) Among the resonance structures, circle the *most contributing* resonance structure.

Structure 4

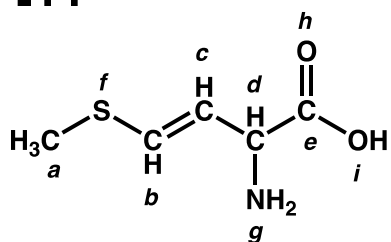
Structure 5

Structure 6

**Question 6** **6 points**

Using the data given below, calculate the **lattice energy** for SrO(s).

	ΔH° (kJ/mol)
$\text{Sr}(s) \longrightarrow \text{Sr}(g)$	+180
$\text{Sr}(g) \longrightarrow \text{Sr}^{2+}(g) + 2e^-$	+1468
$\text{O}_2(g) \longrightarrow 2\text{O}(g)$	+500
$\text{O}(g) + 2e^- \longrightarrow \text{O}^{2-}(g)$	+600
$\text{Sr}(s) + \frac{1}{2}\text{O}_2(g) \longrightarrow \text{SrO}(s)$	-550

**Question 7 10 points**

Consider the molecule given on the left, and answer questions (i) to (vi) using chemical bonding theories that we have learnt in class (valence bond theory, hybridization).

Note that the lone pairs are not shown in the structure:

i. Consider the carbon atoms marked in as ***a***, ***b***, ***c***, ***d***, and ***e*** in the molecule, and determine **hybridization** at each of the carbon centers. (2.5 points)

Atom	Hybridization
Carbon (<i>a</i>)	
Carbon (<i>b</i>)	
Carbon (<i>c</i>)	
Carbon (<i>d</i>)	
Carbon (<i>e</i>)	

ii. Now, consider the sulfur atom (S) marked as ***f***, nitrogen (N) atom (***g***), and oxygen(O) atoms (***h*** and ***i***), and determine their **hybridization**. (2 points)

Atom	Hybridization
Sulfur (<i>f</i>)	
Nitrogen (<i>g</i>)	
Oxygen (<i>h</i>)	
Oxygen (<i>i</i>)	

iii. **Describe** all the bonds formed by the carbon atoms ***a*** and ***b***, using sigma and pi designation as appropriate. (2 points)

Atom	Bonds formed (use sigma and pi as appropriate)
Carbon (<i>a</i>)	
Carbon (<i>b</i>)	

iv. How many total sp^3 -s sigma bonds are in the molecule? (1 point)

v. Determine **electronic** and **molecular geometry** at carbon (C) labeled ***e*** and sulfur (S) labeled ***f*** centres. (2 points)

Atom	Electronic Geometry	Molecular Geometry
Carbon (<i>e</i>)		
Sulfur (<i>f</i>)		

vi. Circle the value from the choices given below, that describes the **C-S-C bond angle at sulfur** (S) marked ***f***: (0.5 points)

>90° to <109.5°

109.5°

>109.5 to <120°

120°

>120° to <180°


Question 8 12 points

Using the **valence shell orbitals only** draw a **complete** molecular orbital (MO) diagram (with no s-p mixing) for the molecule **PS**, in the space provided below. **(6 points)**

Assume that 3s and 3p atomic orbitals will form molecular orbitals like those formed by the combination of 2s and 2p atomic orbitals that we learnt in class, and there is no s-p mixing.

Show **energy arrow**, **atomic orbitals** involved, **atom labels**, the **molecular orbitals** that are formed, **filled electrons** in the MO diagram etc. Make sure to label all the atomic and molecular orbitals.

Using the MO diagram above, answer the following questions:

- (i) Which one of the following will have the highest bond order? Circle the correct choice
 A) PS^{1-} B) PS C) PS^{1+}
- (ii) Which one of the following will have the longest bond length? Circle the correct choice
 A) PS^{1-} B) PS C) PS^{1+}
- (iii) Which of the following will be diamagnetic? Circle the correct choice.
 A) PS^{1-} B) PS C) PS^{1+}
- (iv) Draw Lewis dot structure of the molecule with the shortest bond length (based on the MO diagram) from among the three (PS^{1-} , PS, PS^{1+}), and indicate the bond order calculated from the Lewis dot structure. **(3 points)**

Lewis Dot Structure

Bond Order



2EE8961C-B2AF-459F-BD12-0471EBC6FDC0

midterm-exam-93388

#729 Page 10 of 10

SCRAP PAGE

**CHEMISTRY 110 –Midterm Exam****Thursday November 7, 2024, 6:30 – 8:30 pm**

Examiners: Pallavi Sirjoosingh and Ashok Kakkar

First name (please write as legibly as possible within the boxes)

--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--

Last name

--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--

Student ID number

--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--

This booklet has 8 questions on 10 pages (The last page can be used as a scrap page).

The staple must NOT be removed from the question booklet. Your intact question booklet must be returned at the end of the exam.

A separate booklet contains a periodic table, and datasheet.

Write your answers directly under the questions in the space provided. Do not write on the QR codes or near the top edges (as it may be cut off).

Please write **legibly and dark enough** for the scan to be readable. For students who use English as a second language, dictionaries are permitted. Calculators are allowed.

DO NOT ASK QUESTIONS OF THE INVIGILATORS about interpretations of the exam questions. If you have a concern about a question, please summarize it at the top of this page. All concerns will be evaluated after the exam. GOOD LUCK!

**Question 1 8 points** (*Assume all transitions are possible*)

a. 6 points Calculate the **energy (in kJ/mol)** required to ionize (completely remove) the electron from the **$n=3$** level of **He^+ ion** (He: Helium). Show your work including equations. (3 sig. figs.)

b. 2 points

(I) Select the largest (longest) wavelength from the three choices below:

- i. Wavelength of photon to *completely remove* an electron from *1s orbital of Hydrogen atom*
- ii. Wavelength of photon to *completely remove* an electron from *1s orbital of Boron atom (B)*
- iii. Wavelength of photon to *completely remove* an electron from *1s orbital of Boron cation (B^+)*

i or ii or iii

(II) Select the smallest (shortest) wavelength from the three choices below (*for H atom*):

- i. Wavelength of photon *emitted* for an electron transitioning from *2p to 1s orbital*
- ii. Wavelength of photon *absorbed* for an electron transitioning from *2s to 4p orbital*
- iii. Wavelength of photon *absorbed* for an electron transitioning from *1s to 3p orbital*

i or ii or iii



Question 2 8 points



A Chem110 student plots the graph shown on the left, from data obtained for the kinetic energy (eV) of the ejected electron versus frequency (s^{-1}) of the incident light, for an unknown metal. ($1\text{eV} = 1.6022 \times 10^{-19} \text{ J}$)

Based on the plot, answer the questions **a to c** below:

a. 1 point On the plot above, label the **threshold frequency**.

b. 2 points What is the value of the work potential (in J) of the unknown metal?

c. 3 points What is the *longest* wavelength (in nm) required to eject an electron from this metal's surface? Show your work including equation(s) used. (3 sig figs)

d. 2 points In the space provided below, plot the following

(Assume frequency is *above* threshold frequency):

- Kinetic energy* of an ejected electron versus the *intensity* of the incident light (left)
- Number of electrons* ejected versus the *frequency* of the incident light (right)



**Question 3** **6 points****a. 2 points**

In the ground state of Ge (Germanium; $Z = 32$) atom, **name** the following orbitals:

i. Orbital containing the electron that requires the maximum energy to be removed.

ii. Orbital containing the electron that experiences the minimum effective nuclear charge.

b. 2 points

An unknown anion (A^-) has the following *excited* electronic configuration: $[Ar] 4s^2 3d^3 4p^1$

(Ar: Argon; $Z = 18$)

Give the **condensed ground state electronic configuration** of the atom, **A** and the cation, **A^{2+}** .
Show your work. (Assume all electronic transitions are possible)

Electronic Configuration of A:

Electronic Configuration of A^{2+} :

c. 2 points

In a single atom, determine how many electrons can have the following set of quantum numbers?

(i) $n = 5; l = 3$

In a single atom, determine how many electrons can have the following set of quantum numbers?

(i) $n = 3; l = 2; m_s = +1/2$

**Question 4** **8 points**

Rank the following from 1 to 4 in **increasing** order. Provide the ranking in the boxes provided.

Note: Atomic number (Z) are provided for relevant atoms

a) Increasing (lowest to highest) order of *ionization energy*: (1= lowest; 4 = highest)

Mg

Mg: Z = 12

Rb

Rb: Z = 37

Cs

Cs: Z = 55

K

K : Z = 19

b) Increasing (lowest to highest) order of *bond angle*: (1= lowest; 4 = highest)

H₂OCCl₄XeF₄

HCN

c) Increasing (smallest to largest) order of *ionic size* (1= smallest; 4 = largest)

S²⁻

S: Z = 16

O²⁻

O: Z = 8

Be²⁺

Be: Z = 4

Na⁺

Na: Z = 11

d) Increasing (lowest to highest) order of *bond energy* (1= lowest; 4 = highest)

P-F

P-Cl

P-I

P-Br

**Question 5** **12 points**

a. 9 points For the molecules HClO_4 and TeF_6 , draw the most stable *Lewis structure* showing appropriate shape and *dashed/wedged bonds*, based on VSEPR. Determine the **molecular geometry** around the central atom. For HClO_4 , label the *most polar bond*.



b. 3 points (Note: Lone pairs are not shown in structures below)

(i) For the three *resonance* structures (1, 2 and 3) shown below, answer the following question:



(I) Circle the *most contributing* resonance structure.

Structure 1

Structure 2

Structure 3

(ii) For the three structures given (4, 5, and 6) below, answer the following questions:



(I) Circle all the structures that represent *resonance structures* of each other:

Structure 4

Structure 5

Structure 6

(II) Among the resonance structures, circle the *most contributing* resonance structure.

Structure 4

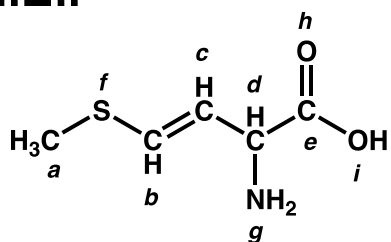
Structure 5

Structure 6

**Question 6** **6 points**

Using the data given below, calculate the **lattice energy** for SrO(s).

	ΔH° (kJ/mol)
$\text{Sr}(s) \longrightarrow \text{Sr}(g)$	+180
$\text{Sr}(g) \longrightarrow \text{Sr}^{2+}(g) + 2e^-$	+1468
$\text{O}_2(g) \longrightarrow 2\text{O}(g)$	+500
$\text{O}(g) + 2e^- \longrightarrow \text{O}^{2-}(g)$	+600
$\text{Sr}(s) + \frac{1}{2}\text{O}_2(g) \longrightarrow \text{SrO}(s)$	-550

**Question 7 10 points**

Consider the molecule given on the left, and answer questions (i) to (vi) using chemical bonding theories that we have learnt in class (valence bond theory, hybridization).

Note that the lone pairs are not shown in the structure:

i. Consider the carbon atoms marked in as ***a***, ***b***, ***c***, ***d***, and ***e*** in the molecule, and determine **hybridization** at each of the carbon centers. (2.5 points)

Atom	Hybridization
Carbon (<i>a</i>)	
Carbon (<i>b</i>)	
Carbon (<i>c</i>)	
Carbon (<i>d</i>)	
Carbon (<i>e</i>)	

ii. Now, consider the sulfur atom (S) marked as ***f***, nitrogen (N) atom (***g***), and oxygen(O) atoms (***h*** and ***i***), and determine their **hybridization**. (2 points)

Atom	Hybridization
Sulfur (<i>f</i>)	
Nitrogen (<i>g</i>)	
Oxygen (<i>h</i>)	
Oxygen (<i>i</i>)	

iii. **Describe** all the bonds formed by the carbon atoms ***a*** and ***b***, using sigma and pi designation as appropriate. (2 points)

Atom	Bonds formed (use sigma and pi as appropriate)
Carbon (<i>a</i>)	
Carbon (<i>b</i>)	

iv. How many total sp^3 -s sigma bonds are in the molecule? (1 point)

v. Determine **electronic** and **molecular geometry** at carbon (C) labeled ***e*** and sulfur (S) labeled ***f*** centres. (2 points)

Atom	Electronic Geometry	Molecular Geometry
Carbon (<i>e</i>)		
Sulfur (<i>f</i>)		

vi. Circle the value from the choices given below, that describes the **C-S-C bond angle at sulfur** (S) marked ***f***: (0.5 points)

>90° to <109.5°

109.5°

>109.5 to <120°

120°

>120° to <180°


Question 8 12 points

Using the **valence shell orbitals only** draw a **complete** molecular orbital (MO) diagram (with no s-p mixing) for the molecule **PS**, in the space provided below. **(6 points)**

Assume that 3s and 3p atomic orbitals will form molecular orbitals like those formed by the combination of 2s and 2p atomic orbitals that we learnt in class, and there is no s-p mixing.

Show **energy arrow**, **atomic orbitals** involved, **atom labels**, the **molecular orbitals** that are formed, **filled electrons** in the MO diagram etc. Make sure to **label all the atomic and molecular orbitals**.

Using the MO diagram above, answer the following questions:

- (i) Which one of the following will have the highest bond order? Circle the correct choice
 A) PS^{1-} B) PS C) PS^{1+}
- (ii) Which one of the following will have the longest bond length? Circle the correct choice
 A) PS^{1-} B) PS C) PS^{1+}
- (iii) Which of the following will be diamagnetic? Circle the correct choice.
 A) PS^{1-} B) PS C) PS^{1+}
- (iv) Draw Lewis dot structure of the molecule with the shortest bond length (based on the MO diagram) from among the three (PS^{1-} , PS, PS^{1+}), and indicate the bond order calculated from the Lewis dot structure. **(3 points)**

Lewis Dot Structure

Bond Order



E77D280D-DDDC-4677-8C5C-A12AF1AC2E94

midterm-exam-93388

#730 Page 10 of 10

SCRAP PAGE

**CHEMISTRY 110 –Midterm Exam****Thursday November 7, 2024, 6:30 – 8:30 pm****Examiners: Pallavi Sirjoosingh and Ashok Kakkar**

First name (please write as legibly as possible within the boxes)

--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--

Last name

--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--

Student ID number

--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--

This booklet has 8 questions on 10 pages (The last page can be used as a scrap page).

The staple must NOT be removed from the question booklet. Your intact question booklet must be returned at the end of the exam.

A separate booklet contains a periodic table, and datasheet.

Write your answers directly under the questions in the space provided. Do not write on the QR codes or near the top edges (as it may be cut off).

Please write **legibly and dark enough** for the scan to be readable. For students who use English as a second language, dictionaries are permitted. Calculators are allowed.

DO NOT ASK QUESTIONS OF THE INVIGILATORS about interpretations of the exam questions. If you have a concern about a question, please summarize it at the top of this page. All concerns will be evaluated after the exam. GOOD LUCK!

**Question 1 8 points** (*Assume all transitions are possible*)

a. 6 points Calculate the **energy (in kJ/mol)** required to ionize (completely remove) the electron from the **n=3** level of **He⁺ ion** (He: Helium). Show your work including equations. (3 sig. figs.)

b. 2 points

(I) Select the largest (longest) wavelength from the three choices below:

- i. Wavelength of photon to *completely remove* an electron from *1s orbital of Hydrogen atom*
- ii. Wavelength of photon to *completely remove* an electron from *1s orbital of Boron atom (B)*
- iii. Wavelength of photon to *completely remove* an electron from *1s orbital of Boron cation (B⁺)*

i or ii or iii

(II) Select the smallest (shortest) wavelength from the three choices below (*for H atom*):

- i. Wavelength of photon *emitted* for an electron transitioning from *2p to 1s orbital*
- ii. Wavelength of photon *absorbed* for an electron transitioning from *2s to 4p orbital*
- iii. Wavelength of photon *absorbed* for an electron transitioning from *1s to 3p orbital*

i or ii or iii



Question 2 8 points



A Chem110 student plots the graph shown on the left, from data obtained for the kinetic energy (eV) of the ejected electron versus frequency (s^{-1}) of the incident light, for an unknown metal. ($1\text{eV} = 1.6022 \times 10^{-19} \text{ J}$)

Based on the plot, answer the questions **a to c** below:

a. 1 point On the plot above, label the **threshold frequency**.

b. 2 points What is the value of the work potential (in J) of the unknown metal?

c. 3 points What is the *longest* wavelength (in nm) required to eject an electron from this metal's surface? Show your work including equation(s) used. (3 sig figs)

d. 2 points In the space provided below, plot the following

(Assume frequency is *above* threshold frequency):

- i. *Kinetic energy* of an ejected electron versus the *intensity* of the incident light (left)
- ii. *Number of electrons* ejected versus the *frequency* of the incident light (right)



**Question 3** **6 points****a. 2 points**

In the ground state of Ge (Germanium; $Z = 32$) atom, **name** the following orbitals:

i. Orbital containing the electron that requires the maximum energy to be removed.

ii. Orbital containing the electron that experiences the minimum effective nuclear charge.

b. 2 points

An unknown anion (A^-) has the following *excited* electronic configuration: $[Ar] 4s^2 3d^3 4p^1$

(Ar: Argon; $Z = 18$)

Give the **condensed ground state electronic configuration** of the atom, **A** and the cation, **A^{2+}** .
Show your work. (Assume all electronic transitions are possible)

Electronic Configuration of A:

Electronic Configuration of A^{2+} :

c. 2 points

In a single atom, determine how many electrons can have the following set of quantum numbers?

(i) $n = 5; l = 3$

In a single atom, determine how many electrons can have the following set of quantum numbers?

(i) $n = 3; l = 2; m_s = +1/2$

**Question 4** **8 points**

Rank the following from 1 to 4 in **increasing** order. Provide the ranking in the boxes provided.

Note: Atomic number (Z) are provided for relevant atoms

a) Increasing (lowest to highest) order of *ionization energy*: (1= lowest; 4 = highest)

Mg

Mg: Z = 12

Rb

Rb: Z = 37

Cs

Cs: Z = 55

K

K : Z = 19

b) Increasing (lowest to highest) order of *bond angle*: (1= lowest; 4 = highest)

H₂OCCl₄XeF₄

HCN

c) Increasing (smallest to largest) order of *ionic size* (1= smallest; 4 = largest)

S²⁻

S: Z = 16

O²⁻

O: Z = 8

Be²⁺

Be: Z = 4

Na⁺

Na: Z = 11

d) Increasing (lowest to highest) order of *bond energy* (1= lowest; 4 = highest)

P-F

P-Cl

P-I

P-Br

**Question 5** **12 points**

a. 9 points For the molecules HClO_4 and TeF_6 , draw the most stable *Lewis structure* showing appropriate shape and *dashed/wedged bonds*, based on VSEPR. Determine the **molecular geometry** around the central atom. For HClO_4 , label the *most polar bond*.

HClO_4 (there are no O-O bonds)

Label the most polar bond

Molecular Geometry around Cl

TeF_6

Molecular Geometry around Te

b. 3 points (Note: Lone pairs are not shown in structures below)

(i) For the three *resonance* structures (1, 2 and 3) shown below, answer the following question:



(I) Circle the *most contributing* resonance structure.

Structure 1

Structure 2

Structure 3

(ii) For the three structures given (4, 5, and 6) below, answer the following questions:



(I) Circle all the structures that represent *resonance structures* of each other:

Structure 4

Structure 5

Structure 6

(II) Among the resonance structures, circle the *most contributing* resonance structure.

Structure 4

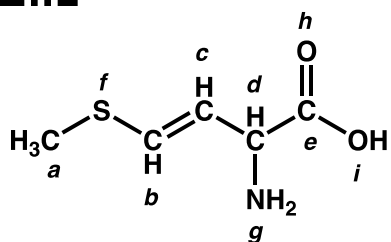
Structure 5

Structure 6

**Question 6** **6 points**

Using the data given below, calculate the **lattice energy** for SrO(s).

	ΔH° (kJ/mol)
$\text{Sr}(s) \longrightarrow \text{Sr}(g)$	+180
$\text{Sr}(g) \longrightarrow \text{Sr}^{2+}(g) + 2e^-$	+1468
$\text{O}_2(g) \longrightarrow 2\text{O}(g)$	+500
$\text{O}(g) + 2e^- \longrightarrow \text{O}^{2-}(g)$	+600
$\text{Sr}(s) + \frac{1}{2}\text{O}_2(g) \longrightarrow \text{SrO}(s)$	-550

**Question 7 10 points**

Consider the molecule given on the left, and answer questions (i) to (vi) using chemical bonding theories that we have learnt in class (valence bond theory, hybridization).

Note that the lone pairs are not shown in the structure:

i. Consider the carbon atoms marked in as ***a***, ***b***, ***c***, ***d***, and ***e*** in the molecule, and determine **hybridization** at each of the carbon centers. (2.5 points)

Atom	Hybridization
Carbon (<i>a</i>)	
Carbon (<i>b</i>)	
Carbon (<i>c</i>)	
Carbon (<i>d</i>)	
Carbon (<i>e</i>)	

ii. Now, consider the sulfur atom (S) marked as ***f***, nitrogen (N) atom (***g***), and oxygen(O) atoms (***h*** and ***i***), and determine their **hybridization**. (2 points)

Atom	Hybridization
Sulfur (<i>f</i>)	
Nitrogen (<i>g</i>)	
Oxygen (<i>h</i>)	
Oxygen (<i>i</i>)	

iii. **Describe** all the bonds formed by the carbon atoms ***a*** and ***b***, using sigma and pi designation as appropriate. (2 points)

Atom	Bonds formed (use sigma and pi as appropriate)
Carbon (<i>a</i>)	
Carbon (<i>b</i>)	

iv. How many total sp^3 -s sigma bonds are in the molecule? (1 point)

v. Determine **electronic** and **molecular geometry** at carbon (C) labeled ***e*** and sulfur (S) labeled ***f*** centres. (2 points)

Atom	Electronic Geometry	Molecular Geometry
Carbon (<i>e</i>)		
Sulfur (<i>f</i>)		

vi. Circle the value from the choices given below, that describes the **C-S-C bond angle at sulfur** (S) marked ***f***: (0.5 points)

$>90^\circ$ to $<109.5^\circ$ 109.5° >109.5 to $<120^\circ$ 120° $>120^\circ$ to $<180^\circ$


Question 8 12 points

Using the **valence shell orbitals only** draw a **complete** molecular orbital (MO) diagram (with no s-p mixing) for the molecule **PS**, in the space provided below. **(6 points)**

Assume that 3s and 3p atomic orbitals will form molecular orbitals like those formed by the combination of 2s and 2p atomic orbitals that we learnt in class, and there is no s-p mixing.

Show **energy arrow**, **atomic orbitals** involved, **atom labels**, the **molecular orbitals** that are formed, **filled electrons** in the MO diagram etc. Make sure to **label all the atomic and molecular orbitals**.

Using the MO diagram above, answer the following questions:

- (i) Which one of the following will have the highest bond order? Circle the correct choice
 A) PS^{1-} B) PS C) PS^{1+}
- (ii) Which one of the following will have the longest bond length? Circle the correct choice
 A) PS^{1-} B) PS C) PS^{1+}
- (iii) Which of the following will be diamagnetic? Circle the correct choice.
 A) PS^{1-} B) PS C) PS^{1+}
- (iv) Draw Lewis dot structure of the molecule with the shortest bond length (based on the MO diagram) from among the three (PS^{1-} , PS, PS^{1+}), and indicate the bond order calculated from the Lewis dot structure. **(3 points)**

Lewis Dot Structure

Bond Order



CC1A105D-D829-4DFB-8F66-440E2B934EF5

midterm-exam-93388

#731 Page 10 of 10

SCRAP PAGE

**CHEMISTRY 110 –Midterm Exam****Thursday November 7, 2024, 6:30 – 8:30 pm****Examiners: Pallavi Sirjoosingh and Ashok Kakkar**

First name (please write as legibly as possible within the boxes)

--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--

Last name

--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--

Student ID number

--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--

This booklet has 8 questions on 10 pages (The last page can be used as a scrap page).

The staple must NOT be removed from the question booklet. Your intact question booklet must be returned at the end of the exam.

A separate booklet contains a periodic table, and datasheet.

Write your answers directly under the questions in the space provided. Do not write on the QR codes or near the top edges (as it may be cut off).

Please write **legibly and dark enough** for the scan to be readable. For students who use English as a second language, dictionaries are permitted. Calculators are allowed.

DO NOT ASK QUESTIONS OF THE INVIGILATORS about interpretations of the exam questions. If you have a concern about a question, please summarize it at the top of this page. All concerns will be evaluated after the exam. GOOD LUCK!

**Question 1 8 points** (*Assume all transitions are possible*)

a. 6 points Calculate the **energy (in kJ/mol)** required to ionize (completely remove) the electron from the **n=3** level of **He⁺ ion** (He: Helium). Show your work including equations. (3 sig. figs.)

b. 2 points

(I) Select the largest (longest) wavelength from the three choices below:

- i. Wavelength of photon to *completely remove* an electron from *1s orbital of Hydrogen atom*
- ii. Wavelength of photon to *completely remove* an electron from *1s orbital of Boron atom (B)*
- iii. Wavelength of photon to *completely remove* an electron from *1s orbital of Boron cation (B⁺)*

i or ii or iii

(II) Select the smallest (shortest) wavelength from the three choices below (*for H atom*):

- i. Wavelength of photon *emitted* for an electron transitioning from *2p to 1s orbital*
- ii. Wavelength of photon *absorbed* for an electron transitioning from *2s to 4p orbital*
- iii. Wavelength of photon *absorbed* for an electron transitioning from *1s to 3p orbital*

i or ii or iii



Question 2 8 points



A Chem110 student plots the graph shown on the left, from data obtained for the kinetic energy (eV) of the ejected electron versus frequency (s^{-1}) of the incident light, for an unknown metal. ($1\text{eV} = 1.6022 \times 10^{-19} \text{ J}$)

Based on the plot, answer the questions **a to c** below:

a. 1 point On the plot above, label the **threshold frequency**.

b. 2 points What is the value of the work potential (in J) of the unknown metal?

c. 3 points What is the *longest* wavelength (in nm) required to eject an electron from this metal's surface? Show your work including equation(s) used. (3 sig figs)

d. 2 points In the space provided below, plot the following

(Assume frequency is *above* threshold frequency):

- i. *Kinetic energy* of an ejected electron versus the *intensity* of the incident light (left)
- ii. *Number of electrons* ejected versus the *frequency* of the incident light (right)



**Question 3** **6 points****a. 2 points**

In the ground state of Ge (Germanium; $Z = 32$) atom, **name** the following orbitals:

i. Orbital containing the electron that requires the maximum energy to be removed.

ii. Orbital containing the electron that experiences the minimum effective nuclear charge.

b. 2 points

An unknown anion (A^-) has the following *excited* electronic configuration: $[Ar] 4s^2 3d^3 4p^1$

(Ar: Argon; $Z = 18$)

Give the **condensed ground state electronic configuration** of the atom, **A** and the cation, **A^{2+}** .
Show your work. (Assume all electronic transitions are possible)

Electronic Configuration of A:

Electronic Configuration of A^{2+} :

c. 2 points

In a single atom, determine how many electrons can have the following set of quantum numbers?

(i) $n = 5; l = 3$

In a single atom, determine how many electrons can have the following set of quantum numbers?

(i) $n = 3; l = 2; m_s = +1/2$

**Question 4** **8 points**

Rank the following from 1 to 4 in **increasing** order. Provide the ranking in the boxes provided.

Note: Atomic number (Z) are provided for relevant atoms

a) Increasing (lowest to highest) order of *ionization energy*: (1= lowest; 4 = highest)

Mg

Mg: Z = 12

Rb

Rb: Z = 37

Cs

Cs: Z = 55

K

K : Z = 19

b) Increasing (lowest to highest) order of *bond angle*: (1= lowest; 4 = highest)

H₂OCCl₄XeF₄

HCN

c) Increasing (smallest to largest) order of *ionic size* (1= smallest; 4 = largest)

S²⁻

S: Z = 16

O²⁻

O: Z = 8

Be²⁺

Be: Z = 4

Na⁺

Na: Z = 11

d) Increasing (lowest to highest) order of *bond energy* (1= lowest; 4 = highest)

P-F

P-Cl

P-I

P-Br

**Question 5** **12 points**

a. 9 points For the molecules HClO_4 and TeF_6 , draw the most stable *Lewis structure* showing appropriate shape and *dashed/wedged bonds*, based on VSEPR. Determine the **molecular geometry** around the central atom. For HClO_4 , label the *most polar bond*.

HClO_4 (there are no O-O bonds)

Label the most polar bond

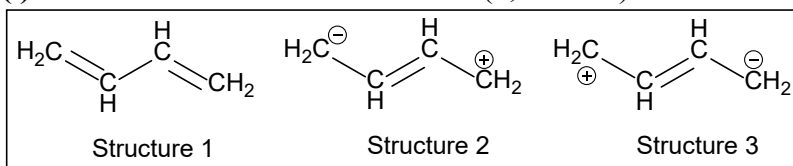
Molecular Geometry around Cl

TeF_6

Molecular Geometry around Te

b. 3 points (Note: Lone pairs are not shown in structures below)

(i) For the three *resonance* structures (1, 2 and 3) shown below, answer the following question:



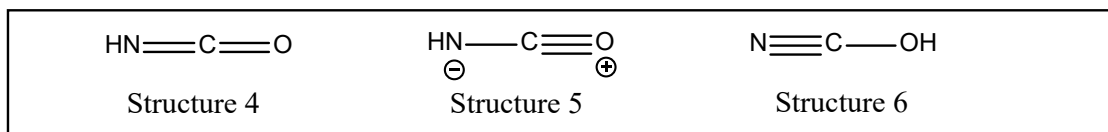
(I) Circle the *most contributing* resonance structure.

Structure 1

Structure 2

Structure 3

(ii) For the three structures given (4, 5, and 6) below, answer the following questions:



(I) Circle all the structures that represent *resonance structures* of each other:

Structure 4

Structure 5

Structure 6

(II) Among the resonance structures, circle the *most contributing* resonance structure.

Structure 4

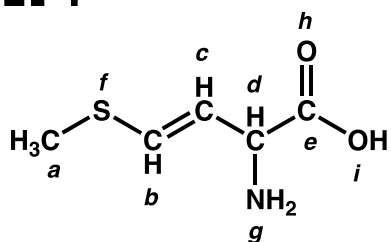
Structure 5

Structure 6

**Question 6** **6 points**

Using the data given below, calculate the **lattice energy** for SrO(s).

	ΔH° (kJ/mol)
$\text{Sr}(s) \longrightarrow \text{Sr}(g)$	+180
$\text{Sr}(g) \longrightarrow \text{Sr}^{2+}(g) + 2e^-$	+1468
$\text{O}_2(g) \longrightarrow 2\text{O}(g)$	+500
$\text{O}(g) + 2e^- \longrightarrow \text{O}^{2-}(g)$	+600
$\text{Sr}(s) + \frac{1}{2}\text{O}_2(g) \longrightarrow \text{SrO}(s)$	-550

**Question 7 10 points**

Consider the molecule given on the left, and answer questions (i) to (vi) using chemical bonding theories that we have learnt in class (valence bond theory, hybridization).

Note that the lone pairs are not shown in the structure:

i. Consider the carbon atoms marked in as ***a***, ***b***, ***c***, ***d***, and ***e*** in the molecule, and determine **hybridization** at each of the carbon centers. (2.5 points)

Atom	Hybridization
Carbon (<i>a</i>)	
Carbon (<i>b</i>)	
Carbon (<i>c</i>)	
Carbon (<i>d</i>)	
Carbon (<i>e</i>)	

ii. Now, consider the sulfur atom (S) marked as ***f***, nitrogen (N) atom (***g***), and oxygen(O) atoms (***h*** and ***i***), and determine their **hybridization**. (2 points)

Atom	Hybridization
Sulfur (<i>f</i>)	
Nitrogen (<i>g</i>)	
Oxygen (<i>h</i>)	
Oxygen (<i>i</i>)	

iii. **Describe** all the bonds formed by the carbon atoms ***a*** and ***b***, using sigma and pi designation as appropriate. (2 points)

Atom	Bonds formed (use sigma and pi as appropriate)
Carbon (<i>a</i>)	
Carbon (<i>b</i>)	

iv. How many total sp^3 -s sigma bonds are in the molecule? (1 point)

v. Determine **electronic** and **molecular geometry** at carbon (C) labeled ***e*** and sulfur (S) labeled ***f*** centres. (2 points)

Atom	Electronic Geometry	Molecular Geometry
Carbon (<i>e</i>)		
Sulfur (<i>f</i>)		

vi. Circle the value from the choices given below, that describes the **C-S-C bond angle at sulfur** (S) marked ***f***: (0.5 points)

>90° to <109.5° 109.5° >109.5 to <120° 120° >120° to <180°


Question 8 12 points

Using the **valence shell orbitals only** draw a **complete** molecular orbital (MO) diagram (with no s-p mixing) for the molecule **PS**, in the space provided below. **(6 points)**

Assume that 3s and 3p atomic orbitals will form molecular orbitals like those formed by the combination of 2s and 2p atomic orbitals that we learnt in class, and there is no s-p mixing.

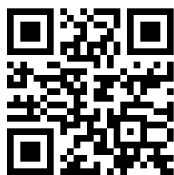
Show **energy arrow**, **atomic orbitals** involved, **atom labels**, the **molecular orbitals** that are formed, **filled electrons** in the MO diagram etc. Make sure to **label all the atomic and molecular orbitals**.

Using the MO diagram above, answer the following questions:

- (i) Which one of the following will have the highest bond order? Circle the correct choice
 A) PS^{1-} B) PS C) PS^{1+}
- (ii) Which one of the following will have the longest bond length? Circle the correct choice
 A) PS^{1-} B) PS C) PS^{1+}
- (iii) Which of the following will be diamagnetic? Circle the correct choice.
 A) PS^{1-} B) PS C) PS^{1+}
- (iv) Draw Lewis dot structure of the molecule with the shortest bond length (based on the MO diagram) from among the three (PS^{1-} , PS, PS^{1+}), and indicate the bond order calculated from the Lewis dot structure. **(3 points)**

Lewis Dot Structure

Bond Order



AA6AFD1B-F192-42FF-AE50-DC4E68A334A8

midterm-exam-93388

#732 Page 10 of 10

SCRAP PAGE

**CHEMISTRY 110 –Midterm Exam****Thursday November 7, 2024, 6:30 – 8:30 pm****Examiners: Pallavi Sirjoosingh and Ashok Kakkar**

First name (please write as legibly as possible within the boxes)

--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--

Last name

--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--

Student ID number

--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--

This booklet has 8 questions on 10 pages (The last page can be used as a scrap page).

The staple must NOT be removed from the question booklet. Your intact question booklet must be returned at the end of the exam.

A separate booklet contains a periodic table, and datasheet.

Write your answers directly under the questions in the space provided. Do not write on the QR codes or near the top edges (as it may be cut off).

Please write **legibly and dark enough** for the scan to be readable. For students who use English as a second language, dictionaries are permitted. Calculators are allowed.

DO NOT ASK QUESTIONS OF THE INVIGILATORS about interpretations of the exam questions. If you have a concern about a question, please summarize it at the top of this page. All concerns will be evaluated after the exam. GOOD LUCK!

**Question 1 8 points** (*Assume all transitions are possible*)

a. 6 points Calculate the **energy (in kJ/mol)** required to ionize (completely remove) the electron from the **n=3** level of **He⁺ ion** (He: Helium). Show your work including equations. (3 sig. figs.)

b. 2 points

(I) Select the largest (longest) wavelength from the three choices below:

- i. Wavelength of photon to *completely remove* an electron from *1s orbital of Hydrogen atom*
- ii. Wavelength of photon to *completely remove* an electron from *1s orbital of Boron atom (B)*
- iii. Wavelength of photon to *completely remove* an electron from *1s orbital of Boron cation (B⁺)*

i or ii or iii

(II) Select the smallest (shortest) wavelength from the three choices below (*for H atom*):

- i. Wavelength of photon *emitted* for an electron transitioning from *2p to 1s orbital*
- ii. Wavelength of photon *absorbed* for an electron transitioning from *2s to 4p orbital*
- iii. Wavelength of photon *absorbed* for an electron transitioning from *1s to 3p orbital*

i or ii or iii



Question 2 8 points



A Chem110 student plots the graph shown on the left, from data obtained for the kinetic energy (eV) of the ejected electron versus frequency (s^{-1}) of the incident light, for an unknown metal. ($1\text{eV} = 1.6022 \times 10^{-19} \text{ J}$)

Based on the plot, answer the questions **a to c** below:

a. 1 point On the plot above, label the **threshold frequency**.

b. 2 points What is the value of the work potential (in J) of the unknown metal?

c. 3 points What is the *longest* wavelength (in nm) required to eject an electron from this metal's surface? Show your work including equation(s) used. (3 sig figs)

d. 2 points In the space provided below, plot the following

(Assume frequency is *above* threshold frequency):

- i. *Kinetic energy* of an ejected electron versus the *intensity* of the incident light (left)
- ii. *Number of electrons* ejected versus the *frequency* of the incident light (right)



**Question 3** **6 points****a. 2 points**

In the ground state of Ge (Germanium; $Z = 32$) atom, **name** the following orbitals:

i. Orbital containing the electron that requires the maximum energy to be removed.

ii. Orbital containing the electron that experiences the minimum effective nuclear charge.

b. 2 points

An unknown anion (A^-) has the following *excited* electronic configuration: $[Ar] 4s^2 3d^3 4p^1$

(Ar: Argon; $Z = 18$)

Give the **condensed ground state electronic configuration** of the atom, **A** and the cation, **A^{2+}** .
Show your work. (Assume all electronic transitions are possible)

Electronic Configuration of A:

Electronic Configuration of A^{2+} :

c. 2 points

In a single atom, determine how many electrons can have the following set of quantum numbers?

(i) $n = 5; l = 3$

In a single atom, determine how many electrons can have the following set of quantum numbers?

(i) $n = 3; l = 2; m_s = +1/2$

**Question 4** **8 points**

Rank the following from 1 to 4 in **increasing** order. Provide the ranking in the boxes provided.

Note: Atomic number (Z) are provided for relevant atoms

a) Increasing (lowest to highest) order of *ionization energy*: (1= lowest; 4 = highest)

Mg

Mg: Z = 12

Rb

Rb: Z = 37

Cs

Cs: Z = 55

K

K : Z = 19

b) Increasing (lowest to highest) order of *bond angle*: (1= lowest; 4 = highest)

H₂OCCl₄XeF₄

HCN

c) Increasing (smallest to largest) order of *ionic size* (1= smallest; 4 = largest)

S²⁻

S: Z = 16

O²⁻

O: Z = 8

Be²⁺

Be: Z = 4

Na⁺

Na: Z = 11

d) Increasing (lowest to highest) order of *bond energy* (1= lowest; 4 = highest)

P-F

P-Cl

P-I

P-Br

**Question 5** **12 points**

a. 9 points For the molecules HClO_4 and TeF_6 , draw the most stable *Lewis structure* showing appropriate shape and *dashed/wedged bonds*, based on VSEPR. Determine the **molecular geometry** around the central atom. For HClO_4 , label the *most polar bond*.



b. 3 points (Note: Lone pairs are not shown in structures below)

(i) For the three *resonance* structures (1, 2 and 3) shown below, answer the following question:



(I) Circle the *most contributing* resonance structure.

Structure 1

Structure 2

Structure 3

(ii) For the three structures given (4, 5, and 6) below, answer the following questions:



(I) Circle all the structures that represent *resonance structures* of each other:

Structure 4

Structure 5

Structure 6

(II) Among the resonance structures, circle the *most contributing* resonance structure.

Structure 4

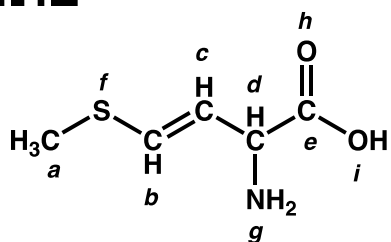
Structure 5

Structure 6

**Question 6** **6 points**

Using the data given below, calculate the **lattice energy** for SrO(s).

	ΔH° (kJ/mol)
$\text{Sr}(s) \longrightarrow \text{Sr}(g)$	+180
$\text{Sr}(g) \longrightarrow \text{Sr}^{2+}(g) + 2e^-$	+1468
$\text{O}_2(g) \longrightarrow 2\text{O}(g)$	+500
$\text{O}(g) + 2e^- \longrightarrow \text{O}^{2-}(g)$	+600
$\text{Sr}(s) + \frac{1}{2}\text{O}_2(g) \longrightarrow \text{SrO}(s)$	-550

**Question 7 10 points**

Consider the molecule given on the left, and answer questions (i) to (vi) using chemical bonding theories that we have learnt in class (valence bond theory, hybridization).

Note that the lone pairs are not shown in the structure:

i. Consider the carbon atoms marked in as ***a***, ***b***, ***c***, ***d***, and ***e*** in the molecule, and determine **hybridization** at each of the carbon centers. (2.5 points)

Atom	Hybridization
Carbon (<i>a</i>)	
Carbon (<i>b</i>)	
Carbon (<i>c</i>)	
Carbon (<i>d</i>)	
Carbon (<i>e</i>)	

ii. Now, consider the sulfur atom (S) marked as ***f***, nitrogen (N) atom (***g***), and oxygen(O) atoms (***h*** and ***i***), and determine their **hybridization**. (2 points)

Atom	Hybridization
Sulfur (<i>f</i>)	
Nitrogen (<i>g</i>)	
Oxygen (<i>h</i>)	
Oxygen (<i>i</i>)	

iii. **Describe** all the bonds formed by the carbon atoms ***a*** and ***b***, using sigma and pi designation as appropriate. (2 points)

Atom	Bonds formed (use sigma and pi as appropriate)
Carbon (<i>a</i>)	
Carbon (<i>b</i>)	

iv. How many total sp^3 -s sigma bonds are in the molecule? (1 point)

v. Determine **electronic** and **molecular geometry** at carbon (C) labeled ***e*** and sulfur (S) labeled ***f*** centres. (2 points)

Atom	Electronic Geometry	Molecular Geometry
Carbon (<i>e</i>)		
Sulfur (<i>f</i>)		

vi. Circle the value from the choices given below, that describes the **C-S-C bond angle at sulfur** (S) marked ***f***: (0.5 points)

>90° to <109.5° 109.5° >109.5 to <120° 120° >120° to <180°

**Question 8 12 points**

Using the **valence shell orbitals only** draw a **complete** molecular orbital (MO) diagram (with no s-p mixing) for the molecule **PS**, in the space provided below. **(6 points)**

Assume that 3s and 3p atomic orbitals will form molecular orbitals like those formed by the combination of 2s and 2p atomic orbitals that we learnt in class, and there is no s-p mixing.

Show **energy arrow**, **atomic orbitals** involved, **atom labels**, the **molecular orbitals** that are formed, **filled electrons** in the MO diagram etc. Make sure to label all the atomic and molecular orbitals.

Using the MO diagram above, answer the following questions:

- (i) Which one of the following will have the highest bond order? Circle the correct choice
A) PS^{1-} B) PS C) PS^{1+}
- (ii) Which one of the following will have the longest bond length? Circle the correct choice
A) PS^{1-} B) PS C) PS^{1+}
- (iii) Which of the following will be diamagnetic? Circle the correct choice.
A) PS^{1-} B) PS C) PS^{1+}
- (iv) Draw Lewis dot structure of the molecule with the shortest bond length (based on the MO diagram) from among the three (PS^{1-} , PS, PS^{1+}), and indicate the bond order calculated from the Lewis dot structure. **(3 points)**

Lewis Dot Structure

Bond Order



EEB3549F-5263-44DC-B7B2-32D4C615C007

midterm-exam-93388

#733 Page 10 of 10

SCRAP PAGE

**CHEMISTRY 110 –Midterm Exam****Thursday November 7, 2024, 6:30 – 8:30 pm****Examiners: Pallavi Sirjoosingh and Ashok Kakkar**

First name (please write as legibly as possible within the boxes)

--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--

Last name

--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--

Student ID number

--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--

This booklet has 8 questions on 10 pages (The last page can be used as a scrap page).

The staple must NOT be removed from the question booklet. Your intact question booklet must be returned at the end of the exam.

A separate booklet contains a periodic table, and datasheet.

Write your answers directly under the questions in the space provided. Do not write on the QR codes or near the top edges (as it may be cut off).

Please write **legibly and dark enough** for the scan to be readable. For students who use English as a second language, dictionaries are permitted. Calculators are allowed.

DO NOT ASK QUESTIONS OF THE INVIGILATORS about interpretations of the exam questions. If you have a concern about a question, please summarize it at the top of this page. All concerns will be evaluated after the exam. GOOD LUCK!

**Question 1 8 points** (*Assume all transitions are possible*)

a. 6 points Calculate the **energy (in kJ/mol)** required to ionize (completely remove) the electron from the **n=3** level of **He⁺ ion** (He: Helium). Show your work including equations. (3 sig. figs.)

b. 2 points

(I) Select the largest (longest) wavelength from the three choices below:

- i. Wavelength of photon to *completely remove* an electron from *1s orbital of Hydrogen atom*
- ii. Wavelength of photon to *completely remove* an electron from *1s orbital of Boron atom (B)*
- iii. Wavelength of photon to *completely remove* an electron from *1s orbital of Boron cation (B⁺)*

i or ii or iii

(II) Select the smallest (shortest) wavelength from the three choices below (*for H atom*):

- i. Wavelength of photon *emitted* for an electron transitioning from *2p to 1s orbital*
- ii. Wavelength of photon *absorbed* for an electron transitioning from *2s to 4p orbital*
- iii. Wavelength of photon *absorbed* for an electron transitioning from *1s to 3p orbital*

i or ii or iii



Question 2 8 points



A Chem110 student plots the graph shown on the left, from data obtained for the kinetic energy (eV) of the ejected electron versus frequency (s^{-1}) of the incident light, for an unknown metal. ($1\text{eV} = 1.6022 \times 10^{-19} \text{ J}$)

Based on the plot, answer the questions **a to c** below:

a. 1 point On the plot above, label the **threshold frequency**.

b. 2 points What is the value of the work potential (in J) of the unknown metal?

c. 3 points What is the *longest* wavelength (in nm) required to eject an electron from this metal's surface? Show your work including equation(s) used. (3 sig figs)

d. 2 points In the space provided below, plot the following

(Assume frequency is *above* threshold frequency):

- i. *Kinetic energy* of an ejected electron versus the *intensity* of the incident light (left)
- ii. *Number of electrons* ejected versus the *frequency* of the incident light (right)



**Question 3** **6 points****a. 2 points**

In the ground state of Ge (Germanium; $Z = 32$) atom, **name** the following orbitals:

i. Orbital containing the electron that requires the maximum energy to be removed.

ii. Orbital containing the electron that experiences the minimum effective nuclear charge.

b. 2 points

An unknown anion (A^-) has the following *excited* electronic configuration: $[Ar] 4s^2 3d^3 4p^1$

(Ar: Argon; $Z = 18$)

Give the **condensed ground state electronic configuration** of the atom, **A** and the cation, **A^{2+}** .
Show your work. (Assume all electronic transitions are possible)

Electronic Configuration of A:

Electronic Configuration of A^{2+} :

c. 2 points

In a single atom, determine how many electrons can have the following set of quantum numbers?

(i) $n = 5; l = 3$

In a single atom, determine how many electrons can have the following set of quantum numbers?

(i) $n = 3; l = 2; m_s = +1/2$

**Question 4** **8 points**

Rank the following from 1 to 4 in **increasing** order. Provide the ranking in the boxes provided.

Note: Atomic number (Z) are provided for relevant atoms

a) Increasing (lowest to highest) order of *ionization energy*: (1= lowest; 4 = highest)

Mg

Mg: Z = 12

Rb

Rb: Z = 37

Cs

Cs: Z = 55

K

K : Z = 19

b) Increasing (lowest to highest) order of *bond angle*: (1= lowest; 4 = highest)

H₂OCCl₄XeF₄

HCN

c) Increasing (smallest to largest) order of *ionic size* (1= smallest; 4 = largest)

S²⁻

S: Z = 16

O²⁻

O: Z = 8

Be²⁺

Be: Z = 4

Na⁺

Na: Z = 11

d) Increasing (lowest to highest) order of *bond energy* (1= lowest; 4 = highest)

P-F

P-Cl

P-I

P-Br

**Question 5** **12 points**

a. 9 points For the molecules HClO_4 and TeF_6 , draw the most stable *Lewis structure* showing appropriate shape and *dashed/wedged bonds*, based on VSEPR. Determine the **molecular geometry** around the central atom. For HClO_4 , label the *most polar bond*.



b. 3 points (Note: Lone pairs are not shown in structures below)

(i) For the three *resonance* structures (1, 2 and 3) shown below, answer the following question:



(I) Circle the *most contributing* resonance structure.

Structure 1

Structure 2

Structure 3

(ii) For the three structures given (4, 5, and 6) below, answer the following questions:



(I) Circle all the structures that represent *resonance structures* of each other:

Structure 4

Structure 5

Structure 6

(II) Among the resonance structures, circle the *most contributing* resonance structure.

Structure 4

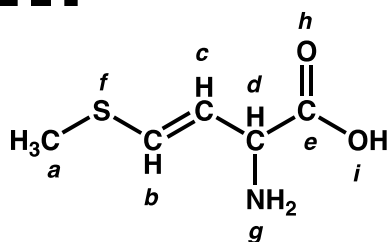
Structure 5

Structure 6

**Question 6** **6 points**

Using the data given below, calculate the **lattice energy** for SrO(s).

	ΔH° (kJ/mol)
$\text{Sr}(s) \longrightarrow \text{Sr}(g)$	+180
$\text{Sr}(g) \longrightarrow \text{Sr}^{2+}(g) + 2e^-$	+1468
$\text{O}_2(g) \longrightarrow 2\text{O}(g)$	+500
$\text{O}(g) + 2e^- \longrightarrow \text{O}^{2-}(g)$	+600
$\text{Sr}(s) + \frac{1}{2}\text{O}_2(g) \longrightarrow \text{SrO}(s)$	-550

**Question 7 10 points**

Consider the molecule given on the left, and answer questions (i) to (vi) using chemical bonding theories that we have learnt in class (valence bond theory, hybridization).

Note that the lone pairs are not shown in the structure:

i. Consider the carbon atoms marked in as ***a***, ***b***, ***c***, ***d***, and ***e*** in the molecule, and determine **hybridization** at each of the carbon centers. (2.5 points)

Atom	Hybridization
Carbon (<i>a</i>)	
Carbon (<i>b</i>)	
Carbon (<i>c</i>)	
Carbon (<i>d</i>)	
Carbon (<i>e</i>)	

ii. Now, consider the sulfur atom (S) marked as ***f***, nitrogen (N) atom (***g***), and oxygen(O) atoms (***h*** and ***i***), and determine their **hybridization**. (2 points)

Atom	Hybridization
Sulfur (<i>f</i>)	
Nitrogen (<i>g</i>)	
Oxygen (<i>h</i>)	
Oxygen (<i>i</i>)	

iii. **Describe** all the bonds formed by the carbon atoms ***a*** and ***b***, using sigma and pi designation as appropriate. (2 points)

Atom	Bonds formed (use sigma and pi as appropriate)
Carbon (<i>a</i>)	
Carbon (<i>b</i>)	

iv. How many total sp^3 -s sigma bonds are in the molecule? (1 point)

v. Determine **electronic** and **molecular geometry** at carbon (C) labeled ***e*** and sulfur (S) labeled ***f*** centres. (2 points)

Atom	Electronic Geometry	Molecular Geometry
Carbon (<i>e</i>)		
Sulfur (<i>f</i>)		

vi. Circle the value from the choices given below, that describes the **C-S-C bond angle at sulfur** (S) marked ***f***: (0.5 points)

>90° to <109.5° 109.5° >109.5 to <120° 120° >120° to <180°


Question 8 12 points

Using the **valence shell orbitals only** draw a **complete** molecular orbital (MO) diagram (with no s-p mixing) for the molecule **PS**, in the space provided below. **(6 points)**

Assume that 3s and 3p atomic orbitals will form molecular orbitals like those formed by the combination of 2s and 2p atomic orbitals that we learnt in class, and there is no s-p mixing.

Show **energy arrow**, **atomic orbitals** involved, **atom labels**, the **molecular orbitals** that are formed, **filled electrons** in the MO diagram etc. Make sure to label all the atomic and molecular orbitals.

Using the MO diagram above, answer the following questions:

- (i) Which one of the following will have the highest bond order? Circle the correct choice
 A) PS^{1-} B) PS C) PS^{1+}
- (ii) Which one of the following will have the longest bond length? Circle the correct choice
 A) PS^{1-} B) PS C) PS^{1+}
- (iii) Which of the following will be diamagnetic? Circle the correct choice.
 A) PS^{1-} B) PS C) PS^{1+}
- (iv) Draw Lewis dot structure of the molecule with the shortest bond length (based on the MO diagram) from among the three (PS^{1-} , PS, PS^{1+}), and indicate the bond order calculated from the Lewis dot structure. **(3 points)**

Lewis Dot Structure

Bond Order



866A3420-F312-4E60-8BD2-DE8B2DD49C8C

midterm-exam-93388

#734 Page 10 of 10

SCRAP PAGE

**CHEMISTRY 110 –Midterm Exam****Thursday November 7, 2024, 6:30 – 8:30 pm**

Examiners: Pallavi Sirjoosingh and Ashok Kakkar

First name (please write as legibly as possible within the boxes)

--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--

Last name

--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--

Student ID number

--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--

This booklet has 8 questions on 10 pages (The last page can be used as a scrap page).

The staple must NOT be removed from the question booklet. Your intact question booklet must be returned at the end of the exam.

A separate booklet contains a periodic table, and datasheet.

Write your answers directly under the questions in the space provided. Do not write on the QR codes or near the top edges (as it may be cut off).

Please write **legibly and dark enough** for the scan to be readable. For students who use English as a second language, dictionaries are permitted. Calculators are allowed.

DO NOT ASK QUESTIONS OF THE INVIGILATORS about interpretations of the exam questions. If you have a concern about a question, please summarize it at the top of this page. All concerns will be evaluated after the exam. GOOD LUCK!

**Question 1 8 points** (*Assume all transitions are possible*)

a. 6 points Calculate the **energy (in kJ/mol)** required to ionize (completely remove) the electron from the **n=3** level of **He⁺ ion** (He: Helium). Show your work including equations. (3 sig. figs.)

b. 2 points

(I) Select the largest (longest) wavelength from the three choices below:

- i. Wavelength of photon to *completely remove* an electron from *1s orbital of Hydrogen atom*
- ii. Wavelength of photon to *completely remove* an electron from *1s orbital of Boron atom (B)*
- iii. Wavelength of photon to *completely remove* an electron from *1s orbital of Boron cation (B⁺)*

i or ii or iii

(II) Select the smallest (shortest) wavelength from the three choices below (*for H atom*):

- i. Wavelength of photon *emitted* for an electron transitioning from *2p to 1s orbital*
- ii. Wavelength of photon *absorbed* for an electron transitioning from *2s to 4p orbital*
- iii. Wavelength of photon *absorbed* for an electron transitioning from *1s to 3p orbital*

i or ii or iii



Question 2 8 points



A Chem110 student plots the graph shown on the left, from data obtained for the kinetic energy (eV) of the ejected electron versus frequency (s^{-1}) of the incident light, for an unknown metal. ($1\text{eV} = 1.6022 \times 10^{-19} \text{ J}$)

Based on the plot, answer the questions **a to c** below:

a. 1 point On the plot above, label the **threshold frequency**.

b. 2 points What is the value of the work potential (in J) of the unknown metal?

c. 3 points What is the *longest* wavelength (in nm) required to eject an electron from this metal's surface? Show your work including equation(s) used. (3 sig figs)

d. 2 points In the space provided below, plot the following

(Assume frequency is *above* threshold frequency):

- Kinetic energy* of an ejected electron versus the *intensity* of the incident light (left)
- Number of electrons* ejected versus the *frequency* of the incident light (right)



**Question 3** **6 points****a. 2 points**

In the ground state of Ge (Germanium; $Z = 32$) atom, **name** the following orbitals:

i. Orbital containing the electron that requires the maximum energy to be removed.

ii. Orbital containing the electron that experiences the minimum effective nuclear charge.

b. 2 points

An unknown anion (A^-) has the following *excited* electronic configuration: $[Ar] 4s^2 3d^3 4p^1$

(Ar: Argon; $Z = 18$)

Give the **condensed ground state electronic configuration** of the atom, **A** and the cation, **A^{2+}** .
Show your work. (Assume all electronic transitions are possible)

Electronic Configuration of A:

Electronic Configuration of A^{2+} :

c. 2 points

In a single atom, determine how many electrons can have the following set of quantum numbers?

(i) $n = 5; l = 3$

In a single atom, determine how many electrons can have the following set of quantum numbers?

(i) $n = 3; l = 2; m_s = +1/2$

**Question 4** **8 points**

Rank the following from 1 to 4 in **increasing** order. Provide the ranking in the boxes provided.

Note: Atomic number (Z) are provided for relevant atoms

a) Increasing (lowest to highest) order of *ionization energy*: (1= lowest; 4 = highest)

Mg

Mg: Z = 12

Rb

Rb: Z = 37

Cs

Cs: Z = 55

K

K : Z = 19

b) Increasing (lowest to highest) order of *bond angle*: (1= lowest; 4 = highest)

H₂OCCl₄XeF₄

HCN

c) Increasing (smallest to largest) order of *ionic size* (1= smallest; 4 = largest)

S²⁻

S: Z = 16

O²⁻

O: Z = 8

Be²⁺

Be: Z = 4

Na⁺

Na: Z = 11

d) Increasing (lowest to highest) order of *bond energy* (1= lowest; 4 = highest)

P-F

P-Cl

P-I

P-Br

**Question 5** **12 points**

a. 9 points For the molecules HClO_4 and TeF_6 , draw the most stable *Lewis structure* showing appropriate shape and *dashed/wedged bonds*, based on VSEPR. Determine the **molecular geometry** around the central atom. For HClO_4 , label the *most polar bond*.



b. 3 points (Note: Lone pairs are not shown in structures below)

(i) For the three *resonance* structures (1, 2 and 3) shown below, answer the following question:



(I) Circle the *most contributing* resonance structure.

Structure 1

Structure 2

Structure 3

(ii) For the three structures given (4, 5, and 6) below, answer the following questions:



(I) Circle all the structures that represent *resonance structures* of each other:

Structure 4

Structure 5

Structure 6

(II) Among the resonance structures, circle the *most contributing* resonance structure.

Structure 4

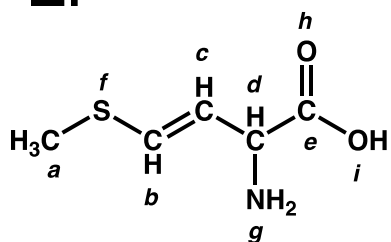
Structure 5

Structure 6

**Question 6** **6 points**

Using the data given below, calculate the **lattice energy** for SrO(s).

	ΔH° (kJ/mol)
$\text{Sr}(s) \longrightarrow \text{Sr}(g)$	+180
$\text{Sr}(g) \longrightarrow \text{Sr}^{2+}(g) + 2e^-$	+1468
$\text{O}_2(g) \longrightarrow 2\text{O}(g)$	+500
$\text{O}(g) + 2e^- \longrightarrow \text{O}^{2-}(g)$	+600
$\text{Sr}(s) + \frac{1}{2}\text{O}_2(g) \longrightarrow \text{SrO}(s)$	-550

**Question 7 10 points**

Consider the molecule given on the left, and answer questions (i) to (vi) using chemical bonding theories that we have learnt in class (valence bond theory, hybridization).

Note that the lone pairs are not shown in the structure:

i. Consider the carbon atoms marked in as ***a***, ***b***, ***c***, ***d***, and ***e*** in the molecule, and determine **hybridization** at each of the carbon centers. (2.5 points)

Atom	Hybridization
Carbon (<i>a</i>)	
Carbon (<i>b</i>)	
Carbon (<i>c</i>)	
Carbon (<i>d</i>)	
Carbon (<i>e</i>)	

ii. Now, consider the sulfur atom (S) marked as ***f***, nitrogen (N) atom (***g***), and oxygen(O) atoms (***h*** and ***i***), and determine their **hybridization**. (2 points)

Atom	Hybridization
Sulfur (<i>f</i>)	
Nitrogen (<i>g</i>)	
Oxygen (<i>h</i>)	
Oxygen (<i>i</i>)	

iii. **Describe** all the bonds formed by the carbon atoms ***a*** and ***b***, using sigma and pi designation as appropriate. (2 points)

Atom	Bonds formed (use sigma and pi as appropriate)
Carbon (<i>a</i>)	
Carbon (<i>b</i>)	

iv. How many total sp^3 -s sigma bonds are in the molecule? (1 point)

v. Determine **electronic** and **molecular geometry** at carbon (C) labeled ***e*** and sulfur (S) labeled ***f*** centres. (2 points)

Atom	Electronic Geometry	Molecular Geometry
Carbon (<i>e</i>)		
Sulfur (<i>f</i>)		

vi. Circle the value from the choices given below, that describes the **C-S-C bond angle at sulfur** (S) marked ***f***: (0.5 points)

>90° to <109.5°

109.5°

>109.5 to <120°

120°

>120° to <180°

**Question 8 12 points**

Using the **valence shell orbitals only** draw a **complete** molecular orbital (MO) diagram (with no s-p mixing) for the molecule **PS**, in the space provided below. **(6 points)**

Assume that 3s and 3p atomic orbitals will form molecular orbitals like those formed by the combination of 2s and 2p atomic orbitals that we learnt in class, and there is no s-p mixing.

Show **energy arrow**, **atomic orbitals** involved, **atom labels**, the **molecular orbitals** that are formed, **filled electrons** in the MO diagram etc. Make sure to label all the atomic and molecular orbitals.

Using the MO diagram above, answer the following questions:

- (i) Which one of the following will have the highest bond order? Circle the correct choice
A) PS^{1-} B) PS C) PS^{1+}
- (ii) Which one of the following will have the longest bond length? Circle the correct choice
A) PS^{1-} B) PS C) PS^{1+}
- (iii) Which of the following will be diamagnetic? Circle the correct choice.
A) PS^{1-} B) PS C) PS^{1+}
- (iv) Draw Lewis dot structure of the molecule with the shortest bond length (based on the MO diagram) from among the three (PS^{1-} , PS, PS^{1+}), and indicate the bond order calculated from the Lewis dot structure. **(3 points)**

Lewis Dot Structure

Bond Order



35643C59-481F-4301-924A-D99D722E0118

midterm-exam-93388

#735 Page 10 of 10

SCRAP PAGE

**CHEMISTRY 110 –Midterm Exam****Thursday November 7, 2024, 6:30 – 8:30 pm****Examiners: Pallavi Sirjoosingh and Ashok Kakkar**

First name (please write as legibly as possible within the boxes)

--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--

Last name

--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--

Student ID number

--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--

This booklet has 8 questions on 10 pages (The last page can be used as a scrap page).

The staple must NOT be removed from the question booklet. Your intact question booklet must be returned at the end of the exam.

A separate booklet contains a periodic table, and datasheet.

Write your answers directly under the questions in the space provided. Do not write on the QR codes or near the top edges (as it may be cut off).

Please write **legibly and dark enough** for the scan to be readable. For students who use English as a second language, dictionaries are permitted. Calculators are allowed.

DO NOT ASK QUESTIONS OF THE INVIGILATORS about interpretations of the exam questions. If you have a concern about a question, please summarize it at the top of this page. All concerns will be evaluated after the exam. GOOD LUCK!

**Question 1 8 points** (*Assume all transitions are possible*)

a. 6 points Calculate the **energy (in kJ/mol)** required to ionize (completely remove) the electron from the **$n=3$** level of **He^+ ion** (He: Helium). Show your work including equations. (3 sig. figs.)

b. 2 points

(I) Select the largest (longest) wavelength from the three choices below:

- i. Wavelength of photon to *completely remove* an electron from *1s orbital of Hydrogen atom*
- ii. Wavelength of photon to *completely remove* an electron from *1s orbital of Boron atom (B)*
- iii. Wavelength of photon to *completely remove* an electron from *1s orbital of Boron cation (B^+)*

i or ii or iii

(II) Select the smallest (shortest) wavelength from the three choices below (*for H atom*):

- i. Wavelength of photon *emitted* for an electron transitioning from *2p to 1s orbital*
- ii. Wavelength of photon *absorbed* for an electron transitioning from *2s to 4p orbital*
- iii. Wavelength of photon *absorbed* for an electron transitioning from *1s to 3p orbital*

i or ii or iii



Question 2 8 points



A Chem110 student plots the graph shown on the left, from data obtained for the kinetic energy (eV) of the ejected electron versus frequency (s^{-1}) of the incident light, for an unknown metal. ($1\text{eV} = 1.6022 \times 10^{-19} \text{ J}$)

Based on the plot, answer the questions **a to c** below:

a. 1 point On the plot above, label the **threshold frequency**.

b. 2 points What is the value of the work potential (in J) of the unknown metal?

c. 3 points What is the *longest* wavelength (in nm) required to eject an electron from this metal's surface? Show your work including equation(s) used. (3 sig figs)

d. 2 points In the space provided below, plot the following

(Assume frequency is *above* threshold frequency):

- i. *Kinetic energy* of an ejected electron versus the *intensity* of the incident light (left)
- ii. *Number of electrons* ejected versus the *frequency* of the incident light (right)



**Question 3** **6 points****a. 2 points**

In the ground state of Ge (Germanium; $Z = 32$) atom, **name** the following orbitals:

i. Orbital containing the electron that requires the maximum energy to be removed.

ii. Orbital containing the electron that experiences the minimum effective nuclear charge.

b. 2 points

An unknown anion (A^-) has the following *excited* electronic configuration: $[Ar] 4s^2 3d^3 4p^1$

(Ar: Argon; $Z = 18$)

Give the **condensed ground state electronic configuration** of the atom, **A** and the cation, **A^{2+}** .
Show your work. (Assume all electronic transitions are possible)

Electronic Configuration of A:

Electronic Configuration of A^{2+} :

c. 2 points

In a single atom, determine how many electrons can have the following set of quantum numbers?

(i) $n = 5; l = 3$

In a single atom, determine how many electrons can have the following set of quantum numbers?

(i) $n = 3; l = 2; m_s = +1/2$

**Question 4** **8 points**

Rank the following from 1 to 4 in **increasing** order. Provide the ranking in the boxes provided.

Note: Atomic number (Z) are provided for relevant atoms

a) Increasing (lowest to highest) order of *ionization energy*: (1= lowest; 4 = highest)

Mg

Mg: Z = 12

Rb

Rb: Z = 37

Cs

Cs: Z = 55

K

K : Z = 19

b) Increasing (lowest to highest) order of *bond angle*: (1= lowest; 4 = highest)

H₂OCCl₄XeF₄

HCN

c) Increasing (smallest to largest) order of *ionic size* (1= smallest; 4 = largest)

S²⁻

S: Z = 16

O²⁻

O: Z = 8

Be²⁺

Be: Z = 4

Na⁺

Na: Z = 11

d) Increasing (lowest to highest) order of *bond energy* (1= lowest; 4 = highest)

P-F

P-Cl

P-I

P-Br

**Question 5** **12 points**

a. 9 points For the molecules HClO_4 and TeF_6 , draw the most stable *Lewis structure* showing appropriate shape and *dashed/wedged bonds*, based on VSEPR. Determine the **molecular geometry** around the central atom. For HClO_4 , label the *most polar bond*.

HClO_4 (there are no O-O bonds)

Label the most polar bond

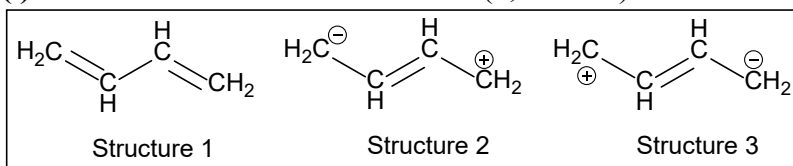
Molecular Geometry around Cl

TeF_6

Molecular Geometry around Te

b. 3 points (Note: Lone pairs are not shown in structures below)

(i) For the three *resonance* structures (1, 2 and 3) shown below, answer the following question:



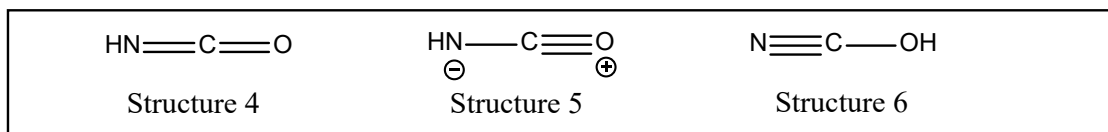
(I) Circle the *most contributing* resonance structure.

Structure 1

Structure 2

Structure 3

(ii) For the three structures given (4, 5, and 6) below, answer the following questions:



(I) Circle all the structures that represent *resonance structures* of each other:

Structure 4

Structure 5

Structure 6

(II) Among the resonance structures, circle the *most contributing* resonance structure.

Structure 4

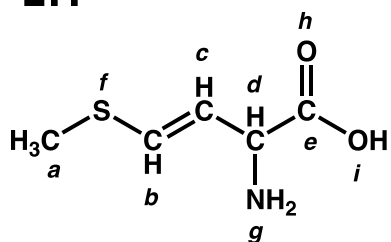
Structure 5

Structure 6

**Question 6** **6 points**

Using the data given below, calculate the **lattice energy** for SrO(s).

	ΔH° (kJ/mol)
$\text{Sr}(s) \longrightarrow \text{Sr}(g)$	+180
$\text{Sr}(g) \longrightarrow \text{Sr}^{2+}(g) + 2e^-$	+1468
$\text{O}_2(g) \longrightarrow 2\text{O}(g)$	+500
$\text{O}(g) + 2e^- \longrightarrow \text{O}^{2-}(g)$	+600
$\text{Sr}(s) + \frac{1}{2}\text{O}_2(g) \longrightarrow \text{SrO}(s)$	-550

**Question 7 10 points**

Consider the molecule given on the left, and answer questions (i) to (vi) using chemical bonding theories that we have learnt in class (valence bond theory, hybridization).

Note that the lone pairs are not shown in the structure:

i. Consider the carbon atoms marked in as ***a***, ***b***, ***c***, ***d***, and ***e*** in the molecule, and determine **hybridization** at each of the carbon centers. (2.5 points)

Atom	Hybridization
Carbon (<i>a</i>)	
Carbon (<i>b</i>)	
Carbon (<i>c</i>)	
Carbon (<i>d</i>)	
Carbon (<i>e</i>)	

ii. Now, consider the sulfur atom (S) marked as ***f***, nitrogen (N) atom (***g***), and oxygen(O) atoms (***h*** and ***i***), and determine their **hybridization**. (2 points)

Atom	Hybridization
Sulfur (<i>f</i>)	
Nitrogen (<i>g</i>)	
Oxygen (<i>h</i>)	
Oxygen (<i>i</i>)	

iii. **Describe** all the bonds formed by the carbon atoms ***a*** and ***b***, using sigma and pi designation as appropriate. (2 points)

Atom	Bonds formed (use sigma and pi as appropriate)
Carbon (<i>a</i>)	
Carbon (<i>b</i>)	

iv. How many total sp^3 -s sigma bonds are in the molecule? (1 point)

v. Determine **electronic** and **molecular geometry** at carbon (C) labeled ***e*** and sulfur (S) labeled ***f*** centres. (2 points)

Atom	Electronic Geometry	Molecular Geometry
Carbon (<i>e</i>)		
Sulfur (<i>f</i>)		

vi. Circle the value from the choices given below, that describes the **C-S-C bond angle at sulfur** (S) marked ***f***: (0.5 points)

$>90^\circ$ to $<109.5^\circ$ 109.5° >109.5 to $<120^\circ$ 120° $>120^\circ$ to $<180^\circ$



Question 8 12 points

Using the **valence shell orbitals only** draw a **complete** molecular orbital (MO) diagram (with no s-p mixing) for the molecule **PS**, in the space provided below. **(6 points)**

Assume that 3s and 3p atomic orbitals will form molecular orbitals like those formed by the combination of 2s and 2p atomic orbitals that we learnt in class, and there is no s-p mixing.

Show **energy arrow**, **atomic orbitals** involved, **atom labels**, the **molecular orbitals** that are formed, **filled electrons** in the MO diagram etc. Make sure to label all the atomic and molecular orbitals.

Using the MO diagram above, answer the following questions:

- (i) Which one of the following will have the highest bond order? Circle the correct choice
 A) PS^{1-} B) PS C) PS^{1+}
- (ii) Which one of the following will have the longest bond length? Circle the correct choice
 A) PS^{1-} B) PS C) PS^{1+}
- (iii) Which of the following will be diamagnetic? Circle the correct choice.
 A) PS^{1-} B) PS C) PS^{1+}
- (iv) Draw Lewis dot structure of the molecule with the shortest bond length (based on the MO diagram) from among the three (PS^{1-} , PS, PS^{1+}), and indicate the bond order calculated from the Lewis dot structure. **(3 points)**

Lewis Dot Structure

Bond Order



76A29C3C-133B-4EDD-B4E7-42F93DF14818

midterm-exam-93388

#736 Page 10 of 10

SCRAP PAGE

**CHEMISTRY 110 –Midterm Exam****Thursday November 7, 2024, 6:30 – 8:30 pm****Examiners: Pallavi Sirjoosingh and Ashok Kakkar**

First name (please write as legibly as possible within the boxes)

--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--

Last name

--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--

Student ID number

--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--

This booklet has 8 questions on 10 pages (The last page can be used as a scrap page).

The staple must NOT be removed from the question booklet. Your intact question booklet must be returned at the end of the exam.

A separate booklet contains a periodic table, and datasheet.

Write your answers directly under the questions in the space provided. Do not write on the QR codes or near the top edges (as it may be cut off).

Please write **legibly and dark enough** for the scan to be readable. For students who use English as a second language, dictionaries are permitted. Calculators are allowed.

DO NOT ASK QUESTIONS OF THE INVIGILATORS about interpretations of the exam questions. If you have a concern about a question, please summarize it at the top of this page. All concerns will be evaluated after the exam. GOOD LUCK!

**Question 1 8 points** (*Assume all transitions are possible*)

a. 6 points Calculate the **energy (in kJ/mol)** required to ionize (completely remove) the electron from the **n=3** level of **He⁺ ion** (He: Helium). Show your work including equations. (3 sig. figs.)

b. 2 points

(I) Select the largest (longest) wavelength from the three choices below:

- i. Wavelength of photon to *completely remove* an electron from *1s orbital of Hydrogen atom*
- ii. Wavelength of photon to *completely remove* an electron from *1s orbital of Boron atom (B)*
- iii. Wavelength of photon to *completely remove* an electron from *1s orbital of Boron cation (B⁺)*

i or ii or iii

(II) Select the smallest (shortest) wavelength from the three choices below (*for H atom*):

- i. Wavelength of photon *emitted* for an electron transitioning from *2p to 1s orbital*
- ii. Wavelength of photon *absorbed* for an electron transitioning from *2s to 4p orbital*
- iii. Wavelength of photon *absorbed* for an electron transitioning from *1s to 3p orbital*

i or ii or iii



Question 2 8 points



A Chem110 student plots the graph shown on the left, from data obtained for the kinetic energy (eV) of the ejected electron versus frequency (s^{-1}) of the incident light, for an unknown metal. ($1\text{eV} = 1.6022 \times 10^{-19} \text{ J}$)

Based on the plot, answer the questions **a to c** below:

a. 1 point On the plot above, label the **threshold frequency**.

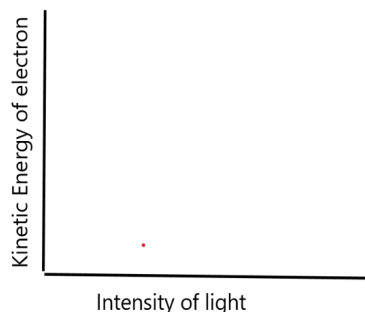
b. 2 points What is the value of the work potential (in J) of the unknown metal?

c. 3 points What is the *longest* wavelength (in nm) required to eject an electron from this metal's surface? Show your work including equation(s) used. (3 sig figs)

d. 2 points In the space provided below, plot the following

(Assume frequency is *above* threshold frequency):

- Kinetic energy* of an ejected electron versus the *intensity* of the incident light (left)
- Number of electrons* ejected versus the *frequency* of the incident light (right)



**Question 3** **6 points****a. 2 points**

In the ground state of Ge (Germanium; $Z = 32$) atom, **name** the following orbitals:

i. Orbital containing the electron that requires the maximum energy to be removed.

ii. Orbital containing the electron that experiences the minimum effective nuclear charge.

b. 2 points

An unknown anion (A^-) has the following *excited* electronic configuration: $[Ar] 4s^2 3d^3 4p^1$

(Ar: Argon; $Z = 18$)

Give the **condensed ground state electronic configuration** of the atom, **A** and the cation, **A^{2+}** .
Show your work. (Assume all electronic transitions are possible)

Electronic Configuration of A:

Electronic Configuration of A^{2+} :

c. 2 points

In a single atom, determine how many electrons can have the following set of quantum numbers?

(i) $n = 5; l = 3$

In a single atom, determine how many electrons can have the following set of quantum numbers?

(i) $n = 3; l = 2; m_s = +1/2$

**Question 4** **8 points**

Rank the following from 1 to 4 in **increasing** order. Provide the ranking in the boxes provided.

Note: Atomic number (Z) are provided for relevant atoms

a) Increasing (lowest to highest) order of *ionization energy*: (1= lowest; 4 = highest)

Mg

Mg: Z = 12

Rb

Rb: Z = 37

Cs

Cs: Z = 55

K

K : Z = 19

b) Increasing (lowest to highest) order of *bond angle*: (1= lowest; 4 = highest)

H₂OCCl₄XeF₄

HCN

c) Increasing (smallest to largest) order of *ionic size* (1= smallest; 4 = largest)

S²⁻

S: Z = 16

O²⁻

O: Z = 8

Be²⁺

Be: Z = 4

Na⁺

Na: Z = 11

d) Increasing (lowest to highest) order of *bond energy* (1= lowest; 4 = highest)

P-F

P-Cl

P-I

P-Br

**Question 5** **12 points**

a. 9 points For the molecules HClO_4 and TeF_6 , draw the most stable *Lewis structure* showing appropriate shape and *dashed/wedged bonds*, based on VSEPR. Determine the **molecular geometry** around the central atom. For HClO_4 , label the *most polar bond*.



b. 3 points (Note: Lone pairs are not shown in structures below)

(i) For the three *resonance* structures (1, 2 and 3) shown below, answer the following question:



(I) Circle the *most contributing* resonance structure.

Structure 1

Structure 2

Structure 3

(ii) For the three structures given (4, 5, and 6) below, answer the following questions:



(I) Circle all the structures that represent *resonance structures* of each other:

Structure 4

Structure 5

Structure 6

(II) Among the resonance structures, circle the *most contributing* resonance structure.

Structure 4

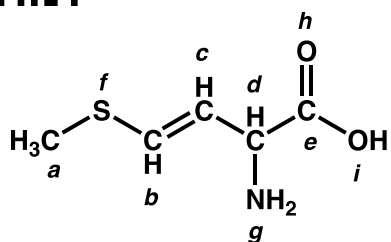
Structure 5

Structure 6

**Question 6** **6 points**

Using the data given below, calculate the **lattice energy** for SrO(s).

	ΔH° (kJ/mol)
$\text{Sr}(s) \longrightarrow \text{Sr}(g)$	+180
$\text{Sr}(g) \longrightarrow \text{Sr}^{2+}(g) + 2e^-$	+1468
$\text{O}_2(g) \longrightarrow 2\text{O}(g)$	+500
$\text{O}(g) + 2e^- \longrightarrow \text{O}^{2-}(g)$	+600
$\text{Sr}(s) + \frac{1}{2}\text{O}_2(g) \longrightarrow \text{SrO}(s)$	-550

**Question 7 10 points**

Consider the molecule given on the left, and answer questions (i) to (vi) using chemical bonding theories that we have learnt in class (valence bond theory, hybridization).

Note that the lone pairs are not shown in the structure:

i. Consider the carbon atoms marked in as ***a***, ***b***, ***c***, ***d***, and ***e*** in the molecule, and determine **hybridization** at each of the carbon centers. (2.5 points)

Atom	Hybridization
Carbon (<i>a</i>)	
Carbon (<i>b</i>)	
Carbon (<i>c</i>)	
Carbon (<i>d</i>)	
Carbon (<i>e</i>)	

ii. Now, consider the sulfur atom (S) marked as ***f***, nitrogen (N) atom (***g***), and oxygen(O) atoms (***h*** and ***i***), and determine their **hybridization**. (2 points)

Atom	Hybridization
Sulfur (<i>f</i>)	
Nitrogen (<i>g</i>)	
Oxygen (<i>h</i>)	
Oxygen (<i>i</i>)	

iii. **Describe** all the bonds formed by the carbon atoms ***a*** and ***b***, using sigma and pi designation as appropriate. (2 points)

Atom	Bonds formed (use sigma and pi as appropriate)
Carbon (<i>a</i>)	
Carbon (<i>b</i>)	

iv. How many total sp^3 -s sigma bonds are in the molecule? (1 point)

v. Determine **electronic** and **molecular geometry** at carbon (C) labeled ***e*** and sulfur (S) labeled ***f*** centres. (2 points)

Atom	Electronic Geometry	Molecular Geometry
Carbon (<i>e</i>)		
Sulfur (<i>f</i>)		

vi. Circle the value from the choices given below, that describes the **C-S-C bond angle at sulfur** (S) marked ***f***: (0.5 points)

>90° to <109.5° 109.5° >109.5 to <120° 120° >120° to <180°


Question 8 12 points

Using the **valence shell orbitals only** draw a **complete** molecular orbital (MO) diagram (with no s-p mixing) for the molecule **PS**, in the space provided below. **(6 points)**

Assume that 3s and 3p atomic orbitals will form molecular orbitals like those formed by the combination of 2s and 2p atomic orbitals that we learnt in class, and there is no s-p mixing.

Show **energy arrow**, **atomic orbitals** involved, **atom labels**, the **molecular orbitals** that are formed, **filled electrons** in the MO diagram etc. Make sure to label all the atomic and molecular orbitals.

Using the MO diagram above, answer the following questions:

- (i) Which one of the following will have the highest bond order? Circle the correct choice
 A) PS^{1-} B) PS C) PS^{1+}
- (ii) Which one of the following will have the longest bond length? Circle the correct choice
 A) PS^{1-} B) PS C) PS^{1+}
- (iii) Which of the following will be diamagnetic? Circle the correct choice.
 A) PS^{1-} B) PS C) PS^{1+}
- (iv) Draw Lewis dot structure of the molecule with the shortest bond length (based on the MO diagram) from among the three (PS^{1-} , PS, PS^{1+}), and indicate the bond order calculated from the Lewis dot structure. **(3 points)**

Lewis Dot Structure

Bond Order



0168BB1C-B90D-4D9B-B5C6-2A8E7CE942AC

midterm-exam-93388

#737 Page 10 of 10

SCRAP PAGE

**CHEMISTRY 110 –Midterm Exam****Thursday November 7, 2024, 6:30 – 8:30 pm****Examiners: Pallavi Sirjoosingh and Ashok Kakkar**

First name (please write as legibly as possible within the boxes)

--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--

Last name

--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--

Student ID number

--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--

This booklet has 8 questions on 10 pages (The last page can be used as a scrap page).

The staple must NOT be removed from the question booklet. Your intact question booklet must be returned at the end of the exam.

A separate booklet contains a periodic table, and datasheet.

Write your answers directly under the questions in the space provided. Do not write on the QR codes or near the top edges (as it may be cut off).

Please write **legibly and dark enough** for the scan to be readable. For students who use English as a second language, dictionaries are permitted. Calculators are allowed.

DO NOT ASK QUESTIONS OF THE INVIGILATORS about interpretations of the exam questions. If you have a concern about a question, please summarize it at the top of this page. All concerns will be evaluated after the exam. GOOD LUCK!

**Question 1 8 points** (*Assume all transitions are possible*)

a. 6 points Calculate the **energy (in kJ/mol)** required to ionize (completely remove) the electron from the **$n=3$** level of **He^+ ion** (He: Helium). Show your work including equations. (3 sig. figs.)

b. 2 points

(I) Select the largest (longest) wavelength from the three choices below:

- i. Wavelength of photon to *completely remove* an electron from *1s orbital of Hydrogen atom*
- ii. Wavelength of photon to *completely remove* an electron from *1s orbital of Boron atom (B)*
- iii. Wavelength of photon to *completely remove* an electron from *1s orbital of Boron cation (B^+)*

i or ii or iii

(II) Select the smallest (shortest) wavelength from the three choices below (*for H atom*):

- i. Wavelength of photon *emitted* for an electron transitioning from *2p to 1s orbital*
- ii. Wavelength of photon *absorbed* for an electron transitioning from *2s to 4p orbital*
- iii. Wavelength of photon *absorbed* for an electron transitioning from *1s to 3p orbital*

i or ii or iii



Question 2 8 points



A Chem110 student plots the graph shown on the left, from data obtained for the kinetic energy (eV) of the ejected electron versus frequency (s^{-1}) of the incident light, for an unknown metal. ($1\text{eV} = 1.6022 \times 10^{-19} \text{ J}$)

Based on the plot, answer the questions **a to c** below:

a. 1 point On the plot above, label the **threshold frequency**.

b. 2 points What is the value of the work potential (in J) of the unknown metal?

c. 3 points What is the *longest* wavelength (in nm) required to eject an electron from this metal's surface? Show your work including equation(s) used. (3 sig figs)

d. 2 points In the space provided below, plot the following

(Assume frequency is *above* threshold frequency):

- Kinetic energy* of an ejected electron versus the *intensity* of the incident light (left)
- Number of electrons* ejected versus the *frequency* of the incident light (right)



**Question 3** **6 points****a. 2 points**

In the ground state of Ge (Germanium; $Z = 32$) atom, **name** the following orbitals:

i. Orbital containing the electron that requires the maximum energy to be removed.

ii. Orbital containing the electron that experiences the minimum effective nuclear charge.

b. 2 points

An unknown anion (A^-) has the following *excited* electronic configuration: $[Ar] 4s^2 3d^3 4p^1$

(Ar: Argon; $Z = 18$)

Give the **condensed ground state electronic configuration** of the atom, **A** and the cation, **A^{2+}** .
Show your work. (Assume all electronic transitions are possible)

Electronic Configuration of A:

Electronic Configuration of A^{2+} :

c. 2 points

In a single atom, determine how many electrons can have the following set of quantum numbers?

(i) $n = 5; l = 3$

In a single atom, determine how many electrons can have the following set of quantum numbers?

(i) $n = 3; l = 2; m_s = +1/2$

**Question 4** **8 points**

Rank the following from 1 to 4 in **increasing** order. Provide the ranking in the boxes provided.

Note: Atomic number (Z) are provided for relevant atoms

a) Increasing (lowest to highest) order of *ionization energy*: (1= lowest; 4 = highest)

Mg

Mg: Z = 12

Rb

Rb: Z = 37

Cs

Cs: Z = 55

K

K : Z = 19

b) Increasing (lowest to highest) order of *bond angle*: (1= lowest; 4 = highest)

H₂OCCl₄XeF₄

HCN

c) Increasing (smallest to largest) order of *ionic size* (1= smallest; 4 = largest)

S²⁻

S: Z = 16

O²⁻

O: Z = 8

Be²⁺

Be: Z = 4

Na⁺

Na: Z = 11

d) Increasing (lowest to highest) order of *bond energy* (1= lowest; 4 = highest)

P-F

P-Cl

P-I

P-Br

**Question 5** **12 points**

a. 9 points For the molecules HClO_4 and TeF_6 , draw the most stable *Lewis structure* showing appropriate shape and *dashed/wedged bonds*, based on VSEPR. Determine the **molecular geometry** around the central atom. For HClO_4 , label the *most polar bond*.



b. 3 points (Note: Lone pairs are not shown in structures below)

(i) For the three *resonance* structures (1, 2 and 3) shown below, answer the following question:



(I) Circle the *most contributing* resonance structure.

Structure 1

Structure 2

Structure 3

(ii) For the three structures given (4, 5, and 6) below, answer the following questions:



(I) Circle all the structures that represent *resonance structures* of each other:

Structure 4

Structure 5

Structure 6

(II) Among the resonance structures, circle the *most contributing* resonance structure.

Structure 4

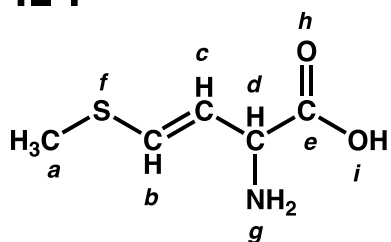
Structure 5

Structure 6

**Question 6** **6 points**

Using the data given below, calculate the **lattice energy** for SrO(s).

	ΔH° (kJ/mol)
$\text{Sr}(s) \longrightarrow \text{Sr}(g)$	+180
$\text{Sr}(g) \longrightarrow \text{Sr}^{2+}(g) + 2e^-$	+1468
$\text{O}_2(g) \longrightarrow 2\text{O}(g)$	+500
$\text{O}(g) + 2e^- \longrightarrow \text{O}^{2-}(g)$	+600
$\text{Sr}(s) + \frac{1}{2}\text{O}_2(g) \longrightarrow \text{SrO}(s)$	-550

**Question 7 10 points**

Consider the molecule given on the left, and answer questions (i) to (vi) using chemical bonding theories that we have learnt in class (valence bond theory, hybridization).

Note that the lone pairs are not shown in the structure:

i. Consider the carbon atoms marked in as ***a***, ***b***, ***c***, ***d***, and ***e*** in the molecule, and determine **hybridization** at each of the carbon centers. (2.5 points)

Atom	Hybridization
Carbon (<i>a</i>)	
Carbon (<i>b</i>)	
Carbon (<i>c</i>)	
Carbon (<i>d</i>)	
Carbon (<i>e</i>)	

ii. Now, consider the sulfur atom (S) marked as ***f***, nitrogen (N) atom (***g***), and oxygen(O) atoms (***h*** and ***i***), and determine their **hybridization**. (2 points)

Atom	Hybridization
Sulfur (<i>f</i>)	
Nitrogen (<i>g</i>)	
Oxygen (<i>h</i>)	
Oxygen (<i>i</i>)	

iii. **Describe** all the bonds formed by the carbon atoms ***a*** and ***b***, using sigma and pi designation as appropriate. (2 points)

Atom	Bonds formed (use sigma and pi as appropriate)
Carbon (<i>a</i>)	
Carbon (<i>b</i>)	

iv. How many total sp^3 -s sigma bonds are in the molecule? (1 point)

v. Determine **electronic** and **molecular geometry** at carbon (C) labeled ***e*** and sulfur (S) labeled ***f*** centres. (2 points)

Atom	Electronic Geometry	Molecular Geometry
Carbon (<i>e</i>)		
Sulfur (<i>f</i>)		

vi. Circle the value from the choices given below, that describes the **C-S-C bond angle at sulfur** (S) marked ***f***: (0.5 points)

>90° to <109.5° 109.5° >109.5 to <120° 120° >120° to <180°


Question 8 12 points

Using the **valence shell orbitals only** draw a **complete** molecular orbital (MO) diagram (with no s-p mixing) for the molecule **PS**, in the space provided below. **(6 points)**

Assume that 3s and 3p atomic orbitals will form molecular orbitals like those formed by the combination of 2s and 2p atomic orbitals that we learnt in class, and there is no s-p mixing.

Show **energy arrow**, **atomic orbitals** involved, **atom labels**, the **molecular orbitals** that are formed, **filled electrons** in the MO diagram etc. Make sure to label all the atomic and molecular orbitals.

Using the MO diagram above, answer the following questions:

- (i) Which one of the following will have the highest bond order? Circle the correct choice
 A) PS^{1-} B) PS C) PS^{1+}
- (ii) Which one of the following will have the longest bond length? Circle the correct choice
 A) PS^{1-} B) PS C) PS^{1+}
- (iii) Which of the following will be diamagnetic? Circle the correct choice.
 A) PS^{1-} B) PS C) PS^{1+}
- (iv) Draw Lewis dot structure of the molecule with the shortest bond length (based on the MO diagram) from among the three (PS^{1-} , PS, PS^{1+}), and indicate the bond order calculated from the Lewis dot structure. **(3 points)**

Lewis Dot Structure

Bond Order



95BC7A76-63C4-40BC-B526-E5363EE8ED16

midterm-exam-93388

#738 Page 10 of 10

SCRAP PAGE

**CHEMISTRY 110 –Midterm Exam****Thursday November 7, 2024, 6:30 – 8:30 pm****Examiners: Pallavi Sirjoosingh and Ashok Kakkar**

First name (please write as legibly as possible within the boxes)

--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--

Last name

--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--

Student ID number

--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--

This booklet has 8 questions on 10 pages (The last page can be used as a scrap page).

The staple must NOT be removed from the question booklet. Your intact question booklet must be returned at the end of the exam.

A separate booklet contains a periodic table, and datasheet.

Write your answers directly under the questions in the space provided. Do not write on the QR codes or near the top edges (as it may be cut off).

Please write **legibly and dark enough** for the scan to be readable. For students who use English as a second language, dictionaries are permitted. Calculators are allowed.

DO NOT ASK QUESTIONS OF THE INVIGILATORS about interpretations of the exam questions. If you have a concern about a question, please summarize it at the top of this page. All concerns will be evaluated after the exam. GOOD LUCK!

**Question 1 8 points** (*Assume all transitions are possible*)

a. 6 points Calculate the **energy (in kJ/mol)** required to ionize (completely remove) the electron from the **$n=3$** level of **He^+ ion** (He: Helium). Show your work including equations. (3 sig. figs.)

b. 2 points

(I) Select the largest (longest) wavelength from the three choices below:

- i. Wavelength of photon to *completely remove* an electron from *1s orbital of Hydrogen atom*
- ii. Wavelength of photon to *completely remove* an electron from *1s orbital of Boron atom (B)*
- iii. Wavelength of photon to *completely remove* an electron from *1s orbital of Boron cation (B^+)*

i or ii or iii

(II) Select the smallest (shortest) wavelength from the three choices below (*for H atom*):

- i. Wavelength of photon *emitted* for an electron transitioning from *2p to 1s orbital*
- ii. Wavelength of photon *absorbed* for an electron transitioning from *2s to 4p orbital*
- iii. Wavelength of photon *absorbed* for an electron transitioning from *1s to 3p orbital*

i or ii or iii



Question 2 8 points



A Chem110 student plots the graph shown on the left, from data obtained for the kinetic energy (eV) of the ejected electron versus frequency (s^{-1}) of the incident light, for an unknown metal. ($1\text{eV} = 1.6022 \times 10^{-19} \text{ J}$)

Based on the plot, answer the questions **a to c** below:

a. 1 point On the plot above, label the **threshold frequency**.

b. 2 points What is the value of the work potential (in J) of the unknown metal?

c. 3 points What is the *longest* wavelength (in nm) required to eject an electron from this metal's surface? Show your work including equation(s) used. (3 sig figs)

d. 2 points In the space provided below, plot the following

(Assume frequency is *above* threshold frequency):

- i. *Kinetic energy* of an ejected electron versus the *intensity* of the incident light (left)
- ii. *Number of electrons* ejected versus the *frequency* of the incident light (right)



**Question 3** **6 points****a. 2 points**

In the ground state of Ge (Germanium; $Z = 32$) atom, **name** the following orbitals:

i. Orbital containing the electron that requires the maximum energy to be removed.

ii. Orbital containing the electron that experiences the minimum effective nuclear charge.

b. 2 points

An unknown anion (A^-) has the following *excited* electronic configuration: $[Ar] 4s^2 3d^3 4p^1$

(Ar: Argon; $Z = 18$)

Give the **condensed ground state electronic configuration** of the atom, **A** and the cation, **A^{2+}** .
Show your work. (Assume all electronic transitions are possible)

Electronic Configuration of A:

Electronic Configuration of A^{2+} :

c. 2 points

In a single atom, determine how many electrons can have the following set of quantum numbers?

(i) $n = 5; l = 3$

In a single atom, determine how many electrons can have the following set of quantum numbers?

(i) $n = 3; l = 2; m_s = +1/2$

**Question 4** **8 points**

Rank the following from 1 to 4 in **increasing** order. Provide the ranking in the boxes provided.

Note: Atomic number (Z) are provided for relevant atoms

a) Increasing (lowest to highest) order of *ionization energy*: (1= lowest; 4 = highest)

Mg

Mg: Z = 12

Rb

Rb: Z = 37

Cs

Cs: Z = 55

K

K : Z = 19

b) Increasing (lowest to highest) order of *bond angle*: (1= lowest; 4 = highest)

H₂OCCl₄XeF₄

HCN

c) Increasing (smallest to largest) order of *ionic size* (1= smallest; 4 = largest)

S²⁻

S: Z = 16

O²⁻

O: Z = 8

Be²⁺

Be: Z = 4

Na⁺

Na: Z = 11

d) Increasing (lowest to highest) order of *bond energy* (1= lowest; 4 = highest)

P-F

P-Cl

P-I

P-Br

**Question 5** **12 points**

a. 9 points For the molecules HClO_4 and TeF_6 , draw the most stable *Lewis structure* showing appropriate shape and *dashed/wedged bonds*, based on VSEPR. Determine the **molecular geometry** around the central atom. For HClO_4 , label the *most polar bond*.



b. 3 points (Note: Lone pairs are not shown in structures below)

(i) For the three *resonance* structures (1, 2 and 3) shown below, answer the following question:



(I) Circle the *most contributing* resonance structure.

Structure 1

Structure 2

Structure 3

(ii) For the three structures given (4, 5, and 6) below, answer the following questions:



(I) Circle all the structures that represent *resonance structures* of each other:

Structure 4

Structure 5

Structure 6

(II) Among the resonance structures, circle the *most contributing* resonance structure.

Structure 4

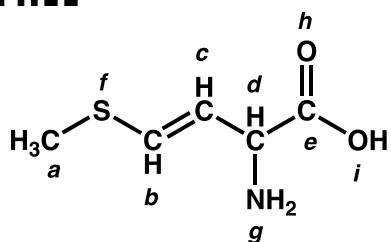
Structure 5

Structure 6

**Question 6** **6 points**

Using the data given below, calculate the **lattice energy** for SrO(s).

	ΔH° (kJ/mol)
$\text{Sr}(s) \longrightarrow \text{Sr}(g)$	+180
$\text{Sr}(g) \longrightarrow \text{Sr}^{2+}(g) + 2e^-$	+1468
$\text{O}_2(g) \longrightarrow 2\text{O}(g)$	+500
$\text{O}(g) + 2e^- \longrightarrow \text{O}^{2-}(g)$	+600
$\text{Sr}(s) + \frac{1}{2}\text{O}_2(g) \longrightarrow \text{SrO}(s)$	-550

**Question 7 10 points**

Consider the molecule given on the left, and answer questions (i) to (vi) using chemical bonding theories that we have learnt in class (valence bond theory, hybridization).

Note that the lone pairs are not shown in the structure:

i. Consider the carbon atoms marked in as ***a***, ***b***, ***c***, ***d***, and ***e*** in the molecule, and determine **hybridization** at each of the carbon centers. (2.5 points)

Atom	Hybridization
Carbon (<i>a</i>)	
Carbon (<i>b</i>)	
Carbon (<i>c</i>)	
Carbon (<i>d</i>)	
Carbon (<i>e</i>)	

ii. Now, consider the sulfur atom (S) marked as ***f***, nitrogen (N) atom (***g***), and oxygen(O) atoms (***h*** and ***i***), and determine their **hybridization**. (2 points)

Atom	Hybridization
Sulfur (<i>f</i>)	
Nitrogen (<i>g</i>)	
Oxygen (<i>h</i>)	
Oxygen (<i>i</i>)	

iii. **Describe** all the bonds formed by the carbon atoms ***a*** and ***b***, using sigma and pi designation as appropriate. (2 points)

Atom	Bonds formed (use sigma and pi as appropriate)
Carbon (<i>a</i>)	
Carbon (<i>b</i>)	

iv. How many total sp^3 -s sigma bonds are in the molecule? (1 point)

v. Determine **electronic** and **molecular geometry** at carbon (C) labeled ***e*** and sulfur (S) labeled ***f*** centres. (2 points)

Atom	Electronic Geometry	Molecular Geometry
Carbon (<i>e</i>)		
Sulfur (<i>f</i>)		

vi. Circle the value from the choices given below, that describes the **C-S-C bond angle at sulfur** (S) marked ***f***: (0.5 points)

>90° to <109.5° 109.5° >109.5 to <120° 120° >120° to <180°


Question 8 12 points

Using the **valence shell orbitals only** draw a **complete** molecular orbital (MO) diagram (with no s-p mixing) for the molecule **PS**, in the space provided below. **(6 points)**

Assume that 3s and 3p atomic orbitals will form molecular orbitals like those formed by the combination of 2s and 2p atomic orbitals that we learnt in class, and there is no s-p mixing.

Show **energy arrow**, **atomic orbitals** involved, **atom labels**, the **molecular orbitals** that are formed, **filled electrons** in the MO diagram etc. Make sure to **label all the atomic and molecular orbitals**.

Using the MO diagram above, answer the following questions:

- (i) Which one of the following will have the highest bond order? Circle the correct choice
 A) PS^{1-} B) PS C) PS^{1+}
- (ii) Which one of the following will have the longest bond length? Circle the correct choice
 A) PS^{1-} B) PS C) PS^{1+}
- (iii) Which of the following will be diamagnetic? Circle the correct choice.
 A) PS^{1-} B) PS C) PS^{1+}
- (iv) Draw Lewis dot structure of the molecule with the shortest bond length (based on the MO diagram) from among the three (PS^{1-} , PS, PS^{1+}), and indicate the bond order calculated from the Lewis dot structure. **(3 points)**

Lewis Dot Structure

Bond Order



8E9FE11F-0E03-4BA6-8BF0-D4FCBF13AFC2

midterm-exam-93388

#739 Page 10 of 10

SCRAP PAGE

**CHEMISTRY 110 –Midterm Exam****Thursday November 7, 2024, 6:30 – 8:30 pm**

Examiners: Pallavi Sirjoosingh and Ashok Kakkar

First name (please write as legibly as possible within the boxes)

--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--

Last name

--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--

Student ID number

--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--

This booklet has 8 questions on 10 pages (The last page can be used as a scrap page).

The staple must NOT be removed from the question booklet. Your intact question booklet must be returned at the end of the exam.

A separate booklet contains a periodic table, and datasheet.

Write your answers directly under the questions in the space provided. Do not write on the QR codes or near the top edges (as it may be cut off).

Please write **legibly and dark enough** for the scan to be readable. For students who use English as a second language, dictionaries are permitted. Calculators are allowed.

DO NOT ASK QUESTIONS OF THE INVIGILATORS about interpretations of the exam questions. If you have a concern about a question, please summarize it at the top of this page. All concerns will be evaluated after the exam. GOOD LUCK!

**Question 1 8 points** (*Assume all transitions are possible*)

a. 6 points Calculate the **energy (in kJ/mol)** required to ionize (completely remove) the electron from the **n=3** level of **He⁺ ion** (He: Helium). Show your work including equations. (3 sig. figs.)

b. 2 points

(I) Select the largest (longest) wavelength from the three choices below:

- i. Wavelength of photon to *completely remove* an electron from *1s orbital of Hydrogen atom*
- ii. Wavelength of photon to *completely remove* an electron from *1s orbital of Boron atom (B)*
- iii. Wavelength of photon to *completely remove* an electron from *1s orbital of Boron cation (B⁺)*

i or ii or iii

(II) Select the smallest (shortest) wavelength from the three choices below (*for H atom*):

- i. Wavelength of photon *emitted* for an electron transitioning from *2p to 1s orbital*
- ii. Wavelength of photon *absorbed* for an electron transitioning from *2s to 4p orbital*
- iii. Wavelength of photon *absorbed* for an electron transitioning from *1s to 3p orbital*

i or ii or iii



Question 2 8 points



A Chem110 student plots the graph shown on the left, from data obtained for the kinetic energy (eV) of the ejected electron versus frequency (s^{-1}) of the incident light, for an unknown metal. ($1\text{eV} = 1.6022 \times 10^{-19} \text{ J}$)

Based on the plot, answer the questions **a to c** below:

a. 1 point On the plot above, label the **threshold frequency**.

b. 2 points What is the value of the work potential (in J) of the unknown metal?

c. 3 points What is the *longest* wavelength (in nm) required to eject an electron from this metal's surface? Show your work including equation(s) used. (3 sig figs)

d. 2 points In the space provided below, plot the following

(Assume frequency is *above* threshold frequency):

- i. *Kinetic energy* of an ejected electron versus the *intensity* of the incident light (left)
- ii. *Number of electrons* ejected versus the *frequency* of the incident light (right)



**Question 3** **6 points****a. 2 points**

In the ground state of Ge (Germanium; $Z = 32$) atom, **name** the following orbitals:

i. Orbital containing the electron that requires the maximum energy to be removed.

ii. Orbital containing the electron that experiences the minimum effective nuclear charge.

b. 2 points

An unknown anion (A^-) has the following *excited* electronic configuration: $[Ar] 4s^2 3d^3 4p^1$

(Ar: Argon; $Z = 18$)

Give the **condensed ground state electronic configuration** of the atom, **A** and the cation, **A^{2+}** .
Show your work. (Assume all electronic transitions are possible)

Electronic Configuration of A:

Electronic Configuration of A^{2+} :

c. 2 points

In a single atom, determine how many electrons can have the following set of quantum numbers?

(i) $n = 5; l = 3$

In a single atom, determine how many electrons can have the following set of quantum numbers?

(i) $n = 3; l = 2; m_s = +1/2$

**Question 4** **8 points**

Rank the following from 1 to 4 in **increasing** order. Provide the ranking in the boxes provided.

Note: Atomic number (Z) are provided for relevant atoms

a) Increasing (lowest to highest) order of *ionization energy*: (1= lowest; 4 = highest)

Mg

Mg: Z = 12

Rb

Rb: Z = 37

Cs

Cs: Z = 55

K

K : Z = 19

b) Increasing (lowest to highest) order of *bond angle*: (1= lowest; 4 = highest)

H₂OCCl₄XeF₄

HCN

c) Increasing (smallest to largest) order of *ionic size* (1= smallest; 4 = largest)

S²⁻

S: Z = 16

O²⁻

O: Z = 8

Be²⁺

Be: Z = 4

Na⁺

Na: Z = 11

d) Increasing (lowest to highest) order of *bond energy* (1= lowest; 4 = highest)

P-F

P-Cl

P-I

P-Br

**Question 5** **12 points**

a. 9 points For the molecules HClO_4 and TeF_6 , draw the most stable *Lewis structure* showing appropriate shape and *dashed/wedged bonds*, based on VSEPR. Determine the **molecular geometry** around the central atom. For HClO_4 , label the *most polar bond*.

HClO_4 (there are no O-O bonds)

Label the most polar bond

Molecular Geometry around Cl

TeF_6

Molecular Geometry around Te

b. 3 points (Note: Lone pairs are not shown in structures below)

(i) For the three *resonance* structures (1, 2 and 3) shown below, answer the following question:



(I) Circle the *most contributing* resonance structure.

Structure 1

Structure 2

Structure 3

(ii) For the three structures given (4, 5, and 6) below, answer the following questions:



(I) Circle all the structures that represent *resonance structures* of each other:

Structure 4

Structure 5

Structure 6

(II) Among the resonance structures, circle the *most contributing* resonance structure.

Structure 4

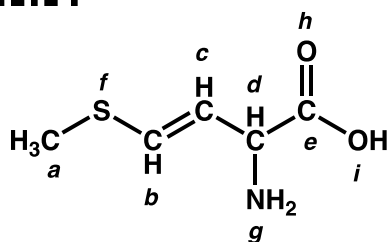
Structure 5

Structure 6

**Question 6** **6 points**

Using the data given below, calculate the **lattice energy** for SrO(s).

	ΔH° (kJ/mol)
$\text{Sr}(s) \longrightarrow \text{Sr}(g)$	+180
$\text{Sr}(g) \longrightarrow \text{Sr}^{2+}(g) + 2e^-$	+1468
$\text{O}_2(g) \longrightarrow 2\text{O}(g)$	+500
$\text{O}(g) + 2e^- \longrightarrow \text{O}^{2-}(g)$	+600
$\text{Sr}(s) + \frac{1}{2}\text{O}_2(g) \longrightarrow \text{SrO}(s)$	-550

**Question 7 10 points**

Consider the molecule given on the left, and answer questions (i) to (vi) using chemical bonding theories that we have learnt in class (valence bond theory, hybridization).

Note that the lone pairs are not shown in the structure:

i. Consider the carbon atoms marked in as ***a***, ***b***, ***c***, ***d***, and ***e*** in the molecule, and determine **hybridization** at each of the carbon centers. (2.5 points)

Atom	Hybridization
Carbon (<i>a</i>)	
Carbon (<i>b</i>)	
Carbon (<i>c</i>)	
Carbon (<i>d</i>)	
Carbon (<i>e</i>)	

ii. Now, consider the sulfur atom (S) marked as ***f***, nitrogen (N) atom (***g***), and oxygen(O) atoms (***h*** and ***i***), and determine their **hybridization**. (2 points)

Atom	Hybridization
Sulfur (<i>f</i>)	
Nitrogen (<i>g</i>)	
Oxygen (<i>h</i>)	
Oxygen (<i>i</i>)	

iii. **Describe** all the bonds formed by the carbon atoms ***a*** and ***b***, using sigma and pi designation as appropriate. (2 points)

Atom	Bonds formed (use sigma and pi as appropriate)
Carbon (<i>a</i>)	
Carbon (<i>b</i>)	

iv. How many total sp^3 -s sigma bonds are in the molecule? (1 point)

v. Determine **electronic** and **molecular geometry** at carbon (C) labeled ***e*** and sulfur (S) labeled ***f*** centres. (2 points)

Atom	Electronic Geometry	Molecular Geometry
Carbon (<i>e</i>)		
Sulfur (<i>f</i>)		

vi. Circle the value from the choices given below, that describes the **C-S-C bond angle at sulfur** (S) marked ***f***: (0.5 points)

>90° to <109.5° 109.5° >109.5 to <120° 120° >120° to <180°

**Question 8 12 points**

Using the **valence shell orbitals only** draw a **complete** molecular orbital (MO) diagram (with no s-p mixing) for the molecule **PS**, in the space provided below. **(6 points)**

Assume that 3s and 3p atomic orbitals will form molecular orbitals like those formed by the combination of 2s and 2p atomic orbitals that we learnt in class, and there is no s-p mixing.

Show **energy arrow**, **atomic orbitals** involved, **atom labels**, the **molecular orbitals** that are formed, **filled electrons** in the MO diagram etc. Make sure to label all the atomic and molecular orbitals.

Using the MO diagram above, answer the following questions:

- (i) Which one of the following will have the highest bond order? Circle the correct choice
A) PS^{1-} B) PS C) PS^{1+}
- (ii) Which one of the following will have the longest bond length? Circle the correct choice
A) PS^{1-} B) PS C) PS^{1+}
- (iii) Which of the following will be diamagnetic? Circle the correct choice.
A) PS^{1-} B) PS C) PS^{1+}
- (iv) Draw Lewis dot structure of the molecule with the shortest bond length (based on the MO diagram) from among the three (PS^{1-} , PS, PS^{1+}), and indicate the bond order calculated from the Lewis dot structure. **(3 points)**

Lewis Dot Structure

Bond Order



50DC9CFA-6B99-4FA4-A8C1-2E24AA559DA0

midterm-exam-93388

#740 Page 10 of 10

SCRAP PAGE

**CHEMISTRY 110 –Midterm Exam****Thursday November 7, 2024, 6:30 – 8:30 pm****Examiners: Pallavi Sirjoosingh and Ashok Kakkar**

First name (please write as legibly as possible within the boxes)

--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--

Last name

--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--

Student ID number

--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--

This booklet has 8 questions on 10 pages (The last page can be used as a scrap page).

The staple must NOT be removed from the question booklet. Your intact question booklet must be returned at the end of the exam.

A separate booklet contains a periodic table, and datasheet.

Write your answers directly under the questions in the space provided. Do not write on the QR codes or near the top edges (as it may be cut off).

Please write **legibly and dark enough** for the scan to be readable. For students who use English as a second language, dictionaries are permitted. Calculators are allowed.

DO NOT ASK QUESTIONS OF THE INVIGILATORS about interpretations of the exam questions. If you have a concern about a question, please summarize it at the top of this page. All concerns will be evaluated after the exam. GOOD LUCK!

**Question 1 8 points** (*Assume all transitions are possible*)

a. 6 points Calculate the **energy (in kJ/mol)** required to ionize (completely remove) the electron from the **n=3** level of **He⁺ ion** (He: Helium). Show your work including equations. (3 sig. figs.)

b. 2 points

(I) Select the largest (longest) wavelength from the three choices below:

- i. Wavelength of photon to *completely remove* an electron from *1s orbital of Hydrogen atom*
- ii. Wavelength of photon to *completely remove* an electron from *1s orbital of Boron atom (B)*
- iii. Wavelength of photon to *completely remove* an electron from *1s orbital of Boron cation (B⁺)*

i or ii or iii

(II) Select the smallest (shortest) wavelength from the three choices below (*for H atom*):

- i. Wavelength of photon *emitted* for an electron transitioning from *2p to 1s orbital*
- ii. Wavelength of photon *absorbed* for an electron transitioning from *2s to 4p orbital*
- iii. Wavelength of photon *absorbed* for an electron transitioning from *1s to 3p orbital*

i or ii or iii



Question 2 8 points



A Chem110 student plots the graph shown on the left, from data obtained for the kinetic energy (eV) of the ejected electron versus frequency (s^{-1}) of the incident light, for an unknown metal. ($1\text{eV} = 1.6022 \times 10^{-19} \text{ J}$)

Based on the plot, answer the questions **a to c** below:

a. 1 point On the plot above, label the **threshold frequency**.

b. 2 points What is the value of the work potential (in J) of the unknown metal?

c. 3 points What is the *longest* wavelength (in nm) required to eject an electron from this metal's surface? Show your work including equation(s) used. (3 sig figs)

d. 2 points In the space provided below, plot the following

(Assume frequency is *above* threshold frequency):

- Kinetic energy* of an ejected electron versus the *intensity* of the incident light (left)
- Number of electrons* ejected versus the *frequency* of the incident light (right)



**Question 3** **6 points****a. 2 points**

In the ground state of Ge (Germanium; $Z = 32$) atom, **name** the following orbitals:

i. Orbital containing the electron that requires the maximum energy to be removed.

ii. Orbital containing the electron that experiences the minimum effective nuclear charge.

b. 2 points

An unknown anion (A^-) has the following *excited* electronic configuration: $[Ar] 4s^2 3d^3 4p^1$

(Ar: Argon; $Z = 18$)

Give the **condensed ground state electronic configuration** of the atom, **A** and the cation, **A^{2+}** .
Show your work. (Assume all electronic transitions are possible)

Electronic Configuration of A:

Electronic Configuration of A^{2+} :

c. 2 points

In a single atom, determine how many electrons can have the following set of quantum numbers?

(i) $n = 5; l = 3$

In a single atom, determine how many electrons can have the following set of quantum numbers?

(i) $n = 3; l = 2; m_s = +1/2$

**Question 4** **8 points**

Rank the following from 1 to 4 in **increasing** order. Provide the ranking in the boxes provided.

Note: Atomic number (Z) are provided for relevant atoms

a) Increasing (lowest to highest) order of *ionization energy*: (1= lowest; 4 = highest)

Mg

Mg: Z = 12

Rb

Rb: Z = 37

Cs

Cs: Z = 55

K

K : Z = 19

b) Increasing (lowest to highest) order of *bond angle*: (1= lowest; 4 = highest)

H₂OCCl₄XeF₄

HCN

c) Increasing (smallest to largest) order of *ionic size* (1= smallest; 4 = largest)

S²⁻

S: Z = 16

O²⁻

O: Z = 8

Be²⁺

Be: Z = 4

Na⁺

Na: Z = 11

d) Increasing (lowest to highest) order of *bond energy* (1= lowest; 4 = highest)

P-F

P-Cl

P-I

P-Br

**Question 5** **12 points**

a. 9 points For the molecules HClO_4 and TeF_6 , draw the most stable *Lewis structure* showing appropriate shape and *dashed/wedged bonds*, based on VSEPR. Determine the **molecular geometry** around the central atom. For HClO_4 , label the *most polar bond*.



b. 3 points (Note: Lone pairs are not shown in structures below)

(i) For the three *resonance* structures (1, 2 and 3) shown below, answer the following question:



(I) Circle the *most contributing* resonance structure.

Structure 1

Structure 2

Structure 3

(ii) For the three structures given (4, 5, and 6) below, answer the following questions:



(I) Circle all the structures that represent *resonance structures* of each other:

Structure 4

Structure 5

Structure 6

(II) Among the resonance structures, circle the *most contributing* resonance structure.

Structure 4

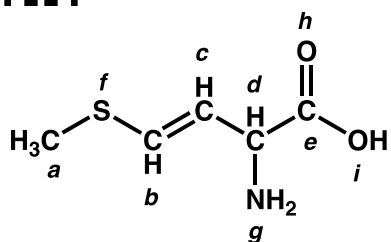
Structure 5

Structure 6

**Question 6** **6 points**

Using the data given below, calculate the **lattice energy** for SrO(s).

	ΔH° (kJ/mol)
$\text{Sr}(s) \longrightarrow \text{Sr}(g)$	+180
$\text{Sr}(g) \longrightarrow \text{Sr}^{2+}(g) + 2e^-$	+1468
$\text{O}_2(g) \longrightarrow 2\text{O}(g)$	+500
$\text{O}(g) + 2e^- \longrightarrow \text{O}^{2-}(g)$	+600
$\text{Sr}(s) + \frac{1}{2}\text{O}_2(g) \longrightarrow \text{SrO}(s)$	-550

**Question 7 10 points**

Consider the molecule given on the left, and answer questions (i) to (vi) using chemical bonding theories that we have learnt in class (valence bond theory, hybridization).

Note that the lone pairs are not shown in the structure:

i. Consider the carbon atoms marked in as ***a***, ***b***, ***c***, ***d***, and ***e*** in the molecule, and determine **hybridization** at each of the carbon centers. (2.5 points)

Atom	Hybridization
Carbon (<i>a</i>)	
Carbon (<i>b</i>)	
Carbon (<i>c</i>)	
Carbon (<i>d</i>)	
Carbon (<i>e</i>)	

ii. Now, consider the sulfur atom (S) marked as ***f***, nitrogen (N) atom (***g***), and oxygen(O) atoms (***h*** and ***i***), and determine their **hybridization**. (2 points)

Atom	Hybridization
Sulfur (<i>f</i>)	
Nitrogen (<i>g</i>)	
Oxygen (<i>h</i>)	
Oxygen (<i>i</i>)	

iii. **Describe** all the bonds formed by the carbon atoms ***a*** and ***b***, using sigma and pi designation as appropriate. (2 points)

Atom	Bonds formed (use sigma and pi as appropriate)
Carbon (<i>a</i>)	
Carbon (<i>b</i>)	

iv. How many total sp^3 -s sigma bonds are in the molecule? (1 point)

v. Determine **electronic** and **molecular geometry** at carbon (C) labeled ***e*** and sulfur (S) labeled ***f*** centres. (2 points)

Atom	Electronic Geometry	Molecular Geometry
Carbon (<i>e</i>)		
Sulfur (<i>f</i>)		

vi. Circle the value from the choices given below, that describes the **C-S-C bond angle at sulfur** (S) marked ***f***: (0.5 points)

>90° to <109.5° 109.5° >109.5 to <120° 120° >120° to <180°

**Question 8 12 points**

Using the **valence shell orbitals only** draw a **complete** molecular orbital (MO) diagram (with no s-p mixing) for the molecule **PS**, in the space provided below. **(6 points)**

Assume that 3s and 3p atomic orbitals will form molecular orbitals like those formed by the combination of 2s and 2p atomic orbitals that we learnt in class, and there is no s-p mixing.

Show **energy arrow**, **atomic orbitals** involved, **atom labels**, the **molecular orbitals** that are formed, **filled electrons** in the MO diagram etc. Make sure to label all the atomic and molecular orbitals.

Using the MO diagram above, answer the following questions:

- (i) Which one of the following will have the highest bond order? Circle the correct choice
A) PS^{1-} B) PS C) PS^{1+}
- (ii) Which one of the following will have the longest bond length? Circle the correct choice
A) PS^{1-} B) PS C) PS^{1+}
- (iii) Which of the following will be diamagnetic? Circle the correct choice.
A) PS^{1-} B) PS C) PS^{1+}
- (iv) Draw Lewis dot structure of the molecule with the shortest bond length (based on the MO diagram) from among the three (PS^{1-} , PS, PS^{1+}), and indicate the bond order calculated from the Lewis dot structure. **(3 points)**

Lewis Dot Structure

Bond Order



67DDEF4F-C55D-4CED-8939-2CD98F01D09C

midterm-exam-93388

#741 Page 10 of 10

SCRAP PAGE

**CHEMISTRY 110 –Midterm Exam****Thursday November 7, 2024, 6:30 – 8:30 pm**

Examiners: Pallavi Sirjoosingh and Ashok Kakkar

First name (please write as legibly as possible within the boxes)

--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--

Last name

--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--

Student ID number

--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--

This booklet has 8 questions on 10 pages (The last page can be used as a scrap page).

The staple must NOT be removed from the question booklet. Your intact question booklet must be returned at the end of the exam.

A separate booklet contains a periodic table, and datasheet.

Write your answers directly under the questions in the space provided. Do not write on the QR codes or near the top edges (as it may be cut off).

Please write **legibly and dark enough** for the scan to be readable. For students who use English as a second language, dictionaries are permitted. Calculators are allowed.

DO NOT ASK QUESTIONS OF THE INVIGILATORS about interpretations of the exam questions. If you have a concern about a question, please summarize it at the top of this page. All concerns will be evaluated after the exam. GOOD LUCK!

**Question 1 8 points** (*Assume all transitions are possible*)

a. 6 points Calculate the **energy (in kJ/mol)** required to ionize (completely remove) the electron from the **n=3** level of **He⁺ ion** (He: Helium). Show your work including equations. (3 sig. figs.)

b. 2 points

(I) Select the largest (longest) wavelength from the three choices below:

- i. Wavelength of photon to *completely remove* an electron from *1s orbital of Hydrogen atom*
- ii. Wavelength of photon to *completely remove* an electron from *1s orbital of Boron atom (B)*
- iii. Wavelength of photon to *completely remove* an electron from *1s orbital of Boron cation (B⁺)*

i or ii or iii

(II) Select the smallest (shortest) wavelength from the three choices below (*for H atom*):

- i. Wavelength of photon *emitted* for an electron transitioning from *2p to 1s orbital*
- ii. Wavelength of photon *absorbed* for an electron transitioning from *2s to 4p orbital*
- iii. Wavelength of photon *absorbed* for an electron transitioning from *1s to 3p orbital*

i or ii or iii



Question 2 8 points



A Chem110 student plots the graph shown on the left, from data obtained for the kinetic energy (eV) of the ejected electron versus frequency (s^{-1}) of the incident light, for an unknown metal. ($1\text{eV} = 1.6022 \times 10^{-19} \text{ J}$)

Based on the plot, answer the questions **a to c** below:

a. 1 point On the plot above, label the **threshold frequency**.

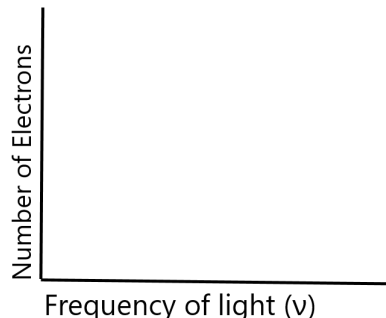
b. 2 points What is the value of the work potential (in J) of the unknown metal?

c. 3 points What is the *longest* wavelength (in nm) required to eject an electron from this metal's surface? Show your work including equation(s) used. (3 sig figs)

d. 2 points In the space provided below, plot the following

(Assume frequency is *above* threshold frequency):

- Kinetic energy* of an ejected electron versus the *intensity* of the incident light (left)
- Number of electrons* ejected versus the *frequency* of the incident light (right)



**Question 3** **6 points****a. 2 points**

In the ground state of Ge (Germanium; $Z = 32$) atom, **name** the following orbitals:

i. Orbital containing the electron that requires the maximum energy to be removed.

ii. Orbital containing the electron that experiences the minimum effective nuclear charge.

b. 2 points

An unknown anion (A^-) has the following *excited* electronic configuration: $[Ar] 4s^2 3d^3 4p^1$

(Ar: Argon; $Z = 18$)

Give the **condensed ground state electronic configuration** of the atom, **A** and the cation, **A^{2+}** .
Show your work. (Assume all electronic transitions are possible)

Electronic Configuration of A:

Electronic Configuration of A^{2+} :

c. 2 points

In a single atom, determine how many electrons can have the following set of quantum numbers?

(i) $n = 5; l = 3$

In a single atom, determine how many electrons can have the following set of quantum numbers?

(i) $n = 3; l = 2; m_s = +1/2$

**Question 4** **8 points**

Rank the following from 1 to 4 in **increasing** order. Provide the ranking in the boxes provided.

Note: Atomic number (Z) are provided for relevant atoms

a) Increasing (lowest to highest) order of *ionization energy*: (1= lowest; 4 = highest)

Mg

Mg: Z = 12

Rb

Rb: Z = 37

Cs

Cs: Z = 55

K

K : Z = 19

b) Increasing (lowest to highest) order of *bond angle*: (1= lowest; 4 = highest)

H₂OCCl₄XeF₄

HCN

c) Increasing (smallest to largest) order of *ionic size* (1= smallest; 4 = largest)

S²⁻

S: Z = 16

O²⁻

O: Z = 8

Be²⁺

Be: Z = 4

Na⁺

Na: Z = 11

d) Increasing (lowest to highest) order of *bond energy* (1= lowest; 4 = highest)

P-F

P-Cl

P-I

P-Br

**Question 5** **12 points**

a. 9 points For the molecules HClO_4 and TeF_6 , draw the most stable *Lewis structure* showing appropriate shape and *dashed/wedged bonds*, based on VSEPR. Determine the **molecular geometry** around the central atom. For HClO_4 , label the *most polar bond*.



b. 3 points (Note: Lone pairs are not shown in structures below)

(i) For the three *resonance* structures (1, 2 and 3) shown below, answer the following question:



(I) Circle the *most contributing* resonance structure.

Structure 1

Structure 2

Structure 3

(ii) For the three structures given (4, 5, and 6) below, answer the following questions:



(I) Circle all the structures that represent *resonance structures* of each other:

Structure 4

Structure 5

Structure 6

(II) Among the resonance structures, circle the *most contributing* resonance structure.

Structure 4

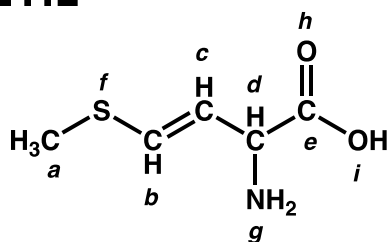
Structure 5

Structure 6

**Question 6** **6 points**

Using the data given below, calculate the **lattice energy** for SrO(s).

	ΔH° (kJ/mol)
$\text{Sr}(s) \longrightarrow \text{Sr}(g)$	+180
$\text{Sr}(g) \longrightarrow \text{Sr}^{2+}(g) + 2e^-$	+1468
$\text{O}_2(g) \longrightarrow 2\text{O}(g)$	+500
$\text{O}(g) + 2e^- \longrightarrow \text{O}^{2-}(g)$	+600
$\text{Sr}(s) + \frac{1}{2}\text{O}_2(g) \longrightarrow \text{SrO}(s)$	-550

**Question 7 10 points**

Consider the molecule given on the left, and answer questions (i) to (vi) using chemical bonding theories that we have learnt in class (valence bond theory, hybridization).

Note that the lone pairs are not shown in the structure:

i. Consider the carbon atoms marked in as ***a***, ***b***, ***c***, ***d***, and ***e*** in the molecule, and determine **hybridization** at each of the carbon centers. (2.5 points)

Atom	Hybridization
Carbon (<i>a</i>)	
Carbon (<i>b</i>)	
Carbon (<i>c</i>)	
Carbon (<i>d</i>)	
Carbon (<i>e</i>)	

ii. Now, consider the sulfur atom (S) marked as ***f***, nitrogen (N) atom (***g***), and oxygen(O) atoms (***h*** and ***i***), and determine their **hybridization**. (2 points)

Atom	Hybridization
Sulfur (<i>f</i>)	
Nitrogen (<i>g</i>)	
Oxygen (<i>h</i>)	
Oxygen (<i>i</i>)	

iii. **Describe** all the bonds formed by the carbon atoms ***a*** and ***b***, using sigma and pi designation as appropriate. (2 points)

Atom	Bonds formed (use sigma and pi as appropriate)
Carbon (<i>a</i>)	
Carbon (<i>b</i>)	

iv. How many total sp^3 -s sigma bonds are in the molecule? (1 point)

v. Determine **electronic** and **molecular geometry** at carbon (C) labeled ***e*** and sulfur (S) labeled ***f*** centres. (2 points)

Atom	Electronic Geometry	Molecular Geometry
Carbon (<i>e</i>)		
Sulfur (<i>f</i>)		

vi. Circle the value from the choices given below, that describes the **C-S-C bond angle at sulfur** (S) marked ***f***: (0.5 points)

>90° to <109.5° 109.5° >109.5 to <120° 120° >120° to <180°


Question 8 12 points

Using the **valence shell orbitals only** draw a **complete** molecular orbital (MO) diagram (with no s-p mixing) for the molecule **PS**, in the space provided below. **(6 points)**

Assume that 3s and 3p atomic orbitals will form molecular orbitals like those formed by the combination of 2s and 2p atomic orbitals that we learnt in class, and there is no s-p mixing.

Show **energy arrow**, **atomic orbitals** involved, **atom labels**, the **molecular orbitals** that are formed, **filled electrons** in the MO diagram etc. Make sure to label all the atomic and molecular orbitals.

Using the MO diagram above, answer the following questions:

- (i) Which one of the following will have the highest bond order? Circle the correct choice
 A) PS^{1-} B) PS C) PS^{1+}
- (ii) Which one of the following will have the longest bond length? Circle the correct choice
 A) PS^{1-} B) PS C) PS^{1+}
- (iii) Which of the following will be diamagnetic? Circle the correct choice.
 A) PS^{1-} B) PS C) PS^{1+}
- (iv) Draw Lewis dot structure of the molecule with the shortest bond length (based on the MO diagram) from among the three (PS^{1-} , PS, PS^{1+}), and indicate the bond order calculated from the Lewis dot structure. **(3 points)**

Lewis Dot Structure

Bond Order



DA13BE0C-316C-486E-AEB2-8F803FC65A85

midterm-exam-93388

#742 Page 10 of 10

SCRAP PAGE

**CHEMISTRY 110 –Midterm Exam****Thursday November 7, 2024, 6:30 – 8:30 pm**

Examiners: Pallavi Sirjoosingh and Ashok Kakkar

First name (please write as legibly as possible within the boxes)

--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--

Last name

--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--

Student ID number

--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--

This booklet has 8 questions on 10 pages (The last page can be used as a scrap page).

The staple must NOT be removed from the question booklet. Your intact question booklet must be returned at the end of the exam.

A separate booklet contains a periodic table, and datasheet.

Write your answers directly under the questions in the space provided. Do not write on the QR codes or near the top edges (as it may be cut off).

Please write **legibly and dark enough** for the scan to be readable. For students who use English as a second language, dictionaries are permitted. Calculators are allowed.

DO NOT ASK QUESTIONS OF THE INVIGILATORS about interpretations of the exam questions. If you have a concern about a question, please summarize it at the top of this page. All concerns will be evaluated after the exam. GOOD LUCK!

**Question 1 8 points** (*Assume all transitions are possible*)

a. 6 points Calculate the **energy (in kJ/mol)** required to ionize (completely remove) the electron from the **n=3** level of **He⁺ ion** (He: Helium). Show your work including equations. (3 sig. figs.)

b. 2 points

(I) Select the largest (longest) wavelength from the three choices below:

- i. Wavelength of photon to *completely remove* an electron from *1s orbital of Hydrogen atom*
- ii. Wavelength of photon to *completely remove* an electron from *1s orbital of Boron atom (B)*
- iii. Wavelength of photon to *completely remove* an electron from *1s orbital of Boron cation (B⁺)*

i or ii or iii

(II) Select the smallest (shortest) wavelength from the three choices below (*for H atom*):

- i. Wavelength of photon *emitted* for an electron transitioning from *2p to 1s orbital*
- ii. Wavelength of photon *absorbed* for an electron transitioning from *2s to 4p orbital*
- iii. Wavelength of photon *absorbed* for an electron transitioning from *1s to 3p orbital*

i or ii or iii



Question 2 8 points



A Chem110 student plots the graph shown on the left, from data obtained for the kinetic energy (eV) of the ejected electron versus frequency (s^{-1}) of the incident light, for an unknown metal. ($1\text{eV} = 1.6022 \times 10^{-19} \text{ J}$)

Based on the plot, answer the questions **a to c** below:

a. 1 point On the plot above, label the **threshold frequency**.

b. 2 points What is the value of the work potential (in J) of the unknown metal?

c. 3 points What is the *longest* wavelength (in nm) required to eject an electron from this metal's surface? Show your work including equation(s) used. (3 sig figs)

d. 2 points In the space provided below, plot the following

(Assume frequency is *above* threshold frequency):

- Kinetic energy* of an ejected electron versus the *intensity* of the incident light (left)
- Number of electrons* ejected versus the *frequency* of the incident light (right)



**Question 3** **6 points****a. 2 points**

In the ground state of Ge (Germanium; $Z = 32$) atom, **name** the following orbitals:

i. Orbital containing the electron that requires the maximum energy to be removed.

ii. Orbital containing the electron that experiences the minimum effective nuclear charge.

b. 2 points

An unknown anion (A^-) has the following *excited* electronic configuration: $[Ar] 4s^2 3d^3 4p^1$

(Ar: Argon; $Z = 18$)

Give the **condensed ground state electronic configuration** of the atom, **A** and the cation, **A^{2+}** .
Show your work. (Assume all electronic transitions are possible)

Electronic Configuration of A:

Electronic Configuration of A^{2+} :

c. 2 points

In a single atom, determine how many electrons can have the following set of quantum numbers?

(i) $n = 5; l = 3$

In a single atom, determine how many electrons can have the following set of quantum numbers?

(i) $n = 3; l = 2; m_s = +1/2$

**Question 4** **8 points**

Rank the following from 1 to 4 in **increasing** order. Provide the ranking in the boxes provided.

Note: Atomic number (Z) are provided for relevant atoms

a) Increasing (lowest to highest) order of *ionization energy*: (1= lowest; 4 = highest)

Mg

Mg: Z = 12

Rb

Rb: Z = 37

Cs

Cs: Z = 55

K

K : Z = 19

b) Increasing (lowest to highest) order of *bond angle*: (1= lowest; 4 = highest)

H₂OCCl₄XeF₄

HCN

c) Increasing (smallest to largest) order of *ionic size* (1= smallest; 4 = largest)

S²⁻

S: Z = 16

O²⁻

O: Z = 8

Be²⁺

Be: Z = 4

Na⁺

Na: Z = 11

d) Increasing (lowest to highest) order of *bond energy* (1= lowest; 4 = highest)

P-F

P-Cl

P-I

P-Br

**Question 5** **12 points**

a. 9 points For the molecules HClO_4 and TeF_6 , draw the most stable *Lewis structure* showing appropriate shape and *dashed/wedged bonds*, based on VSEPR. Determine the **molecular geometry** around the central atom. For HClO_4 , label the *most polar bond*.



b. 3 points (Note: Lone pairs are not shown in structures below)

(i) For the three *resonance* structures (1, 2 and 3) shown below, answer the following question:



(I) Circle the *most contributing* resonance structure.

Structure 1

Structure 2

Structure 3

(ii) For the three structures given (4, 5, and 6) below, answer the following questions:



(I) Circle all the structures that represent *resonance structures* of each other:

Structure 4

Structure 5

Structure 6

(II) Among the resonance structures, circle the *most contributing* resonance structure.

Structure 4

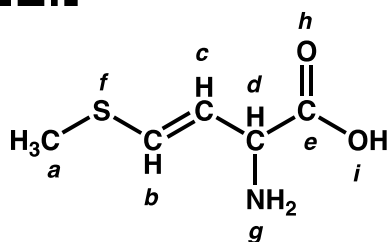
Structure 5

Structure 6

**Question 6** **6 points**

Using the data given below, calculate the **lattice energy** for SrO(s).

	ΔH° (kJ/mol)
$\text{Sr}(s) \longrightarrow \text{Sr}(g)$	+180
$\text{Sr}(g) \longrightarrow \text{Sr}^{2+}(g) + 2e^-$	+1468
$\text{O}_2(g) \longrightarrow 2\text{O}(g)$	+500
$\text{O}(g) + 2e^- \longrightarrow \text{O}^{2-}(g)$	+600
$\text{Sr}(s) + \frac{1}{2}\text{O}_2(g) \longrightarrow \text{SrO}(s)$	-550

**Question 7 10 points**

Consider the molecule given on the left, and answer questions (i) to (vi) using chemical bonding theories that we have learnt in class (valence bond theory, hybridization).

Note that the lone pairs are not shown in the structure:

i. Consider the carbon atoms marked in as ***a***, ***b***, ***c***, ***d***, and ***e*** in the molecule, and determine **hybridization** at each of the carbon centers. (2.5 points)

Atom	Hybridization
Carbon (<i>a</i>)	
Carbon (<i>b</i>)	
Carbon (<i>c</i>)	
Carbon (<i>d</i>)	
Carbon (<i>e</i>)	

ii. Now, consider the sulfur atom (S) marked as ***f***, nitrogen (N) atom (***g***), and oxygen(O) atoms (***h*** and ***i***), and determine their **hybridization**. (2 points)

Atom	Hybridization
Sulfur (<i>f</i>)	
Nitrogen (<i>g</i>)	
Oxygen (<i>h</i>)	
Oxygen (<i>i</i>)	

iii. **Describe** all the bonds formed by the carbon atoms ***a*** and ***b***, using sigma and pi designation as appropriate. (2 points)

Atom	Bonds formed (use sigma and pi as appropriate)
Carbon (<i>a</i>)	
Carbon (<i>b</i>)	

iv. How many total sp^3 -s sigma bonds are in the molecule? (1 point)

v. Determine **electronic** and **molecular geometry** at carbon (C) labeled ***e*** and sulfur (S) labeled ***f*** centres. (2 points)

Atom	Electronic Geometry	Molecular Geometry
Carbon (<i>e</i>)		
Sulfur (<i>f</i>)		

vi. Circle the value from the choices given below, that describes the **C-S-C bond angle at sulfur** (S) marked ***f***: (0.5 points)

>90° to <109.5° 109.5° >109.5 to <120° 120° >120° to <180°


Question 8 12 points

Using the **valence shell orbitals only** draw a **complete** molecular orbital (MO) diagram (with no s-p mixing) for the molecule **PS**, in the space provided below. **(6 points)**

Assume that 3s and 3p atomic orbitals will form molecular orbitals like those formed by the combination of 2s and 2p atomic orbitals that we learnt in class, and there is no s-p mixing.

Show **energy arrow**, **atomic orbitals** involved, **atom labels**, the **molecular orbitals** that are formed, **filled electrons** in the MO diagram etc. Make sure to label all the atomic and molecular orbitals.

Using the MO diagram above, answer the following questions:

- (i) Which one of the following will have the highest bond order? Circle the correct choice
 A) PS^{1-} B) PS C) PS^{1+}
- (ii) Which one of the following will have the longest bond length? Circle the correct choice
 A) PS^{1-} B) PS C) PS^{1+}
- (iii) Which of the following will be diamagnetic? Circle the correct choice.
 A) PS^{1-} B) PS C) PS^{1+}
- (iv) Draw Lewis dot structure of the molecule with the shortest bond length (based on the MO diagram) from among the three (PS^{1-} , PS, PS^{1+}), and indicate the bond order calculated from the Lewis dot structure. **(3 points)**

Lewis Dot Structure

Bond Order



B40A558E-F67E-4C34-8135-161DCFD9E876

midterm-exam-93388

#743 Page 10 of 10

SCRAP PAGE

**CHEMISTRY 110 –Midterm Exam****Thursday November 7, 2024, 6:30 – 8:30 pm**

Examiners: Pallavi Sirjoosingh and Ashok Kakkar

First name (please write as legibly as possible within the boxes)

--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--

Last name

--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--

Student ID number

--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--

This booklet has 8 questions on 10 pages (The last page can be used as a scrap page).

The staple must NOT be removed from the question booklet. Your intact question booklet must be returned at the end of the exam.

A separate booklet contains a periodic table, and datasheet.

Write your answers directly under the questions in the space provided. Do not write on the QR codes or near the top edges (as it may be cut off).

Please write **legibly and dark enough** for the scan to be readable. For students who use English as a second language, dictionaries are permitted. Calculators are allowed.

DO NOT ASK QUESTIONS OF THE INVIGILATORS about interpretations of the exam questions. If you have a concern about a question, please summarize it at the top of this page. All concerns will be evaluated after the exam. GOOD LUCK!

**Question 1 8 points** (*Assume all transitions are possible*)

a. 6 points Calculate the **energy (in kJ/mol)** required to ionize (completely remove) the electron from the **n=3** level of **He⁺ ion** (He: Helium). Show your work including equations. (3 sig. figs.)

b. 2 points

(I) Select the largest (longest) wavelength from the three choices below:

- i. Wavelength of photon to *completely remove* an electron from *1s orbital of Hydrogen atom*
- ii. Wavelength of photon to *completely remove* an electron from *1s orbital of Boron atom (B)*
- iii. Wavelength of photon to *completely remove* an electron from *1s orbital of Boron cation (B⁺)*

i or ii or iii

(II) Select the smallest (shortest) wavelength from the three choices below (*for H atom*):

- i. Wavelength of photon *emitted* for an electron transitioning from *2p to 1s orbital*
- ii. Wavelength of photon *absorbed* for an electron transitioning from *2s to 4p orbital*
- iii. Wavelength of photon *absorbed* for an electron transitioning from *1s to 3p orbital*

i or ii or iii



Question 2 8 points



A Chem110 student plots the graph shown on the left, from data obtained for the kinetic energy (eV) of the ejected electron versus frequency (s^{-1}) of the incident light, for an unknown metal. ($1\text{eV} = 1.6022 \times 10^{-19} \text{ J}$)

Based on the plot, answer the questions **a to c** below:

a. 1 point On the plot above, label the **threshold frequency**.

b. 2 points What is the value of the work potential (in J) of the unknown metal?

c. 3 points What is the *longest* wavelength (in nm) required to eject an electron from this metal's surface? Show your work including equation(s) used. (3 sig figs)

d. 2 points In the space provided below, plot the following

(Assume frequency is *above* threshold frequency):

- i. *Kinetic energy* of an ejected electron versus the *intensity* of the incident light (left)
- ii. *Number of electrons* ejected versus the *frequency* of the incident light (right)



**Question 3** **6 points****a. 2 points**

In the ground state of Ge (Germanium; $Z = 32$) atom, **name** the following orbitals:

i. Orbital containing the electron that requires the maximum energy to be removed.

ii. Orbital containing the electron that experiences the minimum effective nuclear charge.

b. 2 points

An unknown anion (A^-) has the following *excited* electronic configuration: $[Ar] 4s^2 3d^3 4p^1$

(Ar: Argon; $Z = 18$)

Give the **condensed ground state electronic configuration** of the atom, **A** and the cation, **A^{2+}** .
Show your work. (Assume all electronic transitions are possible)

Electronic Configuration of A:

Electronic Configuration of A^{2+} :

c. 2 points

In a single atom, determine how many electrons can have the following set of quantum numbers?

(i) $n = 5; l = 3$

In a single atom, determine how many electrons can have the following set of quantum numbers?

(i) $n = 3; l = 2; m_s = +1/2$

**Question 4** **8 points**

Rank the following from 1 to 4 in **increasing** order. Provide the ranking in the boxes provided.

Note: Atomic number (Z) are provided for relevant atoms

a) Increasing (lowest to highest) order of *ionization energy*: (1= lowest; 4 = highest)

Mg

Mg: Z = 12

Rb

Rb: Z = 37

Cs

Cs: Z = 55

K

K : Z = 19

b) Increasing (lowest to highest) order of *bond angle*: (1= lowest; 4 = highest)

H₂OCCl₄XeF₄

HCN

c) Increasing (smallest to largest) order of *ionic size* (1= smallest; 4 = largest)

S²⁻

S: Z = 16

O²⁻

O: Z = 8

Be²⁺

Be: Z = 4

Na⁺

Na: Z = 11

d) Increasing (lowest to highest) order of *bond energy* (1= lowest; 4 = highest)

P-F

P-Cl

P-I

P-Br

**Question 5** **12 points**

a. 9 points For the molecules HClO_4 and TeF_6 , draw the most stable *Lewis structure* showing appropriate shape and *dashed/wedged bonds*, based on VSEPR. Determine the **molecular geometry** around the central atom. For HClO_4 , label the *most polar bond*.



b. 3 points (Note: Lone pairs are not shown in structures below)

(i) For the three *resonance* structures (1, 2 and 3) shown below, answer the following question:



(I) Circle the *most contributing* resonance structure.

Structure 1

Structure 2

Structure 3

(ii) For the three structures given (4, 5, and 6) below, answer the following questions:



(I) Circle all the structures that represent *resonance structures* of each other:

Structure 4

Structure 5

Structure 6

(II) Among the resonance structures, circle the *most contributing* resonance structure.

Structure 4

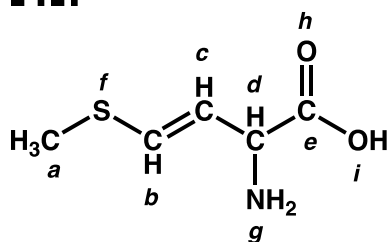
Structure 5

Structure 6

**Question 6** **6 points**

Using the data given below, calculate the **lattice energy** for SrO(s).

	ΔH° (kJ/mol)
$\text{Sr}(s) \longrightarrow \text{Sr}(g)$	+180
$\text{Sr}(g) \longrightarrow \text{Sr}^{2+}(g) + 2e^-$	+1468
$\text{O}_2(g) \longrightarrow 2\text{O}(g)$	+500
$\text{O}(g) + 2e^- \longrightarrow \text{O}^{2-}(g)$	+600
$\text{Sr}(s) + \frac{1}{2}\text{O}_2(g) \longrightarrow \text{SrO}(s)$	-550

**Question 7 10 points**

Consider the molecule given on the left, and answer questions (i) to (vi) using chemical bonding theories that we have learnt in class (valence bond theory, hybridization).

Note that the lone pairs are not shown in the structure:

i. Consider the carbon atoms marked in as ***a***, ***b***, ***c***, ***d***, and ***e*** in the molecule, and determine **hybridization** at each of the carbon centers. (2.5 points)

Atom	Hybridization
Carbon (<i>a</i>)	
Carbon (<i>b</i>)	
Carbon (<i>c</i>)	
Carbon (<i>d</i>)	
Carbon (<i>e</i>)	

ii. Now, consider the sulfur atom (S) marked as ***f***, nitrogen (N) atom (***g***), and oxygen(O) atoms (***h*** and ***i***), and determine their **hybridization**. (2 points)

Atom	Hybridization
Sulfur (<i>f</i>)	
Nitrogen (<i>g</i>)	
Oxygen (<i>h</i>)	
Oxygen (<i>i</i>)	

iii. **Describe** all the bonds formed by the carbon atoms ***a*** and ***b***, using sigma and pi designation as appropriate. (2 points)

Atom	Bonds formed (use sigma and pi as appropriate)
Carbon (<i>a</i>)	
Carbon (<i>b</i>)	

iv. How many total sp^3 -s sigma bonds are in the molecule? (1 point)

v. Determine **electronic** and **molecular geometry** at carbon (C) labeled ***e*** and sulfur (S) labeled ***f*** centres. (2 points)

Atom	Electronic Geometry	Molecular Geometry
Carbon (<i>e</i>)		
Sulfur (<i>f</i>)		

vi. Circle the value from the choices given below, that describes the **C-S-C bond angle at sulfur** (S) marked ***f***: (0.5 points)

>90° to <109.5° 109.5° >109.5 to <120° 120° >120° to <180°


Question 8 12 points

Using the **valence shell orbitals only** draw a **complete** molecular orbital (MO) diagram (with no s-p mixing) for the molecule **PS**, in the space provided below. **(6 points)**

Assume that 3s and 3p atomic orbitals will form molecular orbitals like those formed by the combination of 2s and 2p atomic orbitals that we learnt in class, and there is no s-p mixing.

Show **energy arrow**, **atomic orbitals** involved, **atom labels**, the **molecular orbitals** that are formed, **filled electrons** in the MO diagram etc. Make sure to label all the atomic and molecular orbitals.

Using the MO diagram above, answer the following questions:

- (i) Which one of the following will have the highest bond order? Circle the correct choice
 A) PS^{1-} B) PS C) PS^{1+}
- (ii) Which one of the following will have the longest bond length? Circle the correct choice
 A) PS^{1-} B) PS C) PS^{1+}
- (iii) Which of the following will be diamagnetic? Circle the correct choice.
 A) PS^{1-} B) PS C) PS^{1+}
- (iv) Draw Lewis dot structure of the molecule with the shortest bond length (based on the MO diagram) from among the three (PS^{1-} , PS, PS^{1+}), and indicate the bond order calculated from the Lewis dot structure. **(3 points)**

Lewis Dot Structure

Bond Order



7E08C387-4F14-40CF-8F02-8DF2F2A27C4C

midterm-exam-93388

#744 Page 10 of 10

SCRAP PAGE

**CHEMISTRY 110 –Midterm Exam****Thursday November 7, 2024, 6:30 – 8:30 pm****Examiners: Pallavi Sirjoosingh and Ashok Kakkar**

First name (please write as legibly as possible within the boxes)

--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--

Last name

--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--

Student ID number

--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--

This booklet has 8 questions on 10 pages (The last page can be used as a scrap page).

The staple must NOT be removed from the question booklet. Your intact question booklet must be returned at the end of the exam.

A separate booklet contains a periodic table, and datasheet.

Write your answers directly under the questions in the space provided. Do not write on the QR codes or near the top edges (as it may be cut off).

Please write **legibly and dark enough** for the scan to be readable. For students who use English as a second language, dictionaries are permitted. Calculators are allowed.

DO NOT ASK QUESTIONS OF THE INVIGILATORS about interpretations of the exam questions. If you have a concern about a question, please summarize it at the top of this page. All concerns will be evaluated after the exam. GOOD LUCK!

**Question 1 8 points** (*Assume all transitions are possible*)

a. 6 points Calculate the **energy (in kJ/mol)** required to ionize (completely remove) the electron from the **$n=3$** level of **He^+ ion** (He: Helium). Show your work including equations. (3 sig. figs.)

b. 2 points

(I) Select the largest (longest) wavelength from the three choices below:

- i. Wavelength of photon to *completely remove* an electron from *1s orbital of Hydrogen atom*
- ii. Wavelength of photon to *completely remove* an electron from *1s orbital of Boron atom (B)*
- iii. Wavelength of photon to *completely remove* an electron from *1s orbital of Boron cation (B^+)*

i or ii or iii

(II) Select the smallest (shortest) wavelength from the three choices below (*for H atom*):

- i. Wavelength of photon *emitted* for an electron transitioning from *2p to 1s orbital*
- ii. Wavelength of photon *absorbed* for an electron transitioning from *2s to 4p orbital*
- iii. Wavelength of photon *absorbed* for an electron transitioning from *1s to 3p orbital*

i or ii or iii



Question 2 8 points



A Chem110 student plots the graph shown on the left, from data obtained for the kinetic energy (eV) of the ejected electron versus frequency (s^{-1}) of the incident light, for an unknown metal. ($1\text{eV} = 1.6022 \times 10^{-19} \text{ J}$)

Based on the plot, answer the questions **a to c** below:

a. 1 point On the plot above, label the **threshold frequency**.

b. 2 points What is the value of the work potential (in J) of the unknown metal?

c. 3 points What is the *longest* wavelength (in nm) required to eject an electron from this metal's surface? Show your work including equation(s) used. (3 sig figs)

d. 2 points In the space provided below, plot the following

(Assume frequency is *above* threshold frequency):

- i. *Kinetic energy* of an ejected electron versus the *intensity* of the incident light (left)
- ii. *Number of electrons* ejected versus the *frequency* of the incident light (right)



**Question 3** **6 points****a. 2 points**

In the ground state of Ge (Germanium; $Z = 32$) atom, **name** the following orbitals:

i. Orbital containing the electron that requires the maximum energy to be removed.

ii. Orbital containing the electron that experiences the minimum effective nuclear charge.

b. 2 points

An unknown anion (A^-) has the following *excited* electronic configuration: $[Ar] 4s^2 3d^3 4p^1$

(Ar: Argon; $Z = 18$)

Give the **condensed ground state electronic configuration** of the atom, **A** and the cation, **A^{2+}** .
Show your work. (Assume all electronic transitions are possible)

Electronic Configuration of A:

Electronic Configuration of A^{2+} :

c. 2 points

In a single atom, determine how many electrons can have the following set of quantum numbers?

(i) $n = 5; l = 3$

In a single atom, determine how many electrons can have the following set of quantum numbers?

(i) $n = 3; l = 2; m_s = +1/2$

**Question 4** **8 points**

Rank the following from 1 to 4 in **increasing** order. Provide the ranking in the boxes provided.

Note: Atomic number (Z) are provided for relevant atoms

a) Increasing (lowest to highest) order of *ionization energy*: (1= lowest; 4 = highest)

Mg

Mg: Z = 12

Rb

Rb: Z = 37

Cs

Cs: Z = 55

K

K : Z = 19

b) Increasing (lowest to highest) order of *bond angle*: (1= lowest; 4 = highest)

H₂OCCl₄XeF₄

HCN

c) Increasing (smallest to largest) order of *ionic size* (1= smallest; 4 = largest)

S²⁻

S: Z = 16

O²⁻

O: Z = 8

Be²⁺

Be: Z = 4

Na⁺

Na: Z = 11

d) Increasing (lowest to highest) order of *bond energy* (1= lowest; 4 = highest)

P-F

P-Cl

P-I

P-Br

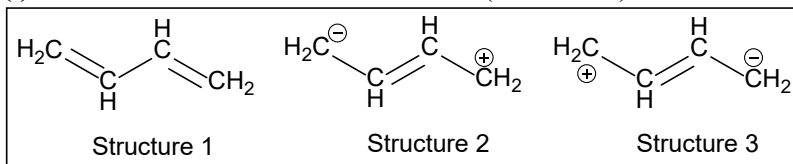
**Question 5** **12 points**

a. 9 points For the molecules HClO_4 and TeF_6 , draw the most stable *Lewis structure* showing appropriate shape and *dashed/wedged bonds*, based on VSEPR. Determine the **molecular geometry** around the central atom. For HClO_4 , label the *most polar bond*.



b. 3 points (Note: Lone pairs are not shown in structures below)

(i) For the three *resonance* structures (1, 2 and 3) shown below, answer the following question:



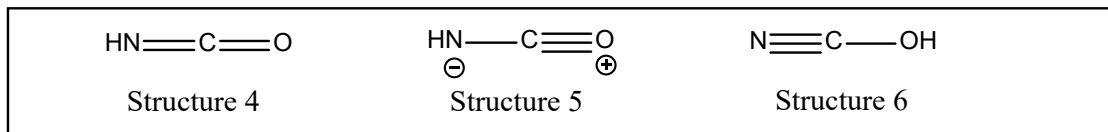
(I) Circle the *most contributing* resonance structure.

Structure 1

Structure 2

Structure 3

(ii) For the three structures given (4, 5, and 6) below, answer the following questions:



(I) Circle all the structures that represent *resonance structures* of each other:

Structure 4

Structure 5

Structure 6

(II) Among the resonance structures, circle the *most contributing* resonance structure.

Structure 4

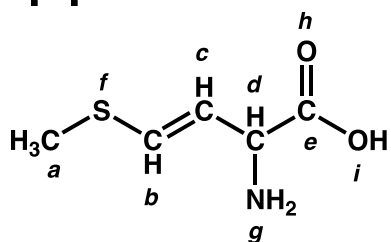
Structure 5

Structure 6

**Question 6** **6 points**

Using the data given below, calculate the **lattice energy** for SrO(s).

	ΔH° (kJ/mol)
$\text{Sr}(s) \longrightarrow \text{Sr}(g)$	+180
$\text{Sr}(g) \longrightarrow \text{Sr}^{2+}(g) + 2e^-$	+1468
$\text{O}_2(g) \longrightarrow 2\text{O}(g)$	+500
$\text{O}(g) + 2e^- \longrightarrow \text{O}^{2-}(g)$	+600
$\text{Sr}(s) + \frac{1}{2}\text{O}_2(g) \longrightarrow \text{SrO}(s)$	-550

**Question 7 10 points**

Consider the molecule given on the left, and answer questions (i) to (vi) using chemical bonding theories that we have learnt in class (valence bond theory, hybridization).

Note that the lone pairs are not shown in the structure:

i. Consider the carbon atoms marked in as ***a***, ***b***, ***c***, ***d***, and ***e*** in the molecule, and determine **hybridization** at each of the carbon centers. (2.5 points)

Atom	Hybridization
Carbon (<i>a</i>)	
Carbon (<i>b</i>)	
Carbon (<i>c</i>)	
Carbon (<i>d</i>)	
Carbon (<i>e</i>)	

ii. Now, consider the sulfur atom (S) marked as ***f***, nitrogen (N) atom (***g***), and oxygen(O) atoms (***h*** and ***i***), and determine their **hybridization**. (2 points)

Atom	Hybridization
Sulfur (<i>f</i>)	
Nitrogen (<i>g</i>)	
Oxygen (<i>h</i>)	
Oxygen (<i>i</i>)	

iii. **Describe** all the bonds formed by the carbon atoms ***a*** and ***b***, using sigma and pi designation as appropriate. (2 points)

Atom	Bonds formed (use sigma and pi as appropriate)
Carbon (<i>a</i>)	
Carbon (<i>b</i>)	

iv. How many total sp^3 -s sigma bonds are in the molecule? (1 point)

v. Determine **electronic** and **molecular geometry** at carbon (C) labeled ***e*** and sulfur (S) labeled ***f*** centres. (2 points)

Atom	Electronic Geometry	Molecular Geometry
Carbon (<i>e</i>)		
Sulfur (<i>f</i>)		

vi. Circle the value from the choices given below, that describes the **C-S-C bond angle at sulfur** (S) marked ***f***: (0.5 points)

>90° to <109.5° 109.5° >109.5 to <120° 120° >120° to <180°


Question 8 12 points

Using the **valence shell orbitals only** draw a **complete** molecular orbital (MO) diagram (with no s-p mixing) for the molecule **PS**, in the space provided below. **(6 points)**

Assume that 3s and 3p atomic orbitals will form molecular orbitals like those formed by the combination of 2s and 2p atomic orbitals that we learnt in class, and there is no s-p mixing.

Show **energy arrow**, **atomic orbitals** involved, **atom labels**, the **molecular orbitals** that are formed, **filled electrons** in the MO diagram etc. Make sure to label all the atomic and molecular orbitals.

Using the MO diagram above, answer the following questions:

- (i) Which one of the following will have the highest bond order? Circle the correct choice
 A) PS^{1-} B) PS C) PS^{1+}
- (ii) Which one of the following will have the longest bond length? Circle the correct choice
 A) PS^{1-} B) PS C) PS^{1+}
- (iii) Which of the following will be diamagnetic? Circle the correct choice.
 A) PS^{1-} B) PS C) PS^{1+}
- (iv) Draw Lewis dot structure of the molecule with the shortest bond length (based on the MO diagram) from among the three (PS^{1-} , PS, PS^{1+}), and indicate the bond order calculated from the Lewis dot structure. **(3 points)**

Lewis Dot Structure

Bond Order



447A1826-0F26-4EAB-85F2-7E158BB9EBAC

midterm-exam-93388

#745 Page 10 of 10

SCRAP PAGE

**CHEMISTRY 110 –Midterm Exam****Thursday November 7, 2024, 6:30 – 8:30 pm****Examiners: Pallavi Sirjoosingh and Ashok Kakkar**

First name (please write as legibly as possible within the boxes)

--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--

Last name

--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--

Student ID number

--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--

This booklet has 8 questions on 10 pages (The last page can be used as a scrap page).

The staple must NOT be removed from the question booklet. Your intact question booklet must be returned at the end of the exam.

A separate booklet contains a periodic table, and datasheet.

Write your answers directly under the questions in the space provided. Do not write on the QR codes or near the top edges (as it may be cut off).

Please write **legibly and dark enough** for the scan to be readable. For students who use English as a second language, dictionaries are permitted. Calculators are allowed.

DO NOT ASK QUESTIONS OF THE INVIGILATORS about interpretations of the exam questions. If you have a concern about a question, please summarize it at the top of this page. All concerns will be evaluated after the exam. GOOD LUCK!

**Question 1 8 points** (*Assume all transitions are possible*)

a. 6 points Calculate the **energy (in kJ/mol)** required to ionize (completely remove) the electron from the **$n=3$** level of **He^+ ion** (He: Helium). Show your work including equations. (3 sig. figs.)

b. 2 points

(I) Select the largest (longest) wavelength from the three choices below:

- i. Wavelength of photon to *completely remove* an electron from *1s orbital of Hydrogen atom*
- ii. Wavelength of photon to *completely remove* an electron from *1s orbital of Boron atom (B)*
- iii. Wavelength of photon to *completely remove* an electron from *1s orbital of Boron cation (B^+)*

i or ii or iii

(II) Select the smallest (shortest) wavelength from the three choices below (*for H atom*):

- i. Wavelength of photon *emitted* for an electron transitioning from *2p to 1s orbital*
- ii. Wavelength of photon *absorbed* for an electron transitioning from *2s to 4p orbital*
- iii. Wavelength of photon *absorbed* for an electron transitioning from *1s to 3p orbital*

i or ii or iii



Question 2 8 points



A Chem110 student plots the graph shown on the left, from data obtained for the kinetic energy (eV) of the ejected electron versus frequency (s^{-1}) of the incident light, for an unknown metal. ($1\text{eV} = 1.6022 \times 10^{-19} \text{ J}$)

Based on the plot, answer the questions **a to c** below:

a. 1 point On the plot above, label the **threshold frequency**.

b. 2 points What is the value of the work potential (in J) of the unknown metal?

c. 3 points What is the *longest* wavelength (in nm) required to eject an electron from this metal's surface? Show your work including equation(s) used. (3 sig figs)

d. 2 points In the space provided below, plot the following

(Assume frequency is *above* threshold frequency):

- Kinetic energy* of an ejected electron versus the *intensity* of the incident light (left)
- Number of electrons* ejected versus the *frequency* of the incident light (right)



**Question 3** **6 points****a. 2 points**

In the ground state of Ge (Germanium; $Z = 32$) atom, **name** the following orbitals:

i. Orbital containing the electron that requires the maximum energy to be removed.

ii. Orbital containing the electron that experiences the minimum effective nuclear charge.

b. 2 points

An unknown anion (A^-) has the following *excited* electronic configuration: $[Ar] 4s^2 3d^3 4p^1$

(Ar: Argon; $Z = 18$)

Give the **condensed ground state electronic configuration** of the atom, **A** and the cation, **A^{2+}** .
Show your work. (Assume all electronic transitions are possible)

Electronic Configuration of A:

Electronic Configuration of A^{2+} :

c. 2 points

In a single atom, determine how many electrons can have the following set of quantum numbers?

(i) $n = 5; l = 3$

In a single atom, determine how many electrons can have the following set of quantum numbers?

(i) $n = 3; l = 2; m_s = +1/2$

**Question 4** **8 points**

Rank the following from 1 to 4 in **increasing** order. Provide the ranking in the boxes provided.

Note: Atomic number (Z) are provided for relevant atoms

a) Increasing (lowest to highest) order of *ionization energy*: (1= lowest; 4 = highest)

Mg

Mg: Z = 12

Rb

Rb: Z = 37

Cs

Cs: Z = 55

K

K : Z = 19

b) Increasing (lowest to highest) order of *bond angle*: (1= lowest; 4 = highest)

H₂OCCl₄XeF₄

HCN

c) Increasing (smallest to largest) order of *ionic size* (1= smallest; 4 = largest)

S²⁻

S: Z = 16

O²⁻

O: Z = 8

Be²⁺

Be: Z = 4

Na⁺

Na: Z = 11

d) Increasing (lowest to highest) order of *bond energy* (1= lowest; 4 = highest)

P-F

P-Cl

P-I

P-Br

**Question 5** **12 points**

a. 9 points For the molecules HClO_4 and TeF_6 , draw the most stable *Lewis structure* showing appropriate shape and *dashed/wedged bonds*, based on VSEPR. Determine the **molecular geometry** around the central atom. For HClO_4 , label the *most polar bond*.



b. 3 points (Note: Lone pairs are not shown in structures below)

(i) For the three *resonance* structures (1, 2 and 3) shown below, answer the following question:



(I) Circle the *most contributing* resonance structure.

Structure 1

Structure 2

Structure 3

(ii) For the three structures given (4, 5, and 6) below, answer the following questions:



(I) Circle all the structures that represent *resonance structures* of each other:

Structure 4

Structure 5

Structure 6

(II) Among the resonance structures, circle the *most contributing* resonance structure.

Structure 4

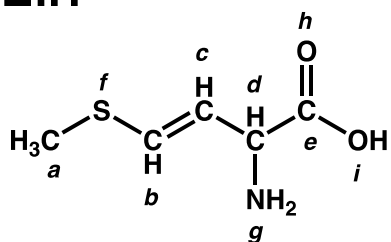
Structure 5

Structure 6

**Question 6** **6 points**

Using the data given below, calculate the **lattice energy** for SrO(s).

	ΔH° (kJ/mol)
$\text{Sr}(s) \longrightarrow \text{Sr}(g)$	+180
$\text{Sr}(g) \longrightarrow \text{Sr}^{2+}(g) + 2e^-$	+1468
$\text{O}_2(g) \longrightarrow 2\text{O}(g)$	+500
$\text{O}(g) + 2e^- \longrightarrow \text{O}^{2-}(g)$	+600
$\text{Sr}(s) + \frac{1}{2}\text{O}_2(g) \longrightarrow \text{SrO}(s)$	-550

**Question 7 10 points**

Consider the molecule given on the left, and answer questions (i) to (vi) using chemical bonding theories that we have learnt in class (valence bond theory, hybridization).

Note that the lone pairs are not shown in the structure:

i. Consider the carbon atoms marked in as ***a***, ***b***, ***c***, ***d***, and ***e*** in the molecule, and determine **hybridization** at each of the carbon centers. (2.5 points)

Atom	Hybridization
Carbon (<i>a</i>)	
Carbon (<i>b</i>)	
Carbon (<i>c</i>)	
Carbon (<i>d</i>)	
Carbon (<i>e</i>)	

ii. Now, consider the sulfur atom (S) marked as ***f***, nitrogen (N) atom (***g***), and oxygen(O) atoms (***h*** and ***i***), and determine their **hybridization**. (2 points)

Atom	Hybridization
Sulfur (<i>f</i>)	
Nitrogen (<i>g</i>)	
Oxygen (<i>h</i>)	
Oxygen (<i>i</i>)	

iii. **Describe** all the bonds formed by the carbon atoms ***a*** and ***b***, using sigma and pi designation as appropriate. (2 points)

Atom	Bonds formed (use sigma and pi as appropriate)
Carbon (<i>a</i>)	
Carbon (<i>b</i>)	

iv. How many total sp^3 -s sigma bonds are in the molecule? (1 point)

v. Determine **electronic** and **molecular geometry** at carbon (C) labeled ***e*** and sulfur (S) labeled ***f*** centres. (2 points)

Atom	Electronic Geometry	Molecular Geometry
Carbon (<i>e</i>)		
Sulfur (<i>f</i>)		

vi. Circle the value from the choices given below, that describes the **C-S-C bond angle at sulfur** (S) marked ***f***: (0.5 points)

>90° to <109.5° 109.5° >109.5 to <120° 120° >120° to <180°


Question 8 12 points

Using the **valence shell orbitals only** draw a **complete** molecular orbital (MO) diagram (with no s-p mixing) for the molecule **PS**, in the space provided below. **(6 points)**

Assume that 3s and 3p atomic orbitals will form molecular orbitals like those formed by the combination of 2s and 2p atomic orbitals that we learnt in class, and there is no s-p mixing.

Show **energy arrow**, **atomic orbitals** involved, **atom labels**, the **molecular orbitals** that are formed, **filled electrons** in the MO diagram etc. Make sure to label all the atomic and molecular orbitals.

Using the MO diagram above, answer the following questions:

- (i) Which one of the following will have the highest bond order? Circle the correct choice
 A) PS^{1-} B) PS C) PS^{1+}
- (ii) Which one of the following will have the longest bond length? Circle the correct choice
 A) PS^{1-} B) PS C) PS^{1+}
- (iii) Which of the following will be diamagnetic? Circle the correct choice.
 A) PS^{1-} B) PS C) PS^{1+}
- (iv) Draw Lewis dot structure of the molecule with the shortest bond length (based on the MO diagram) from among the three (PS^{1-} , PS, PS^{1+}), and indicate the bond order calculated from the Lewis dot structure. **(3 points)**

Lewis Dot Structure

Bond Order



931498BE-A6D9-4C7E-B9A1-762868E27817

midterm-exam-93388

#746 Page 10 of 10

SCRAP PAGE

**CHEMISTRY 110 –Midterm Exam****Thursday November 7, 2024, 6:30 – 8:30 pm**

Examiners: Pallavi Sirjoosingh and Ashok Kakkar

First name (please write as legibly as possible within the boxes)

--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--

Last name

--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--

Student ID number

--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--

This booklet has 8 questions on 10 pages (The last page can be used as a scrap page).

The staple must NOT be removed from the question booklet. Your intact question booklet must be returned at the end of the exam.

A separate booklet contains a periodic table, and datasheet.

Write your answers directly under the questions in the space provided. Do not write on the QR codes or near the top edges (as it may be cut off).

Please write **legibly and dark enough** for the scan to be readable. For students who use English as a second language, dictionaries are permitted. Calculators are allowed.

DO NOT ASK QUESTIONS OF THE INVIGILATORS about interpretations of the exam questions. If you have a concern about a question, please summarize it at the top of this page. All concerns will be evaluated after the exam. GOOD LUCK!

**Question 1 8 points** (*Assume all transitions are possible*)

a. 6 points Calculate the **energy (in kJ/mol)** required to ionize (completely remove) the electron from the **n=3** level of **He⁺ ion** (He: Helium). Show your work including equations. (3 sig. figs.)

b. 2 points

(I) Select the largest (longest) wavelength from the three choices below:

- i. Wavelength of photon to *completely remove* an electron from *1s orbital of Hydrogen atom*
- ii. Wavelength of photon to *completely remove* an electron from *1s orbital of Boron atom (B)*
- iii. Wavelength of photon to *completely remove* an electron from *1s orbital of Boron cation (B⁺)*

i or ii or iii

(II) Select the smallest (shortest) wavelength from the three choices below (*for H atom*):

- i. Wavelength of photon *emitted* for an electron transitioning from *2p to 1s orbital*
- ii. Wavelength of photon *absorbed* for an electron transitioning from *2s to 4p orbital*
- iii. Wavelength of photon *absorbed* for an electron transitioning from *1s to 3p orbital*

i or ii or iii



Question 2 8 points



A Chem110 student plots the graph shown on the left, from data obtained for the kinetic energy (eV) of the ejected electron versus frequency (s^{-1}) of the incident light, for an unknown metal. ($1\text{eV} = 1.6022 \times 10^{-19} \text{ J}$)

Based on the plot, answer the questions **a to c** below:

a. 1 point On the plot above, label the **threshold frequency**.

b. 2 points What is the value of the work potential (in J) of the unknown metal?

c. 3 points What is the *longest* wavelength (in nm) required to eject an electron from this metal's surface? Show your work including equation(s) used. (3 sig figs)

d. 2 points In the space provided below, plot the following

(Assume frequency is *above* threshold frequency):

- i. *Kinetic energy* of an ejected electron versus the *intensity* of the incident light (left)
- ii. *Number of electrons* ejected versus the *frequency* of the incident light (right)



**Question 3** **6 points****a. 2 points**

In the ground state of Ge (Germanium; $Z = 32$) atom, **name** the following orbitals:

i. Orbital containing the electron that requires the maximum energy to be removed.

ii. Orbital containing the electron that experiences the minimum effective nuclear charge.

b. 2 points

An unknown anion (A^-) has the following *excited* electronic configuration: $[Ar] 4s^2 3d^3 4p^1$

(Ar: Argon; $Z = 18$)

Give the **condensed ground state electronic configuration** of the atom, **A** and the cation, **A^{2+}** .
Show your work. (Assume all electronic transitions are possible)

Electronic Configuration of A:

Electronic Configuration of A^{2+} :

c. 2 points

In a single atom, determine how many electrons can have the following set of quantum numbers?

(i) $n = 5; l = 3$

In a single atom, determine how many electrons can have the following set of quantum numbers?

(i) $n = 3; l = 2; m_s = +1/2$

**Question 4** **8 points**

Rank the following from 1 to 4 in **increasing** order. Provide the ranking in the boxes provided.

Note: Atomic number (Z) are provided for relevant atoms

a) Increasing (lowest to highest) order of *ionization energy*: (1= lowest; 4 = highest)

Mg

Mg: Z = 12

Rb

Rb: Z = 37

Cs

Cs: Z = 55

K

K : Z = 19

b) Increasing (lowest to highest) order of *bond angle*: (1= lowest; 4 = highest)

H₂OCCl₄XeF₄

HCN

c) Increasing (smallest to largest) order of *ionic size* (1= smallest; 4 = largest)

S²⁻

S: Z = 16

O²⁻

O: Z = 8

Be²⁺

Be: Z = 4

Na⁺

Na: Z = 11

d) Increasing (lowest to highest) order of *bond energy* (1= lowest; 4 = highest)

P-F

P-Cl

P-I

P-Br

**Question 5** **12 points**

a. 9 points For the molecules HClO_4 and TeF_6 , draw the most stable *Lewis structure* showing appropriate shape and *dashed/wedged bonds*, based on VSEPR. Determine the **molecular geometry** around the central atom. For HClO_4 , label the *most polar bond*.



b. 3 points (Note: Lone pairs are not shown in structures below)

(i) For the three *resonance* structures (1, 2 and 3) shown below, answer the following question:



(I) Circle the *most contributing* resonance structure.

Structure 1

Structure 2

Structure 3

(ii) For the three structures given (4, 5, and 6) below, answer the following questions:



(I) Circle all the structures that represent *resonance structures* of each other:

Structure 4

Structure 5

Structure 6

(II) Among the resonance structures, circle the *most contributing* resonance structure.

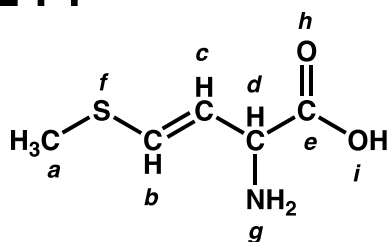
Structure 4

Structure 5

Structure 6

**Question 6** **6 points**Using the data given below, calculate the **lattice energy** for SrO(s).

	ΔH° (kJ/mol)
$\text{Sr}(s) \longrightarrow \text{Sr}(g)$	+180
$\text{Sr}(g) \longrightarrow \text{Sr}^{2+}(g) + 2e^-$	+1468
$\text{O}_2(g) \longrightarrow 2\text{O}(g)$	+500
$\text{O}(g) + 2e^- \longrightarrow \text{O}^{2-}(g)$	+600
$\text{Sr}(s) + \frac{1}{2}\text{O}_2(g) \longrightarrow \text{SrO}(s)$	-550

**Question 7 10 points**

Consider the molecule given on the left, and answer questions (i) to (vi) using chemical bonding theories that we have learnt in class (valence bond theory, hybridization).

Note that the lone pairs are not shown in the structure:

i. Consider the carbon atoms marked in as ***a***, ***b***, ***c***, ***d***, and ***e*** in the molecule, and determine **hybridization** at each of the carbon centers. (2.5 points)

Atom	Hybridization
Carbon (<i>a</i>)	
Carbon (<i>b</i>)	
Carbon (<i>c</i>)	
Carbon (<i>d</i>)	
Carbon (<i>e</i>)	

ii. Now, consider the sulfur atom (S) marked as ***f***, nitrogen (N) atom (***g***), and oxygen(O) atoms (***h*** and ***i***), and determine their **hybridization**. (2 points)

Atom	Hybridization
Sulfur (<i>f</i>)	
Nitrogen (<i>g</i>)	
Oxygen (<i>h</i>)	
Oxygen (<i>i</i>)	

iii. **Describe** all the bonds formed by the carbon atoms ***a*** and ***b***, using sigma and pi designation as appropriate. (2 points)

Atom	Bonds formed (use sigma and pi as appropriate)
Carbon (<i>a</i>)	
Carbon (<i>b</i>)	

iv. How many total sp^3 -s sigma bonds are in the molecule? (1 point)

v. Determine **electronic** and **molecular geometry** at carbon (C) labeled ***e*** and sulfur (S) labeled ***f*** centres. (2 points)

Atom	Electronic Geometry	Molecular Geometry
Carbon (<i>e</i>)		
Sulfur (<i>f</i>)		

vi. Circle the value from the choices given below, that describes the **C-S-C bond angle at sulfur** (S) marked ***f***: (0.5 points)

>90° to <109.5°

109.5°

>109.5 to <120°

120°

>120° to <180°


Question 8 12 points

Using the **valence shell orbitals only** draw a **complete** molecular orbital (MO) diagram (with no s-p mixing) for the molecule **PS**, in the space provided below. **(6 points)**

Assume that 3s and 3p atomic orbitals will form molecular orbitals like those formed by the combination of 2s and 2p atomic orbitals that we learnt in class, and there is no s-p mixing.

Show **energy arrow**, **atomic orbitals** involved, **atom labels**, the **molecular orbitals** that are formed, **filled electrons** in the MO diagram etc. Make sure to label all the atomic and molecular orbitals.

Using the MO diagram above, answer the following questions:

- (i) Which one of the following will have the highest bond order? Circle the correct choice
 A) PS^{1-} B) PS C) PS^{1+}
- (ii) Which one of the following will have the longest bond length? Circle the correct choice
 A) PS^{1-} B) PS C) PS^{1+}
- (iii) Which of the following will be diamagnetic? Circle the correct choice.
 A) PS^{1-} B) PS C) PS^{1+}
- (iv) Draw Lewis dot structure of the molecule with the shortest bond length (based on the MO diagram) from among the three (PS^{1-} , PS, PS^{1+}), and indicate the bond order calculated from the Lewis dot structure. **(3 points)**

Lewis Dot Structure

Bond Order



74A60C88-845B-4042-A07F-269652153977

midterm-exam-93388

#747 Page 10 of 10

SCRAP PAGE

**CHEMISTRY 110 –Midterm Exam****Thursday November 7, 2024, 6:30 – 8:30 pm****Examiners: Pallavi Sirjoosingh and Ashok Kakkar**

First name (please write as legibly as possible within the boxes)

--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--

Last name

--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--

Student ID number

--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--

This booklet has 8 questions on 10 pages (The last page can be used as a scrap page).

The staple must NOT be removed from the question booklet. Your intact question booklet must be returned at the end of the exam.

A separate booklet contains a periodic table, and datasheet.

Write your answers directly under the questions in the space provided. Do not write on the QR codes or near the top edges (as it may be cut off).

Please write **legibly and dark enough** for the scan to be readable. For students who use English as a second language, dictionaries are permitted. Calculators are allowed.

DO NOT ASK QUESTIONS OF THE INVIGILATORS about interpretations of the exam questions. If you have a concern about a question, please summarize it at the top of this page. All concerns will be evaluated after the exam. GOOD LUCK!

**Question 1 8 points** (*Assume all transitions are possible*)

a. 6 points Calculate the **energy (in kJ/mol)** required to ionize (completely remove) the electron from the **n=3** level of **He⁺ ion** (He: Helium). Show your work including equations. (3 sig. figs.)

b. 2 points

(I) Select the largest (longest) wavelength from the three choices below:

- i. Wavelength of photon to *completely remove* an electron from *1s orbital of Hydrogen atom*
- ii. Wavelength of photon to *completely remove* an electron from *1s orbital of Boron atom (B)*
- iii. Wavelength of photon to *completely remove* an electron from *1s orbital of Boron cation (B⁺)*

i or ii or iii

(II) Select the smallest (shortest) wavelength from the three choices below (*for H atom*):

- i. Wavelength of photon *emitted* for an electron transitioning from *2p to 1s orbital*
- ii. Wavelength of photon *absorbed* for an electron transitioning from *2s to 4p orbital*
- iii. Wavelength of photon *absorbed* for an electron transitioning from *1s to 3p orbital*

i or ii or iii



Question 2 8 points



A Chem110 student plots the graph shown on the left, from data obtained for the kinetic energy (eV) of the ejected electron versus frequency (s^{-1}) of the incident light, for an unknown metal. ($1\text{eV} = 1.6022 \times 10^{-19} \text{ J}$)

Based on the plot, answer the questions **a to c** below:

a. 1 point On the plot above, label the **threshold frequency**.

b. 2 points What is the value of the work potential (in J) of the unknown metal?

c. 3 points What is the *longest* wavelength (in nm) required to eject an electron from this metal's surface? Show your work including equation(s) used. (3 sig figs)

d. 2 points In the space provided below, plot the following

(Assume frequency is *above* threshold frequency):

- Kinetic energy* of an ejected electron versus the *intensity* of the incident light (left)
- Number of electrons* ejected versus the *frequency* of the incident light (right)



**Question 3** **6 points****a. 2 points**

In the ground state of Ge (Germanium; $Z = 32$) atom, **name** the following orbitals:

i. Orbital containing the electron that requires the maximum energy to be removed.

ii. Orbital containing the electron that experiences the minimum effective nuclear charge.

b. 2 points

An unknown anion (A^-) has the following *excited* electronic configuration: $[Ar] 4s^2 3d^3 4p^1$

(Ar: Argon; $Z = 18$)

Give the **condensed ground state electronic configuration** of the atom, **A** and the cation, **A^{2+}** .
Show your work. (Assume all electronic transitions are possible)

Electronic Configuration of A:

Electronic Configuration of A^{2+} :

c. 2 points

In a single atom, determine how many electrons can have the following set of quantum numbers?

(i) $n = 5; l = 3$

In a single atom, determine how many electrons can have the following set of quantum numbers?

(i) $n = 3; l = 2; m_s = +1/2$

**Question 4** **8 points**

Rank the following from 1 to 4 in **increasing** order. Provide the ranking in the boxes provided.

Note: Atomic number (Z) are provided for relevant atoms

a) Increasing (lowest to highest) order of *ionization energy*: (1= lowest; 4 = highest)

Mg

Mg: Z = 12

Rb

Rb: Z = 37

Cs

Cs: Z = 55

K

K : Z = 19

b) Increasing (lowest to highest) order of *bond angle*: (1= lowest; 4 = highest)

H₂OCCl₄XeF₄

HCN

c) Increasing (smallest to largest) order of *ionic size* (1= smallest; 4 = largest)

S²⁻

S: Z = 16

O²⁻

O: Z = 8

Be²⁺

Be: Z = 4

Na⁺

Na: Z = 11

d) Increasing (lowest to highest) order of *bond energy* (1= lowest; 4 = highest)

P-F

P-Cl

P-I

P-Br

**Question 5** **12 points**

a. 9 points For the molecules HClO_4 and TeF_6 , draw the most stable *Lewis structure* showing appropriate shape and *dashed/wedged bonds*, based on VSEPR. Determine the **molecular geometry** around the central atom. For HClO_4 , label the *most polar bond*.



b. 3 points (Note: Lone pairs are not shown in structures below)

(i) For the three *resonance* structures (1, 2 and 3) shown below, answer the following question:



(I) Circle the *most contributing* resonance structure.

Structure 1

Structure 2

Structure 3

(ii) For the three structures given (4, 5, and 6) below, answer the following questions:



(I) Circle all the structures that represent *resonance structures* of each other:

Structure 4

Structure 5

Structure 6

(II) Among the resonance structures, circle the *most contributing* resonance structure.

Structure 4

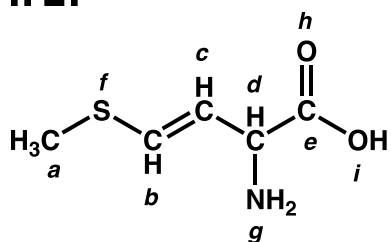
Structure 5

Structure 6

**Question 6** **6 points**

Using the data given below, calculate the **lattice energy** for SrO(s).

	ΔH° (kJ/mol)
$\text{Sr}(s) \longrightarrow \text{Sr}(g)$	+180
$\text{Sr}(g) \longrightarrow \text{Sr}^{2+}(g) + 2e^-$	+1468
$\text{O}_2(g) \longrightarrow 2\text{O}(g)$	+500
$\text{O}(g) + 2e^- \longrightarrow \text{O}^{2-}(g)$	+600
$\text{Sr}(s) + \frac{1}{2}\text{O}_2(g) \longrightarrow \text{SrO}(s)$	-550

**Question 7 10 points**

Consider the molecule given on the left, and answer questions (i) to (vi) using chemical bonding theories that we have learnt in class (valence bond theory, hybridization).

Note that the lone pairs are not shown in the structure:

i. Consider the carbon atoms marked in as ***a***, ***b***, ***c***, ***d***, and ***e*** in the molecule, and determine **hybridization** at each of the carbon centers. (2.5 points)

Atom	Hybridization
Carbon (<i>a</i>)	
Carbon (<i>b</i>)	
Carbon (<i>c</i>)	
Carbon (<i>d</i>)	
Carbon (<i>e</i>)	

ii. Now, consider the sulfur atom (S) marked as ***f***, nitrogen (N) atom (***g***), and oxygen(O) atoms (***h*** and ***i***), and determine their **hybridization**. (2 points)

Atom	Hybridization
Sulfur (<i>f</i>)	
Nitrogen (<i>g</i>)	
Oxygen (<i>h</i>)	
Oxygen (<i>i</i>)	

iii. **Describe** all the bonds formed by the carbon atoms ***a*** and ***b***, using sigma and pi designation as appropriate. (2 points)

Atom	Bonds formed (use sigma and pi as appropriate)
Carbon (<i>a</i>)	
Carbon (<i>b</i>)	

iv. How many total sp^3 -s sigma bonds are in the molecule? (1 point)

v. Determine **electronic** and **molecular geometry** at carbon (C) labeled ***e*** and sulfur (S) labeled ***f*** centres. (2 points)

Atom	Electronic Geometry	Molecular Geometry
Carbon (<i>e</i>)		
Sulfur (<i>f</i>)		

vi. Circle the value from the choices given below, that describes the **C-S-C bond angle at sulfur** (S) marked ***f***: (0.5 points)

$>90^\circ$ to $<109.5^\circ$ 109.5° >109.5 to $<120^\circ$ 120° $>120^\circ$ to $<180^\circ$


Question 8 12 points

Using the **valence shell orbitals only** draw a **complete** molecular orbital (MO) diagram (with no s-p mixing) for the molecule **PS**, in the space provided below. **(6 points)**

Assume that 3s and 3p atomic orbitals will form molecular orbitals like those formed by the combination of 2s and 2p atomic orbitals that we learnt in class, and there is no s-p mixing.

Show **energy arrow**, **atomic orbitals** involved, **atom labels**, the **molecular orbitals** that are formed, **filled electrons** in the MO diagram etc. Make sure to label all the atomic and molecular orbitals.

Using the MO diagram above, answer the following questions:

- (i) Which one of the following will have the highest bond order? Circle the correct choice
 A) PS^{1-} B) PS C) PS^{1+}
- (ii) Which one of the following will have the longest bond length? Circle the correct choice
 A) PS^{1-} B) PS C) PS^{1+}
- (iii) Which of the following will be diamagnetic? Circle the correct choice.
 A) PS^{1-} B) PS C) PS^{1+}
- (iv) Draw Lewis dot structure of the molecule with the shortest bond length (based on the MO diagram) from among the three (PS^{1-} , PS, PS^{1+}), and indicate the bond order calculated from the Lewis dot structure. **(3 points)**

Lewis Dot Structure

Bond Order



0CBC4776-74C9-49B5-9C3F-14440D17559F

midterm-exam-93388

#748 Page 10 of 10

SCRAP PAGE

**CHEMISTRY 110 –Midterm Exam****Thursday November 7, 2024, 6:30 – 8:30 pm****Examiners: Pallavi Sirjoosingh and Ashok Kakkar**

First name (please write as legibly as possible within the boxes)

--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--

Last name

--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--

Student ID number

--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--

This booklet has 8 questions on 10 pages (The last page can be used as a scrap page).

The staple must NOT be removed from the question booklet. Your intact question booklet must be returned at the end of the exam.

A separate booklet contains a periodic table, and datasheet.

Write your answers directly under the questions in the space provided. Do not write on the QR codes or near the top edges (as it may be cut off).

Please write **legibly and dark enough** for the scan to be readable. For students who use English as a second language, dictionaries are permitted. Calculators are allowed.

DO NOT ASK QUESTIONS OF THE INVIGILATORS about interpretations of the exam questions. If you have a concern about a question, please summarize it at the top of this page. All concerns will be evaluated after the exam. GOOD LUCK!

**Question 1 8 points** (*Assume all transitions are possible*)

a. 6 points Calculate the **energy (in kJ/mol)** required to ionize (completely remove) the electron from the **n=3** level of **He⁺ ion** (He: Helium). Show your work including equations. (3 sig. figs.)

b. 2 points

(I) Select the largest (longest) wavelength from the three choices below:

- i. Wavelength of photon to *completely remove* an electron from *1s orbital of Hydrogen atom*
- ii. Wavelength of photon to *completely remove* an electron from *1s orbital of Boron atom (B)*
- iii. Wavelength of photon to *completely remove* an electron from *1s orbital of Boron cation (B⁺)*

i or ii or iii

(II) Select the smallest (shortest) wavelength from the three choices below (*for H atom*):

- i. Wavelength of photon *emitted* for an electron transitioning from *2p to 1s orbital*
- ii. Wavelength of photon *absorbed* for an electron transitioning from *2s to 4p orbital*
- iii. Wavelength of photon *absorbed* for an electron transitioning from *1s to 3p orbital*

i or ii or iii



Question 2 8 points



A Chem110 student plots the graph shown on the left, from data obtained for the kinetic energy (eV) of the ejected electron versus frequency (s^{-1}) of the incident light, for an unknown metal. ($1\text{eV} = 1.6022 \times 10^{-19} \text{ J}$)

Based on the plot, answer the questions **a to c** below:

a. 1 point On the plot above, label the **threshold frequency**.

b. 2 points What is the value of the work potential (in J) of the unknown metal?

c. 3 points What is the *longest* wavelength (in nm) required to eject an electron from this metal's surface? Show your work including equation(s) used. (3 sig figs)

d. 2 points In the space provided below, plot the following

(Assume frequency is *above* threshold frequency):

- Kinetic energy* of an ejected electron versus the *intensity* of the incident light (left)
- Number of electrons* ejected versus the *frequency* of the incident light (right)



**Question 3** **6 points****a. 2 points**

In the ground state of Ge (Germanium; $Z = 32$) atom, **name** the following orbitals:

i. Orbital containing the electron that requires the maximum energy to be removed.

ii. Orbital containing the electron that experiences the minimum effective nuclear charge.

b. 2 points

An unknown anion (A^-) has the following *excited* electronic configuration: $[Ar] 4s^2 3d^3 4p^1$

(Ar: Argon; $Z = 18$)

Give the **condensed ground state electronic configuration** of the atom, **A** and the cation, **A^{2+}** .
Show your work. (Assume all electronic transitions are possible)

Electronic Configuration of A:

Electronic Configuration of A^{2+} :

c. 2 points

In a single atom, determine how many electrons can have the following set of quantum numbers?

(i) $n = 5; l = 3$

In a single atom, determine how many electrons can have the following set of quantum numbers?

(i) $n = 3; l = 2; m_s = +1/2$

**Question 4** **8 points**

Rank the following from 1 to 4 in **increasing** order. Provide the ranking in the boxes provided.

Note: Atomic number (Z) are provided for relevant atoms

a) Increasing (lowest to highest) order of *ionization energy*: (1= lowest; 4 = highest)

Mg

Mg: Z = 12

Rb

Rb: Z = 37

Cs

Cs: Z = 55

K

K : Z = 19

b) Increasing (lowest to highest) order of *bond angle*: (1= lowest; 4 = highest)

H₂OCCl₄XeF₄

HCN

c) Increasing (smallest to largest) order of *ionic size* (1= smallest; 4 = largest)

S²⁻

S: Z = 16

O²⁻

O: Z = 8

Be²⁺

Be: Z = 4

Na⁺

Na: Z = 11

d) Increasing (lowest to highest) order of *bond energy* (1= lowest; 4 = highest)

P-F

P-Cl

P-I

P-Br

**Question 5** **12 points**

a. 9 points For the molecules HClO_4 and TeF_6 , draw the most stable *Lewis structure* showing appropriate shape and *dashed/wedged bonds*, based on VSEPR. Determine the **molecular geometry** around the central atom. For HClO_4 , label the *most polar bond*.



b. 3 points (Note: Lone pairs are not shown in structures below)

(i) For the three *resonance* structures (1, 2 and 3) shown below, answer the following question:



(I) Circle the *most contributing* resonance structure.

Structure 1

Structure 2

Structure 3

(ii) For the three structures given (4, 5, and 6) below, answer the following questions:



(I) Circle all the structures that represent *resonance structures* of each other:

Structure 4

Structure 5

Structure 6

(II) Among the resonance structures, circle the *most contributing* resonance structure.

Structure 4

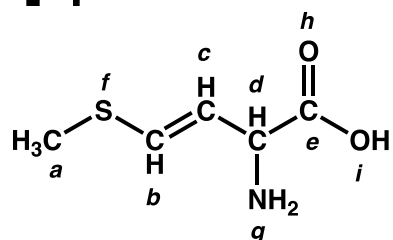
Structure 5

Structure 6

**Question 6** **6 points**

Using the data given below, calculate the **lattice energy** for SrO(s).

	ΔH° (kJ/mol)
$\text{Sr}(s) \longrightarrow \text{Sr}(g)$	+180
$\text{Sr}(g) \longrightarrow \text{Sr}^{2+}(g) + 2e^-$	+1468
$\text{O}_2(g) \longrightarrow 2\text{O}(g)$	+500
$\text{O}(g) + 2e^- \longrightarrow \text{O}^{2-}(g)$	+600
$\text{Sr}(s) + \frac{1}{2}\text{O}_2(g) \longrightarrow \text{SrO}(s)$	-550

**Question 7 10 points**

Consider the molecule given on the left, and answer questions (i) to (vi) using chemical bonding theories that we have learnt in class (valence bond theory, hybridization).

Note that the lone pairs are not shown in the structure:

i. Consider the carbon atoms marked in as ***a***, ***b***, ***c***, ***d***, and ***e*** in the molecule, and determine **hybridization** at each of the carbon centers. (2.5 points)

Atom	Hybridization
Carbon (<i>a</i>)	
Carbon (<i>b</i>)	
Carbon (<i>c</i>)	
Carbon (<i>d</i>)	
Carbon (<i>e</i>)	

ii. Now, consider the sulfur atom (S) marked as ***f***, nitrogen (N) atom (***g***), and oxygen(O) atoms (***h*** and ***i***), and determine their **hybridization**. (2 points)

Atom	Hybridization
Sulfur (<i>f</i>)	
Nitrogen (<i>g</i>)	
Oxygen (<i>h</i>)	
Oxygen (<i>i</i>)	

iii. **Describe** all the bonds formed by the carbon atoms ***a*** and ***b***, using sigma and pi designation as appropriate. (2 points)

Atom	Bonds formed (use sigma and pi as appropriate)
Carbon (<i>a</i>)	
Carbon (<i>b</i>)	

iv. How many total sp^3 -s sigma bonds are in the molecule? (1 point)

v. Determine **electronic** and **molecular geometry** at carbon (C) labeled ***e*** and sulfur (S) labeled ***f*** centres. (2 points)

Atom	Electronic Geometry	Molecular Geometry
Carbon (<i>e</i>)		
Sulfur (<i>f</i>)		

vi. Circle the value from the choices given below, that describes the **C-S-C bond angle at sulfur** (S) marked ***f***: (0.5 points)

>90° to <109.5° 109.5° >109.5 to <120° 120° >120° to <180°


Question 8 12 points

Using the **valence shell orbitals only** draw a **complete** molecular orbital (MO) diagram (with no s-p mixing) for the molecule **PS**, in the space provided below. **(6 points)**

Assume that 3s and 3p atomic orbitals will form molecular orbitals like those formed by the combination of 2s and 2p atomic orbitals that we learnt in class, and there is no s-p mixing.

Show **energy arrow**, **atomic orbitals** involved, **atom labels**, the **molecular orbitals** that are formed, **filled electrons** in the MO diagram etc. Make sure to label all the atomic and molecular orbitals.

Using the MO diagram above, answer the following questions:

- (i) Which one of the following will have the highest bond order? Circle the correct choice
 A) PS^{1-} B) PS C) PS^{1+}
- (ii) Which one of the following will have the longest bond length? Circle the correct choice
 A) PS^{1-} B) PS C) PS^{1+}
- (iii) Which of the following will be diamagnetic? Circle the correct choice.
 A) PS^{1-} B) PS C) PS^{1+}
- (iv) Draw Lewis dot structure of the molecule with the shortest bond length (based on the MO diagram) from among the three (PS^{1-} , PS, PS^{1+}), and indicate the bond order calculated from the Lewis dot structure. **(3 points)**

Lewis Dot Structure

Bond Order



4568D62A-2B5F-486D-9B7C-606FFC83C29F

midterm-exam-93388

#749 Page 10 of 10

SCRAP PAGE

**CHEMISTRY 110 –Midterm Exam****Thursday November 7, 2024, 6:30 – 8:30 pm**

Examiners: Pallavi Sirjoosingh and Ashok Kakkar

First name (please write as legibly as possible within the boxes)

--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--

Last name

--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--

Student ID number

--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--

This booklet has 8 questions on 10 pages (The last page can be used as a scrap page).

The staple must NOT be removed from the question booklet. Your intact question booklet must be returned at the end of the exam.

A separate booklet contains a periodic table, and datasheet.

Write your answers directly under the questions in the space provided. Do not write on the QR codes or near the top edges (as it may be cut off).

Please write **legibly and dark enough** for the scan to be readable. For students who use English as a second language, dictionaries are permitted. Calculators are allowed.

DO NOT ASK QUESTIONS OF THE INVIGILATORS about interpretations of the exam questions. If you have a concern about a question, please summarize it at the top of this page. All concerns will be evaluated after the exam. GOOD LUCK!

**Question 1 8 points** (*Assume all transitions are possible*)

a. 6 points Calculate the **energy (in kJ/mol)** required to ionize (completely remove) the electron from the **n=3** level of **He⁺ ion** (He: Helium). Show your work including equations. (3 sig. figs.)

b. 2 points

(I) Select the largest (longest) wavelength from the three choices below:

- i. Wavelength of photon to *completely remove* an electron from *1s orbital of Hydrogen atom*
- ii. Wavelength of photon to *completely remove* an electron from *1s orbital of Boron atom (B)*
- iii. Wavelength of photon to *completely remove* an electron from *1s orbital of Boron cation (B⁺)*

i or ii or iii

(II) Select the smallest (shortest) wavelength from the three choices below (*for H atom*):

- i. Wavelength of photon *emitted* for an electron transitioning from *2p to 1s orbital*
- ii. Wavelength of photon *absorbed* for an electron transitioning from *2s to 4p orbital*
- iii. Wavelength of photon *absorbed* for an electron transitioning from *1s to 3p orbital*

i or ii or iii



Question 2 8 points



A Chem110 student plots the graph shown on the left, from data obtained for the kinetic energy (eV) of the ejected electron versus frequency (s^{-1}) of the incident light, for an unknown metal. ($1\text{eV} = 1.6022 \times 10^{-19} \text{ J}$)

Based on the plot, answer the questions **a to c** below:

a. 1 point On the plot above, label the **threshold frequency**.

b. 2 points What is the value of the work potential (in J) of the unknown metal?

c. 3 points What is the *longest* wavelength (in nm) required to eject an electron from this metal's surface? Show your work including equation(s) used. (3 sig figs)

d. 2 points In the space provided below, plot the following

(Assume frequency is *above* threshold frequency):

- i. *Kinetic energy* of an ejected electron versus the *intensity* of the incident light (left)
- ii. *Number of electrons* ejected versus the *frequency* of the incident light (right)



**Question 3** **6 points****a. 2 points**

In the ground state of Ge (Germanium; $Z = 32$) atom, **name** the following orbitals:

i. Orbital containing the electron that requires the maximum energy to be removed.

ii. Orbital containing the electron that experiences the minimum effective nuclear charge.

b. 2 points

An unknown anion (A^-) has the following *excited* electronic configuration: $[Ar] 4s^2 3d^3 4p^1$

(Ar: Argon; $Z = 18$)

Give the **condensed ground state electronic configuration** of the atom, **A** and the cation, **A^{2+}** .
Show your work. (Assume all electronic transitions are possible)

Electronic Configuration of A:

Electronic Configuration of A^{2+} :

c. 2 points

In a single atom, determine how many electrons can have the following set of quantum numbers?

(i) $n = 5; l = 3$

In a single atom, determine how many electrons can have the following set of quantum numbers?

(i) $n = 3; l = 2; m_s = +1/2$

**Question 4** **8 points**

Rank the following from 1 to 4 in **increasing** order. Provide the ranking in the boxes provided.

Note: Atomic number (Z) are provided for relevant atoms

a) Increasing (lowest to highest) order of *ionization energy*: (1= lowest; 4 = highest)

Mg

Mg: Z = 12

Rb

Rb: Z = 37

Cs

Cs: Z = 55

K

K : Z = 19

b) Increasing (lowest to highest) order of *bond angle*: (1= lowest; 4 = highest)

H₂OCCl₄XeF₄

HCN

c) Increasing (smallest to largest) order of *ionic size* (1= smallest; 4 = largest)

S²⁻

S: Z = 16

O²⁻

O: Z = 8

Be²⁺

Be: Z = 4

Na⁺

Na: Z = 11

d) Increasing (lowest to highest) order of *bond energy* (1= lowest; 4 = highest)

P-F

P-Cl

P-I

P-Br

**Question 5** **12 points**

a. 9 points For the molecules HClO_4 and TeF_6 , draw the most stable *Lewis structure* showing appropriate shape and *dashed/wedged bonds*, based on VSEPR. Determine the **molecular geometry** around the central atom. For HClO_4 , label the *most polar bond*.



b. 3 points (Note: Lone pairs are not shown in structures below)

(i) For the three *resonance* structures (1, 2 and 3) shown below, answer the following question:



(I) Circle the *most contributing* resonance structure.

Structure 1

Structure 2

Structure 3

(ii) For the three structures given (4, 5, and 6) below, answer the following questions:



(I) Circle all the structures that represent *resonance structures* of each other:

Structure 4

Structure 5

Structure 6

(II) Among the resonance structures, circle the *most contributing* resonance structure.

Structure 4

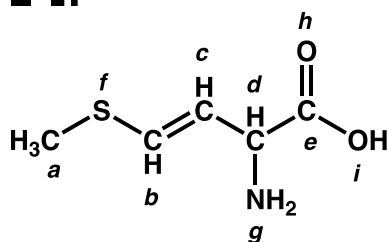
Structure 5

Structure 6

**Question 6** **6 points**

Using the data given below, calculate the **lattice energy** for SrO(s).

	ΔH° (kJ/mol)
$\text{Sr}(s) \longrightarrow \text{Sr}(g)$	+180
$\text{Sr}(g) \longrightarrow \text{Sr}^{2+}(g) + 2e^-$	+1468
$\text{O}_2(g) \longrightarrow 2\text{O}(g)$	+500
$\text{O}(g) + 2e^- \longrightarrow \text{O}^{2-}(g)$	+600
$\text{Sr}(s) + \frac{1}{2}\text{O}_2(g) \longrightarrow \text{SrO}(s)$	-550

**Question 7 10 points**

Consider the molecule given on the left, and answer questions (i) to (vi) using chemical bonding theories that we have learnt in class (valence bond theory, hybridization).

Note that the lone pairs are not shown in the structure:

i. Consider the carbon atoms marked in as ***a***, ***b***, ***c***, ***d***, and ***e*** in the molecule, and determine **hybridization** at each of the carbon centers. (2.5 points)

Atom	Hybridization
Carbon (<i>a</i>)	
Carbon (<i>b</i>)	
Carbon (<i>c</i>)	
Carbon (<i>d</i>)	
Carbon (<i>e</i>)	

ii. Now, consider the sulfur atom (S) marked as ***f***, nitrogen (N) atom (***g***), and oxygen(O) atoms (***h*** and ***i***), and determine their **hybridization**. (2 points)

Atom	Hybridization
Sulfur (<i>f</i>)	
Nitrogen (<i>g</i>)	
Oxygen (<i>h</i>)	
Oxygen (<i>i</i>)	

iii. **Describe** all the bonds formed by the carbon atoms ***a*** and ***b***, using sigma and pi designation as appropriate. (2 points)

Atom	Bonds formed (use sigma and pi as appropriate)
Carbon (<i>a</i>)	
Carbon (<i>b</i>)	

iv. How many total sp^3 -s sigma bonds are in the molecule? (1 point)

v. Determine **electronic** and **molecular geometry** at carbon (C) labeled ***e*** and sulfur (S) labeled ***f*** centres. (2 points)

Atom	Electronic Geometry	Molecular Geometry
Carbon (<i>e</i>)		
Sulfur (<i>f</i>)		

vi. Circle the value from the choices given below, that describes the **C-S-C bond angle at sulfur** (S) marked ***f***: (0.5 points)

>90° to <109.5° 109.5° >109.5 to <120° 120° >120° to <180°


Question 8 12 points

Using the **valence shell orbitals only** draw a **complete** molecular orbital (MO) diagram (with no s-p mixing) for the molecule **PS**, in the space provided below. **(6 points)**

Assume that 3s and 3p atomic orbitals will form molecular orbitals like those formed by the combination of 2s and 2p atomic orbitals that we learnt in class, and there is no s-p mixing.

Show **energy arrow**, **atomic orbitals** involved, **atom labels**, the **molecular orbitals** that are formed, **filled electrons** in the MO diagram etc. Make sure to label all the atomic and molecular orbitals.

Using the MO diagram above, answer the following questions:

- (i) Which one of the following will have the highest bond order? Circle the correct choice
 A) PS^{1-} B) PS C) PS^{1+}
- (ii) Which one of the following will have the longest bond length? Circle the correct choice
 A) PS^{1-} B) PS C) PS^{1+}
- (iii) Which of the following will be diamagnetic? Circle the correct choice.
 A) PS^{1-} B) PS C) PS^{1+}
- (iv) Draw Lewis dot structure of the molecule with the shortest bond length (based on the MO diagram) from among the three (PS^{1-} , PS, PS^{1+}), and indicate the bond order calculated from the Lewis dot structure. **(3 points)**

Lewis Dot Structure

Bond Order



174A673A-08E0-4707-947C-2886B24D4D3A

midterm-exam-93388

#750 Page 10 of 10

SCRAP PAGE

**CHEMISTRY 110 –Midterm Exam****Thursday November 7, 2024, 6:30 – 8:30 pm**

Examiners: Pallavi Sirjoosingh and Ashok Kakkar

First name (please write as legibly as possible within the boxes)

--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--

Last name

--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--

Student ID number

--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--

This booklet has 8 questions on 10 pages (The last page can be used as a scrap page).

The staple must NOT be removed from the question booklet. Your intact question booklet must be returned at the end of the exam.

A separate booklet contains a periodic table, and datasheet.

Write your answers directly under the questions in the space provided. Do not write on the QR codes or near the top edges (as it may be cut off).

Please write **legibly and dark enough** for the scan to be readable. For students who use English as a second language, dictionaries are permitted. Calculators are allowed.

DO NOT ASK QUESTIONS OF THE INVIGILATORS about interpretations of the exam questions. If you have a concern about a question, please summarize it at the top of this page. All concerns will be evaluated after the exam. GOOD LUCK!

**Question 1 8 points** (*Assume all transitions are possible*)

a. 6 points Calculate the **energy (in kJ/mol)** required to ionize (completely remove) the electron from the **$n=3$** level of **He^+ ion** (He: Helium). Show your work including equations. (3 sig. figs.)

b. 2 points

(I) Select the largest (longest) wavelength from the three choices below:

- i. Wavelength of photon to *completely remove* an electron from *1s orbital of Hydrogen atom*
- ii. Wavelength of photon to *completely remove* an electron from *1s orbital of Boron atom (B)*
- iii. Wavelength of photon to *completely remove* an electron from *1s orbital of Boron cation (B^+)*

i or ii or iii

(II) Select the smallest (shortest) wavelength from the three choices below (*for H atom*):

- i. Wavelength of photon *emitted* for an electron transitioning from *2p to 1s orbital*
- ii. Wavelength of photon *absorbed* for an electron transitioning from *2s to 4p orbital*
- iii. Wavelength of photon *absorbed* for an electron transitioning from *1s to 3p orbital*

i or ii or iii



Question 2 8 points



A Chem110 student plots the graph shown on the left, from data obtained for the kinetic energy (eV) of the ejected electron versus frequency (s^{-1}) of the incident light, for an unknown metal. ($1\text{eV} = 1.6022 \times 10^{-19} \text{ J}$)

Based on the plot, answer the questions **a to c** below:

a. 1 point On the plot above, label the **threshold frequency**.

b. 2 points What is the value of the work potential (in J) of the unknown metal?

c. 3 points What is the *longest* wavelength (in nm) required to eject an electron from this metal's surface? Show your work including equation(s) used. (3 sig figs)

d. 2 points In the space provided below, plot the following

(Assume frequency is *above* threshold frequency):

- i. *Kinetic energy* of an ejected electron versus the *intensity* of the incident light (left)
- ii. *Number of electrons* ejected versus the *frequency* of the incident light (right)



**Question 3** **6 points****a. 2 points**

In the ground state of Ge (Germanium; $Z = 32$) atom, **name** the following orbitals:

i. Orbital containing the electron that requires the maximum energy to be removed.

ii. Orbital containing the electron that experiences the minimum effective nuclear charge.

b. 2 points

An unknown anion (A^-) has the following *excited* electronic configuration: $[Ar] 4s^2 3d^3 4p^1$

(Ar: Argon; $Z = 18$)

Give the **condensed ground state electronic configuration** of the atom, **A** and the cation, **A^{2+}** .
Show your work. (Assume all electronic transitions are possible)

Electronic Configuration of A:

Electronic Configuration of A^{2+} :

c. 2 points

In a single atom, determine how many electrons can have the following set of quantum numbers?

(i) $n = 5; l = 3$

In a single atom, determine how many electrons can have the following set of quantum numbers?

(i) $n = 3; l = 2; m_s = +1/2$

**Question 4** **8 points**

Rank the following from 1 to 4 in **increasing** order. Provide the ranking in the boxes provided.

Note: Atomic number (Z) are provided for relevant atoms

a) Increasing (lowest to highest) order of *ionization energy*: (1= lowest; 4 = highest)

Mg

Mg: Z = 12

Rb

Rb: Z = 37

Cs

Cs: Z = 55

K

K : Z = 19

b) Increasing (lowest to highest) order of *bond angle*: (1= lowest; 4 = highest)

H₂OCCl₄XeF₄

HCN

c) Increasing (smallest to largest) order of *ionic size* (1= smallest; 4 = largest)

S²⁻

S: Z = 16

O²⁻

O: Z = 8

Be²⁺

Be: Z = 4

Na⁺

Na: Z = 11

d) Increasing (lowest to highest) order of *bond energy* (1= lowest; 4 = highest)

P-F

P-Cl

P-I

P-Br

**Question 5** **12 points**

a. 9 points For the molecules HClO_4 and TeF_6 , draw the most stable *Lewis structure* showing appropriate shape and *dashed/wedged bonds*, based on VSEPR. Determine the **molecular geometry** around the central atom. For HClO_4 , label the *most polar bond*.



b. 3 points (Note: Lone pairs are not shown in structures below)

(i) For the three *resonance* structures (1, 2 and 3) shown below, answer the following question:



(I) Circle the *most contributing* resonance structure.

Structure 1

Structure 2

Structure 3

(ii) For the three structures given (4, 5, and 6) below, answer the following questions:



(I) Circle all the structures that represent *resonance structures* of each other:

Structure 4

Structure 5

Structure 6

(II) Among the resonance structures, circle the *most contributing* resonance structure.

Structure 4

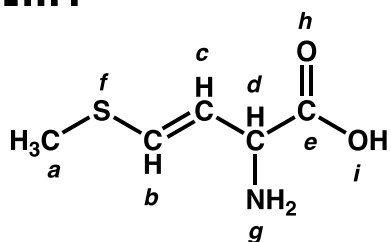
Structure 5

Structure 6

**Question 6** **6 points**

Using the data given below, calculate the **lattice energy** for SrO(s).

	ΔH° (kJ/mol)
$\text{Sr}(s) \longrightarrow \text{Sr}(g)$	+180
$\text{Sr}(g) \longrightarrow \text{Sr}^{2+}(g) + 2e^-$	+1468
$\text{O}_2(g) \longrightarrow 2\text{O}(g)$	+500
$\text{O}(g) + 2e^- \longrightarrow \text{O}^{2-}(g)$	+600
$\text{Sr}(s) + \frac{1}{2}\text{O}_2(g) \longrightarrow \text{SrO}(s)$	-550

**Question 7 10 points**

Consider the molecule given on the left, and answer questions (i) to (vi) using chemical bonding theories that we have learnt in class (valence bond theory, hybridization).

Note that the lone pairs are not shown in the structure:

i. Consider the carbon atoms marked in as ***a***, ***b***, ***c***, ***d***, and ***e*** in the molecule, and determine **hybridization** at each of the carbon centers. (2.5 points)

Atom	Hybridization
Carbon (<i>a</i>)	
Carbon (<i>b</i>)	
Carbon (<i>c</i>)	
Carbon (<i>d</i>)	
Carbon (<i>e</i>)	

ii. Now, consider the sulfur atom (S) marked as ***f***, nitrogen (N) atom (***g***), and oxygen(O) atoms (***h*** and ***i***), and determine their **hybridization**. (2 points)

Atom	Hybridization
Sulfur (<i>f</i>)	
Nitrogen (<i>g</i>)	
Oxygen (<i>h</i>)	
Oxygen (<i>i</i>)	

iii. **Describe** all the bonds formed by the carbon atoms ***a*** and ***b***, using sigma and pi designation as appropriate. (2 points)

Atom	Bonds formed (use sigma and pi as appropriate)
Carbon (<i>a</i>)	
Carbon (<i>b</i>)	

iv. How many total sp^3 -s sigma bonds are in the molecule? (1 point)

v. Determine **electronic** and **molecular geometry** at carbon (C) labeled ***e*** and sulfur (S) labeled ***f*** centres. (2 points)

Atom	Electronic Geometry	Molecular Geometry
Carbon (<i>e</i>)		
Sulfur (<i>f</i>)		

vi. Circle the value from the choices given below, that describes the **C-S-C bond angle at sulfur** (S) marked ***f***: (0.5 points)

>90° to <109.5°

109.5°

>109.5 to <120°

120°

>120° to <180°

**Question 8 12 points**

Using the **valence shell orbitals only** draw a **complete** molecular orbital (MO) diagram (with no s-p mixing) for the molecule **PS**, in the space provided below. **(6 points)**

Assume that 3s and 3p atomic orbitals will form molecular orbitals like those formed by the combination of 2s and 2p atomic orbitals that we learnt in class, and there is no s-p mixing.

Show **energy arrow**, **atomic orbitals** involved, **atom labels**, the **molecular orbitals** that are formed, **filled electrons** in the MO diagram etc. Make sure to label all the atomic and molecular orbitals.

Using the MO diagram above, answer the following questions:

- (i) Which one of the following will have the highest bond order? Circle the correct choice
A) PS^{1-} B) PS C) PS^{1+}
- (ii) Which one of the following will have the longest bond length? Circle the correct choice
A) PS^{1-} B) PS C) PS^{1+}
- (iii) Which of the following will be diamagnetic? Circle the correct choice.
A) PS^{1-} B) PS C) PS^{1+}
- (iv) Draw Lewis dot structure of the molecule with the shortest bond length (based on the MO diagram) from among the three (PS^{1-} , PS, PS^{1+}), and indicate the bond order calculated from the Lewis dot structure. **(3 points)**

Lewis Dot Structure

Bond Order



C8FE6B08-650D-42E3-8E9B-1A9165617C14

midterm-exam-93388

#751 Page 10 of 10

SCRAP PAGE

**CHEMISTRY 110 –Midterm Exam****Thursday November 7, 2024, 6:30 – 8:30 pm**

Examiners: Pallavi Sirjoosingh and Ashok Kakkar

First name (please write as legibly as possible within the boxes)

--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--

Last name

--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--

Student ID number

--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--

This booklet has 8 questions on 10 pages (The last page can be used as a scrap page).

The staple must NOT be removed from the question booklet. Your intact question booklet must be returned at the end of the exam.

A separate booklet contains a periodic table, and datasheet.

Write your answers directly under the questions in the space provided. Do not write on the QR codes or near the top edges (as it may be cut off).

Please write **legibly and dark enough** for the scan to be readable. For students who use English as a second language, dictionaries are permitted. Calculators are allowed.

DO NOT ASK QUESTIONS OF THE INVIGILATORS about interpretations of the exam questions. If you have a concern about a question, please summarize it at the top of this page. All concerns will be evaluated after the exam. GOOD LUCK!

**Question 1 8 points** (*Assume all transitions are possible*)

a. 6 points Calculate the **energy (in kJ/mol)** required to ionize (completely remove) the electron from the **n=3** level of **He⁺ ion** (He: Helium). Show your work including equations. (3 sig. figs.)

b. 2 points

(I) Select the largest (longest) wavelength from the three choices below:

- i. Wavelength of photon to *completely remove* an electron from *1s orbital of Hydrogen atom*
- ii. Wavelength of photon to *completely remove* an electron from *1s orbital of Boron atom (B)*
- iii. Wavelength of photon to *completely remove* an electron from *1s orbital of Boron cation (B⁺)*

i or ii or iii

(II) Select the smallest (shortest) wavelength from the three choices below (*for H atom*):

- i. Wavelength of photon *emitted* for an electron transitioning from *2p to 1s orbital*
- ii. Wavelength of photon *absorbed* for an electron transitioning from *2s to 4p orbital*
- iii. Wavelength of photon *absorbed* for an electron transitioning from *1s to 3p orbital*

i or ii or iii



Question 2 8 points



A Chem110 student plots the graph shown on the left, from data obtained for the kinetic energy (eV) of the ejected electron versus frequency (s^{-1}) of the incident light, for an unknown metal. ($1\text{eV} = 1.6022 \times 10^{-19} \text{ J}$)

Based on the plot, answer the questions **a to c** below:

a. 1 point On the plot above, label the **threshold frequency**.

b. 2 points What is the value of the work potential (in J) of the unknown metal?

c. 3 points What is the *longest* wavelength (in nm) required to eject an electron from this metal's surface? Show your work including equation(s) used. (3 sig figs)

d. 2 points In the space provided below, plot the following

(Assume frequency is *above* threshold frequency):

- Kinetic energy* of an ejected electron versus the *intensity* of the incident light (left)
- Number of electrons* ejected versus the *frequency* of the incident light (right)



**Question 3** **6 points****a. 2 points**

In the ground state of Ge (Germanium; $Z = 32$) atom, **name** the following orbitals:

i. Orbital containing the electron that requires the maximum energy to be removed.

ii. Orbital containing the electron that experiences the minimum effective nuclear charge.

b. 2 points

An unknown anion (A^-) has the following *excited* electronic configuration: $[Ar] 4s^2 3d^3 4p^1$

(Ar: Argon; $Z = 18$)

Give the **condensed ground state electronic configuration** of the atom, **A** and the cation, **A^{2+}** .
Show your work. (Assume all electronic transitions are possible)

Electronic Configuration of A:

Electronic Configuration of A^{2+} :

c. 2 points

In a single atom, determine how many electrons can have the following set of quantum numbers?

(i) $n = 5; l = 3$

In a single atom, determine how many electrons can have the following set of quantum numbers?

(i) $n = 3; l = 2; m_s = +1/2$

**Question 4** **8 points**

Rank the following from 1 to 4 in **increasing** order. Provide the ranking in the boxes provided.

Note: Atomic number (Z) are provided for relevant atoms

a) Increasing (lowest to highest) order of *ionization energy*: (1= lowest; 4 = highest)

Mg

Mg: Z = 12

Rb

Rb: Z = 37

Cs

Cs: Z = 55

K

K : Z = 19

b) Increasing (lowest to highest) order of *bond angle*: (1= lowest; 4 = highest)

H₂OCCl₄XeF₄

HCN

c) Increasing (smallest to largest) order of *ionic size* (1= smallest; 4 = largest)

S²⁻

S: Z = 16

O²⁻

O: Z = 8

Be²⁺

Be: Z = 4

Na⁺

Na: Z = 11

d) Increasing (lowest to highest) order of *bond energy* (1= lowest; 4 = highest)

P-F

P-Cl

P-I

P-Br

**Question 5** **12 points**

a. 9 points For the molecules HClO_4 and TeF_6 , draw the most stable *Lewis structure* showing appropriate shape and *dashed/wedged bonds*, based on VSEPR. Determine the **molecular geometry** around the central atom. For HClO_4 , label the *most polar bond*.

HClO_4 (there are no O-O bonds)

Label the most polar bond

Molecular Geometry around Cl

TeF_6

Molecular Geometry around Te

b. 3 points (Note: Lone pairs are not shown in structures below)

(i) For the three *resonance* structures (1, 2 and 3) shown below, answer the following question:



(I) Circle the *most contributing* resonance structure.

Structure 1

Structure 2

Structure 3

(ii) For the three structures given (4, 5, and 6) below, answer the following questions:



(I) Circle all the structures that represent *resonance structures* of each other:

Structure 4

Structure 5

Structure 6

(II) Among the resonance structures, circle the *most contributing* resonance structure.

Structure 4

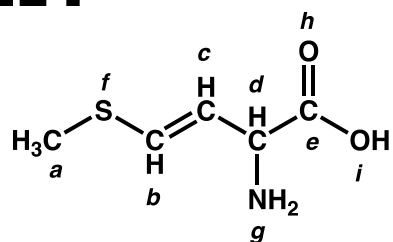
Structure 5

Structure 6

**Question 6** **6 points**

Using the data given below, calculate the **lattice energy** for SrO(s).

	ΔH° (kJ/mol)
$\text{Sr}(s) \longrightarrow \text{Sr}(g)$	+180
$\text{Sr}(g) \longrightarrow \text{Sr}^{2+}(g) + 2e^-$	+1468
$\text{O}_2(g) \longrightarrow 2\text{O}(g)$	+500
$\text{O}(g) + 2e^- \longrightarrow \text{O}^{2-}(g)$	+600
$\text{Sr}(s) + \frac{1}{2}\text{O}_2(g) \longrightarrow \text{SrO}(s)$	-550

**Question 7 10 points**

Consider the molecule given on the left, and answer questions (i) to (vi) using chemical bonding theories that we have learnt in class (valence bond theory, hybridization).

Note that the lone pairs are not shown in the structure:

i. Consider the carbon atoms marked in as ***a***, ***b***, ***c***, ***d***, and ***e*** in the molecule, and determine **hybridization** at each of the carbon centers. (2.5 points)

Atom	Hybridization
Carbon (<i>a</i>)	
Carbon (<i>b</i>)	
Carbon (<i>c</i>)	
Carbon (<i>d</i>)	
Carbon (<i>e</i>)	

ii. Now, consider the sulfur atom (S) marked as ***f***, nitrogen (N) atom (***g***), and oxygen(O) atoms (***h*** and ***i***), and determine their **hybridization**. (2 points)

Atom	Hybridization
Sulfur (<i>f</i>)	
Nitrogen (<i>g</i>)	
Oxygen (<i>h</i>)	
Oxygen (<i>i</i>)	

iii. **Describe** all the bonds formed by the carbon atoms ***a*** and ***b***, using sigma and pi designation as appropriate. (2 points)

Atom	Bonds formed (use sigma and pi as appropriate)
Carbon (<i>a</i>)	
Carbon (<i>b</i>)	

iv. How many total sp^3 -s sigma bonds are in the molecule? (1 point)

v. Determine **electronic** and **molecular geometry** at carbon (C) labeled ***e*** and sulfur (S) labeled ***f*** centres. (2 points)

Atom	Electronic Geometry	Molecular Geometry
Carbon (<i>e</i>)		
Sulfur (<i>f</i>)		

vi. Circle the value from the choices given below, that describes the **C-S-C bond angle at sulfur** (S) marked ***f***: (0.5 points)

>90° to <109.5° 109.5° >109.5 to <120° 120° >120° to <180°


Question 8 12 points

Using the **valence shell orbitals only** draw a **complete** molecular orbital (MO) diagram (with no s-p mixing) for the molecule **PS**, in the space provided below. **(6 points)**

Assume that 3s and 3p atomic orbitals will form molecular orbitals like those formed by the combination of 2s and 2p atomic orbitals that we learnt in class, and there is no s-p mixing.

Show **energy arrow**, **atomic orbitals** involved, **atom labels**, the **molecular orbitals** that are formed, **filled electrons** in the MO diagram etc. Make sure to label all the atomic and molecular orbitals.

Using the MO diagram above, answer the following questions:

- (i) Which one of the following will have the highest bond order? Circle the correct choice
 A) PS^{1-} B) PS C) PS^{1+}
- (ii) Which one of the following will have the longest bond length? Circle the correct choice
 A) PS^{1-} B) PS C) PS^{1+}
- (iii) Which of the following will be diamagnetic? Circle the correct choice.
 A) PS^{1-} B) PS C) PS^{1+}
- (iv) Draw Lewis dot structure of the molecule with the shortest bond length (based on the MO diagram) from among the three (PS^{1-} , PS, PS^{1+}), and indicate the bond order calculated from the Lewis dot structure. **(3 points)**

Lewis Dot Structure

Bond Order



834B03E9-ED0A-44CD-B478-52674717ADDA

midterm-exam-93388

#752 Page 10 of 10

SCRAP PAGE

**CHEMISTRY 110 –Midterm Exam****Thursday November 7, 2024, 6:30 – 8:30 pm**

Examiners: Pallavi Sirjoosingh and Ashok Kakkar

First name (please write as legibly as possible within the boxes)

--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--

Last name

--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--

Student ID number

--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--

This booklet has 8 questions on 10 pages (The last page can be used as a scrap page).

The staple must NOT be removed from the question booklet. Your intact question booklet must be returned at the end of the exam.

A separate booklet contains a periodic table, and datasheet.

Write your answers directly under the questions in the space provided. Do not write on the QR codes or near the top edges (as it may be cut off).

Please write **legibly and dark enough** for the scan to be readable. For students who use English as a second language, dictionaries are permitted. Calculators are allowed.

DO NOT ASK QUESTIONS OF THE INVIGILATORS about interpretations of the exam questions. If you have a concern about a question, please summarize it at the top of this page. All concerns will be evaluated after the exam. GOOD LUCK!

**Question 1 8 points** (*Assume all transitions are possible*)

a. 6 points Calculate the **energy (in kJ/mol)** required to ionize (completely remove) the electron from the **n=3** level of **He⁺ ion** (He: Helium). Show your work including equations. (3 sig. figs.)

b. 2 points

(I) Select the largest (longest) wavelength from the three choices below:

- i. Wavelength of photon to *completely remove* an electron from *1s orbital of Hydrogen atom*
- ii. Wavelength of photon to *completely remove* an electron from *1s orbital of Boron atom (B)*
- iii. Wavelength of photon to *completely remove* an electron from *1s orbital of Boron cation (B⁺)*

i or ii or iii

(II) Select the smallest (shortest) wavelength from the three choices below (*for H atom*):

- i. Wavelength of photon *emitted* for an electron transitioning from *2p to 1s orbital*
- ii. Wavelength of photon *absorbed* for an electron transitioning from *2s to 4p orbital*
- iii. Wavelength of photon *absorbed* for an electron transitioning from *1s to 3p orbital*

i or ii or iii



Question 2 8 points



A Chem110 student plots the graph shown on the left, from data obtained for the kinetic energy (eV) of the ejected electron versus frequency (s^{-1}) of the incident light, for an unknown metal. ($1\text{eV} = 1.6022 \times 10^{-19} \text{ J}$)

Based on the plot, answer the questions **a to c** below:

a. 1 point On the plot above, label the **threshold frequency**.

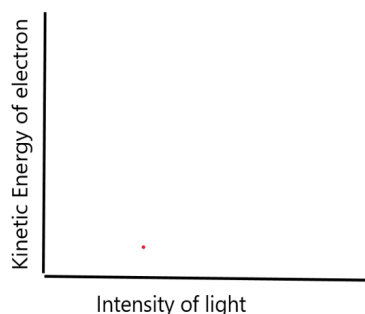
b. 2 points What is the value of the work potential (in J) of the unknown metal?

c. 3 points What is the *longest* wavelength (in nm) required to eject an electron from this metal's surface? Show your work including equation(s) used. (3 sig figs)

d. 2 points In the space provided below, plot the following

(Assume frequency is *above* threshold frequency):

- i. *Kinetic energy* of an ejected electron versus the *intensity* of the incident light (left)
- ii. *Number of electrons* ejected versus the *frequency* of the incident light (right)



**Question 3** **6 points****a. 2 points**

In the ground state of Ge (Germanium; $Z = 32$) atom, **name** the following orbitals:

i. Orbital containing the electron that requires the maximum energy to be removed.

ii. Orbital containing the electron that experiences the minimum effective nuclear charge.

b. 2 points

An unknown anion (A^-) has the following *excited* electronic configuration: $[Ar] 4s^2 3d^3 4p^1$

(Ar: Argon; $Z = 18$)

Give the **condensed ground state electronic configuration** of the atom, **A** and the cation, **A^{2+}** .
Show your work. (Assume all electronic transitions are possible)

Electronic Configuration of A:

Electronic Configuration of A^{2+} :

c. 2 points

In a single atom, determine how many electrons can have the following set of quantum numbers?

(i) $n = 5; l = 3$

In a single atom, determine how many electrons can have the following set of quantum numbers?

(i) $n = 3; l = 2; m_s = +1/2$

**Question 4** **8 points**

Rank the following from 1 to 4 in **increasing** order. Provide the ranking in the boxes provided.

Note: Atomic number (Z) are provided for relevant atoms

a) Increasing (lowest to highest) order of *ionization energy*: (1= lowest; 4 = highest)

Mg

Mg: Z = 12

Rb

Rb: Z = 37

Cs

Cs: Z = 55

K

K : Z = 19

b) Increasing (lowest to highest) order of *bond angle*: (1= lowest; 4 = highest)

H₂OCCl₄XeF₄

HCN

c) Increasing (smallest to largest) order of *ionic size* (1= smallest; 4 = largest)

S²⁻

S: Z = 16

O²⁻

O: Z = 8

Be²⁺

Be: Z = 4

Na⁺

Na: Z = 11

d) Increasing (lowest to highest) order of *bond energy* (1= lowest; 4 = highest)

P-F

P-Cl

P-I

P-Br

**Question 5** **12 points**

a. 9 points For the molecules HClO_4 and TeF_6 , draw the most stable *Lewis structure* showing appropriate shape and *dashed/wedged bonds*, based on VSEPR. Determine the **molecular geometry** around the central atom. For HClO_4 , label the *most polar bond*.



b. 3 points (Note: Lone pairs are not shown in structures below)

(i) For the three *resonance* structures (1, 2 and 3) shown below, answer the following question:



(I) Circle the *most contributing* resonance structure.

Structure 1

Structure 2

Structure 3

(ii) For the three structures given (4, 5, and 6) below, answer the following questions:



(I) Circle all the structures that represent *resonance structures* of each other:

Structure 4

Structure 5

Structure 6

(II) Among the resonance structures, circle the *most contributing* resonance structure.

Structure 4

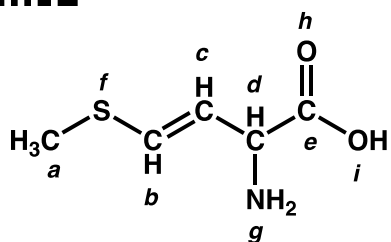
Structure 5

Structure 6

**Question 6** **6 points**

Using the data given below, calculate the **lattice energy** for SrO(s).

	ΔH° (kJ/mol)
$\text{Sr}(s) \longrightarrow \text{Sr}(g)$	+180
$\text{Sr}(g) \longrightarrow \text{Sr}^{2+}(g) + 2e^-$	+1468
$\text{O}_2(g) \longrightarrow 2\text{O}(g)$	+500
$\text{O}(g) + 2e^- \longrightarrow \text{O}^{2-}(g)$	+600
$\text{Sr}(s) + \frac{1}{2}\text{O}_2(g) \longrightarrow \text{SrO}(s)$	-550

**Question 7 10 points**

Consider the molecule given on the left, and answer questions (i) to (vi) using chemical bonding theories that we have learnt in class (valence bond theory, hybridization).

Note that the lone pairs are not shown in the structure:

i. Consider the carbon atoms marked in as ***a***, ***b***, ***c***, ***d***, and ***e*** in the molecule, and determine **hybridization** at each of the carbon centers. (2.5 points)

Atom	Hybridization
Carbon (<i>a</i>)	
Carbon (<i>b</i>)	
Carbon (<i>c</i>)	
Carbon (<i>d</i>)	
Carbon (<i>e</i>)	

ii. Now, consider the sulfur atom (S) marked as ***f***, nitrogen (N) atom (***g***), and oxygen(O) atoms (***h*** and ***i***), and determine their **hybridization**. (2 points)

Atom	Hybridization
Sulfur (<i>f</i>)	
Nitrogen (<i>g</i>)	
Oxygen (<i>h</i>)	
Oxygen (<i>i</i>)	

iii. **Describe** all the bonds formed by the carbon atoms ***a*** and ***b***, using sigma and pi designation as appropriate. (2 points)

Atom	Bonds formed (use sigma and pi as appropriate)
Carbon (<i>a</i>)	
Carbon (<i>b</i>)	

iv. How many total sp^3 -s sigma bonds are in the molecule? (1 point)

v. Determine **electronic** and **molecular geometry** at carbon (C) labeled ***e*** and sulfur (S) labeled ***f*** centres. (2 points)

Atom	Electronic Geometry	Molecular Geometry
Carbon (<i>e</i>)		
Sulfur (<i>f</i>)		

vi. Circle the value from the choices given below, that describes the **C-S-C bond angle at sulfur** (S) marked ***f***: (0.5 points)

>90° to <109.5° 109.5° >109.5 to <120° 120° >120° to <180°


Question 8 12 points

Using the **valence shell orbitals only** draw a **complete** molecular orbital (MO) diagram (with no s-p mixing) for the molecule **PS**, in the space provided below. **(6 points)**

Assume that 3s and 3p atomic orbitals will form molecular orbitals like those formed by the combination of 2s and 2p atomic orbitals that we learnt in class, and there is no s-p mixing.

Show **energy arrow**, **atomic orbitals** involved, **atom labels**, the **molecular orbitals** that are formed, **filled electrons** in the MO diagram etc. Make sure to **label all the atomic and molecular orbitals**.

Using the MO diagram above, answer the following questions:

- (i) Which one of the following will have the highest bond order? Circle the correct choice
 A) PS^{1-} B) PS C) PS^{1+}
- (ii) Which one of the following will have the longest bond length? Circle the correct choice
 A) PS^{1-} B) PS C) PS^{1+}
- (iii) Which of the following will be diamagnetic? Circle the correct choice.
 A) PS^{1-} B) PS C) PS^{1+}
- (iv) Draw Lewis dot structure of the molecule with the shortest bond length (based on the MO diagram) from among the three (PS^{1-} , PS, PS^{1+}), and indicate the bond order calculated from the Lewis dot structure. **(3 points)**

Lewis Dot Structure

Bond Order



C4C788E7-7A7E-431F-962C-A36D16FC33A2

midterm-exam-93388

#753 Page 10 of 10

SCRAP PAGE

**CHEMISTRY 110 –Midterm Exam****Thursday November 7, 2024, 6:30 – 8:30 pm**

Examiners: Pallavi Sirjoosingh and Ashok Kakkar

First name (please write as legibly as possible within the boxes)

--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--

Last name

--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--

Student ID number

--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--

This booklet has 8 questions on 10 pages (The last page can be used as a scrap page).

The staple must NOT be removed from the question booklet. Your intact question booklet must be returned at the end of the exam.

A separate booklet contains a periodic table, and datasheet.

Write your answers directly under the questions in the space provided. Do not write on the QR codes or near the top edges (as it may be cut off).

Please write **legibly and dark enough** for the scan to be readable. For students who use English as a second language, dictionaries are permitted. Calculators are allowed.

DO NOT ASK QUESTIONS OF THE INVIGILATORS about interpretations of the exam questions. If you have a concern about a question, please summarize it at the top of this page. All concerns will be evaluated after the exam. GOOD LUCK!

**Question 1 8 points** (*Assume all transitions are possible*)

a. 6 points Calculate the **energy (in kJ/mol)** required to ionize (completely remove) the electron from the **$n=3$** level of **He^+ ion** (He: Helium). Show your work including equations. (3 sig. figs.)

b. 2 points

(I) Select the largest (longest) wavelength from the three choices below:

- i. Wavelength of photon to *completely remove* an electron from *1s orbital of Hydrogen atom*
- ii. Wavelength of photon to *completely remove* an electron from *1s orbital of Boron atom (B)*
- iii. Wavelength of photon to *completely remove* an electron from *1s orbital of Boron cation (B^+)*

i or ii or iii

(II) Select the smallest (shortest) wavelength from the three choices below (*for H atom*):

- i. Wavelength of photon *emitted* for an electron transitioning from *2p to 1s orbital*
- ii. Wavelength of photon *absorbed* for an electron transitioning from *2s to 4p orbital*
- iii. Wavelength of photon *absorbed* for an electron transitioning from *1s to 3p orbital*

i or ii or iii



Question 2 8 points



A Chem110 student plots the graph shown on the left, from data obtained for the kinetic energy (eV) of the ejected electron versus frequency (s^{-1}) of the incident light, for an unknown metal. ($1\text{eV} = 1.6022 \times 10^{-19} \text{ J}$)

Based on the plot, answer the questions **a to c** below:

a. 1 point On the plot above, label the **threshold frequency**.

b. 2 points What is the value of the work potential (in J) of the unknown metal?

c. 3 points What is the *longest* wavelength (in nm) required to eject an electron from this metal's surface? Show your work including equation(s) used. (3 sig figs)

d. 2 points In the space provided below, plot the following

(Assume frequency is *above* threshold frequency):

- Kinetic energy* of an ejected electron versus the *intensity* of the incident light (left)
- Number of electrons* ejected versus the *frequency* of the incident light (right)



**Question 3** **6 points****a. 2 points**

In the ground state of Ge (Germanium; $Z = 32$) atom, **name** the following orbitals:

i. Orbital containing the electron that requires the maximum energy to be removed.

ii. Orbital containing the electron that experiences the minimum effective nuclear charge.

b. 2 points

An unknown anion (A^-) has the following *excited* electronic configuration: $[Ar] 4s^2 3d^3 4p^1$

(Ar: Argon; $Z = 18$)

Give the **condensed ground state electronic configuration** of the atom, **A** and the cation, **A^{2+}** .
Show your work. (Assume all electronic transitions are possible)

Electronic Configuration of A:

Electronic Configuration of A^{2+} :

c. 2 points

In a single atom, determine how many electrons can have the following set of quantum numbers?

(i) $n = 5; l = 3$

In a single atom, determine how many electrons can have the following set of quantum numbers?

(i) $n = 3; l = 2; m_s = +1/2$

**Question 4** **8 points**

Rank the following from 1 to 4 in **increasing** order. Provide the ranking in the boxes provided.

Note: Atomic number (Z) are provided for relevant atoms

a) Increasing (lowest to highest) order of *ionization energy*: (1= lowest; 4 = highest)

Mg

Mg: Z = 12

Rb

Rb: Z = 37

Cs

Cs: Z = 55

K

K : Z = 19

b) Increasing (lowest to highest) order of *bond angle*: (1= lowest; 4 = highest)

H₂OCCl₄XeF₄

HCN

c) Increasing (smallest to largest) order of *ionic size* (1= smallest; 4 = largest)

S²⁻

S: Z = 16

O²⁻

O: Z = 8

Be²⁺

Be: Z = 4

Na⁺

Na: Z = 11

d) Increasing (lowest to highest) order of *bond energy* (1= lowest; 4 = highest)

P-F

P-Cl

P-I

P-Br

**Question 5** **12 points**

a. 9 points For the molecules HClO_4 and TeF_6 , draw the most stable *Lewis structure* showing appropriate shape and *dashed/wedged bonds*, based on VSEPR. Determine the **molecular geometry** around the central atom. For HClO_4 , label the *most polar bond*.



b. 3 points (Note: Lone pairs are not shown in structures below)

(i) For the three *resonance* structures (1, 2 and 3) shown below, answer the following question:



(I) Circle the *most contributing* resonance structure.

Structure 1

Structure 2

Structure 3

(ii) For the three structures given (4, 5, and 6) below, answer the following questions:



(I) Circle all the structures that represent *resonance structures* of each other:

Structure 4

Structure 5

Structure 6

(II) Among the resonance structures, circle the *most contributing* resonance structure.

Structure 4

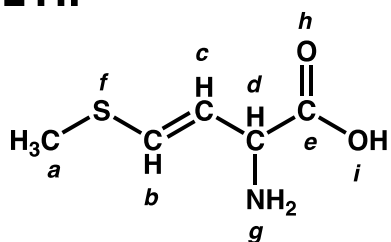
Structure 5

Structure 6

**Question 6** **6 points**

Using the data given below, calculate the **lattice energy** for SrO(s).

	ΔH° (kJ/mol)
$\text{Sr}(s) \longrightarrow \text{Sr}(g)$	+180
$\text{Sr}(g) \longrightarrow \text{Sr}^{2+}(g) + 2e^-$	+1468
$\text{O}_2(g) \longrightarrow 2\text{O}(g)$	+500
$\text{O}(g) + 2e^- \longrightarrow \text{O}^{2-}(g)$	+600
$\text{Sr}(s) + \frac{1}{2}\text{O}_2(g) \longrightarrow \text{SrO}(s)$	-550

**Question 7 10 points**

Consider the molecule given on the left, and answer questions (i) to (vi) using chemical bonding theories that we have learnt in class (valence bond theory, hybridization).

Note that the lone pairs are not shown in the structure:

i. Consider the carbon atoms marked in as ***a***, ***b***, ***c***, ***d***, and ***e*** in the molecule, and determine **hybridization** at each of the carbon centers. (2.5 points)

Atom	Hybridization
Carbon (<i>a</i>)	
Carbon (<i>b</i>)	
Carbon (<i>c</i>)	
Carbon (<i>d</i>)	
Carbon (<i>e</i>)	

ii. Now, consider the sulfur atom (S) marked as ***f***, nitrogen (N) atom (***g***), and oxygen(O) atoms (***h*** and ***i***), and determine their **hybridization**. (2 points)

Atom	Hybridization
Sulfur (<i>f</i>)	
Nitrogen (<i>g</i>)	
Oxygen (<i>h</i>)	
Oxygen (<i>i</i>)	

iii. **Describe** all the bonds formed by the carbon atoms ***a*** and ***b***, using sigma and pi designation as appropriate. (2 points)

Atom	Bonds formed (use sigma and pi as appropriate)
Carbon (<i>a</i>)	
Carbon (<i>b</i>)	

iv. How many total sp^3 -s sigma bonds are in the molecule? (1 point)

v. Determine **electronic** and **molecular geometry** at carbon (C) labeled ***e*** and sulfur (S) labeled ***f*** centres. (2 points)

Atom	Electronic Geometry	Molecular Geometry
Carbon (<i>e</i>)		
Sulfur (<i>f</i>)		

vi. Circle the value from the choices given below, that describes the **C-S-C bond angle at sulfur** (S) marked ***f***: (0.5 points)

>90° to <109.5° 109.5° >109.5 to <120° 120° >120° to <180°


Question 8 12 points

Using the **valence shell orbitals only** draw a **complete** molecular orbital (MO) diagram (with no s-p mixing) for the molecule **PS**, in the space provided below. **(6 points)**

Assume that 3s and 3p atomic orbitals will form molecular orbitals like those formed by the combination of 2s and 2p atomic orbitals that we learnt in class, and there is no s-p mixing.

Show **energy arrow**, **atomic orbitals** involved, **atom labels**, the **molecular orbitals** that are formed, **filled electrons** in the MO diagram etc. Make sure to label all the atomic and molecular orbitals.

Using the MO diagram above, answer the following questions:

- (i) Which one of the following will have the highest bond order? Circle the correct choice
 A) PS^{1-} B) PS C) PS^{1+}
- (ii) Which one of the following will have the longest bond length? Circle the correct choice
 A) PS^{1-} B) PS C) PS^{1+}
- (iii) Which of the following will be diamagnetic? Circle the correct choice.
 A) PS^{1-} B) PS C) PS^{1+}
- (iv) Draw Lewis dot structure of the molecule with the shortest bond length (based on the MO diagram) from among the three (PS^{1-} , PS, PS^{1+}), and indicate the bond order calculated from the Lewis dot structure. **(3 points)**

Lewis Dot Structure

Bond Order



74FC714C-D252-428A-8BC6-D1EF704592E0

midterm-exam-93388

#754 Page 10 of 10

SCRAP PAGE

**CHEMISTRY 110 –Midterm Exam****Thursday November 7, 2024, 6:30 – 8:30 pm**

Examiners: Pallavi Sirjoosingh and Ashok Kakkar

First name (please write as legibly as possible within the boxes)

--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--

Last name

--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--

Student ID number

--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--

This booklet has 8 questions on 10 pages (The last page can be used as a scrap page).

The staple must NOT be removed from the question booklet. Your intact question booklet must be returned at the end of the exam.

A separate booklet contains a periodic table, and datasheet.

Write your answers directly under the questions in the space provided. Do not write on the QR codes or near the top edges (as it may be cut off).

Please write **legibly and dark enough** for the scan to be readable. For students who use English as a second language, dictionaries are permitted. Calculators are allowed.

DO NOT ASK QUESTIONS OF THE INVIGILATORS about interpretations of the exam questions. If you have a concern about a question, please summarize it at the top of this page. All concerns will be evaluated after the exam. GOOD LUCK!

**Question 1 8 points** (*Assume all transitions are possible*)

a. 6 points Calculate the **energy (in kJ/mol)** required to ionize (completely remove) the electron from the **n=3** level of **He⁺ ion** (He: Helium). Show your work including equations. (3 sig. figs.)

b. 2 points

(I) Select the largest (longest) wavelength from the three choices below:

- i. Wavelength of photon to *completely remove* an electron from *1s orbital of Hydrogen atom*
- ii. Wavelength of photon to *completely remove* an electron from *1s orbital of Boron atom (B)*
- iii. Wavelength of photon to *completely remove* an electron from *1s orbital of Boron cation (B⁺)*

i or ii or iii

(II) Select the smallest (shortest) wavelength from the three choices below (*for H atom*):

- i. Wavelength of photon *emitted* for an electron transitioning from *2p to 1s orbital*
- ii. Wavelength of photon *absorbed* for an electron transitioning from *2s to 4p orbital*
- iii. Wavelength of photon *absorbed* for an electron transitioning from *1s to 3p orbital*

i or ii or iii



Question 2 8 points



A Chem110 student plots the graph shown on the left, from data obtained for the kinetic energy (eV) of the ejected electron versus frequency (s^{-1}) of the incident light, for an unknown metal. ($1\text{eV} = 1.6022 \times 10^{-19} \text{ J}$)

Based on the plot, answer the questions **a to c** below:

a. 1 point On the plot above, label the **threshold frequency**.

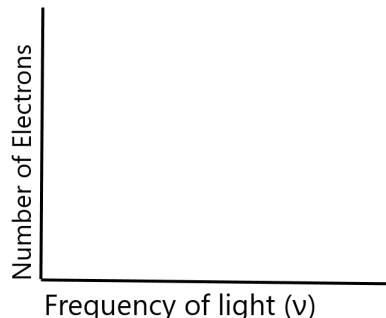
b. 2 points What is the value of the work potential (in J) of the unknown metal?

c. 3 points What is the *longest* wavelength (in nm) required to eject an electron from this metal's surface? Show your work including equation(s) used. (3 sig figs)

d. 2 points In the space provided below, plot the following

(Assume frequency is *above* threshold frequency):

- Kinetic energy* of an ejected electron versus the *intensity* of the incident light (left)
- Number of electrons* ejected versus the *frequency* of the incident light (right)



**Question 3** **6 points****a. 2 points**

In the ground state of Ge (Germanium; $Z = 32$) atom, **name** the following orbitals:

i. Orbital containing the electron that requires the maximum energy to be removed.

ii. Orbital containing the electron that experiences the minimum effective nuclear charge.

b. 2 points

An unknown anion (A^-) has the following *excited* electronic configuration: $[Ar] 4s^2 3d^3 4p^1$

(Ar: Argon; $Z = 18$)

Give the **condensed ground state electronic configuration** of the atom, **A** and the cation, **A^{2+}** .
Show your work. (Assume all electronic transitions are possible)

Electronic Configuration of A:

Electronic Configuration of A^{2+} :

c. 2 points

In a single atom, determine how many electrons can have the following set of quantum numbers?

(i) $n = 5; l = 3$

In a single atom, determine how many electrons can have the following set of quantum numbers?

(i) $n = 3; l = 2; m_s = +1/2$

**Question 4** **8 points**

Rank the following from 1 to 4 in **increasing** order. Provide the ranking in the boxes provided.

Note: Atomic number (Z) are provided for relevant atoms

a) Increasing (lowest to highest) order of *ionization energy*: (1= lowest; 4 = highest)

Mg

Mg: Z = 12

Rb

Rb: Z = 37

Cs

Cs: Z = 55

K

K : Z = 19

b) Increasing (lowest to highest) order of *bond angle*: (1= lowest; 4 = highest)

H₂OCCl₄XeF₄

HCN

c) Increasing (smallest to largest) order of *ionic size* (1= smallest; 4 = largest)

S²⁻

S: Z = 16

O²⁻

O: Z = 8

Be²⁺

Be: Z = 4

Na⁺

Na: Z = 11

d) Increasing (lowest to highest) order of *bond energy* (1= lowest; 4 = highest)

P-F

P-Cl

P-I

P-Br

**Question 5** **12 points**

a. 9 points For the molecules HClO_4 and TeF_6 , draw the most stable *Lewis structure* showing appropriate shape and *dashed/wedged bonds*, based on VSEPR. Determine the **molecular geometry** around the central atom. For HClO_4 , label the *most polar bond*.



b. 3 points (Note: Lone pairs are not shown in structures below)

(i) For the three *resonance* structures (1, 2 and 3) shown below, answer the following question:



(I) Circle the *most contributing* resonance structure.

Structure 1

Structure 2

Structure 3

(ii) For the three structures given (4, 5, and 6) below, answer the following questions:



(I) Circle all the structures that represent *resonance structures* of each other:

Structure 4

Structure 5

Structure 6

(II) Among the resonance structures, circle the *most contributing* resonance structure.

Structure 4

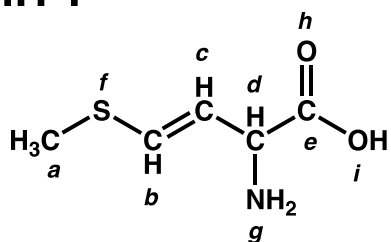
Structure 5

Structure 6

**Question 6** **6 points**

Using the data given below, calculate the **lattice energy** for SrO(s).

	ΔH° (kJ/mol)
$\text{Sr}(s) \longrightarrow \text{Sr}(g)$	+180
$\text{Sr}(g) \longrightarrow \text{Sr}^{2+}(g) + 2e^-$	+1468
$\text{O}_2(g) \longrightarrow 2\text{O}(g)$	+500
$\text{O}(g) + 2e^- \longrightarrow \text{O}^{2-}(g)$	+600
$\text{Sr}(s) + \frac{1}{2}\text{O}_2(g) \longrightarrow \text{SrO}(s)$	-550

**Question 7 10 points**

Consider the molecule given on the left, and answer questions (i) to (vi) using chemical bonding theories that we have learnt in class (valence bond theory, hybridization).

Note that the lone pairs are not shown in the structure:

i. Consider the carbon atoms marked in as ***a***, ***b***, ***c***, ***d***, and ***e*** in the molecule, and determine **hybridization** at each of the carbon centers. (2.5 points)

Atom	Hybridization
Carbon (<i>a</i>)	
Carbon (<i>b</i>)	
Carbon (<i>c</i>)	
Carbon (<i>d</i>)	
Carbon (<i>e</i>)	

ii. Now, consider the sulfur atom (S) marked as ***f***, nitrogen (N) atom (***g***), and oxygen(O) atoms (***h*** and ***i***), and determine their **hybridization**. (2 points)

Atom	Hybridization
Sulfur (<i>f</i>)	
Nitrogen (<i>g</i>)	
Oxygen (<i>h</i>)	
Oxygen (<i>i</i>)	

iii. **Describe** all the bonds formed by the carbon atoms ***a*** and ***b***, using sigma and pi designation as appropriate. (2 points)

Atom	Bonds formed (use sigma and pi as appropriate)
Carbon (<i>a</i>)	
Carbon (<i>b</i>)	

iv. How many total sp^3 -s sigma bonds are in the molecule? (1 point)

v. Determine **electronic** and **molecular geometry** at carbon (C) labeled ***e*** and sulfur (S) labeled ***f*** centres. (2 points)

Atom	Electronic Geometry	Molecular Geometry
Carbon (<i>e</i>)		
Sulfur (<i>f</i>)		

vi. Circle the value from the choices given below, that describes the **C-S-C bond angle at sulfur** (S) marked ***f***: (0.5 points)

>90° to <109.5° 109.5° >109.5 to <120° 120° >120° to <180°


Question 8 12 points

Using the **valence shell orbitals only** draw a **complete** molecular orbital (MO) diagram (with no s-p mixing) for the molecule **PS**, in the space provided below. **(6 points)**

Assume that 3s and 3p atomic orbitals will form molecular orbitals like those formed by the combination of 2s and 2p atomic orbitals that we learnt in class, and there is no s-p mixing.

Show **energy arrow**, **atomic orbitals** involved, **atom labels**, the **molecular orbitals** that are formed, **filled electrons** in the MO diagram etc. Make sure to label all the atomic and molecular orbitals.

Using the MO diagram above, answer the following questions:

- (i) Which one of the following will have the highest bond order? Circle the correct choice
 A) PS^{1-} B) PS C) PS^{1+}
- (ii) Which one of the following will have the longest bond length? Circle the correct choice
 A) PS^{1-} B) PS C) PS^{1+}
- (iii) Which of the following will be diamagnetic? Circle the correct choice.
 A) PS^{1-} B) PS C) PS^{1+}
- (iv) Draw Lewis dot structure of the molecule with the shortest bond length (based on the MO diagram) from among the three (PS^{1-} , PS, PS^{1+}), and indicate the bond order calculated from the Lewis dot structure. **(3 points)**

Lewis Dot Structure

Bond Order



6780F324-0315-407E-91E6-ACFFD12323C3

midterm-exam-93388

#755 Page 10 of 10

SCRAP PAGE

**CHEMISTRY 110 –Midterm Exam****Thursday November 7, 2024, 6:30 – 8:30 pm**

Examiners: Pallavi Sirjoosingh and Ashok Kakkar

First name (please write as legibly as possible within the boxes)

--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--

Last name

--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--

Student ID number

--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--

This booklet has 8 questions on 10 pages (The last page can be used as a scrap page).

The staple must NOT be removed from the question booklet. Your intact question booklet must be returned at the end of the exam.

A separate booklet contains a periodic table, and datasheet.

Write your answers directly under the questions in the space provided. Do not write on the QR codes or near the top edges (as it may be cut off).

Please write **legibly and dark enough** for the scan to be readable. For students who use English as a second language, dictionaries are permitted. Calculators are allowed.

DO NOT ASK QUESTIONS OF THE INVIGILATORS about interpretations of the exam questions. If you have a concern about a question, please summarize it at the top of this page. All concerns will be evaluated after the exam. GOOD LUCK!

**Question 1 8 points** (*Assume all transitions are possible*)

a. 6 points Calculate the **energy (in kJ/mol)** required to ionize (completely remove) the electron from the **n=3** level of **He⁺ ion** (He: Helium). Show your work including equations. (3 sig. figs.)

b. 2 points

(I) Select the largest (longest) wavelength from the three choices below:

- i. Wavelength of photon to *completely remove* an electron from *1s orbital of Hydrogen atom*
- ii. Wavelength of photon to *completely remove* an electron from *1s orbital of Boron atom (B)*
- iii. Wavelength of photon to *completely remove* an electron from *1s orbital of Boron cation (B⁺)*

i or ii or iii

(II) Select the smallest (shortest) wavelength from the three choices below (*for H atom*):

- i. Wavelength of photon *emitted* for an electron transitioning from *2p to 1s orbital*
- ii. Wavelength of photon *absorbed* for an electron transitioning from *2s to 4p orbital*
- iii. Wavelength of photon *absorbed* for an electron transitioning from *1s to 3p orbital*

i or ii or iii



Question 2 8 points



A Chem110 student plots the graph shown on the left, from data obtained for the kinetic energy (eV) of the ejected electron versus frequency (s^{-1}) of the incident light, for an unknown metal. ($1\text{eV} = 1.6022 \times 10^{-19} \text{ J}$)

Based on the plot, answer the questions **a to c** below:

a. 1 point On the plot above, label the **threshold frequency**.

b. 2 points What is the value of the work potential (in J) of the unknown metal?

c. 3 points What is the *longest* wavelength (in nm) required to eject an electron from this metal's surface? Show your work including equation(s) used. (3 sig figs)

d. 2 points In the space provided below, plot the following

(Assume frequency is *above* threshold frequency):

- i. *Kinetic energy* of an ejected electron versus the *intensity* of the incident light (left)
- ii. *Number of electrons* ejected versus the *frequency* of the incident light (right)



**Question 3** **6 points****a. 2 points**

In the ground state of Ge (Germanium; $Z = 32$) atom, **name** the following orbitals:

i. Orbital containing the electron that requires the maximum energy to be removed.

ii. Orbital containing the electron that experiences the minimum effective nuclear charge.

b. 2 points

An unknown anion (A^-) has the following *excited* electronic configuration: $[Ar] 4s^2 3d^3 4p^1$

(Ar: Argon; $Z = 18$)

Give the **condensed ground state electronic configuration** of the atom, **A** and the cation, **A^{2+}** .
Show your work. (Assume all electronic transitions are possible)

Electronic Configuration of A:

Electronic Configuration of A^{2+} :

c. 2 points

In a single atom, determine how many electrons can have the following set of quantum numbers?

(i) $n = 5; l = 3$

In a single atom, determine how many electrons can have the following set of quantum numbers?

(i) $n = 3; l = 2; m_s = +1/2$

**Question 4** **8 points**

Rank the following from 1 to 4 in **increasing** order. Provide the ranking in the boxes provided.

Note: Atomic number (Z) are provided for relevant atoms

a) Increasing (lowest to highest) order of *ionization energy*: (1= lowest; 4 = highest)

Mg

Mg: Z = 12

Rb

Rb: Z = 37

Cs

Cs: Z = 55

K

K : Z = 19

b) Increasing (lowest to highest) order of *bond angle*: (1= lowest; 4 = highest)

H₂OCCl₄XeF₄

HCN

c) Increasing (smallest to largest) order of *ionic size* (1= smallest; 4 = largest)

S²⁻

S: Z = 16

O²⁻

O: Z = 8

Be²⁺

Be: Z = 4

Na⁺

Na: Z = 11

d) Increasing (lowest to highest) order of *bond energy* (1= lowest; 4 = highest)

P-F

P-Cl

P-I

P-Br

**Question 5** **12 points**

a. 9 points For the molecules HClO_4 and TeF_6 , draw the most stable *Lewis structure* showing appropriate shape and *dashed/wedged bonds*, based on VSEPR. Determine the **molecular geometry** around the central atom. For HClO_4 , label the *most polar bond*.



b. 3 points (Note: Lone pairs are not shown in structures below)

(i) For the three *resonance* structures (1, 2 and 3) shown below, answer the following question:



(I) Circle the *most contributing* resonance structure.

Structure 1

Structure 2

Structure 3

(ii) For the three structures given (4, 5, and 6) below, answer the following questions:



(I) Circle all the structures that represent *resonance structures* of each other:

Structure 4

Structure 5

Structure 6

(II) Among the resonance structures, circle the *most contributing* resonance structure.

Structure 4

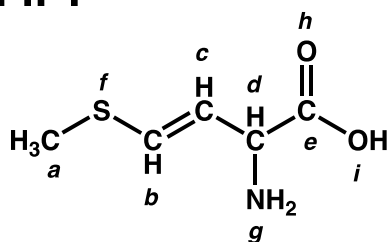
Structure 5

Structure 6

**Question 6** **6 points**

Using the data given below, calculate the **lattice energy** for SrO(s).

	ΔH° (kJ/mol)
$\text{Sr}(s) \longrightarrow \text{Sr}(g)$	+180
$\text{Sr}(g) \longrightarrow \text{Sr}^{2+}(g) + 2e^-$	+1468
$\text{O}_2(g) \longrightarrow 2\text{O}(g)$	+500
$\text{O}(g) + 2e^- \longrightarrow \text{O}^{2-}(g)$	+600
$\text{Sr}(s) + \frac{1}{2}\text{O}_2(g) \longrightarrow \text{SrO}(s)$	-550

**Question 7 10 points**

Consider the molecule given on the left, and answer questions (i) to (vi) using chemical bonding theories that we have learnt in class (valence bond theory, hybridization).

Note that the lone pairs are not shown in the structure:

i. Consider the carbon atoms marked in as ***a***, ***b***, ***c***, ***d***, and ***e*** in the molecule, and determine **hybridization** at each of the carbon centers. (2.5 points)

Atom	Hybridization
Carbon (<i>a</i>)	
Carbon (<i>b</i>)	
Carbon (<i>c</i>)	
Carbon (<i>d</i>)	
Carbon (<i>e</i>)	

ii. Now, consider the sulfur atom (S) marked as ***f***, nitrogen (N) atom (***g***), and oxygen(O) atoms (***h*** and ***i***), and determine their **hybridization**. (2 points)

Atom	Hybridization
Sulfur (<i>f</i>)	
Nitrogen (<i>g</i>)	
Oxygen (<i>h</i>)	
Oxygen (<i>i</i>)	

iii. **Describe** all the bonds formed by the carbon atoms ***a*** and ***b***, using sigma and pi designation as appropriate. (2 points)

Atom	Bonds formed (use sigma and pi as appropriate)
Carbon (<i>a</i>)	
Carbon (<i>b</i>)	

iv. How many total sp^3 -s sigma bonds are in the molecule? (1 point)

v. Determine **electronic** and **molecular geometry** at carbon (C) labeled ***e*** and sulfur (S) labeled ***f*** centres. (2 points)

Atom	Electronic Geometry	Molecular Geometry
Carbon (<i>e</i>)		
Sulfur (<i>f</i>)		

vi. Circle the value from the choices given below, that describes the **C-S-C bond angle at sulfur** (S) marked ***f***: (0.5 points)

>90° to <109.5°

109.5°

>109.5 to <120°

120°

>120° to <180°


Question 8 12 points

Using the **valence shell orbitals only** draw a **complete** molecular orbital (MO) diagram (with no s-p mixing) for the molecule **PS**, in the space provided below. **(6 points)**

Assume that 3s and 3p atomic orbitals will form molecular orbitals like those formed by the combination of 2s and 2p atomic orbitals that we learnt in class, and there is no s-p mixing.

Show **energy arrow**, **atomic orbitals** involved, **atom labels**, the **molecular orbitals** that are formed, **filled electrons** in the MO diagram etc. Make sure to label all the atomic and molecular orbitals.

Using the MO diagram above, answer the following questions:

- (i) Which one of the following will have the highest bond order? Circle the correct choice
 A) PS^{1-} B) PS C) PS^{1+}
- (ii) Which one of the following will have the longest bond length? Circle the correct choice
 A) PS^{1-} B) PS C) PS^{1+}
- (iii) Which of the following will be diamagnetic? Circle the correct choice.
 A) PS^{1-} B) PS C) PS^{1+}
- (iv) Draw Lewis dot structure of the molecule with the shortest bond length (based on the MO diagram) from among the three (PS^{1-} , PS, PS^{1+}), and indicate the bond order calculated from the Lewis dot structure. **(3 points)**

Lewis Dot Structure

Bond Order



2DB0ECB3-6195-49F5-843F-F7D00E680246

midterm-exam-93388

#756 Page 10 of 10

SCRAP PAGE

**CHEMISTRY 110 –Midterm Exam****Thursday November 7, 2024, 6:30 – 8:30 pm****Examiners: Pallavi Sirjoosingh and Ashok Kakkar**

First name (please write as legibly as possible within the boxes)

--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--

Last name

--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--

Student ID number

--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--

This booklet has 8 questions on 10 pages (The last page can be used as a scrap page).

The staple must NOT be removed from the question booklet. Your intact question booklet must be returned at the end of the exam.

A separate booklet contains a periodic table, and datasheet.

Write your answers directly under the questions in the space provided. Do not write on the QR codes or near the top edges (as it may be cut off).

Please write **legibly and dark enough** for the scan to be readable. For students who use English as a second language, dictionaries are permitted. Calculators are allowed.

DO NOT ASK QUESTIONS OF THE INVIGILATORS about interpretations of the exam questions. If you have a concern about a question, please summarize it at the top of this page. All concerns will be evaluated after the exam. GOOD LUCK!

**Question 1 8 points** (*Assume all transitions are possible*)

a. 6 points Calculate the **energy (in kJ/mol)** required to ionize (completely remove) the electron from the **$n=3$** level of **He^+ ion** (He: Helium). Show your work including equations. (3 sig. figs.)

b. 2 points

(I) Select the largest (longest) wavelength from the three choices below:

- i. Wavelength of photon to *completely remove* an electron from *1s orbital of Hydrogen atom*
- ii. Wavelength of photon to *completely remove* an electron from *1s orbital of Boron atom (B)*
- iii. Wavelength of photon to *completely remove* an electron from *1s orbital of Boron cation (B^+)*

i or ii or iii

(II) Select the smallest (shortest) wavelength from the three choices below (*for H atom*):

- i. Wavelength of photon *emitted* for an electron transitioning from *2p to 1s orbital*
- ii. Wavelength of photon *absorbed* for an electron transitioning from *2s to 4p orbital*
- iii. Wavelength of photon *absorbed* for an electron transitioning from *1s to 3p orbital*

i or ii or iii



Question 2 8 points



A Chem110 student plots the graph shown on the left, from data obtained for the kinetic energy (eV) of the ejected electron versus frequency (s^{-1}) of the incident light, for an unknown metal. ($1\text{eV} = 1.6022 \times 10^{-19} \text{ J}$)

Based on the plot, answer the questions **a to c** below:

a. 1 point On the plot above, label the **threshold frequency**.

b. 2 points What is the value of the work potential (in J) of the unknown metal?

c. 3 points What is the *longest* wavelength (in nm) required to eject an electron from this metal's surface? Show your work including equation(s) used. (3 sig figs)

d. 2 points In the space provided below, plot the following

(Assume frequency is *above* threshold frequency):

- Kinetic energy* of an ejected electron versus the *intensity* of the incident light (left)
- Number of electrons* ejected versus the *frequency* of the incident light (right)



**Question 3** **6 points****a. 2 points**

In the ground state of Ge (Germanium; $Z = 32$) atom, **name** the following orbitals:

i. Orbital containing the electron that requires the maximum energy to be removed.

ii. Orbital containing the electron that experiences the minimum effective nuclear charge.

b. 2 points

An unknown anion (A^-) has the following *excited* electronic configuration: $[Ar] 4s^2 3d^3 4p^1$

(Ar: Argon; $Z = 18$)

Give the **condensed ground state electronic configuration** of the atom, **A** and the cation, **A^{2+}** .
Show your work. (Assume all electronic transitions are possible)

Electronic Configuration of A:

Electronic Configuration of A^{2+} :

c. 2 points

In a single atom, determine how many electrons can have the following set of quantum numbers?

(i) $n = 5; l = 3$

In a single atom, determine how many electrons can have the following set of quantum numbers?

(i) $n = 3; l = 2; m_s = +1/2$

**Question 4** **8 points**

Rank the following from 1 to 4 in **increasing** order. Provide the ranking in the boxes provided.

Note: Atomic number (Z) are provided for relevant atoms

a) Increasing (lowest to highest) order of *ionization energy*: (1= lowest; 4 = highest)

Mg

Mg: Z = 12

Rb

Rb: Z = 37

Cs

Cs: Z = 55

K

K : Z = 19

b) Increasing (lowest to highest) order of *bond angle*: (1= lowest; 4 = highest)

H₂OCCl₄XeF₄

HCN

c) Increasing (smallest to largest) order of *ionic size* (1= smallest; 4 = largest)

S²⁻

S: Z = 16

O²⁻

O: Z = 8

Be²⁺

Be: Z = 4

Na⁺

Na: Z = 11

d) Increasing (lowest to highest) order of *bond energy* (1= lowest; 4 = highest)

P-F

P-Cl

P-I

P-Br

**Question 5** **12 points**

a. 9 points For the molecules HClO_4 and TeF_6 , draw the most stable *Lewis structure* showing appropriate shape and *dashed/wedged bonds*, based on VSEPR. Determine the **molecular geometry** around the central atom. For HClO_4 , label the *most polar bond*.



b. 3 points (Note: Lone pairs are not shown in structures below)

(i) For the three *resonance* structures (1, 2 and 3) shown below, answer the following question:



(I) Circle the *most contributing* resonance structure.

Structure 1

Structure 2

Structure 3

(ii) For the three structures given (4, 5, and 6) below, answer the following questions:



(I) Circle all the structures that represent *resonance structures* of each other:

Structure 4

Structure 5

Structure 6

(II) Among the resonance structures, circle the *most contributing* resonance structure.

Structure 4

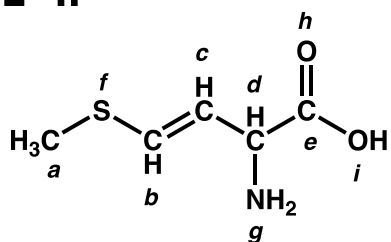
Structure 5

Structure 6

**Question 6** **6 points**

Using the data given below, calculate the **lattice energy** for SrO(s).

	ΔH° (kJ/mol)
$\text{Sr}(s) \longrightarrow \text{Sr}(g)$	+180
$\text{Sr}(g) \longrightarrow \text{Sr}^{2+}(g) + 2e^-$	+1468
$\text{O}_2(g) \longrightarrow 2\text{O}(g)$	+500
$\text{O}(g) + 2e^- \longrightarrow \text{O}^{2-}(g)$	+600
$\text{Sr}(s) + \frac{1}{2}\text{O}_2(g) \longrightarrow \text{SrO}(s)$	-550

**Question 7 10 points**

Consider the molecule given on the left, and answer questions (i) to (vi) using chemical bonding theories that we have learnt in class (valence bond theory, hybridization).

Note that the lone pairs are not shown in the structure:

i. Consider the carbon atoms marked in as ***a***, ***b***, ***c***, ***d***, and ***e*** in the molecule, and determine **hybridization** at each of the carbon centers. (2.5 points)

Atom	Hybridization
Carbon (<i>a</i>)	
Carbon (<i>b</i>)	
Carbon (<i>c</i>)	
Carbon (<i>d</i>)	
Carbon (<i>e</i>)	

ii. Now, consider the sulfur atom (S) marked as ***f***, nitrogen (N) atom (***g***), and oxygen(O) atoms (***h*** and ***i***), and determine their **hybridization**. (2 points)

Atom	Hybridization
Sulfur (<i>f</i>)	
Nitrogen (<i>g</i>)	
Oxygen (<i>h</i>)	
Oxygen (<i>i</i>)	

iii. **Describe** all the bonds formed by the carbon atoms ***a*** and ***b***, using sigma and pi designation as appropriate. (2 points)

Atom	Bonds formed (use sigma and pi as appropriate)
Carbon (<i>a</i>)	
Carbon (<i>b</i>)	

iv. How many total sp^3 -s sigma bonds are in the molecule? (1 point)

v. Determine **electronic** and **molecular geometry** at carbon (C) labeled ***e*** and sulfur (S) labeled ***f*** centres. (2 points)

Atom	Electronic Geometry	Molecular Geometry
Carbon (<i>e</i>)		
Sulfur (<i>f</i>)		

vi. Circle the value from the choices given below, that describes the **C-S-C bond angle at sulfur** (S) marked ***f***: (0.5 points)

>90° to <109.5° 109.5° >109.5 to <120° 120° >120° to <180°


Question 8 12 points

Using the **valence shell orbitals only** draw a **complete** molecular orbital (MO) diagram (with no s-p mixing) for the molecule **PS**, in the space provided below. **(6 points)**

Assume that 3s and 3p atomic orbitals will form molecular orbitals like those formed by the combination of 2s and 2p atomic orbitals that we learnt in class, and there is no s-p mixing.

Show **energy arrow**, **atomic orbitals** involved, **atom labels**, the **molecular orbitals** that are formed, **filled electrons** in the MO diagram etc. Make sure to label all the atomic and molecular orbitals.

Using the MO diagram above, answer the following questions:

- (i) Which one of the following will have the highest bond order? Circle the correct choice
 A) PS^{1-} B) PS C) PS^{1+}
- (ii) Which one of the following will have the longest bond length? Circle the correct choice
 A) PS^{1-} B) PS C) PS^{1+}
- (iii) Which of the following will be diamagnetic? Circle the correct choice.
 A) PS^{1-} B) PS C) PS^{1+}
- (iv) Draw Lewis dot structure of the molecule with the shortest bond length (based on the MO diagram) from among the three (PS^{1-} , PS, PS^{1+}), and indicate the bond order calculated from the Lewis dot structure. **(3 points)**

Lewis Dot Structure

Bond Order



C3B4F108-E9CA-462F-89C3-2B1CD747C1AD

midterm-exam-93388

#757 Page 10 of 10

SCRAP PAGE

**CHEMISTRY 110 –Midterm Exam****Thursday November 7, 2024, 6:30 – 8:30 pm****Examiners: Pallavi Sirjoosingh and Ashok Kakkar**

First name (please write as legibly as possible within the boxes)

--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--

Last name

--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--

Student ID number

--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--

This booklet has 8 questions on 10 pages (The last page can be used as a scrap page).

The staple must NOT be removed from the question booklet. Your intact question booklet must be returned at the end of the exam.

A separate booklet contains a periodic table, and datasheet.

Write your answers directly under the questions in the space provided. Do not write on the QR codes or near the top edges (as it may be cut off).

Please write **legibly and dark enough** for the scan to be readable. For students who use English as a second language, dictionaries are permitted. Calculators are allowed.

DO NOT ASK QUESTIONS OF THE INVIGILATORS about interpretations of the exam questions. If you have a concern about a question, please summarize it at the top of this page. All concerns will be evaluated after the exam. GOOD LUCK!

**Question 1 8 points** (*Assume all transitions are possible*)

a. 6 points Calculate the **energy (in kJ/mol)** required to ionize (completely remove) the electron from the **n=3** level of **He⁺ ion** (He: Helium). Show your work including equations. (3 sig. figs.)

b. 2 points

(I) Select the largest (longest) wavelength from the three choices below:

- i. Wavelength of photon to *completely remove* an electron from *1s orbital of Hydrogen atom*
- ii. Wavelength of photon to *completely remove* an electron from *1s orbital of Boron atom (B)*
- iii. Wavelength of photon to *completely remove* an electron from *1s orbital of Boron cation (B⁺)*

i or ii or iii

(II) Select the smallest (shortest) wavelength from the three choices below (*for H atom*):

- i. Wavelength of photon *emitted* for an electron transitioning from *2p to 1s orbital*
- ii. Wavelength of photon *absorbed* for an electron transitioning from *2s to 4p orbital*
- iii. Wavelength of photon *absorbed* for an electron transitioning from *1s to 3p orbital*

i or ii or iii



Question 2 8 points



A Chem110 student plots the graph shown on the left, from data obtained for the kinetic energy (eV) of the ejected electron versus frequency (s^{-1}) of the incident light, for an unknown metal. ($1\text{eV} = 1.6022 \times 10^{-19} \text{ J}$)

Based on the plot, answer the questions **a to c** below:

a. 1 point On the plot above, label the **threshold frequency**.

b. 2 points What is the value of the work potential (in J) of the unknown metal?

c. 3 points What is the *longest* wavelength (in nm) required to eject an electron from this metal's surface? Show your work including equation(s) used. (3 sig figs)

d. 2 points In the space provided below, plot the following

(Assume frequency is *above* threshold frequency):

- Kinetic energy* of an ejected electron versus the *intensity* of the incident light (left)
- Number of electrons* ejected versus the *frequency* of the incident light (right)



**Question 3** **6 points****a. 2 points**

In the ground state of Ge (Germanium; $Z = 32$) atom, **name** the following orbitals:

i. Orbital containing the electron that requires the maximum energy to be removed.

ii. Orbital containing the electron that experiences the minimum effective nuclear charge.

b. 2 points

An unknown anion (A^-) has the following *excited* electronic configuration: $[Ar] 4s^2 3d^3 4p^1$

(Ar: Argon; $Z = 18$)

Give the **condensed ground state electronic configuration** of the atom, **A** and the cation, **A^{2+}** .
Show your work. (Assume all electronic transitions are possible)

Electronic Configuration of A:

Electronic Configuration of A^{2+} :

c. 2 points

In a single atom, determine how many electrons can have the following set of quantum numbers?

(i) $n = 5; l = 3$

In a single atom, determine how many electrons can have the following set of quantum numbers?

(i) $n = 3; l = 2; m_s = +1/2$

**Question 4** **8 points**

Rank the following from 1 to 4 in **increasing** order. Provide the ranking in the boxes provided.

Note: Atomic number (Z) are provided for relevant atoms

a) Increasing (lowest to highest) order of *ionization energy*: (1= lowest; 4 = highest)

Mg

Mg: Z = 12

Rb

Rb: Z = 37

Cs

Cs: Z = 55

K

K : Z = 19

b) Increasing (lowest to highest) order of *bond angle*: (1= lowest; 4 = highest)

H₂OCCl₄XeF₄

HCN

c) Increasing (smallest to largest) order of *ionic size* (1= smallest; 4 = largest)

S²⁻

S: Z = 16

O²⁻

O: Z = 8

Be²⁺

Be: Z = 4

Na⁺

Na: Z = 11

d) Increasing (lowest to highest) order of *bond energy* (1= lowest; 4 = highest)

P-F

P-Cl

P-I

P-Br

**Question 5** **12 points**

a. 9 points For the molecules HClO_4 and TeF_6 , draw the most stable *Lewis structure* showing appropriate shape and *dashed/wedged bonds*, based on VSEPR. Determine the **molecular geometry** around the central atom. For HClO_4 , label the *most polar bond*.



b. 3 points (Note: Lone pairs are not shown in structures below)

(i) For the three *resonance* structures (1, 2 and 3) shown below, answer the following question:



(I) Circle the *most contributing* resonance structure.

Structure 1

Structure 2

Structure 3

(ii) For the three structures given (4, 5, and 6) below, answer the following questions:



(I) Circle all the structures that represent *resonance structures* of each other:

Structure 4

Structure 5

Structure 6

(II) Among the resonance structures, circle the *most contributing* resonance structure.

Structure 4

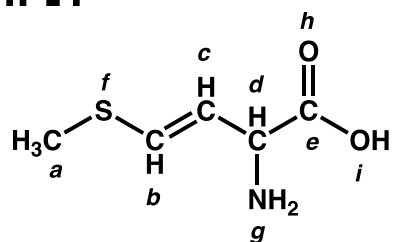
Structure 5

Structure 6

**Question 6** **6 points**

Using the data given below, calculate the **lattice energy** for SrO(s).

	ΔH° (kJ/mol)
$\text{Sr}(s) \longrightarrow \text{Sr}(g)$	+180
$\text{Sr}(g) \longrightarrow \text{Sr}^{2+}(g) + 2e^-$	+1468
$\text{O}_2(g) \longrightarrow 2\text{O}(g)$	+500
$\text{O}(g) + 2e^- \longrightarrow \text{O}^{2-}(g)$	+600
$\text{Sr}(s) + \frac{1}{2}\text{O}_2(g) \longrightarrow \text{SrO}(s)$	-550

**Question 7 10 points**

Consider the molecule given on the left, and answer questions (i) to (vi) using chemical bonding theories that we have learnt in class (valence bond theory, hybridization).

Note that the lone pairs are not shown in the structure:

i. Consider the carbon atoms marked in as ***a***, ***b***, ***c***, ***d***, and ***e*** in the molecule, and determine **hybridization** at each of the carbon centers. (2.5 points)

Atom	Hybridization
Carbon (<i>a</i>)	
Carbon (<i>b</i>)	
Carbon (<i>c</i>)	
Carbon (<i>d</i>)	
Carbon (<i>e</i>)	

ii. Now, consider the sulfur atom (S) marked as ***f***, nitrogen (N) atom (***g***), and oxygen(O) atoms (***h*** and ***i***), and determine their **hybridization**. (2 points)

Atom	Hybridization
Sulfur (<i>f</i>)	
Nitrogen (<i>g</i>)	
Oxygen (<i>h</i>)	
Oxygen (<i>i</i>)	

iii. **Describe** all the bonds formed by the carbon atoms ***a*** and ***b***, using sigma and pi designation as appropriate. (2 points)

Atom	Bonds formed (use sigma and pi as appropriate)
Carbon (<i>a</i>)	
Carbon (<i>b</i>)	

iv. How many total sp^3 -s sigma bonds are in the molecule? (1 point)

v. Determine **electronic** and **molecular geometry** at carbon (C) labeled ***e*** and sulfur (S) labeled ***f*** centres. (2 points)

Atom	Electronic Geometry	Molecular Geometry
Carbon (<i>e</i>)		
Sulfur (<i>f</i>)		

vi. Circle the value from the choices given below, that describes the **C-S-C bond angle at sulfur** (S) marked ***f***: (0.5 points)

$>90^\circ$ to $<109.5^\circ$ 109.5° >109.5 to $<120^\circ$ 120° $>120^\circ$ to $<180^\circ$

**Question 8 12 points**

Using the **valence shell orbitals only** draw a **complete** molecular orbital (MO) diagram (with no s-p mixing) for the molecule **PS**, in the space provided below. **(6 points)**

Assume that 3s and 3p atomic orbitals will form molecular orbitals like those formed by the combination of 2s and 2p atomic orbitals that we learnt in class, and there is no s-p mixing.

Show **energy arrow**, **atomic orbitals** involved, **atom labels**, the **molecular orbitals** that are formed, **filled electrons** in the MO diagram etc. Make sure to label all the atomic and molecular orbitals.

Using the MO diagram above, answer the following questions:

- (i) Which one of the following will have the highest bond order? Circle the correct choice
A) PS^{1-} B) PS C) PS^{1+}
- (ii) Which one of the following will have the longest bond length? Circle the correct choice
A) PS^{1-} B) PS C) PS^{1+}
- (iii) Which of the following will be diamagnetic? Circle the correct choice.
A) PS^{1-} B) PS C) PS^{1+}
- (iv) Draw Lewis dot structure of the molecule with the shortest bond length (based on the MO diagram) from among the three (PS^{1-} , PS, PS^{1+}), and indicate the bond order calculated from the Lewis dot structure. **(3 points)**

Lewis Dot Structure

Bond Order



ED372727-6E70-4774-B180-C1D3ADAFAB95

midterm-exam-93388

#758 Page 10 of 10

SCRAP PAGE

**CHEMISTRY 110 –Midterm Exam****Thursday November 7, 2024, 6:30 – 8:30 pm**

Examiners: Pallavi Sirjoosingh and Ashok Kakkar

First name (please write as legibly as possible within the boxes)

--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--

Last name

--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--

Student ID number

--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--

This booklet has 8 questions on 10 pages (The last page can be used as a scrap page).

The staple must NOT be removed from the question booklet. Your intact question booklet must be returned at the end of the exam.

A separate booklet contains a periodic table, and datasheet.

Write your answers directly under the questions in the space provided. Do not write on the QR codes or near the top edges (as it may be cut off).

Please write **legibly and dark enough** for the scan to be readable. For students who use English as a second language, dictionaries are permitted. Calculators are allowed.

DO NOT ASK QUESTIONS OF THE INVIGILATORS about interpretations of the exam questions. If you have a concern about a question, please summarize it at the top of this page. All concerns will be evaluated after the exam. GOOD LUCK!

**Question 1 8 points** (*Assume all transitions are possible*)

a. 6 points Calculate the **energy (in kJ/mol)** required to ionize (completely remove) the electron from the **$n=3$** level of **He^+ ion** (He: Helium). Show your work including equations. (3 sig. figs.)

b. 2 points

(I) Select the largest (longest) wavelength from the three choices below:

- i. Wavelength of photon to *completely remove* an electron from *1s orbital of Hydrogen atom*
- ii. Wavelength of photon to *completely remove* an electron from *1s orbital of Boron atom (B)*
- iii. Wavelength of photon to *completely remove* an electron from *1s orbital of Boron cation (B^+)*

i or ii or iii

(II) Select the smallest (shortest) wavelength from the three choices below (*for H atom*):

- i. Wavelength of photon *emitted* for an electron transitioning from *2p to 1s orbital*
- ii. Wavelength of photon *absorbed* for an electron transitioning from *2s to 4p orbital*
- iii. Wavelength of photon *absorbed* for an electron transitioning from *1s to 3p orbital*

i or ii or iii



Question 2 8 points



A Chem110 student plots the graph shown on the left, from data obtained for the kinetic energy (eV) of the ejected electron versus frequency (s^{-1}) of the incident light, for an unknown metal. ($1\text{eV} = 1.6022 \times 10^{-19} \text{ J}$)

Based on the plot, answer the questions **a to c** below:

a. 1 point On the plot above, label the **threshold frequency**.

b. 2 points What is the value of the work potential (in J) of the unknown metal?

c. 3 points What is the *longest* wavelength (in nm) required to eject an electron from this metal's surface? Show your work including equation(s) used. (3 sig figs)

d. 2 points In the space provided below, plot the following

(Assume frequency is *above* threshold frequency):

- Kinetic energy* of an ejected electron versus the *intensity* of the incident light (left)
- Number of electrons* ejected versus the *frequency* of the incident light (right)



**Question 3** **6 points****a. 2 points**

In the ground state of Ge (Germanium; $Z = 32$) atom, **name** the following orbitals:

i. Orbital containing the electron that requires the maximum energy to be removed.

ii. Orbital containing the electron that experiences the minimum effective nuclear charge.

b. 2 points

An unknown anion (A^-) has the following *excited* electronic configuration: $[Ar] 4s^2 3d^3 4p^1$

(Ar: Argon; $Z = 18$)

Give the **condensed ground state electronic configuration** of the atom, **A** and the cation, **A^{2+}** .
Show your work. (Assume all electronic transitions are possible)

Electronic Configuration of A:

Electronic Configuration of A^{2+} :

c. 2 points

In a single atom, determine how many electrons can have the following set of quantum numbers?

(i) $n = 5; l = 3$

In a single atom, determine how many electrons can have the following set of quantum numbers?

(i) $n = 3; l = 2; m_s = +1/2$

**Question 4** **8 points**

Rank the following from 1 to 4 in **increasing** order. Provide the ranking in the boxes provided.

Note: Atomic number (Z) are provided for relevant atoms

a) Increasing (lowest to highest) order of *ionization energy*: (1= lowest; 4 = highest)

Mg

Mg: Z = 12

Rb

Rb: Z = 37

Cs

Cs: Z = 55

K

K : Z = 19

b) Increasing (lowest to highest) order of *bond angle*: (1= lowest; 4 = highest)

H₂OCCl₄XeF₄

HCN

c) Increasing (smallest to largest) order of *ionic size* (1= smallest; 4 = largest)

S²⁻

S: Z = 16

O²⁻

O: Z = 8

Be²⁺

Be: Z = 4

Na⁺

Na: Z = 11

d) Increasing (lowest to highest) order of *bond energy* (1= lowest; 4 = highest)

P-F

P-Cl

P-I

P-Br

**Question 5** **12 points**

a. 9 points For the molecules HClO_4 and TeF_6 , draw the most stable *Lewis structure* showing appropriate shape and *dashed/wedged bonds*, based on VSEPR. Determine the **molecular geometry** around the central atom. For HClO_4 , label the *most polar bond*.

HClO_4 (there are no O-O bonds)

Label the most polar bond

Molecular Geometry around Cl

TeF_6

Molecular Geometry around Te

b. 3 points (Note: Lone pairs are not shown in structures below)

(i) For the three *resonance* structures (1, 2 and 3) shown below, answer the following question:



(I) Circle the *most contributing* resonance structure.

Structure 1

Structure 2

Structure 3

(ii) For the three structures given (4, 5, and 6) below, answer the following questions:



(I) Circle all the structures that represent *resonance structures* of each other:

Structure 4

Structure 5

Structure 6

(II) Among the resonance structures, circle the *most contributing* resonance structure.

Structure 4

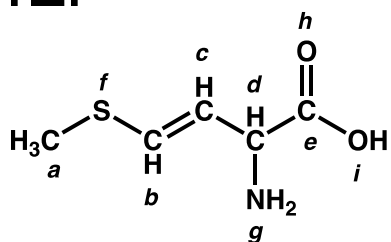
Structure 5

Structure 6

**Question 6** **6 points**

Using the data given below, calculate the **lattice energy** for SrO(s).

	ΔH° (kJ/mol)
$\text{Sr}(s) \longrightarrow \text{Sr}(g)$	+180
$\text{Sr}(g) \longrightarrow \text{Sr}^{2+}(g) + 2e^-$	+1468
$\text{O}_2(g) \longrightarrow 2\text{O}(g)$	+500
$\text{O}(g) + 2e^- \longrightarrow \text{O}^{2-}(g)$	+600
$\text{Sr}(s) + \frac{1}{2}\text{O}_2(g) \longrightarrow \text{SrO}(s)$	-550

**Question 7 10 points**

Consider the molecule given on the left, and answer questions (i) to (vi) using chemical bonding theories that we have learnt in class (valence bond theory, hybridization).

Note that the lone pairs are not shown in the structure:

i. Consider the carbon atoms marked in as ***a***, ***b***, ***c***, ***d***, and ***e*** in the molecule, and determine **hybridization** at each of the carbon centers. (2.5 points)

Atom	Hybridization
Carbon (<i>a</i>)	
Carbon (<i>b</i>)	
Carbon (<i>c</i>)	
Carbon (<i>d</i>)	
Carbon (<i>e</i>)	

ii. Now, consider the sulfur atom (S) marked as ***f***, nitrogen (N) atom (***g***), and oxygen(O) atoms (***h*** and ***i***), and determine their **hybridization**. (2 points)

Atom	Hybridization
Sulfur (<i>f</i>)	
Nitrogen (<i>g</i>)	
Oxygen (<i>h</i>)	
Oxygen (<i>i</i>)	

iii. **Describe** all the bonds formed by the carbon atoms ***a*** and ***b***, using sigma and pi designation as appropriate. (2 points)

Atom	Bonds formed (use sigma and pi as appropriate)
Carbon (<i>a</i>)	
Carbon (<i>b</i>)	

iv. How many total sp^3 -s sigma bonds are in the molecule? (1 point)

v. Determine **electronic** and **molecular geometry** at carbon (C) labeled ***e*** and sulfur (S) labeled ***f*** centres. (2 points)

Atom	Electronic Geometry	Molecular Geometry
Carbon (<i>e</i>)		
Sulfur (<i>f</i>)		

vi. Circle the value from the choices given below, that describes the **C-S-C bond angle at sulfur** (S) marked ***f***: (0.5 points)

>90° to <109.5° 109.5° >109.5 to <120° 120° >120° to <180°

**Question 8 12 points**

Using the **valence shell orbitals only** draw a **complete** molecular orbital (MO) diagram (with no s-p mixing) for the molecule **PS**, in the space provided below. **(6 points)**

Assume that 3s and 3p atomic orbitals will form molecular orbitals like those formed by the combination of 2s and 2p atomic orbitals that we learnt in class, and there is no s-p mixing.

Show **energy arrow**, **atomic orbitals** involved, **atom labels**, the **molecular orbitals** that are formed, **filled electrons** in the MO diagram etc. Make sure to label all the atomic and molecular orbitals.

Using the MO diagram above, answer the following questions:

- (i) Which one of the following will have the highest bond order? Circle the correct choice
A) PS^{1-} B) PS C) PS^{1+}
- (ii) Which one of the following will have the longest bond length? Circle the correct choice
A) PS^{1-} B) PS C) PS^{1+}
- (iii) Which of the following will be diamagnetic? Circle the correct choice.
A) PS^{1-} B) PS C) PS^{1+}
- (iv) Draw Lewis dot structure of the molecule with the shortest bond length (based on the MO diagram) from among the three (PS^{1-} , PS, PS^{1+}), and indicate the bond order calculated from the Lewis dot structure. **(3 points)**

Lewis Dot Structure

Bond Order



65CEEf1F-786F-4594-927B-02BCF553909D

midterm-exam-93388

#759 Page 10 of 10

SCRAP PAGE

**CHEMISTRY 110 –Midterm Exam****Thursday November 7, 2024, 6:30 – 8:30 pm**

Examiners: Pallavi Sirjoosingh and Ashok Kakkar

First name (please write as legibly as possible within the boxes)

--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--

Last name

--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--

Student ID number

--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--

This booklet has 8 questions on 10 pages (The last page can be used as a scrap page).

The staple must NOT be removed from the question booklet. Your intact question booklet must be returned at the end of the exam.

A separate booklet contains a periodic table, and datasheet.

Write your answers directly under the questions in the space provided. Do not write on the QR codes or near the top edges (as it may be cut off).

Please write **legibly and dark enough** for the scan to be readable. For students who use English as a second language, dictionaries are permitted. Calculators are allowed.

DO NOT ASK QUESTIONS OF THE INVIGILATORS about interpretations of the exam questions. If you have a concern about a question, please summarize it at the top of this page. All concerns will be evaluated after the exam. GOOD LUCK!

**Question 1 8 points** (*Assume all transitions are possible*)

a. 6 points Calculate the **energy (in kJ/mol)** required to ionize (completely remove) the electron from the **n=3** level of **He⁺ ion** (He: Helium). Show your work including equations. (3 sig. figs.)

b. 2 points

(I) Select the largest (longest) wavelength from the three choices below:

- i. Wavelength of photon to *completely remove* an electron from *1s orbital of Hydrogen atom*
- ii. Wavelength of photon to *completely remove* an electron from *1s orbital of Boron atom (B)*
- iii. Wavelength of photon to *completely remove* an electron from *1s orbital of Boron cation (B⁺)*

i or ii or iii

(II) Select the smallest (shortest) wavelength from the three choices below (*for H atom*):

- i. Wavelength of photon *emitted* for an electron transitioning from *2p to 1s orbital*
- ii. Wavelength of photon *absorbed* for an electron transitioning from *2s to 4p orbital*
- iii. Wavelength of photon *absorbed* for an electron transitioning from *1s to 3p orbital*

i or ii or iii



Question 2 8 points



A Chem110 student plots the graph shown on the left, from data obtained for the kinetic energy (eV) of the ejected electron versus frequency (s^{-1}) of the incident light, for an unknown metal. ($1\text{eV} = 1.6022 \times 10^{-19} \text{ J}$)

Based on the plot, answer the questions **a to c** below:

a. 1 point On the plot above, label the **threshold frequency**.

b. 2 points What is the value of the work potential (in J) of the unknown metal?

c. 3 points What is the *longest* wavelength (in nm) required to eject an electron from this metal's surface? Show your work including equation(s) used. (3 sig figs)

d. 2 points In the space provided below, plot the following

(Assume frequency is *above* threshold frequency):

- i. *Kinetic energy* of an ejected electron versus the *intensity* of the incident light (left)
- ii. *Number of electrons* ejected versus the *frequency* of the incident light (right)



**Question 3** **6 points****a. 2 points**

In the ground state of Ge (Germanium; $Z = 32$) atom, **name** the following orbitals:

i. Orbital containing the electron that requires the maximum energy to be removed.

ii. Orbital containing the electron that experiences the minimum effective nuclear charge.

b. 2 points

An unknown anion (A^-) has the following *excited* electronic configuration: $[Ar] 4s^2 3d^3 4p^1$

(Ar: Argon; $Z = 18$)

Give the **condensed ground state electronic configuration** of the atom, **A** and the cation, **A^{2+}** .
Show your work. (Assume all electronic transitions are possible)

Electronic Configuration of A:

Electronic Configuration of A^{2+} :

c. 2 points

In a single atom, determine how many electrons can have the following set of quantum numbers?

(i) $n = 5; l = 3$

In a single atom, determine how many electrons can have the following set of quantum numbers?

(i) $n = 3; l = 2; m_s = +1/2$

**Question 4** **8 points**

Rank the following from 1 to 4 in **increasing** order. Provide the ranking in the boxes provided.

Note: Atomic number (Z) are provided for relevant atoms

a) Increasing (lowest to highest) order of *ionization energy*: (1= lowest; 4 = highest)

Mg

Mg: Z = 12

Rb

Rb: Z = 37

Cs

Cs: Z = 55

K

K : Z = 19

b) Increasing (lowest to highest) order of *bond angle*: (1= lowest; 4 = highest)

H₂OCCl₄XeF₄

HCN

c) Increasing (smallest to largest) order of *ionic size* (1= smallest; 4 = largest)

S²⁻

S: Z = 16

O²⁻

O: Z = 8

Be²⁺

Be: Z = 4

Na⁺

Na: Z = 11

d) Increasing (lowest to highest) order of *bond energy* (1= lowest; 4 = highest)

P-F

P-Cl

P-I

P-Br

**Question 5** **12 points**

a. 9 points For the molecules HClO_4 and TeF_6 , draw the most stable *Lewis structure* showing appropriate shape and *dashed/wedged bonds*, based on VSEPR. Determine the **molecular geometry** around the central atom. For HClO_4 , label the *most polar bond*.



b. 3 points (Note: Lone pairs are not shown in structures below)

(i) For the three *resonance* structures (1, 2 and 3) shown below, answer the following question:



(I) Circle the *most contributing* resonance structure.

Structure 1

Structure 2

Structure 3

(ii) For the three structures given (4, 5, and 6) below, answer the following questions:



(I) Circle all the structures that represent *resonance structures* of each other:

Structure 4

Structure 5

Structure 6

(II) Among the resonance structures, circle the *most contributing* resonance structure.

Structure 4

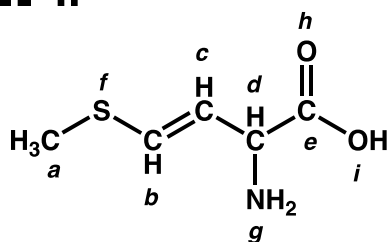
Structure 5

Structure 6

**Question 6** **6 points**

Using the data given below, calculate the **lattice energy** for SrO(s).

	ΔH° (kJ/mol)
$\text{Sr}(s) \longrightarrow \text{Sr}(g)$	+180
$\text{Sr}(g) \longrightarrow \text{Sr}^{2+}(g) + 2e^-$	+1468
$\text{O}_2(g) \longrightarrow 2\text{O}(g)$	+500
$\text{O}(g) + 2e^- \longrightarrow \text{O}^{2-}(g)$	+600
$\text{Sr}(s) + \frac{1}{2}\text{O}_2(g) \longrightarrow \text{SrO}(s)$	-550

**Question 7 10 points**

Consider the molecule given on the left, and answer questions (i) to (vi) using chemical bonding theories that we have learnt in class (valence bond theory, hybridization).

Note that the lone pairs are not shown in the structure:

i. Consider the carbon atoms marked in as ***a***, ***b***, ***c***, ***d***, and ***e*** in the molecule, and determine **hybridization** at each of the carbon centers. (2.5 points)

Atom	Hybridization
Carbon (<i>a</i>)	
Carbon (<i>b</i>)	
Carbon (<i>c</i>)	
Carbon (<i>d</i>)	
Carbon (<i>e</i>)	

ii. Now, consider the sulfur atom (S) marked as ***f***, nitrogen (N) atom (***g***), and oxygen(O) atoms (***h*** and ***i***), and determine their **hybridization**. (2 points)

Atom	Hybridization
Sulfur (<i>f</i>)	
Nitrogen (<i>g</i>)	
Oxygen (<i>h</i>)	
Oxygen (<i>i</i>)	

iii. **Describe** all the bonds formed by the carbon atoms ***a*** and ***b***, using sigma and pi designation as appropriate. (2 points)

Atom	Bonds formed (use sigma and pi as appropriate)
Carbon (<i>a</i>)	
Carbon (<i>b</i>)	

iv. How many total sp^3 -s sigma bonds are in the molecule? (1 point)

v. Determine **electronic** and **molecular geometry** at carbon (C) labeled ***e*** and sulfur (S) labeled ***f*** centres. (2 points)

Atom	Electronic Geometry	Molecular Geometry
Carbon (<i>e</i>)		
Sulfur (<i>f</i>)		

vi. Circle the value from the choices given below, that describes the **C-S-C bond angle at sulfur** (S) marked ***f***: (0.5 points)

>90° to <109.5° 109.5° >109.5 to <120° 120° >120° to <180°


Question 8 12 points

Using the **valence shell orbitals only** draw a **complete** molecular orbital (MO) diagram (with no s-p mixing) for the molecule **PS**, in the space provided below. **(6 points)**

Assume that 3s and 3p atomic orbitals will form molecular orbitals like those formed by the combination of 2s and 2p atomic orbitals that we learnt in class, and there is no s-p mixing.

Show **energy arrow**, **atomic orbitals** involved, **atom labels**, the **molecular orbitals** that are formed, **filled electrons** in the MO diagram etc. Make sure to label all the atomic and molecular orbitals.

Using the MO diagram above, answer the following questions:

- (i) Which one of the following will have the highest bond order? Circle the correct choice
 A) PS^{1-} B) PS C) PS^{1+}
- (ii) Which one of the following will have the longest bond length? Circle the correct choice
 A) PS^{1-} B) PS C) PS^{1+}
- (iii) Which of the following will be diamagnetic? Circle the correct choice.
 A) PS^{1-} B) PS C) PS^{1+}
- (iv) Draw Lewis dot structure of the molecule with the shortest bond length (based on the MO diagram) from among the three (PS^{1-} , PS, PS^{1+}), and indicate the bond order calculated from the Lewis dot structure. **(3 points)**

Lewis Dot Structure

Bond Order



0905BD80-E422-4543-B1F4-56B8311BFF00

midterm-exam-93388

#760 Page 10 of 10

SCRAP PAGE

**CHEMISTRY 110 –Midterm Exam****Thursday November 7, 2024, 6:30 – 8:30 pm****Examiners: Pallavi Sirjoosingh and Ashok Kakkar**

First name (please write as legibly as possible within the boxes)

--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--

Last name

--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--

Student ID number

--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--

This booklet has 8 questions on 10 pages (The last page can be used as a scrap page).

The staple must NOT be removed from the question booklet. Your intact question booklet must be returned at the end of the exam.

A separate booklet contains a periodic table, and datasheet.

Write your answers directly under the questions in the space provided. Do not write on the QR codes or near the top edges (as it may be cut off).

Please write **legibly and dark enough** for the scan to be readable. For students who use English as a second language, dictionaries are permitted. Calculators are allowed.

DO NOT ASK QUESTIONS OF THE INVIGILATORS about interpretations of the exam questions. If you have a concern about a question, please summarize it at the top of this page. All concerns will be evaluated after the exam. GOOD LUCK!

**Question 1 8 points** (*Assume all transitions are possible*)

a. 6 points Calculate the **energy (in kJ/mol)** required to ionize (completely remove) the electron from the **$n=3$** level of **He^+ ion** (He: Helium). Show your work including equations. (3 sig. figs.)

b. 2 points

(I) Select the largest (longest) wavelength from the three choices below:

- i. Wavelength of photon to *completely remove* an electron from *1s orbital of Hydrogen atom*
- ii. Wavelength of photon to *completely remove* an electron from *1s orbital of Boron atom (B)*
- iii. Wavelength of photon to *completely remove* an electron from *1s orbital of Boron cation (B^+)*

i or ii or iii

(II) Select the smallest (shortest) wavelength from the three choices below (*for H atom*):

- i. Wavelength of photon *emitted* for an electron transitioning from *2p to 1s orbital*
- ii. Wavelength of photon *absorbed* for an electron transitioning from *2s to 4p orbital*
- iii. Wavelength of photon *absorbed* for an electron transitioning from *1s to 3p orbital*

i or ii or iii



Question 2 8 points



A Chem110 student plots the graph shown on the left, from data obtained for the kinetic energy (eV) of the ejected electron versus frequency (s^{-1}) of the incident light, for an unknown metal. ($1\text{eV} = 1.6022 \times 10^{-19} \text{ J}$)

Based on the plot, answer the questions **a to c** below:

a. 1 point On the plot above, label the **threshold frequency**.

b. 2 points What is the value of the work potential (in J) of the unknown metal?

c. 3 points What is the *longest* wavelength (in nm) required to eject an electron from this metal's surface? Show your work including equation(s) used. (3 sig figs)

d. 2 points In the space provided below, plot the following

(Assume frequency is *above* threshold frequency):

- Kinetic energy* of an ejected electron versus the *intensity* of the incident light (left)
- Number of electrons* ejected versus the *frequency* of the incident light (right)



**Question 3** **6 points****a. 2 points**

In the ground state of Ge (Germanium; $Z = 32$) atom, **name** the following orbitals:

i. Orbital containing the electron that requires the maximum energy to be removed.

ii. Orbital containing the electron that experiences the minimum effective nuclear charge.

b. 2 points

An unknown anion (A^-) has the following *excited* electronic configuration: $[Ar] 4s^2 3d^3 4p^1$

(Ar: Argon; $Z = 18$)

Give the **condensed ground state electronic configuration** of the atom, **A** and the cation, **A^{2+}** .
Show your work. (Assume all electronic transitions are possible)

Electronic Configuration of A:

Electronic Configuration of A^{2+} :

c. 2 points

In a single atom, determine how many electrons can have the following set of quantum numbers?

(i) $n = 5; l = 3$

In a single atom, determine how many electrons can have the following set of quantum numbers?

(i) $n = 3; l = 2; m_s = +1/2$

**Question 4** **8 points**

Rank the following from 1 to 4 in **increasing** order. Provide the ranking in the boxes provided.

Note: Atomic number (Z) are provided for relevant atoms

a) Increasing (lowest to highest) order of *ionization energy*: (1= lowest; 4 = highest)

Mg

Mg: Z = 12

Rb

Rb: Z = 37

Cs

Cs: Z = 55

K

K : Z = 19

b) Increasing (lowest to highest) order of *bond angle*: (1= lowest; 4 = highest)

H₂OCCl₄XeF₄

HCN

c) Increasing (smallest to largest) order of *ionic size* (1= smallest; 4 = largest)

S²⁻

S: Z = 16

O²⁻

O: Z = 8

Be²⁺

Be: Z = 4

Na⁺

Na: Z = 11

d) Increasing (lowest to highest) order of *bond energy* (1= lowest; 4 = highest)

P-F

P-Cl

P-I

P-Br

**Question 5** **12 points**

a. 9 points For the molecules HClO_4 and TeF_6 , draw the most stable *Lewis structure* showing appropriate shape and *dashed/wedged bonds*, based on VSEPR. Determine the **molecular geometry** around the central atom. For HClO_4 , label the *most polar bond*.



b. 3 points (Note: Lone pairs are not shown in structures below)

(i) For the three *resonance* structures (1, 2 and 3) shown below, answer the following question:



(I) Circle the *most contributing* resonance structure.

Structure 1

Structure 2

Structure 3

(ii) For the three structures given (4, 5, and 6) below, answer the following questions:



(I) Circle all the structures that represent *resonance structures* of each other:

Structure 4

Structure 5

Structure 6

(II) Among the resonance structures, circle the *most contributing* resonance structure.

Structure 4

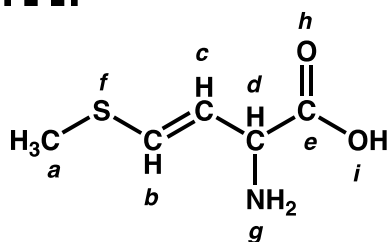
Structure 5

Structure 6

**Question 6** **6 points**

Using the data given below, calculate the **lattice energy** for SrO(s).

	ΔH° (kJ/mol)
$\text{Sr}(s) \longrightarrow \text{Sr}(g)$	+180
$\text{Sr}(g) \longrightarrow \text{Sr}^{2+}(g) + 2e^-$	+1468
$\text{O}_2(g) \longrightarrow 2\text{O}(g)$	+500
$\text{O}(g) + 2e^- \longrightarrow \text{O}^{2-}(g)$	+600
$\text{Sr}(s) + \frac{1}{2}\text{O}_2(g) \longrightarrow \text{SrO}(s)$	-550

**Question 7 10 points**

Consider the molecule given on the left, and answer questions (i) to (vi) using chemical bonding theories that we have learnt in class (valence bond theory, hybridization).

Note that the lone pairs are not shown in the structure:

i. Consider the carbon atoms marked in as ***a***, ***b***, ***c***, ***d***, and ***e*** in the molecule, and determine **hybridization** at each of the carbon centers. (2.5 points)

Atom	Hybridization
Carbon (<i>a</i>)	
Carbon (<i>b</i>)	
Carbon (<i>c</i>)	
Carbon (<i>d</i>)	
Carbon (<i>e</i>)	

ii. Now, consider the sulfur atom (S) marked as ***f***, nitrogen (N) atom (***g***), and oxygen(O) atoms (***h*** and ***i***), and determine their **hybridization**. (2 points)

Atom	Hybridization
Sulfur (<i>f</i>)	
Nitrogen (<i>g</i>)	
Oxygen (<i>h</i>)	
Oxygen (<i>i</i>)	

iii. **Describe** all the bonds formed by the carbon atoms ***a*** and ***b***, using sigma and pi designation as appropriate. (2 points)

Atom	Bonds formed (use sigma and pi as appropriate)
Carbon (<i>a</i>)	
Carbon (<i>b</i>)	

iv. How many total sp^3 -s sigma bonds are in the molecule? (1 point)

v. Determine **electronic** and **molecular geometry** at carbon (C) labeled ***e*** and sulfur (S) labeled ***f*** centres. (2 points)

Atom	Electronic Geometry	Molecular Geometry
Carbon (<i>e</i>)		
Sulfur (<i>f</i>)		

vi. Circle the value from the choices given below, that describes the **C-S-C bond angle at sulfur** (S) marked ***f***: (0.5 points)

>90° to <109.5° 109.5° >109.5 to <120° 120° >120° to <180°


Question 8 12 points

Using the **valence shell orbitals only** draw a **complete** molecular orbital (MO) diagram (with no s-p mixing) for the molecule **PS**, in the space provided below. **(6 points)**

Assume that 3s and 3p atomic orbitals will form molecular orbitals like those formed by the combination of 2s and 2p atomic orbitals that we learnt in class, and there is no s-p mixing.

Show **energy arrow**, **atomic orbitals** involved, **atom labels**, the **molecular orbitals** that are formed, **filled electrons** in the MO diagram etc. Make sure to label all the atomic and molecular orbitals.

Using the MO diagram above, answer the following questions:

- (i) Which one of the following will have the highest bond order? Circle the correct choice
 A) PS^{1-} B) PS C) PS^{1+}
- (ii) Which one of the following will have the longest bond length? Circle the correct choice
 A) PS^{1-} B) PS C) PS^{1+}
- (iii) Which of the following will be diamagnetic? Circle the correct choice.
 A) PS^{1-} B) PS C) PS^{1+}
- (iv) Draw Lewis dot structure of the molecule with the shortest bond length (based on the MO diagram) from among the three (PS^{1-} , PS, PS^{1+}), and indicate the bond order calculated from the Lewis dot structure. **(3 points)**

Lewis Dot Structure

Bond Order



80B64E03-47B1-4330-A805-FC3D2494A9F1

midterm-exam-93388

#761 Page 10 of 10

SCRAP PAGE

**CHEMISTRY 110 –Midterm Exam****Thursday November 7, 2024, 6:30 – 8:30 pm****Examiners: Pallavi Sirjoosingh and Ashok Kakkar**

First name (please write as legibly as possible within the boxes)

--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--

Last name

--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--

Student ID number

--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--

This booklet has 8 questions on 10 pages (The last page can be used as a scrap page).

The staple must NOT be removed from the question booklet. Your intact question booklet must be returned at the end of the exam.

A separate booklet contains a periodic table, and datasheet.

Write your answers directly under the questions in the space provided. Do not write on the QR codes or near the top edges (as it may be cut off).

Please write **legibly and dark enough** for the scan to be readable. For students who use English as a second language, dictionaries are permitted. Calculators are allowed.

DO NOT ASK QUESTIONS OF THE INVIGILATORS about interpretations of the exam questions. If you have a concern about a question, please summarize it at the top of this page. All concerns will be evaluated after the exam. GOOD LUCK!

**Question 1 8 points** (*Assume all transitions are possible*)

a. 6 points Calculate the **energy (in kJ/mol)** required to ionize (completely remove) the electron from the **$n=3$** level of **He^+ ion** (He: Helium). Show your work including equations. (3 sig. figs.)

b. 2 points

(I) Select the largest (longest) wavelength from the three choices below:

- i. Wavelength of photon to *completely remove* an electron from *1s orbital of Hydrogen atom*
- ii. Wavelength of photon to *completely remove* an electron from *1s orbital of Boron atom (B)*
- iii. Wavelength of photon to *completely remove* an electron from *1s orbital of Boron cation (B^+)*

i or ii or iii

(II) Select the smallest (shortest) wavelength from the three choices below (*for H atom*):

- i. Wavelength of photon *emitted* for an electron transitioning from *2p to 1s orbital*
- ii. Wavelength of photon *absorbed* for an electron transitioning from *2s to 4p orbital*
- iii. Wavelength of photon *absorbed* for an electron transitioning from *1s to 3p orbital*

i or ii or iii



Question 2 8 points



A Chem110 student plots the graph shown on the left, from data obtained for the kinetic energy (eV) of the ejected electron versus frequency (s^{-1}) of the incident light, for an unknown metal. ($1\text{eV} = 1.6022 \times 10^{-19} \text{ J}$)

Based on the plot, answer the questions **a to c** below:

a. 1 point On the plot above, label the **threshold frequency**.

b. 2 points What is the value of the work potential (in J) of the unknown metal?

c. 3 points What is the *longest* wavelength (in nm) required to eject an electron from this metal's surface? Show your work including equation(s) used. (3 sig figs)

d. 2 points In the space provided below, plot the following

(Assume frequency is *above* threshold frequency):

- Kinetic energy* of an ejected electron versus the *intensity* of the incident light (left)
- Number of electrons* ejected versus the *frequency* of the incident light (right)



**Question 3** **6 points****a. 2 points**

In the ground state of Ge (Germanium; $Z = 32$) atom, **name** the following orbitals:

i. Orbital containing the electron that requires the maximum energy to be removed.

ii. Orbital containing the electron that experiences the minimum effective nuclear charge.

b. 2 points

An unknown anion (A^-) has the following *excited* electronic configuration: $[Ar] 4s^2 3d^3 4p^1$

(Ar: Argon; $Z = 18$)

Give the **condensed ground state electronic configuration** of the atom, **A** and the cation, **A^{2+}** .
Show your work. (Assume all electronic transitions are possible)

Electronic Configuration of A:

Electronic Configuration of A^{2+} :

c. 2 points

In a single atom, determine how many electrons can have the following set of quantum numbers?

(i) $n = 5; l = 3$

In a single atom, determine how many electrons can have the following set of quantum numbers?

(i) $n = 3; l = 2; m_s = +1/2$

**Question 4** **8 points**

Rank the following from 1 to 4 in **increasing** order. Provide the ranking in the boxes provided.

Note: Atomic number (Z) are provided for relevant atoms

a) Increasing (lowest to highest) order of *ionization energy*: (1= lowest; 4 = highest)

Mg

Mg: Z = 12

Rb

Rb: Z = 37

Cs

Cs: Z = 55

K

K : Z = 19

b) Increasing (lowest to highest) order of *bond angle*: (1= lowest; 4 = highest)

H₂OCCl₄XeF₄

HCN

c) Increasing (smallest to largest) order of *ionic size* (1= smallest; 4 = largest)

S²⁻

S: Z = 16

O²⁻

O: Z = 8

Be²⁺

Be: Z = 4

Na⁺

Na: Z = 11

d) Increasing (lowest to highest) order of *bond energy* (1= lowest; 4 = highest)

P-F

P-Cl

P-I

P-Br

**Question 5** **12 points**

a. 9 points For the molecules HClO_4 and TeF_6 , draw the most stable *Lewis structure* showing appropriate shape and *dashed/wedged bonds*, based on VSEPR. Determine the **molecular geometry** around the central atom. For HClO_4 , label the *most polar bond*.



b. 3 points (Note: Lone pairs are not shown in structures below)

(i) For the three *resonance* structures (1, 2 and 3) shown below, answer the following question:



(I) Circle the *most contributing* resonance structure.

Structure 1

Structure 2

Structure 3

(ii) For the three structures given (4, 5, and 6) below, answer the following questions:



(I) Circle all the structures that represent *resonance structures* of each other:

Structure 4

Structure 5

Structure 6

(II) Among the resonance structures, circle the *most contributing* resonance structure.

Structure 4

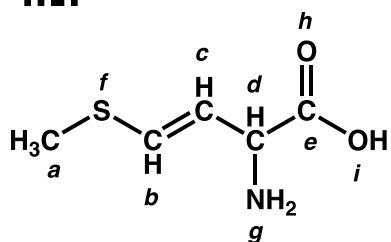
Structure 5

Structure 6

**Question 6** **6 points**

Using the data given below, calculate the **lattice energy** for SrO(s).

	ΔH° (kJ/mol)
$\text{Sr}(s) \longrightarrow \text{Sr}(g)$	+180
$\text{Sr}(g) \longrightarrow \text{Sr}^{2+}(g) + 2e^-$	+1468
$\text{O}_2(g) \longrightarrow 2\text{O}(g)$	+500
$\text{O}(g) + 2e^- \longrightarrow \text{O}^{2-}(g)$	+600
$\text{Sr}(s) + \frac{1}{2}\text{O}_2(g) \longrightarrow \text{SrO}(s)$	-550

**Question 7 10 points**

Consider the molecule given on the left, and answer questions (i) to (vi) using chemical bonding theories that we have learnt in class (valence bond theory, hybridization).

Note that the lone pairs are not shown in the structure:

i. Consider the carbon atoms marked in as ***a***, ***b***, ***c***, ***d***, and ***e*** in the molecule, and determine **hybridization** at each of the carbon centers. (2.5 points)

Atom	Hybridization
Carbon (<i>a</i>)	
Carbon (<i>b</i>)	
Carbon (<i>c</i>)	
Carbon (<i>d</i>)	
Carbon (<i>e</i>)	

ii. Now, consider the sulfur atom (S) marked as ***f***, nitrogen (N) atom (***g***), and oxygen(O) atoms (***h*** and ***i***), and determine their **hybridization**. (2 points)

Atom	Hybridization
Sulfur (<i>f</i>)	
Nitrogen (<i>g</i>)	
Oxygen (<i>h</i>)	
Oxygen (<i>i</i>)	

iii. **Describe** all the bonds formed by the carbon atoms ***a*** and ***b***, using sigma and pi designation as appropriate. (2 points)

Atom	Bonds formed (use sigma and pi as appropriate)
Carbon (<i>a</i>)	
Carbon (<i>b</i>)	

iv. How many total sp^3 -s sigma bonds are in the molecule? (1 point)

v. Determine **electronic** and **molecular geometry** at carbon (C) labeled ***e*** and sulfur (S) labeled ***f*** centres. (2 points)

Atom	Electronic Geometry	Molecular Geometry
Carbon (<i>e</i>)		
Sulfur (<i>f</i>)		

vi. Circle the value from the choices given below, that describes the **C-S-C bond angle at sulfur** (S) marked ***f***: (0.5 points)

>90° to <109.5° 109.5° >109.5 to <120° 120° >120° to <180°


Question 8 12 points

Using the **valence shell orbitals only** draw a **complete** molecular orbital (MO) diagram (with no s-p mixing) for the molecule **PS**, in the space provided below. **(6 points)**

Assume that 3s and 3p atomic orbitals will form molecular orbitals like those formed by the combination of 2s and 2p atomic orbitals that we learnt in class, and there is no s-p mixing.

Show **energy arrow**, **atomic orbitals** involved, **atom labels**, the **molecular orbitals** that are formed, **filled electrons** in the MO diagram etc. Make sure to **label all the atomic and molecular orbitals**.

Using the MO diagram above, answer the following questions:

- (i) Which one of the following will have the highest bond order? Circle the correct choice
 A) PS^{1-} B) PS C) PS^{1+}
- (ii) Which one of the following will have the longest bond length? Circle the correct choice
 A) PS^{1-} B) PS C) PS^{1+}
- (iii) Which of the following will be diamagnetic? Circle the correct choice.
 A) PS^{1-} B) PS C) PS^{1+}
- (iv) Draw Lewis dot structure of the molecule with the shortest bond length (based on the MO diagram) from among the three (PS^{1-} , PS, PS^{1+}), and indicate the bond order calculated from the Lewis dot structure. **(3 points)**

Lewis Dot Structure

Bond Order



7629A903-2602-4D07-BA6D-BBA0C30694D1

midterm-exam-93388

#762 Page 10 of 10

SCRAP PAGE

**CHEMISTRY 110 –Midterm Exam****Thursday November 7, 2024, 6:30 – 8:30 pm**

Examiners: Pallavi Sirjoosingh and Ashok Kakkar

First name (please write as legibly as possible within the boxes)

--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--

Last name

--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--

Student ID number

--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--

This booklet has 8 questions on 10 pages (The last page can be used as a scrap page).

The staple must NOT be removed from the question booklet. Your intact question booklet must be returned at the end of the exam.

A separate booklet contains a periodic table, and datasheet.

Write your answers directly under the questions in the space provided. Do not write on the QR codes or near the top edges (as it may be cut off).

Please write **legibly and dark enough** for the scan to be readable. For students who use English as a second language, dictionaries are permitted. Calculators are allowed.

DO NOT ASK QUESTIONS OF THE INVIGILATORS about interpretations of the exam questions. If you have a concern about a question, please summarize it at the top of this page. All concerns will be evaluated after the exam. GOOD LUCK!

**Question 1 8 points** (*Assume all transitions are possible*)

a. 6 points Calculate the **energy (in kJ/mol)** required to ionize (completely remove) the electron from the **n=3** level of **He⁺ ion** (He: Helium). Show your work including equations. (3 sig. figs.)

b. 2 points

(I) Select the largest (longest) wavelength from the three choices below:

- i. Wavelength of photon to *completely remove* an electron from *1s orbital of Hydrogen atom*
- ii. Wavelength of photon to *completely remove* an electron from *1s orbital of Boron atom (B)*
- iii. Wavelength of photon to *completely remove* an electron from *1s orbital of Boron cation (B⁺)*

i or ii or iii

(II) Select the smallest (shortest) wavelength from the three choices below (*for H atom*):

- i. Wavelength of photon *emitted* for an electron transitioning from *2p to 1s orbital*
- ii. Wavelength of photon *absorbed* for an electron transitioning from *2s to 4p orbital*
- iii. Wavelength of photon *absorbed* for an electron transitioning from *1s to 3p orbital*

i or ii or iii



Question 2 8 points



A Chem110 student plots the graph shown on the left, from data obtained for the kinetic energy (eV) of the ejected electron versus frequency (s^{-1}) of the incident light, for an unknown metal. ($1\text{eV} = 1.6022 \times 10^{-19} \text{ J}$)

Based on the plot, answer the questions **a to c** below:

a. 1 point On the plot above, label the **threshold frequency**.

b. 2 points What is the value of the work potential (in J) of the unknown metal?

c. 3 points What is the *longest* wavelength (in nm) required to eject an electron from this metal's surface? Show your work including equation(s) used. (3 sig figs)

d. 2 points In the space provided below, plot the following

(Assume frequency is *above* threshold frequency):

- Kinetic energy* of an ejected electron versus the *intensity* of the incident light (left)
- Number of electrons* ejected versus the *frequency* of the incident light (right)



**Question 3** **6 points****a. 2 points**

In the ground state of Ge (Germanium; $Z = 32$) atom, **name** the following orbitals:

i. Orbital containing the electron that requires the maximum energy to be removed.

ii. Orbital containing the electron that experiences the minimum effective nuclear charge.

b. 2 points

An unknown anion (A^-) has the following *excited* electronic configuration: $[Ar] 4s^2 3d^3 4p^1$

(Ar: Argon; $Z = 18$)

Give the **condensed ground state electronic configuration** of the atom, **A** and the cation, **A^{2+}** .
Show your work. (Assume all electronic transitions are possible)

Electronic Configuration of A:

Electronic Configuration of A^{2+} :

c. 2 points

In a single atom, determine how many electrons can have the following set of quantum numbers?

(i) $n = 5; l = 3$

In a single atom, determine how many electrons can have the following set of quantum numbers?

(i) $n = 3; l = 2; m_s = +1/2$

**Question 4** **8 points**

Rank the following from 1 to 4 in **increasing** order. Provide the ranking in the boxes provided.

Note: Atomic number (Z) are provided for relevant atoms

a) Increasing (lowest to highest) order of *ionization energy*: (1= lowest; 4 = highest)

Mg

Mg: Z = 12

Rb

Rb: Z = 37

Cs

Cs: Z = 55

K

K : Z = 19

b) Increasing (lowest to highest) order of *bond angle*: (1= lowest; 4 = highest)

H₂OCCl₄XeF₄

HCN

c) Increasing (smallest to largest) order of *ionic size* (1= smallest; 4 = largest)

S²⁻

S: Z = 16

O²⁻

O: Z = 8

Be²⁺

Be: Z = 4

Na⁺

Na: Z = 11

d) Increasing (lowest to highest) order of *bond energy* (1= lowest; 4 = highest)

P-F

P-Cl

P-I

P-Br

**Question 5** **12 points**

a. 9 points For the molecules HClO_4 and TeF_6 , draw the most stable *Lewis structure* showing appropriate shape and *dashed/wedged bonds*, based on VSEPR. Determine the **molecular geometry** around the central atom. For HClO_4 , label the *most polar bond*.



b. 3 points (Note: Lone pairs are not shown in structures below)

(i) For the three *resonance* structures (1, 2 and 3) shown below, answer the following question:



(I) Circle the *most contributing* resonance structure.

Structure 1

Structure 2

Structure 3

(ii) For the three structures given (4, 5, and 6) below, answer the following questions:



(I) Circle all the structures that represent *resonance structures* of each other:

Structure 4

Structure 5

Structure 6

(II) Among the resonance structures, circle the *most contributing* resonance structure.

Structure 4

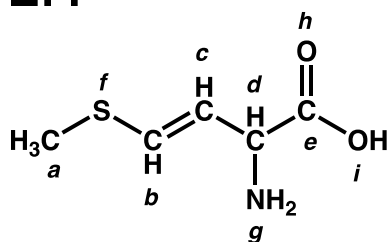
Structure 5

Structure 6

**Question 6** **6 points**

Using the data given below, calculate the **lattice energy** for SrO(s).

	ΔH° (kJ/mol)
$\text{Sr}(s) \longrightarrow \text{Sr}(g)$	+180
$\text{Sr}(g) \longrightarrow \text{Sr}^{2+}(g) + 2e^-$	+1468
$\text{O}_2(g) \longrightarrow 2\text{O}(g)$	+500
$\text{O}(g) + 2e^- \longrightarrow \text{O}^{2-}(g)$	+600
$\text{Sr}(s) + \frac{1}{2}\text{O}_2(g) \longrightarrow \text{SrO}(s)$	-550

**Question 7 10 points**

Consider the molecule given on the left, and answer questions (i) to (vi) using chemical bonding theories that we have learnt in class (valence bond theory, hybridization).

Note that the lone pairs are not shown in the structure:

i. Consider the carbon atoms marked in as ***a***, ***b***, ***c***, ***d***, and ***e*** in the molecule, and determine **hybridization** at each of the carbon centers. (2.5 points)

Atom	Hybridization
Carbon (<i>a</i>)	
Carbon (<i>b</i>)	
Carbon (<i>c</i>)	
Carbon (<i>d</i>)	
Carbon (<i>e</i>)	

ii. Now, consider the sulfur atom (S) marked as ***f***, nitrogen (N) atom (***g***), and oxygen(O) atoms (***h*** and ***i***), and determine their **hybridization**. (2 points)

Atom	Hybridization
Sulfur (<i>f</i>)	
Nitrogen (<i>g</i>)	
Oxygen (<i>h</i>)	
Oxygen (<i>i</i>)	

iii. **Describe** all the bonds formed by the carbon atoms ***a*** and ***b***, using sigma and pi designation as appropriate. (2 points)

Atom	Bonds formed (use sigma and pi as appropriate)
Carbon (<i>a</i>)	
Carbon (<i>b</i>)	

iv. How many total sp^3 -s sigma bonds are in the molecule? (1 point)

v. Determine **electronic** and **molecular geometry** at carbon (C) labeled ***e*** and sulfur (S) labeled ***f*** centres. (2 points)

Atom	Electronic Geometry	Molecular Geometry
Carbon (<i>e</i>)		
Sulfur (<i>f</i>)		

vi. Circle the value from the choices given below, that describes the **C-S-C bond angle at sulfur** (S) marked ***f***: (0.5 points)

>90° to <109.5° 109.5° >109.5 to <120° 120° >120° to <180°


Question 8 12 points

Using the **valence shell orbitals only** draw a **complete** molecular orbital (MO) diagram (with no s-p mixing) for the molecule **PS**, in the space provided below. **(6 points)**

Assume that 3s and 3p atomic orbitals will form molecular orbitals like those formed by the combination of 2s and 2p atomic orbitals that we learnt in class, and there is no s-p mixing.

Show **energy arrow**, **atomic orbitals** involved, **atom labels**, the **molecular orbitals** that are formed, **filled electrons** in the MO diagram etc. Make sure to label all the atomic and molecular orbitals.

Using the MO diagram above, answer the following questions:

- (i) Which one of the following will have the highest bond order? Circle the correct choice
 A) PS^{1-} B) PS C) PS^{1+}
- (ii) Which one of the following will have the longest bond length? Circle the correct choice
 A) PS^{1-} B) PS C) PS^{1+}
- (iii) Which of the following will be diamagnetic? Circle the correct choice.
 A) PS^{1-} B) PS C) PS^{1+}
- (iv) Draw Lewis dot structure of the molecule with the shortest bond length (based on the MO diagram) from among the three (PS^{1-} , PS, PS^{1+}), and indicate the bond order calculated from the Lewis dot structure. **(3 points)**

Lewis Dot Structure

Bond Order



870B027C-E78F-44FE-A8DD-72B3D4609CE4

midterm-exam-93388

#763 Page 10 of 10

SCRAP PAGE

**CHEMISTRY 110 –Midterm Exam****Thursday November 7, 2024, 6:30 – 8:30 pm**

Examiners: Pallavi Sirjoosingh and Ashok Kakkar

First name (please write as legibly as possible within the boxes)

--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--

Last name

--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--

Student ID number

--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--

This booklet has 8 questions on 10 pages (The last page can be used as a scrap page).

The staple must NOT be removed from the question booklet. Your intact question booklet must be returned at the end of the exam.

A separate booklet contains a periodic table, and datasheet.

Write your answers directly under the questions in the space provided. Do not write on the QR codes or near the top edges (as it may be cut off).

Please write **legibly and dark enough** for the scan to be readable. For students who use English as a second language, dictionaries are permitted. Calculators are allowed.

DO NOT ASK QUESTIONS OF THE INVIGILATORS about interpretations of the exam questions. If you have a concern about a question, please summarize it at the top of this page. All concerns will be evaluated after the exam. GOOD LUCK!

**Question 1 8 points** (*Assume all transitions are possible*)

a. 6 points Calculate the **energy (in kJ/mol)** required to ionize (completely remove) the electron from the **n=3** level of **He⁺ ion** (He: Helium). Show your work including equations. (3 sig. figs.)

b. 2 points

(I) Select the largest (longest) wavelength from the three choices below:

- i. Wavelength of photon to *completely remove* an electron from *1s orbital of Hydrogen atom*
- ii. Wavelength of photon to *completely remove* an electron from *1s orbital of Boron atom (B)*
- iii. Wavelength of photon to *completely remove* an electron from *1s orbital of Boron cation (B⁺)*

i or ii or iii

(II) Select the smallest (shortest) wavelength from the three choices below (*for H atom*):

- i. Wavelength of photon *emitted* for an electron transitioning from *2p to 1s orbital*
- ii. Wavelength of photon *absorbed* for an electron transitioning from *2s to 4p orbital*
- iii. Wavelength of photon *absorbed* for an electron transitioning from *1s to 3p orbital*

i or ii or iii



Question 2 8 points



A Chem110 student plots the graph shown on the left, from data obtained for the kinetic energy (eV) of the ejected electron versus frequency (s^{-1}) of the incident light, for an unknown metal. ($1\text{eV} = 1.6022 \times 10^{-19} \text{ J}$)

Based on the plot, answer the questions **a to c** below:

a. 1 point On the plot above, label the **threshold frequency**.

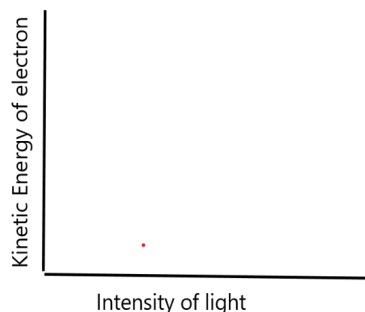
b. 2 points What is the value of the work potential (in J) of the unknown metal?

c. 3 points What is the *longest* wavelength (in nm) required to eject an electron from this metal's surface? Show your work including equation(s) used. (3 sig figs)

d. 2 points In the space provided below, plot the following

(Assume frequency is *above* threshold frequency):

- Kinetic energy* of an ejected electron versus the *intensity* of the incident light (left)
- Number of electrons* ejected versus the *frequency* of the incident light (right)



**Question 3** **6 points****a. 2 points**

In the ground state of Ge (Germanium; $Z = 32$) atom, **name** the following orbitals:

i. Orbital containing the electron that requires the maximum energy to be removed.

ii. Orbital containing the electron that experiences the minimum effective nuclear charge.

b. 2 points

An unknown anion (A^-) has the following *excited* electronic configuration: $[Ar] 4s^2 3d^3 4p^1$

(Ar: Argon; $Z = 18$)

Give the **condensed ground state electronic configuration** of the atom, **A** and the cation, **A^{2+}** .
Show your work. (Assume all electronic transitions are possible)

Electronic Configuration of A:

Electronic Configuration of A^{2+} :

c. 2 points

In a single atom, determine how many electrons can have the following set of quantum numbers?

(i) $n = 5; l = 3$

In a single atom, determine how many electrons can have the following set of quantum numbers?

(i) $n = 3; l = 2; m_s = +1/2$

**Question 4** **8 points**

Rank the following from 1 to 4 in **increasing** order. Provide the ranking in the boxes provided.

Note: Atomic number (Z) are provided for relevant atoms

a) Increasing (lowest to highest) order of *ionization energy*: (1= lowest; 4 = highest)

Mg

Mg: Z = 12

Rb

Rb: Z = 37

Cs

Cs: Z = 55

K

K : Z = 19

b) Increasing (lowest to highest) order of *bond angle*: (1= lowest; 4 = highest)

H₂OCCl₄XeF₄

HCN

c) Increasing (smallest to largest) order of *ionic size* (1= smallest; 4 = largest)

S²⁻

S: Z = 16

O²⁻

O: Z = 8

Be²⁺

Be: Z = 4

Na⁺

Na: Z = 11

d) Increasing (lowest to highest) order of *bond energy* (1= lowest; 4 = highest)

P-F

P-Cl

P-I

P-Br

**Question 5** **12 points**

a. 9 points For the molecules HClO_4 and TeF_6 , draw the most stable *Lewis structure* showing appropriate shape and *dashed/wedged bonds*, based on VSEPR. Determine the **molecular geometry** around the central atom. For HClO_4 , label the *most polar bond*.



b. 3 points (Note: Lone pairs are not shown in structures below)

(i) For the three *resonance* structures (1, 2 and 3) shown below, answer the following question:



(I) Circle the *most contributing* resonance structure.

Structure 1

Structure 2

Structure 3

(ii) For the three structures given (4, 5, and 6) below, answer the following questions:



(I) Circle all the structures that represent *resonance structures* of each other:

Structure 4

Structure 5

Structure 6

(II) Among the resonance structures, circle the *most contributing* resonance structure.

Structure 4

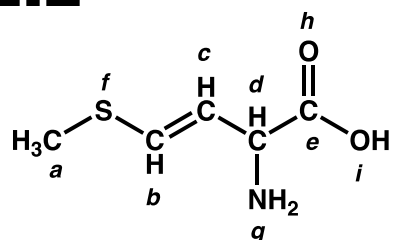
Structure 5

Structure 6

**Question 6** **6 points**

Using the data given below, calculate the **lattice energy** for SrO(s).

	ΔH° (kJ/mol)
$\text{Sr}(s) \longrightarrow \text{Sr}(g)$	+180
$\text{Sr}(g) \longrightarrow \text{Sr}^{2+}(g) + 2e^-$	+1468
$\text{O}_2(g) \longrightarrow 2\text{O}(g)$	+500
$\text{O}(g) + 2e^- \longrightarrow \text{O}^{2-}(g)$	+600
$\text{Sr}(s) + \frac{1}{2}\text{O}_2(g) \longrightarrow \text{SrO}(s)$	-550

**Question 7 10 points**

Consider the molecule given on the left, and answer questions (i) to (vi) using chemical bonding theories that we have learnt in class (valence bond theory, hybridization).

Note that the lone pairs are not shown in the structure:

i. Consider the carbon atoms marked in as ***a***, ***b***, ***c***, ***d***, and ***e*** in the molecule, and determine **hybridization** at each of the carbon centers. (2.5 points)

Atom	Hybridization
Carbon (<i>a</i>)	
Carbon (<i>b</i>)	
Carbon (<i>c</i>)	
Carbon (<i>d</i>)	
Carbon (<i>e</i>)	

ii. Now, consider the sulfur atom (S) marked as ***f***, nitrogen (N) atom (***g***), and oxygen(O) atoms (***h*** and ***i***), and determine their **hybridization**. (2 points)

Atom	Hybridization
Sulfur (<i>f</i>)	
Nitrogen (<i>g</i>)	
Oxygen (<i>h</i>)	
Oxygen (<i>i</i>)	

iii. **Describe** all the bonds formed by the carbon atoms ***a*** and ***b***, using sigma and pi designation as appropriate. (2 points)

Atom	Bonds formed (use sigma and pi as appropriate)
Carbon (<i>a</i>)	
Carbon (<i>b</i>)	

iv. How many total sp^3 -s sigma bonds are in the molecule? (1 point)

v. Determine **electronic** and **molecular geometry** at carbon (C) labeled ***e*** and sulfur (S) labeled ***f*** centres. (2 points)

Atom	Electronic Geometry	Molecular Geometry
Carbon (<i>e</i>)		
Sulfur (<i>f</i>)		

vi. Circle the value from the choices given below, that describes the **C-S-C bond angle at sulfur** (S) marked ***f***: (0.5 points)

>90° to <109.5°

109.5°

>109.5 to <120°

120°

>120° to <180°


Question 8 12 points

Using the **valence shell orbitals only** draw a **complete** molecular orbital (MO) diagram (with no s-p mixing) for the molecule **PS**, in the space provided below. **(6 points)**

Assume that 3s and 3p atomic orbitals will form molecular orbitals like those formed by the combination of 2s and 2p atomic orbitals that we learnt in class, and there is no s-p mixing.

Show **energy arrow**, **atomic orbitals** involved, **atom labels**, the **molecular orbitals** that are formed, **filled electrons** in the MO diagram etc. Make sure to label all the atomic and molecular orbitals.

Using the MO diagram above, answer the following questions:

- (i) Which one of the following will have the highest bond order? Circle the correct choice
 A) PS^{1-} B) PS C) PS^{1+}
- (ii) Which one of the following will have the longest bond length? Circle the correct choice
 A) PS^{1-} B) PS C) PS^{1+}
- (iii) Which of the following will be diamagnetic? Circle the correct choice.
 A) PS^{1-} B) PS C) PS^{1+}
- (iv) Draw Lewis dot structure of the molecule with the shortest bond length (based on the MO diagram) from among the three (PS^{1-} , PS, PS^{1+}), and indicate the bond order calculated from the Lewis dot structure. **(3 points)**

Lewis Dot Structure

Bond Order



9C85D8CD-84E1-4418-9996-E8CC0F7EFFF

midterm-exam-93388

#764 Page 10 of 10

SCRAP PAGE

**CHEMISTRY 110 –Midterm Exam****Thursday November 7, 2024, 6:30 – 8:30 pm**

Examiners: Pallavi Sirjoosingh and Ashok Kakkar

First name (please write as legibly as possible within the boxes)

--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--

Last name

--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--

Student ID number

--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--

This booklet has 8 questions on 10 pages (The last page can be used as a scrap page).

The staple must NOT be removed from the question booklet. Your intact question booklet must be returned at the end of the exam.

A separate booklet contains a periodic table, and datasheet.

Write your answers directly under the questions in the space provided. Do not write on the QR codes or near the top edges (as it may be cut off).

Please write **legibly and dark enough** for the scan to be readable. For students who use English as a second language, dictionaries are permitted. Calculators are allowed.

DO NOT ASK QUESTIONS OF THE INVIGILATORS about interpretations of the exam questions. If you have a concern about a question, please summarize it at the top of this page. All concerns will be evaluated after the exam. GOOD LUCK!

**Question 1 8 points** (*Assume all transitions are possible*)

a. 6 points Calculate the **energy (in kJ/mol)** required to ionize (completely remove) the electron from the **n=3** level of **He⁺ ion** (He: Helium). Show your work including equations. (3 sig. figs.)

b. 2 points

(I) Select the largest (longest) wavelength from the three choices below:

- i. Wavelength of photon to *completely remove* an electron from *1s orbital of Hydrogen atom*
- ii. Wavelength of photon to *completely remove* an electron from *1s orbital of Boron atom (B)*
- iii. Wavelength of photon to *completely remove* an electron from *1s orbital of Boron cation (B⁺)*

i or ii or iii

(II) Select the smallest (shortest) wavelength from the three choices below (*for H atom*):

- i. Wavelength of photon *emitted* for an electron transitioning from *2p to 1s orbital*
- ii. Wavelength of photon *absorbed* for an electron transitioning from *2s to 4p orbital*
- iii. Wavelength of photon *absorbed* for an electron transitioning from *1s to 3p orbital*

i or ii or iii



Question 2 8 points



A Chem110 student plots the graph shown on the left, from data obtained for the kinetic energy (eV) of the ejected electron versus frequency (s^{-1}) of the incident light, for an unknown metal. ($1\text{eV} = 1.6022 \times 10^{-19} \text{ J}$)

Based on the plot, answer the questions **a to c** below:

a. 1 point On the plot above, label the **threshold frequency**.

b. 2 points What is the value of the work potential (in J) of the unknown metal?

c. 3 points What is the *longest* wavelength (in nm) required to eject an electron from this metal's surface? Show your work including equation(s) used. (3 sig figs)

d. 2 points In the space provided below, plot the following

(Assume frequency is *above* threshold frequency):

- i. *Kinetic energy* of an ejected electron versus the *intensity* of the incident light (left)
- ii. *Number of electrons* ejected versus the *frequency* of the incident light (right)



**Question 3** **6 points****a. 2 points**

In the ground state of Ge (Germanium; $Z = 32$) atom, **name** the following orbitals:

i. Orbital containing the electron that requires the maximum energy to be removed.

ii. Orbital containing the electron that experiences the minimum effective nuclear charge.

b. 2 points

An unknown anion (A^-) has the following *excited* electronic configuration: $[Ar] 4s^2 3d^3 4p^1$

(Ar: Argon; $Z = 18$)

Give the **condensed ground state electronic configuration** of the atom, **A** and the cation, **A^{2+}** .
Show your work. (Assume all electronic transitions are possible)

Electronic Configuration of A:

Electronic Configuration of A^{2+} :

c. 2 points

In a single atom, determine how many electrons can have the following set of quantum numbers?

(i) $n = 5; l = 3$

In a single atom, determine how many electrons can have the following set of quantum numbers?

(i) $n = 3; l = 2; m_s = +1/2$

**Question 4** **8 points**

Rank the following from 1 to 4 in **increasing** order. Provide the ranking in the boxes provided.

Note: Atomic number (Z) are provided for relevant atoms

a) Increasing (lowest to highest) order of *ionization energy*: (1= lowest; 4 = highest)

Mg

Mg: Z = 12

Rb

Rb: Z = 37

Cs

Cs: Z = 55

K

K : Z = 19

b) Increasing (lowest to highest) order of *bond angle*: (1= lowest; 4 = highest)

H₂OCCl₄XeF₄

HCN

c) Increasing (smallest to largest) order of *ionic size* (1= smallest; 4 = largest)

S²⁻

S: Z = 16

O²⁻

O: Z = 8

Be²⁺

Be: Z = 4

Na⁺

Na: Z = 11

d) Increasing (lowest to highest) order of *bond energy* (1= lowest; 4 = highest)

P-F

P-Cl

P-I

P-Br

**Question 5** **12 points**

a. 9 points For the molecules HClO_4 and TeF_6 , draw the most stable *Lewis structure* showing appropriate shape and *dashed/wedged bonds*, based on VSEPR. Determine the **molecular geometry** around the central atom. For HClO_4 , label the *most polar bond*.



b. 3 points (Note: Lone pairs are not shown in structures below)

(i) For the three *resonance* structures (1, 2 and 3) shown below, answer the following question:



(I) Circle the *most contributing* resonance structure.

Structure 1

Structure 2

Structure 3

(ii) For the three structures given (4, 5, and 6) below, answer the following questions:



(I) Circle all the structures that represent *resonance structures* of each other:

Structure 4

Structure 5

Structure 6

(II) Among the resonance structures, circle the *most contributing* resonance structure.

Structure 4

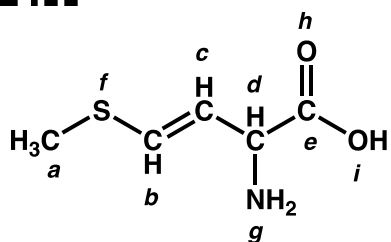
Structure 5

Structure 6

**Question 6** **6 points**

Using the data given below, calculate the **lattice energy** for SrO(s).

	ΔH° (kJ/mol)
$\text{Sr}(s) \longrightarrow \text{Sr}(g)$	+180
$\text{Sr}(g) \longrightarrow \text{Sr}^{2+}(g) + 2e^-$	+1468
$\text{O}_2(g) \longrightarrow 2\text{O}(g)$	+500
$\text{O}(g) + 2e^- \longrightarrow \text{O}^{2-}(g)$	+600
$\text{Sr}(s) + \frac{1}{2}\text{O}_2(g) \longrightarrow \text{SrO}(s)$	-550

**Question 7 10 points**

Consider the molecule given on the left, and answer questions (i) to (vi) using chemical bonding theories that we have learnt in class (valence bond theory, hybridization).

Note that the lone pairs are not shown in the structure:

i. Consider the carbon atoms marked in as ***a***, ***b***, ***c***, ***d***, and ***e*** in the molecule, and determine **hybridization** at each of the carbon centers. (2.5 points)

Atom	Hybridization
Carbon (<i>a</i>)	
Carbon (<i>b</i>)	
Carbon (<i>c</i>)	
Carbon (<i>d</i>)	
Carbon (<i>e</i>)	

ii. Now, consider the sulfur atom (S) marked as ***f***, nitrogen (N) atom (***g***), and oxygen(O) atoms (***h*** and ***i***), and determine their **hybridization**. (2 points)

Atom	Hybridization
Sulfur (<i>f</i>)	
Nitrogen (<i>g</i>)	
Oxygen (<i>h</i>)	
Oxygen (<i>i</i>)	

iii. **Describe** all the bonds formed by the carbon atoms ***a*** and ***b***, using sigma and pi designation as appropriate. (2 points)

Atom	Bonds formed (use sigma and pi as appropriate)
Carbon (<i>a</i>)	
Carbon (<i>b</i>)	

iv. How many total sp^3 -s sigma bonds are in the molecule? (1 point)

v. Determine **electronic** and **molecular geometry** at carbon (C) labeled ***e*** and sulfur (S) labeled ***f*** centres. (2 points)

Atom	Electronic Geometry	Molecular Geometry
Carbon (<i>e</i>)		
Sulfur (<i>f</i>)		

vi. Circle the value from the choices given below, that describes the **C-S-C bond angle at sulfur** (S) marked ***f***: (0.5 points)

>90° to <109.5°

109.5°

>109.5 to <120°

120°

>120° to <180°


Question 8 12 points

Using the **valence shell orbitals only** draw a **complete** molecular orbital (MO) diagram (with no s-p mixing) for the molecule **PS**, in the space provided below. **(6 points)**

Assume that 3s and 3p atomic orbitals will form molecular orbitals like those formed by the combination of 2s and 2p atomic orbitals that we learnt in class, and there is no s-p mixing.

Show **energy arrow**, **atomic orbitals** involved, **atom labels**, the **molecular orbitals** that are formed, **filled electrons** in the MO diagram etc. Make sure to label all the atomic and molecular orbitals.

Using the MO diagram above, answer the following questions:

- (i) Which one of the following will have the highest bond order? Circle the correct choice
 A) PS^{1-} B) PS C) PS^{1+}
- (ii) Which one of the following will have the longest bond length? Circle the correct choice
 A) PS^{1-} B) PS C) PS^{1+}
- (iii) Which of the following will be diamagnetic? Circle the correct choice.
 A) PS^{1-} B) PS C) PS^{1+}
- (iv) Draw Lewis dot structure of the molecule with the shortest bond length (based on the MO diagram) from among the three (PS^{1-} , PS, PS^{1+}), and indicate the bond order calculated from the Lewis dot structure. **(3 points)**

Lewis Dot Structure

Bond Order



02E3593C-5DB6-496D-9927-A4805CFCEA19

midterm-exam-93388

#765 Page 10 of 10

SCRAP PAGE

**CHEMISTRY 110 –Midterm Exam****Thursday November 7, 2024, 6:30 – 8:30 pm**

Examiners: Pallavi Sirjoosingh and Ashok Kakkar

First name (please write as legibly as possible within the boxes)

--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--

Last name

--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--

Student ID number

--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--

This booklet has 8 questions on 10 pages (The last page can be used as a scrap page).

The staple must NOT be removed from the question booklet. Your intact question booklet must be returned at the end of the exam.

A separate booklet contains a periodic table, and datasheet.

Write your answers directly under the questions in the space provided. Do not write on the QR codes or near the top edges (as it may be cut off).

Please write **legibly and dark enough** for the scan to be readable. For students who use English as a second language, dictionaries are permitted. Calculators are allowed.

DO NOT ASK QUESTIONS OF THE INVIGILATORS about interpretations of the exam questions. If you have a concern about a question, please summarize it at the top of this page. All concerns will be evaluated after the exam. GOOD LUCK!

**Question 1 8 points** (*Assume all transitions are possible*)

a. 6 points Calculate the **energy (in kJ/mol)** required to ionize (completely remove) the electron from the **n=3** level of **He⁺ ion** (He: Helium). Show your work including equations. (3 sig. figs.)

b. 2 points

(I) Select the largest (longest) wavelength from the three choices below:

- i. Wavelength of photon to *completely remove* an electron from *1s orbital of Hydrogen atom*
- ii. Wavelength of photon to *completely remove* an electron from *1s orbital of Boron atom (B)*
- iii. Wavelength of photon to *completely remove* an electron from *1s orbital of Boron cation (B⁺)*

i or ii or iii

(II) Select the smallest (shortest) wavelength from the three choices below (*for H atom*):

- i. Wavelength of photon *emitted* for an electron transitioning from *2p to 1s orbital*
- ii. Wavelength of photon *absorbed* for an electron transitioning from *2s to 4p orbital*
- iii. Wavelength of photon *absorbed* for an electron transitioning from *1s to 3p orbital*

i or ii or iii



Question 2 8 points



A Chem110 student plots the graph shown on the left, from data obtained for the kinetic energy (eV) of the ejected electron versus frequency (s^{-1}) of the incident light, for an unknown metal. ($1\text{eV} = 1.6022 \times 10^{-19} \text{ J}$)

Based on the plot, answer the questions **a to c** below:

a. 1 point On the plot above, label the **threshold frequency**.

b. 2 points What is the value of the work potential (in J) of the unknown metal?

c. 3 points What is the *longest* wavelength (in nm) required to eject an electron from this metal's surface? Show your work including equation(s) used. (3 sig figs)

d. 2 points In the space provided below, plot the following

(Assume frequency is *above* threshold frequency):

- Kinetic energy* of an ejected electron versus the *intensity* of the incident light (left)
- Number of electrons* ejected versus the *frequency* of the incident light (right)



**Question 3** **6 points****a. 2 points**

In the ground state of Ge (Germanium; $Z = 32$) atom, **name** the following orbitals:

i. Orbital containing the electron that requires the maximum energy to be removed.

ii. Orbital containing the electron that experiences the minimum effective nuclear charge.

b. 2 points

An unknown anion (A^-) has the following *excited* electronic configuration: $[Ar] 4s^2 3d^3 4p^1$

(Ar: Argon; $Z = 18$)

Give the **condensed ground state electronic configuration** of the atom, **A** and the cation, **A^{2+}** .
Show your work. (Assume all electronic transitions are possible)

Electronic Configuration of A:

Electronic Configuration of A^{2+} :

c. 2 points

In a single atom, determine how many electrons can have the following set of quantum numbers?

(i) $n = 5; l = 3$

In a single atom, determine how many electrons can have the following set of quantum numbers?

(i) $n = 3; l = 2; m_s = +1/2$

**Question 4** **8 points**

Rank the following from 1 to 4 in **increasing** order. Provide the ranking in the boxes provided.

Note: Atomic number (Z) are provided for relevant atoms

a) Increasing (lowest to highest) order of *ionization energy*: (1= lowest; 4 = highest)

Mg

Mg: Z = 12

Rb

Rb: Z = 37

Cs

Cs: Z = 55

K

K : Z = 19

b) Increasing (lowest to highest) order of *bond angle*: (1= lowest; 4 = highest)

H₂OCCl₄XeF₄

HCN

c) Increasing (smallest to largest) order of *ionic size* (1= smallest; 4 = largest)

S²⁻

S: Z = 16

O²⁻

O: Z = 8

Be²⁺

Be: Z = 4

Na⁺

Na: Z = 11

d) Increasing (lowest to highest) order of *bond energy* (1= lowest; 4 = highest)

P-F

P-Cl

P-I

P-Br

**Question 5** **12 points**

a. 9 points For the molecules HClO_4 and TeF_6 , draw the most stable *Lewis structure* showing appropriate shape and *dashed/wedged bonds*, based on VSEPR. Determine the **molecular geometry** around the central atom. For HClO_4 , label the *most polar bond*.



b. 3 points (Note: Lone pairs are not shown in structures below)

(i) For the three *resonance* structures (1, 2 and 3) shown below, answer the following question:



(I) Circle the *most contributing* resonance structure.

Structure 1

Structure 2

Structure 3

(ii) For the three structures given (4, 5, and 6) below, answer the following questions:



(I) Circle all the structures that represent *resonance structures* of each other:

Structure 4

Structure 5

Structure 6

(II) Among the resonance structures, circle the *most contributing* resonance structure.

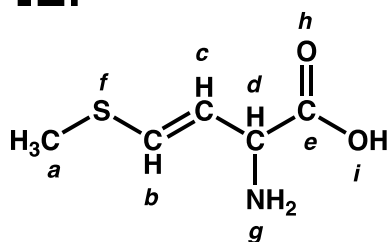
Structure 4

Structure 5

Structure 6

**Question 6** **6 points**Using the data given below, calculate the **lattice energy** for SrO(s).

	ΔH° (kJ/mol)
$\text{Sr}(s) \longrightarrow \text{Sr}(g)$	+180
$\text{Sr}(g) \longrightarrow \text{Sr}^{2+}(g) + 2e^-$	+1468
$\text{O}_2(g) \longrightarrow 2\text{O}(g)$	+500
$\text{O}(g) + 2e^- \longrightarrow \text{O}^{2-}(g)$	+600
$\text{Sr}(s) + \frac{1}{2}\text{O}_2(g) \longrightarrow \text{SrO}(s)$	-550

**Question 7 10 points**

Consider the molecule given on the left, and answer questions (i) to (vi) using chemical bonding theories that we have learnt in class (valence bond theory, hybridization).

Note that the lone pairs are not shown in the structure:

i. Consider the carbon atoms marked in as ***a***, ***b***, ***c***, ***d***, and ***e*** in the molecule, and determine **hybridization** at each of the carbon centers. (2.5 points)

Atom	Hybridization
Carbon (<i>a</i>)	
Carbon (<i>b</i>)	
Carbon (<i>c</i>)	
Carbon (<i>d</i>)	
Carbon (<i>e</i>)	

ii. Now, consider the sulfur atom (S) marked as ***f***, nitrogen (N) atom (***g***), and oxygen(O) atoms (***h*** and ***i***), and determine their **hybridization**. (2 points)

Atom	Hybridization
Sulfur (<i>f</i>)	
Nitrogen (<i>g</i>)	
Oxygen (<i>h</i>)	
Oxygen (<i>i</i>)	

iii. **Describe** all the bonds formed by the carbon atoms ***a*** and ***b***, using sigma and pi designation as appropriate. (2 points)

Atom	Bonds formed (use sigma and pi as appropriate)
Carbon (<i>a</i>)	
Carbon (<i>b</i>)	

iv. How many total sp^3 -s sigma bonds are in the molecule? (1 point)

v. Determine **electronic** and **molecular geometry** at carbon (C) labeled ***e*** and sulfur (S) labeled ***f*** centres. (2 points)

Atom	Electronic Geometry	Molecular Geometry
Carbon (<i>e</i>)		
Sulfur (<i>f</i>)		

vi. Circle the value from the choices given below, that describes the **C-S-C bond angle at sulfur** (S) marked ***f***: (0.5 points)

>90° to <109.5°

109.5°

>109.5 to <120°

120°

>120° to <180°

**Question 8 12 points**

Using the **valence shell orbitals only** draw a **complete** molecular orbital (MO) diagram (with no s-p mixing) for the molecule **PS**, in the space provided below. **(6 points)**

Assume that 3s and 3p atomic orbitals will form molecular orbitals like those formed by the combination of 2s and 2p atomic orbitals that we learnt in class, and there is no s-p mixing.

Show **energy arrow**, **atomic orbitals** involved, **atom labels**, the **molecular orbitals** that are formed, **filled electrons** in the MO diagram etc. Make sure to label all the atomic and molecular orbitals.

Using the MO diagram above, answer the following questions:

- (i) Which one of the following will have the highest bond order? Circle the correct choice
A) PS^{1-} B) PS C) PS^{1+}
- (ii) Which one of the following will have the longest bond length? Circle the correct choice
A) PS^{1-} B) PS C) PS^{1+}
- (iii) Which of the following will be diamagnetic? Circle the correct choice.
A) PS^{1-} B) PS C) PS^{1+}
- (iv) Draw Lewis dot structure of the molecule with the shortest bond length (based on the MO diagram) from among the three (PS^{1-} , PS, PS^{1+}), and indicate the bond order calculated from the Lewis dot structure. **(3 points)**

Lewis Dot Structure

Bond Order



26BD6AB2-312F-4C4C-B6E3-B01D087C34D2

midterm-exam-93388

#766 Page 10 of 10

SCRAP PAGE

**CHEMISTRY 110 –Midterm Exam****Thursday November 7, 2024, 6:30 – 8:30 pm****Examiners: Pallavi Sirjoosingh and Ashok Kakkar**

First name (please write as legibly as possible within the boxes)

--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--

Last name

--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--

Student ID number

--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--

This booklet has 8 questions on 10 pages (The last page can be used as a scrap page).

The staple must NOT be removed from the question booklet. Your intact question booklet must be returned at the end of the exam.

A separate booklet contains a periodic table, and datasheet.

Write your answers directly under the questions in the space provided. Do not write on the QR codes or near the top edges (as it may be cut off).

Please write **legibly and dark enough** for the scan to be readable. For students who use English as a second language, dictionaries are permitted. Calculators are allowed.

DO NOT ASK QUESTIONS OF THE INVIGILATORS about interpretations of the exam questions. If you have a concern about a question, please summarize it at the top of this page. All concerns will be evaluated after the exam. GOOD LUCK!

**Question 1 8 points** (*Assume all transitions are possible*)

a. 6 points Calculate the **energy (in kJ/mol)** required to ionize (completely remove) the electron from the **n=3** level of **He⁺ ion** (He: Helium). Show your work including equations. (3 sig. figs.)

b. 2 points

(I) Select the largest (longest) wavelength from the three choices below:

- i. Wavelength of photon to *completely remove* an electron from *1s orbital of Hydrogen atom*
- ii. Wavelength of photon to *completely remove* an electron from *1s orbital of Boron atom (B)*
- iii. Wavelength of photon to *completely remove* an electron from *1s orbital of Boron cation (B⁺)*

i

or

ii

or

iii

(II) Select the smallest (shortest) wavelength from the three choices below (*for H atom*):

- i. Wavelength of photon *emitted* for an electron transitioning from *2p to 1s orbital*
- ii. Wavelength of photon *absorbed* for an electron transitioning from *2s to 4p orbital*
- iii. Wavelength of photon *absorbed* for an electron transitioning from *1s to 3p orbital*

i

or

ii

or

iii



Question 2 8 points



A Chem110 student plots the graph shown on the left, from data obtained for the kinetic energy (eV) of the ejected electron versus frequency (s^{-1}) of the incident light, for an unknown metal. ($1\text{eV} = 1.6022 \times 10^{-19} \text{ J}$)

Based on the plot, answer the questions **a to c** below:

a. 1 point On the plot above, label the **threshold frequency**.

b. 2 points What is the value of the work potential (in J) of the unknown metal?

c. 3 points What is the *longest* wavelength (in nm) required to eject an electron from this metal's surface? Show your work including equation(s) used. (3 sig figs)

d. 2 points In the space provided below, plot the following

(Assume frequency is *above* threshold frequency):

- i. *Kinetic energy* of an ejected electron versus the *intensity* of the incident light (left)
- ii. *Number of electrons* ejected versus the *frequency* of the incident light (right)



**Question 3** **6 points****a. 2 points**

In the ground state of Ge (Germanium; $Z = 32$) atom, **name** the following orbitals:

i. Orbital containing the electron that requires the maximum energy to be removed.

ii. Orbital containing the electron that experiences the minimum effective nuclear charge.

b. 2 points

An unknown anion (A^-) has the following *excited* electronic configuration: $[Ar] 4s^2 3d^3 4p^1$

(Ar: Argon; $Z = 18$)

Give the **condensed ground state electronic configuration** of the atom, **A** and the cation, **A^{2+}** .
Show your work. (Assume all electronic transitions are possible)

Electronic Configuration of A:

Electronic Configuration of A^{2+} :

c. 2 points

In a single atom, determine how many electrons can have the following set of quantum numbers?

(i) $n = 5; l = 3$

In a single atom, determine how many electrons can have the following set of quantum numbers?

(i) $n = 3; l = 2; m_s = +1/2$

**Question 4** **8 points**

Rank the following from 1 to 4 in **increasing** order. Provide the ranking in the boxes provided.

Note: Atomic number (Z) are provided for relevant atoms

a) Increasing (lowest to highest) order of *ionization energy*: (1= lowest; 4 = highest)

Mg

Mg: Z = 12

Rb

Rb: Z = 37

Cs

Cs: Z = 55

K

K : Z = 19

b) Increasing (lowest to highest) order of *bond angle*: (1= lowest; 4 = highest)

H₂OCCl₄XeF₄

HCN

c) Increasing (smallest to largest) order of *ionic size* (1= smallest; 4 = largest)

S²⁻

S: Z = 16

O²⁻

O: Z = 8

Be²⁺

Be: Z = 4

Na⁺

Na: Z = 11

d) Increasing (lowest to highest) order of *bond energy* (1= lowest; 4 = highest)

P-F

P-Cl

P-I

P-Br

**Question 5** **12 points**

a. 9 points For the molecules HClO_4 and TeF_6 , draw the most stable *Lewis structure* showing appropriate shape and *dashed/wedged bonds*, based on VSEPR. Determine the **molecular geometry** around the central atom. For HClO_4 , label the *most polar bond*.

HClO_4 (there are no O-O bonds)

Label the most polar bond

Molecular Geometry around Cl

TeF_6

Molecular Geometry around Te

b. 3 points (Note: Lone pairs are not shown in structures below)

(i) For the three *resonance* structures (1, 2 and 3) shown below, answer the following question:



(I) Circle the *most contributing* resonance structure.

Structure 1

Structure 2

Structure 3

(ii) For the three structures given (4, 5, and 6) below, answer the following questions:



(I) Circle all the structures that represent *resonance structures* of each other:

Structure 4

Structure 5

Structure 6

(II) Among the resonance structures, circle the *most contributing* resonance structure.

Structure 4

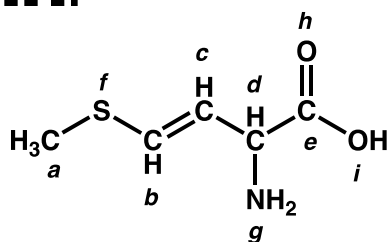
Structure 5

Structure 6

**Question 6** **6 points**

Using the data given below, calculate the **lattice energy** for SrO(s).

	ΔH° (kJ/mol)
$\text{Sr}(s) \longrightarrow \text{Sr}(g)$	+180
$\text{Sr}(g) \longrightarrow \text{Sr}^{2+}(g) + 2e^-$	+1468
$\text{O}_2(g) \longrightarrow 2\text{O}(g)$	+500
$\text{O}(g) + 2e^- \longrightarrow \text{O}^{2-}(g)$	+600
$\text{Sr}(s) + \frac{1}{2}\text{O}_2(g) \longrightarrow \text{SrO}(s)$	-550

**Question 7 10 points**

Consider the molecule given on the left, and answer questions (i) to (vi) using chemical bonding theories that we have learnt in class (valence bond theory, hybridization).

Note that the lone pairs are not shown in the structure:

i. Consider the carbon atoms marked in as ***a***, ***b***, ***c***, ***d***, and ***e*** in the molecule, and determine **hybridization** at each of the carbon centers. (2.5 points)

Atom	Hybridization
Carbon (<i>a</i>)	
Carbon (<i>b</i>)	
Carbon (<i>c</i>)	
Carbon (<i>d</i>)	
Carbon (<i>e</i>)	

ii. Now, consider the sulfur atom (S) marked as ***f***, nitrogen (N) atom (***g***), and oxygen(O) atoms (***h*** and ***i***), and determine their **hybridization**. (2 points)

Atom	Hybridization
Sulfur (<i>f</i>)	
Nitrogen (<i>g</i>)	
Oxygen (<i>h</i>)	
Oxygen (<i>i</i>)	

iii. **Describe** all the bonds formed by the carbon atoms ***a*** and ***b***, using sigma and pi designation as appropriate. (2 points)

Atom	Bonds formed (use sigma and pi as appropriate)
Carbon (<i>a</i>)	
Carbon (<i>b</i>)	

iv. How many total sp^3 -s sigma bonds are in the molecule? (1 point)

v. Determine **electronic** and **molecular geometry** at carbon (C) labeled ***e*** and sulfur (S) labeled ***f*** centres. (2 points)

Atom	Electronic Geometry	Molecular Geometry
Carbon (<i>e</i>)		
Sulfur (<i>f</i>)		

vi. Circle the value from the choices given below, that describes the **C-S-C bond angle at sulfur** (S) marked ***f***: (0.5 points)

>90° to <109.5° 109.5° >109.5 to <120° 120° >120° to <180°


Question 8 12 points

Using the **valence shell orbitals only** draw a **complete** molecular orbital (MO) diagram (with no s-p mixing) for the molecule **PS**, in the space provided below. **(6 points)**

Assume that 3s and 3p atomic orbitals will form molecular orbitals like those formed by the combination of 2s and 2p atomic orbitals that we learnt in class, and there is no s-p mixing.

Show **energy arrow**, **atomic orbitals** involved, **atom labels**, the **molecular orbitals** that are formed, **filled electrons** in the MO diagram etc. Make sure to label all the atomic and molecular orbitals.

Using the MO diagram above, answer the following questions:

- (i) Which one of the following will have the highest bond order? Circle the correct choice
 A) PS^{1-} B) PS C) PS^{1+}
- (ii) Which one of the following will have the longest bond length? Circle the correct choice
 A) PS^{1-} B) PS C) PS^{1+}
- (iii) Which of the following will be diamagnetic? Circle the correct choice.
 A) PS^{1-} B) PS C) PS^{1+}
- (iv) Draw Lewis dot structure of the molecule with the shortest bond length (based on the MO diagram) from among the three (PS^{1-} , PS, PS^{1+}), and indicate the bond order calculated from the Lewis dot structure. **(3 points)**

Lewis Dot Structure

Bond Order



316A03B5-6D5D-45A8-AC48-283645D23A1B

midterm-exam-93388

#767 Page 10 of 10

SCRAP PAGE

**CHEMISTRY 110 –Midterm Exam****Thursday November 7, 2024, 6:30 – 8:30 pm**

Examiners: Pallavi Sirjoosingh and Ashok Kakkar

First name (please write as legibly as possible within the boxes)

--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--

Last name

--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--

Student ID number

--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--

This booklet has 8 questions on 10 pages (The last page can be used as a scrap page).

The staple must NOT be removed from the question booklet. Your intact question booklet must be returned at the end of the exam.

A separate booklet contains a periodic table, and datasheet.

Write your answers directly under the questions in the space provided. Do not write on the QR codes or near the top edges (as it may be cut off).

Please write **legibly and dark enough** for the scan to be readable. For students who use English as a second language, dictionaries are permitted. Calculators are allowed.

DO NOT ASK QUESTIONS OF THE INVIGILATORS about interpretations of the exam questions. If you have a concern about a question, please summarize it at the top of this page. All concerns will be evaluated after the exam. GOOD LUCK!

**Question 1 8 points** (*Assume all transitions are possible*)

a. 6 points Calculate the **energy (in kJ/mol)** required to ionize (completely remove) the electron from the **$n=3$** level of **He^+ ion** (He: Helium). Show your work including equations. (3 sig. figs.)

b. 2 points

(I) Select the largest (longest) wavelength from the three choices below:

- i. Wavelength of photon to *completely remove* an electron from *1s orbital of Hydrogen atom*
- ii. Wavelength of photon to *completely remove* an electron from *1s orbital of Boron atom (B)*
- iii. Wavelength of photon to *completely remove* an electron from *1s orbital of Boron cation (B^+)*

i or ii or iii

(II) Select the smallest (shortest) wavelength from the three choices below (*for H atom*):

- i. Wavelength of photon *emitted* for an electron transitioning from *2p to 1s orbital*
- ii. Wavelength of photon *absorbed* for an electron transitioning from *2s to 4p orbital*
- iii. Wavelength of photon *absorbed* for an electron transitioning from *1s to 3p orbital*

i or ii or iii



Question 2 8 points



A Chem110 student plots the graph shown on the left, from data obtained for the kinetic energy (eV) of the ejected electron versus frequency (s^{-1}) of the incident light, for an unknown metal. ($1\text{eV} = 1.6022 \times 10^{-19} \text{ J}$)

Based on the plot, answer the questions **a to c** below:

a. 1 point On the plot above, label the **threshold frequency**.

b. 2 points What is the value of the work potential (in J) of the unknown metal?

c. 3 points What is the *longest* wavelength (in nm) required to eject an electron from this metal's surface? Show your work including equation(s) used. (3 sig figs)

d. 2 points In the space provided below, plot the following

(Assume frequency is *above* threshold frequency):

- Kinetic energy* of an ejected electron versus the *intensity* of the incident light (left)
- Number of electrons* ejected versus the *frequency* of the incident light (right)



**Question 3** **6 points****a. 2 points**

In the ground state of Ge (Germanium; $Z = 32$) atom, **name** the following orbitals:

i. Orbital containing the electron that requires the maximum energy to be removed.

ii. Orbital containing the electron that experiences the minimum effective nuclear charge.

b. 2 points

An unknown anion (A^-) has the following *excited* electronic configuration: $[Ar] 4s^2 3d^3 4p^1$

(Ar: Argon; $Z = 18$)

Give the **condensed ground state electronic configuration** of the atom, **A** and the cation, **A^{2+}** .
Show your work. (Assume all electronic transitions are possible)

Electronic Configuration of A:

Electronic Configuration of A^{2+} :

c. 2 points

In a single atom, determine how many electrons can have the following set of quantum numbers?

(i) $n = 5; l = 3$

In a single atom, determine how many electrons can have the following set of quantum numbers?

(i) $n = 3; l = 2; m_s = +1/2$

**Question 4** **8 points**

Rank the following from 1 to 4 in **increasing** order. Provide the ranking in the boxes provided.

Note: Atomic number (Z) are provided for relevant atoms

a) Increasing (lowest to highest) order of *ionization energy*: (1= lowest; 4 = highest)

Mg

Mg: Z = 12

Rb

Rb: Z = 37

Cs

Cs: Z = 55

K

K : Z = 19

b) Increasing (lowest to highest) order of *bond angle*: (1= lowest; 4 = highest)

H₂OCCl₄XeF₄

HCN

c) Increasing (smallest to largest) order of *ionic size* (1= smallest; 4 = largest)

S²⁻

S: Z = 16

O²⁻

O: Z = 8

Be²⁺

Be: Z = 4

Na⁺

Na: Z = 11

d) Increasing (lowest to highest) order of *bond energy* (1= lowest; 4 = highest)

P-F

P-Cl

P-I

P-Br

**Question 5** **12 points**

a. 9 points For the molecules HClO_4 and TeF_6 , draw the most stable *Lewis structure* showing appropriate shape and *dashed/wedged bonds*, based on VSEPR. Determine the **molecular geometry** around the central atom. For HClO_4 , label the *most polar bond*.



b. 3 points (Note: Lone pairs are not shown in structures below)

(i) For the three *resonance* structures (1, 2 and 3) shown below, answer the following question:



(I) Circle the *most contributing* resonance structure.

Structure 1

Structure 2

Structure 3

(ii) For the three structures given (4, 5, and 6) below, answer the following questions:



(I) Circle all the structures that represent *resonance structures* of each other:

Structure 4

Structure 5

Structure 6

(II) Among the resonance structures, circle the *most contributing* resonance structure.

Structure 4

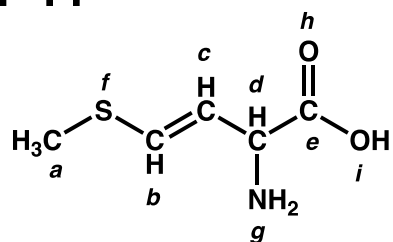
Structure 5

Structure 6

**Question 6** **6 points**

Using the data given below, calculate the **lattice energy** for SrO(s).

	ΔH° (kJ/mol)
$\text{Sr}(s) \longrightarrow \text{Sr}(g)$	+180
$\text{Sr}(g) \longrightarrow \text{Sr}^{2+}(g) + 2e^-$	+1468
$\text{O}_2(g) \longrightarrow 2\text{O}(g)$	+500
$\text{O}(g) + 2e^- \longrightarrow \text{O}^{2-}(g)$	+600
$\text{Sr}(s) + \frac{1}{2}\text{O}_2(g) \longrightarrow \text{SrO}(s)$	-550

**Question 7 10 points**

Consider the molecule given on the left, and answer questions (i) to (vi) using chemical bonding theories that we have learnt in class (valence bond theory, hybridization).

Note that the lone pairs are not shown in the structure:

i. Consider the carbon atoms marked in as ***a***, ***b***, ***c***, ***d***, and ***e*** in the molecule, and determine **hybridization** at each of the carbon centers. (2.5 points)

Atom	Hybridization
Carbon (<i>a</i>)	
Carbon (<i>b</i>)	
Carbon (<i>c</i>)	
Carbon (<i>d</i>)	
Carbon (<i>e</i>)	

ii. Now, consider the sulfur atom (S) marked as ***f***, nitrogen (N) atom (***g***), and oxygen(O) atoms (***h*** and ***i***), and determine their **hybridization**. (2 points)

Atom	Hybridization
Sulfur (<i>f</i>)	
Nitrogen (<i>g</i>)	
Oxygen (<i>h</i>)	
Oxygen (<i>i</i>)	

iii. **Describe** all the bonds formed by the carbon atoms ***a*** and ***b***, using sigma and pi designation as appropriate. (2 points)

Atom	Bonds formed (use sigma and pi as appropriate)
Carbon (<i>a</i>)	
Carbon (<i>b</i>)	

iv. How many total sp^3 -s sigma bonds are in the molecule? (1 point)

v. Determine **electronic** and **molecular geometry** at carbon (C) labeled ***e*** and sulfur (S) labeled ***f*** centres. (2 points)

Atom	Electronic Geometry	Molecular Geometry
Carbon (<i>e</i>)		
Sulfur (<i>f</i>)		

vi. Circle the value from the choices given below, that describes the **C-S-C bond angle at sulfur** (S) marked ***f***: (0.5 points)

>90° to <109.5°

109.5°

>109.5 to <120°

120°

>120° to <180°


Question 8 12 points

Using the **valence shell orbitals only** draw a **complete** molecular orbital (MO) diagram (with no s-p mixing) for the molecule **PS**, in the space provided below. **(6 points)**

Assume that 3s and 3p atomic orbitals will form molecular orbitals like those formed by the combination of 2s and 2p atomic orbitals that we learnt in class, and there is no s-p mixing.

Show **energy arrow**, **atomic orbitals** involved, **atom labels**, the **molecular orbitals** that are formed, **filled electrons** in the MO diagram etc. Make sure to label all the atomic and molecular orbitals.

Using the MO diagram above, answer the following questions:

- (i) Which one of the following will have the highest bond order? Circle the correct choice
 A) PS^{1-} B) PS C) PS^{1+}
- (ii) Which one of the following will have the longest bond length? Circle the correct choice
 A) PS^{1-} B) PS C) PS^{1+}
- (iii) Which of the following will be diamagnetic? Circle the correct choice.
 A) PS^{1-} B) PS C) PS^{1+}
- (iv) Draw Lewis dot structure of the molecule with the shortest bond length (based on the MO diagram) from among the three (PS^{1-} , PS, PS^{1+}), and indicate the bond order calculated from the Lewis dot structure. **(3 points)**

Lewis Dot Structure

Bond Order



071F85AA-1E84-4DDB-A347-905DC4D4C8DE

midterm-exam-93388

#768 Page 10 of 10

SCRAP PAGE

**CHEMISTRY 110 –Midterm Exam****Thursday November 7, 2024, 6:30 – 8:30 pm****Examiners: Pallavi Sirjoosingh and Ashok Kakkar**

First name (please write as legibly as possible within the boxes)

--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--

Last name

--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--

Student ID number

--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--

This booklet has 8 questions on 10 pages (The last page can be used as a scrap page).

The staple must NOT be removed from the question booklet. Your intact question booklet must be returned at the end of the exam.

A separate booklet contains a periodic table, and datasheet.

Write your answers directly under the questions in the space provided. Do not write on the QR codes or near the top edges (as it may be cut off).

Please write **legibly and dark enough** for the scan to be readable. For students who use English as a second language, dictionaries are permitted. Calculators are allowed.

DO NOT ASK QUESTIONS OF THE INVIGILATORS about interpretations of the exam questions. If you have a concern about a question, please summarize it at the top of this page. All concerns will be evaluated after the exam. GOOD LUCK!

**Question 1 8 points** (*Assume all transitions are possible*)

a. 6 points Calculate the **energy (in kJ/mol)** required to ionize (completely remove) the electron from the **n=3** level of **He⁺ ion** (He: Helium). Show your work including equations. (3 sig. figs.)

b. 2 points

(I) Select the largest (longest) wavelength from the three choices below:

- i. Wavelength of photon to *completely remove* an electron from *1s orbital of Hydrogen atom*
- ii. Wavelength of photon to *completely remove* an electron from *1s orbital of Boron atom (B)*
- iii. Wavelength of photon to *completely remove* an electron from *1s orbital of Boron cation (B⁺)*

i or ii or iii

(II) Select the smallest (shortest) wavelength from the three choices below (*for H atom*):

- i. Wavelength of photon *emitted* for an electron transitioning from *2p to 1s orbital*
- ii. Wavelength of photon *absorbed* for an electron transitioning from *2s to 4p orbital*
- iii. Wavelength of photon *absorbed* for an electron transitioning from *1s to 3p orbital*

i or ii or iii



Question 2 8 points



A Chem110 student plots the graph shown on the left, from data obtained for the kinetic energy (eV) of the ejected electron versus frequency (s^{-1}) of the incident light, for an unknown metal. ($1\text{eV} = 1.6022 \times 10^{-19} \text{ J}$)

Based on the plot, answer the questions **a to c** below:

a. 1 point On the plot above, label the **threshold frequency**.

b. 2 points What is the value of the work potential (in J) of the unknown metal?

c. 3 points What is the *longest* wavelength (in nm) required to eject an electron from this metal's surface? Show your work including equation(s) used. (3 sig figs)

d. 2 points In the space provided below, plot the following

(Assume frequency is *above* threshold frequency):

- Kinetic energy* of an ejected electron versus the *intensity* of the incident light (left)
- Number of electrons* ejected versus the *frequency* of the incident light (right)



**Question 3** **6 points****a. 2 points**

In the ground state of Ge (Germanium; $Z = 32$) atom, **name** the following orbitals:

i. Orbital containing the electron that requires the maximum energy to be removed.

ii. Orbital containing the electron that experiences the minimum effective nuclear charge.

b. 2 points

An unknown anion (A^-) has the following *excited* electronic configuration: $[Ar] 4s^2 3d^3 4p^1$

(Ar: Argon; $Z = 18$)

Give the **condensed ground state electronic configuration** of the atom, **A** and the cation, **A^{2+}** .
Show your work. (Assume all electronic transitions are possible)

Electronic Configuration of A:

Electronic Configuration of A^{2+} :

c. 2 points

In a single atom, determine how many electrons can have the following set of quantum numbers?

(i) $n = 5; l = 3$

In a single atom, determine how many electrons can have the following set of quantum numbers?

(i) $n = 3; l = 2; m_s = +1/2$

**Question 4** **8 points**

Rank the following from 1 to 4 in **increasing** order. Provide the ranking in the boxes provided.

Note: Atomic number (Z) are provided for relevant atoms

a) Increasing (lowest to highest) order of *ionization energy*: (1= lowest; 4 = highest)

Mg

Mg: Z = 12

Rb

Rb: Z = 37

Cs

Cs: Z = 55

K

K : Z = 19

b) Increasing (lowest to highest) order of *bond angle*: (1= lowest; 4 = highest)

H₂OCCl₄XeF₄

HCN

c) Increasing (smallest to largest) order of *ionic size* (1= smallest; 4 = largest)

S²⁻

S: Z = 16

O²⁻

O: Z = 8

Be²⁺

Be: Z = 4

Na⁺

Na: Z = 11

d) Increasing (lowest to highest) order of *bond energy* (1= lowest; 4 = highest)

P-F

P-Cl

P-I

P-Br

**Question 5** **12 points**

a. 9 points For the molecules HClO_4 and TeF_6 , draw the most stable *Lewis structure* showing appropriate shape and *dashed/wedged bonds*, based on VSEPR. Determine the **molecular geometry** around the central atom. For HClO_4 , label the *most polar bond*.



b. 3 points (Note: Lone pairs are not shown in structures below)

(i) For the three *resonance* structures (1, 2 and 3) shown below, answer the following question:



(I) Circle the *most contributing* resonance structure.

Structure 1

Structure 2

Structure 3

(ii) For the three structures given (4, 5, and 6) below, answer the following questions:



(I) Circle all the structures that represent *resonance structures* of each other:

Structure 4

Structure 5

Structure 6

(II) Among the resonance structures, circle the *most contributing* resonance structure.

Structure 4

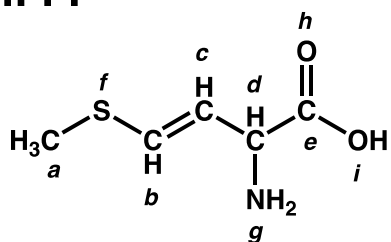
Structure 5

Structure 6

**Question 6** **6 points**

Using the data given below, calculate the **lattice energy** for SrO(s).

	ΔH° (kJ/mol)
$\text{Sr}(s) \longrightarrow \text{Sr}(g)$	+180
$\text{Sr}(g) \longrightarrow \text{Sr}^{2+}(g) + 2e^-$	+1468
$\text{O}_2(g) \longrightarrow 2\text{O}(g)$	+500
$\text{O}(g) + 2e^- \longrightarrow \text{O}^{2-}(g)$	+600
$\text{Sr}(s) + \frac{1}{2}\text{O}_2(g) \longrightarrow \text{SrO}(s)$	-550

**Question 7 10 points**

Consider the molecule given on the left, and answer questions (i) to (vi) using chemical bonding theories that we have learnt in class (valence bond theory, hybridization).

Note that the lone pairs are not shown in the structure:

i. Consider the carbon atoms marked in as ***a***, ***b***, ***c***, ***d***, and ***e*** in the molecule, and determine **hybridization** at each of the carbon centers. (2.5 points)

Atom	Hybridization
Carbon (<i>a</i>)	
Carbon (<i>b</i>)	
Carbon (<i>c</i>)	
Carbon (<i>d</i>)	
Carbon (<i>e</i>)	

ii. Now, consider the sulfur atom (S) marked as ***f***, nitrogen (N) atom (***g***), and oxygen(O) atoms (***h*** and ***i***), and determine their **hybridization**. (2 points)

Atom	Hybridization
Sulfur (<i>f</i>)	
Nitrogen (<i>g</i>)	
Oxygen (<i>h</i>)	
Oxygen (<i>i</i>)	

iii. **Describe** all the bonds formed by the carbon atoms ***a*** and ***b***, using sigma and pi designation as appropriate. (2 points)

Atom	Bonds formed (use sigma and pi as appropriate)
Carbon (<i>a</i>)	
Carbon (<i>b</i>)	

iv. How many total sp^3 -s sigma bonds are in the molecule? (1 point)

v. Determine **electronic** and **molecular geometry** at carbon (C) labeled ***e*** and sulfur (S) labeled ***f*** centres. (2 points)

Atom	Electronic Geometry	Molecular Geometry
Carbon (<i>e</i>)		
Sulfur (<i>f</i>)		

vi. Circle the value from the choices given below, that describes the **C-S-C bond angle at sulfur** (S) marked ***f***: (0.5 points)

>90° to <109.5° 109.5° >109.5 to <120° 120° >120° to <180°


Question 8 12 points

Using the **valence shell orbitals only** draw a **complete** molecular orbital (MO) diagram (with no s-p mixing) for the molecule **PS**, in the space provided below. **(6 points)**

Assume that 3s and 3p atomic orbitals will form molecular orbitals like those formed by the combination of 2s and 2p atomic orbitals that we learnt in class, and there is no s-p mixing.

Show **energy arrow**, **atomic orbitals** involved, **atom labels**, the **molecular orbitals** that are formed, **filled electrons** in the MO diagram etc. Make sure to label all the atomic and molecular orbitals.

Using the MO diagram above, answer the following questions:

- (i) Which one of the following will have the highest bond order? Circle the correct choice
 A) PS^{1-} B) PS C) PS^{1+}
- (ii) Which one of the following will have the longest bond length? Circle the correct choice
 A) PS^{1-} B) PS C) PS^{1+}
- (iii) Which of the following will be diamagnetic? Circle the correct choice.
 A) PS^{1-} B) PS C) PS^{1+}
- (iv) Draw Lewis dot structure of the molecule with the shortest bond length (based on the MO diagram) from among the three (PS^{1-} , PS, PS^{1+}), and indicate the bond order calculated from the Lewis dot structure. **(3 points)**

Lewis Dot Structure

Bond Order



92014642-A2BB-42F2-B010-19EE2528AAB4

midterm-exam-93388

#769 Page 10 of 10

SCRAP PAGE

**CHEMISTRY 110 –Midterm Exam****Thursday November 7, 2024, 6:30 – 8:30 pm**

Examiners: Pallavi Sirjoosingh and Ashok Kakkar

First name (please write as legibly as possible within the boxes)

--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--

Last name

--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--

Student ID number

--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--

This booklet has 8 questions on 10 pages (The last page can be used as a scrap page).

The staple must NOT be removed from the question booklet. Your intact question booklet must be returned at the end of the exam.

A separate booklet contains a periodic table, and datasheet.

Write your answers directly under the questions in the space provided. Do not write on the QR codes or near the top edges (as it may be cut off).

Please write **legibly and dark enough** for the scan to be readable. For students who use English as a second language, dictionaries are permitted. Calculators are allowed.

DO NOT ASK QUESTIONS OF THE INVIGILATORS about interpretations of the exam questions. If you have a concern about a question, please summarize it at the top of this page. All concerns will be evaluated after the exam. GOOD LUCK!

**Question 1 8 points** (*Assume all transitions are possible*)

a. 6 points Calculate the **energy (in kJ/mol)** required to ionize (completely remove) the electron from the **$n=3$** level of **He^+ ion** (He: Helium). Show your work including equations. (3 sig. figs.)

b. 2 points

(I) Select the largest (longest) wavelength from the three choices below:

- i. Wavelength of photon to *completely remove* an electron from *1s orbital of Hydrogen atom*
- ii. Wavelength of photon to *completely remove* an electron from *1s orbital of Boron atom (B)*
- iii. Wavelength of photon to *completely remove* an electron from *1s orbital of Boron cation (B^+)*

i or ii or iii

(II) Select the smallest (shortest) wavelength from the three choices below (*for H atom*):

- i. Wavelength of photon *emitted* for an electron transitioning from *2p to 1s orbital*
- ii. Wavelength of photon *absorbed* for an electron transitioning from *2s to 4p orbital*
- iii. Wavelength of photon *absorbed* for an electron transitioning from *1s to 3p orbital*

i or ii or iii



Question 2 8 points



A Chem110 student plots the graph shown on the left, from data obtained for the kinetic energy (eV) of the ejected electron versus frequency (s^{-1}) of the incident light, for an unknown metal. ($1\text{eV} = 1.6022 \times 10^{-19} \text{ J}$)

Based on the plot, answer the questions **a to c** below:

a. 1 point On the plot above, label the **threshold frequency**.

b. 2 points What is the value of the work potential (in J) of the unknown metal?

c. 3 points What is the *longest* wavelength (in nm) required to eject an electron from this metal's surface? Show your work including equation(s) used. (3 sig figs)

d. 2 points In the space provided below, plot the following

(Assume frequency is *above* threshold frequency):

- i. *Kinetic energy* of an ejected electron versus the *intensity* of the incident light (left)
- ii. *Number of electrons* ejected versus the *frequency* of the incident light (right)



**Question 3** **6 points****a. 2 points**

In the ground state of Ge (Germanium; $Z = 32$) atom, **name** the following orbitals:

i. Orbital containing the electron that requires the maximum energy to be removed.

ii. Orbital containing the electron that experiences the minimum effective nuclear charge.

b. 2 points

An unknown anion (A^-) has the following *excited* electronic configuration: $[Ar] 4s^2 3d^3 4p^1$

(Ar: Argon; $Z = 18$)

Give the **condensed ground state electronic configuration** of the atom, **A** and the cation, **A^{2+}** .
Show your work. (Assume all electronic transitions are possible)

Electronic Configuration of A:

Electronic Configuration of A^{2+} :

c. 2 points

In a single atom, determine how many electrons can have the following set of quantum numbers?

(i) $n = 5; l = 3$

In a single atom, determine how many electrons can have the following set of quantum numbers?

(i) $n = 3; l = 2; m_s = +1/2$

**Question 4** **8 points**

Rank the following from 1 to 4 in **increasing** order. Provide the ranking in the boxes provided.

Note: Atomic number (Z) are provided for relevant atoms

a) Increasing (lowest to highest) order of *ionization energy*: (1= lowest; 4 = highest)

Mg

Mg: Z = 12

Rb

Rb: Z = 37

Cs

Cs: Z = 55

K

K : Z = 19

b) Increasing (lowest to highest) order of *bond angle*: (1= lowest; 4 = highest)

H₂OCCl₄XeF₄

HCN

c) Increasing (smallest to largest) order of *ionic size* (1= smallest; 4 = largest)

S²⁻

S: Z = 16

O²⁻

O: Z = 8

Be²⁺

Be: Z = 4

Na⁺

Na: Z = 11

d) Increasing (lowest to highest) order of *bond energy* (1= lowest; 4 = highest)

P-F

P-Cl

P-I

P-Br

**Question 5** **12 points**

a. 9 points For the molecules HClO_4 and TeF_6 , draw the most stable *Lewis structure* showing appropriate shape and *dashed/wedged bonds*, based on VSEPR. Determine the **molecular geometry** around the central atom. For HClO_4 , label the *most polar bond*.



b. 3 points (Note: Lone pairs are not shown in structures below)

(i) For the three *resonance* structures (1, 2 and 3) shown below, answer the following question:



(I) Circle the *most contributing* resonance structure.

Structure 1

Structure 2

Structure 3

(ii) For the three structures given (4, 5, and 6) below, answer the following questions:



(I) Circle all the structures that represent *resonance structures* of each other:

Structure 4

Structure 5

Structure 6

(II) Among the resonance structures, circle the *most contributing* resonance structure.

Structure 4

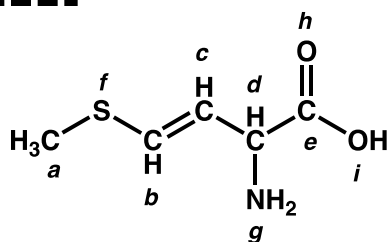
Structure 5

Structure 6

**Question 6** **6 points**

Using the data given below, calculate the **lattice energy** for SrO(s).

	ΔH° (kJ/mol)
$\text{Sr}(s) \longrightarrow \text{Sr}(g)$	+180
$\text{Sr}(g) \longrightarrow \text{Sr}^{2+}(g) + 2e^-$	+1468
$\text{O}_2(g) \longrightarrow 2\text{O}(g)$	+500
$\text{O}(g) + 2e^- \longrightarrow \text{O}^{2-}(g)$	+600
$\text{Sr}(s) + \frac{1}{2}\text{O}_2(g) \longrightarrow \text{SrO}(s)$	-550

**Question 7 10 points**

Consider the molecule given on the left, and answer questions (i) to (vi) using chemical bonding theories that we have learnt in class (valence bond theory, hybridization).

Note that the lone pairs are not shown in the structure:

i. Consider the carbon atoms marked in as ***a***, ***b***, ***c***, ***d***, and ***e*** in the molecule, and determine **hybridization** at each of the carbon centers. (2.5 points)

Atom	Hybridization
Carbon (<i>a</i>)	
Carbon (<i>b</i>)	
Carbon (<i>c</i>)	
Carbon (<i>d</i>)	
Carbon (<i>e</i>)	

ii. Now, consider the sulfur atom (S) marked as ***f***, nitrogen (N) atom (***g***), and oxygen(O) atoms (***h*** and ***i***), and determine their **hybridization**. (2 points)

Atom	Hybridization
Sulfur (<i>f</i>)	
Nitrogen (<i>g</i>)	
Oxygen (<i>h</i>)	
Oxygen (<i>i</i>)	

iii. **Describe** all the bonds formed by the carbon atoms ***a*** and ***b***, using sigma and pi designation as appropriate. (2 points)

Atom	Bonds formed (use sigma and pi as appropriate)
Carbon (<i>a</i>)	
Carbon (<i>b</i>)	

iv. How many total sp^3 -s sigma bonds are in the molecule? (1 point)

v. Determine **electronic** and **molecular geometry** at carbon (C) labeled ***e*** and sulfur (S) labeled ***f*** centres. (2 points)

Atom	Electronic Geometry	Molecular Geometry
Carbon (<i>e</i>)		
Sulfur (<i>f</i>)		

vi. Circle the value from the choices given below, that describes the **C-S-C bond angle at sulfur** (S) marked ***f***: (0.5 points)

>90° to <109.5° 109.5° >109.5 to <120° 120° >120° to <180°


Question 8 12 points

Using the **valence shell orbitals only** draw a **complete** molecular orbital (MO) diagram (with no s-p mixing) for the molecule **PS**, in the space provided below. **(6 points)**

Assume that 3s and 3p atomic orbitals will form molecular orbitals like those formed by the combination of 2s and 2p atomic orbitals that we learnt in class, and there is no s-p mixing.

Show **energy arrow**, **atomic orbitals** involved, **atom labels**, the **molecular orbitals** that are formed, **filled electrons** in the MO diagram etc. Make sure to label all the atomic and molecular orbitals.

Using the MO diagram above, answer the following questions:

- (i) Which one of the following will have the highest bond order? Circle the correct choice
 A) PS^{1-} B) PS C) PS^{1+}
- (ii) Which one of the following will have the longest bond length? Circle the correct choice
 A) PS^{1-} B) PS C) PS^{1+}
- (iii) Which of the following will be diamagnetic? Circle the correct choice.
 A) PS^{1-} B) PS C) PS^{1+}
- (iv) Draw Lewis dot structure of the molecule with the shortest bond length (based on the MO diagram) from among the three (PS^{1-} , PS, PS^{1+}), and indicate the bond order calculated from the Lewis dot structure. **(3 points)**

Lewis Dot Structure

Bond Order



9D832CF3-AD08-444C-9079-E10BEE8CFA60

midterm-exam-93388

#770 Page 10 of 10

SCRAP PAGE

**CHEMISTRY 110 –Midterm Exam****Thursday November 7, 2024, 6:30 – 8:30 pm**

Examiners: Pallavi Sirjoosingh and Ashok Kakkar

First name (please write as legibly as possible within the boxes)

--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--

Last name

--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--

Student ID number

--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--

This booklet has 8 questions on 10 pages (The last page can be used as a scrap page).

The staple must NOT be removed from the question booklet. Your intact question booklet must be returned at the end of the exam.

A separate booklet contains a periodic table, and datasheet.

Write your answers directly under the questions in the space provided. Do not write on the QR codes or near the top edges (as it may be cut off).

Please write **legibly and dark enough** for the scan to be readable. For students who use English as a second language, dictionaries are permitted. Calculators are allowed.

DO NOT ASK QUESTIONS OF THE INVIGILATORS about interpretations of the exam questions. If you have a concern about a question, please summarize it at the top of this page. All concerns will be evaluated after the exam. GOOD LUCK!

**Question 1 8 points** (*Assume all transitions are possible*)

a. 6 points Calculate the **energy (in kJ/mol)** required to ionize (completely remove) the electron from the **$n=3$** level of **He^+ ion** (He: Helium). Show your work including equations. (3 sig. figs.)

b. 2 points

(I) Select the largest (longest) wavelength from the three choices below:

- i. Wavelength of photon to *completely remove* an electron from *1s orbital of Hydrogen atom*
- ii. Wavelength of photon to *completely remove* an electron from *1s orbital of Boron atom (B)*
- iii. Wavelength of photon to *completely remove* an electron from *1s orbital of Boron cation (B^+)*

i or ii or iii

(II) Select the smallest (shortest) wavelength from the three choices below (*for H atom*):

- i. Wavelength of photon *emitted* for an electron transitioning from *2p to 1s orbital*
- ii. Wavelength of photon *absorbed* for an electron transitioning from *2s to 4p orbital*
- iii. Wavelength of photon *absorbed* for an electron transitioning from *1s to 3p orbital*

i or ii or iii



Question 2 8 points



A Chem110 student plots the graph shown on the left, from data obtained for the kinetic energy (eV) of the ejected electron versus frequency (s^{-1}) of the incident light, for an unknown metal. ($1\text{eV} = 1.6022 \times 10^{-19} \text{ J}$)

Based on the plot, answer the questions **a to c** below:

a. 1 point On the plot above, label the **threshold frequency**.

b. 2 points What is the value of the work potential (in J) of the unknown metal?

c. 3 points What is the *longest* wavelength (in nm) required to eject an electron from this metal's surface? Show your work including equation(s) used. (3 sig figs)

d. 2 points In the space provided below, plot the following

(Assume frequency is *above* threshold frequency):

- i. *Kinetic energy* of an ejected electron versus the *intensity* of the incident light (left)
- ii. *Number of electrons* ejected versus the *frequency* of the incident light (right)



**Question 3** **6 points****a. 2 points**

In the ground state of Ge (Germanium; $Z = 32$) atom, **name** the following orbitals:

i. Orbital containing the electron that requires the maximum energy to be removed.

ii. Orbital containing the electron that experiences the minimum effective nuclear charge.

b. 2 points

An unknown anion (A^-) has the following *excited* electronic configuration: $[Ar] 4s^2 3d^3 4p^1$

(Ar: Argon; $Z = 18$)

Give the **condensed ground state electronic configuration** of the atom, **A** and the cation, A^{2+} .
Show your work. (Assume all electronic transitions are possible)

Electronic Configuration of A:

Electronic Configuration of A^{2+} :

c. 2 points

In a single atom, determine how many electrons can have the following set of quantum numbers?

(i) $n = 5; l = 3$

In a single atom, determine how many electrons can have the following set of quantum numbers?

(i) $n = 3; l = 2; m_s = +1/2$

**Question 4** **8 points**

Rank the following from 1 to 4 in **increasing** order. Provide the ranking in the boxes provided.

Note: Atomic number (Z) are provided for relevant atoms

a) Increasing (lowest to highest) order of *ionization energy*: (1= lowest; 4 = highest)

Mg

Mg: Z = 12

Rb

Rb: Z = 37

Cs

Cs: Z = 55

K

K : Z = 19

b) Increasing (lowest to highest) order of *bond angle*: (1= lowest; 4 = highest)

H₂OCCl₄XeF₄

HCN

c) Increasing (smallest to largest) order of *ionic size* (1= smallest; 4 = largest)

S²⁻

S: Z = 16

O²⁻

O: Z = 8

Be²⁺

Be: Z = 4

Na⁺

Na: Z = 11

d) Increasing (lowest to highest) order of *bond energy* (1= lowest; 4 = highest)

P-F

P-Cl

P-I

P-Br

**Question 5** **12 points**

a. 9 points For the molecules HClO_4 and TeF_6 , draw the most stable *Lewis structure* showing appropriate shape and *dashed/wedged bonds*, based on VSEPR. Determine the **molecular geometry** around the central atom. For HClO_4 , label the *most polar bond*.



b. 3 points (Note: Lone pairs are not shown in structures below)

(i) For the three *resonance* structures (1, 2 and 3) shown below, answer the following question:



(I) Circle the *most contributing* resonance structure.

Structure 1

Structure 2

Structure 3

(ii) For the three structures given (4, 5, and 6) below, answer the following questions:



(I) Circle all the structures that represent *resonance structures* of each other:

Structure 4

Structure 5

Structure 6

(II) Among the resonance structures, circle the *most contributing* resonance structure.

Structure 4

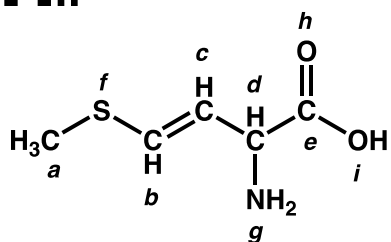
Structure 5

Structure 6

**Question 6** **6 points**

Using the data given below, calculate the **lattice energy** for SrO(s).

	ΔH° (kJ/mol)
$\text{Sr}(s) \longrightarrow \text{Sr}(g)$	+180
$\text{Sr}(g) \longrightarrow \text{Sr}^{2+}(g) + 2e^-$	+1468
$\text{O}_2(g) \longrightarrow 2\text{O}(g)$	+500
$\text{O}(g) + 2e^- \longrightarrow \text{O}^{2-}(g)$	+600
$\text{Sr}(s) + \frac{1}{2}\text{O}_2(g) \longrightarrow \text{SrO}(s)$	-550

**Question 7 10 points**

Consider the molecule given on the left, and answer questions (i) to (vi) using chemical bonding theories that we have learnt in class (valence bond theory, hybridization).

Note that the lone pairs are not shown in the structure:

i. Consider the carbon atoms marked in as ***a***, ***b***, ***c***, ***d***, and ***e*** in the molecule, and determine **hybridization** at each of the carbon centers. (2.5 points)

Atom	Hybridization
Carbon (<i>a</i>)	
Carbon (<i>b</i>)	
Carbon (<i>c</i>)	
Carbon (<i>d</i>)	
Carbon (<i>e</i>)	

ii. Now, consider the sulfur atom (S) marked as ***f***, nitrogen (N) atom (***g***), and oxygen(O) atoms (***h*** and ***i***), and determine their **hybridization**. (2 points)

Atom	Hybridization
Sulfur (<i>f</i>)	
Nitrogen (<i>g</i>)	
Oxygen (<i>h</i>)	
Oxygen (<i>i</i>)	

iii. **Describe** all the bonds formed by the carbon atoms ***a*** and ***b***, using sigma and pi designation as appropriate. (2 points)

Atom	Bonds formed (use sigma and pi as appropriate)
Carbon (<i>a</i>)	
Carbon (<i>b</i>)	

iv. How many total sp^3 -s sigma bonds are in the molecule? (1 point)

v. Determine **electronic** and **molecular geometry** at carbon (C) labeled ***e*** and sulfur (S) labeled ***f*** centres. (2 points)

Atom	Electronic Geometry	Molecular Geometry
Carbon (<i>e</i>)		
Sulfur (<i>f</i>)		

vi. Circle the value from the choices given below, that describes the **C-S-C bond angle at sulfur** (S) marked ***f***: (0.5 points)

>90° to <109.5° 109.5° >109.5 to <120° 120° >120° to <180°


Question 8 12 points

Using the **valence shell orbitals only** draw a **complete** molecular orbital (MO) diagram (with no s-p mixing) for the molecule **PS**, in the space provided below. **(6 points)**

Assume that 3s and 3p atomic orbitals will form molecular orbitals like those formed by the combination of 2s and 2p atomic orbitals that we learnt in class, and there is no s-p mixing.

Show **energy arrow**, **atomic orbitals** involved, **atom labels**, the **molecular orbitals** that are formed, **filled electrons** in the MO diagram etc. Make sure to label all the atomic and molecular orbitals.

Using the MO diagram above, answer the following questions:

- (i) Which one of the following will have the highest bond order? Circle the correct choice
 A) PS^{1-} B) PS C) PS^{1+}
- (ii) Which one of the following will have the longest bond length? Circle the correct choice
 A) PS^{1-} B) PS C) PS^{1+}
- (iii) Which of the following will be diamagnetic? Circle the correct choice.
 A) PS^{1-} B) PS C) PS^{1+}
- (iv) Draw Lewis dot structure of the molecule with the shortest bond length (based on the MO diagram) from among the three (PS^{1-} , PS, PS^{1+}), and indicate the bond order calculated from the Lewis dot structure. **(3 points)**

Lewis Dot Structure

Bond Order



7A8D2857-6B9D-44C9-9EF4-37C47CF6164D

midterm-exam-93388

#771 Page 10 of 10

SCRAP PAGE

**CHEMISTRY 110 –Midterm Exam****Thursday November 7, 2024, 6:30 – 8:30 pm**

Examiners: Pallavi Sirjoosingh and Ashok Kakkar

First name (please write as legibly as possible within the boxes)

--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--

Last name

--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--

Student ID number

--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--

This booklet has 8 questions on 10 pages (The last page can be used as a scrap page).

The staple must NOT be removed from the question booklet. Your intact question booklet must be returned at the end of the exam.

A separate booklet contains a periodic table, and datasheet.

Write your answers directly under the questions in the space provided. Do not write on the QR codes or near the top edges (as it may be cut off).

Please write **legibly and dark enough** for the scan to be readable. For students who use English as a second language, dictionaries are permitted. Calculators are allowed.

DO NOT ASK QUESTIONS OF THE INVIGILATORS about interpretations of the exam questions. If you have a concern about a question, please summarize it at the top of this page. All concerns will be evaluated after the exam. GOOD LUCK!

**Question 1 8 points** (*Assume all transitions are possible*)

a. 6 points Calculate the **energy (in kJ/mol)** required to ionize (completely remove) the electron from the **$n=3$** level of **He^+ ion** (He: Helium). Show your work including equations. (3 sig. figs.)

b. 2 points

(I) Select the largest (longest) wavelength from the three choices below:

- i. Wavelength of photon to *completely remove* an electron from *1s orbital of Hydrogen atom*
- ii. Wavelength of photon to *completely remove* an electron from *1s orbital of Boron atom (B)*
- iii. Wavelength of photon to *completely remove* an electron from *1s orbital of Boron cation (B^+)*

i or ii or iii

(II) Select the smallest (shortest) wavelength from the three choices below (*for H atom*):

- i. Wavelength of photon *emitted* for an electron transitioning from *2p to 1s orbital*
- ii. Wavelength of photon *absorbed* for an electron transitioning from *2s to 4p orbital*
- iii. Wavelength of photon *absorbed* for an electron transitioning from *1s to 3p orbital*

i or ii or iii



Question 2 8 points



A Chem110 student plots the graph shown on the left, from data obtained for the kinetic energy (eV) of the ejected electron versus frequency (s^{-1}) of the incident light, for an unknown metal. ($1\text{eV} = 1.6022 \times 10^{-19} \text{ J}$)

Based on the plot, answer the questions **a to c** below:

a. 1 point On the plot above, label the **threshold frequency**.

b. 2 points What is the value of the work potential (in J) of the unknown metal?

c. 3 points What is the *longest* wavelength (in nm) required to eject an electron from this metal's surface? Show your work including equation(s) used. (3 sig figs)

d. 2 points In the space provided below, plot the following

(Assume frequency is *above* threshold frequency):

- i. *Kinetic energy* of an ejected electron versus the *intensity* of the incident light (left)
- ii. *Number of electrons* ejected versus the *frequency* of the incident light (right)



**Question 3** **6 points****a. 2 points**

In the ground state of Ge (Germanium; $Z = 32$) atom, **name** the following orbitals:

i. Orbital containing the electron that requires the maximum energy to be removed.

ii. Orbital containing the electron that experiences the minimum effective nuclear charge.

b. 2 points

An unknown anion (A^-) has the following *excited* electronic configuration: $[Ar] 4s^2 3d^3 4p^1$

(Ar: Argon; $Z = 18$)

Give the **condensed ground state electronic configuration** of the atom, **A** and the cation, **A^{2+}** .
Show your work. (Assume all electronic transitions are possible)

Electronic Configuration of A:

Electronic Configuration of A^{2+} :

c. 2 points

In a single atom, determine how many electrons can have the following set of quantum numbers?

(i) $n = 5; l = 3$

In a single atom, determine how many electrons can have the following set of quantum numbers?

(i) $n = 3; l = 2; m_s = +1/2$

**Question 4** **8 points**

Rank the following from 1 to 4 in **increasing** order. Provide the ranking in the boxes provided.

Note: Atomic number (Z) are provided for relevant atoms

a) Increasing (lowest to highest) order of *ionization energy*: (1= lowest; 4 = highest)

Mg

Mg: Z = 12

Rb

Rb: Z = 37

Cs

Cs: Z = 55

K

K : Z = 19

b) Increasing (lowest to highest) order of *bond angle*: (1= lowest; 4 = highest)

H₂OCCl₄XeF₄

HCN

c) Increasing (smallest to largest) order of *ionic size* (1= smallest; 4 = largest)

S²⁻

S: Z = 16

O²⁻

O: Z = 8

Be²⁺

Be: Z = 4

Na⁺

Na: Z = 11

d) Increasing (lowest to highest) order of *bond energy* (1= lowest; 4 = highest)

P-F

P-Cl

P-I

P-Br

**Question 5** **12 points**

a. 9 points For the molecules HClO_4 and TeF_6 , draw the most stable *Lewis structure* showing appropriate shape and *dashed/wedged bonds*, based on VSEPR. Determine the **molecular geometry** around the central atom. For HClO_4 , label the *most polar bond*.

HClO_4 (there are no O-O bonds)

Label the most polar bond

Molecular Geometry around Cl

TeF_6

Molecular Geometry around Te

b. 3 points (Note: Lone pairs are not shown in structures below)

(i) For the three *resonance* structures (1, 2 and 3) shown below, answer the following question:



(I) Circle the *most contributing* resonance structure.

Structure 1

Structure 2

Structure 3

(ii) For the three structures given (4, 5, and 6) below, answer the following questions:



(I) Circle all the structures that represent *resonance structures* of each other:

Structure 4

Structure 5

Structure 6

(II) Among the resonance structures, circle the *most contributing* resonance structure.

Structure 4

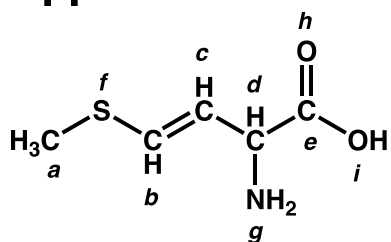
Structure 5

Structure 6

**Question 6** **6 points**

Using the data given below, calculate the **lattice energy** for SrO(s).

	ΔH° (kJ/mol)
$\text{Sr}(s) \longrightarrow \text{Sr}(g)$	+180
$\text{Sr}(g) \longrightarrow \text{Sr}^{2+}(g) + 2e^-$	+1468
$\text{O}_2(g) \longrightarrow 2\text{O}(g)$	+500
$\text{O}(g) + 2e^- \longrightarrow \text{O}^{2-}(g)$	+600
$\text{Sr}(s) + \frac{1}{2}\text{O}_2(g) \longrightarrow \text{SrO}(s)$	-550

**Question 7 10 points**

Consider the molecule given on the left, and answer questions (i) to (vi) using chemical bonding theories that we have learnt in class (valence bond theory, hybridization).

Note that the lone pairs are not shown in the structure:

i. Consider the carbon atoms marked in as ***a***, ***b***, ***c***, ***d***, and ***e*** in the molecule, and determine **hybridization** at each of the carbon centers. (2.5 points)

Atom	Hybridization
Carbon (<i>a</i>)	
Carbon (<i>b</i>)	
Carbon (<i>c</i>)	
Carbon (<i>d</i>)	
Carbon (<i>e</i>)	

ii. Now, consider the sulfur atom (S) marked as ***f***, nitrogen (N) atom (***g***), and oxygen(O) atoms (***h*** and ***i***), and determine their **hybridization**. (2 points)

Atom	Hybridization
Sulfur (<i>f</i>)	
Nitrogen (<i>g</i>)	
Oxygen (<i>h</i>)	
Oxygen (<i>i</i>)	

iii. **Describe** all the bonds formed by the carbon atoms ***a*** and ***b***, using sigma and pi designation as appropriate. (2 points)

Atom	Bonds formed (use sigma and pi as appropriate)
Carbon (<i>a</i>)	
Carbon (<i>b</i>)	

iv. How many total sp^3 -s sigma bonds are in the molecule? (1 point)

v. Determine **electronic** and **molecular geometry** at carbon (C) labeled ***e*** and sulfur (S) labeled ***f*** centres. (2 points)

Atom	Electronic Geometry	Molecular Geometry
Carbon (<i>e</i>)		
Sulfur (<i>f</i>)		

vi. Circle the value from the choices given below, that describes the **C-S-C bond angle at sulfur** (S) marked ***f***: (0.5 points)

>90° to <109.5° 109.5° >109.5 to <120° 120° >120° to <180°


Question 8 12 points

Using the **valence shell orbitals only** draw a **complete** molecular orbital (MO) diagram (with no s-p mixing) for the molecule **PS**, in the space provided below. **(6 points)**

Assume that 3s and 3p atomic orbitals will form molecular orbitals like those formed by the combination of 2s and 2p atomic orbitals that we learnt in class, and there is no s-p mixing.

Show **energy arrow**, **atomic orbitals** involved, **atom labels**, the **molecular orbitals** that are formed, **filled electrons** in the MO diagram etc. Make sure to label all the atomic and molecular orbitals.

Using the MO diagram above, answer the following questions:

- (i) Which one of the following will have the highest bond order? Circle the correct choice
 A) PS^{1-} B) PS C) PS^{1+}
- (ii) Which one of the following will have the longest bond length? Circle the correct choice
 A) PS^{1-} B) PS C) PS^{1+}
- (iii) Which of the following will be diamagnetic? Circle the correct choice.
 A) PS^{1-} B) PS C) PS^{1+}
- (iv) Draw Lewis dot structure of the molecule with the shortest bond length (based on the MO diagram) from among the three (PS^{1-} , PS, PS^{1+}), and indicate the bond order calculated from the Lewis dot structure. **(3 points)**

Lewis Dot Structure

Bond Order



1C20B6CE-5851-4E2B-8202-E2A5226ED8F0

midterm-exam-93388

#772 Page 10 of 10

SCRAP PAGE

**CHEMISTRY 110 –Midterm Exam****Thursday November 7, 2024, 6:30 – 8:30 pm**

Examiners: Pallavi Sirjoosingh and Ashok Kakkar

First name (please write as legibly as possible within the boxes)

--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--

Last name

--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--

Student ID number

--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--

This booklet has 8 questions on 10 pages (The last page can be used as a scrap page).

The staple must NOT be removed from the question booklet. Your intact question booklet must be returned at the end of the exam.

A separate booklet contains a periodic table, and datasheet.

Write your answers directly under the questions in the space provided. Do not write on the QR codes or near the top edges (as it may be cut off).

Please write **legibly and dark enough** for the scan to be readable. For students who use English as a second language, dictionaries are permitted. Calculators are allowed.

DO NOT ASK QUESTIONS OF THE INVIGILATORS about interpretations of the exam questions. If you have a concern about a question, please summarize it at the top of this page. All concerns will be evaluated after the exam. GOOD LUCK!

**Question 1 8 points** (*Assume all transitions are possible*)

a. 6 points Calculate the **energy (in kJ/mol)** required to ionize (completely remove) the electron from the **n=3** level of **He⁺ ion** (He: Helium). Show your work including equations. (3 sig. figs.)

b. 2 points

(I) Select the largest (longest) wavelength from the three choices below:

- i. Wavelength of photon to *completely remove* an electron from *1s orbital of Hydrogen atom*
- ii. Wavelength of photon to *completely remove* an electron from *1s orbital of Boron atom (B)*
- iii. Wavelength of photon to *completely remove* an electron from *1s orbital of Boron cation (B⁺)*

i or ii or iii

(II) Select the smallest (shortest) wavelength from the three choices below (*for H atom*):

- i. Wavelength of photon *emitted* for an electron transitioning from *2p to 1s orbital*
- ii. Wavelength of photon *absorbed* for an electron transitioning from *2s to 4p orbital*
- iii. Wavelength of photon *absorbed* for an electron transitioning from *1s to 3p orbital*

i or ii or iii



Question 2 8 points



A Chem110 student plots the graph shown on the left, from data obtained for the kinetic energy (eV) of the ejected electron versus frequency (s^{-1}) of the incident light, for an unknown metal. ($1\text{eV} = 1.6022 \times 10^{-19} \text{ J}$)

Based on the plot, answer the questions **a to c** below:

a. 1 point On the plot above, label the **threshold frequency**.

b. 2 points What is the value of the work potential (in J) of the unknown metal?

c. 3 points What is the *longest* wavelength (in nm) required to eject an electron from this metal's surface? Show your work including equation(s) used. (3 sig figs)

d. 2 points In the space provided below, plot the following

(Assume frequency is *above* threshold frequency):

- i. *Kinetic energy* of an ejected electron versus the *intensity* of the incident light (left)
- ii. *Number of electrons* ejected versus the *frequency* of the incident light (right)



**Question 3** **6 points****a. 2 points**

In the ground state of Ge (Germanium; $Z = 32$) atom, **name** the following orbitals:

i. Orbital containing the electron that requires the maximum energy to be removed.

ii. Orbital containing the electron that experiences the minimum effective nuclear charge.

b. 2 points

An unknown anion (A^-) has the following *excited* electronic configuration: $[Ar] 4s^2 3d^3 4p^1$

(Ar: Argon; $Z = 18$)

Give the **condensed ground state electronic configuration** of the atom, **A** and the cation, **A^{2+}** .
Show your work. (Assume all electronic transitions are possible)

Electronic Configuration of A:

Electronic Configuration of A^{2+} :

c. 2 points

In a single atom, determine how many electrons can have the following set of quantum numbers?

(i) $n = 5; l = 3$

In a single atom, determine how many electrons can have the following set of quantum numbers?

(i) $n = 3; l = 2; m_s = +1/2$

**Question 4** **8 points**

Rank the following from 1 to 4 in **increasing** order. Provide the ranking in the boxes provided.

Note: Atomic number (Z) are provided for relevant atoms

a) Increasing (lowest to highest) order of *ionization energy*: (1= lowest; 4 = highest)

Mg

Mg: Z = 12

Rb

Rb: Z = 37

Cs

Cs: Z = 55

K

K : Z = 19

b) Increasing (lowest to highest) order of *bond angle*: (1= lowest; 4 = highest)

H₂OCCl₄XeF₄

HCN

c) Increasing (smallest to largest) order of *ionic size* (1= smallest; 4 = largest)

S²⁻

S: Z = 16

O²⁻

O: Z = 8

Be²⁺

Be: Z = 4

Na⁺

Na: Z = 11

d) Increasing (lowest to highest) order of *bond energy* (1= lowest; 4 = highest)

P-F

P-Cl

P-I

P-Br

**Question 5** **12 points**

a. 9 points For the molecules HClO_4 and TeF_6 , draw the most stable *Lewis structure* showing appropriate shape and *dashed/wedged bonds*, based on VSEPR. Determine the **molecular geometry** around the central atom. For HClO_4 , label the *most polar bond*.



b. 3 points (Note: Lone pairs are not shown in structures below)

(i) For the three *resonance* structures (1, 2 and 3) shown below, answer the following question:



(I) Circle the *most contributing* resonance structure.

Structure 1

Structure 2

Structure 3

(ii) For the three structures given (4, 5, and 6) below, answer the following questions:



(I) Circle all the structures that represent *resonance structures* of each other:

Structure 4

Structure 5

Structure 6

(II) Among the resonance structures, circle the *most contributing* resonance structure.

Structure 4

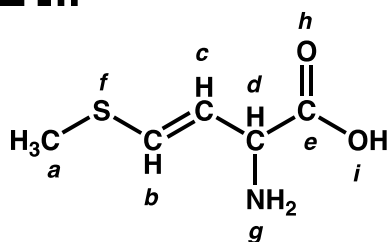
Structure 5

Structure 6

**Question 6** **6 points**

Using the data given below, calculate the **lattice energy** for SrO(s).

	ΔH° (kJ/mol)
$\text{Sr}(s) \longrightarrow \text{Sr}(g)$	+180
$\text{Sr}(g) \longrightarrow \text{Sr}^{2+}(g) + 2e^-$	+1468
$\text{O}_2(g) \longrightarrow 2\text{O}(g)$	+500
$\text{O}(g) + 2e^- \longrightarrow \text{O}^{2-}(g)$	+600
$\text{Sr}(s) + \frac{1}{2}\text{O}_2(g) \longrightarrow \text{SrO}(s)$	-550

**Question 7 10 points**

Consider the molecule given on the left, and answer questions (i) to (vi) using chemical bonding theories that we have learnt in class (valence bond theory, hybridization).

Note that the lone pairs are not shown in the structure:

i. Consider the carbon atoms marked in as ***a***, ***b***, ***c***, ***d***, and ***e*** in the molecule, and determine **hybridization** at each of the carbon centers. (2.5 points)

Atom	Hybridization
Carbon (<i>a</i>)	
Carbon (<i>b</i>)	
Carbon (<i>c</i>)	
Carbon (<i>d</i>)	
Carbon (<i>e</i>)	

ii. Now, consider the sulfur atom (S) marked as ***f***, nitrogen (N) atom (***g***), and oxygen(O) atoms (***h*** and ***i***), and determine their **hybridization**. (2 points)

Atom	Hybridization
Sulfur (<i>f</i>)	
Nitrogen (<i>g</i>)	
Oxygen (<i>h</i>)	
Oxygen (<i>i</i>)	

iii. **Describe** all the bonds formed by the carbon atoms ***a*** and ***b***, using sigma and pi designation as appropriate. (2 points)

Atom	Bonds formed (use sigma and pi as appropriate)
Carbon (<i>a</i>)	
Carbon (<i>b</i>)	

iv. How many total sp^3 -s sigma bonds are in the molecule? (1 point)

v. Determine **electronic** and **molecular geometry** at carbon (C) labeled ***e*** and sulfur (S) labeled ***f*** centres. (2 points)

Atom	Electronic Geometry	Molecular Geometry
Carbon (<i>e</i>)		
Sulfur (<i>f</i>)		

vi. Circle the value from the choices given below, that describes the **C-S-C bond angle at sulfur** (S) marked ***f***: (0.5 points)

$>90^\circ$ to $<109.5^\circ$ 109.5° >109.5 to $<120^\circ$ 120° $>120^\circ$ to $<180^\circ$


Question 8 12 points

Using the **valence shell orbitals only** draw a **complete** molecular orbital (MO) diagram (with no s-p mixing) for the molecule **PS**, in the space provided below. **(6 points)**

Assume that 3s and 3p atomic orbitals will form molecular orbitals like those formed by the combination of 2s and 2p atomic orbitals that we learnt in class, and there is no s-p mixing.

Show **energy arrow**, **atomic orbitals** involved, **atom labels**, the **molecular orbitals** that are formed, **filled electrons** in the MO diagram etc. Make sure to label all the atomic and molecular orbitals.

Using the MO diagram above, answer the following questions:

- (i) Which one of the following will have the highest bond order? Circle the correct choice
 A) PS^{1-} B) PS C) PS^{1+}
- (ii) Which one of the following will have the longest bond length? Circle the correct choice
 A) PS^{1-} B) PS C) PS^{1+}
- (iii) Which of the following will be diamagnetic? Circle the correct choice.
 A) PS^{1-} B) PS C) PS^{1+}
- (iv) Draw Lewis dot structure of the molecule with the shortest bond length (based on the MO diagram) from among the three (PS^{1-} , PS, PS^{1+}), and indicate the bond order calculated from the Lewis dot structure. **(3 points)**

Lewis Dot Structure

Bond Order



F86C83B4-6335-4235-A2F4-C14ABE0DAD6C

midterm-exam-93388

#773 Page 10 of 10

SCRAP PAGE

**CHEMISTRY 110 –Midterm Exam****Thursday November 7, 2024, 6:30 – 8:30 pm**

Examiners: Pallavi Sirjoosingh and Ashok Kakkar

First name (please write as legibly as possible within the boxes)

--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--

Last name

--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--

Student ID number

--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--

This booklet has 8 questions on 10 pages (The last page can be used as a scrap page).

The staple must NOT be removed from the question booklet. Your intact question booklet must be returned at the end of the exam.

A separate booklet contains a periodic table, and datasheet.

Write your answers directly under the questions in the space provided. Do not write on the QR codes or near the top edges (as it may be cut off).

Please write **legibly and dark enough** for the scan to be readable. For students who use English as a second language, dictionaries are permitted. Calculators are allowed.

DO NOT ASK QUESTIONS OF THE INVIGILATORS about interpretations of the exam questions. If you have a concern about a question, please summarize it at the top of this page. All concerns will be evaluated after the exam. GOOD LUCK!

**Question 1 8 points** (*Assume all transitions are possible*)

a. 6 points Calculate the **energy (in kJ/mol)** required to ionize (completely remove) the electron from the **$n=3$** level of **He^+ ion** (He: Helium). Show your work including equations. (3 sig. figs.)

b. 2 points

(I) Select the largest (longest) wavelength from the three choices below:

- i. Wavelength of photon to *completely remove* an electron from *1s orbital of Hydrogen atom*
- ii. Wavelength of photon to *completely remove* an electron from *1s orbital of Boron atom (B)*
- iii. Wavelength of photon to *completely remove* an electron from *1s orbital of Boron cation (B^+)*

i or ii or iii

(II) Select the smallest (shortest) wavelength from the three choices below (*for H atom*):

- i. Wavelength of photon *emitted* for an electron transitioning from *2p to 1s orbital*
- ii. Wavelength of photon *absorbed* for an electron transitioning from *2s to 4p orbital*
- iii. Wavelength of photon *absorbed* for an electron transitioning from *1s to 3p orbital*

i or ii or iii



Question 2 8 points



A Chem110 student plots the graph shown on the left, from data obtained for the kinetic energy (eV) of the ejected electron versus frequency (s^{-1}) of the incident light, for an unknown metal. ($1\text{eV} = 1.6022 \times 10^{-19} \text{ J}$)

Based on the plot, answer the questions **a to c** below:

a. 1 point On the plot above, label the **threshold frequency**.

b. 2 points What is the value of the work potential (in J) of the unknown metal?

c. 3 points What is the *longest* wavelength (in nm) required to eject an electron from this metal's surface? Show your work including equation(s) used. (3 sig figs)

d. 2 points In the space provided below, plot the following

(Assume frequency is *above* threshold frequency):

- Kinetic energy* of an ejected electron versus the *intensity* of the incident light (left)
- Number of electrons* ejected versus the *frequency* of the incident light (right)



**Question 3** **6 points****a. 2 points**

In the ground state of Ge (Germanium; $Z = 32$) atom, **name** the following orbitals:

i. Orbital containing the electron that requires the maximum energy to be removed.

ii. Orbital containing the electron that experiences the minimum effective nuclear charge.

b. 2 points

An unknown anion (A^-) has the following *excited* electronic configuration: $[Ar] 4s^2 3d^3 4p^1$

(Ar: Argon; $Z = 18$)

Give the **condensed ground state electronic configuration** of the atom, **A** and the cation, **A^{2+}** .
Show your work. (Assume all electronic transitions are possible)

Electronic Configuration of A:

Electronic Configuration of A^{2+} :

c. 2 points

In a single atom, determine how many electrons can have the following set of quantum numbers?

(i) $n = 5; l = 3$

In a single atom, determine how many electrons can have the following set of quantum numbers?

(i) $n = 3; l = 2; m_s = +1/2$

**Question 4** **8 points**

Rank the following from 1 to 4 in **increasing** order. Provide the ranking in the boxes provided.

Note: Atomic number (Z) are provided for relevant atoms

a) Increasing (lowest to highest) order of *ionization energy*: (1= lowest; 4 = highest)

Mg

Mg: Z = 12

Rb

Rb: Z = 37

Cs

Cs: Z = 55

K

K : Z = 19

b) Increasing (lowest to highest) order of *bond angle*: (1= lowest; 4 = highest)

H₂OCCl₄XeF₄

HCN

c) Increasing (smallest to largest) order of *ionic size* (1= smallest; 4 = largest)

S²⁻

S: Z = 16

O²⁻

O: Z = 8

Be²⁺

Be: Z = 4

Na⁺

Na: Z = 11

d) Increasing (lowest to highest) order of *bond energy* (1= lowest; 4 = highest)

P-F

P-Cl

P-I

P-Br

**Question 5** **12 points**

a. 9 points For the molecules HClO_4 and TeF_6 , draw the most stable *Lewis structure* showing appropriate shape and *dashed/wedged bonds*, based on VSEPR. Determine the **molecular geometry** around the central atom. For HClO_4 , label the *most polar bond*.



b. 3 points (Note: Lone pairs are not shown in structures below)

(i) For the three *resonance* structures (1, 2 and 3) shown below, answer the following question:



(I) Circle the *most contributing* resonance structure.

Structure 1

Structure 2

Structure 3

(ii) For the three structures given (4, 5, and 6) below, answer the following questions:



(I) Circle all the structures that represent *resonance structures* of each other:

Structure 4

Structure 5

Structure 6

(II) Among the resonance structures, circle the *most contributing* resonance structure.

Structure 4

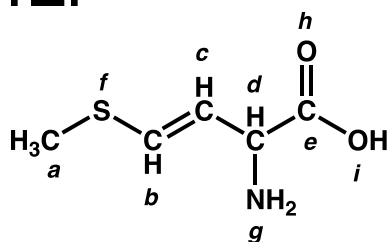
Structure 5

Structure 6

**Question 6** **6 points**

Using the data given below, calculate the **lattice energy** for SrO(s).

	ΔH° (kJ/mol)
$\text{Sr}(s) \longrightarrow \text{Sr}(g)$	+180
$\text{Sr}(g) \longrightarrow \text{Sr}^{2+}(g) + 2e^-$	+1468
$\text{O}_2(g) \longrightarrow 2\text{O}(g)$	+500
$\text{O}(g) + 2e^- \longrightarrow \text{O}^{2-}(g)$	+600
$\text{Sr}(s) + \frac{1}{2}\text{O}_2(g) \longrightarrow \text{SrO}(s)$	-550

**Question 7 10 points**

Consider the molecule given on the left, and answer questions (i) to (vi) using chemical bonding theories that we have learnt in class (valence bond theory, hybridization).

Note that the lone pairs are not shown in the structure:

i. Consider the carbon atoms marked in as ***a***, ***b***, ***c***, ***d***, and ***e*** in the molecule, and determine **hybridization** at each of the carbon centers. (2.5 points)

Atom	Hybridization
Carbon (<i>a</i>)	
Carbon (<i>b</i>)	
Carbon (<i>c</i>)	
Carbon (<i>d</i>)	
Carbon (<i>e</i>)	

ii. Now, consider the sulfur atom (S) marked as ***f***, nitrogen (N) atom (***g***), and oxygen(O) atoms (***h*** and ***i***), and determine their **hybridization**. (2 points)

Atom	Hybridization
Sulfur (<i>f</i>)	
Nitrogen (<i>g</i>)	
Oxygen (<i>h</i>)	
Oxygen (<i>i</i>)	

iii. **Describe** all the bonds formed by the carbon atoms ***a*** and ***b***, using sigma and pi designation as appropriate. (2 points)

Atom	Bonds formed (use sigma and pi as appropriate)
Carbon (<i>a</i>)	
Carbon (<i>b</i>)	

iv. How many total sp^3 -s sigma bonds are in the molecule? (1 point)

v. Determine **electronic** and **molecular geometry** at carbon (C) labeled ***e*** and sulfur (S) labeled ***f*** centres. (2 points)

Atom	Electronic Geometry	Molecular Geometry
Carbon (<i>e</i>)		
Sulfur (<i>f</i>)		

vi. Circle the value from the choices given below, that describes the **C-S-C bond angle at sulfur** (S) marked ***f***: (0.5 points)

>90° to <109.5° 109.5° >109.5 to <120° 120° >120° to <180°


Question 8 12 points

Using the **valence shell orbitals only** draw a **complete** molecular orbital (MO) diagram (with no s-p mixing) for the molecule **PS**, in the space provided below. **(6 points)**

Assume that 3s and 3p atomic orbitals will form molecular orbitals like those formed by the combination of 2s and 2p atomic orbitals that we learnt in class, and there is no s-p mixing.

Show **energy arrow**, **atomic orbitals** involved, **atom labels**, the **molecular orbitals** that are formed, **filled electrons** in the MO diagram etc. Make sure to **label all the atomic and molecular orbitals**.

Using the MO diagram above, answer the following questions:

- (i) Which one of the following will have the highest bond order? Circle the correct choice
 A) PS^{1-} B) PS C) PS^{1+}
- (ii) Which one of the following will have the longest bond length? Circle the correct choice
 A) PS^{1-} B) PS C) PS^{1+}
- (iii) Which of the following will be diamagnetic? Circle the correct choice.
 A) PS^{1-} B) PS C) PS^{1+}
- (iv) Draw Lewis dot structure of the molecule with the shortest bond length (based on the MO diagram) from among the three (PS^{1-} , PS, PS^{1+}), and indicate the bond order calculated from the Lewis dot structure. **(3 points)**

Lewis Dot Structure

Bond Order



E316BBC2-78FF-4D4C-A6DF-81305A8D8148

midterm-exam-93388

#774 Page 10 of 10

SCRAP PAGE

**CHEMISTRY 110 –Midterm Exam****Thursday November 7, 2024, 6:30 – 8:30 pm**

Examiners: Pallavi Sirjoosingh and Ashok Kakkar

First name (please write as legibly as possible within the boxes)

--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--

Last name

--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--

Student ID number

--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--

This booklet has 8 questions on 10 pages (The last page can be used as a scrap page).

The staple must NOT be removed from the question booklet. Your intact question booklet must be returned at the end of the exam.

A separate booklet contains a periodic table, and datasheet.

Write your answers directly under the questions in the space provided. Do not write on the QR codes or near the top edges (as it may be cut off).

Please write **legibly and dark enough** for the scan to be readable. For students who use English as a second language, dictionaries are permitted. Calculators are allowed.

DO NOT ASK QUESTIONS OF THE INVIGILATORS about interpretations of the exam questions. If you have a concern about a question, please summarize it at the top of this page. All concerns will be evaluated after the exam. GOOD LUCK!

**Question 1 8 points** (*Assume all transitions are possible*)

a. 6 points Calculate the **energy (in kJ/mol)** required to ionize (completely remove) the electron from the **n=3** level of **He⁺ ion** (He: Helium). Show your work including equations. (3 sig. figs.)

b. 2 points

(I) Select the largest (longest) wavelength from the three choices below:

- i. Wavelength of photon to *completely remove* an electron from *1s orbital of Hydrogen atom*
- ii. Wavelength of photon to *completely remove* an electron from *1s orbital of Boron atom (B)*
- iii. Wavelength of photon to *completely remove* an electron from *1s orbital of Boron cation (B⁺)*

i or ii or iii

(II) Select the smallest (shortest) wavelength from the three choices below (*for H atom*):

- i. Wavelength of photon *emitted* for an electron transitioning from *2p to 1s orbital*
- ii. Wavelength of photon *absorbed* for an electron transitioning from *2s to 4p orbital*
- iii. Wavelength of photon *absorbed* for an electron transitioning from *1s to 3p orbital*

i or ii or iii



Question 2 8 points



A Chem110 student plots the graph shown on the left, from data obtained for the kinetic energy (eV) of the ejected electron versus frequency (s^{-1}) of the incident light, for an unknown metal. ($1\text{eV} = 1.6022 \times 10^{-19} \text{ J}$)

Based on the plot, answer the questions **a to c** below:

a. 1 point On the plot above, label the **threshold frequency**.

b. 2 points What is the value of the work potential (in J) of the unknown metal?

c. 3 points What is the *longest* wavelength (in nm) required to eject an electron from this metal's surface? Show your work including equation(s) used. (3 sig figs)

d. 2 points In the space provided below, plot the following

(Assume frequency is *above* threshold frequency):

- Kinetic energy* of an ejected electron versus the *intensity* of the incident light (left)
- Number of electrons* ejected versus the *frequency* of the incident light (right)



**Question 3** **6 points****a. 2 points**

In the ground state of Ge (Germanium; $Z = 32$) atom, **name** the following orbitals:

i. Orbital containing the electron that requires the maximum energy to be removed.

ii. Orbital containing the electron that experiences the minimum effective nuclear charge.

b. 2 points

An unknown anion (A^-) has the following *excited* electronic configuration: $[Ar] 4s^2 3d^3 4p^1$

(Ar: Argon; $Z = 18$)

Give the **condensed ground state electronic configuration** of the atom, **A** and the cation, **A^{2+}** .
Show your work. (Assume all electronic transitions are possible)

Electronic Configuration of A:

Electronic Configuration of A^{2+} :

c. 2 points

In a single atom, determine how many electrons can have the following set of quantum numbers?

(i) $n = 5; l = 3$

In a single atom, determine how many electrons can have the following set of quantum numbers?

(i) $n = 3; l = 2; m_s = +1/2$

**Question 4** **8 points**

Rank the following from 1 to 4 in **increasing** order. Provide the ranking in the boxes provided.

Note: Atomic number (Z) are provided for relevant atoms

a) Increasing (lowest to highest) order of *ionization energy*: (1= lowest; 4 = highest)

Mg

Mg: Z = 12

Rb

Rb: Z = 37

Cs

Cs: Z = 55

K

K : Z = 19

b) Increasing (lowest to highest) order of *bond angle*: (1= lowest; 4 = highest)

H₂OCCl₄XeF₄

HCN

c) Increasing (smallest to largest) order of *ionic size* (1= smallest; 4 = largest)

S²⁻

S: Z = 16

O²⁻

O: Z = 8

Be²⁺

Be: Z = 4

Na⁺

Na: Z = 11

d) Increasing (lowest to highest) order of *bond energy* (1= lowest; 4 = highest)

P-F

P-Cl

P-I

P-Br

**Question 5** **12 points**

a. 9 points For the molecules HClO_4 and TeF_6 , draw the most stable *Lewis structure* showing appropriate shape and *dashed/wedged bonds*, based on VSEPR. Determine the **molecular geometry** around the central atom. For HClO_4 , label the *most polar bond*.



b. 3 points (Note: Lone pairs are not shown in structures below)

(i) For the three *resonance* structures (1, 2 and 3) shown below, answer the following question:



(I) Circle the *most contributing* resonance structure.

Structure 1

Structure 2

Structure 3

(ii) For the three structures given (4, 5, and 6) below, answer the following questions:



(I) Circle all the structures that represent *resonance structures* of each other:

Structure 4

Structure 5

Structure 6

(II) Among the resonance structures, circle the *most contributing* resonance structure.

Structure 4

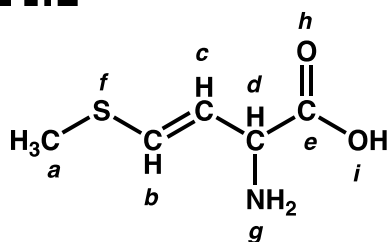
Structure 5

Structure 6

**Question 6** **6 points**

Using the data given below, calculate the **lattice energy** for SrO(s).

	ΔH° (kJ/mol)
$\text{Sr}(s) \longrightarrow \text{Sr}(g)$	+180
$\text{Sr}(g) \longrightarrow \text{Sr}^{2+}(g) + 2e^-$	+1468
$\text{O}_2(g) \longrightarrow 2\text{O}(g)$	+500
$\text{O}(g) + 2e^- \longrightarrow \text{O}^{2-}(g)$	+600
$\text{Sr}(s) + \frac{1}{2}\text{O}_2(g) \longrightarrow \text{SrO}(s)$	-550

**Question 7 10 points**

Consider the molecule given on the left, and answer questions (i) to (vi) using chemical bonding theories that we have learnt in class (valence bond theory, hybridization).

Note that the lone pairs are not shown in the structure:

i. Consider the carbon atoms marked in as ***a***, ***b***, ***c***, ***d***, and ***e*** in the molecule, and determine **hybridization** at each of the carbon centers. (2.5 points)

Atom	Hybridization
Carbon (<i>a</i>)	
Carbon (<i>b</i>)	
Carbon (<i>c</i>)	
Carbon (<i>d</i>)	
Carbon (<i>e</i>)	

ii. Now, consider the sulfur atom (S) marked as ***f***, nitrogen (N) atom (***g***), and oxygen(O) atoms (***h*** and ***i***), and determine their **hybridization**. (2 points)

Atom	Hybridization
Sulfur (<i>f</i>)	
Nitrogen (<i>g</i>)	
Oxygen (<i>h</i>)	
Oxygen (<i>i</i>)	

iii. **Describe** all the bonds formed by the carbon atoms ***a*** and ***b***, using sigma and pi designation as appropriate. (2 points)

Atom	Bonds formed (use sigma and pi as appropriate)
Carbon (<i>a</i>)	
Carbon (<i>b</i>)	

iv. How many total sp^3 -s sigma bonds are in the molecule? (1 point)

v. Determine **electronic** and **molecular geometry** at carbon (C) labeled ***e*** and sulfur (S) labeled ***f*** centres. (2 points)

Atom	Electronic Geometry	Molecular Geometry
Carbon (<i>e</i>)		
Sulfur (<i>f</i>)		

vi. Circle the value from the choices given below, that describes the **C-S-C bond angle at sulfur** (S) marked ***f***: (0.5 points)

>90° to <109.5° 109.5° >109.5 to <120° 120° >120° to <180°



Question 8 12 points

Using the **valence shell orbitals only** draw a **complete** molecular orbital (MO) diagram (with no s-p mixing) for the molecule **PS**, in the space provided below. **(6 points)**

Assume that 3s and 3p atomic orbitals will form molecular orbitals like those formed by the combination of 2s and 2p atomic orbitals that we learnt in class, and there is no s-p mixing.

Show **energy arrow**, **atomic orbitals** involved, **atom labels**, the **molecular orbitals** that are formed, **filled electrons** in the MO diagram etc. Make sure to label all the atomic and molecular orbitals.

Using the MO diagram above, answer the following questions:

- (i) Which one of the following will have the highest bond order? Circle the correct choice
 A) PS^{1-} B) PS C) PS^{1+}
- (ii) Which one of the following will have the longest bond length? Circle the correct choice
 A) PS^{1-} B) PS C) PS^{1+}
- (iii) Which of the following will be diamagnetic? Circle the correct choice.
 A) PS^{1-} B) PS C) PS^{1+}
- (iv) Draw Lewis dot structure of the molecule with the shortest bond length (based on the MO diagram) from among the three (PS^{1-} , PS, PS^{1+}), and indicate the bond order calculated from the Lewis dot structure. **(3 points)**

Lewis Dot Structure

Bond Order



109386AB-F004-440E-A383-319573746175

midterm-exam-93388

#775 Page 10 of 10

SCRAP PAGE

**CHEMISTRY 110 –Midterm Exam****Thursday November 7, 2024, 6:30 – 8:30 pm**

Examiners: Pallavi Sirjoosingh and Ashok Kakkar

First name (please write as legibly as possible within the boxes)

--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--

Last name

--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--

Student ID number

--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--

This booklet has 8 questions on 10 pages (The last page can be used as a scrap page).

The staple must NOT be removed from the question booklet. Your intact question booklet must be returned at the end of the exam.

A separate booklet contains a periodic table, and datasheet.

Write your answers directly under the questions in the space provided. Do not write on the QR codes or near the top edges (as it may be cut off).

Please write **legibly and dark enough** for the scan to be readable. For students who use English as a second language, dictionaries are permitted. Calculators are allowed.

DO NOT ASK QUESTIONS OF THE INVIGILATORS about interpretations of the exam questions. If you have a concern about a question, please summarize it at the top of this page. All concerns will be evaluated after the exam. GOOD LUCK!

**Question 1 8 points** (*Assume all transitions are possible*)

a. 6 points Calculate the **energy (in kJ/mol)** required to ionize (completely remove) the electron from the **n=3** level of **He⁺ ion** (He: Helium). Show your work including equations. (3 sig. figs.)

b. 2 points

(I) Select the largest (longest) wavelength from the three choices below:

- i. Wavelength of photon to *completely remove* an electron from *1s orbital of Hydrogen atom*
- ii. Wavelength of photon to *completely remove* an electron from *1s orbital of Boron atom (B)*
- iii. Wavelength of photon to *completely remove* an electron from *1s orbital of Boron cation (B⁺)*

i or ii or iii

(II) Select the smallest (shortest) wavelength from the three choices below (*for H atom*):

- i. Wavelength of photon *emitted* for an electron transitioning from *2p to 1s orbital*
- ii. Wavelength of photon *absorbed* for an electron transitioning from *2s to 4p orbital*
- iii. Wavelength of photon *absorbed* for an electron transitioning from *1s to 3p orbital*

i or ii or iii



Question 2 8 points



A Chem110 student plots the graph shown on the left, from data obtained for the kinetic energy (eV) of the ejected electron versus frequency (s^{-1}) of the incident light, for an unknown metal. ($1\text{eV} = 1.6022 \times 10^{-19} \text{ J}$)

Based on the plot, answer the questions **a to c** below:

a. 1 point On the plot above, label the **threshold frequency**.

b. 2 points What is the value of the work potential (in J) of the unknown metal?

c. 3 points What is the *longest* wavelength (in nm) required to eject an electron from this metal's surface? Show your work including equation(s) used. (3 sig figs)

d. 2 points In the space provided below, plot the following

(Assume frequency is *above* threshold frequency):

- i. *Kinetic energy* of an ejected electron versus the *intensity* of the incident light (left)
- ii. *Number of electrons* ejected versus the *frequency* of the incident light (right)



**Question 3** **6 points****a. 2 points**

In the ground state of Ge (Germanium; $Z = 32$) atom, **name** the following orbitals:

i. Orbital containing the electron that requires the maximum energy to be removed.

ii. Orbital containing the electron that experiences the minimum effective nuclear charge.

b. 2 points

An unknown anion (A^-) has the following *excited* electronic configuration: $[Ar] 4s^2 3d^3 4p^1$

(Ar: Argon; $Z = 18$)

Give the **condensed ground state electronic configuration** of the atom, **A** and the cation, **A^{2+}** .
Show your work. (Assume all electronic transitions are possible)

Electronic Configuration of A:

Electronic Configuration of A^{2+} :

c. 2 points

In a single atom, determine how many electrons can have the following set of quantum numbers?

(i) $n = 5; l = 3$

In a single atom, determine how many electrons can have the following set of quantum numbers?

(i) $n = 3; l = 2; m_s = +1/2$

**Question 4** **8 points**

Rank the following from 1 to 4 in **increasing** order. Provide the ranking in the boxes provided.

Note: Atomic number (Z) are provided for relevant atoms

a) Increasing (lowest to highest) order of *ionization energy*: (1= lowest; 4 = highest)

Mg

Mg: Z = 12

Rb

Rb: Z = 37

Cs

Cs: Z = 55

K

K : Z = 19

b) Increasing (lowest to highest) order of *bond angle*: (1= lowest; 4 = highest)

H₂OCCl₄XeF₄

HCN

c) Increasing (smallest to largest) order of *ionic size* (1= smallest; 4 = largest)

S²⁻

S: Z = 16

O²⁻

O: Z = 8

Be²⁺

Be: Z = 4

Na⁺

Na: Z = 11

d) Increasing (lowest to highest) order of *bond energy* (1= lowest; 4 = highest)

P-F

P-Cl

P-I

P-Br

**Question 5** **12 points**

a. 9 points For the molecules HClO_4 and TeF_6 , draw the most stable *Lewis structure* showing appropriate shape and *dashed/wedged bonds*, based on VSEPR. Determine the **molecular geometry** around the central atom. For HClO_4 , label the *most polar bond*.



b. 3 points (Note: Lone pairs are not shown in structures below)

(i) For the three *resonance* structures (1, 2 and 3) shown below, answer the following question:



(I) Circle the *most contributing* resonance structure.

Structure 1

Structure 2

Structure 3

(ii) For the three structures given (4, 5, and 6) below, answer the following questions:



(I) Circle all the structures that represent *resonance structures* of each other:

Structure 4

Structure 5

Structure 6

(II) Among the resonance structures, circle the *most contributing* resonance structure.

Structure 4

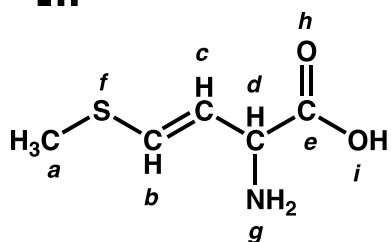
Structure 5

Structure 6

**Question 6** **6 points**

Using the data given below, calculate the **lattice energy** for SrO(s).

	ΔH° (kJ/mol)
$\text{Sr}(s) \longrightarrow \text{Sr}(g)$	+180
$\text{Sr}(g) \longrightarrow \text{Sr}^{2+}(g) + 2e^-$	+1468
$\text{O}_2(g) \longrightarrow 2\text{O}(g)$	+500
$\text{O}(g) + 2e^- \longrightarrow \text{O}^{2-}(g)$	+600
$\text{Sr}(s) + \frac{1}{2}\text{O}_2(g) \longrightarrow \text{SrO}(s)$	-550

**Question 7 10 points**

Consider the molecule given on the left, and answer questions (i) to (vi) using chemical bonding theories that we have learnt in class (valence bond theory, hybridization).

Note that the lone pairs are not shown in the structure:

i. Consider the carbon atoms marked in as ***a***, ***b***, ***c***, ***d***, and ***e*** in the molecule, and determine **hybridization** at each of the carbon centers. (2.5 points)

Atom	Hybridization
Carbon (<i>a</i>)	
Carbon (<i>b</i>)	
Carbon (<i>c</i>)	
Carbon (<i>d</i>)	
Carbon (<i>e</i>)	

ii. Now, consider the sulfur atom (S) marked as ***f***, nitrogen (N) atom (***g***), and oxygen(O) atoms (***h*** and ***i***), and determine their **hybridization**. (2 points)

Atom	Hybridization
Sulfur (<i>f</i>)	
Nitrogen (<i>g</i>)	
Oxygen (<i>h</i>)	
Oxygen (<i>i</i>)	

iii. **Describe** all the bonds formed by the carbon atoms ***a*** and ***b***, using sigma and pi designation as appropriate. (2 points)

Atom	Bonds formed (use sigma and pi as appropriate)
Carbon (<i>a</i>)	
Carbon (<i>b</i>)	

iv. How many total sp^3 -s sigma bonds are in the molecule? (1 point)

v. Determine **electronic** and **molecular geometry** at carbon (C) labeled ***e*** and sulfur (S) labeled ***f*** centres. (2 points)

Atom	Electronic Geometry	Molecular Geometry
Carbon (<i>e</i>)		
Sulfur (<i>f</i>)		

vi. Circle the value from the choices given below, that describes the **C-S-C bond angle at sulfur** (S) marked ***f***: (0.5 points)

>90° to <109.5° 109.5° >109.5 to <120° 120° >120° to <180°


Question 8 12 points

Using the **valence shell orbitals only** draw a **complete** molecular orbital (MO) diagram (with no s-p mixing) for the molecule **PS**, in the space provided below. **(6 points)**

Assume that 3s and 3p atomic orbitals will form molecular orbitals like those formed by the combination of 2s and 2p atomic orbitals that we learnt in class, and there is no s-p mixing.

Show **energy arrow**, **atomic orbitals** involved, **atom labels**, the **molecular orbitals** that are formed, **filled electrons** in the MO diagram etc. Make sure to label all the atomic and molecular orbitals.

Using the MO diagram above, answer the following questions:

- (i) Which one of the following will have the highest bond order? Circle the correct choice
 A) PS^{1-} B) PS C) PS^{1+}
- (ii) Which one of the following will have the longest bond length? Circle the correct choice
 A) PS^{1-} B) PS C) PS^{1+}
- (iii) Which of the following will be diamagnetic? Circle the correct choice.
 A) PS^{1-} B) PS C) PS^{1+}
- (iv) Draw Lewis dot structure of the molecule with the shortest bond length (based on the MO diagram) from among the three (PS^{1-} , PS, PS^{1+}), and indicate the bond order calculated from the Lewis dot structure. **(3 points)**

Lewis Dot Structure

Bond Order



F8456751-70A7-44D7-AFF6-A4C05790A257

midterm-exam-93388

#776 Page 10 of 10

SCRAP PAGE

**CHEMISTRY 110 –Midterm Exam****Thursday November 7, 2024, 6:30 – 8:30 pm****Examiners: Pallavi Sirjoosingh and Ashok Kakkar**

First name (please write as legibly as possible within the boxes)

--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--

Last name

--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--

Student ID number

--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--

This booklet has 8 questions on 10 pages (The last page can be used as a scrap page).

The staple must NOT be removed from the question booklet. Your intact question booklet must be returned at the end of the exam.

A separate booklet contains a periodic table, and datasheet.

Write your answers directly under the questions in the space provided. Do not write on the QR codes or near the top edges (as it may be cut off).

Please write **legibly and dark enough** for the scan to be readable. For students who use English as a second language, dictionaries are permitted. Calculators are allowed.

DO NOT ASK QUESTIONS OF THE INVIGILATORS about interpretations of the exam questions. If you have a concern about a question, please summarize it at the top of this page. All concerns will be evaluated after the exam. GOOD LUCK!

**Question 1 8 points** (*Assume all transitions are possible*)

a. 6 points Calculate the **energy (in kJ/mol)** required to ionize (completely remove) the electron from the **$n=3$** level of **He^+ ion** (He: Helium). Show your work including equations. (3 sig. figs.)

b. 2 points

(I) Select the largest (longest) wavelength from the three choices below:

- i. Wavelength of photon to *completely remove* an electron from *1s orbital of Hydrogen atom*
- ii. Wavelength of photon to *completely remove* an electron from *1s orbital of Boron atom (B)*
- iii. Wavelength of photon to *completely remove* an electron from *1s orbital of Boron cation (B^+)*

i or ii or iii

(II) Select the smallest (shortest) wavelength from the three choices below (*for H atom*):

- i. Wavelength of photon *emitted* for an electron transitioning from *2p to 1s orbital*
- ii. Wavelength of photon *absorbed* for an electron transitioning from *2s to 4p orbital*
- iii. Wavelength of photon *absorbed* for an electron transitioning from *1s to 3p orbital*

i or ii or iii



Question 2 8 points



A Chem110 student plots the graph shown on the left, from data obtained for the kinetic energy (eV) of the ejected electron versus frequency (s^{-1}) of the incident light, for an unknown metal. ($1\text{eV} = 1.6022 \times 10^{-19} \text{ J}$)

Based on the plot, answer the questions **a to c** below:

a. 1 point On the plot above, label the **threshold frequency**.

b. 2 points What is the value of the work potential (in J) of the unknown metal?

c. 3 points What is the *longest* wavelength (in nm) required to eject an electron from this metal's surface? Show your work including equation(s) used. (3 sig figs)

d. 2 points In the space provided below, plot the following

(Assume frequency is *above* threshold frequency):

- i. *Kinetic energy* of an ejected electron versus the *intensity* of the incident light (left)
- ii. *Number of electrons* ejected versus the *frequency* of the incident light (right)



**Question 3** **6 points****a. 2 points**

In the ground state of Ge (Germanium; $Z = 32$) atom, **name** the following orbitals:

i. Orbital containing the electron that requires the maximum energy to be removed.

ii. Orbital containing the electron that experiences the minimum effective nuclear charge.

b. 2 points

An unknown anion (A^-) has the following *excited* electronic configuration: $[Ar] 4s^2 3d^3 4p^1$

(Ar: Argon; $Z = 18$)

Give the **condensed ground state electronic configuration** of the atom, **A** and the cation, **A^{2+}** .
Show your work. (Assume all electronic transitions are possible)

Electronic Configuration of A:

Electronic Configuration of A^{2+} :

c. 2 points

In a single atom, determine how many electrons can have the following set of quantum numbers?

(i) $n = 5; l = 3$

In a single atom, determine how many electrons can have the following set of quantum numbers?

(i) $n = 3; l = 2; m_s = +1/2$

**Question 4** **8 points**

Rank the following from 1 to 4 in **increasing** order. Provide the ranking in the boxes provided.

Note: Atomic number (Z) are provided for relevant atoms

a) Increasing (lowest to highest) order of *ionization energy*: (1= lowest; 4 = highest)

Mg

Mg: Z = 12

Rb

Rb: Z = 37

Cs

Cs: Z = 55

K

K : Z = 19

b) Increasing (lowest to highest) order of *bond angle*: (1= lowest; 4 = highest)

H₂OCCl₄XeF₄

HCN

c) Increasing (smallest to largest) order of *ionic size* (1= smallest; 4 = largest)

S²⁻

S: Z = 16

O²⁻

O: Z = 8

Be²⁺

Be: Z = 4

Na⁺

Na: Z = 11

d) Increasing (lowest to highest) order of *bond energy* (1= lowest; 4 = highest)

P-F

P-Cl

P-I

P-Br

**Question 5** **12 points**

a. 9 points For the molecules HClO_4 and TeF_6 , draw the most stable *Lewis structure* showing appropriate shape and *dashed/wedged bonds*, based on VSEPR. Determine the **molecular geometry** around the central atom. For HClO_4 , label the *most polar bond*.



b. 3 points (Note: Lone pairs are not shown in structures below)

(i) For the three *resonance* structures (1, 2 and 3) shown below, answer the following question:



(I) Circle the *most contributing* resonance structure.

Structure 1

Structure 2

Structure 3

(ii) For the three structures given (4, 5, and 6) below, answer the following questions:



(I) Circle all the structures that represent *resonance structures* of each other:

Structure 4

Structure 5

Structure 6

(II) Among the resonance structures, circle the *most contributing* resonance structure.

Structure 4

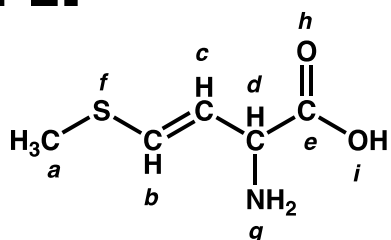
Structure 5

Structure 6

**Question 6** **6 points**

Using the data given below, calculate the **lattice energy** for SrO(s).

	ΔH° (kJ/mol)
$\text{Sr}(s) \longrightarrow \text{Sr}(g)$	+180
$\text{Sr}(g) \longrightarrow \text{Sr}^{2+}(g) + 2e^-$	+1468
$\text{O}_2(g) \longrightarrow 2\text{O}(g)$	+500
$\text{O}(g) + 2e^- \longrightarrow \text{O}^{2-}(g)$	+600
$\text{Sr}(s) + \frac{1}{2}\text{O}_2(g) \longrightarrow \text{SrO}(s)$	-550

**Question 7 10 points**

Consider the molecule given on the left, and answer questions (i) to (vi) using chemical bonding theories that we have learnt in class (valence bond theory, hybridization).

Note that the lone pairs are not shown in the structure:

i. Consider the carbon atoms marked in as ***a***, ***b***, ***c***, ***d***, and ***e*** in the molecule, and determine **hybridization** at each of the carbon centers. (2.5 points)

Atom	Hybridization
Carbon (<i>a</i>)	
Carbon (<i>b</i>)	
Carbon (<i>c</i>)	
Carbon (<i>d</i>)	
Carbon (<i>e</i>)	

ii. Now, consider the sulfur atom (S) marked as ***f***, nitrogen (N) atom (***g***), and oxygen(O) atoms (***h*** and ***i***), and determine their **hybridization**. (2 points)

Atom	Hybridization
Sulfur (<i>f</i>)	
Nitrogen (<i>g</i>)	
Oxygen (<i>h</i>)	
Oxygen (<i>i</i>)	

iii. **Describe** all the bonds formed by the carbon atoms ***a*** and ***b***, using sigma and pi designation as appropriate. (2 points)

Atom	Bonds formed (use sigma and pi as appropriate)
Carbon (<i>a</i>)	
Carbon (<i>b</i>)	

iv. How many total sp^3 -s sigma bonds are in the molecule? (1 point)

v. Determine **electronic** and **molecular geometry** at carbon (C) labeled ***e*** and sulfur (S) labeled ***f*** centres. (2 points)

Atom	Electronic Geometry	Molecular Geometry
Carbon (<i>e</i>)		
Sulfur (<i>f</i>)		

vi. Circle the value from the choices given below, that describes the **C-S-C bond angle at sulfur** (S) marked ***f***: (0.5 points)

>90° to <109.5° 109.5° >109.5 to <120° 120° >120° to <180°


Question 8 12 points

Using the **valence shell orbitals only** draw a **complete** molecular orbital (MO) diagram (with no s-p mixing) for the molecule **PS**, in the space provided below. **(6 points)**

Assume that 3s and 3p atomic orbitals will form molecular orbitals like those formed by the combination of 2s and 2p atomic orbitals that we learnt in class, and there is no s-p mixing.

Show **energy arrow**, **atomic orbitals** involved, **atom labels**, the **molecular orbitals** that are formed, **filled electrons** in the MO diagram etc. Make sure to label all the atomic and molecular orbitals.

Using the MO diagram above, answer the following questions:

- (i) Which one of the following will have the highest bond order? Circle the correct choice
 A) PS^{1-} B) PS C) PS^{1+}
- (ii) Which one of the following will have the longest bond length? Circle the correct choice
 A) PS^{1-} B) PS C) PS^{1+}
- (iii) Which of the following will be diamagnetic? Circle the correct choice.
 A) PS^{1-} B) PS C) PS^{1+}
- (iv) Draw Lewis dot structure of the molecule with the shortest bond length (based on the MO diagram) from among the three (PS^{1-} , PS, PS^{1+}), and indicate the bond order calculated from the Lewis dot structure. **(3 points)**

Lewis Dot Structure

Bond Order



E0C424BD-9CF7-4B0F-A212-35EAC26CEC9

midterm-exam-93388

#777 Page 10 of 10

SCRAP PAGE

**CHEMISTRY 110 –Midterm Exam****Thursday November 7, 2024, 6:30 – 8:30 pm****Examiners: Pallavi Sirjoosingh and Ashok Kakkar**

First name (please write as legibly as possible within the boxes)

--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--

Last name

--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--

Student ID number

--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--

This booklet has 8 questions on 10 pages (The last page can be used as a scrap page).

The staple must NOT be removed from the question booklet. Your intact question booklet must be returned at the end of the exam.

A separate booklet contains a periodic table, and datasheet.

Write your answers directly under the questions in the space provided. Do not write on the QR codes or near the top edges (as it may be cut off).

Please write **legibly and dark enough** for the scan to be readable. For students who use English as a second language, dictionaries are permitted. Calculators are allowed.

DO NOT ASK QUESTIONS OF THE INVIGILATORS about interpretations of the exam questions. If you have a concern about a question, please summarize it at the top of this page. All concerns will be evaluated after the exam. GOOD LUCK!

**Question 1 8 points** (*Assume all transitions are possible*)

a. 6 points Calculate the **energy (in kJ/mol)** required to ionize (completely remove) the electron from the **n=3** level of **He⁺ ion** (He: Helium). Show your work including equations. (3 sig. figs.)

b. 2 points

(I) Select the largest (longest) wavelength from the three choices below:

- i. Wavelength of photon to *completely remove* an electron from *1s orbital of Hydrogen atom*
- ii. Wavelength of photon to *completely remove* an electron from *1s orbital of Boron atom (B)*
- iii. Wavelength of photon to *completely remove* an electron from *1s orbital of Boron cation (B⁺)*

i or ii or iii

(II) Select the smallest (shortest) wavelength from the three choices below (*for H atom*):

- i. Wavelength of photon *emitted* for an electron transitioning from *2p to 1s orbital*
- ii. Wavelength of photon *absorbed* for an electron transitioning from *2s to 4p orbital*
- iii. Wavelength of photon *absorbed* for an electron transitioning from *1s to 3p orbital*

i or ii or iii



Question 2 8 points



A Chem110 student plots the graph shown on the left, from data obtained for the kinetic energy (eV) of the ejected electron versus frequency (s^{-1}) of the incident light, for an unknown metal. ($1\text{eV} = 1.6022 \times 10^{-19} \text{ J}$)

Based on the plot, answer the questions **a to c** below:

a. 1 point On the plot above, label the **threshold frequency**.

b. 2 points What is the value of the work potential (in J) of the unknown metal?

c. 3 points What is the *longest* wavelength (in nm) required to eject an electron from this metal's surface? Show your work including equation(s) used. (3 sig figs)

d. 2 points In the space provided below, plot the following

(Assume frequency is *above* threshold frequency):

- i. *Kinetic energy* of an ejected electron versus the *intensity* of the incident light (left)
- ii. *Number of electrons* ejected versus the *frequency* of the incident light (right)



**Question 3** **6 points****a. 2 points**

In the ground state of Ge (Germanium; $Z = 32$) atom, **name** the following orbitals:

i. Orbital containing the electron that requires the maximum energy to be removed.

ii. Orbital containing the electron that experiences the minimum effective nuclear charge.

b. 2 points

An unknown anion (A^-) has the following *excited* electronic configuration: $[Ar] 4s^2 3d^3 4p^1$

(Ar: Argon; $Z = 18$)

Give the **condensed ground state electronic configuration** of the atom, **A** and the cation, **A^{2+}** .
Show your work. (Assume all electronic transitions are possible)

Electronic Configuration of A:

Electronic Configuration of A^{2+} :

c. 2 points

In a single atom, determine how many electrons can have the following set of quantum numbers?

(i) $n = 5; l = 3$

In a single atom, determine how many electrons can have the following set of quantum numbers?

(i) $n = 3; l = 2; m_s = +1/2$

**Question 4** **8 points**

Rank the following from 1 to 4 in **increasing** order. Provide the ranking in the boxes provided.

Note: Atomic number (Z) are provided for relevant atoms

a) Increasing (lowest to highest) order of *ionization energy*: (1= lowest; 4 = highest)

Mg

Mg: Z = 12

Rb

Rb: Z = 37

Cs

Cs: Z = 55

K

K : Z = 19

b) Increasing (lowest to highest) order of *bond angle*: (1= lowest; 4 = highest)

H₂OCCl₄XeF₄

HCN

c) Increasing (smallest to largest) order of *ionic size* (1= smallest; 4 = largest)

S²⁻

S: Z = 16

O²⁻

O: Z = 8

Be²⁺

Be: Z = 4

Na⁺

Na: Z = 11

d) Increasing (lowest to highest) order of *bond energy* (1= lowest; 4 = highest)

P-F

P-Cl

P-I

P-Br

**Question 5** **12 points**

a. 9 points For the molecules HClO_4 and TeF_6 , draw the most stable *Lewis structure* showing appropriate shape and *dashed/wedged bonds*, based on VSEPR. Determine the **molecular geometry** around the central atom. For HClO_4 , label the *most polar bond*.

HClO_4 (there are no O-O bonds)

Label the most polar bond

Molecular Geometry around Cl

TeF_6

Molecular Geometry around Te

b. 3 points (Note: Lone pairs are not shown in structures below)

(i) For the three *resonance* structures (1, 2 and 3) shown below, answer the following question:



(I) Circle the *most contributing* resonance structure.

Structure 1

Structure 2

Structure 3

(ii) For the three structures given (4, 5, and 6) below, answer the following questions:



(I) Circle all the structures that represent *resonance structures* of each other:

Structure 4

Structure 5

Structure 6

(II) Among the resonance structures, circle the *most contributing* resonance structure.

Structure 4

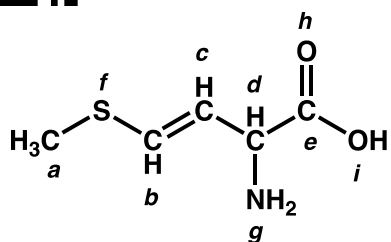
Structure 5

Structure 6

**Question 6** **6 points**

Using the data given below, calculate the **lattice energy** for SrO(s).

	ΔH° (kJ/mol)
$\text{Sr}(s) \longrightarrow \text{Sr}(g)$	+180
$\text{Sr}(g) \longrightarrow \text{Sr}^{2+}(g) + 2e^-$	+1468
$\text{O}_2(g) \longrightarrow 2\text{O}(g)$	+500
$\text{O}(g) + 2e^- \longrightarrow \text{O}^{2-}(g)$	+600
$\text{Sr}(s) + \frac{1}{2}\text{O}_2(g) \longrightarrow \text{SrO}(s)$	-550

**Question 7 10 points**

Consider the molecule given on the left, and answer questions (i) to (vi) using chemical bonding theories that we have learnt in class (valence bond theory, hybridization).

Note that the lone pairs are not shown in the structure:

i. Consider the carbon atoms marked in as ***a***, ***b***, ***c***, ***d***, and ***e*** in the molecule, and determine **hybridization** at each of the carbon centers. (2.5 points)

Atom	Hybridization
Carbon (<i>a</i>)	
Carbon (<i>b</i>)	
Carbon (<i>c</i>)	
Carbon (<i>d</i>)	
Carbon (<i>e</i>)	

ii. Now, consider the sulfur atom (S) marked as ***f***, nitrogen (N) atom (***g***), and oxygen(O) atoms (***h*** and ***i***), and determine their **hybridization**. (2 points)

Atom	Hybridization
Sulfur (<i>f</i>)	
Nitrogen (<i>g</i>)	
Oxygen (<i>h</i>)	
Oxygen (<i>i</i>)	

iii. **Describe** all the bonds formed by the carbon atoms ***a*** and ***b***, using sigma and pi designation as appropriate. (2 points)

Atom	Bonds formed (use sigma and pi as appropriate)
Carbon (<i>a</i>)	
Carbon (<i>b</i>)	

iv. How many total sp^3 -s sigma bonds are in the molecule? (1 point)

v. Determine **electronic** and **molecular geometry** at carbon (C) labeled ***e*** and sulfur (S) labeled ***f*** centres. (2 points)

Atom	Electronic Geometry	Molecular Geometry
Carbon (<i>e</i>)		
Sulfur (<i>f</i>)		

vi. Circle the value from the choices given below, that describes the **C-S-C bond angle at sulfur** (S) marked ***f***: (0.5 points)

$>90^\circ$ to $<109.5^\circ$ 109.5° >109.5 to $<120^\circ$ 120° $>120^\circ$ to $<180^\circ$


Question 8 12 points

Using the **valence shell orbitals only** draw a **complete** molecular orbital (MO) diagram (with no s-p mixing) for the molecule **PS**, in the space provided below. **(6 points)**

Assume that 3s and 3p atomic orbitals will form molecular orbitals like those formed by the combination of 2s and 2p atomic orbitals that we learnt in class, and there is no s-p mixing.

Show **energy arrow**, **atomic orbitals** involved, **atom labels**, the **molecular orbitals** that are formed, **filled electrons** in the MO diagram etc. Make sure to label all the atomic and molecular orbitals.

Using the MO diagram above, answer the following questions:

- (i) Which one of the following will have the highest bond order? Circle the correct choice
 A) PS^{1-} B) PS C) PS^{1+}
- (ii) Which one of the following will have the longest bond length? Circle the correct choice
 A) PS^{1-} B) PS C) PS^{1+}
- (iii) Which of the following will be diamagnetic? Circle the correct choice.
 A) PS^{1-} B) PS C) PS^{1+}
- (iv) Draw Lewis dot structure of the molecule with the shortest bond length (based on the MO diagram) from among the three (PS^{1-} , PS, PS^{1+}), and indicate the bond order calculated from the Lewis dot structure. **(3 points)**

Lewis Dot Structure

Bond Order



9003EB45-DD44-42A7-801C-4495A906F107

midterm-exam-93388

#778 Page 10 of 10

SCRAP PAGE

**CHEMISTRY 110 –Midterm Exam****Thursday November 7, 2024, 6:30 – 8:30 pm**

Examiners: Pallavi Sirjoosingh and Ashok Kakkar

First name (please write as legibly as possible within the boxes)

--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--

Last name

--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--

Student ID number

--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--

This booklet has 8 questions on 10 pages (The last page can be used as a scrap page).

The staple must NOT be removed from the question booklet. Your intact question booklet must be returned at the end of the exam.

A separate booklet contains a periodic table, and datasheet.

Write your answers directly under the questions in the space provided. Do not write on the QR codes or near the top edges (as it may be cut off).

Please write **legibly and dark enough** for the scan to be readable. For students who use English as a second language, dictionaries are permitted. Calculators are allowed.

DO NOT ASK QUESTIONS OF THE INVIGILATORS about interpretations of the exam questions. If you have a concern about a question, please summarize it at the top of this page. All concerns will be evaluated after the exam. GOOD LUCK!

**Question 1 8 points** (*Assume all transitions are possible*)

a. 6 points Calculate the **energy (in kJ/mol)** required to ionize (completely remove) the electron from the **n=3** level of **He⁺ ion** (He: Helium). Show your work including equations. (3 sig. figs.)

b. 2 points

(I) Select the largest (longest) wavelength from the three choices below:

- i. Wavelength of photon to *completely remove* an electron from *1s orbital of Hydrogen atom*
- ii. Wavelength of photon to *completely remove* an electron from *1s orbital of Boron atom (B)*
- iii. Wavelength of photon to *completely remove* an electron from *1s orbital of Boron cation (B⁺)*

i or ii or iii

(II) Select the smallest (shortest) wavelength from the three choices below (*for H atom*):

- i. Wavelength of photon *emitted* for an electron transitioning from *2p to 1s orbital*
- ii. Wavelength of photon *absorbed* for an electron transitioning from *2s to 4p orbital*
- iii. Wavelength of photon *absorbed* for an electron transitioning from *1s to 3p orbital*

i or ii or iii



Question 2 8 points



A Chem110 student plots the graph shown on the left, from data obtained for the kinetic energy (eV) of the ejected electron versus frequency (s^{-1}) of the incident light, for an unknown metal. ($1\text{eV} = 1.6022 \times 10^{-19} \text{ J}$)

Based on the plot, answer the questions **a to c** below:

a. 1 point On the plot above, label the **threshold frequency**.

b. 2 points What is the value of the work potential (in J) of the unknown metal?

c. 3 points What is the *longest* wavelength (in nm) required to eject an electron from this metal's surface? Show your work including equation(s) used. (3 sig figs)

d. 2 points In the space provided below, plot the following

(Assume frequency is *above* threshold frequency):

- i. *Kinetic energy* of an ejected electron versus the *intensity* of the incident light (left)
- ii. *Number of electrons* ejected versus the *frequency* of the incident light (right)



**Question 3** **6 points****a. 2 points**

In the ground state of Ge (Germanium; $Z = 32$) atom, **name** the following orbitals:

i. Orbital containing the electron that requires the maximum energy to be removed.

ii. Orbital containing the electron that experiences the minimum effective nuclear charge.

b. 2 points

An unknown anion (A^-) has the following *excited* electronic configuration: $[Ar] 4s^2 3d^3 4p^1$

(Ar: Argon; $Z = 18$)

Give the **condensed ground state electronic configuration** of the atom, **A** and the cation, **A^{2+}** .
Show your work. (Assume all electronic transitions are possible)

Electronic Configuration of A:

Electronic Configuration of A^{2+} :

c. 2 points

In a single atom, determine how many electrons can have the following set of quantum numbers?

(i) $n = 5; l = 3$

In a single atom, determine how many electrons can have the following set of quantum numbers?

(i) $n = 3; l = 2; m_s = +1/2$

**Question 4** **8 points**

Rank the following from 1 to 4 in **increasing** order. Provide the ranking in the boxes provided.

Note: Atomic number (Z) are provided for relevant atoms

a) Increasing (lowest to highest) order of *ionization energy*: (1= lowest; 4 = highest)

Mg

Mg: Z = 12

Rb

Rb: Z = 37

Cs

Cs: Z = 55

K

K : Z = 19

b) Increasing (lowest to highest) order of *bond angle*: (1= lowest; 4 = highest)

H₂OCCl₄XeF₄

HCN

c) Increasing (smallest to largest) order of *ionic size* (1= smallest; 4 = largest)

S²⁻

S: Z = 16

O²⁻

O: Z = 8

Be²⁺

Be: Z = 4

Na⁺

Na: Z = 11

d) Increasing (lowest to highest) order of *bond energy* (1= lowest; 4 = highest)

P-F

P-Cl

P-I

P-Br

**Question 5** **12 points**

a. 9 points For the molecules HClO_4 and TeF_6 , draw the most stable *Lewis structure* showing appropriate shape and *dashed/wedged bonds*, based on VSEPR. Determine the **molecular geometry** around the central atom. For HClO_4 , label the *most polar bond*.

HClO_4 (there are no O-O bonds)

Label the most polar bond

Molecular Geometry around Cl

TeF_6

Molecular Geometry around Te

b. 3 points (Note: Lone pairs are not shown in structures below)

(i) For the three *resonance* structures (1, 2 and 3) shown below, answer the following question:



(I) Circle the *most contributing* resonance structure.

Structure 1

Structure 2

Structure 3

(ii) For the three structures given (4, 5, and 6) below, answer the following questions:



(I) Circle all the structures that represent *resonance structures* of each other:

Structure 4

Structure 5

Structure 6

(II) Among the resonance structures, circle the *most contributing* resonance structure.

Structure 4

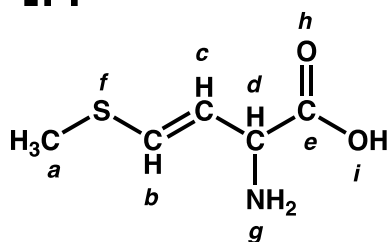
Structure 5

Structure 6

**Question 6** **6 points**

Using the data given below, calculate the **lattice energy** for SrO(s).

	ΔH° (kJ/mol)
$\text{Sr}(s) \longrightarrow \text{Sr}(g)$	+180
$\text{Sr}(g) \longrightarrow \text{Sr}^{2+}(g) + 2e^-$	+1468
$\text{O}_2(g) \longrightarrow 2\text{O}(g)$	+500
$\text{O}(g) + 2e^- \longrightarrow \text{O}^{2-}(g)$	+600
$\text{Sr}(s) + \frac{1}{2}\text{O}_2(g) \longrightarrow \text{SrO}(s)$	-550

**Question 7 10 points**

Consider the molecule given on the left, and answer questions (i) to (vi) using chemical bonding theories that we have learnt in class (valence bond theory, hybridization).

Note that the lone pairs are not shown in the structure:

i. Consider the carbon atoms marked in as ***a***, ***b***, ***c***, ***d***, and ***e*** in the molecule, and determine **hybridization** at each of the carbon centers. (2.5 points)

Atom	Hybridization
Carbon (<i>a</i>)	
Carbon (<i>b</i>)	
Carbon (<i>c</i>)	
Carbon (<i>d</i>)	
Carbon (<i>e</i>)	

ii. Now, consider the sulfur atom (S) marked as ***f***, nitrogen (N) atom (***g***), and oxygen(O) atoms (***h*** and ***i***), and determine their **hybridization**. (2 points)

Atom	Hybridization
Sulfur (<i>f</i>)	
Nitrogen (<i>g</i>)	
Oxygen (<i>h</i>)	
Oxygen (<i>i</i>)	

iii. **Describe** all the bonds formed by the carbon atoms ***a*** and ***b***, using sigma and pi designation as appropriate. (2 points)

Atom	Bonds formed (use sigma and pi as appropriate)
Carbon (<i>a</i>)	
Carbon (<i>b</i>)	

iv. How many total sp^3 -s sigma bonds are in the molecule? (1 point)

v. Determine **electronic** and **molecular geometry** at carbon (C) labeled ***e*** and sulfur (S) labeled ***f*** centres. (2 points)

Atom	Electronic Geometry	Molecular Geometry
Carbon (<i>e</i>)		
Sulfur (<i>f</i>)		

vi. Circle the value from the choices given below, that describes the **C-S-C bond angle at sulfur** (S) marked ***f***: (0.5 points)

>90° to <109.5°

109.5°

>109.5 to <120°

120°

>120° to <180°


Question 8 12 points

Using the **valence shell orbitals only** draw a **complete** molecular orbital (MO) diagram (with no s-p mixing) for the molecule **PS**, in the space provided below. **(6 points)**

Assume that 3s and 3p atomic orbitals will form molecular orbitals like those formed by the combination of 2s and 2p atomic orbitals that we learnt in class, and there is no s-p mixing.

Show **energy arrow**, **atomic orbitals** involved, **atom labels**, the **molecular orbitals** that are formed, **filled electrons** in the MO diagram etc. Make sure to label all the atomic and molecular orbitals.

Using the MO diagram above, answer the following questions:

- (i) Which one of the following will have the highest bond order? Circle the correct choice
 A) PS^{1-} B) PS C) PS^{1+}
- (ii) Which one of the following will have the longest bond length? Circle the correct choice
 A) PS^{1-} B) PS C) PS^{1+}
- (iii) Which of the following will be diamagnetic? Circle the correct choice.
 A) PS^{1-} B) PS C) PS^{1+}
- (iv) Draw Lewis dot structure of the molecule with the shortest bond length (based on the MO diagram) from among the three (PS^{1-} , PS, PS^{1+}), and indicate the bond order calculated from the Lewis dot structure. **(3 points)**

Lewis Dot Structure

Bond Order



190F1F48-0A53-45CE-A0B8-C9837AAABC7B

midterm-exam-93388

#779 Page 10 of 10

SCRAP PAGE

**CHEMISTRY 110 –Midterm Exam****Thursday November 7, 2024, 6:30 – 8:30 pm****Examiners: Pallavi Sirjoosingh and Ashok Kakkar**

First name (please write as legibly as possible within the boxes)

--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--

Last name

--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--

Student ID number

--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--

This booklet has 8 questions on 10 pages (The last page can be used as a scrap page).

The staple must NOT be removed from the question booklet. Your intact question booklet must be returned at the end of the exam.

A separate booklet contains a periodic table, and datasheet.

Write your answers directly under the questions in the space provided. Do not write on the QR codes or near the top edges (as it may be cut off).

Please write **legibly and dark enough** for the scan to be readable. For students who use English as a second language, dictionaries are permitted. Calculators are allowed.

DO NOT ASK QUESTIONS OF THE INVIGILATORS about interpretations of the exam questions. If you have a concern about a question, please summarize it at the top of this page. All concerns will be evaluated after the exam. GOOD LUCK!

**Question 1 8 points** (*Assume all transitions are possible*)

a. 6 points Calculate the **energy (in kJ/mol)** required to ionize (completely remove) the electron from the **n=3** level of **He⁺ ion** (He: Helium). Show your work including equations. (3 sig. figs.)

b. 2 points

(I) Select the largest (longest) wavelength from the three choices below:

- i. Wavelength of photon to *completely remove* an electron from *1s orbital of Hydrogen atom*
- ii. Wavelength of photon to *completely remove* an electron from *1s orbital of Boron atom (B)*
- iii. Wavelength of photon to *completely remove* an electron from *1s orbital of Boron cation (B⁺)*

i or ii or iii

(II) Select the smallest (shortest) wavelength from the three choices below (*for H atom*):

- i. Wavelength of photon *emitted* for an electron transitioning from *2p to 1s orbital*
- ii. Wavelength of photon *absorbed* for an electron transitioning from *2s to 4p orbital*
- iii. Wavelength of photon *absorbed* for an electron transitioning from *1s to 3p orbital*

i or ii or iii



Question 2 8 points



A Chem110 student plots the graph shown on the left, from data obtained for the kinetic energy (eV) of the ejected electron versus frequency (s^{-1}) of the incident light, for an unknown metal. ($1\text{eV} = 1.6022 \times 10^{-19} \text{ J}$)

Based on the plot, answer the questions **a to c** below:

a. 1 point On the plot above, label the **threshold frequency**.

b. 2 points What is the value of the work potential (in J) of the unknown metal?

c. 3 points What is the *longest* wavelength (in nm) required to eject an electron from this metal's surface? Show your work including equation(s) used. (3 sig figs)

d. 2 points In the space provided below, plot the following

(Assume frequency is *above* threshold frequency):

- i. *Kinetic energy* of an ejected electron versus the *intensity* of the incident light (left)
- ii. *Number of electrons* ejected versus the *frequency* of the incident light (right)



**Question 3** **6 points****a. 2 points**

In the ground state of Ge (Germanium; $Z = 32$) atom, **name** the following orbitals:

i. Orbital containing the electron that requires the maximum energy to be removed.

ii. Orbital containing the electron that experiences the minimum effective nuclear charge.

b. 2 points

An unknown anion (A^-) has the following *excited* electronic configuration: $[Ar] 4s^2 3d^3 4p^1$

(Ar: Argon; $Z = 18$)

Give the **condensed ground state electronic configuration** of the atom, **A** and the cation, **A^{2+}** .
Show your work. (Assume all electronic transitions are possible)

Electronic Configuration of A:

Electronic Configuration of A^{2+} :

c. 2 points

In a single atom, determine how many electrons can have the following set of quantum numbers?

(i) $n = 5; l = 3$

In a single atom, determine how many electrons can have the following set of quantum numbers?

(i) $n = 3; l = 2; m_s = +1/2$

**Question 4** **8 points**

Rank the following from 1 to 4 in **increasing** order. Provide the ranking in the boxes provided.

Note: Atomic number (Z) are provided for relevant atoms

a) Increasing (lowest to highest) order of *ionization energy*: (1= lowest; 4 = highest)

Mg

Mg: Z = 12

Rb

Rb: Z = 37

Cs

Cs: Z = 55

K

K : Z = 19

b) Increasing (lowest to highest) order of *bond angle*: (1= lowest; 4 = highest)

H₂OCCl₄XeF₄

HCN

c) Increasing (smallest to largest) order of *ionic size* (1= smallest; 4 = largest)

S²⁻

S: Z = 16

O²⁻

O: Z = 8

Be²⁺

Be: Z = 4

Na⁺

Na: Z = 11

d) Increasing (lowest to highest) order of *bond energy* (1= lowest; 4 = highest)

P-F

P-Cl

P-I

P-Br

**Question 5** **12 points**

a. 9 points For the molecules HClO_4 and TeF_6 , draw the most stable *Lewis structure* showing appropriate shape and *dashed/wedged bonds*, based on VSEPR. Determine the **molecular geometry** around the central atom. For HClO_4 , label the *most polar bond*.



b. 3 points (Note: Lone pairs are not shown in structures below)

(i) For the three *resonance* structures (1, 2 and 3) shown below, answer the following question:



(I) Circle the *most contributing* resonance structure.

Structure 1

Structure 2

Structure 3

(ii) For the three structures given (4, 5, and 6) below, answer the following questions:



(I) Circle all the structures that represent *resonance structures* of each other:

Structure 4

Structure 5

Structure 6

(II) Among the resonance structures, circle the *most contributing* resonance structure.

Structure 4

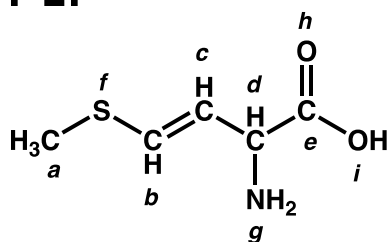
Structure 5

Structure 6

**Question 6** **6 points**

Using the data given below, calculate the **lattice energy** for SrO(s).

	ΔH° (kJ/mol)
$\text{Sr}(s) \longrightarrow \text{Sr}(g)$	+180
$\text{Sr}(g) \longrightarrow \text{Sr}^{2+}(g) + 2e^-$	+1468
$\text{O}_2(g) \longrightarrow 2\text{O}(g)$	+500
$\text{O}(g) + 2e^- \longrightarrow \text{O}^{2-}(g)$	+600
$\text{Sr}(s) + \frac{1}{2}\text{O}_2(g) \longrightarrow \text{SrO}(s)$	-550

**Question 7 10 points**

Consider the molecule given on the left, and answer questions (i) to (vi) using chemical bonding theories that we have learnt in class (valence bond theory, hybridization).

Note that the lone pairs are not shown in the structure:

i. Consider the carbon atoms marked in as ***a***, ***b***, ***c***, ***d***, and ***e*** in the molecule, and determine **hybridization** at each of the carbon centers. (2.5 points)

Atom	Hybridization
Carbon (<i>a</i>)	
Carbon (<i>b</i>)	
Carbon (<i>c</i>)	
Carbon (<i>d</i>)	
Carbon (<i>e</i>)	

ii. Now, consider the sulfur atom (S) marked as ***f***, nitrogen (N) atom (***g***), and oxygen(O) atoms (***h*** and ***i***), and determine their **hybridization**. (2 points)

Atom	Hybridization
Sulfur (<i>f</i>)	
Nitrogen (<i>g</i>)	
Oxygen (<i>h</i>)	
Oxygen (<i>i</i>)	

iii. **Describe** all the bonds formed by the carbon atoms ***a*** and ***b***, using sigma and pi designation as appropriate. (2 points)

Atom	Bonds formed (use sigma and pi as appropriate)
Carbon (<i>a</i>)	
Carbon (<i>b</i>)	

iv. How many total sp^3 -s sigma bonds are in the molecule? (1 point)

v. Determine **electronic** and **molecular geometry** at carbon (C) labeled ***e*** and sulfur (S) labeled ***f*** centres. (2 points)

Atom	Electronic Geometry	Molecular Geometry
Carbon (<i>e</i>)		
Sulfur (<i>f</i>)		

vi. Circle the value from the choices given below, that describes the **C-S-C bond angle at sulfur** (S) marked ***f***: (0.5 points)

>90° to <109.5°

109.5°

>109.5 to <120°

120°

>120° to <180°


Question 8 12 points

Using the **valence shell orbitals only** draw a **complete** molecular orbital (MO) diagram (with no s-p mixing) for the molecule **PS**, in the space provided below. **(6 points)**

Assume that 3s and 3p atomic orbitals will form molecular orbitals like those formed by the combination of 2s and 2p atomic orbitals that we learnt in class, and there is no s-p mixing.

Show **energy arrow**, **atomic orbitals** involved, **atom labels**, the **molecular orbitals** that are formed, **filled electrons** in the MO diagram etc. Make sure to label all the atomic and molecular orbitals.

Using the MO diagram above, answer the following questions:

- (i) Which one of the following will have the highest bond order? Circle the correct choice
 A) PS^{1-} B) PS C) PS^{1+}
- (ii) Which one of the following will have the longest bond length? Circle the correct choice
 A) PS^{1-} B) PS C) PS^{1+}
- (iii) Which of the following will be diamagnetic? Circle the correct choice.
 A) PS^{1-} B) PS C) PS^{1+}
- (iv) Draw Lewis dot structure of the molecule with the shortest bond length (based on the MO diagram) from among the three (PS^{1-} , PS, PS^{1+}), and indicate the bond order calculated from the Lewis dot structure. **(3 points)**

Lewis Dot Structure

Bond Order



F2CF9E17-1B7D-4BC1-90A4-C89B71728733

midterm-exam-93388

#780 Page 10 of 10

SCRAP PAGE

**CHEMISTRY 110 –Midterm Exam****Thursday November 7, 2024, 6:30 – 8:30 pm**

Examiners: Pallavi Sirjoosingh and Ashok Kakkar

First name (please write as legibly as possible within the boxes)

--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--

Last name

--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--

Student ID number

--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--

This booklet has 8 questions on 10 pages (The last page can be used as a scrap page).

The staple must NOT be removed from the question booklet. Your intact question booklet must be returned at the end of the exam.

A separate booklet contains a periodic table, and datasheet.

Write your answers directly under the questions in the space provided. Do not write on the QR codes or near the top edges (as it may be cut off).

Please write **legibly and dark enough** for the scan to be readable. For students who use English as a second language, dictionaries are permitted. Calculators are allowed.

DO NOT ASK QUESTIONS OF THE INVIGILATORS about interpretations of the exam questions. If you have a concern about a question, please summarize it at the top of this page. All concerns will be evaluated after the exam. GOOD LUCK!

**Question 1 8 points** (*Assume all transitions are possible*)

a. 6 points Calculate the **energy (in kJ/mol)** required to ionize (completely remove) the electron from the **n=3** level of **He⁺ ion** (He: Helium). Show your work including equations. (3 sig. figs.)

b. 2 points

(I) Select the largest (longest) wavelength from the three choices below:

- i. Wavelength of photon to *completely remove* an electron from *1s orbital of Hydrogen atom*
- ii. Wavelength of photon to *completely remove* an electron from *1s orbital of Boron atom (B)*
- iii. Wavelength of photon to *completely remove* an electron from *1s orbital of Boron cation (B⁺)*

i or ii or iii

(II) Select the smallest (shortest) wavelength from the three choices below (*for H atom*):

- i. Wavelength of photon *emitted* for an electron transitioning from *2p to 1s orbital*
- ii. Wavelength of photon *absorbed* for an electron transitioning from *2s to 4p orbital*
- iii. Wavelength of photon *absorbed* for an electron transitioning from *1s to 3p orbital*

i or ii or iii



Question 2 8 points



A Chem110 student plots the graph shown on the left, from data obtained for the kinetic energy (eV) of the ejected electron versus frequency (s^{-1}) of the incident light, for an unknown metal. ($1\text{eV} = 1.6022 \times 10^{-19} \text{ J}$)

Based on the plot, answer the questions **a to c** below:

a. 1 point On the plot above, label the **threshold frequency**.

b. 2 points What is the value of the work potential (in J) of the unknown metal?

c. 3 points What is the *longest* wavelength (in nm) required to eject an electron from this metal's surface? Show your work including equation(s) used. (3 sig figs)

d. 2 points In the space provided below, plot the following

(Assume frequency is *above* threshold frequency):

- i. *Kinetic energy* of an ejected electron versus the *intensity* of the incident light (left)
- ii. *Number of electrons* ejected versus the *frequency* of the incident light (right)



**Question 3** **6 points****a. 2 points**

In the ground state of Ge (Germanium; $Z = 32$) atom, **name** the following orbitals:

i. Orbital containing the electron that requires the maximum energy to be removed.

ii. Orbital containing the electron that experiences the minimum effective nuclear charge.

b. 2 points

An unknown anion (A^-) has the following *excited* electronic configuration: $[Ar] 4s^2 3d^3 4p^1$

(Ar: Argon; $Z = 18$)

Give the **condensed ground state electronic configuration** of the atom, **A** and the cation, **A^{2+}** .
Show your work. (Assume all electronic transitions are possible)

Electronic Configuration of A:

Electronic Configuration of A^{2+} :

c. 2 points

In a single atom, determine how many electrons can have the following set of quantum numbers?

(i) $n = 5; l = 3$

In a single atom, determine how many electrons can have the following set of quantum numbers?

(i) $n = 3; l = 2; m_s = +1/2$

**Question 4** **8 points**

Rank the following from 1 to 4 in **increasing** order. Provide the ranking in the boxes provided.

Note: Atomic number (Z) are provided for relevant atoms

a) Increasing (lowest to highest) order of *ionization energy*: (1= lowest; 4 = highest)

Mg

Mg: Z = 12

Rb

Rb: Z = 37

Cs

Cs: Z = 55

K

K : Z = 19

b) Increasing (lowest to highest) order of *bond angle*: (1= lowest; 4 = highest)

H₂OCCl₄XeF₄

HCN

c) Increasing (smallest to largest) order of *ionic size* (1= smallest; 4 = largest)

S²⁻

S: Z = 16

O²⁻

O: Z = 8

Be²⁺

Be: Z = 4

Na⁺

Na: Z = 11

d) Increasing (lowest to highest) order of *bond energy* (1= lowest; 4 = highest)

P-F

P-Cl

P-I

P-Br

**Question 5** **12 points**

a. 9 points For the molecules HClO_4 and TeF_6 , draw the most stable *Lewis structure* showing appropriate shape and *dashed/wedged bonds*, based on VSEPR. Determine the **molecular geometry** around the central atom. For HClO_4 , label the *most polar bond*.



b. 3 points (Note: Lone pairs are not shown in structures below)

(i) For the three *resonance* structures (1, 2 and 3) shown below, answer the following question:



(I) Circle the *most contributing* resonance structure.

Structure 1

Structure 2

Structure 3

(ii) For the three structures given (4, 5, and 6) below, answer the following questions:



(I) Circle all the structures that represent *resonance structures* of each other:

Structure 4

Structure 5

Structure 6

(II) Among the resonance structures, circle the *most contributing* resonance structure.

Structure 4

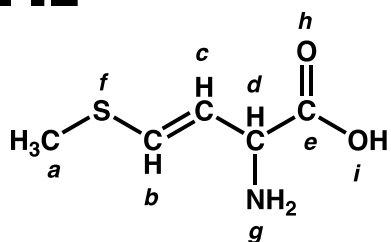
Structure 5

Structure 6

**Question 6** **6 points**

Using the data given below, calculate the **lattice energy** for SrO(s).

	ΔH° (kJ/mol)
$\text{Sr}(s) \longrightarrow \text{Sr}(g)$	+180
$\text{Sr}(g) \longrightarrow \text{Sr}^{2+}(g) + 2e^-$	+1468
$\text{O}_2(g) \longrightarrow 2\text{O}(g)$	+500
$\text{O}(g) + 2e^- \longrightarrow \text{O}^{2-}(g)$	+600
$\text{Sr}(s) + \frac{1}{2}\text{O}_2(g) \longrightarrow \text{SrO}(s)$	-550

**Question 7 10 points**

Consider the molecule given on the left, and answer questions (i) to (vi) using chemical bonding theories that we have learnt in class (valence bond theory, hybridization).

Note that the lone pairs are not shown in the structure:

i. Consider the carbon atoms marked in as ***a***, ***b***, ***c***, ***d***, and ***e*** in the molecule, and determine **hybridization** at each of the carbon centers. (2.5 points)

Atom	Hybridization
Carbon (<i>a</i>)	
Carbon (<i>b</i>)	
Carbon (<i>c</i>)	
Carbon (<i>d</i>)	
Carbon (<i>e</i>)	

ii. Now, consider the sulfur atom (S) marked as ***f***, nitrogen (N) atom (***g***), and oxygen(O) atoms (***h*** and ***i***), and determine their **hybridization**. (2 points)

Atom	Hybridization
Sulfur (<i>f</i>)	
Nitrogen (<i>g</i>)	
Oxygen (<i>h</i>)	
Oxygen (<i>i</i>)	

iii. **Describe** all the bonds formed by the carbon atoms ***a*** and ***b***, using sigma and pi designation as appropriate. (2 points)

Atom	Bonds formed (use sigma and pi as appropriate)
Carbon (<i>a</i>)	
Carbon (<i>b</i>)	

iv. How many total sp^3 -s sigma bonds are in the molecule? (1 point)

v. Determine **electronic** and **molecular geometry** at carbon (C) labeled ***e*** and sulfur (S) labeled ***f*** centres. (2 points)

Atom	Electronic Geometry	Molecular Geometry
Carbon (<i>e</i>)		
Sulfur (<i>f</i>)		

vi. Circle the value from the choices given below, that describes the **C-S-C bond angle at sulfur** (S) marked ***f***: (0.5 points)

>90° to <109.5°

109.5°

>109.5 to <120°

120°

>120° to <180°


Question 8 12 points

Using the **valence shell orbitals only** draw a **complete** molecular orbital (MO) diagram (with no s-p mixing) for the molecule **PS**, in the space provided below. **(6 points)**

Assume that 3s and 3p atomic orbitals will form molecular orbitals like those formed by the combination of 2s and 2p atomic orbitals that we learnt in class, and there is no s-p mixing.

Show **energy arrow**, **atomic orbitals** involved, **atom labels**, the **molecular orbitals** that are formed, **filled electrons** in the MO diagram etc. Make sure to label all the atomic and molecular orbitals.

Using the MO diagram above, answer the following questions:

- (i) Which one of the following will have the highest bond order? Circle the correct choice
 A) PS^{1-} B) PS C) PS^{1+}
- (ii) Which one of the following will have the longest bond length? Circle the correct choice
 A) PS^{1-} B) PS C) PS^{1+}
- (iii) Which of the following will be diamagnetic? Circle the correct choice.
 A) PS^{1-} B) PS C) PS^{1+}
- (iv) Draw Lewis dot structure of the molecule with the shortest bond length (based on the MO diagram) from among the three (PS^{1-} , PS, PS^{1+}), and indicate the bond order calculated from the Lewis dot structure. **(3 points)**

Lewis Dot Structure

Bond Order



8FECFDDC-BE5A-4C16-8E66-EF7368A00984

midterm-exam-93388

#781 Page 10 of 10

SCRAP PAGE

**CHEMISTRY 110 –Midterm Exam****Thursday November 7, 2024, 6:30 – 8:30 pm**

Examiners: Pallavi Sirjoosingh and Ashok Kakkar

First name (please write as legibly as possible within the boxes)

--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--

Last name

--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--

Student ID number

--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--

This booklet has 8 questions on 10 pages (The last page can be used as a scrap page).

The staple must NOT be removed from the question booklet. Your intact question booklet must be returned at the end of the exam.

A separate booklet contains a periodic table, and datasheet.

Write your answers directly under the questions in the space provided. Do not write on the QR codes or near the top edges (as it may be cut off).

Please write **legibly and dark enough** for the scan to be readable. For students who use English as a second language, dictionaries are permitted. Calculators are allowed.

DO NOT ASK QUESTIONS OF THE INVIGILATORS about interpretations of the exam questions. If you have a concern about a question, please summarize it at the top of this page. All concerns will be evaluated after the exam. GOOD LUCK!

**Question 1 8 points** (*Assume all transitions are possible*)

a. 6 points Calculate the **energy (in kJ/mol)** required to ionize (completely remove) the electron from the **n=3** level of **He⁺ ion** (He: Helium). Show your work including equations. (3 sig. figs.)

b. 2 points

(I) Select the largest (longest) wavelength from the three choices below:

- i. Wavelength of photon to *completely remove* an electron from *1s orbital of Hydrogen atom*
- ii. Wavelength of photon to *completely remove* an electron from *1s orbital of Boron atom (B)*
- iii. Wavelength of photon to *completely remove* an electron from *1s orbital of Boron cation (B⁺)*

i or ii or iii

(II) Select the smallest (shortest) wavelength from the three choices below (*for H atom*):

- i. Wavelength of photon *emitted* for an electron transitioning from *2p to 1s orbital*
- ii. Wavelength of photon *absorbed* for an electron transitioning from *2s to 4p orbital*
- iii. Wavelength of photon *absorbed* for an electron transitioning from *1s to 3p orbital*

i or ii or iii



Question 2 8 points



A Chem110 student plots the graph shown on the left, from data obtained for the kinetic energy (eV) of the ejected electron versus frequency (s^{-1}) of the incident light, for an unknown metal. ($1\text{eV} = 1.6022 \times 10^{-19} \text{ J}$)

Based on the plot, answer the questions **a to c** below:

a. 1 point On the plot above, label the **threshold frequency**.

b. 2 points What is the value of the work potential (in J) of the unknown metal?

c. 3 points What is the *longest* wavelength (in nm) required to eject an electron from this metal's surface? Show your work including equation(s) used. (3 sig figs)

d. 2 points In the space provided below, plot the following

(Assume frequency is *above* threshold frequency):

- Kinetic energy* of an ejected electron versus the *intensity* of the incident light (left)
- Number of electrons* ejected versus the *frequency* of the incident light (right)



**Question 3** **6 points****a. 2 points**

In the ground state of Ge (Germanium; $Z = 32$) atom, **name** the following orbitals:

i. Orbital containing the electron that requires the maximum energy to be removed.

ii. Orbital containing the electron that experiences the minimum effective nuclear charge.

b. 2 points

An unknown anion (A^-) has the following *excited* electronic configuration: $[Ar] 4s^2 3d^3 4p^1$

(Ar: Argon; $Z = 18$)

Give the **condensed ground state electronic configuration** of the atom, **A** and the cation, **A^{2+}** .
Show your work. (Assume all electronic transitions are possible)

Electronic Configuration of A:

Electronic Configuration of A^{2+} :

c. 2 points

In a single atom, determine how many electrons can have the following set of quantum numbers?

(i) $n = 5; l = 3$

In a single atom, determine how many electrons can have the following set of quantum numbers?

(i) $n = 3; l = 2; m_s = +1/2$

**Question 4** **8 points**

Rank the following from 1 to 4 in **increasing** order. Provide the ranking in the boxes provided.

Note: Atomic number (Z) are provided for relevant atoms

a) Increasing (lowest to highest) order of *ionization energy*: (1= lowest; 4 = highest)

Mg

Mg: Z = 12

Rb

Rb: Z = 37

Cs

Cs: Z = 55

K

K : Z = 19

b) Increasing (lowest to highest) order of *bond angle*: (1= lowest; 4 = highest)

H₂OCCl₄XeF₄

HCN

c) Increasing (smallest to largest) order of *ionic size* (1= smallest; 4 = largest)

S²⁻

S: Z = 16

O²⁻

O: Z = 8

Be²⁺

Be: Z = 4

Na⁺

Na: Z = 11

d) Increasing (lowest to highest) order of *bond energy* (1= lowest; 4 = highest)

P-F

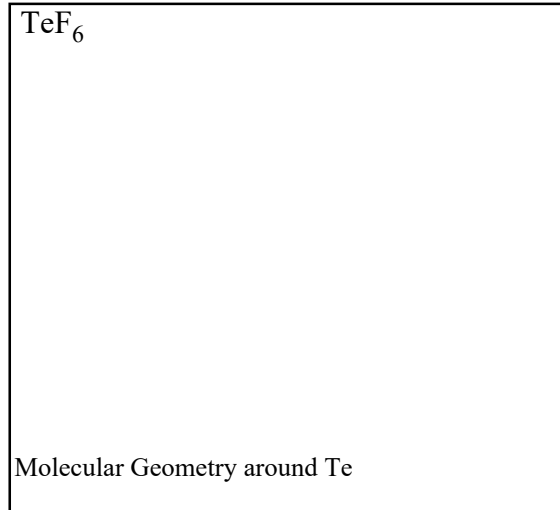
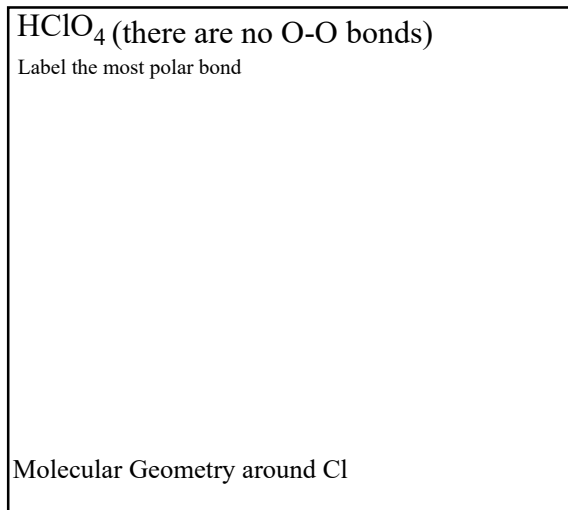
P-Cl

P-I

P-Br

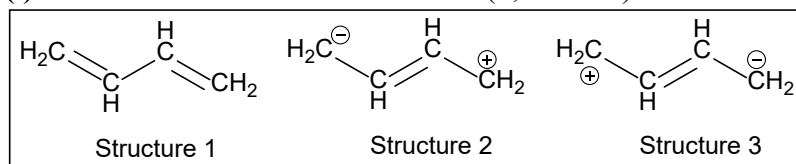
**Question 5** **12 points**

a. 9 points For the molecules HClO_4 and TeF_6 , draw the most stable *Lewis structure* showing appropriate shape and *dashed/wedged bonds*, based on VSEPR. Determine the **molecular geometry** around the central atom. For HClO_4 , label the *most polar bond*.



b. 3 points (Note: Lone pairs are not shown in structures below)

(i) For the three *resonance* structures (1, 2 and 3) shown below, answer the following question:



(I) Circle the *most contributing* resonance structure.

Structure 1

Structure 2

Structure 3

(ii) For the three structures given (4, 5, and 6) below, answer the following questions:



(I) Circle all the structures that represent *resonance structures* of each other:

Structure 4

Structure 5

Structure 6

(II) Among the resonance structures, circle the *most contributing* resonance structure.

Structure 4

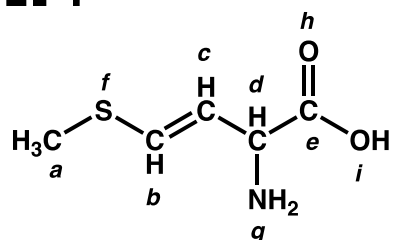
Structure 5

Structure 6

**Question 6** **6 points**

Using the data given below, calculate the **lattice energy** for SrO(s).

	ΔH° (kJ/mol)
$\text{Sr}(s) \longrightarrow \text{Sr}(g)$	+180
$\text{Sr}(g) \longrightarrow \text{Sr}^{2+}(g) + 2e^-$	+1468
$\text{O}_2(g) \longrightarrow 2\text{O}(g)$	+500
$\text{O}(g) + 2e^- \longrightarrow \text{O}^{2-}(g)$	+600
$\text{Sr}(s) + \frac{1}{2}\text{O}_2(g) \longrightarrow \text{SrO}(s)$	-550

**Question 7 10 points**

Consider the molecule given on the left, and answer questions (i) to (vi) using chemical bonding theories that we have learnt in class (valence bond theory, hybridization).

Note that the lone pairs are not shown in the structure:

i. Consider the carbon atoms marked in as ***a***, ***b***, ***c***, ***d***, and ***e*** in the molecule, and determine **hybridization** at each of the carbon centers. (2.5 points)

Atom	Hybridization
Carbon (<i>a</i>)	
Carbon (<i>b</i>)	
Carbon (<i>c</i>)	
Carbon (<i>d</i>)	
Carbon (<i>e</i>)	

ii. Now, consider the sulfur atom (S) marked as ***f***, nitrogen (N) atom (***g***), and oxygen(O) atoms (***h*** and ***i***), and determine their **hybridization**. (2 points)

Atom	Hybridization
Sulfur (<i>f</i>)	
Nitrogen (<i>g</i>)	
Oxygen (<i>h</i>)	
Oxygen (<i>i</i>)	

iii. **Describe** all the bonds formed by the carbon atoms ***a*** and ***b***, using sigma and pi designation as appropriate. (2 points)

Atom	Bonds formed (use sigma and pi as appropriate)
Carbon (<i>a</i>)	
Carbon (<i>b</i>)	

iv. How many total sp^3 -s sigma bonds are in the molecule? (1 point)

v. Determine **electronic** and **molecular geometry** at carbon (C) labeled ***e*** and sulfur (S) labeled ***f*** centres. (2 points)

Atom	Electronic Geometry	Molecular Geometry
Carbon (<i>e</i>)		
Sulfur (<i>f</i>)		

vi. Circle the value from the choices given below, that describes the **C-S-C bond angle at sulfur** (S) marked ***f***: (0.5 points)

$>90^\circ$ to $<109.5^\circ$ 109.5° >109.5 to $<120^\circ$ 120° $>120^\circ$ to $<180^\circ$


Question 8 12 points

Using the **valence shell orbitals only** draw a **complete** molecular orbital (MO) diagram (with no s-p mixing) for the molecule **PS**, in the space provided below. **(6 points)**

Assume that 3s and 3p atomic orbitals will form molecular orbitals like those formed by the combination of 2s and 2p atomic orbitals that we learnt in class, and there is no s-p mixing.

Show **energy arrow**, **atomic orbitals** involved, **atom labels**, the **molecular orbitals** that are formed, **filled electrons** in the MO diagram etc. Make sure to label all the atomic and molecular orbitals.

Using the MO diagram above, answer the following questions:

- (i) Which one of the following will have the highest bond order? Circle the correct choice
 A) PS^{1-} B) PS C) PS^{1+}
- (ii) Which one of the following will have the longest bond length? Circle the correct choice
 A) PS^{1-} B) PS C) PS^{1+}
- (iii) Which of the following will be diamagnetic? Circle the correct choice.
 A) PS^{1-} B) PS C) PS^{1+}
- (iv) Draw Lewis dot structure of the molecule with the shortest bond length (based on the MO diagram) from among the three (PS^{1-} , PS, PS^{1+}), and indicate the bond order calculated from the Lewis dot structure. **(3 points)**

Lewis Dot Structure

Bond Order



16A684E3-24BF-4DD0-B0BD-60A3ED5D0A63

midterm-exam-93388

#782 Page 10 of 10

SCRAP PAGE

**CHEMISTRY 110 –Midterm Exam****Thursday November 7, 2024, 6:30 – 8:30 pm**

Examiners: Pallavi Sirjoosingh and Ashok Kakkar

First name (please write as legibly as possible within the boxes)

--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--

Last name

--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--

Student ID number

--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--

This booklet has 8 questions on 10 pages (The last page can be used as a scrap page).

The staple must NOT be removed from the question booklet. Your intact question booklet must be returned at the end of the exam.

A separate booklet contains a periodic table, and datasheet.

Write your answers directly under the questions in the space provided. Do not write on the QR codes or near the top edges (as it may be cut off).

Please write **legibly and dark enough** for the scan to be readable. For students who use English as a second language, dictionaries are permitted. Calculators are allowed.

DO NOT ASK QUESTIONS OF THE INVIGILATORS about interpretations of the exam questions. If you have a concern about a question, please summarize it at the top of this page. All concerns will be evaluated after the exam. GOOD LUCK!

**Question 1 8 points** (*Assume all transitions are possible*)

a. 6 points Calculate the **energy (in kJ/mol)** required to ionize (completely remove) the electron from the **$n=3$** level of **He^+ ion** (He: Helium). Show your work including equations. (3 sig. figs.)

b. 2 points

(I) Select the largest (longest) wavelength from the three choices below:

- i. Wavelength of photon to *completely remove* an electron from *1s orbital of Hydrogen atom*
- ii. Wavelength of photon to *completely remove* an electron from *1s orbital of Boron atom (B)*
- iii. Wavelength of photon to *completely remove* an electron from *1s orbital of Boron cation (B^+)*

i or ii or iii

(II) Select the smallest (shortest) wavelength from the three choices below (*for H atom*):

- i. Wavelength of photon *emitted* for an electron transitioning from *2p to 1s orbital*
- ii. Wavelength of photon *absorbed* for an electron transitioning from *2s to 4p orbital*
- iii. Wavelength of photon *absorbed* for an electron transitioning from *1s to 3p orbital*

i or ii or iii



Question 2 8 points



A Chem110 student plots the graph shown on the left, from data obtained for the kinetic energy (eV) of the ejected electron versus frequency (s^{-1}) of the incident light, for an unknown metal. ($1\text{eV} = 1.6022 \times 10^{-19} \text{ J}$)

Based on the plot, answer the questions **a to c** below:

a. 1 point On the plot above, label the **threshold frequency**.

b. 2 points What is the value of the work potential (in J) of the unknown metal?

c. 3 points What is the *longest* wavelength (in nm) required to eject an electron from this metal's surface? Show your work including equation(s) used. (3 sig figs)

d. 2 points In the space provided below, plot the following

(Assume frequency is *above* threshold frequency):

- i. *Kinetic energy* of an ejected electron versus the *intensity* of the incident light (left)
- ii. *Number of electrons* ejected versus the *frequency* of the incident light (right)



**Question 3** **6 points****a. 2 points**

In the ground state of Ge (Germanium; $Z = 32$) atom, **name** the following orbitals:

i. Orbital containing the electron that requires the maximum energy to be removed.

ii. Orbital containing the electron that experiences the minimum effective nuclear charge.

b. 2 points

An unknown anion (A^-) has the following *excited* electronic configuration: $[Ar] 4s^2 3d^3 4p^1$

(Ar: Argon; $Z = 18$)

Give the **condensed ground state electronic configuration** of the atom, **A** and the cation, **A^{2+}** .
Show your work. (Assume all electronic transitions are possible)

Electronic Configuration of A:

Electronic Configuration of A^{2+} :

c. 2 points

In a single atom, determine how many electrons can have the following set of quantum numbers?

(i) $n = 5; l = 3$

In a single atom, determine how many electrons can have the following set of quantum numbers?

(i) $n = 3; l = 2; m_s = +1/2$

**Question 4** **8 points**

Rank the following from 1 to 4 in **increasing** order. Provide the ranking in the boxes provided.

Note: Atomic number (Z) are provided for relevant atoms

a) Increasing (lowest to highest) order of *ionization energy*: (1= lowest; 4 = highest)

Mg

Mg: Z = 12

Rb

Rb: Z = 37

Cs

Cs: Z = 55

K

K : Z = 19

b) Increasing (lowest to highest) order of *bond angle*: (1= lowest; 4 = highest)

H₂OCCl₄XeF₄

HCN

c) Increasing (smallest to largest) order of *ionic size* (1= smallest; 4 = largest)

S²⁻

S: Z = 16

O²⁻

O: Z = 8

Be²⁺

Be: Z = 4

Na⁺

Na: Z = 11

d) Increasing (lowest to highest) order of *bond energy* (1= lowest; 4 = highest)

P-F

P-Cl

P-I

P-Br

**Question 5** **12 points**

a. 9 points For the molecules HClO_4 and TeF_6 , draw the most stable *Lewis structure* showing appropriate shape and *dashed/wedged bonds*, based on VSEPR. Determine the **molecular geometry** around the central atom. For HClO_4 , label the *most polar bond*.



b. 3 points (Note: Lone pairs are not shown in structures below)

(i) For the three *resonance* structures (1, 2 and 3) shown below, answer the following question:



(I) Circle the *most contributing* resonance structure.

Structure 1

Structure 2

Structure 3

(ii) For the three structures given (4, 5, and 6) below, answer the following questions:



(I) Circle all the structures that represent *resonance structures* of each other:

Structure 4

Structure 5

Structure 6

(II) Among the resonance structures, circle the *most contributing* resonance structure.

Structure 4

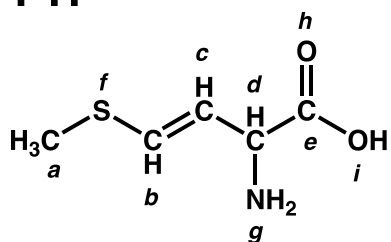
Structure 5

Structure 6

**Question 6** **6 points**

Using the data given below, calculate the **lattice energy** for SrO(s).

	ΔH° (kJ/mol)
$\text{Sr}(s) \longrightarrow \text{Sr}(g)$	+180
$\text{Sr}(g) \longrightarrow \text{Sr}^{2+}(g) + 2e^-$	+1468
$\text{O}_2(g) \longrightarrow 2\text{O}(g)$	+500
$\text{O}(g) + 2e^- \longrightarrow \text{O}^{2-}(g)$	+600
$\text{Sr}(s) + \frac{1}{2}\text{O}_2(g) \longrightarrow \text{SrO}(s)$	-550

**Question 7 10 points**

Consider the molecule given on the left, and answer questions (i) to (vi) using chemical bonding theories that we have learnt in class (valence bond theory, hybridization).

Note that the lone pairs are not shown in the structure:

i. Consider the carbon atoms marked in as ***a***, ***b***, ***c***, ***d***, and ***e*** in the molecule, and determine **hybridization** at each of the carbon centers. (2.5 points)

Atom	Hybridization
Carbon (<i>a</i>)	
Carbon (<i>b</i>)	
Carbon (<i>c</i>)	
Carbon (<i>d</i>)	
Carbon (<i>e</i>)	

ii. Now, consider the sulfur atom (S) marked as ***f***, nitrogen (N) atom (***g***), and oxygen(O) atoms (***h*** and ***i***), and determine their **hybridization**. (2 points)

Atom	Hybridization
Sulfur (<i>f</i>)	
Nitrogen (<i>g</i>)	
Oxygen (<i>h</i>)	
Oxygen (<i>i</i>)	

iii. **Describe** all the bonds formed by the carbon atoms ***a*** and ***b***, using sigma and pi designation as appropriate. (2 points)

Atom	Bonds formed (use sigma and pi as appropriate)
Carbon (<i>a</i>)	
Carbon (<i>b</i>)	

iv. How many total sp^3 -s sigma bonds are in the molecule? (1 point)

v. Determine **electronic** and **molecular geometry** at carbon (C) labeled ***e*** and sulfur (S) labeled ***f*** centres. (2 points)

Atom	Electronic Geometry	Molecular Geometry
Carbon (<i>e</i>)		
Sulfur (<i>f</i>)		

vi. Circle the value from the choices given below, that describes the **C-S-C bond angle at sulfur** (S) marked ***f***: (0.5 points)

>90° to <109.5°

109.5°

>109.5 to <120°

120°

>120° to <180°

**Question 8 12 points**

Using the **valence shell orbitals only** draw a **complete** molecular orbital (MO) diagram (with no s-p mixing) for the molecule **PS**, in the space provided below. **(6 points)**

Assume that 3s and 3p atomic orbitals will form molecular orbitals like those formed by the combination of 2s and 2p atomic orbitals that we learnt in class, and there is no s-p mixing.

Show **energy arrow**, **atomic orbitals** involved, **atom labels**, the **molecular orbitals** that are formed, **filled electrons** in the MO diagram etc. Make sure to label all the atomic and molecular orbitals.

Using the MO diagram above, answer the following questions:

- (i) Which one of the following will have the highest bond order? Circle the correct choice
A) PS^{1-} B) PS C) PS^{1+}
- (ii) Which one of the following will have the longest bond length? Circle the correct choice
A) PS^{1-} B) PS C) PS^{1+}
- (iii) Which of the following will be diamagnetic? Circle the correct choice.
A) PS^{1-} B) PS C) PS^{1+}
- (iv) Draw Lewis dot structure of the molecule with the shortest bond length (based on the MO diagram) from among the three (PS^{1-} , PS, PS^{1+}), and indicate the bond order calculated from the Lewis dot structure. **(3 points)**

Lewis Dot Structure

Bond Order



66855666-35AE-462E-8D71-DC4D8E7A4D3B

midterm-exam-93388

#783 Page 10 of 10

SCRAP PAGE

**CHEMISTRY 110 –Midterm Exam****Thursday November 7, 2024, 6:30 – 8:30 pm**

Examiners: Pallavi Sirjoosingh and Ashok Kakkar

First name (please write as legibly as possible within the boxes)

--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--

Last name

--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--

Student ID number

--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--

This booklet has 8 questions on 10 pages (The last page can be used as a scrap page).

The staple must NOT be removed from the question booklet. Your intact question booklet must be returned at the end of the exam.

A separate booklet contains a periodic table, and datasheet.

Write your answers directly under the questions in the space provided. Do not write on the QR codes or near the top edges (as it may be cut off).

Please write **legibly and dark enough** for the scan to be readable. For students who use English as a second language, dictionaries are permitted. Calculators are allowed.

DO NOT ASK QUESTIONS OF THE INVIGILATORS about interpretations of the exam questions. If you have a concern about a question, please summarize it at the top of this page. All concerns will be evaluated after the exam. GOOD LUCK!

**Question 1 8 points** (*Assume all transitions are possible*)

a. 6 points Calculate the **energy (in kJ/mol)** required to ionize (completely remove) the electron from the **n=3** level of **He⁺ ion** (He: Helium). Show your work including equations. (3 sig. figs.)

b. 2 points

(I) Select the largest (longest) wavelength from the three choices below:

- i. Wavelength of photon to *completely remove* an electron from *1s orbital of Hydrogen atom*
- ii. Wavelength of photon to *completely remove* an electron from *1s orbital of Boron atom (B)*
- iii. Wavelength of photon to *completely remove* an electron from *1s orbital of Boron cation (B⁺)*

i or ii or iii

(II) Select the smallest (shortest) wavelength from the three choices below (*for H atom*):

- i. Wavelength of photon *emitted* for an electron transitioning from *2p to 1s orbital*
- ii. Wavelength of photon *absorbed* for an electron transitioning from *2s to 4p orbital*
- iii. Wavelength of photon *absorbed* for an electron transitioning from *1s to 3p orbital*

i or ii or iii



Question 2 8 points



A Chem110 student plots the graph shown on the left, from data obtained for the kinetic energy (eV) of the ejected electron versus frequency (s^{-1}) of the incident light, for an unknown metal. ($1\text{eV} = 1.6022 \times 10^{-19} \text{ J}$)

Based on the plot, answer the questions **a to c** below:

a. 1 point On the plot above, label the **threshold frequency**.

b. 2 points What is the value of the work potential (in J) of the unknown metal?

c. 3 points What is the *longest* wavelength (in nm) required to eject an electron from this metal's surface? Show your work including equation(s) used. (3 sig figs)

d. 2 points In the space provided below, plot the following

(Assume frequency is *above* threshold frequency):

- i. *Kinetic energy* of an ejected electron versus the *intensity* of the incident light (left)
- ii. *Number of electrons* ejected versus the *frequency* of the incident light (right)



**Question 3** **6 points****a. 2 points**

In the ground state of Ge (Germanium; $Z = 32$) atom, **name** the following orbitals:

i. Orbital containing the electron that requires the maximum energy to be removed.

ii. Orbital containing the electron that experiences the minimum effective nuclear charge.

b. 2 points

An unknown anion (A^-) has the following *excited* electronic configuration: $[Ar] 4s^2 3d^3 4p^1$

(Ar: Argon; $Z = 18$)

Give the **condensed ground state electronic configuration** of the atom, **A** and the cation, **A^{2+}** .
Show your work. (Assume all electronic transitions are possible)

Electronic Configuration of A:

Electronic Configuration of A^{2+} :

c. 2 points

In a single atom, determine how many electrons can have the following set of quantum numbers?

(i) $n = 5; l = 3$

In a single atom, determine how many electrons can have the following set of quantum numbers?

(i) $n = 3; l = 2; m_s = +1/2$

**Question 4** **8 points**

Rank the following from 1 to 4 in **increasing** order. Provide the ranking in the boxes provided.

Note: Atomic number (Z) are provided for relevant atoms

a) Increasing (lowest to highest) order of *ionization energy*: (1= lowest; 4 = highest)

Mg

Mg: Z = 12

Rb

Rb: Z = 37

Cs

Cs: Z = 55

K

K : Z = 19

b) Increasing (lowest to highest) order of *bond angle*: (1= lowest; 4 = highest)

H₂OCCl₄XeF₄

HCN

c) Increasing (smallest to largest) order of *ionic size* (1= smallest; 4 = largest)

S²⁻

S: Z = 16

O²⁻

O: Z = 8

Be²⁺

Be: Z = 4

Na⁺

Na: Z = 11

d) Increasing (lowest to highest) order of *bond energy* (1= lowest; 4 = highest)

P-F

P-Cl

P-I

P-Br

**Question 5** **12 points**

a. 9 points For the molecules HClO_4 and TeF_6 , draw the most stable *Lewis structure* showing appropriate shape and *dashed/wedged bonds*, based on VSEPR. Determine the **molecular geometry** around the central atom. For HClO_4 , label the *most polar bond*.



b. 3 points (Note: Lone pairs are not shown in structures below)

(i) For the three *resonance* structures (1, 2 and 3) shown below, answer the following question:



(I) Circle the *most contributing* resonance structure.

Structure 1

Structure 2

Structure 3

(ii) For the three structures given (4, 5, and 6) below, answer the following questions:



(I) Circle all the structures that represent *resonance structures* of each other:

Structure 4

Structure 5

Structure 6

(II) Among the resonance structures, circle the *most contributing* resonance structure.

Structure 4

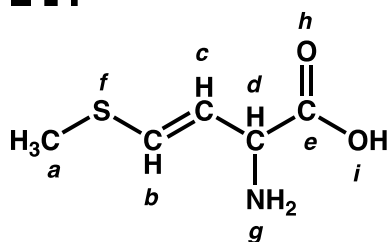
Structure 5

Structure 6

**Question 6** **6 points**

Using the data given below, calculate the **lattice energy** for SrO(s).

	ΔH° (kJ/mol)
$\text{Sr}(s) \longrightarrow \text{Sr}(g)$	+180
$\text{Sr}(g) \longrightarrow \text{Sr}^{2+}(g) + 2e^-$	+1468
$\text{O}_2(g) \longrightarrow 2\text{O}(g)$	+500
$\text{O}(g) + 2e^- \longrightarrow \text{O}^{2-}(g)$	+600
$\text{Sr}(s) + \frac{1}{2}\text{O}_2(g) \longrightarrow \text{SrO}(s)$	-550

**Question 7 10 points**

Consider the molecule given on the left, and answer questions (i) to (vi) using chemical bonding theories that we have learnt in class (valence bond theory, hybridization).

Note that the lone pairs are not shown in the structure:

i. Consider the carbon atoms marked in as ***a***, ***b***, ***c***, ***d***, and ***e*** in the molecule, and determine **hybridization** at each of the carbon centers. (2.5 points)

Atom	Hybridization
Carbon (<i>a</i>)	
Carbon (<i>b</i>)	
Carbon (<i>c</i>)	
Carbon (<i>d</i>)	
Carbon (<i>e</i>)	

ii. Now, consider the sulfur atom (S) marked as ***f***, nitrogen (N) atom (***g***), and oxygen(O) atoms (***h*** and ***i***), and determine their **hybridization**. (2 points)

Atom	Hybridization
Sulfur (<i>f</i>)	
Nitrogen (<i>g</i>)	
Oxygen (<i>h</i>)	
Oxygen (<i>i</i>)	

iii. **Describe** all the bonds formed by the carbon atoms ***a*** and ***b***, using sigma and pi designation as appropriate. (2 points)

Atom	Bonds formed (use sigma and pi as appropriate)
Carbon (<i>a</i>)	
Carbon (<i>b</i>)	

iv. How many total sp^3 -s sigma bonds are in the molecule? (1 point)

v. Determine **electronic** and **molecular geometry** at carbon (C) labeled ***e*** and sulfur (S) labeled ***f*** centres. (2 points)

Atom	Electronic Geometry	Molecular Geometry
Carbon (<i>e</i>)		
Sulfur (<i>f</i>)		

vi. Circle the value from the choices given below, that describes the **C-S-C bond angle at sulfur** (S) marked ***f***: (0.5 points)

>90° to <109.5°

109.5°

>109.5 to <120°

120°

>120° to <180°



Question 8 12 points

Using the **valence shell orbitals only** draw a **complete** molecular orbital (MO) diagram (with no s-p mixing) for the molecule **PS**, in the space provided below. **(6 points)**

Assume that 3s and 3p atomic orbitals will form molecular orbitals like those formed by the combination of 2s and 2p atomic orbitals that we learnt in class, and there is no s-p mixing.

Show **energy arrow**, **atomic orbitals** involved, **atom labels**, the **molecular orbitals** that are formed, **filled electrons** in the MO diagram etc. Make sure to label all the atomic and molecular orbitals.

Using the MO diagram above, answer the following questions:

- (i) Which one of the following will have the highest bond order? Circle the correct choice
 A) PS^{1-} B) PS C) PS^{1+}
- (ii) Which one of the following will have the longest bond length? Circle the correct choice
 A) PS^{1-} B) PS C) PS^{1+}
- (iii) Which of the following will be diamagnetic? Circle the correct choice.
 A) PS^{1-} B) PS C) PS^{1+}
- (iv) Draw Lewis dot structure of the molecule with the shortest bond length (based on the MO diagram) from among the three (PS^{1-} , PS, PS^{1+}), and indicate the bond order calculated from the Lewis dot structure. **(3 points)**

Lewis Dot Structure

Bond Order



9329DC90-A64D-4C41-9FF4-969D85CCD85D

midterm-exam-93388

#784 Page 10 of 10

SCRAP PAGE

**CHEMISTRY 110 –Midterm Exam****Thursday November 7, 2024, 6:30 – 8:30 pm****Examiners: Pallavi Sirjoosingh and Ashok Kakkar**

First name (please write as legibly as possible within the boxes)

--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--

Last name

--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--

Student ID number

--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--

This booklet has 8 questions on 10 pages (The last page can be used as a scrap page).

The staple must NOT be removed from the question booklet. Your intact question booklet must be returned at the end of the exam.

A separate booklet contains a periodic table, and datasheet.

Write your answers directly under the questions in the space provided. Do not write on the QR codes or near the top edges (as it may be cut off).

Please write **legibly and dark enough** for the scan to be readable. For students who use English as a second language, dictionaries are permitted. Calculators are allowed.

DO NOT ASK QUESTIONS OF THE INVIGILATORS about interpretations of the exam questions. If you have a concern about a question, please summarize it at the top of this page. All concerns will be evaluated after the exam. GOOD LUCK!

**Question 1 8 points** (*Assume all transitions are possible*)

a. 6 points Calculate the **energy (in kJ/mol)** required to ionize (completely remove) the electron from the **n=3** level of **He⁺ ion** (He: Helium). Show your work including equations. (3 sig. figs.)

b. 2 points

(I) Select the largest (longest) wavelength from the three choices below:

- i. Wavelength of photon to *completely remove* an electron from *1s orbital of Hydrogen atom*
- ii. Wavelength of photon to *completely remove* an electron from *1s orbital of Boron atom (B)*
- iii. Wavelength of photon to *completely remove* an electron from *1s orbital of Boron cation (B⁺)*

i or ii or iii

(II) Select the smallest (shortest) wavelength from the three choices below (*for H atom*):

- i. Wavelength of photon *emitted* for an electron transitioning from *2p to 1s orbital*
- ii. Wavelength of photon *absorbed* for an electron transitioning from *2s to 4p orbital*
- iii. Wavelength of photon *absorbed* for an electron transitioning from *1s to 3p orbital*

i or ii or iii



Question 2 8 points



A Chem110 student plots the graph shown on the left, from data obtained for the kinetic energy (eV) of the ejected electron versus frequency (s^{-1}) of the incident light, for an unknown metal. ($1\text{eV} = 1.6022 \times 10^{-19} \text{ J}$)

Based on the plot, answer the questions **a to c** below:

a. 1 point On the plot above, label the **threshold frequency**.

b. 2 points What is the value of the work potential (in J) of the unknown metal?

c. 3 points What is the *longest* wavelength (in nm) required to eject an electron from this metal's surface? Show your work including equation(s) used. (3 sig figs)

d. 2 points In the space provided below, plot the following

(Assume frequency is *above* threshold frequency):

- Kinetic energy* of an ejected electron versus the *intensity* of the incident light (left)
- Number of electrons* ejected versus the *frequency* of the incident light (right)



**Question 3** **6 points****a. 2 points**

In the ground state of Ge (Germanium; $Z = 32$) atom, **name** the following orbitals:

i. Orbital containing the electron that requires the maximum energy to be removed.

ii. Orbital containing the electron that experiences the minimum effective nuclear charge.

b. 2 points

An unknown anion (A^-) has the following *excited* electronic configuration: $[Ar] 4s^2 3d^3 4p^1$

(Ar: Argon; $Z = 18$)

Give the **condensed ground state electronic configuration** of the atom, **A** and the cation, **A^{2+}** .
Show your work. (Assume all electronic transitions are possible)

Electronic Configuration of A:

Electronic Configuration of A^{2+} :

c. 2 points

In a single atom, determine how many electrons can have the following set of quantum numbers?

(i) $n = 5; l = 3$

In a single atom, determine how many electrons can have the following set of quantum numbers?

(i) $n = 3; l = 2; m_s = +1/2$

**Question 4** **8 points**

Rank the following from 1 to 4 in **increasing** order. Provide the ranking in the boxes provided.

Note: Atomic number (Z) are provided for relevant atoms

a) Increasing (lowest to highest) order of *ionization energy*: (1= lowest; 4 = highest)

Mg

Mg: Z = 12

Rb

Rb: Z = 37

Cs

Cs: Z = 55

K

K : Z = 19

b) Increasing (lowest to highest) order of *bond angle*: (1= lowest; 4 = highest)

H₂OCCl₄XeF₄

HCN

c) Increasing (smallest to largest) order of *ionic size* (1= smallest; 4 = largest)

S²⁻

S: Z = 16

O²⁻

O: Z = 8

Be²⁺

Be: Z = 4

Na⁺

Na: Z = 11

d) Increasing (lowest to highest) order of *bond energy* (1= lowest; 4 = highest)

P-F

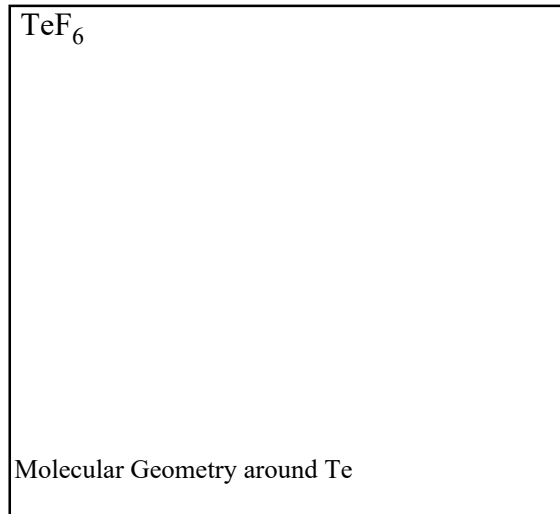
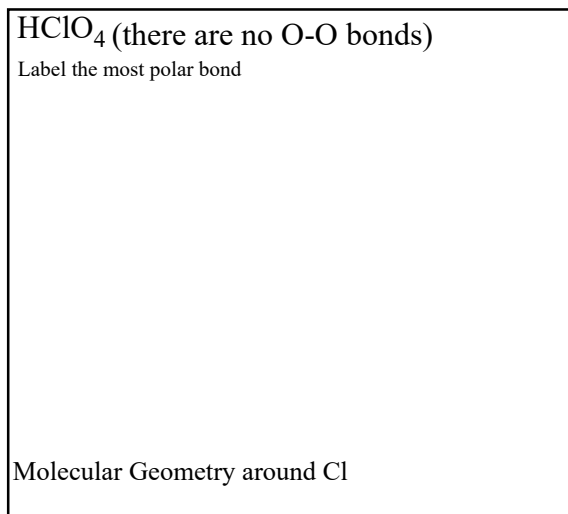
P-Cl

P-I

P-Br

**Question 5** **12 points**

a. 9 points For the molecules HClO_4 and TeF_6 , draw the most stable *Lewis structure* showing appropriate shape and *dashed/wedged bonds*, based on VSEPR. Determine the **molecular geometry** around the central atom. For HClO_4 , label the *most polar bond*.



b. 3 points (Note: Lone pairs are not shown in structures below)

(i) For the three *resonance* structures (1, 2 and 3) shown below, answer the following question:



(I) Circle the *most contributing* resonance structure.

Structure 1

Structure 2

Structure 3

(ii) For the three structures given (4, 5, and 6) below, answer the following questions:



(I) Circle all the structures that represent *resonance structures* of each other:

Structure 4

Structure 5

Structure 6

(II) Among the resonance structures, circle the *most contributing* resonance structure.

Structure 4

Structure 5

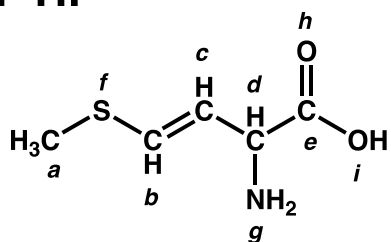
Structure 6



Question 6 6 points

Using the data given below, calculate the **lattice energy** for SrO(s).

	ΔH° (kJ/mol)
$\text{Sr}(s) \longrightarrow \text{Sr}(g)$	+180
$\text{Sr}(g) \longrightarrow \text{Sr}^{2+}(g) + 2e^-$	+1468
$\text{O}_2(g) \longrightarrow 2\text{O}(g)$	+500
$\text{O}(g) + 2e^- \longrightarrow \text{O}^{2-}(g)$	+600
$\text{Sr}(s) + \frac{1}{2}\text{O}_2(g) \longrightarrow \text{SrO}(s)$	-550

**Question 7 10 points**

Consider the molecule given on the left, and answer questions (i) to (vi) using chemical bonding theories that we have learnt in class (valence bond theory, hybridization).

Note that the lone pairs are not shown in the structure:

i. Consider the carbon atoms marked in as ***a***, ***b***, ***c***, ***d***, and ***e*** in the molecule, and determine **hybridization** at each of the carbon centers. (2.5 points)

Atom	Hybridization
Carbon (<i>a</i>)	
Carbon (<i>b</i>)	
Carbon (<i>c</i>)	
Carbon (<i>d</i>)	
Carbon (<i>e</i>)	

ii. Now, consider the sulfur atom (S) marked as ***f***, nitrogen (N) atom (***g***), and oxygen(O) atoms (***h*** and ***i***), and determine their **hybridization**. (2 points)

Atom	Hybridization
Sulfur (<i>f</i>)	
Nitrogen (<i>g</i>)	
Oxygen (<i>h</i>)	
Oxygen (<i>i</i>)	

iii. **Describe** all the bonds formed by the carbon atoms ***a*** and ***b***, using sigma and pi designation as appropriate. (2 points)

Atom	Bonds formed (use sigma and pi as appropriate)
Carbon (<i>a</i>)	
Carbon (<i>b</i>)	

iv. How many total sp^3 -s sigma bonds are in the molecule? (1 point)

v. Determine **electronic** and **molecular geometry** at carbon (C) labeled ***e*** and sulfur (S) labeled ***f*** centres. (2 points)

Atom	Electronic Geometry	Molecular Geometry
Carbon (<i>e</i>)		
Sulfur (<i>f</i>)		

vi. Circle the value from the choices given below, that describes the **C-S-C bond angle at sulfur** (S) marked ***f***: (0.5 points)

>90° to <109.5° 109.5° >109.5 to <120° 120° >120° to <180°


Question 8 12 points

Using the **valence shell orbitals only** draw a **complete** molecular orbital (MO) diagram (with no s-p mixing) for the molecule **PS**, in the space provided below. **(6 points)**

Assume that 3s and 3p atomic orbitals will form molecular orbitals like those formed by the combination of 2s and 2p atomic orbitals that we learnt in class, and there is no s-p mixing.

Show **energy arrow**, **atomic orbitals** involved, **atom labels**, the **molecular orbitals** that are formed, **filled electrons** in the MO diagram etc. Make sure to label all the atomic and molecular orbitals.

Using the MO diagram above, answer the following questions:

- (i) Which one of the following will have the highest bond order? Circle the correct choice
 A) PS^{1-} B) PS C) PS^{1+}
- (ii) Which one of the following will have the longest bond length? Circle the correct choice
 A) PS^{1-} B) PS C) PS^{1+}
- (iii) Which of the following will be diamagnetic? Circle the correct choice.
 A) PS^{1-} B) PS C) PS^{1+}
- (iv) Draw Lewis dot structure of the molecule with the shortest bond length (based on the MO diagram) from among the three (PS^{1-} , PS, PS^{1+}), and indicate the bond order calculated from the Lewis dot structure. **(3 points)**

Lewis Dot Structure

Bond Order



4B793BC3-5F04-4F04-9565-6FFB72DE56D8

midterm-exam-93388

#785 Page 10 of 10

SCRAP PAGE

**CHEMISTRY 110 –Midterm Exam****Thursday November 7, 2024, 6:30 – 8:30 pm**

Examiners: Pallavi Sirjoosingh and Ashok Kakkar

First name (please write as legibly as possible within the boxes)

--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--

Last name

--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--

Student ID number

--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--

This booklet has 8 questions on 10 pages (The last page can be used as a scrap page).

The staple must NOT be removed from the question booklet. Your intact question booklet must be returned at the end of the exam.

A separate booklet contains a periodic table, and datasheet.

Write your answers directly under the questions in the space provided. Do not write on the QR codes or near the top edges (as it may be cut off).

Please write **legibly and dark enough** for the scan to be readable. For students who use English as a second language, dictionaries are permitted. Calculators are allowed.

DO NOT ASK QUESTIONS OF THE INVIGILATORS about interpretations of the exam questions. If you have a concern about a question, please summarize it at the top of this page. All concerns will be evaluated after the exam. GOOD LUCK!

**Question 1 8 points** (*Assume all transitions are possible*)

a. 6 points Calculate the **energy (in kJ/mol)** required to ionize (completely remove) the electron from the **$n=3$** level of **He^+ ion** (He: Helium). Show your work including equations. (3 sig. figs.)

b. 2 points

(I) Select the largest (longest) wavelength from the three choices below:

- i. Wavelength of photon to *completely remove* an electron from *1s orbital of Hydrogen atom*
- ii. Wavelength of photon to *completely remove* an electron from *1s orbital of Boron atom (B)*
- iii. Wavelength of photon to *completely remove* an electron from *1s orbital of Boron cation (B^+)*

i or ii or iii

(II) Select the smallest (shortest) wavelength from the three choices below (*for H atom*):

- i. Wavelength of photon *emitted* for an electron transitioning from *2p to 1s orbital*
- ii. Wavelength of photon *absorbed* for an electron transitioning from *2s to 4p orbital*
- iii. Wavelength of photon *absorbed* for an electron transitioning from *1s to 3p orbital*

i or ii or iii



Question 2 8 points



A Chem110 student plots the graph shown on the left, from data obtained for the kinetic energy (eV) of the ejected electron versus frequency (s^{-1}) of the incident light, for an unknown metal. ($1\text{eV} = 1.6022 \times 10^{-19} \text{ J}$)

Based on the plot, answer the questions **a to c** below:

a. 1 point On the plot above, label the **threshold frequency**.

b. 2 points What is the value of the work potential (in J) of the unknown metal?

c. 3 points What is the *longest* wavelength (in nm) required to eject an electron from this metal's surface? Show your work including equation(s) used. (3 sig figs)

d. 2 points In the space provided below, plot the following

(Assume frequency is *above* threshold frequency):

- Kinetic energy* of an ejected electron versus the *intensity* of the incident light (left)
- Number of electrons* ejected versus the *frequency* of the incident light (right)



**Question 3** **6 points****a. 2 points**

In the ground state of Ge (Germanium; $Z = 32$) atom, **name** the following orbitals:

i. Orbital containing the electron that requires the maximum energy to be removed.

ii. Orbital containing the electron that experiences the minimum effective nuclear charge.

b. 2 points

An unknown anion (A^-) has the following *excited* electronic configuration: $[Ar] 4s^2 3d^3 4p^1$

(Ar: Argon; $Z = 18$)

Give the **condensed ground state electronic configuration** of the atom, **A** and the cation, **A^{2+}** .
Show your work. (Assume all electronic transitions are possible)

Electronic Configuration of A:

Electronic Configuration of A^{2+} :

c. 2 points

In a single atom, determine how many electrons can have the following set of quantum numbers?

(i) $n = 5; l = 3$

In a single atom, determine how many electrons can have the following set of quantum numbers?

(i) $n = 3; l = 2; m_s = +1/2$

**Question 4** **8 points**

Rank the following from 1 to 4 in **increasing** order. Provide the ranking in the boxes provided.

Note: Atomic number (Z) are provided for relevant atoms

a) Increasing (lowest to highest) order of *ionization energy*: (1= lowest; 4 = highest)

Mg

Mg: Z = 12

Rb

Rb: Z = 37

Cs

Cs: Z = 55

K

K : Z = 19

b) Increasing (lowest to highest) order of *bond angle*: (1= lowest; 4 = highest)

H₂OCCl₄XeF₄

HCN

c) Increasing (smallest to largest) order of *ionic size* (1= smallest; 4 = largest)

S²⁻

S: Z = 16

O²⁻

O: Z = 8

Be²⁺

Be: Z = 4

Na⁺

Na: Z = 11

d) Increasing (lowest to highest) order of *bond energy* (1= lowest; 4 = highest)

P-F

P-Cl

P-I

P-Br

**Question 5** **12 points**

a. 9 points For the molecules HClO_4 and TeF_6 , draw the most stable *Lewis structure* showing appropriate shape and *dashed/wedged bonds*, based on VSEPR. Determine the **molecular geometry** around the central atom. For HClO_4 , label the *most polar bond*.



b. 3 points (Note: Lone pairs are not shown in structures below)

(i) For the three *resonance* structures (1, 2 and 3) shown below, answer the following question:



(I) Circle the *most contributing* resonance structure.

Structure 1

Structure 2

Structure 3

(ii) For the three structures given (4, 5, and 6) below, answer the following questions:



(I) Circle all the structures that represent *resonance structures* of each other:

Structure 4

Structure 5

Structure 6

(II) Among the resonance structures, circle the *most contributing* resonance structure.

Structure 4

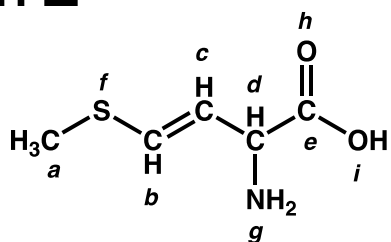
Structure 5

Structure 6

**Question 6** **6 points**

Using the data given below, calculate the **lattice energy** for SrO(s).

	ΔH° (kJ/mol)
$\text{Sr}(s) \longrightarrow \text{Sr}(g)$	+180
$\text{Sr}(g) \longrightarrow \text{Sr}^{2+}(g) + 2e^-$	+1468
$\text{O}_2(g) \longrightarrow 2\text{O}(g)$	+500
$\text{O}(g) + 2e^- \longrightarrow \text{O}^{2-}(g)$	+600
$\text{Sr}(s) + \frac{1}{2}\text{O}_2(g) \longrightarrow \text{SrO}(s)$	-550

**Question 7 10 points**

Consider the molecule given on the left, and answer questions (i) to (vi) using chemical bonding theories that we have learnt in class (valence bond theory, hybridization).

Note that the lone pairs are not shown in the structure:

i. Consider the carbon atoms marked in as ***a***, ***b***, ***c***, ***d***, and ***e*** in the molecule, and determine **hybridization** at each of the carbon centers. (2.5 points)

Atom	Hybridization
Carbon (<i>a</i>)	
Carbon (<i>b</i>)	
Carbon (<i>c</i>)	
Carbon (<i>d</i>)	
Carbon (<i>e</i>)	

ii. Now, consider the sulfur atom (S) marked as ***f***, nitrogen (N) atom (***g***), and oxygen(O) atoms (***h*** and ***i***), and determine their **hybridization**. (2 points)

Atom	Hybridization
Sulfur (<i>f</i>)	
Nitrogen (<i>g</i>)	
Oxygen (<i>h</i>)	
Oxygen (<i>i</i>)	

iii. **Describe** all the bonds formed by the carbon atoms ***a*** and ***b***, using sigma and pi designation as appropriate. (2 points)

Atom	Bonds formed (use sigma and pi as appropriate)
Carbon (<i>a</i>)	
Carbon (<i>b</i>)	

iv. How many total sp^3 -s sigma bonds are in the molecule? (1 point)

v. Determine **electronic** and **molecular geometry** at carbon (C) labeled ***e*** and sulfur (S) labeled ***f*** centres. (2 points)

Atom	Electronic Geometry	Molecular Geometry
Carbon (<i>e</i>)		
Sulfur (<i>f</i>)		

vi. Circle the value from the choices given below, that describes the **C-S-C bond angle at sulfur** (S) marked ***f***: (0.5 points)

>90° to <109.5° 109.5° >109.5 to <120° 120° >120° to <180°

**Question 8 12 points**

Using the **valence shell orbitals only** draw a **complete** molecular orbital (MO) diagram (with no s-p mixing) for the molecule **PS**, in the space provided below. **(6 points)**

Assume that 3s and 3p atomic orbitals will form molecular orbitals like those formed by the combination of 2s and 2p atomic orbitals that we learnt in class, and there is no s-p mixing.

Show **energy arrow**, **atomic orbitals** involved, **atom labels**, the **molecular orbitals** that are formed, **filled electrons** in the MO diagram etc. Make sure to label all the atomic and molecular orbitals.

Using the MO diagram above, answer the following questions:

- (i) Which one of the following will have the highest bond order? Circle the correct choice
A) PS^{1-} B) PS C) PS^{1+}
- (ii) Which one of the following will have the longest bond length? Circle the correct choice
A) PS^{1-} B) PS C) PS^{1+}
- (iii) Which of the following will be diamagnetic? Circle the correct choice.
A) PS^{1-} B) PS C) PS^{1+}
- (iv) Draw Lewis dot structure of the molecule with the shortest bond length (based on the MO diagram) from among the three (PS^{1-} , PS, PS^{1+}), and indicate the bond order calculated from the Lewis dot structure. **(3 points)**

Lewis Dot Structure

Bond Order



D1E99705-7BF5-41BA-8ED8-A3948A04D447

midterm-exam-93388

#786 Page 10 of 10

SCRAP PAGE

**CHEMISTRY 110 –Midterm Exam****Thursday November 7, 2024, 6:30 – 8:30 pm**

Examiners: Pallavi Sirjoosingh and Ashok Kakkar

First name (please write as legibly as possible within the boxes)

--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--

Last name

--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--

Student ID number

--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--

This booklet has 8 questions on 10 pages (The last page can be used as a scrap page).

The staple must NOT be removed from the question booklet. Your intact question booklet must be returned at the end of the exam.

A separate booklet contains a periodic table, and datasheet.

Write your answers directly under the questions in the space provided. Do not write on the QR codes or near the top edges (as it may be cut off).

Please write **legibly and dark enough** for the scan to be readable. For students who use English as a second language, dictionaries are permitted. Calculators are allowed.

DO NOT ASK QUESTIONS OF THE INVIGILATORS about interpretations of the exam questions. If you have a concern about a question, please summarize it at the top of this page. All concerns will be evaluated after the exam. GOOD LUCK!

**Question 1 8 points** (*Assume all transitions are possible*)

a. 6 points Calculate the **energy (in kJ/mol)** required to ionize (completely remove) the electron from the **n=3** level of **He⁺ ion** (He: Helium). Show your work including equations. (3 sig. figs.)

b. 2 points

(I) Select the largest (longest) wavelength from the three choices below:

- i. Wavelength of photon to *completely remove* an electron from *1s orbital of Hydrogen atom*
- ii. Wavelength of photon to *completely remove* an electron from *1s orbital of Boron atom (B)*
- iii. Wavelength of photon to *completely remove* an electron from *1s orbital of Boron cation (B⁺)*

i or ii or iii

(II) Select the smallest (shortest) wavelength from the three choices below (*for H atom*):

- i. Wavelength of photon *emitted* for an electron transitioning from *2p to 1s orbital*
- ii. Wavelength of photon *absorbed* for an electron transitioning from *2s to 4p orbital*
- iii. Wavelength of photon *absorbed* for an electron transitioning from *1s to 3p orbital*

i or ii or iii



Question 2 8 points



A Chem110 student plots the graph shown on the left, from data obtained for the kinetic energy (eV) of the ejected electron versus frequency (s^{-1}) of the incident light, for an unknown metal. ($1\text{eV} = 1.6022 \times 10^{-19} \text{ J}$)

Based on the plot, answer the questions **a to c** below:

a. 1 point On the plot above, label the **threshold frequency**.

b. 2 points What is the value of the work potential (in J) of the unknown metal?

c. 3 points What is the *longest* wavelength (in nm) required to eject an electron from this metal's surface? Show your work including equation(s) used. (3 sig figs)

d. 2 points In the space provided below, plot the following

(Assume frequency is *above* threshold frequency):

- i. *Kinetic energy* of an ejected electron versus the *intensity* of the incident light (left)
- ii. *Number of electrons* ejected versus the *frequency* of the incident light (right)



**Question 3** **6 points****a. 2 points**

In the ground state of Ge (Germanium; $Z = 32$) atom, **name** the following orbitals:

i. Orbital containing the electron that requires the maximum energy to be removed.

ii. Orbital containing the electron that experiences the minimum effective nuclear charge.

b. 2 points

An unknown anion (A^-) has the following *excited* electronic configuration: $[Ar] 4s^2 3d^3 4p^1$

(Ar: Argon; $Z = 18$)

Give the **condensed ground state electronic configuration** of the atom, **A** and the cation, **A^{2+}** .
Show your work. (Assume all electronic transitions are possible)

Electronic Configuration of A:

Electronic Configuration of A^{2+} :

c. 2 points

In a single atom, determine how many electrons can have the following set of quantum numbers?

(i) $n = 5; l = 3$

In a single atom, determine how many electrons can have the following set of quantum numbers?

(i) $n = 3; l = 2; m_s = +1/2$

**Question 4** **8 points**

Rank the following from 1 to 4 in **increasing** order. Provide the ranking in the boxes provided.

Note: Atomic number (Z) are provided for relevant atoms

a) Increasing (lowest to highest) order of *ionization energy*: (1= lowest; 4 = highest)

Mg

Mg: Z = 12

Rb

Rb: Z = 37

Cs

Cs: Z = 55

K

K : Z = 19

b) Increasing (lowest to highest) order of *bond angle*: (1= lowest; 4 = highest)

H₂OCCl₄XeF₄

HCN

c) Increasing (smallest to largest) order of *ionic size* (1= smallest; 4 = largest)

S²⁻

S: Z = 16

O²⁻

O: Z = 8

Be²⁺

Be: Z = 4

Na⁺

Na: Z = 11

d) Increasing (lowest to highest) order of *bond energy* (1= lowest; 4 = highest)

P-F

P-Cl

P-I

P-Br

**Question 5** **12 points**

a. 9 points For the molecules HClO_4 and TeF_6 , draw the most stable *Lewis structure* showing appropriate shape and *dashed/wedged bonds*, based on VSEPR. Determine the **molecular geometry** around the central atom. For HClO_4 , label the *most polar bond*.



b. 3 points (Note: Lone pairs are not shown in structures below)

(i) For the three *resonance* structures (1, 2 and 3) shown below, answer the following question:



(I) Circle the *most contributing* resonance structure.

Structure 1

Structure 2

Structure 3

(ii) For the three structures given (4, 5, and 6) below, answer the following questions:



(I) Circle all the structures that represent *resonance structures* of each other:

Structure 4

Structure 5

Structure 6

(II) Among the resonance structures, circle the *most contributing* resonance structure.

Structure 4

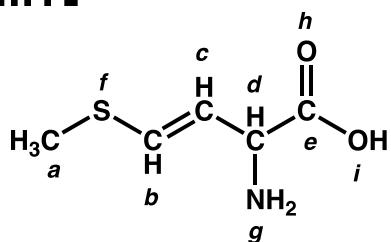
Structure 5

Structure 6

**Question 6** **6 points**

Using the data given below, calculate the **lattice energy** for SrO(s).

	ΔH° (kJ/mol)
$\text{Sr}(s) \longrightarrow \text{Sr}(g)$	+180
$\text{Sr}(g) \longrightarrow \text{Sr}^{2+}(g) + 2e^-$	+1468
$\text{O}_2(g) \longrightarrow 2\text{O}(g)$	+500
$\text{O}(g) + 2e^- \longrightarrow \text{O}^{2-}(g)$	+600
$\text{Sr}(s) + \frac{1}{2}\text{O}_2(g) \longrightarrow \text{SrO}(s)$	-550

**Question 7 10 points**

Consider the molecule given on the left, and answer questions (i) to (vi) using chemical bonding theories that we have learnt in class (valence bond theory, hybridization).

Note that the lone pairs are not shown in the structure:

i. Consider the carbon atoms marked in as ***a***, ***b***, ***c***, ***d***, and ***e*** in the molecule, and determine **hybridization** at each of the carbon centers. (2.5 points)

Atom	Hybridization
Carbon (<i>a</i>)	
Carbon (<i>b</i>)	
Carbon (<i>c</i>)	
Carbon (<i>d</i>)	
Carbon (<i>e</i>)	

ii. Now, consider the sulfur atom (S) marked as ***f***, nitrogen (N) atom (***g***), and oxygen(O) atoms (***h*** and ***i***), and determine their **hybridization**. (2 points)

Atom	Hybridization
Sulfur (<i>f</i>)	
Nitrogen (<i>g</i>)	
Oxygen (<i>h</i>)	
Oxygen (<i>i</i>)	

iii. **Describe** all the bonds formed by the carbon atoms ***a*** and ***b***, using sigma and pi designation as appropriate. (2 points)

Atom	Bonds formed (use sigma and pi as appropriate)
Carbon (<i>a</i>)	
Carbon (<i>b</i>)	

iv. How many total sp^3 -s sigma bonds are in the molecule? (1 point)

v. Determine **electronic** and **molecular geometry** at carbon (C) labeled ***e*** and sulfur (S) labeled ***f*** centres. (2 points)

Atom	Electronic Geometry	Molecular Geometry
Carbon (<i>e</i>)		
Sulfur (<i>f</i>)		

vi. Circle the value from the choices given below, that describes the **C-S-C bond angle at sulfur** (S) marked ***f***: (0.5 points)

>90° to <109.5°

109.5°

>109.5 to <120°

120°

>120° to <180°


Question 8 12 points

Using the **valence shell orbitals only** draw a **complete** molecular orbital (MO) diagram (with no s-p mixing) for the molecule **PS**, in the space provided below. **(6 points)**

Assume that 3s and 3p atomic orbitals will form molecular orbitals like those formed by the combination of 2s and 2p atomic orbitals that we learnt in class, and there is no s-p mixing.

Show **energy arrow**, **atomic orbitals** involved, **atom labels**, the **molecular orbitals** that are formed, **filled electrons** in the MO diagram etc. Make sure to label all the atomic and molecular orbitals.

Using the MO diagram above, answer the following questions:

- (i) Which one of the following will have the highest bond order? Circle the correct choice
 A) PS^{1-} B) PS C) PS^{1+}
- (ii) Which one of the following will have the longest bond length? Circle the correct choice
 A) PS^{1-} B) PS C) PS^{1+}
- (iii) Which of the following will be diamagnetic? Circle the correct choice.
 A) PS^{1-} B) PS C) PS^{1+}
- (iv) Draw Lewis dot structure of the molecule with the shortest bond length (based on the MO diagram) from among the three (PS^{1-} , PS, PS^{1+}), and indicate the bond order calculated from the Lewis dot structure. **(3 points)**

Lewis Dot Structure

Bond Order



6567A18C-C7D8-443C-913E-B17FE1287BDF

midterm-exam-93388

#787 Page 10 of 10

SCRAP PAGE

**CHEMISTRY 110 –Midterm Exam****Thursday November 7, 2024, 6:30 – 8:30 pm****Examiners: Pallavi Sirjoosingh and Ashok Kakkar**

First name (please write as legibly as possible within the boxes)

--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--

Last name

--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--

Student ID number

--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--

This booklet has 8 questions on 10 pages (The last page can be used as a scrap page).

The staple must NOT be removed from the question booklet. Your intact question booklet must be returned at the end of the exam.

A separate booklet contains a periodic table, and datasheet.

Write your answers directly under the questions in the space provided. Do not write on the QR codes or near the top edges (as it may be cut off).

Please write **legibly and dark enough** for the scan to be readable. For students who use English as a second language, dictionaries are permitted. Calculators are allowed.

DO NOT ASK QUESTIONS OF THE INVIGILATORS about interpretations of the exam questions. If you have a concern about a question, please summarize it at the top of this page. All concerns will be evaluated after the exam. GOOD LUCK!

**Question 1 8 points** (*Assume all transitions are possible*)

a. 6 points Calculate the **energy (in kJ/mol)** required to ionize (completely remove) the electron from the **$n=3$** level of **He^+ ion** (He: Helium). Show your work including equations. (3 sig. figs.)

b. 2 points

(I) Select the largest (longest) wavelength from the three choices below:

- i. Wavelength of photon to *completely remove* an electron from *1s orbital of Hydrogen atom*
- ii. Wavelength of photon to *completely remove* an electron from *1s orbital of Boron atom (B)*
- iii. Wavelength of photon to *completely remove* an electron from *1s orbital of Boron cation (B^+)*

i or ii or iii

(II) Select the smallest (shortest) wavelength from the three choices below (*for H atom*):

- i. Wavelength of photon *emitted* for an electron transitioning from *2p to 1s orbital*
- ii. Wavelength of photon *absorbed* for an electron transitioning from *2s to 4p orbital*
- iii. Wavelength of photon *absorbed* for an electron transitioning from *1s to 3p orbital*

i or ii or iii



Question 2 8 points



A Chem110 student plots the graph shown on the left, from data obtained for the kinetic energy (eV) of the ejected electron versus frequency (s^{-1}) of the incident light, for an unknown metal. ($1\text{eV} = 1.6022 \times 10^{-19} \text{ J}$)

Based on the plot, answer the questions **a to c** below:

a. 1 point On the plot above, label the **threshold frequency**.

b. 2 points What is the value of the work potential (in J) of the unknown metal?

c. 3 points What is the *longest* wavelength (in nm) required to eject an electron from this metal's surface? Show your work including equation(s) used. (3 sig figs)

d. 2 points In the space provided below, plot the following

(Assume frequency is *above* threshold frequency):

- Kinetic energy* of an ejected electron versus the *intensity* of the incident light (left)
- Number of electrons* ejected versus the *frequency* of the incident light (right)



**Question 3** **6 points****a. 2 points**

In the ground state of Ge (Germanium; $Z = 32$) atom, **name** the following orbitals:

i. Orbital containing the electron that requires the maximum energy to be removed.

ii. Orbital containing the electron that experiences the minimum effective nuclear charge.

b. 2 points

An unknown anion (A^-) has the following *excited* electronic configuration: $[Ar] 4s^2 3d^3 4p^1$

(Ar: Argon; $Z = 18$)

Give the **condensed ground state electronic configuration** of the atom, **A** and the cation, **A^{2+}** .
Show your work. (Assume all electronic transitions are possible)

Electronic Configuration of A:

Electronic Configuration of A^{2+} :

c. 2 points

In a single atom, determine how many electrons can have the following set of quantum numbers?

(i) $n = 5; l = 3$

In a single atom, determine how many electrons can have the following set of quantum numbers?

(i) $n = 3; l = 2; m_s = +1/2$

**Question 4** **8 points**

Rank the following from 1 to 4 in **increasing** order. Provide the ranking in the boxes provided.

Note: Atomic number (Z) are provided for relevant atoms

a) Increasing (lowest to highest) order of *ionization energy*: (1= lowest; 4 = highest)

Mg

Mg: Z = 12

Rb

Rb: Z = 37

Cs

Cs: Z = 55

K

K : Z = 19

b) Increasing (lowest to highest) order of *bond angle*: (1= lowest; 4 = highest)

H₂OCCl₄XeF₄

HCN

c) Increasing (smallest to largest) order of *ionic size* (1= smallest; 4 = largest)

S²⁻

S: Z = 16

O²⁻

O: Z = 8

Be²⁺

Be: Z = 4

Na⁺

Na: Z = 11

d) Increasing (lowest to highest) order of *bond energy* (1= lowest; 4 = highest)

P-F

P-Cl

P-I

P-Br

**Question 5** **12 points**

a. 9 points For the molecules HClO_4 and TeF_6 , draw the most stable *Lewis structure* showing appropriate shape and *dashed/wedged bonds*, based on VSEPR. Determine the **molecular geometry** around the central atom. For HClO_4 , label the *most polar bond*.



b. 3 points (Note: Lone pairs are not shown in structures below)

(i) For the three *resonance* structures (1, 2 and 3) shown below, answer the following question:



(I) Circle the *most contributing* resonance structure.

Structure 1

Structure 2

Structure 3

(ii) For the three structures given (4, 5, and 6) below, answer the following questions:



(I) Circle all the structures that represent *resonance structures* of each other:

Structure 4

Structure 5

Structure 6

(II) Among the resonance structures, circle the *most contributing* resonance structure.

Structure 4

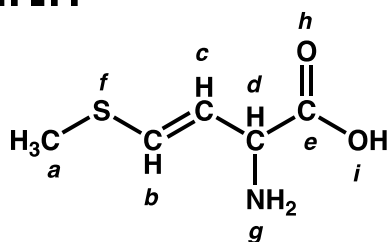
Structure 5

Structure 6

**Question 6** **6 points**

Using the data given below, calculate the **lattice energy** for SrO(s).

	ΔH° (kJ/mol)
$\text{Sr}(s) \longrightarrow \text{Sr}(g)$	+180
$\text{Sr}(g) \longrightarrow \text{Sr}^{2+}(g) + 2e^-$	+1468
$\text{O}_2(g) \longrightarrow 2\text{O}(g)$	+500
$\text{O}(g) + 2e^- \longrightarrow \text{O}^{2-}(g)$	+600
$\text{Sr}(s) + \frac{1}{2}\text{O}_2(g) \longrightarrow \text{SrO}(s)$	-550

**Question 7 10 points**

Consider the molecule given on the left, and answer questions (i) to (vi) using chemical bonding theories that we have learnt in class (valence bond theory, hybridization).

Note that the lone pairs are not shown in the structure:

i. Consider the carbon atoms marked in as ***a***, ***b***, ***c***, ***d***, and ***e*** in the molecule, and determine **hybridization** at each of the carbon centers. (2.5 points)

Atom	Hybridization
Carbon (<i>a</i>)	
Carbon (<i>b</i>)	
Carbon (<i>c</i>)	
Carbon (<i>d</i>)	
Carbon (<i>e</i>)	

ii. Now, consider the sulfur atom (S) marked as ***f***, nitrogen (N) atom (***g***), and oxygen(O) atoms (***h*** and ***i***), and determine their **hybridization**. (2 points)

Atom	Hybridization
Sulfur (<i>f</i>)	
Nitrogen (<i>g</i>)	
Oxygen (<i>h</i>)	
Oxygen (<i>i</i>)	

iii. **Describe** all the bonds formed by the carbon atoms ***a*** and ***b***, using sigma and pi designation as appropriate. (2 points)

Atom	Bonds formed (use sigma and pi as appropriate)
Carbon (<i>a</i>)	
Carbon (<i>b</i>)	

iv. How many total sp^3 -s sigma bonds are in the molecule? (1 point)

v. Determine **electronic** and **molecular geometry** at carbon (C) labeled ***e*** and sulfur (S) labeled ***f*** centres. (2 points)

Atom	Electronic Geometry	Molecular Geometry
Carbon (<i>e</i>)		
Sulfur (<i>f</i>)		

vi. Circle the value from the choices given below, that describes the **C-S-C bond angle at sulfur** (S) marked ***f***: (0.5 points)

>90° to <109.5° 109.5° >109.5 to <120° 120° >120° to <180°


Question 8 12 points

Using the **valence shell orbitals only** draw a **complete** molecular orbital (MO) diagram (with no s-p mixing) for the molecule **PS**, in the space provided below. **(6 points)**

Assume that 3s and 3p atomic orbitals will form molecular orbitals like those formed by the combination of 2s and 2p atomic orbitals that we learnt in class, and there is no s-p mixing.

Show **energy arrow**, **atomic orbitals** involved, **atom labels**, the **molecular orbitals** that are formed, **filled electrons** in the MO diagram etc. Make sure to label all the atomic and molecular orbitals.

Using the MO diagram above, answer the following questions:

- (i) Which one of the following will have the highest bond order? Circle the correct choice
 A) PS^{1-} B) PS C) PS^{1+}
- (ii) Which one of the following will have the longest bond length? Circle the correct choice
 A) PS^{1-} B) PS C) PS^{1+}
- (iii) Which of the following will be diamagnetic? Circle the correct choice.
 A) PS^{1-} B) PS C) PS^{1+}
- (iv) Draw Lewis dot structure of the molecule with the shortest bond length (based on the MO diagram) from among the three (PS^{1-} , PS, PS^{1+}), and indicate the bond order calculated from the Lewis dot structure. **(3 points)**

Lewis Dot Structure

Bond Order



3208789E-DCBE-4BA5-B02E-5026C42C4D16

midterm-exam-93388

#788 Page 10 of 10

SCRAP PAGE

**CHEMISTRY 110 –Midterm Exam****Thursday November 7, 2024, 6:30 – 8:30 pm**

Examiners: Pallavi Sirjoosingh and Ashok Kakkar

First name (please write as legibly as possible within the boxes)

--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--

Last name

--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--

Student ID number

--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--

This booklet has 8 questions on 10 pages (The last page can be used as a scrap page).

The staple must NOT be removed from the question booklet. Your intact question booklet must be returned at the end of the exam.

A separate booklet contains a periodic table, and datasheet.

Write your answers directly under the questions in the space provided. Do not write on the QR codes or near the top edges (as it may be cut off).

Please write **legibly and dark enough** for the scan to be readable. For students who use English as a second language, dictionaries are permitted. Calculators are allowed.

DO NOT ASK QUESTIONS OF THE INVIGILATORS about interpretations of the exam questions. If you have a concern about a question, please summarize it at the top of this page. All concerns will be evaluated after the exam. GOOD LUCK!

**Question 1 8 points** (*Assume all transitions are possible*)

a. 6 points Calculate the **energy (in kJ/mol)** required to ionize (completely remove) the electron from the **$n=3$** level of **He^+ ion** (He: Helium). Show your work including equations. (3 sig. figs.)

b. 2 points

(I) Select the largest (longest) wavelength from the three choices below:

- i. Wavelength of photon to *completely remove* an electron from *1s orbital of Hydrogen atom*
- ii. Wavelength of photon to *completely remove* an electron from *1s orbital of Boron atom (B)*
- iii. Wavelength of photon to *completely remove* an electron from *1s orbital of Boron cation (B^+)*

i or ii or iii

(II) Select the smallest (shortest) wavelength from the three choices below (*for H atom*):

- i. Wavelength of photon *emitted* for an electron transitioning from *2p to 1s orbital*
- ii. Wavelength of photon *absorbed* for an electron transitioning from *2s to 4p orbital*
- iii. Wavelength of photon *absorbed* for an electron transitioning from *1s to 3p orbital*

i or ii or iii



Question 2 8 points



A Chem110 student plots the graph shown on the left, from data obtained for the kinetic energy (eV) of the ejected electron versus frequency (s^{-1}) of the incident light, for an unknown metal. ($1\text{eV} = 1.6022 \times 10^{-19} \text{ J}$)

Based on the plot, answer the questions **a to c** below:

a. 1 point On the plot above, label the **threshold frequency**.

b. 2 points What is the value of the work potential (in J) of the unknown metal?

c. 3 points What is the *longest* wavelength (in nm) required to eject an electron from this metal's surface? Show your work including equation(s) used. (3 sig figs)

d. 2 points In the space provided below, plot the following

(Assume frequency is *above* threshold frequency):

- i. *Kinetic energy* of an ejected electron versus the *intensity* of the incident light (left)
- ii. *Number of electrons* ejected versus the *frequency* of the incident light (right)



**Question 3** **6 points****a. 2 points**

In the ground state of Ge (Germanium; $Z = 32$) atom, **name** the following orbitals:

i. Orbital containing the electron that requires the maximum energy to be removed.

ii. Orbital containing the electron that experiences the minimum effective nuclear charge.

b. 2 points

An unknown anion (A^-) has the following *excited* electronic configuration: $[Ar] 4s^2 3d^3 4p^1$

(Ar: Argon; $Z = 18$)

Give the **condensed ground state electronic configuration** of the atom, **A** and the cation, **A^{2+}** .
Show your work. (Assume all electronic transitions are possible)

Electronic Configuration of A:

Electronic Configuration of A^{2+} :

c. 2 points

In a single atom, determine how many electrons can have the following set of quantum numbers?

(i) $n = 5; l = 3$

In a single atom, determine how many electrons can have the following set of quantum numbers?

(i) $n = 3; l = 2; m_s = +1/2$

**Question 4** **8 points**

Rank the following from 1 to 4 in **increasing** order. Provide the ranking in the boxes provided.

Note: Atomic number (Z) are provided for relevant atoms

a) Increasing (lowest to highest) order of *ionization energy*: (1= lowest; 4 = highest)

Mg

Mg: Z = 12

Rb

Rb: Z = 37

Cs

Cs: Z = 55

K

K : Z = 19

b) Increasing (lowest to highest) order of *bond angle*: (1= lowest; 4 = highest)

H₂OCCl₄XeF₄

HCN

c) Increasing (smallest to largest) order of *ionic size* (1= smallest; 4 = largest)

S²⁻

S: Z = 16

O²⁻

O: Z = 8

Be²⁺

Be: Z = 4

Na⁺

Na: Z = 11

d) Increasing (lowest to highest) order of *bond energy* (1= lowest; 4 = highest)

P-F

P-Cl

P-I

P-Br

**Question 5** **12 points**

a. 9 points For the molecules HClO_4 and TeF_6 , draw the most stable *Lewis structure* showing appropriate shape and *dashed/wedged bonds*, based on VSEPR. Determine the **molecular geometry** around the central atom. For HClO_4 , label the *most polar bond*.



b. 3 points (Note: Lone pairs are not shown in structures below)

(i) For the three *resonance* structures (1, 2 and 3) shown below, answer the following question:



(I) Circle the *most contributing* resonance structure.

Structure 1

Structure 2

Structure 3

(ii) For the three structures given (4, 5, and 6) below, answer the following questions:



(I) Circle all the structures that represent *resonance structures* of each other:

Structure 4

Structure 5

Structure 6

(II) Among the resonance structures, circle the *most contributing* resonance structure.

Structure 4

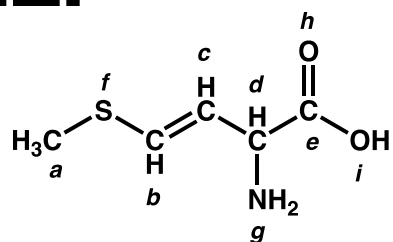
Structure 5

Structure 6

**Question 6** **6 points**

Using the data given below, calculate the **lattice energy** for SrO(s).

	ΔH° (kJ/mol)
$\text{Sr}(s) \longrightarrow \text{Sr}(g)$	+180
$\text{Sr}(g) \longrightarrow \text{Sr}^{2+}(g) + 2e^-$	+1468
$\text{O}_2(g) \longrightarrow 2\text{O}(g)$	+500
$\text{O}(g) + 2e^- \longrightarrow \text{O}^{2-}(g)$	+600
$\text{Sr}(s) + \frac{1}{2}\text{O}_2(g) \longrightarrow \text{SrO}(s)$	-550

**Question 7 10 points**

Consider the molecule given on the left, and answer questions (i) to (vi) using chemical bonding theories that we have learnt in class (valence bond theory, hybridization).

Note that the lone pairs are not shown in the structure:

i. Consider the carbon atoms marked in as ***a***, ***b***, ***c***, ***d***, and ***e*** in the molecule, and determine **hybridization** at each of the carbon centers. (2.5 points)

Atom	Hybridization
Carbon (<i>a</i>)	
Carbon (<i>b</i>)	
Carbon (<i>c</i>)	
Carbon (<i>d</i>)	
Carbon (<i>e</i>)	

ii. Now, consider the sulfur atom (S) marked as ***f***, nitrogen (N) atom (***g***), and oxygen(O) atoms (***h*** and ***i***), and determine their **hybridization**. (2 points)

Atom	Hybridization
Sulfur (<i>f</i>)	
Nitrogen (<i>g</i>)	
Oxygen (<i>h</i>)	
Oxygen (<i>i</i>)	

iii. **Describe** all the bonds formed by the carbon atoms ***a*** and ***b***, using sigma and pi designation as appropriate. (2 points)

Atom	Bonds formed (use sigma and pi as appropriate)
Carbon (<i>a</i>)	
Carbon (<i>b</i>)	

iv. How many total sp^3 -s sigma bonds are in the molecule? (1 point)

v. Determine **electronic** and **molecular geometry** at carbon (C) labeled ***e*** and sulfur (S) labeled ***f*** centres. (2 points)

Atom	Electronic Geometry	Molecular Geometry
Carbon (<i>e</i>)		
Sulfur (<i>f</i>)		

vi. Circle the value from the choices given below, that describes the **C-S-C bond angle at sulfur** (S) marked ***f***: (0.5 points)

>90° to <109.5°

109.5°

>109.5 to <120°

120°

>120° to <180°


Question 8 12 points

Using the **valence shell orbitals only** draw a **complete** molecular orbital (MO) diagram (with no s-p mixing) for the molecule **PS**, in the space provided below. **(6 points)**

Assume that 3s and 3p atomic orbitals will form molecular orbitals like those formed by the combination of 2s and 2p atomic orbitals that we learnt in class, and there is no s-p mixing.

Show **energy arrow**, **atomic orbitals** involved, **atom labels**, the **molecular orbitals** that are formed, **filled electrons** in the MO diagram etc. Make sure to label all the atomic and molecular orbitals.

Using the MO diagram above, answer the following questions:

- (i) Which one of the following will have the highest bond order? Circle the correct choice
 A) PS^{1-} B) PS C) PS^{1+}
- (ii) Which one of the following will have the longest bond length? Circle the correct choice
 A) PS^{1-} B) PS C) PS^{1+}
- (iii) Which of the following will be diamagnetic? Circle the correct choice.
 A) PS^{1-} B) PS C) PS^{1+}
- (iv) Draw Lewis dot structure of the molecule with the shortest bond length (based on the MO diagram) from among the three (PS^{1-} , PS, PS^{1+}), and indicate the bond order calculated from the Lewis dot structure. **(3 points)**

Lewis Dot Structure

Bond Order



EC2DDF64-57CF-40CE-B50B-9E16B37697E5

midterm-exam-93388

#789 Page 10 of 10

SCRAP PAGE

**CHEMISTRY 110 –Midterm Exam****Thursday November 7, 2024, 6:30 – 8:30 pm**

Examiners: Pallavi Sirjoosingh and Ashok Kakkar

First name (please write as legibly as possible within the boxes)

--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--

Last name

--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--

Student ID number

--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--

This booklet has 8 questions on 10 pages (The last page can be used as a scrap page).

The staple must NOT be removed from the question booklet. Your intact question booklet must be returned at the end of the exam.

A separate booklet contains a periodic table, and datasheet.

Write your answers directly under the questions in the space provided. Do not write on the QR codes or near the top edges (as it may be cut off).

Please write **legibly and dark enough** for the scan to be readable. For students who use English as a second language, dictionaries are permitted. Calculators are allowed.

DO NOT ASK QUESTIONS OF THE INVIGILATORS about interpretations of the exam questions. If you have a concern about a question, please summarize it at the top of this page. All concerns will be evaluated after the exam. GOOD LUCK!

**Question 1 8 points** (*Assume all transitions are possible*)

a. 6 points Calculate the **energy (in kJ/mol)** required to ionize (completely remove) the electron from the **n=3** level of **He⁺ ion** (He: Helium). Show your work including equations. (3 sig. figs.)

b. 2 points

(I) Select the largest (longest) wavelength from the three choices below:

- i. Wavelength of photon to *completely remove* an electron from *1s orbital of Hydrogen atom*
- ii. Wavelength of photon to *completely remove* an electron from *1s orbital of Boron atom (B)*
- iii. Wavelength of photon to *completely remove* an electron from *1s orbital of Boron cation (B⁺)*

i or ii or iii

(II) Select the smallest (shortest) wavelength from the three choices below (*for H atom*):

- i. Wavelength of photon *emitted* for an electron transitioning from *2p to 1s orbital*
- ii. Wavelength of photon *absorbed* for an electron transitioning from *2s to 4p orbital*
- iii. Wavelength of photon *absorbed* for an electron transitioning from *1s to 3p orbital*

i or ii or iii



Question 2 8 points



A Chem110 student plots the graph shown on the left, from data obtained for the kinetic energy (eV) of the ejected electron versus frequency (s^{-1}) of the incident light, for an unknown metal. ($1\text{eV} = 1.6022 \times 10^{-19} \text{ J}$)

Based on the plot, answer the questions **a to c** below:

a. 1 point On the plot above, label the **threshold frequency**.

b. 2 points What is the value of the work potential (in J) of the unknown metal?

c. 3 points What is the *longest* wavelength (in nm) required to eject an electron from this metal's surface? Show your work including equation(s) used. (3 sig figs)

d. 2 points In the space provided below, plot the following

(Assume frequency is *above* threshold frequency):

- i. *Kinetic energy* of an ejected electron versus the *intensity* of the incident light (left)
- ii. *Number of electrons* ejected versus the *frequency* of the incident light (right)



**Question 3** **6 points****a. 2 points**

In the ground state of Ge (Germanium; $Z = 32$) atom, **name** the following orbitals:

i. Orbital containing the electron that requires the maximum energy to be removed.

ii. Orbital containing the electron that experiences the minimum effective nuclear charge.

b. 2 points

An unknown anion (A^-) has the following *excited* electronic configuration: $[Ar] 4s^2 3d^3 4p^1$

(Ar: Argon; $Z = 18$)

Give the **condensed ground state electronic configuration** of the atom, **A** and the cation, **A^{2+}** .
Show your work. (Assume all electronic transitions are possible)

Electronic Configuration of A:

Electronic Configuration of A^{2+} :

c. 2 points

In a single atom, determine how many electrons can have the following set of quantum numbers?

(i) $n = 5; l = 3$

In a single atom, determine how many electrons can have the following set of quantum numbers?

(i) $n = 3; l = 2; m_s = +1/2$

**Question 4** **8 points**

Rank the following from 1 to 4 in **increasing** order. Provide the ranking in the boxes provided.

Note: Atomic number (Z) are provided for relevant atoms

a) Increasing (lowest to highest) order of *ionization energy*: (1= lowest; 4 = highest)

Mg

Mg: Z = 12

Rb

Rb: Z = 37

Cs

Cs: Z = 55

K

K : Z = 19

b) Increasing (lowest to highest) order of *bond angle*: (1= lowest; 4 = highest)

H₂OCCl₄XeF₄

HCN

c) Increasing (smallest to largest) order of *ionic size* (1= smallest; 4 = largest)

S²⁻

S: Z = 16

O²⁻

O: Z = 8

Be²⁺

Be: Z = 4

Na⁺

Na: Z = 11

d) Increasing (lowest to highest) order of *bond energy* (1= lowest; 4 = highest)

P-F

P-Cl

P-I

P-Br

**Question 5** **12 points**

a. 9 points For the molecules HClO_4 and TeF_6 , draw the most stable *Lewis structure* showing appropriate shape and *dashed/wedged bonds*, based on VSEPR. Determine the **molecular geometry** around the central atom. For HClO_4 , label the *most polar bond*.

HClO_4 (there are no O-O bonds)

Label the most polar bond

Molecular Geometry around Cl

TeF_6

Molecular Geometry around Te

b. 3 points (Note: Lone pairs are not shown in structures below)

(i) For the three *resonance* structures (1, 2 and 3) shown below, answer the following question:



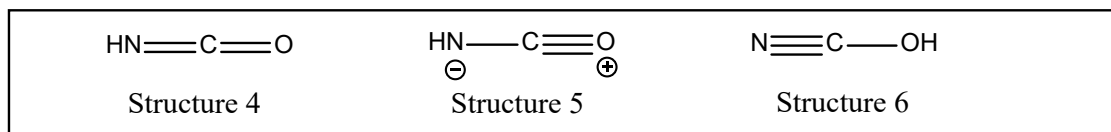
(I) Circle the *most contributing* resonance structure.

Structure 1

Structure 2

Structure 3

(ii) For the three structures given (4, 5, and 6) below, answer the following questions:



(I) Circle all the structures that represent *resonance structures* of each other:

Structure 4

Structure 5

Structure 6

(II) Among the resonance structures, circle the *most contributing* resonance structure.

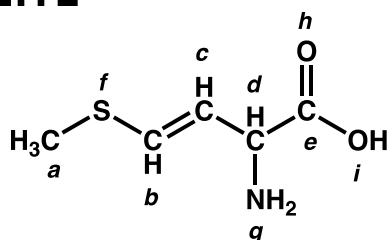
Structure 4

Structure 5

Structure 6

**Question 6** **6 points**Using the data given below, calculate the **lattice energy** for SrO(s).

	ΔH° (kJ/mol)
$\text{Sr}(s) \longrightarrow \text{Sr}(g)$	+180
$\text{Sr}(g) \longrightarrow \text{Sr}^{2+}(g) + 2e^-$	+1468
$\text{O}_2(g) \longrightarrow 2\text{O}(g)$	+500
$\text{O}(g) + 2e^- \longrightarrow \text{O}^{2-}(g)$	+600
$\text{Sr}(s) + \frac{1}{2}\text{O}_2(g) \longrightarrow \text{SrO}(s)$	-550

**Question 7 10 points**

Consider the molecule given on the left, and answer questions (i) to (vi) using chemical bonding theories that we have learnt in class (valence bond theory, hybridization).

Note that the lone pairs are not shown in the structure:

i. Consider the carbon atoms marked in as ***a***, ***b***, ***c***, ***d***, and ***e*** in the molecule, and determine **hybridization** at each of the carbon centers. (2.5 points)

Atom	Hybridization
Carbon (<i>a</i>)	
Carbon (<i>b</i>)	
Carbon (<i>c</i>)	
Carbon (<i>d</i>)	
Carbon (<i>e</i>)	

ii. Now, consider the sulfur atom (S) marked as ***f***, nitrogen (N) atom (***g***), and oxygen(O) atoms (***h*** and ***i***), and determine their **hybridization**. (2 points)

Atom	Hybridization
Sulfur (<i>f</i>)	
Nitrogen (<i>g</i>)	
Oxygen (<i>h</i>)	
Oxygen (<i>i</i>)	

iii. **Describe** all the bonds formed by the carbon atoms ***a*** and ***b***, using sigma and pi designation as appropriate. (2 points)

Atom	Bonds formed (use sigma and pi as appropriate)
Carbon (<i>a</i>)	
Carbon (<i>b</i>)	

iv. How many total sp^3 -s sigma bonds are in the molecule? (1 point)

v. Determine **electronic** and **molecular geometry** at carbon (C) labeled ***e*** and sulfur (S) labeled ***f*** centres. (2 points)

Atom	Electronic Geometry	Molecular Geometry
Carbon (<i>e</i>)		
Sulfur (<i>f</i>)		

vi. Circle the value from the choices given below, that describes the **C-S-C bond angle at sulfur** (S) marked ***f***: (0.5 points)

>90° to <109.5°

109.5°

>109.5 to <120°

120°

>120° to <180°

**Question 8 12 points**

Using the **valence shell orbitals only** draw a **complete** molecular orbital (MO) diagram (with no s-p mixing) for the molecule **PS**, in the space provided below. **(6 points)**

Assume that 3s and 3p atomic orbitals will form molecular orbitals like those formed by the combination of 2s and 2p atomic orbitals that we learnt in class, and there is no s-p mixing.

Show **energy arrow**, **atomic orbitals** involved, **atom labels**, the **molecular orbitals** that are formed, **filled electrons** in the MO diagram etc. Make sure to label all the atomic and molecular orbitals.

Using the MO diagram above, answer the following questions:

- (i) Which one of the following will have the highest bond order? Circle the correct choice
A) PS^{1-} B) PS C) PS^{1+}
- (ii) Which one of the following will have the longest bond length? Circle the correct choice
A) PS^{1-} B) PS C) PS^{1+}
- (iii) Which of the following will be diamagnetic? Circle the correct choice.
A) PS^{1-} B) PS C) PS^{1+}
- (iv) Draw Lewis dot structure of the molecule with the shortest bond length (based on the MO diagram) from among the three (PS^{1-} , PS, PS^{1+}), and indicate the bond order calculated from the Lewis dot structure. **(3 points)**

Lewis Dot Structure

Bond Order



A38E90F5-CEB2-4AF2-B7E9-4CD30A21035B

midterm-exam-93388

#790 Page 10 of 10

SCRAP PAGE

**CHEMISTRY 110 –Midterm Exam****Thursday November 7, 2024, 6:30 – 8:30 pm**

Examiners: Pallavi Sirjoosingh and Ashok Kakkar

First name (please write as legibly as possible within the boxes)

--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--

Last name

--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--

Student ID number

--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--

This booklet has 8 questions on 10 pages (The last page can be used as a scrap page).

The staple must NOT be removed from the question booklet. Your intact question booklet must be returned at the end of the exam.

A separate booklet contains a periodic table, and datasheet.

Write your answers directly under the questions in the space provided. Do not write on the QR codes or near the top edges (as it may be cut off).

Please write **legibly and dark enough** for the scan to be readable. For students who use English as a second language, dictionaries are permitted. Calculators are allowed.

DO NOT ASK QUESTIONS OF THE INVIGILATORS about interpretations of the exam questions. If you have a concern about a question, please summarize it at the top of this page. All concerns will be evaluated after the exam. GOOD LUCK!

**Question 1 8 points** (*Assume all transitions are possible*)

a. 6 points Calculate the **energy (in kJ/mol)** required to ionize (completely remove) the electron from the **n=3** level of **He⁺ ion** (He: Helium). Show your work including equations. (3 sig. figs.)

b. 2 points

(I) Select the largest (longest) wavelength from the three choices below:

- i. Wavelength of photon to *completely remove* an electron from *1s orbital of Hydrogen atom*
- ii. Wavelength of photon to *completely remove* an electron from *1s orbital of Boron atom (B)*
- iii. Wavelength of photon to *completely remove* an electron from *1s orbital of Boron cation (B⁺)*

i or ii or iii

(II) Select the smallest (shortest) wavelength from the three choices below (*for H atom*):

- i. Wavelength of photon *emitted* for an electron transitioning from *2p to 1s orbital*
- ii. Wavelength of photon *absorbed* for an electron transitioning from *2s to 4p orbital*
- iii. Wavelength of photon *absorbed* for an electron transitioning from *1s to 3p orbital*

i or ii or iii



Question 2 8 points



A Chem110 student plots the graph shown on the left, from data obtained for the kinetic energy (eV) of the ejected electron versus frequency (s^{-1}) of the incident light, for an unknown metal. ($1\text{eV} = 1.6022 \times 10^{-19} \text{ J}$)

Based on the plot, answer the questions **a to c** below:

a. 1 point On the plot above, label the **threshold frequency**.

b. 2 points What is the value of the work potential (in J) of the unknown metal?

c. 3 points What is the *longest* wavelength (in nm) required to eject an electron from this metal's surface? Show your work including equation(s) used. (3 sig figs)

d. 2 points In the space provided below, plot the following

(Assume frequency is *above* threshold frequency):

- Kinetic energy* of an ejected electron versus the *intensity* of the incident light (left)
- Number of electrons* ejected versus the *frequency* of the incident light (right)



**Question 3** **6 points****a. 2 points**

In the ground state of Ge (Germanium; $Z = 32$) atom, **name** the following orbitals:

i. Orbital containing the electron that requires the maximum energy to be removed.

ii. Orbital containing the electron that experiences the minimum effective nuclear charge.

b. 2 points

An unknown anion (A^-) has the following *excited* electronic configuration: $[Ar] 4s^2 3d^3 4p^1$

(Ar: Argon; $Z = 18$)

Give the **condensed ground state electronic configuration** of the atom, **A** and the cation, **A^{2+}** .
Show your work. (Assume all electronic transitions are possible)

Electronic Configuration of A:

Electronic Configuration of A^{2+} :

c. 2 points

In a single atom, determine how many electrons can have the following set of quantum numbers?

(i) $n = 5; l = 3$

In a single atom, determine how many electrons can have the following set of quantum numbers?

(i) $n = 3; l = 2; m_s = +1/2$

**Question 4** **8 points**

Rank the following from 1 to 4 in **increasing** order. Provide the ranking in the boxes provided.

Note: Atomic number (Z) are provided for relevant atoms

a) Increasing (lowest to highest) order of *ionization energy*: (1= lowest; 4 = highest)

Mg

Mg: Z = 12

Rb

Rb: Z = 37

Cs

Cs: Z = 55

K

K : Z = 19

b) Increasing (lowest to highest) order of *bond angle*: (1= lowest; 4 = highest)

H₂OCCl₄XeF₄

HCN

c) Increasing (smallest to largest) order of *ionic size* (1= smallest; 4 = largest)

S²⁻

S: Z = 16

O²⁻

O: Z = 8

Be²⁺

Be: Z = 4

Na⁺

Na: Z = 11

d) Increasing (lowest to highest) order of *bond energy* (1= lowest; 4 = highest)

P-F

P-Cl

P-I

P-Br

**Question 5** **12 points**

a. 9 points For the molecules HClO_4 and TeF_6 , draw the most stable *Lewis structure* showing appropriate shape and *dashed/wedged bonds*, based on VSEPR. Determine the **molecular geometry** around the central atom. For HClO_4 , label the *most polar bond*.



b. 3 points (Note: Lone pairs are not shown in structures below)

(i) For the three *resonance* structures (1, 2 and 3) shown below, answer the following question:



(I) Circle the *most contributing* resonance structure.

Structure 1

Structure 2

Structure 3

(ii) For the three structures given (4, 5, and 6) below, answer the following questions:



(I) Circle all the structures that represent *resonance structures* of each other:

Structure 4

Structure 5

Structure 6

(II) Among the resonance structures, circle the *most contributing* resonance structure.

Structure 4

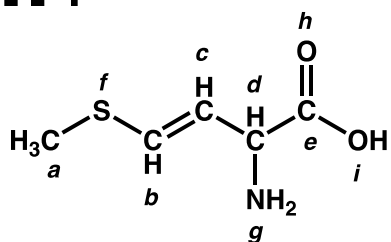
Structure 5

Structure 6

**Question 6** **6 points**

Using the data given below, calculate the **lattice energy** for SrO(s).

	ΔH° (kJ/mol)
$\text{Sr}(s) \longrightarrow \text{Sr}(g)$	+180
$\text{Sr}(g) \longrightarrow \text{Sr}^{2+}(g) + 2e^-$	+1468
$\text{O}_2(g) \longrightarrow 2\text{O}(g)$	+500
$\text{O}(g) + 2e^- \longrightarrow \text{O}^{2-}(g)$	+600
$\text{Sr}(s) + \frac{1}{2}\text{O}_2(g) \longrightarrow \text{SrO}(s)$	-550

**Question 7 10 points**

Consider the molecule given on the left, and answer questions (i) to (vi) using chemical bonding theories that we have learnt in class (valence bond theory, hybridization).

Note that the lone pairs are not shown in the structure:

i. Consider the carbon atoms marked in as ***a***, ***b***, ***c***, ***d***, and ***e*** in the molecule, and determine **hybridization** at each of the carbon centers. (2.5 points)

Atom	Hybridization
Carbon (<i>a</i>)	
Carbon (<i>b</i>)	
Carbon (<i>c</i>)	
Carbon (<i>d</i>)	
Carbon (<i>e</i>)	

ii. Now, consider the sulfur atom (S) marked as ***f***, nitrogen (N) atom (***g***), and oxygen(O) atoms (***h*** and ***i***), and determine their **hybridization**. (2 points)

Atom	Hybridization
Sulfur (<i>f</i>)	
Nitrogen (<i>g</i>)	
Oxygen (<i>h</i>)	
Oxygen (<i>i</i>)	

iii. **Describe** all the bonds formed by the carbon atoms ***a*** and ***b***, using sigma and pi designation as appropriate. (2 points)

Atom	Bonds formed (use sigma and pi as appropriate)
Carbon (<i>a</i>)	
Carbon (<i>b</i>)	

iv. How many total sp^3 -s sigma bonds are in the molecule? (1 point)

v. Determine **electronic** and **molecular geometry** at carbon (C) labeled ***e*** and sulfur (S) labeled ***f*** centres. (2 points)

Atom	Electronic Geometry	Molecular Geometry
Carbon (<i>e</i>)		
Sulfur (<i>f</i>)		

vi. Circle the value from the choices given below, that describes the **C-S-C bond angle at sulfur** (S) marked ***f***: (0.5 points)

>90° to <109.5° 109.5° >109.5 to <120° 120° >120° to <180°

**Question 8 12 points**

Using the **valence shell orbitals only** draw a **complete** molecular orbital (MO) diagram (with no s-p mixing) for the molecule **PS**, in the space provided below. **(6 points)**

Assume that 3s and 3p atomic orbitals will form molecular orbitals like those formed by the combination of 2s and 2p atomic orbitals that we learnt in class, and there is no s-p mixing.

Show **energy arrow**, **atomic orbitals** involved, **atom labels**, the **molecular orbitals** that are formed, **filled electrons** in the MO diagram etc. Make sure to label all the atomic and molecular orbitals.

Using the MO diagram above, answer the following questions:

- (i) Which one of the following will have the highest bond order? Circle the correct choice
A) PS^{1-} B) PS C) PS^{1+}
- (ii) Which one of the following will have the longest bond length? Circle the correct choice
A) PS^{1-} B) PS C) PS^{1+}
- (iii) Which of the following will be diamagnetic? Circle the correct choice.
A) PS^{1-} B) PS C) PS^{1+}
- (iv) Draw Lewis dot structure of the molecule with the shortest bond length (based on the MO diagram) from among the three (PS^{1-} , PS, PS^{1+}), and indicate the bond order calculated from the Lewis dot structure. **(3 points)**

Lewis Dot Structure

Bond Order



CE8065A1-DCAE-43D8-8544-1FF823D56C08

midterm-exam-93388

#791 Page 10 of 10

SCRAP PAGE

**CHEMISTRY 110 –Midterm Exam****Thursday November 7, 2024, 6:30 – 8:30 pm****Examiners: Pallavi Sirjoosingh and Ashok Kakkar**

First name (please write as legibly as possible within the boxes)

--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--

Last name

--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--

Student ID number

--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--

This booklet has 8 questions on 10 pages (The last page can be used as a scrap page).

The staple must NOT be removed from the question booklet. Your intact question booklet must be returned at the end of the exam.

A separate booklet contains a periodic table, and datasheet.

Write your answers directly under the questions in the space provided. Do not write on the QR codes or near the top edges (as it may be cut off).

Please write **legibly and dark enough** for the scan to be readable. For students who use English as a second language, dictionaries are permitted. Calculators are allowed.

DO NOT ASK QUESTIONS OF THE INVIGILATORS about interpretations of the exam questions. If you have a concern about a question, please summarize it at the top of this page. All concerns will be evaluated after the exam. GOOD LUCK!

**Question 1 8 points** (*Assume all transitions are possible*)

a. 6 points Calculate the **energy (in kJ/mol)** required to ionize (completely remove) the electron from the **$n=3$** level of **He^+ ion** (He: Helium). Show your work including equations. (3 sig. figs.)

b. 2 points

(I) Select the largest (longest) wavelength from the three choices below:

- i. Wavelength of photon to *completely remove* an electron from *1s orbital of Hydrogen atom*
- ii. Wavelength of photon to *completely remove* an electron from *1s orbital of Boron atom (B)*
- iii. Wavelength of photon to *completely remove* an electron from *1s orbital of Boron cation (B^+)*

i or ii or iii

(II) Select the smallest (shortest) wavelength from the three choices below (*for H atom*):

- i. Wavelength of photon *emitted* for an electron transitioning from *2p to 1s orbital*
- ii. Wavelength of photon *absorbed* for an electron transitioning from *2s to 4p orbital*
- iii. Wavelength of photon *absorbed* for an electron transitioning from *1s to 3p orbital*

i or ii or iii



Question 2 8 points



A Chem110 student plots the graph shown on the left, from data obtained for the kinetic energy (eV) of the ejected electron versus frequency (s^{-1}) of the incident light, for an unknown metal. ($1\text{eV} = 1.6022 \times 10^{-19} \text{ J}$)

Based on the plot, answer the questions **a to c** below:

a. 1 point On the plot above, label the **threshold frequency**.

b. 2 points What is the value of the work potential (in J) of the unknown metal?

c. 3 points What is the *longest* wavelength (in nm) required to eject an electron from this metal's surface? Show your work including equation(s) used. (3 sig figs)

d. 2 points In the space provided below, plot the following

(Assume frequency is *above* threshold frequency):

- Kinetic energy* of an ejected electron versus the *intensity* of the incident light (left)
- Number of electrons* ejected versus the *frequency* of the incident light (right)



**Question 3** **6 points****a. 2 points**

In the ground state of Ge (Germanium; $Z = 32$) atom, **name** the following orbitals:

i. Orbital containing the electron that requires the maximum energy to be removed.

ii. Orbital containing the electron that experiences the minimum effective nuclear charge.

b. 2 points

An unknown anion (A^-) has the following *excited* electronic configuration: $[Ar] 4s^2 3d^3 4p^1$

(Ar: Argon; $Z = 18$)

Give the **condensed ground state electronic configuration** of the atom, **A** and the cation, **A^{2+}** .
Show your work. (Assume all electronic transitions are possible)

Electronic Configuration of A:

Electronic Configuration of A^{2+} :

c. 2 points

In a single atom, determine how many electrons can have the following set of quantum numbers?

(i) $n = 5; l = 3$

In a single atom, determine how many electrons can have the following set of quantum numbers?

(i) $n = 3; l = 2; m_s = +1/2$

**Question 4** **8 points**

Rank the following from 1 to 4 in **increasing** order. Provide the ranking in the boxes provided.

Note: Atomic number (Z) are provided for relevant atoms

a) Increasing (lowest to highest) order of *ionization energy*: (1= lowest; 4 = highest)

Mg

Mg: Z = 12

Rb

Rb: Z = 37

Cs

Cs: Z = 55

K

K : Z = 19

b) Increasing (lowest to highest) order of *bond angle*: (1= lowest; 4 = highest)

H₂OCCl₄XeF₄

HCN

c) Increasing (smallest to largest) order of *ionic size* (1= smallest; 4 = largest)

S²⁻

S: Z = 16

O²⁻

O: Z = 8

Be²⁺

Be: Z = 4

Na⁺

Na: Z = 11

d) Increasing (lowest to highest) order of *bond energy* (1= lowest; 4 = highest)

P-F

P-Cl

P-I

P-Br

**Question 5** **12 points**

a. 9 points For the molecules HClO_4 and TeF_6 , draw the most stable *Lewis structure* showing appropriate shape and *dashed/wedged bonds*, based on VSEPR. Determine the **molecular geometry** around the central atom. For HClO_4 , label the *most polar bond*.

HClO_4 (there are no O-O bonds)

Label the most polar bond

Molecular Geometry around Cl

TeF_6

Molecular Geometry around Te

b. 3 points (Note: Lone pairs are not shown in structures below)

(i) For the three *resonance* structures (1, 2 and 3) shown below, answer the following question:



(I) Circle the *most contributing* resonance structure.

Structure 1

Structure 2

Structure 3

(ii) For the three structures given (4, 5, and 6) below, answer the following questions:



(I) Circle all the structures that represent *resonance structures* of each other:

Structure 4

Structure 5

Structure 6

(II) Among the resonance structures, circle the *most contributing* resonance structure.

Structure 4

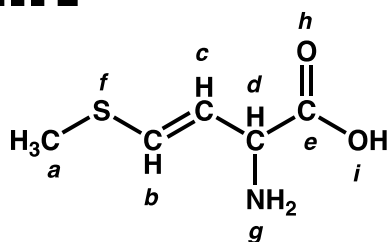
Structure 5

Structure 6

**Question 6** **6 points**

Using the data given below, calculate the **lattice energy** for SrO(s).

	ΔH° (kJ/mol)
$\text{Sr}(s) \longrightarrow \text{Sr}(g)$	+180
$\text{Sr}(g) \longrightarrow \text{Sr}^{2+}(g) + 2e^-$	+1468
$\text{O}_2(g) \longrightarrow 2\text{O}(g)$	+500
$\text{O}(g) + 2e^- \longrightarrow \text{O}^{2-}(g)$	+600
$\text{Sr}(s) + \frac{1}{2}\text{O}_2(g) \longrightarrow \text{SrO}(s)$	-550

**Question 7 10 points**

Consider the molecule given on the left, and answer questions (i) to (vi) using chemical bonding theories that we have learnt in class (valence bond theory, hybridization).

Note that the lone pairs are not shown in the structure:

i. Consider the carbon atoms marked in as ***a***, ***b***, ***c***, ***d***, and ***e*** in the molecule, and determine **hybridization** at each of the carbon centers. (2.5 points)

Atom	Hybridization
Carbon (<i>a</i>)	
Carbon (<i>b</i>)	
Carbon (<i>c</i>)	
Carbon (<i>d</i>)	
Carbon (<i>e</i>)	

ii. Now, consider the sulfur atom (S) marked as ***f***, nitrogen (N) atom (***g***), and oxygen(O) atoms (***h*** and ***i***), and determine their **hybridization**. (2 points)

Atom	Hybridization
Sulfur (<i>f</i>)	
Nitrogen (<i>g</i>)	
Oxygen (<i>h</i>)	
Oxygen (<i>i</i>)	

iii. **Describe** all the bonds formed by the carbon atoms ***a*** and ***b***, using sigma and pi designation as appropriate. (2 points)

Atom	Bonds formed (use sigma and pi as appropriate)
Carbon (<i>a</i>)	
Carbon (<i>b</i>)	

iv. How many total sp^3 -s sigma bonds are in the molecule? (1 point)

v. Determine **electronic** and **molecular geometry** at carbon (C) labeled ***e*** and sulfur (S) labeled ***f*** centres. (2 points)

Atom	Electronic Geometry	Molecular Geometry
Carbon (<i>e</i>)		
Sulfur (<i>f</i>)		

vi. Circle the value from the choices given below, that describes the **C-S-C bond angle at sulfur** (S) marked ***f***: (0.5 points)

>90° to <109.5° 109.5° >109.5 to <120° 120° >120° to <180°


Question 8 12 points

Using the **valence shell orbitals only** draw a **complete** molecular orbital (MO) diagram (with no s-p mixing) for the molecule **PS**, in the space provided below. **(6 points)**

Assume that 3s and 3p atomic orbitals will form molecular orbitals like those formed by the combination of 2s and 2p atomic orbitals that we learnt in class, and there is no s-p mixing.

Show **energy arrow**, **atomic orbitals** involved, **atom labels**, the **molecular orbitals** that are formed, **filled electrons** in the MO diagram etc. Make sure to label all the atomic and molecular orbitals.

Using the MO diagram above, answer the following questions:

- (i) Which one of the following will have the highest bond order? Circle the correct choice
 A) PS^{1-} B) PS C) PS^{1+}
- (ii) Which one of the following will have the longest bond length? Circle the correct choice
 A) PS^{1-} B) PS C) PS^{1+}
- (iii) Which of the following will be diamagnetic? Circle the correct choice.
 A) PS^{1-} B) PS C) PS^{1+}
- (iv) Draw Lewis dot structure of the molecule with the shortest bond length (based on the MO diagram) from among the three (PS^{1-} , PS, PS^{1+}), and indicate the bond order calculated from the Lewis dot structure. **(3 points)**

Lewis Dot Structure

Bond Order



002E7CE4-10CD-46E9-8B79-66BA16B704D0

midterm-exam-93388

#792 Page 10 of 10

SCRAP PAGE

**CHEMISTRY 110 –Midterm Exam****Thursday November 7, 2024, 6:30 – 8:30 pm****Examiners: Pallavi Sirjoosingh and Ashok Kakkar**

First name (please write as legibly as possible within the boxes)

--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--

Last name

--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--

Student ID number

--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--

This booklet has 8 questions on 10 pages (The last page can be used as a scrap page).

The staple must NOT be removed from the question booklet. Your intact question booklet must be returned at the end of the exam.

A separate booklet contains a periodic table, and datasheet.

Write your answers directly under the questions in the space provided. Do not write on the QR codes or near the top edges (as it may be cut off).

Please write **legibly and dark enough** for the scan to be readable. For students who use English as a second language, dictionaries are permitted. Calculators are allowed.

DO NOT ASK QUESTIONS OF THE INVIGILATORS about interpretations of the exam questions. If you have a concern about a question, please summarize it at the top of this page. All concerns will be evaluated after the exam. GOOD LUCK!

**Question 1 8 points** (*Assume all transitions are possible*)

a. 6 points Calculate the **energy (in kJ/mol)** required to ionize (completely remove) the electron from the **n=3** level of **He⁺ ion** (He: Helium). Show your work including equations. (3 sig. figs.)

b. 2 points

(I) Select the largest (longest) wavelength from the three choices below:

- i. Wavelength of photon to *completely remove* an electron from *1s orbital of Hydrogen atom*
- ii. Wavelength of photon to *completely remove* an electron from *1s orbital of Boron atom (B)*
- iii. Wavelength of photon to *completely remove* an electron from *1s orbital of Boron cation (B⁺)*

i or ii or iii

(II) Select the smallest (shortest) wavelength from the three choices below (*for H atom*):

- i. Wavelength of photon *emitted* for an electron transitioning from *2p to 1s orbital*
- ii. Wavelength of photon *absorbed* for an electron transitioning from *2s to 4p orbital*
- iii. Wavelength of photon *absorbed* for an electron transitioning from *1s to 3p orbital*

i or ii or iii



Question 2 8 points



A Chem110 student plots the graph shown on the left, from data obtained for the kinetic energy (eV) of the ejected electron versus frequency (s^{-1}) of the incident light, for an unknown metal. ($1\text{eV} = 1.6022 \times 10^{-19} \text{ J}$)

Based on the plot, answer the questions **a to c** below:

a. 1 point On the plot above, label the **threshold frequency**.

b. 2 points What is the value of the work potential (in J) of the unknown metal?

c. 3 points What is the *longest* wavelength (in nm) required to eject an electron from this metal's surface? Show your work including equation(s) used. (3 sig figs)

d. 2 points In the space provided below, plot the following

(Assume frequency is *above* threshold frequency):

- Kinetic energy* of an ejected electron versus the *intensity* of the incident light (left)
- Number of electrons* ejected versus the *frequency* of the incident light (right)



**Question 3** **6 points****a. 2 points**

In the ground state of Ge (Germanium; $Z = 32$) atom, **name** the following orbitals:

i. Orbital containing the electron that requires the maximum energy to be removed.

ii. Orbital containing the electron that experiences the minimum effective nuclear charge.

b. 2 points

An unknown anion (A^-) has the following *excited* electronic configuration: $[Ar] 4s^2 3d^3 4p^1$

(Ar: Argon; $Z = 18$)

Give the **condensed ground state electronic configuration** of the atom, **A** and the cation, **A^{2+}** .
Show your work. (Assume all electronic transitions are possible)

Electronic Configuration of A:

Electronic Configuration of A^{2+} :

c. 2 points

In a single atom, determine how many electrons can have the following set of quantum numbers?

(i) $n = 5; l = 3$

In a single atom, determine how many electrons can have the following set of quantum numbers?

(i) $n = 3; l = 2; m_s = +1/2$

**Question 4** **8 points**

Rank the following from 1 to 4 in **increasing** order. Provide the ranking in the boxes provided.

Note: Atomic number (Z) are provided for relevant atoms

a) Increasing (lowest to highest) order of *ionization energy*: (1= lowest; 4 = highest)

Mg

Mg: Z = 12

Rb

Rb: Z = 37

Cs

Cs: Z = 55

K

K : Z = 19

b) Increasing (lowest to highest) order of *bond angle*: (1= lowest; 4 = highest)

H₂OCCl₄XeF₄

HCN

c) Increasing (smallest to largest) order of *ionic size* (1= smallest; 4 = largest)

S²⁻

S: Z = 16

O²⁻

O: Z = 8

Be²⁺

Be: Z = 4

Na⁺

Na: Z = 11

d) Increasing (lowest to highest) order of *bond energy* (1= lowest; 4 = highest)

P-F

P-Cl

P-I

P-Br

**Question 5** **12 points**

a. 9 points For the molecules HClO_4 and TeF_6 , draw the most stable *Lewis structure* showing appropriate shape and *dashed/wedged bonds*, based on VSEPR. Determine the **molecular geometry** around the central atom. For HClO_4 , label the *most polar bond*.



b. 3 points (Note: Lone pairs are not shown in structures below)

(i) For the three *resonance* structures (1, 2 and 3) shown below, answer the following question:



(I) Circle the *most contributing* resonance structure.

Structure 1

Structure 2

Structure 3

(ii) For the three structures given (4, 5, and 6) below, answer the following questions:



(I) Circle all the structures that represent *resonance structures* of each other:

Structure 4

Structure 5

Structure 6

(II) Among the resonance structures, circle the *most contributing* resonance structure.

Structure 4

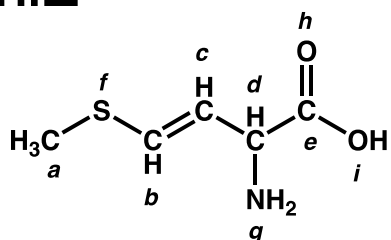
Structure 5

Structure 6

**Question 6** **6 points**

Using the data given below, calculate the **lattice energy** for SrO(s).

	ΔH° (kJ/mol)
$\text{Sr}(s) \longrightarrow \text{Sr}(g)$	+180
$\text{Sr}(g) \longrightarrow \text{Sr}^{2+}(g) + 2e^-$	+1468
$\text{O}_2(g) \longrightarrow 2\text{O}(g)$	+500
$\text{O}(g) + 2e^- \longrightarrow \text{O}^{2-}(g)$	+600
$\text{Sr}(s) + \frac{1}{2}\text{O}_2(g) \longrightarrow \text{SrO}(s)$	-550

**Question 7 10 points**

Consider the molecule given on the left, and answer questions (i) to (vi) using chemical bonding theories that we have learnt in class (valence bond theory, hybridization).

Note that the lone pairs are not shown in the structure:

i. Consider the carbon atoms marked in as ***a***, ***b***, ***c***, ***d***, and ***e*** in the molecule, and determine **hybridization** at each of the carbon centers. (2.5 points)

Atom	Hybridization
Carbon (<i>a</i>)	
Carbon (<i>b</i>)	
Carbon (<i>c</i>)	
Carbon (<i>d</i>)	
Carbon (<i>e</i>)	

ii. Now, consider the sulfur atom (S) marked as ***f***, nitrogen (N) atom (***g***), and oxygen(O) atoms (***h*** and ***i***), and determine their **hybridization**. (2 points)

Atom	Hybridization
Sulfur (<i>f</i>)	
Nitrogen (<i>g</i>)	
Oxygen (<i>h</i>)	
Oxygen (<i>i</i>)	

iii. **Describe** all the bonds formed by the carbon atoms ***a*** and ***b***, using sigma and pi designation as appropriate. (2 points)

Atom	Bonds formed (use sigma and pi as appropriate)
Carbon (<i>a</i>)	
Carbon (<i>b</i>)	

iv. How many total sp^3 -s sigma bonds are in the molecule? (1 point)

v. Determine **electronic** and **molecular geometry** at carbon (C) labeled ***e*** and sulfur (S) labeled ***f*** centres. (2 points)

Atom	Electronic Geometry	Molecular Geometry
Carbon (<i>e</i>)		
Sulfur (<i>f</i>)		

vi. Circle the value from the choices given below, that describes the **C-S-C bond angle at sulfur** (S) marked ***f***: (0.5 points)

>90° to <109.5° 109.5° >109.5 to <120° 120° >120° to <180°


Question 8 12 points

Using the **valence shell orbitals only** draw a **complete** molecular orbital (MO) diagram (with no s-p mixing) for the molecule **PS**, in the space provided below. **(6 points)**

Assume that 3s and 3p atomic orbitals will form molecular orbitals like those formed by the combination of 2s and 2p atomic orbitals that we learnt in class, and there is no s-p mixing.

Show **energy arrow**, **atomic orbitals** involved, **atom labels**, the **molecular orbitals** that are formed, **filled electrons** in the MO diagram etc. Make sure to label all the atomic and molecular orbitals.

Using the MO diagram above, answer the following questions:

- (i) Which one of the following will have the highest bond order? Circle the correct choice
 A) PS^{1-} B) PS C) PS^{1+}
- (ii) Which one of the following will have the longest bond length? Circle the correct choice
 A) PS^{1-} B) PS C) PS^{1+}
- (iii) Which of the following will be diamagnetic? Circle the correct choice.
 A) PS^{1-} B) PS C) PS^{1+}
- (iv) Draw Lewis dot structure of the molecule with the shortest bond length (based on the MO diagram) from among the three (PS^{1-} , PS, PS^{1+}), and indicate the bond order calculated from the Lewis dot structure. **(3 points)**

Lewis Dot Structure

Bond Order



20562A98-1FD5-4863-B06A-66BA4924DE2B

midterm-exam-93388

#793 Page 10 of 10

SCRAP PAGE

**CHEMISTRY 110 –Midterm Exam****Thursday November 7, 2024, 6:30 – 8:30 pm**

Examiners: Pallavi Sirjoosingh and Ashok Kakkar

First name (please write as legibly as possible within the boxes)

--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--

Last name

--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--

Student ID number

--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--

This booklet has 8 questions on 10 pages (The last page can be used as a scrap page).

The staple must NOT be removed from the question booklet. Your intact question booklet must be returned at the end of the exam.

A separate booklet contains a periodic table, and datasheet.

Write your answers directly under the questions in the space provided. Do not write on the QR codes or near the top edges (as it may be cut off).

Please write **legibly and dark enough** for the scan to be readable. For students who use English as a second language, dictionaries are permitted. Calculators are allowed.

DO NOT ASK QUESTIONS OF THE INVIGILATORS about interpretations of the exam questions. If you have a concern about a question, please summarize it at the top of this page. All concerns will be evaluated after the exam. GOOD LUCK!

**Question 1 8 points** (*Assume all transitions are possible*)

a. 6 points Calculate the **energy (in kJ/mol)** required to ionize (completely remove) the electron from the **n=3** level of **He⁺ ion** (He: Helium). Show your work including equations. (3 sig. figs.)

b. 2 points

(I) Select the largest (longest) wavelength from the three choices below:

- i. Wavelength of photon to *completely remove* an electron from *1s orbital of Hydrogen atom*
- ii. Wavelength of photon to *completely remove* an electron from *1s orbital of Boron atom (B)*
- iii. Wavelength of photon to *completely remove* an electron from *1s orbital of Boron cation (B⁺)*

i or ii or iii

(II) Select the smallest (shortest) wavelength from the three choices below (*for H atom*):

- i. Wavelength of photon *emitted* for an electron transitioning from *2p to 1s orbital*
- ii. Wavelength of photon *absorbed* for an electron transitioning from *2s to 4p orbital*
- iii. Wavelength of photon *absorbed* for an electron transitioning from *1s to 3p orbital*

i or ii or iii



Question 2 8 points



A Chem110 student plots the graph shown on the left, from data obtained for the kinetic energy (eV) of the ejected electron versus frequency (s^{-1}) of the incident light, for an unknown metal. ($1\text{eV} = 1.6022 \times 10^{-19} \text{ J}$)

Based on the plot, answer the questions **a to c** below:

a. 1 point On the plot above, label the **threshold frequency**.

b. 2 points What is the value of the work potential (in J) of the unknown metal?

c. 3 points What is the *longest* wavelength (in nm) required to eject an electron from this metal's surface? Show your work including equation(s) used. (3 sig figs)

d. 2 points In the space provided below, plot the following

(Assume frequency is *above* threshold frequency):

- i. *Kinetic energy* of an ejected electron versus the *intensity* of the incident light (left)
- ii. *Number of electrons* ejected versus the *frequency* of the incident light (right)



**Question 3** **6 points****a. 2 points**

In the ground state of Ge (Germanium; $Z = 32$) atom, **name** the following orbitals:

i. Orbital containing the electron that requires the maximum energy to be removed.

ii. Orbital containing the electron that experiences the minimum effective nuclear charge.

b. 2 points

An unknown anion (A^-) has the following *excited* electronic configuration: $[Ar] 4s^2 3d^3 4p^1$

(Ar: Argon; $Z = 18$)

Give the **condensed ground state electronic configuration** of the atom, **A** and the cation, **A^{2+}** .
Show your work. (Assume all electronic transitions are possible)

Electronic Configuration of A:

Electronic Configuration of A^{2+} :

c. 2 points

In a single atom, determine how many electrons can have the following set of quantum numbers?

(i) $n = 5; l = 3$

In a single atom, determine how many electrons can have the following set of quantum numbers?

(i) $n = 3; l = 2; m_s = +1/2$

**Question 4** **8 points**

Rank the following from 1 to 4 in **increasing** order. Provide the ranking in the boxes provided.

Note: Atomic number (Z) are provided for relevant atoms

a) Increasing (lowest to highest) order of *ionization energy*: (1= lowest; 4 = highest)

Mg

Mg: Z = 12

Rb

Rb: Z = 37

Cs

Cs: Z = 55

K

K : Z = 19

b) Increasing (lowest to highest) order of *bond angle*: (1= lowest; 4 = highest)

H₂OCCl₄XeF₄

HCN

c) Increasing (smallest to largest) order of *ionic size* (1= smallest; 4 = largest)

S²⁻

S: Z = 16

O²⁻

O: Z = 8

Be²⁺

Be: Z = 4

Na⁺

Na: Z = 11

d) Increasing (lowest to highest) order of *bond energy* (1= lowest; 4 = highest)

P-F

P-Cl

P-I

P-Br

**Question 5** **12 points**

a. 9 points For the molecules HClO_4 and TeF_6 , draw the most stable *Lewis structure* showing appropriate shape and *dashed/wedged bonds*, based on VSEPR. Determine the **molecular geometry** around the central atom. For HClO_4 , label the *most polar bond*.



b. 3 points (Note: Lone pairs are not shown in structures below)

(i) For the three *resonance* structures (1, 2 and 3) shown below, answer the following question:



(I) Circle the *most contributing* resonance structure.

Structure 1

Structure 2

Structure 3

(ii) For the three structures given (4, 5, and 6) below, answer the following questions:



(I) Circle all the structures that represent *resonance structures* of each other:

Structure 4

Structure 5

Structure 6

(II) Among the resonance structures, circle the *most contributing* resonance structure.

Structure 4

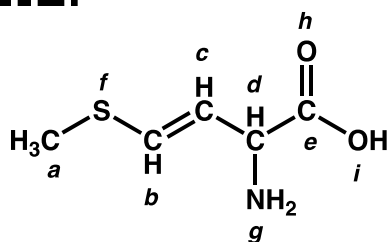
Structure 5

Structure 6

**Question 6** **6 points**

Using the data given below, calculate the **lattice energy** for SrO(s).

	ΔH° (kJ/mol)
$\text{Sr}(s) \longrightarrow \text{Sr}(g)$	+180
$\text{Sr}(g) \longrightarrow \text{Sr}^{2+}(g) + 2e^-$	+1468
$\text{O}_2(g) \longrightarrow 2\text{O}(g)$	+500
$\text{O}(g) + 2e^- \longrightarrow \text{O}^{2-}(g)$	+600
$\text{Sr}(s) + \frac{1}{2}\text{O}_2(g) \longrightarrow \text{SrO}(s)$	-550

**Question 7 10 points**

Consider the molecule given on the left, and answer questions (i) to (vi) using chemical bonding theories that we have learnt in class (valence bond theory, hybridization).

Note that the lone pairs are not shown in the structure:

i. Consider the carbon atoms marked in as ***a***, ***b***, ***c***, ***d***, and ***e*** in the molecule, and determine **hybridization** at each of the carbon centers. (2.5 points)

Atom	Hybridization
Carbon (<i>a</i>)	
Carbon (<i>b</i>)	
Carbon (<i>c</i>)	
Carbon (<i>d</i>)	
Carbon (<i>e</i>)	

ii. Now, consider the sulfur atom (S) marked as ***f***, nitrogen (N) atom (***g***), and oxygen(O) atoms (***h*** and ***i***), and determine their **hybridization**. (2 points)

Atom	Hybridization
Sulfur (<i>f</i>)	
Nitrogen (<i>g</i>)	
Oxygen (<i>h</i>)	
Oxygen (<i>i</i>)	

iii. **Describe** all the bonds formed by the carbon atoms ***a*** and ***b***, using sigma and pi designation as appropriate. (2 points)

Atom	Bonds formed (use sigma and pi as appropriate)
Carbon (<i>a</i>)	
Carbon (<i>b</i>)	

iv. How many total sp^3 -s sigma bonds are in the molecule? (1 point)

v. Determine **electronic** and **molecular geometry** at carbon (C) labeled ***e*** and sulfur (S) labeled ***f*** centres. (2 points)

Atom	Electronic Geometry	Molecular Geometry
Carbon (<i>e</i>)		
Sulfur (<i>f</i>)		

vi. Circle the value from the choices given below, that describes the **C-S-C bond angle at sulfur** (S) marked ***f***: (0.5 points)

>90° to <109.5° 109.5° >109.5 to <120° 120° >120° to <180°


Question 8 12 points

Using the **valence shell orbitals only** draw a **complete** molecular orbital (MO) diagram (with no s-p mixing) for the molecule **PS**, in the space provided below. **(6 points)**

Assume that 3s and 3p atomic orbitals will form molecular orbitals like those formed by the combination of 2s and 2p atomic orbitals that we learnt in class, and there is no s-p mixing.

Show **energy arrow**, **atomic orbitals** involved, **atom labels**, the **molecular orbitals** that are formed, **filled electrons** in the MO diagram etc. Make sure to label all the atomic and molecular orbitals.

Using the MO diagram above, answer the following questions:

- (i) Which one of the following will have the highest bond order? Circle the correct choice
 A) PS^{1-} B) PS C) PS^{1+}
- (ii) Which one of the following will have the longest bond length? Circle the correct choice
 A) PS^{1-} B) PS C) PS^{1+}
- (iii) Which of the following will be diamagnetic? Circle the correct choice.
 A) PS^{1-} B) PS C) PS^{1+}
- (iv) Draw Lewis dot structure of the molecule with the shortest bond length (based on the MO diagram) from among the three (PS^{1-} , PS, PS^{1+}), and indicate the bond order calculated from the Lewis dot structure. **(3 points)**

Lewis Dot Structure

Bond Order



7711254D-C380-4CEA-AE03-1F659D87F7D8

midterm-exam-93388

#794 Page 10 of 10

SCRAP PAGE

**CHEMISTRY 110 –Midterm Exam****Thursday November 7, 2024, 6:30 – 8:30 pm****Examiners: Pallavi Sirjoosingh and Ashok Kakkar**

First name (please write as legibly as possible within the boxes)

--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--

Last name

--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--

Student ID number

--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--

This booklet has 8 questions on 10 pages (The last page can be used as a scrap page).

The staple must NOT be removed from the question booklet. Your intact question booklet must be returned at the end of the exam.

A separate booklet contains a periodic table, and datasheet.

Write your answers directly under the questions in the space provided. Do not write on the QR codes or near the top edges (as it may be cut off).

Please write **legibly and dark enough** for the scan to be readable. For students who use English as a second language, dictionaries are permitted. Calculators are allowed.

DO NOT ASK QUESTIONS OF THE INVIGILATORS about interpretations of the exam questions. If you have a concern about a question, please summarize it at the top of this page. All concerns will be evaluated after the exam. GOOD LUCK!

**Question 1 8 points** (*Assume all transitions are possible*)

a. 6 points Calculate the **energy (in kJ/mol)** required to ionize (completely remove) the electron from the **$n=3$** level of **He^+ ion** (He: Helium). Show your work including equations. (3 sig. figs.)

b. 2 points

(I) Select the largest (longest) wavelength from the three choices below:

- i. Wavelength of photon to *completely remove* an electron from *1s orbital of Hydrogen atom*
- ii. Wavelength of photon to *completely remove* an electron from *1s orbital of Boron atom (B)*
- iii. Wavelength of photon to *completely remove* an electron from *1s orbital of Boron cation (B^+)*

i or ii or iii

(II) Select the smallest (shortest) wavelength from the three choices below (*for H atom*):

- i. Wavelength of photon *emitted* for an electron transitioning from *2p to 1s orbital*
- ii. Wavelength of photon *absorbed* for an electron transitioning from *2s to 4p orbital*
- iii. Wavelength of photon *absorbed* for an electron transitioning from *1s to 3p orbital*

i or ii or iii



Question 2 8 points



A Chem110 student plots the graph shown on the left, from data obtained for the kinetic energy (eV) of the ejected electron versus frequency (s^{-1}) of the incident light, for an unknown metal. ($1\text{eV} = 1.6022 \times 10^{-19} \text{ J}$)

Based on the plot, answer the questions **a to c** below:

a. 1 point On the plot above, label the **threshold frequency**.

b. 2 points What is the value of the work potential (in J) of the unknown metal?

c. 3 points What is the *longest* wavelength (in nm) required to eject an electron from this metal's surface? Show your work including equation(s) used. (3 sig figs)

d. 2 points In the space provided below, plot the following

(Assume frequency is *above* threshold frequency):

- i. *Kinetic energy* of an ejected electron versus the *intensity* of the incident light (left)
- ii. *Number of electrons* ejected versus the *frequency* of the incident light (right)



**Question 3** **6 points****a. 2 points**

In the ground state of Ge (Germanium; $Z = 32$) atom, **name** the following orbitals:

i. Orbital containing the electron that requires the maximum energy to be removed.

ii. Orbital containing the electron that experiences the minimum effective nuclear charge.

b. 2 points

An unknown anion (A^-) has the following *excited* electronic configuration: $[Ar] 4s^2 3d^3 4p^1$

(Ar: Argon; $Z = 18$)

Give the **condensed ground state electronic configuration** of the atom, **A** and the cation, **A^{2+}** .
Show your work. (Assume all electronic transitions are possible)

Electronic Configuration of A:

Electronic Configuration of A^{2+} :

c. 2 points

In a single atom, determine how many electrons can have the following set of quantum numbers?

(i) $n = 5; l = 3$

In a single atom, determine how many electrons can have the following set of quantum numbers?

(i) $n = 3; l = 2; m_s = +1/2$

**Question 4** **8 points**

Rank the following from 1 to 4 in **increasing** order. Provide the ranking in the boxes provided.

Note: Atomic number (Z) are provided for relevant atoms

a) Increasing (lowest to highest) order of *ionization energy*: (1= lowest; 4 = highest)

Mg

Mg: Z = 12

Rb

Rb: Z = 37

Cs

Cs: Z = 55

K

K : Z = 19

b) Increasing (lowest to highest) order of *bond angle*: (1= lowest; 4 = highest)

H₂OCCl₄XeF₄

HCN

c) Increasing (smallest to largest) order of *ionic size* (1= smallest; 4 = largest)

S²⁻

S: Z = 16

O²⁻

O: Z = 8

Be²⁺

Be: Z = 4

Na⁺

Na: Z = 11

d) Increasing (lowest to highest) order of *bond energy* (1= lowest; 4 = highest)

P-F

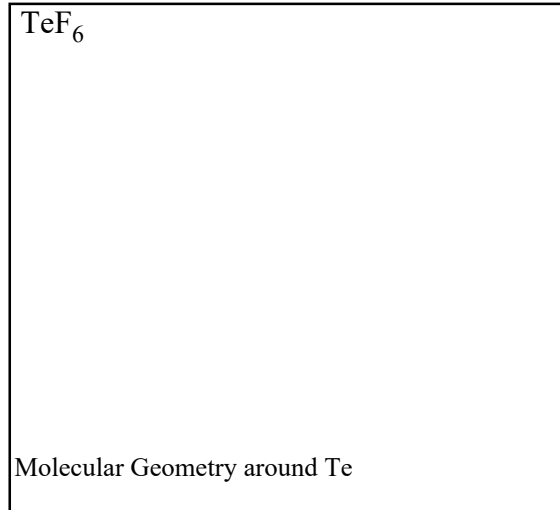
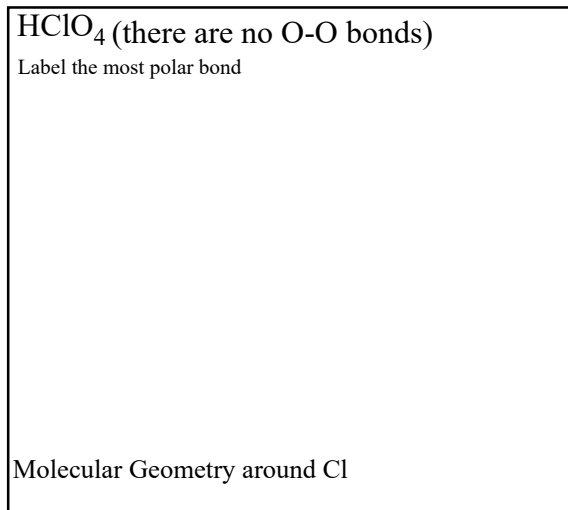
P-Cl

P-I

P-Br

**Question 5** **12 points**

a. 9 points For the molecules HClO_4 and TeF_6 , draw the most stable *Lewis structure* showing appropriate shape and *dashed/wedged bonds*, based on VSEPR. Determine the **molecular geometry** around the central atom. For HClO_4 , label the *most polar bond*.



b. 3 points (Note: Lone pairs are not shown in structures below)

(i) For the three *resonance* structures (1, 2 and 3) shown below, answer the following question:



(I) Circle the *most contributing* resonance structure.

Structure 1

Structure 2

Structure 3

(ii) For the three structures given (4, 5, and 6) below, answer the following questions:



(I) Circle all the structures that represent *resonance structures* of each other:

Structure 4

Structure 5

Structure 6

(II) Among the resonance structures, circle the *most contributing* resonance structure.

Structure 4

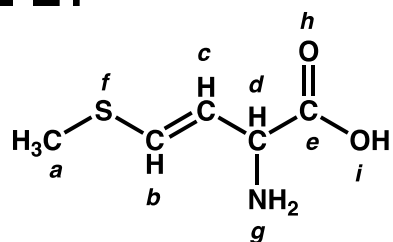
Structure 5

Structure 6

**Question 6** **6 points**

Using the data given below, calculate the **lattice energy** for SrO(s).

	ΔH° (kJ/mol)
$\text{Sr}(s) \longrightarrow \text{Sr}(g)$	+180
$\text{Sr}(g) \longrightarrow \text{Sr}^{2+}(g) + 2e^-$	+1468
$\text{O}_2(g) \longrightarrow 2\text{O}(g)$	+500
$\text{O}(g) + 2e^- \longrightarrow \text{O}^{2-}(g)$	+600
$\text{Sr}(s) + \frac{1}{2}\text{O}_2(g) \longrightarrow \text{SrO}(s)$	-550

**Question 7 10 points**

Consider the molecule given on the left, and answer questions (i) to (vi) using chemical bonding theories that we have learnt in class (valence bond theory, hybridization).

Note that the lone pairs are not shown in the structure:

i. Consider the carbon atoms marked in as ***a***, ***b***, ***c***, ***d***, and ***e*** in the molecule, and determine **hybridization** at each of the carbon centers. (2.5 points)

Atom	Hybridization
Carbon (<i>a</i>)	
Carbon (<i>b</i>)	
Carbon (<i>c</i>)	
Carbon (<i>d</i>)	
Carbon (<i>e</i>)	

ii. Now, consider the sulfur atom (S) marked as ***f***, nitrogen (N) atom (***g***), and oxygen(O) atoms (***h*** and ***i***), and determine their **hybridization**. (2 points)

Atom	Hybridization
Sulfur (<i>f</i>)	
Nitrogen (<i>g</i>)	
Oxygen (<i>h</i>)	
Oxygen (<i>i</i>)	

iii. **Describe** all the bonds formed by the carbon atoms ***a*** and ***b***, using sigma and pi designation as appropriate. (2 points)

Atom	Bonds formed (use sigma and pi as appropriate)
Carbon (<i>a</i>)	
Carbon (<i>b</i>)	

iv. How many total sp^3 -s sigma bonds are in the molecule? (1 point)

v. Determine **electronic** and **molecular geometry** at carbon (C) labeled ***e*** and sulfur (S) labeled ***f*** centres. (2 points)

Atom	Electronic Geometry	Molecular Geometry
Carbon (<i>e</i>)		
Sulfur (<i>f</i>)		

vi. Circle the value from the choices given below, that describes the **C-S-C bond angle at sulfur** (S) marked ***f***: (0.5 points)

>90° to <109.5°

109.5°

>109.5 to <120°

120°

>120° to <180°

**Question 8 12 points**

Using the **valence shell orbitals only** draw a **complete** molecular orbital (MO) diagram (with no s-p mixing) for the molecule **PS**, in the space provided below. **(6 points)**

Assume that 3s and 3p atomic orbitals will form molecular orbitals like those formed by the combination of 2s and 2p atomic orbitals that we learnt in class, and there is no s-p mixing.

Show **energy arrow**, **atomic orbitals** involved, **atom labels**, the **molecular orbitals** that are formed, **filled electrons** in the MO diagram etc. Make sure to label all the atomic and molecular orbitals.

Using the MO diagram above, answer the following questions:

- (i) Which one of the following will have the highest bond order? Circle the correct choice
A) PS^{1-} B) PS C) PS^{1+}
- (ii) Which one of the following will have the longest bond length? Circle the correct choice
A) PS^{1-} B) PS C) PS^{1+}
- (iii) Which of the following will be diamagnetic? Circle the correct choice.
A) PS^{1-} B) PS C) PS^{1+}
- (iv) Draw Lewis dot structure of the molecule with the shortest bond length (based on the MO diagram) from among the three (PS^{1-} , PS, PS^{1+}), and indicate the bond order calculated from the Lewis dot structure. **(3 points)**

Lewis Dot Structure

Bond Order



839F9131-4445-4089-BFD9-988A7925FB86

midterm-exam-93388

#795 Page 10 of 10

SCRAP PAGE

**CHEMISTRY 110 –Midterm Exam****Thursday November 7, 2024, 6:30 – 8:30 pm****Examiners: Pallavi Sirjoosingh and Ashok Kakkar**

First name (please write as legibly as possible within the boxes)

--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--

Last name

--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--

Student ID number

--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--

This booklet has 8 questions on 10 pages (The last page can be used as a scrap page).

The staple must NOT be removed from the question booklet. Your intact question booklet must be returned at the end of the exam.

A separate booklet contains a periodic table, and datasheet.

Write your answers directly under the questions in the space provided. Do not write on the QR codes or near the top edges (as it may be cut off).

Please write **legibly and dark enough** for the scan to be readable. For students who use English as a second language, dictionaries are permitted. Calculators are allowed.

DO NOT ASK QUESTIONS OF THE INVIGILATORS about interpretations of the exam questions. If you have a concern about a question, please summarize it at the top of this page. All concerns will be evaluated after the exam. GOOD LUCK!

**Question 1 8 points** (*Assume all transitions are possible*)

a. 6 points Calculate the **energy (in kJ/mol)** required to ionize (completely remove) the electron from the **n=3** level of **He⁺ ion** (He: Helium). Show your work including equations. (3 sig. figs.)

b. 2 points

(I) Select the largest (longest) wavelength from the three choices below:

- i. Wavelength of photon to *completely remove* an electron from *1s orbital of Hydrogen atom*
- ii. Wavelength of photon to *completely remove* an electron from *1s orbital of Boron atom (B)*
- iii. Wavelength of photon to *completely remove* an electron from *1s orbital of Boron cation (B⁺)*

i or ii or iii

(II) Select the smallest (shortest) wavelength from the three choices below (*for H atom*):

- i. Wavelength of photon *emitted* for an electron transitioning from *2p to 1s orbital*
- ii. Wavelength of photon *absorbed* for an electron transitioning from *2s to 4p orbital*
- iii. Wavelength of photon *absorbed* for an electron transitioning from *1s to 3p orbital*

i or ii or iii



Question 2 8 points



A Chem110 student plots the graph shown on the left, from data obtained for the kinetic energy (eV) of the ejected electron versus frequency (s^{-1}) of the incident light, for an unknown metal. ($1\text{eV} = 1.6022 \times 10^{-19} \text{ J}$)

Based on the plot, answer the questions **a to c** below:

a. 1 point On the plot above, label the **threshold frequency**.

b. 2 points What is the value of the work potential (in J) of the unknown metal?

c. 3 points What is the *longest* wavelength (in nm) required to eject an electron from this metal's surface? Show your work including equation(s) used. (3 sig figs)

d. 2 points In the space provided below, plot the following

(Assume frequency is *above* threshold frequency):

- i. *Kinetic energy* of an ejected electron versus the *intensity* of the incident light (left)
- ii. *Number of electrons* ejected versus the *frequency* of the incident light (right)



**Question 3** **6 points****a. 2 points**

In the ground state of Ge (Germanium; $Z = 32$) atom, **name** the following orbitals:

i. Orbital containing the electron that requires the maximum energy to be removed.

ii. Orbital containing the electron that experiences the minimum effective nuclear charge.

b. 2 points

An unknown anion (A^-) has the following *excited* electronic configuration: $[Ar] 4s^2 3d^3 4p^1$

(Ar: Argon; $Z = 18$)

Give the **condensed ground state electronic configuration** of the atom, **A** and the cation, **A^{2+}** .
Show your work. (Assume all electronic transitions are possible)

Electronic Configuration of A:

Electronic Configuration of A^{2+} :

c. 2 points

In a single atom, determine how many electrons can have the following set of quantum numbers?

(i) $n = 5; l = 3$

In a single atom, determine how many electrons can have the following set of quantum numbers?

(i) $n = 3; l = 2; m_s = +1/2$

**Question 4** **8 points**

Rank the following from 1 to 4 in **increasing** order. Provide the ranking in the boxes provided.

Note: Atomic number (Z) are provided for relevant atoms

a) Increasing (lowest to highest) order of *ionization energy*: (1= lowest; 4 = highest)

Mg

Mg: Z = 12

Rb

Rb: Z = 37

Cs

Cs: Z = 55

K

K : Z = 19

b) Increasing (lowest to highest) order of *bond angle*: (1= lowest; 4 = highest)

H₂OCCl₄XeF₄

HCN

c) Increasing (smallest to largest) order of *ionic size* (1= smallest; 4 = largest)

S²⁻

S: Z = 16

O²⁻

O: Z = 8

Be²⁺

Be: Z = 4

Na⁺

Na: Z = 11

d) Increasing (lowest to highest) order of *bond energy* (1= lowest; 4 = highest)

P-F

P-Cl

P-I

P-Br

**Question 5** **12 points**

a. 9 points For the molecules HClO_4 and TeF_6 , draw the most stable *Lewis structure* showing appropriate shape and *dashed/wedged bonds*, based on VSEPR. Determine the **molecular geometry** around the central atom. For HClO_4 , label the *most polar bond*.

HClO_4 (there are no O-O bonds)

Label the most polar bond

Molecular Geometry around Cl

TeF_6

Molecular Geometry around Te

b. 3 points (Note: Lone pairs are not shown in structures below)

(i) For the three *resonance* structures (1, 2 and 3) shown below, answer the following question:



(I) Circle the *most contributing* resonance structure.

Structure 1

Structure 2

Structure 3

(ii) For the three structures given (4, 5, and 6) below, answer the following questions:



(I) Circle all the structures that represent *resonance structures* of each other:

Structure 4

Structure 5

Structure 6

(II) Among the resonance structures, circle the *most contributing* resonance structure.

Structure 4

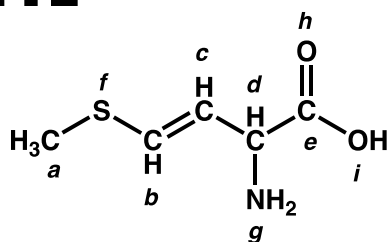
Structure 5

Structure 6

**Question 6** **6 points**

Using the data given below, calculate the **lattice energy** for SrO(s).

	ΔH° (kJ/mol)
$\text{Sr}(s) \longrightarrow \text{Sr}(g)$	+180
$\text{Sr}(g) \longrightarrow \text{Sr}^{2+}(g) + 2e^-$	+1468
$\text{O}_2(g) \longrightarrow 2\text{O}(g)$	+500
$\text{O}(g) + 2e^- \longrightarrow \text{O}^{2-}(g)$	+600
$\text{Sr}(s) + \frac{1}{2}\text{O}_2(g) \longrightarrow \text{SrO}(s)$	-550

**Question 7 10 points**

Consider the molecule given on the left, and answer questions (i) to (vi) using chemical bonding theories that we have learnt in class (valence bond theory, hybridization).

Note that the lone pairs are not shown in the structure:

i. Consider the carbon atoms marked in as ***a***, ***b***, ***c***, ***d***, and ***e*** in the molecule, and determine **hybridization** at each of the carbon centers. (2.5 points)

Atom	Hybridization
Carbon (<i>a</i>)	
Carbon (<i>b</i>)	
Carbon (<i>c</i>)	
Carbon (<i>d</i>)	
Carbon (<i>e</i>)	

ii. Now, consider the sulfur atom (S) marked as ***f***, nitrogen (N) atom (***g***), and oxygen(O) atoms (***h*** and ***i***), and determine their **hybridization**. (2 points)

Atom	Hybridization
Sulfur (<i>f</i>)	
Nitrogen (<i>g</i>)	
Oxygen (<i>h</i>)	
Oxygen (<i>i</i>)	

iii. **Describe** all the bonds formed by the carbon atoms ***a*** and ***b***, using sigma and pi designation as appropriate. (2 points)

Atom	Bonds formed (use sigma and pi as appropriate)
Carbon (<i>a</i>)	
Carbon (<i>b</i>)	

iv. How many total sp^3 -s sigma bonds are in the molecule? (1 point)

v. Determine **electronic** and **molecular geometry** at carbon (C) labeled ***e*** and sulfur (S) labeled ***f*** centres. (2 points)

Atom	Electronic Geometry	Molecular Geometry
Carbon (<i>e</i>)		
Sulfur (<i>f</i>)		

vi. Circle the value from the choices given below, that describes the **C-S-C bond angle at sulfur** (S) marked ***f***: (0.5 points)

>90° to <109.5°

109.5°

>109.5 to <120°

120°

>120° to <180°

**Question 8 12 points**

Using the **valence shell orbitals only** draw a **complete** molecular orbital (MO) diagram (with no s-p mixing) for the molecule **PS**, in the space provided below. **(6 points)**

Assume that 3s and 3p atomic orbitals will form molecular orbitals like those formed by the combination of 2s and 2p atomic orbitals that we learnt in class, and there is no s-p mixing.

Show **energy arrow**, **atomic orbitals** involved, **atom labels**, the **molecular orbitals** that are formed, **filled electrons** in the MO diagram etc. Make sure to label all the atomic and molecular orbitals.

Using the MO diagram above, answer the following questions:

- (i) Which one of the following will have the highest bond order? Circle the correct choice
A) PS^{1-} B) PS C) PS^{1+}
- (ii) Which one of the following will have the longest bond length? Circle the correct choice
A) PS^{1-} B) PS C) PS^{1+}
- (iii) Which of the following will be diamagnetic? Circle the correct choice.
A) PS^{1-} B) PS C) PS^{1+}
- (iv) Draw Lewis dot structure of the molecule with the shortest bond length (based on the MO diagram) from among the three (PS^{1-} , PS, PS^{1+}), and indicate the bond order calculated from the Lewis dot structure. **(3 points)**

Lewis Dot Structure

Bond Order



F32F7C30-3596-4E98-87AF-F627AC2F1CE1

midterm-exam-93388

#796 Page 10 of 10

SCRAP PAGE

**CHEMISTRY 110 –Midterm Exam****Thursday November 7, 2024, 6:30 – 8:30 pm****Examiners: Pallavi Sirjoosingh and Ashok Kakkar**

First name (please write as legibly as possible within the boxes)

--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--

Last name

--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--

Student ID number

--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--

This booklet has 8 questions on 10 pages (The last page can be used as a scrap page).

The staple must NOT be removed from the question booklet. Your intact question booklet must be returned at the end of the exam.

A separate booklet contains a periodic table, and datasheet.

Write your answers directly under the questions in the space provided. Do not write on the QR codes or near the top edges (as it may be cut off).

Please write **legibly and dark enough** for the scan to be readable. For students who use English as a second language, dictionaries are permitted. Calculators are allowed.

DO NOT ASK QUESTIONS OF THE INVIGILATORS about interpretations of the exam questions. If you have a concern about a question, please summarize it at the top of this page. All concerns will be evaluated after the exam. GOOD LUCK!

**Question 1 8 points** (*Assume all transitions are possible*)

a. 6 points Calculate the **energy (in kJ/mol)** required to ionize (completely remove) the electron from the **n=3** level of **He⁺ ion** (He: Helium). Show your work including equations. (3 sig. figs.)

b. 2 points

(I) Select the largest (longest) wavelength from the three choices below:

- i. Wavelength of photon to *completely remove* an electron from *1s orbital of Hydrogen atom*
- ii. Wavelength of photon to *completely remove* an electron from *1s orbital of Boron atom (B)*
- iii. Wavelength of photon to *completely remove* an electron from *1s orbital of Boron cation (B⁺)*

i or ii or iii

(II) Select the smallest (shortest) wavelength from the three choices below (*for H atom*):

- i. Wavelength of photon *emitted* for an electron transitioning from *2p to 1s orbital*
- ii. Wavelength of photon *absorbed* for an electron transitioning from *2s to 4p orbital*
- iii. Wavelength of photon *absorbed* for an electron transitioning from *1s to 3p orbital*

i or ii or iii



Question 2 8 points



A Chem110 student plots the graph shown on the left, from data obtained for the kinetic energy (eV) of the ejected electron versus frequency (s^{-1}) of the incident light, for an unknown metal. ($1\text{eV} = 1.6022 \times 10^{-19} \text{ J}$)

Based on the plot, answer the questions **a to c** below:

a. 1 point On the plot above, label the **threshold frequency**.

b. 2 points What is the value of the work potential (in J) of the unknown metal?

c. 3 points What is the *longest* wavelength (in nm) required to eject an electron from this metal's surface? Show your work including equation(s) used. (3 sig figs)

d. 2 points In the space provided below, plot the following

(Assume frequency is *above* threshold frequency):

- Kinetic energy* of an ejected electron versus the *intensity* of the incident light (left)
- Number of electrons* ejected versus the *frequency* of the incident light (right)



**Question 3** **6 points****a. 2 points**

In the ground state of Ge (Germanium; $Z = 32$) atom, **name** the following orbitals:

i. Orbital containing the electron that requires the maximum energy to be removed.

ii. Orbital containing the electron that experiences the minimum effective nuclear charge.

b. 2 points

An unknown anion (A^-) has the following *excited* electronic configuration: $[Ar] 4s^2 3d^3 4p^1$

(Ar: Argon; $Z = 18$)

Give the **condensed ground state electronic configuration** of the atom, **A** and the cation, **A^{2+}** .
Show your work. (Assume all electronic transitions are possible)

Electronic Configuration of A:

Electronic Configuration of A^{2+} :

c. 2 points

In a single atom, determine how many electrons can have the following set of quantum numbers?

(i) $n = 5; l = 3$

In a single atom, determine how many electrons can have the following set of quantum numbers?

(i) $n = 3; l = 2; m_s = +1/2$

**Question 4** **8 points**

Rank the following from 1 to 4 in **increasing** order. Provide the ranking in the boxes provided.

Note: Atomic number (Z) are provided for relevant atoms

a) Increasing (lowest to highest) order of *ionization energy*: (1= lowest; 4 = highest)

Mg

Mg: Z = 12

Rb

Rb: Z = 37

Cs

Cs: Z = 55

K

K : Z = 19

b) Increasing (lowest to highest) order of *bond angle*: (1= lowest; 4 = highest)

H₂OCCl₄XeF₄

HCN

c) Increasing (smallest to largest) order of *ionic size* (1= smallest; 4 = largest)

S²⁻

S: Z = 16

O²⁻

O: Z = 8

Be²⁺

Be: Z = 4

Na⁺

Na: Z = 11

d) Increasing (lowest to highest) order of *bond energy* (1= lowest; 4 = highest)

P-F

P-Cl

P-I

P-Br

**Question 5** **12 points**

a. 9 points For the molecules HClO_4 and TeF_6 , draw the most stable *Lewis structure* showing appropriate shape and *dashed/wedged bonds*, based on VSEPR. Determine the **molecular geometry** around the central atom. For HClO_4 , label the *most polar bond*.

HClO_4 (there are no O-O bonds)

Label the most polar bond

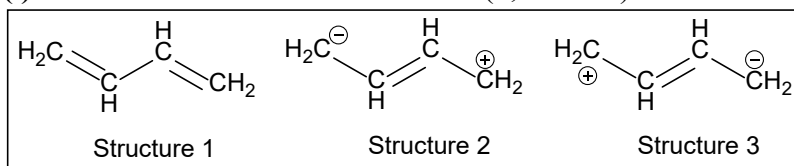
Molecular Geometry around Cl

TeF_6

Molecular Geometry around Te

b. 3 points (Note: Lone pairs are not shown in structures below)

(i) For the three *resonance* structures (1, 2 and 3) shown below, answer the following question:



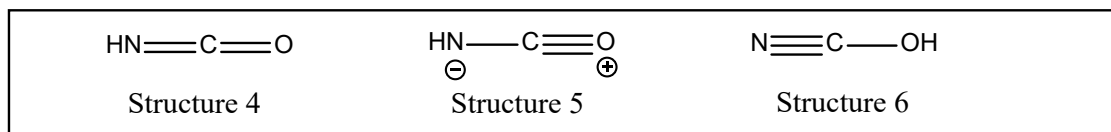
(I) Circle the *most contributing* resonance structure.

Structure 1

Structure 2

Structure 3

(ii) For the three structures given (4, 5, and 6) below, answer the following questions:



(I) Circle all the structures that represent *resonance structures* of each other:

Structure 4

Structure 5

Structure 6

(II) Among the resonance structures, circle the *most contributing* resonance structure.

Structure 4

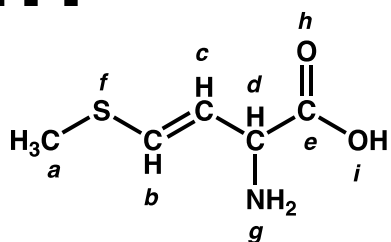
Structure 5

Structure 6

**Question 6** **6 points**

Using the data given below, calculate the **lattice energy** for SrO(s).

	ΔH° (kJ/mol)
$\text{Sr}(s) \longrightarrow \text{Sr}(g)$	+180
$\text{Sr}(g) \longrightarrow \text{Sr}^{2+}(g) + 2e^-$	+1468
$\text{O}_2(g) \longrightarrow 2\text{O}(g)$	+500
$\text{O}(g) + 2e^- \longrightarrow \text{O}^{2-}(g)$	+600
$\text{Sr}(s) + \frac{1}{2}\text{O}_2(g) \longrightarrow \text{SrO}(s)$	-550

**Question 7 10 points**

Consider the molecule given on the left, and answer questions (i) to (vi) using chemical bonding theories that we have learnt in class (valence bond theory, hybridization).

Note that the lone pairs are not shown in the structure:

i. Consider the carbon atoms marked in as ***a***, ***b***, ***c***, ***d***, and ***e*** in the molecule, and determine **hybridization** at each of the carbon centers. (2.5 points)

Atom	Hybridization
Carbon (<i>a</i>)	
Carbon (<i>b</i>)	
Carbon (<i>c</i>)	
Carbon (<i>d</i>)	
Carbon (<i>e</i>)	

ii. Now, consider the sulfur atom (S) marked as ***f***, nitrogen (N) atom (***g***), and oxygen(O) atoms (***h*** and ***i***), and determine their **hybridization**. (2 points)

Atom	Hybridization
Sulfur (<i>f</i>)	
Nitrogen (<i>g</i>)	
Oxygen (<i>h</i>)	
Oxygen (<i>i</i>)	

iii. **Describe** all the bonds formed by the carbon atoms ***a*** and ***b***, using sigma and pi designation as appropriate. (2 points)

Atom	Bonds formed (use sigma and pi as appropriate)
Carbon (<i>a</i>)	
Carbon (<i>b</i>)	

iv. How many total sp^3 -s sigma bonds are in the molecule? (1 point)

v. Determine **electronic** and **molecular geometry** at carbon (C) labeled ***e*** and sulfur (S) labeled ***f*** centres. (2 points)

Atom	Electronic Geometry	Molecular Geometry
Carbon (<i>e</i>)		
Sulfur (<i>f</i>)		

vi. Circle the value from the choices given below, that describes the **C-S-C bond angle at sulfur** (S) marked ***f***: (0.5 points)

>90° to <109.5° 109.5° >109.5 to <120° 120° >120° to <180°


Question 8 12 points

Using the **valence shell orbitals only** draw a **complete** molecular orbital (MO) diagram (with no s-p mixing) for the molecule **PS**, in the space provided below. **(6 points)**

Assume that 3s and 3p atomic orbitals will form molecular orbitals like those formed by the combination of 2s and 2p atomic orbitals that we learnt in class, and there is no s-p mixing.

Show **energy arrow**, **atomic orbitals** involved, **atom labels**, the **molecular orbitals** that are formed, **filled electrons** in the MO diagram etc. Make sure to **label all the atomic and molecular orbitals**.

Using the MO diagram above, answer the following questions:

- (i) Which one of the following will have the highest bond order? Circle the correct choice
 A) PS^{1-} B) PS C) PS^{1+}
- (ii) Which one of the following will have the longest bond length? Circle the correct choice
 A) PS^{1-} B) PS C) PS^{1+}
- (iii) Which of the following will be diamagnetic? Circle the correct choice.
 A) PS^{1-} B) PS C) PS^{1+}
- (iv) Draw Lewis dot structure of the molecule with the shortest bond length (based on the MO diagram) from among the three (PS^{1-} , PS, PS^{1+}), and indicate the bond order calculated from the Lewis dot structure. **(3 points)**

Lewis Dot Structure

Bond Order



047F562B-4F27-431C-988A-DBA3C76020D5

midterm-exam-93388

#797 Page 10 of 10

SCRAP PAGE

**CHEMISTRY 110 –Midterm Exam****Thursday November 7, 2024, 6:30 – 8:30 pm**

Examiners: Pallavi Sirjoosingh and Ashok Kakkar

First name (please write as legibly as possible within the boxes)

--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--

Last name

--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--

Student ID number

--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--

This booklet has 8 questions on 10 pages (The last page can be used as a scrap page).

The staple must NOT be removed from the question booklet. Your intact question booklet must be returned at the end of the exam.

A separate booklet contains a periodic table, and datasheet.

Write your answers directly under the questions in the space provided. Do not write on the QR codes or near the top edges (as it may be cut off).

Please write **legibly and dark enough** for the scan to be readable. For students who use English as a second language, dictionaries are permitted. Calculators are allowed.

DO NOT ASK QUESTIONS OF THE INVIGILATORS about interpretations of the exam questions. If you have a concern about a question, please summarize it at the top of this page. All concerns will be evaluated after the exam. GOOD LUCK!

**Question 1 8 points** (*Assume all transitions are possible*)

a. 6 points Calculate the **energy (in kJ/mol)** required to ionize (completely remove) the electron from the **n=3** level of **He⁺ ion** (He: Helium). Show your work including equations. (3 sig. figs.)

b. 2 points

(I) Select the largest (longest) wavelength from the three choices below:

- i. Wavelength of photon to *completely remove* an electron from *1s orbital of Hydrogen atom*
- ii. Wavelength of photon to *completely remove* an electron from *1s orbital of Boron atom (B)*
- iii. Wavelength of photon to *completely remove* an electron from *1s orbital of Boron cation (B⁺)*

i or ii or iii

(II) Select the smallest (shortest) wavelength from the three choices below (*for H atom*):

- i. Wavelength of photon *emitted* for an electron transitioning from *2p to 1s orbital*
- ii. Wavelength of photon *absorbed* for an electron transitioning from *2s to 4p orbital*
- iii. Wavelength of photon *absorbed* for an electron transitioning from *1s to 3p orbital*

i or ii or iii



Question 2 8 points



A Chem110 student plots the graph shown on the left, from data obtained for the kinetic energy (eV) of the ejected electron versus frequency (s^{-1}) of the incident light, for an unknown metal. ($1\text{eV} = 1.6022 \times 10^{-19} \text{ J}$)

Based on the plot, answer the questions **a to c** below:

a. 1 point On the plot above, label the **threshold frequency**.

b. 2 points What is the value of the work potential (in J) of the unknown metal?

c. 3 points What is the *longest* wavelength (in nm) required to eject an electron from this metal's surface? Show your work including equation(s) used. (3 sig figs)

d. 2 points In the space provided below, plot the following

(Assume frequency is *above* threshold frequency):

- i. *Kinetic energy* of an ejected electron versus the *intensity* of the incident light (left)
- ii. *Number of electrons* ejected versus the *frequency* of the incident light (right)



**Question 3** **6 points****a. 2 points**

In the ground state of Ge (Germanium; $Z = 32$) atom, **name** the following orbitals:

i. Orbital containing the electron that requires the maximum energy to be removed.

ii. Orbital containing the electron that experiences the minimum effective nuclear charge.

b. 2 points

An unknown anion (A^-) has the following *excited* electronic configuration: $[Ar] 4s^2 3d^3 4p^1$

(Ar: Argon; $Z = 18$)

Give the **condensed ground state electronic configuration** of the atom, **A** and the cation, A^{2+} .
Show your work. (Assume all electronic transitions are possible)

Electronic Configuration of A:

Electronic Configuration of A^{2+} :

c. 2 points

In a single atom, determine how many electrons can have the following set of quantum numbers?

(i) $n = 5; l = 3$

In a single atom, determine how many electrons can have the following set of quantum numbers?

(i) $n = 3; l = 2; m_s = +1/2$

**Question 4** **8 points**

Rank the following from 1 to 4 in **increasing** order. Provide the ranking in the boxes provided.

Note: Atomic number (Z) are provided for relevant atoms

a) Increasing (lowest to highest) order of *ionization energy*: (1= lowest; 4 = highest)

Mg

Mg: Z = 12

Rb

Rb: Z = 37

Cs

Cs: Z = 55

K

K : Z = 19

b) Increasing (lowest to highest) order of *bond angle*: (1= lowest; 4 = highest)

H₂OCCl₄XeF₄

HCN

c) Increasing (smallest to largest) order of *ionic size* (1= smallest; 4 = largest)

S²⁻

S: Z = 16

O²⁻

O: Z = 8

Be²⁺

Be: Z = 4

Na⁺

Na: Z = 11

d) Increasing (lowest to highest) order of *bond energy* (1= lowest; 4 = highest)

P-F

P-Cl

P-I

P-Br

**Question 5** **12 points**

a. 9 points For the molecules HClO_4 and TeF_6 , draw the most stable *Lewis structure* showing appropriate shape and *dashed/wedged bonds*, based on VSEPR. Determine the **molecular geometry** around the central atom. For HClO_4 , label the *most polar bond*.

HClO_4 (there are no O-O bonds)

Label the most polar bond

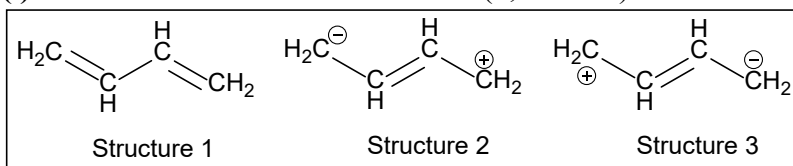
Molecular Geometry around Cl

TeF_6

Molecular Geometry around Te

b. 3 points (Note: Lone pairs are not shown in structures below)

(i) For the three *resonance* structures (1, 2 and 3) shown below, answer the following question:



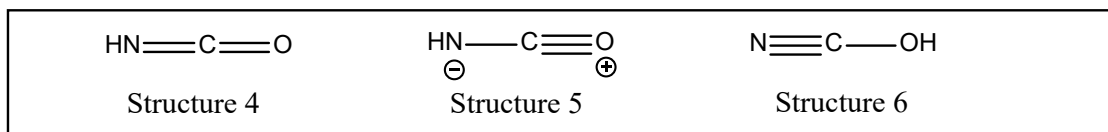
(I) Circle the *most contributing* resonance structure.

Structure 1

Structure 2

Structure 3

(ii) For the three structures given (4, 5, and 6) below, answer the following questions:



(I) Circle all the structures that represent *resonance structures* of each other:

Structure 4

Structure 5

Structure 6

(II) Among the resonance structures, circle the *most contributing* resonance structure.

Structure 4

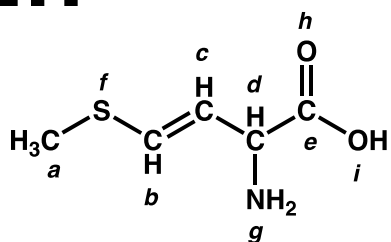
Structure 5

Structure 6

**Question 6** **6 points**

Using the data given below, calculate the **lattice energy** for SrO(s).

	ΔH° (kJ/mol)
$\text{Sr}(s) \longrightarrow \text{Sr}(g)$	+180
$\text{Sr}(g) \longrightarrow \text{Sr}^{2+}(g) + 2e^-$	+1468
$\text{O}_2(g) \longrightarrow 2\text{O}(g)$	+500
$\text{O}(g) + 2e^- \longrightarrow \text{O}^{2-}(g)$	+600
$\text{Sr}(s) + \frac{1}{2}\text{O}_2(g) \longrightarrow \text{SrO}(s)$	-550

**Question 7 10 points**

Consider the molecule given on the left, and answer questions (i) to (vi) using chemical bonding theories that we have learnt in class (valence bond theory, hybridization).

Note that the lone pairs are not shown in the structure:

i. Consider the carbon atoms marked in as ***a***, ***b***, ***c***, ***d***, and ***e*** in the molecule, and determine **hybridization** at each of the carbon centers. (2.5 points)

Atom	Hybridization
Carbon (<i>a</i>)	
Carbon (<i>b</i>)	
Carbon (<i>c</i>)	
Carbon (<i>d</i>)	
Carbon (<i>e</i>)	

ii. Now, consider the sulfur atom (S) marked as ***f***, nitrogen (N) atom (***g***), and oxygen(O) atoms (***h*** and ***i***), and determine their **hybridization**. (2 points)

Atom	Hybridization
Sulfur (<i>f</i>)	
Nitrogen (<i>g</i>)	
Oxygen (<i>h</i>)	
Oxygen (<i>i</i>)	

iii. **Describe** all the bonds formed by the carbon atoms ***a*** and ***b***, using sigma and pi designation as appropriate. (2 points)

Atom	Bonds formed (use sigma and pi as appropriate)
Carbon (<i>a</i>)	
Carbon (<i>b</i>)	

iv. How many total sp^3 -s sigma bonds are in the molecule? (1 point)

v. Determine **electronic** and **molecular geometry** at carbon (C) labeled ***e*** and sulfur (S) labeled ***f*** centres. (2 points)

Atom	Electronic Geometry	Molecular Geometry
Carbon (<i>e</i>)		
Sulfur (<i>f</i>)		

vi. Circle the value from the choices given below, that describes the **C-S-C bond angle at sulfur** (S) marked ***f***: (0.5 points)

>90° to <109.5°

109.5°

>109.5 to <120°

120°

>120° to <180°


Question 8 12 points

Using the **valence shell orbitals only** draw a **complete** molecular orbital (MO) diagram (with no s-p mixing) for the molecule **PS**, in the space provided below. **(6 points)**

Assume that 3s and 3p atomic orbitals will form molecular orbitals like those formed by the combination of 2s and 2p atomic orbitals that we learnt in class, and there is no s-p mixing.

Show **energy arrow**, **atomic orbitals** involved, **atom labels**, the **molecular orbitals** that are formed, **filled electrons** in the MO diagram etc. Make sure to label all the atomic and molecular orbitals.

Using the MO diagram above, answer the following questions:

- (i) Which one of the following will have the highest bond order? Circle the correct choice
 A) PS^{1-} B) PS C) PS^{1+}
- (ii) Which one of the following will have the longest bond length? Circle the correct choice
 A) PS^{1-} B) PS C) PS^{1+}
- (iii) Which of the following will be diamagnetic? Circle the correct choice.
 A) PS^{1-} B) PS C) PS^{1+}
- (iv) Draw Lewis dot structure of the molecule with the shortest bond length (based on the MO diagram) from among the three (PS^{1-} , PS, PS^{1+}), and indicate the bond order calculated from the Lewis dot structure. **(3 points)**

Lewis Dot Structure

Bond Order



E0D6D4CC-7DA9-4E22-A260-FBE3F29819C6

midterm-exam-93388

#798 Page 10 of 10

SCRAP PAGE

**CHEMISTRY 110 –Midterm Exam****Thursday November 7, 2024, 6:30 – 8:30 pm**

Examiners: Pallavi Sirjoosingh and Ashok Kakkar

First name (please write as legibly as possible within the boxes)

--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--

Last name

--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--

Student ID number

--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--

This booklet has 8 questions on 10 pages (The last page can be used as a scrap page).

The staple must NOT be removed from the question booklet. Your intact question booklet must be returned at the end of the exam.

A separate booklet contains a periodic table, and datasheet.

Write your answers directly under the questions in the space provided. Do not write on the QR codes or near the top edges (as it may be cut off).

Please write **legibly and dark enough** for the scan to be readable. For students who use English as a second language, dictionaries are permitted. Calculators are allowed.

DO NOT ASK QUESTIONS OF THE INVIGILATORS about interpretations of the exam questions. If you have a concern about a question, please summarize it at the top of this page. All concerns will be evaluated after the exam. GOOD LUCK!

**Question 1 8 points** (*Assume all transitions are possible*)

a. 6 points Calculate the **energy (in kJ/mol)** required to ionize (completely remove) the electron from the **n=3** level of **He⁺ ion** (He: Helium). Show your work including equations. (3 sig. figs.)

b. 2 points

(I) Select the largest (longest) wavelength from the three choices below:

- i. Wavelength of photon to *completely remove* an electron from *1s orbital of Hydrogen atom*
- ii. Wavelength of photon to *completely remove* an electron from *1s orbital of Boron atom (B)*
- iii. Wavelength of photon to *completely remove* an electron from *1s orbital of Boron cation (B⁺)*

i or ii or iii

(II) Select the smallest (shortest) wavelength from the three choices below (*for H atom*):

- i. Wavelength of photon *emitted* for an electron transitioning from *2p to 1s orbital*
- ii. Wavelength of photon *absorbed* for an electron transitioning from *2s to 4p orbital*
- iii. Wavelength of photon *absorbed* for an electron transitioning from *1s to 3p orbital*

i or ii or iii



Question 2 8 points



A Chem110 student plots the graph shown on the left, from data obtained for the kinetic energy (eV) of the ejected electron versus frequency (s^{-1}) of the incident light, for an unknown metal. ($1\text{eV} = 1.6022 \times 10^{-19} \text{ J}$)

Based on the plot, answer the questions **a to c** below:

a. 1 point On the plot above, label the **threshold frequency**.

b. 2 points What is the value of the work potential (in J) of the unknown metal?

c. 3 points What is the *longest* wavelength (in nm) required to eject an electron from this metal's surface? Show your work including equation(s) used. (3 sig figs)

d. 2 points In the space provided below, plot the following

(Assume frequency is *above* threshold frequency):

- i. *Kinetic energy* of an ejected electron versus the *intensity* of the incident light (left)
- ii. *Number of electrons* ejected versus the *frequency* of the incident light (right)



**Question 3** **6 points****a. 2 points**

In the ground state of Ge (Germanium; $Z = 32$) atom, **name** the following orbitals:

i. Orbital containing the electron that requires the maximum energy to be removed.

ii. Orbital containing the electron that experiences the minimum effective nuclear charge.

b. 2 points

An unknown anion (A^-) has the following *excited* electronic configuration: $[Ar] 4s^2 3d^3 4p^1$

(Ar: Argon; $Z = 18$)

Give the **condensed ground state electronic configuration** of the atom, **A** and the cation, **A^{2+}** .
Show your work. (Assume all electronic transitions are possible)

Electronic Configuration of A:

Electronic Configuration of A^{2+} :

c. 2 points

In a single atom, determine how many electrons can have the following set of quantum numbers?

(i) $n = 5; l = 3$

In a single atom, determine how many electrons can have the following set of quantum numbers?

(i) $n = 3; l = 2; m_s = +1/2$

**Question 4** **8 points**

Rank the following from 1 to 4 in **increasing** order. Provide the ranking in the boxes provided.

Note: Atomic number (Z) are provided for relevant atoms

a) Increasing (lowest to highest) order of *ionization energy*: (1= lowest; 4 = highest)

Mg

Mg: Z = 12

Rb

Rb: Z = 37

Cs

Cs: Z = 55

K

K : Z = 19

b) Increasing (lowest to highest) order of *bond angle*: (1= lowest; 4 = highest)

H₂OCCl₄XeF₄

HCN

c) Increasing (smallest to largest) order of *ionic size* (1= smallest; 4 = largest)

S²⁻

S: Z = 16

O²⁻

O: Z = 8

Be²⁺

Be: Z = 4

Na⁺

Na: Z = 11

d) Increasing (lowest to highest) order of *bond energy* (1= lowest; 4 = highest)

P-F

P-Cl

P-I

P-Br

**Question 5** **12 points**

a. 9 points For the molecules HClO_4 and TeF_6 , draw the most stable *Lewis structure* showing appropriate shape and *dashed/wedged bonds*, based on VSEPR. Determine the **molecular geometry** around the central atom. For HClO_4 , label the *most polar bond*.

HClO_4 (there are no O-O bonds)

Label the most polar bond

Molecular Geometry around Cl

TeF_6

Molecular Geometry around Te

b. 3 points (Note: Lone pairs are not shown in structures below)

(i) For the three *resonance* structures (1, 2 and 3) shown below, answer the following question:



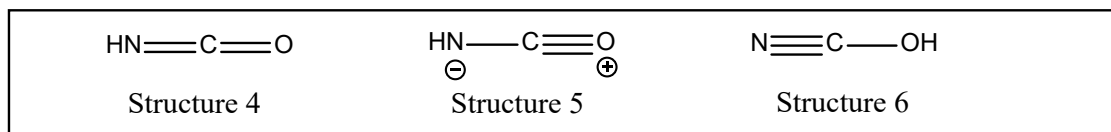
(I) Circle the *most contributing* resonance structure.

Structure 1

Structure 2

Structure 3

(ii) For the three structures given (4, 5, and 6) below, answer the following questions:



(I) Circle all the structures that represent *resonance structures* of each other:

Structure 4

Structure 5

Structure 6

(II) Among the resonance structures, circle the *most contributing* resonance structure.

Structure 4

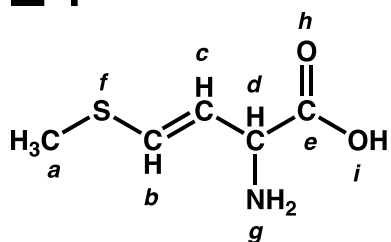
Structure 5

Structure 6

**Question 6** **6 points**

Using the data given below, calculate the **lattice energy** for SrO(s).

	ΔH° (kJ/mol)
$\text{Sr}(s) \longrightarrow \text{Sr}(g)$	+180
$\text{Sr}(g) \longrightarrow \text{Sr}^{2+}(g) + 2e^-$	+1468
$\text{O}_2(g) \longrightarrow 2\text{O}(g)$	+500
$\text{O}(g) + 2e^- \longrightarrow \text{O}^{2-}(g)$	+600
$\text{Sr}(s) + \frac{1}{2}\text{O}_2(g) \longrightarrow \text{SrO}(s)$	-550

**Question 7 10 points**

Consider the molecule given on the left, and answer questions (i) to (vi) using chemical bonding theories that we have learnt in class (valence bond theory, hybridization).

Note that the lone pairs are not shown in the structure:

i. Consider the carbon atoms marked in as ***a***, ***b***, ***c***, ***d***, and ***e*** in the molecule, and determine **hybridization** at each of the carbon centers. (2.5 points)

Atom	Hybridization
Carbon (<i>a</i>)	
Carbon (<i>b</i>)	
Carbon (<i>c</i>)	
Carbon (<i>d</i>)	
Carbon (<i>e</i>)	

ii. Now, consider the sulfur atom (S) marked as ***f***, nitrogen (N) atom (***g***), and oxygen(O) atoms (***h*** and ***i***), and determine their **hybridization**. (2 points)

Atom	Hybridization
Sulfur (<i>f</i>)	
Nitrogen (<i>g</i>)	
Oxygen (<i>h</i>)	
Oxygen (<i>i</i>)	

iii. **Describe** all the bonds formed by the carbon atoms ***a*** and ***b***, using sigma and pi designation as appropriate. (2 points)

Atom	Bonds formed (use sigma and pi as appropriate)
Carbon (<i>a</i>)	
Carbon (<i>b</i>)	

iv. How many total sp^3 -s sigma bonds are in the molecule? (1 point)

v. Determine **electronic** and **molecular geometry** at carbon (C) labeled ***e*** and sulfur (S) labeled ***f*** centres. (2 points)

Atom	Electronic Geometry	Molecular Geometry
Carbon (<i>e</i>)		
Sulfur (<i>f</i>)		

vi. Circle the value from the choices given below, that describes the **C-S-C bond angle at sulfur** (S) marked ***f***: (0.5 points)

>90° to <109.5°

109.5°

>109.5 to <120°

120°

>120° to <180°

**Question 8 12 points**

Using the **valence shell orbitals only** draw a **complete** molecular orbital (MO) diagram (with no s-p mixing) for the molecule **PS**, in the space provided below. **(6 points)**

Assume that 3s and 3p atomic orbitals will form molecular orbitals like those formed by the combination of 2s and 2p atomic orbitals that we learnt in class, and there is no s-p mixing.

Show **energy arrow**, **atomic orbitals** involved, **atom labels**, the **molecular orbitals** that are formed, **filled electrons** in the MO diagram etc. Make sure to label all the atomic and molecular orbitals.

Using the MO diagram above, answer the following questions:

- (i) Which one of the following will have the highest bond order? Circle the correct choice
A) PS^{1-} B) PS C) PS^{1+}
- (ii) Which one of the following will have the longest bond length? Circle the correct choice
A) PS^{1-} B) PS C) PS^{1+}
- (iii) Which of the following will be diamagnetic? Circle the correct choice.
A) PS^{1-} B) PS C) PS^{1+}
- (iv) Draw Lewis dot structure of the molecule with the shortest bond length (based on the MO diagram) from among the three (PS^{1-} , PS, PS^{1+}), and indicate the bond order calculated from the Lewis dot structure. **(3 points)**

Lewis Dot Structure

Bond Order



3F16E3CF-2EDC-4D31-83D8-B6341BC1EED3

midterm-exam-93388

#799 Page 10 of 10

SCRAP PAGE

**CHEMISTRY 110 –Midterm Exam****Thursday November 7, 2024, 6:30 – 8:30 pm****Examiners: Pallavi Sirjoosingh and Ashok Kakkar**

First name (please write as legibly as possible within the boxes)

--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--

Last name

--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--

Student ID number

--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--

This booklet has 8 questions on 10 pages (The last page can be used as a scrap page).

The staple must NOT be removed from the question booklet. Your intact question booklet must be returned at the end of the exam.

A separate booklet contains a periodic table, and datasheet.

Write your answers directly under the questions in the space provided. Do not write on the QR codes or near the top edges (as it may be cut off).

Please write **legibly and dark enough** for the scan to be readable. For students who use English as a second language, dictionaries are permitted. Calculators are allowed.

DO NOT ASK QUESTIONS OF THE INVIGILATORS about interpretations of the exam questions. If you have a concern about a question, please summarize it at the top of this page. All concerns will be evaluated after the exam. GOOD LUCK!

**Question 1 8 points** (*Assume all transitions are possible*)

a. 6 points Calculate the **energy (in kJ/mol)** required to ionize (completely remove) the electron from the **n=3** level of **He⁺ ion** (He: Helium). Show your work including equations. (3 sig. figs.)

b. 2 points

(I) Select the largest (longest) wavelength from the three choices below:

- i. Wavelength of photon to *completely remove* an electron from *1s orbital of Hydrogen atom*
- ii. Wavelength of photon to *completely remove* an electron from *1s orbital of Boron atom (B)*
- iii. Wavelength of photon to *completely remove* an electron from *1s orbital of Boron cation (B⁺)*

i or ii or iii

(II) Select the smallest (shortest) wavelength from the three choices below (*for H atom*):

- i. Wavelength of photon *emitted* for an electron transitioning from *2p to 1s orbital*
- ii. Wavelength of photon *absorbed* for an electron transitioning from *2s to 4p orbital*
- iii. Wavelength of photon *absorbed* for an electron transitioning from *1s to 3p orbital*

i or ii or iii



Question 2 8 points



A Chem110 student plots the graph shown on the left, from data obtained for the kinetic energy (eV) of the ejected electron versus frequency (s^{-1}) of the incident light, for an unknown metal. ($1\text{eV} = 1.6022 \times 10^{-19} \text{ J}$)

Based on the plot, answer the questions **a to c** below:

a. 1 point On the plot above, label the **threshold frequency**.

b. 2 points What is the value of the work potential (in J) of the unknown metal?

c. 3 points What is the *longest* wavelength (in nm) required to eject an electron from this metal's surface? Show your work including equation(s) used. (3 sig figs)

d. 2 points In the space provided below, plot the following

(Assume frequency is *above* threshold frequency):

- i. *Kinetic energy* of an ejected electron versus the *intensity* of the incident light (left)
- ii. *Number of electrons* ejected versus the *frequency* of the incident light (right)



**Question 3** **6 points****a. 2 points**

In the ground state of Ge (Germanium; $Z = 32$) atom, **name** the following orbitals:

i. Orbital containing the electron that requires the maximum energy to be removed.

ii. Orbital containing the electron that experiences the minimum effective nuclear charge.

b. 2 points

An unknown anion (A^-) has the following *excited* electronic configuration: $[Ar] 4s^2 3d^3 4p^1$

(Ar: Argon; $Z = 18$)

Give the **condensed ground state electronic configuration** of the atom, **A** and the cation, **A^{2+}** .
Show your work. (Assume all electronic transitions are possible)

Electronic Configuration of A:

Electronic Configuration of A^{2+} :

c. 2 points

In a single atom, determine how many electrons can have the following set of quantum numbers?

(i) $n = 5; l = 3$

In a single atom, determine how many electrons can have the following set of quantum numbers?

(i) $n = 3; l = 2; m_s = +1/2$

**Question 4** **8 points**

Rank the following from 1 to 4 in **increasing** order. Provide the ranking in the boxes provided.

Note: Atomic number (Z) are provided for relevant atoms

a) Increasing (lowest to highest) order of *ionization energy*: (1= lowest; 4 = highest)

Mg

Mg: Z = 12

Rb

Rb: Z = 37

Cs

Cs: Z = 55

K

K : Z = 19

b) Increasing (lowest to highest) order of *bond angle*: (1= lowest; 4 = highest)

H₂OCCl₄XeF₄

HCN

c) Increasing (smallest to largest) order of *ionic size* (1= smallest; 4 = largest)

S²⁻

S: Z = 16

O²⁻

O: Z = 8

Be²⁺

Be: Z = 4

Na⁺

Na: Z = 11

d) Increasing (lowest to highest) order of *bond energy* (1= lowest; 4 = highest)

P-F

P-Cl

P-I

P-Br

**Question 5** **12 points**

a. 9 points For the molecules HClO_4 and TeF_6 , draw the most stable *Lewis structure* showing appropriate shape and *dashed/wedged bonds*, based on VSEPR. Determine the **molecular geometry** around the central atom. For HClO_4 , label the *most polar bond*.



b. 3 points (Note: Lone pairs are not shown in structures below)

(i) For the three *resonance* structures (1, 2 and 3) shown below, answer the following question:



(I) Circle the *most contributing* resonance structure.

Structure 1

Structure 2

Structure 3

(ii) For the three structures given (4, 5, and 6) below, answer the following questions:



(I) Circle all the structures that represent *resonance structures* of each other:

Structure 4

Structure 5

Structure 6

(II) Among the resonance structures, circle the *most contributing* resonance structure.

Structure 4

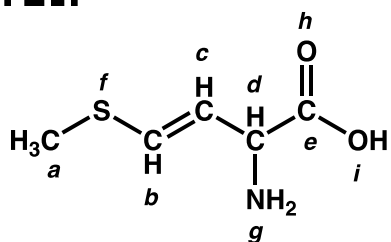
Structure 5

Structure 6

**Question 6** **6 points**

Using the data given below, calculate the **lattice energy** for SrO(s).

	ΔH° (kJ/mol)
$\text{Sr}(s) \longrightarrow \text{Sr}(g)$	+180
$\text{Sr}(g) \longrightarrow \text{Sr}^{2+}(g) + 2e^-$	+1468
$\text{O}_2(g) \longrightarrow 2\text{O}(g)$	+500
$\text{O}(g) + 2e^- \longrightarrow \text{O}^{2-}(g)$	+600
$\text{Sr}(s) + \frac{1}{2}\text{O}_2(g) \longrightarrow \text{SrO}(s)$	-550

**Question 7 10 points**

Consider the molecule given on the left, and answer questions (i) to (vi) using chemical bonding theories that we have learnt in class (valence bond theory, hybridization).

Note that the lone pairs are not shown in the structure:

i. Consider the carbon atoms marked in as ***a***, ***b***, ***c***, ***d***, and ***e*** in the molecule, and determine **hybridization** at each of the carbon centers. (2.5 points)

Atom	Hybridization
Carbon (<i>a</i>)	
Carbon (<i>b</i>)	
Carbon (<i>c</i>)	
Carbon (<i>d</i>)	
Carbon (<i>e</i>)	

ii. Now, consider the sulfur atom (S) marked as ***f***, nitrogen (N) atom (***g***), and oxygen(O) atoms (***h*** and ***i***), and determine their **hybridization**. (2 points)

Atom	Hybridization
Sulfur (<i>f</i>)	
Nitrogen (<i>g</i>)	
Oxygen (<i>h</i>)	
Oxygen (<i>i</i>)	

iii. **Describe** all the bonds formed by the carbon atoms ***a*** and ***b***, using sigma and pi designation as appropriate. (2 points)

Atom	Bonds formed (use sigma and pi as appropriate)
Carbon (<i>a</i>)	
Carbon (<i>b</i>)	

iv. How many total sp^3 -s sigma bonds are in the molecule? (1 point)

v. Determine **electronic** and **molecular geometry** at carbon (C) labeled ***e*** and sulfur (S) labeled ***f*** centres. (2 points)

Atom	Electronic Geometry	Molecular Geometry
Carbon (<i>e</i>)		
Sulfur (<i>f</i>)		

vi. Circle the value from the choices given below, that describes the **C-S-C bond angle at sulfur** (S) marked ***f***: (0.5 points)

>90° to <109.5° 109.5° >109.5 to <120° 120° >120° to <180°

**Question 8 12 points**

Using the **valence shell orbitals only** draw a **complete** molecular orbital (MO) diagram (with no s-p mixing) for the molecule **PS**, in the space provided below. **(6 points)**

Assume that 3s and 3p atomic orbitals will form molecular orbitals like those formed by the combination of 2s and 2p atomic orbitals that we learnt in class, and there is no s-p mixing.

Show **energy arrow**, **atomic orbitals** involved, **atom labels**, the **molecular orbitals** that are formed, **filled electrons** in the MO diagram etc. Make sure to label all the atomic and molecular orbitals.

Using the MO diagram above, answer the following questions:

- (i) Which one of the following will have the highest bond order? Circle the correct choice
A) PS^{1-} B) PS C) PS^{1+}
- (ii) Which one of the following will have the longest bond length? Circle the correct choice
A) PS^{1-} B) PS C) PS^{1+}
- (iii) Which of the following will be diamagnetic? Circle the correct choice.
A) PS^{1-} B) PS C) PS^{1+}
- (iv) Draw Lewis dot structure of the molecule with the shortest bond length (based on the MO diagram) from among the three (PS^{1-} , PS, PS^{1+}), and indicate the bond order calculated from the Lewis dot structure. **(3 points)**

Lewis Dot Structure

Bond Order



FDCC9FA1-3B15-4E99-809C-FC6DD20C53E3

midterm-exam-93388

#800 Page 10 of 10

SCRAP PAGE

**CHEMISTRY 110 –Midterm Exam****Thursday November 7, 2024, 6:30 – 8:30 pm**

Examiners: Pallavi Sirjoosingh and Ashok Kakkar

First name (please write as legibly as possible within the boxes)

--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--

Last name

--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--

Student ID number

--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--

This booklet has 8 questions on 10 pages (The last page can be used as a scrap page).

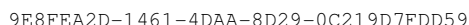
The staple must NOT be removed from the question booklet. Your intact question booklet must be returned at the end of the exam.

A separate booklet contains a periodic table, and datasheet.

Write your answers directly under the questions in the space provided. Do not write on the QR codes or near the top edges (as it may be cut off).

Please write **legibly and dark enough** for the scan to be readable. For students who use English as a second language, dictionaries are permitted. Calculators are allowed.

DO NOT ASK QUESTIONS OF THE INVIGILATORS about interpretations of the exam questions. If you have a concern about a question, please summarize it at the top of this page. All concerns will be evaluated after the exam. GOOD LUCK!



midterm-exam-93388

#801 Page 2 of 10

a. 6 points Calculate the **energy (in kJ/mol)** required to ionize (completely remove) the electron from the **n=3** level of **He⁺ ion** (He: Helium). Show your work including equations. (3 sig. figs.)

(I) Select the largest (longest) wavelength from the three choices below:

- Wavelength of photon to *completely remove* an electron from *1s orbital of Hydrogen atom*
- Wavelength of photon to *completely remove* an electron from *1s orbital of Boron atom (B)*
- Wavelength of photon to *completely remove* an electron from *1s orbital of Boron cation (B⁺)*

i
or
ii
or
iii

(II) Select the smallest (shortest) wavelength from the three choices below (*for H atom*):

- Wavelength of photon *emitted* for an electron transitioning from $2p$ to $1s$ orbital
- Wavelength of photon *absorbed* for an electron transitioning from $2s$ to $4p$ orbital
- Wavelength of photon *absorbed* for an electron transitioning from $1s$ to $3p$ orbital

i or ii or iii



Question 2 8 points



A Chem110 student plots the graph shown on the left, from data obtained for the kinetic energy (eV) of the ejected electron versus frequency (s^{-1}) of the incident light, for an unknown metal. ($1\text{eV} = 1.6022 \times 10^{-19} \text{ J}$)

Based on the plot, answer the questions **a to c** below:

a. 1 point On the plot above, label the **threshold frequency**.

b. 2 points What is the value of the work potential (in J) of the unknown metal?

c. 3 points What is the *longest* wavelength (in nm) required to eject an electron from this metal's surface? Show your work including equation(s) used. (3 sig figs)

d. 2 points In the space provided below, plot the following

(Assume frequency is *above* threshold frequency):

- i. *Kinetic energy* of an ejected electron versus the *intensity* of the incident light (left)
- ii. *Number of electrons* ejected versus the *frequency* of the incident light (right)



**Question 3** **6 points****a. 2 points**

In the ground state of Ge (Germanium; $Z = 32$) atom, **name** the following orbitals:

i. Orbital containing the electron that requires the maximum energy to be removed.

ii. Orbital containing the electron that experiences the minimum effective nuclear charge.

b. 2 points

An unknown anion (A^-) has the following *excited* electronic configuration: $[Ar] 4s^2 3d^3 4p^1$

(Ar: Argon; $Z = 18$)

Give the **condensed ground state electronic configuration** of the atom, **A** and the cation, **A^{2+}** .
Show your work. (Assume all electronic transitions are possible)

Electronic Configuration of A:

Electronic Configuration of A^{2+} :

c. 2 points

In a single atom, determine how many electrons can have the following set of quantum numbers?

(i) $n = 5; l = 3$

In a single atom, determine how many electrons can have the following set of quantum numbers?

(i) $n = 3; l = 2; m_s = +1/2$

**Question 4** **8 points**

Rank the following from 1 to 4 in **increasing** order. Provide the ranking in the boxes provided.

Note: Atomic number (Z) are provided for relevant atoms

a) Increasing (lowest to highest) order of *ionization energy*: (1= lowest; 4 = highest)

Mg

Mg: Z = 12

Rb

Rb: Z = 37

Cs

Cs: Z = 55

K

K : Z = 19

b) Increasing (lowest to highest) order of *bond angle*: (1= lowest; 4 = highest)

H₂OCCl₄XeF₄

HCN

c) Increasing (smallest to largest) order of *ionic size* (1= smallest; 4 = largest)

S²⁻

S: Z = 16

O²⁻

O: Z = 8

Be²⁺

Be: Z = 4

Na⁺

Na: Z = 11

d) Increasing (lowest to highest) order of *bond energy* (1= lowest; 4 = highest)

P-F

P-Cl

P-I

P-Br

**Question 5** **12 points**

a. 9 points For the molecules HClO_4 and TeF_6 , draw the most stable *Lewis structure* showing appropriate shape and *dashed/wedged bonds*, based on VSEPR. Determine the **molecular geometry** around the central atom. For HClO_4 , label the *most polar bond*.



b. 3 points (Note: Lone pairs are not shown in structures below)

(i) For the three *resonance* structures (1, 2 and 3) shown below, answer the following question:



(I) Circle the *most contributing* resonance structure.

Structure 1

Structure 2

Structure 3

(ii) For the three structures given (4, 5, and 6) below, answer the following questions:



(I) Circle all the structures that represent *resonance structures* of each other:

Structure 4

Structure 5

Structure 6

(II) Among the resonance structures, circle the *most contributing* resonance structure.

Structure 4

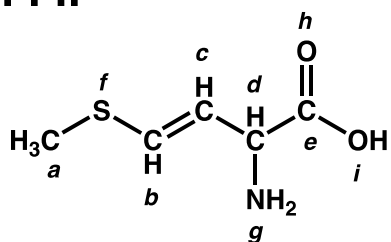
Structure 5

Structure 6

**Question 6** **6 points**

Using the data given below, calculate the **lattice energy** for SrO(s).

	ΔH° (kJ/mol)
$\text{Sr}(s) \longrightarrow \text{Sr}(g)$	+180
$\text{Sr}(g) \longrightarrow \text{Sr}^{2+}(g) + 2e^-$	+1468
$\text{O}_2(g) \longrightarrow 2\text{O}(g)$	+500
$\text{O}(g) + 2e^- \longrightarrow \text{O}^{2-}(g)$	+600
$\text{Sr}(s) + \frac{1}{2}\text{O}_2(g) \longrightarrow \text{SrO}(s)$	-550

**Question 7 10 points**

Consider the molecule given on the left, and answer questions (i) to (vi) using chemical bonding theories that we have learnt in class (valence bond theory, hybridization).

Note that the lone pairs are not shown in the structure:

i. Consider the carbon atoms marked in as ***a***, ***b***, ***c***, ***d***, and ***e*** in the molecule, and determine **hybridization** at each of the carbon centers. (2.5 points)

Atom	Hybridization
Carbon (<i>a</i>)	
Carbon (<i>b</i>)	
Carbon (<i>c</i>)	
Carbon (<i>d</i>)	
Carbon (<i>e</i>)	

ii. Now, consider the sulfur atom (S) marked as ***f***, nitrogen (N) atom (***g***), and oxygen(O) atoms (***h*** and ***i***), and determine their **hybridization**. (2 points)

Atom	Hybridization
Sulfur (<i>f</i>)	
Nitrogen (<i>g</i>)	
Oxygen (<i>h</i>)	
Oxygen (<i>i</i>)	

iii. **Describe** all the bonds formed by the carbon atoms ***a*** and ***b***, using sigma and pi designation as appropriate. (2 points)

Atom	Bonds formed (use sigma and pi as appropriate)
Carbon (<i>a</i>)	
Carbon (<i>b</i>)	

iv. How many total sp^3 -s sigma bonds are in the molecule? (1 point)

v. Determine **electronic** and **molecular geometry** at carbon (C) labeled ***e*** and sulfur (S) labeled ***f*** centres. (2 points)

Atom	Electronic Geometry	Molecular Geometry
Carbon (<i>e</i>)		
Sulfur (<i>f</i>)		

vi. Circle the value from the choices given below, that describes the **C-S-C bond angle at sulfur** (S) marked ***f***: (0.5 points)

>90° to <109.5° 109.5° >109.5 to <120° 120° >120° to <180°


Question 8 12 points

Using the **valence shell orbitals only** draw a **complete** molecular orbital (MO) diagram (with no s-p mixing) for the molecule **PS**, in the space provided below. **(6 points)**

Assume that 3s and 3p atomic orbitals will form molecular orbitals like those formed by the combination of 2s and 2p atomic orbitals that we learnt in class, and there is no s-p mixing.

Show **energy arrow**, **atomic orbitals** involved, **atom labels**, the **molecular orbitals** that are formed, **filled electrons** in the MO diagram etc. Make sure to label all the atomic and molecular orbitals.

Using the MO diagram above, answer the following questions:

- (i) Which one of the following will have the highest bond order? Circle the correct choice
 A) PS^{1-} B) PS C) PS^{1+}
- (ii) Which one of the following will have the longest bond length? Circle the correct choice
 A) PS^{1-} B) PS C) PS^{1+}
- (iii) Which of the following will be diamagnetic? Circle the correct choice.
 A) PS^{1-} B) PS C) PS^{1+}
- (iv) Draw Lewis dot structure of the molecule with the shortest bond length (based on the MO diagram) from among the three (PS^{1-} , PS, PS^{1+}), and indicate the bond order calculated from the Lewis dot structure. **(3 points)**

Lewis Dot Structure

Bond Order



918F9A9E-D987-429F-ADF4-CAF5CE035C2D

midterm-exam-93388

#801 Page 10 of 10

SCRAP PAGE

**CHEMISTRY 110 –Midterm Exam****Thursday November 7, 2024, 6:30 – 8:30 pm**

Examiners: Pallavi Sirjoosingh and Ashok Kakkar

First name (please write as legibly as possible within the boxes)

--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--

Last name

--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--

Student ID number

--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--

This booklet has 8 questions on 10 pages (The last page can be used as a scrap page).

The staple must NOT be removed from the question booklet. Your intact question booklet must be returned at the end of the exam.

A separate booklet contains a periodic table, and datasheet.

Write your answers directly under the questions in the space provided. Do not write on the QR codes or near the top edges (as it may be cut off).

Please write **legibly and dark enough** for the scan to be readable. For students who use English as a second language, dictionaries are permitted. Calculators are allowed.

DO NOT ASK QUESTIONS OF THE INVIGILATORS about interpretations of the exam questions. If you have a concern about a question, please summarize it at the top of this page. All concerns will be evaluated after the exam. GOOD LUCK!

**Question 1 8 points** (*Assume all transitions are possible*)

a. 6 points Calculate the **energy (in kJ/mol)** required to ionize (completely remove) the electron from the **n=3** level of **He⁺ ion** (He: Helium). Show your work including equations. (3 sig. figs.)

b. 2 points

(I) Select the largest (longest) wavelength from the three choices below:

- i. Wavelength of photon to *completely remove* an electron from *1s orbital of Hydrogen atom*
- ii. Wavelength of photon to *completely remove* an electron from *1s orbital of Boron atom (B)*
- iii. Wavelength of photon to *completely remove* an electron from *1s orbital of Boron cation (B⁺)*

i or ii or iii

(II) Select the smallest (shortest) wavelength from the three choices below (*for H atom*):

- i. Wavelength of photon *emitted* for an electron transitioning from *2p to 1s orbital*
- ii. Wavelength of photon *absorbed* for an electron transitioning from *2s to 4p orbital*
- iii. Wavelength of photon *absorbed* for an electron transitioning from *1s to 3p orbital*

i or ii or iii



Question 2 8 points



A Chem110 student plots the graph shown on the left, from data obtained for the kinetic energy (eV) of the ejected electron versus frequency (s^{-1}) of the incident light, for an unknown metal. ($1\text{eV} = 1.6022 \times 10^{-19} \text{ J}$)

Based on the plot, answer the questions **a to c** below:

a. 1 point On the plot above, label the **threshold frequency**.

b. 2 points What is the value of the work potential (in J) of the unknown metal?

c. 3 points What is the *longest* wavelength (in nm) required to eject an electron from this metal's surface? Show your work including equation(s) used. (3 sig figs)

d. 2 points In the space provided below, plot the following

(Assume frequency is *above* threshold frequency):

- i. *Kinetic energy* of an ejected electron versus the *intensity* of the incident light (left)
- ii. *Number of electrons* ejected versus the *frequency* of the incident light (right)



**Question 3** **6 points****a. 2 points**

In the ground state of Ge (Germanium; $Z = 32$) atom, **name** the following orbitals:

i. Orbital containing the electron that requires the maximum energy to be removed.

ii. Orbital containing the electron that experiences the minimum effective nuclear charge.

b. 2 points

An unknown anion (A^-) has the following *excited* electronic configuration: $[Ar] 4s^2 3d^3 4p^1$

(Ar: Argon; $Z = 18$)

Give the **condensed ground state electronic configuration** of the atom, **A** and the cation, **A^{2+}** .
Show your work. (Assume all electronic transitions are possible)

Electronic Configuration of A:

Electronic Configuration of A^{2+} :

c. 2 points

In a single atom, determine how many electrons can have the following set of quantum numbers?

(i) $n = 5; l = 3$

In a single atom, determine how many electrons can have the following set of quantum numbers?

(i) $n = 3; l = 2; m_s = +1/2$

**Question 4** **8 points**

Rank the following from 1 to 4 in **increasing** order. Provide the ranking in the boxes provided.

Note: Atomic number (Z) are provided for relevant atoms

a) Increasing (lowest to highest) order of *ionization energy*: (1= lowest; 4 = highest)

Mg

Mg: Z = 12

Rb

Rb: Z = 37

Cs

Cs: Z = 55

K

K : Z = 19

b) Increasing (lowest to highest) order of *bond angle*: (1= lowest; 4 = highest)

H₂OCCl₄XeF₄

HCN

c) Increasing (smallest to largest) order of *ionic size* (1= smallest; 4 = largest)

S²⁻

S: Z = 16

O²⁻

O: Z = 8

Be²⁺

Be: Z = 4

Na⁺

Na: Z = 11

d) Increasing (lowest to highest) order of *bond energy* (1= lowest; 4 = highest)

P-F

P-Cl

P-I

P-Br

**Question 5** **12 points**

a. 9 points For the molecules HClO_4 and TeF_6 , draw the most stable *Lewis structure* showing appropriate shape and *dashed/wedged bonds*, based on VSEPR. Determine the **molecular geometry** around the central atom. For HClO_4 , label the *most polar bond*.



b. 3 points (Note: Lone pairs are not shown in structures below)

(i) For the three *resonance* structures (1, 2 and 3) shown below, answer the following question:



(I) Circle the *most contributing* resonance structure.

Structure 1

Structure 2

Structure 3

(ii) For the three structures given (4, 5, and 6) below, answer the following questions:



(I) Circle all the structures that represent *resonance structures* of each other:

Structure 4

Structure 5

Structure 6

(II) Among the resonance structures, circle the *most contributing* resonance structure.

Structure 4

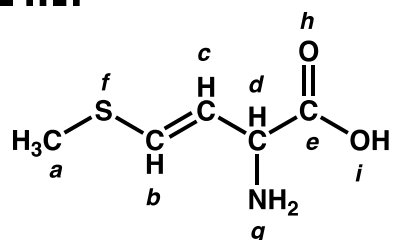
Structure 5

Structure 6

**Question 6** **6 points**

Using the data given below, calculate the **lattice energy** for SrO(s).

	ΔH° (kJ/mol)
$\text{Sr}(s) \longrightarrow \text{Sr}(g)$	+180
$\text{Sr}(g) \longrightarrow \text{Sr}^{2+}(g) + 2e^-$	+1468
$\text{O}_2(g) \longrightarrow 2\text{O}(g)$	+500
$\text{O}(g) + 2e^- \longrightarrow \text{O}^{2-}(g)$	+600
$\text{Sr}(s) + \frac{1}{2}\text{O}_2(g) \longrightarrow \text{SrO}(s)$	-550

**Question 7 10 points**

Consider the molecule given on the left, and answer questions (i) to (vi) using chemical bonding theories that we have learnt in class (valence bond theory, hybridization).

Note that the lone pairs are not shown in the structure:

i. Consider the carbon atoms marked in as ***a***, ***b***, ***c***, ***d***, and ***e*** in the molecule, and determine **hybridization** at each of the carbon centers. (2.5 points)

Atom	Hybridization
Carbon (<i>a</i>)	
Carbon (<i>b</i>)	
Carbon (<i>c</i>)	
Carbon (<i>d</i>)	
Carbon (<i>e</i>)	

ii. Now, consider the sulfur atom (S) marked as ***f***, nitrogen (N) atom (***g***), and oxygen(O) atoms (***h*** and ***i***), and determine their **hybridization**. (2 points)

Atom	Hybridization
Sulfur (<i>f</i>)	
Nitrogen (<i>g</i>)	
Oxygen (<i>h</i>)	
Oxygen (<i>i</i>)	

iii. **Describe** all the bonds formed by the carbon atoms ***a*** and ***b***, using sigma and pi designation as appropriate. (2 points)

Atom	Bonds formed (use sigma and pi as appropriate)
Carbon (<i>a</i>)	
Carbon (<i>b</i>)	

iv. How many total sp^3 -s sigma bonds are in the molecule? (1 point)

v. Determine **electronic** and **molecular geometry** at carbon (C) labeled ***e*** and sulfur (S) labeled ***f*** centres. (2 points)

Atom	Electronic Geometry	Molecular Geometry
Carbon (<i>e</i>)		
Sulfur (<i>f</i>)		

vi. Circle the value from the choices given below, that describes the **C-S-C bond angle at sulfur** (S) marked ***f***: (0.5 points)

>90° to <109.5°

109.5°

>109.5 to <120°

120°

>120° to <180°



Question 8 12 points

Using the **valence shell orbitals only** draw a **complete** molecular orbital (MO) diagram (with no s-p mixing) for the molecule **PS**, in the space provided below. **(6 points)**

Assume that 3s and 3p atomic orbitals will form molecular orbitals like those formed by the combination of 2s and 2p atomic orbitals that we learnt in class, and there is no s-p mixing.

Show **energy arrow**, **atomic orbitals** involved, **atom labels**, the **molecular orbitals** that are formed, **filled electrons** in the MO diagram etc. Make sure to label all the atomic and molecular orbitals.

Using the MO diagram above, answer the following questions:

- (i) Which one of the following will have the highest bond order? Circle the correct choice
 A) PS^{1-} B) PS C) PS^{1+}
- (ii) Which one of the following will have the longest bond length? Circle the correct choice
 A) PS^{1-} B) PS C) PS^{1+}
- (iii) Which of the following will be diamagnetic? Circle the correct choice.
 A) PS^{1-} B) PS C) PS^{1+}
- (iv) Draw Lewis dot structure of the molecule with the shortest bond length (based on the MO diagram) from among the three (PS^{1-} , PS, PS^{1+}), and indicate the bond order calculated from the Lewis dot structure. **(3 points)**

Lewis Dot Structure

Bond Order



F80B93FE-3ACC-451D-8031-2F8099E522B7

midterm-exam-93388

#802 Page 10 of 10

SCRAP PAGE

**CHEMISTRY 110 –Midterm Exam****Thursday November 7, 2024, 6:30 – 8:30 pm**

Examiners: Pallavi Sirjoosingh and Ashok Kakkar

First name (please write as legibly as possible within the boxes)

--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--

Last name

--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--

Student ID number

--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--

This booklet has 8 questions on 10 pages (The last page can be used as a scrap page).

The staple must NOT be removed from the question booklet. Your intact question booklet must be returned at the end of the exam.

A separate booklet contains a periodic table, and datasheet.

Write your answers directly under the questions in the space provided. Do not write on the QR codes or near the top edges (as it may be cut off).

Please write **legibly and dark enough** for the scan to be readable. For students who use English as a second language, dictionaries are permitted. Calculators are allowed.

DO NOT ASK QUESTIONS OF THE INVIGILATORS about interpretations of the exam questions. If you have a concern about a question, please summarize it at the top of this page. All concerns will be evaluated after the exam. GOOD LUCK!

**Question 1 8 points** (*Assume all transitions are possible*)

a. 6 points Calculate the **energy (in kJ/mol)** required to ionize (completely remove) the electron from the **$n=3$** level of **He^+ ion** (He: Helium). Show your work including equations. (3 sig. figs.)

b. 2 points

(I) Select the largest (longest) wavelength from the three choices below:

- i. Wavelength of photon to *completely remove* an electron from *1s orbital of Hydrogen atom*
- ii. Wavelength of photon to *completely remove* an electron from *1s orbital of Boron atom (B)*
- iii. Wavelength of photon to *completely remove* an electron from *1s orbital of Boron cation (B^+)*

i or ii or iii

(II) Select the smallest (shortest) wavelength from the three choices below (*for H atom*):

- i. Wavelength of photon *emitted* for an electron transitioning from *2p to 1s orbital*
- ii. Wavelength of photon *absorbed* for an electron transitioning from *2s to 4p orbital*
- iii. Wavelength of photon *absorbed* for an electron transitioning from *1s to 3p orbital*

i or ii or iii



Question 2 8 points



A Chem110 student plots the graph shown on the left, from data obtained for the kinetic energy (eV) of the ejected electron versus frequency (s^{-1}) of the incident light, for an unknown metal. ($1\text{eV} = 1.6022 \times 10^{-19} \text{ J}$)

Based on the plot, answer the questions **a to c** below:

a. 1 point On the plot above, label the **threshold frequency**.

b. 2 points What is the value of the work potential (in J) of the unknown metal?

c. 3 points What is the *longest* wavelength (in nm) required to eject an electron from this metal's surface? Show your work including equation(s) used. (3 sig figs)

d. 2 points In the space provided below, plot the following

(Assume frequency is *above* threshold frequency):

- Kinetic energy* of an ejected electron versus the *intensity* of the incident light (left)
- Number of electrons* ejected versus the *frequency* of the incident light (right)



**Question 3** **6 points****a. 2 points**

In the ground state of Ge (Germanium; $Z = 32$) atom, **name** the following orbitals:

i. Orbital containing the electron that requires the maximum energy to be removed.

ii. Orbital containing the electron that experiences the minimum effective nuclear charge.

b. 2 points

An unknown anion (A^-) has the following *excited* electronic configuration: $[Ar] 4s^2 3d^3 4p^1$

(Ar: Argon; $Z = 18$)

Give the **condensed ground state electronic configuration** of the atom, **A** and the cation, **A^{2+}** .
Show your work. (Assume all electronic transitions are possible)

Electronic Configuration of A:

Electronic Configuration of A^{2+} :

c. 2 points

In a single atom, determine how many electrons can have the following set of quantum numbers?

(i) $n = 5; l = 3$

In a single atom, determine how many electrons can have the following set of quantum numbers?

(i) $n = 3; l = 2; m_s = +1/2$

**Question 4** **8 points**

Rank the following from 1 to 4 in **increasing** order. Provide the ranking in the boxes provided.

Note: Atomic number (Z) are provided for relevant atoms

a) Increasing (lowest to highest) order of *ionization energy*: (1= lowest; 4 = highest)

Mg

Mg: Z = 12

Rb

Rb: Z = 37

Cs

Cs: Z = 55

K

K : Z = 19

b) Increasing (lowest to highest) order of *bond angle*: (1= lowest; 4 = highest)

H₂OCCl₄XeF₄

HCN

c) Increasing (smallest to largest) order of *ionic size* (1= smallest; 4 = largest)

S²⁻

S: Z = 16

O²⁻

O: Z = 8

Be²⁺

Be: Z = 4

Na⁺

Na: Z = 11

d) Increasing (lowest to highest) order of *bond energy* (1= lowest; 4 = highest)

P-F

P-Cl

P-I

P-Br

**Question 5** **12 points**

a. 9 points For the molecules HClO_4 and TeF_6 , draw the most stable *Lewis structure* showing appropriate shape and *dashed/wedged bonds*, based on VSEPR. Determine the **molecular geometry** around the central atom. For HClO_4 , label the *most polar bond*.



b. 3 points (Note: Lone pairs are not shown in structures below)

(i) For the three *resonance* structures (1, 2 and 3) shown below, answer the following question:



(I) Circle the *most contributing* resonance structure.

Structure 1

Structure 2

Structure 3

(ii) For the three structures given (4, 5, and 6) below, answer the following questions:



(I) Circle all the structures that represent *resonance structures* of each other:

Structure 4

Structure 5

Structure 6

(II) Among the resonance structures, circle the *most contributing* resonance structure.

Structure 4

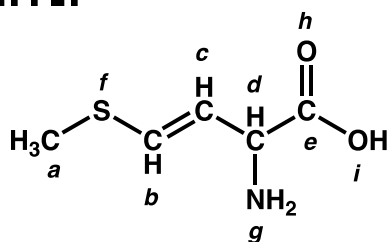
Structure 5

Structure 6

**Question 6** **6 points**

Using the data given below, calculate the **lattice energy** for SrO(s).

	ΔH° (kJ/mol)
$\text{Sr}(s) \longrightarrow \text{Sr}(g)$	+180
$\text{Sr}(g) \longrightarrow \text{Sr}^{2+}(g) + 2e^-$	+1468
$\text{O}_2(g) \longrightarrow 2\text{O}(g)$	+500
$\text{O}(g) + 2e^- \longrightarrow \text{O}^{2-}(g)$	+600
$\text{Sr}(s) + \frac{1}{2}\text{O}_2(g) \longrightarrow \text{SrO}(s)$	-550

**Question 7 10 points**

Consider the molecule given on the left, and answer questions (i) to (vi) using chemical bonding theories that we have learnt in class (valence bond theory, hybridization).

Note that the lone pairs are not shown in the structure:

i. Consider the carbon atoms marked in as ***a***, ***b***, ***c***, ***d***, and ***e*** in the molecule, and determine **hybridization** at each of the carbon centers. (2.5 points)

Atom	Hybridization
Carbon (<i>a</i>)	
Carbon (<i>b</i>)	
Carbon (<i>c</i>)	
Carbon (<i>d</i>)	
Carbon (<i>e</i>)	

ii. Now, consider the sulfur atom (S) marked as ***f***, nitrogen (N) atom (***g***), and oxygen(O) atoms (***h*** and ***i***), and determine their **hybridization**. (2 points)

Atom	Hybridization
Sulfur (<i>f</i>)	
Nitrogen (<i>g</i>)	
Oxygen (<i>h</i>)	
Oxygen (<i>i</i>)	

iii. **Describe** all the bonds formed by the carbon atoms ***a*** and ***b***, using sigma and pi designation as appropriate. (2 points)

Atom	Bonds formed (use sigma and pi as appropriate)
Carbon (<i>a</i>)	
Carbon (<i>b</i>)	

iv. How many total sp^3 -s sigma bonds are in the molecule? (1 point)

v. Determine **electronic** and **molecular geometry** at carbon (C) labeled ***e*** and sulfur (S) labeled ***f*** centres. (2 points)

Atom	Electronic Geometry	Molecular Geometry
Carbon (<i>e</i>)		
Sulfur (<i>f</i>)		

vi. Circle the value from the choices given below, that describes the **C-S-C bond angle at sulfur** (S) marked ***f***: (0.5 points)

>90° to <109.5°

109.5°

>109.5 to <120°

120°

>120° to <180°



Question 8 12 points

Using the **valence shell orbitals only** draw a **complete** molecular orbital (MO) diagram (with no s-p mixing) for the molecule **PS**, in the space provided below. **(6 points)**

Assume that 3s and 3p atomic orbitals will form molecular orbitals like those formed by the combination of 2s and 2p atomic orbitals that we learnt in class, and there is no s-p mixing.

Show **energy arrow**, **atomic orbitals** involved, **atom labels**, the **molecular orbitals** that are formed, **filled electrons** in the MO diagram etc. Make sure to label all the atomic and molecular orbitals.

Using the MO diagram above, answer the following questions:

- (i) Which one of the following will have the highest bond order? Circle the correct choice
 A) PS^{1-} B) PS C) PS^{1+}
- (ii) Which one of the following will have the longest bond length? Circle the correct choice
 A) PS^{1-} B) PS C) PS^{1+}
- (iii) Which of the following will be diamagnetic? Circle the correct choice.
 A) PS^{1-} B) PS C) PS^{1+}
- (iv) Draw Lewis dot structure of the molecule with the shortest bond length (based on the MO diagram) from among the three (PS^{1-} , PS, PS^{1+}), and indicate the bond order calculated from the Lewis dot structure. **(3 points)**

Lewis Dot Structure

Bond Order



5878D5FD-6011-48EF-BF4C-3116635CA7CA

midterm-exam-93388

#803 Page 10 of 10

SCRAP PAGE

**CHEMISTRY 110 –Midterm Exam****Thursday November 7, 2024, 6:30 – 8:30 pm**

Examiners: Pallavi Sirjoosingh and Ashok Kakkar

First name (please write as legibly as possible within the boxes)

--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--

Last name

--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--

Student ID number

--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--

This booklet has 8 questions on 10 pages (The last page can be used as a scrap page).

The staple must NOT be removed from the question booklet. Your intact question booklet must be returned at the end of the exam.

A separate booklet contains a periodic table, and datasheet.

Write your answers directly under the questions in the space provided. Do not write on the QR codes or near the top edges (as it may be cut off).

Please write **legibly and dark enough** for the scan to be readable. For students who use English as a second language, dictionaries are permitted. Calculators are allowed.

DO NOT ASK QUESTIONS OF THE INVIGILATORS about interpretations of the exam questions. If you have a concern about a question, please summarize it at the top of this page. All concerns will be evaluated after the exam. GOOD LUCK!

**Question 1 8 points** (*Assume all transitions are possible*)

a. 6 points Calculate the **energy (in kJ/mol)** required to ionize (completely remove) the electron from the **n=3** level of **He⁺ ion** (He: Helium). Show your work including equations. (3 sig. figs.)

b. 2 points

(I) Select the largest (longest) wavelength from the three choices below:

- i. Wavelength of photon to *completely remove* an electron from *1s orbital of Hydrogen atom*
- ii. Wavelength of photon to *completely remove* an electron from *1s orbital of Boron atom (B)*
- iii. Wavelength of photon to *completely remove* an electron from *1s orbital of Boron cation (B⁺)*

i or ii or iii

(II) Select the smallest (shortest) wavelength from the three choices below (*for H atom*):

- i. Wavelength of photon *emitted* for an electron transitioning from *2p to 1s orbital*
- ii. Wavelength of photon *absorbed* for an electron transitioning from *2s to 4p orbital*
- iii. Wavelength of photon *absorbed* for an electron transitioning from *1s to 3p orbital*

i or ii or iii



Question 2 8 points



A Chem110 student plots the graph shown on the left, from data obtained for the kinetic energy (eV) of the ejected electron versus frequency (s^{-1}) of the incident light, for an unknown metal. ($1\text{eV} = 1.6022 \times 10^{-19} \text{ J}$)

Based on the plot, answer the questions **a to c** below:

a. 1 point On the plot above, label the **threshold frequency**.

b. 2 points What is the value of the work potential (in J) of the unknown metal?

c. 3 points What is the *longest* wavelength (in nm) required to eject an electron from this metal's surface? Show your work including equation(s) used. (3 sig figs)

d. 2 points In the space provided below, plot the following

(Assume frequency is *above* threshold frequency):

- Kinetic energy* of an ejected electron versus the *intensity* of the incident light (left)
- Number of electrons* ejected versus the *frequency* of the incident light (right)



**Question 3** **6 points****a. 2 points**

In the ground state of Ge (Germanium; $Z = 32$) atom, **name** the following orbitals:

i. Orbital containing the electron that requires the maximum energy to be removed.

ii. Orbital containing the electron that experiences the minimum effective nuclear charge.

b. 2 points

An unknown anion (A^-) has the following *excited* electronic configuration: $[Ar] 4s^2 3d^3 4p^1$

(Ar: Argon; $Z = 18$)

Give the **condensed ground state electronic configuration** of the atom, **A** and the cation, **A^{2+}** .
Show your work. (Assume all electronic transitions are possible)

Electronic Configuration of A:

Electronic Configuration of A^{2+} :

c. 2 points

In a single atom, determine how many electrons can have the following set of quantum numbers?

(i) $n = 5; l = 3$

In a single atom, determine how many electrons can have the following set of quantum numbers?

(i) $n = 3; l = 2; m_s = +1/2$

**Question 4** **8 points**

Rank the following from 1 to 4 in **increasing** order. Provide the ranking in the boxes provided.

Note: Atomic number (Z) are provided for relevant atoms

a) Increasing (lowest to highest) order of *ionization energy*: (1= lowest; 4 = highest)

Mg

Mg: Z = 12

Rb

Rb: Z = 37

Cs

Cs: Z = 55

K

K : Z = 19

b) Increasing (lowest to highest) order of *bond angle*: (1= lowest; 4 = highest)

H₂OCCl₄XeF₄

HCN

c) Increasing (smallest to largest) order of *ionic size* (1= smallest; 4 = largest)

S²⁻

S: Z = 16

O²⁻

O: Z = 8

Be²⁺

Be: Z = 4

Na⁺

Na: Z = 11

d) Increasing (lowest to highest) order of *bond energy* (1= lowest; 4 = highest)

P-F

P-Cl

P-I

P-Br

**Question 5** **12 points**

a. 9 points For the molecules HClO_4 and TeF_6 , draw the most stable *Lewis structure* showing appropriate shape and *dashed/wedged bonds*, based on VSEPR. Determine the **molecular geometry** around the central atom. For HClO_4 , label the *most polar bond*.



b. 3 points (Note: Lone pairs are not shown in structures below)

(i) For the three *resonance* structures (1, 2 and 3) shown below, answer the following question:



(I) Circle the *most contributing* resonance structure.

Structure 1

Structure 2

Structure 3

(ii) For the three structures given (4, 5, and 6) below, answer the following questions:



(I) Circle all the structures that represent *resonance structures* of each other:

Structure 4

Structure 5

Structure 6

(II) Among the resonance structures, circle the *most contributing* resonance structure.

Structure 4

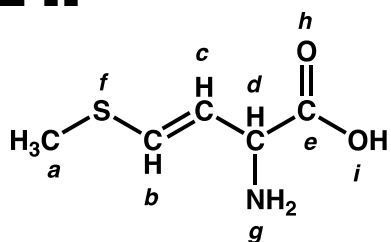
Structure 5

Structure 6

**Question 6** **6 points**

Using the data given below, calculate the **lattice energy** for SrO(s).

	ΔH° (kJ/mol)
$\text{Sr}(s) \longrightarrow \text{Sr}(g)$	+180
$\text{Sr}(g) \longrightarrow \text{Sr}^{2+}(g) + 2e^-$	+1468
$\text{O}_2(g) \longrightarrow 2\text{O}(g)$	+500
$\text{O}(g) + 2e^- \longrightarrow \text{O}^{2-}(g)$	+600
$\text{Sr}(s) + \frac{1}{2}\text{O}_2(g) \longrightarrow \text{SrO}(s)$	-550

**Question 7 10 points**

Consider the molecule given on the left, and answer questions (i) to (vi) using chemical bonding theories that we have learnt in class (valence bond theory, hybridization).

Note that the lone pairs are not shown in the structure:

i. Consider the carbon atoms marked in as ***a***, ***b***, ***c***, ***d***, and ***e*** in the molecule, and determine **hybridization** at each of the carbon centers. (2.5 points)

Atom	Hybridization
Carbon (<i>a</i>)	
Carbon (<i>b</i>)	
Carbon (<i>c</i>)	
Carbon (<i>d</i>)	
Carbon (<i>e</i>)	

ii. Now, consider the sulfur atom (S) marked as ***f***, nitrogen (N) atom (***g***), and oxygen(O) atoms (***h*** and ***i***), and determine their **hybridization**. (2 points)

Atom	Hybridization
Sulfur (<i>f</i>)	
Nitrogen (<i>g</i>)	
Oxygen (<i>h</i>)	
Oxygen (<i>i</i>)	

iii. **Describe** all the bonds formed by the carbon atoms ***a*** and ***b***, using sigma and pi designation as appropriate. (2 points)

Atom	Bonds formed (use sigma and pi as appropriate)
Carbon (<i>a</i>)	
Carbon (<i>b</i>)	

iv. How many total sp^3 -s sigma bonds are in the molecule? (1 point)

v. Determine **electronic** and **molecular geometry** at carbon (C) labeled ***e*** and sulfur (S) labeled ***f*** centres. (2 points)

Atom	Electronic Geometry	Molecular Geometry
Carbon (<i>e</i>)		
Sulfur (<i>f</i>)		

vi. Circle the value from the choices given below, that describes the **C-S-C bond angle at sulfur** (S) marked ***f***: (0.5 points)

>90° to <109.5° 109.5° >109.5 to <120° 120° >120° to <180°

**Question 8 12 points**

Using the **valence shell orbitals only** draw a **complete** molecular orbital (MO) diagram (with no s-p mixing) for the molecule **PS**, in the space provided below. **(6 points)**

Assume that 3s and 3p atomic orbitals will form molecular orbitals like those formed by the combination of 2s and 2p atomic orbitals that we learnt in class, and there is no s-p mixing.

Show **energy arrow**, **atomic orbitals** involved, **atom labels**, the **molecular orbitals** that are formed, **filled electrons** in the MO diagram etc. Make sure to label all the atomic and molecular orbitals.

Using the MO diagram above, answer the following questions:

- (i) Which one of the following will have the highest bond order? Circle the correct choice
A) PS^{1-} B) PS C) PS^{1+}
- (ii) Which one of the following will have the longest bond length? Circle the correct choice
A) PS^{1-} B) PS C) PS^{1+}
- (iii) Which of the following will be diamagnetic? Circle the correct choice.
A) PS^{1-} B) PS C) PS^{1+}
- (iv) Draw Lewis dot structure of the molecule with the shortest bond length (based on the MO diagram) from among the three (PS^{1-} , PS, PS^{1+}), and indicate the bond order calculated from the Lewis dot structure. **(3 points)**

Lewis Dot Structure

Bond Order



CA8F9F29-DFE8-4161-8926-8708BC9F03B3

midterm-exam-93388

#804 Page 10 of 10

SCRAP PAGE

**CHEMISTRY 110 –Midterm Exam****Thursday November 7, 2024, 6:30 – 8:30 pm**

Examiners: Pallavi Sirjoosingh and Ashok Kakkar

First name (please write as legibly as possible within the boxes)

--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--

Last name

--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--

Student ID number

--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--

This booklet has 8 questions on 10 pages (The last page can be used as a scrap page).

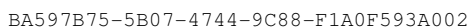
The staple must NOT be removed from the question booklet. Your intact question booklet must be returned at the end of the exam.

A separate booklet contains a periodic table, and datasheet.

Write your answers directly under the questions in the space provided. Do not write on the QR codes or near the top edges (as it may be cut off).

Please write **legibly and dark enough** for the scan to be readable. For students who use English as a second language, dictionaries are permitted. Calculators are allowed.

DO NOT ASK QUESTIONS OF THE INVIGILATORS about interpretations of the exam questions. If you have a concern about a question, please summarize it at the top of this page. All concerns will be evaluated after the exam. GOOD LUCK!



midterm-exam-93388

#805 Page 2 of 10

a. 6 points Calculate the **energy (in kJ/mol)** required to ionize (completely remove) the electron from the **n=3** level of **He⁺ ion** (He: Helium). Show your work including equations. (3 sig. figs.)

(I) Select the largest (longest) wavelength from the three choices below:

- Wavelength of photon to *completely remove* an electron from *1s orbital of Hydrogen atom*
- Wavelength of photon to *completely remove* an electron from *1s orbital of Boron atom (B)*
- Wavelength of photon to *completely remove* an electron from *1s orbital of Boron cation (B⁺)*

i
or
ii
or
iii

(II) Select the smallest (shortest) wavelength from the three choices below (*for H atom*):

- Wavelength of photon *emitted* for an electron transitioning from $2p$ to $1s$ orbital
- Wavelength of photon *absorbed* for an electron transitioning from $2s$ to $4p$ orbital
- Wavelength of photon *absorbed* for an electron transitioning from $1s$ to $3p$ orbital

i or ii or iii



Question 2 8 points



A Chem110 student plots the graph shown on the left, from data obtained for the kinetic energy (eV) of the ejected electron versus frequency (s^{-1}) of the incident light, for an unknown metal. ($1\text{eV} = 1.6022 \times 10^{-19} \text{ J}$)

Based on the plot, answer the questions **a to c** below:

a. 1 point On the plot above, label the **threshold frequency**.

b. 2 points What is the value of the work potential (in J) of the unknown metal?

c. 3 points What is the *longest* wavelength (in nm) required to eject an electron from this metal's surface? Show your work including equation(s) used. (3 sig figs)

d. 2 points In the space provided below, plot the following

(Assume frequency is *above* threshold frequency):

- Kinetic energy* of an ejected electron versus the *intensity* of the incident light (left)
- Number of electrons* ejected versus the *frequency* of the incident light (right)



**Question 3** **6 points****a. 2 points**

In the ground state of Ge (Germanium; $Z = 32$) atom, **name** the following orbitals:

i. Orbital containing the electron that requires the maximum energy to be removed.

ii. Orbital containing the electron that experiences the minimum effective nuclear charge.

b. 2 points

An unknown anion (A^-) has the following *excited* electronic configuration: $[Ar] 4s^2 3d^3 4p^1$

(Ar: Argon; $Z = 18$)

Give the **condensed ground state electronic configuration** of the atom, **A** and the cation, **A^{2+}** .
Show your work. (Assume all electronic transitions are possible)

Electronic Configuration of A:

Electronic Configuration of A^{2+} :

c. 2 points

In a single atom, determine how many electrons can have the following set of quantum numbers?

(i) $n = 5; l = 3$

In a single atom, determine how many electrons can have the following set of quantum numbers?

(i) $n = 3; l = 2; m_s = +1/2$

**Question 4** **8 points**

Rank the following from 1 to 4 in **increasing** order. Provide the ranking in the boxes provided.

Note: Atomic number (Z) are provided for relevant atoms

a) Increasing (lowest to highest) order of *ionization energy*: (1= lowest; 4 = highest)

Mg

Mg: Z = 12

Rb

Rb: Z = 37

Cs

Cs: Z = 55

K

K : Z = 19

b) Increasing (lowest to highest) order of *bond angle*: (1= lowest; 4 = highest)

H₂OCCl₄XeF₄

HCN

c) Increasing (smallest to largest) order of *ionic size* (1= smallest; 4 = largest)

S²⁻

S: Z = 16

O²⁻

O: Z = 8

Be²⁺

Be: Z = 4

Na⁺

Na: Z = 11

d) Increasing (lowest to highest) order of *bond energy* (1= lowest; 4 = highest)

P-F

P-Cl

P-I

P-Br

**Question 5** **12 points**

a. 9 points For the molecules HClO_4 and TeF_6 , draw the most stable *Lewis structure* showing appropriate shape and *dashed/wedged bonds*, based on VSEPR. Determine the **molecular geometry** around the central atom. For HClO_4 , label the *most polar bond*.

HClO_4 (there are no O-O bonds)

Label the most polar bond

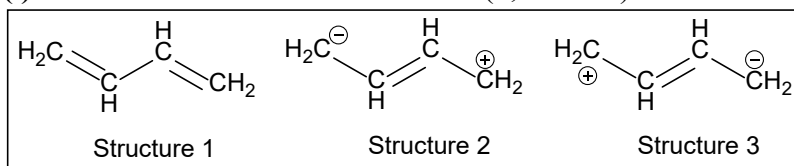
Molecular Geometry around Cl

TeF_6

Molecular Geometry around Te

b. 3 points (Note: Lone pairs are not shown in structures below)

(i) For the three *resonance* structures (1, 2 and 3) shown below, answer the following question:



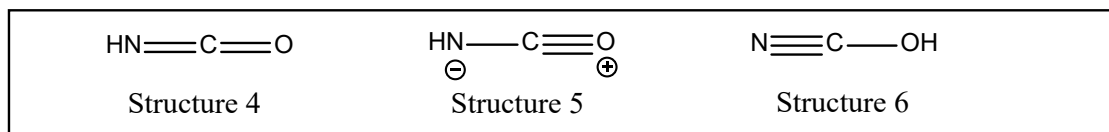
(I) Circle the *most contributing* resonance structure.

Structure 1

Structure 2

Structure 3

(ii) For the three structures given (4, 5, and 6) below, answer the following questions:



(I) Circle all the structures that represent *resonance structures* of each other:

Structure 4

Structure 5

Structure 6

(II) Among the resonance structures, circle the *most contributing* resonance structure.

Structure 4

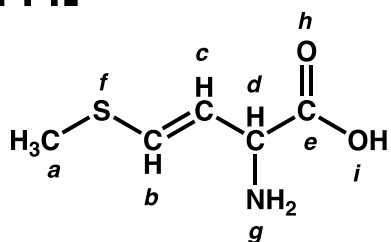
Structure 5

Structure 6

**Question 6** **6 points**

Using the data given below, calculate the **lattice energy** for SrO(s).

	ΔH° (kJ/mol)
$\text{Sr}(s) \longrightarrow \text{Sr}(g)$	+180
$\text{Sr}(g) \longrightarrow \text{Sr}^{2+}(g) + 2e^-$	+1468
$\text{O}_2(g) \longrightarrow 2\text{O}(g)$	+500
$\text{O}(g) + 2e^- \longrightarrow \text{O}^{2-}(g)$	+600
$\text{Sr}(s) + \frac{1}{2}\text{O}_2(g) \longrightarrow \text{SrO}(s)$	-550

**Question 7 10 points**

Consider the molecule given on the left, and answer questions (i) to (vi) using chemical bonding theories that we have learnt in class (valence bond theory, hybridization).

Note that the lone pairs are not shown in the structure:

i. Consider the carbon atoms marked in as ***a***, ***b***, ***c***, ***d***, and ***e*** in the molecule, and determine **hybridization** at each of the carbon centers. (2.5 points)

Atom	Hybridization
Carbon (<i>a</i>)	
Carbon (<i>b</i>)	
Carbon (<i>c</i>)	
Carbon (<i>d</i>)	
Carbon (<i>e</i>)	

ii. Now, consider the sulfur atom (S) marked as ***f***, nitrogen (N) atom (***g***), and oxygen(O) atoms (***h*** and ***i***), and determine their **hybridization**. (2 points)

Atom	Hybridization
Sulfur (<i>f</i>)	
Nitrogen (<i>g</i>)	
Oxygen (<i>h</i>)	
Oxygen (<i>i</i>)	

iii. **Describe** all the bonds formed by the carbon atoms ***a*** and ***b***, using sigma and pi designation as appropriate. (2 points)

Atom	Bonds formed (use sigma and pi as appropriate)
Carbon (<i>a</i>)	
Carbon (<i>b</i>)	

iv. How many total sp^3 -s sigma bonds are in the molecule? (1 point)

v. Determine **electronic** and **molecular geometry** at carbon (C) labeled ***e*** and sulfur (S) labeled ***f*** centres. (2 points)

Atom	Electronic Geometry	Molecular Geometry
Carbon (<i>e</i>)		
Sulfur (<i>f</i>)		

vi. Circle the value from the choices given below, that describes the **C-S-C bond angle at sulfur** (S) marked ***f***: (0.5 points)

>90° to <109.5° 109.5° >109.5 to <120° 120° >120° to <180°


Question 8 12 points

Using the **valence shell orbitals only** draw a **complete** molecular orbital (MO) diagram (with no s-p mixing) for the molecule **PS**, in the space provided below. **(6 points)**

Assume that 3s and 3p atomic orbitals will form molecular orbitals like those formed by the combination of 2s and 2p atomic orbitals that we learnt in class, and there is no s-p mixing.

Show **energy arrow**, **atomic orbitals** involved, **atom labels**, the **molecular orbitals** that are formed, **filled electrons** in the MO diagram etc. Make sure to label all the atomic and molecular orbitals.

Using the MO diagram above, answer the following questions:

- (i) Which one of the following will have the highest bond order? Circle the correct choice
 A) PS^{1-} B) PS C) PS^{1+}
- (ii) Which one of the following will have the longest bond length? Circle the correct choice
 A) PS^{1-} B) PS C) PS^{1+}
- (iii) Which of the following will be diamagnetic? Circle the correct choice.
 A) PS^{1-} B) PS C) PS^{1+}
- (iv) Draw Lewis dot structure of the molecule with the shortest bond length (based on the MO diagram) from among the three (PS^{1-} , PS, PS^{1+}), and indicate the bond order calculated from the Lewis dot structure. **(3 points)**

Lewis Dot Structure

Bond Order



8795CAA5-ED1D-4528-9DC8-D3AC46653175

midterm-exam-93388

#805 Page 10 of 10

SCRAP PAGE

**CHEMISTRY 110 –Midterm Exam****Thursday November 7, 2024, 6:30 – 8:30 pm**

Examiners: Pallavi Sirjoosingh and Ashok Kakkar

First name (please write as legibly as possible within the boxes)

--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--

Last name

--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--

Student ID number

--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--

This booklet has 8 questions on 10 pages (The last page can be used as a scrap page).

The staple must NOT be removed from the question booklet. Your intact question booklet must be returned at the end of the exam.

A separate booklet contains a periodic table, and datasheet.

Write your answers directly under the questions in the space provided. Do not write on the QR codes or near the top edges (as it may be cut off).

Please write **legibly and dark enough** for the scan to be readable. For students who use English as a second language, dictionaries are permitted. Calculators are allowed.

DO NOT ASK QUESTIONS OF THE INVIGILATORS about interpretations of the exam questions. If you have a concern about a question, please summarize it at the top of this page. All concerns will be evaluated after the exam. GOOD LUCK!

**Question 1 8 points** (*Assume all transitions are possible*)

a. 6 points Calculate the **energy (in kJ/mol)** required to ionize (completely remove) the electron from the **n=3** level of **He⁺ ion** (He: Helium). Show your work including equations. (3 sig. figs.)

b. 2 points

(I) Select the largest (longest) wavelength from the three choices below:

- i. Wavelength of photon to *completely remove* an electron from *1s orbital of Hydrogen atom*
- ii. Wavelength of photon to *completely remove* an electron from *1s orbital of Boron atom (B)*
- iii. Wavelength of photon to *completely remove* an electron from *1s orbital of Boron cation (B⁺)*

i or ii or iii

(II) Select the smallest (shortest) wavelength from the three choices below (*for H atom*):

- i. Wavelength of photon *emitted* for an electron transitioning from *2p to 1s orbital*
- ii. Wavelength of photon *absorbed* for an electron transitioning from *2s to 4p orbital*
- iii. Wavelength of photon *absorbed* for an electron transitioning from *1s to 3p orbital*

i or ii or iii



Question 2 8 points



A Chem110 student plots the graph shown on the left, from data obtained for the kinetic energy (eV) of the ejected electron versus frequency (s^{-1}) of the incident light, for an unknown metal. ($1\text{eV} = 1.6022 \times 10^{-19} \text{ J}$)

Based on the plot, answer the questions **a to c** below:

a. 1 point On the plot above, label the **threshold frequency**.

b. 2 points What is the value of the work potential (in J) of the unknown metal?

c. 3 points What is the *longest* wavelength (in nm) required to eject an electron from this metal's surface? Show your work including equation(s) used. (3 sig figs)

d. 2 points In the space provided below, plot the following

(Assume frequency is *above* threshold frequency):

- Kinetic energy* of an ejected electron versus the *intensity* of the incident light (left)
- Number of electrons* ejected versus the *frequency* of the incident light (right)



**Question 3** **6 points****a. 2 points**

In the ground state of Ge (Germanium; $Z = 32$) atom, **name** the following orbitals:

i. Orbital containing the electron that requires the maximum energy to be removed.

ii. Orbital containing the electron that experiences the minimum effective nuclear charge.

b. 2 points

An unknown anion (A^-) has the following *excited* electronic configuration: $[Ar] 4s^2 3d^3 4p^1$

(Ar: Argon; $Z = 18$)

Give the **condensed ground state electronic configuration** of the atom, **A** and the cation, **A^{2+}** .
Show your work. (Assume all electronic transitions are possible)

Electronic Configuration of A:

Electronic Configuration of A^{2+} :

c. 2 points

In a single atom, determine how many electrons can have the following set of quantum numbers?

(i) $n = 5; l = 3$

In a single atom, determine how many electrons can have the following set of quantum numbers?

(i) $n = 3; l = 2; m_s = +1/2$

**Question 4** **8 points**

Rank the following from 1 to 4 in **increasing** order. Provide the ranking in the boxes provided.

Note: Atomic number (Z) are provided for relevant atoms

a) Increasing (lowest to highest) order of *ionization energy*: (1= lowest; 4 = highest)

Mg

Mg: Z = 12

Rb

Rb: Z = 37

Cs

Cs: Z = 55

K

K : Z = 19

b) Increasing (lowest to highest) order of *bond angle*: (1= lowest; 4 = highest)

H₂OCCl₄XeF₄

HCN

c) Increasing (smallest to largest) order of *ionic size* (1= smallest; 4 = largest)

S²⁻

S: Z = 16

O²⁻

O: Z = 8

Be²⁺

Be: Z = 4

Na⁺

Na: Z = 11

d) Increasing (lowest to highest) order of *bond energy* (1= lowest; 4 = highest)

P-F

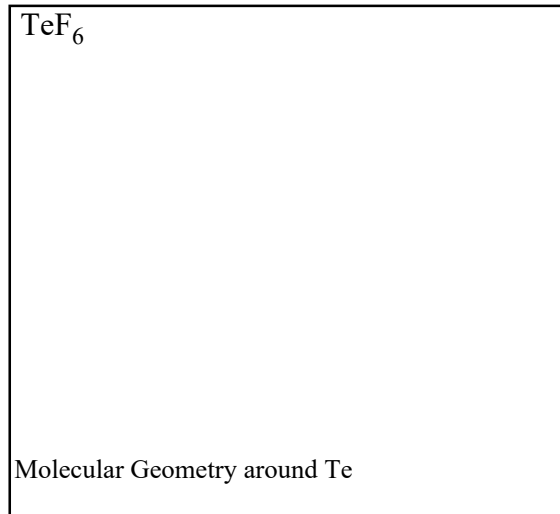
P-Cl

P-I

P-Br

**Question 5** **12 points**

a. 9 points For the molecules HClO_4 and TeF_6 , draw the most stable *Lewis structure* showing appropriate shape and *dashed/wedged bonds*, based on VSEPR. Determine the **molecular geometry** around the central atom. For HClO_4 , label the *most polar bond*.



b. 3 points (Note: Lone pairs are not shown in structures below)

(i) For the three *resonance* structures (1, 2 and 3) shown below, answer the following question:



(I) Circle the *most contributing* resonance structure.

Structure 1

Structure 2

Structure 3

(ii) For the three structures given (4, 5, and 6) below, answer the following questions:



(I) Circle all the structures that represent *resonance structures* of each other:

Structure 4

Structure 5

Structure 6

(II) Among the resonance structures, circle the *most contributing* resonance structure.

Structure 4

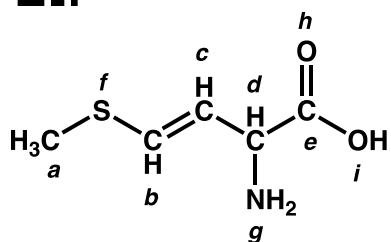
Structure 5

Structure 6

**Question 6** **6 points**

Using the data given below, calculate the **lattice energy** for SrO(s).

	ΔH° (kJ/mol)
$\text{Sr}(s) \longrightarrow \text{Sr}(g)$	+180
$\text{Sr}(g) \longrightarrow \text{Sr}^{2+}(g) + 2e^-$	+1468
$\text{O}_2(g) \longrightarrow 2\text{O}(g)$	+500
$\text{O}(g) + 2e^- \longrightarrow \text{O}^{2-}(g)$	+600
$\text{Sr}(s) + \frac{1}{2}\text{O}_2(g) \longrightarrow \text{SrO}(s)$	-550

**Question 7 10 points**

Consider the molecule given on the left, and answer questions (i) to (vi) using chemical bonding theories that we have learnt in class (valence bond theory, hybridization).

Note that the lone pairs are not shown in the structure:

i. Consider the carbon atoms marked in as ***a***, ***b***, ***c***, ***d***, and ***e*** in the molecule, and determine **hybridization** at each of the carbon centers. (2.5 points)

Atom	Hybridization
Carbon (<i>a</i>)	
Carbon (<i>b</i>)	
Carbon (<i>c</i>)	
Carbon (<i>d</i>)	
Carbon (<i>e</i>)	

ii. Now, consider the sulfur atom (S) marked as ***f***, nitrogen (N) atom (***g***), and oxygen(O) atoms (***h*** and ***i***), and determine their **hybridization**. (2 points)

Atom	Hybridization
Sulfur (<i>f</i>)	
Nitrogen (<i>g</i>)	
Oxygen (<i>h</i>)	
Oxygen (<i>i</i>)	

iii. **Describe** all the bonds formed by the carbon atoms ***a*** and ***b***, using sigma and pi designation as appropriate. (2 points)

Atom	Bonds formed (use sigma and pi as appropriate)
Carbon (<i>a</i>)	
Carbon (<i>b</i>)	

iv. How many total sp^3 -s sigma bonds are in the molecule? (1 point)

v. Determine **electronic** and **molecular geometry** at carbon (C) labeled ***e*** and sulfur (S) labeled ***f*** centres. (2 points)

Atom	Electronic Geometry	Molecular Geometry
Carbon (<i>e</i>)		
Sulfur (<i>f</i>)		

vi. Circle the value from the choices given below, that describes the **C-S-C bond angle at sulfur** (S) marked ***f***: (0.5 points)

>90° to <109.5° 109.5° >109.5 to <120° 120° >120° to <180°


Question 8 12 points

Using the **valence shell orbitals only** draw a **complete** molecular orbital (MO) diagram (with no s-p mixing) for the molecule **PS**, in the space provided below. **(6 points)**

Assume that 3s and 3p atomic orbitals will form molecular orbitals like those formed by the combination of 2s and 2p atomic orbitals that we learnt in class, and there is no s-p mixing.

Show **energy arrow**, **atomic orbitals** involved, **atom labels**, the **molecular orbitals** that are formed, **filled electrons** in the MO diagram etc. Make sure to label all the atomic and molecular orbitals.

Using the MO diagram above, answer the following questions:

- (i) Which one of the following will have the highest bond order? Circle the correct choice
 A) PS^{1-} B) PS C) PS^{1+}
- (ii) Which one of the following will have the longest bond length? Circle the correct choice
 A) PS^{1-} B) PS C) PS^{1+}
- (iii) Which of the following will be diamagnetic? Circle the correct choice.
 A) PS^{1-} B) PS C) PS^{1+}
- (iv) Draw Lewis dot structure of the molecule with the shortest bond length (based on the MO diagram) from among the three (PS^{1-} , PS, PS^{1+}), and indicate the bond order calculated from the Lewis dot structure. **(3 points)**

Lewis Dot Structure

Bond Order



79A29E4E-B561-4468-9640-2DB92E582CEA

midterm-exam-93388

#806 Page 10 of 10

SCRAP PAGE

**CHEMISTRY 110 –Midterm Exam****Thursday November 7, 2024, 6:30 – 8:30 pm**

Examiners: Pallavi Sirjoosingh and Ashok Kakkar

First name (please write as legibly as possible within the boxes)

--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--

Last name

--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--

Student ID number

--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--

This booklet has 8 questions on 10 pages (The last page can be used as a scrap page).

The staple must NOT be removed from the question booklet. Your intact question booklet must be returned at the end of the exam.

A separate booklet contains a periodic table, and datasheet.

Write your answers directly under the questions in the space provided. Do not write on the QR codes or near the top edges (as it may be cut off).

Please write **legibly and dark enough** for the scan to be readable. For students who use English as a second language, dictionaries are permitted. Calculators are allowed.

DO NOT ASK QUESTIONS OF THE INVIGILATORS about interpretations of the exam questions. If you have a concern about a question, please summarize it at the top of this page. All concerns will be evaluated after the exam. GOOD LUCK!

**Question 1 8 points** (*Assume all transitions are possible*)

a. 6 points Calculate the **energy (in kJ/mol)** required to ionize (completely remove) the electron from the **n=3** level of **He⁺ ion** (He: Helium). Show your work including equations. (3 sig. figs.)

b. 2 points

(I) Select the largest (longest) wavelength from the three choices below:

- i. Wavelength of photon to *completely remove* an electron from *1s orbital of Hydrogen atom*
- ii. Wavelength of photon to *completely remove* an electron from *1s orbital of Boron atom (B)*
- iii. Wavelength of photon to *completely remove* an electron from *1s orbital of Boron cation (B⁺)*

i or ii or iii

(II) Select the smallest (shortest) wavelength from the three choices below (*for H atom*):

- i. Wavelength of photon *emitted* for an electron transitioning from *2p to 1s orbital*
- ii. Wavelength of photon *absorbed* for an electron transitioning from *2s to 4p orbital*
- iii. Wavelength of photon *absorbed* for an electron transitioning from *1s to 3p orbital*

i or ii or iii



Question 2 8 points



A Chem110 student plots the graph shown on the left, from data obtained for the kinetic energy (eV) of the ejected electron versus frequency (s^{-1}) of the incident light, for an unknown metal. ($1\text{eV} = 1.6022 \times 10^{-19} \text{ J}$)

Based on the plot, answer the questions **a to c** below:

a. 1 point On the plot above, label the **threshold frequency**.

b. 2 points What is the value of the work potential (in J) of the unknown metal?

c. 3 points What is the *longest* wavelength (in nm) required to eject an electron from this metal's surface? Show your work including equation(s) used. (3 sig figs)

d. 2 points In the space provided below, plot the following

(Assume frequency is *above* threshold frequency):

- i. *Kinetic energy* of an ejected electron versus the *intensity* of the incident light (left)
- ii. *Number of electrons* ejected versus the *frequency* of the incident light (right)



**Question 3** **6 points****a. 2 points**

In the ground state of Ge (Germanium; $Z = 32$) atom, **name** the following orbitals:

i. Orbital containing the electron that requires the maximum energy to be removed.

ii. Orbital containing the electron that experiences the minimum effective nuclear charge.

b. 2 points

An unknown anion (A^-) has the following *excited* electronic configuration: $[Ar] 4s^2 3d^3 4p^1$

(Ar: Argon; $Z = 18$)

Give the **condensed ground state electronic configuration** of the atom, **A** and the cation, **A^{2+}** .
Show your work. (Assume all electronic transitions are possible)

Electronic Configuration of A:

Electronic Configuration of A^{2+} :

c. 2 points

In a single atom, determine how many electrons can have the following set of quantum numbers?

(i) $n = 5; l = 3$

In a single atom, determine how many electrons can have the following set of quantum numbers?

(i) $n = 3; l = 2; m_s = +1/2$

**Question 4** **8 points**

Rank the following from 1 to 4 in **increasing** order. Provide the ranking in the boxes provided.

Note: Atomic number (Z) are provided for relevant atoms

a) Increasing (lowest to highest) order of *ionization energy*: (1= lowest; 4 = highest)

Mg

Mg: Z = 12

Rb

Rb: Z = 37

Cs

Cs: Z = 55

K

K : Z = 19

b) Increasing (lowest to highest) order of *bond angle*: (1= lowest; 4 = highest)

H₂OCCl₄XeF₄

HCN

c) Increasing (smallest to largest) order of *ionic size* (1= smallest; 4 = largest)

S²⁻

S: Z = 16

O²⁻

O: Z = 8

Be²⁺

Be: Z = 4

Na⁺

Na: Z = 11

d) Increasing (lowest to highest) order of *bond energy* (1= lowest; 4 = highest)

P-F

P-Cl

P-I

P-Br

**Question 5** **12 points**

a. 9 points For the molecules HClO_4 and TeF_6 , draw the most stable *Lewis structure* showing appropriate shape and *dashed/wedged bonds*, based on VSEPR. Determine the **molecular geometry** around the central atom. For HClO_4 , label the *most polar bond*.



b. 3 points (Note: Lone pairs are not shown in structures below)

(i) For the three *resonance* structures (1, 2 and 3) shown below, answer the following question:



(I) Circle the *most contributing* resonance structure.

Structure 1

Structure 2

Structure 3

(ii) For the three structures given (4, 5, and 6) below, answer the following questions:



(I) Circle all the structures that represent *resonance structures* of each other:

Structure 4

Structure 5

Structure 6

(II) Among the resonance structures, circle the *most contributing* resonance structure.

Structure 4

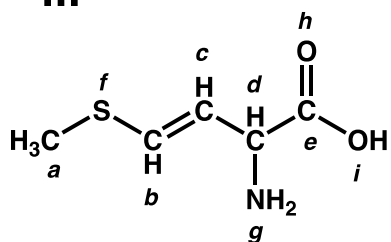
Structure 5

Structure 6

**Question 6** **6 points**

Using the data given below, calculate the **lattice energy** for SrO(s).

	ΔH° (kJ/mol)
$\text{Sr}(s) \longrightarrow \text{Sr}(g)$	+180
$\text{Sr}(g) \longrightarrow \text{Sr}^{2+}(g) + 2e^-$	+1468
$\text{O}_2(g) \longrightarrow 2\text{O}(g)$	+500
$\text{O}(g) + 2e^- \longrightarrow \text{O}^{2-}(g)$	+600
$\text{Sr}(s) + \frac{1}{2}\text{O}_2(g) \longrightarrow \text{SrO}(s)$	-550

**Question 7 10 points**

Consider the molecule given on the left, and answer questions (i) to (vi) using chemical bonding theories that we have learnt in class (valence bond theory, hybridization).

Note that the lone pairs are not shown in the structure:

i. Consider the carbon atoms marked in as ***a***, ***b***, ***c***, ***d***, and ***e*** in the molecule, and determine **hybridization** at each of the carbon centers. (2.5 points)

Atom	Hybridization
Carbon (<i>a</i>)	
Carbon (<i>b</i>)	
Carbon (<i>c</i>)	
Carbon (<i>d</i>)	
Carbon (<i>e</i>)	

ii. Now, consider the sulfur atom (S) marked as ***f***, nitrogen (N) atom (***g***), and oxygen(O) atoms (***h*** and ***i***), and determine their **hybridization**. (2 points)

Atom	Hybridization
Sulfur (<i>f</i>)	
Nitrogen (<i>g</i>)	
Oxygen (<i>h</i>)	
Oxygen (<i>i</i>)	

iii. **Describe** all the bonds formed by the carbon atoms ***a*** and ***b***, using sigma and pi designation as appropriate. (2 points)

Atom	Bonds formed (use sigma and pi as appropriate)
Carbon (<i>a</i>)	
Carbon (<i>b</i>)	

iv. How many total sp^3 -s sigma bonds are in the molecule? (1 point)

v. Determine **electronic** and **molecular geometry** at carbon (C) labeled ***e*** and sulfur (S) labeled ***f*** centres. (2 points)

Atom	Electronic Geometry	Molecular Geometry
Carbon (<i>e</i>)		
Sulfur (<i>f</i>)		

vi. Circle the value from the choices given below, that describes the **C-S-C bond angle at sulfur** (S) marked ***f***: (0.5 points)

>90° to <109.5° 109.5° >109.5 to <120° 120° >120° to <180°


Question 8 12 points

Using the **valence shell orbitals only** draw a **complete** molecular orbital (MO) diagram (with no s-p mixing) for the molecule **PS**, in the space provided below. **(6 points)**

Assume that 3s and 3p atomic orbitals will form molecular orbitals like those formed by the combination of 2s and 2p atomic orbitals that we learnt in class, and there is no s-p mixing.

Show **energy arrow**, **atomic orbitals** involved, **atom labels**, the **molecular orbitals** that are formed, **filled electrons** in the MO diagram etc. Make sure to label all the atomic and molecular orbitals.

Using the MO diagram above, answer the following questions:

- (i) Which one of the following will have the highest bond order? Circle the correct choice
 A) PS^{1-} B) PS C) PS^{1+}
- (ii) Which one of the following will have the longest bond length? Circle the correct choice
 A) PS^{1-} B) PS C) PS^{1+}
- (iii) Which of the following will be diamagnetic? Circle the correct choice.
 A) PS^{1-} B) PS C) PS^{1+}
- (iv) Draw Lewis dot structure of the molecule with the shortest bond length (based on the MO diagram) from among the three (PS^{1-} , PS, PS^{1+}), and indicate the bond order calculated from the Lewis dot structure. **(3 points)**

Lewis Dot Structure

Bond Order



C129B987-D48F-4DFE-995B-A2A0C9C28936

midterm-exam-93388

#807 Page 10 of 10

SCRAP PAGE

**CHEMISTRY 110 –Midterm Exam****Thursday November 7, 2024, 6:30 – 8:30 pm****Examiners: Pallavi Sirjoosingh and Ashok Kakkar**

First name (please write as legibly as possible within the boxes)

--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--

Last name

--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--

Student ID number

--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--

This booklet has 8 questions on 10 pages (The last page can be used as a scrap page).

The staple must NOT be removed from the question booklet. Your intact question booklet must be returned at the end of the exam.

A separate booklet contains a periodic table, and datasheet.

Write your answers directly under the questions in the space provided. Do not write on the QR codes or near the top edges (as it may be cut off).

Please write **legibly and dark enough** for the scan to be readable. For students who use English as a second language, dictionaries are permitted. Calculators are allowed.

DO NOT ASK QUESTIONS OF THE INVIGILATORS about interpretations of the exam questions. If you have a concern about a question, please summarize it at the top of this page. All concerns will be evaluated after the exam. GOOD LUCK!

**Question 1 8 points** (*Assume all transitions are possible*)

a. 6 points Calculate the **energy (in kJ/mol)** required to ionize (completely remove) the electron from the **n=3** level of **He⁺ ion** (He: Helium). Show your work including equations. (3 sig. figs.)

b. 2 points

(I) Select the largest (longest) wavelength from the three choices below:

- i. Wavelength of photon to *completely remove* an electron from *1s orbital of Hydrogen atom*
- ii. Wavelength of photon to *completely remove* an electron from *1s orbital of Boron atom (B)*
- iii. Wavelength of photon to *completely remove* an electron from *1s orbital of Boron cation (B⁺)*

i or ii or iii

(II) Select the smallest (shortest) wavelength from the three choices below (*for H atom*):

- i. Wavelength of photon *emitted* for an electron transitioning from *2p to 1s orbital*
- ii. Wavelength of photon *absorbed* for an electron transitioning from *2s to 4p orbital*
- iii. Wavelength of photon *absorbed* for an electron transitioning from *1s to 3p orbital*

i or ii or iii



Question 2 8 points



A Chem110 student plots the graph shown on the left, from data obtained for the kinetic energy (eV) of the ejected electron versus frequency (s^{-1}) of the incident light, for an unknown metal. ($1\text{eV} = 1.6022 \times 10^{-19} \text{ J}$)

Based on the plot, answer the questions **a to c** below:

a. 1 point On the plot above, label the **threshold frequency**.

b. 2 points What is the value of the work potential (in J) of the unknown metal?

c. 3 points What is the *longest* wavelength (in nm) required to eject an electron from this metal's surface? Show your work including equation(s) used. (3 sig figs)

d. 2 points In the space provided below, plot the following

(Assume frequency is *above* threshold frequency):

- Kinetic energy* of an ejected electron versus the *intensity* of the incident light (left)
- Number of electrons* ejected versus the *frequency* of the incident light (right)



**Question 3** **6 points****a. 2 points**

In the ground state of Ge (Germanium; $Z = 32$) atom, **name** the following orbitals:

i. Orbital containing the electron that requires the maximum energy to be removed.

ii. Orbital containing the electron that experiences the minimum effective nuclear charge.

b. 2 points

An unknown anion (A^-) has the following *excited* electronic configuration: $[Ar] 4s^2 3d^3 4p^1$

(Ar: Argon; $Z = 18$)

Give the **condensed ground state electronic configuration** of the atom, **A** and the cation, **A^{2+}** .
Show your work. (Assume all electronic transitions are possible)

Electronic Configuration of A:

Electronic Configuration of A^{2+} :

c. 2 points

In a single atom, determine how many electrons can have the following set of quantum numbers?

(i) $n = 5; l = 3$

In a single atom, determine how many electrons can have the following set of quantum numbers?

(i) $n = 3; l = 2; m_s = +1/2$

**Question 4** **8 points**

Rank the following from 1 to 4 in **increasing** order. Provide the ranking in the boxes provided.

Note: Atomic number (Z) are provided for relevant atoms

a) Increasing (lowest to highest) order of *ionization energy*: (1= lowest; 4 = highest)

Mg

Mg: Z = 12

Rb

Rb: Z = 37

Cs

Cs: Z = 55

K

K : Z = 19

b) Increasing (lowest to highest) order of *bond angle*: (1= lowest; 4 = highest)

H₂OCCl₄XeF₄

HCN

c) Increasing (smallest to largest) order of *ionic size* (1= smallest; 4 = largest)

S²⁻

S: Z = 16

O²⁻

O: Z = 8

Be²⁺

Be: Z = 4

Na⁺

Na: Z = 11

d) Increasing (lowest to highest) order of *bond energy* (1= lowest; 4 = highest)

P-F

P-Cl

P-I

P-Br

**Question 5** **12 points**

a. 9 points For the molecules HClO_4 and TeF_6 , draw the most stable *Lewis structure* showing appropriate shape and *dashed/wedged bonds*, based on VSEPR. Determine the **molecular geometry** around the central atom. For HClO_4 , label the *most polar bond*.



b. 3 points (Note: Lone pairs are not shown in structures below)

(i) For the three *resonance* structures (1, 2 and 3) shown below, answer the following question:



(I) Circle the *most contributing* resonance structure.

Structure 1

Structure 2

Structure 3

(ii) For the three structures given (4, 5, and 6) below, answer the following questions:



(I) Circle all the structures that represent *resonance structures* of each other:

Structure 4

Structure 5

Structure 6

(II) Among the resonance structures, circle the *most contributing* resonance structure.

Structure 4

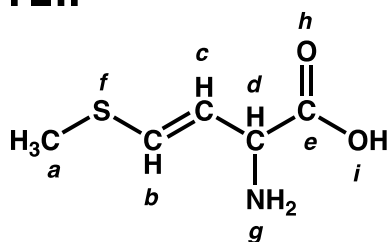
Structure 5

Structure 6

**Question 6** **6 points**

Using the data given below, calculate the **lattice energy** for SrO(s).

	ΔH° (kJ/mol)
$\text{Sr}(s) \longrightarrow \text{Sr}(g)$	+180
$\text{Sr}(g) \longrightarrow \text{Sr}^{2+}(g) + 2e^-$	+1468
$\text{O}_2(g) \longrightarrow 2\text{O}(g)$	+500
$\text{O}(g) + 2e^- \longrightarrow \text{O}^{2-}(g)$	+600
$\text{Sr}(s) + \frac{1}{2}\text{O}_2(g) \longrightarrow \text{SrO}(s)$	-550

**Question 7 10 points**

Consider the molecule given on the left, and answer questions (i) to (vi) using chemical bonding theories that we have learnt in class (valence bond theory, hybridization).

Note that the lone pairs are not shown in the structure:

i. Consider the carbon atoms marked in as ***a***, ***b***, ***c***, ***d***, and ***e*** in the molecule, and determine **hybridization** at each of the carbon centers. (2.5 points)

Atom	Hybridization
Carbon (<i>a</i>)	
Carbon (<i>b</i>)	
Carbon (<i>c</i>)	
Carbon (<i>d</i>)	
Carbon (<i>e</i>)	

ii. Now, consider the sulfur atom (S) marked as ***f***, nitrogen (N) atom (***g***), and oxygen(O) atoms (***h*** and ***i***), and determine their **hybridization**. (2 points)

Atom	Hybridization
Sulfur (<i>f</i>)	
Nitrogen (<i>g</i>)	
Oxygen (<i>h</i>)	
Oxygen (<i>i</i>)	

iii. **Describe** all the bonds formed by the carbon atoms ***a*** and ***b***, using sigma and pi designation as appropriate. (2 points)

Atom	Bonds formed (use sigma and pi as appropriate)
Carbon (<i>a</i>)	
Carbon (<i>b</i>)	

iv. How many total sp^3 -s sigma bonds are in the molecule? (1 point)

v. Determine **electronic** and **molecular geometry** at carbon (C) labeled ***e*** and sulfur (S) labeled ***f*** centres. (2 points)

Atom	Electronic Geometry	Molecular Geometry
Carbon (<i>e</i>)		
Sulfur (<i>f</i>)		

vi. Circle the value from the choices given below, that describes the **C-S-C bond angle at sulfur** (S) marked ***f***: (0.5 points)

>90° to <109.5°

109.5°

>109.5 to <120°

120°

>120° to <180°

**Question 8 12 points**

Using the **valence shell orbitals only** draw a **complete** molecular orbital (MO) diagram (with no s-p mixing) for the molecule **PS**, in the space provided below. **(6 points)**

Assume that 3s and 3p atomic orbitals will form molecular orbitals like those formed by the combination of 2s and 2p atomic orbitals that we learnt in class, and there is no s-p mixing.

Show **energy arrow**, **atomic orbitals** involved, **atom labels**, the **molecular orbitals** that are formed, **filled electrons** in the MO diagram etc. Make sure to label all the atomic and molecular orbitals.

Using the MO diagram above, answer the following questions:

- (i) Which one of the following will have the highest bond order? Circle the correct choice
A) PS^{1-} B) PS C) PS^{1+}
- (ii) Which one of the following will have the longest bond length? Circle the correct choice
A) PS^{1-} B) PS C) PS^{1+}
- (iii) Which of the following will be diamagnetic? Circle the correct choice.
A) PS^{1-} B) PS C) PS^{1+}
- (iv) Draw Lewis dot structure of the molecule with the shortest bond length (based on the MO diagram) from among the three (PS^{1-} , PS, PS^{1+}), and indicate the bond order calculated from the Lewis dot structure. **(3 points)**

Lewis Dot Structure

Bond Order



55629C79-A28F-4262-A4E1-E55CDC58EF0D

midterm-exam-93388

#808 Page 10 of 10

SCRAP PAGE

**CHEMISTRY 110 –Midterm Exam****Thursday November 7, 2024, 6:30 – 8:30 pm****Examiners: Pallavi Sirjoosingh and Ashok Kakkar**

First name (please write as legibly as possible within the boxes)

--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--

Last name

--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--

Student ID number

--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--

This booklet has 8 questions on 10 pages (The last page can be used as a scrap page).

The staple must NOT be removed from the question booklet. Your intact question booklet must be returned at the end of the exam.

A separate booklet contains a periodic table, and datasheet.

Write your answers directly under the questions in the space provided. Do not write on the QR codes or near the top edges (as it may be cut off).

Please write **legibly and dark enough** for the scan to be readable. For students who use English as a second language, dictionaries are permitted. Calculators are allowed.

DO NOT ASK QUESTIONS OF THE INVIGILATORS about interpretations of the exam questions. If you have a concern about a question, please summarize it at the top of this page. All concerns will be evaluated after the exam. GOOD LUCK!

**Question 1 8 points** (*Assume all transitions are possible*)

a. 6 points Calculate the **energy (in kJ/mol)** required to ionize (completely remove) the electron from the **$n=3$** level of **He^+ ion** (He: Helium). Show your work including equations. (3 sig. figs.)

b. 2 points

(I) Select the largest (longest) wavelength from the three choices below:

- i. Wavelength of photon to *completely remove* an electron from *1s orbital of Hydrogen atom*
- ii. Wavelength of photon to *completely remove* an electron from *1s orbital of Boron atom (B)*
- iii. Wavelength of photon to *completely remove* an electron from *1s orbital of Boron cation (B^+)*

i or ii or iii

(II) Select the smallest (shortest) wavelength from the three choices below (*for H atom*):

- i. Wavelength of photon *emitted* for an electron transitioning from *2p to 1s orbital*
- ii. Wavelength of photon *absorbed* for an electron transitioning from *2s to 4p orbital*
- iii. Wavelength of photon *absorbed* for an electron transitioning from *1s to 3p orbital*

i or ii or iii



Question 2 8 points



A Chem110 student plots the graph shown on the left, from data obtained for the kinetic energy (eV) of the ejected electron versus frequency (s^{-1}) of the incident light, for an unknown metal. ($1\text{eV} = 1.6022 \times 10^{-19} \text{ J}$)

Based on the plot, answer the questions **a to c** below:

a. 1 point On the plot above, label the **threshold frequency**.

b. 2 points What is the value of the work potential (in J) of the unknown metal?

c. 3 points What is the *longest* wavelength (in nm) required to eject an electron from this metal's surface? Show your work including equation(s) used. (3 sig figs)

d. 2 points In the space provided below, plot the following

(Assume frequency is *above* threshold frequency):

- i. *Kinetic energy* of an ejected electron versus the *intensity* of the incident light (left)
- ii. *Number of electrons* ejected versus the *frequency* of the incident light (right)



**Question 3** **6 points****a. 2 points**

In the ground state of Ge (Germanium; $Z = 32$) atom, **name** the following orbitals:

i. Orbital containing the electron that requires the maximum energy to be removed.

ii. Orbital containing the electron that experiences the minimum effective nuclear charge.

b. 2 points

An unknown anion (A^-) has the following *excited* electronic configuration: $[Ar] 4s^2 3d^3 4p^1$

(Ar: Argon; $Z = 18$)

Give the **condensed ground state electronic configuration** of the atom, **A** and the cation, **A^{2+}** .
Show your work. (Assume all electronic transitions are possible)

Electronic Configuration of A:

Electronic Configuration of A^{2+} :

c. 2 points

In a single atom, determine how many electrons can have the following set of quantum numbers?

(i) $n = 5; l = 3$

In a single atom, determine how many electrons can have the following set of quantum numbers?

(i) $n = 3; l = 2; m_s = +1/2$

**Question 4** **8 points**

Rank the following from 1 to 4 in **increasing** order. Provide the ranking in the boxes provided.

Note: Atomic number (Z) are provided for relevant atoms

a) Increasing (lowest to highest) order of *ionization energy*: (1= lowest; 4 = highest)

Mg

Mg: Z = 12

Rb

Rb: Z = 37

Cs

Cs: Z = 55

K

K : Z = 19

b) Increasing (lowest to highest) order of *bond angle*: (1= lowest; 4 = highest)

H₂OCCl₄XeF₄

HCN

c) Increasing (smallest to largest) order of *ionic size* (1= smallest; 4 = largest)

S²⁻

S: Z = 16

O²⁻

O: Z = 8

Be²⁺

Be: Z = 4

Na⁺

Na: Z = 11

d) Increasing (lowest to highest) order of *bond energy* (1= lowest; 4 = highest)

P-F

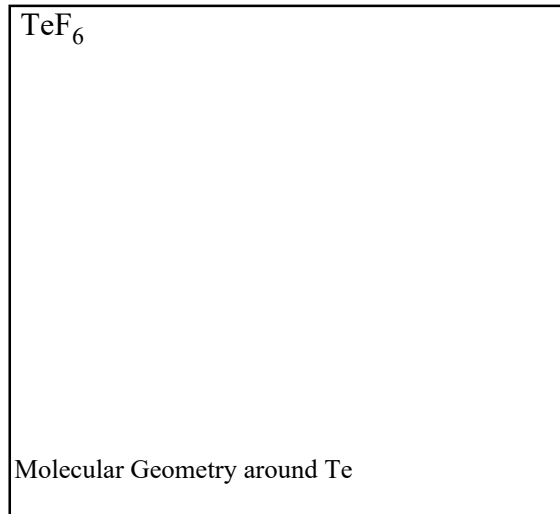
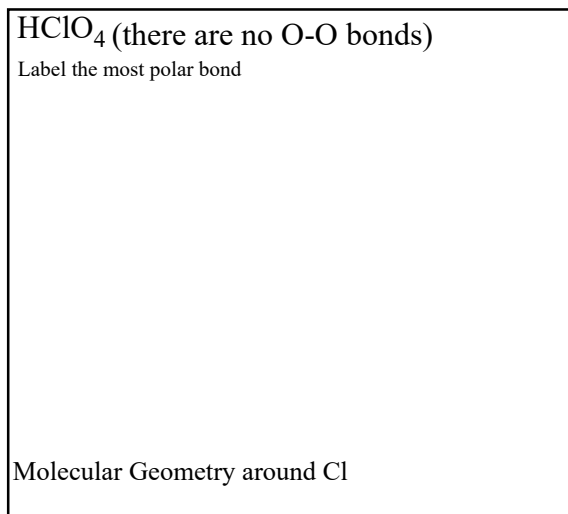
P-Cl

P-I

P-Br

**Question 5** **12 points**

a. 9 points For the molecules HClO_4 and TeF_6 , draw the most stable *Lewis structure* showing appropriate shape and *dashed/wedged bonds*, based on VSEPR. Determine the **molecular geometry** around the central atom. For HClO_4 , label the *most polar bond*.



b. 3 points (Note: Lone pairs are not shown in structures below)

(i) For the three *resonance* structures (1, 2 and 3) shown below, answer the following question:



(I) Circle the *most contributing* resonance structure.

Structure 1

Structure 2

Structure 3

(ii) For the three structures given (4, 5, and 6) below, answer the following questions:



(I) Circle all the structures that represent *resonance structures* of each other:

Structure 4

Structure 5

Structure 6

(II) Among the resonance structures, circle the *most contributing* resonance structure.

Structure 4

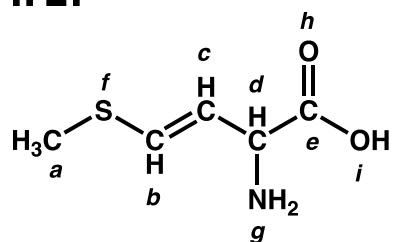
Structure 5

Structure 6

**Question 6** **6 points**

Using the data given below, calculate the **lattice energy** for SrO(s).

	ΔH° (kJ/mol)
$\text{Sr}(s) \longrightarrow \text{Sr}(g)$	+180
$\text{Sr}(g) \longrightarrow \text{Sr}^{2+}(g) + 2e^-$	+1468
$\text{O}_2(g) \longrightarrow 2\text{O}(g)$	+500
$\text{O}(g) + 2e^- \longrightarrow \text{O}^{2-}(g)$	+600
$\text{Sr}(s) + \frac{1}{2}\text{O}_2(g) \longrightarrow \text{SrO}(s)$	-550

**Question 7 10 points**

Consider the molecule given on the left, and answer questions (i) to (vi) using chemical bonding theories that we have learnt in class (valence bond theory, hybridization).

Note that the lone pairs are not shown in the structure:

i. Consider the carbon atoms marked in as ***a***, ***b***, ***c***, ***d***, and ***e*** in the molecule, and determine **hybridization** at each of the carbon centers. (2.5 points)

Atom	Hybridization
Carbon (<i>a</i>)	
Carbon (<i>b</i>)	
Carbon (<i>c</i>)	
Carbon (<i>d</i>)	
Carbon (<i>e</i>)	

ii. Now, consider the sulfur atom (S) marked as ***f***, nitrogen (N) atom (***g***), and oxygen(O) atoms (***h*** and ***i***), and determine their **hybridization**. (2 points)

Atom	Hybridization
Sulfur (<i>f</i>)	
Nitrogen (<i>g</i>)	
Oxygen (<i>h</i>)	
Oxygen (<i>i</i>)	

iii. **Describe** all the bonds formed by the carbon atoms ***a*** and ***b***, using sigma and pi designation as appropriate. (2 points)

Atom	Bonds formed (use sigma and pi as appropriate)
Carbon (<i>a</i>)	
Carbon (<i>b</i>)	

iv. How many total sp^3 -s sigma bonds are in the molecule? (1 point)

v. Determine **electronic** and **molecular geometry** at carbon (C) labeled ***e*** and sulfur (S) labeled ***f*** centres. (2 points)

Atom	Electronic Geometry	Molecular Geometry
Carbon (<i>e</i>)		
Sulfur (<i>f</i>)		

vi. Circle the value from the choices given below, that describes the **C-S-C bond angle at sulfur** (S) marked ***f***: (0.5 points)

>90° to <109.5°

109.5°

>109.5 to <120°

120°

>120° to <180°



Question 8 12 points

Using the **valence shell orbitals only** draw a **complete** molecular orbital (MO) diagram (with no s-p mixing) for the molecule **PS**, in the space provided below. **(6 points)**

Assume that 3s and 3p atomic orbitals will form molecular orbitals like those formed by the combination of 2s and 2p atomic orbitals that we learnt in class, and there is no s-p mixing.

Show **energy arrow**, **atomic orbitals** involved, **atom labels**, the **molecular orbitals** that are formed, **filled electrons** in the MO diagram etc. Make sure to label all the atomic and molecular orbitals.

Using the MO diagram above, answer the following questions:

- (i) Which one of the following will have the highest bond order? Circle the correct choice
 A) PS^{1-} B) PS C) PS^{1+}
- (ii) Which one of the following will have the longest bond length? Circle the correct choice
 A) PS^{1-} B) PS C) PS^{1+}
- (iii) Which of the following will be diamagnetic? Circle the correct choice.
 A) PS^{1-} B) PS C) PS^{1+}
- (iv) Draw Lewis dot structure of the molecule with the shortest bond length (based on the MO diagram) from among the three (PS^{1-} , PS, PS^{1+}), and indicate the bond order calculated from the Lewis dot structure. **(3 points)**

Lewis Dot Structure

Bond Order



1D51E935-5D61-4699-BF53-06C7CC5A2393

midterm-exam-93388

#809 Page 10 of 10

SCRAP PAGE

**CHEMISTRY 110 –Midterm Exam****Thursday November 7, 2024, 6:30 – 8:30 pm**

Examiners: Pallavi Sirjoosingh and Ashok Kakkar

First name (please write as legibly as possible within the boxes)

--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--

Last name

--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--

Student ID number

--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--

This booklet has 8 questions on 10 pages (The last page can be used as a scrap page).

The staple must NOT be removed from the question booklet. Your intact question booklet must be returned at the end of the exam.

A separate booklet contains a periodic table, and datasheet.

Write your answers directly under the questions in the space provided. Do not write on the QR codes or near the top edges (as it may be cut off).

Please write **legibly and dark enough** for the scan to be readable. For students who use English as a second language, dictionaries are permitted. Calculators are allowed.

DO NOT ASK QUESTIONS OF THE INVIGILATORS about interpretations of the exam questions. If you have a concern about a question, please summarize it at the top of this page. All concerns will be evaluated after the exam. GOOD LUCK!

**Question 1 8 points** (*Assume all transitions are possible*)

a. 6 points Calculate the **energy (in kJ/mol)** required to ionize (completely remove) the electron from the **n=3** level of **He⁺ ion** (He: Helium). Show your work including equations. (3 sig. figs.)

b. 2 points

(I) Select the largest (longest) wavelength from the three choices below:

- i. Wavelength of photon to *completely remove* an electron from *1s orbital of Hydrogen atom*
- ii. Wavelength of photon to *completely remove* an electron from *1s orbital of Boron atom (B)*
- iii. Wavelength of photon to *completely remove* an electron from *1s orbital of Boron cation (B⁺)*

i or ii or iii

(II) Select the smallest (shortest) wavelength from the three choices below (*for H atom*):

- i. Wavelength of photon *emitted* for an electron transitioning from *2p to 1s orbital*
- ii. Wavelength of photon *absorbed* for an electron transitioning from *2s to 4p orbital*
- iii. Wavelength of photon *absorbed* for an electron transitioning from *1s to 3p orbital*

i or ii or iii



Question 2 8 points



A Chem110 student plots the graph shown on the left, from data obtained for the kinetic energy (eV) of the ejected electron versus frequency (s^{-1}) of the incident light, for an unknown metal. ($1\text{eV} = 1.6022 \times 10^{-19} \text{ J}$)

Based on the plot, answer the questions **a to c** below:

a. 1 point On the plot above, label the **threshold frequency**.

b. 2 points What is the value of the work potential (in J) of the unknown metal?

c. 3 points What is the *longest* wavelength (in nm) required to eject an electron from this metal's surface? Show your work including equation(s) used. (3 sig figs)

d. 2 points In the space provided below, plot the following

(Assume frequency is *above* threshold frequency):

- i. *Kinetic energy* of an ejected electron versus the *intensity* of the incident light (left)
- ii. *Number of electrons* ejected versus the *frequency* of the incident light (right)



**Question 3** **6 points****a. 2 points**

In the ground state of Ge (Germanium; $Z = 32$) atom, **name** the following orbitals:

i. Orbital containing the electron that requires the maximum energy to be removed.

ii. Orbital containing the electron that experiences the minimum effective nuclear charge.

b. 2 points

An unknown anion (A^-) has the following *excited* electronic configuration: $[Ar] 4s^2 3d^3 4p^1$

(Ar: Argon; $Z = 18$)

Give the **condensed ground state electronic configuration** of the atom, **A** and the cation, **A^{2+}** .
Show your work. (Assume all electronic transitions are possible)

Electronic Configuration of A:

Electronic Configuration of A^{2+} :

c. 2 points

In a single atom, determine how many electrons can have the following set of quantum numbers?

(i) $n = 5; l = 3$

In a single atom, determine how many electrons can have the following set of quantum numbers?

(i) $n = 3; l = 2; m_s = +1/2$

**Question 4** **8 points**

Rank the following from 1 to 4 in **increasing** order. Provide the ranking in the boxes provided.

Note: Atomic number (Z) are provided for relevant atoms

a) Increasing (lowest to highest) order of *ionization energy*: (1= lowest; 4 = highest)

Mg

Mg: Z = 12

Rb

Rb: Z = 37

Cs

Cs: Z = 55

K

K : Z = 19

b) Increasing (lowest to highest) order of *bond angle*: (1= lowest; 4 = highest)

H₂OCCl₄XeF₄

HCN

c) Increasing (smallest to largest) order of *ionic size* (1= smallest; 4 = largest)

S²⁻

S: Z = 16

O²⁻

O: Z = 8

Be²⁺

Be: Z = 4

Na⁺

Na: Z = 11

d) Increasing (lowest to highest) order of *bond energy* (1= lowest; 4 = highest)

P-F

P-Cl

P-I

P-Br

**Question 5** **12 points**

a. 9 points For the molecules HClO_4 and TeF_6 , draw the most stable *Lewis structure* showing appropriate shape and *dashed/wedged bonds*, based on VSEPR. Determine the **molecular geometry** around the central atom. For HClO_4 , label the *most polar bond*.



b. 3 points (Note: Lone pairs are not shown in structures below)

(i) For the three *resonance* structures (1, 2 and 3) shown below, answer the following question:



(I) Circle the *most contributing* resonance structure.

Structure 1

Structure 2

Structure 3

(ii) For the three structures given (4, 5, and 6) below, answer the following questions:



(I) Circle all the structures that represent *resonance structures* of each other:

Structure 4

Structure 5

Structure 6

(II) Among the resonance structures, circle the *most contributing* resonance structure.

Structure 4

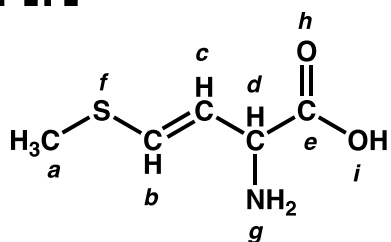
Structure 5

Structure 6

**Question 6** **6 points**

Using the data given below, calculate the **lattice energy** for SrO(s).

	ΔH° (kJ/mol)
$\text{Sr}(s) \longrightarrow \text{Sr}(g)$	+180
$\text{Sr}(g) \longrightarrow \text{Sr}^{2+}(g) + 2e^-$	+1468
$\text{O}_2(g) \longrightarrow 2\text{O}(g)$	+500
$\text{O}(g) + 2e^- \longrightarrow \text{O}^{2-}(g)$	+600
$\text{Sr}(s) + \frac{1}{2}\text{O}_2(g) \longrightarrow \text{SrO}(s)$	-550

**Question 7 10 points**

Consider the molecule given on the left, and answer questions (i) to (vi) using chemical bonding theories that we have learnt in class (valence bond theory, hybridization).

Note that the lone pairs are not shown in the structure:

i. Consider the carbon atoms marked in as ***a***, ***b***, ***c***, ***d***, and ***e*** in the molecule, and determine **hybridization** at each of the carbon centers. (2.5 points)

Atom	Hybridization
Carbon (<i>a</i>)	
Carbon (<i>b</i>)	
Carbon (<i>c</i>)	
Carbon (<i>d</i>)	
Carbon (<i>e</i>)	

ii. Now, consider the sulfur atom (S) marked as ***f***, nitrogen (N) atom (***g***), and oxygen(O) atoms (***h*** and ***i***), and determine their **hybridization**. (2 points)

Atom	Hybridization
Sulfur (<i>f</i>)	
Nitrogen (<i>g</i>)	
Oxygen (<i>h</i>)	
Oxygen (<i>i</i>)	

iii. **Describe** all the bonds formed by the carbon atoms ***a*** and ***b***, using sigma and pi designation as appropriate. (2 points)

Atom	Bonds formed (use sigma and pi as appropriate)
Carbon (<i>a</i>)	
Carbon (<i>b</i>)	

iv. How many total sp^3 -s sigma bonds are in the molecule? (1 point)

v. Determine **electronic** and **molecular geometry** at carbon (C) labeled ***e*** and sulfur (S) labeled ***f*** centres. (2 points)

Atom	Electronic Geometry	Molecular Geometry
Carbon (<i>e</i>)		
Sulfur (<i>f</i>)		

vi. Circle the value from the choices given below, that describes the **C-S-C bond angle at sulfur** (S) marked ***f***: (0.5 points)

>90° to <109.5° 109.5° >109.5 to <120° 120° >120° to <180°


Question 8 12 points

Using the **valence shell orbitals only** draw a **complete** molecular orbital (MO) diagram (with no s-p mixing) for the molecule **PS**, in the space provided below. **(6 points)**

Assume that 3s and 3p atomic orbitals will form molecular orbitals like those formed by the combination of 2s and 2p atomic orbitals that we learnt in class, and there is no s-p mixing.

Show **energy arrow**, **atomic orbitals** involved, **atom labels**, the **molecular orbitals** that are formed, **filled electrons** in the MO diagram etc. Make sure to label all the atomic and molecular orbitals.

Using the MO diagram above, answer the following questions:

- (i) Which one of the following will have the highest bond order? Circle the correct choice
 A) PS^{1-} B) PS C) PS^{1+}
- (ii) Which one of the following will have the longest bond length? Circle the correct choice
 A) PS^{1-} B) PS C) PS^{1+}
- (iii) Which of the following will be diamagnetic? Circle the correct choice.
 A) PS^{1-} B) PS C) PS^{1+}
- (iv) Draw Lewis dot structure of the molecule with the shortest bond length (based on the MO diagram) from among the three (PS^{1-} , PS, PS^{1+}), and indicate the bond order calculated from the Lewis dot structure. **(3 points)**

Lewis Dot Structure

Bond Order



E59E04F1-6480-40E6-B2EA-3EA740C4339E

midterm-exam-93388

#810 Page 10 of 10

SCRAP PAGE

**CHEMISTRY 110 –Midterm Exam****Thursday November 7, 2024, 6:30 – 8:30 pm**

Examiners: Pallavi Sirjoosingh and Ashok Kakkar

First name (please write as legibly as possible within the boxes)

--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--

Last name

--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--

Student ID number

--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--

This booklet has 8 questions on 10 pages (The last page can be used as a scrap page).

The staple must NOT be removed from the question booklet. Your intact question booklet must be returned at the end of the exam.

A separate booklet contains a periodic table, and datasheet.

Write your answers directly under the questions in the space provided. Do not write on the QR codes or near the top edges (as it may be cut off).

Please write **legibly and dark enough** for the scan to be readable. For students who use English as a second language, dictionaries are permitted. Calculators are allowed.

DO NOT ASK QUESTIONS OF THE INVIGILATORS about interpretations of the exam questions. If you have a concern about a question, please summarize it at the top of this page. All concerns will be evaluated after the exam. GOOD LUCK!

**Question 1 8 points** (*Assume all transitions are possible*)

a. 6 points Calculate the **energy (in kJ/mol)** required to ionize (completely remove) the electron from the **n=3** level of **He⁺ ion** (He: Helium). Show your work including equations. (3 sig. figs.)

b. 2 points

(I) Select the largest (longest) wavelength from the three choices below:

- i. Wavelength of photon to *completely remove* an electron from *1s orbital of Hydrogen atom*
- ii. Wavelength of photon to *completely remove* an electron from *1s orbital of Boron atom (B)*
- iii. Wavelength of photon to *completely remove* an electron from *1s orbital of Boron cation (B⁺)*

i or ii or iii

(II) Select the smallest (shortest) wavelength from the three choices below (*for H atom*):

- i. Wavelength of photon *emitted* for an electron transitioning from *2p to 1s orbital*
- ii. Wavelength of photon *absorbed* for an electron transitioning from *2s to 4p orbital*
- iii. Wavelength of photon *absorbed* for an electron transitioning from *1s to 3p orbital*

i or ii or iii



Question 2 8 points



A Chem110 student plots the graph shown on the left, from data obtained for the kinetic energy (eV) of the ejected electron versus frequency (s^{-1}) of the incident light, for an unknown metal. ($1\text{eV} = 1.6022 \times 10^{-19} \text{ J}$)

Based on the plot, answer the questions **a to c** below:

a. 1 point On the plot above, label the **threshold frequency**.

b. 2 points What is the value of the work potential (in J) of the unknown metal?

c. 3 points What is the *longest* wavelength (in nm) required to eject an electron from this metal's surface? Show your work including equation(s) used. (3 sig figs)

d. 2 points In the space provided below, plot the following

(Assume frequency is *above* threshold frequency):

- i. *Kinetic energy* of an ejected electron versus the *intensity* of the incident light (left)
- ii. *Number of electrons* ejected versus the *frequency* of the incident light (right)



**Question 3** **6 points****a. 2 points**

In the ground state of Ge (Germanium; $Z = 32$) atom, **name** the following orbitals:

i. Orbital containing the electron that requires the maximum energy to be removed.

ii. Orbital containing the electron that experiences the minimum effective nuclear charge.

b. 2 points

An unknown anion (A^-) has the following *excited* electronic configuration: $[Ar] 4s^2 3d^3 4p^1$

(Ar: Argon; $Z = 18$)

Give the **condensed ground state electronic configuration** of the atom, **A** and the cation, **A^{2+}** .
Show your work. (Assume all electronic transitions are possible)

Electronic Configuration of A:

Electronic Configuration of A^{2+} :

c. 2 points

In a single atom, determine how many electrons can have the following set of quantum numbers?

(i) $n = 5; l = 3$

In a single atom, determine how many electrons can have the following set of quantum numbers?

(i) $n = 3; l = 2; m_s = +1/2$

**Question 4** **8 points**

Rank the following from 1 to 4 in **increasing** order. Provide the ranking in the boxes provided.

Note: Atomic number (Z) are provided for relevant atoms

a) Increasing (lowest to highest) order of *ionization energy*: (1= lowest; 4 = highest)

Mg

Mg: Z = 12

Rb

Rb: Z = 37

Cs

Cs: Z = 55

K

K : Z = 19

b) Increasing (lowest to highest) order of *bond angle*: (1= lowest; 4 = highest)

H₂OCCl₄XeF₄

HCN

c) Increasing (smallest to largest) order of *ionic size* (1= smallest; 4 = largest)

S²⁻

S: Z = 16

O²⁻

O: Z = 8

Be²⁺

Be: Z = 4

Na⁺

Na: Z = 11

d) Increasing (lowest to highest) order of *bond energy* (1= lowest; 4 = highest)

P-F

P-Cl

P-I

P-Br

**Question 5** **12 points**

a. 9 points For the molecules HClO_4 and TeF_6 , draw the most stable *Lewis structure* showing appropriate shape and *dashed/wedged bonds*, based on VSEPR. Determine the **molecular geometry** around the central atom. For HClO_4 , label the *most polar bond*.



b. 3 points (Note: Lone pairs are not shown in structures below)

(i) For the three *resonance* structures (1, 2 and 3) shown below, answer the following question:



(I) Circle the *most contributing* resonance structure.

Structure 1

Structure 2

Structure 3

(ii) For the three structures given (4, 5, and 6) below, answer the following questions:



(I) Circle all the structures that represent *resonance structures* of each other:

Structure 4

Structure 5

Structure 6

(II) Among the resonance structures, circle the *most contributing* resonance structure.

Structure 4

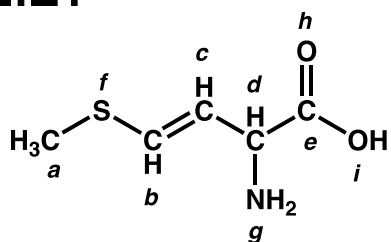
Structure 5

Structure 6

**Question 6** **6 points**

Using the data given below, calculate the **lattice energy** for SrO(s).

	ΔH° (kJ/mol)
$\text{Sr}(s) \longrightarrow \text{Sr}(g)$	+180
$\text{Sr}(g) \longrightarrow \text{Sr}^{2+}(g) + 2e^-$	+1468
$\text{O}_2(g) \longrightarrow 2\text{O}(g)$	+500
$\text{O}(g) + 2e^- \longrightarrow \text{O}^{2-}(g)$	+600
$\text{Sr}(s) + \frac{1}{2}\text{O}_2(g) \longrightarrow \text{SrO}(s)$	-550

**Question 7 10 points**

Consider the molecule given on the left, and answer questions (i) to (vi) using chemical bonding theories that we have learnt in class (valence bond theory, hybridization).

Note that the lone pairs are not shown in the structure:

i. Consider the carbon atoms marked in as ***a***, ***b***, ***c***, ***d***, and ***e*** in the molecule, and determine **hybridization** at each of the carbon centers. (2.5 points)

Atom	Hybridization
Carbon (<i>a</i>)	
Carbon (<i>b</i>)	
Carbon (<i>c</i>)	
Carbon (<i>d</i>)	
Carbon (<i>e</i>)	

ii. Now, consider the sulfur atom (S) marked as ***f***, nitrogen (N) atom (***g***), and oxygen(O) atoms (***h*** and ***i***), and determine their **hybridization**. (2 points)

Atom	Hybridization
Sulfur (<i>f</i>)	
Nitrogen (<i>g</i>)	
Oxygen (<i>h</i>)	
Oxygen (<i>i</i>)	

iii. **Describe** all the bonds formed by the carbon atoms ***a*** and ***b***, using sigma and pi designation as appropriate. (2 points)

Atom	Bonds formed (use sigma and pi as appropriate)
Carbon (<i>a</i>)	
Carbon (<i>b</i>)	

iv. How many total sp^3 -s sigma bonds are in the molecule? (1 point)

v. Determine **electronic** and **molecular geometry** at carbon (C) labeled ***e*** and sulfur (S) labeled ***f*** centres. (2 points)

Atom	Electronic Geometry	Molecular Geometry
Carbon (<i>e</i>)		
Sulfur (<i>f</i>)		

vi. Circle the value from the choices given below, that describes the **C-S-C bond angle at sulfur** (S) marked ***f***: (0.5 points)

>90° to <109.5°

109.5°

>109.5 to <120°

120°

>120° to <180°

**Question 8 12 points**

Using the **valence shell orbitals only** draw a **complete** molecular orbital (MO) diagram (with no s-p mixing) for the molecule **PS**, in the space provided below. **(6 points)**

Assume that 3s and 3p atomic orbitals will form molecular orbitals like those formed by the combination of 2s and 2p atomic orbitals that we learnt in class, and there is no s-p mixing.

Show **energy arrow**, **atomic orbitals** involved, **atom labels**, the **molecular orbitals** that are formed, **filled electrons** in the MO diagram etc. Make sure to label all the atomic and molecular orbitals.

Using the MO diagram above, answer the following questions:

- (i) Which one of the following will have the highest bond order? Circle the correct choice
A) PS^{1-} B) PS C) PS^{1+}
- (ii) Which one of the following will have the longest bond length? Circle the correct choice
A) PS^{1-} B) PS C) PS^{1+}
- (iii) Which of the following will be diamagnetic? Circle the correct choice.
A) PS^{1-} B) PS C) PS^{1+}
- (iv) Draw Lewis dot structure of the molecule with the shortest bond length (based on the MO diagram) from among the three (PS^{1-} , PS, PS^{1+}), and indicate the bond order calculated from the Lewis dot structure. **(3 points)**

Lewis Dot Structure

Bond Order



AB7FEC49-1F62-47D9-818A-91535E09F0FD

midterm-exam-93388

#811 Page 10 of 10

SCRAP PAGE

**CHEMISTRY 110 –Midterm Exam****Thursday November 7, 2024, 6:30 – 8:30 pm**

Examiners: Pallavi Sirjoosingh and Ashok Kakkar

First name (please write as legibly as possible within the boxes)

--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--

Last name

--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--

Student ID number

--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--

This booklet has 8 questions on 10 pages (The last page can be used as a scrap page).

The staple must NOT be removed from the question booklet. Your intact question booklet must be returned at the end of the exam.

A separate booklet contains a periodic table, and datasheet.

Write your answers directly under the questions in the space provided. Do not write on the QR codes or near the top edges (as it may be cut off).

Please write **legibly and dark enough** for the scan to be readable. For students who use English as a second language, dictionaries are permitted. Calculators are allowed.

DO NOT ASK QUESTIONS OF THE INVIGILATORS about interpretations of the exam questions. If you have a concern about a question, please summarize it at the top of this page. All concerns will be evaluated after the exam. GOOD LUCK!

**Question 1 8 points** (*Assume all transitions are possible*)

a. 6 points Calculate the **energy (in kJ/mol)** required to ionize (completely remove) the electron from the **$n=3$** level of **He^+ ion** (He: Helium). Show your work including equations. (3 sig. figs.)

b. 2 points

(I) Select the largest (longest) wavelength from the three choices below:

- i. Wavelength of photon to *completely remove* an electron from *1s orbital of Hydrogen atom*
- ii. Wavelength of photon to *completely remove* an electron from *1s orbital of Boron atom (B)*
- iii. Wavelength of photon to *completely remove* an electron from *1s orbital of Boron cation (B^+)*

i or ii or iii

(II) Select the smallest (shortest) wavelength from the three choices below (*for H atom*):

- i. Wavelength of photon *emitted* for an electron transitioning from *2p to 1s orbital*
- ii. Wavelength of photon *absorbed* for an electron transitioning from *2s to 4p orbital*
- iii. Wavelength of photon *absorbed* for an electron transitioning from *1s to 3p orbital*

i or ii or iii



Question 2 8 points



A Chem110 student plots the graph shown on the left, from data obtained for the kinetic energy (eV) of the ejected electron versus frequency (s^{-1}) of the incident light, for an unknown metal. ($1\text{eV} = 1.6022 \times 10^{-19} \text{ J}$)

Based on the plot, answer the questions **a to c** below:

a. 1 point On the plot above, label the **threshold frequency**.

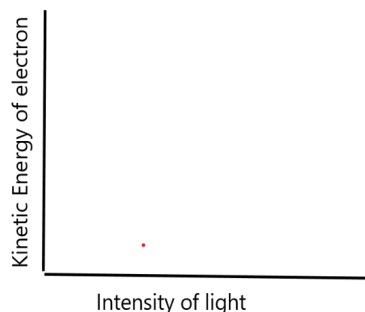
b. 2 points What is the value of the work potential (in J) of the unknown metal?

c. 3 points What is the *longest* wavelength (in nm) required to eject an electron from this metal's surface? Show your work including equation(s) used. (3 sig figs)

d. 2 points In the space provided below, plot the following

(Assume frequency is *above* threshold frequency):

- Kinetic energy* of an ejected electron versus the *intensity* of the incident light (left)
- Number of electrons* ejected versus the *frequency* of the incident light (right)



**Question 3** **6 points****a. 2 points**

In the ground state of Ge (Germanium; $Z = 32$) atom, **name** the following orbitals:

i. Orbital containing the electron that requires the maximum energy to be removed.

ii. Orbital containing the electron that experiences the minimum effective nuclear charge.

b. 2 points

An unknown anion (A^-) has the following *excited* electronic configuration: $[Ar] 4s^2 3d^3 4p^1$

(Ar: Argon; $Z = 18$)

Give the **condensed ground state electronic configuration** of the atom, **A** and the cation, **A^{2+}** .
Show your work. (Assume all electronic transitions are possible)

Electronic Configuration of A:

Electronic Configuration of A^{2+} :

c. 2 points

In a single atom, determine how many electrons can have the following set of quantum numbers?

(i) $n = 5; l = 3$

In a single atom, determine how many electrons can have the following set of quantum numbers?

(i) $n = 3; l = 2; m_s = +1/2$

**Question 4** **8 points**

Rank the following from 1 to 4 in **increasing** order. Provide the ranking in the boxes provided.

Note: Atomic number (Z) are provided for relevant atoms

a) Increasing (lowest to highest) order of *ionization energy*: (1= lowest; 4 = highest)

Mg

Mg: Z = 12

Rb

Rb: Z = 37

Cs

Cs: Z = 55

K

K : Z = 19

b) Increasing (lowest to highest) order of *bond angle*: (1= lowest; 4 = highest)

H₂OCCl₄XeF₄

HCN

c) Increasing (smallest to largest) order of *ionic size* (1= smallest; 4 = largest)

S²⁻

S: Z = 16

O²⁻

O: Z = 8

Be²⁺

Be: Z = 4

Na⁺

Na: Z = 11

d) Increasing (lowest to highest) order of *bond energy* (1= lowest; 4 = highest)

P-F

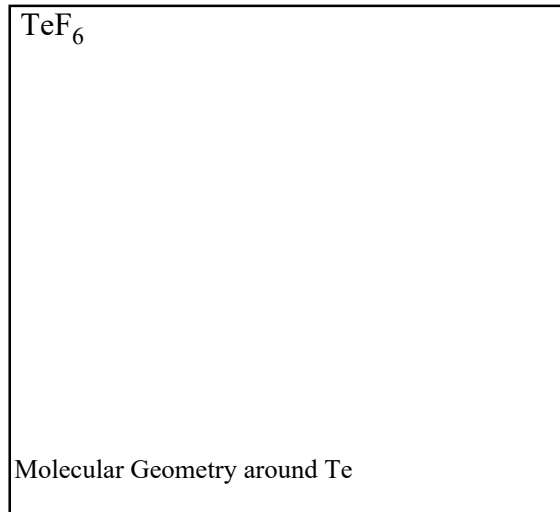
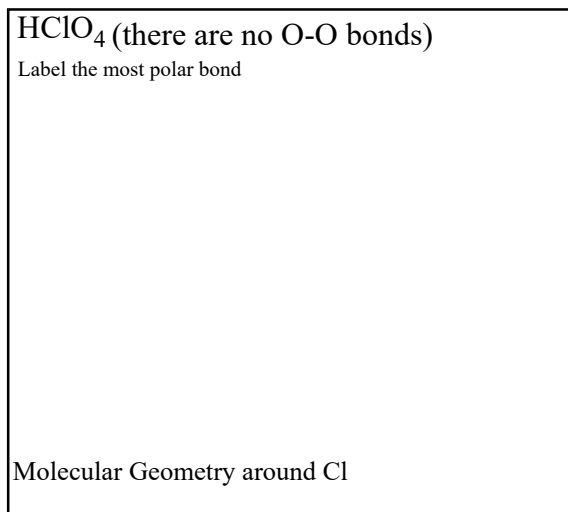
P-Cl

P-I

P-Br

**Question 5** **12 points**

a. 9 points For the molecules HClO_4 and TeF_6 , draw the most stable *Lewis structure* showing appropriate shape and *dashed/wedged bonds*, based on VSEPR. Determine the **molecular geometry** around the central atom. For HClO_4 , label the *most polar bond*.



b. 3 points (Note: Lone pairs are not shown in structures below)

(i) For the three *resonance* structures (1, 2 and 3) shown below, answer the following question:



(I) Circle the *most contributing* resonance structure.

Structure 1

Structure 2

Structure 3

(ii) For the three structures given (4, 5, and 6) below, answer the following questions:



(I) Circle all the structures that represent *resonance structures* of each other:

Structure 4

Structure 5

Structure 6

(II) Among the resonance structures, circle the *most contributing* resonance structure.

Structure 4

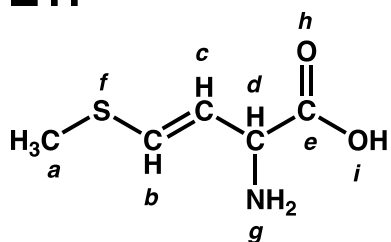
Structure 5

Structure 6

**Question 6** **6 points**

Using the data given below, calculate the **lattice energy** for SrO(s).

	ΔH° (kJ/mol)
$\text{Sr}(s) \longrightarrow \text{Sr}(g)$	+180
$\text{Sr}(g) \longrightarrow \text{Sr}^{2+}(g) + 2e^-$	+1468
$\text{O}_2(g) \longrightarrow 2\text{O}(g)$	+500
$\text{O}(g) + 2e^- \longrightarrow \text{O}^{2-}(g)$	+600
$\text{Sr}(s) + \frac{1}{2}\text{O}_2(g) \longrightarrow \text{SrO}(s)$	-550

**Question 7 10 points**

Consider the molecule given on the left, and answer questions (i) to (vi) using chemical bonding theories that we have learnt in class (valence bond theory, hybridization).

Note that the lone pairs are not shown in the structure:

i. Consider the carbon atoms marked in as ***a***, ***b***, ***c***, ***d***, and ***e*** in the molecule, and determine **hybridization** at each of the carbon centers. (2.5 points)

Atom	Hybridization
Carbon (<i>a</i>)	
Carbon (<i>b</i>)	
Carbon (<i>c</i>)	
Carbon (<i>d</i>)	
Carbon (<i>e</i>)	

ii. Now, consider the sulfur atom (S) marked as ***f***, nitrogen (N) atom (***g***), and oxygen(O) atoms (***h*** and ***i***), and determine their **hybridization**. (2 points)

Atom	Hybridization
Sulfur (<i>f</i>)	
Nitrogen (<i>g</i>)	
Oxygen (<i>h</i>)	
Oxygen (<i>i</i>)	

iii. **Describe** all the bonds formed by the carbon atoms ***a*** and ***b***, using sigma and pi designation as appropriate. (2 points)

Atom	Bonds formed (use sigma and pi as appropriate)
Carbon (<i>a</i>)	
Carbon (<i>b</i>)	

iv. How many total sp^3 -s sigma bonds are in the molecule? (1 point)

v. Determine **electronic** and **molecular geometry** at carbon (C) labeled ***e*** and sulfur (S) labeled ***f*** centres. (2 points)

Atom	Electronic Geometry	Molecular Geometry
Carbon (<i>e</i>)		
Sulfur (<i>f</i>)		

vi. Circle the value from the choices given below, that describes the **C-S-C bond angle at sulfur** (S) marked ***f***: (0.5 points)

>90° to <109.5° 109.5° >109.5 to <120° 120° >120° to <180°


Question 8 12 points

Using the **valence shell orbitals only** draw a **complete** molecular orbital (MO) diagram (with no s-p mixing) for the molecule **PS**, in the space provided below. **(6 points)**

Assume that 3s and 3p atomic orbitals will form molecular orbitals like those formed by the combination of 2s and 2p atomic orbitals that we learnt in class, and there is no s-p mixing.

Show **energy arrow**, **atomic orbitals** involved, **atom labels**, the **molecular orbitals** that are formed, **filled electrons** in the MO diagram etc. Make sure to label all the atomic and molecular orbitals.

Using the MO diagram above, answer the following questions:

- (i) Which one of the following will have the highest bond order? Circle the correct choice
 A) PS^{1-} B) PS C) PS^{1+}
- (ii) Which one of the following will have the longest bond length? Circle the correct choice
 A) PS^{1-} B) PS C) PS^{1+}
- (iii) Which of the following will be diamagnetic? Circle the correct choice.
 A) PS^{1-} B) PS C) PS^{1+}
- (iv) Draw Lewis dot structure of the molecule with the shortest bond length (based on the MO diagram) from among the three (PS^{1-} , PS, PS^{1+}), and indicate the bond order calculated from the Lewis dot structure. **(3 points)**

Lewis Dot Structure

Bond Order



F375D81C-11DF-4553-8B57-D39AB917AA78

midterm-exam-93388

#812 Page 10 of 10

SCRAP PAGE

**CHEMISTRY 110 –Midterm Exam****Thursday November 7, 2024, 6:30 – 8:30 pm**

Examiners: Pallavi Sirjoosingh and Ashok Kakkar

First name (please write as legibly as possible within the boxes)

--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--

Last name

--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--

Student ID number

--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--

This booklet has 8 questions on 10 pages (The last page can be used as a scrap page).

The staple must NOT be removed from the question booklet. Your intact question booklet must be returned at the end of the exam.

A separate booklet contains a periodic table, and datasheet.

Write your answers directly under the questions in the space provided. Do not write on the QR codes or near the top edges (as it may be cut off).

Please write **legibly and dark enough** for the scan to be readable. For students who use English as a second language, dictionaries are permitted. Calculators are allowed.

DO NOT ASK QUESTIONS OF THE INVIGILATORS about interpretations of the exam questions. If you have a concern about a question, please summarize it at the top of this page. All concerns will be evaluated after the exam. GOOD LUCK!

**Question 1 8 points** (*Assume all transitions are possible*)

a. 6 points Calculate the **energy (in kJ/mol)** required to ionize (completely remove) the electron from the **$n=3$** level of **He^+ ion** (He: Helium). Show your work including equations. (3 sig. figs.)

b. 2 points

(I) Select the largest (longest) wavelength from the three choices below:

- i. Wavelength of photon to *completely remove* an electron from *1s orbital of Hydrogen atom*
- ii. Wavelength of photon to *completely remove* an electron from *1s orbital of Boron atom (B)*
- iii. Wavelength of photon to *completely remove* an electron from *1s orbital of Boron cation (B^+)*

i or ii or iii

(II) Select the smallest (shortest) wavelength from the three choices below (*for H atom*):

- i. Wavelength of photon *emitted* for an electron transitioning from *2p to 1s orbital*
- ii. Wavelength of photon *absorbed* for an electron transitioning from *2s to 4p orbital*
- iii. Wavelength of photon *absorbed* for an electron transitioning from *1s to 3p orbital*

i or ii or iii



Question 2 8 points



A Chem110 student plots the graph shown on the left, from data obtained for the kinetic energy (eV) of the ejected electron versus frequency (s^{-1}) of the incident light, for an unknown metal. ($1\text{eV} = 1.6022 \times 10^{-19} \text{ J}$)

Based on the plot, answer the questions **a to c** below:

a. 1 point On the plot above, label the **threshold frequency**.

b. 2 points What is the value of the work potential (in J) of the unknown metal?

c. 3 points What is the *longest* wavelength (in nm) required to eject an electron from this metal's surface? Show your work including equation(s) used. (3 sig figs)

d. 2 points In the space provided below, plot the following

(Assume frequency is *above* threshold frequency):

- i. *Kinetic energy* of an ejected electron versus the *intensity* of the incident light (left)
- ii. *Number of electrons* ejected versus the *frequency* of the incident light (right)



**Question 3** **6 points****a. 2 points**

In the ground state of Ge (Germanium; $Z = 32$) atom, **name** the following orbitals:

i. Orbital containing the electron that requires the maximum energy to be removed.

ii. Orbital containing the electron that experiences the minimum effective nuclear charge.

b. 2 points

An unknown anion (A^-) has the following *excited* electronic configuration: $[Ar] 4s^2 3d^3 4p^1$

(Ar: Argon; $Z = 18$)

Give the **condensed ground state electronic configuration** of the atom, **A** and the cation, **A^{2+}** .
Show your work. (Assume all electronic transitions are possible)

Electronic Configuration of A:

Electronic Configuration of A^{2+} :

c. 2 points

In a single atom, determine how many electrons can have the following set of quantum numbers?

(i) $n = 5; l = 3$

In a single atom, determine how many electrons can have the following set of quantum numbers?

(i) $n = 3; l = 2; m_s = +1/2$

**Question 4** **8 points**

Rank the following from 1 to 4 in **increasing** order. Provide the ranking in the boxes provided.

Note: Atomic number (Z) are provided for relevant atoms

a) Increasing (lowest to highest) order of *ionization energy*: (1= lowest; 4 = highest)

Mg

Mg: Z = 12

Rb

Rb: Z = 37

Cs

Cs: Z = 55

K

K : Z = 19

b) Increasing (lowest to highest) order of *bond angle*: (1= lowest; 4 = highest)

H₂OCCl₄XeF₄

HCN

c) Increasing (smallest to largest) order of *ionic size* (1= smallest; 4 = largest)

S²⁻

S: Z = 16

O²⁻

O: Z = 8

Be²⁺

Be: Z = 4

Na⁺

Na: Z = 11

d) Increasing (lowest to highest) order of *bond energy* (1= lowest; 4 = highest)

P-F

P-Cl

P-I

P-Br

**Question 5** **12 points**

a. 9 points For the molecules HClO_4 and TeF_6 , draw the most stable *Lewis structure* showing appropriate shape and *dashed/wedged bonds*, based on VSEPR. Determine the **molecular geometry** around the central atom. For HClO_4 , label the *most polar bond*.



b. 3 points (Note: Lone pairs are not shown in structures below)

(i) For the three *resonance* structures (1, 2 and 3) shown below, answer the following question:



(I) Circle the *most contributing* resonance structure.

Structure 1

Structure 2

Structure 3

(ii) For the three structures given (4, 5, and 6) below, answer the following questions:



(I) Circle all the structures that represent *resonance structures* of each other:

Structure 4

Structure 5

Structure 6

(II) Among the resonance structures, circle the *most contributing* resonance structure.

Structure 4

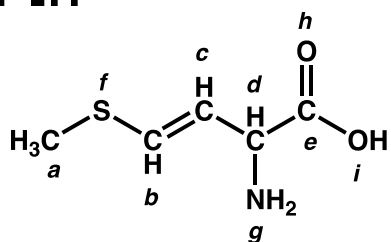
Structure 5

Structure 6

**Question 6** **6 points**

Using the data given below, calculate the **lattice energy** for SrO(s).

	ΔH° (kJ/mol)
$\text{Sr}(s) \longrightarrow \text{Sr}(g)$	+180
$\text{Sr}(g) \longrightarrow \text{Sr}^{2+}(g) + 2e^-$	+1468
$\text{O}_2(g) \longrightarrow 2\text{O}(g)$	+500
$\text{O}(g) + 2e^- \longrightarrow \text{O}^{2-}(g)$	+600
$\text{Sr}(s) + \frac{1}{2}\text{O}_2(g) \longrightarrow \text{SrO}(s)$	-550

**Question 7 10 points**

Consider the molecule given on the left, and answer questions (i) to (vi) using chemical bonding theories that we have learnt in class (valence bond theory, hybridization).

Note that the lone pairs are not shown in the structure:

i. Consider the carbon atoms marked in as ***a***, ***b***, ***c***, ***d***, and ***e*** in the molecule, and determine **hybridization** at each of the carbon centers. (2.5 points)

Atom	Hybridization
Carbon (<i>a</i>)	
Carbon (<i>b</i>)	
Carbon (<i>c</i>)	
Carbon (<i>d</i>)	
Carbon (<i>e</i>)	

ii. Now, consider the sulfur atom (S) marked as ***f***, nitrogen (N) atom (***g***), and oxygen(O) atoms (***h*** and ***i***), and determine their **hybridization**. (2 points)

Atom	Hybridization
Sulfur (<i>f</i>)	
Nitrogen (<i>g</i>)	
Oxygen (<i>h</i>)	
Oxygen (<i>i</i>)	

iii. **Describe** all the bonds formed by the carbon atoms ***a*** and ***b***, using sigma and pi designation as appropriate. (2 points)

Atom	Bonds formed (use sigma and pi as appropriate)
Carbon (<i>a</i>)	
Carbon (<i>b</i>)	

iv. How many total sp^3 -s sigma bonds are in the molecule? (1 point)

v. Determine **electronic** and **molecular geometry** at carbon (C) labeled ***e*** and sulfur (S) labeled ***f*** centres. (2 points)

Atom	Electronic Geometry	Molecular Geometry
Carbon (<i>e</i>)		
Sulfur (<i>f</i>)		

vi. Circle the value from the choices given below, that describes the **C-S-C bond angle at sulfur** (S) marked ***f***: (0.5 points)

>90° to <109.5° 109.5° >109.5 to <120° 120° >120° to <180°


Question 8 12 points

Using the **valence shell orbitals only** draw a **complete** molecular orbital (MO) diagram (with no s-p mixing) for the molecule **PS**, in the space provided below. **(6 points)**

Assume that 3s and 3p atomic orbitals will form molecular orbitals like those formed by the combination of 2s and 2p atomic orbitals that we learnt in class, and there is no s-p mixing.

Show **energy arrow**, **atomic orbitals** involved, **atom labels**, the **molecular orbitals** that are formed, **filled electrons** in the MO diagram etc. Make sure to label all the atomic and molecular orbitals.

Using the MO diagram above, answer the following questions:

- (i) Which one of the following will have the highest bond order? Circle the correct choice
 A) PS^{1-} B) PS C) PS^{1+}
- (ii) Which one of the following will have the longest bond length? Circle the correct choice
 A) PS^{1-} B) PS C) PS^{1+}
- (iii) Which of the following will be diamagnetic? Circle the correct choice.
 A) PS^{1-} B) PS C) PS^{1+}
- (iv) Draw Lewis dot structure of the molecule with the shortest bond length (based on the MO diagram) from among the three (PS^{1-} , PS, PS^{1+}), and indicate the bond order calculated from the Lewis dot structure. **(3 points)**

Lewis Dot Structure

Bond Order



52557F3A-83AA-47C4-9E5C-A2C0B29129E7

midterm-exam-93388

#813 Page 10 of 10

SCRAP PAGE

**CHEMISTRY 110 –Midterm Exam****Thursday November 7, 2024, 6:30 – 8:30 pm****Examiners: Pallavi Sirjoosingh and Ashok Kakkar**

First name (please write as legibly as possible within the boxes)

--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--

Last name

--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--

Student ID number

--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--

This booklet has 8 questions on 10 pages (The last page can be used as a scrap page).

The staple must NOT be removed from the question booklet. Your intact question booklet must be returned at the end of the exam.

A separate booklet contains a periodic table, and datasheet.

Write your answers directly under the questions in the space provided. Do not write on the QR codes or near the top edges (as it may be cut off).

Please write **legibly and dark enough** for the scan to be readable. For students who use English as a second language, dictionaries are permitted. Calculators are allowed.

DO NOT ASK QUESTIONS OF THE INVIGILATORS about interpretations of the exam questions. If you have a concern about a question, please summarize it at the top of this page. All concerns will be evaluated after the exam. GOOD LUCK!

**Question 1 8 points** (*Assume all transitions are possible*)

a. 6 points Calculate the **energy (in kJ/mol)** required to ionize (completely remove) the electron from the **n=3** level of **He⁺ ion** (He: Helium). Show your work including equations. (3 sig. figs.)

b. 2 points

(I) Select the largest (longest) wavelength from the three choices below:

- i. Wavelength of photon to *completely remove* an electron from *1s orbital of Hydrogen atom*
- ii. Wavelength of photon to *completely remove* an electron from *1s orbital of Boron atom (B)*
- iii. Wavelength of photon to *completely remove* an electron from *1s orbital of Boron cation (B⁺)*

i or ii or iii

(II) Select the smallest (shortest) wavelength from the three choices below (*for H atom*):

- i. Wavelength of photon *emitted* for an electron transitioning from *2p to 1s orbital*
- ii. Wavelength of photon *absorbed* for an electron transitioning from *2s to 4p orbital*
- iii. Wavelength of photon *absorbed* for an electron transitioning from *1s to 3p orbital*

i or ii or iii



Question 2 8 points



A Chem110 student plots the graph shown on the left, from data obtained for the kinetic energy (eV) of the ejected electron versus frequency (s^{-1}) of the incident light, for an unknown metal. ($1\text{eV} = 1.6022 \times 10^{-19} \text{ J}$)

Based on the plot, answer the questions **a to c** below:

a. 1 point On the plot above, label the **threshold frequency**.

b. 2 points What is the value of the work potential (in J) of the unknown metal?

c. 3 points What is the *longest* wavelength (in nm) required to eject an electron from this metal's surface? Show your work including equation(s) used. (3 sig figs)

d. 2 points In the space provided below, plot the following

(Assume frequency is *above* threshold frequency):

- Kinetic energy* of an ejected electron versus the *intensity* of the incident light (left)
- Number of electrons* ejected versus the *frequency* of the incident light (right)



**Question 3** **6 points****a. 2 points**

In the ground state of Ge (Germanium; $Z = 32$) atom, **name** the following orbitals:

i. Orbital containing the electron that requires the maximum energy to be removed.

ii. Orbital containing the electron that experiences the minimum effective nuclear charge.

b. 2 points

An unknown anion (A^-) has the following *excited* electronic configuration: $[Ar] 4s^2 3d^3 4p^1$

(Ar: Argon; $Z = 18$)

Give the **condensed ground state electronic configuration** of the atom, **A** and the cation, **A^{2+}** .
Show your work. (Assume all electronic transitions are possible)

Electronic Configuration of A:

Electronic Configuration of A^{2+} :

c. 2 points

In a single atom, determine how many electrons can have the following set of quantum numbers?

(i) $n = 5; l = 3$

In a single atom, determine how many electrons can have the following set of quantum numbers?

(i) $n = 3; l = 2; m_s = +1/2$

**Question 4** **8 points**

Rank the following from 1 to 4 in **increasing** order. Provide the ranking in the boxes provided.

Note: Atomic number (Z) are provided for relevant atoms

a) Increasing (lowest to highest) order of *ionization energy*: (1= lowest; 4 = highest)

Mg

Mg: Z = 12

Rb

Rb: Z = 37

Cs

Cs: Z = 55

K

K : Z = 19

b) Increasing (lowest to highest) order of *bond angle*: (1= lowest; 4 = highest)

H₂OCCl₄XeF₄

HCN

c) Increasing (smallest to largest) order of *ionic size* (1= smallest; 4 = largest)

S²⁻

S: Z = 16

O²⁻

O: Z = 8

Be²⁺

Be: Z = 4

Na⁺

Na: Z = 11

d) Increasing (lowest to highest) order of *bond energy* (1= lowest; 4 = highest)

P-F

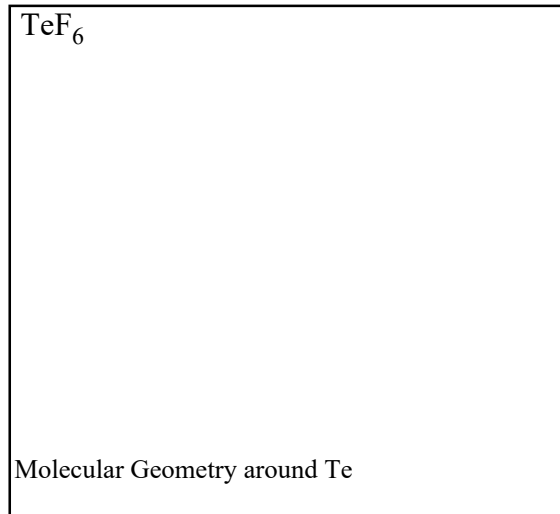
P-Cl

P-I

P-Br

**Question 5** **12 points**

a. 9 points For the molecules HClO_4 and TeF_6 , draw the most stable *Lewis structure* showing appropriate shape and *dashed/wedged bonds*, based on VSEPR. Determine the **molecular geometry** around the central atom. For HClO_4 , label the *most polar bond*.



b. 3 points (Note: Lone pairs are not shown in structures below)

(i) For the three *resonance* structures (1, 2 and 3) shown below, answer the following question:



(I) Circle the *most contributing* resonance structure.

Structure 1

Structure 2

Structure 3

(ii) For the three structures given (4, 5, and 6) below, answer the following questions:



(I) Circle all the structures that represent *resonance structures* of each other:

Structure 4

Structure 5

Structure 6

(II) Among the resonance structures, circle the *most contributing* resonance structure.

Structure 4

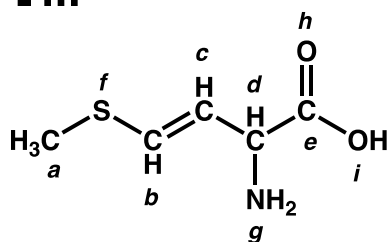
Structure 5

Structure 6

**Question 6** **6 points**

Using the data given below, calculate the **lattice energy** for SrO(s).

	ΔH° (kJ/mol)
$\text{Sr}(s) \longrightarrow \text{Sr}(g)$	+180
$\text{Sr}(g) \longrightarrow \text{Sr}^{2+}(g) + 2e^-$	+1468
$\text{O}_2(g) \longrightarrow 2\text{O}(g)$	+500
$\text{O}(g) + 2e^- \longrightarrow \text{O}^{2-}(g)$	+600
$\text{Sr}(s) + \frac{1}{2}\text{O}_2(g) \longrightarrow \text{SrO}(s)$	-550

**Question 7 10 points**

Consider the molecule given on the left, and answer questions (i) to (vi) using chemical bonding theories that we have learnt in class (valence bond theory, hybridization).

Note that the lone pairs are not shown in the structure:

i. Consider the carbon atoms marked in as ***a***, ***b***, ***c***, ***d***, and ***e*** in the molecule, and determine **hybridization** at each of the carbon centers. (2.5 points)

Atom	Hybridization
Carbon (<i>a</i>)	
Carbon (<i>b</i>)	
Carbon (<i>c</i>)	
Carbon (<i>d</i>)	
Carbon (<i>e</i>)	

ii. Now, consider the sulfur atom (S) marked as ***f***, nitrogen (N) atom (***g***), and oxygen(O) atoms (***h*** and ***i***), and determine their **hybridization**. (2 points)

Atom	Hybridization
Sulfur (<i>f</i>)	
Nitrogen (<i>g</i>)	
Oxygen (<i>h</i>)	
Oxygen (<i>i</i>)	

iii. **Describe** all the bonds formed by the carbon atoms ***a*** and ***b***, using sigma and pi designation as appropriate. (2 points)

Atom	Bonds formed (use sigma and pi as appropriate)
Carbon (<i>a</i>)	
Carbon (<i>b</i>)	

iv. How many total sp^3 -s sigma bonds are in the molecule? (1 point)

v. Determine **electronic** and **molecular geometry** at carbon (C) labeled ***e*** and sulfur (S) labeled ***f*** centres. (2 points)

Atom	Electronic Geometry	Molecular Geometry
Carbon (<i>e</i>)		
Sulfur (<i>f</i>)		

vi. Circle the value from the choices given below, that describes the **C-S-C bond angle at sulfur** (S) marked ***f***: (0.5 points)

$>90^\circ$ to $<109.5^\circ$ 109.5° >109.5 to $<120^\circ$ 120° $>120^\circ$ to $<180^\circ$

**Question 8 12 points**

Using the **valence shell orbitals only** draw a **complete** molecular orbital (MO) diagram (with no s-p mixing) for the molecule **PS**, in the space provided below. **(6 points)**

Assume that 3s and 3p atomic orbitals will form molecular orbitals like those formed by the combination of 2s and 2p atomic orbitals that we learnt in class, and there is no s-p mixing.

Show **energy arrow**, **atomic orbitals** involved, **atom labels**, the **molecular orbitals** that are formed, **filled electrons** in the MO diagram etc. Make sure to label all the atomic and molecular orbitals.

Using the MO diagram above, answer the following questions:

- (i) Which one of the following will have the highest bond order? Circle the correct choice
A) PS^{1-} B) PS C) PS^{1+}
- (ii) Which one of the following will have the longest bond length? Circle the correct choice
A) PS^{1-} B) PS C) PS^{1+}
- (iii) Which of the following will be diamagnetic? Circle the correct choice.
A) PS^{1-} B) PS C) PS^{1+}
- (iv) Draw Lewis dot structure of the molecule with the shortest bond length (based on the MO diagram) from among the three (PS^{1-} , PS, PS^{1+}), and indicate the bond order calculated from the Lewis dot structure. **(3 points)**

Lewis Dot Structure

Bond Order



AEE8AA3B-EFAB-4B3F-9BD7-B5DA08742796

midterm-exam-93388

#814 Page 10 of 10

SCRAP PAGE

**CHEMISTRY 110 –Midterm Exam****Thursday November 7, 2024, 6:30 – 8:30 pm**

Examiners: Pallavi Sirjoosingh and Ashok Kakkar

First name (please write as legibly as possible within the boxes)

--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--

Last name

--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--

Student ID number

--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--

This booklet has 8 questions on 10 pages (The last page can be used as a scrap page).

The staple must NOT be removed from the question booklet. Your intact question booklet must be returned at the end of the exam.

A separate booklet contains a periodic table, and datasheet.

Write your answers directly under the questions in the space provided. Do not write on the QR codes or near the top edges (as it may be cut off).

Please write **legibly and dark enough** for the scan to be readable. For students who use English as a second language, dictionaries are permitted. Calculators are allowed.

DO NOT ASK QUESTIONS OF THE INVIGILATORS about interpretations of the exam questions. If you have a concern about a question, please summarize it at the top of this page. All concerns will be evaluated after the exam. GOOD LUCK!

**Question 1 8 points** (*Assume all transitions are possible*)

a. 6 points Calculate the **energy (in kJ/mol)** required to ionize (completely remove) the electron from the **n=3** level of **He⁺ ion** (He: Helium). Show your work including equations. (3 sig. figs.)

b. 2 points

(I) Select the largest (longest) wavelength from the three choices below:

- i. Wavelength of photon to *completely remove* an electron from *1s orbital of Hydrogen atom*
- ii. Wavelength of photon to *completely remove* an electron from *1s orbital of Boron atom (B)*
- iii. Wavelength of photon to *completely remove* an electron from *1s orbital of Boron cation (B⁺)*

i or ii or iii

(II) Select the smallest (shortest) wavelength from the three choices below (*for H atom*):

- i. Wavelength of photon *emitted* for an electron transitioning from *2p to 1s orbital*
- ii. Wavelength of photon *absorbed* for an electron transitioning from *2s to 4p orbital*
- iii. Wavelength of photon *absorbed* for an electron transitioning from *1s to 3p orbital*

i or ii or iii



Question 2 8 points



A Chem110 student plots the graph shown on the left, from data obtained for the kinetic energy (eV) of the ejected electron versus frequency (s^{-1}) of the incident light, for an unknown metal. ($1\text{eV} = 1.6022 \times 10^{-19} \text{ J}$)

Based on the plot, answer the questions **a to c** below:

a. 1 point On the plot above, label the **threshold frequency**.

b. 2 points What is the value of the work potential (in J) of the unknown metal?

c. 3 points What is the *longest* wavelength (in nm) required to eject an electron from this metal's surface? Show your work including equation(s) used. (3 sig figs)

d. 2 points In the space provided below, plot the following

(Assume frequency is *above* threshold frequency):

- i. *Kinetic energy* of an ejected electron versus the *intensity* of the incident light (left)
- ii. *Number of electrons* ejected versus the *frequency* of the incident light (right)



**Question 3** **6 points****a. 2 points**

In the ground state of Ge (Germanium; $Z = 32$) atom, **name** the following orbitals:

i. Orbital containing the electron that requires the maximum energy to be removed.

ii. Orbital containing the electron that experiences the minimum effective nuclear charge.

b. 2 points

An unknown anion (A^-) has the following *excited* electronic configuration: $[Ar] 4s^2 3d^3 4p^1$

(Ar: Argon; $Z = 18$)

Give the **condensed ground state electronic configuration** of the atom, **A** and the cation, **A^{2+}** .
Show your work. (Assume all electronic transitions are possible)

Electronic Configuration of A:

Electronic Configuration of A^{2+} :

c. 2 points

In a single atom, determine how many electrons can have the following set of quantum numbers?

(i) $n = 5; l = 3$

In a single atom, determine how many electrons can have the following set of quantum numbers?

(i) $n = 3; l = 2; m_s = +1/2$

**Question 4** **8 points**

Rank the following from 1 to 4 in **increasing** order. Provide the ranking in the boxes provided.

Note: Atomic number (Z) are provided for relevant atoms

a) Increasing (lowest to highest) order of *ionization energy*: (1= lowest; 4 = highest)

Mg

Mg: Z = 12

Rb

Rb: Z = 37

Cs

Cs: Z = 55

K

K : Z = 19

b) Increasing (lowest to highest) order of *bond angle*: (1= lowest; 4 = highest)

H₂OCCl₄XeF₄

HCN

c) Increasing (smallest to largest) order of *ionic size* (1= smallest; 4 = largest)

S²⁻

S: Z = 16

O²⁻

O: Z = 8

Be²⁺

Be: Z = 4

Na⁺

Na: Z = 11

d) Increasing (lowest to highest) order of *bond energy* (1= lowest; 4 = highest)

P-F

P-Cl

P-I

P-Br

**Question 5** **12 points**

a. 9 points For the molecules HClO_4 and TeF_6 , draw the most stable *Lewis structure* showing appropriate shape and *dashed/wedged bonds*, based on VSEPR. Determine the **molecular geometry** around the central atom. For HClO_4 , label the *most polar bond*.

HClO_4 (there are no O-O bonds)

Label the most polar bond

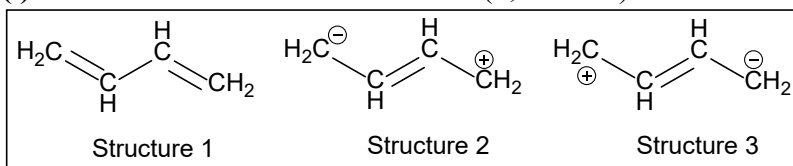
Molecular Geometry around Cl

TeF_6

Molecular Geometry around Te

b. 3 points (Note: Lone pairs are not shown in structures below)

(i) For the three *resonance* structures (1, 2 and 3) shown below, answer the following question:



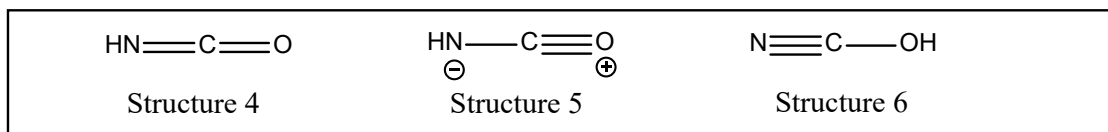
(I) Circle the *most contributing* resonance structure.

Structure 1

Structure 2

Structure 3

(ii) For the three structures given (4, 5, and 6) below, answer the following questions:



(I) Circle all the structures that represent *resonance structures* of each other:

Structure 4

Structure 5

Structure 6

(II) Among the resonance structures, circle the *most contributing* resonance structure.

Structure 4

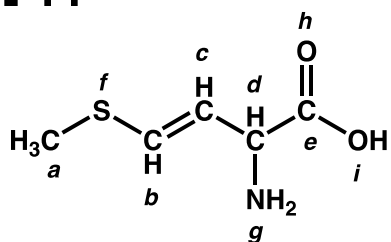
Structure 5

Structure 6

**Question 6** **6 points**

Using the data given below, calculate the **lattice energy** for SrO(s).

	ΔH° (kJ/mol)
$\text{Sr}(s) \longrightarrow \text{Sr}(g)$	+180
$\text{Sr}(g) \longrightarrow \text{Sr}^{2+}(g) + 2e^-$	+1468
$\text{O}_2(g) \longrightarrow 2\text{O}(g)$	+500
$\text{O}(g) + 2e^- \longrightarrow \text{O}^{2-}(g)$	+600
$\text{Sr}(s) + \frac{1}{2}\text{O}_2(g) \longrightarrow \text{SrO}(s)$	-550

**Question 7 10 points**

Consider the molecule given on the left, and answer questions (i) to (vi) using chemical bonding theories that we have learnt in class (valence bond theory, hybridization).

Note that the lone pairs are not shown in the structure:

i. Consider the carbon atoms marked in as ***a***, ***b***, ***c***, ***d***, and ***e*** in the molecule, and determine **hybridization** at each of the carbon centers. (2.5 points)

Atom	Hybridization
Carbon (<i>a</i>)	
Carbon (<i>b</i>)	
Carbon (<i>c</i>)	
Carbon (<i>d</i>)	
Carbon (<i>e</i>)	

ii. Now, consider the sulfur atom (S) marked as ***f***, nitrogen (N) atom (***g***), and oxygen(O) atoms (***h*** and ***i***), and determine their **hybridization**. (2 points)

Atom	Hybridization
Sulfur (<i>f</i>)	
Nitrogen (<i>g</i>)	
Oxygen (<i>h</i>)	
Oxygen (<i>i</i>)	

iii. **Describe** all the bonds formed by the carbon atoms ***a*** and ***b***, using sigma and pi designation as appropriate. (2 points)

Atom	Bonds formed (use sigma and pi as appropriate)
Carbon (<i>a</i>)	
Carbon (<i>b</i>)	

iv. How many total sp^3 -s sigma bonds are in the molecule? (1 point)

v. Determine **electronic** and **molecular geometry** at carbon (C) labeled ***e*** and sulfur (S) labeled ***f*** centres. (2 points)

Atom	Electronic Geometry	Molecular Geometry
Carbon (<i>e</i>)		
Sulfur (<i>f</i>)		

vi. Circle the value from the choices given below, that describes the **C-S-C bond angle at sulfur** (S) marked ***f***: (0.5 points)

>90° to <109.5° 109.5° >109.5 to <120° 120° >120° to <180°



Question 8 12 points

Using the **valence shell orbitals only** draw a **complete** molecular orbital (MO) diagram (with no s-p mixing) for the molecule **PS**, in the space provided below. **(6 points)**

Assume that 3s and 3p atomic orbitals will form molecular orbitals like those formed by the combination of 2s and 2p atomic orbitals that we learnt in class, and there is no s-p mixing.

Show **energy arrow**, **atomic orbitals** involved, **atom labels**, the **molecular orbitals** that are formed, **filled electrons** in the MO diagram etc. Make sure to label all the atomic and molecular orbitals.

Using the MO diagram above, answer the following questions:

- (i) Which one of the following will have the highest bond order? Circle the correct choice
 A) PS^{1-} B) PS C) PS^{1+}
- (ii) Which one of the following will have the longest bond length? Circle the correct choice
 A) PS^{1-} B) PS C) PS^{1+}
- (iii) Which of the following will be diamagnetic? Circle the correct choice.
 A) PS^{1-} B) PS C) PS^{1+}
- (iv) Draw Lewis dot structure of the molecule with the shortest bond length (based on the MO diagram) from among the three (PS^{1-} , PS, PS^{1+}), and indicate the bond order calculated from the Lewis dot structure. **(3 points)**

Lewis Dot Structure

Bond Order



A05E41FB-CDAC-46D4-A77B-163BAEB84004

midterm-exam-93388

#815 Page 10 of 10

SCRAP PAGE

**CHEMISTRY 110 –Midterm Exam****Thursday November 7, 2024, 6:30 – 8:30 pm**

Examiners: Pallavi Sirjoosingh and Ashok Kakkar

First name (please write as legibly as possible within the boxes)

--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--

Last name

--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--

Student ID number

--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--

This booklet has 8 questions on 10 pages (The last page can be used as a scrap page).

The staple must NOT be removed from the question booklet. Your intact question booklet must be returned at the end of the exam.

A separate booklet contains a periodic table, and datasheet.

Write your answers directly under the questions in the space provided. Do not write on the QR codes or near the top edges (as it may be cut off).

Please write **legibly and dark enough** for the scan to be readable. For students who use English as a second language, dictionaries are permitted. Calculators are allowed.

DO NOT ASK QUESTIONS OF THE INVIGILATORS about interpretations of the exam questions. If you have a concern about a question, please summarize it at the top of this page. All concerns will be evaluated after the exam. GOOD LUCK!

**Question 1 8 points** (*Assume all transitions are possible*)

a. 6 points Calculate the **energy (in kJ/mol)** required to ionize (completely remove) the electron from the **n=3** level of **He⁺ ion** (He: Helium). Show your work including equations. (3 sig. figs.)

b. 2 points

(I) Select the largest (longest) wavelength from the three choices below:

- i. Wavelength of photon to *completely remove* an electron from *1s orbital of Hydrogen atom*
- ii. Wavelength of photon to *completely remove* an electron from *1s orbital of Boron atom (B)*
- iii. Wavelength of photon to *completely remove* an electron from *1s orbital of Boron cation (B⁺)*

i or ii or iii

(II) Select the smallest (shortest) wavelength from the three choices below (*for H atom*):

- i. Wavelength of photon *emitted* for an electron transitioning from *2p to 1s orbital*
- ii. Wavelength of photon *absorbed* for an electron transitioning from *2s to 4p orbital*
- iii. Wavelength of photon *absorbed* for an electron transitioning from *1s to 3p orbital*

i or ii or iii



Question 2 8 points



A Chem110 student plots the graph shown on the left, from data obtained for the kinetic energy (eV) of the ejected electron versus frequency (s^{-1}) of the incident light, for an unknown metal. ($1\text{eV} = 1.6022 \times 10^{-19} \text{ J}$)

Based on the plot, answer the questions **a to c** below:

a. 1 point On the plot above, label the **threshold frequency**.

b. 2 points What is the value of the work potential (in J) of the unknown metal?

c. 3 points What is the *longest* wavelength (in nm) required to eject an electron from this metal's surface? Show your work including equation(s) used. (3 sig figs)

d. 2 points In the space provided below, plot the following

(Assume frequency is *above* threshold frequency):

- i. *Kinetic energy* of an ejected electron versus the *intensity* of the incident light (left)
- ii. *Number of electrons* ejected versus the *frequency* of the incident light (right)



**Question 3** **6 points****a. 2 points**

In the ground state of Ge (Germanium; $Z = 32$) atom, **name** the following orbitals:

i. Orbital containing the electron that requires the maximum energy to be removed.

ii. Orbital containing the electron that experiences the minimum effective nuclear charge.

b. 2 points

An unknown anion (A^-) has the following *excited* electronic configuration: $[Ar] 4s^2 3d^3 4p^1$

(Ar: Argon; $Z = 18$)

Give the **condensed ground state electronic configuration** of the atom, **A** and the cation, **A^{2+}** .
Show your work. (Assume all electronic transitions are possible)

Electronic Configuration of A:

Electronic Configuration of A^{2+} :

c. 2 points

In a single atom, determine how many electrons can have the following set of quantum numbers?

(i) $n = 5; l = 3$

In a single atom, determine how many electrons can have the following set of quantum numbers?

(i) $n = 3; l = 2; m_s = +1/2$

**Question 4** **8 points**

Rank the following from 1 to 4 in **increasing** order. Provide the ranking in the boxes provided.

Note: Atomic number (Z) are provided for relevant atoms

a) Increasing (lowest to highest) order of *ionization energy*: (1= lowest; 4 = highest)

Mg

Mg: Z = 12

Rb

Rb: Z = 37

Cs

Cs: Z = 55

K

K : Z = 19

b) Increasing (lowest to highest) order of *bond angle*: (1= lowest; 4 = highest)

H₂OCCl₄XeF₄

HCN

c) Increasing (smallest to largest) order of *ionic size* (1= smallest; 4 = largest)

S²⁻

S: Z = 16

O²⁻

O: Z = 8

Be²⁺

Be: Z = 4

Na⁺

Na: Z = 11

d) Increasing (lowest to highest) order of *bond energy* (1= lowest; 4 = highest)

P-F

P-Cl

P-I

P-Br

**Question 5** **12 points**

a. 9 points For the molecules HClO_4 and TeF_6 , draw the most stable *Lewis structure* showing appropriate shape and *dashed/wedged bonds*, based on VSEPR. Determine the **molecular geometry** around the central atom. For HClO_4 , label the *most polar bond*.

HClO_4 (there are no O-O bonds)

Label the most polar bond

Molecular Geometry around Cl

TeF_6

Molecular Geometry around Te

b. 3 points (Note: Lone pairs are not shown in structures below)

(i) For the three *resonance* structures (1, 2 and 3) shown below, answer the following question:



(I) Circle the *most contributing* resonance structure.

Structure 1

Structure 2

Structure 3

(ii) For the three structures given (4, 5, and 6) below, answer the following questions:



(I) Circle all the structures that represent *resonance structures* of each other:

Structure 4

Structure 5

Structure 6

(II) Among the resonance structures, circle the *most contributing* resonance structure.

Structure 4

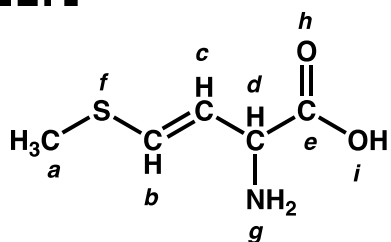
Structure 5

Structure 6

**Question 6** **6 points**

Using the data given below, calculate the **lattice energy** for SrO(s).

	ΔH° (kJ/mol)
$\text{Sr}(s) \longrightarrow \text{Sr}(g)$	+180
$\text{Sr}(g) \longrightarrow \text{Sr}^{2+}(g) + 2e^-$	+1468
$\text{O}_2(g) \longrightarrow 2\text{O}(g)$	+500
$\text{O}(g) + 2e^- \longrightarrow \text{O}^{2-}(g)$	+600
$\text{Sr}(s) + \frac{1}{2}\text{O}_2(g) \longrightarrow \text{SrO}(s)$	-550

**Question 7 10 points**

Consider the molecule given on the left, and answer questions (i) to (vi) using chemical bonding theories that we have learnt in class (valence bond theory, hybridization).

Note that the lone pairs are not shown in the structure:

i. Consider the carbon atoms marked in as ***a***, ***b***, ***c***, ***d***, and ***e*** in the molecule, and determine **hybridization** at each of the carbon centers. (2.5 points)

Atom	Hybridization
Carbon (<i>a</i>)	
Carbon (<i>b</i>)	
Carbon (<i>c</i>)	
Carbon (<i>d</i>)	
Carbon (<i>e</i>)	

ii. Now, consider the sulfur atom (S) marked as ***f***, nitrogen (N) atom (***g***), and oxygen(O) atoms (***h*** and ***i***), and determine their **hybridization**. (2 points)

Atom	Hybridization
Sulfur (<i>f</i>)	
Nitrogen (<i>g</i>)	
Oxygen (<i>h</i>)	
Oxygen (<i>i</i>)	

iii. **Describe** all the bonds formed by the carbon atoms ***a*** and ***b***, using sigma and pi designation as appropriate. (2 points)

Atom	Bonds formed (use sigma and pi as appropriate)
Carbon (<i>a</i>)	
Carbon (<i>b</i>)	

iv. How many total sp^3 -s sigma bonds are in the molecule? (1 point)

v. Determine **electronic** and **molecular geometry** at carbon (C) labeled ***e*** and sulfur (S) labeled ***f*** centres. (2 points)

Atom	Electronic Geometry	Molecular Geometry
Carbon (<i>e</i>)		
Sulfur (<i>f</i>)		

vi. Circle the value from the choices given below, that describes the **C-S-C bond angle at sulfur** (S) marked ***f***: (0.5 points)

>90° to <109.5° 109.5° >109.5 to <120° 120° >120° to <180°

**Question 8 12 points**

Using the **valence shell orbitals only** draw a **complete** molecular orbital (MO) diagram (with no s-p mixing) for the molecule **PS**, in the space provided below. **(6 points)**

Assume that 3s and 3p atomic orbitals will form molecular orbitals like those formed by the combination of 2s and 2p atomic orbitals that we learnt in class, and there is no s-p mixing.

Show **energy arrow**, **atomic orbitals** involved, **atom labels**, the **molecular orbitals** that are formed, **filled electrons** in the MO diagram etc. Make sure to label all the atomic and molecular orbitals.

Using the MO diagram above, answer the following questions:

- (i) Which one of the following will have the highest bond order? Circle the correct choice
A) PS^{1-} B) PS C) PS^{1+}
- (ii) Which one of the following will have the longest bond length? Circle the correct choice
A) PS^{1-} B) PS C) PS^{1+}
- (iii) Which of the following will be diamagnetic? Circle the correct choice.
A) PS^{1-} B) PS C) PS^{1+}
- (iv) Draw Lewis dot structure of the molecule with the shortest bond length (based on the MO diagram) from among the three (PS^{1-} , PS, PS^{1+}), and indicate the bond order calculated from the Lewis dot structure. **(3 points)**

Lewis Dot Structure

Bond Order



9AFC6BC6-BEF4-4DEC-AD44-DF3E13E305F5

midterm-exam-93388

#816 Page 10 of 10

SCRAP PAGE

**CHEMISTRY 110 –Midterm Exam****Thursday November 7, 2024, 6:30 – 8:30 pm**

Examiners: Pallavi Sirjoosingh and Ashok Kakkar

First name (please write as legibly as possible within the boxes)

--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--

Last name

--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--

Student ID number

--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--

This booklet has 8 questions on 10 pages (The last page can be used as a scrap page).

The staple must NOT be removed from the question booklet. Your intact question booklet must be returned at the end of the exam.

A separate booklet contains a periodic table, and datasheet.

Write your answers directly under the questions in the space provided. Do not write on the QR codes or near the top edges (as it may be cut off).

Please write **legibly and dark enough** for the scan to be readable. For students who use English as a second language, dictionaries are permitted. Calculators are allowed.

DO NOT ASK QUESTIONS OF THE INVIGILATORS about interpretations of the exam questions. If you have a concern about a question, please summarize it at the top of this page. All concerns will be evaluated after the exam. GOOD LUCK!

**Question 1 8 points** (*Assume all transitions are possible*)

a. 6 points Calculate the **energy (in kJ/mol)** required to ionize (completely remove) the electron from the **n=3** level of **He⁺ ion** (He: Helium). Show your work including equations. (3 sig. figs.)

b. 2 points

(I) Select the largest (longest) wavelength from the three choices below:

- i. Wavelength of photon to *completely remove* an electron from *1s orbital of Hydrogen atom*
- ii. Wavelength of photon to *completely remove* an electron from *1s orbital of Boron atom (B)*
- iii. Wavelength of photon to *completely remove* an electron from *1s orbital of Boron cation (B⁺)*

i or ii or iii

(II) Select the smallest (shortest) wavelength from the three choices below (*for H atom*):

- i. Wavelength of photon *emitted* for an electron transitioning from *2p to 1s orbital*
- ii. Wavelength of photon *absorbed* for an electron transitioning from *2s to 4p orbital*
- iii. Wavelength of photon *absorbed* for an electron transitioning from *1s to 3p orbital*

i or ii or iii



Question 2 8 points



A Chem110 student plots the graph shown on the left, from data obtained for the kinetic energy (eV) of the ejected electron versus frequency (s^{-1}) of the incident light, for an unknown metal. ($1\text{eV} = 1.6022 \times 10^{-19} \text{ J}$)

Based on the plot, answer the questions **a to c** below:

a. 1 point On the plot above, label the **threshold frequency**.

b. 2 points What is the value of the work potential (in J) of the unknown metal?

c. 3 points What is the *longest* wavelength (in nm) required to eject an electron from this metal's surface? Show your work including equation(s) used. (3 sig figs)

d. 2 points In the space provided below, plot the following

(Assume frequency is *above* threshold frequency):

- i. *Kinetic energy* of an ejected electron versus the *intensity* of the incident light (left)
- ii. *Number of electrons* ejected versus the *frequency* of the incident light (right)



**Question 3** **6 points****a. 2 points**

In the ground state of Ge (Germanium; $Z = 32$) atom, **name** the following orbitals:

i. Orbital containing the electron that requires the maximum energy to be removed.

ii. Orbital containing the electron that experiences the minimum effective nuclear charge.

b. 2 points

An unknown anion (A^-) has the following *excited* electronic configuration: $[Ar] 4s^2 3d^3 4p^1$

(Ar: Argon; $Z = 18$)

Give the **condensed ground state electronic configuration** of the atom, **A** and the cation, **A^{2+}** .
Show your work. (Assume all electronic transitions are possible)

Electronic Configuration of A:

Electronic Configuration of A^{2+} :

c. 2 points

In a single atom, determine how many electrons can have the following set of quantum numbers?

(i) $n = 5; l = 3$

In a single atom, determine how many electrons can have the following set of quantum numbers?

(i) $n = 3; l = 2; m_s = +1/2$

**Question 4** **8 points**

Rank the following from 1 to 4 in **increasing** order. Provide the ranking in the boxes provided.

Note: Atomic number (Z) are provided for relevant atoms

a) Increasing (lowest to highest) order of *ionization energy*: (1= lowest; 4 = highest)

Mg

Mg: Z = 12

Rb

Rb: Z = 37

Cs

Cs: Z = 55

K

K : Z = 19

b) Increasing (lowest to highest) order of *bond angle*: (1= lowest; 4 = highest)

H₂OCCl₄XeF₄

HCN

c) Increasing (smallest to largest) order of *ionic size* (1= smallest; 4 = largest)

S²⁻

S: Z = 16

O²⁻

O: Z = 8

Be²⁺

Be: Z = 4

Na⁺

Na: Z = 11

d) Increasing (lowest to highest) order of *bond energy* (1= lowest; 4 = highest)

P-F

P-Cl

P-I

P-Br

**Question 5** **12 points**

a. 9 points For the molecules HClO_4 and TeF_6 , draw the most stable *Lewis structure* showing appropriate shape and *dashed/wedged bonds*, based on VSEPR. Determine the **molecular geometry** around the central atom. For HClO_4 , label the *most polar bond*.



b. 3 points (Note: Lone pairs are not shown in structures below)

(i) For the three *resonance* structures (1, 2 and 3) shown below, answer the following question:



(I) Circle the *most contributing* resonance structure.

Structure 1

Structure 2

Structure 3

(ii) For the three structures given (4, 5, and 6) below, answer the following questions:



(I) Circle all the structures that represent *resonance structures* of each other:

Structure 4

Structure 5

Structure 6

(II) Among the resonance structures, circle the *most contributing* resonance structure.

Structure 4

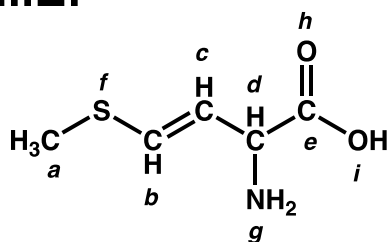
Structure 5

Structure 6

**Question 6** **6 points**

Using the data given below, calculate the **lattice energy** for SrO(s).

	ΔH° (kJ/mol)
$\text{Sr}(s) \longrightarrow \text{Sr}(g)$	+180
$\text{Sr}(g) \longrightarrow \text{Sr}^{2+}(g) + 2e^-$	+1468
$\text{O}_2(g) \longrightarrow 2\text{O}(g)$	+500
$\text{O}(g) + 2e^- \longrightarrow \text{O}^{2-}(g)$	+600
$\text{Sr}(s) + \frac{1}{2}\text{O}_2(g) \longrightarrow \text{SrO}(s)$	-550

**Question 7 10 points**

Consider the molecule given on the left, and answer questions (i) to (vi) using chemical bonding theories that we have learnt in class (valence bond theory, hybridization).

Note that the lone pairs are not shown in the structure:

i. Consider the carbon atoms marked in as ***a***, ***b***, ***c***, ***d***, and ***e*** in the molecule, and determine **hybridization** at each of the carbon centers. (2.5 points)

Atom	Hybridization
Carbon (<i>a</i>)	
Carbon (<i>b</i>)	
Carbon (<i>c</i>)	
Carbon (<i>d</i>)	
Carbon (<i>e</i>)	

ii. Now, consider the sulfur atom (S) marked as ***f***, nitrogen (N) atom (***g***), and oxygen(O) atoms (***h*** and ***i***), and determine their **hybridization**. (2 points)

Atom	Hybridization
Sulfur (<i>f</i>)	
Nitrogen (<i>g</i>)	
Oxygen (<i>h</i>)	
Oxygen (<i>i</i>)	

iii. **Describe** all the bonds formed by the carbon atoms ***a*** and ***b***, using sigma and pi designation as appropriate. (2 points)

Atom	Bonds formed (use sigma and pi as appropriate)
Carbon (<i>a</i>)	
Carbon (<i>b</i>)	

iv. How many total sp^3 -s sigma bonds are in the molecule? (1 point)

v. Determine **electronic** and **molecular geometry** at carbon (C) labeled ***e*** and sulfur (S) labeled ***f*** centres. (2 points)

Atom	Electronic Geometry	Molecular Geometry
Carbon (<i>e</i>)		
Sulfur (<i>f</i>)		

vi. Circle the value from the choices given below, that describes the **C-S-C bond angle at sulfur** (S) marked ***f***: (0.5 points)

>90° to <109.5° 109.5° >109.5 to <120° 120° >120° to <180°


Question 8 12 points

Using the **valence shell orbitals only** draw a **complete** molecular orbital (MO) diagram (with no s-p mixing) for the molecule **PS**, in the space provided below. **(6 points)**

Assume that 3s and 3p atomic orbitals will form molecular orbitals like those formed by the combination of 2s and 2p atomic orbitals that we learnt in class, and there is no s-p mixing.

Show **energy arrow**, **atomic orbitals** involved, **atom labels**, the **molecular orbitals** that are formed, **filled electrons** in the MO diagram etc. Make sure to label all the atomic and molecular orbitals.

Using the MO diagram above, answer the following questions:

- (i) Which one of the following will have the highest bond order? Circle the correct choice
 A) PS^{1-} B) PS C) PS^{1+}
- (ii) Which one of the following will have the longest bond length? Circle the correct choice
 A) PS^{1-} B) PS C) PS^{1+}
- (iii) Which of the following will be diamagnetic? Circle the correct choice.
 A) PS^{1-} B) PS C) PS^{1+}
- (iv) Draw Lewis dot structure of the molecule with the shortest bond length (based on the MO diagram) from among the three (PS^{1-} , PS, PS^{1+}), and indicate the bond order calculated from the Lewis dot structure. **(3 points)**

Lewis Dot Structure

Bond Order



35E22BE4-C70B-4E03-86B4-FA1B0C8E184D

midterm-exam-93388

#817 Page 10 of 10

SCRAP PAGE

**CHEMISTRY 110 –Midterm Exam****Thursday November 7, 2024, 6:30 – 8:30 pm**

Examiners: Pallavi Sirjoosingh and Ashok Kakkar

First name (please write as legibly as possible within the boxes)

--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--

Last name

--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--

Student ID number

--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--

This booklet has 8 questions on 10 pages (The last page can be used as a scrap page).

The staple must NOT be removed from the question booklet. Your intact question booklet must be returned at the end of the exam.

A separate booklet contains a periodic table, and datasheet.

Write your answers directly under the questions in the space provided. Do not write on the QR codes or near the top edges (as it may be cut off).

Please write **legibly and dark enough** for the scan to be readable. For students who use English as a second language, dictionaries are permitted. Calculators are allowed.

DO NOT ASK QUESTIONS OF THE INVIGILATORS about interpretations of the exam questions. If you have a concern about a question, please summarize it at the top of this page. All concerns will be evaluated after the exam. GOOD LUCK!

**Question 1 8 points** (*Assume all transitions are possible*)

a. 6 points Calculate the **energy (in kJ/mol)** required to ionize (completely remove) the electron from the **$n=3$** level of **He^+ ion** (He: Helium). Show your work including equations. (3 sig. figs.)

b. 2 points

(I) Select the largest (longest) wavelength from the three choices below:

- i. Wavelength of photon to *completely remove* an electron from *1s orbital of Hydrogen atom*
- ii. Wavelength of photon to *completely remove* an electron from *1s orbital of Boron atom (B)*
- iii. Wavelength of photon to *completely remove* an electron from *1s orbital of Boron cation (B^+)*

i or ii or iii

(II) Select the smallest (shortest) wavelength from the three choices below (*for H atom*):

- i. Wavelength of photon *emitted* for an electron transitioning from *2p to 1s orbital*
- ii. Wavelength of photon *absorbed* for an electron transitioning from *2s to 4p orbital*
- iii. Wavelength of photon *absorbed* for an electron transitioning from *1s to 3p orbital*

i or ii or iii



Question 2 8 points



A Chem110 student plots the graph shown on the left, from data obtained for the kinetic energy (eV) of the ejected electron versus frequency (s^{-1}) of the incident light, for an unknown metal. ($1\text{eV} = 1.6022 \times 10^{-19} \text{ J}$)

Based on the plot, answer the questions **a to c** below:

a. 1 point On the plot above, label the **threshold frequency**.

b. 2 points What is the value of the work potential (in J) of the unknown metal?

c. 3 points What is the *longest* wavelength (in nm) required to eject an electron from this metal's surface? Show your work including equation(s) used. (3 sig figs)

d. 2 points In the space provided below, plot the following

(Assume frequency is *above* threshold frequency):

- Kinetic energy* of an ejected electron versus the *intensity* of the incident light (left)
- Number of electrons* ejected versus the *frequency* of the incident light (right)



**Question 3** **6 points****a. 2 points**

In the ground state of Ge (Germanium; $Z = 32$) atom, **name** the following orbitals:

i. Orbital containing the electron that requires the maximum energy to be removed.

ii. Orbital containing the electron that experiences the minimum effective nuclear charge.

b. 2 points

An unknown anion (A^-) has the following *excited* electronic configuration: $[Ar] 4s^2 3d^3 4p^1$

(Ar: Argon; $Z = 18$)

Give the **condensed ground state electronic configuration** of the atom, **A** and the cation, **A^{2+}** .
Show your work. (Assume all electronic transitions are possible)

Electronic Configuration of A:

Electronic Configuration of A^{2+} :

c. 2 points

In a single atom, determine how many electrons can have the following set of quantum numbers?

(i) $n = 5; l = 3$

In a single atom, determine how many electrons can have the following set of quantum numbers?

(i) $n = 3; l = 2; m_s = +1/2$

**Question 4** **8 points**

Rank the following from 1 to 4 in **increasing** order. Provide the ranking in the boxes provided.

Note: Atomic number (Z) are provided for relevant atoms

a) Increasing (lowest to highest) order of *ionization energy*: (1= lowest; 4 = highest)

Mg

Mg: Z = 12

Rb

Rb: Z = 37

Cs

Cs: Z = 55

K

K : Z = 19

b) Increasing (lowest to highest) order of *bond angle*: (1= lowest; 4 = highest)

H₂OCCl₄XeF₄

HCN

c) Increasing (smallest to largest) order of *ionic size* (1= smallest; 4 = largest)

S²⁻

S: Z = 16

O²⁻

O: Z = 8

Be²⁺

Be: Z = 4

Na⁺

Na: Z = 11

d) Increasing (lowest to highest) order of *bond energy* (1= lowest; 4 = highest)

P-F

P-Cl

P-I

P-Br

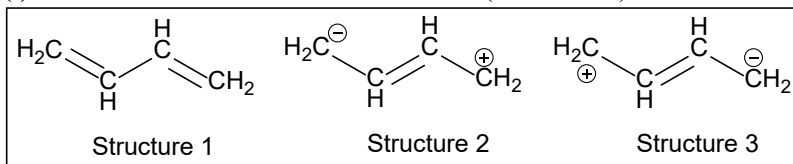
**Question 5** **12 points**

a. 9 points For the molecules HClO_4 and TeF_6 , draw the most stable *Lewis structure* showing appropriate shape and *dashed/wedged bonds*, based on VSEPR. Determine the **molecular geometry** around the central atom. For HClO_4 , label the *most polar bond*.



b. 3 points (Note: Lone pairs are not shown in structures below)

(i) For the three *resonance* structures (1, 2 and 3) shown below, answer the following question:



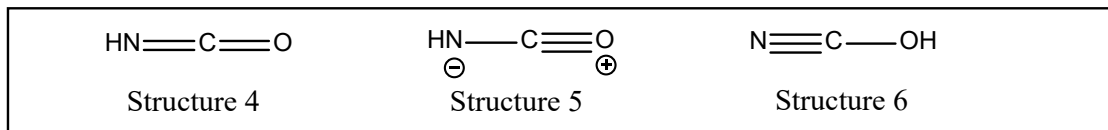
(I) Circle the *most contributing* resonance structure.

Structure 1

Structure 2

Structure 3

(ii) For the three structures given (4, 5, and 6) below, answer the following questions:



(I) Circle all the structures that represent *resonance structures* of each other:

Structure 4

Structure 5

Structure 6

(II) Among the resonance structures, circle the *most contributing* resonance structure.

Structure 4

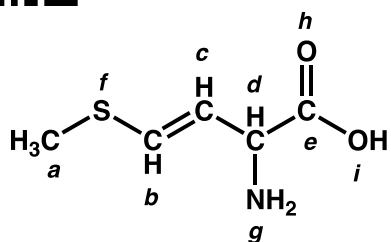
Structure 5

Structure 6

**Question 6** **6 points**

Using the data given below, calculate the **lattice energy** for SrO(s).

	ΔH° (kJ/mol)
$\text{Sr}(s) \longrightarrow \text{Sr}(g)$	+180
$\text{Sr}(g) \longrightarrow \text{Sr}^{2+}(g) + 2e^-$	+1468
$\text{O}_2(g) \longrightarrow 2\text{O}(g)$	+500
$\text{O}(g) + 2e^- \longrightarrow \text{O}^{2-}(g)$	+600
$\text{Sr}(s) + \frac{1}{2}\text{O}_2(g) \longrightarrow \text{SrO}(s)$	-550

**Question 7 10 points**

Consider the molecule given on the left, and answer questions (i) to (vi) using chemical bonding theories that we have learnt in class (valence bond theory, hybridization).

Note that the lone pairs are not shown in the structure:

i. Consider the carbon atoms marked in as ***a***, ***b***, ***c***, ***d***, and ***e*** in the molecule, and determine **hybridization** at each of the carbon centers. (2.5 points)

Atom	Hybridization
Carbon (<i>a</i>)	
Carbon (<i>b</i>)	
Carbon (<i>c</i>)	
Carbon (<i>d</i>)	
Carbon (<i>e</i>)	

ii. Now, consider the sulfur atom (S) marked as ***f***, nitrogen (N) atom (***g***), and oxygen(O) atoms (***h*** and ***i***), and determine their **hybridization**. (2 points)

Atom	Hybridization
Sulfur (<i>f</i>)	
Nitrogen (<i>g</i>)	
Oxygen (<i>h</i>)	
Oxygen (<i>i</i>)	

iii. **Describe** all the bonds formed by the carbon atoms ***a*** and ***b***, using sigma and pi designation as appropriate. (2 points)

Atom	Bonds formed (use sigma and pi as appropriate)
Carbon (<i>a</i>)	
Carbon (<i>b</i>)	

iv. How many total sp^3 -s sigma bonds are in the molecule? (1 point)

v. Determine **electronic** and **molecular geometry** at carbon (C) labeled ***e*** and sulfur (S) labeled ***f*** centres. (2 points)

Atom	Electronic Geometry	Molecular Geometry
Carbon (<i>e</i>)		
Sulfur (<i>f</i>)		

vi. Circle the value from the choices given below, that describes the **C-S-C bond angle at sulfur** (S) marked ***f***: (0.5 points)

>90° to <109.5° 109.5° >109.5 to <120° 120° >120° to <180°


Question 8 12 points

Using the **valence shell orbitals only** draw a **complete** molecular orbital (MO) diagram (with no s-p mixing) for the molecule **PS**, in the space provided below. **(6 points)**

Assume that 3s and 3p atomic orbitals will form molecular orbitals like those formed by the combination of 2s and 2p atomic orbitals that we learnt in class, and there is no s-p mixing.

Show **energy arrow**, **atomic orbitals** involved, **atom labels**, the **molecular orbitals** that are formed, **filled electrons** in the MO diagram etc. Make sure to label all the atomic and molecular orbitals.

Using the MO diagram above, answer the following questions:

- (i) Which one of the following will have the highest bond order? Circle the correct choice
 A) PS^{1-} B) PS C) PS^{1+}
- (ii) Which one of the following will have the longest bond length? Circle the correct choice
 A) PS^{1-} B) PS C) PS^{1+}
- (iii) Which of the following will be diamagnetic? Circle the correct choice.
 A) PS^{1-} B) PS C) PS^{1+}
- (iv) Draw Lewis dot structure of the molecule with the shortest bond length (based on the MO diagram) from among the three (PS^{1-} , PS, PS^{1+}), and indicate the bond order calculated from the Lewis dot structure. **(3 points)**

Lewis Dot Structure

Bond Order



FB46DE79-880B-445A-ABC6-CC7D42EDFC05

midterm-exam-93388

#818 Page 10 of 10

SCRAP PAGE

**CHEMISTRY 110 –Midterm Exam****Thursday November 7, 2024, 6:30 – 8:30 pm**

Examiners: Pallavi Sirjoosingh and Ashok Kakkar

First name (please write as legibly as possible within the boxes)

--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--

Last name

--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--

Student ID number

--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--

This booklet has 8 questions on 10 pages (The last page can be used as a scrap page).

The staple must NOT be removed from the question booklet. Your intact question booklet must be returned at the end of the exam.

A separate booklet contains a periodic table, and datasheet.

Write your answers directly under the questions in the space provided. Do not write on the QR codes or near the top edges (as it may be cut off).

Please write **legibly and dark enough** for the scan to be readable. For students who use English as a second language, dictionaries are permitted. Calculators are allowed.

DO NOT ASK QUESTIONS OF THE INVIGILATORS about interpretations of the exam questions. If you have a concern about a question, please summarize it at the top of this page. All concerns will be evaluated after the exam. GOOD LUCK!

**Question 1 8 points** (*Assume all transitions are possible*)

a. 6 points Calculate the **energy (in kJ/mol)** required to ionize (completely remove) the electron from the **n=3** level of **He⁺ ion** (He: Helium). Show your work including equations. (3 sig. figs.)

b. 2 points

(I) Select the largest (longest) wavelength from the three choices below:

- i. Wavelength of photon to *completely remove* an electron from *1s orbital of Hydrogen atom*
- ii. Wavelength of photon to *completely remove* an electron from *1s orbital of Boron atom (B)*
- iii. Wavelength of photon to *completely remove* an electron from *1s orbital of Boron cation (B⁺)*

i or ii or iii

(II) Select the smallest (shortest) wavelength from the three choices below (*for H atom*):

- i. Wavelength of photon *emitted* for an electron transitioning from *2p to 1s orbital*
- ii. Wavelength of photon *absorbed* for an electron transitioning from *2s to 4p orbital*
- iii. Wavelength of photon *absorbed* for an electron transitioning from *1s to 3p orbital*

i or ii or iii



Question 2 8 points



A Chem110 student plots the graph shown on the left, from data obtained for the kinetic energy (eV) of the ejected electron versus frequency (s^{-1}) of the incident light, for an unknown metal. ($1\text{eV} = 1.6022 \times 10^{-19} \text{ J}$)

Based on the plot, answer the questions **a to c** below:

a. 1 point On the plot above, label the **threshold frequency**.

b. 2 points What is the value of the work potential (in J) of the unknown metal?

c. 3 points What is the *longest* wavelength (in nm) required to eject an electron from this metal's surface? Show your work including equation(s) used. (3 sig figs)

d. 2 points In the space provided below, plot the following

(Assume frequency is *above* threshold frequency):

- i. *Kinetic energy* of an ejected electron versus the *intensity* of the incident light (left)
- ii. *Number of electrons* ejected versus the *frequency* of the incident light (right)



**Question 3** **6 points****a. 2 points**

In the ground state of Ge (Germanium; $Z = 32$) atom, **name** the following orbitals:

i. Orbital containing the electron that requires the maximum energy to be removed.

ii. Orbital containing the electron that experiences the minimum effective nuclear charge.

b. 2 points

An unknown anion (A^-) has the following *excited* electronic configuration: $[Ar] 4s^2 3d^3 4p^1$

(Ar: Argon; $Z = 18$)

Give the **condensed ground state electronic configuration** of the atom, **A** and the cation, **A^{2+}** .
Show your work. (Assume all electronic transitions are possible)

Electronic Configuration of A:

Electronic Configuration of A^{2+} :

c. 2 points

In a single atom, determine how many electrons can have the following set of quantum numbers?

(i) $n = 5; l = 3$

In a single atom, determine how many electrons can have the following set of quantum numbers?

(i) $n = 3; l = 2; m_s = +1/2$

**Question 4** **8 points**

Rank the following from 1 to 4 in **increasing** order. Provide the ranking in the boxes provided.

Note: Atomic number (Z) are provided for relevant atoms

a) Increasing (lowest to highest) order of *ionization energy*: (1= lowest; 4 = highest)

Mg

Mg: Z = 12

Rb

Rb: Z = 37

Cs

Cs: Z = 55

K

K : Z = 19

b) Increasing (lowest to highest) order of *bond angle*: (1= lowest; 4 = highest)

H₂OCCl₄XeF₄

HCN

c) Increasing (smallest to largest) order of *ionic size* (1= smallest; 4 = largest)

S²⁻

S: Z = 16

O²⁻

O: Z = 8

Be²⁺

Be: Z = 4

Na⁺

Na: Z = 11

d) Increasing (lowest to highest) order of *bond energy* (1= lowest; 4 = highest)

P-F

P-Cl

P-I

P-Br

**Question 5** **12 points**

a. 9 points For the molecules HClO_4 and TeF_6 , draw the most stable *Lewis structure* showing appropriate shape and *dashed/wedged bonds*, based on VSEPR. Determine the **molecular geometry** around the central atom. For HClO_4 , label the *most polar bond*.



b. 3 points (Note: Lone pairs are not shown in structures below)

(i) For the three *resonance* structures (1, 2 and 3) shown below, answer the following question:



(I) Circle the *most contributing* resonance structure.

Structure 1

Structure 2

Structure 3

(ii) For the three structures given (4, 5, and 6) below, answer the following questions:



(I) Circle all the structures that represent *resonance structures* of each other:

Structure 4

Structure 5

Structure 6

(II) Among the resonance structures, circle the *most contributing* resonance structure.

Structure 4

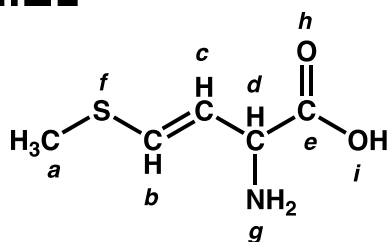
Structure 5

Structure 6

**Question 6** **6 points**

Using the data given below, calculate the **lattice energy** for SrO(s).

	ΔH° (kJ/mol)
$\text{Sr}(s) \longrightarrow \text{Sr}(g)$	+180
$\text{Sr}(g) \longrightarrow \text{Sr}^{2+}(g) + 2e^-$	+1468
$\text{O}_2(g) \longrightarrow 2\text{O}(g)$	+500
$\text{O}(g) + 2e^- \longrightarrow \text{O}^{2-}(g)$	+600
$\text{Sr}(s) + \frac{1}{2}\text{O}_2(g) \longrightarrow \text{SrO}(s)$	-550

**Question 7 10 points**

Consider the molecule given on the left, and answer questions (i) to (vi) using chemical bonding theories that we have learnt in class (valence bond theory, hybridization).

Note that the lone pairs are not shown in the structure:

i. Consider the carbon atoms marked in as ***a***, ***b***, ***c***, ***d***, and ***e*** in the molecule, and determine **hybridization** at each of the carbon centers. (2.5 points)

Atom	Hybridization
Carbon (<i>a</i>)	
Carbon (<i>b</i>)	
Carbon (<i>c</i>)	
Carbon (<i>d</i>)	
Carbon (<i>e</i>)	

ii. Now, consider the sulfur atom (S) marked as ***f***, nitrogen (N) atom (***g***), and oxygen(O) atoms (***h*** and ***i***), and determine their **hybridization**. (2 points)

Atom	Hybridization
Sulfur (<i>f</i>)	
Nitrogen (<i>g</i>)	
Oxygen (<i>h</i>)	
Oxygen (<i>i</i>)	

iii. **Describe** all the bonds formed by the carbon atoms ***a*** and ***b***, using sigma and pi designation as appropriate. (2 points)

Atom	Bonds formed (use sigma and pi as appropriate)
Carbon (<i>a</i>)	
Carbon (<i>b</i>)	

iv. How many total sp^3 -s sigma bonds are in the molecule? (1 point)

v. Determine **electronic** and **molecular geometry** at carbon (C) labeled ***e*** and sulfur (S) labeled ***f*** centres. (2 points)

Atom	Electronic Geometry	Molecular Geometry
Carbon (<i>e</i>)		
Sulfur (<i>f</i>)		

vi. Circle the value from the choices given below, that describes the **C-S-C bond angle at sulfur** (S) marked ***f***: (0.5 points)

>90° to <109.5° 109.5° >109.5 to <120° 120° >120° to <180°


Question 8 12 points

Using the **valence shell orbitals only** draw a **complete** molecular orbital (MO) diagram (with no s-p mixing) for the molecule **PS**, in the space provided below. **(6 points)**

Assume that 3s and 3p atomic orbitals will form molecular orbitals like those formed by the combination of 2s and 2p atomic orbitals that we learnt in class, and there is no s-p mixing.

Show **energy arrow**, **atomic orbitals** involved, **atom labels**, the **molecular orbitals** that are formed, **filled electrons** in the MO diagram etc. Make sure to label all the atomic and molecular orbitals.

Using the MO diagram above, answer the following questions:

- (i) Which one of the following will have the highest bond order? Circle the correct choice
 A) PS^{1-} B) PS C) PS^{1+}
- (ii) Which one of the following will have the longest bond length? Circle the correct choice
 A) PS^{1-} B) PS C) PS^{1+}
- (iii) Which of the following will be diamagnetic? Circle the correct choice.
 A) PS^{1-} B) PS C) PS^{1+}
- (iv) Draw Lewis dot structure of the molecule with the shortest bond length (based on the MO diagram) from among the three (PS^{1-} , PS, PS^{1+}), and indicate the bond order calculated from the Lewis dot structure. **(3 points)**

Lewis Dot Structure

Bond Order



11BBF36A-9623-4122-8EC6-065031F32DD0

midterm-exam-93388

#819 Page 10 of 10

SCRAP PAGE

**CHEMISTRY 110 –Midterm Exam****Thursday November 7, 2024, 6:30 – 8:30 pm**

Examiners: Pallavi Sirjoosingh and Ashok Kakkar

First name (please write as legibly as possible within the boxes)

--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--

Last name

--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--

Student ID number

--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--

This booklet has 8 questions on 10 pages (The last page can be used as a scrap page).

The staple must NOT be removed from the question booklet. Your intact question booklet must be returned at the end of the exam.

A separate booklet contains a periodic table, and datasheet.

Write your answers directly under the questions in the space provided. Do not write on the QR codes or near the top edges (as it may be cut off).

Please write **legibly and dark enough** for the scan to be readable. For students who use English as a second language, dictionaries are permitted. Calculators are allowed.

DO NOT ASK QUESTIONS OF THE INVIGILATORS about interpretations of the exam questions. If you have a concern about a question, please summarize it at the top of this page. All concerns will be evaluated after the exam. GOOD LUCK!

**Question 1 8 points** (*Assume all transitions are possible*)

a. 6 points Calculate the **energy (in kJ/mol)** required to ionize (completely remove) the electron from the **n=3** level of **He⁺ ion** (He: Helium). Show your work including equations. (3 sig. figs.)

b. 2 points

(I) Select the largest (longest) wavelength from the three choices below:

- i. Wavelength of photon to *completely remove* an electron from *1s orbital of Hydrogen atom*
- ii. Wavelength of photon to *completely remove* an electron from *1s orbital of Boron atom (B)*
- iii. Wavelength of photon to *completely remove* an electron from *1s orbital of Boron cation (B⁺)*

i or ii or iii

(II) Select the smallest (shortest) wavelength from the three choices below (*for H atom*):

- i. Wavelength of photon *emitted* for an electron transitioning from *2p to 1s orbital*
- ii. Wavelength of photon *absorbed* for an electron transitioning from *2s to 4p orbital*
- iii. Wavelength of photon *absorbed* for an electron transitioning from *1s to 3p orbital*

i or ii or iii



Question 2 8 points



A Chem110 student plots the graph shown on the left, from data obtained for the kinetic energy (eV) of the ejected electron versus frequency (s^{-1}) of the incident light, for an unknown metal. ($1\text{eV} = 1.6022 \times 10^{-19} \text{ J}$)

Based on the plot, answer the questions **a to c** below:

a. 1 point On the plot above, label the **threshold frequency**.

b. 2 points What is the value of the work potential (in J) of the unknown metal?

c. 3 points What is the *longest* wavelength (in nm) required to eject an electron from this metal's surface? Show your work including equation(s) used. (3 sig figs)

d. 2 points In the space provided below, plot the following

(Assume frequency is *above* threshold frequency):

- Kinetic energy* of an ejected electron versus the *intensity* of the incident light (left)
- Number of electrons* ejected versus the *frequency* of the incident light (right)



**Question 3** **6 points****a. 2 points**

In the ground state of Ge (Germanium; $Z = 32$) atom, **name** the following orbitals:

i. Orbital containing the electron that requires the maximum energy to be removed.

ii. Orbital containing the electron that experiences the minimum effective nuclear charge.

b. 2 points

An unknown anion (A^-) has the following *excited* electronic configuration: $[Ar] 4s^2 3d^3 4p^1$

(Ar: Argon; $Z = 18$)

Give the **condensed ground state electronic configuration** of the atom, **A** and the cation, **A^{2+}** .
Show your work. (Assume all electronic transitions are possible)

Electronic Configuration of A:

Electronic Configuration of A^{2+} :

c. 2 points

In a single atom, determine how many electrons can have the following set of quantum numbers?

(i) $n = 5; l = 3$

In a single atom, determine how many electrons can have the following set of quantum numbers?

(i) $n = 3; l = 2; m_s = +1/2$

**Question 4** **8 points**

Rank the following from 1 to 4 in **increasing** order. Provide the ranking in the boxes provided.

Note: Atomic number (Z) are provided for relevant atoms

a) Increasing (lowest to highest) order of *ionization energy*: (1= lowest; 4 = highest)

Mg

Mg: Z = 12

Rb

Rb: Z = 37

Cs

Cs: Z = 55

K

K : Z = 19

b) Increasing (lowest to highest) order of *bond angle*: (1= lowest; 4 = highest)

H₂OCCl₄XeF₄

HCN

c) Increasing (smallest to largest) order of *ionic size* (1= smallest; 4 = largest)

S²⁻

S: Z = 16

O²⁻

O: Z = 8

Be²⁺

Be: Z = 4

Na⁺

Na: Z = 11

d) Increasing (lowest to highest) order of *bond energy* (1= lowest; 4 = highest)

P-F

P-Cl

P-I

P-Br

**Question 5** **12 points**

a. 9 points For the molecules HClO_4 and TeF_6 , draw the most stable *Lewis structure* showing appropriate shape and *dashed/wedged bonds*, based on VSEPR. Determine the **molecular geometry** around the central atom. For HClO_4 , label the *most polar bond*.



b. 3 points (Note: Lone pairs are not shown in structures below)

(i) For the three *resonance* structures (1, 2 and 3) shown below, answer the following question:



(I) Circle the *most contributing* resonance structure.

Structure 1

Structure 2

Structure 3

(ii) For the three structures given (4, 5, and 6) below, answer the following questions:



(I) Circle all the structures that represent *resonance structures* of each other:

Structure 4

Structure 5

Structure 6

(II) Among the resonance structures, circle the *most contributing* resonance structure.

Structure 4

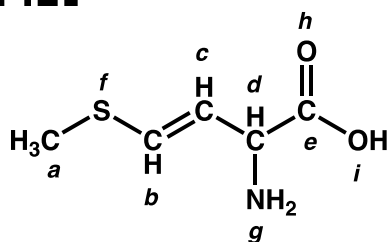
Structure 5

Structure 6

**Question 6** **6 points**

Using the data given below, calculate the **lattice energy** for SrO(s).

	ΔH° (kJ/mol)
$\text{Sr}(s) \longrightarrow \text{Sr}(g)$	+180
$\text{Sr}(g) \longrightarrow \text{Sr}^{2+}(g) + 2e^-$	+1468
$\text{O}_2(g) \longrightarrow 2\text{O}(g)$	+500
$\text{O}(g) + 2e^- \longrightarrow \text{O}^{2-}(g)$	+600
$\text{Sr}(s) + \frac{1}{2}\text{O}_2(g) \longrightarrow \text{SrO}(s)$	-550

**Question 7 10 points**

Consider the molecule given on the left, and answer questions (i) to (vi) using chemical bonding theories that we have learnt in class (valence bond theory, hybridization).

Note that the lone pairs are not shown in the structure:

i. Consider the carbon atoms marked in as ***a***, ***b***, ***c***, ***d***, and ***e*** in the molecule, and determine **hybridization** at each of the carbon centers. (2.5 points)

Atom	Hybridization
Carbon (<i>a</i>)	
Carbon (<i>b</i>)	
Carbon (<i>c</i>)	
Carbon (<i>d</i>)	
Carbon (<i>e</i>)	

ii. Now, consider the sulfur atom (S) marked as ***f***, nitrogen (N) atom (***g***), and oxygen(O) atoms (***h*** and ***i***), and determine their **hybridization**. (2 points)

Atom	Hybridization
Sulfur (<i>f</i>)	
Nitrogen (<i>g</i>)	
Oxygen (<i>h</i>)	
Oxygen (<i>i</i>)	

iii. **Describe** all the bonds formed by the carbon atoms ***a*** and ***b***, using sigma and pi designation as appropriate. (2 points)

Atom	Bonds formed (use sigma and pi as appropriate)
Carbon (<i>a</i>)	
Carbon (<i>b</i>)	

iv. How many total sp^3 -s sigma bonds are in the molecule? (1 point)

v. Determine **electronic** and **molecular geometry** at carbon (C) labeled ***e*** and sulfur (S) labeled ***f*** centres. (2 points)

Atom	Electronic Geometry	Molecular Geometry
Carbon (<i>e</i>)		
Sulfur (<i>f</i>)		

vi. Circle the value from the choices given below, that describes the **C-S-C bond angle at sulfur** (S) marked ***f***: (0.5 points)

>90° to <109.5° 109.5° >109.5 to <120° 120° >120° to <180°

**Question 8 12 points**

Using the **valence shell orbitals only** draw a **complete** molecular orbital (MO) diagram (with no s-p mixing) for the molecule **PS**, in the space provided below. **(6 points)**

Assume that 3s and 3p atomic orbitals will form molecular orbitals like those formed by the combination of 2s and 2p atomic orbitals that we learnt in class, and there is no s-p mixing.

Show **energy arrow**, **atomic orbitals** involved, **atom labels**, the **molecular orbitals** that are formed, **filled electrons** in the MO diagram etc. Make sure to label all the atomic and molecular orbitals.

Using the MO diagram above, answer the following questions:

- (i) Which one of the following will have the highest bond order? Circle the correct choice
A) PS^{1-} B) PS C) PS^{1+}
- (ii) Which one of the following will have the longest bond length? Circle the correct choice
A) PS^{1-} B) PS C) PS^{1+}
- (iii) Which of the following will be diamagnetic? Circle the correct choice.
A) PS^{1-} B) PS C) PS^{1+}
- (iv) Draw Lewis dot structure of the molecule with the shortest bond length (based on the MO diagram) from among the three (PS^{1-} , PS, PS^{1+}), and indicate the bond order calculated from the Lewis dot structure. **(3 points)**

Lewis Dot Structure

Bond Order



E56EC2A3-1E3C-4DAD-A1E3-4920903432D8

midterm-exam-93388

#820 Page 10 of 10

SCRAP PAGE

**CHEMISTRY 110 –Midterm Exam****Thursday November 7, 2024, 6:30 – 8:30 pm****Examiners: Pallavi Sirjoosingh and Ashok Kakkar**

First name (please write as legibly as possible within the boxes)

--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--

Last name

--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--

Student ID number

--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--

This booklet has 8 questions on 10 pages (The last page can be used as a scrap page).

The staple must NOT be removed from the question booklet. Your intact question booklet must be returned at the end of the exam.

A separate booklet contains a periodic table, and datasheet.

Write your answers directly under the questions in the space provided. Do not write on the QR codes or near the top edges (as it may be cut off).

Please write **legibly and dark enough** for the scan to be readable. For students who use English as a second language, dictionaries are permitted. Calculators are allowed.

DO NOT ASK QUESTIONS OF THE INVIGILATORS about interpretations of the exam questions. If you have a concern about a question, please summarize it at the top of this page. All concerns will be evaluated after the exam. GOOD LUCK!

**Question 1 8 points** (*Assume all transitions are possible*)

a. 6 points Calculate the **energy (in kJ/mol)** required to ionize (completely remove) the electron from the **$n=3$** level of **He^+ ion** (He: Helium). Show your work including equations. (3 sig. figs.)

b. 2 points

(I) Select the largest (longest) wavelength from the three choices below:

- i. Wavelength of photon to *completely remove* an electron from *1s orbital of Hydrogen atom*
- ii. Wavelength of photon to *completely remove* an electron from *1s orbital of Boron atom (B)*
- iii. Wavelength of photon to *completely remove* an electron from *1s orbital of Boron cation (B^+)*

i or ii or iii

(II) Select the smallest (shortest) wavelength from the three choices below (*for H atom*):

- i. Wavelength of photon *emitted* for an electron transitioning from *2p to 1s orbital*
- ii. Wavelength of photon *absorbed* for an electron transitioning from *2s to 4p orbital*
- iii. Wavelength of photon *absorbed* for an electron transitioning from *1s to 3p orbital*

i or ii or iii



Question 2 8 points



A Chem110 student plots the graph shown on the left, from data obtained for the kinetic energy (eV) of the ejected electron versus frequency (s^{-1}) of the incident light, for an unknown metal. ($1\text{eV} = 1.6022 \times 10^{-19} \text{ J}$)

Based on the plot, answer the questions **a to c** below:

a. 1 point On the plot above, label the **threshold frequency**.

b. 2 points What is the value of the work potential (in J) of the unknown metal?

c. 3 points What is the *longest* wavelength (in nm) required to eject an electron from this metal's surface? Show your work including equation(s) used. (3 sig figs)

d. 2 points In the space provided below, plot the following

(Assume frequency is *above* threshold frequency):

- Kinetic energy* of an ejected electron versus the *intensity* of the incident light (left)
- Number of electrons* ejected versus the *frequency* of the incident light (right)



**Question 3** **6 points****a. 2 points**

In the ground state of Ge (Germanium; $Z = 32$) atom, **name** the following orbitals:

i. Orbital containing the electron that requires the maximum energy to be removed.

ii. Orbital containing the electron that experiences the minimum effective nuclear charge.

b. 2 points

An unknown anion (A^-) has the following *excited* electronic configuration: $[Ar] 4s^2 3d^3 4p^1$

(Ar: Argon; $Z = 18$)

Give the **condensed ground state electronic configuration** of the atom, **A** and the cation, **A^{2+}** .
Show your work. (Assume all electronic transitions are possible)

Electronic Configuration of A:

Electronic Configuration of A^{2+} :

c. 2 points

In a single atom, determine how many electrons can have the following set of quantum numbers?

(i) $n = 5; l = 3$

In a single atom, determine how many electrons can have the following set of quantum numbers?

(i) $n = 3; l = 2; m_s = +1/2$

**Question 4** **8 points**

Rank the following from 1 to 4 in **increasing** order. Provide the ranking in the boxes provided.

Note: Atomic number (Z) are provided for relevant atoms

a) Increasing (lowest to highest) order of *ionization energy*: (1= lowest; 4 = highest)

Mg

Mg: Z = 12

Rb

Rb: Z = 37

Cs

Cs: Z = 55

K

K : Z = 19

b) Increasing (lowest to highest) order of *bond angle*: (1= lowest; 4 = highest)

H₂OCCl₄XeF₄

HCN

c) Increasing (smallest to largest) order of *ionic size* (1= smallest; 4 = largest)

S²⁻

S: Z = 16

O²⁻

O: Z = 8

Be²⁺

Be: Z = 4

Na⁺

Na: Z = 11

d) Increasing (lowest to highest) order of *bond energy* (1= lowest; 4 = highest)

P-F

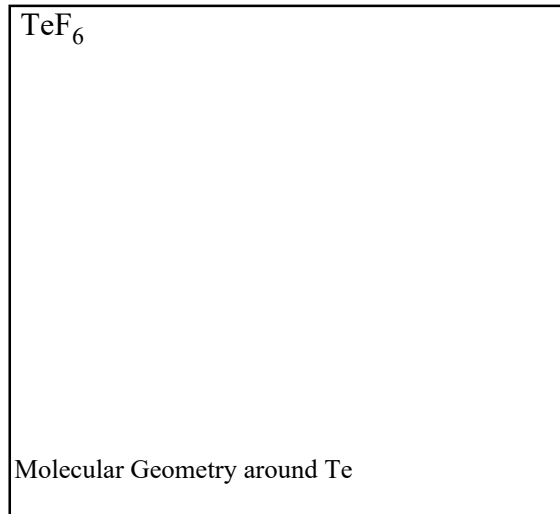
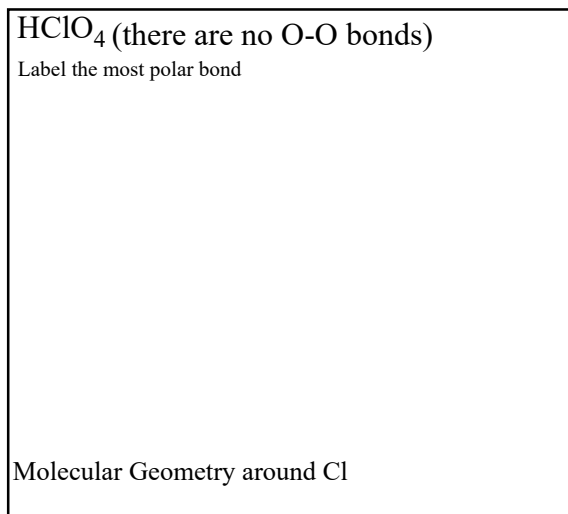
P-Cl

P-I

P-Br

**Question 5** **12 points**

a. 9 points For the molecules HClO_4 and TeF_6 , draw the most stable *Lewis structure* showing appropriate shape and *dashed/wedged bonds*, based on VSEPR. Determine the **molecular geometry** around the central atom. For HClO_4 , label the *most polar bond*.



b. 3 points (Note: Lone pairs are not shown in structures below)

(i) For the three *resonance* structures (1, 2 and 3) shown below, answer the following question:



(I) Circle the *most contributing* resonance structure.

Structure 1

Structure 2

Structure 3

(ii) For the three structures given (4, 5, and 6) below, answer the following questions:



(I) Circle all the structures that represent *resonance structures* of each other:

Structure 4

Structure 5

Structure 6

(II) Among the resonance structures, circle the *most contributing* resonance structure.

Structure 4

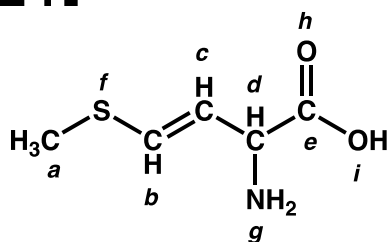
Structure 5

Structure 6

**Question 6** **6 points**

Using the data given below, calculate the **lattice energy** for SrO(s).

	ΔH° (kJ/mol)
$\text{Sr}(s) \longrightarrow \text{Sr}(g)$	+180
$\text{Sr}(g) \longrightarrow \text{Sr}^{2+}(g) + 2e^-$	+1468
$\text{O}_2(g) \longrightarrow 2\text{O}(g)$	+500
$\text{O}(g) + 2e^- \longrightarrow \text{O}^{2-}(g)$	+600
$\text{Sr}(s) + \frac{1}{2}\text{O}_2(g) \longrightarrow \text{SrO}(s)$	-550

**Question 7 10 points**

Consider the molecule given on the left, and answer questions (i) to (vi) using chemical bonding theories that we have learnt in class (valence bond theory, hybridization).

Note that the lone pairs are not shown in the structure:

i. Consider the carbon atoms marked in as ***a***, ***b***, ***c***, ***d***, and ***e*** in the molecule, and determine **hybridization** at each of the carbon centers. (2.5 points)

Atom	Hybridization
Carbon (<i>a</i>)	
Carbon (<i>b</i>)	
Carbon (<i>c</i>)	
Carbon (<i>d</i>)	
Carbon (<i>e</i>)	

ii. Now, consider the sulfur atom (S) marked as ***f***, nitrogen (N) atom (***g***), and oxygen(O) atoms (***h*** and ***i***), and determine their **hybridization**. (2 points)

Atom	Hybridization
Sulfur (<i>f</i>)	
Nitrogen (<i>g</i>)	
Oxygen (<i>h</i>)	
Oxygen (<i>i</i>)	

iii. **Describe** all the bonds formed by the carbon atoms ***a*** and ***b***, using sigma and pi designation as appropriate. (2 points)

Atom	Bonds formed (use sigma and pi as appropriate)
Carbon (<i>a</i>)	
Carbon (<i>b</i>)	

iv. How many total sp^3 -s sigma bonds are in the molecule? (1 point)

v. Determine **electronic** and **molecular geometry** at carbon (C) labeled ***e*** and sulfur (S) labeled ***f*** centres. (2 points)

Atom	Electronic Geometry	Molecular Geometry
Carbon (<i>e</i>)		
Sulfur (<i>f</i>)		

vi. Circle the value from the choices given below, that describes the **C-S-C bond angle at sulfur** (S) marked ***f***: (0.5 points)

>90° to <109.5° 109.5° >109.5 to <120° 120° >120° to <180°


Question 8 12 points

Using the **valence shell orbitals only** draw a **complete** molecular orbital (MO) diagram (with no s-p mixing) for the molecule **PS**, in the space provided below. **(6 points)**

Assume that 3s and 3p atomic orbitals will form molecular orbitals like those formed by the combination of 2s and 2p atomic orbitals that we learnt in class, and there is no s-p mixing.

Show **energy arrow**, **atomic orbitals** involved, **atom labels**, the **molecular orbitals** that are formed, **filled electrons** in the MO diagram etc. Make sure to label all the atomic and molecular orbitals.

Using the MO diagram above, answer the following questions:

- (i) Which one of the following will have the highest bond order? Circle the correct choice
 A) PS^{1-} B) PS C) PS^{1+}
- (ii) Which one of the following will have the longest bond length? Circle the correct choice
 A) PS^{1-} B) PS C) PS^{1+}
- (iii) Which of the following will be diamagnetic? Circle the correct choice.
 A) PS^{1-} B) PS C) PS^{1+}
- (iv) Draw Lewis dot structure of the molecule with the shortest bond length (based on the MO diagram) from among the three (PS^{1-} , PS, PS^{1+}), and indicate the bond order calculated from the Lewis dot structure. **(3 points)**

Lewis Dot Structure

Bond Order



7B87E497-4DCB-47A0-8F36-867E23219F70

midterm-exam-93388

#821 Page 10 of 10

SCRAP PAGE

**CHEMISTRY 110 –Midterm Exam****Thursday November 7, 2024, 6:30 – 8:30 pm**

Examiners: Pallavi Sirjoosingh and Ashok Kakkar

First name (please write as legibly as possible within the boxes)

--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--

Last name

--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--

Student ID number

--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--

This booklet has 8 questions on 10 pages (The last page can be used as a scrap page).

The staple must NOT be removed from the question booklet. Your intact question booklet must be returned at the end of the exam.

A separate booklet contains a periodic table, and datasheet.

Write your answers directly under the questions in the space provided. Do not write on the QR codes or near the top edges (as it may be cut off).

Please write **legibly and dark enough** for the scan to be readable. For students who use English as a second language, dictionaries are permitted. Calculators are allowed.

DO NOT ASK QUESTIONS OF THE INVIGILATORS about interpretations of the exam questions. If you have a concern about a question, please summarize it at the top of this page. All concerns will be evaluated after the exam. GOOD LUCK!

**Question 1 8 points** (*Assume all transitions are possible*)

a. 6 points Calculate the **energy (in kJ/mol)** required to ionize (completely remove) the electron from the **n=3** level of **He⁺ ion** (He: Helium). Show your work including equations. (3 sig. figs.)

b. 2 points

(I) Select the largest (longest) wavelength from the three choices below:

- i. Wavelength of photon to *completely remove* an electron from *1s orbital of Hydrogen atom*
- ii. Wavelength of photon to *completely remove* an electron from *1s orbital of Boron atom (B)*
- iii. Wavelength of photon to *completely remove* an electron from *1s orbital of Boron cation (B⁺)*

i or ii or iii

(II) Select the smallest (shortest) wavelength from the three choices below (*for H atom*):

- i. Wavelength of photon *emitted* for an electron transitioning from *2p to 1s orbital*
- ii. Wavelength of photon *absorbed* for an electron transitioning from *2s to 4p orbital*
- iii. Wavelength of photon *absorbed* for an electron transitioning from *1s to 3p orbital*

i or ii or iii



Question 2 8 points



A Chem110 student plots the graph shown on the left, from data obtained for the kinetic energy (eV) of the ejected electron versus frequency (s^{-1}) of the incident light, for an unknown metal. ($1\text{eV} = 1.6022 \times 10^{-19} \text{ J}$)

Based on the plot, answer the questions **a to c** below:

a. 1 point On the plot above, label the **threshold frequency**.

b. 2 points What is the value of the work potential (in J) of the unknown metal?

c. 3 points What is the *longest* wavelength (in nm) required to eject an electron from this metal's surface? Show your work including equation(s) used. (3 sig figs)

d. 2 points In the space provided below, plot the following

(Assume frequency is *above* threshold frequency):

- Kinetic energy* of an ejected electron versus the *intensity* of the incident light (left)
- Number of electrons* ejected versus the *frequency* of the incident light (right)



**Question 3** **6 points****a. 2 points**

In the ground state of Ge (Germanium; $Z = 32$) atom, **name** the following orbitals:

i. Orbital containing the electron that requires the maximum energy to be removed.

ii. Orbital containing the electron that experiences the minimum effective nuclear charge.

b. 2 points

An unknown anion (A^-) has the following *excited* electronic configuration: $[Ar] 4s^2 3d^3 4p^1$

(Ar: Argon; $Z = 18$)

Give the **condensed ground state electronic configuration** of the atom, **A** and the cation, **A^{2+}** .
Show your work. (Assume all electronic transitions are possible)

Electronic Configuration of A:

Electronic Configuration of A^{2+} :

c. 2 points

In a single atom, determine how many electrons can have the following set of quantum numbers?

(i) $n = 5; l = 3$

In a single atom, determine how many electrons can have the following set of quantum numbers?

(i) $n = 3; l = 2; m_s = +1/2$

**Question 4** **8 points**

Rank the following from 1 to 4 in **increasing** order. Provide the ranking in the boxes provided.

Note: Atomic number (Z) are provided for relevant atoms

a) Increasing (lowest to highest) order of *ionization energy*: (1= lowest; 4 = highest)

Mg

Mg: Z = 12

Rb

Rb: Z = 37

Cs

Cs: Z = 55

K

K : Z = 19

b) Increasing (lowest to highest) order of *bond angle*: (1= lowest; 4 = highest)

H₂OCCl₄XeF₄

HCN

c) Increasing (smallest to largest) order of *ionic size* (1= smallest; 4 = largest)

S²⁻

S: Z = 16

O²⁻

O: Z = 8

Be²⁺

Be: Z = 4

Na⁺

Na: Z = 11

d) Increasing (lowest to highest) order of *bond energy* (1= lowest; 4 = highest)

P-F

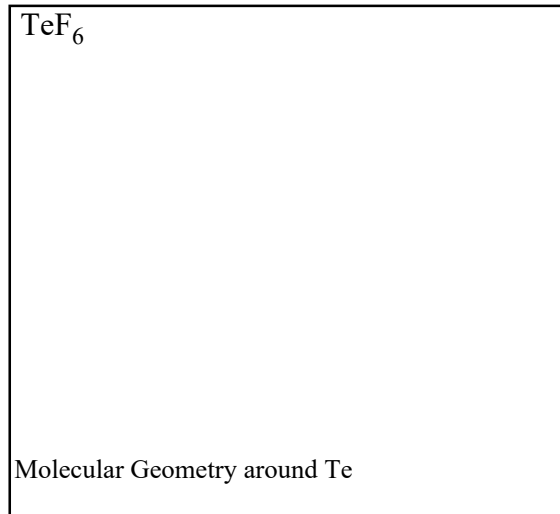
P-Cl

P-I

P-Br

**Question 5** **12 points**

a. 9 points For the molecules HClO_4 and TeF_6 , draw the most stable *Lewis structure* showing appropriate shape and *dashed/wedged bonds*, based on VSEPR. Determine the **molecular geometry** around the central atom. For HClO_4 , label the *most polar bond*.



b. 3 points (Note: Lone pairs are not shown in structures below)

(i) For the three *resonance* structures (1, 2 and 3) shown below, answer the following question:



(I) Circle the *most contributing* resonance structure.

Structure 1

Structure 2

Structure 3

(ii) For the three structures given (4, 5, and 6) below, answer the following questions:



(I) Circle all the structures that represent *resonance structures* of each other:

Structure 4

Structure 5

Structure 6

(II) Among the resonance structures, circle the *most contributing* resonance structure.

Structure 4

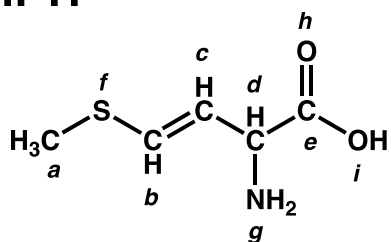
Structure 5

Structure 6

**Question 6** **6 points**

Using the data given below, calculate the **lattice energy** for SrO(s).

	ΔH° (kJ/mol)
$\text{Sr}(s) \longrightarrow \text{Sr}(g)$	+180
$\text{Sr}(g) \longrightarrow \text{Sr}^{2+}(g) + 2e^-$	+1468
$\text{O}_2(g) \longrightarrow 2\text{O}(g)$	+500
$\text{O}(g) + 2e^- \longrightarrow \text{O}^{2-}(g)$	+600
$\text{Sr}(s) + \frac{1}{2}\text{O}_2(g) \longrightarrow \text{SrO}(s)$	-550

**Question 7 10 points**

Consider the molecule given on the left, and answer questions (i) to (vi) using chemical bonding theories that we have learnt in class (valence bond theory, hybridization).

Note that the lone pairs are not shown in the structure:

i. Consider the carbon atoms marked in as ***a***, ***b***, ***c***, ***d***, and ***e*** in the molecule, and determine **hybridization** at each of the carbon centers. (2.5 points)

Atom	Hybridization
Carbon (<i>a</i>)	
Carbon (<i>b</i>)	
Carbon (<i>c</i>)	
Carbon (<i>d</i>)	
Carbon (<i>e</i>)	

ii. Now, consider the sulfur atom (S) marked as ***f***, nitrogen (N) atom (***g***), and oxygen(O) atoms (***h*** and ***i***), and determine their **hybridization**. (2 points)

Atom	Hybridization
Sulfur (<i>f</i>)	
Nitrogen (<i>g</i>)	
Oxygen (<i>h</i>)	
Oxygen (<i>i</i>)	

iii. **Describe** all the bonds formed by the carbon atoms ***a*** and ***b***, using sigma and pi designation as appropriate. (2 points)

Atom	Bonds formed (use sigma and pi as appropriate)
Carbon (<i>a</i>)	
Carbon (<i>b</i>)	

iv. How many total sp^3 -s sigma bonds are in the molecule? (1 point)

v. Determine **electronic** and **molecular geometry** at carbon (C) labeled ***e*** and sulfur (S) labeled ***f*** centres. (2 points)

Atom	Electronic Geometry	Molecular Geometry
Carbon (<i>e</i>)		
Sulfur (<i>f</i>)		

vi. Circle the value from the choices given below, that describes the **C-S-C bond angle at sulfur** (S) marked ***f***: (0.5 points)

>90° to <109.5° 109.5° >109.5 to <120° 120° >120° to <180°

**Question 8 12 points**

Using the **valence shell orbitals only** draw a **complete** molecular orbital (MO) diagram (with no s-p mixing) for the molecule **PS**, in the space provided below. **(6 points)**

Assume that 3s and 3p atomic orbitals will form molecular orbitals like those formed by the combination of 2s and 2p atomic orbitals that we learnt in class, and there is no s-p mixing.

Show **energy arrow**, **atomic orbitals** involved, **atom labels**, the **molecular orbitals** that are formed, **filled electrons** in the MO diagram etc. Make sure to label all the atomic and molecular orbitals.

Using the MO diagram above, answer the following questions:

- (i) Which one of the following will have the highest bond order? Circle the correct choice
A) PS^{1-} B) PS C) PS^{1+}
- (ii) Which one of the following will have the longest bond length? Circle the correct choice
A) PS^{1-} B) PS C) PS^{1+}
- (iii) Which of the following will be diamagnetic? Circle the correct choice.
A) PS^{1-} B) PS C) PS^{1+}
- (iv) Draw Lewis dot structure of the molecule with the shortest bond length (based on the MO diagram) from among the three (PS^{1-} , PS, PS^{1+}), and indicate the bond order calculated from the Lewis dot structure. **(3 points)**

Lewis Dot Structure

Bond Order



A8BB8CF8-297C-49F3-BB06-C6A5BAEC260F

midterm-exam-93388

#822 Page 10 of 10

SCRAP PAGE

**CHEMISTRY 110 –Midterm Exam****Thursday November 7, 2024, 6:30 – 8:30 pm**

Examiners: Pallavi Sirjoosingh and Ashok Kakkar

First name (please write as legibly as possible within the boxes)

--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--

Last name

--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--

Student ID number

--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--

This booklet has 8 questions on 10 pages (The last page can be used as a scrap page).

The staple must NOT be removed from the question booklet. Your intact question booklet must be returned at the end of the exam.

A separate booklet contains a periodic table, and datasheet.

Write your answers directly under the questions in the space provided. Do not write on the QR codes or near the top edges (as it may be cut off).

Please write **legibly and dark enough** for the scan to be readable. For students who use English as a second language, dictionaries are permitted. Calculators are allowed.

DO NOT ASK QUESTIONS OF THE INVIGILATORS about interpretations of the exam questions. If you have a concern about a question, please summarize it at the top of this page. All concerns will be evaluated after the exam. GOOD LUCK!

**Question 1 8 points** (*Assume all transitions are possible*)

a. 6 points Calculate the **energy (in kJ/mol)** required to ionize (completely remove) the electron from the **n=3** level of **He⁺ ion** (He: Helium). Show your work including equations. (3 sig. figs.)

b. 2 points

(I) Select the largest (longest) wavelength from the three choices below:

- i. Wavelength of photon to *completely remove* an electron from *1s orbital of Hydrogen atom*
- ii. Wavelength of photon to *completely remove* an electron from *1s orbital of Boron atom (B)*
- iii. Wavelength of photon to *completely remove* an electron from *1s orbital of Boron cation (B⁺)*

i or ii or iii

(II) Select the smallest (shortest) wavelength from the three choices below (*for H atom*):

- i. Wavelength of photon *emitted* for an electron transitioning from *2p to 1s orbital*
- ii. Wavelength of photon *absorbed* for an electron transitioning from *2s to 4p orbital*
- iii. Wavelength of photon *absorbed* for an electron transitioning from *1s to 3p orbital*

i or ii or iii



Question 2 8 points



A Chem110 student plots the graph shown on the left, from data obtained for the kinetic energy (eV) of the ejected electron versus frequency (s^{-1}) of the incident light, for an unknown metal. ($1\text{eV} = 1.6022 \times 10^{-19} \text{ J}$)

Based on the plot, answer the questions **a to c** below:

a. 1 point On the plot above, label the **threshold frequency**.

b. 2 points What is the value of the work potential (in J) of the unknown metal?

c. 3 points What is the *longest* wavelength (in nm) required to eject an electron from this metal's surface? Show your work including equation(s) used. (3 sig figs)

d. 2 points In the space provided below, plot the following

(Assume frequency is *above* threshold frequency):

- i. *Kinetic energy* of an ejected electron versus the *intensity* of the incident light (left)
- ii. *Number of electrons* ejected versus the *frequency* of the incident light (right)



**Question 3** **6 points****a. 2 points**

In the ground state of Ge (Germanium; $Z = 32$) atom, **name** the following orbitals:

i. Orbital containing the electron that requires the maximum energy to be removed.

ii. Orbital containing the electron that experiences the minimum effective nuclear charge.

b. 2 points

An unknown anion (A^-) has the following *excited* electronic configuration: $[Ar] 4s^2 3d^3 4p^1$

(Ar: Argon; $Z = 18$)

Give the **condensed ground state electronic configuration** of the atom, **A** and the cation, **A^{2+}** .
Show your work. (Assume all electronic transitions are possible)

Electronic Configuration of A:

Electronic Configuration of A^{2+} :

c. 2 points

In a single atom, determine how many electrons can have the following set of quantum numbers?

(i) $n = 5; l = 3$

In a single atom, determine how many electrons can have the following set of quantum numbers?

(i) $n = 3; l = 2; m_s = +1/2$

**Question 4** **8 points**

Rank the following from 1 to 4 in **increasing** order. Provide the ranking in the boxes provided.

Note: Atomic number (Z) are provided for relevant atoms

a) Increasing (lowest to highest) order of *ionization energy*: (1= lowest; 4 = highest)

Mg

Mg: Z = 12

Rb

Rb: Z = 37

Cs

Cs: Z = 55

K

K : Z = 19

b) Increasing (lowest to highest) order of *bond angle*: (1= lowest; 4 = highest)

H₂OCCl₄XeF₄

HCN

c) Increasing (smallest to largest) order of *ionic size* (1= smallest; 4 = largest)

S²⁻

S: Z = 16

O²⁻

O: Z = 8

Be²⁺

Be: Z = 4

Na⁺

Na: Z = 11

d) Increasing (lowest to highest) order of *bond energy* (1= lowest; 4 = highest)

P-F

P-Cl

P-I

P-Br

**Question 5** **12 points**

a. 9 points For the molecules HClO_4 and TeF_6 , draw the most stable *Lewis structure* showing appropriate shape and *dashed/wedged bonds*, based on VSEPR. Determine the **molecular geometry** around the central atom. For HClO_4 , label the *most polar bond*.



b. 3 points (Note: Lone pairs are not shown in structures below)

(i) For the three *resonance* structures (1, 2 and 3) shown below, answer the following question:



(I) Circle the *most contributing* resonance structure.

Structure 1

Structure 2

Structure 3

(ii) For the three structures given (4, 5, and 6) below, answer the following questions:



(I) Circle all the structures that represent *resonance structures* of each other:

Structure 4

Structure 5

Structure 6

(II) Among the resonance structures, circle the *most contributing* resonance structure.

Structure 4

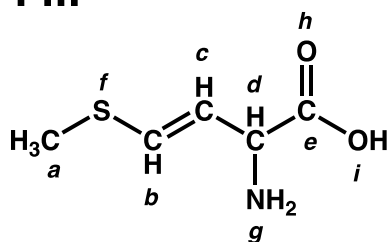
Structure 5

Structure 6

**Question 6** **6 points**

Using the data given below, calculate the **lattice energy** for SrO(s).

	ΔH° (kJ/mol)
$\text{Sr}(s) \longrightarrow \text{Sr}(g)$	+180
$\text{Sr}(g) \longrightarrow \text{Sr}^{2+}(g) + 2e^-$	+1468
$\text{O}_2(g) \longrightarrow 2\text{O}(g)$	+500
$\text{O}(g) + 2e^- \longrightarrow \text{O}^{2-}(g)$	+600
$\text{Sr}(s) + \frac{1}{2}\text{O}_2(g) \longrightarrow \text{SrO}(s)$	-550

**Question 7 10 points**

Consider the molecule given on the left, and answer questions (i) to (vi) using chemical bonding theories that we have learnt in class (valence bond theory, hybridization).

Note that the lone pairs are not shown in the structure:

i. Consider the carbon atoms marked in as ***a***, ***b***, ***c***, ***d***, and ***e*** in the molecule, and determine **hybridization** at each of the carbon centers. (2.5 points)

Atom	Hybridization
Carbon (<i>a</i>)	
Carbon (<i>b</i>)	
Carbon (<i>c</i>)	
Carbon (<i>d</i>)	
Carbon (<i>e</i>)	

ii. Now, consider the sulfur atom (S) marked as ***f***, nitrogen (N) atom (***g***), and oxygen(O) atoms (***h*** and ***i***), and determine their **hybridization**. (2 points)

Atom	Hybridization
Sulfur (<i>f</i>)	
Nitrogen (<i>g</i>)	
Oxygen (<i>h</i>)	
Oxygen (<i>i</i>)	

iii. **Describe** all the bonds formed by the carbon atoms ***a*** and ***b***, using sigma and pi designation as appropriate. (2 points)

Atom	Bonds formed (use sigma and pi as appropriate)
Carbon (<i>a</i>)	
Carbon (<i>b</i>)	

iv. How many total sp^3 -s sigma bonds are in the molecule? (1 point)

v. Determine **electronic** and **molecular geometry** at carbon (C) labeled ***e*** and sulfur (S) labeled ***f*** centres. (2 points)

Atom	Electronic Geometry	Molecular Geometry
Carbon (<i>e</i>)		
Sulfur (<i>f</i>)		

vi. Circle the value from the choices given below, that describes the **C-S-C bond angle at sulfur** (S) marked ***f***: (0.5 points)

>90° to <109.5° 109.5° >109.5 to <120° 120° >120° to <180°


Question 8 12 points

Using the **valence shell orbitals only** draw a **complete** molecular orbital (MO) diagram (with no s-p mixing) for the molecule **PS**, in the space provided below. **(6 points)**

Assume that 3s and 3p atomic orbitals will form molecular orbitals like those formed by the combination of 2s and 2p atomic orbitals that we learnt in class, and there is no s-p mixing.

Show **energy arrow**, **atomic orbitals** involved, **atom labels**, the **molecular orbitals** that are formed, **filled electrons** in the MO diagram etc. Make sure to label all the atomic and molecular orbitals.

Using the MO diagram above, answer the following questions:

- (i) Which one of the following will have the highest bond order? Circle the correct choice
 A) PS^{1-} B) PS C) PS^{1+}
- (ii) Which one of the following will have the longest bond length? Circle the correct choice
 A) PS^{1-} B) PS C) PS^{1+}
- (iii) Which of the following will be diamagnetic? Circle the correct choice.
 A) PS^{1-} B) PS C) PS^{1+}
- (iv) Draw Lewis dot structure of the molecule with the shortest bond length (based on the MO diagram) from among the three (PS^{1-} , PS, PS^{1+}), and indicate the bond order calculated from the Lewis dot structure. **(3 points)**

Lewis Dot Structure

Bond Order



BBEA6A35-84A2-4B4A-9A82-4724F9A5CC0D

midterm-exam-93388

#823 Page 10 of 10

SCRAP PAGE

**CHEMISTRY 110 –Midterm Exam****Thursday November 7, 2024, 6:30 – 8:30 pm****Examiners: Pallavi Sirjoosingh and Ashok Kakkar**

First name (please write as legibly as possible within the boxes)

--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--

Last name

--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--

Student ID number

--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--

This booklet has 8 questions on 10 pages (The last page can be used as a scrap page).

The staple must NOT be removed from the question booklet. Your intact question booklet must be returned at the end of the exam.

A separate booklet contains a periodic table, and datasheet.

Write your answers directly under the questions in the space provided. Do not write on the QR codes or near the top edges (as it may be cut off).

Please write **legibly and dark enough** for the scan to be readable. For students who use English as a second language, dictionaries are permitted. Calculators are allowed.

DO NOT ASK QUESTIONS OF THE INVIGILATORS about interpretations of the exam questions. If you have a concern about a question, please summarize it at the top of this page. All concerns will be evaluated after the exam. GOOD LUCK!

**Question 1 8 points** (*Assume all transitions are possible*)

a. 6 points Calculate the **energy (in kJ/mol)** required to ionize (completely remove) the electron from the **$n=3$** level of **He^+ ion** (He: Helium). Show your work including equations. (3 sig. figs.)

b. 2 points

(I) Select the largest (longest) wavelength from the three choices below:

- i. Wavelength of photon to *completely remove* an electron from *1s orbital of Hydrogen atom*
- ii. Wavelength of photon to *completely remove* an electron from *1s orbital of Boron atom (B)*
- iii. Wavelength of photon to *completely remove* an electron from *1s orbital of Boron cation (B^+)*

i or ii or iii

(II) Select the smallest (shortest) wavelength from the three choices below (*for H atom*):

- i. Wavelength of photon *emitted* for an electron transitioning from *2p to 1s orbital*
- ii. Wavelength of photon *absorbed* for an electron transitioning from *2s to 4p orbital*
- iii. Wavelength of photon *absorbed* for an electron transitioning from *1s to 3p orbital*

i or ii or iii



Question 2 8 points



A Chem110 student plots the graph shown on the left, from data obtained for the kinetic energy (eV) of the ejected electron versus frequency (s^{-1}) of the incident light, for an unknown metal. ($1\text{eV} = 1.6022 \times 10^{-19} \text{ J}$)

Based on the plot, answer the questions **a to c** below:

a. 1 point On the plot above, label the **threshold frequency**.

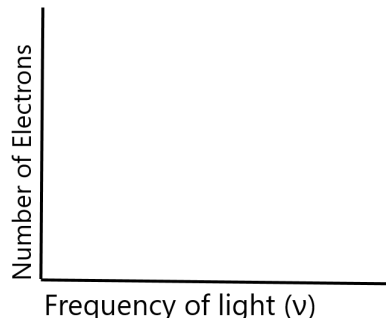
b. 2 points What is the value of the work potential (in J) of the unknown metal?

c. 3 points What is the *longest* wavelength (in nm) required to eject an electron from this metal's surface? Show your work including equation(s) used. (3 sig figs)

d. 2 points In the space provided below, plot the following

(Assume frequency is *above* threshold frequency):

- i. *Kinetic energy* of an ejected electron versus the *intensity* of the incident light (left)
- ii. *Number of electrons* ejected versus the *frequency* of the incident light (right)



**Question 3** **6 points****a. 2 points**

In the ground state of Ge (Germanium; $Z = 32$) atom, **name** the following orbitals:

i. Orbital containing the electron that requires the maximum energy to be removed.

ii. Orbital containing the electron that experiences the minimum effective nuclear charge.

b. 2 points

An unknown anion (A^-) has the following *excited* electronic configuration: $[Ar] 4s^2 3d^3 4p^1$

(Ar: Argon; $Z = 18$)

Give the **condensed ground state electronic configuration** of the atom, **A** and the cation, **A^{2+}** .
Show your work. (Assume all electronic transitions are possible)

Electronic Configuration of A:

Electronic Configuration of A^{2+} :

c. 2 points

In a single atom, determine how many electrons can have the following set of quantum numbers?

(i) $n = 5; l = 3$

In a single atom, determine how many electrons can have the following set of quantum numbers?

(i) $n = 3; l = 2; m_s = +1/2$

**Question 4** **8 points**

Rank the following from 1 to 4 in **increasing** order. Provide the ranking in the boxes provided.

Note: Atomic number (Z) are provided for relevant atoms

a) Increasing (lowest to highest) order of *ionization energy*: (1= lowest; 4 = highest)

Mg

Mg: Z = 12

Rb

Rb: Z = 37

Cs

Cs: Z = 55

K

K : Z = 19

b) Increasing (lowest to highest) order of *bond angle*: (1= lowest; 4 = highest)

H₂OCCl₄XeF₄

HCN

c) Increasing (smallest to largest) order of *ionic size* (1= smallest; 4 = largest)

S²⁻

S: Z = 16

O²⁻

O: Z = 8

Be²⁺

Be: Z = 4

Na⁺

Na: Z = 11

d) Increasing (lowest to highest) order of *bond energy* (1= lowest; 4 = highest)

P-F

P-Cl

P-I

P-Br

**Question 5** **12 points**

a. 9 points For the molecules HClO_4 and TeF_6 , draw the most stable *Lewis structure* showing appropriate shape and *dashed/wedged bonds*, based on VSEPR. Determine the **molecular geometry** around the central atom. For HClO_4 , label the *most polar bond*.



b. 3 points (Note: Lone pairs are not shown in structures below)

(i) For the three *resonance* structures (1, 2 and 3) shown below, answer the following question:



(I) Circle the *most contributing* resonance structure.

Structure 1

Structure 2

Structure 3

(ii) For the three structures given (4, 5, and 6) below, answer the following questions:



(I) Circle all the structures that represent *resonance structures* of each other:

Structure 4

Structure 5

Structure 6

(II) Among the resonance structures, circle the *most contributing* resonance structure.

Structure 4

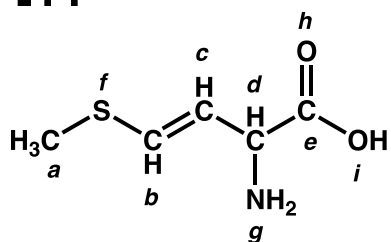
Structure 5

Structure 6

**Question 6** **6 points**

Using the data given below, calculate the **lattice energy** for SrO(s).

	ΔH° (kJ/mol)
$\text{Sr}(s) \longrightarrow \text{Sr}(g)$	+180
$\text{Sr}(g) \longrightarrow \text{Sr}^{2+}(g) + 2e^-$	+1468
$\text{O}_2(g) \longrightarrow 2\text{O}(g)$	+500
$\text{O}(g) + 2e^- \longrightarrow \text{O}^{2-}(g)$	+600
$\text{Sr}(s) + \frac{1}{2}\text{O}_2(g) \longrightarrow \text{SrO}(s)$	-550

**Question 7 10 points**

Consider the molecule given on the left, and answer questions (i) to (vi) using chemical bonding theories that we have learnt in class (valence bond theory, hybridization).

Note that the lone pairs are not shown in the structure:

i. Consider the carbon atoms marked in as ***a***, ***b***, ***c***, ***d***, and ***e*** in the molecule, and determine **hybridization** at each of the carbon centers. (2.5 points)

Atom	Hybridization
Carbon (<i>a</i>)	
Carbon (<i>b</i>)	
Carbon (<i>c</i>)	
Carbon (<i>d</i>)	
Carbon (<i>e</i>)	

ii. Now, consider the sulfur atom (S) marked as ***f***, nitrogen (N) atom (***g***), and oxygen(O) atoms (***h*** and ***i***), and determine their **hybridization**. (2 points)

Atom	Hybridization
Sulfur (<i>f</i>)	
Nitrogen (<i>g</i>)	
Oxygen (<i>h</i>)	
Oxygen (<i>i</i>)	

iii. **Describe** all the bonds formed by the carbon atoms ***a*** and ***b***, using sigma and pi designation as appropriate. (2 points)

Atom	Bonds formed (use sigma and pi as appropriate)
Carbon (<i>a</i>)	
Carbon (<i>b</i>)	

iv. How many total sp^3 -s sigma bonds are in the molecule? (1 point)

v. Determine **electronic** and **molecular geometry** at carbon (C) labeled ***e*** and sulfur (S) labeled ***f*** centres. (2 points)

Atom	Electronic Geometry	Molecular Geometry
Carbon (<i>e</i>)		
Sulfur (<i>f</i>)		

vi. Circle the value from the choices given below, that describes the **C-S-C bond angle at sulfur** (S) marked ***f***: (0.5 points)

>90° to <109.5° 109.5° >109.5 to <120° 120° >120° to <180°


Question 8 12 points

Using the **valence shell orbitals only** draw a **complete** molecular orbital (MO) diagram (with no s-p mixing) for the molecule **PS**, in the space provided below. **(6 points)**

Assume that 3s and 3p atomic orbitals will form molecular orbitals like those formed by the combination of 2s and 2p atomic orbitals that we learnt in class, and there is no s-p mixing.

Show **energy arrow**, **atomic orbitals** involved, **atom labels**, the **molecular orbitals** that are formed, **filled electrons** in the MO diagram etc. Make sure to label all the atomic and molecular orbitals.

Using the MO diagram above, answer the following questions:

- (i) Which one of the following will have the highest bond order? Circle the correct choice
 A) PS^{1-} B) PS C) PS^{1+}
- (ii) Which one of the following will have the longest bond length? Circle the correct choice
 A) PS^{1-} B) PS C) PS^{1+}
- (iii) Which of the following will be diamagnetic? Circle the correct choice.
 A) PS^{1-} B) PS C) PS^{1+}
- (iv) Draw Lewis dot structure of the molecule with the shortest bond length (based on the MO diagram) from among the three (PS^{1-} , PS, PS^{1+}), and indicate the bond order calculated from the Lewis dot structure. **(3 points)**

Lewis Dot Structure

Bond Order



DCF8D1D0-7D08-4F4C-B209-F449039A1B1E

midterm-exam-93388

#824 Page 10 of 10

SCRAP PAGE

**CHEMISTRY 110 –Midterm Exam****Thursday November 7, 2024, 6:30 – 8:30 pm****Examiners: Pallavi Sirjoosingh and Ashok Kakkar**

First name (please write as legibly as possible within the boxes)

--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--

Last name

--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--

Student ID number

--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--

This booklet has 8 questions on 10 pages (The last page can be used as a scrap page).

The staple must NOT be removed from the question booklet. Your intact question booklet must be returned at the end of the exam.

A separate booklet contains a periodic table, and datasheet.

Write your answers directly under the questions in the space provided. Do not write on the QR codes or near the top edges (as it may be cut off).

Please write **legibly and dark enough** for the scan to be readable. For students who use English as a second language, dictionaries are permitted. Calculators are allowed.

DO NOT ASK QUESTIONS OF THE INVIGILATORS about interpretations of the exam questions. If you have a concern about a question, please summarize it at the top of this page. All concerns will be evaluated after the exam. GOOD LUCK!

**Question 1 8 points** (*Assume all transitions are possible*)

a. 6 points Calculate the **energy (in kJ/mol)** required to ionize (completely remove) the electron from the **n=3** level of **He⁺ ion** (He: Helium). Show your work including equations. (3 sig. figs.)

b. 2 points

(I) Select the largest (longest) wavelength from the three choices below:

- i. Wavelength of photon to *completely remove* an electron from *1s orbital of Hydrogen atom*
- ii. Wavelength of photon to *completely remove* an electron from *1s orbital of Boron atom (B)*
- iii. Wavelength of photon to *completely remove* an electron from *1s orbital of Boron cation (B⁺)*

i or ii or iii

(II) Select the smallest (shortest) wavelength from the three choices below (*for H atom*):

- i. Wavelength of photon *emitted* for an electron transitioning from *2p to 1s orbital*
- ii. Wavelength of photon *absorbed* for an electron transitioning from *2s to 4p orbital*
- iii. Wavelength of photon *absorbed* for an electron transitioning from *1s to 3p orbital*

i or ii or iii



Question 2 8 points



A Chem110 student plots the graph shown on the left, from data obtained for the kinetic energy (eV) of the ejected electron versus frequency (s^{-1}) of the incident light, for an unknown metal. ($1\text{eV} = 1.6022 \times 10^{-19} \text{ J}$)

Based on the plot, answer the questions **a to c** below:

a. 1 point On the plot above, label the **threshold frequency**.

b. 2 points What is the value of the work potential (in J) of the unknown metal?

c. 3 points What is the *longest* wavelength (in nm) required to eject an electron from this metal's surface? Show your work including equation(s) used. (3 sig figs)

d. 2 points In the space provided below, plot the following

(Assume frequency is *above* threshold frequency):

- i. *Kinetic energy* of an ejected electron versus the *intensity* of the incident light (left)
- ii. *Number of electrons* ejected versus the *frequency* of the incident light (right)



**Question 3** **6 points****a. 2 points**

In the ground state of Ge (Germanium; $Z = 32$) atom, **name** the following orbitals:

i. Orbital containing the electron that requires the maximum energy to be removed.

ii. Orbital containing the electron that experiences the minimum effective nuclear charge.

b. 2 points

An unknown anion (A^-) has the following *excited* electronic configuration: $[Ar] 4s^2 3d^3 4p^1$

(Ar: Argon; $Z = 18$)

Give the **condensed ground state electronic configuration** of the atom, **A** and the cation, **A^{2+}** .
Show your work. (Assume all electronic transitions are possible)

Electronic Configuration of A:

Electronic Configuration of A^{2+} :

c. 2 points

In a single atom, determine how many electrons can have the following set of quantum numbers?

(i) $n = 5; l = 3$

In a single atom, determine how many electrons can have the following set of quantum numbers?

(i) $n = 3; l = 2; m_s = +1/2$

**Question 4** **8 points**

Rank the following from 1 to 4 in **increasing** order. Provide the ranking in the boxes provided.

Note: Atomic number (Z) are provided for relevant atoms

a) Increasing (lowest to highest) order of *ionization energy*: (1= lowest; 4 = highest)

Mg

Mg: Z = 12

Rb

Rb: Z = 37

Cs

Cs: Z = 55

K

K : Z = 19

b) Increasing (lowest to highest) order of *bond angle*: (1= lowest; 4 = highest)

H₂OCCl₄XeF₄

HCN

c) Increasing (smallest to largest) order of *ionic size* (1= smallest; 4 = largest)

S²⁻

S: Z = 16

O²⁻

O: Z = 8

Be²⁺

Be: Z = 4

Na⁺

Na: Z = 11

d) Increasing (lowest to highest) order of *bond energy* (1= lowest; 4 = highest)

P-F

P-Cl

P-I

P-Br

**Question 5** **12 points**

a. 9 points For the molecules HClO_4 and TeF_6 , draw the most stable *Lewis structure* showing appropriate shape and *dashed/wedged bonds*, based on VSEPR. Determine the **molecular geometry** around the central atom. For HClO_4 , label the *most polar bond*.



b. 3 points (Note: Lone pairs are not shown in structures below)

(i) For the three *resonance* structures (1, 2 and 3) shown below, answer the following question:



(I) Circle the *most contributing* resonance structure.

Structure 1

Structure 2

Structure 3

(ii) For the three structures given (4, 5, and 6) below, answer the following questions:



(I) Circle all the structures that represent *resonance structures* of each other:

Structure 4

Structure 5

Structure 6

(II) Among the resonance structures, circle the *most contributing* resonance structure.

Structure 4

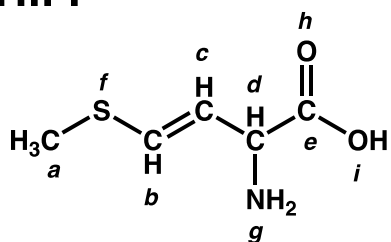
Structure 5

Structure 6

**Question 6** **6 points**

Using the data given below, calculate the **lattice energy** for SrO(s).

	ΔH° (kJ/mol)
$\text{Sr}(s) \longrightarrow \text{Sr}(g)$	+180
$\text{Sr}(g) \longrightarrow \text{Sr}^{2+}(g) + 2e^-$	+1468
$\text{O}_2(g) \longrightarrow 2\text{O}(g)$	+500
$\text{O}(g) + 2e^- \longrightarrow \text{O}^{2-}(g)$	+600
$\text{Sr}(s) + \frac{1}{2}\text{O}_2(g) \longrightarrow \text{SrO}(s)$	-550

**Question 7 10 points**

Consider the molecule given on the left, and answer questions (i) to (vi) using chemical bonding theories that we have learnt in class (valence bond theory, hybridization).

Note that the lone pairs are not shown in the structure:

i. Consider the carbon atoms marked in as ***a***, ***b***, ***c***, ***d***, and ***e*** in the molecule, and determine **hybridization** at each of the carbon centers. (2.5 points)

Atom	Hybridization
Carbon (<i>a</i>)	
Carbon (<i>b</i>)	
Carbon (<i>c</i>)	
Carbon (<i>d</i>)	
Carbon (<i>e</i>)	

ii. Now, consider the sulfur atom (S) marked as ***f***, nitrogen (N) atom (***g***), and oxygen(O) atoms (***h*** and ***i***), and determine their **hybridization**. (2 points)

Atom	Hybridization
Sulfur (<i>f</i>)	
Nitrogen (<i>g</i>)	
Oxygen (<i>h</i>)	
Oxygen (<i>i</i>)	

iii. **Describe** all the bonds formed by the carbon atoms ***a*** and ***b***, using sigma and pi designation as appropriate. (2 points)

Atom	Bonds formed (use sigma and pi as appropriate)
Carbon (<i>a</i>)	
Carbon (<i>b</i>)	

iv. How many total sp^3 -s sigma bonds are in the molecule? (1 point)

v. Determine **electronic** and **molecular geometry** at carbon (C) labeled ***e*** and sulfur (S) labeled ***f*** centres. (2 points)

Atom	Electronic Geometry	Molecular Geometry
Carbon (<i>e</i>)		
Sulfur (<i>f</i>)		

vi. Circle the value from the choices given below, that describes the **C-S-C bond angle at sulfur** (S) marked ***f***: (0.5 points)

>90° to <109.5° 109.5° >109.5 to <120° 120° >120° to <180°


Question 8 12 points

Using the **valence shell orbitals only** draw a **complete** molecular orbital (MO) diagram (with no s-p mixing) for the molecule **PS**, in the space provided below. **(6 points)**

Assume that 3s and 3p atomic orbitals will form molecular orbitals like those formed by the combination of 2s and 2p atomic orbitals that we learnt in class, and there is no s-p mixing.

Show **energy arrow**, **atomic orbitals** involved, **atom labels**, the **molecular orbitals** that are formed, **filled electrons** in the MO diagram etc. Make sure to label all the atomic and molecular orbitals.

Using the MO diagram above, answer the following questions:

- (i) Which one of the following will have the highest bond order? Circle the correct choice
 A) PS^{1-} B) PS C) PS^{1+}
- (ii) Which one of the following will have the longest bond length? Circle the correct choice
 A) PS^{1-} B) PS C) PS^{1+}
- (iii) Which of the following will be diamagnetic? Circle the correct choice.
 A) PS^{1-} B) PS C) PS^{1+}
- (iv) Draw Lewis dot structure of the molecule with the shortest bond length (based on the MO diagram) from among the three (PS^{1-} , PS, PS^{1+}), and indicate the bond order calculated from the Lewis dot structure. **(3 points)**

Lewis Dot Structure

Bond Order



03F72EEE-B7F9-4B33-A7B6-AAF5DBCAA7A8

midterm-exam-93388

#825 Page 10 of 10

SCRAP PAGE

**CHEMISTRY 110 –Midterm Exam****Thursday November 7, 2024, 6:30 – 8:30 pm****Examiners: Pallavi Sirjoosingh and Ashok Kakkar**

First name (please write as legibly as possible within the boxes)

--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--

Last name

--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--

Student ID number

--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--

This booklet has 8 questions on 10 pages (The last page can be used as a scrap page).

The staple must NOT be removed from the question booklet. Your intact question booklet must be returned at the end of the exam.

A separate booklet contains a periodic table, and datasheet.

Write your answers directly under the questions in the space provided. Do not write on the QR codes or near the top edges (as it may be cut off).

Please write **legibly and dark enough** for the scan to be readable. For students who use English as a second language, dictionaries are permitted. Calculators are allowed.

DO NOT ASK QUESTIONS OF THE INVIGILATORS about interpretations of the exam questions. If you have a concern about a question, please summarize it at the top of this page. All concerns will be evaluated after the exam. GOOD LUCK!

**Question 1 8 points** (*Assume all transitions are possible*)

a. 6 points Calculate the **energy (in kJ/mol)** required to ionize (completely remove) the electron from the **n=3** level of **He⁺ ion** (He: Helium). Show your work including equations. (3 sig. figs.)

b. 2 points

(I) Select the largest (longest) wavelength from the three choices below:

- i. Wavelength of photon to *completely remove* an electron from *1s orbital of Hydrogen atom*
- ii. Wavelength of photon to *completely remove* an electron from *1s orbital of Boron atom (B)*
- iii. Wavelength of photon to *completely remove* an electron from *1s orbital of Boron cation (B⁺)*

i or ii or iii

(II) Select the smallest (shortest) wavelength from the three choices below (*for H atom*):

- i. Wavelength of photon *emitted* for an electron transitioning from *2p to 1s orbital*
- ii. Wavelength of photon *absorbed* for an electron transitioning from *2s to 4p orbital*
- iii. Wavelength of photon *absorbed* for an electron transitioning from *1s to 3p orbital*

i or ii or iii



Question 2 8 points



A Chem110 student plots the graph shown on the left, from data obtained for the kinetic energy (eV) of the ejected electron versus frequency (s^{-1}) of the incident light, for an unknown metal. ($1\text{eV} = 1.6022 \times 10^{-19} \text{ J}$)

Based on the plot, answer the questions **a to c** below:

a. 1 point On the plot above, label the **threshold frequency**.

b. 2 points What is the value of the work potential (in J) of the unknown metal?

c. 3 points What is the *longest* wavelength (in nm) required to eject an electron from this metal's surface? Show your work including equation(s) used. (3 sig figs)

d. 2 points In the space provided below, plot the following

(Assume frequency is *above* threshold frequency):

- i. *Kinetic energy* of an ejected electron versus the *intensity* of the incident light (left)
- ii. *Number of electrons* ejected versus the *frequency* of the incident light (right)



**Question 3** **6 points****a. 2 points**

In the ground state of Ge (Germanium; $Z = 32$) atom, **name** the following orbitals:

i. Orbital containing the electron that requires the maximum energy to be removed.

ii. Orbital containing the electron that experiences the minimum effective nuclear charge.

b. 2 points

An unknown anion (A^-) has the following *excited* electronic configuration: $[Ar] 4s^2 3d^3 4p^1$

(Ar: Argon; $Z = 18$)

Give the **condensed ground state electronic configuration** of the atom, **A** and the cation, **A^{2+}** .
Show your work. (Assume all electronic transitions are possible)

Electronic Configuration of A:

Electronic Configuration of A^{2+} :

c. 2 points

In a single atom, determine how many electrons can have the following set of quantum numbers?

(i) $n = 5; l = 3$

In a single atom, determine how many electrons can have the following set of quantum numbers?

(i) $n = 3; l = 2; m_s = +1/2$

**Question 4** **8 points**

Rank the following from 1 to 4 in **increasing** order. Provide the ranking in the boxes provided.

Note: Atomic number (Z) are provided for relevant atoms

a) Increasing (lowest to highest) order of *ionization energy*: (1= lowest; 4 = highest)

Mg

Mg: Z = 12

Rb

Rb: Z = 37

Cs

Cs: Z = 55

K

K : Z = 19

b) Increasing (lowest to highest) order of *bond angle*: (1= lowest; 4 = highest)

H₂OCCl₄XeF₄

HCN

c) Increasing (smallest to largest) order of *ionic size* (1= smallest; 4 = largest)

S²⁻

S: Z = 16

O²⁻

O: Z = 8

Be²⁺

Be: Z = 4

Na⁺

Na: Z = 11

d) Increasing (lowest to highest) order of *bond energy* (1= lowest; 4 = highest)

P-F

P-Cl

P-I

P-Br

**Question 5** **12 points**

a. 9 points For the molecules HClO_4 and TeF_6 , draw the most stable *Lewis structure* showing appropriate shape and *dashed/wedged bonds*, based on VSEPR. Determine the **molecular geometry** around the central atom. For HClO_4 , label the *most polar bond*.



b. 3 points (Note: Lone pairs are not shown in structures below)

(i) For the three *resonance* structures (1, 2 and 3) shown below, answer the following question:



(I) Circle the *most contributing* resonance structure.

Structure 1

Structure 2

Structure 3

(ii) For the three structures given (4, 5, and 6) below, answer the following questions:



(I) Circle all the structures that represent *resonance structures* of each other:

Structure 4

Structure 5

Structure 6

(II) Among the resonance structures, circle the *most contributing* resonance structure.

Structure 4

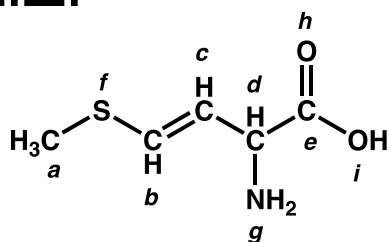
Structure 5

Structure 6

**Question 6** **6 points**

Using the data given below, calculate the **lattice energy** for SrO(s).

	ΔH° (kJ/mol)
$\text{Sr}(s) \longrightarrow \text{Sr}(g)$	+180
$\text{Sr}(g) \longrightarrow \text{Sr}^{2+}(g) + 2e^-$	+1468
$\text{O}_2(g) \longrightarrow 2\text{O}(g)$	+500
$\text{O}(g) + 2e^- \longrightarrow \text{O}^{2-}(g)$	+600
$\text{Sr}(s) + \frac{1}{2}\text{O}_2(g) \longrightarrow \text{SrO}(s)$	-550

**Question 7 10 points**

Consider the molecule given on the left, and answer questions (i) to (vi) using chemical bonding theories that we have learnt in class (valence bond theory, hybridization).

Note that the lone pairs are not shown in the structure:

i. Consider the carbon atoms marked in as ***a***, ***b***, ***c***, ***d***, and ***e*** in the molecule, and determine **hybridization** at each of the carbon centers. (2.5 points)

Atom	Hybridization
Carbon (<i>a</i>)	
Carbon (<i>b</i>)	
Carbon (<i>c</i>)	
Carbon (<i>d</i>)	
Carbon (<i>e</i>)	

ii. Now, consider the sulfur atom (S) marked as ***f***, nitrogen (N) atom (***g***), and oxygen(O) atoms (***h*** and ***i***), and determine their **hybridization**. (2 points)

Atom	Hybridization
Sulfur (<i>f</i>)	
Nitrogen (<i>g</i>)	
Oxygen (<i>h</i>)	
Oxygen (<i>i</i>)	

iii. **Describe** all the bonds formed by the carbon atoms ***a*** and ***b***, using sigma and pi designation as appropriate. (2 points)

Atom	Bonds formed (use sigma and pi as appropriate)
Carbon (<i>a</i>)	
Carbon (<i>b</i>)	

iv. How many total sp^3 -s sigma bonds are in the molecule? (1 point)

v. Determine **electronic** and **molecular geometry** at carbon (C) labeled ***e*** and sulfur (S) labeled ***f*** centres. (2 points)

Atom	Electronic Geometry	Molecular Geometry
Carbon (<i>e</i>)		
Sulfur (<i>f</i>)		

vi. Circle the value from the choices given below, that describes the **C-S-C bond angle at sulfur** (S) marked ***f***: (0.5 points)

$>90^\circ$ to $<109.5^\circ$ 109.5° >109.5 to $<120^\circ$ 120° $>120^\circ$ to $<180^\circ$


Question 8 12 points

Using the **valence shell orbitals only** draw a **complete** molecular orbital (MO) diagram (with no s-p mixing) for the molecule **PS**, in the space provided below. **(6 points)**

Assume that 3s and 3p atomic orbitals will form molecular orbitals like those formed by the combination of 2s and 2p atomic orbitals that we learnt in class, and there is no s-p mixing.

Show **energy arrow**, **atomic orbitals** involved, **atom labels**, the **molecular orbitals** that are formed, **filled electrons** in the MO diagram etc. Make sure to label all the atomic and molecular orbitals.

Using the MO diagram above, answer the following questions:

- (i) Which one of the following will have the highest bond order? Circle the correct choice
 A) PS^{1-} B) PS C) PS^{1+}
- (ii) Which one of the following will have the longest bond length? Circle the correct choice
 A) PS^{1-} B) PS C) PS^{1+}
- (iii) Which of the following will be diamagnetic? Circle the correct choice.
 A) PS^{1-} B) PS C) PS^{1+}
- (iv) Draw Lewis dot structure of the molecule with the shortest bond length (based on the MO diagram) from among the three (PS^{1-} , PS, PS^{1+}), and indicate the bond order calculated from the Lewis dot structure. **(3 points)**

Lewis Dot Structure

Bond Order



7C88D87C-4D70-42C9-B83D-5F007E54BDCC

midterm-exam-93388

#826 Page 10 of 10

SCRAP PAGE

**CHEMISTRY 110 –Midterm Exam****Thursday November 7, 2024, 6:30 – 8:30 pm****Examiners: Pallavi Sirjoosingh and Ashok Kakkar**

First name (please write as legibly as possible within the boxes)

--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--

Last name

--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--

Student ID number

--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--

This booklet has 8 questions on 10 pages (The last page can be used as a scrap page).

The staple must NOT be removed from the question booklet. Your intact question booklet must be returned at the end of the exam.

A separate booklet contains a periodic table, and datasheet.

Write your answers directly under the questions in the space provided. Do not write on the QR codes or near the top edges (as it may be cut off).

Please write **legibly and dark enough** for the scan to be readable. For students who use English as a second language, dictionaries are permitted. Calculators are allowed.

DO NOT ASK QUESTIONS OF THE INVIGILATORS about interpretations of the exam questions. If you have a concern about a question, please summarize it at the top of this page. All concerns will be evaluated after the exam. GOOD LUCK!

**Question 1 8 points** (*Assume all transitions are possible*)

a. 6 points Calculate the **energy (in kJ/mol)** required to ionize (completely remove) the electron from the **n=3** level of **He⁺ ion** (He: Helium). Show your work including equations. (3 sig. figs.)

b. 2 points

(I) Select the largest (longest) wavelength from the three choices below:

- i. Wavelength of photon to *completely remove* an electron from *1s orbital of Hydrogen atom*
- ii. Wavelength of photon to *completely remove* an electron from *1s orbital of Boron atom (B)*
- iii. Wavelength of photon to *completely remove* an electron from *1s orbital of Boron cation (B⁺)*

i or ii or iii

(II) Select the smallest (shortest) wavelength from the three choices below (*for H atom*):

- i. Wavelength of photon *emitted* for an electron transitioning from *2p to 1s orbital*
- ii. Wavelength of photon *absorbed* for an electron transitioning from *2s to 4p orbital*
- iii. Wavelength of photon *absorbed* for an electron transitioning from *1s to 3p orbital*

i or ii or iii



Question 2 8 points



A Chem110 student plots the graph shown on the left, from data obtained for the kinetic energy (eV) of the ejected electron versus frequency (s^{-1}) of the incident light, for an unknown metal. ($1\text{eV} = 1.6022 \times 10^{-19} \text{ J}$)

Based on the plot, answer the questions **a to c** below:

a. 1 point On the plot above, label the **threshold frequency**.

b. 2 points What is the value of the work potential (in J) of the unknown metal?

c. 3 points What is the *longest* wavelength (in nm) required to eject an electron from this metal's surface? Show your work including equation(s) used. (3 sig figs)

d. 2 points In the space provided below, plot the following

(Assume frequency is *above* threshold frequency):

- i. *Kinetic energy* of an ejected electron versus the *intensity* of the incident light (left)
- ii. *Number of electrons* ejected versus the *frequency* of the incident light (right)



**Question 3** **6 points****a. 2 points**

In the ground state of Ge (Germanium; $Z = 32$) atom, **name** the following orbitals:

i. Orbital containing the electron that requires the maximum energy to be removed.

ii. Orbital containing the electron that experiences the minimum effective nuclear charge.

b. 2 points

An unknown anion (A^-) has the following *excited* electronic configuration: $[Ar] 4s^2 3d^3 4p^1$

(Ar: Argon; $Z = 18$)

Give the **condensed ground state electronic configuration** of the atom, **A** and the cation, **A^{2+}** .
Show your work. (Assume all electronic transitions are possible)

Electronic Configuration of A:

Electronic Configuration of A^{2+} :

c. 2 points

In a single atom, determine how many electrons can have the following set of quantum numbers?

(i) $n = 5; l = 3$

In a single atom, determine how many electrons can have the following set of quantum numbers?

(i) $n = 3; l = 2; m_s = +1/2$

**Question 4** **8 points**

Rank the following from 1 to 4 in **increasing** order. Provide the ranking in the boxes provided.

Note: Atomic number (Z) are provided for relevant atoms

a) Increasing (lowest to highest) order of *ionization energy*: (1= lowest; 4 = highest)

Mg

Mg: Z = 12

Rb

Rb: Z = 37

Cs

Cs: Z = 55

K

K : Z = 19

b) Increasing (lowest to highest) order of *bond angle*: (1= lowest; 4 = highest)

H₂OCCl₄XeF₄

HCN

c) Increasing (smallest to largest) order of *ionic size* (1= smallest; 4 = largest)

S²⁻

S: Z = 16

O²⁻

O: Z = 8

Be²⁺

Be: Z = 4

Na⁺

Na: Z = 11

d) Increasing (lowest to highest) order of *bond energy* (1= lowest; 4 = highest)

P-F

P-Cl

P-I

P-Br

**Question 5** **12 points**

a. 9 points For the molecules HClO_4 and TeF_6 , draw the most stable *Lewis structure* showing appropriate shape and *dashed/wedged bonds*, based on VSEPR. Determine the **molecular geometry** around the central atom. For HClO_4 , label the *most polar bond*.



b. 3 points (Note: Lone pairs are not shown in structures below)

(i) For the three *resonance* structures (1, 2 and 3) shown below, answer the following question:



(I) Circle the *most contributing* resonance structure.

Structure 1

Structure 2

Structure 3

(ii) For the three structures given (4, 5, and 6) below, answer the following questions:



(I) Circle all the structures that represent *resonance structures* of each other:

Structure 4

Structure 5

Structure 6

(II) Among the resonance structures, circle the *most contributing* resonance structure.

Structure 4

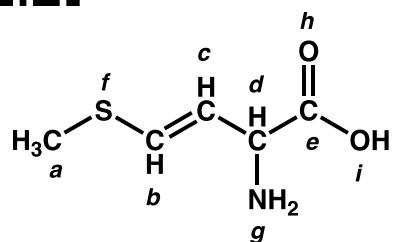
Structure 5

Structure 6

**Question 6** **6 points**

Using the data given below, calculate the **lattice energy** for SrO(s).

	ΔH° (kJ/mol)
$\text{Sr}(s) \longrightarrow \text{Sr}(g)$	+180
$\text{Sr}(g) \longrightarrow \text{Sr}^{2+}(g) + 2e^-$	+1468
$\text{O}_2(g) \longrightarrow 2\text{O}(g)$	+500
$\text{O}(g) + 2e^- \longrightarrow \text{O}^{2-}(g)$	+600
$\text{Sr}(s) + \frac{1}{2}\text{O}_2(g) \longrightarrow \text{SrO}(s)$	-550

**Question 7 10 points**

Consider the molecule given on the left, and answer questions (i) to (vi) using chemical bonding theories that we have learnt in class (valence bond theory, hybridization).

Note that the lone pairs are not shown in the structure:

i. Consider the carbon atoms marked in as ***a***, ***b***, ***c***, ***d***, and ***e*** in the molecule, and determine **hybridization** at each of the carbon centers. (2.5 points)

Atom	Hybridization
Carbon (<i>a</i>)	
Carbon (<i>b</i>)	
Carbon (<i>c</i>)	
Carbon (<i>d</i>)	
Carbon (<i>e</i>)	

ii. Now, consider the sulfur atom (S) marked as ***f***, nitrogen (N) atom (***g***), and oxygen(O) atoms (***h*** and ***i***), and determine their **hybridization**. (2 points)

Atom	Hybridization
Sulfur (<i>f</i>)	
Nitrogen (<i>g</i>)	
Oxygen (<i>h</i>)	
Oxygen (<i>i</i>)	

iii. **Describe** all the bonds formed by the carbon atoms ***a*** and ***b***, using sigma and pi designation as appropriate. (2 points)

Atom	Bonds formed (use sigma and pi as appropriate)
Carbon (<i>a</i>)	
Carbon (<i>b</i>)	

iv. How many total sp^3 -s sigma bonds are in the molecule? (1 point)

v. Determine **electronic** and **molecular geometry** at carbon (C) labeled ***e*** and sulfur (S) labeled ***f*** centres. (2 points)

Atom	Electronic Geometry	Molecular Geometry
Carbon (<i>e</i>)		
Sulfur (<i>f</i>)		

vi. Circle the value from the choices given below, that describes the **C-S-C bond angle at sulfur** (S) marked ***f***: (0.5 points)

>90° to <109.5° 109.5° >109.5 to <120° 120° >120° to <180°

**Question 8 12 points**

Using the **valence shell orbitals only** draw a **complete** molecular orbital (MO) diagram (with no s-p mixing) for the molecule **PS**, in the space provided below. **(6 points)**

Assume that 3s and 3p atomic orbitals will form molecular orbitals like those formed by the combination of 2s and 2p atomic orbitals that we learnt in class, and there is no s-p mixing.

Show **energy arrow**, **atomic orbitals** involved, **atom labels**, the **molecular orbitals** that are formed, **filled electrons** in the MO diagram etc. Make sure to label all the atomic and molecular orbitals.

Using the MO diagram above, answer the following questions:

- (i) Which one of the following will have the highest bond order? Circle the correct choice
A) PS^{1-} B) PS C) PS^{1+}
- (ii) Which one of the following will have the longest bond length? Circle the correct choice
A) PS^{1-} B) PS C) PS^{1+}
- (iii) Which of the following will be diamagnetic? Circle the correct choice.
A) PS^{1-} B) PS C) PS^{1+}
- (iv) Draw Lewis dot structure of the molecule with the shortest bond length (based on the MO diagram) from among the three (PS^{1-} , PS, PS^{1+}), and indicate the bond order calculated from the Lewis dot structure. **(3 points)**

Lewis Dot Structure

Bond Order



A978B3FB-51AA-4570-9D4D-CAE0E9A95C2C

midterm-exam-93388

#827 Page 10 of 10

SCRAP PAGE

**CHEMISTRY 110 –Midterm Exam****Thursday November 7, 2024, 6:30 – 8:30 pm**

Examiners: Pallavi Sirjoosingh and Ashok Kakkar

First name (please write as legibly as possible within the boxes)

--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--

Last name

--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--

Student ID number

--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--

This booklet has 8 questions on 10 pages (The last page can be used as a scrap page).

The staple must NOT be removed from the question booklet. Your intact question booklet must be returned at the end of the exam.

A separate booklet contains a periodic table, and datasheet.

Write your answers directly under the questions in the space provided. Do not write on the QR codes or near the top edges (as it may be cut off).

Please write **legibly and dark enough** for the scan to be readable. For students who use English as a second language, dictionaries are permitted. Calculators are allowed.

DO NOT ASK QUESTIONS OF THE INVIGILATORS about interpretations of the exam questions. If you have a concern about a question, please summarize it at the top of this page. All concerns will be evaluated after the exam. GOOD LUCK!

**Question 1 8 points** (*Assume all transitions are possible*)

a. 6 points Calculate the **energy (in kJ/mol)** required to ionize (completely remove) the electron from the **$n=3$** level of **He^+ ion** (He: Helium). Show your work including equations. (3 sig. figs.)

b. 2 points

(I) Select the largest (longest) wavelength from the three choices below:

- i. Wavelength of photon to *completely remove* an electron from *1s orbital of Hydrogen atom*
- ii. Wavelength of photon to *completely remove* an electron from *1s orbital of Boron atom (B)*
- iii. Wavelength of photon to *completely remove* an electron from *1s orbital of Boron cation (B^+)*

i or ii or iii

(II) Select the smallest (shortest) wavelength from the three choices below (*for H atom*):

- i. Wavelength of photon *emitted* for an electron transitioning from *2p to 1s orbital*
- ii. Wavelength of photon *absorbed* for an electron transitioning from *2s to 4p orbital*
- iii. Wavelength of photon *absorbed* for an electron transitioning from *1s to 3p orbital*

i or ii or iii



Question 2 8 points



A Chem110 student plots the graph shown on the left, from data obtained for the kinetic energy (eV) of the ejected electron versus frequency (s^{-1}) of the incident light, for an unknown metal. ($1\text{eV} = 1.6022 \times 10^{-19} \text{ J}$)

Based on the plot, answer the questions **a to c** below:

a. 1 point On the plot above, label the **threshold frequency**.

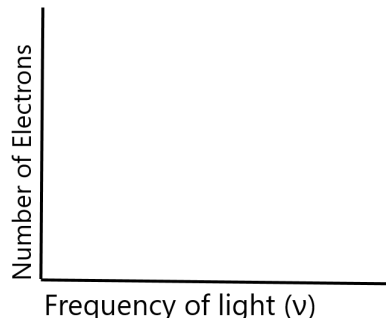
b. 2 points What is the value of the work potential (in J) of the unknown metal?

c. 3 points What is the *longest* wavelength (in nm) required to eject an electron from this metal's surface? Show your work including equation(s) used. (3 sig figs)

d. 2 points In the space provided below, plot the following

(Assume frequency is *above* threshold frequency):

- i. *Kinetic energy* of an ejected electron versus the *intensity* of the incident light (left)
- ii. *Number of electrons* ejected versus the *frequency* of the incident light (right)



**Question 3** **6 points****a. 2 points**

In the ground state of Ge (Germanium; $Z = 32$) atom, **name** the following orbitals:

i. Orbital containing the electron that requires the maximum energy to be removed.

ii. Orbital containing the electron that experiences the minimum effective nuclear charge.

b. 2 points

An unknown anion (A^-) has the following *excited* electronic configuration: $[Ar] 4s^2 3d^3 4p^1$

(Ar: Argon; $Z = 18$)

Give the **condensed ground state electronic configuration** of the atom, **A** and the cation, **A^{2+}** .
Show your work. (Assume all electronic transitions are possible)

Electronic Configuration of A:

Electronic Configuration of A^{2+} :

c. 2 points

In a single atom, determine how many electrons can have the following set of quantum numbers?

(i) $n = 5; l = 3$

In a single atom, determine how many electrons can have the following set of quantum numbers?

(i) $n = 3; l = 2; m_s = +1/2$

**Question 4** **8 points**

Rank the following from 1 to 4 in **increasing** order. Provide the ranking in the boxes provided.

Note: Atomic number (Z) are provided for relevant atoms

a) Increasing (lowest to highest) order of *ionization energy*: (1= lowest; 4 = highest)

Mg

Mg: Z = 12

Rb

Rb: Z = 37

Cs

Cs: Z = 55

K

K : Z = 19

b) Increasing (lowest to highest) order of *bond angle*: (1= lowest; 4 = highest)

H₂OCCl₄XeF₄

HCN

c) Increasing (smallest to largest) order of *ionic size* (1= smallest; 4 = largest)

S²⁻

S: Z = 16

O²⁻

O: Z = 8

Be²⁺

Be: Z = 4

Na⁺

Na: Z = 11

d) Increasing (lowest to highest) order of *bond energy* (1= lowest; 4 = highest)

P-F

P-Cl

P-I

P-Br

**Question 5** **12 points**

a. 9 points For the molecules HClO_4 and TeF_6 , draw the most stable *Lewis structure* showing appropriate shape and *dashed/wedged bonds*, based on VSEPR. Determine the **molecular geometry** around the central atom. For HClO_4 , label the *most polar bond*.

HClO_4 (there are no O-O bonds)

Label the most polar bond

Molecular Geometry around Cl

TeF_6

Molecular Geometry around Te

b. 3 points (Note: Lone pairs are not shown in structures below)

(i) For the three *resonance* structures (1, 2 and 3) shown below, answer the following question:



(I) Circle the *most contributing* resonance structure.

Structure 1

Structure 2

Structure 3

(ii) For the three structures given (4, 5, and 6) below, answer the following questions:



(I) Circle all the structures that represent *resonance structures* of each other:

Structure 4

Structure 5

Structure 6

(II) Among the resonance structures, circle the *most contributing* resonance structure.

Structure 4

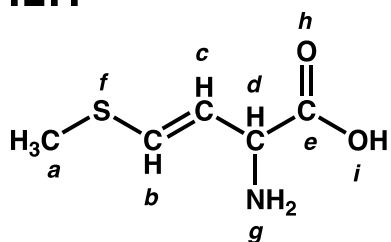
Structure 5

Structure 6

**Question 6** **6 points**

Using the data given below, calculate the **lattice energy** for SrO(s).

	ΔH° (kJ/mol)
$\text{Sr}(s) \longrightarrow \text{Sr}(g)$	+180
$\text{Sr}(g) \longrightarrow \text{Sr}^{2+}(g) + 2e^-$	+1468
$\text{O}_2(g) \longrightarrow 2\text{O}(g)$	+500
$\text{O}(g) + 2e^- \longrightarrow \text{O}^{2-}(g)$	+600
$\text{Sr}(s) + \frac{1}{2}\text{O}_2(g) \longrightarrow \text{SrO}(s)$	-550

**Question 7 10 points**

Consider the molecule given on the left, and answer questions (i) to (vi) using chemical bonding theories that we have learnt in class (valence bond theory, hybridization).

Note that the lone pairs are not shown in the structure:

i. Consider the carbon atoms marked in as ***a***, ***b***, ***c***, ***d***, and ***e*** in the molecule, and determine **hybridization** at each of the carbon centers. (2.5 points)

Atom	Hybridization
Carbon (<i>a</i>)	
Carbon (<i>b</i>)	
Carbon (<i>c</i>)	
Carbon (<i>d</i>)	
Carbon (<i>e</i>)	

ii. Now, consider the sulfur atom (S) marked as ***f***, nitrogen (N) atom (***g***), and oxygen(O) atoms (***h*** and ***i***), and determine their **hybridization**. (2 points)

Atom	Hybridization
Sulfur (<i>f</i>)	
Nitrogen (<i>g</i>)	
Oxygen (<i>h</i>)	
Oxygen (<i>i</i>)	

iii. **Describe** all the bonds formed by the carbon atoms ***a*** and ***b***, using sigma and pi designation as appropriate. (2 points)

Atom	Bonds formed (use sigma and pi as appropriate)
Carbon (<i>a</i>)	
Carbon (<i>b</i>)	

iv. How many total sp^3 -s sigma bonds are in the molecule? (1 point)

v. Determine **electronic** and **molecular geometry** at carbon (C) labeled ***e*** and sulfur (S) labeled ***f*** centres. (2 points)

Atom	Electronic Geometry	Molecular Geometry
Carbon (<i>e</i>)		
Sulfur (<i>f</i>)		

vi. Circle the value from the choices given below, that describes the **C-S-C bond angle at sulfur** (S) marked ***f***: (0.5 points)

$>90^\circ$ to $<109.5^\circ$ 109.5° >109.5 to $<120^\circ$ 120° $>120^\circ$ to $<180^\circ$


Question 8 12 points

Using the **valence shell orbitals only** draw a **complete** molecular orbital (MO) diagram (with no s-p mixing) for the molecule **PS**, in the space provided below. **(6 points)**

Assume that 3s and 3p atomic orbitals will form molecular orbitals like those formed by the combination of 2s and 2p atomic orbitals that we learnt in class, and there is no s-p mixing.

Show **energy arrow**, **atomic orbitals** involved, **atom labels**, the **molecular orbitals** that are formed, **filled electrons** in the MO diagram etc. Make sure to label all the atomic and molecular orbitals.

Using the MO diagram above, answer the following questions:

- (i) Which one of the following will have the highest bond order? Circle the correct choice
 A) PS^{1-} B) PS C) PS^{1+}
- (ii) Which one of the following will have the longest bond length? Circle the correct choice
 A) PS^{1-} B) PS C) PS^{1+}
- (iii) Which of the following will be diamagnetic? Circle the correct choice.
 A) PS^{1-} B) PS C) PS^{1+}
- (iv) Draw Lewis dot structure of the molecule with the shortest bond length (based on the MO diagram) from among the three (PS^{1-} , PS, PS^{1+}), and indicate the bond order calculated from the Lewis dot structure. **(3 points)**

Lewis Dot Structure

Bond Order



B0B0CE67-40D6-490D-9812-12A846DC7FBE

midterm-exam-93388

#828 Page 10 of 10

SCRAP PAGE

**CHEMISTRY 110 –Midterm Exam****Thursday November 7, 2024, 6:30 – 8:30 pm****Examiners: Pallavi Sirjoosingh and Ashok Kakkar**

First name (please write as legibly as possible within the boxes)

--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--

Last name

--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--

Student ID number

--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--

This booklet has 8 questions on 10 pages (The last page can be used as a scrap page).

The staple must NOT be removed from the question booklet. Your intact question booklet must be returned at the end of the exam.

A separate booklet contains a periodic table, and datasheet.

Write your answers directly under the questions in the space provided. Do not write on the QR codes or near the top edges (as it may be cut off).

Please write **legibly and dark enough** for the scan to be readable. For students who use English as a second language, dictionaries are permitted. Calculators are allowed.

DO NOT ASK QUESTIONS OF THE INVIGILATORS about interpretations of the exam questions. If you have a concern about a question, please summarize it at the top of this page. All concerns will be evaluated after the exam. GOOD LUCK!

**Question 1 8 points** (*Assume all transitions are possible*)

a. 6 points Calculate the **energy (in kJ/mol)** required to ionize (completely remove) the electron from the **$n=3$** level of **He^+ ion** (He: Helium). Show your work including equations. (3 sig. figs.)

b. 2 points

(I) Select the largest (longest) wavelength from the three choices below:

- i. Wavelength of photon to *completely remove* an electron from *1s orbital of Hydrogen atom*
- ii. Wavelength of photon to *completely remove* an electron from *1s orbital of Boron atom (B)*
- iii. Wavelength of photon to *completely remove* an electron from *1s orbital of Boron cation (B^+)*

i or ii or iii

(II) Select the smallest (shortest) wavelength from the three choices below (*for H atom*):

- i. Wavelength of photon *emitted* for an electron transitioning from *2p to 1s orbital*
- ii. Wavelength of photon *absorbed* for an electron transitioning from *2s to 4p orbital*
- iii. Wavelength of photon *absorbed* for an electron transitioning from *1s to 3p orbital*

i or ii or iii



Question 2 8 points



A Chem110 student plots the graph shown on the left, from data obtained for the kinetic energy (eV) of the ejected electron versus frequency (s^{-1}) of the incident light, for an unknown metal. ($1\text{eV} = 1.6022 \times 10^{-19} \text{ J}$)

Based on the plot, answer the questions **a to c** below:

a. 1 point On the plot above, label the **threshold frequency**.

b. 2 points What is the value of the work potential (in J) of the unknown metal?

c. 3 points What is the *longest* wavelength (in nm) required to eject an electron from this metal's surface? Show your work including equation(s) used. (3 sig figs)

d. 2 points In the space provided below, plot the following

(Assume frequency is *above* threshold frequency):

- i. *Kinetic energy* of an ejected electron versus the *intensity* of the incident light (left)
- ii. *Number of electrons* ejected versus the *frequency* of the incident light (right)



**Question 3** **6 points****a. 2 points**

In the ground state of Ge (Germanium; $Z = 32$) atom, **name** the following orbitals:

i. Orbital containing the electron that requires the maximum energy to be removed.

ii. Orbital containing the electron that experiences the minimum effective nuclear charge.

b. 2 points

An unknown anion (A^-) has the following *excited* electronic configuration: $[Ar] 4s^2 3d^3 4p^1$

(Ar: Argon; $Z = 18$)

Give the **condensed ground state electronic configuration** of the atom, **A** and the cation, **A^{2+}** .
Show your work. (Assume all electronic transitions are possible)

Electronic Configuration of A:

Electronic Configuration of A^{2+} :

c. 2 points

In a single atom, determine how many electrons can have the following set of quantum numbers?

(i) $n = 5; l = 3$

In a single atom, determine how many electrons can have the following set of quantum numbers?

(i) $n = 3; l = 2; m_s = +1/2$

**Question 4** **8 points**

Rank the following from 1 to 4 in **increasing** order. Provide the ranking in the boxes provided.

Note: Atomic number (Z) are provided for relevant atoms

a) Increasing (lowest to highest) order of *ionization energy*: (1= lowest; 4 = highest)

Mg

Mg: Z = 12

Rb

Rb: Z = 37

Cs

Cs: Z = 55

K

K : Z = 19

b) Increasing (lowest to highest) order of *bond angle*: (1= lowest; 4 = highest)

H₂OCCl₄XeF₄

HCN

c) Increasing (smallest to largest) order of *ionic size* (1= smallest; 4 = largest)

S²⁻

S: Z = 16

O²⁻

O: Z = 8

Be²⁺

Be: Z = 4

Na⁺

Na: Z = 11

d) Increasing (lowest to highest) order of *bond energy* (1= lowest; 4 = highest)

P-F

P-Cl

P-I

P-Br

**Question 5** **12 points**

a. 9 points For the molecules HClO_4 and TeF_6 , draw the most stable *Lewis structure* showing appropriate shape and *dashed/wedged bonds*, based on VSEPR. Determine the **molecular geometry** around the central atom. For HClO_4 , label the *most polar bond*.



b. 3 points (Note: Lone pairs are not shown in structures below)

(i) For the three *resonance* structures (1, 2 and 3) shown below, answer the following question:



(I) Circle the *most contributing* resonance structure.

Structure 1

Structure 2

Structure 3

(ii) For the three structures given (4, 5, and 6) below, answer the following questions:



(I) Circle all the structures that represent *resonance structures* of each other:

Structure 4

Structure 5

Structure 6

(II) Among the resonance structures, circle the *most contributing* resonance structure.

Structure 4

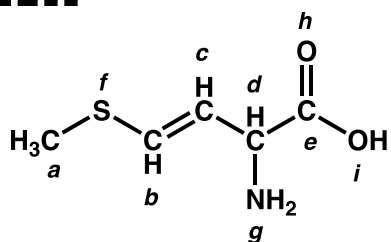
Structure 5

Structure 6

**Question 6** **6 points**

Using the data given below, calculate the **lattice energy** for SrO(s).

	ΔH° (kJ/mol)
$\text{Sr}(s) \longrightarrow \text{Sr}(g)$	+180
$\text{Sr}(g) \longrightarrow \text{Sr}^{2+}(g) + 2e^-$	+1468
$\text{O}_2(g) \longrightarrow 2\text{O}(g)$	+500
$\text{O}(g) + 2e^- \longrightarrow \text{O}^{2-}(g)$	+600
$\text{Sr}(s) + \frac{1}{2}\text{O}_2(g) \longrightarrow \text{SrO}(s)$	-550

**Question 7 10 points**

Consider the molecule given on the left, and answer questions (i) to (vi) using chemical bonding theories that we have learnt in class (valence bond theory, hybridization).

Note that the lone pairs are not shown in the structure:

i. Consider the carbon atoms marked in as ***a***, ***b***, ***c***, ***d***, and ***e*** in the molecule, and determine **hybridization** at each of the carbon centers. (2.5 points)

Atom	Hybridization
Carbon (<i>a</i>)	
Carbon (<i>b</i>)	
Carbon (<i>c</i>)	
Carbon (<i>d</i>)	
Carbon (<i>e</i>)	

ii. Now, consider the sulfur atom (S) marked as ***f***, nitrogen (N) atom (***g***), and oxygen(O) atoms (***h*** and ***i***), and determine their **hybridization**. (2 points)

Atom	Hybridization
Sulfur (<i>f</i>)	
Nitrogen (<i>g</i>)	
Oxygen (<i>h</i>)	
Oxygen (<i>i</i>)	

iii. **Describe** all the bonds formed by the carbon atoms ***a*** and ***b***, using sigma and pi designation as appropriate. (2 points)

Atom	Bonds formed (use sigma and pi as appropriate)
Carbon (<i>a</i>)	
Carbon (<i>b</i>)	

iv. How many total sp^3 -s sigma bonds are in the molecule? (1 point)

v. Determine **electronic** and **molecular geometry** at carbon (C) labeled ***e*** and sulfur (S) labeled ***f*** centres. (2 points)

Atom	Electronic Geometry	Molecular Geometry
Carbon (<i>e</i>)		
Sulfur (<i>f</i>)		

vi. Circle the value from the choices given below, that describes the **C-S-C bond angle at sulfur** (S) marked ***f***: (0.5 points)

>90° to <109.5°

109.5°

>109.5 to <120°

120°

>120° to <180°


Question 8 12 points

Using the **valence shell orbitals only** draw a **complete** molecular orbital (MO) diagram (with no s-p mixing) for the molecule **PS**, in the space provided below. **(6 points)**

Assume that 3s and 3p atomic orbitals will form molecular orbitals like those formed by the combination of 2s and 2p atomic orbitals that we learnt in class, and there is no s-p mixing.

Show **energy arrow**, **atomic orbitals** involved, **atom labels**, the **molecular orbitals** that are formed, **filled electrons** in the MO diagram etc. Make sure to label all the atomic and molecular orbitals.

Using the MO diagram above, answer the following questions:

- (i) Which one of the following will have the highest bond order? Circle the correct choice
 A) PS^{1-} B) PS C) PS^{1+}
- (ii) Which one of the following will have the longest bond length? Circle the correct choice
 A) PS^{1-} B) PS C) PS^{1+}
- (iii) Which of the following will be diamagnetic? Circle the correct choice.
 A) PS^{1-} B) PS C) PS^{1+}
- (iv) Draw Lewis dot structure of the molecule with the shortest bond length (based on the MO diagram) from among the three (PS^{1-} , PS, PS^{1+}), and indicate the bond order calculated from the Lewis dot structure. **(3 points)**

Lewis Dot Structure

Bond Order



36685B88-740A-4819-91CC-F4A09187FD6B

midterm-exam-93388

#829 Page 10 of 10

SCRAP PAGE

**CHEMISTRY 110 –Midterm Exam****Thursday November 7, 2024, 6:30 – 8:30 pm**

Examiners: Pallavi Sirjoosingh and Ashok Kakkar

First name (please write as legibly as possible within the boxes)

--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--

Last name

--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--

Student ID number

--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--

This booklet has 8 questions on 10 pages (The last page can be used as a scrap page).

The staple must NOT be removed from the question booklet. Your intact question booklet must be returned at the end of the exam.

A separate booklet contains a periodic table, and datasheet.

Write your answers directly under the questions in the space provided. Do not write on the QR codes or near the top edges (as it may be cut off).

Please write **legibly and dark enough** for the scan to be readable. For students who use English as a second language, dictionaries are permitted. Calculators are allowed.

DO NOT ASK QUESTIONS OF THE INVIGILATORS about interpretations of the exam questions. If you have a concern about a question, please summarize it at the top of this page. All concerns will be evaluated after the exam. GOOD LUCK!

**Question 1 8 points** (*Assume all transitions are possible*)

a. 6 points Calculate the **energy (in kJ/mol)** required to ionize (completely remove) the electron from the **$n=3$** level of **He^+ ion** (He: Helium). Show your work including equations. (3 sig. figs.)

b. 2 points

(I) Select the largest (longest) wavelength from the three choices below:

- i. Wavelength of photon to *completely remove* an electron from *1s orbital of Hydrogen atom*
- ii. Wavelength of photon to *completely remove* an electron from *1s orbital of Boron atom (B)*
- iii. Wavelength of photon to *completely remove* an electron from *1s orbital of Boron cation (B^+)*

i or ii or iii

(II) Select the smallest (shortest) wavelength from the three choices below (*for H atom*):

- i. Wavelength of photon *emitted* for an electron transitioning from *2p to 1s orbital*
- ii. Wavelength of photon *absorbed* for an electron transitioning from *2s to 4p orbital*
- iii. Wavelength of photon *absorbed* for an electron transitioning from *1s to 3p orbital*

i or ii or iii



Question 2 8 points



A Chem110 student plots the graph shown on the left, from data obtained for the kinetic energy (eV) of the ejected electron versus frequency (s^{-1}) of the incident light, for an unknown metal. ($1\text{eV} = 1.6022 \times 10^{-19} \text{ J}$)

Based on the plot, answer the questions **a to c** below:

a. 1 point On the plot above, label the **threshold frequency**.

b. 2 points What is the value of the work potential (in J) of the unknown metal?

c. 3 points What is the *longest* wavelength (in nm) required to eject an electron from this metal's surface? Show your work including equation(s) used. (3 sig figs)

d. 2 points In the space provided below, plot the following

(Assume frequency is *above* threshold frequency):

- i. *Kinetic energy* of an ejected electron versus the *intensity* of the incident light (left)
- ii. *Number of electrons* ejected versus the *frequency* of the incident light (right)



**Question 3** **6 points****a. 2 points**

In the ground state of Ge (Germanium; $Z = 32$) atom, **name** the following orbitals:

i. Orbital containing the electron that requires the maximum energy to be removed.

ii. Orbital containing the electron that experiences the minimum effective nuclear charge.

b. 2 points

An unknown anion (A^-) has the following *excited* electronic configuration: $[Ar] 4s^2 3d^3 4p^1$

(Ar: Argon; $Z = 18$)

Give the **condensed ground state electronic configuration** of the atom, **A** and the cation, **A^{2+}** .
Show your work. (Assume all electronic transitions are possible)

Electronic Configuration of A:

Electronic Configuration of A^{2+} :

c. 2 points

In a single atom, determine how many electrons can have the following set of quantum numbers?

(i) $n = 5; l = 3$

In a single atom, determine how many electrons can have the following set of quantum numbers?

(i) $n = 3; l = 2; m_s = +1/2$

**Question 4** **8 points**

Rank the following from 1 to 4 in **increasing** order. Provide the ranking in the boxes provided.

Note: Atomic number (Z) are provided for relevant atoms

a) Increasing (lowest to highest) order of *ionization energy*: (1= lowest; 4 = highest)

Mg

Mg: Z = 12

Rb

Rb: Z = 37

Cs

Cs: Z = 55

K

K : Z = 19

b) Increasing (lowest to highest) order of *bond angle*: (1= lowest; 4 = highest)

H₂OCCl₄XeF₄

HCN

c) Increasing (smallest to largest) order of *ionic size* (1= smallest; 4 = largest)

S²⁻

S: Z = 16

O²⁻

O: Z = 8

Be²⁺

Be: Z = 4

Na⁺

Na: Z = 11

d) Increasing (lowest to highest) order of *bond energy* (1= lowest; 4 = highest)

P-F

P-Cl

P-I

P-Br

**Question 5** **12 points**

a. 9 points For the molecules HClO_4 and TeF_6 , draw the most stable *Lewis structure* showing appropriate shape and *dashed/wedged bonds*, based on VSEPR. Determine the **molecular geometry** around the central atom. For HClO_4 , label the *most polar bond*.



b. 3 points (Note: Lone pairs are not shown in structures below)

(i) For the three *resonance* structures (1, 2 and 3) shown below, answer the following question:



(I) Circle the *most contributing* resonance structure.

Structure 1

Structure 2

Structure 3

(ii) For the three structures given (4, 5, and 6) below, answer the following questions:



(I) Circle all the structures that represent *resonance structures* of each other:

Structure 4

Structure 5

Structure 6

(II) Among the resonance structures, circle the *most contributing* resonance structure.

Structure 4

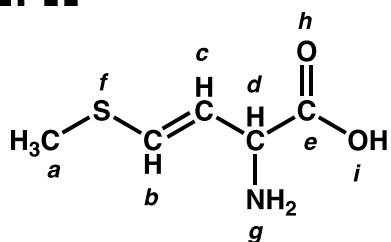
Structure 5

Structure 6

**Question 6** **6 points**

Using the data given below, calculate the **lattice energy** for SrO(s).

	ΔH° (kJ/mol)
$\text{Sr}(s) \longrightarrow \text{Sr}(g)$	+180
$\text{Sr}(g) \longrightarrow \text{Sr}^{2+}(g) + 2e^-$	+1468
$\text{O}_2(g) \longrightarrow 2\text{O}(g)$	+500
$\text{O}(g) + 2e^- \longrightarrow \text{O}^{2-}(g)$	+600
$\text{Sr}(s) + \frac{1}{2}\text{O}_2(g) \longrightarrow \text{SrO}(s)$	-550

**Question 7 10 points**

Consider the molecule given on the left, and answer questions (i) to (vi) using chemical bonding theories that we have learnt in class (valence bond theory, hybridization).

Note that the lone pairs are not shown in the structure:

i. Consider the carbon atoms marked in as ***a***, ***b***, ***c***, ***d***, and ***e*** in the molecule, and determine **hybridization** at each of the carbon centers. (2.5 points)

Atom	Hybridization
Carbon (<i>a</i>)	
Carbon (<i>b</i>)	
Carbon (<i>c</i>)	
Carbon (<i>d</i>)	
Carbon (<i>e</i>)	

ii. Now, consider the sulfur atom (S) marked as ***f***, nitrogen (N) atom (***g***), and oxygen(O) atoms (***h*** and ***i***), and determine their **hybridization**. (2 points)

Atom	Hybridization
Sulfur (<i>f</i>)	
Nitrogen (<i>g</i>)	
Oxygen (<i>h</i>)	
Oxygen (<i>i</i>)	

iii. **Describe** all the bonds formed by the carbon atoms ***a*** and ***b***, using sigma and pi designation as appropriate. (2 points)

Atom	Bonds formed (use sigma and pi as appropriate)
Carbon (<i>a</i>)	
Carbon (<i>b</i>)	

iv. How many total sp^3 -s sigma bonds are in the molecule? (1 point)

v. Determine **electronic** and **molecular geometry** at carbon (C) labeled ***e*** and sulfur (S) labeled ***f*** centres. (2 points)

Atom	Electronic Geometry	Molecular Geometry
Carbon (<i>e</i>)		
Sulfur (<i>f</i>)		

vi. Circle the value from the choices given below, that describes the **C-S-C bond angle at sulfur** (S) marked ***f***: (0.5 points)

>90° to <109.5° 109.5° >109.5 to <120° 120° >120° to <180°



Question 8 12 points

Using the **valence shell orbitals only** draw a **complete** molecular orbital (MO) diagram (with no s-p mixing) for the molecule **PS**, in the space provided below. **(6 points)**

Assume that 3s and 3p atomic orbitals will form molecular orbitals like those formed by the combination of 2s and 2p atomic orbitals that we learnt in class, and there is no s-p mixing.

Show **energy arrow**, **atomic orbitals** involved, **atom labels**, the **molecular orbitals** that are formed, **filled electrons** in the MO diagram etc. Make sure to label all the atomic and molecular orbitals.

Using the MO diagram above, answer the following questions:

- (i) Which one of the following will have the highest bond order? Circle the correct choice
 A) PS^{1-} B) PS C) PS^{1+}
- (ii) Which one of the following will have the longest bond length? Circle the correct choice
 A) PS^{1-} B) PS C) PS^{1+}
- (iii) Which of the following will be diamagnetic? Circle the correct choice.
 A) PS^{1-} B) PS C) PS^{1+}
- (iv) Draw Lewis dot structure of the molecule with the shortest bond length (based on the MO diagram) from among the three (PS^{1-} , PS, PS^{1+}), and indicate the bond order calculated from the Lewis dot structure. **(3 points)**

Lewis Dot Structure

Bond Order



7A3462BA-B3EB-406C-B19E-5D2888A82815

midterm-exam-93388

#830 Page 10 of 10

SCRAP PAGE

**CHEMISTRY 110 –Midterm Exam****Thursday November 7, 2024, 6:30 – 8:30 pm**

Examiners: Pallavi Sirjoosingh and Ashok Kakkar

First name (please write as legibly as possible within the boxes)

--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--

Last name

--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--

Student ID number

--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--

This booklet has 8 questions on 10 pages (The last page can be used as a scrap page).

The staple must NOT be removed from the question booklet. Your intact question booklet must be returned at the end of the exam.

A separate booklet contains a periodic table, and datasheet.

Write your answers directly under the questions in the space provided. Do not write on the QR codes or near the top edges (as it may be cut off).

Please write **legibly and dark enough** for the scan to be readable. For students who use English as a second language, dictionaries are permitted. Calculators are allowed.

DO NOT ASK QUESTIONS OF THE INVIGILATORS about interpretations of the exam questions. If you have a concern about a question, please summarize it at the top of this page. All concerns will be evaluated after the exam. GOOD LUCK!

**Question 1 8 points** (*Assume all transitions are possible*)

a. 6 points Calculate the **energy (in kJ/mol)** required to ionize (completely remove) the electron from the **n=3** level of **He⁺ ion** (He: Helium). Show your work including equations. (3 sig. figs.)

b. 2 points

(I) Select the largest (longest) wavelength from the three choices below:

- i. Wavelength of photon to *completely remove* an electron from *1s orbital of Hydrogen atom*
- ii. Wavelength of photon to *completely remove* an electron from *1s orbital of Boron atom (B)*
- iii. Wavelength of photon to *completely remove* an electron from *1s orbital of Boron cation (B⁺)*

i or ii or iii

(II) Select the smallest (shortest) wavelength from the three choices below (*for H atom*):

- i. Wavelength of photon *emitted* for an electron transitioning from *2p to 1s orbital*
- ii. Wavelength of photon *absorbed* for an electron transitioning from *2s to 4p orbital*
- iii. Wavelength of photon *absorbed* for an electron transitioning from *1s to 3p orbital*

i or ii or iii



Question 2 8 points



A Chem110 student plots the graph shown on the left, from data obtained for the kinetic energy (eV) of the ejected electron versus frequency (s^{-1}) of the incident light, for an unknown metal. ($1\text{eV} = 1.6022 \times 10^{-19} \text{ J}$)

Based on the plot, answer the questions **a to c** below:

a. 1 point On the plot above, label the **threshold frequency**.

b. 2 points What is the value of the work potential (in J) of the unknown metal?

c. 3 points What is the *longest* wavelength (in nm) required to eject an electron from this metal's surface? Show your work including equation(s) used. (3 sig figs)

d. 2 points In the space provided below, plot the following

(Assume frequency is *above* threshold frequency):

- Kinetic energy* of an ejected electron versus the *intensity* of the incident light (left)
- Number of electrons* ejected versus the *frequency* of the incident light (right)



**Question 3** **6 points****a. 2 points**

In the ground state of Ge (Germanium; $Z = 32$) atom, **name** the following orbitals:

i. Orbital containing the electron that requires the maximum energy to be removed.

ii. Orbital containing the electron that experiences the minimum effective nuclear charge.

b. 2 points

An unknown anion (A^-) has the following *excited* electronic configuration: $[Ar] 4s^2 3d^3 4p^1$

(Ar: Argon; $Z = 18$)

Give the **condensed ground state electronic configuration** of the atom, **A** and the cation, **A^{2+}** .
Show your work. (Assume all electronic transitions are possible)

Electronic Configuration of A:

Electronic Configuration of A^{2+} :

c. 2 points

In a single atom, determine how many electrons can have the following set of quantum numbers?

(i) $n = 5; l = 3$

In a single atom, determine how many electrons can have the following set of quantum numbers?

(i) $n = 3; l = 2; m_s = +1/2$

**Question 4** **8 points**

Rank the following from 1 to 4 in **increasing** order. Provide the ranking in the boxes provided.

Note: Atomic number (Z) are provided for relevant atoms

a) Increasing (lowest to highest) order of *ionization energy*: (1= lowest; 4 = highest)

Mg

Mg: Z = 12

Rb

Rb: Z = 37

Cs

Cs: Z = 55

K

K : Z = 19

b) Increasing (lowest to highest) order of *bond angle*: (1= lowest; 4 = highest)

H₂OCCl₄XeF₄

HCN

c) Increasing (smallest to largest) order of *ionic size* (1= smallest; 4 = largest)

S²⁻

S: Z = 16

O²⁻

O: Z = 8

Be²⁺

Be: Z = 4

Na⁺

Na: Z = 11

d) Increasing (lowest to highest) order of *bond energy* (1= lowest; 4 = highest)

P-F

P-Cl

P-I

P-Br

**Question 5** **12 points**

a. 9 points For the molecules HClO_4 and TeF_6 , draw the most stable *Lewis structure* showing appropriate shape and *dashed/wedged bonds*, based on VSEPR. Determine the **molecular geometry** around the central atom. For HClO_4 , label the *most polar bond*.



b. 3 points (Note: Lone pairs are not shown in structures below)

(i) For the three *resonance* structures (1, 2 and 3) shown below, answer the following question:



(I) Circle the *most contributing* resonance structure.

Structure 1

Structure 2

Structure 3

(ii) For the three structures given (4, 5, and 6) below, answer the following questions:



(I) Circle all the structures that represent *resonance structures* of each other:

Structure 4

Structure 5

Structure 6

(II) Among the resonance structures, circle the *most contributing* resonance structure.

Structure 4

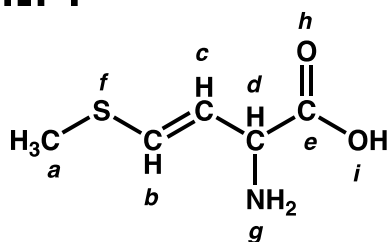
Structure 5

Structure 6

**Question 6** **6 points**

Using the data given below, calculate the **lattice energy** for SrO(s).

	ΔH° (kJ/mol)
$\text{Sr}(s) \longrightarrow \text{Sr}(g)$	+180
$\text{Sr}(g) \longrightarrow \text{Sr}^{2+}(g) + 2e^-$	+1468
$\text{O}_2(g) \longrightarrow 2\text{O}(g)$	+500
$\text{O}(g) + 2e^- \longrightarrow \text{O}^{2-}(g)$	+600
$\text{Sr}(s) + \frac{1}{2}\text{O}_2(g) \longrightarrow \text{SrO}(s)$	-550

**Question 7 10 points**

Consider the molecule given on the left, and answer questions (i) to (vi) using chemical bonding theories that we have learnt in class (valence bond theory, hybridization).

Note that the lone pairs are not shown in the structure:

i. Consider the carbon atoms marked in as ***a***, ***b***, ***c***, ***d***, and ***e*** in the molecule, and determine **hybridization** at each of the carbon centers. (2.5 points)

Atom	Hybridization
Carbon (<i>a</i>)	
Carbon (<i>b</i>)	
Carbon (<i>c</i>)	
Carbon (<i>d</i>)	
Carbon (<i>e</i>)	

ii. Now, consider the sulfur atom (S) marked as ***f***, nitrogen (N) atom (***g***), and oxygen(O) atoms (***h*** and ***i***), and determine their **hybridization**. (2 points)

Atom	Hybridization
Sulfur (<i>f</i>)	
Nitrogen (<i>g</i>)	
Oxygen (<i>h</i>)	
Oxygen (<i>i</i>)	

iii. **Describe** all the bonds formed by the carbon atoms ***a*** and ***b***, using sigma and pi designation as appropriate. (2 points)

Atom	Bonds formed (use sigma and pi as appropriate)
Carbon (<i>a</i>)	
Carbon (<i>b</i>)	

iv. How many total sp^3 -s sigma bonds are in the molecule? (1 point)

v. Determine **electronic** and **molecular geometry** at carbon (C) labeled ***e*** and sulfur (S) labeled ***f*** centres. (2 points)

Atom	Electronic Geometry	Molecular Geometry
Carbon (<i>e</i>)		
Sulfur (<i>f</i>)		

vi. Circle the value from the choices given below, that describes the **C-S-C bond angle at sulfur** (S) marked ***f***: (0.5 points)

>90° to <109.5°

109.5°

>109.5 to <120°

120°

>120° to <180°

**Question 8 12 points**

Using the **valence shell orbitals only** draw a **complete** molecular orbital (MO) diagram (with no s-p mixing) for the molecule **PS**, in the space provided below. **(6 points)**

Assume that 3s and 3p atomic orbitals will form molecular orbitals like those formed by the combination of 2s and 2p atomic orbitals that we learnt in class, and there is no s-p mixing.

Show **energy arrow**, **atomic orbitals** involved, **atom labels**, the **molecular orbitals** that are formed, **filled electrons** in the MO diagram etc. Make sure to label all the atomic and molecular orbitals.

Using the MO diagram above, answer the following questions:

- (i) Which one of the following will have the highest bond order? Circle the correct choice
A) PS^{1-} B) PS C) PS^{1+}
- (ii) Which one of the following will have the longest bond length? Circle the correct choice
A) PS^{1-} B) PS C) PS^{1+}
- (iii) Which of the following will be diamagnetic? Circle the correct choice.
A) PS^{1-} B) PS C) PS^{1+}
- (iv) Draw Lewis dot structure of the molecule with the shortest bond length (based on the MO diagram) from among the three (PS^{1-} , PS, PS^{1+}), and indicate the bond order calculated from the Lewis dot structure. **(3 points)**

Lewis Dot Structure

Bond Order



F76527B2-3C0F-475B-B60A-6946DAB167CB

midterm-exam-93388

#831 Page 10 of 10

SCRAP PAGE

**CHEMISTRY 110 –Midterm Exam****Thursday November 7, 2024, 6:30 – 8:30 pm****Examiners: Pallavi Sirjoosingh and Ashok Kakkar**

First name (please write as legibly as possible within the boxes)

--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--

Last name

--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--

Student ID number

--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--

This booklet has 8 questions on 10 pages (The last page can be used as a scrap page).

The staple must NOT be removed from the question booklet. Your intact question booklet must be returned at the end of the exam.

A separate booklet contains a periodic table, and datasheet.

Write your answers directly under the questions in the space provided. Do not write on the QR codes or near the top edges (as it may be cut off).

Please write **legibly and dark enough** for the scan to be readable. For students who use English as a second language, dictionaries are permitted. Calculators are allowed.

DO NOT ASK QUESTIONS OF THE INVIGILATORS about interpretations of the exam questions. If you have a concern about a question, please summarize it at the top of this page. All concerns will be evaluated after the exam. GOOD LUCK!

**Question 1 8 points** (*Assume all transitions are possible*)

a. 6 points Calculate the **energy (in kJ/mol)** required to ionize (completely remove) the electron from the **$n=3$** level of **He^+ ion** (He: Helium). Show your work including equations. (3 sig. figs.)

b. 2 points

(I) Select the largest (longest) wavelength from the three choices below:

- i. Wavelength of photon to *completely remove* an electron from *1s orbital of Hydrogen atom*
- ii. Wavelength of photon to *completely remove* an electron from *1s orbital of Boron atom (B)*
- iii. Wavelength of photon to *completely remove* an electron from *1s orbital of Boron cation (B^+)*

i or ii or iii

(II) Select the smallest (shortest) wavelength from the three choices below (*for H atom*):

- i. Wavelength of photon *emitted* for an electron transitioning from *2p to 1s orbital*
- ii. Wavelength of photon *absorbed* for an electron transitioning from *2s to 4p orbital*
- iii. Wavelength of photon *absorbed* for an electron transitioning from *1s to 3p orbital*

i or ii or iii



Question 2 8 points



A Chem110 student plots the graph shown on the left, from data obtained for the kinetic energy (eV) of the ejected electron versus frequency (s^{-1}) of the incident light, for an unknown metal. ($1\text{eV} = 1.6022 \times 10^{-19} \text{ J}$)

Based on the plot, answer the questions **a to c** below:

a. 1 point On the plot above, label the **threshold frequency**.

b. 2 points What is the value of the work potential (in J) of the unknown metal?

c. 3 points What is the *longest* wavelength (in nm) required to eject an electron from this metal's surface? Show your work including equation(s) used. (3 sig figs)

d. 2 points In the space provided below, plot the following

(Assume frequency is *above* threshold frequency):

- Kinetic energy* of an ejected electron versus the *intensity* of the incident light (left)
- Number of electrons* ejected versus the *frequency* of the incident light (right)



**Question 3** **6 points****a. 2 points**

In the ground state of Ge (Germanium; $Z = 32$) atom, **name** the following orbitals:

i. Orbital containing the electron that requires the maximum energy to be removed.

ii. Orbital containing the electron that experiences the minimum effective nuclear charge.

b. 2 points

An unknown anion (A^-) has the following *excited* electronic configuration: $[Ar] 4s^2 3d^3 4p^1$

(Ar: Argon; $Z = 18$)

Give the **condensed ground state electronic configuration** of the atom, **A** and the cation, **A^{2+}** .
Show your work. (Assume all electronic transitions are possible)

Electronic Configuration of A:

Electronic Configuration of A^{2+} :

c. 2 points

In a single atom, determine how many electrons can have the following set of quantum numbers?

(i) $n = 5; l = 3$

In a single atom, determine how many electrons can have the following set of quantum numbers?

(i) $n = 3; l = 2; m_s = +1/2$

**Question 4** **8 points**

Rank the following from 1 to 4 in **increasing** order. Provide the ranking in the boxes provided.

Note: Atomic number (Z) are provided for relevant atoms

a) Increasing (lowest to highest) order of *ionization energy*: (1= lowest; 4 = highest)

Mg

Mg: Z = 12

Rb

Rb: Z = 37

Cs

Cs: Z = 55

K

K : Z = 19

b) Increasing (lowest to highest) order of *bond angle*: (1= lowest; 4 = highest)

H₂OCCl₄XeF₄

HCN

c) Increasing (smallest to largest) order of *ionic size* (1= smallest; 4 = largest)

S²⁻

S: Z = 16

O²⁻

O: Z = 8

Be²⁺

Be: Z = 4

Na⁺

Na: Z = 11

d) Increasing (lowest to highest) order of *bond energy* (1= lowest; 4 = highest)

P-F

P-Cl

P-I

P-Br

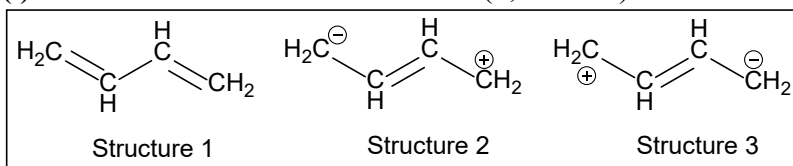
**Question 5** **12 points**

a. 9 points For the molecules HClO_4 and TeF_6 , draw the most stable *Lewis structure* showing appropriate shape and *dashed/wedged bonds*, based on VSEPR. Determine the **molecular geometry** around the central atom. For HClO_4 , label the *most polar bond*.



b. 3 points (Note: Lone pairs are not shown in structures below)

(i) For the three *resonance* structures (1, 2 and 3) shown below, answer the following question:



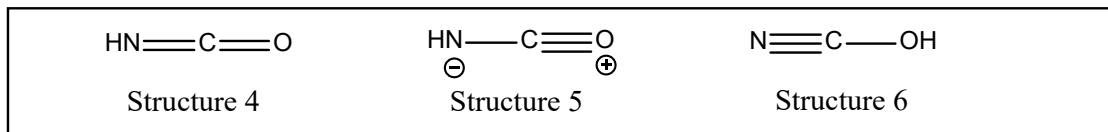
(I) Circle the *most contributing* resonance structure.

Structure 1

Structure 2

Structure 3

(ii) For the three structures given (4, 5, and 6) below, answer the following questions:



(I) Circle all the structures that represent *resonance structures* of each other:

Structure 4

Structure 5

Structure 6

(II) Among the resonance structures, circle the *most contributing* resonance structure.

Structure 4

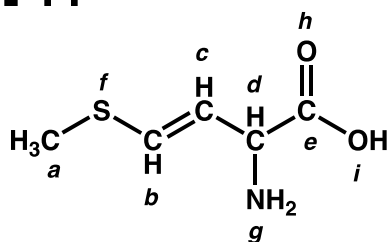
Structure 5

Structure 6

**Question 6** **6 points**

Using the data given below, calculate the **lattice energy** for SrO(s).

	ΔH° (kJ/mol)
$\text{Sr}(s) \longrightarrow \text{Sr}(g)$	+180
$\text{Sr}(g) \longrightarrow \text{Sr}^{2+}(g) + 2e^-$	+1468
$\text{O}_2(g) \longrightarrow 2\text{O}(g)$	+500
$\text{O}(g) + 2e^- \longrightarrow \text{O}^{2-}(g)$	+600
$\text{Sr}(s) + \frac{1}{2}\text{O}_2(g) \longrightarrow \text{SrO}(s)$	-550

**Question 7 10 points**

Consider the molecule given on the left, and answer questions (i) to (vi) using chemical bonding theories that we have learnt in class (valence bond theory, hybridization).

Note that the lone pairs are not shown in the structure:

i. Consider the carbon atoms marked in as ***a***, ***b***, ***c***, ***d***, and ***e*** in the molecule, and determine **hybridization** at each of the carbon centers. (2.5 points)

Atom	Hybridization
Carbon (<i>a</i>)	
Carbon (<i>b</i>)	
Carbon (<i>c</i>)	
Carbon (<i>d</i>)	
Carbon (<i>e</i>)	

ii. Now, consider the sulfur atom (S) marked as ***f***, nitrogen (N) atom (***g***), and oxygen(O) atoms (***h*** and ***i***), and determine their **hybridization**. (2 points)

Atom	Hybridization
Sulfur (<i>f</i>)	
Nitrogen (<i>g</i>)	
Oxygen (<i>h</i>)	
Oxygen (<i>i</i>)	

iii. **Describe** all the bonds formed by the carbon atoms ***a*** and ***b***, using sigma and pi designation as appropriate. (2 points)

Atom	Bonds formed (use sigma and pi as appropriate)
Carbon (<i>a</i>)	
Carbon (<i>b</i>)	

iv. How many total sp^3 -s sigma bonds are in the molecule? (1 point)

v. Determine **electronic** and **molecular geometry** at carbon (C) labeled ***e*** and sulfur (S) labeled ***f*** centres. (2 points)

Atom	Electronic Geometry	Molecular Geometry
Carbon (<i>e</i>)		
Sulfur (<i>f</i>)		

vi. Circle the value from the choices given below, that describes the **C-S-C bond angle at sulfur** (S) marked ***f***: (0.5 points)

$>90^\circ$ to $<109.5^\circ$ 109.5° >109.5 to $<120^\circ$ 120° $>120^\circ$ to $<180^\circ$

**Question 8 12 points**

Using the **valence shell orbitals only** draw a **complete** molecular orbital (MO) diagram (with no s-p mixing) for the molecule **PS**, in the space provided below. **(6 points)**

Assume that 3s and 3p atomic orbitals will form molecular orbitals like those formed by the combination of 2s and 2p atomic orbitals that we learnt in class, and there is no s-p mixing.

Show **energy arrow**, **atomic orbitals** involved, **atom labels**, the **molecular orbitals** that are formed, **filled electrons** in the MO diagram etc. Make sure to label all the atomic and molecular orbitals.

Using the MO diagram above, answer the following questions:

- (i) Which one of the following will have the highest bond order? Circle the correct choice
A) PS^{1-} B) PS C) PS^{1+}
- (ii) Which one of the following will have the longest bond length? Circle the correct choice
A) PS^{1-} B) PS C) PS^{1+}
- (iii) Which of the following will be diamagnetic? Circle the correct choice.
A) PS^{1-} B) PS C) PS^{1+}
- (iv) Draw Lewis dot structure of the molecule with the shortest bond length (based on the MO diagram) from among the three (PS^{1-} , PS, PS^{1+}), and indicate the bond order calculated from the Lewis dot structure. **(3 points)**

Lewis Dot Structure

Bond Order



D1F152A0-BD3D-48A1-BAC4-924781326060

midterm-exam-93388

#832 Page 10 of 10

SCRAP PAGE

**CHEMISTRY 110 –Midterm Exam****Thursday November 7, 2024, 6:30 – 8:30 pm**

Examiners: Pallavi Sirjoosingh and Ashok Kakkar

First name (please write as legibly as possible within the boxes)

--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--

Last name

--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--

Student ID number

--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--

This booklet has 8 questions on 10 pages (The last page can be used as a scrap page).

The staple must NOT be removed from the question booklet. Your intact question booklet must be returned at the end of the exam.

A separate booklet contains a periodic table, and datasheet.

Write your answers directly under the questions in the space provided. Do not write on the QR codes or near the top edges (as it may be cut off).

Please write **legibly and dark enough** for the scan to be readable. For students who use English as a second language, dictionaries are permitted. Calculators are allowed.

DO NOT ASK QUESTIONS OF THE INVIGILATORS about interpretations of the exam questions. If you have a concern about a question, please summarize it at the top of this page. All concerns will be evaluated after the exam. GOOD LUCK!

**Question 1 8 points** (*Assume all transitions are possible*)

a. 6 points Calculate the **energy (in kJ/mol)** required to ionize (completely remove) the electron from the **n=3** level of **He⁺ ion** (He: Helium). Show your work including equations. (3 sig. figs.)

b. 2 points

(I) Select the largest (longest) wavelength from the three choices below:

- i. Wavelength of photon to *completely remove* an electron from *1s orbital of Hydrogen atom*
- ii. Wavelength of photon to *completely remove* an electron from *1s orbital of Boron atom (B)*
- iii. Wavelength of photon to *completely remove* an electron from *1s orbital of Boron cation (B⁺)*

i or ii or iii

(II) Select the smallest (shortest) wavelength from the three choices below (*for H atom*):

- i. Wavelength of photon *emitted* for an electron transitioning from *2p to 1s orbital*
- ii. Wavelength of photon *absorbed* for an electron transitioning from *2s to 4p orbital*
- iii. Wavelength of photon *absorbed* for an electron transitioning from *1s to 3p orbital*

i or ii or iii



Question 2 8 points



A Chem110 student plots the graph shown on the left, from data obtained for the kinetic energy (eV) of the ejected electron versus frequency (s^{-1}) of the incident light, for an unknown metal. ($1\text{eV} = 1.6022 \times 10^{-19} \text{ J}$)

Based on the plot, answer the questions **a to c** below:

a. 1 point On the plot above, label the **threshold frequency**.

b. 2 points What is the value of the work potential (in J) of the unknown metal?

c. 3 points What is the *longest* wavelength (in nm) required to eject an electron from this metal's surface? Show your work including equation(s) used. (3 sig figs)

d. 2 points In the space provided below, plot the following

(Assume frequency is *above* threshold frequency):

- Kinetic energy* of an ejected electron versus the *intensity* of the incident light (left)
- Number of electrons* ejected versus the *frequency* of the incident light (right)



**Question 3** **6 points****a. 2 points**

In the ground state of Ge (Germanium; $Z = 32$) atom, **name** the following orbitals:

i. Orbital containing the electron that requires the maximum energy to be removed.

ii. Orbital containing the electron that experiences the minimum effective nuclear charge.

b. 2 points

An unknown anion (A^-) has the following *excited* electronic configuration: $[Ar] 4s^2 3d^3 4p^1$

(Ar: Argon; $Z = 18$)

Give the **condensed ground state electronic configuration** of the atom, **A** and the cation, **A^{2+}** .
Show your work. (Assume all electronic transitions are possible)

Electronic Configuration of A:

Electronic Configuration of A^{2+} :

c. 2 points

In a single atom, determine how many electrons can have the following set of quantum numbers?

(i) $n = 5; l = 3$

In a single atom, determine how many electrons can have the following set of quantum numbers?

(i) $n = 3; l = 2; m_s = +1/2$

**Question 4** **8 points**

Rank the following from 1 to 4 in **increasing** order. Provide the ranking in the boxes provided.

Note: Atomic number (Z) are provided for relevant atoms

a) Increasing (lowest to highest) order of *ionization energy*: (1= lowest; 4 = highest)

Mg

Mg: Z = 12

Rb

Rb: Z = 37

Cs

Cs: Z = 55

K

K : Z = 19

b) Increasing (lowest to highest) order of *bond angle*: (1= lowest; 4 = highest)

H₂OCCl₄XeF₄

HCN

c) Increasing (smallest to largest) order of *ionic size* (1= smallest; 4 = largest)

S²⁻

S: Z = 16

O²⁻

O: Z = 8

Be²⁺

Be: Z = 4

Na⁺

Na: Z = 11

d) Increasing (lowest to highest) order of *bond energy* (1= lowest; 4 = highest)

P-F

P-Cl

P-I

P-Br

**Question 5** **12 points**

a. 9 points For the molecules HClO_4 and TeF_6 , draw the most stable *Lewis structure* showing appropriate shape and *dashed/wedged bonds*, based on VSEPR. Determine the **molecular geometry** around the central atom. For HClO_4 , label the *most polar bond*.



b. 3 points (Note: Lone pairs are not shown in structures below)

(i) For the three *resonance* structures (1, 2 and 3) shown below, answer the following question:



(I) Circle the *most contributing* resonance structure.

Structure 1

Structure 2

Structure 3

(ii) For the three structures given (4, 5, and 6) below, answer the following questions:



(I) Circle all the structures that represent *resonance structures* of each other:

Structure 4

Structure 5

Structure 6

(II) Among the resonance structures, circle the *most contributing* resonance structure.

Structure 4

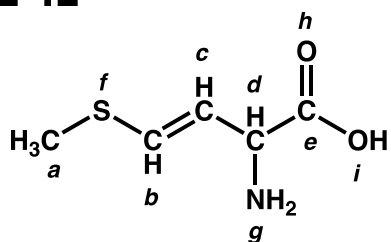
Structure 5

Structure 6

**Question 6** **6 points**

Using the data given below, calculate the **lattice energy** for SrO(s).

	ΔH° (kJ/mol)
$\text{Sr}(s) \longrightarrow \text{Sr}(g)$	+180
$\text{Sr}(g) \longrightarrow \text{Sr}^{2+}(g) + 2e^-$	+1468
$\text{O}_2(g) \longrightarrow 2\text{O}(g)$	+500
$\text{O}(g) + 2e^- \longrightarrow \text{O}^{2-}(g)$	+600
$\text{Sr}(s) + \frac{1}{2}\text{O}_2(g) \longrightarrow \text{SrO}(s)$	-550

**Question 7 10 points**

Consider the molecule given on the left, and answer questions (i) to (vi) using chemical bonding theories that we have learnt in class (valence bond theory, hybridization).

Note that the lone pairs are not shown in the structure:

i. Consider the carbon atoms marked in as ***a***, ***b***, ***c***, ***d***, and ***e*** in the molecule, and determine **hybridization** at each of the carbon centers. (2.5 points)

Atom	Hybridization
Carbon (<i>a</i>)	
Carbon (<i>b</i>)	
Carbon (<i>c</i>)	
Carbon (<i>d</i>)	
Carbon (<i>e</i>)	

ii. Now, consider the sulfur atom (S) marked as ***f***, nitrogen (N) atom (***g***), and oxygen(O) atoms (***h*** and ***i***), and determine their **hybridization**. (2 points)

Atom	Hybridization
Sulfur (<i>f</i>)	
Nitrogen (<i>g</i>)	
Oxygen (<i>h</i>)	
Oxygen (<i>i</i>)	

iii. **Describe** all the bonds formed by the carbon atoms ***a*** and ***b***, using sigma and pi designation as appropriate. (2 points)

Atom	Bonds formed (use sigma and pi as appropriate)
Carbon (<i>a</i>)	
Carbon (<i>b</i>)	

iv. How many total sp^3 -s sigma bonds are in the molecule? (1 point)

v. Determine **electronic** and **molecular geometry** at carbon (C) labeled ***e*** and sulfur (S) labeled ***f*** centres. (2 points)

Atom	Electronic Geometry	Molecular Geometry
Carbon (<i>e</i>)		
Sulfur (<i>f</i>)		

vi. Circle the value from the choices given below, that describes the **C-S-C bond angle at sulfur** (S) marked ***f***: (0.5 points)

>90° to <109.5°

109.5°

>109.5 to <120°

120°

>120° to <180°

**Question 8 12 points**

Using the **valence shell orbitals only** draw a **complete** molecular orbital (MO) diagram (with no s-p mixing) for the molecule **PS**, in the space provided below. **(6 points)**

Assume that 3s and 3p atomic orbitals will form molecular orbitals like those formed by the combination of 2s and 2p atomic orbitals that we learnt in class, and there is no s-p mixing.

Show **energy arrow**, **atomic orbitals** involved, **atom labels**, the **molecular orbitals** that are formed, **filled electrons** in the MO diagram etc. Make sure to label all the atomic and molecular orbitals.

Using the MO diagram above, answer the following questions:

- (i) Which one of the following will have the highest bond order? Circle the correct choice
A) PS^{1-} B) PS C) PS^{1+}
- (ii) Which one of the following will have the longest bond length? Circle the correct choice
A) PS^{1-} B) PS C) PS^{1+}
- (iii) Which of the following will be diamagnetic? Circle the correct choice.
A) PS^{1-} B) PS C) PS^{1+}
- (iv) Draw Lewis dot structure of the molecule with the shortest bond length (based on the MO diagram) from among the three (PS^{1-} , PS, PS^{1+}), and indicate the bond order calculated from the Lewis dot structure. **(3 points)**

Lewis Dot Structure

Bond Order



B785421E-1821-4324-8AE4-F13C4BCC9BF4

midterm-exam-93388

#833 Page 10 of 10

SCRAP PAGE

**CHEMISTRY 110 –Midterm Exam****Thursday November 7, 2024, 6:30 – 8:30 pm**

Examiners: Pallavi Sirjoosingh and Ashok Kakkar

First name (please write as legibly as possible within the boxes)

--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--

Last name

--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--

Student ID number

--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--

This booklet has 8 questions on 10 pages (The last page can be used as a scrap page).

The staple must NOT be removed from the question booklet. Your intact question booklet must be returned at the end of the exam.

A separate booklet contains a periodic table, and datasheet.

Write your answers directly under the questions in the space provided. Do not write on the QR codes or near the top edges (as it may be cut off).

Please write **legibly and dark enough** for the scan to be readable. For students who use English as a second language, dictionaries are permitted. Calculators are allowed.

DO NOT ASK QUESTIONS OF THE INVIGILATORS about interpretations of the exam questions. If you have a concern about a question, please summarize it at the top of this page. All concerns will be evaluated after the exam. GOOD LUCK!

**Question 1 8 points** (*Assume all transitions are possible*)

a. 6 points Calculate the **energy (in kJ/mol)** required to ionize (completely remove) the electron from the **$n=3$** level of **He^+ ion** (He: Helium). Show your work including equations. (3 sig. figs.)

b. 2 points

(I) Select the largest (longest) wavelength from the three choices below:

- i. Wavelength of photon to *completely remove* an electron from *1s orbital of Hydrogen atom*
- ii. Wavelength of photon to *completely remove* an electron from *1s orbital of Boron atom (B)*
- iii. Wavelength of photon to *completely remove* an electron from *1s orbital of Boron cation (B^+)*

i or ii or iii

(II) Select the smallest (shortest) wavelength from the three choices below (*for H atom*):

- i. Wavelength of photon *emitted* for an electron transitioning from *2p to 1s orbital*
- ii. Wavelength of photon *absorbed* for an electron transitioning from *2s to 4p orbital*
- iii. Wavelength of photon *absorbed* for an electron transitioning from *1s to 3p orbital*

i or ii or iii



Question 2 8 points



A Chem110 student plots the graph shown on the left, from data obtained for the kinetic energy (eV) of the ejected electron versus frequency (s^{-1}) of the incident light, for an unknown metal. ($1\text{eV} = 1.6022 \times 10^{-19} \text{ J}$)

Based on the plot, answer the questions **a to c** below:

a. 1 point On the plot above, label the **threshold frequency**.

b. 2 points What is the value of the work potential (in J) of the unknown metal?

c. 3 points What is the *longest* wavelength (in nm) required to eject an electron from this metal's surface? Show your work including equation(s) used. (3 sig figs)

d. 2 points In the space provided below, plot the following

(Assume frequency is *above* threshold frequency):

- Kinetic energy* of an ejected electron versus the *intensity* of the incident light (left)
- Number of electrons* ejected versus the *frequency* of the incident light (right)



**Question 3** **6 points****a. 2 points**

In the ground state of Ge (Germanium; $Z = 32$) atom, **name** the following orbitals:

i. Orbital containing the electron that requires the maximum energy to be removed.

ii. Orbital containing the electron that experiences the minimum effective nuclear charge.

b. 2 points

An unknown anion (A^-) has the following *excited* electronic configuration: $[Ar] 4s^2 3d^3 4p^1$

(Ar: Argon; $Z = 18$)

Give the **condensed ground state electronic configuration** of the atom, **A** and the cation, **A^{2+}** .
Show your work. (Assume all electronic transitions are possible)

Electronic Configuration of A:

Electronic Configuration of A^{2+} :

c. 2 points

In a single atom, determine how many electrons can have the following set of quantum numbers?

(i) $n = 5; l = 3$

In a single atom, determine how many electrons can have the following set of quantum numbers?

(i) $n = 3; l = 2; m_s = +1/2$

**Question 4** **8 points**

Rank the following from 1 to 4 in **increasing** order. Provide the ranking in the boxes provided.

Note: Atomic number (Z) are provided for relevant atoms

a) Increasing (lowest to highest) order of *ionization energy*: (1= lowest; 4 = highest)

Mg

Mg: Z = 12

Rb

Rb: Z = 37

Cs

Cs: Z = 55

K

K : Z = 19

b) Increasing (lowest to highest) order of *bond angle*: (1= lowest; 4 = highest)

H₂OCCl₄XeF₄

HCN

c) Increasing (smallest to largest) order of *ionic size* (1= smallest; 4 = largest)

S²⁻

S: Z = 16

O²⁻

O: Z = 8

Be²⁺

Be: Z = 4

Na⁺

Na: Z = 11

d) Increasing (lowest to highest) order of *bond energy* (1= lowest; 4 = highest)

P-F

P-Cl

P-I

P-Br

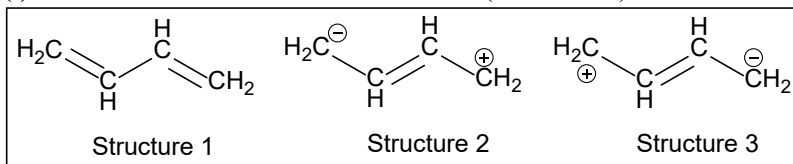
**Question 5** **12 points**

a. 9 points For the molecules HClO_4 and TeF_6 , draw the most stable *Lewis structure* showing appropriate shape and *dashed/wedged bonds*, based on VSEPR. Determine the **molecular geometry** around the central atom. For HClO_4 , label the *most polar bond*.



b. 3 points (Note: Lone pairs are not shown in structures below)

(i) For the three *resonance* structures (1, 2 and 3) shown below, answer the following question:



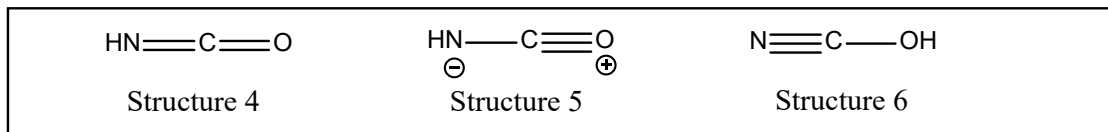
(I) Circle the *most contributing* resonance structure.

Structure 1

Structure 2

Structure 3

(ii) For the three structures given (4, 5, and 6) below, answer the following questions:



(I) Circle all the structures that represent *resonance structures* of each other:

Structure 4

Structure 5

Structure 6

(II) Among the resonance structures, circle the *most contributing* resonance structure.

Structure 4

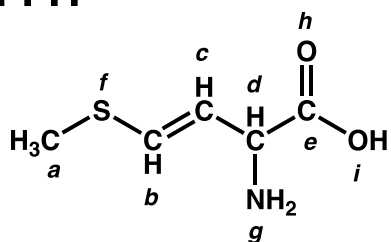
Structure 5

Structure 6

**Question 6** **6 points**

Using the data given below, calculate the **lattice energy** for SrO(s).

	ΔH° (kJ/mol)
$\text{Sr}(s) \longrightarrow \text{Sr}(g)$	+180
$\text{Sr}(g) \longrightarrow \text{Sr}^{2+}(g) + 2e^-$	+1468
$\text{O}_2(g) \longrightarrow 2\text{O}(g)$	+500
$\text{O}(g) + 2e^- \longrightarrow \text{O}^{2-}(g)$	+600
$\text{Sr}(s) + \frac{1}{2}\text{O}_2(g) \longrightarrow \text{SrO}(s)$	-550

**Question 7 10 points**

Consider the molecule given on the left, and answer questions (i) to (vi) using chemical bonding theories that we have learnt in class (valence bond theory, hybridization).

Note that the lone pairs are not shown in the structure:

i. Consider the carbon atoms marked in as ***a***, ***b***, ***c***, ***d***, and ***e*** in the molecule, and determine **hybridization** at each of the carbon centers. (2.5 points)

Atom	Hybridization
Carbon (<i>a</i>)	
Carbon (<i>b</i>)	
Carbon (<i>c</i>)	
Carbon (<i>d</i>)	
Carbon (<i>e</i>)	

ii. Now, consider the sulfur atom (S) marked as ***f***, nitrogen (N) atom (***g***), and oxygen(O) atoms (***h*** and ***i***), and determine their **hybridization**. (2 points)

Atom	Hybridization
Sulfur (<i>f</i>)	
Nitrogen (<i>g</i>)	
Oxygen (<i>h</i>)	
Oxygen (<i>i</i>)	

iii. **Describe** all the bonds formed by the carbon atoms ***a*** and ***b***, using sigma and pi designation as appropriate. (2 points)

Atom	Bonds formed (use sigma and pi as appropriate)
Carbon (<i>a</i>)	
Carbon (<i>b</i>)	

iv. How many total sp^3 -s sigma bonds are in the molecule? (1 point)

v. Determine **electronic** and **molecular geometry** at carbon (C) labeled ***e*** and sulfur (S) labeled ***f*** centres. (2 points)

Atom	Electronic Geometry	Molecular Geometry
Carbon (<i>e</i>)		
Sulfur (<i>f</i>)		

vi. Circle the value from the choices given below, that describes the **C-S-C bond angle at sulfur** (S) marked ***f***: (0.5 points)

>90° to <109.5° 109.5° >109.5 to <120° 120° >120° to <180°

**Question 8 12 points**

Using the **valence shell orbitals only** draw a **complete** molecular orbital (MO) diagram (with no s-p mixing) for the molecule **PS**, in the space provided below. **(6 points)**

Assume that 3s and 3p atomic orbitals will form molecular orbitals like those formed by the combination of 2s and 2p atomic orbitals that we learnt in class, and there is no s-p mixing.

Show **energy arrow**, **atomic orbitals** involved, **atom labels**, the **molecular orbitals** that are formed, **filled electrons** in the MO diagram etc. Make sure to label all the atomic and molecular orbitals.

Using the MO diagram above, answer the following questions:

- (i) Which one of the following will have the highest bond order? Circle the correct choice
A) PS^{1-} B) PS C) PS^{1+}
- (ii) Which one of the following will have the longest bond length? Circle the correct choice
A) PS^{1-} B) PS C) PS^{1+}
- (iii) Which of the following will be diamagnetic? Circle the correct choice.
A) PS^{1-} B) PS C) PS^{1+}
- (iv) Draw Lewis dot structure of the molecule with the shortest bond length (based on the MO diagram) from among the three (PS^{1-} , PS, PS^{1+}), and indicate the bond order calculated from the Lewis dot structure. **(3 points)**

Lewis Dot Structure

Bond Order



293E195D-77E5-45EF-B8D6-CF233721442C

midterm-exam-93388

#834 Page 10 of 10

SCRAP PAGE

**CHEMISTRY 110 –Midterm Exam****Thursday November 7, 2024, 6:30 – 8:30 pm****Examiners: Pallavi Sirjoosingh and Ashok Kakkar**

First name (please write as legibly as possible within the boxes)

--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--

Last name

--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--

Student ID number

--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--

This booklet has 8 questions on 10 pages (The last page can be used as a scrap page).

The staple must NOT be removed from the question booklet. Your intact question booklet must be returned at the end of the exam.

A separate booklet contains a periodic table, and datasheet.

Write your answers directly under the questions in the space provided. Do not write on the QR codes or near the top edges (as it may be cut off).

Please write **legibly and dark enough** for the scan to be readable. For students who use English as a second language, dictionaries are permitted. Calculators are allowed.

DO NOT ASK QUESTIONS OF THE INVIGILATORS about interpretations of the exam questions. If you have a concern about a question, please summarize it at the top of this page. All concerns will be evaluated after the exam. GOOD LUCK!

**Question 1 8 points** (*Assume all transitions are possible*)

a. 6 points Calculate the **energy (in kJ/mol)** required to ionize (completely remove) the electron from the **n=3** level of **He⁺ ion** (He: Helium). Show your work including equations. (3 sig. figs.)

b. 2 points

(I) Select the largest (longest) wavelength from the three choices below:

- i. Wavelength of photon to *completely remove* an electron from *1s orbital of Hydrogen atom*
- ii. Wavelength of photon to *completely remove* an electron from *1s orbital of Boron atom (B)*
- iii. Wavelength of photon to *completely remove* an electron from *1s orbital of Boron cation (B⁺)*

i or ii or iii

(II) Select the smallest (shortest) wavelength from the three choices below (*for H atom*):

- i. Wavelength of photon *emitted* for an electron transitioning from *2p to 1s orbital*
- ii. Wavelength of photon *absorbed* for an electron transitioning from *2s to 4p orbital*
- iii. Wavelength of photon *absorbed* for an electron transitioning from *1s to 3p orbital*

i or ii or iii



Question 2 8 points



A Chem110 student plots the graph shown on the left, from data obtained for the kinetic energy (eV) of the ejected electron versus frequency (s^{-1}) of the incident light, for an unknown metal. ($1\text{eV} = 1.6022 \times 10^{-19} \text{ J}$)

Based on the plot, answer the questions **a to c** below:

a. 1 point On the plot above, label the **threshold frequency**.

b. 2 points What is the value of the work potential (in J) of the unknown metal?

c. 3 points What is the *longest* wavelength (in nm) required to eject an electron from this metal's surface? Show your work including equation(s) used. (3 sig figs)

d. 2 points In the space provided below, plot the following

(Assume frequency is *above* threshold frequency):

- i. *Kinetic energy* of an ejected electron versus the *intensity* of the incident light (left)
- ii. *Number of electrons* ejected versus the *frequency* of the incident light (right)



**Question 3** **6 points****a. 2 points**

In the ground state of Ge (Germanium; $Z = 32$) atom, **name** the following orbitals:

i. Orbital containing the electron that requires the maximum energy to be removed.

ii. Orbital containing the electron that experiences the minimum effective nuclear charge.

b. 2 points

An unknown anion (A^-) has the following *excited* electronic configuration: $[Ar] 4s^2 3d^3 4p^1$

(Ar: Argon; $Z = 18$)

Give the **condensed ground state electronic configuration** of the atom, **A** and the cation, **A^{2+}** .
Show your work. (Assume all electronic transitions are possible)

Electronic Configuration of A:

Electronic Configuration of A^{2+} :

c. 2 points

In a single atom, determine how many electrons can have the following set of quantum numbers?

(i) $n = 5; l = 3$

In a single atom, determine how many electrons can have the following set of quantum numbers?

(i) $n = 3; l = 2; m_s = +1/2$

**Question 4** **8 points**

Rank the following from 1 to 4 in **increasing** order. Provide the ranking in the boxes provided.

Note: Atomic number (Z) are provided for relevant atoms

a) Increasing (lowest to highest) order of *ionization energy*: (1= lowest; 4 = highest)

Mg

Mg: Z = 12

Rb

Rb: Z = 37

Cs

Cs: Z = 55

K

K : Z = 19

b) Increasing (lowest to highest) order of *bond angle*: (1= lowest; 4 = highest)

H₂OCCl₄XeF₄

HCN

c) Increasing (smallest to largest) order of *ionic size* (1= smallest; 4 = largest)

S²⁻

S: Z = 16

O²⁻

O: Z = 8

Be²⁺

Be: Z = 4

Na⁺

Na: Z = 11

d) Increasing (lowest to highest) order of *bond energy* (1= lowest; 4 = highest)

P-F

P-Cl

P-I

P-Br

**Question 5** **12 points**

a. 9 points For the molecules HClO_4 and TeF_6 , draw the most stable *Lewis structure* showing appropriate shape and *dashed/wedged bonds*, based on VSEPR. Determine the **molecular geometry** around the central atom. For HClO_4 , label the *most polar bond*.

HClO_4 (there are no O-O bonds)

Label the most polar bond

Molecular Geometry around Cl

TeF_6

Molecular Geometry around Te

b. 3 points (Note: Lone pairs are not shown in structures below)

(i) For the three *resonance* structures (1, 2 and 3) shown below, answer the following question:



(I) Circle the *most contributing* resonance structure.

Structure 1

Structure 2

Structure 3

(ii) For the three structures given (4, 5, and 6) below, answer the following questions:



(I) Circle all the structures that represent *resonance structures* of each other:

Structure 4

Structure 5

Structure 6

(II) Among the resonance structures, circle the *most contributing* resonance structure.

Structure 4

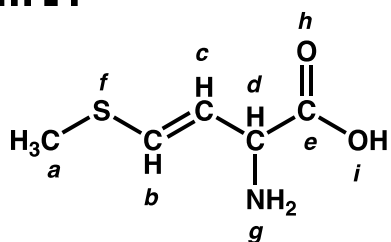
Structure 5

Structure 6

**Question 6** **6 points**

Using the data given below, calculate the **lattice energy** for SrO(s).

	ΔH° (kJ/mol)
$\text{Sr}(s) \longrightarrow \text{Sr}(g)$	+180
$\text{Sr}(g) \longrightarrow \text{Sr}^{2+}(g) + 2e^-$	+1468
$\text{O}_2(g) \longrightarrow 2\text{O}(g)$	+500
$\text{O}(g) + 2e^- \longrightarrow \text{O}^{2-}(g)$	+600
$\text{Sr}(s) + \frac{1}{2}\text{O}_2(g) \longrightarrow \text{SrO}(s)$	-550

**Question 7 10 points**

Consider the molecule given on the left, and answer questions (i) to (vi) using chemical bonding theories that we have learnt in class (valence bond theory, hybridization).

Note that the lone pairs are not shown in the structure:

i. Consider the carbon atoms marked in as ***a***, ***b***, ***c***, ***d***, and ***e*** in the molecule, and determine **hybridization** at each of the carbon centers. (2.5 points)

Atom	Hybridization
Carbon (<i>a</i>)	
Carbon (<i>b</i>)	
Carbon (<i>c</i>)	
Carbon (<i>d</i>)	
Carbon (<i>e</i>)	

ii. Now, consider the sulfur atom (S) marked as ***f***, nitrogen (N) atom (***g***), and oxygen(O) atoms (***h*** and ***i***), and determine their **hybridization**. (2 points)

Atom	Hybridization
Sulfur (<i>f</i>)	
Nitrogen (<i>g</i>)	
Oxygen (<i>h</i>)	
Oxygen (<i>i</i>)	

iii. **Describe** all the bonds formed by the carbon atoms ***a*** and ***b***, using sigma and pi designation as appropriate. (2 points)

Atom	Bonds formed (use sigma and pi as appropriate)
Carbon (<i>a</i>)	
Carbon (<i>b</i>)	

iv. How many total sp^3 -s sigma bonds are in the molecule? (1 point)

v. Determine **electronic** and **molecular geometry** at carbon (C) labeled ***e*** and sulfur (S) labeled ***f*** centres. (2 points)

Atom	Electronic Geometry	Molecular Geometry
Carbon (<i>e</i>)		
Sulfur (<i>f</i>)		

vi. Circle the value from the choices given below, that describes the **C-S-C bond angle at sulfur** (S) marked ***f***: (0.5 points)

>90° to <109.5° 109.5° >109.5 to <120° 120° >120° to <180°



Question 8 12 points

Using the **valence shell orbitals only** draw a **complete** molecular orbital (MO) diagram (with no s-p mixing) for the molecule **PS**, in the space provided below. **(6 points)**

Assume that 3s and 3p atomic orbitals will form molecular orbitals like those formed by the combination of 2s and 2p atomic orbitals that we learnt in class, and there is no s-p mixing.

Show **energy arrow**, **atomic orbitals** involved, **atom labels**, the **molecular orbitals** that are formed, **filled electrons** in the MO diagram etc. Make sure to label all the atomic and molecular orbitals.

Using the MO diagram above, answer the following questions:

- (i) Which one of the following will have the highest bond order? Circle the correct choice
 A) PS^{1-} B) PS C) PS^{1+}
- (ii) Which one of the following will have the longest bond length? Circle the correct choice
 A) PS^{1-} B) PS C) PS^{1+}
- (iii) Which of the following will be diamagnetic? Circle the correct choice.
 A) PS^{1-} B) PS C) PS^{1+}
- (iv) Draw Lewis dot structure of the molecule with the shortest bond length (based on the MO diagram) from among the three (PS^{1-} , PS, PS^{1+}), and indicate the bond order calculated from the Lewis dot structure. **(3 points)**

Lewis Dot Structure

Bond Order



1E7562CA-7766-46B6-BAE5-DD41D965C229

midterm-exam-93388

#835 Page 10 of 10

SCRAP PAGE

**CHEMISTRY 110 –Midterm Exam****Thursday November 7, 2024, 6:30 – 8:30 pm**

Examiners: Pallavi Sirjoosingh and Ashok Kakkar

First name (please write as legibly as possible within the boxes)

--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--

Last name

--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--

Student ID number

--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--

This booklet has 8 questions on 10 pages (The last page can be used as a scrap page).

The staple must NOT be removed from the question booklet. Your intact question booklet must be returned at the end of the exam.

A separate booklet contains a periodic table, and datasheet.

Write your answers directly under the questions in the space provided. Do not write on the QR codes or near the top edges (as it may be cut off).

Please write **legibly and dark enough** for the scan to be readable. For students who use English as a second language, dictionaries are permitted. Calculators are allowed.

DO NOT ASK QUESTIONS OF THE INVIGILATORS about interpretations of the exam questions. If you have a concern about a question, please summarize it at the top of this page. All concerns will be evaluated after the exam. GOOD LUCK!

**Question 1 8 points** (*Assume all transitions are possible*)

a. 6 points Calculate the **energy (in kJ/mol)** required to ionize (completely remove) the electron from the **n=3** level of **He⁺ ion** (He: Helium). Show your work including equations. (3 sig. figs.)

b. 2 points

(I) Select the largest (longest) wavelength from the three choices below:

- i. Wavelength of photon to *completely remove* an electron from *1s orbital of Hydrogen atom*
- ii. Wavelength of photon to *completely remove* an electron from *1s orbital of Boron atom (B)*
- iii. Wavelength of photon to *completely remove* an electron from *1s orbital of Boron cation (B⁺)*

i or ii or iii

(II) Select the smallest (shortest) wavelength from the three choices below (*for H atom*):

- i. Wavelength of photon *emitted* for an electron transitioning from *2p to 1s orbital*
- ii. Wavelength of photon *absorbed* for an electron transitioning from *2s to 4p orbital*
- iii. Wavelength of photon *absorbed* for an electron transitioning from *1s to 3p orbital*

i or ii or iii



Question 2 8 points



A Chem110 student plots the graph shown on the left, from data obtained for the kinetic energy (eV) of the ejected electron versus frequency (s^{-1}) of the incident light, for an unknown metal. ($1\text{eV} = 1.6022 \times 10^{-19} \text{ J}$)

Based on the plot, answer the questions **a to c** below:

a. 1 point On the plot above, label the **threshold frequency**.

b. 2 points What is the value of the work potential (in J) of the unknown metal?

c. 3 points What is the *longest* wavelength (in nm) required to eject an electron from this metal's surface? Show your work including equation(s) used. (3 sig figs)

d. 2 points In the space provided below, plot the following

(Assume frequency is *above* threshold frequency):

- i. *Kinetic energy* of an ejected electron versus the *intensity* of the incident light (left)
- ii. *Number of electrons* ejected versus the *frequency* of the incident light (right)



**Question 3** **6 points****a. 2 points**

In the ground state of Ge (Germanium; $Z = 32$) atom, **name** the following orbitals:

i. Orbital containing the electron that requires the maximum energy to be removed.

ii. Orbital containing the electron that experiences the minimum effective nuclear charge.

b. 2 points

An unknown anion (A^-) has the following *excited* electronic configuration: $[Ar] 4s^2 3d^3 4p^1$

(Ar: Argon; $Z = 18$)

Give the **condensed ground state electronic configuration** of the atom, **A** and the cation, **A^{2+}** .
Show your work. (Assume all electronic transitions are possible)

Electronic Configuration of A:

Electronic Configuration of A^{2+} :

c. 2 points

In a single atom, determine how many electrons can have the following set of quantum numbers?

(i) $n = 5; l = 3$

In a single atom, determine how many electrons can have the following set of quantum numbers?

(i) $n = 3; l = 2; m_s = +1/2$

**Question 4** **8 points**

Rank the following from 1 to 4 in **increasing** order. Provide the ranking in the boxes provided.

Note: Atomic number (Z) are provided for relevant atoms

a) Increasing (lowest to highest) order of *ionization energy*: (1= lowest; 4 = highest)

Mg

Mg: Z = 12

Rb

Rb: Z = 37

Cs

Cs: Z = 55

K

K : Z = 19

b) Increasing (lowest to highest) order of *bond angle*: (1= lowest; 4 = highest)

H₂OCCl₄XeF₄

HCN

c) Increasing (smallest to largest) order of *ionic size* (1= smallest; 4 = largest)

S²⁻

S: Z = 16

O²⁻

O: Z = 8

Be²⁺

Be: Z = 4

Na⁺

Na: Z = 11

d) Increasing (lowest to highest) order of *bond energy* (1= lowest; 4 = highest)

P-F

P-Cl

P-I

P-Br

**Question 5** **12 points**

a. 9 points For the molecules HClO_4 and TeF_6 , draw the most stable *Lewis structure* showing appropriate shape and *dashed/wedged bonds*, based on VSEPR. Determine the **molecular geometry** around the central atom. For HClO_4 , label the *most polar bond*.



b. 3 points (Note: Lone pairs are not shown in structures below)

(i) For the three *resonance* structures (1, 2 and 3) shown below, answer the following question:



(I) Circle the *most contributing* resonance structure.

Structure 1

Structure 2

Structure 3

(ii) For the three structures given (4, 5, and 6) below, answer the following questions:



(I) Circle all the structures that represent *resonance structures* of each other:

Structure 4

Structure 5

Structure 6

(II) Among the resonance structures, circle the *most contributing* resonance structure.

Structure 4

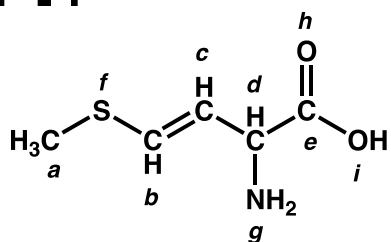
Structure 5

Structure 6

**Question 6** **6 points**

Using the data given below, calculate the **lattice energy** for SrO(s).

	ΔH° (kJ/mol)
$\text{Sr}(s) \longrightarrow \text{Sr}(g)$	+180
$\text{Sr}(g) \longrightarrow \text{Sr}^{2+}(g) + 2e^-$	+1468
$\text{O}_2(g) \longrightarrow 2\text{O}(g)$	+500
$\text{O}(g) + 2e^- \longrightarrow \text{O}^{2-}(g)$	+600
$\text{Sr}(s) + \frac{1}{2}\text{O}_2(g) \longrightarrow \text{SrO}(s)$	-550

**Question 7 10 points**

Consider the molecule given on the left, and answer questions (i) to (vi) using chemical bonding theories that we have learnt in class (valence bond theory, hybridization).

Note that the lone pairs are not shown in the structure:

i. Consider the carbon atoms marked in as ***a***, ***b***, ***c***, ***d***, and ***e*** in the molecule, and determine **hybridization** at each of the carbon centers. (2.5 points)

Atom	Hybridization
Carbon (<i>a</i>)	
Carbon (<i>b</i>)	
Carbon (<i>c</i>)	
Carbon (<i>d</i>)	
Carbon (<i>e</i>)	

ii. Now, consider the sulfur atom (S) marked as ***f***, nitrogen (N) atom (***g***), and oxygen(O) atoms (***h*** and ***i***), and determine their **hybridization**. (2 points)

Atom	Hybridization
Sulfur (<i>f</i>)	
Nitrogen (<i>g</i>)	
Oxygen (<i>h</i>)	
Oxygen (<i>i</i>)	

iii. **Describe** all the bonds formed by the carbon atoms ***a*** and ***b***, using sigma and pi designation as appropriate. (2 points)

Atom	Bonds formed (use sigma and pi as appropriate)
Carbon (<i>a</i>)	
Carbon (<i>b</i>)	

iv. How many total sp^3 -s sigma bonds are in the molecule? (1 point)

v. Determine **electronic** and **molecular geometry** at carbon (C) labeled ***e*** and sulfur (S) labeled ***f*** centres. (2 points)

Atom	Electronic Geometry	Molecular Geometry
Carbon (<i>e</i>)		
Sulfur (<i>f</i>)		

vi. Circle the value from the choices given below, that describes the **C-S-C bond angle at sulfur** (S) marked ***f***: (0.5 points)

>90° to <109.5° 109.5° >109.5 to <120° 120° >120° to <180°


Question 8 12 points

Using the **valence shell orbitals only** draw a **complete** molecular orbital (MO) diagram (with no s-p mixing) for the molecule **PS**, in the space provided below. **(6 points)**

Assume that 3s and 3p atomic orbitals will form molecular orbitals like those formed by the combination of 2s and 2p atomic orbitals that we learnt in class, and there is no s-p mixing.

Show **energy arrow**, **atomic orbitals** involved, **atom labels**, the **molecular orbitals** that are formed, **filled electrons** in the MO diagram etc. Make sure to **label all the atomic and molecular orbitals**.

Using the MO diagram above, answer the following questions:

- (i) Which one of the following will have the highest bond order? Circle the correct choice
 A) PS^{1-} B) PS C) PS^{1+}
- (ii) Which one of the following will have the longest bond length? Circle the correct choice
 A) PS^{1-} B) PS C) PS^{1+}
- (iii) Which of the following will be diamagnetic? Circle the correct choice.
 A) PS^{1-} B) PS C) PS^{1+}
- (iv) Draw Lewis dot structure of the molecule with the shortest bond length (based on the MO diagram) from among the three (PS^{1-} , PS, PS^{1+}), and indicate the bond order calculated from the Lewis dot structure. **(3 points)**

Lewis Dot Structure

Bond Order



5921E05F-0BDE-4662-BA33-3134980186A3

midterm-exam-93388

#836 Page 10 of 10

SCRAP PAGE

**CHEMISTRY 110 –Midterm Exam****Thursday November 7, 2024, 6:30 – 8:30 pm**

Examiners: Pallavi Sirjoosingh and Ashok Kakkar

First name (please write as legibly as possible within the boxes)

--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--

Last name

--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--

Student ID number

--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--

This booklet has 8 questions on 10 pages (The last page can be used as a scrap page).

The staple must NOT be removed from the question booklet. Your intact question booklet must be returned at the end of the exam.

A separate booklet contains a periodic table, and datasheet.

Write your answers directly under the questions in the space provided. Do not write on the QR codes or near the top edges (as it may be cut off).

Please write **legibly and dark enough** for the scan to be readable. For students who use English as a second language, dictionaries are permitted. Calculators are allowed.

DO NOT ASK QUESTIONS OF THE INVIGILATORS about interpretations of the exam questions. If you have a concern about a question, please summarize it at the top of this page. All concerns will be evaluated after the exam. GOOD LUCK!

**Question 1 8 points** (*Assume all transitions are possible*)

a. 6 points Calculate the **energy (in kJ/mol)** required to ionize (completely remove) the electron from the **$n=3$** level of **He^+ ion** (He: Helium). Show your work including equations. (3 sig. figs.)

b. 2 points

(I) Select the largest (longest) wavelength from the three choices below:

- i. Wavelength of photon to *completely remove* an electron from *1s orbital of Hydrogen atom*
- ii. Wavelength of photon to *completely remove* an electron from *1s orbital of Boron atom (B)*
- iii. Wavelength of photon to *completely remove* an electron from *1s orbital of Boron cation (B^+)*

i or ii or iii

(II) Select the smallest (shortest) wavelength from the three choices below (*for H atom*):

- i. Wavelength of photon *emitted* for an electron transitioning from *2p to 1s orbital*
- ii. Wavelength of photon *absorbed* for an electron transitioning from *2s to 4p orbital*
- iii. Wavelength of photon *absorbed* for an electron transitioning from *1s to 3p orbital*

i or ii or iii



Question 2 8 points



A Chem110 student plots the graph shown on the left, from data obtained for the kinetic energy (eV) of the ejected electron versus frequency (s^{-1}) of the incident light, for an unknown metal. ($1\text{eV} = 1.6022 \times 10^{-19} \text{ J}$)

Based on the plot, answer the questions **a to c** below:

a. 1 point On the plot above, label the **threshold frequency**.

b. 2 points What is the value of the work potential (in J) of the unknown metal?

c. 3 points What is the *longest* wavelength (in nm) required to eject an electron from this metal's surface? Show your work including equation(s) used. (3 sig figs)

d. 2 points In the space provided below, plot the following

(Assume frequency is *above* threshold frequency):

- Kinetic energy* of an ejected electron versus the *intensity* of the incident light (left)
- Number of electrons* ejected versus the *frequency* of the incident light (right)



**Question 3** **6 points****a. 2 points**

In the ground state of Ge (Germanium; $Z = 32$) atom, **name** the following orbitals:

i. Orbital containing the electron that requires the maximum energy to be removed.

ii. Orbital containing the electron that experiences the minimum effective nuclear charge.

b. 2 points

An unknown anion (A^-) has the following *excited* electronic configuration: $[Ar] 4s^2 3d^3 4p^1$

(Ar: Argon; $Z = 18$)

Give the **condensed ground state electronic configuration** of the atom, **A** and the cation, **A^{2+}** .
Show your work. (Assume all electronic transitions are possible)

Electronic Configuration of A:

Electronic Configuration of A^{2+} :

c. 2 points

In a single atom, determine how many electrons can have the following set of quantum numbers?

(i) $n = 5; l = 3$

In a single atom, determine how many electrons can have the following set of quantum numbers?

(i) $n = 3; l = 2; m_s = +1/2$


Question 4 **8 points**

Rank the following from 1 to 4 in **increasing** order. Provide the ranking in the boxes provided.

Note: Atomic number (Z) are provided for relevant atoms

a) Increasing (lowest to highest) order of *ionization energy*: **(1= lowest; 4 = highest)**

Mg

Mg: Z = 12

Rb

Rb: Z = 37

Cs

Cs: Z = 55

K

K : Z = 19

b) Increasing (lowest to highest) order of *bond angle*: **(1= lowest; 4 = highest)**

H₂O
CCl₄

XeF₄

HCN

c) Increasing (smallest to largest) order of *ionic size* **(1= smallest; 4 = largest)**

S²⁻

S: Z = 16

O²⁻

O: Z = 8

Be²⁺

Be: Z = 4

Na⁺

Na: Z = 11

d) Increasing (lowest to highest) order of *bond energy* **(1= lowest; 4 = highest)**

P-F

P-Cl

P-I

P-Br

**Question 5** **12 points**

a. 9 points For the molecules HClO_4 and TeF_6 , draw the most stable *Lewis structure* showing appropriate shape and *dashed/wedged bonds*, based on VSEPR. Determine the **molecular geometry** around the central atom. For HClO_4 , label the *most polar bond*.



b. 3 points (Note: Lone pairs are not shown in structures below)

(i) For the three *resonance* structures (1, 2 and 3) shown below, answer the following question:



(I) Circle the *most contributing* resonance structure.

Structure 1

Structure 2

Structure 3

(ii) For the three structures given (4, 5, and 6) below, answer the following questions:



(I) Circle all the structures that represent *resonance structures* of each other:

Structure 4

Structure 5

Structure 6

(II) Among the resonance structures, circle the *most contributing* resonance structure.

Structure 4

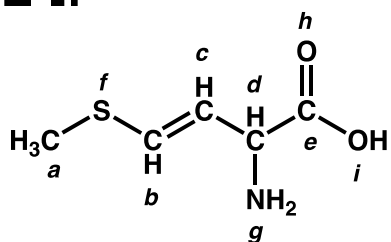
Structure 5

Structure 6

**Question 6** **6 points**

Using the data given below, calculate the **lattice energy** for SrO(s).

	ΔH° (kJ/mol)
$\text{Sr}(s) \longrightarrow \text{Sr}(g)$	+180
$\text{Sr}(g) \longrightarrow \text{Sr}^{2+}(g) + 2e^-$	+1468
$\text{O}_2(g) \longrightarrow 2\text{O}(g)$	+500
$\text{O}(g) + 2e^- \longrightarrow \text{O}^{2-}(g)$	+600
$\text{Sr}(s) + \frac{1}{2}\text{O}_2(g) \longrightarrow \text{SrO}(s)$	-550

**Question 7 10 points**

Consider the molecule given on the left, and answer questions (i) to (vi) using chemical bonding theories that we have learnt in class (valence bond theory, hybridization).

Note that the lone pairs are not shown in the structure:

i. Consider the carbon atoms marked in as ***a***, ***b***, ***c***, ***d***, and ***e*** in the molecule, and determine **hybridization** at each of the carbon centers. (2.5 points)

Atom	Hybridization
Carbon (<i>a</i>)	
Carbon (<i>b</i>)	
Carbon (<i>c</i>)	
Carbon (<i>d</i>)	
Carbon (<i>e</i>)	

ii. Now, consider the sulfur atom (S) marked as ***f***, nitrogen (N) atom (***g***), and oxygen(O) atoms (***h*** and ***i***), and determine their **hybridization**. (2 points)

Atom	Hybridization
Sulfur (<i>f</i>)	
Nitrogen (<i>g</i>)	
Oxygen (<i>h</i>)	
Oxygen (<i>i</i>)	

iii. **Describe** all the bonds formed by the carbon atoms ***a*** and ***b***, using sigma and pi designation as appropriate. (2 points)

Atom	Bonds formed (use sigma and pi as appropriate)
Carbon (<i>a</i>)	
Carbon (<i>b</i>)	

iv. How many total sp^3 -s sigma bonds are in the molecule? (1 point)

v. Determine **electronic** and **molecular geometry** at carbon (C) labeled ***e*** and sulfur (S) labeled ***f*** centres. (2 points)

Atom	Electronic Geometry	Molecular Geometry
Carbon (<i>e</i>)		
Sulfur (<i>f</i>)		

vi. Circle the value from the choices given below, that describes the **C-S-C bond angle at sulfur** (S) marked ***f***: (0.5 points)

$>90^\circ$ to $<109.5^\circ$

109.5°

>109.5 to $<120^\circ$

120°

$>120^\circ$ to $<180^\circ$


Question 8 12 points

Using the **valence shell orbitals only** draw a **complete** molecular orbital (MO) diagram (with no s-p mixing) for the molecule **PS**, in the space provided below. **(6 points)**

Assume that 3s and 3p atomic orbitals will form molecular orbitals like those formed by the combination of 2s and 2p atomic orbitals that we learnt in class, and there is no s-p mixing.

Show **energy arrow**, **atomic orbitals** involved, **atom labels**, the **molecular orbitals** that are formed, **filled electrons** in the MO diagram etc. Make sure to label all the atomic and molecular orbitals.

Using the MO diagram above, answer the following questions:

- (i) Which one of the following will have the highest bond order? Circle the correct choice
 A) PS^{1-} B) PS C) PS^{1+}
- (ii) Which one of the following will have the longest bond length? Circle the correct choice
 A) PS^{1-} B) PS C) PS^{1+}
- (iii) Which of the following will be diamagnetic? Circle the correct choice.
 A) PS^{1-} B) PS C) PS^{1+}
- (iv) Draw Lewis dot structure of the molecule with the shortest bond length (based on the MO diagram) from among the three (PS^{1-} , PS, PS^{1+}), and indicate the bond order calculated from the Lewis dot structure. **(3 points)**

Lewis Dot Structure

Bond Order



5FB3A194-ED8B-4908-8BBF-02E7557746EE

midterm-exam-93388

#837 Page 10 of 10

SCRAP PAGE

**CHEMISTRY 110 –Midterm Exam****Thursday November 7, 2024, 6:30 – 8:30 pm****Examiners: Pallavi Sirjoosingh and Ashok Kakkar**

First name (please write as legibly as possible within the boxes)

--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--

Last name

--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--

Student ID number

--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--

This booklet has 8 questions on 10 pages (The last page can be used as a scrap page).

The staple must NOT be removed from the question booklet. Your intact question booklet must be returned at the end of the exam.

A separate booklet contains a periodic table, and datasheet.

Write your answers directly under the questions in the space provided. Do not write on the QR codes or near the top edges (as it may be cut off).

Please write **legibly and dark enough** for the scan to be readable. For students who use English as a second language, dictionaries are permitted. Calculators are allowed.

DO NOT ASK QUESTIONS OF THE INVIGILATORS about interpretations of the exam questions. If you have a concern about a question, please summarize it at the top of this page. All concerns will be evaluated after the exam. GOOD LUCK!

**Question 1 8 points** (*Assume all transitions are possible*)

a. 6 points Calculate the **energy (in kJ/mol)** required to ionize (completely remove) the electron from the **n=3** level of **He⁺ ion** (He: Helium). Show your work including equations. (3 sig. figs.)

b. 2 points

(I) Select the largest (longest) wavelength from the three choices below:

- i. Wavelength of photon to *completely remove* an electron from *1s orbital of Hydrogen atom*
- ii. Wavelength of photon to *completely remove* an electron from *1s orbital of Boron atom (B)*
- iii. Wavelength of photon to *completely remove* an electron from *1s orbital of Boron cation (B⁺)*

i or ii or iii

(II) Select the smallest (shortest) wavelength from the three choices below (*for H atom*):

- i. Wavelength of photon *emitted* for an electron transitioning from *2p to 1s orbital*
- ii. Wavelength of photon *absorbed* for an electron transitioning from *2s to 4p orbital*
- iii. Wavelength of photon *absorbed* for an electron transitioning from *1s to 3p orbital*

i or ii or iii



Question 2 8 points



A Chem110 student plots the graph shown on the left, from data obtained for the kinetic energy (eV) of the ejected electron versus frequency (s^{-1}) of the incident light, for an unknown metal. ($1\text{eV} = 1.6022 \times 10^{-19} \text{ J}$)

Based on the plot, answer the questions **a to c** below:

a. 1 point On the plot above, label the **threshold frequency**.

b. 2 points What is the value of the work potential (in J) of the unknown metal?

c. 3 points What is the *longest* wavelength (in nm) required to eject an electron from this metal's surface? Show your work including equation(s) used. (3 sig figs)

d. 2 points In the space provided below, plot the following

(Assume frequency is *above* threshold frequency):

- i. *Kinetic energy* of an ejected electron versus the *intensity* of the incident light (left)
- ii. *Number of electrons* ejected versus the *frequency* of the incident light (right)



**Question 3** **6 points****a. 2 points**

In the ground state of Ge (Germanium; $Z = 32$) atom, **name** the following orbitals:

i. Orbital containing the electron that requires the maximum energy to be removed.

ii. Orbital containing the electron that experiences the minimum effective nuclear charge.

b. 2 points

An unknown anion (A^-) has the following *excited* electronic configuration: $[Ar] 4s^2 3d^3 4p^1$

(Ar: Argon; $Z = 18$)

Give the **condensed ground state electronic configuration** of the atom, **A** and the cation, **A^{2+}** .
Show your work. (Assume all electronic transitions are possible)

Electronic Configuration of A:

Electronic Configuration of A^{2+} :

c. 2 points

In a single atom, determine how many electrons can have the following set of quantum numbers?

(i) $n = 5; l = 3$

In a single atom, determine how many electrons can have the following set of quantum numbers?

(i) $n = 3; l = 2; m_s = +1/2$

**Question 4** **8 points**

Rank the following from 1 to 4 in **increasing** order. Provide the ranking in the boxes provided.

Note: Atomic number (Z) are provided for relevant atoms

a) Increasing (lowest to highest) order of *ionization energy*: (1= lowest; 4 = highest)

Mg

Mg: Z = 12

Rb

Rb: Z = 37

Cs

Cs: Z = 55

K

K : Z = 19

b) Increasing (lowest to highest) order of *bond angle*: (1= lowest; 4 = highest)

H₂OCCl₄XeF₄

HCN

c) Increasing (smallest to largest) order of *ionic size* (1= smallest; 4 = largest)

S²⁻

S: Z = 16

O²⁻

O: Z = 8

Be²⁺

Be: Z = 4

Na⁺

Na: Z = 11

d) Increasing (lowest to highest) order of *bond energy* (1= lowest; 4 = highest)

P-F

P-Cl

P-I

P-Br

**Question 5** **12 points**

a. 9 points For the molecules HClO_4 and TeF_6 , draw the most stable *Lewis structure* showing appropriate shape and *dashed/wedged bonds*, based on VSEPR. Determine the **molecular geometry** around the central atom. For HClO_4 , label the *most polar bond*.



b. 3 points (Note: Lone pairs are not shown in structures below)

(i) For the three *resonance* structures (1, 2 and 3) shown below, answer the following question:



(I) Circle the *most contributing* resonance structure.

Structure 1

Structure 2

Structure 3

(ii) For the three structures given (4, 5, and 6) below, answer the following questions:



(I) Circle all the structures that represent *resonance structures* of each other:

Structure 4

Structure 5

Structure 6

(II) Among the resonance structures, circle the *most contributing* resonance structure.

Structure 4

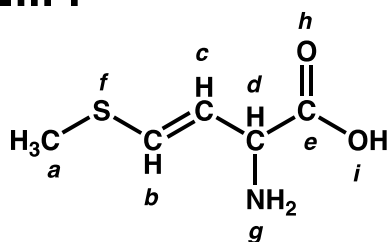
Structure 5

Structure 6

**Question 6** **6 points**

Using the data given below, calculate the **lattice energy** for SrO(s).

	ΔH° (kJ/mol)
$\text{Sr}(s) \longrightarrow \text{Sr}(g)$	+180
$\text{Sr}(g) \longrightarrow \text{Sr}^{2+}(g) + 2e^-$	+1468
$\text{O}_2(g) \longrightarrow 2\text{O}(g)$	+500
$\text{O}(g) + 2e^- \longrightarrow \text{O}^{2-}(g)$	+600
$\text{Sr}(s) + \frac{1}{2}\text{O}_2(g) \longrightarrow \text{SrO}(s)$	-550

**Question 7 10 points**

Consider the molecule given on the left, and answer questions (i) to (vi) using chemical bonding theories that we have learnt in class (valence bond theory, hybridization).

Note that the lone pairs are not shown in the structure:

i. Consider the carbon atoms marked in as ***a***, ***b***, ***c***, ***d***, and ***e*** in the molecule, and determine **hybridization** at each of the carbon centers. (2.5 points)

Atom	Hybridization
Carbon (<i>a</i>)	
Carbon (<i>b</i>)	
Carbon (<i>c</i>)	
Carbon (<i>d</i>)	
Carbon (<i>e</i>)	

ii. Now, consider the sulfur atom (S) marked as ***f***, nitrogen (N) atom (***g***), and oxygen(O) atoms (***h*** and ***i***), and determine their **hybridization**. (2 points)

Atom	Hybridization
Sulfur (<i>f</i>)	
Nitrogen (<i>g</i>)	
Oxygen (<i>h</i>)	
Oxygen (<i>i</i>)	

iii. **Describe** all the bonds formed by the carbon atoms ***a*** and ***b***, using sigma and pi designation as appropriate. (2 points)

Atom	Bonds formed (use sigma and pi as appropriate)
Carbon (<i>a</i>)	
Carbon (<i>b</i>)	

iv. How many total sp^3 -s sigma bonds are in the molecule? (1 point)

v. Determine **electronic** and **molecular geometry** at carbon (C) labeled ***e*** and sulfur (S) labeled ***f*** centres. (2 points)

Atom	Electronic Geometry	Molecular Geometry
Carbon (<i>e</i>)		
Sulfur (<i>f</i>)		

vi. Circle the value from the choices given below, that describes the **C-S-C bond angle at sulfur** (S) marked ***f***: (0.5 points)

$>90^\circ$ to $<109.5^\circ$ 109.5° >109.5 to $<120^\circ$ 120° $>120^\circ$ to $<180^\circ$

**Question 8 12 points**

Using the **valence shell orbitals only** draw a **complete** molecular orbital (MO) diagram (with no s-p mixing) for the molecule **PS**, in the space provided below. **(6 points)**

Assume that 3s and 3p atomic orbitals will form molecular orbitals like those formed by the combination of 2s and 2p atomic orbitals that we learnt in class, and there is no s-p mixing.

Show **energy arrow**, **atomic orbitals** involved, **atom labels**, the **molecular orbitals** that are formed, **filled electrons** in the MO diagram etc. Make sure to label all the atomic and molecular orbitals.

Using the MO diagram above, answer the following questions:

- (i) Which one of the following will have the highest bond order? Circle the correct choice
A) PS^{1-} B) PS C) PS^{1+}
- (ii) Which one of the following will have the longest bond length? Circle the correct choice
A) PS^{1-} B) PS C) PS^{1+}
- (iii) Which of the following will be diamagnetic? Circle the correct choice.
A) PS^{1-} B) PS C) PS^{1+}
- (iv) Draw Lewis dot structure of the molecule with the shortest bond length (based on the MO diagram) from among the three (PS^{1-} , PS, PS^{1+}), and indicate the bond order calculated from the Lewis dot structure. **(3 points)**

Lewis Dot Structure

Bond Order



E6F5CD7F-4182-45EA-9778-643E7A63BC6A

midterm-exam-93388

#838 Page 10 of 10

SCRAP PAGE

**CHEMISTRY 110 –Midterm Exam****Thursday November 7, 2024, 6:30 – 8:30 pm**

Examiners: Pallavi Sirjoosingh and Ashok Kakkar

First name (please write as legibly as possible within the boxes)

--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--

Last name

--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--

Student ID number

--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--

This booklet has 8 questions on 10 pages (The last page can be used as a scrap page).

The staple must NOT be removed from the question booklet. Your intact question booklet must be returned at the end of the exam.

A separate booklet contains a periodic table, and datasheet.

Write your answers directly under the questions in the space provided. Do not write on the QR codes or near the top edges (as it may be cut off).

Please write **legibly and dark enough** for the scan to be readable. For students who use English as a second language, dictionaries are permitted. Calculators are allowed.

DO NOT ASK QUESTIONS OF THE INVIGILATORS about interpretations of the exam questions. If you have a concern about a question, please summarize it at the top of this page. All concerns will be evaluated after the exam. GOOD LUCK!

**Question 1 8 points** (*Assume all transitions are possible*)

a. 6 points Calculate the **energy (in kJ/mol)** required to ionize (completely remove) the electron from the **n=3** level of **He⁺ ion** (He: Helium). Show your work including equations. (3 sig. figs.)

b. 2 points

(I) Select the largest (longest) wavelength from the three choices below:

- i. Wavelength of photon to *completely remove* an electron from *1s orbital of Hydrogen atom*
- ii. Wavelength of photon to *completely remove* an electron from *1s orbital of Boron atom (B)*
- iii. Wavelength of photon to *completely remove* an electron from *1s orbital of Boron cation (B⁺)*

i or ii or iii

(II) Select the smallest (shortest) wavelength from the three choices below (*for H atom*):

- i. Wavelength of photon *emitted* for an electron transitioning from *2p to 1s orbital*
- ii. Wavelength of photon *absorbed* for an electron transitioning from *2s to 4p orbital*
- iii. Wavelength of photon *absorbed* for an electron transitioning from *1s to 3p orbital*

i or ii or iii



Question 2 8 points



A Chem110 student plots the graph shown on the left, from data obtained for the kinetic energy (eV) of the ejected electron versus frequency (s^{-1}) of the incident light, for an unknown metal. ($1\text{eV} = 1.6022 \times 10^{-19} \text{ J}$)

Based on the plot, answer the questions **a to c** below:

a. 1 point On the plot above, label the **threshold frequency**.

b. 2 points What is the value of the work potential (in J) of the unknown metal?

c. 3 points What is the *longest* wavelength (in nm) required to eject an electron from this metal's surface? Show your work including equation(s) used. (3 sig figs)

d. 2 points In the space provided below, plot the following

(Assume frequency is *above* threshold frequency):

- i. *Kinetic energy* of an ejected electron versus the *intensity* of the incident light (left)
- ii. *Number of electrons* ejected versus the *frequency* of the incident light (right)



**Question 3** **6 points****a. 2 points**

In the ground state of Ge (Germanium; $Z = 32$) atom, **name** the following orbitals:

i. Orbital containing the electron that requires the maximum energy to be removed.

ii. Orbital containing the electron that experiences the minimum effective nuclear charge.

b. 2 points

An unknown anion (A^-) has the following *excited* electronic configuration: $[Ar] 4s^2 3d^3 4p^1$

(Ar: Argon; $Z = 18$)

Give the **condensed ground state electronic configuration** of the atom, **A** and the cation, **A^{2+}** .
Show your work. (Assume all electronic transitions are possible)

Electronic Configuration of A:

Electronic Configuration of A^{2+} :

c. 2 points

In a single atom, determine how many electrons can have the following set of quantum numbers?

(i) $n = 5; l = 3$

In a single atom, determine how many electrons can have the following set of quantum numbers?

(i) $n = 3; l = 2; m_s = +1/2$

**Question 4** **8 points**

Rank the following from 1 to 4 in **increasing** order. Provide the ranking in the boxes provided.

Note: Atomic number (Z) are provided for relevant atoms

a) Increasing (lowest to highest) order of *ionization energy*: (1= lowest; 4 = highest)

Mg

Mg: Z = 12

Rb

Rb: Z = 37

Cs

Cs: Z = 55

K

K : Z = 19

b) Increasing (lowest to highest) order of *bond angle*: (1= lowest; 4 = highest)

H₂OCCl₄XeF₄

HCN

c) Increasing (smallest to largest) order of *ionic size* (1= smallest; 4 = largest)

S²⁻

S: Z = 16

O²⁻

O: Z = 8

Be²⁺

Be: Z = 4

Na⁺

Na: Z = 11

d) Increasing (lowest to highest) order of *bond energy* (1= lowest; 4 = highest)

P-F

P-Cl

P-I

P-Br

**Question 5** **12 points**

a. 9 points For the molecules HClO_4 and TeF_6 , draw the most stable *Lewis structure* showing appropriate shape and *dashed/wedged bonds*, based on VSEPR. Determine the **molecular geometry** around the central atom. For HClO_4 , label the *most polar bond*.



b. 3 points (Note: Lone pairs are not shown in structures below)

(i) For the three *resonance* structures (1, 2 and 3) shown below, answer the following question:



(I) Circle the *most contributing* resonance structure.

Structure 1

Structure 2

Structure 3

(ii) For the three structures given (4, 5, and 6) below, answer the following questions:



(I) Circle all the structures that represent *resonance structures* of each other:

Structure 4

Structure 5

Structure 6

(II) Among the resonance structures, circle the *most contributing* resonance structure.

Structure 4

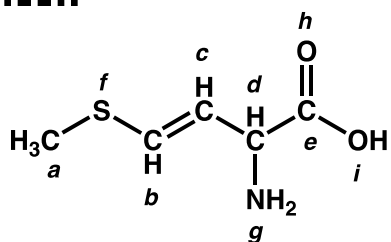
Structure 5

Structure 6

**Question 6** **6 points**

Using the data given below, calculate the **lattice energy** for SrO(s).

	ΔH° (kJ/mol)
$\text{Sr}(s) \longrightarrow \text{Sr}(g)$	+180
$\text{Sr}(g) \longrightarrow \text{Sr}^{2+}(g) + 2e^-$	+1468
$\text{O}_2(g) \longrightarrow 2\text{O}(g)$	+500
$\text{O}(g) + 2e^- \longrightarrow \text{O}^{2-}(g)$	+600
$\text{Sr}(s) + \frac{1}{2}\text{O}_2(g) \longrightarrow \text{SrO}(s)$	-550

**Question 7 10 points**

Consider the molecule given on the left, and answer questions (i) to (vi) using chemical bonding theories that we have learnt in class (valence bond theory, hybridization).

Note that the lone pairs are not shown in the structure:

i. Consider the carbon atoms marked in as ***a***, ***b***, ***c***, ***d***, and ***e*** in the molecule, and determine **hybridization** at each of the carbon centers. (2.5 points)

Atom	Hybridization
Carbon (<i>a</i>)	
Carbon (<i>b</i>)	
Carbon (<i>c</i>)	
Carbon (<i>d</i>)	
Carbon (<i>e</i>)	

ii. Now, consider the sulfur atom (S) marked as ***f***, nitrogen (N) atom (***g***), and oxygen(O) atoms (***h*** and ***i***), and determine their **hybridization**. (2 points)

Atom	Hybridization
Sulfur (<i>f</i>)	
Nitrogen (<i>g</i>)	
Oxygen (<i>h</i>)	
Oxygen (<i>i</i>)	

iii. **Describe** all the bonds formed by the carbon atoms ***a*** and ***b***, using sigma and pi designation as appropriate. (2 points)

Atom	Bonds formed (use sigma and pi as appropriate)
Carbon (<i>a</i>)	
Carbon (<i>b</i>)	

iv. How many total sp^3 -s sigma bonds are in the molecule? (1 point)

v. Determine **electronic** and **molecular geometry** at carbon (C) labeled ***e*** and sulfur (S) labeled ***f*** centres. (2 points)

Atom	Electronic Geometry	Molecular Geometry
Carbon (<i>e</i>)		
Sulfur (<i>f</i>)		

vi. Circle the value from the choices given below, that describes the **C-S-C bond angle at sulfur** (S) marked ***f***: (0.5 points)

$>90^\circ$ to $<109.5^\circ$ 109.5° >109.5 to $<120^\circ$ 120° $>120^\circ$ to $<180^\circ$

**Question 8 12 points**

Using the **valence shell orbitals only** draw a **complete** molecular orbital (MO) diagram (with no s-p mixing) for the molecule **PS**, in the space provided below. **(6 points)**

Assume that 3s and 3p atomic orbitals will form molecular orbitals like those formed by the combination of 2s and 2p atomic orbitals that we learnt in class, and there is no s-p mixing.

Show **energy arrow**, **atomic orbitals** involved, **atom labels**, the **molecular orbitals** that are formed, **filled electrons** in the MO diagram etc. Make sure to label all the atomic and molecular orbitals.

Using the MO diagram above, answer the following questions:

- (i) Which one of the following will have the highest bond order? Circle the correct choice
A) PS^{1-} B) PS C) PS^{1+}
- (ii) Which one of the following will have the longest bond length? Circle the correct choice
A) PS^{1-} B) PS C) PS^{1+}
- (iii) Which of the following will be diamagnetic? Circle the correct choice.
A) PS^{1-} B) PS C) PS^{1+}
- (iv) Draw Lewis dot structure of the molecule with the shortest bond length (based on the MO diagram) from among the three (PS^{1-} , PS, PS^{1+}), and indicate the bond order calculated from the Lewis dot structure. **(3 points)**

Lewis Dot Structure

Bond Order



5DA7AAE2-29AF-4906-B42B-FA6A692EEE24

midterm-exam-93388

#839 Page 10 of 10

SCRAP PAGE

**CHEMISTRY 110 –Midterm Exam****Thursday November 7, 2024, 6:30 – 8:30 pm**

Examiners: Pallavi Sirjoosingh and Ashok Kakkar

First name (please write as legibly as possible within the boxes)

--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--

Last name

--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--

Student ID number

--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--

This booklet has 8 questions on 10 pages (The last page can be used as a scrap page).

The staple must NOT be removed from the question booklet. Your intact question booklet must be returned at the end of the exam.

A separate booklet contains a periodic table, and datasheet.

Write your answers directly under the questions in the space provided. Do not write on the QR codes or near the top edges (as it may be cut off).

Please write **legibly and dark enough** for the scan to be readable. For students who use English as a second language, dictionaries are permitted. Calculators are allowed.

DO NOT ASK QUESTIONS OF THE INVIGILATORS about interpretations of the exam questions. If you have a concern about a question, please summarize it at the top of this page. All concerns will be evaluated after the exam. GOOD LUCK!

**Question 1 8 points** (*Assume all transitions are possible*)

a. 6 points Calculate the **energy (in kJ/mol)** required to ionize (completely remove) the electron from the **n=3** level of **He⁺ ion** (He: Helium). Show your work including equations. (3 sig. figs.)

b. 2 points

(I) Select the largest (longest) wavelength from the three choices below:

- i. Wavelength of photon to *completely remove* an electron from *1s orbital of Hydrogen atom*
- ii. Wavelength of photon to *completely remove* an electron from *1s orbital of Boron atom (B)*
- iii. Wavelength of photon to *completely remove* an electron from *1s orbital of Boron cation (B⁺)*

i or ii or iii

(II) Select the smallest (shortest) wavelength from the three choices below (*for H atom*):

- i. Wavelength of photon *emitted* for an electron transitioning from *2p to 1s orbital*
- ii. Wavelength of photon *absorbed* for an electron transitioning from *2s to 4p orbital*
- iii. Wavelength of photon *absorbed* for an electron transitioning from *1s to 3p orbital*

i or ii or iii



Question 2 8 points



A Chem110 student plots the graph shown on the left, from data obtained for the kinetic energy (eV) of the ejected electron versus frequency (s^{-1}) of the incident light, for an unknown metal. ($1\text{eV} = 1.6022 \times 10^{-19} \text{ J}$)

Based on the plot, answer the questions **a to c** below:

a. 1 point On the plot above, label the **threshold frequency**.

b. 2 points What is the value of the work potential (in J) of the unknown metal?

c. 3 points What is the *longest* wavelength (in nm) required to eject an electron from this metal's surface? Show your work including equation(s) used. (3 sig figs)

d. 2 points In the space provided below, plot the following

(Assume frequency is *above* threshold frequency):

- i. *Kinetic energy* of an ejected electron versus the *intensity* of the incident light (left)
- ii. *Number of electrons* ejected versus the *frequency* of the incident light (right)



**Question 3** **6 points****a. 2 points**

In the ground state of Ge (Germanium; $Z = 32$) atom, **name** the following orbitals:

i. Orbital containing the electron that requires the maximum energy to be removed.

ii. Orbital containing the electron that experiences the minimum effective nuclear charge.

b. 2 points

An unknown anion (A^-) has the following *excited* electronic configuration: $[Ar] 4s^2 3d^3 4p^1$

(Ar: Argon; $Z = 18$)

Give the **condensed ground state electronic configuration** of the atom, **A** and the cation, **A^{2+}** .
Show your work. (Assume all electronic transitions are possible)

Electronic Configuration of A:

Electronic Configuration of A^{2+} :

c. 2 points

In a single atom, determine how many electrons can have the following set of quantum numbers?

(i) $n = 5; l = 3$

In a single atom, determine how many electrons can have the following set of quantum numbers?

(i) $n = 3; l = 2; m_s = +1/2$

**Question 4** **8 points**

Rank the following from 1 to 4 in **increasing** order. Provide the ranking in the boxes provided.

Note: Atomic number (Z) are provided for relevant atoms

a) Increasing (lowest to highest) order of *ionization energy*: (1= lowest; 4 = highest)

Mg

Mg: Z = 12

Rb

Rb: Z = 37

Cs

Cs: Z = 55

K

K : Z = 19

b) Increasing (lowest to highest) order of *bond angle*: (1= lowest; 4 = highest)

H₂OCCl₄XeF₄

HCN

c) Increasing (smallest to largest) order of *ionic size* (1= smallest; 4 = largest)

S²⁻

S: Z = 16

O²⁻

O: Z = 8

Be²⁺

Be: Z = 4

Na⁺

Na: Z = 11

d) Increasing (lowest to highest) order of *bond energy* (1= lowest; 4 = highest)

P-F

P-Cl

P-I

P-Br

**Question 5** **12 points**

a. 9 points For the molecules HClO_4 and TeF_6 , draw the most stable *Lewis structure* showing appropriate shape and *dashed/wedged bonds*, based on VSEPR. Determine the **molecular geometry** around the central atom. For HClO_4 , label the *most polar bond*.



b. 3 points (Note: Lone pairs are not shown in structures below)

(i) For the three *resonance* structures (1, 2 and 3) shown below, answer the following question:



(I) Circle the *most contributing* resonance structure.

Structure 1

Structure 2

Structure 3

(ii) For the three structures given (4, 5, and 6) below, answer the following questions:



(I) Circle all the structures that represent *resonance structures* of each other:

Structure 4

Structure 5

Structure 6

(II) Among the resonance structures, circle the *most contributing* resonance structure.

Structure 4

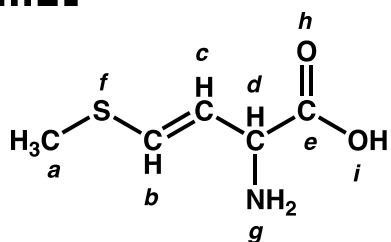
Structure 5

Structure 6

**Question 6** **6 points**

Using the data given below, calculate the **lattice energy** for SrO(s).

	ΔH° (kJ/mol)
$\text{Sr}(s) \longrightarrow \text{Sr}(g)$	+180
$\text{Sr}(g) \longrightarrow \text{Sr}^{2+}(g) + 2e^-$	+1468
$\text{O}_2(g) \longrightarrow 2\text{O}(g)$	+500
$\text{O}(g) + 2e^- \longrightarrow \text{O}^{2-}(g)$	+600
$\text{Sr}(s) + \frac{1}{2}\text{O}_2(g) \longrightarrow \text{SrO}(s)$	-550

**Question 7 10 points**

Consider the molecule given on the left, and answer questions (i) to (vi) using chemical bonding theories that we have learnt in class (valence bond theory, hybridization).

Note that the lone pairs are not shown in the structure:

i. Consider the carbon atoms marked in as ***a***, ***b***, ***c***, ***d***, and ***e*** in the molecule, and determine **hybridization** at each of the carbon centers. (2.5 points)

Atom	Hybridization
Carbon (<i>a</i>)	
Carbon (<i>b</i>)	
Carbon (<i>c</i>)	
Carbon (<i>d</i>)	
Carbon (<i>e</i>)	

ii. Now, consider the sulfur atom (S) marked as ***f***, nitrogen (N) atom (***g***), and oxygen(O) atoms (***h*** and ***i***), and determine their **hybridization**. (2 points)

Atom	Hybridization
Sulfur (<i>f</i>)	
Nitrogen (<i>g</i>)	
Oxygen (<i>h</i>)	
Oxygen (<i>i</i>)	

iii. **Describe** all the bonds formed by the carbon atoms ***a*** and ***b***, using sigma and pi designation as appropriate. (2 points)

Atom	Bonds formed (use sigma and pi as appropriate)
Carbon (<i>a</i>)	
Carbon (<i>b</i>)	

iv. How many total sp^3 -s sigma bonds are in the molecule? (1 point)

v. Determine **electronic** and **molecular geometry** at carbon (C) labeled ***e*** and sulfur (S) labeled ***f*** centres. (2 points)

Atom	Electronic Geometry	Molecular Geometry
Carbon (<i>e</i>)		
Sulfur (<i>f</i>)		

vi. Circle the value from the choices given below, that describes the **C-S-C bond angle at sulfur** (S) marked ***f***: (0.5 points)

>90° to <109.5° 109.5° >109.5 to <120° 120° >120° to <180°

**Question 8 12 points**

Using the **valence shell orbitals only** draw a **complete** molecular orbital (MO) diagram (with no s-p mixing) for the molecule **PS**, in the space provided below. **(6 points)**

Assume that 3s and 3p atomic orbitals will form molecular orbitals like those formed by the combination of 2s and 2p atomic orbitals that we learnt in class, and there is no s-p mixing.

Show **energy arrow**, **atomic orbitals** involved, **atom labels**, the **molecular orbitals** that are formed, **filled electrons** in the MO diagram etc. Make sure to label all the atomic and molecular orbitals.

Using the MO diagram above, answer the following questions:

- (i) Which one of the following will have the highest bond order? Circle the correct choice
A) PS^{1-} B) PS C) PS^{1+}
- (ii) Which one of the following will have the longest bond length? Circle the correct choice
A) PS^{1-} B) PS C) PS^{1+}
- (iii) Which of the following will be diamagnetic? Circle the correct choice.
A) PS^{1-} B) PS C) PS^{1+}
- (iv) Draw Lewis dot structure of the molecule with the shortest bond length (based on the MO diagram) from among the three (PS^{1-} , PS, PS^{1+}), and indicate the bond order calculated from the Lewis dot structure. **(3 points)**

Lewis Dot Structure

Bond Order



803BCF29-0E9D-4455-8471-B10EABD17BEE

midterm-exam-93388

#840 Page 10 of 10

SCRAP PAGE

**CHEMISTRY 110 –Midterm Exam****Thursday November 7, 2024, 6:30 – 8:30 pm**

Examiners: Pallavi Sirjoosingh and Ashok Kakkar

First name (please write as legibly as possible within the boxes)

--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--

Last name

--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--

Student ID number

--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--

This booklet has 8 questions on 10 pages (The last page can be used as a scrap page).

The staple must NOT be removed from the question booklet. Your intact question booklet must be returned at the end of the exam.

A separate booklet contains a periodic table, and datasheet.

Write your answers directly under the questions in the space provided. Do not write on the QR codes or near the top edges (as it may be cut off).

Please write **legibly and dark enough** for the scan to be readable. For students who use English as a second language, dictionaries are permitted. Calculators are allowed.

DO NOT ASK QUESTIONS OF THE INVIGILATORS about interpretations of the exam questions. If you have a concern about a question, please summarize it at the top of this page. All concerns will be evaluated after the exam. GOOD LUCK!

**Question 1 8 points** (*Assume all transitions are possible*)

a. 6 points Calculate the **energy (in kJ/mol)** required to ionize (completely remove) the electron from the **n=3** level of **He⁺ ion** (He: Helium). Show your work including equations. (3 sig. figs.)

b. 2 points

(I) Select the largest (longest) wavelength from the three choices below:

- i. Wavelength of photon to *completely remove* an electron from *1s orbital of Hydrogen atom*
- ii. Wavelength of photon to *completely remove* an electron from *1s orbital of Boron atom (B)*
- iii. Wavelength of photon to *completely remove* an electron from *1s orbital of Boron cation (B⁺)*

i or ii or iii

(II) Select the smallest (shortest) wavelength from the three choices below (*for H atom*):

- i. Wavelength of photon *emitted* for an electron transitioning from *2p to 1s orbital*
- ii. Wavelength of photon *absorbed* for an electron transitioning from *2s to 4p orbital*
- iii. Wavelength of photon *absorbed* for an electron transitioning from *1s to 3p orbital*

i or ii or iii



Question 2 8 points



A Chem110 student plots the graph shown on the left, from data obtained for the kinetic energy (eV) of the ejected electron versus frequency (s^{-1}) of the incident light, for an unknown metal. ($1\text{eV} = 1.6022 \times 10^{-19} \text{ J}$)

Based on the plot, answer the questions **a to c** below:

a. 1 point On the plot above, label the **threshold frequency**.

b. 2 points What is the value of the work potential (in J) of the unknown metal?

c. 3 points What is the *longest* wavelength (in nm) required to eject an electron from this metal's surface? Show your work including equation(s) used. (3 sig figs)

d. 2 points In the space provided below, plot the following

(Assume frequency is *above* threshold frequency):

- Kinetic energy* of an ejected electron versus the *intensity* of the incident light (left)
- Number of electrons* ejected versus the *frequency* of the incident light (right)



**Question 3** **6 points****a. 2 points**

In the ground state of Ge (Germanium; $Z = 32$) atom, **name** the following orbitals:

i. Orbital containing the electron that requires the maximum energy to be removed.

ii. Orbital containing the electron that experiences the minimum effective nuclear charge.

b. 2 points

An unknown anion (A^-) has the following *excited* electronic configuration: $[Ar] 4s^2 3d^3 4p^1$

(Ar: Argon; $Z = 18$)

Give the **condensed ground state electronic configuration** of the atom, **A** and the cation, **A^{2+}** .
Show your work. (Assume all electronic transitions are possible)

Electronic Configuration of A:

Electronic Configuration of A^{2+} :

c. 2 points

In a single atom, determine how many electrons can have the following set of quantum numbers?

(i) $n = 5; l = 3$

In a single atom, determine how many electrons can have the following set of quantum numbers?

(i) $n = 3; l = 2; m_s = +1/2$

**Question 4** **8 points**

Rank the following from 1 to 4 in **increasing** order. Provide the ranking in the boxes provided.

Note: Atomic number (Z) are provided for relevant atoms

a) Increasing (lowest to highest) order of *ionization energy*: (1= lowest; 4 = highest)

Mg

Mg: Z = 12

Rb

Rb: Z = 37

Cs

Cs: Z = 55

K

K : Z = 19

b) Increasing (lowest to highest) order of *bond angle*: (1= lowest; 4 = highest)

H₂OCCl₄XeF₄

HCN

c) Increasing (smallest to largest) order of *ionic size* (1= smallest; 4 = largest)

S²⁻

S: Z = 16

O²⁻

O: Z = 8

Be²⁺

Be: Z = 4

Na⁺

Na: Z = 11

d) Increasing (lowest to highest) order of *bond energy* (1= lowest; 4 = highest)

P-F

P-Cl

P-I

P-Br

**Question 5** **12 points**

a. 9 points For the molecules HClO_4 and TeF_6 , draw the most stable *Lewis structure* showing appropriate shape and *dashed/wedged bonds*, based on VSEPR. Determine the **molecular geometry** around the central atom. For HClO_4 , label the *most polar bond*.

HClO_4 (there are no O-O bonds)

Label the most polar bond

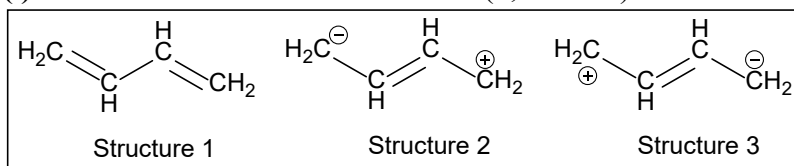
Molecular Geometry around Cl

TeF_6

Molecular Geometry around Te

b. 3 points (Note: Lone pairs are not shown in structures below)

(i) For the three *resonance* structures (1, 2 and 3) shown below, answer the following question:



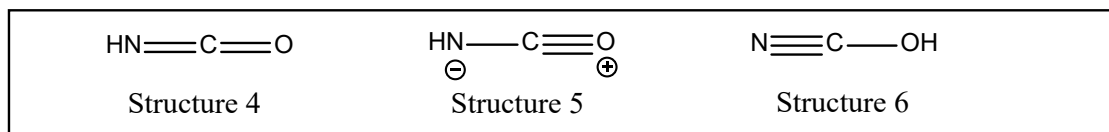
(I) Circle the *most contributing* resonance structure.

Structure 1

Structure 2

Structure 3

(ii) For the three structures given (4, 5, and 6) below, answer the following questions:



(I) Circle all the structures that represent *resonance structures* of each other:

Structure 4

Structure 5

Structure 6

(II) Among the resonance structures, circle the *most contributing* resonance structure.

Structure 4

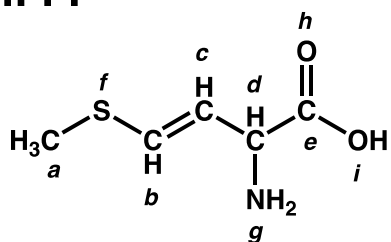
Structure 5

Structure 6

**Question 6** **6 points**

Using the data given below, calculate the **lattice energy** for SrO(s).

	ΔH° (kJ/mol)
$\text{Sr}(s) \longrightarrow \text{Sr}(g)$	+180
$\text{Sr}(g) \longrightarrow \text{Sr}^{2+}(g) + 2e^-$	+1468
$\text{O}_2(g) \longrightarrow 2\text{O}(g)$	+500
$\text{O}(g) + 2e^- \longrightarrow \text{O}^{2-}(g)$	+600
$\text{Sr}(s) + \frac{1}{2}\text{O}_2(g) \longrightarrow \text{SrO}(s)$	-550

**Question 7 10 points**

Consider the molecule given on the left, and answer questions (i) to (vi) using chemical bonding theories that we have learnt in class (valence bond theory, hybridization).

Note that the lone pairs are not shown in the structure:

i. Consider the carbon atoms marked in as ***a***, ***b***, ***c***, ***d***, and ***e*** in the molecule, and determine **hybridization** at each of the carbon centers. (2.5 points)

Atom	Hybridization
Carbon (<i>a</i>)	
Carbon (<i>b</i>)	
Carbon (<i>c</i>)	
Carbon (<i>d</i>)	
Carbon (<i>e</i>)	

ii. Now, consider the sulfur atom (S) marked as ***f***, nitrogen (N) atom (***g***), and oxygen(O) atoms (***h*** and ***i***), and determine their **hybridization**. (2 points)

Atom	Hybridization
Sulfur (<i>f</i>)	
Nitrogen (<i>g</i>)	
Oxygen (<i>h</i>)	
Oxygen (<i>i</i>)	

iii. **Describe** all the bonds formed by the carbon atoms ***a*** and ***b***, using sigma and pi designation as appropriate. (2 points)

Atom	Bonds formed (use sigma and pi as appropriate)
Carbon (<i>a</i>)	
Carbon (<i>b</i>)	

iv. How many total sp^3 -s sigma bonds are in the molecule? (1 point)

v. Determine **electronic** and **molecular geometry** at carbon (C) labeled ***e*** and sulfur (S) labeled ***f*** centres. (2 points)

Atom	Electronic Geometry	Molecular Geometry
Carbon (<i>e</i>)		
Sulfur (<i>f</i>)		

vi. Circle the value from the choices given below, that describes the **C-S-C bond angle at sulfur** (S) marked ***f***: (0.5 points)

>90° to <109.5° 109.5° >109.5 to <120° 120° >120° to <180°


Question 8 12 points

Using the **valence shell orbitals only** draw a **complete** molecular orbital (MO) diagram (with no s-p mixing) for the molecule **PS**, in the space provided below. **(6 points)**

Assume that 3s and 3p atomic orbitals will form molecular orbitals like those formed by the combination of 2s and 2p atomic orbitals that we learnt in class, and there is no s-p mixing.

Show **energy arrow**, **atomic orbitals** involved, **atom labels**, the **molecular orbitals** that are formed, **filled electrons** in the MO diagram etc. Make sure to label all the atomic and molecular orbitals.

Using the MO diagram above, answer the following questions:

- (i) Which one of the following will have the highest bond order? Circle the correct choice
 A) PS^{1-} B) PS C) PS^{1+}
- (ii) Which one of the following will have the longest bond length? Circle the correct choice
 A) PS^{1-} B) PS C) PS^{1+}
- (iii) Which of the following will be diamagnetic? Circle the correct choice.
 A) PS^{1-} B) PS C) PS^{1+}
- (iv) Draw Lewis dot structure of the molecule with the shortest bond length (based on the MO diagram) from among the three (PS^{1-} , PS, PS^{1+}), and indicate the bond order calculated from the Lewis dot structure. **(3 points)**

Lewis Dot Structure

Bond Order



F6E9346E-A558-4838-8AE3-21780B89CDC5

midterm-exam-93388

#841 Page 10 of 10

SCRAP PAGE

**CHEMISTRY 110 –Midterm Exam****Thursday November 7, 2024, 6:30 – 8:30 pm**

Examiners: Pallavi Sirjoosingh and Ashok Kakkar

First name (please write as legibly as possible within the boxes)

--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--

Last name

--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--

Student ID number

--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--

This booklet has 8 questions on 10 pages (The last page can be used as a scrap page).

The staple must NOT be removed from the question booklet. Your intact question booklet must be returned at the end of the exam.

A separate booklet contains a periodic table, and datasheet.

Write your answers directly under the questions in the space provided. Do not write on the QR codes or near the top edges (as it may be cut off).

Please write **legibly and dark enough** for the scan to be readable. For students who use English as a second language, dictionaries are permitted. Calculators are allowed.

DO NOT ASK QUESTIONS OF THE INVIGILATORS about interpretations of the exam questions. If you have a concern about a question, please summarize it at the top of this page. All concerns will be evaluated after the exam. GOOD LUCK!

**Question 1 8 points** (*Assume all transitions are possible*)

a. 6 points Calculate the **energy (in kJ/mol)** required to ionize (completely remove) the electron from the **n=3** level of **He⁺ ion** (He: Helium). Show your work including equations. (3 sig. figs.)

b. 2 points

(I) Select the largest (longest) wavelength from the three choices below:

- i. Wavelength of photon to *completely remove* an electron from *1s orbital of Hydrogen atom*
- ii. Wavelength of photon to *completely remove* an electron from *1s orbital of Boron atom (B)*
- iii. Wavelength of photon to *completely remove* an electron from *1s orbital of Boron cation (B⁺)*

i or ii or iii

(II) Select the smallest (shortest) wavelength from the three choices below (*for H atom*):

- i. Wavelength of photon *emitted* for an electron transitioning from *2p to 1s orbital*
- ii. Wavelength of photon *absorbed* for an electron transitioning from *2s to 4p orbital*
- iii. Wavelength of photon *absorbed* for an electron transitioning from *1s to 3p orbital*

i or ii or iii



Question 2 8 points



A Chem110 student plots the graph shown on the left, from data obtained for the kinetic energy (eV) of the ejected electron versus frequency (s^{-1}) of the incident light, for an unknown metal. ($1\text{eV} = 1.6022 \times 10^{-19} \text{ J}$)

Based on the plot, answer the questions **a to c** below:

a. 1 point On the plot above, label the **threshold frequency**.

b. 2 points What is the value of the work potential (in J) of the unknown metal?

c. 3 points What is the *longest* wavelength (in nm) required to eject an electron from this metal's surface? Show your work including equation(s) used. (3 sig figs)

d. 2 points In the space provided below, plot the following

(Assume frequency is *above* threshold frequency):

- i. *Kinetic energy* of an ejected electron versus the *intensity* of the incident light (left)
- ii. *Number of electrons* ejected versus the *frequency* of the incident light (right)



**Question 3** **6 points****a. 2 points**

In the ground state of Ge (Germanium; $Z = 32$) atom, **name** the following orbitals:

i. Orbital containing the electron that requires the maximum energy to be removed.

ii. Orbital containing the electron that experiences the minimum effective nuclear charge.

b. 2 points

An unknown anion (A^-) has the following *excited* electronic configuration: $[Ar] 4s^2 3d^3 4p^1$

(Ar: Argon; $Z = 18$)

Give the **condensed ground state electronic configuration** of the atom, **A** and the cation, **A^{2+}** .
Show your work. (Assume all electronic transitions are possible)

Electronic Configuration of A:

Electronic Configuration of A^{2+} :

c. 2 points

In a single atom, determine how many electrons can have the following set of quantum numbers?

(i) $n = 5; l = 3$

In a single atom, determine how many electrons can have the following set of quantum numbers?

(i) $n = 3; l = 2; m_s = +1/2$

**Question 4** **8 points**

Rank the following from 1 to 4 in **increasing** order. Provide the ranking in the boxes provided.

Note: Atomic number (Z) are provided for relevant atoms

a) Increasing (lowest to highest) order of *ionization energy*: (1= lowest; 4 = highest)

Mg

Mg: Z = 12

Rb

Rb: Z = 37

Cs

Cs: Z = 55

K

K : Z = 19

b) Increasing (lowest to highest) order of *bond angle*: (1= lowest; 4 = highest)

H₂OCCl₄XeF₄

HCN

c) Increasing (smallest to largest) order of *ionic size* (1= smallest; 4 = largest)

S²⁻

S: Z = 16

O²⁻

O: Z = 8

Be²⁺

Be: Z = 4

Na⁺

Na: Z = 11

d) Increasing (lowest to highest) order of *bond energy* (1= lowest; 4 = highest)

P-F

P-Cl

P-I

P-Br

**Question 5** **12 points**

a. 9 points For the molecules HClO_4 and TeF_6 , draw the most stable *Lewis structure* showing appropriate shape and *dashed/wedged bonds*, based on VSEPR. Determine the **molecular geometry** around the central atom. For HClO_4 , label the *most polar bond*.



b. 3 points (Note: Lone pairs are not shown in structures below)

(i) For the three *resonance* structures (1, 2 and 3) shown below, answer the following question:



(I) Circle the *most contributing* resonance structure.

Structure 1

Structure 2

Structure 3

(ii) For the three structures given (4, 5, and 6) below, answer the following questions:



(I) Circle all the structures that represent *resonance structures* of each other:

Structure 4

Structure 5

Structure 6

(II) Among the resonance structures, circle the *most contributing* resonance structure.

Structure 4

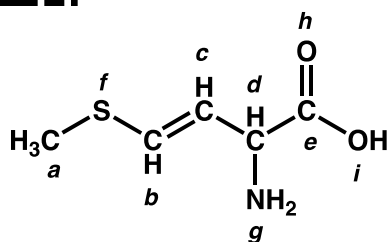
Structure 5

Structure 6

**Question 6** **6 points**

Using the data given below, calculate the **lattice energy** for SrO(s).

	ΔH° (kJ/mol)
$\text{Sr}(s) \longrightarrow \text{Sr}(g)$	+180
$\text{Sr}(g) \longrightarrow \text{Sr}^{2+}(g) + 2e^-$	+1468
$\text{O}_2(g) \longrightarrow 2\text{O}(g)$	+500
$\text{O}(g) + 2e^- \longrightarrow \text{O}^{2-}(g)$	+600
$\text{Sr}(s) + \frac{1}{2}\text{O}_2(g) \longrightarrow \text{SrO}(s)$	-550

**Question 7 10 points**

Consider the molecule given on the left, and answer questions (i) to (vi) using chemical bonding theories that we have learnt in class (valence bond theory, hybridization).

Note that the lone pairs are not shown in the structure:

i. Consider the carbon atoms marked in as ***a***, ***b***, ***c***, ***d***, and ***e*** in the molecule, and determine **hybridization** at each of the carbon centers. (2.5 points)

Atom	Hybridization
Carbon (<i>a</i>)	
Carbon (<i>b</i>)	
Carbon (<i>c</i>)	
Carbon (<i>d</i>)	
Carbon (<i>e</i>)	

ii. Now, consider the sulfur atom (S) marked as ***f***, nitrogen (N) atom (***g***), and oxygen(O) atoms (***h*** and ***i***), and determine their **hybridization**. (2 points)

Atom	Hybridization
Sulfur (<i>f</i>)	
Nitrogen (<i>g</i>)	
Oxygen (<i>h</i>)	
Oxygen (<i>i</i>)	

iii. **Describe** all the bonds formed by the carbon atoms ***a*** and ***b***, using sigma and pi designation as appropriate. (2 points)

Atom	Bonds formed (use sigma and pi as appropriate)
Carbon (<i>a</i>)	
Carbon (<i>b</i>)	

iv. How many total sp^3 -s sigma bonds are in the molecule? (1 point)

v. Determine **electronic** and **molecular geometry** at carbon (C) labeled ***e*** and sulfur (S) labeled ***f*** centres. (2 points)

Atom	Electronic Geometry	Molecular Geometry
Carbon (<i>e</i>)		
Sulfur (<i>f</i>)		

vi. Circle the value from the choices given below, that describes the **C-S-C bond angle at sulfur** (S) marked ***f***: (0.5 points)

>90° to <109.5° 109.5° >109.5 to <120° 120° >120° to <180°

**Question 8 12 points**

Using the **valence shell orbitals only** draw a **complete** molecular orbital (MO) diagram (with no s-p mixing) for the molecule **PS**, in the space provided below. **(6 points)**

Assume that 3s and 3p atomic orbitals will form molecular orbitals like those formed by the combination of 2s and 2p atomic orbitals that we learnt in class, and there is no s-p mixing.

Show **energy arrow**, **atomic orbitals** involved, **atom labels**, the **molecular orbitals** that are formed, **filled electrons** in the MO diagram etc. Make sure to label all the atomic and molecular orbitals.

Using the MO diagram above, answer the following questions:

- (i) Which one of the following will have the highest bond order? Circle the correct choice
A) PS^{1-} B) PS C) PS^{1+}
- (ii) Which one of the following will have the longest bond length? Circle the correct choice
A) PS^{1-} B) PS C) PS^{1+}
- (iii) Which of the following will be diamagnetic? Circle the correct choice.
A) PS^{1-} B) PS C) PS^{1+}
- (iv) Draw Lewis dot structure of the molecule with the shortest bond length (based on the MO diagram) from among the three (PS^{1-} , PS, PS^{1+}), and indicate the bond order calculated from the Lewis dot structure. **(3 points)**

Lewis Dot Structure

Bond Order



6427AA99-3956-446A-8FC8-2527108B61EA

midterm-exam-93388

#842 Page 10 of 10

SCRAP PAGE

**CHEMISTRY 110 –Midterm Exam****Thursday November 7, 2024, 6:30 – 8:30 pm****Examiners: Pallavi Sirjoosingh and Ashok Kakkar**

First name (please write as legibly as possible within the boxes)

--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--

Last name

--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--

Student ID number

--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--

This booklet has 8 questions on 10 pages (The last page can be used as a scrap page).

The staple must NOT be removed from the question booklet. Your intact question booklet must be returned at the end of the exam.

A separate booklet contains a periodic table, and datasheet.

Write your answers directly under the questions in the space provided. Do not write on the QR codes or near the top edges (as it may be cut off).

Please write **legibly and dark enough** for the scan to be readable. For students who use English as a second language, dictionaries are permitted. Calculators are allowed.

DO NOT ASK QUESTIONS OF THE INVIGILATORS about interpretations of the exam questions. If you have a concern about a question, please summarize it at the top of this page. All concerns will be evaluated after the exam. GOOD LUCK!

**Question 1 8 points** (*Assume all transitions are possible*)

a. 6 points Calculate the **energy (in kJ/mol)** required to ionize (completely remove) the electron from the **$n=3$** level of **He^+ ion** (He: Helium). Show your work including equations. (3 sig. figs.)

b. 2 points

(I) Select the largest (longest) wavelength from the three choices below:

- i. Wavelength of photon to *completely remove* an electron from *1s orbital of Hydrogen atom*
- ii. Wavelength of photon to *completely remove* an electron from *1s orbital of Boron atom (B)*
- iii. Wavelength of photon to *completely remove* an electron from *1s orbital of Boron cation (B^+)*

i or ii or iii

(II) Select the smallest (shortest) wavelength from the three choices below (*for H atom*):

- i. Wavelength of photon *emitted* for an electron transitioning from *2p to 1s orbital*
- ii. Wavelength of photon *absorbed* for an electron transitioning from *2s to 4p orbital*
- iii. Wavelength of photon *absorbed* for an electron transitioning from *1s to 3p orbital*

i or ii or iii



Question 2 8 points



A Chem110 student plots the graph shown on the left, from data obtained for the kinetic energy (eV) of the ejected electron versus frequency (s^{-1}) of the incident light, for an unknown metal. ($1\text{eV} = 1.6022 \times 10^{-19} \text{ J}$)

Based on the plot, answer the questions **a to c** below:

a. 1 point On the plot above, label the **threshold frequency**.

b. 2 points What is the value of the work potential (in J) of the unknown metal?

c. 3 points What is the *longest* wavelength (in nm) required to eject an electron from this metal's surface? Show your work including equation(s) used. (3 sig figs)

d. 2 points In the space provided below, plot the following

(Assume frequency is *above* threshold frequency):

- i. *Kinetic energy* of an ejected electron versus the *intensity* of the incident light (left)
- ii. *Number of electrons* ejected versus the *frequency* of the incident light (right)



**Question 3** **6 points****a. 2 points**

In the ground state of Ge (Germanium; $Z = 32$) atom, **name** the following orbitals:

i. Orbital containing the electron that requires the maximum energy to be removed.

ii. Orbital containing the electron that experiences the minimum effective nuclear charge.

b. 2 points

An unknown anion (A^-) has the following *excited* electronic configuration: $[Ar] 4s^2 3d^3 4p^1$

(Ar: Argon; $Z = 18$)

Give the **condensed ground state electronic configuration** of the atom, **A** and the cation, **A^{2+}** .
Show your work. (Assume all electronic transitions are possible)

Electronic Configuration of A:

Electronic Configuration of A^{2+} :

c. 2 points

In a single atom, determine how many electrons can have the following set of quantum numbers?

(i) $n = 5; l = 3$

In a single atom, determine how many electrons can have the following set of quantum numbers?

(i) $n = 3; l = 2; m_s = +1/2$

**Question 4** **8 points**

Rank the following from 1 to 4 in **increasing** order. Provide the ranking in the boxes provided.

Note: Atomic number (Z) are provided for relevant atoms

a) Increasing (lowest to highest) order of *ionization energy*: (1= lowest; 4 = highest)

Mg

Mg: Z = 12

Rb

Rb: Z = 37

Cs

Cs: Z = 55

K

K : Z = 19

b) Increasing (lowest to highest) order of *bond angle*: (1= lowest; 4 = highest)

H₂OCCl₄XeF₄

HCN

c) Increasing (smallest to largest) order of *ionic size* (1= smallest; 4 = largest)

S²⁻

S: Z = 16

O²⁻

O: Z = 8

Be²⁺

Be: Z = 4

Na⁺

Na: Z = 11

d) Increasing (lowest to highest) order of *bond energy* (1= lowest; 4 = highest)

P-F

P-Cl

P-I

P-Br

**Question 5** **12 points**

a. 9 points For the molecules HClO_4 and TeF_6 , draw the most stable *Lewis structure* showing appropriate shape and *dashed/wedged bonds*, based on VSEPR. Determine the **molecular geometry** around the central atom. For HClO_4 , label the *most polar bond*.



b. 3 points (Note: Lone pairs are not shown in structures below)

(i) For the three *resonance* structures (1, 2 and 3) shown below, answer the following question:



(I) Circle the *most contributing* resonance structure.

Structure 1

Structure 2

Structure 3

(ii) For the three structures given (4, 5, and 6) below, answer the following questions:



(I) Circle all the structures that represent *resonance structures* of each other:

Structure 4

Structure 5

Structure 6

(II) Among the resonance structures, circle the *most contributing* resonance structure.

Structure 4

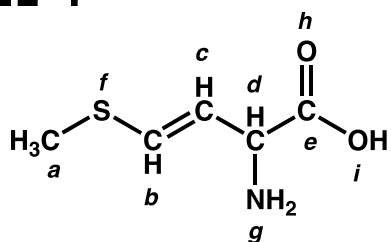
Structure 5

Structure 6

**Question 6** **6 points**

Using the data given below, calculate the **lattice energy** for SrO(s).

	ΔH° (kJ/mol)
$\text{Sr}(s) \longrightarrow \text{Sr}(g)$	+180
$\text{Sr}(g) \longrightarrow \text{Sr}^{2+}(g) + 2e^-$	+1468
$\text{O}_2(g) \longrightarrow 2\text{O}(g)$	+500
$\text{O}(g) + 2e^- \longrightarrow \text{O}^{2-}(g)$	+600
$\text{Sr}(s) + \frac{1}{2}\text{O}_2(g) \longrightarrow \text{SrO}(s)$	-550

**Question 7 10 points**

Consider the molecule given on the left, and answer questions (i) to (vi) using chemical bonding theories that we have learnt in class (valence bond theory, hybridization).

Note that the lone pairs are not shown in the structure:

i. Consider the carbon atoms marked in as ***a***, ***b***, ***c***, ***d***, and ***e*** in the molecule, and determine **hybridization** at each of the carbon centers. (2.5 points)

Atom	Hybridization
Carbon (<i>a</i>)	
Carbon (<i>b</i>)	
Carbon (<i>c</i>)	
Carbon (<i>d</i>)	
Carbon (<i>e</i>)	

ii. Now, consider the sulfur atom (S) marked as ***f***, nitrogen (N) atom (***g***), and oxygen(O) atoms (***h*** and ***i***), and determine their **hybridization**. (2 points)

Atom	Hybridization
Sulfur (<i>f</i>)	
Nitrogen (<i>g</i>)	
Oxygen (<i>h</i>)	
Oxygen (<i>i</i>)	

iii. **Describe** all the bonds formed by the carbon atoms ***a*** and ***b***, using sigma and pi designation as appropriate. (2 points)

Atom	Bonds formed (use sigma and pi as appropriate)
Carbon (<i>a</i>)	
Carbon (<i>b</i>)	

iv. How many total sp^3 -s sigma bonds are in the molecule? (1 point)

v. Determine **electronic** and **molecular geometry** at carbon (C) labeled ***e*** and sulfur (S) labeled ***f*** centres. (2 points)

Atom	Electronic Geometry	Molecular Geometry
Carbon (<i>e</i>)		
Sulfur (<i>f</i>)		

vi. Circle the value from the choices given below, that describes the **C-S-C bond angle at sulfur** (S) marked ***f***: (0.5 points)

>90° to <109.5° 109.5° >109.5 to <120° 120° >120° to <180°

**Question 8 12 points**

Using the **valence shell orbitals only** draw a **complete** molecular orbital (MO) diagram (with no s-p mixing) for the molecule **PS**, in the space provided below. **(6 points)**

Assume that 3s and 3p atomic orbitals will form molecular orbitals like those formed by the combination of 2s and 2p atomic orbitals that we learnt in class, and there is no s-p mixing.

Show **energy arrow**, **atomic orbitals** involved, **atom labels**, the **molecular orbitals** that are formed, **filled electrons** in the MO diagram etc. Make sure to label all the atomic and molecular orbitals.

Using the MO diagram above, answer the following questions:

- (i) Which one of the following will have the highest bond order? Circle the correct choice
A) PS^{1-} B) PS C) PS^{1+}
- (ii) Which one of the following will have the longest bond length? Circle the correct choice
A) PS^{1-} B) PS C) PS^{1+}
- (iii) Which of the following will be diamagnetic? Circle the correct choice.
A) PS^{1-} B) PS C) PS^{1+}
- (iv) Draw Lewis dot structure of the molecule with the shortest bond length (based on the MO diagram) from among the three (PS^{1-} , PS, PS^{1+}), and indicate the bond order calculated from the Lewis dot structure. **(3 points)**

Lewis Dot Structure

Bond Order



966A755D-EB01-4244-AE27-8EF7A50C7333

midterm-exam-93388

#843 Page 10 of 10

SCRAP PAGE

**CHEMISTRY 110 –Midterm Exam****Thursday November 7, 2024, 6:30 – 8:30 pm**

Examiners: Pallavi Sirjoosingh and Ashok Kakkar

First name (please write as legibly as possible within the boxes)

--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--

Last name

--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--

Student ID number

--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--

This booklet has 8 questions on 10 pages (The last page can be used as a scrap page).

The staple must NOT be removed from the question booklet. Your intact question booklet must be returned at the end of the exam.

A separate booklet contains a periodic table, and datasheet.

Write your answers directly under the questions in the space provided. Do not write on the QR codes or near the top edges (as it may be cut off).

Please write **legibly and dark enough** for the scan to be readable. For students who use English as a second language, dictionaries are permitted. Calculators are allowed.

DO NOT ASK QUESTIONS OF THE INVIGILATORS about interpretations of the exam questions. If you have a concern about a question, please summarize it at the top of this page. All concerns will be evaluated after the exam. GOOD LUCK!

**Question 1 8 points** (*Assume all transitions are possible*)

a. 6 points Calculate the **energy (in kJ/mol)** required to ionize (completely remove) the electron from the **n=3** level of **He⁺ ion** (He: Helium). Show your work including equations. (3 sig. figs.)

b. 2 points

(I) Select the largest (longest) wavelength from the three choices below:

- i. Wavelength of photon to *completely remove* an electron from *1s orbital of Hydrogen atom*
- ii. Wavelength of photon to *completely remove* an electron from *1s orbital of Boron atom (B)*
- iii. Wavelength of photon to *completely remove* an electron from *1s orbital of Boron cation (B⁺)*

i or ii or iii

(II) Select the smallest (shortest) wavelength from the three choices below (*for H atom*):

- i. Wavelength of photon *emitted* for an electron transitioning from *2p to 1s orbital*
- ii. Wavelength of photon *absorbed* for an electron transitioning from *2s to 4p orbital*
- iii. Wavelength of photon *absorbed* for an electron transitioning from *1s to 3p orbital*

i or ii or iii



Question 2 8 points



A Chem110 student plots the graph shown on the left, from data obtained for the kinetic energy (eV) of the ejected electron versus frequency (s^{-1}) of the incident light, for an unknown metal. ($1\text{eV} = 1.6022 \times 10^{-19} \text{ J}$)

Based on the plot, answer the questions **a to c** below:

a. 1 point On the plot above, label the **threshold frequency**.

b. 2 points What is the value of the work potential (in J) of the unknown metal?

c. 3 points What is the *longest* wavelength (in nm) required to eject an electron from this metal's surface? Show your work including equation(s) used. (3 sig figs)

d. 2 points In the space provided below, plot the following

(Assume frequency is *above* threshold frequency):

- i. *Kinetic energy* of an ejected electron versus the *intensity* of the incident light (left)
- ii. *Number of electrons* ejected versus the *frequency* of the incident light (right)



**Question 3** **6 points****a. 2 points**

In the ground state of Ge (Germanium; $Z = 32$) atom, **name** the following orbitals:

i. Orbital containing the electron that requires the maximum energy to be removed.

ii. Orbital containing the electron that experiences the minimum effective nuclear charge.

b. 2 points

An unknown anion (A^-) has the following *excited* electronic configuration: $[Ar] 4s^2 3d^3 4p^1$

(Ar: Argon; $Z = 18$)

Give the **condensed ground state electronic configuration** of the atom, **A** and the cation, **A^{2+}** .
Show your work. (Assume all electronic transitions are possible)

Electronic Configuration of A:

Electronic Configuration of A^{2+} :

c. 2 points

In a single atom, determine how many electrons can have the following set of quantum numbers?

(i) $n = 5; l = 3$

In a single atom, determine how many electrons can have the following set of quantum numbers?

(i) $n = 3; l = 2; m_s = +1/2$

**Question 4** **8 points**

Rank the following from 1 to 4 in **increasing** order. Provide the ranking in the boxes provided.

Note: Atomic number (Z) are provided for relevant atoms

a) Increasing (lowest to highest) order of *ionization energy*: (1= lowest; 4 = highest)

Mg

Mg: Z = 12

Rb

Rb: Z = 37

Cs

Cs: Z = 55

K

K : Z = 19

b) Increasing (lowest to highest) order of *bond angle*: (1= lowest; 4 = highest)

H₂OCCl₄XeF₄

HCN

c) Increasing (smallest to largest) order of *ionic size* (1= smallest; 4 = largest)

S²⁻

S: Z = 16

O²⁻

O: Z = 8

Be²⁺

Be: Z = 4

Na⁺

Na: Z = 11

d) Increasing (lowest to highest) order of *bond energy* (1= lowest; 4 = highest)

P-F

P-Cl

P-I

P-Br

**Question 5** **12 points**

a. 9 points For the molecules HClO_4 and TeF_6 , draw the most stable *Lewis structure* showing appropriate shape and *dashed/wedged bonds*, based on VSEPR. Determine the **molecular geometry** around the central atom. For HClO_4 , label the *most polar bond*.



b. 3 points (Note: Lone pairs are not shown in structures below)

(i) For the three *resonance* structures (1, 2 and 3) shown below, answer the following question:



(I) Circle the *most contributing* resonance structure.

Structure 1

Structure 2

Structure 3

(ii) For the three structures given (4, 5, and 6) below, answer the following questions:



(I) Circle all the structures that represent *resonance structures* of each other:

Structure 4

Structure 5

Structure 6

(II) Among the resonance structures, circle the *most contributing* resonance structure.

Structure 4

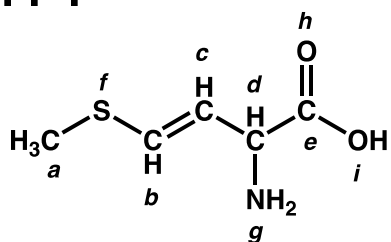
Structure 5

Structure 6

**Question 6** **6 points**

Using the data given below, calculate the **lattice energy** for SrO(s).

	ΔH° (kJ/mol)
$\text{Sr}(s) \longrightarrow \text{Sr}(g)$	+180
$\text{Sr}(g) \longrightarrow \text{Sr}^{2+}(g) + 2e^-$	+1468
$\text{O}_2(g) \longrightarrow 2\text{O}(g)$	+500
$\text{O}(g) + 2e^- \longrightarrow \text{O}^{2-}(g)$	+600
$\text{Sr}(s) + \frac{1}{2}\text{O}_2(g) \longrightarrow \text{SrO}(s)$	-550

**Question 7 10 points**

Consider the molecule given on the left, and answer questions (i) to (vi) using chemical bonding theories that we have learnt in class (valence bond theory, hybridization).

Note that the lone pairs are not shown in the structure:

i. Consider the carbon atoms marked in as ***a***, ***b***, ***c***, ***d***, and ***e*** in the molecule, and determine **hybridization** at each of the carbon centers. (2.5 points)

Atom	Hybridization
Carbon (<i>a</i>)	
Carbon (<i>b</i>)	
Carbon (<i>c</i>)	
Carbon (<i>d</i>)	
Carbon (<i>e</i>)	

ii. Now, consider the sulfur atom (S) marked as ***f***, nitrogen (N) atom (***g***), and oxygen(O) atoms (***h*** and ***i***), and determine their **hybridization**. (2 points)

Atom	Hybridization
Sulfur (<i>f</i>)	
Nitrogen (<i>g</i>)	
Oxygen (<i>h</i>)	
Oxygen (<i>i</i>)	

iii. **Describe** all the bonds formed by the carbon atoms ***a*** and ***b***, using sigma and pi designation as appropriate. (2 points)

Atom	Bonds formed (use sigma and pi as appropriate)
Carbon (<i>a</i>)	
Carbon (<i>b</i>)	

iv. How many total sp^3 -s sigma bonds are in the molecule? (1 point)

v. Determine **electronic** and **molecular geometry** at carbon (C) labeled ***e*** and sulfur (S) labeled ***f*** centres. (2 points)

Atom	Electronic Geometry	Molecular Geometry
Carbon (<i>e</i>)		
Sulfur (<i>f</i>)		

vi. Circle the value from the choices given below, that describes the **C-S-C bond angle at sulfur** (S) marked ***f***: (0.5 points)

>90° to <109.5°

109.5°

>109.5 to <120°

120°

>120° to <180°



Question 8 12 points

Using the **valence shell orbitals only** draw a **complete** molecular orbital (MO) diagram (with no s-p mixing) for the molecule **PS**, in the space provided below. **(6 points)**

Assume that 3s and 3p atomic orbitals will form molecular orbitals like those formed by the combination of 2s and 2p atomic orbitals that we learnt in class, and there is no s-p mixing.

Show **energy arrow**, **atomic orbitals** involved, **atom labels**, the **molecular orbitals** that are formed, **filled electrons** in the MO diagram etc. Make sure to label all the atomic and molecular orbitals.

Using the MO diagram above, answer the following questions:

- (i) Which one of the following will have the highest bond order? Circle the correct choice
 A) PS^{1-} B) PS C) PS^{1+}
- (ii) Which one of the following will have the longest bond length? Circle the correct choice
 A) PS^{1-} B) PS C) PS^{1+}
- (iii) Which of the following will be diamagnetic? Circle the correct choice.
 A) PS^{1-} B) PS C) PS^{1+}
- (iv) Draw Lewis dot structure of the molecule with the shortest bond length (based on the MO diagram) from among the three (PS^{1-} , PS, PS^{1+}), and indicate the bond order calculated from the Lewis dot structure. **(3 points)**

Lewis Dot Structure

Bond Order



1F5248A6-58EE-4166-9E08-0706EC53A05C

midterm-exam-93388

#844 Page 10 of 10

SCRAP PAGE

**CHEMISTRY 110 –Midterm Exam****Thursday November 7, 2024, 6:30 – 8:30 pm****Examiners: Pallavi Sirjoosingh and Ashok Kakkar**

First name (please write as legibly as possible within the boxes)

--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--

Last name

--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--

Student ID number

--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--

This booklet has 8 questions on 10 pages (The last page can be used as a scrap page).

The staple must NOT be removed from the question booklet. Your intact question booklet must be returned at the end of the exam.

A separate booklet contains a periodic table, and datasheet.

Write your answers directly under the questions in the space provided. Do not write on the QR codes or near the top edges (as it may be cut off).

Please write **legibly and dark enough** for the scan to be readable. For students who use English as a second language, dictionaries are permitted. Calculators are allowed.

DO NOT ASK QUESTIONS OF THE INVIGILATORS about interpretations of the exam questions. If you have a concern about a question, please summarize it at the top of this page. All concerns will be evaluated after the exam. GOOD LUCK!

**Question 1 8 points** (*Assume all transitions are possible*)

a. 6 points Calculate the **energy (in kJ/mol)** required to ionize (completely remove) the electron from the **$n=3$** level of **He^+ ion** (He: Helium). Show your work including equations. (3 sig. figs.)

b. 2 points

(I) Select the largest (longest) wavelength from the three choices below:

- i. Wavelength of photon to *completely remove* an electron from *1s orbital of Hydrogen atom*
- ii. Wavelength of photon to *completely remove* an electron from *1s orbital of Boron atom (B)*
- iii. Wavelength of photon to *completely remove* an electron from *1s orbital of Boron cation (B^+)*

i or ii or iii

(II) Select the smallest (shortest) wavelength from the three choices below (*for H atom*):

- i. Wavelength of photon *emitted* for an electron transitioning from *2p to 1s orbital*
- ii. Wavelength of photon *absorbed* for an electron transitioning from *2s to 4p orbital*
- iii. Wavelength of photon *absorbed* for an electron transitioning from *1s to 3p orbital*

i or ii or iii



Question 2 8 points



A Chem110 student plots the graph shown on the left, from data obtained for the kinetic energy (eV) of the ejected electron versus frequency (s^{-1}) of the incident light, for an unknown metal. ($1\text{eV} = 1.6022 \times 10^{-19} \text{ J}$)

Based on the plot, answer the questions **a to c** below:

a. 1 point On the plot above, label the **threshold frequency**.

b. 2 points What is the value of the work potential (in J) of the unknown metal?

c. 3 points What is the *longest* wavelength (in nm) required to eject an electron from this metal's surface? Show your work including equation(s) used. (3 sig figs)

d. 2 points In the space provided below, plot the following

(Assume frequency is *above* threshold frequency):

- Kinetic energy* of an ejected electron versus the *intensity* of the incident light (left)
- Number of electrons* ejected versus the *frequency* of the incident light (right)



**Question 3** **6 points****a. 2 points**

In the ground state of Ge (Germanium; $Z = 32$) atom, **name** the following orbitals:

i. Orbital containing the electron that requires the maximum energy to be removed.

ii. Orbital containing the electron that experiences the minimum effective nuclear charge.

b. 2 points

An unknown anion (A^-) has the following *excited* electronic configuration: $[Ar] 4s^2 3d^3 4p^1$

(Ar: Argon; $Z = 18$)

Give the **condensed ground state electronic configuration** of the atom, **A** and the cation, A^{2+} .
Show your work. (Assume all electronic transitions are possible)

Electronic Configuration of A:

Electronic Configuration of A^{2+} :

c. 2 points

In a single atom, determine how many electrons can have the following set of quantum numbers?

(i) $n = 5; l = 3$

In a single atom, determine how many electrons can have the following set of quantum numbers?

(i) $n = 3; l = 2; m_s = +1/2$

**Question 4** **8 points**

Rank the following from 1 to 4 in **increasing** order. Provide the ranking in the boxes provided.

Note: Atomic number (Z) are provided for relevant atoms

a) Increasing (lowest to highest) order of *ionization energy*: (1= lowest; 4 = highest)

Mg

Mg: Z = 12

Rb

Rb: Z = 37

Cs

Cs: Z = 55

K

K : Z = 19

b) Increasing (lowest to highest) order of *bond angle*: (1= lowest; 4 = highest)

H₂OCCl₄XeF₄

HCN

c) Increasing (smallest to largest) order of *ionic size* (1= smallest; 4 = largest)

S²⁻

S: Z = 16

O²⁻

O: Z = 8

Be²⁺

Be: Z = 4

Na⁺

Na: Z = 11

d) Increasing (lowest to highest) order of *bond energy* (1= lowest; 4 = highest)

P-F

P-Cl

P-I

P-Br

**Question 5** **12 points**

a. 9 points For the molecules HClO_4 and TeF_6 , draw the most stable *Lewis structure* showing appropriate shape and *dashed/wedged bonds*, based on VSEPR. Determine the **molecular geometry** around the central atom. For HClO_4 , label the *most polar bond*.



b. 3 points (Note: Lone pairs are not shown in structures below)

(i) For the three *resonance* structures (1, 2 and 3) shown below, answer the following question:



(I) Circle the *most contributing* resonance structure.

Structure 1

Structure 2

Structure 3

(ii) For the three structures given (4, 5, and 6) below, answer the following questions:



(I) Circle all the structures that represent *resonance structures* of each other:

Structure 4

Structure 5

Structure 6

(II) Among the resonance structures, circle the *most contributing* resonance structure.

Structure 4

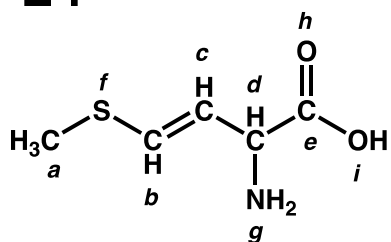
Structure 5

Structure 6

**Question 6** **6 points**

Using the data given below, calculate the **lattice energy** for SrO(s).

	ΔH° (kJ/mol)
$\text{Sr}(s) \longrightarrow \text{Sr}(g)$	+180
$\text{Sr}(g) \longrightarrow \text{Sr}^{2+}(g) + 2e^-$	+1468
$\text{O}_2(g) \longrightarrow 2\text{O}(g)$	+500
$\text{O}(g) + 2e^- \longrightarrow \text{O}^{2-}(g)$	+600
$\text{Sr}(s) + \frac{1}{2}\text{O}_2(g) \longrightarrow \text{SrO}(s)$	-550

**Question 7 10 points**

Consider the molecule given on the left, and answer questions (i) to (vi) using chemical bonding theories that we have learnt in class (valence bond theory, hybridization).

Note that the lone pairs are not shown in the structure:

i. Consider the carbon atoms marked in as ***a***, ***b***, ***c***, ***d***, and ***e*** in the molecule, and determine **hybridization** at each of the carbon centers. (2.5 points)

Atom	Hybridization
Carbon (<i>a</i>)	
Carbon (<i>b</i>)	
Carbon (<i>c</i>)	
Carbon (<i>d</i>)	
Carbon (<i>e</i>)	

ii. Now, consider the sulfur atom (S) marked as ***f***, nitrogen (N) atom (***g***), and oxygen(O) atoms (***h*** and ***i***), and determine their **hybridization**. (2 points)

Atom	Hybridization
Sulfur (<i>f</i>)	
Nitrogen (<i>g</i>)	
Oxygen (<i>h</i>)	
Oxygen (<i>i</i>)	

iii. **Describe** all the bonds formed by the carbon atoms ***a*** and ***b***, using sigma and pi designation as appropriate. (2 points)

Atom	Bonds formed (use sigma and pi as appropriate)
Carbon (<i>a</i>)	
Carbon (<i>b</i>)	

iv. How many total sp^3 -s sigma bonds are in the molecule? (1 point)

v. Determine **electronic** and **molecular geometry** at carbon (C) labeled ***e*** and sulfur (S) labeled ***f*** centres. (2 points)

Atom	Electronic Geometry	Molecular Geometry
Carbon (<i>e</i>)		
Sulfur (<i>f</i>)		

vi. Circle the value from the choices given below, that describes the **C-S-C bond angle at sulfur** (S) marked ***f***: (0.5 points)

>90° to <109.5° 109.5° >109.5 to <120° 120° >120° to <180°

**Question 8 12 points**

Using the **valence shell orbitals only** draw a **complete** molecular orbital (MO) diagram (with no s-p mixing) for the molecule **PS**, in the space provided below. **(6 points)**

Assume that 3s and 3p atomic orbitals will form molecular orbitals like those formed by the combination of 2s and 2p atomic orbitals that we learnt in class, and there is no s-p mixing.

Show **energy arrow**, **atomic orbitals** involved, **atom labels**, the **molecular orbitals** that are formed, **filled electrons** in the MO diagram etc. Make sure to label all the atomic and molecular orbitals.

Using the MO diagram above, answer the following questions:

- (i) Which one of the following will have the highest bond order? Circle the correct choice
A) PS^{1-} B) PS C) PS^{1+}
- (ii) Which one of the following will have the longest bond length? Circle the correct choice
A) PS^{1-} B) PS C) PS^{1+}
- (iii) Which of the following will be diamagnetic? Circle the correct choice.
A) PS^{1-} B) PS C) PS^{1+}
- (iv) Draw Lewis dot structure of the molecule with the shortest bond length (based on the MO diagram) from among the three (PS^{1-} , PS, PS^{1+}), and indicate the bond order calculated from the Lewis dot structure. **(3 points)**

Lewis Dot Structure

Bond Order



70C71BA8-7DA1-472B-8A60-0ED272EBBE1D

midterm-exam-93388

#845 Page 10 of 10

SCRAP PAGE

**CHEMISTRY 110 –Midterm Exam****Thursday November 7, 2024, 6:30 – 8:30 pm****Examiners: Pallavi Sirjoosingh and Ashok Kakkar**

First name (please write as legibly as possible within the boxes)

--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--

Last name

--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--

Student ID number

--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--

This booklet has 8 questions on 10 pages (The last page can be used as a scrap page).

The staple must NOT be removed from the question booklet. Your intact question booklet must be returned at the end of the exam.

A separate booklet contains a periodic table, and datasheet.

Write your answers directly under the questions in the space provided. Do not write on the QR codes or near the top edges (as it may be cut off).

Please write **legibly and dark enough** for the scan to be readable. For students who use English as a second language, dictionaries are permitted. Calculators are allowed.

DO NOT ASK QUESTIONS OF THE INVIGILATORS about interpretations of the exam questions. If you have a concern about a question, please summarize it at the top of this page. All concerns will be evaluated after the exam. GOOD LUCK!

**Question 1 8 points** (*Assume all transitions are possible*)

a. 6 points Calculate the **energy (in kJ/mol)** required to ionize (completely remove) the electron from the **n=3** level of **He⁺ ion** (He: Helium). Show your work including equations. (3 sig. figs.)

b. 2 points

(I) Select the largest (longest) wavelength from the three choices below:

- i. Wavelength of photon to *completely remove* an electron from *1s orbital of Hydrogen atom*
- ii. Wavelength of photon to *completely remove* an electron from *1s orbital of Boron atom (B)*
- iii. Wavelength of photon to *completely remove* an electron from *1s orbital of Boron cation (B⁺)*

i or ii or iii

(II) Select the smallest (shortest) wavelength from the three choices below (*for H atom*):

- i. Wavelength of photon *emitted* for an electron transitioning from *2p to 1s orbital*
- ii. Wavelength of photon *absorbed* for an electron transitioning from *2s to 4p orbital*
- iii. Wavelength of photon *absorbed* for an electron transitioning from *1s to 3p orbital*

i or ii or iii



Question 2 8 points



A Chem110 student plots the graph shown on the left, from data obtained for the kinetic energy (eV) of the ejected electron versus frequency (s^{-1}) of the incident light, for an unknown metal. ($1\text{eV} = 1.6022 \times 10^{-19} \text{ J}$)

Based on the plot, answer the questions **a to c** below:

a. 1 point On the plot above, label the **threshold frequency**.

b. 2 points What is the value of the work potential (in J) of the unknown metal?

c. 3 points What is the *longest* wavelength (in nm) required to eject an electron from this metal's surface? Show your work including equation(s) used. (3 sig figs)

d. 2 points In the space provided below, plot the following

(Assume frequency is *above* threshold frequency):

- i. *Kinetic energy* of an ejected electron versus the *intensity* of the incident light (left)
- ii. *Number of electrons* ejected versus the *frequency* of the incident light (right)



**Question 3** **6 points****a. 2 points**

In the ground state of Ge (Germanium; $Z = 32$) atom, **name** the following orbitals:

i. Orbital containing the electron that requires the maximum energy to be removed.

ii. Orbital containing the electron that experiences the minimum effective nuclear charge.

b. 2 points

An unknown anion (A^-) has the following *excited* electronic configuration: $[Ar] 4s^2 3d^3 4p^1$

(Ar: Argon; $Z = 18$)

Give the **condensed ground state electronic configuration** of the atom, **A** and the cation, **A^{2+}** .
Show your work. (Assume all electronic transitions are possible)

Electronic Configuration of A:

Electronic Configuration of A^{2+} :

c. 2 points

In a single atom, determine how many electrons can have the following set of quantum numbers?

(i) $n = 5; l = 3$

In a single atom, determine how many electrons can have the following set of quantum numbers?

(i) $n = 3; l = 2; m_s = +1/2$

**Question 4** **8 points**

Rank the following from 1 to 4 in **increasing** order. Provide the ranking in the boxes provided.

Note: Atomic number (Z) are provided for relevant atoms

a) Increasing (lowest to highest) order of *ionization energy*: (1= lowest; 4 = highest)

Mg

Mg: Z = 12

Rb

Rb: Z = 37

Cs

Cs: Z = 55

K

K : Z = 19

b) Increasing (lowest to highest) order of *bond angle*: (1= lowest; 4 = highest)

H₂OCCl₄XeF₄

HCN

c) Increasing (smallest to largest) order of *ionic size* (1= smallest; 4 = largest)

S²⁻

S: Z = 16

O²⁻

O: Z = 8

Be²⁺

Be: Z = 4

Na⁺

Na: Z = 11

d) Increasing (lowest to highest) order of *bond energy* (1= lowest; 4 = highest)

P-F

P-Cl

P-I

P-Br

**Question 5** **12 points**

a. 9 points For the molecules HClO_4 and TeF_6 , draw the most stable *Lewis structure* showing appropriate shape and *dashed/wedged bonds*, based on VSEPR. Determine the **molecular geometry** around the central atom. For HClO_4 , label the *most polar bond*.



b. 3 points (Note: Lone pairs are not shown in structures below)

(i) For the three *resonance* structures (1, 2 and 3) shown below, answer the following question:



(I) Circle the *most contributing* resonance structure.

Structure 1

Structure 2

Structure 3

(ii) For the three structures given (4, 5, and 6) below, answer the following questions:



(I) Circle all the structures that represent *resonance structures* of each other:

Structure 4

Structure 5

Structure 6

(II) Among the resonance structures, circle the *most contributing* resonance structure.

Structure 4

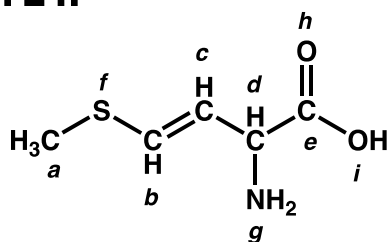
Structure 5

Structure 6

**Question 6** **6 points**

Using the data given below, calculate the **lattice energy** for SrO(s).

	ΔH° (kJ/mol)
$\text{Sr}(s) \longrightarrow \text{Sr}(g)$	+180
$\text{Sr}(g) \longrightarrow \text{Sr}^{2+}(g) + 2e^-$	+1468
$\text{O}_2(g) \longrightarrow 2\text{O}(g)$	+500
$\text{O}(g) + 2e^- \longrightarrow \text{O}^{2-}(g)$	+600
$\text{Sr}(s) + \frac{1}{2}\text{O}_2(g) \longrightarrow \text{SrO}(s)$	-550

**Question 7 10 points**

Consider the molecule given on the left, and answer questions (i) to (vi) using chemical bonding theories that we have learnt in class (valence bond theory, hybridization).

Note that the lone pairs are not shown in the structure:

i. Consider the carbon atoms marked in as ***a***, ***b***, ***c***, ***d***, and ***e*** in the molecule, and determine **hybridization** at each of the carbon centers. (2.5 points)

Atom	Hybridization
Carbon (<i>a</i>)	
Carbon (<i>b</i>)	
Carbon (<i>c</i>)	
Carbon (<i>d</i>)	
Carbon (<i>e</i>)	

ii. Now, consider the sulfur atom (S) marked as ***f***, nitrogen (N) atom (***g***), and oxygen(O) atoms (***h*** and ***i***), and determine their **hybridization**. (2 points)

Atom	Hybridization
Sulfur (<i>f</i>)	
Nitrogen (<i>g</i>)	
Oxygen (<i>h</i>)	
Oxygen (<i>i</i>)	

iii. **Describe** all the bonds formed by the carbon atoms ***a*** and ***b***, using sigma and pi designation as appropriate. (2 points)

Atom	Bonds formed (use sigma and pi as appropriate)
Carbon (<i>a</i>)	
Carbon (<i>b</i>)	

iv. How many total sp^3 -s sigma bonds are in the molecule? (1 point)

v. Determine **electronic** and **molecular geometry** at carbon (C) labeled ***e*** and sulfur (S) labeled ***f*** centres. (2 points)

Atom	Electronic Geometry	Molecular Geometry
Carbon (<i>e</i>)		
Sulfur (<i>f</i>)		

vi. Circle the value from the choices given below, that describes the **C-S-C bond angle at sulfur** (S) marked ***f***: (0.5 points)

$>90^\circ$ to $<109.5^\circ$ 109.5° >109.5 to $<120^\circ$ 120° $>120^\circ$ to $<180^\circ$


Question 8 12 points

Using the **valence shell orbitals only** draw a **complete** molecular orbital (MO) diagram (with no s-p mixing) for the molecule **PS**, in the space provided below. **(6 points)**

Assume that 3s and 3p atomic orbitals will form molecular orbitals like those formed by the combination of 2s and 2p atomic orbitals that we learnt in class, and there is no s-p mixing.

Show **energy arrow**, **atomic orbitals** involved, **atom labels**, the **molecular orbitals** that are formed, **filled electrons** in the MO diagram etc. Make sure to label all the atomic and molecular orbitals.

Using the MO diagram above, answer the following questions:

- (i) Which one of the following will have the highest bond order? Circle the correct choice
 A) PS^{1-} B) PS C) PS^{1+}
- (ii) Which one of the following will have the longest bond length? Circle the correct choice
 A) PS^{1-} B) PS C) PS^{1+}
- (iii) Which of the following will be diamagnetic? Circle the correct choice.
 A) PS^{1-} B) PS C) PS^{1+}
- (iv) Draw Lewis dot structure of the molecule with the shortest bond length (based on the MO diagram) from among the three (PS^{1-} , PS, PS^{1+}), and indicate the bond order calculated from the Lewis dot structure. **(3 points)**

Lewis Dot Structure

Bond Order



267AAD3F-EB64-40D2-B3BD-49B43A75CA75

midterm-exam-93388

#846 Page 10 of 10

SCRAP PAGE

**CHEMISTRY 110 –Midterm Exam****Thursday November 7, 2024, 6:30 – 8:30 pm**

Examiners: Pallavi Sirjoosingh and Ashok Kakkar

First name (please write as legibly as possible within the boxes)

--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--

Last name

--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--

Student ID number

--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--

This booklet has 8 questions on 10 pages (The last page can be used as a scrap page).

The staple must NOT be removed from the question booklet. Your intact question booklet must be returned at the end of the exam.

A separate booklet contains a periodic table, and datasheet.

Write your answers directly under the questions in the space provided. Do not write on the QR codes or near the top edges (as it may be cut off).

Please write **legibly and dark enough** for the scan to be readable. For students who use English as a second language, dictionaries are permitted. Calculators are allowed.

DO NOT ASK QUESTIONS OF THE INVIGILATORS about interpretations of the exam questions. If you have a concern about a question, please summarize it at the top of this page. All concerns will be evaluated after the exam. GOOD LUCK!

**Question 1 8 points** (*Assume all transitions are possible*)

a. 6 points Calculate the **energy (in kJ/mol)** required to ionize (completely remove) the electron from the **n=3** level of **He⁺ ion** (He: Helium). Show your work including equations. (3 sig. figs.)

b. 2 points

(I) Select the largest (longest) wavelength from the three choices below:

- i. Wavelength of photon to *completely remove* an electron from *1s orbital of Hydrogen atom*
- ii. Wavelength of photon to *completely remove* an electron from *1s orbital of Boron atom (B)*
- iii. Wavelength of photon to *completely remove* an electron from *1s orbital of Boron cation (B⁺)*

i or ii or iii

(II) Select the smallest (shortest) wavelength from the three choices below (*for H atom*):

- i. Wavelength of photon *emitted* for an electron transitioning from *2p to 1s orbital*
- ii. Wavelength of photon *absorbed* for an electron transitioning from *2s to 4p orbital*
- iii. Wavelength of photon *absorbed* for an electron transitioning from *1s to 3p orbital*

i or ii or iii



Question 2 8 points



A Chem110 student plots the graph shown on the left, from data obtained for the kinetic energy (eV) of the ejected electron versus frequency (s^{-1}) of the incident light, for an unknown metal. ($1\text{eV} = 1.6022 \times 10^{-19} \text{ J}$)

Based on the plot, answer the questions **a to c** below:

a. 1 point On the plot above, label the **threshold frequency**.

b. 2 points What is the value of the work potential (in J) of the unknown metal?

c. 3 points What is the *longest* wavelength (in nm) required to eject an electron from this metal's surface? Show your work including equation(s) used. (3 sig figs)

d. 2 points In the space provided below, plot the following

(Assume frequency is *above* threshold frequency):

- i. *Kinetic energy* of an ejected electron versus the *intensity* of the incident light (left)
- ii. *Number of electrons* ejected versus the *frequency* of the incident light (right)



**Question 3** **6 points****a. 2 points**

In the ground state of Ge (Germanium; $Z = 32$) atom, **name** the following orbitals:

i. Orbital containing the electron that requires the maximum energy to be removed.

ii. Orbital containing the electron that experiences the minimum effective nuclear charge.

b. 2 points

An unknown anion (A^-) has the following *excited* electronic configuration: $[Ar] 4s^2 3d^3 4p^1$

(Ar: Argon; $Z = 18$)

Give the **condensed ground state electronic configuration** of the atom, **A** and the cation, **A^{2+}** .
Show your work. (Assume all electronic transitions are possible)

Electronic Configuration of A:

Electronic Configuration of A^{2+} :

c. 2 points

In a single atom, determine how many electrons can have the following set of quantum numbers?

(i) $n = 5; l = 3$

In a single atom, determine how many electrons can have the following set of quantum numbers?

(i) $n = 3; l = 2; m_s = +1/2$

**Question 4** **8 points**

Rank the following from 1 to 4 in **increasing** order. Provide the ranking in the boxes provided.

Note: Atomic number (Z) are provided for relevant atoms

a) Increasing (lowest to highest) order of *ionization energy*: (1= lowest; 4 = highest)

Mg

Mg: Z = 12

Rb

Rb: Z = 37

Cs

Cs: Z = 55

K

K : Z = 19

b) Increasing (lowest to highest) order of *bond angle*: (1= lowest; 4 = highest)

H₂OCCl₄XeF₄

HCN

c) Increasing (smallest to largest) order of *ionic size* (1= smallest; 4 = largest)

S²⁻

S: Z = 16

O²⁻

O: Z = 8

Be²⁺

Be: Z = 4

Na⁺

Na: Z = 11

d) Increasing (lowest to highest) order of *bond energy* (1= lowest; 4 = highest)

P-F

P-Cl

P-I

P-Br

**Question 5** **12 points**

a. 9 points For the molecules HClO_4 and TeF_6 , draw the most stable *Lewis structure* showing appropriate shape and *dashed/wedged bonds*, based on VSEPR. Determine the **molecular geometry** around the central atom. For HClO_4 , label the *most polar bond*.

HClO_4 (there are no O-O bonds)

Label the most polar bond

Molecular Geometry around Cl

TeF_6

Molecular Geometry around Te

b. 3 points (Note: Lone pairs are not shown in structures below)

(i) For the three *resonance* structures (1, 2 and 3) shown below, answer the following question:



(I) Circle the *most contributing* resonance structure.

Structure 1

Structure 2

Structure 3

(ii) For the three structures given (4, 5, and 6) below, answer the following questions:



(I) Circle all the structures that represent *resonance structures* of each other:

Structure 4

Structure 5

Structure 6

(II) Among the resonance structures, circle the *most contributing* resonance structure.

Structure 4

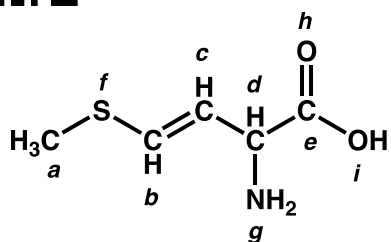
Structure 5

Structure 6

**Question 6** **6 points**

Using the data given below, calculate the **lattice energy** for SrO(s).

	ΔH° (kJ/mol)
$\text{Sr}(s) \longrightarrow \text{Sr}(g)$	+180
$\text{Sr}(g) \longrightarrow \text{Sr}^{2+}(g) + 2e^-$	+1468
$\text{O}_2(g) \longrightarrow 2\text{O}(g)$	+500
$\text{O}(g) + 2e^- \longrightarrow \text{O}^{2-}(g)$	+600
$\text{Sr}(s) + \frac{1}{2}\text{O}_2(g) \longrightarrow \text{SrO}(s)$	-550

**Question 7 10 points**

Consider the molecule given on the left, and answer questions (i) to (vi) using chemical bonding theories that we have learnt in class (valence bond theory, hybridization).

Note that the lone pairs are not shown in the structure:

i. Consider the carbon atoms marked in as ***a***, ***b***, ***c***, ***d***, and ***e*** in the molecule, and determine **hybridization** at each of the carbon centers. (2.5 points)

Atom	Hybridization
Carbon (<i>a</i>)	
Carbon (<i>b</i>)	
Carbon (<i>c</i>)	
Carbon (<i>d</i>)	
Carbon (<i>e</i>)	

ii. Now, consider the sulfur atom (S) marked as ***f***, nitrogen (N) atom (***g***), and oxygen(O) atoms (***h*** and ***i***), and determine their **hybridization**. (2 points)

Atom	Hybridization
Sulfur (<i>f</i>)	
Nitrogen (<i>g</i>)	
Oxygen (<i>h</i>)	
Oxygen (<i>i</i>)	

iii. **Describe** all the bonds formed by the carbon atoms ***a*** and ***b***, using sigma and pi designation as appropriate. (2 points)

Atom	Bonds formed (use sigma and pi as appropriate)
Carbon (<i>a</i>)	
Carbon (<i>b</i>)	

iv. How many total sp^3 -s sigma bonds are in the molecule? (1 point)

v. Determine **electronic** and **molecular geometry** at carbon (C) labeled ***e*** and sulfur (S) labeled ***f*** centres. (2 points)

Atom	Electronic Geometry	Molecular Geometry
Carbon (<i>e</i>)		
Sulfur (<i>f</i>)		

vi. Circle the value from the choices given below, that describes the **C-S-C bond angle at sulfur** (S) marked ***f***: (0.5 points)

>90° to <109.5° 109.5° >109.5 to <120° 120° >120° to <180°

**Question 8 12 points**

Using the **valence shell orbitals only** draw a **complete** molecular orbital (MO) diagram (with no s-p mixing) for the molecule **PS**, in the space provided below. **(6 points)**

Assume that 3s and 3p atomic orbitals will form molecular orbitals like those formed by the combination of 2s and 2p atomic orbitals that we learnt in class, and there is no s-p mixing.

Show **energy arrow**, **atomic orbitals** involved, **atom labels**, the **molecular orbitals** that are formed, **filled electrons** in the MO diagram etc. Make sure to label all the atomic and molecular orbitals.

Using the MO diagram above, answer the following questions:

- (i) Which one of the following will have the highest bond order? Circle the correct choice
A) PS^{1-} B) PS C) PS^{1+}
- (ii) Which one of the following will have the longest bond length? Circle the correct choice
A) PS^{1-} B) PS C) PS^{1+}
- (iii) Which of the following will be diamagnetic? Circle the correct choice.
A) PS^{1-} B) PS C) PS^{1+}
- (iv) Draw Lewis dot structure of the molecule with the shortest bond length (based on the MO diagram) from among the three (PS^{1-} , PS, PS^{1+}), and indicate the bond order calculated from the Lewis dot structure. **(3 points)**

Lewis Dot Structure

Bond Order



E4994C75-488C-45B3-84EE-1B2267E301A9

midterm-exam-93388

#847 Page 10 of 10

SCRAP PAGE

**CHEMISTRY 110 –Midterm Exam****Thursday November 7, 2024, 6:30 – 8:30 pm**

Examiners: Pallavi Sirjoosingh and Ashok Kakkar

First name (please write as legibly as possible within the boxes)

--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--

Last name

--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--

Student ID number

--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--

This booklet has 8 questions on 10 pages (The last page can be used as a scrap page).

The staple must NOT be removed from the question booklet. Your intact question booklet must be returned at the end of the exam.

A separate booklet contains a periodic table, and datasheet.

Write your answers directly under the questions in the space provided. Do not write on the QR codes or near the top edges (as it may be cut off).

Please write **legibly and dark enough** for the scan to be readable. For students who use English as a second language, dictionaries are permitted. Calculators are allowed.

DO NOT ASK QUESTIONS OF THE INVIGILATORS about interpretations of the exam questions. If you have a concern about a question, please summarize it at the top of this page. All concerns will be evaluated after the exam. GOOD LUCK!

**Question 1 8 points** (*Assume all transitions are possible*)

a. 6 points Calculate the **energy (in kJ/mol)** required to ionize (completely remove) the electron from the **$n=3$** level of **He^+ ion** (He: Helium). Show your work including equations. (3 sig. figs.)

b. 2 points

(I) Select the largest (longest) wavelength from the three choices below:

- i. Wavelength of photon to *completely remove* an electron from *1s orbital of Hydrogen atom*
- ii. Wavelength of photon to *completely remove* an electron from *1s orbital of Boron atom (B)*
- iii. Wavelength of photon to *completely remove* an electron from *1s orbital of Boron cation (B^+)*

i or ii or iii

(II) Select the smallest (shortest) wavelength from the three choices below (*for H atom*):

- i. Wavelength of photon *emitted* for an electron transitioning from *2p to 1s orbital*
- ii. Wavelength of photon *absorbed* for an electron transitioning from *2s to 4p orbital*
- iii. Wavelength of photon *absorbed* for an electron transitioning from *1s to 3p orbital*

i or ii or iii



Question 2 8 points



A Chem110 student plots the graph shown on the left, from data obtained for the kinetic energy (eV) of the ejected electron versus frequency (s^{-1}) of the incident light, for an unknown metal. ($1\text{eV} = 1.6022 \times 10^{-19} \text{ J}$)

Based on the plot, answer the questions **a to c** below:

a. 1 point On the plot above, label the **threshold frequency**.

b. 2 points What is the value of the work potential (in J) of the unknown metal?

c. 3 points What is the *longest* wavelength (in nm) required to eject an electron from this metal's surface? Show your work including equation(s) used. (3 sig figs)

d. 2 points In the space provided below, plot the following

(Assume frequency is *above* threshold frequency):

- Kinetic energy* of an ejected electron versus the *intensity* of the incident light (left)
- Number of electrons* ejected versus the *frequency* of the incident light (right)



**Question 3** **6 points****a. 2 points**

In the ground state of Ge (Germanium; $Z = 32$) atom, **name** the following orbitals:

i. Orbital containing the electron that requires the maximum energy to be removed.

ii. Orbital containing the electron that experiences the minimum effective nuclear charge.

b. 2 points

An unknown anion (A^-) has the following *excited* electronic configuration: $[Ar] 4s^2 3d^3 4p^1$

(Ar: Argon; $Z = 18$)

Give the **condensed ground state electronic configuration** of the atom, **A** and the cation, **A^{2+}** .
Show your work. (Assume all electronic transitions are possible)

Electronic Configuration of A:

Electronic Configuration of A^{2+} :

c. 2 points

In a single atom, determine how many electrons can have the following set of quantum numbers?

(i) $n = 5; l = 3$

In a single atom, determine how many electrons can have the following set of quantum numbers?

(i) $n = 3; l = 2; m_s = +1/2$

**Question 4** **8 points**

Rank the following from 1 to 4 in **increasing** order. Provide the ranking in the boxes provided.

Note: Atomic number (Z) are provided for relevant atoms

a) Increasing (lowest to highest) order of *ionization energy*: (1= lowest; 4 = highest)

Mg

Mg: Z = 12

Rb

Rb: Z = 37

Cs

Cs: Z = 55

K

K : Z = 19

b) Increasing (lowest to highest) order of *bond angle*: (1= lowest; 4 = highest)

H₂OCCl₄XeF₄

HCN

c) Increasing (smallest to largest) order of *ionic size* (1= smallest; 4 = largest)

S²⁻

S: Z = 16

O²⁻

O: Z = 8

Be²⁺

Be: Z = 4

Na⁺

Na: Z = 11

d) Increasing (lowest to highest) order of *bond energy* (1= lowest; 4 = highest)

P-F

P-Cl

P-I

P-Br

**Question 5** **12 points**

a. 9 points For the molecules HClO_4 and TeF_6 , draw the most stable *Lewis structure* showing appropriate shape and *dashed/wedged bonds*, based on VSEPR. Determine the **molecular geometry** around the central atom. For HClO_4 , label the *most polar bond*.



b. 3 points (Note: Lone pairs are not shown in structures below)

(i) For the three *resonance* structures (1, 2 and 3) shown below, answer the following question:



(I) Circle the *most contributing* resonance structure.

Structure 1

Structure 2

Structure 3

(ii) For the three structures given (4, 5, and 6) below, answer the following questions:



(I) Circle all the structures that represent *resonance structures* of each other:

Structure 4

Structure 5

Structure 6

(II) Among the resonance structures, circle the *most contributing* resonance structure.

Structure 4

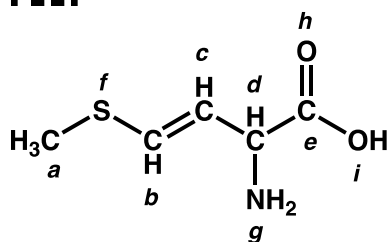
Structure 5

Structure 6

**Question 6** **6 points**

Using the data given below, calculate the **lattice energy** for SrO(s).

	ΔH° (kJ/mol)
$\text{Sr}(s) \longrightarrow \text{Sr}(g)$	+180
$\text{Sr}(g) \longrightarrow \text{Sr}^{2+}(g) + 2e^-$	+1468
$\text{O}_2(g) \longrightarrow 2\text{O}(g)$	+500
$\text{O}(g) + 2e^- \longrightarrow \text{O}^{2-}(g)$	+600
$\text{Sr}(s) + \frac{1}{2}\text{O}_2(g) \longrightarrow \text{SrO}(s)$	-550

**Question 7 10 points**

Consider the molecule given on the left, and answer questions (i) to (vi) using chemical bonding theories that we have learnt in class (valence bond theory, hybridization).

Note that the lone pairs are not shown in the structure:

i. Consider the carbon atoms marked in as ***a***, ***b***, ***c***, ***d***, and ***e*** in the molecule, and determine **hybridization** at each of the carbon centers. (2.5 points)

Atom	Hybridization
Carbon (<i>a</i>)	
Carbon (<i>b</i>)	
Carbon (<i>c</i>)	
Carbon (<i>d</i>)	
Carbon (<i>e</i>)	

ii. Now, consider the sulfur atom (S) marked as ***f***, nitrogen (N) atom (***g***), and oxygen(O) atoms (***h*** and ***i***), and determine their **hybridization**. (2 points)

Atom	Hybridization
Sulfur (<i>f</i>)	
Nitrogen (<i>g</i>)	
Oxygen (<i>h</i>)	
Oxygen (<i>i</i>)	

iii. **Describe** all the bonds formed by the carbon atoms ***a*** and ***b***, using sigma and pi designation as appropriate. (2 points)

Atom	Bonds formed (use sigma and pi as appropriate)
Carbon (<i>a</i>)	
Carbon (<i>b</i>)	

iv. How many total sp^3 -s sigma bonds are in the molecule? (1 point)

v. Determine **electronic** and **molecular geometry** at carbon (C) labeled ***e*** and sulfur (S) labeled ***f*** centres. (2 points)

Atom	Electronic Geometry	Molecular Geometry
Carbon (<i>e</i>)		
Sulfur (<i>f</i>)		

vi. Circle the value from the choices given below, that describes the **C-S-C bond angle at sulfur** (S) marked ***f***: (0.5 points)

>90° to <109.5° 109.5° >109.5 to <120° 120° >120° to <180°


Question 8 12 points

Using the **valence shell orbitals only** draw a **complete** molecular orbital (MO) diagram (with no s-p mixing) for the molecule **PS**, in the space provided below. **(6 points)**

Assume that 3s and 3p atomic orbitals will form molecular orbitals like those formed by the combination of 2s and 2p atomic orbitals that we learnt in class, and there is no s-p mixing.

Show **energy arrow**, **atomic orbitals** involved, **atom labels**, the **molecular orbitals** that are formed, **filled electrons** in the MO diagram etc. Make sure to label all the atomic and molecular orbitals.

Using the MO diagram above, answer the following questions:

- (i) Which one of the following will have the highest bond order? Circle the correct choice
 A) PS^{1-} B) PS C) PS^{1+}
- (ii) Which one of the following will have the longest bond length? Circle the correct choice
 A) PS^{1-} B) PS C) PS^{1+}
- (iii) Which of the following will be diamagnetic? Circle the correct choice.
 A) PS^{1-} B) PS C) PS^{1+}
- (iv) Draw Lewis dot structure of the molecule with the shortest bond length (based on the MO diagram) from among the three (PS^{1-} , PS, PS^{1+}), and indicate the bond order calculated from the Lewis dot structure. **(3 points)**

Lewis Dot Structure

Bond Order



77FE241D-0C49-41E2-85DD-F53DBCDD8408

midterm-exam-93388

#848 Page 10 of 10

SCRAP PAGE

**CHEMISTRY 110 –Midterm Exam****Thursday November 7, 2024, 6:30 – 8:30 pm****Examiners: Pallavi Sirjoosingh and Ashok Kakkar**

First name (please write as legibly as possible within the boxes)

--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--

Last name

--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--

Student ID number

--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--

This booklet has 8 questions on 10 pages (The last page can be used as a scrap page).

The staple must NOT be removed from the question booklet. Your intact question booklet must be returned at the end of the exam.

A separate booklet contains a periodic table, and datasheet.

Write your answers directly under the questions in the space provided. Do not write on the QR codes or near the top edges (as it may be cut off).

Please write **legibly and dark enough** for the scan to be readable. For students who use English as a second language, dictionaries are permitted. Calculators are allowed.

DO NOT ASK QUESTIONS OF THE INVIGILATORS about interpretations of the exam questions. If you have a concern about a question, please summarize it at the top of this page. All concerns will be evaluated after the exam. GOOD LUCK!

**Question 1 8 points** (*Assume all transitions are possible*)

a. 6 points Calculate the **energy (in kJ/mol)** required to ionize (completely remove) the electron from the **$n=3$** level of **He^+ ion** (He: Helium). Show your work including equations. (3 sig. figs.)

b. 2 points

(I) Select the largest (longest) wavelength from the three choices below:

- i. Wavelength of photon to *completely remove* an electron from *1s orbital of Hydrogen atom*
- ii. Wavelength of photon to *completely remove* an electron from *1s orbital of Boron atom (B)*
- iii. Wavelength of photon to *completely remove* an electron from *1s orbital of Boron cation (B^+)*

i or ii or iii

(II) Select the smallest (shortest) wavelength from the three choices below (*for H atom*):

- i. Wavelength of photon *emitted* for an electron transitioning from *2p to 1s orbital*
- ii. Wavelength of photon *absorbed* for an electron transitioning from *2s to 4p orbital*
- iii. Wavelength of photon *absorbed* for an electron transitioning from *1s to 3p orbital*

i or ii or iii



Question 2 8 points



A Chem110 student plots the graph shown on the left, from data obtained for the kinetic energy (eV) of the ejected electron versus frequency (s^{-1}) of the incident light, for an unknown metal. ($1\text{eV} = 1.6022 \times 10^{-19} \text{ J}$)

Based on the plot, answer the questions **a to c** below:

a. 1 point On the plot above, label the **threshold frequency**.

b. 2 points What is the value of the work potential (in J) of the unknown metal?

c. 3 points What is the *longest* wavelength (in nm) required to eject an electron from this metal's surface? Show your work including equation(s) used. (3 sig figs)

d. 2 points In the space provided below, plot the following

(Assume frequency is *above* threshold frequency):

- Kinetic energy* of an ejected electron versus the *intensity* of the incident light (left)
- Number of electrons* ejected versus the *frequency* of the incident light (right)



**Question 3** **6 points****a. 2 points**

In the ground state of Ge (Germanium; $Z = 32$) atom, **name** the following orbitals:

i. Orbital containing the electron that requires the maximum energy to be removed.

ii. Orbital containing the electron that experiences the minimum effective nuclear charge.

b. 2 points

An unknown anion (A^-) has the following *excited* electronic configuration: $[Ar] 4s^2 3d^3 4p^1$

(Ar: Argon; $Z = 18$)

Give the **condensed ground state electronic configuration** of the atom, **A** and the cation, **A^{2+}** .
Show your work. (Assume all electronic transitions are possible)

Electronic Configuration of A:

Electronic Configuration of A^{2+} :

c. 2 points

In a single atom, determine how many electrons can have the following set of quantum numbers?

(i) $n = 5; l = 3$

In a single atom, determine how many electrons can have the following set of quantum numbers?

(i) $n = 3; l = 2; m_s = +1/2$

**Question 4** **8 points**

Rank the following from 1 to 4 in **increasing** order. Provide the ranking in the boxes provided.

Note: Atomic number (Z) are provided for relevant atoms

a) Increasing (lowest to highest) order of *ionization energy*: (1= lowest; 4 = highest)

Mg

Mg: Z = 12

Rb

Rb: Z = 37

Cs

Cs: Z = 55

K

K : Z = 19

b) Increasing (lowest to highest) order of *bond angle*: (1= lowest; 4 = highest)

H₂OCCl₄XeF₄

HCN

c) Increasing (smallest to largest) order of *ionic size* (1= smallest; 4 = largest)

S²⁻

S: Z = 16

O²⁻

O: Z = 8

Be²⁺

Be: Z = 4

Na⁺

Na: Z = 11

d) Increasing (lowest to highest) order of *bond energy* (1= lowest; 4 = highest)

P-F

P-Cl

P-I

P-Br

**Question 5** **12 points**

a. 9 points For the molecules HClO_4 and TeF_6 , draw the most stable *Lewis structure* showing appropriate shape and *dashed/wedged bonds*, based on VSEPR. Determine the **molecular geometry** around the central atom. For HClO_4 , label the *most polar bond*.



b. 3 points (Note: Lone pairs are not shown in structures below)

(i) For the three *resonance* structures (1, 2 and 3) shown below, answer the following question:



(I) Circle the *most contributing* resonance structure.

Structure 1

Structure 2

Structure 3

(ii) For the three structures given (4, 5, and 6) below, answer the following questions:



(I) Circle all the structures that represent *resonance structures* of each other:

Structure 4

Structure 5

Structure 6

(II) Among the resonance structures, circle the *most contributing* resonance structure.

Structure 4

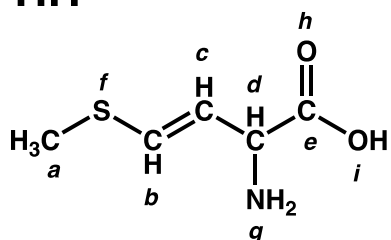
Structure 5

Structure 6

**Question 6** **6 points**

Using the data given below, calculate the **lattice energy** for SrO(s).

	ΔH° (kJ/mol)
$\text{Sr}(s) \longrightarrow \text{Sr}(g)$	+180
$\text{Sr}(g) \longrightarrow \text{Sr}^{2+}(g) + 2e^-$	+1468
$\text{O}_2(g) \longrightarrow 2\text{O}(g)$	+500
$\text{O}(g) + 2e^- \longrightarrow \text{O}^{2-}(g)$	+600
$\text{Sr}(s) + \frac{1}{2}\text{O}_2(g) \longrightarrow \text{SrO}(s)$	-550

**Question 7 10 points**

Consider the molecule given on the left, and answer questions (i) to (vi) using chemical bonding theories that we have learnt in class (valence bond theory, hybridization).

Note that the lone pairs are not shown in the structure:

i. Consider the carbon atoms marked in as ***a***, ***b***, ***c***, ***d***, and ***e*** in the molecule, and determine **hybridization** at each of the carbon centers. (2.5 points)

Atom	Hybridization
Carbon (<i>a</i>)	
Carbon (<i>b</i>)	
Carbon (<i>c</i>)	
Carbon (<i>d</i>)	
Carbon (<i>e</i>)	

ii. Now, consider the sulfur atom (S) marked as ***f***, nitrogen (N) atom (***g***), and oxygen(O) atoms (***h*** and ***i***), and determine their **hybridization**. (2 points)

Atom	Hybridization
Sulfur (<i>f</i>)	
Nitrogen (<i>g</i>)	
Oxygen (<i>h</i>)	
Oxygen (<i>i</i>)	

iii. **Describe** all the bonds formed by the carbon atoms ***a*** and ***b***, using sigma and pi designation as appropriate. (2 points)

Atom	Bonds formed (use sigma and pi as appropriate)
Carbon (<i>a</i>)	
Carbon (<i>b</i>)	

iv. How many total sp^3 -s sigma bonds are in the molecule? (1 point)

v. Determine **electronic** and **molecular geometry** at carbon (C) labeled ***e*** and sulfur (S) labeled ***f*** centres. (2 points)

Atom	Electronic Geometry	Molecular Geometry
Carbon (<i>e</i>)		
Sulfur (<i>f</i>)		

vi. Circle the value from the choices given below, that describes the **C-S-C bond angle at sulfur** (S) marked ***f***: (0.5 points)

>90° to <109.5° 109.5° >109.5 to <120° 120° >120° to <180°



Question 8 12 points

Using the **valence shell orbitals only** draw a **complete** molecular orbital (MO) diagram (with no s-p mixing) for the molecule **PS**, in the space provided below. **(6 points)**

Assume that 3s and 3p atomic orbitals will form molecular orbitals like those formed by the combination of 2s and 2p atomic orbitals that we learnt in class, and there is no s-p mixing.

Show **energy arrow**, **atomic orbitals** involved, **atom labels**, the **molecular orbitals** that are formed, **filled electrons** in the MO diagram etc. Make sure to label all the atomic and molecular orbitals.

Using the MO diagram above, answer the following questions:

- (i) Which one of the following will have the highest bond order? Circle the correct choice
 A) PS^{1-} B) PS C) PS^{1+}
- (ii) Which one of the following will have the longest bond length? Circle the correct choice
 A) PS^{1-} B) PS C) PS^{1+}
- (iii) Which of the following will be diamagnetic? Circle the correct choice.
 A) PS^{1-} B) PS C) PS^{1+}
- (iv) Draw Lewis dot structure of the molecule with the shortest bond length (based on the MO diagram) from among the three (PS^{1-} , PS, PS^{1+}), and indicate the bond order calculated from the Lewis dot structure. **(3 points)**

Lewis Dot Structure

Bond Order



F41ED2CA-6AF5-4FBB-9ACC-25A8E55FB346

midterm-exam-93388

#849 Page 10 of 10

SCRAP PAGE

**CHEMISTRY 110 –Midterm Exam****Thursday November 7, 2024, 6:30 – 8:30 pm**

Examiners: Pallavi Sirjoosingh and Ashok Kakkar

First name (please write as legibly as possible within the boxes)

--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--

Last name

--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--

Student ID number

--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--

This booklet has 8 questions on 10 pages (The last page can be used as a scrap page).

The staple must NOT be removed from the question booklet. Your intact question booklet must be returned at the end of the exam.

A separate booklet contains a periodic table, and datasheet.

Write your answers directly under the questions in the space provided. Do not write on the QR codes or near the top edges (as it may be cut off).

Please write **legibly and dark enough** for the scan to be readable. For students who use English as a second language, dictionaries are permitted. Calculators are allowed.

DO NOT ASK QUESTIONS OF THE INVIGILATORS about interpretations of the exam questions. If you have a concern about a question, please summarize it at the top of this page. All concerns will be evaluated after the exam. GOOD LUCK!

**Question 1 8 points** (*Assume all transitions are possible*)

a. 6 points Calculate the **energy (in kJ/mol)** required to ionize (completely remove) the electron from the **$n=3$** level of **He^+ ion** (He: Helium). Show your work including equations. (3 sig. figs.)

b. 2 points

(I) Select the largest (longest) wavelength from the three choices below:

- i. Wavelength of photon to *completely remove* an electron from *1s orbital of Hydrogen atom*
- ii. Wavelength of photon to *completely remove* an electron from *1s orbital of Boron atom (B)*
- iii. Wavelength of photon to *completely remove* an electron from *1s orbital of Boron cation (B^+)*

i or ii or iii

(II) Select the smallest (shortest) wavelength from the three choices below (*for H atom*):

- i. Wavelength of photon *emitted* for an electron transitioning from *2p to 1s orbital*
- ii. Wavelength of photon *absorbed* for an electron transitioning from *2s to 4p orbital*
- iii. Wavelength of photon *absorbed* for an electron transitioning from *1s to 3p orbital*

i or ii or iii



Question 2 8 points



A Chem110 student plots the graph shown on the left, from data obtained for the kinetic energy (eV) of the ejected electron versus frequency (s^{-1}) of the incident light, for an unknown metal. ($1\text{eV} = 1.6022 \times 10^{-19} \text{ J}$)

Based on the plot, answer the questions **a to c** below:

a. 1 point On the plot above, label the **threshold frequency**.

b. 2 points What is the value of the work potential (in J) of the unknown metal?

c. 3 points What is the *longest* wavelength (in nm) required to eject an electron from this metal's surface? Show your work including equation(s) used. (3 sig figs)

d. 2 points In the space provided below, plot the following

(Assume frequency is *above* threshold frequency):

- Kinetic energy* of an ejected electron versus the *intensity* of the incident light (left)
- Number of electrons* ejected versus the *frequency* of the incident light (right)



**Question 3** **6 points****a. 2 points**

In the ground state of Ge (Germanium; $Z = 32$) atom, **name** the following orbitals:

i. Orbital containing the electron that requires the maximum energy to be removed.

ii. Orbital containing the electron that experiences the minimum effective nuclear charge.

b. 2 points

An unknown anion (A^-) has the following *excited* electronic configuration: $[Ar] 4s^2 3d^3 4p^1$

(Ar: Argon; $Z = 18$)

Give the **condensed ground state electronic configuration** of the atom, **A** and the cation, **A^{2+}** .
Show your work. (Assume all electronic transitions are possible)

Electronic Configuration of A:

Electronic Configuration of A^{2+} :

c. 2 points

In a single atom, determine how many electrons can have the following set of quantum numbers?

(i) $n = 5; l = 3$

In a single atom, determine how many electrons can have the following set of quantum numbers?

(i) $n = 3; l = 2; m_s = +1/2$

**Question 4** **8 points**

Rank the following from 1 to 4 in **increasing** order. Provide the ranking in the boxes provided.

Note: Atomic number (Z) are provided for relevant atoms

a) Increasing (lowest to highest) order of *ionization energy*: (1= lowest; 4 = highest)

Mg

Mg: Z = 12

Rb

Rb: Z = 37

Cs

Cs: Z = 55

K

K : Z = 19

b) Increasing (lowest to highest) order of *bond angle*: (1= lowest; 4 = highest)

H₂OCCl₄XeF₄

HCN

c) Increasing (smallest to largest) order of *ionic size* (1= smallest; 4 = largest)

S²⁻

S: Z = 16

O²⁻

O: Z = 8

Be²⁺

Be: Z = 4

Na⁺

Na: Z = 11

d) Increasing (lowest to highest) order of *bond energy* (1= lowest; 4 = highest)

P-F

P-Cl

P-I

P-Br

**Question 5** **12 points**

a. 9 points For the molecules HClO_4 and TeF_6 , draw the most stable *Lewis structure* showing appropriate shape and *dashed/wedged bonds*, based on VSEPR. Determine the **molecular geometry** around the central atom. For HClO_4 , label the *most polar bond*.



b. 3 points (Note: Lone pairs are not shown in structures below)

(i) For the three *resonance* structures (1, 2 and 3) shown below, answer the following question:



(I) Circle the *most contributing* resonance structure.

Structure 1

Structure 2

Structure 3

(ii) For the three structures given (4, 5, and 6) below, answer the following questions:



(I) Circle all the structures that represent *resonance structures* of each other:

Structure 4

Structure 5

Structure 6

(II) Among the resonance structures, circle the *most contributing* resonance structure.

Structure 4

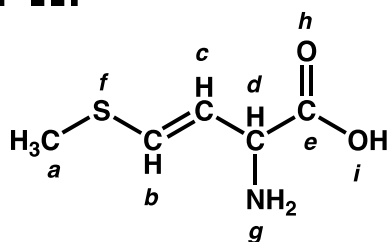
Structure 5

Structure 6

**Question 6** **6 points**

Using the data given below, calculate the **lattice energy** for SrO(s).

	ΔH° (kJ/mol)
$\text{Sr}(s) \longrightarrow \text{Sr}(g)$	+180
$\text{Sr}(g) \longrightarrow \text{Sr}^{2+}(g) + 2e^-$	+1468
$\text{O}_2(g) \longrightarrow 2\text{O}(g)$	+500
$\text{O}(g) + 2e^- \longrightarrow \text{O}^{2-}(g)$	+600
$\text{Sr}(s) + \frac{1}{2}\text{O}_2(g) \longrightarrow \text{SrO}(s)$	-550

**Question 7 10 points**

Consider the molecule given on the left, and answer questions (i) to (vi) using chemical bonding theories that we have learnt in class (valence bond theory, hybridization).

Note that the lone pairs are not shown in the structure:

i. Consider the carbon atoms marked in as ***a***, ***b***, ***c***, ***d***, and ***e*** in the molecule, and determine **hybridization** at each of the carbon centers. (2.5 points)

Atom	Hybridization
Carbon (<i>a</i>)	
Carbon (<i>b</i>)	
Carbon (<i>c</i>)	
Carbon (<i>d</i>)	
Carbon (<i>e</i>)	

ii. Now, consider the sulfur atom (S) marked as ***f***, nitrogen (N) atom (***g***), and oxygen(O) atoms (***h*** and ***i***), and determine their **hybridization**. (2 points)

Atom	Hybridization
Sulfur (<i>f</i>)	
Nitrogen (<i>g</i>)	
Oxygen (<i>h</i>)	
Oxygen (<i>i</i>)	

iii. **Describe** all the bonds formed by the carbon atoms ***a*** and ***b***, using sigma and pi designation as appropriate. (2 points)

Atom	Bonds formed (use sigma and pi as appropriate)
Carbon (<i>a</i>)	
Carbon (<i>b</i>)	

iv. How many total sp^3 -s sigma bonds are in the molecule? (1 point)

v. Determine **electronic** and **molecular geometry** at carbon (C) labeled ***e*** and sulfur (S) labeled ***f*** centres. (2 points)

Atom	Electronic Geometry	Molecular Geometry
Carbon (<i>e</i>)		
Sulfur (<i>f</i>)		

vi. Circle the value from the choices given below, that describes the **C-S-C bond angle at sulfur** (S) marked ***f***: (0.5 points)

>90° to <109.5° 109.5° >109.5 to <120° 120° >120° to <180°

**Question 8 12 points**

Using the **valence shell orbitals only** draw a **complete** molecular orbital (MO) diagram (with no s-p mixing) for the molecule **PS**, in the space provided below. **(6 points)**

Assume that 3s and 3p atomic orbitals will form molecular orbitals like those formed by the combination of 2s and 2p atomic orbitals that we learnt in class, and there is no s-p mixing.

Show **energy arrow**, **atomic orbitals** involved, **atom labels**, the **molecular orbitals** that are formed, **filled electrons** in the MO diagram etc. Make sure to label all the atomic and molecular orbitals.

Using the MO diagram above, answer the following questions:

- (i) Which one of the following will have the highest bond order? Circle the correct choice
A) PS^{1-} B) PS C) PS^{1+}
- (ii) Which one of the following will have the longest bond length? Circle the correct choice
A) PS^{1-} B) PS C) PS^{1+}
- (iii) Which of the following will be diamagnetic? Circle the correct choice.
A) PS^{1-} B) PS C) PS^{1+}
- (iv) Draw Lewis dot structure of the molecule with the shortest bond length (based on the MO diagram) from among the three (PS^{1-} , PS, PS^{1+}), and indicate the bond order calculated from the Lewis dot structure. **(3 points)**

Lewis Dot Structure

Bond Order



E3AF153D-66DB-4BB8-AC3C-74202AD4AE63

midterm-exam-93388

#850 Page 10 of 10

SCRAP PAGE

**CHEMISTRY 110 –Midterm Exam****Thursday November 7, 2024, 6:30 – 8:30 pm**

Examiners: Pallavi Sirjoosingh and Ashok Kakkar

First name (please write as legibly as possible within the boxes)

--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--

Last name

--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--

Student ID number

--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--

This booklet has 8 questions on 10 pages (The last page can be used as a scrap page).

The staple must NOT be removed from the question booklet. Your intact question booklet must be returned at the end of the exam.

A separate booklet contains a periodic table, and datasheet.

Write your answers directly under the questions in the space provided. Do not write on the QR codes or near the top edges (as it may be cut off).

Please write **legibly and dark enough** for the scan to be readable. For students who use English as a second language, dictionaries are permitted. Calculators are allowed.

DO NOT ASK QUESTIONS OF THE INVIGILATORS about interpretations of the exam questions. If you have a concern about a question, please summarize it at the top of this page. All concerns will be evaluated after the exam. GOOD LUCK!

**Question 1 8 points** (*Assume all transitions are possible*)

a. 6 points Calculate the **energy (in kJ/mol)** required to ionize (completely remove) the electron from the **n=3** level of **He⁺ ion** (He: Helium). Show your work including equations. (3 sig. figs.)

b. 2 points

(I) Select the largest (longest) wavelength from the three choices below:

- i. Wavelength of photon to *completely remove* an electron from *1s orbital of Hydrogen atom*
- ii. Wavelength of photon to *completely remove* an electron from *1s orbital of Boron atom (B)*
- iii. Wavelength of photon to *completely remove* an electron from *1s orbital of Boron cation (B⁺)*

i or ii or iii

(II) Select the smallest (shortest) wavelength from the three choices below (*for H atom*):

- i. Wavelength of photon *emitted* for an electron transitioning from *2p to 1s orbital*
- ii. Wavelength of photon *absorbed* for an electron transitioning from *2s to 4p orbital*
- iii. Wavelength of photon *absorbed* for an electron transitioning from *1s to 3p orbital*

i or ii or iii



Question 2 8 points



A Chem110 student plots the graph shown on the left, from data obtained for the kinetic energy (eV) of the ejected electron versus frequency (s^{-1}) of the incident light, for an unknown metal. ($1\text{eV} = 1.6022 \times 10^{-19} \text{ J}$)

Based on the plot, answer the questions **a to c** below:

a. 1 point On the plot above, label the **threshold frequency**.

b. 2 points What is the value of the work potential (in J) of the unknown metal?

c. 3 points What is the *longest* wavelength (in nm) required to eject an electron from this metal's surface? Show your work including equation(s) used. (3 sig figs)

d. 2 points In the space provided below, plot the following

(Assume frequency is *above* threshold frequency):

- i. *Kinetic energy* of an ejected electron versus the *intensity* of the incident light (left)
- ii. *Number of electrons* ejected versus the *frequency* of the incident light (right)



**Question 3** **6 points****a. 2 points**

In the ground state of Ge (Germanium; $Z = 32$) atom, **name** the following orbitals:

i. Orbital containing the electron that requires the maximum energy to be removed.

ii. Orbital containing the electron that experiences the minimum effective nuclear charge.

b. 2 points

An unknown anion (A^-) has the following *excited* electronic configuration: $[Ar] 4s^2 3d^3 4p^1$

(Ar: Argon; $Z = 18$)

Give the **condensed ground state electronic configuration** of the atom, **A** and the cation, **A^{2+}** .
Show your work. (Assume all electronic transitions are possible)

Electronic Configuration of A:

Electronic Configuration of A^{2+} :

c. 2 points

In a single atom, determine how many electrons can have the following set of quantum numbers?

(i) $n = 5; l = 3$

In a single atom, determine how many electrons can have the following set of quantum numbers?

(i) $n = 3; l = 2; m_s = +1/2$

**Question 4** **8 points**

Rank the following from 1 to 4 in **increasing** order. Provide the ranking in the boxes provided.

Note: Atomic number (Z) are provided for relevant atoms

a) Increasing (lowest to highest) order of *ionization energy*: (1= lowest; 4 = highest)

Mg

Mg: Z = 12

Rb

Rb: Z = 37

Cs

Cs: Z = 55

K

K : Z = 19

b) Increasing (lowest to highest) order of *bond angle*: (1= lowest; 4 = highest)

H₂OCCl₄XeF₄

HCN

c) Increasing (smallest to largest) order of *ionic size* (1= smallest; 4 = largest)

S²⁻

S: Z = 16

O²⁻

O: Z = 8

Be²⁺

Be: Z = 4

Na⁺

Na: Z = 11

d) Increasing (lowest to highest) order of *bond energy* (1= lowest; 4 = highest)

P-F

P-Cl

P-I

P-Br

**Question 5** **12 points**

a. 9 points For the molecules HClO_4 and TeF_6 , draw the most stable *Lewis structure* showing appropriate shape and *dashed/wedged bonds*, based on VSEPR. Determine the **molecular geometry** around the central atom. For HClO_4 , label the *most polar bond*.



b. 3 points (Note: Lone pairs are not shown in structures below)

(i) For the three *resonance* structures (1, 2 and 3) shown below, answer the following question:



(I) Circle the *most contributing* resonance structure.

Structure 1

Structure 2

Structure 3

(ii) For the three structures given (4, 5, and 6) below, answer the following questions:



(I) Circle all the structures that represent *resonance structures* of each other:

Structure 4

Structure 5

Structure 6

(II) Among the resonance structures, circle the *most contributing* resonance structure.

Structure 4

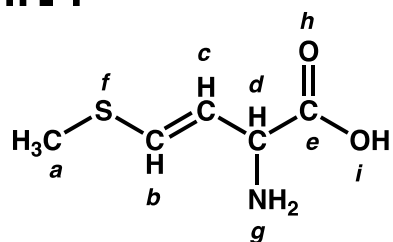
Structure 5

Structure 6

**Question 6** **6 points**

Using the data given below, calculate the **lattice energy** for SrO(s).

	ΔH° (kJ/mol)
$\text{Sr}(s) \longrightarrow \text{Sr}(g)$	+180
$\text{Sr}(g) \longrightarrow \text{Sr}^{2+}(g) + 2e^-$	+1468
$\text{O}_2(g) \longrightarrow 2\text{O}(g)$	+500
$\text{O}(g) + 2e^- \longrightarrow \text{O}^{2-}(g)$	+600
$\text{Sr}(s) + \frac{1}{2}\text{O}_2(g) \longrightarrow \text{SrO}(s)$	-550

**Question 7 10 points**

Consider the molecule given on the left, and answer questions (i) to (vi) using chemical bonding theories that we have learnt in class (valence bond theory, hybridization).

Note that the lone pairs are not shown in the structure:

i. Consider the carbon atoms marked in as ***a***, ***b***, ***c***, ***d***, and ***e*** in the molecule, and determine **hybridization** at each of the carbon centers. (2.5 points)

Atom	Hybridization
Carbon (<i>a</i>)	
Carbon (<i>b</i>)	
Carbon (<i>c</i>)	
Carbon (<i>d</i>)	
Carbon (<i>e</i>)	

ii. Now, consider the sulfur atom (S) marked as ***f***, nitrogen (N) atom (***g***), and oxygen(O) atoms (***h*** and ***i***), and determine their **hybridization**. (2 points)

Atom	Hybridization
Sulfur (<i>f</i>)	
Nitrogen (<i>g</i>)	
Oxygen (<i>h</i>)	
Oxygen (<i>i</i>)	

iii. **Describe** all the bonds formed by the carbon atoms ***a*** and ***b***, using sigma and pi designation as appropriate. (2 points)

Atom	Bonds formed (use sigma and pi as appropriate)
Carbon (<i>a</i>)	
Carbon (<i>b</i>)	

iv. How many total sp^3 -s sigma bonds are in the molecule? (1 point)

v. Determine **electronic** and **molecular geometry** at carbon (C) labeled ***e*** and sulfur (S) labeled ***f*** centres. (2 points)

Atom	Electronic Geometry	Molecular Geometry
Carbon (<i>e</i>)		
Sulfur (<i>f</i>)		

vi. Circle the value from the choices given below, that describes the **C-S-C bond angle at sulfur** (S) marked ***f***: (0.5 points)

>90° to <109.5° 109.5° >109.5 to <120° 120° >120° to <180°


Question 8 12 points

Using the **valence shell orbitals only** draw a **complete** molecular orbital (MO) diagram (with no s-p mixing) for the molecule **PS**, in the space provided below. **(6 points)**

Assume that 3s and 3p atomic orbitals will form molecular orbitals like those formed by the combination of 2s and 2p atomic orbitals that we learnt in class, and there is no s-p mixing.

Show **energy arrow**, **atomic orbitals** involved, **atom labels**, the **molecular orbitals** that are formed, **filled electrons** in the MO diagram etc. Make sure to label all the atomic and molecular orbitals.

Using the MO diagram above, answer the following questions:

- (i) Which one of the following will have the highest bond order? Circle the correct choice
 A) PS^{1-} B) PS C) PS^{1+}
- (ii) Which one of the following will have the longest bond length? Circle the correct choice
 A) PS^{1-} B) PS C) PS^{1+}
- (iii) Which of the following will be diamagnetic? Circle the correct choice.
 A) PS^{1-} B) PS C) PS^{1+}
- (iv) Draw Lewis dot structure of the molecule with the shortest bond length (based on the MO diagram) from among the three (PS^{1-} , PS, PS^{1+}), and indicate the bond order calculated from the Lewis dot structure. **(3 points)**

Lewis Dot Structure

Bond Order



E03CBC72-42C4-4B52-A8C6-61D5C92144D2

midterm-exam-93388

#851 Page 10 of 10

SCRAP PAGE

**CHEMISTRY 110 –Midterm Exam****Thursday November 7, 2024, 6:30 – 8:30 pm****Examiners: Pallavi Sirjoosingh and Ashok Kakkar**

First name (please write as legibly as possible within the boxes)

--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--

Last name

--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--

Student ID number

--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--

This booklet has 8 questions on 10 pages (The last page can be used as a scrap page).

The staple must NOT be removed from the question booklet. Your intact question booklet must be returned at the end of the exam.

A separate booklet contains a periodic table, and datasheet.

Write your answers directly under the questions in the space provided. Do not write on the QR codes or near the top edges (as it may be cut off).

Please write **legibly and dark enough** for the scan to be readable. For students who use English as a second language, dictionaries are permitted. Calculators are allowed.

DO NOT ASK QUESTIONS OF THE INVIGILATORS about interpretations of the exam questions. If you have a concern about a question, please summarize it at the top of this page. All concerns will be evaluated after the exam. GOOD LUCK!

**Question 1 8 points** (*Assume all transitions are possible*)

a. 6 points Calculate the **energy (in kJ/mol)** required to ionize (completely remove) the electron from the **n=3** level of **He⁺ ion** (He: Helium). Show your work including equations. (3 sig. figs.)

b. 2 points

(I) Select the largest (longest) wavelength from the three choices below:

- i. Wavelength of photon to *completely remove* an electron from *1s orbital of Hydrogen atom*
- ii. Wavelength of photon to *completely remove* an electron from *1s orbital of Boron atom (B)*
- iii. Wavelength of photon to *completely remove* an electron from *1s orbital of Boron cation (B⁺)*

i or ii or iii

(II) Select the smallest (shortest) wavelength from the three choices below (*for H atom*):

- i. Wavelength of photon *emitted* for an electron transitioning from *2p to 1s orbital*
- ii. Wavelength of photon *absorbed* for an electron transitioning from *2s to 4p orbital*
- iii. Wavelength of photon *absorbed* for an electron transitioning from *1s to 3p orbital*

i or ii or iii



Question 2 8 points



A Chem110 student plots the graph shown on the left, from data obtained for the kinetic energy (eV) of the ejected electron versus frequency (s^{-1}) of the incident light, for an unknown metal. ($1\text{eV} = 1.6022 \times 10^{-19} \text{ J}$)

Based on the plot, answer the questions **a to c** below:

a. 1 point On the plot above, label the **threshold frequency**.

b. 2 points What is the value of the work potential (in J) of the unknown metal?

c. 3 points What is the *longest* wavelength (in nm) required to eject an electron from this metal's surface? Show your work including equation(s) used. (3 sig figs)

d. 2 points In the space provided below, plot the following

(Assume frequency is *above* threshold frequency):

- Kinetic energy* of an ejected electron versus the *intensity* of the incident light (left)
- Number of electrons* ejected versus the *frequency* of the incident light (right)



**Question 3** **6 points****a. 2 points**

In the ground state of Ge (Germanium; $Z = 32$) atom, **name** the following orbitals:

i. Orbital containing the electron that requires the maximum energy to be removed.

ii. Orbital containing the electron that experiences the minimum effective nuclear charge.

b. 2 points

An unknown anion (A^-) has the following *excited* electronic configuration: $[Ar] 4s^2 3d^3 4p^1$

(Ar: Argon; $Z = 18$)

Give the **condensed ground state electronic configuration** of the atom, **A** and the cation, **A^{2+}** .
Show your work. (Assume all electronic transitions are possible)

Electronic Configuration of A:

Electronic Configuration of A^{2+} :

c. 2 points

In a single atom, determine how many electrons can have the following set of quantum numbers?

(i) $n = 5; l = 3$

In a single atom, determine how many electrons can have the following set of quantum numbers?

(i) $n = 3; l = 2; m_s = +1/2$

**Question 4** **8 points**

Rank the following from 1 to 4 in **increasing** order. Provide the ranking in the boxes provided.

Note: Atomic number (Z) are provided for relevant atoms

a) Increasing (lowest to highest) order of *ionization energy*: (1= lowest; 4 = highest)

Mg

Mg: Z = 12

Rb

Rb: Z = 37

Cs

Cs: Z = 55

K

K : Z = 19

b) Increasing (lowest to highest) order of *bond angle*: (1= lowest; 4 = highest)

H₂OCCl₄XeF₄

HCN

c) Increasing (smallest to largest) order of *ionic size* (1= smallest; 4 = largest)

S²⁻

S: Z = 16

O²⁻

O: Z = 8

Be²⁺

Be: Z = 4

Na⁺

Na: Z = 11

d) Increasing (lowest to highest) order of *bond energy* (1= lowest; 4 = highest)

P-F

P-Cl

P-I

P-Br

**Question 5** **12 points**

a. 9 points For the molecules HClO_4 and TeF_6 , draw the most stable *Lewis structure* showing appropriate shape and *dashed/wedged bonds*, based on VSEPR. Determine the **molecular geometry** around the central atom. For HClO_4 , label the *most polar bond*.



b. 3 points (Note: Lone pairs are not shown in structures below)

(i) For the three *resonance* structures (1, 2 and 3) shown below, answer the following question:



(I) Circle the *most contributing* resonance structure.

Structure 1

Structure 2

Structure 3

(ii) For the three structures given (4, 5, and 6) below, answer the following questions:



(I) Circle all the structures that represent *resonance structures* of each other:

Structure 4

Structure 5

Structure 6

(II) Among the resonance structures, circle the *most contributing* resonance structure.

Structure 4

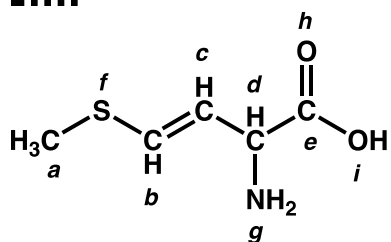
Structure 5

Structure 6

**Question 6** **6 points**

Using the data given below, calculate the **lattice energy** for SrO(s).

	ΔH° (kJ/mol)
$\text{Sr}(s) \longrightarrow \text{Sr}(g)$	+180
$\text{Sr}(g) \longrightarrow \text{Sr}^{2+}(g) + 2e^-$	+1468
$\text{O}_2(g) \longrightarrow 2\text{O}(g)$	+500
$\text{O}(g) + 2e^- \longrightarrow \text{O}^{2-}(g)$	+600
$\text{Sr}(s) + \frac{1}{2}\text{O}_2(g) \longrightarrow \text{SrO}(s)$	-550

**Question 7 10 points**

Consider the molecule given on the left, and answer questions (i) to (vi) using chemical bonding theories that we have learnt in class (valence bond theory, hybridization).

Note that the lone pairs are not shown in the structure:

i. Consider the carbon atoms marked in as ***a***, ***b***, ***c***, ***d***, and ***e*** in the molecule, and determine **hybridization** at each of the carbon centers. (2.5 points)

Atom	Hybridization
Carbon (<i>a</i>)	
Carbon (<i>b</i>)	
Carbon (<i>c</i>)	
Carbon (<i>d</i>)	
Carbon (<i>e</i>)	

ii. Now, consider the sulfur atom (S) marked as ***f***, nitrogen (N) atom (***g***), and oxygen(O) atoms (***h*** and ***i***), and determine their **hybridization**. (2 points)

Atom	Hybridization
Sulfur (<i>f</i>)	
Nitrogen (<i>g</i>)	
Oxygen (<i>h</i>)	
Oxygen (<i>i</i>)	

iii. **Describe** all the bonds formed by the carbon atoms ***a*** and ***b***, using sigma and pi designation as appropriate. (2 points)

Atom	Bonds formed (use sigma and pi as appropriate)
Carbon (<i>a</i>)	
Carbon (<i>b</i>)	

iv. How many total sp^3 -s sigma bonds are in the molecule? (1 point)

v. Determine **electronic** and **molecular geometry** at carbon (C) labeled ***e*** and sulfur (S) labeled ***f*** centres. (2 points)

Atom	Electronic Geometry	Molecular Geometry
Carbon (<i>e</i>)		
Sulfur (<i>f</i>)		

vi. Circle the value from the choices given below, that describes the **C-S-C bond angle at sulfur** (S) marked ***f***: (0.5 points)

>90° to <109.5° 109.5° >109.5 to <120° 120° >120° to <180°


Question 8 12 points

Using the **valence shell orbitals only** draw a **complete** molecular orbital (MO) diagram (with no s-p mixing) for the molecule **PS**, in the space provided below. **(6 points)**

Assume that 3s and 3p atomic orbitals will form molecular orbitals like those formed by the combination of 2s and 2p atomic orbitals that we learnt in class, and there is no s-p mixing.

Show **energy arrow**, **atomic orbitals** involved, **atom labels**, the **molecular orbitals** that are formed, **filled electrons** in the MO diagram etc. Make sure to label all the atomic and molecular orbitals.

Using the MO diagram above, answer the following questions:

- (i) Which one of the following will have the highest bond order? Circle the correct choice
 A) PS^{1-} B) PS C) PS^{1+}
- (ii) Which one of the following will have the longest bond length? Circle the correct choice
 A) PS^{1-} B) PS C) PS^{1+}
- (iii) Which of the following will be diamagnetic? Circle the correct choice.
 A) PS^{1-} B) PS C) PS^{1+}
- (iv) Draw Lewis dot structure of the molecule with the shortest bond length (based on the MO diagram) from among the three (PS^{1-} , PS, PS^{1+}), and indicate the bond order calculated from the Lewis dot structure. **(3 points)**

Lewis Dot Structure

Bond Order



4D70487C-5D30-4C6A-B836-58056E253DB8

midterm-exam-93388

#852 Page 10 of 10

SCRAP PAGE

**CHEMISTRY 110 –Midterm Exam****Thursday November 7, 2024, 6:30 – 8:30 pm****Examiners: Pallavi Sirjoosingh and Ashok Kakkar**

First name (please write as legibly as possible within the boxes)

--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--

Last name

--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--

Student ID number

--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--

This booklet has 8 questions on 10 pages (The last page can be used as a scrap page).

The staple must NOT be removed from the question booklet. Your intact question booklet must be returned at the end of the exam.

A separate booklet contains a periodic table, and datasheet.

Write your answers directly under the questions in the space provided. Do not write on the QR codes or near the top edges (as it may be cut off).

Please write **legibly and dark enough** for the scan to be readable. For students who use English as a second language, dictionaries are permitted. Calculators are allowed.

DO NOT ASK QUESTIONS OF THE INVIGILATORS about interpretations of the exam questions. If you have a concern about a question, please summarize it at the top of this page. All concerns will be evaluated after the exam. GOOD LUCK!

**Question 1 8 points** (*Assume all transitions are possible*)

a. 6 points Calculate the **energy (in kJ/mol)** required to ionize (completely remove) the electron from the **n=3** level of **He⁺ ion** (He: Helium). Show your work including equations. (3 sig. figs.)

b. 2 points

(I) Select the largest (longest) wavelength from the three choices below:

- i. Wavelength of photon to *completely remove* an electron from *1s orbital of Hydrogen atom*
- ii. Wavelength of photon to *completely remove* an electron from *1s orbital of Boron atom (B)*
- iii. Wavelength of photon to *completely remove* an electron from *1s orbital of Boron cation (B⁺)*

i

or

ii

or

iii

(II) Select the smallest (shortest) wavelength from the three choices below (*for H atom*):

- i. Wavelength of photon *emitted* for an electron transitioning from *2p to 1s orbital*
- ii. Wavelength of photon *absorbed* for an electron transitioning from *2s to 4p orbital*
- iii. Wavelength of photon *absorbed* for an electron transitioning from *1s to 3p orbital*

i

or

ii

or

iii



Question 2 8 points



A Chem110 student plots the graph shown on the left, from data obtained for the kinetic energy (eV) of the ejected electron versus frequency (s^{-1}) of the incident light, for an unknown metal. ($1\text{eV} = 1.6022 \times 10^{-19} \text{ J}$)

Based on the plot, answer the questions **a to c** below:

a. 1 point On the plot above, label the **threshold frequency**.

b. 2 points What is the value of the work potential (in J) of the unknown metal?

c. 3 points What is the *longest* wavelength (in nm) required to eject an electron from this metal's surface? Show your work including equation(s) used. (3 sig figs)

d. 2 points In the space provided below, plot the following

(Assume frequency is *above* threshold frequency):

- i. *Kinetic energy* of an ejected electron versus the *intensity* of the incident light (left)
- ii. *Number of electrons* ejected versus the *frequency* of the incident light (right)



**Question 3** **6 points****a. 2 points**

In the ground state of Ge (Germanium; $Z = 32$) atom, **name** the following orbitals:

i. Orbital containing the electron that requires the maximum energy to be removed.

ii. Orbital containing the electron that experiences the minimum effective nuclear charge.

b. 2 points

An unknown anion (A^-) has the following *excited* electronic configuration: $[Ar] 4s^2 3d^3 4p^1$

(Ar: Argon; $Z = 18$)

Give the **condensed ground state electronic configuration** of the atom, **A** and the cation, **A^{2+}** .
Show your work. (Assume all electronic transitions are possible)

Electronic Configuration of A:

Electronic Configuration of A^{2+} :

c. 2 points

In a single atom, determine how many electrons can have the following set of quantum numbers?

(i) $n = 5; l = 3$

In a single atom, determine how many electrons can have the following set of quantum numbers?

(i) $n = 3; l = 2; m_s = +1/2$

**Question 4** **8 points**

Rank the following from 1 to 4 in **increasing** order. Provide the ranking in the boxes provided.

Note: Atomic number (Z) are provided for relevant atoms

a) Increasing (lowest to highest) order of *ionization energy*: (1= lowest; 4 = highest)

Mg

Mg: Z = 12

Rb

Rb: Z = 37

Cs

Cs: Z = 55

K

K : Z = 19

b) Increasing (lowest to highest) order of *bond angle*: (1= lowest; 4 = highest)

H₂OCCl₄XeF₄

HCN

c) Increasing (smallest to largest) order of *ionic size* (1= smallest; 4 = largest)

S²⁻

S: Z = 16

O²⁻

O: Z = 8

Be²⁺

Be: Z = 4

Na⁺

Na: Z = 11

d) Increasing (lowest to highest) order of *bond energy* (1= lowest; 4 = highest)

P-F

P-Cl

P-I

P-Br

**Question 5** **12 points**

a. 9 points For the molecules HClO_4 and TeF_6 , draw the most stable *Lewis structure* showing appropriate shape and *dashed/wedged bonds*, based on VSEPR. Determine the **molecular geometry** around the central atom. For HClO_4 , label the *most polar bond*.



b. 3 points (Note: Lone pairs are not shown in structures below)

(i) For the three *resonance* structures (1, 2 and 3) shown below, answer the following question:



(I) Circle the *most contributing* resonance structure.

Structure 1

Structure 2

Structure 3

(ii) For the three structures given (4, 5, and 6) below, answer the following questions:



(I) Circle all the structures that represent *resonance structures* of each other:

Structure 4

Structure 5

Structure 6

(II) Among the resonance structures, circle the *most contributing* resonance structure.

Structure 4

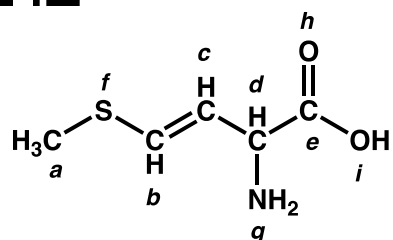
Structure 5

Structure 6

**Question 6** **6 points**

Using the data given below, calculate the **lattice energy** for SrO(s).

	ΔH° (kJ/mol)
$\text{Sr}(s) \longrightarrow \text{Sr}(g)$	+180
$\text{Sr}(g) \longrightarrow \text{Sr}^{2+}(g) + 2e^-$	+1468
$\text{O}_2(g) \longrightarrow 2\text{O}(g)$	+500
$\text{O}(g) + 2e^- \longrightarrow \text{O}^{2-}(g)$	+600
$\text{Sr}(s) + \frac{1}{2}\text{O}_2(g) \longrightarrow \text{SrO}(s)$	-550

**Question 7 10 points**

Consider the molecule given on the left, and answer questions (i) to (vi) using chemical bonding theories that we have learnt in class (valence bond theory, hybridization).

Note that the lone pairs are not shown in the structure:

i. Consider the carbon atoms marked in as ***a***, ***b***, ***c***, ***d***, and ***e*** in the molecule, and determine **hybridization** at each of the carbon centers. (2.5 points)

Atom	Hybridization
Carbon (<i>a</i>)	
Carbon (<i>b</i>)	
Carbon (<i>c</i>)	
Carbon (<i>d</i>)	
Carbon (<i>e</i>)	

ii. Now, consider the sulfur atom (S) marked as ***f***, nitrogen (N) atom (***g***), and oxygen(O) atoms (***h*** and ***i***), and determine their **hybridization**. (2 points)

Atom	Hybridization
Sulfur (<i>f</i>)	
Nitrogen (<i>g</i>)	
Oxygen (<i>h</i>)	
Oxygen (<i>i</i>)	

iii. **Describe** all the bonds formed by the carbon atoms ***a*** and ***b***, using sigma and pi designation as appropriate. (2 points)

Atom	Bonds formed (use sigma and pi as appropriate)
Carbon (<i>a</i>)	
Carbon (<i>b</i>)	

iv. How many total sp^3 -s sigma bonds are in the molecule? (1 point)

v. Determine **electronic** and **molecular geometry** at carbon (C) labeled ***e*** and sulfur (S) labeled ***f*** centres. (2 points)

Atom	Electronic Geometry	Molecular Geometry
Carbon (<i>e</i>)		
Sulfur (<i>f</i>)		

vi. Circle the value from the choices given below, that describes the **C-S-C bond angle at sulfur** (S) marked ***f***: (0.5 points)

>90° to <109.5° 109.5° >109.5 to <120° 120° >120° to <180°

**Question 8 12 points**

Using the **valence shell orbitals only** draw a **complete** molecular orbital (MO) diagram (with no s-p mixing) for the molecule **PS**, in the space provided below. **(6 points)**

Assume that 3s and 3p atomic orbitals will form molecular orbitals like those formed by the combination of 2s and 2p atomic orbitals that we learnt in class, and there is no s-p mixing.

Show **energy arrow**, **atomic orbitals** involved, **atom labels**, the **molecular orbitals** that are formed, **filled electrons** in the MO diagram etc. Make sure to label all the atomic and molecular orbitals.

Using the MO diagram above, answer the following questions:

- (i) Which one of the following will have the highest bond order? Circle the correct choice
A) PS^{1-} B) PS C) PS^{1+}
- (ii) Which one of the following will have the longest bond length? Circle the correct choice
A) PS^{1-} B) PS C) PS^{1+}
- (iii) Which of the following will be diamagnetic? Circle the correct choice.
A) PS^{1-} B) PS C) PS^{1+}
- (iv) Draw Lewis dot structure of the molecule with the shortest bond length (based on the MO diagram) from among the three (PS^{1-} , PS, PS^{1+}), and indicate the bond order calculated from the Lewis dot structure. **(3 points)**

Lewis Dot Structure

Bond Order



1AE5BBE8-2C78-42AB-90F5-88FF5610EAD3

midterm-exam-93388

#853 Page 10 of 10

SCRAP PAGE

**CHEMISTRY 110 –Midterm Exam****Thursday November 7, 2024, 6:30 – 8:30 pm**

Examiners: Pallavi Sirjoosingh and Ashok Kakkar

First name (please write as legibly as possible within the boxes)

--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--

Last name

--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--

Student ID number

--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--

This booklet has 8 questions on 10 pages (The last page can be used as a scrap page).

The staple must NOT be removed from the question booklet. Your intact question booklet must be returned at the end of the exam.

A separate booklet contains a periodic table, and datasheet.

Write your answers directly under the questions in the space provided. Do not write on the QR codes or near the top edges (as it may be cut off).

Please write **legibly and dark enough** for the scan to be readable. For students who use English as a second language, dictionaries are permitted. Calculators are allowed.

DO NOT ASK QUESTIONS OF THE INVIGILATORS about interpretations of the exam questions. If you have a concern about a question, please summarize it at the top of this page. All concerns will be evaluated after the exam. GOOD LUCK!

**Question 1 8 points** (*Assume all transitions are possible*)

a. 6 points Calculate the **energy (in kJ/mol)** required to ionize (completely remove) the electron from the **n=3** level of **He⁺ ion** (He: Helium). Show your work including equations. (3 sig. figs.)

b. 2 points

(I) Select the largest (longest) wavelength from the three choices below:

- i. Wavelength of photon to *completely remove* an electron from *1s orbital of Hydrogen atom*
- ii. Wavelength of photon to *completely remove* an electron from *1s orbital of Boron atom (B)*
- iii. Wavelength of photon to *completely remove* an electron from *1s orbital of Boron cation (B⁺)*

i or ii or iii

(II) Select the smallest (shortest) wavelength from the three choices below (*for H atom*):

- i. Wavelength of photon *emitted* for an electron transitioning from *2p to 1s orbital*
- ii. Wavelength of photon *absorbed* for an electron transitioning from *2s to 4p orbital*
- iii. Wavelength of photon *absorbed* for an electron transitioning from *1s to 3p orbital*

i or ii or iii



Question 2 8 points



A Chem110 student plots the graph shown on the left, from data obtained for the kinetic energy (eV) of the ejected electron versus frequency (s^{-1}) of the incident light, for an unknown metal. ($1\text{eV} = 1.6022 \times 10^{-19} \text{ J}$)

Based on the plot, answer the questions **a to c** below:

a. 1 point On the plot above, label the **threshold frequency**.

b. 2 points What is the value of the work potential (in J) of the unknown metal?

c. 3 points What is the *longest* wavelength (in nm) required to eject an electron from this metal's surface? Show your work including equation(s) used. (3 sig figs)

d. 2 points In the space provided below, plot the following

(Assume frequency is *above* threshold frequency):

- i. *Kinetic energy* of an ejected electron versus the *intensity* of the incident light (left)
- ii. *Number of electrons* ejected versus the *frequency* of the incident light (right)



**Question 3** **6 points****a. 2 points**

In the ground state of Ge (Germanium; $Z = 32$) atom, **name** the following orbitals:

i. Orbital containing the electron that requires the maximum energy to be removed.

ii. Orbital containing the electron that experiences the minimum effective nuclear charge.

b. 2 points

An unknown anion (A^-) has the following *excited* electronic configuration: $[Ar] 4s^2 3d^3 4p^1$

(Ar: Argon; $Z = 18$)

Give the **condensed ground state electronic configuration** of the atom, **A** and the cation, **A^{2+}** .
Show your work. (Assume all electronic transitions are possible)

Electronic Configuration of A:

Electronic Configuration of A^{2+} :

c. 2 points

In a single atom, determine how many electrons can have the following set of quantum numbers?

(i) $n = 5; l = 3$

In a single atom, determine how many electrons can have the following set of quantum numbers?

(i) $n = 3; l = 2; m_s = +1/2$

**Question 4** **8 points**

Rank the following from 1 to 4 in **increasing** order. Provide the ranking in the boxes provided.

Note: Atomic number (Z) are provided for relevant atoms

a) Increasing (lowest to highest) order of *ionization energy*: (1= lowest; 4 = highest)

Mg

Mg: Z = 12

Rb

Rb: Z = 37

Cs

Cs: Z = 55

K

K : Z = 19

b) Increasing (lowest to highest) order of *bond angle*: (1= lowest; 4 = highest)

H₂OCCl₄XeF₄

HCN

c) Increasing (smallest to largest) order of *ionic size* (1= smallest; 4 = largest)

S²⁻

S: Z = 16

O²⁻

O: Z = 8

Be²⁺

Be: Z = 4

Na⁺

Na: Z = 11

d) Increasing (lowest to highest) order of *bond energy* (1= lowest; 4 = highest)

P-F

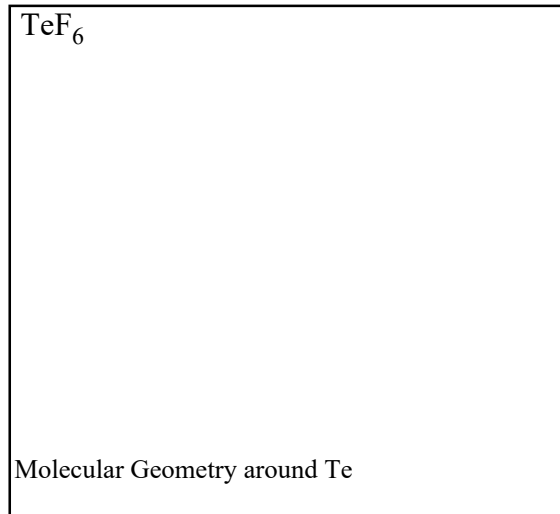
P-Cl

P-I

P-Br

**Question 5** **12 points**

a. 9 points For the molecules HClO_4 and TeF_6 , draw the most stable *Lewis structure* showing appropriate shape and *dashed/wedged bonds*, based on VSEPR. Determine the **molecular geometry** around the central atom. For HClO_4 , label the *most polar bond*.



b. 3 points (Note: Lone pairs are not shown in structures below)

(i) For the three *resonance* structures (1, 2 and 3) shown below, answer the following question:



(I) Circle the *most contributing* resonance structure.

Structure 1

Structure 2

Structure 3

(ii) For the three structures given (4, 5, and 6) below, answer the following questions:



(I) Circle all the structures that represent *resonance structures* of each other:

Structure 4

Structure 5

Structure 6

(II) Among the resonance structures, circle the *most contributing* resonance structure.

Structure 4

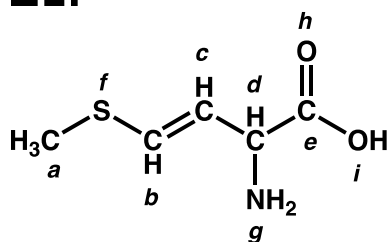
Structure 5

Structure 6

**Question 6** **6 points**

Using the data given below, calculate the **lattice energy** for SrO(s).

	ΔH° (kJ/mol)
$\text{Sr}(s) \longrightarrow \text{Sr}(g)$	+180
$\text{Sr}(g) \longrightarrow \text{Sr}^{2+}(g) + 2e^-$	+1468
$\text{O}_2(g) \longrightarrow 2\text{O}(g)$	+500
$\text{O}(g) + 2e^- \longrightarrow \text{O}^{2-}(g)$	+600
$\text{Sr}(s) + \frac{1}{2}\text{O}_2(g) \longrightarrow \text{SrO}(s)$	-550

**Question 7 10 points**

Consider the molecule given on the left, and answer questions (i) to (vi) using chemical bonding theories that we have learnt in class (valence bond theory, hybridization).

Note that the lone pairs are not shown in the structure:

i. Consider the carbon atoms marked in as ***a***, ***b***, ***c***, ***d***, and ***e*** in the molecule, and determine **hybridization** at each of the carbon centers. (2.5 points)

Atom	Hybridization
Carbon (<i>a</i>)	
Carbon (<i>b</i>)	
Carbon (<i>c</i>)	
Carbon (<i>d</i>)	
Carbon (<i>e</i>)	

ii. Now, consider the sulfur atom (S) marked as ***f***, nitrogen (N) atom (***g***), and oxygen(O) atoms (***h*** and ***i***), and determine their **hybridization**. (2 points)

Atom	Hybridization
Sulfur (<i>f</i>)	
Nitrogen (<i>g</i>)	
Oxygen (<i>h</i>)	
Oxygen (<i>i</i>)	

iii. **Describe** all the bonds formed by the carbon atoms ***a*** and ***b***, using sigma and pi designation as appropriate. (2 points)

Atom	Bonds formed (use sigma and pi as appropriate)
Carbon (<i>a</i>)	
Carbon (<i>b</i>)	

iv. How many total sp^3 -s sigma bonds are in the molecule? (1 point)

v. Determine **electronic** and **molecular geometry** at carbon (C) labeled ***e*** and sulfur (S) labeled ***f*** centres. (2 points)

Atom	Electronic Geometry	Molecular Geometry
Carbon (<i>e</i>)		
Sulfur (<i>f</i>)		

vi. Circle the value from the choices given below, that describes the **C-S-C bond angle at sulfur** (S) marked ***f***: (0.5 points)

>90° to <109.5° 109.5° >109.5 to <120° 120° >120° to <180°


Question 8 12 points

Using the **valence shell orbitals only** draw a **complete** molecular orbital (MO) diagram (with no s-p mixing) for the molecule **PS**, in the space provided below. **(6 points)**

Assume that 3s and 3p atomic orbitals will form molecular orbitals like those formed by the combination of 2s and 2p atomic orbitals that we learnt in class, and there is no s-p mixing.

Show **energy arrow**, **atomic orbitals** involved, **atom labels**, the **molecular orbitals** that are formed, **filled electrons** in the MO diagram etc. Make sure to label all the atomic and molecular orbitals.

Using the MO diagram above, answer the following questions:

- (i) Which one of the following will have the highest bond order? Circle the correct choice
 A) PS^{1-} B) PS C) PS^{1+}
- (ii) Which one of the following will have the longest bond length? Circle the correct choice
 A) PS^{1-} B) PS C) PS^{1+}
- (iii) Which of the following will be diamagnetic? Circle the correct choice.
 A) PS^{1-} B) PS C) PS^{1+}
- (iv) Draw Lewis dot structure of the molecule with the shortest bond length (based on the MO diagram) from among the three (PS^{1-} , PS, PS^{1+}), and indicate the bond order calculated from the Lewis dot structure. **(3 points)**

Lewis Dot Structure

Bond Order



D4CA1F8B-430F-4C89-9CFC-818D68BB7684

midterm-exam-93388

#854 Page 10 of 10

SCRAP PAGE

**CHEMISTRY 110 –Midterm Exam****Thursday November 7, 2024, 6:30 – 8:30 pm**

Examiners: Pallavi Sirjoosingh and Ashok Kakkar

First name (please write as legibly as possible within the boxes)

--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--

Last name

--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--

Student ID number

--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--

This booklet has 8 questions on 10 pages (The last page can be used as a scrap page).

The staple must NOT be removed from the question booklet. Your intact question booklet must be returned at the end of the exam.

A separate booklet contains a periodic table, and datasheet.

Write your answers directly under the questions in the space provided. Do not write on the QR codes or near the top edges (as it may be cut off).

Please write **legibly and dark enough** for the scan to be readable. For students who use English as a second language, dictionaries are permitted. Calculators are allowed.

DO NOT ASK QUESTIONS OF THE INVIGILATORS about interpretations of the exam questions. If you have a concern about a question, please summarize it at the top of this page. All concerns will be evaluated after the exam. GOOD LUCK!

**Question 1 8 points** (*Assume all transitions are possible*)

a. 6 points Calculate the **energy (in kJ/mol)** required to ionize (completely remove) the electron from the **$n=3$** level of **He^+ ion** (He: Helium). Show your work including equations. (3 sig. figs.)

b. 2 points

(I) Select the largest (longest) wavelength from the three choices below:

- i. Wavelength of photon to *completely remove* an electron from *1s orbital of Hydrogen atom*
- ii. Wavelength of photon to *completely remove* an electron from *1s orbital of Boron atom (B)*
- iii. Wavelength of photon to *completely remove* an electron from *1s orbital of Boron cation (B^+)*

i or ii or iii

(II) Select the smallest (shortest) wavelength from the three choices below (*for H atom*):

- i. Wavelength of photon *emitted* for an electron transitioning from *2p to 1s orbital*
- ii. Wavelength of photon *absorbed* for an electron transitioning from *2s to 4p orbital*
- iii. Wavelength of photon *absorbed* for an electron transitioning from *1s to 3p orbital*

i or ii or iii



Question 2 8 points



A Chem110 student plots the graph shown on the left, from data obtained for the kinetic energy (eV) of the ejected electron versus frequency (s^{-1}) of the incident light, for an unknown metal. ($1\text{eV} = 1.6022 \times 10^{-19} \text{ J}$)

Based on the plot, answer the questions **a to c** below:

a. 1 point On the plot above, label the **threshold frequency**.

b. 2 points What is the value of the work potential (in J) of the unknown metal?

c. 3 points What is the *longest* wavelength (in nm) required to eject an electron from this metal's surface? Show your work including equation(s) used. (3 sig figs)

d. 2 points In the space provided below, plot the following

(Assume frequency is *above* threshold frequency):

- Kinetic energy* of an ejected electron versus the *intensity* of the incident light (left)
- Number of electrons* ejected versus the *frequency* of the incident light (right)



**Question 3** **6 points****a. 2 points**

In the ground state of Ge (Germanium; $Z = 32$) atom, **name** the following orbitals:

i. Orbital containing the electron that requires the maximum energy to be removed.

ii. Orbital containing the electron that experiences the minimum effective nuclear charge.

b. 2 points

An unknown anion (A^-) has the following *excited* electronic configuration: $[Ar] 4s^2 3d^3 4p^1$

(Ar: Argon; $Z = 18$)

Give the **condensed ground state electronic configuration** of the atom, **A** and the cation, **A^{2+}** .
Show your work. (Assume all electronic transitions are possible)

Electronic Configuration of A:

Electronic Configuration of A^{2+} :

c. 2 points

In a single atom, determine how many electrons can have the following set of quantum numbers?

(i) $n = 5; l = 3$

In a single atom, determine how many electrons can have the following set of quantum numbers?

(i) $n = 3; l = 2; m_s = +1/2$

**Question 4** **8 points**

Rank the following from 1 to 4 in **increasing** order. Provide the ranking in the boxes provided.

Note: Atomic number (Z) are provided for relevant atoms

a) Increasing (lowest to highest) order of *ionization energy*: (1= lowest; 4 = highest)

Mg

Mg: Z = 12

Rb

Rb: Z = 37

Cs

Cs: Z = 55

K

K : Z = 19

b) Increasing (lowest to highest) order of *bond angle*: (1= lowest; 4 = highest)

H₂OCCl₄XeF₄

HCN

c) Increasing (smallest to largest) order of *ionic size* (1= smallest; 4 = largest)

S²⁻

S: Z = 16

O²⁻

O: Z = 8

Be²⁺

Be: Z = 4

Na⁺

Na: Z = 11

d) Increasing (lowest to highest) order of *bond energy* (1= lowest; 4 = highest)

P-F

P-Cl

P-I

P-Br

**Question 5** **12 points**

a. 9 points For the molecules HClO_4 and TeF_6 , draw the most stable *Lewis structure* showing appropriate shape and *dashed/wedged bonds*, based on VSEPR. Determine the **molecular geometry** around the central atom. For HClO_4 , label the *most polar bond*.



b. 3 points (Note: Lone pairs are not shown in structures below)

(i) For the three *resonance* structures (1, 2 and 3) shown below, answer the following question:



(I) Circle the *most contributing* resonance structure.

Structure 1

Structure 2

Structure 3

(ii) For the three structures given (4, 5, and 6) below, answer the following questions:



(I) Circle all the structures that represent *resonance structures* of each other:

Structure 4

Structure 5

Structure 6

(II) Among the resonance structures, circle the *most contributing* resonance structure.

Structure 4

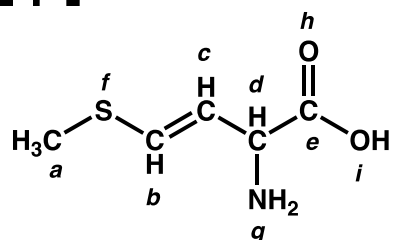
Structure 5

Structure 6

**Question 6** **6 points**

Using the data given below, calculate the **lattice energy** for SrO(s).

	ΔH° (kJ/mol)
$\text{Sr}(s) \longrightarrow \text{Sr}(g)$	+180
$\text{Sr}(g) \longrightarrow \text{Sr}^{2+}(g) + 2e^-$	+1468
$\text{O}_2(g) \longrightarrow 2\text{O}(g)$	+500
$\text{O}(g) + 2e^- \longrightarrow \text{O}^{2-}(g)$	+600
$\text{Sr}(s) + \frac{1}{2}\text{O}_2(g) \longrightarrow \text{SrO}(s)$	-550

**Question 7 10 points**

Consider the molecule given on the left, and answer questions (i) to (vi) using chemical bonding theories that we have learnt in class (valence bond theory, hybridization).

Note that the lone pairs are not shown in the structure:

i. Consider the carbon atoms marked in as ***a***, ***b***, ***c***, ***d***, and ***e*** in the molecule, and determine **hybridization** at each of the carbon centers. (2.5 points)

Atom	Hybridization
Carbon (<i>a</i>)	
Carbon (<i>b</i>)	
Carbon (<i>c</i>)	
Carbon (<i>d</i>)	
Carbon (<i>e</i>)	

ii. Now, consider the sulfur atom (S) marked as ***f***, nitrogen (N) atom (***g***), and oxygen(O) atoms (***h*** and ***i***), and determine their **hybridization**. (2 points)

Atom	Hybridization
Sulfur (<i>f</i>)	
Nitrogen (<i>g</i>)	
Oxygen (<i>h</i>)	
Oxygen (<i>i</i>)	

iii. **Describe** all the bonds formed by the carbon atoms ***a*** and ***b***, using sigma and pi designation as appropriate. (2 points)

Atom	Bonds formed (use sigma and pi as appropriate)
Carbon (<i>a</i>)	
Carbon (<i>b</i>)	

iv. How many total sp^3 -s sigma bonds are in the molecule? (1 point)

v. Determine **electronic** and **molecular geometry** at carbon (C) labeled ***e*** and sulfur (S) labeled ***f*** centres. (2 points)

Atom	Electronic Geometry	Molecular Geometry
Carbon (<i>e</i>)		
Sulfur (<i>f</i>)		

vi. Circle the value from the choices given below, that describes the **C-S-C bond angle at sulfur** (S) marked ***f***: (0.5 points)

$>90^\circ$ to $<109.5^\circ$

109.5°

>109.5 to $<120^\circ$

120°

$>120^\circ$ to $<180^\circ$

**Question 8 12 points**

Using the **valence shell orbitals only** draw a **complete** molecular orbital (MO) diagram (with no s-p mixing) for the molecule **PS**, in the space provided below. **(6 points)**

Assume that 3s and 3p atomic orbitals will form molecular orbitals like those formed by the combination of 2s and 2p atomic orbitals that we learnt in class, and there is no s-p mixing.

Show **energy arrow**, **atomic orbitals** involved, **atom labels**, the **molecular orbitals** that are formed, **filled electrons** in the MO diagram etc. Make sure to label all the atomic and molecular orbitals.

Using the MO diagram above, answer the following questions:

- (i) Which one of the following will have the highest bond order? Circle the correct choice
A) PS^{1-} B) PS C) PS^{1+}
- (ii) Which one of the following will have the longest bond length? Circle the correct choice
A) PS^{1-} B) PS C) PS^{1+}
- (iii) Which of the following will be diamagnetic? Circle the correct choice.
A) PS^{1-} B) PS C) PS^{1+}
- (iv) Draw Lewis dot structure of the molecule with the shortest bond length (based on the MO diagram) from among the three (PS^{1-} , PS, PS^{1+}), and indicate the bond order calculated from the Lewis dot structure. **(3 points)**

Lewis Dot Structure

Bond Order



FF306D17-E3C9-4448-A773-BFD44F6FB0FB

midterm-exam-93388

#855 Page 10 of 10

SCRAP PAGE

**CHEMISTRY 110 –Midterm Exam****Thursday November 7, 2024, 6:30 – 8:30 pm****Examiners: Pallavi Sirjoosingh and Ashok Kakkar**

First name (please write as legibly as possible within the boxes)

--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--

Last name

--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--

Student ID number

--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--

This booklet has 8 questions on 10 pages (The last page can be used as a scrap page).

The staple must NOT be removed from the question booklet. Your intact question booklet must be returned at the end of the exam.

A separate booklet contains a periodic table, and datasheet.

Write your answers directly under the questions in the space provided. Do not write on the QR codes or near the top edges (as it may be cut off).

Please write **legibly and dark enough** for the scan to be readable. For students who use English as a second language, dictionaries are permitted. Calculators are allowed.

DO NOT ASK QUESTIONS OF THE INVIGILATORS about interpretations of the exam questions. If you have a concern about a question, please summarize it at the top of this page. All concerns will be evaluated after the exam. GOOD LUCK!

**Question 1 8 points** (*Assume all transitions are possible*)

a. 6 points Calculate the **energy (in kJ/mol)** required to ionize (completely remove) the electron from the **$n=3$** level of **He^+ ion** (He: Helium). Show your work including equations. (3 sig. figs.)

b. 2 points

(I) Select the largest (longest) wavelength from the three choices below:

- i. Wavelength of photon to *completely remove* an electron from *1s orbital of Hydrogen atom*
- ii. Wavelength of photon to *completely remove* an electron from *1s orbital of Boron atom (B)*
- iii. Wavelength of photon to *completely remove* an electron from *1s orbital of Boron cation (B^+)*

i or ii or iii

(II) Select the smallest (shortest) wavelength from the three choices below (*for H atom*):

- i. Wavelength of photon *emitted* for an electron transitioning from *2p to 1s orbital*
- ii. Wavelength of photon *absorbed* for an electron transitioning from *2s to 4p orbital*
- iii. Wavelength of photon *absorbed* for an electron transitioning from *1s to 3p orbital*

i or ii or iii



Question 2 8 points



A Chem110 student plots the graph shown on the left, from data obtained for the kinetic energy (eV) of the ejected electron versus frequency (s^{-1}) of the incident light, for an unknown metal. ($1\text{eV} = 1.6022 \times 10^{-19} \text{ J}$)

Based on the plot, answer the questions **a to c** below:

a. 1 point On the plot above, label the **threshold frequency**.

b. 2 points What is the value of the work potential (in J) of the unknown metal?

c. 3 points What is the *longest* wavelength (in nm) required to eject an electron from this metal's surface? Show your work including equation(s) used. (3 sig figs)

d. 2 points In the space provided below, plot the following

(Assume frequency is *above* threshold frequency):

- Kinetic energy* of an ejected electron versus the *intensity* of the incident light (left)
- Number of electrons* ejected versus the *frequency* of the incident light (right)



**Question 3** **6 points****a. 2 points**

In the ground state of Ge (Germanium; $Z = 32$) atom, **name** the following orbitals:

i. Orbital containing the electron that requires the maximum energy to be removed.

ii. Orbital containing the electron that experiences the minimum effective nuclear charge.

b. 2 points

An unknown anion (A^-) has the following *excited* electronic configuration: $[Ar] 4s^2 3d^3 4p^1$

(Ar: Argon; $Z = 18$)

Give the **condensed ground state electronic configuration** of the atom, **A** and the cation, **A^{2+}** .
Show your work. (Assume all electronic transitions are possible)

Electronic Configuration of A:

Electronic Configuration of A^{2+} :

c. 2 points

In a single atom, determine how many electrons can have the following set of quantum numbers?

(i) $n = 5; l = 3$

In a single atom, determine how many electrons can have the following set of quantum numbers?

(i) $n = 3; l = 2; m_s = +1/2$

**Question 4** **8 points**

Rank the following from 1 to 4 in **increasing** order. Provide the ranking in the boxes provided.

Note: Atomic number (Z) are provided for relevant atoms

a) Increasing (lowest to highest) order of *ionization energy*: (1= lowest; 4 = highest)

Mg

Mg: Z = 12

Rb

Rb: Z = 37

Cs

Cs: Z = 55

K

K : Z = 19

b) Increasing (lowest to highest) order of *bond angle*: (1= lowest; 4 = highest)

H₂OCCl₄XeF₄

HCN

c) Increasing (smallest to largest) order of *ionic size* (1= smallest; 4 = largest)

S²⁻

S: Z = 16

O²⁻

O: Z = 8

Be²⁺

Be: Z = 4

Na⁺

Na: Z = 11

d) Increasing (lowest to highest) order of *bond energy* (1= lowest; 4 = highest)

P-F

P-Cl

P-I

P-Br

**Question 5** **12 points**

a. 9 points For the molecules HClO_4 and TeF_6 , draw the most stable *Lewis structure* showing appropriate shape and *dashed/wedged bonds*, based on VSEPR. Determine the **molecular geometry** around the central atom. For HClO_4 , label the *most polar bond*.



b. 3 points (Note: Lone pairs are not shown in structures below)

(i) For the three *resonance* structures (1, 2 and 3) shown below, answer the following question:



(I) Circle the *most contributing* resonance structure.

Structure 1

Structure 2

Structure 3

(ii) For the three structures given (4, 5, and 6) below, answer the following questions:



(I) Circle all the structures that represent *resonance structures* of each other:

Structure 4

Structure 5

Structure 6

(II) Among the resonance structures, circle the *most contributing* resonance structure.

Structure 4

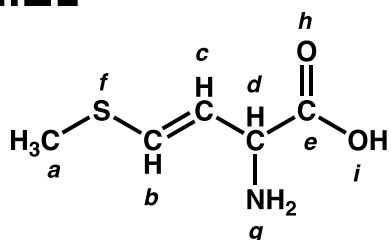
Structure 5

Structure 6

**Question 6** **6 points**

Using the data given below, calculate the **lattice energy** for SrO(s).

	ΔH° (kJ/mol)
$\text{Sr}(s) \longrightarrow \text{Sr}(g)$	+180
$\text{Sr}(g) \longrightarrow \text{Sr}^{2+}(g) + 2e^-$	+1468
$\text{O}_2(g) \longrightarrow 2\text{O}(g)$	+500
$\text{O}(g) + 2e^- \longrightarrow \text{O}^{2-}(g)$	+600
$\text{Sr}(s) + \frac{1}{2}\text{O}_2(g) \longrightarrow \text{SrO}(s)$	-550

**Question 7 10 points**

Consider the molecule given on the left, and answer questions (i) to (vi) using chemical bonding theories that we have learnt in class (valence bond theory, hybridization).

Note that the lone pairs are not shown in the structure:

i. Consider the carbon atoms marked in as ***a***, ***b***, ***c***, ***d***, and ***e*** in the molecule, and determine **hybridization** at each of the carbon centers. (2.5 points)

Atom	Hybridization
Carbon (<i>a</i>)	
Carbon (<i>b</i>)	
Carbon (<i>c</i>)	
Carbon (<i>d</i>)	
Carbon (<i>e</i>)	

ii. Now, consider the sulfur atom (S) marked as ***f***, nitrogen (N) atom (***g***), and oxygen(O) atoms (***h*** and ***i***), and determine their **hybridization**. (2 points)

Atom	Hybridization
Sulfur (<i>f</i>)	
Nitrogen (<i>g</i>)	
Oxygen (<i>h</i>)	
Oxygen (<i>i</i>)	

iii. **Describe** all the bonds formed by the carbon atoms ***a*** and ***b***, using sigma and pi designation as appropriate. (2 points)

Atom	Bonds formed (use sigma and pi as appropriate)
Carbon (<i>a</i>)	
Carbon (<i>b</i>)	

iv. How many total sp^3 -s sigma bonds are in the molecule? (1 point)

v. Determine **electronic** and **molecular geometry** at carbon (C) labeled ***e*** and sulfur (S) labeled ***f*** centres. (2 points)

Atom	Electronic Geometry	Molecular Geometry
Carbon (<i>e</i>)		
Sulfur (<i>f</i>)		

vi. Circle the value from the choices given below, that describes the **C-S-C bond angle at sulfur** (S) marked ***f***: (0.5 points)

>90° to <109.5°

109.5°

>109.5 to <120°

120°

>120° to <180°

**Question 8 12 points**

Using the **valence shell orbitals only** draw a **complete** molecular orbital (MO) diagram (with no s-p mixing) for the molecule **PS**, in the space provided below. **(6 points)**

Assume that 3s and 3p atomic orbitals will form molecular orbitals like those formed by the combination of 2s and 2p atomic orbitals that we learnt in class, and there is no s-p mixing.

Show **energy arrow**, **atomic orbitals** involved, **atom labels**, the **molecular orbitals** that are formed, **filled electrons** in the MO diagram etc. Make sure to label all the atomic and molecular orbitals.

Using the MO diagram above, answer the following questions:

- (i) Which one of the following will have the highest bond order? Circle the correct choice
A) PS^{1-} B) PS C) PS^{1+}
- (ii) Which one of the following will have the longest bond length? Circle the correct choice
A) PS^{1-} B) PS C) PS^{1+}
- (iii) Which of the following will be diamagnetic? Circle the correct choice.
A) PS^{1-} B) PS C) PS^{1+}
- (iv) Draw Lewis dot structure of the molecule with the shortest bond length (based on the MO diagram) from among the three (PS^{1-} , PS, PS^{1+}), and indicate the bond order calculated from the Lewis dot structure. **(3 points)**

Lewis Dot Structure

Bond Order



794A6FDC-D8A1-43EA-9F1A-BE6EC9BA460C

midterm-exam-93388

#856 Page 10 of 10

SCRAP PAGE

**CHEMISTRY 110 –Midterm Exam****Thursday November 7, 2024, 6:30 – 8:30 pm**

Examiners: Pallavi Sirjoosingh and Ashok Kakkar

First name (please write as legibly as possible within the boxes)

--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--

Last name

--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--

Student ID number

--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--

This booklet has 8 questions on 10 pages (The last page can be used as a scrap page).

The staple must NOT be removed from the question booklet. Your intact question booklet must be returned at the end of the exam.

A separate booklet contains a periodic table, and datasheet.

Write your answers directly under the questions in the space provided. Do not write on the QR codes or near the top edges (as it may be cut off).

Please write **legibly and dark enough** for the scan to be readable. For students who use English as a second language, dictionaries are permitted. Calculators are allowed.

DO NOT ASK QUESTIONS OF THE INVIGILATORS about interpretations of the exam questions. If you have a concern about a question, please summarize it at the top of this page. All concerns will be evaluated after the exam. GOOD LUCK!

**Question 1 8 points** (*Assume all transitions are possible*)

a. 6 points Calculate the **energy (in kJ/mol)** required to ionize (completely remove) the electron from the **$n=3$** level of **He^+ ion** (He: Helium). Show your work including equations. (3 sig. figs.)

b. 2 points

(I) Select the largest (longest) wavelength from the three choices below:

- i. Wavelength of photon to *completely remove* an electron from *1s orbital of Hydrogen atom*
- ii. Wavelength of photon to *completely remove* an electron from *1s orbital of Boron atom (B)*
- iii. Wavelength of photon to *completely remove* an electron from *1s orbital of Boron cation (B^+)*

i or ii or iii

(II) Select the smallest (shortest) wavelength from the three choices below (*for H atom*):

- i. Wavelength of photon *emitted* for an electron transitioning from *2p to 1s orbital*
- ii. Wavelength of photon *absorbed* for an electron transitioning from *2s to 4p orbital*
- iii. Wavelength of photon *absorbed* for an electron transitioning from *1s to 3p orbital*

i or ii or iii



Question 2 8 points



A Chem110 student plots the graph shown on the left, from data obtained for the kinetic energy (eV) of the ejected electron versus frequency (s^{-1}) of the incident light, for an unknown metal. ($1\text{eV} = 1.6022 \times 10^{-19} \text{ J}$)

Based on the plot, answer the questions **a to c** below:

a. 1 point On the plot above, label the **threshold frequency**.

b. 2 points What is the value of the work potential (in J) of the unknown metal?

c. 3 points What is the *longest* wavelength (in nm) required to eject an electron from this metal's surface? Show your work including equation(s) used. (3 sig figs)

d. 2 points In the space provided below, plot the following

(Assume frequency is *above* threshold frequency):

- i. *Kinetic energy* of an ejected electron versus the *intensity* of the incident light (left)
- ii. *Number of electrons* ejected versus the *frequency* of the incident light (right)



**Question 3** **6 points****a. 2 points**

In the ground state of Ge (Germanium; $Z = 32$) atom, **name** the following orbitals:

i. Orbital containing the electron that requires the maximum energy to be removed.

ii. Orbital containing the electron that experiences the minimum effective nuclear charge.

b. 2 points

An unknown anion (A^-) has the following *excited* electronic configuration: $[Ar] 4s^2 3d^3 4p^1$

(Ar: Argon; $Z = 18$)

Give the **condensed ground state electronic configuration** of the atom, **A** and the cation, **A^{2+}** .
Show your work. (Assume all electronic transitions are possible)

Electronic Configuration of A:

Electronic Configuration of A^{2+} :

c. 2 points

In a single atom, determine how many electrons can have the following set of quantum numbers?

(i) $n = 5; l = 3$

In a single atom, determine how many electrons can have the following set of quantum numbers?

(i) $n = 3; l = 2; m_s = +1/2$

**Question 4** **8 points**

Rank the following from 1 to 4 in **increasing** order. Provide the ranking in the boxes provided.

Note: Atomic number (Z) are provided for relevant atoms

a) Increasing (lowest to highest) order of *ionization energy*: (1= lowest; 4 = highest)

Mg

Mg: Z = 12

Rb

Rb: Z = 37

Cs

Cs: Z = 55

K

K : Z = 19

b) Increasing (lowest to highest) order of *bond angle*: (1= lowest; 4 = highest)

H₂OCCl₄XeF₄

HCN

c) Increasing (smallest to largest) order of *ionic size* (1= smallest; 4 = largest)

S²⁻

S: Z = 16

O²⁻

O: Z = 8

Be²⁺

Be: Z = 4

Na⁺

Na: Z = 11

d) Increasing (lowest to highest) order of *bond energy* (1= lowest; 4 = highest)

P-F

P-Cl

P-I

P-Br

**Question 5** **12 points**

a. 9 points For the molecules HClO_4 and TeF_6 , draw the most stable *Lewis structure* showing appropriate shape and *dashed/wedged bonds*, based on VSEPR. Determine the **molecular geometry** around the central atom. For HClO_4 , label the *most polar bond*.



b. 3 points (Note: Lone pairs are not shown in structures below)

(i) For the three *resonance* structures (1, 2 and 3) shown below, answer the following question:



(I) Circle the *most contributing* resonance structure.

Structure 1

Structure 2

Structure 3

(ii) For the three structures given (4, 5, and 6) below, answer the following questions:



(I) Circle all the structures that represent *resonance structures* of each other:

Structure 4

Structure 5

Structure 6

(II) Among the resonance structures, circle the *most contributing* resonance structure.

Structure 4

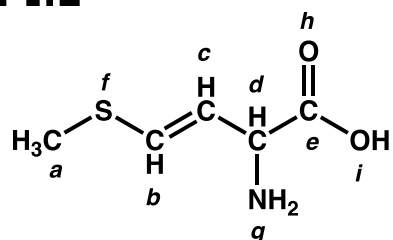
Structure 5

Structure 6

**Question 6** **6 points**

Using the data given below, calculate the **lattice energy** for SrO(s).

	ΔH° (kJ/mol)
$\text{Sr}(s) \longrightarrow \text{Sr}(g)$	+180
$\text{Sr}(g) \longrightarrow \text{Sr}^{2+}(g) + 2e^-$	+1468
$\text{O}_2(g) \longrightarrow 2\text{O}(g)$	+500
$\text{O}(g) + 2e^- \longrightarrow \text{O}^{2-}(g)$	+600
$\text{Sr}(s) + \frac{1}{2}\text{O}_2(g) \longrightarrow \text{SrO}(s)$	-550

**Question 7 10 points**

Consider the molecule given on the left, and answer questions (i) to (vi) using chemical bonding theories that we have learnt in class (valence bond theory, hybridization).

Note that the lone pairs are not shown in the structure:

i. Consider the carbon atoms marked in as ***a***, ***b***, ***c***, ***d***, and ***e*** in the molecule, and determine **hybridization** at each of the carbon centers. (2.5 points)

Atom	Hybridization
Carbon (<i>a</i>)	
Carbon (<i>b</i>)	
Carbon (<i>c</i>)	
Carbon (<i>d</i>)	
Carbon (<i>e</i>)	

ii. Now, consider the sulfur atom (S) marked as ***f***, nitrogen (N) atom (***g***), and oxygen(O) atoms (***h*** and ***i***), and determine their **hybridization**. (2 points)

Atom	Hybridization
Sulfur (<i>f</i>)	
Nitrogen (<i>g</i>)	
Oxygen (<i>h</i>)	
Oxygen (<i>i</i>)	

iii. **Describe** all the bonds formed by the carbon atoms ***a*** and ***b***, using sigma and pi designation as appropriate. (2 points)

Atom	Bonds formed (use sigma and pi as appropriate)
Carbon (<i>a</i>)	
Carbon (<i>b</i>)	

iv. How many total sp^3 -s sigma bonds are in the molecule? (1 point)

v. Determine **electronic** and **molecular geometry** at carbon (C) labeled ***e*** and sulfur (S) labeled ***f*** centres. (2 points)

Atom	Electronic Geometry	Molecular Geometry
Carbon (<i>e</i>)		
Sulfur (<i>f</i>)		

vi. Circle the value from the choices given below, that describes the **C-S-C bond angle at sulfur** (S) marked ***f***: (0.5 points)

>90° to <109.5° 109.5° >109.5 to <120° 120° >120° to <180°


Question 8 12 points

Using the **valence shell orbitals only** draw a **complete** molecular orbital (MO) diagram (with no s-p mixing) for the molecule **PS**, in the space provided below. **(6 points)**

Assume that 3s and 3p atomic orbitals will form molecular orbitals like those formed by the combination of 2s and 2p atomic orbitals that we learnt in class, and there is no s-p mixing.

Show **energy arrow**, **atomic orbitals** involved, **atom labels**, the **molecular orbitals** that are formed, **filled electrons** in the MO diagram etc. Make sure to label all the atomic and molecular orbitals.

Using the MO diagram above, answer the following questions:

- (i) Which one of the following will have the highest bond order? Circle the correct choice
 A) PS^{1-} B) PS C) PS^{1+}
- (ii) Which one of the following will have the longest bond length? Circle the correct choice
 A) PS^{1-} B) PS C) PS^{1+}
- (iii) Which of the following will be diamagnetic? Circle the correct choice.
 A) PS^{1-} B) PS C) PS^{1+}
- (iv) Draw Lewis dot structure of the molecule with the shortest bond length (based on the MO diagram) from among the three (PS^{1-} , PS, PS^{1+}), and indicate the bond order calculated from the Lewis dot structure. **(3 points)**

Lewis Dot Structure

Bond Order



38825330-F754-4996-AF0E-9E3807ED24DF

midterm-exam-93388

#857 Page 10 of 10

SCRAP PAGE

**CHEMISTRY 110 –Midterm Exam****Thursday November 7, 2024, 6:30 – 8:30 pm**

Examiners: Pallavi Sirjoosingh and Ashok Kakkar

First name (please write as legibly as possible within the boxes)

--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--

Last name

--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--

Student ID number

--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--

This booklet has 8 questions on 10 pages (The last page can be used as a scrap page).

The staple must NOT be removed from the question booklet. Your intact question booklet must be returned at the end of the exam.

A separate booklet contains a periodic table, and datasheet.

Write your answers directly under the questions in the space provided. Do not write on the QR codes or near the top edges (as it may be cut off).

Please write **legibly and dark enough** for the scan to be readable. For students who use English as a second language, dictionaries are permitted. Calculators are allowed.

DO NOT ASK QUESTIONS OF THE INVIGILATORS about interpretations of the exam questions. If you have a concern about a question, please summarize it at the top of this page. All concerns will be evaluated after the exam. GOOD LUCK!

**Question 1 8 points** (*Assume all transitions are possible*)

a. 6 points Calculate the **energy (in kJ/mol)** required to ionize (completely remove) the electron from the **n=3** level of **He⁺ ion** (He: Helium). Show your work including equations. (3 sig. figs.)

b. 2 points

(I) Select the largest (longest) wavelength from the three choices below:

- i. Wavelength of photon to *completely remove* an electron from *1s orbital of Hydrogen atom*
- ii. Wavelength of photon to *completely remove* an electron from *1s orbital of Boron atom (B)*
- iii. Wavelength of photon to *completely remove* an electron from *1s orbital of Boron cation (B⁺)*

i or ii or iii

(II) Select the smallest (shortest) wavelength from the three choices below (*for H atom*):

- i. Wavelength of photon *emitted* for an electron transitioning from *2p to 1s orbital*
- ii. Wavelength of photon *absorbed* for an electron transitioning from *2s to 4p orbital*
- iii. Wavelength of photon *absorbed* for an electron transitioning from *1s to 3p orbital*

i or ii or iii



Question 2 8 points



A Chem110 student plots the graph shown on the left, from data obtained for the kinetic energy (eV) of the ejected electron versus frequency (s^{-1}) of the incident light, for an unknown metal. ($1\text{eV} = 1.6022 \times 10^{-19} \text{ J}$)

Based on the plot, answer the questions **a to c** below:

a. 1 point On the plot above, label the **threshold frequency**.

b. 2 points What is the value of the work potential (in J) of the unknown metal?

c. 3 points What is the *longest* wavelength (in nm) required to eject an electron from this metal's surface? Show your work including equation(s) used. (3 sig figs)

d. 2 points In the space provided below, plot the following

(Assume frequency is *above* threshold frequency):

- Kinetic energy* of an ejected electron versus the *intensity* of the incident light (left)
- Number of electrons* ejected versus the *frequency* of the incident light (right)



**Question 3** **6 points****a. 2 points**

In the ground state of Ge (Germanium; $Z = 32$) atom, **name** the following orbitals:

i. Orbital containing the electron that requires the maximum energy to be removed.

ii. Orbital containing the electron that experiences the minimum effective nuclear charge.

b. 2 points

An unknown anion (A^-) has the following *excited* electronic configuration: $[Ar] 4s^2 3d^3 4p^1$

(Ar: Argon; $Z = 18$)

Give the **condensed ground state electronic configuration** of the atom, **A** and the cation, **A^{2+}** .
Show your work. (Assume all electronic transitions are possible)

Electronic Configuration of A:

Electronic Configuration of A^{2+} :

c. 2 points

In a single atom, determine how many electrons can have the following set of quantum numbers?

(i) $n = 5; l = 3$

In a single atom, determine how many electrons can have the following set of quantum numbers?

(i) $n = 3; l = 2; m_s = +1/2$

**Question 4** **8 points**

Rank the following from 1 to 4 in **increasing** order. Provide the ranking in the boxes provided.

Note: Atomic number (Z) are provided for relevant atoms

a) Increasing (lowest to highest) order of *ionization energy*: (1= lowest; 4 = highest)

Mg

Mg: Z = 12

Rb

Rb: Z = 37

Cs

Cs: Z = 55

K

K : Z = 19

b) Increasing (lowest to highest) order of *bond angle*: (1= lowest; 4 = highest)

H₂OCCl₄XeF₄

HCN

c) Increasing (smallest to largest) order of *ionic size* (1= smallest; 4 = largest)

S²⁻

S: Z = 16

O²⁻

O: Z = 8

Be²⁺

Be: Z = 4

Na⁺

Na: Z = 11

d) Increasing (lowest to highest) order of *bond energy* (1= lowest; 4 = highest)

P-F

P-Cl

P-I

P-Br

**Question 5** **12 points**

a. 9 points For the molecules HClO_4 and TeF_6 , draw the most stable *Lewis structure* showing appropriate shape and *dashed/wedged bonds*, based on VSEPR. Determine the **molecular geometry** around the central atom. For HClO_4 , label the *most polar bond*.



b. 3 points (Note: Lone pairs are not shown in structures below)

(i) For the three *resonance* structures (1, 2 and 3) shown below, answer the following question:



(I) Circle the *most contributing* resonance structure.

Structure 1

Structure 2

Structure 3

(ii) For the three structures given (4, 5, and 6) below, answer the following questions:



(I) Circle all the structures that represent *resonance structures* of each other:

Structure 4

Structure 5

Structure 6

(II) Among the resonance structures, circle the *most contributing* resonance structure.

Structure 4

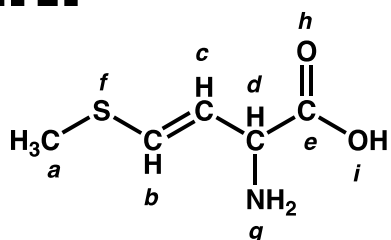
Structure 5

Structure 6

**Question 6** **6 points**

Using the data given below, calculate the **lattice energy** for SrO(s).

	ΔH° (kJ/mol)
$\text{Sr}(s) \longrightarrow \text{Sr}(g)$	+180
$\text{Sr}(g) \longrightarrow \text{Sr}^{2+}(g) + 2e^-$	+1468
$\text{O}_2(g) \longrightarrow 2\text{O}(g)$	+500
$\text{O}(g) + 2e^- \longrightarrow \text{O}^{2-}(g)$	+600
$\text{Sr}(s) + \frac{1}{2}\text{O}_2(g) \longrightarrow \text{SrO}(s)$	-550

**Question 7 10 points**

Consider the molecule given on the left, and answer questions (i) to (vi) using chemical bonding theories that we have learnt in class (valence bond theory, hybridization).

Note that the lone pairs are not shown in the structure:

i. Consider the carbon atoms marked in as ***a***, ***b***, ***c***, ***d***, and ***e*** in the molecule, and determine **hybridization** at each of the carbon centers. (2.5 points)

Atom	Hybridization
Carbon (<i>a</i>)	
Carbon (<i>b</i>)	
Carbon (<i>c</i>)	
Carbon (<i>d</i>)	
Carbon (<i>e</i>)	

ii. Now, consider the sulfur atom (S) marked as ***f***, nitrogen (N) atom (***g***), and oxygen(O) atoms (***h*** and ***i***), and determine their **hybridization**. (2 points)

Atom	Hybridization
Sulfur (<i>f</i>)	
Nitrogen (<i>g</i>)	
Oxygen (<i>h</i>)	
Oxygen (<i>i</i>)	

iii. **Describe** all the bonds formed by the carbon atoms ***a*** and ***b***, using sigma and pi designation as appropriate. (2 points)

Atom	Bonds formed (use sigma and pi as appropriate)
Carbon (<i>a</i>)	
Carbon (<i>b</i>)	

iv. How many total sp^3 -s sigma bonds are in the molecule? (1 point)

v. Determine **electronic** and **molecular geometry** at carbon (C) labeled ***e*** and sulfur (S) labeled ***f*** centres. (2 points)

Atom	Electronic Geometry	Molecular Geometry
Carbon (<i>e</i>)		
Sulfur (<i>f</i>)		

vi. Circle the value from the choices given below, that describes the **C-S-C bond angle at sulfur** (S) marked ***f***: (0.5 points)

>90° to <109.5°

109.5°

>109.5 to <120°

120°

>120° to <180°

**Question 8 12 points**

Using the **valence shell orbitals only** draw a **complete** molecular orbital (MO) diagram (with no s-p mixing) for the molecule **PS**, in the space provided below. **(6 points)**

Assume that 3s and 3p atomic orbitals will form molecular orbitals like those formed by the combination of 2s and 2p atomic orbitals that we learnt in class, and there is no s-p mixing.

Show **energy arrow**, **atomic orbitals** involved, **atom labels**, the **molecular orbitals** that are formed, **filled electrons** in the MO diagram etc. Make sure to label all the atomic and molecular orbitals.

Using the MO diagram above, answer the following questions:

- (i) Which one of the following will have the highest bond order? Circle the correct choice
A) PS^{1-} B) PS C) PS^{1+}
- (ii) Which one of the following will have the longest bond length? Circle the correct choice
A) PS^{1-} B) PS C) PS^{1+}
- (iii) Which of the following will be diamagnetic? Circle the correct choice.
A) PS^{1-} B) PS C) PS^{1+}
- (iv) Draw Lewis dot structure of the molecule with the shortest bond length (based on the MO diagram) from among the three (PS^{1-} , PS, PS^{1+}), and indicate the bond order calculated from the Lewis dot structure. **(3 points)**

Lewis Dot Structure

Bond Order



E8019B87-EE60-47A5-AF63-429ED0777215

midterm-exam-93388

#858 Page 10 of 10

SCRAP PAGE

**CHEMISTRY 110 –Midterm Exam****Thursday November 7, 2024, 6:30 – 8:30 pm**

Examiners: Pallavi Sirjoosingh and Ashok Kakkar

First name (please write as legibly as possible within the boxes)

--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--

Last name

--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--

Student ID number

--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--

This booklet has 8 questions on 10 pages (The last page can be used as a scrap page).

The staple must NOT be removed from the question booklet. Your intact question booklet must be returned at the end of the exam.

A separate booklet contains a periodic table, and datasheet.

Write your answers directly under the questions in the space provided. Do not write on the QR codes or near the top edges (as it may be cut off).

Please write **legibly and dark enough** for the scan to be readable. For students who use English as a second language, dictionaries are permitted. Calculators are allowed.

DO NOT ASK QUESTIONS OF THE INVIGILATORS about interpretations of the exam questions. If you have a concern about a question, please summarize it at the top of this page. All concerns will be evaluated after the exam. GOOD LUCK!

**Question 1 8 points** (*Assume all transitions are possible*)

a. 6 points Calculate the **energy (in kJ/mol)** required to ionize (completely remove) the electron from the **n=3** level of **He⁺ ion** (He: Helium). Show your work including equations. (3 sig. figs.)

b. 2 points

(I) Select the largest (longest) wavelength from the three choices below:

- i. Wavelength of photon to *completely remove* an electron from *1s orbital of Hydrogen atom*
- ii. Wavelength of photon to *completely remove* an electron from *1s orbital of Boron atom (B)*
- iii. Wavelength of photon to *completely remove* an electron from *1s orbital of Boron cation (B⁺)*

i or ii or iii

(II) Select the smallest (shortest) wavelength from the three choices below (*for H atom*):

- i. Wavelength of photon *emitted* for an electron transitioning from *2p to 1s orbital*
- ii. Wavelength of photon *absorbed* for an electron transitioning from *2s to 4p orbital*
- iii. Wavelength of photon *absorbed* for an electron transitioning from *1s to 3p orbital*

i or ii or iii



Question 2 8 points



A Chem110 student plots the graph shown on the left, from data obtained for the kinetic energy (eV) of the ejected electron versus frequency (s^{-1}) of the incident light, for an unknown metal. ($1\text{eV} = 1.6022 \times 10^{-19} \text{ J}$)

Based on the plot, answer the questions **a to c** below:

a. 1 point On the plot above, label the **threshold frequency**.

b. 2 points What is the value of the work potential (in J) of the unknown metal?

c. 3 points What is the *longest* wavelength (in nm) required to eject an electron from this metal's surface? Show your work including equation(s) used. (3 sig figs)

d. 2 points In the space provided below, plot the following

(Assume frequency is *above* threshold frequency):

- Kinetic energy* of an ejected electron versus the *intensity* of the incident light (left)
- Number of electrons* ejected versus the *frequency* of the incident light (right)



**Question 3** **6 points****a. 2 points**

In the ground state of Ge (Germanium; $Z = 32$) atom, **name** the following orbitals:

i. Orbital containing the electron that requires the maximum energy to be removed.

ii. Orbital containing the electron that experiences the minimum effective nuclear charge.

b. 2 points

An unknown anion (A^-) has the following *excited* electronic configuration: $[Ar] 4s^2 3d^3 4p^1$

(Ar: Argon; $Z = 18$)

Give the **condensed ground state electronic configuration** of the atom, **A** and the cation, **A^{2+}** .
Show your work. (Assume all electronic transitions are possible)

Electronic Configuration of A:

Electronic Configuration of A^{2+} :

c. 2 points

In a single atom, determine how many electrons can have the following set of quantum numbers?

(i) $n = 5; l = 3$

In a single atom, determine how many electrons can have the following set of quantum numbers?

(i) $n = 3; l = 2; m_s = +1/2$

**Question 4** **8 points**

Rank the following from 1 to 4 in **increasing** order. Provide the ranking in the boxes provided.

Note: Atomic number (Z) are provided for relevant atoms

a) Increasing (lowest to highest) order of *ionization energy*: (1= lowest; 4 = highest)

Mg

Mg: Z = 12

Rb

Rb: Z = 37

Cs

Cs: Z = 55

K

K : Z = 19

b) Increasing (lowest to highest) order of *bond angle*: (1= lowest; 4 = highest)

H₂OCCl₄XeF₄

HCN

c) Increasing (smallest to largest) order of *ionic size* (1= smallest; 4 = largest)

S²⁻

S: Z = 16

O²⁻

O: Z = 8

Be²⁺

Be: Z = 4

Na⁺

Na: Z = 11

d) Increasing (lowest to highest) order of *bond energy* (1= lowest; 4 = highest)

P-F

P-Cl

P-I

P-Br

**Question 5** **12 points**

a. 9 points For the molecules HClO_4 and TeF_6 , draw the most stable *Lewis structure* showing appropriate shape and *dashed/wedged bonds*, based on VSEPR. Determine the **molecular geometry** around the central atom. For HClO_4 , label the *most polar bond*.



b. 3 points (Note: Lone pairs are not shown in structures below)

(i) For the three *resonance* structures (1, 2 and 3) shown below, answer the following question:



(I) Circle the *most contributing* resonance structure.

Structure 1

Structure 2

Structure 3

(ii) For the three structures given (4, 5, and 6) below, answer the following questions:



(I) Circle all the structures that represent *resonance structures* of each other:

Structure 4

Structure 5

Structure 6

(II) Among the resonance structures, circle the *most contributing* resonance structure.

Structure 4

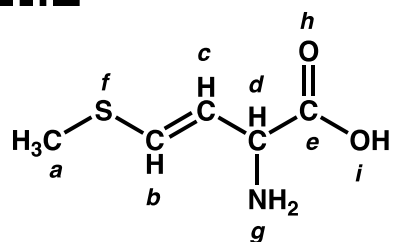
Structure 5

Structure 6

**Question 6** **6 points**

Using the data given below, calculate the **lattice energy** for SrO(s).

	ΔH° (kJ/mol)
$\text{Sr}(s) \longrightarrow \text{Sr}(g)$	+180
$\text{Sr}(g) \longrightarrow \text{Sr}^{2+}(g) + 2e^-$	+1468
$\text{O}_2(g) \longrightarrow 2\text{O}(g)$	+500
$\text{O}(g) + 2e^- \longrightarrow \text{O}^{2-}(g)$	+600
$\text{Sr}(s) + \frac{1}{2}\text{O}_2(g) \longrightarrow \text{SrO}(s)$	-550

**Question 7 10 points**

Consider the molecule given on the left, and answer questions (i) to (vi) using chemical bonding theories that we have learnt in class (valence bond theory, hybridization).

Note that the lone pairs are not shown in the structure:

i. Consider the carbon atoms marked in as ***a***, ***b***, ***c***, ***d***, and ***e*** in the molecule, and determine **hybridization** at each of the carbon centers. (2.5 points)

Atom	Hybridization
Carbon (<i>a</i>)	
Carbon (<i>b</i>)	
Carbon (<i>c</i>)	
Carbon (<i>d</i>)	
Carbon (<i>e</i>)	

ii. Now, consider the sulfur atom (S) marked as ***f***, nitrogen (N) atom (***g***), and oxygen(O) atoms (***h*** and ***i***), and determine their **hybridization**. (2 points)

Atom	Hybridization
Sulfur (<i>f</i>)	
Nitrogen (<i>g</i>)	
Oxygen (<i>h</i>)	
Oxygen (<i>i</i>)	

iii. **Describe** all the bonds formed by the carbon atoms ***a*** and ***b***, using sigma and pi designation as appropriate. (2 points)

Atom	Bonds formed (use sigma and pi as appropriate)
Carbon (<i>a</i>)	
Carbon (<i>b</i>)	

iv. How many total sp^3 -s sigma bonds are in the molecule? (1 point)

v. Determine **electronic** and **molecular geometry** at carbon (C) labeled ***e*** and sulfur (S) labeled ***f*** centres. (2 points)

Atom	Electronic Geometry	Molecular Geometry
Carbon (<i>e</i>)		
Sulfur (<i>f</i>)		

vi. Circle the value from the choices given below, that describes the **C-S-C bond angle at sulfur** (S) marked ***f***: (0.5 points)

>90° to <109.5°

109.5°

>109.5 to <120°

120°

>120° to <180°


Question 8 12 points

Using the **valence shell orbitals only** draw a **complete** molecular orbital (MO) diagram (with no s-p mixing) for the molecule **PS**, in the space provided below. **(6 points)**

Assume that 3s and 3p atomic orbitals will form molecular orbitals like those formed by the combination of 2s and 2p atomic orbitals that we learnt in class, and there is no s-p mixing.

Show **energy arrow**, **atomic orbitals** involved, **atom labels**, the **molecular orbitals** that are formed, **filled electrons** in the MO diagram etc. Make sure to label all the atomic and molecular orbitals.

Using the MO diagram above, answer the following questions:

- (i) Which one of the following will have the highest bond order? Circle the correct choice
 A) PS^{1-} B) PS C) PS^{1+}
- (ii) Which one of the following will have the longest bond length? Circle the correct choice
 A) PS^{1-} B) PS C) PS^{1+}
- (iii) Which of the following will be diamagnetic? Circle the correct choice.
 A) PS^{1-} B) PS C) PS^{1+}
- (iv) Draw Lewis dot structure of the molecule with the shortest bond length (based on the MO diagram) from among the three (PS^{1-} , PS, PS^{1+}), and indicate the bond order calculated from the Lewis dot structure. **(3 points)**

Lewis Dot Structure

Bond Order



4D42A61E-DC1B-4892-82A7-7449709DE06E

midterm-exam-93388

#859 Page 10 of 10

SCRAP PAGE

**CHEMISTRY 110 –Midterm Exam****Thursday November 7, 2024, 6:30 – 8:30 pm****Examiners: Pallavi Sirjoosingh and Ashok Kakkar**

First name (please write as legibly as possible within the boxes)

--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--

Last name

--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--

Student ID number

--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--

This booklet has 8 questions on 10 pages (The last page can be used as a scrap page).

The staple must NOT be removed from the question booklet. Your intact question booklet must be returned at the end of the exam.

A separate booklet contains a periodic table, and datasheet.

Write your answers directly under the questions in the space provided. Do not write on the QR codes or near the top edges (as it may be cut off).

Please write **legibly and dark enough** for the scan to be readable. For students who use English as a second language, dictionaries are permitted. Calculators are allowed.

DO NOT ASK QUESTIONS OF THE INVIGILATORS about interpretations of the exam questions. If you have a concern about a question, please summarize it at the top of this page. All concerns will be evaluated after the exam. GOOD LUCK!

**Question 1 8 points** (*Assume all transitions are possible*)

a. 6 points Calculate the **energy (in kJ/mol)** required to ionize (completely remove) the electron from the **n=3** level of **He⁺ ion** (He: Helium). Show your work including equations. (3 sig. figs.)

b. 2 points

(I) Select the largest (longest) wavelength from the three choices below:

- i. Wavelength of photon to *completely remove* an electron from *1s orbital of Hydrogen atom*
- ii. Wavelength of photon to *completely remove* an electron from *1s orbital of Boron atom (B)*
- iii. Wavelength of photon to *completely remove* an electron from *1s orbital of Boron cation (B⁺)*

i or ii or iii

(II) Select the smallest (shortest) wavelength from the three choices below (*for H atom*):

- i. Wavelength of photon *emitted* for an electron transitioning from *2p to 1s orbital*
- ii. Wavelength of photon *absorbed* for an electron transitioning from *2s to 4p orbital*
- iii. Wavelength of photon *absorbed* for an electron transitioning from *1s to 3p orbital*

i or ii or iii



Question 2 8 points



A Chem110 student plots the graph shown on the left, from data obtained for the kinetic energy (eV) of the ejected electron versus frequency (s^{-1}) of the incident light, for an unknown metal. ($1\text{eV} = 1.6022 \times 10^{-19} \text{ J}$)

Based on the plot, answer the questions **a to c** below:

a. 1 point On the plot above, label the **threshold frequency**.

b. 2 points What is the value of the work potential (in J) of the unknown metal?

c. 3 points What is the *longest* wavelength (in nm) required to eject an electron from this metal's surface? Show your work including equation(s) used. (3 sig figs)

d. 2 points In the space provided below, plot the following

(Assume frequency is *above* threshold frequency):

- Kinetic energy* of an ejected electron versus the *intensity* of the incident light (left)
- Number of electrons* ejected versus the *frequency* of the incident light (right)



**Question 3** **6 points****a. 2 points**

In the ground state of Ge (Germanium; $Z = 32$) atom, **name** the following orbitals:

i. Orbital containing the electron that requires the maximum energy to be removed.

ii. Orbital containing the electron that experiences the minimum effective nuclear charge.

b. 2 points

An unknown anion (A^-) has the following *excited* electronic configuration: $[Ar] 4s^2 3d^3 4p^1$

(Ar: Argon; $Z = 18$)

Give the **condensed ground state electronic configuration** of the atom, **A** and the cation, **A^{2+}** .
Show your work. (Assume all electronic transitions are possible)

Electronic Configuration of A:

Electronic Configuration of A^{2+} :

c. 2 points

In a single atom, determine how many electrons can have the following set of quantum numbers?

(i) $n = 5; l = 3$

In a single atom, determine how many electrons can have the following set of quantum numbers?

(i) $n = 3; l = 2; m_s = +1/2$

**Question 4** **8 points**

Rank the following from 1 to 4 in **increasing** order. Provide the ranking in the boxes provided.

Note: Atomic number (Z) are provided for relevant atoms

a) Increasing (lowest to highest) order of *ionization energy*: (1= lowest; 4 = highest)

Mg

Mg: Z = 12

Rb

Rb: Z = 37

Cs

Cs: Z = 55

K

K : Z = 19

b) Increasing (lowest to highest) order of *bond angle*: (1= lowest; 4 = highest)

H₂OCCl₄XeF₄

HCN

c) Increasing (smallest to largest) order of *ionic size* (1= smallest; 4 = largest)

S²⁻

S: Z = 16

O²⁻

O: Z = 8

Be²⁺

Be: Z = 4

Na⁺

Na: Z = 11

d) Increasing (lowest to highest) order of *bond energy* (1= lowest; 4 = highest)

P-F

P-Cl

P-I

P-Br

**Question 5** **12 points**

a. 9 points For the molecules HClO_4 and TeF_6 , draw the most stable *Lewis structure* showing appropriate shape and *dashed/wedged bonds*, based on VSEPR. Determine the **molecular geometry** around the central atom. For HClO_4 , label the *most polar bond*.



b. 3 points (Note: Lone pairs are not shown in structures below)

(i) For the three *resonance* structures (1, 2 and 3) shown below, answer the following question:



(I) Circle the *most contributing* resonance structure.

Structure 1

Structure 2

Structure 3

(ii) For the three structures given (4, 5, and 6) below, answer the following questions:



(I) Circle all the structures that represent *resonance structures* of each other:

Structure 4

Structure 5

Structure 6

(II) Among the resonance structures, circle the *most contributing* resonance structure.

Structure 4

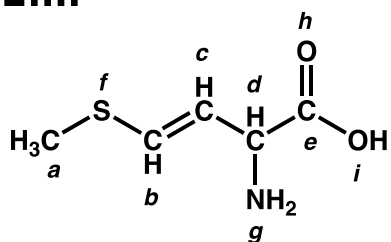
Structure 5

Structure 6

**Question 6** **6 points**

Using the data given below, calculate the **lattice energy** for SrO(s).

	ΔH° (kJ/mol)
$\text{Sr}(s) \longrightarrow \text{Sr}(g)$	+180
$\text{Sr}(g) \longrightarrow \text{Sr}^{2+}(g) + 2e^-$	+1468
$\text{O}_2(g) \longrightarrow 2\text{O}(g)$	+500
$\text{O}(g) + 2e^- \longrightarrow \text{O}^{2-}(g)$	+600
$\text{Sr}(s) + \frac{1}{2}\text{O}_2(g) \longrightarrow \text{SrO}(s)$	-550

**Question 7 10 points**

Consider the molecule given on the left, and answer questions (i) to (vi) using chemical bonding theories that we have learnt in class (valence bond theory, hybridization).

Note that the lone pairs are not shown in the structure:

i. Consider the carbon atoms marked in as ***a***, ***b***, ***c***, ***d***, and ***e*** in the molecule, and determine **hybridization** at each of the carbon centers. (2.5 points)

Atom	Hybridization
Carbon (<i>a</i>)	
Carbon (<i>b</i>)	
Carbon (<i>c</i>)	
Carbon (<i>d</i>)	
Carbon (<i>e</i>)	

ii. Now, consider the sulfur atom (S) marked as ***f***, nitrogen (N) atom (***g***), and oxygen(O) atoms (***h*** and ***i***), and determine their **hybridization**. (2 points)

Atom	Hybridization
Sulfur (<i>f</i>)	
Nitrogen (<i>g</i>)	
Oxygen (<i>h</i>)	
Oxygen (<i>i</i>)	

iii. **Describe** all the bonds formed by the carbon atoms ***a*** and ***b***, using sigma and pi designation as appropriate. (2 points)

Atom	Bonds formed (use sigma and pi as appropriate)
Carbon (<i>a</i>)	
Carbon (<i>b</i>)	

iv. How many total sp^3 -s sigma bonds are in the molecule? (1 point)

v. Determine **electronic** and **molecular geometry** at carbon (C) labeled ***e*** and sulfur (S) labeled ***f*** centres. (2 points)

Atom	Electronic Geometry	Molecular Geometry
Carbon (<i>e</i>)		
Sulfur (<i>f</i>)		

vi. Circle the value from the choices given below, that describes the **C-S-C bond angle at sulfur** (S) marked ***f***: (0.5 points)

$>90^\circ$ to $<109.5^\circ$ 109.5° >109.5 to $<120^\circ$ 120° $>120^\circ$ to $<180^\circ$


Question 8 12 points

Using the **valence shell orbitals only** draw a **complete** molecular orbital (MO) diagram (with no s-p mixing) for the molecule **PS**, in the space provided below. **(6 points)**

Assume that 3s and 3p atomic orbitals will form molecular orbitals like those formed by the combination of 2s and 2p atomic orbitals that we learnt in class, and there is no s-p mixing.

Show **energy arrow**, **atomic orbitals** involved, **atom labels**, the **molecular orbitals** that are formed, **filled electrons** in the MO diagram etc. Make sure to label all the atomic and molecular orbitals.

Using the MO diagram above, answer the following questions:

- (i) Which one of the following will have the highest bond order? Circle the correct choice
 A) PS^{1-} B) PS C) PS^{1+}
- (ii) Which one of the following will have the longest bond length? Circle the correct choice
 A) PS^{1-} B) PS C) PS^{1+}
- (iii) Which of the following will be diamagnetic? Circle the correct choice.
 A) PS^{1-} B) PS C) PS^{1+}
- (iv) Draw Lewis dot structure of the molecule with the shortest bond length (based on the MO diagram) from among the three (PS^{1-} , PS, PS^{1+}), and indicate the bond order calculated from the Lewis dot structure. **(3 points)**

Lewis Dot Structure

Bond Order



E24AC3F5-D9E6-466C-BB9D-405FC72A57E8

midterm-exam-93388

#860 Page 10 of 10

SCRAP PAGE

**CHEMISTRY 110 –Midterm Exam****Thursday November 7, 2024, 6:30 – 8:30 pm**

Examiners: Pallavi Sirjoosingh and Ashok Kakkar

First name (please write as legibly as possible within the boxes)

--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--

Last name

--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--

Student ID number

--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--

This booklet has 8 questions on 10 pages (The last page can be used as a scrap page).

The staple must NOT be removed from the question booklet. Your intact question booklet must be returned at the end of the exam.

A separate booklet contains a periodic table, and datasheet.

Write your answers directly under the questions in the space provided. Do not write on the QR codes or near the top edges (as it may be cut off).

Please write **legibly and dark enough** for the scan to be readable. For students who use English as a second language, dictionaries are permitted. Calculators are allowed.

DO NOT ASK QUESTIONS OF THE INVIGILATORS about interpretations of the exam questions. If you have a concern about a question, please summarize it at the top of this page. All concerns will be evaluated after the exam. GOOD LUCK!

**Question 1 8 points** (*Assume all transitions are possible*)

a. 6 points Calculate the **energy (in kJ/mol)** required to ionize (completely remove) the electron from the **n=3** level of **He⁺ ion** (He: Helium). Show your work including equations. (3 sig. figs.)

b. 2 points

(I) Select the largest (longest) wavelength from the three choices below:

- i. Wavelength of photon to *completely remove* an electron from *1s orbital of Hydrogen atom*
- ii. Wavelength of photon to *completely remove* an electron from *1s orbital of Boron atom (B)*
- iii. Wavelength of photon to *completely remove* an electron from *1s orbital of Boron cation (B⁺)*

i or ii or iii

(II) Select the smallest (shortest) wavelength from the three choices below (*for H atom*):

- i. Wavelength of photon *emitted* for an electron transitioning from *2p to 1s orbital*
- ii. Wavelength of photon *absorbed* for an electron transitioning from *2s to 4p orbital*
- iii. Wavelength of photon *absorbed* for an electron transitioning from *1s to 3p orbital*

i or ii or iii



Question 2 8 points



A Chem110 student plots the graph shown on the left, from data obtained for the kinetic energy (eV) of the ejected electron versus frequency (s^{-1}) of the incident light, for an unknown metal. ($1\text{eV} = 1.6022 \times 10^{-19} \text{ J}$)

Based on the plot, answer the questions **a to c** below:

a. 1 point On the plot above, label the **threshold frequency**.

b. 2 points What is the value of the work potential (in J) of the unknown metal?

c. 3 points What is the *longest* wavelength (in nm) required to eject an electron from this metal's surface? Show your work including equation(s) used. (3 sig figs)

d. 2 points In the space provided below, plot the following

(Assume frequency is *above* threshold frequency):

- i. *Kinetic energy* of an ejected electron versus the *intensity* of the incident light (left)
- ii. *Number of electrons* ejected versus the *frequency* of the incident light (right)



**Question 3** **6 points****a. 2 points**

In the ground state of Ge (Germanium; $Z = 32$) atom, **name** the following orbitals:

i. Orbital containing the electron that requires the maximum energy to be removed.

ii. Orbital containing the electron that experiences the minimum effective nuclear charge.

b. 2 points

An unknown anion (A^-) has the following *excited* electronic configuration: $[Ar] 4s^2 3d^3 4p^1$

(Ar: Argon; $Z = 18$)

Give the **condensed ground state electronic configuration** of the atom, **A** and the cation, **A^{2+}** .
Show your work. (Assume all electronic transitions are possible)

Electronic Configuration of A:

Electronic Configuration of A^{2+} :

c. 2 points

In a single atom, determine how many electrons can have the following set of quantum numbers?

(i) $n = 5; l = 3$

In a single atom, determine how many electrons can have the following set of quantum numbers?

(i) $n = 3; l = 2; m_s = +1/2$

**Question 4** **8 points**

Rank the following from 1 to 4 in **increasing** order. Provide the ranking in the boxes provided.

Note: Atomic number (Z) are provided for relevant atoms

a) Increasing (lowest to highest) order of *ionization energy*: (1= lowest; 4 = highest)

Mg

Mg: Z = 12

Rb

Rb: Z = 37

Cs

Cs: Z = 55

K

K : Z = 19

b) Increasing (lowest to highest) order of *bond angle*: (1= lowest; 4 = highest)

H₂OCCl₄XeF₄

HCN

c) Increasing (smallest to largest) order of *ionic size* (1= smallest; 4 = largest)

S²⁻

S: Z = 16

O²⁻

O: Z = 8

Be²⁺

Be: Z = 4

Na⁺

Na: Z = 11

d) Increasing (lowest to highest) order of *bond energy* (1= lowest; 4 = highest)

P-F

P-Cl

P-I

P-Br

**Question 5** **12 points**

a. 9 points For the molecules HClO_4 and TeF_6 , draw the most stable *Lewis structure* showing appropriate shape and *dashed/wedged bonds*, based on VSEPR. Determine the **molecular geometry** around the central atom. For HClO_4 , label the *most polar bond*.



b. 3 points (Note: Lone pairs are not shown in structures below)

(i) For the three *resonance* structures (1, 2 and 3) shown below, answer the following question:



(I) Circle the *most contributing* resonance structure.

Structure 1

Structure 2

Structure 3

(ii) For the three structures given (4, 5, and 6) below, answer the following questions:



(I) Circle all the structures that represent *resonance structures* of each other:

Structure 4

Structure 5

Structure 6

(II) Among the resonance structures, circle the *most contributing* resonance structure.

Structure 4

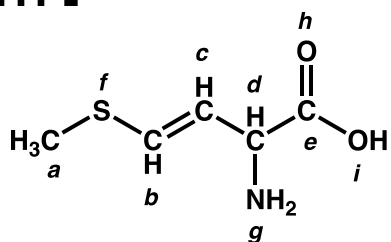
Structure 5

Structure 6

**Question 6** **6 points**

Using the data given below, calculate the **lattice energy** for SrO(s).

	ΔH° (kJ/mol)
$\text{Sr}(s) \longrightarrow \text{Sr}(g)$	+180
$\text{Sr}(g) \longrightarrow \text{Sr}^{2+}(g) + 2e^-$	+1468
$\text{O}_2(g) \longrightarrow 2\text{O}(g)$	+500
$\text{O}(g) + 2e^- \longrightarrow \text{O}^{2-}(g)$	+600
$\text{Sr}(s) + \frac{1}{2}\text{O}_2(g) \longrightarrow \text{SrO}(s)$	-550

**Question 7 10 points**

Consider the molecule given on the left, and answer questions (i) to (vi) using chemical bonding theories that we have learnt in class (valence bond theory, hybridization).

Note that the lone pairs are not shown in the structure:

i. Consider the carbon atoms marked in as ***a***, ***b***, ***c***, ***d***, and ***e*** in the molecule, and determine **hybridization** at each of the carbon centers. (2.5 points)

Atom	Hybridization
Carbon (<i>a</i>)	
Carbon (<i>b</i>)	
Carbon (<i>c</i>)	
Carbon (<i>d</i>)	
Carbon (<i>e</i>)	

ii. Now, consider the sulfur atom (S) marked as ***f***, nitrogen (N) atom (***g***), and oxygen(O) atoms (***h*** and ***i***), and determine their **hybridization**. (2 points)

Atom	Hybridization
Sulfur (<i>f</i>)	
Nitrogen (<i>g</i>)	
Oxygen (<i>h</i>)	
Oxygen (<i>i</i>)	

iii. **Describe** all the bonds formed by the carbon atoms ***a*** and ***b***, using sigma and pi designation as appropriate. (2 points)

Atom	Bonds formed (use sigma and pi as appropriate)
Carbon (<i>a</i>)	
Carbon (<i>b</i>)	

iv. How many total sp^3 -s sigma bonds are in the molecule? (1 point)

v. Determine **electronic** and **molecular geometry** at carbon (C) labeled ***e*** and sulfur (S) labeled ***f*** centres. (2 points)

Atom	Electronic Geometry	Molecular Geometry
Carbon (<i>e</i>)		
Sulfur (<i>f</i>)		

vi. Circle the value from the choices given below, that describes the **C-S-C bond angle at sulfur** (S) marked ***f***: (0.5 points)

$>90^\circ$ to $<109.5^\circ$ 109.5° >109.5 to $<120^\circ$ 120° $>120^\circ$ to $<180^\circ$


Question 8 12 points

Using the **valence shell orbitals only** draw a **complete** molecular orbital (MO) diagram (with no s-p mixing) for the molecule **PS**, in the space provided below. **(6 points)**

Assume that 3s and 3p atomic orbitals will form molecular orbitals like those formed by the combination of 2s and 2p atomic orbitals that we learnt in class, and there is no s-p mixing.

Show **energy arrow**, **atomic orbitals** involved, **atom labels**, the **molecular orbitals** that are formed, **filled electrons** in the MO diagram etc. Make sure to label all the atomic and molecular orbitals.

Using the MO diagram above, answer the following questions:

- (i) Which one of the following will have the highest bond order? Circle the correct choice
 A) PS^{1-} B) PS C) PS^{1+}
- (ii) Which one of the following will have the longest bond length? Circle the correct choice
 A) PS^{1-} B) PS C) PS^{1+}
- (iii) Which of the following will be diamagnetic? Circle the correct choice.
 A) PS^{1-} B) PS C) PS^{1+}
- (iv) Draw Lewis dot structure of the molecule with the shortest bond length (based on the MO diagram) from among the three (PS^{1-} , PS, PS^{1+}), and indicate the bond order calculated from the Lewis dot structure. **(3 points)**

Lewis Dot Structure

Bond Order



387980C7-6A1B-408D-BE48-6007352A3C59

midterm-exam-93388

#861 Page 10 of 10

SCRAP PAGE

**CHEMISTRY 110 –Midterm Exam****Thursday November 7, 2024, 6:30 – 8:30 pm****Examiners: Pallavi Sirjoosingh and Ashok Kakkar**

First name (please write as legibly as possible within the boxes)

--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--

Last name

--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--

Student ID number

--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--

This booklet has 8 questions on 10 pages (The last page can be used as a scrap page).

The staple must NOT be removed from the question booklet. Your intact question booklet must be returned at the end of the exam.

A separate booklet contains a periodic table, and datasheet.

Write your answers directly under the questions in the space provided. Do not write on the QR codes or near the top edges (as it may be cut off).

Please write **legibly and dark enough** for the scan to be readable. For students who use English as a second language, dictionaries are permitted. Calculators are allowed.

DO NOT ASK QUESTIONS OF THE INVIGILATORS about interpretations of the exam questions. If you have a concern about a question, please summarize it at the top of this page. All concerns will be evaluated after the exam. GOOD LUCK!

**Question 1 8 points** (*Assume all transitions are possible*)

a. 6 points Calculate the **energy (in kJ/mol)** required to ionize (completely remove) the electron from the **n=3** level of **He⁺ ion** (He: Helium). Show your work including equations. (3 sig. figs.)

b. 2 points

(I) Select the largest (longest) wavelength from the three choices below:

- i. Wavelength of photon to *completely remove* an electron from *1s orbital of Hydrogen atom*
- ii. Wavelength of photon to *completely remove* an electron from *1s orbital of Boron atom (B)*
- iii. Wavelength of photon to *completely remove* an electron from *1s orbital of Boron cation (B⁺)*

i or ii or iii

(II) Select the smallest (shortest) wavelength from the three choices below (*for H atom*):

- i. Wavelength of photon *emitted* for an electron transitioning from *2p to 1s orbital*
- ii. Wavelength of photon *absorbed* for an electron transitioning from *2s to 4p orbital*
- iii. Wavelength of photon *absorbed* for an electron transitioning from *1s to 3p orbital*

i or ii or iii



Question 2 8 points



A Chem110 student plots the graph shown on the left, from data obtained for the kinetic energy (eV) of the ejected electron versus frequency (s^{-1}) of the incident light, for an unknown metal. ($1\text{eV} = 1.6022 \times 10^{-19} \text{ J}$)

Based on the plot, answer the questions **a to c** below:

a. 1 point On the plot above, label the **threshold frequency**.

b. 2 points What is the value of the work potential (in J) of the unknown metal?

c. 3 points What is the *longest* wavelength (in nm) required to eject an electron from this metal's surface? Show your work including equation(s) used. (3 sig figs)

d. 2 points In the space provided below, plot the following

(Assume frequency is *above* threshold frequency):

- i. *Kinetic energy* of an ejected electron versus the *intensity* of the incident light (left)
- ii. *Number of electrons* ejected versus the *frequency* of the incident light (right)



**Question 3** **6 points****a. 2 points**

In the ground state of Ge (Germanium; $Z = 32$) atom, **name** the following orbitals:

i. Orbital containing the electron that requires the maximum energy to be removed.

ii. Orbital containing the electron that experiences the minimum effective nuclear charge.

b. 2 points

An unknown anion (A^-) has the following *excited* electronic configuration: $[Ar] 4s^2 3d^3 4p^1$

(Ar: Argon; $Z = 18$)

Give the **condensed ground state electronic configuration** of the atom, **A** and the cation, **A^{2+}** .
Show your work. (Assume all electronic transitions are possible)

Electronic Configuration of A:

Electronic Configuration of A^{2+} :

c. 2 points

In a single atom, determine how many electrons can have the following set of quantum numbers?

(i) $n = 5; l = 3$

In a single atom, determine how many electrons can have the following set of quantum numbers?

(i) $n = 3; l = 2; m_s = +1/2$

**Question 4** **8 points**

Rank the following from 1 to 4 in **increasing** order. Provide the ranking in the boxes provided.

Note: Atomic number (Z) are provided for relevant atoms

a) Increasing (lowest to highest) order of *ionization energy*: (1= lowest; 4 = highest)

Mg

Mg: Z = 12

Rb

Rb: Z = 37

Cs

Cs: Z = 55

K

K : Z = 19

b) Increasing (lowest to highest) order of *bond angle*: (1= lowest; 4 = highest)

H₂OCCl₄XeF₄

HCN

c) Increasing (smallest to largest) order of *ionic size* (1= smallest; 4 = largest)

S²⁻

S: Z = 16

O²⁻

O: Z = 8

Be²⁺

Be: Z = 4

Na⁺

Na: Z = 11

d) Increasing (lowest to highest) order of *bond energy* (1= lowest; 4 = highest)

P-F

P-Cl

P-I

P-Br

**Question 5** **12 points**

a. 9 points For the molecules HClO_4 and TeF_6 , draw the most stable *Lewis structure* showing appropriate shape and *dashed/wedged bonds*, based on VSEPR. Determine the **molecular geometry** around the central atom. For HClO_4 , label the *most polar bond*.



b. 3 points (Note: Lone pairs are not shown in structures below)

(i) For the three *resonance* structures (1, 2 and 3) shown below, answer the following question:



(I) Circle the *most contributing* resonance structure.

Structure 1

Structure 2

Structure 3

(ii) For the three structures given (4, 5, and 6) below, answer the following questions:



(I) Circle all the structures that represent *resonance structures* of each other:

Structure 4

Structure 5

Structure 6

(II) Among the resonance structures, circle the *most contributing* resonance structure.

Structure 4

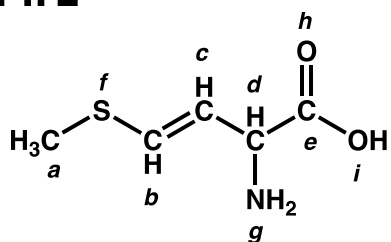
Structure 5

Structure 6

**Question 6** **6 points**

Using the data given below, calculate the **lattice energy** for SrO(s).

	ΔH° (kJ/mol)
$\text{Sr}(s) \longrightarrow \text{Sr}(g)$	+180
$\text{Sr}(g) \longrightarrow \text{Sr}^{2+}(g) + 2e^-$	+1468
$\text{O}_2(g) \longrightarrow 2\text{O}(g)$	+500
$\text{O}(g) + 2e^- \longrightarrow \text{O}^{2-}(g)$	+600
$\text{Sr}(s) + \frac{1}{2}\text{O}_2(g) \longrightarrow \text{SrO}(s)$	-550

**Question 7 10 points**

Consider the molecule given on the left, and answer questions (i) to (vi) using chemical bonding theories that we have learnt in class (valence bond theory, hybridization).

Note that the lone pairs are not shown in the structure:

i. Consider the carbon atoms marked in as ***a***, ***b***, ***c***, ***d***, and ***e*** in the molecule, and determine **hybridization** at each of the carbon centers. (2.5 points)

Atom	Hybridization
Carbon (<i>a</i>)	
Carbon (<i>b</i>)	
Carbon (<i>c</i>)	
Carbon (<i>d</i>)	
Carbon (<i>e</i>)	

ii. Now, consider the sulfur atom (S) marked as ***f***, nitrogen (N) atom (***g***), and oxygen(O) atoms (***h*** and ***i***), and determine their **hybridization**. (2 points)

Atom	Hybridization
Sulfur (<i>f</i>)	
Nitrogen (<i>g</i>)	
Oxygen (<i>h</i>)	
Oxygen (<i>i</i>)	

iii. **Describe** all the bonds formed by the carbon atoms ***a*** and ***b***, using sigma and pi designation as appropriate. (2 points)

Atom	Bonds formed (use sigma and pi as appropriate)
Carbon (<i>a</i>)	
Carbon (<i>b</i>)	

iv. How many total sp^3 -s sigma bonds are in the molecule? (1 point)

v. Determine **electronic** and **molecular geometry** at carbon (C) labeled ***e*** and sulfur (S) labeled ***f*** centres. (2 points)

Atom	Electronic Geometry	Molecular Geometry
Carbon (<i>e</i>)		
Sulfur (<i>f</i>)		

vi. Circle the value from the choices given below, that describes the **C-S-C bond angle at sulfur** (S) marked ***f***: (0.5 points)

>90° to <109.5° 109.5° >109.5 to <120° 120° >120° to <180°


Question 8 12 points

Using the **valence shell orbitals only** draw a **complete** molecular orbital (MO) diagram (with no s-p mixing) for the molecule **PS**, in the space provided below. **(6 points)**

Assume that 3s and 3p atomic orbitals will form molecular orbitals like those formed by the combination of 2s and 2p atomic orbitals that we learnt in class, and there is no s-p mixing.

Show **energy arrow**, **atomic orbitals** involved, **atom labels**, the **molecular orbitals** that are formed, **filled electrons** in the MO diagram etc. Make sure to label all the atomic and molecular orbitals.

Using the MO diagram above, answer the following questions:

- (i) Which one of the following will have the highest bond order? Circle the correct choice
 A) PS^{1-} B) PS C) PS^{1+}
- (ii) Which one of the following will have the longest bond length? Circle the correct choice
 A) PS^{1-} B) PS C) PS^{1+}
- (iii) Which of the following will be diamagnetic? Circle the correct choice.
 A) PS^{1-} B) PS C) PS^{1+}
- (iv) Draw Lewis dot structure of the molecule with the shortest bond length (based on the MO diagram) from among the three (PS^{1-} , PS, PS^{1+}), and indicate the bond order calculated from the Lewis dot structure. **(3 points)**

Lewis Dot Structure

Bond Order



C7FCC733-BEA5-46F2-8E20-9041F4B8AABC

midterm-exam-93388

#862 Page 10 of 10

SCRAP PAGE

**CHEMISTRY 110 –Midterm Exam****Thursday November 7, 2024, 6:30 – 8:30 pm****Examiners: Pallavi Sirjoosingh and Ashok Kakkar**

First name (please write as legibly as possible within the boxes)

--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--

Last name

--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--

Student ID number

--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--

This booklet has 8 questions on 10 pages (The last page can be used as a scrap page).

The staple must NOT be removed from the question booklet. Your intact question booklet must be returned at the end of the exam.

A separate booklet contains a periodic table, and datasheet.

Write your answers directly under the questions in the space provided. Do not write on the QR codes or near the top edges (as it may be cut off).

Please write **legibly and dark enough** for the scan to be readable. For students who use English as a second language, dictionaries are permitted. Calculators are allowed.

DO NOT ASK QUESTIONS OF THE INVIGILATORS about interpretations of the exam questions. If you have a concern about a question, please summarize it at the top of this page. All concerns will be evaluated after the exam. GOOD LUCK!

**Question 1 8 points** (*Assume all transitions are possible*)

a. 6 points Calculate the **energy (in kJ/mol)** required to ionize (completely remove) the electron from the **n=3** level of **He⁺ ion** (He: Helium). Show your work including equations. (3 sig. figs.)

b. 2 points

(I) Select the largest (longest) wavelength from the three choices below:

- i. Wavelength of photon to *completely remove* an electron from *1s orbital of Hydrogen atom*
- ii. Wavelength of photon to *completely remove* an electron from *1s orbital of Boron atom (B)*
- iii. Wavelength of photon to *completely remove* an electron from *1s orbital of Boron cation (B⁺)*

i or ii or iii

(II) Select the smallest (shortest) wavelength from the three choices below (*for H atom*):

- i. Wavelength of photon *emitted* for an electron transitioning from *2p to 1s orbital*
- ii. Wavelength of photon *absorbed* for an electron transitioning from *2s to 4p orbital*
- iii. Wavelength of photon *absorbed* for an electron transitioning from *1s to 3p orbital*

i or ii or iii



Question 2 8 points



A Chem110 student plots the graph shown on the left, from data obtained for the kinetic energy (eV) of the ejected electron versus frequency (s^{-1}) of the incident light, for an unknown metal. ($1\text{eV} = 1.6022 \times 10^{-19} \text{ J}$)

Based on the plot, answer the questions **a to c** below:

a. 1 point On the plot above, label the **threshold frequency**.

b. 2 points What is the value of the work potential (in J) of the unknown metal?

c. 3 points What is the *longest* wavelength (in nm) required to eject an electron from this metal's surface? Show your work including equation(s) used. (3 sig figs)

d. 2 points In the space provided below, plot the following

(Assume frequency is *above* threshold frequency):

- Kinetic energy* of an ejected electron versus the *intensity* of the incident light (left)
- Number of electrons* ejected versus the *frequency* of the incident light (right)



**Question 3** **6 points****a. 2 points**

In the ground state of Ge (Germanium; $Z = 32$) atom, **name** the following orbitals:

i. Orbital containing the electron that requires the maximum energy to be removed.

ii. Orbital containing the electron that experiences the minimum effective nuclear charge.

b. 2 points

An unknown anion (A^-) has the following *excited* electronic configuration: $[Ar] 4s^2 3d^3 4p^1$

(Ar: Argon; $Z = 18$)

Give the **condensed ground state electronic configuration** of the atom, **A** and the cation, **A^{2+}** .
Show your work. (Assume all electronic transitions are possible)

Electronic Configuration of A:

Electronic Configuration of A^{2+} :

c. 2 points

In a single atom, determine how many electrons can have the following set of quantum numbers?

(i) $n = 5; l = 3$

In a single atom, determine how many electrons can have the following set of quantum numbers?

(i) $n = 3; l = 2; m_s = +1/2$

**Question 4** **8 points**

Rank the following from 1 to 4 in **increasing** order. Provide the ranking in the boxes provided.

Note: Atomic number (Z) are provided for relevant atoms

a) Increasing (lowest to highest) order of *ionization energy*: (1= lowest; 4 = highest)

Mg

Mg: Z = 12

Rb

Rb: Z = 37

Cs

Cs: Z = 55

K

K : Z = 19

b) Increasing (lowest to highest) order of *bond angle*: (1= lowest; 4 = highest)

H₂OCCl₄XeF₄

HCN

c) Increasing (smallest to largest) order of *ionic size* (1= smallest; 4 = largest)

S²⁻

S: Z = 16

O²⁻

O: Z = 8

Be²⁺

Be: Z = 4

Na⁺

Na: Z = 11

d) Increasing (lowest to highest) order of *bond energy* (1= lowest; 4 = highest)

P-F

P-Cl

P-I

P-Br

**Question 5** **12 points**

a. 9 points For the molecules HClO_4 and TeF_6 , draw the most stable *Lewis structure* showing appropriate shape and *dashed/wedged bonds*, based on VSEPR. Determine the **molecular geometry** around the central atom. For HClO_4 , label the *most polar bond*.



b. 3 points (Note: Lone pairs are not shown in structures below)

(i) For the three *resonance* structures (1, 2 and 3) shown below, answer the following question:



(I) Circle the *most contributing* resonance structure.

Structure 1

Structure 2

Structure 3

(ii) For the three structures given (4, 5, and 6) below, answer the following questions:



(I) Circle all the structures that represent *resonance structures* of each other:

Structure 4

Structure 5

Structure 6

(II) Among the resonance structures, circle the *most contributing* resonance structure.

Structure 4

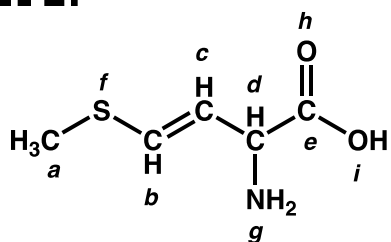
Structure 5

Structure 6

**Question 6** **6 points**

Using the data given below, calculate the **lattice energy** for SrO(s).

	ΔH° (kJ/mol)
$\text{Sr}(s) \longrightarrow \text{Sr}(g)$	+180
$\text{Sr}(g) \longrightarrow \text{Sr}^{2+}(g) + 2e^-$	+1468
$\text{O}_2(g) \longrightarrow 2\text{O}(g)$	+500
$\text{O}(g) + 2e^- \longrightarrow \text{O}^{2-}(g)$	+600
$\text{Sr}(s) + \frac{1}{2}\text{O}_2(g) \longrightarrow \text{SrO}(s)$	-550

**Question 7 10 points**

Consider the molecule given on the left, and answer questions (i) to (vi) using chemical bonding theories that we have learnt in class (valence bond theory, hybridization).

Note that the lone pairs are not shown in the structure:

i. Consider the carbon atoms marked in as ***a***, ***b***, ***c***, ***d***, and ***e*** in the molecule, and determine **hybridization** at each of the carbon centers. (2.5 points)

Atom	Hybridization
Carbon (<i>a</i>)	
Carbon (<i>b</i>)	
Carbon (<i>c</i>)	
Carbon (<i>d</i>)	
Carbon (<i>e</i>)	

ii. Now, consider the sulfur atom (S) marked as ***f***, nitrogen (N) atom (***g***), and oxygen(O) atoms (***h*** and ***i***), and determine their **hybridization**. (2 points)

Atom	Hybridization
Sulfur (<i>f</i>)	
Nitrogen (<i>g</i>)	
Oxygen (<i>h</i>)	
Oxygen (<i>i</i>)	

iii. **Describe** all the bonds formed by the carbon atoms ***a*** and ***b***, using sigma and pi designation as appropriate. (2 points)

Atom	Bonds formed (use sigma and pi as appropriate)
Carbon (<i>a</i>)	
Carbon (<i>b</i>)	

iv. How many total sp^3 -s sigma bonds are in the molecule? (1 point)

v. Determine **electronic** and **molecular geometry** at carbon (C) labeled ***e*** and sulfur (S) labeled ***f*** centres. (2 points)

Atom	Electronic Geometry	Molecular Geometry
Carbon (<i>e</i>)		
Sulfur (<i>f</i>)		

vi. Circle the value from the choices given below, that describes the **C-S-C bond angle at sulfur** (S) marked ***f***: (0.5 points)

$>90^\circ$ to $<109.5^\circ$ 109.5° >109.5 to $<120^\circ$ 120° $>120^\circ$ to $<180^\circ$


Question 8 12 points

Using the **valence shell orbitals only** draw a **complete** molecular orbital (MO) diagram (with no s-p mixing) for the molecule **PS**, in the space provided below. **(6 points)**

Assume that 3s and 3p atomic orbitals will form molecular orbitals like those formed by the combination of 2s and 2p atomic orbitals that we learnt in class, and there is no s-p mixing.

Show **energy arrow**, **atomic orbitals** involved, **atom labels**, the **molecular orbitals** that are formed, **filled electrons** in the MO diagram etc. Make sure to label all the atomic and molecular orbitals.

Using the MO diagram above, answer the following questions:

- (i) Which one of the following will have the highest bond order? Circle the correct choice
 A) PS^{1-} B) PS C) PS^{1+}
- (ii) Which one of the following will have the longest bond length? Circle the correct choice
 A) PS^{1-} B) PS C) PS^{1+}
- (iii) Which of the following will be diamagnetic? Circle the correct choice.
 A) PS^{1-} B) PS C) PS^{1+}
- (iv) Draw Lewis dot structure of the molecule with the shortest bond length (based on the MO diagram) from among the three (PS^{1-} , PS, PS^{1+}), and indicate the bond order calculated from the Lewis dot structure. **(3 points)**

Lewis Dot Structure

Bond Order



3F8675BE-80F2-4E40-8507-38A8294ACED5

midterm-exam-93388

#863 Page 10 of 10

SCRAP PAGE

**CHEMISTRY 110 –Midterm Exam****Thursday November 7, 2024, 6:30 – 8:30 pm**

Examiners: Pallavi Sirjoosingh and Ashok Kakkar

First name (please write as legibly as possible within the boxes)

--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--

Last name

--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--

Student ID number

--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--

This booklet has 8 questions on 10 pages (The last page can be used as a scrap page).

The staple must NOT be removed from the question booklet. Your intact question booklet must be returned at the end of the exam.

A separate booklet contains a periodic table, and datasheet.

Write your answers directly under the questions in the space provided. Do not write on the QR codes or near the top edges (as it may be cut off).

Please write **legibly and dark enough** for the scan to be readable. For students who use English as a second language, dictionaries are permitted. Calculators are allowed.

DO NOT ASK QUESTIONS OF THE INVIGILATORS about interpretations of the exam questions. If you have a concern about a question, please summarize it at the top of this page. All concerns will be evaluated after the exam. GOOD LUCK!

**Question 1 8 points** (*Assume all transitions are possible*)

a. 6 points Calculate the **energy (in kJ/mol)** required to ionize (completely remove) the electron from the **n=3** level of **He⁺ ion** (He: Helium). Show your work including equations. (3 sig. figs.)

b. 2 points

(I) Select the largest (longest) wavelength from the three choices below:

- i. Wavelength of photon to *completely remove* an electron from *1s orbital of Hydrogen atom*
- ii. Wavelength of photon to *completely remove* an electron from *1s orbital of Boron atom (B)*
- iii. Wavelength of photon to *completely remove* an electron from *1s orbital of Boron cation (B⁺)*

i or ii or iii

(II) Select the smallest (shortest) wavelength from the three choices below (*for H atom*):

- i. Wavelength of photon *emitted* for an electron transitioning from *2p to 1s orbital*
- ii. Wavelength of photon *absorbed* for an electron transitioning from *2s to 4p orbital*
- iii. Wavelength of photon *absorbed* for an electron transitioning from *1s to 3p orbital*

i or ii or iii



Question 2 8 points



A Chem110 student plots the graph shown on the left, from data obtained for the kinetic energy (eV) of the ejected electron versus frequency (s^{-1}) of the incident light, for an unknown metal. ($1\text{eV} = 1.6022 \times 10^{-19} \text{ J}$)

Based on the plot, answer the questions **a to c** below:

a. 1 point On the plot above, label the **threshold frequency**.

b. 2 points What is the value of the work potential (in J) of the unknown metal?

c. 3 points What is the *longest* wavelength (in nm) required to eject an electron from this metal's surface? Show your work including equation(s) used. (3 sig figs)

d. 2 points In the space provided below, plot the following

(Assume frequency is *above* threshold frequency):

- i. *Kinetic energy* of an ejected electron versus the *intensity* of the incident light (left)
- ii. *Number of electrons* ejected versus the *frequency* of the incident light (right)



**Question 3** **6 points****a. 2 points**

In the ground state of Ge (Germanium; $Z = 32$) atom, **name** the following orbitals:

i. Orbital containing the electron that requires the maximum energy to be removed.

ii. Orbital containing the electron that experiences the minimum effective nuclear charge.

b. 2 points

An unknown anion (A^-) has the following *excited* electronic configuration: $[Ar] 4s^2 3d^3 4p^1$

(Ar: Argon; $Z = 18$)

Give the **condensed ground state electronic configuration** of the atom, **A** and the cation, **A^{2+}** .
Show your work. (Assume all electronic transitions are possible)

Electronic Configuration of A:

Electronic Configuration of A^{2+} :

c. 2 points

In a single atom, determine how many electrons can have the following set of quantum numbers?

(i) $n = 5; l = 3$

In a single atom, determine how many electrons can have the following set of quantum numbers?

(i) $n = 3; l = 2; m_s = +1/2$

**Question 4** **8 points**

Rank the following from 1 to 4 in **increasing** order. Provide the ranking in the boxes provided.

Note: Atomic number (Z) are provided for relevant atoms

a) Increasing (lowest to highest) order of *ionization energy*: (1= lowest; 4 = highest)

Mg

Mg: Z = 12

Rb

Rb: Z = 37

Cs

Cs: Z = 55

K

K : Z = 19

b) Increasing (lowest to highest) order of *bond angle*: (1= lowest; 4 = highest)

H₂OCCl₄XeF₄

HCN

c) Increasing (smallest to largest) order of *ionic size* (1= smallest; 4 = largest)

S²⁻

S: Z = 16

O²⁻

O: Z = 8

Be²⁺

Be: Z = 4

Na⁺

Na: Z = 11

d) Increasing (lowest to highest) order of *bond energy* (1= lowest; 4 = highest)

P-F

P-Cl

P-I

P-Br

**Question 5** **12 points**

a. 9 points For the molecules HClO_4 and TeF_6 , draw the most stable *Lewis structure* showing appropriate shape and *dashed/wedged bonds*, based on VSEPR. Determine the **molecular geometry** around the central atom. For HClO_4 , label the *most polar bond*.



b. 3 points (Note: Lone pairs are not shown in structures below)

(i) For the three *resonance* structures (1, 2 and 3) shown below, answer the following question:



(I) Circle the *most contributing* resonance structure.

Structure 1

Structure 2

Structure 3

(ii) For the three structures given (4, 5, and 6) below, answer the following questions:



(I) Circle all the structures that represent *resonance structures* of each other:

Structure 4

Structure 5

Structure 6

(II) Among the resonance structures, circle the *most contributing* resonance structure.

Structure 4

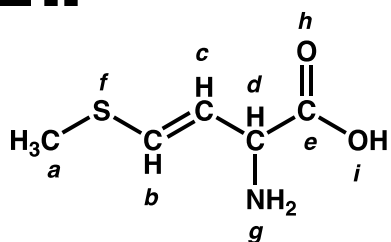
Structure 5

Structure 6

**Question 6** **6 points**

Using the data given below, calculate the **lattice energy** for SrO(s).

	ΔH° (kJ/mol)
$\text{Sr}(s) \longrightarrow \text{Sr}(g)$	+180
$\text{Sr}(g) \longrightarrow \text{Sr}^{2+}(g) + 2e^-$	+1468
$\text{O}_2(g) \longrightarrow 2\text{O}(g)$	+500
$\text{O}(g) + 2e^- \longrightarrow \text{O}^{2-}(g)$	+600
$\text{Sr}(s) + \frac{1}{2}\text{O}_2(g) \longrightarrow \text{SrO}(s)$	-550

**Question 7 10 points**

Consider the molecule given on the left, and answer questions (i) to (vi) using chemical bonding theories that we have learnt in class (valence bond theory, hybridization).

Note that the lone pairs are not shown in the structure:

i. Consider the carbon atoms marked in as ***a***, ***b***, ***c***, ***d***, and ***e*** in the molecule, and determine **hybridization** at each of the carbon centers. (2.5 points)

Atom	Hybridization
Carbon (<i>a</i>)	
Carbon (<i>b</i>)	
Carbon (<i>c</i>)	
Carbon (<i>d</i>)	
Carbon (<i>e</i>)	

ii. Now, consider the sulfur atom (S) marked as ***f***, nitrogen (N) atom (***g***), and oxygen(O) atoms (***h*** and ***i***), and determine their **hybridization**. (2 points)

Atom	Hybridization
Sulfur (<i>f</i>)	
Nitrogen (<i>g</i>)	
Oxygen (<i>h</i>)	
Oxygen (<i>i</i>)	

iii. **Describe** all the bonds formed by the carbon atoms ***a*** and ***b***, using sigma and pi designation as appropriate. (2 points)

Atom	Bonds formed (use sigma and pi as appropriate)
Carbon (<i>a</i>)	
Carbon (<i>b</i>)	

iv. How many total sp^3 -s sigma bonds are in the molecule? (1 point)

v. Determine **electronic** and **molecular geometry** at carbon (C) labeled ***e*** and sulfur (S) labeled ***f*** centres. (2 points)

Atom	Electronic Geometry	Molecular Geometry
Carbon (<i>e</i>)		
Sulfur (<i>f</i>)		

vi. Circle the value from the choices given below, that describes the **C-S-C bond angle at sulfur** (S) marked ***f***: (0.5 points)

>90° to <109.5°

109.5°

>109.5 to <120°

120°

>120° to <180°


Question 8 12 points

Using the **valence shell orbitals only** draw a **complete** molecular orbital (MO) diagram (with no s-p mixing) for the molecule **PS**, in the space provided below. **(6 points)**

Assume that 3s and 3p atomic orbitals will form molecular orbitals like those formed by the combination of 2s and 2p atomic orbitals that we learnt in class, and there is no s-p mixing.

Show **energy arrow**, **atomic orbitals** involved, **atom labels**, the **molecular orbitals** that are formed, **filled electrons** in the MO diagram etc. Make sure to label all the atomic and molecular orbitals.

Using the MO diagram above, answer the following questions:

- (i) Which one of the following will have the highest bond order? Circle the correct choice
 A) PS^{1-} B) PS C) PS^{1+}
- (ii) Which one of the following will have the longest bond length? Circle the correct choice
 A) PS^{1-} B) PS C) PS^{1+}
- (iii) Which of the following will be diamagnetic? Circle the correct choice.
 A) PS^{1-} B) PS C) PS^{1+}
- (iv) Draw Lewis dot structure of the molecule with the shortest bond length (based on the MO diagram) from among the three (PS^{1-} , PS, PS^{1+}), and indicate the bond order calculated from the Lewis dot structure. **(3 points)**

Lewis Dot Structure

Bond Order



1B212D58-CE66-4A96-82B1-9615F8E2392A

midterm-exam-93388

#864 Page 10 of 10

SCRAP PAGE

**CHEMISTRY 110 –Midterm Exam****Thursday November 7, 2024, 6:30 – 8:30 pm****Examiners: Pallavi Sirjoosingh and Ashok Kakkar**

First name (please write as legibly as possible within the boxes)

--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--

Last name

--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--

Student ID number

--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--

This booklet has 8 questions on 10 pages (The last page can be used as a scrap page).

The staple must NOT be removed from the question booklet. Your intact question booklet must be returned at the end of the exam.

A separate booklet contains a periodic table, and datasheet.

Write your answers directly under the questions in the space provided. Do not write on the QR codes or near the top edges (as it may be cut off).

Please write **legibly and dark enough** for the scan to be readable. For students who use English as a second language, dictionaries are permitted. Calculators are allowed.

DO NOT ASK QUESTIONS OF THE INVIGILATORS about interpretations of the exam questions. If you have a concern about a question, please summarize it at the top of this page. All concerns will be evaluated after the exam. GOOD LUCK!

**Question 1 8 points** (*Assume all transitions are possible*)

a. 6 points Calculate the **energy (in kJ/mol)** required to ionize (completely remove) the electron from the **n=3** level of **He⁺ ion** (He: Helium). Show your work including equations. (3 sig. figs.)

b. 2 points

(I) Select the largest (longest) wavelength from the three choices below:

- i. Wavelength of photon to *completely remove* an electron from *1s orbital of Hydrogen atom*
- ii. Wavelength of photon to *completely remove* an electron from *1s orbital of Boron atom (B)*
- iii. Wavelength of photon to *completely remove* an electron from *1s orbital of Boron cation (B⁺)*

i or ii or iii

(II) Select the smallest (shortest) wavelength from the three choices below (*for H atom*):

- i. Wavelength of photon *emitted* for an electron transitioning from *2p to 1s orbital*
- ii. Wavelength of photon *absorbed* for an electron transitioning from *2s to 4p orbital*
- iii. Wavelength of photon *absorbed* for an electron transitioning from *1s to 3p orbital*

i or ii or iii



Question 2 8 points



A Chem110 student plots the graph shown on the left, from data obtained for the kinetic energy (eV) of the ejected electron versus frequency (s^{-1}) of the incident light, for an unknown metal. ($1\text{eV} = 1.6022 \times 10^{-19} \text{ J}$)

Based on the plot, answer the questions **a to c** below:

a. 1 point On the plot above, label the **threshold frequency**.

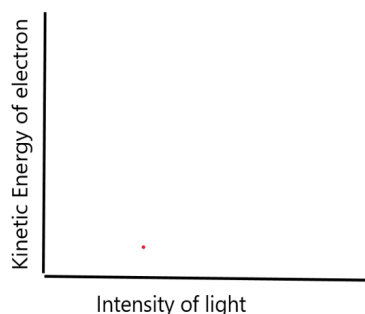
b. 2 points What is the value of the work potential (in J) of the unknown metal?

c. 3 points What is the *longest* wavelength (in nm) required to eject an electron from this metal's surface? Show your work including equation(s) used. (3 sig figs)

d. 2 points In the space provided below, plot the following

(Assume frequency is *above* threshold frequency):

- Kinetic energy* of an ejected electron versus the *intensity* of the incident light (left)
- Number of electrons* ejected versus the *frequency* of the incident light (right)



**Question 3** **6 points****a. 2 points**

In the ground state of Ge (Germanium; $Z = 32$) atom, **name** the following orbitals:

i. Orbital containing the electron that requires the maximum energy to be removed.

ii. Orbital containing the electron that experiences the minimum effective nuclear charge.

b. 2 points

An unknown anion (A^-) has the following *excited* electronic configuration: $[Ar] 4s^2 3d^3 4p^1$

(Ar: Argon; $Z = 18$)

Give the **condensed ground state electronic configuration** of the atom, **A** and the cation, **A^{2+}** .
Show your work. (Assume all electronic transitions are possible)

Electronic Configuration of A:

Electronic Configuration of A^{2+} :

c. 2 points

In a single atom, determine how many electrons can have the following set of quantum numbers?

(i) $n = 5; l = 3$

In a single atom, determine how many electrons can have the following set of quantum numbers?

(i) $n = 3; l = 2; m_s = +1/2$

**Question 4** **8 points**

Rank the following from 1 to 4 in **increasing** order. Provide the ranking in the boxes provided.

Note: Atomic number (Z) are provided for relevant atoms

a) Increasing (lowest to highest) order of *ionization energy*: (1= lowest; 4 = highest)

Mg

Mg: Z = 12

Rb

Rb: Z = 37

Cs

Cs: Z = 55

K

K : Z = 19

b) Increasing (lowest to highest) order of *bond angle*: (1= lowest; 4 = highest)

H₂OCCl₄XeF₄

HCN

c) Increasing (smallest to largest) order of *ionic size* (1= smallest; 4 = largest)

S²⁻

S: Z = 16

O²⁻

O: Z = 8

Be²⁺

Be: Z = 4

Na⁺

Na: Z = 11

d) Increasing (lowest to highest) order of *bond energy* (1= lowest; 4 = highest)

P-F

P-Cl

P-I

P-Br

**Question 5** **12 points**

a. 9 points For the molecules HClO_4 and TeF_6 , draw the most stable *Lewis structure* showing appropriate shape and *dashed/wedged bonds*, based on VSEPR. Determine the **molecular geometry** around the central atom. For HClO_4 , label the *most polar bond*.



b. 3 points (Note: Lone pairs are not shown in structures below)

(i) For the three *resonance* structures (1, 2 and 3) shown below, answer the following question:



(I) Circle the *most contributing* resonance structure.

Structure 1

Structure 2

Structure 3

(ii) For the three structures given (4, 5, and 6) below, answer the following questions:



(I) Circle all the structures that represent *resonance structures* of each other:

Structure 4

Structure 5

Structure 6

(II) Among the resonance structures, circle the *most contributing* resonance structure.

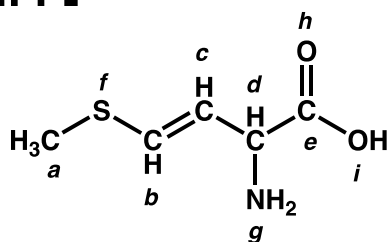
Structure 4

Structure 5

Structure 6

**Question 6** **6 points**Using the data given below, calculate the **lattice energy** for SrO(s).

	ΔH° (kJ/mol)
$\text{Sr}(s) \longrightarrow \text{Sr}(g)$	+180
$\text{Sr}(g) \longrightarrow \text{Sr}^{2+}(g) + 2e^-$	+1468
$\text{O}_2(g) \longrightarrow 2\text{O}(g)$	+500
$\text{O}(g) + 2e^- \longrightarrow \text{O}^{2-}(g)$	+600
$\text{Sr}(s) + \frac{1}{2}\text{O}_2(g) \longrightarrow \text{SrO}(s)$	-550

**Question 7 10 points**

Consider the molecule given on the left, and answer questions (i) to (vi) using chemical bonding theories that we have learnt in class (valence bond theory, hybridization).

Note that the lone pairs are not shown in the structure:

i. Consider the carbon atoms marked in as ***a***, ***b***, ***c***, ***d***, and ***e*** in the molecule, and determine **hybridization** at each of the carbon centers. (2.5 points)

Atom	Hybridization
Carbon (<i>a</i>)	
Carbon (<i>b</i>)	
Carbon (<i>c</i>)	
Carbon (<i>d</i>)	
Carbon (<i>e</i>)	

ii. Now, consider the sulfur atom (S) marked as ***f***, nitrogen (N) atom (***g***), and oxygen(O) atoms (***h*** and ***i***), and determine their **hybridization**. (2 points)

Atom	Hybridization
Sulfur (<i>f</i>)	
Nitrogen (<i>g</i>)	
Oxygen (<i>h</i>)	
Oxygen (<i>i</i>)	

iii. **Describe** all the bonds formed by the carbon atoms ***a*** and ***b***, using sigma and pi designation as appropriate. (2 points)

Atom	Bonds formed (use sigma and pi as appropriate)
Carbon (<i>a</i>)	
Carbon (<i>b</i>)	

iv. How many total sp^3 -s sigma bonds are in the molecule? (1 point)

v. Determine **electronic** and **molecular geometry** at carbon (C) labeled ***e*** and sulfur (S) labeled ***f*** centres. (2 points)

Atom	Electronic Geometry	Molecular Geometry
Carbon (<i>e</i>)		
Sulfur (<i>f</i>)		

vi. Circle the value from the choices given below, that describes the **C-S-C bond angle at sulfur** (S) marked ***f***: (0.5 points)

>90° to <109.5° 109.5° >109.5 to <120° 120° >120° to <180°

**Question 8 12 points**

Using the **valence shell orbitals only** draw a **complete** molecular orbital (MO) diagram (with no s-p mixing) for the molecule **PS**, in the space provided below. **(6 points)**

Assume that 3s and 3p atomic orbitals will form molecular orbitals like those formed by the combination of 2s and 2p atomic orbitals that we learnt in class, and there is no s-p mixing.

Show **energy arrow**, **atomic orbitals** involved, **atom labels**, the **molecular orbitals** that are formed, **filled electrons** in the MO diagram etc. Make sure to label all the atomic and molecular orbitals.

Using the MO diagram above, answer the following questions:

- (i) Which one of the following will have the highest bond order? Circle the correct choice
A) PS^{1-} B) PS C) PS^{1+}
- (ii) Which one of the following will have the longest bond length? Circle the correct choice
A) PS^{1-} B) PS C) PS^{1+}
- (iii) Which of the following will be diamagnetic? Circle the correct choice.
A) PS^{1-} B) PS C) PS^{1+}
- (iv) Draw Lewis dot structure of the molecule with the shortest bond length (based on the MO diagram) from among the three (PS^{1-} , PS, PS^{1+}), and indicate the bond order calculated from the Lewis dot structure. **(3 points)**

Lewis Dot Structure

Bond Order



AAAED2B9-3873-4329-AB1D-FCFB05AE479C

midterm-exam-93388

#865 Page 10 of 10

SCRAP PAGE

**CHEMISTRY 110 –Midterm Exam****Thursday November 7, 2024, 6:30 – 8:30 pm**

Examiners: Pallavi Sirjoosingh and Ashok Kakkar

First name (please write as legibly as possible within the boxes)

--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--

Last name

--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--

Student ID number

--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--

This booklet has 8 questions on 10 pages (The last page can be used as a scrap page).

The staple must NOT be removed from the question booklet. Your intact question booklet must be returned at the end of the exam.

A separate booklet contains a periodic table, and datasheet.

Write your answers directly under the questions in the space provided. Do not write on the QR codes or near the top edges (as it may be cut off).

Please write **legibly and dark enough** for the scan to be readable. For students who use English as a second language, dictionaries are permitted. Calculators are allowed.

DO NOT ASK QUESTIONS OF THE INVIGILATORS about interpretations of the exam questions. If you have a concern about a question, please summarize it at the top of this page. All concerns will be evaluated after the exam. GOOD LUCK!

**Question 1 8 points** (*Assume all transitions are possible*)

a. 6 points Calculate the **energy (in kJ/mol)** required to ionize (completely remove) the electron from the **n=3** level of **He⁺ ion** (He: Helium). Show your work including equations. (3 sig. figs.)

b. 2 points

(I) Select the largest (longest) wavelength from the three choices below:

- i. Wavelength of photon to *completely remove* an electron from *1s orbital of Hydrogen atom*
- ii. Wavelength of photon to *completely remove* an electron from *1s orbital of Boron atom (B)*
- iii. Wavelength of photon to *completely remove* an electron from *1s orbital of Boron cation (B⁺)*

i or ii or iii

(II) Select the smallest (shortest) wavelength from the three choices below (*for H atom*):

- i. Wavelength of photon *emitted* for an electron transitioning from *2p to 1s orbital*
- ii. Wavelength of photon *absorbed* for an electron transitioning from *2s to 4p orbital*
- iii. Wavelength of photon *absorbed* for an electron transitioning from *1s to 3p orbital*

i or ii or iii



Question 2 8 points



A Chem110 student plots the graph shown on the left, from data obtained for the kinetic energy (eV) of the ejected electron versus frequency (s^{-1}) of the incident light, for an unknown metal. ($1\text{eV} = 1.6022 \times 10^{-19} \text{ J}$)

Based on the plot, answer the questions **a to c** below:

a. 1 point On the plot above, label the **threshold frequency**.

b. 2 points What is the value of the work potential (in J) of the unknown metal?

c. 3 points What is the *longest* wavelength (in nm) required to eject an electron from this metal's surface? Show your work including equation(s) used. (3 sig figs)

d. 2 points In the space provided below, plot the following

(Assume frequency is *above* threshold frequency):

- Kinetic energy* of an ejected electron versus the *intensity* of the incident light (left)
- Number of electrons* ejected versus the *frequency* of the incident light (right)



**Question 3** **6 points****a. 2 points**

In the ground state of Ge (Germanium; $Z = 32$) atom, **name** the following orbitals:

i. Orbital containing the electron that requires the maximum energy to be removed.

ii. Orbital containing the electron that experiences the minimum effective nuclear charge.

b. 2 points

An unknown anion (A^-) has the following *excited* electronic configuration: $[Ar] 4s^2 3d^3 4p^1$

(Ar: Argon; $Z = 18$)

Give the **condensed ground state electronic configuration** of the atom, **A** and the cation, **A^{2+}** .
Show your work. (Assume all electronic transitions are possible)

Electronic Configuration of A:

Electronic Configuration of A^{2+} :

c. 2 points

In a single atom, determine how many electrons can have the following set of quantum numbers?

(i) $n = 5; l = 3$

In a single atom, determine how many electrons can have the following set of quantum numbers?

(i) $n = 3; l = 2; m_s = +1/2$

**Question 4** **8 points**

Rank the following from 1 to 4 in **increasing** order. Provide the ranking in the boxes provided.

Note: Atomic number (Z) are provided for relevant atoms

a) Increasing (lowest to highest) order of *ionization energy*: (1= lowest; 4 = highest)

Mg

Mg: Z = 12

Rb

Rb: Z = 37

Cs

Cs: Z = 55

K

K : Z = 19

b) Increasing (lowest to highest) order of *bond angle*: (1= lowest; 4 = highest)

H₂OCCl₄XeF₄

HCN

c) Increasing (smallest to largest) order of *ionic size* (1= smallest; 4 = largest)

S²⁻

S: Z = 16

O²⁻

O: Z = 8

Be²⁺

Be: Z = 4

Na⁺

Na: Z = 11

d) Increasing (lowest to highest) order of *bond energy* (1= lowest; 4 = highest)

P-F

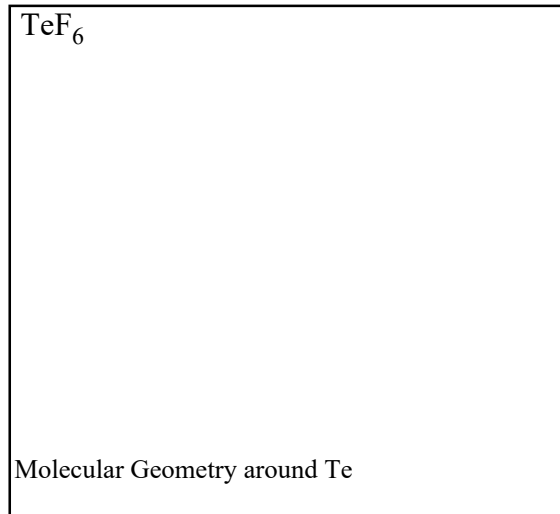
P-Cl

P-I

P-Br

**Question 5** **12 points**

a. 9 points For the molecules HClO_4 and TeF_6 , draw the most stable *Lewis structure* showing appropriate shape and *dashed/wedged bonds*, based on VSEPR. Determine the **molecular geometry** around the central atom. For HClO_4 , label the *most polar bond*.



b. 3 points (Note: Lone pairs are not shown in structures below)

(i) For the three *resonance* structures (1, 2 and 3) shown below, answer the following question:



(I) Circle the *most contributing* resonance structure.

Structure 1

Structure 2

Structure 3

(ii) For the three structures given (4, 5, and 6) below, answer the following questions:



(I) Circle all the structures that represent *resonance structures* of each other:

Structure 4

Structure 5

Structure 6

(II) Among the resonance structures, circle the *most contributing* resonance structure.

Structure 4

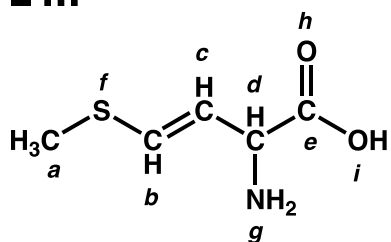
Structure 5

Structure 6

**Question 6** **6 points**

Using the data given below, calculate the **lattice energy** for SrO(s).

	ΔH° (kJ/mol)
$\text{Sr}(s) \longrightarrow \text{Sr}(g)$	+180
$\text{Sr}(g) \longrightarrow \text{Sr}^{2+}(g) + 2e^-$	+1468
$\text{O}_2(g) \longrightarrow 2\text{O}(g)$	+500
$\text{O}(g) + 2e^- \longrightarrow \text{O}^{2-}(g)$	+600
$\text{Sr}(s) + \frac{1}{2}\text{O}_2(g) \longrightarrow \text{SrO}(s)$	-550

**Question 7 10 points**

Consider the molecule given on the left, and answer questions (i) to (vi) using chemical bonding theories that we have learnt in class (valence bond theory, hybridization).

Note that the lone pairs are not shown in the structure:

i. Consider the carbon atoms marked in as ***a***, ***b***, ***c***, ***d***, and ***e*** in the molecule, and determine **hybridization** at each of the carbon centers. (2.5 points)

Atom	Hybridization
Carbon (<i>a</i>)	
Carbon (<i>b</i>)	
Carbon (<i>c</i>)	
Carbon (<i>d</i>)	
Carbon (<i>e</i>)	

ii. Now, consider the sulfur atom (S) marked as ***f***, nitrogen (N) atom (***g***), and oxygen(O) atoms (***h*** and ***i***), and determine their **hybridization**. (2 points)

Atom	Hybridization
Sulfur (<i>f</i>)	
Nitrogen (<i>g</i>)	
Oxygen (<i>h</i>)	
Oxygen (<i>i</i>)	

iii. **Describe** all the bonds formed by the carbon atoms ***a*** and ***b***, using sigma and pi designation as appropriate. (2 points)

Atom	Bonds formed (use sigma and pi as appropriate)
Carbon (<i>a</i>)	
Carbon (<i>b</i>)	

iv. How many total sp^3 -s sigma bonds are in the molecule? (1 point)

v. Determine **electronic** and **molecular geometry** at carbon (C) labeled ***e*** and sulfur (S) labeled ***f*** centres. (2 points)

Atom	Electronic Geometry	Molecular Geometry
Carbon (<i>e</i>)		
Sulfur (<i>f</i>)		

vi. Circle the value from the choices given below, that describes the **C-S-C bond angle at sulfur** (S) marked ***f***: (0.5 points)

$>90^\circ$ to $<109.5^\circ$ 109.5° >109.5 to $<120^\circ$ 120° $>120^\circ$ to $<180^\circ$

**Question 8 12 points**

Using the **valence shell orbitals only** draw a **complete** molecular orbital (MO) diagram (with no s-p mixing) for the molecule **PS**, in the space provided below. **(6 points)**

Assume that 3s and 3p atomic orbitals will form molecular orbitals like those formed by the combination of 2s and 2p atomic orbitals that we learnt in class, and there is no s-p mixing.

Show **energy arrow**, **atomic orbitals** involved, **atom labels**, the **molecular orbitals** that are formed, **filled electrons** in the MO diagram etc. Make sure to label all the atomic and molecular orbitals.

Using the MO diagram above, answer the following questions:

- (i) Which one of the following will have the highest bond order? Circle the correct choice
A) PS^{1-} B) PS C) PS^{1+}
- (ii) Which one of the following will have the longest bond length? Circle the correct choice
A) PS^{1-} B) PS C) PS^{1+}
- (iii) Which of the following will be diamagnetic? Circle the correct choice.
A) PS^{1-} B) PS C) PS^{1+}
- (iv) Draw Lewis dot structure of the molecule with the shortest bond length (based on the MO diagram) from among the three (PS^{1-} , PS, PS^{1+}), and indicate the bond order calculated from the Lewis dot structure. **(3 points)**

Lewis Dot Structure

Bond Order



6FD29A44-DB57-487D-AF03-7D2474968DED

midterm-exam-93388

#866 Page 10 of 10

SCRAP PAGE

**CHEMISTRY 110 –Midterm Exam****Thursday November 7, 2024, 6:30 – 8:30 pm****Examiners: Pallavi Sirjoosingh and Ashok Kakkar**

First name (please write as legibly as possible within the boxes)

--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--

Last name

--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--

Student ID number

--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--

This booklet has 8 questions on 10 pages (The last page can be used as a scrap page).

The staple must NOT be removed from the question booklet. Your intact question booklet must be returned at the end of the exam.

A separate booklet contains a periodic table, and datasheet.

Write your answers directly under the questions in the space provided. Do not write on the QR codes or near the top edges (as it may be cut off).

Please write **legibly and dark enough** for the scan to be readable. For students who use English as a second language, dictionaries are permitted. Calculators are allowed.

DO NOT ASK QUESTIONS OF THE INVIGILATORS about interpretations of the exam questions. If you have a concern about a question, please summarize it at the top of this page. All concerns will be evaluated after the exam. GOOD LUCK!

**Question 1 8 points** (*Assume all transitions are possible*)

a. 6 points Calculate the **energy (in kJ/mol)** required to ionize (completely remove) the electron from the **n=3** level of **He⁺ ion** (He: Helium). Show your work including equations. (3 sig. figs.)

b. 2 points

(I) Select the largest (longest) wavelength from the three choices below:

- i. Wavelength of photon to *completely remove* an electron from *1s orbital of Hydrogen atom*
- ii. Wavelength of photon to *completely remove* an electron from *1s orbital of Boron atom (B)*
- iii. Wavelength of photon to *completely remove* an electron from *1s orbital of Boron cation (B⁺)*

i or ii or iii

(II) Select the smallest (shortest) wavelength from the three choices below (*for H atom*):

- i. Wavelength of photon *emitted* for an electron transitioning from *2p to 1s orbital*
- ii. Wavelength of photon *absorbed* for an electron transitioning from *2s to 4p orbital*
- iii. Wavelength of photon *absorbed* for an electron transitioning from *1s to 3p orbital*

i or ii or iii



Question 2 8 points



A Chem110 student plots the graph shown on the left, from data obtained for the kinetic energy (eV) of the ejected electron versus frequency (s^{-1}) of the incident light, for an unknown metal. ($1\text{eV} = 1.6022 \times 10^{-19} \text{ J}$)

Based on the plot, answer the questions **a to c** below:

a. 1 point On the plot above, label the **threshold frequency**.

b. 2 points What is the value of the work potential (in J) of the unknown metal?

c. 3 points What is the *longest* wavelength (in nm) required to eject an electron from this metal's surface? Show your work including equation(s) used. (3 sig figs)

d. 2 points In the space provided below, plot the following

(Assume frequency is *above* threshold frequency):

- i. *Kinetic energy* of an ejected electron versus the *intensity* of the incident light (left)
- ii. *Number of electrons* ejected versus the *frequency* of the incident light (right)



**Question 3** **6 points****a. 2 points**

In the ground state of Ge (Germanium; $Z = 32$) atom, **name** the following orbitals:

i. Orbital containing the electron that requires the maximum energy to be removed.

ii. Orbital containing the electron that experiences the minimum effective nuclear charge.

b. 2 points

An unknown anion (A^-) has the following *excited* electronic configuration: $[Ar] 4s^2 3d^3 4p^1$

(Ar: Argon; $Z = 18$)

Give the **condensed ground state electronic configuration** of the atom, **A** and the cation, **A^{2+}** .
Show your work. (Assume all electronic transitions are possible)

Electronic Configuration of A:

Electronic Configuration of A^{2+} :

c. 2 points

In a single atom, determine how many electrons can have the following set of quantum numbers?

(i) $n = 5; l = 3$

In a single atom, determine how many electrons can have the following set of quantum numbers?

(i) $n = 3; l = 2; m_s = +1/2$

**Question 4** **8 points**

Rank the following from 1 to 4 in **increasing** order. Provide the ranking in the boxes provided.

Note: Atomic number (Z) are provided for relevant atoms

a) Increasing (lowest to highest) order of *ionization energy*: (1= lowest; 4 = highest)

Mg

Mg: Z = 12

Rb

Rb: Z = 37

Cs

Cs: Z = 55

K

K : Z = 19

b) Increasing (lowest to highest) order of *bond angle*: (1= lowest; 4 = highest)

H₂OCCl₄XeF₄

HCN

c) Increasing (smallest to largest) order of *ionic size* (1= smallest; 4 = largest)

S²⁻

S: Z = 16

O²⁻

O: Z = 8

Be²⁺

Be: Z = 4

Na⁺

Na: Z = 11

d) Increasing (lowest to highest) order of *bond energy* (1= lowest; 4 = highest)

P-F

P-Cl

P-I

P-Br

**Question 5** **12 points**

a. 9 points For the molecules HClO_4 and TeF_6 , draw the most stable *Lewis structure* showing appropriate shape and *dashed/wedged bonds*, based on VSEPR. Determine the **molecular geometry** around the central atom. For HClO_4 , label the *most polar bond*.



b. 3 points (Note: Lone pairs are not shown in structures below)

(i) For the three *resonance* structures (1, 2 and 3) shown below, answer the following question:



(I) Circle the *most contributing* resonance structure.

Structure 1

Structure 2

Structure 3

(ii) For the three structures given (4, 5, and 6) below, answer the following questions:



(I) Circle all the structures that represent *resonance structures* of each other:

Structure 4

Structure 5

Structure 6

(II) Among the resonance structures, circle the *most contributing* resonance structure.

Structure 4

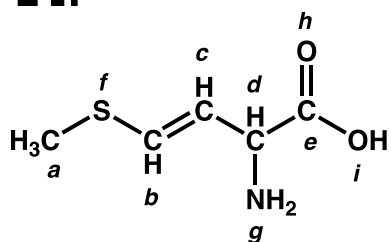
Structure 5

Structure 6

**Question 6** **6 points**

Using the data given below, calculate the **lattice energy** for SrO(s).

	ΔH° (kJ/mol)
$\text{Sr}(s) \longrightarrow \text{Sr}(g)$	+180
$\text{Sr}(g) \longrightarrow \text{Sr}^{2+}(g) + 2e^-$	+1468
$\text{O}_2(g) \longrightarrow 2\text{O}(g)$	+500
$\text{O}(g) + 2e^- \longrightarrow \text{O}^{2-}(g)$	+600
$\text{Sr}(s) + \frac{1}{2}\text{O}_2(g) \longrightarrow \text{SrO}(s)$	-550

**Question 7 10 points**

Consider the molecule given on the left, and answer questions (i) to (vi) using chemical bonding theories that we have learnt in class (valence bond theory, hybridization).

Note that the lone pairs are not shown in the structure:

i. Consider the carbon atoms marked in as ***a***, ***b***, ***c***, ***d***, and ***e*** in the molecule, and determine **hybridization** at each of the carbon centers. (2.5 points)

Atom	Hybridization
Carbon (<i>a</i>)	
Carbon (<i>b</i>)	
Carbon (<i>c</i>)	
Carbon (<i>d</i>)	
Carbon (<i>e</i>)	

ii. Now, consider the sulfur atom (S) marked as ***f***, nitrogen (N) atom (***g***), and oxygen(O) atoms (***h*** and ***i***), and determine their **hybridization**. (2 points)

Atom	Hybridization
Sulfur (<i>f</i>)	
Nitrogen (<i>g</i>)	
Oxygen (<i>h</i>)	
Oxygen (<i>i</i>)	

iii. **Describe** all the bonds formed by the carbon atoms ***a*** and ***b***, using sigma and pi designation as appropriate. (2 points)

Atom	Bonds formed (use sigma and pi as appropriate)
Carbon (<i>a</i>)	
Carbon (<i>b</i>)	

iv. How many total sp^3 -s sigma bonds are in the molecule? (1 point)

v. Determine **electronic** and **molecular geometry** at carbon (C) labeled ***e*** and sulfur (S) labeled ***f*** centres. (2 points)

Atom	Electronic Geometry	Molecular Geometry
Carbon (<i>e</i>)		
Sulfur (<i>f</i>)		

vi. Circle the value from the choices given below, that describes the **C-S-C bond angle at sulfur** (S) marked ***f***: (0.5 points)

>90° to <109.5° 109.5° >109.5 to <120° 120° >120° to <180°


Question 8 12 points

Using the **valence shell orbitals only** draw a **complete** molecular orbital (MO) diagram (with no s-p mixing) for the molecule **PS**, in the space provided below. **(6 points)**

Assume that 3s and 3p atomic orbitals will form molecular orbitals like those formed by the combination of 2s and 2p atomic orbitals that we learnt in class, and there is no s-p mixing.

Show **energy arrow**, **atomic orbitals** involved, **atom labels**, the **molecular orbitals** that are formed, **filled electrons** in the MO diagram etc. Make sure to label all the atomic and molecular orbitals.

Using the MO diagram above, answer the following questions:

- (i) Which one of the following will have the highest bond order? Circle the correct choice
 A) PS^{1-} B) PS C) PS^{1+}
- (ii) Which one of the following will have the longest bond length? Circle the correct choice
 A) PS^{1-} B) PS C) PS^{1+}
- (iii) Which of the following will be diamagnetic? Circle the correct choice.
 A) PS^{1-} B) PS C) PS^{1+}
- (iv) Draw Lewis dot structure of the molecule with the shortest bond length (based on the MO diagram) from among the three (PS^{1-} , PS, PS^{1+}), and indicate the bond order calculated from the Lewis dot structure. **(3 points)**

Lewis Dot Structure

Bond Order



E1D9FFA8-C062-4565-8775-9D5E5DB041B0

midterm-exam-93388

#867 Page 10 of 10

SCRAP PAGE

**CHEMISTRY 110 –Midterm Exam****Thursday November 7, 2024, 6:30 – 8:30 pm**

Examiners: Pallavi Sirjoosingh and Ashok Kakkar

First name (please write as legibly as possible within the boxes)

--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--

Last name

--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--

Student ID number

--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--

This booklet has 8 questions on 10 pages (The last page can be used as a scrap page).

The staple must NOT be removed from the question booklet. Your intact question booklet must be returned at the end of the exam.

A separate booklet contains a periodic table, and datasheet.

Write your answers directly under the questions in the space provided. Do not write on the QR codes or near the top edges (as it may be cut off).

Please write **legibly and dark enough** for the scan to be readable. For students who use English as a second language, dictionaries are permitted. Calculators are allowed.

DO NOT ASK QUESTIONS OF THE INVIGILATORS about interpretations of the exam questions. If you have a concern about a question, please summarize it at the top of this page. All concerns will be evaluated after the exam. GOOD LUCK!

**Question 1 8 points** (*Assume all transitions are possible*)

a. 6 points Calculate the **energy (in kJ/mol)** required to ionize (completely remove) the electron from the **n=3** level of **He⁺ ion** (He: Helium). Show your work including equations. (3 sig. figs.)

b. 2 points

(I) Select the largest (longest) wavelength from the three choices below:

- i. Wavelength of photon to *completely remove* an electron from *1s orbital of Hydrogen atom*
- ii. Wavelength of photon to *completely remove* an electron from *1s orbital of Boron atom (B)*
- iii. Wavelength of photon to *completely remove* an electron from *1s orbital of Boron cation (B⁺)*

i or ii or iii

(II) Select the smallest (shortest) wavelength from the three choices below (*for H atom*):

- i. Wavelength of photon *emitted* for an electron transitioning from *2p to 1s orbital*
- ii. Wavelength of photon *absorbed* for an electron transitioning from *2s to 4p orbital*
- iii. Wavelength of photon *absorbed* for an electron transitioning from *1s to 3p orbital*

i or ii or iii



Question 2 8 points



A Chem110 student plots the graph shown on the left, from data obtained for the kinetic energy (eV) of the ejected electron versus frequency (s^{-1}) of the incident light, for an unknown metal. ($1\text{eV} = 1.6022 \times 10^{-19} \text{ J}$)

Based on the plot, answer the questions **a to c** below:

a. 1 point On the plot above, label the **threshold frequency**.

b. 2 points What is the value of the work potential (in J) of the unknown metal?

c. 3 points What is the *longest* wavelength (in nm) required to eject an electron from this metal's surface? Show your work including equation(s) used. (3 sig figs)

d. 2 points In the space provided below, plot the following

(Assume frequency is *above* threshold frequency):

- Kinetic energy* of an ejected electron versus the *intensity* of the incident light (left)
- Number of electrons* ejected versus the *frequency* of the incident light (right)



**Question 3** **6 points****a. 2 points**

In the ground state of Ge (Germanium; $Z = 32$) atom, **name** the following orbitals:

i. Orbital containing the electron that requires the maximum energy to be removed.

ii. Orbital containing the electron that experiences the minimum effective nuclear charge.

b. 2 points

An unknown anion (A^-) has the following *excited* electronic configuration: $[Ar] 4s^2 3d^3 4p^1$

(Ar: Argon; $Z = 18$)

Give the **condensed ground state electronic configuration** of the atom, **A** and the cation, **A^{2+}** .
Show your work. (Assume all electronic transitions are possible)

Electronic Configuration of A:

Electronic Configuration of A^{2+} :

c. 2 points

In a single atom, determine how many electrons can have the following set of quantum numbers?

(i) $n = 5; l = 3$

In a single atom, determine how many electrons can have the following set of quantum numbers?

(i) $n = 3; l = 2; m_s = +1/2$

**Question 4** **8 points**

Rank the following from 1 to 4 in **increasing** order. Provide the ranking in the boxes provided.

Note: Atomic number (Z) are provided for relevant atoms

a) Increasing (lowest to highest) order of *ionization energy*: (1= lowest; 4 = highest)

Mg

Mg: Z = 12

Rb

Rb: Z = 37

Cs

Cs: Z = 55

K

K : Z = 19

b) Increasing (lowest to highest) order of *bond angle*: (1= lowest; 4 = highest)

H₂OCCl₄XeF₄

HCN

c) Increasing (smallest to largest) order of *ionic size* (1= smallest; 4 = largest)

S²⁻

S: Z = 16

O²⁻

O: Z = 8

Be²⁺

Be: Z = 4

Na⁺

Na: Z = 11

d) Increasing (lowest to highest) order of *bond energy* (1= lowest; 4 = highest)

P-F

P-Cl

P-I

P-Br

**Question 5** **12 points**

a. 9 points For the molecules HClO_4 and TeF_6 , draw the most stable *Lewis structure* showing appropriate shape and *dashed/wedged bonds*, based on VSEPR. Determine the **molecular geometry** around the central atom. For HClO_4 , label the *most polar bond*.



b. 3 points (Note: Lone pairs are not shown in structures below)

(i) For the three *resonance* structures (1, 2 and 3) shown below, answer the following question:



(I) Circle the *most contributing* resonance structure.

Structure 1

Structure 2

Structure 3

(ii) For the three structures given (4, 5, and 6) below, answer the following questions:



(I) Circle all the structures that represent *resonance structures* of each other:

Structure 4

Structure 5

Structure 6

(II) Among the resonance structures, circle the *most contributing* resonance structure.

Structure 4

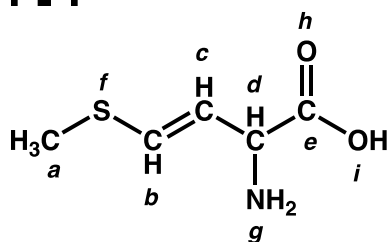
Structure 5

Structure 6

**Question 6** **6 points**

Using the data given below, calculate the **lattice energy** for SrO(s).

	ΔH° (kJ/mol)
$\text{Sr}(s) \longrightarrow \text{Sr}(g)$	+180
$\text{Sr}(g) \longrightarrow \text{Sr}^{2+}(g) + 2e^-$	+1468
$\text{O}_2(g) \longrightarrow 2\text{O}(g)$	+500
$\text{O}(g) + 2e^- \longrightarrow \text{O}^{2-}(g)$	+600
$\text{Sr}(s) + \frac{1}{2}\text{O}_2(g) \longrightarrow \text{SrO}(s)$	-550

**Question 7 10 points**

Consider the molecule given on the left, and answer questions (i) to (vi) using chemical bonding theories that we have learnt in class (valence bond theory, hybridization).

Note that the lone pairs are not shown in the structure:

i. Consider the carbon atoms marked in as ***a***, ***b***, ***c***, ***d***, and ***e*** in the molecule, and determine **hybridization** at each of the carbon centers. (2.5 points)

Atom	Hybridization
Carbon (<i>a</i>)	
Carbon (<i>b</i>)	
Carbon (<i>c</i>)	
Carbon (<i>d</i>)	
Carbon (<i>e</i>)	

ii. Now, consider the sulfur atom (S) marked as ***f***, nitrogen (N) atom (***g***), and oxygen(O) atoms (***h*** and ***i***), and determine their **hybridization**. (2 points)

Atom	Hybridization
Sulfur (<i>f</i>)	
Nitrogen (<i>g</i>)	
Oxygen (<i>h</i>)	
Oxygen (<i>i</i>)	

iii. **Describe** all the bonds formed by the carbon atoms ***a*** and ***b***, using sigma and pi designation as appropriate. (2 points)

Atom	Bonds formed (use sigma and pi as appropriate)
Carbon (<i>a</i>)	
Carbon (<i>b</i>)	

iv. How many total sp^3 -s sigma bonds are in the molecule? (1 point)

v. Determine **electronic** and **molecular geometry** at carbon (C) labeled ***e*** and sulfur (S) labeled ***f*** centres. (2 points)

Atom	Electronic Geometry	Molecular Geometry
Carbon (<i>e</i>)		
Sulfur (<i>f</i>)		

vi. Circle the value from the choices given below, that describes the **C-S-C bond angle at sulfur** (S) marked ***f***: (0.5 points)

>90° to <109.5° 109.5° >109.5 to <120° 120° >120° to <180°



Question 8 12 points

Using the **valence shell orbitals only** draw a **complete** molecular orbital (MO) diagram (with no s-p mixing) for the molecule **PS**, in the space provided below. **(6 points)**

Assume that 3s and 3p atomic orbitals will form molecular orbitals like those formed by the combination of 2s and 2p atomic orbitals that we learnt in class, and there is no s-p mixing.

Show **energy arrow**, **atomic orbitals** involved, **atom labels**, the **molecular orbitals** that are formed, **filled electrons** in the MO diagram etc. Make sure to label all the atomic and molecular orbitals.

Using the MO diagram above, answer the following questions:

- (i) Which one of the following will have the highest bond order? Circle the correct choice
 A) PS^{1-} B) PS C) PS^{1+}
- (ii) Which one of the following will have the longest bond length? Circle the correct choice
 A) PS^{1-} B) PS C) PS^{1+}
- (iii) Which of the following will be diamagnetic? Circle the correct choice.
 A) PS^{1-} B) PS C) PS^{1+}
- (iv) Draw Lewis dot structure of the molecule with the shortest bond length (based on the MO diagram) from among the three (PS^{1-} , PS, PS^{1+}), and indicate the bond order calculated from the Lewis dot structure. **(3 points)**

Lewis Dot Structure

Bond Order



98C18284-9EEF-47E4-8071-37E6D1F90364

midterm-exam-93388

#868 Page 10 of 10

SCRAP PAGE

**CHEMISTRY 110 –Midterm Exam****Thursday November 7, 2024, 6:30 – 8:30 pm**

Examiners: Pallavi Sirjoosingh and Ashok Kakkar

First name (please write as legibly as possible within the boxes)

--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--

Last name

--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--

Student ID number

--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--

This booklet has 8 questions on 10 pages (The last page can be used as a scrap page).

The staple must NOT be removed from the question booklet. Your intact question booklet must be returned at the end of the exam.

A separate booklet contains a periodic table, and datasheet.

Write your answers directly under the questions in the space provided. Do not write on the QR codes or near the top edges (as it may be cut off).

Please write **legibly and dark enough** for the scan to be readable. For students who use English as a second language, dictionaries are permitted. Calculators are allowed.

DO NOT ASK QUESTIONS OF THE INVIGILATORS about interpretations of the exam questions. If you have a concern about a question, please summarize it at the top of this page. All concerns will be evaluated after the exam. GOOD LUCK!

**Question 1 8 points** (*Assume all transitions are possible*)

a. 6 points Calculate the **energy (in kJ/mol)** required to ionize (completely remove) the electron from the **$n=3$** level of **He^+ ion** (He: Helium). Show your work including equations. (3 sig. figs.)

b. 2 points

(I) Select the largest (longest) wavelength from the three choices below:

- i. Wavelength of photon to *completely remove* an electron from *1s orbital of Hydrogen atom*
- ii. Wavelength of photon to *completely remove* an electron from *1s orbital of Boron atom (B)*
- iii. Wavelength of photon to *completely remove* an electron from *1s orbital of Boron cation (B^+)*

i or ii or iii

(II) Select the smallest (shortest) wavelength from the three choices below (*for H atom*):

- i. Wavelength of photon *emitted* for an electron transitioning from *2p to 1s orbital*
- ii. Wavelength of photon *absorbed* for an electron transitioning from *2s to 4p orbital*
- iii. Wavelength of photon *absorbed* for an electron transitioning from *1s to 3p orbital*

i or ii or iii



Question 2 8 points



A Chem110 student plots the graph shown on the left, from data obtained for the kinetic energy (eV) of the ejected electron versus frequency (s^{-1}) of the incident light, for an unknown metal. ($1\text{eV} = 1.6022 \times 10^{-19} \text{ J}$)

Based on the plot, answer the questions **a to c** below:

a. 1 point On the plot above, label the **threshold frequency**.

b. 2 points What is the value of the work potential (in J) of the unknown metal?

c. 3 points What is the *longest* wavelength (in nm) required to eject an electron from this metal's surface? Show your work including equation(s) used. (3 sig figs)

d. 2 points In the space provided below, plot the following

(Assume frequency is *above* threshold frequency):

- Kinetic energy* of an ejected electron versus the *intensity* of the incident light (left)
- Number of electrons* ejected versus the *frequency* of the incident light (right)



**Question 3** **6 points****a. 2 points**

In the ground state of Ge (Germanium; $Z = 32$) atom, **name** the following orbitals:

i. Orbital containing the electron that requires the maximum energy to be removed.

ii. Orbital containing the electron that experiences the minimum effective nuclear charge.

b. 2 points

An unknown anion (A^-) has the following *excited* electronic configuration: $[Ar] 4s^2 3d^3 4p^1$

(Ar: Argon; $Z = 18$)

Give the **condensed ground state electronic configuration** of the atom, **A** and the cation, **A^{2+}** .
Show your work. (Assume all electronic transitions are possible)

Electronic Configuration of A:

Electronic Configuration of A^{2+} :

c. 2 points

In a single atom, determine how many electrons can have the following set of quantum numbers?

(i) $n = 5; l = 3$

In a single atom, determine how many electrons can have the following set of quantum numbers?

(i) $n = 3; l = 2; m_s = +1/2$

**Question 4** **8 points**

Rank the following from 1 to 4 in **increasing** order. Provide the ranking in the boxes provided.

Note: Atomic number (Z) are provided for relevant atoms

a) Increasing (lowest to highest) order of *ionization energy*: (1= lowest; 4 = highest)

Mg

Mg: Z = 12

Rb

Rb: Z = 37

Cs

Cs: Z = 55

K

K : Z = 19

b) Increasing (lowest to highest) order of *bond angle*: (1= lowest; 4 = highest)

H₂OCCl₄XeF₄

HCN

c) Increasing (smallest to largest) order of *ionic size* (1= smallest; 4 = largest)

S²⁻

S: Z = 16

O²⁻

O: Z = 8

Be²⁺

Be: Z = 4

Na⁺

Na: Z = 11

d) Increasing (lowest to highest) order of *bond energy* (1= lowest; 4 = highest)

P-F

P-Cl

P-I

P-Br

**Question 5** **12 points**

a. 9 points For the molecules HClO_4 and TeF_6 , draw the most stable *Lewis structure* showing appropriate shape and *dashed/wedged bonds*, based on VSEPR. Determine the **molecular geometry** around the central atom. For HClO_4 , label the *most polar bond*.



b. 3 points (Note: Lone pairs are not shown in structures below)

(i) For the three *resonance* structures (1, 2 and 3) shown below, answer the following question:



(I) Circle the *most contributing* resonance structure.

Structure 1

Structure 2

Structure 3

(ii) For the three structures given (4, 5, and 6) below, answer the following questions:



(I) Circle all the structures that represent *resonance structures* of each other:

Structure 4

Structure 5

Structure 6

(II) Among the resonance structures, circle the *most contributing* resonance structure.

Structure 4

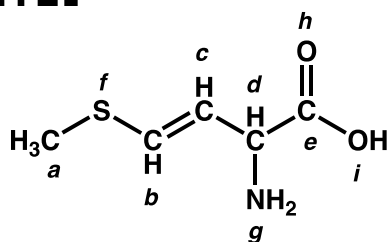
Structure 5

Structure 6

**Question 6** **6 points**

Using the data given below, calculate the **lattice energy** for SrO(s).

	ΔH° (kJ/mol)
$\text{Sr}(s) \longrightarrow \text{Sr}(g)$	+180
$\text{Sr}(g) \longrightarrow \text{Sr}^{2+}(g) + 2e^-$	+1468
$\text{O}_2(g) \longrightarrow 2\text{O}(g)$	+500
$\text{O}(g) + 2e^- \longrightarrow \text{O}^{2-}(g)$	+600
$\text{Sr}(s) + \frac{1}{2}\text{O}_2(g) \longrightarrow \text{SrO}(s)$	-550

**Question 7 10 points**

Consider the molecule given on the left, and answer questions (i) to (vi) using chemical bonding theories that we have learnt in class (valence bond theory, hybridization).

Note that the lone pairs are not shown in the structure:

i. Consider the carbon atoms marked in as ***a***, ***b***, ***c***, ***d***, and ***e*** in the molecule, and determine **hybridization** at each of the carbon centers. (2.5 points)

Atom	Hybridization
Carbon (<i>a</i>)	
Carbon (<i>b</i>)	
Carbon (<i>c</i>)	
Carbon (<i>d</i>)	
Carbon (<i>e</i>)	

ii. Now, consider the sulfur atom (S) marked as ***f***, nitrogen (N) atom (***g***), and oxygen(O) atoms (***h*** and ***i***), and determine their **hybridization**. (2 points)

Atom	Hybridization
Sulfur (<i>f</i>)	
Nitrogen (<i>g</i>)	
Oxygen (<i>h</i>)	
Oxygen (<i>i</i>)	

iii. **Describe** all the bonds formed by the carbon atoms ***a*** and ***b***, using sigma and pi designation as appropriate. (2 points)

Atom	Bonds formed (use sigma and pi as appropriate)
Carbon (<i>a</i>)	
Carbon (<i>b</i>)	

iv. How many total sp^3 -s sigma bonds are in the molecule? (1 point)

v. Determine **electronic** and **molecular geometry** at carbon (C) labeled ***e*** and sulfur (S) labeled ***f*** centres. (2 points)

Atom	Electronic Geometry	Molecular Geometry
Carbon (<i>e</i>)		
Sulfur (<i>f</i>)		

vi. Circle the value from the choices given below, that describes the **C-S-C bond angle at sulfur** (S) marked ***f***: (0.5 points)

>90° to <109.5° 109.5° >109.5 to <120° 120° >120° to <180°


Question 8 12 points

Using the **valence shell orbitals only** draw a **complete** molecular orbital (MO) diagram (with no s-p mixing) for the molecule **PS**, in the space provided below. **(6 points)**

Assume that 3s and 3p atomic orbitals will form molecular orbitals like those formed by the combination of 2s and 2p atomic orbitals that we learnt in class, and there is no s-p mixing.

Show **energy arrow**, **atomic orbitals** involved, **atom labels**, the **molecular orbitals** that are formed, **filled electrons** in the MO diagram etc. Make sure to label all the atomic and molecular orbitals.

Using the MO diagram above, answer the following questions:

- (i) Which one of the following will have the highest bond order? Circle the correct choice
 A) PS^{1-} B) PS C) PS^{1+}
- (ii) Which one of the following will have the longest bond length? Circle the correct choice
 A) PS^{1-} B) PS C) PS^{1+}
- (iii) Which of the following will be diamagnetic? Circle the correct choice.
 A) PS^{1-} B) PS C) PS^{1+}
- (iv) Draw Lewis dot structure of the molecule with the shortest bond length (based on the MO diagram) from among the three (PS^{1-} , PS, PS^{1+}), and indicate the bond order calculated from the Lewis dot structure. **(3 points)**

Lewis Dot Structure

Bond Order



35FF0CF5-E956-44AB-BD99-BCFDA2EAB033

midterm-exam-93388

#869 Page 10 of 10

SCRAP PAGE

**CHEMISTRY 110 –Midterm Exam****Thursday November 7, 2024, 6:30 – 8:30 pm****Examiners: Pallavi Sirjoosingh and Ashok Kakkar**

First name (please write as legibly as possible within the boxes)

--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--

Last name

--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--

Student ID number

--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--

This booklet has 8 questions on 10 pages (The last page can be used as a scrap page).

The staple must NOT be removed from the question booklet. Your intact question booklet must be returned at the end of the exam.

A separate booklet contains a periodic table, and datasheet.

Write your answers directly under the questions in the space provided. Do not write on the QR codes or near the top edges (as it may be cut off).

Please write **legibly and dark enough** for the scan to be readable. For students who use English as a second language, dictionaries are permitted. Calculators are allowed.

DO NOT ASK QUESTIONS OF THE INVIGILATORS about interpretations of the exam questions. If you have a concern about a question, please summarize it at the top of this page. All concerns will be evaluated after the exam. GOOD LUCK!

**Question 1 8 points** (*Assume all transitions are possible*)

a. 6 points Calculate the **energy (in kJ/mol)** required to ionize (completely remove) the electron from the **$n=3$** level of **He^+ ion** (He: Helium). Show your work including equations. (3 sig. figs.)

b. 2 points

(I) Select the largest (longest) wavelength from the three choices below:

- i. Wavelength of photon to *completely remove* an electron from *1s orbital of Hydrogen atom*
- ii. Wavelength of photon to *completely remove* an electron from *1s orbital of Boron atom (B)*
- iii. Wavelength of photon to *completely remove* an electron from *1s orbital of Boron cation (B^+)*

i or ii or iii

(II) Select the smallest (shortest) wavelength from the three choices below (*for H atom*):

- i. Wavelength of photon *emitted* for an electron transitioning from *2p to 1s orbital*
- ii. Wavelength of photon *absorbed* for an electron transitioning from *2s to 4p orbital*
- iii. Wavelength of photon *absorbed* for an electron transitioning from *1s to 3p orbital*

i or ii or iii



Question 2 8 points



A Chem110 student plots the graph shown on the left, from data obtained for the kinetic energy (eV) of the ejected electron versus frequency (s^{-1}) of the incident light, for an unknown metal. ($1\text{eV} = 1.6022 \times 10^{-19} \text{ J}$)

Based on the plot, answer the questions **a to c** below:

a. 1 point On the plot above, label the **threshold frequency**.

b. 2 points What is the value of the work potential (in J) of the unknown metal?

c. 3 points What is the *longest* wavelength (in nm) required to eject an electron from this metal's surface? Show your work including equation(s) used. (3 sig figs)

d. 2 points In the space provided below, plot the following

(Assume frequency is *above* threshold frequency):

- i. *Kinetic energy* of an ejected electron versus the *intensity* of the incident light (left)
- ii. *Number of electrons* ejected versus the *frequency* of the incident light (right)



**Question 3** **6 points****a. 2 points**

In the ground state of Ge (Germanium; $Z = 32$) atom, **name** the following orbitals:

i. Orbital containing the electron that requires the maximum energy to be removed.

ii. Orbital containing the electron that experiences the minimum effective nuclear charge.

b. 2 points

An unknown anion (A^-) has the following *excited* electronic configuration: $[Ar] 4s^2 3d^3 4p^1$

(Ar: Argon; $Z = 18$)

Give the **condensed ground state electronic configuration** of the atom, **A** and the cation, **A^{2+}** .
Show your work. (Assume all electronic transitions are possible)

Electronic Configuration of A:

Electronic Configuration of A^{2+} :

c. 2 points

In a single atom, determine how many electrons can have the following set of quantum numbers?

(i) $n = 5; l = 3$

In a single atom, determine how many electrons can have the following set of quantum numbers?

(i) $n = 3; l = 2; m_s = +1/2$

**Question 4** **8 points**

Rank the following from 1 to 4 in **increasing** order. Provide the ranking in the boxes provided.

Note: Atomic number (Z) are provided for relevant atoms

a) Increasing (lowest to highest) order of *ionization energy*: (1= lowest; 4 = highest)

Mg

Mg: Z = 12

Rb

Rb: Z = 37

Cs

Cs: Z = 55

K

K : Z = 19

b) Increasing (lowest to highest) order of *bond angle*: (1= lowest; 4 = highest)

H₂OCCl₄XeF₄

HCN

c) Increasing (smallest to largest) order of *ionic size* (1= smallest; 4 = largest)

S²⁻

S: Z = 16

O²⁻

O: Z = 8

Be²⁺

Be: Z = 4

Na⁺

Na: Z = 11

d) Increasing (lowest to highest) order of *bond energy* (1= lowest; 4 = highest)

P-F

P-Cl

P-I

P-Br

**Question 5** **12 points**

a. 9 points For the molecules HClO_4 and TeF_6 , draw the most stable *Lewis structure* showing appropriate shape and *dashed/wedged bonds*, based on VSEPR. Determine the **molecular geometry** around the central atom. For HClO_4 , label the *most polar bond*.



b. 3 points (Note: Lone pairs are not shown in structures below)

(i) For the three *resonance* structures (1, 2 and 3) shown below, answer the following question:



(I) Circle the *most contributing* resonance structure.

Structure 1

Structure 2

Structure 3

(ii) For the three structures given (4, 5, and 6) below, answer the following questions:



(I) Circle all the structures that represent *resonance structures* of each other:

Structure 4

Structure 5

Structure 6

(II) Among the resonance structures, circle the *most contributing* resonance structure.

Structure 4

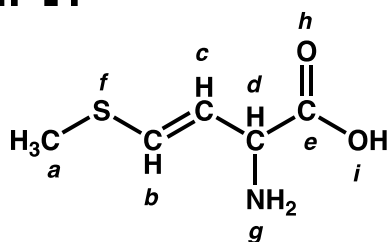
Structure 5

Structure 6

**Question 6** **6 points**

Using the data given below, calculate the **lattice energy** for SrO(s).

	ΔH° (kJ/mol)
$\text{Sr}(s) \longrightarrow \text{Sr}(g)$	+180
$\text{Sr}(g) \longrightarrow \text{Sr}^{2+}(g) + 2e^-$	+1468
$\text{O}_2(g) \longrightarrow 2\text{O}(g)$	+500
$\text{O}(g) + 2e^- \longrightarrow \text{O}^{2-}(g)$	+600
$\text{Sr}(s) + \frac{1}{2}\text{O}_2(g) \longrightarrow \text{SrO}(s)$	-550

**Question 7 10 points**

Consider the molecule given on the left, and answer questions (i) to (vi) using chemical bonding theories that we have learnt in class (valence bond theory, hybridization).

Note that the lone pairs are not shown in the structure:

i. Consider the carbon atoms marked in as ***a***, ***b***, ***c***, ***d***, and ***e*** in the molecule, and determine **hybridization** at each of the carbon centers. (2.5 points)

Atom	Hybridization
Carbon (<i>a</i>)	
Carbon (<i>b</i>)	
Carbon (<i>c</i>)	
Carbon (<i>d</i>)	
Carbon (<i>e</i>)	

ii. Now, consider the sulfur atom (S) marked as ***f***, nitrogen (N) atom (***g***), and oxygen(O) atoms (***h*** and ***i***), and determine their **hybridization**. (2 points)

Atom	Hybridization
Sulfur (<i>f</i>)	
Nitrogen (<i>g</i>)	
Oxygen (<i>h</i>)	
Oxygen (<i>i</i>)	

iii. **Describe** all the bonds formed by the carbon atoms ***a*** and ***b***, using sigma and pi designation as appropriate. (2 points)

Atom	Bonds formed (use sigma and pi as appropriate)
Carbon (<i>a</i>)	
Carbon (<i>b</i>)	

iv. How many total sp^3 -s sigma bonds are in the molecule? (1 point)

v. Determine **electronic** and **molecular geometry** at carbon (C) labeled ***e*** and sulfur (S) labeled ***f*** centres. (2 points)

Atom	Electronic Geometry	Molecular Geometry
Carbon (<i>e</i>)		
Sulfur (<i>f</i>)		

vi. Circle the value from the choices given below, that describes the **C-S-C bond angle at sulfur** (S) marked ***f***: (0.5 points)

>90° to <109.5°

109.5°

>109.5 to <120°

120°

>120° to <180°



Question 8 12 points

Using the **valence shell orbitals only** draw a **complete** molecular orbital (MO) diagram (with no s-p mixing) for the molecule **PS**, in the space provided below. **(6 points)**

Assume that 3s and 3p atomic orbitals will form molecular orbitals like those formed by the combination of 2s and 2p atomic orbitals that we learnt in class, and there is no s-p mixing.

Show **energy arrow**, **atomic orbitals** involved, **atom labels**, the **molecular orbitals** that are formed, **filled electrons** in the MO diagram etc. Make sure to label all the atomic and molecular orbitals.

Using the MO diagram above, answer the following questions:

- (i) Which one of the following will have the highest bond order? Circle the correct choice
 A) PS^{1-} B) PS C) PS^{1+}
- (ii) Which one of the following will have the longest bond length? Circle the correct choice
 A) PS^{1-} B) PS C) PS^{1+}
- (iii) Which of the following will be diamagnetic? Circle the correct choice.
 A) PS^{1-} B) PS C) PS^{1+}
- (iv) Draw Lewis dot structure of the molecule with the shortest bond length (based on the MO diagram) from among the three (PS^{1-} , PS, PS^{1+}), and indicate the bond order calculated from the Lewis dot structure. **(3 points)**

Lewis Dot Structure

Bond Order



9450E847-5504-4C9B-883A-EF31E20B9100

midterm-exam-93388

#870 Page 10 of 10

SCRAP PAGE

**CHEMISTRY 110 –Midterm Exam****Thursday November 7, 2024, 6:30 – 8:30 pm**

Examiners: Pallavi Sirjoosingh and Ashok Kakkar

First name (please write as legibly as possible within the boxes)

--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--

Last name

--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--

Student ID number

--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--

This booklet has 8 questions on 10 pages (The last page can be used as a scrap page).

The staple must NOT be removed from the question booklet. Your intact question booklet must be returned at the end of the exam.

A separate booklet contains a periodic table, and datasheet.

Write your answers directly under the questions in the space provided. Do not write on the QR codes or near the top edges (as it may be cut off).

Please write **legibly and dark enough** for the scan to be readable. For students who use English as a second language, dictionaries are permitted. Calculators are allowed.

DO NOT ASK QUESTIONS OF THE INVIGILATORS about interpretations of the exam questions. If you have a concern about a question, please summarize it at the top of this page. All concerns will be evaluated after the exam. GOOD LUCK!

**Question 1 8 points** (*Assume all transitions are possible*)

a. 6 points Calculate the **energy (in kJ/mol)** required to ionize (completely remove) the electron from the **$n=3$** level of **He^+ ion** (He: Helium). Show your work including equations. (3 sig. figs.)

b. 2 points

(I) Select the largest (longest) wavelength from the three choices below:

- i. Wavelength of photon to *completely remove* an electron from *1s orbital of Hydrogen atom*
- ii. Wavelength of photon to *completely remove* an electron from *1s orbital of Boron atom (B)*
- iii. Wavelength of photon to *completely remove* an electron from *1s orbital of Boron cation (B^+)*

i or ii or iii

(II) Select the smallest (shortest) wavelength from the three choices below (*for H atom*):

- i. Wavelength of photon *emitted* for an electron transitioning from *2p to 1s orbital*
- ii. Wavelength of photon *absorbed* for an electron transitioning from *2s to 4p orbital*
- iii. Wavelength of photon *absorbed* for an electron transitioning from *1s to 3p orbital*

i or ii or iii



Question 2 8 points



A Chem110 student plots the graph shown on the left, from data obtained for the kinetic energy (eV) of the ejected electron versus frequency (s^{-1}) of the incident light, for an unknown metal. ($1\text{eV} = 1.6022 \times 10^{-19} \text{ J}$)

Based on the plot, answer the questions **a to c** below:

a. 1 point On the plot above, label the **threshold frequency**.

b. 2 points What is the value of the work potential (in J) of the unknown metal?

c. 3 points What is the *longest* wavelength (in nm) required to eject an electron from this metal's surface? Show your work including equation(s) used. (3 sig figs)

d. 2 points In the space provided below, plot the following

(Assume frequency is *above* threshold frequency):

- Kinetic energy* of an ejected electron versus the *intensity* of the incident light (left)
- Number of electrons* ejected versus the *frequency* of the incident light (right)



**Question 3** **6 points****a. 2 points**

In the ground state of Ge (Germanium; $Z = 32$) atom, **name** the following orbitals:

i. Orbital containing the electron that requires the maximum energy to be removed.

ii. Orbital containing the electron that experiences the minimum effective nuclear charge.

b. 2 points

An unknown anion (A^-) has the following *excited* electronic configuration: $[Ar] 4s^2 3d^3 4p^1$

(Ar: Argon; $Z = 18$)

Give the **condensed ground state electronic configuration** of the atom, **A** and the cation, A^{2+} .
Show your work. (Assume all electronic transitions are possible)

Electronic Configuration of A:

Electronic Configuration of A^{2+} :

c. 2 points

In a single atom, determine how many electrons can have the following set of quantum numbers?

(i) $n = 5; l = 3$

In a single atom, determine how many electrons can have the following set of quantum numbers?

(i) $n = 3; l = 2; m_s = +1/2$

**Question 4** **8 points**

Rank the following from 1 to 4 in **increasing** order. Provide the ranking in the boxes provided.

Note: Atomic number (Z) are provided for relevant atoms

a) Increasing (lowest to highest) order of *ionization energy*: (1= lowest; 4 = highest)

Mg

Mg: Z = 12

Rb

Rb: Z = 37

Cs

Cs: Z = 55

K

K : Z = 19

b) Increasing (lowest to highest) order of *bond angle*: (1= lowest; 4 = highest)

H₂OCCl₄XeF₄

HCN

c) Increasing (smallest to largest) order of *ionic size* (1= smallest; 4 = largest)

S²⁻

S: Z = 16

O²⁻

O: Z = 8

Be²⁺

Be: Z = 4

Na⁺

Na: Z = 11

d) Increasing (lowest to highest) order of *bond energy* (1= lowest; 4 = highest)

P-F

P-Cl

P-I

P-Br

**Question 5** **12 points**

a. 9 points For the molecules HClO_4 and TeF_6 , draw the most stable *Lewis structure* showing appropriate shape and *dashed/wedged bonds*, based on VSEPR. Determine the **molecular geometry** around the central atom. For HClO_4 , label the *most polar bond*.

HClO_4 (there are no O-O bonds)

Label the most polar bond

Molecular Geometry around Cl

TeF_6

Molecular Geometry around Te

b. 3 points (Note: Lone pairs are not shown in structures below)

(i) For the three *resonance* structures (1, 2 and 3) shown below, answer the following question:



(I) Circle the *most contributing* resonance structure.

Structure 1

Structure 2

Structure 3

(ii) For the three structures given (4, 5, and 6) below, answer the following questions:



(I) Circle all the structures that represent *resonance structures* of each other:

Structure 4

Structure 5

Structure 6

(II) Among the resonance structures, circle the *most contributing* resonance structure.

Structure 4

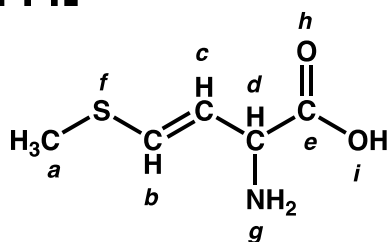
Structure 5

Structure 6

**Question 6** **6 points**

Using the data given below, calculate the **lattice energy** for SrO(s).

	ΔH° (kJ/mol)
$\text{Sr}(s) \longrightarrow \text{Sr}(g)$	+180
$\text{Sr}(g) \longrightarrow \text{Sr}^{2+}(g) + 2e^-$	+1468
$\text{O}_2(g) \longrightarrow 2\text{O}(g)$	+500
$\text{O}(g) + 2e^- \longrightarrow \text{O}^{2-}(g)$	+600
$\text{Sr}(s) + \frac{1}{2}\text{O}_2(g) \longrightarrow \text{SrO}(s)$	-550

**Question 7 10 points**

Consider the molecule given on the left, and answer questions (i) to (vi) using chemical bonding theories that we have learnt in class (valence bond theory, hybridization).

Note that the lone pairs are not shown in the structure:

i. Consider the carbon atoms marked in as ***a***, ***b***, ***c***, ***d***, and ***e*** in the molecule, and determine **hybridization** at each of the carbon centers. (2.5 points)

Atom	Hybridization
Carbon (<i>a</i>)	
Carbon (<i>b</i>)	
Carbon (<i>c</i>)	
Carbon (<i>d</i>)	
Carbon (<i>e</i>)	

ii. Now, consider the sulfur atom (S) marked as ***f***, nitrogen (N) atom (***g***), and oxygen(O) atoms (***h*** and ***i***), and determine their **hybridization**. (2 points)

Atom	Hybridization
Sulfur (<i>f</i>)	
Nitrogen (<i>g</i>)	
Oxygen (<i>h</i>)	
Oxygen (<i>i</i>)	

iii. **Describe** all the bonds formed by the carbon atoms ***a*** and ***b***, using sigma and pi designation as appropriate. (2 points)

Atom	Bonds formed (use sigma and pi as appropriate)
Carbon (<i>a</i>)	
Carbon (<i>b</i>)	

iv. How many total sp^3 -s sigma bonds are in the molecule? (1 point)

v. Determine **electronic** and **molecular geometry** at carbon (C) labeled ***e*** and sulfur (S) labeled ***f*** centres. (2 points)

Atom	Electronic Geometry	Molecular Geometry
Carbon (<i>e</i>)		
Sulfur (<i>f</i>)		

vi. Circle the value from the choices given below, that describes the **C-S-C bond angle at sulfur** (S) marked ***f***: (0.5 points)

>90° to <109.5° 109.5° >109.5 to <120° 120° >120° to <180°



Question 8 12 points

Using the **valence shell orbitals only** draw a **complete** molecular orbital (MO) diagram (with no s-p mixing) for the molecule **PS**, in the space provided below. **(6 points)**

Assume that 3s and 3p atomic orbitals will form molecular orbitals like those formed by the combination of 2s and 2p atomic orbitals that we learnt in class, and there is no s-p mixing.

Show **energy arrow**, **atomic orbitals** involved, **atom labels**, the **molecular orbitals** that are formed, **filled electrons** in the MO diagram etc. Make sure to label all the atomic and molecular orbitals.

Using the MO diagram above, answer the following questions:

- (i) Which one of the following will have the highest bond order? Circle the correct choice
 A) PS^{1-} B) PS C) PS^{1+}
- (ii) Which one of the following will have the longest bond length? Circle the correct choice
 A) PS^{1-} B) PS C) PS^{1+}
- (iii) Which of the following will be diamagnetic? Circle the correct choice.
 A) PS^{1-} B) PS C) PS^{1+}
- (iv) Draw Lewis dot structure of the molecule with the shortest bond length (based on the MO diagram) from among the three (PS^{1-} , PS, PS^{1+}), and indicate the bond order calculated from the Lewis dot structure. **(3 points)**

Lewis Dot Structure

Bond Order



666D6267-0710-4E9F-9C65-41F6E8A6BF45

midterm-exam-93388

#871 Page 10 of 10

SCRAP PAGE

**CHEMISTRY 110 –Midterm Exam****Thursday November 7, 2024, 6:30 – 8:30 pm****Examiners: Pallavi Sirjoosingh and Ashok Kakkar**

First name (please write as legibly as possible within the boxes)

--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--

Last name

--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--

Student ID number

--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--

This booklet has 8 questions on 10 pages (The last page can be used as a scrap page).

The staple must NOT be removed from the question booklet. Your intact question booklet must be returned at the end of the exam.

A separate booklet contains a periodic table, and datasheet.

Write your answers directly under the questions in the space provided. Do not write on the QR codes or near the top edges (as it may be cut off).

Please write **legibly and dark enough** for the scan to be readable. For students who use English as a second language, dictionaries are permitted. Calculators are allowed.

DO NOT ASK QUESTIONS OF THE INVIGILATORS about interpretations of the exam questions. If you have a concern about a question, please summarize it at the top of this page. All concerns will be evaluated after the exam. GOOD LUCK!

**Question 1 8 points** (*Assume all transitions are possible*)

a. 6 points Calculate the **energy (in kJ/mol)** required to ionize (completely remove) the electron from the **n=3** level of **He⁺ ion** (He: Helium). Show your work including equations. (3 sig. figs.)

b. 2 points

(I) Select the largest (longest) wavelength from the three choices below:

- i. Wavelength of photon to *completely remove* an electron from *1s orbital of Hydrogen atom*
- ii. Wavelength of photon to *completely remove* an electron from *1s orbital of Boron atom (B)*
- iii. Wavelength of photon to *completely remove* an electron from *1s orbital of Boron cation (B⁺)*

i or ii or iii

(II) Select the smallest (shortest) wavelength from the three choices below (*for H atom*):

- i. Wavelength of photon *emitted* for an electron transitioning from *2p to 1s orbital*
- ii. Wavelength of photon *absorbed* for an electron transitioning from *2s to 4p orbital*
- iii. Wavelength of photon *absorbed* for an electron transitioning from *1s to 3p orbital*

i or ii or iii



Question 2 8 points



A Chem110 student plots the graph shown on the left, from data obtained for the kinetic energy (eV) of the ejected electron versus frequency (s^{-1}) of the incident light, for an unknown metal. ($1\text{eV} = 1.6022 \times 10^{-19} \text{ J}$)

Based on the plot, answer the questions **a to c** below:

a. 1 point On the plot above, label the **threshold frequency**.

b. 2 points What is the value of the work potential (in J) of the unknown metal?

c. 3 points What is the *longest* wavelength (in nm) required to eject an electron from this metal's surface? Show your work including equation(s) used. (3 sig figs)

d. 2 points In the space provided below, plot the following

(Assume frequency is *above* threshold frequency):

- i. *Kinetic energy* of an ejected electron versus the *intensity* of the incident light (left)
- ii. *Number of electrons* ejected versus the *frequency* of the incident light (right)



**Question 3** **6 points****a. 2 points**

In the ground state of Ge (Germanium; $Z = 32$) atom, **name** the following orbitals:

i. Orbital containing the electron that requires the maximum energy to be removed.

ii. Orbital containing the electron that experiences the minimum effective nuclear charge.

b. 2 points

An unknown anion (A^-) has the following *excited* electronic configuration: $[Ar] 4s^2 3d^3 4p^1$

(Ar: Argon; $Z = 18$)

Give the **condensed ground state electronic configuration** of the atom, **A** and the cation, **A^{2+}** .
Show your work. (Assume all electronic transitions are possible)

Electronic Configuration of A:

Electronic Configuration of A^{2+} :

c. 2 points

In a single atom, determine how many electrons can have the following set of quantum numbers?

(i) $n = 5; l = 3$

In a single atom, determine how many electrons can have the following set of quantum numbers?

(i) $n = 3; l = 2; m_s = +1/2$

**Question 4** **8 points**

Rank the following from 1 to 4 in **increasing** order. Provide the ranking in the boxes provided.

Note: Atomic number (Z) are provided for relevant atoms

a) Increasing (lowest to highest) order of *ionization energy*: (1= lowest; 4 = highest)

Mg

Mg: Z = 12

Rb

Rb: Z = 37

Cs

Cs: Z = 55

K

K : Z = 19

b) Increasing (lowest to highest) order of *bond angle*: (1= lowest; 4 = highest)

H₂OCCl₄XeF₄

HCN

c) Increasing (smallest to largest) order of *ionic size* (1= smallest; 4 = largest)

S²⁻

S: Z = 16

O²⁻

O: Z = 8

Be²⁺

Be: Z = 4

Na⁺

Na: Z = 11

d) Increasing (lowest to highest) order of *bond energy* (1= lowest; 4 = highest)

P-F

P-Cl

P-I

P-Br

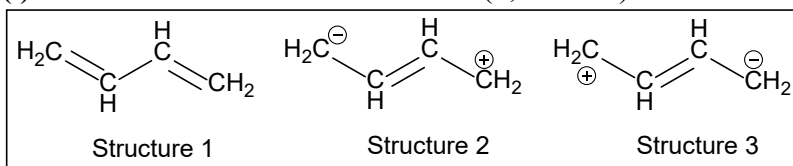
**Question 5** **12 points**

a. 9 points For the molecules HClO_4 and TeF_6 , draw the most stable *Lewis structure* showing appropriate shape and *dashed/wedged bonds*, based on VSEPR. Determine the **molecular geometry** around the central atom. For HClO_4 , label the *most polar bond*.



b. 3 points (Note: Lone pairs are not shown in structures below)

(i) For the three *resonance* structures (1, 2 and 3) shown below, answer the following question:



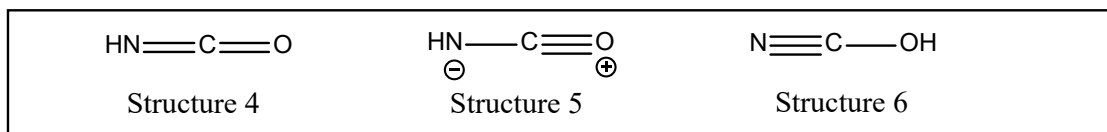
(I) Circle the *most contributing* resonance structure.

Structure 1

Structure 2

Structure 3

(ii) For the three structures given (4, 5, and 6) below, answer the following questions:



(I) Circle all the structures that represent *resonance structures* of each other:

Structure 4

Structure 5

Structure 6

(II) Among the resonance structures, circle the *most contributing* resonance structure.

Structure 4

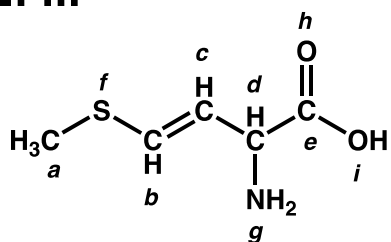
Structure 5

Structure 6

**Question 6** **6 points**

Using the data given below, calculate the **lattice energy** for SrO(s).

	ΔH° (kJ/mol)
$\text{Sr}(s) \longrightarrow \text{Sr}(g)$	+180
$\text{Sr}(g) \longrightarrow \text{Sr}^{2+}(g) + 2e^-$	+1468
$\text{O}_2(g) \longrightarrow 2\text{O}(g)$	+500
$\text{O}(g) + 2e^- \longrightarrow \text{O}^{2-}(g)$	+600
$\text{Sr}(s) + \frac{1}{2}\text{O}_2(g) \longrightarrow \text{SrO}(s)$	-550

**Question 7 10 points**

Consider the molecule given on the left, and answer questions (i) to (vi) using chemical bonding theories that we have learnt in class (valence bond theory, hybridization).

Note that the lone pairs are not shown in the structure:

i. Consider the carbon atoms marked in as ***a***, ***b***, ***c***, ***d***, and ***e*** in the molecule, and determine **hybridization** at each of the carbon centers. (2.5 points)

Atom	Hybridization
Carbon (<i>a</i>)	
Carbon (<i>b</i>)	
Carbon (<i>c</i>)	
Carbon (<i>d</i>)	
Carbon (<i>e</i>)	

ii. Now, consider the sulfur atom (S) marked as ***f***, nitrogen (N) atom (***g***), and oxygen(O) atoms (***h*** and ***i***), and determine their **hybridization**. (2 points)

Atom	Hybridization
Sulfur (<i>f</i>)	
Nitrogen (<i>g</i>)	
Oxygen (<i>h</i>)	
Oxygen (<i>i</i>)	

iii. **Describe** all the bonds formed by the carbon atoms ***a*** and ***b***, using sigma and pi designation as appropriate. (2 points)

Atom	Bonds formed (use sigma and pi as appropriate)
Carbon (<i>a</i>)	
Carbon (<i>b</i>)	

iv. How many total sp^3 -s sigma bonds are in the molecule? (1 point)

v. Determine **electronic** and **molecular geometry** at carbon (C) labeled ***e*** and sulfur (S) labeled ***f*** centres. (2 points)

Atom	Electronic Geometry	Molecular Geometry
Carbon (<i>e</i>)		
Sulfur (<i>f</i>)		

vi. Circle the value from the choices given below, that describes the **C-S-C bond angle at sulfur** (S) marked ***f***: (0.5 points)

$>90^\circ$ to $<109.5^\circ$ 109.5° >109.5 to $<120^\circ$ 120° $>120^\circ$ to $<180^\circ$


Question 8 12 points

Using the **valence shell orbitals only** draw a **complete** molecular orbital (MO) diagram (with no s-p mixing) for the molecule **PS**, in the space provided below. **(6 points)**

Assume that 3s and 3p atomic orbitals will form molecular orbitals like those formed by the combination of 2s and 2p atomic orbitals that we learnt in class, and there is no s-p mixing.

Show **energy arrow**, **atomic orbitals** involved, **atom labels**, the **molecular orbitals** that are formed, **filled electrons** in the MO diagram etc. Make sure to label all the atomic and molecular orbitals.

Using the MO diagram above, answer the following questions:

- (i) Which one of the following will have the highest bond order? Circle the correct choice
 A) PS^{1-} B) PS C) PS^{1+}
- (ii) Which one of the following will have the longest bond length? Circle the correct choice
 A) PS^{1-} B) PS C) PS^{1+}
- (iii) Which of the following will be diamagnetic? Circle the correct choice.
 A) PS^{1-} B) PS C) PS^{1+}
- (iv) Draw Lewis dot structure of the molecule with the shortest bond length (based on the MO diagram) from among the three (PS^{1-} , PS, PS^{1+}), and indicate the bond order calculated from the Lewis dot structure. **(3 points)**

Lewis Dot Structure

Bond Order



064CEC44-0ADC-4C13-AD60-135D4F5F6970

midterm-exam-93388

#872 Page 10 of 10

SCRAP PAGE

**CHEMISTRY 110 –Midterm Exam****Thursday November 7, 2024, 6:30 – 8:30 pm****Examiners: Pallavi Sirjoosingh and Ashok Kakkar**

First name (please write as legibly as possible within the boxes)

--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--

Last name

--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--

Student ID number

--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--

This booklet has 8 questions on 10 pages (The last page can be used as a scrap page).

The staple must NOT be removed from the question booklet. Your intact question booklet must be returned at the end of the exam.

A separate booklet contains a periodic table, and datasheet.

Write your answers directly under the questions in the space provided. Do not write on the QR codes or near the top edges (as it may be cut off).

Please write **legibly and dark enough** for the scan to be readable. For students who use English as a second language, dictionaries are permitted. Calculators are allowed.

DO NOT ASK QUESTIONS OF THE INVIGILATORS about interpretations of the exam questions. If you have a concern about a question, please summarize it at the top of this page. All concerns will be evaluated after the exam. GOOD LUCK!

**Question 1 8 points** (*Assume all transitions are possible*)

a. 6 points Calculate the **energy (in kJ/mol)** required to ionize (completely remove) the electron from the **$n=3$** level of **He^+ ion** (He: Helium). Show your work including equations. (3 sig. figs.)

b. 2 points

(I) Select the largest (longest) wavelength from the three choices below:

- i. Wavelength of photon to *completely remove* an electron from *1s orbital of Hydrogen atom*
- ii. Wavelength of photon to *completely remove* an electron from *1s orbital of Boron atom (B)*
- iii. Wavelength of photon to *completely remove* an electron from *1s orbital of Boron cation (B^+)*

i or ii or iii

(II) Select the smallest (shortest) wavelength from the three choices below (*for H atom*):

- i. Wavelength of photon *emitted* for an electron transitioning from *2p to 1s orbital*
- ii. Wavelength of photon *absorbed* for an electron transitioning from *2s to 4p orbital*
- iii. Wavelength of photon *absorbed* for an electron transitioning from *1s to 3p orbital*

i or ii or iii



Question 2 8 points



A Chem110 student plots the graph shown on the left, from data obtained for the kinetic energy (eV) of the ejected electron versus frequency (s^{-1}) of the incident light, for an unknown metal. ($1\text{eV} = 1.6022 \times 10^{-19} \text{ J}$)

Based on the plot, answer the questions **a to c** below:

a. 1 point On the plot above, label the **threshold frequency**.

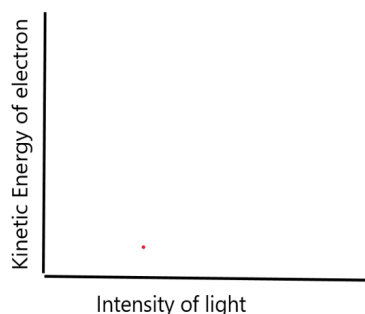
b. 2 points What is the value of the work potential (in J) of the unknown metal?

c. 3 points What is the *longest* wavelength (in nm) required to eject an electron from this metal's surface? Show your work including equation(s) used. (3 sig figs)

d. 2 points In the space provided below, plot the following

(Assume frequency is *above* threshold frequency):

- i. *Kinetic energy* of an ejected electron versus the *intensity* of the incident light (left)
- ii. *Number of electrons* ejected versus the *frequency* of the incident light (right)



**Question 3** **6 points****a. 2 points**

In the ground state of Ge (Germanium; $Z = 32$) atom, **name** the following orbitals:

i. Orbital containing the electron that requires the maximum energy to be removed.

ii. Orbital containing the electron that experiences the minimum effective nuclear charge.

b. 2 points

An unknown anion (A^-) has the following *excited* electronic configuration: $[Ar] 4s^2 3d^3 4p^1$

(Ar: Argon; $Z = 18$)

Give the **condensed ground state electronic configuration** of the atom, **A** and the cation, **A^{2+}** .
Show your work. (Assume all electronic transitions are possible)

Electronic Configuration of A:

Electronic Configuration of A^{2+} :

c. 2 points

In a single atom, determine how many electrons can have the following set of quantum numbers?

(i) $n = 5; l = 3$

In a single atom, determine how many electrons can have the following set of quantum numbers?

(i) $n = 3; l = 2; m_s = +1/2$

**Question 4** **8 points**

Rank the following from 1 to 4 in **increasing** order. Provide the ranking in the boxes provided.

Note: Atomic number (Z) are provided for relevant atoms

a) Increasing (lowest to highest) order of *ionization energy*: (1= lowest; 4 = highest)

Mg

Mg: Z = 12

Rb

Rb: Z = 37

Cs

Cs: Z = 55

K

K : Z = 19

b) Increasing (lowest to highest) order of *bond angle*: (1= lowest; 4 = highest)

H₂OCCl₄XeF₄

HCN

c) Increasing (smallest to largest) order of *ionic size* (1= smallest; 4 = largest)

S²⁻

S: Z = 16

O²⁻

O: Z = 8

Be²⁺

Be: Z = 4

Na⁺

Na: Z = 11

d) Increasing (lowest to highest) order of *bond energy* (1= lowest; 4 = highest)

P-F

P-Cl

P-I

P-Br

**Question 5** **12 points**

a. 9 points For the molecules HClO_4 and TeF_6 , draw the most stable *Lewis structure* showing appropriate shape and *dashed/wedged bonds*, based on VSEPR. Determine the **molecular geometry** around the central atom. For HClO_4 , label the *most polar bond*.



b. 3 points (Note: Lone pairs are not shown in structures below)

(i) For the three *resonance* structures (1, 2 and 3) shown below, answer the following question:



(I) Circle the *most contributing* resonance structure.

Structure 1

Structure 2

Structure 3

(ii) For the three structures given (4, 5, and 6) below, answer the following questions:



(I) Circle all the structures that represent *resonance structures* of each other:

Structure 4

Structure 5

Structure 6

(II) Among the resonance structures, circle the *most contributing* resonance structure.

Structure 4

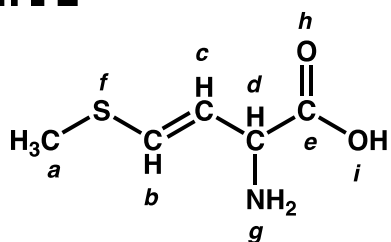
Structure 5

Structure 6

**Question 6** **6 points**

Using the data given below, calculate the **lattice energy** for SrO(s).

	ΔH° (kJ/mol)
$\text{Sr}(s) \longrightarrow \text{Sr}(g)$	+180
$\text{Sr}(g) \longrightarrow \text{Sr}^{2+}(g) + 2e^-$	+1468
$\text{O}_2(g) \longrightarrow 2\text{O}(g)$	+500
$\text{O}(g) + 2e^- \longrightarrow \text{O}^{2-}(g)$	+600
$\text{Sr}(s) + \frac{1}{2}\text{O}_2(g) \longrightarrow \text{SrO}(s)$	-550

**Question 7 10 points**

Consider the molecule given on the left, and answer questions (i) to (vi) using chemical bonding theories that we have learnt in class (valence bond theory, hybridization).

Note that the lone pairs are not shown in the structure:

i. Consider the carbon atoms marked in as ***a***, ***b***, ***c***, ***d***, and ***e*** in the molecule, and determine **hybridization** at each of the carbon centers. (2.5 points)

Atom	Hybridization
Carbon (<i>a</i>)	
Carbon (<i>b</i>)	
Carbon (<i>c</i>)	
Carbon (<i>d</i>)	
Carbon (<i>e</i>)	

ii. Now, consider the sulfur atom (S) marked as ***f***, nitrogen (N) atom (***g***), and oxygen(O) atoms (***h*** and ***i***), and determine their **hybridization**. (2 points)

Atom	Hybridization
Sulfur (<i>f</i>)	
Nitrogen (<i>g</i>)	
Oxygen (<i>h</i>)	
Oxygen (<i>i</i>)	

iii. **Describe** all the bonds formed by the carbon atoms ***a*** and ***b***, using sigma and pi designation as appropriate. (2 points)

Atom	Bonds formed (use sigma and pi as appropriate)
Carbon (<i>a</i>)	
Carbon (<i>b</i>)	

iv. How many total sp^3 -s sigma bonds are in the molecule? (1 point)

v. Determine **electronic** and **molecular geometry** at carbon (C) labeled ***e*** and sulfur (S) labeled ***f*** centres. (2 points)

Atom	Electronic Geometry	Molecular Geometry
Carbon (<i>e</i>)		
Sulfur (<i>f</i>)		

vi. Circle the value from the choices given below, that describes the **C-S-C bond angle at sulfur** (S) marked ***f***: (0.5 points)

>90° to <109.5° 109.5° >109.5 to <120° 120° >120° to <180°

**Question 8 12 points**

Using the **valence shell orbitals only** draw a **complete** molecular orbital (MO) diagram (with no s-p mixing) for the molecule **PS**, in the space provided below. **(6 points)**

Assume that 3s and 3p atomic orbitals will form molecular orbitals like those formed by the combination of 2s and 2p atomic orbitals that we learnt in class, and there is no s-p mixing.

Show **energy arrow**, **atomic orbitals** involved, **atom labels**, the **molecular orbitals** that are formed, **filled electrons** in the MO diagram etc. Make sure to label all the atomic and molecular orbitals.

Using the MO diagram above, answer the following questions:

- (i) Which one of the following will have the highest bond order? Circle the correct choice
A) PS^{1-} B) PS C) PS^{1+}
- (ii) Which one of the following will have the longest bond length? Circle the correct choice
A) PS^{1-} B) PS C) PS^{1+}
- (iii) Which of the following will be diamagnetic? Circle the correct choice.
A) PS^{1-} B) PS C) PS^{1+}
- (iv) Draw Lewis dot structure of the molecule with the shortest bond length (based on the MO diagram) from among the three (PS^{1-} , PS, PS^{1+}), and indicate the bond order calculated from the Lewis dot structure. **(3 points)**

Lewis Dot Structure

Bond Order



F9E4CBD9-AAE1-40A2-9EBF-6CA99E837C3C

midterm-exam-93388

#873 Page 10 of 10

SCRAP PAGE

**CHEMISTRY 110 –Midterm Exam****Thursday November 7, 2024, 6:30 – 8:30 pm**

Examiners: Pallavi Sirjoosingh and Ashok Kakkar

First name (please write as legibly as possible within the boxes)

--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--

Last name

--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--

Student ID number

--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--

This booklet has 8 questions on 10 pages (The last page can be used as a scrap page).

The staple must NOT be removed from the question booklet. Your intact question booklet must be returned at the end of the exam.

A separate booklet contains a periodic table, and datasheet.

Write your answers directly under the questions in the space provided. Do not write on the QR codes or near the top edges (as it may be cut off).

Please write **legibly and dark enough** for the scan to be readable. For students who use English as a second language, dictionaries are permitted. Calculators are allowed.

DO NOT ASK QUESTIONS OF THE INVIGILATORS about interpretations of the exam questions. If you have a concern about a question, please summarize it at the top of this page. All concerns will be evaluated after the exam. GOOD LUCK!

**Question 1 8 points** (*Assume all transitions are possible*)

a. 6 points Calculate the **energy (in kJ/mol)** required to ionize (completely remove) the electron from the **$n=3$** level of **He^+ ion** (He: Helium). Show your work including equations. (3 sig. figs.)

b. 2 points

(I) Select the largest (longest) wavelength from the three choices below:

- i. Wavelength of photon to *completely remove* an electron from *1s orbital of Hydrogen atom*
- ii. Wavelength of photon to *completely remove* an electron from *1s orbital of Boron atom (B)*
- iii. Wavelength of photon to *completely remove* an electron from *1s orbital of Boron cation (B^+)*

i or ii or iii

(II) Select the smallest (shortest) wavelength from the three choices below (*for H atom*):

- i. Wavelength of photon *emitted* for an electron transitioning from *2p to 1s orbital*
- ii. Wavelength of photon *absorbed* for an electron transitioning from *2s to 4p orbital*
- iii. Wavelength of photon *absorbed* for an electron transitioning from *1s to 3p orbital*

i or ii or iii



Question 2 8 points



A Chem110 student plots the graph shown on the left, from data obtained for the kinetic energy (eV) of the ejected electron versus frequency (s^{-1}) of the incident light, for an unknown metal. ($1\text{eV} = 1.6022 \times 10^{-19} \text{ J}$)

Based on the plot, answer the questions **a to c** below:

a. 1 point On the plot above, label the **threshold frequency**.

b. 2 points What is the value of the work potential (in J) of the unknown metal?

c. 3 points What is the *longest* wavelength (in nm) required to eject an electron from this metal's surface? Show your work including equation(s) used. (3 sig figs)

d. 2 points In the space provided below, plot the following

(Assume frequency is *above* threshold frequency):

- i. *Kinetic energy* of an ejected electron versus the *intensity* of the incident light (left)
- ii. *Number of electrons* ejected versus the *frequency* of the incident light (right)



**Question 3** **6 points****a. 2 points**

In the ground state of Ge (Germanium; $Z = 32$) atom, **name** the following orbitals:

i. Orbital containing the electron that requires the maximum energy to be removed.

ii. Orbital containing the electron that experiences the minimum effective nuclear charge.

b. 2 points

An unknown anion (A^-) has the following *excited* electronic configuration: $[Ar] 4s^2 3d^3 4p^1$

(Ar: Argon; $Z = 18$)

Give the **condensed ground state electronic configuration** of the atom, **A** and the cation, A^{2+} .
Show your work. (Assume all electronic transitions are possible)

Electronic Configuration of A:

Electronic Configuration of A^{2+} :

c. 2 points

In a single atom, determine how many electrons can have the following set of quantum numbers?

(i) $n = 5; l = 3$

In a single atom, determine how many electrons can have the following set of quantum numbers?

(i) $n = 3; l = 2; m_s = +1/2$

**Question 4** **8 points**

Rank the following from 1 to 4 in **increasing** order. Provide the ranking in the boxes provided.

Note: Atomic number (Z) are provided for relevant atoms

a) Increasing (lowest to highest) order of *ionization energy*: (1= lowest; 4 = highest)

Mg

Mg: Z = 12

Rb

Rb: Z = 37

Cs

Cs: Z = 55

K

K : Z = 19

b) Increasing (lowest to highest) order of *bond angle*: (1= lowest; 4 = highest)

H₂OCCl₄XeF₄

HCN

c) Increasing (smallest to largest) order of *ionic size* (1= smallest; 4 = largest)

S²⁻

S: Z = 16

O²⁻

O: Z = 8

Be²⁺

Be: Z = 4

Na⁺

Na: Z = 11

d) Increasing (lowest to highest) order of *bond energy* (1= lowest; 4 = highest)

P-F

P-Cl

P-I

P-Br

**Question 5** **12 points**

a. 9 points For the molecules HClO_4 and TeF_6 , draw the most stable *Lewis structure* showing appropriate shape and *dashed/wedged bonds*, based on VSEPR. Determine the **molecular geometry** around the central atom. For HClO_4 , label the *most polar bond*.



b. 3 points (Note: Lone pairs are not shown in structures below)

(i) For the three *resonance* structures (1, 2 and 3) shown below, answer the following question:



(I) Circle the *most contributing* resonance structure.

Structure 1

Structure 2

Structure 3

(ii) For the three structures given (4, 5, and 6) below, answer the following questions:



(I) Circle all the structures that represent *resonance structures* of each other:

Structure 4

Structure 5

Structure 6

(II) Among the resonance structures, circle the *most contributing* resonance structure.

Structure 4

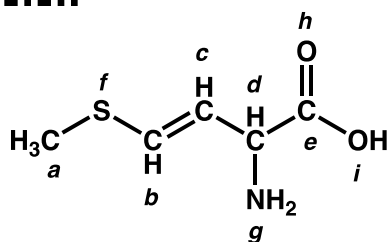
Structure 5

Structure 6

**Question 6** **6 points**

Using the data given below, calculate the **lattice energy** for SrO(s).

	ΔH° (kJ/mol)
$\text{Sr}(s) \longrightarrow \text{Sr}(g)$	+180
$\text{Sr}(g) \longrightarrow \text{Sr}^{2+}(g) + 2e^-$	+1468
$\text{O}_2(g) \longrightarrow 2\text{O}(g)$	+500
$\text{O}(g) + 2e^- \longrightarrow \text{O}^{2-}(g)$	+600
$\text{Sr}(s) + \frac{1}{2}\text{O}_2(g) \longrightarrow \text{SrO}(s)$	-550

**Question 7 10 points**

Consider the molecule given on the left, and answer questions (i) to (vi) using chemical bonding theories that we have learnt in class (valence bond theory, hybridization).

Note that the lone pairs are not shown in the structure:

i. Consider the carbon atoms marked in as ***a***, ***b***, ***c***, ***d***, and ***e*** in the molecule, and determine **hybridization** at each of the carbon centers. (2.5 points)

Atom	Hybridization
Carbon (<i>a</i>)	
Carbon (<i>b</i>)	
Carbon (<i>c</i>)	
Carbon (<i>d</i>)	
Carbon (<i>e</i>)	

ii. Now, consider the sulfur atom (S) marked as ***f***, nitrogen (N) atom (***g***), and oxygen(O) atoms (***h*** and ***i***), and determine their **hybridization**. (2 points)

Atom	Hybridization
Sulfur (<i>f</i>)	
Nitrogen (<i>g</i>)	
Oxygen (<i>h</i>)	
Oxygen (<i>i</i>)	

iii. **Describe** all the bonds formed by the carbon atoms ***a*** and ***b***, using sigma and pi designation as appropriate. (2 points)

Atom	Bonds formed (use sigma and pi as appropriate)
Carbon (<i>a</i>)	
Carbon (<i>b</i>)	

iv. How many total sp^3 -s sigma bonds are in the molecule? (1 point)

v. Determine **electronic** and **molecular geometry** at carbon (C) labeled ***e*** and sulfur (S) labeled ***f*** centres. (2 points)

Atom	Electronic Geometry	Molecular Geometry
Carbon (<i>e</i>)		
Sulfur (<i>f</i>)		

vi. Circle the value from the choices given below, that describes the **C-S-C bond angle at sulfur** (S) marked ***f***: (0.5 points)

>90° to <109.5° 109.5° >109.5 to <120° 120° >120° to <180°


Question 8 12 points

Using the **valence shell orbitals only** draw a **complete** molecular orbital (MO) diagram (with no s-p mixing) for the molecule **PS**, in the space provided below. **(6 points)**

Assume that 3s and 3p atomic orbitals will form molecular orbitals like those formed by the combination of 2s and 2p atomic orbitals that we learnt in class, and there is no s-p mixing.

Show **energy arrow**, **atomic orbitals** involved, **atom labels**, the **molecular orbitals** that are formed, **filled electrons** in the MO diagram etc. Make sure to label all the atomic and molecular orbitals.

Using the MO diagram above, answer the following questions:

- (i) Which one of the following will have the highest bond order? Circle the correct choice
 A) PS^{1-} B) PS C) PS^{1+}
- (ii) Which one of the following will have the longest bond length? Circle the correct choice
 A) PS^{1-} B) PS C) PS^{1+}
- (iii) Which of the following will be diamagnetic? Circle the correct choice.
 A) PS^{1-} B) PS C) PS^{1+}
- (iv) Draw Lewis dot structure of the molecule with the shortest bond length (based on the MO diagram) from among the three (PS^{1-} , PS, PS^{1+}), and indicate the bond order calculated from the Lewis dot structure. **(3 points)**

Lewis Dot Structure

Bond Order



FE16CD00-5943-42EA-BC6C-81A929733D79

midterm-exam-93388

#874 Page 10 of 10

SCRAP PAGE

**CHEMISTRY 110 –Midterm Exam****Thursday November 7, 2024, 6:30 – 8:30 pm****Examiners: Pallavi Sirjoosingh and Ashok Kakkar**

First name (please write as legibly as possible within the boxes)

--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--

Last name

--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--

Student ID number

--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--

This booklet has 8 questions on 10 pages (The last page can be used as a scrap page).

The staple must NOT be removed from the question booklet. Your intact question booklet must be returned at the end of the exam.

A separate booklet contains a periodic table, and datasheet.

Write your answers directly under the questions in the space provided. Do not write on the QR codes or near the top edges (as it may be cut off).

Please write **legibly and dark enough** for the scan to be readable. For students who use English as a second language, dictionaries are permitted. Calculators are allowed.

DO NOT ASK QUESTIONS OF THE INVIGILATORS about interpretations of the exam questions. If you have a concern about a question, please summarize it at the top of this page. All concerns will be evaluated after the exam. GOOD LUCK!

**Question 1 8 points** (*Assume all transitions are possible*)

a. 6 points Calculate the **energy (in kJ/mol)** required to ionize (completely remove) the electron from the **$n=3$** level of **He^+ ion** (He: Helium). Show your work including equations. (3 sig. figs.)

b. 2 points

(I) Select the largest (longest) wavelength from the three choices below:

- i. Wavelength of photon to *completely remove* an electron from *1s orbital of Hydrogen atom*
- ii. Wavelength of photon to *completely remove* an electron from *1s orbital of Boron atom (B)*
- iii. Wavelength of photon to *completely remove* an electron from *1s orbital of Boron cation (B^+)*

i or ii or iii

(II) Select the smallest (shortest) wavelength from the three choices below (*for H atom*):

- i. Wavelength of photon *emitted* for an electron transitioning from *2p to 1s orbital*
- ii. Wavelength of photon *absorbed* for an electron transitioning from *2s to 4p orbital*
- iii. Wavelength of photon *absorbed* for an electron transitioning from *1s to 3p orbital*

i or ii or iii



Question 2 8 points



A Chem110 student plots the graph shown on the left, from data obtained for the kinetic energy (eV) of the ejected electron versus frequency (s^{-1}) of the incident light, for an unknown metal. ($1\text{eV} = 1.6022 \times 10^{-19} \text{ J}$)

Based on the plot, answer the questions **a to c** below:

a. 1 point On the plot above, label the **threshold frequency**.

b. 2 points What is the value of the work potential (in J) of the unknown metal?

c. 3 points What is the *longest* wavelength (in nm) required to eject an electron from this metal's surface? Show your work including equation(s) used. (3 sig figs)

d. 2 points In the space provided below, plot the following

(Assume frequency is *above* threshold frequency):

- i. *Kinetic energy* of an ejected electron versus the *intensity* of the incident light (left)
- ii. *Number of electrons* ejected versus the *frequency* of the incident light (right)



**Question 3** **6 points****a. 2 points**

In the ground state of Ge (Germanium; $Z = 32$) atom, **name** the following orbitals:

i. Orbital containing the electron that requires the maximum energy to be removed.

ii. Orbital containing the electron that experiences the minimum effective nuclear charge.

b. 2 points

An unknown anion (A^-) has the following *excited* electronic configuration: $[Ar] 4s^2 3d^3 4p^1$

(Ar: Argon; $Z = 18$)

Give the **condensed ground state electronic configuration** of the atom, **A** and the cation, **A^{2+}** .
Show your work. (Assume all electronic transitions are possible)

Electronic Configuration of A:

Electronic Configuration of A^{2+} :

c. 2 points

In a single atom, determine how many electrons can have the following set of quantum numbers?

(i) $n = 5; l = 3$

In a single atom, determine how many electrons can have the following set of quantum numbers?

(i) $n = 3; l = 2; m_s = +1/2$

**Question 4** **8 points**

Rank the following from 1 to 4 in **increasing** order. Provide the ranking in the boxes provided.

Note: Atomic number (Z) are provided for relevant atoms

a) Increasing (lowest to highest) order of *ionization energy*: (1= lowest; 4 = highest)

Mg

Mg: Z = 12

Rb

Rb: Z = 37

Cs

Cs: Z = 55

K

K : Z = 19

b) Increasing (lowest to highest) order of *bond angle*: (1= lowest; 4 = highest)

H₂OCCl₄XeF₄

HCN

c) Increasing (smallest to largest) order of *ionic size* (1= smallest; 4 = largest)

S²⁻

S: Z = 16

O²⁻

O: Z = 8

Be²⁺

Be: Z = 4

Na⁺

Na: Z = 11

d) Increasing (lowest to highest) order of *bond energy* (1= lowest; 4 = highest)

P-F

P-Cl

P-I

P-Br

**Question 5** **12 points**

a. 9 points For the molecules HClO_4 and TeF_6 , draw the most stable *Lewis structure* showing appropriate shape and *dashed/wedged bonds*, based on VSEPR. Determine the **molecular geometry** around the central atom. For HClO_4 , label the *most polar bond*.



b. 3 points (Note: Lone pairs are not shown in structures below)

(i) For the three *resonance* structures (1, 2 and 3) shown below, answer the following question:



(I) Circle the *most contributing* resonance structure.

Structure 1

Structure 2

Structure 3

(ii) For the three structures given (4, 5, and 6) below, answer the following questions:



(I) Circle all the structures that represent *resonance structures* of each other:

Structure 4

Structure 5

Structure 6

(II) Among the resonance structures, circle the *most contributing* resonance structure.

Structure 4

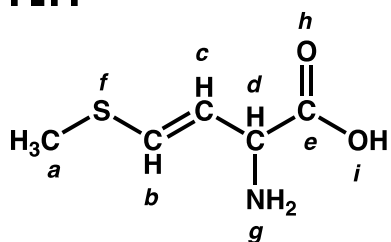
Structure 5

Structure 6

**Question 6** **6 points**

Using the data given below, calculate the **lattice energy** for SrO(s).

	ΔH° (kJ/mol)
$\text{Sr}(s) \longrightarrow \text{Sr}(g)$	+180
$\text{Sr}(g) \longrightarrow \text{Sr}^{2+}(g) + 2e^-$	+1468
$\text{O}_2(g) \longrightarrow 2\text{O}(g)$	+500
$\text{O}(g) + 2e^- \longrightarrow \text{O}^{2-}(g)$	+600
$\text{Sr}(s) + \frac{1}{2}\text{O}_2(g) \longrightarrow \text{SrO}(s)$	-550

**Question 7 10 points**

Consider the molecule given on the left, and answer questions (i) to (vi) using chemical bonding theories that we have learnt in class (valence bond theory, hybridization).

Note that the lone pairs are not shown in the structure:

i. Consider the carbon atoms marked in as ***a***, ***b***, ***c***, ***d***, and ***e*** in the molecule, and determine **hybridization** at each of the carbon centers. (2.5 points)

Atom	Hybridization
Carbon (<i>a</i>)	
Carbon (<i>b</i>)	
Carbon (<i>c</i>)	
Carbon (<i>d</i>)	
Carbon (<i>e</i>)	

ii. Now, consider the sulfur atom (S) marked as ***f***, nitrogen (N) atom (***g***), and oxygen(O) atoms (***h*** and ***i***), and determine their **hybridization**. (2 points)

Atom	Hybridization
Sulfur (<i>f</i>)	
Nitrogen (<i>g</i>)	
Oxygen (<i>h</i>)	
Oxygen (<i>i</i>)	

iii. **Describe** all the bonds formed by the carbon atoms ***a*** and ***b***, using sigma and pi designation as appropriate. (2 points)

Atom	Bonds formed (use sigma and pi as appropriate)
Carbon (<i>a</i>)	
Carbon (<i>b</i>)	

iv. How many total sp^3 -s sigma bonds are in the molecule? (1 point)

v. Determine **electronic** and **molecular geometry** at carbon (C) labeled ***e*** and sulfur (S) labeled ***f*** centres. (2 points)

Atom	Electronic Geometry	Molecular Geometry
Carbon (<i>e</i>)		
Sulfur (<i>f</i>)		

vi. Circle the value from the choices given below, that describes the **C-S-C bond angle at sulfur** (S) marked ***f***: (0.5 points)

>90° to <109.5° 109.5° >109.5 to <120° 120° >120° to <180°


Question 8 12 points

Using the **valence shell orbitals only** draw a **complete** molecular orbital (MO) diagram (with no s-p mixing) for the molecule **PS**, in the space provided below. **(6 points)**

Assume that 3s and 3p atomic orbitals will form molecular orbitals like those formed by the combination of 2s and 2p atomic orbitals that we learnt in class, and there is no s-p mixing.

Show **energy arrow**, **atomic orbitals** involved, **atom labels**, the **molecular orbitals** that are formed, **filled electrons** in the MO diagram etc. Make sure to label all the atomic and molecular orbitals.

Using the MO diagram above, answer the following questions:

- (i) Which one of the following will have the highest bond order? Circle the correct choice
 A) PS^{1-} B) PS C) PS^{1+}
- (ii) Which one of the following will have the longest bond length? Circle the correct choice
 A) PS^{1-} B) PS C) PS^{1+}
- (iii) Which of the following will be diamagnetic? Circle the correct choice.
 A) PS^{1-} B) PS C) PS^{1+}
- (iv) Draw Lewis dot structure of the molecule with the shortest bond length (based on the MO diagram) from among the three (PS^{1-} , PS, PS^{1+}), and indicate the bond order calculated from the Lewis dot structure. **(3 points)**

Lewis Dot Structure

Bond Order



43A12553-ED3E-4F00-B9DB-897E31F17CB8

midterm-exam-93388

#875 Page 10 of 10

SCRAP PAGE

**CHEMISTRY 110 –Midterm Exam****Thursday November 7, 2024, 6:30 – 8:30 pm**

Examiners: Pallavi Sirjoosingh and Ashok Kakkar

First name (please write as legibly as possible within the boxes)

--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--

Last name

--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--

Student ID number

--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--

This booklet has 8 questions on 10 pages (The last page can be used as a scrap page).

The staple must NOT be removed from the question booklet. Your intact question booklet must be returned at the end of the exam.

A separate booklet contains a periodic table, and datasheet.

Write your answers directly under the questions in the space provided. Do not write on the QR codes or near the top edges (as it may be cut off).

Please write **legibly and dark enough** for the scan to be readable. For students who use English as a second language, dictionaries are permitted. Calculators are allowed.

DO NOT ASK QUESTIONS OF THE INVIGILATORS about interpretations of the exam questions. If you have a concern about a question, please summarize it at the top of this page. All concerns will be evaluated after the exam. GOOD LUCK!

**Question 1 8 points** (*Assume all transitions are possible*)

a. 6 points Calculate the **energy (in kJ/mol)** required to ionize (completely remove) the electron from the **$n=3$** level of **He^+ ion** (He: Helium). Show your work including equations. (3 sig. figs.)

b. 2 points

(I) Select the largest (longest) wavelength from the three choices below:

- i. Wavelength of photon to *completely remove* an electron from *1s orbital of Hydrogen atom*
- ii. Wavelength of photon to *completely remove* an electron from *1s orbital of Boron atom (B)*
- iii. Wavelength of photon to *completely remove* an electron from *1s orbital of Boron cation (B^+)*

i or ii or iii

(II) Select the smallest (shortest) wavelength from the three choices below (*for H atom*):

- i. Wavelength of photon *emitted* for an electron transitioning from *2p to 1s orbital*
- ii. Wavelength of photon *absorbed* for an electron transitioning from *2s to 4p orbital*
- iii. Wavelength of photon *absorbed* for an electron transitioning from *1s to 3p orbital*

i or ii or iii



Question 2 8 points



A Chem110 student plots the graph shown on the left, from data obtained for the kinetic energy (eV) of the ejected electron versus frequency (s^{-1}) of the incident light, for an unknown metal. ($1\text{eV} = 1.6022 \times 10^{-19} \text{ J}$)

Based on the plot, answer the questions **a to c** below:

a. 1 point On the plot above, label the **threshold frequency**.

b. 2 points What is the value of the work potential (in J) of the unknown metal?

c. 3 points What is the *longest* wavelength (in nm) required to eject an electron from this metal's surface? Show your work including equation(s) used. (3 sig figs)

d. 2 points In the space provided below, plot the following

(Assume frequency is *above* threshold frequency):

- Kinetic energy* of an ejected electron versus the *intensity* of the incident light (left)
- Number of electrons* ejected versus the *frequency* of the incident light (right)



**Question 3** **6 points****a. 2 points**

In the ground state of Ge (Germanium; $Z = 32$) atom, **name** the following orbitals:

i. Orbital containing the electron that requires the maximum energy to be removed.

ii. Orbital containing the electron that experiences the minimum effective nuclear charge.

b. 2 points

An unknown anion (A^-) has the following *excited* electronic configuration: $[Ar] 4s^2 3d^3 4p^1$

(Ar: Argon; $Z = 18$)

Give the **condensed ground state electronic configuration** of the atom, **A** and the cation, **A^{2+}** .
Show your work. (Assume all electronic transitions are possible)

Electronic Configuration of A:

Electronic Configuration of A^{2+} :

c. 2 points

In a single atom, determine how many electrons can have the following set of quantum numbers?

(i) $n = 5; l = 3$

In a single atom, determine how many electrons can have the following set of quantum numbers?

(i) $n = 3; l = 2; m_s = +1/2$

**Question 4** **8 points**

Rank the following from 1 to 4 in **increasing** order. Provide the ranking in the boxes provided.

Note: Atomic number (Z) are provided for relevant atoms

a) Increasing (lowest to highest) order of *ionization energy*: (1= lowest; 4 = highest)

Mg

Mg: Z = 12

Rb

Rb: Z = 37

Cs

Cs: Z = 55

K

K : Z = 19

b) Increasing (lowest to highest) order of *bond angle*: (1= lowest; 4 = highest)

H₂OCCl₄XeF₄

HCN

c) Increasing (smallest to largest) order of *ionic size* (1= smallest; 4 = largest)

S²⁻

S: Z = 16

O²⁻

O: Z = 8

Be²⁺

Be: Z = 4

Na⁺

Na: Z = 11

d) Increasing (lowest to highest) order of *bond energy* (1= lowest; 4 = highest)

P-F

P-Cl

P-I

P-Br

**Question 5** **12 points**

a. 9 points For the molecules HClO_4 and TeF_6 , draw the most stable *Lewis structure* showing appropriate shape and *dashed/wedged bonds*, based on VSEPR. Determine the **molecular geometry** around the central atom. For HClO_4 , label the *most polar bond*.



b. 3 points (Note: Lone pairs are not shown in structures below)

(i) For the three *resonance* structures (1, 2 and 3) shown below, answer the following question:



(I) Circle the *most contributing* resonance structure.

Structure 1

Structure 2

Structure 3

(ii) For the three structures given (4, 5, and 6) below, answer the following questions:



(I) Circle all the structures that represent *resonance structures* of each other:

Structure 4

Structure 5

Structure 6

(II) Among the resonance structures, circle the *most contributing* resonance structure.

Structure 4

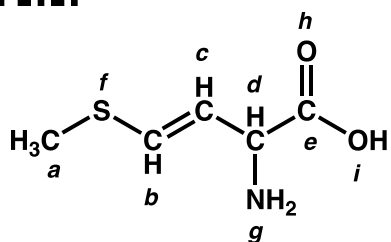
Structure 5

Structure 6

**Question 6** **6 points**

Using the data given below, calculate the **lattice energy** for SrO(s).

	ΔH° (kJ/mol)
$\text{Sr}(s) \longrightarrow \text{Sr}(g)$	+180
$\text{Sr}(g) \longrightarrow \text{Sr}^{2+}(g) + 2e^-$	+1468
$\text{O}_2(g) \longrightarrow 2\text{O}(g)$	+500
$\text{O}(g) + 2e^- \longrightarrow \text{O}^{2-}(g)$	+600
$\text{Sr}(s) + \frac{1}{2}\text{O}_2(g) \longrightarrow \text{SrO}(s)$	-550

**Question 7 10 points**

Consider the molecule given on the left, and answer questions (i) to (vi) using chemical bonding theories that we have learnt in class (valence bond theory, hybridization).

Note that the lone pairs are not shown in the structure:

i. Consider the carbon atoms marked in as ***a***, ***b***, ***c***, ***d***, and ***e*** in the molecule, and determine **hybridization** at each of the carbon centers. (2.5 points)

Atom	Hybridization
Carbon (<i>a</i>)	
Carbon (<i>b</i>)	
Carbon (<i>c</i>)	
Carbon (<i>d</i>)	
Carbon (<i>e</i>)	

ii. Now, consider the sulfur atom (S) marked as ***f***, nitrogen (N) atom (***g***), and oxygen(O) atoms (***h*** and ***i***), and determine their **hybridization**. (2 points)

Atom	Hybridization
Sulfur (<i>f</i>)	
Nitrogen (<i>g</i>)	
Oxygen (<i>h</i>)	
Oxygen (<i>i</i>)	

iii. **Describe** all the bonds formed by the carbon atoms ***a*** and ***b***, using sigma and pi designation as appropriate. (2 points)

Atom	Bonds formed (use sigma and pi as appropriate)
Carbon (<i>a</i>)	
Carbon (<i>b</i>)	

iv. How many total sp^3 -s sigma bonds are in the molecule? (1 point)

v. Determine **electronic** and **molecular geometry** at carbon (C) labeled ***e*** and sulfur (S) labeled ***f*** centres. (2 points)

Atom	Electronic Geometry	Molecular Geometry
Carbon (<i>e</i>)		
Sulfur (<i>f</i>)		

vi. Circle the value from the choices given below, that describes the **C-S-C bond angle at sulfur** (S) marked ***f***: (0.5 points)

>90° to <109.5° 109.5° >109.5 to <120° 120° >120° to <180°


Question 8 12 points

Using the **valence shell orbitals only** draw a **complete** molecular orbital (MO) diagram (with no s-p mixing) for the molecule **PS**, in the space provided below. **(6 points)**

Assume that 3s and 3p atomic orbitals will form molecular orbitals like those formed by the combination of 2s and 2p atomic orbitals that we learnt in class, and there is no s-p mixing.

Show **energy arrow**, **atomic orbitals** involved, **atom labels**, the **molecular orbitals** that are formed, **filled electrons** in the MO diagram etc. Make sure to label all the atomic and molecular orbitals.

Using the MO diagram above, answer the following questions:

- (i) Which one of the following will have the highest bond order? Circle the correct choice
 A) PS^{1-} B) PS C) PS^{1+}
- (ii) Which one of the following will have the longest bond length? Circle the correct choice
 A) PS^{1-} B) PS C) PS^{1+}
- (iii) Which of the following will be diamagnetic? Circle the correct choice.
 A) PS^{1-} B) PS C) PS^{1+}
- (iv) Draw Lewis dot structure of the molecule with the shortest bond length (based on the MO diagram) from among the three (PS^{1-} , PS, PS^{1+}), and indicate the bond order calculated from the Lewis dot structure. **(3 points)**

Lewis Dot Structure

Bond Order



9A4C57A7-EF19-4333-8A1A-544B71D9BFC0

midterm-exam-93388

#876 Page 10 of 10

SCRAP PAGE

**CHEMISTRY 110 –Midterm Exam****Thursday November 7, 2024, 6:30 – 8:30 pm****Examiners: Pallavi Sirjoosingh and Ashok Kakkar**

First name (please write as legibly as possible within the boxes)

--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--

Last name

--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--

Student ID number

--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--

This booklet has 8 questions on 10 pages (The last page can be used as a scrap page).

The staple must NOT be removed from the question booklet. Your intact question booklet must be returned at the end of the exam.

A separate booklet contains a periodic table, and datasheet.

Write your answers directly under the questions in the space provided. Do not write on the QR codes or near the top edges (as it may be cut off).

Please write **legibly and dark enough** for the scan to be readable. For students who use English as a second language, dictionaries are permitted. Calculators are allowed.

DO NOT ASK QUESTIONS OF THE INVIGILATORS about interpretations of the exam questions. If you have a concern about a question, please summarize it at the top of this page. All concerns will be evaluated after the exam. GOOD LUCK!

**Question 1 8 points** (*Assume all transitions are possible*)

a. 6 points Calculate the **energy (in kJ/mol)** required to ionize (completely remove) the electron from the **n=3** level of **He⁺ ion** (He: Helium). Show your work including equations. (3 sig. figs.)

b. 2 points

(I) Select the largest (longest) wavelength from the three choices below:

- i. Wavelength of photon to *completely remove* an electron from *1s orbital of Hydrogen atom*
- ii. Wavelength of photon to *completely remove* an electron from *1s orbital of Boron atom (B)*
- iii. Wavelength of photon to *completely remove* an electron from *1s orbital of Boron cation (B⁺)*

i or ii or iii

(II) Select the smallest (shortest) wavelength from the three choices below (*for H atom*):

- i. Wavelength of photon *emitted* for an electron transitioning from *2p to 1s orbital*
- ii. Wavelength of photon *absorbed* for an electron transitioning from *2s to 4p orbital*
- iii. Wavelength of photon *absorbed* for an electron transitioning from *1s to 3p orbital*

i or ii or iii



Question 2 8 points



A Chem110 student plots the graph shown on the left, from data obtained for the kinetic energy (eV) of the ejected electron versus frequency (s^{-1}) of the incident light, for an unknown metal. ($1\text{eV} = 1.6022 \times 10^{-19} \text{ J}$)

Based on the plot, answer the questions **a to c** below:

a. 1 point On the plot above, label the **threshold frequency**.

b. 2 points What is the value of the work potential (in J) of the unknown metal?

c. 3 points What is the *longest* wavelength (in nm) required to eject an electron from this metal's surface? Show your work including equation(s) used. (3 sig figs)

d. 2 points In the space provided below, plot the following

(Assume frequency is *above* threshold frequency):

- i. *Kinetic energy* of an ejected electron versus the *intensity* of the incident light (left)
- ii. *Number of electrons* ejected versus the *frequency* of the incident light (right)



**Question 3** **6 points****a. 2 points**

In the ground state of Ge (Germanium; $Z = 32$) atom, **name** the following orbitals:

i. Orbital containing the electron that requires the maximum energy to be removed.

ii. Orbital containing the electron that experiences the minimum effective nuclear charge.

b. 2 points

An unknown anion (A^-) has the following *excited* electronic configuration: $[Ar] 4s^2 3d^3 4p^1$

(Ar: Argon; $Z = 18$)

Give the **condensed ground state electronic configuration** of the atom, **A** and the cation, **A^{2+}** .
Show your work. (Assume all electronic transitions are possible)

Electronic Configuration of A:

Electronic Configuration of A^{2+} :

c. 2 points

In a single atom, determine how many electrons can have the following set of quantum numbers?

(i) $n = 5; l = 3$

In a single atom, determine how many electrons can have the following set of quantum numbers?

(i) $n = 3; l = 2; m_s = +1/2$

**Question 4** **8 points**

Rank the following from 1 to 4 in **increasing** order. Provide the ranking in the boxes provided.

Note: Atomic number (Z) are provided for relevant atoms

a) Increasing (lowest to highest) order of *ionization energy*: (1= lowest; 4 = highest)

Mg

Mg: Z = 12

Rb

Rb: Z = 37

Cs

Cs: Z = 55

K

K : Z = 19

b) Increasing (lowest to highest) order of *bond angle*: (1= lowest; 4 = highest)

H₂OCCl₄XeF₄

HCN

c) Increasing (smallest to largest) order of *ionic size* (1= smallest; 4 = largest)

S²⁻

S: Z = 16

O²⁻

O: Z = 8

Be²⁺

Be: Z = 4

Na⁺

Na: Z = 11

d) Increasing (lowest to highest) order of *bond energy* (1= lowest; 4 = highest)

P-F

P-Cl

P-I

P-Br

**Question 5** **12 points**

a. 9 points For the molecules HClO_4 and TeF_6 , draw the most stable *Lewis structure* showing appropriate shape and *dashed/wedged bonds*, based on VSEPR. Determine the **molecular geometry** around the central atom. For HClO_4 , label the *most polar bond*.



b. 3 points (Note: Lone pairs are not shown in structures below)

(i) For the three *resonance* structures (1, 2 and 3) shown below, answer the following question:



(I) Circle the *most contributing* resonance structure.

Structure 1

Structure 2

Structure 3

(ii) For the three structures given (4, 5, and 6) below, answer the following questions:



(I) Circle all the structures that represent *resonance structures* of each other:

Structure 4

Structure 5

Structure 6

(II) Among the resonance structures, circle the *most contributing* resonance structure.

Structure 4

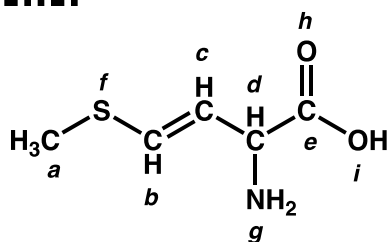
Structure 5

Structure 6

**Question 6** **6 points**

Using the data given below, calculate the **lattice energy** for SrO(s).

	ΔH° (kJ/mol)
$\text{Sr}(s) \longrightarrow \text{Sr}(g)$	+180
$\text{Sr}(g) \longrightarrow \text{Sr}^{2+}(g) + 2e^-$	+1468
$\text{O}_2(g) \longrightarrow 2\text{O}(g)$	+500
$\text{O}(g) + 2e^- \longrightarrow \text{O}^{2-}(g)$	+600
$\text{Sr}(s) + \frac{1}{2}\text{O}_2(g) \longrightarrow \text{SrO}(s)$	-550

**Question 7 10 points**

Consider the molecule given on the left, and answer questions (i) to (vi) using chemical bonding theories that we have learnt in class (valence bond theory, hybridization).

Note that the lone pairs are not shown in the structure:

i. Consider the carbon atoms marked in as ***a***, ***b***, ***c***, ***d***, and ***e*** in the molecule, and determine **hybridization** at each of the carbon centers. (2.5 points)

Atom	Hybridization
Carbon (<i>a</i>)	
Carbon (<i>b</i>)	
Carbon (<i>c</i>)	
Carbon (<i>d</i>)	
Carbon (<i>e</i>)	

ii. Now, consider the sulfur atom (S) marked as ***f***, nitrogen (N) atom (***g***), and oxygen(O) atoms (***h*** and ***i***), and determine their **hybridization**. (2 points)

Atom	Hybridization
Sulfur (<i>f</i>)	
Nitrogen (<i>g</i>)	
Oxygen (<i>h</i>)	
Oxygen (<i>i</i>)	

iii. **Describe** all the bonds formed by the carbon atoms ***a*** and ***b***, using sigma and pi designation as appropriate. (2 points)

Atom	Bonds formed (use sigma and pi as appropriate)
Carbon (<i>a</i>)	
Carbon (<i>b</i>)	

iv. How many total sp^3 -s sigma bonds are in the molecule? (1 point)

v. Determine **electronic** and **molecular geometry** at carbon (C) labeled ***e*** and sulfur (S) labeled ***f*** centres. (2 points)

Atom	Electronic Geometry	Molecular Geometry
Carbon (<i>e</i>)		
Sulfur (<i>f</i>)		

vi. Circle the value from the choices given below, that describes the **C-S-C bond angle at sulfur** (S) marked ***f***: (0.5 points)

>90° to <109.5° 109.5° >109.5 to <120° 120° >120° to <180°

**Question 8 12 points**

Using the **valence shell orbitals only** draw a **complete** molecular orbital (MO) diagram (with no s-p mixing) for the molecule **PS**, in the space provided below. **(6 points)**

Assume that 3s and 3p atomic orbitals will form molecular orbitals like those formed by the combination of 2s and 2p atomic orbitals that we learnt in class, and there is no s-p mixing.

Show **energy arrow**, **atomic orbitals** involved, **atom labels**, the **molecular orbitals** that are formed, **filled electrons** in the MO diagram etc. Make sure to label all the atomic and molecular orbitals.

Using the MO diagram above, answer the following questions:

- (i) Which one of the following will have the highest bond order? Circle the correct choice
A) PS^{1-} B) PS C) PS^{1+}
- (ii) Which one of the following will have the longest bond length? Circle the correct choice
A) PS^{1-} B) PS C) PS^{1+}
- (iii) Which of the following will be diamagnetic? Circle the correct choice.
A) PS^{1-} B) PS C) PS^{1+}
- (iv) Draw Lewis dot structure of the molecule with the shortest bond length (based on the MO diagram) from among the three (PS^{1-} , PS, PS^{1+}), and indicate the bond order calculated from the Lewis dot structure. **(3 points)**

Lewis Dot Structure

Bond Order



2B042A8A-C983-4D74-A1F8-7F260902AB97

midterm-exam-93388

#877 Page 10 of 10

SCRAP PAGE

**CHEMISTRY 110 –Midterm Exam****Thursday November 7, 2024, 6:30 – 8:30 pm**

Examiners: Pallavi Sirjoosingh and Ashok Kakkar

First name (please write as legibly as possible within the boxes)

--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--

Last name

--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--

Student ID number

--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--

This booklet has 8 questions on 10 pages (The last page can be used as a scrap page).

The staple must NOT be removed from the question booklet. Your intact question booklet must be returned at the end of the exam.

A separate booklet contains a periodic table, and datasheet.

Write your answers directly under the questions in the space provided. Do not write on the QR codes or near the top edges (as it may be cut off).

Please write **legibly and dark enough** for the scan to be readable. For students who use English as a second language, dictionaries are permitted. Calculators are allowed.

DO NOT ASK QUESTIONS OF THE INVIGILATORS about interpretations of the exam questions. If you have a concern about a question, please summarize it at the top of this page. All concerns will be evaluated after the exam. GOOD LUCK!

**Question 1 8 points** (*Assume all transitions are possible*)

a. 6 points Calculate the **energy (in kJ/mol)** required to ionize (completely remove) the electron from the **n=3** level of **He⁺ ion** (He: Helium). Show your work including equations. (3 sig. figs.)

b. 2 points

(I) Select the largest (longest) wavelength from the three choices below:

- i. Wavelength of photon to *completely remove* an electron from *1s orbital of Hydrogen atom*
- ii. Wavelength of photon to *completely remove* an electron from *1s orbital of Boron atom (B)*
- iii. Wavelength of photon to *completely remove* an electron from *1s orbital of Boron cation (B⁺)*

i or ii or iii

(II) Select the smallest (shortest) wavelength from the three choices below (*for H atom*):

- i. Wavelength of photon *emitted* for an electron transitioning from *2p to 1s orbital*
- ii. Wavelength of photon *absorbed* for an electron transitioning from *2s to 4p orbital*
- iii. Wavelength of photon *absorbed* for an electron transitioning from *1s to 3p orbital*

i or ii or iii



Question 2 8 points



A Chem110 student plots the graph shown on the left, from data obtained for the kinetic energy (eV) of the ejected electron versus frequency (s^{-1}) of the incident light, for an unknown metal. ($1\text{eV} = 1.6022 \times 10^{-19} \text{ J}$)

Based on the plot, answer the questions **a to c** below:

a. 1 point On the plot above, label the **threshold frequency**.

b. 2 points What is the value of the work potential (in J) of the unknown metal?

c. 3 points What is the *longest* wavelength (in nm) required to eject an electron from this metal's surface? Show your work including equation(s) used. (3 sig figs)

d. 2 points In the space provided below, plot the following

(Assume frequency is *above* threshold frequency):

- i. *Kinetic energy* of an ejected electron versus the *intensity* of the incident light (left)
- ii. *Number of electrons* ejected versus the *frequency* of the incident light (right)



**Question 3** **6 points****a. 2 points**

In the ground state of Ge (Germanium; $Z = 32$) atom, **name** the following orbitals:

i. Orbital containing the electron that requires the maximum energy to be removed.

ii. Orbital containing the electron that experiences the minimum effective nuclear charge.

b. 2 points

An unknown anion (A^-) has the following *excited* electronic configuration: $[Ar] 4s^2 3d^3 4p^1$

(Ar: Argon; $Z = 18$)

Give the **condensed ground state electronic configuration** of the atom, **A** and the cation, **A^{2+}** .
Show your work. (Assume all electronic transitions are possible)

Electronic Configuration of A:

Electronic Configuration of A^{2+} :

c. 2 points

In a single atom, determine how many electrons can have the following set of quantum numbers?

(i) $n = 5; l = 3$

In a single atom, determine how many electrons can have the following set of quantum numbers?

(i) $n = 3; l = 2; m_s = +1/2$

**Question 4** **8 points**

Rank the following from 1 to 4 in **increasing** order. Provide the ranking in the boxes provided.

Note: Atomic number (Z) are provided for relevant atoms

a) Increasing (lowest to highest) order of *ionization energy*: (1= lowest; 4 = highest)

Mg

Mg: Z = 12

Rb

Rb: Z = 37

Cs

Cs: Z = 55

K

K : Z = 19

b) Increasing (lowest to highest) order of *bond angle*: (1= lowest; 4 = highest)

H₂OCCl₄XeF₄

HCN

c) Increasing (smallest to largest) order of *ionic size* (1= smallest; 4 = largest)

S²⁻

S: Z = 16

O²⁻

O: Z = 8

Be²⁺

Be: Z = 4

Na⁺

Na: Z = 11

d) Increasing (lowest to highest) order of *bond energy* (1= lowest; 4 = highest)

P-F

P-Cl

P-I

P-Br

**Question 5** **12 points**

a. 9 points For the molecules HClO_4 and TeF_6 , draw the most stable *Lewis structure* showing appropriate shape and *dashed/wedged bonds*, based on VSEPR. Determine the **molecular geometry** around the central atom. For HClO_4 , label the *most polar bond*.



b. 3 points (Note: Lone pairs are not shown in structures below)

(i) For the three *resonance* structures (1, 2 and 3) shown below, answer the following question:



(I) Circle the *most contributing* resonance structure.

Structure 1

Structure 2

Structure 3

(ii) For the three structures given (4, 5, and 6) below, answer the following questions:



(I) Circle all the structures that represent *resonance structures* of each other:

Structure 4

Structure 5

Structure 6

(II) Among the resonance structures, circle the *most contributing* resonance structure.

Structure 4

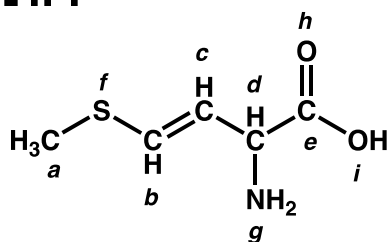
Structure 5

Structure 6

**Question 6** **6 points**

Using the data given below, calculate the **lattice energy** for SrO(s).

	ΔH° (kJ/mol)
$\text{Sr}(s) \longrightarrow \text{Sr}(g)$	+180
$\text{Sr}(g) \longrightarrow \text{Sr}^{2+}(g) + 2e^-$	+1468
$\text{O}_2(g) \longrightarrow 2\text{O}(g)$	+500
$\text{O}(g) + 2e^- \longrightarrow \text{O}^{2-}(g)$	+600
$\text{Sr}(s) + \frac{1}{2}\text{O}_2(g) \longrightarrow \text{SrO}(s)$	-550

**Question 7 10 points**

Consider the molecule given on the left, and answer questions (i) to (vi) using chemical bonding theories that we have learnt in class (valence bond theory, hybridization).

Note that the lone pairs are not shown in the structure:

i. Consider the carbon atoms marked in as ***a***, ***b***, ***c***, ***d***, and ***e*** in the molecule, and determine **hybridization** at each of the carbon centers. (2.5 points)

Atom	Hybridization
Carbon (<i>a</i>)	
Carbon (<i>b</i>)	
Carbon (<i>c</i>)	
Carbon (<i>d</i>)	
Carbon (<i>e</i>)	

ii. Now, consider the sulfur atom (S) marked as ***f***, nitrogen (N) atom (***g***), and oxygen(O) atoms (***h*** and ***i***), and determine their **hybridization**. (2 points)

Atom	Hybridization
Sulfur (<i>f</i>)	
Nitrogen (<i>g</i>)	
Oxygen (<i>h</i>)	
Oxygen (<i>i</i>)	

iii. **Describe** all the bonds formed by the carbon atoms ***a*** and ***b***, using sigma and pi designation as appropriate. (2 points)

Atom	Bonds formed (use sigma and pi as appropriate)
Carbon (<i>a</i>)	
Carbon (<i>b</i>)	

iv. How many total sp^3 -s sigma bonds are in the molecule? (1 point)

v. Determine **electronic** and **molecular geometry** at carbon (C) labeled ***e*** and sulfur (S) labeled ***f*** centres. (2 points)

Atom	Electronic Geometry	Molecular Geometry
Carbon (<i>e</i>)		
Sulfur (<i>f</i>)		

vi. Circle the value from the choices given below, that describes the **C-S-C bond angle at sulfur** (S) marked ***f***: (0.5 points)

$>90^\circ$ to $<109.5^\circ$ 109.5° >109.5 to $<120^\circ$ 120° $>120^\circ$ to $<180^\circ$



Question 8 12 points

Using the **valence shell orbitals only** draw a **complete** molecular orbital (MO) diagram (with no s-p mixing) for the molecule **PS**, in the space provided below. **(6 points)**

Assume that 3s and 3p atomic orbitals will form molecular orbitals like those formed by the combination of 2s and 2p atomic orbitals that we learnt in class, and there is no s-p mixing.

Show **energy arrow**, **atomic orbitals** involved, **atom labels**, the **molecular orbitals** that are formed, **filled electrons** in the MO diagram etc. Make sure to label all the atomic and molecular orbitals.

Using the MO diagram above, answer the following questions:

- (i) Which one of the following will have the highest bond order? Circle the correct choice
 A) PS^{1-} B) PS C) PS^{1+}
- (ii) Which one of the following will have the longest bond length? Circle the correct choice
 A) PS^{1-} B) PS C) PS^{1+}
- (iii) Which of the following will be diamagnetic? Circle the correct choice.
 A) PS^{1-} B) PS C) PS^{1+}
- (iv) Draw Lewis dot structure of the molecule with the shortest bond length (based on the MO diagram) from among the three (PS^{1-} , PS, PS^{1+}), and indicate the bond order calculated from the Lewis dot structure. **(3 points)**

Lewis Dot Structure

Bond Order



1C47928C-268A-47F6-8703-4F044D15D1FD

midterm-exam-93388

#878 Page 10 of 10

SCRAP PAGE

**CHEMISTRY 110 –Midterm Exam****Thursday November 7, 2024, 6:30 – 8:30 pm****Examiners: Pallavi Sirjoosingh and Ashok Kakkar**

First name (please write as legibly as possible within the boxes)

--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--

Last name

--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--

Student ID number

--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--

This booklet has 8 questions on 10 pages (The last page can be used as a scrap page).

The staple must NOT be removed from the question booklet. Your intact question booklet must be returned at the end of the exam.

A separate booklet contains a periodic table, and datasheet.

Write your answers directly under the questions in the space provided. Do not write on the QR codes or near the top edges (as it may be cut off).

Please write **legibly and dark enough** for the scan to be readable. For students who use English as a second language, dictionaries are permitted. Calculators are allowed.

DO NOT ASK QUESTIONS OF THE INVIGILATORS about interpretations of the exam questions. If you have a concern about a question, please summarize it at the top of this page. All concerns will be evaluated after the exam. GOOD LUCK!

**Question 1 8 points** (*Assume all transitions are possible*)

a. 6 points Calculate the **energy (in kJ/mol)** required to ionize (completely remove) the electron from the **$n=3$** level of **He^+ ion** (He: Helium). Show your work including equations. (3 sig. figs.)

b. 2 points

(I) Select the largest (longest) wavelength from the three choices below:

- i. Wavelength of photon to *completely remove* an electron from *1s orbital of Hydrogen atom*
- ii. Wavelength of photon to *completely remove* an electron from *1s orbital of Boron atom (B)*
- iii. Wavelength of photon to *completely remove* an electron from *1s orbital of Boron cation (B^+)*

i or ii or iii

(II) Select the smallest (shortest) wavelength from the three choices below (*for H atom*):

- i. Wavelength of photon *emitted* for an electron transitioning from *2p to 1s orbital*
- ii. Wavelength of photon *absorbed* for an electron transitioning from *2s to 4p orbital*
- iii. Wavelength of photon *absorbed* for an electron transitioning from *1s to 3p orbital*

i or ii or iii



Question 2 8 points



A Chem110 student plots the graph shown on the left, from data obtained for the kinetic energy (eV) of the ejected electron versus frequency (s^{-1}) of the incident light, for an unknown metal. ($1\text{eV} = 1.6022 \times 10^{-19} \text{ J}$)

Based on the plot, answer the questions **a to c** below:

a. 1 point On the plot above, label the **threshold frequency**.

b. 2 points What is the value of the work potential (in J) of the unknown metal?

c. 3 points What is the *longest* wavelength (in nm) required to eject an electron from this metal's surface? Show your work including equation(s) used. (3 sig figs)

d. 2 points In the space provided below, plot the following

(Assume frequency is *above* threshold frequency):

- i. *Kinetic energy* of an ejected electron versus the *intensity* of the incident light (left)
- ii. *Number of electrons* ejected versus the *frequency* of the incident light (right)



**Question 3** **6 points****a. 2 points**

In the ground state of Ge (Germanium; $Z = 32$) atom, **name** the following orbitals:

i. Orbital containing the electron that requires the maximum energy to be removed.

ii. Orbital containing the electron that experiences the minimum effective nuclear charge.

b. 2 points

An unknown anion (A^-) has the following *excited* electronic configuration: $[Ar] 4s^2 3d^3 4p^1$

(Ar: Argon; $Z = 18$)

Give the **condensed ground state electronic configuration** of the atom, **A** and the cation, **A^{2+}** .
Show your work. (Assume all electronic transitions are possible)

Electronic Configuration of A:

Electronic Configuration of A^{2+} :

c. 2 points

In a single atom, determine how many electrons can have the following set of quantum numbers?

(i) $n = 5; l = 3$

In a single atom, determine how many electrons can have the following set of quantum numbers?

(i) $n = 3; l = 2; m_s = +1/2$

**Question 4** **8 points**

Rank the following from 1 to 4 in **increasing** order. Provide the ranking in the boxes provided.

Note: Atomic number (Z) are provided for relevant atoms

a) Increasing (lowest to highest) order of *ionization energy*: (1= lowest; 4 = highest)

Mg

Mg: Z = 12

Rb

Rb: Z = 37

Cs

Cs: Z = 55

K

K : Z = 19

b) Increasing (lowest to highest) order of *bond angle*: (1= lowest; 4 = highest)

H₂OCCl₄XeF₄

HCN

c) Increasing (smallest to largest) order of *ionic size* (1= smallest; 4 = largest)

S²⁻

S: Z = 16

O²⁻

O: Z = 8

Be²⁺

Be: Z = 4

Na⁺

Na: Z = 11

d) Increasing (lowest to highest) order of *bond energy* (1= lowest; 4 = highest)

P-F

P-Cl

P-I

P-Br

**Question 5** **12 points**

a. 9 points For the molecules HClO_4 and TeF_6 , draw the most stable *Lewis structure* showing appropriate shape and *dashed/wedged bonds*, based on VSEPR. Determine the **molecular geometry** around the central atom. For HClO_4 , label the *most polar bond*.



b. 3 points (Note: Lone pairs are not shown in structures below)

(i) For the three *resonance* structures (1, 2 and 3) shown below, answer the following question:



(I) Circle the *most contributing* resonance structure.

Structure 1

Structure 2

Structure 3

(ii) For the three structures given (4, 5, and 6) below, answer the following questions:



(I) Circle all the structures that represent *resonance structures* of each other:

Structure 4

Structure 5

Structure 6

(II) Among the resonance structures, circle the *most contributing* resonance structure.

Structure 4

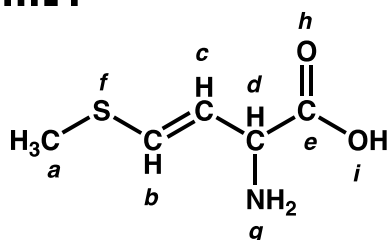
Structure 5

Structure 6

**Question 6** **6 points**

Using the data given below, calculate the **lattice energy** for SrO(s).

	ΔH° (kJ/mol)
$\text{Sr}(s) \longrightarrow \text{Sr}(g)$	+180
$\text{Sr}(g) \longrightarrow \text{Sr}^{2+}(g) + 2e^-$	+1468
$\text{O}_2(g) \longrightarrow 2\text{O}(g)$	+500
$\text{O}(g) + 2e^- \longrightarrow \text{O}^{2-}(g)$	+600
$\text{Sr}(s) + \frac{1}{2}\text{O}_2(g) \longrightarrow \text{SrO}(s)$	-550

**Question 7 10 points**

Consider the molecule given on the left, and answer questions (i) to (vi) using chemical bonding theories that we have learnt in class (valence bond theory, hybridization).

Note that the lone pairs are not shown in the structure:

i. Consider the carbon atoms marked in as ***a***, ***b***, ***c***, ***d***, and ***e*** in the molecule, and determine **hybridization** at each of the carbon centers. (2.5 points)

Atom	Hybridization
Carbon (<i>a</i>)	
Carbon (<i>b</i>)	
Carbon (<i>c</i>)	
Carbon (<i>d</i>)	
Carbon (<i>e</i>)	

ii. Now, consider the sulfur atom (S) marked as ***f***, nitrogen (N) atom (***g***), and oxygen(O) atoms (***h*** and ***i***), and determine their **hybridization**. (2 points)

Atom	Hybridization
Sulfur (<i>f</i>)	
Nitrogen (<i>g</i>)	
Oxygen (<i>h</i>)	
Oxygen (<i>i</i>)	

iii. **Describe** all the bonds formed by the carbon atoms ***a*** and ***b***, using sigma and pi designation as appropriate. (2 points)

Atom	Bonds formed (use sigma and pi as appropriate)
Carbon (<i>a</i>)	
Carbon (<i>b</i>)	

iv. How many total sp^3 -s sigma bonds are in the molecule? (1 point)

v. Determine **electronic** and **molecular geometry** at carbon (C) labeled ***e*** and sulfur (S) labeled ***f*** centres. (2 points)

Atom	Electronic Geometry	Molecular Geometry
Carbon (<i>e</i>)		
Sulfur (<i>f</i>)		

vi. Circle the value from the choices given below, that describes the **C-S-C bond angle at sulfur** (S) marked ***f***: (0.5 points)

>90° to <109.5° 109.5° >109.5 to <120° 120° >120° to <180°


Question 8 12 points

Using the **valence shell orbitals only** draw a **complete** molecular orbital (MO) diagram (with no s-p mixing) for the molecule **PS**, in the space provided below. **(6 points)**

Assume that 3s and 3p atomic orbitals will form molecular orbitals like those formed by the combination of 2s and 2p atomic orbitals that we learnt in class, and there is no s-p mixing.

Show **energy arrow**, **atomic orbitals** involved, **atom labels**, the **molecular orbitals** that are formed, **filled electrons** in the MO diagram etc. Make sure to label all the atomic and molecular orbitals.

Using the MO diagram above, answer the following questions:

- (i) Which one of the following will have the highest bond order? Circle the correct choice
 A) PS^{1-} B) PS C) PS^{1+}
- (ii) Which one of the following will have the longest bond length? Circle the correct choice
 A) PS^{1-} B) PS C) PS^{1+}
- (iii) Which of the following will be diamagnetic? Circle the correct choice.
 A) PS^{1-} B) PS C) PS^{1+}
- (iv) Draw Lewis dot structure of the molecule with the shortest bond length (based on the MO diagram) from among the three (PS^{1-} , PS, PS^{1+}), and indicate the bond order calculated from the Lewis dot structure. **(3 points)**

Lewis Dot Structure

Bond Order



21A69A0D-674B-4A84-9148-CF0C7421101C

midterm-exam-93388

#879 Page 10 of 10

SCRAP PAGE

**CHEMISTRY 110 –Midterm Exam****Thursday November 7, 2024, 6:30 – 8:30 pm**

Examiners: Pallavi Sirjoosingh and Ashok Kakkar

First name (please write as legibly as possible within the boxes)

--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--

Last name

--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--

Student ID number

--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--

This booklet has 8 questions on 10 pages (The last page can be used as a scrap page).

The staple must NOT be removed from the question booklet. Your intact question booklet must be returned at the end of the exam.

A separate booklet contains a periodic table, and datasheet.

Write your answers directly under the questions in the space provided. Do not write on the QR codes or near the top edges (as it may be cut off).

Please write **legibly and dark enough** for the scan to be readable. For students who use English as a second language, dictionaries are permitted. Calculators are allowed.

DO NOT ASK QUESTIONS OF THE INVIGILATORS about interpretations of the exam questions. If you have a concern about a question, please summarize it at the top of this page. All concerns will be evaluated after the exam. GOOD LUCK!

**Question 1 8 points** (*Assume all transitions are possible*)

a. 6 points Calculate the **energy (in kJ/mol)** required to ionize (completely remove) the electron from the **$n=3$** level of **He^+ ion** (He: Helium). Show your work including equations. (3 sig. figs.)

b. 2 points

(I) Select the largest (longest) wavelength from the three choices below:

- i. Wavelength of photon to *completely remove* an electron from *1s orbital of Hydrogen atom*
- ii. Wavelength of photon to *completely remove* an electron from *1s orbital of Boron atom (B)*
- iii. Wavelength of photon to *completely remove* an electron from *1s orbital of Boron cation (B^+)*

i or ii or iii

(II) Select the smallest (shortest) wavelength from the three choices below (*for H atom*):

- i. Wavelength of photon *emitted* for an electron transitioning from *2p to 1s orbital*
- ii. Wavelength of photon *absorbed* for an electron transitioning from *2s to 4p orbital*
- iii. Wavelength of photon *absorbed* for an electron transitioning from *1s to 3p orbital*

i or ii or iii



Question 2 8 points



A Chem110 student plots the graph shown on the left, from data obtained for the kinetic energy (eV) of the ejected electron versus frequency (s^{-1}) of the incident light, for an unknown metal. ($1\text{eV} = 1.6022 \times 10^{-19} \text{ J}$)

Based on the plot, answer the questions **a to c** below:

a. 1 point On the plot above, label the **threshold frequency**.

b. 2 points What is the value of the work potential (in J) of the unknown metal?

c. 3 points What is the *longest* wavelength (in nm) required to eject an electron from this metal's surface? Show your work including equation(s) used. (3 sig figs)

d. 2 points In the space provided below, plot the following

(Assume frequency is *above* threshold frequency):

- Kinetic energy* of an ejected electron versus the *intensity* of the incident light (left)
- Number of electrons* ejected versus the *frequency* of the incident light (right)



**Question 3** **6 points****a. 2 points**

In the ground state of Ge (Germanium; $Z = 32$) atom, **name** the following orbitals:

i. Orbital containing the electron that requires the maximum energy to be removed.

ii. Orbital containing the electron that experiences the minimum effective nuclear charge.

b. 2 points

An unknown anion (A^-) has the following *excited* electronic configuration: $[Ar] 4s^2 3d^3 4p^1$

(Ar: Argon; $Z = 18$)

Give the **condensed ground state electronic configuration** of the atom, **A** and the cation, **A^{2+}** .
Show your work. (Assume all electronic transitions are possible)

Electronic Configuration of A:

Electronic Configuration of A^{2+} :

c. 2 points

In a single atom, determine how many electrons can have the following set of quantum numbers?

(i) $n = 5; l = 3$

In a single atom, determine how many electrons can have the following set of quantum numbers?

(i) $n = 3; l = 2; m_s = +1/2$

**Question 4** **8 points**

Rank the following from 1 to 4 in **increasing** order. Provide the ranking in the boxes provided.

Note: Atomic number (Z) are provided for relevant atoms

a) Increasing (lowest to highest) order of *ionization energy*: (1= lowest; 4 = highest)

Mg

Mg: Z = 12

Rb

Rb: Z = 37

Cs

Cs: Z = 55

K

K : Z = 19

b) Increasing (lowest to highest) order of *bond angle*: (1= lowest; 4 = highest)

H₂OCCl₄XeF₄

HCN

c) Increasing (smallest to largest) order of *ionic size* (1= smallest; 4 = largest)

S²⁻

S: Z = 16

O²⁻

O: Z = 8

Be²⁺

Be: Z = 4

Na⁺

Na: Z = 11

d) Increasing (lowest to highest) order of *bond energy* (1= lowest; 4 = highest)

P-F

P-Cl

P-I

P-Br

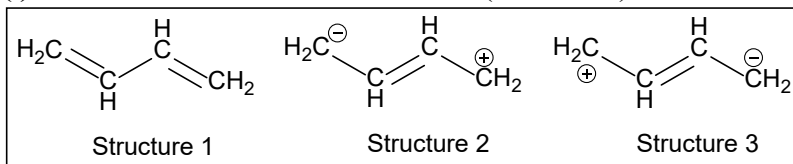
**Question 5** **12 points**

a. 9 points For the molecules HClO_4 and TeF_6 , draw the most stable *Lewis structure* showing appropriate shape and *dashed/wedged bonds*, based on VSEPR. Determine the **molecular geometry** around the central atom. For HClO_4 , label the *most polar bond*.



b. 3 points (Note: Lone pairs are not shown in structures below)

(i) For the three *resonance* structures (1, 2 and 3) shown below, answer the following question:



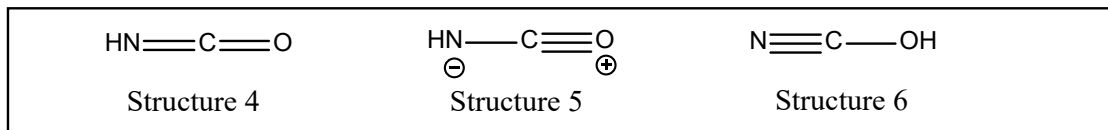
(I) Circle the *most contributing* resonance structure.

Structure 1

Structure 2

Structure 3

(ii) For the three structures given (4, 5, and 6) below, answer the following questions:



(I) Circle all the structures that represent *resonance structures* of each other:

Structure 4

Structure 5

Structure 6

(II) Among the resonance structures, circle the *most contributing* resonance structure.

Structure 4

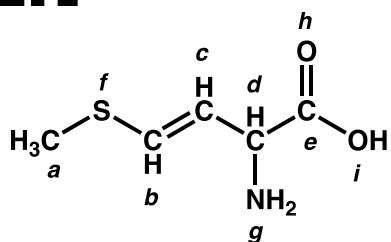
Structure 5

Structure 6

**Question 6** **6 points**

Using the data given below, calculate the **lattice energy** for SrO(s).

	ΔH° (kJ/mol)
$\text{Sr}(s) \longrightarrow \text{Sr}(g)$	+180
$\text{Sr}(g) \longrightarrow \text{Sr}^{2+}(g) + 2e^-$	+1468
$\text{O}_2(g) \longrightarrow 2\text{O}(g)$	+500
$\text{O}(g) + 2e^- \longrightarrow \text{O}^{2-}(g)$	+600
$\text{Sr}(s) + \frac{1}{2}\text{O}_2(g) \longrightarrow \text{SrO}(s)$	-550

**Question 7 10 points**

Consider the molecule given on the left, and answer questions (i) to (vi) using chemical bonding theories that we have learnt in class (valence bond theory, hybridization).

Note that the lone pairs are not shown in the structure:

i. Consider the carbon atoms marked in as ***a***, ***b***, ***c***, ***d***, and ***e*** in the molecule, and determine **hybridization** at each of the carbon centers. (2.5 points)

Atom	Hybridization
Carbon (<i>a</i>)	
Carbon (<i>b</i>)	
Carbon (<i>c</i>)	
Carbon (<i>d</i>)	
Carbon (<i>e</i>)	

ii. Now, consider the sulfur atom (S) marked as ***f***, nitrogen (N) atom (***g***), and oxygen(O) atoms (***h*** and ***i***), and determine their **hybridization**. (2 points)

Atom	Hybridization
Sulfur (<i>f</i>)	
Nitrogen (<i>g</i>)	
Oxygen (<i>h</i>)	
Oxygen (<i>i</i>)	

iii. **Describe** all the bonds formed by the carbon atoms ***a*** and ***b***, using sigma and pi designation as appropriate. (2 points)

Atom	Bonds formed (use sigma and pi as appropriate)
Carbon (<i>a</i>)	
Carbon (<i>b</i>)	

iv. How many total sp^3 -s sigma bonds are in the molecule? (1 point)

v. Determine **electronic** and **molecular geometry** at carbon (C) labeled ***e*** and sulfur (S) labeled ***f*** centres. (2 points)

Atom	Electronic Geometry	Molecular Geometry
Carbon (<i>e</i>)		
Sulfur (<i>f</i>)		

vi. Circle the value from the choices given below, that describes the **C-S-C bond angle at sulfur** (S) marked ***f***: (0.5 points)

>90° to <109.5° 109.5° >109.5 to <120° 120° >120° to <180°


Question 8 12 points

Using the **valence shell orbitals only** draw a **complete** molecular orbital (MO) diagram (with no s-p mixing) for the molecule **PS**, in the space provided below. **(6 points)**

Assume that 3s and 3p atomic orbitals will form molecular orbitals like those formed by the combination of 2s and 2p atomic orbitals that we learnt in class, and there is no s-p mixing.

Show **energy arrow**, **atomic orbitals** involved, **atom labels**, the **molecular orbitals** that are formed, **filled electrons** in the MO diagram etc. Make sure to label all the atomic and molecular orbitals.

Using the MO diagram above, answer the following questions:

- (i) Which one of the following will have the highest bond order? Circle the correct choice
 A) PS^{1-} B) PS C) PS^{1+}
- (ii) Which one of the following will have the longest bond length? Circle the correct choice
 A) PS^{1-} B) PS C) PS^{1+}
- (iii) Which of the following will be diamagnetic? Circle the correct choice.
 A) PS^{1-} B) PS C) PS^{1+}
- (iv) Draw Lewis dot structure of the molecule with the shortest bond length (based on the MO diagram) from among the three (PS^{1-} , PS, PS^{1+}), and indicate the bond order calculated from the Lewis dot structure. **(3 points)**

Lewis Dot Structure

Bond Order



81F94765-882D-4943-BEB8-EA4801DFED86

midterm-exam-93388

#880 Page 10 of 10

SCRAP PAGE

**CHEMISTRY 110 –Midterm Exam****Thursday November 7, 2024, 6:30 – 8:30 pm**

Examiners: Pallavi Sirjoosingh and Ashok Kakkar

First name (please write as legibly as possible within the boxes)

--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--

Last name

--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--

Student ID number

--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--

This booklet has 8 questions on 10 pages (The last page can be used as a scrap page).

The staple must NOT be removed from the question booklet. Your intact question booklet must be returned at the end of the exam.

A separate booklet contains a periodic table, and datasheet.

Write your answers directly under the questions in the space provided. Do not write on the QR codes or near the top edges (as it may be cut off).

Please write **legibly and dark enough** for the scan to be readable. For students who use English as a second language, dictionaries are permitted. Calculators are allowed.

DO NOT ASK QUESTIONS OF THE INVIGILATORS about interpretations of the exam questions. If you have a concern about a question, please summarize it at the top of this page. All concerns will be evaluated after the exam. GOOD LUCK!

**Question 1 8 points** (*Assume all transitions are possible*)

a. 6 points Calculate the **energy (in kJ/mol)** required to ionize (completely remove) the electron from the **n=3** level of **He⁺ ion** (He: Helium). Show your work including equations. (3 sig. figs.)

b. 2 points

(I) Select the largest (longest) wavelength from the three choices below:

- i. Wavelength of photon to *completely remove* an electron from *1s orbital of Hydrogen atom*
- ii. Wavelength of photon to *completely remove* an electron from *1s orbital of Boron atom (B)*
- iii. Wavelength of photon to *completely remove* an electron from *1s orbital of Boron cation (B⁺)*

i or ii or iii

(II) Select the smallest (shortest) wavelength from the three choices below (*for H atom*):

- i. Wavelength of photon *emitted* for an electron transitioning from *2p to 1s orbital*
- ii. Wavelength of photon *absorbed* for an electron transitioning from *2s to 4p orbital*
- iii. Wavelength of photon *absorbed* for an electron transitioning from *1s to 3p orbital*

i or ii or iii



Question 2 8 points



A Chem110 student plots the graph shown on the left, from data obtained for the kinetic energy (eV) of the ejected electron versus frequency (s^{-1}) of the incident light, for an unknown metal. ($1\text{eV} = 1.6022 \times 10^{-19} \text{ J}$)

Based on the plot, answer the questions **a to c** below:

a. 1 point On the plot above, label the **threshold frequency**.

b. 2 points What is the value of the work potential (in J) of the unknown metal?

c. 3 points What is the *longest* wavelength (in nm) required to eject an electron from this metal's surface? Show your work including equation(s) used. (3 sig figs)

d. 2 points In the space provided below, plot the following

(Assume frequency is *above* threshold frequency):

- i. *Kinetic energy* of an ejected electron versus the *intensity* of the incident light (left)
- ii. *Number of electrons* ejected versus the *frequency* of the incident light (right)



**Question 3** **6 points****a. 2 points**

In the ground state of Ge (Germanium; $Z = 32$) atom, **name** the following orbitals:

i. Orbital containing the electron that requires the maximum energy to be removed.

ii. Orbital containing the electron that experiences the minimum effective nuclear charge.

b. 2 points

An unknown anion (A^-) has the following *excited* electronic configuration: $[Ar] 4s^2 3d^3 4p^1$

(Ar: Argon; $Z = 18$)

Give the **condensed ground state electronic configuration** of the atom, **A** and the cation, **A^{2+}** .
Show your work. (Assume all electronic transitions are possible)

Electronic Configuration of A:

Electronic Configuration of A^{2+} :

c. 2 points

In a single atom, determine how many electrons can have the following set of quantum numbers?

(i) $n = 5; l = 3$

In a single atom, determine how many electrons can have the following set of quantum numbers?

(i) $n = 3; l = 2; m_s = +1/2$

**Question 4** **8 points**

Rank the following from 1 to 4 in **increasing** order. Provide the ranking in the boxes provided.

Note: Atomic number (Z) are provided for relevant atoms

a) Increasing (lowest to highest) order of *ionization energy*: (1= lowest; 4 = highest)

Mg

Mg: Z = 12

Rb

Rb: Z = 37

Cs

Cs: Z = 55

K

K : Z = 19

b) Increasing (lowest to highest) order of *bond angle*: (1= lowest; 4 = highest)

H₂OCCl₄XeF₄

HCN

c) Increasing (smallest to largest) order of *ionic size* (1= smallest; 4 = largest)

S²⁻

S: Z = 16

O²⁻

O: Z = 8

Be²⁺

Be: Z = 4

Na⁺

Na: Z = 11

d) Increasing (lowest to highest) order of *bond energy* (1= lowest; 4 = highest)

P-F

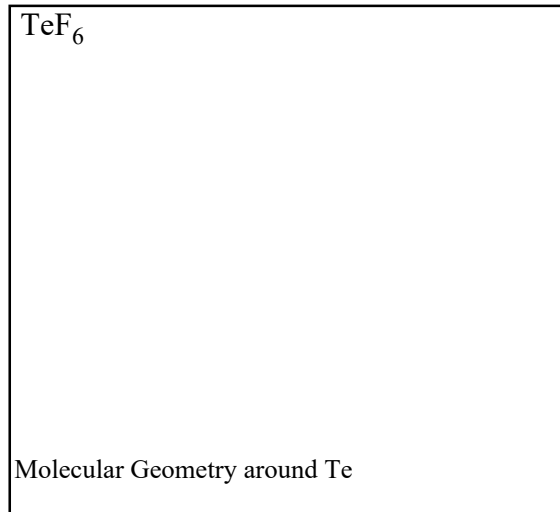
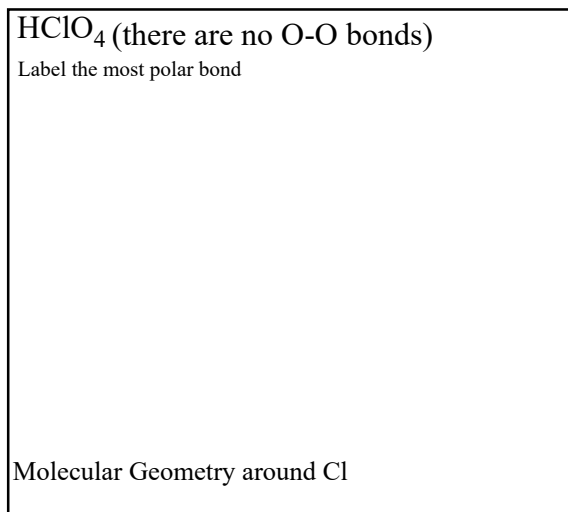
P-Cl

P-I

P-Br

**Question 5** **12 points**

a. 9 points For the molecules HClO_4 and TeF_6 , draw the most stable *Lewis structure* showing appropriate shape and *dashed/wedged bonds*, based on VSEPR. Determine the **molecular geometry** around the central atom. For HClO_4 , label the *most polar bond*.



b. 3 points (Note: Lone pairs are not shown in structures below)

(i) For the three *resonance* structures (1, 2 and 3) shown below, answer the following question:



(I) Circle the *most contributing* resonance structure.

Structure 1

Structure 2

Structure 3

(ii) For the three structures given (4, 5, and 6) below, answer the following questions:



(I) Circle all the structures that represent *resonance structures* of each other:

Structure 4

Structure 5

Structure 6

(II) Among the resonance structures, circle the *most contributing* resonance structure.

Structure 4

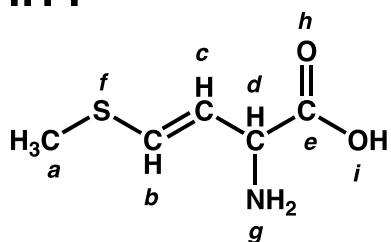
Structure 5

Structure 6

**Question 6** **6 points**

Using the data given below, calculate the **lattice energy** for SrO(s).

	ΔH° (kJ/mol)
$\text{Sr}(s) \longrightarrow \text{Sr}(g)$	+180
$\text{Sr}(g) \longrightarrow \text{Sr}^{2+}(g) + 2e^-$	+1468
$\text{O}_2(g) \longrightarrow 2\text{O}(g)$	+500
$\text{O}(g) + 2e^- \longrightarrow \text{O}^{2-}(g)$	+600
$\text{Sr}(s) + \frac{1}{2}\text{O}_2(g) \longrightarrow \text{SrO}(s)$	-550

**Question 7 10 points**

Consider the molecule given on the left, and answer questions (i) to (vi) using chemical bonding theories that we have learnt in class (valence bond theory, hybridization).

Note that the lone pairs are not shown in the structure:

i. Consider the carbon atoms marked in as ***a***, ***b***, ***c***, ***d***, and ***e*** in the molecule, and determine **hybridization** at each of the carbon centers. (2.5 points)

Atom	Hybridization
Carbon (<i>a</i>)	
Carbon (<i>b</i>)	
Carbon (<i>c</i>)	
Carbon (<i>d</i>)	
Carbon (<i>e</i>)	

ii. Now, consider the sulfur atom (S) marked as ***f***, nitrogen (N) atom (***g***), and oxygen(O) atoms (***h*** and ***i***), and determine their **hybridization**. (2 points)

Atom	Hybridization
Sulfur (<i>f</i>)	
Nitrogen (<i>g</i>)	
Oxygen (<i>h</i>)	
Oxygen (<i>i</i>)	

iii. **Describe** all the bonds formed by the carbon atoms ***a*** and ***b***, using sigma and pi designation as appropriate. (2 points)

Atom	Bonds formed (use sigma and pi as appropriate)
Carbon (<i>a</i>)	
Carbon (<i>b</i>)	

iv. How many total sp^3 -s sigma bonds are in the molecule? (1 point)

v. Determine **electronic** and **molecular geometry** at carbon (C) labeled ***e*** and sulfur (S) labeled ***f*** centres. (2 points)

Atom	Electronic Geometry	Molecular Geometry
Carbon (<i>e</i>)		
Sulfur (<i>f</i>)		

vi. Circle the value from the choices given below, that describes the **C-S-C bond angle at sulfur** (S) marked ***f***: (0.5 points)

>90° to <109.5° 109.5° >109.5 to <120° 120° >120° to <180°


Question 8 12 points

Using the **valence shell orbitals only** draw a **complete** molecular orbital (MO) diagram (with no s-p mixing) for the molecule **PS**, in the space provided below. **(6 points)**

Assume that 3s and 3p atomic orbitals will form molecular orbitals like those formed by the combination of 2s and 2p atomic orbitals that we learnt in class, and there is no s-p mixing.

Show **energy arrow**, **atomic orbitals** involved, **atom labels**, the **molecular orbitals** that are formed, **filled electrons** in the MO diagram etc. Make sure to label all the atomic and molecular orbitals.

Using the MO diagram above, answer the following questions:

- (i) Which one of the following will have the highest bond order? Circle the correct choice
 A) PS^{1-} B) PS C) PS^{1+}
- (ii) Which one of the following will have the longest bond length? Circle the correct choice
 A) PS^{1-} B) PS C) PS^{1+}
- (iii) Which of the following will be diamagnetic? Circle the correct choice.
 A) PS^{1-} B) PS C) PS^{1+}
- (iv) Draw Lewis dot structure of the molecule with the shortest bond length (based on the MO diagram) from among the three (PS^{1-} , PS, PS^{1+}), and indicate the bond order calculated from the Lewis dot structure. **(3 points)**

Lewis Dot Structure

Bond Order



60A59D73-5063-478F-A465-CD93B3B32DA8

midterm-exam-93388

#881 Page 10 of 10

SCRAP PAGE

**CHEMISTRY 110 –Midterm Exam****Thursday November 7, 2024, 6:30 – 8:30 pm**

Examiners: Pallavi Sirjoosingh and Ashok Kakkar

First name (please write as legibly as possible within the boxes)

--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--

Last name

--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--

Student ID number

--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--

This booklet has 8 questions on 10 pages (The last page can be used as a scrap page).

The staple must NOT be removed from the question booklet. Your intact question booklet must be returned at the end of the exam.

A separate booklet contains a periodic table, and datasheet.

Write your answers directly under the questions in the space provided. Do not write on the QR codes or near the top edges (as it may be cut off).

Please write **legibly and dark enough** for the scan to be readable. For students who use English as a second language, dictionaries are permitted. Calculators are allowed.

DO NOT ASK QUESTIONS OF THE INVIGILATORS about interpretations of the exam questions. If you have a concern about a question, please summarize it at the top of this page. All concerns will be evaluated after the exam. GOOD LUCK!

**Question 1 8 points** (*Assume all transitions are possible*)

a. 6 points Calculate the **energy (in kJ/mol)** required to ionize (completely remove) the electron from the **n=3** level of **He⁺ ion** (He: Helium). Show your work including equations. (3 sig. figs.)

b. 2 points

(I) Select the largest (longest) wavelength from the three choices below:

- i. Wavelength of photon to *completely remove* an electron from *1s orbital of Hydrogen atom*
- ii. Wavelength of photon to *completely remove* an electron from *1s orbital of Boron atom (B)*
- iii. Wavelength of photon to *completely remove* an electron from *1s orbital of Boron cation (B⁺)*

i or ii or iii

(II) Select the smallest (shortest) wavelength from the three choices below (*for H atom*):

- i. Wavelength of photon *emitted* for an electron transitioning from *2p to 1s orbital*
- ii. Wavelength of photon *absorbed* for an electron transitioning from *2s to 4p orbital*
- iii. Wavelength of photon *absorbed* for an electron transitioning from *1s to 3p orbital*

i or ii or iii



Question 2 8 points



A Chem110 student plots the graph shown on the left, from data obtained for the kinetic energy (eV) of the ejected electron versus frequency (s^{-1}) of the incident light, for an unknown metal. ($1\text{eV} = 1.6022 \times 10^{-19} \text{ J}$)

Based on the plot, answer the questions **a to c** below:

a. 1 point On the plot above, label the **threshold frequency**.

b. 2 points What is the value of the work potential (in J) of the unknown metal?

c. 3 points What is the *longest* wavelength (in nm) required to eject an electron from this metal's surface? Show your work including equation(s) used. (3 sig figs)

d. 2 points In the space provided below, plot the following

(Assume frequency is *above* threshold frequency):

- i. *Kinetic energy* of an ejected electron versus the *intensity* of the incident light (left)
- ii. *Number of electrons* ejected versus the *frequency* of the incident light (right)



**Question 3** **6 points****a. 2 points**

In the ground state of Ge (Germanium; $Z = 32$) atom, **name** the following orbitals:

i. Orbital containing the electron that requires the maximum energy to be removed.

ii. Orbital containing the electron that experiences the minimum effective nuclear charge.

b. 2 points

An unknown anion (A^-) has the following *excited* electronic configuration: $[Ar] 4s^2 3d^3 4p^1$

(Ar: Argon; $Z = 18$)

Give the **condensed ground state electronic configuration** of the atom, **A** and the cation, **A^{2+}** .
Show your work. (Assume all electronic transitions are possible)

Electronic Configuration of A:

Electronic Configuration of A^{2+} :

c. 2 points

In a single atom, determine how many electrons can have the following set of quantum numbers?

(i) $n = 5; l = 3$

In a single atom, determine how many electrons can have the following set of quantum numbers?

(i) $n = 3; l = 2; m_s = +1/2$

**Question 4** **8 points**

Rank the following from 1 to 4 in **increasing** order. Provide the ranking in the boxes provided.

Note: Atomic number (Z) are provided for relevant atoms

a) Increasing (lowest to highest) order of *ionization energy*: (1= lowest; 4 = highest)

Mg

Mg: Z = 12

Rb

Rb: Z = 37

Cs

Cs: Z = 55

K

K : Z = 19

b) Increasing (lowest to highest) order of *bond angle*: (1= lowest; 4 = highest)

H₂OCCl₄XeF₄

HCN

c) Increasing (smallest to largest) order of *ionic size* (1= smallest; 4 = largest)

S²⁻

S: Z = 16

O²⁻

O: Z = 8

Be²⁺

Be: Z = 4

Na⁺

Na: Z = 11

d) Increasing (lowest to highest) order of *bond energy* (1= lowest; 4 = highest)

P-F

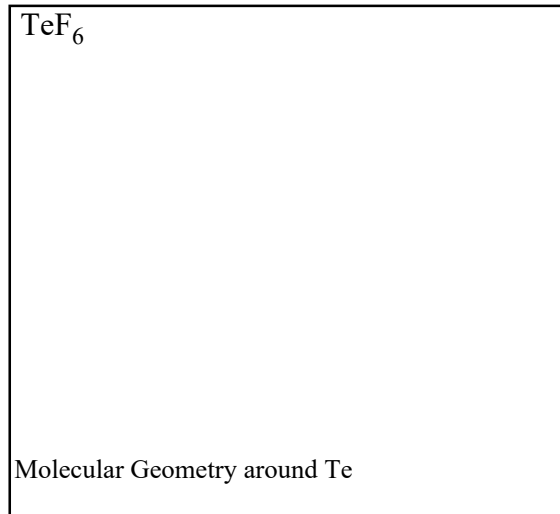
P-Cl

P-I

P-Br

**Question 5** **12 points**

a. 9 points For the molecules HClO_4 and TeF_6 , draw the most stable *Lewis structure* showing appropriate shape and *dashed/wedged bonds*, based on VSEPR. Determine the **molecular geometry** around the central atom. For HClO_4 , label the *most polar bond*.



b. 3 points (Note: Lone pairs are not shown in structures below)

(i) For the three *resonance* structures (1, 2 and 3) shown below, answer the following question:



(I) Circle the *most contributing* resonance structure.

Structure 1

Structure 2

Structure 3

(ii) For the three structures given (4, 5, and 6) below, answer the following questions:



(I) Circle all the structures that represent *resonance structures* of each other:

Structure 4

Structure 5

Structure 6

(II) Among the resonance structures, circle the *most contributing* resonance structure.

Structure 4

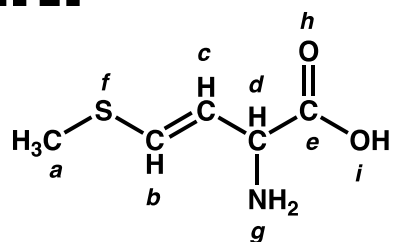
Structure 5

Structure 6

**Question 6** **6 points**

Using the data given below, calculate the **lattice energy** for SrO(s).

	ΔH° (kJ/mol)
$\text{Sr}(s) \longrightarrow \text{Sr}(g)$	+180
$\text{Sr}(g) \longrightarrow \text{Sr}^{2+}(g) + 2e^-$	+1468
$\text{O}_2(g) \longrightarrow 2\text{O}(g)$	+500
$\text{O}(g) + 2e^- \longrightarrow \text{O}^{2-}(g)$	+600
$\text{Sr}(s) + \frac{1}{2}\text{O}_2(g) \longrightarrow \text{SrO}(s)$	-550

**Question 7 10 points**

Consider the molecule given on the left, and answer questions (i) to (vi) using chemical bonding theories that we have learnt in class (valence bond theory, hybridization).

Note that the lone pairs are not shown in the structure:

i. Consider the carbon atoms marked in as ***a***, ***b***, ***c***, ***d***, and ***e*** in the molecule, and determine **hybridization** at each of the carbon centers. (2.5 points)

Atom	Hybridization
Carbon (<i>a</i>)	
Carbon (<i>b</i>)	
Carbon (<i>c</i>)	
Carbon (<i>d</i>)	
Carbon (<i>e</i>)	

ii. Now, consider the sulfur atom (S) marked as ***f***, nitrogen (N) atom (***g***), and oxygen(O) atoms (***h*** and ***i***), and determine their **hybridization**. (2 points)

Atom	Hybridization
Sulfur (<i>f</i>)	
Nitrogen (<i>g</i>)	
Oxygen (<i>h</i>)	
Oxygen (<i>i</i>)	

iii. **Describe** all the bonds formed by the carbon atoms ***a*** and ***b***, using sigma and pi designation as appropriate. (2 points)

Atom	Bonds formed (use sigma and pi as appropriate)
Carbon (<i>a</i>)	
Carbon (<i>b</i>)	

iv. How many total sp^3 -s sigma bonds are in the molecule? (1 point)

v. Determine **electronic** and **molecular geometry** at carbon (C) labeled ***e*** and sulfur (S) labeled ***f*** centres. (2 points)

Atom	Electronic Geometry	Molecular Geometry
Carbon (<i>e</i>)		
Sulfur (<i>f</i>)		

vi. Circle the value from the choices given below, that describes the **C-S-C bond angle at sulfur** (S) marked ***f***: (0.5 points)

$>90^\circ$ to $<109.5^\circ$ 109.5° >109.5 to $<120^\circ$ 120° $>120^\circ$ to $<180^\circ$


Question 8 12 points

Using the **valence shell orbitals only** draw a **complete** molecular orbital (MO) diagram (with no s-p mixing) for the molecule **PS**, in the space provided below. **(6 points)**

Assume that 3s and 3p atomic orbitals will form molecular orbitals like those formed by the combination of 2s and 2p atomic orbitals that we learnt in class, and there is no s-p mixing.

Show **energy arrow**, **atomic orbitals** involved, **atom labels**, the **molecular orbitals** that are formed, **filled electrons** in the MO diagram etc. Make sure to label all the atomic and molecular orbitals.

Using the MO diagram above, answer the following questions:

- (i) Which one of the following will have the highest bond order? Circle the correct choice
 A) PS^{1-} B) PS C) PS^{1+}
- (ii) Which one of the following will have the longest bond length? Circle the correct choice
 A) PS^{1-} B) PS C) PS^{1+}
- (iii) Which of the following will be diamagnetic? Circle the correct choice.
 A) PS^{1-} B) PS C) PS^{1+}
- (iv) Draw Lewis dot structure of the molecule with the shortest bond length (based on the MO diagram) from among the three (PS^{1-} , PS, PS^{1+}), and indicate the bond order calculated from the Lewis dot structure. **(3 points)**

Lewis Dot Structure

Bond Order



66B94101-034E-4B11-B383-5C608FAAB168

midterm-exam-93388

#882 Page 10 of 10

SCRAP PAGE

**CHEMISTRY 110 –Midterm Exam****Thursday November 7, 2024, 6:30 – 8:30 pm****Examiners: Pallavi Sirjoosingh and Ashok Kakkar**

First name (please write as legibly as possible within the boxes)

--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--

Last name

--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--

Student ID number

--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--

This booklet has 8 questions on 10 pages (The last page can be used as a scrap page).

The staple must NOT be removed from the question booklet. Your intact question booklet must be returned at the end of the exam.

A separate booklet contains a periodic table, and datasheet.

Write your answers directly under the questions in the space provided. Do not write on the QR codes or near the top edges (as it may be cut off).

Please write **legibly and dark enough** for the scan to be readable. For students who use English as a second language, dictionaries are permitted. Calculators are allowed.

DO NOT ASK QUESTIONS OF THE INVIGILATORS about interpretations of the exam questions. If you have a concern about a question, please summarize it at the top of this page. All concerns will be evaluated after the exam. GOOD LUCK!

**Question 1 8 points** (*Assume all transitions are possible*)

a. 6 points Calculate the **energy (in kJ/mol)** required to ionize (completely remove) the electron from the **$n=3$** level of **He^+ ion** (He: Helium). Show your work including equations. (3 sig. figs.)

b. 2 points

(I) Select the largest (longest) wavelength from the three choices below:

- i. Wavelength of photon to *completely remove* an electron from *1s orbital of Hydrogen atom*
- ii. Wavelength of photon to *completely remove* an electron from *1s orbital of Boron atom (B)*
- iii. Wavelength of photon to *completely remove* an electron from *1s orbital of Boron cation (B^+)*

i or ii or iii

(II) Select the smallest (shortest) wavelength from the three choices below (*for H atom*):

- i. Wavelength of photon *emitted* for an electron transitioning from *2p to 1s orbital*
- ii. Wavelength of photon *absorbed* for an electron transitioning from *2s to 4p orbital*
- iii. Wavelength of photon *absorbed* for an electron transitioning from *1s to 3p orbital*

i or ii or iii



Question 2 8 points



A Chem110 student plots the graph shown on the left, from data obtained for the kinetic energy (eV) of the ejected electron versus frequency (s^{-1}) of the incident light, for an unknown metal. ($1\text{eV} = 1.6022 \times 10^{-19} \text{ J}$)

Based on the plot, answer the questions **a to c** below:

a. 1 point On the plot above, label the **threshold frequency**.

b. 2 points What is the value of the work potential (in J) of the unknown metal?

c. 3 points What is the *longest* wavelength (in nm) required to eject an electron from this metal's surface? Show your work including equation(s) used. (3 sig figs)

d. 2 points In the space provided below, plot the following

(Assume frequency is *above* threshold frequency):

- Kinetic energy* of an ejected electron versus the *intensity* of the incident light (left)
- Number of electrons* ejected versus the *frequency* of the incident light (right)



**Question 3** **6 points****a. 2 points**

In the ground state of Ge (Germanium; $Z = 32$) atom, **name** the following orbitals:

i. Orbital containing the electron that requires the maximum energy to be removed.

ii. Orbital containing the electron that experiences the minimum effective nuclear charge.

b. 2 points

An unknown anion (A^-) has the following *excited* electronic configuration: $[Ar] 4s^2 3d^3 4p^1$

(Ar: Argon; $Z = 18$)

Give the **condensed ground state electronic configuration** of the atom, **A** and the cation, **A^{2+}** .
Show your work. (Assume all electronic transitions are possible)

Electronic Configuration of A:

Electronic Configuration of A^{2+} :

c. 2 points

In a single atom, determine how many electrons can have the following set of quantum numbers?

(i) $n = 5; l = 3$

In a single atom, determine how many electrons can have the following set of quantum numbers?

(i) $n = 3; l = 2; m_s = +1/2$

**Question 4** **8 points**

Rank the following from 1 to 4 in **increasing** order. Provide the ranking in the boxes provided.

Note: Atomic number (Z) are provided for relevant atoms

a) Increasing (lowest to highest) order of *ionization energy*: (1= lowest; 4 = highest)

Mg

Mg: Z = 12

Rb

Rb: Z = 37

Cs

Cs: Z = 55

K

K : Z = 19

b) Increasing (lowest to highest) order of *bond angle*: (1= lowest; 4 = highest)

H₂OCCl₄XeF₄

HCN

c) Increasing (smallest to largest) order of *ionic size* (1= smallest; 4 = largest)

S²⁻

S: Z = 16

O²⁻

O: Z = 8

Be²⁺

Be: Z = 4

Na⁺

Na: Z = 11

d) Increasing (lowest to highest) order of *bond energy* (1= lowest; 4 = highest)

P-F

P-Cl

P-I

P-Br

**Question 5** **12 points**

a. 9 points For the molecules HClO_4 and TeF_6 , draw the most stable *Lewis structure* showing appropriate shape and *dashed/wedged bonds*, based on VSEPR. Determine the **molecular geometry** around the central atom. For HClO_4 , label the *most polar bond*.



b. 3 points (Note: Lone pairs are not shown in structures below)

(i) For the three *resonance* structures (1, 2 and 3) shown below, answer the following question:



(I) Circle the *most contributing* resonance structure.

Structure 1

Structure 2

Structure 3

(ii) For the three structures given (4, 5, and 6) below, answer the following questions:



(I) Circle all the structures that represent *resonance structures* of each other:

Structure 4

Structure 5

Structure 6

(II) Among the resonance structures, circle the *most contributing* resonance structure.

Structure 4

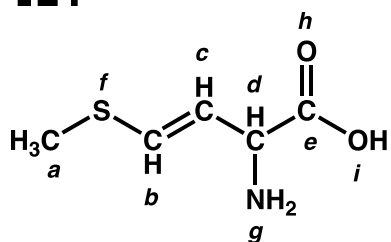
Structure 5

Structure 6

**Question 6** **6 points**

Using the data given below, calculate the **lattice energy** for SrO(s).

	ΔH° (kJ/mol)
$\text{Sr}(s) \longrightarrow \text{Sr}(g)$	+180
$\text{Sr}(g) \longrightarrow \text{Sr}^{2+}(g) + 2e^-$	+1468
$\text{O}_2(g) \longrightarrow 2\text{O}(g)$	+500
$\text{O}(g) + 2e^- \longrightarrow \text{O}^{2-}(g)$	+600
$\text{Sr}(s) + \frac{1}{2}\text{O}_2(g) \longrightarrow \text{SrO}(s)$	-550

**Question 7 10 points**

Consider the molecule given on the left, and answer questions (i) to (vi) using chemical bonding theories that we have learnt in class (valence bond theory, hybridization).

Note that the lone pairs are not shown in the structure:

i. Consider the carbon atoms marked in as ***a***, ***b***, ***c***, ***d***, and ***e*** in the molecule, and determine **hybridization** at each of the carbon centers. (2.5 points)

Atom	Hybridization
Carbon (<i>a</i>)	
Carbon (<i>b</i>)	
Carbon (<i>c</i>)	
Carbon (<i>d</i>)	
Carbon (<i>e</i>)	

ii. Now, consider the sulfur atom (S) marked as ***f***, nitrogen (N) atom (***g***), and oxygen(O) atoms (***h*** and ***i***), and determine their **hybridization**. (2 points)

Atom	Hybridization
Sulfur (<i>f</i>)	
Nitrogen (<i>g</i>)	
Oxygen (<i>h</i>)	
Oxygen (<i>i</i>)	

iii. **Describe** all the bonds formed by the carbon atoms ***a*** and ***b***, using sigma and pi designation as appropriate. (2 points)

Atom	Bonds formed (use sigma and pi as appropriate)
Carbon (<i>a</i>)	
Carbon (<i>b</i>)	

iv. How many total sp^3 -s sigma bonds are in the molecule? (1 point)

v. Determine **electronic** and **molecular geometry** at carbon (C) labeled ***e*** and sulfur (S) labeled ***f*** centres. (2 points)

Atom	Electronic Geometry	Molecular Geometry
Carbon (<i>e</i>)		
Sulfur (<i>f</i>)		

vi. Circle the value from the choices given below, that describes the **C-S-C bond angle at sulfur** (S) marked ***f***: (0.5 points)

>90° to <109.5° 109.5° >109.5 to <120° 120° >120° to <180°



Question 8 12 points

Using the **valence shell orbitals only** draw a **complete** molecular orbital (MO) diagram (with no s-p mixing) for the molecule **PS**, in the space provided below. **(6 points)**

Assume that 3s and 3p atomic orbitals will form molecular orbitals like those formed by the combination of 2s and 2p atomic orbitals that we learnt in class, and there is no s-p mixing.

Show **energy arrow**, **atomic orbitals** involved, **atom labels**, the **molecular orbitals** that are formed, **filled electrons** in the MO diagram etc. Make sure to label all the atomic and molecular orbitals.

Using the MO diagram above, answer the following questions:

- (i) Which one of the following will have the highest bond order? Circle the correct choice
 A) PS^{1-} B) PS C) PS^{1+}
- (ii) Which one of the following will have the longest bond length? Circle the correct choice
 A) PS^{1-} B) PS C) PS^{1+}
- (iii) Which of the following will be diamagnetic? Circle the correct choice.
 A) PS^{1-} B) PS C) PS^{1+}
- (iv) Draw Lewis dot structure of the molecule with the shortest bond length (based on the MO diagram) from among the three (PS^{1-} , PS, PS^{1+}), and indicate the bond order calculated from the Lewis dot structure. **(3 points)**

Lewis Dot Structure

Bond Order



DC4165C8-2082-49AA-AC89-C996605BD453

midterm-exam-93388

#883 Page 10 of 10

SCRAP PAGE

**CHEMISTRY 110 –Midterm Exam****Thursday November 7, 2024, 6:30 – 8:30 pm**

Examiners: Pallavi Sirjoosingh and Ashok Kakkar

First name (please write as legibly as possible within the boxes)

--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--

Last name

--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--

Student ID number

--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--

This booklet has 8 questions on 10 pages (The last page can be used as a scrap page).

The staple must NOT be removed from the question booklet. Your intact question booklet must be returned at the end of the exam.

A separate booklet contains a periodic table, and datasheet.

Write your answers directly under the questions in the space provided. Do not write on the QR codes or near the top edges (as it may be cut off).

Please write **legibly and dark enough** for the scan to be readable. For students who use English as a second language, dictionaries are permitted. Calculators are allowed.

DO NOT ASK QUESTIONS OF THE INVIGILATORS about interpretations of the exam questions. If you have a concern about a question, please summarize it at the top of this page. All concerns will be evaluated after the exam. GOOD LUCK!

**Question 1 8 points** (*Assume all transitions are possible*)

a. 6 points Calculate the **energy (in kJ/mol)** required to ionize (completely remove) the electron from the **n=3** level of **He⁺ ion** (He: Helium). Show your work including equations. (3 sig. figs.)

b. 2 points

(I) Select the largest (longest) wavelength from the three choices below:

- i. Wavelength of photon to *completely remove* an electron from *1s orbital of Hydrogen atom*
- ii. Wavelength of photon to *completely remove* an electron from *1s orbital of Boron atom (B)*
- iii. Wavelength of photon to *completely remove* an electron from *1s orbital of Boron cation (B⁺)*

i or ii or iii

(II) Select the smallest (shortest) wavelength from the three choices below (*for H atom*):

- i. Wavelength of photon *emitted* for an electron transitioning from *2p to 1s orbital*
- ii. Wavelength of photon *absorbed* for an electron transitioning from *2s to 4p orbital*
- iii. Wavelength of photon *absorbed* for an electron transitioning from *1s to 3p orbital*

i or ii or iii



Question 2 8 points



A Chem110 student plots the graph shown on the left, from data obtained for the kinetic energy (eV) of the ejected electron versus frequency (s^{-1}) of the incident light, for an unknown metal. ($1\text{eV} = 1.6022 \times 10^{-19} \text{ J}$)

Based on the plot, answer the questions **a to c** below:

a. 1 point On the plot above, label the **threshold frequency**.

b. 2 points What is the value of the work potential (in J) of the unknown metal?

c. 3 points What is the *longest* wavelength (in nm) required to eject an electron from this metal's surface? Show your work including equation(s) used. (3 sig figs)

d. 2 points In the space provided below, plot the following

(Assume frequency is *above* threshold frequency):

- i. *Kinetic energy* of an ejected electron versus the *intensity* of the incident light (left)
- ii. *Number of electrons* ejected versus the *frequency* of the incident light (right)



**Question 3** **6 points****a. 2 points**

In the ground state of Ge (Germanium; $Z = 32$) atom, **name** the following orbitals:

i. Orbital containing the electron that requires the maximum energy to be removed.

ii. Orbital containing the electron that experiences the minimum effective nuclear charge.

b. 2 points

An unknown anion (A^-) has the following *excited* electronic configuration: $[Ar] 4s^2 3d^3 4p^1$

(Ar: Argon; $Z = 18$)

Give the **condensed ground state electronic configuration** of the atom, **A** and the cation, **A^{2+}** .
Show your work. (Assume all electronic transitions are possible)

Electronic Configuration of A:

Electronic Configuration of A^{2+} :

c. 2 points

In a single atom, determine how many electrons can have the following set of quantum numbers?

(i) $n = 5; l = 3$

In a single atom, determine how many electrons can have the following set of quantum numbers?

(i) $n = 3; l = 2; m_s = +1/2$

**Question 4** **8 points**

Rank the following from 1 to 4 in **increasing** order. Provide the ranking in the boxes provided.

Note: Atomic number (Z) are provided for relevant atoms

a) Increasing (lowest to highest) order of *ionization energy*: (1= lowest; 4 = highest)

Mg

Mg: Z = 12

Rb

Rb: Z = 37

Cs

Cs: Z = 55

K

K : Z = 19

b) Increasing (lowest to highest) order of *bond angle*: (1= lowest; 4 = highest)

H₂OCCl₄XeF₄

HCN

c) Increasing (smallest to largest) order of *ionic size* (1= smallest; 4 = largest)

S²⁻

S: Z = 16

O²⁻

O: Z = 8

Be²⁺

Be: Z = 4

Na⁺

Na: Z = 11

d) Increasing (lowest to highest) order of *bond energy* (1= lowest; 4 = highest)

P-F

P-Cl

P-I

P-Br

**Question 5** **12 points**

a. 9 points For the molecules HClO_4 and TeF_6 , draw the most stable *Lewis structure* showing appropriate shape and *dashed/wedged bonds*, based on VSEPR. Determine the **molecular geometry** around the central atom. For HClO_4 , label the *most polar bond*.



b. 3 points (Note: Lone pairs are not shown in structures below)

(i) For the three *resonance* structures (1, 2 and 3) shown below, answer the following question:



(I) Circle the *most contributing* resonance structure.

Structure 1

Structure 2

Structure 3

(ii) For the three structures given (4, 5, and 6) below, answer the following questions:



(I) Circle all the structures that represent *resonance structures* of each other:

Structure 4

Structure 5

Structure 6

(II) Among the resonance structures, circle the *most contributing* resonance structure.

Structure 4

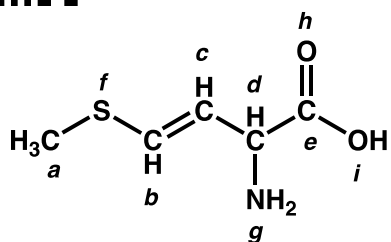
Structure 5

Structure 6

**Question 6** **6 points**

Using the data given below, calculate the **lattice energy** for SrO(s).

	ΔH° (kJ/mol)
$\text{Sr}(s) \longrightarrow \text{Sr}(g)$	+180
$\text{Sr}(g) \longrightarrow \text{Sr}^{2+}(g) + 2e^-$	+1468
$\text{O}_2(g) \longrightarrow 2\text{O}(g)$	+500
$\text{O}(g) + 2e^- \longrightarrow \text{O}^{2-}(g)$	+600
$\text{Sr}(s) + \frac{1}{2}\text{O}_2(g) \longrightarrow \text{SrO}(s)$	-550

**Question 7 10 points**

Consider the molecule given on the left, and answer questions (i) to (vi) using chemical bonding theories that we have learnt in class (valence bond theory, hybridization).

Note that the lone pairs are not shown in the structure:

i. Consider the carbon atoms marked in as ***a***, ***b***, ***c***, ***d***, and ***e*** in the molecule, and determine **hybridization** at each of the carbon centers. (2.5 points)

Atom	Hybridization
Carbon (<i>a</i>)	
Carbon (<i>b</i>)	
Carbon (<i>c</i>)	
Carbon (<i>d</i>)	
Carbon (<i>e</i>)	

ii. Now, consider the sulfur atom (S) marked as ***f***, nitrogen (N) atom (***g***), and oxygen(O) atoms (***h*** and ***i***), and determine their **hybridization**. (2 points)

Atom	Hybridization
Sulfur (<i>f</i>)	
Nitrogen (<i>g</i>)	
Oxygen (<i>h</i>)	
Oxygen (<i>i</i>)	

iii. **Describe** all the bonds formed by the carbon atoms ***a*** and ***b***, using sigma and pi designation as appropriate. (2 points)

Atom	Bonds formed (use sigma and pi as appropriate)
Carbon (<i>a</i>)	
Carbon (<i>b</i>)	

iv. How many total sp^3 -s sigma bonds are in the molecule? (1 point)

v. Determine **electronic** and **molecular geometry** at carbon (C) labeled ***e*** and sulfur (S) labeled ***f*** centres. (2 points)

Atom	Electronic Geometry	Molecular Geometry
Carbon (<i>e</i>)		
Sulfur (<i>f</i>)		

vi. Circle the value from the choices given below, that describes the **C-S-C bond angle at sulfur** (S) marked ***f***: (0.5 points)

>90° to <109.5° 109.5° >109.5 to <120° 120° >120° to <180°


Question 8 12 points

Using the **valence shell orbitals only** draw a **complete** molecular orbital (MO) diagram (with no s-p mixing) for the molecule **PS**, in the space provided below. **(6 points)**

Assume that 3s and 3p atomic orbitals will form molecular orbitals like those formed by the combination of 2s and 2p atomic orbitals that we learnt in class, and there is no s-p mixing.

Show **energy arrow**, **atomic orbitals** involved, **atom labels**, the **molecular orbitals** that are formed, **filled electrons** in the MO diagram etc. Make sure to label all the atomic and molecular orbitals.

Using the MO diagram above, answer the following questions:

- (i) Which one of the following will have the highest bond order? Circle the correct choice
 A) PS^{1-} B) PS C) PS^{1+}
- (ii) Which one of the following will have the longest bond length? Circle the correct choice
 A) PS^{1-} B) PS C) PS^{1+}
- (iii) Which of the following will be diamagnetic? Circle the correct choice.
 A) PS^{1-} B) PS C) PS^{1+}
- (iv) Draw Lewis dot structure of the molecule with the shortest bond length (based on the MO diagram) from among the three (PS^{1-} , PS, PS^{1+}), and indicate the bond order calculated from the Lewis dot structure. **(3 points)**

Lewis Dot Structure

Bond Order



26807A95-A4FB-4CC1-8952-7FD885A67639

midterm-exam-93388

#884 Page 10 of 10

SCRAP PAGE

**CHEMISTRY 110 –Midterm Exam****Thursday November 7, 2024, 6:30 – 8:30 pm****Examiners: Pallavi Sirjoosingh and Ashok Kakkar**

First name (please write as legibly as possible within the boxes)

--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--

Last name

--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--

Student ID number

--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--

This booklet has 8 questions on 10 pages (The last page can be used as a scrap page).

The staple must NOT be removed from the question booklet. Your intact question booklet must be returned at the end of the exam.

A separate booklet contains a periodic table, and datasheet.

Write your answers directly under the questions in the space provided. Do not write on the QR codes or near the top edges (as it may be cut off).

Please write **legibly and dark enough** for the scan to be readable. For students who use English as a second language, dictionaries are permitted. Calculators are allowed.

DO NOT ASK QUESTIONS OF THE INVIGILATORS about interpretations of the exam questions. If you have a concern about a question, please summarize it at the top of this page. All concerns will be evaluated after the exam. GOOD LUCK!

**Question 1 8 points** (*Assume all transitions are possible*)

a. 6 points Calculate the **energy (in kJ/mol)** required to ionize (completely remove) the electron from the **$n=3$** level of **He^+ ion** (He: Helium). Show your work including equations. (3 sig. figs.)

b. 2 points

(I) Select the largest (longest) wavelength from the three choices below:

- i. Wavelength of photon to *completely remove* an electron from *1s orbital of Hydrogen atom*
- ii. Wavelength of photon to *completely remove* an electron from *1s orbital of Boron atom (B)*
- iii. Wavelength of photon to *completely remove* an electron from *1s orbital of Boron cation (B^+)*

i or ii or iii

(II) Select the smallest (shortest) wavelength from the three choices below (*for H atom*):

- i. Wavelength of photon *emitted* for an electron transitioning from *2p to 1s orbital*
- ii. Wavelength of photon *absorbed* for an electron transitioning from *2s to 4p orbital*
- iii. Wavelength of photon *absorbed* for an electron transitioning from *1s to 3p orbital*

i or ii or iii



Question 2 8 points



A Chem110 student plots the graph shown on the left, from data obtained for the kinetic energy (eV) of the ejected electron versus frequency (s^{-1}) of the incident light, for an unknown metal. ($1\text{eV} = 1.6022 \times 10^{-19} \text{ J}$)

Based on the plot, answer the questions **a to c** below:

a. 1 point On the plot above, label the **threshold frequency**.

b. 2 points What is the value of the work potential (in J) of the unknown metal?

c. 3 points What is the *longest* wavelength (in nm) required to eject an electron from this metal's surface? Show your work including equation(s) used. (3 sig figs)

d. 2 points In the space provided below, plot the following

(Assume frequency is *above* threshold frequency):

- i. *Kinetic energy* of an ejected electron versus the *intensity* of the incident light (left)
- ii. *Number of electrons* ejected versus the *frequency* of the incident light (right)



**Question 3** **6 points****a. 2 points**

In the ground state of Ge (Germanium; $Z = 32$) atom, **name** the following orbitals:

i. Orbital containing the electron that requires the maximum energy to be removed.

ii. Orbital containing the electron that experiences the minimum effective nuclear charge.

b. 2 points

An unknown anion (A^-) has the following *excited* electronic configuration: $[Ar] 4s^2 3d^3 4p^1$

(Ar: Argon; $Z = 18$)

Give the **condensed ground state electronic configuration** of the atom, **A** and the cation, **A^{2+}** .
Show your work. (Assume all electronic transitions are possible)

Electronic Configuration of A:

Electronic Configuration of A^{2+} :

c. 2 points

In a single atom, determine how many electrons can have the following set of quantum numbers?

(i) $n = 5; l = 3$

In a single atom, determine how many electrons can have the following set of quantum numbers?

(i) $n = 3; l = 2; m_s = +1/2$

**Question 4** **8 points**

Rank the following from 1 to 4 in **increasing** order. Provide the ranking in the boxes provided.

Note: Atomic number (Z) are provided for relevant atoms

a) Increasing (lowest to highest) order of *ionization energy*: (1= lowest; 4 = highest)

Mg

Mg: Z = 12

Rb

Rb: Z = 37

Cs

Cs: Z = 55

K

K : Z = 19

b) Increasing (lowest to highest) order of *bond angle*: (1= lowest; 4 = highest)

H₂OCCl₄XeF₄

HCN

c) Increasing (smallest to largest) order of *ionic size* (1= smallest; 4 = largest)

S²⁻

S: Z = 16

O²⁻

O: Z = 8

Be²⁺

Be: Z = 4

Na⁺

Na: Z = 11

d) Increasing (lowest to highest) order of *bond energy* (1= lowest; 4 = highest)

P-F

P-Cl

P-I

P-Br

**Question 5** **12 points**

a. 9 points For the molecules HClO_4 and TeF_6 , draw the most stable *Lewis structure* showing appropriate shape and *dashed/wedged bonds*, based on VSEPR. Determine the **molecular geometry** around the central atom. For HClO_4 , label the *most polar bond*.



b. 3 points (Note: Lone pairs are not shown in structures below)

(i) For the three *resonance* structures (1, 2 and 3) shown below, answer the following question:



(I) Circle the *most contributing* resonance structure.

Structure 1

Structure 2

Structure 3

(ii) For the three structures given (4, 5, and 6) below, answer the following questions:



(I) Circle all the structures that represent *resonance structures* of each other:

Structure 4

Structure 5

Structure 6

(II) Among the resonance structures, circle the *most contributing* resonance structure.

Structure 4

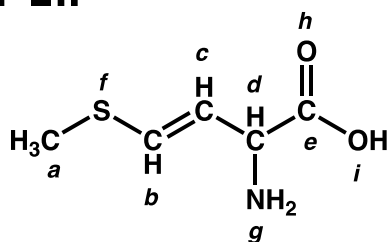
Structure 5

Structure 6

**Question 6** **6 points**

Using the data given below, calculate the **lattice energy** for SrO(s).

	ΔH° (kJ/mol)
$\text{Sr}(s) \longrightarrow \text{Sr}(g)$	+180
$\text{Sr}(g) \longrightarrow \text{Sr}^{2+}(g) + 2e^-$	+1468
$\text{O}_2(g) \longrightarrow 2\text{O}(g)$	+500
$\text{O}(g) + 2e^- \longrightarrow \text{O}^{2-}(g)$	+600
$\text{Sr}(s) + \frac{1}{2}\text{O}_2(g) \longrightarrow \text{SrO}(s)$	-550

**Question 7 10 points**

Consider the molecule given on the left, and answer questions (i) to (vi) using chemical bonding theories that we have learnt in class (valence bond theory, hybridization).

Note that the lone pairs are not shown in the structure:

i. Consider the carbon atoms marked in as ***a***, ***b***, ***c***, ***d***, and ***e*** in the molecule, and determine **hybridization** at each of the carbon centers. (2.5 points)

Atom	Hybridization
Carbon (<i>a</i>)	
Carbon (<i>b</i>)	
Carbon (<i>c</i>)	
Carbon (<i>d</i>)	
Carbon (<i>e</i>)	

ii. Now, consider the sulfur atom (S) marked as ***f***, nitrogen (N) atom (***g***), and oxygen(O) atoms (***h*** and ***i***), and determine their **hybridization**. (2 points)

Atom	Hybridization
Sulfur (<i>f</i>)	
Nitrogen (<i>g</i>)	
Oxygen (<i>h</i>)	
Oxygen (<i>i</i>)	

iii. **Describe** all the bonds formed by the carbon atoms ***a*** and ***b***, using sigma and pi designation as appropriate. (2 points)

Atom	Bonds formed (use sigma and pi as appropriate)
Carbon (<i>a</i>)	
Carbon (<i>b</i>)	

iv. How many total sp^3 -s sigma bonds are in the molecule? (1 point)

v. Determine **electronic** and **molecular geometry** at carbon (C) labeled ***e*** and sulfur (S) labeled ***f*** centres. (2 points)

Atom	Electronic Geometry	Molecular Geometry
Carbon (<i>e</i>)		
Sulfur (<i>f</i>)		

vi. Circle the value from the choices given below, that describes the **C-S-C bond angle at sulfur** (S) marked ***f***: (0.5 points)

>90° to <109.5° 109.5° >109.5 to <120° 120° >120° to <180°


Question 8 12 points

Using the **valence shell orbitals only** draw a **complete** molecular orbital (MO) diagram (with no s-p mixing) for the molecule **PS**, in the space provided below. **(6 points)**

Assume that 3s and 3p atomic orbitals will form molecular orbitals like those formed by the combination of 2s and 2p atomic orbitals that we learnt in class, and there is no s-p mixing.

Show **energy arrow**, **atomic orbitals** involved, **atom labels**, the **molecular orbitals** that are formed, **filled electrons** in the MO diagram etc. Make sure to label all the atomic and molecular orbitals.

Using the MO diagram above, answer the following questions:

- (i) Which one of the following will have the highest bond order? Circle the correct choice
 A) PS^{1-} B) PS C) PS^{1+}
- (ii) Which one of the following will have the longest bond length? Circle the correct choice
 A) PS^{1-} B) PS C) PS^{1+}
- (iii) Which of the following will be diamagnetic? Circle the correct choice.
 A) PS^{1-} B) PS C) PS^{1+}
- (iv) Draw Lewis dot structure of the molecule with the shortest bond length (based on the MO diagram) from among the three (PS^{1-} , PS, PS^{1+}), and indicate the bond order calculated from the Lewis dot structure. **(3 points)**

Lewis Dot Structure

Bond Order



96CCFB4D-A632-400C-922E-32584B9A2A42

midterm-exam-93388

#885 Page 10 of 10

SCRAP PAGE

**CHEMISTRY 110 –Midterm Exam****Thursday November 7, 2024, 6:30 – 8:30 pm**

Examiners: Pallavi Sirjoosingh and Ashok Kakkar

First name (please write as legibly as possible within the boxes)

--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--

Last name

--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--

Student ID number

--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--

This booklet has 8 questions on 10 pages (The last page can be used as a scrap page).

The staple must NOT be removed from the question booklet. Your intact question booklet must be returned at the end of the exam.

A separate booklet contains a periodic table, and datasheet.

Write your answers directly under the questions in the space provided. Do not write on the QR codes or near the top edges (as it may be cut off).

Please write **legibly and dark enough** for the scan to be readable. For students who use English as a second language, dictionaries are permitted. Calculators are allowed.

DO NOT ASK QUESTIONS OF THE INVIGILATORS about interpretations of the exam questions. If you have a concern about a question, please summarize it at the top of this page. All concerns will be evaluated after the exam. GOOD LUCK!

**Question 1 8 points** (*Assume all transitions are possible*)

a. 6 points Calculate the **energy (in kJ/mol)** required to ionize (completely remove) the electron from the **n=3** level of **He⁺ ion** (He: Helium). Show your work including equations. (3 sig. figs.)

b. 2 points

(I) Select the largest (longest) wavelength from the three choices below:

- i. Wavelength of photon to *completely remove* an electron from *1s orbital of Hydrogen atom*
- ii. Wavelength of photon to *completely remove* an electron from *1s orbital of Boron atom (B)*
- iii. Wavelength of photon to *completely remove* an electron from *1s orbital of Boron cation (B⁺)*

i or ii or iii

(II) Select the smallest (shortest) wavelength from the three choices below (*for H atom*):

- i. Wavelength of photon *emitted* for an electron transitioning from *2p to 1s orbital*
- ii. Wavelength of photon *absorbed* for an electron transitioning from *2s to 4p orbital*
- iii. Wavelength of photon *absorbed* for an electron transitioning from *1s to 3p orbital*

i or ii or iii



Question 2 8 points



A Chem110 student plots the graph shown on the left, from data obtained for the kinetic energy (eV) of the ejected electron versus frequency (s^{-1}) of the incident light, for an unknown metal. ($1\text{eV} = 1.6022 \times 10^{-19} \text{ J}$)

Based on the plot, answer the questions **a to c** below:

a. 1 point On the plot above, label the **threshold frequency**.

b. 2 points What is the value of the work potential (in J) of the unknown metal?

c. 3 points What is the *longest* wavelength (in nm) required to eject an electron from this metal's surface? Show your work including equation(s) used. (3 sig figs)

d. 2 points In the space provided below, plot the following

(Assume frequency is *above* threshold frequency):

- i. *Kinetic energy* of an ejected electron versus the *intensity* of the incident light (left)
- ii. *Number of electrons* ejected versus the *frequency* of the incident light (right)



**Question 3** **6 points****a. 2 points**

In the ground state of Ge (Germanium; $Z = 32$) atom, **name** the following orbitals:

i. Orbital containing the electron that requires the maximum energy to be removed.

ii. Orbital containing the electron that experiences the minimum effective nuclear charge.

b. 2 points

An unknown anion (A^-) has the following *excited* electronic configuration: $[Ar] 4s^2 3d^3 4p^1$

(Ar: Argon; $Z = 18$)

Give the **condensed ground state electronic configuration** of the atom, **A** and the cation, **A^{2+}** .
Show your work. (Assume all electronic transitions are possible)

Electronic Configuration of A:

Electronic Configuration of A^{2+} :

c. 2 points

In a single atom, determine how many electrons can have the following set of quantum numbers?

(i) $n = 5; l = 3$

In a single atom, determine how many electrons can have the following set of quantum numbers?

(i) $n = 3; l = 2; m_s = +1/2$

**Question 4** **8 points**

Rank the following from 1 to 4 in **increasing** order. Provide the ranking in the boxes provided.

Note: Atomic number (Z) are provided for relevant atoms

a) Increasing (lowest to highest) order of *ionization energy*: (1= lowest; 4 = highest)

Mg

Mg: Z = 12

Rb

Rb: Z = 37

Cs

Cs: Z = 55

K

K : Z = 19

b) Increasing (lowest to highest) order of *bond angle*: (1= lowest; 4 = highest)

H₂OCCl₄XeF₄

HCN

c) Increasing (smallest to largest) order of *ionic size* (1= smallest; 4 = largest)

S²⁻

S: Z = 16

O²⁻

O: Z = 8

Be²⁺

Be: Z = 4

Na⁺

Na: Z = 11

d) Increasing (lowest to highest) order of *bond energy* (1= lowest; 4 = highest)

P-F

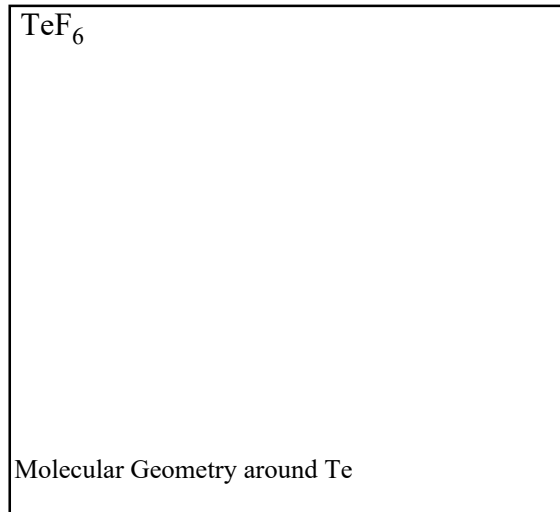
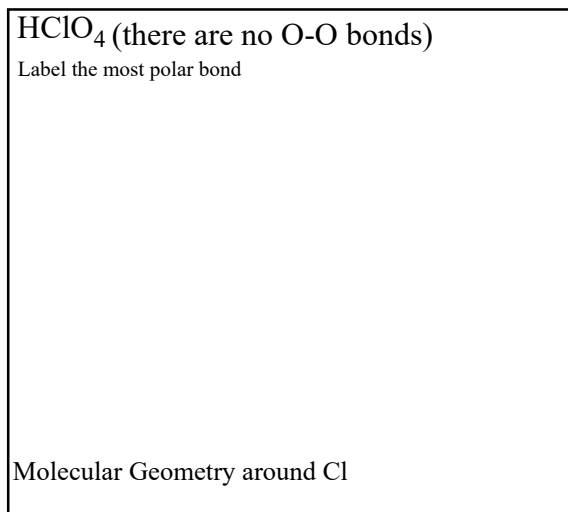
P-Cl

P-I

P-Br

**Question 5** **12 points**

a. 9 points For the molecules HClO_4 and TeF_6 , draw the most stable *Lewis structure* showing appropriate shape and *dashed/wedged bonds*, based on VSEPR. Determine the **molecular geometry** around the central atom. For HClO_4 , label the *most polar bond*.



b. 3 points (Note: Lone pairs are not shown in structures below)

(i) For the three *resonance* structures (1, 2 and 3) shown below, answer the following question:



(I) Circle the *most contributing* resonance structure.

Structure 1

Structure 2

Structure 3

(ii) For the three structures given (4, 5, and 6) below, answer the following questions:



(I) Circle all the structures that represent *resonance structures* of each other:

Structure 4

Structure 5

Structure 6

(II) Among the resonance structures, circle the *most contributing* resonance structure.

Structure 4

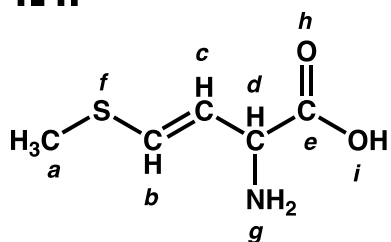
Structure 5

Structure 6

**Question 6** **6 points**

Using the data given below, calculate the **lattice energy** for SrO(s).

	ΔH° (kJ/mol)
$\text{Sr}(s) \longrightarrow \text{Sr}(g)$	+180
$\text{Sr}(g) \longrightarrow \text{Sr}^{2+}(g) + 2e^-$	+1468
$\text{O}_2(g) \longrightarrow 2\text{O}(g)$	+500
$\text{O}(g) + 2e^- \longrightarrow \text{O}^{2-}(g)$	+600
$\text{Sr}(s) + \frac{1}{2}\text{O}_2(g) \longrightarrow \text{SrO}(s)$	-550

**Question 7 10 points**

Consider the molecule given on the left, and answer questions (i) to (vi) using chemical bonding theories that we have learnt in class (valence bond theory, hybridization).

Note that the lone pairs are not shown in the structure:

i. Consider the carbon atoms marked in as ***a***, ***b***, ***c***, ***d***, and ***e*** in the molecule, and determine **hybridization** at each of the carbon centers. (2.5 points)

Atom	Hybridization
Carbon (<i>a</i>)	
Carbon (<i>b</i>)	
Carbon (<i>c</i>)	
Carbon (<i>d</i>)	
Carbon (<i>e</i>)	

ii. Now, consider the sulfur atom (S) marked as ***f***, nitrogen (N) atom (***g***), and oxygen(O) atoms (***h*** and ***i***), and determine their **hybridization**. (2 points)

Atom	Hybridization
Sulfur (<i>f</i>)	
Nitrogen (<i>g</i>)	
Oxygen (<i>h</i>)	
Oxygen (<i>i</i>)	

iii. **Describe** all the bonds formed by the carbon atoms ***a*** and ***b***, using sigma and pi designation as appropriate. (2 points)

Atom	Bonds formed (use sigma and pi as appropriate)
Carbon (<i>a</i>)	
Carbon (<i>b</i>)	

iv. How many total sp^3 -s sigma bonds are in the molecule? (1 point)

v. Determine **electronic** and **molecular geometry** at carbon (C) labeled ***e*** and sulfur (S) labeled ***f*** centres. (2 points)

Atom	Electronic Geometry	Molecular Geometry
Carbon (<i>e</i>)		
Sulfur (<i>f</i>)		

vi. Circle the value from the choices given below, that describes the **C-S-C bond angle at sulfur** (S) marked ***f***: (0.5 points)

>90° to <109.5° 109.5° >109.5 to <120° 120° >120° to <180°


Question 8 12 points

Using the **valence shell orbitals only** draw a **complete** molecular orbital (MO) diagram (with no s-p mixing) for the molecule **PS**, in the space provided below. **(6 points)**

Assume that 3s and 3p atomic orbitals will form molecular orbitals like those formed by the combination of 2s and 2p atomic orbitals that we learnt in class, and there is no s-p mixing.

Show **energy arrow**, **atomic orbitals** involved, **atom labels**, the **molecular orbitals** that are formed, **filled electrons** in the MO diagram etc. Make sure to label all the atomic and molecular orbitals.

Using the MO diagram above, answer the following questions:

- (i) Which one of the following will have the highest bond order? Circle the correct choice
 A) PS^{1-} B) PS C) PS^{1+}
- (ii) Which one of the following will have the longest bond length? Circle the correct choice
 A) PS^{1-} B) PS C) PS^{1+}
- (iii) Which of the following will be diamagnetic? Circle the correct choice.
 A) PS^{1-} B) PS C) PS^{1+}
- (iv) Draw Lewis dot structure of the molecule with the shortest bond length (based on the MO diagram) from among the three (PS^{1-} , PS, PS^{1+}), and indicate the bond order calculated from the Lewis dot structure. **(3 points)**

Lewis Dot Structure

Bond Order



D2F7F7F3-A56F-40D7-B58C-7DFC87EC74BA

midterm-exam-93388

#886 Page 10 of 10

SCRAP PAGE

**CHEMISTRY 110 –Midterm Exam****Thursday November 7, 2024, 6:30 – 8:30 pm****Examiners: Pallavi Sirjoosingh and Ashok Kakkar**

First name (please write as legibly as possible within the boxes)

--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--

Last name

--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--

Student ID number

--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--

This booklet has 8 questions on 10 pages (The last page can be used as a scrap page).

The staple must NOT be removed from the question booklet. Your intact question booklet must be returned at the end of the exam.

A separate booklet contains a periodic table, and datasheet.

Write your answers directly under the questions in the space provided. Do not write on the QR codes or near the top edges (as it may be cut off).

Please write **legibly and dark enough** for the scan to be readable. For students who use English as a second language, dictionaries are permitted. Calculators are allowed.

DO NOT ASK QUESTIONS OF THE INVIGILATORS about interpretations of the exam questions. If you have a concern about a question, please summarize it at the top of this page. All concerns will be evaluated after the exam. GOOD LUCK!

**Question 1 8 points** (*Assume all transitions are possible*)

a. 6 points Calculate the **energy (in kJ/mol)** required to ionize (completely remove) the electron from the **n=3** level of **He⁺ ion** (He: Helium). Show your work including equations. (3 sig. figs.)

b. 2 points

(I) Select the largest (longest) wavelength from the three choices below:

- i. Wavelength of photon to *completely remove* an electron from *1s orbital of Hydrogen atom*
- ii. Wavelength of photon to *completely remove* an electron from *1s orbital of Boron atom (B)*
- iii. Wavelength of photon to *completely remove* an electron from *1s orbital of Boron cation (B⁺)*

i or ii or iii

(II) Select the smallest (shortest) wavelength from the three choices below (*for H atom*):

- i. Wavelength of photon *emitted* for an electron transitioning from *2p to 1s orbital*
- ii. Wavelength of photon *absorbed* for an electron transitioning from *2s to 4p orbital*
- iii. Wavelength of photon *absorbed* for an electron transitioning from *1s to 3p orbital*

i or ii or iii



Question 2 8 points



A Chem110 student plots the graph shown on the left, from data obtained for the kinetic energy (eV) of the ejected electron versus frequency (s^{-1}) of the incident light, for an unknown metal. ($1\text{eV} = 1.6022 \times 10^{-19} \text{ J}$)

Based on the plot, answer the questions **a to c** below:

a. 1 point On the plot above, label the **threshold frequency**.

b. 2 points What is the value of the work potential (in J) of the unknown metal?

c. 3 points What is the *longest* wavelength (in nm) required to eject an electron from this metal's surface? Show your work including equation(s) used. (3 sig figs)

d. 2 points In the space provided below, plot the following

(Assume frequency is *above* threshold frequency):

- Kinetic energy* of an ejected electron versus the *intensity* of the incident light (left)
- Number of electrons* ejected versus the *frequency* of the incident light (right)



**Question 3** **6 points****a. 2 points**

In the ground state of Ge (Germanium; $Z = 32$) atom, **name** the following orbitals:

i. Orbital containing the electron that requires the maximum energy to be removed.

ii. Orbital containing the electron that experiences the minimum effective nuclear charge.

b. 2 points

An unknown anion (A^-) has the following *excited* electronic configuration: $[Ar] 4s^2 3d^3 4p^1$

(Ar: Argon; $Z = 18$)

Give the **condensed ground state electronic configuration** of the atom, **A** and the cation, **A^{2+}** .
Show your work. (Assume all electronic transitions are possible)

Electronic Configuration of A:

Electronic Configuration of A^{2+} :

c. 2 points

In a single atom, determine how many electrons can have the following set of quantum numbers?

(i) $n = 5; l = 3$

In a single atom, determine how many electrons can have the following set of quantum numbers?

(i) $n = 3; l = 2; m_s = +1/2$

**Question 4** **8 points**

Rank the following from 1 to 4 in **increasing** order. Provide the ranking in the boxes provided.

Note: Atomic number (Z) are provided for relevant atoms

a) Increasing (lowest to highest) order of *ionization energy*: (1= lowest; 4 = highest)

Mg

Mg: Z = 12

Rb

Rb: Z = 37

Cs

Cs: Z = 55

K

K : Z = 19

b) Increasing (lowest to highest) order of *bond angle*: (1= lowest; 4 = highest)

H₂OCCl₄XeF₄

HCN

c) Increasing (smallest to largest) order of *ionic size* (1= smallest; 4 = largest)

S²⁻

S: Z = 16

O²⁻

O: Z = 8

Be²⁺

Be: Z = 4

Na⁺

Na: Z = 11

d) Increasing (lowest to highest) order of *bond energy* (1= lowest; 4 = highest)

P-F

P-Cl

P-I

P-Br

**Question 5** **12 points**

a. 9 points For the molecules HClO_4 and TeF_6 , draw the most stable *Lewis structure* showing appropriate shape and *dashed/wedged bonds*, based on VSEPR. Determine the **molecular geometry** around the central atom. For HClO_4 , label the *most polar bond*.



b. 3 points (Note: Lone pairs are not shown in structures below)

(i) For the three *resonance* structures (1, 2 and 3) shown below, answer the following question:



(I) Circle the *most contributing* resonance structure.

Structure 1

Structure 2

Structure 3

(ii) For the three structures given (4, 5, and 6) below, answer the following questions:



(I) Circle all the structures that represent *resonance structures* of each other:

Structure 4

Structure 5

Structure 6

(II) Among the resonance structures, circle the *most contributing* resonance structure.

Structure 4

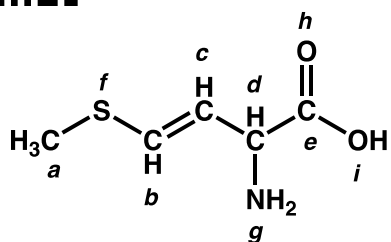
Structure 5

Structure 6

**Question 6** **6 points**

Using the data given below, calculate the **lattice energy** for SrO(s).

	ΔH° (kJ/mol)
$\text{Sr}(s) \longrightarrow \text{Sr}(g)$	+180
$\text{Sr}(g) \longrightarrow \text{Sr}^{2+}(g) + 2e^-$	+1468
$\text{O}_2(g) \longrightarrow 2\text{O}(g)$	+500
$\text{O}(g) + 2e^- \longrightarrow \text{O}^{2-}(g)$	+600
$\text{Sr}(s) + \frac{1}{2}\text{O}_2(g) \longrightarrow \text{SrO}(s)$	-550

**Question 7 10 points**

Consider the molecule given on the left, and answer questions (i) to (vi) using chemical bonding theories that we have learnt in class (valence bond theory, hybridization).

Note that the lone pairs are not shown in the structure:

i. Consider the carbon atoms marked in as ***a***, ***b***, ***c***, ***d***, and ***e*** in the molecule, and determine **hybridization** at each of the carbon centers. (2.5 points)

Atom	Hybridization
Carbon (<i>a</i>)	
Carbon (<i>b</i>)	
Carbon (<i>c</i>)	
Carbon (<i>d</i>)	
Carbon (<i>e</i>)	

ii. Now, consider the sulfur atom (S) marked as ***f***, nitrogen (N) atom (***g***), and oxygen(O) atoms (***h*** and ***i***), and determine their **hybridization**. (2 points)

Atom	Hybridization
Sulfur (<i>f</i>)	
Nitrogen (<i>g</i>)	
Oxygen (<i>h</i>)	
Oxygen (<i>i</i>)	

iii. **Describe** all the bonds formed by the carbon atoms ***a*** and ***b***, using sigma and pi designation as appropriate. (2 points)

Atom	Bonds formed (use sigma and pi as appropriate)
Carbon (<i>a</i>)	
Carbon (<i>b</i>)	

iv. How many total sp^3 -s sigma bonds are in the molecule? (1 point)

v. Determine **electronic** and **molecular geometry** at carbon (C) labeled ***e*** and sulfur (S) labeled ***f*** centres. (2 points)

Atom	Electronic Geometry	Molecular Geometry
Carbon (<i>e</i>)		
Sulfur (<i>f</i>)		

vi. Circle the value from the choices given below, that describes the **C-S-C bond angle at sulfur** (S) marked ***f***: (0.5 points)

>90° to <109.5° 109.5° >109.5 to <120° 120° >120° to <180°


Question 8 12 points

Using the **valence shell orbitals only** draw a **complete** molecular orbital (MO) diagram (with no s-p mixing) for the molecule **PS**, in the space provided below. **(6 points)**

Assume that 3s and 3p atomic orbitals will form molecular orbitals like those formed by the combination of 2s and 2p atomic orbitals that we learnt in class, and there is no s-p mixing.

Show **energy arrow**, **atomic orbitals** involved, **atom labels**, the **molecular orbitals** that are formed, **filled electrons** in the MO diagram etc. Make sure to label all the atomic and molecular orbitals.

Using the MO diagram above, answer the following questions:

- (i) Which one of the following will have the highest bond order? Circle the correct choice
 A) PS^{1-} B) PS C) PS^{1+}
- (ii) Which one of the following will have the longest bond length? Circle the correct choice
 A) PS^{1-} B) PS C) PS^{1+}
- (iii) Which of the following will be diamagnetic? Circle the correct choice.
 A) PS^{1-} B) PS C) PS^{1+}
- (iv) Draw Lewis dot structure of the molecule with the shortest bond length (based on the MO diagram) from among the three (PS^{1-} , PS, PS^{1+}), and indicate the bond order calculated from the Lewis dot structure. **(3 points)**

Lewis Dot Structure

Bond Order



248AE8F4-BCAB-4AEE-9841-07DB751818BF

midterm-exam-93388

#887 Page 10 of 10

SCRAP PAGE

**CHEMISTRY 110 –Midterm Exam****Thursday November 7, 2024, 6:30 – 8:30 pm****Examiners: Pallavi Sirjoosingh and Ashok Kakkar**

First name (please write as legibly as possible within the boxes)

--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--

Last name

--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--

Student ID number

--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--

This booklet has 8 questions on 10 pages (The last page can be used as a scrap page).

The staple must NOT be removed from the question booklet. Your intact question booklet must be returned at the end of the exam.

A separate booklet contains a periodic table, and datasheet.

Write your answers directly under the questions in the space provided. Do not write on the QR codes or near the top edges (as it may be cut off).

Please write **legibly and dark enough** for the scan to be readable. For students who use English as a second language, dictionaries are permitted. Calculators are allowed.

DO NOT ASK QUESTIONS OF THE INVIGILATORS about interpretations of the exam questions. If you have a concern about a question, please summarize it at the top of this page. All concerns will be evaluated after the exam. GOOD LUCK!

**Question 1 8 points** (*Assume all transitions are possible*)

a. 6 points Calculate the **energy (in kJ/mol)** required to ionize (completely remove) the electron from the **n=3** level of **He⁺ ion** (He: Helium). Show your work including equations. (3 sig. figs.)

b. 2 points

(I) Select the largest (longest) wavelength from the three choices below:

- i. Wavelength of photon to *completely remove* an electron from *1s orbital of Hydrogen atom*
- ii. Wavelength of photon to *completely remove* an electron from *1s orbital of Boron atom (B)*
- iii. Wavelength of photon to *completely remove* an electron from *1s orbital of Boron cation (B⁺)*

i or ii or iii

(II) Select the smallest (shortest) wavelength from the three choices below (*for H atom*):

- i. Wavelength of photon *emitted* for an electron transitioning from *2p to 1s orbital*
- ii. Wavelength of photon *absorbed* for an electron transitioning from *2s to 4p orbital*
- iii. Wavelength of photon *absorbed* for an electron transitioning from *1s to 3p orbital*

i or ii or iii



Question 2 8 points



A Chem110 student plots the graph shown on the left, from data obtained for the kinetic energy (eV) of the ejected electron versus frequency (s^{-1}) of the incident light, for an unknown metal. ($1\text{eV} = 1.6022 \times 10^{-19} \text{ J}$)

Based on the plot, answer the questions **a to c** below:

a. 1 point On the plot above, label the **threshold frequency**.

b. 2 points What is the value of the work potential (in J) of the unknown metal?

c. 3 points What is the *longest* wavelength (in nm) required to eject an electron from this metal's surface? Show your work including equation(s) used. (3 sig figs)

d. 2 points In the space provided below, plot the following

(Assume frequency is *above* threshold frequency):

- i. *Kinetic energy* of an ejected electron versus the *intensity* of the incident light (left)
- ii. *Number of electrons* ejected versus the *frequency* of the incident light (right)



**Question 3** **6 points****a. 2 points**

In the ground state of Ge (Germanium; $Z = 32$) atom, **name** the following orbitals:

i. Orbital containing the electron that requires the maximum energy to be removed.

ii. Orbital containing the electron that experiences the minimum effective nuclear charge.

b. 2 points

An unknown anion (A^-) has the following *excited* electronic configuration: $[Ar] 4s^2 3d^3 4p^1$

(Ar: Argon; $Z = 18$)

Give the **condensed ground state electronic configuration** of the atom, **A** and the cation, **A^{2+}** .
Show your work. (Assume all electronic transitions are possible)

Electronic Configuration of A:

Electronic Configuration of A^{2+} :

c. 2 points

In a single atom, determine how many electrons can have the following set of quantum numbers?

(i) $n = 5; l = 3$

In a single atom, determine how many electrons can have the following set of quantum numbers?

(i) $n = 3; l = 2; m_s = +1/2$

**Question 4** **8 points**

Rank the following from 1 to 4 in **increasing** order. Provide the ranking in the boxes provided.

Note: Atomic number (Z) are provided for relevant atoms

a) Increasing (lowest to highest) order of *ionization energy*: (1= lowest; 4 = highest)

Mg

Mg: Z = 12

Rb

Rb: Z = 37

Cs

Cs: Z = 55

K

K : Z = 19

b) Increasing (lowest to highest) order of *bond angle*: (1= lowest; 4 = highest)

H₂OCCl₄XeF₄

HCN

c) Increasing (smallest to largest) order of *ionic size* (1= smallest; 4 = largest)

S²⁻

S: Z = 16

O²⁻

O: Z = 8

Be²⁺

Be: Z = 4

Na⁺

Na: Z = 11

d) Increasing (lowest to highest) order of *bond energy* (1= lowest; 4 = highest)

P-F

P-Cl

P-I

P-Br

**Question 5** **12 points**

a. 9 points For the molecules HClO_4 and TeF_6 , draw the most stable *Lewis structure* showing appropriate shape and *dashed/wedged bonds*, based on VSEPR. Determine the **molecular geometry** around the central atom. For HClO_4 , label the *most polar bond*.



b. 3 points (Note: Lone pairs are not shown in structures below)

(i) For the three *resonance* structures (1, 2 and 3) shown below, answer the following question:



(I) Circle the *most contributing* resonance structure.

Structure 1

Structure 2

Structure 3

(ii) For the three structures given (4, 5, and 6) below, answer the following questions:



(I) Circle all the structures that represent *resonance structures* of each other:

Structure 4

Structure 5

Structure 6

(II) Among the resonance structures, circle the *most contributing* resonance structure.

Structure 4

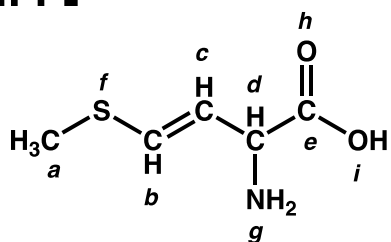
Structure 5

Structure 6

**Question 6** **6 points**

Using the data given below, calculate the **lattice energy** for SrO(s).

	ΔH° (kJ/mol)
$\text{Sr}(s) \longrightarrow \text{Sr}(g)$	+180
$\text{Sr}(g) \longrightarrow \text{Sr}^{2+}(g) + 2e^-$	+1468
$\text{O}_2(g) \longrightarrow 2\text{O}(g)$	+500
$\text{O}(g) + 2e^- \longrightarrow \text{O}^{2-}(g)$	+600
$\text{Sr}(s) + \frac{1}{2}\text{O}_2(g) \longrightarrow \text{SrO}(s)$	-550

**Question 7 10 points**

Consider the molecule given on the left, and answer questions (i) to (vi) using chemical bonding theories that we have learnt in class (valence bond theory, hybridization).

Note that the lone pairs are not shown in the structure:

i. Consider the carbon atoms marked in as ***a***, ***b***, ***c***, ***d***, and ***e*** in the molecule, and determine **hybridization** at each of the carbon centers. (2.5 points)

Atom	Hybridization
Carbon (<i>a</i>)	
Carbon (<i>b</i>)	
Carbon (<i>c</i>)	
Carbon (<i>d</i>)	
Carbon (<i>e</i>)	

ii. Now, consider the sulfur atom (S) marked as ***f***, nitrogen (N) atom (***g***), and oxygen(O) atoms (***h*** and ***i***), and determine their **hybridization**. (2 points)

Atom	Hybridization
Sulfur (<i>f</i>)	
Nitrogen (<i>g</i>)	
Oxygen (<i>h</i>)	
Oxygen (<i>i</i>)	

iii. **Describe** all the bonds formed by the carbon atoms ***a*** and ***b***, using sigma and pi designation as appropriate. (2 points)

Atom	Bonds formed (use sigma and pi as appropriate)
Carbon (<i>a</i>)	
Carbon (<i>b</i>)	

iv. How many total sp^3 -s sigma bonds are in the molecule? (1 point)

v. Determine **electronic** and **molecular geometry** at carbon (C) labeled ***e*** and sulfur (S) labeled ***f*** centres. (2 points)

Atom	Electronic Geometry	Molecular Geometry
Carbon (<i>e</i>)		
Sulfur (<i>f</i>)		

vi. Circle the value from the choices given below, that describes the **C-S-C bond angle at sulfur** (S) marked ***f***: (0.5 points)

>90° to <109.5° 109.5° >109.5 to <120° 120° >120° to <180°


Question 8 12 points

Using the **valence shell orbitals only** draw a **complete** molecular orbital (MO) diagram (with no s-p mixing) for the molecule **PS**, in the space provided below. **(6 points)**

Assume that 3s and 3p atomic orbitals will form molecular orbitals like those formed by the combination of 2s and 2p atomic orbitals that we learnt in class, and there is no s-p mixing.

Show **energy arrow**, **atomic orbitals** involved, **atom labels**, the **molecular orbitals** that are formed, **filled electrons** in the MO diagram etc. Make sure to label all the atomic and molecular orbitals.

Using the MO diagram above, answer the following questions:

- (i) Which one of the following will have the highest bond order? Circle the correct choice
 A) PS^{1-} B) PS C) PS^{1+}
- (ii) Which one of the following will have the longest bond length? Circle the correct choice
 A) PS^{1-} B) PS C) PS^{1+}
- (iii) Which of the following will be diamagnetic? Circle the correct choice.
 A) PS^{1-} B) PS C) PS^{1+}
- (iv) Draw Lewis dot structure of the molecule with the shortest bond length (based on the MO diagram) from among the three (PS^{1-} , PS, PS^{1+}), and indicate the bond order calculated from the Lewis dot structure. **(3 points)**

Lewis Dot Structure

Bond Order



2FEAE468-6943-46D3-8773-B091B0B1A57C

midterm-exam-93388

#888 Page 10 of 10

SCRAP PAGE

**CHEMISTRY 110 –Midterm Exam****Thursday November 7, 2024, 6:30 – 8:30 pm**

Examiners: Pallavi Sirjoosingh and Ashok Kakkar

First name (please write as legibly as possible within the boxes)

--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--

Last name

--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--

Student ID number

--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--

This booklet has 8 questions on 10 pages (The last page can be used as a scrap page).

The staple must NOT be removed from the question booklet. Your intact question booklet must be returned at the end of the exam.

A separate booklet contains a periodic table, and datasheet.

Write your answers directly under the questions in the space provided. Do not write on the QR codes or near the top edges (as it may be cut off).

Please write **legibly and dark enough** for the scan to be readable. For students who use English as a second language, dictionaries are permitted. Calculators are allowed.

DO NOT ASK QUESTIONS OF THE INVIGILATORS about interpretations of the exam questions. If you have a concern about a question, please summarize it at the top of this page. All concerns will be evaluated after the exam. GOOD LUCK!

**Question 1 8 points** (*Assume all transitions are possible*)

a. 6 points Calculate the **energy (in kJ/mol)** required to ionize (completely remove) the electron from the **n=3** level of **He⁺ ion** (He: Helium). Show your work including equations. (3 sig. figs.)

b. 2 points

(I) Select the largest (longest) wavelength from the three choices below:

- i. Wavelength of photon to *completely remove* an electron from *1s orbital of Hydrogen atom*
- ii. Wavelength of photon to *completely remove* an electron from *1s orbital of Boron atom (B)*
- iii. Wavelength of photon to *completely remove* an electron from *1s orbital of Boron cation (B⁺)*

i or ii or iii

(II) Select the smallest (shortest) wavelength from the three choices below (*for H atom*):

- i. Wavelength of photon *emitted* for an electron transitioning from *2p to 1s orbital*
- ii. Wavelength of photon *absorbed* for an electron transitioning from *2s to 4p orbital*
- iii. Wavelength of photon *absorbed* for an electron transitioning from *1s to 3p orbital*

i or ii or iii



Question 2 8 points



A Chem110 student plots the graph shown on the left, from data obtained for the kinetic energy (eV) of the ejected electron versus frequency (s^{-1}) of the incident light, for an unknown metal. ($1\text{eV} = 1.6022 \times 10^{-19} \text{ J}$)

Based on the plot, answer the questions **a to c** below:

a. 1 point On the plot above, label the **threshold frequency**.

b. 2 points What is the value of the work potential (in J) of the unknown metal?

c. 3 points What is the *longest* wavelength (in nm) required to eject an electron from this metal's surface? Show your work including equation(s) used. (3 sig figs)

d. 2 points In the space provided below, plot the following

(Assume frequency is *above* threshold frequency):

- Kinetic energy* of an ejected electron versus the *intensity* of the incident light (left)
- Number of electrons* ejected versus the *frequency* of the incident light (right)



**Question 3** **6 points****a. 2 points**

In the ground state of Ge (Germanium; $Z = 32$) atom, **name** the following orbitals:

i. Orbital containing the electron that requires the maximum energy to be removed.

ii. Orbital containing the electron that experiences the minimum effective nuclear charge.

b. 2 points

An unknown anion (A^-) has the following *excited* electronic configuration: $[Ar] 4s^2 3d^3 4p^1$

(Ar: Argon; $Z = 18$)

Give the **condensed ground state electronic configuration** of the atom, **A** and the cation, **A^{2+}** .
Show your work. (Assume all electronic transitions are possible)

Electronic Configuration of A:

Electronic Configuration of A^{2+} :

c. 2 points

In a single atom, determine how many electrons can have the following set of quantum numbers?

(i) $n = 5; l = 3$

In a single atom, determine how many electrons can have the following set of quantum numbers?

(i) $n = 3; l = 2; m_s = +1/2$

**Question 4** **8 points**

Rank the following from 1 to 4 in **increasing** order. Provide the ranking in the boxes provided.

Note: Atomic number (Z) are provided for relevant atoms

a) Increasing (lowest to highest) order of *ionization energy*: (1= lowest; 4 = highest)

Mg

Mg: Z = 12

Rb

Rb: Z = 37

Cs

Cs: Z = 55

K

K : Z = 19

b) Increasing (lowest to highest) order of *bond angle*: (1= lowest; 4 = highest)

H₂OCCl₄XeF₄

HCN

c) Increasing (smallest to largest) order of *ionic size* (1= smallest; 4 = largest)

S²⁻

S: Z = 16

O²⁻

O: Z = 8

Be²⁺

Be: Z = 4

Na⁺

Na: Z = 11

d) Increasing (lowest to highest) order of *bond energy* (1= lowest; 4 = highest)

P-F

P-Cl

P-I

P-Br

**Question 5** **12 points**

a. 9 points For the molecules HClO_4 and TeF_6 , draw the most stable *Lewis structure* showing appropriate shape and *dashed/wedged bonds*, based on VSEPR. Determine the **molecular geometry** around the central atom. For HClO_4 , label the *most polar bond*.



b. 3 points (Note: Lone pairs are not shown in structures below)

(i) For the three *resonance* structures (1, 2 and 3) shown below, answer the following question:



(I) Circle the *most contributing* resonance structure.

Structure 1

Structure 2

Structure 3

(ii) For the three structures given (4, 5, and 6) below, answer the following questions:



(I) Circle all the structures that represent *resonance structures* of each other:

Structure 4

Structure 5

Structure 6

(II) Among the resonance structures, circle the *most contributing* resonance structure.

Structure 4

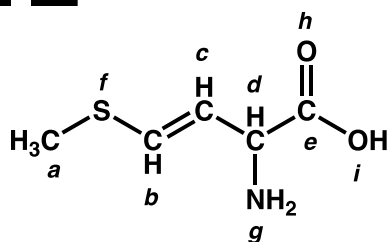
Structure 5

Structure 6

**Question 6** **6 points**

Using the data given below, calculate the **lattice energy** for SrO(s).

	ΔH° (kJ/mol)
$\text{Sr}(s) \longrightarrow \text{Sr}(g)$	+180
$\text{Sr}(g) \longrightarrow \text{Sr}^{2+}(g) + 2e^-$	+1468
$\text{O}_2(g) \longrightarrow 2\text{O}(g)$	+500
$\text{O}(g) + 2e^- \longrightarrow \text{O}^{2-}(g)$	+600
$\text{Sr}(s) + \frac{1}{2}\text{O}_2(g) \longrightarrow \text{SrO}(s)$	-550

**Question 7 10 points**

Consider the molecule given on the left, and answer questions (i) to (vi) using chemical bonding theories that we have learnt in class (valence bond theory, hybridization).

Note that the lone pairs are not shown in the structure:

i. Consider the carbon atoms marked in as ***a***, ***b***, ***c***, ***d***, and ***e*** in the molecule, and determine **hybridization** at each of the carbon centers. (2.5 points)

Atom	Hybridization
Carbon (<i>a</i>)	
Carbon (<i>b</i>)	
Carbon (<i>c</i>)	
Carbon (<i>d</i>)	
Carbon (<i>e</i>)	

ii. Now, consider the sulfur atom (S) marked as ***f***, nitrogen (N) atom (***g***), and oxygen(O) atoms (***h*** and ***i***), and determine their **hybridization**. (2 points)

Atom	Hybridization
Sulfur (<i>f</i>)	
Nitrogen (<i>g</i>)	
Oxygen (<i>h</i>)	
Oxygen (<i>i</i>)	

iii. **Describe** all the bonds formed by the carbon atoms ***a*** and ***b***, using sigma and pi designation as appropriate. (2 points)

Atom	Bonds formed (use sigma and pi as appropriate)
Carbon (<i>a</i>)	
Carbon (<i>b</i>)	

iv. How many total sp^3 -s sigma bonds are in the molecule? (1 point)

v. Determine **electronic** and **molecular geometry** at carbon (C) labeled ***e*** and sulfur (S) labeled ***f*** centres. (2 points)

Atom	Electronic Geometry	Molecular Geometry
Carbon (<i>e</i>)		
Sulfur (<i>f</i>)		

vi. Circle the value from the choices given below, that describes the **C-S-C bond angle at sulfur** (S) marked ***f***: (0.5 points)

>90° to <109.5°

109.5°

>109.5 to <120°

120°

>120° to <180°


Question 8 12 points

Using the **valence shell orbitals only** draw a **complete** molecular orbital (MO) diagram (with no s-p mixing) for the molecule **PS**, in the space provided below. **(6 points)**

Assume that 3s and 3p atomic orbitals will form molecular orbitals like those formed by the combination of 2s and 2p atomic orbitals that we learnt in class, and there is no s-p mixing.

Show **energy arrow**, **atomic orbitals** involved, **atom labels**, the **molecular orbitals** that are formed, **filled electrons** in the MO diagram etc. Make sure to label all the atomic and molecular orbitals.

Using the MO diagram above, answer the following questions:

- (i) Which one of the following will have the highest bond order? Circle the correct choice
 A) PS^{1-} B) PS C) PS^{1+}
- (ii) Which one of the following will have the longest bond length? Circle the correct choice
 A) PS^{1-} B) PS C) PS^{1+}
- (iii) Which of the following will be diamagnetic? Circle the correct choice.
 A) PS^{1-} B) PS C) PS^{1+}
- (iv) Draw Lewis dot structure of the molecule with the shortest bond length (based on the MO diagram) from among the three (PS^{1-} , PS, PS^{1+}), and indicate the bond order calculated from the Lewis dot structure. **(3 points)**

Lewis Dot Structure

Bond Order



F6046A24-1285-42ED-8A3D-A5B0DDE1BCFE

midterm-exam-93388

#889 Page 10 of 10

SCRAP PAGE

**CHEMISTRY 110 –Midterm Exam****Thursday November 7, 2024, 6:30 – 8:30 pm****Examiners: Pallavi Sirjoosingh and Ashok Kakkar**

First name (please write as legibly as possible within the boxes)

--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--

Last name

--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--

Student ID number

--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--

This booklet has 8 questions on 10 pages (The last page can be used as a scrap page).

The staple must NOT be removed from the question booklet. Your intact question booklet must be returned at the end of the exam.

A separate booklet contains a periodic table, and datasheet.

Write your answers directly under the questions in the space provided. Do not write on the QR codes or near the top edges (as it may be cut off).

Please write **legibly and dark enough** for the scan to be readable. For students who use English as a second language, dictionaries are permitted. Calculators are allowed.

DO NOT ASK QUESTIONS OF THE INVIGILATORS about interpretations of the exam questions. If you have a concern about a question, please summarize it at the top of this page. All concerns will be evaluated after the exam. GOOD LUCK!

**Question 1 8 points** (*Assume all transitions are possible*)

a. 6 points Calculate the **energy (in kJ/mol)** required to ionize (completely remove) the electron from the **n=3** level of **He⁺ ion** (He: Helium). Show your work including equations. (3 sig. figs.)

b. 2 points

(I) Select the largest (longest) wavelength from the three choices below:

- i. Wavelength of photon to *completely remove* an electron from *1s orbital of Hydrogen atom*
- ii. Wavelength of photon to *completely remove* an electron from *1s orbital of Boron atom (B)*
- iii. Wavelength of photon to *completely remove* an electron from *1s orbital of Boron cation (B⁺)*

i or ii or iii

(II) Select the smallest (shortest) wavelength from the three choices below (*for H atom*):

- i. Wavelength of photon *emitted* for an electron transitioning from *2p to 1s orbital*
- ii. Wavelength of photon *absorbed* for an electron transitioning from *2s to 4p orbital*
- iii. Wavelength of photon *absorbed* for an electron transitioning from *1s to 3p orbital*

i or ii or iii



Question 2 8 points



A Chem110 student plots the graph shown on the left, from data obtained for the kinetic energy (eV) of the ejected electron versus frequency (s^{-1}) of the incident light, for an unknown metal. ($1\text{eV} = 1.6022 \times 10^{-19} \text{ J}$)

Based on the plot, answer the questions **a to c** below:

a. 1 point On the plot above, label the **threshold frequency**.

b. 2 points What is the value of the work potential (in J) of the unknown metal?

c. 3 points What is the *longest* wavelength (in nm) required to eject an electron from this metal's surface? Show your work including equation(s) used. (3 sig figs)

d. 2 points In the space provided below, plot the following

(Assume frequency is *above* threshold frequency):

- i. *Kinetic energy* of an ejected electron versus the *intensity* of the incident light (left)
- ii. *Number of electrons* ejected versus the *frequency* of the incident light (right)



**Question 3** **6 points****a. 2 points**

In the ground state of Ge (Germanium; $Z = 32$) atom, **name** the following orbitals:

i. Orbital containing the electron that requires the maximum energy to be removed.

ii. Orbital containing the electron that experiences the minimum effective nuclear charge.

b. 2 points

An unknown anion (A^-) has the following *excited* electronic configuration: $[Ar] 4s^2 3d^3 4p^1$

(Ar: Argon; $Z = 18$)

Give the **condensed ground state electronic configuration** of the atom, **A** and the cation, **A^{2+}** .
Show your work. (Assume all electronic transitions are possible)

Electronic Configuration of A:

Electronic Configuration of A^{2+} :

c. 2 points

In a single atom, determine how many electrons can have the following set of quantum numbers?

(i) $n = 5; l = 3$

In a single atom, determine how many electrons can have the following set of quantum numbers?

(i) $n = 3; l = 2; m_s = +1/2$

**Question 4** **8 points**

Rank the following from 1 to 4 in **increasing** order. Provide the ranking in the boxes provided.

Note: Atomic number (Z) are provided for relevant atoms

a) Increasing (lowest to highest) order of *ionization energy*: (1= lowest; 4 = highest)

Mg

Mg: Z = 12

Rb

Rb: Z = 37

Cs

Cs: Z = 55

K

K : Z = 19

b) Increasing (lowest to highest) order of *bond angle*: (1= lowest; 4 = highest)

H₂OCCl₄XeF₄

HCN

c) Increasing (smallest to largest) order of *ionic size* (1= smallest; 4 = largest)

S²⁻

S: Z = 16

O²⁻

O: Z = 8

Be²⁺

Be: Z = 4

Na⁺

Na: Z = 11

d) Increasing (lowest to highest) order of *bond energy* (1= lowest; 4 = highest)

P-F

P-Cl

P-I

P-Br

**Question 5** **12 points**

a. 9 points For the molecules HClO_4 and TeF_6 , draw the most stable *Lewis structure* showing appropriate shape and *dashed/wedged bonds*, based on VSEPR. Determine the **molecular geometry** around the central atom. For HClO_4 , label the *most polar bond*.



b. 3 points (Note: Lone pairs are not shown in structures below)

(i) For the three *resonance* structures (1, 2 and 3) shown below, answer the following question:



(I) Circle the *most contributing* resonance structure.

Structure 1

Structure 2

Structure 3

(ii) For the three structures given (4, 5, and 6) below, answer the following questions:



(I) Circle all the structures that represent *resonance structures* of each other:

Structure 4

Structure 5

Structure 6

(II) Among the resonance structures, circle the *most contributing* resonance structure.

Structure 4

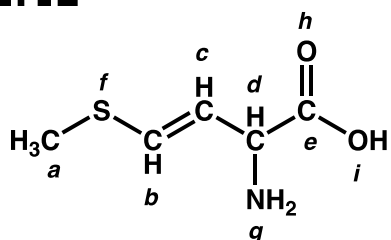
Structure 5

Structure 6

**Question 6** **6 points**

Using the data given below, calculate the **lattice energy** for SrO(s).

	ΔH° (kJ/mol)
$\text{Sr}(s) \longrightarrow \text{Sr}(g)$	+180
$\text{Sr}(g) \longrightarrow \text{Sr}^{2+}(g) + 2e^-$	+1468
$\text{O}_2(g) \longrightarrow 2\text{O}(g)$	+500
$\text{O}(g) + 2e^- \longrightarrow \text{O}^{2-}(g)$	+600
$\text{Sr}(s) + \frac{1}{2}\text{O}_2(g) \longrightarrow \text{SrO}(s)$	-550

**Question 7 10 points**

Consider the molecule given on the left, and answer questions (i) to (vi) using chemical bonding theories that we have learnt in class (valence bond theory, hybridization).

Note that the lone pairs are not shown in the structure:

i. Consider the carbon atoms marked in as ***a***, ***b***, ***c***, ***d***, and ***e*** in the molecule, and determine **hybridization** at each of the carbon centers. (2.5 points)

Atom	Hybridization
Carbon (<i>a</i>)	
Carbon (<i>b</i>)	
Carbon (<i>c</i>)	
Carbon (<i>d</i>)	
Carbon (<i>e</i>)	

ii. Now, consider the sulfur atom (S) marked as ***f***, nitrogen (N) atom (***g***), and oxygen(O) atoms (***h*** and ***i***), and determine their **hybridization**. (2 points)

Atom	Hybridization
Sulfur (<i>f</i>)	
Nitrogen (<i>g</i>)	
Oxygen (<i>h</i>)	
Oxygen (<i>i</i>)	

iii. **Describe** all the bonds formed by the carbon atoms ***a*** and ***b***, using sigma and pi designation as appropriate. (2 points)

Atom	Bonds formed (use sigma and pi as appropriate)
Carbon (<i>a</i>)	
Carbon (<i>b</i>)	

iv. How many total sp^3 -s sigma bonds are in the molecule? (1 point)

v. Determine **electronic** and **molecular geometry** at carbon (C) labeled ***e*** and sulfur (S) labeled ***f*** centres. (2 points)

Atom	Electronic Geometry	Molecular Geometry
Carbon (<i>e</i>)		
Sulfur (<i>f</i>)		

vi. Circle the value from the choices given below, that describes the **C-S-C bond angle at sulfur** (S) marked ***f***: (0.5 points)

>90° to <109.5°

109.5°

>109.5 to <120°

120°

>120° to <180°


Question 8 12 points

Using the **valence shell orbitals only** draw a **complete** molecular orbital (MO) diagram (with no s-p mixing) for the molecule **PS**, in the space provided below. **(6 points)**

Assume that 3s and 3p atomic orbitals will form molecular orbitals like those formed by the combination of 2s and 2p atomic orbitals that we learnt in class, and there is no s-p mixing.

Show **energy arrow**, **atomic orbitals** involved, **atom labels**, the **molecular orbitals** that are formed, **filled electrons** in the MO diagram etc. Make sure to label all the atomic and molecular orbitals.

Using the MO diagram above, answer the following questions:

- (i) Which one of the following will have the highest bond order? Circle the correct choice
 A) PS^{1-} B) PS C) PS^{1+}
- (ii) Which one of the following will have the longest bond length? Circle the correct choice
 A) PS^{1-} B) PS C) PS^{1+}
- (iii) Which of the following will be diamagnetic? Circle the correct choice.
 A) PS^{1-} B) PS C) PS^{1+}
- (iv) Draw Lewis dot structure of the molecule with the shortest bond length (based on the MO diagram) from among the three (PS^{1-} , PS, PS^{1+}), and indicate the bond order calculated from the Lewis dot structure. **(3 points)**

Lewis Dot Structure

Bond Order



20C0DA2A-B158-44A8-BBD2-41F3C8623EF0

midterm-exam-93388

#890 Page 10 of 10

SCRAP PAGE

**CHEMISTRY 110 –Midterm Exam****Thursday November 7, 2024, 6:30 – 8:30 pm****Examiners: Pallavi Sirjoosingh and Ashok Kakkar**

First name (please write as legibly as possible within the boxes)

--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--

Last name

--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--

Student ID number

--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--

This booklet has 8 questions on 10 pages (The last page can be used as a scrap page).

The staple must NOT be removed from the question booklet. Your intact question booklet must be returned at the end of the exam.

A separate booklet contains a periodic table, and datasheet.

Write your answers directly under the questions in the space provided. Do not write on the QR codes or near the top edges (as it may be cut off).

Please write **legibly and dark enough** for the scan to be readable. For students who use English as a second language, dictionaries are permitted. Calculators are allowed.

DO NOT ASK QUESTIONS OF THE INVIGILATORS about interpretations of the exam questions. If you have a concern about a question, please summarize it at the top of this page. All concerns will be evaluated after the exam. GOOD LUCK!

**Question 1 8 points** (*Assume all transitions are possible*)

a. 6 points Calculate the **energy (in kJ/mol)** required to ionize (completely remove) the electron from the **$n=3$** level of **He^+ ion** (He: Helium). Show your work including equations. (3 sig. figs.)

b. 2 points

(I) Select the largest (longest) wavelength from the three choices below:

- i. Wavelength of photon to *completely remove* an electron from *1s orbital of Hydrogen atom*
- ii. Wavelength of photon to *completely remove* an electron from *1s orbital of Boron atom (B)*
- iii. Wavelength of photon to *completely remove* an electron from *1s orbital of Boron cation (B^+)*

i or ii or iii

(II) Select the smallest (shortest) wavelength from the three choices below (*for H atom*):

- i. Wavelength of photon *emitted* for an electron transitioning from *2p to 1s orbital*
- ii. Wavelength of photon *absorbed* for an electron transitioning from *2s to 4p orbital*
- iii. Wavelength of photon *absorbed* for an electron transitioning from *1s to 3p orbital*

i or ii or iii



Question 2 8 points



A Chem110 student plots the graph shown on the left, from data obtained for the kinetic energy (eV) of the ejected electron versus frequency (s^{-1}) of the incident light, for an unknown metal. ($1\text{eV} = 1.6022 \times 10^{-19} \text{ J}$)

Based on the plot, answer the questions **a to c** below:

a. 1 point On the plot above, label the **threshold frequency**.

b. 2 points What is the value of the work potential (in J) of the unknown metal?

c. 3 points What is the *longest* wavelength (in nm) required to eject an electron from this metal's surface? Show your work including equation(s) used. (3 sig figs)

d. 2 points In the space provided below, plot the following

(Assume frequency is *above* threshold frequency):

- i. *Kinetic energy* of an ejected electron versus the *intensity* of the incident light (left)
- ii. *Number of electrons* ejected versus the *frequency* of the incident light (right)



**Question 3** **6 points****a. 2 points**

In the ground state of Ge (Germanium; $Z = 32$) atom, **name** the following orbitals:

i. Orbital containing the electron that requires the maximum energy to be removed.

ii. Orbital containing the electron that experiences the minimum effective nuclear charge.

b. 2 points

An unknown anion (A^-) has the following *excited* electronic configuration: $[Ar] 4s^2 3d^3 4p^1$

(Ar: Argon; $Z = 18$)

Give the **condensed ground state electronic configuration** of the atom, **A** and the cation, **A^{2+}** .
Show your work. (Assume all electronic transitions are possible)

Electronic Configuration of A:

Electronic Configuration of A^{2+} :

c. 2 points

In a single atom, determine how many electrons can have the following set of quantum numbers?

(i) $n = 5; l = 3$

In a single atom, determine how many electrons can have the following set of quantum numbers?

(i) $n = 3; l = 2; m_s = +1/2$

**Question 4** **8 points**

Rank the following from 1 to 4 in **increasing** order. Provide the ranking in the boxes provided.

Note: Atomic number (Z) are provided for relevant atoms

a) Increasing (lowest to highest) order of *ionization energy*: (1= lowest; 4 = highest)

Mg

Mg: Z = 12

Rb

Rb: Z = 37

Cs

Cs: Z = 55

K

K : Z = 19

b) Increasing (lowest to highest) order of *bond angle*: (1= lowest; 4 = highest)

H₂OCCl₄XeF₄

HCN

c) Increasing (smallest to largest) order of *ionic size* (1= smallest; 4 = largest)

S²⁻

S: Z = 16

O²⁻

O: Z = 8

Be²⁺

Be: Z = 4

Na⁺

Na: Z = 11

d) Increasing (lowest to highest) order of *bond energy* (1= lowest; 4 = highest)

P-F

P-Cl

P-I

P-Br

**Question 5** **12 points**

a. 9 points For the molecules HClO_4 and TeF_6 , draw the most stable *Lewis structure* showing appropriate shape and *dashed/wedged bonds*, based on VSEPR. Determine the **molecular geometry** around the central atom. For HClO_4 , label the *most polar bond*.



b. 3 points (Note: Lone pairs are not shown in structures below)

(i) For the three *resonance* structures (1, 2 and 3) shown below, answer the following question:



(I) Circle the *most contributing* resonance structure.

Structure 1

Structure 2

Structure 3

(ii) For the three structures given (4, 5, and 6) below, answer the following questions:



(I) Circle all the structures that represent *resonance structures* of each other:

Structure 4

Structure 5

Structure 6

(II) Among the resonance structures, circle the *most contributing* resonance structure.

Structure 4

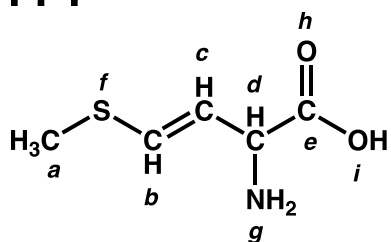
Structure 5

Structure 6

**Question 6** **6 points**

Using the data given below, calculate the **lattice energy** for SrO(s).

	ΔH° (kJ/mol)
$\text{Sr}(s) \longrightarrow \text{Sr}(g)$	+180
$\text{Sr}(g) \longrightarrow \text{Sr}^{2+}(g) + 2e^-$	+1468
$\text{O}_2(g) \longrightarrow 2\text{O}(g)$	+500
$\text{O}(g) + 2e^- \longrightarrow \text{O}^{2-}(g)$	+600
$\text{Sr}(s) + \frac{1}{2}\text{O}_2(g) \longrightarrow \text{SrO}(s)$	-550

**Question 7 10 points**

Consider the molecule given on the left, and answer questions (i) to (vi) using chemical bonding theories that we have learnt in class (valence bond theory, hybridization).

Note that the lone pairs are not shown in the structure:

i. Consider the carbon atoms marked in as ***a***, ***b***, ***c***, ***d***, and ***e*** in the molecule, and determine **hybridization** at each of the carbon centers. (2.5 points)

Atom	Hybridization
Carbon (<i>a</i>)	
Carbon (<i>b</i>)	
Carbon (<i>c</i>)	
Carbon (<i>d</i>)	
Carbon (<i>e</i>)	

ii. Now, consider the sulfur atom (S) marked as ***f***, nitrogen (N) atom (***g***), and oxygen(O) atoms (***h*** and ***i***), and determine their **hybridization**. (2 points)

Atom	Hybridization
Sulfur (<i>f</i>)	
Nitrogen (<i>g</i>)	
Oxygen (<i>h</i>)	
Oxygen (<i>i</i>)	

iii. **Describe** all the bonds formed by the carbon atoms ***a*** and ***b***, using sigma and pi designation as appropriate. (2 points)

Atom	Bonds formed (use sigma and pi as appropriate)
Carbon (<i>a</i>)	
Carbon (<i>b</i>)	

iv. How many total sp^3 -s sigma bonds are in the molecule? (1 point)

v. Determine **electronic** and **molecular geometry** at carbon (C) labeled ***e*** and sulfur (S) labeled ***f*** centres. (2 points)

Atom	Electronic Geometry	Molecular Geometry
Carbon (<i>e</i>)		
Sulfur (<i>f</i>)		

vi. Circle the value from the choices given below, that describes the **C-S-C bond angle at sulfur** (S) marked ***f***: (0.5 points)

>90° to <109.5° 109.5° >109.5 to <120° 120° >120° to <180°

**Question 8 12 points**

Using the **valence shell orbitals only** draw a **complete** molecular orbital (MO) diagram (with no s-p mixing) for the molecule **PS**, in the space provided below. **(6 points)**

Assume that 3s and 3p atomic orbitals will form molecular orbitals like those formed by the combination of 2s and 2p atomic orbitals that we learnt in class, and there is no s-p mixing.

Show **energy arrow**, **atomic orbitals** involved, **atom labels**, the **molecular orbitals** that are formed, **filled electrons** in the MO diagram etc. Make sure to label all the atomic and molecular orbitals.

Using the MO diagram above, answer the following questions:

- (i) Which one of the following will have the highest bond order? Circle the correct choice
A) PS^{1-} B) PS C) PS^{1+}
- (ii) Which one of the following will have the longest bond length? Circle the correct choice
A) PS^{1-} B) PS C) PS^{1+}
- (iii) Which of the following will be diamagnetic? Circle the correct choice.
A) PS^{1-} B) PS C) PS^{1+}
- (iv) Draw Lewis dot structure of the molecule with the shortest bond length (based on the MO diagram) from among the three (PS^{1-} , PS, PS^{1+}), and indicate the bond order calculated from the Lewis dot structure. **(3 points)**

Lewis Dot Structure

Bond Order



E9B7308D-BA1A-48D9-8D91-0501BACE6F11

midterm-exam-93388

#891 Page 10 of 10

SCRAP PAGE

**CHEMISTRY 110 –Midterm Exam****Thursday November 7, 2024, 6:30 – 8:30 pm**

Examiners: Pallavi Sirjoosingh and Ashok Kakkar

First name (please write as legibly as possible within the boxes)

--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--

Last name

--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--

Student ID number

--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--

This booklet has 8 questions on 10 pages (The last page can be used as a scrap page).

The staple must NOT be removed from the question booklet. Your intact question booklet must be returned at the end of the exam.

A separate booklet contains a periodic table, and datasheet.

Write your answers directly under the questions in the space provided. Do not write on the QR codes or near the top edges (as it may be cut off).

Please write **legibly and dark enough** for the scan to be readable. For students who use English as a second language, dictionaries are permitted. Calculators are allowed.

DO NOT ASK QUESTIONS OF THE INVIGILATORS about interpretations of the exam questions. If you have a concern about a question, please summarize it at the top of this page. All concerns will be evaluated after the exam. GOOD LUCK!

**Question 1 8 points** (*Assume all transitions are possible*)

a. 6 points Calculate the **energy (in kJ/mol)** required to ionize (completely remove) the electron from the **$n=3$** level of **He^+ ion** (He: Helium). Show your work including equations. (3 sig. figs.)

b. 2 points

(I) Select the largest (longest) wavelength from the three choices below:

- i. Wavelength of photon to *completely remove* an electron from *1s orbital of Hydrogen atom*
- ii. Wavelength of photon to *completely remove* an electron from *1s orbital of Boron atom (B)*
- iii. Wavelength of photon to *completely remove* an electron from *1s orbital of Boron cation (B^+)*

i or ii or iii

(II) Select the smallest (shortest) wavelength from the three choices below (*for H atom*):

- i. Wavelength of photon *emitted* for an electron transitioning from *2p to 1s orbital*
- ii. Wavelength of photon *absorbed* for an electron transitioning from *2s to 4p orbital*
- iii. Wavelength of photon *absorbed* for an electron transitioning from *1s to 3p orbital*

i or ii or iii



Question 2 8 points



A Chem110 student plots the graph shown on the left, from data obtained for the kinetic energy (eV) of the ejected electron versus frequency (s^{-1}) of the incident light, for an unknown metal. ($1\text{eV} = 1.6022 \times 10^{-19} \text{ J}$)

Based on the plot, answer the questions **a to c** below:

a. 1 point On the plot above, label the **threshold frequency**.

b. 2 points What is the value of the work potential (in J) of the unknown metal?

c. 3 points What is the *longest* wavelength (in nm) required to eject an electron from this metal's surface? Show your work including equation(s) used. (3 sig figs)

d. 2 points In the space provided below, plot the following

(Assume frequency is *above* threshold frequency):

- Kinetic energy* of an ejected electron versus the *intensity* of the incident light (left)
- Number of electrons* ejected versus the *frequency* of the incident light (right)



**Question 3** **6 points****a. 2 points**

In the ground state of Ge (Germanium; $Z = 32$) atom, **name** the following orbitals:

i. Orbital containing the electron that requires the maximum energy to be removed.

ii. Orbital containing the electron that experiences the minimum effective nuclear charge.

b. 2 points

An unknown anion (A^-) has the following *excited* electronic configuration: $[Ar] 4s^2 3d^3 4p^1$

(Ar: Argon; $Z = 18$)

Give the **condensed ground state electronic configuration** of the atom, **A** and the cation, **A^{2+}** .
Show your work. (Assume all electronic transitions are possible)

Electronic Configuration of A:

Electronic Configuration of A^{2+} :

c. 2 points

In a single atom, determine how many electrons can have the following set of quantum numbers?

(i) $n = 5; l = 3$

In a single atom, determine how many electrons can have the following set of quantum numbers?

(i) $n = 3; l = 2; m_s = +1/2$

**Question 4** **8 points**

Rank the following from 1 to 4 in **increasing** order. Provide the ranking in the boxes provided.

Note: Atomic number (Z) are provided for relevant atoms

a) Increasing (lowest to highest) order of *ionization energy*: (1= lowest; 4 = highest)

Mg

Mg: Z = 12

Rb

Rb: Z = 37

Cs

Cs: Z = 55

K

K : Z = 19

b) Increasing (lowest to highest) order of *bond angle*: (1= lowest; 4 = highest)

H₂OCCl₄XeF₄

HCN

c) Increasing (smallest to largest) order of *ionic size* (1= smallest; 4 = largest)

S²⁻

S: Z = 16

O²⁻

O: Z = 8

Be²⁺

Be: Z = 4

Na⁺

Na: Z = 11

d) Increasing (lowest to highest) order of *bond energy* (1= lowest; 4 = highest)

P-F

P-Cl

P-I

P-Br

**Question 5** **12 points**

a. 9 points For the molecules HClO_4 and TeF_6 , draw the most stable *Lewis structure* showing appropriate shape and *dashed/wedged bonds*, based on VSEPR. Determine the **molecular geometry** around the central atom. For HClO_4 , label the *most polar bond*.



b. 3 points (Note: Lone pairs are not shown in structures below)

(i) For the three *resonance* structures (1, 2 and 3) shown below, answer the following question:



(I) Circle the *most contributing* resonance structure.

Structure 1

Structure 2

Structure 3

(ii) For the three structures given (4, 5, and 6) below, answer the following questions:



(I) Circle all the structures that represent *resonance structures* of each other:

Structure 4

Structure 5

Structure 6

(II) Among the resonance structures, circle the *most contributing* resonance structure.

Structure 4

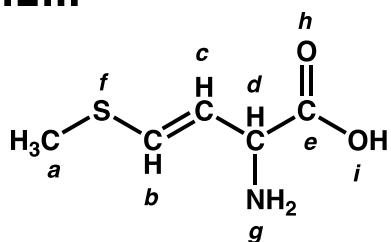
Structure 5

Structure 6

**Question 6** **6 points**

Using the data given below, calculate the **lattice energy** for SrO(s).

	ΔH° (kJ/mol)
$\text{Sr}(s) \longrightarrow \text{Sr}(g)$	+180
$\text{Sr}(g) \longrightarrow \text{Sr}^{2+}(g) + 2e^-$	+1468
$\text{O}_2(g) \longrightarrow 2\text{O}(g)$	+500
$\text{O}(g) + 2e^- \longrightarrow \text{O}^{2-}(g)$	+600
$\text{Sr}(s) + \frac{1}{2}\text{O}_2(g) \longrightarrow \text{SrO}(s)$	-550

**Question 7 10 points**

Consider the molecule given on the left, and answer questions (i) to (vi) using chemical bonding theories that we have learnt in class (valence bond theory, hybridization).

Note that the lone pairs are not shown in the structure:

i. Consider the carbon atoms marked in as ***a***, ***b***, ***c***, ***d***, and ***e*** in the molecule, and determine **hybridization** at each of the carbon centers. (2.5 points)

Atom	Hybridization
Carbon (<i>a</i>)	
Carbon (<i>b</i>)	
Carbon (<i>c</i>)	
Carbon (<i>d</i>)	
Carbon (<i>e</i>)	

ii. Now, consider the sulfur atom (S) marked as ***f***, nitrogen (N) atom (***g***), and oxygen(O) atoms (***h*** and ***i***), and determine their **hybridization**. (2 points)

Atom	Hybridization
Sulfur (<i>f</i>)	
Nitrogen (<i>g</i>)	
Oxygen (<i>h</i>)	
Oxygen (<i>i</i>)	

iii. **Describe** all the bonds formed by the carbon atoms ***a*** and ***b***, using sigma and pi designation as appropriate. (2 points)

Atom	Bonds formed (use sigma and pi as appropriate)
Carbon (<i>a</i>)	
Carbon (<i>b</i>)	

iv. How many total sp^3 -s sigma bonds are in the molecule? (1 point)

v. Determine **electronic** and **molecular geometry** at carbon (C) labeled ***e*** and sulfur (S) labeled ***f*** centres. (2 points)

Atom	Electronic Geometry	Molecular Geometry
Carbon (<i>e</i>)		
Sulfur (<i>f</i>)		

vi. Circle the value from the choices given below, that describes the **C-S-C bond angle at sulfur** (S) marked ***f***: (0.5 points)

>90° to <109.5°

109.5°

>109.5 to <120°

120°

>120° to <180°

**Question 8 12 points**

Using the **valence shell orbitals only** draw a **complete** molecular orbital (MO) diagram (with no s-p mixing) for the molecule **PS**, in the space provided below. **(6 points)**

Assume that 3s and 3p atomic orbitals will form molecular orbitals like those formed by the combination of 2s and 2p atomic orbitals that we learnt in class, and there is no s-p mixing.

Show **energy arrow**, **atomic orbitals** involved, **atom labels**, the **molecular orbitals** that are formed, **filled electrons** in the MO diagram etc. Make sure to label all the atomic and molecular orbitals.

Using the MO diagram above, answer the following questions:

- (i) Which one of the following will have the highest bond order? Circle the correct choice
A) PS^{1-} B) PS C) PS^{1+}
- (ii) Which one of the following will have the longest bond length? Circle the correct choice
A) PS^{1-} B) PS C) PS^{1+}
- (iii) Which of the following will be diamagnetic? Circle the correct choice.
A) PS^{1-} B) PS C) PS^{1+}
- (iv) Draw Lewis dot structure of the molecule with the shortest bond length (based on the MO diagram) from among the three (PS^{1-} , PS, PS^{1+}), and indicate the bond order calculated from the Lewis dot structure. **(3 points)**

Lewis Dot Structure

Bond Order



1CA07AF0-254E-4AC9-BEA9-65F665CA63B6

midterm-exam-93388

#892 Page 10 of 10

SCRAP PAGE

**CHEMISTRY 110 –Midterm Exam****Thursday November 7, 2024, 6:30 – 8:30 pm****Examiners: Pallavi Sirjoosingh and Ashok Kakkar**

First name (please write as legibly as possible within the boxes)

--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--

Last name

--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--

Student ID number

--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--

This booklet has 8 questions on 10 pages (The last page can be used as a scrap page).

The staple must NOT be removed from the question booklet. Your intact question booklet must be returned at the end of the exam.

A separate booklet contains a periodic table, and datasheet.

Write your answers directly under the questions in the space provided. Do not write on the QR codes or near the top edges (as it may be cut off).

Please write **legibly and dark enough** for the scan to be readable. For students who use English as a second language, dictionaries are permitted. Calculators are allowed.

DO NOT ASK QUESTIONS OF THE INVIGILATORS about interpretations of the exam questions. If you have a concern about a question, please summarize it at the top of this page. All concerns will be evaluated after the exam. GOOD LUCK!

**Question 1 8 points** (*Assume all transitions are possible*)

a. 6 points Calculate the **energy (in kJ/mol)** required to ionize (completely remove) the electron from the **$n=3$** level of **He^+ ion** (He: Helium). Show your work including equations. (3 sig. figs.)

b. 2 points

(I) Select the largest (longest) wavelength from the three choices below:

- i. Wavelength of photon to *completely remove* an electron from *1s orbital of Hydrogen atom*
- ii. Wavelength of photon to *completely remove* an electron from *1s orbital of Boron atom (B)*
- iii. Wavelength of photon to *completely remove* an electron from *1s orbital of Boron cation (B^+)*

i

or

ii

or

iii

(II) Select the smallest (shortest) wavelength from the three choices below (*for H atom*):

- i. Wavelength of photon *emitted* for an electron transitioning from *2p to 1s orbital*
- ii. Wavelength of photon *absorbed* for an electron transitioning from *2s to 4p orbital*
- iii. Wavelength of photon *absorbed* for an electron transitioning from *1s to 3p orbital*

i

or

ii

or

iii



Question 2 8 points



A Chem110 student plots the graph shown on the left, from data obtained for the kinetic energy (eV) of the ejected electron versus frequency (s^{-1}) of the incident light, for an unknown metal. ($1\text{eV} = 1.6022 \times 10^{-19} \text{ J}$)

Based on the plot, answer the questions **a to c** below:

a. 1 point On the plot above, label the **threshold frequency**.

b. 2 points What is the value of the work potential (in J) of the unknown metal?

c. 3 points What is the *longest* wavelength (in nm) required to eject an electron from this metal's surface? Show your work including equation(s) used. (3 sig figs)

d. 2 points In the space provided below, plot the following

(Assume frequency is *above* threshold frequency):

- i. *Kinetic energy* of an ejected electron versus the *intensity* of the incident light (left)
- ii. *Number of electrons* ejected versus the *frequency* of the incident light (right)



**Question 3** **6 points****a. 2 points**

In the ground state of Ge (Germanium; $Z = 32$) atom, **name** the following orbitals:

i. Orbital containing the electron that requires the maximum energy to be removed.

ii. Orbital containing the electron that experiences the minimum effective nuclear charge.

b. 2 points

An unknown anion (A^-) has the following *excited* electronic configuration: $[Ar] 4s^2 3d^3 4p^1$

(Ar: Argon; $Z = 18$)

Give the **condensed ground state electronic configuration** of the atom, **A** and the cation, **A^{2+}** .
Show your work. (Assume all electronic transitions are possible)

Electronic Configuration of A:

Electronic Configuration of A^{2+} :

c. 2 points

In a single atom, determine how many electrons can have the following set of quantum numbers?

(i) $n = 5; l = 3$

In a single atom, determine how many electrons can have the following set of quantum numbers?

(i) $n = 3; l = 2; m_s = +1/2$

**Question 4** **8 points**

Rank the following from 1 to 4 in **increasing** order. Provide the ranking in the boxes provided.

Note: Atomic number (Z) are provided for relevant atoms

a) Increasing (lowest to highest) order of *ionization energy*: (1= lowest; 4 = highest)

Mg

Mg: Z = 12

Rb

Rb: Z = 37

Cs

Cs: Z = 55

K

K : Z = 19

b) Increasing (lowest to highest) order of *bond angle*: (1= lowest; 4 = highest)

H₂OCCl₄XeF₄

HCN

c) Increasing (smallest to largest) order of *ionic size* (1= smallest; 4 = largest)

S²⁻

S: Z = 16

O²⁻

O: Z = 8

Be²⁺

Be: Z = 4

Na⁺

Na: Z = 11

d) Increasing (lowest to highest) order of *bond energy* (1= lowest; 4 = highest)

P-F

P-Cl

P-I

P-Br

**Question 5** **12 points**

a. 9 points For the molecules HClO_4 and TeF_6 , draw the most stable *Lewis structure* showing appropriate shape and *dashed/wedged bonds*, based on VSEPR. Determine the **molecular geometry** around the central atom. For HClO_4 , label the *most polar bond*.

HClO_4 (there are no O-O bonds)

Label the most polar bond

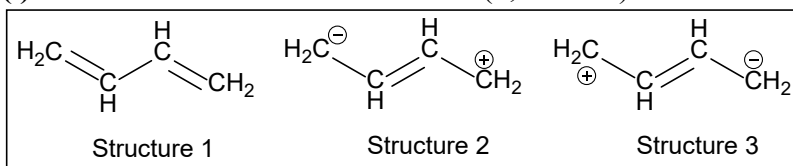
Molecular Geometry around Cl

TeF_6

Molecular Geometry around Te

b. 3 points (Note: Lone pairs are not shown in structures below)

(i) For the three *resonance* structures (1, 2 and 3) shown below, answer the following question:



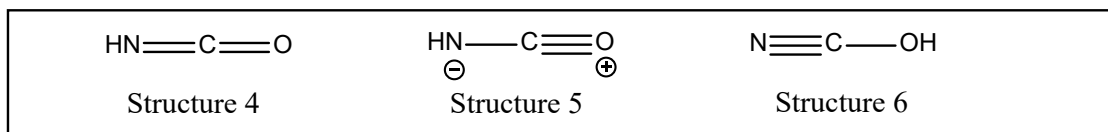
(I) Circle the *most contributing* resonance structure.

Structure 1

Structure 2

Structure 3

(ii) For the three structures given (4, 5, and 6) below, answer the following questions:



(I) Circle all the structures that represent *resonance structures* of each other:

Structure 4

Structure 5

Structure 6

(II) Among the resonance structures, circle the *most contributing* resonance structure.

Structure 4

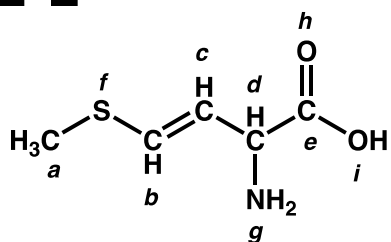
Structure 5

Structure 6

**Question 6** **6 points**

Using the data given below, calculate the **lattice energy** for SrO(s).

	ΔH° (kJ/mol)
$\text{Sr}(s) \longrightarrow \text{Sr}(g)$	+180
$\text{Sr}(g) \longrightarrow \text{Sr}^{2+}(g) + 2e^-$	+1468
$\text{O}_2(g) \longrightarrow 2\text{O}(g)$	+500
$\text{O}(g) + 2e^- \longrightarrow \text{O}^{2-}(g)$	+600
$\text{Sr}(s) + \frac{1}{2}\text{O}_2(g) \longrightarrow \text{SrO}(s)$	-550

**Question 7 10 points**

Consider the molecule given on the left, and answer questions (i) to (vi) using chemical bonding theories that we have learnt in class (valence bond theory, hybridization).

Note that the lone pairs are not shown in the structure:

i. Consider the carbon atoms marked in as ***a***, ***b***, ***c***, ***d***, and ***e*** in the molecule, and determine **hybridization** at each of the carbon centers. (2.5 points)

Atom	Hybridization
Carbon (<i>a</i>)	
Carbon (<i>b</i>)	
Carbon (<i>c</i>)	
Carbon (<i>d</i>)	
Carbon (<i>e</i>)	

ii. Now, consider the sulfur atom (S) marked as ***f***, nitrogen (N) atom (***g***), and oxygen(O) atoms (***h*** and ***i***), and determine their **hybridization**. (2 points)

Atom	Hybridization
Sulfur (<i>f</i>)	
Nitrogen (<i>g</i>)	
Oxygen (<i>h</i>)	
Oxygen (<i>i</i>)	

iii. **Describe** all the bonds formed by the carbon atoms ***a*** and ***b***, using sigma and pi designation as appropriate. (2 points)

Atom	Bonds formed (use sigma and pi as appropriate)
Carbon (<i>a</i>)	
Carbon (<i>b</i>)	

iv. How many total sp^3 -s sigma bonds are in the molecule? (1 point)

v. Determine **electronic** and **molecular geometry** at carbon (C) labeled ***e*** and sulfur (S) labeled ***f*** centres. (2 points)

Atom	Electronic Geometry	Molecular Geometry
Carbon (<i>e</i>)		
Sulfur (<i>f</i>)		

vi. Circle the value from the choices given below, that describes the **C-S-C bond angle at sulfur** (S) marked ***f***: (0.5 points)

>90° to <109.5° 109.5° >109.5 to <120° 120° >120° to <180°


Question 8 12 points

Using the **valence shell orbitals only** draw a **complete** molecular orbital (MO) diagram (with no s-p mixing) for the molecule **PS**, in the space provided below. **(6 points)**

Assume that 3s and 3p atomic orbitals will form molecular orbitals like those formed by the combination of 2s and 2p atomic orbitals that we learnt in class, and there is no s-p mixing.

Show **energy arrow**, **atomic orbitals** involved, **atom labels**, the **molecular orbitals** that are formed, **filled electrons** in the MO diagram etc. Make sure to label all the atomic and molecular orbitals.

Using the MO diagram above, answer the following questions:

- (i) Which one of the following will have the highest bond order? Circle the correct choice
 A) PS^{1-} B) PS C) PS^{1+}
- (ii) Which one of the following will have the longest bond length? Circle the correct choice
 A) PS^{1-} B) PS C) PS^{1+}
- (iii) Which of the following will be diamagnetic? Circle the correct choice.
 A) PS^{1-} B) PS C) PS^{1+}
- (iv) Draw Lewis dot structure of the molecule with the shortest bond length (based on the MO diagram) from among the three (PS^{1-} , PS, PS^{1+}), and indicate the bond order calculated from the Lewis dot structure. **(3 points)**

Lewis Dot Structure

Bond Order



8C47E914-B875-403F-88FA-59F0C61D7EC7

midterm-exam-93388

#893 Page 10 of 10

SCRAP PAGE

**CHEMISTRY 110 –Midterm Exam****Thursday November 7, 2024, 6:30 – 8:30 pm****Examiners: Pallavi Sirjoosingh and Ashok Kakkar**

First name (please write as legibly as possible within the boxes)

--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--

Last name

--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--

Student ID number

--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--

This booklet has 8 questions on 10 pages (The last page can be used as a scrap page).

The staple must NOT be removed from the question booklet. Your intact question booklet must be returned at the end of the exam.

A separate booklet contains a periodic table, and datasheet.

Write your answers directly under the questions in the space provided. Do not write on the QR codes or near the top edges (as it may be cut off).

Please write **legibly and dark enough** for the scan to be readable. For students who use English as a second language, dictionaries are permitted. Calculators are allowed.

DO NOT ASK QUESTIONS OF THE INVIGILATORS about interpretations of the exam questions. If you have a concern about a question, please summarize it at the top of this page. All concerns will be evaluated after the exam. GOOD LUCK!

**Question 1 8 points** (*Assume all transitions are possible*)

a. 6 points Calculate the **energy (in kJ/mol)** required to ionize (completely remove) the electron from the **$n=3$** level of **He^+ ion** (He: Helium). Show your work including equations. (3 sig. figs.)

b. 2 points

(I) Select the largest (longest) wavelength from the three choices below:

- i. Wavelength of photon to *completely remove* an electron from *1s orbital of Hydrogen atom*
- ii. Wavelength of photon to *completely remove* an electron from *1s orbital of Boron atom (B)*
- iii. Wavelength of photon to *completely remove* an electron from *1s orbital of Boron cation (B^+)*

i or ii or iii

(II) Select the smallest (shortest) wavelength from the three choices below (*for H atom*):

- i. Wavelength of photon *emitted* for an electron transitioning from *2p to 1s orbital*
- ii. Wavelength of photon *absorbed* for an electron transitioning from *2s to 4p orbital*
- iii. Wavelength of photon *absorbed* for an electron transitioning from *1s to 3p orbital*

i or ii or iii



Question 2 8 points



A Chem110 student plots the graph shown on the left, from data obtained for the kinetic energy (eV) of the ejected electron versus frequency (s^{-1}) of the incident light, for an unknown metal. ($1\text{eV} = 1.6022 \times 10^{-19} \text{ J}$)

Based on the plot, answer the questions **a to c** below:

a. 1 point On the plot above, label the **threshold frequency**.

b. 2 points What is the value of the work potential (in J) of the unknown metal?

c. 3 points What is the *longest* wavelength (in nm) required to eject an electron from this metal's surface? Show your work including equation(s) used. (3 sig figs)

d. 2 points In the space provided below, plot the following

(Assume frequency is *above* threshold frequency):

- Kinetic energy* of an ejected electron versus the *intensity* of the incident light (left)
- Number of electrons* ejected versus the *frequency* of the incident light (right)



**Question 3** **6 points****a. 2 points**

In the ground state of Ge (Germanium; $Z = 32$) atom, **name** the following orbitals:

i. Orbital containing the electron that requires the maximum energy to be removed.

ii. Orbital containing the electron that experiences the minimum effective nuclear charge.

b. 2 points

An unknown anion (A^-) has the following *excited* electronic configuration: $[Ar] 4s^2 3d^3 4p^1$

(Ar: Argon; $Z = 18$)

Give the **condensed ground state electronic configuration** of the atom, **A** and the cation, A^{2+} .
Show your work. (Assume all electronic transitions are possible)

Electronic Configuration of A:

Electronic Configuration of A^{2+} :

c. 2 points

In a single atom, determine how many electrons can have the following set of quantum numbers?

(i) $n = 5; l = 3$

In a single atom, determine how many electrons can have the following set of quantum numbers?

(i) $n = 3; l = 2; m_s = +1/2$

**Question 4** **8 points**

Rank the following from 1 to 4 in **increasing** order. Provide the ranking in the boxes provided.

Note: Atomic number (Z) are provided for relevant atoms

a) Increasing (lowest to highest) order of *ionization energy*: (1= lowest; 4 = highest)

Mg

Mg: Z = 12

Rb

Rb: Z = 37

Cs

Cs: Z = 55

K

K : Z = 19

b) Increasing (lowest to highest) order of *bond angle*: (1= lowest; 4 = highest)

H₂OCCl₄XeF₄

HCN

c) Increasing (smallest to largest) order of *ionic size* (1= smallest; 4 = largest)

S²⁻

S: Z = 16

O²⁻

O: Z = 8

Be²⁺

Be: Z = 4

Na⁺

Na: Z = 11

d) Increasing (lowest to highest) order of *bond energy* (1= lowest; 4 = highest)

P-F

P-Cl

P-I

P-Br

**Question 5** **12 points**

a. 9 points For the molecules HClO_4 and TeF_6 , draw the most stable *Lewis structure* showing appropriate shape and *dashed/wedged bonds*, based on VSEPR. Determine the **molecular geometry** around the central atom. For HClO_4 , label the *most polar bond*.



b. 3 points (Note: Lone pairs are not shown in structures below)

(i) For the three *resonance* structures (1, 2 and 3) shown below, answer the following question:



(I) Circle the *most contributing* resonance structure.

Structure 1

Structure 2

Structure 3

(ii) For the three structures given (4, 5, and 6) below, answer the following questions:



(I) Circle all the structures that represent *resonance structures* of each other:

Structure 4

Structure 5

Structure 6

(II) Among the resonance structures, circle the *most contributing* resonance structure.

Structure 4

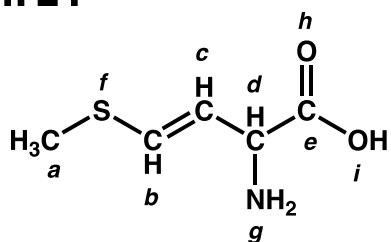
Structure 5

Structure 6

**Question 6** **6 points**

Using the data given below, calculate the **lattice energy** for SrO(s).

	ΔH° (kJ/mol)
$\text{Sr}(s) \longrightarrow \text{Sr}(g)$	+180
$\text{Sr}(g) \longrightarrow \text{Sr}^{2+}(g) + 2e^-$	+1468
$\text{O}_2(g) \longrightarrow 2\text{O}(g)$	+500
$\text{O}(g) + 2e^- \longrightarrow \text{O}^{2-}(g)$	+600
$\text{Sr}(s) + \frac{1}{2}\text{O}_2(g) \longrightarrow \text{SrO}(s)$	-550

**Question 7 10 points**

Consider the molecule given on the left, and answer questions (i) to (vi) using chemical bonding theories that we have learnt in class (valence bond theory, hybridization).

Note that the lone pairs are not shown in the structure:

i. Consider the carbon atoms marked in as ***a***, ***b***, ***c***, ***d***, and ***e*** in the molecule, and determine **hybridization** at each of the carbon centers. (2.5 points)

Atom	Hybridization
Carbon (<i>a</i>)	
Carbon (<i>b</i>)	
Carbon (<i>c</i>)	
Carbon (<i>d</i>)	
Carbon (<i>e</i>)	

ii. Now, consider the sulfur atom (S) marked as ***f***, nitrogen (N) atom (***g***), and oxygen(O) atoms (***h*** and ***i***), and determine their **hybridization**. (2 points)

Atom	Hybridization
Sulfur (<i>f</i>)	
Nitrogen (<i>g</i>)	
Oxygen (<i>h</i>)	
Oxygen (<i>i</i>)	

iii. **Describe** all the bonds formed by the carbon atoms ***a*** and ***b***, using sigma and pi designation as appropriate. (2 points)

Atom	Bonds formed (use sigma and pi as appropriate)
Carbon (<i>a</i>)	
Carbon (<i>b</i>)	

iv. How many total sp^3 -s sigma bonds are in the molecule? (1 point)

v. Determine **electronic** and **molecular geometry** at carbon (C) labeled ***e*** and sulfur (S) labeled ***f*** centres. (2 points)

Atom	Electronic Geometry	Molecular Geometry
Carbon (<i>e</i>)		
Sulfur (<i>f</i>)		

vi. Circle the value from the choices given below, that describes the **C-S-C bond angle at sulfur** (S) marked ***f***: (0.5 points)

$>90^\circ$ to $<109.5^\circ$ 109.5° >109.5 to $<120^\circ$ 120° $>120^\circ$ to $<180^\circ$


Question 8 12 points

Using the **valence shell orbitals only** draw a **complete** molecular orbital (MO) diagram (with no s-p mixing) for the molecule **PS**, in the space provided below. **(6 points)**

Assume that 3s and 3p atomic orbitals will form molecular orbitals like those formed by the combination of 2s and 2p atomic orbitals that we learnt in class, and there is no s-p mixing.

Show **energy arrow**, **atomic orbitals** involved, **atom labels**, the **molecular orbitals** that are formed, **filled electrons** in the MO diagram etc. Make sure to label all the atomic and molecular orbitals.

Using the MO diagram above, answer the following questions:

- (i) Which one of the following will have the highest bond order? Circle the correct choice
 A) PS^{1-} B) PS C) PS^{1+}
- (ii) Which one of the following will have the longest bond length? Circle the correct choice
 A) PS^{1-} B) PS C) PS^{1+}
- (iii) Which of the following will be diamagnetic? Circle the correct choice.
 A) PS^{1-} B) PS C) PS^{1+}
- (iv) Draw Lewis dot structure of the molecule with the shortest bond length (based on the MO diagram) from among the three (PS^{1-} , PS, PS^{1+}), and indicate the bond order calculated from the Lewis dot structure. **(3 points)**

Lewis Dot Structure

Bond Order



7EE32BCC-B805-431B-AF59-552341F31159

midterm-exam-93388

#894 Page 10 of 10

SCRAP PAGE

**CHEMISTRY 110 –Midterm Exam****Thursday November 7, 2024, 6:30 – 8:30 pm****Examiners: Pallavi Sirjoosingh and Ashok Kakkar**

First name (please write as legibly as possible within the boxes)

--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--

Last name

--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--

Student ID number

--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--

This booklet has 8 questions on 10 pages (The last page can be used as a scrap page).

The staple must NOT be removed from the question booklet. Your intact question booklet must be returned at the end of the exam.

A separate booklet contains a periodic table, and datasheet.

Write your answers directly under the questions in the space provided. Do not write on the QR codes or near the top edges (as it may be cut off).

Please write **legibly and dark enough** for the scan to be readable. For students who use English as a second language, dictionaries are permitted. Calculators are allowed.

DO NOT ASK QUESTIONS OF THE INVIGILATORS about interpretations of the exam questions. If you have a concern about a question, please summarize it at the top of this page. All concerns will be evaluated after the exam. GOOD LUCK!

**Question 1 8 points** (*Assume all transitions are possible*)

a. 6 points Calculate the **energy (in kJ/mol)** required to ionize (completely remove) the electron from the **$n=3$** level of **He^+ ion** (He: Helium). Show your work including equations. (3 sig. figs.)

b. 2 points

(I) Select the largest (longest) wavelength from the three choices below:

- i. Wavelength of photon to *completely remove* an electron from *1s orbital of Hydrogen atom*
- ii. Wavelength of photon to *completely remove* an electron from *1s orbital of Boron atom (B)*
- iii. Wavelength of photon to *completely remove* an electron from *1s orbital of Boron cation (B^+)*

i or ii or iii

(II) Select the smallest (shortest) wavelength from the three choices below (*for H atom*):

- i. Wavelength of photon *emitted* for an electron transitioning from *2p to 1s orbital*
- ii. Wavelength of photon *absorbed* for an electron transitioning from *2s to 4p orbital*
- iii. Wavelength of photon *absorbed* for an electron transitioning from *1s to 3p orbital*

i or ii or iii



Question 2 8 points



A Chem110 student plots the graph shown on the left, from data obtained for the kinetic energy (eV) of the ejected electron versus frequency (s^{-1}) of the incident light, for an unknown metal. ($1\text{eV} = 1.6022 \times 10^{-19} \text{ J}$)

Based on the plot, answer the questions **a to c** below:

a. 1 point On the plot above, label the **threshold frequency**.

b. 2 points What is the value of the work potential (in J) of the unknown metal?

c. 3 points What is the *longest* wavelength (in nm) required to eject an electron from this metal's surface? Show your work including equation(s) used. (3 sig figs)

d. 2 points In the space provided below, plot the following

(Assume frequency is *above* threshold frequency):

- i. *Kinetic energy* of an ejected electron versus the *intensity* of the incident light (left)
- ii. *Number of electrons* ejected versus the *frequency* of the incident light (right)



**Question 3** **6 points****a. 2 points**

In the ground state of Ge (Germanium; $Z = 32$) atom, **name** the following orbitals:

i. Orbital containing the electron that requires the maximum energy to be removed.

ii. Orbital containing the electron that experiences the minimum effective nuclear charge.

b. 2 points

An unknown anion (A^-) has the following *excited* electronic configuration: $[Ar] 4s^2 3d^3 4p^1$

(Ar: Argon; $Z = 18$)

Give the **condensed ground state electronic configuration** of the atom, **A** and the cation, **A^{2+}** .
Show your work. (Assume all electronic transitions are possible)

Electronic Configuration of A:

Electronic Configuration of A^{2+} :

c. 2 points

In a single atom, determine how many electrons can have the following set of quantum numbers?

(i) $n = 5; l = 3$

In a single atom, determine how many electrons can have the following set of quantum numbers?

(i) $n = 3; l = 2; m_s = +1/2$

**Question 4** **8 points**

Rank the following from 1 to 4 in **increasing** order. Provide the ranking in the boxes provided.

Note: Atomic number (Z) are provided for relevant atoms

a) Increasing (lowest to highest) order of *ionization energy*: (1= lowest; 4 = highest)

Mg

Mg: Z = 12

Rb

Rb: Z = 37

Cs

Cs: Z = 55

K

K : Z = 19

b) Increasing (lowest to highest) order of *bond angle*: (1= lowest; 4 = highest)

H₂OCCl₄XeF₄

HCN

c) Increasing (smallest to largest) order of *ionic size* (1= smallest; 4 = largest)

S²⁻

S: Z = 16

O²⁻

O: Z = 8

Be²⁺

Be: Z = 4

Na⁺

Na: Z = 11

d) Increasing (lowest to highest) order of *bond energy* (1= lowest; 4 = highest)

P-F

P-Cl

P-I

P-Br

**Question 5** **12 points**

a. 9 points For the molecules HClO_4 and TeF_6 , draw the most stable *Lewis structure* showing appropriate shape and *dashed/wedged bonds*, based on VSEPR. Determine the **molecular geometry** around the central atom. For HClO_4 , label the *most polar bond*.



b. 3 points (Note: Lone pairs are not shown in structures below)

(i) For the three *resonance* structures (1, 2 and 3) shown below, answer the following question:



(I) Circle the *most contributing* resonance structure.

Structure 1

Structure 2

Structure 3

(ii) For the three structures given (4, 5, and 6) below, answer the following questions:



(I) Circle all the structures that represent *resonance structures* of each other:

Structure 4

Structure 5

Structure 6

(II) Among the resonance structures, circle the *most contributing* resonance structure.

Structure 4

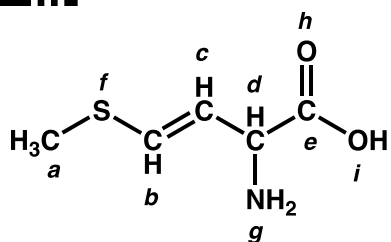
Structure 5

Structure 6

**Question 6** **6 points**

Using the data given below, calculate the **lattice energy** for SrO(s).

	ΔH° (kJ/mol)
$\text{Sr}(s) \longrightarrow \text{Sr}(g)$	+180
$\text{Sr}(g) \longrightarrow \text{Sr}^{2+}(g) + 2e^-$	+1468
$\text{O}_2(g) \longrightarrow 2\text{O}(g)$	+500
$\text{O}(g) + 2e^- \longrightarrow \text{O}^{2-}(g)$	+600
$\text{Sr}(s) + \frac{1}{2}\text{O}_2(g) \longrightarrow \text{SrO}(s)$	-550

**Question 7 10 points**

Consider the molecule given on the left, and answer questions (i) to (vi) using chemical bonding theories that we have learnt in class (valence bond theory, hybridization).

Note that the lone pairs are not shown in the structure:

i. Consider the carbon atoms marked in as ***a***, ***b***, ***c***, ***d***, and ***e*** in the molecule, and determine **hybridization** at each of the carbon centers. (2.5 points)

Atom	Hybridization
Carbon (<i>a</i>)	
Carbon (<i>b</i>)	
Carbon (<i>c</i>)	
Carbon (<i>d</i>)	
Carbon (<i>e</i>)	

ii. Now, consider the sulfur atom (S) marked as ***f***, nitrogen (N) atom (***g***), and oxygen(O) atoms (***h*** and ***i***), and determine their **hybridization**. (2 points)

Atom	Hybridization
Sulfur (<i>f</i>)	
Nitrogen (<i>g</i>)	
Oxygen (<i>h</i>)	
Oxygen (<i>i</i>)	

iii. **Describe** all the bonds formed by the carbon atoms ***a*** and ***b***, using sigma and pi designation as appropriate. (2 points)

Atom	Bonds formed (use sigma and pi as appropriate)
Carbon (<i>a</i>)	
Carbon (<i>b</i>)	

iv. How many total sp^3 -s sigma bonds are in the molecule? (1 point)

v. Determine **electronic** and **molecular geometry** at carbon (C) labeled ***e*** and sulfur (S) labeled ***f*** centres. (2 points)

Atom	Electronic Geometry	Molecular Geometry
Carbon (<i>e</i>)		
Sulfur (<i>f</i>)		

vi. Circle the value from the choices given below, that describes the **C-S-C bond angle at sulfur** (S) marked ***f***: (0.5 points)

>90° to <109.5° 109.5° >109.5 to <120° 120° >120° to <180°


Question 8 12 points

Using the **valence shell orbitals only** draw a **complete** molecular orbital (MO) diagram (with no s-p mixing) for the molecule **PS**, in the space provided below. **(6 points)**

Assume that 3s and 3p atomic orbitals will form molecular orbitals like those formed by the combination of 2s and 2p atomic orbitals that we learnt in class, and there is no s-p mixing.

Show **energy arrow**, **atomic orbitals** involved, **atom labels**, the **molecular orbitals** that are formed, **filled electrons** in the MO diagram etc. Make sure to label all the atomic and molecular orbitals.

Using the MO diagram above, answer the following questions:

- (i) Which one of the following will have the highest bond order? Circle the correct choice
 A) PS^{1-} B) PS C) PS^{1+}
- (ii) Which one of the following will have the longest bond length? Circle the correct choice
 A) PS^{1-} B) PS C) PS^{1+}
- (iii) Which of the following will be diamagnetic? Circle the correct choice.
 A) PS^{1-} B) PS C) PS^{1+}
- (iv) Draw Lewis dot structure of the molecule with the shortest bond length (based on the MO diagram) from among the three (PS^{1-} , PS, PS^{1+}), and indicate the bond order calculated from the Lewis dot structure. **(3 points)**

Lewis Dot Structure

Bond Order



B7D784C4-3737-4BDC-8785-E6215B58F338

midterm-exam-93388

#895 Page 10 of 10

SCRAP PAGE

**CHEMISTRY 110 –Midterm Exam****Thursday November 7, 2024, 6:30 – 8:30 pm**

Examiners: Pallavi Sirjoosingh and Ashok Kakkar

First name (please write as legibly as possible within the boxes)

--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--

Last name

--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--

Student ID number

--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--

This booklet has 8 questions on 10 pages (The last page can be used as a scrap page).

The staple must NOT be removed from the question booklet. Your intact question booklet must be returned at the end of the exam.

A separate booklet contains a periodic table, and datasheet.

Write your answers directly under the questions in the space provided. Do not write on the QR codes or near the top edges (as it may be cut off).

Please write **legibly and dark enough** for the scan to be readable. For students who use English as a second language, dictionaries are permitted. Calculators are allowed.

DO NOT ASK QUESTIONS OF THE INVIGILATORS about interpretations of the exam questions. If you have a concern about a question, please summarize it at the top of this page. All concerns will be evaluated after the exam. GOOD LUCK!

**Question 1 8 points** (*Assume all transitions are possible*)

a. 6 points Calculate the **energy (in kJ/mol)** required to ionize (completely remove) the electron from the **$n=3$** level of **He^+ ion** (He: Helium). Show your work including equations. (3 sig. figs.)

b. 2 points

(I) Select the largest (longest) wavelength from the three choices below:

- i. Wavelength of photon to *completely remove* an electron from *1s orbital of Hydrogen atom*
- ii. Wavelength of photon to *completely remove* an electron from *1s orbital of Boron atom (B)*
- iii. Wavelength of photon to *completely remove* an electron from *1s orbital of Boron cation (B^+)*

i or ii or iii

(II) Select the smallest (shortest) wavelength from the three choices below (*for H atom*):

- i. Wavelength of photon *emitted* for an electron transitioning from *2p to 1s orbital*
- ii. Wavelength of photon *absorbed* for an electron transitioning from *2s to 4p orbital*
- iii. Wavelength of photon *absorbed* for an electron transitioning from *1s to 3p orbital*

i or ii or iii



Question 2 8 points



A Chem110 student plots the graph shown on the left, from data obtained for the kinetic energy (eV) of the ejected electron versus frequency (s^{-1}) of the incident light, for an unknown metal. ($1\text{eV} = 1.6022 \times 10^{-19} \text{ J}$)

Based on the plot, answer the questions **a to c** below:

a. 1 point On the plot above, label the **threshold frequency**.

b. 2 points What is the value of the work potential (in J) of the unknown metal?

c. 3 points What is the *longest* wavelength (in nm) required to eject an electron from this metal's surface? Show your work including equation(s) used. (3 sig figs)

d. 2 points In the space provided below, plot the following

(Assume frequency is *above* threshold frequency):

- Kinetic energy* of an ejected electron versus the *intensity* of the incident light (left)
- Number of electrons* ejected versus the *frequency* of the incident light (right)



**Question 3** **6 points****a. 2 points**

In the ground state of Ge (Germanium; $Z = 32$) atom, **name** the following orbitals:

i. Orbital containing the electron that requires the maximum energy to be removed.

ii. Orbital containing the electron that experiences the minimum effective nuclear charge.

b. 2 points

An unknown anion (A^-) has the following *excited* electronic configuration: $[Ar] 4s^2 3d^3 4p^1$

(Ar: Argon; $Z = 18$)

Give the **condensed ground state electronic configuration** of the atom, **A** and the cation, **A^{2+}** .
Show your work. (Assume all electronic transitions are possible)

Electronic Configuration of A:

Electronic Configuration of A^{2+} :

c. 2 points

In a single atom, determine how many electrons can have the following set of quantum numbers?

(i) $n = 5; l = 3$

In a single atom, determine how many electrons can have the following set of quantum numbers?

(i) $n = 3; l = 2; m_s = +1/2$

**Question 4** **8 points**

Rank the following from 1 to 4 in **increasing** order. Provide the ranking in the boxes provided.

Note: Atomic number (Z) are provided for relevant atoms

a) Increasing (lowest to highest) order of *ionization energy*: (1= lowest; 4 = highest)

Mg

Mg: Z = 12

Rb

Rb: Z = 37

Cs

Cs: Z = 55

K

K : Z = 19

b) Increasing (lowest to highest) order of *bond angle*: (1= lowest; 4 = highest)

H₂OCCl₄XeF₄

HCN

c) Increasing (smallest to largest) order of *ionic size* (1= smallest; 4 = largest)

S²⁻

S: Z = 16

O²⁻

O: Z = 8

Be²⁺

Be: Z = 4

Na⁺

Na: Z = 11

d) Increasing (lowest to highest) order of *bond energy* (1= lowest; 4 = highest)

P-F

P-Cl

P-I

P-Br

**Question 5** **12 points**

a. 9 points For the molecules HClO_4 and TeF_6 , draw the most stable *Lewis structure* showing appropriate shape and *dashed/wedged bonds*, based on VSEPR. Determine the **molecular geometry** around the central atom. For HClO_4 , label the *most polar bond*.



b. 3 points (Note: Lone pairs are not shown in structures below)

(i) For the three *resonance* structures (1, 2 and 3) shown below, answer the following question:



(I) Circle the *most contributing* resonance structure.

Structure 1

Structure 2

Structure 3

(ii) For the three structures given (4, 5, and 6) below, answer the following questions:



(I) Circle all the structures that represent *resonance structures* of each other:

Structure 4

Structure 5

Structure 6

(II) Among the resonance structures, circle the *most contributing* resonance structure.

Structure 4

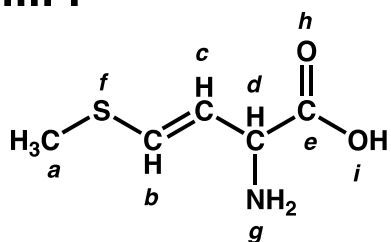
Structure 5

Structure 6

**Question 6** **6 points**

Using the data given below, calculate the **lattice energy** for SrO(s).

	ΔH° (kJ/mol)
$\text{Sr}(s) \longrightarrow \text{Sr}(g)$	+180
$\text{Sr}(g) \longrightarrow \text{Sr}^{2+}(g) + 2e^-$	+1468
$\text{O}_2(g) \longrightarrow 2\text{O}(g)$	+500
$\text{O}(g) + 2e^- \longrightarrow \text{O}^{2-}(g)$	+600
$\text{Sr}(s) + \frac{1}{2}\text{O}_2(g) \longrightarrow \text{SrO}(s)$	-550

**Question 7 10 points**

Consider the molecule given on the left, and answer questions (i) to (vi) using chemical bonding theories that we have learnt in class (valence bond theory, hybridization).

Note that the lone pairs are not shown in the structure:

i. Consider the carbon atoms marked in as ***a***, ***b***, ***c***, ***d***, and ***e*** in the molecule, and determine **hybridization** at each of the carbon centers. (2.5 points)

Atom	Hybridization
Carbon (<i>a</i>)	
Carbon (<i>b</i>)	
Carbon (<i>c</i>)	
Carbon (<i>d</i>)	
Carbon (<i>e</i>)	

ii. Now, consider the sulfur atom (S) marked as ***f***, nitrogen (N) atom (***g***), and oxygen(O) atoms (***h*** and ***i***), and determine their **hybridization**. (2 points)

Atom	Hybridization
Sulfur (<i>f</i>)	
Nitrogen (<i>g</i>)	
Oxygen (<i>h</i>)	
Oxygen (<i>i</i>)	

iii. **Describe** all the bonds formed by the carbon atoms ***a*** and ***b***, using sigma and pi designation as appropriate. (2 points)

Atom	Bonds formed (use sigma and pi as appropriate)
Carbon (<i>a</i>)	
Carbon (<i>b</i>)	

iv. How many total sp^3 -s sigma bonds are in the molecule? (1 point)

v. Determine **electronic** and **molecular geometry** at carbon (C) labeled ***e*** and sulfur (S) labeled ***f*** centres. (2 points)

Atom	Electronic Geometry	Molecular Geometry
Carbon (<i>e</i>)		
Sulfur (<i>f</i>)		

vi. Circle the value from the choices given below, that describes the **C-S-C bond angle at sulfur** (S) marked ***f***: (0.5 points)

$>90^\circ$ to $<109.5^\circ$ 109.5° >109.5 to $<120^\circ$ 120° $>120^\circ$ to $<180^\circ$


Question 8 12 points

Using the **valence shell orbitals only** draw a **complete** molecular orbital (MO) diagram (with no s-p mixing) for the molecule **PS**, in the space provided below. **(6 points)**

Assume that 3s and 3p atomic orbitals will form molecular orbitals like those formed by the combination of 2s and 2p atomic orbitals that we learnt in class, and there is no s-p mixing.

Show **energy arrow**, **atomic orbitals** involved, **atom labels**, the **molecular orbitals** that are formed, **filled electrons** in the MO diagram etc. Make sure to label all the atomic and molecular orbitals.

Using the MO diagram above, answer the following questions:

- (i) Which one of the following will have the highest bond order? Circle the correct choice
 A) PS^{1-} B) PS C) PS^{1+}
- (ii) Which one of the following will have the longest bond length? Circle the correct choice
 A) PS^{1-} B) PS C) PS^{1+}
- (iii) Which of the following will be diamagnetic? Circle the correct choice.
 A) PS^{1-} B) PS C) PS^{1+}
- (iv) Draw Lewis dot structure of the molecule with the shortest bond length (based on the MO diagram) from among the three (PS^{1-} , PS, PS^{1+}), and indicate the bond order calculated from the Lewis dot structure. **(3 points)**

Lewis Dot Structure

Bond Order



3C103E05-276A-40FD-A05F-7EF43D9BB4D7

midterm-exam-93388

#896 Page 10 of 10

SCRAP PAGE

**CHEMISTRY 110 –Midterm Exam****Thursday November 7, 2024, 6:30 – 8:30 pm****Examiners: Pallavi Sirjoosingh and Ashok Kakkar**

First name (please write as legibly as possible within the boxes)

--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--

Last name

--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--

Student ID number

--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--

This booklet has 8 questions on 10 pages (The last page can be used as a scrap page).

The staple must NOT be removed from the question booklet. Your intact question booklet must be returned at the end of the exam.

A separate booklet contains a periodic table, and datasheet.

Write your answers directly under the questions in the space provided. Do not write on the QR codes or near the top edges (as it may be cut off).

Please write **legibly and dark enough** for the scan to be readable. For students who use English as a second language, dictionaries are permitted. Calculators are allowed.

DO NOT ASK QUESTIONS OF THE INVIGILATORS about interpretations of the exam questions. If you have a concern about a question, please summarize it at the top of this page. All concerns will be evaluated after the exam. GOOD LUCK!

**Question 1 8 points** (*Assume all transitions are possible*)

a. 6 points Calculate the **energy (in kJ/mol)** required to ionize (completely remove) the electron from the **n=3** level of **He⁺ ion** (He: Helium). Show your work including equations. (3 sig. figs.)

b. 2 points

(I) Select the largest (longest) wavelength from the three choices below:

- i. Wavelength of photon to *completely remove* an electron from *1s orbital of Hydrogen atom*
- ii. Wavelength of photon to *completely remove* an electron from *1s orbital of Boron atom (B)*
- iii. Wavelength of photon to *completely remove* an electron from *1s orbital of Boron cation (B⁺)*

i or ii or iii

(II) Select the smallest (shortest) wavelength from the three choices below (*for H atom*):

- i. Wavelength of photon *emitted* for an electron transitioning from *2p to 1s orbital*
- ii. Wavelength of photon *absorbed* for an electron transitioning from *2s to 4p orbital*
- iii. Wavelength of photon *absorbed* for an electron transitioning from *1s to 3p orbital*

i or ii or iii



Question 2 8 points



A Chem110 student plots the graph shown on the left, from data obtained for the kinetic energy (eV) of the ejected electron versus frequency (s^{-1}) of the incident light, for an unknown metal. ($1\text{eV} = 1.6022 \times 10^{-19} \text{ J}$)

Based on the plot, answer the questions **a to c** below:

a. 1 point On the plot above, label the **threshold frequency**.

b. 2 points What is the value of the work potential (in J) of the unknown metal?

c. 3 points What is the *longest* wavelength (in nm) required to eject an electron from this metal's surface? Show your work including equation(s) used. (3 sig figs)

d. 2 points In the space provided below, plot the following

(Assume frequency is *above* threshold frequency):

- i. *Kinetic energy* of an ejected electron versus the *intensity* of the incident light (left)
- ii. *Number of electrons* ejected versus the *frequency* of the incident light (right)



**Question 3** **6 points****a. 2 points**

In the ground state of Ge (Germanium; $Z = 32$) atom, **name** the following orbitals:

i. Orbital containing the electron that requires the maximum energy to be removed.

ii. Orbital containing the electron that experiences the minimum effective nuclear charge.

b. 2 points

An unknown anion (A^-) has the following *excited* electronic configuration: $[Ar] 4s^2 3d^3 4p^1$

(Ar: Argon; $Z = 18$)

Give the **condensed ground state electronic configuration** of the atom, **A** and the cation, **A^{2+}** .
Show your work. (Assume all electronic transitions are possible)

Electronic Configuration of A:

Electronic Configuration of A^{2+} :

c. 2 points

In a single atom, determine how many electrons can have the following set of quantum numbers?

(i) $n = 5; l = 3$

In a single atom, determine how many electrons can have the following set of quantum numbers?

(i) $n = 3; l = 2; m_s = +1/2$

**Question 4** **8 points**

Rank the following from 1 to 4 in **increasing** order. Provide the ranking in the boxes provided.

Note: Atomic number (Z) are provided for relevant atoms

a) Increasing (lowest to highest) order of *ionization energy*: (1= lowest; 4 = highest)

Mg

Mg: Z = 12

Rb

Rb: Z = 37

Cs

Cs: Z = 55

K

K : Z = 19

b) Increasing (lowest to highest) order of *bond angle*: (1= lowest; 4 = highest)

H₂OCCl₄XeF₄

HCN

c) Increasing (smallest to largest) order of *ionic size* (1= smallest; 4 = largest)

S²⁻

S: Z = 16

O²⁻

O: Z = 8

Be²⁺

Be: Z = 4

Na⁺

Na: Z = 11

d) Increasing (lowest to highest) order of *bond energy* (1= lowest; 4 = highest)

P-F

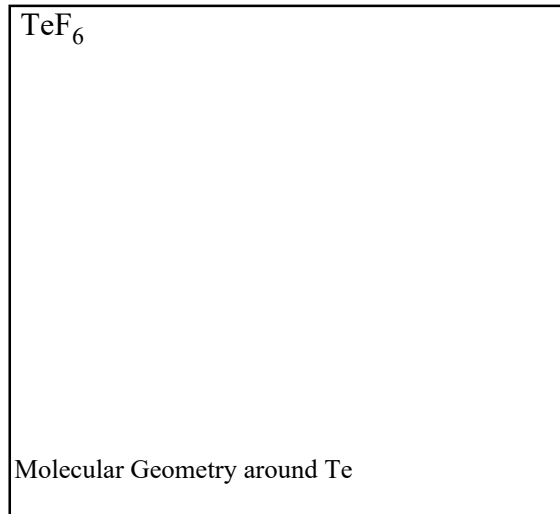
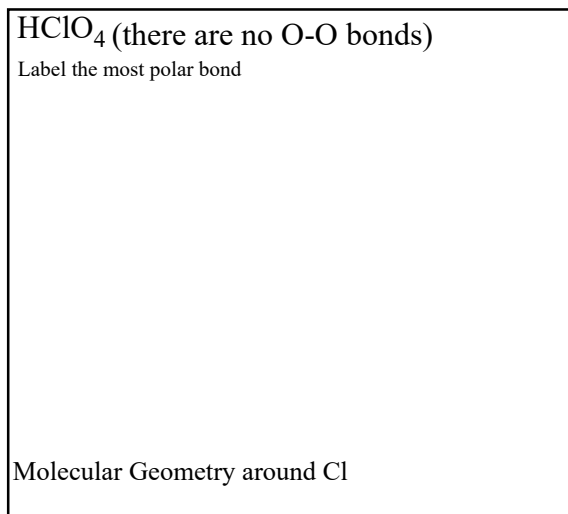
P-Cl

P-I

P-Br

**Question 5** **12 points**

a. 9 points For the molecules HClO_4 and TeF_6 , draw the most stable *Lewis structure* showing appropriate shape and *dashed/wedged bonds*, based on VSEPR. Determine the **molecular geometry** around the central atom. For HClO_4 , label the *most polar bond*.



b. 3 points (Note: Lone pairs are not shown in structures below)

(i) For the three *resonance* structures (1, 2 and 3) shown below, answer the following question:



(I) Circle the *most contributing* resonance structure.

Structure 1

Structure 2

Structure 3

(ii) For the three structures given (4, 5, and 6) below, answer the following questions:



(I) Circle all the structures that represent *resonance structures* of each other:

Structure 4

Structure 5

Structure 6

(II) Among the resonance structures, circle the *most contributing* resonance structure.

Structure 4

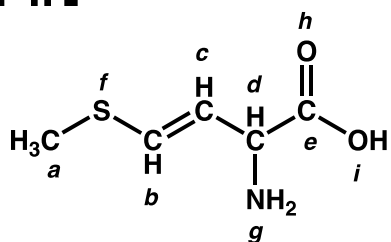
Structure 5

Structure 6

**Question 6** **6 points**

Using the data given below, calculate the **lattice energy** for SrO(s).

	ΔH° (kJ/mol)
$\text{Sr}(s) \longrightarrow \text{Sr}(g)$	+180
$\text{Sr}(g) \longrightarrow \text{Sr}^{2+}(g) + 2e^-$	+1468
$\text{O}_2(g) \longrightarrow 2\text{O}(g)$	+500
$\text{O}(g) + 2e^- \longrightarrow \text{O}^{2-}(g)$	+600
$\text{Sr}(s) + \frac{1}{2}\text{O}_2(g) \longrightarrow \text{SrO}(s)$	-550

**Question 7 10 points**

Consider the molecule given on the left, and answer questions (i) to (vi) using chemical bonding theories that we have learnt in class (valence bond theory, hybridization).

Note that the lone pairs are not shown in the structure:

i. Consider the carbon atoms marked in as ***a***, ***b***, ***c***, ***d***, and ***e*** in the molecule, and determine **hybridization** at each of the carbon centers. (2.5 points)

Atom	Hybridization
Carbon (<i>a</i>)	
Carbon (<i>b</i>)	
Carbon (<i>c</i>)	
Carbon (<i>d</i>)	
Carbon (<i>e</i>)	

ii. Now, consider the sulfur atom (S) marked as ***f***, nitrogen (N) atom (***g***), and oxygen(O) atoms (***h*** and ***i***), and determine their **hybridization**. (2 points)

Atom	Hybridization
Sulfur (<i>f</i>)	
Nitrogen (<i>g</i>)	
Oxygen (<i>h</i>)	
Oxygen (<i>i</i>)	

iii. **Describe** all the bonds formed by the carbon atoms ***a*** and ***b***, using sigma and pi designation as appropriate. (2 points)

Atom	Bonds formed (use sigma and pi as appropriate)
Carbon (<i>a</i>)	
Carbon (<i>b</i>)	

iv. How many total sp^3 -s sigma bonds are in the molecule? (1 point)

v. Determine **electronic** and **molecular geometry** at carbon (C) labeled ***e*** and sulfur (S) labeled ***f*** centres. (2 points)

Atom	Electronic Geometry	Molecular Geometry
Carbon (<i>e</i>)		
Sulfur (<i>f</i>)		

vi. Circle the value from the choices given below, that describes the **C-S-C bond angle at sulfur** (S) marked ***f***: (0.5 points)

>90° to <109.5° 109.5° >109.5 to <120° 120° >120° to <180°

**Question 8 12 points**

Using the **valence shell orbitals only** draw a **complete** molecular orbital (MO) diagram (with no s-p mixing) for the molecule **PS**, in the space provided below. **(6 points)**

Assume that 3s and 3p atomic orbitals will form molecular orbitals like those formed by the combination of 2s and 2p atomic orbitals that we learnt in class, and there is no s-p mixing.

Show **energy arrow**, **atomic orbitals** involved, **atom labels**, the **molecular orbitals** that are formed, **filled electrons** in the MO diagram etc. Make sure to label all the atomic and molecular orbitals.

Using the MO diagram above, answer the following questions:

- (i) Which one of the following will have the highest bond order? Circle the correct choice
A) PS^{1-} B) PS C) PS^{1+}
- (ii) Which one of the following will have the longest bond length? Circle the correct choice
A) PS^{1-} B) PS C) PS^{1+}
- (iii) Which of the following will be diamagnetic? Circle the correct choice.
A) PS^{1-} B) PS C) PS^{1+}
- (iv) Draw Lewis dot structure of the molecule with the shortest bond length (based on the MO diagram) from among the three (PS^{1-} , PS, PS^{1+}), and indicate the bond order calculated from the Lewis dot structure. **(3 points)**

Lewis Dot Structure

Bond Order



67A3C4E3-C4BC-47E5-83D8-223AAE5D4F5F

midterm-exam-93388

#897 Page 10 of 10

SCRAP PAGE

**CHEMISTRY 110 –Midterm Exam****Thursday November 7, 2024, 6:30 – 8:30 pm**

Examiners: Pallavi Sirjoosingh and Ashok Kakkar

First name (please write as legibly as possible within the boxes)

--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--

Last name

--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--

Student ID number

--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--

This booklet has 8 questions on 10 pages (The last page can be used as a scrap page).

The staple must NOT be removed from the question booklet. Your intact question booklet must be returned at the end of the exam.

A separate booklet contains a periodic table, and datasheet.

Write your answers directly under the questions in the space provided. Do not write on the QR codes or near the top edges (as it may be cut off).

Please write **legibly and dark enough** for the scan to be readable. For students who use English as a second language, dictionaries are permitted. Calculators are allowed.

DO NOT ASK QUESTIONS OF THE INVIGILATORS about interpretations of the exam questions. If you have a concern about a question, please summarize it at the top of this page. All concerns will be evaluated after the exam. GOOD LUCK!

**Question 1 8 points** (*Assume all transitions are possible*)

a. 6 points Calculate the **energy (in kJ/mol)** required to ionize (completely remove) the electron from the **$n=3$** level of **He^+ ion** (He: Helium). Show your work including equations. (3 sig. figs.)

b. 2 points

(I) Select the largest (longest) wavelength from the three choices below:

- i. Wavelength of photon to *completely remove* an electron from *1s orbital of Hydrogen atom*
- ii. Wavelength of photon to *completely remove* an electron from *1s orbital of Boron atom (B)*
- iii. Wavelength of photon to *completely remove* an electron from *1s orbital of Boron cation (B^+)*

i or ii or iii

(II) Select the smallest (shortest) wavelength from the three choices below (*for H atom*):

- i. Wavelength of photon *emitted* for an electron transitioning from *2p to 1s orbital*
- ii. Wavelength of photon *absorbed* for an electron transitioning from *2s to 4p orbital*
- iii. Wavelength of photon *absorbed* for an electron transitioning from *1s to 3p orbital*

i or ii or iii



Question 2 8 points



A Chem110 student plots the graph shown on the left, from data obtained for the kinetic energy (eV) of the ejected electron versus frequency (s^{-1}) of the incident light, for an unknown metal. ($1\text{eV} = 1.6022 \times 10^{-19} \text{ J}$)

Based on the plot, answer the questions **a to c** below:

a. 1 point On the plot above, label the **threshold frequency**.

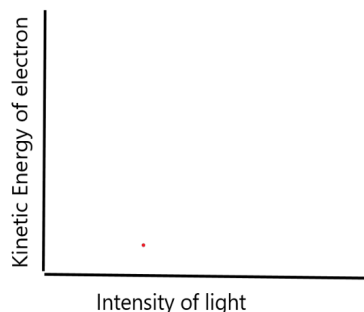
b. 2 points What is the value of the work potential (in J) of the unknown metal?

c. 3 points What is the *longest* wavelength (in nm) required to eject an electron from this metal's surface? Show your work including equation(s) used. (3 sig figs)

d. 2 points In the space provided below, plot the following

(Assume frequency is *above* threshold frequency):

- Kinetic energy* of an ejected electron versus the *intensity* of the incident light (left)
- Number of electrons* ejected versus the *frequency* of the incident light (right)



**Question 3** **6 points****a. 2 points**

In the ground state of Ge (Germanium; $Z = 32$) atom, **name** the following orbitals:

i. Orbital containing the electron that requires the maximum energy to be removed.

ii. Orbital containing the electron that experiences the minimum effective nuclear charge.

b. 2 points

An unknown anion (A^-) has the following *excited* electronic configuration: $[Ar] 4s^2 3d^3 4p^1$

(Ar: Argon; $Z = 18$)

Give the **condensed ground state electronic configuration** of the atom, **A** and the cation, **A^{2+}** .
Show your work. (Assume all electronic transitions are possible)

Electronic Configuration of A:

Electronic Configuration of A^{2+} :

c. 2 points

In a single atom, determine how many electrons can have the following set of quantum numbers?

(i) $n = 5; l = 3$

In a single atom, determine how many electrons can have the following set of quantum numbers?

(i) $n = 3; l = 2; m_s = +1/2$

**Question 4** **8 points**

Rank the following from 1 to 4 in **increasing** order. Provide the ranking in the boxes provided.

Note: Atomic number (Z) are provided for relevant atoms

a) Increasing (lowest to highest) order of *ionization energy*: (1= lowest; 4 = highest)

Mg

Mg: Z = 12

Rb

Rb: Z = 37

Cs

Cs: Z = 55

K

K : Z = 19

b) Increasing (lowest to highest) order of *bond angle*: (1= lowest; 4 = highest)

H₂OCCl₄XeF₄

HCN

c) Increasing (smallest to largest) order of *ionic size* (1= smallest; 4 = largest)

S²⁻

S: Z = 16

O²⁻

O: Z = 8

Be²⁺

Be: Z = 4

Na⁺

Na: Z = 11

d) Increasing (lowest to highest) order of *bond energy* (1= lowest; 4 = highest)

P-F

P-Cl

P-I

P-Br

**Question 5** **12 points**

a. 9 points For the molecules HClO_4 and TeF_6 , draw the most stable *Lewis structure* showing appropriate shape and *dashed/wedged bonds*, based on VSEPR. Determine the **molecular geometry** around the central atom. For HClO_4 , label the *most polar bond*.

HClO_4 (there are no O-O bonds)

Label the most polar bond

Molecular Geometry around Cl

TeF_6

Molecular Geometry around Te

b. 3 points (Note: Lone pairs are not shown in structures below)

(i) For the three *resonance* structures (1, 2 and 3) shown below, answer the following question:



(I) Circle the *most contributing* resonance structure.

Structure 1

Structure 2

Structure 3

(ii) For the three structures given (4, 5, and 6) below, answer the following questions:



(I) Circle all the structures that represent *resonance structures* of each other:

Structure 4

Structure 5

Structure 6

(II) Among the resonance structures, circle the *most contributing* resonance structure.

Structure 4

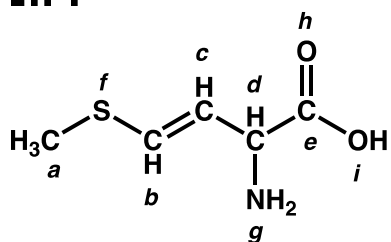
Structure 5

Structure 6

**Question 6** **6 points**

Using the data given below, calculate the **lattice energy** for SrO(s).

	ΔH° (kJ/mol)
$\text{Sr}(s) \longrightarrow \text{Sr}(g)$	+180
$\text{Sr}(g) \longrightarrow \text{Sr}^{2+}(g) + 2e^-$	+1468
$\text{O}_2(g) \longrightarrow 2\text{O}(g)$	+500
$\text{O}(g) + 2e^- \longrightarrow \text{O}^{2-}(g)$	+600
$\text{Sr}(s) + \frac{1}{2}\text{O}_2(g) \longrightarrow \text{SrO}(s)$	-550

**Question 7 10 points**

Consider the molecule given on the left, and answer questions (i) to (vi) using chemical bonding theories that we have learnt in class (valence bond theory, hybridization).

Note that the lone pairs are not shown in the structure:

i. Consider the carbon atoms marked in as ***a***, ***b***, ***c***, ***d***, and ***e*** in the molecule, and determine **hybridization** at each of the carbon centers. (2.5 points)

Atom	Hybridization
Carbon (<i>a</i>)	
Carbon (<i>b</i>)	
Carbon (<i>c</i>)	
Carbon (<i>d</i>)	
Carbon (<i>e</i>)	

ii. Now, consider the sulfur atom (S) marked as ***f***, nitrogen (N) atom (***g***), and oxygen(O) atoms (***h*** and ***i***), and determine their **hybridization**. (2 points)

Atom	Hybridization
Sulfur (<i>f</i>)	
Nitrogen (<i>g</i>)	
Oxygen (<i>h</i>)	
Oxygen (<i>i</i>)	

iii. **Describe** all the bonds formed by the carbon atoms ***a*** and ***b***, using sigma and pi designation as appropriate. (2 points)

Atom	Bonds formed (use sigma and pi as appropriate)
Carbon (<i>a</i>)	
Carbon (<i>b</i>)	

iv. How many total sp^3 -s sigma bonds are in the molecule? (1 point)

v. Determine **electronic** and **molecular geometry** at carbon (C) labeled ***e*** and sulfur (S) labeled ***f*** centres. (2 points)

Atom	Electronic Geometry	Molecular Geometry
Carbon (<i>e</i>)		
Sulfur (<i>f</i>)		

vi. Circle the value from the choices given below, that describes the **C-S-C bond angle at sulfur** (S) marked ***f***: (0.5 points)

>90° to <109.5° 109.5° >109.5 to <120° 120° >120° to <180°


Question 8 12 points

Using the **valence shell orbitals only** draw a **complete** molecular orbital (MO) diagram (with no s-p mixing) for the molecule **PS**, in the space provided below. **(6 points)**

Assume that 3s and 3p atomic orbitals will form molecular orbitals like those formed by the combination of 2s and 2p atomic orbitals that we learnt in class, and there is no s-p mixing.

Show **energy arrow**, **atomic orbitals** involved, **atom labels**, the **molecular orbitals** that are formed, **filled electrons** in the MO diagram etc. Make sure to label all the atomic and molecular orbitals.

Using the MO diagram above, answer the following questions:

- (i) Which one of the following will have the highest bond order? Circle the correct choice
 A) PS^{1-} B) PS C) PS^{1+}
- (ii) Which one of the following will have the longest bond length? Circle the correct choice
 A) PS^{1-} B) PS C) PS^{1+}
- (iii) Which of the following will be diamagnetic? Circle the correct choice.
 A) PS^{1-} B) PS C) PS^{1+}
- (iv) Draw Lewis dot structure of the molecule with the shortest bond length (based on the MO diagram) from among the three (PS^{1-} , PS, PS^{1+}), and indicate the bond order calculated from the Lewis dot structure. **(3 points)**

Lewis Dot Structure

Bond Order



560C58F8-8401-42E8-8397-54600E0A21C2

midterm-exam-93388

#898 Page 10 of 10

SCRAP PAGE

**CHEMISTRY 110 –Midterm Exam****Thursday November 7, 2024, 6:30 – 8:30 pm**

Examiners: Pallavi Sirjoosingh and Ashok Kakkar

First name (please write as legibly as possible within the boxes)

--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--

Last name

--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--

Student ID number

--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--

This booklet has 8 questions on 10 pages (The last page can be used as a scrap page).

The staple must NOT be removed from the question booklet. Your intact question booklet must be returned at the end of the exam.

A separate booklet contains a periodic table, and datasheet.

Write your answers directly under the questions in the space provided. Do not write on the QR codes or near the top edges (as it may be cut off).

Please write **legibly and dark enough** for the scan to be readable. For students who use English as a second language, dictionaries are permitted. Calculators are allowed.

DO NOT ASK QUESTIONS OF THE INVIGILATORS about interpretations of the exam questions. If you have a concern about a question, please summarize it at the top of this page. All concerns will be evaluated after the exam. GOOD LUCK!

**Question 1 8 points** (*Assume all transitions are possible*)

a. 6 points Calculate the **energy (in kJ/mol)** required to ionize (completely remove) the electron from the **$n=3$** level of **He^+ ion** (He: Helium). Show your work including equations. (3 sig. figs.)

b. 2 points

(I) Select the largest (longest) wavelength from the three choices below:

- i. Wavelength of photon to *completely remove* an electron from *1s orbital of Hydrogen atom*
- ii. Wavelength of photon to *completely remove* an electron from *1s orbital of Boron atom (B)*
- iii. Wavelength of photon to *completely remove* an electron from *1s orbital of Boron cation (B^+)*

i or ii or iii

(II) Select the smallest (shortest) wavelength from the three choices below (*for H atom*):

- i. Wavelength of photon *emitted* for an electron transitioning from *2p to 1s orbital*
- ii. Wavelength of photon *absorbed* for an electron transitioning from *2s to 4p orbital*
- iii. Wavelength of photon *absorbed* for an electron transitioning from *1s to 3p orbital*

i or ii or iii



Question 2 8 points



A Chem110 student plots the graph shown on the left, from data obtained for the kinetic energy (eV) of the ejected electron versus frequency (s^{-1}) of the incident light, for an unknown metal. ($1\text{eV} = 1.6022 \times 10^{-19} \text{ J}$)

Based on the plot, answer the questions **a to c** below:

a. 1 point On the plot above, label the **threshold frequency**.

b. 2 points What is the value of the work potential (in J) of the unknown metal?

c. 3 points What is the *longest* wavelength (in nm) required to eject an electron from this metal's surface? Show your work including equation(s) used. (3 sig figs)

d. 2 points In the space provided below, plot the following

(Assume frequency is *above* threshold frequency):

- Kinetic energy* of an ejected electron versus the *intensity* of the incident light (left)
- Number of electrons* ejected versus the *frequency* of the incident light (right)



**Question 3** **6 points****a. 2 points**

In the ground state of Ge (Germanium; $Z = 32$) atom, **name** the following orbitals:

i. Orbital containing the electron that requires the maximum energy to be removed.

ii. Orbital containing the electron that experiences the minimum effective nuclear charge.

b. 2 points

An unknown anion (A^-) has the following *excited* electronic configuration: $[Ar] 4s^2 3d^3 4p^1$

(Ar: Argon; $Z = 18$)

Give the **condensed ground state electronic configuration** of the atom, **A** and the cation, **A^{2+}** .
Show your work. (Assume all electronic transitions are possible)

Electronic Configuration of A:

Electronic Configuration of A^{2+} :

c. 2 points

In a single atom, determine how many electrons can have the following set of quantum numbers?

(i) $n = 5; l = 3$

In a single atom, determine how many electrons can have the following set of quantum numbers?

(i) $n = 3; l = 2; m_s = +1/2$

**Question 4** **8 points**

Rank the following from 1 to 4 in **increasing** order. Provide the ranking in the boxes provided.

Note: Atomic number (Z) are provided for relevant atoms

a) Increasing (lowest to highest) order of *ionization energy*: (1= lowest; 4 = highest)

Mg

Mg: Z = 12

Rb

Rb: Z = 37

Cs

Cs: Z = 55

K

K : Z = 19

b) Increasing (lowest to highest) order of *bond angle*: (1= lowest; 4 = highest)

H₂OCCl₄XeF₄

HCN

c) Increasing (smallest to largest) order of *ionic size* (1= smallest; 4 = largest)

S²⁻

S: Z = 16

O²⁻

O: Z = 8

Be²⁺

Be: Z = 4

Na⁺

Na: Z = 11

d) Increasing (lowest to highest) order of *bond energy* (1= lowest; 4 = highest)

P-F

P-Cl

P-I

P-Br

**Question 5** **12 points**

a. 9 points For the molecules HClO_4 and TeF_6 , draw the most stable *Lewis structure* showing appropriate shape and *dashed/wedged bonds*, based on VSEPR. Determine the **molecular geometry** around the central atom. For HClO_4 , label the *most polar bond*.



b. 3 points (Note: Lone pairs are not shown in structures below)

(i) For the three *resonance* structures (1, 2 and 3) shown below, answer the following question:



(I) Circle the *most contributing* resonance structure.

Structure 1

Structure 2

Structure 3

(ii) For the three structures given (4, 5, and 6) below, answer the following questions:



(I) Circle all the structures that represent *resonance structures* of each other:

Structure 4

Structure 5

Structure 6

(II) Among the resonance structures, circle the *most contributing* resonance structure.

Structure 4

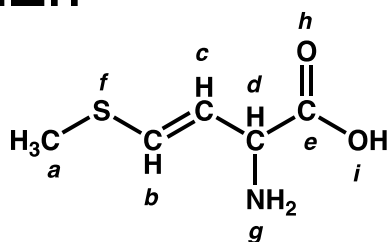
Structure 5

Structure 6

**Question 6** **6 points**

Using the data given below, calculate the **lattice energy** for SrO(s).

	ΔH° (kJ/mol)
$\text{Sr}(s) \longrightarrow \text{Sr}(g)$	+180
$\text{Sr}(g) \longrightarrow \text{Sr}^{2+}(g) + 2e^-$	+1468
$\text{O}_2(g) \longrightarrow 2\text{O}(g)$	+500
$\text{O}(g) + 2e^- \longrightarrow \text{O}^{2-}(g)$	+600
$\text{Sr}(s) + \frac{1}{2}\text{O}_2(g) \longrightarrow \text{SrO}(s)$	-550

**Question 7 10 points**

Consider the molecule given on the left, and answer questions (i) to (vi) using chemical bonding theories that we have learnt in class (valence bond theory, hybridization).

Note that the lone pairs are not shown in the structure:

i. Consider the carbon atoms marked in as ***a***, ***b***, ***c***, ***d***, and ***e*** in the molecule, and determine **hybridization** at each of the carbon centers. (2.5 points)

Atom	Hybridization
Carbon (<i>a</i>)	
Carbon (<i>b</i>)	
Carbon (<i>c</i>)	
Carbon (<i>d</i>)	
Carbon (<i>e</i>)	

ii. Now, consider the sulfur atom (S) marked as ***f***, nitrogen (N) atom (***g***), and oxygen(O) atoms (***h*** and ***i***), and determine their **hybridization**. (2 points)

Atom	Hybridization
Sulfur (<i>f</i>)	
Nitrogen (<i>g</i>)	
Oxygen (<i>h</i>)	
Oxygen (<i>i</i>)	

iii. **Describe** all the bonds formed by the carbon atoms ***a*** and ***b***, using sigma and pi designation as appropriate. (2 points)

Atom	Bonds formed (use sigma and pi as appropriate)
Carbon (<i>a</i>)	
Carbon (<i>b</i>)	

iv. How many total sp^3 -s sigma bonds are in the molecule? (1 point)

v. Determine **electronic** and **molecular geometry** at carbon (C) labeled ***e*** and sulfur (S) labeled ***f*** centres. (2 points)

Atom	Electronic Geometry	Molecular Geometry
Carbon (<i>e</i>)		
Sulfur (<i>f</i>)		

vi. Circle the value from the choices given below, that describes the **C-S-C bond angle at sulfur** (S) marked ***f***: (0.5 points)

>90° to <109.5° 109.5° >109.5 to <120° 120° >120° to <180°


Question 8 12 points

Using the **valence shell orbitals only** draw a **complete** molecular orbital (MO) diagram (with no s-p mixing) for the molecule **PS**, in the space provided below. **(6 points)**

Assume that 3s and 3p atomic orbitals will form molecular orbitals like those formed by the combination of 2s and 2p atomic orbitals that we learnt in class, and there is no s-p mixing.

Show **energy arrow**, **atomic orbitals** involved, **atom labels**, the **molecular orbitals** that are formed, **filled electrons** in the MO diagram etc. Make sure to label all the atomic and molecular orbitals.

Using the MO diagram above, answer the following questions:

- (i) Which one of the following will have the highest bond order? Circle the correct choice
 A) PS^{1-} B) PS C) PS^{1+}
- (ii) Which one of the following will have the longest bond length? Circle the correct choice
 A) PS^{1-} B) PS C) PS^{1+}
- (iii) Which of the following will be diamagnetic? Circle the correct choice.
 A) PS^{1-} B) PS C) PS^{1+}
- (iv) Draw Lewis dot structure of the molecule with the shortest bond length (based on the MO diagram) from among the three (PS^{1-} , PS, PS^{1+}), and indicate the bond order calculated from the Lewis dot structure. **(3 points)**

Lewis Dot Structure

Bond Order



496AAC81-3A9C-41DE-AB38-38308F7E0D86

midterm-exam-93388

#899 Page 10 of 10

SCRAP PAGE

**CHEMISTRY 110 –Midterm Exam****Thursday November 7, 2024, 6:30 – 8:30 pm**

Examiners: Pallavi Sirjoosingh and Ashok Kakkar

First name (please write as legibly as possible within the boxes)

--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--

Last name

--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--

Student ID number

--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--

This booklet has 8 questions on 10 pages (The last page can be used as a scrap page).

The staple must NOT be removed from the question booklet. Your intact question booklet must be returned at the end of the exam.

A separate booklet contains a periodic table, and datasheet.

Write your answers directly under the questions in the space provided. Do not write on the QR codes or near the top edges (as it may be cut off).

Please write **legibly and dark enough** for the scan to be readable. For students who use English as a second language, dictionaries are permitted. Calculators are allowed.

DO NOT ASK QUESTIONS OF THE INVIGILATORS about interpretations of the exam questions. If you have a concern about a question, please summarize it at the top of this page. All concerns will be evaluated after the exam. GOOD LUCK!

**Question 1 8 points** (*Assume all transitions are possible*)

a. 6 points Calculate the **energy (in kJ/mol)** required to ionize (completely remove) the electron from the **$n=3$** level of **He^+ ion** (He: Helium). Show your work including equations. (3 sig. figs.)

b. 2 points

(I) Select the largest (longest) wavelength from the three choices below:

- i. Wavelength of photon to *completely remove* an electron from *1s orbital of Hydrogen atom*
- ii. Wavelength of photon to *completely remove* an electron from *1s orbital of Boron atom (B)*
- iii. Wavelength of photon to *completely remove* an electron from *1s orbital of Boron cation (B^+)*

i or ii or iii

(II) Select the smallest (shortest) wavelength from the three choices below (*for H atom*):

- i. Wavelength of photon *emitted* for an electron transitioning from *2p to 1s orbital*
- ii. Wavelength of photon *absorbed* for an electron transitioning from *2s to 4p orbital*
- iii. Wavelength of photon *absorbed* for an electron transitioning from *1s to 3p orbital*

i or ii or iii



Question 2 8 points



A Chem110 student plots the graph shown on the left, from data obtained for the kinetic energy (eV) of the ejected electron versus frequency (s^{-1}) of the incident light, for an unknown metal. ($1\text{eV} = 1.6022 \times 10^{-19} \text{ J}$)

Based on the plot, answer the questions **a to c** below:

a. 1 point On the plot above, label the **threshold frequency**.

b. 2 points What is the value of the work potential (in J) of the unknown metal?

c. 3 points What is the *longest* wavelength (in nm) required to eject an electron from this metal's surface? Show your work including equation(s) used. (3 sig figs)

d. 2 points In the space provided below, plot the following

(Assume frequency is *above* threshold frequency):

- i. *Kinetic energy* of an ejected electron versus the *intensity* of the incident light (left)
- ii. *Number of electrons* ejected versus the *frequency* of the incident light (right)



**Question 3** **6 points****a. 2 points**

In the ground state of Ge (Germanium; $Z = 32$) atom, **name** the following orbitals:

i. Orbital containing the electron that requires the maximum energy to be removed.

ii. Orbital containing the electron that experiences the minimum effective nuclear charge.

b. 2 points

An unknown anion (A^-) has the following *excited* electronic configuration: $[Ar] 4s^2 3d^3 4p^1$

(Ar: Argon; $Z = 18$)

Give the **condensed ground state electronic configuration** of the atom, **A** and the cation, **A^{2+}** .
Show your work. (Assume all electronic transitions are possible)

Electronic Configuration of A:

Electronic Configuration of A^{2+} :

c. 2 points

In a single atom, determine how many electrons can have the following set of quantum numbers?

(i) $n = 5; l = 3$

In a single atom, determine how many electrons can have the following set of quantum numbers?

(i) $n = 3; l = 2; m_s = +1/2$

**Question 4** **8 points**

Rank the following from 1 to 4 in **increasing** order. Provide the ranking in the boxes provided.

Note: Atomic number (Z) are provided for relevant atoms

a) Increasing (lowest to highest) order of *ionization energy*: (1= lowest; 4 = highest)

Mg

Mg: Z = 12

Rb

Rb: Z = 37

Cs

Cs: Z = 55

K

K : Z = 19

b) Increasing (lowest to highest) order of *bond angle*: (1= lowest; 4 = highest)

H₂OCCl₄XeF₄

HCN

c) Increasing (smallest to largest) order of *ionic size* (1= smallest; 4 = largest)

S²⁻

S: Z = 16

O²⁻

O: Z = 8

Be²⁺

Be: Z = 4

Na⁺

Na: Z = 11

d) Increasing (lowest to highest) order of *bond energy* (1= lowest; 4 = highest)

P-F

P-Cl

P-I

P-Br

**Question 5** **12 points**

a. 9 points For the molecules HClO_4 and TeF_6 , draw the most stable *Lewis structure* showing appropriate shape and *dashed/wedged bonds*, based on VSEPR. Determine the **molecular geometry** around the central atom. For HClO_4 , label the *most polar bond*.

HClO_4 (there are no O-O bonds)

Label the most polar bond

Molecular Geometry around Cl

TeF_6

Molecular Geometry around Te

b. 3 points (Note: Lone pairs are not shown in structures below)

(i) For the three *resonance* structures (1, 2 and 3) shown below, answer the following question:



(I) Circle the *most contributing* resonance structure.

Structure 1

Structure 2

Structure 3

(ii) For the three structures given (4, 5, and 6) below, answer the following questions:



(I) Circle all the structures that represent *resonance structures* of each other:

Structure 4

Structure 5

Structure 6

(II) Among the resonance structures, circle the *most contributing* resonance structure.

Structure 4

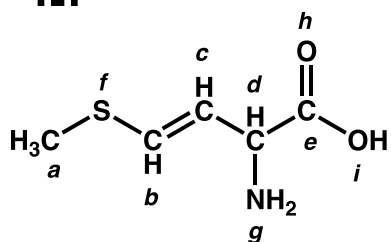
Structure 5

Structure 6

**Question 6** **6 points**

Using the data given below, calculate the **lattice energy** for SrO(s).

	ΔH° (kJ/mol)
$\text{Sr}(s) \longrightarrow \text{Sr}(g)$	+180
$\text{Sr}(g) \longrightarrow \text{Sr}^{2+}(g) + 2e^-$	+1468
$\text{O}_2(g) \longrightarrow 2\text{O}(g)$	+500
$\text{O}(g) + 2e^- \longrightarrow \text{O}^{2-}(g)$	+600
$\text{Sr}(s) + \frac{1}{2}\text{O}_2(g) \longrightarrow \text{SrO}(s)$	-550

**Question 7 10 points**

Consider the molecule given on the left, and answer questions (i) to (vi) using chemical bonding theories that we have learnt in class (valence bond theory, hybridization).

Note that the lone pairs are not shown in the structure:

i. Consider the carbon atoms marked in as ***a***, ***b***, ***c***, ***d***, and ***e*** in the molecule, and determine **hybridization** at each of the carbon centers. (2.5 points)

Atom	Hybridization
Carbon (<i>a</i>)	
Carbon (<i>b</i>)	
Carbon (<i>c</i>)	
Carbon (<i>d</i>)	
Carbon (<i>e</i>)	

ii. Now, consider the sulfur atom (S) marked as ***f***, nitrogen (N) atom (***g***), and oxygen(O) atoms (***h*** and ***i***), and determine their **hybridization**. (2 points)

Atom	Hybridization
Sulfur (<i>f</i>)	
Nitrogen (<i>g</i>)	
Oxygen (<i>h</i>)	
Oxygen (<i>i</i>)	

iii. **Describe** all the bonds formed by the carbon atoms ***a*** and ***b***, using sigma and pi designation as appropriate. (2 points)

Atom	Bonds formed (use sigma and pi as appropriate)
Carbon (<i>a</i>)	
Carbon (<i>b</i>)	

iv. How many total sp^3 -s sigma bonds are in the molecule? (1 point)

v. Determine **electronic** and **molecular geometry** at carbon (C) labeled ***e*** and sulfur (S) labeled ***f*** centres. (2 points)

Atom	Electronic Geometry	Molecular Geometry
Carbon (<i>e</i>)		
Sulfur (<i>f</i>)		

vi. Circle the value from the choices given below, that describes the **C-S-C bond angle at sulfur** (S) marked ***f***: (0.5 points)

>90° to <109.5°

109.5°

>109.5 to <120°

120°

>120° to <180°


Question 8 12 points

Using the **valence shell orbitals only** draw a **complete** molecular orbital (MO) diagram (with no s-p mixing) for the molecule **PS**, in the space provided below. **(6 points)**

Assume that 3s and 3p atomic orbitals will form molecular orbitals like those formed by the combination of 2s and 2p atomic orbitals that we learnt in class, and there is no s-p mixing.

Show **energy arrow**, **atomic orbitals** involved, **atom labels**, the **molecular orbitals** that are formed, **filled electrons** in the MO diagram etc. Make sure to label all the atomic and molecular orbitals.

Using the MO diagram above, answer the following questions:

- (i) Which one of the following will have the highest bond order? Circle the correct choice
 A) PS^{1-} B) PS C) PS^{1+}
- (ii) Which one of the following will have the longest bond length? Circle the correct choice
 A) PS^{1-} B) PS C) PS^{1+}
- (iii) Which of the following will be diamagnetic? Circle the correct choice.
 A) PS^{1-} B) PS C) PS^{1+}
- (iv) Draw Lewis dot structure of the molecule with the shortest bond length (based on the MO diagram) from among the three (PS^{1-} , PS, PS^{1+}), and indicate the bond order calculated from the Lewis dot structure. **(3 points)**

Lewis Dot Structure

Bond Order



DAC109AD-305A-4C27-BCE3-F4DA5DA7D5A1

midterm-exam-93388

#900 Page 10 of 10

SCRAP PAGE